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# Preface

Linear Algebra is often the first place where a student encounters the true power of mathematical abstraction. In an introductory course, one learns to manipulate matrices, solve systems of linear equations, and perhaps calculate eigenvalues. But *Advanced* Linear Algebra is about something deeper: it is the study of structures.

## The Philosophy of this Book

The central theme of these notes is the transition from concrete calculation to abstract reasoning. We begin not with numbers, but with axioms. By choosing to define mathematical objects—be they vectors, linear transformations, or inner products—entirely by how they behave rather than what they are, we unlock a universal language.

A theorem proved for an abstract vector space is a theorem proved simultaneously for geometric arrows, for polynomials, for continuous functions, and for quantum states. This “solve once, apply everywhere” philosophy is the heartbeat of modern mathematics.

## How to Use These Notes

This book is designed to be a rigorous companion for a second course in linear algebra. Each chapter is structured to build a solid theoretical foundation:

- **Rigorous Definitions:** We start with precise mathematical statements to ensure no ambiguity in our logic.
- **Complete Proofs:** Every major result is accompanied by a full proof. Mathematics is not a spectator sport; I encourage you to attempt the proofs yourself before reading the solutions.
- **Illustrative Examples:** Abstract theory is balanced with concrete examples to build intuition.
- **Exercises:** Practice problems are provided at the end of sections to test your understanding.

## Prerequisites

To get the most out of these notes, you should have a solid grasp of:

- **Introductory Linear Algebra:** You should be comfortable with matrix multiplication, determinants, and Gaussian elimination.

- **Calculus:** Familiarity with derivatives and integrals is assumed for some of our more advanced examples.
  - **Mathematical Maturity:** A basic understanding of logic and proof techniques (induction, contradiction, contrapositive) is essential. These are reviewed briefly in Chapter 0.
- 

Welcome to the world of abstract linear algebra. I hope you find as much beauty in these structures as I do.

# Chapter 0

## Preliminaries

Most people meet linear algebra as a way of computing: row reduction, determinants, maybe eigenvalues. This book treats it as a subject you **prove**. Every statement comes with a reason, and those reasons are built from a small set of ideas: sets, functions, equivalence relations, fields, polynomials and matrices. This chapter collects that toolkit in one place.

You have probably seen some of this before. Some of it, such as negating a statement with several quantifiers, or checking that a rule on equivalence classes is well defined, is exactly where careful readers later stumble. So each section spends its time on the delicate points and moves quickly past the rest.

**How to use this chapter.** Skim the sections you know, but try the *A. Check your understanding* exercises at the end of each one. If you can do them without looking back, move on. If not, that section is worth a slower read.

**What you need.** Nothing from this book. School algebra is enough, plus some familiarity with numbers such as  $\mathbb{Z}$ ,  $\mathbb{Q}$  and  $\mathbb{R}$ . One convention to note now: in this book the natural numbers  $\mathbb{N}$  start at 0.

### 0.0.1 Roadmap

- **Statements, implications and quantifiers.** How mathematical statements are built, and how to negate them correctly.
- **How proofs are built.** Direct proof, contrapositive, contradiction, cases, uniqueness, counterexamples and induction, and the conventions this book's proofs follow.
- **Sets.** Subsets, operations, indexed families, and the habit of proving two sets equal by double inclusion.
- **Functions.** Injective, surjective and bijective maps; composition; left, right and two-sided inverses.
- **Relations, equivalence classes and quotients.** Sorting a set into classes, and defining functions on classes without ambiguity.
- **Complex numbers.** Arithmetic, conjugates, the modulus, polar form and roots of unity.
- **Fields.** The number systems in which linear algebra works, from  $\mathbb{Q}$ ,  $\mathbb{R}$  and  $\mathbb{C}$  to the finite fields  $\mathbb{F}_p$ .
- **Polynomials.** Degree, evaluation, and why a polynomial is not the same thing as a function.

- **Matrices.** Addition, the three ways to read a matrix product, transpose, trace and inverses.
- **Groups and permutations.** One structure shared by invertible matrices, the non-zero elements of a field, and rearrangements of a set.

## 0.0.2 Named moves

These proof techniques appear throughout the book. Each is introduced in this chapter.

- **Contrapositive.** To prove “if  $P$  then  $Q$ ”, prove “if not  $Q$  then not  $P$ ”.
- **Assume two.** To prove something is unique, suppose two objects have the property and show they are equal.
- **Double inclusion.** To prove  $A = B$ , prove  $A \subseteq B$  and  $B \subseteq A$ .
- **Suppose  $f(a) = f(b)$ .** The standard first line when proving that a function is injective.
- **Check well-definedness.** A rule given through a representative must not depend on which representative is chosen.
- **Induction on  $n$ .** Find a smaller instance of the problem inside the larger one.

# 0.1 Statements, Implications and Quantifiers

Linear algebra is a subject of proofs. Every theorem in this book has the shape “if these hypotheses hold, then this conclusion holds”, and most definitions hide a “for every” or a “there exists”. Before we can prove anything, we need to read such sentences exactly: when they are true, what they claim, and what their opposite says. This section builds that reading skill. It is the grammar that the rest of the book is written in.

## 0.1.1 Statements and connectives

A **statement** is a sentence that is either true or false, but not both. “7 is prime” is a statement, and it is true. “ $2 + 2 = 5$ ” is a statement, and it is false. “Is 7 prime?” is not a statement, and neither is “ $x > 3$ ” on its own: its truth depends on  $x$ , which nobody has chosen yet. Sentences with a free variable are handled by quantifiers later in this section.

We build longer statements from shorter ones with a few fixed words. Everyday language uses these words loosely, so mathematics pins each one down by saying exactly when the combined statement is true.

*A connective builds a new statement whose truth depends only on the truth of its parts.*

### Definition 0.1.1 — Negation, conjunction, disjunction

Let  $P$  and  $Q$  be statements.

- The **negation**  $\neg P$  (“not  $P$ ”) is true when  $P$  is false, and false when  $P$  is true.
- The **conjunction**  $P \wedge Q$  (“ $P$  and  $Q$ ”) is true when **both**  $P$  and  $Q$  are true, and false otherwise.
- The **disjunction**  $P \vee Q$  (“ $P$  or  $Q$ ”) is true when **at least one** of  $P$ ,  $Q$  is true,

and false only when both are false.

A **truth table** lists every combination of truth values of the parts, one per row, and records the value of the compound statement. With T for true and F for false:

| $P$ | $Q$ | $\neg P$ | $P \wedge Q$ | $P \vee Q$ |
|-----|-----|----------|--------------|------------|
| T   | T   | F        | T            | T          |
| T   | F   | F        | F            | T          |
| F   | T   | T        | F            | T          |
| F   | F   | T        | F            | F          |

Two statement forms are **logically equivalent**, written  $\equiv$ , if they have the same truth value in every row of the truth table. For example,  $\neg(\neg P) \equiv P$ : negating twice returns the original value in both rows.

A few examples, checked against the definition:

- “7 is prime **and** 7 is even” is false, because its second part is false.
- “7 is prime **or** 7 is even” is true, because its first part is true.
- “ $2 < 3$  **or**  $3 < 4$ ” is true. Both parts are true, and that is allowed.

*Warning.* In mathematics “or” is **inclusive**:  $P \vee Q$  is true when both parts are true. The menu phrase “soup or salad”, which forbids having both, is a different connective. So “6 is even or 6 is divisible by 3” is true, even though both halves hold.

## 0.1.2 Implication

Almost every theorem is an implication: “if  $P$ , then  $Q$ ”. It is worth being completely precise about when such a sentence counts as true, because the answer surprises many readers in one row of the table.

Think of an implication as a **promise**. A parent says: “If you finish your homework, then you may watch a film.” When has the parent broken the promise? Only when the homework is finished and the film is still refused. If the homework is not finished, the promise says nothing about what happens, so whatever happens, the promise was not broken.

*An implication is false only when the hypothesis holds and the conclusion fails.*

### Definition 0.1.2 — Implication, converse, contrapositive

Let  $P$  and  $Q$  be statements.

- The **implication**  $P \Rightarrow Q$  (“if  $P$  then  $Q$ ”) is false when  $P$  is true and  $Q$  is false, and it is true in **every** other case. We call  $P$  the **hypothesis** and  $Q$  the **conclusion**.
- The **converse** of  $P \Rightarrow Q$  is the implication  $Q \Rightarrow P$ .
- The **contrapositive** of  $P \Rightarrow Q$  is the implication  $\neg Q \Rightarrow \neg P$ .

In words: the first clause says an implication can only fail in one way, by a true hypothesis leading to a false conclusion. The converse swaps the roles of hypothesis and conclusion. The

contrapositive swaps them **and** negates both.

| $P$ | $Q$ | $P \Rightarrow Q$ |
|-----|-----|-------------------|
| T   | T   | T                 |
| T   | F   | F                 |
| F   | T   | T                 |
| F   | F   | T                 |

The implication  $P \Rightarrow Q$  is read in many ways, and all of them mean the same thing:

- “ $P$  implies  $Q$ ”, “ $Q$  whenever  $P$ ”, “ $Q$  if  $P$ ”;
- “ $P$  only if  $Q$ ”;
- “ $P$  is **sufficient** for  $Q$ ”, “ $Q$  is **necessary** for  $P$ ”.

The phrase “only if” trips people up. “ $P$  only if  $Q$ ” says that  $P$  cannot happen without  $Q$ , which is exactly  $P \Rightarrow Q$ .

The last two rows of the table, where the hypothesis is false, are called **vacuous** cases. An implication with a false hypothesis is **vacuously true**.

**Example. (A vacuously true implication)**

Let  $n$  be an integer. Is the statement “if  $n$  is odd and  $n$  is even, then  $n = 7$ ” true?

*Solution.* Yes. No integer is both odd and even, so the hypothesis “ $n$  is odd and  $n$  is even” is false for every integer  $n$ . By [Definition 0.1.2](#), an implication with a false hypothesis is true, whatever the conclusion says. So the statement is true for every integer  $n$ , including  $n = 7$  and  $n = 0$ . It is true, but it tells us nothing.

**Why this definition.** The two vacuous rows may look like a convention picked for convenience. They are forced once we want implications to work inside “for every” statements. Consider the true fact

for every integer  $n$ , if  $n > 2$  then  $n^2 > 4$ .

For this to be true, the implication must be true for **every** integer  $n$ . Try  $n = -3$ : the hypothesis  $-3 > 2$  is false, and the conclusion  $9 > 4$  is true. Try  $n = 1$ : the hypothesis  $1 > 2$  is false, and the conclusion  $1 > 4$  is false. If either row were declared false, this perfectly good theorem would become false. So both vacuous rows must be T. In the same way, a theorem with hypotheses says nothing about objects that fail them, and that is exactly how we want to read it.

*Remark.* The implication  $P \Rightarrow Q$  does **not** claim that  $P$  causes  $Q$ , or that the two are related at all. “If  $2 + 2 = 4$ , then 7 is prime” is a true implication, because both parts are true. In practice we only ever prove implications where the hypothesis is used, but the truth value does not depend on that.

The contrapositive is the most useful rewriting of an implication, because it is always interchangeable with the original.

**Theorem 0.1.1 — Contrapositive equivalence**

For all statements  $P$  and  $Q$ , the implication  $P \Rightarrow Q$  is logically equivalent to its contrapositive  $\neg Q \Rightarrow \neg P$ .

*Proof.* Let  $P$  and  $Q$  be statements. We compute both sides in each of the four rows, using Definition 0.1.1 for the negations and Definition 0.1.2 for the implications.

| $P$ | $Q$ | $\neg Q$ | $\neg P$ | $P \Rightarrow Q$ | $\neg Q \Rightarrow \neg P$ |
|-----|-----|----------|----------|-------------------|-----------------------------|
| T   | T   | F        | F        | T                 | T                           |
| T   | F   | T        | F        | F                 | F                           |
| F   | T   | F        | T        | T                 | T                           |
| F   | F   | T        | T        | T                 | T                           |

In the second row,  $\neg Q$  is true and  $\neg P$  is false, so  $\neg Q \Rightarrow \neg P$  is false. In every other row the hypothesis  $\neg Q$  is false or the conclusion  $\neg P$  is true, so the contrapositive is true. The last two columns agree in every row. This proves that the two statements are logically equivalent. ■

So to prove “if  $P$  then  $Q$ ” we may instead prove “if  $Q$  fails then  $P$  fails”. The next section turns this into a proof technique.

The converse is a different matter. It is a separate statement, and it can be false while the original is true.

**Warning. The converse is not the contrapositive.** “If  $x = 2$ , then  $x^2 = 4$ ” is true for every real number  $x$ . Its converse “if  $x^2 = 4$ , then  $x = 2$ ” is false:  $x = -2$  satisfies the hypothesis and fails the conclusion. Proving  $P \Rightarrow Q$  never proves  $Q \Rightarrow P$ .

Later in the book the converse question comes up constantly. In Chapter 2 we will see that if a square matrix  $\mathbf{A}$  is invertible, then  $\mathbf{Ax} = \mathbf{0}$  has only the solution  $\mathbf{x} = \mathbf{0}$ . The converse, “if  $\mathbf{Ax} = \mathbf{0}$  has only the solution  $\mathbf{x} = \mathbf{0}$ , then  $\mathbf{A}$  is invertible”, also turns out to be true for square matrices. But it is a separate theorem, it needs its own proof, and that proof takes real work.

*Quick check.* Consider “if an integer  $n$  is divisible by 6, then  $n$  is even”. Write its converse and its contrapositive, and decide which of the three statements are true for every integer  $n$ .

*Solution.* The converse is “if  $n$  is even, then  $n$  is divisible by 6”. The contrapositive is “if  $n$  is not even, then  $n$  is not divisible by 6”, that is, “if  $n$  is odd, then  $6 \nmid n$ ”.

The original is true: if  $n = 6k$  then  $n = 2(3k)$  is even. The contrapositive is true, by Theorem 0.1.1. The converse is false:  $n = 2$  is even but not divisible by 6.

When an implication and its converse are both true, we say the two statements are equivalent.

The **biconditional**  $P \Leftrightarrow Q$  (“ $P$  if and only if  $Q$ ”, often written “ $P$  iff  $Q$ ”) is the statement  $(P \Rightarrow Q) \wedge (Q \Rightarrow P)$ . It is true exactly when  $P$  and  $Q$  have the same truth value:



| $P$ | $Q$ | $P \Rightarrow Q$ | $Q \Rightarrow P$ | $P \Leftrightarrow Q$ |
|-----|-----|-------------------|-------------------|-----------------------|
| T   | T   | T                 | T                 | T                     |
| T   | F   | F                 | T                 | F                     |
| F   | T   | T                 | F                 | F                     |
| F   | F   | T                 | T                 | T                     |

The phrase splits in two: “ $P$  if  $Q$ ” is  $Q \Rightarrow P$ , and “ $P$  only if  $Q$ ” is  $P \Rightarrow Q$ . Other phrasings are “ $P$  is necessary and sufficient for  $Q$ ” and “ $P$  and  $Q$  are equivalent”. Proving an “iff” therefore always means proving two implications.

### 0.1.3 Quantifiers

The sentence “ $x^2 \geq 0$ ” is not yet a statement, because  $x$  is not specified. A sentence  $P(x)$  that becomes a statement once a value of  $x$  is substituted is called a **predicate** (or open sentence) in the variable  $x$ . There are two standard ways to turn a predicate into a statement: claim it for every value, or claim it for at least one.

“For every” makes a claim about each element; “there exists” asks for a single witness.

#### Definition 0.1.3 — Quantifiers

Let  $P(x)$  be a predicate whose variable  $x$  ranges over a fixed collection  $S$ .

- The **universal** statement  $\forall x \in S : P(x)$  (“for every  $x$  in  $S$ ,  $P(x)$ ”) is true when  $P(x)$  is true for **every** element  $x$  of  $S$ , and false otherwise.
- The **existential** statement  $\exists x \in S : P(x)$  (“there exists  $x$  in  $S$  such that  $P(x)$ ”) is true when  $P(x)$  is true for **at least one** element  $x$  of  $S$ , and false otherwise.

The symbols  $\forall$  and  $\exists$  are the **universal** and **existential quantifiers**.

In words: a universal statement is a promise about every element, so a single element where  $P$  fails makes it false. An existential statement needs one element that works. It does not say which one, and it does not say there is only one.

The collection  $S$  is part of the statement. Changing  $S$  can change the truth value, so always ask: *ranging over what?*

#### Example. (Quantified statements)

Decide whether each statement is true.

- $\forall x \in \mathbb{R} : x^2 \geq 0$ .
- $\exists x \in \mathbb{R} : x^2 = -1$ .
- $\exists n \in \mathbb{Z} : n^2 = 2$ .
- $\forall n \in \mathbb{Z} : n^2 \neq 2$ .

*Solution.*

- (a) True. Every real number  $x$  is either  $\geq 0$  or  $< 0$ , and in both cases  $x^2 = x \cdot x \geq 0$ .
- (b) False. By (a),  $x^2 \geq 0 > -1$  for every real  $x$ , so no real  $x$  is a witness. Over the complex numbers (introduced later in this chapter),  $x = i$  is a witness, so the same predicate becomes true when the range changes from  $\mathbb{R}$  to  $\mathbb{C}$ .
- (c) False. If  $|n| \leq 1$  then  $n^2 \leq 1 < 2$ , and if  $|n| \geq 2$  then  $n^2 \geq 4 > 2$ . So no integer squares to 2.
- (d) True. This is the statement shown in (c), said the other way round: every integer fails to square to 2.

Parts (c) and (d) already show the pattern of the next subsection: “there is no witness” is the same as “everything fails”.

**A degenerate case.** What if the range contains no elements at all? Take “every integer  $n$  with  $n^2 < 0$  satisfies  $n = 7$ ”. There are no such integers, so there is nothing that could fail, and the statement is true. It is the quantifier version of vacuous truth: it reads  $\forall n \in \mathbb{Z} : (n^2 < 0 \Rightarrow n = 7)$ , and every instance has a false hypothesis. By contrast, “there exists an integer  $n$  with  $n^2 < 0$ ” is false, because there is no witness. This case matters later: facts about “every element of an empty list” will be vacuously true, and that is what makes certain edge cases of definitions work.

*Remark.* We write  $\exists! x \in S : P(x)$  for “there exists a **unique**  $x$  in  $S$  with  $P(x)$ ”. It means two things: some  $x$  satisfies  $P$ , and any two elements satisfying  $P$  are equal. The next section shows how each half is proved.

## 0.1.4 The order of quantifiers

Most definitions in this book use several quantifiers in a row. Their order is part of the meaning.

### Example. (Swapping two quantifiers)

Compare the two statements

$$(i) \quad \forall x \in \mathbb{R} \exists y \in \mathbb{R} : y > x, \quad (ii) \quad \exists y \in \mathbb{R} \forall x \in \mathbb{R} : y > x.$$

Decide which is true.

*Solution.* Statement (i) says: whichever real  $x$  you are given, you can find a real  $y$  larger than it. This is true. Given  $x$ , take  $y = x + 1$ , and then  $y > x$ .

Statement (ii) says: there is one real number  $y$  that is larger than every real  $x$ . This is false. Whatever  $y$  is proposed, the choice  $x = y + 1$  gives  $y > x$  false. So no  $y$  is a witness.

The difference is **who chooses first**. In (i),  $x$  is chosen first and  $y$  may depend on it:  $y = x + 1$  changes with  $x$ . In (ii),  $y$  must be fixed before  $x$  is known, so it has to work for all  $x$  at once. Reading quantifiers as a game helps:  $\forall$  is a move by an opponent,  $\exists$  is your reply, and you may only use information already on the table.

*Warning.* Never swap a  $\forall$  and an  $\exists$  without a proof. The implication “ $\exists y \forall x \Rightarrow \forall x \exists y$ ” is always valid, since the single  $y$  works for each  $x$ . The reverse direction fails, as [Example \(Swapping two quantifiers\)](#) shows. Two quantifiers of the **same** kind, such as  $\forall x \forall y$ , may be swapped freely.

*Quick check.* Over the positive real numbers, statement (A) is “ $\forall \varepsilon > 0 \exists \delta > 0 : \delta < \varepsilon$ ”. Statement (B) is obtained by swapping the quantifiers: “ $\exists \delta > 0 \forall \varepsilon > 0 : \delta < \varepsilon$ ”. Are (A) and (B) both true?

*Solution.* No. (A) is true: given  $\varepsilon > 0$ , take  $\delta = \varepsilon/2$ , which is positive and smaller than  $\varepsilon$ . (B) is false: whatever  $\delta > 0$  is proposed, the choice  $\varepsilon = \delta$  makes  $\delta < \varepsilon$  false. In (A) the number  $\delta$  may depend on  $\varepsilon$ ; in (B) it may not.

## 0.1.5 Negation

Much of mathematics consists of showing that something **fails**: a function is not injective, a list is not independent, a claimed theorem is false. To do this we need the negation of the statement written in a usable form. Putting “it is not the case that” in front is correct but useless; we want the negation with the “not” pushed all the way inside, next to a simple condition we can check.

First, the connectives.

### Theorem 0.1.2 — De Morgan’s laws for statements

For all statements  $P$  and  $Q$ :

- (a)  $\neg(P \wedge Q) \equiv \neg P \vee \neg Q$ ;
- (b)  $\neg(P \vee Q) \equiv \neg P \wedge \neg Q$ ;
- (c)  $\neg(P \Rightarrow Q) \equiv P \wedge \neg Q$ .

*Proof.* Let  $P$  and  $Q$  be statements. We compare truth tables, using [Definition 0.1.1](#) and [Definition 0.1.2](#).

| $P$ | $Q$ | $\neg(P \wedge Q)$ | $\neg P \vee \neg Q$ | $\neg(P \vee Q)$ | $\neg P \wedge \neg Q$ | $\neg(P \Rightarrow Q)$ | $P \wedge \neg Q$ |
|-----|-----|--------------------|----------------------|------------------|------------------------|-------------------------|-------------------|
| T   | T   | F                  | F                    | F                | F                      | F                       | F                 |
| T   | F   | T                  | T                    | F                | F                      | T                       | T                 |
| F   | T   | T                  | T                    | F                | F                      | F                       | F                 |
| F   | F   | T                  | T                    | T                | T                      | F                       | F                 |

For (a),  $P \wedge Q$  is true only in the first row, so its negation is false only there. The statement  $\neg P \vee \neg Q$  is false only when  $\neg P$  and  $\neg Q$  are both false, which is again only the first row. For (b),  $\neg(P \vee Q)$  is true only in the last row, where both parts are false, and so is  $\neg P \wedge \neg Q$ . For (c),  $P \Rightarrow Q$  is false only in the second row, so  $\neg(P \Rightarrow Q)$  is true only there, and  $P \wedge \neg Q$  is also true only there. In each case the two columns agree in all four rows, as claimed. ■

Law (c) deserves a sentence. The negation of an implication is **not** another implication. To refute “if  $P$  then  $Q$ ”, you must exhibit a situation where  $P$  holds **and**  $Q$  fails.

Now the quantifiers. A universal statement fails exactly when some element fails it, and an existential statement fails exactly when every element fails it. We record this as the rule we will use every time.

*To negate, flip each quantifier and negate what is inside.*

### Definition 0.1.4 — Negation of quantifiers

Let  $P(x)$  be a predicate with  $x$  ranging over  $S$ . Then

$$\neg(\forall x \in S : P(x)) \equiv \exists x \in S : \neg P(x), \quad \neg(\exists x \in S : P(x)) \equiv \forall x \in S : \neg P(x).$$

For a statement with several quantifiers, apply these rules from the outside in: each  $\forall$  becomes  $\exists$ , each  $\exists$  becomes  $\forall$ , the ranges stay the same, and the innermost condition is negated using [Theorem 0.1.2](#).

In words: “not every  $x$  works” means “some  $x$  fails”, and “no  $x$  works” means “every  $x$  fails”. Both rules are just [Definition 0.1.3](#) read carefully. The universal statement is false precisely when it is not the case that every element satisfies  $P$ , which is precisely when at least one element satisfies  $\neg P$ . The **ranges do not change**: the negation of “every real number squared is non-negative” talks about real numbers, not about some other collection.

### Example. (Negating simple quantified statements)

Negate each statement, and say which of the statement and its negation is true.

- (a)  $\forall x \in \mathbb{R} : x^2 \geq 0$ .
- (b)  $\exists n \in \mathbb{Z} : n$  is even and  $n > 10$ .
- (c)  $\forall x \in \mathbb{R} \exists y \in \mathbb{R} : y > x$ .

*Solution.*

- (a) The negation is  $\exists x \in \mathbb{R} : x^2 < 0$ . The original is true, by [Example \(Quantified statements\)](#), so the negation is false.
- (b) By [Definition 0.1.4](#) and law (a) of [Theorem 0.1.2](#), the negation is  $\forall n \in \mathbb{Z} : n$  is odd or  $n \leq 10$ . The original is true, with witness  $n = 12$ , so the negation

is false:  $n = 12$  is neither odd nor  $\leq 10$ .

- (c) Flip both quantifiers and negate the inside:  $\exists x \in \mathbb{R} \forall y \in \mathbb{R} : y \leq x$ . This says that some real number is at least as large as every real number. The original is true by [Example \(Swapping two quantifiers\)](#), so the negation is false.

The next two examples negate definitions that will appear over and over. We use the word **function** informally here, as a rule  $f$  that assigns to each  $a$  in a collection  $A$  one value  $f(a)$  in a collection  $B$ ; functions are treated properly later in this chapter.

**Example. (What “not injective” means)**

A function  $f$  from  $A$  to  $B$  is called **injective** if

$$\forall a_1 \in A \forall a_2 \in A : (f(a_1) = f(a_2) \Rightarrow a_1 = a_2).$$

Write down, with no “not” in front, what it means for  $f$  to be **not** injective. Then show that  $f(x) = x^2$ , from  $\mathbb{R}$  to  $\mathbb{R}$ , is not injective.

*Solution.* By [Definition 0.1.4](#), both  $\forall$  become  $\exists$ , and the inside implication is negated. By law (c) of [Theorem 0.1.2](#),  $\neg(f(a_1) = f(a_2) \Rightarrow a_1 = a_2) \equiv f(a_1) = f(a_2) \wedge a_1 \neq a_2$ . So  $f$  is not injective if and only if

$$\exists a_1 \in A \exists a_2 \in A : f(a_1) = f(a_2) \text{ and } a_1 \neq a_2.$$

In words, two **different** inputs have the same output. For  $f(x) = x^2$ , take  $a_1 = 1$  and  $a_2 = -1$ . Then  $f(1) = 1 = f(-1)$  and  $1 \neq -1$ , so  $f$  is not injective.

**Example. (What “not surjective” means)**

A function  $f$  from  $A$  to  $B$  is called **surjective** if

$$\forall b \in B \exists a \in A : f(a) = b.$$

Negate this, and show that  $f(x) = x^2$ , from  $\mathbb{R}$  to  $\mathbb{R}$ , is not surjective.

*Solution.* Flipping the two quantifiers and negating the equation gives:  $f$  is not surjective if and only if

$$\exists b \in B \forall a \in A : f(a) \neq b.$$

In words, **one** value in  $B$  is never reached. For  $f(x) = x^2$ , take  $b = -1$ . For every real  $a$  we have  $f(a) = a^2 \geq 0 > -1$  by [Example \(Quantified statements\)](#), so  $f(a) \neq -1$ . Hence  $f$  is not surjective.

The most important negation in the early chapters is that of a statement ending in “all zero”. We meet it here without any vector spaces, using pairs of real numbers added and scaled entry

by entry.

**Example. (Negating “only the zero combination”)**

Consider the pairs  $(1, 2)$ ,  $(2, 4)$ ,  $(0, 1)$ , and the statement

$$\forall a, b, c \in \mathbb{R} : (a(1, 2) + b(2, 4) + c(0, 1) = (0, 0) \Rightarrow a = 0 \wedge b = 0 \wedge c = 0).$$

Negate it, and decide which of the statement and its negation is true.

*Solution.* Flip the quantifier, and use law (c) of [Theorem 0.1.2](#) for the implication and law (a) for the conjunction. The negation is

$$\exists a, b, c \in \mathbb{R} : a(1, 2) + b(2, 4) + c(0, 1) = (0, 0) \text{ and } (a \neq 0 \vee b \neq 0 \vee c \neq 0).$$

The last condition says that  $a, b, c$  are **not all zero**: at least one of them is non-zero.

The negation is true. Take  $a = 2$ ,  $b = -1$ ,  $c = 0$ . Then

$$2(1, 2) - (2, 4) + 0(0, 1) = (2 - 2 + 0, 4 - 4 + 0) = (0, 0),$$

and  $a = 2 \neq 0$ . So the original statement is false.

In Chapter 1 the original statement is what “the list is linearly independent” means, and its negation is “linearly dependent”. The small words in the negation carry the weight.

*Warning.* **“Not all zero” is not “all non-zero”.** The negation of “ $a = b = c = 0$ ” is “at least one of  $a, b, c$  is non-zero”. In [Example \(Negating “only the zero combination”\)](#) the witness is  $(2, -1, 0)$ , which has a zero entry. In fact **no** witness has all three entries non-zero: comparing entries,  $a + 2b = 0$  and  $2a + 4b + c = 0$ , so  $c = -2(a + 2b) = 0$ . Negating with “all non-zero” would wrongly conclude that the original statement is true.

*Quick check.* Negate “every student passed some exam”, with no “not” in front of the whole sentence.

*Solution.* Write it as “ $\forall$  students  $s \exists$  exam  $e : s$  passed  $e$ ”. Flipping both quantifiers and negating the inside gives “ $\exists$  student  $s \forall$  exams  $e : s$  did not pass  $e$ ”. In words: **some student passed no exam**. The tempting answer “no student passed any exam” is the negation of “some student passed some exam”, a different statement.

*Remark.* Each form of statement suggests how to prove or refute it. To prove  $\forall x \in S : P(x)$ , take an arbitrary  $x \in S$  and show  $P(x)$ . To prove  $\exists x \in S : P(x)$ , produce a witness. To refute a  $\forall$ , produce one counterexample. To refute an  $\exists$ , show every candidate fails. The next section develops these moves into proof techniques.

## 0.1.6 Exercises

### 0.1.6.1 A. Check your understanding

#### Exercise A1.

- (a) Decide whether each statement is true: (i) “ $3 < 2$  or 3 is odd”; (ii) “if  $3 < 2$ , then  $3 = 5$ ”; (iii) “if 3 is odd, then  $3 < 2$ ”.
- (b) Write the converse and the contrapositive of “if an integer  $n$  is divisible by 4, then  $n$  is even”. Which of the three statements are true for every integer  $n$ ?
- (c) Negate  $\forall x \in \mathbb{R} \exists n \in \mathbb{Z} : n > x$ , with no  $\neg$  in front. Which of the statement and its negation is true?
- (d) True or false:  $\neg(P \wedge Q) \equiv \neg P \wedge \neg Q$  for all statements  $P, Q$ . Give a reason.

*Solution.*

- (a) (i) True: the second part is true, and a disjunction needs only one true part. (ii) True: the hypothesis  $3 < 2$  is false, so the implication is vacuously true. (iii) False: the hypothesis is true and the conclusion is false, which is the one case where an implication fails.
- (b) The converse is “if  $n$  is even, then  $n$  is divisible by 4”. The contrapositive is “if  $n$  is odd, then  $n$  is not divisible by 4”. The original is true:  $n = 4k$  gives  $n = 2(2k)$ . The contrapositive is true by [Theorem 0.1.1](#). The converse is false:  $n = 2$  is even and not divisible by 4.
- (c) By [Definition 0.1.4](#), the negation is  $\exists x \in \mathbb{R} \forall n \in \mathbb{Z} : n \leq x$ . The original is true: given  $x$ , some integer exceeds  $x$  (for instance the integer part of  $|x|$  plus 1). So the negation is false.
- (d) False. Take  $P$  true and  $Q$  false. Then  $P \wedge Q$  is false, so  $\neg(P \wedge Q)$  is true, while  $\neg P \wedge \neg Q$  is false because  $\neg P$  is false. The correct law is  $\neg(P \wedge Q) \equiv \neg P \vee \neg Q$ , by [Theorem 0.1.2](#).

### 0.1.6.2 B. Practice

#### Exercise B1: Negating definitions.

Write the negation of each definition as a quantified statement with no  $\neg$  in front, and then say it in words.

- (a) A function  $f$  from  $\mathbb{R}$  to  $\mathbb{R}$  is **bounded** if  $\exists M \in \mathbb{R} \forall x \in \mathbb{R} : |f(x)| \leq M$ .
- (b) A function  $f$  from  $A$  to  $B$  is injective if  $\forall a_1, a_2 \in A : (a_1 \neq a_2 \Rightarrow f(a_1) \neq f(a_2))$ . Compare your answer with [Example \(What “not injective” means\)](#).
- (c) “Every pair of reals is a combination of  $(1, 0)$  and  $(1, 1)$ ”:  $\forall p, q \in \mathbb{R} \exists a, b \in \mathbb{R} : (p, q) = a(1, 0) + b(1, 1)$ , where pairs are added and scaled entry by entry.

*Solution.*

- (a) By [Definition 0.1.4](#),  $f$  is not bounded if and only if  $\forall M \in \mathbb{R} \exists x \in \mathbb{R} : |f(x)| > M$ . In words: whatever bound  $M$  is proposed, some value of  $f$  exceeds it in absolute value.

- (b) Flip both quantifiers and negate the implication using law (c) of [Theorem 0.1.2](#):  $\exists a_1, a_2 \in A : a_1 \neq a_2 \wedge f(a_1) = f(a_2)$ . In words: two different inputs share an output. This is the same condition as in [Example \(What “not injective” means\)](#). That is expected: the given implication is the contrapositive of  $f(a_1) = f(a_2) \Rightarrow a_1 = a_2$ , so by [Theorem 0.1.1](#) the two definitions agree, and so do their negations.
- (c) Flip all the quantifiers and negate the equation:  $\exists p, q \in \mathbb{R} \forall a, b \in \mathbb{R} : (p, q) \neq a(1, 0) + b(1, 1)$ . In words: some pair is not a combination of  $(1, 0)$  and  $(1, 1)$ , whatever coefficients are used. (The original is in fact true, since  $(p - q)(1, 0) + q(1, 1) = (p, q)$ , so this negation is false.)

### Exercise B2: Integers versus reals.

For each statement, decide whether it is true when  $x$  and  $y$  range over  $\mathbb{Z}$ , and whether it is true when they range over  $\mathbb{R}$ . Justify your answer.

- (a)  $\forall x \exists y : x < y < x + 1$ .
- (b)  $\forall x (x \neq 0 \Rightarrow \exists y : xy = 1)$ .
- (c)  $\exists x \forall y : xy = y$ .
- (d)  $\forall x \exists y : y^2 = x$ .

*Solution.*

- (a) Over  $\mathbb{Z}$ : false. Take  $x = 0$ . No integer  $y$  satisfies  $0 < y < 1$ . Over  $\mathbb{R}$ : true. Given  $x$ , take  $y = x + \frac{1}{2}$ ; then  $x < y < x + 1$ .
- (b) Over  $\mathbb{Z}$ : false. Take  $x = 2$ , which is non-zero. If  $2y = 1$  for an integer  $y$ , then 1 would be even, which is false. Over  $\mathbb{R}$ : true. Given  $x \neq 0$ , take  $y = 1/x$ , which exists because  $x \neq 0$ , and then  $xy = 1$ .
- (c) True over both. Take  $x = 1$ , which lies in  $\mathbb{Z}$  and in  $\mathbb{R}$ . Then  $xy = y$  for every  $y$ . Here one  $x$  works for all  $y$ , so the order  $\exists x \forall y$  causes no difficulty.
- (d) False over both. Take  $x = -1$ . For every  $y$ , integer or real,  $y^2 \geq 0 > -1$  by [Example \(Quantified statements\)](#), so  $y^2 \neq -1$ .

### 0.1.6.3 C. Going deeper

#### Exercise C1: Exporting a hypothesis.

Prove that  $P \Rightarrow (Q \Rightarrow R)$  is logically equivalent to  $(P \wedge Q) \Rightarrow R$  for all statements  $P, Q, R$ . Explain in one sentence why this lets a theorem with two hypotheses be stated either way.

*Solution.* Let  $P, Q, R$  be statements. By [Definition 0.1.2](#), an implication is false exactly when its hypothesis is true and its conclusion is false.

The statement  $(P \wedge Q) \Rightarrow R$  is false exactly when  $P \wedge Q$  is true and  $R$  is false, that is, when  $P$  is true,  $Q$  is true and  $R$  is false.

The statement  $P \Rightarrow (Q \Rightarrow R)$  is false exactly when  $P$  is true and  $Q \Rightarrow R$  is false. By the same definition,  $Q \Rightarrow R$  is false exactly when  $Q$  is true and  $R$  is false. So  $P \Rightarrow (Q \Rightarrow R)$  is false exactly when  $P$  is true,  $Q$  is true and  $R$  is false.



Both statements are false in exactly the same row of the eight-row truth table, and true in the other seven. Hence they are logically equivalent.

Consequently “if  $P$ , then (if  $Q$ , then  $R$ )” and “if  $P$  and  $Q$ , then  $R$ ” say the same thing, so a theorem may list its hypotheses one after another or all together.

### Exercise C2: When can a converse fail too?.

- (a) Determine whether there exist statements  $P$  and  $Q$  such that both  $P \Rightarrow Q$  and its converse  $Q \Rightarrow P$  are false. Justify your answer.
- (b) Give predicates  $P(n)$  and  $Q(n)$  about integers such that both  $\forall n \in \mathbb{Z} : (P(n) \Rightarrow Q(n))$  and  $\forall n \in \mathbb{Z} : (Q(n) \Rightarrow P(n))$  are false.
- (c) Explain why (a) and (b) do not contradict each other.

*Solution.*

- (a) No such statements exist. Suppose  $P \Rightarrow Q$  is false. By [Definition 0.1.2](#),  $P$  is true and  $Q$  is false. Then  $Q \Rightarrow P$  has a false hypothesis, so it is true. Hence an implication and its converse can never both be false.
- (b) Let  $P(n)$  be “ $n$  is even” and  $Q(n)$  be “ $n$  is divisible by 3”. The first universal statement is false:  $n = 2$  is even and not divisible by 3. The second is false:  $n = 3$  is divisible by 3 and not even.
- (c) In (a),  $P$  and  $Q$  are single statements, each with one fixed truth value. In (b), each universal statement is refuted by its own counterexample, and the two counterexamples are **different** integers. At  $n = 2$  the converse  $Q(2) \Rightarrow P(2)$  is vacuously true, and at  $n = 3$  the original  $P(3) \Rightarrow Q(3)$  is vacuously true, in line with (a). A “for every” statement can fail at one place while its converse fails somewhere else.

## 0.2 How Proofs Are Built

The previous section taught us to read statements: what an implication claims, what the quantifiers demand, and how to negate. Now we write proofs of them. A proof is a chain of statements, each one justified by a definition, a hypothesis or an earlier result, that ends in the statement we want. This section collects the handful of shapes that almost every proof in this book takes, shows each on a small fact about integers, and ends with the conventions that the book’s proofs follow.

Throughout, a theorem will usually come with two blocks. The **Idea** says how the argument was found, often by working backwards from the goal. The **Proof** then presents the argument forwards, one justified step at a time. Read the Idea first, then check the Proof line by line.

### 0.2.1 Direct proof

Most statements have the form “for every  $x$ , if  $P(x)$  then  $Q(x)$ ”. The direct way to prove this follows the shape of the statement:

1. **Let**  $x$  be arbitrary. By [Definition 0.1.3](#), a universal statement is a claim about each element, so we must not pick a convenient one.
2. **Suppose**  $P(x)$ . By [Definition 0.1.2](#), the implication can only fail when  $P(x)$  is true, so that is the only case we need to handle.
3. **Unwind**  $P(x)$  and  $Q(x)$  into their definitions. Then push forward from what we have towards what we want, often working backwards from the goal on scrap paper first.

Before we can prove anything about integers, we need definitions to unwind.

### Definition 0.2.1 — Even and odd

An integer  $n$  is **even** if there **exists** an integer  $k$  with  $n = 2k$ . An integer  $n$  is **odd** if there **exists** an integer  $k$  with  $n = 2k + 1$ .

So  $0 = 2 \cdot 0$  is even,  $-6 = 2 \cdot (-3)$  is even, and  $7 = 2 \cdot 3 + 1$  is odd. Each definition is an existential statement: to show a number is even, we must produce the integer  $k$ . We use the school fact that **every integer is even or odd, and not both**; an exercise at the end of this section proves a more general fact about remainders.

### Proposition 0.2.1 — Sum of even integers

For all integers  $m$  and  $n$ , if  $m$  and  $n$  are even, then  $m + n$  is even.

*Idea.* Givens:  $m = 2a$  and  $n = 2b$  for some integers  $a, b$ . Goal:  $m + n = 2k$  for some integer  $k$ . So we must *find*  $k$ . Adding the givens gives  $m + n = 2a + 2b = 2(a + b)$ , and  $k = a + b$  is the witness.

*Proof.* Let  $m$  and  $n$  be even integers. By [Definition 0.2.1](#), there exist integers  $a$  and  $b$  with  $m = 2a$  and  $n = 2b$ . Then

$$m + n = 2a + 2b = 2(a + b).$$

Since  $a + b$  is an integer, [Definition 0.2.1](#) shows that  $m + n$  is even. This proves the claim. ■

The pattern is worth naming. The hypothesis “ $m$  is even” is an **existence** statement, so using it hands us an integer  $a$  to work with. The goal “ $m + n$  is even” is also an existence statement, so proving it means **producing** an integer, here  $a + b$ . The two letters  $a$  and  $b$  must be different:  $m$  and  $n$  are arbitrary, so their halves have nothing to do with each other.

*Warning.* A **proof of a “for every” statement cannot start with a specific example**. Writing “ $4 + 6 = 10$ , which is even” checks one case and proves nothing about all even integers. Equally, writing  $m = 2k$  and  $n = 2k$  with the **same**  $k$  silently assumes  $m = n$ , so it only proves that  $2m$  is even.

## 0.2.2 Proof by contrapositive

Sometimes the direct route stalls. Consider the statement “if  $n^2$  is even, then  $n$  is even”. Going directly, we would suppose  $n^2 = 2k$  and try to say something about  $n$ . But  $n = \pm\sqrt{2k}$  is not visibly of the form  $2 \cdot (\text{integer})$ , and we are stuck.

By [Theorem 0.1.1](#), “if  $P$  then  $Q$ ” is equivalent to “if not  $Q$  then not  $P$ ”. Here the contrapositive is “if  $n$  is not even, then  $n^2$  is not even”, that is, “if  $n$  is odd, then  $n^2$  is odd”. Now the hypothesis hands us a formula for  $n$ , and squaring a formula is easy.

### Proposition 0.2.2 — Even squares

For every integer  $n$ , if  $n^2$  is even, then  $n$  is even.

*Idea.* The hypothesis  $n^2$  even says little about  $n$ , but “ $n$  odd” gives  $n = 2k + 1$  explicitly. So prove the contrapositive: from  $n = 2k + 1$ , expand  $n^2$  and look for the form  $2(\dots) + 1$ .

*Proof.* Let  $n$  be an integer. By [Theorem 0.1.1](#), it suffices to prove that if  $n$  is not even, then  $n^2$  is not even. Suppose  $n$  is not even. Since every integer is even or odd,  $n$  is odd, so by [Definition 0.2.1](#) there is an integer  $k$  with  $n = 2k + 1$ . Then

$$n^2 = 4k^2 + 4k + 1 = 2(2k^2 + 2k) + 1.$$

Since  $2k^2 + 2k$  is an integer,  $n^2$  is odd by [Definition 0.2.1](#). Since no integer is both even and odd,  $n^2$  is not even. This proves the contrapositive, and hence the statement. ■

A contrapositive proof is a direct proof of a different, equivalent implication. Use it when the negated conclusion gives more to work with than the hypothesis does.

## 0.2.3 Proof by contradiction

To prove a statement  $S$  by **contradiction**, suppose that  $S$  is false and deduce something impossible: a statement together with its negation. Why does this work? We have shown that the implication  $\neg S \Rightarrow (R \wedge \neg R)$  is true, for some statement  $R$ . Its conclusion  $R \wedge \neg R$  is false in every row, so by [Definition 0.1.2](#) the implication can only be true if its hypothesis  $\neg S$  is false. Hence  $S$  is true.

Contradiction is the natural tool for statements that say something **cannot** happen, or that there is **no** object of some kind. Such statements give nothing to start from directly, while their negation hands us an object to study.

Statements about fractions and primes need the language of divisibility.

### Definition 0.2.2 — Divisibility and primes

Let  $a$  and  $b$  be integers. We say  $a$  **divides**  $b$ , written  $a \mid b$ , if there **exists** an integer  $k$  with  $b = ak$ . An integer  $p$  is **prime** if  $p \geq 2$  and the only **positive** divisors of  $p$  are 1

and  $p$ .

For example,  $3 \mid 12$  since  $12 = 3 \cdot 4$ , and  $3 \nmid 7$ . The first primes are 2, 3, 5, 7, 11. The integer 1 is **not** prime, because the definition requires  $p \geq 2$ ; this convention is what makes factorizations into primes unique. In this language,  $n$  is even exactly when  $2 \mid n$ .

A real number is **rational** if it equals  $a/b$  for some integers  $a$  and  $b$  with  $b \neq 0$ , and **irrational** otherwise. We use the school fact that every rational number can be written **in lowest terms**: as  $a/b$  with  $b \geq 1$ , where no integer  $d \geq 2$  divides both  $a$  and  $b$ . (We justify this fact at the end of this section.)

### Theorem 0.2.3 — Irrationality of $\sqrt{2}$

There is no rational number whose square is 2. That is,  $\sqrt{2}$  is irrational.

*Idea.* “There is no rational  $x$  with  $x^2 = 2$ ” gives nothing to hold, so suppose there is one, and write it in lowest terms  $a/b$ . Clearing denominators gives  $a^2 = 2b^2$ , an integer equation. Then Proposition 0.2.2 applies twice: first  $a$  is even, then (after substituting  $a = 2c$ )  $b$  is even. That contradicts lowest terms.

*Proof.* Suppose, for a contradiction, that some rational number  $x$  satisfies  $x^2 = 2$ . Write  $x = a/b$  in lowest terms, so  $b \geq 1$  and no integer  $d \geq 2$  divides both  $a$  and  $b$ . Then  $a^2/b^2 = 2$ , and multiplying by  $b^2$ , which is non-zero, gives

$$a^2 = 2b^2.$$

Therefore  $a^2$  is even by Definition 0.2.1, and hence  $a$  is even by Proposition 0.2.2. Write  $a = 2c$  with  $c$  an integer. Substituting gives  $4c^2 = 2b^2$ , so  $b^2 = 2c^2$  is even. By Proposition 0.2.2 again,  $b$  is even. Thus 2 divides both  $a$  and  $b$ , which contradicts the choice of  $a/b$  in lowest terms. Hence no rational number has square 2. ■

The contradiction came from the lowest-terms condition, which is the hypothesis we chose to impose. That is typical: in a contradiction proof, look for the assumption you have not yet used, because the contradiction usually comes from there.

*Remark.* If a contradiction proof of “ $P \Rightarrow Q$ ” assumes  $P$  and  $\neg Q$  and then only ever derives  $\neg P$ , it is really a contrapositive proof in disguise. Write it as a contrapositive; it is shorter and clearer.

A second classic contradiction proof, that there are infinitely many primes, needs one more tool and appears at the end of this section.

## 0.2.4 Proof by cases

When a hypothesis splits naturally into possibilities, handle each one separately. If  $P_1 \vee P_2$  is true, and both  $P_1 \Rightarrow Q$  and  $P_2 \Rightarrow Q$  are true, then  $Q$  is true: whichever of  $P_1, P_2$  holds, its implication delivers  $Q$ . The cases must **cover every possibility**. They are allowed to overlap.

**Proposition 0.2.4** —  $n^2 + n$  is even

For every integer  $n$ , the integer  $n^2 + n$  is even.

*Idea.* Nothing about  $n$  is given, but every integer is even or odd, and each case gives a formula for  $n$ . Factoring first,  $n^2 + n = n(n+1)$ , makes each case a one-line computation.

*Proof.* Let  $n$  be an integer. Then  $n^2 + n = n(n+1)$ . Since every integer is even or odd, there are two cases.

*Case 1:  $n$  is even.* By [Definition 0.2.1](#),  $n = 2k$  for some integer  $k$ . Then  $n(n+1) = 2(k(n+1))$ , and  $k(n+1)$  is an integer.

*Case 2:  $n$  is odd.* By [Definition 0.2.1](#),  $n = 2k+1$  for some integer  $k$ . Then  $n+1 = 2(k+1)$ , so  $n(n+1) = 2(n(k+1))$ , and  $n(k+1)$  is an integer.

In both cases  $n^2 + n$  is even by [Definition 0.2.1](#). This shows the claim for every integer  $n$ . ■

**0.2.5 Proving “if and only if”**

By the definition of the biconditional in the previous section,  $P \Leftrightarrow Q$  means  $(P \Rightarrow Q) \wedge (Q \Rightarrow P)$ . So a proof of an “iff” has **two directions**, labeled  $(\Rightarrow)$  and  $(\Leftarrow)$ , and each is proved by any method we like. Often one direction is already known.

**Proposition 0.2.5** — Parity of a square

For every integer  $n$ ,  $n$  is even if and only if  $n^2$  is even.

*Proof.* Let  $n$  be an integer.

$(\Rightarrow)$  Suppose  $n$  is even, so  $n = 2k$  for some integer  $k$  by [Definition 0.2.1](#). Then  $n^2 = 4k^2 = 2(2k^2)$ , and  $2k^2$  is an integer, so  $n^2$  is even.

$(\Leftarrow)$  This is [Proposition 0.2.2](#).

Both directions hold, so  $n$  is even if and only if  $n^2$  is even. ■

**Warning. Reversing the steps is not automatic.** A common slip is to “prove the converse” by rewriting the same argument backwards without checking that every step reverses. For instance, from  $x = 2$  we get  $x^2 = 4$ ; but from  $x^2 = 4$  we only get  $x = 2$  **or**  $x = -2$ . Each direction needs its own justification.

## 0.2.6 Existence and uniqueness

To prove “there exists  $x$  with  $P(x)$ ”, the most direct method is to **produce a witness** and check it. For example, “there exists an integer  $n$  with  $n^2 = n + 2$ ” is proved by  $n = 2$ : indeed  $2^2 = 4 = 2 + 2$ .

To prove “there is **at most one**  $x$  with  $P(x)$ ”, the standard move is: *suppose  $x$  and  $x'$  both satisfy  $P$ , and show  $x = x'$* . Notice that we do not assume  $x \neq x'$ ; we simply take two objects with the property and prove they are the same. “There exists a **unique**  $x$ ” means both existence and at most one, so it takes both arguments.

Here is an example that we will meet again when we study fields and groups. A **binary operation**  $*$  on a collection  $S$  is a rule that assigns to each ordered pair  $a, b$  of elements of  $S$  an element  $a * b$  of  $S$ . Addition and multiplication of integers are binary operations on  $\mathbb{Z}$ . An **identity element** for  $*$  is an element  $e$  of  $S$  such that  $e * a = a$  and  $a * e = a$  for **every**  $a$  in  $S$ .

### Proposition 0.2.6 — Uniqueness of the identity

Let  $*$  be a binary operation on  $S$ . Then  $*$  has at most one identity element.

*Idea.* Suppose  $e$  and  $e'$  are both identities. The only thing we know about them is how they behave in a product, so look at the one product that involves both,  $e * e'$ , and evaluate it in two ways.

*Proof.* Suppose  $e$  and  $e'$  are both identity elements for  $*$ . Since  $e'$  is an identity,  $e * e' = e$ , taking  $a = e$  in its defining property. Since  $e$  is an identity,  $e * e' = e'$ , taking  $a = e'$  in its defining property. Therefore  $e = e * e' = e'$ . This shows that any two identity elements are equal. ■

So 0 is **the** identity for addition on  $\mathbb{Z}$ , and 1 is **the** identity for multiplication. The result says nothing about existence. Subtraction on  $\mathbb{Z}$  has no identity at all:  $a - e = a$  for every  $a$  forces  $e = 0$ , but then  $e - a = -a$ , which is not  $a$  when  $a = 1$ . Uniqueness proofs of exactly this shape will recur for zero vectors, negatives, inverse matrices and inverse functions.

## 0.2.7 Counterexamples

By [Definition 0.1.4](#), the negation of “ $\forall x : P(x)$ ” is “ $\exists x : \neg P(x)$ ”. So to **refute** a universal statement, one witness suffices: a single  $x$  where  $P(x)$  fails. Such an  $x$  is a **counterexample**. Conversely, no number of examples can prove a universal statement over an infinite range.

### Example. (Refuting “every prime is odd”)

Decide whether every prime number is odd. Justify your answer.

*Solution.* False. The statement is  $\forall p \text{ prime} : p \text{ is odd}$ , so by [Definition 0.1.4](#) one prime that is not odd refutes it. Take  $p = 2$ . It is prime, since  $2 \geq 2$  and a positive divisor of 2 is at most 2, so it is 1 or 2. It is even, since  $2 = 2 \cdot 1$ , and hence not odd. So 2 is a

counterexample.

*Warning. Many examples prove nothing.* The number  $n^2 + n + 41$  is prime for every integer  $n$  from 0 to 39. Yet the statement “ $n^2 + n + 41$  is prime for every  $n \geq 0$ ” is false: at  $n = 40$  we get  $40^2 + 40 + 41 = 1681 = 41^2$ , which is not prime.

## 0.2.8 Mathematical induction

Some statements are really infinitely many statements, one for each natural number: “for every  $n \geq 1$ ,  $1 + 2 + \cdots + n = n(n+1)/2$ ”. Checking cases one at a time never finishes. Induction replaces the infinite list of checks by two finite tasks, like knocking over a row of dominoes: push the first one, and make sure each domino knocks over the next.

We take this as a basic property of the integers, not something to prove.

### Theorem 0.2.7 — Principle of Mathematical Induction

Let  $n_0$  be an integer, and for each integer  $n \geq n_0$  let  $P(n)$  be a statement. Suppose

- (i) (**base case**)  $P(n_0)$  is true, and
- (ii) (**inductive step**) for **every** integer  $n \geq n_0$ , if  $P(n)$  is true then  $P(n+1)$  is true.

Then  $P(n)$  is true for every integer  $n \geq n_0$ .

In the inductive step, the assumption “ $P(n)$  is true” is called the **induction hypothesis**. It is not circular: we do not assume the whole conclusion, only that the statement holds at one arbitrary  $n$ , and we prove that it then holds at  $n+1$ .

### Proposition 0.2.8 — Sum of the first $n$ integers

For every integer  $n \geq 1$ ,

$$1 + 2 + \cdots + n = \frac{n(n+1)}{2}.$$

*Idea.* Passing from  $n$  to  $n+1$  adds exactly one term,  $n+1$ , to the left side. So the step reduces to the algebra identity  $\frac{n(n+1)}{2} + (n+1) = \frac{(n+1)(n+2)}{2}$ , which holds after taking out the factor  $n+1$ .

*Proof.* We prove this by induction on  $n$ , using Theorem 0.2.7 with  $n_0 = 1$ . Let  $P(n)$  be the statement  $1 + 2 + \cdots + n = n(n+1)/2$ .

*Base case.* For  $n = 1$  the left side is 1 and the right side is  $1 \cdot 2/2 = 1$ . So  $P(1)$  is true.

*Inductive step.* Let  $n \geq 1$  and suppose  $P(n)$  is true. Then

$$\begin{aligned} 1 + 2 + \cdots + n + (n + 1) &= \frac{n(n + 1)}{2} + (n + 1) \\ &= (n + 1) \left( \frac{n}{2} + 1 \right) \\ &= \frac{(n + 1)(n + 2)}{2}, \end{aligned}$$

where the first equality uses the induction hypothesis  $P(n)$ . This is  $P(n + 1)$ .

By [Theorem 0.2.7](#),  $P(n)$  is true for every integer  $n \geq 1$ , as claimed. ■

The base case does not have to be 1, and the step does not have to be an equation. In this book the **natural numbers** are  $\mathbb{N} = \{0, 1, 2, \dots\}$ : they **include** 0. (Some books start at 1; the next section fixes our convention formally.)

### Proposition 0.2.9 — $2^n > n$

For every natural number  $n$ , we have  $2^n > n$ .

*Proof.* We prove this by induction on  $n$ , with  $n_0 = 0$ . For  $n = 0$ ,  $2^0 = 1 > 0$ . Now let  $n \geq 0$  and suppose  $2^n > n$ . Then

$$2^{n+1} = 2^n + 2^n > n + 2^n \geq n + 1,$$

where the strict inequality uses the induction hypothesis and the last step uses  $2^n \geq 1$ . Hence  $2^{n+1} > n + 1$ . By [Theorem 0.2.7](#),  $2^n > n$  for every  $n \in \mathbb{N}$ . ■

The inductive step must work for **every**  $n \geq n_0$ , including the smallest ones. The following fake proof fails exactly there.

*Warning.* The “**all horses are the same color**” fallacy. *Claim:* in every group of  $n \geq 1$  horses, all horses have the same color. *“Proof”:* for  $n = 1$  there is nothing to show. Suppose the claim holds for groups of  $n$  horses, and take a group of  $n + 1$  horses  $h_1, \dots, h_{n+1}$ . The group  $h_1, \dots, h_n$  is single-colored by the induction hypothesis, and so is  $h_2, \dots, h_{n+1}$ . The two groups share a horse, so all  $n + 1$  horses have its color. The claim is plainly false, so this “proof” has a gap. Before opening the check below, try to find the exact value of  $n$  where the step breaks.

*Quick check.* Where does the inductive step in the horse argument fail?

*Solution.* It fails when passing from  $n = 1$  to  $n + 1 = 2$ . The two groups are then  $\{h_1\}$  and  $\{h_2\}$ , which share **no** horse, so nothing links the color of  $h_1$  to that of  $h_2$ . The sentence “the two groups share a horse” is true only when  $n \geq 2$ . Since the step  $P(1) \Rightarrow P(2)$  is not proved, the chain of dominoes breaks at the very first push, and the base case alone gives nothing.



## 0.2.9 Strong induction and well-ordering

In some arguments, knowing  $P(n)$  alone is not enough to reach  $P(n+1)$ : we need the statement at some **earlier** value, and we do not know in advance which one. Strong induction allows the hypothesis at all earlier values at once.

### Theorem 0.2.10 — Strong induction

Let  $n_0$  be an integer, and for each integer  $n \geq n_0$  let  $P(n)$  be a statement. Suppose that for **every** integer  $n \geq n_0$ ,

if  $P(k)$  is true for all integers  $k$  with  $n_0 \leq k < n$ , then  $P(n)$  is true.

Then  $P(n)$  is true for every integer  $n \geq n_0$ .

*Idea.* Ordinary induction applies to the stronger statement  $Q(n)$ : “ $P(k)$  holds for all  $k$  from  $n_0$  to  $n$ ”. The hypothesis is exactly what pushes  $Q(n)$  to  $Q(n+1)$ . At  $n = n_0$  the condition “for all  $k$  with  $n_0 \leq k < n_0$ ” is vacuous, so the hypothesis already contains the base case.

*Proof.* For each integer  $n \geq n_0$ , let  $Q(n)$  be the statement “ $P(k)$  is true for all integers  $k$  with  $n_0 \leq k \leq n$ ”. We prove  $Q(n)$  for all  $n \geq n_0$  by Theorem 0.2.7.

*Base case.* There is no integer  $k$  with  $n_0 \leq k < n_0$ , so the condition “ $P(k)$  is true for all such  $k$ ” is vacuously true. By the hypothesis with  $n = n_0$ ,  $P(n_0)$  is true. Hence  $Q(n_0)$  is true.

*Inductive step.* Let  $n \geq n_0$  and suppose  $Q(n)$  is true, so  $P(k)$  holds for all  $k$  with  $n_0 \leq k < n+1$ . By the hypothesis applied to  $n+1$ ,  $P(n+1)$  is true. Together with  $Q(n)$ , this gives  $Q(n+1)$ .

By Theorem 0.2.7,  $Q(n)$  holds for all  $n \geq n_0$ . In particular  $P(n)$  holds for all  $n \geq n_0$ , since  $Q(n)$  includes  $k = n$ . This proves the theorem. ■

The same property of the natural numbers can be phrased without any statements  $P(n)$  at all: there is no endless strictly decreasing sequence of natural numbers, so every search for a smallest example ends.

### Theorem 0.2.11 — Well-Ordering Principle

Every non-empty collection of natural numbers has a least element: an element  $m$  of the collection with  $m \leq s$  for every  $s$  in the collection.

We take this as a basic property too. It is equivalent to the Principle of Mathematical Induction, in the sense that each can be proved from the other, but we will not need that. Well-ordering is the natural language for “take a smallest counterexample” arguments, which are contradiction proofs in disguise.

Here is a result that needs strong induction, because the smaller number we reach is not  $n - 1$ .

### Proposition 0.2.12 — Existence of a prime factor

Every integer  $n \geq 2$  has a prime divisor.

*Idea.* If  $n$  is prime,  $n$  itself works. If not,  $n = ab$  with  $1 < a < n$ , and a prime dividing  $a$  also divides  $n$ . The factor  $a$  can be any size below  $n$ , so we need the statement for all smaller values, which is strong induction.

*Proof.* We use [Theorem 0.2.10](#) with  $n_0 = 2$ . Let  $n \geq 2$ , and suppose every integer  $k$  with  $2 \leq k < n$  has a prime divisor.

*Case 1:  $n$  is prime.* Then  $n \mid n$ , so  $n$  is a prime divisor of  $n$ .

*Case 2:  $n$  is not prime.* Since  $n \geq 2$ , [Definition 0.2.2](#) gives a positive divisor  $a$  of  $n$  with  $a \neq 1$  and  $a \neq n$ . Write  $n = ab$  with  $b$  an integer. Since  $a$  and  $n$  are positive,  $b \geq 1$ , and  $b \neq 1$  because  $a \neq n$ ; so  $b \geq 2$  and  $a = n/b < n$ . Thus  $2 \leq a < n$ , and by the hypothesis  $a$  has a prime divisor  $p$ , say  $a = pc$ . Then  $n = p(cb)$ , so  $p \mid n$ .

In both cases  $n$  has a prime divisor. By [Theorem 0.2.10](#), this holds for every  $n \geq 2$ . ■

Now the promised contradiction proof.

### Theorem 0.2.13 — Infinitely many primes

There are infinitely many prime numbers.

*Idea.* Suppose the primes form a finite list. Build a number that no prime on the list can divide: multiply them all and add 1. Dividing by any  $p_i$  leaves remainder 1. But [Proposition 0.2.12](#) says this number has some prime divisor, and that prime must be on the list.

*Proof.* Suppose, for a contradiction, that there are only finitely many primes, and list them all as  $p_1, p_2, \dots, p_k$ . Let

$$N = p_1 p_2 \cdots p_k + 1.$$

The list is non-empty, since 2 is prime, and each  $p_i \geq 2$ ; hence  $N \geq 3$ . By [Proposition 0.2.12](#),  $N$  has a prime divisor  $p$ . Since the list contains every prime,  $p = p_i$  for some  $i$ . Then  $p_i \mid N$  and  $p_i \mid p_1 p_2 \cdots p_k$ , so there are integers  $s, t$  with  $N = p_i s$  and  $p_1 \cdots p_k = p_i t$ . Subtracting gives  $1 = p_i(s - t)$ . Since  $s - t$  is an integer and  $p_i \geq 2$ , this is impossible:  $p_i(s - t)$  is 0 when  $s = t$ , and has absolute value at least 2 otherwise. This contradiction shows that there are infinitely many primes. ■

*Remark.* Well-ordering also justifies the “lowest terms” fact used in [Theorem 0.2.3](#). Let  $x$  be rational. There is at least one way of writing  $x = a/b$  with  $b \geq 1$ : if  $x = a/b$  with  $b < 0$ ,

then also  $x = (-a)/(-b)$  with  $-b \geq 1$ . Among all such ways, the collection of possible denominators  $b$  is a non-empty collection of natural numbers, so by [Theorem 0.2.11](#) it has a least element. Take a representation  $a/b$  with that least  $b$ . If some  $d \geq 2$  divided both, say  $a = da'$  and  $b = db'$ , then  $x = a'/b'$  with  $1 \leq b' < b$ , contradicting minimality. So  $a/b$  is in lowest terms.

## 0.2.10 Writing proofs in this book

The proofs in this book follow a few conventions. Knowing them makes proofs faster to read, and copying them makes your own proofs easier to check.

- **The first sentence does mathematics.** Proofs open with “Let ...”, “Suppose ...”, “Since ...” or “By ...”. The motivation has already been given in the Idea block.
- **Every step names its reason.** Each line cites a hypothesis, a definition, or an earlier result by name, as in “by [Proposition 0.2.2](#)”. When a display hides a reason, a clause such as “where the first equality uses the induction hypothesis” points to it.
- **Structure is marked.** Two directions of an iff are labeled  $(\Rightarrow)$  and  $(\Leftarrow)$ , cases are labeled *Case 1* and *Case 2*, and inductions name their base case and inductive step.
- **No hidden gaps.** Words like “obviously” and “clearly” never stand in for an argument. If a step is short, it is written out; if it is routine and has been done before, the proof says so.
- **The ending restates the result:** “This shows ...”, “This proves ...”, “as claimed”.

**How to read a proof here.** First read the statement and identify the givens and the goal. Then read the Idea, which says why the argument has the shape it has. Then read the Proof with a pencil. At each line, ask which earlier line or result justifies it; if you cannot answer, that is the line to reread. Finally, ask which hypothesis each step used. If a hypothesis was never used, either it was not needed or you have missed a step.

## 0.2.11 Exercises

### 0.2.11.1 A. Check your understanding

#### Exercise A1.

For each statement, name the proof technique you would use first (direct, contrapositive, contradiction, cases, induction, or counterexample), and give a one-sentence reason. Then carry out the proof or refutation.

- The sum of two odd integers is even.
- For every integer  $n$ , if  $3n + 2$  is odd, then  $n$  is odd.
- For every integer  $n \geq 1$ ,  $3 \mid 4^n - 1$ .
- For every integer  $n \geq 1$ ,  $n^2 \geq 2n$ .
- There is no largest integer.

*Solution.*

- Direct.* The hypothesis gives formulas to add. Let  $m = 2a + 1$  and  $n = 2b + 1$  with

- $a, b$  integers, by [Definition 0.2.1](#). Then  $m + n = 2(a + b + 1)$ , which is even.
- (b) *Contrapositive*. The conclusion's negation " $n$  even" gives a formula, while " $3n + 2$  odd" does not. By [Theorem 0.1.1](#) it suffices to show: if  $n$  is even then  $3n + 2$  is even. If  $n = 2k$ , then  $3n + 2 = 2(3k + 1)$ , which is even, hence not odd.
- (c) *Induction*. It is a statement for every  $n \geq 1$ , and  $4^{n+1} - 1$  is built from  $4^n - 1$ . For  $n = 1$ ,  $4 - 1 = 3 = 3 \cdot 1$ . If  $4^n - 1 = 3k$ , then  $4^{n+1} - 1 = 4(4^n - 1) + 3 = 3(4k + 1)$ . By [Theorem 0.2.7](#) the claim holds for all  $n \geq 1$ .
- (d) *Counterexample*. The statement is false, so a single witness refutes it. At  $n = 1$ ,  $n^2 = 1 < 2 = 2n$ .
- (e) *Contradiction*. It says an object does not exist, so assume it does. Suppose  $m$  is a largest integer. Then  $m + 1$  is an integer with  $m + 1 > m$ , contradicting the choice of  $m$ . Hence there is no largest integer.

### 0.2.11.2 B. Practice

#### Exercise B1: Contrapositive.

Prove that for every integer  $n$ , if  $n^2$  is odd, then  $n$  is odd.

*Solution*. Let  $n$  be an integer. By [Theorem 0.1.1](#), it suffices to prove that if  $n$  is not odd, then  $n^2$  is not odd. Suppose  $n$  is not odd. Since every integer is even or odd,  $n$  is even, so  $n = 2k$  for some integer  $k$  by [Definition 0.2.1](#). Then  $n^2 = 4k^2 = 2(2k^2)$ , so  $n^2$  is even. Since no integer is both even and odd,  $n^2$  is not odd. This proves the statement.

#### Exercise B2: Induction.

Prove that for every integer  $n \geq 1$ , the sum of the first  $n$  odd numbers is  $n^2$ :

$$1 + 3 + 5 + \cdots + (2n - 1) = n^2.$$

*Solution*. We prove this by induction on  $n$ , using [Theorem 0.2.7](#) with  $n_0 = 1$ . Let  $P(n)$  be the displayed statement.

*Base case*. For  $n = 1$  the left side is 1 and the right side is  $1^2 = 1$ , so  $P(1)$  is true.

*Inductive step*. Let  $n \geq 1$  and suppose  $P(n)$  is true. The  $(n + 1)$ -st odd number is  $2(n + 1) - 1 = 2n + 1$ . Then

$$1 + 3 + \cdots + (2n - 1) + (2n + 1) = n^2 + 2n + 1 = (n + 1)^2,$$

where the first equality uses the induction hypothesis. This is  $P(n + 1)$ .

By [Theorem 0.2.7](#),  $P(n)$  holds for every  $n \geq 1$ .

#### Exercise B3: $\sqrt{3}$ is irrational.

You may use the school fact that every integer has exactly one of the forms  $3k$ ,  $3k + 1$ ,  $3k + 2$  with  $k$  an integer.

- (a) Prove that for every integer  $n$ , if  $3 \mid n^2$  then  $3 \mid n$ .

- (b) Hence prove that there is no rational number whose square is 3.

*Solution.*

- (a) Let  $n$  be an integer. By [Theorem 0.1.1](#), it suffices to show that if  $3 \nmid n$  then  $3 \nmid n^2$ . Suppose  $3 \nmid n$ . Then  $n = 3k + 1$  or  $n = 3k + 2$  for some integer  $k$ .  
*Case 1:*  $n = 3k + 1$ . Then  $n^2 = 9k^2 + 6k + 1 = 3(3k^2 + 2k) + 1$ .  
*Case 2:*  $n = 3k + 2$ . Then  $n^2 = 9k^2 + 12k + 4 = 3(3k^2 + 4k + 1) + 1$ .  
 In both cases  $n^2$  has the form  $3j + 1$  with  $j$  an integer. Since an integer has only one of the three forms,  $n^2$  is not of the form  $3j$ , so  $3 \nmid n^2$ . This proves (a).  
 (b) Suppose, for a contradiction, that a rational number  $x$  satisfies  $x^2 = 3$ . Write  $x = a/b$  in lowest terms, so  $b \geq 1$  and no integer  $d \geq 2$  divides both  $a$  and  $b$ . Multiplying  $a^2/b^2 = 3$  by  $b^2 \neq 0$  gives  $a^2 = 3b^2$ . Therefore  $3 \mid a^2$ , and by (a),  $3 \mid a$ . Write  $a = 3c$ . Then  $9c^2 = 3b^2$ , so  $b^2 = 3c^2$  and  $3 \mid b^2$ . By (a) again,  $3 \mid b$ . Thus 3 divides both  $a$  and  $b$ , contradicting lowest terms. Hence no rational number has square 3.

### 0.2.11.3 C. Going deeper

#### Exercise C1: Stamps of 4 and 5.

- (a) Show that each of 12, 13, 14, 15 can be written as  $4a + 5b$  with  $a, b \in \mathbb{N}$ .  
 (b) Prove by strong induction that every integer  $n \geq 12$  can be written as  $4a + 5b$  with  $a, b \in \mathbb{N}$ .  
 (c) Show that 11 cannot be written in this form, so the bound 12 in (b) cannot be lowered to 11.

*Hint: for  $n \geq 16$ , look at  $n - 4$ .*

*Solution.*

- (a)  $12 = 4 \cdot 3 + 5 \cdot 0$ ,  $13 = 4 \cdot 2 + 5 \cdot 1$ ,  $14 = 4 \cdot 1 + 5 \cdot 2$ , and  $15 = 4 \cdot 0 + 5 \cdot 3$ .  
 (b) We use [Theorem 0.2.10](#) with  $n_0 = 12$ . Let  $n \geq 12$ , and suppose every integer  $k$  with  $12 \leq k < n$  can be written as  $4a + 5b$  with  $a, b \in \mathbb{N}$ .  
*Case 1:*  $12 \leq n \leq 15$ . Then  $n$  has the required form by (a).  
*Case 2:*  $n \geq 16$ . Then  $k = n - 4$  satisfies  $12 \leq k < n$ , so by the hypothesis  $n - 4 = 4a + 5b$  with  $a, b \in \mathbb{N}$ . Hence  $n = 4(a + 1) + 5b$ , and  $a + 1 \in \mathbb{N}$ .  
 In both cases  $n$  has the required form. By [Theorem 0.2.10](#), every integer  $n \geq 12$  does.  
 (c) Suppose  $11 = 4a + 5b$  with  $a, b \in \mathbb{N}$ . Since  $5b \leq 11$ ,  $b \in \{0, 1, 2\}$ . If  $b = 0$  then  $4a = 11$ ; if  $b = 1$  then  $4a = 6$ ; if  $b = 2$  then  $4a = 1$ . None of 11, 6, 1 is a multiple of 4, so no such  $a$  exists. Hence 11 is not of the form  $4a + 5b$ .

#### Exercise C2: Division with remainder.

Let  $a$  and  $b$  be integers with  $b \geq 1$ . Let  $R$  be the collection of all **non-negative** integers of the form  $a - qb$  with  $q$  an integer.

- (a) Show that  $R$  is non-empty.

- (b) Let  $r$  be the least element of  $R$ , which exists by [Theorem 0.2.11](#), and write  $r = a - qb$ . Prove that  $0 \leq r < b$ .
- (c) Prove that the pair  $(q, r)$  is unique: if  $a = qb + r = q'b + r'$  with  $0 \leq r < b$  and  $0 \leq r' < b$ , then  $q = q'$  and  $r = r'$ .

*Hint for (a): try  $q = -|a|$ .*

*Solution.*

- (a) Take  $q = -|a|$ . Then  $a - qb = a + |a|b$ . Since  $b \geq 1$  and  $|a| \geq 0$ , we have  $|a|b \geq |a|$ , so  $a + |a|b \geq a + |a| \geq 0$ , where the last step uses  $|a| \geq -a$ . Hence  $a + |a|b$  belongs to  $R$ , and  $R$  is non-empty.
- (b) By (a) and [Theorem 0.2.11](#),  $R$  has a least element  $r = a - qb$  with  $q \in \mathbb{Z}$ . Since  $r$  belongs to  $R$ ,  $r \geq 0$ . Suppose, for a contradiction, that  $r \geq b$ . Then  $r - b = a - (q + 1)b \geq 0$ , and  $q + 1 \in \mathbb{Z}$ , so  $r - b$  belongs to  $R$ . But  $b \geq 1$  gives  $r - b < r$ , contradicting the choice of  $r$  as the least element of  $R$ . Hence  $0 \leq r < b$ .
- (c) Suppose  $qb + r = q'b + r'$  with  $0 \leq r, r' < b$ . Then  $r' - r = (q - q')b$ , so  $b \mid r' - r$ . From the bounds,  $-b < r' - r < b$ . Suppose  $q \neq q'$ . Then  $|q - q'| \geq 1$ , so  $|r' - r| = |q - q'|b \geq b$ , contradicting  $-b < r' - r < b$ . Hence  $q = q'$ , and then  $r' - r = 0$ , so  $r = r'$ . This proves uniqueness.

Together, (a)–(c) show that for integers  $a$  and  $b \geq 1$  there are unique integers  $q, r$  with  $a = qb + r$  and  $0 \leq r < b$ . With  $b = 2$ , this is the fact that every integer is even or odd, and not both.

## 0.3 Sets

Every object in linear algebra is a set with some extra structure. A vector space is a set on which we can add and scale, a subspace is a subset of it, and a linear map sends elements of one set to elements of another. Before the structure arrives, we need the plain language of sets: how to describe them, when two of them are equal, and how to combine them. The most important move in this section is proving that two sets are equal by proving two inclusions, a move we will use for subspaces, spans, kernels and images throughout the book.

### 0.3.1 Sets, elements and set-builder notation

A **set** is a collection of objects, called its **elements**. We write  $x \in A$  for “ $x$  is an element of  $A$ ” (also “ $x$  belongs to  $A$ ”, “ $x$  lies in  $A$ ”), and  $x \notin A$  for its negation. A set is determined by its elements and nothing else: it has no order and no repetition. So  $\{1, 2\}$ ,  $\{2, 1\}$  and  $\{1, 1, 2\}$  all describe the same set, whose elements are 1 and 2. We make this precise below.

Small sets can be listed between braces. Infinite sets we meet constantly have fixed names.

#### Definition 0.3.1 — Common number sets

We write

- $\mathbb{N} = \{0, 1, 2, 3, \dots\}$  for the **natural numbers**; in this book  $\mathbb{N}$  **includes** 0;
- $\mathbb{Z} = \{\dots, -2, -1, 0, 1, 2, \dots\}$  for the **integers**;
- $\mathbb{Q}$  for the **rational numbers**, the numbers  $a/b$  with  $a, b \in \mathbb{Z}$  and  $b \neq 0$ ;
- $\mathbb{R}$  for the **real numbers**;
- $\mathbb{C}$  for the **complex numbers**, studied later in this chapter.

Books disagree about whether  $0 \in \mathbb{N}$ . With our convention, the Well-Ordering Principle ([Theorem 0.2.11](#)) applies to collections that contain 0, and “ $n \in \mathbb{N}$ ” means  $n \geq 0$ . When we need positive integers only, we say “ $n \geq 1$ ”. For real numbers  $a \leq b$  we also use **intervals**:  $[a, b]$  is the set of real  $x$  with  $a \leq x \leq b$ ,  $(a, b)$  the set with  $a < x < b$ , and  $[a, b)$ ,  $(a, b]$  are defined in the same way.

Listing elements fails for most sets we care about, such as “the real numbers whose square is less than 4”. Instead we describe a set by a property, and select from a set we already have.

### Definition 0.3.2 — Set-builder notation

Let  $S$  be a set and let  $P(x)$  be a predicate for  $x \in S$ . The set

$$\{x \in S : P(x)\}$$

is the set whose elements are **exactly** those  $x \in S$  for which  $P(x)$  is true.

In words: an object  $y$  belongs to  $\{x \in S : P(x)\}$  if and only if  $y \in S$  **and**  $P(y)$  is true. Both parts matter. The ambient set  $S$  says where we are looking, and the predicate says what we keep. The colon is read “such that”; some books use a vertical bar instead. When it is plain which sets the elements come from, as in the unions and intersections below, we sometimes leave the ambient set out and write  $\{x : P(x)\}$ .

A second form describes the elements by a formula:  $\{2k : k \in \mathbb{Z}\}$  is the set of all numbers of the form  $2k$  with  $k$  an integer. An object  $y$  belongs to it exactly when  $y = 2k$  for **some**  $k \in \mathbb{Z}$ . This form will describe sets such as “all combinations of given vectors” in Chapter 1.

### Example. (Reading set-builder notation)

Describe each set more simply.

- $\{x \in \mathbb{R} : x^2 < 4\}$ .
- $\{x \in \mathbb{Z} : x^2 < 4\}$ .
- $\{n \in \mathbb{Z} : n \text{ is even}\}$  and  $\{2k : k \in \mathbb{Z}\}$ .
- $\{(x, y) : x, y \in \mathbb{R}, x + y = 1\}$ , a set of ordered pairs (made precise under Cartesian products below).

*Solution.*

- A real number satisfies  $x^2 < 4$  exactly when  $-2 < x < 2$ , so the set is the interval  $(-2, 2)$ .
- The same predicate, selected from  $\mathbb{Z}$  instead, keeps only integers: the set is  $\{-1, 0, 1\}$ . Changing the ambient set changes the set.
- Both are the set  $\{\dots, -4, -2, 0, 2, 4, \dots\}$  of even integers. The first selects the

integers with a property; the second produces each element by the formula  $2k$ . By the definition of “even” (Definition 0.2.1), an integer is even exactly when it equals  $2k$  for some  $k \in \mathbb{Z}$ .

- (d) The pairs of reals whose coordinates add to 1: the line in the plane through  $(1, 0)$  and  $(0, 1)$ .

What if no element of  $S$  satisfies the predicate? We still want a set, and it has no elements.

### Definition 0.3.3 — Empty set

The **empty set**  $\emptyset$  is the set with **no** elements:  $x \notin \emptyset$  for every object  $x$ .

For example,  $\{x \in \mathbb{R} : x^2 = -1\} = \emptyset$ , since  $x^2 \geq 0$  for every real  $x$  (Example (Quantified statements)). Also  $\{n \in \mathbb{Z} : 0 < n < 1\} = \emptyset$ . We will see shortly that there is only **one** empty set, so these two descriptions give the same set.

Statements about all elements of  $\emptyset$  are vacuously true: “every element of  $\emptyset$  is a prime number” has no element that could fail, just as in the degenerate case discussed after Example (Quantified statements).

*Warning.*  $\emptyset$ ,  $\{\emptyset\}$  and  $\{0\}$  are three different sets. The set  $\emptyset$  has no elements. The set  $\{\emptyset\}$  has exactly one element, namely the empty set. The set  $\{0\}$  has exactly one element, namely the number 0. A box containing an empty box is not an empty box. In Chapter 1 this matters:  $\{0\}$  will be a subspace, and  $\emptyset$  never is.

## 0.3.2 Subsets

Most sets in this book live inside a bigger set: the even integers inside  $\mathbb{Z}$ , a line inside the plane, and later a subspace inside a vector space. The first thing to check about a candidate subspace will be that it sits inside the space at all. We need a word for “sits inside”.

*A is a subset of B when everything in A is also in B.*

### Definition 0.3.4 — Subset

Let  $A$  and  $B$  be sets. We say  $A$  is a **subset** of  $B$ , and write  $A \subseteq B$ , if

$$\forall x : (x \in A \Rightarrow x \in B).$$

If  $A \subseteq B$  and  $A \neq B$ , then  $A$  is a **proper** subset of  $B$ , written  $A \subsetneq B$ .

In words:  $A \subseteq B$  says that **every** element of  $A$  is an element of  $B$ . It says nothing about elements of  $B$ ;  $B$  may contain many more. The definition is an implication inside a “for every”, so the proof template of the previous section applies: to show  $A \subseteq B$ , **let**  $x \in A$  and show  $x \in B$ .

By Definition 0.1.4 and Theorem 0.1.2, the negation reads

$$A \not\subseteq B \iff \exists x : x \in A \text{ and } x \notin B.$$



So to show  $A$  is **not** a subset of  $B$ , exhibit one element of  $A$  outside  $B$ .

**Example. (Subsets)**

Decide whether each inclusion holds.

- (a)  $\mathbb{N} \subseteq \mathbb{Z} \subseteq \mathbb{Q} \subseteq \mathbb{R}$ .
- (b)  $\{1, 2\} \subseteq \{1, 2, 3\}$  and  $\{1, 4\} \subseteq \{1, 2, 3\}$ .
- (c)  $A \subseteq A$  and  $\emptyset \subseteq A$ , for any set  $A$ .
- (d)  $\{x \in \mathbb{R} : x^2 = 1\} \subseteq \mathbb{Z}$ .

*Solution.*

- (a) All hold. Every natural number is an integer; every integer  $n$  equals  $n/1$ , so it is rational; every rational number is real.
- (b) The first holds: each of  $1, 2$  is an element of  $\{1, 2, 3\}$ . The second fails, although it differs from the first in only one element:  $1$  passes, but  $4 \in \{1, 4\}$  and  $4 \notin \{1, 2, 3\}$ , so  $4$  is a witness for  $\{1, 4\} \not\subseteq \{1, 2, 3\}$ .
- (c) Both hold. For  $A \subseteq A$ : let  $x \in A$ ; then  $x \in A$ . For  $\emptyset \subseteq A$ : the implication  $x \in \emptyset \Rightarrow x \in A$  has a false hypothesis for every  $x$  by [Definition 0.3.3](#), so it is vacuously true ([Definition 0.1.2](#)). The empty set is a subset of every set, including itself.
- (d) Holds. Let  $x \in \mathbb{R}$  with  $x^2 = 1$ . Then  $(x - 1)(x + 1) = 0$ , so  $x = 1$  or  $x = -1$ , and both are integers.

*Warning.*  $\in$  and  $\subseteq$  are different relations.  $x \in A$  says the object  $x$  is one of the elements of  $A$ .  $A \subseteq B$  compares two sets, element by element. So  $1 \in \{1, 2\}$  is true while  $1 \subseteq \{1, 2\}$  makes no sense ( $1$  is not a set here); and  $\{1\} \subseteq \{1, 2\}$  is true while  $\{1\} \in \{1, 2\}$  is false.

*Quick check.* Let  $X = \{\{1, 2\}, 3\}$ . Is  $\{1, 2\} \subseteq X$ ? Is  $\{1, 2\} \in X$ ?

*Solution.* The set  $X$  has exactly two elements: the set  $\{1, 2\}$  and the number  $3$ . So  $\{1, 2\} \in X$  is **true**. But  $\{1, 2\} \subseteq X$  is **false**:  $1 \in \{1, 2\}$ , and  $1$  is neither the set  $\{1, 2\}$  nor the number  $3$ , so  $1 \notin X$ .

### 0.3.3 Equality of sets and double inclusion

We said a set is determined by its elements. Here is the precise version.

**Definition 0.3.5 — Set equality**

Two sets  $A$  and  $B$  are **equal**, written  $A = B$ , if they have exactly the same elements:

$$\forall x : (x \in A \Leftrightarrow x \in B).$$

So  $\{1, 2\} = \{2, 1\} = \{1, 1, 2\}$ : each object belongs to one of these sets exactly when it is  $1$  or  $2$ .

Order and repetition in a list are features of how we **write** a set, not of the set.

Proving a biconditional for every  $x$  at once is awkward. It is much easier to prove two implications separately, and that is exactly what the next result allows.

### Theorem 0.3.1 — Double inclusion

Let  $A$  and  $B$  be sets. Then  $A = B$  if and only if  $A \subseteq B$  and  $B \subseteq A$ .

*Proof.* By Definition 0.3.5,  $A = B$  means that for every  $x$ , both  $x \in A \Rightarrow x \in B$  and  $x \in B \Rightarrow x \in A$  hold, since a biconditional is the conjunction of an implication and its converse. A statement of the form “for every  $x$ ,  $P(x)$  and  $Q(x)$ ” holds exactly when “for every  $x$ ,  $P(x)$ ” and “for every  $x$ ,  $Q(x)$ ” both hold, by Definition 0.1.3. Hence  $A = B$  holds exactly when  $\forall x (x \in A \Rightarrow x \in B)$  and  $\forall x (x \in B \Rightarrow x \in A)$  both hold. By Definition 0.3.4, these are  $A \subseteq B$  and  $B \subseteq A$ . This proves the theorem. ■

This is the move “**prove set equality by double inclusion**”. The proof is written in two labeled halves,  $(\subseteq)$  and  $(\supseteq)$ , each starting “Let  $x \in \dots$ ”. Here is a first consequence.

*Remark.* There is only one empty set. If  $E$  and  $E'$  are sets with no elements, then  $E \subseteq E'$  and  $E' \subseteq E$  hold vacuously, as in Example (Subsets) (c), so  $E = E'$  by Theorem 0.3.1. This is why we may say **the** empty set.

### Example. (A solution set by double inclusion)

Prove that  $\{x \in \mathbb{R} : x^2 - 3x + 2 = 0\} = \{1, 2\}$ .

*Solution.* Write  $S = \{x \in \mathbb{R} : x^2 - 3x + 2 = 0\}$ . By Theorem 0.3.1 it suffices to prove  $S \subseteq \{1, 2\}$  and  $\{1, 2\} \subseteq S$ .

$(\subseteq)$  Let  $x \in S$ . Then  $x$  is real and  $x^2 - 3x + 2 = 0$ . Factoring,  $(x - 1)(x - 2) = 0$ . A product of two real numbers is zero only if one factor is zero, so  $x = 1$  or  $x = 2$ . Hence  $x \in \{1, 2\}$ .

$(\supseteq)$  Let  $x \in \{1, 2\}$ , so  $x = 1$  or  $x = 2$ . Both are real. If  $x = 1$ , then  $1 - 3 + 2 = 0$ ; if  $x = 2$ , then  $4 - 6 + 2 = 0$ . In either case  $x \in S$ .

This shows  $S = \{1, 2\}$ .

Solving an equation “at school” usually produces only the  $(\subseteq)$  half: “if  $x$  is a solution, then  $x$  is 1 or 2”. The  $(\supseteq)$  half, checking that the candidates really are solutions, is the part that gets forgotten. It matters whenever a step in the solving is not reversible, such as squaring both sides.

## 0.3.4 Union, intersection and difference

Given two sets, there are three basic ways to form a new one: keep what is in either, keep what is in both, or keep what is in the first but not the second. Each is a set-builder description

whose predicate uses a connective from the section on logic.

### Definition 0.3.6 — Union

The **union** of sets  $A$  and  $B$  is

$$A \cup B = \{x : x \in A \text{ or } x \in B\}.$$

### Definition 0.3.7 — Intersection

The **intersection** of sets  $A$  and  $B$  is

$$A \cap B = \{x : x \in A \text{ and } x \in B\}.$$

We call  $A$  and  $B$  **disjoint** if  $A \cap B = \emptyset$ .

### Definition 0.3.8 — Set difference and complement

The **difference** of sets  $A$  and  $B$  is

$$A \setminus B = \{x \in A : x \notin B\}.$$

When all sets under discussion are subsets of one fixed set  $U$ , the **complement** of  $B \subseteq U$  is  $B^c = U \setminus B$ .

In words: the union keeps an element that lies in **at least one** of the sets (the “or” is inclusive, as always); the intersection keeps an element that lies in **both**; the difference  $A \setminus B$  keeps the elements of  $A$  that are **not** in  $B$ . The complement depends on the choice of  $U$ : the complement of  $\mathbb{N}$  is the set of negative integers inside  $\mathbb{Z}$ , but a much bigger set inside  $\mathbb{R}$ .

#### Example. (Operations on finite sets)

Let  $A = \{1, 2, 3, 4\}$  and  $B = \{3, 4, 5, 6\}$ . Compute  $A \cup B$ ,  $A \cap B$ ,  $A \setminus B$  and  $B \setminus A$ .

*Solution.* Checking each element against the definitions:  $A \cup B = \{1, 2, 3, 4, 5, 6\}$ ;  $A \cap B = \{3, 4\}$ ;  $A \setminus B = \{1, 2\}$ ; and  $B \setminus A = \{5, 6\}$ . Since  $A \setminus B \neq B \setminus A$ , the difference is not symmetric in its two arguments, unlike union and intersection.

#### Example. (Operations on intervals)

Let  $A = [0, 2]$  and  $B = [1, 3]$ . Compute  $A \cup B$ ,  $A \cap B$  and  $A \setminus B$ .

*Solution.* A real  $x$  lies in  $A \cup B$  when  $0 \leq x \leq 2$  or  $1 \leq x \leq 3$ , which happens exactly when  $0 \leq x \leq 3$ ; so  $A \cup B = [0, 3]$ . It lies in  $A \cap B$  when both hold, that is, when  $1 \leq x \leq 2$ ; so  $A \cap B = [1, 2]$ . It lies in  $A \setminus B$  when  $0 \leq x \leq 2$  and not  $1 \leq x \leq 3$ . Given  $x \leq 2$ , “not  $1 \leq x \leq 3$ ” means  $x < 1$ ; so  $A \setminus B = [0, 1)$ . The endpoint 1 is excluded because  $1 \in B$ .

Unions, intersections and differences interact through laws that mirror those for “or”, “and” and “not”. The first pair is De Morgan’s laws, the set version of [Theorem 0.1.2](#).

### Theorem 0.3.2 — De Morgan’s laws for sets

Let  $A$ ,  $B$  and  $C$  be sets. Then

- (a)  $A \setminus (B \cup C) = (A \setminus B) \cap (A \setminus C)$ ;
- (b)  $A \setminus (B \cap C) = (A \setminus B) \cup (A \setminus C)$ .

In particular, if  $B, C \subseteq U$ , then  $(B \cup C)^c = B^c \cap C^c$  and  $(B \cap C)^c = B^c \cup C^c$ .

*Idea.* Unwind membership into connectives. “ $x \notin B \cup C$ ” is “not ( $x \in B$  or  $x \in C$ )”, and De Morgan for statements turns that into “ $x \notin B$  and  $x \notin C$ ”. So each law is its logical counterpart, applied one element at a time. We prove (a) by double inclusion; (b) is an exercise.

*Proof.* We prove (a) using [Theorem 0.3.1](#).

( $\subseteq$ ) Let  $x \in A \setminus (B \cup C)$ . By [Definition 0.3.8](#),  $x \in A$  and  $x \notin B \cup C$ . By [Definition 0.3.6](#), the second says that “ $x \in B$  or  $x \in C$ ” is false, so by [Theorem 0.1.2](#),  $x \notin B$  and  $x \notin C$ . Since  $x \in A$  and  $x \notin B$ , we have  $x \in A \setminus B$ ; since  $x \in A$  and  $x \notin C$ , we have  $x \in A \setminus C$ . Hence  $x \in (A \setminus B) \cap (A \setminus C)$  by [Definition 0.3.7](#).

( $\supseteq$ ) Let  $x \in (A \setminus B) \cap (A \setminus C)$ . By [Definition 0.3.7](#) and [Definition 0.3.8](#),  $x \in A$ ,  $x \notin B$  and  $x \notin C$ . By [Theorem 0.1.2](#), “ $x \in B$  or  $x \in C$ ” is false, so  $x \notin B \cup C$ . Together with  $x \in A$ , this gives  $x \in A \setminus (B \cup C)$ .

This proves (a). For the complement form, take  $A = U$  and use [Definition 0.3.8](#):  $(B \cup C)^c = U \setminus (B \cup C) = (U \setminus B) \cap (U \setminus C) = B^c \cap C^c$ , and similarly for (b). ■

The second pair of laws says each of  $\cap$  and  $\cup$  distributes over the other, like multiplication over addition, except that here **both** directions hold.

### Theorem 0.3.3 — Distributive laws for sets

Let  $A$ ,  $B$  and  $C$  be sets. Then

- (a)  $A \cap (B \cup C) = (A \cap B) \cup (A \cap C)$ ;
- (b)  $A \cup (B \cap C) = (A \cup B) \cap (A \cup C)$ .

*Idea.* Double inclusion again, and in each half the “or” in a union forces a split into cases. For (b) ( $\supseteq$ ), an element of both  $A \cup B$  and  $A \cup C$  is either in  $A$ , and then done, or not in  $A$ , and then it must be in  $B$  and in  $C$ . We prove (b); (a) is an exercise.

*Proof.* We prove (b) using [Theorem 0.3.1](#).

( $\subseteq$ ) Let  $x \in A \cup (B \cap C)$ . By [Definition 0.3.6](#),  $x \in A$  or  $x \in B \cap C$ .

*Case 1:*  $x \in A$ . Then  $x \in A \cup B$  and  $x \in A \cup C$  by Definition 0.3.6.

*Case 2:*  $x \in B \cap C$ . Then  $x \in B$  and  $x \in C$  by Definition 0.3.7, so again  $x \in A \cup B$  and  $x \in A \cup C$ .

In both cases  $x \in (A \cup B) \cap (A \cup C)$  by Definition 0.3.7.

( $\supseteq$ ) Let  $x \in (A \cup B) \cap (A \cup C)$ , so  $x \in A \cup B$  and  $x \in A \cup C$ .

*Case 1:*  $x \in A$ . Then  $x \in A \cup (B \cap C)$ .

*Case 2:*  $x \notin A$ . Since  $x \in A \cup B$  and  $x \notin A$ , we get  $x \in B$ . Since  $x \in A \cup C$  and  $x \notin A$ , we get  $x \in C$ . Hence  $x \in B \cap C$ , so  $x \in A \cup (B \cap C)$ .

In both cases  $x \in A \cup (B \cap C)$ . This proves (b). ■

*Remark.* In Chapter 1 we will see that the intersection of two subspaces is always a subspace, while their union usually is not: the two axes in the plane are subspaces, but  $(1, 0) + (0, 1) = (1, 1)$  lies on neither. The union is replaced there by a different construction, the sum of subspaces.

## 0.3.5 Indexed families of sets

Two sets are not always enough. In Chapter 1 we will need “the intersection of **all** subspaces that contain a given set”, and there are usually infinitely many of them. So we need unions and intersections of any number of sets at once.

*An indexed family is a list of sets with one set for each label; its union keeps what lies in some member, its intersection what lies in every member.*

### Definition 0.3.9 — Unions and intersections of indexed families

Let  $I$  be a set, and for each  $i \in I$  let  $A_i$  be a set. The collection  $(A_i)_{i \in I}$  is an **indexed family** of sets, with **index set**  $I$ . Its **union** is

$$\bigcup_{i \in I} A_i = \{x : \exists i \in I : x \in A_i\},$$

and, if  $I$  is **non-empty**, its **intersection** is

$$\bigcap_{i \in I} A_i = \{x : \forall i \in I : x \in A_i\}.$$

In words:  $x$  lies in the union when it lies in **at least one**  $A_i$ , and in the intersection when it lies in **every**  $A_i$ . When  $I = \{1, 2\}$ , these are  $A_1 \cup A_2$  and  $A_1 \cap A_2$ . When  $I = \{1, 2, 3, \dots\}$ , we also write  $\bigcup_{n=1}^{\infty} A_n$  and  $\bigcap_{n=1}^{\infty} A_n$ .

**Why  $I$  must be non-empty for the intersection.** If  $I = \emptyset$ , the condition “ $\forall i \in I : x \in A_i$ ” is vacuously true for **every** object  $x$ , so the “intersection” would contain everything, which is not a set we want. (When all sets live inside a fixed  $U$ , one sometimes defines the empty intersection

to be  $U$ .) The empty union causes no such trouble: “ $\exists i \in \emptyset$ ” is always false, so  $\bigcup_{i \in \emptyset} A_i = \emptyset$ .

**Example. (Indexed unions and intersections of intervals)**

For each integer  $n \geq 1$ , let  $A_n = [1/n, 1]$  and  $B_n = (0, 1/n)$ . Show that

$$\bigcup_{n=1}^{\infty} A_n = (0, 1] \quad \text{and} \quad \bigcap_{n=1}^{\infty} B_n = \emptyset.$$

You may use the fact that for every real  $x > 0$  there is an integer  $n \geq 1$  with  $1/n < x$ .

*Solution. The union.* ( $\subseteq$ ) Let  $x \in \bigcup_n A_n$ . By [Definition 0.3.9](#),  $x \in A_n$  for some  $n \geq 1$ , so  $0 < 1/n \leq x \leq 1$ . Hence  $x \in (0, 1]$ . ( $\supseteq$ ) Let  $x \in (0, 1]$ . Since  $x > 0$ , there is an integer  $n \geq 1$  with  $1/n < x$ . Then  $1/n \leq x \leq 1$ , so  $x \in A_n$ , and hence  $x$  lies in the union. By [Theorem 0.3.1](#), the union is  $(0, 1]$ .

*The intersection.* Suppose, for a contradiction, that some  $x$  lies in  $\bigcap_n B_n$ . Then  $x \in B_1 = (0, 1)$ , so  $x > 0$ . Choose an integer  $n \geq 1$  with  $1/n < x$ . Then  $x \notin (0, 1/n) = B_n$ , contradicting  $x \in B_n$  for every  $n$ . Hence the intersection has no elements, so it equals  $\emptyset$ .

Each  $B_n$  is non-empty (it contains  $1/(2n)$ ), and each is contained in the one before. Yet their intersection is empty. Infinite intersections can shrink to nothing even when no single step does.

De Morgan’s laws hold for indexed families with the same proofs: for a non-empty index set  $I$ ,  $A \setminus \bigcup_i B_i = \bigcap_i (A \setminus B_i)$  and  $A \setminus \bigcap_i B_i = \bigcup_i (A \setminus B_i)$ . The “or” of the finite version becomes “there exists”, and the “and” becomes “for every”, exactly as in [Definition 0.1.4](#).

## 0.3.6 Cartesian products and $n$ -tuples

A set forgets order:  $\{1, 2\} = \{2, 1\}$ . But a point in the plane is not like that. The point  $(1, 2)$  is different from  $(2, 1)$ . We need objects that remember order.

An **ordered pair**  $(a, b)$  has a first entry  $a$  and a second entry  $b$ , and two ordered pairs are equal exactly when their entries agree in order:

$$(a, b) = (a', b') \iff a = a' \text{ and } b = b'.$$

### Definition 0.3.10 — Cartesian product

Let  $A$  and  $B$  be sets. The **Cartesian product** of  $A$  and  $B$  is the set of ordered pairs

$$A \times B = \{(a, b) : a \in A, b \in B\}.$$

More generally, for sets  $A_1, \dots, A_n$ , the product  $A_1 \times \dots \times A_n$  is the set of all ordered lists  $(a_1, \dots, a_n)$  with  $a_i \in A_i$  for each  $i$ .

For example,  $\{1, 2\} \times \{x, y\} = \{(1, x), (1, y), (2, x), (2, y)\}$ . The order of the factors matters:

$(1, x) \in \{1, 2\} \times \{x, y\}$ , but  $(1, x) \notin \{x, y\} \times \{1, 2\}$ , whose pairs have a letter first. A degenerate case:  $A \times \emptyset = \emptyset$  for every set  $A$ , since no pair can have a second entry from  $\emptyset$ .

### Definition 0.3.11 — $n$ -tuples

Let  $A$  be a set and  $n \geq 1$  an integer. An  **$n$ -tuple** of elements of  $A$  is an ordered list  $(a_1, \dots, a_n)$  with each  $a_i \in A$ . The set of all of them is

$$A^n = \underbrace{A \times A \times \cdots \times A}_{n \text{ factors}}.$$

Two  $n$ -tuples are equal exactly when  $a_i = b_i$  for **every**  $i = 1, \dots, n$ .

### Example. (Sets of tuples)

Describe some elements of  $\mathbb{R}^2$ ,  $\mathbb{Z}^3$  and  $\{0, 1\}^3$ , and count the elements of  $\{0, 1\}^3$ .

*Solution.* The set  $\mathbb{R}^2$  is the plane: it contains  $(0, 0)$ ,  $(1, -2)$  and  $(\sqrt{2}, \pi)$ . The set  $\mathbb{Z}^3$  contains  $(1, 0, -4)$  but not  $(1, \frac{1}{2}, 0)$ , whose second entry is not an integer. The set  $\{0, 1\}^3$  consists of the lists of three bits:

$$\{0, 1\}^3 = \{(0, 0, 0), (0, 0, 1), (0, 1, 0), (0, 1, 1), (1, 0, 0), (1, 0, 1), (1, 1, 0), (1, 1, 1)\}.$$

There are two choices for each of the three entries, giving  $2 \cdot 2 \cdot 2 = 8$  elements. In the same way, if  $A$  has  $k$  elements then  $A^n$  has  $k^n$  elements.

In Chapter 1,  $\mathbb{R}^n$  and more generally  $F^n$  become the first examples of vector spaces, with  $n$ -tuples added and scaled entry by entry.

## 0.3.7 Power sets

Sets can themselves be elements of sets, as the warning about  $\{\emptyset\}$  showed. One such set collects every subset of a given set.

### Definition 0.3.12 — Power set

The **power set** of a set  $S$  is the set of all subsets of  $S$ :

$$\mathcal{P}(S) = \{X : X \subseteq S\}.$$

So  $X \in \mathcal{P}(S)$  means exactly  $X \subseteq S$ . For example,

$$\mathcal{P}(\{1, 2\}) = \{\emptyset, \{1\}, \{2\}, \{1, 2\}\},$$

which has 4 elements. The degenerate case is instructive:  $\mathcal{P}(\emptyset) = \{\emptyset\}$  has **one** element, because the empty set has exactly one subset, itself. For a set with  $n$  elements the power set has  $2^n$  elements; the last exercise proves this by induction.

## 0.3.8 Exercises

### 0.3.8.1 A. Check your understanding

#### Exercise A1.

Decide whether each statement is true or false. Give a one-line reason.

- (a)  $0 \in \mathbb{N}$ .
- (b)  $\emptyset \in \emptyset$ .
- (c)  $\emptyset \subseteq \emptyset$ .
- (d)  $\{0\} = \emptyset$ .
- (e)  $\{1, 2\} = \{2, 1, 1\}$  and  $(1, 2) = (2, 1)$ .
- (f)  $\{1\} \in \mathcal{P}(\{1, 2\})$  and  $\{1\} \subseteq \mathcal{P}(\{1, 2\})$ .

*Solution.*

- (a) True. By [Definition 0.3.1](#),  $\mathbb{N}$  includes 0 in this book.
- (b) False. By [Definition 0.3.3](#), nothing is an element of  $\emptyset$ , not even  $\emptyset$ .
- (c) True. The empty set is a subset of every set, vacuously, as in [Example \(Subsets\)](#).
- (d) False.  $0 \in \{0\}$  but  $0 \notin \emptyset$ , so the two sets do not have the same elements ([Definition 0.3.5](#)).
- (e) The first is true: both sets have exactly the elements 1 and 2. The second is false: ordered pairs are equal only when their first entries agree, and  $1 \neq 2$ .
- (f) The first is true:  $\{1\} \subseteq \{1, 2\}$ , so  $\{1\} \in \mathcal{P}(\{1, 2\})$  by [Definition 0.3.12](#). The second is false:  $1 \in \{1\}$ , but 1 is a number, not a subset of  $\{1, 2\}$ , so  $1 \notin \mathcal{P}(\{1, 2\})$ .

### 0.3.8.2 B. Practice

#### Exercise B1: The second De Morgan law.

Let  $A, B, C$  be sets. Prove that  $A \setminus (B \cap C) = (A \setminus B) \cup (A \setminus C)$ , which is part (b) of [Theorem 0.3.2](#).

*Solution.* We use [Theorem 0.3.1](#).

( $\subseteq$ ) Let  $x \in A \setminus (B \cap C)$ . By [Definition 0.3.8](#),  $x \in A$  and  $x \notin B \cap C$ . By [Definition 0.3.7](#), “ $x \in B$  and  $x \in C$ ” is false, so by [Theorem 0.1.2](#),  $x \notin B$  or  $x \notin C$ . If  $x \notin B$ , then  $x \in A \setminus B$ ; if  $x \notin C$ , then  $x \in A \setminus C$ . In either case  $x \in (A \setminus B) \cup (A \setminus C)$  by [Definition 0.3.6](#).

( $\supseteq$ ) Let  $x \in (A \setminus B) \cup (A \setminus C)$ . By [Definition 0.3.6](#),  $x \in A \setminus B$  or  $x \in A \setminus C$ . In the first case  $x \in A$  and  $x \notin B$ ; in the second,  $x \in A$  and  $x \notin C$ . In both cases  $x \in A$ , and “ $x \notin B$  or  $x \notin C$ ” is true, so by [Theorem 0.1.2](#) “ $x \in B$  and  $x \in C$ ” is false, that is,  $x \notin B \cap C$ . Hence  $x \in A \setminus (B \cap C)$ .

This proves the equality.



**Exercise B2: A distributive law.**

Let  $A, B, C$  be sets. Prove that  $A \cap (B \cup C) = (A \cap B) \cup (A \cap C)$ , which is part (a) of [Theorem 0.3.3](#).

*Solution.* We use [Theorem 0.3.1](#).

( $\subseteq$ ) Let  $x \in A \cap (B \cup C)$ . Then  $x \in A$  and  $x \in B \cup C$ , so  $x \in B$  or  $x \in C$ . If  $x \in B$ , then  $x \in A \cap B$ ; if  $x \in C$ , then  $x \in A \cap C$ . In either case  $x \in (A \cap B) \cup (A \cap C)$ .

( $\supseteq$ ) Let  $x \in (A \cap B) \cup (A \cap C)$ . If  $x \in A \cap B$ , then  $x \in A$  and  $x \in B \subseteq B \cup C$ . If  $x \in A \cap C$ , then  $x \in A$  and  $x \in C \subseteq B \cup C$ . In either case  $x \in A$  and  $x \in B \cup C$ , so  $x \in A \cap (B \cup C)$ .

This proves the equality.

**Exercise B3: An indexed intersection.**

Prove that

$$\bigcap_{n=1}^{\infty} \left(-\frac{1}{n}, \frac{1}{n}\right) = \{0\}.$$

You may use the fact that for every real  $x > 0$  there is an integer  $n \geq 1$  with  $1/n < x$ .

*Solution.* Write  $B_n = (-1/n, 1/n)$  and  $B = \bigcap_{n \geq 1} B_n$ . We use [Theorem 0.3.1](#).

( $\supseteq$ ) For every  $n \geq 1$ ,  $-1/n < 0 < 1/n$ , so  $0 \in B_n$ . By [Definition 0.3.9](#),  $0 \in B$ .

( $\subseteq$ ) Let  $x \in B$ . Suppose, for a contradiction, that  $x \neq 0$ . Then  $|x| > 0$ , so there is an integer  $n \geq 1$  with  $1/n < |x|$ . Then  $x \geq 1/n$  or  $x \leq -1/n$ , so  $x \notin B_n$ . This contradicts  $x \in B$ , which requires  $x \in B_n$  for **every**  $n$ . Hence  $x = 0$ , that is,  $x \in \{0\}$ .

This shows  $B = \{0\}$ .

**0.3.8.3 C. Going deeper****Exercise C1: When union equals intersection.**

Let  $A$  and  $B$  be sets. Prove that  $A \cup B = A \cap B$  if and only if  $A = B$ .

*Solution.* ( $\Leftarrow$ ) Suppose  $A = B$ . Then  $A \cup B = A \cup A$  and  $A \cap B = A \cap A$ . By [Definition 0.3.6](#) and [Definition 0.3.7](#), “ $x \in A$  or  $x \in A$ ” and “ $x \in A$  and  $x \in A$ ” are both equivalent to  $x \in A$ , so  $A \cup A = A = A \cap A$ . Hence  $A \cup B = A \cap B$ .

( $\Rightarrow$ ) Suppose  $A \cup B = A \cap B$ . We show  $A = B$  using [Theorem 0.3.1](#).

( $\subseteq$ ) Let  $x \in A$ . Then  $x \in A \cup B$  by [Definition 0.3.6](#). Since  $A \cup B = A \cap B$ , we get  $x \in A \cap B$ , and in particular  $x \in B$ .

( $\supseteq$ ) Swapping the roles of  $A$  and  $B$  in the previous paragraph, which is allowed since  $A \cup B = B \cup A$  and  $A \cap B = B \cap A$ , gives  $B \subseteq A$ .

Hence  $A = B$ . This proves the equivalence.

**Exercise C2: Counting subsets.**

For a finite set  $S$ , write  $|S|$  for its number of elements. The aim is to prove that  $|\mathcal{P}(S)| = 2^{|S|}$  for every finite set  $S$ .

- (a) Check the claim when  $S = \emptyset$ .
- (b) Let  $S$  have  $n+1$  elements, fix  $s \in S$ , and let  $S' = S \setminus \{s\}$ . Show that every subset  $X$  of  $S$  belongs to exactly one of the two collections: the subsets of  $S$  with  $s \notin X$ , and the subsets of  $S$  with  $s \in X$ .
- (c) Show that the subsets of  $S$  with  $s \notin X$  are exactly the subsets of  $S'$ , and that  $X \mapsto X \setminus \{s\}$  and  $T \mapsto T \cup \{s\}$  pair off the subsets of  $S$  containing  $s$  with the subsets of  $S'$ , each undoing the other.
- (d) Hence prove the claim by induction on  $n = |S|$ .

*Hint: in (c), use double inclusion to check that  $(X \setminus \{s\}) \cup \{s\} = X$  when  $s \in X$ .*

*Solution.*

- (a) The only subset of  $\emptyset$  is  $\emptyset$  itself, since any  $X \subseteq \emptyset$  has no elements and so equals  $\emptyset$  by [Theorem 0.3.1](#). So  $\mathcal{P}(\emptyset) = \{\emptyset\}$  has  $1 = 2^0$  element.
- (b) Let  $X \subseteq S$ . The statement  $s \in X$  is either true or false, and not both. So  $X$  lies in exactly one of the two collections.
- (c) *First collection.* If  $X \subseteq S$  and  $s \notin X$ , then every  $x \in X$  lies in  $S$  and is not  $s$ , so  $x \in S'$ ; hence  $X \subseteq S'$ . Conversely, if  $T \subseteq S'$ , then  $T \subseteq S$  and  $s \notin T$ , since no element of  $S'$  equals  $s$ . So the first collection is exactly  $\mathcal{P}(S')$ .  
*Second collection.* If  $X \subseteq S$  with  $s \in X$ , then  $X \setminus \{s\} \subseteq S'$ . If  $T \subseteq S'$ , then  $T \cup \{s\} \subseteq S$  contains  $s$ . Now we check that the two operations undo each other. For  $X$  containing  $s$ : an element lies in  $(X \setminus \{s\}) \cup \{s\}$  exactly when it is in  $X$  and not  $s$ , or it is  $s$ ; since  $s \in X$ , this is exactly when it is in  $X$ . So  $(X \setminus \{s\}) \cup \{s\} = X$ . For  $T \subseteq S'$ : an element lies in  $(T \cup \{s\}) \setminus \{s\}$  exactly when it is in  $T$  or equal to  $s$ , and not equal to  $s$ ; since  $s \notin T$ , this is exactly when it is in  $T$ . So  $(T \cup \{s\}) \setminus \{s\} = T$ . Hence the two operations pair the subsets of  $S$  containing  $s$  one-to-one with the subsets of  $S'$ , and the two collections have the same number of elements.
- (d) Let  $P(n)$  be the statement “every set with  $n$  elements has exactly  $2^n$  subsets”. We use [Theorem 0.2.7](#) with  $n_0 = 0$ . The base case  $P(0)$  is (a), since the only set with no elements is  $\emptyset$ . Let  $n \geq 0$ , suppose  $P(n)$ , and let  $S$  have  $n+1$  elements. Fix  $s \in S$ , which exists since  $n+1 \geq 1$ , and let  $S' = S \setminus \{s\}$ , a set with  $n$  elements. By (b), the number of subsets of  $S$  is the size of the first collection plus the size of the second. By (c) and the induction hypothesis, each of these sizes equals  $|\mathcal{P}(S')| = 2^n$ . Hence  $S$  has  $2^n + 2^n = 2^{n+1}$  subsets, which is  $P(n+1)$ . By [Theorem 0.2.7](#),  $P(n)$  holds for every  $n \in \mathbb{N}$ , as claimed.

## 0.4 Functions

Sets are the objects of this chapter. The next question is how to move between them, and the tool is a function. Almost everything in linear algebra is a question about a function: a linear map is a function, a matrix will turn out to encode one, and “does  $\mathbf{A}\mathbf{x} = \mathbf{b}$  have a solution, and

is it unique?” is a question about whether a function is surjective and injective. This section fixes the language of functions, composition and inverses, and proves the handful of facts about them that the rest of the book uses without comment.

## 0.4.1 What a function is

At school a function is usually a formula, such as  $x^2 + 1$ . A formula alone is not enough. The same formula  $x^2$  behaves differently depending on which numbers we feed it and where we allow the answers to land, and whether an equation  $x^2 = c$  can be solved depends on exactly that information. So a function carries three pieces of data, not one.

*A function is a machine with a fixed input set and a fixed target set that turns each input into exactly one output.*

### Definition 0.4.1 — Function

Let  $X$  and  $Y$  be sets. A **function** (or **map**)  $f: X \rightarrow Y$  is a rule that assigns to **every** element  $x \in X$  **exactly one** element  $f(x) \in Y$ . The set  $X$  is the **domain** of  $f$ , the set  $Y$  is its **codomain**, and  $f(x)$  is the **value** of  $f$  at  $x$ . We also write  $x \mapsto f(x)$ .

Two functions  $f: X \rightarrow Y$  and  $g: X' \rightarrow Y'$  are **equal** if  $X = X'$ ,  $Y = Y'$ , and  $f(x) = g(x)$  for every  $x \in X$ .

In words: the first clause says every input gets an output, so nothing in the domain is left out. The second says the output is unique, so the rule never hesitates between two values. The codomain is where outputs are *allowed* to land; nothing says every element of  $Y$  must be hit.

If you want a definition that avoids the vague word “rule”, use the Cartesian product ([Definition 0.3.10](#)): a function  $X \rightarrow Y$  is a subset  $G \subseteq X \times Y$  such that for every  $x \in X$  there is exactly one  $y \in Y$  with  $(x, y) \in G$ . The set  $G$  is the **graph** of  $f$ , and  $f(x)$  is that unique  $y$ . Nothing in this book depends on which version you prefer.

### Example. (First examples)

Check each of the following against the two clauses of [Definition 0.4.1](#).

- (a)  $f: \mathbb{R} \rightarrow \mathbb{R}$ ,  $f(x) = 3x - 5$ . Every real  $x$  gives one real number  $3x - 5$ .
- (b)  $s: \mathbb{R}^2 \rightarrow \mathbb{R}$ ,  $s(x, y) = x + y$ . The domain is a Cartesian product; each pair gives one sum.
- (c)  $\ell: \{\text{finite strings of the letters } a, b\} \rightarrow \mathbb{N}$ , sending a string to its length. The empty string has length 0, which is why it matters that  $\mathbb{N}$  contains 0.
- (d) For any set  $X$ , the rule  $x \mapsto x$  is a function  $X \rightarrow X$ . It looks too simple to matter, but it will play the role that the number 1 plays for multiplication.
- (e) For any set  $Y$ , there is exactly one function  $\emptyset \rightarrow Y$ , the **empty function**. Both clauses are about elements of the domain, and there are none, so they hold vacuously. It is a degenerate case that makes statements like “for every set  $X$ ” true without exceptions.

The failures are just as instructive. The rule “ $x \mapsto$  the real number  $y$  with  $y^2 = x$ ” is **not** a

function  $\mathbb{R} \rightarrow \mathbb{R}$ . At  $x = -1$  there is no such  $y$ , so the first clause fails. At  $x = 4$  there are two,  $y = 2$  and  $y = -2$ , so the second clause fails. Repairing both problems gives a genuine function  $[0, \infty) \rightarrow \mathbb{R}$ ,  $x \mapsto \sqrt{x}$ , where  $\sqrt{x}$  means the non-negative root.

The equality clause says the codomain is part of the function. The functions  $\mathbb{R} \rightarrow \mathbb{R}$ ,  $x \mapsto x^2$  and  $\mathbb{R} \rightarrow [0, \infty)$ ,  $x \mapsto x^2$  have the same values but are **different** functions. This is not pedantry: we will see in a moment that one of them is surjective and the other is not.

*Quick check.* Let  $f, g: \mathbb{R} \rightarrow \mathbb{R}$  be  $f(x) = |x|$  and  $g(x) = \sqrt{x^2}$ . Are  $f$  and  $g$  equal?

*Solution.* Yes. They have the same domain and the same codomain, and for every real  $x$  we have  $\sqrt{x^2} = |x|$ , because  $|x|$  is the non-negative number whose square is  $x^2$ . Different formulas can describe the same function; what counts is the values.

## 0.4.2 Images and preimages

Often we want to push a whole set through a function, or ask which inputs land in a given region. Both operations get a name.

### Definition 0.4.2 — Image and preimage

Let  $f: X \rightarrow Y$  be a function,  $A \subseteq X$  and  $B \subseteq Y$ .

- The **image** of  $A$  under  $f$  is  $f(A) := \{f(a) : a \in A\} \subseteq Y$ . The set  $f(X)$  is called the **image of  $f$** .
- The **preimage** of  $B$  under  $f$  is  $f^{-1}(B) := \{x \in X : f(x) \in B\} \subseteq X$ .

In words:  $f(A)$  collects the outputs of the inputs in  $A$ . The preimage  $f^{-1}(B)$  collects every input whose output lies in  $B$ . So  $y \in f(A)$  means “ $y = f(a)$  for **some**  $a \in A$ ”, while  $x \in f^{-1}(B)$  means “ $f(x) \in B$ ”, with no existential quantifier at all.

### Example. (Images and preimages of the squaring map)

Let  $q: \mathbb{R} \rightarrow \mathbb{R}$ ,  $q(x) = x^2$ . Compute  $q([-1, 2])$ ,  $q^{-1}([1, 4])$  and  $q^{-1}([-5, -1])$ .

*Solution.* As  $x$  runs over  $[-1, 2]$ , the square  $x^2$  takes every value in  $[0, 4]$ : values in  $[0, 1]$  come from  $[0, 1]$ , values in  $[1, 4]$  from  $[1, 2]$ , and no square is negative or exceeds  $2^2 = 4$  on this interval. Hence  $q([-1, 2]) = [0, 4]$ .

Next,  $x^2 \in [1, 4]$  means  $1 \leq |x| \leq 2$ . Hence  $q^{-1}([1, 4]) = [-2, -1] \cup [1, 2]$ .

Finally, no real square lies in  $[-5, -1]$ , so  $q^{-1}([-5, -1]) = \emptyset$ . A preimage may be empty; that is not an error.

*Warning.* The symbol  $f^{-1}(B)$  is defined for **every** function, invertible or not; it does not mean that an inverse function exists. For the same reason, pushing forward and pulling back need not undo each other. With  $q(x) = x^2$  on  $\mathbb{R}$ : for  $B = [-1, 4]$  we get  $q^{-1}(B) = [-2, 2]$  and  $q(q^{-1}(B)) = [0, 4] \neq B$ ; for  $A = [0, 1]$  we get  $q(A) = [0, 1]$  and  $q^{-1}(q(A)) = [-1, 1] \neq A$ .

What does survive is one inclusion in each direction.

### Proposition 0.4.1 — Image and preimage inclusions

Let  $f: X \rightarrow Y$ ,  $A \subseteq X$  and  $B \subseteq Y$ . Then

$$f(f^{-1}(B)) \subseteq B \quad \text{and} \quad A \subseteq f^{-1}(f(A)).$$

*Proof.* Let  $y \in f(f^{-1}(B))$ . By Definition 0.4.2,  $y = f(x)$  for some  $x \in f^{-1}(B)$ , and  $x \in f^{-1}(B)$  means  $f(x) \in B$ . Hence  $y \in B$ .

Let  $a \in A$ . Then  $f(a) \in f(A)$  by definition of the image, so  $a \in f^{-1}(f(A))$  by definition of the preimage. This proves both inclusions. ■

The warning shows both inclusions can be strict. In [Exercise \(C3: When images and preimages undo each other\)](#) you will find exactly which property of  $f$  turns each one into an equality.

## 0.4.3 Injective, surjective, bijective

Given  $f: X \rightarrow Y$  and a target  $y \in Y$ , the most basic question is how many solutions the equation  $f(x) = y$  has. There are two separate issues: whether there is **at least one** solution, and whether there is **at most one**. In linear algebra these become “does  $A\mathbf{x} = \mathbf{b}$  have a solution?” and “is the solution unique?”, and they are answered by different tools. So the two halves deserve separate names.

*Injective means no two inputs collide; surjective means every target is hit; bijective means both, so every equation  $f(x) = y$  has exactly one solution.*

### Definition 0.4.3 — Injective, surjective, bijective

Let  $f: X \rightarrow Y$  be a function.

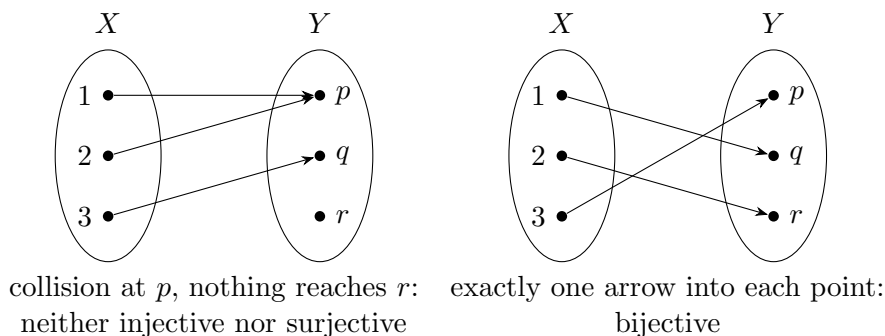
- $f$  is **injective** (one-to-one) if **for all**  $x_1, x_2 \in X$ ,  $f(x_1) = f(x_2)$  implies  $x_1 = x_2$ .
- $f$  is **surjective** (onto) if **for every**  $y \in Y$  **there exists**  $x \in X$  with  $f(x) = y$ .  
Equivalently,  $f(X) = Y$ .
- $f$  is **bijective** if it is both injective and surjective.

In words: injectivity says that if two inputs give the same output, they were the same input. Equivalently, by the contrapositive ([Theorem 0.1.1](#)), distinct inputs give distinct outputs. Surjectivity says the image of  $f$  fills the whole codomain. The quantifiers matter: in surjectivity the  $x$  is allowed to depend on  $y$ .

Negating these, as in [Example \(What “not injective” means\)](#) and [Example \(What “not surjective” means\)](#), tells us what a counterexample looks like. The map  $f$  is **not injective** when there exist  $x_1 \neq x_2$  with  $f(x_1) = f(x_2)$ : one collision is enough. It is **not surjective** when there exists  $y \in Y$  that equals  $f(x)$  for no  $x \in X$ : one missed target is enough.

**The standard moves.** To prove  $f$  injective, write “Suppose  $f(x_1) = f(x_2)$ ” and work towards

The picture below shows maps between three-element sets, drawn as arrows. A collision is two arrows landing on one point; a missed target is a point with no arrow.



Determine which of the following functions are injective, surjective or bijective.

- (a)  $f: \mathbb{R} \rightarrow \mathbb{R}$ ,  $f(x) = 3x - 5$ .  
 (b)  $g: \mathbb{Z} \rightarrow \mathbb{Z}$ ,  $g(n) = 2n$ .  
 (c)  $s: \mathbb{R}^2 \rightarrow \mathbb{R}$ ,  $s(x, y) = x + y$ .  
 (d)  $\text{id}_X: X \rightarrow X$ ,  $x \mapsto x$ , for any set  $X$ .

*Solution.*

- (a) Bijective. Suppose  $f(x_1) = f(x_2)$ . Then  $3x_1 - 5 = 3x_2 - 5$ , so  $3x_1 = 3x_2$ , and dividing by  $3 \neq 0$  gives  $x_1 = x_2$ . Hence  $f$  is injective. Let  $y \in \mathbb{R}$ . Solving  $3x - 5 = y$  suggests  $x = (y + 5)/3$ , and indeed  $f((y + 5)/3) = (y + 5) - 5 = y$ . Hence  $f$  is surjective.
- (b) Injective, not surjective. Suppose  $2n_1 = 2n_2$ ; then  $n_1 = n_2$ . But 1 is not hit:  $2n = 1$  has no integer solution, since  $2n$  is even. Note that the same formula  $x \mapsto 2x$  on  $\mathbb{R} \rightarrow \mathbb{R}$  is surjective; the failure comes from the domain and codomain, not the formula.
- (c) Surjective, not injective. Let  $t \in \mathbb{R}$ ; then  $s(t, 0) = t$ , so  $s$  is surjective. But  $s(1, 0) = 1 = s(0, 1)$  while  $(1, 0) \neq (0, 1)$ , a collision, so  $s$  is not injective.
- (d) Bijective. If  $\text{id}_X(x_1) = \text{id}_X(x_2)$  then  $x_1 = x_2$  directly, and every  $x \in X$  is hit by itself. This degenerate case is the model of a bijection and reappears as the identity function below.

The cleanest way to see that domain and codomain are part of the question is to keep the formula  $x \mapsto x^2$  fixed and change only the sets. Write  $[0, \infty)$  for the non-negative reals.

| Function                             | Injective?         | Surjective?              |
|--------------------------------------|--------------------|--------------------------|
| $\mathbb{R} \rightarrow \mathbb{R}$  | no: $(-1)^2 = 1^2$ | no: $-1$ is not a square |
| $\mathbb{R} \rightarrow [0, \infty)$ | no: $(-1)^2 = 1^2$ | yes: $y = (\sqrt{y})^2$  |

| Function                              | Injective? | Surjective?              |
|---------------------------------------|------------|--------------------------|
| $[0, \infty) \rightarrow \mathbb{R}$  | yes        | no: $-1$ is not a square |
| $[0, \infty) \rightarrow [0, \infty)$ | yes        | yes                      |

For the injective rows: if  $x_1, x_2 \geq 0$  and  $x_1^2 = x_2^2$ , then  $(x_1 - x_2)(x_1 + x_2) = 0$ . Either  $x_1 = x_2$ , or  $x_1 + x_2 = 0$ , which for non-negative numbers forces  $x_1 = x_2 = 0$ . In both cases  $x_1 = x_2$ .

Read the table as a set of minimal changes. Shrinking the domain from  $\mathbb{R}$  to  $[0, \infty)$  removes the collision between  $-1$  and  $1$ , so the injectivity clause now holds. Shrinking the codomain removes the missed targets, so the surjectivity clause now holds. The formula never changed.

*Warning.* Do not confuse injectivity with the statement “ $x_1 = x_2 \Rightarrow f(x_1) = f(x_2)$ ”. That implication is true for **every** function; it is the second clause of [Definition 0.4.1](#). Injectivity is its converse, “ $f(x_1) = f(x_2) \Rightarrow x_1 = x_2$ ”, and  $x \mapsto x^2$  on  $\mathbb{R}$  satisfies the first but not the second.

**Why these definitions.** Surjectivity depends on the codomain, and injectivity on the domain, so neither is a property of a formula alone; this is why [Definition 0.4.1](#) insists on both sets. The names are descriptive: “in-jective” throws the domain *into* the codomain without collisions, and “sur-jective” (French *sur*, onto) lands *onto* all of it. In Chapter 3 we will meet a fast test for linear maps: a linear map is injective exactly when the only input sent to  $\mathbf{0}$  is  $\mathbf{0}$ . That test is special to linear maps and fails for general functions, which is one reason the general definitions come first.

## 0.4.4 Composition

If the outputs of one function can be fed into another, we can run them in sequence. This is the operation that matrix multiplication will describe in Chapter 3.

### Definition 0.4.4 — Composition

Let  $f: X \rightarrow Y$  and  $g: Y \rightarrow Z$  be functions. The **composition**  $g \circ f: X \rightarrow Z$  is defined by

$$(g \circ f)(x) := g(f(x)) \quad \text{for every } x \in X.$$

Read  $g \circ f$  as “ $g$  after  $f$ ”: the function written on the **right** acts **first**. The definition needs the codomain of  $f$  to equal the domain of  $g$ , so that  $g(f(x))$  makes sense.

### Example. (Order matters)

- (a) Let  $f, g: \mathbb{R} \rightarrow \mathbb{R}$  be  $f(x) = x + 3$  and  $g(x) = 2x$ . Compute  $g \circ f$  and  $f \circ g$ .
- (b) Let  $q: \mathbb{R} \rightarrow [0, \infty)$ ,  $q(x) = x^2$ , and  $r: [0, \infty) \rightarrow \mathbb{R}$ ,  $r(y) = \sqrt{y}$ . Compute  $r \circ q$  and  $q \circ r$ .

*Solution.*

- (a)  $(g \circ f)(x) = g(x + 3) = 2x + 6$ , while  $(f \circ g)(x) = f(2x) = 2x + 3$ . At  $x = 0$  these

are  $6 \neq 3$ , so  $g \circ f \neq f \circ g$ . Composition is not commutative, even when both orders make sense.

- (b)  $r \circ q: \mathbb{R} \rightarrow \mathbb{R}$  is  $x \mapsto \sqrt{x^2} = |x|$ . In the other order,  $q \circ r: [0, \infty) \rightarrow [0, \infty)$  is  $y \mapsto (\sqrt{y})^2 = y$ . So  $q \circ r$  does nothing, while  $r \circ q$  forgets the sign of  $x$ . The two orders do not even have the same domain. We will return to this pair when we discuss one-sided inverses.

Although the order matters, the grouping does not. This is what lets us write  $h \circ g \circ f$  without brackets.

### Theorem 0.4.2 — Associativity of composition

Let  $f: X \rightarrow Y$ ,  $g: Y \rightarrow Z$  and  $h: Z \rightarrow W$  be functions. Then  $(h \circ g) \circ f = h \circ (g \circ f)$ .

*Proof.* Both sides are functions  $X \rightarrow W$ . Let  $x \in X$ . By Definition 0.4.4 applied twice on each side,

$$((h \circ g) \circ f)(x) = (h \circ g)(f(x)) = h(g(f(x))) = h((g \circ f)(x)) = (h \circ (g \circ f))(x).$$

Since the values agree at every  $x$ , the two functions are equal by Definition 0.4.1. ■

Next, the properties of the previous subsection pass through composition.

### Theorem 0.4.3 — Composition preserves injectivity and surjectivity

Let  $f: X \rightarrow Y$  and  $g: Y \rightarrow Z$  be functions.

- (a) If  $f$  and  $g$  are injective, then  $g \circ f$  is injective.
- (b) If  $f$  and  $g$  are surjective, then  $g \circ f$  is surjective.
- (c) If  $f$  and  $g$  are bijective, then  $g \circ f$  is bijective.

*Idea.* For (a), start from a collision of  $g \circ f$  and peel off one function at a time, outermost first: injectivity of  $g$  removes  $g$ , then injectivity of  $f$  removes  $f$ . For (b), run backwards: reach  $z$  from some  $y$  using  $g$ , then reach that  $y$  from some  $x$  using  $f$ .

*Proof.*

- (a) Suppose  $(g \circ f)(x_1) = (g \circ f)(x_2)$ , that is,  $g(f(x_1)) = g(f(x_2))$ . Since  $g$  is injective,  $f(x_1) = f(x_2)$ . Since  $f$  is injective,  $x_1 = x_2$ . Hence  $g \circ f$  is injective.
- (b) Let  $z \in Z$ . Since  $g$  is surjective, there is  $y \in Y$  with  $g(y) = z$ . Since  $f$  is surjective, there is  $x \in X$  with  $f(x) = y$ . Then  $(g \circ f)(x) = g(y) = z$ . Hence  $g \circ f$  is surjective.
- (c) If  $f$  and  $g$  are bijective, then  $g \circ f$  is injective by (a) and surjective by (b), so it is bijective. ■



The converse of (a) is false, but half of it survives, and that half is surprisingly useful. Suppose  $g \circ f$  is injective. A collision of  $f$  would be a collision of  $g \circ f$ , so  $f$  must be injective; but  $g$  is only used on the image  $f(X)$ , so  $g$  may still collide outside it. The surjective case is dual.

### Theorem 0.4.4 — Partial converse for compositions

Let  $f: X \rightarrow Y$  and  $g: Y \rightarrow Z$  be functions.

- (a) If  $g \circ f$  is injective, then  $f$  is injective.
- (b) If  $g \circ f$  is surjective, then  $g$  is surjective.

*Proof.*

- (a) Suppose  $f(x_1) = f(x_2)$ . Applying  $g$  to both sides gives  $(g \circ f)(x_1) = (g \circ f)(x_2)$ . Since  $g \circ f$  is injective,  $x_1 = x_2$ . Hence  $f$  is injective.
- (b) Let  $z \in Z$ . Since  $g \circ f$  is surjective, there is  $x \in X$  with  $g(f(x)) = z$ . Put  $y = f(x) \in Y$ ; then  $g(y) = z$ . Hence  $g$  is surjective.

■

*Remark.* The other halves really fail. Take  $X = Z = \{1\}$ ,  $Y = \{1, 2\}$ ,  $f(1) = 1$  and  $g(1) = g(2) = 1$ . Then  $g \circ f = \text{id}_{\{1\}}$  is bijective, yet  $g$  is not injective ( $g(1) = g(2)$ ) and  $f$  is not surjective (nothing maps to 2).

## 0.4.5 Identity and one-sided inverses

The function that does nothing deserves a name, because “undoing  $f$ ” means “composing with  $f$  gives the function that does nothing”.

### Definition 0.4.5 — Identity function

For a set  $X$ , the **identity function** on  $X$  is  $\text{id}_X: X \rightarrow X$ ,  $\text{id}_X(x) = x$ .

For every  $f: X \rightarrow Y$  we have  $f \circ \text{id}_X = f$  and  $\text{id}_Y \circ f = f$ , since both sides send  $x$  to  $f(x)$ . Notice the subscripts: on the right of  $f$  the identity lives on the domain, on the left on the codomain.

Now look again at [Example \(Order matters\)](#) (b). There  $q \circ r = \text{id}_{[0, \infty)}$ , but  $r \circ q = |\cdot| \neq \text{id}_{\mathbb{R}}$ . So  $r$  undoes  $q$  from one side only. This one-sided behavior is common enough to need a word.

### Definition 0.4.6 — Left and right inverses

Let  $f: X \rightarrow Y$ . A function  $g: Y \rightarrow X$  is a **left inverse** of  $f$  if  $g \circ f = \text{id}_X$ , and a **right inverse** of  $f$  if  $f \circ g = \text{id}_Y$ .

“Left” and “right” refer to where  $g$  is written in the composition. A left inverse recovers the

input from the output:  $g(f(x)) = x$ . A right inverse picks, for each target  $y$ , an input that reaches it:  $f(g(y)) = y$ . In [Example \(Order matters\)](#) (b),  $r$  is a right inverse of  $q$ , and  $q$  is a left inverse of  $r$ . But  $r$  is **not** a left inverse of  $q$ , because  $r(q(-1)) = 1 \neq -1$ .

The descriptions “recovers the input” and “reaches every target” should remind you of injective and surjective. They are exactly that.

### Theorem 0.4.5 — Left inverses and injectivity

Let  $f: X \rightarrow Y$  with  $X$  **non-empty**. Then  $f$  has a left inverse if and only if  $f$  is injective.

*Idea.*  $(\Rightarrow)$  is [Theorem 0.4.4](#) with  $g \circ f = \text{id}_X$ , which is injective. For  $(\Leftarrow)$  we must define  $g(y)$  for every  $y \in Y$ . If  $y$  is in the image, injectivity gives a **unique**  $x$  with  $f(x) = y$ , and  $g(y)$  must be that  $x$ . If  $y$  is not in the image, the condition  $g \circ f = \text{id}_X$  says nothing about  $g(y)$ , so we send  $y$  anywhere in  $X$ . That is where non-emptiness is used.

*Proof.*  $(\Rightarrow)$  Suppose  $g \circ f = \text{id}_X$ . The identity  $\text{id}_X$  is injective, so  $f$  is injective by [Theorem 0.4.4](#) (a).

$(\Leftarrow)$  Suppose  $f$  is injective. Since  $X \neq \emptyset$ , we may fix some  $x_0 \in X$ . Define  $g: Y \rightarrow X$  as follows. If  $y \in f(X)$ , there is some  $x \in X$  with  $f(x) = y$ , and this  $x$  is unique because  $f$  is injective; set  $g(y) = x$ . If  $y \notin f(X)$ , set  $g(y) = x_0$ . Every  $y$  receives exactly one value, so  $g$  is a function. For  $x \in X$ , the element  $y = f(x)$  lies in  $f(X)$ , and the unique input sent to it is  $x$ , so  $g(f(x)) = x$ . Hence  $g \circ f = \text{id}_X$ , as claimed. ■

The hypothesis  $X \neq \emptyset$  is not decoration. The empty function  $\emptyset \rightarrow \{1\}$  is injective (there are no two inputs to collide), but there is no function  $\{1\} \rightarrow \emptyset$  at all, since 1 would have nowhere to go.

### Theorem 0.4.6 — Right inverses and surjectivity

Let  $f: X \rightarrow Y$ . Then  $f$  has a right inverse if and only if  $f$  is surjective.

*Proof.*  $(\Rightarrow)$  Suppose  $f \circ g = \text{id}_Y$ . The identity  $\text{id}_Y$  is surjective, so  $f$  is surjective by [Theorem 0.4.4](#) (b).

$(\Leftarrow)$  Suppose  $f$  is surjective. For each  $y \in Y$  the set  $f^{-1}(\{y\})$  is non-empty, so we may choose one element of it and call it  $g(y)$ . This defines  $g: Y \rightarrow X$  with  $f(g(y)) = y$  for every  $y \in Y$ , that is,  $f \circ g = \text{id}_Y$ . ■

When  $Y$  is infinite, the step “choose one element from each of infinitely many non-empty sets at once” is an axiom of set theory, the **axiom of choice**; we use it freely and never mention it again in this chapter.

Unlike an inverse in the usual sense, one-sided inverses are rarely unique. For  $q: \mathbb{R} \rightarrow [0, \infty)$ ,  $x \mapsto x^2$ , both  $y \mapsto \sqrt{y}$  and  $y \mapsto -\sqrt{y}$  are right inverses: each picks one of the two square roots.

*Quick check.* Give a function  $f: \mathbb{N} \rightarrow \mathbb{N}$  that has a left inverse but no right inverse.

*Solution.* Take  $f(n) = n + 1$ . It is injective, since  $n_1 + 1 = n_2 + 1$  gives  $n_1 = n_2$ , so by [Theorem 0.4.5](#) it has a left inverse; explicitly,  $g(m) = m - 1$  for  $m \geq 1$  and  $g(0) = 0$  satisfies  $g(f(n)) = n$ . It is not surjective, since  $n + 1 = 0$  has no solution in  $\mathbb{N}$ . So by [Theorem 0.4.6](#) it has no right inverse.

## 0.4.6 Inverses

When a function has an inverse on both sides, the two one-sided inverses turn out to coincide, and we get the notion of inverse we know from  $e^x$  and  $\ln x$ .

### Definition 0.4.7 — Inverse function

Let  $f: X \rightarrow Y$ . A function  $g: Y \rightarrow X$  is an **inverse** of  $f$  if

$$g \circ f = \text{id}_X \quad \text{and} \quad f \circ g = \text{id}_Y.$$

If  $f$  has an inverse,  $f$  is **invertible**.

Both conditions are needed: [Example \(Order matters\)](#) (b) shows a pair with one but not the other. The next theorem says which functions are invertible and that “the” inverse is justified.

### Theorem 0.4.7 — Bijective if and only if invertible

A function  $f: X \rightarrow Y$  is invertible if and only if it is bijective. Moreover, an inverse of  $f$ , when it exists, is **unique**; we denote it  $f^{-1}: Y \rightarrow X$ .

*Idea.* For  $(\Leftarrow)$  one could combine the two one-sided theorems, but that route needs  $X \neq \emptyset$  and a choice. A bijection makes both unnecessary: every  $y$  has **exactly one** preimage, so  $f^{-1}(y)$  is forced. For uniqueness, the standard move is to assume two inverses  $g$  and  $h$ , and sandwich:  $g = g \circ (f \circ h) = (g \circ f) \circ h = h$ .

*Proof.*  $(\Rightarrow)$  Suppose  $g$  is an inverse of  $f$ . Since  $g \circ f = \text{id}_X$  is injective,  $f$  is injective by [Theorem 0.4.4](#) (a). Since  $f \circ g = \text{id}_Y$  is surjective,  $f$  is surjective by [Theorem 0.4.4](#) (b). Hence  $f$  is bijective.

$(\Leftarrow)$  Suppose  $f$  is bijective, and let  $y \in Y$ . Since  $f$  is surjective, some  $x \in X$  has  $f(x) = y$ , and since  $f$  is injective, this  $x$  is unique. Define  $g(y)$  to be this  $x$ . Then  $f(g(y)) = y$  for every  $y \in Y$ , so  $f \circ g = \text{id}_Y$ . For  $x \in X$ , the unique input sent to  $f(x)$  is  $x$  itself, so  $g(f(x)) = x$ , and  $g \circ f = \text{id}_X$ . Hence  $g$  is an inverse of  $f$ .

(Uniqueness) Suppose  $g$  and  $h$  are both inverses of  $f$ . Then

$$g = g \circ \text{id}_Y = g \circ (f \circ h) = (g \circ f) \circ h = \text{id}_X \circ h = h,$$

where the third equality is [Theorem 0.4.2](#). This proves the theorem. ■

The uniqueness computation used only  $g \circ f = \text{id}_X$  and  $f \circ h = \text{id}_Y$ . So it proves more: **if  $f$  has a left inverse  $g$  and a right inverse  $h$ , then  $g = h$** , and this common function is the inverse. The same sandwich will prove that a matrix has at most one inverse.

*Warning.* The notation  $f^{-1}$  now has two meanings. The preimage  $f^{-1}(B)$  of a **set**  $B \subseteq Y$  exists for every  $f$ . The inverse function  $f^{-1}$ , applied to an **element**  $y \in Y$ , exists only for bijective  $f$ . Writing  $q^{-1}(4)$  for  $q(x) = x^2$  on  $\mathbb{R}$  is meaningless, while  $q^{-1}(\{4\}) = \{-2, 2\}$  is fine. When  $f$  is bijective the two agree:  $f^{-1}(\{y\}) = \{f^{-1}(y)\}$ .

### Example. (Finding inverses)

For each function, decide whether it is invertible, and if so find its inverse.

- (a)  $f: \mathbb{R} \rightarrow \mathbb{R}, f(x) = 5 - 2x$ .
- (b)  $t: \mathbb{R}^2 \rightarrow \mathbb{R}^2, t(x, y) = (y, x)$ .
- (c)  $g: \mathbb{R} \rightarrow \mathbb{R}, g(x) = x^2$ .
- (d)  $h: [0, \infty) \rightarrow [0, \infty), h(x) = x^2$ .

*Solution.*

- (a) Solving  $5 - 2x = y$  gives  $x = (5 - y)/2$ . Put  $k(y) = (5 - y)/2$ . Check:  $k(f(x)) = (5 - (5 - 2x))/2 = x$  and  $f(k(y)) = 5 - (5 - y) = y$ . So  $k$  is an inverse of  $f$ , and  $f^{-1}(y) = (5 - y)/2$ .
- (b)  $t(t(x, y)) = t(y, x) = (x, y)$ , so  $t \circ t = \text{id}_{\mathbb{R}^2}$ . Hence  $t$  is its own inverse:  $t^{-1} = t$ .
- (c) Not invertible.  $g(-1) = g(1)$ , so  $g$  is not injective, hence not bijective, and [Theorem 0.4.7](#) says it has no inverse.
- (d) Invertible. By the table in the previous subsection  $h$  is bijective, and the inverse is  $h^{-1}(y) = \sqrt{y}$ : indeed  $\sqrt{x^2} = x$  for  $x \geq 0$  and  $(\sqrt{y})^2 = y$  for  $y \geq 0$ .

To undo two steps, undo the last one first. When you dress, the shirt goes on before the jacket; to undress, the jacket comes off first.

### Theorem 0.4.8 — Inverse of a composition

Let  $f: X \rightarrow Y$  and  $g: Y \rightarrow Z$  be bijective. Then  $g \circ f$  is bijective and

$$(g \circ f)^{-1} = f^{-1} \circ g^{-1}.$$

*Proof.* Using [Theorem 0.4.2](#) to regroup,

$$(f^{-1} \circ g^{-1}) \circ (g \circ f) = f^{-1} \circ (g^{-1} \circ g) \circ f = f^{-1} \circ \text{id}_Y \circ f = f^{-1} \circ f = \text{id}_X,$$

and in the same way  $(g \circ f) \circ (f^{-1} \circ g^{-1}) = g \circ (f \circ f^{-1}) \circ g^{-1} = g \circ \text{id}_X \circ g^{-1} = g \circ g^{-1} = \text{id}_Z$ . Hence  $f^{-1} \circ g^{-1}$  is an inverse of  $g \circ f$ . By [Theorem 0.4.7](#),  $g \circ f$  is bijective and its unique inverse is  $f^{-1} \circ g^{-1}$ , as claimed. ■

The same reversal will appear for matrices as  $(\mathbf{AB})^{-1} = \mathbf{B}^{-1}\mathbf{A}^{-1}$ , and for the same reason.

## 0.4.7 Maps between finite sets

For functions between two finite sets **of the same size**, one of the two conditions comes for free. This is the pigeonhole principle, and it is the ancestor of a move used constantly from Chapter 3 on: instead of checking both injectivity and surjectivity, count, and check only one.

We write  $|X|$  for the number of elements of a finite set  $X$ . We use two counting facts about lists without proof: a list of  $n$  entries with a repeated entry contains at most  $n - 1$  distinct elements, and a subset of a finite set  $Y$  with  $|Y|$  elements is all of  $Y$ .

### Theorem 0.4.9 — Injective iff surjective for finite sets of equal size

Let  $X$  and  $Y$  be finite sets with  $|X| = |Y|$ , and let  $f: X \rightarrow Y$ . Then  $f$  is injective if and only if  $f$  is surjective. In that case  $f$  is bijective.

*Idea.* List the elements of  $X$  as  $x_1, \dots, x_n$ ; the image  $f(X)$  is then listed by  $f(x_1), \dots, f(x_n)$ . Injective means this list has no repeats, so it has  $n$  distinct entries and must fill  $Y$ . Surjective means the list fills  $Y$ , which has  $n$  elements, so there is no room for a repeat.

*Proof.* Let  $n = |X| = |Y|$ , and write  $X = \{x_1, \dots, x_n\}$  with the  $x_i$  distinct. By Definition 0.4.2,  $f(X)$  consists of the entries of the list  $f(x_1), \dots, f(x_n)$ .

( $\Rightarrow$ ) Suppose  $f$  is injective. Since the  $x_i$  are distinct, the entries  $f(x_1), \dots, f(x_n)$  are distinct. Hence  $f(X)$  is a subset of  $Y$  with  $n = |Y|$  elements, so  $f(X) = Y$ , and  $f$  is surjective.

( $\Leftarrow$ ) We prove the contrapositive (Theorem 0.1.1). Suppose  $f$  is not injective. Then  $f(x_i) = f(x_j)$  for some  $i \neq j$ , so the list  $f(x_1), \dots, f(x_n)$  has a repeated entry and  $f(X)$  has at most  $n - 1$  elements. Hence  $f(X) \neq Y$ , and  $f$  is not surjective.

So if  $f$  is injective or surjective, the two directions show that it is both, that is, bijective. ■

For a finite set  $X$ , this applies to every  $f: X \rightarrow X$ : such a map is injective if and only if it is surjective. In Chapter 3 the same shape of statement returns for linear maps  $V \rightarrow V$  on a finite-dimensional space, with dimension playing the role of the number of elements.

*Warning.* The theorem needs finiteness. The map  $g: \mathbb{N} \rightarrow \mathbb{N}$ ,  $g(n) = 2n$ , is injective but misses every odd number, so it is not surjective, even though domain and codomain are the same set. Infinite sets have room for an injection into a proper part of themselves.

## 0.4.8 Exercises

### 0.4.8.1 A. Check your understanding

#### Exercise A1.

- (a) State what it means for  $f: X \rightarrow Y$  to be injective, and what it means to be surjective.
- (b) True or false: the functions  $\mathbb{R} \rightarrow \mathbb{R}$ ,  $x \mapsto x^3$  and  $\mathbb{R} \rightarrow \mathbb{R}$ ,  $x \mapsto x \cdot x \cdot x$  are equal. Give a reason.
- (c) True or false: if  $g \circ f$  is injective, then  $g$  is injective. Give a reason.
- (d) Write down the first line of a proof that a function  $f$  is injective, and the first line of a proof that it is surjective.
- (e) True or false: the preimage  $f^{-1}(B)$  is defined only when  $f$  is bijective.
- (f) True or false: every injective function  $\{1, 2, 3\} \rightarrow \{1, 2, 3\}$  is bijective. Name the result you use.

*Solution.*

- (a)  $f$  is injective if for all  $x_1, x_2 \in X$ ,  $f(x_1) = f(x_2)$  implies  $x_1 = x_2$ . It is surjective if for every  $y \in Y$  there exists  $x \in X$  with  $f(x) = y$ .
- (b) True. They have the same domain and codomain, and  $x^3 = x \cdot x \cdot x$  for every real  $x$ . Equality of functions is about values, not formulas.
- (c) False. In the remark after [Theorem 0.4.4](#),  $g \circ f = \text{id}_{\{1\}}$  is injective but  $g(1) = g(2)$ . Only  $f$  is forced to be injective.
- (d) Injective: “Suppose  $x_1, x_2 \in X$  satisfy  $f(x_1) = f(x_2)$ .” Surjective: “Let  $y \in Y$ .”
- (e) False.  $f^{-1}(B) = \{x \in X : f(x) \in B\}$  makes sense for every function; see [Definition 0.4.2](#) and the warning after it.
- (f) True, by [Theorem 0.4.9](#): domain and codomain are finite with the same number of elements, so an injective map is surjective, hence bijective.

### 0.4.8.2 B. Practice

#### Exercise B1: Injective or surjective?.

Determine whether each function is injective, and whether it is surjective. Justify your answers.

- (a)  $f: \mathbb{Z} \rightarrow \mathbb{Z}$ ,  $f(n) = 3n + 1$ .
- (b)  $g: \mathbb{R}^2 \rightarrow \mathbb{R}^2$ ,  $g(x, y) = (x + y, x - y)$ .
- (c)  $h: \mathbb{N} \rightarrow \mathbb{N}$ ,  $h(n) = \lfloor n/2 \rfloor$ , the largest integer not exceeding  $n/2$ .
- (d)  $p: \mathbb{R} \rightarrow \mathbb{R}^2$ ,  $p(t) = (t, t^2)$ .

*Solution.*

- (a) Injective, not surjective. Suppose  $3n_1 + 1 = 3n_2 + 1$ . Then  $3n_1 = 3n_2$ , so  $n_1 = n_2$ . But 0 is not hit:  $3n + 1 = 0$  would give  $n = -1/3 \notin \mathbb{Z}$ .
- (b) Bijective. Suppose  $g(x_1, y_1) = g(x_2, y_2)$ . Then  $x_1 + y_1 = x_2 + y_2$  and  $x_1 - y_1 = x_2 - y_2$ . Adding the two equations gives  $2x_1 = 2x_2$ , and subtracting gives  $2y_1 = 2y_2$ , so  $(x_1, y_1) = (x_2, y_2)$ . Hence  $g$  is injective. Let  $(u, v) \in \mathbb{R}^2$ . Then

$$g\left(\frac{u+v}{2}, \frac{u-v}{2}\right) = \left(\frac{u+v}{2} + \frac{u-v}{2}, \frac{u+v}{2} - \frac{u-v}{2}\right) = (u, v).$$

Hence  $g$  is surjective.

- (c) Surjective, not injective. Let  $m \in \mathbb{N}$ . Then  $h(2m) = \lfloor m \rfloor = m$ , so  $h$  is surjective. But  $h(0) = 0 = h(1)$ , a collision, so  $h$  is not injective.
- (d) Injective, not surjective. Suppose  $(t_1, t_1^2) = (t_2, t_2^2)$ . Comparing first coordinates gives  $t_1 = t_2$ . But  $(0, 1)$  is not hit:  $p(t) = (0, 1)$  forces  $t = 0$  and then  $t^2 = 0 \neq 1$ .

### Exercise B2: Inverses.

- (a) Show that  $f: \mathbb{R} \rightarrow \mathbb{R}$ ,  $f(x) = 7 - 4x$ , is invertible and find  $f^{-1}$ .
- (b) Let  $D = \mathbb{R} \setminus \{1\}$  and define  $k: D \rightarrow D$  by  $k(x) = \frac{x+1}{x-1}$ . Show that  $k$  really takes values in  $D$ , and that  $k \circ k = \text{id}_D$ . Hence deduce that  $k$  is bijective and find  $k^{-1}$ .
- (c) With  $g$  as in B1(b), find  $g^{-1}$ . Hence find the inverse of  $g \circ g$  without computing  $g \circ g$  first.

*Solution.*

- (a) Put  $e(y) = (7 - y)/4$ . Then  $e(f(x)) = (7 - (7 - 4x))/4 = x$  and  $f(e(y)) = 7 - (7 - y) = y$ . So  $e$  is an inverse of  $f$ , and by [Theorem 0.4.7](#)  $f$  is invertible with  $f^{-1}(y) = (7 - y)/4$ .
- (b) For  $x \in D$  the denominator  $x - 1$  is non-zero, so  $k(x)$  is a real number. If  $k(x) = 1$ , then  $x + 1 = x - 1$ , that is,  $1 = -1$ , which is false. Hence  $k(x) \in D$ . Next, for  $x \in D$ ,

$$k(k(x)) = \frac{\frac{x+1}{x-1} + 1}{\frac{x+1}{x-1} - 1} = \frac{(x+1) + (x-1)}{(x+1) - (x-1)} = \frac{2x}{2} = x,$$

where the second equality multiplies numerator and denominator by  $x - 1 \neq 0$ . So  $k \circ k = \text{id}_D$ . This says  $k$  is an inverse of itself, so  $k$  is invertible, hence bijective by [Theorem 0.4.7](#), and  $k^{-1} = k$ .

- (c) The solution of B1(b) shows that  $e(u, v) = (\frac{u+v}{2}, \frac{u-v}{2})$  satisfies  $g \circ e = \text{id}_{\mathbb{R}^2}$ . Also  $e(g(x, y)) = e(x + y, x - y) = (\frac{2x}{2}, \frac{2y}{2}) = (x, y)$ , so  $e \circ g = \text{id}_{\mathbb{R}^2}$ . Hence  $g^{-1} = e$ . By [Theorem 0.4.8](#),

$$(g \circ g)^{-1} = g^{-1} \circ g^{-1}, \quad g^{-1}(g^{-1}(u, v)) = g^{-1}\left(\frac{u+v}{2}, \frac{u-v}{2}\right) = \left(\frac{u}{2}, \frac{v}{2}\right).$$

(As a check,  $g(g(x, y)) = g(x + y, x - y) = (2x, 2y)$ , which  $(u, v) \mapsto (u/2, v/2)$  indeed undoes.)

### Exercise B3: Images of unions and intersections.

Let  $f: X \rightarrow Y$  and  $A_1, A_2 \subseteq X$ .

- (a) Prove that  $f(A_1 \cup A_2) = f(A_1) \cup f(A_2)$ .
- (b) Prove that  $f(A_1 \cap A_2) \subseteq f(A_1) \cap f(A_2)$ .
- (c) Using  $q(x) = x^2$  on  $\mathbb{R}$ , give an example where the inclusion in (b) is strict.

*Solution.*

- (a) We use double inclusion ([Theorem 0.3.1](#)).
- ( $\subseteq$ ) Let  $y \in f(A_1 \cup A_2)$ . Then  $y = f(a)$  for some  $a \in A_1 \cup A_2$ . If  $a \in A_1$ , then  $y \in f(A_1)$ ; if  $a \in A_2$ , then  $y \in f(A_2)$ . In either case  $y \in f(A_1) \cup f(A_2)$ .
- ( $\supseteq$ ) Let  $y \in f(A_1) \cup f(A_2)$ . If  $y \in f(A_1)$ , then  $y = f(a)$  for some  $a \in A_1 \subseteq A_1 \cup A_2$ , so  $y \in f(A_1 \cup A_2)$ . The case  $y \in f(A_2)$  is the same with the roles of  $A_1$  and  $A_2$  swapped. This proves the equality.
- (b) Let  $y \in f(A_1 \cap A_2)$ . Then  $y = f(a)$  for some  $a \in A_1 \cap A_2$ . Since  $a \in A_1$ ,  $y \in f(A_1)$ , and since  $a \in A_2$ ,  $y \in f(A_2)$ . Hence  $y \in f(A_1) \cap f(A_2)$ .
- (c) Take  $A_1 = \{-1\}$  and  $A_2 = \{1\}$ . Then  $A_1 \cap A_2 = \emptyset$ , so  $q(A_1 \cap A_2) = \emptyset$ , while  $q(A_1) \cap q(A_2) = \{1\} \cap \{1\} = \{1\}$ . The inclusion is strict. The collision  $q(-1) = q(1)$  is what goes wrong; for an injective  $f$ , equality holds in (b).

### 0.4.8.3 C. Going deeper

#### Exercise C1: Left inverses are not unique.

- (a) Let  $f: X \rightarrow Y$  be injective but **not** surjective, with  $X$  having at least two elements. Prove that  $f$  has at least two different left inverses.
- (b) Find two different left inverses of  $f: \mathbb{N} \rightarrow \mathbb{N}$ ,  $f(n) = n + 1$ .
- (c) Explain why a bijective function has exactly one left inverse.

*Hint for (a): look at the proof of [Theorem 0.4.5](#) and ask where it made a free choice.*

*Solution.*

- (a) Since  $f$  is not surjective, there is  $y_0 \in Y \setminus f(X)$ . Since  $X$  has at least two elements, there are  $x_0 \neq x_1$  in  $X$ . As in the proof of [Theorem 0.4.5](#), for  $y \in f(X)$  let  $\bar{y}$  be the unique  $x \in X$  with  $f(x) = y$ . Define  $g_0, g_1: Y \rightarrow X$  by  $g_i(y) = \bar{y}$  for  $y \in f(X)$ , and  $g_i(y) = x_i$  for  $y \notin f(X)$ . The proof of [Theorem 0.4.5](#) shows that each  $g_i$  is a left inverse of  $f$ , since  $g_i(f(x)) = x$  uses only the values on  $f(X)$ . But  $g_0(y_0) = x_0 \neq x_1 = g_1(y_0)$ , so  $g_0 \neq g_1$ .
- (b) The only element missed by  $f$  is 0. Define  $g_0(m) = m - 1$  for  $m \geq 1$  and  $g_0(0) = 0$ ; define  $g_1(m) = m - 1$  for  $m \geq 1$  and  $g_1(0) = 5$ . Both satisfy  $g_i(n + 1) = n$  for every  $n \in \mathbb{N}$ , and  $g_0(0) \neq g_1(0)$ .
- (c) Let  $f$  be bijective, and let  $g$  be any left inverse. By [Theorem 0.4.7](#),  $f$  has an inverse  $f^{-1}$ , which in particular is a right inverse. By the observation following that theorem, a left inverse and a right inverse of the same function are equal, so  $g = f^{-1}$ . Hence every left inverse equals  $f^{-1}$ . Conversely,  $f^{-1}$  is itself a left inverse, since  $f^{-1} \circ f = \text{id}_X$ . So  $f$  has exactly one left inverse.

#### Exercise C2: Self-maps of finite and infinite sets.

- (a) Let  $X$  be a finite set and  $f: X \rightarrow X$  injective. Prove that  $f$  is bijective.
- (b) Show that (a) fails for  $X = \mathbb{N}$ : give an injective  $f: \mathbb{N} \rightarrow \mathbb{N}$  that is not bijective.
- (c) Give a surjective  $h: \mathbb{N} \rightarrow \mathbb{N}$  that is not bijective.
- (d) Let  $X$  be finite and  $f, g: X \rightarrow X$  with  $g \circ f = \text{id}_X$ . Prove that  $f \circ g = \text{id}_X$  as well.



*Hint for (d): first show that  $f$  is bijective.*

*Solution.*

- (a) The domain and codomain are the same finite set, so  $|X| = |X|$ . By [Theorem 0.4.9](#), the injective map  $f$  is bijective.
- (b)  $f(n) = n + 1$  is injective, since  $n_1 + 1 = n_2 + 1$  gives  $n_1 = n_2$ , but 0 is not of the form  $n + 1$  with  $n \in \mathbb{N}$ . Hence  $f$  is not surjective, so not bijective.
- (c) Take  $h(n) = \lfloor n/2 \rfloor$ . By B1(c) it is surjective but not injective.
- (d) Since  $g \circ f = \text{id}_X$  is injective,  $f$  is injective by [Theorem 0.4.4](#) (a). By (a),  $f$  is bijective, so it has an inverse  $f^{-1}$  by [Theorem 0.4.7](#). Then

$$g = g \circ \text{id}_X = g \circ (f \circ f^{-1}) = (g \circ f) \circ f^{-1} = \text{id}_X \circ f^{-1} = f^{-1},$$

using [Theorem 0.4.2](#). Hence  $f \circ g = f \circ f^{-1} = \text{id}_X$ , as claimed. (For  $X = \mathbb{N}$  this fails: with  $f(n) = n + 1$  and  $g_0$  from C1(b),  $g_0 \circ f = \text{id}_{\mathbb{N}}$  but  $f(g_0(0)) = 1 \neq 0$ . The matrix version of (d), “ $\mathbf{AB} = \mathbf{I}$  implies  $\mathbf{BA} = \mathbf{I}$  for square matrices”, is proved in Chapter 2.)

### Exercise C3: When images and preimages undo each other.

Let  $f: X \rightarrow Y$ .

- (a) Prove that  $f(f^{-1}(B)) = B$  for **every**  $B \subseteq Y$  if and only if  $f$  is surjective.
- (b) Prove that  $f^{-1}(f(A)) = A$  for **every**  $A \subseteq X$  if and only if  $f$  is injective.

*Hint: for each direction you need only one well-chosen set; try a one-element set.*

*Solution.*

- (a) ( $\Rightarrow$ ) Suppose the equality holds for every  $B$ , and let  $y \in Y$ . Taking  $B = \{y\}$  gives  $f(f^{-1}(\{y\})) = \{y\}$ , so  $y \in f(f^{-1}(\{y\}))$ . Hence  $y = f(x)$  for some  $x$ , and  $f$  is surjective.
- ( $\Leftarrow$ ) Suppose  $f$  is surjective, and let  $B \subseteq Y$ . The inclusion  $f(f^{-1}(B)) \subseteq B$  is [Proposition 0.4.1](#). For the reverse, let  $y \in B$ . Since  $f$  is surjective,  $y = f(x)$  for some  $x \in X$ . Then  $f(x) \in B$ , so  $x \in f^{-1}(B)$ , and hence  $y = f(x) \in f(f^{-1}(B))$ . By [Theorem 0.3.1](#),  $f(f^{-1}(B)) = B$ .
- (b) ( $\Rightarrow$ ) Suppose the equality holds for every  $A$ , and suppose  $f(x_1) = f(x_2)$ . Taking  $A = \{x_1\}$  gives  $f^{-1}(\{f(x_1)\}) = \{x_1\}$ . Since  $f(x_2) = f(x_1)$ , we have  $x_2 \in f^{-1}(\{f(x_1)\}) = \{x_1\}$ , so  $x_2 = x_1$ . Hence  $f$  is injective.
- ( $\Leftarrow$ ) Suppose  $f$  is injective, and let  $A \subseteq X$ . The inclusion  $A \subseteq f^{-1}(f(A))$  is [Proposition 0.4.1](#). For the reverse, let  $x \in f^{-1}(f(A))$ . Then  $f(x) \in f(A)$ , so  $f(x) = f(a)$  for some  $a \in A$ . Since  $f$  is injective,  $x = a \in A$ . By [Theorem 0.3.1](#),  $f^{-1}(f(A)) = A$ .

## 0.5 Relations, Equivalence Classes and Quotients

Mathematics constantly declares different-looking things to be “the same”:  $\frac{1}{2}$  and  $\frac{2}{4}$  are the same number, 14:00 and 2 p.m. are the same time, and in Chapter 3 two vectors will count as the same when their difference lies in a fixed subspace. This section makes “the same, for present purposes” precise. The notion is an equivalence relation. Gluing equivalent things together produces a new set, the quotient, and the one real danger of working with quotients is defining a function on them that secretly depends on a choice. We will learn to check for that.

### 0.5.1 Relations

Before we can say what a good notion of sameness is, we need a word for any way of comparing two elements of a set, good or bad. Examples are  $\leq$  on numbers, “divides” on integers, and “has the same birthday as” on people. What such a comparison really amounts to is the list of pairs that are related, and a list of pairs is a subset of the Cartesian product (Definition 0.3.10).

#### Definition 0.5.1 — Relation

A **relation** on a set  $X$  is a subset  $R \subseteq X \times X$ . When  $(x, y) \in R$  we say  $x$  **is related to**  $y$  and write  $x R y$ , or  $x \sim y$  when the relation is fixed.

#### Example. (Relations)

- (a) On  $\mathbb{Z}$ :  $a \leq b$ . The pairs  $(1, 2)$  and  $(2, 2)$  belong to the relation,  $(2, 1)$  does not.
- (b) On  $\mathbb{Z}$ :  $a \mid b$  (“ $a$  divides  $b$ ”), meaning  $b = ka$  for some  $k \in \mathbb{Z}$ . So  $3 \mid 12$  and  $3 \mid 0$ , but  $3 \nmid 5$ .
- (c) On  $\mathbb{R}$ :  $x \sim y$  if  $x^2 + y^2 = 1$ . As a subset of  $\mathbb{R} \times \mathbb{R} = \mathbb{R}^2$  this relation is the unit circle, so a relation can literally be a picture.
- (d) On any set  $X$ : the **empty relation**  $R = \emptyset$ , in which nothing is related to anything, and the **full relation**  $R = X \times X$ , in which everything is related to everything. These two extremes are useful test cases for every property below.

Every subset of  $X \times X$  is a relation, so the definition has no non-examples. The interesting question is which properties a given relation has.

### 0.5.2 Equivalence relations

What must any reasonable notion of “ $x$  is the same as  $y$ ” satisfy? Think of “has the same birthday as”. Everyone has the same birthday as themselves. If you share a birthday with me, I share one with you. If  $x$  shares a birthday with  $y$ , and  $y$  with  $z$ , then  $x$  shares one with  $z$ . Those three features are exactly what we ask for.

*An equivalence relation is a way of declaring elements “the same” that behaves like equality:*

everything is the same as itself, sameness goes both ways, and it can be chained.

### Definition 0.5.2 — Equivalence relation

A relation  $\sim$  on a set  $X$  is an **equivalence relation** if it satisfies the following three conditions.

- (E1) **Reflexive:**  $x \sim x$  for every  $x \in X$ .
- (E2) **Symmetric:** for all  $x, y \in X$ , if  $x \sim y$  then  $y \sim x$ .
- (E3) **Transitive:** for all  $x, y, z \in X$ , if  $x \sim y$  and  $y \sim z$  then  $x \sim z$ .

In words: (E1) asks every single element to be related to itself, with no exceptions. (E2) says the relation cannot be one-way. (E3) says a chain of two links can be shortened to one link, and so, by applying it repeatedly, can any finite chain. When  $x \sim y$  we say  $x$  and  $y$  are **equivalent**.

The standard move for checking an equivalence relation is to write three short paragraphs, one per condition, each starting from its hypothesis (“Let  $x \in X$ .”, “Suppose  $x \sim y$ .”, “Suppose  $x \sim y$  and  $y \sim z$ .”).

### Example. (Equivalence relations)

Show that each of the following is an equivalence relation.

- (a) Equality on any set  $X$ :  $x \sim y$  if  $x = y$ .
- (b) **Congruence modulo  $n$**  on  $\mathbb{Z}$ , for a fixed integer  $n \geq 1$ :  $a \equiv b \pmod{n}$  if  $n \mid a - b$ .
- (c) “Same length” on the set  $S$  of finite strings of the letters  $a, b$ :  $u \sim v$  if  $u$  and  $v$  have the same number of letters.
- (d) “Same value”: for a function  $f: X \rightarrow Y$ , put  $x \sim_f x'$  if  $f(x) = f(x')$ .
- (e) Fractions: on  $P = \mathbb{Z} \times (\mathbb{Z} \setminus \{0\})$ , put  $(a, b) \sim (c, d)$  if  $ad = bc$ .

*Solution.*

- (a) Every  $x$  equals itself;  $x = y$  gives  $y = x$ ;  $x = y$  and  $y = z$  give  $x = z$ . Equality is the degenerate case: it is the equivalence relation that identifies nothing new, and the model the other examples imitate.
- (b) *Reflexive.* Let  $a \in \mathbb{Z}$ . Then  $a - a = 0 = 0 \cdot n$ , so  $n \mid a - a$ . *Symmetric.* Suppose  $a - b = kn$  with  $k \in \mathbb{Z}$ . Then  $b - a = (-k)n$ , so  $n \mid b - a$ . *Transitive.* Suppose  $a - b = kn$  and  $b - c = \ell n$  with  $k, \ell \in \mathbb{Z}$ . Then  $a - c = (a - b) + (b - c) = (k + \ell)n$ , so  $n \mid a - c$ . For  $n = 2$  this is “same parity”:  $a \equiv b \pmod{2}$  says  $a$  and  $b$  are both even or both odd.
- (c) This is the special case of (d) in which  $f: S \rightarrow \mathbb{N}$  sends a string to its length, so it follows from (d), whose proof does not use (c).
- (d) *Reflexive:*  $f(x) = f(x)$ . *Symmetric:*  $f(x) = f(x')$  gives  $f(x') = f(x)$ . *Transitive:*  $f(x) = f(x')$  and  $f(x') = f(x'')$  give  $f(x) = f(x'')$ . Each condition is inherited from the same property of equality in  $Y$ . Many equivalence relations are of this form: “same distance from the origin” on  $\mathbb{R}^2$  is  $\sim_f$  for  $f(x, y) = x^2 + y^2$ .
- (e) *Reflexive:*  $ab = ba$ , so  $(a, b) \sim (a, b)$ . *Symmetric:*  $ad = bc$  gives  $cb = da$ , which says  $(c, d) \sim (a, b)$ . *Transitive:* suppose  $ad = bc$  and  $cf = de$ . Multiplying the first

equation by  $f$  and the second by  $b$  gives  $adf = bcf = bde$ . Hence  $d(af - be) = 0$ . Since  $d \neq 0$ , we get  $af = be$ , that is,  $(a, b) \sim (e, f)$ . Transitivity is exactly where second entries must be non-zero: with  $d = 0$  allowed,  $(1, 1) \sim (0, 0) \sim (2, 1)$  but  $(1, 1) \not\sim (2, 1)$ .

Minimal changes to these examples show that each condition can fail on its own.

**Example. (Relations that are not equivalence relations)**

Decide which of (E1)–(E3) each relation satisfies.

- (a)  $\leq$  on  $\mathbb{Z}$ .
- (b) On  $\mathbb{R}$ :  $x \sim y$  if  $|x - y| \leq 1$ .
- (c) On  $\mathbb{R}$ :  $x \sim y$  if  $xy > 0$ .

*Solution.*

- (a) Change “=” to “ $\leq$ ”. Reflexivity survives ( $a \leq a$ ) and so does transitivity ( $a \leq b \leq c$  gives  $a \leq c$ ). **Symmetry fails:**  $1 \leq 2$  but  $2 \not\leq 1$ .
- (b) Change “ $|x - y| = 0$ ” (which is equality) to “ $|x - y| \leq 1$ ”. Reflexivity survives, since  $|x - x| = 0$ , and so does symmetry, since  $|y - x| = |x - y|$ . **Transitivity fails:**  $0 \sim 1$  and  $1 \sim 2$ , but  $|0 - 2| = 2 > 1$ . “Close to” cannot be chained; many small steps make a big one.
- (c) Symmetry holds, since  $yx = xy$ . Transitivity holds: if  $xy > 0$  and  $yz > 0$ , then  $xy^2z > 0$ , and since  $y^2 > 0$  this gives  $xz > 0$ . **Reflexivity fails, at exactly one point:**  $0 \cdot 0 = 0$ , so  $0 \not\sim 0$ . On  $\mathbb{R} \setminus \{0\}$  the same rule is an equivalence relation (“same sign”).

*Warning.* A tempting argument claims that (E1) follows from (E2) and (E3): “if  $x \sim y$ , then  $y \sim x$  by symmetry, so  $x \sim x$  by transitivity.” The gap is the word “if”. The argument needs some  $y$  with  $x \sim y$ , and there may be none. In [Example \(Relations that are not equivalence relations\)](#) (c), the element 0 is related to nothing, which is exactly why it escapes. Reflexivity must always be checked on its own.

*Quick check.* On  $\mathbb{Z}$ , define  $a \sim b$  if  $ab \geq 0$ . Is  $\sim$  an equivalence relation?

*Solution.* No. It is reflexive ( $a^2 \geq 0$ ) and symmetric, but not transitive:  $1 \sim 0$  and  $0 \sim -1$  since both products are 0, yet  $1 \cdot (-1) = -1 < 0$ , so  $1 \not\sim -1$ . Compare [Example \(Relations that are not equivalence relations\)](#) (c): moving from  $>$  to  $\geq$  repairs reflexivity at 0 but lets 0 link the positives to the negatives.

**Why these three conditions.** They are precisely what the next subsection needs. Reflexivity will guarantee that every element lies in its own class, so no class is empty and nothing is lost. Symmetry and transitivity together will guarantee that two classes are either identical or disjoint. Drop any one of them and the picture of  $X$  cut cleanly into pieces falls apart, as the non-examples above show.

*Remark.* Two equivalence relations on matrices are central later: *row equivalence*, which records that one matrix can be reached from another by row operations, and *similarity*, which records that two square matrices describe the same linear map in different

coordinates. Both are introduced once matrices are available, later in this chapter, and studied in Chapters 2 and 3.

### 0.5.3 Equivalence classes and partitions

Once we have decided which elements count as the same, it is natural to collect everything that is the same as a given element into one set.

#### Definition 0.5.3 — Equivalence class

Let  $\sim$  be an equivalence relation on  $X$ , and let  $x \in X$ . The **equivalence class** of  $x$  is

$$[x] := \{y \in X : y \sim x\}.$$

Any element of  $[x]$  is called a **representative** of the class.

The word “representative” is chosen carefully. A class is one set, but it usually has many names:  $[x] = [y]$  is possible with  $x \neq y$ .

#### Example. (Equivalence classes)

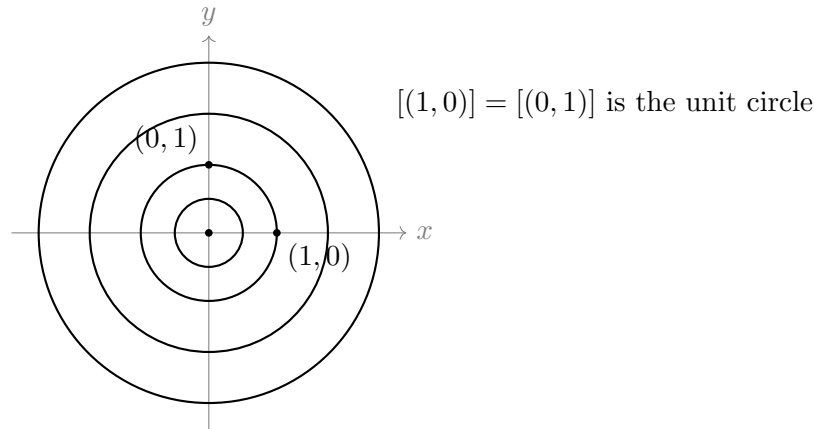
- (a) For equality on  $X$ ,  $[x] = \{x\}$ . Every class has one element.
- (b) For congruence modulo 3 on  $\mathbb{Z}$ :

$$\begin{aligned} [0] &= \{\dots, -6, -3, 0, 3, 6, \dots\}, \\ [1] &= \{\dots, -5, -2, 1, 4, 7, \dots\}, \\ [2] &= \{\dots, -4, -1, 2, 5, 8, \dots\}. \end{aligned}$$

Here  $[0] = [3] = [-3]$  and  $[2] = [-1]$ : one class, many names.

- (c) For “same length” on strings of  $a, b$ , the class of  $ab$  is  $\{aa, ab, ba, bb\}$ , and the class of the empty string contains only the empty string.
- (d) For the full relation  $X \times X$  on a non-empty set  $X$ , there is a single class,  $[x] = X$ .
- (e) For “same distance from the origin” on  $\mathbb{R}^2$ , the class of  $(x, y)$  is the circle of radius  $\sqrt{x^2 + y^2}$  centered at the origin. The class of the origin is the one-point set  $\{(0, 0)\}$ .

In example (e), the classes look like this. Each circle is one class; the origin on its own is another.



The picture suggests two facts: every point lies on exactly one circle, and two different circles never meet. Both hold for every equivalence relation, and the proof is where the three conditions do their work.

### Lemma 0.5.1 — Classes are equal or disjoint

Let  $\sim$  be an equivalence relation on  $X$ , and let  $a, b \in X$ .

- (a)  $[a] = [b]$  if and only if  $a \sim b$ .
- (b) Either  $[a] = [b]$  or  $[a] \cap [b] = \emptyset$ .

*Proof.*

- (a)  $(\Rightarrow)$  Suppose  $[a] = [b]$ . By reflexivity  $a \sim a$ , so  $a \in [a] = [b]$ , which means  $a \sim b$ .  
 $(\Leftarrow)$  Suppose  $a \sim b$ . Let  $x \in [a]$ , so  $x \sim a$ . Since  $a \sim b$ , transitivity gives  $x \sim b$ , so  $x \in [b]$ . Hence  $[a] \subseteq [b]$ . By symmetry  $b \sim a$ , and swapping the roles of  $a$  and  $b$  in the argument gives  $[b] \subseteq [a]$ . By [Theorem 0.3.1](#),  $[a] = [b]$ .
- (b) Suppose  $[a] \cap [b] \neq \emptyset$ , and let  $c \in [a] \cap [b]$ . Then  $c \sim a$  and  $c \sim b$ . By symmetry  $a \sim c$ , and by transitivity  $a \sim b$ . By (a),  $[a] = [b]$ .

■

Part (a) is the working form of the lemma: to show two classes are equal, show one representative is equivalent to the other. The picture of non-overlapping pieces that cover everything has its own name.

### Definition 0.5.4 — Partition

A **partition** of a set  $X$  is a collection  $\mathcal{P}$  of subsets of  $X$ , called **blocks**, such that:

- (P1) every block is **non-empty**;
- (P2) **distinct** blocks are disjoint: if  $B, B' \in \mathcal{P}$  and  $B \neq B'$ , then  $B \cap B' = \emptyset$ ;
- (P3) the blocks cover  $X$ : every  $x \in X$  lies in some block.

By (P2) and (P3), every element of  $X$  lies in **exactly one** block. For example,  $\{\{1, 3\}, \{2\}\}$  is a partition of  $\{1, 2, 3\}$ , while  $\{\{1, 2\}, \{2, 3\}\}$  is not, because the distinct blocks share 2 and (P2)

fails.

*Quick check.* Is  $\{\{1, 2, 3\}, \emptyset\}$  a partition of  $\{1, 2, 3\}$ ? Is  $\{\{1\}, \{2\}\}$ ?

*Solution.* Neither. The first contains the empty block, so (P1) fails, even though (P2) and (P3) hold. The second does not cover 3, so (P3) fails.

The main theorem says that equivalence relations and partitions are two descriptions of the same thing: “which elements are the same” and “how  $X$  is cut into pieces”.

### Theorem 0.5.2 — Equivalence relations and partitions

Let  $X$  be a set.

- (a) If  $\sim$  is an equivalence relation on  $X$ , then the set of its equivalence classes is a partition of  $X$ .
- (b) If  $\mathcal{P}$  is a partition of  $X$ , then the relation “ $x \sim_{\mathcal{P}} y$  if  $x$  and  $y$  lie in the same block” is an equivalence relation on  $X$ , and its equivalence classes are exactly the blocks of  $\mathcal{P}$ .
- (c) The two constructions undo each other: starting from  $\sim$ , the relation defined in (b) by the partition in (a) is  $\sim$  again.

*Idea.* Part (a) is the lemma plus reflexivity: (P1) and (P3) come from  $x \in [x]$ , and (P2) is Lemma 0.5.1 (b). Part (b) runs the other way, and each block condition pays for one equivalence condition: covering gives reflexivity, and disjointness gives transitivity, because a middle element  $y$  can lie in only one block. Part (c) is Lemma 0.5.1 (a) read as a sentence about blocks.

*Proof.*

- (a) Let  $\mathcal{P} = \{[x] : x \in X\}$ . For every  $x \in X$ , reflexivity gives  $x \in [x]$ . Hence every class is non-empty, which is (P1), and every element lies in some class, which is (P3). If  $[a] \neq [b]$ , then  $[a] \cap [b] = \emptyset$  by Lemma 0.5.1 (b), which is (P2). Hence  $\mathcal{P}$  is a partition of  $X$ .
- (b) *Reflexive.* Let  $x \in X$ . By (P3),  $x$  lies in some block, so  $x \sim_{\mathcal{P}} x$ . *Symmetric.* “ $x$  and  $y$  lie in a common block” does not depend on the order of  $x$  and  $y$ . *Transitive.* Suppose  $x, y \in B$  and  $y, z \in B'$  for blocks  $B, B'$ . Then  $y \in B \cap B'$ , so  $B = B'$  by (P2). Hence  $x, z \in B$ , and  $x \sim_{\mathcal{P}} z$ .  
Now let  $B \in \mathcal{P}$  and  $x \in B$ ; we show  $[x] = B$ , where the class is taken for  $\sim_{\mathcal{P}}$ . ( $\subseteq$ ) If  $y \in [x]$ , then  $y$  and  $x$  lie in a common block  $B'$ ; since  $x \in B \cap B'$ , (P2) gives  $B' = B$ , so  $y \in B$ . ( $\supseteq$ ) If  $y \in B$ , then  $y$  and  $x$  both lie in  $B$ , so  $y \sim_{\mathcal{P}} x$  and  $y \in [x]$ . Hence  $[x] = B$  by Theorem 0.3.1. Every class  $[x]$  is therefore the block containing  $x$ , which exists by (P3); and every block  $B$  is a class, namely  $[x]$  for any  $x \in B$ , which exists by (P1). So the classes are exactly the blocks.
- (c) Let  $\sim$  be an equivalence relation, and let  $\approx$  be the relation built by (b) from the classes of  $\sim$ . If  $x \sim y$ , then  $x, y \in [y]$ , so  $x \approx y$ . Conversely, if  $x \approx y$ , then  $x, y \in [z]$  for some  $z$ ; thus  $x \sim z$  and  $y \sim z$ , and symmetry and transitivity give  $x \sim y$ . Hence  $\approx$  and  $\sim$  are the same relation. This proves the theorem. ■

So whenever you meet an equivalence relation, you may think of it as a way of cutting the set into non-overlapping pieces, and vice versa. Exercise C2 uses this to count equivalence relations by counting partitions instead.

## 0.5.4 Quotient sets

Now we take the decisive step: treat each piece as a single new element. The time of day is the classic case. Hours counted from some starting moment are integers, but a clock only records the class of the hour modulo 12.

### Definition 0.5.5 — Quotient set

Let  $\sim$  be an equivalence relation on  $X$ . The **quotient set** of  $X$  by  $\sim$  is the set of all equivalence classes,

$$X/\sim := \{[x] : x \in X\}.$$

The function  $\pi: X \rightarrow X/\sim$ ,  $\pi(x) = [x]$ , is the **quotient map**.

The elements of  $X/\sim$  are sets, but once the quotient is formed we stop looking inside them and treat each class as a single point. The quotient map is surjective, since every class is  $[x] = \pi(x)$  for some  $x$ . By Lemma 0.5.1 (a),  $\pi(x) = \pi(y)$  exactly when  $x \sim y$ : the quotient map makes equivalent elements literally equal.

### Example. (Quotient sets)

- (a) For an integer  $n \geq 1$ , the quotient of  $\mathbb{Z}$  by congruence modulo  $n$  is written  $\mathbb{Z}/n\mathbb{Z}$ . Show that it has exactly  $n$  elements, namely  $[0], [1], \dots, [n-1]$ .
- (b) Describe the quotient of the set  $S$  of strings in  $a, b$  by “same length”, and of  $\mathbb{R}^2$  by “same distance from the origin”.

*Solution.*

- (a) Let  $a \in \mathbb{Z}$ . By division with remainder (Exercise C2: Division with remainder),  $a = qn + r$  with  $q, r \in \mathbb{Z}$  and  $0 \leq r < n$ . Then  $a - r = qn$ , so  $a \equiv r \pmod{n}$ , and  $[a] = [r]$  by Lemma 0.5.1 (a). Hence every class is one of  $[0], \dots, [n-1]$ . These are distinct: if  $0 \leq r < s < n$ , then  $0 < s - r < n$ , so  $s - r$  is not a multiple of  $n$ , hence  $r \not\equiv s$  and  $[r] \neq [s]$  by Lemma 0.5.1 (a). So  $\mathbb{Z}/n\mathbb{Z} = \{[0], [1], \dots, [n-1]\}$  has exactly  $n$  elements.
- (b) Two strings are identified exactly when they have the same length, so there is one class for each length  $k \in \mathbb{N}$ : the set of all  $2^k$  strings of length  $k$ . Similarly,  $\mathbb{R}^2$  modulo “same distance” has one class for each radius  $r \in [0, \infty)$ : the circle of radius  $r$ , or the origin when  $r = 0$ . In both cases the quotient is “the same as” the set of values  $\mathbb{N}$  or  $[0, \infty)$ , and the next subsection makes that precise.



## 0.5.5 Functions on a quotient

We want to compute with classes. On a clock, “three hours after 11 o’clock” is 2 o’clock, which is really addition on  $\mathbb{Z}/12\mathbb{Z}$ . The natural way to define such an operation is through representatives: to add two classes, pick an element of each, add them, and take the class of the result. The danger is that a class has many representatives, and different choices might give different answers. When they do, the “definition” does not define a function at all.

Here is that failure in its simplest form.

### Example. (The numerator is not well defined)

Let  $P = \mathbb{Z} \times (\mathbb{Z} \setminus \{0\})$  with  $(a, b) \sim (c, d)$  if  $ad = bc$ , as in [Example \(Equivalence relations\)](#) (e). Its classes are the rational numbers: the class of  $(a, b)$  is the fraction  $a/b$ , with all its other names. Is “ $[(a, b)] \mapsto a$ ” a function  $P/\sim \rightarrow \mathbb{Z}$ ?

*Solution.* No. Since  $1 \cdot 4 = 2 \cdot 2$ , we have  $(1, 2) \sim (2, 4)$ , so  $[(1, 2)] = [(2, 4)]$  by [Lemma 0.5.1](#) (a). The rule assigns to this one class the value 1 through one name and 2 through the other. So it does not assign **exactly one** value to each element of the domain, and [Definition 0.4.1](#) fails. “The numerator of  $\frac{1}{2}$ ” is a statement about how a fraction is written, not about the number.

By contrast, “ $[(a, b)] \mapsto 1$  if  $a = 0$ , and 0 otherwise” is fine. If  $(a, b) \sim (c, d)$  and  $a = 0$ , then  $bc = ad = 0$ , so  $c = 0$  because  $b \neq 0$ . By symmetry,  $c = 0$  forces  $a = 0$ . So equivalent pairs always get the same value.

The contrast in that example is the whole story, and the following theorem turns it into a test. A function  $f$  on  $X$  passes to the quotient exactly when it cannot tell equivalent elements apart.

### Theorem 0.5.3 — Functions on a quotient

Let  $\sim$  be an equivalence relation on  $X$ , let  $\pi: X \rightarrow X/\sim$  be the quotient map, and let  $f: X \rightarrow Y$  be a function.

- (a) If  $f(a) = f(b)$  **whenever**  $a \sim b$ , then there is **exactly one** function  $\bar{f}: X/\sim \rightarrow Y$  with  $\bar{f}([x]) = f(x)$  for every  $x \in X$ , that is,  $\bar{f} \circ \pi = f$ .
- (b) Conversely, if some function  $\bar{f}: X/\sim \rightarrow Y$  satisfies  $\bar{f} \circ \pi = f$ , then  $f(a) = f(b)$  whenever  $a \sim b$ .

*Idea.* The formula  $\bar{f}([x]) = f(x)$  leaves no freedom: if  $\bar{f}$  exists, this is its value. The only question is whether the formula makes sense, since the class  $C = [x]$  may also be called  $[x']$ . So look at **all** the values  $f$  takes on  $C$  at once, the image  $f(C)$ . A function needs exactly one value;  $f(C)$  is non-empty because  $C$  is, and has at most one element because all elements of  $C$  are equivalent.

*Proof.*

- (a) Let  $C \in X/\sim$ . By [Definition 0.5.5](#),  $C = [x]$  for some  $x \in X$ , so  $C$  is non-empty, and hence the image  $f(C)$  is non-empty. If  $x', x'' \in C$ , then  $[x'] = C = [x'']$  by [Lemma 0.5.1](#) (a) applied to  $x' \sim x$  and  $x'' \sim x$ ; hence  $x' \sim x''$  by the same lemma,

and  $f(x') = f(x'')$  by hypothesis. Therefore  $f(C)$  has exactly one element, and we define  $\bar{f}(C)$  to be that element. This assigns exactly one value to each class, so  $\bar{f}$  is a function. For  $x \in X$ , we have  $x \in [x]$  by reflexivity, so  $\bar{f}([x]) \in f([x]) = \{f(x)\}$ , that is,  $\bar{f}([x]) = f(x)$ .

For uniqueness, suppose  $g: X/\sim \rightarrow Y$  also satisfies  $g([x]) = f(x)$  for every  $x$ . Every class is  $[x]$  for some  $x$ , and there  $g([x]) = f(x) = \bar{f}([x])$ . Hence  $g = \bar{f}$ .

- (b) Suppose  $\bar{f} \circ \pi = f$  and  $a \sim b$ . By [Lemma 0.5.1](#) (a),  $[a] = [b]$ . Hence  $f(a) = \bar{f}([a]) = \bar{f}([b]) = f(b)$ , as claimed. ■

In practice we rarely name  $f$  and  $\bar{f}$  separately. We write down a rule “ $[x] \mapsto (\text{expression in } x)$ ” and check that equivalent representatives give the same value. That check is what people mean when they say a function is **well defined**. For example, “same value” in [Example \(Equivalence relations\)](#) (d) satisfies the hypothesis by its very definition, so every  $f: X \rightarrow Y$  induces a function  $X/\sim_f \rightarrow Y$ . This  $\bar{f}$  is injective:  $\bar{f}([x]) = \bar{f}([x'])$  means  $f(x) = f(x')$ , which is  $x \sim_f x'$ , so  $[x] = [x']$ . This makes precise the remark that  $\mathbb{R}^2$  modulo “same distance” is “the same as”  $[0, \infty)$ : the induced map  $[(x, y)] \mapsto \sqrt{x^2 + y^2}$  is a bijection onto  $[0, \infty)$ .

The most important application in this chapter is arithmetic modulo  $n$ . The operation takes two classes, so we check both representatives at once.

**Example. (Addition modulo  $n$  is well defined)**

Fix an integer  $n \geq 1$ . Show that the rule  $[a] + [b] := [a + b]$  defines a function  $\mathbb{Z}/n\mathbb{Z} \times \mathbb{Z}/n\mathbb{Z} \rightarrow \mathbb{Z}/n\mathbb{Z}$ . Then compute  $[3] + [2]$  in  $\mathbb{Z}/4\mathbb{Z}$ .

*Solution.* We must show that the value  $[a + b]$  depends only on the classes  $[a]$  and  $[b]$ , not on the representatives  $a$  and  $b$ . Suppose  $[a] = [a']$  and  $[b] = [b']$ . By [Lemma 0.5.1](#) (a),  $a \equiv a' \pmod{n}$  and  $b \equiv b' \pmod{n}$ , so  $a - a' = kn$  and  $b - b' = \ell n$  for some  $k, \ell \in \mathbb{Z}$ . Then

$$(a + b) - (a' + b') = (a - a') + (b - b') = (k + \ell)n,$$

so  $a + b \equiv a' + b' \pmod{n}$ , and  $[a + b] = [a' + b']$  by the same lemma. Hence every choice of representatives gives the same class, and the rule defines a function.

In  $\mathbb{Z}/4\mathbb{Z}$ ,  $[3] + [2] = [5] = [1]$ , since  $5 - 1 = 4$ . Choosing other names gives the same answer, as it must:  $[7] + [-2] = [5] = [1]$ .

This is the same principle as [Theorem 0.5.3](#), applied to a function of two variables. The same check ([Exercise \(B3: Multiplication modulo  \$n\$ \)](#)) shows multiplication  $[a][b] := [ab]$  is well defined. With these operations,  $\mathbb{Z}/p\mathbb{Z}$  for a prime  $p$  will become our first finite field later in this chapter, and the identical pattern, now with “ $\mathbf{v} \sim \mathbf{w}$  if  $\mathbf{v} - \mathbf{w} \in U$ ”, will build quotient vector spaces in Chapter 3.

*Warning.* A rule written in terms of a representative is not automatically a function on classes. Define it, then check it. For example, “ $[a] \mapsto a$ ” from  $\mathbb{Z}/5\mathbb{Z}$  to  $\mathbb{Z}$  is not a function, because  $[0] = [5]$  would be sent to both 0 and 5. The check is not a formality: forgetting it is the most common error in any argument that uses quotients.

*Quick check.* Which of these rules defines a function? (i)  $\mathbb{Z}/6\mathbb{Z} \rightarrow \mathbb{Z}/3\mathbb{Z}$ ,  $[a]_6 \mapsto [a]_3$ . (ii)  $\mathbb{Z}/6\mathbb{Z} \rightarrow \mathbb{Z}/4\mathbb{Z}$ ,  $[a]_6 \mapsto [a]_4$ . Here the subscript records the modulus.

*Solution.*

- (i) Yes. If  $[a]_6 = [a']_6$ , then  $a - a' = 6k = 3(2k)$ , so  $[a]_3 = [a']_3$ .
- (ii) No.  $[0]_6 = [6]_6$ , but  $[0]_4 \neq [6]_4 = [2]_4$ , since  $6 - 0 = 6$  is not a multiple of 4. The first works because  $3 \mid 6$ ; the second fails because  $4 \nmid 6$ .

## 0.5.6 Exercises

### 0.5.6.1 A. Check your understanding

#### Exercise A1.

- (a) State the three conditions for a relation  $\sim$  on  $X$  to be an equivalence relation.
- (b) True or false: a symmetric and transitive relation is automatically reflexive. Give a reason.
- (c) True or false: if two equivalence classes  $[a]$  and  $[b]$  have an element in common, then  $a \sim b$ .
- (d) You want to define a function on  $X/\sim$  by a formula in a representative  $x$ . What must you check?
- (e) How many elements does  $\mathbb{Z}/5\mathbb{Z}$  have? Is  $[12] = [-3]$  in  $\mathbb{Z}/5\mathbb{Z}$ ?

*Solution.*

- (a) Reflexive:  $x \sim x$  for every  $x \in X$ . Symmetric:  $x \sim y$  implies  $y \sim x$ . Transitive:  $x \sim y$  and  $y \sim z$  imply  $x \sim z$ .
- (b) False. On  $\mathbb{R}$ ,  $x \sim y$  if  $xy > 0$  is symmetric and transitive, but  $0 \not\sim 0$ ; see [Example \(Relations that are not equivalence relations\)](#) (c) and the warning after it.
- (c) True. By [Lemma 0.5.1](#) (b) the classes are equal, and by part (a) of the same lemma  $a \sim b$ .
- (d) That equivalent representatives give the same value: if  $x \sim x'$ , the formula gives the same result for  $x$  and  $x'$ . By [Theorem 0.5.3](#) this is exactly what is needed.
- (e) Five, by [Example \(Quotient sets\)](#) (a). Yes:  $12 - (-3) = 15 = 3 \cdot 5$ , so  $12 \equiv -3 \pmod{5}$ , and both classes equal  $[2]$ .

### 0.5.6.2 B. Practice

#### Exercise B1: Which are equivalence relations?.

Determine which of the following are equivalence relations. For those that are, describe the equivalence classes. For those that are not, name a condition that fails with a specific witness.

- (a) On  $\mathbb{Z}$ :  $a \sim b$  if  $a + b$  is even.

- (b) On  $\mathbb{Z}$ :  $a \sim b$  if  $a - b$  is odd.
- (c) On  $\mathbb{R}$ :  $x \sim y$  if  $x - y \in \mathbb{Z}$ .
- (d) On  $\{1, 2, 3, \dots\}$ :  $m \sim n$  if  $m$  and  $n$  have a common divisor greater than 1.
- (e) On the set of all subsets of  $\{1, 2, 3\}$ :  $A \sim B$  if  $A$  and  $B$  have the same number of elements.

*Solution.*

- (a) Equivalence relation. *Reflexive*:  $a + a = 2a$  is even. *Symmetric*:  $b + a = a + b$ . *Transitive*: if  $a + b$  and  $b + c$  are even, then  $a + c = (a + b) + (b + c) - 2b$  is a sum of even integers, hence even. There are two classes: the even integers and the odd integers. (In fact  $a + b$  is even exactly when  $a - b = (a + b) - 2b$  is even, so this is congruence modulo 2.)
- (b) Not an equivalence relation. Reflexivity fails:  $0 - 0 = 0$  is even, so  $0 \not\sim 0$ . (Transitivity also fails:  $0 \sim 1$  and  $1 \sim 2$ , but  $0 - 2$  is even.)
- (c) Equivalence relation. *Reflexive*:  $x - x = 0 \in \mathbb{Z}$ . *Symmetric*: if  $x - y = k \in \mathbb{Z}$ , then  $y - x = -k \in \mathbb{Z}$ . *Transitive*: if  $x - y = k$  and  $y - z = \ell$  with  $k, \ell \in \mathbb{Z}$ , then  $x - z = k + \ell \in \mathbb{Z}$ . The class of  $x$  is  $\{x + k : k \in \mathbb{Z}\}$ , the points at integer distance from  $x$  in either direction. Each class contains exactly one number in  $[0, 1)$ , namely  $x - \lfloor x \rfloor$ .
- (d) Not an equivalence relation. Reflexivity fails at 1: the only positive divisor of 1 is 1. Transitivity also fails:  $2 \sim 6$  (common divisor 2) and  $6 \sim 3$  (common divisor 3), but 2 and 3 have no common divisor greater than 1.
- (e) Equivalence relation, since it is “same value” (Example (Equivalence relations) (d)) for the function  $A \mapsto |A|$ . There are four classes:  $\{\emptyset\}$ ;  $\{\{1\}, \{2\}, \{3\}\}$ ;  $\{\{1, 2\}, \{1, 3\}, \{2, 3\}\}$ ;  $\{\{1, 2, 3\}\}$ .

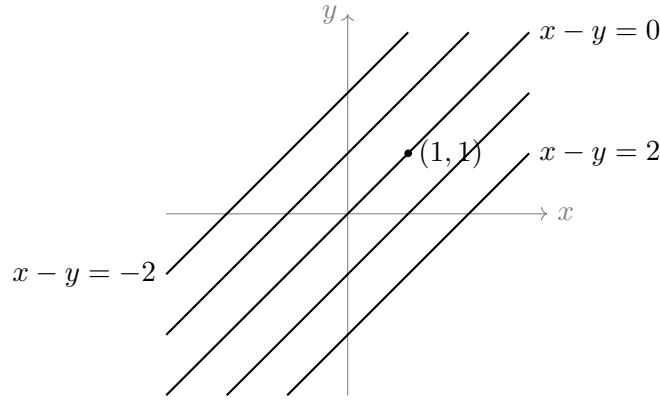
### Exercise B2: Parallel lines as classes.

On  $\mathbb{R}^2$ , define  $(x, y) \sim (x', y')$  if  $x - y = x' - y'$ .

- (a) Prove that  $\sim$  is an equivalence relation.
- (b) Describe the class of  $(x_0, y_0)$  geometrically, and draw several classes.
- (c) Show that  $[(x, y)] \mapsto x - y$  is a well-defined function  $\mathbb{R}^2/\sim \rightarrow \mathbb{R}$ , and that it is a bijection.

*Solution.*

- (a) This is “same value” for  $f: \mathbb{R}^2 \rightarrow \mathbb{R}$ ,  $f(x, y) = x - y$ , so it is an equivalence relation by Example (Equivalence relations) (d). Directly:  $x - y = x - y$ ; equality of the numbers  $x - y$  and  $x' - y'$  is symmetric and transitive.
- (b) Put  $c = x_0 - y_0$ . Then  $[(x_0, y_0)] = \{(x, y) \in \mathbb{R}^2 : x - y = c\} = \{(x, y) : y = x - c\}$ , the line of slope 1 through  $(x_0, y_0)$ , which meets the  $y$ -axis at  $(0, -c)$ . The classes are all the lines parallel to  $y = x$ : each point of the plane lies on exactly one of them, as Theorem 0.5.2 predicts.



- (c) Let  $f(x, y) = x - y$ . If  $(x, y) \sim (x', y')$ , then  $f(x, y) = f(x', y')$  by the definition of  $\sim$ . By [Theorem 0.5.3](#) (a), there is a function  $\bar{f}: \mathbb{R}^2/\sim \rightarrow \mathbb{R}$  with  $\bar{f}([(x, y)]) = x - y$ .

*Injective.* Suppose  $\bar{f}([(x, y)]) = \bar{f}([(x', y')])$ . Then  $x - y = x' - y'$ , so  $(x, y) \sim (x', y')$ , and  $[(x, y)] = [(x', y')]$  by [Lemma 0.5.1](#) (a).

*Surjective.* Let  $c \in \mathbb{R}$ . Then  $\bar{f}([(c, 0)]) = c - 0 = c$ .

Hence  $\bar{f}$  is a bijection: the set of parallel lines is labeled, one to one, by the real number  $c$ .

### Exercise B3: Multiplication modulo $n$ .

Fix an integer  $n \geq 1$ .

- Prove that  $[a][b] := [ab]$  is a well-defined operation on  $\mathbb{Z}/n\mathbb{Z}$ .
- Write out the multiplication table of  $\mathbb{Z}/4\mathbb{Z}$ , using the representatives 0, 1, 2, 3.
- Hence find a class  $[a] \neq [0]$  in  $\mathbb{Z}/4\mathbb{Z}$  with  $[a][a] = [0]$ .

*Solution.*

- (a) Suppose  $[a] = [a']$  and  $[b] = [b']$ . By [Lemma 0.5.1](#) (a),  $a - a' = kn$  and  $b - b' = \ell n$  for some  $k, \ell \in \mathbb{Z}$ . Then

$$ab - a'b' = ab - a'b + a'b - a'b' = (a - a')b + a'(b - b') = (kb + a'\ell)n,$$

so  $ab \equiv a'b' \pmod{n}$  and  $[ab] = [a'b']$  by the same lemma. Hence the value does not depend on the representatives, and the operation is well defined.

- (b) Reducing each product modulo 4:

| $\cdot$ | [0] | [1] | [2] | [3] |
|---------|-----|-----|-----|-----|
| [0]     | [0] | [0] | [0] | [0] |
| [1]     | [0] | [1] | [2] | [3] |
| [2]     | [0] | [2] | [0] | [2] |
| [3]     | [0] | [3] | [2] | [1] |

For instance  $[2][3] = [6] = [2]$  and  $[3][3] = [9] = [1]$ .

- (c) From the table,  $[2][2] = [4] = [0]$ , while  $[2] \neq [0]$  since  $4 \nmid 2$ . So a product of non-zero classes can be zero in  $\mathbb{Z}/4\mathbb{Z}$ , something that never happens with ordinary numbers.

### 0.5.6.3 C. Going deeper

#### Exercise C1: Squares and powers modulo $n$ .

- (a) Let  $n \geq 1$ . Prove that  $[x] \mapsto [x^2]$  is a well-defined function  $\mathbb{Z}/n\mathbb{Z} \rightarrow \mathbb{Z}/n\mathbb{Z}$ .
- (b) On  $\mathbb{N}$ , let  $\equiv_3$  be congruence modulo 3. Show that the rule  $[x] \mapsto [2^x]_3$  does **not** define a function from  $\mathbb{N}/\equiv_3$  to  $\mathbb{Z}/3\mathbb{Z}$ .
- (c) On  $\mathbb{N}$ , let  $\equiv_2$  be congruence modulo 2. Prove that  $[x] \mapsto [2^x]_3$  **does** define a function from  $\mathbb{N}/\equiv_2$  to  $\mathbb{Z}/3\mathbb{Z}$ .

*Hint for (c): compute  $2^2$  modulo 3.*

*Solution.*

- (a) Suppose  $[x] = [x']$ . By [Exercise \(B3: Multiplication modulo  \$n\$ \)](#) (a) with  $a = b = x$  and  $a' = b' = x'$ , we get  $[x][x] = [x'][x']$ , that is,  $[x^2] = [x'^2]$ . Hence the value depends only on the class, and the function is well defined.
- (b) In  $\mathbb{N}$ ,  $0 \equiv 3 \pmod{3}$ , so  $[0] = [3]$ . The rule sends this class to  $[2^0]_3 = [1]_3$  through the representative 0, and to  $[2^3]_3 = [8]_3 = [2]_3$  through the representative 3. Since  $8 - 1 = 7$  is not a multiple of 3,  $[1]_3 \neq [2]_3$ . So one class receives two different values, and the rule is not a function.
- (c) By [Theorem 0.5.3](#) (a), it suffices to show that  $x \equiv x' \pmod{2}$  implies  $2^x \equiv 2^{x'} \pmod{3}$ . By symmetry of both relations we may assume  $x \leq x'$ , so  $x' = x + 2k$  with  $k \in \mathbb{N}$ . First,  $4^k \equiv 1 \pmod{3}$  for every  $k \in \mathbb{N}$ , by induction on  $k$  ([Theorem 0.2.7](#)):  $4^0 = 1$ , and if  $4^k - 1 = 3m$ , then  $4^{k+1} - 1 = 4(4^k - 1) + 3 = 3(4m + 1)$ . Hence

$$2^{x'} - 2^x = 2^x(4^k - 1)$$

is a multiple of 3, so  $[2^{x'}]_3 = [2^x]_3$ , as claimed. The difference between (b) and (c) is that the powers of 2 modulo 3 repeat with period 2, and 3 is not a multiple of 2.

#### Exercise C2: Counting equivalence relations.

- (a) List all partitions of  $\{1, 2, 3\}$ .
- (b) Hence deduce that there are exactly 5 equivalence relations on  $\{1, 2, 3\}$ .
- (c) Write the equivalence relation corresponding to the partition  $\{\{1, 3\}, \{2\}\}$  as a subset of  $\{1, 2, 3\} \times \{1, 2, 3\}$ .

*Solution.*

- (a) Sort by the number of blocks. One block:  $\{\{1, 2, 3\}\}$ . Two blocks: one block is a single element and the other the remaining pair, giving  $\{\{1\}, \{2, 3\}\}$ ,  $\{\{2\}, \{1, 3\}\}$ ,  $\{\{3\}, \{1, 2\}\}$ . Three blocks:  $\{\{1\}, \{2\}, \{3\}\}$ . Four or more non-empty disjoint blocks are impossible in a three-element set. That is 5 partitions.

- (b) By [Theorem 0.5.2](#) (a), every equivalence relation gives a partition, its set of classes. By [Theorem 0.5.2](#) (c), the relation can be recovered from that partition, so distinct equivalence relations give distinct partitions. By [Theorem 0.5.2](#) (b), every partition arises this way. Hence equivalence relations on  $\{1, 2, 3\}$  correspond one to one with partitions, and there are exactly 5 of them.
- (c) Two elements are related exactly when they lie in a common block:

$$\{(1, 1), (2, 2), (3, 3), (1, 3), (3, 1)\}.$$

### Exercise C3: Intersections and unions of equivalence relations.

Let  $R_1$  and  $R_2$  be equivalence relations on a set  $X$ , viewed as subsets of  $X \times X$ .

- (a) Prove that  $R_1 \cap R_2$  is an equivalence relation on  $X$ .
- (b) Show by an example that  $R_1 \cup R_2$  need not be an equivalence relation. Which condition fails?
- (c) For congruence modulo 2 and modulo 3 on  $\mathbb{Z}$ , identify  $R_1 \cap R_2$  as a familiar relation.

*Hint for (b): try congruence modulo 2 and modulo 3 on  $\mathbb{Z}$ .*

*Solution.*

- (a) *Reflexive.* Let  $x \in X$ . Then  $(x, x) \in R_1$  and  $(x, x) \in R_2$ , since both are reflexive, so  $(x, x) \in R_1 \cap R_2$ . *Symmetric.* Suppose  $(x, y) \in R_1 \cap R_2$ . Since  $R_1$  and  $R_2$  are symmetric,  $(y, x) \in R_1$  and  $(y, x) \in R_2$ , so  $(y, x) \in R_1 \cap R_2$ . *Transitive.* Suppose  $(x, y), (y, z) \in R_1 \cap R_2$ . Both pairs lie in  $R_1$ , which is transitive, so  $(x, z) \in R_1$ ; likewise  $(x, z) \in R_2$ . Hence  $(x, z) \in R_1 \cap R_2$ .
- (b) Let  $R_1$  be congruence modulo 2 and  $R_2$  congruence modulo 3 on  $\mathbb{Z}$ . The union is still reflexive and symmetric, since each of  $R_1, R_2$  is. But transitivity fails:  $(0, 2) \in R_1$  since  $2 \mid 2$ , and  $(2, 5) \in R_2$  since  $3 \mid 3$ , yet  $(0, 5)$  lies in neither, because 5 is not a multiple of 2 or of 3.
- (c)  $(a, b) \in R_1 \cap R_2$  means  $2 \mid a - b$  and  $3 \mid a - b$ . This holds exactly when  $6 \mid a - b$ . Indeed, if  $a - b = 2k$  and  $a - b = 3m$  with  $k, m \in \mathbb{Z}$ , then  $a - b = 3(a - b) - 2(a - b) = 3 \cdot 2k - 2 \cdot 3m = 6(k - m)$ . Conversely, if  $a - b = 6j$ , then  $a - b = 2(3j) = 3(2j)$ . So  $R_1 \cap R_2$  is congruence modulo 6. The same phenomenon, where intersections behave well and unions do not, will reappear for subspaces in Chapter 1.

## 0.6 Complex Numbers

The equation  $x^2 + 1 = 0$  has no real solution, because the square of a real number is never negative. For most of school mathematics this gap is harmless. In linear algebra it is not: later in the book we meet real matrices, such as a rotation of the plane, whose most useful data cannot be found among the real numbers. This section builds the complex numbers, the number system that closes the gap, and collects the facts about them that the rest of the book uses.

## 0.6.1 The complex numbers as pairs of reals

The quick way to introduce complex numbers is to say “invent a symbol  $i$  with  $i^2 = -1$  and compute as usual”. That raises a fair worry. How do we know that such a symbol does not lead to a contradiction, the way “invent a number  $z$  with  $0 \cdot z = 1$ ” does? The answer is to build the new numbers out of objects we already trust: pairs of real numbers.

Here is the recurring expression that suggests the definition. If we pretend  $i^2 = -1$  and expand, we get

$$(a + bi)(c + di) = ac + adi + bci + bd i^2 = (ac - bd) + (ad + bc)i.$$

So a “number  $a + bi$ ” is really the pair  $(a, b)$ , and the product should be the pair  $(ac - bd, ad + bc)$ . We take this formula as the definition, so nothing is pretended.

*A complex number is a pair of real numbers, multiplied as if  $i^2 = -1$ .*

### Definition 0.6.1 — Complex Numbers

The set of **complex numbers** is  $\mathbb{C} := \mathbb{R}^2 = \{(a, b) : a, b \in \mathbb{R}\}$ , with addition and multiplication defined **for every**  $(a, b), (c, d) \in \mathbb{C}$  by

$$(a, b) + (c, d) := (a + c, b + d), \quad (a, b)(c, d) := (ac - bd, ad + bc).$$

We write  $a$  for  $(a, 0)$  and  $i$  for  $(0, 1)$ , so that  $(a, b) = a + bi$ . For  $z = a + bi$  with  $a, b \in \mathbb{R}$ , the **real part** of  $z$  is  $\operatorname{Re} z := a$  and the **imaginary part** of  $z$  is  $\operatorname{Im} z := b$ .

In words: addition is coordinate by coordinate, as for points of the plane. Multiplication is the formula found by expanding. The last sentence is a naming convention: the real number  $a$  is identified with the point  $(a, 0)$  on the horizontal axis, and  $i$  is the point one unit up. Note that the imaginary part  $b$  is a **real** number; it is not  $bi$ .

Two things need checking before this convention is safe.

- **The notation  $a + bi$  is honest.** By the definitions,  $a + bi = (a, 0) + (b, 0)(0, 1) = (a, 0) + (0, b) = (a, b)$ . Two complex numbers are equal exactly when they are equal as pairs, so  $a + bi = c + di$  with  $a, b, c, d \in \mathbb{R}$  means  $a = c$  **and**  $b = d$ . This is how we “compare real and imaginary parts”.
- **Identifying  $a$  with  $(a, 0)$  does not change real arithmetic.** We have  $(a, 0) + (c, 0) = (a + c, 0)$  and  $(a, 0)(c, 0) = (ac - 0, 0 + 0) = (ac, 0)$ . So adding or multiplying two real numbers inside  $\mathbb{C}$  gives the same answer as in  $\mathbb{R}$ .

With these in place, the equation that motivated everything is now a computation:  $i^2 = (0, 1)(0, 1) = (0 \cdot 0 - 1 \cdot 1, 0 \cdot 1 + 1 \cdot 0) = (-1, 0) = -1$ .

### Example. (First Computations in $\mathbb{C}$ )

Compute  $(2 + 3i)(1 - i)$ , the powers  $i^0, i^1, \dots, i^7$ , and the product  $(a + bi)(a - bi)$  for real  $a, b$ .

*Solution.* By the multiplication rule with  $a = 2, b = 3, c = 1, d = -1$ ,

$$(2 + 3i)(1 - i) = (2 \cdot 1 - 3 \cdot (-1)) + (2 \cdot (-1) + 3 \cdot 1)i = 5 + i.$$



For the powers,  $i^0 = 1$ ,  $i^1 = i$ ,  $i^2 = -1$ ,  $i^3 = i^2 \cdot i = -i$  and  $i^4 = (-i)i = -i^2 = 1$ . From then on the pattern repeats with period 4:  $i^5 = i$ ,  $i^6 = -1$ ,  $i^7 = -i$ .

Finally,  $(a + bi)(a - bi) = (a^2 - b(-b)) + (a(-b) + ba)i = a^2 + b^2$ . This product is a real number, and it is **non-negative**. We will meet it again as  $|z|^2$ .

The degenerate case deserves a look too. A real number  $a = a + 0i$  is a complex number whose imaginary part is zero, and the number  $0 = 0 + 0i$  is the only complex number with both parts zero. Every computation with real numbers is also a computation in  $\mathbb{C}$ , by the second check above.

**A non-example by minimal change.** Change one sign in the multiplication rule: define  $(a, b) * (c, d) := (ac + bd, ad + bc)$ , which is what expanding gives if we pretend  $j^2 = +1$  instead. Addition is unchanged, and this product is still commutative and has  $(1, 0)$  as a unit. But

$$(1, 1) * (1, -1) = (1 - 1, -1 + 1) = (0, 0),$$

so two non-zero elements multiply to zero. Division also breaks down:  $(1, 1) * (c, d) = (c + d, c + d)$  can never equal the unit  $(1, 0)$ , because that would need  $c + d = 1$  and  $c + d = 0$  at once. So the non-zero element  $(1, 1)$  has no inverse. The single sign in  $ac - bd$  is what makes the complex numbers a number system in which we can divide, as we are about to see.

*Quick check.* Using the definition, compute  $i^{10}$  and  $(1 + i)^2$ . Is  $\text{Im}(3 - 4i)$  equal to  $-4i$ ?

*Solution.* Since the powers of  $i$  repeat with period 4 and  $10 = 8 + 2$ , we get  $i^{10} = i^2 = -1$ . Next,  $(1 + i)^2 = (1 - 1) + (1 + 1)i = 2i$ . No: the imaginary part is the real number  $-4$ , not  $-4i$ .

**The usual rules of arithmetic hold.** Before computing freely, we should confirm that rearranging, regrouping and expanding brackets are legal in  $\mathbb{C}$ . The definition does not promise this; it has to be checked.

### Proposition 0.6.1 — Arithmetic Laws in $\mathbb{C}$

For all  $z, w, u \in \mathbb{C}$ :

- (a)  $z + w = w + z$  and  $zw = wz$ ;
- (b)  $(z + w) + u = z + (w + u)$  and  $(zw)u = z(wu)$ ;
- (c)  $z(w + u) = zw + zu$ ;
- (d)  $z + 0 = z$ ,  $z \cdot 1 = z$ , and  $z + (-z) = 0$ , where  $-(a + bi) := (-a) + (-b)i$ .

*Idea.* Every law reduces to the corresponding law for real numbers once both sides are written as pairs. Addition is coordinate by coordinate, so its laws are immediate. For multiplication the only real work is associativity: expand both sides and see that they produce the same four terms in each coordinate.

*Proof.* Let  $z = a + bi$ ,  $w = c + di$  and  $u = e + fi$  with  $a, b, c, d, e, f \in \mathbb{R}$ . Since addition acts coordinate by coordinate, the laws for addition in (a), (b) and (d) follow from the same laws in  $\mathbb{R}$ , applied to each coordinate.

- (a) By definition  $zw = (ac - bd) + (ad + bc)i$  and  $wz = (ca - db) + (cb + da)i$ . These

are equal because real multiplication and addition are commutative.

(b) Expanding with the definition twice gives

$$(zw)u = ((ac - bd)e - (ad + bc)f) + ((ac - bd)f + (ad + bc)e)i$$

and

$$z(wu) = (a(ce - df) - b(cf + de)) + (a(cf + de) + b(ce - df))i.$$

Multiplying out, both real parts equal  $ace - adf - bcf - bde$  and both imaginary parts equal  $acf + ade + bce - bdf$ . Hence  $(zw)u = z(wu)$ .

(c) We have  $w + u = (c + e) + (d + f)i$ , so

$$\begin{aligned} z(w + u) &= (a(c + e) - b(d + f)) + (a(d + f) + b(c + e))i \\ &= ((ac - bd) + (ae - bf)) + ((ad + bc) + (af + be))i, \end{aligned}$$

which is  $zw + zu$  by the definitions of the product and the sum.

(d) With  $1 = 1 + 0i$ , we get  $z \cdot 1 = (a \cdot 1 - b \cdot 0) + (a \cdot 0 + b \cdot 1)i = a + bi = z$ . Also  $z + (-z) = (a - a) + (b - b)i = 0$ . This proves the proposition. ■

From now on we compute in  $\mathbb{C}$  exactly as in  $\mathbb{R}$ : expand brackets, collect terms, and replace  $i^2$  by  $-1$ . The proposition is what makes that habit legitimate.

## 0.6.2 Conjugate and modulus

Division is still missing. To divide by  $3 + 4i$  we want some number to multiply by that lands in the real numbers, where division is familiar. The computation  $(a + bi)(a - bi) = a^2 + b^2$  in [Example \(First Computations in  \$\mathbb{C}\$ \)](#) shows exactly which number does the job. Both ingredients get names.

*Conjugating reflects a point in the real axis; the modulus is its distance from the origin.*

### Definition 0.6.2 — Conjugate and Modulus

Let  $z = a + bi \in \mathbb{C}$  with  $a, b \in \mathbb{R}$ . The **complex conjugate** of  $z$  is  $\bar{z} := a - bi$ . The **modulus** (or absolute value) of  $z$  is the real number  $|z| := \sqrt{a^2 + b^2} \geq 0$ , the non-negative square root.

In words: the conjugate keeps the real part and flips the sign of the imaginary part. The modulus is the length of the segment from 0 to the point  $(a, b)$ , by Pythagoras. For a real number  $a = a + 0i$ , we get  $\bar{a} = a$  and  $|a| = \sqrt{a^2}$ , the usual absolute value, so the notation does not clash with its real meaning. For example,  $\overline{3 + 4i} = 3 - 4i$ ,  $|3 + 4i| = 5$ ,  $\bar{i} = -i$  and  $|i| = 1$ .

The following rules are used constantly, in this section and much later for complex inner products.

### Theorem 0.6.2 — Properties of Conjugation and Modulus

For all  $z, w \in \mathbb{C}$ :

- (a)  $\overline{z + w} = \bar{z} + \bar{w}$  and  $\overline{\bar{z}} = z$ ;
- (b)  $\overline{zw} = \bar{z}\bar{w}$ ;
- (c)  $z\bar{z} = |z|^2$  and  $z + \bar{z} = 2\operatorname{Re} z$ ;
- (d)  $|zw| = |z||w|$  and  $|\bar{z}| = |z|$ .

*Idea.* Parts (a)–(c) are direct computations with real and imaginary parts. For (d), expanding  $|zw|$  directly is legitimate but messy. It is cleaner to square: by (c) a squared modulus is a product with a conjugate, and (b) lets the conjugate split across the product.

*Proof.* Let  $z = a + bi$  and  $w = c + di$  with  $a, b, c, d \in \mathbb{R}$ .

- (a) We have  $\overline{z + w} = (a + c) - (b + d)i = (a - bi) + (c - di) = \bar{z} + \bar{w}$ , and  $\overline{\bar{z}} = \overline{a - bi} = a + bi = z$ .
- (b) Since  $zw = (ac - bd) + (ad + bc)i$ , we get  $\overline{zw} = (ac - bd) - (ad + bc)i$ . On the other hand,

$$\begin{aligned}\bar{z}\bar{w} &= (a - bi)(c - di) \\ &= (ac - (-b)(-d)) + (a(-d) + (-b)c)i \\ &= (ac - bd) - (ad + bc)i.\end{aligned}$$

- (c) By the computation in [Example \(First Computations in  \$\mathbb{C}\$ \)](#),  $z\bar{z} = a^2 + b^2 = |z|^2$ . Also  $z + \bar{z} = 2a = 2\operatorname{Re} z$ .
- (d) By (c), then (b), then [Proposition 0.6.1](#) to reorder the four factors,

$$|zw|^2 = (zw)\overline{zw} = zw\bar{z}\bar{w} = (z\bar{z})(w\bar{w}) = |z|^2|w|^2.$$

Both  $|zw|$  and  $|z||w|$  are non-negative real numbers with the same square, so they are equal. Finally  $|\bar{z}| = \sqrt{a^2 + (-b)^2} = |z|$ . This proves the theorem. ■

The first payoff is division. If  $z \neq 0$ , then  $|z|^2 = a^2 + b^2$  is a **strictly** positive real number, so we may divide by it.

### Proposition 0.6.3 — Inverse of a Complex Number

Let  $z \in \mathbb{C}$  with  $z \neq 0$ . Then  $w := \frac{1}{|z|^2} \bar{z}$  satisfies  $zw = 1$ . We write  $z^{-1}$  or  $\frac{1}{z}$  for this  $w$ .

*Proof.* Write  $z = a + bi$ . Since  $z \neq 0$ , at least one of  $a, b$  is non-zero, so  $|z|^2 = a^2 + b^2 > 0$ . By [Proposition 0.6.1](#) and [Theorem 0.6.2](#) (c),

$$zw = \frac{1}{|z|^2} (z\bar{z}) = \frac{1}{|z|^2} |z|^2 = 1,$$

as claimed. ■

In practice, “multiply top and bottom by the conjugate”. For instance

$$\begin{aligned}\frac{1}{3+4i} &= \frac{3-4i}{(3+4i)(3-4i)} = \frac{3-4i}{25}, \\ \frac{5+i}{1-i} &= \frac{(5+i)(1+i)}{(1-i)(1+i)} = \frac{4+6i}{2} = 2+3i,\end{aligned}$$

which agrees with  $(2+3i)(1-i) = 5+i$  from [Example \(First Computations in  \$\mathbb{C}\$ \)](#). Only one inverse exists; this is a special case of a general fact about fields proved in the next section.

**The triangle inequality.** The modulus measures distance:  $|z-w|$  is the distance between the points  $z$  and  $w$  of the plane. The side of a triangle is never longer than the other two sides together, and this is the algebraic form of that fact.

### Theorem 0.6.4 — Triangle Inequality

For all  $z, w \in \mathbb{C}$ ,

$$|z+w| \leq |z| + |w|.$$

*Idea.* Moduli are square roots, which are awkward to add. Both sides are non-negative, so it suffices to compare their squares. Expand  $|z+w|^2$  using  $|u|^2 = u\bar{u}$ : the cross terms combine into  $2\operatorname{Re}(z\bar{w})$ . A real part is never bigger than the modulus, and that single estimate finishes the proof.

*Proof.* First, for any  $u = x + yi$  with  $x, y \in \mathbb{R}$ , we have  $\operatorname{Re} u = x \leq |x| = \sqrt{x^2} \leq \sqrt{x^2 + y^2} = |u|$ , since the square root is increasing.

Now let  $z, w \in \mathbb{C}$ . By [Theorem 0.6.2](#) (c), (a) and [Proposition 0.6.1](#),

$$|z+w|^2 = (z+w)(\bar{z}+\bar{w}) = z\bar{z} + z\bar{w} + w\bar{z} + w\bar{w} = |z|^2 + (z\bar{w} + \overline{z\bar{w}}) + |w|^2,$$

where we used  $w\bar{z} = \overline{z\bar{w}}$ , which follows from [Theorem 0.6.2](#) (b) and (a). By [Theorem 0.6.2](#) (c) the bracket equals  $2\operatorname{Re}(z\bar{w})$ . Therefore, using  $\operatorname{Re} u \leq |u|$  and then [Theorem 0.6.2](#) (d),

$$|z+w|^2 \leq |z|^2 + 2|z\bar{w}| + |w|^2 = |z|^2 + 2|z||w| + |w|^2 = (|z| + |w|)^2.$$

Both  $|z+w|$  and  $|z|+|w|$  are non-negative, so taking square roots preserves the inequality. This proves the triangle inequality. ■

The route “square, expand, bound the cross term” is worth remembering: it is exactly the route we will take for norms in inner product spaces. The inequality is often strict. For  $z = 1$  and  $w = -1$ , the left side is 0 and the right side is 2.

## 0.6.3 Two things that do not carry over from the reals

Complex numbers obey the rules of arithmetic, but not every habit from the real line survives.

The first loss is order. On  $\mathbb{R}$  we use two facts about positivity: for every  $x$ , exactly one of  $x > 0$ ,  $x = 0$ ,  $-x > 0$  holds, and sums and products of positive numbers are positive. Suppose some relation  $>$  on  $\mathbb{C}$  had both properties. Since  $i \neq 0$ , either  $i > 0$  or  $-i > 0$ , and in both cases the product rule gives  $i^2 = (-i)^2 = -1 > 0$ . The same argument applied to  $1 \neq 0$  gives  $1 = 1^2 = (-1)^2 > 0$ . Then both 1 and  $-1$  are positive, so their sum 0 is positive, which contradicts “exactly one”.

*Warning.* There is no ordering on  $\mathbb{C}$  compatible with addition and multiplication, so an inequality such as  $z < w$  between complex numbers means nothing. Inequalities in this book compare **real** numbers: moduli  $|z|$ , real parts, or quantities known to be real. Writing  $i > 0$  leads, as shown above, to  $0 > 0$ .

The second loss is the square-root rule  $\sqrt{x}\sqrt{y} = \sqrt{xy}$ , which on  $\mathbb{R}$  is only claimed for  $x, y \geq 0$ .

*Warning.* Every non-zero complex number has **two** square roots, and no choice of “the” square root makes  $\sqrt{z}\sqrt{w} = \sqrt{zw}$  true for all  $z, w$ . With  $\sqrt{-1} = i$ :  $\sqrt{-1} \cdot \sqrt{-1} = i^2 = -1$ , but  $\sqrt{(-1)(-1)} = \sqrt{1} = 1$ . Avoid the symbol  $\sqrt{z}$  for non-real  $z$ ; say “a square root of  $z$ ” and solve  $w^2 = z$  instead.

Solving  $w^2 = z$  is a finite computation, done by comparing real and imaginary parts.

#### Example. (Square Roots of $5 - 12i$ )

Find all  $w \in \mathbb{C}$  with  $w^2 = 5 - 12i$ .

*Solution.* Write  $w = x + yi$  with  $x, y \in \mathbb{R}$ . Then  $w^2 = (x^2 - y^2) + 2xyi$ , so comparing real and imaginary parts,  $w^2 = 5 - 12i$  holds exactly when

$$x^2 - y^2 = 5 \quad \text{and} \quad xy = -6.$$

Taking moduli in  $w^2 = 5 - 12i$  and using [Theorem 0.6.2](#) (d) gives  $|w|^2 = |5 - 12i| = \sqrt{25 + 144} = 13$ , that is  $x^2 + y^2 = 13$ . Adding and subtracting this and the first equation gives  $x^2 = 9$  and  $y^2 = 4$ , so  $x = \pm 3$  and  $y = \pm 2$ . The condition  $xy = -6 < 0$  says  $x$  and  $y$  have opposite signs. Hence  $w = 3 - 2i$  or  $w = -3 + 2i$ . Both work:  $(3 - 2i)^2 = 9 - 12i + 4i^2 = 5 - 12i$ , and  $(-3 + 2i)^2 = (3 - 2i)^2$ .

## 0.6.4 Polar form: multiplication rotates and scales

Addition in  $\mathbb{C}$  is the familiar addition of arrows in the plane. Multiplication has a geometric meaning too, and it only becomes visible when we describe a point by its distance from 0 and its angle, instead of by its coordinates.

Let  $z \neq 0$  and put  $r = |z| > 0$ . The number  $z/r$  has modulus 1 by [Theorem 0.6.2](#) (d), so it lies on the unit circle. From trigonometry, every point  $(x, y)$  with  $x^2 + y^2 = 1$  is  $(\cos \theta, \sin \theta)$  for some real  $\theta$ . We abbreviate

$$e^{i\theta} := \cos \theta + i \sin \theta \quad (\theta \in \mathbb{R}),$$

and then every non-zero complex number can be written as

$$z = re^{i\theta}, \quad r = |z| > 0, \quad \theta \in \mathbb{R}.$$

This is the **polar form** of  $z$ , and  $\theta$  is **an** argument of  $z$ . The word “an” matters:  $\theta$  and  $\theta + 2\pi k$  give the same point for every integer  $k$ . Conversely, if  $re^{i\theta} = se^{i\varphi}$  with  $r, s > 0$ , then taking moduli gives  $r = s$  (as  $|e^{i\theta}| = \sqrt{\cos^2 \theta + \sin^2 \theta} = 1$ ). Comparing real and imaginary parts then gives  $\cos \theta = \cos \varphi$  and  $\sin \theta = \sin \varphi$ , which from trigonometry means  $\theta - \varphi$  is an integer multiple of  $2\pi$ .

For now  $e^{i\theta}$  is only a name for  $\cos \theta + i \sin \theta$ . The next result is the reason the name is a good one: these numbers multiply the way exponentials do.

### Theorem 0.6.5 — Multiplication in Polar Form

For all  $\theta, \varphi \in \mathbb{R}$ ,

$$e^{i\theta} e^{i\varphi} = e^{i(\theta+\varphi)}.$$

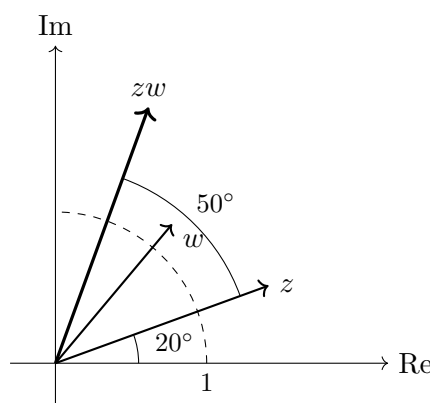
Consequently, if  $z = re^{i\theta}$  and  $w = se^{i\varphi}$  with  $r, s > 0$ , then  $zw = rs e^{i(\theta+\varphi)}$ .

*Proof.* By the multiplication rule,

$$\begin{aligned} e^{i\theta} e^{i\varphi} &= (\cos \theta \cos \varphi - \sin \theta \sin \varphi) + (\cos \theta \sin \varphi + \sin \theta \cos \varphi)i \\ &= \cos(\theta + \varphi) + i \sin(\theta + \varphi), \end{aligned}$$

where the last equality uses the addition formulas for cosine and sine. This is  $e^{i(\theta+\varphi)}$ . For the second statement,  $zw = (rs)(e^{i\theta} e^{i\varphi})$  by [Proposition 0.6.1](#), and the first statement finishes the proof. ■

In words: **to multiply two complex numbers, multiply their moduli and add their arguments**. Multiplying by a fixed  $w = se^{i\varphi}$  is therefore a rotation of the plane about 0 by the angle  $\varphi$ , combined with a stretch by the factor  $s$ . The picture shows  $z$ ,  $w$  and  $zw$  for  $|z| = 1.5$ ,  $|w| = 1.2$  and arguments  $20^\circ$  and  $50^\circ$ .



The arrow for  $zw$  has length  $1.5 \times 1.2 = 1.8$  and makes the angle  $20^\circ + 50^\circ = 70^\circ$  with the positive real axis: it is the arrow for  $z$ , turned by  $w$ 's angle and stretched by  $w$ 's length. The small arc marks the argument of  $z$ , the large arc marks the extra turn by the argument of  $w$ , and the dashed arc is the unit circle, where the numbers  $e^{i\theta}$  live.

Two special cases are worth isolating. Multiplying by  $i = e^{i\pi/2}$  is a quarter turn counterclockwise, which explains  $i^2 = -1$ : two quarter turns send 1 to  $-1$ . And  $e^{i\theta} e^{-i\theta} = e^0 = 1$ , so

$$(e^{i\theta})^{-1} = e^{-i\theta} = \overline{e^{i\theta}}.$$

Repeating [Theorem 0.6.5](#) gives powers.

### Theorem 0.6.6 — De Moivre's Formula

For every  $\theta \in \mathbb{R}$  and every  $n \in \mathbb{N}$ ,

$$(e^{i\theta})^n = e^{in\theta}, \quad \text{that is,} \quad (\cos \theta + i \sin \theta)^n = \cos n\theta + i \sin n\theta.$$

Consequently  $(re^{i\theta})^n = r^n e^{in\theta}$  for every real  $r > 0$ .

*Proof.* We use induction on  $n$  ([Theorem 0.2.7](#)). For  $n = 0$ , both sides equal 1, since  $e^{i \cdot 0} = \cos 0 + i \sin 0 = 1$ . Suppose  $(e^{i\theta})^n = e^{in\theta}$  for some  $n \in \mathbb{N}$ . Then  $(e^{i\theta})^{n+1} = (e^{i\theta})^n e^{i\theta} = e^{in\theta} e^{i\theta} = e^{i(n+1)\theta}$  by the induction hypothesis and [Theorem 0.6.5](#). The consequence follows by the same induction, since  $(re^{i\theta})^{n+1} = (r^n e^{in\theta})(re^{i\theta}) = r^{n+1} e^{i(n+1)\theta}$  by [Theorem 0.6.5](#). ■

Powers that are painful to expand become one line in polar form.

#### Example. (A Power via Polar Form)

Compute  $(\sqrt{3} + i)^6$ .

*Solution.* The modulus is  $|\sqrt{3} + i| = \sqrt{3+1} = 2$ , and  $\sqrt{3} + i = 2\left(\frac{\sqrt{3}}{2} + \frac{1}{2}i\right) = 2\left(\cos \frac{\pi}{6} + i \sin \frac{\pi}{6}\right) = 2e^{i\pi/6}$ . By [Theorem 0.6.6](#),

$$(\sqrt{3} + i)^6 = 2^6 e^{i \cdot 6\pi/6} = 64(\cos \pi + i \sin \pi) = -64.$$

Expanding  $(\sqrt{3} + i)^6$  with the binomial theorem gives the same answer after seven terms and some cancellation.

*Quick check.* Use polar form to compute  $(1 + i)^8$ .

*Solution.* We have  $|1 + i| = \sqrt{2}$  and  $1 + i = \sqrt{2}\left(\frac{1}{\sqrt{2}} + \frac{1}{\sqrt{2}}i\right) = \sqrt{2}e^{i\pi/4}$ . By [Theorem 0.6.6](#),  $(1 + i)^8 = (\sqrt{2})^8 e^{i \cdot 8\pi/4} = 16e^{2\pi i} = 16$ . As a cross-check,  $(1 + i)^2 = 2i$ , so  $(1 + i)^8 = (2i)^4 = 16i^4 = 16$ .

## 0.6.5 Roots of unity

Polar form also turns the equation  $z^n = c$  into a question about angles. The most important case is  $c = 1$ .

#### Example. (Roots of Unity)

Let  $n \geq 1$  be an integer. Find all  $z \in \mathbb{C}$  with  $z^n = 1$ , and draw them for  $n = 6$ .

*Solution.* Since  $0^n = 0 \neq 1$ , every solution is non-zero, so we may write  $z = re^{i\theta}$  with  $r > 0$ . By [Theorem 0.6.6](#),  $z^n = r^n e^{in\theta}$ , and  $1 = 1 \cdot e^{i0}$ . By uniqueness of the polar form up to multiples of  $2\pi$ ,  $z^n = 1$  holds exactly when  $r^n = 1$  and  $n\theta = 2\pi k$  for some integer  $k$ . Since  $r > 0$  is real,  $r^n = 1$  forces  $r = 1$ . Hence the solutions are

$$e^{2\pi i k/n}, \quad k \in \mathbb{Z}.$$

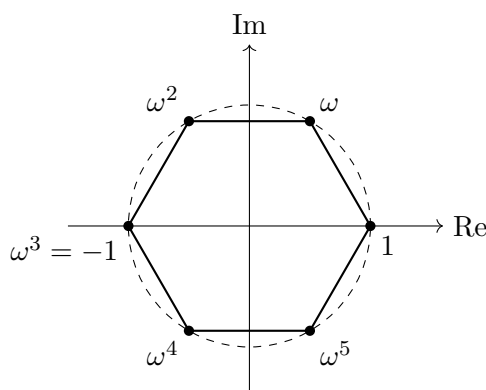
Put  $\omega := e^{2\pi i/n}$ . By [Theorem 0.6.6](#),  $e^{2\pi i k/n} = \omega^k$  for  $k \geq 0$ . Two integers  $k, l$  give the same number exactly when  $2\pi(k-l)/n$  is a multiple of  $2\pi$ , that is, when  $n$  divides  $k-l$ . So there are exactly  $n$  distinct solutions,

$$1, \omega, \omega^2, \dots, \omega^{n-1},$$

the  **$n$ -th roots of unity**. They sit on the unit circle at equally spaced angles. For  $n \geq 3$  they are the corners of a regular polygon with  $n$  corners, one of them at 1; for  $n = 2$  they are  $\pm 1$ .

For  $n = 6$ , the angles are multiples of  $60^\circ$ , and the roots are

$$1, \quad \frac{1}{2} + \frac{\sqrt{3}}{2}i, \quad -\frac{1}{2} + \frac{\sqrt{3}}{2}i, \quad -1, \quad -\frac{1}{2} - \frac{\sqrt{3}}{2}i, \quad \frac{1}{2} - \frac{\sqrt{3}}{2}i.$$



The six roots sum to 0: since  $\omega^3 = e^{i\pi} = -1$ , we have  $\omega^{k+3} = -\omega^k$ , so the roots cancel in the opposite pairs  $\{1, \omega^3\}$ ,  $\{\omega, \omega^4\}$ ,  $\{\omega^2, \omega^5\}$ . In the picture, the hexagon is symmetric about its center 0. The sum of the  $n$ -th roots of unity is 0 for every  $n \geq 2$ ; see [Exercise \(C2: Roots of Unity Sum to Zero\)](#).

## 0.6.6 The Fundamental Theorem of Algebra

We built  $\mathbb{C}$  so that one polynomial equation,  $z^2 + 1 = 0$ , has a solution. It turns out that nothing more ever needs to be added: every polynomial equation already has a solution in  $\mathbb{C}$ . Here a **complex polynomial of degree  $n$**  means an expression  $a_n z^n + \dots + a_1 z + a_0$  with  $a_0, \dots, a_n \in \mathbb{C}$  and  $a_n \neq 0$ ; polynomials are studied properly in a later section of this chapter.

### Theorem 0.6.7 — Fundamental Theorem of Algebra, Statement



Let  $n \geq 1$  and let  $a_0, a_1, \dots, a_n \in \mathbb{C}$  with  $a_n \neq 0$ . Then there exists  $z \in \mathbb{C}$  with

$$a_n z^n + a_{n-1} z^{n-1} + \dots + a_1 z + a_0 = 0.$$

We do not prove this theorem here. Its proof needs tools we have not built, and it is proved in Chapter 5. Nothing in Chapters 0–4 depends on it.

*Remark.* Combined with division of polynomials (also in Chapter 5), the theorem gives the form in which it is usually applied: every complex polynomial of degree  $n \geq 1$  **splits into linear factors**,  $a_n z^n + \dots + a_0 = a_n(z - z_1)(z - z_2) \dots (z - z_n)$  for some  $z_1, \dots, z_n \in \mathbb{C}$ , not necessarily distinct. For linear algebra this is the fact that every complex square matrix has an eigenvalue, the starting point of the theory of canonical forms.

One companion fact needs no heavy machinery. If the coefficients  $a_0, \dots, a_n$  are **real** and  $p(z) = a_n z^n + \dots + a_0$  has  $p(z_0) = 0$ , then applying [Theorem 0.6.2](#) (a) and (b) term by term, and  $\overline{a_k} = a_k$ , gives  $p(\overline{z_0}) = \overline{p(z_0)} = \overline{0} = 0$ . So the non-real roots of a real polynomial come in conjugate pairs. For example,  $z^2 - 2z + 5$  has the roots  $1 + 2i$  and  $1 - 2i$ : indeed  $(1 + 2i)^2 - 2(1 + 2i) + 5 = (-3 + 4i) - (2 + 4i) + 5 = 0$ .

## 0.6.7 Exercises

### 0.6.7.1 A. Check your understanding

#### Exercise A1.

- Define the complex conjugate  $\bar{z}$  and the modulus  $|z|$  of  $z = a + bi$ , where  $a, b \in \mathbb{R}$ .
- State the triangle inequality for complex numbers.
- True or false:  $|z + w| = |z| + |w|$  for all  $z, w \in \mathbb{C}$ . Justify your answer.
- True or false:  $z\bar{z}$  is a non-negative real number for every  $z \in \mathbb{C}$ . Justify your answer.
- True or false: if  $z, w \in \mathbb{C}$  and  $z^2 = w^2$ , then  $z = w$ . Justify your answer.
- State the Fundamental Theorem of Algebra.

*Solution.*

- See [Definition 0.6.2](#):  $\bar{z} = a - bi$  and  $|z| = \sqrt{a^2 + b^2} \geq 0$ .
- See [Theorem 0.6.4](#):  $|z + w| \leq |z| + |w|$  for all  $z, w \in \mathbb{C}$ .
- False. For  $z = 1$  and  $w = -1$ ,  $|z + w| = 0$  but  $|z| + |w| = 2$ .
- True. By [Theorem 0.6.2](#) (c),  $z\bar{z} = |z|^2 = a^2 + b^2$ , which is real and non-negative.
- False. For  $z = 1$  and  $w = -1$ ,  $z^2 = w^2 = 1$  but  $z \neq w$ . (What is true is that  $z^2 = w^2$  implies  $z = w$  or  $z = -w$ .)
- See [Theorem 0.6.7](#): every complex polynomial  $a_n z^n + \dots + a_0$  with  $n \geq 1$  and  $a_n \neq 0$  has a root in  $\mathbb{C}$ .

### 0.6.7.2 B. Practice

**Exercise B1: Arithmetic.**

Write each of the following in the form  $a + bi$  with  $a, b \in \mathbb{R}$ , or as a real number where indicated.

- (a)  $(3 - 2i)(4 + i)$ .
- (b)  $\frac{1 + 2i}{2 - i}$ .
- (c)  $|(3 + 4i)(5 - 12i)|$ , without expanding the product.
- (d)  $\operatorname{Re} \frac{1}{1 + i}$  and  $\operatorname{Im} \frac{1}{1 + i}$ .

*Solution.*

- (a) Expanding and using  $i^2 = -1$ :  $(3 - 2i)(4 + i) = 12 + 3i - 8i - 2i^2 = 14 - 5i$ .
- (b) Multiplying top and bottom by  $\overline{2 - i} = 2 + i$  ([Proposition 0.6.3](#)),

$$\frac{1 + 2i}{2 - i} = \frac{(1 + 2i)(2 + i)}{(2 - i)(2 + i)} = \frac{2 + i + 4i + 2i^2}{4 + 1} = \frac{5i}{5} = i.$$

- (c) By [Theorem 0.6.2](#) (d),  $|(3 + 4i)(5 - 12i)| = |3 + 4i| |5 - 12i| = \sqrt{25} \cdot \sqrt{169} = 5 \cdot 13 = 65$ .
- (d) By [Proposition 0.6.3](#),  $\frac{1}{1 + i} = \frac{1 - i}{|1 + i|^2} = \frac{1 - i}{2} = \frac{1}{2} - \frac{1}{2}i$ . Hence the real part is  $\frac{1}{2}$  and the imaginary part is  $-\frac{1}{2}$ .

**Exercise B2: A Square Root.**

- (a) Find all  $w \in \mathbb{C}$  with  $w^2 = -3 + 4i$ .
- (b) Hence solve  $z^2 + 2z + 4 - 4i = 0$ .

*Hint: for (b), complete the square.*

*Solution.*

- (a) Write  $w = x + yi$  with  $x, y \in \mathbb{R}$ . Comparing real and imaginary parts in  $(x^2 - y^2) + 2xyi = -3 + 4i$  gives  $x^2 - y^2 = -3$  and  $xy = 2$ . Taking moduli, by [Theorem 0.6.2](#) (d),  $x^2 + y^2 = |w|^2 = |-3 + 4i| = 5$ . Adding and subtracting,  $x^2 = 1$  and  $y^2 = 4$ . Since  $xy = 2 > 0$ ,  $x$  and  $y$  have the same sign. Therefore  $w = 1 + 2i$  or  $w = -1 - 2i$ . Check:  $(1 + 2i)^2 = 1 + 4i - 4 = -3 + 4i$ , and  $(-1 - 2i)^2 = (1 + 2i)^2$ .
- (b) Completing the square,  $z^2 + 2z + 4 - 4i = (z + 1)^2 - 1 + 4 - 4i = (z + 1)^2 - (-3 + 4i)$ . So the equation holds exactly when  $(z + 1)^2 = -3 + 4i$ . By (a), this means  $z + 1 = 1 + 2i$  or  $z + 1 = -1 - 2i$ . Hence the solutions are  $z = 2i$  and  $z = -2 - 2i$ .

**Exercise B3.**

Let  $z \in \mathbb{C}$  with  $z \neq 0$ . Prove that  $\frac{1}{z} = \bar{z}$  if and only if  $|z| = 1$ .

*Solution.* By [Proposition 0.6.3](#),  $\frac{1}{z} = \frac{1}{|z|^2} \bar{z}$ .

( $\Leftarrow$ ) Suppose  $|z| = 1$ . Then  $\frac{1}{z} = \frac{1}{1} \bar{z} = \bar{z}$ .

( $\Rightarrow$ ) Suppose  $\frac{1}{z} = \bar{z}$ . Then  $\frac{1}{|z|^2} \bar{z} = \bar{z}$ . Since  $z \neq 0$ , also  $\bar{z} \neq 0$ , so multiplying both sides by  $(\bar{z})^{-1}$  gives  $\frac{1}{|z|^2} = 1$ , that is,  $|z|^2 = 1$ . As  $|z| \geq 0$ , this forces  $|z| = 1$ . This proves the equivalence.

#### Exercise B4: Polar Form.

- (a) Compute  $(1 - \sqrt{3}i)^6$ .
- (b) Find all  $z \in \mathbb{C}$  with  $z^3 = -8i$ , in the form  $a + bi$ .

*Solution.*

- (a) We have  $|1 - \sqrt{3}i| = \sqrt{1+3} = 2$  and  $1 - \sqrt{3}i = 2(\cos(-\frac{\pi}{3}) + i \sin(-\frac{\pi}{3})) = 2e^{-i\pi/3}$ . By [Theorem 0.6.6](#),  $(1 - \sqrt{3}i)^6 = 64e^{-2\pi i} = 64(\cos 2\pi - i \sin 2\pi) = 64$ .
- (b) Since  $0^3 \neq -8i$ , write  $z = re^{i\theta}$  with  $r > 0$ . Also  $-8i = 8e^{-i\pi/2}$ . By [Theorem 0.6.6](#),  $z^3 = r^3 e^{3i\theta}$ . By uniqueness of polar form up to multiples of  $2\pi$ ,  $z^3 = -8i$  holds exactly when  $r^3 = 8$  and  $3\theta = -\frac{\pi}{2} + 2\pi k$  for some integer  $k$ . Therefore  $r = 2$  and  $\theta = -\frac{\pi}{6} + \frac{2\pi k}{3}$ . The values  $k = 0, 1, 2$  give  $\theta = -\frac{\pi}{6}, \frac{\pi}{2}, \frac{7\pi}{6}$ , and every other  $k$  differs from one of these by a multiple of 3, which changes  $\theta$  by a multiple of  $2\pi$ . Hence the solutions are

$$2e^{-i\pi/6} = \sqrt{3} - i, \quad 2e^{i\pi/2} = 2i, \quad 2e^{7i\pi/6} = -\sqrt{3} - i.$$

Check:  $(2i)^3 = 8i^3 = -8i$ .

#### 0.6.7.3 C. Going deeper

##### Exercise C1: Reverse Triangle Inequality.

Prove that  $||z| - |w|| \leq |z - w|$  for all  $z, w \in \mathbb{C}$ .

*Hint:* write  $z = (z - w) + w$ .

*Solution.* Let  $z, w \in \mathbb{C}$ . Since  $z = (z - w) + w$ , the triangle inequality ([Theorem 0.6.4](#)) gives  $|z| \leq |z - w| + |w|$ , so  $|z| - |w| \leq |z - w|$ . Swapping the roles of  $z$  and  $w$  gives  $|w| - |z| \leq |w - z|$ . Moreover  $|w - z| = |(-1)(z - w)| = |-1||z - w| = |z - w|$  by [Theorem 0.6.2](#) (d). The real number  $||z| - |w||$  equals one of  $|z| - |w|$  and  $|w| - |z|$ , and both are at most  $|z - w|$ . This proves the inequality.

##### Exercise C2: Roots of Unity Sum to Zero.

Let  $n \geq 2$  and  $\omega = e^{2\pi i/n}$ .

- (a) Prove that  $1 + \omega + \omega^2 + \cdots + \omega^{n-1} = 0$ .
- (b) Deduce that  $\sum_{k=0}^{n-1} \cos \frac{2\pi k}{n} = 0$  and  $\sum_{k=0}^{n-1} \sin \frac{2\pi k}{n} = 0$ .

*Hint: compare  $S$  with  $\omega S$ , where  $S$  is the sum in (a).*

*Solution.*

- (a) Let  $S = 1 + \omega + \cdots + \omega^{n-1}$ . By [Theorem 0.6.6](#),  $\omega^n = e^{2\pi i} = 1$ . Therefore

$$\omega S = \omega + \omega^2 + \cdots + \omega^{n-1} + \omega^n = \omega + \omega^2 + \cdots + \omega^{n-1} + 1 = S,$$

so  $(\omega - 1)S = 0$ . Since  $n \geq 2$ , we have  $0 < 2\pi/n \leq \pi$ , so  $\operatorname{Re} \omega = \cos(2\pi/n) < 1$  and  $\omega \neq 1$ . Hence  $\omega - 1 \neq 0$ , and multiplying  $(\omega - 1)S = 0$  by  $(\omega - 1)^{-1}$  ([Proposition 0.6.3](#)) gives  $S = 0$ .

- (b) By [Theorem 0.6.6](#),  $\omega^k = \cos \frac{2\pi k}{n} + i \sin \frac{2\pi k}{n}$  for each  $k$ . Since addition in  $\mathbb{C}$  is coordinate by coordinate, the real part of  $S$  is  $\sum_{k=0}^{n-1} \cos \frac{2\pi k}{n}$  and the imaginary part is  $\sum_{k=0}^{n-1} \sin \frac{2\pi k}{n}$ . By (a),  $S = 0$ , so both sums are 0.

### Exercise C3: Symmetries of $\mathbb{C}$ Fixing $\mathbb{R}$ .

Let  $\sigma: \mathbb{C} \rightarrow \mathbb{C}$  be a function with  $\sigma(z + w) = \sigma(z) + \sigma(w)$  and  $\sigma(zw) = \sigma(z)\sigma(w)$  for all  $z, w \in \mathbb{C}$ , and  $\sigma(a) = a$  for every  $a \in \mathbb{R}$ .

- (a) Prove that  $\sigma(i)^2 = -1$ .
- (b) Prove that the only complex numbers  $u$  with  $u^2 = -1$  are  $i$  and  $-i$ .
- (c) Deduce that either  $\sigma(z) = z$  for every  $z \in \mathbb{C}$ , or  $\sigma(z) = \bar{z}$  for every  $z \in \mathbb{C}$ .
- (d) Show that  $z \mapsto \bar{z}$  does satisfy all three conditions on  $\sigma$ .

*Hint: for (b), write  $u = x + yi$  and compare real and imaginary parts.*

*Solution.*

- (a) Since  $\sigma$  respects products and fixes the real number  $-1$ , we get  $\sigma(i)^2 = \sigma(i \cdot i) = \sigma(-1) = -1$ .
- (b) Let  $u = x + yi$  with  $x, y \in \mathbb{R}$  and  $u^2 = -1$ . Comparing real and imaginary parts in  $(x^2 - y^2) + 2xyi = -1$  gives  $x^2 - y^2 = -1$  and  $xy = 0$ . If  $y = 0$ , then  $x^2 = -1$ , which is impossible for real  $x$ . Hence  $x = 0$ , so  $y^2 = 1$  and  $y = \pm 1$ . Therefore  $u = i$  or  $u = -i$ , and both satisfy  $u^2 = -1$ .
- (c) By (a) and (b),  $\sigma(i) = i$  or  $\sigma(i) = -i$ . Let  $z = a + bi$  with  $a, b \in \mathbb{R}$ . Since  $\sigma$  respects sums and products and fixes  $a$  and  $b$ ,

$$\sigma(z) = \sigma(a) + \sigma(b)\sigma(i) = a + b\sigma(i).$$

If  $\sigma(i) = i$ , this gives  $\sigma(z) = z$  for every  $z$ . If  $\sigma(i) = -i$ , it gives  $\sigma(z) = a - bi = \bar{z}$  for every  $z$ .

- (d) By [Theorem 0.6.2](#) (a) and (b), conjugation respects sums and products. For  $a \in \mathbb{R}$ ,  $\bar{a} = \overline{a + 0i} = a - 0i = a$ . So conjugation satisfies all three conditions.

## 0.7 Fields

Linear algebra is about equations like  $2x + 3y = 7$ , and so far we have met several number systems in which such equations make sense:  $\mathbb{Q}$ ,  $\mathbb{R}$ ,  $\mathbb{C}$ , and the arithmetic of remainders  $\mathbb{Z}/n\mathbb{Z}$ . Which of them can serve as the numbers of linear algebra? This section answers by isolating the rules that solving linear equations actually uses. A number system obeying those rules is called a field, and fields are the scalars of every later chapter.

### 0.7.1 What solving linear equations needs

Here is a small system, solved the way you learned at school:

$$\begin{aligned} 2x + 3y &= 7, \\ x - y &= 1. \end{aligned}$$

Subtracting twice the second equation from the first gives  $5y = 5$ . Dividing by 5 gives  $y = 1$ , and then  $x = 1 + y = 2$ . Check:  $2 \cdot 2 + 3 \cdot 1 = 7$  and  $2 - 1 = 1$ .

Look at what we used. We added and subtracted equations, multiplied an equation by a number, and divided by a **non-zero** number. Silently we also rearranged terms, regrouped them and expanded brackets. Every step of Gaussian elimination is of this kind, and every one of them works in  $\mathbb{Q}$ ,  $\mathbb{R}$  and  $\mathbb{C}$ .

Now try the integers. The equation  $2x = 1$  involves only integers, yet it has no integer solution, because the step “divide by 2” leaves  $\mathbb{Z}$ . Elimination stalls as soon as a pivot has no inverse. So what separates  $\mathbb{Z}$  from  $\mathbb{Q}$  is division, and division is exactly what we must demand.

*A field is a number system where you can add, subtract, multiply, and divide by anything non-zero, with the usual rules.*

### 0.7.2 The definition

#### Definition 0.7.1 — Field

A **field** is a set  $F$  together with two functions, **addition**  $+: F \times F \rightarrow F$ ,  $(a, b) \mapsto a + b$ , and **multiplication**  $\cdot: F \times F \rightarrow F$ ,  $(a, b) \mapsto a \cdot b = ab$ , satisfying the following axioms.

- (F1)  $a + b = b + a$  **for all**  $a, b \in F$ .
- (F2)  $(a + b) + c = a + (b + c)$  **for all**  $a, b, c \in F$ .
- (F3) There exists an element  $0 \in F$  such that  $a + 0 = a$  **for every**  $a \in F$ .
- (F4) **For every**  $a \in F$  there exists an element  $-a \in F$  such that  $a + (-a) = 0$ .
- (F5)  $(ab)c = a(bc)$  **for all**  $a, b, c \in F$ .
- (F6)  $ab = ba$  **for all**  $a, b \in F$ .
- (F7) There exists an element  $1 \in F$  with  $1 \neq 0$  such that  $a \cdot 1 = a$  **for every**  $a \in F$ .
- (F8) **For every non-zero**  $a \in F$  there exists an element  $a^{-1} \in F$  such that  $aa^{-1} = 1$ .
- (F9)  $a(b + c) = ab + ac$  **for all**  $a, b, c \in F$ .

Read the axioms in groups.

- **(F1)–(F4)** are about addition alone. It is commutative and associative, it has a neutral element 0, and every element has a negative. Subtraction is then defined by  $a - b := a + (-b)$ .
- **(F5)–(F8)** say the same things about multiplication, with one change: only **non-zero** elements are required to have inverses. Division is defined by  $a/b := ab^{-1}$  for  $b \neq 0$ . We never divide by 0, and the axioms never ask us to.
- **(F7)** also insists that  $1 \neq 0$ : the two neutral elements are different.
- **(F9)** is the one rule linking the two operations. It is what lets us expand brackets.

Note the order of quantifiers in (F3): **one** element 0 works **for every**  $a$ . This is stronger than “for every  $a$  there is some element  $z_a$  with  $a + z_a = a$ ”. In (F4) the order is reversed, as it must be: the negative  $-a$  depends on  $a$ .

We often say “ $F$  is a field” when the operations are clear, and write  $(F, +, \cdot)$  when they need to be named. Since addition and multiplication are functions into  $F$ , sums and products of elements of  $F$  stay in  $F$ ; when we build a field inside a bigger number system, this “closure” is the first thing to check.

### 0.7.3 First consequences of the axioms

The notation in the definition hides a question. We wrote “0”, “ $-a$ ” and “ $a^{-1}$ ” as if each were a single well-defined element. The axioms only say that such elements **exist**. If a field could contain two different elements behaving like 0, the symbol 0 would be ambiguous, and so would  $a - b$  and  $a/b$ . The next theorem removes this worry and proves the familiar rules of algebra from the nine axioms.

#### Theorem 0.7.1 — Basic Properties of Fields

Let  $F$  be a field and let  $a, b, c \in F$ .

- The element 0 in (F3) is unique, and the element 1 in (F7) is unique.
- (Cancellation) If  $a + b = a + c$ , then  $b = c$ . If  $ab = ac$  and  $a \neq 0$ , then  $b = c$ .
- The negative  $-a$  in (F4) is unique, and for  $a \neq 0$  the inverse  $a^{-1}$  in (F8) is unique.
- $0 \cdot a = 0$ .
- $(-1)a = -a$ .
- If  $ab = 0$ , then  $a = 0$  or  $b = 0$ .

*Idea.* Each part is a short chain of axioms, and the chain is found backwards from the goal. For uniqueness we use the standard move: suppose two elements both have the property, and show they are equal. For (d) there is no axiom about  $0 \cdot a$  at all, so we must create a situation where one applies. The trick is to rewrite 0 in a fancy way as  $0 + 0$ , expand with (F9), and cancel. For (f), the hypothesis  $a \neq 0$  is there to be spent on (F8).

We prove the uniqueness of 0, additive cancellation, the uniqueness of negatives, (d) and (f). The remaining statements are [Exercise \(B1: The Remaining Basic Properties\)](#).

*Proof.*

- (a) Suppose  $0$  and  $0'$  both satisfy (F3). Applying (F3) for  $0'$  to the element  $0$ , and (F3) for  $0$  to the element  $0'$ , and using (F1),

$$0 = 0 + 0' = 0' + 0 = 0'.$$

So the zero element is unique.

- (b) Suppose  $a + b = a + c$ . Adding  $-a$  on the left of both sides and using (F2),

$$((-a) + a) + b = ((-a) + a) + c.$$

By (F1) and (F4),  $(-a) + a = a + (-a) = 0$ . Hence  $0 + b = 0 + c$ , and by (F1) and (F3) this says  $b = c$ .

- (c) Suppose  $b$  and  $c$  both satisfy (F4) for  $a$ , that is,  $a + b = 0$  and  $a + c = 0$ . Then  $a + b = a + c$ , so  $b = c$  by additive cancellation in (b).  
 (d) By (F3),  $0 + 0 = 0$ . Therefore, by (F9),

$$a \cdot 0 = a(0 + 0) = a \cdot 0 + a \cdot 0.$$

By (F3), also  $a \cdot 0 = a \cdot 0 + 0$ . Comparing,  $a \cdot 0 + 0 = a \cdot 0 + a \cdot 0$ , and additive cancellation (b) gives  $0 = a \cdot 0$ . Finally  $0 \cdot a = a \cdot 0$  by (F6), so  $0 \cdot a = 0$ .

- (e) Suppose  $ab = 0$ . If  $a = 0$  there is nothing to prove, so suppose  $a \neq 0$ . By (F8),  $a^{-1}$  exists. Then, by (F7), (F8), (F6) and (F5),

$$b = b \cdot 1 = b(aa^{-1}) = (ba)a^{-1} = (ab)a^{-1} = 0 \cdot a^{-1} = 0,$$

where the last equality is (d). This shows that  $a = 0$  or  $b = 0$ . ■

Two small consequences are used without comment from now on:  $0 + a = a$  and  $1 \cdot a = a$ , by (F1), (F3), (F6) and (F7). More importantly, every identity of school algebra that is proved from these rules, such as  $(a + b)^2 = a^2 + 2ab + b^2$  or  $(-a)(-b) = ab$ , now holds in every field. Part (f) is the property we will use most: a product of non-zero elements is non-zero.

## 0.7.4 Examples

We check the standard examples first, then an exotic one, then a degenerate case.

### Example. (The Fields $\mathbb{Q}$ , $\mathbb{R}$ and $\mathbb{C}$ )

Explain why  $\mathbb{Q}$ ,  $\mathbb{R}$  and  $\mathbb{C}$ , with their usual addition and multiplication, are fields.

*Solution.* For  $\mathbb{R}$ , the axioms (F1)–(F9) are the rules of arithmetic for real numbers that we take as familiar, and  $1 \neq 0$ .

For  $\mathbb{Q}$ , the sum, product and negative of fractions are fractions, and so is the inverse  $\frac{q}{p}$  of a non-zero fraction  $\frac{p}{q}$ . So the operations of  $\mathbb{R}$  restrict to functions  $\mathbb{Q} \times \mathbb{Q} \rightarrow \mathbb{Q}$ , and  $0, 1, -a, a^{-1}$  lie in  $\mathbb{Q}$  whenever  $a$  does. The identities in (F1), (F2), (F5), (F6) and (F9)

hold for all real numbers, so in particular for rational ones. Hence  $\mathbb{Q}$  is a field.

For  $\mathbb{C}$ , the axioms (F1)–(F6) and (F9), together with the existence of  $1 = 1 + 0i$ , are [Proposition 0.6.1](#). We have  $1 = (1, 0) \neq (0, 0) = 0$ . Finally (F8) is [Proposition 0.6.3](#): a non-zero  $z$  has the inverse  $\bar{z}/|z|^2$ . Hence  $\mathbb{C}$  is a field.

The next example is a field strictly between  $\mathbb{Q}$  and  $\mathbb{R}$ . It shows that the inverse axiom (F8) can take real work to check.

**Example. ( $\mathbb{Q}(\sqrt{2})$  Is a Field)**

Let  $\mathbb{Q}(\sqrt{2}) := \{a + b\sqrt{2} : a, b \in \mathbb{Q}\} \subseteq \mathbb{R}$ , with the addition and multiplication of  $\mathbb{R}$ . Show that  $\mathbb{Q}(\sqrt{2})$  is a field, and find the inverse of  $1 + 2\sqrt{2}$ .

*Solution.* Let  $x = a + b\sqrt{2}$  and  $y = c + d\sqrt{2}$  with  $a, b, c, d \in \mathbb{Q}$ .

*Closure.* We have  $x + y = (a + c) + (b + d)\sqrt{2}$  and  $xy = (ac + 2bd) + (ad + bc)\sqrt{2}$ , using  $\sqrt{2} \cdot \sqrt{2} = 2$ . The coefficients are rational, so  $x + y, xy \in \mathbb{Q}(\sqrt{2})$ .

*Axioms that are identities.* (F1), (F2), (F5), (F6) and (F9) hold for all real numbers, hence for the elements of  $\mathbb{Q}(\sqrt{2})$ .

*Neutral elements and negatives.*  $0 = 0 + 0\sqrt{2}$  and  $1 = 1 + 0\sqrt{2}$  lie in  $\mathbb{Q}(\sqrt{2})$ , and  $1 \neq 0$ . The negative  $-x = (-a) + (-b)\sqrt{2}$  lies in  $\mathbb{Q}(\sqrt{2})$ .

*Inverses.* Suppose  $x \neq 0$ . We first claim  $a - b\sqrt{2} \neq 0$ . If  $b \neq 0$  and  $a - b\sqrt{2} = 0$ , then  $\sqrt{2} = a/b$  would be rational, contradicting [Theorem 0.2.3](#). If  $b = 0$ , then  $a = x \neq 0$ , so  $a - b\sqrt{2} = a \neq 0$ . Therefore  $a - b\sqrt{2} \neq 0$ , and by [Theorem 0.7.1](#) (f) in  $\mathbb{R}$ ,

$$N := (a + b\sqrt{2})(a - b\sqrt{2}) = a^2 - 2b^2 \neq 0.$$

Since  $N$  is a non-zero rational number, the element

$$x' := \frac{a}{N} + \frac{-b}{N}\sqrt{2} \in \mathbb{Q}(\sqrt{2})$$

satisfies  $xx' = \frac{1}{N}(a + b\sqrt{2})(a - b\sqrt{2}) = 1$ . So (F8) holds, and  $\mathbb{Q}(\sqrt{2})$  is a field.

For  $x = 1 + 2\sqrt{2}$  we get  $N = 1 - 8 = -7$ , so  $x^{-1} = -\frac{1}{7} + \frac{2}{7}\sqrt{2} = \frac{-1+2\sqrt{2}}{7}$ . Check:  $(1 + 2\sqrt{2})(-1 + 2\sqrt{2}) = -1 + 2\sqrt{2} - 2\sqrt{2} + 8 = 7$ .

The method is “rationalize the denominator”, the same move as multiplying by the conjugate in  $\mathbb{C}$ . The irrationality of  $\sqrt{2}$  is exactly what guarantees the new denominator is not zero.

*Quick check.* Is  $\{a + b\sqrt{3} : a, b \in \mathbb{Q}\}$ , with the operations of  $\mathbb{R}$ , a field?

*Solution.* Yes, by the same argument as [Example \( \$\mathbb{Q}\(\sqrt{2}\)\$  Is a Field\)](#). Products stay in the set because  $\sqrt{3} \cdot \sqrt{3} = 3$  is rational. For inverses, the number  $a^2 - 3b^2$  is non-zero whenever  $a + b\sqrt{3} \neq 0$ , because  $\sqrt{3}$  is irrational ([Exercise \(B3:  \$\sqrt{3}\$  is irrational\)](#)). The inverse is  $\frac{a-b\sqrt{3}}{a^2-3b^2}$ .

**The degenerate case.** Let  $F = \{0\}$  with  $0 + 0 = 0$  and  $0 \cdot 0 = 0$ . Every axiom except one holds: (F1), (F2), (F5), (F6) and (F9) are identities between elements that all equal 0, (F3) and



(F4) hold with  $-0 = 0$ , and (F8) holds vacuously because there is no non-zero element. The only failure is (F7): the one element that could serve as 1 is 0, and the axiom demands  $1 \neq 0$ . So  $\{0\}$  is not a field, and every field has at least the two elements 0 and 1. This one-element system is the reason the clause  $1 \neq 0$  is there.

## 0.7.5 Finite fields

Every field so far is infinite. The arithmetic of remainders gives finite ones, and they are genuinely useful: computer science and coding theory do linear algebra over them all the time.

Recall from [Definition 0.5.2](#) and [Definition 0.5.5](#) that for an integer  $n \geq 1$ , congruence modulo  $n$  is an equivalence relation on  $\mathbb{Z}$ . We write  $[a]$  for the class of  $a$  and  $\mathbb{Z}/n\mathbb{Z}$  for the quotient set. By division with remainder ([Exercise \(C2: Division with remainder\)](#)),  $\mathbb{Z}/n\mathbb{Z} = \{[0], [1], \dots, [n-1]\}$ , and these  $n$  classes are distinct. Also  $[a] = [b]$  exactly when  $n$  divides  $a - b$ .

### Example. (Arithmetic on $\mathbb{Z}/n\mathbb{Z}$ and the Field $\mathbb{F}_2$ )

Let  $n \geq 1$ . Define  $[a] + [b] := [a + b]$  and  $[a][b] := [ab]$ . Show that these operations are well defined, that they satisfy every field axiom except possibly (F8) and the clause  $1 \neq 0$ , and that  $1 \neq 0$  holds exactly when  $n \geq 2$ . Write out the tables for  $n = 2$ .

*Solution. Well-definedness.* The formulas use representatives, so we must check that the answer does not depend on which representatives we pick. For addition this is [Example \(Addition modulo  \$n\$  is well defined\)](#). For multiplication, suppose  $[a] = [a']$  and  $[b] = [b']$ , so  $n$  divides  $a - a'$  and  $b - b'$ . Then

$$ab - a'b' = a(b - b') + (a - a')b'$$

is divisible by  $n$ , so  $[ab] = [a'b']$ .

*Axioms.* Every identity in (F1), (F2), (F5), (F6), (F9) transfers from  $\mathbb{Z}$ , because both sides can be computed on representatives. For instance, for (F9),

$$[a]([b] + [c]) = [a][b + c] = [a(b + c)] = [ab + ac] = [ab] + [ac] = [a][b] + [a][c].$$

The same one-line computation works for the other four. The class  $[0]$  satisfies (F3) since  $[a] + [0] = [a]$ , the class  $[-a]$  is a negative of  $[a]$  as in (F4), and  $[1]$  satisfies  $[a][1] = [a]$ .

*The clause  $1 \neq 0$ .* We have  $[1] = [0]$  exactly when  $n$  divides 1, that is, when  $n = 1$ . So  $[1] \neq [0]$  exactly when  $n \geq 2$ .

*Tables for  $n = 2$ .* Writing 0, 1 for  $[0], [1]$ , the set  $\mathbb{F}_2 := \mathbb{Z}/2\mathbb{Z} = \{0, 1\}$  has the tables

| $+$ | 0 | 1 |
|-----|---|---|
| 0   | 0 | 1 |
| 1   | 1 | 0 |

| $\cdot$ | 0 | 1 |
|---------|---|---|
| 0       | 0 | 0 |
| 1       | 0 | 1 |

The only surprise is  $1 + 1 = 0$ , since 2 is even. Here (F8) holds as well, since the only non-zero element is 1 and  $1 \cdot 1 = 1$ . Hence  $\mathbb{F}_2$  is a field.

So for  $n \geq 2$ , whether  $\mathbb{Z}/n\mathbb{Z}$  is a field comes down to one question: does every non-zero class have an inverse? To find an inverse of  $[a]$  we need an integer  $x$  with  $[ax] = [1]$ , that is,  $ax + ny = 1$

for some integer  $y$ . The following lemma produces such  $x$  and  $y$ .

### Lemma 0.7.2 — Bézout's Lemma for Integers

Let  $a, n \in \mathbb{Z}$  with  $n \geq 1$ , and suppose that the only positive integer dividing both  $a$  and  $n$  is 1. Then there exist  $x, y \in \mathbb{Z}$  with

$$ax + ny = 1.$$

*Idea.* Consider all positive integers of the form  $ax + ny$ . There is a smallest one,  $d$ , by well-ordering. The plan: ① show  $d$  divides  $a$  and  $n$ ; ② conclude  $d = 1$  from the hypothesis. For ①, divide  $a$  by  $d$ : the remainder is again of the form  $ax + ny$  and is smaller than  $d$ , so minimality forces it to be 0.

*Proof.* Let  $S = \{ax + ny : x, y \in \mathbb{Z}\} \cap \{1, 2, 3, \dots\}$ . Since  $n = a \cdot 0 + n \cdot 1 \geq 1$ , we have  $n \in S$ , so  $S$  is a non-empty subset of  $\mathbb{N}$ . By the well-ordering principle (Theorem 0.2.11),  $S$  has a smallest element  $d = ax_0 + ny_0$ .

By division with remainder (Exercise (C2: Division with remainder)),  $a = qd + r$  for some integers  $q, r$  with  $0 \leq r < d$ . Then

$$r = a - q(ax_0 + ny_0) = a(1 - qx_0) + n(-qy_0)$$

has the form  $ax + ny$ . If  $r \geq 1$ , then  $r \in S$  and  $r < d$ , contradicting the minimality of  $d$ . Hence  $r = 0$ , and  $d$  divides  $a$ . The same argument with  $n$  in place of  $a$  shows that  $d$  divides  $n$ .

Therefore  $d$  is a positive integer dividing both  $a$  and  $n$ . By hypothesis  $d = 1$ , so  $ax_0 + ny_0 = 1$ , as claimed. ■

Now we can decide exactly when  $\mathbb{Z}/n\mathbb{Z}$  is a field.

### Theorem 0.7.3 — $\mathbb{Z}/n\mathbb{Z}$ Is a Field Exactly for Prime $n$

Let  $n \geq 2$  be an integer. Then  $\mathbb{Z}/n\mathbb{Z}$  is a field if and only if  $n$  is prime.

*Idea.* By Example (Arithmetic on  $\mathbb{Z}/n\mathbb{Z}$  and the Field  $\mathbb{F}_2$ ), only (F8) is in question. If  $n$  is prime, a non-zero class  $[a]$  has  $a$  not divisible by  $n$ , so  $a$  and  $n$  share no factor, and Bézout hands us the inverse. If  $n = rs$  factors, then  $[r][s] = [n] = [0]$  with both factors non-zero, which no field allows.

*Proof.* By Example (Arithmetic on  $\mathbb{Z}/n\mathbb{Z}$  and the Field  $\mathbb{F}_2$ ),  $\mathbb{Z}/n\mathbb{Z}$  satisfies all field axioms except possibly (F8), including  $[1] \neq [0]$  since  $n \geq 2$ .

( $\Leftarrow$ ) Suppose  $n = p$  is prime, and let  $[a] \neq [0]$ . Then  $p$  does not divide  $a$ . A positive integer dividing  $p$  is 1 or  $p$ , and  $p$  does not divide  $a$ , so the only positive common divisor of  $a$  and  $p$  is 1. By Lemma 0.7.2 there are  $x, y \in \mathbb{Z}$  with  $ax + py = 1$ . Since  $ax - 1 = -py$  is divisible by  $p$ ,

$$[a][x] = [ax] = [1].$$

So  $[x]$  is an inverse of  $[a]$ , and (F8) holds. Hence  $\mathbb{Z}/p\mathbb{Z}$  is a field.

( $\Rightarrow$ ) We prove the contrapositive. Suppose  $n \geq 2$  is not prime. Then  $n = rs$  for some integers  $r, s$  with  $1 < r < n$  and  $1 < s < n$ . Since  $0 < r < n$ ,  $n$  does not divide  $r$ , so  $[r] \neq [0]$ ; likewise  $[s] \neq [0]$ . But  $[r][s] = [rs] = [n] = [0]$ . By [Theorem 0.7.1](#) (f), this cannot happen in a field. Hence  $\mathbb{Z}/n\mathbb{Z}$  is not a field. This proves the theorem. ■

For a prime  $p$  we write  $\mathbb{F}_p := \mathbb{Z}/p\mathbb{Z}$ , and usually drop the brackets, writing its elements as  $0, 1, \dots, p-1$  and computing “mod  $p$ ”. So  $\mathbb{F}_2, \mathbb{F}_3, \mathbb{F}_5, \mathbb{F}_7, \dots$  are fields, while  $\mathbb{Z}/4\mathbb{Z}$ ,  $\mathbb{Z}/6\mathbb{Z}$  and  $\mathbb{Z}/9\mathbb{Z}$  are not.

*Quick check.* Find the inverse of 3 in  $\mathbb{F}_7$ .

*Solution.* We need  $x$  with  $3x$  leaving remainder 1 on division by 7. Trying  $x = 1, 2, \dots$ :  $3 \cdot 5 = 15 = 2 \cdot 7 + 1$ . So  $3^{-1} = 5$  in  $\mathbb{F}_7$ . (By [Theorem 0.7.1](#) (c) this is the only inverse.)

Because  $\mathbb{F}_p$  is a field, the elimination from the start of the section runs unchanged in it. Only the arithmetic is new.

### Example. (Elimination over $\mathbb{F}_5$ )

Solve the system  $2x + y = 3$ ,  $3x + 2y = 4$  over  $\mathbb{F}_5$ .

*Solution.* In  $\mathbb{F}_5$  the inverses are  $1^{-1} = 1$ ,  $2^{-1} = 3$ ,  $3^{-1} = 2$ ,  $4^{-1} = 4$ , since  $2 \cdot 3 = 6 = 5 + 1$  and  $4 \cdot 4 = 16 = 15 + 1$ . Multiply the first equation by  $2^{-1} = 3$ : since  $3 \cdot 2 = 1$ ,  $3 \cdot 1 = 3$  and  $3 \cdot 3 = 9 = 4$ , it becomes  $x + 3y = 4$ . Subtract 3 times this from the second equation: the  $x$ -terms cancel, and we get

$$(2 - 9)y = 4 - 12, \quad \text{that is,} \quad 3y = 2,$$

since  $-7 = -10 + 3$  and  $-8 = -10 + 2$ . Multiply by  $3^{-1} = 2$ :  $y = 4$ . Then  $x = 4 - 3y = 4 - 12 = -8 = 2$ . Check:  $2 \cdot 2 + 4 = 8 = 3$  and  $3 \cdot 2 + 2 \cdot 4 = 14 = 4$  in  $\mathbb{F}_5$ . Hence  $(x, y) = (2, 4)$  is the unique solution.

## 0.7.6 Non-examples

Each of these fails some part of the slogan, and it pays to name the exact clause that fails.

- $\mathbb{Z}$ . Everything but (F8) holds:  $\mathbb{Z}$  is closed under  $+$  and  $\cdot$ , contains 0, 1 and negatives, and inherits the identities from  $\mathbb{R}$ . But  $2 \neq 0$  has no inverse in  $\mathbb{Z}$ , since  $2m$  is even and never equals 1. Changing  $\mathbb{Z}$  to  $\mathbb{Q}$  is the minimal change that repairs it.
- $\mathbb{Z}/6\mathbb{Z}$ . This is  $\mathbb{F}_5$  with the modulus changed from 5 to 6. All axioms but (F8) still hold ([Example \(Arithmetic on  \$\mathbb{Z}/n\mathbb{Z}\$  and the Field  \$\mathbb{F}\_2\$ \)](#)), but  $[2][3] = [0]$  with  $[2], [3] \neq [0]$ . By [Theorem 0.7.1](#) (f),  $[2]$  cannot have an inverse; indeed  $[2][x]$  is always one of  $[0], [2], [4]$ .
- $2 \times 2$  **real matrices**, with the matrix operations you know from school (they are treated properly later in this chapter). The identity matrix plays the role of 1. Two axioms fail.

(F6) fails:

$$\begin{pmatrix} 0 & 1 \\ 0 & 0 \end{pmatrix} \begin{pmatrix} 0 & 0 \\ 1 & 0 \end{pmatrix} = \begin{pmatrix} 1 & 0 \\ 0 & 0 \end{pmatrix}, \quad \begin{pmatrix} 0 & 0 \\ 1 & 0 \end{pmatrix} \begin{pmatrix} 0 & 1 \\ 0 & 0 \end{pmatrix} = \begin{pmatrix} 0 & 0 \\ 0 & 1 \end{pmatrix}.$$

(F8) fails too:  $\mathbf{E} = \begin{pmatrix} 1 & 0 \\ 0 & 0 \end{pmatrix}$  is not the zero matrix, but every product  $\mathbf{E}\mathbf{X}$  has second row zero, so  $\mathbf{E}\mathbf{X}$  is never the identity.

*Warning.* “[ $a$ ]  $\neq$  [ $0$ ]” does not mean [ $a$ ] can be divided by in  $\mathbb{Z}/n\mathbb{Z}$ . In  $\mathbb{Z}/6\mathbb{Z}$  the equation  $[2]x = [1]$  has no solution, and  $[2]x = [4]$  has two,  $x = [2]$  and  $x = [5]$ . Before dividing modulo  $n$ , check that  $n$  is prime, or at least that  $a$  and  $n$  share no factor.

## 0.7.7 Why these axioms

Each clause is there because something we rely on breaks without it.

- **Without (F8)**, elimination stalls, as  $2x = 1$  over  $\mathbb{Z}$  showed. Worse, the theorems of the next chapters fail: over  $\mathbb{Z}$ , the analogues of vector spaces (called modules) need not have bases, and “a set of  $n$  independent vectors in an  $n$ -dimensional space spans it” becomes false. This book always works over a field.
- **Without (F6)**, scalars would not commute with each other. The matrices above show such systems exist, and there are even systems with division but without commutativity. Linear algebra constantly swaps scalars, as in  $a(b\mathbf{v}) = b(a\mathbf{v})$  for a vector  $\mathbf{v}$  (Chapter 1), so we keep commutativity.
- **Without  $1 \neq 0$** , the one-element system  $\{0\}$  would count, and every statement of the form “a field has a non-zero element” would fail. Keeping it costs nothing.
- **The name** “field” is historical and carries no extra meaning.

*Remark.* Many identities we are used to are consequences of the axioms and hold in **every** field, including  $\mathbb{F}_2$ . But properties that involve **order** or **size**, such as “squares are non-negative” or “ $1 + 1 + \cdots + 1 \neq 0$ ”, are not among the axioms. The first already fails in  $\mathbb{C}$ , where  $i^2 = -1$ , as the warning on ordering in the previous section explains. The second fails in  $\mathbb{F}_p$ , which is the subject of the last part of this section.

## 0.7.8 The characteristic of a field

In  $\mathbb{Q}$  we can add 1 to itself as often as we like without reaching 0. In  $\mathbb{F}_2$  two copies already give 0, and in  $\mathbb{F}_p$  it takes  $p$  copies. This number is the most basic invariant of a field, and it decides whether familiar arguments such as “ $2x = 0$  implies  $x = 0$ ” are valid.

First some notation. For  $a \in F$  and  $m \in \mathbb{N}$ , define the **integer multiple**  $m \cdot a$  by  $0 \cdot a := 0$  and  $(m+1) \cdot a := m \cdot a + a$ . So  $m \cdot a = a + a + \cdots + a$  with  $m$  terms. This agrees with the field product when  $m = 0$ , by [Theorem 0.7.1](#) (d).

### Lemma 0.7.4 — Integer Multiples

Let  $F$  be a field,  $a \in F$  and  $m, k \in \mathbb{N}$ . Then

$$\begin{aligned}(m \cdot 1) + (k \cdot 1) &= (m + k) \cdot 1, \\ (m \cdot 1)(k \cdot 1) &= (mk) \cdot 1, \\ (m \cdot 1)a &= m \cdot a.\end{aligned}$$

*Proof.* All three follow by induction on one variable ([Theorem 0.2.7](#)). For the first, induct on  $k$ : the case  $k = 0$  is  $m \cdot 1 + 0 = m \cdot 1$  by (F3), and if it holds for  $k$ , then by (F2) and the definition,

$$m \cdot 1 + (k + 1) \cdot 1 = (m \cdot 1 + k \cdot 1) + 1 = (m + k) \cdot 1 + 1 = (m + k + 1) \cdot 1.$$

For the third, induct on  $m$ : the case  $m = 0$  is  $0 \cdot a = 0$  by [Theorem 0.7.1](#) (d), and if it holds for  $m$ , then by (F6), (F9) and (F7),

$$((m + 1) \cdot 1)a = (m \cdot 1 + 1)a = (m \cdot 1)a + 1 \cdot a = m \cdot a + a = (m + 1) \cdot a.$$

The second is the third with  $a = k \cdot 1$ , combined with  $m \cdot (k \cdot 1) = (mk) \cdot 1$ , which follows from the first by induction on  $m$ . ■

### Definition 0.7.2 — Characteristic

Let  $F$  be a field. If there is a **positive** integer  $n$  with  $n \cdot 1 = 0$ , the **characteristic** of  $F$  is the **smallest** such  $n$ . If there is no such  $n$ , the characteristic of  $F$  is 0.

In words: count how many copies of 1 must be added to reach 0. If it never happens, the convention is to say “characteristic 0”, not “characteristic  $\infty$ ”: in that case the only integer  $m \geq 0$  with  $m \cdot 1 = 0$  is  $m = 0$ .

- $\mathbb{Q}, \mathbb{R}, \mathbb{C}$  have characteristic 0, because  $n \cdot 1 = n \neq 0$  for every positive integer  $n$ .
- $\mathbb{F}_p$  has characteristic  $p$ :  $n \cdot [1] = [n]$ , which is  $[0]$  exactly when  $p$  divides  $n$ , and the smallest positive such  $n$  is  $p$ .
- $\mathbb{F}_2$  is the smallest case: characteristic 2, because  $1 + 1 = 0$ .

Can a field have characteristic 6? In  $\mathbb{Z}/6\mathbb{Z}$ , six copies of  $[1]$  give  $[0]$ , but  $\mathbb{Z}/6\mathbb{Z}$  is not a field. This is no accident.

### Theorem 0.7.5 — The Characteristic Is 0 or Prime

The characteristic of a field is either 0 or a prime number.

*Idea.* Suppose the characteristic  $n$  factors as  $rs$  with smaller factors. Then the elements  $r \cdot 1$  and  $s \cdot 1$  are non-zero, by minimality of  $n$ , but their product is  $n \cdot 1 = 0$ . A field has no such pairs. This is the same argument as in [Theorem 0.7.3](#), run inside an arbitrary field.

*Proof.* Let  $F$  be a field of characteristic  $n \neq 0$ . Since the integer multiple  $1 \cdot 1 = 0 + 1 = 1$  is non-zero by (F7), we have  $n \neq 1$ , so  $n \geq 2$ . Suppose, for a contradiction, that  $n$  is not

prime. Then  $n = rs$  with integers  $1 < r < n$  and  $1 < s < n$ . By [Lemma 0.7.4](#),

$$(r \cdot 1)(s \cdot 1) = (rs) \cdot 1 = n \cdot 1 = 0.$$

By [Theorem 0.7.1](#) (f),  $r \cdot 1 = 0$  or  $s \cdot 1 = 0$ . Either way we have a positive integer smaller than  $n$  whose multiple of 1 is 0, contradicting the minimality of  $n$ . Hence  $n$  is prime. ■

So the characteristic is 0, 2, 3, 5, 7, and so on. Together with [Exercise \(C2: Finite Fields Have Prime Characteristic\)](#), this says a finite field always has prime characteristic.

*Warning.* In a field of characteristic 2, such as  $\mathbb{F}_2$ , we have  $1 + 1 = 0$ , so  $-1 = 1$  and  $x = -x$  for **every**  $x$ . The familiar argument “ $x = -x$ , so  $2x = 0$ , so  $x = 0$ ” divides by  $2 = 1 + 1$ , which is 0 there. For example, in  $\mathbb{F}_2$  the element 1 satisfies  $1 = -1$  but is not 0. This is why later results, notably on symmetric bilinear forms in Chapter 13, carry the hypothesis “characteristic  $\neq 2$ ”.

With fields in hand, the stage for linear algebra is set: in Chapter 1, a vector space is defined **over a field**  $F$ , and every result proved there holds at once for  $\mathbb{Q}$ ,  $\mathbb{R}$ ,  $\mathbb{C}$ ,  $\mathbb{Q}(\sqrt{2})$  and  $\mathbb{F}_p$ .

## 0.7.9 Exercises

### 0.7.9.1 A. Check your understanding

#### Exercise A1.

- (a) Define what it means for a set  $F$  with two operations to be a field.
- (b) True or false: in every field,  $a^2 = 0$  implies  $a = 0$ . Justify your answer.
- (c) True or false:  $\mathbb{Z}/8\mathbb{Z}$  is a field. Justify your answer.
- (d) True or false: in every field,  $1 + 1 \neq 0$ . Justify your answer.
- (e) What are the characteristics of  $\mathbb{C}$  and of  $\mathbb{F}_7$ ?
- (f) The set  $\mathbb{N}$  with the usual addition and multiplication is not a field. Name two axioms that fail, each with a witness.

*Solution.*

- (a) See [Definition 0.7.1](#): two operations satisfying (F1)–(F9), including  $1 \neq 0$ .
- (b) True. If  $a \cdot a = 0$ , then  $a = 0$  or  $a = 0$  by [Theorem 0.7.1](#) (f).
- (c) False. Since  $8 = 2 \cdot 4$  is not prime,  $\mathbb{Z}/8\mathbb{Z}$  is not a field by [Theorem 0.7.3](#); concretely  $[2][4] = [0]$  with  $[2], [4] \neq [0]$ .
- (d) False. In  $\mathbb{F}_2$ ,  $1 + 1 = 0$ .
- (e)  $\mathbb{C}$  has characteristic 0, and  $\mathbb{F}_7$  has characteristic 7 (see the examples after [Definition 0.7.2](#)).
- (f) (F4) fails:  $1 \in \mathbb{N}$ , but no  $b \in \mathbb{N}$  has  $1 + b = 0$ . (F8) fails:  $2 \neq 0$ , but no  $b \in \mathbb{N}$  has  $2b = 1$ .

## 0.7.9.2 B. Practice

**Exercise B1: The Remaining Basic Properties.**

Let  $F$  be a field and  $a, b, c \in F$ . Using only the axioms and the parts of [Theorem 0.7.1](#) proved in the text, prove:

- (a) the element 1 in (F7) is unique;
- (b) if  $ab = ac$  and  $a \neq 0$ , then  $b = c$ ;
- (c) for  $a \neq 0$ , the inverse  $a^{-1}$  is unique;
- (d)  $(-1)a = -a$ .

*Hint: for (d), show that  $a + (-1)a = 0$ .*

*Solution.*

- (a) Suppose 1 and  $1'$  both satisfy (F7). Then, by (F7) for  $1'$ , (F6), and (F7) for 1,  $1 = 1 \cdot 1' = 1' \cdot 1 = 1'$ .
- (b) Suppose  $ab = ac$  and  $a \neq 0$ . By (F8),  $a^{-1}$  exists, and multiplying both sides by it gives  $(ab)a^{-1} = (ac)a^{-1}$ . By (F6) and (F5),  $(ab)a^{-1} = (ba)a^{-1} = b(aa^{-1}) = b \cdot 1 = b$ , using (F8) and (F7) at the end. In the same way  $(ac)a^{-1} = c$ . Hence  $b = c$ .
- (c) Suppose  $b$  and  $c$  both satisfy (F8) for  $a \neq 0$ :  $ab = 1 = ac$ . By (b),  $b = c$ .
- (d) By (F7) and (F6),  $a = 1 \cdot a$ . Therefore, using (F6) and (F9),

$$a + (-1)a = 1 \cdot a + (-1)a = a \cdot 1 + a(-1) = a(1 + (-1)) = a \cdot 0 = 0,$$

where the last two equalities use (F4) and [Theorem 0.7.1](#) (d). So  $(-1)a$  satisfies (F4) for  $a$ . By uniqueness of negatives ([Theorem 0.7.1](#) (c)),  $(-1)a = -a$ .

**Exercise B2: Solving Equations in  $\mathbb{F}_7$ .**

- (a) Solve  $3x + 5 = 2$  in  $\mathbb{F}_7$ .
- (b) Solve the system  $x + 2y = 3$ ,  $3x + y = 1$  over  $\mathbb{F}_7$ .

*Solution.*

- (a) Adding  $-5$  to both sides gives  $3x = 2 - 5 = -3 = 4$  in  $\mathbb{F}_7$ . Since  $3 \cdot 5 = 15 = 1$  in  $\mathbb{F}_7$ , we have  $3^{-1} = 5$ . Multiplying by 5,  $x = 20 = 6$ . Check:  $3 \cdot 6 + 5 = 23 = 2$  in  $\mathbb{F}_7$ . Hence  $x = 6$ .
- (b) Subtract 3 times the first equation from the second:  $(1 - 6)y = 1 - 9$ , that is,  $-5y = -8$ , or  $2y = 6$  in  $\mathbb{F}_7$ . Since  $2 \cdot 4 = 8 = 1$ ,  $2^{-1} = 4$ , so  $y = 24 = 3$ . Then  $x = 3 - 2y = 3 - 6 = -3 = 4$ . Check:  $4 + 2 \cdot 3 = 10 = 3$  and  $3 \cdot 4 + 3 = 15 = 1$  in  $\mathbb{F}_7$ . Hence  $(x, y) = (4, 3)$ .

**Exercise B3: Which Are Fields?**

Determine which of the following are fields, with the stated operations. Justify your answer.

- (a)  $\mathbb{Z}/9\mathbb{Z}$ , with the operations of [Example \(Arithmetic on  \$\mathbb{Z}/n\mathbb{Z}\$  and the Field  \$\mathbb{F}\_2\$ \)](#).
- (b)  $\mathbb{Q}(i) := \{a + bi : a, b \in \mathbb{Q}\}$ , with the operations of  $\mathbb{C}$ .
- (c)  $\{a + bi : a, b \in \mathbb{Z}\}$ , with the operations of  $\mathbb{C}$ .
- (d)  $\mathbb{F}_5$ .

*Solution.*

- (a) Not a field. Since  $9 = 3 \cdot 3$  is not prime, [Theorem 0.7.3](#) applies; concretely  $[3][3] = [0]$  with  $[3] \neq [0]$ , which is impossible in a field by [Theorem 0.7.1 \(f\)](#).
- (b) A field. Let  $x = a + bi$  and  $y = c + di$  with  $a, b, c, d \in \mathbb{Q}$ . Then  $x + y = (a + c) + (b + d)i$ ,  $xy = (ac - bd) + (ad + bc)i$  and  $-x = -a - bi$  have rational parts, and  $0, 1 \in \mathbb{Q}(i)$  with  $1 \neq 0$ . The identities (F1), (F2), (F5), (F6), (F9) hold because they hold in  $\mathbb{C}$  ([Example \(The Fields  \$\mathbb{Q}\$ ,  \$\mathbb{R}\$  and  \$\mathbb{C}\$ \)](#)). If  $x \neq 0$ , then  $a^2 + b^2$  is a non-zero rational number, and by [Proposition 0.6.3](#)  $x^{-1} = \frac{a}{a^2 + b^2} - \frac{b}{a^2 + b^2}i$ , which lies in  $\mathbb{Q}(i)$ . So (F8) holds.
- (c) Not a field: (F8) fails for 2. Suppose  $u = a + bi$  with  $a, b \in \mathbb{Z}$  satisfied  $2u = 1$ . This is also an equation in the field  $\mathbb{C}$ , where  $\frac{1}{2}$  is an inverse of 2; by uniqueness of inverses in  $\mathbb{C}$  ([Theorem 0.7.1 \(c\)](#)),  $u = \frac{1}{2}$ . But  $\frac{1}{2}$  has real part  $\frac{1}{2} \notin \mathbb{Z}$ , a contradiction.
- (d) A field, by [Theorem 0.7.3](#), since 5 is prime.

### 0.7.9.3 C. Going deeper

#### Exercise C1: A Field with Four Elements.

Let  $F = \{a + b\alpha : a, b \in \mathbb{F}_2\}$ , a set of four formal symbols  $0, 1, \alpha, \beta$ , where  $\beta := 1 + \alpha$ . Define

$$\begin{aligned}(a + b\alpha) + (c + d\alpha) &:= (a + c) + (b + d)\alpha, \\ (a + b\alpha)(c + d\alpha) &:= (ac + bd) + (ad + bc + bd)\alpha,\end{aligned}$$

with all coefficient arithmetic in  $\mathbb{F}_2$ . (The product is what expanding gives if we replace  $\alpha^2$  by  $\alpha + 1$ .)

- (a) Write out the addition and multiplication tables of  $F$ .
- (b) Prove that  $F$  is a field.
- (c) Explain why there is no bijection  $\varphi: F \rightarrow \mathbb{Z}/4\mathbb{Z}$  with  $\varphi(x + y) = \varphi(x) + \varphi(y)$  and  $\varphi(xy) = \varphi(x)\varphi(y)$  for all  $x, y \in F$ . So this four-element field is not “ $\mathbb{Z}/4\mathbb{Z}$  in disguise”.

*Hint: for (c), look for a non-zero element of  $\mathbb{Z}/4\mathbb{Z}$  whose square is zero.*

*Solution.*



- (a) Using the formulas, with  $1 + 1 = 0$  in  $\mathbb{F}_2$ :

| +        | 0        | 1        | $\alpha$ | $\beta$  | ·        | 0 | 1        | $\alpha$ | $\beta$  |
|----------|----------|----------|----------|----------|----------|---|----------|----------|----------|
| 0        | 0        | 1        | $\alpha$ | $\beta$  | 0        | 0 | 0        | 0        | 0        |
| 1        | 1        | 0        | $\beta$  | $\alpha$ | 1        | 0 | 1        | $\alpha$ | $\beta$  |
| $\alpha$ | $\alpha$ | $\beta$  | 0        | 1        | $\alpha$ | 0 | $\alpha$ | $\beta$  | 1        |
| $\beta$  | $\beta$  | $\alpha$ | 1        | 0        | $\beta$  | 0 | $\beta$  | 1        | $\alpha$ |

For instance,  $\alpha \cdot \alpha = (0 + 1\alpha)(0 + 1\alpha) = (0 + 1) + (0 + 0 + 1)\alpha = 1 + \alpha = \beta$ , and  $\alpha\beta = (0 + 1\alpha)(1 + 1\alpha) = (0 + 1) + (0 + 1 + 1)\alpha = 1$ .

- (b) Let  $x = a + b\alpha$ ,  $y = c + d\alpha$ ,  $z = e + f\alpha$  with  $a, \dots, f \in \mathbb{F}_2$ . Since  $\mathbb{F}_2$  is a field (Example (Arithmetic on  $\mathbb{Z}/n\mathbb{Z}$  and the Field  $\mathbb{F}_2$ )), we may compute with the coefficients using the rules of algebra.

*Addition.* It acts coordinate by coordinate, so (F1) and (F2) follow from the same laws in  $\mathbb{F}_2$ , the element  $0 = 0 + 0\alpha$  satisfies (F3), and  $-x = x$  satisfies (F4) because  $a + a = 0$  and  $b + b = 0$  in  $\mathbb{F}_2$ .

(F6). The formula for  $xy$  is unchanged when  $(a, b)$  and  $(c, d)$  are swapped:  $ac + bd = ca + db$  and  $ad + bc + bd = cb + da + db$ .

(F7). With  $1 = 1 + 0\alpha$ ,  $x \cdot 1 = (a + 0) + (0 + b + 0)\alpha = x$ , and  $1 \neq 0$  since the coefficients differ.

(F9). Since  $y + z = (c + e) + (d + f)\alpha$ ,

$$x(y + z) = (a(c + e) + b(d + f)) + (a(d + f) + b(c + e) + b(d + f))\alpha,$$

and expanding the coefficients shows this is  $((ac + bd) + (ae + bf)) + ((ad + bc + bd) + (af + be + bf))\alpha = xy + xz$ .

(F5). Write  $xy = g + h\alpha$  with  $g = ac + bd$ ,  $h = ad + bc + bd$ . Then  $(xy)z = (ge + hf) + (gf + he + hf)\alpha$ . Expanding,

$$\begin{aligned} ge + hf &= ace + adf + bcf + bde + bdf, \\ gf + he + hf &= acf + ade + adf + bce + bcf + bde, \end{aligned}$$

where in the second we used  $bdf + bdf = 0$  in  $\mathbb{F}_2$ . Similarly  $yz = k + l\alpha$  with  $k = ce + df$ ,  $l = cf + de + df$ , and  $x(yz) = (ak + bl) + (al + bk + bl)\alpha$ . Expanding gives  $ak + bl = ace + adf + bcf + bde + bdf$  and  $al + bk + bl = acf + ade + adf + bce + bcf + bde$ , again using  $bdf + bdf = 0$ . Hence  $(xy)z = x(yz)$ .

(F8). From the table,  $1 \cdot 1 = 1$ ,  $\alpha\beta = 1$  and  $\beta\alpha = 1$ , so every non-zero element has an inverse.

This proves that  $F$  is a field.

- (c) Suppose such a bijection  $\varphi$  existed. First,  $\varphi(0) = \varphi(0 + 0) = \varphi(0) + \varphi(0)$ , so additive cancellation in  $\mathbb{Z}/4\mathbb{Z}$  (adding  $-\varphi(0)$ , which only uses (F1)–(F4), valid there by Example (Arithmetic on  $\mathbb{Z}/n\mathbb{Z}$  and the Field  $\mathbb{F}_2$ )) gives  $\varphi(0) = [0]$ . Since  $\varphi$  is surjective, there is  $x \in F$  with  $\varphi(x) = [2]$ . As  $[2] \neq [0] = \varphi(0)$ , we have  $x \neq 0$ . But

$$\varphi(x \cdot x) = \varphi(x)\varphi(x) = [2][2] = [4] = [0] = \varphi(0),$$

so  $x \cdot x = 0$  because  $\varphi$  is injective. This contradicts Theorem 0.7.1 (f) in the field  $F$  (from (b)). Hence no such  $\varphi$  exists.

### Exercise C2: Finite Fields Have Prime Characteristic.

- (a) Let  $F$  be a finite field. Prove that the characteristic of  $F$  is a prime  $p$ .
- (b) Let  $F$  be a field of characteristic  $p > 0$ . Prove that  $p \cdot a = 0$  for every  $a \in F$ .

*Hint: for (a), the elements  $1 \cdot 1, 2 \cdot 1, 3 \cdot 1, \dots$  cannot all be different.*

*Solution.*

- (a) The elements  $m \cdot 1$  for  $m = 1, 2, 3, \dots$  all lie in the finite set  $F$ , so they cannot all be distinct. Hence there are positive integers  $k < m$  with  $m \cdot 1 = k \cdot 1$ . By Lemma 0.7.4,  $m \cdot 1 = k \cdot 1 + (m - k) \cdot 1$ , so  $k \cdot 1 + (m - k) \cdot 1 = k \cdot 1 + 0$ , and additive cancellation (Theorem 0.7.1 (b)) gives  $(m - k) \cdot 1 = 0$  with  $m - k \geq 1$ . Therefore the characteristic of  $F$  is not 0. By Theorem 0.7.5, it is a prime  $p$ .
- (b) Let  $a \in F$ . By Lemma 0.7.4 and the definition of characteristic,  $p \cdot a = (p \cdot 1) a = 0 \cdot a = 0$ , where the last step is Theorem 0.7.1 (d).

### Exercise C3: The Frobenius Identity.

Let  $p$  be a prime.

- (a) Let  $a, b \in \mathbb{Z}$  with  $p$  dividing  $ab$ . Prove that  $p$  divides  $a$  or  $p$  divides  $b$ .
- (b) Deduce that  $p$  divides the binomial coefficient  $\binom{p}{k}$  for  $0 < k < p$ .
- (c) Let  $F$  be a field of characteristic  $p$ . Prove that  $(x + y)^p = x^p + y^p$  for all  $x, y \in F$ .

You may use the binomial theorem  $(x + y)^n = \sum_{k=0}^n \binom{n}{k} \cdot (x^k y^{n-k})$  in any field, with integer multiples as in Lemma 0.7.4; its inductive proof uses only the field axioms.

*Hint: for (a), use Lemma 0.7.2. For (b), use  $p! = \binom{p}{k} k! (p - k)!$ .*

*Solution.*

- (a) Suppose  $p$  does not divide  $a$ . A positive integer dividing  $p$  is 1 or  $p$ , so the only positive common divisor of  $a$  and  $p$  is 1. By Lemma 0.7.2 there are  $x, y \in \mathbb{Z}$  with  $ax + py = 1$ . Multiplying by  $b$ ,  $b = (ab)x + p(by)$ . Both terms on the right are divisible by  $p$ , the first because  $p$  divides  $ab$ . Hence  $p$  divides  $b$ .
- (b) First, by (a) and induction on the number of factors, if  $p$  divides a product of integers then it divides one of the factors. The integers  $1, 2, \dots, p - 1$  are not divisible by  $p$ , so  $p$  divides neither  $k!$  nor  $(p - k)!$  for  $0 < k < p$ , and hence, by (a) once more, not their product. Now  $p$  divides  $p! = \binom{p}{k} \cdot (k! (p - k)!)$ . By (a),  $p$  divides  $\binom{p}{k}$ .
- (c) Let  $x, y \in F$ . By the binomial theorem,

$$(x + y)^p = x^p + y^p + \sum_{k=1}^{p-1} \binom{p}{k} \cdot (x^k y^{p-k}),$$

since  $\binom{p}{0} = \binom{p}{p} = 1$ . Fix  $0 < k < p$ . By (b),  $\binom{p}{k} = pm$  for some  $m \in \mathbb{N}$ . By Lemma 0.7.4,

$$\binom{p}{k} \cdot (x^k y^{p-k}) = ((pm) \cdot 1)(x^k y^{p-k}) = (p \cdot 1)(m \cdot 1)(x^k y^{p-k}) = 0,$$

because  $p \cdot 1 = 0$  in characteristic  $p$  and 0 times anything is 0 ([Theorem 0.7.1 \(d\)](#)). So every middle term vanishes, and  $(x + y)^p = x^p + y^p$ , as claimed.

## 0.8 Polynomials

We now have fields: number systems in which we can add, subtract, multiply and divide by anything non-zero. The next objects built on top of a field are polynomials. They will be everywhere in this book: as examples of vectors in Chapter 1, as the characteristic and minimal polynomials of a matrix later on, and as the reason some fields behave differently from others. This section says precisely what a polynomial is, introduces its degree, and explains why a polynomial is **not** the same thing as the function it defines.

### 0.8.1 What a polynomial is

At school, a polynomial is “an expression like  $3x^2 - x + 2$ ”. That is a good picture, but it leaves one question open: what is  $x$ ? One tempting answer is that  $x$  is a number to be plugged in, so a polynomial is really a function. This answer breaks down as soon as we leave the familiar number systems.

Work over the field  $\mathbb{F}_2 = \{0, 1\}$  of [Example \(Arithmetic on  \$\mathbb{Z}/n\mathbb{Z}\$  and the Field  \$\mathbb{F}\_2\$ \)](#), where  $1 + 1 = 0$ . The expressions  $x^2 + x$  and 0 look different. But if we plug in the two elements of  $\mathbb{F}_2$ , we get  $0^2 + 0 = 0$  and  $1^2 + 1 = 1 + 1 = 0$ . As functions on  $\mathbb{F}_2$ , the two are the same. If a polynomial *were* its function, then  $x^2 + x$  would be the zero polynomial, and it would have no sensible degree. We want  $x^2 + x$  to be a genuine polynomial of degree 2, so we must define polynomials by their coefficients alone.

*A polynomial is its list of coefficients, and nothing else.*

#### Definition 0.8.1 — Polynomial

Let  $F$  be a field. A **polynomial over  $F$**  in the indeterminate  $x$  is a sequence  $(a_0, a_1, a_2, \dots)$  of elements of  $F$ , indexed by  $\mathbb{N}$ , such that **only finitely many**  $a_k$  are non-zero. That is, there is some  $N \in \mathbb{N}$  with  $a_k = 0$  **for every**  $k > N$ .

The element  $a_k$  is the **coefficient of  $x^k$** . We write the polynomial as

$$p = a_0 + a_1x + a_2x^2 + \cdots + a_Nx^N = \sum_{k=0}^N a_kx^k.$$

Two polynomials  $p = \sum a_kx^k$  and  $q = \sum b_kx^k$  are **equal** if and only if  $a_k = b_k$  **for every**  $k \in \mathbb{N}$ .

In words: a polynomial is an infinite list of coefficients that is eventually all zero. The expression  $\sum_{k=0}^N a_kx^k$  is only a way of displaying the list:  $x^k$  marks the position  $k$ , and we stop writing once the remaining coefficients are all zero. Two polynomials are equal exactly when they agree position by position. A sequence indexed by  $\mathbb{N}$  is a function  $\mathbb{N} \rightarrow F$ , so this is just equality of functions in the sense of [Definition 0.4.1](#).

There is one thing to check about the notation. The number  $N$  in the display is not unique: if  $a_k = 0$  for every  $k > N$ , the same is true for every  $k > N + 1$ . Writing  $\sum_{k=0}^N a_k x^k$  or  $\sum_{k=0}^{N+1} a_k x^k$  only adds a term  $0 \cdot x^{N+1}$ , which we agree to drop. So the display names the same sequence whichever admissible  $N$  we use. We also drop terms with zero coefficient in the middle, write  $x$  for  $1 \cdot x^1$ , and write  $a_0$  for  $a_0 x^0$ .

**Example. (Polynomials over various fields)**

Read off the coefficient sequence of each polynomial, and check it against [Definition 0.8.1](#).

$$\begin{aligned} 3x^2 - x + 2 &\in \mathbb{R}[x], & x^5 + \pi x^3 - \sqrt{2} &\in \mathbb{R}[x], \\ (1+i)x - i &\in \mathbb{C}[x], & x^2 + x &\in \mathbb{F}_2[x], \\ 7, & & 0. \end{aligned}$$

(The symbol  $F[x]$  for the set of all polynomials over  $F$  is defined formally in [Definition 0.8.2](#) below.)

*Solution.* The first is the sequence  $(2, -1, 3, 0, 0, \dots)$  of real numbers. It has finitely many non-zero terms, so it is a polynomial over  $\mathbb{R}$ .

The second is  $(-\sqrt{2}, 0, 0, \pi, 0, 1, 0, \dots)$ . Its coefficients are real but irrational; the definition only asks that they lie in the field.

The third is  $(-i, 1+i, 0, \dots)$ , a polynomial over  $\mathbb{C}$ .

The fourth is  $(0, 1, 1, 0, \dots)$  with entries in  $\mathbb{F}_2$ . It is **not** the zero sequence, so it is not the zero polynomial, even though it vanishes at both points of  $\mathbb{F}_2$ .

The constant 7 is the sequence  $(7, 0, 0, \dots)$ . It is a polynomial over  $\mathbb{Q}$ ,  $\mathbb{R}$  or  $\mathbb{C}$ . Over  $\mathbb{F}_2$  it would be  $(1, 0, 0, \dots)$ , because  $7 = 1$  there.

The last,  $0 = (0, 0, 0, \dots)$ , is the **zero polynomial**. It is the degenerate case: every coefficient is zero, and the finiteness condition holds with  $N = 0$ . It matters because it is the one polynomial whose degree needs a special convention, as we will see.

Now the non-example, by a minimal change. Take the polynomial  $1 + x + x^2$ , whose sequence is  $(1, 1, 1, 0, 0, \dots)$ , and change every later 0 to 1. The result  $(1, 1, 1, 1, \dots)$ , informally “ $1 + x + x^2 + x^3 + \dots$ ”, is still a sequence of elements of  $\mathbb{R}$ . What fails is the clause **only finitely many** coefficients are non-zero, so this is **not** a polynomial. (It is a *formal power series*, which we will not need.) Expressions such as  $x^{-1} + 1$  and  $\sqrt{x}$  are not polynomials either: they are not of the form  $\sum a_k x^k$  with  $k \in \mathbb{N}$  at all.

Why insist on finitely many non-zero coefficients? Because we want to plug elements of the field into polynomials, which means forming the sum  $\sum a_k c^k$ , and in a field only finite sums make sense. Finiteness is also what gives every non-zero polynomial a highest power, its degree.

## 0.8.2 Adding and multiplying polynomials

The coefficient picture tells us how to add and multiply. Adding  $3x^2 - x + 2$  and  $x + 5$  at school means collecting like powers, so we add coefficients position by position. Multiplying means expanding every product  $a_i x^i \cdot b_j x^j = a_i b_j x^{i+j}$  and collecting like powers, so the coefficient of

$x^k$  collects all pairs  $(i, j)$  with  $i + j = k$ .

### Definition 0.8.2 — The polynomial ring $F[x]$

Let  $F$  be a field. The set of **all** polynomials over  $F$  is denoted  $F[x]$ . For  $p = \sum a_k x^k$  and  $q = \sum b_k x^k$  in  $F[x]$ , define the **sum**  $p + q$  and the **product**  $pq$  by their coefficients:

$$p + q = \sum_k (a_k + b_k) x^k, \quad pq = \sum_k \left( \sum_{i=0}^k a_i b_{k-i} \right) x^k.$$

We must check that  $p + q$  and  $pq$  are again polynomials, that is, that their coefficient sequences are eventually zero. Suppose  $a_i = 0$  for every  $i > m$  and  $b_j = 0$  for every  $j > n$ . If  $k > \max(m, n)$ , then  $a_k + b_k = 0 + 0 = 0$ . If  $k > m + n$ , then in each product  $a_i b_{k-i}$  either  $i > m$  or  $k - i > n$ , because otherwise  $k = i + (k - i) \leq m + n$ . So every term is 0, and the coefficient of  $x^k$  in  $pq$  is 0. Hence both are polynomials, and addition and multiplication are well defined on  $F[x]$ .

With this multiplication,  $x \cdot x$  has coefficient sequence  $(0, 0, 1, 0, \dots)$ , and by induction on  $k$  the  $k$ -th power of the polynomial  $x$  is the sequence with a 1 in position  $k$  and 0 elsewhere. So the notation  $x^k$  in Definition 0.8.1 agrees with the actual power, and  $\sum a_k x^k$  is an honest sum of products.

### Example. (A product, coefficient by coefficient)

In  $\mathbb{R}[x]$ , compute  $(3x^2 - x + 2)(x + 1)$  from the definition.

*Solution.* Here  $(a_0, a_1, a_2) = (2, -1, 3)$  and  $(b_0, b_1) = (1, 1)$ , with all other coefficients 0. The coefficient of  $x^k$  in the product is  $\sum_{i=0}^k a_i b_{k-i}$ :

$$\begin{aligned} k = 0 : a_0 b_0 &= 2, \\ k = 1 : a_0 b_1 + a_1 b_0 &= 2 - 1 = 1, \\ k = 2 : a_0 b_2 + a_1 b_1 + a_2 b_0 &= 0 - 1 + 3 = 2, \\ k = 3 : a_0 b_3 + a_1 b_2 + a_2 b_1 + a_3 b_0 &= 0 + 0 + 3 + 0 = 3, \end{aligned}$$

and every coefficient with  $k \geq 4$  is 0 by the argument above (here  $m + n = 3$ ). Hence  $(3x^2 - x + 2)(x + 1) = 3x^3 + 2x^2 + x + 2$ , which is what expanding by hand gives.

The operations obey the rules we expect. We record them once, so that later sections can use them freely.

### Theorem 0.8.1 — Arithmetic in $F[x]$

Let  $F$  be a field and  $p, q, r \in F[x]$ . Then:

1.  $p + q = q + p$  and  $(p + q) + r = p + (q + r)$ ;
2.  $p + 0 = p$ , and  $p + (-p) = 0$ , where  $-p = \sum (-a_k) x^k$ ;
3.  $pq = qp$  and  $(pq)r = p(qr)$ ;
4.  $1 \cdot p = p$ , where 1 is the constant polynomial  $(1, 0, 0, \dots)$ ;
5.  $p(q + r) = pq + pr$ .

*Idea.* Two polynomials are equal when their coefficients agree, so each identity reduces to an identity between coefficients, which the field axioms supply. For products it helps to write the coefficient of  $x^k$  in  $pq$  symmetrically, as a sum over all pairs  $(i, j)$  of natural numbers with  $i + j = k$ .

*Proof.* Let  $p = \sum a_k x^k$ ,  $q = \sum b_k x^k$  and  $r = \sum c_k x^k$ . By [Definition 0.8.1](#), it suffices to compare the coefficient of  $x^k$  on both sides, for each  $k \in \mathbb{N}$ .

Parts 1, 2 and 4 hold because in each position they are the corresponding field axioms of [Definition 0.7.1](#): for example, the coefficient of  $x^k$  in  $p + q$  is  $a_k + b_k = b_k + a_k$ , the coefficient in  $q + p$ . For 4, the coefficient of  $x^k$  in  $1 \cdot p$  is  $\sum_{i+j=k} e_i a_j$  with  $e_0 = 1$  and  $e_i = 0$  for  $i \geq 1$ , which equals  $a_k$ .

For 3, the coefficient of  $x^k$  in  $pq$  is  $\sum_{i+j=k} a_i b_j$ . Since multiplication in  $F$  is commutative, this equals  $\sum_{i+j=k} b_j a_i$ , the coefficient in  $qp$ . Next, the coefficient of  $x^k$  in  $(pq)r$  is

$$\sum_{s+l=k} \left( \sum_{i+j=s} a_i b_j \right) c_l = \sum_{i+j+l=k} a_i b_j c_l,$$

where the equality uses distributivity and associativity in  $F$ . The same computation for  $p(qr)$  gives  $\sum_{i+s=k} a_i \left( \sum_{j+l=s} b_j c_l \right) = \sum_{i+j+l=k} a_i b_j c_l$ . Therefore  $(pq)r = p(qr)$ .

For 5, the coefficient of  $x^k$  in  $p(q + r)$  is  $\sum_{i+j=k} a_i (b_j + c_j) = \sum_{i+j=k} a_i b_j + \sum_{i+j=k} a_i c_j$  by distributivity in  $F$ , which is the coefficient in  $pq + pr$ . This proves the theorem. ■

A set with an addition and a multiplication satisfying these rules is called a **commutative ring**; this is where the name “polynomial ring” comes from. We will not need the word again in this chapter. What matters is that we may now calculate with polynomials exactly as at school.

*Quick check.* Over  $\mathbb{R}$ , are  $(x + 1)^2$  and  $x^2 + 2x + 1$  equal as polynomials? Over  $\mathbb{F}_2$ , are  $(x + 1)^2$  and  $x^2 + 1$  equal as polynomials?

*Solution.* Both answers are yes, but for a reason that depends on the field. Expanding gives  $(x + 1)^2 = x^2 + 2x + 1$ , with coefficient sequence  $(1, 2, 1, 0, \dots)$ . Over  $\mathbb{R}$  this is exactly the sequence of  $x^2 + 2x + 1$ . Over  $\mathbb{F}_2$  we have  $2 = 1 + 1 = 0$ , so the sequence is  $(1, 0, 1, 0, \dots)$ , which is the sequence of  $x^2 + 1$ .

### 0.8.3 Degree

The most useful single number attached to a polynomial is the highest power of  $x$  that actually occurs. Every non-zero polynomial has one: it has at least one non-zero coefficient and only finitely many, so there is a largest position holding a non-zero coefficient. The zero polynomial has no such position, and we must decide what to do with it.

#### Definition 0.8.3 — Degree

Let  $p = \sum a_k x^k \in F[x]$ .

- If  $p \neq 0$ , the **degree** of  $p$ , written  $\deg p$ , is the **largest**  $n \in \mathbb{N}$  with  $a_n \neq 0$ . The coefficient  $a_n$  is the **leading coefficient** of  $p$ , and  $p$  is **monic** if its leading coefficient is 1.
- The zero polynomial has degree  $\deg 0 = -\infty$ .

We use the conventions  $-\infty < n$  and  $-\infty + n = n + (-\infty) = -\infty$  for every  $n \in \mathbb{N}$ , and  $-\infty + (-\infty) = -\infty$ .

Examples, simplest first. A non-zero constant  $c$  has degree 0; its leading coefficient is  $c$ . The polynomial  $x + 5 \in \mathbb{R}[x]$  is monic of degree 1. The polynomial  $3x^3 + 2x^2 + x + 2$  from [Example \(A product, coefficient by coefficient\)](#) has degree 3 and leading coefficient 3. Over  $\mathbb{F}_2$ ,  $x^2 + x$  has degree 2, even though it vanishes at every point of  $\mathbb{F}_2$ .

Why  $-\infty$ , and not 0 or  $-1$ ? Look at the formula we expect: the degree of a product is the sum of the degrees. With  $\deg 0 = 0$  it fails at once, since  $0 \cdot x = 0$  would give  $0 = \deg 0 = \deg 0 + \deg x = 1$ . With  $\deg 0 = -1$  it fails too:  $-1 = \deg(0 \cdot x) \neq \deg 0 + \deg x = 0$ . The formula needs  $\deg 0 + n = \deg 0$  for every  $n$ , and  $-\infty$  is exactly the symbol with that property. The convention keeps a formula true, and that is its only job.

It turns out that the degree of a sum is controlled by the larger degree, and the degree of a product is the sum of the degrees:

### Theorem 0.8.2 — Degree of a sum

Let  $F$  be a field and  $p, q \in F[x]$ . Then

$$\deg(p + q) \leq \max(\deg p, \deg q),$$

with **equality** whenever  $\deg p \neq \deg q$ .

*Proof.* Let  $p = \sum a_k x^k$ ,  $q = \sum b_k x^k$  and  $M = \max(\deg p, \deg q)$ . If  $M = -\infty$ , then  $p = q = 0$ , so  $p + q = 0$  and both sides are  $-\infty$ . Suppose  $M \in \mathbb{N}$ . For every  $k > M$  we have  $k > \deg p$  and  $k > \deg q$ , so  $a_k = b_k = 0$  by [Definition 0.8.3](#). Hence the coefficient  $a_k + b_k$  of  $x^k$  in  $p + q$  is 0 for every  $k > M$ , and therefore  $\deg(p + q) \leq M$ .

Now suppose  $\deg p \neq \deg q$ ; by symmetry we may assume  $\deg p > \deg q$ , so  $M = \deg p \in \mathbb{N}$ . Then  $a_M \neq 0$  and  $b_M = 0$ , because  $M > \deg q$ . The coefficient of  $x^M$  in  $p + q$  is  $a_M + 0 = a_M \neq 0$ , so  $\deg(p + q) \geq M$ . Together with the first part,  $\deg(p + q) = M$ . This proves the theorem. ■

The inequality can be strict: over any field,  $\deg(x + (-x)) = \deg 0 = -\infty < 1$ . The leading terms cancel. That is the only way the degree of a sum can drop.

### Theorem 0.8.3 — Degree of a product

Let  $F$  be a field and  $p, q \in F[x]$ . Then

$$\deg(pq) = \deg p + \deg q.$$

If  $p$  and  $q$  are non-zero, the leading coefficient of  $pq$  is the product of their leading coefficients.

*Idea.* Expanding  $(a_mx^m + \text{lower terms})(b_nx^n + \text{lower terms})$ , the only way to reach  $x^{m+n}$  is  $a_mx^m \cdot b_nx^n$ , and nothing reaches beyond. So the top coefficient is  $a_mb_n$ . The one real question is whether  $a_mb_n$  could be 0. In a field it cannot, and this is exactly where the field axioms enter.

*Proof.* If  $p = 0$  or  $q = 0$ , then  $pq = 0$  by the definition of the product, so both sides equal  $-\infty$  by the conventions in Definition 0.8.3. Suppose now  $p = \sum a_kx^k$  and  $q = \sum b_kx^k$  are non-zero, with  $m = \deg p$  and  $n = \deg q$ . Then  $a_m \neq 0$ ,  $b_n \neq 0$ ,  $a_i = 0$  for  $i > m$  and  $b_j = 0$  for  $j > n$ .

We showed after Definition 0.8.2 that the coefficient of  $x^k$  in  $pq$  is 0 for every  $k > m+n$ . For  $k = m+n$ , the coefficient is  $\sum_{i+j=m+n} a_ib_j$ . A term with  $i > m$  has  $a_i = 0$ , and a term with  $i < m$  has  $j = m+n-i > n$ , so  $b_j = 0$ . Hence only  $i = m$ ,  $j = n$  survives, and the coefficient is  $a_mb_n$ .

Since  $a_m \neq 0$  and  $b_n \neq 0$ , Theorem 0.7.1 gives  $a_mb_n \neq 0$ , because a product of non-zero elements of a field is non-zero. Therefore  $\deg(pq) = m+n$  and the leading coefficient of  $pq$  is  $a_mb_n$ , as claimed. ■

*Warning.* The field axioms are doing real work here. The same definitions make sense with coefficients in  $\mathbb{Z}/6\mathbb{Z}$ , which is **not** a field. There  $(2x+1)(3x^2+x) = 6x^3+5x^2+x = 5x^2+x$ , because  $2 \cdot 3 = 6 = 0$ . The factors have degrees 1 and 2, but the product has degree 2, not 3.

The first consequence is that polynomials over a field behave like integers: a product can only be zero if a factor is, and non-zero factors can be canceled.

### Corollary 0.8.4 — No zero divisors in $F[x]$

Let  $F$  be a field and  $p, q, r \in F[x]$ .

1. If  $pq = 0$ , then  $p = 0$  or  $q = 0$ .
2. If  $pq = pr$  and  $p \neq 0$ , then  $q = r$ .

*Proof.* For 1, we prove the contrapositive. Suppose  $p \neq 0$  and  $q \neq 0$ . Then  $\deg p, \deg q \in \mathbb{N}$ , so by Theorem 0.8.3,  $\deg(pq) = \deg p + \deg q \in \mathbb{N}$ . In particular  $\deg(pq) \neq -\infty$ , so  $pq \neq 0$ .

For 2, suppose  $pq = pr$  and  $p \neq 0$ . By Theorem 0.8.1,  $p(q-r) = pq - pr = 0$ . By part 1 and  $p \neq 0$ , we get  $q-r = 0$ , that is,  $q = r$ . This proves the corollary. ■

*Quick check.* Let  $p, q \in \mathbb{R}[x]$  with  $\deg p = 3$  and  $\deg q = 2$ . What are  $\deg(pq)$ ,  $\deg(p+q)$  and  $\deg(p^2)$ ? Can you say what  $\deg(p-x^3)$  is?



*Solution.* By Theorem 0.8.3,  $\deg(pq) = 5$  and  $\deg(p^2) = 6$ . Since the degrees differ, Theorem 0.8.2 gives  $\deg(p + q) = 3$ . The degree of  $p - x^3$  is not determined: if the leading coefficient of  $p$  is 1, the  $x^3$  terms cancel and the degree is at most 2 (it could even be  $-\infty$  if  $p = x^3$ ); otherwise it is 3.

## 0.8.4 Polynomials of bounded degree

In linear algebra we constantly need “all polynomials up to a given size”. For instance, the quadratics  $ax^2 + bx + c$  are described by three numbers, much like points of  $\mathbb{R}^3$ . So we give such collections names.

### Definition 0.8.4 — Polynomials of degree at most $n$

Let  $F$  be a field and  $n \in \mathbb{N}$ . The set of polynomials over  $F$  of degree **at most**  $n$  is

$$F[x]_{\leq n} = \{p \in F[x] : \deg p \leq n\} = \{a_0 + a_1x + \cdots + a_nx^n : a_0, \dots, a_n \in F\}.$$

The two descriptions agree:  $\deg p \leq n$  says exactly that every coefficient beyond position  $n$  is zero, and the coefficients  $a_0, \dots, a_n$  are then free, any of them allowed to be zero. In particular the zero polynomial lies in  $F[x]_{\leq n}$ , since  $-\infty \leq n$ .

#### Example. (Inside $\mathbb{R}[x]_{\leq 2}$ )

Which of  $3x^2 - 2x + 1$ ,  $5x + 7$ ,  $-4$ ,  $0$  and  $x^3 - x^3 + x$  lie in  $\mathbb{R}[x]_{\leq 2}$ ? Which lie in  $\mathbb{R}[x]_{\leq 0}$ ?

*Solution.* All five lie in  $\mathbb{R}[x]_{\leq 2}$ . Their degrees are 2, 1, 0,  $-\infty$ , and 1 (the last is the polynomial  $x$ , since  $x^3 - x^3 = 0$  as polynomials). Only  $-4$  and  $0$  lie in  $\mathbb{R}[x]_{\leq 0}$ , which is the set of constant polynomials, 0 included.

Sometimes we need the degree to be exactly  $n$ , for instance when we speak of “a quadratic” and mean that the  $x^2$  term is really there.

### Definition 0.8.5 — Polynomials of degree exactly $n$

Let  $F$  be a field and  $n \in \mathbb{N}$ . The set of polynomials over  $F$  of degree **exactly**  $n$  is

$$\begin{aligned} F[x]_{=n} &= \{p \in F[x] : \deg p = n\} \\ &= \{a_0 + a_1x + \cdots + a_nx^n : a_0, \dots, a_n \in F, a_n \neq 0\}. \end{aligned}$$

#### Example. (Comparing the two notations)

For each of  $x^2 + 1$ ,  $3x - 2$  and  $0$ , decide whether it lies in  $\mathbb{R}[x]_{\leq 2}$  and whether it lies in  $\mathbb{R}[x]_{=2}$ .

*Solution.* The polynomial  $x^2 + 1$  has degree 2, so it lies in both. The polynomial  $3x - 2$  has degree  $1 \leq 2$ , so it lies in  $\mathbb{R}[x]_{\leq 2}$  but **not** in  $\mathbb{R}[x]_{=2}$ . The zero polynomial has

degree  $-\infty \leq 2$ , so it lies in  $\mathbb{R}[x]_{\leq 2}$ , but  $-\infty \neq 2$ , so it is **not** in  $\mathbb{R}[x]_{=2}$ . In general  $F[x]_{\leq n} = F[x]_{=n} \cup F[x]_{\leq n-1}$  for  $n \geq 1$ , and the union is disjoint.

The two sets differ in one small clause,  $a_n \neq 0$ , and that clause changes their behavior completely.

*Warning.*  $F[x]_{\leq n}$  is closed under addition, but  $F[x]_{=n}$  is **not**. For example  $x^2 + x$  and  $-x^2 + 1$  both lie in  $\mathbb{R}[x]_{=2}$ , but their sum  $x + 1$  has degree 1. The leading terms cancel. Also  $F[x]_{=n}$  does not contain 0. In Chapter 1 this is why  $F[x]_{\leq n}$  is a subspace and  $F[x]_{=n}$  is not.

The positive half is [Theorem 0.8.2](#) at work: if  $\deg p \leq n$  and  $\deg q \leq n$ , then  $\deg(p + q) \leq \max(\deg p, \deg q) \leq n$ . Multiplying by a constant  $c \in F$  also cannot raise the degree, by [Theorem 0.8.3](#).

## 0.8.5 Evaluation and roots

A polynomial is not a function, but every polynomial *defines* a function: plug an element of the field in for  $x$ .

### Definition 0.8.6 — Evaluation and roots

Let  $F$  be a field,  $p = \sum_{k=0}^N a_k x^k \in F[x]$  and  $c \in F$ . The **value** of  $p$  at  $c$  is the element

$$p(c) = \sum_{k=0}^N a_k c^k = a_0 + a_1 c + \cdots + a_N c^N \in F,$$

where  $c^0 = 1$ . An element  $c \in F$  is a **root** of  $p$  if  $p(c) = 0$ . The function  $F \rightarrow F$ ,  $c \mapsto p(c)$ , is the **polynomial function** of  $p$ .

The value does not depend on the choice of  $N$ : enlarging  $N$  adds terms  $0 \cdot c^k = 0$  (by [Theorem 0.7.1](#)), which do not change the sum. So  $p(c)$  is well defined.

For example,  $p = x^2 - 3x + 2 \in \mathbb{R}[x]$  has  $p(0) = 2$  and  $p(1) = 1 - 3 + 2 = 0$ , so 1 is a root. Over  $\mathbb{F}_3 = \{0, 1, 2\}$ , the polynomial  $x^3 + 2$  has  $1^3 + 2 = 3 = 0$ , so 1 is a root; the other two values are  $0 + 2 = 2$  and  $8 + 2 = 10 = 1$ , so 1 is its only root in  $\mathbb{F}_3$ .

Evaluation turns the operations of  $F[x]$  into the operations of  $F$ :

### Theorem 0.8.5 — Evaluation respects sums and products

Let  $F$  be a field,  $p, q \in F[x]$  and  $c \in F$ . Then

$$(p + q)(c) = p(c) + q(c), \quad (pq)(c) = p(c)q(c),$$

and the constant polynomial  $a$  has value  $a$  at  $c$ .

*Proof.* Let  $p = \sum_{i=0}^m a_i x^i$  and  $q = \sum_{j=0}^n b_j x^j$ , where we choose  $m$  and  $n$  large enough that all non-zero coefficients are included. The statement about constants is immediate from [Definition 0.8.6](#). For sums, taking  $N = \max(m, n)$  and padding with zero coefficients,

$(p + q)(c) = \sum_{k=0}^N (a_k + b_k)c^k = \sum_{k=0}^N a_k c^k + \sum_{k=0}^N b_k c^k$  by distributivity in  $F$ .

For products, all non-zero coefficients of  $pq$  occur at positions  $k \leq m + n$ , so

$$\begin{aligned} (pq)(c) &= \sum_{k=0}^{m+n} \left( \sum_{i+j=k} a_i b_j \right) c^k \\ &= \sum_{i=0}^m \sum_{j=0}^n a_i b_j c^i c^j = \sum_{i=0}^m \sum_{j=0}^n (a_i c^i) (b_j c^j) \\ &= \left( \sum_{i=0}^m a_i c^i \right) \left( \sum_{j=0}^n b_j c^j \right), \end{aligned}$$

where the first equality is [Definition 0.8.2](#), the second uses distributivity and  $c^{i+j} = c^i c^j$  and regroups the pairs  $(i, j)$ , the third uses **commutativity** of multiplication in  $F$  to move  $c^i$  past  $b_j$ , and the last is distributivity again. This shows  $(pq)(c) = p(c)q(c)$ . ■

Note which step used commutativity. In the next section we will plug a square matrix  $\mathbf{A}$  into a polynomial. Products of matrices do not commute in general, but the proof only moves **scalars** past powers of  $\mathbf{A}$ , and multiplies powers of the same  $\mathbf{A}$ ; both are still allowed for matrices, so the same identity will hold there.

As a first use, a root of a factor is a root of the product: if  $q(c) = 0$ , then  $(pq)(c) = p(c) \cdot 0 = 0$ . Conversely, if  $(pq)(c) = 0$  then  $p(c)q(c) = 0$ , so  $p(c) = 0$  or  $q(c) = 0$  by [Theorem 0.7.1](#). This is how one solves  $x^2 - 3x + 2 = (x - 1)(x - 2) = 0$  at school.

Now we can return to the question from the start of the section.

### Example. (Different polynomials, same function)

Over  $\mathbb{F}_2$ , compare the polynomials  $x^2 + x$  and 0, and the polynomials  $x^2$  and  $x$ , both as polynomials and as functions  $\mathbb{F}_2 \rightarrow \mathbb{F}_2$ .

*Solution.* As polynomials,  $x^2 + x$  has coefficient sequence  $(0, 1, 1, 0, \dots)$  and 0 has  $(0, 0, 0, \dots)$ . They differ in position 1, so  $x^2 + x \neq 0$  by [Definition 0.8.1](#). Its degree is 2.

As functions,  $\mathbb{F}_2 = \{0, 1\}$  has two elements, and

$$0^2 + 0 = 0, \quad 1^2 + 1 = 1 + 1 = 0.$$

So the polynomial function of  $x^2 + x$  is the zero function, the same as that of 0. Both elements of  $\mathbb{F}_2$  are roots of a polynomial of degree 2 that is not the zero polynomial.

Similarly  $x^2 \neq x$  as polynomials, but  $0^2 = 0$  and  $1^2 = 1$ , so they define the same function. Indeed, by [Theorem 0.8.5](#), two polynomials define the same function exactly when their difference is a polynomial whose function is zero, and  $x^2 - x = x^2 + x$  in  $\mathbb{F}_2[x]$ .

**Warning. A polynomial is not a function.** Two different polynomials can define the same polynomial function, as  $x^2 + x$  and 0 do over  $\mathbb{F}_2$ . So “ $p(c) = 0$  for every  $c \in F$ ” does **not** imply  $p = 0$  in general, and we may never read off coefficients or a degree

from the values alone. This happens over every finite field: if  $F = \{c_1, \dots, c_q\}$ , then  $(x - c_1)(x - c_2) \cdots (x - c_q)$  has degree  $q$  by [Theorem 0.8.3](#), yet vanishes at every  $c_i$  by [Theorem 0.8.5](#). Over  $\mathbb{F}_p$ , [Exercise \(C2: A non-zero polynomial vanishing everywhere\)](#) below gives the shorter example  $x^p - x$ .

Over an infinite field such as  $\mathbb{Q}$ ,  $\mathbb{R}$  or  $\mathbb{C}$ , the problem disappears: a polynomial whose function is zero is the zero polynomial, so polynomials and polynomial functions can be identified. We prove this in Chapter 2 (in the section on applications of linear systems), once we can count roots. Until then, whenever we say two polynomials are equal we mean their coefficients agree.

## 0.8.6 Looking ahead

Two tools will be built on this section later.

*Remark. Division with remainder.* Over a field, if  $f, g \in F[x]$  and  $g \neq 0$ , there are unique  $q, r \in F[x]$  with  $f = qg + r$  and  $\deg r < \deg g$ . This is long division of polynomials, and the convention  $\deg 0 = -\infty$  makes “ $\deg r < \deg g$ ” cover the case  $r = 0$ . We state it here only for orientation; it is proved in Chapter 5 and is not used before then.

*Remark. Polynomials in a matrix.* In the next section, once matrices and their products are available, we substitute a square matrix  $\mathbf{A}$  for  $x$ , replacing the constant term  $a_0$  by  $a_0\mathbf{I}$ . This is how the characteristic and minimal polynomials of later chapters act on matrices.

## 0.8.7 Exercises

### 0.8.7.1 A. Check your understanding

#### Exercise A1.

- Define the degree of a polynomial over a field  $F$ , including the zero polynomial. State the formula for  $\deg(pq)$ .
- Is  $1 + x + x^2 + x^3 + \cdots$ , with every coefficient equal to 1, a polynomial over  $\mathbb{R}$ ? Justify your answer.
- Determine whether the following statement is true: “for all  $p, q \in F[x]$ ,  $\deg(p+q) = \max(\deg p, \deg q)$ .” Justify your answer.
- Determine whether the following statement is true: “if  $p \in \mathbb{F}_2[x]$  and  $p(c) = 0$  for every  $c \in \mathbb{F}_2$ , then  $p = 0$ .” Justify your answer.
- Explain why  $F[x]_{=2}$  is **not** closed under addition, while  $F[x]_{\leq 2}$  is.

*Solution.*

- For  $p = \sum a_k x^k \neq 0$ ,  $\deg p$  is the largest  $n$  with  $a_n \neq 0$ ; and  $\deg 0 = -\infty$ . For all  $p, q \in F[x]$ ,  $\deg(pq) = \deg p + \deg q$  ([Theorem 0.8.3](#)).
- No. Its coefficient sequence  $(1, 1, 1, \dots)$  has infinitely many non-zero terms, so it fails the clause “only finitely many  $a_k$  are non-zero” of [Definition 0.8.1](#).
- False. Take  $p = x$  and  $q = -x$ . Then  $p + q = 0$  has degree  $-\infty$ , while

$\max(\deg p, \deg q) = 1$ . Only the inequality  $\leq$  holds in general ([Theorem 0.8.2](#)).

- (d) False. By [Example \(Different polynomials, same function\)](#),  $p = x^2 + x$  satisfies  $p(0) = p(1) = 0$ , but  $p \neq 0$ .
- (e) The sum of  $x^2$  and  $-x^2 + 1$ , both of degree exactly 2, is 1, of degree 0. For  $F[x]_{\leq 2}$ : if  $\deg p \leq 2$  and  $\deg q \leq 2$ , then  $\deg(p + q) \leq \max(\deg p, \deg q) \leq 2$  by [Theorem 0.8.2](#).

### 0.8.7.2 B. Practice

#### Exercise B1: Sums, products and degrees.

- (a) In  $\mathbb{R}[x]$ , let  $p = 2x^3 - x + 4$  and  $q = -2x^3 + x^2 + 3$ . Compute  $p + q$  and  $pq$ , and give their degrees. Which inequality or formula from this section does each degree illustrate?
- (b) In  $\mathbb{F}_3[x]$ , compute  $(x + 2)(x^2 + x + 1)$ . Hence find all roots in  $\mathbb{F}_3$  of the product.

*Solution.*

- (a) Adding coefficients gives  $p + q = x^2 - x + 7$ , of degree 2. Here  $\deg p = \deg q = 3$  and the  $x^3$  terms cancel, so the inequality of [Theorem 0.8.2](#) is strict. Expanding,

$$pq = -4x^6 + 2x^5 + 2x^4 - 3x^3 + 4x^2 - 3x + 12,$$

of degree  $6 = 3 + 3$ , as [Theorem 0.8.3](#) predicts; its leading coefficient is  $2 \cdot (-2) = -4$ .

- (b) Expanding,  $(x + 2)(x^2 + x + 1) = x^3 + 3x^2 + 3x + 2$ . In  $\mathbb{F}_3$  we have  $3 = 0$ , so the product is  $x^3 + 2$ . By [Theorem 0.8.5](#) and [Theorem 0.7.1](#),  $c$  is a root of the product if and only if  $c + 2 = 0$  or  $c^2 + c + 1 = 0$ . The first gives  $c = 1$ . For the second, the values at 0, 1, 2 are 1,  $3 = 0$  and  $7 = 1$ . Hence the only root in  $\mathbb{F}_3$  is  $c = 1$ , which occurs in both factors.

#### Exercise B2: Degree of a composition.

For  $p = \sum_{k=0}^m a_k x^k$  and  $q$  in  $F[x]$ , define the **composition**  $p \circ q = \sum_{k=0}^m a_k q^k \in F[x]$ , where  $q^0 = 1$ .

- (a) Prove that if  $p$  and  $q$  are non-constant, then  $\deg(p \circ q) = \deg p \cdot \deg q$ .
- (b) Give an example showing the formula fails when  $q$  is constant.

*Hint: first find  $\deg(q^k)$ .*

*Solution.*

- (a) Let  $m = \deg p \geq 1$  and  $d = \deg q \geq 1$ , so  $a_m \neq 0$ . By [Theorem 0.8.3](#) and induction on  $k$  ([Theorem 0.2.7](#)),  $\deg(q^k) = kd$  for every  $k \in \mathbb{N}$ : the case  $k = 0$  is  $\deg 1 = 0$ , and  $\deg(q^{k+1}) = \deg(q^k) + \deg q = kd + d$ . Hence, again by [Theorem 0.8.3](#),  $\deg(a_m q^m) = 0 + md = md$ , and for  $k < m$ ,  $\deg(a_k q^k) \leq kd < md$  (it is  $kd$  or  $-\infty$ ). Let  $s = \sum_{k=0}^{m-1} a_k q^k$ . Applying [Theorem 0.8.2](#)  $m - 1$  times gives  $\deg s \leq \max_{k < m} \deg(a_k q^k) < md$ . Since  $\deg s \neq \deg(a_m q^m)$ , the equality case of [Theorem 0.8.2](#) gives  $\deg(p \circ q) = \deg(s + a_m q^m) = md$ . This proves  $\deg(p \circ q) = \deg p \cdot \deg q$ .

- (b) Take  $p = x - 1$  and  $q = 1$ . Then  $p \circ q = 1 - 1 = 0$  has degree  $-\infty$ , while  $\deg p \cdot \deg q = 1 \cdot 0 = 0$ .

### Exercise B3: All small polynomials over $\mathbb{F}_2$ .

List all polynomials in  $\mathbb{F}_2[x]_{\leq 2}$ . For each, give its degree and its polynomial function (the values at 0 and 1). Hence determine which of them define the zero function, and explain why every function  $\mathbb{F}_2 \rightarrow \mathbb{F}_2$  is the polynomial function of exactly two of them.

*Solution.* A polynomial  $a_0 + a_1x + a_2x^2$  with  $a_i \in \mathbb{F}_2$  has value  $a_0$  at 0 and  $a_0 + a_1 + a_2$  at 1. With  $2^3 = 8$  choices of coefficients:

| polynomial    | degree    | value at 0 | value at 1 |
|---------------|-----------|------------|------------|
| 0             | $-\infty$ | 0          | 0          |
| 1             | 0         | 1          | 1          |
| $x$           | 1         | 0          | 1          |
| $x + 1$       | 1         | 1          | 0          |
| $x^2$         | 2         | 0          | 1          |
| $x^2 + 1$     | 2         | 1          | 0          |
| $x^2 + x$     | 2         | 0          | 0          |
| $x^2 + x + 1$ | 2         | 1          | 1          |

The zero function requires  $a_0 = 0$  and  $a_0 + a_1 + a_2 = 0$ , that is,  $a_0 = 0$  and  $a_1 = a_2$ . Hence exactly 0 and  $x^2 + x$  define the zero function.

A function  $f: \mathbb{F}_2 \rightarrow \mathbb{F}_2$  is determined by the pair  $(f(0), f(1))$ . Given such a pair  $(u, v)$ , the polynomial  $a_0 + a_1x + a_2x^2$  defines  $f$  exactly when  $a_0 = u$  and  $a_1 + a_2 = v - u$ . The first fixes  $a_0$ , and for each of the two choices of  $a_2$  the second fixes  $a_1 = v - u - a_2$ . Therefore exactly two of the eight polynomials define  $f$ , and the table confirms this.

### 0.8.7.3 C. Going deeper

#### Exercise C1: Roots of a quadratic.

Let  $F$  be a field.

- Let  $p = ax^2 + bx + c \in F[x]$  and  $r \in F$  with  $p(r) = 0$ . Prove that  $p = (x - r)(ax + (b + ar))$ .
- Deduce that a non-zero polynomial in  $F[x]_{\leq 2}$  has at most 2 roots in  $F$ .
- The same definitions make sense with coefficients in  $\mathbb{Z}/8\mathbb{Z}$ . Show that  $x^2 - 1$  has four roots there, and name the step of (b) that breaks.

*Hint for (b): use (a) together with [Theorem 0.8.5](#).*

*Solution.*

- Since  $p(r) = 0$ , we have  $ar^2 + br + c = 0$ , so  $c = -ar^2 - br$ . Expanding with

**Theorem 0.8.1,**

$$\begin{aligned}(x-r)(ax+(b+ar)) &= ax^2 + (b+ar)x - arx - r(b+ar) \\ &= ax^2 + bx - (br+ar^2) = ax^2 + bx + c = p.\end{aligned}$$

- (b) Let  $p \neq 0$  with  $\deg p \leq 2$ . *Case 1.*  $\deg p = 0$ . Then  $p = c \neq 0$  is constant and  $p(r) = c \neq 0$  for every  $r$ , so  $p$  has no roots. *Case 2.*  $\deg p = 1$ , say  $p = bx + c$  with  $b \neq 0$ . Then  $p(r) = 0$  if and only if  $r = -b^{-1}c$ , so  $p$  has exactly one root. *Case 3.*  $\deg p = 2$ , say  $p = ax^2 + bx + c$  with  $a \neq 0$ . If  $p$  has no root we are done. Otherwise let  $r$  be a root. By (a),  $p = (x-r)q$  with  $q = ax + (b+ar)$ , of degree 1 since  $a \neq 0$ . Let  $s$  be any root of  $p$ . By [Theorem 0.8.5](#),  $0 = p(s) = (s-r)q(s)$ , so  $s = r$  or  $q(s) = 0$  by [Theorem 0.7.1](#). By Case 2,  $q$  has exactly one root. Hence every root of  $p$  is  $r$  or the root of  $q$ , and  $p$  has at most 2 roots. This proves the claim.
- (c) In  $\mathbb{Z}/8\mathbb{Z}$ ,  $1^2 = 1$ ,  $3^2 = 9 = 1$ ,  $5^2 = 25 = 1$  and  $7^2 = 49 = 1$ , so 1, 3, 5, 7 are four roots of  $x^2 - 1$ . The argument in (b) breaks at “ $(s-r)q(s) = 0$  implies  $s-r=0$  or  $q(s)=0$ ”: for  $r=1$ ,  $s=3$  we get  $(3-1)(3+1) = 2 \cdot 4 = 0$  with both factors non-zero. That step needs the field property that a product of non-zero elements is non-zero.

### Exercise C2: A non-zero polynomial vanishing everywhere.

Let  $p$  be a prime and  $F = \mathbb{F}_p$ .

- (a) Fix  $a \in F \setminus \{0\}$ . Prove that the map  $F \setminus \{0\} \rightarrow F \setminus \{0\}$ ,  $b \mapsto ab$ , is a bijection.
- (b) Deduce that  $a^{p-1} = 1$  for every  $a \in F \setminus \{0\}$ .
- (c) Hence show that the polynomial  $x^p - x \in \mathbb{F}_p[x]$  is not the zero polynomial but that its polynomial function is the zero function.

*Hint for (b): multiply together all the elements of  $F \setminus \{0\}$  in two ways.*

*Solution.*

- (a) The map lands in  $F \setminus \{0\}$ , since a product of non-zero elements of a field is non-zero ([Theorem 0.7.1](#)). It is injective: if  $ab = ab'$ , multiplying by  $a^{-1}$  gives  $b = b'$ . The set  $F \setminus \{0\}$  is finite with  $p-1$  elements, so an injective map from it to itself is surjective by [Theorem 0.4.9](#), hence bijective.
- (b) Let  $P$  be the product of all  $p-1$  elements of  $F \setminus \{0\}$ ; it is non-zero by [Theorem 0.7.1](#). Since  $b \mapsto ab$  is a bijection by (a), the elements  $ab$ , as  $b$  runs over  $F \setminus \{0\}$ , are the same elements in a different order. Multiplication in  $F$  is commutative and associative, so the product does not depend on the order, and

$$P = \prod_{b \neq 0} (ab) = a^{p-1} \prod_{b \neq 0} b = a^{p-1} P.$$

Multiplying by  $P^{-1}$  gives  $a^{p-1} = 1$ .

- (c) The coefficient of  $x^p$  in  $x^p - x$  is  $1 \neq 0$ , so the polynomial is non-zero, of degree  $p$ . For its function:  $0^p - 0 = 0$ , and for  $a \neq 0$ , (b) gives  $a^p - a = a \cdot a^{p-1} - a = a - a = 0$ . Hence every element of  $\mathbb{F}_p$  is a root, and the polynomial function of  $x^p - x$  is the zero function. For  $p = 2$  this is  $x^2 - x = x^2 + x$ , the polynomial of [Example](#)

(Different polynomials, same function).

### Exercise C3: Which polynomials have inverses?.

- (a) Let  $F$  be a field and  $p, q \in F[x]$  with  $pq = 1$ . Prove that  $p$  and  $q$  are non-zero constants.
- (b) Show that (a) fails with coefficients in  $\mathbb{Z}/4\mathbb{Z}$  by computing  $(2x + 1)^2$  there.

*Solution.*

- (a) Since  $pq = 1 \neq 0$ , neither  $p$  nor  $q$  is the zero polynomial, so  $\deg p, \deg q \in \mathbb{N}$ . By [Theorem 0.8.3](#),  $\deg p + \deg q = \deg 1 = 0$ . Two natural numbers with sum 0 are both 0, so  $\deg p = \deg q = 0$ . Therefore  $p$  and  $q$  are non-zero constants.
- (b) In  $\mathbb{Z}/4\mathbb{Z}$ ,  $(2x + 1)^2 = 4x^2 + 4x + 1 = 1$ , because  $4 = 0$ . So  $2x + 1$  has degree 1 and is its own inverse. The proof of (a) breaks at [Theorem 0.8.3](#), which fails here because the leading coefficient  $2 \cdot 2 = 4 = 0$ .

## 0.9 Matrices

You have probably met matrices as tables of numbers that are multiplied by a strange rule. This section rebuilds them over an arbitrary field  $F$  and answers the questions a computational course leaves open. Why is the product defined the way it is? Which rules of school algebra survive, and which fail? What does it mean to divide by a matrix? Everything later in the book, from Gaussian elimination in Chapter 2 to the matrix of a linear map in Chapter 3, rests on the answers.

Throughout,  $F$  is a field ([Definition 0.7.1](#)). Nothing here uses more about  $F$  than the field axioms, so every statement holds over  $\mathbb{Q}$ ,  $\mathbb{R}$ ,  $\mathbb{C}$  and  $\mathbb{F}_p$  alike.

### 0.9.1 Matrices and their entries

#### Definition 0.9.1 — Matrix

Let  $F$  be a field and  $m, n \geq 1$  integers. An  $m \times n$  **matrix over**  $F$  is a rectangular array

$$\mathbf{A} = \begin{pmatrix} a_{11} & a_{12} & \cdots & a_{1n} \\ a_{21} & a_{22} & \cdots & a_{2n} \\ \vdots & \vdots & \ddots & \vdots \\ a_{m1} & a_{m2} & \cdots & a_{mn} \end{pmatrix}$$

with  $m$  rows and  $n$  columns, whose **entries**  $a_{ij}$  lie in  $F$ . Formally, it is a function  $\{1, \dots, m\} \times \{1, \dots, n\} \rightarrow F$ ,  $(i, j) \mapsto a_{ij}$ . The set of all such matrices is  $M_{m \times n}(F)$ , and we write  $M_n(F) = M_{n \times n}(F)$  for the **square** matrices.

The entry  $a_{ij}$  sits in row  $i$  and column  $j$ : **row first, column second**. We write  $\mathbf{A} = (a_{ij})$ ,



and  $(\mathbf{A})_{ij}$  for the  $(i, j)$ -entry when the matrix has no letter of its own. Two matrices are **equal** when they have the **same size** and the same entry in every position; this is equality of functions (Definition 0.4.1).

Two sizes deserve their own names. A matrix in  $M_{n \times 1}(F)$  is a **column vector**, and we write

$$F^n = M_{n \times 1}(F), \quad \mathbf{x} = \begin{pmatrix} x_1 \\ \vdots \\ x_n \end{pmatrix}.$$

In running text we write  $\mathbf{x} = (x_1, \dots, x_n)$  and still mean a column. Columns are printed in bold. A matrix in  $M_{1 \times n}(F)$  is a **row vector**. For  $j = 1, \dots, n$ , the column  $\mathbf{e}_j \in F^n$  has 1 in position  $j$  and 0 elsewhere.

We will often cut a matrix into its columns. If  $\mathbf{a}_j = (a_{1j}, \dots, a_{mj}) \in F^m$  is the  $j$ -th column of  $\mathbf{A}$ , we write

$$\mathbf{A} = (\mathbf{a}_1 \quad \mathbf{a}_2 \quad \cdots \quad \mathbf{a}_n).$$

The  $i$ -th row of  $\mathbf{A}$  is the row vector  $(a_{i1} \quad \cdots \quad a_{in}) \in M_{1 \times n}(F)$ .

### Example. (Matrices over different fields)

Identify the size, the field and a few entries of each matrix:

$$\mathbf{A} = \begin{pmatrix} 1 & -2 & 0 \\ 3 & 1 & 4 \end{pmatrix}, \quad \mathbf{B} = \begin{pmatrix} i & 1+i \\ 0 & 2-i \end{pmatrix}, \quad \mathbf{C} = \begin{pmatrix} 1 & 0 & 1 \\ 1 & 1 & 0 \\ 0 & 1 & 1 \end{pmatrix}, \quad \mathbf{D} = (5).$$

Here  $\mathbf{C}$  is meant over  $\mathbb{F}_2$ . Also write out the  $2 \times 2$  real matrix  $\mathbf{E}$  with  $e_{ij} = i + 2j$ .

*Solution.*  $\mathbf{A} \in M_{2 \times 3}(\mathbb{R})$ , with  $a_{12} = -2$  and  $a_{23} = 4$ ; its second column is  $\mathbf{a}_2 = (-2, 1)$ .  $\mathbf{B} \in M_2(\mathbb{C})$ , with  $b_{12} = 1+i$  and  $b_{21} = 0$ .  $\mathbf{C} \in M_3(\mathbb{F}_2)$ ; every entry is 0 or 1.  $\mathbf{D} \in M_1(\mathbb{R})$  is the degenerate case, a  $1 \times 1$  matrix. It behaves exactly like the number 5, and  $1 \times 1$  matrices are how numbers will appear inside matrix formulas. Finally  $e_{11} = 3$ ,  $e_{12} = 5$ ,  $e_{21} = 4$ ,  $e_{22} = 6$ , so  $\mathbf{E} = \begin{pmatrix} 3 & 5 \\ 4 & 6 \end{pmatrix}$ . Swapping the roles of  $i$  and  $j$  would give a different matrix, which is why the order “row, column” matters.

## 0.9.2 Special matrices

A few shapes of matrix occur so often that they have names. The **main diagonal** of a square matrix  $\mathbf{A} \in M_n(F)$  consists of the entries  $a_{11}, a_{22}, \dots, a_{nn}$ .

### Definition 0.9.2 — Zero matrix

The **zero matrix**  $0 = 0_{m \times n} \in M_{m \times n}(F)$  is the matrix all of whose entries are 0.

### Definition 0.9.3 — Identity matrix

The **identity matrix**  $\mathbf{I}_n \in M_n(F)$  has entries

$$(\mathbf{I}_n)_{ij} = \delta_{ij} = \begin{cases} 1 & \text{if } i = j, \\ 0 & \text{if } i \neq j. \end{cases}$$

That is, 1 on the main diagonal and 0 elsewhere. We write  $\mathbf{I}$  when the size is clear. The symbol  $\delta_{ij}$  is the **Kronecker delta**.

The columns of  $\mathbf{I}_n$  are exactly  $\mathbf{e}_1, \dots, \mathbf{e}_n$ .

### Definition 0.9.4 — Diagonal matrix

A square matrix  $\mathbf{A} \in M_n(F)$  is **diagonal** if  $a_{ij} = 0$  whenever  $i \neq j$ . We write  $\text{diag}(d_1, \dots, d_n)$  for the diagonal matrix with diagonal entries  $d_1, \dots, d_n$ .

### Definition 0.9.5 — Upper triangular matrix

A square matrix  $\mathbf{A} \in M_n(F)$  is **upper triangular** if  $a_{ij} = 0$  whenever  $i > j$ , that is, every entry **strictly below** the main diagonal is zero. It is **lower triangular** if  $a_{ij} = 0$  whenever  $i < j$ .

#### Example. (Zero and identity matrices)

Write out  $0_{2 \times 3}$ ,  $\mathbf{I}_2$  and  $\mathbf{I}_3$ . Which of them are diagonal?

*Solution.*

$$0_{2 \times 3} = \begin{pmatrix} 0 & 0 & 0 \\ 0 & 0 & 0 \end{pmatrix}, \quad \mathbf{I}_2 = \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix}, \quad \mathbf{I}_3 = \begin{pmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{pmatrix}.$$

$\mathbf{I}_2 = \text{diag}(1, 1)$  and  $\mathbf{I}_3 = \text{diag}(1, 1, 1)$  are diagonal. The matrix  $0_{2 \times 3}$  is **not** diagonal, because the definition only applies to square matrices; the square zero matrix  $0_{n \times n} = \text{diag}(0, \dots, 0)$  is diagonal.

#### Example. (Diagonal and triangular)

Classify each matrix as diagonal, upper triangular, lower triangular, or none of these:

$$\mathbf{P} = \begin{pmatrix} 3 & 0 & 0 \\ 0 & -1 & 0 \\ 0 & 0 & 0 \end{pmatrix}, \quad \mathbf{Q} = \begin{pmatrix} 1 & 2 & 3 \\ 0 & 4 & 5 \\ 0 & 0 & 6 \end{pmatrix}, \quad \mathbf{R} = \begin{pmatrix} 1 & 2 & 3 \\ 0 & 4 & 5 \\ 0 & 7 & 6 \end{pmatrix}.$$

*Solution.*  $\mathbf{P} = \text{diag}(3, -1, 0)$  is diagonal. A zero on the diagonal is allowed: the definition only constrains off-diagonal entries.  $\mathbf{P}$  is also both upper and lower triangular, and in general a matrix is diagonal exactly when it is both.  $\mathbf{Q}$  is upper triangular but not lower triangular, since  $q_{12} = 2 \neq 0$ .  $\mathbf{R}$  differs from  $\mathbf{Q}$  in one entry,  $r_{32} = 7$ . All other entries below the diagonal are still zero, but the clause “ $a_{ij} = 0$  whenever  $i > j$ ” fails at  $(i, j) = (3, 2)$ , so  $\mathbf{R}$  is not upper triangular. It is not lower triangular either.

### 0.9.3 Addition and scalar multiplication

Matrices of the same size are added, and multiplied by scalars, one entry at a time.

#### Definition 0.9.6 — Matrix addition

Let  $\mathbf{A} = (a_{ij})$  and  $\mathbf{B} = (b_{ij})$  be in  $M_{m \times n}(F)$ , of the **same size**. Their **sum** is  $\mathbf{A} + \mathbf{B} \in M_{m \times n}(F)$  with  $(\mathbf{A} + \mathbf{B})_{ij} = a_{ij} + b_{ij}$ .

#### Definition 0.9.7 — Scalar multiplication

Let  $\mathbf{A} = (a_{ij}) \in M_{m \times n}(F)$  and  $c \in F$ . The **scalar multiple**  $c\mathbf{A} \in M_{m \times n}(F)$  has entries  $(c\mathbf{A})_{ij} = c a_{ij}$ .

#### Definition 0.9.8 — Additive inverse

The **additive inverse** of  $\mathbf{A} \in M_{m \times n}(F)$  is  $-\mathbf{A} = (-1)\mathbf{A}$ , with entries  $-a_{ij}$ . We write  $\mathbf{A} - \mathbf{B} = \mathbf{A} + (-\mathbf{B})$ .

Matrices of different sizes are never added:  $\begin{pmatrix} 1 & 2 \end{pmatrix} + \begin{pmatrix} 1 \\ 2 \end{pmatrix}$  is not defined.

#### Example. (Adding and scaling)

Let  $\mathbf{A} = \begin{pmatrix} 1 & -2 & 0 \\ 3 & 1 & 4 \end{pmatrix}$  and  $\mathbf{B} = \begin{pmatrix} 2 & 2 & -1 \\ 0 & -5 & 1 \end{pmatrix}$  in  $M_{2 \times 3}(\mathbb{R})$ . Compute  $\mathbf{A} + \mathbf{B}$ ,  $3\mathbf{A}$  and  $\mathbf{A} - 2\mathbf{B}$ .

*Solution.* Entry by entry,

$$\begin{aligned} \mathbf{A} + \mathbf{B} &= \begin{pmatrix} 3 & 0 & -1 \\ 3 & -4 & 5 \end{pmatrix}, \\ 3\mathbf{A} &= \begin{pmatrix} 3 & -6 & 0 \\ 9 & 3 & 12 \end{pmatrix}, \\ \mathbf{A} - 2\mathbf{B} &= \begin{pmatrix} 1-4 & -2-4 & 0+2 \\ 3-0 & 1+10 & 4-2 \end{pmatrix} = \begin{pmatrix} -3 & -6 & 2 \\ 3 & 11 & 2 \end{pmatrix}. \end{aligned}$$

Because both operations work entry by entry, they inherit the rules of  $F$ :

#### Theorem 0.9.1 — Properties of addition and scalar multiplication

Let  $\mathbf{A}, \mathbf{B}, \mathbf{C} \in M_{m \times n}(F)$  and  $c, d \in F$ . Then:

1.  $\mathbf{A} + \mathbf{B} = \mathbf{B} + \mathbf{A}$ ;
2.  $(\mathbf{A} + \mathbf{B}) + \mathbf{C} = \mathbf{A} + (\mathbf{B} + \mathbf{C})$ ;
3.  $\mathbf{A} + \mathbf{0} = \mathbf{A}$ ;

4.  $\mathbf{A} + (-\mathbf{A}) = \mathbf{0}$ ;
5.  $c(\mathbf{A} + \mathbf{B}) = c\mathbf{A} + c\mathbf{B}$  and  $(c + d)\mathbf{A} = c\mathbf{A} + d\mathbf{A}$ ;
6.  $(cd)\mathbf{A} = c(d\mathbf{A})$  and  $1\mathbf{A} = \mathbf{A}$ .

*Proof.* Two matrices of the same size are equal when their entries agree, so it suffices to compare  $(i, j)$ -entries. For 1,  $(\mathbf{A} + \mathbf{B})_{ij} = a_{ij} + b_{ij} = b_{ij} + a_{ij} = (\mathbf{B} + \mathbf{A})_{ij}$ , where the middle equality is commutativity of addition in  $F$ . For 5,  $(c(\mathbf{A} + \mathbf{B}))_{ij} = c(a_{ij} + b_{ij}) = ca_{ij} + cb_{ij} = (c\mathbf{A} + c\mathbf{B})_{ij}$  by distributivity in  $F$ . Each remaining identity reduces in the same way to the corresponding field axiom of Definition 0.7.1 in each entry. This proves the theorem. ■

These eight rules will reappear in Chapter 1 as the axioms of a vector space:  $M_{m \times n}(F)$  is one of our standard examples.

## 0.9.4 Matrix multiplication

Addition was entrywise. The obvious guess for multiplication is also entrywise, but that product is almost useless. The product that matters is the one that records **substitution**, and it looks very different.

Suppose new variables  $y_1, y_2$  are given in terms of  $x_1, x_2$ , and  $z_1, z_2$  in terms of  $y_1, y_2$ :

$$\begin{aligned} y_1 &= 2x_1 + x_2 & z_1 &= y_1 + 2y_2 \\ y_2 &= x_1 + 4x_2 & z_2 &= 3y_1 - y_2 \end{aligned}$$

Record each substitution by its table of coefficients:  $\mathbf{B} = \begin{pmatrix} 2 & 1 \\ 1 & 4 \end{pmatrix}$  for  $\mathbf{y}$  in terms of  $\mathbf{x}$ , and

$\mathbf{A} = \begin{pmatrix} 1 & 2 \\ 3 & -1 \end{pmatrix}$  for  $\mathbf{z}$  in terms of  $\mathbf{y}$ . Substituting,

$$\begin{aligned} z_1 &= (2x_1 + x_2) + 2(x_1 + 4x_2) \\ &= (1 \cdot 2 + 2 \cdot 1)x_1 + (1 \cdot 1 + 2 \cdot 4)x_2 = 4x_1 + 9x_2, \\ z_2 &= 3(2x_1 + x_2) - (x_1 + 4x_2) \\ &= (3 \cdot 2 + (-1) \cdot 1)x_1 + (3 \cdot 1 + (-1) \cdot 4)x_2 = 5x_1 - x_2. \end{aligned}$$

So  $\mathbf{z}$  in terms of  $\mathbf{x}$  has coefficient table  $\begin{pmatrix} 4 & 9 \\ 5 & -1 \end{pmatrix}$ . Look at how each entry arose: the coefficient

of  $x_k$  in  $z_i$  is  $a_{i1}b_{1k} + a_{i2}b_{2k}$ , row  $i$  of  $\mathbf{A}$  against column  $k$  of  $\mathbf{B}$ . The entrywise product  $\begin{pmatrix} 2 & 2 \\ 3 & -4 \end{pmatrix}$  has nothing to do with the substitution. This forces the definition.

*To multiply matrices, run each row of the first against each column of the second: the product of two matrices is the table of the combined substitution.*

### Definition 0.9.9 — Matrix multiplication

Let  $\mathbf{A} = (a_{ij}) \in M_{m \times n}(F)$  and  $\mathbf{B} = (b_{jk}) \in M_{n \times p}(F)$ , so that **the number of columns of  $\mathbf{A}$  equals the number of rows of  $\mathbf{B}$** . Their **product** is the matrix  $\mathbf{AB} \in M_{m \times p}(F)$

with entries

$$(\mathbf{AB})_{ik} = \sum_{j=1}^n a_{ij}b_{jk} = a_{i1}b_{1k} + a_{i2}b_{2k} + \cdots + a_{in}b_{nk}$$

for  $1 \leq i \leq m$  and  $1 \leq k \leq p$ . If the sizes do not match,  $\mathbf{AB}$  is **not defined**.

In words: to get the  $(i, k)$ -entry of  $\mathbf{AB}$ , walk along row  $i$  of  $\mathbf{A}$  and down column  $k$  of  $\mathbf{B}$  at the same time, multiply the entries you meet in pairs, and add. The size condition says that the row and the column have the same length  $n$ , so the pairing makes sense. The product has as many rows as  $\mathbf{A}$  and as many columns as  $\mathbf{B}$ :

$$\begin{matrix} \mathbf{A} & \mathbf{B} & = & \mathbf{AB} \\ m \times n & n \times p & & m \times p \end{matrix}$$

In particular, for  $\mathbf{A} \in M_{m \times n}(F)$  and a column  $\mathbf{x} \in F^n$ , the product  $\mathbf{Ax} \in F^m$  is defined, with  $i$ -th entry  $a_{i1}x_1 + \cdots + a_{in}x_n$ . A system of  $m$  linear equations in  $n$  unknowns is therefore a single equation  $\mathbf{Ax} = \mathbf{b}$ .

**Example. (A product of square matrices)**

Let  $\mathbf{A} = \begin{pmatrix} 2 & -1 \\ 0 & 3 \end{pmatrix}$  and  $\mathbf{B} = \begin{pmatrix} 1 & 4 \\ -2 & 1 \end{pmatrix}$  in  $M_2(\mathbb{R})$ . Compute  $\mathbf{AB}$  and  $\mathbf{BA}$ .

*Solution.* Row  $i$  of  $\mathbf{A}$  against column  $k$  of  $\mathbf{B}$ :

$$\begin{aligned} \mathbf{AB} &= \begin{pmatrix} 2 \cdot 1 + (-1)(-2) & 2 \cdot 4 + (-1) \cdot 1 \\ 0 \cdot 1 + 3 \cdot (-2) & 0 \cdot 4 + 3 \cdot 1 \end{pmatrix} = \begin{pmatrix} 4 & 7 \\ -6 & 3 \end{pmatrix}, \\ \mathbf{BA} &= \begin{pmatrix} 1 \cdot 2 + 4 \cdot 0 & 1 \cdot (-1) + 4 \cdot 3 \\ -2 \cdot 2 + 1 \cdot 0 & (-2)(-1) + 1 \cdot 3 \end{pmatrix} = \begin{pmatrix} 2 & 11 \\ -4 & 5 \end{pmatrix}. \end{aligned}$$

Both products are defined, and they are different.

**Example. (Products of rectangular matrices)**

Let  $\mathbf{A} = \begin{pmatrix} 1 & 0 & 2 \\ -1 & 3 & 1 \end{pmatrix} \in M_{2 \times 3}(\mathbb{R})$  and  $\mathbf{B} = \begin{pmatrix} 3 & 1 \\ 2 & 1 \\ 1 & 0 \end{pmatrix} \in M_{3 \times 2}(\mathbb{R})$ . Determine the sizes of  $\mathbf{AB}$  and  $\mathbf{BA}$  and compute them. Then compute  $(1 \ 2) \begin{pmatrix} 3 \\ 4 \end{pmatrix}$  and  $\begin{pmatrix} 3 \\ 4 \end{pmatrix} (1 \ 2)$ .

*Solution.*  $\mathbf{AB}$  is  $(2 \times 3)(3 \times 2)$ , so it is  $2 \times 2$ ;  $\mathbf{BA}$  is  $(3 \times 2)(2 \times 3)$ , so it is  $3 \times 3$ . Computing row against column,

$$\mathbf{AB} = \begin{pmatrix} 3 + 0 + 2 & 1 + 0 + 0 \\ -3 + 6 + 1 & -1 + 3 + 0 \end{pmatrix} = \begin{pmatrix} 5 & 1 \\ 4 & 2 \end{pmatrix}, \quad \mathbf{BA} = \begin{pmatrix} 2 & 3 & 7 \\ 1 & 3 & 5 \\ 1 & 0 & 2 \end{pmatrix}.$$

For instance  $(\mathbf{BA})_{13} = 3 \cdot 2 + 1 \cdot 1 = 7$ . The last two products show the extreme cases: a row times a column is  $1 \times 1$ , namely  $(1 \cdot 3 + 2 \cdot 4) = (11)$ , while a column times a row

is  $2 \times 2$ , namely  $\begin{pmatrix} 3 & 6 \\ 4 & 8 \end{pmatrix}$ .

Here is a non-example by minimal change. Take the same six entries as  $\mathbf{B}$ , but arrange them in two rows:  $\mathbf{C} = \begin{pmatrix} 3 & 2 & 1 \\ 1 & 1 & 0 \end{pmatrix} \in M_{2 \times 3}(\mathbb{R})$ . The entries are still real numbers and  $\mathbf{A}$  is unchanged. But  $\mathbf{A}$  has 3 columns and  $\mathbf{C}$  has 2 rows, so forming  $\mathbf{AC}$  would pair a row of length 3 with a column of length 2. The clause “the number of columns of  $\mathbf{A}$  equals the number of rows of  $\mathbf{B}$ ” fails, and  $\mathbf{AC}$  is **not defined**.

Why this definition, and not the entrywise one? Because it is the only rule under which “the table of a combined substitution is the product of the tables”. In Chapter 3 the same fact becomes: the matrix of a composition of linear maps is the product of their matrices. Every property of matrix multiplication, good or bad, comes from this.

## 0.9.5 Three views of a product

The entry formula is how one computes a product by hand, but it hides the structure. Two other descriptions, by columns and by rows, are what we will actually think with. It turns out that multiplying by  $\mathbf{A}$  on the left acts on each column of  $\mathbf{B}$  separately, and multiplying by  $\mathbf{B}$  on the right acts on each row of  $\mathbf{A}$  separately:

### Theorem 0.9.2 — Three views of a product

Let  $\mathbf{A} \in M_{m \times n}(F)$  and  $\mathbf{B} \in M_{n \times p}(F)$ , and write  $\mathbf{B} = (\mathbf{b}_1 \ \cdots \ \mathbf{b}_p)$  with columns  $\mathbf{b}_k \in F^n$ . Then:

1. **(Entries)**  $(\mathbf{AB})_{ik} = \sum_{j=1}^n a_{ij}b_{jk}$  for all  $i, k$ ;
2. **(Columns)** the  $k$ -th column of  $\mathbf{AB}$  is  $\mathbf{Ab}_k$ , that is,

$$\mathbf{AB} = (\mathbf{Ab}_1 \ \mathbf{Ab}_2 \ \cdots \ \mathbf{Ab}_p);$$

3. **(Rows)** the  $i$ -th row of  $\mathbf{AB}$  is  $(\text{row } i \text{ of } \mathbf{A})\mathbf{B}$ .

*Proof.* Part 1 is Definition 0.9.9. For part 2, fix  $k$ . The column  $\mathbf{b}_k$  has entries  $b_{1k}, \dots, b_{nk}$ , so by Definition 0.9.9 the  $i$ -th entry of  $\mathbf{Ab}_k \in F^m$  is  $\sum_{j=1}^n a_{ij}b_{jk}$ . By part 1 this is  $(\mathbf{AB})_{ik}$ , the  $i$ -th entry of the  $k$ -th column of  $\mathbf{AB}$ . Hence the  $k$ -th column of  $\mathbf{AB}$  is  $\mathbf{Ab}_k$ .

For part 3, fix  $i$ , and let  $\mathbf{r} = (a_{i1} \ \cdots \ a_{in})$  be row  $i$  of  $\mathbf{A}$ . Then  $\mathbf{rB}$  is a  $1 \times p$  matrix whose  $k$ -th entry is  $\sum_{j=1}^n a_{ij}b_{jk} = (\mathbf{AB})_{ik}$ , again by Definition 0.9.9. This is the  $k$ -th entry of row  $i$  of  $\mathbf{AB}$ . This proves the theorem. ■

The column view reduces every matrix product to products  $\mathbf{Ax}$  of a matrix with a single column. So it pays to understand  $\mathbf{Ax}$  well, and here is the fact to understand:

### Theorem 0.9.3 — $\mathbf{Ax}$ is a combination of the columns of $\mathbf{A}$

Let  $\mathbf{A} = (\mathbf{a}_1 \ \cdots \ \mathbf{a}_n) \in M_{m \times n}(F)$  with columns  $\mathbf{a}_j \in F^m$ , and let  $\mathbf{x} = (x_1, \dots, x_n) \in F^n$ . Then

$$\mathbf{Ax} = x_1\mathbf{a}_1 + x_2\mathbf{a}_2 + \cdots + x_n\mathbf{a}_n.$$

In particular  $\mathbf{A}\mathbf{e}_j = \mathbf{a}_j$  for each  $j$ . Similarly, if  $\mathbf{r}_1, \dots, \mathbf{r}_m$  are the rows of  $\mathbf{A}$  and  $\mathbf{y} = (y_1 \ \cdots \ y_m)$  is a row vector, then  $\mathbf{yA} = y_1\mathbf{r}_1 + \cdots + y_m\mathbf{r}_m$ .

*Proof.* Both sides are in  $F^m$ , so we compare  $i$ -th entries. By Definition 0.9.9, the  $i$ -th entry of  $\mathbf{Ax}$  is  $\sum_{j=1}^n a_{ij}x_j$ . On the right, the  $i$ -th entry of  $x_j\mathbf{a}_j$  is  $x_ja_{ij}$  by Definition 0.9.7, so by Definition 0.9.6 the  $i$ -th entry of  $\sum_j x_j\mathbf{a}_j$  is  $\sum_{j=1}^n x_ja_{ij}$ . These agree because multiplication in  $F$  is commutative. Taking  $\mathbf{x} = \mathbf{e}_j$ , whose only non-zero entry is  $x_j = 1$ , gives  $\mathbf{A}\mathbf{e}_j = \mathbf{a}_j$ . The statement for rows follows in the same way, comparing the  $k$ -th entries  $\sum_i y_ia_{ik}$ . This proves the theorem. ■

Read the formula as a sentence: **the entries of  $\mathbf{x}$  are the weights, and the columns of  $\mathbf{A}$  are what gets weighted.** So  $\mathbf{Ax} = \mathbf{b}$  has a solution exactly when  $\mathbf{b}$  can be built from the columns of  $\mathbf{A}$ . This single observation drives much of Chapters 2 and 3. Combined with Theorem 0.9.2, it also says that **every column of  $\mathbf{AB}$  is a combination of the columns of  $\mathbf{A}$** , with weights taken from the corresponding column of  $\mathbf{B}$ , and **every row of  $\mathbf{AB}$  is a combination of the rows of  $\mathbf{B}$ .**

**Example. (One product, three ways)**

Let

$$\mathbf{A} = \begin{pmatrix} 1 & 2 \\ 3 & -1 \\ 0 & 4 \end{pmatrix} \in M_{3 \times 2}(\mathbb{R}), \quad \mathbf{B} = \begin{pmatrix} 2 & 1 \\ -1 & 1 \end{pmatrix} \in M_2(\mathbb{R}), \quad \mathbf{x} = \begin{pmatrix} 2 \\ -1 \end{pmatrix}.$$

Compute  $\mathbf{Ax}$  as a combination of the columns of  $\mathbf{A}$ . Then compute  $\mathbf{AB}$  by columns, check one row by the row view, and check one entry by the entry formula.

*Solution.* The columns of  $\mathbf{A}$  are  $\mathbf{a}_1 = (1, 3, 0)$  and  $\mathbf{a}_2 = (2, -1, 4)$ . By Theorem 0.9.3,

$$\mathbf{Ax} = 2\mathbf{a}_1 - \mathbf{a}_2 = \begin{pmatrix} 2 \\ 6 \\ 0 \end{pmatrix} - \begin{pmatrix} 2 \\ -1 \\ 4 \end{pmatrix} = \begin{pmatrix} 0 \\ 7 \\ -4 \end{pmatrix}.$$

**Columns.** The first column of  $\mathbf{B}$  is  $\mathbf{b}_1 = (2, -1) = \mathbf{x}$ , so the first column of  $\mathbf{AB}$  is  $\mathbf{Ab}_1 = (0, 7, -4)$ , just computed. The second column is  $\mathbf{b}_2 = (1, 1)$ , so  $\mathbf{Ab}_2 = \mathbf{a}_1 + \mathbf{a}_2 = (3, 2, 4)$ . Hence

$$\mathbf{AB} = \begin{pmatrix} 0 & 3 \\ 7 & 2 \\ -4 & 4 \end{pmatrix}.$$

**Rows.** Row 2 of  $\mathbf{A}$  is  $(3 \ -1)$ , and by Theorem 0.9.3 (rows version)  $(3 \ -1)\mathbf{B} = 3(2 \ 1) - (-1 \ 1) = (7 \ 2)$ , which is row 2 of  $\mathbf{AB}$ .

**Entries.**  $(\mathbf{AB})_{32} = a_{31}b_{12} + a_{32}b_{22} = 0 \cdot 1 + 4 \cdot 1 = 4$ , as in the answer.

The column view needed nothing beyond scaling and adding columns of  $\mathbf{A}$ . When the

columns of  $\mathbf{B}$  are simple, it is usually the fastest route.

*Quick check.* Without multiplying out, what is the second column of  $\begin{pmatrix} 5 & -1 & 7 \\ 2 & 0 & 3 \end{pmatrix} \begin{pmatrix} 1 & 0 \\ 0 & 0 \\ 4 & 1 \end{pmatrix}$ ?

*Solution.* The second column of the right factor is  $\mathbf{e}_3 = (0, 0, 1)$ . By [Theorem 0.9.2](#) and [Theorem 0.9.3](#), the second column of the product is  $\mathbf{A}\mathbf{e}_3$ , the third column of the left factor:  $(7, 3)$ .

## 0.9.6 The rules of multiplication

Matrix multiplication keeps the most important rules of arithmetic: associativity, both distributive laws, and an identity.

### Theorem 0.9.4 — Properties of matrix multiplication

Let  $c \in F$ , and let  $\mathbf{A}, \mathbf{A}' \in M_{m \times n}(F)$ ,  $\mathbf{B}, \mathbf{B}' \in M_{n \times p}(F)$  and  $\mathbf{C} \in M_{p \times q}(F)$ . Then:

1. **(Associativity)**  $(\mathbf{AB})\mathbf{C} = \mathbf{A}(\mathbf{BC})$ ;
2. **(Distributivity)**  $\mathbf{A}(\mathbf{B} + \mathbf{B}') = \mathbf{AB} + \mathbf{AB}'$  and  $(\mathbf{A} + \mathbf{A}')\mathbf{B} = \mathbf{AB} + \mathbf{A'B}$ ;
3. **(Identity)**  $\mathbf{I}_m\mathbf{A} = \mathbf{A} = \mathbf{A}\mathbf{I}_n$ ;
4. **(Scalars)**  $c(\mathbf{AB}) = (c\mathbf{A})\mathbf{B} = \mathbf{A}(c\mathbf{B})$ ;
5. **(Zero)**  $0_{\ell \times m}\mathbf{A} = 0_{\ell \times n}$  and  $\mathbf{A}0_{n \times r} = 0_{m \times r}$  for all  $\ell, r \geq 1$ .

*Idea.* Each identity is checked entry by entry. The only one with content is associativity. Both sides have  $(i, l)$ -entry equal to the double sum  $\sum_j \sum_k a_{ij}b_{jk}c_{kl}$ , added up in two different orders, so the proof is “expand, swap the order of the finite sums, and fold back up”. In the substitution picture it is no surprise: substituting in two stages gives the same result whichever pair of stages we combine first.

*Proof.* For 1, both sides lie in  $M_{m \times q}(F)$ . Fix  $i$  and  $l$ . By [Definition 0.9.9](#) applied twice,

$$((\mathbf{AB})\mathbf{C})_{il} = \sum_{k=1}^p (\mathbf{AB})_{ik} c_{kl} = \sum_{k=1}^p \left( \sum_{j=1}^n a_{ij}b_{jk} \right) c_{kl} = \sum_{k=1}^p \sum_{j=1}^n a_{ij}b_{jk}c_{kl},$$

where the last step uses distributivity in  $F$ . The same computation gives

$$(\mathbf{A}(\mathbf{BC}))_{il} = \sum_{j=1}^n a_{ij}(\mathbf{BC})_{jl} = \sum_{j=1}^n a_{ij} \left( \sum_{k=1}^p b_{jk}c_{kl} \right) = \sum_{j=1}^n \sum_{k=1}^p a_{ij}b_{jk}c_{kl}.$$

The two double sums add the same  $np$  terms in different orders, so they are equal, by commutativity and associativity of addition in  $F$ . Hence  $(\mathbf{AB})\mathbf{C} = \mathbf{A}(\mathbf{BC})$ .

For 2,  $(\mathbf{A}(\mathbf{B} + \mathbf{B}'))_{ik} = \sum_j a_{ij}(b_{jk} + b'_{jk}) = \sum_j a_{ij}b_{jk} + \sum_j a_{ij}b'_{jk} = (\mathbf{AB} + \mathbf{AB}')_{ik}$  by distributivity in  $F$ ; the other law is the same with the roles of the factors swapped.



For 3,  $(\mathbf{I}_m \mathbf{A})_{ik} = \sum_{j=1}^m \delta_{ij} a_{jk} = a_{ik}$ , because by [Definition 0.9.3](#) only the term  $j = i$  survives. Likewise  $(\mathbf{A} \mathbf{I}_n)_{ik} = \sum_j a_{ij} \delta_{jk} = a_{ik}$ .

For 4,  $(c(\mathbf{A}\mathbf{B}))_{ik} = c \sum_j a_{ij} b_{jk} = \sum_j (ca_{ij}) b_{jk} = ((c\mathbf{A})\mathbf{B})_{ik}$ , and moving  $c$  next to  $b_{jk}$  instead, using commutativity in  $F$ , gives  $(\mathbf{A}(c\mathbf{B}))_{ik}$ . For 5, every term  $0 \cdot a_{jk}$  or  $a_{ij} \cdot 0$  is 0 by [Theorem 0.7.1](#). This proves the theorem. ■

Associativity means we may write  $\mathbf{ABC}$  without brackets, and more generally a product of any number of matrices, as long as consecutive sizes match.

Just as important is what is **missing** from the list. There is no commutative law, and no cancellation law. Each of the following failures will cost you a wrong proof if you forget it.

*Warning.*  **$\mathbf{AB} \neq \mathbf{BA}$  in general.** Take  $\mathbf{A} = \begin{pmatrix} 0 & 1 \\ 0 & 0 \end{pmatrix}$  and  $\mathbf{B} = \begin{pmatrix} 0 & 0 \\ 1 & 0 \end{pmatrix}$ . Then  $\mathbf{AB} = \begin{pmatrix} 1 & 0 \\ 0 & 0 \end{pmatrix}$  but  $\mathbf{BA} = \begin{pmatrix} 0 & 0 \\ 0 & 1 \end{pmatrix}$ . Before swapping two factors in a proof, you must know that they commute. (And if  $\mathbf{A}$  is  $2 \times 3$  and  $\mathbf{B}$  is  $3 \times 2$ , then  $\mathbf{AB}$  and  $\mathbf{BA}$  do not even have the same size.)

*Warning.* **A product can be zero with both factors non-zero.** In  $M_2(\mathbb{R})$ ,

$$\begin{pmatrix} 1 & 1 \\ 1 & 1 \end{pmatrix} \begin{pmatrix} 1 & -1 \\ -1 & 1 \end{pmatrix} = \begin{pmatrix} 0 & 0 \\ 0 & 0 \end{pmatrix}.$$

So  $M_n(F)$  is not a field for  $n \geq 2$ , and “ $\mathbf{AB} = 0$  implies  $\mathbf{A} = 0$  or  $\mathbf{B} = 0$ ” is false.

*Warning.* **Cancellation fails:  $\mathbf{AC} = \mathbf{BC}$  does not imply  $\mathbf{A} = \mathbf{B}$ .** With

$$\mathbf{A} = \begin{pmatrix} 1 & 2 \\ 3 & 4 \end{pmatrix}, \quad \mathbf{B} = \begin{pmatrix} 1 & 2 \\ 3 & 9 \end{pmatrix}, \quad \mathbf{C} = \begin{pmatrix} 1 & 1 \\ 0 & 0 \end{pmatrix},$$

we get  $\mathbf{AC} = \mathbf{BC} = \begin{pmatrix} 1 & 1 \\ 3 & 3 \end{pmatrix}$  although  $\mathbf{A} \neq \mathbf{B}$ . By the column view, both columns of  $\mathbf{C}$  are  $\mathbf{e}_1$ , so  $\mathbf{AC}$  and  $\mathbf{BC}$  only see the first columns of  $\mathbf{A}$  and  $\mathbf{B}$ , which agree. Canceling  $\mathbf{C}$  is legal only when  $\mathbf{C}$  is invertible, as we will see below.

*Quick check.* Find a non-zero  $\mathbf{A} \in M_2(\mathbb{R})$  with  $\mathbf{A}^2 = \mathbf{AA} = 0$ .

*Solution.* Take  $\mathbf{A} = \begin{pmatrix} 0 & 1 \\ 0 & 0 \end{pmatrix}$ . By the column view,  $\mathbf{A}\mathbf{e}_1 = \mathbf{0}$  and  $\mathbf{A}\mathbf{e}_2 = \mathbf{e}_1$ , so the columns of  $\mathbf{A}^2$  are  $\mathbf{A}\mathbf{0} = \mathbf{0}$  and  $\mathbf{A}\mathbf{e}_1 = \mathbf{0}$ . Hence  $\mathbf{A}^2 = 0$  although  $\mathbf{A} \neq 0$ . This small matrix is the source of a remarkable number of counterexamples in this book; it will reappear often.

## 0.9.7 The transpose

Rows and columns play symmetric roles in a matrix, and the transpose exchanges them.

**Definition 0.9.10 — Transpose**

Let  $\mathbf{A} = (a_{ij}) \in M_{m \times n}(F)$ . The **transpose** of  $\mathbf{A}$  is the matrix  $\mathbf{A}^\top \in M_{n \times m}(F)$  with entries

$$(\mathbf{A}^\top)_{ij} = a_{ji}.$$

In words: row  $i$  of  $\mathbf{A}$  becomes column  $i$  of  $\mathbf{A}^\top$ , and the size flips from  $m \times n$  to  $n \times m$ . A square matrix is reflected in its main diagonal.

**Definition 0.9.11 — Symmetric and skew-symmetric matrices**

A square matrix  $\mathbf{A} \in M_n(F)$  is **symmetric** if  $\mathbf{A}^\top = \mathbf{A}$ , that is,  $a_{ij} = a_{ji}$  for all  $i, j$ . It is **skew-symmetric** if  $\mathbf{A}^\top = -\mathbf{A}$ , that is,  $a_{ij} = -a_{ji}$  for all  $i, j$ .

**Example. (Transposes, symmetric and skew-symmetric matrices)**

Compute the transpose of  $\mathbf{A} = \begin{pmatrix} 1 & -2 & 0 \\ 3 & 1 & 4 \end{pmatrix}$ . Decide whether

$$\mathbf{S} = \begin{pmatrix} 1 & 2 & 3 \\ 2 & 4 & 5 \\ 3 & 5 & 6 \end{pmatrix}, \quad \mathbf{K} = \begin{pmatrix} 0 & 2 & -1 \\ -2 & 0 & 3 \\ 1 & -3 & 0 \end{pmatrix}, \quad \mathbf{S}' = \begin{pmatrix} 1 & 2 & 3 \\ 2 & 4 & 5 \\ 3 & 6 & 6 \end{pmatrix}$$

over  $\mathbb{R}$  are symmetric or skew-symmetric.

*Solution.* The rows of  $\mathbf{A}$  become columns:  $\mathbf{A}^\top = \begin{pmatrix} 1 & 3 \\ -2 & 1 \\ 0 & 4 \end{pmatrix} \in M_{3 \times 2}(\mathbb{R})$ .

$\mathbf{S}$  is symmetric:  $s_{12} = s_{21} = 2$ ,  $s_{13} = s_{31} = 3$ ,  $s_{23} = s_{32} = 5$ .  $\mathbf{K}$  is skew-symmetric:  $k_{12} = 2 = -k_{21}$ ,  $k_{13} = -1 = -k_{31}$ ,  $k_{23} = 3 = -k_{32}$ , and each diagonal entry satisfies  $k_{ii} = -k_{ii}$ , so it is 0. The matrix  $\mathbf{S}'$  differs from  $\mathbf{S}$  only in the entry  $(3, 2)$ . Every other symmetric pair still matches, but  $s'_{23} = 5 \neq 6 = s'_{32}$ , so  $\mathbf{S}'$  is **not** symmetric. A  $1 \times 1$  matrix is always symmetric, and the identity  $\mathbf{I}_n$  and every diagonal matrix are symmetric.

Over  $\mathbb{R}$ , a skew-symmetric matrix has zero diagonal, since  $a_{ii} = -a_{ii}$  forces  $2a_{ii} = 0$ . Over  $\mathbb{F}_2$ , where  $-1 = 1$ , “skew-symmetric” and “symmetric” mean the same thing, and  $\mathbf{I}_2$  is skew-symmetric. The characteristic of the field (Definition 0.7.2) matters.

The transpose is compatible with the operations, except that it **reverses** products:

**Theorem 0.9.5 — Properties of the transpose**

Let  $c \in F$ . For matrices of sizes for which each expression is defined:

1.  $(\mathbf{A}^\top)^\top = \mathbf{A}$ ;
2.  $(\mathbf{A} + \mathbf{B})^\top = \mathbf{A}^\top + \mathbf{B}^\top$ ;

3.  $(c\mathbf{A})^\top = c\mathbf{A}^\top$ ;
4.  $(\mathbf{AB})^\top = \mathbf{B}^\top \mathbf{A}^\top$  for  $\mathbf{A} \in M_{m \times n}(F)$  and  $\mathbf{B} \in M_{n \times p}(F)$ .

*Idea.* The order reverses for a size reason before any computation:  $(\mathbf{AB})^\top$  is  $p \times m$ , and  $\mathbf{B}^\top \mathbf{A}^\top$  is  $(p \times n)(n \times m) = p \times m$ , while  $\mathbf{A}^\top \mathbf{B}^\top$  is  $(n \times m)(p \times n)$ , usually not even defined. Then compare entries.

*Proof.* For 1,  $((\mathbf{A}^\top)^\top)_{ij} = (\mathbf{A}^\top)_{ji} = a_{ij}$  by Definition 0.9.10. For 2 and 3,  $((\mathbf{A}+\mathbf{B})^\top)_{ij} = a_{ji} + b_{ji} = (\mathbf{A}^\top + \mathbf{B}^\top)_{ij}$  and  $((c\mathbf{A})^\top)_{ij} = ca_{ji} = (c\mathbf{A}^\top)_{ij}$ .

For 4, both sides lie in  $M_{p \times m}(F)$ . Fix  $1 \leq i \leq p$  and  $1 \leq k \leq m$ . By Definition 0.9.10 and Definition 0.9.9,

$$\begin{aligned} ((\mathbf{AB})^\top)_{ik} &= (\mathbf{AB})_{ki} = \sum_{j=1}^n a_{kj} b_{ji}, \\ (\mathbf{B}^\top \mathbf{A}^\top)_{ik} &= \sum_{j=1}^n (\mathbf{B}^\top)_{ij} (\mathbf{A}^\top)_{jk} = \sum_{j=1}^n b_{ji} a_{kj}. \end{aligned}$$

The two sums agree term by term because multiplication in  $F$  is commutative. Hence  $(\mathbf{AB})^\top = \mathbf{B}^\top \mathbf{A}^\top$ . This proves the theorem. ■

For example, for every  $\mathbf{A} \in M_{m \times n}(F)$ , the square matrix  $\mathbf{A}^\top \mathbf{A}$  is symmetric: by parts 4 and 1,  $(\mathbf{A}^\top \mathbf{A})^\top = \mathbf{A}^\top (\mathbf{A}^\top)^\top = \mathbf{A}^\top \mathbf{A}$ . Matrices of this form are central to least squares and the singular value decomposition much later on.

## 0.9.8 The trace

The trace is the simplest number attached to a square matrix, and it has one surprising property.

### Definition 0.9.12 — Trace

The **trace** of a square matrix  $\mathbf{A} = (a_{ij}) \in M_n(F)$  is the sum of its diagonal entries:

$$\operatorname{tr} \mathbf{A} = \sum_{i=1}^n a_{ii} = a_{11} + a_{22} + \cdots + a_{nn} \in F.$$

#### Example. (Computing traces)

Compute  $\operatorname{tr} \mathbf{A}$  for  $\mathbf{A} = \begin{pmatrix} 2 & 7 & -1 \\ 0 & -3 & 5 \\ 4 & 1 & 6 \end{pmatrix} \in M_3(\mathbb{R})$ , and  $\operatorname{tr} \mathbf{I}_n$  over  $\mathbb{R}$  and over  $\mathbb{F}_2$ .

*Solution.*  $\operatorname{tr} \mathbf{A} = 2 + (-3) + 6 = 5$ ; the off-diagonal entries play no role.  $\operatorname{tr} \mathbf{I}_n =$

$1 + \cdots + 1 = n \cdot 1$ . Over  $\mathbb{R}$  this is  $n$ . Over  $\mathbb{F}_2$  it is 0 when  $n$  is even and 1 when  $n$  is odd. The trace is **not** defined for a non-square matrix.

### Theorem 0.9.6 — Properties of the trace

Let  $\mathbf{A}, \mathbf{B} \in M_n(F)$  and  $c \in F$ . Then:

1.  $\text{tr}(\mathbf{A} + \mathbf{B}) = \text{tr} \mathbf{A} + \text{tr} \mathbf{B}$  and  $\text{tr}(c\mathbf{A}) = c \text{tr} \mathbf{A}$ ;
2.  $\text{tr}(\mathbf{A}^\top) = \text{tr} \mathbf{A}$ ;
3.  $\text{tr}(\mathbf{AB}) = \text{tr}(\mathbf{BA})$ . More generally, this holds for all  $\mathbf{A} \in M_{m \times n}(F)$  and  $\mathbf{B} \in M_{n \times m}(F)$ , where  $\mathbf{AB}$  is  $m \times m$  and  $\mathbf{BA}$  is  $n \times n$ .

*Idea.* Part 3 is surprising because  $\mathbf{AB} \neq \mathbf{BA}$  in general. But the trace only sees the diagonal, and  $\text{tr}(\mathbf{AB}) = \sum_i \sum_j a_{ij} b_{ji}$  is a sum over all pairs  $(i, j)$  of the product of  $a_{ij}$  with the “mirror” entry  $b_{ji}$ . That description does not care which matrix is written first.

*Proof.* Parts 1 and 2 follow from the definitions: the diagonal entries of  $\mathbf{A} + \mathbf{B}$ ,  $c\mathbf{A}$  and  $\mathbf{A}^\top$  are  $a_{ii} + b_{ii}$ ,  $ca_{ii}$  and  $a_{ii}$ .

For 3, let  $\mathbf{A} \in M_{m \times n}(F)$  and  $\mathbf{B} \in M_{n \times m}(F)$ . By Definition 0.9.12 and Definition 0.9.9,

$$\text{tr}(\mathbf{AB}) = \sum_{i=1}^m (\mathbf{AB})_{ii} = \sum_{i=1}^m \sum_{j=1}^n a_{ij} b_{ji} = \sum_{j=1}^n \sum_{i=1}^m b_{ji} a_{ij} = \sum_{j=1}^n (\mathbf{BA})_{jj} = \text{tr}(\mathbf{BA}),$$

where the middle equality swaps the order of the finite double sum and uses commutativity of multiplication in  $F$ . This proves the theorem. ■

A first payoff: for all  $\mathbf{A}, \mathbf{B} \in M_n(F)$ , the matrix  $\mathbf{AB} - \mathbf{BA}$  has trace  $\text{tr}(\mathbf{AB}) - \text{tr}(\mathbf{BA}) = 0$ . So over  $\mathbb{R}$  it can never equal  $\mathbf{I}_n$ , whose trace is  $n$  (Exercise (C1:  $\mathbf{AB} - \mathbf{BA} = \mathbf{I}$  is impossible, or is it?)). The trace will also turn out to be one of the quantities that does not change when we change coordinates.

It is tempting to conclude that the trace of a product ignores the order of **all** the factors. It does not.

*Warning.*  $\text{tr}(\mathbf{ABC}) \neq \text{tr}(\mathbf{BAC})$  in general. Take  $\mathbf{A} = \begin{pmatrix} 1 & 0 \\ 0 & 0 \end{pmatrix}$ ,  $\mathbf{B} = \begin{pmatrix} 0 & 1 \\ 0 & 0 \end{pmatrix}$  and  $\mathbf{C} = \begin{pmatrix} 0 & 0 \\ 1 & 0 \end{pmatrix}$ . Then  $\mathbf{ABC} = \begin{pmatrix} 1 & 0 \\ 0 & 0 \end{pmatrix}$  has trace 1, but  $\mathbf{BA} = 0$ , so  $\mathbf{BAC} = 0$  has trace 0. What **is** true is that a **cyclic** rotation preserves the trace:  $\text{tr}(\mathbf{ABC}) = \text{tr}((\mathbf{AB})\mathbf{C}) = \text{tr}(\mathbf{C}(\mathbf{AB})) = \text{tr}(\mathbf{CAB})$  by part 3.

## 0.9.9 Inverse matrices

In a field we solve  $ax = b$  by multiplying by  $a^{-1}$ , and this works exactly when  $a \neq 0$ . We would like to solve a system  $\mathbf{Ax} = \mathbf{b}$  the same way, by “dividing by  $\mathbf{A}$ ”. What should that mean, and

for which  $\mathbf{A}$  is it possible?

The naive answer, “every non-zero matrix”, fails. Let  $\mathbf{N} = \begin{pmatrix} 0 & 1 \\ 0 & 0 \end{pmatrix} \neq \mathbf{0}$ . For **any**  $\mathbf{B} \in M_2(F)$ , the first column of  $\mathbf{BN}$  is  $\mathbf{B}$  times the first column of  $\mathbf{N}$ , that is  $\mathbf{B}\mathbf{0} = \mathbf{0}$ , by [Theorem 0.9.2](#). The first column of  $\mathbf{I}_2$  is  $\mathbf{e}_1 \neq \mathbf{0}$ , so  $\mathbf{BN} \neq \mathbf{I}_2$ : no matrix undoes  $\mathbf{N}$  from the left. Dividing needs a genuine two-sided inverse, and not every non-zero matrix has one.

*An invertible matrix is a square matrix whose effect can be undone by multiplying, on either side.*

### Definition 0.9.13 — Invertible matrix

A square matrix  $\mathbf{A} \in M_n(F)$  is **invertible** if there exists  $\mathbf{B} \in M_n(F)$  such that

$$\mathbf{AB} = \mathbf{I}_n \quad \text{and} \quad \mathbf{BA} = \mathbf{I}_n.$$

Such a  $\mathbf{B}$  is called **an inverse** of  $\mathbf{A}$ . A square matrix that is not invertible is called **singular**.

In words:  $\mathbf{A}$  is invertible when some matrix  $\mathbf{B}$  of the same size cancels it **from the right** ( $\mathbf{AB} = \mathbf{I}_n$ ) **and** from the left ( $\mathbf{BA} = \mathbf{I}_n$ ). The definition asks for one  $\mathbf{B}$  doing both jobs, and it only speaks about square matrices.

Is “an inverse” really “the inverse”? Yes: an inverse, if it exists, is unique. This is part 1 of the next theorem, and it lets us write  $\mathbf{A}^{-1}$ . The remaining parts are what inverses are for.

### Theorem 0.9.7 — Properties of inverses

Let  $\mathbf{A}, \mathbf{B} \in M_n(F)$ .

1. **(Uniqueness)** If  $\mathbf{B}$  and  $\mathbf{B}'$  are both inverses of  $\mathbf{A}$ , then  $\mathbf{B} = \mathbf{B}'$ . We write  $\mathbf{A}^{-1}$  for **the** inverse of an invertible  $\mathbf{A}$ .
2. If  $\mathbf{A}$  is invertible, then so is  $\mathbf{A}^{-1}$ , and  $(\mathbf{A}^{-1})^{-1} = \mathbf{A}$ .
3. If  $\mathbf{A}$  and  $\mathbf{B}$  are invertible, then so is  $\mathbf{AB}$ , and  $(\mathbf{AB})^{-1} = \mathbf{B}^{-1}\mathbf{A}^{-1}$ .
4. If  $\mathbf{A}$  is invertible, then so is  $\mathbf{A}^\top$ , and  $(\mathbf{A}^\top)^{-1} = (\mathbf{A}^{-1})^\top$ .
5. If  $\mathbf{A}$  is invertible and  $c \in F$  is **non-zero**, then  $c\mathbf{A}$  is invertible, and  $(c\mathbf{A})^{-1} = c^{-1}\mathbf{A}^{-1}$ .
6. **(Cancellation)** If  $\mathbf{A}$  is invertible, then  $\mathbf{AX} = \mathbf{AY}$  implies  $\mathbf{X} = \mathbf{Y}$  for all  $\mathbf{X}, \mathbf{Y} \in M_{n \times p}(F)$ , and  $\mathbf{XA} = \mathbf{YA}$  implies  $\mathbf{X} = \mathbf{Y}$  for all  $\mathbf{X}, \mathbf{Y} \in M_{p \times n}(F)$ .
7. **(Solving)** If  $\mathbf{A}$  is invertible, then for every  $\mathbf{b} \in F^n$  the equation  $\mathbf{Ax} = \mathbf{b}$  has **exactly one** solution  $\mathbf{x} \in F^n$ , namely  $\mathbf{x} = \mathbf{A}^{-1}\mathbf{b}$ .

*Idea.* Uniqueness is the standard trick for “prove unique by assuming two”: sandwich  $\mathbf{B}'$  between  $\mathbf{B}$  and  $\mathbf{A}$ , and use associativity to collapse the product both ways. For the other parts the candidate inverse is written in the statement, so we only multiply it out on **both** sides, as the definition demands. The order in part 3 is the “socks and shoes” rule: to undo “first  $\mathbf{B}$ , then  $\mathbf{A}$ ”, undo  $\mathbf{A}$  first.

*Proof.* For 1, suppose  $\mathbf{AB} = \mathbf{BA} = \mathbf{I}_n$  and  $\mathbf{AB}' = \mathbf{B}'\mathbf{A} = \mathbf{I}_n$ . By [Theorem 0.9.4](#)

(identity and associativity),

$$\mathbf{B} = \mathbf{B}\mathbf{I}_n = \mathbf{B}(\mathbf{A}\mathbf{B}') = (\mathbf{B}\mathbf{A})\mathbf{B}' = \mathbf{I}_n\mathbf{B}' = \mathbf{B}'.$$

For 2, the equations  $\mathbf{A}\mathbf{A}^{-1} = \mathbf{I}_n$  and  $\mathbf{A}^{-1}\mathbf{A} = \mathbf{I}_n$  say precisely that  $\mathbf{A}$  is an inverse of  $\mathbf{A}^{-1}$ . By part 1 it is the inverse.

For 3, by associativity,

$$(\mathbf{A}\mathbf{B})(\mathbf{B}^{-1}\mathbf{A}^{-1}) = \mathbf{A}(\mathbf{B}\mathbf{B}^{-1})\mathbf{A}^{-1} = \mathbf{A}\mathbf{I}_n\mathbf{A}^{-1} = \mathbf{A}\mathbf{A}^{-1} = \mathbf{I}_n,$$

and similarly  $(\mathbf{B}^{-1}\mathbf{A}^{-1})(\mathbf{A}\mathbf{B}) = \mathbf{B}^{-1}(\mathbf{A}^{-1}\mathbf{A})\mathbf{B} = \mathbf{B}^{-1}\mathbf{B} = \mathbf{I}_n$ . Hence  $\mathbf{B}^{-1}\mathbf{A}^{-1}$  is an inverse of  $\mathbf{A}\mathbf{B}$ , and by part 1 it is  $(\mathbf{A}\mathbf{B})^{-1}$ .

For 4, by [Theorem 0.9.5](#),  $\mathbf{A}^\top(\mathbf{A}^{-1})^\top = (\mathbf{A}^{-1}\mathbf{A})^\top = \mathbf{I}_n^\top = \mathbf{I}_n$  and  $(\mathbf{A}^{-1})^\top\mathbf{A}^\top = (\mathbf{A}\mathbf{A}^{-1})^\top = \mathbf{I}_n$ , where  $\mathbf{I}_n^\top = \mathbf{I}_n$  since  $\mathbf{I}_n$  is symmetric. Hence  $(\mathbf{A}^{-1})^\top$  is the inverse of  $\mathbf{A}^\top$ .

For 5, since  $c \neq 0$ , the inverse  $c^{-1}$  exists in  $F$ . By [Theorem 0.9.4](#) (scalars),  $(c\mathbf{A})(c^{-1}\mathbf{A}^{-1}) = (cc^{-1})(\mathbf{A}\mathbf{A}^{-1}) = \mathbf{I}_n$ , and similarly in the other order.

For 6, suppose  $\mathbf{A}\mathbf{X} = \mathbf{A}\mathbf{Y}$ . Multiplying on the left by  $\mathbf{A}^{-1}$  and using associativity,  $\mathbf{X} = \mathbf{I}_n\mathbf{X} = (\mathbf{A}^{-1}\mathbf{A})\mathbf{X} = \mathbf{A}^{-1}(\mathbf{A}\mathbf{X}) = \mathbf{A}^{-1}(\mathbf{A}\mathbf{Y}) = (\mathbf{A}^{-1}\mathbf{A})\mathbf{Y} = \mathbf{Y}$ . The statement for  $\mathbf{X}\mathbf{A} = \mathbf{Y}\mathbf{A}$  follows by multiplying on the right by  $\mathbf{A}^{-1}$ .

For 7,  $\mathbf{x} = \mathbf{A}^{-1}\mathbf{b}$  is a solution, since  $\mathbf{A}(\mathbf{A}^{-1}\mathbf{b}) = (\mathbf{A}\mathbf{A}^{-1})\mathbf{b} = \mathbf{b}$ . If  $\mathbf{x}$  is any solution, then  $\mathbf{A}\mathbf{x} = \mathbf{b} = \mathbf{A}(\mathbf{A}^{-1}\mathbf{b})$ , and part 6 (with  $p = 1$ ) gives  $\mathbf{x} = \mathbf{A}^{-1}\mathbf{b}$ . This proves the theorem. ■

Part 6 is the repair of the failed cancellation law from the warnings above: **you may cancel an invertible matrix**, on the side where it stands, and nothing else.

### Example. (Some invertible matrices)

Show that each of the following is invertible and find its inverse:  $\mathbf{I}_n$ ; a diagonal matrix  $\mathbf{D} = \text{diag}(d_1, \dots, d_n)$  with **all**  $d_i \neq 0$ ;  $\mathbf{A} = \begin{pmatrix} 2 & 1 \\ 5 & 3 \end{pmatrix} \in M_2(\mathbb{R})$ ; and  $\mathbf{U} = \begin{pmatrix} 1 & 1 \\ 0 & 1 \end{pmatrix} \in M_2(\mathbb{F}_2)$ .

*Solution.*  $\mathbf{I}_n\mathbf{I}_n = \mathbf{I}_n$ , so  $\mathbf{I}_n$  is its own inverse. This is the degenerate case:  $\mathbf{I}_n$  plays the role that 1 plays among numbers.

For  $\mathbf{D}$ , let  $\mathbf{E} = \text{diag}(d_1^{-1}, \dots, d_n^{-1})$ , which exists since every  $d_i \neq 0$ . Then  $(\mathbf{D}\mathbf{E})_{ik} = \sum_j d_{ij}e_{jk}$ , and only  $j = i = k$  can contribute, so  $\mathbf{D}\mathbf{E} = \text{diag}(d_1d_1^{-1}, \dots, d_nd_n^{-1}) = \mathbf{I}_n$ ; likewise  $\mathbf{E}\mathbf{D} = \mathbf{I}_n$ . Hence  $\mathbf{D}^{-1} = \mathbf{E}$ . For  $n = 1$  this says the  $1 \times 1$  matrix  $(a)$  is invertible when  $a \neq 0$ , with inverse  $(a^{-1})$ , exactly as in the field.

For  $\mathbf{A}$ , let  $\mathbf{B} = \begin{pmatrix} 3 & -1 \\ -5 & 2 \end{pmatrix}$ . Then

$$\mathbf{AB} = \begin{pmatrix} 6-5 & -2+2 \\ 15-15 & -5+6 \end{pmatrix} = \mathbf{I}_2, \quad \mathbf{BA} = \begin{pmatrix} 6-5 & 3-3 \\ -10+10 & -5+6 \end{pmatrix} = \mathbf{I}_2,$$

so  $\mathbf{A}^{-1} = \mathbf{B}$ . (Where  $\mathbf{B}$  came from is explained by [Theorem 0.9.8](#) below.)

For  $\mathbf{U}$  over  $\mathbb{F}_2$ ,  $\mathbf{U}^2 = \begin{pmatrix} 1 & 1+1 \\ 0 & 1 \end{pmatrix} = \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix}$ , since  $1+1=0$ . So  $\mathbf{U}$  is its own inverse.

Over  $\mathbb{R}$  the same array would have inverse  $\begin{pmatrix} 1 & -1 \\ 0 & 1 \end{pmatrix}$ : the answer depends on the field.

Now a non-example by minimal change. The matrix  $\text{diag}(1, 2)$  is invertible, with inverse  $\text{diag}(1, \frac{1}{2})$ . Change the entry 2 to 0:  $\mathbf{A} = \text{diag}(1, 0)$  is still diagonal and still non-zero. But for any  $\mathbf{B} \in M_2(F)$ , the second column of  $\mathbf{BA}$  is  $\mathbf{B}$  times the second column of  $\mathbf{A}$ , which is  $\mathbf{B}\mathbf{0} = \mathbf{0}$ , while the second column of  $\mathbf{I}_2$  is  $\mathbf{e}_2$ . So  $\mathbf{BA} \neq \mathbf{I}_2$  for every  $\mathbf{B}$ : the clause “**there exists**  $\mathbf{B}$  with ...  $\mathbf{BA} = \mathbf{I}_n$ ” fails, and  $\text{diag}(1, 0)$  is singular. The same argument shows that any square matrix with a zero column is singular, and in particular so is every zero matrix  $0_{n \times n}$ .

Why does the definition insist on a **square** matrix and on **both** equations? For non-square matrices one-sided inverses can exist without the other side: with  $\mathbf{C} = \begin{pmatrix} 1 & 0 \end{pmatrix} \in M_{1 \times 2}(\mathbb{R})$  and  $\mathbf{D} = \begin{pmatrix} 1 \\ 0 \end{pmatrix}$ , we get  $\mathbf{CD} = (1) = \mathbf{I}_1$  but  $\mathbf{DC} = \begin{pmatrix} 1 & 0 \\ 0 & 0 \end{pmatrix} \neq \mathbf{I}_2$ . Demanding both equations makes the inverse unique (part 1) and makes both cancellation laws (part 6) hold.

*Warning.* For square matrices,  $\mathbf{AB} = \mathbf{I}_n$  already implies  $\mathbf{BA} = \mathbf{I}_n$ . This is true, but it is not obvious, and we do not know it yet. Nothing in this chapter proves it: the entries of  $\mathbf{AB}$  and  $\mathbf{BA}$  are different sums. It is proved in Chapter 2, using row reduction. Until then, to show that  $\mathbf{B}$  is the inverse of  $\mathbf{A}$ , check **both**  $\mathbf{AB} = \mathbf{I}_n$  and  $\mathbf{BA} = \mathbf{I}_n$ . (The non-square example above shows that some argument is needed.)

For  $2 \times 2$  matrices, invertibility is decided by one number.

### Theorem 0.9.8 — Inverse of a $2 \times 2$ matrix

Let  $\mathbf{A} = \begin{pmatrix} a & b \\ c & d \end{pmatrix} \in M_2(F)$ . Then  $\mathbf{A}$  is invertible **if and only if**  $ad - bc \neq 0$ . In that case,

$$\mathbf{A}^{-1} = \frac{1}{ad - bc} \begin{pmatrix} d & -b \\ -c & a \end{pmatrix}.$$

*Idea.* Where does the formula come from? Try to kill the off-diagonal entries of  $\mathbf{A}$  by multiplying with a matrix built from its own entries. Swapping  $a$  and  $d$  and negating  $b$  and  $c$  gives  $\begin{pmatrix} a & b \\ c & d \end{pmatrix} \begin{pmatrix} d & -b \\ -c & a \end{pmatrix} = (ad - bc)\mathbf{I}_2$ . If  $ad - bc \neq 0$ , divide by it. If  $ad - bc = 0$ , the same computation shows that  $\mathbf{A}$  sends a non-zero column to  $\mathbf{0}$ , and an invertible matrix cannot do that, since its inverse would have to send  $\mathbf{0}$  back to a non-zero column.

*Proof.* Write  $\Delta = ad - bc$  and  $\tilde{\mathbf{A}} = \begin{pmatrix} d & -b \\ -c & a \end{pmatrix}$ . By [Definition 0.9.9](#) and commutativity

of multiplication in  $F$ ,

$$\begin{aligned}\mathbf{A}\tilde{\mathbf{A}} &= \begin{pmatrix} ad - bc & -ab + ba \\ cd - dc & -cb + da \end{pmatrix} = \Delta \mathbf{I}_2, \\ \tilde{\mathbf{A}}\mathbf{A} &= \begin{pmatrix} da - bc & db - bd \\ -ca + ac & -cb + ad \end{pmatrix} = \Delta \mathbf{I}_2.\end{aligned}$$

( $\Leftarrow$ ) Suppose  $\Delta \neq 0$ , so  $\Delta^{-1}$  exists in  $F$ . By [Theorem 0.9.4](#) (scalars),  $\mathbf{A}(\Delta^{-1}\tilde{\mathbf{A}}) = \Delta^{-1}(\mathbf{A}\tilde{\mathbf{A}}) = \Delta^{-1}\Delta\mathbf{I}_2 = \mathbf{I}_2$ , and likewise  $(\Delta^{-1}\tilde{\mathbf{A}})\mathbf{A} = \mathbf{I}_2$ . Hence  $\mathbf{A}$  is invertible with  $\mathbf{A}^{-1} = \Delta^{-1}\tilde{\mathbf{A}}$ , which is the stated formula.

( $\Rightarrow$ ) We prove the contrapositive. Suppose  $\Delta = 0$ . We first find  $\mathbf{x} \in F^2$  with  $\mathbf{x} \neq \mathbf{0}$  and  $\mathbf{A}\mathbf{x} = \mathbf{0}$ .

*Case 1.*  $\mathbf{A} = \mathbf{0}$ . Take  $\mathbf{x} = \mathbf{e}_1$ .

*Case 2.*  $c \neq 0$  or  $d \neq 0$ . Take  $\mathbf{x} = (d, -c) \neq \mathbf{0}$ . Then  $\mathbf{A}\mathbf{x} = (ad - bc, cd - dc) = (\Delta, 0) = \mathbf{0}$ .

*Case 3.*  $c = d = 0$  but  $a \neq 0$  or  $b \neq 0$ . Take  $\mathbf{x} = (b, -a) \neq \mathbf{0}$ . Then  $\mathbf{A}\mathbf{x} = (ab - ba, cb - da) = (0, 0) = \mathbf{0}$ , since  $c = d = 0$ .

These cases cover every  $\mathbf{A}$ . Now suppose, for a contradiction, that  $\mathbf{A}$  is invertible. Then by [Theorem 0.9.4](#),

$$\mathbf{x} = \mathbf{I}_2\mathbf{x} = (\mathbf{A}^{-1}\mathbf{A})\mathbf{x} = \mathbf{A}^{-1}(\mathbf{A}\mathbf{x}) = \mathbf{A}^{-1}\mathbf{0} = \mathbf{0},$$

contradicting  $\mathbf{x} \neq \mathbf{0}$ . Hence  $\mathbf{A}$  is not invertible. This proves the theorem. ■

The number  $ad - bc$  will return in Chapter 6 as the determinant of  $\mathbf{A}$ . Here it arose from a single multiplication, and the proof used nothing beyond the field axioms, so the theorem holds over every field. For instance,  $\begin{pmatrix} 2 & 1 \\ 5 & 3 \end{pmatrix}$  has  $ad - bc = 6 - 5 = 1$ , which explains the inverse found in [Example \(Some invertible matrices\)](#); and  $\begin{pmatrix} 1 & 2 \\ 2 & 4 \end{pmatrix}$  has  $ad - bc = 0$ , so it is singular, with  $\mathbf{A}(2, -1) = \mathbf{0}$ .

## 0.9.10 Powers and polynomials of a matrix

A square matrix can be multiplied by itself, so we can form powers, and then plug the matrix into a polynomial ([Definition 0.8.1](#)). This is how polynomials act on matrices throughout the book.

### Definition 0.9.14 — Powers and polynomials of a matrix

Let  $\mathbf{A} \in M_n(F)$ . The **powers** of  $\mathbf{A}$  are defined recursively by

$$\mathbf{A}^0 = \mathbf{I}_n, \quad \mathbf{A}^{k+1} = \mathbf{A}^k \mathbf{A} \quad (k \in \mathbb{N}).$$



For a polynomial  $p = a_0 + a_1x + \cdots + a_Nx^N \in F[x]$ , the matrix  $p(\mathbf{A})$  is

$$p(\mathbf{A}) = a_0\mathbf{I}_n + a_1\mathbf{A} + a_2\mathbf{A}^2 + \cdots + a_N\mathbf{A}^N \in M_n(F).$$

Two points in this definition deserve attention. The **constant term**  $a_0$  becomes  $a_0\mathbf{I}_n$ , because  $a_0$  alone is a number and cannot be added to matrices; this matches  $x^0 = 1$  with  $\mathbf{A}^0 = \mathbf{I}_n$ . And  $p(\mathbf{A})$  is well defined, because enlarging  $N$  only adds terms  $0 \cdot \mathbf{A}^k = 0$ , exactly as for evaluation at a number (Definition 0.8.6).

The next theorem says that plugging in a matrix respects sums and products, just as plugging in a number does (Theorem 0.8.5).

### Theorem 0.9.9 — Polynomials in one matrix

Let  $\mathbf{A} \in M_n(F)$  and  $p, q \in F[x]$ . Then:

1.  $\mathbf{A}^j \mathbf{A}^k = \mathbf{A}^{j+k}$  for all  $j, k \in \mathbb{N}$ ;
2.  $(p + q)(\mathbf{A}) = p(\mathbf{A}) + q(\mathbf{A})$  and  $(pq)(\mathbf{A}) = p(\mathbf{A})q(\mathbf{A})$ ;
3.  $p(\mathbf{A})q(\mathbf{A}) = q(\mathbf{A})p(\mathbf{A})$ .

*Idea.* In the proof of Theorem 0.8.5, the only delicate step moved  $c^i$  past a coefficient  $b_j$ . Here the coefficients are scalars, and scalars can be moved freely through matrix products. The other step,  $c^i c^j = c^{i+j}$ , becomes part 1, which is associativity plus induction. Part 3 is then free, because  $pq = qp$  in  $F[x]$ .

*Proof.* For 1, fix  $j$  and use induction on  $k$  (Theorem 0.2.7). For  $k = 0$ ,  $\mathbf{A}^j \mathbf{A}^0 = \mathbf{A}^j \mathbf{I}_n = \mathbf{A}^j$ . If  $\mathbf{A}^j \mathbf{A}^k = \mathbf{A}^{j+k}$ , then by Definition 0.9.14 and associativity (Theorem 0.9.4),

$$\mathbf{A}^j \mathbf{A}^{k+1} = \mathbf{A}^j (\mathbf{A}^k \mathbf{A}) = (\mathbf{A}^j \mathbf{A}^k) \mathbf{A} = \mathbf{A}^{j+k} \mathbf{A} = \mathbf{A}^{j+k+1}.$$

For 2, let  $p = \sum_{i=0}^m a_i x^i$  and  $q = \sum_{j=0}^l b_j x^j$ . The statement for sums follows from Theorem 0.9.1, since  $(a_i + b_i)\mathbf{A}^i = a_i\mathbf{A}^i + b_i\mathbf{A}^i$ . For products, by Definition 0.8.2 the coefficient of  $x^k$  in  $pq$  is  $\sum_{i+j=k} a_i b_j$ , so

$$\begin{aligned} (pq)(\mathbf{A}) &= \sum_k \left( \sum_{i+j=k} a_i b_j \right) \mathbf{A}^k = \sum_{i=0}^m \sum_{j=0}^l (a_i b_j) \mathbf{A}^i \mathbf{A}^j \\ &= \left( \sum_{i=0}^m a_i \mathbf{A}^i \right) \left( \sum_{j=0}^l b_j \mathbf{A}^j \right) = p(\mathbf{A}) q(\mathbf{A}), \end{aligned}$$

where the second equality uses part 1 and Theorem 0.9.1, and the third uses distributivity and the scalar rule  $(a_i \mathbf{A}^i)(b_j \mathbf{A}^j) = (a_i b_j) \mathbf{A}^i \mathbf{A}^j$  from Theorem 0.9.4.

For 3,  $pq = qp$  in  $F[x]$  by Theorem 0.8.1. Hence by part 2,  $p(\mathbf{A})q(\mathbf{A}) = (pq)(\mathbf{A}) = (qp)(\mathbf{A}) = q(\mathbf{A})p(\mathbf{A})$ . This proves the theorem. ■

**Example. (Plugging a matrix into a polynomial)**

Let  $\mathbf{A} = \begin{pmatrix} 1 & 2 \\ 0 & 3 \end{pmatrix} \in M_2(\mathbb{R})$  and  $p = x^2 - 4x + 3$ . Compute  $p(\mathbf{A})$  directly, and again using the factorization  $p = (x - 1)(x - 3)$ .

*Solution.* First  $\mathbf{A}^2 = \begin{pmatrix} 1 & 2 \\ 0 & 3 \end{pmatrix} \begin{pmatrix} 1 & 2 \\ 0 & 3 \end{pmatrix} = \begin{pmatrix} 1 & 8 \\ 0 & 9 \end{pmatrix}$ . Hence

$$p(\mathbf{A}) = \mathbf{A}^2 - 4\mathbf{A} + 3\mathbf{I}_2 = \begin{pmatrix} 1 & 8 \\ 0 & 9 \end{pmatrix} - \begin{pmatrix} 4 & 8 \\ 0 & 12 \end{pmatrix} + \begin{pmatrix} 3 & 0 \\ 0 & 3 \end{pmatrix} = \begin{pmatrix} 0 & 0 \\ 0 & 0 \end{pmatrix}.$$

The constant term contributed  $3\mathbf{I}_2$ , not “3”. By [Theorem 0.9.9](#),  $p(\mathbf{A}) = (\mathbf{A} - \mathbf{I}_2)(\mathbf{A} - 3\mathbf{I}_2)$ , and indeed

$$(\mathbf{A} - \mathbf{I}_2)(\mathbf{A} - 3\mathbf{I}_2) = \begin{pmatrix} 0 & 2 \\ 0 & 2 \end{pmatrix} \begin{pmatrix} -2 & 2 \\ 0 & 0 \end{pmatrix} = \begin{pmatrix} 0 & 0 \\ 0 & 0 \end{pmatrix}.$$

Note that neither factor is zero; this is the second warning of the multiplication part in action. That the diagonal entries 1, 3 of  $\mathbf{A}$  are the roots of  $p$  is no accident, as we will see with eigenvalues.

[Theorem 0.9.9](#) is about polynomials in **one** matrix. Identities with **two** different matrices are another matter.

*Warning. School identities fail for matrices that do not commute.* Expanding with distributivity,  $(\mathbf{A} + \mathbf{B})^2 = \mathbf{A}^2 + \mathbf{AB} + \mathbf{BA} + \mathbf{B}^2$ , and this equals  $\mathbf{A}^2 + 2\mathbf{AB} + \mathbf{B}^2$  only when  $\mathbf{AB} = \mathbf{BA}$ . With  $\mathbf{A} = \begin{pmatrix} 0 & 1 \\ 0 & 0 \end{pmatrix}$  and  $\mathbf{B} = \begin{pmatrix} 0 & 0 \\ 1 & 0 \end{pmatrix}$ :  $(\mathbf{A} + \mathbf{B})^2 = \mathbf{I}_2$ , but  $\mathbf{A}^2 + 2\mathbf{AB} + \mathbf{B}^2 = \begin{pmatrix} 2 & 0 \\ 0 & 0 \end{pmatrix}$ . Before using a polynomial identity valid for numbers, check that all the matrices involved commute; polynomials in a single  $\mathbf{A}$  always do.

## 0.9.11 Two equivalence relations on matrices

We close with two relations on matrices that the rest of the book studies in depth. Both are equivalence relations ([Definition 0.5.2](#)), and the proofs are short exercises in using [Theorem 0.9.7](#).

**Example. (Similarity)**

For  $\mathbf{A}, \mathbf{B} \in M_n(F)$ , say  $\mathbf{A}$  is **similar** to  $\mathbf{B}$ , written  $\mathbf{A} \sim \mathbf{B}$ , if there is an **invertible**  $\mathbf{P} \in M_n(F)$  with  $\mathbf{B} = \mathbf{P}^{-1}\mathbf{A}\mathbf{P}$ . Prove that similarity is an equivalence relation on  $M_n(F)$ . Then check that  $\mathbf{A} = \begin{pmatrix} 1 & 2 \\ 0 & 3 \end{pmatrix}$  is similar to  $\text{diag}(1, 3)$  using  $\mathbf{P} = \begin{pmatrix} 1 & 1 \\ 0 & 1 \end{pmatrix}$ , and show that  $\text{diag}(1, 2)$  is **not** similar to  $\mathbf{I}_2$  over  $\mathbb{R}$ .

*Solution.* We check the three conditions of [Definition 0.5.2](#).

*Reflexive.* Let  $\mathbf{A} \in M_n(F)$ . The identity  $\mathbf{I}_n$  is invertible with  $\mathbf{I}_n^{-1} = \mathbf{I}_n$  ([Example \(Some invertible matrices\)](#)), and  $\mathbf{I}_n^{-1}\mathbf{A}\mathbf{I}_n = \mathbf{A}$ . Hence  $\mathbf{A} \sim \mathbf{A}$ .

*Symmetric.* Suppose  $\mathbf{A} \sim \mathbf{B}$ , so  $\mathbf{B} = \mathbf{P}^{-1}\mathbf{A}\mathbf{P}$  with  $\mathbf{P}$  invertible. Multiplying on the left

by  $\mathbf{P}$  and on the right by  $\mathbf{P}^{-1}$ , and using associativity,  $\mathbf{PBP}^{-1} = (\mathbf{PP}^{-1})\mathbf{A}(\mathbf{PP}^{-1}) = \mathbf{A}$ . By [Theorem 0.9.7](#) (part 2),  $\mathbf{Q} = \mathbf{P}^{-1}$  is invertible with  $\mathbf{Q}^{-1} = \mathbf{P}$ , so  $\mathbf{A} = \mathbf{Q}^{-1}\mathbf{BQ}$ . Hence  $\mathbf{B} \sim \mathbf{A}$ .

*Transitive.* Suppose  $\mathbf{A} \sim \mathbf{B}$  and  $\mathbf{B} \sim \mathbf{C}$ , so  $\mathbf{B} = \mathbf{P}^{-1}\mathbf{AP}$  and  $\mathbf{C} = \mathbf{Q}^{-1}\mathbf{BQ}$  with  $\mathbf{P}, \mathbf{Q}$  invertible. Then

$$\mathbf{C} = \mathbf{Q}^{-1}(\mathbf{P}^{-1}\mathbf{AP})\mathbf{Q} = (\mathbf{Q}^{-1}\mathbf{P}^{-1})\mathbf{A}(\mathbf{PQ}) = (\mathbf{PQ})^{-1}\mathbf{A}(\mathbf{PQ}),$$

where the last step is [Theorem 0.9.7](#) (part 3), which also says  $\mathbf{PQ}$  is invertible. Hence  $\mathbf{A} \sim \mathbf{C}$ . This shows that similarity is an equivalence relation.

For the numerical part,  $\mathbf{P}^{-1} = \begin{pmatrix} 1 & -1 \\ 0 & 1 \end{pmatrix}$  by [Theorem 0.9.8](#) (here  $ad - bc = 1$ ), and

$$\mathbf{P}^{-1}\mathbf{AP} = \begin{pmatrix} 1 & -1 \\ 0 & 1 \end{pmatrix} \begin{pmatrix} 1 & 2 \\ 0 & 3 \end{pmatrix} \begin{pmatrix} 1 & 1 \\ 0 & 1 \end{pmatrix} = \begin{pmatrix} 1 & -1 \\ 0 & 3 \end{pmatrix} \begin{pmatrix} 1 & 1 \\ 0 & 1 \end{pmatrix} = \begin{pmatrix} 1 & 0 \\ 0 & 3 \end{pmatrix}.$$

Finally, if  $\mathbf{P}^{-1}\mathbf{I}_2\mathbf{P} = \text{diag}(1, 2)$  for some invertible  $\mathbf{P}$ , then  $\text{diag}(1, 2) = \mathbf{P}^{-1}\mathbf{P} = \mathbf{I}_2$ , which is false. (Alternatively: similar matrices have equal traces by [Theorem 0.9.6](#), and  $\text{tr } \text{diag}(1, 2) = 3 \neq 2 = \text{tr } \mathbf{I}_2$ .) Hence  $\text{diag}(1, 2) \not\sim \mathbf{I}_2$ .

In Chapter 3 we will see that similar matrices describe the same linear map in two different coordinate systems. That is why a quantity like the trace, which similar matrices share, is really a property of the map and not of the matrix. Deciding when two matrices are similar is one of the main problems of this book.

### Example. (Row equivalence)

For  $\mathbf{A}, \mathbf{B} \in M_{m \times n}(F)$ , say  $\mathbf{A}$  is **row equivalent** to  $\mathbf{B}$  if there is an **invertible**  $\mathbf{E} \in M_m(F)$  with  $\mathbf{B} = \mathbf{EA}$ . Prove that row equivalence is an equivalence relation on  $M_{m \times n}(F)$ . Then, with  $\mathbf{E} = \begin{pmatrix} 1 & 0 \\ -2 & 1 \end{pmatrix}$ , compute  $\mathbf{EA}$  for  $\mathbf{A} = \begin{pmatrix} 1 & 2 & 1 \\ 2 & 5 & 0 \end{pmatrix}$  and describe what  $\mathbf{E}$  did to the rows of  $\mathbf{A}$ .

*Solution. Reflexive.*  $\mathbf{A} = \mathbf{I}_m\mathbf{A}$  and  $\mathbf{I}_m$  is invertible, so  $\mathbf{A}$  is row equivalent to itself.

*Symmetric.* Suppose  $\mathbf{B} = \mathbf{EA}$  with  $\mathbf{E}$  invertible. Multiplying on the left by  $\mathbf{E}^{-1}$ ,  $\mathbf{E}^{-1}\mathbf{B} = (\mathbf{E}^{-1}\mathbf{E})\mathbf{A} = \mathbf{A}$ , and  $\mathbf{E}^{-1}$  is invertible by [Theorem 0.9.7](#) (part 2). So  $\mathbf{B}$  is row equivalent to  $\mathbf{A}$ .

*Transitive.* Suppose  $\mathbf{B} = \mathbf{EA}$  and  $\mathbf{C} = \mathbf{E}'\mathbf{B}$  with  $\mathbf{E}, \mathbf{E}'$  invertible. Then  $\mathbf{C} = \mathbf{E}'(\mathbf{EA}) = (\mathbf{E}'\mathbf{E})\mathbf{A}$  by associativity, and  $\mathbf{E}'\mathbf{E}$  is invertible by [Theorem 0.9.7](#) (part 3). So  $\mathbf{A}$  is row equivalent to  $\mathbf{C}$ . This shows that row equivalence is an equivalence relation.

For the computation,  $\mathbf{E}$  is invertible, with inverse  $\begin{pmatrix} 1 & 0 \\ 2 & 1 \end{pmatrix}$  (check both products, or use [Theorem 0.9.8](#) with  $ad - bc = 1$ ). By the row view of [Theorem 0.9.2](#) and [Theorem 0.9.3](#), row 1 of  $\mathbf{EA}$  is  $1 \cdot (\text{row 1 of } \mathbf{A})$  and row 2 is  $-2 \cdot (\text{row 1}) + 1 \cdot (\text{row 2})$ :

$$\mathbf{EA} = \begin{pmatrix} 1 & 2 & 1 \\ 2-2 & 5-4 & 0-2 \end{pmatrix} = \begin{pmatrix} 1 & 2 & 1 \\ 0 & 1 & -2 \end{pmatrix}.$$

So  $\mathbf{E}$  subtracted twice row 1 from row 2, which is the first step of Gaussian elimination

on  $\mathbf{A}$ .

In Chapter 2 we will see that every elementary row operation is multiplication on the left by an invertible matrix, and that row equivalence is exactly “reachable by a sequence of row operations”. This is why Gaussian elimination never changes the solutions of a linear system.

## 0.9.12 Exercises

### 0.9.12.1 A. Check your understanding

#### Exercise A1.

- Let  $\mathbf{A} \in M_{2 \times 3}(F)$  and  $\mathbf{B} \in M_{3 \times 4}(F)$ . Which of  $\mathbf{AB}$ ,  $\mathbf{BA}$ ,  $\mathbf{A}^\top \mathbf{A}$  and  $\mathbf{B}^\top \mathbf{A}^\top$  are defined? Give the size of each one that is.
- Define what it means for  $\mathbf{A} \in M_n(F)$  to be invertible.
- Determine whether the following statement is true: “if  $\mathbf{A}, \mathbf{B} \in M_n(F)$  and  $\mathbf{AB} = 0$ , then  $\mathbf{A} = 0$  or  $\mathbf{B} = 0$ .” Justify your answer.
- Determine whether the following statement is true: “ $(\mathbf{A} + \mathbf{B})(\mathbf{A} - \mathbf{B}) = \mathbf{A}^2 - \mathbf{B}^2$  for all  $\mathbf{A}, \mathbf{B} \in M_2(\mathbb{R})$ .” Justify your answer.
- Let  $\mathbf{A} \in M_3(F)$  have columns  $\mathbf{a}_1, \mathbf{a}_2, \mathbf{a}_3$ . Express  $\mathbf{A}(2, 0, -1)$  and the second column of  $\mathbf{A}^2$  in terms of the columns of  $\mathbf{A}$  and  $\mathbf{A}$  itself.

*Solution.*

- $\mathbf{AB}$  is  $(2 \times 3)(3 \times 4)$ , defined, of size  $2 \times 4$ .  $\mathbf{BA}$  would pair rows of length 4 with columns of length 2, so it is not defined.  $\mathbf{A}^\top \mathbf{A}$  is  $(3 \times 2)(2 \times 3)$ , defined, of size  $3 \times 3$ .  $\mathbf{B}^\top \mathbf{A}^\top$  is  $(4 \times 3)(3 \times 2)$ , defined, of size  $4 \times 2$ ; it equals  $(\mathbf{AB})^\top$  by [Theorem 0.9.5](#).
- $\mathbf{A}$  is invertible if there exists  $\mathbf{B} \in M_n(F)$  with  $\mathbf{AB} = \mathbf{I}_n$  and  $\mathbf{BA} = \mathbf{I}_n$ .
- False.  $\begin{pmatrix} 1 & 1 \\ 1 & 1 \end{pmatrix} \begin{pmatrix} 1 & -1 \\ -1 & 1 \end{pmatrix} = 0$  with both factors non-zero.
- False. Expanding gives  $(\mathbf{A} + \mathbf{B})(\mathbf{A} - \mathbf{B}) = \mathbf{A}^2 - \mathbf{AB} + \mathbf{BA} - \mathbf{B}^2$ , which differs from  $\mathbf{A}^2 - \mathbf{B}^2$  unless  $\mathbf{AB} = \mathbf{BA}$ . For  $\mathbf{A} = \begin{pmatrix} 0 & 1 \\ 0 & 0 \end{pmatrix}$  and  $\mathbf{B} = \begin{pmatrix} 0 & 0 \\ 1 & 0 \end{pmatrix}$ , we have  $\mathbf{A}^2 = \mathbf{B}^2 = 0$ , but  $(\mathbf{A} + \mathbf{B})(\mathbf{A} - \mathbf{B}) = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix} \begin{pmatrix} 0 & 1 \\ -1 & 0 \end{pmatrix} = \begin{pmatrix} -1 & 0 \\ 0 & 1 \end{pmatrix} \neq 0$ .
- By [Theorem 0.9.3](#),  $\mathbf{A}(2, 0, -1) = 2\mathbf{a}_1 - \mathbf{a}_3$ . By [Theorem 0.9.2](#), the second column of  $\mathbf{A}^2 = \mathbf{AA}$  is  $\mathbf{Aa}_2$ .

### 0.9.12.2 B. Practice

#### Exercise B1: Products and sizes.

Let

$$\mathbf{A} = \begin{pmatrix} 1 & -1 & 2 \\ 0 & 3 & 1 \end{pmatrix}, \quad \mathbf{B} = \begin{pmatrix} 2 & 1 \\ 1 & 0 \\ 0 & 4 \end{pmatrix}, \quad \mathbf{C} = \begin{pmatrix} 1 & 2 \\ 3 & 4 \end{pmatrix}$$

over  $\mathbb{R}$ .

- (a) Compute  $\mathbf{AB}$  and  $\mathbf{BA}$ .
- (b) Determine which of  $\mathbf{AC}$  and  $\mathbf{CA}$  is defined, and compute it.
- (c) Compute  $\mathbf{Ax}$  for  $\mathbf{x} = (2, 0, -1)$  as a combination of the columns of  $\mathbf{A}$ .

*Solution.*

- (a)  $\mathbf{AB}$  is  $2 \times 2$  and  $\mathbf{BA}$  is  $3 \times 3$ . Row against column,

$$\mathbf{AB} = \begin{pmatrix} 2 - 1 + 0 & 1 + 0 + 8 \\ 0 + 3 + 0 & 0 + 0 + 4 \end{pmatrix} = \begin{pmatrix} 1 & 9 \\ 3 & 4 \end{pmatrix}, \quad \mathbf{BA} = \begin{pmatrix} 2 & 1 & 5 \\ 1 & -1 & 2 \\ 0 & 12 & 4 \end{pmatrix}.$$

For instance  $(\mathbf{BA})_{12} = 2 \cdot (-1) + 1 \cdot 3 = 1$  and  $(\mathbf{BA})_{32} = 0 \cdot (-1) + 4 \cdot 3 = 12$ .

- (b)  $\mathbf{AC}$  would be  $(2 \times 3)(2 \times 2)$ , and  $3 \neq 2$ , so it is not defined.  $\mathbf{CA}$  is  $(2 \times 2)(2 \times 3)$ , of size  $2 \times 3$ :

$$\mathbf{CA} = \begin{pmatrix} 1 & -1 + 6 & 2 + 2 \\ 3 & -3 + 12 & 6 + 4 \end{pmatrix} = \begin{pmatrix} 1 & 5 & 4 \\ 3 & 9 & 10 \end{pmatrix}.$$

- (c) By [Theorem 0.9.3](#),  $\mathbf{Ax} = 2\mathbf{a}_1 + 0\mathbf{a}_2 - \mathbf{a}_3 = 2(1, 0) - (2, 1) = (0, -1)$ .

### Exercise B2: Traces of products.

- (a) With  $\mathbf{A}$  and  $\mathbf{B}$  as in [Exercise \(B1: Products and sizes\)](#), verify that  $\text{tr}(\mathbf{AB}) = \text{tr}(\mathbf{BA})$ , although  $\mathbf{AB}$  and  $\mathbf{BA}$  have different sizes.
- (b) Let  $\mathbf{A} = \text{diag}(1, 2)$ ,  $\mathbf{B} = \begin{pmatrix} 0 & 1 \\ 0 & 0 \end{pmatrix}$  and  $\mathbf{C} = \begin{pmatrix} 0 & 0 \\ 1 & 0 \end{pmatrix}$ . Show that  $\text{tr}(\mathbf{ABC}) \neq \text{tr}(\mathbf{BAC})$ , and verify that  $\text{tr}(\mathbf{ABC}) = \text{tr}(\mathbf{CAB})$ .

*Solution.*

- (a) From [Exercise \(B1: Products and sizes\)](#),  $\text{tr}(\mathbf{AB}) = 1 + 4 = 5$  and  $\text{tr}(\mathbf{BA}) = 2 + (-1) + 4 = 5$ , as [Theorem 0.9.6](#) predicts.

- (b) By the column view,  $\mathbf{BC}$  has columns  $\mathbf{Be}_2 = \mathbf{e}_1$  and  $\mathbf{B}\mathbf{0} = \mathbf{0}$ , so  $\mathbf{BC} = \begin{pmatrix} 1 & 0 \\ 0 & 0 \end{pmatrix}$ , and  $\mathbf{ABC} = \mathbf{A}(\mathbf{BC}) = \begin{pmatrix} 1 & 0 \\ 0 & 0 \end{pmatrix}$ , with trace 1. Next,  $\mathbf{BA} = \begin{pmatrix} 0 & 2 \\ 0 & 0 \end{pmatrix}$ , so  $\mathbf{BAC} = \begin{pmatrix} 2 & 0 \\ 0 & 0 \end{pmatrix}$ , with trace 2. Hence  $\text{tr}(\mathbf{ABC}) = 1 \neq 2 = \text{tr}(\mathbf{BAC})$ . Finally  $\mathbf{AB} = \begin{pmatrix} 0 & 1 \\ 0 & 0 \end{pmatrix}$  and  $\mathbf{CAB} = \begin{pmatrix} 0 & 0 \\ 0 & 1 \end{pmatrix}$ , with trace  $1 = \text{tr}(\mathbf{ABC})$ , as the cyclic rule predicts.

### Exercise B3: Inverses of $2 \times 2$ matrices.

- (a) Find the inverse of  $\begin{pmatrix} 3 & 4 \\ 2 & 3 \end{pmatrix} \in M_2(\mathbb{R})$ , and check both products.
- (b) Show that  $\mathbf{A} = \begin{pmatrix} 2 & 6 \\ 1 & 3 \end{pmatrix}$  is not invertible by finding a non-zero  $\mathbf{x}$  with  $\mathbf{Ax} = \mathbf{0}$ .

- (c) Find the inverse of  $\begin{pmatrix} 1 & 2 \\ 3 & 4 \end{pmatrix} \in M_2(\mathbb{F}_5)$ .
- (d) For which  $t \in \mathbb{R}$  is  $\begin{pmatrix} t & 2 \\ 2 & t \end{pmatrix}$  invertible?

*Solution.*

- (a) Here  $ad - bc = 9 - 8 = 1 \neq 0$ , so by [Theorem 0.9.8](#) the inverse is  $\mathbf{B} = \begin{pmatrix} 3 & -4 \\ -2 & 3 \end{pmatrix}$ .  
 Check:  $\begin{pmatrix} 3 & 4 \\ 2 & 3 \end{pmatrix} \mathbf{B} = \begin{pmatrix} 9-8 & -12+12 \\ 6-6 & -8+9 \end{pmatrix} = \mathbf{I}_2$  and  $\mathbf{B} \begin{pmatrix} 3 & 4 \\ 2 & 3 \end{pmatrix} = \begin{pmatrix} 9-8 & 12-12 \\ -6+6 & -8+9 \end{pmatrix} = \mathbf{I}_2$ .
- (b) Here  $ad - bc = 6 - 6 = 0$ . Following the proof of [Theorem 0.9.8](#), take  $\mathbf{x} = (d, -c) = (3, -1) \neq \mathbf{0}$ . Then  $\mathbf{Ax} = (6-6, 3-3) = \mathbf{0}$ . If  $\mathbf{A}$  were invertible, then  $\mathbf{x} = \mathbf{A}^{-1}(\mathbf{Ax}) = \mathbf{0}$ , a contradiction. Hence  $\mathbf{A}$  is not invertible.
- (c) In  $\mathbb{F}_5$ ,  $ad - bc = 4 - 6 = -2 = 3 \neq 0$ , and  $3^{-1} = 2$  since  $3 \cdot 2 = 6 = 1$ .  
 By [Theorem 0.9.8](#) the inverse is  $2 \begin{pmatrix} 4 & -2 \\ -3 & 1 \end{pmatrix} = \begin{pmatrix} 8 & -4 \\ -6 & 2 \end{pmatrix} = \begin{pmatrix} 3 & 1 \\ 4 & 2 \end{pmatrix}$ . Check:  
 $\begin{pmatrix} 1 & 2 \\ 3 & 4 \end{pmatrix} \begin{pmatrix} 3 & 1 \\ 4 & 2 \end{pmatrix} = \begin{pmatrix} 11 & 5 \\ 25 & 11 \end{pmatrix} = \mathbf{I}_2$  and  $\begin{pmatrix} 3 & 1 \\ 4 & 2 \end{pmatrix} \begin{pmatrix} 1 & 2 \\ 3 & 4 \end{pmatrix} = \begin{pmatrix} 6 & 10 \\ 10 & 16 \end{pmatrix} = \mathbf{I}_2$  in  $\mathbb{F}_5$ .
- (d) By [Theorem 0.9.8](#), it is invertible if and only if  $t^2 - 4 \neq 0$ , that is,  $t \neq 2$  and  $t \neq -2$ .

#### Exercise B4: Transposes of products.

- (a) Prove that  $(\mathbf{A}_1 \mathbf{A}_2 \cdots \mathbf{A}_k)^\top = \mathbf{A}_k^\top \cdots \mathbf{A}_2^\top \mathbf{A}_1^\top$  for all  $k \geq 1$  and matrices  $\mathbf{A}_1, \dots, \mathbf{A}_k$  whose sizes allow the product.
- (b) Let  $\mathbf{A} \in M_n(F)$ . Prove that  $\mathbf{A} + \mathbf{A}^\top$  is symmetric and  $\mathbf{A} - \mathbf{A}^\top$  is skew-symmetric.

*Solution.*

- (a) We use induction on  $k$  ([Theorem 0.2.7](#)). For  $k = 1$  there is nothing to prove. Suppose the formula holds for products of  $k$  matrices, and let  $\mathbf{A}_1, \dots, \mathbf{A}_{k+1}$  be given. By associativity,  $\mathbf{A}_1 \cdots \mathbf{A}_{k+1} = (\mathbf{A}_1 \cdots \mathbf{A}_k) \mathbf{A}_{k+1}$ . By [Theorem 0.9.5](#) (part 4) and then the induction hypothesis,

$$(\mathbf{A}_1 \cdots \mathbf{A}_{k+1})^\top = \mathbf{A}_{k+1}^\top (\mathbf{A}_1 \cdots \mathbf{A}_k)^\top = \mathbf{A}_{k+1}^\top \mathbf{A}_k^\top \cdots \mathbf{A}_1^\top.$$

This proves the formula for all  $k \geq 1$ .

- (b) By [Theorem 0.9.5](#) (parts 1–3),  $(\mathbf{A} + \mathbf{A}^\top)^\top = \mathbf{A}^\top + (\mathbf{A}^\top)^\top = \mathbf{A}^\top + \mathbf{A} = \mathbf{A} + \mathbf{A}^\top$ , so  $\mathbf{A} + \mathbf{A}^\top$  is symmetric. Similarly  $(\mathbf{A} - \mathbf{A}^\top)^\top = \mathbf{A}^\top - \mathbf{A} = -(\mathbf{A} - \mathbf{A}^\top)$ , so  $\mathbf{A} - \mathbf{A}^\top$  is skew-symmetric.

#### 0.9.12.3 C. Going deeper

##### Exercise C1: $\mathbf{AB} - \mathbf{BA} = \mathbf{I}$ is impossible, or is it?.

- (a) Prove that there are no  $\mathbf{A}, \mathbf{B} \in M_n(\mathbb{R})$ , for any  $n \geq 1$ , with  $\mathbf{AB} - \mathbf{BA} = \mathbf{I}_n$ .

- (b) Show that the argument of (a) breaks down in  $M_2(\mathbb{F}_2)$ , and that  $\mathbf{A} = \begin{pmatrix} 0 & 0 \\ 1 & 0 \end{pmatrix}$ ,  $\mathbf{B} = \begin{pmatrix} 0 & 1 \\ 0 & 0 \end{pmatrix}$  satisfy  $\mathbf{AB} - \mathbf{BA} = \mathbf{I}_2$  over  $\mathbb{F}_2$ .

*Hint: consider the trace of both sides.*

*Solution.*

- (a) Suppose, for a contradiction, that  $\mathbf{AB} - \mathbf{BA} = \mathbf{I}_n$ . Taking traces and using [Theorem 0.9.6](#) (parts 1 and 3),

$$n = \text{tr } \mathbf{I}_n = \text{tr}(\mathbf{AB} - \mathbf{BA}) = \text{tr}(\mathbf{AB}) - \text{tr}(\mathbf{BA}) = 0.$$

This contradicts  $n \geq 1$ . Hence no such  $\mathbf{A}, \mathbf{B}$  exist.

- (b) Over  $\mathbb{F}_2$  the same computation gives  $\text{tr } \mathbf{I}_2 = 1 + 1 = 0$ , so taking traces yields  $0 = 0$  and no contradiction. For the given matrices, the column view gives  $\mathbf{AB} = \begin{pmatrix} 0 & 0 \\ 0 & 1 \end{pmatrix}$  and  $\mathbf{BA} = \begin{pmatrix} 1 & 0 \\ 0 & 0 \end{pmatrix}$ , so  $\mathbf{AB} - \mathbf{BA} = \begin{pmatrix} -1 & 0 \\ 0 & 1 \end{pmatrix} = \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix} = \mathbf{I}_2$ , because  $-1 = 1$  in  $\mathbb{F}_2$ .

### Exercise C2: Triangular matrices with non-zero diagonal.

Let  $F$  be a field.

- (a) Let  $\mathbf{U} = \begin{pmatrix} a & b \\ 0 & d \end{pmatrix}$  with  $a, d \neq 0$ . Show that  $\mathbf{U}$  is invertible and that  $\mathbf{U}^{-1}$  is upper triangular.
- (b) Let  $\mathbf{U} = \begin{pmatrix} a & b & c \\ 0 & d & e \\ 0 & 0 & f \end{pmatrix}$  with  $a, d, f \neq 0$ . Show that

$$\mathbf{V} = \begin{pmatrix} a^{-1} & -b(ad)^{-1} & (be - cd)(adf)^{-1} \\ 0 & d^{-1} & -e(df)^{-1} \\ 0 & 0 & f^{-1} \end{pmatrix}$$

is the inverse of  $\mathbf{U}$ .

- (c) Explain how the entries of  $\mathbf{V}$  can be found, column by column, by solving  $\mathbf{U}\mathbf{v}_k = \mathbf{e}_k$ .

*The general case, for all  $n$ , follows from the methods of Chapter 2.*

*Solution.*

- (a) Here  $ad - b \cdot 0 = ad \neq 0$ , since a product of non-zero elements of a field is non-zero ([Theorem 0.7.1](#)). By [Theorem 0.9.8](#),  $\mathbf{U}$  is invertible with  $\mathbf{U}^{-1} = (ad)^{-1} \begin{pmatrix} d & -b \\ 0 & a \end{pmatrix} = \begin{pmatrix} a^{-1} & -b(ad)^{-1} \\ 0 & d^{-1} \end{pmatrix}$ , which is upper triangular.
- (b) The inverses  $a^{-1}, d^{-1}, f^{-1}$  exist, and  $ad, df, adf \neq 0$  by [Theorem 0.7.1](#), so  $\mathbf{V}$  is

defined. We check  $\mathbf{UV} = \mathbf{I}_3$  entry by entry. The diagonal entries are  $a \cdot a^{-1} = 1$ ,  $d \cdot d^{-1} = 1$ ,  $f \cdot f^{-1} = 1$ . Below the diagonal both matrices are zero, so  $(\mathbf{UV})_{21} = (\mathbf{UV})_{31} = (\mathbf{UV})_{32} = 0$ . Above the diagonal,

$$\begin{aligned}(\mathbf{UV})_{12} &= a \cdot (-b(ad)^{-1}) + bd^{-1} = -bd^{-1} + bd^{-1} = 0, \\(\mathbf{UV})_{23} &= d \cdot (-e(df)^{-1}) + ef^{-1} = -ef^{-1} + ef^{-1} = 0, \\(\mathbf{UV})_{13} &= a(be - cd)(adf)^{-1} + b(-e(df)^{-1}) + cf^{-1} \\&= (be - cd)(df)^{-1} - be(df)^{-1} + cd(df)^{-1} = 0.\end{aligned}$$

Hence  $\mathbf{UV} = \mathbf{I}_3$ . For  $\mathbf{VU}$ , again the diagonal entries are 1 and the entries below the diagonal are 0, and

$$\begin{aligned}(\mathbf{VU})_{12} &= a^{-1}b - b(ad)^{-1}d = a^{-1}b - a^{-1}b = 0, \\(\mathbf{VU})_{23} &= d^{-1}e - e(df)^{-1}f = 0, \\(\mathbf{VU})_{13} &= a^{-1}c - b(ad)^{-1}e + (be - cd)(adf)^{-1}f \\&= a^{-1}c - be(ad)^{-1} + be(ad)^{-1} - cd(ad)^{-1} = 0,\end{aligned}$$

using  $cd(ad)^{-1} = ca^{-1}$ . Hence  $\mathbf{VU} = \mathbf{I}_3$ , and  $\mathbf{V} = \mathbf{U}^{-1}$ . Both products had to be checked, since we may not yet use that one implies the other.

- (c) By [Theorem 0.9.2](#),  $\mathbf{UV} = \mathbf{I}_3$  says exactly that the columns  $\mathbf{v}_1, \mathbf{v}_2, \mathbf{v}_3$  of  $\mathbf{V}$  satisfy  $\mathbf{U}\mathbf{v}_k = \mathbf{e}_k$ . Because  $\mathbf{U}$  is upper triangular, each system can be solved from the bottom row up. For example,  $\mathbf{U}\mathbf{v}_3 = \mathbf{e}_3$  reads  $fx_3 = 1$ ,  $dx_2 + ex_3 = 0$ ,  $ax_1 + bx_2 + cx_3 = 0$ , giving  $x_3 = f^{-1}$ , then  $x_2 = -e(df)^{-1}$ , then  $x_1 = (be - cd)(adf)^{-1}$ , which is the third column of  $\mathbf{V}$ . (This only finds a candidate satisfying  $\mathbf{UV} = \mathbf{I}_3$ ; the check  $\mathbf{VU} = \mathbf{I}_3$  in (b) is still needed.)

### Exercise C3: Matrices that commute.

Let  $\mathbf{N} = \begin{pmatrix} 0 & 1 \\ 0 & 0 \end{pmatrix} \in M_2(\mathbb{R})$ .

- Find all  $\mathbf{X} \in M_2(\mathbb{R})$  with  $\mathbf{XN} = \mathbf{NX}$ .
- Deduce that the only matrices  $\mathbf{X} \in M_2(\mathbb{R})$  with  $\mathbf{XY} = \mathbf{YX}$  for every  $\mathbf{Y} \in M_2(\mathbb{R})$  are the scalar multiples  $c\mathbf{I}_2$ ,  $c \in \mathbb{R}$ .

*Solution.*

- Let  $\mathbf{X} = \begin{pmatrix} p & q \\ r & s \end{pmatrix}$ . By the column view,  $\mathbf{XN} = (\mathbf{X}\mathbf{0} \quad \mathbf{X}\mathbf{e}_1) = \begin{pmatrix} 0 & p \\ 0 & r \end{pmatrix}$ , and by the row view,  $\mathbf{NX}$  has rows (row 2 of  $\mathbf{X}$ ) and  $\mathbf{0}$ :  $\mathbf{NX} = \begin{pmatrix} r & s \\ 0 & 0 \end{pmatrix}$ . Comparing entries,  $\mathbf{XN} = \mathbf{NX}$  if and only if  $r = 0$  and  $p = s$ , that is,  $\mathbf{X} = \begin{pmatrix} p & q \\ 0 & p \end{pmatrix} = p\mathbf{I}_2 + q\mathbf{N}$  with  $p, q \in \mathbb{R}$  arbitrary.
- Every  $c\mathbf{I}_2$  commutes with every  $\mathbf{Y}$ , by [Theorem 0.9.4](#):  $(c\mathbf{I}_2)\mathbf{Y} = c\mathbf{Y} = \mathbf{Y}(c\mathbf{I}_2)$ . Conversely, suppose  $\mathbf{XY} = \mathbf{YX}$  for every  $\mathbf{Y}$ . Taking  $\mathbf{Y} = \mathbf{N}$ , part (a) gives  $\mathbf{X} = \begin{pmatrix} p & q \\ 0 & p \end{pmatrix}$ . Taking  $\mathbf{Y} = \mathbf{N}^\top = \begin{pmatrix} 0 & 0 \\ 1 & 0 \end{pmatrix}$ , we compute  $\mathbf{XN}^\top = \begin{pmatrix} q & 0 \\ p & 0 \end{pmatrix}$  and



$\mathbf{N}^\top \mathbf{X} = \begin{pmatrix} 0 & 0 \\ p & q \end{pmatrix}$ . Equality forces  $q = 0$ . Hence  $\mathbf{X} = p\mathbf{I}_2$ , as claimed.

## 0.10 Groups and Permutations

Several structures in this chapter have quietly behaved alike. The invertible  $n \times n$  matrices, the non-zero elements of a field, and the bijections of a set can each be combined, have an element that does nothing, and can be undone. This short section gives that common structure a name, proves its basic rules once, and introduces the permutations that the determinant in Chapter 6 is built from. We keep it brief: groups are a language for us here, not a subject.

### 0.10.1 Groups

Look at four situations side by side. In  $\text{GL}_n(F)$ , the set of invertible matrices in  $M_n(F)$ , we multiply matrices,  $\mathbf{I}_n$  does nothing, and each  $\mathbf{A}$  is undone by  $\mathbf{A}^{-1}$  (Theorem 0.9.7). In a field  $F$ , we multiply non-zero elements, 1 does nothing, and each  $a$  is undone by  $a^{-1}$ . The four rotations of a square about its center (by  $0^\circ, 90^\circ, 180^\circ, 270^\circ$ ) can be performed one after another, the rotation by  $0^\circ$  does nothing, and each rotation is undone by rotating back. The bijections  $X \rightarrow X$  of a set are composed,  $\text{id}_X$  does nothing, and each is undone by its inverse function (Theorem 0.4.7). The objects differ completely, but the rules we use are the same three. So we give the pattern a name.

*A group is a set with one way of combining elements, in which combining is associative, some element does nothing, and every element can be undone.*

#### Definition 0.10.1 — Group

A **group** is a set  $G$  together with a **binary operation**  $G \times G \rightarrow G$ ,  $(a, b) \mapsto ab$ , such that:

- (G1) (**Associativity**)  $(ab)c = a(bc)$  for all  $a, b, c \in G$ ;
- (G2) (**Identity**) there is an element  $e \in G$  with  $ea = ae = a$  for every  $a \in G$ ;
- (G3) (**Inverses**) for every  $a \in G$  there is  $b \in G$  with  $ab = ba = e$ , where  $e$  is the element of (G2).

The group is **abelian** if moreover  $ab = ba$  for all  $a, b \in G$ .

In words: the operation is a function on pairs, so the combination of two elements of  $G$  lies in  $G$  again; this “closure” is built into the phrase “binary operation  $G \times G \rightarrow G$ ”. (G1) lets us drop brackets. (G2) asks for an element that changes nothing, from **both** sides. (G3) asks that each element can be undone, again from both sides, back to that same  $e$ . When the operation is addition we write  $a + b$ , 0 and  $-a$  instead of  $ab$ ,  $e$  and  $a^{-1}$ ; this notation is only used for abelian groups.

**Example. (Groups)**

Check that each of the following is a group, and decide which are abelian:  $(\mathbb{Z}, +)$ ;  $(F, +)$  and  $(F \setminus \{0\}, \cdot)$  for a field  $F$ ;  $\text{GL}_n(F)$  under matrix multiplication; the set  $S_n$  of bijections  $\{1, \dots, n\} \rightarrow \{1, \dots, n\}$  under composition; and a one-element set  $\{e\}$  with  $ee = e$ .

*Solution.*  $(\mathbb{Z}, +)$ : the sum of integers is an integer, addition is associative, 0 is an identity, and  $-a$  undoes  $a$ . It is abelian.  $(F, +)$  is a group for the same reasons, by the field axioms (Definition 0.7.1); it is abelian.

$(F \setminus \{0\}, \cdot)$ : the product of two non-zero elements is non-zero by Theorem 0.7.1, so the operation lands in  $F \setminus \{0\}$ . Associativity and the identity  $1 \neq 0$  come from the field axioms, and each  $a \neq 0$  has an inverse  $a^{-1}$ , which is non-zero since  $aa^{-1} = 1 \neq 0$ . It is abelian.

$\text{GL}_n(F)$ : by Theorem 0.9.7 (part 3), a product of invertible matrices is invertible, so the operation lands in  $\text{GL}_n(F)$ . Associativity holds by Theorem 0.9.4,  $\mathbf{I}_n$  is invertible and is an identity, and  $\mathbf{A}^{-1} \in \text{GL}_n(F)$  undoes  $\mathbf{A}$  by part 2. For  $n \geq 2$  it is **not** abelian: for  $n = 2$ ,  $\begin{pmatrix} 1 & 1 \\ 0 & 1 \end{pmatrix} \begin{pmatrix} 1 & 0 \\ 1 & 1 \end{pmatrix} = \begin{pmatrix} 2 & 1 \\ 1 & 1 \end{pmatrix}$ , while the product in the other order is  $\begin{pmatrix} 1 & 1 \\ 1 & 2 \end{pmatrix}$ . (Over  $\mathbb{F}_2$  read 2 as 0; the two products still differ.) For  $n \geq 3$ , put these blocks in the top-left corner and 1's further down the diagonal.  $\text{GL}_1(F)$  is just  $F \setminus \{0\}$ , and is abelian.

$S_n$ : a composition of bijections is a bijection (Theorem 0.4.3), composition is associative (Theorem 0.4.2), id is an identity, and each bijection has an inverse function (Theorem 0.4.7), which is again a bijection. We will see below that  $S_n$  is not abelian for  $n \geq 3$ .

$\{e\}$ : all three axioms reduce to  $ee = e$ . This **trivial group** is the degenerate case; it reappears as the smallest subgroup of every group.

Now non-examples, each a minimal change of an example. Replace addition on  $\mathbb{Z}$  by multiplication. The operation still lands in  $\mathbb{Z}$ , it is associative, and 1 is an identity. But (G3) fails: there is no integer  $b$  with  $2b = 1$ . So  $(\mathbb{Z}, \cdot)$  is **not** a group. Similarly, replace  $\text{GL}_n(F)$  by all of  $M_n(F)$  under multiplication: (G1) and (G2) still hold with  $\mathbf{I}_n$ , but the zero matrix has no inverse, so (G3) fails. Keep  $\mathbb{N}$  under  $+$ : the identity 0 is there, but 1 has no inverse in  $\mathbb{N}$ .

*Warning.* **The operation must land in the set.**  $\text{GL}_n(F)$  under matrix **addition** is not a group:  $\mathbf{I}_n$  and  $-\mathbf{I}_n$  are invertible, but  $\mathbf{I}_n + (-\mathbf{I}_n) = 0$  is not, so addition is not even a binary operation on  $\text{GL}_n(F)$ . Before checking (G1)–(G3), check that combining two elements of the set gives an element of the set.

The rules we proved separately for fields and for matrices follow from the axioms alone. It is worth proving them once:

**Theorem 0.10.1 — Basic properties of groups**

Let  $G$  be a group and  $a, b, c \in G$ .

1. The identity element of  $G$  is unique.
2. The inverse of  $a$  is unique. We denote it by  $a^{-1}$ .

3.  $(a^{-1})^{-1} = a$  and  $(ab)^{-1} = b^{-1}a^{-1}$ .
4. **(Cancellation)** If  $ab = ac$ , then  $b = c$ . If  $ba = ca$ , then  $b = c$ .

*Proof.* For 1, suppose  $e$  and  $e'$  both satisfy (G2). Then  $e = ee'$ , using that  $e'$  is an identity, and  $ee' = e'$ , using that  $e$  is one. Hence  $e = e'$ .

For 2, suppose  $b$  and  $b'$  both satisfy  $ab = ba = e$  and  $ab' = b'a = e$ . By (G2) and (G1),

$$b = be = b(ab') = (ba)b' = eb' = b'.$$

For 3, the equations  $aa^{-1} = a^{-1}a = e$  say that  $a$  is an inverse of  $a^{-1}$ , so  $(a^{-1})^{-1} = a$  by part 2. Next, by (G1),

$$(ab)(b^{-1}a^{-1}) = a(bb^{-1})a^{-1} = aea^{-1} = aa^{-1} = e,$$

and similarly  $(b^{-1}a^{-1})(ab) = b^{-1}(a^{-1}a)b = e$ . By part 2,  $(ab)^{-1} = b^{-1}a^{-1}$ .

For 4, suppose  $ab = ac$ . Multiplying on the left by  $a^{-1}$  and using (G1),  $b = (a^{-1}a)b = a^{-1}(ab) = a^{-1}(ac) = (a^{-1}a)c = c$ . The second statement follows by multiplying on the right by  $a^{-1}$ . This proves the theorem. ■

These are the arguments of [Theorem 0.9.7](#) with matrices erased. That is the point of the definition: a proof that uses only (G1)–(G3) holds in every group at once.

## 0.10.2 Subgroups

After a structure comes the substructure. The even integers sit inside  $(\mathbb{Z}, +)$  and form a group on their own, with the same operation.

### Definition 0.10.2 — Subgroup

Let  $G$  be a group. A subset  $H \subseteq G$  is a **subgroup** of  $G$  if  $e \in H$ , and  $ab \in H$  and  $a^{-1} \in H$  for all  $a, b \in H$ .

Then the operation of  $G$  restricts to a binary operation on  $H$ , and  $H$  is a group with it: (G1) holds in  $H$  because it holds in  $G$ , and (G2), (G3) hold because  $e$  and the inverses lie in  $H$ . In practice one checks a single condition:

### Theorem 0.10.2 — Subgroup test

Let  $G$  be a group and  $H \subseteq G$ . Then  $H$  is a subgroup of  $G$  if and only if  $H \neq \emptyset$  and  $ab^{-1} \in H$  for all  $a, b \in H$ .

*Proof.* ( $\Rightarrow$ ) If  $H$  is a subgroup, then  $e \in H$ , so  $H \neq \emptyset$ . For  $a, b \in H$  we have  $b^{-1} \in H$ , and then  $ab^{-1} \in H$ .

( $\Leftarrow$ ) Suppose  $H \neq \emptyset$  and  $ab^{-1} \in H$  for all  $a, b \in H$ . Pick  $h \in H$ . Taking  $a = b = h$  gives  $e = hh^{-1} \in H$ . For  $b \in H$ , taking  $a = e$  gives  $b^{-1} = eb^{-1} \in H$ . For  $a, b \in H$ , we now know  $b^{-1} \in H$ , so  $ab = a(b^{-1})^{-1} \in H$  by [Theorem 0.10.1](#). Hence  $H$  is a subgroup. This proves the theorem. ■

For example,  $2\mathbb{Z} = \{2k : k \in \mathbb{Z}\}$  is a subgroup of  $(\mathbb{Z}, +)$ . In additive notation the test reads “ $a - b \in H$ ”, and  $2k - 2l = 2(k - l) \in 2\mathbb{Z}$ , while  $0 \in 2\mathbb{Z}$ . By contrast  $\mathbb{N} \subseteq \mathbb{Z}$  is non-empty and closed under  $+$ , but  $0 - 1 = -1 \notin \mathbb{N}$ : the test fails, because  $\mathbb{N}$  lacks inverses. The set of odd integers fails more badly, since it does not contain 0. In  $\text{GL}_n(F)$ , the invertible diagonal matrices form a subgroup:  $\mathbf{I}_n$  is one, and by [Example \(Some invertible matrices\)](#) the inverse of  $\text{diag}(d_1, \dots, d_n)$  is  $\text{diag}(d_1^{-1}, \dots, d_n^{-1})$ , so  $\mathbf{D}\mathbf{E}^{-1}$  is again diagonal with non-zero diagonal entries. Every group  $G$  has the subgroups  $\{e\}$  and  $G$ .

### 0.10.3 Homomorphisms

After objects come the maps between them that respect the structure.

#### Definition 0.10.3 — Group homomorphism

Let  $G$  and  $H$  be groups. A function  $\varphi: G \rightarrow H$  is a **homomorphism** if  $\varphi(ab) = \varphi(a)\varphi(b)$  for all  $a, b \in G$ . A **bijective** homomorphism is an **isomorphism**. The **kernel** of  $\varphi$  is  $\ker \varphi = \{g \in G : \varphi(g) = e_H\}$ , where  $e_H$  is the identity of  $H$ .

The product  $ab$  on the left is taken in  $G$ , and  $\varphi(a)\varphi(b)$  on the right in  $H$ . For example,  $F \setminus \{0\} \rightarrow \text{GL}_2(F)$ ,  $a \mapsto \text{diag}(a, 1)$ , is a homomorphism, since  $\text{diag}(ab, 1) = \text{diag}(a, 1) \text{diag}(b, 1)$ ; its kernel is  $\{1\}$ . The map  $\mathbb{Z} \rightarrow \mathbb{Z}/n\mathbb{Z}$ ,  $a \mapsto [a]$ , is a homomorphism of additive groups, because  $[a + b] = [a] + [b]$  is exactly how addition on  $\mathbb{Z}/n\mathbb{Z}$  was defined ([Example \(Addition modulo  \$n\$  is well defined\)](#)); its kernel is  $n\mathbb{Z}$ . On the other hand,  $x \mapsto x + 1$  from  $(\mathbb{R}, +)$  to itself is not a homomorphism: writing  $\varphi(x) = x + 1$ , we get  $\varphi(0 + 0) = 1$  but  $\varphi(0) + \varphi(0) = 2$ .

#### Theorem 0.10.3 — Basic properties of homomorphisms

Let  $\varphi: G \rightarrow H$  be a homomorphism of groups. Then  $\varphi(e_G) = e_H$ ,  $\varphi(a^{-1}) = \varphi(a)^{-1}$  for every  $a \in G$ , and  $\ker \varphi$  is a subgroup of  $G$ .

*Proof.* Since  $\varphi(e_G)\varphi(e_G) = \varphi(e_G e_G) = \varphi(e_G) = \varphi(e_G)e_H$ , cancellation ([Theorem 0.10.1](#)) gives  $\varphi(e_G) = e_H$ . Next,  $\varphi(a)\varphi(a^{-1}) = \varphi(aa^{-1}) = \varphi(e_G) = e_H$ , and similarly  $\varphi(a^{-1})\varphi(a) = e_H$ , so  $\varphi(a^{-1}) = \varphi(a)^{-1}$  by uniqueness of inverses. Finally,  $e_G \in \ker \varphi$ , and for  $a, b \in \ker \varphi$ ,

$$\varphi(ab^{-1}) = \varphi(a)\varphi(b)^{-1} = e_H e_H^{-1} = e_H,$$

so  $ab^{-1} \in \ker \varphi$ . By [Theorem 0.10.2](#),  $\ker \varphi$  is a subgroup. This proves the theorem. ■

Two homomorphisms will matter later: the determinant  $\text{GL}_n(F) \rightarrow F \setminus \{0\}$  and the sign  $S_n \rightarrow$

$\{1, -1\}$ , both in Chapter 6. Linear maps in Chapter 3 are, among other things, homomorphisms of additive groups, and their kernels will be one of our main tools.

## 0.10.4 Permutations

A **permutation** of  $\{1, \dots, n\}$  is a bijection  $\{1, \dots, n\} \rightarrow \{1, \dots, n\}$ , and  $S_n$  is the group of all of them, the **symmetric group**. It has  $n!$  elements: there are  $n$  choices for  $\sigma(1)$ , then  $n-1$  for  $\sigma(2)$ , and so on.

A permutation  $\sigma$  can be written in **two-line notation**, with  $\sigma(i)$  below  $i$ . A more compact notation lists what happens along a loop. For **distinct**  $i_1, \dots, i_k \in \{1, \dots, n\}$ , the **cycle**  $(i_1 i_2 \cdots i_k)$  is the permutation sending

$$i_1 \mapsto i_2 \mapsto \cdots \mapsto i_k \mapsto i_1$$

and fixing every other element. A cycle of length 2 is a **transposition**:  $(i j)$  swaps  $i$  and  $j$ . The same cycle can be written starting at any of its entries:  $(1 2 3) = (2 3 1) = (3 1 2)$ , but  $(1 3 2)$  is different.

The product  $\sigma\tau$  in  $S_n$  means the composition  $\sigma \circ \tau$ : **first apply  $\tau$ , then  $\sigma$** . So products of cycles are read **from right to left**.

### Example. (Computing with cycles)

In  $S_3$ , compute  $(1 3)(1 2)$  and  $(1 2)(1 3)$ . In  $S_5$ , write  $\sigma = \begin{pmatrix} 1 & 2 & 3 & 4 & 5 \\ 3 & 5 & 1 & 2 & 4 \end{pmatrix}$  (two-line notation) as a product of cycles.

*Solution.* For  $(1 3)(1 2)$ , follow each element through  $(1 2)$  first, then  $(1 3)$ :  $1 \mapsto 2 \mapsto 2$ ,  $2 \mapsto 1 \mapsto 3$ ,  $3 \mapsto 3 \mapsto 1$ . So  $(1 3)(1 2) = (1 2 3)$ . For  $(1 2)(1 3)$ :  $1 \mapsto 3 \mapsto 3$ ,  $3 \mapsto 1 \mapsto 2$ ,  $2 \mapsto 2 \mapsto 1$ . So  $(1 2)(1 3) = (1 3 2)$ . The two products differ, so  $S_3$  is not abelian; the same two transpositions show that  $S_n$  is not abelian for every  $n \geq 3$ .

For  $\sigma$ , start at 1 and follow:  $1 \mapsto 3 \mapsto 1$ , a loop  $(1 3)$ . The smallest element not yet used is 2:  $2 \mapsto 5 \mapsto 4 \mapsto 2$ , a loop  $(2 5 4)$ . Every element is now used, so  $\sigma = (1 3)(2 5 4)$ . These two cycles move disjoint sets of elements, so they can be applied in either order.

*Warning. Mind the order.*  $\sigma\tau$  applies  $\tau$  first, exactly as for functions, and as for matrices:  $(\mathbf{AB})\mathbf{x} = \mathbf{A}(\mathbf{Bx})$  applies  $\mathbf{B}$  first. Reading a product of cycles from left to right gives the wrong answer, as the two different products  $(1 3)(1 2) \neq (1 2)(1 3)$  above show.

*Quick check.* Compute  $(1 2 3)(1 2)$  in  $S_3$ .

*Solution.* Apply  $(1 2)$  first:  $1 \mapsto 2 \mapsto 3$ ,  $2 \mapsto 1 \mapsto 2$ ,  $3 \mapsto 3 \mapsto 1$ . So 1 and 3 are swapped and 2 is fixed:  $(1 2 3)(1 2) = (1 3)$ .

The example showed  $(1 2 3) = (1 3)(1 2)$ : a 3-cycle is a product of transpositions. Every permutation is, and the proof is a model induction on  $n$ .

**Theorem 0.10.4 — Transpositions generate  $S_n$** 

Let  $n \geq 2$ . Every permutation in  $S_n$  is a product of transpositions.

*Idea.* Induction on  $n$ : what is the smaller object inside a permutation of  $n+1$  elements? If  $\sigma$  fixes  $n+1$ , it is really a permutation of  $\{1, \dots, n\}$ . If not, one transposition repairs that: swap  $\sigma(n+1)$  back to  $n+1$ . The result fixes  $n+1$ , so the induction hypothesis applies, and we undo the swap.

*Proof.* We use induction on  $n \geq 2$  (Theorem 0.2.7). For  $n = 2$ ,  $S_2 = \{\text{id}, (1\ 2)\}$ , and  $\text{id} = (1\ 2)(1\ 2)$ .

Suppose every permutation in  $S_n$  is a product of transpositions, and let  $\sigma \in S_{n+1}$ .

*Case 1.*  $\sigma(n+1) = n+1$ . Since  $\sigma$  is injective,  $\sigma(i) \neq n+1$  for  $i \leq n$ , so  $\sigma$  restricts to a function  $\sigma': \{1, \dots, n\} \rightarrow \{1, \dots, n\}$ . It is injective because  $\sigma$  is, and hence bijective by Theorem 0.4.9. By the induction hypothesis,  $\sigma' = \tau_1 \cdots \tau_r$  with transpositions  $\tau_j \in S_n$ . Regard each  $\tau_j$  as a transposition in  $S_{n+1}$  fixing  $n+1$ . Then  $\tau_1 \cdots \tau_r$  agrees with  $\sigma' = \sigma$  on  $\{1, \dots, n\}$  and fixes  $n+1$ , as  $\sigma$  does. Hence  $\sigma = \tau_1 \cdots \tau_r$ .

*Case 2.*  $\sigma(n+1) = k \neq n+1$ . Let  $\tau = (k\ n+1)$ . Then  $(\tau\sigma)(n+1) = \tau(k) = n+1$ , so by Case 1,  $\tau\sigma = \tau_1 \cdots \tau_r$  for some transpositions  $\tau_j$ . Since  $\tau\tau = \text{id}$ , multiplying on the left by  $\tau$  gives  $\sigma = \tau\tau_1 \cdots \tau_r$ , a product of transpositions.

This completes the induction, and proves the theorem. ■

The product is not unique:  $(1\ 2\ 3) = (1\ 3)(1\ 2) = (1\ 2)(2\ 3)$ , and one can always insert  $(1\ 2)(1\ 2) = \text{id}$ . What turns out to be unique is whether the number of transpositions is even or odd. That fact defines the **sign** of a permutation, which we do not define here; it is the subject of Chapter 6, where it builds the determinant.

**0.10.5 Exercises****0.10.5.1 A. Check your understanding****Exercise A1.**

- (a) State the three axioms of a group.
- (b) Is  $\mathbb{Z}$  with the operation  $(a, b) \mapsto a - b$  a group? Justify your answer.
- (c) Determine whether the following statement is true: “in every group,  $(ab)^{-1} = a^{-1}b^{-1}$ .” Justify your answer.
- (d) Compute  $(1\ 3)(1\ 2\ 3)$  in  $S_3$ .
- (e) Explain why  $\{\mathbf{A} \in M_2(\mathbb{R}) : \mathbf{A} \text{ is not invertible}\}$  is **not** a subgroup of  $\text{GL}_2(\mathbb{R})$ .

*Solution.*

- (a) (G1)  $(ab)c = a(bc)$  for all  $a, b, c$ ; (G2) there is  $e$  with  $ea = ae = a$  for all  $a$ ; (G3)

for every  $a$  there is  $b$  with  $ab = ba = e$ . The operation must be a binary operation  $G \times G \rightarrow G$ .

- (b) No. Associativity fails:  $(1 - 1) - 1 = -1$ , but  $1 - (1 - 1) = 1$ .
- (c) False. By [Theorem 0.10.1](#),  $(ab)^{-1} = b^{-1}a^{-1}$ , and this can differ from  $a^{-1}b^{-1}$ . In  $S_3$  take  $a = (1\ 2)$  and  $b = (1\ 3)$ , which are their own inverses. By [Example \(Computing with cycles\)](#),  $ab = (1\ 3\ 2)$ , whose inverse is  $(1\ 2\ 3)$ , while  $a^{-1}b^{-1} = ab = (1\ 3\ 2)$ . The statement is true in abelian groups.
- (d) Apply  $(1\ 2\ 3)$  first:  $1 \mapsto 2 \mapsto 2$ ,  $2 \mapsto 3 \mapsto 1$ ,  $3 \mapsto 1 \mapsto 3$ . Hence  $(1\ 3)(1\ 2\ 3) = (1\ 2)$ .
- (e) It is not even a subset of  $\text{GL}_2(\mathbb{R})$ , since its elements are not invertible; for instance it does not contain the identity  $\mathbf{I}_2$ .

### 0.10.5.2 B. Practice

#### Exercise B1: The table of $S_3$ .

List the six elements of  $S_3$  in cycle notation and write out the table whose entry in row  $x$  and column  $y$  is  $xy$ . Hence find all elements  $x \in S_3$  with  $x^2 = \text{id}$ , and all  $x$  that commute with every element of  $S_3$ .

*Solution.* The elements are  $\text{id}, (1\ 2), (1\ 3), (2\ 3), (1\ 2\ 3), (1\ 3\ 2)$ . Computing each product right to left, as in [Example \(Computing with cycles\)](#), the entry in row  $x$ , column  $y$  is  $xy$ :

| $xy$        | $\text{id}$ | $(1\ 2)$    | $(1\ 3)$    | $(2\ 3)$    | $(1\ 2\ 3)$ | $(1\ 3\ 2)$ |
|-------------|-------------|-------------|-------------|-------------|-------------|-------------|
| $\text{id}$ | $\text{id}$ | $(1\ 2)$    | $(1\ 3)$    | $(2\ 3)$    | $(1\ 2\ 3)$ | $(1\ 3\ 2)$ |
| $(1\ 2)$    | $(1\ 2)$    | $\text{id}$ | $(1\ 3\ 2)$ | $(1\ 2\ 3)$ | $(2\ 3)$    | $(1\ 3)$    |
| $(1\ 3)$    | $(1\ 3)$    | $(1\ 2\ 3)$ | $\text{id}$ | $(1\ 3\ 2)$ | $(1\ 2)$    | $(2\ 3)$    |
| $(2\ 3)$    | $(2\ 3)$    | $(1\ 3\ 2)$ | $(1\ 2\ 3)$ | $\text{id}$ | $(1\ 3)$    | $(1\ 2)$    |
| $(1\ 2\ 3)$ | $(1\ 2\ 3)$ | $(1\ 3)$    | $(2\ 3)$    | $(1\ 2)$    | $(1\ 3\ 2)$ | $\text{id}$ |
| $(1\ 3\ 2)$ | $(1\ 3\ 2)$ | $(2\ 3)$    | $(1\ 2)$    | $(1\ 3)$    | $\text{id}$ | $(1\ 2\ 3)$ |

For example, the entry in row  $(1\ 2)$ , column  $(2\ 3)$  is  $(1\ 2)(2\ 3)$ :  $1 \mapsto 1 \mapsto 2$ ,  $2 \mapsto 3 \mapsto 3$ ,  $3 \mapsto 2 \mapsto 1$ , which is  $(1\ 2\ 3)$ .

Reading the diagonal,  $x^2 = \text{id}$  exactly for  $x = \text{id}, (1\ 2), (1\ 3), (2\ 3)$ . An element  $x$  commutes with every element exactly when row  $x$  equals column  $x$ . This holds for  $\text{id}$ . It fails for every other element:  $(1\ 2)(1\ 3) = (1\ 3\ 2) \neq (1\ 2\ 3) = (1\ 3)(1\ 2)$  handles  $(1\ 2)$  and  $(1\ 3)$ ;  $(2\ 3)(1\ 2) = (1\ 3\ 2) \neq (1\ 2\ 3) = (1\ 2)(2\ 3)$  handles  $(2\ 3)$ ; and  $(1\ 2\ 3)(1\ 2) = (1\ 3) \neq (2\ 3) = (1\ 2)(1\ 2\ 3)$  handles  $(1\ 2\ 3)$  and, since  $(1\ 3\ 2)(1\ 2) = (2\ 3) \neq (1\ 3) = (1\ 2)(1\ 3\ 2)$ , also  $(1\ 3\ 2)$ . Hence only  $\text{id}$  commutes with everything.

#### Exercise B2: Invertible upper triangular matrices.

Let  $F$  be a field and  $T = \left\{ \begin{pmatrix} a & b \\ 0 & d \end{pmatrix} : a, b, d \in F, a \neq 0, d \neq 0 \right\}$ .

- (a) Prove that  $T$  is a subgroup of  $\text{GL}_2(F)$ .  
 (b) For which fields  $F$  is  $T$  abelian? Justify your answer.

*Solution.*

- (a) Let  $\mathbf{X} = \begin{pmatrix} a & b \\ 0 & d \end{pmatrix} \in T$ . Since  $a, d \neq 0$ ,  $ad - b \cdot 0 = ad \neq 0$  by [Theorem 0.7.1](#), so  $\mathbf{X}$  is invertible by [Theorem 0.9.8](#). Hence  $T \subseteq \text{GL}_2(F)$ , and  $T \neq \emptyset$  since  $\mathbf{I}_2 \in T$ . Now let  $\mathbf{Y} = \begin{pmatrix} a' & b' \\ 0 & d' \end{pmatrix} \in T$ . By [Theorem 0.9.8](#),  $\mathbf{Y}^{-1} = (a'd')^{-1} \begin{pmatrix} d' & -b' \\ 0 & a' \end{pmatrix} = \begin{pmatrix} a'^{-1} & -b'(a'd')^{-1} \\ 0 & d'^{-1} \end{pmatrix}$ . Therefore

$$\mathbf{XY}^{-1} = \begin{pmatrix} aa'^{-1} & -ab'(a'd')^{-1} + bd'^{-1} \\ 0 & dd'^{-1} \end{pmatrix},$$

whose  $(2, 1)$ -entry is 0 and whose diagonal entries  $aa'^{-1}$ ,  $dd'^{-1}$  are non-zero by [Theorem 0.7.1](#). So  $\mathbf{XY}^{-1} \in T$ , and  $T$  is a subgroup by [Theorem 0.10.2](#).

- (b) It depends on the field. *Case 1:*  $F$  has an element  $c$  with  $c \neq 0$  and  $c \neq 1$  (for example  $c = 2$  in  $\mathbb{R}$  or  $\mathbb{F}_3$ ). Let  $\mathbf{X} = \begin{pmatrix} 1 & 1 \\ 0 & 1 \end{pmatrix}$  and  $\mathbf{Y} = \text{diag}(1, c)$ , both in  $T$ . Then  $\mathbf{XY} = \begin{pmatrix} 1 & c \\ 0 & c \end{pmatrix}$  and  $\mathbf{YX} = \begin{pmatrix} 1 & 1 \\ 0 & c \end{pmatrix}$ , which differ in the  $(1, 2)$ -entry since  $c \neq 1$ . So  $T$  is **not** abelian. *Case 2:*  $F = \{0, 1\}$ , that is,  $F = \mathbb{F}_2$ . Then  $a = d = 1$  is forced, so  $T = \left\{ \begin{pmatrix} 1 & b \\ 0 & 1 \end{pmatrix} : b \in \mathbb{F}_2 \right\}$ , and  $\begin{pmatrix} 1 & b \\ 0 & 1 \end{pmatrix} \begin{pmatrix} 1 & b' \\ 0 & 1 \end{pmatrix} = \begin{pmatrix} 1 & b+b' \\ 0 & 1 \end{pmatrix}$  is symmetric in  $b, b'$ . So  $T$  is abelian over  $\mathbb{F}_2$ , and over every other field it is not.

### Exercise B3: The exponential is an isomorphism.

Let  $\mathbb{R}_{>0} = \{x \in \mathbb{R} : x > 0\}$ . Show that  $(\mathbb{R}_{>0}, \cdot)$  is a group, and that  $\exp: (\mathbb{R}, +) \rightarrow (\mathbb{R}_{>0}, \cdot)$ ,  $x \mapsto e^x$ , is an isomorphism. What is its kernel? You may use the familiar facts  $e^{x+y} = e^x e^y$ ,  $e^x > 0$ , and that  $\ln: \mathbb{R}_{>0} \rightarrow \mathbb{R}$  satisfies  $\ln(e^x) = x$  and  $e^{\ln y} = y$ .

*Solution.* A product of positive reals is positive, so multiplication is a binary operation on  $\mathbb{R}_{>0}$ . It is associative,  $1 \in \mathbb{R}_{>0}$  is an identity, and for  $y > 0$  the inverse  $1/y$  is positive. Hence  $(\mathbb{R}_{>0}, \cdot)$  is a group.

Since  $e^x > 0$ ,  $\exp$  maps  $\mathbb{R}$  into  $\mathbb{R}_{>0}$ , and  $e^{x+y} = e^x e^y$  says exactly that it is a homomorphism from  $(\mathbb{R}, +)$  to  $(\mathbb{R}_{>0}, \cdot)$ . The function  $\ln$  satisfies  $\ln \circ \exp = \text{id}_{\mathbb{R}}$  and  $\exp \circ \ln = \text{id}_{\mathbb{R}_{>0}}$ , so  $\exp$  is bijective by [Theorem 0.4.7](#). Hence  $\exp$  is an isomorphism. Its kernel is  $\{x \in \mathbb{R} : e^x = 1\} = \{0\}$ , since  $\exp$  is injective and  $e^0 = 1$ .

### 0.10.5.3 C. Going deeper

#### Exercise C1: Every element its own inverse.

- (a) Let  $G$  be a group in which  $g^2 = gg = e$  for every  $g \in G$ . Prove that  $G$  is abelian.  
 (b) Show that  $(M_2(\mathbb{F}_2), +)$  is an example of such a group, written additively.



*Solution.*

- (a) Let  $a, b \in G$ . The hypothesis  $gg = e$  says that every  $g$  is its own inverse, by uniqueness of inverses ([Theorem 0.10.1](#)). Applying this to  $ab$ ,  $a$  and  $b$ , and using [Theorem 0.10.1](#),

$$ab = (ab)^{-1} = b^{-1}a^{-1} = ba.$$

Hence  $G$  is abelian.

- (b)  $(M_2(\mathbb{F}_2), +)$  is a group by [Theorem 0.9.1](#): addition is associative,  $0$  is an identity, and  $-\mathbf{A}$  is an inverse. In additive notation, “ $g^2 = e$ ” reads  $\mathbf{A} + \mathbf{A} = 0$ , and indeed  $(\mathbf{A} + \mathbf{A})_{ij} = a_{ij} + a_{ij} = (1 + 1)a_{ij} = 0$  in  $\mathbb{F}_2$ . (This group is abelian anyway, consistent with (a).)

### Exercise C2: The order of a product of disjoint cycles.

The **order** of  $\sigma \in S_n$  is the smallest integer  $m \geq 1$  with  $\sigma^m = \text{id}$ , where  $\sigma^m$  is the product of  $m$  copies of  $\sigma$ . Two cycles are **disjoint** if no element appears in both.

- (a) Let  $\sigma = (i_1 \cdots i_k)$  be a  $k$ -cycle. Prove that  $\sigma^m = \text{id}$  if and only if  $k$  divides  $m$ . Deduce that  $\sigma$  has order  $k$ .
- (b) Prove that disjoint cycles  $\sigma$  and  $\tau$  commute, and deduce  $(\sigma\tau)^m = \sigma^m\tau^m$  for all  $m \geq 1$ .
- (c) Let  $\sigma$  be a  $k$ -cycle and  $\tau$  an  $l$ -cycle, disjoint. Prove that  $\sigma\tau$  has order  $\text{lcm}(k, l)$ .
- (d) Find the order of  $(1\ 2)(3\ 4\ 5) \in S_5$ .

*Hint for (c):  $\sigma^m$  moves only elements that appear in  $\sigma$ .*

*Solution.*

- (a) Index the entries of the cycle modulo  $k$ , so  $\sigma(i_j) = i_{j+1}$  with  $i_{k+1} = i_1$ . By induction on  $m$ ,  $\sigma^m(i_j) = i_{j+m}$ , indices taken modulo  $k$ , and  $\sigma^m$  fixes every element not in the cycle. Hence  $\sigma^m = \text{id}$  if and only if  $i_{j+m} = i_j$  for all  $j$ . Since  $i_1, \dots, i_k$  are distinct, this holds if and only if  $j + m \equiv j \pmod{k}$ , that is,  $k \mid m$ . The smallest  $m \geq 1$  with  $k \mid m$  is  $k$ , so  $\sigma$  has order  $k$ .
- (b) Let  $A$  and  $B$  be the disjoint sets of elements appearing in  $\sigma$  and  $\tau$ . Note that  $\sigma$  maps  $A$  to  $A$  and fixes everything outside  $A$ , and  $\tau$  does the same with  $B$ . Let  $i \in \{1, \dots, n\}$ . If  $i \in A$ , then  $i \notin B$  and  $\sigma(i) \in A$ , so  $\sigma(i) \notin B$ ; hence  $\sigma\tau(i) = \sigma(i) = \tau\sigma(i)$ . The case  $i \in B$  is the same with the roles swapped. If  $i \notin A \cup B$ , both sides give  $i$ . Hence  $\sigma\tau = \tau\sigma$ . Then  $(\sigma\tau)^m = \sigma^m\tau^m$  follows by induction on  $m$ :  $(\sigma\tau)^{m+1} = \sigma^m\tau^m\sigma\tau = \sigma^m\sigma\tau^m\tau = \sigma^{m+1}\tau^{m+1}$ , moving  $\sigma$  past  $\tau$  one factor at a time, which is allowed because they commute.
- (c) By (b),  $(\sigma\tau)^m = \sigma^m\tau^m$ . We claim this is  $\text{id}$  if and only if  $\sigma^m = \text{id}$  and  $\tau^m = \text{id}$ . One direction is clear. Conversely, suppose  $\sigma^m\tau^m = \text{id}$ , and let  $i \in A$ . Then  $i \notin B$ , so  $\tau^m(i) = i$  by the hint, and  $\sigma^m(i) = \sigma^m\tau^m(i) = i$ . Since  $\sigma^m$  also fixes every  $i \notin A$ ,  $\sigma^m = \text{id}$ , and then  $\tau^m = \sigma^m\tau^m = \text{id}$  as well. By (a),  $(\sigma\tau)^m = \text{id}$  if and only if  $k \mid m$  and  $l \mid m$ . The smallest such  $m \geq 1$  is  $\text{lcm}(k, l)$ , which is therefore the order of  $\sigma\tau$ .
- (d)  $(1\ 2)$  and  $(3\ 4\ 5)$  are disjoint cycles of lengths 2 and 3, so by (c) the order is  $\text{lcm}(2, 3) = 6$ .

# Chapter 1

## Vector Spaces

Arrows in the plane can be added and stretched. So can solutions of a homogeneous system of equations, polynomials, matrices, and real-valued functions. In each case the same handful of rules governs the arithmetic, and none of the arguments we care about uses anything beyond those rules. This chapter names the structure, a **vector space**, so that every theorem proved once holds for all of these examples at the same time.

The second half of the chapter answers a question that sounds innocent: how big is a vector space? The spaces  $\mathbb{R}^3$ ,  $\mathbb{R}[x]_{\leq 2}$  and the symmetric  $2 \times 2$  matrices are all infinite sets, yet each is “three-dimensional” in a precise sense. Making that sense precise, and proving that it does not depend on any choices, takes the Exchange Theorem, the most important proof in the chapter. Everything later in the book measures with the dimension defined here.

**What you need.** From Chapter 0: fields ([Definition 0.7.1](#)) and their basic arithmetic ([Theorem 0.7.1](#)), sets and double inclusion ([Theorem 0.3.1](#)), functions, and induction ([Theorem 0.2.7](#)). Polynomials ([Definition 0.8.1](#)) and matrices ([Definition 0.9.1](#)) supply many of the examples.

### 1.0.1 Roadmap

- **Vector spaces.** The eight axioms, a gallery of examples and non-examples, and the first consequences of the axioms.
- **Subspaces.** A subset that is a vector space in its own right, recognized by three checks.
- **Linear combinations and span.** Everything that can be built from a list of vectors, and the smallest subspace containing them.
- **Linear independence.** When a list contains no redundant vector, and the Linear Dependence Lemma that finds the redundant one when it does.
- **Bases and coordinates.** Lists that span without waste; every vector gets unique coordinates, and every finite spanning list can be trimmed to a basis.
- **The Exchange Theorem and dimension.** Independent lists are never longer than spanning lists, so all bases have the same length; extending bases and counting instead of checking.
- **Sums and direct sums.** Combining subspaces, the dimension formula, and splitting a space into independent pieces.
- **Infinite-dimensional spaces.** Every vector space has a basis (via Zorn’s lemma), and which finite-dimensional facts fail without finiteness.
- **Changing the field.** The same set viewed over a smaller or larger field: restriction of

scalars, the tower law, and complexification.

## 1.0.2 Named moves

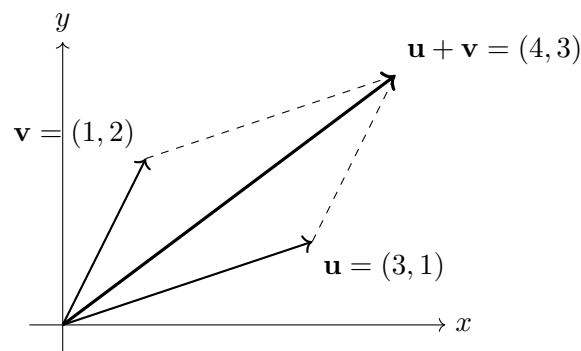
- **The three-check subspace test.** Zero vector, closed under addition, closed under scaling, each check ending with its own sentence.
- **“Let  $a_1\mathbf{v}_1 + \cdots + a_k\mathbf{v}_k = \mathbf{0}$ .”** The first line of every independence proof; the goal is then to show every  $a_i$  is zero.
- **Exchange one vector at a time.** Swap a vector of an independent list into a spanning list without losing the span.
- **Count instead of check.** In a space of dimension  $n$ , an independent list of length  $n$  is automatically a basis, and so is a spanning list of length  $n$ .
- **Basis of the smallest space, then extend.** For any dimension formula, start from a basis of the smallest subspace in sight, extend it, and count.

# 1.1 Vector Spaces

Chapter 0 met several kinds of objects that can be added and scaled: pairs of real numbers, polynomials, matrices. Each time we checked the same handful of rules by hand. This section collects those rules into one definition, the vector space, and proves the first consequences once, so that they hold in every example at the same time.

## 1.1.1 One set of rules in three places

Start with the plane  $\mathbb{R}^2$ . A pair  $\mathbf{u} = (3, 1)$  can be pictured as an arrow from the origin to the point  $(3, 1)$ . Two arrows are added tip to tail, which is the same as taking the diagonal of the parallelogram they span, and an arrow is scaled by stretching it.



The picture shows one rule at once: going along  $\mathbf{u}$  and then  $\mathbf{v}$  ends at the same corner as going along  $\mathbf{v}$  and then  $\mathbf{u}$ , so  $\mathbf{u} + \mathbf{v} = \mathbf{v} + \mathbf{u}$ . Other rules are just as familiar. Scaling a sum is the same as adding the scaled arrows,  $2(\mathbf{u} + \mathbf{v}) = 2\mathbf{u} + 2\mathbf{v}$ . Adding the zero arrow changes nothing, and every arrow is undone by its reverse.

Now look at polynomials of degree at most 2. We add them coefficient by coefficient,

$$(1 + 2x - x^2) + (3 - x + 4x^2) = 4 + x + 3x^2,$$

and scale them the same way,  $2(1 + 2x - x^2) = 2 + 4x - 2x^2$ . There is no picture of arrows here, yet the rules are identical: order does not matter in a sum, the zero polynomial changes nothing, and  $-1 - 2x + x^2$  undoes  $1 + 2x - x^2$ .

The same happens for  $2 \times 2$  real matrices. [Theorem 0.9.1](#) lists the rules for adding matrices and multiplying them by numbers, and they are again the same rules.

Arrows, polynomials and matrices are very different objects, but many arguments about them only use these rules. If we prove a statement from the rules alone, it holds for arrows, polynomials and matrices at once, and for every other system obeying the rules. So we list the rules once and give the systems that obey them a name.

*A vector space is a set whose elements can be added and scaled by numbers from a field, with the usual rules of arithmetic.*

## 1.1.2 The definition

### Definition 1.1.1 — Vector Space

Let  $F$  be a field. A **vector space over  $F$**  is a set  $V$  together with two functions

$$+ : V \times V \rightarrow V, (\mathbf{u}, \mathbf{v}) \mapsto \mathbf{u} + \mathbf{v}, \quad \cdot : F \times V \rightarrow V, (a, \mathbf{v}) \mapsto a\mathbf{v},$$

called **addition** and **scalar multiplication**, such that the following axioms hold.

- (VS1)  $\mathbf{u} + \mathbf{v} = \mathbf{v} + \mathbf{u}$  for all  $\mathbf{u}, \mathbf{v} \in V$ .
- (VS2)  $(\mathbf{u} + \mathbf{v}) + \mathbf{w} = \mathbf{u} + (\mathbf{v} + \mathbf{w})$  for all  $\mathbf{u}, \mathbf{v}, \mathbf{w} \in V$ .
- (VS3) There exists an element  $\mathbf{0} \in V$  such that  $\mathbf{v} + \mathbf{0} = \mathbf{v}$  for every  $\mathbf{v} \in V$ .
- (VS4) For every  $\mathbf{v} \in V$  there exists an element  $\mathbf{v}' \in V$  such that  $\mathbf{v} + \mathbf{v}' = \mathbf{0}$ .
- (VS5)  $a(\mathbf{u} + \mathbf{v}) = a\mathbf{u} + a\mathbf{v}$  for all  $a \in F$  and  $\mathbf{u}, \mathbf{v} \in V$ .
- (VS6)  $(a + b)\mathbf{v} = a\mathbf{v} + b\mathbf{v}$  for all  $a, b \in F$  and  $\mathbf{v} \in V$ .
- (VS7)  $a(b\mathbf{v}) = (ab)\mathbf{v}$  for all  $a, b \in F$  and  $\mathbf{v} \in V$ .
- (VS8)  $1\mathbf{v} = \mathbf{v}$  for every  $\mathbf{v} \in V$ , where 1 is the identity element of  $F$ .

Read the definition clause by clause.

- **The two functions.** Addition takes two elements of  $V$  and returns an element of  $V$ . Scalar multiplication takes a number from  $F$  and an element of  $V$  and returns an element of  $V$ . So “the sum stays in the set” and “a multiple stays in the set” are not extra axioms: they are built into the word “function into  $V$ ”. When we test a candidate, this closure is the first thing to check.
- **(VS1)–(VS4)** concern addition alone. It is commutative and associative, there is a vector  $\mathbf{0}$  that changes nothing, and every vector can be undone. These are the axioms (F1)–(F4) of [Definition 0.7.1](#), now for vectors.
- **(VS5) and (VS6)** are the two distributive laws. In (VS6) the symbol  $+$  appears twice with two meanings: on the left it adds scalars in  $F$ , on the right it adds vectors in  $V$ .
- **(VS7)** says scaling by  $b$  and then by  $a$  is scaling by the product  $ab$ , computed in  $F$ .
- **(VS8)** says the scalar 1 acts as it should. It looks too obvious to state, but we will see below that it cannot be dropped.

Note the order of quantifiers in (VS3): **one** vector  $\mathbf{0}$  works for **every**  $\mathbf{v}$ . In (VS4) the vector  $\mathbf{v}'$  is allowed to depend on  $\mathbf{v}$ .

### Definition 1.1.2 — Vectors and Scalars

Let  $V$  be a vector space over  $F$ . The elements of  $V$  are called **vectors**, and the elements of  $F$  are called **scalars**. The element  $\mathbf{0}$  in (VS3) is a **zero vector** of  $V$ , and an element  $\mathbf{v}'$  as in (VS4) is an **additive inverse** of  $\mathbf{v}$ .

A vector space over  $\mathbb{R}$  is also called a **real vector space**, and one over  $\mathbb{C}$  a **complex vector space**. When the operations are clear from context, we say “ $V$  is a vector space over  $F$ ” instead of naming the triple  $(V, +, \cdot)$ . The field is part of the data, and we will see shortly that changing it changes the space.

We wrote “**a** zero vector” and “**an** additive inverse” on purpose. The axioms only say these exist. That there is only one of each is a theorem, proved later in this section.

*Warning. Two different zeros.* The scalar  $0 \in F$  and the zero vector  $\mathbf{0} \in V$  are different objects. In  $M_2(\mathbb{R})$ , the scalar is the number 0 and the zero vector is the zero matrix; in  $\mathbb{R}^3$ , the zero vector is  $(0, 0, 0)$ . This book writes the zero vector in bold. An equation such as  $0\mathbf{v} = \mathbf{0}$  has a scalar on the left and a vector on the right, and mixing them up produces statements that make no sense, such as “ $\mathbf{v} = 0$ ” for a matrix  $\mathbf{v}$ .

## 1.1.3 Examples

We start with the families we will use throughout the book. For each, the work is to say what the operations are and why the axioms hold. We check some axioms in full and point out the pattern for the rest.

### Example. (The Standard Vector Spaces)

Let  $F$  be a field. Explain why each of the following is a vector space over  $F$ .

- (a)  $F^n$ , with entrywise addition and scalar multiplication.
- (b)  $M_{m \times n}(F)$ , with the matrix operations of Definition 0.9.6 and Definition 0.9.7.
- (c)  $F[x]$ , with the addition of Definition 0.8.2 and  $c \sum a_k x^k := \sum (ca_k) x^k$ .
- (d) For a set  $X$ , the set  $F^X$  of **all** functions  $X \rightarrow F$ , with the **pointwise** operations

$$(f + g)(t) := f(t) + g(t), \quad (af)(t) := af(t) \quad \text{for all } t \in X.$$

- (e) The set  $F^{\mathbb{N}}$  of all sequences  $(s_0, s_1, s_2, \dots)$  of elements of  $F$ , with termwise operations.

*Solution.*

- (b) The sum and scalar multiple of  $m \times n$  matrices are  $m \times n$  matrices, so both operations are functions into  $M_{m \times n}(F)$ . The eight axioms are exactly the rules of Theorem 0.9.1, with the zero matrix as  $\mathbf{0}$  and  $-\mathbf{A}$  as an additive inverse of  $\mathbf{A}$ .
- (c) Chapter 0 defines  $F^n = M_{n \times 1}(F)$ , the column vectors, and the entrywise opera-

tions are the matrix operations. So (a) is the case of (b) with  $n$  rows and 1 column. The zero vector is  $(0, \dots, 0)$ .

- (d) The formulas define functions  $f + g$  and  $af$  from  $X$  to  $F$ , so the operations land in  $F^X$ . Two functions  $X \rightarrow F$  are equal when they take the same value at every  $t \in X$  (Definition 0.4.1), so each axiom is checked one point at a time. For (VS3), let  $\mathbf{0}$  be the zero function,  $\mathbf{0}(t) = 0$  for all  $t$ . Then  $(f + \mathbf{0})(t) = f(t) + 0 = f(t)$  for every  $t$ , by (F3), so  $f + \mathbf{0} = f$ . For (VS4), let  $f'(t) := -f(t)$ . Then  $(f + f')(t) = f(t) + (-f(t)) = 0 = \mathbf{0}(t)$  by (F4), so  $f + f' = \mathbf{0}$ . For (VS6), let  $a, b \in F$ . For every  $t \in X$ ,

$$((a + b)f)(t) = (a + b)f(t) = af(t) + bf(t) = (af)(t) + (bf)(t) = (af + bf)(t),$$

where the second equality is (F9) together with (F6) in  $F$ . Hence  $(a + b)f = af + bf$ . The other five axioms reduce in the same way to a field axiom at each point: (VS1) to (F1), (VS2) to (F2), (VS5) to (F9), (VS7) to (F5), and (VS8) to (F7) with (F6).

- (e) A sequence  $(s_0, s_1, \dots)$  is a function  $\mathbb{N} \rightarrow F$ ,  $k \mapsto s_k$ , and the termwise operations are the pointwise ones. So (e) is (d) with  $X = \mathbb{N}$ .
- (f) Let  $p = \sum a_k x^k \in F[x]$  and  $c \in F$ . Only finitely many  $a_k$  are non-zero, and  $a_k = 0$  forces  $ca_k = 0$ , so  $cp$  is again a polynomial. The sum of polynomials is a polynomial by Definition 0.8.2. By Theorem 0.8.1 (parts 1 and 2), addition satisfies (VS1)–(VS4), with the zero polynomial as  $\mathbf{0}$  and  $-p$  as an additive inverse. Since polynomials are equal when their coefficients agree (Definition 0.8.1), (VS5)–(VS8) are checked coefficient by coefficient. For instance, the coefficient of  $x^k$  in  $(c + d)p$  is  $(c + d)a_k = ca_k + da_k$ , which is the coefficient of  $x^k$  in  $cp + dp$ . The others reduce in the same way: (VS5) to (F9), (VS7) to (F5), and (VS8) to (F7) with (F6).

The pattern is worth naming. Whenever the operations are defined **entry by entry**, **point by point** or **coefficient by coefficient**, each axiom for vectors is a field axiom applied in each slot. In fact (a), (b) and (e) are all instances of (d): a column in  $F^n$  is a function  $\{1, \dots, n\} \rightarrow F$ , and a matrix in  $M_{m \times n}(F)$  is a function on the set of positions  $\{1, \dots, m\} \times \{1, \dots, n\}$ .

A **subset** of a known space often works just as well, provided the operations do not lead out of it.

The set  $F[x]_{\leq n}$  of Definition 0.8.4 is a vector space over  $F$ , with the operations of  $F[x]$ . Closure is the only real issue. If  $\deg p \leq n$  and  $\deg q \leq n$ , then  $\deg(p + q) \leq \max(\deg p, \deg q) \leq n$  by Theorem 0.8.2. If  $a_k = 0$  for all  $k > n$ , then  $ca_k = 0$  for all  $k > n$ , so  $\deg(cp) \leq n$ . The zero polynomial has degree  $-\infty \leq n$ , and  $-p$  has the same coefficients as  $p$  up to sign, so it stays in  $F[x]_{\leq n}$ . The six axioms (VS1), (VS2), (VS5)–(VS8) are equations that hold for **all** polynomials, so in particular for those of degree at most  $n$ .

The next example applies the same pattern to a set of real numbers, with a smaller field of scalars.

### Example. $(\mathbb{Q}(\sqrt{2})$ over $\mathbb{Q}$ )

Let  $\mathbb{Q}(\sqrt{2}) = \{a + b\sqrt{2} : a, b \in \mathbb{Q}\}$ , with the addition of  $\mathbb{R}$  and with scalar multiplication  $c(a + b\sqrt{2}) := ca + cb\sqrt{2}$  for  $c \in \mathbb{Q}$ . Show that  $\mathbb{Q}(\sqrt{2})$  is a vector space over  $\mathbb{Q}$ .

*Solution.* Let  $\mathbf{u} = a + b\sqrt{2}$  and  $\mathbf{v} = a' + b'\sqrt{2}$  with  $a, b, a', b' \in \mathbb{Q}$ , and let  $c \in \mathbb{Q}$ .

*Closure.* We have  $\mathbf{u} + \mathbf{v} = (a + a') + (b + b')\sqrt{2}$  and  $c\mathbf{u} = ca + cb\sqrt{2}$ . Sums and

products of rationals are rational, so both lie in  $\mathbb{Q}(\sqrt{2})$ . Hence addition is a function  $\mathbb{Q}(\sqrt{2}) \times \mathbb{Q}(\sqrt{2}) \rightarrow \mathbb{Q}(\sqrt{2})$ , and scalar multiplication is a function  $\mathbb{Q} \times \mathbb{Q}(\sqrt{2}) \rightarrow \mathbb{Q}(\sqrt{2})$ .

*Zero and additive inverses.* The number  $0 = 0 + 0\sqrt{2}$  lies in  $\mathbb{Q}(\sqrt{2})$  and satisfies  $\mathbf{u} + 0 = \mathbf{u}$ , so (VS3) holds. The number  $-\mathbf{u} = (-a) + (-b)\sqrt{2}$  lies in  $\mathbb{Q}(\sqrt{2})$  and satisfies  $\mathbf{u} + (-\mathbf{u}) = 0$ , so (VS4) holds.

*The identities.* Scalar multiplication by  $c \in \mathbb{Q}$  is the multiplication of real numbers, since  $c(a + b\sqrt{2}) = ca + cb\sqrt{2}$  in  $\mathbb{R}$ . So each of (VS1), (VS2) and (VS5)–(VS8) is an identity between real numbers, and it holds by the field axioms of  $\mathbb{R}$ : (VS1) is (F1), (VS2) is (F2), (VS5) is (F9), (VS6) is (F9) with (F6), (VS7) is (F5), and (VS8) is (F7) with (F6). Hence  $\mathbb{Q}(\sqrt{2})$  is a vector space over  $\mathbb{Q}$ .

We saw in [Example \( \$\mathbb{Q}\(\sqrt{2}\)\$  Is a Field\)](#) that  $\mathbb{Q}(\sqrt{2})$  is even a field. As a vector space over  $\mathbb{Q}$ , we simply ignore how to multiply two of its elements together and keep only multiplication by rationals.

The same set can carry more than one field of scalars. The next example makes this concrete.

### Example. (The Same Set over Different Fields)

Explain why  $\mathbb{C}$  is a vector space over  $\mathbb{C}$  and also over  $\mathbb{R}$ , and why  $\mathbb{R}$  is a vector space over  $\mathbb{R}$  and also over  $\mathbb{Q}$ . In each case the addition is the usual one and the scalar multiplication is the usual product.

*Solution.*  $\mathbb{C}$  over  $\mathbb{C}$  is  $F^1$  for  $F = \mathbb{C}$ , covered by [Example \(The Standard Vector Spaces\)](#) (a). For  $\mathbb{C}$  over  $\mathbb{R}$ , a real number  $r$  is the complex number  $r + 0i$ , and  $rz$  is their complex product, which lies in  $\mathbb{C}$ . Each axiom is an identity among complex numbers in which some of the numbers happen to be real, so it holds by [Proposition 0.6.1](#): (VS5) and (VS6) by distributivity together with commutativity, (VS7) by associativity of multiplication, and (VS8) since  $1 \cdot z = z$ . The argument for  $\mathbb{R}$  over  $\mathbb{Q}$  is the same, with the field axioms of  $\mathbb{R}$  in place of the laws of  $\mathbb{C}$ .

As sets,  $\mathbb{C}$  over  $\mathbb{C}$  and  $\mathbb{C}$  over  $\mathbb{R}$  are the same. As vector spaces they differ, because the scalars available differ. In  $\mathbb{C}$  over  $\mathbb{C}$ , the vector  $i$  is a scalar multiple of the vector  $1$ . In  $\mathbb{C}$  over  $\mathbb{R}$  it is not, since  $r \cdot 1 = i$  has no real solution  $r$ . Later in this chapter this difference becomes a difference in dimension. That is why we always say **over which field**.

**The degenerate case.** The set  $\{\mathbf{0}\}$  with one element, with  $\mathbf{0} + \mathbf{0} = \mathbf{0}$  and  $a\mathbf{0} = \mathbf{0}$  for every  $a \in F$ , is a vector space over any field. Every axiom is an equation between elements that all equal  $\mathbf{0}$ , and  $\mathbf{0}$  serves as the zero vector and as its own additive inverse. It is called the **zero space**. It matters because it turns up constantly as an answer: the only solution of a system, or the smallest vector space inside every other one. Unlike a field, which must contain  $1 \neq 0$  ([Definition 0.7.1](#)), a vector space is allowed to have a single element, since no axiom asks for a vector different from  $\mathbf{0}$ .

A final example looks strange, but it satisfies every axiom.

### Example. (Positive Reals with Exotic Operations)

Let  $V = \mathbb{R}_{>0} = \{t \in \mathbb{R} : t > 0\}$ . For  $u, v \in V$  and  $c \in \mathbb{R}$ , define

$$u \boxplus v := uv, \quad c \boxminus v := v^c.$$

Show that these operations are functions into  $V$ , find the zero vector and the additive inverse of  $v$ , and check (VS6) and (VS8).

*Solution. Closure.* For  $u, v > 0$  and  $c \in \mathbb{R}$ , the product  $uv$  is positive and the power  $v^c$  is positive. So  $\boxplus: V \times V \rightarrow V$  and  $\boxminus: \mathbb{R} \times V \rightarrow V$ .

(VS3). We need  $z \in V$  with  $v \boxplus z = v$ , that is,  $vz = v$ , for every  $v > 0$ . The number  $z = 1$  works, since  $v \cdot 1 = v$ . So the zero vector of  $V$  is the real number 1.

(VS4). For  $v \in V$ , we need  $v' \in V$  with  $v \boxplus v' = 1$ , that is,  $vv' = 1$ . The number  $v' = 1/v$  is positive and works. So the additive inverse of  $v$  is  $1/v$ .

(VS6). For  $a, b \in \mathbb{R}$  and  $v \in V$ , by the laws of exponents,

$$(a + b) \boxminus v = v^{a+b} = v^a v^b = (a \boxminus v) \boxplus (b \boxminus v).$$

(VS8).  $1 \boxminus v = v^1 = v$ .

The remaining axioms (VS1), (VS2), (VS5) and (VS7) are [Exercise \(B3: Finishing the Positive Reals\)](#). So  $V$  is a real vector space.

This example earns its place for one reason. The words “zero vector” and “addition” describe **roles** fixed by the axioms, not the familiar number 0 or the familiar  $+$ . Here the vector playing the role of  $\mathbf{0}$  is the number 1, and “adding” means multiplying.

*Quick check.* Explain why each of the following, with the usual operations, is a vector space over  $\mathbb{R}$ .

- (a) The set  $S$  of twice-differentiable functions  $f: \mathbb{R} \rightarrow \mathbb{R}$  with  $f'' + f = 0$ .
- (b) The set of symmetric matrices  $\mathbf{A} \in M_2(\mathbb{R})$ , that is, those with  $\mathbf{A}^\top = \mathbf{A}$ .

*Solution.* Both are subsets of known spaces, so we use the pattern after [Example \(The Standard Vector Spaces\)](#): check closure, the zero vector and additive inverses; the other six axioms are identities inherited from the big space.

- (a)  $S \subseteq \mathbb{R}^{\mathbb{R}}$ . If  $f, g \in S$  and  $c \in \mathbb{R}$ , then by the sum and constant multiple rules of calculus,  $f + g$  and  $cf$  are twice differentiable, with  $(f + g)'' + (f + g) = (f'' + f) + (g'' + g) = 0$  and  $(cf)'' + cf = c(f'' + f) = 0$ . So both lie in  $S$ . The zero function  $\mathbf{0}$  satisfies  $\mathbf{0}'' + \mathbf{0} = \mathbf{0}$ , so  $\mathbf{0} \in S$ . If  $f \in S$ , its additive inverse in  $\mathbb{R}^{\mathbb{R}}$  is  $t \mapsto -f(t)$ , and  $-f(t) = (-1)f(t)$  by [Theorem 0.7.1](#) (e); so it is the function  $(-1)f$ , which lies in  $S$  by closure under scaling. The identities hold in  $\mathbb{R}^{\mathbb{R}}$  ([Example \(The Standard Vector Spaces\)](#) (d)), hence in  $S$ . For instance  $\sin$  and  $\cos$  lie in  $S$ .
- (b) By [Theorem 0.9.5](#),  $(\mathbf{A} + \mathbf{B})^\top = \mathbf{A}^\top + \mathbf{B}^\top = \mathbf{A} + \mathbf{B}$  and  $(c\mathbf{A})^\top = c\mathbf{A}^\top = c\mathbf{A}$  when  $\mathbf{A}^\top = \mathbf{A}$  and  $\mathbf{B}^\top = \mathbf{B}$ . The zero matrix is symmetric, and the additive inverse  $-\mathbf{A}$ , which Chapter 0 defines as  $(-1)\mathbf{A}$ , is symmetric by closure under scaling. The identities hold in  $M_2(\mathbb{R})$ , hence for symmetric matrices.



### 1.1.4 Non-examples

Each non-example below is a small change to an example. We say what still works and name the exact clause that fails.

- **Change scalar multiplication on  $\mathbb{R}^2$  to  $c(x, y) := (cx, 0)$** , keeping the usual addition. Addition is untouched, so (VS1)–(VS4) hold. (VS5) holds:  $c((x, y) + (x', y')) = (c(x + x'), 0) = (cx, 0) + (cx', 0)$ . In the same way (VS6) and (VS7) hold. But  $1(0, 1) = (0, 0) \neq (0, 1)$ , so **(VS8) fails**.
- **Change the line  $x + 2y = 0$  to  $x + 2y = 3$** . Let  $L = \{(x, y) \in \mathbb{R}^2 : x + 2y = 3\}$  with the operations of  $\mathbb{R}^2$ . The points  $(3, 0)$  and  $(1, 1)$  lie on  $L$ , but their sum  $(4, 1)$  has  $4 + 2 \cdot 1 = 6 \neq 3$ . So addition is **not a function**  $L \times L \rightarrow L$ , and  $L$  is not a vector space with these operations.
- **Change “degree at most 2” to “degree exactly 2”**. In  $\mathbb{R}[x]_{=2}$  (Definition 0.8.5),  $x^2 + 2x$  and  $-x^2 + 3$  have degree 2, but their sum  $2x + 3$  has degree 1. Again addition leaves the set. There is no zero vector either: the zero polynomial has degree  $-\infty$ , not 2.
- **Change the entries from real to integer**, keeping  $\mathbb{R}$  as the field. For  $\mathbb{Z}^2 \subseteq \mathbb{R}^2$ , addition is fine, but  $\frac{1}{2}(1, 0) = (\frac{1}{2}, 0) \notin \mathbb{Z}^2$ . So scalar multiplication is **not a function**  $\mathbb{R} \times \mathbb{Z}^2 \rightarrow \mathbb{Z}^2$ .

### 1.1.5 Why these axioms

Each axiom is there for a reason, and a few conventions deserve a comment.

- **(VS8) cannot be dropped**. The first non-example satisfies all other axioms. Without (VS8), the “scalar multiplication”  $c\mathbf{v} = \mathbf{0}$  for all  $c$  and  $\mathbf{v}$  would also be allowed, and scaling would carry no information at all.
- **(VS1) is in fact redundant**. Commutativity of addition follows from the other seven axioms, by expanding  $(1 + 1)(\mathbf{u} + \mathbf{v})$  in two ways (Exercise (C1: Commutativity Comes for Free)). It is kept because the proof is fiddly and nobody wants to repeat it.
- **Closure is not optional**. Two of the non-examples fail only because an operation leads out of the set. That is why the definition asks for **functions into  $V$** .
- **The scalars form a field**. We use division by non-zero scalars in the zero product law at the end of the next subsection. The warning after it shows what goes wrong without division.
- **The name**. “Vector” originally meant an arrow in physics. The word is kept for elements of any vector space, even when they are polynomials or functions.

### 1.1.6 First consequences of the axioms

The definition says a zero vector and additive inverses **exist**, not that they are unique. Before writing  $\mathbf{0}$  and  $-\mathbf{v}$  as if each named one thing, we prove uniqueness. We also prove that familiar rules such as  $0\mathbf{v} = \mathbf{0}$  follow from the axioms. None of this is obvious from the axioms: in Example (Positive Reals with Exotic Operations),  $0 \boxplus v = v^0 = 1$ , which happens to be the zero vector there, and it takes a proof to see that this is no coincidence.

Throughout, we use one fact about functions without further comment: if two inputs are equal, the outputs are equal. So from  $\mathbf{v} = \mathbf{w}$  we may conclude  $\mathbf{u} + \mathbf{v} = \mathbf{u} + \mathbf{w}$  and  $a\mathbf{v} = a\mathbf{w}$ .

The first tool lets us cancel a vector from both sides of an equation.

**Theorem 1.1.1 — Left Cancellation Law**

Let  $V$  be a vector space over  $F$  and let  $\mathbf{u}, \mathbf{v}, \mathbf{w} \in V$ . If  $\mathbf{u} + \mathbf{v} = \mathbf{u} + \mathbf{w}$ , then  $\mathbf{v} = \mathbf{w}$ .

*Proof.* Suppose  $\mathbf{u} + \mathbf{v} = \mathbf{u} + \mathbf{w}$ . By (VS4) there is  $\mathbf{u}' \in V$  with  $\mathbf{u} + \mathbf{u}' = \mathbf{0}$ . Adding  $\mathbf{u}'$  on the left of both sides gives  $\mathbf{u}' + (\mathbf{u} + \mathbf{v}) = \mathbf{u}' + (\mathbf{u} + \mathbf{w})$ , and by (VS2),

$$(\mathbf{u}' + \mathbf{u}) + \mathbf{v} = (\mathbf{u}' + \mathbf{u}) + \mathbf{w}.$$

By (VS1) and the choice of  $\mathbf{u}'$ ,  $\mathbf{u}' + \mathbf{u} = \mathbf{u} + \mathbf{u}' = \mathbf{0}$ . Hence  $\mathbf{0} + \mathbf{v} = \mathbf{0} + \mathbf{w}$ . By (VS1) this is  $\mathbf{v} + \mathbf{0} = \mathbf{w} + \mathbf{0}$ , and by (VS3),  $\mathbf{v} = \mathbf{w}$ . ■

**Theorem 1.1.2 — Right Cancellation Law**

Let  $V$  be a vector space over  $F$  and let  $\mathbf{u}, \mathbf{v}, \mathbf{w} \in V$ . If  $\mathbf{v} + \mathbf{u} = \mathbf{w} + \mathbf{u}$ , then  $\mathbf{v} = \mathbf{w}$ .

*Proof.* By (VS1), the hypothesis says  $\mathbf{u} + \mathbf{v} = \mathbf{u} + \mathbf{w}$ . By [Theorem 1.1.1](#),  $\mathbf{v} = \mathbf{w}$ . ■

Uniqueness uses the standard move from Chapter 0: suppose there are two, and show they are equal.

**Theorem 1.1.3 — Zero Vector Is Unique**

Let  $V$  be a vector space over  $F$ . There is exactly one element  $\mathbf{0} \in V$  satisfying (VS3).

*Proof.* Existence is (VS3). Suppose  $\mathbf{0}$  and  $\mathbf{0}'$  both satisfy (VS3). Applying (VS3) for  $\mathbf{0}'$  to the vector  $\mathbf{0}$ , then (VS1), then (VS3) for  $\mathbf{0}$  to the vector  $\mathbf{0}'$ ,

$$\mathbf{0} = \mathbf{0} + \mathbf{0}' = \mathbf{0}' + \mathbf{0} = \mathbf{0}'.$$

This shows the zero vector is unique. ■

**Theorem 1.1.4 — Additive Inverse Is Unique**

Let  $V$  be a vector space over  $F$  and  $\mathbf{v} \in V$ . There is exactly one  $\mathbf{v}' \in V$  with  $\mathbf{v} + \mathbf{v}' = \mathbf{0}$ .

*Proof.* Existence is (VS4). Suppose  $\mathbf{v} + \mathbf{v}' = \mathbf{0}$  and  $\mathbf{v} + \mathbf{v}'' = \mathbf{0}$ . Then  $\mathbf{v} + \mathbf{v}' = \mathbf{v} + \mathbf{v}''$ , and [Theorem 1.1.1](#) gives  $\mathbf{v}' = \mathbf{v}''$ . ■

Since the inverse is unique, it deserves a name.

### Definition 1.1.3 — Notation for the Additive Inverse

Let  $V$  be a vector space over  $F$  and  $\mathbf{v} \in V$ . The unique additive inverse of  $\mathbf{v}$  is denoted  $-\mathbf{v}$ . For  $\mathbf{u}, \mathbf{v} \in V$  we write  $\mathbf{u} - \mathbf{v} := \mathbf{u} + (-\mathbf{v})$ .

For example,  $\mathbf{v} + (-\mathbf{v}) = \mathbf{0}$  and  $(-\mathbf{v}) + \mathbf{v} = \mathbf{0}$  by (VS1), so  $\mathbf{v}$  is the additive inverse of  $-\mathbf{v}$ . By Theorem 1.1.4,  $-(-\mathbf{v}) = \mathbf{v}$ .

*Remark.* Axioms (VS1)–(VS4) say that  $V$  with addition is an abelian group (Definition 0.10.1). So the four results above are also special cases of Theorem 0.10.1. We proved them directly to practice citing the axioms.

Now to scalars. No axiom mentions  $0\mathbf{v}$ , so we must create a situation in which one applies. The move is to write the scalar  $0$  in a fancy way, as  $0 + 0$ , expand with a distributive law, and cancel. We will call it **expand zero and cancel**.

### Theorem 1.1.5 — Zero Scalar Gives the Zero Vector

Let  $V$  be a vector space over  $F$ . Then  $0\mathbf{v} = \mathbf{0}$  for every  $\mathbf{v} \in V$ .

*Proof.* Let  $\mathbf{v} \in V$ . Since  $0 + 0 = 0$  in  $F$  by (F3), (VS6) gives

$$0\mathbf{v} + 0\mathbf{v} = (0 + 0)\mathbf{v} = 0\mathbf{v} = 0\mathbf{v} + \mathbf{0},$$

where the last equality is (VS3). By Theorem 1.1.1,  $0\mathbf{v} = \mathbf{0}$ . ■

The same move, with the zero vector in place of the zero scalar, handles  $a\mathbf{0}$ .

### Theorem 1.1.6 — Scalar Multiple of the Zero Vector

Let  $V$  be a vector space over  $F$ . Then  $a\mathbf{0} = \mathbf{0}$  for every  $a \in F$ .

*Proof.* Let  $a \in F$ . Since  $\mathbf{0} + \mathbf{0} = \mathbf{0}$  by (VS3), (VS5) gives

$$a\mathbf{0} + a\mathbf{0} = a(\mathbf{0} + \mathbf{0}) = a\mathbf{0} = a\mathbf{0} + \mathbf{0},$$

where the last equality is (VS3). By Theorem 1.1.1,  $a\mathbf{0} = \mathbf{0}$ . ■

To identify a vector as  $-\mathbf{v}$ , it is enough to show it adds with  $\mathbf{v}$  to  $\mathbf{0}$ , by Theorem 1.1.4. This gives the next two results.

**Theorem 1.1.7 — Negation as Scalar Multiplication**

Let  $V$  be a vector space over  $F$ . Then  $(-1)\mathbf{v} = -\mathbf{v}$  for every  $\mathbf{v} \in V$ .

*Proof.* Let  $\mathbf{v} \in V$ . By (VS8), (VS6), (F4) and [Theorem 1.1.5](#),

$$\mathbf{v} + (-1)\mathbf{v} = 1\mathbf{v} + (-1)\mathbf{v} = (1 + (-1))\mathbf{v} = 0\mathbf{v} = \mathbf{0}.$$

So  $(-1)\mathbf{v}$  is an additive inverse of  $\mathbf{v}$ , and by [Theorem 1.1.4](#) it equals  $-\mathbf{v}$ . ■

**Theorem 1.1.8 — Negative Scalars**

Let  $V$  be a vector space over  $F$ . Then  $(-a)\mathbf{v} = -(a\mathbf{v})$  for every  $a \in F$  and  $\mathbf{v} \in V$ .

*Proof.* Let  $a \in F$  and  $\mathbf{v} \in V$ . By (VS6), (F4) and [Theorem 1.1.5](#),

$$a\mathbf{v} + (-a)\mathbf{v} = (a + (-a))\mathbf{v} = 0\mathbf{v} = \mathbf{0}.$$

So  $(-a)\mathbf{v}$  is an additive inverse of  $a\mathbf{v}$ , and by [Theorem 1.1.4](#) it equals  $-(a\mathbf{v})$ . ■

The companion rules  $a(-\mathbf{v}) = -(a\mathbf{v})$  and  $a(\mathbf{u} - \mathbf{v}) = a\mathbf{u} - a\mathbf{v}$  are [Exercise \(B2: More Rules for Signs\)](#). From now on, we compute with  $+$ ,  $-$  and scalars as in school algebra, citing these results when a step is not a single axiom.

The last result is the one we will use most. In a field, a product of non-zero numbers is non-zero ([Theorem 0.7.1 \(f\)](#)). The same holds for a scalar times a vector.

**Theorem 1.1.9 — Zero Product Law**

Let  $V$  be a vector space over  $F$ , let  $a \in F$  and  $\mathbf{v} \in V$ . Then  $a\mathbf{v} = \mathbf{0}$  if and only if  $a = 0$  or  $\mathbf{v} = \mathbf{0}$ .

*Idea.* The direction  $(\Leftarrow)$  is the two zero results above, [Theorem 1.1.5](#) and [Theorem 1.1.6](#). For  $(\Rightarrow)$ , the only way to get rid of  $a$  is to multiply by  $a^{-1}$ , and that is exactly where the hypothesis  $a \neq 0$  and the field axiom (F8) are spent.

*Proof.*  $(\Leftarrow)$  If  $a = 0$ , then  $a\mathbf{v} = \mathbf{0}$  by [Theorem 1.1.5](#). If  $\mathbf{v} = \mathbf{0}$ , then  $a\mathbf{v} = \mathbf{0}$  by [Theorem 1.1.6](#).

$(\Rightarrow)$  Suppose  $a\mathbf{v} = \mathbf{0}$ . If  $a = 0$  there is nothing to prove, so suppose  $a \neq 0$ . By (F8),  $a$  has an inverse  $a^{-1} \in F$ . Then

$$\mathbf{v} = 1\mathbf{v} = (a^{-1}a)\mathbf{v} = a^{-1}(a\mathbf{v}) = a^{-1}\mathbf{0} = \mathbf{0},$$

by (VS8), (F8) with (F6), (VS7), the hypothesis, and [Theorem 1.1.6](#). This shows  $a = 0$  or  $\mathbf{v} = \mathbf{0}$ .



**Warning. The zero product law needs division by scalars.** Suppose we allowed integer “scalars” and let  $V = \mathbb{F}_2 = \{0, 1\}$  with its own addition, where  $n \cdot v$  is  $v$  for odd  $n$  and  $0$  for even  $n$  (for  $n \geq 0$  this is  $v + \cdots + v$  with  $n$  terms). Every rule of the shape (VS1)–(VS8) still holds. But  $2 \cdot 1 = 1 + 1 = 0$ , although  $2 \neq 0$  in  $\mathbb{Z}$  and  $1 \neq 0$  in  $V$ . The proof above breaks exactly where it multiplies by  $a^{-1}$ : the integer  $2$  has no inverse in  $\mathbb{Z}$ . This is one reason vector spaces are defined **over a field**.

As a first use, here is a fact about sizes. Let  $\mathbf{v} \neq \mathbf{0}$  be a vector in a space over  $\mathbb{R}$ . If  $a\mathbf{v} = b\mathbf{v}$ , then  $(a - b)\mathbf{v} = a\mathbf{v} - b\mathbf{v} = \mathbf{0}$  by (VS6) and Theorem 1.1.8, so  $a = b$  by Theorem 1.1.9. So the multiples  $a\mathbf{v}$ ,  $a \in \mathbb{R}$ , are all different, and a real vector space other than  $\{\mathbf{0}\}$  has infinitely many elements (Exercise (C2: How Big Is a Vector Space?) treats a general field). This is our first glimpse of the question at the heart of this chapter: all these spaces are infinite sets, so how should we measure their size? The next section starts on the answer by looking at the vector spaces that sit inside a given one.

## 1.1.7 Exercises

### 1.1.7.1 A. Check your understanding

#### Exercise A1.

- State the eight axioms of a vector space over a field  $F$ , and say which of them involve scalars.
- Identify the zero vector in each of:  $M_{2 \times 3}(\mathbb{R})$ ;  $\mathbb{R}[x]_{\leq 2}$ ;  $\mathbb{R}^{\mathbb{R}}$ ; the space  $\mathbb{R}_{>0}$  of Example (Positive Reals with Exotic Operations).
- True or false: a vector space can have exactly two elements. Justify your answer.
- True or false: in every vector space,  $\mathbf{v} + \mathbf{v} = \mathbf{0}$  implies  $\mathbf{v} = \mathbf{0}$ . Justify your answer.
- True or false: if  $a\mathbf{v} = b\mathbf{v}$  and  $\mathbf{v} \neq \mathbf{0}$ , then  $a = b$ . Justify your answer.

*Solution.*

- See Definition 1.1.1. The axioms (VS5)–(VS8) involve scalars; (VS1)–(VS4) involve only addition.
- The zero matrix of size  $2 \times 3$ ; the zero polynomial; the zero function  $t \mapsto 0$ ; the number  $1$  (Example (Positive Reals with Exotic Operations)).
- True. Let  $F = \mathbb{F}_2$ . By Example (The Standard Vector Spaces) (a),  $\mathbb{F}_2^1 = \mathbb{F}_2 = \{0, 1\}$  is a vector space over  $\mathbb{F}_2$ , and it has exactly two elements.
- False. In the vector space  $\mathbb{F}_2$  over  $\mathbb{F}_2$  from (c), the vector  $1 \neq 0$  satisfies  $1 + 1 = 0$ . (Over  $\mathbb{R}$  the statement is true:  $\mathbf{v} + \mathbf{v} = (1 + 1)\mathbf{v} = 2\mathbf{v}$  by (VS8) and (VS6), and  $2 \neq 0$ , so  $\mathbf{v} = \mathbf{0}$  by Theorem 1.1.9.)
- True. By (VS6) and Theorem 1.1.8,  $(a - b)\mathbf{v} = a\mathbf{v} + (-b)\mathbf{v} = a\mathbf{v} - b\mathbf{v} = \mathbf{0}$ . Since  $\mathbf{v} \neq \mathbf{0}$ , Theorem 1.1.9 gives  $a - b = 0$ , so  $a = b$ .

### 1.1.7.2 B. Practice

**Exercise B1: Which Are Vector Spaces?.**

Determine which of the following are vector spaces. Justify your answer.

- (a)  $\mathbb{R}^2$  over  $\mathbb{R}$ , with the usual scalar multiplication but with addition  $(x, y) + (x', y') := (x + x', 0)$ .
- (b) The set of upper triangular matrices in  $M_2(\mathbb{R})$ , over  $\mathbb{R}$ , with the matrix operations.
- (c)  $\{p \in \mathbb{R}[x] : p(1) = 0\}$ , over  $\mathbb{R}$ , with the operations of  $\mathbb{R}[x]$ .
- (d)  $\{p \in \mathbb{R}[x] : p(1) = 1\}$ , over  $\mathbb{R}$ , with the operations of  $\mathbb{R}[x]$ .
- (e)  $\mathbb{R}^2$  over  $\mathbb{C}$ , with  $c(x, y) := (cx, cy)$  and the usual addition.

*Solution.*

- (a) Not a vector space: (VS3) fails. Suppose  $\mathbf{0} = (z_1, z_2)$  satisfied (VS3). Then  $(0, 1) + (z_1, z_2) = (z_1, 0)$  would equal  $(0, 1)$ , which is impossible since the second entries differ.
- (b) A vector space. Let  $\mathbf{A} = \begin{pmatrix} a & b \\ 0 & d \end{pmatrix}$  and  $\mathbf{B} = \begin{pmatrix} a' & b' \\ 0 & d' \end{pmatrix}$  be upper triangular and  $c \in \mathbb{R}$ . Then  $\mathbf{A} + \mathbf{B} = \begin{pmatrix} a + a' & b + b' \\ 0 & d + d' \end{pmatrix}$  and  $c\mathbf{A} = \begin{pmatrix} ca & cb \\ 0 & cd \end{pmatrix}$  are upper triangular, so the operations land in the set. The zero matrix is upper triangular, and so is  $-\mathbf{A} = (-1)\mathbf{A}$ . The identities (VS1), (VS2), (VS5)–(VS8) hold for all matrices in  $M_2(\mathbb{R})$  by [Example \(The Standard Vector Spaces\)](#) (b), hence for upper triangular ones.
- (c) A vector space. Let  $U = \{p \in \mathbb{R}[x] : p(1) = 0\}$ ,  $p, q \in U$  and  $c \in \mathbb{R}$ . By [Theorem 0.8.5](#),  $(p + q)(1) = p(1) + q(1) = 0$ . Also  $cp$  is the product of the constant polynomial  $c$  with  $p$ , since by the product formula of [Definition 0.8.2](#) that product has coefficients  $ca_k$ . So  $(cp)(1) = cp(1) = 0$ . Hence  $p + q, cp \in U$ . The zero polynomial has value 0 at 1, and  $-p = (-1)p \in U$  by closure under scaling. The identities hold in  $\mathbb{R}[x]$  by [Example \(The Standard Vector Spaces\)](#) (c), hence in  $U$ .
- (d) Not a vector space: addition leaves the set. The constant polynomial 1 lies in the set, but  $(1 + 1)(1) = 2 \neq 1$ , so  $1 + 1$  does not.
- (e) Not a vector space: scalar multiplication leaves the set. We have  $i \in \mathbb{C}$  and  $(1, 0) \in \mathbb{R}^2$ , but  $i(1, 0) = (i, 0) \notin \mathbb{R}^2$ . So the rule is not a function  $\mathbb{C} \times \mathbb{R}^2 \rightarrow \mathbb{R}^2$ .

**Exercise B2: More Rules for Signs.**

Let  $V$  be a vector space over  $F$ , let  $a, b \in F$  and  $\mathbf{u}, \mathbf{v} \in V$ . Prove that:

- (a)  $a(-\mathbf{v}) = -(a\mathbf{v})$ ;
- (b)  $(-a)(-\mathbf{v}) = a\mathbf{v}$ ;
- (c)  $a(\mathbf{u} - \mathbf{v}) = a\mathbf{u} - a\mathbf{v}$  and  $(a - b)\mathbf{v} = a\mathbf{v} - b\mathbf{v}$ .

*Hint: to show a vector equals  $-(a\mathbf{v})$ , use [Theorem 1.1.4](#).*

*Solution.*

- (a) By (VS5), (VS4) and [Theorem 1.1.6](#),

$$a\mathbf{v} + a(-\mathbf{v}) = a(\mathbf{v} + (-\mathbf{v})) = a\mathbf{0} = \mathbf{0}.$$

So  $a(-\mathbf{v})$  is an additive inverse of  $a\mathbf{v}$ . By [Theorem 1.1.4](#),  $a(-\mathbf{v}) = -(a\mathbf{v})$ .

- (b) By (a) applied with the scalar  $-a$ , then [Theorem 1.1.8](#), then the remark after [Definition 1.1.3](#),

$$(-a)(-\mathbf{v}) = -((-a)\mathbf{v}) = -(-(a\mathbf{v})) = a\mathbf{v}.$$

- (c) By [Definition 1.1.3](#), (VS5) and (a),

$$a(\mathbf{u} - \mathbf{v}) = a(\mathbf{u} + (-\mathbf{v})) = a\mathbf{u} + a(-\mathbf{v}) = a\mathbf{u} + (-(a\mathbf{v})) = a\mathbf{u} - a\mathbf{v}.$$

Similarly, since  $a - b = a + (-b)$  in  $F$ , (VS6) and [Theorem 1.1.8](#) give  $(a - b)\mathbf{v} = a\mathbf{v} + (-b)\mathbf{v} = a\mathbf{v} + (-(b\mathbf{v})) = a\mathbf{v} - b\mathbf{v}$ .

### Exercise B3: Finishing the Positive Reals.

Let  $V = \mathbb{R}_{>0}$  with the operations  $\boxplus$  and  $\boxminus$  of [Example \(Positive Reals with Exotic Operations\)](#).

- (a) Verify (VS1), (VS2), (VS5) and (VS7). Hence  $V$  is a vector space over  $\mathbb{R}$ .
- (b) Compute the vector  $(2 \boxminus 3) \boxplus ((-1) \boxminus 6)$ .
- (c) Find the vector  $-(2 \boxminus 3)$ , and check your answer against [Theorem 1.1.8](#).

*Solution.*

- (a) Let  $u, v, w \in V$  and  $a, b \in \mathbb{R}$ . By the laws of real multiplication and of exponents for positive bases:
  - (VS1):  $u \boxplus v = uv = vu = v \boxplus u$ .
  - (VS2):  $(u \boxplus v) \boxplus w = (uv)w = u(vw) = u \boxplus (v \boxplus w)$ .
  - (VS5):  $a \boxminus (u \boxplus v) = (uv)^a = u^a v^a = (a \boxminus u) \boxplus (a \boxminus v)$ .
  - (VS7):  $a \boxminus (b \boxminus v) = (v^b)^a = v^{ab} = (ab) \boxminus v$ .

Together with closure, (VS3), (VS4), (VS6) and (VS8) from [Example \(Positive Reals with Exotic Operations\)](#), all eight axioms hold. Hence  $V$  is a vector space over  $\mathbb{R}$ .

- (b)  $2 \boxminus 3 = 3^2 = 9$  and  $(-1) \boxminus 6 = 6^{-1} = \frac{1}{6}$ . So the vector is  $9 \boxplus \frac{1}{6} = 9 \cdot \frac{1}{6} = \frac{3}{2}$ .
- (c) By [Example \(Positive Reals with Exotic Operations\)](#), the additive inverse of a vector  $v$  is  $1/v$ . Since  $2 \boxminus 3 = 9$ , we get  $-(2 \boxminus 3) = \frac{1}{9}$ . By [Theorem 1.1.8](#), this should equal  $(-2) \boxminus 3 = 3^{-2} = \frac{1}{9}$ , and it does.

### 1.1.7.3 C. Going deeper

#### Exercise C1: Commutativity Comes for Free.

Let  $V$  be a set with operations  $+: V \times V \rightarrow V$  and  $\cdot: F \times V \rightarrow V$  satisfying (VS2)–(VS8), but **not necessarily** (VS1). Fix  $\mathbf{0}$  as in (VS3).

- (a) Let  $\mathbf{v} \in V$  and  $\mathbf{v}'$  as in (VS4). Put  $\mathbf{w} = \mathbf{v}' + \mathbf{v}$ . Show that  $\mathbf{w} + \mathbf{w} = \mathbf{w}$ , and deduce that  $\mathbf{w} = \mathbf{0}$ .
- (b) Show that  $\mathbf{0} + \mathbf{v} = \mathbf{v}$  for every  $\mathbf{v} \in V$ .
- (c) Deduce that both cancellation laws hold:  $\mathbf{u} + \mathbf{v} = \mathbf{u} + \mathbf{w}$  implies  $\mathbf{v} = \mathbf{w}$ , and  $\mathbf{v} + \mathbf{u} = \mathbf{w} + \mathbf{u}$  implies  $\mathbf{v} = \mathbf{w}$ .
- (d) By expanding  $(1+1)(\mathbf{u} + \mathbf{v})$  in two ways, prove that  $\mathbf{u} + \mathbf{v} = \mathbf{v} + \mathbf{u}$  for all  $\mathbf{u}, \mathbf{v} \in V$ .

*Hint: in (a), add an additive inverse of  $\mathbf{w}$  on the right. The proof of Theorem 1.1.1 used (VS1), so it cannot be quoted in (c).*

*Solution.* Throughout we use only (VS2)–(VS8), and we drop brackets in sums whenever (VS2) allows it.

- (a) Since  $\mathbf{v} + \mathbf{v}' = \mathbf{0}$  and  $\mathbf{v}' + \mathbf{0} = \mathbf{v}'$  by (VS3),

$$\mathbf{w} + \mathbf{w} = \mathbf{v}' + (\mathbf{v} + \mathbf{v}') + \mathbf{v} = \mathbf{v}' + \mathbf{0} + \mathbf{v} = \mathbf{v}' + \mathbf{v} = \mathbf{w}.$$

By (VS4) choose  $\mathbf{w}'$  with  $\mathbf{w} + \mathbf{w}' = \mathbf{0}$ . Adding  $\mathbf{w}'$  on the right of  $\mathbf{w} + \mathbf{w} = \mathbf{w}$  gives  $\mathbf{w} + (\mathbf{w} + \mathbf{w}') = \mathbf{w} + \mathbf{w}'$ , that is,  $\mathbf{w} + \mathbf{0} = \mathbf{0}$ . By (VS3),  $\mathbf{w} = \mathbf{0}$ . So  $\mathbf{v}' + \mathbf{v} = \mathbf{0}$  as well.

- (b) Let  $\mathbf{v} \in V$  and  $\mathbf{v}'$  as in (VS4). By (a), (VS2) and (VS3),

$$\mathbf{0} + \mathbf{v} = (\mathbf{v} + \mathbf{v}') + \mathbf{v} = \mathbf{v} + (\mathbf{v}' + \mathbf{v}) = \mathbf{v} + \mathbf{0} = \mathbf{v}.$$

- (c) Suppose  $\mathbf{u} + \mathbf{v} = \mathbf{u} + \mathbf{w}$ , and let  $\mathbf{u}'$  be as in (VS4). Adding  $\mathbf{u}'$  on the left and using (VS2),  $(\mathbf{u}' + \mathbf{u}) + \mathbf{v} = (\mathbf{u}' + \mathbf{u}) + \mathbf{w}$ . By (a),  $\mathbf{u}' + \mathbf{u} = \mathbf{0}$ , so  $\mathbf{0} + \mathbf{v} = \mathbf{0} + \mathbf{w}$ , and (b) gives  $\mathbf{v} = \mathbf{w}$ . Now suppose  $\mathbf{v} + \mathbf{u} = \mathbf{w} + \mathbf{u}$ . Adding  $\mathbf{u}'$  on the right and using (VS2),  $\mathbf{v} + (\mathbf{u} + \mathbf{u}') = \mathbf{w} + (\mathbf{u} + \mathbf{u}')$ , so  $\mathbf{v} + \mathbf{0} = \mathbf{w} + \mathbf{0}$ , and (VS3) gives  $\mathbf{v} = \mathbf{w}$ .
- (d) Let  $\mathbf{u}, \mathbf{v} \in V$ . On the one hand, by (VS6) and (VS8),

$$(1+1)(\mathbf{u} + \mathbf{v}) = 1(\mathbf{u} + \mathbf{v}) + 1(\mathbf{u} + \mathbf{v}) = \mathbf{u} + \mathbf{v} + \mathbf{u} + \mathbf{v}.$$

On the other hand, by (VS5), then (VS6) and (VS8),

$$(1+1)(\mathbf{u} + \mathbf{v}) = (1+1)\mathbf{u} + (1+1)\mathbf{v} = \mathbf{u} + \mathbf{u} + \mathbf{v} + \mathbf{v}.$$

Hence  $\mathbf{u} + (\mathbf{v} + \mathbf{u}) + \mathbf{v} = \mathbf{u} + (\mathbf{u} + \mathbf{v}) + \mathbf{v}$ , grouping by (VS2). Canceling  $\mathbf{u}$  on the left and then  $\mathbf{v}$  on the right, by (c), gives  $\mathbf{v} + \mathbf{u} = \mathbf{u} + \mathbf{v}$ . This proves (VS1).

### Exercise C2: How Big Is a Vector Space?.

- (a) Let  $F$  be an infinite field and  $V \neq \{\mathbf{0}\}$  a vector space over  $F$ . Prove that  $V$  is an infinite set.
- (b) Let  $p$  be a prime and  $n \geq 1$ . How many elements does the vector space  $\mathbb{F}_p^n$  over  $\mathbb{F}_p$  have? Explain why this does not contradict (a).

*Hint: for (a), fix  $\mathbf{v} \neq \mathbf{0}$  and consider the function  $F \rightarrow V$ ,  $a \mapsto a\mathbf{v}$ .*

*Solution.*



- (a) Since  $V \neq \{\mathbf{0}\}$  and  $\mathbf{0} \in V$ , there is  $\mathbf{v} \in V$  with  $\mathbf{v} \neq \mathbf{0}$ . Consider  $\varphi: F \rightarrow V$ ,  $\varphi(a) = a\mathbf{v}$ . Suppose  $\varphi(a) = \varphi(b)$ , so  $a\mathbf{v} = b\mathbf{v}$ . By [Exercise \(B2: More Rules for Signs\)](#) (c),  $(a - b)\mathbf{v} = a\mathbf{v} - b\mathbf{v} = \mathbf{0}$ . Since  $\mathbf{v} \neq \mathbf{0}$ , [Theorem 1.1.9](#) gives  $a - b = 0$ , so  $a = b$ . Therefore  $\varphi$  is injective. An injective function from an infinite set has an infinite image, so  $V$ , which contains the image of  $\varphi$ , is infinite.
- (b) An element of  $\mathbb{F}_p^n$  is a list  $(x_1, \dots, x_n)$  with each  $x_i$  chosen freely from the  $p$  elements of  $\mathbb{F}_p$ . So  $\mathbb{F}_p^n$  has  $p^n$  elements. There is no contradiction, because (a) assumes the field is infinite and  $\mathbb{F}_p$  is finite. The proof of (a) still shows that  $a \mapsto a\mathbf{v}$  is injective, so a non-zero space over  $\mathbb{F}_p$  has at least  $p$  elements.

### Exercise C3: Shrinking the Field.

Let  $F$  be a field and let  $K \subseteq F$  be a subset with  $0, 1 \in K$  that is closed under addition, multiplication, negatives, and inverses of non-zero elements. (So  $K$  is itself a field with the operations of  $F$ ; for example  $\mathbb{Q} \subseteq \mathbb{R} \subseteq \mathbb{C}$ .)

- (a) Let  $V$  be a vector space over  $F$ . Prove that  $V$ , with the same addition and with scalar multiplication restricted to  $K \times V$ , is a vector space over  $K$ .
- (b) Deduce that  $\mathbb{C}^n$  is a vector space over  $\mathbb{R}$  and over  $\mathbb{Q}$ .
- (c) Explain why  $\mathbb{R}$ , with its usual addition and multiplication, is **not** a vector space over  $\mathbb{C}$ .

*Solution.*

- (a) The addition is unchanged, so it is a function  $V \times V \rightarrow V$  satisfying (VS1)–(VS4). Scalar multiplication  $F \times V \rightarrow V$  restricts to a function  $K \times V \rightarrow V$ , since  $K \subseteq F$ . The axioms (VS5)–(VS7) are statements about **all** scalars in  $F$ , so they hold in particular for all scalars in  $K$ ; the operations  $a + b$  and  $ab$  appearing in them are the same in  $K$  as in  $F$ . For (VS8), the identity of  $K$  is the identity 1 of  $F$ , and  $1\mathbf{v} = \mathbf{v}$  holds in  $V$  over  $F$ . Hence  $V$  is a vector space over  $K$ .
- (b)  $\mathbb{C}^n$  is a vector space over  $\mathbb{C}$  by [Example \(The Standard Vector Spaces\)](#) (a). The subsets  $\mathbb{R} \subseteq \mathbb{C}$  and  $\mathbb{Q} \subseteq \mathbb{C}$  contain 0 and 1 and are closed under the four operations, so (a) applies to both.
- (c) The only candidate for scalar multiplication is  $(c, t) \mapsto ct$ , the product in  $\mathbb{C}$ . But it does not land in  $\mathbb{R}$ :  $i \in \mathbb{C}$  and  $1 \in \mathbb{R}$ , while  $i \cdot 1 = i \notin \mathbb{R}$ . So there is no function  $\mathbb{C} \times \mathbb{R} \rightarrow \mathbb{R}$  given by this rule, and  $\mathbb{R}$  is not a vector space over  $\mathbb{C}$  with these operations.

## 1.2 Subspaces

After defining a structure, the next step is to look at the smaller copies of it that sit inside. In the previous section we showed three times, for  $F[x]_{\leq n}$ ,  $\mathbb{Q}(\sqrt{2})$  and the solutions of  $f'' + f = 0$ , that a subset of a known vector space is itself a vector space, and each time the same few checks did the work. This section turns that pattern into a theorem, the subspace test, and makes its three checks a routine we will use for the rest of the book.

## 1.2.1 Spaces inside spaces

Consider the set

$$P = \{(x, y, z) \in \mathbb{R}^3 : x + y - z = 0\},$$

a plane through the origin in  $\mathbb{R}^3$ . It contains  $(1, 0, 1)$  and  $(0, 1, 1)$ , and also their sum  $(1, 1, 2)$ , since  $1 + 1 - 2 = 0$ . Stretching a vector of  $P$  keeps it in the plane. So  $P$ , with the addition and scalar multiplication of  $\mathbb{R}^3$ , looks like a vector space in its own right.

To prove it we could check all eight axioms again. That is wasteful, and the previous section already showed why. The axioms (VS1), (VS2) and (VS5)–(VS8) are equations that hold for **all** vectors of  $\mathbb{R}^3$ , so they hold in particular for the vectors of  $P$ . What can go wrong is different: a sum or a multiple might leave  $P$ , and the zero vector or an additive inverse might not lie in  $P$ . The question of this section is which of these checks are really needed.

*A subspace is a subset that is a vector space in its own right, using the same addition and scalar multiplication.*

### Definition 1.2.1 — Subspace

Let  $V$  be a vector space over  $F$ . A subset  $U \subseteq V$  is a **subspace** of  $V$  if  $U$  is **itself a vector space over  $F$**  when it is given the addition and the scalar multiplication of  $V$ .

For that sentence to make sense,  $U$  must first be closed under those two operations:

$$\mathbf{u} + \mathbf{w} \in U \quad \text{and} \quad a\mathbf{u} \in U \quad \text{for all } \mathbf{u}, \mathbf{w} \in U \text{ and all } a \in F.$$

In words: take the plus and the dot that  $V$  already has, use them only on vectors of  $U$ , and ask whether  $U$  is a vector space with them. Nothing new is defined;  $U$  borrows everything from  $V$ . The displayed condition is what makes the borrowing possible, because an operation on  $U$  must land **in**  $U$ : if  $\mathbf{u}, \mathbf{w} \in U$  but  $\mathbf{u} + \mathbf{w} \notin U$ , then addition is not an operation on  $U$  at all, and the question of the axioms never arises. Once the condition holds, the eight axioms are asked of  $U$  exactly as they were asked of  $V$  in the previous section. In particular  $U$  must contain **a** zero vector of its own, and every vector of  $U$  must have an additive inverse **lying in**  $U$ . The field does not change: a subspace of a space over  $F$  is again a space over  $F$ .

Two words of the definition carry all the weight. **Itself a vector space** is the whole content. **The same operations** is what distinguishes a subspace from a subset that merely happens to be a vector space under some other rule:  $\mathbb{R}_{>0}$  with the operations of [Example \(Positive Reals with Exotic Operations\)](#) is a vector space and a subset of  $\mathbb{R}$ , but it is not a subspace of  $\mathbb{R}$ , because those are not the operations of  $\mathbb{R}$ .

There is a question hidden in “a zero vector of its own”. Could  $U$  have a zero vector different from the zero vector of  $V$ ? The proof of the subspace test below shows that it cannot.

The two extreme cases come first.

#### Example. (Trivial Subspaces)

Let  $V$  be a vector space over  $F$ . Show that  $V$  and  $\{\mathbf{0}\}$  are subspaces of  $V$ , and that the empty set  $\emptyset$  is not.

*Solution.* For  $V$  itself, the restricted operations are the operations of  $V$ , and  $V$  is a vector space by assumption.

For  $\{\mathbf{0}\}$ , the only sum is  $\mathbf{0} + \mathbf{0} = \mathbf{0}$  by (VS3), and  $a\mathbf{0} = \mathbf{0}$  for every  $a \in F$  by [Theorem 1.1.6](#). So both operations stay in  $\{\mathbf{0}\}$ , and  $\{\mathbf{0}\}$  with them is the zero space, a vector space as noted in the previous section.

For  $\emptyset$ , axiom (VS3) demands that some element  $\mathbf{0}$  exists in the set. The empty set has no elements, so (VS3) fails and  $\emptyset$  is not a subspace.

The subspace  $\{\mathbf{0}\}$  is the **zero subspace**. A subspace other than  $V$  is called **proper**. The zero subspace is small, but it is the answer to many questions later: the solutions of a system with only the trivial solution, or the intersection of two lines through the origin that are not equal.

## 1.2.2 The subspace test

The following theorem says that three checks suffice. Two of them are the closure conditions of the definition, and the third replaces all of the axioms.

### Theorem 1.2.1 — Subspace Test

Let  $V$  be a vector space over  $F$  and let  $U \subseteq V$ . Then  $U$  is a subspace of  $V$  if and only if the following three conditions hold.

- (1)  $\mathbf{0} \in U$ , where  $\mathbf{0}$  is the zero vector of  $V$ .
- (2)  $U$  is **closed under addition**:  $\mathbf{u} + \mathbf{w} \in U$  for all  $\mathbf{u}, \mathbf{w} \in U$ .
- (3)  $U$  is **closed under scalar multiplication**:  $a\mathbf{u} \in U$  for all  $a \in F$  and  $\mathbf{u} \in U$ .

*Idea.* Sort the eight axioms into two kinds. Six of them, (VS1), (VS2) and (VS5)–(VS8), say “an equation holds **for all** vectors”; they are true in  $V$ , so they are automatically true for the fewer vectors in  $U$ . The other two, (VS3) and (VS4), say “**there exists** a vector”, and a witness must be found **inside**  $U$ . Condition (1) supplies the zero. For inverses, the results of the previous section give  $-\mathbf{u} = (-1)\mathbf{u}$ , so closure under scaling supplies them for free. For  $(\Rightarrow)$ , the only non-obvious point is that the zero vector of  $U$  is the zero vector of  $V$ ; cancellation in  $V$  settles it.

*Proof.*  $(\Rightarrow)$  Suppose  $U$  is a subspace of  $V$ . Conditions (2) and (3) are the closure condition of [Definition 1.2.1](#). By (VS3) in  $U$ , there is  $\mathbf{z} \in U$  with  $\mathbf{u} + \mathbf{z} = \mathbf{u}$  for every  $\mathbf{u} \in U$ ; in particular  $\mathbf{z} + \mathbf{z} = \mathbf{z}$ . By (VS3) in  $V$ , also  $\mathbf{z} + \mathbf{0} = \mathbf{z}$ . Hence  $\mathbf{z} + \mathbf{z} = \mathbf{z} + \mathbf{0}$ , and [Theorem 1.1.1](#) in  $V$  gives  $\mathbf{z} = \mathbf{0}$ . Therefore  $\mathbf{0} \in U$ , which is (1).

$(\Leftarrow)$  Suppose (1), (2) and (3) hold. By (2) and (3), addition and scalar multiplication of  $V$  restrict to functions  $U \times U \rightarrow U$  and  $F \times U \rightarrow U$ . We check the axioms for  $U$  with these operations.

- (VS1), (VS2), (VS5), (VS6), (VS7) and (VS8) state that certain equations hold for all vectors and scalars. They hold for all vectors of  $V$ , since  $V$  is a vector space, so in particular for all vectors of  $U$ .
- (VS3): by (1),  $\mathbf{0} \in U$ , and  $\mathbf{u} + \mathbf{0} = \mathbf{u}$  for every  $\mathbf{u} \in U$  by (VS3) in  $V$ .

- (VS4): let  $\mathbf{u} \in U$ . By (3),  $(-1)\mathbf{u} \in U$ . By Theorem 1.1.7,  $(-1)\mathbf{u} = -\mathbf{u}$ , so  $\mathbf{u} + (-1)\mathbf{u} = \mathbf{0}$ . Thus  $\mathbf{u}$  has an additive inverse in  $U$ .

Hence  $U$  is a vector space over  $F$  with the operations of  $V$ , that is, a subspace of  $V$ . ■

From now on we never check eight axioms for a subset of a known vector space. We check three things, always in the same order and with the same wording. This is the first named move of the chapter.

**The three-check move.** To show that a subset  $U$  of a vector space  $V$  is a subspace:

- make sure  $U$  is a subset of  $V$  (what are its elements, and do they lie in  $V$ ?);
- **(1) Zero vector.** Show  $\mathbf{0} \in U$  by checking that  $\mathbf{0}$  satisfies the condition defining  $U$ . End with “Therefore  $\mathbf{0} \in U$ .”
- **(2) Closed under addition.** Let  $\mathbf{u}, \mathbf{w} \in U$ . Write down what membership in  $U$  means for each, then regroup  $\mathbf{u} + \mathbf{w}$  so that those conditions appear. End with “This shows that  $U$  is closed under addition.”
- **(3) Closed under scalar multiplication.** Let  $a \in F$  and  $\mathbf{u} \in U$ , and do the same for  $a\mathbf{u}$ . End with “This shows that  $U$  is closed under scalar multiplication.”
- Conclude: “Hence  $U$  is a subspace of  $V$  by the subspace test (Theorem 1.2.1).”

To show that  $U$  is **not** a subspace, one failure suffices. Check (1) first, since it is the cheapest, and if it holds, look for a concrete pair of vectors or a concrete scalar that leads out of  $U$ .

*Remark.* Condition (1) may be replaced by “ $U$  is non-empty”. Indeed, if  $U$  contains some  $\mathbf{u}$ , then (3) gives  $0\mathbf{u} \in U$ , and  $0\mathbf{u} = \mathbf{0}$  by Theorem 1.1.5. We still prefer (1): the zero vector is the easiest element to test, and when it is missing we have a refutation in one line.

*Warning.* **The empty set is never a subspace.** It is tempting to check only (2) and (3). But  $\emptyset$  satisfies both, vacuously: there are no vectors in it to add or scale. It fails only (1). So whichever version of the test you use, do not skip the first condition, and remember that “ $\mathbf{0} \in U$ ” and “ $U$  is non-empty” are interchangeable only in the presence of (3).

### 1.2.3 Examples: the three-check move

Every example below follows the move exactly. The first is the model.

#### Example. (A Line through the Origin)

Show that  $U = \{(x, y) \in \mathbb{R}^2 : x + 2y = 0\}$  is a subspace of  $\mathbb{R}^2$ .

*Solution.* By definition,  $U$  is a subset of  $\mathbb{R}^2$ .

**(1) Zero vector.** The zero vector  $(0, 0)$  satisfies  $0 + 2 \cdot 0 = 0$ . Therefore  $\mathbf{0} \in U$ .

**(2) Closed under addition.** Let  $\mathbf{u} = (x_1, y_1)$  and  $\mathbf{w} = (x_2, y_2)$  be in  $U$ , so that

$x_1 + 2y_1 = 0$  and  $x_2 + 2y_2 = 0$ . Then  $\mathbf{u} + \mathbf{w} = (x_1 + x_2, y_1 + y_2)$ , and

$$(x_1 + x_2) + 2(y_1 + y_2) = (x_1 + 2y_1) + (x_2 + 2y_2) = 0 + 0 = 0.$$

Therefore  $\mathbf{u} + \mathbf{w} \in U$ . This shows that  $U$  is closed under addition.

**(3) Closed under scalar multiplication.** Let  $a \in \mathbb{R}$  and  $\mathbf{u} = (x, y) \in U$ , so that  $x + 2y = 0$ . Then  $a\mathbf{u} = (ax, ay)$ , and

$$ax + 2(ay) = a(x + 2y) = a \cdot 0 = 0.$$

Therefore  $a\mathbf{u} \in U$ . This shows that  $U$  is closed under scalar multiplication.

Hence  $U$  is a subspace of  $\mathbb{R}^2$  by the subspace test ([Theorem 1.2.1](#)).

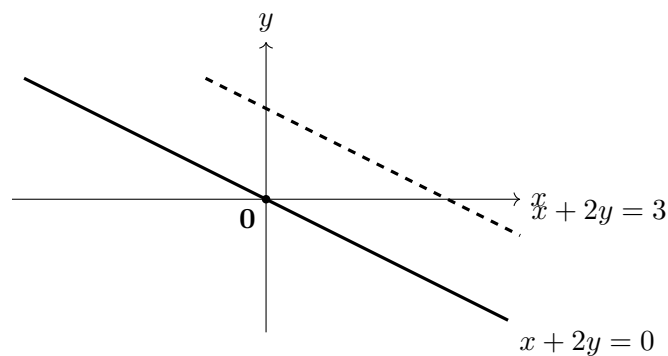
Change the right-hand side from 0 to 3, and everything breaks.

**Example. (A Line Missing the Origin)**

Show that  $L = \{(x, y) \in \mathbb{R}^2 : x + 2y = 3\}$  is **not** a subspace of  $\mathbb{R}^2$ .

*Solution.* Condition (1) fails: the zero vector has  $0 + 2 \cdot 0 = 0 \neq 3$ , so  $\mathbf{0} \notin L$ . By the subspace test ([Theorem 1.2.1](#)),  $L$  is not a subspace of  $\mathbb{R}^2$ .

The other two conditions fail as well, as we saw in the previous section:  $(3, 0) + (1, 1) = (4, 1) \notin L$ , and  $2(3, 0) = (6, 0) \notin L$ . One failure is enough, though, and (1) is the cheapest to find.



The solid line  $U$  passes through the origin; the dashed line  $L$  is parallel to it and misses the origin. The line  $L$  is a translate of a subspace, a kind of set we will study in Chapter 3, but it is not a subspace.

The line  $U$  is the set of solutions of one homogeneous equation. The same three checks work for any number of homogeneous equations, written as one matrix equation.

**Example. (Solutions of a Homogeneous System)**

Let  $\mathbf{A} \in M_{m \times n}(F)$ . Show that  $N = \{\mathbf{x} \in F^n : \mathbf{A}\mathbf{x} = \mathbf{0}\}$  is a subspace of  $F^n$ .

*Solution.* By definition,  $N$  is a subset of  $F^n$ . Recall that  $F^n = M_{n \times 1}(F)$ , so  $\mathbf{A}\mathbf{x} \in F^m$  is a matrix product, and we may use [Theorem 0.9.4](#).

**(1) Zero vector.** By the zero rule of [Theorem 0.9.4](#),  $\mathbf{A}\mathbf{0} = \mathbf{0}$ , where the first  $\mathbf{0}$  is the zero column in  $F^n$  and the second is the zero column in  $F^m$ . Therefore  $\mathbf{0} \in N$ .

**(2) Closed under addition.** Let  $\mathbf{x}, \mathbf{y} \in N$ , so that  $\mathbf{A}\mathbf{x} = \mathbf{0}$  and  $\mathbf{A}\mathbf{y} = \mathbf{0}$ . By distributivity ([Theorem 0.9.4](#)),

$$\mathbf{A}(\mathbf{x} + \mathbf{y}) = \mathbf{A}\mathbf{x} + \mathbf{A}\mathbf{y} = \mathbf{0} + \mathbf{0} = \mathbf{0}.$$

Therefore  $\mathbf{x} + \mathbf{y} \in N$ . This shows that  $N$  is closed under addition.

**(3) Closed under scalar multiplication.** Let  $a \in F$  and  $\mathbf{x} \in N$ . By the scalar rule of [Theorem 0.9.4](#) and [Theorem 1.1.6](#) in  $F^m$ ,

$$\mathbf{A}(a\mathbf{x}) = a(\mathbf{A}\mathbf{x}) = a\mathbf{0} = \mathbf{0}.$$

Therefore  $a\mathbf{x} \in N$ . This shows that  $N$  is closed under scalar multiplication.

Hence  $N$  is a subspace of  $F^n$  by the subspace test ([Theorem 1.2.1](#)).

With  $\mathbf{A} = \begin{pmatrix} 1 & 2 \end{pmatrix}$  and  $F = \mathbb{R}$ , this is [Example \(A Line through the Origin\)](#) again, and with  $\mathbf{A} = \begin{pmatrix} 1 & 1 & -1 \end{pmatrix}$  it is the plane  $P$  from the start of the section. The word **homogeneous** (right-hand side  $\mathbf{0}$ ) is essential: [Example \(A Line Missing the Origin\)](#) is a system with right-hand side 3. This subspace will be called the null space of  $\mathbf{A}$  in Chapter 2.

Next, polynomials. The pair “degree at most  $n$ ” and “degree exactly  $n$ ” is another minimal change. The previous section checked the first of them by hand; here it is as a three-check move.

### Example. (Polynomials of Bounded Degree)

Let  $n \in \mathbb{N}$ . Show that  $F[x]_{\leq n}$  is a subspace of  $F[x]$ .

*Solution.* By [Definition 0.8.4](#),  $F[x]_{\leq n}$  is a subset of  $F[x]$ .

**(1) Zero vector.** The zero polynomial has degree  $-\infty \leq n$  ([Definition 0.8.3](#)). Therefore  $\mathbf{0} \in F[x]_{\leq n}$ .

**(2) Closed under addition.** Let  $p, q \in F[x]_{\leq n}$ , so that  $\deg p \leq n$  and  $\deg q \leq n$ . By [Theorem 0.8.2](#),  $\deg(p + q) \leq \max(\deg p, \deg q) \leq n$ . Therefore  $p + q \in F[x]_{\leq n}$ . This shows that  $F[x]_{\leq n}$  is closed under addition.

**(3) Closed under scalar multiplication.** Let  $c \in F$  and  $p = a_0 + a_1x + \cdots + a_nx^n \in F[x]_{\leq n}$ . Then  $cp = ca_0 + ca_1x + \cdots + ca_nx^n$ , whose coefficients of  $x^k$  for  $k > n$  are all 0. So  $\deg(cp) \leq n$ , and therefore  $cp \in F[x]_{\leq n}$ . This shows that  $F[x]_{\leq n}$  is closed under scalar multiplication.

Hence  $F[x]_{\leq n}$  is a subspace of  $F[x]$  by the subspace test ([Theorem 1.2.1](#)).

### Example. (Polynomials of Exact Degree)

Let  $n \in \mathbb{N}$ . Show that  $F[x]_{=n}$  is **not** a subspace of  $F[x]$ .

*Solution.* Condition (1) fails: the zero polynomial has degree  $-\infty \neq n$ , so  $\mathbf{0} \notin F[x]_{=n}$ . By the subspace test ([Theorem 1.2.1](#)),  $F[x]_{=n}$  is not a subspace of  $F[x]$ .

Even if we added the zero polynomial by hand, condition (2) would still fail for  $n \geq 1$ : the polynomials  $x^n + 1$  and  $-x^n$  have degree exactly  $n$ , but their sum is the constant 1, of degree 0. The leading terms cancel, and nothing in “degree exactly  $n$ ” prevents it.

The same move works in function spaces. Here the facts we need come from calculus.

### Example. (Differentiable Functions)

Let  $D$  be the set of differentiable functions  $\mathbb{R} \rightarrow \mathbb{R}$ . Show that  $D$  is a subspace of  $\mathbb{R}^{\mathbb{R}}$ .

*Solution.* Every element of  $D$  is a function  $\mathbb{R} \rightarrow \mathbb{R}$ , so  $D \subseteq \mathbb{R}^{\mathbb{R}}$ .

**(1) Zero vector.** The zero vector of  $\mathbb{R}^{\mathbb{R}}$  is the zero function ([Example \(The Standard Vector Spaces\)](#)), which is constant and hence differentiable. Therefore  $\mathbf{0} \in D$ .

**(2) Closed under addition.** Let  $f, g \in D$ . By the sum rule of calculus,  $f + g$  is differentiable, with  $(f + g)' = f' + g'$ . Therefore  $f + g \in D$ . This shows that  $D$  is closed under addition.

**(3) Closed under scalar multiplication.** Let  $a \in \mathbb{R}$  and  $f \in D$ . By the constant multiple rule,  $af$  is differentiable, with  $(af)' = af'$ . Therefore  $af \in D$ . This shows that  $D$  is closed under scalar multiplication.

Hence  $D$  is a subspace of  $\mathbb{R}^{\mathbb{R}}$  by the subspace test ([Theorem 1.2.1](#)).

The same three checks show that the continuous functions  $\mathbb{R} \rightarrow \mathbb{R}$  form a subspace of  $\mathbb{R}^{\mathbb{R}}$ , and that  $D$  is a subspace of it. A subspace of a subspace is a subspace, since the operations are the same throughout.

Finally, two families of matrices. Each uses one rule from Chapter 0 for all three checks.

### Example. (Symmetric Matrices)

Show that the set  $S = \{\mathbf{A} \in M_n(F) : \mathbf{A}^\top = \mathbf{A}\}$  of symmetric matrices ([Definition 0.9.11](#)) is a subspace of  $M_n(F)$ .

*Solution.* By definition,  $S$  is a subset of  $M_n(F)$ .

**(1) Zero vector.** Every entry of the zero matrix is 0, so  $(0)_{ji} = 0 = (0)_{ij}$  for all  $i, j$ , and the zero matrix is symmetric. Therefore  $\mathbf{0} \in S$ .

**(2) Closed under addition.** Let  $\mathbf{A}, \mathbf{B} \in S$ , so that  $\mathbf{A}^\top = \mathbf{A}$  and  $\mathbf{B}^\top = \mathbf{B}$ . By [Theorem 0.9.5](#),

$$(\mathbf{A} + \mathbf{B})^\top = \mathbf{A}^\top + \mathbf{B}^\top = \mathbf{A} + \mathbf{B}.$$

Therefore  $\mathbf{A} + \mathbf{B} \in S$ . This shows that  $S$  is closed under addition.

**(3) Closed under scalar multiplication.** Let  $c \in F$  and  $\mathbf{A} \in S$ . By [Theorem 0.9.5](#),  $(c\mathbf{A})^\top = c\mathbf{A}^\top = c\mathbf{A}$ . Therefore  $c\mathbf{A} \in S$ . This shows that  $S$  is closed under scalar multiplication.

Hence  $S$  is a subspace of  $M_n(F)$  by the subspace test (Theorem 1.2.1).

### Example. (Trace-Free Matrices)

Show that the set  $T = \{\mathbf{A} \in M_n(F) : \operatorname{tr} \mathbf{A} = 0\}$  of trace-free matrices is a subspace of  $M_n(F)$ .

*Solution.* By definition,  $T$  is a subset of  $M_n(F)$ .

(1) **Zero vector.** The zero matrix has trace  $0 + \cdots + 0 = 0$ . Therefore  $\mathbf{0} \in T$ .

(2) **Closed under addition.** Let  $\mathbf{A}, \mathbf{B} \in T$ , so that  $\operatorname{tr} \mathbf{A} = \operatorname{tr} \mathbf{B} = 0$ . By Theorem 0.9.6,  $\operatorname{tr}(\mathbf{A} + \mathbf{B}) = \operatorname{tr} \mathbf{A} + \operatorname{tr} \mathbf{B} = 0 + 0 = 0$ . Therefore  $\mathbf{A} + \mathbf{B} \in T$ . This shows that  $T$  is closed under addition.

(3) **Closed under scalar multiplication.** Let  $c \in F$  and  $\mathbf{A} \in T$ . By Theorem 0.9.6,  $\operatorname{tr}(c\mathbf{A}) = c \operatorname{tr} \mathbf{A} = c \cdot 0 = 0$ . Therefore  $c\mathbf{A} \in T$ . This shows that  $T$  is closed under scalar multiplication.

Hence  $T$  is a subspace of  $M_n(F)$  by the subspace test (Theorem 1.2.1).

The pattern in the last two examples, and in Example (Solutions of a Homogeneous System), is the same: the condition defining the set is “some rule that respects sums and scalar multiples sends the element to zero”. Chapter 3 gives such rules a name, linear maps, and this pattern becomes a single theorem about their kernels.

## 1.2.4 Non-examples

Here are two more subsets of  $\mathbb{R}^2$  that fail, each in a different way.

- **The unit circle**  $\{(x, y) : x^2 + y^2 = 1\}$  fails (1), since  $0^2 + 0^2 \neq 1$ . It fails (2) and (3) as well:  $(1, 0) + (0, 1) = (1, 1)$  and  $2(1, 0) = (2, 0)$  are not on the circle.
- **The set**  $Q = \{(x, y) : xy \geq 0\}$ , the first and third quadrants together with both axes. Here (1) holds, since  $0 \cdot 0 = 0 \geq 0$ . Condition (3) holds too: if  $xy \geq 0$  and  $a \in \mathbb{R}$ , then  $(ax)(ay) = a^2xy \geq 0$ . But (2) fails:  $(1, 0)$  and  $(0, -1)$  lie in  $Q$ , while their sum  $(1, -1)$  has  $1 \cdot (-1) = -1 < 0$ . So  $Q$  is not a subspace.

The set  $Q$  shows that (3) does not imply (2). The next check shows that (2) does not imply (3).

*Quick check.* Find a subset of  $\mathbb{R}^2$  that contains  $\mathbf{0}$  and is closed under addition, but is not a subspace. Which condition of the subspace test fails?

*Solution.* Take the closed first quadrant  $U = \{(x, y) : x \geq 0 \text{ and } y \geq 0\}$ . It contains  $(0, 0)$ , and if  $x_1, y_1, x_2, y_2 \geq 0$ , then  $x_1 + x_2 \geq 0$  and  $y_1 + y_2 \geq 0$ , so it is closed under addition. Condition (3) fails:  $(1, 0) \in U$  but  $(-1)(1, 0) = (-1, 0) \notin U$ . Another answer is  $\mathbb{Z}^2$ : it contains  $\mathbf{0}$  and is closed under addition, but  $\frac{1}{2}(1, 0) \notin \mathbb{Z}^2$ .



## 1.2.5 Why this definition

The definition insists on **the same** operations. This is not a formality. The positive reals  $\mathbb{R}_{>0}$  form a vector space over  $\mathbb{R}$  ([Example \(Positive Reals with Exotic Operations\)](#)), and  $\mathbb{R}_{>0}$  is a subset of the vector space  $\mathbb{R}$ . But  $\mathbb{R}_{>0}$  is **not** a subspace of  $\mathbb{R}$ : with the operations of  $\mathbb{R}$ , it does not contain the zero vector 0, and  $(-1) \cdot 1 = -1$  leaves it. Its vector space structure uses different operations, multiplication and powers, and those do not count.

The definition also keeps the field. For example,  $\mathbb{R}^2$  is a subset of  $\mathbb{C}^2$ , and it is closed under addition and under multiplication by **real** scalars. But it is not a subspace of  $\mathbb{C}^2$  over  $\mathbb{C}$ , because  $i(1, 0) = (i, 0) \notin \mathbb{R}^2$ . Condition (3) quantifies over **every** scalar of the field.

Finally, why “(1)  $\mathbf{0} \in U$ ” rather than “ $U$  is non-empty”? The two are equivalent given (3), as the remark after the test explains. We choose the version that is quicker to check and quicker to refute.

## 1.2.6 Intersections of subspaces

Given several subspaces, can we build new ones? The first way is to take the vectors they have in common. Recall from [Definition 0.3.9](#) that the intersection of a **non-empty** family of sets  $(U_i)_{i \in I}$  consists of the elements lying in every  $U_i$ .

### Theorem 1.2.2 — Intersection of Subspaces

Let  $V$  be a vector space over  $F$ , and let  $(U_i)_{i \in I}$  be a family of subspaces of  $V$  with  $I$  non-empty. Then  $\bigcap_{i \in I} U_i$  is a subspace of  $V$ .

*Proof.* Let  $U = \bigcap_{i \in I} U_i$ . Each  $U_i$  is a subset of  $V$ , so  $U \subseteq V$ . We use the subspace test ([Theorem 1.2.1](#)) for  $U$ , and for each  $U_i$  its direction ( $\Rightarrow$ ).

- (1) For every  $i \in I$ ,  $\mathbf{0} \in U_i$ , since  $U_i$  is a subspace. Therefore  $\mathbf{0} \in U$ .
- (2) Let  $\mathbf{u}, \mathbf{w} \in U$ . Then  $\mathbf{u}, \mathbf{w} \in U_i$  for every  $i \in I$ . Since each  $U_i$  is closed under addition,  $\mathbf{u} + \mathbf{w} \in U_i$  for every  $i \in I$ . Therefore  $\mathbf{u} + \mathbf{w} \in U$ . This shows that  $U$  is closed under addition.
- (3) Let  $a \in F$  and  $\mathbf{u} \in U$ . Then  $\mathbf{u} \in U_i$  for every  $i \in I$ , and since each  $U_i$  is closed under scalar multiplication,  $a\mathbf{u} \in U_i$  for every  $i \in I$ . Therefore  $a\mathbf{u} \in U$ . This shows that  $U$  is closed under scalar multiplication.

Hence  $U$  is a subspace of  $V$  by the subspace test ([Theorem 1.2.1](#)). ■

The index set may be infinite, and this matters. In the next section we will build the smallest subspace containing a given set of vectors, and one way to describe it is as the intersection of **all** subspaces containing that set.

For two subspaces, the theorem gives familiar pictures: two different planes through the origin in  $\mathbb{R}^3$  meet in a line through the origin. It also explains [Example \(Solutions of a Homogeneous System\)](#) from a new angle. The solution set of  $\mathbf{Ax} = \mathbf{0}$  is the intersection of the solution sets of

its  $m$  individual equations, each of which is a subspace of  $F^n$ . An algebraic example: in  $M_2(F)$ , the matrices that are both symmetric and trace-free ([Example \(Symmetric Matrices\)](#), [Example \(Trace-Free Matrices\)](#)) are those of the form  $\begin{pmatrix} a & b \\ b & -a \end{pmatrix}$ , and they form a subspace.

## 1.2.7 Unions of subspaces

Intersections behave well. Unions do not, and the reason is visible in the simplest picture.

### Example. (The Union of the Two Axes)

In  $\mathbb{R}^2$ , let  $U = \{(x, 0) : x \in \mathbb{R}\}$  be the  $x$ -axis and  $W = \{(0, y) : y \in \mathbb{R}\}$  the  $y$ -axis. Show that  $U$  and  $W$  are subspaces of  $\mathbb{R}^2$ , but  $U \cup W$  is not.

*Solution.* The  $x$ -axis is the solution set of the homogeneous equation  $y = 0$ , that is, of  $\mathbf{A}\mathbf{x} = \mathbf{0}$  with  $\mathbf{A} = \begin{pmatrix} 0 & 1 \end{pmatrix}$ . By [Example \(Solutions of a Homogeneous System\)](#),  $U$  is a subspace. In the same way  $W$  is the solution set of  $x = 0$ , so it is a subspace.

For the union, condition (2) fails. We have  $(1, 0) \in U \subseteq U \cup W$  and  $(0, 1) \in W \subseteq U \cup W$ , but

$$(1, 0) + (0, 1) = (1, 1),$$

which is on neither axis, since both of its entries are non-zero. So  $(1, 1) \notin U \cup W$ , and  $U \cup W$  is not a subspace of  $\mathbb{R}^2$ .

The union passes (1) and (3): it contains  $\mathbf{0}$ , and scaling a vector on an axis keeps it on that axis. Only addition breaks it, by mixing a vector from each axis. That observation is the whole proof of the general fact.

### Proposition 1.2.3 — Union of Two Subspaces

Let  $V$  be a vector space over  $F$ , and let  $U$  and  $W$  be subspaces of  $V$ . Then  $U \cup W$  is a subspace of  $V$  if and only if  $U \subseteq W$  or  $W \subseteq U$ .

*Idea.* The direction  $(\Leftarrow)$  is immediate, because the union is then just the bigger of the two. For  $(\Rightarrow)$ , a statement of the form “one of two inclusions holds” invites contradiction. If neither inclusion holds, there is a vector  $\mathbf{u}$  of  $U$  outside  $W$  and a vector  $\mathbf{w}$  of  $W$  outside  $U$ , exactly like  $(1, 0)$  and  $(0, 1)$  for the axes. Look at  $\mathbf{u} + \mathbf{w}$ . It must lie in  $U$  or in  $W$ . If it lies in  $U$ , subtracting  $\mathbf{u}$  keeps us in  $U$  and leaves  $\mathbf{w}$ , which is not in  $U$ .

*Proof.*  $(\Leftarrow)$  If  $U \subseteq W$ , then  $U \cup W = W$ , which is a subspace. If  $W \subseteq U$ , then  $U \cup W = U$ , which is a subspace.

$(\Rightarrow)$  Suppose  $U \cup W$  is a subspace of  $V$ , and suppose, for a contradiction, that  $U \not\subseteq W$  and  $W \not\subseteq U$ . Then there exist  $\mathbf{u} \in U$  with  $\mathbf{u} \notin W$ , and  $\mathbf{w} \in W$  with  $\mathbf{w} \notin U$ . Both lie in  $U \cup W$ , so  $\mathbf{u} + \mathbf{w} \in U \cup W$  by condition (2) of the subspace test ([Theorem 1.2.1](#)).

*Case 1:*  $\mathbf{u} + \mathbf{w} \in U$ . Since  $U$  is a subspace and  $\mathbf{u} \in U$ , condition (3) gives  $(-1)\mathbf{u} \in U$ , and  $(-1)\mathbf{u} = -\mathbf{u}$  by [Theorem 1.1.7](#). By condition (2) for  $U$ ,  $(\mathbf{u} + \mathbf{w}) + (-\mathbf{u}) \in U$ . By

(VS1), (VS2), (VS4) and (VS3),

$$(\mathbf{u} + \mathbf{w}) + (-\mathbf{u}) = \mathbf{w} + (\mathbf{u} + (-\mathbf{u})) = \mathbf{w} + \mathbf{0} = \mathbf{w}.$$

So  $\mathbf{w} \in U$ , contradicting the choice of  $\mathbf{w}$ .

*Case 2:*  $\mathbf{u} + \mathbf{w} \in W$ . Swapping the roles of  $U$  and  $W$ , and of  $\mathbf{u}$  and  $\mathbf{w}$ , in Case 1 gives  $\mathbf{u} \in W$ , contradicting the choice of  $\mathbf{u}$ .

Both cases lead to a contradiction. Hence  $U \subseteq W$  or  $W \subseteq U$ .

■

**Warning.** Do not combine subspaces by taking their union. Apart from the case where one subspace already contains the other, the union of two subspaces is never a subspace, by Proposition 1.2.3. The two axes in  $\mathbb{R}^2$  are the standard failing case: their union misses  $(1, 1)$ , and in fact misses every vector with two non-zero entries.

The right way to combine two subspaces  $U$  and  $W$  is to take all sums  $\mathbf{u} + \mathbf{w}$  with  $\mathbf{u} \in U$  and  $\mathbf{w} \in W$ . For the two axes this gives all of  $\mathbb{R}^2$ . This construction, the sum of subspaces, is the subject of a later section of this chapter; Exercise (C2: The Sum of Two Subspaces) gives a preview.

We now have a supply of vector spaces for free: every subspace of a known space is one, and every theorem of the previous section applies to it. In the next section we ask a sharper question. Given a few vectors, what is the smallest subspace that contains them? Its answer, the span, is the first step toward measuring the size of a vector space.

## 1.2.8 Exercises

### 1.2.8.1 A. Check your understanding

#### Exercise A1.

- Define what it means for a subset  $U$  of a vector space  $V$  over  $F$  to be a subspace.
- State the subspace test.
- True or false:  $\emptyset$  is a subspace of every vector space. Justify your answer.
- True or false: if  $U$  is a subspace of  $V$  and  $W$  is a subspace of  $U$ , then  $W$  is a subspace of  $V$ . Justify your answer.
- True or false: the union of two subspaces is never a subspace. Justify your answer.
- Which condition of the subspace test would you check first to show that  $\{(x, y, z) \in \mathbb{R}^3 : x + y + z = 1\}$  is not a subspace of  $\mathbb{R}^3$ ? Carry out the check.

*Solution.*

- See Definition 1.2.1:  $U$  is closed under the addition and scalar multiplication of  $V$ , and with these operations it is a vector space over  $F$ .
- See Theorem 1.2.1:  $U \subseteq V$  is a subspace if and only if  $\mathbf{0} \in U$ ,  $U$  is closed under addition, and  $U$  is closed under scalar multiplication.

- (c) False. The empty set does not contain  $\mathbf{0}$ , so condition (1) fails ([Example \(Trivial Subspaces\)](#)).
- (d) True. Since  $W$  is a subspace of  $U$ , it contains the zero vector of  $U$ , which is  $\mathbf{0}$  (the proof of [Theorem 1.2.1](#) shows the zero vector of a subspace is the zero vector of the big space). The operations of  $U$  are those of  $V$ , so closure of  $W$  under the operations of  $U$  is closure under the operations of  $V$ . By the subspace test ([Theorem 1.2.1](#)),  $W$  is a subspace of  $V$ .
- (e) False. If  $U \subseteq W$ , then  $U \cup W = W$  is a subspace ([Proposition 1.2.3](#)). For example,  $\{\mathbf{0}\} \cup V = V$ .
- (f) Condition (1). The zero vector has  $0 + 0 + 0 = 0 \neq 1$ , so it is not in the set, and the set is not a subspace.

### 1.2.8.2 B. Practice

#### Exercise B1: Which Are Subspaces?.

Determine which of the following are subspaces of the given vector space over  $\mathbb{R}$ . Justify your answer.

- (a)  $\{(x, y, z) \in \mathbb{R}^3 : x - 2y + z = 0\}$  in  $\mathbb{R}^3$ .
- (b)  $\{(x, y, z) \in \mathbb{R}^3 : x = y^2\}$  in  $\mathbb{R}^3$ .
- (c)  $\{\mathbf{A} \in M_2(\mathbb{R}) : \mathbf{A} \text{ is invertible}\}$  in  $M_2(\mathbb{R})$ .
- (d)  $\{\mathbf{A} \in M_2(\mathbb{R}) : a_{12} = 2a_{21}\}$  in  $M_2(\mathbb{R})$ .
- (e)  $\{p \in \mathbb{R}[x] : p(2) = 0 \text{ and } p'(2) = 0\}$  in  $\mathbb{R}[x]$ , where  $p'$  is the derivative of  $p$ .
- (f) The even functions  $\{f \in \mathbb{R}^{\mathbb{R}} : f(-t) = f(t) \text{ for all } t \in \mathbb{R}\}$  in  $\mathbb{R}^{\mathbb{R}}$ .
- (g)  $\{f \in \mathbb{R}^{\mathbb{R}} : f(0) = 1\}$  in  $\mathbb{R}^{\mathbb{R}}$ .
- (h) The continuous functions  $\mathbb{R} \rightarrow \mathbb{R}$ , in  $\mathbb{R}^{\mathbb{R}}$ .

*Solution.*

- (a) A subspace. It is  $\{\mathbf{x} \in \mathbb{R}^3 : \mathbf{A}\mathbf{x} = \mathbf{0}\}$  with  $\mathbf{A} = \begin{pmatrix} 1 & -2 & 1 \end{pmatrix}$ , a subspace by [Example \(Solutions of a Homogeneous System\)](#).
- (b) Not a subspace: (3) fails. The vector  $(1, 1, 0)$  lies in the set, since  $1 = 1^2$ . But  $2(1, 1, 0) = (2, 2, 0)$  does not, since  $2 \neq 2^2$ .
- (c) Not a subspace: (1) fails. The zero matrix is not invertible, because  $0 \cdot \mathbf{B} = 0 \neq \mathbf{I}_2$  for every  $\mathbf{B} \in M_2(\mathbb{R})$ .
- (d) A subspace. Call the set  $U$ .
  - (1) The zero matrix has  $a_{12} = 0 = 2 \cdot 0 = 2a_{21}$ . Therefore  $\mathbf{0} \in U$ .
  - (2) Let  $\mathbf{A}, \mathbf{B} \in U$ , so  $a_{12} = 2a_{21}$  and  $b_{12} = 2b_{21}$ . The  $(1, 2)$ -entry of  $\mathbf{A} + \mathbf{B}$  is  $a_{12} + b_{12} = 2a_{21} + 2b_{21} = 2(a_{21} + b_{21})$ , which is twice its  $(2, 1)$ -entry. Therefore  $\mathbf{A} + \mathbf{B} \in U$ . This shows that  $U$  is closed under addition.
  - (3) Let  $c \in \mathbb{R}$  and  $\mathbf{A} \in U$ . The  $(1, 2)$ -entry of  $c\mathbf{A}$  is  $ca_{12} = c(2a_{21}) = 2(ca_{21})$ , twice its  $(2, 1)$ -entry. Therefore  $c\mathbf{A} \in U$ . This shows that  $U$  is closed under scalar multiplication.
 Hence  $U$  is a subspace of  $M_2(\mathbb{R})$  by the subspace test ([Theorem 1.2.1](#)).
- (e) A subspace. Call the set  $U$ . We use [Theorem 0.8.5](#) for values, and the sum and constant multiple rules for derivatives.

- (1) The zero polynomial has value 0 at 2, and its derivative is the zero polynomial, also with value 0 at 2. Therefore  $\mathbf{0} \in U$ .
- (2) Let  $p, q \in U$ . Then  $(p+q)(2) = p(2) + q(2) = 0$  and  $(p+q)'(2) = p'(2) + q'(2) = 0$ . Therefore  $p+q \in U$ . This shows that  $U$  is closed under addition.
- (3) Let  $c \in \mathbb{R}$  and  $p \in U$ . Then  $(cp)(2) = cp(2) = 0$  and  $(cp)'(2) = cp'(2) = 0$ . Therefore  $cp \in U$ . This shows that  $U$  is closed under scalar multiplication.
- Hence  $U$  is a subspace of  $\mathbb{R}[x]$  by the subspace test (Theorem 1.2.1).
- (f) A subspace. Call the set  $E$ .
- (1) The zero function satisfies  $\mathbf{0}(-t) = 0 = \mathbf{0}(t)$  for all  $t$ . Therefore  $\mathbf{0} \in E$ .
- (2) Let  $f, g \in E$ . For every  $t \in \mathbb{R}$ ,  $(f+g)(-t) = f(-t) + g(-t) = f(t) + g(t) = (f+g)(t)$ . Therefore  $f+g \in E$ . This shows that  $E$  is closed under addition.
- (3) Let  $a \in \mathbb{R}$  and  $f \in E$ . For every  $t \in \mathbb{R}$ ,  $(af)(-t) = af(-t) = af(t) = (af)(t)$ . Therefore  $af \in E$ . This shows that  $E$  is closed under scalar multiplication.
- Hence  $E$  is a subspace of  $\mathbb{R}^{\mathbb{R}}$  by the subspace test (Theorem 1.2.1).
- (g) Not a subspace: (1) fails. The zero function has value  $0 \neq 1$  at 0.
- (h) A subspace. (1) The zero function is constant, hence continuous. (2) A sum of continuous functions is continuous. (3) A constant multiple of a continuous function is continuous. Hence the continuous functions form a subspace of  $\mathbb{R}^{\mathbb{R}}$  by the subspace test (Theorem 1.2.1).

### Exercise B2: Matrices Commuting with a Given Matrix.

Fix  $\mathbf{B} \in M_n(F)$ , and let  $C(\mathbf{B}) = \{\mathbf{A} \in M_n(F) : \mathbf{AB} = \mathbf{BA}\}$ .

- (a) Prove that  $C(\mathbf{B})$  is a subspace of  $M_n(F)$ .
- (b) Let  $n = 2$ ,  $F = \mathbb{R}$  and  $\mathbf{B} = \begin{pmatrix} 1 & 0 \\ 0 & 2 \end{pmatrix}$ . Find all  $\mathbf{A} \in C(\mathbf{B})$ . Hence describe  $C(\mathbf{B})$  in words.

*Solution.*

- (a) By definition,  $C(\mathbf{B}) \subseteq M_n(F)$ . We use Theorem 0.9.4 throughout.
- (1) By the zero rule,  $\mathbf{0B} = \mathbf{0} = \mathbf{B0}$ . Therefore  $\mathbf{0} \in C(\mathbf{B})$ .
- (2) Let  $\mathbf{A}_1, \mathbf{A}_2 \in C(\mathbf{B})$ , so  $\mathbf{A}_1\mathbf{B} = \mathbf{BA}_1$  and  $\mathbf{A}_2\mathbf{B} = \mathbf{BA}_2$ . By distributivity,

$$(\mathbf{A}_1 + \mathbf{A}_2)\mathbf{B} = \mathbf{A}_1\mathbf{B} + \mathbf{A}_2\mathbf{B} = \mathbf{BA}_1 + \mathbf{BA}_2 = \mathbf{B}(\mathbf{A}_1 + \mathbf{A}_2).$$

Therefore  $\mathbf{A}_1 + \mathbf{A}_2 \in C(\mathbf{B})$ . This shows that  $C(\mathbf{B})$  is closed under addition.

(3) Let  $c \in F$  and  $\mathbf{A} \in C(\mathbf{B})$ . By the scalar rule,  $(c\mathbf{A})\mathbf{B} = c(\mathbf{AB}) = c(\mathbf{BA}) = \mathbf{B}(c\mathbf{A})$ . Therefore  $c\mathbf{A} \in C(\mathbf{B})$ . This shows that  $C(\mathbf{B})$  is closed under scalar multiplication.

Hence  $C(\mathbf{B})$  is a subspace of  $M_n(F)$  by the subspace test (Theorem 1.2.1).

- (b) Let  $\mathbf{A} = \begin{pmatrix} a & b \\ c & d \end{pmatrix}$ . Then

$$\mathbf{AB} = \begin{pmatrix} a & 2b \\ c & 2d \end{pmatrix}, \quad \mathbf{BA} = \begin{pmatrix} a & b \\ 2c & 2d \end{pmatrix}.$$

So  $\mathbf{AB} = \mathbf{BA}$  if and only if  $2b = b$  and  $c = 2c$ , that is,  $b = 0$  and  $c = 0$ . Hence

$C(\mathbf{B})$  is the set of diagonal matrices  $\begin{pmatrix} a & 0 \\ 0 & d \end{pmatrix}$  with  $a, d \in \mathbb{R}$ .

### Exercise B3: Intersection and Union of Two Planes.

In  $\mathbb{R}^3$ , let  $U = \{(x, y, z) : x + y + z = 0\}$  and  $W = \{(x, y, z) : x - y = 0\}$ .

- (a) Explain why  $U$  and  $W$  are subspaces of  $\mathbb{R}^3$ , and find  $U \cap W$ .
- (b) Find  $\mathbf{u} \in U$  with  $\mathbf{u} \notin W$  and  $\mathbf{w} \in W$  with  $\mathbf{w} \notin U$ . Hence show directly that  $U \cup W$  is not a subspace of  $\mathbb{R}^3$ .

*Solution.*

- (a)  $U$  and  $W$  are the solution sets of  $\mathbf{A}\mathbf{x} = \mathbf{0}$  for  $\mathbf{A} = \begin{pmatrix} 1 & 1 & 1 \end{pmatrix}$  and  $\mathbf{A} = \begin{pmatrix} 1 & -1 & 0 \end{pmatrix}$ , so they are subspaces by [Example \(Solutions of a Homogeneous System\)](#). A vector  $(x, y, z)$  lies in  $U \cap W$  exactly when  $x + y + z = 0$  and  $x = y$ . Substituting  $y = x$  gives  $z = -2x$ . Hence

$$U \cap W = \{(t, t, -2t) : t \in \mathbb{R}\},$$

a line through the origin, which is a subspace by [Theorem 1.2.2](#).

- (b) Take  $\mathbf{u} = (1, -1, 0)$ , which lies in  $U$  since  $1 - 1 + 0 = 0$ , but not in  $W$  since  $1 - (-1) = 2 \neq 0$ . Take  $\mathbf{w} = (1, 1, 0)$ , which lies in  $W$ , but not in  $U$  since  $1 + 1 + 0 = 2 \neq 0$ . Both lie in  $U \cup W$ . Their sum is  $(2, 0, 0)$ , which is not in  $U$  (as  $2 + 0 + 0 \neq 0$ ) and not in  $W$  (as  $2 - 0 \neq 0$ ). Hence  $U \cup W$  is not closed under addition, and it is not a subspace of  $\mathbb{R}^3$ .

### 1.2.8.3 C. Going deeper

#### Exercise C1: Covering a Space by Subspaces.

- (a) Let  $V$  be a vector space over any field  $F$ . Prove that  $V$  is not the union of two proper subspaces.
- (b) Show that  $\mathbb{F}_2^2$  is the union of three proper subspaces.

*Hint: for (a), use [Proposition 1.2.3](#).*

*Solution.*

- (a) Suppose, for a contradiction, that  $V = U \cup W$  with  $U$  and  $W$  proper subspaces of  $V$ . Then  $U \cup W = V$  is a subspace of  $V$ , so by [Proposition 1.2.3](#),  $U \subseteq W$  or  $W \subseteq U$ . In the first case  $V = U \cup W = W$ , and in the second  $V = U$ . Either way one of the subspaces equals  $V$ , contradicting that both are proper. Hence  $V$  is not the union of two proper subspaces.
- (b) The space  $\mathbb{F}_2^2$  has the four elements  $(0, 0), (1, 0), (0, 1), (1, 1)$ . Let  $\mathbf{v}$  be one of the three non-zero elements, and let  $U_{\mathbf{v}} = \{\mathbf{0}, \mathbf{v}\}$ . We check that  $U_{\mathbf{v}}$  is a subspace with the three-check move.

- (1)  $\mathbf{0} \in U_{\mathbf{v}}$  by construction.

(2) The possible sums are  $\mathbf{0} + \mathbf{0} = \mathbf{0}$ ,  $\mathbf{0} + \mathbf{v} = \mathbf{v} + \mathbf{0} = \mathbf{v}$ , and  $\mathbf{v} + \mathbf{v} = (1+1)\mathbf{v} = 0\mathbf{v} = \mathbf{0}$ , using (VS8), (VS6),  $1+1=0$  in  $\mathbb{F}_2$ , and Theorem 1.1.5. All lie in  $U_{\mathbf{v}}$ , so  $U_{\mathbf{v}}$  is closed under addition.

(3) The only scalars are 0 and 1. For  $\mathbf{u} \in U_{\mathbf{v}}$ ,  $0\mathbf{u} = \mathbf{0}$  by Theorem 1.1.5 and  $1\mathbf{u} = \mathbf{u}$  by (VS8), both in  $U_{\mathbf{v}}$ . So  $U_{\mathbf{v}}$  is closed under scalar multiplication.

Hence each  $U_{\mathbf{v}}$  is a subspace. Each has two elements, so it is proper. Every element of  $\mathbb{F}_2^2$  is  $\mathbf{0}$  or one of the three non-zero vectors, so

$$\mathbb{F}_2^2 = U_{(1,0)} \cup U_{(0,1)} \cup U_{(1,1)}.$$

So three proper subspaces can cover a vector space, although (a) shows two never can.

### Exercise C2: The Sum of Two Subspaces.

Let  $U$  and  $W$  be subspaces of a vector space  $V$  over  $F$ , and let

$$U + W = \{\mathbf{u} + \mathbf{w} : \mathbf{u} \in U, \mathbf{w} \in W\}.$$

- (a) Prove that  $U + W$  is a subspace of  $V$  containing both  $U$  and  $W$ .
- (b) Prove that if  $X$  is a subspace of  $V$  with  $U \cup W \subseteq X$ , then  $U + W \subseteq X$ .
- (c) For the two axes  $U$  and  $W$  of Example (The Union of the Two Axes), show that  $U + W = \mathbb{R}^2$ .

*Solution.*

- (a) Every  $\mathbf{u} + \mathbf{w}$  is a vector of  $V$ , so  $U + W \subseteq V$ .
  - (1) Since  $\mathbf{0} \in U$  and  $\mathbf{0} \in W$ , and  $\mathbf{0} + \mathbf{0} = \mathbf{0}$  by (VS3), we get  $\mathbf{0} \in U + W$ .
  - (2) Let  $\mathbf{u}_1 + \mathbf{w}_1$  and  $\mathbf{u}_2 + \mathbf{w}_2$  be in  $U + W$ , with  $\mathbf{u}_1, \mathbf{u}_2 \in U$  and  $\mathbf{w}_1, \mathbf{w}_2 \in W$ . By (VS1) and (VS2),

$$(\mathbf{u}_1 + \mathbf{w}_1) + (\mathbf{u}_2 + \mathbf{w}_2) = (\mathbf{u}_1 + \mathbf{u}_2) + (\mathbf{w}_1 + \mathbf{w}_2).$$

Since  $U$  and  $W$  are closed under addition,  $\mathbf{u}_1 + \mathbf{u}_2 \in U$  and  $\mathbf{w}_1 + \mathbf{w}_2 \in W$ . Therefore the sum lies in  $U + W$ . This shows that  $U + W$  is closed under addition.

(3) Let  $a \in F$  and  $\mathbf{u} + \mathbf{w} \in U + W$ . By (VS5),  $a(\mathbf{u} + \mathbf{w}) = a\mathbf{u} + a\mathbf{w}$ , with  $a\mathbf{u} \in U$  and  $a\mathbf{w} \in W$  by closure. Therefore  $a(\mathbf{u} + \mathbf{w}) \in U + W$ . This shows that  $U + W$  is closed under scalar multiplication.

Hence  $U + W$  is a subspace of  $V$ . For  $\mathbf{u} \in U$ , we have  $\mathbf{u} = \mathbf{u} + \mathbf{0}$  with  $\mathbf{0} \in W$ , so  $\mathbf{u} \in U + W$ ; thus  $U \subseteq U + W$ . Swapping the roles of  $U$  and  $W$  gives  $W \subseteq U + W$ , using (VS1).

- (b) Let  $\mathbf{u} + \mathbf{w} \in U + W$  with  $\mathbf{u} \in U$  and  $\mathbf{w} \in W$ . Then  $\mathbf{u}, \mathbf{w} \in U \cup W \subseteq X$ . Since  $X$  is a subspace, it is closed under addition, so  $\mathbf{u} + \mathbf{w} \in X$ . Hence  $U + W \subseteq X$ .
- (c) By (a),  $U + W \subseteq \mathbb{R}^2$ . Conversely, let  $(x, y) \in \mathbb{R}^2$ . Then  $(x, y) = (x, 0) + (0, y)$  with  $(x, 0) \in U$  and  $(0, y) \in W$ , so  $(x, y) \in U + W$ . Hence  $U + W = \mathbb{R}^2$ , whereas  $U \cup W$  is only the two axes.

### Exercise C3: A One-Line Test, and a Test over $\mathbb{F}_2$ .

Let  $V$  be a vector space over  $F$  and  $U \subseteq V$ .

- (a) Prove that  $U$  is a subspace of  $V$  if and only if  $U \neq \emptyset$  and  $a\mathbf{u} + \mathbf{w} \in U$  for all  $a \in F$  and  $\mathbf{u}, \mathbf{w} \in U$ .
- (b) Suppose  $F = \mathbb{F}_2$ . Prove that  $U$  is a subspace of  $V$  if and only if  $\mathbf{0} \in U$  and  $U$  is closed under addition.
- (c) Show by an example that the statement in (b) is false over  $\mathbb{R}$ .

*Solution.*

- (a)  $(\Rightarrow)$  Suppose  $U$  is a subspace. By the subspace test (Theorem 1.2.1),  $\mathbf{0} \in U$ , so  $U \neq \emptyset$ . For  $a \in F$  and  $\mathbf{u}, \mathbf{w} \in U$ , condition (3) gives  $a\mathbf{u} \in U$ , and then condition (2) gives  $a\mathbf{u} + \mathbf{w} \in U$ .
- $(\Leftarrow)$  Suppose  $U \neq \emptyset$  and  $a\mathbf{u} + \mathbf{w} \in U$  for all  $a \in F$ ,  $\mathbf{u}, \mathbf{w} \in U$ . Pick  $\mathbf{u}_0 \in U$ . Taking  $a = -1$  and  $\mathbf{u} = \mathbf{w} = \mathbf{u}_0$ , and using Theorem 1.1.7 and (VS4),

$$(-1)\mathbf{u}_0 + \mathbf{u}_0 = -\mathbf{u}_0 + \mathbf{u}_0 = \mathbf{0} \in U,$$

where  $-\mathbf{u}_0 + \mathbf{u}_0 = \mathbf{u}_0 + (-\mathbf{u}_0)$  by (VS1). This is condition (1). For  $\mathbf{u}, \mathbf{w} \in U$ , taking  $a = 1$  and using (VS8) gives  $\mathbf{u} + \mathbf{w} = 1\mathbf{u} + \mathbf{w} \in U$ , which is (2). For  $a \in F$  and  $\mathbf{u} \in U$ , taking  $\mathbf{w} = \mathbf{0} \in U$  and using (VS3) gives  $a\mathbf{u} = a\mathbf{u} + \mathbf{0} \in U$ , which is (3). By the subspace test (Theorem 1.2.1),  $U$  is a subspace.

- (b)  $(\Rightarrow)$  This is part of the subspace test (Theorem 1.2.1).  $(\Leftarrow)$  Suppose  $\mathbf{0} \in U$  and  $U$  is closed under addition. The only scalars in  $\mathbb{F}_2$  are 0 and 1. For  $\mathbf{u} \in U$ ,  $0\mathbf{u} = \mathbf{0} \in U$  by Theorem 1.1.5, and  $1\mathbf{u} = \mathbf{u} \in U$  by (VS8). So  $U$  is closed under scalar multiplication, and by the subspace test (Theorem 1.2.1)  $U$  is a subspace.
- (c) Take  $V = \mathbb{R}^2$  and  $U = \mathbb{Z}^2$ . Then  $\mathbf{0} \in U$ , and the sum of two integer vectors is an integer vector. But  $\frac{1}{2}(1, 0) = (\frac{1}{2}, 0) \notin U$ , so  $U$  is not a subspace. Over  $\mathbb{R}$  there are many more scalars to test than 0 and 1.

## 1.3 Linear Combinations and Span

We now have vector spaces and, inside them, subspaces. So far every subspace has been handed to us by a condition, such as “ $x + 2y = 0$ ” or “ $\mathbf{A}^\top = \mathbf{A}$ ”. This section goes the other way. We start from a few vectors, build everything that can be built from them, and show that the result is always a subspace: the smallest one containing the vectors we started with. Along the way we meet the first way of measuring a space, by asking whether finitely many vectors are enough to build all of it.

### 1.3.1 Linear combinations

Take two vectors in  $\mathbb{R}^3$ , say  $\mathbf{v}_1 = (1, 0, 1)$  and  $\mathbf{v}_2 = (0, 1, 1)$ . Which vectors can we build from them, using only the two operations a vector space gives us? Scaling gives  $a\mathbf{v}_1$  and  $b\mathbf{v}_2$ , and adding gives

$$a\mathbf{v}_1 + b\mathbf{v}_2 = (a, b, a + b).$$

Adding or scaling again produces nothing new: for instance  $2(a\mathbf{v}_1 + b\mathbf{v}_2) + (c\mathbf{v}_1 + d\mathbf{v}_2) = (2a + c)\mathbf{v}_1 + (2b + d)\mathbf{v}_2$  has the same shape. So the vectors we can reach are exactly those of the



form  $(a, b, a + b)$ : the plane  $z = x + y$  through the origin. The expression  $a\mathbf{v}_1 + b\mathbf{v}_2$  will appear in almost every argument from now on, so it deserves a name.

The vectors we combine come as a **list**  $(\mathbf{v}_1, \dots, \mathbf{v}_k)$ : a finite sequence of vectors, in order, in which repeats are allowed. A list of length  $k = 0$  is the **empty list**  $()$ . Why lists rather than sets becomes important in the next section; for now it costs nothing.

### Definition 1.3.1 — Linear Combination

Let  $V$  be a vector space over  $F$ , and let  $(\mathbf{v}_1, \dots, \mathbf{v}_k)$  be a list of vectors in  $V$ , where  $k \geq 0$ . A **linear combination** of  $(\mathbf{v}_1, \dots, \mathbf{v}_k)$  is a vector of the form

$$a_1\mathbf{v}_1 + a_2\mathbf{v}_2 + \cdots + a_k\mathbf{v}_k, \quad \text{where } a_1, \dots, a_k \in F.$$

The scalars  $a_i$  are the **coefficients** of the combination. A linear combination of the empty list is, by convention, the **empty sum**  $\mathbf{0}$ .

In words: scale each vector of the list by a scalar from  $F$ , then add up the results. Two small points hide in the notation. First, a sum of  $k$  vectors needs no brackets, because addition in  $V$  is associative (VS2); so the expression names a single vector. Second, the coefficients must lie in the field  $F$  we are working over. Changing the field can change what counts as a combination, as [Exercise \(C2: The Span Depends on the Field\)](#) shows.

The empty sum convention is the additive twin of  $0! = 1$  and  $x^0 = 1$ , where a product with no factors is the neutral element 1 for multiplication: a sum with no terms is the neutral element  $\mathbf{0}$  for addition. It will make several statements true without exceptions.

### Example. (Linear Combinations in Four Families)

Show that each vector is a linear combination of the given list.

- (a)  $(3, -2, 1) \in \mathbb{R}^3$ , of the list  $((1, 0, 1), (0, 1, 1))$ .
- (b)  $x^2 - 1 \in \mathbb{R}[x]$ , of the list  $((x + 1)^2, x + 1)$ .
- (c) The function  $\sin^2$  in  $\mathbb{R}^{\mathbb{R}}$ , of the list  $(1, \cos 2x)$ , where 1 is the constant function.
- (d) The zero vector of any vector space  $V$ , of any list  $(\mathbf{v}_1, \dots, \mathbf{v}_k)$  in  $V$ .

*Solution.*

- (a) We need  $a(1, 0, 1) + b(0, 1, 1) = (a, b, a + b)$  to equal  $(3, -2, 1)$ . The first two entries force  $a = 3$  and  $b = -2$ , and then the third entry is  $a + b = 1$ , as required. Hence  $(3, -2, 1) = 3(1, 0, 1) - 2(0, 1, 1)$ .
- (b) Expanding,  $(x + 1)^2 - 2(x + 1) = x^2 + 2x + 1 - 2x - 2 = x^2 - 1$ . So the coefficients are 1 and  $-2$ .
- (c) The double-angle formula  $\cos 2x = 1 - 2\sin^2 x$  holds for every real  $x$ . Hence  $\sin^2 x = \frac{1}{2} \cdot 1 + (-\frac{1}{2})\cos 2x$  for every  $x$ , which is an equality of functions by [Definition 0.4.1](#). The coefficients  $\frac{1}{2}$  and  $-\frac{1}{2}$  are real, as they must be.
- (d) Take every coefficient to be 0. By [Theorem 1.1.5](#),  $0\mathbf{v}_i = \mathbf{0}$  for each  $i$ , so  $0\mathbf{v}_1 + \cdots + 0\mathbf{v}_k = \mathbf{0}$ . For the empty list,  $\mathbf{0}$  is the empty sum. This degenerate case matters: it says the zero vector can always be built, from anything, even from nothing.

Notice that the question in (a) was a small system of equations in the unknown coefficients. That is typical: deciding whether a vector is a combination of a list means deciding whether a system of linear equations has a solution. Here we solve such systems by hand; Chapter 2 gives a systematic method.

## 1.3.2 The span

*The span of some vectors is everything you can build from them by scaling and adding.*

We have two kinds of input to cover. Usually we start from a list, as above. But sometimes the natural input is a set, possibly infinite, such as all the powers  $1, x, x^2, \dots$  in  $F[x]$ . A linear combination only ever uses finitely many vectors, so for a set we allow every finite list drawn from it.

### Definition 1.3.2 — Span

Let  $V$  be a vector space over  $F$ .

- The **span** of a list  $(\mathbf{v}_1, \dots, \mathbf{v}_k)$  of vectors in  $V$  is the set of **all** its linear combinations:

$$\text{span}(\mathbf{v}_1, \dots, \mathbf{v}_k) = \{a_1\mathbf{v}_1 + \dots + a_k\mathbf{v}_k : a_1, \dots, a_k \in F\}.$$

- The **span** of a subset  $S \subseteq V$ , possibly infinite, is the set of all vectors that are linear combinations of **some finite** list of vectors in  $S$ :

$$\text{span}(S) = \{a_1\mathbf{s}_1 + \dots + a_k\mathbf{s}_k : k \geq 0, \mathbf{s}_1, \dots, \mathbf{s}_k \in S, a_1, \dots, a_k \in F\}.$$

In particular  $\text{span}() = \text{span}(\emptyset) = \{\mathbf{0}\}$ . When the field matters we write  $\text{span}_F$ .

In words: for a list, the span collects every choice of coefficients. For a set, we may first pick any finite number of vectors from  $S$  (the same vector more than once is allowed), and then any coefficients. The value  $k = 0$  is allowed, and it contributes the empty sum  $\mathbf{0}$ . So  $\mathbf{0} \in \text{span}(S)$  for **every**  $S$ , including  $S = \emptyset$ .

**One thing to check.** A list  $(\mathbf{v}_1, \dots, \mathbf{v}_k)$  now has two spans: the span of the list, and the span of the set  $\{\mathbf{v}_1, \dots, \mathbf{v}_k\}$  of its entries. They agree. A combination of the list is a combination of a finite list from the set, so the first is contained in the second. Conversely, take a combination  $b_1\mathbf{w}_1 + \dots + b_m\mathbf{w}_m$  where each  $\mathbf{w}_r$  is one of the  $\mathbf{v}_i$ ; for each  $r$ , choose one index  $i(r)$  with  $\mathbf{w}_r = \mathbf{v}_{i(r)}$ . Group the terms with the same index together, using (VS1), (VS2) and  $b\mathbf{v} + b'\mathbf{v} = (b + b')\mathbf{v}$  (VS6). The result is  $a_1\mathbf{v}_1 + \dots + a_k\mathbf{v}_k$ , where  $a_i$  is the sum of the  $b_r$  with  $i(r) = i$  (and  $a_i = 0$  if there are none, by [Theorem 1.1.5](#)). So the second is contained in the first. In particular the span of a list does not change if we reorder the list or repeat an entry.

**Why  $\text{span}(\emptyset) = \{\mathbf{0}\}$ ?** The definition forces it through the empty sum, but it is also the only sensible choice. We will prove that the span of  $S$  is the smallest subspace containing  $S$ . The smallest subspace containing nothing in particular is  $\{\mathbf{0}\}$ , since every subspace contains  $\mathbf{0}$ . Any other convention would make that theorem false for  $S = \emptyset$ , and the empty list will turn out to be a basis of the zero space.

**Example. (Computing Spans)**

Describe each span.

- (a)  $\text{span}((1, 0, 1), (0, 1, 1))$  in  $\mathbb{R}^3$ . Is  $(1, 2, 3)$  in it? Is  $(1, 2, 4)$ ?
- (b)  $\text{span}(\mathbf{v})$  for a single vector  $\mathbf{v}$  in a vector space  $V$ .
- (c)  $\text{span}(1, x^2)$  in  $\mathbb{R}[x]$ .
- (d)  $\text{span}(\{1, x, x^2, x^3, \dots\})$  in  $F[x]$ .

*Solution.*

- (a) As computed at the start of the section, the combinations are  $a(1, 0, 1) + b(0, 1, 1) = (a, b, a + b)$ . So the span is  $\{(x, y, z) \in \mathbb{R}^3 : z = x + y\}$ , a plane through the origin. For membership, a vector  $(x, y, z)$  is in the span exactly when the entries fit: the first two entries force  $a = x$  and  $b = y$ , and then we need  $z = x + y$ . Since  $3 = 1 + 2$ , the vector  $(1, 2, 3) = 1(1, 0, 1) + 2(0, 1, 1)$  is in the span. Since  $4 \neq 1 + 2$ , the vector  $(1, 2, 4)$  is not.
- (b)  $\text{span}(\mathbf{v}) = \{a\mathbf{v} : a \in F\}$ . If  $\mathbf{v} \neq \mathbf{0}$  and  $V = \mathbb{R}^2$  or  $\mathbb{R}^3$ , this is the line through the origin in the direction of  $\mathbf{v}$ . If  $\mathbf{v} = \mathbf{0}$ , then  $a\mathbf{0} = \mathbf{0}$  for every  $a$  ([Theorem 1.1.6](#)), so  $\text{span}(\mathbf{0}) = \{\mathbf{0}\}$ . This degenerate case shows that a span can be as small as the zero space even when the list is not empty.
- (c) The combinations are  $a + bx^2$  with  $a, b \in \mathbb{R}$ : the polynomials of degree at most 2 with no  $x$  term. For example  $x^2 + x$  is **not** in the span, because its coefficient of  $x$  is  $1 \neq 0$  and polynomials are equal only when all coefficients agree ([Definition 0.8.1](#)).
- (d) Every polynomial  $a_0 + a_1x + \dots + a_Nx^N$  is a combination of the finite list  $(1, x, \dots, x^N)$  of vectors in the set. Conversely every combination of finitely many powers of  $x$  is a polynomial. So the span is all of  $F[x]$ . No single polynomial here is a combination of *infinitely* many powers: each one uses only the powers up to its degree.

*Quick check.* Is  $\text{span}((1, 2))$  a subset of  $\text{span}((2, 4))$  in  $\mathbb{R}^2$ ? Are they equal?

*Solution.* Yes to both. Since  $(2, 4) = 2(1, 2)$ , every  $a(2, 4) = (2a)(1, 2)$  lies in  $\text{span}((1, 2))$ . Since  $(1, 2) = \frac{1}{2}(2, 4)$ , every  $a(1, 2) = \frac{a}{2}(2, 4)$  lies in  $\text{span}((2, 4))$ . Both spans are the line  $y = 2x$ . Different lists can have the same span.

### 1.3.3 The span is the smallest subspace

The plane in [Example \(Computing Spans\)](#) (a) was a subspace. That was no accident. Adding two combinations gives a combination, and so does scaling one. And any subspace that contains the vectors must contain all their combinations, because a subspace is closed under exactly the operations that build them. So the span is both a subspace and the smallest one available:

#### Theorem 1.3.1 — Span Is the Smallest Subspace

Let  $V$  be a vector space over  $F$  and let  $S \subseteq V$ . Then:

- (1)  $\text{span}(S)$  is a subspace of  $V$ ;

- (2)  $S \subseteq \text{span}(S)$ ;
- (3) if  $U$  is a subspace of  $V$  with  $S \subseteq U$ , then  $\text{span}(S) \subseteq U$ .

The same holds for the span of a list, which equals the span of the set of its entries.

*Idea.* Part (1) is the three-check subspace test. Part (2) is the combination 1s. Part (3) is “a subspace is closed under the operations that build combinations”; since a combination has  $k$  terms for an arbitrary  $k$ , we add one term at a time, by induction on  $k$ .

*Proof.*

- (1) We apply [Theorem 1.2.1](#).

- *Zero.* The empty list is a finite list of vectors in  $S$ , and its only combination is  $\mathbf{0}$ . Hence  $\mathbf{0} \in \text{span}(S)$ .
- *Addition.* Let  $\mathbf{x}, \mathbf{y} \in \text{span}(S)$ . Then  $\mathbf{x} = a_1\mathbf{s}_1 + \cdots + a_k\mathbf{s}_k$  and  $\mathbf{y} = b_1\mathbf{t}_1 + \cdots + b_m\mathbf{t}_m$  for some  $\mathbf{s}_i, \mathbf{t}_j \in S$  and scalars  $a_i, b_j \in F$ . Therefore

$$\mathbf{x} + \mathbf{y} = a_1\mathbf{s}_1 + \cdots + a_k\mathbf{s}_k + b_1\mathbf{t}_1 + \cdots + b_m\mathbf{t}_m,$$

which is a combination of the list  $(\mathbf{s}_1, \dots, \mathbf{s}_k, \mathbf{t}_1, \dots, \mathbf{t}_m)$  of vectors in  $S$ . (If some  $\mathbf{t}_j$  equals some  $\mathbf{s}_i$ , the list has a repeat, which lists allow.) Hence  $\mathbf{x} + \mathbf{y} \in \text{span}(S)$ . This shows  $\text{span}(S)$  is closed under addition.

- *Scaling.* Let  $\mathbf{x} = a_1\mathbf{s}_1 + \cdots + a_k\mathbf{s}_k \in \text{span}(S)$  and  $c \in F$ . By (VS5), applied  $k - 1$  times, and by (VS7) for each term,

$$c\mathbf{x} = (ca_1)\mathbf{s}_1 + \cdots + (ca_k)\mathbf{s}_k,$$

which is a combination of the same list. Hence  $c\mathbf{x} \in \text{span}(S)$ . This shows  $\text{span}(S)$  is closed under scalar multiplication.

Therefore  $\text{span}(S)$  is a subspace of  $V$ .

- (2) Let  $\mathbf{s} \in S$ . By (VS8),  $\mathbf{s} = 1\mathbf{s}$ , which is a combination of the list  $(\mathbf{s})$ . Hence  $\mathbf{s} \in \text{span}(S)$ .
- (3) Let  $U$  be a subspace with  $S \subseteq U$ . By [Theorem 1.2.1](#),  $U$  contains  $\mathbf{0}$  and is closed under addition and scalar multiplication. We prove by induction on  $k \geq 0$  ([Theorem 0.2.7](#)) that every combination  $a_1\mathbf{s}_1 + \cdots + a_k\mathbf{s}_k$  with  $\mathbf{s}_i \in S$  lies in  $U$ . For  $k = 0$  the combination is  $\mathbf{0} \in U$ . Suppose the claim holds for  $k$ , and consider  $a_1\mathbf{s}_1 + \cdots + a_{k+1}\mathbf{s}_{k+1}$ . By the induction hypothesis  $\mathbf{x} = a_1\mathbf{s}_1 + \cdots + a_k\mathbf{s}_k \in U$ . Since  $\mathbf{s}_{k+1} \in S \subseteq U$  and  $U$  is closed under scaling,  $a_{k+1}\mathbf{s}_{k+1} \in U$ . Since  $U$  is closed under addition,  $\mathbf{x} + a_{k+1}\mathbf{s}_{k+1} \in U$ . This completes the induction, and shows  $\text{span}(S) \subseteq U$ .

■

In one phrase:  $\text{span}(S)$  is a subspace containing  $S$ , and it sits inside every other one. So “the smallest subspace containing  $S$ ” makes sense, and it is  $\text{span}(S)$ . This gives a second, top-down description of the span, which does not mention combinations at all.

### Corollary 1.3.2 — Span as an Intersection

Let  $V$  be a vector space and  $S \subseteq V$ . Let  $\mathcal{U}$  be the set of all subspaces  $U$  of  $V$  with

$S \subseteq U$ . Then

$$\text{span}(S) = \bigcap_{U \in \mathcal{U}} U.$$

*Proof.* The index set  $\mathcal{U}$  is non-empty, because  $V$  is a subspace of itself containing  $S$ ; so the intersection is defined (Definition 0.3.9).

( $\subseteq$ ) By Theorem 1.3.1 (3),  $\text{span}(S) \subseteq U$  for every  $U \in \mathcal{U}$ . Hence  $\text{span}(S)$  is contained in the intersection.

( $\supseteq$ ) By Theorem 1.3.1 (1) and (2),  $\text{span}(S)$  is itself a member of  $\mathcal{U}$ . An intersection is contained in each of its members, so the intersection is contained in  $\text{span}(S)$ .

By Theorem 0.3.1, the two sets are equal. ■

By Theorem 1.2.2, an intersection of subspaces is a subspace, so the right-hand side is a subspace for a reason that never mentions combinations. The two descriptions complement each other. The bottom-up one (combinations) tells us what the elements look like. The top-down one (intersection) is convenient for proving that the span lies inside something.

*Warning.* The span of  $S$  is **not**  $S$ , and it is a subspace **even when**  $S$  is **not**. For example  $S = \{(1, 0), (0, 1)\}$  is a two-element set, and not a subspace of  $\mathbb{R}^2$  (it does not contain  $(0, 0)$ ). Its span is the whole plane  $\mathbb{R}^2$ , since  $(x, y) = x(1, 0) + y(0, 1)$ . By Theorem 1.3.1,  $\text{span}(S) = S$  holds exactly when  $S$  is already a subspace. Also keep the two uses of the word apart: “the span of  $S$ ” is a noun, a subspace; “ $S$  spans  $V$ ” is a verb, defined next.

## 1.3.4 Spanning lists and finite-dimensional spaces

The most important case is when the span is everything. Then the list is a complete toolkit for the space: every vector can be built from it.

### Definition 1.3.3 — Spanning List, Spanning Set

Let  $V$  be a vector space over  $F$ . A list  $(\mathbf{v}_1, \dots, \mathbf{v}_k)$  in  $V$  **spans**  $V$ , or is a **spanning list** of  $V$ , if  $\text{span}(\mathbf{v}_1, \dots, \mathbf{v}_k) = V$ . A subset  $S \subseteq V$  **spans**  $V$ , or is a **spanning set** of  $V$ , if  $\text{span}(S) = V$ .

Since every span is contained in  $V$ , to show that a list spans  $V$  we must show one thing: **every**  $\mathbf{v} \in V$  can be written as  $a_1\mathbf{v}_1 + \dots + a_k\mathbf{v}_k$  for **some** scalars. The scalars may depend on  $\mathbf{v}$ , and they need not be unique.

#### Example. (Spanning Lists)

Show that each list spans the given space.

- (a)  $(\mathbf{e}_1, \dots, \mathbf{e}_n)$  in  $F^n$ , where  $\mathbf{e}_i$  has 1 in position  $i$  and 0 elsewhere.

- (b)  $(1, x, x^2)$  in  $F[x]_{\leq 2}$ .
- (c)  $((1, 1), (1, -1))$  in  $\mathbb{R}^2$ .
- (d)  $(\mathbf{E}_{11}, \mathbf{E}_{12}, \mathbf{E}_{21}, \mathbf{E}_{22})$  in  $M_2(F)$ , where  $\mathbf{E}_{ij}$  has 1 in entry  $(i, j)$  and 0 elsewhere.
- (e) The empty list, in the zero space  $\{\mathbf{0}\}$ .

*Solution.*

- (a) Let  $(x_1, \dots, x_n) \in F^n$ . Adding entry by entry,  $x_1\mathbf{e}_1 + \dots + x_n\mathbf{e}_n = (x_1, \dots, x_n)$ . So every vector is a combination, with its own entries as coefficients.
- (b) Every  $p \in F[x]_{\leq 2}$  has the form  $a_0 + a_1x + a_2x^2$  (Definition 0.8.4), which is the combination  $a_0 \cdot 1 + a_1 \cdot x + a_2 \cdot x^2$ .
- (c) Let  $(p, q) \in \mathbb{R}^2$ . We look for  $a, b$  with  $a(1, 1) + b(1, -1) = (p, q)$ , that is,

$$a + b = p, \quad a - b = q.$$

Adding the equations gives  $2a = p + q$ , and subtracting gives  $2b = p - q$ . So  $a = \frac{p+q}{2}$  and  $b = \frac{p-q}{2}$ . Check:  $\frac{p+q}{2} + \frac{p-q}{2} = p$  and  $\frac{p+q}{2} - \frac{p-q}{2} = q$ . Hence  $(p, q) = \frac{p+q}{2}(1, 1) + \frac{p-q}{2}(1, -1)$ , and the list spans  $\mathbb{R}^2$ . For instance  $(3, 5) = 4(1, 1) - (1, -1)$ .

- (d) For any matrix,  $\begin{pmatrix} a & b \\ c & d \end{pmatrix} = a\mathbf{E}_{11} + b\mathbf{E}_{12} + c\mathbf{E}_{21} + d\mathbf{E}_{22}$ , by adding entry by entry.
- (e)  $\text{span}() = \{\mathbf{0}\}$  by Definition 1.3.2. So the empty list spans the zero space. (So does the list  $(\mathbf{0})$ .)

The solution of (3) divided by 2, which is fine in  $\mathbb{R}$ . Over  $\mathbb{F}_2$ , where  $2 = 0$ , the same list fails: there  $1 = -1$ , so both vectors equal  $(1, 1)$ , and their span is  $\{(0, 0), (1, 1)\} \neq \mathbb{F}_2^2$ . Whether a list spans depends on the field.

Now the non-example, by a minimal change of (3). Replace the list by  $((1, 2), (2, 4))$ . It still has two vectors, and neither is zero. But every combination is

$$a(1, 2) + b(2, 4) = (a + 2b)(1, 2),$$

a vector whose second entry is twice its first. So  $(0, 1)$  is **not** reached, and the list does **not** span  $\mathbb{R}^2$ . The clause that fails is “**every**  $\mathbf{v} \in V$  is a combination”. The two vectors point along the same line, so their span is only the line  $y = 2x$ .

Some spaces are spanned by a short list; others need infinitely many vectors. That distinction is the first coarse measurement of size in this book.

### Definition 1.3.4 — Finite-Dimensional

A vector space  $V$  over  $F$  is **finite-dimensional** if **some** finite list of vectors in  $V$  spans  $V$ . Otherwise  $V$  is **infinite-dimensional**.

By Example (Spanning Lists),  $F^n$ ,  $F[x]_{\leq 2}$ ,  $M_2(F)$  and  $\{\mathbf{0}\}$  are finite-dimensional. The same argument as in (2) and (4) shows that  $F[x]_{\leq n}$  and  $M_{m \times n}(F)$  are finite-dimensional too. The word “dimension” is used before we have defined a number called dimension; that number, and the fact that it is well defined, are the subject of section 6 of this chapter.

For the whole polynomial space no finite list is enough. The reason is degree: finitely many polynomials have a largest degree, and combining them never climbs above it.

**Theorem 1.3.3 —  $F[x]$  Is Infinite-Dimensional**

For every field  $F$ , the vector space  $F[x]$  is infinite-dimensional.

*Proof.* Let  $(p_1, \dots, p_k)$  be any finite list in  $F[x]$ . We show it does not span  $F[x]$ . Each  $\deg p_i$  is either a natural number or  $-\infty$ , and there are finitely many, so we can choose  $N \in \mathbb{N}$  with  $\deg p_i \leq N$  for every  $i$ .

Let  $a_1, \dots, a_k \in F$ . The coefficients of  $a_i p_i$  are  $a_i$  times those of  $p_i$  (Example (The Standard Vector Spaces) (c)). Every coefficient of  $x^m$  with  $m > N$  in  $p_i$  is 0, by Definition 0.8.3, and  $a_i \cdot 0 = 0$  by (F6) and Theorem 0.7.1 (d); hence  $\deg(a_i p_i) \leq N$ . By Theorem 0.8.2,  $\deg(p + q) \leq \max(\deg p, \deg q)$ , so a sum of two polynomials of degree at most  $N$  has degree at most  $N$ . Applying this  $k - 1$  times,  $\deg(a_1 p_1 + \dots + a_k p_k) \leq N$ . (For  $k = 0$  the combination is 0, of degree  $-\infty \leq N$ .)

Hence every element of  $\text{span}(p_1, \dots, p_k)$  has degree at most  $N$ . But  $x^{N+1} \in F[x]$  has degree  $N + 1 > N$ , so  $x^{N+1} \notin \text{span}(p_1, \dots, p_k)$ . Since the list was arbitrary, no finite list spans  $F[x]$ . This shows  $F[x]$  is infinite-dimensional. ■

(In the proof we wrote polynomials in plain italic,  $p$ , as in Chapter 0; they are vectors of  $F[x]$  all the same.) The proof is the move “find a quantity that combinations cannot increase”. The space  $\mathbb{R}^{\mathbb{R}}$  of all functions is infinite-dimensional too, and so are the sequence spaces; we return to such spaces in section 8 of this chapter.

*Remark.* For an arbitrary subset  $S \subseteq V$ , finite or infinite, “ $S$  spans  $V$ ” means that **every** vector of  $V$  is a combination of **some finite** list of vectors in  $S$ . Different vectors may use different finite lists, as in Example (Computing Spans) (d). We never form infinite sums: a vector space only lets us add two vectors at a time, so only finite sums have a meaning. The same “every finite list” reading will be used for independence of an arbitrary set in the next section.

**1.3.5 Working with spans**

Three facts make spans easy to manipulate. Each follows from Theorem 1.3.1 without writing a single combination.

**Proposition 1.3.4 — Basic Properties of Span**

Let  $V$  be a vector space and  $S, T \subseteq V$ .

- (1) If  $T \subseteq \text{span}(S)$ , then  $\text{span}(T) \subseteq \text{span}(S)$ .
- (2) (Monotonicity) If  $T \subseteq S$ , then  $\text{span}(T) \subseteq \text{span}(S)$ .
- (3)  $\text{span}(\text{span}(S)) = \text{span}(S)$ .

*Proof.*

- (1) By [Theorem 1.3.1](#) (1),  $\text{span}(S)$  is a subspace, and by hypothesis it contains  $T$ . By [Theorem 1.3.1](#) (3) applied to  $T$ ,  $\text{span}(T) \subseteq \text{span}(S)$ .
- (2) By [Theorem 1.3.1](#) (2),  $T \subseteq S \subseteq \text{span}(S)$ . Now apply (1).
- (3) Taking  $T = \text{span}(S)$  in (1) gives  $\text{span}(\text{span}(S)) \subseteq \text{span}(S)$ . The reverse inclusion is [Theorem 1.3.1](#) (2) applied to the set  $\text{span}(S)$ .

■

Part (1) is the workhorse. To show  $\text{span}(T) \subseteq \text{span}(S)$ , it is enough to check that each vector of  $T$  is a combination of vectors of  $S$ ; we do not have to handle every combination of  $T$ .

The next fact is the one later sections lean on most. If a vector can already be built from  $S$ , then adding it to  $S$  builds nothing new.

### Theorem 1.3.5 — Absorbing a Vector into a Span

Let  $V$  be a vector space,  $S \subseteq V$  and  $\mathbf{v} \in V$ . If  $\mathbf{v} \in \text{span}(S)$ , then

$$\text{span}(S \cup \{\mathbf{v}\}) = \text{span}(S).$$

In particular, for a list: if  $\mathbf{v} \in \text{span}(\mathbf{v}_1, \dots, \mathbf{v}_k)$ , then  $\text{span}(\mathbf{v}_1, \dots, \mathbf{v}_k, \mathbf{v}) = \text{span}(\mathbf{v}_1, \dots, \mathbf{v}_k)$ , and the same holds with  $\mathbf{v}$  inserted at any position of the list.

*Proof.* ( $\supseteq$ ) Since  $S \subseteq S \cup \{\mathbf{v}\}$ , monotonicity ([Proposition 1.3.4](#) (2)) gives  $\text{span}(S) \subseteq \text{span}(S \cup \{\mathbf{v}\})$ .

( $\subseteq$ ) By [Theorem 1.3.1](#) (2),  $S \subseteq \text{span}(S)$ , and  $\mathbf{v} \in \text{span}(S)$  by hypothesis. Hence  $S \cup \{\mathbf{v}\} \subseteq \text{span}(S)$ , and [Proposition 1.3.4](#) (1) gives  $\text{span}(S \cup \{\mathbf{v}\}) \subseteq \text{span}(S)$ .

For the list version, the span of a list equals the span of the set of its entries, whatever the order. The entries of  $(\mathbf{v}_1, \dots, \mathbf{v}_k, \mathbf{v})$  form the set  $\{\mathbf{v}_1, \dots, \mathbf{v}_k\} \cup \{\mathbf{v}\}$ , so the first part applies.

■

Concretely, the proof substitutes. If  $\mathbf{v} = c_1\mathbf{v}_1 + \dots + c_k\mathbf{v}_k$ , then any combination  $a_1\mathbf{v}_1 + \dots + a_k\mathbf{v}_k + b\mathbf{v}$  equals  $(a_1 + bc_1)\mathbf{v}_1 + \dots + (a_k + bc_k)\mathbf{v}_k$ , which no longer uses  $\mathbf{v}$ . Read backwards, the lemma says that a vector lying in the span of the others can be **deleted** from a list without shrinking its span. The next section turns this into the idea of a redundant vector, and section 6 of this chapter uses it at every step of the Exchange Theorem.

### Example. (Two Lists with the Same Span)

Show that  $\text{span}(1+x, 1-x) = \text{span}(1, x)$  in  $\mathbb{R}[x]$ .

*Solution.* ( $\subseteq$ ) Both  $1+x = 1 \cdot 1 + 1 \cdot x$  and  $1-x = 1 \cdot 1 + (-1) \cdot x$  lie in  $\text{span}(1, x)$ . By [Proposition 1.3.4](#) (1),  $\text{span}(1+x, 1-x) \subseteq \text{span}(1, x)$ .

( $\supseteq$ ) Conversely,

$$1 = \frac{1}{2}(1+x) + \frac{1}{2}(1-x), \quad x = \frac{1}{2}(1+x) - \frac{1}{2}(1-x),$$



so both 1 and  $x$  lie in  $\text{span}(1+x, 1-x)$ . By [Proposition 1.3.4](#) (1) again,  $\text{span}(1, x) \subseteq \text{span}(1+x, 1-x)$ .

Hence the spans are equal; both are  $\mathbb{R}[x]_{\leq 1}$ . Notice that we only checked four vectors, not all combinations. We also used  $\frac{1}{2}$ , so the argument needs  $2 \neq 0$  in the field.

*Quick check.* Is  $x^2 + 1 \in \text{span}(x^2 + x, x + 1)$  in  $\mathbb{R}[x]$ ?

*Solution.* No. Suppose  $a(x^2 + x) + b(x + 1) = x^2 + 1$ . The left side is  $ax^2 + (a+b)x + b$ . Comparing coefficients ([Definition 0.8.1](#)):  $a = 1$  from  $x^2$ ,  $b = 1$  from the constant term, and  $a + b = 0$  from  $x$ . But then  $a + b = 2 \neq 0$ , a contradiction. So no such  $a, b$  exist.

## 1.3.6 Exercises

### 1.3.6.1 A. Check your understanding

#### Exercise A1.

- Define the span of a subset  $S$  of a vector space  $V$ . What is  $\text{span}(\emptyset)$ , and why?
- Determine whether the following statement is true: “for a subset  $S \subseteq V$ ,  $\text{span}(S) = S$  if and only if  $S$  is a subspace of  $V$ .” Justify your answer.
- Determine whether the following statement is true: “if  $\text{span}(S) = \text{span}(T)$ , then  $S = T$ .” Justify your answer.
- Is the zero vector in the span of every list? Is it in the span of the empty list?
- State what it means for  $V$  to be finite-dimensional, and name the quantity used to show that  $F[x]$  is not.

*Solution.*

- $\text{span}(S)$  is the set of all linear combinations  $a_1\mathbf{s}_1 + \cdots + a_k\mathbf{s}_k$  with  $k \geq 0$ ,  $\mathbf{s}_i \in S$  and  $a_i \in F$ .  $\text{span}(\emptyset) = \{\mathbf{0}\}$ : the only combination with no vectors is the empty sum  $\mathbf{0}$ . This is also the smallest subspace containing  $\emptyset$ , as [Theorem 1.3.1](#) requires.
- True. If  $\text{span}(S) = S$ , then  $S$  is a subspace by [Theorem 1.3.1](#) (1). Conversely, if  $S$  is a subspace, then  $\text{span}(S) \subseteq S$  by [Theorem 1.3.1](#) (3) with  $U = S$ , and  $S \subseteq \text{span}(S)$  by (2).
- False. In  $\mathbb{R}^2$ , take  $S = \{(1, 0), (0, 1)\}$  and  $T = \{(1, 1), (1, -1)\}$ . Both span  $\mathbb{R}^2$  ([Example \(Spanning Lists\)](#)), but  $S \neq T$ .
- Yes to both: take all coefficients 0, or, for the empty list, the empty sum. See [Example \(Linear Combinations in Four Families\)](#) (d).
- $V$  is finite-dimensional if some finite list of vectors in  $V$  spans  $V$ . For  $F[x]$  the quantity is the degree: combinations of a finite list have degree at most the largest degree in the list ([Theorem 1.3.3](#)).

### 1.3.6.2 B. Practice

**Exercise B1: Membership in a Span.**

Determine whether each vector lies in the given span. If it does, write it as a linear combination. Justify your answer.

- (a)  $(2, -1, 5)$  and  $(3, 4, 1)$ , in  $\text{span}((1, 1, 0), (0, 1, 1)) \subseteq \mathbb{R}^3$ .  
 (b)  $2x^2 - 3x + 1$  and  $x^2 + x + 1$ , in  $\text{span}(x^2 - x, x - 1) \subseteq \mathbb{R}[x]$ .

*Solution.*

- (a) A combination is  $a(1, 1, 0) + b(0, 1, 1) = (a, a + b, b)$ . For  $(2, -1, 5)$ , the first and third entries force  $a = 2$  and  $b = 5$ ; then the middle entry would be  $7 \neq -1$ . Hence  $(2, -1, 5)$  is **not** in the span. For  $(3, 4, 1)$ , we get  $a = 3$ ,  $b = 1$ , and the middle entry is 4, as required. Hence  $(3, 4, 1) = 3(1, 1, 0) + (0, 1, 1)$  is in the span.
- (b) A combination is  $a(x^2 - x) + b(x - 1) = ax^2 + (b - a)x - b$ . Compare coefficients (Definition 0.8.1). For  $2x^2 - 3x + 1$ :  $a = 2$ ,  $-b = 1$  so  $b = -1$ , and then  $b - a = -3$ , as required. Hence  $2x^2 - 3x + 1 = 2(x^2 - x) - (x - 1)$  is in the span. For  $x^2 + x + 1$ :  $a = 1$ ,  $b = -1$ , and then  $b - a = -2 \neq 1$ . Hence  $x^2 + x + 1$  is **not** in the span. (Every element  $p$  of this span has  $p(1) = a - a + b - b = 0$ , and indeed  $x^2 + x + 1$  has value 3 at 1.)

**Exercise B2: Two Lists with the Same Span.**

Prove that  $\text{span}((1, 1, 0), (0, 1, 1)) = \text{span}((1, 2, 1), (1, 0, -1))$  in  $\mathbb{R}^3$ .

*Solution.* Write  $\mathbf{u} = (1, 1, 0)$ ,  $\mathbf{w} = (0, 1, 1)$ .

( $\subseteq$ ) We have  $\mathbf{u} = \frac{1}{2}(1, 2, 1) + \frac{1}{2}(1, 0, -1)$  and  $\mathbf{w} = \frac{1}{2}(1, 2, 1) - \frac{1}{2}(1, 0, -1)$ : indeed  $\frac{1}{2}(2, 2, 0) = (1, 1, 0)$  and  $\frac{1}{2}(0, 2, 2) = (0, 1, 1)$ . So  $\mathbf{u}, \mathbf{w} \in \text{span}((1, 2, 1), (1, 0, -1))$ , and Proposition 1.3.4 (1) gives the inclusion.

( $\supseteq$ ) We have  $(1, 2, 1) = \mathbf{u} + \mathbf{w}$  and  $(1, 0, -1) = \mathbf{u} - \mathbf{w}$ . So both lie in  $\text{span}(\mathbf{u}, \mathbf{w})$ , and Proposition 1.3.4 (1) gives the reverse inclusion.

Hence the two spans are equal.

**Exercise B3: Spans in  $\mathbb{R}^3$ , Geometrically.**

Describe each span as the origin, a line through the origin, a plane through the origin (give an equation), or all of  $\mathbb{R}^3$ . Justify your answer.

- (a)  $\text{span}((1, 2, 3))$ .  
 (b)  $\text{span}((1, 0, 0), (0, 1, 0), (1, 1, 0))$ .  
 (c)  $\text{span}((1, -1, 0), (0, 1, -1))$ .  
 (d)  $\text{span}((0, 0, 0))$ .

*Solution.*

- (a) The span is  $\{(t, 2t, 3t) : t \in \mathbb{R}\}$ : the line through the origin and  $(1, 2, 3)$ .

- (b) Since  $(1, 1, 0) = (1, 0, 0) + (0, 1, 0)$ , [Theorem 1.3.5](#) gives  $\text{span}((1, 0, 0), (0, 1, 0), (1, 1, 0)) = \text{span}((1, 0, 0), (0, 1, 0)) = \{(a, b, 0) : a, b \in \mathbb{R}\}$ : the plane  $z = 0$ .
- (c) A combination is  $a(1, -1, 0) + b(0, 1, -1) = (a, b - a, -b)$ , whose entries sum to 0. Conversely, if  $x + y + z = 0$ , take  $a = x$  and  $b = -z$ ; then  $b - a = -z - x = y$ , so  $(x, y, z) = x(1, -1, 0) - z(0, 1, -1)$ . Hence the span is the plane  $x + y + z = 0$ .
- (d)  $\text{span}((0, 0, 0)) = \{\mathbf{0}\}$ , by [Example \(Computing Spans\)](#) (b): the origin alone.

### 1.3.6.3 C. Going deeper

#### Exercise C1: Spans of Intersections and Unions.

Let  $S, T$  be subsets of a vector space  $V$ .

- (a) Prove that  $\text{span}(S \cap T) \subseteq \text{span}(S) \cap \text{span}(T)$ .
- (b) Give an example in  $\mathbb{R}^2$  where the inclusion in (a) is strict.
- (c) Prove that  $\text{span}(S \cup T) = \{\mathbf{s} + \mathbf{t} : \mathbf{s} \in \text{span}(S), \mathbf{t} \in \text{span}(T)\}$ .

*Hint for (c): in a combination of vectors of  $S \cup T$ , sort the terms into two groups.*

*Solution.*

- (a) Since  $S \cap T \subseteq S$ , monotonicity ([Proposition 1.3.4](#) (2)) gives  $\text{span}(S \cap T) \subseteq \text{span}(S)$ . By symmetry,  $\text{span}(S \cap T) \subseteq \text{span}(T)$ . Hence  $\text{span}(S \cap T) \subseteq \text{span}(S) \cap \text{span}(T)$ .
- (b) Take  $S = \{(1, 0), (0, 1)\}$  and  $T = \{(1, 1)\}$ . Then  $S \cap T = \emptyset$ , so  $\text{span}(S \cap T) = \{\mathbf{0}\}$ . But  $\text{span}(S) = \mathbb{R}^2$  ([Example \(Spanning Lists\)](#)), so  $\text{span}(S) \cap \text{span}(T) = \text{span}(T)$ , the line  $y = x$ , which contains  $(1, 1) \neq \mathbf{0}$ . The inclusion is strict.
- (c) Write  $W = \{\mathbf{s} + \mathbf{t} : \mathbf{s} \in \text{span}(S), \mathbf{t} \in \text{span}(T)\}$ .  
 $(\subseteq)$  Let  $\mathbf{v} \in \text{span}(S \cup T)$ , say  $\mathbf{v} = a_1 \mathbf{u}_1 + \cdots + a_k \mathbf{u}_k$  with each  $\mathbf{u}_i \in S \cup T$ . Let  $\mathbf{s}$  be the sum of the terms  $a_i \mathbf{u}_i$  with  $\mathbf{u}_i \in S$ , and  $\mathbf{t}$  the sum of the remaining terms, whose vectors lie in  $T$ . Since addition is commutative and associative,  $\mathbf{v} = \mathbf{s} + \mathbf{t}$ . Now  $\mathbf{s} \in \text{span}(S)$  and  $\mathbf{t} \in \text{span}(T)$ ; if a group is empty, its sum is  $\mathbf{0}$ , which lies in every span. Hence  $\mathbf{v} \in W$ .  
 $(\supseteq)$  Let  $\mathbf{s} \in \text{span}(S)$  and  $\mathbf{t} \in \text{span}(T)$ . By monotonicity, both lie in  $\text{span}(S \cup T)$ , which is closed under addition by [Theorem 1.3.1](#) (1). Hence  $\mathbf{s} + \mathbf{t} \in \text{span}(S \cup T)$ . By [Theorem 0.3.1](#),  $\text{span}(S \cup T) = W$ . (So  $W$  is the set  $\text{span}(S) + \text{span}(T)$  of [Exercise \(C2: The Sum of Two Subspaces\)](#), the **sum** of the two subspaces, studied in section 7 of this chapter.)

#### Exercise C2: The Span Depends on the Field.

The set  $\mathbb{C}^2$  is a vector space over  $\mathbb{C}$ . With the same addition, and scalar multiplication restricted to real scalars, it is also a vector space over  $\mathbb{R}$ , by [Exercise \(C3: Shrinking the Field\)](#): every axiom holds for all complex scalars, so in particular for real ones.

- (a) Show that  $((1, 0), (0, 1))$  spans  $\mathbb{C}^2$  over  $\mathbb{C}$ .
- (b) Explain why  $((1, 0), (0, 1))$  does **not** span  $\mathbb{C}^2$  over  $\mathbb{R}$ .
- (c) Find a finite list that spans  $\mathbb{C}^2$  over  $\mathbb{R}$ . Hence  $\mathbb{C}^2$  is finite-dimensional over  $\mathbb{R}$  as well.

*Solution.*

- (a) For  $(z, w) \in \mathbb{C}^2$ ,  $(z, w) = z(1, 0) + w(0, 1)$  with  $z, w \in \mathbb{C}$ .
- (b) Over  $\mathbb{R}$  the coefficients must be real, and  $a(1, 0) + b(0, 1) = (a, b)$  with  $a, b \in \mathbb{R}$  has real entries. So  $(i, 0)$  is not in  $\text{span}_{\mathbb{R}}((1, 0), (0, 1))$ .
- (c) Take  $((1, 0), (i, 0), (0, 1), (0, i))$ . Write  $z = a + bi$  and  $w = c + di$  with  $a, b, c, d \in \mathbb{R}$  (Definition 0.6.1). Then

$$(z, w) = a(1, 0) + b(i, 0) + c(0, 1) + d(0, i),$$

a combination with real coefficients. So this list spans  $\mathbb{C}^2$  over  $\mathbb{R}$ , and  $\mathbb{C}^2$  is finite-dimensional over  $\mathbb{R}$  by Definition 1.3.4.

## 1.4 Linear Independence

A spanning list tells us that every vector of a space can be built. It says nothing about waste. The list  $((1, 0), (0, 1), (1, 1))$  spans  $\mathbb{R}^2$ , but its third vector adds nothing that the first two did not already reach. To measure a space by the length of a list, we need lists with no such waste. This section makes “no vector is wasted” precise, proves the lemma that finds a wasted vector whenever there is one, and shows that lists without waste give every vector in their span a unique description.

### 1.4.1 Redundant vectors

The previous section gave a way to shorten a list: if a vector lies in the span of the others, we may delete it without changing the span (Theorem 1.3.5). Call such a vector **redundant**. We want a test for “this list has no redundant vector”.

A first guess is that a vector is wasted when it repeats the direction of another one, so the test would be “no vector is a scalar multiple of another”. This guess fails. In  $\mathbb{R}^3$ , take

$$\mathbf{u} = (1, 0, 1), \quad \mathbf{v} = (0, 1, 1), \quad \mathbf{w} = (1, 1, 2).$$

No one of them is a multiple of another: a multiple of  $\mathbf{u}$  has second entry 0, a multiple of  $\mathbf{v}$  has first entry 0, a multiple  $(c, c, 2c)$  of  $\mathbf{w}$  has equal first and second entries, and in each case neither of the other two vectors has that shape. Yet  $\mathbf{w} = \mathbf{u} + \mathbf{v}$ . All three lie in the plane  $z = x + y$ , and by Theorem 1.3.5,  $\text{span}(\mathbf{u}, \mathbf{v}, \mathbf{w}) = \text{span}(\mathbf{u}, \mathbf{v})$ . The vector  $\mathbf{w}$  is redundant even though it is not parallel to anything.

A second guess is correct but clumsy: “no vector is a linear combination of the others”. For a list of length  $k$  this is  $k$  separate conditions, one for each vector. Here is a way to fold them into one. Move everything in  $\mathbf{w} = \mathbf{u} + \mathbf{v}$  to one side:

$$1\mathbf{u} + 1\mathbf{v} + (-1)\mathbf{w} = \mathbf{0}.$$

This is a way to build  $\mathbf{0}$  from the list with coefficients that are **not all zero**. Conversely, suppose  $a\mathbf{u} + b\mathbf{v} + c\mathbf{w} = \mathbf{0}$  with, say,  $c \neq 0$ . Then  $\mathbf{w} = -\frac{a}{c}\mathbf{u} - \frac{b}{c}\mathbf{v}$ , so  $\mathbf{w}$  is redundant. The same works whichever coefficient is non-zero. So a redundant vector exists exactly when  $\mathbf{0}$  can be built in a non-trivial way. Building  $\mathbf{0}$  trivially, with every coefficient 0, is always possible, so the condition we want is that this is the only way.

A list is independent when the only way to build the zero vector from it is the trivial way.

## 1.4.2 Definition and examples

### Definition 1.4.1 — Linear Independence

Let  $V$  be a vector space over  $F$  and let  $(\mathbf{v}_1, \dots, \mathbf{v}_k)$  be a list of vectors in  $V$ , where  $k \geq 0$ . The list is **linearly independent** (over  $F$ ) if, for all  $a_1, \dots, a_k \in F$ ,

$$a_1\mathbf{v}_1 + a_2\mathbf{v}_2 + \dots + a_k\mathbf{v}_k = \mathbf{0} \implies a_1 = a_2 = \dots = a_k = 0.$$

The list is **linearly dependent** if it is not linearly independent: that is, if there exist  $a_1, \dots, a_k \in F$ , **not all zero**, with  $a_1\mathbf{v}_1 + \dots + a_k\mathbf{v}_k = \mathbf{0}$ . Such an equation is a **linear relation** among the vectors. The empty list is linearly independent.

In words: the hypothesis is an arbitrary linear combination that happens to equal  $\mathbf{0}$ . The conclusion says that **every** coefficient in it is 0. Dependence is the negation, obtained by the rules of Definition 0.1.4: “for all  $a$ , (combination =  $\mathbf{0} \Rightarrow$  all  $a_i = 0$ )” becomes “there exist  $a$  with combination =  $\mathbf{0}$  and **not** all  $a_i = 0$ ”. This is exactly the negation worked out in Example (Negating “only the zero combination”). The empty list is independent because the condition is vacuous: there are no coefficients, so “all of them are 0” holds automatically.

The definition also dictates how every independence proof starts. It is a “for all” statement about coefficients, so we take arbitrary coefficients and assume the hypothesis:

**Let**  $a_1\mathbf{v}_1 + \dots + a_k\mathbf{v}_k = \mathbf{0}$ . Then ... hence all  $a_i = 0$ .

Between “Let” and “hence”, the work is to turn one vector equation into scalar equations. How we do that depends on what  $\mathbf{0}$  means in the space: in  $F^n$  we compare entries, in  $F[x]$  we compare coefficients, in a function space we evaluate at points, and in  $M_{m \times n}(F)$  we compare entries.

### Example. (Independent Lists in Four Families)

Show that each list is linearly independent.

- (a)  $(\mathbf{e}_1, \dots, \mathbf{e}_n)$  in  $F^n$ .
- (b)  $(1, x, x^2)$  in  $F[x]$ .
- (c)  $((1, 1), (1, -1))$  in  $\mathbb{R}^2$ .
- (d)  $(\sin, \cos)$  in the space  $\mathbb{R}^{\mathbb{R}}$  of all functions  $\mathbb{R} \rightarrow \mathbb{R}$ .

*Solution.*

- (a) Let  $a_1\mathbf{e}_1 + \dots + a_n\mathbf{e}_n = \mathbf{0}$ . The left side is the vector  $(a_1, \dots, a_n)$ , and  $\mathbf{0} = (0, \dots, 0)$ . Comparing entries,  $a_1 = \dots = a_n = 0$ . Hence the list is independent.
- (b) Let  $a + bx + cx^2 = 0$ , where 0 is the **zero polynomial**, the polynomial with every coefficient 0. Two polynomials are equal exactly when all their coefficients agree (Definition 0.8.1). Comparing the coefficients of  $1, x, x^2$  gives  $a = b = c = 0$ . Hence the list is independent.
- (c) Let  $a(1, 1) + b(1, -1) = (0, 0)$ . Comparing entries,  $a + b = 0$  and  $a - b = 0$ . Adding

the two equations gives  $2a = 0$ , so  $a = 0$  since  $2 \neq 0$  in  $\mathbb{R}$ ; then  $b = -a = 0$ . Hence the list is independent.

- (d) Let  $a \sin t + b \cos t = 0$ , where  $0$  is the **zero function**. Equality of functions means equality at every point (Definition 0.4.1), so  $a \sin t + b \cos t = 0$  for **every**  $t \in \mathbb{R}$ . We may choose convenient points. At  $t = 0$ :  $a \cdot 0 + b \cdot 1 = 0$ , so  $b = 0$ . At  $t = \pi/2$ :  $a \cdot 1 + b \cdot 0 = 0$ , so  $a = 0$ . Hence the list is independent.

In (d) we did not need the equation at every point, only at two well-chosen ones. That is the general move for functions: **choose good points**, where most terms vanish. Note the logic. Evaluating at a point can only prove independence; if the equations at our chosen points had allowed non-zero solutions, we would have learned nothing, and would have to try other points or another method.

The degenerate cases are worth settling once, because they come up inside every later argument.

### Example. (Degenerate Lists)

Let  $V$  be a vector space over  $F$ . Decide whether each list is independent.

- (a) A list  $(\mathbf{v}_1, \dots, \mathbf{v}_k)$  with some  $\mathbf{v}_j = \mathbf{0}$ .
- (b) A list  $(\mathbf{v})$  of length one.
- (c) A list  $(\mathbf{v}_1, \dots, \mathbf{v}_k)$  with a repeat:  $\mathbf{v}_i = \mathbf{v}_j$  for some  $i \neq j$ .

*Solution.*

- (a) Dependent. Take  $a_j = 1$  and every other coefficient 0. By Theorem 1.1.5 and Theorem 1.1.6, every term is  $\mathbf{0}$ , so the combination is  $\mathbf{0}$ , and the coefficient  $a_j = 1$  is not zero.
- (b) Independent if and only if  $\mathbf{v} \neq \mathbf{0}$ . If  $\mathbf{v} = \mathbf{0}$ , use (a). If  $\mathbf{v} \neq \mathbf{0}$ , let  $a\mathbf{v} = \mathbf{0}$ ; by Theorem 1.1.9,  $a = 0$  or  $\mathbf{v} = \mathbf{0}$ , and the second is false, so  $a = 0$ .
- (c) Dependent. Take  $a_i = 1$ ,  $a_j = -1$ , and every other coefficient 0. The combination is  $\mathbf{v}_i + (-1)\mathbf{v}_j = \mathbf{v}_i - \mathbf{v}_j = \mathbf{0}$  by Theorem 1.1.7, since  $\mathbf{v}_i = \mathbf{v}_j$ , and  $a_i = 1 \neq 0$ .

Now the non-example, by a minimal change. Add one vector to the independent list  $((1, 0), (0, 1))$  of Example (Independent Lists in Four Families) (a), to get  $((1, 0), (0, 1), (1, 1))$ . Each vector is still non-zero, and any two of the three still form an independent list. But

$$1(1, 0) + 1(0, 1) + (-1)(1, 1) = (0, 0)$$

with coefficients not all zero, so the clause “ $\implies a_1 = \dots = a_k = 0$ ” fails, and the list is **dependent**. The polynomial version behaves the same way:  $(1, x)$  is independent, but  $(1, x, 1 + x)$  is dependent, because  $1 \cdot 1 + 1 \cdot x + (-1)(1 + x) = 0$ .

*Quick check.* Is the list  $((1, 2, 0), (0, 1, 1), (1, 0, -2))$  in  $\mathbb{R}^3$  linearly independent?

*Solution.* No. Let  $a(1, 2, 0) + b(0, 1, 1) + c(1, 0, -2) = \mathbf{0}$ . Comparing entries:  $a + c = 0$ ,  $2a + b = 0$ ,  $b - 2c = 0$ . The first two give  $c = -a$  and  $b = -2a$ , and then the third reads  $-2a + 2a = 0$ , which holds for every  $a$ . Taking  $a = 1$  gives the relation  $(1, 2, 0) - 2(0, 1, 1) - (1, 0, -2) = \mathbf{0}$ , with coefficients not all zero.

Two more kinds of question come up constantly: lists that depend on a parameter, and lists of

matrices. The method does not change.

**Example. (A Parameter and a List of Matrices)**

- (a) For which  $k \in \mathbb{R}$  is the list  $((1, 1, 0), (1, 0, 1), (0, 1, k))$  in  $\mathbb{R}^3$  linearly independent?
- (b) Is the list  $\left(\begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix}, \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}, \begin{pmatrix} 0 & 1 \\ 0 & 0 \end{pmatrix}\right)$  in  $M_2(\mathbb{R})$  linearly independent?

*Solution.*

- (a) Let  $a(1, 1, 0) + b(1, 0, 1) + c(0, 1, k) = \mathbf{0}$ . Comparing entries gives  $a + b = 0$ ,  $a + c = 0$  and  $b + kc = 0$ . The first two give  $b = -a$  and  $c = -a$ ; substituting into the third,  $-a - ka = -(1 + k)a = 0$ .

If  $k \neq -1$ , then  $1 + k \neq 0$ , so  $a = 0$  by [Theorem 0.7.1](#), and then  $b = c = 0$ . The list is independent.

If  $k = -1$ , the equations hold for every  $a$ ; taking  $a = 1$  gives  $(1, 1, 0) - (1, 0, 1) - (0, 1, -1) = \mathbf{0}$  with coefficients not all zero. The list is dependent.

So the list is independent exactly when  $k \neq -1$ . Notice the shape of the argument: the parameter ends up as a factor multiplying one unknown, and the only question is when that factor can be zero.

- (b) Let  $a \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix} + b \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix} + c \begin{pmatrix} 0 & 1 \\ 0 & 0 \end{pmatrix} = \begin{pmatrix} 0 & 0 \\ 0 & 0 \end{pmatrix}$ . The left side is  $\begin{pmatrix} a & b + c \\ b & a \end{pmatrix}$ . Two matrices are equal when their entries agree ([Definition 0.9.1](#)), so  $a = 0$ ,  $b + c = 0$  and  $b = 0$ ; hence  $c = 0$  too. The list is independent.

For matrices, “compare entries” turns one matrix equation into four scalar equations, one per entry.

### 1.4.3 Why lists, and why this definition

*Remark. Why lists, not sets.* In a set, repeats disappear:  $\{\mathbf{v}, \mathbf{v}\} = \{\mathbf{v}\}$ . If independence were defined for sets, the “list”  $(\mathbf{v}, \mathbf{v})$ , which has the obvious relation  $\mathbf{v} - \mathbf{v} = \mathbf{0}$ , would silently become the independent set  $\{\mathbf{v}\}$  when  $\mathbf{v} \neq \mathbf{0}$ . The repeat can also be hidden:  $x + 1$  and  $1 + x$  look different on the page but are the same vector. A list keeps both entries and so records the dependence. The second reason is order. Coordinates with respect to a basis (next section) are read off in order, so  $(\mathbf{e}_1, \mathbf{e}_2)$  and  $(\mathbf{e}_2, \mathbf{e}_1)$  must be different objects. The Linear Dependence Lemma below also refers to “the vectors before  $\mathbf{v}_j$ ”, which needs an order. Reordering a list does not change whether it is independent, since a relation can be rearranged term by term.

For an arbitrary subset  $S \subseteq V$ , finite or infinite, we say  $S$  is **linearly independent** if **every** finite list of **distinct** vectors of  $S$  is linearly independent. The word “distinct” is essential: without it, the list  $(\mathbf{s}, \mathbf{s})$  would make every non-empty set dependent. For a finite set  $S = \{\mathbf{v}_1, \dots, \mathbf{v}_k\}$  with the  $\mathbf{v}_i$  distinct, this agrees with independence of the list  $(\mathbf{v}_1, \dots, \mathbf{v}_k)$ : every list of distinct vectors of  $S$  is a reordered sub-list, and a sub-list of an independent list is independent ([Exercise \(B3: Sub-Lists\)](#)). Together with “ $S$  spans  $V$ ” from the previous section, this is the language section 8 of this chapter uses for spaces with no finite spanning list.

**Why “all  $a_i = 0$ ”, and why is the empty list independent?** Suppose we only asked that **some**  $a_i$  be 0. The list  $((1, 0), (0, 1), (0, 2))$  has the redundant vector  $(0, 2) = 2(0, 1)$ . But in any

relation  $a(1, 0) + b(0, 1) + c(0, 2) = \mathbf{0}$ , comparing first entries forces  $a = 0$ , so every relation has a zero coefficient, and the list would pass the weaker test. Only “**all**  $a_i = 0$ ” rules out every non-trivial relation. The empty list is independent because it wastes nothing, and this convention makes the empty list a basis of the zero space in the next section, just as  $\text{span}() = \{\mathbf{0}\}$  makes it a spanning list.

*Warning.* “**Not all zero**” is not “**all non-zero**”. A dependence relation needs **one** non-zero coefficient, not every coefficient non-zero. The list  $((1, 0), (0, 1), (0, 2))$  is dependent, by  $0(1, 0) + 2(0, 1) - (0, 2) = \mathbf{0}$ . But no relation has all three coefficients non-zero: comparing first entries in  $a(1, 0) + b(0, 1) + c(0, 2) = \mathbf{0}$  forces  $a = 0$ . Hunting for a relation with no zero coefficient would wrongly conclude that the list is independent.

*Warning.* **Pairwise independence is not independence.** Checking pairs is not enough. In the list  $((1, 0, 1), (0, 1, 1), (1, 1, 2))$  from the start of the section, every two of the three vectors form an independent list, but the three together are dependent, since  $(1, 0, 1) + (0, 1, 1) - (1, 1, 2) = \mathbf{0}$ . For three or more vectors, independence is a condition on all of them at once.

## 1.4.4 The Linear Dependence Lemma

The definition says that a dependent list has a relation. It does not say which vector is redundant, and the discussion at the start of the section only found a redundant vector where the coefficient was non-zero. The next lemma makes a definite choice, and chooses it so that the redundant vector depends only on vectors **before** it in the list. This one-directional form is what later proofs need, because they build lists from left to right.

### Theorem 1.4.1 — Linear Dependence Lemma

Let  $V$  be a vector space over  $F$ , and let  $(\mathbf{v}_1, \dots, \mathbf{v}_k)$  be a linearly dependent list in  $V$ . Then:

- (1) there is an index  $j \in \{1, \dots, k\}$  with  $\mathbf{v}_j \in \text{span}(\mathbf{v}_1, \dots, \mathbf{v}_{j-1})$ . For  $j = 1$  this means  $\mathbf{v}_1 = \mathbf{0}$ , so if  $\mathbf{v}_1 \neq \mathbf{0}$ , then  $j \geq 2$ ;
- (2) for any such  $j$ , removing  $\mathbf{v}_j$  does not change the span:

$$\text{span}(\mathbf{v}_1, \dots, \mathbf{v}_{j-1}, \mathbf{v}_{j+1}, \dots, \mathbf{v}_k) = \text{span}(\mathbf{v}_1, \dots, \mathbf{v}_k).$$

*Idea.* Take a relation with coefficients not all zero. We want to solve for one vector in terms of the vectors before it, so we need a vector whose coefficient is non-zero and which comes **after** every other vector with a non-zero coefficient. That is the **largest** index  $j$  with  $a_j \neq 0$ . Everything after it has coefficient zero and drops out, and we divide by  $a_j$ . Part (2) is then [Theorem 1.3.5](#) read backwards.

*Proof.*

- (1) Since the list is dependent, there are  $a_1, \dots, a_k \in F$ , not all zero, with  $a_1\mathbf{v}_1 + \dots + a_k\mathbf{v}_k = \mathbf{0}$ . The set  $\{i : a_i \neq 0\}$  is non-empty and finite, so it has a largest element; call it  $j$ . Then  $a_j \neq 0$ , and  $a_i = 0$  for every  $i > j$ . By [Theorem 1.1.5](#) the terms



with  $i > j$  are  $\mathbf{0}$ , so

$$a_1\mathbf{v}_1 + \cdots + a_{j-1}\mathbf{v}_{j-1} + a_j\mathbf{v}_j = \mathbf{0}.$$

Adding the negative of  $a_1\mathbf{v}_1 + \cdots + a_{j-1}\mathbf{v}_{j-1}$  to both sides gives  $a_j\mathbf{v}_j = -(a_1\mathbf{v}_1 + \cdots + a_{j-1}\mathbf{v}_{j-1})$ . Since  $a_j \neq 0$  and  $F$  is a field,  $a_j$  has an inverse  $a_j^{-1} \in F$  (Definition 0.7.1). Multiplying both sides by  $a_j^{-1}$ , and using  $a_j^{-1}(a_j\mathbf{v}_j) = (a_j^{-1}a_j)\mathbf{v}_j = 1\mathbf{v}_j = \mathbf{v}_j$  on the left, and  $-\mathbf{x} = (-1)\mathbf{x}$  (Theorem 1.1.7) with the distributive laws on the right, we get

$$\mathbf{v}_j = (-a_j^{-1}a_1)\mathbf{v}_1 + \cdots + (-a_j^{-1}a_{j-1})\mathbf{v}_{j-1}.$$

Hence  $\mathbf{v}_j \in \text{span}(\mathbf{v}_1, \dots, \mathbf{v}_{j-1})$ . When  $j = 1$  the right side is the empty sum, so  $\mathbf{v}_1 = \mathbf{0}$ ; therefore if  $\mathbf{v}_1 \neq \mathbf{0}$ , then  $j \geq 2$ .

- (2) Let  $j$  be any index with  $\mathbf{v}_j \in \text{span}(\mathbf{v}_1, \dots, \mathbf{v}_{j-1})$ , and let  $L = (\mathbf{v}_1, \dots, \mathbf{v}_{j-1}, \mathbf{v}_{j+1}, \dots, \mathbf{v}_k)$ . Each of  $\mathbf{v}_1, \dots, \mathbf{v}_{j-1}$  is an entry of  $L$ , so by monotonicity (Proposition 1.3.4 (2))  $\text{span}(\mathbf{v}_1, \dots, \mathbf{v}_{j-1}) \subseteq \text{span}(L)$ , and hence  $\mathbf{v}_j \in \text{span}(L)$ . By Theorem 1.3.5, inserting  $\mathbf{v}_j$  back into  $L$  at position  $j$  does not change the span. That is,  $\text{span}(\mathbf{v}_1, \dots, \mathbf{v}_k) = \text{span}(L)$ , as claimed. ■

The index  $j$  need not be unique, and the lemma does **not** say that every vector depends on earlier ones. A small example shows both.

#### Example. (The Lemma on a Concrete List)

In  $\mathbb{R}^2$ , let  $\mathbf{v}_1 = (1, 1)$ ,  $\mathbf{v}_2 = (1, -1)$ ,  $\mathbf{v}_3 = (2, 0)$ ,  $\mathbf{v}_4 = (0, 1)$ . Find two different linear relations, and the index  $j$  that the proof of Theorem 1.4.1 extracts from each. Which indices  $j$  satisfy  $\mathbf{v}_j \in \text{span}(\mathbf{v}_1, \dots, \mathbf{v}_{j-1})$ ?

*Solution.* First relation:  $\mathbf{v}_1 + \mathbf{v}_2 - \mathbf{v}_3 = (1 + 1 - 2, 1 - 1 - 0) = (0, 0)$ . The largest index with a non-zero coefficient is  $j = 3$ , and solving gives  $\mathbf{v}_3 = \mathbf{v}_1 + \mathbf{v}_2 \in \text{span}(\mathbf{v}_1, \mathbf{v}_2)$ .

Second relation:  $\mathbf{v}_1 - \mathbf{v}_2 - 2\mathbf{v}_4 = (1 - 1 - 0, 1 + 1 - 2) = (0, 0)$ . Here the largest index is  $j = 4$ , and  $\mathbf{v}_4 = \frac{1}{2}\mathbf{v}_1 - \frac{1}{2}\mathbf{v}_2 \in \text{span}(\mathbf{v}_1, \mathbf{v}_2, \mathbf{v}_3)$ .

So  $j = 3$  and  $j = 4$  both qualify, and by part (2) of the lemma either  $\mathbf{v}_3$  or  $\mathbf{v}_4$  can be removed without changing the span. On the other hand,  $j = 2$  does not qualify:  $\mathbf{v}_2 = (1, -1)$  is not a multiple of  $\mathbf{v}_1 = (1, 1)$ , because a multiple  $(c, c)$  has equal entries. And  $j = 1$  does not occur since  $\mathbf{v}_1 \neq \mathbf{0}$ . Removing  $\mathbf{v}_3$  and then (by the lemma applied to  $(\mathbf{v}_1, \mathbf{v}_2, \mathbf{v}_4)$ , which is still dependent by the second relation)  $\mathbf{v}_4$  leaves  $(\mathbf{v}_1, \mathbf{v}_2)$ , whose span is  $\mathbb{R}^2$  by Example (Spanning Lists).

The lemma only locates vectors that depend on **earlier** ones. Here  $\mathbf{v}_1$  and  $\mathbf{v}_2$  could also be removed without changing the span, since  $\mathbf{v}_1 = \mathbf{v}_3 - \mathbf{v}_2$  and  $\mathbf{v}_2 = \mathbf{v}_3 - \mathbf{v}_1$ , but the lemma never points at them.

Read the other way round, part (1) is a test for independence that works left to right.

*Remark. Independence, one vector at a time.* A list  $(\mathbf{v}_1, \dots, \mathbf{v}_k)$  is linearly independent **if and only if** no  $\mathbf{v}_j$  lies in  $\text{span}(\mathbf{v}_1, \dots, \mathbf{v}_{j-1})$ . The “if” direction is the

contrapositive of part (1). For “only if”, suppose  $\mathbf{v}_j = c_1\mathbf{v}_1 + \cdots + c_{j-1}\mathbf{v}_{j-1}$ . Then  $c_1\mathbf{v}_1 + \cdots + c_{j-1}\mathbf{v}_{j-1} + (-1)\mathbf{v}_j = \mathbf{0}$ , with zero coefficients on  $\mathbf{v}_{j+1}, \dots, \mathbf{v}_k$ , is a relation whose coefficient  $-1$  is non-zero. So the list is dependent. In words: a list is independent exactly when each vector reaches something new beyond the vectors before it. For instance  $(1, x, x^2)$  passes:  $1 \neq 0$ ,  $x$  is not a constant, and  $x^2$  has degree too high to be in  $\text{span}(1, x)$ .

Combining the two parts of the lemma gives a clean statement about spans alone: a list is dependent exactly when it contains a vector that can be thrown away for free.

### Theorem 1.4.2 — Span Preservation

Let  $V$  be a vector space and  $(\mathbf{v}_1, \dots, \mathbf{v}_k)$  a list in  $V$  with  $k \geq 1$ . Then the list is linearly dependent **if and only if** there is an index  $j$  such that

$$\text{span}(\mathbf{v}_1, \dots, \mathbf{v}_{j-1}, \mathbf{v}_{j+1}, \dots, \mathbf{v}_k) = \text{span}(\mathbf{v}_1, \dots, \mathbf{v}_k).$$

*Proof.* ( $\Rightarrow$ ) By Theorem 1.4.1 (1) there is an index  $j$  with  $\mathbf{v}_j \in \text{span}(\mathbf{v}_1, \dots, \mathbf{v}_{j-1})$ , and by part (2) removing that  $\mathbf{v}_j$  does not change the span.

( $\Leftarrow$ ) Let  $L$  be the list with  $\mathbf{v}_j$  removed, and suppose  $\text{span}(L) = \text{span}(\mathbf{v}_1, \dots, \mathbf{v}_k)$ . By Theorem 1.3.1 (2),  $\mathbf{v}_j \in \text{span}(\mathbf{v}_1, \dots, \mathbf{v}_k) = \text{span}(L)$ . So  $\mathbf{v}_j = \sum_{i \neq j} c_i \mathbf{v}_i$  for some scalars  $c_i$ , and therefore  $\sum_{i \neq j} c_i \mathbf{v}_i + (-1)\mathbf{v}_j = \mathbf{0}$ . The coefficient of  $\mathbf{v}_j$  is  $-1 \neq 0$ , so the list is dependent. ■

So a spanning list that is dependent can always be shortened and still span. Repeating this until the list becomes independent is the idea behind the next section’s proof that every finite spanning list contains a basis.

## 1.4.5 Unique representations

Independence was introduced to rule out waste. Its most useful consequence is about uniqueness. If a vector could be built from a list in two different ways, subtracting the two recipes would give a non-trivial way to build  $\mathbf{0}$ . So for an independent list, every vector in the span has exactly one recipe.

### Theorem 1.4.3 — Independence and Uniqueness of Coefficients

Let  $V$  be a vector space over  $F$  and  $(\mathbf{v}_1, \dots, \mathbf{v}_k)$  a list in  $V$ . The following are equivalent:

- (a)  $(\mathbf{v}_1, \dots, \mathbf{v}_k)$  is linearly independent;
- (b) every  $\mathbf{v} \in \text{span}(\mathbf{v}_1, \dots, \mathbf{v}_k)$  has **exactly one** representation as a linear combination: if  $a_1\mathbf{v}_1 + \cdots + a_k\mathbf{v}_k = b_1\mathbf{v}_1 + \cdots + b_k\mathbf{v}_k$ , then  $a_i = b_i$  for every  $i$ .

*Proof.* (a  $\Rightarrow$  b) Suppose the list is independent, and let  $a_1\mathbf{v}_1 + \cdots + a_k\mathbf{v}_k = b_1\mathbf{v}_1 + \cdots + b_k\mathbf{v}_k$ . Subtracting the right side from both sides and using  $a\mathbf{v} - b\mathbf{v} = (a - b)\mathbf{v}$

(Theorem 1.1.8 and Definition 1.1.1) gives

$$(a_1 - b_1)\mathbf{v}_1 + \cdots + (a_k - b_k)\mathbf{v}_k = \mathbf{0}.$$

By independence,  $a_i - b_i = 0$  for every  $i$ , that is,  $a_i = b_i$ .

(b  $\Rightarrow$  a) Suppose (b) holds, and let  $a_1\mathbf{v}_1 + \cdots + a_k\mathbf{v}_k = \mathbf{0}$ . By Theorem 1.1.5, also  $0\mathbf{v}_1 + \cdots + 0\mathbf{v}_k = \mathbf{0}$ . These are two representations of  $\mathbf{0} \in \text{span}(\mathbf{v}_1, \dots, \mathbf{v}_k)$ , so by (b),  $a_i = 0$  for every  $i$ . Hence the list is independent. ■

Every vector of a span has **at least** one representation, by definition of the span; independence is what makes it **at most** one. For a dependent list uniqueness fails: with the list  $((1, 0), (0, 1), (1, 1))$ ,

$$(1, 1) = 1(1, 0) + 1(0, 1) + 0(1, 1) = 0(1, 0) + 0(0, 1) + 1(1, 1).$$

In the next section, a list that is independent **and** spans is called a basis. By this theorem every vector then has a unique list of coefficients, its coordinates.

## 1.4.6 Polynomials of distinct degrees

For polynomials there is a test that needs no equations at all. The list  $(x^2 + x, x - 3, 5)$  is independent, and the reason is visible: the degrees are 2, 1 and 0, and combining lower-degree polynomials can never produce a higher degree.

### Theorem 1.4.4 — Polynomials of Distinct Degrees Are Independent

Let  $F$  be a field and let  $p_1, \dots, p_k \in F[x]$  be **non-zero** polynomials whose degrees are **pairwise distinct**. Then the list  $(p_1, \dots, p_k)$  is linearly independent.

*Idea.* Put the polynomials in order of increasing degree. If the list were dependent, the Linear Dependence Lemma would give a polynomial lying in the span of the ones before it. Those all have smaller degree, so their span consists of polynomials of smaller degree, and cannot contain it.

*Proof.* A linear relation among the  $p_i$  can be rearranged term by term, so reordering the list does not affect independence. Hence we may assume  $\deg p_1 < \deg p_2 < \cdots < \deg p_k$ .

Suppose, for a contradiction, that the list is dependent. Since  $p_1 \neq 0$ , Theorem 1.4.1 gives an index  $j \geq 2$  with  $p_j \in \text{span}(p_1, \dots, p_{j-1})$ . Let  $n = \deg p_{j-1}$ , a natural number since  $p_{j-1} \neq 0$ . For every  $i \leq j-1$  we have  $\deg p_i \leq n$ , so  $p_1, \dots, p_{j-1}$  lie in  $F[x]_{\leq n}$ , which is a subspace of  $F[x]$  by Example (Polynomials of Bounded Degree). By Theorem 1.3.1 (3),  $\text{span}(p_1, \dots, p_{j-1}) \subseteq F[x]_{\leq n}$ . Hence  $\deg p_j \leq n = \deg p_{j-1}$ , contradicting  $\deg p_{j-1} < \deg p_j$ . Therefore the list is linearly independent. ■

Both hypotheses matter. The zero polynomial must be excluded, because a list containing 0 is

dependent (Example (Degenerate Lists)). And the converse fails: distinct degrees are sufficient, not necessary. The list  $(x, x+1)$  has equal degrees but is independent, since  $ax + b(x+1) = (a+b)x + b = 0$  forces  $b = 0$  and then  $a = 0$ .

## 1.4.7 Independence depends on the field

The same set can be a vector space over two fields, and the answer to “independent?” can change with the field, because the field decides which coefficients are allowed.

**Warning. Always ask: independent over which field?** In  $\mathbb{C}^2$ , let  $\mathbf{v}_1 = (1, i)$  and  $\mathbf{v}_2 = (i, -1)$ . Over  $\mathbb{C}$  the list  $(\mathbf{v}_1, \mathbf{v}_2)$  is **dependent**:  $i\mathbf{v}_1 = (i, i^2) = (i, -1) = \mathbf{v}_2$ , so  $i\mathbf{v}_1 + (-1)\mathbf{v}_2 = \mathbf{0}$  with coefficient  $i \neq 0$ . Over  $\mathbb{R}$  (the same addition, but only real scalars, as in Exercise (C2: The Span Depends on the Field)) the list is **independent**: if  $a\mathbf{v}_1 + b\mathbf{v}_2 = (a + bi, ai - b) = (0, 0)$  with  $a, b \in \mathbb{R}$ , then the first entry  $a + bi = 0$  forces  $a = b = 0$ , since a complex number is zero only when its real and imaginary parts are. The relation over  $\mathbb{C}$  used the coefficient  $i$ , which is not available over  $\mathbb{R}$ .

A second example of the same phenomenon:  $((1, 1), (1, -1))$  is independent in  $\mathbb{R}^2$  (Example (Independent Lists in Four Families) (c)), but in  $\mathbb{F}_2^2$ , where  $-1 = 1$ , it is the list  $((1, 1), (1, 1))$  with a repeat, hence dependent. The proof over  $\mathbb{R}$  divided by 2, which is 0 in  $\mathbb{F}_2$ .

## 1.4.8 Exercises

### 1.4.8.1 A. Check your understanding

#### Exercise A1.

- Define what it means for a list  $(\mathbf{v}_1, \dots, \mathbf{v}_k)$  in a vector space over  $F$  to be linearly independent, and what it means for it to be linearly dependent.
- Determine whether the following statement is true: “a list of vectors is linearly dependent if and only if one of its vectors is a scalar multiple of another.” Justify your answer.
- Is the empty list linearly independent? Is the list  $(\mathbf{0})$ ?
- Determine whether the following statement is true: “if  $(\mathbf{v}_1, \mathbf{v}_2, \mathbf{v}_3)$  is linearly dependent, then  $\mathbf{v}_3 \in \text{span}(\mathbf{v}_1, \mathbf{v}_2)$ .” Justify your answer.
- To prove that a list of functions  $\mathbb{R} \rightarrow \mathbb{R}$  is independent, what does “ $= 0$ ” mean, and what is a standard way to extract scalar equations?

*Solution.*

- It is independent if for all  $a_1, \dots, a_k \in F$ ,  $a_1\mathbf{v}_1 + \dots + a_k\mathbf{v}_k = \mathbf{0}$  implies  $a_1 = \dots = a_k = 0$ . It is dependent if there are  $a_1, \dots, a_k \in F$ , not all zero, with  $a_1\mathbf{v}_1 + \dots + a_k\mathbf{v}_k = \mathbf{0}$ .
- False. The list  $((1, 0, 1), (0, 1, 1), (1, 1, 2))$  in  $\mathbb{R}^3$  is dependent, since the third vector is the sum of the first two, but no vector is a multiple of another. (The “if” direction is true:  $\mathbf{v}_i = c\mathbf{v}_j$  with  $i \neq j$  gives the relation  $\mathbf{v}_i - c\mathbf{v}_j = \mathbf{0}$  with coefficient  $1 \neq 0$ .)
- The empty list is independent: there are no coefficients, so the condition holds

vacuously. The list  $(\mathbf{0})$  is dependent:  $1 \cdot \mathbf{0} = \mathbf{0}$  with coefficient  $1 \neq 0$  (Example (Degenerate Lists)).

- (d) False. Take  $\mathbf{v}_1 = (1, 0)$ ,  $\mathbf{v}_2 = (2, 0)$ ,  $\mathbf{v}_3 = (0, 1)$  in  $\mathbb{R}^2$ . The list is dependent, since  $2\mathbf{v}_1 - \mathbf{v}_2 = \mathbf{0}$ . But every combination of  $\mathbf{v}_1, \mathbf{v}_2$  has second entry 0, so  $\mathbf{v}_3 \notin \text{span}(\mathbf{v}_1, \mathbf{v}_2)$ . Theorem 1.4.1 only promises **some**  $j$ ; here  $j = 2$ .
- (e) It means the zero function: the combination takes the value 0 at **every** point. Evaluating at well-chosen points turns it into scalar equations for the coefficients.

### 1.4.8.2 B. Practice

#### Exercise B1: Independent or Not?

Determine which of the following lists are linearly independent. Justify your answer.

- (a)  $((1, 0, 2), (0, 1, 1), (1, 1, 3))$  in  $\mathbb{R}^3$ .  
 (b)  $((1, 1, 0), (0, 1, 1), (1, 0, 1))$  in  $\mathbb{R}^3$ .  
 (c)  $(x^2 + 1, x^2 - 1, x)$  in  $\mathbb{R}[x]$ .  
 (d)  $\left( \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix}, \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}, \begin{pmatrix} 1 & 1 \\ 1 & 1 \end{pmatrix} \right)$  in  $M_2(\mathbb{R})$ .  
 (e)  $(f, g)$  in  $\mathbb{R}^{\mathbb{R}}$ , where  $f(t) = e^t$  and  $g(t) = e^{2t}$ .

*Solution.*

- (a) Dependent. Since  $(1, 0, 2) + (0, 1, 1) = (1, 1, 3)$ , we have  $(1, 0, 2) + (0, 1, 1) - (1, 1, 3) = \mathbf{0}$ , with coefficients not all zero.
- (b) Independent. Let  $a(1, 1, 0) + b(0, 1, 1) + c(1, 0, 1) = \mathbf{0}$ . Comparing entries:  $a + c = 0$ ,  $a + b = 0$ ,  $b + c = 0$ . The first two give  $c = -a$  and  $b = -a$ ; then the third gives  $-2a = 0$ , so  $a = 0$ , and hence  $b = c = 0$ .
- (c) Independent. Let  $a(x^2 + 1) + b(x^2 - 1) + cx = 0$ . The left side is  $(a+b)x^2 + cx + (a-b)$ . Comparing coefficients (Definition 0.8.1):  $a + b = 0$ ,  $c = 0$ ,  $a - b = 0$ . Adding the first and third gives  $2a = 0$ , so  $a = 0$ , and then  $b = 0$ .
- (d) Dependent. The first two matrices add up to the third, so  $1 \cdot \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix} + 1 \cdot \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix} + (-1) \begin{pmatrix} 1 & 1 \\ 1 & 1 \end{pmatrix}$  is the zero matrix.
- (e) Independent. Let  $af + bg = 0$ , the zero function. Then  $ae^t + be^{2t} = 0$  for every  $t \in \mathbb{R}$ . At  $t = 0$ :  $a + b = 0$ . At  $t = \ln 2$ :  $2a + 4b = 0$ , so  $a + 2b = 0$ . Subtracting the first equation gives  $b = 0$ , and then  $a = 0$ .

#### Exercise B2: A Parameter.

For which  $t \in \mathbb{R}$  is the list  $((1, t), (t, 1))$  in  $\mathbb{R}^2$  linearly independent? Justify your answer.

*Solution.* Let  $a(1, t) + b(t, 1) = (0, 0)$ , that is,  $a + tb = 0$  and  $ta + b = 0$ . The first equation gives  $a = -tb$ . Substituting into the second,  $-t^2b + b = (1 - t^2)b = 0$ .

If  $t \neq \pm 1$ , then  $1 - t^2 \neq 0$ , so  $b = 0$  by Theorem 0.7.1, and then  $a = -tb = 0$ . The list is independent.

If  $t = 1$ , the list is  $((1, 1), (1, 1))$ , which has a repeat and is dependent. If  $t = -1$ , the list is  $((1, -1), (-1, 1))$ , and  $1 \cdot (1, -1) + 1 \cdot (-1, 1) = (0, 0)$ , so it is dependent.

Hence the list is linearly independent exactly when  $t \neq 1$  and  $t \neq -1$ .

### Exercise B3: Sub-Lists.

A **sub-list** of  $(\mathbf{v}_1, \dots, \mathbf{v}_k)$  is a list  $(\mathbf{v}_{i_1}, \dots, \mathbf{v}_{i_m})$  with  $1 \leq i_1 < i_2 < \dots < i_m \leq k$ .

- (a) Prove that every sub-list of a linearly independent list is linearly independent.
- (b) Deduce that if some sub-list of a list is linearly dependent, then the whole list is linearly dependent.

*Solution.*

- (a) Let  $(\mathbf{v}_1, \dots, \mathbf{v}_k)$  be independent, and let  $b_1\mathbf{v}_{i_1} + \dots + b_m\mathbf{v}_{i_m} = \mathbf{0}$ . Define  $a_i = b_r$  if  $i = i_r$  for some  $r$  (the indices  $i_r$  are distinct, so this is well defined), and  $a_i = 0$  otherwise. By [Theorem 1.1.5](#) the extra terms are  $\mathbf{0}$ , so  $a_1\mathbf{v}_1 + \dots + a_k\mathbf{v}_k = b_1\mathbf{v}_{i_1} + \dots + b_m\mathbf{v}_{i_m} = \mathbf{0}$ . By independence, every  $a_i = 0$ ; in particular every  $b_r = a_{i_r} = 0$ . Hence the sub-list is independent.
- (b) This is the contrapositive of (a): if the whole list were independent, every sub-list would be independent by (a).

### 1.4.8.3 C. Going deeper

#### Exercise C1: Sums of Pairs, and the Field $\mathbb{F}_2$ .

Let  $V$  be a vector space over  $F$  and  $(\mathbf{u}, \mathbf{v}, \mathbf{w})$  a linearly independent list in  $V$ .

- (a) Suppose  $F = \mathbb{R}$ . Prove that  $(\mathbf{u} + \mathbf{v}, \mathbf{v} + \mathbf{w}, \mathbf{w} + \mathbf{u})$  is linearly independent.
- (b) Suppose  $F = \mathbb{F}_2$ . Prove that  $(\mathbf{u} + \mathbf{v}, \mathbf{v} + \mathbf{w}, \mathbf{w} + \mathbf{u})$  is linearly dependent, and give an example of an independent list  $(\mathbf{u}, \mathbf{v}, \mathbf{w})$  over  $\mathbb{F}_2$ , so that the statement is not vacuous.
- (c) Which property of  $\mathbb{R}$  did (a) use that  $\mathbb{F}_2$  lacks? Deduce that (a) holds over every field in which  $1 + 1 \neq 0$ .

*Solution.*

- (a) Let  $a(\mathbf{u} + \mathbf{v}) + b(\mathbf{v} + \mathbf{w}) + c(\mathbf{w} + \mathbf{u}) = \mathbf{0}$ . Expanding and regrouping with the axioms of [Definition 1.1.1](#) gives  $(a + c)\mathbf{u} + (a + b)\mathbf{v} + (b + c)\mathbf{w} = \mathbf{0}$ . By independence of  $(\mathbf{u}, \mathbf{v}, \mathbf{w})$ ,  $a + c = 0$ ,  $a + b = 0$  and  $b + c = 0$ . The first two give  $c = -a$  and  $b = -a$ , and then the third gives  $-2a = 0$ . Since  $2 \neq 0$  in  $\mathbb{R}$ ,  $a = 0$ , and hence  $b = c = 0$ . This proves independence.
- (b) Over  $\mathbb{F}_2$ ,

$$\begin{aligned} 1(\mathbf{u} + \mathbf{v}) + 1(\mathbf{v} + \mathbf{w}) + 1(\mathbf{w} + \mathbf{u}) &= (1 + 1)\mathbf{u} + (1 + 1)\mathbf{v} + (1 + 1)\mathbf{w} \\ &= 0\mathbf{u} + 0\mathbf{v} + 0\mathbf{w} = \mathbf{0}, \end{aligned}$$

using  $1 + 1 = 0$  in  $\mathbb{F}_2$  and [Theorem 1.1.5](#). The coefficients are  $1 \neq 0$ , so the list is

dependent. For an example,  $(\mathbf{e}_1, \mathbf{e}_2, \mathbf{e}_3)$  in  $\mathbb{F}_2^3$  is independent by the argument of [Example \(Independent Lists in Four Families\)](#) (a), which works over any field.

- (c) Part (a) divided by  $2 = 1 + 1$ , that is, used  $2 \neq 0$ , to pass from  $-2a = 0$  to  $a = 0$ . In  $\mathbb{F}_2$ ,  $2 = 0$ . In any field with  $1 + 1 \neq 0$ , the element 2 has an inverse, and the proof of (a) goes through word for word.

### Exercise C2: Adding a Vector outside the Span.

Let  $(\mathbf{v}_1, \dots, \mathbf{v}_k)$  be a linearly independent list in  $V$ , and let  $\mathbf{v} \in V$  with  $\mathbf{v} \notin \text{span}(\mathbf{v}_1, \dots, \mathbf{v}_k)$ . Prove that  $(\mathbf{v}_1, \dots, \mathbf{v}_k, \mathbf{v})$  is linearly independent. Give an example showing that the hypothesis  $\mathbf{v} \notin \text{span}(\mathbf{v}_1, \dots, \mathbf{v}_k)$  cannot be dropped.

*Solution.* Let  $a_1\mathbf{v}_1 + \dots + a_k\mathbf{v}_k + a\mathbf{v} = \mathbf{0}$ . Suppose  $a \neq 0$ . Then, as in the proof of [Theorem 1.4.1](#),  $\mathbf{v} = (-a^{-1}a_1)\mathbf{v}_1 + \dots + (-a^{-1}a_k)\mathbf{v}_k \in \text{span}(\mathbf{v}_1, \dots, \mathbf{v}_k)$ , contradicting the hypothesis. Hence  $a = 0$ , and the relation becomes  $a_1\mathbf{v}_1 + \dots + a_k\mathbf{v}_k = \mathbf{0}$ . By independence of  $(\mathbf{v}_1, \dots, \mathbf{v}_k)$ ,  $a_1 = \dots = a_k = 0$ . So all coefficients are zero, and the longer list is independent.

For the example, take  $(\mathbf{e}_1)$  in  $\mathbb{R}^2$  and  $\mathbf{v} = 2\mathbf{e}_1 \in \text{span}(\mathbf{e}_1)$ . The list  $(\mathbf{e}_1)$  is independent, but  $(\mathbf{e}_1, 2\mathbf{e}_1)$  is dependent, by  $2\mathbf{e}_1 - 2\mathbf{e}_1 = \mathbf{0}$ .

### Exercise C3: $1, \sqrt{2}, \sqrt{3}$ over $\mathbb{Q}$ .

Regard  $\mathbb{R}$  as a vector space over  $\mathbb{Q}$ . Prove that the list  $(1, \sqrt{2}, \sqrt{3})$  is linearly independent over  $\mathbb{Q}$ . Is it linearly independent over  $\mathbb{R}$ ?

*Hint: you may use without proof that  $\sqrt{3}$  and  $\sqrt{6}$  are irrational (the proof of [Theorem 0.2.3](#) adapts). Isolate one square root and square.*

*Solution.* Let  $a + b\sqrt{2} + c\sqrt{3} = 0$  with  $a, b, c \in \mathbb{Q}$ . Then  $a + b\sqrt{2} = -c\sqrt{3}$ , and squaring both sides gives  $a^2 + 2b^2 + 2ab\sqrt{2} = 3c^2$ .

If  $ab \neq 0$ , then  $\sqrt{2} = \frac{3c^2 - a^2 - 2b^2}{2ab} \in \mathbb{Q}$ , contradicting [Theorem 0.2.3](#). Hence  $ab = 0$ , so  $a = 0$  or  $b = 0$ .

*Case 1:*  $b = 0$ . Then  $a + c\sqrt{3} = 0$ . If  $c \neq 0$ , then  $\sqrt{3} = -a/c \in \mathbb{Q}$ , a contradiction. So  $c = 0$ , and then  $a = 0$ .

*Case 2:*  $a = 0$ . Then  $b\sqrt{2} + c\sqrt{3} = 0$ . Multiplying by  $\sqrt{2}$  gives  $2b + c\sqrt{6} = 0$ . If  $c \neq 0$ , then  $\sqrt{6} = -2b/c \in \mathbb{Q}$ , a contradiction. So  $c = 0$ , then  $b\sqrt{2} = 0$ , and  $b = 0$  since  $\sqrt{2} \neq 0$ .

In both cases  $a = b = c = 0$ , so the list is linearly independent over  $\mathbb{Q}$ .

Over  $\mathbb{R}$  it is dependent:  $\sqrt{2} \cdot 1 + (-1) \cdot \sqrt{2} + 0 \cdot \sqrt{3} = 0$ , and the coefficient  $\sqrt{2}$  is a non-zero real number. Over  $\mathbb{R}$  any two non-zero real numbers form a dependent list.

## 1.5 Bases and Coordinates

Spanning lists and independent lists each solve half of a problem. A spanning list reaches every vector of the space, but it may reach some vectors in more than one way. An independent list never wastes a vector, but it may not reach everything. This section puts the two halves together. The result, a basis, turns an abstract vector space into columns of numbers, and we prove that every space spanned by a finite list has one.

### 1.5.1 What makes coordinates work

Start with the plane. Every vector of  $\mathbb{R}^2$  can be written as

$$(x, y) = x\mathbf{e}_1 + y\mathbf{e}_2,$$

and the pair  $(x, y)$  is its address. Two separate facts make this addressing system work. First, **every** vector has an address: the list  $(\mathbf{e}_1, \mathbf{e}_2)$  spans  $\mathbb{R}^2$ . Second, every vector has **only one** address: if  $x\mathbf{e}_1 + y\mathbf{e}_2 = x'\mathbf{e}_1 + y'\mathbf{e}_2$ , then  $(x, y) = (x', y')$ , so  $x = x'$  and  $y = y'$ .

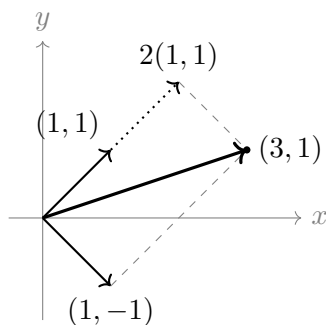
Nothing about this is special to  $\mathbf{e}_1$  and  $\mathbf{e}_2$ . Take instead the list  $((1, 1), (1, -1))$ . To find an address for  $(x, y)$  we solve

$$a(1, 1) + b(1, -1) = (x, y), \quad \text{that is,} \quad a + b = x, \quad a - b = y.$$

Adding and subtracting the two equations gives  $2a = x + y$  and  $2b = x - y$ . So there is a solution, and it is the only one:

$$a = \frac{x + y}{2}, \quad b = \frac{x - y}{2}.$$

For example  $(3, 1) = 2(1, 1) + 1(1, -1)$ , and no other pair of scalars works. The grid is slanted, but every point still has exactly one address.



Look at which property did which job. **Existence** of an address is spanning. **Uniqueness** of the address is independence: by [Theorem 1.4.3](#), a list is independent exactly when every vector in its span has only one expression as a combination of it. A coordinate system is a list that does both jobs at once.

*A basis is a list with enough vectors to build everything and no vector to spare.*

### 1.5.2 Bases



**Definition 1.5.1 — Basis**

Let  $V$  be a vector space over  $F$ . A **basis** of  $V$  is a **list**  $(\mathbf{v}_1, \dots, \mathbf{v}_n)$  of vectors in  $V$  such that

- (B1)  $(\mathbf{v}_1, \dots, \mathbf{v}_n)$  is **linearly independent**, and
- (B2)  $(\mathbf{v}_1, \dots, \mathbf{v}_n)$  **spans**  $V$ , that is,  $\text{span}(\mathbf{v}_1, \dots, \mathbf{v}_n) = V$ .

We usually name a basis with a calligraphic letter,  $\mathcal{B} = (\mathbf{v}_1, \dots, \mathbf{v}_n)$ , and call  $n$  its **length**. The plural of basis is **bases**.

In words: (B1) says that the only combination of the  $\mathbf{v}_i$  equal to  $\mathbf{0}$  is the one with **all** coefficients zero, so no vector in the list is wasted. (B2) says that **every** vector of  $V$  is some combination of the  $\mathbf{v}_i$ , so nothing is out of reach. The basis is a list, so its entries come in a definite order, and a basis is always a basis **of** a particular space:  $(\mathbf{e}_1, \mathbf{e}_2)$  is a basis of  $\mathbb{R}^2$ , but not of  $\mathbb{R}^3$ .

We start with the bases that come with the running examples.

**Example. (Standard Bases)**

Show that each of the following lists is a basis. They are called the **standard bases**.

- (a)  $(\mathbf{e}_1, \dots, \mathbf{e}_n)$  in  $F^n$ , where  $\mathbf{e}_i$  has 1 in position  $i$  and 0 elsewhere.
- (b)  $(1, x, x^2, \dots, x^n)$  in  $F[x]_{\leq n}$ .
- (c)  $(\mathbf{E}_{11}, \mathbf{E}_{12}, \dots, \mathbf{E}_{1n}, \mathbf{E}_{21}, \dots, \mathbf{E}_{mn})$  in  $M_{m \times n}(F)$ , where the **matrix unit**  $\mathbf{E}_{ij}$  has 1 in entry  $(i, j)$  and 0 elsewhere, listed row by row.

*Solution.*

- (a) (B2): every  $(x_1, \dots, x_n) \in F^n$  equals  $x_1\mathbf{e}_1 + \dots + x_n\mathbf{e}_n$ . (B1): let  $a_1\mathbf{e}_1 + \dots + a_n\mathbf{e}_n = \mathbf{0}$ . The left side is the column  $(a_1, \dots, a_n)$ , so comparing entries gives  $a_1 = \dots = a_n = 0$ .
- (b) (B2): by [Definition 0.8.4](#), every element of  $F[x]_{\leq n}$  has the form  $a_0 + a_1x + \dots + a_nx^n$ , which is a combination of  $1, x, \dots, x^n$ . (B1): let  $a_0 \cdot 1 + a_1x + \dots + a_nx^n = 0$ . Here 0 is the zero polynomial, all of whose coefficients are 0. Two polynomials are equal exactly when their coefficients agree ([Definition 0.8.1](#)), so  $a_0 = a_1 = \dots = a_n = 0$ .
- (c) (B2): a matrix  $\mathbf{A} = (a_{ij})$  equals  $\sum_{i,j} a_{ij}\mathbf{E}_{ij}$ , since both sides have  $a_{ij}$  in entry  $(i, j)$ . (B1): if  $\sum_{i,j} c_{ij}\mathbf{E}_{ij}$  is the zero matrix, then its  $(i, j)$  entry  $c_{ij}$  is 0 for every  $i, j$ .

The next examples show that a space has other bases, that the field matters, and what happens in the smallest space.

**Example. (Three More Bases)**

Show that each list is a basis.

- (a)  $((1, 1), (1, -1))$  in  $\mathbb{R}^2$ .
- (b)  $(1, i)$  in  $\mathbb{C}$ , regarded as a vector space **over**  $\mathbb{R}$ .

(c) The empty list  $()$  in the zero space  $\{\mathbf{0}\}$ .

*Solution.*

- (a) We showed above that for every  $(x, y) \in \mathbb{R}^2$  the equation  $a(1, 1) + b(1, -1) = (x, y)$  has a solution, so (B2) holds. Taking  $(x, y) = (0, 0)$ , the only solution is  $a = \frac{0+0}{2} = 0$  and  $b = 0$ , so (B1) holds.
- (b) The scalars are now real. (B2): every complex number is  $a + bi = a \cdot 1 + b \cdot i$  with  $a, b \in \mathbb{R}$  (Definition 0.6.1). (B1): let  $a \cdot 1 + b \cdot i = 0$  with  $a, b \in \mathbb{R}$ . A complex number is 0 exactly when its real and imaginary parts are 0, so  $a = b = 0$ .
- (c) The empty list is linearly independent: there are no coefficients, so the condition “all coefficients are zero” holds vacuously. Its span is  $\{\mathbf{0}\}$  by the convention for the empty span. So both (B1) and (B2) hold.

The degenerate case (c) matters more than it looks. Without it, the zero space would be the one space with no basis, and every theorem below would need an exception for it.

**Non-examples by minimal change.** Add one vector to the standard basis of  $\mathbb{R}^2$ , making  $((1, 0), (0, 1), (1, 1))$ . Spanning (B2) still holds, since the first two vectors already span. But (B1) fails:

$$1 \cdot (1, 0) + 1 \cdot (0, 1) + (-1) \cdot (1, 1) = (0, 0),$$

with coefficients **not all zero**. The effect is exactly the loss of unique addresses:  $(3, 1) = 3(1, 0) + 1(0, 1) + 0(1, 1) = 2(1, 0) + 0(0, 1) + 1(1, 1)$ .

Now remove a vector instead. In  $\mathbb{R}^3$ , the list  $((1, 0, 0), (0, 1, 0))$  is still independent, since  $a(1, 0, 0) + b(0, 1, 0) = (a, b, 0) = \mathbf{0}$  forces  $a = b = 0$ . But (B2) fails: every combination  $(a, b, 0)$  has third entry 0, so  $(0, 0, 1)$  is out of reach.

*Quick check.* Is  $((1, 2), (2, 1), (0, 0))$  a basis of  $\mathbb{R}^2$ ? If not, which of (B1) and (B2) fails?

*Solution.* No. (B2) holds:  $(1, 2)$  and  $(2, 1)$  already span  $\mathbb{R}^2$ , because solving  $a(1, 2) + b(2, 1) = (x, y)$  gives  $a = \frac{2y-x}{3}$  and  $b = \frac{2x-y}{3}$ . But (B1) fails, because the list contains the zero vector:  $0(1, 2) + 0(2, 1) + 1(0, 0) = (0, 0)$  with a non-zero coefficient.

**Why this definition.** Drop (B1), and addresses stop being unique, as the list  $((1, 0), (0, 1), (1, 1))$  showed. Drop (B2), and some vectors have no address at all. Keeping both is exactly what an addressing system needs. The name comes from the everyday word: the basis is what everything else is built on.

*Remark.* Why a list and not a set? Two reasons. Coordinates, defined below, are a column of numbers, and a column needs to know which scalar comes first. And a set would hide repeats: the list  $((1, 0), (1, 0), (0, 1))$  is dependent, since  $1(1, 0) - 1(1, 0) = \mathbf{0}$ , but as a set  $\{(1, 0), (1, 0), (0, 1)\} = \{(1, 0), (0, 1)\}$  it would look like the standard basis.

*Warning.* Say **a** basis, not **the** basis. A space usually has many bases; a non-zero space over an infinite field such as  $\mathbb{R}$  has infinitely many. For instance  $((1, 0), (c, 1))$  is a basis of  $\mathbb{R}^2$  for **every** real number  $c$ , since  $(x, y) = (x - cy)(1, 0) + y(c, 1)$ , and this is the only solution. The standard basis is a convenient choice, not a canonical one.

### 1.5.3 Unique representation and coordinates

We found the definition by asking for addresses that exist and are unique. The first result confirms that this is exactly what a basis delivers, and nothing more.

#### Theorem 1.5.1 — Unique Representation

Let  $V$  be a vector space over  $F$ , and let  $\mathcal{B} = (\mathbf{v}_1, \dots, \mathbf{v}_n)$  be a list in  $V$ . Then  $\mathcal{B}$  is a basis of  $V$  if and only if every  $\mathbf{v} \in V$  can be written as

$$\mathbf{v} = a_1\mathbf{v}_1 + \cdots + a_n\mathbf{v}_n \quad (a_1, \dots, a_n \in F)$$

in **exactly one** way.

*Proof.* ( $\Rightarrow$ ) Let  $\mathbf{v} \in V$ . Since  $\mathcal{B}$  spans  $V$ , there are scalars with  $\mathbf{v} = a_1\mathbf{v}_1 + \cdots + a_n\mathbf{v}_n$ . Suppose also  $\mathbf{v} = b_1\mathbf{v}_1 + \cdots + b_n\mathbf{v}_n$ . Subtracting the two expressions and regrouping with the vector space axioms,

$$(a_1 - b_1)\mathbf{v}_1 + \cdots + (a_n - b_n)\mathbf{v}_n = \mathbf{v} - \mathbf{v} = \mathbf{0}.$$

Since  $\mathcal{B}$  is linearly independent,  $a_i - b_i = 0$  for every  $i$ . Hence the two expressions are the same.

( $\Leftarrow$ ) Every  $\mathbf{v} \in V$  is a combination of  $\mathcal{B}$ , so  $\mathcal{B}$  spans  $V$ . For independence, let  $a_1\mathbf{v}_1 + \cdots + a_n\mathbf{v}_n = \mathbf{0}$ . We also have  $\mathbf{0} = 0\mathbf{v}_1 + \cdots + 0\mathbf{v}_n$ . By uniqueness of the expression for the vector  $\mathbf{0}$ , all  $a_i = 0$ . Hence  $\mathcal{B}$  is linearly independent, and  $\mathcal{B}$  is a basis. ■

This is [Theorem 1.4.3](#) in the special case where the span is the whole space. The theorem also gives a second way to check that a list is a basis: solve for the coefficients of a general vector, and see that there is always exactly one solution. We did exactly that for  $((1, 1), (1, -1))$ .

Whenever something exists and is unique, it deserves a name.

#### Definition 1.5.2 — Coordinates

Let  $\mathcal{B} = (\mathbf{v}_1, \dots, \mathbf{v}_n)$  be a basis of a vector space  $V$  over  $F$ , and let  $\mathbf{v} \in V$ . The unique scalars  $a_1, \dots, a_n$  with  $\mathbf{v} = a_1\mathbf{v}_1 + \cdots + a_n\mathbf{v}_n$  are the **coordinates** of  $\mathbf{v}$  with respect to  $\mathcal{B}$ . The column

$$[\mathbf{v}]_{\mathcal{B}} := \begin{pmatrix} a_1 \\ \vdots \\ a_n \end{pmatrix} \in F^n$$

is the **coordinate vector** of  $\mathbf{v}$  with respect to  $\mathcal{B}$ .

In the standard basis of  $F^n$ , the coordinate vector of  $(x_1, \dots, x_n)$  is the column  $(x_1, \dots, x_n)$  itself. That is why we rarely notice coordinates in  $F^n$ . In other bases and other spaces they carry real information.

**Example. (Coordinates of a Polynomial)**

Show that  $\mathcal{B} = (1, 1+x, 1+x+x^2)$  is a basis of  $\mathbb{R}[x]_{\leq 2}$ , and find  $[p]_{\mathcal{B}}$  for  $p = 3 - x + 2x^2$ .

*Solution.* We use [Theorem 1.5.1](#). Let  $q = q_0 + q_1x + q_2x^2 \in \mathbb{R}[x]_{\leq 2}$  be arbitrary. For scalars  $a, b, c$ ,

$$a \cdot 1 + b(1+x) + c(1+x+x^2) = (a+b+c) + (b+c)x + cx^2.$$

Two polynomials are equal exactly when their coefficients agree, so this equals  $q$  if and only if

$$c = q_2, \quad b + c = q_1, \quad a + b + c = q_0.$$

Solving from the top,  $c = q_2$ , then  $b = q_1 - q_2$ , then  $a = q_0 - q_1$ . So every  $q$  has exactly one expression, and  $\mathcal{B}$  is a basis of  $\mathbb{R}[x]_{\leq 2}$ .

For  $p = 3 - x + 2x^2$  we have  $q_0 = 3$ ,  $q_1 = -1$ ,  $q_2 = 2$ . Hence  $c = 2$ ,  $b = -1 - 2 = -3$ , and  $a = 3 - (-1) = 4$ :

$$[p]_{\mathcal{B}} = \begin{pmatrix} 4 \\ -3 \\ 2 \end{pmatrix}.$$

Check:  $4 - 3(1+x) + 2(1+x+x^2) = (4-3+2) + (-3+2)x + 2x^2 = 3 - x + 2x^2$ . In the standard basis  $(1, x, x^2)$ , the same polynomial has coordinate vector  $(3, -1, 2)$ .

*Warning.* Coordinates depend on the basis **and on its order**. Take  $\mathbf{v} = (3, 1) \in \mathbb{R}^2$ . With respect to the standard basis  $\mathcal{E} = (\mathbf{e}_1, \mathbf{e}_2)$ ,  $[\mathbf{v}]_{\mathcal{E}} = (3, 1)$ . With respect to  $\mathcal{B} = ((1, 1), (1, -1))$ ,  $[\mathbf{v}]_{\mathcal{B}} = (2, 1)$ . With respect to  $\mathcal{B}' = ((1, -1), (1, 1))$ , the same two vectors in the other order,  $[\mathbf{v}]_{\mathcal{B}'} = (1, 2)$ . A column of numbers means nothing until you say which ordered basis it refers to.

The point of coordinates is that they turn computations in  $V$  into computations with columns. That works because they respect the two operations.

**Theorem 1.5.2 — Coordinates Respect the Operations**

Let  $\mathcal{B} = (\mathbf{v}_1, \dots, \mathbf{v}_n)$  be a basis of a vector space  $V$  over  $F$ . Then for all  $\mathbf{u}, \mathbf{v} \in V$  and  $c \in F$ ,

$$[\mathbf{u} + \mathbf{v}]_{\mathcal{B}} = [\mathbf{u}]_{\mathcal{B}} + [\mathbf{v}]_{\mathcal{B}}, \quad [c\mathbf{v}]_{\mathcal{B}} = c[\mathbf{v}]_{\mathcal{B}}.$$

Moreover, every column  $(a_1, \dots, a_n) \in F^n$  is the coordinate vector of exactly one vector of  $V$ , namely  $a_1\mathbf{v}_1 + \dots + a_n\mathbf{v}_n$ .

*Proof.* Let  $[\mathbf{u}]_{\mathcal{B}} = (a_1, \dots, a_n)$  and  $[\mathbf{v}]_{\mathcal{B}} = (b_1, \dots, b_n)$ . By the vector space axioms,

$$\mathbf{u} + \mathbf{v} = \sum_{i=1}^n a_i \mathbf{v}_i + \sum_{i=1}^n b_i \mathbf{v}_i = \sum_{i=1}^n (a_i + b_i) \mathbf{v}_i, \quad c\mathbf{v} = c \sum_{i=1}^n b_i \mathbf{v}_i = \sum_{i=1}^n (cb_i) \mathbf{v}_i.$$

By [Theorem 1.5.1](#) these are the only expressions of  $\mathbf{u} + \mathbf{v}$  and  $c\mathbf{v}$  in  $\mathcal{B}$ , so their coefficients are the coordinates:  $[\mathbf{u} + \mathbf{v}]_{\mathcal{B}} = (a_i + b_i)_i$  and  $[c\mathbf{v}]_{\mathcal{B}} = (cb_i)_i$ , as claimed.

For the last statement, let  $(a_1, \dots, a_n) \in F^n$  and  $\mathbf{w} = a_1\mathbf{v}_1 + \dots + a_n\mathbf{v}_n$ . By [Def-](#)

inition 1.5.2,  $[\mathbf{w}]_{\mathcal{B}} = (a_1, \dots, a_n)$ . If also  $[\mathbf{w}']_{\mathcal{B}} = (a_1, \dots, a_n)$ , then by definition  $\mathbf{w}' = a_1 \mathbf{v}_1 + \dots + a_n \mathbf{v}_n = \mathbf{w}$ . ■

So a basis of length  $n$  sets up a one-to-one correspondence  $\mathbf{v} \mapsto [\mathbf{v}]_{\mathcal{B}}$  between  $V$  and  $F^n$  that respects addition and scaling. Any question about sums and multiples in  $V$  can be answered in  $F^n$  instead. In Chapter 3 we will call such a correspondence an **isomorphism**.

## 1.5.4 Every finite spanning list contains a basis

Where do bases come from? A finite-dimensional space is, by [Definition 1.3.4](#), a space spanned by some finite list. That list reaches everything but may contain waste, so the natural plan is to throw the waste away.

Care is needed about **what** to throw away. In  $((1, 0), (0, 1), (1, 1))$  each vector is a combination of the other two. Any single one can go, but not two of them. So redundancy is not a property of one vector alone: it depends on which other vectors stay. The fix is to decide one vector at a time, reading from left to right, and to compare each vector only with the vectors already kept.

First, a small lemma that tells us when a kept list stays independent.

### Lemma 1.5.3 — Appending a Vector to an Independent List

Let  $(\mathbf{u}_1, \dots, \mathbf{u}_k)$  be a linearly independent list in a vector space  $V$ , and let  $\mathbf{v} \in V$ . Then  $(\mathbf{u}_1, \dots, \mathbf{u}_k, \mathbf{v})$  is linearly independent **if and only if**  $\mathbf{v} \notin \text{span}(\mathbf{u}_1, \dots, \mathbf{u}_k)$ .

*Proof.*  $(\Rightarrow)$  Suppose  $\mathbf{v} \in \text{span}(\mathbf{u}_1, \dots, \mathbf{u}_k)$ , say  $\mathbf{v} = c_1 \mathbf{u}_1 + \dots + c_k \mathbf{u}_k$ . Then  $c_1 \mathbf{u}_1 + \dots + c_k \mathbf{u}_k + (-1)\mathbf{v} = \mathbf{0}$ , with the coefficient  $-1 \neq 0$ . So the longer list is dependent. This proves  $(\Rightarrow)$  by contraposition.

$(\Leftarrow)$  Suppose  $\mathbf{v} \notin \text{span}(\mathbf{u}_1, \dots, \mathbf{u}_k)$ , and let  $a_1 \mathbf{u}_1 + \dots + a_k \mathbf{u}_k + b\mathbf{v} = \mathbf{0}$ . If  $b \neq 0$ , then  $b^{-1}$  exists and

$$\mathbf{v} = -b^{-1}a_1 \mathbf{u}_1 - \dots - b^{-1}a_k \mathbf{u}_k \in \text{span}(\mathbf{u}_1, \dots, \mathbf{u}_k),$$

a contradiction. Hence  $b = 0$ . Then  $a_1 \mathbf{u}_1 + \dots + a_k \mathbf{u}_k = \mathbf{0}$ , and the independence of  $(\mathbf{u}_1, \dots, \mathbf{u}_k)$  gives  $a_1 = \dots = a_k = 0$ . So all coefficients are zero, and the longer list is independent. ■

The direction  $(\Leftarrow)$  is [Exercise \(C2: Adding a Vector outside the Span\)](#). When  $k = 0$ , the lemma says that the one-vector list  $(\mathbf{v})$  is independent exactly when  $\mathbf{v} \notin \text{span}() = \{\mathbf{0}\}$ , that is, when  $\mathbf{v} \neq \mathbf{0}$ , in agreement with [Example \(Degenerate Lists\)](#).

Now the procedure. Given a list  $(\mathbf{v}_1, \dots, \mathbf{v}_m)$  in  $V$ , **sifting** it means the following.

*Algorithm.* Start with the empty list  $S_0 = ()$ . For  $r = 1, 2, \dots, m$  in turn:

- if  $\mathbf{v}_r \notin \text{span}(S_{r-1})$ , **keep** it:  $S_r = (S_{r-1}, \mathbf{v}_r)$ ;
- if  $\mathbf{v}_r \in \text{span}(S_{r-1})$ , **discard** it:  $S_r = S_{r-1}$ .

The final list  $S_m$  is the **sifted list** of  $(\mathbf{v}_1, \dots, \mathbf{v}_m)$ .

Here  $(S_{r-1}, \mathbf{v}_r)$  means the list  $S_{r-1}$  with  $\mathbf{v}_r$  appended at the end. Each  $S_r$  is a **sub-list** of  $(\mathbf{v}_1, \dots, \mathbf{v}_m)$ : it is obtained by deleting some entries and keeping the rest in their original order.

### Theorem 1.5.4 — Sifting Theorem

Let  $(\mathbf{v}_1, \dots, \mathbf{v}_m)$  be a list in a vector space  $V$ , and let  $S_m$  be its sifted list. Then  $S_m$  is linearly independent and

$$\text{span}(S_m) = \text{span}(\mathbf{v}_1, \dots, \mathbf{v}_m).$$

In particular, every finite list that spans  $V$  contains a sub-list that is a basis of  $V$ .

*Idea.* Sifting protects two things at every step. A vector is kept only when it lies outside the span of what was kept before, so by [Lemma 1.5.3](#) the kept list stays independent. A vector is discarded only when it is already in that span, so by [Theorem 1.3.5](#) discarding it loses nothing. We turn “at every step” into an induction on  $r$ , the number of vectors read so far, with the invariant:  $S_r$  is independent and has the same span as  $\mathbf{v}_1, \dots, \mathbf{v}_r$ .

*Proof.* We prove by induction on  $r \in \{0, 1, \dots, m\}$  the statement

$$P(r): \text{the list } S_r \text{ is linearly independent and } \text{span}(S_r) = \text{span}(\mathbf{v}_1, \dots, \mathbf{v}_r).$$

*Base case.*  $S_0$  is the empty list, which is linearly independent, and both sides of the span equation are the span of the empty list. So  $P(0)$  holds.

*Inductive step.* Let  $1 \leq r \leq m$  and suppose  $P(r-1)$  holds. Put  $W = \text{span}(S_{r-1})$ , so that  $W = \text{span}(\mathbf{v}_1, \dots, \mathbf{v}_{r-1})$  by  $P(r-1)$ .

*Case 1:*  $\mathbf{v}_r \in W$ . Then  $S_r = S_{r-1}$ , which is independent. Since  $\mathbf{v}_r \in \text{span}(\mathbf{v}_1, \dots, \mathbf{v}_{r-1})$ , [Theorem 1.3.5](#) gives  $\text{span}(\mathbf{v}_1, \dots, \mathbf{v}_r) = \text{span}(\mathbf{v}_1, \dots, \mathbf{v}_{r-1}) = W = \text{span}(S_r)$ .

*Case 2:*  $\mathbf{v}_r \notin W$ . Then  $S_r = (S_{r-1}, \mathbf{v}_r)$ . Since  $S_{r-1}$  is independent and  $\mathbf{v}_r \notin \text{span}(S_{r-1})$ , [Lemma 1.5.3](#) shows that  $S_r$  is independent. For the spans, recall from [Theorem 1.3.1](#) that the span of a list is a subspace, and is contained in every subspace containing the entries of the list. Every entry of  $S_r$  is one of  $\mathbf{v}_1, \dots, \mathbf{v}_r$ , so  $\text{span}(S_r) \subseteq \text{span}(\mathbf{v}_1, \dots, \mathbf{v}_r)$ . Conversely,  $\text{span}(S_r)$  is a subspace containing the entries of  $S_{r-1}$ , so it contains  $W$ , and with it  $\mathbf{v}_1, \dots, \mathbf{v}_{r-1}$ ; it also contains  $\mathbf{v}_r$ . Hence  $\text{span}(\mathbf{v}_1, \dots, \mathbf{v}_r) \subseteq \text{span}(S_r)$ .

In both cases  $P(r)$  holds. By induction  $P(m)$  holds, which is the first statement.

Finally, suppose  $(\mathbf{v}_1, \dots, \mathbf{v}_m)$  spans  $V$ . Then  $S_m$  is a sub-list of it that is independent and satisfies  $\text{span}(S_m) = V$ . So  $S_m$  is a basis of  $V$ . ■

Alternatively, one can delete instead of keep. As long as the list is dependent, the Linear

Dependence Lemma ([Theorem 1.4.1](#)) locates an entry lying in the span of the entries before it, and deleting that entry does not change the span. Each deletion shortens the list, so the process stops after at most  $m$  steps. Sifting performs the same idea in a single left-to-right pass.

The decision “is  $\mathbf{v}_r$  in the span of the kept vectors?” is a small system of equations. For now we solve these systems by hand; Chapter 2 gives a systematic method.

**Example. (Sifting a List in  $\mathbb{R}^3$ )**

Sift the list  $(\mathbf{v}_1, \dots, \mathbf{v}_6)$  in  $\mathbb{R}^3$ , where

$$\begin{aligned}\mathbf{v}_1 &= (1, 1, 1), & \mathbf{v}_2 &= (2, 2, 2), & \mathbf{v}_3 &= (1, 0, 0), \\ \mathbf{v}_4 &= (3, 2, 2), & \mathbf{v}_5 &= (1, 1, 0), & \mathbf{v}_6 &= (0, 0, 1).\end{aligned}$$

Hence find a basis of  $\mathbb{R}^3$  inside the list.

*Solution.* We go through the list once, comparing each vector with the vectors kept so far.

- $\mathbf{v}_1$ : the kept list is empty, with span  $\{\mathbf{0}\}$ . Since  $\mathbf{v}_1 \neq \mathbf{0}$ , **keep**  $\mathbf{v}_1$ .
- $\mathbf{v}_2$ :  $\mathbf{v}_2 = 2\mathbf{v}_1 \in \text{span}(\mathbf{v}_1)$ . **Discard.**
- $\mathbf{v}_3$ : if  $(1, 0, 0) = a(1, 1, 1) = (a, a, a)$ , the first entry gives  $a = 1$  and the second gives  $a = 0$ , a contradiction. So  $\mathbf{v}_3 \notin \text{span}(\mathbf{v}_1)$ . **Keep.**
- $\mathbf{v}_4$ :  $2\mathbf{v}_1 + \mathbf{v}_3 = (2, 2, 2) + (1, 0, 0) = (3, 2, 2) = \mathbf{v}_4$ , so  $\mathbf{v}_4 \in \text{span}(\mathbf{v}_1, \mathbf{v}_3)$ . **Discard.**
- $\mathbf{v}_5$ : suppose  $(1, 1, 0) = a(1, 1, 1) + b(1, 0, 0) = (a + b, a, a)$ . The second entry gives  $a = 1$  and the third gives  $a = 0$ , a contradiction. So  $\mathbf{v}_5 \notin \text{span}(\mathbf{v}_1, \mathbf{v}_3)$ . **Keep.**
- $\mathbf{v}_6$ :  $\mathbf{v}_1 - \mathbf{v}_5 = (1, 1, 1) - (1, 1, 0) = (0, 0, 1) = \mathbf{v}_6$ , so  $\mathbf{v}_6 \in \text{span}(\mathbf{v}_1, \mathbf{v}_3, \mathbf{v}_5)$ . **Discard.**

The sifted list is  $(\mathbf{v}_1, \mathbf{v}_3, \mathbf{v}_5) = ((1, 1, 1), (1, 0, 0), (1, 1, 0))$ . By [Theorem 1.5.4](#) it is independent and has the same span as the original list. That span is all of  $\mathbb{R}^3$ , since it contains  $\mathbf{e}_1 = \mathbf{v}_3$ ,  $\mathbf{e}_2 = \mathbf{v}_5 - \mathbf{v}_3$  and  $\mathbf{e}_3 = \mathbf{v}_6$ , and hence every combination of them. So  $(\mathbf{v}_1, \mathbf{v}_3, \mathbf{v}_5)$  is a basis of  $\mathbb{R}^3$ .

Sifting has no memory of the future, so its output depends on the order of the input.

**Example. (The Order Matters)**

Sift  $((1, 0), (0, 1), (1, 1))$  and  $((1, 1), (1, 0), (0, 1))$  in  $\mathbb{R}^2$ .

*Solution.* For the first list:  $(1, 0) \neq \mathbf{0}$ , so keep it. Every multiple of  $(1, 0)$  has second entry 0, so  $(0, 1) \notin \text{span}((1, 0))$ ; keep it. Then  $(1, 1) = (1, 0) + (0, 1)$ ; discard it. The result is  $((1, 0), (0, 1))$ .

For the second list:  $(1, 1) \neq \mathbf{0}$ , so keep it. If  $(1, 0) = a(1, 1) = (a, a)$ , then  $a = 1$  and  $a = 0$ , which is impossible; keep  $(1, 0)$ . Then  $(0, 1) = (1, 1) - (1, 0)$ ; discard it. The result is  $((1, 1), (1, 0))$ .

Both results are bases of  $\mathbb{R}^2$  (the lists span  $\mathbb{R}^2$  because they contain  $\mathbf{e}_1$  and  $\mathbf{e}_2$ ), but they are different bases. A zero vector, by contrast, is discarded wherever it stands, because  $\mathbf{0}$  lies in every span.

*Quick check.* Sift the list  $(x, 2x, 1 + x, 1)$  in  $\mathbb{R}[x]_{\leq 1}$ . Which vectors are kept?

*Solution.* Keep  $x$ , since  $x \neq 0$ . Discard  $2x = 2 \cdot x$ . Keep  $1 + x$ : every multiple  $ax$  has constant term 0, but  $1 + x$  has constant term 1. Discard  $1 = (1 + x) - x$ . The sifted list is  $(x, 1 + x)$ .

The theorem answers the question we started with.

### Corollary 1.5.5 — Existence of Bases

Every finite-dimensional vector space has a basis.

*Proof.* Let  $V$  be finite-dimensional. By [Definition 1.3.4](#),  $V$  is spanned by some finite list. By [Theorem 1.5.4](#), that list contains a sub-list that is a basis of  $V$ . ■

For  $V = \{\mathbf{0}\}$  the proof still works:  $\{\mathbf{0}\}$  is spanned by the empty list, and sifting it returns the empty list, which is the basis found in [Example \(Three More Bases\)](#).

We now know that bases exist, and we can find one inside any finite spanning list. But a space has many bases. The next section proves the fact that makes them measurable: all bases of the same space have the same length.

## 1.5.5 Exercises

### 1.5.5.1 A. Check your understanding

#### Exercise A1.

- Define what it means for a list  $(\mathbf{v}_1, \dots, \mathbf{v}_n)$  to be a basis of a vector space  $V$  over  $F$ .
- True or false: every list that spans  $\mathbb{R}^2$  is a basis of  $\mathbb{R}^2$ . Justify your answer.
- True or false:  $((1, 0), (0, 1))$  and  $((0, 1), (1, 0))$  are the same basis of  $\mathbb{R}^2$ . Justify your answer.
- What is a basis of the zero space  $\{\mathbf{0}\}$ ?
- When sifting a list, under what condition is the vector  $\mathbf{v}_r$  discarded?
- True or false: if  $\mathcal{B}$  is a basis of  $V$  and  $[\mathbf{u}]_{\mathcal{B}} = [\mathbf{v}]_{\mathcal{B}}$ , then  $\mathbf{u} = \mathbf{v}$ . Justify your answer.

*Solution.*

- See [Definition 1.5.1](#): the list is linearly independent and spans  $V$ .
- False.  $((1, 0), (0, 1), (1, 1))$  spans  $\mathbb{R}^2$ , but it is dependent, since  $(1, 0) + (0, 1) - (1, 1) = \mathbf{0}$ .
- False. A basis is a list, and the two lists have their entries in different orders. The difference is visible in coordinates:  $(3, 1)$  has coordinate vector  $(3, 1)$  in the first



basis and  $(1, 3)$  in the second.

- (d) The empty list  $()$ , by [Example \(Three More Bases\)](#) (c).
- (e) When  $\mathbf{v}_r$  lies in the span of the vectors kept before it, that is,  $\mathbf{v}_r \in \text{span}(S_{r-1})$ .
- (f) True. If both coordinate vectors equal  $(a_1, \dots, a_n)$ , then  $\mathbf{u} = a_1\mathbf{v}_1 + \dots + a_n\mathbf{v}_n = \mathbf{v}$  by [Definition 1.5.2](#), where  $\mathcal{B} = (\mathbf{v}_1, \dots, \mathbf{v}_n)$ .

### 1.5.5.2 B. Practice

#### Exercise B1: A Basis of $\mathbb{R}^3$ and Coordinates.

Let  $\mathcal{B} = ((1, 1, 0), (0, 1, 1), (1, 0, 1))$ . Show that  $\mathcal{B}$  is a basis of  $\mathbb{R}^3$ . Hence find  $[\mathbf{v}]_{\mathcal{B}}$  for  $\mathbf{v} = (2, 3, 5)$ .

*Solution.* Let  $(p, q, r) \in \mathbb{R}^3$ . For scalars  $a, b, c$ ,

$$a(1, 1, 0) + b(0, 1, 1) + c(1, 0, 1) = (a + c, a + b, b + c),$$

so this equals  $(p, q, r)$  if and only if  $a + c = p$ ,  $a + b = q$  and  $b + c = r$ . Adding the three equations gives  $2(a + b + c) = p + q + r$ , so  $a + b + c = \frac{p+q+r}{2}$ . Subtracting each equation from this one,

$$b = \frac{-p + q + r}{2}, \quad c = \frac{p - q + r}{2}, \quad a = \frac{p + q - r}{2}.$$

Conversely, these values satisfy the three equations: for instance  $a + c = \frac{2p}{2} = p$ , and the other two are checked the same way. So every vector of  $\mathbb{R}^3$  has exactly one expression in  $\mathcal{B}$ , and  $\mathcal{B}$  is a basis by [Theorem 1.5.1](#).

For  $(p, q, r) = (2, 3, 5)$ :  $a = \frac{2+3-5}{2} = 0$ ,  $b = \frac{-2+3+5}{2} = 3$ ,  $c = \frac{2-3+5}{2} = 2$ . Check:  $0(1, 1, 0) + 3(0, 1, 1) + 2(1, 0, 1) = (2, 3, 5)$ . Hence  $[\mathbf{v}]_{\mathcal{B}} = (0, 3, 2)$ .

#### Exercise B2: Sifting Polynomials.

Sift the list  $(1 + x, 2 + 2x, x^2, 1 + x + x^2, x, 1)$  in  $\mathbb{R}[x]_{\leq 2}$ , justifying each decision. Hence find a basis of  $\mathbb{R}[x]_{\leq 2}$  contained in the list.

*Solution.* We read the list from left to right, comparing coefficients.

- $1 + x \neq 0$ : **keep**.
- $2 + 2x = 2(1 + x)$ : **discard**.
- $x^2$ : every multiple  $a(1 + x)$  has  $x^2$ -coefficient 0, while  $x^2$  has  $x^2$ -coefficient 1. So  $x^2 \notin \text{span}(1 + x)$ : **keep**.
- $1 + x + x^2 = (1 + x) + x^2$ : **discard**.
- $x$ : suppose  $x = a(1 + x) + bx^2 = a + ax + bx^2$ . The constant terms give  $a = 0$  and the  $x$ -coefficients give  $a = 1$ , a contradiction. So  $x \notin \text{span}(1 + x, x^2)$ : **keep**.
- $1 = (1 + x) - x$ : **discard**.

The sifted list is  $(1 + x, x^2, x)$ . The original list contains 1,  $x$  and  $x^2$ , so its span contains every combination of them, which is all of  $\mathbb{R}[x]_{\leq 2}$  by [Example \(Standard Bases\)](#). Hence the list spans  $\mathbb{R}[x]_{\leq 2}$ , and by [Theorem 1.5.4](#) the sifted list  $(1 + x, x^2, x)$  is a basis of

$\mathbb{R}[x]_{\leq 2}$ .

### Exercise B3: Symmetric Matrices.

Let  $U = \{\mathbf{A} \in M_2(\mathbb{R}) : \mathbf{A}^\top = \mathbf{A}\}$ . Show that  $(\mathbf{E}_{11}, \mathbf{E}_{12} + \mathbf{E}_{21}, \mathbf{E}_{22})$  is a basis of  $U$ , and find the coordinate vector of  $\begin{pmatrix} 2 & -1 \\ -1 & 5 \end{pmatrix}$  with respect to it.

*Solution.* Each of the three matrices is symmetric, so the list lies in  $U$ .

*Spanning.* Let  $\mathbf{A} \in U$ . Writing  $\mathbf{A} = \begin{pmatrix} a & b \\ c & d \end{pmatrix}$ , the condition  $\mathbf{A}^\top = \mathbf{A}$  says  $c = b$ . Hence

$$\mathbf{A} = \begin{pmatrix} a & b \\ b & d \end{pmatrix} = a\mathbf{E}_{11} + b(\mathbf{E}_{12} + \mathbf{E}_{21}) + d\mathbf{E}_{22}.$$

*Independence.* Let  $a\mathbf{E}_{11} + b(\mathbf{E}_{12} + \mathbf{E}_{21}) + d\mathbf{E}_{22} = \mathbf{0}$ . The left side is  $\begin{pmatrix} a & b \\ b & d \end{pmatrix}$ , so comparing entries with the zero matrix gives  $a = b = d = 0$ .

Hence the list is a basis of  $U$ . Reading off  $a = 2$ ,  $b = -1$ ,  $d = 5$ , the coordinate vector is  $(2, -1, 5)$ .

### 1.5.5.3 C. Going deeper

#### Exercise C1: Polynomials Vanishing at 1.

Let  $F$  be a field,  $n \geq 1$ , and  $U = \{p \in F[x]_{\leq n} : p(1) = 0\}$ .

- (a) Prove that  $(x - 1, x^2 - 1, \dots, x^n - 1)$  is a basis of  $U$ .
- (b) For  $n = 3$ , find the coordinate vector of  $x^3 - 2x + 1$  with respect to this basis.

*Hint:* for  $p = a_0 + a_1x + \dots + a_nx^n$ , what does  $p(1) = 0$  say about  $a_0$ ?

*Solution.*

- (a) For  $p = a_0 + a_1x + \dots + a_nx^n$ , evaluation gives  $p(1) = a_0 + a_1 + \dots + a_n$ , since every power of 1 is 1. In particular each  $x^k - 1$  lies in  $U$ , because  $1 - 1 = 0$ .

*Spanning.* Let  $p \in U$ . Then  $a_0 = -(a_1 + \dots + a_n)$ , so

$$\begin{aligned} p &= a_1x + \dots + a_nx^n - (a_1 + \dots + a_n) \\ &= a_1(x - 1) + a_2(x^2 - 1) + \dots + a_n(x^n - 1). \end{aligned}$$

*Independence.* Let  $c_1(x - 1) + \dots + c_n(x^n - 1) = 0$ . For  $1 \leq k \leq n$ , the coefficient of  $x^k$  on the left is  $c_k$ , since  $x^k$  appears in no other term. The zero polynomial has all coefficients 0, so  $c_k = 0$  for every  $k$ .

Hence the list is a basis of  $U$ .

- (b) For  $p = x^3 - 2x + 1$  we have  $p(1) = 0$ , and  $(a_1, a_2, a_3) = (-2, 0, 1)$ . By the formula

in (a),  $p = -2(x-1) + 0(x^2-1) + 1(x^3-1)$ . Check:  $-2x + 2 + x^3 - 1 = x^3 - 2x + 1$ . The coordinate vector is  $(-2, 0, 1)$ .

### Exercise C2: The Same List over a Different Field.

- (a) Explain why  $((1, 1), (1, -1))$  is **not** a basis of  $\mathbb{F}_2^2$ .
- (b) Let  $F$  be any field. Prove that  $((1, 1), (1, -1))$  is a basis of  $F^2$  if and only if  $1 + 1 \neq 0$  in  $F$ .

*Solution.*

- (a) In  $\mathbb{F}_2$  we have  $-1 = 1$ , so the list is  $((1, 1), (1, 1))$ . It is dependent:  $1 \cdot (1, 1) + 1 \cdot (1, 1) = (1 + 1, 1 + 1) = (0, 0)$  with non-zero coefficients. (It also fails to span: every combination is  $(c, c)$ , so  $(1, 0)$  is out of reach.)
- (b) Write  $2 = 1 + 1 \in F$ .

( $\Leftarrow$ ) Suppose  $2 \neq 0$ , so  $2^{-1}$  exists. Let  $(x, y) \in F^2$ . The equation  $a(1, 1) + b(1, -1) = (x, y)$  says  $a + b = x$  and  $a - b = y$ . If it holds, adding and subtracting gives  $2a = x + y$  and  $2b = x - y$ , so  $a = 2^{-1}(x + y)$  and  $b = 2^{-1}(x - y)$ . Conversely these values satisfy  $a + b = 2^{-1} \cdot 2x = x$  and  $a - b = 2^{-1} \cdot 2y = y$ . So every vector has exactly one expression, and the list is a basis by [Theorem 1.5.1](#).

( $\Rightarrow$ ) Suppose  $2 = 0$ . Then  $1 = -1$ , and the argument of (a) applies word for word:  $1 \cdot (1, 1) + 1 \cdot (1, -1) = (2, 0) = (0, 0)$ , so the list is dependent and is not a basis. This proves ( $\Rightarrow$ ) by contraposition.

### Exercise C3: Rescaling a Basis.

Let  $\mathcal{B} = (\mathbf{v}_1, \dots, \mathbf{v}_n)$  be a basis of  $V$  over  $F$ , and let  $c_1, \dots, c_n \in F$  be **non-zero**.

- (a) Prove that  $\mathcal{B}' = (c_1\mathbf{v}_1, \dots, c_n\mathbf{v}_n)$  is a basis of  $V$ .
- (b) Express  $[\mathbf{v}]_{\mathcal{B}'}$  in terms of  $[\mathbf{v}]_{\mathcal{B}}$ .
- (c) Explain why the conclusion of (a) can fail if some  $c_i = 0$ .

*Solution.*

- (a) Let  $\mathbf{v} \in V$  and  $[\mathbf{v}]_{\mathcal{B}} = (a_1, \dots, a_n)$ . Since each  $c_i \neq 0$ ,  $c_i^{-1}$  exists, and

$$\mathbf{v} = \sum_{i=1}^n a_i \mathbf{v}_i = \sum_{i=1}^n (a_i c_i^{-1})(c_i \mathbf{v}_i).$$

So  $\mathbf{v}$  has an expression in  $\mathcal{B}'$ . Suppose  $\mathbf{v} = \sum_i b_i (c_i \mathbf{v}_i) = \sum_i (b_i c_i) \mathbf{v}_i$ . By uniqueness of the expression in  $\mathcal{B}$  ([Theorem 1.5.1](#)),  $b_i c_i = a_i$  for each  $i$ , so  $b_i = a_i c_i^{-1}$ . Hence the expression in  $\mathcal{B}'$  is unique, and  $\mathcal{B}'$  is a basis by [Theorem 1.5.1](#).

- (b) By (a),  $[\mathbf{v}]_{\mathcal{B}'} = (a_1 c_1^{-1}, \dots, a_n c_n^{-1})$ : divide the  $i$ -th coordinate by  $c_i$ .
- (c) If  $c_1 = 0$ , then  $\mathcal{B}'$  contains  $0\mathbf{v}_1 = \mathbf{0}$ , and  $1 \cdot \mathbf{0} + 0 \cdot c_2 \mathbf{v}_2 + \dots + 0 \cdot c_n \mathbf{v}_n = \mathbf{0}$  is a combination with a non-zero coefficient. So  $\mathcal{B}'$  is dependent and is not a basis.

## 1.6 The Exchange Theorem and Dimension

Every finite-dimensional space has a basis, and usually a great many of them. This section proves that they all have the same length, so that the length is a property of the space and not of the basis we happened to pick. That number is the dimension. The key step is the Steinitz Exchange Theorem, the most important proof of the chapter. With it we build the everyday tools of dimension counting: size bounds, “count instead of check”, extension of independent lists to bases, and dimensions of subspaces.

### 1.6.1 How big is a vector space?

We would like to say that a plane is smaller than space, and that  $M_2(\mathbb{R})$  and  $\mathbb{R}[x]_{\leq 3}$  are, in some sense, the same size. The naive measure is the number of elements. It fails at once:  $\mathbb{R}^2$ ,  $M_2(\mathbb{R})$  and  $\mathbb{R}[x]_{\leq 3}$  are all infinite sets, and counting elements cannot tell them apart.

A better measure counts **how many vectors it takes to build everything, without waste**, which is the length of a basis. The standard bases give

$$\mathbb{R}^2 : 2, \quad M_2(\mathbb{R}) : 4 \text{ } (\mathbf{E}_{11}, \mathbf{E}_{12}, \mathbf{E}_{21}, \mathbf{E}_{22}), \quad \mathbb{R}[x]_{\leq 3} : 4 \text{ } (1, x, x^2, x^3).$$

This matches intuition: a matrix  $\begin{pmatrix} a & b \\ c & d \end{pmatrix}$  and a polynomial  $a + bx + cx^2 + dx^3$  are both described by four free numbers.

But there is a gap. A space has many bases. We counted the standard basis of  $\mathbb{R}^2$ ; the basis  $((1, 1), (1, -1))$  also has length 2, but what guarantees that no basis of  $\mathbb{R}^2$  has length 3? If some space had bases of two different lengths, “the length of a basis” would be meaningless. So before we can define anything, we need the following.

**Goal.** Any two bases of a finite-dimensional vector space have the same length.

A basis is independent **and** spanning, and the goal follows from a single inequality about the two properties separately: *an independent list is never longer than a spanning list*. Granting this, take two bases  $\mathcal{B}$  and  $\mathcal{C}$ . Since  $\mathcal{B}$  is independent and  $\mathcal{C}$  spans,  $\mathcal{B}$  is not longer than  $\mathcal{C}$ . Swapping the roles,  $\mathcal{C}$  is not longer than  $\mathcal{B}$ . So they have the same length. Everything rests on the inequality.

### 1.6.2 The Steinitz Exchange Theorem

Why should an independent list be no longer than a spanning list? In  $\mathbb{R}^2$  the spanning list  $(\mathbf{e}_1, \mathbf{e}_2)$  has length 2, and three vectors in the plane always satisfy a linear relation, but “the plane is two-dimensional” is what we are trying to define, so we cannot use it. The idea is instead to feed the independent vectors into the spanning list one at a time, pushing out one spanning vector each time, while keeping the list spanning. If the independent vectors outnumbered the spanning ones, we would run out of vectors to push out.

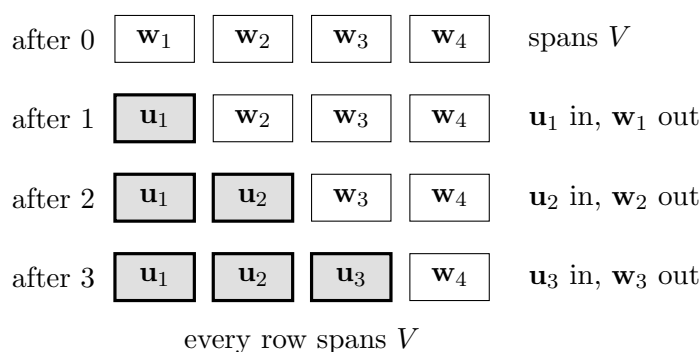
### Theorem 1.6.1 — Steinitz Exchange Theorem

Let  $V$  be a vector space over  $F$ . Suppose that  $(\mathbf{u}_1, \dots, \mathbf{u}_m)$  is a **linearly independent** list in  $V$  and that  $(\mathbf{w}_1, \dots, \mathbf{w}_n)$  is a list that **spans**  $V$ . Then

- (a)  $m \leq n$ , and
- (b) after renumbering  $\mathbf{w}_1, \dots, \mathbf{w}_n$  suitably, the list  $(\mathbf{u}_1, \dots, \mathbf{u}_m, \mathbf{w}_{m+1}, \dots, \mathbf{w}_n)$  spans  $V$ .

In words: an independent list is at most as long as a spanning list, and the independent vectors can **replace**  $m$  of the spanning vectors without losing the spanning property. “Renumbering” means listing the  $\mathbf{w}$ ’s in a different order; which  $\mathbf{w}$ ’s get replaced depends on the  $\mathbf{u}$ ’s.

*Idea.* Picture a row of  $n$  slots holding  $\mathbf{w}_1, \dots, \mathbf{w}_n$ ; the row spans  $V$ . We push  $\mathbf{u}_1$  into the row and push one  $\mathbf{w}$  out, then  $\mathbf{u}_2$ , and so on, keeping the row spanning after every exchange. Here is the picture for  $m = 3$ ,  $n = 4$ , with the  $\mathbf{w}$ ’s numbered in the order they leave.



Which  $\mathbf{w}$  may leave? Write the incoming  $\mathbf{u}_{k+1}$  as a combination of the current row. Any  $\mathbf{w}$  with a **non-zero** coefficient can be solved for, so it lies in the span of the other entries together with  $\mathbf{u}_{k+1}$ , and removing it costs nothing. Such a  $\mathbf{w}$  must exist: otherwise  $\mathbf{u}_{k+1}$  would be a combination of  $\mathbf{u}_1, \dots, \mathbf{u}_k$  alone, which independence forbids. In particular, while  $\mathbf{u}$ ’s remain to be inserted, a  $\mathbf{w}$  remains to be removed, and that is why  $m \leq n$ .

*Step roadmap.* Induction on the number  $k$  of exchanges, with the invariant

$P(k)$ :  $k \leq n$ , and after renumbering the  $\mathbf{w}$ ’s, the list  $(\mathbf{u}_1, \dots, \mathbf{u}_k, \mathbf{w}_{k+1}, \dots, \mathbf{w}_n)$  spans  $V$ .

**Step 1** is the base case  $k = 0$ . **Step 2** writes  $\mathbf{u}_{k+1}$  in the current list. **Step 3** is the Claim that some remaining  $\mathbf{w}$  has a non-zero coefficient. **Step 4** exchanges that  $\mathbf{w}$  for  $\mathbf{u}_{k+1}$  and checks the span.

*Proof.* We prove  $P(k)$  for  $k = 0, 1, \dots, m$  by induction on  $k$ . The case  $k = m$  is the theorem.

**Step 1 (base case).** For  $k = 0$  the list in  $P(0)$  is  $(\mathbf{w}_1, \dots, \mathbf{w}_n)$ , which spans  $V$  by hypothesis, and  $0 \leq n$ . So  $P(0)$  holds.

**Step 2.** Let  $0 \leq k < m$  and suppose  $P(k)$  holds. Renumber the  $\mathbf{w}$ 's as  $P(k)$  allows, so that

$$L_k = (\mathbf{u}_1, \dots, \mathbf{u}_k, \mathbf{w}_{k+1}, \dots, \mathbf{w}_n)$$

spans  $V$ . Since  $k < m$ , the vector  $\mathbf{u}_{k+1}$  exists, and  $\mathbf{u}_{k+1} \in V = \text{span}(L_k)$ . Hence there are scalars  $a_1, \dots, a_k, b_{k+1}, \dots, b_n \in F$  with

$$\mathbf{u}_{k+1} = a_1\mathbf{u}_1 + \dots + a_k\mathbf{u}_k + b_{k+1}\mathbf{w}_{k+1} + \dots + b_n\mathbf{w}_n. \quad (*)$$

(If  $k = n$ , the list  $L_k$  contains no  $\mathbf{w}$ 's and the second group of terms is empty.)

**Step 3.**

*Claim.* We have  $k+1 \leq n$ , and  $b_j \neq 0$  for some  $j$  with  $k+1 \leq j \leq n$ .

*Proof.* Suppose not. Then either  $k = n$ , so that there are no  $b$ 's at all, or  $k < n$  and  $b_{k+1} = \dots = b_n = 0$ . In both cases  $(*)$  reads  $\mathbf{u}_{k+1} = a_1\mathbf{u}_1 + \dots + a_k\mathbf{u}_k$ . Therefore

$$a_1\mathbf{u}_1 + \dots + a_k\mathbf{u}_k + (-1)\mathbf{u}_{k+1} + 0\mathbf{u}_{k+2} + \dots + 0\mathbf{u}_m = \mathbf{0}$$

is a linear relation among  $\mathbf{u}_1, \dots, \mathbf{u}_m$  in which the coefficient of  $\mathbf{u}_{k+1}$  is  $-1 \neq 0$ . This contradicts the linear independence of  $(\mathbf{u}_1, \dots, \mathbf{u}_m)$ . Hence there is at least one  $\mathbf{w}$  in  $L_k$ , so  $k+1 \leq n$ , and some  $b_j$  with  $j \geq k+1$  is non-zero. ■

**Step 4.** Renumber  $\mathbf{w}_{k+1}, \dots, \mathbf{w}_n$  among themselves so that  $b_{k+1} \neq 0$ . This only changes the order of entries of  $L_k$ , so  $L_k$  still spans  $V$  (the span of a list does not depend on the order of its entries), and  $(*)$  still holds with the coefficients renumbered in the same way. Put

$$L_{k+1} = (\mathbf{u}_1, \dots, \mathbf{u}_{k+1}, \mathbf{w}_{k+2}, \dots, \mathbf{w}_n).$$

Since  $b_{k+1} \neq 0$ , the scalar  $b_{k+1}^{-1}$  exists, and solving  $(*)$  for  $\mathbf{w}_{k+1}$  gives

$$\mathbf{w}_{k+1} = b_{k+1}^{-1}(\mathbf{u}_{k+1} - a_1\mathbf{u}_1 - \dots - a_k\mathbf{u}_k - b_{k+2}\mathbf{w}_{k+2} - \dots - b_n\mathbf{w}_n).$$

The right side is a combination of the entries of  $L_{k+1}$ , so  $\mathbf{w}_{k+1} \in \text{span}(L_{k+1})$ . By [Theorem 1.3.5](#), inserting  $\mathbf{w}_{k+1}$  into  $L_{k+1}$  does not change the span:

$$\text{span}(L_{k+1}) = \text{span}(\mathbf{u}_1, \dots, \mathbf{u}_{k+1}, \mathbf{w}_{k+1}, \mathbf{w}_{k+2}, \dots, \mathbf{w}_n).$$

The list on the right contains every entry of  $L_k$ , so its span is a subspace containing all entries of  $L_k$ , and hence contains  $\text{span}(L_k) = V$  by [Theorem 1.3.1](#). Therefore  $\text{span}(L_{k+1}) = V$ . Together with  $k+1 \leq n$  from the Claim, this is  $P(k+1)$ .

By induction,  $P(m)$  holds:  $m \leq n$ , and after renumbering the  $\mathbf{w}$ 's,  $(\mathbf{u}_1, \dots, \mathbf{u}_m, \mathbf{w}_{m+1}, \dots, \mathbf{w}_n)$  spans  $V$ . This proves the theorem. ■

Part (a) is the inequality our goal needed. Part (b), the exchange itself, is a bonus that we will spend later in this section, to extend independent lists to bases.

*Warning.* The  $\mathbf{w}$  that leaves must have a **non-zero** coefficient; an arbitrary  $\mathbf{w}$  will not do. In  $\mathbb{R}^2$ , take the spanning list  $(\mathbf{w}_1, \mathbf{w}_2) = ((1, 0), (0, 1))$  and the independent vector  $\mathbf{u}_1 = (1, 0) = 1\mathbf{w}_1 + 0\mathbf{w}_2$ . Exchanging  $\mathbf{u}_1$  for  $\mathbf{w}_1$  gives  $((1, 0), (0, 1))$ , which spans. Exchanging  $\mathbf{u}_1$  for  $\mathbf{w}_2$ , whose coefficient is 0, gives  $((1, 0), (1, 0))$ , which does not reach  $(0, 1)$ .

The theorem already settles questions that would otherwise need a computation.

*Quick check.* Can the four polynomials  $1 + x$ ,  $x + x^2$ ,  $x^2 + 1$ ,  $1 + x + x^2$  be linearly independent in  $\mathbb{R}[x]_{\leq 2}$ ?

*Solution.* No. The list  $(1, x, x^2)$  spans  $\mathbb{R}[x]_{\leq 2}$  and has length 3, so by [Theorem 1.6.1](#) every independent list there has length at most  $3 < 4$ . Indeed  $(1 + x) + (x + x^2) + (x^2 + 1) - 2(1 + x + x^2) = 0$ .

### 1.6.3 Dimension

The goal of the section is now one line away.

#### Theorem 1.6.2 — Basis Theorem

Let  $V$  be a finite-dimensional vector space. Then  $V$  has a basis, and any two bases of  $V$  have the same length.

*Proof.* By [Corollary 1.5.5](#),  $V$  has a basis. Let  $(\mathbf{v}_1, \dots, \mathbf{v}_n)$  and  $(\mathbf{w}_1, \dots, \mathbf{w}_m)$  be bases of  $V$ . The first is linearly independent and the second spans  $V$ , so  $n \leq m$  by [Theorem 1.6.1](#). Swapping the roles of the two bases gives  $m \leq n$ . Hence  $m = n$ . ■

We started from a question: how big is a vector space? The Basis Theorem says that one number, the length of a basis, answers it without depending on any choice. So we give that number a name.

*The dimension of a space is the number of vectors in any basis: how many independent directions the space has.*

#### Definition 1.6.1 — Dimension

Let  $V$  be a finite-dimensional vector space over  $F$ . The **dimension** of  $V$  over  $F$ , written  $\dim_F V$  or simply  $\dim V$ , is the length of **any** basis of  $V$ .

In words: pick a basis of  $V$ , any basis at all, and count its vectors. The dimension is defined only for finite-dimensional spaces ([Definition 1.3.4](#)); a space spanned by no finite list is infinite-dimensional, and we do not assign it a number here. The subscript  $F$  records the field of scalars, which, as we will see, matters.

**Well-definedness.** Two things could go wrong with “the length of any basis”: there might

be no basis to count, or different bases might give different counts. The first is ruled out by [Corollary 1.5.5](#) and the second by [Theorem 1.6.2](#). That is exactly what the word **any** in the definition needs.

**Example. (Dimensions of the Running Examples)**

Find the dimension of each space.

- (a)  $F^n$ ,  $M_{m \times n}(F)$  and  $F[x]_{\leq n}$ .
- (b) The zero space  $\{\mathbf{0}\}$ .
- (c)  $\mathbb{C}$  over  $\mathbb{C}$ , and  $\mathbb{C}$  over  $\mathbb{R}$ .
- (d)  $U = \{\mathbf{A} \in M_2(\mathbb{R}) : \mathbf{A}^\top = \mathbf{A}\}$ , the symmetric  $2 \times 2$  real matrices.
- (e)  $\mathbb{Q}(\sqrt{2}) = \{a + b\sqrt{2} : a, b \in \mathbb{Q}\}$  over  $\mathbb{Q}$  ([Example \( \$\mathbb{Q}\(\sqrt{2}\)\$  over  \$\mathbb{Q}\$ \)](#)).

*Solution.* In each case we exhibit one basis and count it.

- (a) The standard bases of [Example \(Standard Bases\)](#) have lengths  $n$ ,  $mn$  and  $n + 1$ . So  $\dim F^n = n$ ,  $\dim M_{m \times n}(F) = mn$ , and  $\dim F[x]_{\leq n} = n + 1$ .
- (b) The empty list is a basis of  $\{\mathbf{0}\}$  ([Example \(Three More Bases\)](#)), so  $\dim\{\mathbf{0}\} = 0$ .
- (c) Over  $\mathbb{C}$ , the list  $(1)$  is a basis of  $\mathbb{C}$ : every  $z$  equals  $z \cdot 1$ , and  $z \cdot 1 = 0$  forces  $z = 0$ . So  $\dim_{\mathbb{C}} \mathbb{C} = 1$ . Over  $\mathbb{R}$ , the list  $(1, i)$  is a basis ([Example \(Three More Bases\)](#)), so  $\dim_{\mathbb{R}} \mathbb{C} = 2$ .
- (d) A symmetric matrix has the form  $\begin{pmatrix} a & b \\ b & d \end{pmatrix} = a\mathbf{E}_{11} + b(\mathbf{E}_{12} + \mathbf{E}_{21}) + d\mathbf{E}_{22}$ , and this combination is the zero matrix only when  $a = b = d = 0$ . So  $(\mathbf{E}_{11}, \mathbf{E}_{12} + \mathbf{E}_{21}, \mathbf{E}_{22})$  is a basis of  $U$ , as in [Exercise \(B3: Symmetric Matrices\)](#), and  $\dim U = 3$ .
- (e) Every element is  $a \cdot 1 + b\sqrt{2}$  with  $a, b \in \mathbb{Q}$ , so  $(1, \sqrt{2})$  spans. Let  $a + b\sqrt{2} = 0$  with  $a, b \in \mathbb{Q}$ . If  $b \neq 0$ , then  $\sqrt{2} = -a/b \in \mathbb{Q}$ , contradicting [Theorem 0.2.3](#). So  $b = 0$ , and then  $a = 0$ . Hence  $(1, \sqrt{2})$  is a basis, and  $\dim_{\mathbb{Q}} \mathbb{Q}(\sqrt{2}) = 2$ .

The zero space in (b) is the degenerate case: it has dimension 0 because its basis is empty, not because it has no elements. It has one element,  $\mathbf{0}$ .

**Non-example by minimal change.** Remove the degree bound from  $F[x]_{\leq n}$ . The space  $F[x]$  is not spanned by any finite list ([Theorem 1.3.3](#)), so it has no finite basis and no dimension in the sense of [Definition 1.6.1](#). The hypothesis “finite-dimensional” in the Basis Theorem is exactly what fails.

**Why this definition.** Why a basis, and not some other list? Spanning lists can be made as long as we like by repeating vectors, and independent lists can be made as short as we like by deleting vectors, so neither length alone is an invariant. A basis sits exactly where the two meet: it is the longest possible independent list and the shortest possible spanning list, as we prove next. And why not count elements? Over  $\mathbb{R}$  every non-zero space is infinite, so the count says nothing. Over a finite field the count does carry information, but it is determined by the dimension; see [Exercise \(C3: Counting Elements over a Finite Field\)](#).

*Warning.* The dimension of  $F[x]_{\leq n}$  is  $n + 1$ , **not**  $n$ . The basis  $(1, x, \dots, x^n)$  has one vector for each exponent  $0, 1, \dots, n$ , and the constant 1 is easy to forget. For instance  $\dim \mathbb{R}[x]_{\leq 2} = 3$ .



*Warning.* Dimension depends on the field. The set  $\mathbb{C}$  has dimension 1 over  $\mathbb{C}$  but dimension 2 over  $\mathbb{R}$ , because over  $\mathbb{R}$  the scalar  $i$  is no longer available and  $i$  must be counted as a separate basis vector. When more than one field is in sight, write  $\dim_F$ .

*Quick check.* What are  $\dim M_{2 \times 3}(F)$  and  $\dim F[x]_{\leq 5}$ ? Which of them equals  $\dim F^6$ ?

*Solution.*  $\dim M_{2 \times 3}(F) = 2 \cdot 3 = 6$  and  $\dim F[x]_{\leq 5} = 5 + 1 = 6$ . Both equal  $\dim F^6 = 6$ .

## 1.6.4 Size bounds

The first payoff turns the Exchange Theorem into statements about the number  $\dim V$ .

### Theorem 1.6.3 — Size Bounds

Let  $V$  be a vector space over  $F$  with  $\dim V = n$ .

- (a) Every linearly independent list in  $V$  has length at most  $n$ . Hence every list in  $V$  of length **greater than**  $n$  is linearly dependent.
- (b) Every list that spans  $V$  has length at least  $n$ . Hence no list of length **less than**  $n$  spans  $V$ .

*Proof.* Let  $\mathcal{B}$  be a basis of  $V$ ; it has length  $n$ .

- (a) Let  $\mathcal{L}$  be an independent list of length  $\ell$ . Since  $\mathcal{B}$  spans  $V$ , Theorem 1.6.1 gives  $\ell \leq n$ . The second sentence is the contrapositive.
- (b) Let  $\mathcal{L}$  be a list of length  $\ell$  that spans  $V$ . Since  $\mathcal{B}$  is independent, Theorem 1.6.1 gives  $n \leq \ell$ . The second sentence is the contrapositive.

■

So in a space of dimension  $n$ ,

$$\text{length of an independent list} \leq n \leq \text{length of a spanning list}.$$

This settles many questions with no computation. The four vectors  $(1, 2, 3), (4, 5, 6), (7, 8, 9), (1, 0, 0)$  in  $\mathbb{R}^3$  are dependent, because  $4 > 3$ . Three matrices never span  $M_2(F)$ , because  $3 < 4$ .

## 1.6.5 Count instead of check

To show a list is a basis we normally check two things, independence and spanning. When the length of the list equals the dimension, one check is enough. This is one of the most used facts in the book.

### Theorem 1.6.4 — Count Instead of Check

Let  $V$  be a vector space with  $\dim V = n$ , and let  $\mathcal{L} = (\mathbf{v}_1, \dots, \mathbf{v}_n)$  be a list in  $V$  of length exactly  $n$ .

- (a) If  $\mathcal{L}$  is linearly independent, then  $\mathcal{L}$  is a basis of  $V$ .
- (b) If  $\mathcal{L}$  spans  $V$ , then  $\mathcal{L}$  is a basis of  $V$ .

*Idea.* For (a), the exchange part of [Theorem 1.6.1](#) does the work: we can push all  $n$  vectors of  $\mathcal{L}$  into a basis of length  $n$ , and then no old vectors are left, so  $\mathcal{L}$  alone spans. For (b), sift  $\mathcal{L}$ : the result is a basis, so it has length  $n$ , so nothing was discarded.

*Proof.*

- (a) Let  $(\mathbf{w}_1, \dots, \mathbf{w}_n)$  be a basis of  $V$ ; in particular it spans  $V$ . Since  $\mathcal{L}$  is independent of length  $n$ , [Theorem 1.6.1](#) (b) with  $m = n$  says that, after renumbering the  $\mathbf{w}$ 's, the list  $(\mathbf{v}_1, \dots, \mathbf{v}_n, \mathbf{w}_{n+1}, \dots, \mathbf{w}_n)$  spans  $V$ . This list contains no  $\mathbf{w}$ 's: it is  $\mathcal{L}$ . So  $\mathcal{L}$  is independent and spans  $V$ , and it is a basis.
- (b) By [Theorem 1.5.4](#),  $\mathcal{L}$  contains a sub-list that is a basis of  $V$ . By [Theorem 1.6.2](#), that sub-list has length  $\dim V = n$ . The only sub-list of length  $n$  of a list of length  $n$  is the list itself. Hence  $\mathcal{L}$  is a basis.

■

The theorem lets us choose the easier check, and independence is almost always easier: it is a single equation equal to  $\mathbf{0}$ , while spanning asks us to solve for an arbitrary vector.

### Example. (A Basis by Counting)

Show that  $(1, 1 + x, (1 + x)^2)$  is a basis of  $\mathbb{R}[x]_{\leq 2}$ .

*Solution.* By [Example \(Dimensions of the Running Examples\)](#),  $\dim \mathbb{R}[x]_{\leq 2} = 3$ , and the list has length 3. So by [Theorem 1.6.4](#) (a) it suffices to check independence. Let

$$a \cdot 1 + b(1 + x) + c(1 + x)^2 = 0.$$

Expanding, the left side is  $(a + b + c) + (b + 2c)x + cx^2$ . Comparing coefficients with the zero polynomial:  $c = 0$ ; then  $b + 2c = 0$  gives  $b = 0$ ; then  $a + b + c = 0$  gives  $a = 0$ . Hence the list is independent, with  $3 = \dim \mathbb{R}[x]_{\leq 2}$  elements, and therefore it is a basis.

We never had to show that every polynomial of degree at most 2 is a combination of  $1, 1 + x, (1 + x)^2$ . The dimension count did it for us.

## 1.6.6 Extending to a basis

Sifting finds a basis **inside** a spanning list. The opposite direction starts from an independent list and grows it into a basis. This is the tool behind every dimension formula later in the book: start from a basis of the smallest space, then extend.

### Theorem 1.6.5 — Basis Extension Theorem

Let  $V$  be a finite-dimensional vector space with  $\dim V = n$ , and let  $(\mathbf{u}_1, \dots, \mathbf{u}_m)$  be a linearly independent list in  $V$ . Then  $m \leq n$ , and there are vectors  $\mathbf{v}_{m+1}, \dots, \mathbf{v}_n \in V$

such that

$$(\mathbf{u}_1, \dots, \mathbf{u}_m, \mathbf{v}_{m+1}, \dots, \mathbf{v}_n)$$

is a basis of  $V$ . Moreover, the  $\mathbf{v}$ 's can be chosen from any given basis of  $V$ .

*Proof.* Let  $(\mathbf{w}_1, \dots, \mathbf{w}_n)$  be a basis of  $V$ . It spans  $V$ , so by [Theorem 1.6.1](#),  $m \leq n$ , and after renumbering the  $\mathbf{w}$ 's the list  $(\mathbf{u}_1, \dots, \mathbf{u}_m, \mathbf{w}_{m+1}, \dots, \mathbf{w}_n)$  spans  $V$ . This list has length  $m + (n - m) = n = \dim V$ , so it is a basis by [Theorem 1.6.4](#) (b). Take  $\mathbf{v}_j = \mathbf{w}_j$  for  $m < j \leq n$ . ■

Alternatively, sift the list  $(\mathbf{u}_1, \dots, \mathbf{u}_m, \mathbf{w}_1, \dots, \mathbf{w}_n)$ . It spans  $V$ , and sifting keeps every  $\mathbf{u}_k$ : the sub-list  $(\mathbf{u}_1, \dots, \mathbf{u}_k)$  is independent ([Exercise \(B3: Sub-Lists\)](#)), so  $\mathbf{u}_k \notin \text{span}(\mathbf{u}_1, \dots, \mathbf{u}_{k-1})$  by [Lemma 1.5.3](#). Sifting has the advantage of telling us **which**  $\mathbf{w}$ 's to add.

### Example. (Extending an Independent List)

Extend the list  $((1, 1, 0), (0, 1, 1))$  to a basis of  $\mathbb{R}^3$ .

*Solution.* The list is independent:  $a(1, 1, 0) + b(0, 1, 1) = (a, a+b, b) = \mathbf{0}$  forces  $a = b = 0$ . By [Theorem 1.6.5](#), one vector from the standard basis will do, so we try  $\mathbf{e}_1$ . Let

$$a(1, 1, 0) + b(0, 1, 1) + c(1, 0, 0) = (a + c, a + b, b) = (0, 0, 0).$$

Then  $b = 0$ , so  $a = 0$  from the second entry, so  $c = 0$  from the first. Hence  $((1, 1, 0), (0, 1, 1), (1, 0, 0))$  is independent, with  $3 = \dim \mathbb{R}^3$  elements, and therefore a basis by [Theorem 1.6.4](#) (a).

The extension is far from unique. The same computation with  $\mathbf{e}_2$  gives  $(a, a + b + c, b) = \mathbf{0}$ , and with  $\mathbf{e}_3$  gives  $(a, a + b, b + c) = \mathbf{0}$ ; both force  $a = b = c = 0$ . So any one of  $\mathbf{e}_1, \mathbf{e}_2, \mathbf{e}_3$  completes this list to a basis.

## 1.6.7 Largest independent lists and smallest spanning lists

We claimed that a basis is where independence and spanning meet. Here is the precise statement. It does not need finite dimension, and it is the form of “basis” that generalizes to infinite-dimensional spaces later in this chapter.

Call an independent list  $(\mathbf{v}_1, \dots, \mathbf{v}_n)$  in  $V$  **maximal independent** if for **every**  $\mathbf{v} \in V$  the longer list  $(\mathbf{v}_1, \dots, \mathbf{v}_n, \mathbf{v})$  is dependent. Call a spanning list **minimal spanning** if deleting **any one** of its entries gives a list that does not span  $V$ .

### Theorem 1.6.6 — Maximal Independent and Minimal Spanning Lists

Let  $V$  be a vector space and  $\mathcal{L} = (\mathbf{v}_1, \dots, \mathbf{v}_n)$  a list in  $V$ .

- (a)  $\mathcal{L}$  is a basis of  $V$  if and only if  $\mathcal{L}$  is maximal independent.
- (b)  $\mathcal{L}$  is a basis of  $V$  if and only if  $\mathcal{L}$  is minimal spanning.

*Proof.*

- (a) ( $\Rightarrow$ ) Let  $\mathcal{L}$  be a basis and  $\mathbf{v} \in V$ . Then  $\mathbf{v} \in \text{span}(\mathcal{L}) = V$ , so  $(\mathbf{v}_1, \dots, \mathbf{v}_n, \mathbf{v})$  is dependent by Lemma 1.5.3. Hence  $\mathcal{L}$  is maximal independent.  
 ( $\Leftarrow$ ) Let  $\mathcal{L}$  be maximal independent and  $\mathbf{v} \in V$ . The list  $(\mathbf{v}_1, \dots, \mathbf{v}_n, \mathbf{v})$  is dependent while  $\mathcal{L}$  is independent, so  $\mathbf{v} \in \text{span}(\mathcal{L})$  by Lemma 1.5.3. Hence  $\mathcal{L}$  spans  $V$ , and  $\mathcal{L}$  is a basis.
- (b) ( $\Rightarrow$ ) Let  $\mathcal{L}$  be a basis, and suppose that deleting  $\mathbf{v}_j$  gives a list that still spans  $V$ . Then  $\mathbf{v}_j = \sum_{i \neq j} c_i \mathbf{v}_i$  for some scalars, so  $\sum_{i \neq j} c_i \mathbf{v}_i + (-1)\mathbf{v}_j = \mathbf{0}$ , a linear relation with the coefficient  $-1 \neq 0$ . This contradicts independence. Hence  $\mathcal{L}$  is minimal spanning.  
 ( $\Leftarrow$ ) Let  $\mathcal{L}$  be minimal spanning. By Theorem 1.5.4, some sub-list  $S$  of  $\mathcal{L}$  is a basis of  $V$ . Suppose  $S \neq \mathcal{L}$ . Then  $S$  omits some entry  $\mathbf{v}_j$ , so every entry of  $S$  is an entry of the list  $\mathcal{L}'$  obtained by deleting  $\mathbf{v}_j$ . By Theorem 1.3.1,  $\text{span}(\mathcal{L}')$  contains  $\text{span}(S) = V$ , so  $\mathcal{L}'$  spans  $V$ , contradicting minimality. Hence  $S = \mathcal{L}$ , and  $\mathcal{L}$  is a basis. ■

“Maximal” means “cannot be enlarged”, which is different from “longest possible”. In a finite-dimensional space the two agree: by Theorem 1.6.3 a longest independent list has length  $\dim V$ , and by Theorem 1.6.4 every independent list of that length is a basis.

## 1.6.8 Dimensions of subspaces

A subspace should be no bigger than the space containing it. Proving this needs care, because of a tempting circular argument.

### Theorem 1.6.7 — Dimension of a Subspace

Let  $V$  be a finite-dimensional vector space and  $U$  a subspace of  $V$ . Then  $U$  is finite-dimensional, and  $\dim U \leq \dim V$ .

*Idea.* The tempting proof is: “take a basis of  $U$ ; it is independent in  $V$ , so its length is at most  $\dim V$ ”. But we do not yet know that  $U$  has a basis. Corollary 1.5.5 needs a finite spanning list of  $U$ , and that is exactly what we are trying to prove. So we build from below instead. Independent lists inside  $U$  are independent in  $V$ , so their lengths are bounded by  $\dim V$ . Take one of the largest possible length. If it did not span  $U$ , we could append a vector of  $U$  outside its span and get a longer one.

*Proof.* Let  $n = \dim V$ . Independence of a list depends only on its vectors and the operations of  $V$ , which  $U$  shares, so every independent list in  $U$  is an independent list in  $V$ , and has length at most  $n$  by Theorem 1.6.3. The empty list is independent in  $U$ . Hence the set of lengths of independent lists in  $U$  is a non-empty set of integers in  $\{0, 1, \dots, n\}$ , and it has a largest element  $k \leq n$ . Choose an independent list  $(\mathbf{u}_1, \dots, \mathbf{u}_k)$

in  $U$  of this length.

We claim  $\text{span}(\mathbf{u}_1, \dots, \mathbf{u}_k) = U$ . Since  $U$  is a subspace containing every  $\mathbf{u}_i$ , [Theorem 1.3.1](#) gives  $\text{span}(\mathbf{u}_1, \dots, \mathbf{u}_k) \subseteq U$ . Conversely, let  $\mathbf{u} \in U$ , and suppose  $\mathbf{u} \notin \text{span}(\mathbf{u}_1, \dots, \mathbf{u}_k)$ . By [Lemma 1.5.3](#),  $(\mathbf{u}_1, \dots, \mathbf{u}_k, \mathbf{u})$  is an independent list in  $U$  of length  $k + 1$ , contradicting the maximality of  $k$ . Hence  $\mathbf{u} \in \text{span}(\mathbf{u}_1, \dots, \mathbf{u}_k)$ .

Therefore  $(\mathbf{u}_1, \dots, \mathbf{u}_k)$  spans  $U$ , so  $U$  is finite-dimensional ([Definition 1.3.4](#)), and the list is a basis of  $U$ . Hence  $\dim U = k \leq n = \dim V$ . ■

When the dimensions are equal, there is no room left.

### Theorem 1.6.8 — Equal Dimension Forces Equality

Let  $V$  be a finite-dimensional vector space and  $U$  a subspace of  $V$ . If  $\dim U = \dim V$ , then  $U = V$ .

*Proof.* Let  $n = \dim U = \dim V$ ;  $U$  is finite-dimensional by [Theorem 1.6.7](#). Let  $(\mathbf{u}_1, \dots, \mathbf{u}_n)$  be a basis of  $U$  ([Corollary 1.5.5](#)). It is an independent list in  $V$  of length  $n = \dim V$ , so it is a basis of  $V$  by [Theorem 1.6.4](#) (a). Hence  $V = \text{span}(\mathbf{u}_1, \dots, \mathbf{u}_n) \subseteq U$ , where the inclusion is [Theorem 1.3.1](#) (3) applied to the subspace  $U$ . Since also  $U \subseteq V$ , we get  $U = V$ . ■

This is the dimension version of “count instead of check”: to prove two subspaces are equal, prove **one** inclusion and compare dimensions. It is usually much less work than proving both inclusions.

*Quick check.* Let  $U$  be a subspace of  $\mathbb{R}[x]_{\leq 2}$  that contains  $1$ ,  $1 + x$  and  $(1 + x)^2$ . Must  $U = \mathbb{R}[x]_{\leq 2}$ ?

*Solution.* Yes. By [Example \(A Basis by Counting\)](#),  $(1, 1 + x, (1 + x)^2)$  is independent, and it lies in  $U$ , so  $\dim U \geq 3$  by [Theorem 1.6.3](#) applied in  $U$  (which is finite-dimensional by [Theorem 1.6.7](#)). Also  $\dim U \leq \dim \mathbb{R}[x]_{\leq 2} = 3$ . So  $\dim U = 3$ , and [Theorem 1.6.8](#) gives  $U = \mathbb{R}[x]_{\leq 2}$ .

*Warning.* Both hypotheses of [Theorem 1.6.8](#) matter. Containment is needed: the  $xy$ -plane and the  $xz$ -plane in  $\mathbb{R}^3$  both have dimension 2, but neither contains the other, and they are different. Without finite dimension it fails too. In  $F[x]$ , the subspace  $U$  of polynomials with constant coefficient 0 has the basis-like list  $x, x^2, x^3, \dots$ , which matches  $1, x, x^2, \dots$  one for one; yet  $1 \notin U$ . Later in this chapter we make this precise for infinite-dimensional spaces.

## 1.6.9 Exercises

### 1.6.9.1 A. Check your understanding

**Exercise A1.**

- (a) State the Steinitz Exchange Theorem.
- (b) True or false:  $\dim \mathbb{R}[x]_{\leq 3} = 3$ . Justify your answer.
- (c) True or false: in a vector space of dimension 3, every spanning list of length 3 is a basis. Justify your answer.
- (d) True or false:  $M_2(\mathbb{R})$  contains a linearly independent list of length 5. Justify your answer.
- (e) True or false: if  $U$  and  $W$  are subspaces of  $\mathbb{R}^4$  with  $\dim U = \dim W = 2$ , then  $U = W$ . Justify your answer.
- (f) What are  $\dim_{\mathbb{C}} \mathbb{C}^2$  and  $\dim_{\mathbb{R}} \mathbb{C}^2$ ?

*Solution.*

- (a) See [Theorem 1.6.1](#): if  $(\mathbf{u}_1, \dots, \mathbf{u}_m)$  is independent and  $(\mathbf{w}_1, \dots, \mathbf{w}_n)$  spans  $V$ , then  $m \leq n$ , and after renumbering the  $\mathbf{w}$ 's,  $(\mathbf{u}_1, \dots, \mathbf{u}_m, \mathbf{w}_{m+1}, \dots, \mathbf{w}_n)$  spans  $V$ .
- (b) False. The basis  $(1, x, x^2, x^3)$  has length 4, so  $\dim \mathbb{R}[x]_{\leq 3} = 4$ .
- (c) True, by [Theorem 1.6.4](#) (b).
- (d) False.  $\dim M_2(\mathbb{R}) = 4$ , and by [Theorem 1.6.3](#) (a) every list of length  $5 > 4$  is dependent.
- (e) False. The subspaces  $\text{span}(\mathbf{e}_1, \mathbf{e}_2)$  and  $\text{span}(\mathbf{e}_3, \mathbf{e}_4)$  both have dimension 2, but  $\mathbf{e}_1$  lies in the first and not in the second. [Theorem 1.6.8](#) needs one subspace to be contained in the other.
- (f)  $\dim_{\mathbb{C}} \mathbb{C}^2 = 2$ , with basis  $(\mathbf{e}_1, \mathbf{e}_2)$ . Over  $\mathbb{R}$ , the list  $(\mathbf{e}_1, i\mathbf{e}_1, \mathbf{e}_2, i\mathbf{e}_2)$  is a basis: every  $(a + bi, c + di)$  with  $a, b, c, d \in \mathbb{R}$  equals  $a\mathbf{e}_1 + b(i\mathbf{e}_1) + c\mathbf{e}_2 + d(i\mathbf{e}_2)$ , and this is  $\mathbf{0}$  only if  $a + bi = 0$  and  $c + di = 0$ , that is,  $a = b = c = d = 0$ . So  $\dim_{\mathbb{R}} \mathbb{C}^2 = 4$ .

**1.6.9.2 B. Practice****Exercise B1: Dimension and a Basis of a Subspace.**

Find a basis and the dimension of each subspace.

- (a)  $U_1 = \{(x_1, x_2, x_3, x_4) \in \mathbb{R}^4 : x_1 + x_2 = 0 \text{ and } x_3 - x_4 = 0\}$ .
- (b)  $U_2 = \{\mathbf{A} \in M_3(\mathbb{R}) : \text{tr } \mathbf{A} = 0\}$ .
- (c)  $U_3 = \{p \in \mathbb{R}[x]_{\leq 3} : p(1) = p(-1) = 0\}$ .

*Solution.*

- (a) A vector lies in  $U_1$  exactly when  $x_2 = -x_1$  and  $x_3 = x_4$ , that is, when it equals

$$(x_1, -x_1, x_4, x_4) = x_1(1, -1, 0, 0) + x_4(0, 0, 1, 1).$$

So  $((1, -1, 0, 0), (0, 0, 1, 1))$  spans  $U_1$ . It is independent:  $a(1, -1, 0, 0) + b(0, 0, 1, 1) = (a, -a, b, b) = \mathbf{0}$  forces  $a = b = 0$ . Hence it is a basis, and  $\dim U_1 = 2$ .

(b) Write  $\mathbf{A} = (a_{ij})$ . The condition  $a_{11} + a_{22} + a_{33} = 0$  says  $a_{33} = -a_{11} - a_{22}$ , so

$$\mathbf{A} = \sum_{i \neq j} a_{ij} \mathbf{E}_{ij} + a_{11}(\mathbf{E}_{11} - \mathbf{E}_{33}) + a_{22}(\mathbf{E}_{22} - \mathbf{E}_{33}).$$

All eight matrices on the right have trace 0, so the list of the six  $\mathbf{E}_{ij}$  with  $i \neq j$ , followed by  $\mathbf{E}_{11} - \mathbf{E}_{33}$  and  $\mathbf{E}_{22} - \mathbf{E}_{33}$ , spans  $U_2$ . For independence, let  $\sum_{i \neq j} c_{ij} \mathbf{E}_{ij} + d_1(\mathbf{E}_{11} - \mathbf{E}_{33}) + d_2(\mathbf{E}_{22} - \mathbf{E}_{33}) = 0$ . The entry  $(i, j)$  with  $i \neq j$  of the left side is  $c_{ij}$ , the entry  $(1, 1)$  is  $d_1$ , and the entry  $(2, 2)$  is  $d_2$ ; all must be 0. Hence the list is a basis, and  $\dim U_2 = 8$ .

(c) Let  $p = a_0 + a_1x + a_2x^2 + a_3x^3$ . Then  $p(1) = a_0 + a_1 + a_2 + a_3$  and  $p(-1) = a_0 - a_1 + a_2 - a_3$ . Adding and subtracting,  $p \in U_3$  exactly when  $a_0 + a_2 = 0$  and  $a_1 + a_3 = 0$ . Hence  $p \in U_3$  exactly when

$$p = -a_2 - a_3x + a_2x^2 + a_3x^3 = a_2(x^2 - 1) + a_3(x^3 - x).$$

So  $(x^2 - 1, x^3 - x)$  spans  $U_3$ . It is independent:  $a(x^2 - 1) + b(x^3 - x) = 0$  has  $x^3$ -coefficient  $b$  and  $x^2$ -coefficient  $a$ , so  $a = b = 0$ . Hence it is a basis, and  $\dim U_3 = 2$ .

### Exercise B2: Extending to a Basis of $\mathbb{R}^4$ .

Show that  $((1, 1, 0, 0), (0, 1, 1, 0))$  is linearly independent, and extend it to a basis of  $\mathbb{R}^4$  using vectors from the standard basis.

*Solution.* Let  $\mathbf{u}_1 = (1, 1, 0, 0)$  and  $\mathbf{u}_2 = (0, 1, 1, 0)$ . If  $a\mathbf{u}_1 + b\mathbf{u}_2 = (a, a + b, b, 0) = \mathbf{0}$ , then  $a = b = 0$ , so the list is independent. By [Theorem 1.6.5](#), two standard basis vectors complete it; we sift  $(\mathbf{u}_1, \mathbf{u}_2, \mathbf{e}_1, \mathbf{e}_2, \mathbf{e}_3, \mathbf{e}_4)$  to find which.

- $\mathbf{u}_1 \neq \mathbf{0}$ : keep.  $\mathbf{u}_2 \notin \text{span}(\mathbf{u}_1)$ , since a multiple  $(a, a, 0, 0)$  of  $\mathbf{u}_1$  has third entry 0: keep.
- $\mathbf{e}_1$ : if  $(a, a + b, b, 0) = (1, 0, 0, 0)$ , then  $a = 1$  and  $b = 0$ , but then  $a + b = 1 \neq 0$ . So  $\mathbf{e}_1 \notin \text{span}(\mathbf{u}_1, \mathbf{u}_2)$ : keep.
- $\mathbf{e}_2 = \mathbf{u}_1 - \mathbf{e}_1$ : discard.
- $\mathbf{e}_3 = -\mathbf{u}_1 + \mathbf{u}_2 + \mathbf{e}_1$ , since  $-(1, 1, 0, 0) + (0, 1, 1, 0) + (1, 0, 0, 0) = (0, 0, 1, 0)$ : discard.
- $\mathbf{e}_4$ : every combination of  $\mathbf{u}_1, \mathbf{u}_2, \mathbf{e}_1$  has fourth entry 0: keep.

The sifted list  $(\mathbf{u}_1, \mathbf{u}_2, \mathbf{e}_1, \mathbf{e}_4)$  is independent by [Theorem 1.5.4](#). It has  $4 = \dim \mathbb{R}^4$  entries, so it is a basis of  $\mathbb{R}^4$  by [Theorem 1.6.4](#) (a).

### Exercise B3: Count Instead of Check.

Show that  $(1, x - 1, (x - 1)^2)$  is a basis of  $\mathbb{R}[x]_{\leq 2}$ . Hence find the coordinate vector of  $x^2$  with respect to this basis.

*Solution.* Let  $a + b(x - 1) + c(x - 1)^2 = 0$ . Expanding,  $(a - b + c) + (b - 2c)x + cx^2 = 0$ . Comparing coefficients,  $c = 0$ , then  $b = 2c = 0$ , then  $a = b - c = 0$ . So the list is independent, with  $3 = \dim \mathbb{R}[x]_{\leq 2}$  elements, and it is a basis by [Theorem 1.6.4](#) (a).

Writing  $x = (x - 1) + 1$ ,

$$x^2 = ((x - 1) + 1)^2 = 1 + 2(x - 1) + (x - 1)^2.$$

Hence the coordinate vector of  $x^2$  is  $(1, 2, 1)$ .

### 1.6.9.3 C. Going deeper

#### Exercise C1: Symmetric and Skew-Symmetric Matrices.

Let  $n \geq 1$ . For a field  $F$ , let  $U_+$  and  $U_-$  be the subspaces of symmetric and of skew-symmetric matrices in  $M_n(F)$  (Definition 0.9.11):  $U_+ = \{\mathbf{A} : \mathbf{A}^\top = \mathbf{A}\}$  and  $U_- = \{\mathbf{A} : \mathbf{A}^\top = -\mathbf{A}\}$ .

- (a) For  $F = \mathbb{R}$ , prove that  $\dim U_+ = \frac{n(n+1)}{2}$ .
- (b) For  $F = \mathbb{R}$ , prove that  $\dim U_- = \frac{n(n-1)}{2}$ .
- (c) For  $F = \mathbb{F}_2$ , show that  $U_- = U_+$ , and find its dimension. Which step of (b) fails?

*Solution.*

- (a) Consider the list  $\mathcal{S}$  consisting of  $\mathbf{E}_{ii}$  for  $1 \leq i \leq n$  and  $\mathbf{E}_{ij} + \mathbf{E}_{ji}$  for  $1 \leq i < j \leq n$ . Each is symmetric. If  $\mathbf{A} = (a_{ij})$  is symmetric, then  $a_{ji} = a_{ij}$ , so

$$\mathbf{A} = \sum_i a_{ii} \mathbf{E}_{ii} + \sum_{i < j} a_{ij} (\mathbf{E}_{ij} + \mathbf{E}_{ji}),$$

because both sides have  $a_{ij}$  in entry  $(i, j)$  and  $a_{ij} = a_{ji}$  in entry  $(j, i)$ . So  $\mathcal{S}$  spans  $U_+$ . If  $\sum_i c_{ii} \mathbf{E}_{ii} + \sum_{i < j} c_{ij} (\mathbf{E}_{ij} + \mathbf{E}_{ji}) = 0$ , then for  $i \leq j$  the entry  $(i, j)$  of the left side is  $c_{ij}$ , so every coefficient is 0. Hence  $\mathcal{S}$  is a basis, of length  $n + \frac{n(n-1)}{2} = \frac{n(n+1)}{2}$ .

- (b) If  $\mathbf{A}^\top = -\mathbf{A}$ , then comparing diagonal entries gives  $a_{ii} = -a_{ii}$ , so  $2a_{ii} = 0$ , and  $a_{ii} = 0$  because  $2 \neq 0$  in  $\mathbb{R}$ . Off the diagonal,  $a_{ji} = -a_{ij}$ . Hence

$$\mathbf{A} = \sum_{i < j} a_{ij} (\mathbf{E}_{ij} - \mathbf{E}_{ji}).$$

Each  $\mathbf{E}_{ij} - \mathbf{E}_{ji}$  is skew-symmetric, so these matrices span  $U_-$ , and as in (a) the entry  $(i, j)$  with  $i < j$  of a combination is its coefficient, so they are independent. There are  $\frac{n(n-1)}{2}$  pairs  $i < j$ , so  $\dim U_- = \frac{n(n-1)}{2}$ .

- (c) In  $\mathbb{F}_2$ ,  $-1 = 1$ , so  $-\mathbf{A} = \mathbf{A}$  for every matrix  $\mathbf{A}$ , and the conditions  $\mathbf{A}^\top = -\mathbf{A}$  and  $\mathbf{A}^\top = \mathbf{A}$  coincide. Hence  $U_- = U_+$  over  $\mathbb{F}_2$ , which has dimension  $\frac{n(n+1)}{2}$  by the argument of (a), which works over any field. The step of (b) that fails is “ $2a_{ii} = 0$  implies  $a_{ii} = 0$ ”, which divides by  $2 = 0$ . For example,  $\mathbf{I}_n \in U_-$  over  $\mathbb{F}_2$ , although its diagonal entries are not 0.

#### Exercise C2: A complex space viewed over the reals.

Let  $V$  be a vector space over  $\mathbb{C}$  with  $\dim_{\mathbb{C}} V = n$ . Restricting the scalars to real numbers makes  $V$  a vector space over  $\mathbb{R}$ . Prove that  $\dim_{\mathbb{R}} V = 2n$ .



*Hint: start from a basis  $(\mathbf{v}_1, \dots, \mathbf{v}_n)$  over  $\mathbb{C}$ , and consider the vectors  $i\mathbf{v}_k$ .*

*Solution.* Let  $(\mathbf{v}_1, \dots, \mathbf{v}_n)$  be a basis of  $V$  over  $\mathbb{C}$ . We show that  $\mathcal{B} = (\mathbf{v}_1, i\mathbf{v}_1, \dots, \mathbf{v}_n, i\mathbf{v}_n)$  is a basis of  $V$  over  $\mathbb{R}$ .

*Spanning.* Let  $\mathbf{v} \in V$ . Then  $\mathbf{v} = \sum_k z_k \mathbf{v}_k$  with  $z_k \in \mathbb{C}$ . Write  $z_k = a_k + b_k i$  with  $a_k, b_k \in \mathbb{R}$ . By the distributive and associative axioms of scalar multiplication,  $z_k \mathbf{v}_k = a_k \mathbf{v}_k + (b_k i) \mathbf{v}_k = a_k \mathbf{v}_k + b_k (i \mathbf{v}_k)$ . Hence  $\mathbf{v} = \sum_k (a_k \mathbf{v}_k + b_k (i \mathbf{v}_k))$ , a real combination of  $\mathcal{B}$ .

*Independence.* Let  $\sum_k (a_k \mathbf{v}_k + b_k (i \mathbf{v}_k)) = \mathbf{0}$  with  $a_k, b_k \in \mathbb{R}$ . Reading the same axioms backwards,  $\sum_k (a_k + b_k i) \mathbf{v}_k = \mathbf{0}$ . Since  $(\mathbf{v}_1, \dots, \mathbf{v}_n)$  is independent over  $\mathbb{C}$ , every  $a_k + b_k i = 0$ , so  $a_k = b_k = 0$ .

Hence  $\mathcal{B}$  is a basis over  $\mathbb{R}$  of length  $2n$ , and  $\dim_{\mathbb{R}} V = 2n$ .

### Exercise C3: Counting Elements over a Finite Field.

Let  $p$  be a prime.

- (a) Let  $V$  be a vector space over  $\mathbb{F}_p$  with  $\dim V = k$ . Prove that  $V$  has exactly  $p^k$  elements.
- (b) Deduce that no vector space over  $\mathbb{F}_2$  has exactly 6 elements.
- (c) Let  $K$  be a finite field that contains  $\mathbb{F}_p$ , with the same addition and multiplication. Explain why  $K$  is a finite-dimensional vector space over  $\mathbb{F}_p$ , and deduce that  $K$  has  $p^k$  elements for some integer  $k \geq 1$ .

*Hint: for (a), use coordinates.*

*Solution.*

- (a) Let  $\mathcal{B}$  be a basis of  $V$ , of length  $k$ . By Theorem 1.5.2,  $\mathbf{v} \mapsto [\mathbf{v}]_{\mathcal{B}}$  is a bijection from  $V$  to  $\mathbb{F}_p^k$ : every column is the coordinate vector of exactly one vector. So  $V$  has as many elements as  $\mathbb{F}_p^k$ . A column in  $\mathbb{F}_p^k$  is a choice of  $k$  entries, each from the  $p$  elements of  $\mathbb{F}_p$ , so there are  $p^k$  columns. Hence  $V$  has  $p^k$  elements.
- (b) A vector space over  $\mathbb{F}_2$  with finitely many elements is spanned by the finite list of all its elements, so it is finite-dimensional, of some dimension  $k$ . By (a) it has  $2^k$  elements. Since 6 is not a power of 2, no such space has 6 elements.
- (c) Define scalar multiplication  $\mathbb{F}_p \times K \rightarrow K$  by the multiplication of  $K$ . The vector space axioms are then instances of the field axioms of  $K$ : addition in  $K$  is commutative and associative with zero and negatives, multiplication is associative,  $1 \cdot a = a$ , and multiplication distributes over addition. So  $K$  is a vector space over  $\mathbb{F}_p$ . It is spanned by the finite list of all its elements, so it is finite-dimensional; let  $k = \dim_{\mathbb{F}_p} K$ . By (a),  $K$  has  $p^k$  elements. Since  $K$  contains  $0 \neq 1$ , it is not the zero space, so  $k \geq 1$ .

## 1.7 Sums and Direct Sums

We can now measure a single subspace: take a basis and count. This section asks how two or more subspaces combine. The intersection of two subspaces is again a subspace, but the union usually is not, so we need a better way to join them. The answer is the sum  $U + W$ . We then prove the dimension formula for sums and single out the best case, the direct sum, in which every vector splits into pieces in exactly one way.

### 1.7.1 The sum of subspaces

Recall from [Theorem 1.2.2](#) that  $U \cap W$  is a subspace whenever  $U$  and  $W$  are. It is the largest subspace inside both. The natural companion would be the smallest subspace containing both, and the first guess is the union  $U \cup W$ .

That guess fails. In  $\mathbb{R}^2$ , let  $U$  be the  $x$ -axis and  $W$  the  $y$ -axis. Then  $(1, 0) \in U$  and  $(0, 1) \in W$ , but their sum  $(1, 1)$  lies on neither axis, so  $U \cup W$  is not closed under addition ([Example \(The Union of the Two Axes\)](#)). In fact [Proposition 1.2.3](#) shows that  $U \cup W$  is a subspace only in the dull case  $U \subseteq W$  or  $W \subseteq U$ .

The failure tells us what is missing. Any subspace containing  $U$  and  $W$  must contain  $\mathbf{u} + \mathbf{w}$  for every  $\mathbf{u} \in U$  and  $\mathbf{w} \in W$ , because subspaces are closed under addition. So we put all these sums in from the start.

*The sum of two subspaces is everything you can reach by adding one vector from each.*

#### Definition 1.7.1 — Sum of Subspaces

Let  $V$  be a vector space over  $F$ , and let  $U$  and  $W$  be subspaces of  $V$ . The **sum** of  $U$  and  $W$  is the subset

$$U + W := \{\mathbf{u} + \mathbf{w} : \mathbf{u} \in U \text{ and } \mathbf{w} \in W\}.$$

More generally, for subspaces  $U_1, \dots, U_k$  of  $V$  (with  $k \geq 1$ ), their **sum** is

$$U_1 + \dots + U_k := \{\mathbf{u}_1 + \dots + \mathbf{u}_k : \mathbf{u}_i \in U_i \text{ for each } i = 1, \dots, k\}.$$

In words: a vector  $\mathbf{v}$  lies in  $U + W$  when **there exist** some  $\mathbf{u} \in U$  and some  $\mathbf{w} \in W$  with  $\mathbf{v} = \mathbf{u} + \mathbf{w}$ . The definition asks for one such pair. It does not say the pair is unique, and in general it is not. For  $k$  subspaces we pick **one** vector from **each**  $U_i$  and add them. We also write  $\sum_{i=1}^k U_i$ , and for  $k = 1$  the sum is just  $U_1$ .

Here are some first examples.

- **The two axes.** In  $\mathbb{R}^2$ , with  $U$  the  $x$ -axis and  $W$  the  $y$ -axis, every  $(a, b)$  equals  $(a, 0) + (0, b)$ , so  $U + W = \mathbb{R}^2$ . The union is two lines; the sum is the whole plane.
- **Two planes in  $\mathbb{R}^3$ .** Let  $P_{xy} = \{(x, y, 0)\}$  and  $P_{yz} = \{(0, y, z)\}$ . Every  $(a, b, c) = (a, b, 0) + (0, 0, c)$  is a sum of a vector in  $P_{xy}$  and one in  $P_{yz}$ , so  $P_{xy} + P_{yz} = \mathbb{R}^3$ .
- **Polynomials.** In  $F[x]_{\leq 2}$ , let  $U$  be the constant polynomials and  $W = \{bx + cx^2 : b, c \in F\}$ . Then  $a + bx + cx^2 = a + (bx + cx^2)$ , so  $U + W = F[x]_{\leq 2}$ .
- **Matrices.** In  $M_2(F)$ , the upper triangular matrices plus the lower triangular matrices

give everything:

$$\begin{pmatrix} a & b \\ c & d \end{pmatrix} = \begin{pmatrix} a & b \\ 0 & d \end{pmatrix} + \begin{pmatrix} 0 & 0 \\ c & 0 \end{pmatrix}.$$

- **Degenerate cases.** For any subspace  $U$ , we have  $U + \{\mathbf{0}\} = U$  and  $U + U = U$ . In both, “ $\supseteq$ ” holds because  $\mathbf{u} = \mathbf{u} + \mathbf{0}$ , and “ $\subseteq$ ” holds because  $U$  is closed under addition. So adding a subspace to itself produces nothing new: a sum is not a count.

**Non-example by minimal change.** The definition makes sense for any two subsets  $S, T \subseteq V$ , but the result need not be a subspace. Take  $S = \{(1, 0)\}$  and  $T = \{(0, 1)\}$  in  $\mathbb{R}^2$ . Then  $S + T = \{(1, 1)\}$ , which does not contain  $\mathbf{0}$ . What changed is that  $S$  and  $T$  are not subspaces: the hypothesis “subspace” is exactly what makes the next theorem work.

*Warning.* The sum  $U + W$  is **not** the union  $U \cup W$ . A vector of  $U + W$  need not lie in  $U$  or in  $W$ :  $(1, 1)$  lies in the sum of the two axes but on neither axis. Conversely,  $U + W$  is usually far bigger than  $U \cup W$ .

The sum does what we built it to do.

### Theorem 1.7.1 — Sum of Subspaces Theorem

Let  $U_1, \dots, U_k$  be subspaces of a vector space  $V$ . Then  $U_1 + \dots + U_k$  is a subspace of  $V$  that contains each  $U_i$ . Moreover, if  $X$  is any subspace of  $V$  containing every  $U_i$ , then  $U_1 + \dots + U_k \subseteq X$ . In other words,  $U_1 + \dots + U_k$  is the **smallest** subspace of  $V$  containing  $U_1, \dots, U_k$ .

*Proof.* Write  $S = U_1 + \dots + U_k$ . We apply the Subspace Test (Theorem 1.2.1).

- (1) Each  $U_i$  contains  $\mathbf{0}$ , so  $\mathbf{0} = \mathbf{0} + \dots + \mathbf{0} \in S$ .
- (2) Let  $\mathbf{u}_1 + \dots + \mathbf{u}_k$  and  $\mathbf{u}'_1 + \dots + \mathbf{u}'_k$  be in  $S$ , with  $\mathbf{u}_i, \mathbf{u}'_i \in U_i$ . Regrouping with (VS1) and (VS2),

$$(\mathbf{u}_1 + \dots + \mathbf{u}_k) + (\mathbf{u}'_1 + \dots + \mathbf{u}'_k) = (\mathbf{u}_1 + \mathbf{u}'_1) + \dots + (\mathbf{u}_k + \mathbf{u}'_k),$$

and  $\mathbf{u}_i + \mathbf{u}'_i \in U_i$  because  $U_i$  is closed under addition. Hence  $S$  is closed under addition.

- (3) Let  $c \in F$ . By the distributive axiom,  $c(\mathbf{u}_1 + \dots + \mathbf{u}_k) = c\mathbf{u}_1 + \dots + c\mathbf{u}_k$ , and  $c\mathbf{u}_i \in U_i$  because  $U_i$  is closed under scaling. Hence  $S$  is closed under scalar multiplication.

Therefore  $S$  is a subspace. It contains  $U_j$ : for  $\mathbf{u} \in U_j$ , take  $\mathbf{u}_j = \mathbf{u}$  and  $\mathbf{u}_i = \mathbf{0} \in U_i$  for  $i \neq j$ . Finally, let  $X$  be a subspace containing every  $U_i$ . An element  $\mathbf{u}_1 + \dots + \mathbf{u}_k$  of  $S$  is a sum of vectors  $\mathbf{u}_i \in U_i \subseteq X$ , so it lies in  $X$  since  $X$  is closed under addition. This shows  $S \subseteq X$ . ■

So the sum plays the role that the union could not. It is the subspace analogue of span, and the two notions fit together exactly. Recall from Theorem 1.3.1 that  $\text{span}(S)$  is the smallest subspace containing  $S$ .

### Proposition 1.7.2 — Sum of Spans

Let  $L_1, \dots, L_k$  be finite lists of vectors in  $V$ , and let  $L$  be the list obtained by writing  $L_1, \dots, L_k$  one after another. Then

$$\text{span}(L_1) + \cdots + \text{span}(L_k) = \text{span}(L).$$

*Proof.* ( $\subseteq$ ) The subspace  $\text{span}(L)$  contains every vector of each  $L_i$ , so by Theorem 1.3.1 it contains each  $\text{span}(L_i)$ . By Theorem 1.7.1 it therefore contains  $\text{span}(L_1) + \cdots + \text{span}(L_k)$ .

( $\supseteq$ ) The sum  $\text{span}(L_1) + \cdots + \text{span}(L_k)$  is a subspace by Theorem 1.7.1, and it contains each  $\text{span}(L_i)$ , hence every vector of  $L$ . By Theorem 1.3.1 it contains  $\text{span}(L)$ . ■

In practice this means: **to find a spanning list of a sum, concatenate spanning lists of the summands.** That list may be dependent, and the dimension formula below tells us by how much.

*Quick check.* In  $\mathbb{R}^3$ , what is  $\text{span}((1, 0, 0)) + \text{span}((0, 1, 0))$ ? Is it equal to  $\text{span}((1, 0, 0)) \cup \text{span}((0, 1, 0))$ ?

*Solution.* By Proposition 1.7.2 the sum is  $\text{span}((1, 0, 0), (0, 1, 0)) = \{(a, b, 0) : a, b \in \mathbb{R}\}$ , the  $xy$ -plane. It is not the union:  $(1, 1, 0)$  lies in the plane but on neither axis.

## 1.7.2 The dimension formula

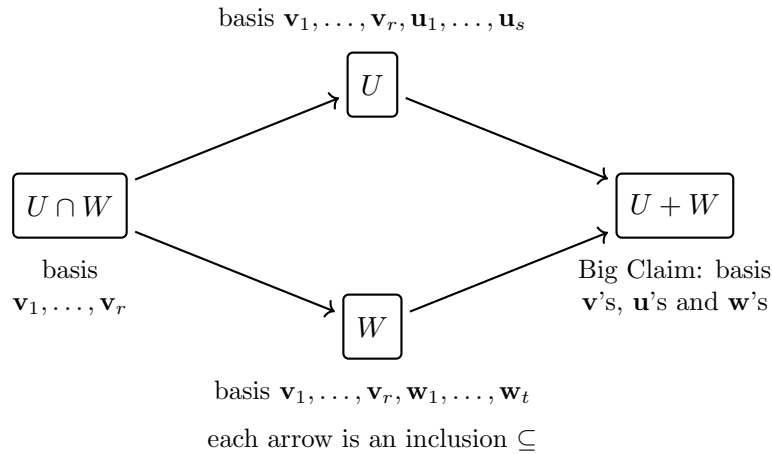
If  $U$  and  $W$  are finite-dimensional, how big is  $U + W$ ? Concatenating a basis of  $U$  and a basis of  $W$  gives a spanning list of  $U + W$  of length  $\dim U + \dim W$  (Proposition 1.7.2). That count is too large when  $U$  and  $W$  overlap. For the two planes  $P_{xy}$  and  $P_{yz}$  in  $\mathbb{R}^3$ , it would give  $2 + 2 = 4$ , but  $P_{xy} + P_{yz} = \mathbb{R}^3$  has dimension 3. The overlap is the  $y$ -axis, of dimension 1, and it has been counted twice. The theorem says this is the whole story.

### Theorem 1.7.3 — Dimension Formula for Sums

Let  $U$  and  $W$  be finite-dimensional subspaces of a vector space  $V$ . Then  $U + W$  is finite-dimensional, and

$$\dim(U + W) = \dim U + \dim W - \dim(U \cap W).$$

The picture behind the proof is the inclusion diagram. The smallest space,  $U \cap W$ , sits inside both  $U$  and  $W$ , and both sit inside  $U + W$ . We build bases from the bottom up.



*Idea.* This is a dimension formula, so we start from the **smallest** space and extend outward.

① Take a basis  $(\mathbf{v}_1, \dots, \mathbf{v}_r)$  of  $U \cap W$ . ② Extend it to a basis of  $U$  by adding  $\mathbf{u}_1, \dots, \mathbf{u}_s$ , and separately to a basis of  $W$  by adding  $\mathbf{w}_1, \dots, \mathbf{w}_t$ . ③ **Big Claim:** the combined list of all  $r + s + t$  vectors is a basis of  $U + W$ . Spanning is easy. For independence, move the  $\mathbf{w}$ -part to the other side: the common vector lies in  $U$  and in  $W$ , hence in  $U \cap W$ , where the  $\mathbf{v}$ 's are a basis. ④ Count:  $r + s + t = (r + s) + (r + t) - r$ .

*Proof. Step 1: bases.* By [Theorem 1.2.2](#),  $U \cap W$  is a subspace of  $V$ ; it lies inside  $U$  and uses the same operations, so it is also a subspace of  $U$ . Since  $U$  is finite-dimensional, so is  $U \cap W$  ([Theorem 1.6.7](#)), and it has a basis  $(\mathbf{v}_1, \dots, \mathbf{v}_r)$ , where  $r = \dim(U \cap W)$  ([Corollary 1.5.5](#)). This list is independent and lies in  $U$ , so by the Basis Extension Theorem ([Theorem 1.6.5](#)) there are vectors  $\mathbf{u}_1, \dots, \mathbf{u}_s \in U$  such that  $(\mathbf{v}_1, \dots, \mathbf{v}_r, \mathbf{u}_1, \dots, \mathbf{u}_s)$  is a basis of  $U$ . In the same way, since the list also lies in  $W$ , there are  $\mathbf{w}_1, \dots, \mathbf{w}_t \in W$  such that  $(\mathbf{v}_1, \dots, \mathbf{v}_r, \mathbf{w}_1, \dots, \mathbf{w}_t)$  is a basis of  $W$ . Thus  $\dim U = r + s$  and  $\dim W = r + t$ .

**Step 2: the Big Claim.**

*Claim.* The list  $\mathcal{B} = (\mathbf{v}_1, \dots, \mathbf{v}_r, \mathbf{u}_1, \dots, \mathbf{u}_s, \mathbf{w}_1, \dots, \mathbf{w}_t)$  is a basis of  $U + W$ .

*Proof. Spanning.* By [Proposition 1.7.2](#),  $U + W = \text{span}(\mathbf{v}_1, \dots, \mathbf{v}_r, \mathbf{u}_1, \dots, \mathbf{u}_s) + \text{span}(\mathbf{v}_1, \dots, \mathbf{v}_r, \mathbf{w}_1, \dots, \mathbf{w}_t)$  is the span of the concatenated list. That list contains each  $\mathbf{v}_i$  twice; removing the second copies does not change the span, since every removed vector already lies in the span of what remains ([Theorem 1.3.5](#)). Hence  $\mathcal{B}$  spans  $U + W$ .

*Independence.* Let  $a_i, b_j, c_k \in F$  satisfy

$$a_1\mathbf{v}_1 + \dots + a_r\mathbf{v}_r + b_1\mathbf{u}_1 + \dots + b_s\mathbf{u}_s + c_1\mathbf{w}_1 + \dots + c_t\mathbf{w}_t = \mathbf{0}.$$

Move the  $\mathbf{w}$ -part across and name the common vector:

$$\mathbf{x} := a_1\mathbf{v}_1 + \dots + a_r\mathbf{v}_r + b_1\mathbf{u}_1 + \dots + b_s\mathbf{u}_s = -c_1\mathbf{w}_1 - \dots - c_t\mathbf{w}_t.$$

The middle expression lies in  $U$ , because all the  $\mathbf{v}_i$  and  $\mathbf{u}_j$  do. The right-hand expression lies in  $W$ , because all the  $\mathbf{w}_k$  do. Hence  $\mathbf{x} \in U \cap W$ . Since  $(\mathbf{v}_1, \dots, \mathbf{v}_r)$  spans  $U \cap W$ , there are  $d_1, \dots, d_r \in F$  with  $\mathbf{x} = d_1\mathbf{v}_1 + \dots + d_r\mathbf{v}_r$ . Comparing with the right-hand expression for  $\mathbf{x}$ ,

$$d_1\mathbf{v}_1 + \dots + d_r\mathbf{v}_r + c_1\mathbf{w}_1 + \dots + c_t\mathbf{w}_t = \mathbf{0}.$$

This is a combination of the basis  $(\mathbf{v}_1, \dots, \mathbf{v}_r, \mathbf{w}_1, \dots, \mathbf{w}_t)$  of  $W$ , which is independent, so all  $d_i = 0$  and all  $c_k = 0$ . Putting  $c_1 = \dots = c_t = 0$  into the original equation leaves

$$a_1\mathbf{v}_1 + \dots + a_r\mathbf{v}_r + b_1\mathbf{u}_1 + \dots + b_s\mathbf{u}_s = \mathbf{0},$$

a combination of the basis  $(\mathbf{v}_1, \dots, \mathbf{v}_r, \mathbf{u}_1, \dots, \mathbf{u}_s)$  of  $U$ . By its independence, all  $a_i = 0$  and all  $b_j = 0$ . Hence  $\mathcal{B}$  is linearly independent ([Definition 1.4.1](#)). ■

**Step 3: count.** By the claim,  $U + W$  has a finite basis of length  $r + s + t$ , so it is finite-dimensional with  $\dim(U + W) = r + s + t$  ([Definition 1.6.1](#)). Therefore

$$\begin{aligned} \dim(U + W) &= r + s + t = (r + s) + (r + t) - r \\ &= \dim U + \dim W - \dim(U \cap W), \end{aligned}$$

as claimed. ■

The proof also works when some of  $r, s, t$  are 0: an empty list is independent and spans  $\{\mathbf{0}\}$ , and every step reads correctly with empty sums. The move “basis of the smallest space, extend, Big Claim, count” is worth learning as a unit. It proves Rank–Nullity in Chapter 3 with the same four blocks.

The formula is often used to find one of the four numbers from the other three. Here are two uses, one by counting and one by computing.

**Example. (Two planes in  $\mathbb{R}^3$ )**

Let  $U = \{(x, y, z) \in \mathbb{R}^3 : x + y + z = 0\}$  and  $W = \{(x, y, z) \in \mathbb{R}^3 : x = z\}$ . Show that  $U + W = \mathbb{R}^3$ , and find a basis of  $\mathbb{R}^3$  of the shape used in the proof.

*Solution.* Solving  $x + y + z = 0$  for  $y$ , every vector of  $U$  is  $(x, -x - z, z) = x(1, -1, 0) + z(0, -1, 1)$ . The two vectors are independent (look at the first and third entries), so  $\dim U = 2$ . Similarly every vector of  $W$  is  $(x, y, x) = x(1, 0, 1) + y(0, 1, 0)$ , and  $\dim W = 2$ .

A vector lies in  $U \cap W$  when  $x = z$  and  $x + y + z = 0$ , that is,  $y = -2x$ . So  $U \cap W = \{(x, -2x, x)\} = \text{span}((1, -2, 1))$ , which has dimension 1. By the dimension formula,

$$\dim(U + W) = 2 + 2 - 1 = 3 = \dim \mathbb{R}^3.$$

Since  $U + W \subseteq \mathbb{R}^3$  and the dimensions agree,  $U + W = \mathbb{R}^3$  by [Theorem 1.6.8](#).

Following the proof:  $(1, -2, 1)$  extends to a basis of  $U$  by  $(1, -1, 0) \in U$ . Indeed  $(1, -1, 0)$  is not a multiple  $(c, -2c, c)$  of  $(1, -2, 1)$ , since its first and third entries differ, so the list  $((1, -2, 1), (1, -1, 0))$  is independent by [Lemma 1.5.3](#); it has length  $2 = \dim U$ , so it is a basis of  $U$  ([Theorem 1.6.4](#)). It extends to a basis of  $W$  by  $(0, 1, 0) \in W$ , for the same reason: a multiple of  $(1, -2, 1)$  with first entry 0 is  $\mathbf{0}$ . The Big Claim says  $((1, -2, 1), (1, -1, 0), (0, 1, 0))$  is a basis of  $U + W = \mathbb{R}^3$ .

**Example. (Two planes in  $\mathbb{R}^4$  from spanning lists)**

Let  $U = \text{span}((1, 1, 0, 0), (0, 0, 1, 1))$  and  $W = \text{span}((1, 0, 1, 0), (0, 1, 0, 1))$  in  $\mathbb{R}^4$ . Find  $\dim(U \cap W)$  and  $\dim(U + W)$ , and hence a basis of  $U + W$ .

*Solution.* Each spanning list is independent: in  $a(1, 1, 0, 0) + b(0, 0, 1, 1) = (a, a, b, b) = \mathbf{0}$ , both  $a$  and  $b$  are 0, and similarly  $c(1, 0, 1, 0) + d(0, 1, 0, 1) = (c, d, c, d)$ . So  $\dim U = \dim W = 2$ .

A vector lies in  $U \cap W$  exactly when it can be written in both forms,  $(a, a, b, b) = (c, d, c, d)$ . Comparing entries gives  $a = c$ ,  $a = d$ ,  $b = c$ ,  $b = d$ , so  $a = b = c = d$ . Hence  $U \cap W = \text{span}((1, 1, 1, 1))$  and  $\dim(U \cap W) = 1$ . By the dimension formula,  $\dim(U + W) = 2 + 2 - 1 = 3$ .

For a basis, follow the proof. The vector  $(1, 1, 1, 1)$  together with  $(1, 1, 0, 0)$  is a list in  $U$ , and it is independent by [Lemma 1.5.3](#), because a multiple  $(c, c, c, c)$  of  $(1, 1, 1, 1)$  never equals  $(1, 1, 0, 0)$ . It has length  $2 = \dim U$ , hence it is a basis of  $U$  ([Theorem 1.6.4](#)). Likewise  $((1, 1, 1, 1), (1, 0, 1, 0))$  is a basis of  $W$ . By the Big Claim,  $((1, 1, 1, 1), (1, 1, 0, 0), (1, 0, 1, 0))$  is a basis of  $U + W$ . In particular  $U + W \neq \mathbb{R}^4$ .

### 1.7.3 Direct sums

In a basis, every vector has **exactly one** expression as a combination ([Theorem 1.5.1](#)). Read that for  $F^n$  with the standard basis:  $F^n = \text{span}(\mathbf{e}_1) + \cdots + \text{span}(\mathbf{e}_n)$ , and each vector splits into pieces from these lines in exactly one way. We want the same property for subspaces that are not lines.

Sums do not always have it. In  $\mathbb{R}^3 = P_{xy} + P_{yz}$ ,

$$(1, 1, 1) = (1, 1, 0) + (0, 0, 1) = (1, 0, 0) + (0, 1, 1),$$

two different splittings, because the planes share the  $y$ -axis and the  $y$ -component can be handed to either side. If we replace  $P_{yz}$  by the  $z$ -axis  $Z$ , then  $(a, b, c) = (a, b, 0) + (0, 0, c)$  is the **only** splitting, since the  $Z$ -piece must have first two entries 0. We give this situation a name.

*A sum is direct when every vector in it breaks into one piece from each subspace in exactly one way.*

### Definition 1.7.2 — Direct Sum

Let  $U_1, \dots, U_k$  be subspaces of a vector space  $V$  over  $F$ . The sum  $U_1 + \dots + U_k$  is **direct** if **every** vector  $\mathbf{v} \in U_1 + \dots + U_k$  can be written in **exactly one** way as

$$\mathbf{v} = \mathbf{u}_1 + \dots + \mathbf{u}_k \quad \text{with } \mathbf{u}_i \in U_i \text{ for each } i.$$

In that case we write the sum as  $U_1 \oplus \dots \oplus U_k$ . We write  $V = U_1 \oplus \dots \oplus U_k$  to mean that  $V = U_1 + \dots + U_k$  **and** the sum is direct.

In words: existence of a splitting is automatic, since that is what membership in the sum means. The content of “direct” is **uniqueness**: if  $\mathbf{u}_1 + \dots + \mathbf{u}_k = \mathbf{u}'_1 + \dots + \mathbf{u}'_k$  with  $\mathbf{u}_i, \mathbf{u}'_i \in U_i$ , then  $\mathbf{u}_i = \mathbf{u}'_i$  for **every**  $i$ . The symbol  $\oplus$  does not build a new space. It is the same subspace  $U_1 + \dots + U_k$ , with a property recorded in the notation. (Chapter 3 builds an “external” direct sum out of spaces that do not sit inside a common  $V$ .)

#### Examples.

- **Standard basis.**  $F^n = \text{span}(\mathbf{e}_1) \oplus \dots \oplus \text{span}(\mathbf{e}_n)$ . More generally, if  $(\mathbf{v}_1, \dots, \mathbf{v}_n)$  is a basis of  $V$ , then  $V = \text{span}(\mathbf{v}_1) \oplus \dots \oplus \text{span}(\mathbf{v}_n)$ . A splitting  $\mathbf{v} = a_1\mathbf{v}_1 + \dots + a_n\mathbf{v}_n$  is a choice of coefficients, and [Theorem 1.5.1](#) says there is exactly one.
- **Two splittings of the plane.** Let  $X$  be the  $x$ -axis,  $Y$  the  $y$ -axis and  $D = \{(t, t) : t \in \mathbb{R}\}$  the line  $y = x$ . Then  $\mathbb{R}^2 = X \oplus Y$ , with  $(a, b) = (a, 0) + (0, b)$  the only splitting. Also  $\mathbb{R}^2 = X \oplus D$ : if  $(a, b) = (s, 0) + (t, t)$ , the second entry forces  $t = b$ , and then the first forces  $s = a - b$ . So  $(a, b) = (a - b, 0) + (b, b)$ , and there is no other choice.
- **Polynomials.**  $F[x]_{\leq 2} = \text{span}(1) \oplus \text{span}(x, x^2)$ . If  $a + bx + cx^2 = a' + (b'x + c'x^2)$ , comparing coefficients gives  $a = a'$ ,  $b = b'$ ,  $c = c'$ .
- **Degenerate.**  $U \oplus \{\mathbf{0}\} = U$  for every subspace  $U$ : the only possible  $\{\mathbf{0}\}$ -piece is  $\mathbf{0}$ , so the  $U$ -piece of  $\mathbf{v}$  must be  $\mathbf{v}$ .

**Non-example by minimal change.** Replace the  $z$ -axis in  $\mathbb{R}^3 = P_{xy} \oplus Z$  by the plane  $P_{yz}$ . The sum is still all of  $\mathbb{R}^3$ , so existence still holds. What fails is uniqueness, as the two splittings of  $(1, 1, 1)$  above show. So  $\mathbb{R}^3 = P_{xy} + P_{yz}$ , but the sum is **not** direct. Likewise upper plus lower triangular  $2 \times 2$  matrices give  $M_2(F)$ , but a diagonal matrix can be put in either summand, so that sum is not direct.

Checking uniqueness for **every** vector looks like a lot of work. For two subspaces it reduces to one condition you can check in a line.

### Theorem 1.7.4 — Direct Sum Criteria for Two Subspaces

Let  $U$  and  $W$  be subspaces of a vector space  $V$ . The following are equivalent:



- (a) the sum  $U + W$  is direct;
- (b)  $U \cap W = \{\mathbf{0}\}$ .

If moreover  $U$  and  $W$  are finite-dimensional, they are also equivalent to

- (c)  $\dim(U + W) = \dim U + \dim W$ .

*Idea.* Uniqueness questions go through differences. If  $\mathbf{u} + \mathbf{w} = \mathbf{u}' + \mathbf{w}'$ , then  $\mathbf{u} - \mathbf{u}' = \mathbf{w}' - \mathbf{w}$  is a single vector lying in both  $U$  and  $W$ . So “no overlap” is exactly “no freedom in splitting”. For (c), the dimension formula says the overcount  $\dim U + \dim W - \dim(U + W)$  is  $\dim(U \cap W)$ , which is 0 exactly when  $U \cap W = \{\mathbf{0}\}$ .

*Proof.*

- (a)  $\Rightarrow$  (b). Suppose the sum is direct and let  $\mathbf{v} \in U \cap W$ . Then  $\mathbf{v} = \mathbf{v} + \mathbf{0}$  with  $\mathbf{v} \in U$ ,  $\mathbf{0} \in W$ , and  $\mathbf{v} = \mathbf{0} + \mathbf{v}$  with  $\mathbf{0} \in U$ ,  $\mathbf{v} \in W$ . These are two splittings of  $\mathbf{v} \in U + W$ , so by uniqueness their  $U$ -pieces agree:  $\mathbf{v} = \mathbf{0}$ . Hence  $U \cap W = \{\mathbf{0}\}$ .
- (b)  $\Rightarrow$  (a). Suppose  $U \cap W = \{\mathbf{0}\}$ , and let  $\mathbf{u} + \mathbf{w} = \mathbf{u}' + \mathbf{w}'$  with  $\mathbf{u}, \mathbf{u}' \in U$  and  $\mathbf{w}, \mathbf{w}' \in W$ . Then  $\mathbf{u} - \mathbf{u}' = \mathbf{w}' - \mathbf{w}$ . The left side lies in  $U$  and the right side in  $W$ , since both are subspaces. So this common vector lies in  $U \cap W = \{\mathbf{0}\}$ , which gives  $\mathbf{u} = \mathbf{u}'$  and  $\mathbf{w} = \mathbf{w}'$ . Hence every vector of  $U + W$  has only one splitting.
- (c)  $\Leftrightarrow$  (c). Now let  $U$  and  $W$  be finite-dimensional. By the dimension formula ([Theorem 1.7.3](#)), (c) holds if and only if  $\dim(U \cap W) = 0$ . A space has dimension 0 exactly when the empty list is a basis of it, that is, when it equals the span of the empty list, which is  $\{\mathbf{0}\}$ . So (c) holds if and only if  $U \cap W = \{\mathbf{0}\}$ .

■

This is the everyday test: to show  $V = U \oplus W$ , show  $V = U + W$  and  $U \cap W = \{\mathbf{0}\}$ . In finite dimension there is a shortcut that avoids checking  $V = U + W$  at all: if  $U \cap W = \{\mathbf{0}\}$  and  $\dim U + \dim W = \dim V$ , then  $\dim(U + W) = \dim V$  by (c), so  $U + W = V$  by [Theorem 1.6.8](#). Count instead of check.

### Example. (A plane and a line in $\mathbb{R}^3$ )

Let  $U = \{(x, y, z) \in \mathbb{R}^3 : x + y + z = 0\}$  and  $L = \text{span}((1, 1, 1))$ . Show that  $\mathbb{R}^3 = U \oplus L$ , and split  $(a, b, c)$  into its two pieces.

*Solution.* From [Example \(Two planes in  \$\mathbb{R}^3\$ \)](#),  $\dim U = 2$ , and  $\dim L = 1$  since  $(1, 1, 1) \neq \mathbf{0}$ . If  $t(1, 1, 1) \in U$ , then  $t + t + t = 3t = 0$ , so  $t = 0$ . Hence  $U \cap L = \{\mathbf{0}\}$ . By [Theorem 1.7.4](#),  $\dim(U + L) = 2 + 1 = 3$ , so  $U + L = \mathbb{R}^3$  by [Theorem 1.6.8](#). Therefore  $\mathbb{R}^3 = U \oplus L$ .

To split  $(a, b, c)$ , look for  $t$  with  $(a, b, c) - t(1, 1, 1) \in U$ , that is,  $(a - t) + (b - t) + (c - t) = 0$ . This gives  $t = \frac{a+b+c}{3}$ , and

$$(a, b, c) = \left(a - \frac{a+b+c}{3}, b - \frac{a+b+c}{3}, c - \frac{a+b+c}{3}\right) + \frac{a+b+c}{3}(1, 1, 1).$$

The  $L$ -piece is the average of the entries times  $(1, 1, 1)$ ; the  $U$ -piece is what is left over.

The same splitting trick, “average with a reflected copy”, works for functions and matrices.

**Example. (Even and odd functions)**

Let  $V = \mathbb{R}^{\mathbb{R}}$  be the space of all functions  $\mathbb{R} \rightarrow \mathbb{R}$ , with  $E = \{f : f(-x) = f(x) \text{ for all } x\}$  and  $O = \{f : f(-x) = -f(x) \text{ for all } x\}$ . Then  $V = E \oplus O$ .

*Solution.* Both are subspaces by the Subspace Test ([Theorem 1.2.1](#)): the zero function is even and odd, and the defining equations are preserved by sums and scalar multiples. For instance, if  $f, g \in O$ , then  $(f + g)(-x) = -f(x) - g(x) = -(f + g)(x)$ .

*Sum.* For any  $f \in V$ , put  $f_E(x) = \frac{f(x) + f(-x)}{2}$  and  $f_O(x) = \frac{f(x) - f(-x)}{2}$ . Then  $f = f_E + f_O$ , and  $f_E(-x) = \frac{f(-x) + f(x)}{2} = f_E(x)$ ,  $f_O(-x) = \frac{f(-x) - f(x)}{2} = -f_O(x)$ . So  $V = E + O$ .

*Intersection.* If  $f \in E \cap O$ , then  $f(x) = f(-x) = -f(x)$  for every  $x$ , so  $2f(x) = 0$  and  $f(x) = 0$ . Hence  $E \cap O = \{0\}$ , and  $V = E \oplus O$  by [Theorem 1.7.4](#).

For example,  $e^x = \cosh x + \sinh x$  is the splitting of the exponential function.

**Example. (Symmetric and skew-symmetric matrices)**

Let  $F$  be a field of characteristic not 2 (for instance  $\mathbb{R}$  or  $\mathbb{C}$ ), and let  $n \geq 1$ . In  $M_n(F)$ , let  $U_+ = \{\mathbf{A} : \mathbf{A}^\top = \mathbf{A}\}$  be the symmetric matrices and  $U_- = \{\mathbf{A} : \mathbf{A}^\top = -\mathbf{A}\}$  the skew-symmetric ones ([Definition 0.9.11](#)). Then  $M_n(F) = U_+ \oplus U_-$ .

*Solution.* Both are subspaces:  $0^\top = 0 = -0$ , and by [Theorem 0.9.5](#),  $(\mathbf{A} + \mathbf{B})^\top = \mathbf{A}^\top + \mathbf{B}^\top$  and  $(c\mathbf{A})^\top = c\mathbf{A}^\top$ , so the equations  $\mathbf{A}^\top = \pm\mathbf{A}$  survive sums and scalar multiples.

Since the characteristic is not 2, we have  $2 = 1 + 1 \neq 0$  in  $F$ , so  $\frac{1}{2}$  exists ([Definition 0.7.2](#)). For  $\mathbf{A} \in M_n(F)$ ,

$$\mathbf{A} = \frac{1}{2}(\mathbf{A} + \mathbf{A}^\top) + \frac{1}{2}(\mathbf{A} - \mathbf{A}^\top).$$

By [Theorem 0.9.5](#),  $(\frac{1}{2}(\mathbf{A} + \mathbf{A}^\top))^\top = \frac{1}{2}(\mathbf{A}^\top + \mathbf{A})$  and  $(\frac{1}{2}(\mathbf{A} - \mathbf{A}^\top))^\top = \frac{1}{2}(\mathbf{A}^\top - \mathbf{A}) = -\frac{1}{2}(\mathbf{A} - \mathbf{A}^\top)$ . So the first piece is in  $U_+$ , the second in  $U_-$ , and  $M_n(F) = U_+ + U_-$ .

If  $\mathbf{A} \in U_+ \cap U_-$ , then  $\mathbf{A} = \mathbf{A}^\top = -\mathbf{A}$ , so  $2\mathbf{A} = 0$ . Multiplying by  $\frac{1}{2}$  gives  $\mathbf{A} = 0$ . Hence  $U_+ \cap U_- = \{0\}$ , and  $M_n(F) = U_+ \oplus U_-$  by [Theorem 1.7.4](#). The pattern is the same as even plus odd: transpose plays the role of  $x \mapsto -x$ .

The hypothesis on the characteristic is used twice, both times to divide by 2. [Exercise \(C1: Symmetric Plus Skew over  \$\mathbb{F}\_2\$ \)](#) shows what goes wrong over  $\mathbb{F}_2$ .

*Quick check.* Can two 2-dimensional subspaces of  $\mathbb{R}^3$  form a direct sum?

*Solution.* No. If the sum  $U + W$  were direct, [Theorem 1.7.4](#) (c) would give  $\dim(U + W) = 2 + 2 = 4$ . But  $U + W \subseteq \mathbb{R}^3$ , so  $\dim(U + W) \leq 3$  by [Theorem 1.6.7](#). So two planes through  $\mathbf{0}$  in  $\mathbb{R}^3$  always share a non-zero vector.

## 1.7.4 Direct sums of several subspaces

For  $k \geq 3$  subspaces, the tidy condition  $U \cap W = \{\mathbf{0}\}$  has no direct replacement by pairwise intersections; we will see why in a moment. What does carry over is the zero vector test, which looks just like linear independence, and a condition that compares each subspace with the sum of **all the others**.

### Theorem 1.7.5 — Direct Sum Criteria for Several Subspaces

Let  $U_1, \dots, U_k$  be subspaces of a vector space  $V$ , and write  $S = U_1 + \dots + U_k$ . The following are equivalent:

- (a) the sum  $S$  is direct;
- (b) the only way to write  $\mathbf{0} = \mathbf{u}_1 + \dots + \mathbf{u}_k$  with  $\mathbf{u}_i \in U_i$  is  $\mathbf{u}_1 = \dots = \mathbf{u}_k = \mathbf{0}$ ;
- (c) for **every**  $j = 1, \dots, k$ ,

$$U_j \cap \sum_{i \neq j} U_i = \{\mathbf{0}\}.$$

If moreover each  $U_i$  is finite-dimensional with a basis  $\mathcal{B}_i$ , and  $\mathcal{B}$  is the list obtained by writing  $\mathcal{B}_1, \dots, \mathcal{B}_k$  one after another, then (a)–(c) are also equivalent to

- (d)  $\mathcal{B}$  is a basis of  $S$ ;
- (e)  $\dim S = \dim U_1 + \dots + \dim U_k$ .

Here  $\sum_{i \neq j} U_i$  is the sum of the  $k - 1$  subspaces other than  $U_j$  (and  $\{\mathbf{0}\}$  if  $k = 1$ ).

*Idea.* Condition (b) is uniqueness for the single vector  $\mathbf{0}$ , and just as for independence, uniqueness for  $\mathbf{0}$  gives uniqueness everywhere by taking differences. For (b)  $\Leftrightarrow$  (c), a non-trivial way of writing  $\mathbf{0}$  is the same thing as a non-zero vector  $\mathbf{u}_j$  that equals minus the sum of the other pieces. For (d), a combination of  $\mathcal{B}$  groups into one piece per  $U_i$ ; independence of each  $\mathcal{B}_i$  turns “piece is  $\mathbf{0}$ ” into “coefficients are 0”. Finally (d)  $\Leftrightarrow$  (e) is count instead of check:  $\mathcal{B}$  always spans  $S$ , so it is a basis exactly when its length is right.

*Proof.*

- (a)  $\Rightarrow$  (b). The vector  $\mathbf{0} \in S$  has the splitting  $\mathbf{0} = \mathbf{0} + \dots + \mathbf{0}$ . If the sum is direct, it has no other, which is (b).
- (b)  $\Rightarrow$  (a). Assume (b), and let  $\mathbf{u}_1 + \dots + \mathbf{u}_k = \mathbf{u}'_1 + \dots + \mathbf{u}'_k$  with  $\mathbf{u}_i, \mathbf{u}'_i \in U_i$ . Subtracting and regrouping,

$$(\mathbf{u}_1 - \mathbf{u}'_1) + \dots + (\mathbf{u}_k - \mathbf{u}'_k) = \mathbf{0},$$

where  $\mathbf{u}_i - \mathbf{u}'_i \in U_i$  because  $U_i$  is a subspace. By (b),  $\mathbf{u}_i - \mathbf{u}'_i = \mathbf{0}$ , that is,  $\mathbf{u}_i = \mathbf{u}'_i$ , for every  $i$ . Hence every vector of  $S$  has only one splitting.

- (c)  $\Rightarrow$  (c). Assume (b), fix  $j$ , and let  $\mathbf{v} \in U_j \cap \sum_{i \neq j} U_i$ . Then  $\mathbf{v} = \sum_{i \neq j} \mathbf{u}_i$  for some  $\mathbf{u}_i \in U_i$ , and so

$$\mathbf{u}_1 + \dots + \mathbf{u}_{j-1} + (-\mathbf{v}) + \mathbf{u}_{j+1} + \dots + \mathbf{u}_k = \mathbf{0},$$

with  $-\mathbf{v} \in U_j$  in the  $j$ -th slot. By (b), every piece is  $\mathbf{0}$ ; in particular  $-\mathbf{v} = \mathbf{0}$ , so  $\mathbf{v} = \mathbf{0}$ .

(d)  $\Rightarrow$  (b). Assume (c), and let  $\mathbf{u}_1 + \cdots + \mathbf{u}_k = \mathbf{0}$  with  $\mathbf{u}_i \in U_i$ . Fix  $j$ . Then

$$\mathbf{u}_j = -\sum_{i \neq j} \mathbf{u}_i = \sum_{i \neq j} (-\mathbf{u}_i).$$

The left side lies in  $U_j$ , and the right side lies in  $\sum_{i \neq j} U_i$  since each  $-\mathbf{u}_i \in U_i$ . By (c),  $\mathbf{u}_j = \mathbf{0}$ . As  $j$  was arbitrary, all pieces are  $\mathbf{0}$ .

For the rest, assume each  $U_i$  is finite-dimensional with basis  $\mathcal{B}_i = (\mathbf{b}_{i1}, \dots, \mathbf{b}_{im_i})$ , where  $m_i = \dim U_i$ . By [Proposition 1.7.2](#),  $\mathcal{B}$  spans  $S = \text{span}(\mathcal{B}_1) + \cdots + \text{span}(\mathcal{B}_k)$ . In particular  $S$  is finite-dimensional.

(b)  $\Rightarrow$  (d). Since  $\mathcal{B}$  spans  $S$ , it remains to prove independence. Let  $\sum_{i=1}^k \sum_{r=1}^{m_i} a_{ir} \mathbf{b}_{ir} = \mathbf{0}$ , and group the terms by subspace: put  $\mathbf{u}_i = \sum_{r=1}^{m_i} a_{ir} \mathbf{b}_{ir} \in U_i$ . Then  $\mathbf{u}_1 + \cdots + \mathbf{u}_k = \mathbf{0}$ , so each  $\mathbf{u}_i = \mathbf{0}$  by (b). Since  $\mathcal{B}_i$  is independent,  $a_{i1} = \cdots = a_{im_i} = 0$  for each  $i$ . Hence  $\mathcal{B}$  is independent, and so a basis of  $S$ .

(c)  $\Rightarrow$  (b). Assume (d), and let  $\mathbf{u}_1 + \cdots + \mathbf{u}_k = \mathbf{0}$  with  $\mathbf{u}_i \in U_i$ . Since  $\mathcal{B}_i$  spans  $U_i$ , write  $\mathbf{u}_i = \sum_{r=1}^{m_i} a_{ir} \mathbf{b}_{ir}$ . Then  $\sum_{i=1}^k \sum_{r=1}^{m_i} a_{ir} \mathbf{b}_{ir} = \mathbf{0}$  is a combination of the independent list  $\mathcal{B}$ , so every  $a_{ir} = 0$ , and therefore every  $\mathbf{u}_i = \mathbf{0}$ .

(d)  $\Rightarrow$  (e). The list  $\mathcal{B}$  has length  $m_1 + \cdots + m_k$ , so if it is a basis of  $S$ , then  $\dim S = \dim U_1 + \cdots + \dim U_k$  by [Definition 1.6.1](#).

(e)  $\Rightarrow$  (d). The list  $\mathcal{B}$  spans  $S$  and has length  $m_1 + \cdots + m_k$ , which equals  $\dim S$  by (e). A spanning list whose length is the dimension is a basis ([Theorem 1.6.4](#)). This proves the theorem. ■

For  $k = 2$ , condition (c) reads  $U_1 \cap U_2 = \{\mathbf{0}\}$  and  $U_2 \cap U_1 = \{\mathbf{0}\}$ , so we recover [Theorem 1.7.4](#). The criteria (d) and (e) say that **bases of the pieces assemble into a basis of a direct sum, and dimensions add**.

For  $k \geq 3$ , the tempting shortcut is to check only the intersections of the subspaces two at a time. It does not work.

*Warning.* **Pairwise intersections  $\{\mathbf{0}\}$  do not make a sum direct.** In  $\mathbb{R}^2$ , take the three lines  $X$  (the  $x$ -axis),  $Y$  (the  $y$ -axis) and  $D$  (the line  $y = x$ ). Any two of them meet only in  $\mathbf{0}$ . But

$$(1, 0) + (0, 1) + (-1, -1) = (0, 0)$$

writes  $\mathbf{0}$  with non-zero pieces from  $X$ ,  $Y$ ,  $D$ , so (b) fails and  $X + Y + D$  is **not** direct. Condition (c) sees the problem:  $D \cap (X + Y) = D \cap \mathbb{R}^2 = D \neq \{\mathbf{0}\}$ .

The same example breaks another tempting generalization. For finite sets,  $|A \cup B \cup C|$  is given by inclusion–exclusion, and the dimension formula looks like its two-set case. For three subspaces the analogous formula would be

$$\dim(U_1 + U_2 + U_3) \stackrel{?}{=} \sum_i \dim U_i - \sum_{i < j} \dim(U_i \cap U_j) + \dim(U_1 \cap U_2 \cap U_3).$$

For  $X, Y, D$  the left side is  $\dim \mathbb{R}^2 = 2$ , and the right side is  $1 + 1 + 1 - 0 - 0 - 0 + 0 = 3$ . So the formula is **false**. The reason is that  $U + W$  is not a union: the dimension of  $(U_1 + U_2) \cap U_3$  is

not determined by the pairwise intersections. [Exercise \(C3: Three Subspaces\)](#) gives the correct replacement and a second failure.

## 1.7.5 Complements

Given a subspace  $U$  of  $V$ , can we always find a subspace  $W$  with  $V = U \oplus W$ ? Such a  $W$  is called a **complement** of  $U$  in  $V$ . In  $\mathbb{R}^3$ , the plane  $U$  of [Example \(A plane and a line in  \$\mathbb{R}^3\$ \)](#) has the complement  $\text{span}((1, 1, 1))$ . In finite dimension, complements always exist, and the proof is again “basis of the smaller space, then extend”.

### Theorem 1.7.6 — Existence of Complements

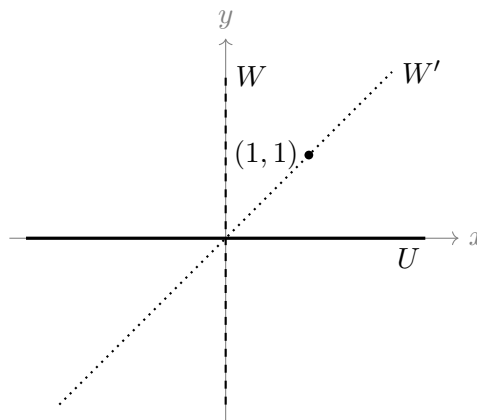
Let  $V$  be a finite-dimensional vector space and  $U$  a subspace of  $V$ . Then there is a subspace  $W$  of  $V$  with  $V = U \oplus W$ . Every such  $W$  satisfies  $\dim W = \dim V - \dim U$ .

*Proof.* By [Theorem 1.6.7](#),  $U$  is finite-dimensional; let  $(\mathbf{u}_1, \dots, \mathbf{u}_m)$  be a basis of  $U$  ([Corollary 1.5.5](#)). This list is independent in  $V$ , so by the Basis Extension Theorem ([Theorem 1.6.5](#)) there are  $\mathbf{w}_1, \dots, \mathbf{w}_r \in V$  such that  $(\mathbf{u}_1, \dots, \mathbf{u}_m, \mathbf{w}_1, \dots, \mathbf{w}_r)$  is a basis of  $V$ . Put  $W = \text{span}(\mathbf{w}_1, \dots, \mathbf{w}_r)$ , a subspace by [Theorem 1.3.1](#).

By [Proposition 1.7.2](#),  $U + W = \text{span}(\mathbf{u}_1, \dots, \mathbf{u}_m, \mathbf{w}_1, \dots, \mathbf{w}_r) = V$ . The list  $(\mathbf{w}_1, \dots, \mathbf{w}_r)$  is independent, since a relation  $c_1 \mathbf{w}_1 + \dots + c_r \mathbf{w}_r = \mathbf{0}$  is also a relation  $0\mathbf{u}_1 + \dots + 0\mathbf{u}_m + c_1 \mathbf{w}_1 + \dots + c_r \mathbf{w}_r = \mathbf{0}$  among the basis vectors of  $V$ , forcing all  $c_k = 0$ . So  $(\mathbf{w}_1, \dots, \mathbf{w}_r)$  is a basis of  $W$ . The concatenation of the bases of  $U$  and  $W$  is a basis of  $U + W = V$ , so the sum is direct by [Theorem 1.7.5](#) ((d)  $\Rightarrow$  (a)). Hence  $V = U \oplus W$ .

Finally, if  $W$  is any subspace with  $V = U \oplus W$ , then  $W$  is finite-dimensional ([Theorem 1.6.7](#)), and [Theorem 1.7.4](#) (c) gives  $\dim V = \dim U + \dim W$ , as claimed. ■

The proof made a choice: the vectors  $\mathbf{w}_1, \dots, \mathbf{w}_r$  that extend the basis. Different choices give different complements.



**Warning. A complement is not unique.** In  $\mathbb{R}^2$ , the  $x$ -axis  $U$  (solid) has the complement  $W$ , the  $y$ -axis (dashed), and also the complement  $W'$ , the line  $y = x$  (dotted), as shown in the examples after [Definition 1.7.2](#). So “the complement of  $U$ ” is meaningless,

and  $U \oplus W = U \oplus W'$  does **not** allow us to cancel  $U$  and conclude  $W = W'$ . What all complements share is their dimension. Also, a complement is not the set complement  $V \setminus U$ , which does not even contain  $\mathbf{0}$ .

Complements matter because a splitting  $V = U \oplus W$  lets us study  $V$  one piece at a time. In Chapter 3 each such splitting gives a projection of  $V$  onto  $U$  along  $W$ , and the warning above becomes visible there: different complements give different projections. In Chapter 8 the eigenspaces of a diagonalizable operator split  $V$  as a direct sum, and that is what makes such operators easy to understand.

## 1.7.6 Exercises

### 1.7.6.1 A. Check your understanding

#### Exercise A1.

- Define the sum  $U_1 + \cdots + U_k$  of subspaces of  $V$ , and what it means for the sum to be direct.
- State the dimension formula for  $\dim(U + W)$ , with its hypotheses.
- True or false: if  $V = U + W$  and  $U \cap W = \{\mathbf{0}\}$ , then  $V = U \oplus W$ . Justify your answer.
- True or false: if  $U_1, U_2, U_3$  are subspaces with  $U_i \cap U_j = \{\mathbf{0}\}$  for all  $i \neq j$ , then the sum  $U_1 + U_2 + U_3$  is direct. Justify your answer.
- True or false: if  $U$  is a subspace of  $V$  with  $U \neq V$ , then  $V \setminus U$  is a complement of  $U$ . Justify your answer.
- Name the method used to prove the dimension formula, in four steps.

*Solution.*

- See [Definition 1.7.1](#):  $U_1 + \cdots + U_k = \{\mathbf{u}_1 + \cdots + \mathbf{u}_k : \mathbf{u}_i \in U_i\}$ . By [Definition 1.7.2](#), the sum is direct if every vector in it can be written as  $\mathbf{u}_1 + \cdots + \mathbf{u}_k$  with  $\mathbf{u}_i \in U_i$  in exactly one way.
- If  $U$  and  $W$  are finite-dimensional subspaces of a vector space  $V$ , then  $\dim(U + W) = \dim U + \dim W - \dim(U \cap W)$  ([Theorem 1.7.3](#)).
- True. By [Theorem 1.7.4](#),  $U \cap W = \{\mathbf{0}\}$  means the sum  $U + W$  is direct, and it equals  $V$ .
- False. The three lines  $X, Y, D$  in  $\mathbb{R}^2$  (the axes and the line  $y = x$ ) meet pairwise only in  $\mathbf{0}$ , but  $(1, 0) + (0, 1) + (-1, -1) = \mathbf{0}$  is a non-trivial way to write  $\mathbf{0}$ , so the sum is not direct by [Theorem 1.7.5](#).
- False.  $V \setminus U$  does not contain  $\mathbf{0}$ , because  $\mathbf{0} \in U$ . So it is not a subspace, and cannot be a complement.
- Take a basis of the smallest space  $U \cap W$ ; extend it to bases of  $U$  and of  $W$  (Basis Extension); prove the Big Claim that all the vectors together form a basis of  $U + W$  (spanning, then independence by moving one block across and noting the common vector lies in  $U \cap W$ ); count.

## 1.7.6.2 B. Practice

**Exercise B1: Two Planes in  $\mathbb{R}^4$ .**

Let  $U = \text{span}((1, 2, 0, 1), (0, 1, 1, 0))$  and  $W = \text{span}((1, 3, 1, 1), (0, 0, 1, 1))$  in  $\mathbb{R}^4$ .

- (a) Show that  $\dim U = \dim W = 2$ .
- (b) Find  $U \cap W$  and its dimension.
- (c) Hence find  $\dim(U + W)$  and a basis of  $U + W$ .

*Solution.*

- (a) If  $a(1, 2, 0, 1) + b(0, 1, 1, 0) = (a, 2a + b, b, a) = \mathbf{0}$ , then  $a = 0$  and  $b = 0$  from the first and third entries. So the spanning list of  $U$  is independent, hence a basis, and  $\dim U = 2$ . Similarly  $c(1, 3, 1, 1) + d(0, 0, 1, 1) = (c, 3c, c + d, c + d) = \mathbf{0}$  forces  $c = 0$ , then  $d = 0$ , so  $\dim W = 2$ .
- (b) A vector lies in  $U \cap W$  exactly when  $(a, 2a + b, b, a) = (c, 3c, c + d, c + d)$  for some  $a, b, c, d \in \mathbb{R}$ . The first entries give  $a = c$ . The second gives  $2a + b = 3a$ , so  $b = a$ . The third gives  $a = a + d$ , so  $d = 0$ ; the fourth,  $a = a + 0$ , then holds. So the common vectors are  $(a, 3a, a, a) = a(1, 3, 1, 1)$ , and  $U \cap W = \text{span}((1, 3, 1, 1))$ , of dimension 1.
- (c) By the dimension formula ([Theorem 1.7.3](#)),  $\dim(U + W) = 2 + 2 - 1 = 3$ . Following its proof:  $((1, 3, 1, 1), (0, 1, 1, 0))$  is a list in  $U$  (by (b),  $(1, 3, 1, 1) \in U \cap W$ ). It is independent by [Lemma 1.5.3](#), since a multiple  $c(1, 3, 1, 1)$  with first entry 0 is  $\mathbf{0} \neq (0, 1, 1, 0)$ . It has length  $2 = \dim U$ , hence it is a basis of  $U$  by [Theorem 1.6.4](#). Likewise  $((1, 3, 1, 1), (0, 0, 1, 1))$  is a basis of  $W$ . By the Big Claim in the proof,  $((1, 3, 1, 1), (0, 1, 1, 0), (0, 0, 1, 1))$  is a basis of  $U + W$ .

**Exercise B2: A Plane and a Line.**

Let  $U = \{(x, y, z) \in \mathbb{R}^3 : x - 2y + z = 0\}$  and  $L = \text{span}((1, 0, 1))$ .

- (a) Prove that  $\mathbb{R}^3 = U \oplus L$ .
- (b) Write  $(2, 0, 4)$  as  $\mathbf{u} + \mathbf{w}$  with  $\mathbf{u} \in U$  and  $\mathbf{w} \in L$ .

*Solution.*

- (a) Solving for  $x$ , every vector of  $U$  is  $(2y - z, y, z) = y(2, 1, 0) + z(-1, 0, 1)$ . These two vectors are independent (look at the second and third entries), so  $\dim U = 2$ . Since  $(1, 0, 1) \neq \mathbf{0}$ ,  $\dim L = 1$ . If  $t(1, 0, 1) \in U$ , then  $t - 0 + t = 2t = 0$ , so  $t = 0$ ; hence  $U \cap L = \{\mathbf{0}\}$ . By [Theorem 1.7.4](#),  $\dim(U + L) = 2 + 1 = 3$ , so  $U + L = \mathbb{R}^3$  by [Theorem 1.6.8](#). Therefore  $\mathbb{R}^3 = U \oplus L$ .
- (b) We need  $t$  with  $(2, 0, 4) - t(1, 0, 1) = (2 - t, 0, 4 - t) \in U$ , that is,  $(2 - t) - 0 + (4 - t) = 0$ . This gives  $t = 3$ . So

$$(2, 0, 4) = (-1, 0, 1) + 3(1, 0, 1),$$

and indeed  $-1 - 0 + 1 = 0$ , so  $(-1, 0, 1) \in U$ . By (a) this is the only such splitting.

**Exercise B3: Direct or Not?.**

Determine which of the following sums are direct. Justify your answer.

- (a)  $\text{span}(1, x^2) + \text{span}(x, x^3)$  in  $\mathbb{R}[x]_{\leq 3}$ .
- (b)  $T_{\text{up}} + T_{\text{low}}$  in  $M_2(\mathbb{R})$ , where  $T_{\text{up}}$  and  $T_{\text{low}}$  are the upper and lower triangular matrices.
- (c)  $\text{span}((1, 0, 0)) + \text{span}((0, 1, 0)) + \text{span}((1, 1, 1))$  in  $\mathbb{R}^3$ .
- (d)  $\text{span}((1, 1, 0)) + \text{span}((0, 1, 1)) + \text{span}((1, 0, -1))$  in  $\mathbb{R}^3$ .

*Solution.*

- (a) Direct. If  $p \in \text{span}(1, x^2) \cap \text{span}(x, x^3)$ , then  $p = a + bx^2 = cx + dx^3$ . Comparing coefficients of  $1, x, x^2, x^3$  gives  $a = c = b = d = 0$ , so  $p = 0$ . By [Theorem 1.7.4](#) the sum is direct.
- (b) Not direct. The identity matrix  $\mathbf{I}_2$  is both upper and lower triangular, so  $\mathbf{I}_2 \in T_{\text{up}} \cap T_{\text{low}} \neq \{0\}$ , and [Theorem 1.7.4](#) applies.
- (c) Direct. Each line has the basis given by its spanning vector, and we check that the concatenated list  $((1, 0, 0), (0, 1, 0), (1, 1, 1))$  is independent. If  $a(1, 0, 0) + b(0, 1, 0) + c(1, 1, 1) = (a + c, b + c, c) = \mathbf{0}$ , then  $c = 0$ , hence  $a = 0$  and  $b = 0$ . The list spans the sum by [Proposition 1.7.2](#), so it is a basis of the sum, and the sum is direct by [Theorem 1.7.5](#) ((d)  $\Rightarrow$  (a)).
- (d) Not direct, although any two of the three lines meet only in  $\mathbf{0}$  (no spanning vector is a multiple of another). Indeed

$$(1, 1, 0) + (-1)(0, 1, 1) + (-1)(1, 0, -1) = (0, 0, 0)$$

writes  $\mathbf{0}$  as a sum of **non-zero** pieces from the three lines, so condition (b) of [Theorem 1.7.5](#) fails.

**1.7.6.3 C. Going deeper****Exercise C1: Symmetric Plus Skew over  $\mathbb{F}_2$ .**

Let  $U_+$  and  $U_-$  be the symmetric and skew-symmetric matrices in  $M_2(\mathbb{F}_2)$ .

- (a) Show that  $U_+ = U_-$ .
- (b) Deduce that  $M_2(\mathbb{F}_2) \neq U_+ \oplus U_-$ . Which step of [Example \(Symmetric and skew-symmetric matrices\)](#) fails, and why?

*Solution.*

- (a) In  $\mathbb{F}_2$  we have  $1 + 1 = 0$ , so  $-a = a$  for every  $a \in \mathbb{F}_2$ , and hence  $-\mathbf{A} = \mathbf{A}$  for every  $\mathbf{A} \in M_2(\mathbb{F}_2)$ . Therefore  $\mathbf{A}^\top = -\mathbf{A}$  holds if and only if  $\mathbf{A}^\top = \mathbf{A}$ , which says  $U_- = U_+$ .
- (b) By (a),  $U_+ \cap U_- = U_+$ , which contains  $\mathbf{I}_2 \neq 0$ . So the sum is not direct ([Theorem 1.7.4](#)). It is not even all of  $M_2(\mathbb{F}_2)$ :  $U_+ + U_- = U_+ + U_+ = U_+$ , and  $\begin{pmatrix} 0 & 1 \\ 0 & 0 \end{pmatrix}$  is not symmetric. In [Example \(Symmetric and skew-symmetric matrices\)](#) both



halves of the proof divide by 2: the splitting  $\mathbf{A} = \frac{1}{2}(\mathbf{A} + \mathbf{A}^\top) + \frac{1}{2}(\mathbf{A} - \mathbf{A}^\top)$  and the step from  $2\mathbf{A} = 0$  to  $\mathbf{A} = 0$ . In  $\mathbb{F}_2$ ,  $2 = 1 + 1 = 0$  has no inverse, so neither step is available. This is exactly the case of characteristic 2 excluded by the hypothesis.

### Exercise C2: Subspaces That Must Meet.

- (a) Let  $V$  be a vector space of dimension  $n$ , and let  $U, W$  be subspaces with  $\dim U + \dim W > n$ . Prove that  $U \cap W \neq \{\mathbf{0}\}$ .
- (b) Deduce that any two 3-dimensional subspaces of  $\mathbb{R}[x]_{\leq 4}$  contain a common non-zero polynomial.
- (c) Give an example showing that (a) fails if “ $>$ ” is replaced by “ $=$ ”.

*Solution.*

- (a) By [Theorem 1.6.7](#),  $U, W$  and  $U + W$  are finite-dimensional, and  $\dim(U + W) \leq \dim V = n$ . By the dimension formula ([Theorem 1.7.3](#)),

$$\dim(U \cap W) = \dim U + \dim W - \dim(U + W) \geq \dim U + \dim W - n > 0.$$

A subspace of positive dimension has a basis with at least one vector, which is non-zero because a list containing  $\mathbf{0}$  is dependent. Hence  $U \cap W \neq \{\mathbf{0}\}$ .

- (b)  $\dim \mathbb{R}[x]_{\leq 4} = 5$  and  $3 + 3 = 6 > 5$ . By (a), the two subspaces meet in a non-zero polynomial.
- (c) In  $\mathbb{R}^2$  (with  $n = 2$ ), the  $x$ -axis and the  $y$ -axis have  $1 + 1 = 2 = n$ , but their intersection is  $\{\mathbf{0}\}$ .

### Exercise C3: Three Subspaces.

- (a) Let  $U_1, U_2, U_3$  be subspaces of a vector space  $V$ . Show that  $(U_1 + U_2) + U_3 = U_1 + U_2 + U_3$ .
- (b) Now let  $U_1, U_2, U_3$  be finite-dimensional. Prove that

$$\begin{aligned} \dim(U_1 + U_2 + U_3) &= \dim U_1 + \dim U_2 + \dim U_3 \\ &\quad - \dim(U_1 \cap U_2) - \dim((U_1 + U_2) \cap U_3). \end{aligned}$$

- (c) In  $\mathbb{R}^3$ , let  $U_1 = \{z = 0\}$ ,  $U_2 = \{y = 0\}$  and  $U_3 = \{y = z\}$ . Compute  $\dim(U_1 + U_2 + U_3)$ , every  $\dim(U_i \cap U_j)$  and  $\dim(U_1 \cap U_2 \cap U_3)$ . Show that the inclusion-exclusion formula fails, and compare with (b).

*Hint: for (b), apply the dimension formula twice.*

*Solution.*

- (a) Both sides are subspaces by [Theorem 1.7.1](#). The right side contains  $U_1$  and  $U_2$ , hence  $U_1 + U_2$  ([Theorem 1.7.1](#)), and it contains  $U_3$ ; so it contains  $(U_1 + U_2) + U_3$  by [Theorem 1.7.1](#). Conversely, an element  $\mathbf{u}_1 + \mathbf{u}_2 + \mathbf{u}_3$  of the right side is the sum of  $\mathbf{u}_1 + \mathbf{u}_2 \in U_1 + U_2$  and  $\mathbf{u}_3 \in U_3$ , so it lies in the left side.
- (b) By [Theorem 1.7.3](#),  $U_1 + U_2$  is finite-dimensional with  $\dim(U_1 + U_2) = \dim U_1 +$

$\dim U_2 - \dim(U_1 \cap U_2)$ . Applying the formula again to  $U_1 + U_2$  and  $U_3$ , and using (a),

$$\dim(U_1 + U_2 + U_3) = \dim(U_1 + U_2) + \dim U_3 - \dim((U_1 + U_2) \cap U_3).$$

Substituting the first equation into the second gives the formula.

(c) Each  $U_i$  is a plane, so  $\dim U_i = 2$ :  $U_1 = \text{span}(\mathbf{e}_1, \mathbf{e}_2)$ ,  $U_2 = \text{span}(\mathbf{e}_1, \mathbf{e}_3)$ ,  $U_3 = \text{span}(\mathbf{e}_1, \mathbf{e}_2 + \mathbf{e}_3)$ , each with an evidently independent spanning list. All three contain the  $x$ -axis  $\text{span}(\mathbf{e}_1)$ .

- $U_1 \cap U_2 = \{y = z = 0\} = \text{span}(\mathbf{e}_1)$ .
- $U_1 \cap U_3 = \{z = 0, y = z\} = \{y = z = 0\} = \text{span}(\mathbf{e}_1)$ .
- $U_2 \cap U_3 = \{y = 0, y = z\} = \text{span}(\mathbf{e}_1)$ .
- $U_1 \cap U_2 \cap U_3 = \text{span}(\mathbf{e}_1)$ .

So each of these has dimension 1. Since  $U_1 + U_2$  contains  $\mathbf{e}_1, \mathbf{e}_2, \mathbf{e}_3$ , it is  $\mathbb{R}^3$ , and so  $U_1 + U_2 + U_3 = \mathbb{R}^3$  has dimension 3. The inclusion–exclusion formula would give

$$2 + 2 + 2 - 1 - 1 - 1 + 1 = 4 \neq 3,$$

so it fails. In (b), the last term is  $(U_1 + U_2) \cap U_3 = \mathbb{R}^3 \cap U_3 = U_3$ , of dimension 2, and indeed  $2 + 2 + 2 - 1 - 2 = 3$ . The inclusion–exclusion guess replaces this 2 by  $\dim(U_1 \cap U_3) + \dim(U_2 \cap U_3) - \dim(U_1 \cap U_2 \cap U_3) = 1 + 1 - 1 = 1$ , which is wrong because  $(U_1 + U_2) \cap U_3$  is much bigger than  $(U_1 \cap U_3) + (U_2 \cap U_3) = \text{span}(\mathbf{e}_1)$ .

## 1.8 Infinite-Dimensional Spaces

Every theorem about bases so far has assumed a finite spanning list. Yet several of our running examples have none:  $F[x]$  is not finite-dimensional ([Theorem 1.3.3](#)), and neither are the sequences or the functions  $\mathbb{R} \rightarrow \mathbb{R}$ . Do such spaces have a basis at all? This section shows that they always do, using a principle from set theory called Zorn’s Lemma, and then looks honestly at the price. The basis exists, but often nobody can write it down, and several finite-dimensional facts stop being true.

### 1.8.1 Bases of infinite sets of vectors

An infinite-dimensional space cannot have a finite basis, so a basis now has to be an infinite collection of vectors. We recall the conventions for arbitrary subsets, set up with the span ([Definition 1.3.2](#)) and with independence ([Definition 1.4.1](#)). A linear combination always involves **finitely many** vectors, because the axioms only let us add two vectors at a time.

Let  $S \subseteq V$  be any subset, finite or infinite.

- $\text{span}(S)$  is the set of all linear combinations  $a_1 \mathbf{v}_1 + \cdots + a_k \mathbf{v}_k$  with  $k \in \mathbb{N}$ ,  $a_i \in F$  and  $\mathbf{v}_i \in S$ .
- $S$  is **linearly independent** if **every finite list of distinct vectors** of  $S$  is linearly independent in the sense of [Definition 1.4.1](#).
- $S$  is a **basis** of  $V$  if  $S$  is linearly independent and  $\text{span}(S) = V$ .

For a finite set  $S = \{\mathbf{v}_1, \dots, \mathbf{v}_k\}$  of distinct vectors, these agree with the notions for the list  $(\mathbf{v}_1, \dots, \mathbf{v}_k)$ , since a sub-list of an independent list is independent. The empty set is independent: its only finite list of distinct vectors is the empty list. In the infinite case we lose the ordering of a basis, and we will not need it.

**Example. (A Basis of  $F[x]$ )**

Show that  $\{x^k : k \in \mathbb{N}\} = \{1, x, x^2, \dots\}$  is a basis of  $F[x]$ .

*Solution. Spanning.* By [Definition 0.8.1](#), every polynomial can be written as  $a_0 + a_1x + \dots + a_Nx^N$  for some  $N \in \mathbb{N}$ . This is a finite linear combination of  $1, x, \dots, x^N$ .

*Independence.* Let  $x^{k_1}, \dots, x^{k_m}$  be distinct monomials and suppose  $a_1x^{k_1} + \dots + a_mx^{k_m} = 0$ . The left side is the polynomial whose coefficient of  $x^{k_j}$  is  $a_j$ , because the exponents  $k_j$  are distinct. The zero polynomial has every coefficient equal to 0, so by the definition of equality of polynomials,  $a_j = 0$  for every  $j$ .

Hence  $\{x^k : k \in \mathbb{N}\}$  is a basis of  $F[x]$ .

The sequence space behaves differently. Let  $F^{\mathbb{N}}$  be the space of all sequences  $(a_0, a_1, a_2, \dots)$  with entries in  $F$ , with entrywise addition and scaling. For  $i \in \mathbb{N}$  let  $\mathbf{e}_i$  be the sequence with 1 in position  $i$  and 0 everywhere else.

**Example. (The Vectors  $\mathbf{e}_i$  Do Not Span  $F^{\mathbb{N}}$ )**

Show that  $\{\mathbf{e}_i : i \in \mathbb{N}\}$  is linearly independent in  $F^{\mathbb{N}}$ , but is **not** a basis of  $F^{\mathbb{N}}$ .

*Solution. Independence.* Let  $i_1, \dots, i_m$  be distinct and suppose  $a_1\mathbf{e}_{i_1} + \dots + a_m\mathbf{e}_{i_m} = \mathbf{0}$ . Position  $i_j$  of the left side is  $a_j$ , since only  $\mathbf{e}_{i_j}$  has a non-zero entry there. Position  $i_j$  of the zero sequence is 0, so  $a_j = 0$  for every  $j$ .

*Not spanning.* A linear combination  $a_1\mathbf{e}_{i_1} + \dots + a_m\mathbf{e}_{i_m}$  has non-zero entries only in the  $m$  positions  $i_1, \dots, i_m$ . So every vector in  $\text{span}\{\mathbf{e}_i : i \in \mathbb{N}\}$  has only finitely many non-zero entries. The sequence  $(1, 1, 1, \dots)$  has infinitely many, so it is not in the span.

The same example shows that  $F^{\mathbb{N}}$  is infinite-dimensional. For each  $n$ , the list  $(\mathbf{e}_0, \dots, \mathbf{e}_n)$  is independent of length  $n + 1$ . If a list of length  $m$  spanned  $F^{\mathbb{N}}$ , then  $n + 1 \leq m$  for every  $n$  by [Theorem 1.6.1](#), which is impossible.

*Warning.* It is tempting to write  $(1, 1, 1, \dots) = \mathbf{e}_0 + \mathbf{e}_1 + \mathbf{e}_2 + \dots$ . In a vector space this sum **means nothing**: the axioms define sums of two vectors, hence of finitely many, and an infinite sum would need a notion of limit. So “the  $\mathbf{e}_i$  span” is false, even though every sequence is “made of” its entries. Infinite sums return only in spaces with a notion of distance, much later.

*Quick check.* Let  $V$  be the space of formal power series  $a_0 + a_1x + a_2x^2 + \dots$  over  $F$ , that is,  $F^{\mathbb{N}}$  with  $x^k$  standing for  $\mathbf{e}_k$ . Is  $\{1, x, x^2, \dots\}$  a basis of  $V$ ?

*Solution.* No. It is independent, by the argument of [Example \(The Vectors  \$\mathbf{e}\_i\$  Do Not Span  \$F^{\mathbb{N}}\$ \)](#). But its span consists of the series with only finitely many non-zero coefficients, which are the polynomials. The series  $1 + x + x^2 + \dots$  is not among them, so the set

does not span  $V$ .

So does  $F^{\mathbb{N}}$  have a basis? In finite dimension we built bases by sifting (Theorem 1.5.4) or by extending one vector at a time (Theorem 1.6.5). Both constructions stop after finitely many steps. For  $F^{\mathbb{N}}$  no step-by-step process can stop, and we need a different kind of existence argument.

## 1.8.2 Zorn's Lemma

Look again at what a basis is in finite dimension: an independent list that cannot be enlarged (Theorem 1.6.6). “Cannot be enlarged” is a statement about the family of all independent sets, ordered by inclusion. Zorn's Lemma is a general principle that produces objects which cannot be enlarged. To state it we need words for orderings.

*A partial order is a way of comparing elements in which some pairs may be incomparable.*

### Definition 1.8.1 — Partial Order, Chain, Upper Bound, Maximal Element

Let  $P$  be a set. A **partial order** on  $P$  is a relation  $\leq$  on  $P$  (Definition 0.5.1) such that, for all  $p, q, r \in P$ :

- (P1)  $p \leq p$  (reflexive);
- (P2) if  $p \leq q$  and  $q \leq p$ , then  $p = q$  (antisymmetric);
- (P3) if  $p \leq q$  and  $q \leq r$ , then  $p \leq r$  (transitive).

Let  $(P, \leq)$  be a partially ordered set.

- A **chain** in  $P$  is a subset  $C \subseteq P$  such that for all  $p, q \in C$ ,  $p \leq q$  or  $q \leq p$ .
- An **upper bound** of a subset  $C \subseteq P$  is an element  $u \in P$  with  $c \leq u$  for every  $c \in C$ .
- A **maximal element** of  $P$  is an element  $m \in P$  such that no  $p \in P$  satisfies  $m \leq p$  and  $p \neq m$ .

In words: a chain is a subset in which any two elements can be compared. An upper bound of  $C$  sits above everything in  $C$ , but it only has to lie in  $P$ , **not** in  $C$ . A maximal element has nothing strictly above it. It is **not** required to lie above everything else.

### Example. (Inclusion as a Partial Order)

Let  $X$  be a set and  $P$  the power set of  $X$  (Definition 0.3.12), ordered by  $\subseteq$ . Check that this is a partial order, and describe chains, upper bounds and maximal elements when  $X = \{1, 2\}$ . Then do the same for the subsets of  $\{1, 2\}$  with **at most one** element.

*Solution.* Every set satisfies  $A \subseteq A$ , which is (P1). If  $A \subseteq B$  and  $B \subseteq A$ , then  $A = B$  by Theorem 0.3.1, which is (P2). Inclusion is transitive, which is (P3).

For  $X = \{1, 2\}$ , the set  $C = \{\emptyset, \{1\}, \{1, 2\}\}$  is a chain, while  $\{1\}$  and  $\{2\}$  are incomparable, so  $\{\{1\}, \{2\}\}$  is not a chain. The set  $\{1, 2\}$  is an upper bound of every subset of

$P$ , and it is the only maximal element.

Now let  $Q = \{\emptyset, \{1\}, \{2\}\}$ , still ordered by  $\subseteq$ . Both  $\{1\}$  and  $\{2\}$  are maximal: nothing in  $Q$  strictly contains either. Neither is above the other. The subset  $\{\{1\}, \{2\}\}$  has no upper bound in  $Q$ .

### Theorem 1.8.1 — Zorn's Lemma

Let  $(P, \leq)$  be a **non-empty** partially ordered set in which **every non-empty chain** has an upper bound in  $P$ . Then  $P$  has a maximal element.

We do not prove Zorn's Lemma, and no proof is possible from the other everyday rules of set theory. It is equivalent to the **axiom of choice**, the principle already used to build right inverses of surjections in Theorem 0.4.6: from any family of non-empty sets one may choose one element from each. Mathematicians accept it as an axiom, and we do too. In this section we point out each place where it is used.

*Warning.* “Maximal” is not “largest”. The partially ordered set  $Q$  in Example (Inclusion as a Partial Order) has two maximal elements and no largest one. For bases this is the familiar fact that a space has many bases: in  $\mathbb{R}^2$ , both  $\{\mathbf{e}_1, \mathbf{e}_2\}$  and  $\{(1, 1), (1, -1)\}$  are maximal independent sets, and neither contains the other.

## 1.8.3 Every vector space has a basis

We prove the more general extension statement first, since the existence of a basis is the case  $S = \emptyset$ .

### Theorem 1.8.2 — Basis Extension, General Form

Let  $V$  be a vector space over  $F$ , and let  $S \subseteq V$  be a linearly independent subset. Then there is a basis  $B$  of  $V$  with  $S \subseteq B$ .

*Idea.* ① Let  $P$  be the family of independent sets containing  $S$ , ordered by  $\subseteq$ . ② A chain has the union as an upper bound, because an independence test uses only finitely many vectors, and finitely many members of a chain all fit inside one of them. ③ Zorn gives a maximal  $B$ . ④ If some  $\mathbf{v}$  were outside  $\text{span}(B)$ , then  $B \cup \{\mathbf{v}\}$  would be a bigger independent set, so  $B$  spans.

*Proof.* Let  $P$  be the set of all linearly independent subsets  $T \subseteq V$  with  $S \subseteq T$ , partially ordered by  $\subseteq$  (see Example (Inclusion as a Partial Order)). Since  $S$  is independent,  $S \in P$ , so  $P$  is non-empty.

**Step 1: every non-empty chain has an upper bound.** Let  $C \subseteq P$  be a non-empty chain and let  $U = \bigcup_{T \in C} T$ . Every  $T \in C$  contains  $S$  and  $C$  is non-empty, so  $S \subseteq U$ , and  $T \subseteq U$  for every  $T \in C$ .

*Claim.* If  $T_1, \dots, T_k \in C$  with  $k \geq 1$ , then one of them contains all the others.

*Proof.* We use induction on  $k$ . For  $k = 1$  there is nothing to prove. Suppose the claim holds for  $k$ , and let  $T_1, \dots, T_{k+1} \in C$ . By the induction hypothesis some  $T_j$  contains  $T_1, \dots, T_k$ . Since  $C$  is a chain,  $T_j \subseteq T_{k+1}$  or  $T_{k+1} \subseteq T_j$ . In the first case  $T_{k+1}$  contains all of  $T_1, \dots, T_{k+1}$ , and in the second case  $T_j$  does. ■

Now let  $(\mathbf{u}_1, \dots, \mathbf{u}_k)$  be a list of distinct vectors of  $U$ . If  $k = 0$ , the empty list is independent. Otherwise, each  $\mathbf{u}_i$  lies in some  $T_i \in C$ , and by the Claim one of  $T_1, \dots, T_k$ , say  $T_j$ , contains all the  $\mathbf{u}_i$ . Since  $T_j$  is independent, the list  $(\mathbf{u}_1, \dots, \mathbf{u}_k)$  is independent. Hence  $U$  is independent, so  $U \in P$ , and  $U$  is an upper bound of  $C$ .

**Step 2: a maximal element.** By Step 1 and Theorem 1.8.1,  $P$  has a maximal element  $B$ . Thus  $B$  is independent and  $S \subseteq B$ .

**Step 3:  $B$  spans  $V$ .** Suppose not, and let  $\mathbf{v} \in V$  with  $\mathbf{v} \notin \text{span}(B)$ . Then  $\mathbf{v} \notin B$ , since  $B \subseteq \text{span}(B)$ . We show  $B \cup \{\mathbf{v}\}$  is independent. Take a list of distinct vectors of  $B \cup \{\mathbf{v}\}$ . If  $\mathbf{v}$  is not in the list, the list lies in  $B$  and is independent. Otherwise, after reordering (which does not affect independence, as addition is commutative), the list is  $(\mathbf{v}, \mathbf{w}_1, \dots, \mathbf{w}_m)$  with distinct  $\mathbf{w}_i \in B$ . Let

$$a\mathbf{v} + b_1\mathbf{w}_1 + \dots + b_m\mathbf{w}_m = \mathbf{0}.$$

If  $a \neq 0$ , then  $\mathbf{v} = -a^{-1}(b_1\mathbf{w}_1 + \dots + b_m\mathbf{w}_m) \in \text{span}(B)$ , a contradiction. So  $a = 0$ , and then  $b_1 = \dots = b_m = 0$  because  $B$  is independent. Hence  $B \cup \{\mathbf{v}\} \in P$ , it contains  $B$ , and it is not equal to  $B$ . This contradicts the maximality of  $B$ . Therefore  $\text{span}(B) = V$ , and  $B$  is a basis of  $V$  containing  $S$ . ■

Notice where finiteness went. Steps 1 and 3 each look at only finitely many vectors at a time, which is all that linear algebra ever does. The one infinite step, “keep enlarging until you cannot”, is handed to Zorn’s Lemma.

### Theorem 1.8.3 — Every Vector Space Has a Basis

Every vector space  $V$  over a field  $F$  has a basis.

*Proof.* The empty set is linearly independent. By Theorem 1.8.2 with  $S = \emptyset$ ,  $V$  has a basis. For  $V = \{\mathbf{0}\}$  that basis is  $\emptyset$ , since  $\text{span}(\emptyset) = \{\mathbf{0}\}$ . ■

So  $F^{\mathbb{N}}$  has a basis, and so do  $\mathbb{R}^{\mathbb{R}}$  and the continuous functions on  $[0, 1]$ . The theorem does not tell us what these bases are.

## 1.8.4 Existence without construction: $\mathbb{R}$ over $\mathbb{Q}$

The real numbers form a vector space over  $\mathbb{Q}$ : add reals as usual, and multiply a real by a rational. By Theorem 1.8.3 this space has a basis, often called a **Hamel basis**. Nobody has

ever written one down, and nobody will. It is known that the existence of such a basis cannot be proved without some form of the axiom of choice. What we **can** prove is that this basis is infinite. The tool is counting.

*A countable set is one whose elements can be listed as a sequence, possibly with repeats.*

### Definition 1.8.2 — Countable Set

A set  $A$  is **countable** if  $A = \emptyset$  or there is a **surjective** function  $\mathbb{N} \rightarrow A$ . Otherwise  $A$  is **uncountable**.

A surjection  $\sigma: \mathbb{N} \rightarrow A$  lists  $A$  as  $\sigma(0), \sigma(1), \sigma(2), \dots$ , and every element of  $A$  appears somewhere. Repeats are allowed, so every finite non-empty set is countable. The next lemma collects what we need.

**Lemma 1.8.4 — Countable Sets** (a) If  $A$  is countable and  $f: A \rightarrow A'$  is surjective, then  $A'$  is countable.  
 (b)  $\mathbb{N} \times \mathbb{N}$  is countable.  
 (c) If  $A$  and  $A'$  are countable, then  $A \times A'$  is countable.  
 (d)  $\mathbb{Q}$  is countable, and  $\mathbb{Q}^n$  is countable for every  $n \geq 1$ .

*Proof.*

- (a) If  $A = \emptyset$ , then  $A' = f(A) = \emptyset$ . Otherwise take a surjection  $\sigma: \mathbb{N} \rightarrow A$ . Then  $f \circ \sigma: \mathbb{N} \rightarrow A'$  is surjective by [Theorem 0.4.3](#) (a composite of surjections is surjective).
- (b) For  $n \in \mathbb{N}$ , let  $2^a$  be the largest power of 2 dividing  $n+1$ ; it exists because  $2^a \leq n+1$  bounds  $a$ . Then  $(n+1)/2^a$  is odd, so it equals  $2b+1$  for a unique  $b \in \mathbb{N}$ . Put  $g(n) = (a, b)$ . Given  $(a, b) \in \mathbb{N} \times \mathbb{N}$ , let  $n = 2^a(2b+1) - 1$ . The largest power of 2 dividing  $2^a(2b+1)$  is  $2^a$ , because  $2^{a+1} \mid 2^a(2b+1)$  would force  $2 \mid 2b+1$ . Hence  $g(n) = (a, b)$ , and  $g$  is surjective.
- (c) If  $A$  or  $A'$  is empty, so is  $A \times A'$ . Otherwise take surjections  $\sigma: \mathbb{N} \rightarrow A$  and  $\tau: \mathbb{N} \rightarrow A'$ . The map  $\mathbb{N} \times \mathbb{N} \rightarrow A \times A'$ ,  $(m, n) \mapsto (\sigma(m), \tau(n))$ , is surjective. By (b) and (a),  $A \times A'$  is countable.
- (d) The map  $\mathbb{N} \rightarrow \mathbb{Z}$  sending an even  $m$  to  $m/2$  and an odd  $m$  to  $-(m+1)/2$  is surjective, so  $\mathbb{Z}$  is countable. The map  $\mathbb{Z} \times \mathbb{N} \rightarrow \mathbb{Q}$ ,  $(p, q) \mapsto p/(q+1)$ , is surjective, so  $\mathbb{Q}$  is countable by (c) and (a). Finally  $\mathbb{Q}^{n+1}$  is  $\mathbb{Q}^n \times \mathbb{Q}$  after identifying  $(q_1, \dots, q_{n+1})$  with  $((q_1, \dots, q_n), q_{n+1})$ , so induction on  $n$  with (c) shows that  $\mathbb{Q}^n$  is countable for every  $n \geq 1$ .

■

The real numbers are not countable. This is a fact about  $\mathbb{R}$ , not about linear algebra, and its proof needs one property of  $\mathbb{R}$  that we take from analysis: the **nested interval property**. If  $I_0 \supseteq I_1 \supseteq I_2 \supseteq \dots$  are closed bounded intervals  $[a_n, b_n]$  with  $a_n \leq b_n$ , then some real number lies in every  $I_n$ .

**Lemma 1.8.5 —  $\mathbb{R}$  Is Uncountable**

Assuming the nested interval property, there is no surjection  $\mathbb{N} \rightarrow \mathbb{R}$ .

*Proof.* Let  $r: \mathbb{N} \rightarrow \mathbb{R}$  be any function. We build closed intervals  $I_0 \supseteq I_1 \supseteq \cdots$  of positive length with  $r(n) \notin I_n$ . Split  $[0, 1]$  into the thirds  $[0, \frac{1}{3}]$  and  $[\frac{2}{3}, 1]$ . They are disjoint, so  $r(0)$  misses at least one of them; let  $I_0$  be such a third. Given  $I_n = [a, b]$  with  $a < b$ , the intervals  $[a, a + \frac{b-a}{3}]$  and  $[b - \frac{b-a}{3}, b]$  are disjoint and lie in  $I_n$ , so  $r(n+1)$  misses one of them; let  $I_{n+1}$  be that one. By the nested interval property some  $x$  lies in every  $I_n$ . For each  $n$ ,  $x \in I_n$  and  $r(n) \notin I_n$ , so  $x \neq r(n)$ . Hence  $x$  is not a value of  $r$ , and  $r$  is not surjective. ■

**Theorem 1.8.6 —  $\mathbb{R}$  Is Infinite-Dimensional over  $\mathbb{Q}$** 

The vector space  $\mathbb{R}$  over  $\mathbb{Q}$  is not finite-dimensional.

*Proof.* Suppose a finite list spans  $\mathbb{R}$  over  $\mathbb{Q}$ . Appending 1 keeps it spanning, so we may take a spanning list  $(b_1, \dots, b_n)$  with  $n \geq 1$ . Then the map

$$\mathbb{Q}^n \rightarrow \mathbb{R}, \quad (q_1, \dots, q_n) \mapsto q_1 b_1 + \cdots + q_n b_n$$

is surjective, since every real number is a rational linear combination of  $b_1, \dots, b_n$ . By Lemma 1.8.4 (d) and (a),  $\mathbb{R}$  is countable. This contradicts Lemma 1.8.5. Hence  $\mathbb{R}$  is not finite-dimensional over  $\mathbb{Q}$ . ■

The proof really shows more: **every** finite-dimensional vector space over  $\mathbb{Q}$  is countable. It is honest about its inputs. Linear algebra supplied only “a spanning list gives a surjection from  $\mathbb{Q}^n$ ”; the rest is counting and one property of the real line.

*Quick check.* Is  $\mathbb{C}$  finite-dimensional as a vector space over  $\mathbb{Q}$ ?

*Solution.* No. If  $\mathbb{C}$  were countable, then  $\mathbb{R} \subseteq \mathbb{C}$  would be countable too: pick any  $y_0 \in \mathbb{R}$  and send  $z \in \mathbb{C}$  to  $z$  if  $z \in \mathbb{R}$  and to  $y_0$  otherwise, a surjection  $\mathbb{C} \rightarrow \mathbb{R}$ , and apply Lemma 1.8.4 (a). That contradicts Lemma 1.8.5. So  $\mathbb{C}$  is uncountable, and the proof of Theorem 1.8.6, with  $\mathbb{C}$  in place of  $\mathbb{R}$ , shows it is not finite-dimensional over  $\mathbb{Q}$ .

**1.8.5 What changes in infinite dimension**

Several finite-dimensional theorems use a count, and counting is exactly what fails here.

*Warning.* **A proper subspace can be as big as the whole space.** In finite dimension,  $U \subseteq V$  with  $\dim U = \dim V$  forces  $U = V$  (Theorem 1.6.8). Now let  $U = \{p \in F[x] : \text{the constant coefficient of } p \text{ is } 0\}$ . By the argument of Example (A Basis of  $F[x]$ ),  $U$  has the basis  $\{x, x^2, x^3, \dots\}$ , and  $x^k \leftrightarrow x^{k+1}$  matches it one-to-one



with the basis  $\{1, x, x^2, \dots\}$  of  $F[x]$ . Yet  $1 \notin U$ , so  $U \neq F[x]$ . The same happens in  $F^{\mathbb{N}}$ : the sequences with first entry 0 form a proper subspace, and deleting that first entry matches it one-to-one with all of  $F^{\mathbb{N}}$ , respecting addition and scaling. In Chapter 3 this becomes the shift map, which is injective but not surjective.

*Remark.* In an infinite-dimensional space, “ $\dim V = \infty$ ” hides real differences of size:  $F[x]$  has a countable basis, while every basis of  $\mathbb{R}$  over  $\mathbb{Q}$  is uncountable ([Exercise \(C2: Bases of  \$\mathbb{R}\$  over  \$\mathbb{Q}\$  Are Uncountable\)](#)), and set theory shows that all bases of one space have the same cardinality, a fact we will not develop.

**Scope.** From here on the book is mostly about finite-dimensional spaces, where bases are finite lists and counting works. Every theorem that needs finite dimension says so in its hypotheses. Infinite-dimensional spaces return with more structure, such as inner products and limits, where infinite sums can finally be given a meaning.

## 1.8.6 Exercises

### 1.8.6.1 A. Check your understanding

#### Exercise A1.

- (a) Define what it means for an arbitrary subset  $S$  of a vector space  $V$  to be a basis of  $V$ .
- (b) State Zorn’s Lemma, including the meaning of “chain” and “upper bound”.
- (c) True or false:  $\{\mathbf{e}_i : i \in \mathbb{N}\}$  is a basis of  $F^{\mathbb{N}}$ . Justify your answer.
- (d) True or false: every linearly independent subset of an infinite-dimensional space is infinite. Justify your answer.
- (e) True or false: in a partially ordered set, a maximal element  $m$  satisfies  $p \leq m$  for every  $p$ . Justify your answer.

*Solution.*

- (a)  $S$  is a basis if every finite list of distinct vectors of  $S$  is linearly independent, and every vector of  $V$  is a linear combination of finitely many vectors of  $S$ .
- (b) See [Theorem 1.8.1](#) and [Definition 1.8.1](#): a non-empty partially ordered set in which every non-empty chain (a subset whose elements are pairwise comparable) has an upper bound (an element of  $P$  above every element of the chain) has a maximal element.
- (c) False. The set is independent, but  $(1, 1, 1, \dots)$  is not a finite linear combination of the  $\mathbf{e}_i$  ([Example \(The Vectors  \$\mathbf{e}\_i\$  Do Not Span  \$F^{\mathbb{N}}\$ \)](#)).
- (d) False. The empty set and  $\{x\} \subseteq F[x]$  are finite independent subsets of the infinite-dimensional space  $F[x]$ .
- (e) False. In  $Q = \{\emptyset, \{1\}, \{2\}\}$  ordered by  $\subseteq$ , the element  $\{1\}$  is maximal, but  $\{2\} \not\subseteq \{1\}$  ([Example \(Inclusion as a Partial Order\)](#)).

### 1.8.6.2 B. Practice

**Exercise B1: Finitely Supported Sequences.**

Let  $F^{(\mathbb{N})} \subseteq F^{\mathbb{N}}$  be the set of sequences with only finitely many non-zero entries.

- (a) Prove that  $F^{(\mathbb{N})}$  is a subspace of  $F^{\mathbb{N}}$ .
- (b) Prove that  $\{\mathbf{e}_i : i \in \mathbb{N}\}$  is a basis of  $F^{(\mathbb{N})}$ .

*Solution.*

- (a) We use [Theorem 1.2.1](#). (1) The zero sequence has no non-zero entries, so  $\mathbf{0} \in F^{(\mathbb{N})}$ . (2) Let  $\mathbf{x}, \mathbf{y} \in F^{(\mathbb{N})}$ , with non-zero entries only in finite sets of positions  $A$  and  $A'$ . If  $i \notin A \cup A'$ , then  $x_i + y_i = 0 + 0 = 0$ . So  $\mathbf{x} + \mathbf{y}$  has non-zero entries only in the finite set  $A \cup A'$ , and  $\mathbf{x} + \mathbf{y} \in F^{(\mathbb{N})}$ . (3) Let  $c \in F$ . If  $i \notin A$ , then  $cx_i = c \cdot 0 = 0$ , so  $c\mathbf{x} \in F^{(\mathbb{N})}$ . Hence  $F^{(\mathbb{N})}$  is a subspace.
- (b) Each  $\mathbf{e}_i$  lies in  $F^{(\mathbb{N})}$ , and the set is independent by [Example \(The Vectors  \$\mathbf{e}\_i\$  Do Not Span  \$F^{\mathbb{N}}\$ \)](#). For spanning, let  $\mathbf{x} \in F^{(\mathbb{N})}$  have non-zero entries only in positions  $i_1, \dots, i_m$ . Then  $\mathbf{x}$  and  $x_{i_1}\mathbf{e}_{i_1} + \dots + x_{i_m}\mathbf{e}_{i_m}$  agree in positions  $i_1, \dots, i_m$  and are both 0 elsewhere, so they are equal. Hence  $\{\mathbf{e}_i : i \in \mathbb{N}\}$  is a basis of  $F^{(\mathbb{N})}$ .

**Exercise B2: Exponentials Are Independent.**

For  $k \in \mathbb{N}$  let  $f_k \in \mathbb{R}^{\mathbb{R}}$  be  $f_k(x) = e^{kx}$ . Prove that  $\{f_k : k \in \mathbb{N}\}$  is linearly independent. Hence deduce that  $\mathbb{R}^{\mathbb{R}}$  is infinite-dimensional.

*Hint: divide by the fastest-growing term.*

*Solution.* Suppose not. Then there are distinct  $k_1 < k_2 < \dots < k_m$  and scalars  $a_1, \dots, a_m$ , not all zero, with  $a_1e^{k_1x} + \dots + a_me^{k_mx} = 0$  for every  $x \in \mathbb{R}$ . Discarding terms with zero coefficient, we may assume  $a_m \neq 0$ . Dividing by  $e^{k_mx} > 0$  gives, for every  $x \in \mathbb{R}$ ,

$$a_m = - \sum_{j=1}^{m-1} a_j e^{(k_j - k_m)x}.$$

Since  $k_j - k_m < 0$  for  $j < m$ , each  $e^{(k_j - k_m)x} \rightarrow 0$  as  $x \rightarrow \infty$ , so the right side tends to 0. The left side is the constant  $a_m$ , so  $a_m = 0$ , a contradiction. Hence the set is independent.

For each  $n$ , the list  $(f_0, \dots, f_n)$  is independent of length  $n + 1$ . If a list of length  $m$  spanned  $\mathbb{R}^{\mathbb{R}}$ , [Theorem 1.6.1](#) would give  $n + 1 \leq m$  for every  $n$ , which is impossible. Hence  $\mathbb{R}^{\mathbb{R}}$  is infinite-dimensional.

**Exercise B3: Continuous Functions.**

Let  $C[0, 1]$  be the space of continuous functions  $[0, 1] \rightarrow \mathbb{R}$ . Prove that  $C[0, 1]$  is infinite-dimensional.

*Hint: use “tent” functions supported on disjoint intervals, and choose good points.*

*Solution.* For  $k \geq 1$ , let  $m_k = \frac{1}{2} \left( \frac{1}{k+1} + \frac{1}{k} \right)$  and  $r_k = \frac{1}{2} \left( \frac{1}{k} - \frac{1}{k+1} \right) > 0$ , the midpoint

and half-length of  $J_k = [\frac{1}{k+1}, \frac{1}{k}]$ . Define

$$t_k(x) = \max(0, r_k - |x - m_k|).$$

This is continuous, as built from continuous functions by  $|\cdot|$  and  $\max$ . Also  $t_k(x) > 0$  exactly when  $|x - m_k| < r_k$ , that is, when  $x$  lies in the open interval  $(\frac{1}{k+1}, \frac{1}{k})$ . These open intervals are pairwise disjoint, and  $m_j$  lies in the  $j$ -th one. Hence  $t_k(m_j) = 0$  for  $j \neq k$ , while  $t_k(m_k) = r_k$ .

Let  $a_1 t_1 + \cdots + a_n t_n = 0$ . Evaluating at  $m_j$  gives  $a_j r_j = 0$ , so  $a_j = 0$  since  $r_j \neq 0$ . Hence  $(t_1, \dots, t_n)$  is independent for every  $n$ . As in [Exercise \(B2: Exponentials Are Independent\)](#), [Theorem 1.6.1](#) shows that no finite list spans  $C[0, 1]$ .

### 1.8.6.3 C. Going deeper

#### Exercise C1: Complements in Any Dimension.

Let  $U$  be a subspace of a vector space  $V$ , of any dimension. Prove that there is a subspace  $W$  of  $V$  with  $V = U \oplus W$ , that is, every  $\mathbf{v} \in V$  can be written **uniquely** as  $\mathbf{u} + \mathbf{w}$  with  $\mathbf{u} \in U$ ,  $\mathbf{w} \in W$ .

*Hint:* [Theorem 1.8.2](#).

*Solution.* By [Theorem 1.8.3](#),  $U$  has a basis  $B_U$ . It is independent in  $V$ , so by [Theorem 1.8.2](#) there is a basis  $B$  of  $V$  with  $B_U \subseteq B$ . Let  $W = \text{span}(B \setminus B_U)$ , a subspace by [Theorem 1.3.1](#).

*Existence.* Let  $\mathbf{v} \in V$ . Since  $B$  spans,  $\mathbf{v} = \sum_i a_i \mathbf{u}_i + \sum_j b_j \mathbf{w}_j$ , a finite sum with  $\mathbf{u}_i \in B_U$  and  $\mathbf{w}_j \in B \setminus B_U$ . The first sum lies in  $U$  and the second in  $W$ .

$U \cap W = \{\mathbf{0}\}$ . Let  $\mathbf{v} \in U \cap W$ . Since  $B_U$  spans  $U$  and  $B \setminus B_U$  spans  $W$ , we have  $\mathbf{v} = \sum_i a_i \mathbf{u}_i = \sum_j b_j \mathbf{w}_j$  with distinct  $\mathbf{u}_i \in B_U$  and distinct  $\mathbf{w}_j \in B \setminus B_U$ . Then  $\sum_i a_i \mathbf{u}_i - \sum_j b_j \mathbf{w}_j = \mathbf{0}$  is a combination of a list of distinct vectors of  $B$ , because  $B_U$  and  $B \setminus B_U$  are disjoint. Since  $B$  is independent, every  $a_i$  and  $b_j$  is 0, so  $\mathbf{v} = \mathbf{0}$ .

*Uniqueness.* If  $\mathbf{u} + \mathbf{w} = \mathbf{u}' + \mathbf{w}'$  with  $\mathbf{u}, \mathbf{u}' \in U$  and  $\mathbf{w}, \mathbf{w}' \in W$ , then  $\mathbf{u} - \mathbf{u}' = \mathbf{w}' - \mathbf{w}$  lies in  $U \cap W = \{\mathbf{0}\}$ . Hence  $\mathbf{u} = \mathbf{u}'$  and  $\mathbf{w} = \mathbf{w}'$ . This proves  $V = U \oplus W$ .

#### Exercise C2: Bases of $\mathbb{R}$ over $\mathbb{Q}$ Are Uncountable.

- (a) Let  $A_0, A_1, A_2, \dots$  be countable sets. Prove that  $\bigcup_{n \in \mathbb{N}} A_n$  is countable. (You may use the axiom of choice.)
- (b) Let  $V$  be a vector space over  $\mathbb{Q}$  with a countable basis. Prove that  $V$  is countable.
- (c) Deduce that every basis of  $\mathbb{R}$  over  $\mathbb{Q}$  is uncountable.

*Hint:* for (b), write  $V$  as a union of spans of finite lists.

*Solution.*

- (a) Let  $A = \bigcup_n A_n$ . If  $A = \emptyset$  there is nothing to prove; otherwise fix  $a^* \in A$ . For each  $n$  with  $A_n \neq \emptyset$ , choose a surjection  $\sigma_n: \mathbb{N} \rightarrow A_n$  (this is where choice is used). Define  $h: \mathbb{N} \times \mathbb{N} \rightarrow A$  by  $h(n, m) = \sigma_n(m)$  if  $A_n \neq \emptyset$ , and  $h(n, m) = a^*$  otherwise.

Every  $a \in A$  lies in some non-empty  $A_n$  and equals  $\sigma_n(m)$  for some  $m$ , so  $h$  is surjective. By Lemma 1.8.4 (b) and (a),  $A$  is countable.

- (b) If the basis is empty,  $V = \{\mathbf{0}\}$  is countable. Otherwise list the basis as  $\sigma(0), \sigma(1), \dots$  for a surjection  $\sigma: \mathbb{N} \rightarrow B$ . Every  $\mathbf{v} \in V$  is a combination of finitely many basis vectors, all among  $\sigma(0), \dots, \sigma(n)$  for some  $n$ . Hence  $V = \bigcup_n V_n$  with  $V_n = \text{span}(\sigma(0), \dots, \sigma(n))$ . The map  $\mathbb{Q}^{n+1} \rightarrow V_n$ ,  $(q_0, \dots, q_n) \mapsto \sum_i q_i \sigma(i)$ , is surjective, so each  $V_n$  is countable by Lemma 1.8.4 (d) and (a). By (a),  $V$  is countable.
- (c) By Theorem 1.8.3,  $\mathbb{R}$  has a basis over  $\mathbb{Q}$ . If some basis were countable, then  $\mathbb{R}$  would be countable by (b), contradicting Lemma 1.8.5. Hence every basis is uncountable.

## 1.9 Changing the Field

Every vector space comes with its field, and we have already seen that the field matters:  $\dim_{\mathbb{C}} \mathbb{C} = 1$  but  $\dim_{\mathbb{R}} \mathbb{C} = 2$  (Example (Dimensions of the Running Examples)). This section explains the pattern behind that example. Shrinking the field multiplies the dimension by a fixed factor, the **tower law**. Going the other way, every real vector space can be enlarged to a complex one, its **complexification**, without changing the dimension. Both moves are used later, most of all when real matrices need complex eigenvalues.

### 1.9.1 Restricting the scalars

Start with the standard example. The space  $\mathbb{C}^2$  over  $\mathbb{C}$  has the basis  $(\mathbf{e}_1, \mathbf{e}_2)$ , so its dimension is 2. But nothing stops us from multiplying vectors of  $\mathbb{C}^2$  only by **real** numbers. The addition is the same, and the axioms still hold, because they hold for all complex scalars. So  $\mathbb{C}^2$  is also a vector space over  $\mathbb{R}$ . What is its dimension now?

#### Example. ( $\mathbb{C}^2$ as a Real Vector Space)

Show that  $(\mathbf{e}_1, i\mathbf{e}_1, \mathbf{e}_2, i\mathbf{e}_2)$  is a basis of  $\mathbb{C}^2$  over  $\mathbb{R}$ , and write  $(1 + 2i, 3 - i)$  in it.

*Solution. Spanning.* Let  $(z_1, z_2) \in \mathbb{C}^2$  and write  $z_k = a_k + b_k i$  with  $a_k, b_k \in \mathbb{R}$ . Then

$$(z_1, z_2) = a_1 \mathbf{e}_1 + b_1 (i\mathbf{e}_1) + a_2 \mathbf{e}_2 + b_2 (i\mathbf{e}_2),$$

a combination with **real** coefficients. For example,  $(1 + 2i, 3 - i) = 1\mathbf{e}_1 + 2(i\mathbf{e}_1) + 3\mathbf{e}_2 + (-1)(i\mathbf{e}_2)$ .

*Independence.* Let  $a_1, b_1, a_2, b_2 \in \mathbb{R}$  with  $a_1 \mathbf{e}_1 + b_1 (i\mathbf{e}_1) + a_2 \mathbf{e}_2 + b_2 (i\mathbf{e}_2) = \mathbf{0}$ . The left side is  $(a_1 + b_1 i, a_2 + b_2 i)$ . Each entry is 0, and comparing real and imaginary parts (Definition 0.6.1) gives  $a_1 = b_1 = a_2 = b_2 = 0$ .

Hence  $\dim_{\mathbb{R}} \mathbb{C}^2 = 4 = 2 \dim_{\mathbb{C}} \mathbb{C}^2$ .

The general setting has a big field  $L$ , a smaller field  $K$  inside it, and a vector space over  $L$ .

**Definition 1.9.1 — Subfield, Restriction of Scalars, Degree**

Let  $L$  be a field. A **subfield** of  $L$  is a subset  $K \subseteq L$  that contains 0 and 1 and is closed under addition, multiplication, negatives, and inverses of **non-zero** elements.

Let  $K$  be a subfield of  $L$ , and let  $V$  be a vector space over  $L$ . Keeping the addition of  $V$  and using scalar multiplication only for scalars in  $K$  makes  $V$  a vector space **over**  $K$ . We say  $V$  is obtained by **restriction of scalars**.

In particular  $L$  itself, a vector space over  $L$ , is a vector space over  $K$ . If it is finite-dimensional, its dimension is the **degree**  $[L : K] := \dim_K L$ .

In words: a subfield is a field sitting inside  $L$  with the same operations. Restriction of scalars forgets how to multiply by elements of  $L$  outside  $K$ .

There are two things to check. First,  $K$  with the operations of  $L$  is a field. Closure makes the operations land in  $K$ , the elements 0, 1, negatives and inverses are in  $K$  by assumption, and the remaining axioms are identities that hold for all elements of  $L$ , so in particular for those of  $K$ . Second,  $V$  over  $K$  is a vector space. Scalar multiplication  $K \times V \rightarrow V$  lands in  $V$ , and each axiom of [Definition 1.1.1](#) holds **for all** scalars in  $L$ , so in particular for all scalars in  $K$ .

We always write  $\dim_K V$  or  $\dim_L V$  when both fields are in play.

**Example. (Degrees of Some Fields)**

Find  $[\mathbb{C} : \mathbb{R}]$ ,  $[\mathbb{Q}(\sqrt{2}) : \mathbb{Q}]$  and  $[K : K]$  for any field  $K$ . Is  $[\mathbb{R} : \mathbb{Q}]$  defined?

*Solution.*  $[\mathbb{C} : \mathbb{R}] = 2$ . Every  $z \in \mathbb{C}$  is  $a \cdot 1 + b \cdot i$  with  $a, b \in \mathbb{R}$ , and  $a + bi = 0$  forces  $a = b = 0$  by [Definition 0.6.1](#). So  $(1, i)$  is a basis of  $\mathbb{C}$  over  $\mathbb{R}$ .

$[\mathbb{Q}(\sqrt{2}) : \mathbb{Q}] = 2$ . The set  $\mathbb{Q}(\sqrt{2}) = \{a + b\sqrt{2} : a, b \in \mathbb{Q}\}$  is a subfield of  $\mathbb{R}$  by [Example  \$\(\mathbb{Q}\(\sqrt{2}\) \text{ Is a Field}\)\$](#) . The list  $(1, \sqrt{2})$  spans it over  $\mathbb{Q}$  by definition. For independence, let  $a + b\sqrt{2} = 0$  with  $a, b \in \mathbb{Q}$ . If  $b \neq 0$ , then  $\sqrt{2} = -a/b \in \mathbb{Q}$ , contradicting [Theorem 0.2.3](#). So  $b = 0$ , and then  $a = 0$ .

$[K : K] = 1$ . The list  $(1)$  is a basis of  $K$  over  $K$ : every  $a$  equals  $a \cdot 1$ , and  $a \cdot 1 = 0$  forces  $a = 0$ . This degenerate case says that restricting to the same field changes nothing.

$[\mathbb{R} : \mathbb{Q}]$  is not defined, because  $\mathbb{R}$  is not finite-dimensional over  $\mathbb{Q}$  ([Theorem 1.8.6](#)).

Restriction only goes **down**. The set  $\mathbb{R}$  with the operations of  $\mathbb{C}$  is **not** a vector space over  $\mathbb{C}$ : everything about addition works, but scalar multiplication leaves the set, since  $i \cdot 1 = i \notin \mathbb{R}$ . Enlarging the field needs a construction, which is the second half of this section.

**1.9.2 The tower law**

In [Example  \$\(\mathbb{C}^2 \text{ as a Real Vector Space}\)\$](#)  we took a vector, wrote it in coordinates over  $\mathbb{C}$ , and then split each complex coordinate into real coordinates. The same two-stage process works for any fields.

**Theorem 1.9.1 — Tower Law**

Let  $K$  be a subfield of  $L$ , and let  $V$  be a vector space over  $L$ . Suppose  $(a_1, \dots, a_m)$  is a basis of  $L$  over  $K$  and  $(\mathbf{v}_1, \dots, \mathbf{v}_n)$  is a basis of  $V$  over  $L$ . Then the list of the  $mn$  vectors

$$(a_1\mathbf{v}_1, \dots, a_m\mathbf{v}_1, a_1\mathbf{v}_2, \dots, a_m\mathbf{v}_2, \dots, a_1\mathbf{v}_n, \dots, a_m\mathbf{v}_n)$$

is a basis of  $V$  over  $K$ . In particular,

$$\dim_K V = [L : K] \cdot \dim_L V.$$

*Idea.* Coordinates in two stages. A vector is an  $L$ -combination of the  $\mathbf{v}_j$ ; each  $L$ -coefficient is a  $K$ -combination of the  $a_i$ . Substituting gives a double sum  $\sum_j \sum_i b_{ij} a_i \mathbf{v}_j$  with  $b_{ij} \in K$ . For independence, run the same two stages backwards: regroup the double sum by  $j$ , use independence over  $L$ , then independence over  $K$ .

*Proof.* Throughout we use two axioms of Definition 1.1.1 for scalars in  $L$ :  $(c + c')\mathbf{x} = c\mathbf{x} + c'\mathbf{x}$  and  $c(c'\mathbf{x}) = (cc')\mathbf{x}$ .

*Spanning.* Let  $\mathbf{x} \in V$ . Since  $(\mathbf{v}_1, \dots, \mathbf{v}_n)$  spans  $V$  over  $L$ , there are  $c_1, \dots, c_n \in L$  with  $\mathbf{x} = \sum_{j=1}^n c_j \mathbf{v}_j$ . Since  $(a_1, \dots, a_m)$  spans  $L$  over  $K$ , for each  $j$  there are  $b_{1j}, \dots, b_{mj} \in K$  with  $c_j = \sum_{i=1}^m b_{ij} a_i$ . Therefore

$$\mathbf{x} = \sum_{j=1}^n \left( \sum_{i=1}^m b_{ij} a_i \right) \mathbf{v}_j = \sum_{j=1}^n \sum_{i=1}^m (b_{ij} a_i) \mathbf{v}_j = \sum_{j=1}^n \sum_{i=1}^m b_{ij} (a_i \mathbf{v}_j),$$

where the second equality uses  $(c + c')\mathbf{x} = c\mathbf{x} + c'\mathbf{x}$  and the third uses  $c(c'\mathbf{x}) = (cc')\mathbf{x}$ . This is a  $K$ -linear combination of the vectors  $a_i \mathbf{v}_j$ .

*Independence.* Let  $b_{ij} \in K$  with  $\sum_{j=1}^n \sum_{i=1}^m b_{ij} (a_i \mathbf{v}_j) = \mathbf{0}$ . Reading the display above from right to left,

$$\sum_{j=1}^n \left( \sum_{i=1}^m b_{ij} a_i \right) \mathbf{v}_j = \mathbf{0}.$$

Each coefficient  $\sum_i b_{ij} a_i$  lies in  $L$ , and  $(\mathbf{v}_1, \dots, \mathbf{v}_n)$  is independent over  $L$ . Hence  $\sum_{i=1}^m b_{ij} a_i = 0$  for every  $j$ . For each fixed  $j$ , this is a  $K$ -combination of  $(a_1, \dots, a_m)$  equal to 0, and that list is independent over  $K$ . Hence  $b_{ij} = 0$  for all  $i$  and  $j$ .

So the  $mn$  vectors  $a_i \mathbf{v}_j$  form a basis of  $V$  over  $K$ , and  $\dim_K V = mn = [L : K] \dim_L V$ . ■

The name comes from the picture of fields stacked in a tower,  $K \subseteq L$ , with  $V$  on top: dimensions multiply as you go down. The most important case is  $\mathbb{R} \subseteq \mathbb{C}$ .

**Theorem 1.9.2 — Complex Spaces as Real Spaces**

Let  $V$  be a vector space over  $\mathbb{C}$  with basis  $(\mathbf{v}_1, \dots, \mathbf{v}_n)$ . Then  $(\mathbf{v}_1, i\mathbf{v}_1, \dots, \mathbf{v}_n, i\mathbf{v}_n)$  is a basis of  $V$  over  $\mathbb{R}$ . In particular  $\dim_{\mathbb{R}} V = 2 \dim_{\mathbb{C}} V$ .

*Proof.* By [Example \(Degrees of Some Fields\)](#),  $(1, i)$  is a basis of  $\mathbb{C}$  over  $\mathbb{R}$ . By [Theorem 1.9.1](#) with  $a_1 = 1$  and  $a_2 = i$ , the list  $(1\mathbf{v}_1, i\mathbf{v}_1, \dots, 1\mathbf{v}_n, i\mathbf{v}_n)$  is a basis of  $V$  over  $\mathbb{R}$ , and  $1\mathbf{v}_j = \mathbf{v}_j$  by [Definition 1.1.1](#). The dimension is  $2n$ . ■

This is the result of [Exercise \(C2: A complex space viewed over the reals\)](#), now as a case of the tower law. For example,  $\dim_{\mathbb{R}} \mathbb{C}^n = 2n$ , and  $\dim_{\mathbb{R}} \mathbb{C}[x]_{\leq n} = 2(n+1)$ , with basis  $(1, i, x, ix, \dots, x^n, ix^n)$ .

*Warning.* A subset of  $\mathbb{C}^n$  can be a subspace over  $\mathbb{R}$  without being a subspace over  $\mathbb{C}$ . The real vectors  $\mathbb{R}^n \subseteq \mathbb{C}^n$  contain  $\mathbf{0}$  and are closed under addition and under multiplication by **real** scalars, so they form an  $\mathbb{R}$ -subspace of dimension  $n$ . But  $\mathbf{e}_1 \in \mathbb{R}^n$  and  $i\mathbf{e}_1 \notin \mathbb{R}^n$ , so  $\mathbb{R}^n$  is **not** a  $\mathbb{C}$ -subspace. Always ask: a subspace over which field?

*Quick check.* What is  $\dim_{\mathbb{R}} M_{2 \times 3}(\mathbb{C})$ ? What is  $\dim_{\mathbb{Q}} \mathbb{Q}(\sqrt{2})^2$ ?

*Solution.*  $M_{2 \times 3}(\mathbb{C})$  has dimension 6 over  $\mathbb{C}$  ([Example \(Dimensions of the Running Examples\)](#)), so  $\dim_{\mathbb{R}} = 2 \cdot 6 = 12$  by [Theorem 1.9.2](#). The space  $\mathbb{Q}(\sqrt{2})^2$  has dimension 2 over the field  $\mathbb{Q}(\sqrt{2})$ , and  $[\mathbb{Q}(\sqrt{2}) : \mathbb{Q}] = 2$  by [Example \(Degrees of Some Fields\)](#), so  $\dim_{\mathbb{Q}} = 2 \cdot 2 = 4$  by [Theorem 1.9.1](#).

## 1.9.3 Complexification

Now go up. A real matrix such as the rotation  $\begin{pmatrix} 0 & -1 \\ 1 & 0 \end{pmatrix}$  moves every non-zero vector of  $\mathbb{R}^2$  off its own line. In  $\mathbb{C}^2$ , however, it sends  $(1, -i)$  to  $(i, 1) = i(1, -i)$ . To use such complex vectors for an arbitrary real space  $V$ , not just  $\mathbb{R}^n$ , we need a complex space that contains  $V$  the way  $\mathbb{C}^n$  contains  $\mathbb{R}^n$ .

The model is  $\mathbb{C}^n$  itself. Every  $\mathbf{z} \in \mathbb{C}^n$  splits as  $\mathbf{z} = \mathbf{x} + i\mathbf{y}$  with  $\mathbf{x}, \mathbf{y} \in \mathbb{R}^n$ , its real and imaginary parts, taken entry by entry. Multiplying by  $a + bi$  with  $a, b \in \mathbb{R}$  gives

$$(a + bi)(\mathbf{x} + i\mathbf{y}) = (a\mathbf{x} - b\mathbf{y}) + i(a\mathbf{y} + b\mathbf{x}).$$

The right side only uses real scalars acting on real vectors. So we can copy it for any real space, exactly as  $\mathbb{C}$  was built from pairs of real numbers in [Definition 0.6.1](#).

*The complexification of  $V$  is the space of formal expressions  $\mathbf{u} + i\mathbf{v}$  with  $\mathbf{u}, \mathbf{v} \in V$ , multiplied by complex numbers the way the rules of algebra dictate.*

### Definition 1.9.2 — Complexification

Let  $V$  be a vector space over  $\mathbb{R}$ . The **complexification** of  $V$  is the set  $V_{\mathbb{C}} := V \times V$  with addition and scalar multiplication by  $\mathbb{C}$  defined, **for all**  $(\mathbf{u}, \mathbf{v}), (\mathbf{u}', \mathbf{v}') \in V_{\mathbb{C}}$  and  $a, b \in \mathbb{R}$ , by

$$(\mathbf{u}, \mathbf{v}) + (\mathbf{u}', \mathbf{v}') := (\mathbf{u} + \mathbf{u}', \mathbf{v} + \mathbf{v}'), \quad (a + bi)(\mathbf{u}, \mathbf{v}) := (a\mathbf{u} - b\mathbf{v}, a\mathbf{v} + b\mathbf{u}).$$

We write  $\mathbf{u} + i\mathbf{v}$  for  $(\mathbf{u}, \mathbf{v})$ , and  $\mathbf{u}$  for  $(\mathbf{u}, \mathbf{0})$ .

In words: addition is entry by entry. Multiplication by  $a + bi$  is the formula forced by  $i^2 = -1$ . The notation is honest:  $i(\mathbf{v}, \mathbf{0}) = (0\mathbf{v} - 1\mathbf{0}, 0\mathbf{0} + 1\mathbf{v}) = (\mathbf{0}, \mathbf{v})$ , by [Theorem 1.1.5](#) and [Theorem 1.1.6](#), so  $(\mathbf{u}, \mathbf{0}) + i(\mathbf{v}, \mathbf{0}) = (\mathbf{u}, \mathbf{v})$ . Also a **real** scalar acts entrywise:  $a(\mathbf{u}, \mathbf{v}) = (a\mathbf{u}, a\mathbf{v})$ . So  $V_{\mathbb{C}}$ , restricted to real scalars, is just two copies of  $V$ .

**Why  $V_{\mathbb{C}}$  is a vector space over  $\mathbb{C}$ .** The operations land in  $V \times V$ . Addition is entrywise, so (commutativity, associativity, zero, negatives) follow from the same axioms in each entry of  $V$ , with zero  $(\mathbf{0}, \mathbf{0})$  and negative  $(-\mathbf{u}, -\mathbf{v})$ . We check the scalar axioms that involve the multiplication of  $\mathbb{C}$ ; below,  $a\mathbf{u} - b\mathbf{v}$  means  $a\mathbf{u} + (-b)\mathbf{v}$ , which is the same by [Theorem 1.1.8](#). Let  $\alpha = a + bi$  and  $\beta = c + di$  with  $a, b, c, d \in \mathbb{R}$ , and  $\mathbf{z} = (\mathbf{u}, \mathbf{v})$ .

- $1\mathbf{z} = \mathbf{z}$ : we get  $(1\mathbf{u} - 0\mathbf{v}, 1\mathbf{v} + 0\mathbf{u}) = (\mathbf{u}, \mathbf{v})$ , since  $0\mathbf{v} = \mathbf{0}$  by [Theorem 1.1.5](#).
- $\alpha(\beta\mathbf{z}) = (\alpha\beta)\mathbf{z}$ : since  $\beta\mathbf{z} = (c\mathbf{u} - d\mathbf{v}, c\mathbf{v} + d\mathbf{u})$ , the axioms of  $V$  give

$$\begin{aligned}\alpha(\beta\mathbf{z}) &= (a(c\mathbf{u} - d\mathbf{v}) - b(c\mathbf{v} + d\mathbf{u}), a(c\mathbf{v} + d\mathbf{u}) + b(c\mathbf{u} - d\mathbf{v})) \\ &= ((ac - bd)\mathbf{u} - (ad + bc)\mathbf{v}, (ac - bd)\mathbf{v} + (ad + bc)\mathbf{u}).\end{aligned}$$

Since  $\alpha\beta = (ac - bd) + (ad + bc)i$ , this is  $(\alpha\beta)\mathbf{z}$ .

- $(\alpha + \beta)\mathbf{z} = \alpha\mathbf{z} + \beta\mathbf{z}$ : with  $\alpha + \beta = (a + c) + (b + d)i$ ,

$$\begin{aligned}(\alpha + \beta)\mathbf{z} &= ((a + c)\mathbf{u} - (b + d)\mathbf{v}, (a + c)\mathbf{v} + (b + d)\mathbf{u}) \\ &= (a\mathbf{u} - b\mathbf{v}, a\mathbf{v} + b\mathbf{u}) + (c\mathbf{u} - d\mathbf{v}, c\mathbf{v} + d\mathbf{u}) \\ &= \alpha\mathbf{z} + \beta\mathbf{z}.\end{aligned}$$

The remaining axiom,  $\alpha(\mathbf{z} + \mathbf{w}) = \alpha\mathbf{z} + \alpha\mathbf{w}$ , is [Exercise \(B3: Working in a Complexification\)](#).

Here is the payoff: complexifying keeps the dimension, and even keeps the basis.

### Theorem 1.9.3 — A Real Basis Is a Complex Basis of the Complexification

Let  $V$  be a vector space over  $\mathbb{R}$  with basis  $(\mathbf{v}_1, \dots, \mathbf{v}_n)$ . Then  $(\mathbf{v}_1, \dots, \mathbf{v}_n)$ , that is,  $((\mathbf{v}_1, \mathbf{0}), \dots, (\mathbf{v}_n, \mathbf{0}))$ , is a basis of  $V_{\mathbb{C}}$  over  $\mathbb{C}$ . In particular  $\dim_{\mathbb{C}} V_{\mathbb{C}} = \dim_{\mathbb{R}} V$ .

*Idea.* A complex coefficient  $a_j + b_j i$  on  $\mathbf{v}_j$  puts  $a_j \mathbf{v}_j$  into the real part and  $b_j \mathbf{v}_j$  into the imaginary part. So one complex combination of the  $\mathbf{v}_j$  is the same as two real combinations, one for each part, and both questions (spanning, independence) split into two real questions about  $(\mathbf{v}_1, \dots, \mathbf{v}_n)$ .

*Proof.* For  $a, b \in \mathbb{R}$  and  $\mathbf{x} \in V$ , [Definition 1.9.2](#) and [Theorem 1.1.6](#) give

$$(a + bi)(\mathbf{x}, \mathbf{0}) = (a\mathbf{x} - b\mathbf{0}, a\mathbf{0} + b\mathbf{x}) = (a\mathbf{x}, b\mathbf{x}).$$

Call this identity (\*).

*Spanning.* Let  $(\mathbf{u}, \mathbf{v}) \in V_{\mathbb{C}}$ . Since  $(\mathbf{v}_1, \dots, \mathbf{v}_n)$  spans  $V$ , there are  $a_j, b_j \in \mathbb{R}$  with  $\mathbf{u} = \sum_j a_j \mathbf{v}_j$  and  $\mathbf{v} = \sum_j b_j \mathbf{v}_j$ . By (\*) and entrywise addition,

$$\sum_{j=1}^n (a_j + b_j i)(\mathbf{v}_j, \mathbf{0}) = \sum_{j=1}^n (a_j \mathbf{v}_j, b_j \mathbf{v}_j) = \left( \sum_{j=1}^n a_j \mathbf{v}_j, \sum_{j=1}^n b_j \mathbf{v}_j \right) = (\mathbf{u}, \mathbf{v}).$$



*Independence.* Let  $\alpha_1, \dots, \alpha_n \in \mathbb{C}$  with  $\sum_j \alpha_j(\mathbf{v}_j, \mathbf{0}) = (\mathbf{0}, \mathbf{0})$ , and write  $\alpha_j = a_j + b_j i$  with  $a_j, b_j \in \mathbb{R}$ . By the same computation,  $(\sum_j a_j \mathbf{v}_j, \sum_j b_j \mathbf{v}_j) = (\mathbf{0}, \mathbf{0})$ . Since  $(\mathbf{v}_1, \dots, \mathbf{v}_n)$  is independent over  $\mathbb{R}$ , all  $a_j = 0$  and all  $b_j = 0$ . Hence every  $\alpha_j = 0$ .

Therefore  $(\mathbf{v}_1, \dots, \mathbf{v}_n)$  is a basis of  $V_{\mathbb{C}}$ , and  $\dim_{\mathbb{C}} V_{\mathbb{C}} = n$ . ■

As a consistency check, restricting  $V_{\mathbb{C}}$  back to  $\mathbb{R}$  gives  $\dim_{\mathbb{R}} V_{\mathbb{C}} = 2n$  by [Theorem 1.9.2](#), which matches the fact that  $V_{\mathbb{C}}$  over  $\mathbb{R}$  is two copies of  $V$ . The same proof works word for word for an infinite basis set, since every combination involves finitely many vectors.

### Example. (The Complexification of $\mathbb{R}^n$ Is $\mathbb{C}^n$ )

Show that the map  $(\mathbf{x}, \mathbf{y}) \mapsto \mathbf{x} + i\mathbf{y}$ , from  $(\mathbb{R}^n)_{\mathbb{C}}$  to  $\mathbb{C}^n$ , is a bijection that respects addition and multiplication by complex scalars.

*Solution.* Here  $\mathbf{x} + i\mathbf{y}$  is computed in  $\mathbb{C}^n$ : its  $k$ -th entry is  $x_k + y_k i$ . Every  $\mathbf{z} \in \mathbb{C}^n$  has unique real and imaginary parts entry by entry ([Definition 0.6.1](#)), so exactly one pair  $(\mathbf{x}, \mathbf{y})$  of real vectors maps to  $\mathbf{z}$ . Hence the map is a bijection.

Addition is entrywise on both sides, so it is respected. For scaling, let  $a, b \in \mathbb{R}$ . In  $(\mathbb{R}^n)_{\mathbb{C}}$ ,  $(a + bi)(\mathbf{x}, \mathbf{y}) = (a\mathbf{x} - b\mathbf{y}, a\mathbf{y} + b\mathbf{x})$ , which maps to the vector with entries  $(ax_k - by_k) + (ay_k + bx_k)i$ . In  $\mathbb{C}^n$ , the  $k$ -th entry of  $(a + bi)(\mathbf{x} + i\mathbf{y})$  is

$$(a + bi)(x_k + y_k i) = (ax_k - by_k) + (ay_k + bx_k)i.$$

The two agree, so the map respects scalar multiplication. It sends the basis  $(\mathbf{e}_1, \dots, \mathbf{e}_n)$  of  $(\mathbb{R}^n)_{\mathbb{C}}$  from [Theorem 1.9.3](#) to the standard basis of  $\mathbb{C}^n$ .

So  $(\mathbb{R}^n)_{\mathbb{C}}$  is  $\mathbb{C}^n$  in disguise; in Chapter 3 such a bijection will be called an isomorphism. In the same way  $(M_n(\mathbb{R}))_{\mathbb{C}}$  is  $M_n(\mathbb{C})$ , and  $(\mathbb{R}[x]_{\leq n})_{\mathbb{C}}$  is  $\mathbb{C}[x]_{\leq n}$ . The construction earns its keep in Chapter 8, where real matrices without real eigenvalues, like the rotation above, get complex ones, and in Chapter 11, in the spectral theory of real operators.

## 1.9.4 Exercises

### 1.9.4.1 A. Check your understanding

#### Exercise A1.

- Define the degree  $[L : K]$  of a subfield  $K \subseteq L$ .
- State the Tower Law.
- What is  $\dim_{\mathbb{R}} \mathbb{C}^3$ ? Give a basis.
- True or false: every  $\mathbb{C}$ -subspace of  $\mathbb{C}^n$  is an  $\mathbb{R}$ -subspace. Justify your answer.
- What is  $\dim_{\mathbb{C}}(M_2(\mathbb{R}))_{\mathbb{C}}$ ?

*Solution.*

- (a)  $[L : K] = \dim_K L$ , where  $L$  is regarded as a vector space over  $K$  by restriction of scalars (Definition 1.9.1), when this dimension is finite.
- (b) See Theorem 1.9.1: if  $(a_i)$  is a basis of  $L$  over  $K$  and  $(\mathbf{v}_j)$  is a basis of  $V$  over  $L$ , then  $(a_i \mathbf{v}_j)$  is a basis of  $V$  over  $K$ , so  $\dim_K V = [L : K] \dim_L V$ .
- (c) 6, with basis  $(\mathbf{e}_1, i\mathbf{e}_1, \mathbf{e}_2, i\mathbf{e}_2, \mathbf{e}_3, i\mathbf{e}_3)$ , by Theorem 1.9.2.
- (d) True. A  $\mathbb{C}$ -subspace contains  $\mathbf{0}$ , is closed under addition, and is closed under multiplication by every complex number, in particular by every real number.
- (e) 4:  $\dim_{\mathbb{R}} M_2(\mathbb{R}) = 4$  (Example (Dimensions of the Running Examples)), and Theorem 1.9.3 keeps the dimension.

### 1.9.4.2 B. Practice

#### Exercise B1: A Tower over $\mathbb{Q}$ .

Find  $\dim_{\mathbb{Q}} \mathbb{Q}(\sqrt{2})^3$  and a basis of  $\mathbb{Q}(\sqrt{2})^3$  over  $\mathbb{Q}$ .

*Solution.* By Example ( $\mathbb{Q}(\sqrt{2})$  Is a Field),  $\mathbb{Q}(\sqrt{2})$  is a field, so  $\mathbb{Q}(\sqrt{2})^3$  is a vector space over it with basis  $(\mathbf{e}_1, \mathbf{e}_2, \mathbf{e}_3)$  and dimension 3 (Example (Dimensions of the Running Examples)). By Example (Degrees of Some Fields),  $(1, \sqrt{2})$  is a basis of  $\mathbb{Q}(\sqrt{2})$  over the subfield  $\mathbb{Q}$ . By Theorem 1.9.1,

$$(\mathbf{e}_1, \sqrt{2}\mathbf{e}_1, \mathbf{e}_2, \sqrt{2}\mathbf{e}_2, \mathbf{e}_3, \sqrt{2}\mathbf{e}_3)$$

is a basis of  $\mathbb{Q}(\sqrt{2})^3$  over  $\mathbb{Q}$ . Hence  $\dim_{\mathbb{Q}} \mathbb{Q}(\sqrt{2})^3 = 2 \cdot 3 = 6$ .

#### Exercise B2: Complex Matrices over $\mathbb{R}$ .

Let  $\mathbf{E}_{11}, \mathbf{E}_{12}, \mathbf{E}_{21}, \mathbf{E}_{22}$  be the matrix units of  $M_2(\mathbb{C})$ . Find a basis of  $M_2(\mathbb{C})$  over  $\mathbb{R}$ , and write  $\mathbf{A} = \begin{pmatrix} 1+i & 2 \\ -i & 3i \end{pmatrix}$  in it.

*Solution.* The matrix units form a basis of  $M_2(\mathbb{C})$  over  $\mathbb{C}$  (Example (Standard Bases)). By Theorem 1.9.2,

$$(\mathbf{E}_{11}, i\mathbf{E}_{11}, \mathbf{E}_{12}, i\mathbf{E}_{12}, \mathbf{E}_{21}, i\mathbf{E}_{21}, \mathbf{E}_{22}, i\mathbf{E}_{22})$$

is a basis of  $M_2(\mathbb{C})$  over  $\mathbb{R}$ , so  $\dim_{\mathbb{R}} M_2(\mathbb{C}) = 8$ . Splitting each entry of  $\mathbf{A}$  into real and imaginary parts,

$$\mathbf{A} = 1\mathbf{E}_{11} + 1(i\mathbf{E}_{11}) + 2\mathbf{E}_{12} + 0(i\mathbf{E}_{12}) + 0\mathbf{E}_{21} + (-1)(i\mathbf{E}_{21}) + 0\mathbf{E}_{22} + 3(i\mathbf{E}_{22}).$$

#### Exercise B3: Working in a Complexification.

Let  $V$  be a vector space over  $\mathbb{R}$ .

- (a) Verify that  $\alpha(\mathbf{z} + \mathbf{w}) = \alpha\mathbf{z} + \alpha\mathbf{w}$  for all  $\alpha \in \mathbb{C}$  and  $\mathbf{z}, \mathbf{w} \in V_{\mathbb{C}}$ .
- (b) In  $(\mathbb{R}[x]_{\leq 1})_{\mathbb{C}}$ , compute  $(2-i)((1+x)+ix)$  from Definition 1.9.2, and compare with the product  $(2-i)(1+x+ix)$  computed in  $\mathbb{C}[x]$ .

*Solution.*

- (a) Let  $\alpha = a + bi$  with  $a, b \in \mathbb{R}$ ,  $\mathbf{z} = (\mathbf{u}, \mathbf{v})$  and  $\mathbf{w} = (\mathbf{u}', \mathbf{v}')$ . Then  $\mathbf{z} + \mathbf{w} = (\mathbf{u} + \mathbf{u}', \mathbf{v} + \mathbf{v}')$ , and by the distributive law in  $V$  and [Theorem 1.1.8](#),

$$\begin{aligned}\alpha(\mathbf{z} + \mathbf{w}) &= (a(\mathbf{u} + \mathbf{u}') - b(\mathbf{v} + \mathbf{v}'), a(\mathbf{v} + \mathbf{v}') + b(\mathbf{u} + \mathbf{u}')) \\ &= ((a\mathbf{u} - b\mathbf{v}) + (a\mathbf{u}' - b\mathbf{v}'), (a\mathbf{v} + b\mathbf{u}) + (a\mathbf{v}' + b\mathbf{u}')), \end{aligned}$$

where we also rearranged the sums using commutativity and associativity of addition in  $V$ . The right side is  $(a\mathbf{u} - b\mathbf{v}, a\mathbf{v} + b\mathbf{u}) + (a\mathbf{u}' - b\mathbf{v}', a\mathbf{v}' + b\mathbf{u}') = \alpha\mathbf{z} + \alpha\mathbf{w}$ .

- (b) Here  $a = 2$ ,  $b = -1$ ,  $\mathbf{u} = 1 + x$  and  $\mathbf{v} = x$ . Then  $a\mathbf{u} - b\mathbf{v} = 2 + 2x + x = 2 + 3x$  and  $a\mathbf{v} + b\mathbf{u} = 2x - (1 + x) = -1 + x$ . So the product is  $(2 + 3x) + i(-1 + x)$ . In  $\mathbb{C}[x]$ ,  $(2 - i)(1 + x + ix) = 2 + 2x + 2ix - i - ix - i^2x = (2 + 3x) + i(-1 + x)$ , the same answer.

### 1.9.4.3 C. Going deeper

#### Exercise C1: When Is a Real Subspace Complex?.

Let  $W \subseteq \mathbb{C}^n$  be a subspace over  $\mathbb{R}$ .

- Prove that  $W$  is a subspace over  $\mathbb{C}$  if and only if  $i\mathbf{w} \in W$  for every  $\mathbf{w} \in W$ .
- Deduce that if  $W$  is a subspace over  $\mathbb{C}$ , then  $\dim_{\mathbb{R}} W$  is even.
- Show that  $W_0 = \{(z, \bar{z}) : z \in \mathbb{C}\} \subseteq \mathbb{C}^2$  is a subspace over  $\mathbb{R}$  of even real dimension that is **not** a subspace over  $\mathbb{C}$ . So the converse of (b) fails.

*Solution.*

- ( $\Rightarrow$ ) If  $W$  is a  $\mathbb{C}$ -subspace, it is closed under multiplication by  $i$ .  
( $\Leftarrow$ ) Suppose  $i\mathbf{w} \in W$  for every  $\mathbf{w} \in W$ . We use [Theorem 1.2.1](#) over  $\mathbb{C}$ . Since  $W$  is an  $\mathbb{R}$ -subspace,  $\mathbf{0} \in W$  and  $W$  is closed under addition. Let  $\mathbf{w} \in W$  and  $a + bi \in \mathbb{C}$  with  $a, b \in \mathbb{R}$ . Then  $a\mathbf{w} \in W$  and  $b(i\mathbf{w}) \in W$ , because  $W$  is closed under real scalars, and  $(a + bi)\mathbf{w} = a\mathbf{w} + b(i\mathbf{w})$  by the axioms of  $\mathbb{C}^n$ . This lies in  $W$  by closure under addition. Hence  $W$  is a  $\mathbb{C}$ -subspace.
- $W$  is a subspace of the finite-dimensional  $\mathbb{C}$ -space  $\mathbb{C}^n$ , so by [Theorem 1.6.7](#) it is finite-dimensional over  $\mathbb{C}$ , and by [Corollary 1.5.5](#) it has a basis over  $\mathbb{C}$ , of some length  $k$ . Restricting the scalars of  $W$  to  $\mathbb{R}$  gives the same real space as before, so  $\dim_{\mathbb{R}} W = 2k$  by [Theorem 1.9.2](#).
- Write  $z = a + bi$  with  $a, b \in \mathbb{R}$ . Then  $(z, \bar{z}) = (a + bi, a - bi) = a(1, 1) + b(i, -i)$ , so  $W_0$  is the  $\mathbb{R}$ -span of  $((1, 1), (i, -i))$  and is an  $\mathbb{R}$ -subspace by [Theorem 1.3.1](#). The list is independent over  $\mathbb{R}$ :  $a(1, 1) + b(i, -i) = \mathbf{0}$  gives  $a + bi = 0$ , so  $a = b = 0$ . Hence  $\dim_{\mathbb{R}} W_0 = 2$ . But  $(1, 1) \in W_0$  while  $i(1, 1) = (i, i) \notin W_0$ , because  $\bar{i} = -i \neq i$ . By (a),  $W_0$  is not a  $\mathbb{C}$ -subspace.

#### Exercise C2: A Tower of Degree Four.

Let  $V = \{x + y\sqrt{3} : x, y \in \mathbb{Q}(\sqrt{2})\} \subseteq \mathbb{R}$ .

- Prove that  $\sqrt{3} \notin \mathbb{Q}(\sqrt{2})$ .

- (b) Regard  $\mathbb{R}$  as a vector space over its subfield  $\mathbb{Q}(\sqrt{2})$ . Prove that  $V$  is a subspace with basis  $(1, \sqrt{3})$ .
- (c) Deduce that  $\dim_{\mathbb{Q}} V = 4$ , with basis  $(1, \sqrt{2}, \sqrt{3}, \sqrt{6})$ .

*Hint for (a): square both sides, and argue with remainders on division by 3.*

*Solution.*

- (a) First, there are no integers  $p, q$  with  $q \neq 0$ , not both divisible by 3, and  $cp^2 = 3q^2$  for  $c \in \{1, 2\}$ . Indeed, if  $3 \nmid p$ , then  $p = 3t \pm 1$  and  $cp^2 = c(9t^2 \pm 6t + 1)$  leaves remainder  $c \in \{1, 2\}$  on division by 3, while  $3q^2$  leaves remainder 0. So  $3 \mid p$ , say  $p = 3s$ . Then  $9cs^2 = 3q^2$ , so  $q^2 = 3cs^2$ , and the same remainder argument (a square not divisible by 3 leaves remainder 1) gives  $3 \mid q$ , a contradiction.
- Now suppose  $\sqrt{3} = a + b\sqrt{2}$  with  $a, b \in \mathbb{Q}$ . Squaring,  $3 = a^2 + 2b^2 + 2ab\sqrt{2}$ . If  $ab \neq 0$ , then  $\sqrt{2} = (3 - a^2 - 2b^2)/(2ab) \in \mathbb{Q}$ , contradicting [Theorem 0.2.3](#). If  $b = 0$ , then  $a^2 = 3$ ; writing  $a = p/q$  in lowest terms gives  $p^2 = 3q^2$  with  $p, q$  not both divisible by 3, which is impossible. If  $a = 0$ , then  $2b^2 = 3$ , and  $b = p/q$  in lowest terms gives  $2p^2 = 3q^2$ , also impossible. Hence  $\sqrt{3} \notin \mathbb{Q}(\sqrt{2})$ .
- (b)  $\mathbb{Q}(\sqrt{2})$  is a subfield of  $\mathbb{R}$  by [Example \( \$\mathbb{Q}\(\sqrt{2}\)\$  Is a Field\)](#), so  $\mathbb{R}$  is a vector space over it by restriction of scalars. We use [Theorem 1.2.1](#). (1)  $0 = 0 + 0\sqrt{3} \in V$ . (2)  $(x + y\sqrt{3}) + (x' + y'\sqrt{3}) = (x + x') + (y + y')\sqrt{3} \in V$ , since  $\mathbb{Q}(\sqrt{2})$  is closed under addition. (3) For  $c \in \mathbb{Q}(\sqrt{2})$ ,  $c(x + y\sqrt{3}) = cx + (cy)\sqrt{3} \in V$ , since  $\mathbb{Q}(\sqrt{2})$  is closed under multiplication. So  $V$  is a subspace.
- The list  $(1, \sqrt{3})$  spans  $V$  by the definition of  $V$ . For independence, let  $x + y\sqrt{3} = 0$  with  $x, y \in \mathbb{Q}(\sqrt{2})$ . If  $y \neq 0$ , then  $y^{-1} \in \mathbb{Q}(\sqrt{2})$  and  $\sqrt{3} = -xy^{-1} \in \mathbb{Q}(\sqrt{2})$ , contradicting (a). So  $y = 0$ , and then  $x = 0$ . Hence  $(1, \sqrt{3})$  is a basis and  $\dim_{\mathbb{Q}(\sqrt{2})} V = 2$ .
- (c) By [Example \(Degrees of Some Fields\)](#),  $(1, \sqrt{2})$  is a basis of  $\mathbb{Q}(\sqrt{2})$  over  $\mathbb{Q}$ . By [Theorem 1.9.1](#) applied to  $\mathbb{Q} \subseteq \mathbb{Q}(\sqrt{2})$  and the  $\mathbb{Q}(\sqrt{2})$ -space  $V$ , the list  $(1 \cdot 1, \sqrt{2} \cdot 1, 1 \cdot \sqrt{3}, \sqrt{2} \cdot \sqrt{3}) = (1, \sqrt{2}, \sqrt{3}, \sqrt{6})$  is a basis of  $V$  over  $\mathbb{Q}$ . Hence  $\dim_{\mathbb{Q}} V = 2 \cdot 2 = 4$ .

# Chapter 2

## Linear Systems and Matrices

Chapter 1 kept asking questions that end in a system of equations. Is this vector in the span of those? Is this list independent? Extend this list to a basis. Each time we solved a small system by hand and promised a systematic method. This chapter keeps that promise. Every question about finitely many vectors in  $F^n$  turns into a linear system, and linear systems have a complete algorithm: Gaussian elimination.

We do not treat the algorithm as a recipe. We prove that row operations never change the answer, that the fully reduced form of a matrix does not depend on the choices made along the way, and that the numbers it produces, pivots and free variables, measure exactly the dimensions of Chapter 1. Along the way the invertible matrices get their first full characterization, and we prove a fact Chapter 0 had to postpone: for square matrices,  $\mathbf{AB} = \mathbf{I}$  already forces  $\mathbf{BA} = \mathbf{I}$ .

**What you need.** From Chapter 0, matrix multiplication read column by column ([Theorem 0.9.3](#)) and invertible matrices ([Definition 0.9.13](#)). From Chapter 1, span ([Definition 1.3.2](#)), linear independence ([Definition 1.4.1](#)), bases ([Definition 1.5.1](#)) and dimension ([Definition 1.6.1](#)).

### 2.0.1 Roadmap

- **Three views of a linear system.** Intersecting lines and planes, combinations of columns, and a single matrix equation; the shape of the solution set as a particular solution plus the homogeneous solutions.
- **Gaussian elimination.** Row operations, echelon and reduced echelon forms, and how to read the solutions off the reduced form.
- **Uniqueness of the reduced form.** Different sequences of row operations always reach the same reduced matrix, so pivot columns are well defined.
- **Elementary matrices and inverses.** Row operations as left multiplication, the invertible matrix theorem, and computing inverses by row reduction.
- **Rank.** Row space, column space and null space; row rank equals column rank; the Rouché–Capelli theorem; and a toolkit for the questions of Chapter 1.
- **LU factorization.** Storing elimination as a product of triangular matrices, with and without row swaps.
- **Applications.** Polynomial interpolation, balancing chemical equations, networks, and a puzzle over  $\mathbb{F}_2$ .

## 2.0.2 Named moves

- **Row operations do not change the solution set.** So simplify first, then read the answer.
- **Put it in reduced form, then read off.** Consistency, free variables, bases of the column, row and null spaces all come from the reduced row echelon form.
- **A question about vectors is a question about a matrix.** Put the vectors in as columns; independence, spanning and membership become statements about pivots.
- **Pivots count dimensions.** The number of pivots is the rank, and the free variables count the dimension of the null space.

# 2.1 Three Views of a Linear System

Chapter 1 kept running into the same kind of question. Is this vector in the span of those? Is this list independent? Every time, the question turned into a handful of equations in unknown coefficients, and we solved them by hand. This chapter gives those equations a name, looks at them from three sides, and turns solving them into an algorithm with a proof behind it. This first section sets up the language and answers a structural question before any computing: what can the set of all solutions look like?

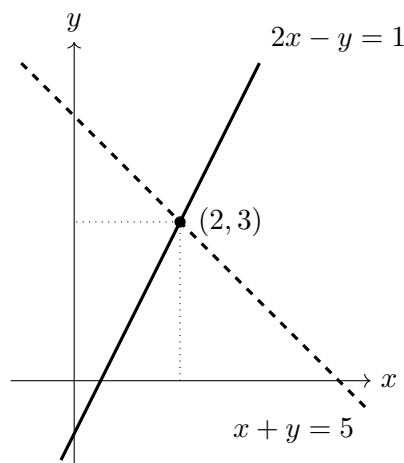
## 2.1.1 One system, three pictures

Here is a system of two equations in two unknowns, over  $\mathbb{R}$ :

$$\begin{aligned} 2x - y &= 1, \\ x + y &= 5. \end{aligned}$$

Adding the equations gives  $3x = 6$ , so  $x = 2$ , and then  $y = 5 - x = 3$ . We can read this one small system in three different ways, and each will be useful.

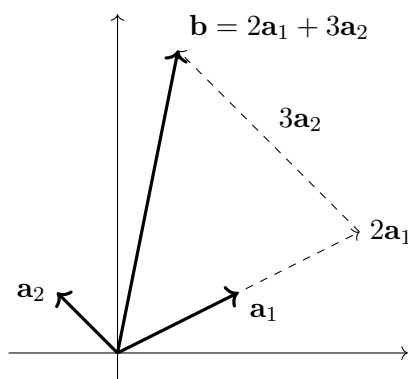
**The row picture.** Each equation describes a line in the plane:  $y = 2x - 1$  and  $y = 5 - x$ . A solution is a point on **both** lines, so solving the system means intersecting them. The lines meet at  $(2, 3)$ .



**The column picture.** Stack the coefficients of  $x$  into one column and those of  $y$  into another. The system says

$$x \begin{pmatrix} 2 \\ 1 \end{pmatrix} + y \begin{pmatrix} -1 \\ 1 \end{pmatrix} = \begin{pmatrix} 1 \\ 5 \end{pmatrix}.$$

Now the unknowns are **weights**. We are asked how much of the vector  $\mathbf{a}_1 = (2, 1)$  and how much of  $\mathbf{a}_2 = (-1, 1)$  to combine to reach  $\mathbf{b} = (1, 5)$ . The answer  $x = 2$ ,  $y = 3$  says: walk twice along  $\mathbf{a}_1$ , then three times along  $\mathbf{a}_2$ .



**The matrix picture.** Collect the coefficients into a matrix and the unknowns into a column. By the definition of matrix multiplication (Definition 0.9.9), the system is the single equation

$$\begin{pmatrix} 2 & -1 \\ 1 & 1 \end{pmatrix} \begin{pmatrix} x \\ y \end{pmatrix} = \begin{pmatrix} 1 \\ 5 \end{pmatrix}, \quad \text{that is,} \quad \mathbf{Ax} = \mathbf{b}.$$

The three pictures say the same thing, and they are tied together by one fact from Chapter 0:  $\mathbf{Ax} = x_1\mathbf{a}_1 + \cdots + x_n\mathbf{a}_n$ , a combination of the columns of  $\mathbf{A}$  weighted by the entries of  $\mathbf{x}$  (Theorem 0.9.3). The row picture is the one we draw. The column picture connects to Chapter 1, since it asks whether  $\mathbf{b}$  lies in a span. The matrix picture is the one we compute and prove things with.

*Remark.* With more unknowns the pictures can no longer be drawn, but they do not change. An equation in three unknowns whose coefficients are not all zero describes a plane in  $\mathbb{R}^3$ , and a system asks where several planes meet. In the column picture, three unknowns mean three columns, and we ask whether a combination of them hits  $\mathbf{b}$ .

## 2.1.2 Linear systems

The example had two equations and two unknowns over  $\mathbb{R}$ . The general notion allows any numbers of each, and any field. Allowing any field is not generality for its own sake. Systems over  $\mathbb{F}_2$  describe parity checks and switching puzzles, and systems over  $\mathbb{Q}$  are what a computer can solve exactly.

*A linear system is a finite list of equations, each saying that a fixed combination of the unknowns equals a fixed scalar.*

**Definition 2.1.1 — Linear system**

Let  $F$  be a field and let  $m, n \geq 1$ . A **system of  $m$  linear equations in  $n$  unknowns**  $x_1, \dots, x_n$  **over**  $F$  is a list of equations

$$\begin{aligned} a_{11}x_1 + a_{12}x_2 + \cdots + a_{1n}x_n &= b_1, \\ a_{21}x_1 + a_{22}x_2 + \cdots + a_{2n}x_n &= b_2, \\ &\vdots \\ a_{m1}x_1 + a_{m2}x_2 + \cdots + a_{mn}x_n &= b_m, \end{aligned}$$

where all  $a_{ij}$  and  $b_i$  lie in  $F$ .

- The **coefficient matrix** is  $\mathbf{A} = (a_{ij}) \in M_{m \times n}(F)$ , the **right-hand side** is  $\mathbf{b} = (b_1, \dots, b_m) \in F^m$ , and the system is written  $\mathbf{Ax} = \mathbf{b}$ .
- The **augmented matrix** is the  $m \times (n+1)$  matrix  $[\mathbf{A} \mid \mathbf{b}]$  obtained by appending  $\mathbf{b}$  to  $\mathbf{A}$  as a last column.
- A **solution** is a vector  $\mathbf{s} = (s_1, \dots, s_n) \in F^n$  such that **every** equation holds when  $x_j = s_j$  for all  $j$ ; equivalently,  $\mathbf{As} = \mathbf{b}$ . The **solution set** is the set of all solutions, a subset of  $F^n$ .
- The system is **consistent** if it has **at least one** solution, and **inconsistent** if its solution set is empty.

In words: the coefficient matrix records the left-hand sides, row  $i$  for equation  $i$  and column  $j$  for unknown  $x_j$ . The augmented matrix records the whole system, and it is the object we will manipulate in the next section. A solution is a single vector of  $F^n$ , and it must satisfy all  $m$  equations at once, not just some of them. “Equivalently,  $\mathbf{As} = \mathbf{b}$ ” holds because the  $i$ -th entry of  $\mathbf{As}$  is  $a_{i1}s_1 + \cdots + a_{in}s_n$  by [Definition 0.9.9](#), and two columns are equal exactly when all their entries agree.

Two small words deserve attention. A solution lives in  $F^n$ , with entries in **the field**  $F$ : the equation  $2x = 1$  has the solution  $\frac{1}{2}$  over  $\mathbb{Q}$ , but over  $\mathbb{F}_2$  it reads  $0 = 1$  and has none. And “consistent” means **at least one** solution, not exactly one.

**Example. (None, one, or a whole line)**

Find the solution set of each system over  $\mathbb{R}$ , and describe it in the row picture.

- (a)  $x + 2y = 1, 2x + 4y = 5$ .
- (b)  $2x - y = 1, x + y = 5$ .
- (c)  $x + 2y = 1, 2x + 4y = 2$ .

*Solution.*

- (a) Suppose  $(x, y)$  is a solution. Doubling the first equation gives  $2x + 4y = 2$ , while the second says  $2x + 4y = 5$ . Then  $2 = 5$ , which is false. So there is no solution, and the system is **inconsistent**. In the row picture, the lines  $y = \frac{1}{2} - \frac{1}{2}x$  and  $y = \frac{5}{4} - \frac{1}{2}x$  have the same slope and different intercepts: they are parallel and never meet.
- (b) As computed at the start of the section, any solution has  $x = 2$  and  $y = 3$ ;



conversely  $2 \cdot 2 - 3 = 1$  and  $2 + 3 = 5$ . The solution set is  $\{(2, 3)\}$ , the single point where two non-parallel lines cross.

- (c) The second equation is twice the first, so a pair  $(x, y)$  satisfies both exactly when it satisfies  $x + 2y = 1$ , that is, when  $x = 1 - 2y$ . Writing  $t$  for  $y$ , the solution set is  $\{(1 - 2t, t) : t \in \mathbb{R}\}$ . Both equations describe the **same** line, and every point on it is a solution.

The three answers are the three things two lines can do: miss, cross, or coincide. Over  $\mathbb{R}$ , and over any infinite field, no other sizes are possible: once we know the shape of the solution set (Theorem 2.1.3 below), this is a short argument, which you will give in Exercise (C1: How many solutions?). For instance, no system over  $\mathbb{R}$  has exactly two solutions. Over a finite field the picture is different.

**Example. (Exactly two solutions over  $\mathbb{F}_2$ )**

Find all solutions of the system  $x + y + z = 1$ ,  $y + z = 0$  over  $\mathbb{F}_2$ . Then compare with the same system over  $\mathbb{R}$ .

*Solution.* In  $\mathbb{F}_2$  we have  $-1 = 1$ , so the second equation says  $y = -z = z$ . Substituting into the first gives  $x + 2z = 1$ , and  $2z = 0$  in  $\mathbb{F}_2$ , so  $x = 1$ . Conversely, every  $(1, z, z)$  satisfies both equations:  $1 + z + z = 1 + 2z = 1$  and  $z + z = 0$ . Since  $z$  can only be 0 or 1, the solution set is

$$\{(1, 0, 0), (1, 1, 1)\},$$

with exactly two elements.

Over  $\mathbb{R}$  the second equation gives  $y = -z$ , and then the first gives  $x = 1$ . The solution set is  $\{(1, -t, t) : t \in \mathbb{R}\}$ , a line in  $\mathbb{R}^3$  with infinitely many points. The equations are the same, but the field decides how many values the free unknown can take.

Now a non-example, by a minimal change. The equation  $x + y^2 = 1$  looks like one of the equations above, and it still has a left side built from the unknowns and a constant on the right. But the unknown  $y$  appears squared, so the left side is not of the form  $a_1x_1 + a_2x_2$  with **fixed scalars**  $a_1, a_2$ . The clause that fails is “a fixed combination of the unknowns”. Its solution set in  $\mathbb{R}^2$  is a parabola, not a line, and nothing in this chapter applies to it.

*Warning.* “**Consistent**” does not mean “has exactly one solution”. The system  $x + 2y = 1$ ,  $2x + 4y = 2$  of Example (None, one, or a whole line) (c) is consistent, and it has infinitely many solutions. Consistency only rules out the empty solution set. Whether the solution is unique is a separate question, answered by Corollary 2.1.4 below.

Why record the augmented matrix at all, when  $\mathbf{Ax} = \mathbf{b}$  is shorter? Because solving a system never touches the names of the unknowns. Adding two equations adds their coefficients and their right-hand sides, and nothing else. The augmented matrix keeps exactly this data and throws away the symbols  $x_1, \dots, x_n$  and the plus signs. In the next section every manipulation is performed on it.

One kind of system is always consistent, because it has an obvious solution.

**Definition 2.1.2 — Homogeneous system**

A linear system  $\mathbf{Ax} = \mathbf{b}$  over  $F$  is **homogeneous** if  $\mathbf{b} = \mathbf{0}$ . The vector  $\mathbf{0} \in F^n$  is a solution of every homogeneous system  $\mathbf{Ax} = \mathbf{0}$ , called the **trivial solution**. For any system  $\mathbf{Ax} = \mathbf{b}$ , the system  $\mathbf{Ax} = \mathbf{0}$ , with the same coefficient matrix, is its **associated homogeneous system**.

The trivial solution exists by the zero rule of [Theorem 0.9.4](#):  $\mathbf{A}\mathbf{0} = \mathbf{0}$ . So for a homogeneous system the interesting question is never “is there a solution?” but “is there a **non-trivial** one?”. For example,  $x + 2y = 0$ ,  $2x + 4y = 0$  has the non-trivial solution  $(-2, 1)$ , while  $2x - y = 0$ ,  $x + y = 0$  has only the trivial one: adding the equations gives  $3x = 0$ , so  $x = 0$ , and then  $y = -x = 0$ .

*Quick check.* Over  $\mathbb{R}$ , write the system  $x_1 - x_2 + 3x_3 = 4$ ,  $2x_2 + x_3 = 0$  in the matrix picture and the column picture, and give its augmented matrix. Is  $(4, 0, 0)$  a solution? Is  $(1, 0, 1)$ ?

*Solution.* With  $\mathbf{A} = \begin{pmatrix} 1 & -1 & 3 \\ 0 & 2 & 1 \end{pmatrix}$  and  $\mathbf{b} = (4, 0)$ , the system is  $\mathbf{Ax} = \mathbf{b}$ . In the column picture it reads  $x_1(1, 0) + x_2(-1, 2) + x_3(3, 1) = (4, 0)$ . The augmented matrix is

$$[\mathbf{A} \mid \mathbf{b}] = \left[ \begin{array}{ccc|c} 1 & -1 & 3 & 4 \\ 0 & 2 & 1 & 0 \end{array} \right].$$

For  $(4, 0, 0)$ :  $4 - 0 + 0 = 4$  and  $0 + 0 = 0$ , so it is a solution. For  $(1, 0, 1)$ : the first equation gives  $1 - 0 + 3 = 4$ , but the second gives  $0 + 1 = 1 \neq 0$ . One failing equation is enough, so it is not a solution.

### 2.1.3 Consistency is a question about span

The column picture turns “does this system have a solution?” into a question from Chapter 1.

#### Theorem 2.1.1 — Consistency and the span of the columns

Let  $\mathbf{A} \in M_{m \times n}(F)$  have columns  $\mathbf{a}_1, \dots, \mathbf{a}_n \in F^m$ , and let  $\mathbf{b} \in F^m$ . Then the system  $\mathbf{Ax} = \mathbf{b}$  is consistent **if and only if**  $\mathbf{b} \in \text{span}(\mathbf{a}_1, \dots, \mathbf{a}_n)$ . More precisely,  $\mathbf{s} = (s_1, \dots, s_n)$  is a solution exactly when  $\mathbf{b} = s_1\mathbf{a}_1 + \dots + s_n\mathbf{a}_n$ .

*Proof.* By [Theorem 0.9.3](#),  $\mathbf{As} = s_1\mathbf{a}_1 + \dots + s_n\mathbf{a}_n$  for every  $\mathbf{s} \in F^n$ . Hence  $\mathbf{s}$  is a solution, that is  $\mathbf{As} = \mathbf{b}$ , exactly when  $\mathbf{b} = s_1\mathbf{a}_1 + \dots + s_n\mathbf{a}_n$ . This proves the second statement.

( $\Rightarrow$ ) If the system is consistent, it has a solution  $\mathbf{s}$ , and then  $\mathbf{b} = s_1\mathbf{a}_1 + \dots + s_n\mathbf{a}_n$  is a linear combination of the columns. By [Definition 1.3.2](#),  $\mathbf{b} \in \text{span}(\mathbf{a}_1, \dots, \mathbf{a}_n)$ .

( $\Leftarrow$ ) If  $\mathbf{b} \in \text{span}(\mathbf{a}_1, \dots, \mathbf{a}_n)$ , then  $\mathbf{b} = s_1\mathbf{a}_1 + \dots + s_n\mathbf{a}_n$  for some  $s_1, \dots, s_n \in F$ , and  $\mathbf{s} = (s_1, \dots, s_n)$  is a solution by the first paragraph. Hence the system is consistent. ■

Read backwards, the theorem is a recipe for Chapter 1 questions: to decide whether  $\mathbf{b}$  lies in the span of some vectors of  $F^m$ , put the vectors as the columns of a matrix and ask whether the system is consistent.

**Example. (Span membership as a system)**

In  $\mathbb{R}^3$ , let  $\mathbf{v}_1 = (1, 2, 3)$  and  $\mathbf{v}_2 = (0, 1, 2)$ . Decide whether  $(1, 4, 7)$  and  $(1, 4, 8)$  lie in  $\text{span}(\mathbf{v}_1, \mathbf{v}_2)$ .

*Solution.* Let  $\mathbf{A}$  be the  $3 \times 2$  matrix with columns  $\mathbf{v}_1, \mathbf{v}_2$ . By [Theorem 2.1.1](#),  $(1, 4, 7) \in \text{span}(\mathbf{v}_1, \mathbf{v}_2)$  if and only if the system

$$a = 1, \quad 2a + b = 4, \quad 3a + 2b = 7$$

in the unknowns  $a, b$  is consistent. The first equation gives  $a = 1$ , and then the second gives  $b = 2$ . The third equation must also hold, and it does:  $3 + 4 = 7$ . Hence  $(1, 4, 7) = \mathbf{v}_1 + 2\mathbf{v}_2$  lies in the span.

For  $(1, 4, 8)$ , the first two equations are the same, so any solution has  $a = 1$  and  $b = 2$ . The third equation now demands  $3 + 4 = 8$ , which is false. The system is inconsistent, so  $(1, 4, 8) \notin \text{span}(\mathbf{v}_1, \mathbf{v}_2)$ .

Geometrically,  $\text{span}(\mathbf{v}_1, \mathbf{v}_2)$  is a plane through  $\mathbf{0}$  in  $\mathbb{R}^3$ . The first right-hand side lies on that plane and the second does not.

## 2.1.4 The shape of the solution set

We now describe the solution set of an arbitrary system, starting with the homogeneous case. Chapter 1 already did the work there: in [Example \(Solutions of a Homogeneous System\)](#) we checked, with the three conditions of the subspace test, that the solutions of  $\mathbf{A}\mathbf{x} = \mathbf{0}$  form a subspace.

### Theorem 2.1.2 — Homogeneous solutions form a subspace

Let  $\mathbf{A} \in M_{m \times n}(F)$ . The solution set  $N = \{\mathbf{x} \in F^n : \mathbf{A}\mathbf{x} = \mathbf{0}\}$  of the homogeneous system  $\mathbf{A}\mathbf{x} = \mathbf{0}$  is a subspace of  $F^n$ .

*Proof.* This is [Example \(Solutions of a Homogeneous System\)](#):  $\mathbf{A}\mathbf{0} = \mathbf{0}$ , and for  $\mathbf{x}, \mathbf{y} \in N$  and  $c \in F$ , the rules of [Theorem 0.9.4](#) give  $\mathbf{A}(\mathbf{x} + \mathbf{y}) = \mathbf{A}\mathbf{x} + \mathbf{A}\mathbf{y} = \mathbf{0}$  and  $\mathbf{A}(c\mathbf{x}) = c(\mathbf{A}\mathbf{x}) = \mathbf{0}$ . By the subspace test ([Theorem 1.2.1](#)),  $N$  is a subspace of  $F^n$ . ■

The solutions of  $\mathbf{A}\mathbf{x} = \mathbf{b}$  with  $\mathbf{b} \neq \mathbf{0}$  behave differently.

**Warning.** The solution set of  $\mathbf{A}\mathbf{x} = \mathbf{b}$  with  $\mathbf{b} \neq \mathbf{0}$  is never a subspace. A subspace must contain  $\mathbf{0}$ , but  $\mathbf{A}\mathbf{0} = \mathbf{0} \neq \mathbf{b}$ , so  $\mathbf{0}$  is not a solution. For example, the solutions of  $x - y = 1$  in  $\mathbb{R}^2$  form a line that misses the origin. The sum of the solutions  $(1, 0)$  and  $(2, 1)$  is  $(3, 1)$ , and  $3 - 1 = 2 \neq 1$ : adding two solutions does not give a solution.

What does survive is this: the **difference** of two solutions of  $\mathbf{A}\mathbf{x} = \mathbf{b}$  solves the homogeneous system, since  $\mathbf{A}(\mathbf{s} - \mathbf{s}') = \mathbf{b} - \mathbf{b} = \mathbf{0}$ . So once one solution is known, every other one differs from it by a homogeneous solution. That is the whole structure.

**Theorem 2.1.3 — Structure of the solution set**

Let  $\mathbf{A} \in M_{m \times n}(F)$  and  $\mathbf{b} \in F^m$ , let  $N$  be the solution set of  $\mathbf{A}\mathbf{x} = \mathbf{0}$ , and suppose  $\mathbf{p} \in F^n$  is **one** solution of  $\mathbf{A}\mathbf{x} = \mathbf{b}$ . Then the solution set of  $\mathbf{A}\mathbf{x} = \mathbf{b}$  is

$$\mathbf{p} + N = \{\mathbf{p} + \mathbf{h} : \mathbf{h} \in N\}.$$

*Idea.* The theorem is “particular plus homogeneous”. To prove equality of two sets, prove both inclusions. Going from a solution  $\mathbf{s}$  to  $\mathbf{p} + N$  uses the difference  $\mathbf{s} - \mathbf{p}$ , as observed above; going back uses the sum  $\mathbf{A}(\mathbf{p} + \mathbf{h}) = \mathbf{A}\mathbf{p} + \mathbf{A}\mathbf{h}$ .

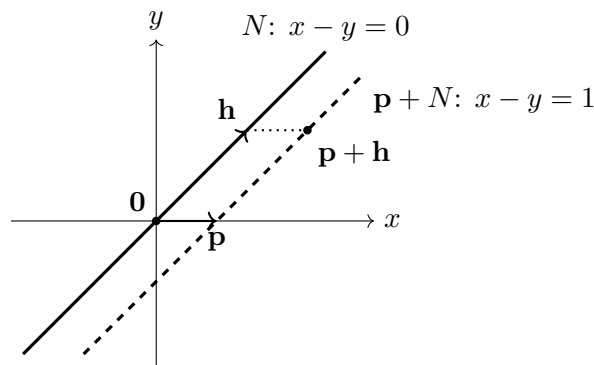
*Proof.* Let  $S$  be the solution set of  $\mathbf{A}\mathbf{x} = \mathbf{b}$ . We use [Theorem 0.3.1](#).

( $\subseteq$ ) Let  $\mathbf{s} \in S$ , and put  $\mathbf{h} = \mathbf{s} - \mathbf{p}$ . By distributivity ([Theorem 0.9.4](#)),  $\mathbf{A}\mathbf{h} = \mathbf{A}\mathbf{s} - \mathbf{A}\mathbf{p} = \mathbf{b} - \mathbf{b} = \mathbf{0}$ , so  $\mathbf{h} \in N$ . Hence  $\mathbf{s} = \mathbf{p} + \mathbf{h} \in \mathbf{p} + N$ .

( $\supseteq$ ) Let  $\mathbf{h} \in N$ . Then  $\mathbf{A}(\mathbf{p} + \mathbf{h}) = \mathbf{A}\mathbf{p} + \mathbf{A}\mathbf{h} = \mathbf{b} + \mathbf{0} = \mathbf{b}$ , again by distributivity, so  $\mathbf{p} + \mathbf{h} \in S$ .

Therefore  $S = \mathbf{p} + N$ , as claimed. ■

The theorem splits solving  $\mathbf{A}\mathbf{x} = \mathbf{b}$  into two independent jobs: find **one** solution, and find **all** solutions of the homogeneous system. The set  $\mathbf{p} + N$  is a translate of the subspace  $N$  by the vector  $\mathbf{p}$ , the kind of set that the picture in [Example \(A Line Missing the Origin\)](#) showed. For  $x - y = 1$  in  $\mathbb{R}^2$ , the homogeneous solutions form the line  $N = \text{span}((1, 1))$  through  $\mathbf{0}$ , one solution is  $\mathbf{p} = (1, 0)$ , and the solution set is the parallel line through  $\mathbf{p}$ .



A system solved by substitution fits the same pattern once we collect the free parameter.

**Example. (Particular plus homogeneous)**

Solve  $x_1 + 2x_2 - x_3 = 4$ ,  $x_2 + x_3 = 1$  over  $\mathbb{R}$ , and write the solution set in the form  $\mathbf{p} + \text{span}(\mathbf{h})$ .

*Solution.* The second equation gives  $x_2 = 1 - x_3$ . Substituting into the first,  $x_1 = 4 - 2(1 - x_3) + x_3 = 2 + 3x_3$ . The value of  $x_3$  is not constrained, so write  $x_3 = t$ . Conversely, for every  $t \in \mathbb{R}$  the vector  $(2 + 3t, 1 - t, t)$  satisfies both equations, since

$(2 + 3t) + 2(1 - t) - t = 4$  and  $(1 - t) + t = 1$ . Hence the solution set is

$$\{(2 + 3t, 1 - t, t) : t \in \mathbb{R}\} = \{(2, 1, 0) + t(3, -1, 1) : t \in \mathbb{R}\} = \mathbf{p} + \text{span}(\mathbf{h}),$$

with  $\mathbf{p} = (2, 1, 0)$  and  $\mathbf{h} = (3, -1, 1)$ . As [Theorem 2.1.3](#) predicts,  $\mathbf{p}$  is a solution (take  $t = 0$ ), and  $\mathbf{h}$  solves the homogeneous system:  $3 - 2 - 1 = 0$  and  $-1 + 1 = 0$ . In the row picture, two planes in  $\mathbb{R}^3$  meet in a line, and this line does not pass through  $\mathbf{0}$ .

## 2.1.5 When is the solution unique?

A consistent system has a unique solution exactly when there is nothing to add to the particular solution. The theorem just proved turns this into a statement about the homogeneous system, and the column picture turns that into independence.

### Corollary 2.1.4 — Uniqueness of solutions

Let  $\mathbf{A} \in M_{m \times n}(F)$  have columns  $\mathbf{a}_1, \dots, \mathbf{a}_n$ , and let  $\mathbf{b} \in F^m$  be such that  $\mathbf{Ax} = \mathbf{b}$  is **consistent**. The following are equivalent:

- (a)  $\mathbf{Ax} = \mathbf{b}$  has exactly one solution;
- (b)  $\mathbf{Ax} = \mathbf{0}$  has only the trivial solution;
- (c) the list  $(\mathbf{a}_1, \dots, \mathbf{a}_n)$  is linearly independent.

*Proof.* Let  $N$  be the solution set of  $\mathbf{Ax} = \mathbf{0}$ . Since the system is consistent, it has a solution  $\mathbf{p}$ , and by [Theorem 2.1.3](#) its solution set is  $\mathbf{p} + N$ .

(a  $\Leftrightarrow$  b) The function  $N \rightarrow \mathbf{p} + N$ ,  $\mathbf{h} \mapsto \mathbf{p} + \mathbf{h}$ , is surjective by definition of  $\mathbf{p} + N$ , and injective because  $\mathbf{p} + \mathbf{h} = \mathbf{p} + \mathbf{h}'$  implies  $\mathbf{h} = \mathbf{h}'$  by adding  $-\mathbf{p}$  to both sides. So it is a bijection, and  $\mathbf{p} + N$  has exactly one element if and only if  $N$  does. Since  $\mathbf{0} \in N$  always,  $N$  has exactly one element if and only if  $N = \{\mathbf{0}\}$ , which is (b).

(b  $\Leftrightarrow$  c) By [Theorem 0.9.3](#),  $\mathbf{Ax} = x_1\mathbf{a}_1 + \dots + x_n\mathbf{a}_n$  for every  $\mathbf{x} \in F^n$ . So (b) says: for all  $x_1, \dots, x_n \in F$ ,  $x_1\mathbf{a}_1 + \dots + x_n\mathbf{a}_n = \mathbf{0}$  implies  $x_1 = \dots = x_n = 0$ . This is word for word [Definition 1.4.1](#) for the list  $(\mathbf{a}_1, \dots, \mathbf{a}_n)$ . ■

Two remarks on the hypotheses. The equivalence of (b) and (c) does not need consistency; only (a) does. And consistency cannot be dropped from (a  $\Leftrightarrow$  b): the system  $x = 0$ ,  $x = 1$  in one unknown has only the trivial solution in its homogeneous version ( $x = 0$ ,  $x = 0$ ), but it has no solution at all, so it certainly does not have exactly one.

So the three pictures answer the two basic questions about  $\mathbf{Ax} = \mathbf{b}$  in the language of Chapter 1: a solution **exists** when  $\mathbf{b}$  is in the span of the columns, and it is **unique** when the columns are independent. What we still lack is a reliable way to decide either question for a given matrix. That is the job of Gaussian elimination, in the next section.

## 2.1.6 Exercises

### 2.1.6.1 A. Check your understanding

#### Exercise A1.

- Define what it means for a linear system  $\mathbf{Ax} = \mathbf{b}$  over  $F$  to be consistent, and what its augmented matrix is. What is the size of the augmented matrix of a system of  $m$  equations in  $n$  unknowns?
- Determine whether the following statement is true: “the solution set of  $\mathbf{Ax} = \mathbf{b}$  is a subspace of  $F^n$ .” Justify your answer.
- Determine whether the following statement is true: “if  $\mathbf{Ax} = \mathbf{b}$  has two different solutions, then  $\mathbf{Ax} = \mathbf{0}$  has a non-trivial solution.” Justify your answer.
- Let  $\mathbf{A} \in M_{3 \times 5}(\mathbb{R})$ . In which spaces do the unknown vector  $\mathbf{x}$  and the right-hand side  $\mathbf{b}$  of  $\mathbf{Ax} = \mathbf{b}$  live?
- Name the theorem that turns “is  $\mathbf{b}$  in the span of  $\mathbf{a}_1, \dots, \mathbf{a}_n$ ?” into a question about a linear system, and state which system.

*Solution.*

- It is consistent if it has at least one solution  $\mathbf{s} \in F^n$ , that is, some  $\mathbf{s}$  with  $\mathbf{As} = \mathbf{b}$ . The augmented matrix is  $[\mathbf{A} \mid \mathbf{b}]$ , the matrix  $\mathbf{A}$  with  $\mathbf{b}$  appended as an extra last column. It has size  $m \times (n + 1)$ .
- False in general. If  $\mathbf{b} \neq \mathbf{0}$ , then  $\mathbf{A}\mathbf{0} = \mathbf{0} \neq \mathbf{b}$ , so  $\mathbf{0}$  is not a solution and the solution set is not a subspace. (If  $\mathbf{b} = \mathbf{0}$ , it is a subspace, by [Theorem 2.1.2](#).)
- True. If  $\mathbf{s} \neq \mathbf{s}'$  are solutions, then  $\mathbf{A}(\mathbf{s} - \mathbf{s}') = \mathbf{b} - \mathbf{b} = \mathbf{0}$  by distributivity, and  $\mathbf{s} - \mathbf{s}' \neq \mathbf{0}$ .
- $\mathbf{x} \in \mathbb{R}^5$ , one entry per column of  $\mathbf{A}$ , and  $\mathbf{b} \in \mathbb{R}^3$ , one entry per row.
- [Theorem 2.1.1](#):  $\mathbf{b} \in \text{span}(\mathbf{a}_1, \dots, \mathbf{a}_n)$  if and only if  $\mathbf{Ax} = \mathbf{b}$  is consistent, where  $\mathbf{A}$  is the matrix with columns  $\mathbf{a}_1, \dots, \mathbf{a}_n$ .

### 2.1.6.2 B. Practice

#### Exercise B1: Three views.

- Write the system  $x_1 - x_2 + 2x_3 = 3$ ,  $2x_1 + x_3 = 1$  over  $\mathbb{R}$  in the matrix picture and in the column picture, and give its augmented matrix.
- Write out, as a list of equations, the system with augmented matrix 
$$\left[ \begin{array}{cc|c} 1 & 4 & 0 \\ 0 & 1 & 2 \\ 3 & -1 & 5 \end{array} \right]$$
 over  $\mathbb{R}$ . How many equations and unknowns does it have? Hence decide whether it is consistent.

*Solution.*

- (a) The matrix picture is

$$\begin{pmatrix} 1 & -1 & 2 \\ 2 & 0 & 1 \end{pmatrix} \begin{pmatrix} x_1 \\ x_2 \\ x_3 \end{pmatrix} = \begin{pmatrix} 3 \\ 1 \end{pmatrix}.$$

The column picture is  $x_1(1, 2) + x_2(-1, 0) + x_3(2, 1) = (3, 1)$ . The augmented matrix is  $\left[ \begin{array}{ccc|c} 1 & -1 & 2 & 3 \\ 2 & 0 & 1 & 1 \end{array} \right]$ .

- (b) The system is  $x_1 + 4x_2 = 0$ ,  $x_2 = 2$ ,  $3x_1 - x_2 = 5$ : three equations in two unknowns. The second equation gives  $x_2 = 2$ , and then the first gives  $x_1 = -8$ . The third equation would require  $-24 - 2 = 5$ , which is false. Hence no vector satisfies all three equations, and the system is inconsistent.

### Exercise B2: In the span or not?

Determine whether the vector  $\mathbf{b}$  lies in the span of the given vectors. Justify your answer, and when it does, write  $\mathbf{b}$  as a linear combination of them.

- (a)  $\mathbf{b} = (2, 1, 4)$  and  $(1, 0, 1), (0, 1, 2)$  in  $\mathbb{R}^3$ .
- (b)  $\mathbf{b} = (1, 1, 1)$  and  $(1, 2, 0), (2, 4, 1)$  in  $\mathbb{R}^3$ .
- (c)  $\mathbf{b} = (1, 1, 0)$  and  $(1, 0, 1), (0, 1, 1)$  in  $\mathbb{F}_2^3$ . Is the answer the same in  $\mathbb{R}^3$ ?

*Solution.* By [Theorem 2.1.1](#), in each case we decide whether the system  $a\mathbf{v}_1 + b\mathbf{v}_2 = \mathbf{b}$  in the unknowns  $a, b$  is consistent.

- (a) The system is  $a = 2$ ,  $b = 1$ ,  $a + 2b = 4$ . The first two equations force  $a = 2$ ,  $b = 1$ , and then  $a + 2b = 4$  holds. Hence  $(2, 1, 4) = 2(1, 0, 1) + (0, 1, 2)$  is in the span.
- (b) The system is  $a + 2b = 1$ ,  $2a + 4b = 1$ ,  $b = 1$ . Doubling the first equation gives  $2a + 4b = 2$ , contradicting the second, since  $2 \neq 1$ . Hence the system is inconsistent and  $(1, 1, 1)$  is not in the span.
- (c) The system is  $a = 1$ ,  $b = 1$ ,  $a + b = 0$ . The first two force  $a = b = 1$ , and then  $a + b = 1 + 1 = 0$  in  $\mathbb{F}_2$ . Hence  $(1, 1, 0) = (1, 0, 1) + (0, 1, 1)$  is in the span over  $\mathbb{F}_2$ . In  $\mathbb{R}^3$  the same first two equations force  $a = b = 1$ , but  $1 + 1 = 2 \neq 0$ , so the system is inconsistent and  $\mathbf{b}$  is **not** in the span. The answer depends on the field.

### Exercise B3: Solve and describe.

Solve each system over  $\mathbb{R}$  by substitution, and write its solution set in the form  $\mathbf{p} + \text{span}(\dots)$ . Check that your  $\mathbf{p}$  is a solution and that your spanning vectors solve the associated homogeneous system.

- (a)  $x_1 + x_2 + x_3 = 2$ ,  $x_2 - x_3 = 1$ .
- (b)  $x_1 - 2x_2 + x_3 - x_4 = 1$ ,  $x_3 + 2x_4 = 3$ .

*Solution.*

- (a) The second equation gives  $x_2 = 1 + x_3$ , and then the first gives  $x_1 = 2 - (1 + x_3) - x_3 = 1 - 2x_3$ . With  $x_3 = t$  free, every solution is  $(1 - 2t, 1 + t, t)$ , and conversely each such vector satisfies both equations. So the solution set is  $(1, 1, 0) + \text{span}((-2, 1, 1))$ .

Check:  $1 + 1 + 0 = 2$  and  $1 - 0 = 1$ ; and  $-2 + 1 + 1 = 0$ ,  $1 - 1 = 0$ .

- (b) The second equation gives  $x_3 = 3 - 2x_4$ . The first then gives  $x_1 = 1 + 2x_2 - (3 - 2x_4) + x_4 = -2 + 2x_2 + 3x_4$ . The unknowns  $x_2 = s$  and  $x_4 = t$  are free, and every solution is

$$(-2 + 2s + 3t, s, 3 - 2t, t) = (-2, 0, 3, 0) + s(2, 1, 0, 0) + t(3, 0, -2, 1),$$

and conversely each such vector is a solution. So the solution set is  $(-2, 0, 3, 0) + \text{span}((2, 1, 0, 0), (3, 0, -2, 1))$ . Check:  $-2 - 0 + 3 - 0 = 1$  and  $3 + 0 = 3$ ; for  $(2, 1, 0, 0)$ :  $2 - 2 + 0 - 0 = 0$  and  $0 + 0 = 0$ ; for  $(3, 0, -2, 1)$ :  $3 - 0 - 2 - 1 = 0$  and  $-2 + 2 = 0$ .

### 2.1.6.3 C. Going deeper

#### Exercise C1: How many solutions?

Let  $\mathbf{A} \in M_{m \times n}(F)$  and  $\mathbf{b} \in F^m$ , and let  $N$  be the solution set of  $\mathbf{Ax} = \mathbf{0}$ .

- Suppose  $F$  is **infinite**. Prove that  $\mathbf{Ax} = \mathbf{b}$  has either no solution, exactly one solution, or infinitely many solutions.
- Suppose  $F$  is **finite**. Prove that  $\mathbf{Ax} = \mathbf{b}$  has either no solution or exactly  $|N|$  solutions.
- Give a system over  $\mathbb{F}_3$  with exactly three solutions.

*Hint: for (a), if  $N$  contains a non-zero vector  $\mathbf{h}$ , look at its multiples.*

*Solution.*

- Suppose the system is consistent, with a solution  $\mathbf{p}$ . By Theorem 2.1.3 its solution set is  $\mathbf{p} + N$ , and as in the proof of Corollary 2.1.4 the map  $\mathbf{h} \mapsto \mathbf{p} + \mathbf{h}$  is a bijection  $N \rightarrow \mathbf{p} + N$ . If  $N = \{\mathbf{0}\}$ , there is exactly one solution. Otherwise, let  $\mathbf{h} \in N$  with  $\mathbf{h} \neq \mathbf{0}$ . Since  $N$  is a subspace (Theorem 2.1.2),  $c\mathbf{h} \in N$  for every  $c \in F$ . If  $c\mathbf{h} = c'\mathbf{h}$ , then  $(c - c')\mathbf{h} = \mathbf{0}$ , and since  $\mathbf{h} \neq \mathbf{0}$ , Theorem 1.1.9 gives  $c = c'$ . So  $c \mapsto c\mathbf{h}$  is injective, and  $N$  contains infinitely many vectors because  $F$  is infinite. Hence  $\mathbf{p} + N$  is infinite.
- If the system is consistent with solution  $\mathbf{p}$ , the bijection  $N \rightarrow \mathbf{p} + N$  from (a) shows that the solution set has exactly  $|N|$  elements. Otherwise it has none.
- The single equation  $x + y = 1$  over  $\mathbb{F}_3$  (a system with  $m = 1$ ,  $n = 2$ ). For each of the three values of  $y \in \mathbb{F}_3$  there is exactly one  $x = 1 - y$ , so the solutions are  $(1, 0)$ ,  $(0, 1)$ ,  $(2, 2)$ . Consistently with (b), the homogeneous equation  $x + y = 0$  has the three solutions  $(0, 0)$ ,  $(2, 1)$ ,  $(1, 2)$ .

#### Exercise C2: The right-hand sides that work.

Let  $\mathbf{A} \in M_{m \times n}(F)$  have columns  $\mathbf{a}_1, \dots, \mathbf{a}_n$ .

- Prove that the set  $C = \{\mathbf{b} \in F^m : \mathbf{Ax} = \mathbf{b} \text{ is consistent}\}$  is a subspace of  $F^m$ .
- Deduce that if  $n < m$ , then there is some  $\mathbf{b} \in F^m$  for which  $\mathbf{Ax} = \mathbf{b}$  is inconsistent.



- (c) For  $\mathbf{A} = \begin{pmatrix} 1 & 2 \\ 2 & 4 \\ 0 & 1 \end{pmatrix}$  over  $\mathbb{R}$ , find one such  $\mathbf{b}$ .

*Solution.*

- (a) By Theorem 2.1.1,  $\mathbf{b} \in C$  if and only if  $\mathbf{b} \in \text{span}(\mathbf{a}_1, \dots, \mathbf{a}_n)$ . So  $C = \text{span}(\mathbf{a}_1, \dots, \mathbf{a}_n)$ , which is a subspace of  $F^m$  by Theorem 1.3.1.
- (b) By (a),  $C$  is spanned by the  $n$  vectors  $\mathbf{a}_1, \dots, \mathbf{a}_n$ . Since  $\dim F^m = m$  (Example (Dimensions of the Running Examples)), Theorem 1.6.3 says no list of fewer than  $m$  vectors spans  $F^m$ . As  $n < m$ ,  $C \neq F^m$ , so some  $\mathbf{b} \in F^m$  is not in  $C$ , and for that  $\mathbf{b}$  the system is inconsistent.
- (c) Take  $\mathbf{b} = (1, 0, 0)$ . A solution would satisfy  $x + 2y = 1$  and  $2x + 4y = 0$ . Doubling the first gives  $2x + 4y = 2 \neq 0$ , a contradiction. So  $\mathbf{Ax} = (1, 0, 0)$  is inconsistent.

## 2.2 Gaussian Elimination

Every reader of this book has solved a system of equations by eliminating unknowns: subtract a multiple of one equation from another, and a variable disappears. Done by hand on small systems, this feels like common sense. Done on a system of 50 equations, or over  $\mathbb{F}_2$ , or in a proof about all systems at once, it needs to be an algorithm, and we need to know that it never changes the answer. This section makes elimination precise, proves that it always reaches a standard shape, and shows how to read every solution from that shape.

### 2.2.1 Three moves that do not change the answer

Here is a system over  $\mathbb{R}$ , solved the way one would solve it on paper:

$$\begin{aligned} x + 2y + z &= 4, \\ 2x + 4y + 3z &= 9, \\ x + 3y + z &= 5. \end{aligned}$$

Subtract twice the first equation from the second:  $z = 1$ . Subtract the first from the third:  $y = 1$ . Then the first equation gives  $x = 4 - 2 - 1 = 1$ .

Only three kinds of move were used, or could reasonably be used: **reorder** the equations, **multiply** an equation by a non-zero number, and **add a multiple** of one equation to another. None of them looks at the names  $x, y, z$ . Each acts on the coefficients and the right-hand side of whole equations, that is, on the rows of the augmented matrix. The first step above, for instance, replaces row 2,  $(2, 4, 3 \mid 9)$ , by row 2 minus twice row 1,  $(0, 0, 1 \mid 1)$ .

Not every move that looks harmless is safe. Multiplying an equation by 0 turns it into  $0 = 0$ , and a constraint silently disappears. So we list the allowed moves carefully.

*An elementary row operation is a reversible move on the rows of a matrix: swap two rows, rescale one row by a non-zero scalar, or add a multiple of one row to a different row.*

**Definition 2.2.1 — Elementary row operations**

Let  $F$  be a field and let  $\mathbf{M} \in M_{m \times k}(F)$  have rows  $\mathbf{r}_1, \dots, \mathbf{r}_m$ . An **elementary row operation** on  $\mathbf{M}$  is one of the following.

- (1) **Swap** two rows: for  $i \neq j$ , exchange  $\mathbf{r}_i$  and  $\mathbf{r}_j$ . Notation:  $R_i \leftrightarrow R_j$ .
- (2) **Scale** a row: for an index  $i$  and a scalar  $c \in F$  with  $c \neq 0$ , replace  $\mathbf{r}_i$  by  $c\mathbf{r}_i$ . Notation:  $R_i \rightarrow cR_i$ .
- (3) **Replace** a row by adding a multiple of **another** row: for indices  $i \neq j$  and any  $c \in F$ , replace  $\mathbf{r}_i$  by  $\mathbf{r}_i + c\mathbf{r}_j$ . Notation:  $R_i \rightarrow R_i + cR_j$ .

In each case the rows not mentioned are left unchanged.

In words: (1) reorders, (2) rescales one row by a **non-zero** scalar, and (3) changes **one** row, row  $i$ , using a **different** row  $j$ , which itself stays as it was. We write  $R_i \rightarrow R_i - cR_j$  for  $R_i \rightarrow R_i + (-c)R_j$ . The notation  $R_i$  refers to the current row  $i$  at the moment the operation is performed.

Take  $\mathbf{M} = \begin{pmatrix} 1 & 2 & 0 \\ 3 & 1 & 1 \end{pmatrix}$  over  $\mathbb{R}$ . Then  $R_1 \leftrightarrow R_2$  gives  $\begin{pmatrix} 3 & 1 & 1 \\ 1 & 2 & 0 \end{pmatrix}$ ,  $R_1 \rightarrow -2R_1$  gives  $\begin{pmatrix} -2 & -4 & 0 \\ 3 & 1 & 1 \end{pmatrix}$ , and  $R_2 \rightarrow R_2 - 3R_1$  gives  $\begin{pmatrix} 1 & 2 & 0 \\ 0 & -5 & 1 \end{pmatrix}$ . The degenerate case  $m = 1$  is worth a glance: a one-row matrix admits no swap and no replacement, since both need two different rows, so its only elementary row operations are the scalings.

The minimal change that breaks the definition is to drop “ $c \neq 0$ ”. The move  $R_2 \rightarrow 0R_2$  still replaces one row by a multiple of itself, but it sends  $\begin{pmatrix} 1 & 2 & 0 \\ 3 & 1 & 1 \end{pmatrix}$  to  $\begin{pmatrix} 1 & 2 & 0 \\ 0 & 0 & 0 \end{pmatrix}$ , and no move of any kind can recover the lost row  $(3, 1, 1)$  from that matrix. The failing clause is  $c \neq 0$ , and the damage is exactly irreversibility.

*Warning. Perform operations one at a time.* It is tempting to write “ $R_1 \rightarrow R_1 - R_2$  and  $R_2 \rightarrow R_2 - R_1$ ” as one step, using the old rows in both. That is not a sequence of elementary row operations, and it can destroy information. On the system  $x = 1$ ,  $x = 1$  it produces  $0 = 0$ ,  $0 = 0$ , whose solution set is all of  $F$ , not  $\{1\}$ . Done in order, the second operation uses the **new** row 1,  $(0 \mid 0)$ , and leaves row 2 unchanged, so no information is lost.

Why these three and no others? They are exactly the moves of hand elimination, and each can be undone by a move of the same kind, which is what the theorem below needs. The condition  $i \neq j$  in (3) is not fussiness: “add  $c$  times row  $i$  to row  $i$ ” would be scaling by  $1 + c$ , which is illegal when  $c = -1$  (Exercise C1).

An elementary row operation acts on the rows, but it treats each column separately: the new entry in position  $(i, l)$  is computed only from entries of column  $l$ . We record this, because it will be used again and again.

**Lemma 2.2.1 — Row operations act column by column**

Let  $\mathbf{M} = [\mathbf{M}_1 \mid \mathbf{M}_2] \in M_{m \times k}(F)$  be split into its first  $k_1$  columns  $\mathbf{M}_1$  and its remaining columns  $\mathbf{M}_2$ . If an elementary row operation turns  $\mathbf{M}$  into  $\mathbf{M}' = [\mathbf{M}'_1 \mid \mathbf{M}'_2]$  (split

in the same place), then the same operation turns  $\mathbf{M}_1$  into  $\mathbf{M}'_1$  and  $\mathbf{M}_2$  into  $\mathbf{M}'_2$ . In particular, an elementary row operation turns a zero column into a zero column.

*Proof.* Let  $m_{hl}$  be the entries of  $\mathbf{M}$  and  $m'_{hl}$  those of  $\mathbf{M}'$ . For a swap  $R_i \leftrightarrow R_j$ :  $m'_{il} = m_{jl}$ ,  $m'_{jl} = m_{il}$ , and  $m'_{hl} = m_{hl}$  for  $h \neq i, j$ . For a scaling  $R_i \rightarrow cR_i$ :  $m'_{il} = cm_{il}$ , and other rows are unchanged. For a replacement  $R_i \rightarrow R_i + cR_j$ :  $m'_{il} = m_{il} + cm_{jl}$ , and other rows are unchanged. In each case, every entry of column  $l$  of  $\mathbf{M}'$  is computed from column  $l$  of  $\mathbf{M}$  alone, by the same rule for every  $l$ . Hence applying the operation to  $\mathbf{M}_1$  or to  $\mathbf{M}_2$  separately gives  $\mathbf{M}'_1$  and  $\mathbf{M}'_2$ . If column  $l$  of  $\mathbf{M}$  is zero, each formula gives 0, since  $c \cdot 0 = 0$  and  $0 + c \cdot 0 = 0$  (Theorem 0.7.1). ■

For an augmented matrix, the lemma says: operating on  $[\mathbf{A} \mid \mathbf{b}]$  is the same as operating on  $\mathbf{A}$  and on  $\mathbf{b}$  with the same moves. Now we name what sequences of moves produce.

### Definition 2.2.2 — Row equivalence

Let  $\mathbf{A}, \mathbf{B} \in M_{m \times k}(F)$ . We say  $\mathbf{A}$  is **row equivalent** to  $\mathbf{B}$  if  $\mathbf{B}$  can be obtained from  $\mathbf{A}$  by a **finite** sequence of elementary row operations. The empty sequence is allowed, so every matrix is row equivalent to itself.

In Example (Row equivalence) of Chapter 0 we met a definition that looks different:  $\mathbf{B} = \mathbf{EA}$  for an **invertible** matrix  $\mathbf{E}$ . In section 4 of this chapter we prove that the two definitions agree (Theorem 2.4.7), by writing each elementary row operation as multiplication by an invertible matrix. Until then, “row equivalent” means the definition above.

## 2.2.2 Row operations preserve solutions

Here is the promise from the start of the section, now as a theorem.

### Theorem 2.2.2 — Row operations preserve solutions

Let  $F$  be a field.

- (a) (**Reversibility**) If an elementary row operation turns  $\mathbf{M}$  into  $\mathbf{M}'$ , then an elementary row operation **of the same type** turns  $\mathbf{M}'$  back into  $\mathbf{M}$ .
- (b) Row equivalence is an equivalence relation on  $M_{m \times k}(F)$ .
- (c) If  $[\mathbf{A} \mid \mathbf{b}]$  and  $[\mathbf{A}' \mid \mathbf{b}']$  in  $M_{m \times (n+1)}(F)$  are row equivalent, then the systems  $\mathbf{Ax} = \mathbf{b}$  and  $\mathbf{A'x} = \mathbf{b'}$  have the **same** solution set.

*Idea.* Reversibility is checked type by type, and the check for scaling is the only place where  $c \neq 0$  is used. For (c), every equation of the new system is a combination of equations of the old one, so an old solution satisfies the new system. That gives one inclusion. Instead of repeating the argument backwards, reversibility gives the other inclusion for free.

*Proof.*

- (a) A swap  $R_i \leftrightarrow R_j$  is undone by the same swap. A scaling  $R_i \rightarrow cR_i$  is undone by  $R_i \rightarrow c^{-1}R_i$ , which is a legal scaling because  $c \neq 0$  has an inverse  $c^{-1} \neq 0$  in  $F$  (Definition 0.7.1): the row becomes  $c^{-1}(c\mathbf{r}_i) = \mathbf{r}_i$ . A replacement  $R_i \rightarrow R_i + cR_j$  is undone by  $R_i \rightarrow R_i - cR_j$ . Here we use  $i \neq j$ : row  $j$  was not changed by the first operation, so the second one gives  $(\mathbf{r}_i + c\mathbf{r}_j) - c\mathbf{r}_j = \mathbf{r}_i$ .
- (b) Reflexivity holds via the empty sequence. For symmetry, suppose operations  $O_1, \dots, O_s$ , in this order, turn  $\mathbf{A}$  into  $\mathbf{B}$ . By (a), each  $O_t$  has an undoing operation  $O'_t$ . Then  $O'_s, O'_{s-1}, \dots, O'_1$ , in this order, turn  $\mathbf{B}$  back into  $\mathbf{A}$ . For transitivity, if a sequence turns  $\mathbf{A}$  into  $\mathbf{B}$  and another turns  $\mathbf{B}$  into  $\mathbf{C}$ , performing the first and then the second turns  $\mathbf{A}$  into  $\mathbf{C}$ , and this is still a finite sequence.
- (c) First suppose a **single** operation turns  $[\mathbf{A} \mid \mathbf{b}]$  into  $[\mathbf{A}' \mid \mathbf{b}']$ . Let  $S$  and  $S'$  be the two solution sets. Write  $\mathbf{r}_h$  for row  $h$  of  $\mathbf{A}$ , so that equation  $h$  says  $\mathbf{r}_h\mathbf{x} = b_h$ , where  $\mathbf{r}_h\mathbf{x}$  is a  $1 \times 1$  product. Let  $\mathbf{s} \in S$ , so  $\mathbf{r}_h\mathbf{s} = b_h$  for every  $h$ . Each equation of the new system holds at  $\mathbf{s}$ :
- after  $R_i \leftrightarrow R_j$ , the new equations are the old ones in another order;
  - after  $R_i \rightarrow cR_i$ , the new equation  $i$  is  $(c\mathbf{r}_i)\mathbf{x} = cb_i$ , and  $(c\mathbf{r}_i)\mathbf{s} = c(\mathbf{r}_i\mathbf{s}) = cb_i$ ;
  - after  $R_i \rightarrow R_i + cR_j$ , the new equation  $i$  is  $(\mathbf{r}_i + c\mathbf{r}_j)\mathbf{x} = b_i + cb_j$ , and  $(\mathbf{r}_i + c\mathbf{r}_j)\mathbf{s} = \mathbf{r}_i\mathbf{s} + c(\mathbf{r}_j\mathbf{s}) = b_i + cb_j$ , by Theorem 0.9.4.

The other equations are unchanged. Hence  $\mathbf{s} \in S'$ , and  $S \subseteq S'$ . By (a), a single operation turns  $[\mathbf{A}' \mid \mathbf{b}']$  back into  $[\mathbf{A} \mid \mathbf{b}]$ , so the same argument gives  $S' \subseteq S$ . Therefore  $S = S'$ .

For a sequence of  $s$  operations, the intermediate augmented matrices all have the same solution set, by the single-operation case applied  $s$  times (formally, by induction on  $s$ ). This proves (c). ■

This theorem is the license for everything that follows: we may transform a system as much as we like by row operations, and the solutions of the final system are the solutions of the original one. What it does **not** license is just as important.

**Warning. Scaling by 0 is not a row operation, and it changes the solution set.** Over  $\mathbb{R}$ , the system  $x + y = 2$ ,  $x - y = 0$  has exactly one solution,  $(1, 1)$ . Applying the illegal move  $R_2 \rightarrow 0R_2$  gives  $x + y = 2$ ,  $0 = 0$ , whose solutions  $(2 - t, t)$  form a whole line. The inclusion  $S \subseteq S'$  in the proof above still holds; what fails is reversibility, and with it  $S' \subseteq S$ .

**Warning. Column operations are not allowed.** Operations on the columns of  $[\mathbf{A} \mid \mathbf{b}]$  mix up the unknowns. The system  $x + y = 3$ ,  $x - y = 1$  has the unique solution  $(2, 1)$ . Adding column 1 to column 2 of its augmented matrix gives  $\left[ \begin{array}{cc|c} 1 & 2 & 3 \\ 1 & 0 & 1 \end{array} \right]$ , the system  $x + 2y = 3$ ,  $x = 1$ , whose only solution is  $(1, 1)$ . Even swapping two columns of  $\mathbf{A}$  swaps the corresponding entries of every solution.

## 2.2.3 Echelon forms

Row operations can change a system a great deal. We now choose a target: a shape of matrix from which the solutions can be read off with no further work. Look at the finished computation from the start of the section. After eliminating, the augmented matrix could be brought to

$$\left[ \begin{array}{ccc|c} 1 & 0 & 0 & 1 \\ 0 & 1 & 0 & 1 \\ 0 & 0 & 1 & 1 \end{array} \right],$$

which says  $x = 1$ ,  $y = 1$ ,  $z = 1$  outright. Not every system can be brought to this shape, since not every system has exactly one solution. The general target is a staircase.

For a non-zero row of a matrix, its **leading entry** is its first non-zero entry, reading from the left.

*A matrix is in row echelon form when its non-zero rows come first and their leading entries step strictly to the right as we go down.*

### Definition 2.2.3 — Row echelon form

A matrix  $\mathbf{R} \in M_{m \times k}(F)$  is in **row echelon form** if:

- (E1) there is an integer  $r$  with  $0 \leq r \leq m$  such that rows  $1, \dots, r$  are non-zero and rows  $r+1, \dots, m$  are zero; and
- (E2) if the leading entry of row  $i$  lies in column  $j_i$ , for  $i = 1, \dots, r$ , then  $j_1 < j_2 < \dots < j_r$ , **strictly**.

The leading entries of rows  $1, \dots, r$  are the **pivots** of  $\mathbf{R}$ , and the columns  $j_1, \dots, j_r$  are its **pivot columns**.

In words: (E1) says zero rows sit at the bottom, and (E2) says each pivot lies strictly to the right of the pivot above it. Two consequences are used constantly. Each row contains at most one pivot, so there are at most  $m$  pivots. And **every entry below a pivot is zero**: if  $h > i$ , then row  $h$  is zero or has its leading entry in column  $j_h > j_i$ , so its entry in column  $j_i$  is 0.

### Definition 2.2.4 — Reduced row echelon form

A matrix  $\mathbf{R} \in M_{m \times k}(F)$  is in **reduced row echelon form (RREF)** if it is in row echelon form, with pivot columns  $j_1 < \dots < j_r$ , and moreover:

- (E3) every pivot equals 1:  $r_{ij_i} = 1$  for  $i = 1, \dots, r$ ; and
- (E4) every pivot is the **only** non-zero entry in its column:  $r_{hj_i} = 0$  for all  $h \neq i$ .

In words: (E3) normalizes the pivots, and (E4) clears the entries **above** each pivot as well as below. Together they say that the pivot column  $j_i$  is the standard basis vector  $\mathbf{e}_i \in F^m$ .

#### Example. (Echelon or not?)

Decide which of the following real matrices are in row echelon form, and which are in

reduced row echelon form. Give the pivot columns where they exist.

$$\mathbf{P} = \begin{pmatrix} 2 & 1 & 3 \\ 0 & 0 & -1 \\ 0 & 0 & 0 \end{pmatrix}, \quad \mathbf{Q} = \begin{pmatrix} 1 & 5 & 0 & 2 \\ 0 & 0 & 1 & 3 \end{pmatrix}, \quad \mathbf{T} = \begin{pmatrix} 1 & 0 & 2 \\ 0 & 0 & 1 \\ 0 & 1 & 0 \end{pmatrix}, \quad \mathbf{Z} = \begin{pmatrix} 0 & 0 \\ 0 & 0 \end{pmatrix}.$$

*Solution.*  $\mathbf{P}$  is in row echelon form: rows 1 and 2 are non-zero, row 3 is zero, and the leading entries lie in columns  $1 < 3$ . The pivot columns are 1 and 3. It is **not** reduced: the pivot 2 is not 1, failing (E3), and column 3 has the non-zero entry 3 above the pivot  $-1$ , failing (E4).

$\mathbf{Q}$  is in reduced row echelon form, with pivot columns 1 and 3. Both pivots are 1, and columns 1 and 3 are  $\mathbf{e}_1$  and  $\mathbf{e}_2$ . The entries 5, 2, 3 in the non-pivot columns 2 and 4 are allowed to be anything.

$\mathbf{T}$  is **not** in row echelon form. Its leading entries lie in columns 1, 3, 2, and  $3 < 2$  fails, so (E2) fails. Swapping rows 2 and 3 would repair it.

$\mathbf{Z}$  is in reduced row echelon form, with  $r = 0$ : there are no pivots, and (E2)–(E4) hold vacuously. This degenerate case matters, because the zero matrix must have an RREF too, and it is itself.

For  $\mathbf{Q}$ , a minimal change gives a non-example: replacing the 0 in position  $(1, 3)$  by 4 keeps (E1)–(E3), but column 3 now has the non-zero entry 4 above its pivot, so (E4) fails.

Why insist on the reduced form, when row echelon form already allows back substitution? Because a system in reduced form needs **no** substitution: each pivot variable appears in exactly one equation. More importantly, the reduced form turns out to be **unique**: a matrix is row equivalent to exactly one matrix in RREF, however the operations are chosen. That is proved in the next section, and it is what makes later definitions, such as the rank, independent of choices. Row echelon form is not unique, as the warning after the worked example shows.

*Quick check.* Is  $\begin{pmatrix} 1 & 2 & 0 & -1 \\ 0 & 0 & 1 & 3 \\ 0 & 0 & 0 & 0 \end{pmatrix}$  in reduced row echelon form? If it is the coefficient matrix of a homogeneous system in  $x_1, x_2, x_3, x_4$ , which unknowns correspond to pivot columns?

*Solution.* Yes. Row 3 is the only zero row and it is last; the leading entries lie in columns  $1 < 3$ ; both are 1; and columns 1 and 3 are  $\mathbf{e}_1$  and  $\mathbf{e}_2$ . The pivot columns are 1 and 3, so the unknowns  $x_1$  and  $x_3$  correspond to pivots, and  $x_2, x_4$  do not.

## 2.2.4 Gauss–Jordan elimination

We now show that every matrix can be brought to reduced row echelon form. The procedure works through the columns from left to right, keeping a count  $r$  of the pivots found so far.

*Algorithm.* **Gauss–Jordan elimination.** Input:  $\mathbf{M} \in M_{m \times k}(F)$ . Set  $r = 0$ . For  $l = 1, 2, \dots, k$  in turn:

1. Look for a row  $h$  with  $h > r$  whose entry in column  $l$  is non-zero. If there is none, column  $l$  gets no pivot; go on to the next  $l$ .
2. Otherwise choose such an  $h$ . If  $h \neq r + 1$ , perform  $R_{r+1} \leftrightarrow R_h$ .
3. Let  $a \neq 0$  be the entry now in position  $(r + 1, l)$ . Perform  $R_{r+1} \rightarrow a^{-1}R_{r+1}$ , so that this entry becomes 1.
4. For every row  $g \neq r + 1$  whose entry  $c$  in column  $l$  is non-zero, perform  $R_g \rightarrow R_g - cR_{r+1}$ .
5. Increase  $r$  by 1.

Output: the final matrix, which is in reduced row echelon form, with  $r$  pivots.

Step 2 involves a choice: any non-zero entry will do. By hand, we choose one that avoids fractions. If we skip the scaling in step 3 and clear only the entries **below** the pivot  $a$ , by  $R_g \rightarrow R_g - ca^{-1}R_{r+1}$  for the rows  $g > r + 1$ , the output is a row echelon form instead; this cheaper variant is **Gaussian elimination**, and it is followed by back substitution.

### Theorem 2.2.3 — Existence of the reduced row echelon form

Let  $F$  be a field. Every matrix  $\mathbf{M} \in M_{m \times k}(F)$  is row equivalent to a matrix in reduced row echelon form.

*Idea.* We prove that the algorithm works, by induction on the number  $l$  of columns processed. The invariant is: the first  $l$  columns of the current matrix form a matrix in RREF. When a new column is processed, all the operations involve only rows below the pivots found so far, or add multiples of the new pivot row, which is zero in the first  $l$  columns. So the part already built is never disturbed. After all  $k$  columns, the whole matrix is in RREF.

*Proof.* For  $0 \leq l \leq k$ , let  $P(l)$  be the statement: there is a matrix  $\mathbf{M}_l$  row equivalent to  $\mathbf{M}$  and an integer  $r \geq 0$  such that the  $m \times l$  matrix  $\mathbf{L}$  formed by the first  $l$  columns of  $\mathbf{M}_l$  is in reduced row echelon form with exactly  $r$  non-zero rows. We prove  $P(l)$  for all  $l$  by induction (Theorem 0.2.7). For  $l = 0$  take  $\mathbf{M}_0 = \mathbf{M}$  and  $r = 0$ ; there are no columns, so there is nothing to check.

Suppose  $P(l)$  holds for some  $l < k$ , with  $\mathbf{M}_l$ ,  $\mathbf{L}$  and  $r$  as described, and let  $j_1 < \dots < j_r \leq l$  be the pivot columns of  $\mathbf{L}$ . By (E1) for  $\mathbf{L}$ , rows  $r + 1, \dots, m$  of  $\mathbf{M}_l$  are zero in columns  $1, \dots, l$ . Let  $\mathbf{L}'$  denote the first  $l + 1$  columns of the matrix we are about to construct.

*Case 1: every entry of column  $l + 1$  of  $\mathbf{M}_l$  in rows  $h > r$  is zero* (this includes the case  $r = m$ ). Put  $\mathbf{M}_{l+1} = \mathbf{M}_l$ . Rows  $r + 1, \dots, m$  of  $\mathbf{L}'$  are zero, and rows  $1, \dots, r$  are non-zero with the same leading entries as in  $\mathbf{L}$ , which lie in columns  $\leq l$ . So (E1)–(E3) hold for  $\mathbf{L}'$  with the same  $r$  and pivot columns, and so does (E4), since the pivot columns of  $\mathbf{L}'$  are those of  $\mathbf{L}$  and are unchanged. Hence  $\mathbf{L}'$  is in RREF and  $P(l + 1)$  holds.

*Case 2: some row  $h > r$  has a non-zero entry in column  $l + 1$ .* Then  $r + 1 \leq h \leq m$ . Perform steps 2, 3 and 4 of the algorithm with this  $h$  and with column  $l + 1$ ; call the result  $\mathbf{M}_{l+1}$ . It is row equivalent to  $\mathbf{M}_l$ , hence to  $\mathbf{M}$ , by Theorem 2.2.2 (b).

These operations do not change columns  $1, \dots, l$ . Indeed, the swap in step 2 and the scaling in step 3 only move or rescale rows  $h$  and  $r + 1$ , which are both zero in those columns. Each replacement  $R_g \rightarrow R_g - cR_{r+1}$  in step 4 adds a multiple of row  $r + 1$ ,

which is still zero in columns  $1, \dots, l$ . So the first  $l$  columns of  $\mathbf{M}_{l+1}$  form the matrix  $\mathbf{L}$ . In column  $l+1$ , step 3 makes the entry in row  $r+1$  equal to 1, and each replacement in step 4 makes the entry of row  $g$  equal to  $c - c \cdot 1 = 0$  without changing any other row. So column  $l+1$  of  $\mathbf{M}_{l+1}$  is  $\mathbf{e}_{r+1}$ .

We check that  $\mathbf{L}'$  is in RREF with  $r+1$  non-zero rows. Rows  $1, \dots, r$  of  $\mathbf{L}'$  have the same leading entries as in  $\mathbf{L}$ , in columns  $j_1 < \dots < j_r \leq l$ . Row  $r+1$  is zero in columns  $1, \dots, l$  and has a 1 in column  $l+1$ , so its leading entry is in column  $j_{r+1} := l+1 > j_r$ . Rows  $r+2, \dots, m$  are zero in columns  $1, \dots, l$  and in column  $l+1$ , so they are zero. This gives (E1) and (E2). The pivots are 1, by (E3) for  $\mathbf{L}$  and the construction, so (E3) holds. For (E4), the columns  $j_1, \dots, j_r$  are unchanged from  $\mathbf{L}$ , and column  $l+1$  is  $\mathbf{e}_{r+1}$ . Hence  $P(l+1)$  holds.

By induction  $P(k)$  holds:  $\mathbf{M}_k$  is row equivalent to  $\mathbf{M}$ , and its first  $k$  columns, which are all of  $\mathbf{M}_k$ , form a matrix in reduced row echelon form. This proves the theorem. ■

The proof is also the correctness proof of the algorithm: the matrix  $\mathbf{M}_l$  is the state of the computation after column  $l$ . Note where the field axioms were used. Step 3 divides by  $a \neq 0$ , which needs (F8), and nothing else in the argument depends on which field we work over.

### Example. (Gauss–Jordan on a $3 \times 4$ system)

Solve the following system over  $\mathbb{R}$  by Gauss–Jordan elimination, naming each operation:

$$\begin{aligned} 2x_2 - 2x_3 + x_4 &= 3, \\ x_1 + x_2 + x_3 &= 3, \\ 2x_1 + x_2 + 3x_3 + x_4 &= 3. \end{aligned}$$

*Solution.* We work on the augmented matrix. **Column 1.** Row 1 has a 0 there, so we bring up row 2, whose entry is already 1. Then we clear the rest of column 1.

$$\begin{aligned} \left[ \begin{array}{cccc|c} 0 & 2 & -2 & 1 & 3 \\ 1 & 1 & 1 & 0 & 3 \\ 2 & 1 & 3 & 1 & 3 \end{array} \right] &\xrightarrow{R_1 \leftrightarrow R_2} \left[ \begin{array}{cccc|c} 1 & 1 & 1 & 0 & 3 \\ 0 & 2 & -2 & 1 & 3 \\ 2 & 1 & 3 & 1 & 3 \end{array} \right] \\ &\xrightarrow{R_3 \rightarrow R_3 - 2R_1} \left[ \begin{array}{cccc|c} 1 & 1 & 1 & 0 & 3 \\ 0 & 2 & -2 & 1 & 3 \\ 0 & -1 & 1 & 1 & -3 \end{array} \right] \end{aligned}$$

**Column 2.** The candidates below row 1 are 2 and  $-1$ . Choosing  $-1$  avoids fractions, so we swap it up and scale.

$$\xrightarrow{R_2 \leftrightarrow R_3} \left[ \begin{array}{cccc|c} 1 & 1 & 1 & 0 & 3 \\ 0 & -1 & 1 & 1 & -3 \\ 0 & 2 & -2 & 1 & 3 \end{array} \right] \xrightarrow{R_2 \rightarrow -R_2} \left[ \begin{array}{cccc|c} 1 & 1 & 1 & 0 & 3 \\ 0 & 1 & -1 & -1 & 3 \\ 0 & 2 & -2 & 1 & 3 \end{array} \right]$$

Now clear column 2 above and below the pivot.

$$\xrightarrow{R_1 \rightarrow R_1 - R_2} \left[ \begin{array}{cccc|c} 1 & 0 & 2 & 1 & 0 \\ 0 & 1 & -1 & -1 & 3 \\ 0 & 2 & -2 & 1 & 3 \end{array} \right] \xrightarrow{R_3 \rightarrow R_3 - 2R_2} \left[ \begin{array}{cccc|c} 1 & 0 & 2 & 1 & 0 \\ 0 & 1 & -1 & -1 & 3 \\ 0 & 0 & 0 & 3 & -3 \end{array} \right]$$



**Column 3.** The only entry below row 2 is 0, so column 3 gets no pivot. **Column 4.** The entry 3 in row 3 becomes the pivot.

$$\begin{aligned} &\xrightarrow{R_3 \rightarrow \frac{1}{3}R_3} \left[ \begin{array}{cccc|c} 1 & 0 & 2 & 1 & 0 \\ 0 & 1 & -1 & -1 & 3 \\ 0 & 0 & 0 & 1 & -1 \end{array} \right] \\ &\xrightarrow{R_1 \rightarrow R_1 - R_3} \left[ \begin{array}{cccc|c} 1 & 0 & 2 & 0 & 1 \\ 0 & 1 & -1 & -1 & 3 \\ 0 & 0 & 0 & 1 & -1 \end{array} \right] \\ &\xrightarrow{R_2 \rightarrow R_2 + R_3} \left[ \begin{array}{cccc|c} 1 & 0 & 2 & 0 & 1 \\ 0 & 1 & -1 & 0 & 2 \\ 0 & 0 & 0 & 1 & -1 \end{array} \right] \end{aligned}$$

**Column 5.** No rows remain below row 3, so we stop. The final matrix is in RREF, with pivot columns 1, 2 and 4. It is the system

$$x_1 + 2x_3 = 1, \quad x_2 - x_3 = 2, \quad x_4 = -1,$$

which by [Theorem 2.2.2](#) has the same solutions as the original. The unknown  $x_3$  is unconstrained; writing  $x_3 = t$ , the solutions are

$$(1 - 2t, 2 + t, t, -1) = (1, 2, 0, -1) + t(-2, 1, 1, 0), \quad t \in \mathbb{R}.$$

Check:  $(1, 2, 0, -1)$  gives  $4 + 0 - 1 = 3$ ,  $1 + 2 + 0 = 3$  and  $2 + 2 + 0 - 1 = 3$ ; and  $(-2, 1, 1, 0)$  gives  $2 - 2 + 0 = 0$ ,  $-2 + 1 + 1 = 0$  and  $-4 + 1 + 3 + 0 = 0$ . This is the “particular plus homogeneous” form of [Theorem 2.1.3](#).

A computer algebra system does the same computation exactly. In the online version the cell below can be edited and run, which is a good way to check a hand computation.

```
from sympy import Matrix

M = Matrix([
    [0, 2, -2, 1, 3],
    [1, 1, 1, 0, 3],
    [2, 1, 3, 1, 3],
])
R, pivots = M.rref() # the RREF and the (0-based) pivot columns
print(R)
print("pivot columns:", [j + 1 for j in pivots])
```

**Warning. Echelon forms are not unique; only the reduced form is.** Different choices in step 2 give different intermediate matrices. For  $\begin{pmatrix} 1 & 2 \\ 3 & 4 \end{pmatrix}$  over  $\mathbb{R}$ , the operation  $R_2 \rightarrow R_2 - 3R_1$  gives the row echelon form  $\begin{pmatrix} 1 & 2 \\ 0 & -2 \end{pmatrix}$ . Starting instead with  $R_1 \leftrightarrow R_2$  and then  $R_2 \rightarrow R_2 - \frac{1}{3}R_1$  gives the row echelon form  $\begin{pmatrix} 3 & 4 \\ 0 & \frac{2}{3} \end{pmatrix}$ . Both continue to the same RREF,  $I_2$ . That this always happens is the theorem of the next section; until it is proved, “the” RREF of a matrix is not yet justified.

## 2.2.5 Reading off the solutions

Once  $[\mathbf{A} \mid \mathbf{b}]$  is in reduced row echelon form, the example shows what to do: a pivot in the last column signals a contradiction, and otherwise the unknowns without pivots can be chosen freely. We name the two kinds of unknowns.

Suppose  $[\mathbf{A} \mid \mathbf{b}]$ , with  $\mathbf{A} \in M_{m \times n}(F)$ , is row equivalent to a matrix  $[\mathbf{R} \mid \mathbf{c}]$  in reduced row echelon form, where  $\mathbf{R}$  is  $m \times n$ . For  $l \leq n$ , the unknown  $x_l$  is a **pivot variable** if  $l$  is a pivot column of  $[\mathbf{R} \mid \mathbf{c}]$ , and a **free variable** otherwise. (By the uniqueness theorem of the next section, this does not depend on which sequence of operations was used.)

### Theorem 2.2.4 — Reading solutions from the RREF

Let  $\mathbf{A} \in M_{m \times n}(F)$  and  $\mathbf{b} \in F^m$ , and suppose  $[\mathbf{A} \mid \mathbf{b}]$  is row equivalent to a matrix  $[\mathbf{R} \mid \mathbf{c}]$  in reduced row echelon form. Let  $r_{il}$  be the entries of  $\mathbf{R}$  and  $\mathbf{c} = (c_1, \dots, c_m)$ .

- (a)  $\mathbf{Ax} = \mathbf{b}$  is inconsistent **if and only if** the last column, column  $n + 1$ , is a pivot column of  $[\mathbf{R} \mid \mathbf{c}]$ , that is, some row of  $[\mathbf{R} \mid \mathbf{c}]$  is  $(0 \ \cdots \ 0 \mid 1)$ .
- (b) Suppose there is no pivot in the last column. Let  $j_1 < \cdots < j_r$  be the pivot columns. Then for **every** choice of scalars  $t_l \in F$ , one for each free variable  $x_l$ , there is **exactly one** solution with  $x_l = t_l$  for all free  $l$ , namely the one with

$$x_{j_i} = c_i - \sum_{l \text{ free}} r_{il} t_l \quad (i = 1, \dots, r).$$

In particular, setting every free variable to 0 gives the solution  $\mathbf{p}$  with  $p_{j_i} = c_i$  and all other entries 0.

- (c) A consistent system  $\mathbf{Ax} = \mathbf{b}$  has exactly one solution **if and only if** it has no free variables, that is, each of the columns  $1, \dots, n$  is a pivot column.

*Proof.* By Theorem 2.2.2 (c),  $\mathbf{Ax} = \mathbf{b}$  has the same solution set as  $\mathbf{Rx} = \mathbf{c}$ , so we work with the latter.

(a,  $\Leftarrow$ ) If row  $i$  of  $[\mathbf{R} \mid \mathbf{c}]$  has its pivot in column  $n + 1$ , then its first  $n$  entries are 0 and  $c_i = 1$  by (E3). Equation  $i$  reads  $0 = 1$ , which is false in any field (Definition 0.7.1). So there is no solution.

- (b) Suppose no pivot lies in column  $n + 1$ . Then all pivots of  $[\mathbf{R} \mid \mathbf{c}]$  lie in columns  $j_1, \dots, j_r \leq n$ . By (E1), rows  $r + 1, \dots, m$  of  $[\mathbf{R} \mid \mathbf{c}]$  are zero, so the corresponding equations read  $0 = 0$  and hold for every  $\mathbf{x}$ . For  $i \leq r$ , row  $i$  of  $\mathbf{R}$  has  $r_{ij_i} = 1$  by (E3) and  $r_{ij_h} = 0$  for  $h \neq i$  by (E4). So equation  $i$  reads

$$x_{j_i} + \sum_{l \text{ free}} r_{il} x_l = c_i.$$

Hence  $\mathbf{x}$  is a solution if and only if  $x_{j_i} = c_i - \sum_{l \text{ free}} r_{il} x_l$  for  $i = 1, \dots, r$ . Each unknown is either a pivot variable or a free variable, not both. So once the free variables are given the values  $t_l$ , these equations determine every pivot variable uniquely, and the resulting vector is a solution. This proves (b).

(a,  $\Rightarrow$ ) This is the contrapositive of what (b) just showed: if there is no pivot in the last column, then choosing all  $t_l = 0$  produces a solution, so the system is consistent.

- (c) By (a), a consistent system has no pivot in the last column, so (b) applies. If there are no free variables, (b) gives exactly one solution. If  $x_l$  is free, the choices  $t_l = 0$  and  $t_l = 1$ , with all other free variables 0, give two solutions whose  $l$ -th entries differ, since  $0 \neq 1$ . This proves (c). ■

In [Example \(Gauss–Jordan on a  \$3 \times 4\$  system\)](#) the pivot columns were 1, 2, 4, the free variable was  $x_3$ , and (b) gave  $x_1 = 1 - 2t$ ,  $x_2 = 2 + t$ ,  $x_4 = -1$ , with  $\mathbf{p} = (1, 2, 0, -1)$ . The same recipe works over any field, and over a finite field the free variables can take only finitely many values.

**Example. (Elimination over  $\mathbb{F}_2$ )**

Solve the following system over  $\mathbb{F}_2$ :

$$x_1 + x_2 + x_4 = 1, \quad x_2 + x_3 + x_4 = 0, \quad x_1 + x_3 + x_4 = 1.$$

How many solutions are there? What changes if the same equations are read over  $\mathbb{R}$ ?

*Solution.* In  $\mathbb{F}_2$ ,  $-1 = 1$ , so  $R_i \rightarrow R_i - R_j$  is the same operation as  $R_i \rightarrow R_i + R_j$ , and every non-zero pivot is already 1. **Column 1** has its pivot in row 1; clear row 3. **Column 2** has its pivot in row 2; clear rows 3 and 1.

$$\begin{aligned} \left[ \begin{array}{cccc|c} 1 & 1 & 0 & 1 & 1 \\ 0 & 1 & 1 & 1 & 0 \\ 1 & 0 & 1 & 1 & 1 \end{array} \right] &\xrightarrow{R_3 \rightarrow R_3 + R_1} \left[ \begin{array}{cccc|c} 1 & 1 & 0 & 1 & 1 \\ 0 & 1 & 1 & 1 & 0 \\ 0 & 1 & 1 & 0 & 0 \end{array} \right] &\xrightarrow{R_3 \rightarrow R_3 + R_2} \left[ \begin{array}{cccc|c} 1 & 1 & 0 & 1 & 1 \\ 0 & 1 & 1 & 1 & 0 \\ 0 & 0 & 0 & 1 & 0 \end{array} \right] \\ &\xrightarrow{R_1 \rightarrow R_1 + R_2} \left[ \begin{array}{cccc|c} 1 & 0 & 1 & 0 & 1 \\ 0 & 1 & 1 & 1 & 0 \\ 0 & 0 & 0 & 1 & 0 \end{array} \right] \end{aligned}$$

In the row  $R_3 + R_1$ , the entries  $1 + 1$  became 0, and in  $R_3 + R_2$  the entry in column 3 is  $1 + 1 = 0$ . **Column 3** has only a 0 below row 2, so no pivot. **Column 4** has its pivot in row 3; row 1 already has a 0 there, so only row 2 needs clearing.

$$\xrightarrow{R_2 \rightarrow R_2 + R_3} \left[ \begin{array}{cccc|c} 1 & 0 & 1 & 0 & 1 \\ 0 & 1 & 1 & 0 & 0 \\ 0 & 0 & 0 & 1 & 0 \end{array} \right]$$

This is in RREF, with pivot columns 1, 2, 4 and no pivot in the last column, so the system is consistent by [Theorem 2.2.4](#) (a). The free variable is  $x_3 = t$ , and part (b) gives  $x_1 = 1 - t = 1 + t$ ,  $x_2 = -t = t$ ,  $x_4 = 0$ . Since  $t \in \mathbb{F}_2 = \{0, 1\}$ , there are exactly two solutions:

$$(1, 0, 0, 0) \quad \text{and} \quad (0, 1, 1, 0).$$

Check the second:  $0 + 1 + 0 = 1$ ,  $1 + 1 + 0 = 0$ ,  $0 + 1 + 0 = 1$  in  $\mathbb{F}_2$ .

Over  $\mathbb{R}$  the first step is  $R_3 \rightarrow R_3 - R_1$ , giving row 3 =  $(0, -1, 1, 0 \mid 0)$ , and adding row 2 then gives  $(0, 0, 2, 1 \mid 0)$ . The entry 2 is non-zero in  $\mathbb{R}$ , so column 3 now gets a pivot. Completing the reduction gives pivot columns 1, 2, 3, free variable  $x_4$ , and infinitely many real solutions. The same equations lead to different pivot patterns over different fields, because  $1 + 1 = 0$  in one and not in the other.

When a coefficient depends on a parameter, whether a column gets a pivot can depend on the parameter too. Then the reduction splits into cases, one for each answer to “is this entry zero?”.

**Example. (A system with a parameter)**

For which  $k \in \mathbb{R}$  does the system  $x + ky = 1$ ,  $kx + y = 1$  have no solution, exactly one solution, or infinitely many? Find the solutions when they exist.

*Solution.* The entry in position  $(1, 1)$  is 1 for every  $k$ , so it is a pivot with no case distinction:

$$\left[ \begin{array}{cc|c} 1 & k & 1 \\ k & 1 & 1 \end{array} \right] \xrightarrow{R_2 \rightarrow R_2 - kR_1} \left[ \begin{array}{cc|c} 1 & k & 1 \\ 0 & 1 - k^2 & 1 - k \end{array} \right].$$

(For  $k = 0$  this operation changes nothing, which is allowed.) Column 2 gets a pivot in row 2 exactly when  $1 - k^2 = (1 - k)(1 + k)$  is non-zero, and only then may we scale by its inverse. So we split into cases.

*Case  $k \neq \pm 1$ .* Then  $1 - k^2 \neq 0$ , and  $R_2 \rightarrow (1 - k^2)^{-1}R_2$  turns row 2 into  $(0, 1 \mid \frac{1}{1+k})$ , since  $\frac{1-k}{(1-k)(1+k)} = \frac{1}{1+k}$ . Then  $R_1 \rightarrow R_1 - kR_2$  turns row 1 into  $(1, 0 \mid 1 - \frac{k}{1+k}) = (1, 0 \mid \frac{1}{1+k})$ . The RREF is  $[\mathbf{I}_2 \mid \mathbf{c}]$  with  $\mathbf{c} = (\frac{1}{1+k}, \frac{1}{1+k})$ , and by [Theorem 2.2.4](#) (c) this is the only solution. Check:  $\frac{1}{1+k} + \frac{k}{1+k} = 1$  for both equations.

*Case  $k = 1$ .* Row 2 is  $(0, 0 \mid 0)$ , and  $\left[ \begin{array}{cc|c} 1 & 1 & 1 \\ 0 & 0 & 0 \end{array} \right]$  is in RREF with pivot column 1. The free variable is  $y = t$ , and [Theorem 2.2.4](#) (b) gives the solutions  $(1 - t, t)$ ,  $t \in \mathbb{R}$ : infinitely many.

*Case  $k = -1$ .* Row 2 is  $(0, 0 \mid 2)$ , and  $R_2 \rightarrow \frac{1}{2}R_2$  makes it  $(0, 0 \mid 1)$ . The last column is a pivot column, so the system is inconsistent by [Theorem 2.2.4](#) (a). Directly:  $x - y = 1$  and  $-x + y = 1$  add up to  $0 = 2$ .

Hence there is no solution for  $k = -1$ , there are infinitely many for  $k = 1$ , and there is exactly one otherwise. The trap is to scale by  $(1 - k^2)^{-1}$  before asking whether  $1 - k^2$  is zero: that step silently assumes  $k \neq \pm 1$ , and loses exactly the two interesting cases.

## 2.2.6 More unknowns than equations

Counting pivots gives a first general theorem. A matrix with  $m$  rows has at most  $m$  pivots, one per non-zero row. If there are more unknowns than that, some unknown must be free.

### Corollary 2.2.5 — More unknowns than equations

Let  $\mathbf{A} \in M_{m \times n}(F)$  with  $n > m$ .

- (a) The homogeneous system  $\mathbf{Ax} = \mathbf{0}$  has a **non-trivial** solution.
- (b) Consequently, every list of  $n$  vectors in  $F^m$  with  $n > m$  is linearly dependent.

*Proof.*

- (a) By [Theorem 2.2.3](#),  $[\mathbf{A} \mid \mathbf{0}]$  is row equivalent to a matrix  $[\mathbf{R} \mid \mathbf{c}]$  in RREF. By

**Lemma 2.2.1**, each operation keeps the last column zero, so  $\mathbf{c} = \mathbf{0}$ , and the last column is not a pivot column. Each pivot is the leading entry of a different row, so there are at most  $m$  pivots, hence at most  $m < n$  pivot columns among the first  $n$  columns. So some  $x_l$  is a free variable. By **Theorem 2.2.4** (b), with  $t_l = 1$  and every other free variable 0, there is a solution whose  $l$ -th entry is  $1 \neq 0$ . This solution is non-trivial.

- (b) Let  $\mathbf{v}_1, \dots, \mathbf{v}_n \in F^m$  and let  $\mathbf{A} \in M_{m \times n}(F)$  have these columns. By (a) there is  $\mathbf{x} \neq \mathbf{0}$  with  $\mathbf{A}\mathbf{x} = \mathbf{0}$ , and by **Theorem 0.9.3** this says  $x_1\mathbf{v}_1 + \dots + x_n\mathbf{v}_n = \mathbf{0}$  with  $x_1, \dots, x_n$  not all zero. So the list is linearly dependent. ■

Part (b) is not new: since  $\dim F^m = m$ , it is also a case of **Theorem 1.6.3**, which rests on the Steinitz Exchange Theorem (**Theorem 1.6.1**). The proof here is independent of that one. It uses nothing from Chapter 1 beyond the definition of independence, only elimination and a count of pivots. Two proofs by different routes give some confidence that the dimension theory of Chapter 1 and the computations of this chapter will fit together, and the rest of the chapter makes that fit exact.

## 2.2.7 Exercises

### 2.2.7.1 A. Check your understanding

#### Exercise A1.

- List the three types of elementary row operation, including the conditions on the scalar and on the indices.
- Determine whether the following statement is true: “every matrix in row echelon form is in reduced row echelon form.” Justify your answer.
- Determine whether the following statement is true: “if the RREF of  $[\mathbf{A} \mid \mathbf{b}]$  has a zero row, then  $\mathbf{A}\mathbf{x} = \mathbf{b}$  is inconsistent.” Justify your answer.
- Determine whether the following statement is true: “a homogeneous system of 3 equations in 5 unknowns over  $\mathbb{R}$  has infinitely many solutions.” Justify your answer.
- Explain why swapping two columns of  $\mathbf{A}$  in  $[\mathbf{A} \mid \mathbf{b}]$  is **not** a safe step when solving  $\mathbf{A}\mathbf{x} = \mathbf{b}$ .

*Solution.*

- $R_i \leftrightarrow R_j$  with  $i \neq j$ ;  $R_i \rightarrow cR_i$  with  $c \neq 0$ ;  $R_i \rightarrow R_i + cR_j$  with  $i \neq j$  and any  $c \in F$ .
- False.  $\begin{pmatrix} 2 & 1 \\ 0 & 1 \end{pmatrix}$  is in row echelon form, but its first pivot is  $2 \neq 1$ , so (E3) fails; its second column also has a non-zero entry above the pivot, so (E4) fails.
- False. A zero row is the equation  $0 = 0$ , which every vector satisfies. Inconsistency means a row  $(0 \ \cdots \ 0 \mid 1)$  (**Theorem 2.2.4** (a)). For example  $\left[ \begin{array}{c|c} 1 & 1 \\ 0 & 0 \end{array} \right]$  is the consistent system  $x = 1, 0 = 0$ .

- (d) True. By [Corollary 2.2.5](#) ( $5 > 3$ ) there is a non-trivial solution  $\mathbf{h}$ , and since  $\mathbb{R}$  is infinite, the multiples  $c\mathbf{h}$  ( $c \in \mathbb{R}$ ) are infinitely many distinct solutions, as in [Exercise \(C1: How many solutions?\)](#).
- (e) It swaps the roles of two unknowns:  $\mathbf{s}$  solves the new system exactly when the vector obtained from  $\mathbf{s}$  by swapping those two entries solves the old one. So the solution set changes, unless the swap happens to fix it. For instance  $x = 1, y = 2$  becomes  $y = 1, x = 2$ .

### 2.2.7.2 B. Practice

#### Exercise B1: Reduce and solve.

Reduce the augmented matrix of each system over  $\mathbb{R}$  to reduced row echelon form, naming each operation. Hence decide whether the system is consistent, and if so, describe all solutions.

- (a)  $x_1 + 2x_2 - x_3 = 1, 2x_1 + 5x_2 + x_3 = 4, x_1 + 3x_2 + 2x_3 = 4.$   
 (b)  $x_1 + x_2 + 2x_3 = 3, 2x_1 + 3x_2 + 5x_3 = 7, x_1 + 2x_2 + 4x_3 = 5.$   
 (c)  $x_1 - x_2 + 2x_3 + x_4 = 2, 2x_1 - 2x_2 + 5x_3 + 4x_4 = 5, -x_1 + x_2 - x_3 + x_4 = -1.$

*Solution.*

(a)

$$\begin{aligned} \left[ \begin{array}{ccc|c} 1 & 2 & -1 & 1 \\ 2 & 5 & 1 & 4 \\ 1 & 3 & 2 & 4 \end{array} \right] &\xrightarrow{R_2 \rightarrow R_2 - 2R_1} \left[ \begin{array}{ccc|c} 1 & 2 & -1 & 1 \\ 0 & 1 & 3 & 2 \\ 1 & 3 & 2 & 4 \end{array} \right] \xrightarrow{R_3 \rightarrow R_3 - R_1} \left[ \begin{array}{ccc|c} 1 & 2 & -1 & 1 \\ 0 & 1 & 3 & 2 \\ 0 & 1 & 3 & 3 \end{array} \right] \\ &\xrightarrow{R_1 \rightarrow R_1 - 2R_2} \left[ \begin{array}{ccc|c} 1 & 0 & -7 & -3 \\ 0 & 1 & 3 & 2 \\ 0 & 1 & 3 & 3 \end{array} \right] \xrightarrow{R_3 \rightarrow R_3 - R_2} \left[ \begin{array}{ccc|c} 1 & 0 & -7 & -3 \\ 0 & 1 & 3 & 2 \\ 0 & 0 & 0 & 1 \end{array} \right] \end{aligned}$$

Column 3 gets no pivot. Column 4 has its pivot in row 3:

$$\xrightarrow{R_1 \rightarrow R_1 + 3R_3} \left[ \begin{array}{ccc|c} 1 & 0 & -7 & 0 \\ 0 & 1 & 3 & 2 \\ 0 & 0 & 0 & 1 \end{array} \right] \xrightarrow{R_2 \rightarrow R_2 - 2R_3} \left[ \begin{array}{ccc|c} 1 & 0 & -7 & 0 \\ 0 & 1 & 3 & 0 \\ 0 & 0 & 0 & 1 \end{array} \right].$$

The last column is a pivot column, so by [Theorem 2.2.4](#) (a) the system is inconsistent. (Row 3 already said  $0 = 1$  before the last two operations.)

(b)

$$\left[ \begin{array}{ccc|c} 1 & 1 & 2 & 3 \\ 2 & 3 & 5 & 7 \\ 1 & 2 & 4 & 5 \end{array} \right] \xrightarrow{R_2 \rightarrow R_2 - 2R_1} \left[ \begin{array}{ccc|c} 1 & 1 & 2 & 3 \\ 0 & 1 & 1 & 1 \\ 1 & 2 & 4 & 5 \end{array} \right] \xrightarrow{R_3 \rightarrow R_3 - R_1} \left[ \begin{array}{ccc|c} 1 & 1 & 2 & 3 \\ 0 & 1 & 1 & 1 \\ 0 & 1 & 2 & 2 \end{array} \right]$$

$$\begin{aligned}
&\xrightarrow{R_1 \rightarrow R_1 - R_2} \left[ \begin{array}{ccc|c} 1 & 0 & 1 & 2 \\ 0 & 1 & 1 & 1 \\ 0 & 1 & 2 & 2 \end{array} \right] \\
&\xrightarrow{R_3 \rightarrow R_3 - R_2} \left[ \begin{array}{ccc|c} 1 & 0 & 1 & 2 \\ 0 & 1 & 1 & 1 \\ 0 & 0 & 1 & 1 \end{array} \right] \\
&\xrightarrow{R_1 \rightarrow R_1 - R_3} \left[ \begin{array}{ccc|c} 1 & 0 & 0 & 1 \\ 0 & 1 & 1 & 1 \\ 0 & 0 & 1 & 1 \end{array} \right] \\
&\xrightarrow{R_2 \rightarrow R_2 - R_3} \left[ \begin{array}{ccc|c} 1 & 0 & 0 & 1 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 1 \end{array} \right].
\end{aligned}$$

Columns 1, 2, 3 are pivot columns and the last column is not, so by [Theorem 2.2.4](#) (c) there is exactly one solution,  $(1, 0, 1)$ . Check:  $1 + 0 + 2 = 3$ ,  $2 + 0 + 5 = 7$ ,  $1 + 0 + 4 = 5$ .

(c)

$$\begin{aligned}
&\left[ \begin{array}{cccc|c} 1 & -1 & 2 & 1 & 2 \\ 2 & -2 & 5 & 4 & 5 \\ -1 & 1 & -1 & 1 & -1 \end{array} \right] \xrightarrow{R_2 \rightarrow R_2 - 2R_1} \left[ \begin{array}{cccc|c} 1 & -1 & 2 & 1 & 2 \\ 0 & 0 & 1 & 2 & 1 \\ -1 & 1 & -1 & 1 & -1 \end{array} \right] \\
&\xrightarrow{R_3 \rightarrow R_3 + R_1} \left[ \begin{array}{cccc|c} 1 & -1 & 2 & 1 & 2 \\ 0 & 0 & 1 & 2 & 1 \\ 0 & 0 & 1 & 2 & 1 \end{array} \right]
\end{aligned}$$

Column 2 has only zeros below row 1, so no pivot. Column 3 has its pivot in row 2:

$$\xrightarrow{R_1 \rightarrow R_1 - 2R_2} \left[ \begin{array}{cccc|c} 1 & -1 & 0 & -3 & 0 \\ 0 & 0 & 1 & 2 & 1 \\ 0 & 0 & 1 & 2 & 1 \end{array} \right] \xrightarrow{R_3 \rightarrow R_3 - R_2} \left[ \begin{array}{cccc|c} 1 & -1 & 0 & -3 & 0 \\ 0 & 0 & 1 & 2 & 1 \\ 0 & 0 & 0 & 0 & 0 \end{array} \right].$$

Columns 4 and 5 have only a 0 in row 3, so this is the RREF. The pivot columns are 1 and 3; the free variables are  $x_2 = s$  and  $x_4 = t$ . By [Theorem 2.2.4](#) (b),  $x_1 = s + 3t$  and  $x_3 = 1 - 2t$ . Hence the solution set is

$$\{(0, 0, 1, 0) + s(1, 1, 0, 0) + t(3, 0, -2, 1) : s, t \in \mathbb{R}\}.$$

### Exercise B2: A parameter.

For which  $k \in \mathbb{R}$  is the system

$$x + y - z = 2, \quad x + 2y + z = 3, \quad x + y + (k^2 - 5)z = k$$

inconsistent, which give exactly one solution, and which give infinitely many? Justify your answer, and find the solutions when they exist.

*Solution.*

$$\begin{aligned} \left[ \begin{array}{ccc|c} 1 & 1 & -1 & 2 \\ 1 & 2 & 1 & 3 \\ 1 & 1 & k^2 - 5 & k \end{array} \right] &\xrightarrow{R_2 \rightarrow R_2 - R_1} \left[ \begin{array}{ccc|c} 1 & 1 & -1 & 2 \\ 0 & 1 & 2 & 1 \\ 1 & 1 & k^2 - 5 & k \end{array} \right] \\ &\xrightarrow{R_3 \rightarrow R_3 - R_1} \left[ \begin{array}{ccc|c} 1 & 1 & -1 & 2 \\ 0 & 1 & 2 & 1 \\ 0 & 0 & k^2 - 4 & k - 2 \end{array} \right] \\ &\xrightarrow{R_1 \rightarrow R_1 - R_2} \left[ \begin{array}{ccc|c} 1 & 0 & -3 & 1 \\ 0 & 1 & 2 & 1 \\ 0 & 0 & k^2 - 4 & k - 2 \end{array} \right]. \end{aligned}$$

Whether column 3 gets a pivot depends on whether  $k^2 - 4 = (k - 2)(k + 2)$  is zero.

*Case  $k \neq \pm 2$ .* Then  $k^2 - 4 \neq 0$ , and  $R_3 \rightarrow (k^2 - 4)^{-1}R_3$  turns row 3 into  $(0, 0, 1 \mid \frac{1}{k+2})$ , since  $\frac{k-2}{(k-2)(k+2)} = \frac{1}{k+2}$ . Then  $R_1 \rightarrow R_1 + 3R_3$  and  $R_2 \rightarrow R_2 - 2R_3$  give the RREF  $[\mathbf{I}_3 \mid \mathbf{c}]$  with

$$\mathbf{c} = \left( \frac{k+5}{k+2}, \frac{k}{k+2}, \frac{1}{k+2} \right),$$

using  $1 + \frac{3}{k+2} = \frac{k+5}{k+2}$  and  $1 - \frac{2}{k+2} = \frac{k}{k+2}$ . By [Theorem 2.2.4](#) (c) there is exactly one solution,  $\mathbf{c}$ .

*Case  $k = 2$ .* Row 3 is zero, and the matrix is in RREF with pivot columns 1, 2 and free variable  $z = t$ . The solutions are  $(1 + 3t, 1 - 2t, t)$ ,  $t \in \mathbb{R}$ : infinitely many.

*Case  $k = -2$ .* Row 3 is  $(0, 0, 0 \mid -4)$ , and  $R_3 \rightarrow -\frac{1}{4}R_3$  makes it  $(0, 0, 0 \mid 1)$ . The last column is a pivot column, so the system is inconsistent.

Hence: inconsistent for  $k = -2$ , infinitely many solutions for  $k = 2$ , and exactly one solution otherwise.

### Exercise B3: Over $\mathbb{F}_3$ .

Solve each system over  $\mathbb{F}_3$  by reducing to RREF, naming each operation, and list all solutions.

- (a)  $x + 2y + z = 1$ ,  $2x + y + z = 0$ ,  $x + z = 2$ .
- (b)  $x + y + 2z = 1$ ,  $2x + z = 2$ .

*Solution.* In  $\mathbb{F}_3$  the inverses are  $1^{-1} = 1$  and  $2^{-1} = 2$ , since  $2 \cdot 2 = 4 = 1$ , and  $-1 = 2$ ,  $-2 = 1$ .

(a)

$$\left[ \begin{array}{ccc|c} 1 & 2 & 1 & 1 \\ 2 & 1 & 1 & 0 \\ 1 & 0 & 1 & 2 \end{array} \right] \xrightarrow{R_2 \rightarrow R_2 - 2R_1} \left[ \begin{array}{ccc|c} 1 & 2 & 1 & 1 \\ 0 & 0 & 2 & 1 \\ 1 & 0 & 1 & 2 \end{array} \right] \xrightarrow{R_3 \rightarrow R_3 - R_1} \left[ \begin{array}{ccc|c} 1 & 2 & 1 & 1 \\ 0 & 0 & 2 & 1 \\ 0 & 1 & 0 & 1 \end{array} \right]$$

Here  $1 - 4 = -3 = 0$ ,  $1 - 2 = -1 = 2$ ,  $0 - 2 = 1$ , and  $0 - 2 = 1$ ,  $2 - 1 = 1$ . Column



2 has entries 0, 1 below row 1, so we swap:

$$\begin{aligned} &\xrightarrow{R_2 \leftrightarrow R_3} \left[ \begin{array}{ccc|c} 1 & 2 & 1 & 1 \\ 0 & 1 & 0 & 1 \\ 0 & 0 & 2 & 1 \end{array} \right] \\ &\xrightarrow{R_1 \rightarrow R_1 - 2R_2} \left[ \begin{array}{ccc|c} 1 & 0 & 1 & 2 \\ 0 & 1 & 0 & 1 \\ 0 & 0 & 2 & 1 \end{array} \right] \\ &\xrightarrow{R_3 \rightarrow 2R_3} \left[ \begin{array}{ccc|c} 1 & 0 & 1 & 2 \\ 0 & 1 & 0 & 1 \\ 0 & 0 & 1 & 2 \end{array} \right] \\ &\xrightarrow{R_1 \rightarrow R_1 - R_3} \left[ \begin{array}{ccc|c} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 1 \\ 0 & 0 & 1 & 2 \end{array} \right]. \end{aligned}$$

There are no free variables, so the unique solution is  $(0, 1, 2)$ . Check:  $0 + 2 + 2 = 4 = 1$ ,  $0 + 1 + 2 = 3 = 0$ ,  $0 + 2 = 2$ .

(b)

$$\left[ \begin{array}{ccc|c} 1 & 1 & 2 & 1 \\ 2 & 0 & 1 & 2 \end{array} \right] \xrightarrow{R_2 \rightarrow R_2 - 2R_1} \left[ \begin{array}{ccc|c} 1 & 1 & 2 & 1 \\ 0 & 1 & 0 & 0 \end{array} \right] \xrightarrow{R_1 \rightarrow R_1 - R_2} \left[ \begin{array}{ccc|c} 1 & 0 & 2 & 1 \\ 0 & 1 & 0 & 0 \end{array} \right],$$

using  $0 - 2 = 1$ ,  $1 - 4 = 0$ ,  $2 - 2 = 0$ . The free variable is  $z = t$ , and  $x = 1 - 2t = 1 + t$ ,  $y = 0$ . As  $t$  runs through  $0, 1, 2$ , the solutions are  $(1, 0, 0)$ ,  $(2, 0, 1)$ ,  $(0, 0, 2)$ : exactly three.

### 2.2.7.3 C. Going deeper

#### Exercise C1: Why $i \neq j$ ?

Consider the move “ $R_i \rightarrow R_i + cR_i$ ”, which the definition does **not** allow.

- Show that for  $c = -1$  this move is not reversible: give a matrix on which its effect cannot be undone by any sequence of elementary row operations and moves of this kind, and a system whose solution set it changes.
- Show that for  $c \neq -1$  the move is an elementary row operation of another type. Explain why the condition  $i \neq j$  in type (3) therefore loses nothing.

*Solution.*

- The move replaces  $\mathbf{r}_i$  by  $\mathbf{r}_i + (-1)\mathbf{r}_i = \mathbf{0}$ . Apply it to the  $1 \times 2$  augmented matrix  $[1 \mid 1]$  of the equation  $x = 1$  over any field  $F$ : the result is  $[0 \mid 0]$ , the equation  $0 = 0$ . The solution set changes from  $\{1\}$  to all of  $F$ . No sequence of moves can return to  $[1 \mid 1]$ : every elementary row operation, and every move “ $R_i \rightarrow R_i + cR_i$ ”, sends a zero matrix to a zero matrix, by the formulas in the proof of [Lemma 2.2.1](#) and  $c \cdot 0 = 0$ .
- The move replaces  $\mathbf{r}_i$  by  $(1 + c)\mathbf{r}_i$ . If  $c \neq -1$ , then  $1 + c \neq 0$ , and this is the scaling  $R_i \rightarrow (1 + c)R_i$ , an operation of type (2). So allowing  $i = j$  in type (3) would add only moves that are already available (when  $c \neq -1$ ) and one move that destroys information ( $c = -1$ ). Requiring  $i \neq j$  excludes exactly the bad case.

**Exercise C2: The cost of elimination.**

Let  $\mathbf{A} \in M_n(F)$  and  $\mathbf{b} \in F^n$ , and suppose that Gauss–Jordan elimination on  $[\mathbf{A} \mid \mathbf{b}]$  finds, for each  $l = 1, \dots, n$ , a non-zero entry in position  $(l, l)$  and so never swaps. Count as one **operation** each multiplication or division of two scalars, under these conventions: scaling row  $l$  costs one division for each entry in columns  $l + 1, \dots, n + 1$  (the pivot is simply set to 1), and each replacement  $R_g \rightarrow R_g - cR_l$  costs one multiplication for each entry in columns  $l + 1, \dots, n + 1$  (the entry in column  $l$  is simply set to 0).

- Show that processing column  $l$  costs at most  $n(n + 1 - l)$  operations.
- Hence show that the whole elimination costs at most  $\frac{1}{2}(n^3 + n^2)$  operations.
- Roughly how many times more work is a system of  $2n$  equations than one of  $n$  equations, for large  $n$ ?

*Solution.*

- Scaling row  $l$  costs  $n + 1 - l$  divisions. At most  $n - 1$  other rows need a replacement, each costing  $n + 1 - l$  multiplications. The total is at most  $(n + 1 - l) + (n - 1)(n + 1 - l) = n(n + 1 - l)$ .
- Summing (a) over  $l = 1, \dots, n$ , and substituting  $q = n + 1 - l$ , which runs through  $1, \dots, n$ ,

$$\sum_{l=1}^n n(n + 1 - l) = n \sum_{q=1}^n q = n \cdot \frac{n(n + 1)}{2} = \frac{n^3 + n^2}{2},$$

using the formula for  $1 + \dots + n$  (Proposition 0.2.8).

- The bound is dominated by  $\frac{1}{2}n^3$  for large  $n$ , since  $\frac{1}{2}n^2$  is small compared with it. Replacing  $n$  by  $2n$  multiplies  $n^3$  by 8, so doubling the size of the system costs roughly eight times as much.

**Exercise C3: Solvable for every right-hand side.**

Let  $\mathbf{A} \in M_{m \times n}(F)$ , and suppose that  $\mathbf{Ax} = \mathbf{b}$  is consistent for **every**  $\mathbf{b} \in F^m$ . Prove that  $m \leq n$ .

*Hint:* suppose  $m > n$ , reduce  $\mathbf{A}$  to its RREF  $\mathbf{R}$ , and run the operations backwards on  $[\mathbf{R} \mid \mathbf{e}_m]$ .

*Solution.* Suppose, for a contradiction, that  $m > n$ . By Theorem 2.2.3, some sequence of operations  $O_1, \dots, O_s$  turns  $\mathbf{A}$  into a matrix  $\mathbf{R}$  in RREF. By (E2),  $\mathbf{R}$  has at most  $n$  pivots, one in each pivot column, so at most  $n < m$  non-zero rows, and by (E1) its last row is zero. Therefore the last row of  $[\mathbf{R} \mid \mathbf{e}_m]$  is  $(0 \ \cdots \ 0 \mid 1)$ , so the system  $\mathbf{Rx} = \mathbf{e}_m$  has no solution.

By Theorem 2.2.2 (a), each  $O_t$  is undone by an operation  $O'_t$ , and  $O'_s, \dots, O'_1$  turn  $\mathbf{R}$  back into  $\mathbf{A}$ . Apply them to  $[\mathbf{R} \mid \mathbf{e}_m]$ . By Lemma 2.2.1 they act on the two blocks separately, so the result is  $[\mathbf{A} \mid \mathbf{b}]$  for some  $\mathbf{b} \in F^m$ . By Theorem 2.2.2 (c),  $\mathbf{Ax} = \mathbf{b}$  has the same solutions as  $\mathbf{Rx} = \mathbf{e}_m$ , namely none. This contradicts the hypothesis, so  $m \leq n$ .

## 2.3 Uniqueness of the Reduced Form

Gauss–Jordan elimination involves choices: which non-zero entry to use as a pivot, and in what order to perform the operations. Different choices give different intermediate matrices, and different row echelon forms. The previous section claimed that they always end at the **same** reduced row echelon form. This matters more than it seems. Pivot columns, free variables, and later the rank of a matrix are all read off from the reduced form; if two people could reduce the same matrix to two different answers, none of these would be well defined. This section proves the claim.

### 2.3.1 What row operations remember

Row operations change the columns of a matrix. What do they keep? Take

$$\mathbf{A} = \begin{pmatrix} 1 & 2 & 1 & 3 \\ 2 & 4 & 0 & 2 \\ 3 & 6 & 2 & 7 \end{pmatrix} \quad \text{and its RREF} \quad \mathbf{R} = \begin{pmatrix} 1 & 2 & 0 & 1 \\ 0 & 0 & 1 & 2 \\ 0 & 0 & 0 & 0 \end{pmatrix}$$

over  $\mathbb{R}$  (the computation is carried out in an example later in this section). The columns of  $\mathbf{R}$  look nothing like those of  $\mathbf{A}$ . But look at how the columns depend on each other. In  $\mathbf{R}$ , column 2 is twice column 1, and column 4 is column 1 plus twice column 3. In  $\mathbf{A}$ , with columns  $\mathbf{a}_1, \dots, \mathbf{a}_4$ ,

$$\mathbf{a}_2 = (2, 4, 6) = 2\mathbf{a}_1, \quad \mathbf{a}_4 = (3, 2, 7) = (1, 2, 3) + 2(1, 0, 2) = \mathbf{a}_1 + 2\mathbf{a}_3.$$

The same relations, with the same coefficients. This is no accident. A relation  $x_1\mathbf{a}_1 + \dots + x_n\mathbf{a}_n = \mathbf{0}$  among the columns says  $\mathbf{A}\mathbf{x} = \mathbf{0}$  (Theorem 0.9.3), and row operations do not change the solutions of a homogeneous system.

#### Lemma 2.3.1 — Row equivalent matrices have the same homogeneous solutions

Let  $\mathbf{A}, \mathbf{B} \in M_{m \times n}(F)$  be row equivalent. Then for every  $\mathbf{x} \in F^n$ ,  $\mathbf{A}\mathbf{x} = \mathbf{0}$  if and only if  $\mathbf{B}\mathbf{x} = \mathbf{0}$ .

*Proof.* Let a sequence of elementary row operations turn  $\mathbf{A}$  into  $\mathbf{B}$ . By Lemma 2.2.1, the same sequence turns  $[\mathbf{A} \mid \mathbf{0}]$  into  $[\mathbf{B} \mid \mathbf{0}']$ , where  $\mathbf{0}'$  is obtained from the zero column by row operations and so is again the zero column. Hence  $[\mathbf{A} \mid \mathbf{0}]$  and  $[\mathbf{B} \mid \mathbf{0}]$  are row equivalent, and by Theorem 2.2.2 (c) the systems  $\mathbf{A}\mathbf{x} = \mathbf{0}$  and  $\mathbf{B}\mathbf{x} = \mathbf{0}$  have the same solution set. ■

Rewriting  $\mathbf{A}\mathbf{x}$  as a combination of columns turns the lemma into the statement we observed.

#### Lemma 2.3.2 — Row operations preserve column relations

Let  $\mathbf{A}, \mathbf{B} \in M_{m \times n}(F)$  have columns  $\mathbf{a}_1, \dots, \mathbf{a}_n$  and  $\mathbf{b}_1, \dots, \mathbf{b}_n$ , and suppose that  $\mathbf{A}\mathbf{x} = \mathbf{0} \Leftrightarrow \mathbf{B}\mathbf{x} = \mathbf{0}$  for every  $\mathbf{x} \in F^n$ . This holds, for instance, if  $\mathbf{A}$  and  $\mathbf{B}$  are row equivalent (Lemma 2.3.1). Then:

- (a) for all  $x_1, \dots, x_n \in F$ :  $x_1 \mathbf{a}_1 + \dots + x_n \mathbf{a}_n = \mathbf{0}$  if and only if  $x_1 \mathbf{b}_1 + \dots + x_n \mathbf{b}_n = \mathbf{0}$ ;
- (b) for every index  $l$  and all scalars  $d_1, \dots, d_{l-1}$ :  $\mathbf{a}_l = d_1 \mathbf{a}_1 + \dots + d_{l-1} \mathbf{a}_{l-1}$  if and only if  $\mathbf{b}_l = d_1 \mathbf{b}_1 + \dots + d_{l-1} \mathbf{b}_{l-1}$ ;
- (c) for every index  $l$ :  $\mathbf{a}_l \in \text{span}(\mathbf{a}_1, \dots, \mathbf{a}_{l-1})$  if and only if  $\mathbf{b}_l \in \text{span}(\mathbf{b}_1, \dots, \mathbf{b}_{l-1})$ .

*Proof.*

- (a) By [Theorem 0.9.3](#),  $x_1 \mathbf{a}_1 + \dots + x_n \mathbf{a}_n = \mathbf{A}\mathbf{x}$  and  $x_1 \mathbf{b}_1 + \dots + x_n \mathbf{b}_n = \mathbf{B}\mathbf{x}$ , where  $\mathbf{x} = (x_1, \dots, x_n)$ . The hypothesis says  $\mathbf{A}\mathbf{x} = \mathbf{0} \Leftrightarrow \mathbf{B}\mathbf{x} = \mathbf{0}$ .
- (b) The equation  $\mathbf{a}_l = d_1 \mathbf{a}_1 + \dots + d_{l-1} \mathbf{a}_{l-1}$  holds if and only if  $d_1 \mathbf{a}_1 + \dots + d_{l-1} \mathbf{a}_{l-1} + (-1)\mathbf{a}_l + 0\mathbf{a}_{l+1} + \dots + 0\mathbf{a}_n = \mathbf{0}$ . By (a), with these coefficients, this holds if and only if the same combination of the  $\mathbf{b}_j$  is  $\mathbf{0}$ , that is,  $\mathbf{b}_l = d_1 \mathbf{b}_1 + \dots + d_{l-1} \mathbf{b}_{l-1}$ .
- (c) By [Definition 1.3.2](#),  $\mathbf{a}_l \in \text{span}(\mathbf{a}_1, \dots, \mathbf{a}_{l-1})$  means that the left side of (b) holds for **some** scalars  $d_1, \dots, d_{l-1}$ . By (b) this happens if and only if the right side holds for some scalars, which means  $\mathbf{b}_l \in \text{span}(\mathbf{b}_1, \dots, \mathbf{b}_{l-1})$ .

■

*Warning.* **Row operations preserve column relations, not columns or their span.** For  $\mathbf{A} = \begin{pmatrix} 1 & 2 \\ 1 & 2 \end{pmatrix}$  over  $\mathbb{R}$ , the operation  $R_2 \rightarrow R_2 - R_1$  gives  $\begin{pmatrix} 1 & 2 \\ 0 & 0 \end{pmatrix}$ . The relation “column 2 = 2 · column 1” holds in both. But every column of  $\mathbf{A}$  lies on the line  $\text{span}((1, 1))$ , and every column of the new matrix lies on the different line  $\text{span}((1, 0))$ . So the span of the columns changed. To answer a question about the columns of  $\mathbf{A}$  themselves, find the relations from the reduced form, and then use them on the columns of  $\mathbf{A}$ .

## 2.3.2 The columns of a reduced matrix

In a matrix in RREF, the column relations can be read off at a glance: pivot columns are standard basis vectors, and a non-pivot column **displays its own coefficients**. We record this precisely, together with a description of the pivot columns that does not mention pivots at all.

### Lemma 2.3.3 — Columns of a matrix in RREF

Let  $\mathbf{R} \in M_{m \times n}(F)$  be in reduced row echelon form, with entries  $r_{il}$ , columns  $\mathbf{u}_1, \dots, \mathbf{u}_n$ , and pivot columns  $j_1 < \dots < j_r$ .

- (a) For each  $i = 1, \dots, r$ , the pivot column  $\mathbf{u}_{j_i}$  is  $\mathbf{e}_i$ .
- (b) If  $l$  is **not** a pivot column, then

$$\mathbf{u}_l = \sum_{i: j_i < l} r_{il} \mathbf{e}_i = \sum_{i: j_i < l} r_{il} \mathbf{u}_{j_i}.$$

- (c) Column  $l$  is a pivot column **if and only if**  $\mathbf{u}_l \notin \text{span}(\mathbf{u}_1, \dots, \mathbf{u}_{l-1})$ .

*Proof.*

- (a) By (E3) the entry of  $\mathbf{u}_{j_i}$  in row  $i$  is 1, and by (E4) its other entries are 0.
- (b) Let  $l$  be a non-pivot column, and look at the entry  $r_{il}$  in each row  $i$ . If  $i > r$ , row  $i$  is zero by (E1), so  $r_{il} = 0$ . If  $i \leq r$  and  $j_i > l$ , then  $r_{il}$  lies strictly before the leading entry of row  $i$ , so  $r_{il} = 0$ . The case  $j_i = l$  does not occur, since  $l$  is not a pivot column. So the only possibly non-zero entries of  $\mathbf{u}_l$  are the  $r_{il}$  with  $j_i < l$ , which gives the first expression; the second follows from (a).
- (c) If  $l$  is not a pivot column, (b) writes  $\mathbf{u}_l$  as a combination of the columns  $\mathbf{u}_{j_i}$  with  $j_i < l$ , so  $\mathbf{u}_l \in \text{span}(\mathbf{u}_1, \dots, \mathbf{u}_{l-1})$ . Conversely, let  $l = j_i$  be a pivot column. For each  $l' < j_i$ , the entry  $r_{il'}$  lies before the leading entry of row  $i$ , so it is 0. Hence every vector of  $\text{span}(\mathbf{u}_1, \dots, \mathbf{u}_{l-1})$ , being a combination of vectors with  $i$ -th entry 0, has  $i$ -th entry 0. But  $\mathbf{u}_l = \mathbf{e}_i$  has  $i$ -th entry 1  $\neq 0$ , so  $\mathbf{u}_l \notin \text{span}(\mathbf{u}_1, \dots, \mathbf{u}_{l-1})$ .

■

Part (c) says: reading the columns from left to right, a column gets a pivot exactly when it is **new**, not a combination of the columns before it. That condition is about column relations only, and column relations survive row operations. This is the whole idea of the uniqueness proof.

### 2.3.3 The uniqueness theorem

We can now prove that the reduced form depends only on the homogeneous solutions, and hence not on the operations used to reach it.

#### Theorem 2.3.4 — Uniqueness of the reduced row echelon form

Let  $\mathbf{R}, \mathbf{R}' \in M_{m \times n}(F)$  be in reduced row echelon form, and suppose that  $\mathbf{R}\mathbf{x} = \mathbf{0} \Leftrightarrow \mathbf{R}'\mathbf{x} = \mathbf{0}$  for every  $\mathbf{x} \in F^n$ . Then  $\mathbf{R} = \mathbf{R}'$ .

Consequently, every matrix  $\mathbf{A} \in M_{m \times n}(F)$  is row equivalent to **exactly one** matrix in reduced row echelon form, called **the RREF of A**.

*Idea.* A column dependence  $\sum_j x_j \mathbf{u}_j = \mathbf{0}$  is exactly a null vector  $\mathbf{x}$ , so  $\mathbf{R}$  and  $\mathbf{R}'$  have the same column relations. The pivot columns are exactly the columns that are not combinations of earlier columns (Lemma 2.3.3 (c)), so  $\mathbf{R}$  and  $\mathbf{R}'$  have the same pivot columns. A pivot column is a standard basis vector in both. A non-pivot column of an RREF matrix is determined by how it is built from the pivot columns before it (Lemma 2.3.3 (b)), and that recipe is a column relation, so it is the same in both. Hence every column agrees.

*Proof.* Let  $\mathbf{u}_1, \dots, \mathbf{u}_n$  and  $\mathbf{u}'_1, \dots, \mathbf{u}'_n$  be the columns of  $\mathbf{R}$  and  $\mathbf{R}'$ , and  $r_{il}$  the entries of  $\mathbf{R}$ . By hypothesis, Lemma 2.3.2 applies to  $\mathbf{R}$  and  $\mathbf{R}'$ .

*Same pivot columns.* For each  $l$ , by Lemma 2.3.2 (c),  $\mathbf{u}_l \in \text{span}(\mathbf{u}_1, \dots, \mathbf{u}_{l-1})$  if and only if  $\mathbf{u}'_l \in \text{span}(\mathbf{u}'_1, \dots, \mathbf{u}'_{l-1})$ . By Lemma 2.3.3 (c) applied to  $\mathbf{R}$  and to  $\mathbf{R}'$ , column  $l$  is a pivot column of  $\mathbf{R}$  if and only if it is a pivot column of  $\mathbf{R}'$ . So both matrices have the same pivot columns  $j_1 < \dots < j_r$ .

*Pivot columns agree.* By Lemma 2.3.3 (a),  $\mathbf{u}_{j_i} = \mathbf{e}_i = \mathbf{u}'_{j_i}$  for  $i = 1, \dots, r$ .

*Same non-pivot columns.* Let  $l$  be a non-pivot column. By [Lemma 2.3.3](#) (b) for  $\mathbf{R}$ ,

$$\mathbf{u}_l = \sum_{i: j_i < l} r_{il} \mathbf{u}_{j_i}.$$

This is an equation of the form in [Lemma 2.3.2](#) (b), with coefficient  $r_{il}$  on column  $j_i$  and 0 on the other earlier columns. By that lemma, the same equation holds for  $\mathbf{R}'$ :

$$\mathbf{u}'_l = \sum_{i: j_i < l} r_{il} \mathbf{u}'_{j_i} = \sum_{i: j_i < l} r_{il} \mathbf{e}_i = \mathbf{u}_l,$$

where the middle equality uses [Lemma 2.3.3](#) (a) for  $\mathbf{R}'$ , and the last uses [Lemma 2.3.3](#) (b) for  $\mathbf{R}$ .

Hence  $\mathbf{u}_l = \mathbf{u}'_l$  for every  $l$ , and  $\mathbf{R} = \mathbf{R}'$ .

For the consequence, let  $\mathbf{A} \in M_{m \times n}(F)$ . By [Theorem 2.2.3](#),  $\mathbf{A}$  is row equivalent to at least one matrix in RREF. Suppose  $\mathbf{A}$  is row equivalent to  $\mathbf{R}$  and to  $\mathbf{R}'$ , both in RREF. Row equivalence is an equivalence relation ([Theorem 2.2.2](#) (b)), so  $\mathbf{R}$  is row equivalent to  $\mathbf{R}'$ . By [Lemma 2.3.1](#),  $\mathbf{R}\mathbf{x} = \mathbf{0} \Leftrightarrow \mathbf{R}'\mathbf{x} = \mathbf{0}$  for every  $\mathbf{x}$ , and by the first part  $\mathbf{R} = \mathbf{R}'$ . This proves the theorem. ■

Notice what the proof did **not** do: it never compared the two sequences of operations. A direct comparison of two runs of the algorithm would have to track every choice, and quickly becomes unmanageable. The proof instead found a description of the RREF, “pivots where a new column appears, and coefficients of the old columns elsewhere”, that refers only to the null space, which the operations cannot change.

The theorem also gives a practical test: two matrices of the same size are row equivalent exactly when their RREFs are equal (you will write out the two-line proof in [Exercise \(B2: Row equivalence via the RREF\)](#)). For example, over  $\mathbb{R}$ ,  $\begin{pmatrix} 1 & 2 \\ 2 & 4 \end{pmatrix}$  and  $\begin{pmatrix} 2 & 4 \\ 0 & 0 \end{pmatrix}$  both reduce to  $\begin{pmatrix} 1 & 2 \\ 0 & 0 \end{pmatrix}$ , by  $R_2 \rightarrow R_2 - 2R_1$  and by  $R_1 \rightarrow \frac{1}{2}R_1$  respectively, so they are row equivalent. But  $\begin{pmatrix} 1 & 3 \\ 2 & 6 \end{pmatrix}$  reduces to  $\begin{pmatrix} 1 & 3 \\ 0 & 0 \end{pmatrix} \neq \begin{pmatrix} 1 & 2 \\ 0 & 0 \end{pmatrix}$ , so it is row equivalent to neither, even though all three matrices have the same pivot columns.

**Example. (Two echelon forms, one reduced form)**

Let  $\mathbf{A} = \begin{pmatrix} 1 & 2 & 1 & 3 \\ 2 & 4 & 0 & 2 \\ 3 & 6 & 2 & 7 \end{pmatrix}$  over  $\mathbb{R}$ .

- (a) Reduce  $\mathbf{A}$  to a row echelon form in two ways: first using row 1 as the first pivot row, then starting with  $R_1 \leftrightarrow R_2$ .
- (b) Continue both to reduced row echelon form.
- (c) Read the column relations of  $\mathbf{A}$  off the RREF.

*Solution.*

(a) First route:

$$\begin{aligned} \mathbf{A} &\xrightarrow{R_2 \rightarrow R_2 - 2R_1} \begin{pmatrix} 1 & 2 & 1 & 3 \\ 0 & 0 & -2 & -4 \\ 3 & 6 & 2 & 7 \end{pmatrix} \xrightarrow{R_3 \rightarrow R_3 - 3R_1} \begin{pmatrix} 1 & 2 & 1 & 3 \\ 0 & 0 & -2 & -4 \\ 0 & 0 & -1 & -2 \end{pmatrix} \\ &\xrightarrow{R_3 \rightarrow R_3 - \frac{1}{2}R_2} \begin{pmatrix} 1 & 2 & 1 & 3 \\ 0 & 0 & -2 & -4 \\ 0 & 0 & 0 & 0 \end{pmatrix} = \mathbf{E}_1. \end{aligned}$$

Second route:

$$\begin{aligned} \mathbf{A} &\xrightarrow{R_1 \leftrightarrow R_2} \begin{pmatrix} 2 & 4 & 0 & 2 \\ 1 & 2 & 1 & 3 \\ 3 & 6 & 2 & 7 \end{pmatrix} \xrightarrow{R_2 \rightarrow R_2 - \frac{1}{2}R_1} \begin{pmatrix} 2 & 4 & 0 & 2 \\ 0 & 0 & 1 & 2 \\ 3 & 6 & 2 & 7 \end{pmatrix} \\ &\xrightarrow{R_3 \rightarrow R_3 - \frac{3}{2}R_1} \begin{pmatrix} 2 & 4 & 0 & 2 \\ 0 & 0 & 1 & 2 \\ 0 & 0 & 2 & 4 \end{pmatrix} \\ &\xrightarrow{R_3 \rightarrow R_3 - 2R_2} \begin{pmatrix} 2 & 4 & 0 & 2 \\ 0 & 0 & 1 & 2 \\ 0 & 0 & 0 & 0 \end{pmatrix} = \mathbf{E}_2. \end{aligned}$$

Both  $\mathbf{E}_1$  and  $\mathbf{E}_2$  are in row echelon form, and  $\mathbf{E}_1 \neq \mathbf{E}_2$ . Both have pivot columns 1 and 3.

(b) From  $\mathbf{E}_1$ :

$$\mathbf{E}_1 \xrightarrow{R_2 \rightarrow -\frac{1}{2}R_2} \begin{pmatrix} 1 & 2 & 1 & 3 \\ 0 & 0 & 1 & 2 \\ 0 & 0 & 0 & 0 \end{pmatrix} \xrightarrow{R_1 \rightarrow R_1 - R_2} \begin{pmatrix} 1 & 2 & 0 & 1 \\ 0 & 0 & 1 & 2 \\ 0 & 0 & 0 & 0 \end{pmatrix}.$$

From  $\mathbf{E}_2$ :

$$\mathbf{E}_2 \xrightarrow{R_1 \rightarrow \frac{1}{2}R_1} \begin{pmatrix} 1 & 2 & 0 & 1 \\ 0 & 0 & 1 & 2 \\ 0 & 0 & 0 & 0 \end{pmatrix}.$$

The two routes end at the same matrix  $\mathbf{R}$ , as [Theorem 2.3.4](#) guarantees.

(c) By [Lemma 2.3.3](#) (b), the non-pivot columns of  $\mathbf{R}$  are  $\mathbf{u}_2 = 2\mathbf{u}_1$  and  $\mathbf{u}_4 = 1 \cdot \mathbf{u}_1 + 2\mathbf{u}_3$ , reading the coefficients from the entries of each column. By [Lemma 2.3.2](#) (b), the same relations hold for the columns of  $\mathbf{A}$ :  $\mathbf{a}_2 = 2\mathbf{a}_1$  and  $\mathbf{a}_4 = \mathbf{a}_1 + 2\mathbf{a}_3$ . Check:  $2(1, 2, 3) = (2, 4, 6)$  and  $(1, 2, 3) + 2(1, 0, 2) = (3, 2, 7)$ .

Both echelon forms in the example had the same pivot columns, although they were different matrices. That is also true in general, and it is the practical version of the theorem.

## 2.3.4 Pivot columns are well defined

### Corollary 2.3.5 — Pivot columns are well defined

Let  $\mathbf{A} \in M_{m \times n}(F)$  have columns  $\mathbf{a}_1, \dots, \mathbf{a}_n$ . Define the **pivot columns** of  $\mathbf{A}$  to be the

pivot columns of the RREF of  $\mathbf{A}$ .

- (a) Column  $l$  is a pivot column of  $\mathbf{A}$  **if and only if**  $\mathbf{a}_l \notin \text{span}(\mathbf{a}_1, \dots, \mathbf{a}_{l-1})$ .
- (b) If  $\mathbf{E}$  is **any** matrix in row echelon form that is row equivalent to  $\mathbf{A}$ , then the pivot columns of  $\mathbf{E}$  are the pivot columns of  $\mathbf{A}$ . In particular, every row echelon form of  $\mathbf{A}$  has the same number of non-zero rows.

*Idea.* Part (a) combines Lemma 2.3.3 (c) with the fact that column relations pass from the RREF back to  $\mathbf{A}$ . For (b), finish the reduction of  $\mathbf{E}$ : scale each pivot to 1 and clear the entries above the pivots. These operations never move a leading entry, so they produce a matrix in RREF with the same pivot columns as  $\mathbf{E}$ , and by uniqueness it is the RREF of  $\mathbf{A}$ .

*Proof.* The RREF of  $\mathbf{A}$  is well defined by Theorem 2.3.4; let  $\mathbf{u}_1, \dots, \mathbf{u}_n$  be its columns.

- (a)  $\mathbf{A}$  and its RREF are row equivalent, so by Lemma 2.3.2 (c),  $\mathbf{a}_l \in \text{span}(\mathbf{a}_1, \dots, \mathbf{a}_{l-1})$  if and only if  $\mathbf{u}_l \in \text{span}(\mathbf{u}_1, \dots, \mathbf{u}_{l-1})$ . By Lemma 2.3.3 (c), the latter fails exactly when  $l$  is a pivot column.
- (b) Let  $\mathbf{E}$  have entries  $e_{hl}$  and pivot columns  $j_1 < \dots < j_s$ . First perform  $R_i \rightarrow e_{ij_i}^{-1} R_i$  for  $i = 1, \dots, s$ , which is legal since each pivot  $e_{ij_i}$  is non-zero. Scaling a row by a non-zero scalar does not change which of its entries are zero (Theorem 0.7.1 (d), (f)), so the result is still in row echelon form with the same pivot columns, and now every pivot is 1. Next, for  $i = 1, 2, \dots, s$  in turn, and for each  $h < i$ , perform  $R_h \rightarrow R_h - cR_i$ , where  $c$  is the current entry of row  $h$  in column  $j_i$ . Row  $i$  is zero in all columns before  $j_i$ , so this changes row  $h$  only in columns  $\geq j_i > j_h$ . Hence the leading entry of row  $h$  is still a 1 in column  $j_h$ , no zero row is touched, and the entries below every pivot stay 0. The operation makes the entry in position  $(h, j_i)$  equal to 0, and it does not spoil the entries  $(h, j_{i'})$  with  $i' < i$  cleared earlier, because row  $i$  has a 0 in column  $j_{i'} < j_i$ . At the end, every pivot column is a standard basis vector, so the matrix is in RREF, with pivot columns  $j_1, \dots, j_s$ .

This matrix is row equivalent to  $\mathbf{E}$ , hence to  $\mathbf{A}$  (Theorem 2.2.2 (b)). By Theorem 2.3.4 it is the RREF of  $\mathbf{A}$ . Therefore the pivot columns of  $\mathbf{E}$  are those of  $\mathbf{A}$ . Since the non-zero rows of  $\mathbf{E}$  correspond one-to-one to its pivots, their number  $s$  is the number of pivot columns of  $\mathbf{A}$ , whichever  $\mathbf{E}$  we chose. ■

In practice, (b) means we may stop at any row echelon form to find the pivot columns, and (a) means we can sometimes find them with no elimination at all. Part (a) also says that the pivot columns of  $\mathbf{A}$  are exactly the columns kept when the list  $(\mathbf{a}_1, \dots, \mathbf{a}_n)$  is sifted as in Theorem 1.5.4: a column is kept exactly when it is not in the span of the columns before it. The number of pivot columns is the first candidate for a measure of “how much a matrix contains”; later in this chapter it becomes the rank.

*Quick check.* Without computing any row operations, find the pivot columns of  $\mathbf{A} = \begin{pmatrix} 1 & 2 & 0 & 1 \\ 2 & 4 & 1 & 3 \end{pmatrix}$  over  $\mathbb{R}$ .

*Solution.* Use Corollary 2.3.5 (a) column by column.  $\mathbf{a}_1 = (1, 2) \neq \mathbf{0}$ , so it is not in  $\text{span}() = \{\mathbf{0}\}$ : pivot.  $\mathbf{a}_2 = (2, 4) = 2\mathbf{a}_1$ : not a pivot.  $\mathbf{a}_3 = (0, 1)$  is not a multiple of



$(1, 2)$ , and  $\text{span}(\mathbf{a}_1, \mathbf{a}_2) = \text{span}((1, 2))$ : pivot.  $\mathbf{a}_4 = (1, 3) = \mathbf{a}_1 + \mathbf{a}_3$ : not a pivot. The pivot columns are 1 and 3.

*Warning.* **Uniqueness is a property of the reduced form only.** A matrix has many row echelon forms (Example (Two echelon forms, one reduced form) gave two), and it is only their pivot positions that agree. Also, two different matrices in RREF with the **same** pivot columns need not be row equivalent:  $\begin{pmatrix} 1 & 2 \\ 0 & 0 \end{pmatrix}$  and  $\begin{pmatrix} 1 & 3 \\ 0 & 0 \end{pmatrix}$  are both in RREF with pivot column 1, but they are different RREF matrices, so by Theorem 2.3.4 neither is row equivalent to the other.

## 2.3.5 Exercises

### 2.3.5.1 A. Check your understanding

#### Exercise A1.

- State the uniqueness theorem for the reduced row echelon form.
- Determine whether the following statement is true: “if  $\mathbf{A}$  and  $\mathbf{B}$  are row equivalent, then the columns of  $\mathbf{A}$  and the columns of  $\mathbf{B}$  span the same subspace.” Justify your answer.
- Determine whether the following statement is true: “if  $\mathbf{A}$  and  $\mathbf{B}$  are row equivalent and  $\mathbf{a}_3 = \mathbf{a}_1 - \mathbf{a}_2$  for the columns of  $\mathbf{A}$ , then  $\mathbf{b}_3 = \mathbf{b}_1 - \mathbf{b}_2$  for the columns of  $\mathbf{B}$ .” Justify your answer.
- Determine whether the following statement is true: “any two row echelon forms of the same matrix have the same number of zero rows.” Justify your answer.
- In the RREF of a matrix, what do the entries of a non-pivot column tell you about the original matrix?

*Solution.*

- If  $\mathbf{R}, \mathbf{R}' \in M_{m \times n}(F)$  are in RREF and  $\mathbf{R}\mathbf{x} = \mathbf{0} \Leftrightarrow \mathbf{R}'\mathbf{x} = \mathbf{0}$  for all  $\mathbf{x} \in F^n$ , then  $\mathbf{R} = \mathbf{R}'$ . Hence every matrix is row equivalent to exactly one matrix in RREF.
- False.  $\begin{pmatrix} 1 & 2 \\ 1 & 2 \end{pmatrix}$  and  $\begin{pmatrix} 1 & 2 \\ 0 & 0 \end{pmatrix}$  are row equivalent, but their columns span  $\text{span}((1, 1))$  and  $\text{span}((1, 0))$ , which are different lines in  $\mathbb{R}^2$ .
- True, by Lemma 2.3.2 (b) with  $l = 3$ ,  $d_1 = 1$ ,  $d_2 = -1$ .
- True. By Corollary 2.3.5 (b), every row echelon form of an  $m \times n$  matrix  $\mathbf{A}$  has exactly as many non-zero rows as  $\mathbf{A}$  has pivot columns, so the number of zero rows is  $m$  minus that number.
- If column  $l$  is not a pivot column, its entries  $r_{il}$  (for the pivot rows  $i$  with  $j_i < l$ ) are the coefficients expressing column  $l$  of the original matrix as a combination of its pivot columns:  $\mathbf{a}_l = \sum_{j_i < l} r_{il} \mathbf{a}_{j_i}$ .

### 2.3.5.2 B. Practice

**Exercise B1: Column relations.**

Let  $\mathbf{A} = \begin{pmatrix} 1 & -1 & 0 & 2 & 1 \\ 2 & -2 & 1 & 7 & 0 \\ 1 & -1 & 1 & 5 & 1 \end{pmatrix}$  over  $\mathbb{R}$ . Find the RREF of  $\mathbf{A}$ , naming each operation.

Hence find the pivot columns of  $\mathbf{A}$ , and express each non-pivot column of  $\mathbf{A}$  as a linear combination of the pivot columns before it. Check your answer on the columns of  $\mathbf{A}$ .

*Solution.*

$$\begin{aligned} \mathbf{A} &\xrightarrow{R_2 \rightarrow R_2 - 2R_1} \begin{pmatrix} 1 & -1 & 0 & 2 & 1 \\ 0 & 0 & 1 & 3 & -2 \\ 1 & -1 & 1 & 5 & 1 \end{pmatrix} \xrightarrow{R_3 \rightarrow R_3 - R_1} \begin{pmatrix} 1 & -1 & 0 & 2 & 1 \\ 0 & 0 & 1 & 3 & -2 \\ 0 & 0 & 1 & 3 & 0 \end{pmatrix} \\ &\xrightarrow{R_3 \rightarrow R_3 - R_2} \begin{pmatrix} 1 & -1 & 0 & 2 & 1 \\ 0 & 0 & 1 & 3 & -2 \\ 0 & 0 & 0 & 0 & 2 \end{pmatrix} \\ &\xrightarrow{R_3 \rightarrow \frac{1}{2}R_3} \begin{pmatrix} 1 & -1 & 0 & 2 & 1 \\ 0 & 0 & 1 & 3 & -2 \\ 0 & 0 & 0 & 0 & 1 \end{pmatrix} \\ &\xrightarrow{R_1 \rightarrow R_1 - R_3} \begin{pmatrix} 1 & -1 & 0 & 2 & 0 \\ 0 & 0 & 1 & 3 & -2 \\ 0 & 0 & 0 & 0 & 1 \end{pmatrix} \\ &\xrightarrow{R_2 \rightarrow R_2 + 2R_3} \begin{pmatrix} 1 & -1 & 0 & 2 & 0 \\ 0 & 0 & 1 & 3 & 0 \\ 0 & 0 & 0 & 0 & 1 \end{pmatrix}. \end{aligned}$$

The pivot columns are 1, 3 and 5. By [Lemma 2.3.3](#) (b), in the RREF column 2 is  $-1$  times column 1, and column 4 is 2 times column 1 plus 3 times column 3. By [Lemma 2.3.2](#) (b), the columns of  $\mathbf{A}$  satisfy

$$\mathbf{a}_2 = -\mathbf{a}_1, \quad \mathbf{a}_4 = 2\mathbf{a}_1 + 3\mathbf{a}_3.$$

Check:  $-(1, 2, 1) = (-1, -2, -1) = \mathbf{a}_2$ , and  $2(1, 2, 1) + 3(0, 1, 1) = (2, 7, 5) = \mathbf{a}_4$ .

**Exercise B2: Row equivalence via the RREF.**

Prove that two matrices  $\mathbf{A}, \mathbf{B} \in M_{m \times n}(F)$  are row equivalent if and only if they have the same RREF.

*Solution.* ( $\Rightarrow$ ) Suppose  $\mathbf{A}$  and  $\mathbf{B}$  are row equivalent, and let  $\mathbf{R}$  be the RREF of  $\mathbf{B}$ . Then  $\mathbf{A}$  is row equivalent to  $\mathbf{B}$ , and  $\mathbf{B}$  to  $\mathbf{R}$ , so  $\mathbf{A}$  is row equivalent to  $\mathbf{R}$  by transitivity ([Theorem 2.2.2](#) (b)). By the uniqueness in [Theorem 2.3.4](#),  $\mathbf{R}$  is the RREF of  $\mathbf{A}$ .

( $\Leftarrow$ ) Suppose both have the RREF  $\mathbf{R}$ . Then  $\mathbf{A}$  is row equivalent to  $\mathbf{R}$ , and by symmetry  $\mathbf{R}$  is row equivalent to  $\mathbf{B}$  ([Theorem 2.2.2](#) (b)). By transitivity  $\mathbf{A}$  is row equivalent to  $\mathbf{B}$ .

**Exercise B3: Row equivalent or not?**

Determine whether the matrices in each pair are row equivalent over  $\mathbb{R}$ . Justify your answer.

- (a)  $\mathbf{P} = \begin{pmatrix} 1 & 2 & 3 \\ 2 & 5 & 7 \end{pmatrix}$  and  $\mathbf{Q} = \begin{pmatrix} 1 & 1 & 2 \\ 3 & 4 & 7 \end{pmatrix}$ .
- (b)  $\mathbf{P} = \begin{pmatrix} 1 & 1 & 1 \\ 1 & 2 & 3 \\ 0 & 1 & 2 \end{pmatrix}$  and  $\mathbf{Q} = \begin{pmatrix} 1 & 0 & -1 \\ 0 & 1 & 2 \\ 1 & 1 & 2 \end{pmatrix}$ .

*Solution.* By [Exercise \(B2: Row equivalence via the RREF\)](#), two matrices of the same size are row equivalent exactly when their RREFs are equal. So we compute both RREFs.

$$(a) \mathbf{P} \xrightarrow{R_2 \rightarrow R_2 - 2R_1} \begin{pmatrix} 1 & 2 & 3 \\ 0 & 1 & 1 \end{pmatrix} \xrightarrow{R_1 \rightarrow R_1 - 2R_2} \begin{pmatrix} 1 & 0 & 1 \\ 0 & 1 & 1 \end{pmatrix}, \text{ and } \mathbf{Q} \xrightarrow{R_2 \rightarrow R_2 - 3R_1} \begin{pmatrix} 1 & 1 & 2 \\ 0 & 1 & 1 \end{pmatrix} \xrightarrow{R_1 \rightarrow R_1 - R_2} \begin{pmatrix} 1 & 0 & 1 \\ 0 & 1 & 1 \end{pmatrix}.$$

The RREFs are equal, so  $\mathbf{P}$  and  $\mathbf{Q}$  are row equivalent.

$$(b) \mathbf{P} \xrightarrow{R_2 \rightarrow R_2 - R_1} \begin{pmatrix} 1 & 1 & 1 \\ 0 & 1 & 2 \\ 0 & 1 & 2 \end{pmatrix} \xrightarrow{R_1 \rightarrow R_1 - R_2} \begin{pmatrix} 1 & 0 & -1 \\ 0 & 1 & 2 \\ 0 & 1 & 2 \end{pmatrix} \xrightarrow{R_3 \rightarrow R_3 - R_2} \begin{pmatrix} 1 & 0 & -1 \\ 0 & 1 & 2 \\ 0 & 0 & 0 \end{pmatrix}.$$

$$\mathbf{Q} \xrightarrow{R_3 \rightarrow R_3 - R_1} \begin{pmatrix} 1 & 0 & -1 \\ 0 & 1 & 2 \\ 0 & 1 & 3 \end{pmatrix} \xrightarrow{R_3 \rightarrow R_3 - R_2} \begin{pmatrix} 1 & 0 & -1 \\ 0 & 1 & 2 \\ 0 & 0 & 1 \end{pmatrix} \xrightarrow{R_1 \rightarrow R_1 + R_3} \begin{pmatrix} 1 & 0 & 0 \\ 0 & 1 & 2 \\ 0 & 0 & 1 \end{pmatrix} \xrightarrow{R_2 \rightarrow R_2 - 2R_3} \begin{pmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{pmatrix}.$$

$\mathbf{I}_3$ .

The RREFs differ (one has a zero row, the other is  $\mathbf{I}_3$ ), so  $\mathbf{P}$  and  $\mathbf{Q}$  are **not** row equivalent. Consistently with [Corollary 2.3.5](#),  $\mathbf{P}$  has pivot columns 1, 2 while  $\mathbf{Q}$  has 1, 2, 3.

### 2.3.5.3 C. Going deeper

#### Exercise C1: Counting reduced forms.

- (a) List, by pivot columns, all  $2 \times 3$  matrices over  $\mathbb{F}_2$  in reduced row echelon form, and show that there are exactly 15 of them.
- (b) Deduce that  $M_{2 \times 3}(\mathbb{F}_2)$ , which has 64 elements, splits into exactly 15 classes of row equivalent matrices.
- (c) Show that over a field with  $q$  elements there are exactly  $2q^2 + 2q + 3$  such matrices.

*Solution.* We count over a field  $F$  with  $q$  elements, sorting by the list of pivot columns. In a matrix in RREF, a pivot column is forced (it is  $\mathbf{e}_i$ ), an entry before a leading entry or in a zero row is forced to be 0, and every other entry, one lying in a non-pivot column to the right of its row's pivot, is free.

- (a) With  $q = 2$ , writing  $*$  for a free entry:

- no pivots:  $\begin{pmatrix} 0 & 0 & 0 \\ 0 & 0 & 0 \end{pmatrix}$ , 1 matrix;
- pivot column 1 only:  $\begin{pmatrix} 1 & * & * \\ 0 & 0 & 0 \end{pmatrix}$ , 4 matrices;
- pivot column 2 only:  $\begin{pmatrix} 0 & 1 & * \\ 0 & 0 & 0 \end{pmatrix}$ , 2 matrices;
- pivot column 3 only:  $\begin{pmatrix} 0 & 0 & 1 \\ 0 & 0 & 0 \end{pmatrix}$ , 1 matrix;

- pivot columns 1, 2:  $\begin{pmatrix} 1 & 0 & * \\ 0 & 1 & * \end{pmatrix}$ , 4 matrices;
- pivot columns 1, 3:  $\begin{pmatrix} 1 & * & 0 \\ 0 & 0 & 1 \end{pmatrix}$ , 2 matrices;
- pivot columns 2, 3:  $\begin{pmatrix} 0 & 1 & 0 \\ 0 & 0 & 1 \end{pmatrix}$ , 1 matrix.

A  $2 \times 3$  matrix has at most two pivots, so these are all the cases. The total is  $1 + 4 + 2 + 1 + 4 + 2 + 1 = 15$ .

- (b) Row equivalence is an equivalence relation (Theorem 2.2.2 (b)), so  $M_{2 \times 3}(\mathbb{F}_2)$  is partitioned into equivalence classes (Theorem 0.5.2). By Theorem 2.3.4, each class contains exactly one matrix in RREF: at least one by Theorem 2.2.3, and at most one since two RREF matrices in the same class would be row equivalent, hence equal. So the classes correspond one-to-one to the 15 matrices of (a).
- (c) The same list, with each free entry taking  $q$  values, gives  $1 + q^2 + q + 1 + q^2 + q + 1 = 2q^2 + 2q + 3$ . For  $q = 2$  this is  $8 + 4 + 3 = 15$ , as in (a).

### Exercise C2: Deleting a column.

Let  $\mathbf{A} \in M_{m \times n}(F)$  with  $n \geq 2$ , and let  $\mathbf{R}$  be its RREF.

- (a) Let  $\mathbf{A}'$  and  $\mathbf{R}'$  be obtained from  $\mathbf{A}$  and  $\mathbf{R}$  by deleting the **last** column. Prove that  $\mathbf{R}'$  is the RREF of  $\mathbf{A}'$ .
- (b) Show by an example that the corresponding statement for deleting the **first** column is false.

*Solution.*

- (a) Let a sequence of row operations turn  $\mathbf{A}$  into  $\mathbf{R}$ . By Lemma 2.2.1, the same sequence turns  $\mathbf{A}'$  into  $\mathbf{R}'$ , so  $\mathbf{A}'$  is row equivalent to  $\mathbf{R}'$ . By Theorem 2.3.4 it suffices to show that  $\mathbf{R}'$  is in RREF. Let  $j_1 < \cdots < j_r$  be the pivot columns of  $\mathbf{R}$ . If  $j_r < n$ , then deleting column  $n$  does not change any leading entry or any pivot column, and the zero rows of  $\mathbf{R}$  stay zero, so (E1)–(E4) still hold. If  $j_r = n$ , then row  $r$  of  $\mathbf{R}$  is zero except for its pivot in column  $n$ , so in  $\mathbf{R}'$  rows  $r, \dots, m$  are zero and rows  $1, \dots, r-1$  keep their leading entries in columns  $j_1 < \cdots < j_{r-1}$  and their pivot columns. Again (E1)–(E4) hold, with  $r-1$  pivots. Hence  $\mathbf{R}'$  is in RREF, and it is the RREF of  $\mathbf{A}'$ .
- (b) Let  $\mathbf{A} = \begin{pmatrix} 1 & 1 \\ 0 & 1 \end{pmatrix}$  over  $\mathbb{R}$ . Then  $\mathbf{A} \xrightarrow{R_1 \rightarrow R_1 - R_2} \mathbf{I}_2$ , so its RREF is  $\mathbf{I}_2$ , and deleting the first column of  $\mathbf{I}_2$  gives  $\begin{pmatrix} 0 \\ 1 \end{pmatrix}$ . But deleting the first column of  $\mathbf{A}$  gives  $\begin{pmatrix} 1 \\ 1 \end{pmatrix}$ , whose RREF is  $\begin{pmatrix} 1 \\ 0 \end{pmatrix}$ , via  $R_2 \rightarrow R_2 - R_1$ . These differ. The pivot of a later column depends on the columns before it, so removing an early column can move pivots.

## 2.4 Elementary Matrices and Inverses

Gaussian elimination is a list of instructions: swap these rows, scale that one, add a multiple of one row to another. So far it lives outside algebra, as something we do to a matrix. This section turns every row operation into a matrix, so that “apply an operation” becomes “multiply on the left”. With that one observation the algorithm becomes a tool for proofs. We use it to prove the main theorem about invertible matrices, to show that a one-sided inverse of a square matrix is automatically two-sided, which Chapter 0 stated after [Definition 0.9.13](#) but could not prove, and to compute inverses.

Throughout,  $F$  is a field, and we use the three elementary row operations of [Definition 2.2.1](#) with their notation: the swap  $R_i \leftrightarrow R_j$  ( $i \neq j$ ), the scaling  $R_i \rightarrow cR_i$  ( $c$  **non-zero**), and the replacement  $R_i \rightarrow R_i + cR_j$  ( $i \neq j$ , any  $c \in F$ ).

### 2.4.1 Row operations are matrix multiplications

We have already seen one row operation disguised as a product. In [Example \(Row equivalence\)](#) of Chapter 0, multiplying  $\mathbf{A} = \begin{pmatrix} 1 & 2 & 1 \\ 2 & 5 & 0 \end{pmatrix}$  on the left by  $\mathbf{E} = \begin{pmatrix} 1 & 0 \\ -2 & 1 \end{pmatrix}$  replaced row 2 by row 2 minus twice row 1, that is, it performed  $R_2 \rightarrow R_2 - 2R_1$ . Look at  $\mathbf{E}$  itself: it is what that same operation does to the identity matrix  $\mathbf{I}_2$ . This is no accident, and it deserves a name.

*An elementary matrix is the identity matrix after one row operation.*

#### Definition 2.4.1 — Elementary matrix

Let  $m \geq 1$ . A matrix  $\mathbf{E} \in M_m(F)$  is an **elementary matrix** if it is obtained from  $\mathbf{I}_m$  by performing **exactly one** elementary row operation. Writing  $\mathbf{E}_{ij}$  for the matrix unit (1 in entry  $(i, j)$ , 0 elsewhere; see [Example \(Standard Bases\)](#)), the three kinds are:

1.  $\mathbf{P}_{ij}$ , obtained by  $R_i \leftrightarrow R_j$  with  $i \neq j$ : the identity with rows  $i$  and  $j$  swapped;
2.  $\mathbf{D}_i(c) = \text{diag}(1, \dots, 1, c, 1, \dots, 1)$ , with  $c$  in position  $i$ , obtained by  $R_i \rightarrow cR_i$  with  $c \neq 0$ ;
3.  $\mathbf{I}_m + c\mathbf{E}_{ij}$ , obtained by  $R_i \rightarrow R_i + cR_j$  with  $i \neq j$  and  $c \in F$ : the identity with an extra  $c$  in entry  $(i, j)$ .

In words: start from  $\mathbf{I}_m$ , do **one** operation, and record the result. The size  $m$  is the number of **rows** of the matrices we plan to operate on. Type 3 puts  $c$  in row  $i$ , column  $j$ : the row that changes is  $i$ , and the row that gets copied is  $j$ .

**Examples.** In  $M_3(F)$ ,

$$\mathbf{P}_{13} = \begin{pmatrix} 0 & 0 & 1 \\ 0 & 1 & 0 \\ 1 & 0 & 0 \end{pmatrix}, \quad \mathbf{D}_2(5) = \begin{pmatrix} 1 & 0 & 0 \\ 0 & 5 & 0 \\ 0 & 0 & 1 \end{pmatrix}, \quad \mathbf{I}_3 - 3\mathbf{E}_{31} = \begin{pmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ -3 & 0 & 1 \end{pmatrix}$$

come from  $R_1 \leftrightarrow R_3$ ,  $R_2 \rightarrow 5R_2$  (over a field where  $5 \neq 0$ ), and  $R_3 \rightarrow R_3 - 3R_1$ . The degenerate case is worth a look:  $\mathbf{D}_1(1) = \mathbf{I}_m$  is elementary, since “multiply row 1 by 1” is a legal operation, and so is  $\mathbf{I}_m + 0\mathbf{E}_{12}$ . The identity is an elementary matrix. For  $m = 1$  the elementary matrices are exactly the  $1 \times 1$  matrices  $(c)$  with  $c \neq 0$ , since there are no two distinct rows to swap or

combine.

**Non-example by minimal change.** Change  $c = 5$  in  $\mathbf{D}_2(5)$  to  $c = 0$ . The matrix  $\text{diag}(1, 0, 1)$  is still diagonal and still differs from  $\mathbf{I}_3$  in one entry, but “multiply row 2 by 0” is not an elementary row operation: the clause  $c \neq 0$  fails. Likewise  $\begin{pmatrix} 0 & 0 & 1 \\ 1 & 0 & 0 \\ 0 & 1 & 0 \end{pmatrix}$  is a rearrangement of the rows of  $\mathbf{I}_3$ , but every one of its rows has moved, and a single swap moves only two rows. So it is not elementary. It is, however, the **product**  $\mathbf{P}_{12}\mathbf{P}_{23}$  of two elementary matrices, as Exercise A1 at the end of this section checks.

The whole point of the definition is the next theorem. To perform a row operation on **any** matrix, multiply on the left by what that operation does to the identity.

### Theorem 2.4.1 — Row operations are left multiplications

Let  $\rho$  be an elementary row operation on matrices with  $m$  rows, and let  $\mathbf{E} = \rho(\mathbf{I}_m)$  be the corresponding elementary matrix. Then for **every**  $n \geq 1$  and every  $\mathbf{A} \in M_{m \times n}(F)$ ,

$$\rho(\mathbf{A}) = \mathbf{E}\mathbf{A}.$$

*Idea.* Each operation builds every new row as a combination of the old rows, with coefficients that do not depend on the matrix: for  $R_i \rightarrow R_i + cR_j$ , new row  $i$  is  $1 \cdot (\text{row } i) + c \cdot (\text{row } j)$ , and every other row is copied. Feed in  $\mathbf{I}_m$ , whose rows are the standard row vectors, and the coefficients appear as the entries of  $\mathbf{E}$ . Then the row view of a product says  $\mathbf{E}\mathbf{A}$  forms exactly those combinations of the rows of  $\mathbf{A}$ .

*Proof.* Let  $\mathbf{r}_1, \dots, \mathbf{r}_m$  be the rows of  $\mathbf{A}$ , and write  $\mathbf{e}_l^\top$  for the  $l$ -th row of  $\mathbf{I}_m$ . By Definition 2.2.1, for each  $k$  there are scalars  $c_{k1}, \dots, c_{km}$ , depending on  $\rho$  but **not** on  $\mathbf{A}$ , such that row  $k$  of  $\rho(\mathbf{A})$  is  $c_{k1}\mathbf{r}_1 + \dots + c_{km}\mathbf{r}_m$ . Explicitly, every row that the operation does not touch has  $c_{kk} = 1$  and all other  $c_{kl} = 0$ , and the touched rows are:

- for  $R_i \leftrightarrow R_j$ : row  $i$  is  $\mathbf{r}_j$  and row  $j$  is  $\mathbf{r}_i$ ;
- for  $R_i \rightarrow cR_i$ : row  $i$  is  $c\mathbf{r}_i$ ;
- for  $R_i \rightarrow R_i + cR_j$ : row  $i$  is  $\mathbf{r}_i + c\mathbf{r}_j$ .

Applying  $\rho$  to  $\mathbf{I}_m$ , row  $k$  of  $\mathbf{E}$  is  $c_{k1}\mathbf{e}_1^\top + \dots + c_{km}\mathbf{e}_m^\top = (c_{k1} \ \dots \ c_{km})$ . So the  $(k, l)$ -entry of  $\mathbf{E}$  is  $c_{kl}$ . By Theorem 0.9.2 (rows), row  $k$  of  $\mathbf{E}\mathbf{A}$  is (row  $k$  of  $\mathbf{E}$ )  $\mathbf{A}$ , and by Theorem 0.9.3 (rows version) this equals  $c_{k1}\mathbf{r}_1 + \dots + c_{km}\mathbf{r}_m$ , which is row  $k$  of  $\rho(\mathbf{A})$ . Since every row agrees,  $\rho(\mathbf{A}) = \mathbf{E}\mathbf{A}$ . This proves the theorem. ■

So a sequence of operations  $\rho_1, \rho_2, \dots, \rho_k$ , performed in that order, turns  $\mathbf{A}$  into

$$\rho_k(\dots \rho_2(\rho_1(\mathbf{A}))\dots) = \mathbf{E}_k \dots \mathbf{E}_2 \mathbf{E}_1 \mathbf{A}, \quad \mathbf{E}_i = \rho_i(\mathbf{I}_m),$$

by applying the theorem  $k$  times and using associativity. The whole of Gaussian elimination is a single matrix  $\mathbf{E}_k \dots \mathbf{E}_1$  standing on the left.

**Warning.** The **first** operation stands **rightmost**. Doing  $R_1 \leftrightarrow R_2$  and then  $R_1 \rightarrow R_1 + R_2$  on  $\mathbf{I}_2$  gives  $(\mathbf{I}_2 + \mathbf{E}_{12})\mathbf{P}_{12} = \begin{pmatrix} 1 & 1 \\ 1 & 0 \end{pmatrix}$ , while the product in the other order is

$\mathbf{P}_{12}(\mathbf{I}_2 + \mathbf{E}_{12}) = \begin{pmatrix} 0 & 1 \\ 1 & 1 \end{pmatrix}$ . Also, multiplying by  $\mathbf{E}$  on the **right** performs a **column** operation:  $\mathbf{A}(\mathbf{I}_2 + \mathbf{E}_{12})$  adds column 1 to column 2, and column operations change solution sets.

**Example. (Three operations, three products)**

Let  $\mathbf{A} = \begin{pmatrix} 1 & 2 \\ 0 & 1 \\ 3 & 4 \end{pmatrix} \in M_{3 \times 2}(\mathbb{R})$ . Compute  $\mathbf{P}_{13}\mathbf{A}$ ,  $\mathbf{D}_2(5)\mathbf{A}$  and  $(\mathbf{I}_3 - 3\mathbf{E}_{31})\mathbf{A}$ , and check that each is the result of the corresponding row operation.

*Solution.* By the row view, row  $k$  of  $\mathbf{EA}$  is the combination of the rows of  $\mathbf{A}$  whose weights are row  $k$  of  $\mathbf{E}$ .

- Rows of  $\mathbf{P}_{13}$  are  $\mathbf{e}_3^\top, \mathbf{e}_2^\top, \mathbf{e}_1^\top$ , so  $\mathbf{P}_{13}\mathbf{A} = \begin{pmatrix} 3 & 4 \\ 0 & 1 \\ 1 & 2 \end{pmatrix}$ : rows 1 and 3 are swapped.
- Row 2 of  $\mathbf{D}_2(5)$  is  $5\mathbf{e}_2^\top$ , so  $\mathbf{D}_2(5)\mathbf{A} = \begin{pmatrix} 1 & 2 \\ 0 & 5 \\ 3 & 4 \end{pmatrix}$ : row 2 is multiplied by 5.
- Row 3 of  $\mathbf{I}_3 - 3\mathbf{E}_{31}$  is  $(-3 \ 0 \ 1)$ , so row 3 of the product is  $-3(1, 2) + (3, 4) = (0, -2)$ , and  $(\mathbf{I}_3 - 3\mathbf{E}_{31})\mathbf{A} = \begin{pmatrix} 1 & 2 \\ 0 & 1 \\ 0 & -2 \end{pmatrix}$ : this is  $R_3 \rightarrow R_3 - 3R_1$ .

In each case the product agrees with the operation, as [Theorem 2.4.1](#) predicts.

*Quick check.* Which  $3 \times 3$  elementary matrix performs  $R_2 \rightarrow R_2 + 4R_3$ ? Which operation undoes it, and what is its matrix?

*Solution.* Apply the operation to  $\mathbf{I}_3$ : row 2 becomes  $\mathbf{e}_2^\top + 4\mathbf{e}_3^\top$ , so the matrix is  $\mathbf{I}_3 + 4\mathbf{E}_{23}$ , with 4 in entry  $(2, 3)$ . The operation  $R_2 \rightarrow R_2 - 4R_3$  undoes it, since row 3 is unchanged by both; its matrix is  $\mathbf{I}_3 - 4\mathbf{E}_{23}$ .

Every row operation can be undone by another row operation, and the matching statement for matrices is:

**Theorem 2.4.2 — Elementary matrices are invertible**

Every elementary matrix is invertible, and its inverse is an elementary matrix of the same kind:

$$\mathbf{P}_{ij}^{-1} = \mathbf{P}_{ij}, \quad \mathbf{D}_i(c)^{-1} = \mathbf{D}_i(c^{-1}), \quad (\mathbf{I}_m + c\mathbf{E}_{ij})^{-1} = \mathbf{I}_m - c\mathbf{E}_{ij} \quad (i \neq j).$$

*Proof.* Let  $\mathbf{E} = \rho(\mathbf{I}_m)$ , and let  $\rho'$  be the operation on the same line of the display:  $\rho' = \rho$  for a swap,  $R_i \rightarrow c^{-1}R_i$  for  $R_i \rightarrow cR_i$  (legal, since  $c \neq 0$  gives  $c^{-1} \neq 0$ ), and  $R_i \rightarrow R_i - cR_j$  for  $R_i \rightarrow R_i + cR_j$ . In each case  $\rho'(\rho(\mathbf{A})) = \mathbf{A}$  and  $\rho(\rho'(\mathbf{A})) = \mathbf{A}$  for every  $\mathbf{A}$ , as in [Theorem 2.2.2](#) (a): swapping twice restores the order; scaling by  $c$

and  $c^{-1}$  multiplies row  $i$  by  $cc^{-1} = 1$ ; and adding  $c\mathbf{r}_j$  and then  $-c\mathbf{r}_j$  to row  $i$  restores row  $i$ , because  $i \neq j$  means row  $j$  is not changed in between. Let  $\mathbf{E}' = \rho'(\mathbf{I}_m)$ . By [Theorem 2.4.1](#),

$$\mathbf{E}'\mathbf{E} = \rho'(\mathbf{E}) = \rho'(\rho(\mathbf{I}_m)) = \mathbf{I}_m \quad \text{and} \quad \mathbf{E}\mathbf{E}' = \rho(\mathbf{E}') = \rho(\rho'(\mathbf{I}_m)) = \mathbf{I}_m.$$

Hence  $\mathbf{E}$  is invertible with  $\mathbf{E}^{-1} = \mathbf{E}'$ , which is the matrix in the display. This proves the theorem. ■

The hypothesis  $i \neq j$  did real work: “add  $c$  times row  $i$  to row  $i$ ” is scaling by  $1 + c$ , which destroys row  $i$  when  $c = -1$  and cannot be undone. That is why [Definition 2.2.1](#) excludes it.

## 2.4.2 The invertible matrix theorem

Chapter 0 defined an invertible matrix by the existence of a two-sided inverse, and gave only one practical test,  $ad - bc \neq 0$ , for  $2 \times 2$  matrices ([Theorem 0.9.8](#)). For larger matrices, searching for  $\mathbf{B}$  with  $\mathbf{AB} = \mathbf{BA} = \mathbf{I}_n$  directly is hopeless. What we want is a list of conditions, each of which we can check by a method we already have, that are all equivalent to invertibility. Then we may test whichever is cheapest.

We need one small fact about square matrices in reduced form.

### Lemma 2.4.3 — Square reduced matrices

Let  $\mathbf{R} \in M_n(F)$  be in reduced row echelon form. Then either  $\mathbf{R} = \mathbf{I}_n$ , or the last row of  $\mathbf{R}$  is zero.

*Proof.* Suppose the last row of  $\mathbf{R}$  is not zero. Since zero rows of a matrix in echelon form come after all non-zero rows ([Definition 2.2.3](#)), no row of  $\mathbf{R}$  is zero, so each of the  $n$  rows has a pivot. Let the pivot of row  $k$  be in column  $c_k$ . Pivots move strictly to the right as we go down, so  $1 \leq c_1 < c_2 < \cdots < c_n \leq n$ . A strictly increasing list of  $n$  integers in  $\{1, \dots, n\}$  must be  $1, 2, \dots, n$ , so  $c_k = k$ . By [Definition 2.2.4](#), each pivot equals 1 and is the only non-zero entry of its column, so column  $k$  of  $\mathbf{R}$  is  $\mathbf{e}_k$  for every  $k$ . Hence  $\mathbf{R} = \mathbf{I}_n$ . ■

Here is the theorem. It collects, for a square matrix, facts about solving systems, about row reduction, and about the columns as vectors in the sense of Chapter 1.

### Theorem 2.4.4 — Invertible Matrix Theorem

Let  $\mathbf{A} \in M_n(F)$ , with columns  $\mathbf{a}_1, \dots, \mathbf{a}_n \in F^n$ . The following are equivalent.

- (a)  $\mathbf{A}$  is invertible.
- (b) The homogeneous system  $\mathbf{Ax} = \mathbf{0}$  has only the trivial solution  $\mathbf{x} = \mathbf{0}$ .
- (c) The reduced row echelon form of  $\mathbf{A}$  is  $\mathbf{I}_n$ .



- (d)  $\mathbf{A}$  is a product of elementary matrices.
- (e) For **every**  $\mathbf{b} \in F^n$ , the system  $\mathbf{A}\mathbf{x} = \mathbf{b}$  has a solution.
- (f) The columns  $(\mathbf{a}_1, \dots, \mathbf{a}_n)$  are linearly independent.
- (g) The columns  $(\mathbf{a}_1, \dots, \mathbf{a}_n)$  span  $F^n$ .
- (h) The columns  $(\mathbf{a}_1, \dots, \mathbf{a}_n)$  form a basis of  $F^n$ .

*Idea.* Eight conditions would need 56 implications if proved pairwise; a cycle through them needs far fewer. The plan:

- ① the core cycle  $(a) \Rightarrow (b) \Rightarrow (c) \Rightarrow (d) \Rightarrow (a)$ , where each arrow uses one earlier tool (cancel by  $\mathbf{A}^{-1}$ ; row reduction; elementary matrices; products of invertible matrices);
- ②  $(b) \Leftrightarrow (f)$ , which is the column view  $\mathbf{A}\mathbf{x} = x_1\mathbf{a}_1 + \dots + x_n\mathbf{a}_n$ ;
- ③  $(f) \Leftrightarrow (h) \Leftrightarrow (g)$ , which is “count instead of check” in the  $n$ -dimensional space  $F^n$ ;
- ④  $(g) \Leftrightarrow (e)$ , which is “consistent means  $\mathbf{b}$  is in the column span”.

The only arrow that needs a new argument is  $(b) \Rightarrow (c)$ ; the others quote a result we already have. An RREF that is not  $\mathbf{I}_n$  has a zero row, so it really has fewer equations than unknowns, and such a homogeneous system has a non-trivial solution.

*Proof.* **(a)  $\Rightarrow$  (b).** Suppose  $\mathbf{A}$  is invertible and  $\mathbf{A}\mathbf{x} = \mathbf{0}$ . Multiplying on the left by  $\mathbf{A}^{-1}$  and using [Theorem 0.9.4](#),  $\mathbf{x} = (\mathbf{A}^{-1}\mathbf{A})\mathbf{x} = \mathbf{A}^{-1}(\mathbf{A}\mathbf{x}) = \mathbf{A}^{-1}\mathbf{0} = \mathbf{0}$ .

**(b)  $\Rightarrow$  (c).** Suppose (b) holds. By [Theorem 2.2.3](#),  $\mathbf{A}$  is row equivalent to a matrix  $\mathbf{R}$  in reduced row echelon form, and  $\mathbf{R}$  is the RREF of  $\mathbf{A}$  by [Theorem 2.3.4](#). By [Lemma 2.3.1](#),  $\mathbf{R}\mathbf{x} = \mathbf{0}$  has the same solutions as  $\mathbf{A}\mathbf{x} = \mathbf{0}$ , so it too has only the trivial solution. Suppose, for a contradiction, that  $\mathbf{R} \neq \mathbf{I}_n$ . By [Lemma 2.4.3](#), the last row of  $\mathbf{R}$  is zero. If  $n = 1$ , then  $\mathbf{R} = (0)$  and  $\mathbf{x} = (1)$  is a non-trivial solution. If  $n \geq 2$ , the last equation of  $\mathbf{R}\mathbf{x} = \mathbf{0}$  reads  $0 = 0$ , so  $\mathbf{R}\mathbf{x} = \mathbf{0}$  has the same solutions as the homogeneous system  $\mathbf{R}'\mathbf{x} = \mathbf{0}$  formed by the first  $n - 1$  rows. That system has  $n - 1$  equations in  $n > n - 1$  unknowns, so by [Corollary 2.2.5](#) it has a non-trivial solution. Either way we contradict (b). Hence  $\mathbf{R} = \mathbf{I}_n$ .

**(c)  $\Rightarrow$  (d).** Suppose the RREF of  $\mathbf{A}$  is  $\mathbf{I}_n$ . By [Definition 2.2.2](#) there are row operations  $\rho_1, \dots, \rho_k$  taking  $\mathbf{A}$  to  $\mathbf{I}_n$ . If  $k = 0$ , then  $\mathbf{A} = \mathbf{I}_n = \mathbf{D}_1(1)$  is elementary. Otherwise, with  $\mathbf{E}_i = \rho_i(\mathbf{I}_n)$ , [Theorem 2.4.1](#) gives  $\mathbf{E}_k \cdots \mathbf{E}_1 \mathbf{A} = \mathbf{I}_n$ . Each  $\mathbf{E}_i$  is invertible by [Theorem 2.4.2](#), so multiplying on the left by  $\mathbf{E}_k^{-1}$ , then  $\mathbf{E}_{k-1}^{-1}$ , and so on, gives

$$\mathbf{A} = \mathbf{E}_1^{-1} \mathbf{E}_2^{-1} \cdots \mathbf{E}_k^{-1},$$

and each  $\mathbf{E}_i^{-1}$  is elementary by [Theorem 2.4.2](#).

**(d)  $\Rightarrow$  (a).** Suppose  $\mathbf{A} = \mathbf{E}_1 \cdots \mathbf{E}_k$  with each  $\mathbf{E}_i$  elementary and  $k \geq 1$ . Each  $\mathbf{E}_i$  is invertible by [Theorem 2.4.2](#), and a product of two invertible matrices is invertible by [Theorem 0.9.7](#) (part 3). By induction on  $k$ ,  $\mathbf{A}$  is invertible.

So far (a), (b), (c), (d) are equivalent.

**(b)  $\Leftrightarrow$  (f).** By [Theorem 0.9.3](#),  $\mathbf{A}\mathbf{x} = x_1\mathbf{a}_1 + \dots + x_n\mathbf{a}_n$ . So “ $\mathbf{A}\mathbf{x} = \mathbf{0}$  only for  $\mathbf{x} = \mathbf{0}$ ” says exactly that  $x_1\mathbf{a}_1 + \dots + x_n\mathbf{a}_n = \mathbf{0}$  forces  $x_1 = \dots = x_n = 0$ , which is [Definition 1.4.1](#) for the columns.

(f)  $\Leftrightarrow$  (h)  $\Leftrightarrow$  (g). The columns form a list of exactly  $n$  vectors in  $F^n$ , and  $\dim F^n = n$  (Example (Dimensions of the Running Examples)). By Theorem 1.6.4, such a list is a basis if it is linearly independent, and also if it spans. Conversely a basis is independent and spanning by Definition 1.5.1. Hence (f)  $\Rightarrow$  (h)  $\Rightarrow$  (g)  $\Rightarrow$  (h)  $\Rightarrow$  (f).

(g)  $\Leftrightarrow$  (e). By Theorem 2.1.1,  $\mathbf{Ax} = \mathbf{b}$  has a solution if and only if  $\mathbf{b} \in \text{span}(\mathbf{a}_1, \dots, \mathbf{a}_n)$ . So (e) says every  $\mathbf{b} \in F^n$  lies in this span, that is,  $F^n \subseteq \text{span}(\mathbf{a}_1, \dots, \mathbf{a}_n)$ . The reverse inclusion always holds, so (e) is equivalent to (g).

Combining the four parts, all eight conditions are equivalent. This proves the theorem. ■

In practice the theorem is a menu. To show a matrix **is** invertible, pick the most convenient item, very often (b): take a solution of  $\mathbf{Ax} = \mathbf{0}$  and show it is  $\mathbf{0}$ . To show a matrix is **not** invertible, exhibit one non-zero solution of  $\mathbf{Ax} = \mathbf{0}$ , or one linear relation among the columns. In later chapters the list grows: kernels and images of linear maps in Chapter 3, determinants in Chapter 6, eigenvalues in Chapter 8, and singular values in Chapter 12.

*Quick check.* Is  $\mathbf{A} = \begin{pmatrix} 1 & 2 & 3 \\ 0 & 1 & 1 \\ 1 & 3 & 4 \end{pmatrix} \in M_3(\mathbb{R})$  invertible? Use whichever item of Theorem 2.4.4 is quickest.

*Solution.* No. The third column is the sum of the first two:  $(1, 0, 1) + (2, 1, 3) = (3, 1, 4)$ . So  $\mathbf{a}_1 + \mathbf{a}_2 - \mathbf{a}_3 = \mathbf{0}$  is a linear relation with coefficients not all zero, item (f) fails, and hence (a) fails. Equivalently,  $\mathbf{x} = (1, 1, -1)$  is a non-trivial solution of  $\mathbf{Ax} = \mathbf{0}$ .

*Warning.* The theorem is about **square** matrices. For  $\mathbf{A} = \begin{pmatrix} 1 & 0 & 1 \\ 0 & 1 & 1 \end{pmatrix} \in M_{2 \times 3}(\mathbb{R})$ , the columns span  $\mathbb{R}^2$  (the first two are  $\mathbf{e}_1, \mathbf{e}_2$ ), so (g) holds, but  $\mathbf{x} = (1, 1, -1)$  solves  $\mathbf{Ax} = \mathbf{0}$ , so (b) fails. The proof of (f)  $\Leftrightarrow$  (g) used that the number of columns equals  $\dim F^n$ , and that is exactly what breaks.

### 2.4.3 One-sided inverses are two-sided

Chapter 0 left a gap on purpose. The definition of invertibility (Definition 0.9.13) asks for **both**  $\mathbf{AB} = \mathbf{I}_n$  and  $\mathbf{BA} = \mathbf{I}_n$ . Chapter 0 then stated that, for square matrices, the first equation already implies the second, but that nothing there could prove it: the entries of  $\mathbf{AB}$  and  $\mathbf{BA}$  are different sums. Now we can prove it, and the proof uses the menu above instead of entries.

#### Theorem 2.4.5 — One-sided inverses of square matrices

Let  $\mathbf{A}, \mathbf{B} \in M_n(F)$ . If  $\mathbf{AB} = \mathbf{I}_n$ , then  $\mathbf{BA} = \mathbf{I}_n$ . Hence  $\mathbf{A}$  and  $\mathbf{B}$  are both invertible,  $\mathbf{B} = \mathbf{A}^{-1}$  and  $\mathbf{A} = \mathbf{B}^{-1}$ .

*Idea.* We cannot touch  $\mathbf{BA}$  directly, so we aim at invertibility of one factor instead. Which one? From  $\mathbf{AB} = \mathbf{I}_n$ , a vector killed by  $\mathbf{B}$  is killed by  $\mathbf{AB} = \mathbf{I}_n$ , so it is  $\mathbf{0}$ . That is item (b) for  $\mathbf{B}$ . Once  $\mathbf{B}^{-1}$  exists, we may cancel it:  $\mathbf{A} = (\mathbf{AB})\mathbf{B}^{-1} = \mathbf{B}^{-1}$ .

*Proof.* Suppose  $\mathbf{AB} = \mathbf{I}_n$ , and let  $\mathbf{x} \in F^n$  with  $\mathbf{Bx} = \mathbf{0}$ . Then, by [Theorem 0.9.4](#),

$$\mathbf{x} = \mathbf{I}_n \mathbf{x} = (\mathbf{AB})\mathbf{x} = \mathbf{A}(\mathbf{Bx}) = \mathbf{A}\mathbf{0} = \mathbf{0}.$$

So  $\mathbf{Bx} = \mathbf{0}$  has only the trivial solution, and  $\mathbf{B}$  is invertible by [Theorem 2.4.4](#) ((b)  $\Rightarrow$  (a)). Multiplying  $\mathbf{AB} = \mathbf{I}_n$  on the right by  $\mathbf{B}^{-1}$  and using associativity,

$$\mathbf{A} = \mathbf{A}(\mathbf{BB}^{-1}) = (\mathbf{AB})\mathbf{B}^{-1} = \mathbf{I}_n \mathbf{B}^{-1} = \mathbf{B}^{-1}.$$

Therefore  $\mathbf{BA} = \mathbf{BB}^{-1} = \mathbf{I}_n$ . Since  $\mathbf{A} = \mathbf{B}^{-1}$  is invertible with inverse  $\mathbf{B}$  by [Theorem 0.9.7](#) (part 2), we get  $\mathbf{A}^{-1} = \mathbf{B}$ . This proves the theorem. ■

By symmetry,  $\mathbf{BA} = \mathbf{I}_n$  also implies  $\mathbf{AB} = \mathbf{I}_n$ : apply the theorem with the roles of  $\mathbf{A}$  and  $\mathbf{B}$  swapped. So from now on, **to verify that  $\mathbf{B}$  is the inverse of a square matrix  $\mathbf{A}$ , one product suffices**. We also get a useful consequence of the same kind: if  $\mathbf{A}, \mathbf{B} \in M_n(F)$  and  $\mathbf{AB}$  is invertible, then  $\mathbf{A}$  and  $\mathbf{B}$  are invertible (Exercise B3 at the end of this section).

*Warning.* For **non-square** matrices a one-sided inverse need not be two-sided. Let

$$\mathbf{A} = \begin{pmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \end{pmatrix}, \quad \mathbf{B} = \begin{pmatrix} 1 & 0 \\ 0 & 1 \\ 0 & 0 \end{pmatrix}.$$

Then  $\mathbf{AB} = \mathbf{I}_2$ , but  $\mathbf{BA} = \text{diag}(1, 1, 0) \neq \mathbf{I}_3$ . In fact no  $\mathbf{C} \in M_{3 \times 2}(F)$  has  $\mathbf{CA} = \mathbf{I}_3$ :  $\mathbf{Ae}_3 = \mathbf{0}$ , so  $\mathbf{CAe}_3 = \mathbf{0} \neq \mathbf{e}_3 = \mathbf{I}_3 \mathbf{e}_3$ . The proof above breaks because  $\mathbf{Bx} = \mathbf{0} \Rightarrow \mathbf{x} = \mathbf{0}$  no longer gives invertibility:  $\mathbf{B}$  is not square, so [Theorem 2.4.4](#) does not apply.

## 2.4.4 Computing the inverse

[Theorem 2.4.4](#) (c) says  $\mathbf{A}$  is invertible exactly when it row reduces to  $\mathbf{I}_n$ . The proof of (c)  $\Rightarrow$  (d) even produced a matrix: if  $\mathbf{E}_k \cdots \mathbf{E}_1 \mathbf{A} = \mathbf{I}_n$ , then  $\mathbf{E}_k \cdots \mathbf{E}_1$  is a left inverse, hence **the** inverse by [Theorem 2.4.5](#). We do not want to multiply out the  $\mathbf{E}_i$  by hand. The trick is to let the operations act on  $\mathbf{I}_n$  at the same time as on  $\mathbf{A}$ : since  $\mathbf{E}_k \cdots \mathbf{E}_1 \mathbf{I}_n = \mathbf{E}_k \cdots \mathbf{E}_1$ , the operations that turn  $\mathbf{A}$  into  $\mathbf{I}_n$  turn  $\mathbf{I}_n$  into  $\mathbf{A}^{-1}$ . We write  $[\mathbf{A} \mid \mathbf{I}_n]$  for the  $n \times 2n$  matrix whose first  $n$  columns are those of  $\mathbf{A}$  and whose last  $n$  columns are those of  $\mathbf{I}_n$ .

*Algorithm.* **Input:**  $\mathbf{A} \in M_n(F)$ .

1. Form the  $n \times 2n$  matrix  $[\mathbf{A} \mid \mathbf{I}_n]$ .
2. Row reduce it to reduced row echelon form  $[\mathbf{R} \mid \mathbf{C}]$ , with  $\mathbf{R}, \mathbf{C} \in M_n(F)$ .
3. If  $\mathbf{R} = \mathbf{I}_n$ , output  $\mathbf{A}^{-1} = \mathbf{C}$ . Otherwise, output “ $\mathbf{A}$  is not invertible”.

In practice, stop as soon as a zero row appears in the left half:  $\mathbf{A}$  is then not invertible (see the remark after the theorem).

### Theorem 2.4.6 — Inverse by row reduction

Let  $\mathbf{A} \in M_n(F)$ , and let  $[\mathbf{R} \mid \mathbf{C}]$  be the reduced row echelon form of  $[\mathbf{A} \mid \mathbf{I}_n]$ , with  $\mathbf{R}, \mathbf{C} \in M_n(F)$ . Then  $\mathbf{R}$  is the reduced row echelon form of  $\mathbf{A}$ . Consequently,  $\mathbf{A}$  is invertible if and only if  $\mathbf{R} = \mathbf{I}_n$ , and in that case  $\mathbf{A}^{-1} = \mathbf{C}$ .

*Idea.* Two things need checking. First, the operations act on the two halves independently, so the left half is transformed exactly as  $\mathbf{A}$  alone would be. Second, the left half of a reduced matrix is itself reduced, so it really is the RREF of  $\mathbf{A}$ . The first is the column view of a product; the second is a small check on the staircase shape. After that,  $\mathbf{R} = \mathbf{EA}$  and  $\mathbf{C} = \mathbf{EI}_n = \mathbf{E}$ , and [Theorem 2.4.5](#) finishes.

*Proof.* Let  $\rho_1, \dots, \rho_k$  be row operations taking  $[\mathbf{A} \mid \mathbf{I}_n]$  to  $[\mathbf{R} \mid \mathbf{C}]$ , and put  $\mathbf{E} = \mathbf{E}_k \cdots \mathbf{E}_1$  with  $\mathbf{E}_i = \rho_i(\mathbf{I}_n)$  (and  $\mathbf{E} = \mathbf{I}_n$  if  $k = 0$ ). By [Theorem 2.4.1](#),  $[\mathbf{R} \mid \mathbf{C}] = \mathbf{E}[\mathbf{A} \mid \mathbf{I}_n]$ . By [Theorem 0.9.2](#) (columns), each column of  $\mathbf{E}[\mathbf{A} \mid \mathbf{I}_n]$  is  $\mathbf{E}$  times the corresponding column, so

$$\mathbf{R} = \mathbf{EA} \quad \text{and} \quad \mathbf{C} = \mathbf{EI}_n = \mathbf{E}.$$

*Claim.*  $\mathbf{R}$  is in reduced row echelon form.

*Proof.* If row  $k$  of  $\mathbf{R}$  is non-zero, its first non-zero entry is also the first non-zero entry of row  $k$  of  $[\mathbf{R} \mid \mathbf{C}]$ , so the pivots of the non-zero rows of  $\mathbf{R}$  are pivots of  $[\mathbf{R} \mid \mathbf{C}]$ , in the same positions. Suppose row  $k$  of  $\mathbf{R}$  is zero but some later row  $l > k$  of  $\mathbf{R}$  is non-zero. Row  $k$  of  $[\mathbf{R} \mid \mathbf{C}]$  cannot be zero, since zero rows of  $[\mathbf{R} \mid \mathbf{C}]$  come after its non-zero rows and row  $l$  is non-zero. So its pivot lies in a column  $> n$ , while the pivot of row  $l$  lies in a column  $\leq n$ , to the left: this contradicts the staircase condition of [Definition 2.2.3](#). Hence the zero rows of  $\mathbf{R}$  come last, the pivots of  $\mathbf{R}$  move strictly right, and each pivot of  $\mathbf{R}$  is 1 and the only non-zero entry in its column, because this holds in  $[\mathbf{R} \mid \mathbf{C}]$ . So  $\mathbf{R}$  satisfies [Definition 2.2.4](#). ■

Now  $\mathbf{R} = \mathbf{E}_k \cdots \mathbf{E}_1 \mathbf{A}$  is obtained from  $\mathbf{A}$  by the operations  $\rho_1, \dots, \rho_k$  ([Theorem 2.4.1](#)), so  $\mathbf{A}$  is row equivalent to  $\mathbf{R}$ , and  $\mathbf{R}$  is in reduced form. By [Theorem 2.3.4](#),  $\mathbf{R}$  is the RREF of  $\mathbf{A}$ . By [Theorem 2.4.4](#) ((a)  $\Leftrightarrow$  (c)),  $\mathbf{A}$  is invertible if and only if  $\mathbf{R} = \mathbf{I}_n$ . If  $\mathbf{R} = \mathbf{I}_n$ , then  $\mathbf{EA} = \mathbf{I}_n$ , and [Theorem 2.4.5](#) gives  $\mathbf{A}^{-1} = \mathbf{E} = \mathbf{C}$ . This proves the theorem. ■

*Remark.* The early stop is justified the same way. If, after some operations,  $[\mathbf{A} \mid \mathbf{I}_n]$  has become  $[\mathbf{A}' \mid \mathbf{E}']$  with a zero row in  $\mathbf{A}'$ , then, exactly as in the proof,  $\mathbf{E}'$  is the product of the elementary matrices of the operations used so far and  $\mathbf{A}' = \mathbf{E}'\mathbf{A}$ . So  $\mathbf{E}'$  is invertible by [Theorem 2.4.2](#) and [Theorem 0.9.7](#) (part 3). A matrix with a zero row  $k$  is not invertible, since row  $k$  of  $\mathbf{A}'\mathbf{X}$  is zero for every  $\mathbf{X}$ , so  $\mathbf{A}'\mathbf{X} \neq \mathbf{I}_n$ . If  $\mathbf{A}$  were invertible,  $\mathbf{A}' = \mathbf{E}'\mathbf{A}$  would be invertible by [Theorem 0.9.7](#) (part 3). Hence  $\mathbf{A}$  is not invertible.

**Example.** ( $\mathbf{A}$   $3 \times 3$  inverse)

Find the inverse of  $\mathbf{A} = \begin{pmatrix} 1 & 2 & 1 \\ 2 & 5 & 3 \\ 1 & 3 & 3 \end{pmatrix} \in M_3(\mathbb{R})$ , or show that it does not exist.

*Solution.* Row reduce  $[\mathbf{A} \mid \mathbf{I}_3]$ , naming each operation.

$$\begin{aligned} \left[ \begin{array}{ccc|ccc} 1 & 2 & 1 & 1 & 0 & 0 \\ 2 & 5 & 3 & 0 & 1 & 0 \\ 1 & 3 & 3 & 0 & 0 & 1 \end{array} \right] &\xrightarrow[\substack{R_2 \rightarrow R_2 - 2R_1 \\ R_3 \rightarrow R_3 - R_1}]{} \left[ \begin{array}{ccc|ccc} 1 & 2 & 1 & 1 & 0 & 0 \\ 0 & 1 & 1 & -2 & 1 & 0 \\ 0 & 1 & 2 & -1 & 0 & 1 \end{array} \right] \\ &\xrightarrow{R_3 \rightarrow R_3 - R_2} \left[ \begin{array}{ccc|ccc} 1 & 2 & 1 & 1 & 0 & 0 \\ 0 & 1 & 1 & -2 & 1 & 0 \\ 0 & 0 & 1 & 1 & -1 & 1 \end{array} \right] \end{aligned}$$

The left half now has three pivots, so no zero row will appear. Clear above the pivots, from the bottom up:

$$\xrightarrow[\substack{R_2 \rightarrow R_2 - R_3 \\ R_1 \rightarrow R_1 - R_3}]{} \left[ \begin{array}{ccc|ccc} 1 & 2 & 0 & 0 & 1 & -1 \\ 0 & 1 & 0 & -3 & 2 & -1 \\ 0 & 0 & 1 & 1 & -1 & 1 \end{array} \right] \xrightarrow{R_1 \rightarrow R_1 - 2R_2} \left[ \begin{array}{ccc|ccc} 1 & 0 & 0 & 6 & -3 & 1 \\ 0 & 1 & 0 & -3 & 2 & -1 \\ 0 & 0 & 1 & 1 & -1 & 1 \end{array} \right].$$

The left half is  $\mathbf{I}_3$ , so by [Theorem 2.4.6](#)  $\mathbf{A}$  is invertible and

$$\mathbf{A}^{-1} = \begin{pmatrix} 6 & -3 & 1 \\ -3 & 2 & -1 \\ 1 & -1 & 1 \end{pmatrix}.$$

Check one product: row 1 of  $\mathbf{A}$  times the columns of  $\mathbf{A}^{-1}$  gives  $6 - 6 + 1 = 1$ ,  $-3 + 4 - 1 = 0$ ,  $1 - 2 + 1 = 0$ ; rows 2 and 3 give  $(0, 1, 0)$  and  $(0, 0, 1)$  in the same way. So  $\mathbf{A}\mathbf{A}^{-1} = \mathbf{I}_3$ , and by [Theorem 2.4.5](#) this single product is enough.

**Example. (When the algorithm stops)**

Decide whether  $\mathbf{A} = \begin{pmatrix} 1 & 2 & 3 \\ 2 & 5 & 7 \\ 1 & 3 & 4 \end{pmatrix} \in M_3(\mathbb{R})$  is invertible.

*Solution.* Row reduce  $[\mathbf{A} \mid \mathbf{I}_3]$ :

$$\begin{aligned} \left[ \begin{array}{ccc|ccc} 1 & 2 & 3 & 1 & 0 & 0 \\ 2 & 5 & 7 & 0 & 1 & 0 \\ 1 & 3 & 4 & 0 & 0 & 1 \end{array} \right] &\xrightarrow[\substack{R_2 \rightarrow R_2 - 2R_1 \\ R_3 \rightarrow R_3 - R_1}]{} \left[ \begin{array}{ccc|ccc} 1 & 2 & 3 & 1 & 0 & 0 \\ 0 & 1 & 1 & -2 & 1 & 0 \\ 0 & 1 & 1 & -1 & 0 & 1 \end{array} \right] \\ &\xrightarrow{R_3 \rightarrow R_3 - R_2} \left[ \begin{array}{ccc|ccc} 1 & 2 & 3 & 1 & 0 & 0 \\ 0 & 1 & 1 & -2 & 1 & 0 \\ 0 & 0 & 0 & 1 & -1 & 1 \end{array} \right]. \end{aligned}$$

A zero row has appeared in the left half, so by the remark after [Theorem 2.4.6](#),  $\mathbf{A}$  is not invertible. The right half of that row is not wasted: it records the combination of rows that produced zero. Row 3 of the left half is  $1 \cdot (\text{row } 1) - 1 \cdot (\text{row } 2) + 1 \cdot (\text{row } 3)$  of  $\mathbf{A}$ , and indeed  $(1, 2, 3) - (2, 5, 7) + (1, 3, 4) = (0, 0, 0)$ .

Finally, the method writes an invertible matrix as a product of elementary matrices, as in

[Theorem 2.4.4](#) (d).

**Example. (Factoring into elementary matrices)**

Write  $\mathbf{A} = \begin{pmatrix} 2 & 1 \\ 5 & 3 \end{pmatrix} \in M_2(\mathbb{R})$  as a product of elementary matrices, and read off  $\mathbf{A}^{-1}$ .

*Solution.* Reduce  $\mathbf{A}$  to  $\mathbf{I}_2$ , keeping integers:

$$\begin{pmatrix} 2 & 1 \\ 5 & 3 \end{pmatrix} \xrightarrow{R_2 \rightarrow R_2 - 2R_1} \begin{pmatrix} 2 & 1 \\ 1 & 1 \end{pmatrix} \xrightarrow{R_1 \rightarrow R_1 - R_2} \begin{pmatrix} 1 & 0 \\ 1 & 1 \end{pmatrix} \xrightarrow{R_2 \rightarrow R_2 - R_1} \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix}.$$

The elementary matrices of these operations are  $\mathbf{E}_1 = \mathbf{I}_2 - 2\mathbf{E}_{21}$ ,  $\mathbf{E}_2 = \mathbf{I}_2 - \mathbf{E}_{12}$ ,  $\mathbf{E}_3 = \mathbf{I}_2 - \mathbf{E}_{21}$ , and by [Theorem 2.4.1](#),  $\mathbf{E}_3\mathbf{E}_2\mathbf{E}_1\mathbf{A} = \mathbf{I}_2$ . Multiplying on the left by  $\mathbf{E}_3^{-1}$ , then  $\mathbf{E}_2^{-1}$ , then  $\mathbf{E}_1^{-1}$ , and using [Theorem 2.4.2](#),

$$\mathbf{A} = \mathbf{E}_1^{-1}\mathbf{E}_2^{-1}\mathbf{E}_3^{-1} = \begin{pmatrix} 1 & 0 \\ 2 & 1 \end{pmatrix} \begin{pmatrix} 1 & 1 \\ 0 & 1 \end{pmatrix} \begin{pmatrix} 1 & 0 \\ 1 & 1 \end{pmatrix}.$$

Multiplying out confirms it: the last two factors give  $\begin{pmatrix} 2 & 1 \\ 1 & 1 \end{pmatrix}$ , and then the first factor adds twice row 1 to row 2, giving  $\begin{pmatrix} 2 & 1 \\ 5 & 3 \end{pmatrix}$ . Also  $\mathbf{E}_3\mathbf{E}_2\mathbf{E}_1$  is a left inverse of  $\mathbf{A}$ , hence  $\mathbf{A}^{-1}$  by [Theorem 2.4.5](#):

$$\mathbf{A}^{-1} = \mathbf{E}_3\mathbf{E}_2\mathbf{E}_1 = \begin{pmatrix} 3 & -1 \\ -5 & 2 \end{pmatrix},$$

which agrees with [Theorem 0.9.8](#), since  $ad - bc = 1$ . The factorization is not unique: a different sequence of operations gives different factors.

## 2.4.5 Row equivalence, revisited

Two descriptions of “row equivalent” are now on the table. [Example \(Row equivalence\)](#) of Chapter 0 said  $\mathbf{B} = \mathbf{E}\mathbf{A}$  for some invertible  $\mathbf{E}$ . [Definition 2.2.2](#) says  $\mathbf{B}$  is reachable from  $\mathbf{A}$  by finitely many row operations. Chapter 0 promised that they agree, and now it is one line each way.

### **Theorem 2.4.7 — Row equivalence is left multiplication by an invertible matrix**

Let  $\mathbf{A}, \mathbf{B} \in M_{m \times n}(F)$ . Then  $\mathbf{A}$  is row equivalent to  $\mathbf{B}$  (that is,  $\mathbf{B}$  can be reached from  $\mathbf{A}$  by row operations) if and only if  $\mathbf{B} = \mathbf{E}\mathbf{A}$  for some invertible  $\mathbf{E} \in M_m(F)$ .

*Proof.* ( $\Rightarrow$ ) Suppose row operations  $\rho_1, \dots, \rho_k$  take  $\mathbf{A}$  to  $\mathbf{B}$ . By [Theorem 2.4.1](#),  $\mathbf{B} = \mathbf{E}_k \cdots \mathbf{E}_1 \mathbf{A}$  with  $\mathbf{E}_i = \rho_i(\mathbf{I}_m)$ , and  $\mathbf{E} = \mathbf{E}_k \cdots \mathbf{E}_1$  is invertible by [Theorem 2.4.2](#) and [Theorem 0.9.7](#) (part 3). (If  $k = 0$ , take  $\mathbf{E} = \mathbf{I}_m$ .)

( $\Leftarrow$ ) Suppose  $\mathbf{B} = \mathbf{E}\mathbf{A}$  with  $\mathbf{E}$  invertible. By [Theorem 2.4.4](#) ((a)  $\Rightarrow$  (d)),  $\mathbf{E} = \mathbf{E}_k \cdots \mathbf{E}_1$  for some elementary matrices  $\mathbf{E}_i$ . Let  $\rho_i$  be the operation with  $\rho_i(\mathbf{I}_m) = \mathbf{E}_i$ . By [Theorem 2.4.1](#), performing  $\rho_1$ , then  $\rho_2$ ,  $\dots$ , then  $\rho_k$  on  $\mathbf{A}$  produces  $\mathbf{E}_k \cdots \mathbf{E}_1 \mathbf{A} = \mathbf{B}$ . Hence

$\mathbf{A}$  is row equivalent to  $\mathbf{B}$ .

■

So the equivalence relation of [Example \(Row equivalence\)](#) is the one Gaussian elimination moves along, and “two matrices have the same RREF” is the same as “each is an invertible matrix times the other”. In the next section this form,  $\mathbf{B} = \mathbf{E}\mathbf{A}$ , is what makes row spaces and ranks easy to compare.

## 2.4.6 Exercises

### 2.4.6.1 A. Check your understanding

#### Exercise A1.

- Define an elementary matrix, and write down the  $3 \times 3$  elementary matrix that performs  $R_3 \rightarrow R_3 + 7R_1$ .
- State four of the conditions in [Theorem 2.4.4](#).
- True or false: the product of two elementary matrices is always an elementary matrix. Justify your answer.
- True or false: if  $\mathbf{A} \in M_{2 \times 3}(F)$ ,  $\mathbf{B} \in M_{3 \times 2}(F)$  and  $\mathbf{AB} = \mathbf{I}_2$ , then  $\mathbf{BA} = \mathbf{I}_3$ . Justify your answer.
- True or false: if  $\mathbf{A} \in M_n(F)$  and  $\mathbf{Ax} = \mathbf{e}_j$  has a solution for each  $j = 1, \dots, n$ , then  $\mathbf{A}$  is invertible. Justify your answer.
- Describe the method for computing the inverse of a square matrix, and how it detects a matrix that is not invertible.

*Solution.*

- An elementary matrix in  $M_m(F)$  is one obtained from  $\mathbf{I}_m$  by exactly one elementary row operation ([Definition 2.4.1](#)). Applying  $R_3 \rightarrow R_3 + 7R_1$  to  $\mathbf{I}_3$  gives  $\mathbf{I}_3 + 7\mathbf{E}_{31} = \begin{pmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 7 & 0 & 1 \end{pmatrix}$ .
- For  $\mathbf{A} \in M_n(F)$ , each of the following is equivalent to  $\mathbf{A}$  being invertible:  $\mathbf{Ax} = \mathbf{0}$  has only the trivial solution; the RREF of  $\mathbf{A}$  is  $\mathbf{I}_n$ ;  $\mathbf{A}$  is a product of elementary matrices; the columns of  $\mathbf{A}$  form a basis of  $F^n$ .
- False. A single row operation changes at most two rows of  $\mathbf{I}_3$  (a swap changes two, a scaling or an addition changes one), so every elementary  $3 \times 3$  matrix has at least one row equal to the corresponding row of  $\mathbf{I}_3$ . But  $\mathbf{P}_{12}\mathbf{P}_{23} = \begin{pmatrix} 0 & 0 & 1 \\ 1 & 0 & 0 \\ 0 & 1 & 0 \end{pmatrix}$  (by the row view, its rows are rows 2, 1, 3 of  $\mathbf{P}_{23}$ ) differs from  $\mathbf{I}_3$  in every row. Hence it is a product of two elementary matrices that is not elementary.
- False. With  $\mathbf{A} = \begin{pmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \end{pmatrix}$  and  $\mathbf{B} = \begin{pmatrix} 1 & 0 \\ 0 & 1 \\ 0 & 0 \end{pmatrix}$ ,  $\mathbf{AB} = \mathbf{I}_2$  but  $\mathbf{BA} = \text{diag}(1, 1, 0) \neq \mathbf{I}_3$ . [Theorem 2.4.5](#) needs square matrices.

- (e) True. Let  $\mathbf{x}_j$  solve  $\mathbf{A}\mathbf{x}_j = \mathbf{e}_j$ , and let  $\mathbf{B}$  have columns  $\mathbf{x}_1, \dots, \mathbf{x}_n$ . By [Theorem 0.9.2](#),  $\mathbf{AB}$  has columns  $\mathbf{e}_1, \dots, \mathbf{e}_n$ , so  $\mathbf{AB} = \mathbf{I}_n$ , and  $\mathbf{A}$  is invertible by [Theorem 2.4.5](#).
- (f) Row reduce  $[\mathbf{A} \mid \mathbf{I}_n]$  to reduced form  $[\mathbf{R} \mid \mathbf{C}]$ . If  $\mathbf{R} = \mathbf{I}_n$ , then  $\mathbf{A}^{-1} = \mathbf{C}$ ; if  $\mathbf{R} \neq \mathbf{I}_n$ , in particular if a zero row appears in the left half at any stage,  $\mathbf{A}$  is not invertible ([Theorem 2.4.6](#)).

### 2.4.6.2 B. Practice

#### Exercise B1: Inverses by row reduction.

For each matrix, find the inverse or show that it is not invertible.

(a)  $\mathbf{A} = \begin{pmatrix} 1 & 0 & 2 \\ 2 & 1 & 3 \\ 4 & 1 & 8 \end{pmatrix} \in M_3(\mathbb{R})$ .

(b)  $\mathbf{B} = \begin{pmatrix} 1 & 2 & -1 \\ 0 & 1 & 1 \\ 2 & 5 & -1 \end{pmatrix} \in M_3(\mathbb{R})$ .

(c)  $\mathbf{C} = \begin{pmatrix} 1 & 1 & 0 \\ 0 & 1 & 1 \\ 1 & 1 & 1 \end{pmatrix} \in M_3(\mathbb{F}_2)$ .

*Solution.*

(a) Row reduce  $[\mathbf{A} \mid \mathbf{I}_3]$ :

$$\left[ \begin{array}{ccc|ccc} 1 & 0 & 2 & 1 & 0 & 0 \\ 2 & 1 & 3 & 0 & 1 & 0 \\ 4 & 1 & 8 & 0 & 0 & 1 \end{array} \right] \xrightarrow[R_3 \rightarrow R_3 - 4R_1]{R_2 \rightarrow R_2 - 2R_1} \left[ \begin{array}{ccc|ccc} 1 & 0 & 2 & 1 & 0 & 0 \\ 0 & 1 & -1 & -2 & 1 & 0 \\ 0 & 1 & 0 & -4 & 0 & 1 \end{array} \right]$$

$$\xrightarrow{R_3 \rightarrow R_3 - R_2} \left[ \begin{array}{ccc|ccc} 1 & 0 & 2 & 1 & 0 & 0 \\ 0 & 1 & -1 & -2 & 1 & 0 \\ 0 & 0 & 1 & -2 & -1 & 1 \end{array} \right]$$

$$\xrightarrow[R_1 \rightarrow R_1 - 2R_3]{R_2 \rightarrow R_2 + R_3} \left[ \begin{array}{ccc|ccc} 1 & 0 & 0 & 5 & 2 & -2 \\ 0 & 1 & 0 & -4 & 0 & 1 \\ 0 & 0 & 1 & -2 & -1 & 1 \end{array} \right].$$

The left half is  $\mathbf{I}_3$ , so by [Theorem 2.4.6](#),  $\mathbf{A}^{-1} = \begin{pmatrix} 5 & 2 & -2 \\ -4 & 0 & 1 \\ -2 & -1 & 1 \end{pmatrix}$ . Check: row 1

of  $\mathbf{A}$  against the columns of  $\mathbf{A}^{-1}$  gives  $5 - 4 = 1$ ,  $2 - 2 = 0$ ,  $-2 + 2 = 0$ , and rows 2 and 3 give  $(0, 1, 0)$  and  $(0, 0, 1)$ , so  $\mathbf{AA}^{-1} = \mathbf{I}_3$ , which suffices by [Theorem 2.4.5](#).



(b) Row reduce  $[\mathbf{B} \mid \mathbf{I}_3]$ :

$$\begin{aligned} \left[ \begin{array}{ccc|ccc} 1 & 2 & -1 & 1 & 0 & 0 \\ 0 & 1 & 1 & 0 & 1 & 0 \\ 2 & 5 & -1 & 0 & 0 & 1 \end{array} \right] &\xrightarrow{R_3 \rightarrow R_3 - 2R_1} \left[ \begin{array}{ccc|ccc} 1 & 2 & -1 & 1 & 0 & 0 \\ 0 & 1 & 1 & 0 & 1 & 0 \\ 0 & 1 & 1 & -2 & 0 & 1 \end{array} \right] \\ &\xrightarrow{R_3 \rightarrow R_3 - R_2} \left[ \begin{array}{ccc|ccc} 1 & 2 & -1 & 1 & 0 & 0 \\ 0 & 1 & 1 & 0 & 1 & 0 \\ 0 & 0 & 0 & -2 & -1 & 1 \end{array} \right]. \end{aligned}$$

A zero row appears in the left half, so  $\mathbf{B}$  is not invertible (remark after [Theorem 2.4.6](#)). The right half of that row says  $-2(\text{row } 1) - (\text{row } 2) + (\text{row } 3) = \mathbf{0}$ , that is, row 3 of  $\mathbf{B}$  is  $2(\text{row } 1) + (\text{row } 2)$ .

(c) Over  $\mathbb{F}_2$  we have  $-1 = 1$ , so subtracting a row is adding it. Row reduce  $[\mathbf{C} \mid \mathbf{I}_3]$ :

$$\begin{aligned} \left[ \begin{array}{ccc|ccc} 1 & 1 & 0 & 1 & 0 & 0 \\ 0 & 1 & 1 & 0 & 1 & 0 \\ 1 & 1 & 1 & 0 & 0 & 1 \end{array} \right] &\xrightarrow{R_3 \rightarrow R_3 + R_1} \left[ \begin{array}{ccc|ccc} 1 & 1 & 0 & 1 & 0 & 0 \\ 0 & 1 & 1 & 0 & 1 & 0 \\ 0 & 0 & 1 & 1 & 0 & 1 \end{array} \right] \\ &\xrightarrow{R_2 \rightarrow R_2 + R_3} \left[ \begin{array}{ccc|ccc} 1 & 1 & 0 & 1 & 0 & 0 \\ 0 & 1 & 0 & 1 & 1 & 1 \\ 0 & 0 & 1 & 1 & 0 & 1 \end{array} \right] \\ &\xrightarrow{R_1 \rightarrow R_1 + R_2} \left[ \begin{array}{ccc|ccc} 1 & 0 & 0 & 0 & 1 & 1 \\ 0 & 1 & 0 & 1 & 1 & 1 \\ 0 & 0 & 1 & 1 & 0 & 1 \end{array} \right]. \end{aligned}$$

Hence  $\mathbf{C}^{-1} = \begin{pmatrix} 0 & 1 & 1 \\ 1 & 1 & 1 \\ 1 & 0 & 1 \end{pmatrix}$ . Check in  $\mathbb{F}_2$ : row 1 of  $\mathbf{C}$  is  $(1, 1, 0)$ , and against the columns of  $\mathbf{C}^{-1}$  it gives  $0 + 1 = 1$ ,  $1 + 1 = 0$ ,  $1 + 1 = 0$ ; rows 2 and 3 give  $(0, 1, 0)$  and  $(0, 0, 1)$  similarly.

### Exercise B2: A product of elementary matrices.

Let  $\mathbf{A} = \begin{pmatrix} 1 & 3 \\ 2 & 5 \end{pmatrix} \in M_2(\mathbb{R})$ .

- Write  $\mathbf{A}$  as a product of elementary matrices.
- Hence write  $\mathbf{A}^{-1}$  as a product of elementary matrices, and compute it.

*Solution.*

(a) Reduce  $\mathbf{A}$  to  $\mathbf{I}_2$ :

$$\begin{pmatrix} 1 & 3 \\ 2 & 5 \end{pmatrix} \xrightarrow{R_2 \rightarrow R_2 - 2R_1} \begin{pmatrix} 1 & 3 \\ 0 & -1 \end{pmatrix} \xrightarrow{R_2 \rightarrow -R_2} \begin{pmatrix} 1 & 3 \\ 0 & 1 \end{pmatrix} \xrightarrow{R_1 \rightarrow R_1 - 3R_2} \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix}.$$

With  $\mathbf{E}_1 = \mathbf{I}_2 - 2\mathbf{E}_{21}$ ,  $\mathbf{E}_2 = \mathbf{D}_2(-1)$ ,  $\mathbf{E}_3 = \mathbf{I}_2 - 3\mathbf{E}_{12}$ , [Theorem 2.4.1](#) gives  $\mathbf{E}_3\mathbf{E}_2\mathbf{E}_1\mathbf{A} = \mathbf{I}_2$ . Multiplying on the left by  $\mathbf{E}_3^{-1}, \mathbf{E}_2^{-1}, \mathbf{E}_1^{-1}$  in turn and using

Theorem 2.4.2,

$$\mathbf{A} = \mathbf{E}_1^{-1} \mathbf{E}_2^{-1} \mathbf{E}_3^{-1} = \begin{pmatrix} 1 & 0 \\ 2 & 1 \end{pmatrix} \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix} \begin{pmatrix} 1 & 3 \\ 0 & 1 \end{pmatrix}.$$

Check: the last two factors give  $\begin{pmatrix} 1 & 3 \\ 0 & -1 \end{pmatrix}$ , and adding twice row 1 to row 2 gives  $\begin{pmatrix} 1 & 3 \\ 2 & 5 \end{pmatrix}$ .

(b) Since  $\mathbf{E}_3 \mathbf{E}_2 \mathbf{E}_1 \mathbf{A} = \mathbf{I}_2$ , Theorem 2.4.5 gives

$$\mathbf{A}^{-1} = \mathbf{E}_3 \mathbf{E}_2 \mathbf{E}_1 = \begin{pmatrix} 1 & -3 \\ 0 & 1 \end{pmatrix} \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix} \begin{pmatrix} 1 & 0 \\ -2 & 1 \end{pmatrix} = \begin{pmatrix} -5 & 3 \\ 2 & -1 \end{pmatrix}.$$

(The last two factors give  $\begin{pmatrix} 1 & 0 \\ 2 & -1 \end{pmatrix}$ , and  $R_1 \rightarrow R_1 - 3R_2$  then gives the answer.)

This agrees with Theorem 0.9.8, since  $ad - bc = -1$ .

### Exercise B3: Invertible products.

Let  $\mathbf{A}, \mathbf{B} \in M_n(F)$ , and suppose  $\mathbf{AB}$  is invertible. Prove that  $\mathbf{A}$  and  $\mathbf{B}$  are both invertible.

*Solution.* Let  $\mathbf{C} = (\mathbf{AB})^{-1}$ . Then  $\mathbf{A}(\mathbf{BC}) = (\mathbf{AB})\mathbf{C} = \mathbf{I}_n$  by associativity, so  $\mathbf{A}$  has the right inverse  $\mathbf{BC}$ , and by Theorem 2.4.5  $\mathbf{A}$  is invertible. Similarly  $(\mathbf{CA})\mathbf{B} = \mathbf{C}(\mathbf{AB}) = \mathbf{I}_n$ , so  $\mathbf{B}$  has the left inverse  $\mathbf{CA}$ . By Theorem 2.4.5 with the roles of the two matrices swapped ( $\mathbf{CA}$  times  $\mathbf{B}$  is  $\mathbf{I}_n$ ),  $\mathbf{B}$  is invertible. This proves the claim.

### 2.4.6.3 C. Going deeper

#### Exercise C1: Triangular matrices.

Let  $\mathbf{U} = (u_{ij}) \in M_n(F)$  be upper triangular (Definition 0.9.5).

- (a) Prove that  $\mathbf{U}$  is invertible if and only if  $u_{ii} \neq 0$  for every  $i$ .
- (b) Prove that if  $\mathbf{U}$  is invertible, then  $\mathbf{U}^{-1}$  is upper triangular.

*Hint:* for (a), use item (b) or item (f) of Theorem 2.4.4; for (b), consider the systems  $\mathbf{U}\mathbf{x} = \mathbf{e}_k$ .

*Solution.*

- (a) ( $\Leftarrow$ ) Suppose every  $u_{ii} \neq 0$ , and let  $\mathbf{U}\mathbf{x} = \mathbf{0}$ . Row  $i$  of this system reads  $u_{ii}x_i + u_{i,i+1}x_{i+1} + \cdots + u_{in}x_n = 0$ , since  $u_{ij} = 0$  for  $j < i$ . We show  $x_n = x_{n-1} = \cdots = x_1 = 0$  by downward induction. Row  $n$  is  $u_{nn}x_n = 0$ , and  $u_{nn} \neq 0$ , so  $x_n = 0$ . If  $x_{i+1} = \cdots = x_n = 0$ , row  $i$  becomes  $u_{ii}x_i = 0$ , so  $x_i = 0$  since  $u_{ii} \neq 0$ . Hence  $\mathbf{x} = \mathbf{0}$ , and  $\mathbf{U}$  is invertible by Theorem 2.4.4 ((b)  $\Rightarrow$  (a)).
- ( $\Rightarrow$ ) Suppose  $u_{kk} = 0$  for some  $k$ . Every column  $j \leq k$  of  $\mathbf{U}$  has zero entries in rows  $k, k+1, \dots, n$ : below the diagonal by triangularity, and in row  $k$  of column  $k$  by assumption. So the  $k$  columns  $\mathbf{u}_1, \dots, \mathbf{u}_k$  all lie in  $\text{span}(\mathbf{e}_1, \dots, \mathbf{e}_{k-1})$ , a subspace

of dimension  $k - 1$  (its spanning list is a sub-list of the standard basis, hence independent). By [Theorem 1.6.3](#) (a), a list of  $k > k - 1$  vectors there is linearly dependent. A non-trivial relation among  $\mathbf{u}_1, \dots, \mathbf{u}_k$ , extended by zero coefficients on  $\mathbf{u}_{k+1}, \dots, \mathbf{u}_n$ , is a non-trivial relation among all the columns. So the columns of  $\mathbf{U}$  are dependent, and  $\mathbf{U}$  is not invertible by [Theorem 2.4.4](#) ((a)  $\Rightarrow$  (f)). (For  $k = 1$ , the span is  $\{\mathbf{0}\}$  and  $\mathbf{u}_1 = \mathbf{0}$ , so the same conclusion holds.)

- (b) Suppose  $\mathbf{U}$  is invertible, so every  $u_{ii} \neq 0$  by (a). Fix  $k$ , and let  $\mathbf{x}$  be the  $k$ -th column of  $\mathbf{U}^{-1}$ . By [Theorem 0.9.2](#),  $\mathbf{U}\mathbf{x}$  is the  $k$ -th column of  $\mathbf{U}\mathbf{U}^{-1} = \mathbf{I}_n$ , so  $\mathbf{U}\mathbf{x} = \mathbf{e}_k$ . For  $i > k$ , row  $i$  of this system reads  $u_{ii}x_i + \dots + u_{in}x_n = 0$ . By the same downward induction as in (a), running only over  $i = n, n-1, \dots, k+1$ , we get  $x_n = \dots = x_{k+1} = 0$ . So the  $k$ -th column of  $\mathbf{U}^{-1}$  has zeros below position  $k$ . As  $k$  was arbitrary,  $\mathbf{U}^{-1}$  is upper triangular.

### Exercise C2: All right inverses.

Let  $\mathbf{A} = \begin{pmatrix} 1 & 0 & 1 \\ 0 & 1 & 1 \end{pmatrix} \in M_{2 \times 3}(\mathbb{R})$ .

- (a) Find all  $\mathbf{B} \in M_{3 \times 2}(\mathbb{R})$  with  $\mathbf{AB} = \mathbf{I}_2$ .  
 (b) Show that there is no  $\mathbf{C} \in M_{3 \times 2}(\mathbb{R})$  with  $\mathbf{CA} = \mathbf{I}_3$ .

*Solution.*

- (a) By [Theorem 0.9.2](#),  $\mathbf{AB} = \mathbf{I}_2$  means that the columns  $\mathbf{b}_1, \mathbf{b}_2$  of  $\mathbf{B}$  satisfy  $\mathbf{Ab}_1 = \mathbf{e}_1$  and  $\mathbf{Ab}_2 = \mathbf{e}_2$ . The matrix  $\mathbf{A}$  is already in reduced form with pivots in columns 1 and 2 and  $x_3$  free, so  $\mathbf{Ax} = \mathbf{0}$  means  $x_1 = -x_3$ ,  $x_2 = -x_3$ , and its solutions are the multiples of  $(-1, -1, 1)$ . One solution of  $\mathbf{Ax} = \mathbf{e}_1$  is  $(1, 0, 0)$ , and one solution of  $\mathbf{Ax} = \mathbf{e}_2$  is  $(0, 1, 0)$ . By [Theorem 2.1.3](#),

$$\mathbf{b}_1 = (1, 0, 0) + s(-1, -1, 1), \quad \mathbf{b}_2 = (0, 1, 0) + t(-1, -1, 1) \quad (s, t \in \mathbb{R}).$$

Hence the right inverses of  $\mathbf{A}$  are exactly

$$\mathbf{B} = \begin{pmatrix} 1-s & -t \\ -s & 1-t \\ s & t \end{pmatrix}, \quad s, t \in \mathbb{R},$$

infinitely many of them.

- (b) Suppose  $\mathbf{CA} = \mathbf{I}_3$ , and let  $\mathbf{x} = (-1, -1, 1)$ , so  $\mathbf{Ax} = \mathbf{0}$ . Then  $\mathbf{x} = \mathbf{I}_3\mathbf{x} = \mathbf{C}(\mathbf{Ax}) = \mathbf{C}\mathbf{0} = \mathbf{0}$ , a contradiction since  $\mathbf{x} \neq \mathbf{0}$ . Hence  $\mathbf{A}$  has no left inverse.

### Exercise C3: Nilpotent perturbations of the identity.

Let  $\mathbf{A} \in M_n(F)$  and suppose  $\mathbf{A}^k = \mathbf{0}$  for some integer  $k \geq 1$ .

- (a) Prove that  $\mathbf{I}_n + \mathbf{A}$  is invertible, by using item (b) of [Theorem 2.4.4](#).  
 (b) Verify that  $(\mathbf{I}_n + \mathbf{A})(\mathbf{I}_n - \mathbf{A} + \mathbf{A}^2 - \dots + (-1)^{k-1}\mathbf{A}^{k-1}) = \mathbf{I}_n$ , and deduce a formula for  $(\mathbf{I}_n + \mathbf{A})^{-1}$ .

*Solution.*

- (a) Let  $\mathbf{x} \in F^n$  with  $(\mathbf{I}_n + \mathbf{A})\mathbf{x} = \mathbf{0}$ . Then  $\mathbf{A}\mathbf{x} = -\mathbf{x}$ . Multiplying on the left by  $\mathbf{A}$  and using this again,  $\mathbf{A}^2\mathbf{x} = \mathbf{A}(-\mathbf{x}) = -\mathbf{A}\mathbf{x} = (-1)^2\mathbf{x}$ , and inductively  $\mathbf{A}^j\mathbf{x} = (-1)^j\mathbf{x}$  for every  $j \geq 1$ . Taking  $j = k$ ,  $\mathbf{0} = \mathbf{A}^k\mathbf{x} = (-1)^k\mathbf{x}$ , so  $\mathbf{x} = \mathbf{0}$  (multiply by  $(-1)^k$ ). By Theorem 2.4.4 ((b)  $\Rightarrow$  (a)),  $\mathbf{I}_n + \mathbf{A}$  is invertible.
- (b) Expanding with Theorem 0.9.4,

$$(\mathbf{I}_n + \mathbf{A}) \sum_{j=0}^{k-1} (-1)^j \mathbf{A}^j = \sum_{j=0}^{k-1} (-1)^j \mathbf{A}^j + \sum_{j=0}^{k-1} (-1)^j \mathbf{A}^{j+1} = \mathbf{I}_n + (-1)^{k-1} \mathbf{A}^k = \mathbf{I}_n,$$

since the terms  $(-1)^j \mathbf{A}^j$  for  $1 \leq j \leq k-1$  in the first sum cancel against the terms  $(-1)^{j-1} \mathbf{A}^j$  of the second, and  $\mathbf{A}^k = \mathbf{0}$ . By Theorem 2.4.5,  $(\mathbf{I}_n + \mathbf{A})^{-1} = \mathbf{I}_n - \mathbf{A} + \mathbf{A}^2 - \cdots + (-1)^{k-1} \mathbf{A}^{k-1}$ .

## 2.5 Row Space, Column Space, Null Space and Rank

We can now reduce any matrix and read off the solutions of any linear system. This section asks a coarser question: how much information does a matrix carry? Three subspaces come attached to every matrix  $\mathbf{A}$ : the vectors  $\mathbf{A}\mathbf{x}$  it can produce, the combinations of its rows, and the vectors it sends to zero. Their dimensions are governed by a single number, the rank, which is the number of pivots. With it we settle when a system is solvable and how many solutions it has, and we turn the questions of Chapter 1 about vectors in  $F^m$  (independent? spanning? a basis of a sum or an intersection?) into a routine computation, which is the “systematic method” that Chapter 1 promised.

Throughout,  $F$  is a field and  $\mathbf{A} \in M_{m \times n}(F)$  has columns  $\mathbf{a}_1, \dots, \mathbf{a}_n \in F^m$ .

### 2.5.1 Three subspaces of a matrix

Here is the matrix we will follow through the section:

$$\mathbf{A} = \begin{pmatrix} 1 & 2 & 0 & 0 & -1 \\ 2 & 4 & 1 & 0 & 0 \\ 1 & 2 & 1 & 1 & 2 \\ 0 & 0 & 1 & 1 & 3 \end{pmatrix} \in M_{4 \times 5}(\mathbb{R}).$$

Three questions about it have come up in this chapter, and each answer is a set. For which  $\mathbf{b} \in \mathbb{R}^4$  can we solve  $\mathbf{A}\mathbf{x} = \mathbf{b}$ ? By Theorem 2.1.1, exactly for  $\mathbf{b}$  in the span of the columns. Which  $\mathbf{x} \in \mathbb{R}^5$  solve  $\mathbf{A}\mathbf{x} = \mathbf{0}$ ? By Theorem 2.1.2, a subspace of  $\mathbb{R}^5$ . And which new equations can we derive from the four by adding multiples of them? Their left-hand sides are exactly the combinations of the rows. These three sets recur so often that they get names.

*The column space is everything  $\mathbf{A}$  can produce, the null space is everything  $\mathbf{A}$  kills, and the row space is every left-hand side we can build from the equations of  $\mathbf{A}\mathbf{x} = \mathbf{b}$ .*

**Definition 2.5.1 — Column space**

The **column space** of  $\mathbf{A} \in M_{m \times n}(F)$  is the span of its columns,

$$\text{col}(\mathbf{A}) := \text{span}(\mathbf{a}_1, \dots, \mathbf{a}_n) = \{\mathbf{A}\mathbf{x} : \mathbf{x} \in F^n\} \subseteq F^m.$$

**Definition 2.5.2 — Row space**

The **row space** of  $\mathbf{A} \in M_{m \times n}(F)$  is the span of its rows,

$$\text{row}(\mathbf{A}) := \text{span}(\text{row } 1, \dots, \text{row } m) \subseteq M_{1 \times n}(F).$$

**Definition 2.5.3 — Null space**

The **null space** of  $\mathbf{A} \in M_{m \times n}(F)$  is the solution set of the homogeneous system,

$$\text{nul}(\mathbf{A}) := \{\mathbf{x} \in F^n : \mathbf{A}\mathbf{x} = \mathbf{0}\} \subseteq F^n.$$

In words:  $\text{col}(\mathbf{A})$  lives among the **outputs**, in  $F^m$  (one entry per row);  $\text{nul}(\mathbf{A})$  lives among the **inputs**, in  $F^n$  (one entry per column); and  $\text{row}(\mathbf{A})$  consists of row vectors of length  $n$ . The two descriptions of  $\text{col}(\mathbf{A})$  agree by [Theorem 0.9.3](#):  $\mathbf{A}\mathbf{x} = x_1\mathbf{a}_1 + \dots + x_n\mathbf{a}_n$ , and as  $\mathbf{x}$  runs over  $F^n$  these are all the combinations of the columns.

**Well-definedness.** All three are subspaces.  $\text{col}(\mathbf{A})$  and  $\text{row}(\mathbf{A})$  are spans, hence subspaces by [Theorem 1.3.1](#), of  $F^m$  and of  $M_{1 \times n}(F)$  respectively.  $\text{nul}(\mathbf{A})$  is a subspace of  $F^n$  by [Theorem 2.1.2](#). The row vectors  $M_{1 \times n}(F)$  form an  $n$ -dimensional space, with the standard basis  $\mathbf{e}_1^\top, \dots, \mathbf{e}_n^\top$  ([Example \(Dimensions of the Running Examples\)](#)). Transposing turns row vectors into columns and respects sums and scalar multiples ([Theorem 0.9.5](#)), so it sends a combination of rows to the same combination of their transposes, and only the zero row transposes to  $\mathbf{0}$ . The transposes of the rows of  $\mathbf{A}$  are the columns of  $\mathbf{A}^\top$ . Hence transposing sends  $\text{row}(\mathbf{A})$  onto  $\text{col}(\mathbf{A}^\top)$ , a list of rows is independent exactly when the list of their transposes is, and transposing a basis of  $\text{row}(\mathbf{A})$  gives a basis of  $\text{col}(\mathbf{A}^\top)$ . Therefore

$$\dim \text{row}(\mathbf{A}) = \dim \text{col}(\mathbf{A}^\top). \quad (*)$$

**Examples.**

- **The identity.**  $\text{col}(\mathbf{I}_n) = F^n$ , since the columns  $\mathbf{e}_1, \dots, \mathbf{e}_n$  span;  $\text{row}(\mathbf{I}_n) = M_{1 \times n}(F)$ ; and  $\text{nul}(\mathbf{I}_n) = \{\mathbf{0}\}$ , since  $\mathbf{I}_n\mathbf{x} = \mathbf{x}$ .
- **The zero matrix** is the degenerate case. For  $\mathbf{0} = \mathbf{0}_{m \times n}$ ,  $\text{col}(\mathbf{0}) = \text{span}(\mathbf{0}, \dots, \mathbf{0}) = \{\mathbf{0}\}$ ,  $\text{row}(\mathbf{0}) = \{\mathbf{0}\}$ , and  $\text{nul}(\mathbf{0}) = F^n$ , since every  $\mathbf{x}$  is killed. It carries no information, and its column space is as small as possible while its null space is as large as possible. We will see that this trade-off is exact.
- **A single row.** For  $\mathbf{A} = \begin{pmatrix} 1 & 1 & -1 \end{pmatrix} \in M_{1 \times 3}(\mathbb{R})$ , the columns are the scalars  $1, 1, -1 \in \mathbb{R}^1$ , so  $\text{col}(\mathbf{A}) = \mathbb{R}^1$ ,  $\text{row}(\mathbf{A}) = \text{span}(\begin{pmatrix} 1 & 1 & -1 \end{pmatrix})$  is a line of row vectors, and  $\text{nul}(\mathbf{A})$  is the plane  $x + y - z = 0$  in  $\mathbb{R}^3$ .

**Non-example by minimal change.** Replace  $\mathbf{0}$  by a non-zero right-hand side: the solution set  $\{\mathbf{x} \in F^n : \mathbf{A}\mathbf{x} = \mathbf{b}\}$  with  $\mathbf{b} \neq \mathbf{0}$ . It is described by the same equations, and it is still of the form

$\mathbf{p} + \text{nul}(\mathbf{A})$  when non-empty (Theorem 2.1.3). But it does not contain  $\mathbf{0}$ , since  $\mathbf{A}\mathbf{0} = \mathbf{0} \neq \mathbf{b}$ , so it fails the first condition of Theorem 1.2.1.

*Warning.* The three spaces live in **different places**. For  $\mathbf{A} \in M_{4 \times 5}(\mathbb{R})$  above,  $\text{col}(\mathbf{A}) \subseteq \mathbb{R}^4$  but  $\text{nul}(\mathbf{A}) \subseteq \mathbb{R}^5$ , so a statement such as “ $\text{nul}(\mathbf{A}) \subseteq \text{col}(\mathbf{A})$ ” does not even make sense. Keep track by size:  $\mathbf{A}\mathbf{x}$  needs  $\mathbf{x}$  with  $n$  entries and returns a vector with  $m$  entries.

## 2.5.2 Rank and nullity

The column space measures how much  $\mathbf{A}$  can reach, and the null space how much it collapses. We measure each by its dimension. Both are subspaces of finite-dimensional spaces, so both have a dimension by Theorem 1.6.7.

*The rank counts the independent directions that  $\mathbf{A}$  can produce.*

### Definition 2.5.4 — Rank of a matrix

The **rank** of  $\mathbf{A} \in M_{m \times n}(F)$  is  $\text{rank } \mathbf{A} := \dim \text{col}(\mathbf{A})$ .

### Definition 2.5.5 — Nullity of a matrix

The **nullity** of  $\mathbf{A} \in M_{m \times n}(F)$  is  $\text{nullity } \mathbf{A} := \dim \text{nul}(\mathbf{A})$ .

From the examples above:  $\text{rank } \mathbf{I}_n = n$  and  $\text{nullity } \mathbf{I}_n = 0$ ;  $\text{rank } \mathbf{0}_{m \times n} = 0$  and  $\text{nullity } \mathbf{0}_{m \times n} = n$ ; and  $\text{rank } \begin{pmatrix} 1 & 1 & -1 \end{pmatrix} = 1$ , with nullity 2. In each case rank plus nullity equals the number of columns. Two simple bounds hold at once:  $\text{rank } \mathbf{A} \leq m$ , because  $\text{col}(\mathbf{A}) \subseteq F^m$ , and  $\text{rank } \mathbf{A} \leq n$ , because  $\text{col}(\mathbf{A})$  is spanned by  $n$  vectors (Theorem 1.6.3). Why define rank through columns and not rows? We could have, and one of the main theorems of this section says the answer would be the same.

## 2.5.3 What row operations preserve

To compute these spaces we will row reduce. So we first ask what a row operation does to each of them. Recall from Theorem 2.4.7 that  $\mathbf{A}$  is row equivalent to  $\mathbf{B}$  exactly when  $\mathbf{B} = \mathbf{E}\mathbf{A}$  with  $\mathbf{E} \in M_m(F)$  invertible; since row equivalence is symmetric (Theorem 2.2.2 (b)), we may simply say that  $\mathbf{A}$  and  $\mathbf{B}$  are row equivalent.

### Theorem 2.5.1 — Row operations and the three spaces

Let  $\mathbf{A}, \mathbf{B} \in M_{m \times n}(F)$  be row equivalent. Then:

- (a)  $\text{row}(\mathbf{B}) = \text{row}(\mathbf{A})$ ;
- (b)  $\text{nul}(\mathbf{B}) = \text{nul}(\mathbf{A})$ . Equivalently, writing  $\mathbf{b}_1, \dots, \mathbf{b}_n$  for the columns of  $\mathbf{B}$ : for all  $c_1, \dots, c_n \in F$ , the columns of  $\mathbf{A}$  satisfy  $c_1\mathbf{a}_1 + \dots + c_n\mathbf{a}_n = \mathbf{0}$  if and only if the columns of  $\mathbf{B}$  satisfy  $c_1\mathbf{b}_1 + \dots + c_n\mathbf{b}_n = \mathbf{0}$ ;

(c) but  $\text{col}(\mathbf{B})$  and  $\text{col}(\mathbf{A})$  **need not** be equal.

*Proof.*

- (a) Write  $\mathbf{B} = \mathbf{E}\mathbf{A}$  with  $\mathbf{E}$  invertible, by Theorem 2.4.7. By Theorem 0.9.2, row  $i$  of  $\mathbf{E}\mathbf{A}$  is (row  $i$  of  $\mathbf{E}$ ) $\mathbf{A}$ , which by Theorem 0.9.3 (rows version) is a combination of the rows of  $\mathbf{A}$ . So every row of  $\mathbf{B}$  lies in  $\text{row}(\mathbf{A})$ , and  $\text{row}(\mathbf{B}) \subseteq \text{row}(\mathbf{A})$  by Proposition 1.3.4 (1). Since  $\mathbf{A} = \mathbf{E}^{-1}\mathbf{B}$  with  $\mathbf{E}^{-1}$  invertible, the same argument gives  $\text{row}(\mathbf{A}) \subseteq \text{row}(\mathbf{B})$ .
- (b) The equality  $\text{nul}(\mathbf{B}) = \text{nul}(\mathbf{A})$  is Lemma 2.3.1. By Theorem 0.9.3,  $c_1\mathbf{a}_1 + \cdots + c_n\mathbf{a}_n = \mathbf{A}\mathbf{c}$  and  $c_1\mathbf{b}_1 + \cdots + c_n\mathbf{b}_n = \mathbf{B}\mathbf{c}$  with  $\mathbf{c} = (c_1, \dots, c_n)$ , so the second statement is the first one written out (this is Lemma 2.3.2).
- (c) Let  $\mathbf{A} = \begin{pmatrix} 1 & 1 \\ 1 & 1 \end{pmatrix} \in M_2(\mathbb{R})$  and  $\mathbf{B} = \begin{pmatrix} 1 & 1 \\ 0 & 0 \end{pmatrix}$ , obtained by  $R_2 \rightarrow R_2 - R_1$ . Then  $\text{col}(\mathbf{A}) = \text{span}((1, 1))$  and  $\text{col}(\mathbf{B}) = \text{span}((1, 0))$ . These lines differ, since  $(1, 0)$  is not a multiple of  $(1, 1)$ .

■

So row reduction keeps the row space and the **relations among the columns**, but it moves the columns themselves.

*Warning. Do not read  $\text{col}(\mathbf{A})$  off the reduced form.* In (c) above, the RREF of  $\mathbf{A}$  is  $\mathbf{B}$ , whose column space is the  $x$ -axis, while  $\text{col}(\mathbf{A})$  is the line  $y = x$ . What survives row reduction is the *pattern* of dependence: in both matrices column 2 equals column 1. So we use the RREF to decide **which** columns to take, and then take those columns from  $\mathbf{A}$ .

## 2.5.4 Bases from the reduced form

Fix the notation for the rest of the section. Let  $\mathbf{R} = (r_{kj})$  be the reduced row echelon form of  $\mathbf{A}$  (Theorem 2.2.3, Theorem 2.3.4), with columns  $\mathbf{r}_1, \dots, \mathbf{r}_n \in F^m$ . Let  $\mathbf{R}$  have  $r$  non-zero rows, and let the pivot of row  $k$  lie in column  $j_k$ , so that  $j_1 < j_2 < \cdots < j_r$ . These are the **pivot columns** of  $\mathbf{A}$ , well defined by Corollary 2.3.5. The remaining columns are the **free columns**. We use three facts from Definition 2.2.4:

- rows  $r + 1, \dots, m$  of  $\mathbf{R}$  are zero;
- column  $j_k$  of  $\mathbf{R}$  is  $\mathbf{e}_k$  (the pivot is 1 and the only non-zero entry of its column);
- in row  $k$ , every entry to the left of the pivot is 0.

From the last two, if column  $j$  of  $\mathbf{R}$  has exactly  $t$  pivot columns to its left, then **every entry of  $\mathbf{r}_j$  in rows  $t + 1, \dots, m$  is zero, except the pivot entry of row  $t + 1$  when  $j = j_{t+1}$** . Indeed, for  $k \geq t + 1$ , row  $k$  is either zero or has its pivot in column  $j_k \geq j$ , and entries to the left of a pivot vanish. In the running example,

$$\mathbf{R} = \begin{pmatrix} 1 & 2 & 0 & 0 & -1 \\ 0 & 0 & 1 & 0 & 2 \\ 0 & 0 & 0 & 1 & 1 \\ 0 & 0 & 0 & 0 & 0 \end{pmatrix},$$

with  $r = 3$ , pivot columns 1, 3, 4 and free columns 2, 5 (the reduction is carried out in Example

(Bases for the running example) below).

The column space comes first. In  $\mathbf{R}$  the answer is visible: the pivot columns are  $\mathbf{e}_1, \dots, \mathbf{e}_r$ , and every column has zeros below row  $r$ . Row reduction does not preserve columns, but by [Theorem 2.5.1](#) (b) it preserves every linear relation among them, and “is a basis” and “equals this combination” are statements about relations.

### Theorem 2.5.2 — Pivot columns form a basis of the column space

Let  $\mathbf{A} \in M_{m \times n}(F)$  have pivot columns  $j_1 < \dots < j_r$ , and let  $\mathbf{R} = (r_{kj})$  be its RREF. Then:

- (a) the pivot columns of  $\mathbf{A}$ ,  $(\mathbf{a}_{j_1}, \dots, \mathbf{a}_{j_r})$ , form a basis of  $\text{col}(\mathbf{A})$ ;
- (b) if column  $j$  is free and  $t$  pivot columns lie to its left, then  $\mathbf{a}_j = r_{1j}\mathbf{a}_{j_1} + \dots + r_{tj}\mathbf{a}_{j_t}$ ;
- (c) the pivot columns of  $\mathbf{A}$  are exactly the vectors kept when  $(\mathbf{a}_1, \dots, \mathbf{a}_n)$  is sifted as in [Theorem 1.5.4](#).

*Idea.* Both claims are statements about linear relations among columns, so by [Theorem 2.5.1](#) (b) it suffices to prove them for  $\mathbf{R}$ , where they can be seen by inspection. The pivot columns of  $\mathbf{R}$  are  $\mathbf{e}_1, \dots, \mathbf{e}_r$ , which are independent. A free column of  $\mathbf{R}$  is  $\sum_k r_{kj}\mathbf{e}_k$ , a combination of the pivot columns to its left with its own entries as coefficients. So we transfer two relations back to  $\mathbf{A}$ : “ $\sum c_k \mathbf{a}_{j_k} = \mathbf{0}$  forces  $c_1 = \dots = c_r = 0$ ” and “ $\mathbf{a}_j - \sum_k r_{kj}\mathbf{a}_{j_k} = \mathbf{0}$ ”. Part (c) is a rereading of [Corollary 2.3.5](#).

*Proof.* Throughout, we use [Theorem 2.5.1](#) (b) in the form: for scalars  $c_1, \dots, c_n$ ,

$$c_1 \mathbf{a}_1 + \dots + c_n \mathbf{a}_n = \mathbf{0} \iff c_1 \mathbf{r}_1 + \dots + c_n \mathbf{r}_n = \mathbf{0}. \quad (\dagger)$$

- (b) Let column  $j$  be free, with  $t$  pivot columns  $j_1, \dots, j_t$  to its left. By the fact in bold before the theorem, the entries of  $\mathbf{r}_j$  in rows  $t+1, \dots, m$  are zero, so  $\mathbf{r}_j = r_{1j}\mathbf{e}_1 + \dots + r_{tj}\mathbf{e}_t = r_{1j}\mathbf{r}_{j_1} + \dots + r_{tj}\mathbf{r}_{j_t}$ , since  $\mathbf{r}_{j_k} = \mathbf{e}_k$ . This is a linear relation among the columns of  $\mathbf{R}$  with coefficient 1 on  $\mathbf{r}_j$ ,  $-r_{kj}$  on  $\mathbf{r}_{j_k}$ , and 0 elsewhere. By  $(\dagger)$ , the columns of  $\mathbf{A}$  satisfy the same relation, which is (b).
- (c) *Spanning.* Each pivot column  $\mathbf{a}_{j_k}$  lies in  $\text{span}(\mathbf{a}_{j_1}, \dots, \mathbf{a}_{j_r})$ , and each free column does by (b). Hence  $\text{col}(\mathbf{A}) = \text{span}(\mathbf{a}_1, \dots, \mathbf{a}_n) \subseteq \text{span}(\mathbf{a}_{j_1}, \dots, \mathbf{a}_{j_r})$  by [Proposition 1.3.4](#) (1). The reverse inclusion holds by [Proposition 1.3.4](#) (2), since the pivot columns are among the columns.

*Independence.* Let  $c_1 \mathbf{a}_{j_1} + \dots + c_r \mathbf{a}_{j_r} = \mathbf{0}$ . This is a relation among all the columns of  $\mathbf{A}$  with coefficient 0 on the free columns. By  $(\dagger)$ ,  $c_1 \mathbf{r}_{j_1} + \dots + c_r \mathbf{r}_{j_r} = c_1 \mathbf{e}_1 + \dots + c_r \mathbf{e}_r = \mathbf{0}$ . The  $k$ -th entry of the left side is  $c_k$ , so  $c_1 = \dots = c_r = 0$ . Hence the pivot columns of  $\mathbf{A}$  are linearly independent, and with spanning they form a basis of  $\text{col}(\mathbf{A})$  by [Definition 1.5.1](#).

- (c) By [Corollary 2.3.5](#) (a), column  $j$  is a pivot column if and only if  $\mathbf{a}_j \notin \text{span}(\mathbf{a}_1, \dots, \mathbf{a}_{j-1})$ . Sifting keeps  $\mathbf{a}_j$  exactly when  $\mathbf{a}_j$  is not in the span of the vectors kept before it, and by [Theorem 1.5.4](#) (applied to  $(\mathbf{a}_1, \dots, \mathbf{a}_{j-1})$ ) that span is  $\text{span}(\mathbf{a}_1, \dots, \mathbf{a}_{j-1})$ . Hence the kept vectors are exactly the pivot columns. This proves the theorem. ■



This is sifting ([Theorem 1.5.4](#)) done by elimination. In Chapter 1, each step “is  $\mathbf{v}_j$  in the span of the vectors kept so far?” was a separate small system. One row reduction answers all of them at once, and part (b) even gives the coefficients.

The row space and the null space are read off  $\mathbf{R}$  directly.

### Theorem 2.5.3 — Basis of the row space

The non-zero rows of the RREF  $\mathbf{R}$  of  $\mathbf{A}$  form a basis of  $\text{row}(\mathbf{A})$ .

*Proof.* By [Theorem 2.5.1](#) (a),  $\text{row}(\mathbf{A}) = \text{row}(\mathbf{R})$ . The zero rows of  $\mathbf{R}$  contribute nothing to a combination, so the  $r$  non-zero rows span  $\text{row}(\mathbf{R})$ . For independence, let  $c_1(\text{row } 1) + \cdots + c_r(\text{row } r) = 0$ . Look at the entry in column  $j_k$ : since column  $j_k$  of  $\mathbf{R}$  is  $\mathbf{e}_k$ , only row  $k$  has a non-zero entry there, equal to 1. So that entry of the combination is  $c_k$ , and  $c_k = 0$  for each  $k$ . Hence the non-zero rows are a basis of  $\text{row}(\mathbf{A})$ . ■

For the null space, each free variable gets one basis vector: set that free variable to 1, the other free variables to 0, and solve for the pivot variables.

### Theorem 2.5.4 — Basis of the null space

Let  $\mathbf{A} \in M_{m \times n}(F)$  have RREF  $\mathbf{R} = (r_{kj})$  with pivot columns  $j_1 < \cdots < j_r$ . For each free column  $f$ , define  $\mathbf{s}_f \in F^n$  by

$$\begin{aligned} (\mathbf{s}_f)_f &= 1, \\ (\mathbf{s}_f)_{f'} &= 0 \text{ for every other free } f', \\ (\mathbf{s}_f)_{j_k} &= -r_{kf} \text{ for } k = 1, \dots, r. \end{aligned}$$

Then the vectors  $\mathbf{s}_f$ , one for each free column  $f$ , form a basis of  $\text{nul}(\mathbf{A})$ . In particular  $\text{nullity } \mathbf{A} = n - r$ , the number of free columns.

*Proof.* By [Theorem 2.5.1](#) (b),  $\text{nul}(\mathbf{A}) = \text{nul}(\mathbf{R})$ . For  $k \leq r$ , row  $k$  of  $\mathbf{R}$  has entry 1 in column  $j_k$ , entry 0 in every other pivot column (since column  $j_{k'}$  is  $\mathbf{e}_{k'}$ ), and entries  $r_{kf}$  in the free columns. Rows  $k > r$  are zero. Hence  $\mathbf{R}\mathbf{x} = \mathbf{0}$  is equivalent to the  $r$  equations

$$x_{j_k} = - \sum_{f \text{ free}} r_{kf} x_f \quad (k = 1, \dots, r). \quad (\ddagger)$$

*Spanning.* Let  $\mathbf{x} \in \text{nul}(\mathbf{A})$ , and put  $\mathbf{y} = \sum_{f \text{ free}} x_f \mathbf{s}_f$ . For a free column  $f$ , the  $f$ -th entry of  $\mathbf{y}$  is  $x_f$ , since  $\mathbf{s}_f$  has a 1 there and the other  $\mathbf{s}_{f'}$  have 0. For a pivot column, the  $j_k$ -th entry of  $\mathbf{y}$  is  $\sum_f x_f (-r_{kf})$ , which equals  $x_{j_k}$  by  $(\ddagger)$ . So  $\mathbf{x} = \mathbf{y}$  lies in the span of the  $\mathbf{s}_f$ . Each  $\mathbf{s}_f$  itself satisfies  $(\ddagger)$ , so it lies in  $\text{nul}(\mathbf{A})$ .

*Independence.* If  $\sum_f c_f \mathbf{s}_f = \mathbf{0}$ , then the  $f$ -th entry of the left side is  $c_f$ , so every  $c_f = 0$ .

Hence the  $\mathbf{s}_f$  form a basis of  $\text{nul}(\mathbf{A})$ , and there are  $n - r$  of them. ■

**Example. (Bases for the running example)**

Find bases of  $\text{col}(\mathbf{A})$ ,  $\text{row}(\mathbf{A})$  and  $\text{nul}(\mathbf{A})$ , and the rank and nullity, for

$$\mathbf{A} = \begin{pmatrix} 1 & 2 & 0 & 0 & -1 \\ 2 & 4 & 1 & 0 & 0 \\ 1 & 2 & 1 & 1 & 2 \\ 0 & 0 & 1 & 1 & 3 \end{pmatrix} \in M_{4 \times 5}(\mathbb{R}).$$

*Solution.* Row reduce:

$$\begin{aligned} \mathbf{A} &\xrightarrow[\substack{R_3 \rightarrow R_3 - R_1}]{R_2 \rightarrow R_2 - 2R_1} \begin{pmatrix} 1 & 2 & 0 & 0 & -1 \\ 0 & 0 & 1 & 0 & 2 \\ 0 & 0 & 1 & 1 & 3 \\ 0 & 0 & 1 & 1 & 3 \end{pmatrix} \xrightarrow[\substack{R_4 \rightarrow R_4 - R_2}]{R_3 \rightarrow R_3 - R_2} \begin{pmatrix} 1 & 2 & 0 & 0 & -1 \\ 0 & 0 & 1 & 0 & 2 \\ 0 & 0 & 0 & 1 & 1 \\ 0 & 0 & 0 & 1 & 1 \end{pmatrix} \\ &\xrightarrow{R_4 \rightarrow R_4 - R_3} \mathbf{R} = \begin{pmatrix} 1 & 2 & 0 & 0 & -1 \\ 0 & 0 & 1 & 0 & 2 \\ 0 & 0 & 0 & 1 & 1 \\ 0 & 0 & 0 & 0 & 0 \end{pmatrix}. \end{aligned}$$

The pivot columns are 1, 3, 4, and the free columns are 2, 5.

**Column space.** By [Theorem 2.5.2](#) (a), the pivot columns of  $\mathbf{A}$ ,

$$\mathbf{a}_1 = (1, 2, 1, 0), \quad \mathbf{a}_3 = (0, 1, 1, 1), \quad \mathbf{a}_4 = (0, 0, 1, 1),$$

form a basis of  $\text{col}(\mathbf{A})$ , so  $\text{rank } \mathbf{A} = 3$ . By part (b), the free columns are  $\mathbf{a}_2 = 2\mathbf{a}_1$  (from column 2 of  $\mathbf{R}$ ) and  $\mathbf{a}_5 = -\mathbf{a}_1 + 2\mathbf{a}_3 + \mathbf{a}_4$  (from column 5). Check:  $-(1, 2, 1, 0) + 2(0, 1, 1, 1) + (0, 0, 1, 1) = (-1, 0, 2, 3) = \mathbf{a}_5$ . Note that  $\text{col}(\mathbf{A}) \neq \text{col}(\mathbf{R})$ : every column of  $\mathbf{R}$  has last entry 0, but  $\mathbf{a}_3$  does not.

**Row space.** By [Theorem 2.5.3](#), a basis is

$$(1 \ 2 \ 0 \ 0 \ -1), \quad (0 \ 0 \ 1 \ 0 \ 2), \quad (0 \ 0 \ 0 \ 1 \ 1),$$

so  $\dim \text{row}(\mathbf{A}) = 3$ .

**Null space.** The equations  $(\dagger)$  read  $x_1 = -2x_2 + x_5$ ,  $x_3 = -2x_5$ ,  $x_4 = -x_5$ . By [Theorem 2.5.4](#), a basis is

$$\mathbf{s}_2 = (-2, 1, 0, 0, 0), \quad \mathbf{s}_5 = (1, 0, -2, -1, 1),$$

so  $\text{nullity } \mathbf{A} = 2$ . Check:  $\mathbf{A}\mathbf{s}_2 = -2\mathbf{a}_1 + \mathbf{a}_2 = \mathbf{0}$  and  $\mathbf{A}\mathbf{s}_5 = \mathbf{a}_1 - 2\mathbf{a}_3 - \mathbf{a}_4 + \mathbf{a}_5 = \mathbf{0}$ , which are the two relations found above.

So  $\text{rank } \mathbf{A} = \dim \text{row}(\mathbf{A}) = 3$  and  $\text{rank } \mathbf{A} + \text{nullity } \mathbf{A} = 5$ , the number of columns.

*Quick check.* Without computing anything, what is the largest possible rank of a  $3 \times 5$  matrix, and the smallest possible nullity?

*Solution.*  $\text{rank } \mathbf{A} = \dim \text{col}(\mathbf{A}) \leq 3$ , since  $\text{col}(\mathbf{A}) \subseteq F^3$ . By [Theorem 2.5.4](#), the nullity is the number of free columns,  $5 - r$ , where  $r \leq 3$  is the number of pivots (one per non-zero row). So the nullity is at least 2. For example  $(\mathbf{I}_3 \ 0_{3 \times 2})$  attains rank 3 and nullity 2.

## 2.5.5 Row rank equals column rank

The example came out with  $\dim \text{row}(\mathbf{A}) = \dim \text{col}(\mathbf{A})$ . This is not luck, and it is surprising: the row space and the column space live in different spaces, and the rows and columns of a  $4 \times 5$  matrix have nothing obvious in common. Counting pivots explains it.

### Theorem 2.5.5 — Row rank equals column rank

Let  $\mathbf{A} \in M_{m \times n}(F)$ , and let  $r$  be the number of pivots of its RREF. Then

$$\dim \text{row}(\mathbf{A}) = \dim \text{col}(\mathbf{A}) = r.$$

Consequently  $\text{rank } \mathbf{A} = \text{rank } \mathbf{A}^\top$ .

*Proof.* By Theorem 2.5.2 (a),  $\text{col}(\mathbf{A})$  has a basis consisting of the  $r$  pivot columns, so  $\dim \text{col}(\mathbf{A}) = r$ . By Theorem 2.5.3,  $\text{row}(\mathbf{A})$  has a basis consisting of the non-zero rows of the RREF. Each non-zero row contains exactly one pivot and each pivot lies in a non-zero row, so there are  $r$  of them, and  $\dim \text{row}(\mathbf{A}) = r$ . Finally, by (\*) at the start of this section,  $\text{rank } \mathbf{A}^\top = \dim \text{col}(\mathbf{A}^\top) = \dim \text{row}(\mathbf{A}) = r = \text{rank } \mathbf{A}$ . ■

The theorem makes the rank independent of **which** echelon form we compute. Gaussian elimination without the final clearing-up phase produces an echelon form that is not reduced, and different sequences of operations produce different echelon forms. They all give the same count.

### Corollary 2.5.6 — Rank from any echelon form

Let  $\mathbf{A} \in M_{m \times n}(F)$ , and let  $\mathbf{G}$  be a matrix in row echelon form that is row equivalent to  $\mathbf{A}$ . Then the number of non-zero rows of  $\mathbf{G}$  is  $\text{rank } \mathbf{A}$ , and the columns of  $\mathbf{G}$  containing its pivots are exactly the pivot columns of  $\mathbf{A}$ .

*Proof.* By Corollary 2.3.5 (b), the pivot columns of  $\mathbf{G}$  are the pivot columns of  $\mathbf{A}$ . Each non-zero row of  $\mathbf{G}$  has exactly one pivot, in a different column (Definition 2.2.3), so the number of non-zero rows of  $\mathbf{G}$  equals the number of pivot columns of  $\mathbf{A}$ , which is  $\text{rank } \mathbf{A}$  by Theorem 2.5.5. ■

*Warning.* Rank is a property of  $\mathbf{A}$ , not of a computation, but it must be counted from a row echelon form. The number of non-zero rows of  $\mathbf{A}$  itself is not the rank:  $\begin{pmatrix} 1 & 2 \\ 2 & 4 \end{pmatrix}$  has two non-zero rows and rank 1. Count non-zero rows only after reaching an echelon form. And the rank depends on the field when the entries are read in different fields:  $\begin{pmatrix} 1 & 1 \\ 1 & -1 \end{pmatrix}$  has rank 2 over  $\mathbb{R}$ , but over  $\mathbb{F}_2$ , where  $-1 = 1$ , both rows are  $(1 \ 1)$  and the rank is 1.

## 2.5.6 Rank–nullity and the Rouché–Capelli theorem

The trade-off between the zero matrix and the identity was exact, and now we can say why.

### Theorem 2.5.7 — Rank–Nullity for Matrices

For every  $\mathbf{A} \in M_{m \times n}(F)$ ,

$$\text{rank } \mathbf{A} + \text{nullity } \mathbf{A} = n,$$

the number of **columns** of  $\mathbf{A}$ .

*Proof.* Let  $r$  be the number of pivot columns of  $\mathbf{A}$ . By Theorem 2.5.5,  $\text{rank } \mathbf{A} = r$ . By Theorem 2.5.4,  $\text{nullity } \mathbf{A} = n - r$ . Adding gives  $n$ . ■

Each column of  $\mathbf{A}$  is either a pivot column, contributing a direction to  $\text{col}(\mathbf{A})$ , or a free column, contributing a direction to  $\text{nul}(\mathbf{A})$ . Nothing is lost and nothing is counted twice. In Chapter 3 the same equation becomes the Rank–Nullity Theorem for linear maps, with a proof that uses no matrices at all.

Now the question this chapter started with. When does  $\mathbf{Ax} = \mathbf{b}$  have a solution, and how many solutions are there? The first answer was  $\mathbf{b} \in \text{col}(\mathbf{A})$  (Theorem 2.1.1); here is a version that is a pure count.

### Theorem 2.5.8 — Rouché–Capelli Theorem

Let  $\mathbf{A} \in M_{m \times n}(F)$ ,  $\mathbf{b} \in F^m$ , and let  $[\mathbf{A} \mid \mathbf{b}] \in M_{m \times (n+1)}(F)$  be the augmented matrix.

- (a) The system  $\mathbf{Ax} = \mathbf{b}$  is consistent if and only if  $\text{rank } \mathbf{A} = \text{rank } [\mathbf{A} \mid \mathbf{b}]$ .
- (b) Suppose it is consistent, let  $\mathbf{p}$  be one solution, let  $r = \text{rank } \mathbf{A}$ , and let  $\mathbf{s}_1, \dots, \mathbf{s}_{n-r}$  be a basis of  $\text{nul}(\mathbf{A})$ . Then every solution can be written as

$$\mathbf{x} = \mathbf{p} + t_1 \mathbf{s}_1 + \dots + t_{n-r} \mathbf{s}_{n-r}$$

for **exactly one** choice of  $(t_1, \dots, t_{n-r}) \in F^{n-r}$ . In words: the solutions form an  $(n - r)$ -parameter family.

- (c) If moreover  $F$  is a finite field with  $q$  elements (for instance  $\mathbb{F}_p$ , with  $q = p$ ), a consistent system has exactly  $q^{n - \text{rank } \mathbf{A}}$  solutions.

*Idea.* For (a), adding the column  $\mathbf{b}$  can only enlarge the column space, and the dimension goes up exactly when  $\mathbf{b}$  is new, that is, when  $\mathbf{b} \notin \text{col}(\mathbf{A})$ . For (b), the structure theorem says solutions are  $\mathbf{p}$  plus null vectors, and null vectors have unique coordinates in a basis. For (c), unique coordinates mean the solutions are in bijection with  $F^{n-r}$ , which has  $q^{n-r}$  elements.

*Proof.*

- (a) By [Definition 2.5.1](#),  $\text{col}([\mathbf{A} \mid \mathbf{b}]) = \text{span}(\mathbf{a}_1, \dots, \mathbf{a}_n, \mathbf{b})$ , which contains  $\text{col}(\mathbf{A})$  by [Proposition 1.3.4](#) (2).  
 $(\Rightarrow)$  Suppose the system is consistent. By [Theorem 2.1.1](#),  $\mathbf{b} \in \text{span}(\mathbf{a}_1, \dots, \mathbf{a}_n)$ , so by [Theorem 1.3.5](#)  $\text{col}([\mathbf{A} \mid \mathbf{b}]) = \text{col}(\mathbf{A})$ , and the ranks are equal.  
 $(\Leftarrow)$  Suppose  $\text{rank } \mathbf{A} = \text{rank } [\mathbf{A} \mid \mathbf{b}]$ . Then  $\text{col}(\mathbf{A})$  is a subspace of  $\text{col}([\mathbf{A} \mid \mathbf{b}])$  of the same dimension, so  $\text{col}(\mathbf{A}) = \text{col}([\mathbf{A} \mid \mathbf{b}])$  by [Theorem 1.6.8](#). In particular  $\mathbf{b} \in \text{col}(\mathbf{A})$ , and the system is consistent by [Theorem 2.1.1](#).
- (b) By [Theorem 2.5.7](#), nullity  $\mathbf{A} = n - r$ , so a basis  $\mathbf{s}_1, \dots, \mathbf{s}_{n-r}$  of  $\text{nul}(\mathbf{A})$  exists ([Theorem 2.5.4](#)). By [Theorem 2.1.3](#),  $\mathbf{x}$  is a solution if and only if  $\mathbf{x} - \mathbf{p} \in \text{nul}(\mathbf{A})$ . By [Theorem 1.5.1](#), applied in the vector space  $\text{nul}(\mathbf{A})$  with this basis, each  $\mathbf{x} - \mathbf{p} \in \text{nul}(\mathbf{A})$  equals  $t_1\mathbf{s}_1 + \dots + t_{n-r}\mathbf{s}_{n-r}$  for exactly one  $(t_1, \dots, t_{n-r})$ . Conversely every such expression is in  $\text{nul}(\mathbf{A})$ , so  $\mathbf{p}$  plus it is a solution. This proves (b).
- (c) By (b), the rule  $(t_1, \dots, t_{n-r}) \mapsto \mathbf{p} + t_1\mathbf{s}_1 + \dots + t_{n-r}\mathbf{s}_{n-r}$  is a bijection from  $F^{n-r}$  onto the solution set: every solution is hit, and by exactly one tuple. A tuple in  $F^{n-r}$  is a choice of  $n-r$  entries, each one of  $q$  elements, so  $F^{n-r}$  has  $q^{n-r}$  elements. (When  $r = n$  there is exactly one tuple, the empty one, and the basis of  $\text{nul}(\mathbf{A})$  is empty, so  $\mathbf{p}$  is the only solution and  $q^0 = 1$ .) Hence the solution set has  $q^{n-r}$  elements. ■

Over an infinite field such as  $\mathbb{Q}$ ,  $\mathbb{R}$  or  $\mathbb{C}$ , part (b) gives the familiar trichotomy: no solution, exactly one solution (when  $\text{rank } \mathbf{A} = n$ ), or infinitely many (when  $\text{rank } \mathbf{A} < n$ , since distinct values of  $t_1$  give distinct solutions). Over a finite field “infinitely many” is replaced by an exact power of  $q$ . For example, over  $\mathbb{F}_2$  a system can never have exactly 3 or 6 solutions.

**Example. (Counting solutions over  $\mathbb{F}_3$ )**

Over  $\mathbb{F}_3$ , consider

$$\mathbf{A} = \begin{pmatrix} 1 & 1 & 0 & 2 \\ 1 & 2 & 1 & 0 \\ 2 & 0 & 1 & 2 \end{pmatrix}, \quad \mathbf{b} = \begin{pmatrix} 1 \\ 0 \\ 1 \end{pmatrix}, \quad \mathbf{b}' = \begin{pmatrix} 1 \\ 0 \\ 0 \end{pmatrix}.$$

Decide whether  $\mathbf{Ax} = \mathbf{b}$  and  $\mathbf{Ax} = \mathbf{b}'$  are consistent, and count their solutions.

*Solution.* We reduce both augmented matrices at once, carrying two right-hand columns; arithmetic is modulo 3, so  $-1 = 2$  and  $-2 = 1$ .

$$\left[ \begin{array}{cccc|cc} 1 & 1 & 0 & 2 & 1 & 1 \\ 1 & 2 & 1 & 0 & 0 & 0 \\ 2 & 0 & 1 & 2 & 1 & 0 \end{array} \right] \xrightarrow[\substack{R_2 \rightarrow R_2 - R_1 \\ R_3 \rightarrow R_3 - 2R_1}]{\substack{R_2 \rightarrow R_2 - R_1 \\ R_3 \rightarrow R_3 - 2R_1}} \left[ \begin{array}{cccc|cc} 1 & 1 & 0 & 2 & 1 & 1 \\ 0 & 1 & 1 & 1 & 2 & 2 \\ 0 & 1 & 1 & 1 & 2 & 1 \end{array} \right]$$

$$\xrightarrow[\substack{R_3 \rightarrow R_3 - R_2 \\ R_1 \rightarrow R_1 - R_2}]{\substack{R_3 \rightarrow R_3 - R_2 \\ R_1 \rightarrow R_1 - R_2}} \left[ \begin{array}{cccc|cc} 1 & 0 & 2 & 1 & 2 & 2 \\ 0 & 1 & 1 & 1 & 2 & 2 \\ 0 & 0 & 0 & 0 & 0 & 2 \end{array} \right].$$

Row operations act on each column separately, so deleting either right-hand column gives a reduction of the corresponding augmented matrix. For instance, in row 2,  $(1, 2, 1, 0) - (1, 1, 0, 2) = (0, 1, 1, -2) = (0, 1, 1, 1)$ , and in row 3,  $(2, 0, 1, 2) - 2(1, 1, 0, 2) = (0, -2, 1, -2) = (0, 1, 1, 1)$ .

The left block is an echelon form of  $\mathbf{A}$  with two non-zero rows, so  $\text{rank } \mathbf{A} = 2$  by [Corollary 2.5.6](#). With  $\mathbf{b}$ , the augmented matrix  $[\mathbf{A} \mid \mathbf{b}]$  has the same two non-zero rows (the last is zero), so  $\text{rank } [\mathbf{A} \mid \mathbf{b}] = 2 = \text{rank } \mathbf{A}$ , and by [Theorem 2.5.8](#) (a) the system is consistent. By part (c) it has exactly  $3^{4-2} = 9$  solutions. Explicitly, the reduced system is  $x_1 = 2 - 2x_3 - x_4 = 2 + x_3 + 2x_4$  and  $x_2 = 2 - x_3 - x_4 = 2 + 2x_3 + 2x_4$ , with  $x_3, x_4 \in \mathbb{F}_3$  free: nine choices.

With  $\mathbf{b}'$ , the last row  $(0, 0, 0, 0 \mid 2)$  is a non-zero row with its pivot in the last column, so  $[\mathbf{A} \mid \mathbf{b}']$  has three non-zero rows in echelon form and  $\text{rank } [\mathbf{A} \mid \mathbf{b}'] = 3 \neq 2$ . By [Theorem 2.5.8](#) (a),  $\mathbf{A}\mathbf{x} = \mathbf{b}'$  has no solution. Indeed, row 3 of  $\mathbf{A}$  is the sum of rows 1 and 2, so any solution would need  $b'_3 = b'_1 + b'_2 = 1$ , but  $b'_3 = 0$ .

## 2.5.7 Rank of a product

Multiplying matrices cannot create information: every column of  $\mathbf{AB}$  is a combination of the columns of  $\mathbf{A}$ , and every row is a combination of the rows of  $\mathbf{B}$ . The rank records this.

### Theorem 2.5.9 — Rank of a product

Let  $\mathbf{A} \in M_{m \times n}(F)$  and  $\mathbf{B} \in M_{n \times p}(F)$ . Then

$$\text{rank}(\mathbf{AB}) \leq \min(\text{rank } \mathbf{A}, \text{rank } \mathbf{B}).$$

Moreover, if  $\mathbf{P} \in M_m(F)$  and  $\mathbf{Q} \in M_n(F)$  are invertible, then  $\text{rank}(\mathbf{PA}) = \text{rank } \mathbf{A} = \text{rank}(\mathbf{AQ})$ .

*Proof.* Let  $\mathbf{b}_1, \dots, \mathbf{b}_p \in F^n$  be the columns of  $\mathbf{B}$ . By [Theorem 0.9.2](#), the columns of  $\mathbf{AB}$  are  $\mathbf{Ab}_1, \dots, \mathbf{Ab}_p$ , which lie in  $\text{col}(\mathbf{A})$  by [Definition 2.5.1](#). By [Proposition 1.3.4](#) (1),  $\text{col}(\mathbf{AB}) \subseteq \text{col}(\mathbf{A})$ , so  $\text{rank}(\mathbf{AB}) \leq \text{rank } \mathbf{A}$  by [Theorem 1.6.7](#). Similarly, row  $i$  of  $\mathbf{AB}$  is  $(\text{row } i \text{ of } \mathbf{A})\mathbf{B}$ , a combination of the rows of  $\mathbf{B}$  by [Theorem 0.9.3](#) (rows version), so  $\text{row}(\mathbf{AB}) \subseteq \text{row}(\mathbf{B})$ . By [Theorem 2.5.5](#),  $\text{rank}(\mathbf{AB}) = \dim \text{row}(\mathbf{AB}) \leq \dim \text{row}(\mathbf{B}) = \text{rank } \mathbf{B}$ .

For the second statement, apply the inequality twice:  $\text{rank } \mathbf{A} = \text{rank}(\mathbf{P}^{-1}(\mathbf{PA})) \leq \text{rank}(\mathbf{PA}) \leq \text{rank } \mathbf{A}$ , so equality holds throughout. The same sandwich with  $\mathbf{A} = (\mathbf{AQ})\mathbf{Q}^{-1}$  gives  $\text{rank}(\mathbf{AQ}) = \text{rank } \mathbf{A}$ . ■

The equality case recovers something we already knew in a new form: row operations (left multiplication by an invertible  $\mathbf{P}$ ) do not change the rank. The inequality is often strict. For

$\mathbf{A} = \begin{pmatrix} 0 & 1 \\ 0 & 0 \end{pmatrix}$ , both factors of  $\mathbf{A} \cdot \mathbf{A}$  have rank 1, but  $\mathbf{A}^2 = \mathbf{0}$  has rank 0.

## 2.5.8 A toolkit for questions from Chapter 1

In Chapter 1 we asked, again and again, whether some vectors in  $F^m$  are independent, whether they span, how to extend them to a basis, and what the sum and intersection of two subspaces are. Each was answered by solving a small system by hand. Everything now reduces to one

move: put the vectors as columns of a matrix, row reduce, and read off pivots.

### Theorem 2.5.10 — Independence and spanning by rank

Let  $\mathbf{v}_1, \dots, \mathbf{v}_k \in F^m$ , and let  $\mathbf{A} = (\mathbf{v}_1 \ \cdots \ \mathbf{v}_k) \in M_{m \times k}(F)$ . Then:

- (a)  $(\mathbf{v}_1, \dots, \mathbf{v}_k)$  is linearly independent if and only if  $\text{rank } \mathbf{A} = k$ , that is, every column is a pivot column;
- (b)  $(\mathbf{v}_1, \dots, \mathbf{v}_k)$  spans  $F^m$  if and only if  $\text{rank } \mathbf{A} = m$ , that is, every row of the RREF has a pivot;
- (c)  $(\mathbf{v}_1, \dots, \mathbf{v}_k)$  is a basis of  $F^m$  if and only if  $k = m = \text{rank } \mathbf{A}$ .

*Proof.*

- (a) By Theorem 0.9.3,  $c_1\mathbf{v}_1 + \cdots + c_k\mathbf{v}_k = \mathbf{A}\mathbf{c}$  with  $\mathbf{c} = (c_1, \dots, c_k)$ , so the list is independent if and only if  $\text{nul}(\mathbf{A}) = \{\mathbf{0}\}$ , that is, nullity  $\mathbf{A} = 0$ . By Theorem 2.5.7 this means  $\text{rank } \mathbf{A} = k$ .
- (b) The list spans  $F^m$  if and only if  $\text{col}(\mathbf{A}) = F^m$ . Since  $\text{col}(\mathbf{A}) \subseteq F^m$  and  $\dim F^m = m$ , this holds if and only if  $\text{rank } \mathbf{A} = m$ , by Theorem 1.6.8 for one direction and because equal spaces have equal dimension for the other. The number of pivots is  $\text{rank } \mathbf{A}$  by Theorem 2.5.5, and the RREF has  $m$  rows, each with at most one pivot.
- (c) Combine (a) and (b) with Definition 1.5.1.

■

### Example. (Testing independence)

Decide whether  $\mathbf{v}_1 = (1, 2, 0, 1)$ ,  $\mathbf{v}_2 = (2, 4, 1, 1)$ ,  $\mathbf{v}_3 = (0, 0, 1, -1)$  are linearly independent in  $\mathbb{R}^4$ . If not, find a linear relation.

*Solution.* Row reduce the matrix with these columns:

$$\begin{aligned} \begin{pmatrix} 1 & 2 & 0 \\ 2 & 4 & 0 \\ 0 & 1 & 1 \\ 1 & 1 & -1 \end{pmatrix} &\xrightarrow[\substack{R_2 \rightarrow R_2 - 2R_1 \\ R_4 \rightarrow R_4 - R_1}]{\phantom{R_2 \rightarrow R_2 - 2R_1}} \begin{pmatrix} 1 & 2 & 0 \\ 0 & 0 & 0 \\ 0 & 1 & 1 \\ 0 & -1 & -1 \end{pmatrix} \\ &\xrightarrow{R_2 \leftrightarrow R_3} \begin{pmatrix} 1 & 2 & 0 \\ 0 & 1 & 1 \\ 0 & 0 & 0 \\ 0 & -1 & -1 \end{pmatrix} \\ &\xrightarrow[\substack{R_4 \rightarrow R_4 + R_2 \\ R_1 \rightarrow R_1 - 2R_2}]{\phantom{R_4 \rightarrow R_4 + R_2}} \begin{pmatrix} 1 & 0 & -2 \\ 0 & 1 & 1 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{pmatrix}. \end{aligned}$$

Only columns 1 and 2 are pivot columns, so the rank is  $2 < 3$ , and the vectors are **dependent** by Theorem 2.5.10 (a). Column 3 of the RREF gives the relation, by Theorem 2.5.2 (b):  $\mathbf{v}_3 = -2\mathbf{v}_1 + \mathbf{v}_2$ . Check:  $-2(1, 2, 0, 1) + (2, 4, 1, 1) = (0, 0, 1, -1)$ .

Moreover  $(\mathbf{v}_1, \mathbf{v}_2)$  is a basis of  $\text{span}(\mathbf{v}_1, \mathbf{v}_2, \mathbf{v}_3)$  by [Theorem 2.5.2](#) (a).

To **extend** an independent list to a basis of  $F^m$ , append the standard basis and keep the pivot columns. The appended vectors guarantee that the columns span  $F^m$ . The original vectors all survive, because by [Theorem 2.5.2](#) (c) a column is discarded only if it lies in the span of the columns before it, and an independent list has no such vector ([Lemma 1.5.3](#)).

**Example. (Extending to a basis by elimination)**

Extend the independent list  $(\mathbf{v}_1, \mathbf{v}_2) = ((1, 2, 0, 1), (2, 4, 1, 1))$  from [Example \(Testing independence\)](#) to a basis of  $\mathbb{R}^4$ .

*Solution.* Form  $\mathbf{M} = (\mathbf{v}_1 \ \mathbf{v}_2 \ \mathbf{e}_1 \ \mathbf{e}_2 \ \mathbf{e}_3 \ \mathbf{e}_4)$  and row reduce:

$$\begin{aligned} \begin{pmatrix} 1 & 2 & 1 & 0 & 0 & 0 \\ 2 & 4 & 0 & 1 & 0 & 0 \\ 0 & 1 & 0 & 0 & 1 & 0 \\ 1 & 1 & 0 & 0 & 0 & 1 \end{pmatrix} &\xrightarrow[R_4 \rightarrow R_4 - R_1]{R_2 \rightarrow R_2 - 2R_1} \begin{pmatrix} 1 & 2 & 1 & 0 & 0 & 0 \\ 0 & 0 & -2 & 1 & 0 & 0 \\ 0 & 1 & 0 & 0 & 1 & 0 \\ 0 & -1 & -1 & 0 & 0 & 1 \end{pmatrix} \\ &\xrightarrow[\text{then } R_4 \rightarrow R_4 + R_2]{R_2 \leftrightarrow R_3} \begin{pmatrix} 1 & 2 & 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 & 1 & 0 \\ 0 & 0 & -2 & 1 & 0 & 0 \\ 0 & 0 & -1 & 0 & 1 & 1 \end{pmatrix} \\ &\xrightarrow[\text{then } R_3 \rightarrow -R_3]{R_3 \leftrightarrow R_4} \begin{pmatrix} 1 & 2 & 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 & 1 & 0 \\ 0 & 0 & 1 & 0 & -1 & -1 \\ 0 & 0 & -2 & 1 & 0 & 0 \end{pmatrix} \xrightarrow{R_4 \rightarrow R_4 + 2R_3} \begin{pmatrix} 1 & 2 & 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 & 1 & 0 \\ 0 & 0 & 1 & 0 & -1 & -1 \\ 0 & 0 & 0 & 1 & -2 & -2 \end{pmatrix}. \end{aligned}$$

This is an echelon form with pivots in columns 1, 2, 3, 4. By [Corollary 2.5.6](#) these are the pivot columns of  $\mathbf{M}$ , so by [Theorem 2.5.2](#) (a) the corresponding columns of  $\mathbf{M}$  form a basis of  $\text{col}(\mathbf{M}) = \mathbb{R}^4$ :

$$(\mathbf{v}_1, \mathbf{v}_2, \mathbf{e}_1, \mathbf{e}_2).$$

(Clearing above the pivots gives the RREF, whose column 5 says  $\mathbf{e}_3 = -\mathbf{v}_1 + \mathbf{v}_2 - \mathbf{e}_1 - 2\mathbf{e}_2$ ; check:  $-(1, 2, 0, 1) + (2, 4, 1, 1) - (1, 0, 0, 0) - 2(0, 1, 0, 0) = (0, 0, 1, 0)$ .)

For sums and intersections, one reduction does both jobs.

**Proposition 2.5.11 — Sum and intersection by elimination**

Let  $U, W \subseteq F^m$  be subspaces with bases  $(\mathbf{u}_1, \dots, \mathbf{u}_k)$  and  $(\mathbf{w}_1, \dots, \mathbf{w}_l)$ , and let  $\mathbf{M} = (\mathbf{u}_1 \ \cdots \ \mathbf{u}_k \ \mathbf{w}_1 \ \cdots \ \mathbf{w}_l) \in M_{m \times (k+l)}(F)$ . Then:

- (a) the pivot columns of  $\mathbf{M}$  form a basis of  $U + W$ ;
- (b) for  $\mathbf{x} = (x_1, \dots, x_{k+l}) \in F^{k+l}$  write  $\mathbf{u}_{\mathbf{x}} := x_1\mathbf{u}_1 + \cdots + x_k\mathbf{u}_k$ , the combination of the  $\mathbf{u}$ 's with the first  $k$  entries of  $\mathbf{x}$  as coefficients. If  $(\mathbf{n}_1, \dots, \mathbf{n}_d)$  is a basis of  $\text{nul}(\mathbf{M})$ , then  $(\mathbf{u}_{\mathbf{n}_1}, \dots, \mathbf{u}_{\mathbf{n}_d})$  is a basis of  $U \cap W$ .

In particular  $\dim(U + W) = \text{rank } \mathbf{M}$  and  $\dim(U \cap W) = \text{nullity } \mathbf{M}$ .



*Proof.*

- (a) By [Proposition 1.7.2](#),  $U + W = \text{span}(\mathbf{u}_1, \dots, \mathbf{u}_k, \mathbf{w}_1, \dots, \mathbf{w}_l) = \text{col}(\mathbf{M})$ , and [Theorem 2.5.2](#) (a) applies.
- (b) By [Theorem 0.9.3](#),  $\mathbf{x} \in \text{nul}(\mathbf{M})$  means  $x_1\mathbf{u}_1 + \dots + x_k\mathbf{u}_k = -(x_{k+1}\mathbf{w}_1 + \dots + x_{k+l}\mathbf{w}_l)$ , so  $\mathbf{u}_{\mathbf{x}}$  lies in  $U$  and in  $W$ . Also  $\mathbf{u}_{\mathbf{x}+\mathbf{y}} = \mathbf{u}_{\mathbf{x}} + \mathbf{u}_{\mathbf{y}}$  and  $\mathbf{u}_{c\mathbf{x}} = c\mathbf{u}_{\mathbf{x}}$ , by the vector space axioms in  $F^m$ .

*Spanning.* Let  $\mathbf{v} \in U \cap W$ . Since the  $\mathbf{u}$ 's span  $U$  and the  $\mathbf{w}$ 's span  $W$ ,  $\mathbf{v} = a_1\mathbf{u}_1 + \dots + a_k\mathbf{u}_k = b_1\mathbf{w}_1 + \dots + b_l\mathbf{w}_l$ . Then  $\mathbf{x} = (a_1, \dots, a_k, -b_1, \dots, -b_l) \in \text{nul}(\mathbf{M})$  and  $\mathbf{u}_{\mathbf{x}} = \mathbf{v}$ . Writing  $\mathbf{x} = \sum_i t_i \mathbf{n}_i$  gives  $\mathbf{v} = \sum_i t_i \mathbf{u}_{\mathbf{n}_i}$ .

*Independence.* Let  $\sum_i t_i \mathbf{u}_{\mathbf{n}_i} = \mathbf{0}$ , and put  $\mathbf{x} = \sum_i t_i \mathbf{n}_i \in \text{nul}(\mathbf{M})$ . Then  $\mathbf{u}_{\mathbf{x}} = x_1\mathbf{u}_1 + \dots + x_k\mathbf{u}_k = \mathbf{0}$ , so  $x_1 = \dots = x_k = 0$  because the  $\mathbf{u}$ 's are independent. Since  $\mathbf{M}\mathbf{x} = \mathbf{0}$ , also  $x_{k+1}\mathbf{w}_1 + \dots + x_{k+l}\mathbf{w}_l = \mathbf{0}$ , so the remaining entries vanish because the  $\mathbf{w}$ 's are independent. Hence  $\mathbf{x} = \mathbf{0}$ , and  $t_1 = \dots = t_d = 0$  because the  $\mathbf{n}_i$  are independent.

The dimension statements follow by counting the two bases, using [Theorem 2.5.5](#) for rank  $\mathbf{M}$ . ■

Adding the two counts and using [Theorem 2.5.7](#) gives  $\dim(U + W) + \dim(U \cap W) = \text{rank } \mathbf{M} + \text{nullity } \mathbf{M} = k + l = \dim U + \dim W$ . This is the Dimension Formula for Sums ([Theorem 1.7.3](#)) again, now with a second, computational proof for subspaces of  $F^m$ . (Some texts use the matrix with columns  $\mathbf{u}_1, \dots, \mathbf{u}_k, -\mathbf{w}_1, \dots, -\mathbf{w}_l$  instead of  $\mathbf{M}$ , so that a null vector gives equal combinations  $\sum a_i \mathbf{u}_i = \sum b_j \mathbf{w}_j$  with no sign change. The two null spaces differ only by the signs of the last  $l$  entries.)

#### Example. (A basis of a sum)

Let  $U = \text{span}(\mathbf{u}_1, \mathbf{u}_2)$  and  $W = \text{span}(\mathbf{w}_1, \mathbf{w}_2)$  in  $\mathbb{R}^4$ , where

$$\mathbf{u}_1 = (1, 0, 1, 2), \quad \mathbf{u}_2 = (0, 1, 1, 1), \quad \mathbf{w}_1 = (1, 1, 0, 1), \quad \mathbf{w}_2 = (1, 3, 0, 1).$$

Find a basis and the dimension of  $U + W$ .

*Solution.* Each pair is independent, since neither vector is a multiple of the other (compare the first two entries), so  $\dim U = \dim W = 2$ . Row reduce  $\mathbf{M} = (\mathbf{u}_1 \quad \mathbf{u}_2 \quad \mathbf{w}_1 \quad \mathbf{w}_2)$ :

$$\begin{aligned} \begin{pmatrix} 1 & 0 & 1 & 1 \\ 0 & 1 & 1 & 3 \\ 1 & 1 & 0 & 0 \\ 2 & 1 & 1 & 1 \end{pmatrix} &\xrightarrow[R_4 \rightarrow R_4 - 2R_1]{R_3 \rightarrow R_3 - R_1} \begin{pmatrix} 1 & 0 & 1 & 1 \\ 0 & 1 & 1 & 3 \\ 0 & 1 & -1 & -1 \\ 0 & 1 & -1 & -1 \end{pmatrix} \xrightarrow[R_4 \rightarrow R_4 - R_2]{R_3 \rightarrow R_3 - R_2} \begin{pmatrix} 1 & 0 & 1 & 1 \\ 0 & 1 & 1 & 3 \\ 0 & 0 & -2 & -4 \\ 0 & 0 & -2 & -4 \end{pmatrix} \\ &\xrightarrow[\text{then } R_3 \rightarrow -\frac{1}{2}R_3]{R_4 \rightarrow R_4 - R_3} \begin{pmatrix} 1 & 0 & 1 & 1 \\ 0 & 1 & 1 & 3 \\ 0 & 0 & 1 & 2 \\ 0 & 0 & 0 & 0 \end{pmatrix} \xrightarrow[R_2 \rightarrow R_2 - R_3]{R_1 \rightarrow R_1 - R_3} \mathbf{R} = \begin{pmatrix} 1 & 0 & 0 & -1 \\ 0 & 1 & 0 & 1 \\ 0 & 0 & 1 & 2 \\ 0 & 0 & 0 & 0 \end{pmatrix}. \end{aligned}$$

The pivot columns are 1, 2, 3. By [Proposition 2.5.11](#) (a),  $(\mathbf{u}_1, \mathbf{u}_2, \mathbf{w}_1)$  is a basis of  $U + W$ , and  $\dim(U + W) = 3$ . Column 4 of  $\mathbf{R}$  says  $\mathbf{w}_2 = -\mathbf{u}_1 + \mathbf{u}_2 + 2\mathbf{w}_1$ ; check:  $-(1, 0, 1, 2) +$

$$(0, 1, 1, 1) + 2(1, 1, 0, 1) = (1, 3, 0, 1).$$

**Example. (A basis of an intersection)**

For  $U$  and  $W$  as in [Example \(A basis of a sum\)](#), find a basis of  $U \cap W$ , and check the Dimension Formula for Sums.

*Solution.* We reuse the RREF  $\mathbf{R}$  of  $\mathbf{M}$ . Its only free column is column 4, so by [Theorem 2.5.4](#)  $\text{nul}(\mathbf{M})$  has the basis  $\mathbf{n} = (1, -1, -2, 1)$  (set  $x_4 = 1$ ; then  $x_1 = 1$ ,  $x_2 = -1$ ,  $x_3 = -2$ ). The relation  $\mathbf{Mn} = \mathbf{0}$  reads  $\mathbf{u}_1 - \mathbf{u}_2 - 2\mathbf{w}_1 + \mathbf{w}_2 = \mathbf{0}$ , that is,

$$\mathbf{u}_1 - \mathbf{u}_2 = 2\mathbf{w}_1 - \mathbf{w}_2.$$

By [Proposition 2.5.11](#) (b),  $\mathbf{u}_\mathbf{n} = \mathbf{u}_1 - \mathbf{u}_2 = (1, -1, 0, 1)$  is a basis of  $U \cap W$ . Check that it lies in  $W$ :  $2(1, 1, 0, 1) - (1, 3, 0, 1) = (1, -1, 0, 1)$ . So  $\dim(U \cap W) = 1$ , and

$$\dim U + \dim W - \dim(U \cap W) = 2 + 2 - 1 = 3 = \dim(U + W),$$

in agreement with [Theorem 1.7.3](#).

## 2.5.9 Exercises

### 2.5.9.1 A. Check your understanding

**Exercise A1.**

- Define  $\text{col}(\mathbf{A})$ ,  $\text{row}(\mathbf{A})$ ,  $\text{nul}(\mathbf{A})$  and  $\text{rank } \mathbf{A}$  for  $\mathbf{A} \in M_{m \times n}(F)$ , and say which space each subspace lives in.
- State the Rank–Nullity Theorem for matrices and the Rouché–Capelli Theorem.
- True or false: if  $\mathbf{R}$  is the RREF of  $\mathbf{A}$ , then  $\text{col}(\mathbf{R}) = \text{col}(\mathbf{A})$ . Justify your answer.
- True or false: there is a  $4 \times 6$  real matrix with nullity 1. Justify your answer.
- True or false: for every  $\mathbf{A} \in M_{m \times n}(F)$ ,  $\text{rank } \mathbf{A} = \text{rank } \mathbf{A}^\top$ . Justify your answer.
- Describe how to find a basis of  $U \cap W$  for subspaces  $U, W \subseteq F^m$  given by bases.

*Solution.*

- $\text{col}(\mathbf{A})$  is the span of the columns, a subspace of  $F^m$ ;  $\text{row}(\mathbf{A})$  is the span of the rows, a subspace of  $M_{1 \times n}(F)$ ;  $\text{nul}(\mathbf{A}) = \{\mathbf{x} \in F^n : \mathbf{Ax} = \mathbf{0}\}$ , a subspace of  $F^n$ ; and  $\text{rank } \mathbf{A} = \dim \text{col}(\mathbf{A})$  ([Definition 2.5.1](#), [Definition 2.5.2](#), [Definition 2.5.3](#), [Definition 2.5.4](#)).
- For  $\mathbf{A} \in M_{m \times n}(F)$ ,  $\text{rank } \mathbf{A} + \text{nullity } \mathbf{A} = n$  ([Theorem 2.5.7](#)). For  $\mathbf{b} \in F^m$ ,  $\mathbf{Ax} = \mathbf{b}$  is consistent if and only if  $\text{rank } \mathbf{A} = \text{rank } [\mathbf{A} \mid \mathbf{b}]$ ; then the solutions are  $\mathbf{p} + \text{nul}(\mathbf{A})$ , an  $(n - \text{rank } \mathbf{A})$ -parameter family, with exactly  $q^{n - \text{rank } \mathbf{A}}$  solutions over a field with  $q$  elements ([Theorem 2.5.8](#)).
- False. For  $\mathbf{A} = \begin{pmatrix} 1 & 1 \\ 1 & 1 \end{pmatrix}$ ,  $\mathbf{R} = \begin{pmatrix} 1 & 1 \\ 0 & 0 \end{pmatrix}$ , and  $(1, 1) \in \text{col}(\mathbf{A})$  but  $(1, 1) \notin \text{col}(\mathbf{R}) = \text{span}((1, 0))$ . Row operations preserve the row space and the column relations, not

the column space ([Theorem 2.5.1](#)).

- (d) False. Its rank is at most 4, since the column space lies in  $\mathbb{R}^4$ , so by [Theorem 2.5.7](#) its nullity is  $6 - \text{rank } \mathbf{A} \geq 2$ .
- (e) True. This is [Theorem 2.5.5](#).
- (f) Put the basis vectors of  $U$  and then of  $W$  as the columns of  $\mathbf{M}$ , find a basis of  $\text{nul}(\mathbf{M})$  from the RREF, and for each basis vector  $\mathbf{n}$  take the combination of the  $\mathbf{u}$ 's with the first  $k$  entries of  $\mathbf{n}$  as coefficients ([Proposition 2.5.11](#)).

### 2.5.9.2 B. Practice

#### Exercise B1: The three subspaces of a $3 \times 5$ matrix.

Let

$$\mathbf{A} = \begin{pmatrix} 1 & -1 & 2 & 0 & 3 \\ 2 & -2 & 5 & 1 & 4 \\ -1 & 1 & -1 & 1 & -5 \end{pmatrix} \in M_{3 \times 5}(\mathbb{R}).$$

Find bases of  $\text{col}(\mathbf{A})$ ,  $\text{row}(\mathbf{A})$  and  $\text{nul}(\mathbf{A})$ . Hence verify  $\text{rank } \mathbf{A} + \text{nullity } \mathbf{A} = 5$ .

*Solution.* Row reduce:

$$\mathbf{A} \xrightarrow[\substack{R_2 \rightarrow R_2 - 2R_1 \\ R_3 \rightarrow R_3 + R_1}]{\begin{pmatrix} 1 & -1 & 2 & 0 & 3 \\ 0 & 0 & 1 & 1 & -2 \\ 0 & 0 & 1 & 1 & -2 \end{pmatrix}} \xrightarrow[\substack{R_3 \rightarrow R_3 - R_2 \\ R_1 \rightarrow R_1 - 2R_2}]{\mathbf{R}} = \begin{pmatrix} 1 & -1 & 0 & -2 & 7 \\ 0 & 0 & 1 & 1 & -2 \\ 0 & 0 & 0 & 0 & 0 \end{pmatrix}.$$

The pivot columns are 1 and 3; the free columns are 2, 4, 5.

By [Theorem 2.5.2](#), the pivot columns of  $\mathbf{A}$ ,  $(1, 2, -1)$  and  $(2, 5, -1)$ , form a basis of  $\text{col}(\mathbf{A})$ . By [Theorem 2.5.3](#),  $(1 \ -1 \ 0 \ -2 \ 7)$  and  $(0 \ 0 \ 1 \ 1 \ -2)$  form a basis of  $\text{row}(\mathbf{A})$ . The reduced equations are  $x_1 = x_2 + 2x_4 - 7x_5$  and  $x_3 = -x_4 + 2x_5$ , so by [Theorem 2.5.4](#) a basis of  $\text{nul}(\mathbf{A})$  is

$$\mathbf{s}_2 = (1, 1, 0, 0, 0), \quad \mathbf{s}_4 = (2, 0, -1, 1, 0), \quad \mathbf{s}_5 = (-7, 0, 2, 0, 1).$$

For example  $\mathbf{A}\mathbf{s}_4 = 2\mathbf{a}_1 - \mathbf{a}_3 + \mathbf{a}_4 = (2, 4, -2) - (2, 5, -1) + (0, 1, 1) = \mathbf{0}$ . Hence  $\text{rank } \mathbf{A} = 2$  and  $\text{nullity } \mathbf{A} = 3$ , and  $2 + 3 = 5$ , as [Theorem 2.5.7](#) requires.

#### Exercise B2: Independence and extension in $\mathbb{R}^4$ .

Let  $\mathbf{v}_1 = (1, 0, 2, 1)$ ,  $\mathbf{v}_2 = (2, 1, 3, 1)$ ,  $\mathbf{v}_3 = (0, 1, -1, -1)$  in  $\mathbb{R}^4$ .

- (a) Determine whether  $(\mathbf{v}_1, \mathbf{v}_2, \mathbf{v}_3)$  is linearly independent. Justify your answer.
- (b) Find a basis of  $\text{span}(\mathbf{v}_1, \mathbf{v}_2, \mathbf{v}_3)$ , and extend it to a basis of  $\mathbb{R}^4$  using standard basis vectors.

*Solution.*

- (a) Row reduce the matrix with columns
- $\mathbf{v}_1, \mathbf{v}_2, \mathbf{v}_3$
- :

$$\begin{pmatrix} 1 & 2 & 0 \\ 0 & 1 & 1 \\ 2 & 3 & -1 \\ 1 & 1 & -1 \end{pmatrix} \xrightarrow[R_4 \rightarrow R_4 - R_1]{R_3 \rightarrow R_3 - 2R_1} \begin{pmatrix} 1 & 2 & 0 \\ 0 & 1 & 1 \\ 0 & -1 & -1 \\ 0 & -1 & -1 \end{pmatrix} \xrightarrow[R_4 \rightarrow R_4 + R_2, R_1 \rightarrow R_1 - 2R_2]{R_3 \rightarrow R_3 + R_2} \begin{pmatrix} 1 & 0 & -2 \\ 0 & 1 & 1 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{pmatrix}.$$

The rank is  $2 < 3$ , so the list is dependent by [Theorem 2.5.10](#) (a). Column 3 gives  $\mathbf{v}_3 = -2\mathbf{v}_1 + \mathbf{v}_2$  ([Theorem 2.5.2](#) (b)); check:  $(-2, 0, -4, -2) + (2, 1, 3, 1) = (0, 1, -1, -1)$ .

- (b) By
- [Theorem 2.5.2](#)
- (a),
- $(\mathbf{v}_1, \mathbf{v}_2)$
- is a basis of
- $\text{span}(\mathbf{v}_1, \mathbf{v}_2, \mathbf{v}_3)$
- . To extend it, row reduce
- $(\mathbf{v}_1 \ \mathbf{v}_2 \ \mathbf{e}_1 \ \mathbf{e}_2 \ \mathbf{e}_3 \ \mathbf{e}_4)$
- :

$$\begin{pmatrix} 1 & 2 & 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 1 & 0 & 0 \\ 2 & 3 & 0 & 0 & 1 & 0 \\ 1 & 1 & 0 & 0 & 0 & 1 \end{pmatrix} \xrightarrow[R_4 \rightarrow R_4 - R_1]{R_3 \rightarrow R_3 - 2R_1} \begin{pmatrix} 1 & 2 & 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 1 & 0 & 0 \\ 0 & -1 & -2 & 0 & 1 & 0 \\ 0 & -1 & -1 & 0 & 0 & 1 \end{pmatrix}$$

$$\xrightarrow[R_4 \rightarrow R_4 + R_2]{R_3 \rightarrow R_3 + R_2} \begin{pmatrix} 1 & 2 & 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 1 & 0 & 0 \\ 0 & 0 & -2 & 1 & 1 & 0 \\ 0 & 0 & -1 & 1 & 0 & 1 \end{pmatrix}$$

$$\xrightarrow[\text{then } R_3 \rightarrow -R_3]{R_3 \leftrightarrow R_4} \begin{pmatrix} 1 & 2 & 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & -1 & 0 & -1 \\ 0 & 0 & -2 & 1 & 1 & 0 \end{pmatrix} \xrightarrow{R_4 \rightarrow R_4 + 2R_3} \begin{pmatrix} 1 & 2 & 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & -1 & 0 & -1 \\ 0 & 0 & 0 & -1 & 1 & -2 \end{pmatrix}.$$

This echelon form has pivots in columns 1, 2, 3, 4, which are the pivot columns by [Corollary 2.5.6](#). By [Theorem 2.5.2](#) (a) the corresponding columns form a basis of the column space, which is  $\mathbb{R}^4$  since it contains  $\mathbf{e}_1, \dots, \mathbf{e}_4$ . Hence  $(\mathbf{v}_1, \mathbf{v}_2, \mathbf{e}_1, \mathbf{e}_2)$  is a basis of  $\mathbb{R}^4$  extending  $(\mathbf{v}_1, \mathbf{v}_2)$ .

### Exercise B3: Sum and intersection in $\mathbb{R}^5$ .

Let  $U = \text{span}(\mathbf{u}_1, \mathbf{u}_2, \mathbf{u}_3)$  and  $W = \text{span}(\mathbf{w}_1, \mathbf{w}_2)$  in  $\mathbb{R}^5$ , where

$$\begin{aligned} \mathbf{u}_1 &= (1, 1, 0, 0, 1), & \mathbf{u}_2 &= (0, 1, 1, 0, 0), & \mathbf{u}_3 &= (1, 0, 0, 1, 0), \\ \mathbf{w}_1 &= (1, 2, 2, 1, 2), & \mathbf{w}_2 &= (0, 0, 1, 1, 1). \end{aligned}$$

Find  $\dim U$ ,  $\dim W$ , a basis of  $U + W$  and a basis of  $U \cap W$ . Hence verify [Theorem 1.7.3](#).

*Solution.* Row reduce  $\mathbf{M} = (\mathbf{u}_1 \ \mathbf{u}_2 \ \mathbf{u}_3 \ \mathbf{w}_1 \ \mathbf{w}_2)$ :

$$\begin{aligned}
 \begin{pmatrix} 1 & 0 & 1 & 1 & 0 \\ 1 & 1 & 0 & 2 & 0 \\ 0 & 1 & 0 & 2 & 1 \\ 0 & 0 & 1 & 1 & 1 \\ 1 & 0 & 0 & 2 & 1 \end{pmatrix} &\xrightarrow[R_5 \rightarrow R_5 - R_1]{R_2 \rightarrow R_2 - R_1} \begin{pmatrix} 1 & 0 & 1 & 1 & 0 \\ 0 & 1 & -1 & 1 & 0 \\ 0 & 1 & 0 & 2 & 1 \\ 0 & 0 & 1 & 1 & 1 \\ 0 & 0 & -1 & 1 & 1 \end{pmatrix} \\
 &\xrightarrow{R_3 \rightarrow R_3 - R_2} \begin{pmatrix} 1 & 0 & 1 & 1 & 0 \\ 0 & 1 & -1 & 1 & 0 \\ 0 & 0 & 1 & 1 & 1 \\ 0 & 0 & 1 & 1 & 1 \\ 0 & 0 & -1 & 1 & 1 \end{pmatrix} \\
 &\xrightarrow[R_5 \rightarrow R_5 + R_3]{R_4 \rightarrow R_4 - R_3} \begin{pmatrix} 1 & 0 & 1 & 1 & 0 \\ 0 & 1 & -1 & 1 & 0 \\ 0 & 0 & 1 & 1 & 1 \\ 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 2 & 2 \end{pmatrix} \xrightarrow[\text{then } R_4 \rightarrow \frac{1}{2}R_4]{R_4 \leftrightarrow R_5} \begin{pmatrix} 1 & 0 & 1 & 1 & 0 \\ 0 & 1 & -1 & 1 & 0 \\ 0 & 0 & 1 & 1 & 1 \\ 0 & 0 & 0 & 1 & 1 \\ 0 & 0 & 0 & 0 & 0 \end{pmatrix}.
 \end{aligned}$$

Clearing above the pivots ( $R_3 \rightarrow R_3 - R_4$ ,  $R_2 \rightarrow R_2 - R_4$ ,  $R_1 \rightarrow R_1 - R_4$ , then  $R_2 \rightarrow R_2 + R_3$ ,  $R_1 \rightarrow R_1 - R_3$ ) gives

$$\mathbf{R} = \begin{pmatrix} 1 & 0 & 0 & 0 & -1 \\ 0 & 1 & 0 & 0 & -1 \\ 0 & 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 1 & 1 \\ 0 & 0 & 0 & 0 & 0 \end{pmatrix}.$$

The pivot columns are 1, 2, 3, 4, so  $(\mathbf{u}_1, \mathbf{u}_2, \mathbf{u}_3, \mathbf{w}_1)$  is independent by [Theorem 2.5.2](#) (a), and so are its sub-lists  $(\mathbf{u}_1, \mathbf{u}_2, \mathbf{u}_3)$  and  $(\mathbf{w}_1)$ ; also  $\mathbf{w}_2$  is not a multiple of  $\mathbf{w}_1$  (compare first entries). Hence  $\dim U = 3$  and  $\dim W = 2$ , with the given spanning lists as bases. By [Proposition 2.5.11](#) (a),  $(\mathbf{u}_1, \mathbf{u}_2, \mathbf{u}_3, \mathbf{w}_1)$  is a basis of  $U + W$ , so  $\dim(U + W) = 4$ .

The only free column is column 5. Setting  $x_5 = 1$  gives  $\mathbf{n} = (1, 1, 0, -1, 1)$ , a basis of  $\text{nul}(\mathbf{M})$  by [Theorem 2.5.4](#). So  $\mathbf{u}_1 + \mathbf{u}_2 - \mathbf{w}_1 + \mathbf{w}_2 = \mathbf{0}$ , and by [Proposition 2.5.11](#) (b),  $\mathbf{u}_1 + \mathbf{u}_2 = (1, 2, 1, 0, 1)$  is a basis of  $U \cap W$ . Check:  $\mathbf{w}_1 - \mathbf{w}_2 = (1, 2, 1, 0, 1)$ . Hence  $\dim(U \cap W) = 1$ , and

$$\dim U + \dim W - \dim(U \cap W) = 3 + 2 - 1 = 4 = \dim(U + W).$$

### 2.5.9.3 C. Going deeper

#### Exercise C1: Rank of a sum.

Let  $\mathbf{A}, \mathbf{B} \in M_{m \times n}(F)$ . Prove that  $\text{rank}(\mathbf{A} + \mathbf{B}) \leq \text{rank } \mathbf{A} + \text{rank } \mathbf{B}$ . Give an example where the inequality is strict.

*Solution.* The  $j$ -th column of  $\mathbf{A} + \mathbf{B}$  is  $\mathbf{a}_j + \mathbf{b}_j$ , which lies in  $\text{col}(\mathbf{A}) + \text{col}(\mathbf{B})$  ([Definition 1.7.1](#)). This sum is a subspace ([Theorem 1.7.1](#)), so by [Theorem 1.3.1](#),  $\text{col}(\mathbf{A} + \mathbf{B}) \subseteq$

$\text{col}(\mathbf{A}) + \text{col}(\mathbf{B})$ . By [Theorem 1.6.7](#) and [Theorem 1.7.3](#),

$$\begin{aligned}\text{rank}(\mathbf{A} + \mathbf{B}) &\leq \dim(\text{col}(\mathbf{A}) + \text{col}(\mathbf{B})) \\ &= \text{rank } \mathbf{A} + \text{rank } \mathbf{B} - \dim(\text{col}(\mathbf{A}) \cap \text{col}(\mathbf{B})) \\ &\leq \text{rank } \mathbf{A} + \text{rank } \mathbf{B}.\end{aligned}$$

For strictness, take  $\mathbf{A} = \mathbf{I}_2$  and  $\mathbf{B} = -\mathbf{I}_2$  over  $\mathbb{R}$ :  $\text{rank}(\mathbf{A} + \mathbf{B}) = \text{rank } \mathbf{0} = 0 < 2 + 2$ .

### Exercise C2: Matrices of rank one.

Let  $\mathbf{A} \in M_{m \times n}(F)$ . Prove that  $\text{rank } \mathbf{A} = 1$  if and only if  $\mathbf{A} = \mathbf{xy}^\top$  for some **non-zero**  $\mathbf{x} \in F^m$  and  $\mathbf{y} \in F^n$ .

*Solution.* ( $\Leftarrow$ ) Suppose  $\mathbf{A} = \mathbf{xy}^\top$  with  $\mathbf{x}, \mathbf{y} \neq \mathbf{0}$ , and write  $\mathbf{y} = (y_1, \dots, y_n)$ . By [Theorem 0.9.2](#), the  $j$ -th column of  $\mathbf{xy}^\top$  is  $\mathbf{x}$  times the  $j$ -th column of  $\mathbf{y}^\top$ , the  $1 \times 1$  matrix  $(y_j)$ ; that is,  $y_j \mathbf{x}$ . So every column lies in  $\text{span}(\mathbf{x})$ , and  $\text{col}(\mathbf{A}) \subseteq \text{span}(\mathbf{x})$ . Since  $\mathbf{y} \neq \mathbf{0}$ , some  $y_j \neq 0$ , and then  $y_j \mathbf{x} \neq \mathbf{0}$  because  $\mathbf{x} \neq \mathbf{0}$  ([Theorem 1.1.9](#)). The list  $(\mathbf{x})$  is independent, since  $a\mathbf{x} = \mathbf{0}$  forces  $a = 0$  ([Theorem 1.1.9](#)), so  $\text{span}(\mathbf{x})$  has dimension 1, and  $\text{rank } \mathbf{A} \leq 1$  by [Theorem 1.6.7](#). On the other hand  $\text{rank } \mathbf{A} \neq 0$ : a space of dimension 0 has the empty list as a basis, so it equals  $\text{span}() = \{\mathbf{0}\}$ , but  $\text{col}(\mathbf{A})$  contains the non-zero vector  $y_j \mathbf{x}$ . Hence  $\text{rank } \mathbf{A} = 1$ .

( $\Rightarrow$ ) Suppose  $\text{rank } \mathbf{A} = 1$ , and let  $(\mathbf{x})$  be a basis of  $\text{col}(\mathbf{A})$ ; then  $\mathbf{x} \neq \mathbf{0}$ . Each column  $\mathbf{a}_j \in \text{col}(\mathbf{A})$  equals  $y_j \mathbf{x}$  for some  $y_j \in F$ . Put  $\mathbf{y} = (y_1, \dots, y_n)$ . By the computation in ( $\Leftarrow$ ), the  $j$ -th column of  $\mathbf{xy}^\top$  is  $y_j \mathbf{x} = \mathbf{a}_j$ , so  $\mathbf{A} = \mathbf{xy}^\top$ . If  $\mathbf{y} = \mathbf{0}$ , then  $\mathbf{A} = \mathbf{0}$  would have rank 0, so  $\mathbf{y} \neq \mathbf{0}$ . This proves the claim.

### Exercise C3: The matrix $i + j$ .

For  $n \geq 1$ , let  $\mathbf{A}_n \in M_n(\mathbb{R})$  have entries  $a_{ij} = i + j$ . Prove that  $\text{rank } \mathbf{A}_1 = 1$  and  $\text{rank } \mathbf{A}_n = 2$  for every  $n \geq 2$ . Hence find nullity  $\mathbf{A}_n$ .

*Hint: write each row as a combination of two fixed row vectors.*

*Solution.* For  $n = 1$ ,  $\mathbf{A}_1 = (2) \neq 0$ , so  $\text{rank } \mathbf{A}_1 = 1$ . Let  $n \geq 2$ , and put  $\mathbf{1} = (1 \ 1 \ \dots \ 1)$  and  $\mathbf{d} = (1 \ 2 \ \dots \ n)$  in  $M_{1 \times n}(\mathbb{R})$ . Row  $i$  of  $\mathbf{A}_n$  is  $(i+1 \ \dots \ i+n) = i\mathbf{1} + \mathbf{d}$ . So  $\text{row}(\mathbf{A}_n) \subseteq \text{span}(\mathbf{1}, \mathbf{d})$ , and  $\dim \text{row}(\mathbf{A}_n) \leq 2$  by [Theorem 1.6.3](#).

Rows 1 and 2 are  $(2 \ 3 \ \dots)$  and  $(3 \ 4 \ \dots)$ . If  $c_1(\text{row } 1) + c_2(\text{row } 2) = \mathbf{0}$ , the first two entries give  $2c_1 + 3c_2 = 0$  and  $3c_1 + 4c_2 = 0$ . Subtracting the first from the second gives  $c_1 + c_2 = 0$ , so  $c_1 = -c_2$ , and then  $-2c_2 + 3c_2 = c_2 = 0$ , so  $c_1 = 0$ . Hence  $\text{row}(\mathbf{A}_n)$  contains two independent vectors, and  $\dim \text{row}(\mathbf{A}_n) \geq 2$  by [Theorem 1.6.3](#).

So  $\dim \text{row}(\mathbf{A}_n) = 2$ , and  $\text{rank } \mathbf{A}_n = 2$  by [Theorem 2.5.5](#). By [Theorem 2.5.7](#), nullity  $\mathbf{A}_n = n - 2$  for  $n \geq 2$ , and nullity  $\mathbf{A}_1 = 0$ .

### Exercise C4: Counting right-hand sides over a finite field.

Let  $F$  be a finite field with  $q$  elements, and let  $\mathbf{A} \in M_{m \times n}(F)$  have rank  $r$ .

- (a) Prove that exactly  $q^r$  vectors  $\mathbf{b} \in F^m$  make  $\mathbf{Ax} = \mathbf{b}$  consistent.
- (b) By counting the pairs  $(\mathbf{x}, \mathbf{b}) \in F^n \times F^m$  with  $\mathbf{Ax} = \mathbf{b}$  in two ways, and using [Theorem 2.1.3](#) but **not** [Theorem 2.5.7](#), prove that  $\text{nul}(\mathbf{A})$  has exactly  $q^{n-r}$  elements. Deduce nullity  $\mathbf{A} = n - r$ : a counting proof of Rank–Nullity over a finite field.
- (c) Over  $\mathbb{F}_2$ , let  $\mathbf{A} = \begin{pmatrix} 1 & 1 & 0 \\ 0 & 1 & 1 \\ 1 & 0 & 1 \end{pmatrix}$ . How many  $\mathbf{b} \in \mathbb{F}_2^3$  give a consistent system, and how many solutions does each such system have?

*Solution.*

- (a) By [Theorem 2.1.1](#),  $\mathbf{Ax} = \mathbf{b}$  is consistent exactly when  $\mathbf{b} \in \text{col}(\mathbf{A})$ . Let  $(\mathbf{c}_1, \dots, \mathbf{c}_r)$  be a basis of  $\text{col}(\mathbf{A})$ . By [Theorem 1.5.1](#),  $(t_1, \dots, t_r) \mapsto t_1\mathbf{c}_1 + \dots + t_r\mathbf{c}_r$  is a bijection  $F^r \rightarrow \text{col}(\mathbf{A})$ : every vector of  $\text{col}(\mathbf{A})$  is hit by exactly one tuple. Since  $F^r$  has  $q^r$  elements, so does  $\text{col}(\mathbf{A})$ .
- (b) Let  $N = |\text{nul}(\mathbf{A})|$ . Each  $\mathbf{x} \in F^n$  determines exactly one  $\mathbf{b} = \mathbf{Ax}$ , so there are  $q^n$  pairs. Now group the pairs by  $\mathbf{b}$ . If  $\mathbf{b} \notin \text{col}(\mathbf{A})$ , there are none, by [Theorem 2.1.1](#). If  $\mathbf{b} \in \text{col}(\mathbf{A})$ , pick one solution  $\mathbf{p}$ ; by [Theorem 2.1.3](#) the solutions are  $\mathbf{p} + \text{nul}(\mathbf{A})$ , and  $\mathbf{h} \mapsto \mathbf{p} + \mathbf{h}$  is a bijection from  $\text{nul}(\mathbf{A})$  onto this set (it is onto by definition, and  $\mathbf{p} + \mathbf{h} = \mathbf{p} + \mathbf{h}'$  gives  $\mathbf{h} = \mathbf{h}'$ ). So each of the  $q^r$  vectors  $\mathbf{b}$  of (a) accounts for exactly  $N$  pairs, and there are  $q^r N$  pairs in total. Hence  $q^r N = q^n$ , and  $N = q^{n-r}$ . Let  $d = \text{nullity } \mathbf{A}$ . The argument of (a), applied to a basis of  $\text{nul}(\mathbf{A})$  of length  $d$ , gives  $N = q^d$ . So  $q^d = q^{n-r}$ . Since  $q \geq 2$ , distinct exponents give distinct powers of  $q$ , so  $d = n - r$ . This is [Theorem 2.5.7](#), proved here by counting alone.
- (c) Over  $\mathbb{F}_2$ ,  $R_3 \rightarrow R_3 + R_1$  gives row 3 equal to  $(0, 1, 1)$ , and then  $R_3 \rightarrow R_3 + R_2$  makes it zero. So an echelon form is  $\begin{pmatrix} 1 & 1 & 0 \\ 0 & 1 & 1 \\ 0 & 0 & 0 \end{pmatrix}$ , and rank  $\mathbf{A} = 2$  by [Corollary 2.5.6](#). By (a), exactly  $2^2 = 4$  of the 8 vectors  $\mathbf{b}$  give a consistent system, and by [Theorem 2.5.8](#) (c) each has  $2^{3-2} = 2$  solutions. (The consistent  $\mathbf{b}$  are those with  $b_1 + b_2 + b_3 = 0$ . The rows of  $\mathbf{A}$  add up to zero, so every consistent  $\mathbf{b}$  satisfies this equation; exactly 4 vectors of  $\mathbb{F}_2^3$  satisfy it, and 4 are consistent, so the two sets coincide.)

## 2.6 LU and PLU Factorizations

Gaussian elimination solves one system  $\mathbf{Ax} = \mathbf{b}$ . In practice the matrix often stays fixed while the right-hand side changes: the same network under different loads, the same model fed different data. The row operations depend only on  $\mathbf{A}$ , so repeating them for every  $\mathbf{b}$  wastes almost all of the work. This section stores the elimination as a factorization  $\mathbf{A} = \mathbf{LU}$ , a lower triangular matrix that remembers the operations times an upper triangular matrix that is their result. We prove when it exists and when it is unique, and we repair its failures with row swaps.

Throughout,  $F$  is a field and all matrices are square, in  $M_n(F)$ .

## 2.6.1 Triangular systems are cheap

Some systems need no elimination at all. Take

$$\mathbf{U} = \begin{pmatrix} 2 & 1 & -1 \\ 0 & 3 & 1 \\ 0 & 0 & -2 \end{pmatrix}, \quad \mathbf{U}\mathbf{x} = \begin{pmatrix} -1 \\ -1 \\ -4 \end{pmatrix}.$$

The last equation mentions only  $x_3$ :  $-2x_3 = -4$ , so  $x_3 = 2$ . The second mentions  $x_2$  and the known  $x_3$ :  $3x_2 + 2 = -1$ , so  $x_2 = -1$ . The first then gives  $2x_1 - 1 - 2 = -1$ , so  $x_1 = 1$ . This is **back substitution**: in an upper triangular system with non-zero diagonal entries, each equation introduces exactly one new unknown, read from the bottom up. A lower triangular system is solved the same way from the top down, by **forward substitution**.

Now suppose a matrix  $\mathbf{A}$  factors as  $\mathbf{A} = \mathbf{L}\mathbf{U}$  with  $\mathbf{L}$  lower triangular and  $\mathbf{U}$  upper triangular. Then  $\mathbf{A}\mathbf{x} = \mathbf{b}$  reads  $\mathbf{L}(\mathbf{U}\mathbf{x}) = \mathbf{b}$ . Naming the inner vector  $\mathbf{y} = \mathbf{U}\mathbf{x}$  splits the problem in two:

$$\text{first solve } \mathbf{L}\mathbf{y} = \mathbf{b} \text{ for } \mathbf{y}, \quad \text{then solve } \mathbf{U}\mathbf{x} = \mathbf{y} \text{ for } \mathbf{x}.$$

Both are triangular, so both are cheap. The expensive part, finding  $\mathbf{L}$  and  $\mathbf{U}$ , is done once for all right-hand sides.

*An LU factorization is Gaussian elimination written down once, so that it never has to be done again.*

## 2.6.2 LU factorizations

### Definition 2.6.1 — LU factorization

A matrix  $\mathbf{L} \in M_n(F)$  is **unit lower triangular** if it is lower triangular and **every** diagonal entry equals 1. Let  $\mathbf{A} \in M_n(F)$ . An **LU factorization** of  $\mathbf{A}$  is a pair  $(\mathbf{L}, \mathbf{U})$  of matrices in  $M_n(F)$  such that

$$\mathbf{A} = \mathbf{L}\mathbf{U}, \quad \mathbf{L} \text{ is unit lower triangular,} \quad \mathbf{U} \text{ is upper triangular.}$$

In words:  $\mathbf{L}$  has 1's on the diagonal and zeros above it, and the entries below the diagonal are unrestricted.  $\mathbf{U}$  has zeros strictly below the diagonal, and its diagonal entries are unrestricted; some of them may be 0. The definition says **an** factorization: whether there is only one is a theorem, and it needs a hypothesis.

### Example. (An LU factorization from elimination)

Let

$$\mathbf{A} = \begin{pmatrix} 2 & 1 & -1 \\ 4 & 5 & -1 \\ -2 & 8 & 2 \end{pmatrix} \in M_3(\mathbb{Q}).$$

Reduce  $\mathbf{A}$  to upper triangular form using only operations “subtract a multiple of a higher row from a lower row”, and record each multiplier. Compare the result with the matrix  $\mathbf{U}$  of the previous subsection.



*Solution.* The first pivot is 2. To clear column 1 below it we subtract  $\ell_{21} = 4/2 = 2$  times row 1 from row 2, and  $\ell_{31} = -2/2 = -1$  times row 1 from row 3:

$$\begin{pmatrix} 2 & 1 & -1 \\ 4 & 5 & -1 \\ -2 & 8 & 2 \end{pmatrix} \xrightarrow[R_3 \rightarrow R_3 - (-1)R_1]{R_2 \rightarrow R_2 - 2R_1} \begin{pmatrix} 2 & 1 & -1 \\ 0 & 3 & 1 \\ 0 & 9 & 1 \end{pmatrix}.$$

The second pivot is 3. We subtract  $\ell_{32} = 9/3 = 3$  times row 2 from row 3:

$$\begin{pmatrix} 2 & 1 & -1 \\ 0 & 3 & 1 \\ 0 & 9 & 1 \end{pmatrix} \xrightarrow{R_3 \rightarrow R_3 - 3R_2} \begin{pmatrix} 2 & 1 & -1 \\ 0 & 3 & 1 \\ 0 & 0 & -2 \end{pmatrix} = \mathbf{U}.$$

Put the multipliers below the diagonal of a unit lower triangular matrix, each in the position of the entry it cleared:

$$\mathbf{L} = \begin{pmatrix} 1 & 0 & 0 \\ \ell_{21} & 1 & 0 \\ \ell_{31} & \ell_{32} & 1 \end{pmatrix} = \begin{pmatrix} 1 & 0 & 0 \\ 2 & 1 & 0 \\ -1 & 3 & 1 \end{pmatrix}.$$

Multiplying out, row by row,

$$\mathbf{LU} = \begin{pmatrix} 2 & 1 & -1 \\ 2 \cdot 2 & 2 \cdot 1 + 3 & 2 \cdot (-1) + 1 \\ -2 & -1 + 9 & 1 + 3 - 2 \end{pmatrix} = \begin{pmatrix} 2 & 1 & -1 \\ 4 & 5 & -1 \\ -2 & 8 & 2 \end{pmatrix} = \mathbf{A}.$$

So  $(\mathbf{L}, \mathbf{U})$  is an LU factorization of  $\mathbf{A}$ , and  $\mathbf{U}$  is exactly the triangular matrix of the back substitution above.

The example suggests the general statement: elimination produces  $\mathbf{U}$ , and the multipliers, **with their signs as subtracted**, fill in  $\mathbf{L}$ . We prove this below. First, the smallest cases.

- **Upper triangular  $\mathbf{A}$ .**  $\mathbf{A} = \mathbf{I} \cdot \mathbf{A}$ , and  $\mathbf{I}$  is unit lower triangular. So every upper triangular matrix, including 0 and  $\mathbf{I}$ , has the LU factorization  $(\mathbf{I}, \mathbf{A})$ .
- $1 \times 1$ . Every  $(a) \in M_1(F)$  is upper triangular, so  $(a) = (1)(a)$ . Nothing can go wrong here, and that is exactly why the  $2 \times 2$  failure below is instructive.

**Non-example by minimal change.** The matrix  $\begin{pmatrix} 1 & 1 \\ 1 & 0 \end{pmatrix}$  has the LU factorization

$$\begin{pmatrix} 1 & 1 \\ 1 & 0 \end{pmatrix} = \begin{pmatrix} 1 & 0 \\ 1 & 1 \end{pmatrix} \begin{pmatrix} 1 & 1 \\ 0 & -1 \end{pmatrix}.$$

Change the top-left entry from 1 to 0, and everything collapses.

**Example. (An invertible matrix with no LU factorization)**

Show that  $\mathbf{A} = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix} \in M_2(F)$  has **no** LU factorization, over any field  $F$ , although  $\mathbf{A}$  is invertible.

*Solution.*  $\mathbf{A}$  is invertible, since  $\mathbf{AA} = \mathbf{I}_2$ . Suppose, for a contradiction, that  $\mathbf{A} = \mathbf{LU}$

with

$$\mathbf{L} = \begin{pmatrix} 1 & 0 \\ \ell & 1 \end{pmatrix}, \quad \mathbf{U} = \begin{pmatrix} u & v \\ 0 & w \end{pmatrix}.$$

Then

$$\mathbf{LU} = \begin{pmatrix} u & v \\ \ell u & \ell v + w \end{pmatrix}.$$

Comparing  $(1, 1)$ -entries with  $\mathbf{A}$  gives  $u = 0$ . Comparing  $(2, 1)$ -entries gives  $\ell u = 1$ . But  $\ell u = \ell \cdot 0 = 0$  by [Theorem 0.7.1](#), and  $0 \neq 1$  in a field. This contradiction shows that no LU factorization exists.

What failed is visible: the  $(1, 1)$ -entry of any product  $\mathbf{LU}$  is  $u_{11}$  alone, so a zero there forces the whole first column of  $\mathbf{LU}$  to vanish. In elimination terms, the first pivot is 0 while the entry below it is not, and the only cure is to **swap** rows, which “subtract a multiple of a higher row from a lower row” cannot do. We return to this with permutation matrices.

**Why “unit”.** Without the 1’s on the diagonal of  $\mathbf{L}$ , uniqueness would fail even for invertible matrices. If  $\mathbf{A} = \mathbf{LU}$  and  $\mathbf{D}$  is any invertible diagonal matrix, then also  $\mathbf{A} = (\mathbf{LD})(\mathbf{D}^{-1}\mathbf{U})$ , where  $\mathbf{LD}$  is still lower triangular and  $\mathbf{D}^{-1}\mathbf{U}$  still upper triangular. When  $\mathbf{L}$  is invertible and  $\mathbf{D} \neq \mathbf{I}$ , this is a different pair, and over any field with more than two elements such a  $\mathbf{D}$  exists. For instance  $(2) = (1) \ (2) = (2) \ (1) = (4) \ (1/2)$  over  $\mathbb{Q}$ . Fixing the diagonal of  $\mathbf{L}$  removes this freedom and matches what elimination does: it never rescales a row.

*Warning.* **Invertible does not mean “has an LU factorization”.** The matrix of [Example \(An invertible matrix with no LU factorization\)](#) is invertible and has none. Conversely, a singular matrix can have one:  $0 = \mathbf{I} \cdot 0$ . Existence of an LU factorization is a property of the **order of the rows**, not of invertibility.

### 2.6.3 Unit lower triangular matrices

Elimination builds  $\mathbf{L}$  out of elementary matrices, and the uniqueness proof below divides one factorization by another. Both need the same facts: products and inverses of unit lower triangular matrices stay unit lower triangular. We also need to know exactly when a triangular matrix is invertible.

The key observation is a statement about where the non-zero entries of a product can sit.

#### Lemma 2.6.1 — Unit lower triangular matrices are closed

Let  $\mathbf{L}, \mathbf{L}' \in M_n(F)$  be unit lower triangular. Then:

- (a)  $\mathbf{LL}'$  is unit lower triangular;
- (b)  $\mathbf{L}$  is invertible, and  $\mathbf{L}^{-1}$  is unit lower triangular.

*Idea.* Write  $\mathbf{L} = \mathbf{I} + \mathbf{N}$ , where  $\mathbf{N}$  has non-zero entries only strictly below the diagonal. Multiplying two such matrices pushes the non-zero entries **further** below the diagonal, so  $\mathbf{N}^n = 0$ . Then the geometric series  $1/(1+t) = 1 - t + t^2 - \cdots$  stops after  $n$  terms and gives the inverse exactly.

*Proof.* Say that  $\mathbf{X} \in M_n(F)$  is  $p$ -low if  $x_{ij} = 0$  whenever  $i < j + p$ . So 0-low means lower triangular, and 1-low means zero on and above the diagonal.

*Claim.* If  $\mathbf{X}$  is  $p$ -low and  $\mathbf{Y}$  is  $q$ -low, then  $\mathbf{XY}$  is  $(p + q)$ -low. A matrix that is  $n$ -low is the zero matrix.

*Proof.* By Definition 0.9.9,  $(\mathbf{XY})_{ij} = \sum_{l=1}^n x_{il}y_{lj}$ . A term can be non-zero only if  $i \geq l + p$  and  $l \geq j + q$ , which together give  $i \geq j + p + q$ . So if  $i < j + p + q$ , every term is 0 and  $(\mathbf{XY})_{ij} = 0$ . For the second statement, every pair of indices satisfies  $i \leq n < 1 + n \leq j + n$ , so every entry of an  $n$ -low matrix is 0. ■

Write  $\mathbf{L} = \mathbf{I} + \mathbf{N}$  and  $\mathbf{L}' = \mathbf{I} + \mathbf{N}'$ . Since  $\mathbf{L}$  and  $\mathbf{L}'$  are unit lower triangular,  $\mathbf{N}$  and  $\mathbf{N}'$  are 1-low.

For (a),  $\mathbf{LL}' = \mathbf{I} + \mathbf{N} + \mathbf{N}' + \mathbf{NN}'$  by Theorem 0.9.4. Here  $\mathbf{N} + \mathbf{N}'$  is 1-low, since a sum of entries that are both 0 is 0, and  $\mathbf{NN}'$  is 2-low, hence 1-low, by the Claim. So  $\mathbf{LL}' - \mathbf{I}$  is 1-low, which says that  $\mathbf{LL}'$  is unit lower triangular.

For (b), let  $\mathbf{K} = \mathbf{I} - \mathbf{N} + \mathbf{N}^2 - \cdots + (-1)^{n-1}\mathbf{N}^{n-1}$ . Multiplying out, the sum telescopes:

$$(\mathbf{I} + \mathbf{N})\mathbf{K} = \mathbf{I} + (-1)^{n-1}\mathbf{N}^n = \mathbf{I},$$

where  $\mathbf{N}^n = 0$  because  $\mathbf{N}^n$  is  $n$ -low by the Claim applied  $n - 1$  times. The same computation gives  $\mathbf{K}(\mathbf{I} + \mathbf{N}) = \mathbf{I}$ , since  $\mathbf{N}$  commutes with its own powers. Hence  $\mathbf{L}$  is invertible with  $\mathbf{L}^{-1} = \mathbf{K}$ . Finally  $\mathbf{K} - \mathbf{I} = \sum_{p=1}^{n-1} (-1)^p \mathbf{N}^p$  is a sum of 1-low matrices, by the Claim, so  $\mathbf{K}$  is unit lower triangular. This proves the lemma. ■

Transposing everything, the same statements hold for **unit upper triangular** matrices (upper triangular with 1's on the diagonal).

The next lemma decides invertibility of a triangular matrix from its diagonal, without any computation beyond back substitution.

### Lemma 2.6.2 — Invertibility of a triangular matrix

An upper triangular matrix  $\mathbf{U} \in M_n(F)$  is invertible if and only if **every** diagonal entry  $u_{11}, \dots, u_{nn}$  is non-zero.

*Proof.* ( $\Leftarrow$ ) Suppose every  $u_{kk} \neq 0$ , and let  $\mathbf{U}\mathbf{x} = \mathbf{0}$ . We show  $x_k = 0$  for  $k = n, n-1, \dots, 1$  in turn. Suppose  $x_{k+1} = \cdots = x_n = 0$  (nothing is assumed when  $k = n$ ). Row  $k$  of  $\mathbf{U}\mathbf{x} = \mathbf{0}$  reads  $u_{kk}x_k + \sum_{j>k} u_{kj}x_j = 0$ , because  $u_{kj} = 0$  for  $j < k$ . The sum vanishes, so  $u_{kk}x_k = 0$ , and  $x_k = 0$  since  $u_{kk} \neq 0$ . Hence  $\mathbf{x} = \mathbf{0}$  is the only solution of  $\mathbf{U}\mathbf{x} = \mathbf{0}$ , and  $\mathbf{U}$  is invertible by Theorem 2.4.4.

( $\Rightarrow$ ) We prove the contrapositive. Suppose  $u_{kk} = 0$  for some  $k$ . For each  $j \leq k$ , the entries of column  $j$  in rows  $i \geq k$  are zero: if  $i > j$  this is upper triangularity, and the only remaining case  $i = j = k$  is  $u_{kk} = 0$ . So the first  $k$  columns of  $\mathbf{U}$  lie in  $W = \text{span}(\mathbf{e}_1, \dots, \mathbf{e}_{k-1})$ . The list  $(\mathbf{e}_1, \dots, \mathbf{e}_{k-1})$  is independent and spans  $W$ , so

$\dim W = k - 1$  (for  $k = 1$ ,  $W = \{\mathbf{0}\}$  has dimension 0). By [Theorem 1.6.3](#) (a), these  $k$  columns are linearly dependent, so the columns of  $\mathbf{U}$  are dependent, and  $\mathbf{U}$  is not invertible by [Theorem 2.4.4](#). This proves the lemma. ■

## 2.6.4 Elimination without row swaps

To prove that elimination always produces an LU factorization, we first need to say exactly which elimination we mean. It is Gaussian elimination restricted to one kind of operation: subtracting a multiple of a row from a row **below** it.

*Algorithm. Elimination without row swaps.* Input:  $\mathbf{A} \in M_n(F)$ . Put  $\mathbf{A}^{(0)} = \mathbf{A}$ . For  $k = 1, 2, \dots, n - 1$ , write  $\mathbf{A}^{(k-1)} = (a_{ij})$  and do the following.

- If  $a_{kk} \neq 0$ , set  $\ell_{ik} = a_{ik}/a_{kk}$  for  $i = k + 1, \dots, n$ .
- If  $a_{kk} = 0$  and  $a_{ik} = 0$  for **every**  $i > k$ , set  $\ell_{ik} = 0$  for  $i = k + 1, \dots, n$ .
- If  $a_{kk} = 0$  but  $a_{ik} \neq 0$  for **some**  $i > k$ , **stop**: a row swap is needed.

If we did not stop, obtain  $\mathbf{A}^{(k)}$  from  $\mathbf{A}^{(k-1)}$  by replacing row  $i$  with (row  $i$ )  $-\ell_{ik}$  · (row  $k$ ), for each  $i = k + 1, \dots, n$ .

If the loop finishes without stopping, we say **elimination without row swaps succeeds** on  $\mathbf{A}$ . The numbers  $\ell_{ik}$  ( $i > k$ ) are its **multipliers**, and  $\mathbf{U} = \mathbf{A}^{(n-1)}$  is its **result**.

In both non-stopping cases the new  $(i, k)$ -entry is  $a_{ik} - \ell_{ik}a_{kk} = 0$ : in the first case by the choice of  $\ell_{ik}$ , in the second because both terms are already 0. The multipliers are well defined: row  $k$  is not changed during step  $k$ , so it does not matter in which order the rows  $i > k$  are updated.

The proof will write each step as a matrix. By [Definition 2.4.1](#), the elementary matrix of the operation  $R_i \rightarrow R_i + cR_k$  ( $i \neq k$ ) is  $\mathbf{I} + c\mathbf{E}_{ik}$ , where  $\mathbf{E}_{ik}$  is the matrix unit with 1 in entry  $(i, k)$ . We will write the matrix unit as a product of a column and a row,

$$\mathbf{E}_{ik} = \mathbf{e}_i \mathbf{e}_k^\top,$$

which holds because  $\mathbf{e}_i \mathbf{e}_k^\top$  has  $(p, q)$ -entry  $(\mathbf{e}_i)_p (\mathbf{e}_k)_q$ , equal to 1 if  $p = i$  and  $q = k$ , and 0 otherwise. We will use one more rule constantly: for a vector  $\mathbf{v} \in F^n$ , the  $1 \times 1$  matrix  $\mathbf{e}_k^\top \mathbf{v}$  is the  $k$ -th entry  $v_k$ , which we treat as a scalar. In particular  $\mathbf{e}_k^\top \mathbf{e}_j = 0$  for  $j \neq k$ .

For  $1 \leq k \leq n$  and a vector  $\mathbf{v} \in F^n$  whose first  $k$  entries are zero, put

$$\mathbf{G}_k(\mathbf{v}) = \mathbf{I} + \mathbf{v} \mathbf{e}_k^\top.$$

Its only non-identity entries are  $v_i$  in position  $(i, k)$ , for  $i > k$ : it is the identity with the column vector  $\mathbf{v}$  written into column  $k$  below the diagonal. So  $\mathbf{G}_k(\mathbf{v})$  is unit lower triangular.

### Lemma 2.6.3 — Multiplier matrices

Let  $1 \leq k \leq n$ , and let  $\mathbf{v}, \mathbf{v}_1, \dots, \mathbf{v}_m \in F^n$  be vectors such that the first  $k$  entries of  $\mathbf{v}$  are zero and, for each  $j$ , the first  $j$  entries of  $\mathbf{v}_j$  are zero ( $m \leq n$ ). Then:

- (a)  $\mathbf{G}_k(\mathbf{v})$  is the product, in any order, of the elementary matrices  $\mathbf{I} + v_i \mathbf{E}_{ik}$  for  $i = k+1, \dots, n$ ;
- (b)  $\mathbf{G}_k(\mathbf{v})\mathbf{G}_k(-\mathbf{v}) = \mathbf{G}_k(-\mathbf{v})\mathbf{G}_k(\mathbf{v}) = \mathbf{I}$ ;
- (c)  $\mathbf{G}_1(\mathbf{v}_1)\mathbf{G}_2(\mathbf{v}_2) \cdots \mathbf{G}_m(\mathbf{v}_m) = \mathbf{I} + \mathbf{v}_1 \mathbf{e}_1^\top + \mathbf{v}_2 \mathbf{e}_2^\top + \cdots + \mathbf{v}_m \mathbf{e}_m^\top$ .

*Proof.* For (a), let  $i, j > k$ . By [Theorem 0.9.4](#),

$$(\mathbf{I} + a \mathbf{e}_i \mathbf{e}_k^\top)(\mathbf{I} + b \mathbf{e}_j \mathbf{e}_k^\top) = \mathbf{I} + a \mathbf{e}_i \mathbf{e}_k^\top + b \mathbf{e}_j \mathbf{e}_k^\top + ab \mathbf{e}_i (\mathbf{e}_k^\top \mathbf{e}_j) \mathbf{e}_k^\top,$$

and the last term is zero because  $\mathbf{e}_k^\top \mathbf{e}_j = 0$  for  $j \neq k$ . Repeating this, one factor at a time, the product of the  $\mathbf{I} + v_i \mathbf{E}_{ik} = \mathbf{I} + v_i \mathbf{e}_i \mathbf{e}_k^\top$  in any order is  $\mathbf{I} + \sum_{i>k} v_i \mathbf{e}_i \mathbf{e}_k^\top = \mathbf{I} + \mathbf{v} \mathbf{e}_k^\top$ , since  $\mathbf{v} = \sum_{i>k} v_i \mathbf{e}_i$ .

For (b),  $(\mathbf{I} + \mathbf{v} \mathbf{e}_k^\top)(\mathbf{I} - \mathbf{v} \mathbf{e}_k^\top) = \mathbf{I} - \mathbf{v}(\mathbf{e}_k^\top \mathbf{v}) \mathbf{e}_k^\top = \mathbf{I}$ , because  $\mathbf{e}_k^\top \mathbf{v} = v_k = 0$ . Swapping the signs gives the other order.

For (c), we use induction on  $m$ ; the case  $m = 1$  is the definition. Suppose the formula holds for  $m-1$ . Then

$$\left(\mathbf{I} + \sum_{j<m} \mathbf{v}_j \mathbf{e}_j^\top\right)(\mathbf{I} + \mathbf{v}_m \mathbf{e}_m^\top) = \mathbf{I} + \sum_{j<m} \mathbf{v}_j \mathbf{e}_j^\top + \mathbf{v}_m \mathbf{e}_m^\top + \sum_{j<m} \mathbf{v}_j (\mathbf{e}_j^\top \mathbf{v}_m) \mathbf{e}_m^\top.$$

For  $j < m$ ,  $\mathbf{e}_j^\top \mathbf{v}_m$  is entry  $j$  of  $\mathbf{v}_m$ , which is zero because the first  $m$  entries of  $\mathbf{v}_m$  are zero. So the last sum vanishes, and the formula holds for  $m$ . This proves the lemma. ■

Part (c) is the heart of the matter: when the factors come in the order  $\mathbf{G}_1, \mathbf{G}_2, \dots$ , they **do not interfere**, and each vector  $\mathbf{v}_j$  lands untouched in column  $j$ .

### Theorem 2.6.4 — LU from elimination

Let  $\mathbf{A} \in M_n(F)$ , and suppose elimination without row swaps succeeds on  $\mathbf{A}$ , with multipliers  $\ell_{ik}$  and result  $\mathbf{U}$ . Let  $\mathbf{L}$  be the unit lower triangular matrix whose  $(i, k)$ -entry is  $\ell_{ik}$  for every  $i > k$ . Then  $\mathbf{U}$  is upper triangular and

$$\mathbf{A} = \mathbf{L}\mathbf{U}.$$

In particular  $\mathbf{A}$  has an LU factorization.

*Idea.* Each step of elimination is left multiplication by a product of elementary matrices, which collapses to one matrix  $\mathbf{G}_k(-\ell_k)$ . So  $\mathbf{U} = \mathbf{G}_{n-1}(-\ell_{n-1}) \cdots \mathbf{G}_1(-\ell_1)\mathbf{A}$ . To get  $\mathbf{A}$  back, undo the steps in reverse order. Their inverses  $\mathbf{G}_k(\ell_k)$  then appear in the order  $\mathbf{G}_1, \mathbf{G}_2, \dots$ , which is exactly the order in which, by [Lemma 2.6.3](#) (c), the multipliers drop into place.

*Proof.* Let  $\mathbf{A}^{(0)}, \dots, \mathbf{A}^{(n-1)} = \mathbf{U}$  be the matrices produced by the algorithm. For each  $k$ , let  $\ell_k = \sum_{i>k} \ell_{ik} \mathbf{e}_i \in F^n$ , a vector whose first  $k$  entries are zero.

**Step 1:  $\mathbf{U}$  is upper triangular.** We show by induction on  $k$  that in  $\mathbf{A}^{(k)}$  every entry below the diagonal in columns  $1, \dots, k$  is zero. For  $k = 0$  there is nothing to

prove. Suppose it holds for  $\mathbf{A}^{(k-1)}$ . Step  $k$  changes only rows  $i > k$ , replacing row  $i$  by row  $i$  minus a multiple of row  $k$ . In columns  $j < k$ , the entry of row  $k$  lies below the diagonal ( $k > j$ ), so it is 0 by the induction hypothesis, and the entries of row  $i$  in these columns stay 0. In column  $k$ , the new entries below the diagonal are 0, as noted after the algorithm. This completes the induction, and for  $k = n - 1$  it says that  $\mathbf{U}$  is upper triangular.

**Step 2: each step is a matrix.** Step  $k$  performs the operations “add  $-\ell_{ik}$  times row  $k$  to row  $i$ ” for  $i = k + 1, \dots, n$ . By [Theorem 2.4.1](#), applying them one after another multiplies  $\mathbf{A}^{(k-1)}$  on the left by the corresponding elementary matrices  $\mathbf{I} - \ell_{ik}\mathbf{E}_{ik}$ . By [Lemma 2.6.3](#) (a), their product is  $\mathbf{G}_k(-\ell_k)$ . Hence  $\mathbf{A}^{(k)} = \mathbf{G}_k(-\ell_k)\mathbf{A}^{(k-1)}$ , and therefore

$$\mathbf{U} = \mathbf{G}_{n-1}(-\ell_{n-1}) \cdots \mathbf{G}_2(-\ell_2) \mathbf{G}_1(-\ell_1) \mathbf{A}.$$

**Step 3: undo the steps.** Multiply both sides on the left by  $\mathbf{G}_{n-1}(\ell_{n-1})$ , then by  $\mathbf{G}_{n-2}(\ell_{n-2})$ , and so on down to  $\mathbf{G}_1(\ell_1)$ . By [Lemma 2.6.3](#) (b), each one cancels the factor next to it, and we get

$$\mathbf{G}_1(\ell_1) \mathbf{G}_2(\ell_2) \cdots \mathbf{G}_{n-1}(\ell_{n-1}) \mathbf{U} = \mathbf{A}.$$

By [Lemma 2.6.3](#) (c), the product on the left of  $\mathbf{U}$  is  $\mathbf{I} + \sum_{k=1}^{n-1} \ell_k \mathbf{e}_k^\top$ . Its  $(i, k)$ -entry for  $i > k$  is  $\ell_{ik}$ ; its diagonal entries are 1; its entries above the diagonal are 0. That is, it is  $\mathbf{L}$ . Hence  $\mathbf{A} = \mathbf{L}\mathbf{U}$  with  $\mathbf{L}$  unit lower triangular and  $\mathbf{U}$  upper triangular, as claimed. ■

So the multipliers are not something to be remembered separately: they **are** the lower triangular factor. When a computer performs elimination it stores each  $\ell_{ik}$  in the slot  $(i, k)$  that it has just cleared, and at the end the one array holds  $\mathbf{L}$  below the diagonal and  $\mathbf{U}$  on and above it.

**Warning.** The matrix that performs elimination,  $\mathbf{L}^{-1}$ , is not  $\mathbf{L}$  with its multipliers negated. In [Example \(An LU factorization from elimination\)](#) the operations multiply  $\mathbf{A}$  on the left by

$$\mathbf{G}_2(-\ell_2) \mathbf{G}_1(-\ell_1) = \begin{pmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & -3 & 1 \end{pmatrix} \begin{pmatrix} 1 & 0 & 0 \\ -2 & 1 & 0 \\ 1 & 0 & 1 \end{pmatrix} = \begin{pmatrix} 1 & 0 & 0 \\ -2 & 1 & 0 \\ 7 & -3 & 1 \end{pmatrix},$$

and the entry  $7 = 1 + (-3)(-2)$  mixes two multipliers. This matrix is  $\mathbf{L}^{-1}$ , by Step 3 of the proof. It is  $\mathbf{L}$ , whose factors come in the order  $\mathbf{G}_1, \mathbf{G}_2$ , that has the multipliers cleanly in place.

**Quick check.** Find the LU factorization of  $\mathbf{A} = \begin{pmatrix} 2 & 4 \\ 6 & 5 \end{pmatrix}$  over  $\mathbb{Q}$  by elimination, and check it by multiplying.

**Solution.** The pivot is 2 and the multiplier is  $\ell_{21} = 6/2 = 3$ . Subtracting 3 times row 1 from row 2 gives  $\mathbf{U} = \begin{pmatrix} 2 & 4 \\ 0 & 5 - 12 \end{pmatrix} = \begin{pmatrix} 2 & 4 \\ 0 & -7 \end{pmatrix}$ , and  $\mathbf{L} = \begin{pmatrix} 1 & 0 \\ 3 & 1 \end{pmatrix}$ . Check:  
 $\mathbf{L}\mathbf{U} = \begin{pmatrix} 2 & 4 \\ 6 & 12 - 7 \end{pmatrix} = \mathbf{A}.$

## 2.6.5 Solving with an LU factorization

Once  $\mathbf{A} = \mathbf{LU}$  is known, each system  $\mathbf{Ax} = \mathbf{b}$  costs two substitutions.

**Example. (Two right-hand sides, one factorization)**

Let  $\mathbf{A} = \begin{pmatrix} 2 & 1 & -1 \\ 4 & 5 & -1 \\ -2 & 8 & 2 \end{pmatrix}$  with the factorization  $\mathbf{A} = \mathbf{LU}$  of [Example \(An LU factorization from elimination\)](#). Solve  $\mathbf{Ax} = \mathbf{b}$  for

$$\mathbf{b} = \begin{pmatrix} -1 \\ -3 \\ -6 \end{pmatrix} \quad \text{and} \quad \mathbf{b}' = \begin{pmatrix} 3 \\ 7 \\ -2 \end{pmatrix}.$$

*Solution. First right-hand side.* Forward substitution in  $\mathbf{Ly} = \mathbf{b}$ :

$$\begin{aligned} y_1 &= -1, \\ 2y_1 + y_2 &= -3 \Rightarrow y_2 = -1, \\ -y_1 + 3y_2 + y_3 &= -6 \Rightarrow y_3 = -6 - 1 + 3 = -4. \end{aligned}$$

Back substitution in  $\mathbf{Ux} = \mathbf{y}$ :

$$\begin{aligned} -2x_3 &= -4 \Rightarrow x_3 = 2, \\ 3x_2 + x_3 &= -1 \Rightarrow x_2 = -1, \\ 2x_1 + x_2 - x_3 &= -1 \Rightarrow x_1 = 1. \end{aligned}$$

So  $\mathbf{x} = (1, -1, 2)$ . Check:  $\mathbf{Ax} = (2 - 1 - 2, 4 - 5 - 2, -2 - 8 + 4) = (-1, -3, -6) = \mathbf{b}$ .

*Second right-hand side.*  $\mathbf{Ly} = \mathbf{b}'$ :  $y_1 = 3$ ,  $y_2 = 7 - 2 \cdot 3 = 1$ ,  $y_3 = -2 + y_1 - 3y_2 = -2 + 3 - 3 = -2$ . Then  $\mathbf{Ux} = \mathbf{y}$ :  $-2x_3 = -2$ , so  $x_3 = 1$ ;  $3x_2 + 1 = 1$ , so  $x_2 = 0$ ;  $2x_1 + 0 - 1 = 3$ , so  $x_1 = 2$ . So  $\mathbf{x}' = (2, 0, 1)$ , and indeed  $\mathbf{Ax}' = (4 + 0 - 1, 8 + 0 - 1, -4 + 0 + 2) = (3, 7, -2)$ .

Since the diagonal of  $\mathbf{U}$  is  $2, 3, -2$ , all non-zero,  $\mathbf{U}$  is invertible by [Lemma 2.6.2](#),  $\mathbf{L}$  is invertible by [Lemma 2.6.1](#), and so  $\mathbf{A} = \mathbf{LU}$  is invertible. Each solution found is therefore the only one.

How much is saved? Count the arithmetic operations  $(+, -, \times, \div)$  for an  $n \times n$  matrix. At step  $k$  of elimination there are  $n - k$  rows below the pivot. Each needs one division for its multiplier, then  $n - k$  multiplications and  $n - k$  subtractions in the columns to the right of column  $k$ . With  $j = n - k$ , the total is

$$\sum_{j=1}^{n-1} j(2j+1) = \frac{n(n-1)(4n+1)}{6} \approx \frac{2}{3}n^3.$$

Forward substitution with a unit lower triangular  $\mathbf{L}$  uses  $n(n-1)$  operations, and back substitution  $n^2$ , so a pair of triangular solves costs  $2n^2 - n$ . For  $n = 1000$  that is about  $6.7 \times 10^8$  operations for the factorization, done once, against about  $2 \times 10^6$  for each new right-hand side.

## 2.6.6 Uniqueness

The factorization of [Example \(An LU factorization from elimination\)](#) came from one particular procedure. Could another procedure give a different pair  $(L, U)$ ? For invertible matrices, no.

### Theorem 2.6.5 — Uniqueness of the LU factorization

Let  $\mathbf{A} \in M_n(F)$  be **invertible**. If  $\mathbf{A} = L_1 U_1$  and  $\mathbf{A} = L_2 U_2$  are LU factorizations, then  $L_1 = L_2$  and  $U_1 = U_2$ .

*Idea.* Take the difference in multiplicative form: from  $L_1 U_1 = L_2 U_2$  we get  $(L_2^{-1} L_1) U_1 = U_2$ . The matrix  $\mathbf{M} = L_2^{-1} L_1$  is unit lower triangular, and multiplying  $U_1$  by it must leave an upper triangular matrix. Clearing below the diagonal column by column, the non-zero pivots of  $U_1$  force  $\mathbf{M}$  to have nothing below its diagonal. Invertibility of  $\mathbf{A}$  is what provides those non-zero pivots.

*Proof.* By [Lemma 2.6.1](#) (b),  $L_1$  and  $L_2$  are invertible. Hence  $U_1 = L_1^{-1} \mathbf{A}$  is a product of invertible matrices, so it is invertible by [Theorem 0.9.7](#), and by [Lemma 2.6.2](#) its diagonal entries  $u_{11}, \dots, u_{nn}$  are all non-zero.

Let  $\mathbf{M} = L_2^{-1} L_1$ . By [Lemma 2.6.1](#),  $\mathbf{M} = (m_{ij})$  is unit lower triangular. Multiplying  $L_1 U_1 = L_2 U_2$  on the left by  $L_2^{-1}$  gives  $\mathbf{M} U_1 = U_2$ , so  $\mathbf{M} U_1$  is upper triangular. We show  $m_{ij} = 0$  for all  $i > j$ , by induction on the column  $j$ .

Fix  $j$ , and suppose  $m_{ik} = 0$  whenever  $k < j$  and  $i > k$ . Let  $i > j$ . Since  $\mathbf{M} U_1$  is upper triangular and  $i > j$ ,

$$0 = (\mathbf{M} U_1)_{ij} = \sum_{k=1}^n m_{ik} u_{kj} = \sum_{k \leq j} m_{ik} u_{kj} = m_{ij} u_{jj},$$

where the third equality uses  $u_{kj} = 0$  for  $k > j$ , and the last uses  $m_{ik} = 0$  for  $k < j$  (here  $i > j > k$ ). Since  $u_{jj} \neq 0$ , we get  $m_{ij} = 0$ . This completes the induction, so  $\mathbf{M}$  is lower triangular with 1's on the diagonal and nothing below it:  $\mathbf{M} = \mathbf{I}$ .

Hence  $L_2^{-1} L_1 = \mathbf{I}$ , and multiplying by  $L_2$  gives  $L_1 = L_2$ . Then  $U_1 = L_1^{-1} \mathbf{A} = L_2^{-1} \mathbf{A} = U_2$ . This proves the theorem. ■

Two consequences are worth saying. The LU factorization of an invertible matrix does not depend on how it was found, so it is legitimate to speak of **the** factors  $L$  and  $U$ . And if elimination without row swaps succeeds on an invertible  $\mathbf{A}$ , then every pivot is non-zero: row  $k$  is not changed after step  $k$ , so the pivot of step  $k$  is  $u_{kk}$ , which is non-zero by the first paragraph of the proof. So the second case of the algorithm (a zero pivot over a zero column) never occurs.

*Remark.* The hypothesis “invertible” cannot be dropped. For every  $c \in F$ ,

$$\begin{pmatrix} 0 & 0 \\ 0 & 1 \end{pmatrix} = \begin{pmatrix} 1 & 0 \\ c & 1 \end{pmatrix} \begin{pmatrix} 0 & 0 \\ 0 & 1 \end{pmatrix},$$

since the first column of the right factor is zero. Taking  $c = 0$  and  $c = 1$  gives two



different LU factorizations. The proof breaks exactly at “ $u_{jj} \neq 0$ ”, here for  $j = 1$ .

## 2.6.7 Row swaps and the PLU factorization

**Example (An invertible matrix with no LU factorization)** showed that some matrices, even invertible ones, have no LU factorization, because elimination meets a zero pivot with a non-zero entry below it. The cure in Gaussian elimination is to swap rows. We now show that the swaps can all be collected in front, as a single matrix that rearranges the rows of  $\mathbf{A}$  **before** elimination starts. After that rearrangement, no swaps are needed.

Matrices that rearrange rows have a name.

### Definition 2.6.2 — Permutation matrix

Let  $\sigma \in S_n$  be a permutation of  $\{1, \dots, n\}$ . The **permutation matrix** of  $\sigma$  is

$$\mathbf{P}_\sigma = (\mathbf{e}_{\sigma(1)} \quad \mathbf{e}_{\sigma(2)} \quad \cdots \quad \mathbf{e}_{\sigma(n)}) \in M_n(F),$$

the matrix whose  $j$ -th column is  $\mathbf{e}_{\sigma(j)}$ . A matrix is a **permutation matrix** if it equals  $\mathbf{P}_\sigma$  for some  $\sigma \in S_n$ .

Equivalently, a permutation matrix has exactly one 1 in each row and in each column, and zeros elsewhere. The identity is  $\mathbf{P}_{\text{id}}$ . The matrix  $\begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}$  of **Example (An invertible matrix with no LU factorization)** is  $\mathbf{P}_\tau$  for the transposition  $\tau = (1 \ 2)$ . By contrast  $\begin{pmatrix} 1 & 1 \\ 0 & 1 \end{pmatrix}$  is **not** a permutation matrix: each column is still non-zero, but its first row contains two 1's, and its second column  $(1, 1)$  is not a standard basis vector.

### Lemma 2.6.6 — Permutation matrices

Let  $\sigma, \tau \in S_n$ .

- (a)  $\mathbf{P}_\sigma \mathbf{P}_\tau = \mathbf{P}_{\sigma \circ \tau}$ . In particular a product of permutation matrices is a permutation matrix.
- (b)  $\mathbf{P}_\sigma$  is invertible and  $\mathbf{P}_\sigma^{-1} = \mathbf{P}_\sigma^\top$ .
- (c) If  $\tau = (r \ s)$  is a transposition, then  $\mathbf{P}_\tau$  is the elementary matrix  $\mathbf{P}_{rs}$  that swaps rows  $r$  and  $s$ , and  $\mathbf{P}_\tau^2 = \mathbf{I}$ .
- (d) Let  $1 \leq k < r < s \leq n$ , let  $\tau = (r \ s)$ , and let  $\mathbf{v} \in F^n$  have its first  $k$  entries zero. Then  $\mathbf{P}_\tau \mathbf{v}$  has its first  $k$  entries zero, and

$$\mathbf{P}_\tau \mathbf{G}_k(\mathbf{v}) \mathbf{P}_\tau = \mathbf{G}_k(\mathbf{P}_\tau \mathbf{v}).$$

*Proof.*

- (a) By **Theorem 0.9.3**,  $\mathbf{P}_\tau \mathbf{e}_j = \mathbf{e}_{\tau(j)}$  and so  $\mathbf{P}_\sigma \mathbf{P}_\tau \mathbf{e}_j = \mathbf{P}_\sigma \mathbf{e}_{\tau(j)} = \mathbf{e}_{\sigma(\tau(j))}$ . So the  $j$ -th column of  $\mathbf{P}_\sigma \mathbf{P}_\tau$  is  $\mathbf{e}_{(\sigma \circ \tau)(j)}$ , for every  $j$ , which is the claim.
- (b) By **Theorem 0.9.2**,  $(\mathbf{P}_\sigma^\top \mathbf{P}_\sigma)_{ij} = \mathbf{e}_{\sigma(i)}^\top \mathbf{e}_{\sigma(j)}$ , which is 1 if  $\sigma(i) = \sigma(j)$  and 0 otherwise.

Since  $\sigma$  is injective,  $\sigma(i) = \sigma(j)$  exactly when  $i = j$ . Hence  $\mathbf{P}_\sigma^\top \mathbf{P}_\sigma = \mathbf{I}$ , and by Theorem 2.4.5,  $\mathbf{P}_\sigma$  is invertible with inverse  $\mathbf{P}_\sigma^\top$ .

- (c) Swapping rows  $r$  and  $s$  of  $\mathbf{I}$  produces a matrix whose column  $r$  is  $\mathbf{e}_s$ , whose column  $s$  is  $\mathbf{e}_r$ , and whose other columns  $j$  are  $\mathbf{e}_j$ . These are the columns  $\mathbf{e}_{\tau(j)}$ , so the swap matrix is  $\mathbf{P}_\tau$ . Since  $\tau \circ \tau = \text{id}$ , part (a) gives  $\mathbf{P}_\tau^2 = \mathbf{P}_{\text{id}} = \mathbf{I}$ .
- (d) Write  $\mathbf{P} = \mathbf{P}_\tau$ . Its effect on a vector is to exchange entries  $r$  and  $s$ , which are both beyond position  $k$ , so  $\mathbf{P}\mathbf{v}$  still has its first  $k$  entries zero. Since  $k \neq r, s$ , we have  $\mathbf{P}\mathbf{e}_k = \mathbf{e}_{\tau(k)} = \mathbf{e}_k$ . By (b) and (c),  $\mathbf{P}^\top = \mathbf{P}^{-1} = \mathbf{P}$ , so  $\mathbf{e}_k^\top \mathbf{P} = \mathbf{e}_k^\top \mathbf{P}^\top = (\mathbf{P}\mathbf{e}_k)^\top = \mathbf{e}_k^\top$  by Theorem 0.9.5. Therefore

$$\mathbf{P}(\mathbf{I} + \mathbf{v} \mathbf{e}_k^\top) \mathbf{P} = \mathbf{P}^2 + (\mathbf{P}\mathbf{v})(\mathbf{e}_k^\top \mathbf{P}) = \mathbf{I} + (\mathbf{P}\mathbf{v}) \mathbf{e}_k^\top = \mathbf{G}_k(\mathbf{P}\mathbf{v}),$$

using  $\mathbf{P}^2 = \mathbf{I}$  from (c). This proves the lemma. ■

By (c) and Theorem 2.4.1,  $\mathbf{P}_\tau \mathbf{A}$  is  $\mathbf{A}$  with rows  $r$  and  $s$  swapped, and by (a) every permutation matrix is a product of such swaps applied in turn, since for  $n \geq 2$  every permutation is a product of transpositions (Theorem 0.10.4). So  $\mathbf{P}\mathbf{A}$  is always  $\mathbf{A}$  with its rows rearranged.

### Definition 2.6.3 — PLU factorization

Let  $\mathbf{A} \in M_n(F)$ . A **PLU factorization** of  $\mathbf{A}$  is a triple  $(\mathbf{P}, \mathbf{L}, \mathbf{U})$  of matrices in  $M_n(F)$  such that

$$\begin{aligned} \mathbf{P}\mathbf{A} &= \mathbf{L}\mathbf{U}, & \mathbf{P} &\text{ is a permutation matrix,} \\ \mathbf{L} &\text{ is unit lower triangular,} & \mathbf{U} &\text{ is upper triangular.} \end{aligned}$$

In words: after rearranging the rows of  $\mathbf{A}$  by  $\mathbf{P}$ , the matrix has an LU factorization. By Lemma 2.6.6 (b) this can also be written  $\mathbf{A} = \mathbf{P}^\top \mathbf{L}\mathbf{U}$ , which is why some books write the factorization as  $\mathbf{A} = \mathbf{P}\mathbf{L}\mathbf{U}$  with their  $\mathbf{P}$  our  $\mathbf{P}^\top$ . For example,  $\mathbf{A} = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}$  has the PLU factorization  $(\mathbf{A}, \mathbf{I}, \mathbf{I})$ , since  $\mathbf{A}\mathbf{A} = \mathbf{I} = \mathbf{I} \cdot \mathbf{I}$ . An LU factorization is the special case  $\mathbf{P} = \mathbf{I}$ .

To produce  $\mathbf{P}$ , run elimination and allow a swap whenever a zero pivot has a non-zero entry below it.

*Algorithm. Elimination with row swaps.* Input:  $\mathbf{A} \in M_n(F)$ . Put  $\mathbf{A}^{(0)} = \mathbf{A}$ . For  $k = 1, 2, \dots, n-1$ , with  $\mathbf{A}^{(k-1)} = (a_{ij})$ :

- If  $a_{ik} = 0$  for **every**  $i \geq k$ , set  $\mathbf{P}_k = \mathbf{I}$  and all  $\ell_{ik} = 0$ , and let  $\mathbf{A}^{(k)} = \mathbf{A}^{(k-1)}$ .
- Otherwise choose some  $r_k \geq k$  with  $a_{r_k k} \neq 0$ . Let  $\mathbf{P}_k$  be the matrix swapping rows  $k$  and  $r_k$  (so  $\mathbf{P}_k = \mathbf{I}$  if  $r_k = k$ ), and let  $\mathbf{B} = \mathbf{P}_k \mathbf{A}^{(k-1)}$ . Set  $\ell_{ik} = b_{ik}/b_{kk}$  for  $i > k$ , and obtain  $\mathbf{A}^{(k)}$  from  $\mathbf{B}$  by replacing row  $i$  with (row  $i$ )  $-\ell_{ik} \cdot$  (row  $k$ ) for each  $i > k$ .

This version never stops. Here is the result it delivers.

**Theorem 2.6.7 — Existence of a PLU factorization**

**Every**  $\mathbf{A} \in M_n(F)$  has a PLU factorization.

*Idea. Step roadmap.* ① As before, each step is a matrix:  $\mathbf{A}^{(k)} = \mathbf{G}_k(-\ell_k)\mathbf{P}_k\mathbf{A}^{(k-1)}$ , and the final matrix  $\mathbf{U}$  is upper triangular. ② The swaps are tangled with the multiplier matrices. A swap of two rows below row  $k$  can be pushed to the right past  $\mathbf{G}_k$  at the price of swapping the entries of the multiplier vector: this is Lemma 2.6.6 (d). ③ Once all swaps sit next to  $\mathbf{A}$ , the rest is the LU proof word for word. In practice, ② says: when you swap two rows, also swap the multipliers already stored in those rows.

*Proof.* Run elimination with row swaps on  $\mathbf{A}$ , producing  $\mathbf{A}^{(0)}, \dots, \mathbf{A}^{(n-1)} = \mathbf{U}$ , swap matrices  $\mathbf{P}_1, \dots, \mathbf{P}_{n-1}$  and multipliers  $\ell_{ik}$ . Let  $\ell_k = \sum_{i>k} \ell_{ik}\mathbf{e}_i$ , whose first  $k$  entries are zero.

**Step 1.** Exactly as in Step 1 of the proof of Theorem 2.6.4, every entry of  $\mathbf{A}^{(k)}$  below the diagonal in columns  $1, \dots, k$  is zero. The only new point is the swap in step  $k$ : it exchanges rows  $k$  and  $r_k \geq k$ , whose entries in columns  $j < k$  are below the diagonal and hence zero, so the swap does not disturb these columns. After the swap the pivot  $b_{kk}$  is non-zero, and the new entries below it are  $b_{ik} - \ell_{ik}b_{kk} = 0$ . Hence  $\mathbf{U}$  is upper triangular. By Lemma 2.6.6 (c), Theorem 2.4.1 and Lemma 2.6.3 (a), as in Step 2 of that proof,

$$\mathbf{A}^{(k)} = \mathbf{G}_k(-\ell_k)\mathbf{P}_k\mathbf{A}^{(k-1)} \quad (k = 1, \dots, n-1).$$

This also holds in the first case of the algorithm, where  $\mathbf{P}_k = \mathbf{I}$  and  $\mathbf{G}_k(\mathbf{0}) = \mathbf{I}$ .

**Step 2.** For  $1 \leq k \leq m \leq n-1$  put  $\ell_k^{(m)} = \mathbf{P}_m\mathbf{P}_{m-1} \cdots \mathbf{P}_{k+1}\ell_k$ , with  $\ell_m^{(m)} = \ell_m$ . We claim that the first  $k$  entries of  $\ell_k^{(m)}$  are zero and that

$$\mathbf{G}_m(-\ell_m)\mathbf{P}_m \cdots \mathbf{G}_1(-\ell_1)\mathbf{P}_1 = \mathbf{G}_m(-\ell_m^{(m)}) \cdots \mathbf{G}_1(-\ell_1^{(m)}) \mathbf{P}_m \cdots \mathbf{P}_1.$$

For  $m = 1$  both sides are  $\mathbf{G}_1(-\ell_1)\mathbf{P}_1$ . Suppose the claim holds for  $m$ , and multiply it on the left by  $\mathbf{G}_{m+1}(-\ell_{m+1})\mathbf{P}_{m+1}$ . The matrix  $\mathbf{P}_{m+1}$  is  $\mathbf{I}$  or swaps rows  $m+1$  and  $r_{m+1} > m+1$ , both beyond every  $k \leq m$ . So by Lemma 2.6.6 (c) and (d), for each  $k \leq m$  and each vector  $\mathbf{v}$  with its first  $k$  entries zero,

$$\mathbf{P}_{m+1} \mathbf{G}_k(\mathbf{v}) = \mathbf{P}_{m+1} \mathbf{G}_k(\mathbf{v}) \mathbf{P}_{m+1} \mathbf{P}_{m+1} = \mathbf{G}_k(\mathbf{P}_{m+1}\mathbf{v}) \mathbf{P}_{m+1},$$

and  $\mathbf{P}_{m+1}\mathbf{v}$  again has its first  $k$  entries zero. Applying this for  $k = m, m-1, \dots, 1$  moves  $\mathbf{P}_{m+1}$  to the right past every factor  $\mathbf{G}_k$ , replacing  $\ell_k^{(m)}$  by  $\mathbf{P}_{m+1}\ell_k^{(m)} = \ell_k^{(m+1)}$  (note  $\mathbf{P}_{m+1}(-\mathbf{v}) = -\mathbf{P}_{m+1}\mathbf{v}$ ). This is the claim for  $m+1$ .

**Step 3.** By Step 1,  $\mathbf{U} = \mathbf{G}_{n-1}(-\ell_{n-1})\mathbf{P}_{n-1} \cdots \mathbf{G}_1(-\ell_1)\mathbf{P}_1\mathbf{A}$ . Write  $\ell'_k = \ell_k^{(n-1)}$  and  $\mathbf{P} = \mathbf{P}_{n-1} \cdots \mathbf{P}_1$ . By Step 2,

$$\mathbf{U} = \mathbf{G}_{n-1}(-\ell'_{n-1}) \cdots \mathbf{G}_1(-\ell'_1) \mathbf{P} \mathbf{A}.$$

$\mathbf{P}$  is a permutation matrix by Lemma 2.6.6 (a) and (c). Multiplying on the left by  $\mathbf{G}_{n-1}(\ell'_{n-1})$ , then  $\mathbf{G}_{n-2}(\ell'_{n-2})$ , and so on, and using Lemma 2.6.3 (b) and (c) as in Step

3 of the proof of [Theorem 2.6.4](#),

$$\mathbf{PA} = \mathbf{LU}, \quad \mathbf{L} = \mathbf{I} + \sum_{k=1}^{n-1} \ell'_k \mathbf{e}_k^\top,$$

where  $\mathbf{L}$  is unit lower triangular because the first  $k$  entries of each  $\ell'_k$  are zero. So  $(\mathbf{P}, \mathbf{L}, \mathbf{U})$  is a PLU factorization of  $\mathbf{A}$ . This proves the theorem. ■

### Example. (A swap in the middle)

Find a PLU factorization of

$$\mathbf{A} = \begin{pmatrix} 1 & 2 & 1 \\ 2 & 4 & 3 \\ 3 & 7 & 2 \end{pmatrix} \in M_3(\mathbb{Q}).$$

*Solution. Step 1.* The pivot 1 is non-zero, so no swap. The multipliers are  $\ell_{21} = 2$  and  $\ell_{31} = 3$ :

$$\mathbf{A} \xrightarrow[R_3 \rightarrow R_3 - 3R_1]{R_2 \rightarrow R_2 - 2R_1} \begin{pmatrix} 1 & 2 & 1 \\ 0 & 0 & 1 \\ 0 & 1 & -1 \end{pmatrix}, \quad \ell_1 = \begin{pmatrix} 0 \\ 2 \\ 3 \end{pmatrix}.$$

*Step 2.* The pivot position (2, 2) holds 0, but the entry below it is 1. Elimination without row swaps would stop here. We swap rows 2 and 3, so  $\mathbf{P}_2 = \mathbf{P}_{23}$ :

$$\xrightarrow{R_2 \leftrightarrow R_3} \begin{pmatrix} 1 & 2 & 1 \\ 0 & 1 & -1 \\ 0 & 0 & 1 \end{pmatrix} = \mathbf{U}.$$

The entry below the new pivot is already 0, so  $\ell_{32} = 0$ , and  $\mathbf{U}$  is upper triangular.

By Step 2 of the proof, the stored multipliers of step 1 are swapped too:  $\ell'_1 = \mathbf{P}_2 \ell_1 = (0, 3, 2)$ . Hence

$$\mathbf{P} = \begin{pmatrix} 1 & 0 & 0 \\ 0 & 0 & 1 \\ 0 & 1 & 0 \end{pmatrix}, \quad \mathbf{L} = \begin{pmatrix} 1 & 0 & 0 \\ 3 & 1 & 0 \\ 2 & 0 & 1 \end{pmatrix}.$$

Check:  $\mathbf{PA}$  is  $\mathbf{A}$  with rows 2 and 3 swapped, and

$$\mathbf{LU} = \begin{pmatrix} 1 & 2 & 1 \\ 3 & 6+1 & 3-1 \\ 2 & 4 & 2+1 \end{pmatrix} = \begin{pmatrix} 1 & 2 & 1 \\ 3 & 7 & 2 \\ 2 & 4 & 3 \end{pmatrix} = \mathbf{PA}.$$

Notice where the multipliers ended up: 3 in row 2 and 2 in row 3, traveling with the rows they belong to. To solve  $\mathbf{Ax} = \mathbf{b}$ , multiply by  $\mathbf{P}$  and solve  $\mathbf{LUx} = \mathbf{Pb}$  by two substitutions; since  $\mathbf{P}$  is invertible, this changes no solutions. For  $\mathbf{b} = (2, 3, 8)$ ,  $\mathbf{Pb} = (2, 8, 3)$ . Forward substitution in  $\mathbf{Ly} = \mathbf{Pb}$  gives  $y_1 = 2$ ,  $y_2 = 8 - 3 \cdot 2 = 2$ ,  $y_3 = 3 - 2 \cdot 2 = -1$ . Back substitution in  $\mathbf{Ux} = \mathbf{y}$  gives  $x_3 = -1$ ,  $x_2 = 2 + x_3 = 1$ ,  $x_1 = 2 - 2x_2 - x_3 = 1$ . Check:  $\mathbf{A}(1, 1, -1) = (1 + 2 - 1, 2 + 4 - 3, 3 + 7 - 2) = (2, 3, 8)$ .

*Remark.* Over  $\mathbb{R}$ , computers do not choose just any non-zero pivot. They choose the entry of **largest absolute value** in the column, which is called **partial pivoting**. The reason is rounding. For  $\mathbf{A} = \begin{pmatrix} \varepsilon & 1 \\ 1 & 1 \end{pmatrix}$  with  $\varepsilon = 10^{-20}$ , elimination without swaps gives the multiplier  $10^{20}$  and  $u_{22} = 1 - 10^{20}$ , which in standard double-precision arithmetic is stored as  $-10^{20}$ . Multiplying back, the computed LU has  $(2, 2)$ -entry  $10^{20} - 10^{20} = 0$  instead of 1. Swapping the rows first gives the harmless multiplier  $\varepsilon$ . Chapter 23 makes this precise.

## 2.6.8 Symmetric matrices: the $\mathbf{LDL}^\top$ factorization

When  $\mathbf{A}$  is symmetric, the two factors of  $\mathbf{A} = \mathbf{LU}$  are secretly related: up to a diagonal matrix,  $\mathbf{U}$  is the transpose of  $\mathbf{L}$ . In [Example \(An LU factorization from elimination\)](#) the factors look unrelated, but that matrix is not symmetric. Try a symmetric one:

$$\begin{pmatrix} 1 & 3 & -2 \\ 3 & 7 & -8 \\ -2 & -8 & 5 \end{pmatrix} \xrightarrow[R_3 \rightarrow R_3 + 2R_1]{R_2 \rightarrow R_2 - 3R_1} \begin{pmatrix} 1 & 3 & -2 \\ 0 & -2 & -2 \\ 0 & -2 & 1 \end{pmatrix} \xrightarrow{R_3 \rightarrow R_3 - R_2} \begin{pmatrix} 1 & 3 & -2 \\ 0 & -2 & -2 \\ 0 & 0 & 3 \end{pmatrix} = \mathbf{U},$$

with multipliers  $\ell_{21} = 3$ ,  $\ell_{31} = -2$ ,  $\ell_{32} = 1$ . Divide each row of  $\mathbf{U}$  by its diagonal entry: the rows become  $(1, 3, -2)$ ,  $(0, 1, 1)$ ,  $(0, 0, 1)$ . These are the columns of  $\mathbf{L} = \begin{pmatrix} 1 & 0 & 0 \\ 3 & 1 & 0 \\ -2 & 1 & 1 \end{pmatrix}$ , read as rows. So  $\mathbf{U} = \mathbf{DL}^\top$  with  $\mathbf{D} = \text{diag}(1, -2, 3)$ , and  $\mathbf{A} = \mathbf{LDL}^\top$ .

### Theorem 2.6.8 — $\mathbf{LDL}^\top$ factorization

Let  $\mathbf{A} \in M_n(F)$  be **symmetric** and **invertible**, and suppose  $\mathbf{A} = \mathbf{LU}$  is an LU factorization. Then the diagonal entries  $u_{11}, \dots, u_{nn}$  of  $\mathbf{U}$  are non-zero, and with  $\mathbf{D} = \text{diag}(u_{11}, \dots, u_{nn})$ ,

$$\mathbf{U} = \mathbf{DL}^\top \quad \text{and} \quad \mathbf{A} = \mathbf{LDL}^\top.$$

*Idea.* Pull the diagonal out of  $\mathbf{U}$ , writing  $\mathbf{U} = \mathbf{DV}$  with  $\mathbf{V}$  unit upper triangular, so  $\mathbf{A} = \mathbf{LDV}$ . Transposing,  $\mathbf{A} = \mathbf{A}^\top = \mathbf{V}^\top(\mathbf{DL}^\top)$ , which is a second LU factorization of the same invertible matrix. Uniqueness forces the two to agree, so  $\mathbf{L} = \mathbf{V}^\top$ .

*Proof.* As in the proof of [Theorem 2.6.5](#),  $\mathbf{U} = \mathbf{L}^{-1}\mathbf{A}$  is invertible, so its diagonal entries are non-zero by [Lemma 2.6.2](#). Hence  $\mathbf{D}$  is invertible, with  $\mathbf{D}^{-1} = \text{diag}(u_{11}^{-1}, \dots, u_{nn}^{-1})$ . Let  $\mathbf{V} = \mathbf{D}^{-1}\mathbf{U}$ . Row  $i$  of  $\mathbf{V}$  is  $u_{ii}^{-1}$  times row  $i$  of  $\mathbf{U}$ , so  $\mathbf{V}$  is upper triangular with  $v_{ii} = u_{ii}^{-1}u_{ii} = 1$ , and  $\mathbf{A} = \mathbf{LDV}$ .

Since  $\mathbf{A}$  is symmetric and  $\mathbf{D}^\top = \mathbf{D}$ , [Theorem 0.9.5](#) gives

$$\mathbf{A} = \mathbf{A}^\top = (\mathbf{LDV})^\top = \mathbf{V}^\top \mathbf{DL}^\top = \mathbf{V}^\top (\mathbf{DL}^\top).$$

Here  $\mathbf{V}^\top$  is unit lower triangular, because  $\mathbf{V}$  is unit upper triangular. And  $\mathbf{DL}^\top$  is upper triangular: its  $(i, j)$ -entry is  $u_{ii}\ell_{ji}$ , and  $\ell_{ji} = 0$  when  $j < i$ , since  $\mathbf{L}$  is lower triangular. So  $(\mathbf{V}^\top, \mathbf{DL}^\top)$  is an LU factorization of the invertible matrix  $\mathbf{A}$ . By [Theorem 2.6.5](#),

$\mathbf{V}^\top = \mathbf{L}$  and  $\mathbf{D}\mathbf{L}^\top = \mathbf{U}$ . Therefore  $\mathbf{A} = \mathbf{L}\mathbf{U} = \mathbf{L}\mathbf{D}\mathbf{L}^\top$ , as claimed. ■

The factorization stores about half as much:  $\mathbf{L}$  and the diagonal  $\mathbf{D}$  determine everything. Over  $\mathbb{R}$ , when every entry of  $\mathbf{D}$  is positive, one can split  $\mathbf{D} = \mathbf{D}^{1/2}\mathbf{D}^{1/2}$  and write  $\mathbf{A} = (\mathbf{L}\mathbf{D}^{1/2})(\mathbf{L}\mathbf{D}^{1/2})^\top$ ; this is the **Cholesky factorization**, which Chapter 12 connects with positive definite matrices.

*Remark.* Symmetry alone does not give an LU factorization:  $\begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}$  is symmetric and invertible, and has none by [Example \(An invertible matrix with no LU factorization\)](#). The theorem assumes the factorization exists and then improves it.

## 2.6.9 Exercises

### 2.6.9.1 A. Check your understanding

#### Exercise A1.

- Define an LU factorization of a matrix  $\mathbf{A} \in M_n(F)$ , and a PLU factorization.
- True or false: every invertible matrix has an LU factorization. Justify your answer.
- Suppose  $\mathbf{A} = \mathbf{L}\mathbf{U}$ . In which order are the two triangular systems solved to find a solution of  $\mathbf{A}\mathbf{x} = \mathbf{b}$ , and what is each one?
- True or false: the product of two unit lower triangular  $n \times n$  matrices is unit lower triangular. Justify your answer.
- True or false: the zero matrix in  $M_2(F)$  has more than one LU factorization. Justify your answer.
- What is the inverse of a permutation matrix  $\mathbf{P}$ ?

*Solution.*

- See [Definition 2.6.1](#):  $\mathbf{A} = \mathbf{L}\mathbf{U}$  with  $\mathbf{L}$  unit lower triangular and  $\mathbf{U}$  upper triangular. See [Definition 2.6.3](#):  $\mathbf{P}\mathbf{A} = \mathbf{L}\mathbf{U}$  with  $\mathbf{P}$  a permutation matrix,  $\mathbf{L}$  unit lower triangular and  $\mathbf{U}$  upper triangular.
- False.  $\begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}$  is invertible but has no LU factorization, by [Example \(An invertible matrix with no LU factorization\)](#).
- First  $\mathbf{L}\mathbf{y} = \mathbf{b}$ , by forward substitution; then  $\mathbf{U}\mathbf{x} = \mathbf{y}$ , by back substitution. Then  $\mathbf{A}\mathbf{x} = \mathbf{L}(\mathbf{U}\mathbf{x}) = \mathbf{L}\mathbf{y} = \mathbf{b}$ .
- True, by [Lemma 2.6.1](#) (a).
- True. For every  $c \in F$ ,  $\begin{pmatrix} 1 & 0 \\ c & 1 \end{pmatrix} \begin{pmatrix} 0 \\ 0 \end{pmatrix} = \begin{pmatrix} 0 \\ 0 \end{pmatrix}$ , so  $\left(\begin{pmatrix} 1 & 0 \\ c & 1 \end{pmatrix}, \begin{pmatrix} 0 \\ 0 \end{pmatrix}\right)$  is an LU factorization;  $c = 0$  and  $c = 1$  give two different ones. This does not contradict [Theorem 2.6.5](#), because 0 is not invertible.
- $\mathbf{P}^{-1} = \mathbf{P}^\top$ , by [Lemma 2.6.6](#) (b).

## 2.6.9.2 B. Practice

**Exercise B1: An LU factorization.**

Find the LU factorization of

$$\mathbf{A} = \begin{pmatrix} 1 & 2 & 0 \\ -3 & -7 & 4 \\ 2 & 6 & -3 \end{pmatrix} \in M_3(\mathbb{Q})$$

by elimination without row swaps. Hence show that  $\mathbf{A}$  is invertible.

*Solution. Step 1.* The pivot is 1, with multipliers  $\ell_{21} = -3$  and  $\ell_{31} = 2$ :

$$\mathbf{A} \xrightarrow[R_3 \rightarrow R_3 - 2R_1]{R_2 \rightarrow R_2 - (-3)R_1} \begin{pmatrix} 1 & 2 & 0 \\ 0 & -1 & 4 \\ 0 & 2 & -3 \end{pmatrix}.$$

*Step 2.* The pivot is  $-1$ , with multiplier  $\ell_{32} = 2/(-1) = -2$ :

$$\xrightarrow{R_3 \rightarrow R_3 - (-2)R_2} \begin{pmatrix} 1 & 2 & 0 \\ 0 & -1 & 4 \\ 0 & 0 & 5 \end{pmatrix} = \mathbf{U}.$$

By [Theorem 2.6.4](#),

$$\mathbf{L} = \begin{pmatrix} 1 & 0 & 0 \\ -3 & 1 & 0 \\ 2 & -2 & 1 \end{pmatrix}, \quad \mathbf{A} = \mathbf{LU}.$$

$$\text{Check: } \mathbf{LU} = \begin{pmatrix} 1 & 2 & 0 \\ -3 & -6-1 & 4 \\ 2 & 4+2 & -8+5 \end{pmatrix} = \mathbf{A}.$$

The diagonal entries  $1, -1, 5$  of  $\mathbf{U}$  are non-zero, so  $\mathbf{U}$  is invertible by [Lemma 2.6.2](#), and  $\mathbf{L}$  is invertible by [Lemma 2.6.1](#). Hence  $\mathbf{A} = \mathbf{LU}$  is invertible by [Theorem 0.9.7](#).

**Exercise B2: Two systems, one factorization.**

With  $\mathbf{A} = \mathbf{LU}$  as in [Exercise \(B1: An LU factorization\)](#), solve  $\mathbf{Ax} = \mathbf{b}$  for  $\mathbf{b} = (3, -6, 5)$  and for  $\mathbf{b} = (0, 1, -2)$ , using only forward and back substitution.

*Solution. For  $\mathbf{b} = (3, -6, 5)$ .*  $\mathbf{Ly} = \mathbf{b}$ :  $y_1 = 3$ ;  $-3y_1 + y_2 = -6$ , so  $y_2 = 3$ ;  $2y_1 - 2y_2 + y_3 = 5$ , so  $y_3 = 5 - 6 + 6 = 5$ . Then  $\mathbf{Ux} = \mathbf{y}$ :  $5x_3 = 5$ , so  $x_3 = 1$ ;  $-x_2 + 4x_3 = 3$ , so  $x_2 = 1$ ;  $x_1 + 2x_2 = 3$ , so  $x_1 = 1$ . Hence  $\mathbf{x} = (1, 1, 1)$ . Check:  $\mathbf{Ax} = (1+2, -3-7+4, 2+6-3) = (3, -6, 5)$ .

*For  $\mathbf{b} = (0, 1, -2)$ .*  $\mathbf{Ly} = \mathbf{b}$ :  $y_1 = 0$ ;  $y_2 = 1 + 3y_1 = 1$ ;  $y_3 = -2 - 2y_1 + 2y_2 = 0$ . Then  $\mathbf{Ux} = \mathbf{y}$ :  $5x_3 = 0$ , so  $x_3 = 0$ ;  $-x_2 + 0 = 1$ , so  $x_2 = -1$ ;  $x_1 + 2(-1) = 0$ , so  $x_1 = 2$ . Hence  $\mathbf{x} = (2, -1, 0)$ . Check:  $\mathbf{Ax} = (2-2, -6+7, 4-6) = (0, 1, -2)$ .

Since  $\mathbf{A}$  is invertible ([Exercise \(B1: An LU factorization\)](#)), each solution is the only one, by [Theorem 0.9.7](#).

**Exercise B3: A PLU factorization.**

Let  $\mathbf{A} = \begin{pmatrix} 0 & 1 & 2 \\ 1 & 1 & 1 \\ 2 & 3 & 1 \end{pmatrix} \in M_3(\mathbb{Q})$ .

- (a) Explain why  $\mathbf{A}$  has no LU factorization.
- (b) Find a PLU factorization of  $\mathbf{A}$ .
- (c) Hence solve  $\mathbf{Ax} = (0, 2, 7)$ .

*Solution.*

- (a) If  $\mathbf{A} = \mathbf{LU}$ , then the  $(1, 1)$ -entry gives  $u_{11} = a_{11} = 0$ , and the  $(2, 1)$ -entry gives  $a_{21} = \ell_{21}u_{11} = 0$ . But  $a_{21} = 1 \neq 0$ , a contradiction, exactly as in [Example \(An invertible matrix with no LU factorization\)](#).
- (b) The first pivot position holds 0 and the entry below is 1, so swap rows 1 and 2,  $\mathbf{P}_1 = \mathbf{P}_{12}$ :

$$\begin{aligned} \mathbf{A} &\xrightarrow{R_1 \leftrightarrow R_2} \begin{pmatrix} 1 & 1 & 1 \\ 0 & 1 & 2 \\ 2 & 3 & 1 \end{pmatrix} \xrightarrow[R_3 \rightarrow R_3 - 2R_1]{R_2 \rightarrow R_2 - 0R_1} \begin{pmatrix} 1 & 1 & 1 \\ 0 & 1 & 2 \\ 0 & 1 & -1 \end{pmatrix} \\ &\xrightarrow{R_3 \rightarrow R_3 - 1R_2} \begin{pmatrix} 1 & 1 & 1 \\ 0 & 1 & 2 \\ 0 & 0 & -3 \end{pmatrix} = \mathbf{U}. \end{aligned}$$

The multipliers are  $\ell_{21} = 0$ ,  $\ell_{31} = 2$ ,  $\ell_{32} = 1$ , and no swap happens after step 1, so they need no rearranging. Hence

$$\mathbf{P} = \begin{pmatrix} 0 & 1 & 0 \\ 1 & 0 & 0 \\ 0 & 0 & 1 \end{pmatrix}, \quad \mathbf{L} = \begin{pmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 2 & 1 & 1 \end{pmatrix}.$$

$$\text{Check: } \mathbf{PA} = \begin{pmatrix} 1 & 1 & 1 \\ 0 & 1 & 2 \\ 2 & 3 & 1 \end{pmatrix}, \text{ and } \mathbf{LU} = \begin{pmatrix} 1 & 1 & 1 \\ 0 & 1 & 2 \\ 2 & 2+1 & 2+2-3 \end{pmatrix} = \mathbf{PA}.$$

- (c)  $\mathbf{Ax} = \mathbf{b}$  holds if and only if  $\mathbf{PAx} = \mathbf{Pb}$ , since  $\mathbf{P}$  is invertible ([Lemma 2.6.6](#) (b)). Swapping the first two entries of  $\mathbf{b} = (0, 2, 7)$ , we solve  $\mathbf{LUx} = \mathbf{Pb} = (2, 0, 7)$ .  $\mathbf{Ly} = (2, 0, 7)$ :  $y_1 = 2$ ,  $y_2 = 0$ ,  $y_3 = 7 - 2 \cdot 2 - 0 = 3$ .  $\mathbf{Ux} = \mathbf{y}$ :  $-3x_3 = 3$ , so  $x_3 = -1$ ;  $x_2 + 2x_3 = 0$ , so  $x_2 = 2$ ;  $x_1 + x_2 + x_3 = 2$ , so  $x_1 = 1$ . Hence  $\mathbf{x} = (1, 2, -1)$ . Check:  $\mathbf{Ax} = (0 + 2 - 2, 1 + 2 - 1, 2 + 6 - 1) = (0, 2, 7)$ .

**2.6.9.3 C. Going deeper****Exercise C1: Leading principal submatrices.**

For  $\mathbf{A} \in M_n(F)$  and  $1 \leq k \leq n$ , the **leading principal submatrix**  $\mathbf{A}_k \in M_k(F)$  is the top-left  $k \times k$  corner of  $\mathbf{A}$ , with entries  $a_{ij}$  for  $i, j \leq k$ .

- (a) Let  $\mathbf{L}$  be lower triangular and  $\mathbf{U}$  upper triangular in  $M_n(F)$ . Show that  $(\mathbf{LU})_k = \mathbf{L}_k \mathbf{U}_k$  for every  $k$ .
- (b) Deduce that if  $\mathbf{A}$  has an LU factorization  $\mathbf{A} = \mathbf{LU}$  with  $\mathbf{U}$  invertible, then every



leading principal submatrix  $\mathbf{A}_1, \dots, \mathbf{A}_n$  is invertible.

- (c) Prove the converse: if  $\mathbf{A}_1, \dots, \mathbf{A}_n$  are all invertible, then  $\mathbf{A}$  has an LU factorization with  $\mathbf{U}$  invertible.

*Hint for (c): induction on  $n$ . Write  $\mathbf{A}$  with  $\mathbf{A}_{n-1}$  in its top-left corner, a column  $\mathbf{b}$ , a row  $\mathbf{c}^\top$  and a corner entry  $d$ , and look for  $\mathbf{L}$  and  $\mathbf{U}$  of the same shape.*

*Solution.*

- (a) Let  $i, j \leq k$ . By [Definition 0.9.9](#),  $(\mathbf{LU})_{ij} = \sum_{l=1}^n \ell_{il} u_{lj}$ . If  $l > k$ , then  $l > i$ , so  $\ell_{il} = 0$  because  $\mathbf{L}$  is lower triangular. Hence  $(\mathbf{LU})_{ij} = \sum_{l=1}^k \ell_{il} u_{lj} = (\mathbf{L}_k \mathbf{U}_k)_{ij}$ , as claimed.
- (b) Suppose  $\mathbf{A} = \mathbf{LU}$  with  $\mathbf{U}$  invertible. By (a),  $\mathbf{A}_k = \mathbf{L}_k \mathbf{U}_k$ . The matrix  $\mathbf{L}_k$  is unit lower triangular, so it is invertible by [Lemma 2.6.1](#). By [Lemma 2.6.2](#),  $u_{11}, \dots, u_{nn}$  are non-zero;  $\mathbf{U}_k$  is upper triangular with diagonal  $u_{11}, \dots, u_{kk}$ , so  $\mathbf{U}_k$  is invertible by the same lemma. Hence  $\mathbf{A}_k$  is invertible by [Theorem 0.9.7](#).
- (c) We prove by induction on  $n$  that every  $\mathbf{A} \in M_n(F)$  whose leading principal submatrices are all invertible has an LU factorization with  $\mathbf{U}$  invertible. For  $n = 1$ ,  $\mathbf{A} = (a_{11})$  with  $a_{11} \neq 0$ , and  $\mathbf{A} = (1)(a_{11})$  works.

Let  $n \geq 2$  and suppose the statement holds for  $n - 1$ . Write

$$\mathbf{A} = \begin{pmatrix} \mathbf{A}_{n-1} & \mathbf{b} \\ \mathbf{c}^\top & d \end{pmatrix}, \quad \mathbf{b}, \mathbf{c} \in F^{n-1}, d \in F.$$

The leading principal submatrices of  $\mathbf{A}_{n-1}$  are  $\mathbf{A}_1, \dots, \mathbf{A}_{n-1}$ , all invertible, so by the induction hypothesis  $\mathbf{A}_{n-1} = \mathbf{L}' \mathbf{U}'$  with  $\mathbf{L}'$  unit lower triangular and  $\mathbf{U}'$  upper triangular and invertible. Put  $\mathbf{y} = \mathbf{L}'^{-1} \mathbf{b}$ ,  $\mathbf{x}^\top = \mathbf{c}^\top \mathbf{U}'^{-1}$  and  $z = d - \mathbf{x}^\top \mathbf{y}$ , and let

$$\mathbf{L} = \begin{pmatrix} \mathbf{L}' & \mathbf{0} \\ \mathbf{x}^\top & 1 \end{pmatrix}, \quad \mathbf{U} = \begin{pmatrix} \mathbf{U}' & \mathbf{y} \\ \mathbf{0}^\top & z \end{pmatrix}.$$

Then  $\mathbf{L}$  is unit lower triangular and  $\mathbf{U}$  is upper triangular. Computing  $\mathbf{LU}$  entry by entry, splitting each sum  $\sum_{l=1}^n$  into  $l \leq n - 1$  and  $l = n$ : the top-left block is  $\mathbf{L}' \mathbf{U}' + \mathbf{0} \mathbf{0}^\top = \mathbf{A}_{n-1}$ ; the top-right column is  $\mathbf{L}' \mathbf{y} + \mathbf{0} z = \mathbf{b}$ ; the bottom-left row is  $\mathbf{x}^\top \mathbf{U}' + 1 \cdot \mathbf{0}^\top = \mathbf{c}^\top$ ; the corner is  $\mathbf{x}^\top \mathbf{y} + z = d$ . Hence  $\mathbf{A} = \mathbf{LU}$ .

Finally  $\mathbf{A} = \mathbf{A}_n$  is invertible by hypothesis, and  $\mathbf{L}$  is invertible by [Lemma 2.6.1](#), so  $\mathbf{U} = \mathbf{L}^{-1} \mathbf{A}$  is invertible by [Theorem 0.9.7](#). This completes the induction.

In the language of elimination:  $z$  is the  $n$ -th pivot, and (c) shows that invertible leading corners are exactly what keeps every pivot non-zero.

### Exercise C2: An $\mathbf{LDL}^\top$ factorization.

Let  $\mathbf{A} = \begin{pmatrix} 4 & -4 & 8 \\ -4 & 5 & -11 \\ 8 & -11 & 27 \end{pmatrix} \in M_3(\mathbb{R})$ .

- (a) Find the LU factorization of  $\mathbf{A}$ , show that  $\mathbf{A}$  is invertible, and write  $\mathbf{A} = \mathbf{LDL}^\top$ .
- (b) Deduce that  $\mathbf{x}^\top \mathbf{A} \mathbf{x} > 0$  for every non-zero  $\mathbf{x} \in \mathbb{R}^3$ .

*Solution.*

- (a) Step 1: pivot 4, multipliers  $\ell_{21} = -1$ ,  $\ell_{31} = 2$ . Row 2 becomes  $(-4, 5, -11) + (4, -4, 8) = (0, 1, -3)$ , row 3 becomes  $(8, -11, 27) - 2(4, -4, 8) = (0, -3, 11)$ . Step 2: pivot 1, multiplier  $\ell_{32} = -3$ ; row 3 becomes  $(0, -3, 11) + 3(0, 1, -3) = (0, 0, 2)$ . Hence

$$\mathbf{L} = \begin{pmatrix} 1 & 0 & 0 \\ -1 & 1 & 0 \\ 2 & -3 & 1 \end{pmatrix}, \quad \mathbf{U} = \begin{pmatrix} 4 & -4 & 8 \\ 0 & 1 & -3 \\ 0 & 0 & 2 \end{pmatrix},$$

and  $\mathbf{A} = \mathbf{LU}$  by [Theorem 2.6.4](#). The diagonal 4, 1, 2 of  $\mathbf{U}$  is non-zero, so  $\mathbf{U}$  is invertible ([Lemma 2.6.2](#)),  $\mathbf{L}$  is invertible ([Lemma 2.6.1](#)), and  $\mathbf{A} = \mathbf{LU}$  is invertible.  $\mathbf{A}$  is symmetric, so by [Theorem 2.6.8](#), with  $\mathbf{D} = \text{diag}(4, 1, 2)$ ,

$$\mathbf{A} = \mathbf{LDL}^\top, \quad \mathbf{DL}^\top = \begin{pmatrix} 4 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 2 \end{pmatrix} \begin{pmatrix} 1 & -1 & 2 \\ 0 & 1 & -3 \\ 0 & 0 & 1 \end{pmatrix} = \begin{pmatrix} 4 & -4 & 8 \\ 0 & 1 & -3 \\ 0 & 0 & 2 \end{pmatrix} = \mathbf{U},$$

which confirms the theorem directly.

- (b) Let  $\mathbf{x} \neq \mathbf{0}$  and put  $\mathbf{y} = \mathbf{L}^\top \mathbf{x}$ . By [Lemma 2.6.1](#),  $\mathbf{L}$  is invertible, so  $\mathbf{L}^\top$  is invertible by [Theorem 0.9.7](#) (4), so  $\mathbf{y} \neq \mathbf{0}$  (otherwise  $\mathbf{x} = (\mathbf{L}^\top)^{-1} \mathbf{y} = \mathbf{0}$ ). By [Theorem 0.9.5](#),  $\mathbf{y}^\top = \mathbf{x}^\top \mathbf{L}$ , so

$$\mathbf{x}^\top \mathbf{A} \mathbf{x} = \mathbf{x}^\top \mathbf{LDL}^\top \mathbf{x} = \mathbf{y}^\top \mathbf{D} \mathbf{y} = 4y_1^2 + y_2^2 + 2y_3^2,$$

reading the  $1 \times 1$  matrix as a real number. Each term is  $\geq 0$ , and at least one  $y_i \neq 0$ , which makes its term strictly positive. Hence  $\mathbf{x}^\top \mathbf{A} \mathbf{x} > 0$ .

### Exercise C3: Counting matrices without an LU factorization.

- (a) Let  $\mathbf{A} = \begin{pmatrix} a & b \\ c & d \end{pmatrix} \in M_2(F)$ . Prove that  $\mathbf{A}$  has an LU factorization if and only if  $a \neq 0$  or  $c = 0$ .
- (b) Let  $q$  be a prime. How many matrices in  $M_2(\mathbb{F}_q)$  have **no** LU factorization? List them for  $q = 2$ .

*Solution.*

- (a)  $(\Rightarrow)$  Suppose  $\mathbf{A} = \mathbf{LU}$  with  $\mathbf{L} = \begin{pmatrix} 1 & 0 \\ \ell & 1 \end{pmatrix}$  and  $\mathbf{U} = \begin{pmatrix} u & v \\ 0 & w \end{pmatrix}$ . As in [Example \(An invertible matrix with no LU factorization\)](#),  $\mathbf{LU} = \begin{pmatrix} u & v \\ \ell u & \ell v + w \end{pmatrix}$ , so  $a = u$  and  $c = \ell u = \ell a$ . If  $a = 0$ , then  $c = 0$ . Hence  $a \neq 0$  or  $c = 0$ .
- $(\Leftarrow)$  If  $a \neq 0$ , elimination without row swaps succeeds with multiplier  $\ell = c/a$ , and indeed

$$\begin{pmatrix} 1 & 0 \\ c/a & 1 \end{pmatrix} \begin{pmatrix} a & b \\ 0 & d - cb/a \end{pmatrix} = \begin{pmatrix} a & b \\ c & cb/a + d - cb/a \end{pmatrix} = \mathbf{A}.$$

If  $c = 0$ , then  $\mathbf{A}$  is upper triangular and  $\mathbf{A} = \mathbf{IA}$ .

- (b) By (a),  $\mathbf{A}$  has no LU factorization exactly when  $a = 0$  and  $c \neq 0$ . There are  $q - 1$  choices of  $c$ , and  $q$  choices each of  $b$  and  $d$ , so there are  $(q - 1)q^2$  such matrices.

For  $q = 2$  this gives 4 of the 16 matrices:  $a = 0$ ,  $c = 1$ , and  $b, d \in \{0, 1\}$ , that is,

$$\begin{pmatrix} 0 & 0 \\ 1 & 0 \end{pmatrix}, \quad \begin{pmatrix} 0 & 0 \\ 1 & 1 \end{pmatrix}, \quad \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}, \quad \begin{pmatrix} 0 & 1 \\ 1 & 1 \end{pmatrix}.$$

## 2.7 Applications of Linear Systems

We now have a complete engine: elimination decides whether  $\mathbf{Ax} = \mathbf{b}$  is consistent, rank-nullity counts the free parameters, and the null space describes the freedom. This section runs the engine on four problems from outside linear algebra. In each one the modeling step is where the thinking happens: which quantities are unknown, why the constraints are linear, and over which field. Each model is also honest about what it leaves out.

### 2.7.1 Polynomial interpolation

Suppose we know the values of an unknown polynomial at a few points, and want the polynomial. Through two points with different  $x$ -values there is exactly one line. Through three there is exactly one parabola, possibly degenerate. The general question: given points  $(x_0, y_0), \dots, (x_n, y_n)$  in  $F^2$ , is there a polynomial  $p \in F[x]_{\leq n}$  with

$$p(x_0) = y_0, \quad p(x_1) = y_1, \quad \dots, \quad p(x_n) = y_n,$$

and is it unique?

The unknowns are the coefficients. Writing  $p = c_0 + c_1x + \dots + c_nx^n$ , the condition  $p(x_i) = y_i$  reads  $c_0 + c_1x_i + \dots + c_nx_i^n = y_i$ , by [Definition 0.8.6](#). The  $x_i$  and  $y_i$  are known numbers, so this is a **linear** equation in  $c_0, \dots, c_n$ , even though  $p$  itself is not linear. The  $n + 1$  conditions form a square system

$$\mathbf{V}\mathbf{c} = \mathbf{y}, \quad \mathbf{V} = \begin{pmatrix} 1 & x_0 & x_0^2 & \cdots & x_0^n \\ 1 & x_1 & x_1^2 & \cdots & x_1^n \\ \vdots & \vdots & \vdots & & \vdots \\ 1 & x_n & x_n^2 & \cdots & x_n^n \end{pmatrix}, \quad \mathbf{c} = \begin{pmatrix} c_0 \\ \vdots \\ c_n \end{pmatrix}, \quad \mathbf{y} = \begin{pmatrix} y_0 \\ \vdots \\ y_n \end{pmatrix}.$$

By [Theorem 2.4.4](#), this has exactly one solution for **every**  $\mathbf{y}$  as soon as  $\mathbf{V}\mathbf{c} = \mathbf{0}$  has only the trivial solution. And  $\mathbf{V}\mathbf{c} = \mathbf{0}$  says that the polynomial with coefficients  $\mathbf{c}$  has the  $n + 1$  roots  $x_0, \dots, x_n$ . So everything hinges on a fact about polynomials: a non-zero polynomial of degree at most  $n$  cannot have  $n + 1$  distinct roots. Chapter 0 proved this only for degree at most 2, in [Exercise \(C1: Roots of a quadratic\)](#), so we prove it now in general. It takes two lemmas.

The first is long division by  $x - c$ , carried out one leading term at a time.

#### Lemma 2.7.1 — Factor Theorem for a Linear Divisor

Let  $p \in F[x]$  and  $c \in F$ . Then there is  $q \in F[x]$  such that

$$p = (x - c)q + p(c).$$

In particular, if  $c$  is a root of  $p$ , then  $p = (x - c)q$ .

*Idea.* Divide as at school. If  $p$  has leading term  $a_d x^d$ , subtract  $a_d x^{d-1}(x - c)$ , which kills the leading term and leaves a polynomial of smaller degree. Repeat until only a constant remains. To turn “repeat” into a proof, use induction on the degree. The constant left over is found by evaluating at  $c$ , where the factor  $x - c$  vanishes.

*Proof.* We first show that  $p = (x - c)q + r$  for some  $q \in F[x]$  and some constant  $r \in F$ , by strong induction on  $\deg p$  ([Theorem 0.2.10](#)), treating  $p = 0$  together with the constants. If  $p = 0$  or  $\deg p = 0$ , then  $p$  is a constant, and  $q = 0$ ,  $r = p$  work.

Let  $d \geq 1$ , and suppose the statement holds for every polynomial of degree less than  $d$ , and for 0. Let  $\deg p = d$  with leading coefficient  $a_d$ , and put

$$p_1 = p - a_d x^{d-1}(x - c) = p - a_d x^d + c a_d x^{d-1}.$$

The coefficient of  $x^d$  in  $p_1$  is  $a_d - a_d = 0$ , and all higher coefficients are 0, so  $p_1 = 0$  or  $\deg p_1 < d$ . By the induction hypothesis,  $p_1 = (x - c)q_1 + r$  with  $r \in F$ . Hence

$$p = (x - c)(a_d x^{d-1} + q_1) + r,$$

which is the statement for  $p$ , with  $q = a_d x^{d-1} + q_1$ .

Now evaluate at  $c$ . By [Theorem 0.8.5](#),  $p(c) = (c - c)q(c) + r = 0 \cdot q(c) + r = r$ . So  $r = p(c)$ , and if  $p(c) = 0$ , then  $p = (x - c)q$ . This proves the lemma. ■

### Lemma 2.7.2 — Root bound

Let  $n \in \mathbb{N}$ , and let  $p \in F[x]$  be **non-zero** with  $\deg p \leq n$ . Then  $p$  has **at most**  $n$  distinct roots in  $F$ .

*Proof.* We use induction on  $n$ . If  $n = 0$ , then  $p = a$  is a non-zero constant, and  $p(c) = a \neq 0$  for every  $c \in F$ , so  $p$  has no roots.

Let  $n \geq 1$ , suppose the statement holds for  $n - 1$ , and let  $p \neq 0$  with  $\deg p \leq n$ . If  $p$  has no root, we are done. Otherwise let  $c$  be a root. By [Lemma 2.7.1](#),  $p = (x - c)q$  for some  $q \in F[x]$ . Then  $q \neq 0$ , since otherwise  $p = 0$ , and by [Theorem 0.8.3](#),  $\deg p = 1 + \deg q$ , so  $\deg q \leq n - 1$ . By the induction hypothesis,  $q$  has at most  $n - 1$  distinct roots.

Let  $s \in F$  be any root of  $p$ . By [Theorem 0.8.5](#),  $0 = p(s) = (s - c)q(s)$ , so  $s - c = 0$  or  $q(s) = 0$  by [Theorem 0.7.1](#). Hence every root of  $p$  is  $c$  or a root of  $q$ , and  $p$  has at most  $1 + (n - 1) = n$  distinct roots. This completes the induction. ■

The field axioms did real work in the last step: a product of two non-zero elements is non-zero. Over  $\mathbb{Z}/8\mathbb{Z}$ , which is not a field,  $x^2 - 1$  has the four roots 1, 3, 5, 7, as [Exercise \(C1: Roots of a quadratic\)](#) showed. And the lemma is sharp: over  $\mathbb{F}_2$ , the polynomial  $x^2 + x$  of [Example \(Different polynomials, same function\)](#) has degree 2 and the two roots 0 and 1.

The lemma also settles a question left open in Chapter 0. If  $F$  is **infinite** and the polynomial function of  $p$  is zero, then every element of  $F$  is a root of  $p$ . If  $p$  were non-zero, this would be more than  $\deg p$  roots, so  $p = 0$ . Over an infinite field, a polynomial is therefore determined by its values: if  $p(c) = q(c)$  for every  $c \in F$ , then  $p - q$  has zero polynomial function by [Theorem 0.8.5](#), so  $p - q = 0$ .

### Theorem 2.7.3 — Polynomial interpolation

Let  $x_0, x_1, \dots, x_n \in F$  be **distinct**, and let  $y_0, y_1, \dots, y_n \in F$  be arbitrary. Then there is **exactly one** polynomial  $p \in F[x]_{\leq n}$  with  $p(x_i) = y_i$  for  $i = 0, 1, \dots, n$ .

*Proof.* A polynomial  $p \in F[x]_{\leq n}$  is the same thing as its coefficient vector  $\mathbf{c} = (c_0, \dots, c_n) \in F^{n+1}$ , since two polynomials are equal exactly when their coefficients agree ([Definition 0.8.1](#)). As computed above,  $p(x_i) = y_i$  for all  $i$  if and only if  $\mathbf{V}\mathbf{c} = \mathbf{y}$ , where  $\mathbf{V} \in M_{n+1}(F)$  has  $(i, j)$ -entry  $x_i^j$  (rows and columns numbered from 0 to  $n$ ).

Let  $\mathbf{V}\mathbf{c} = \mathbf{0}$ , and let  $q = c_0 + c_1x + \dots + c_nx^n$ . Then  $q(x_i) = 0$  for every  $i$ , so  $q$  has the  $n+1$  distinct roots  $x_0, \dots, x_n$ , while  $\deg q \leq n$ . By [Lemma 2.7.2](#),  $q$  cannot be non-zero, so  $q = 0$  and  $\mathbf{c} = \mathbf{0}$ . Hence  $\mathbf{V}\mathbf{c} = \mathbf{0}$  has only the trivial solution, and  $\mathbf{V}$  is invertible by [Theorem 2.4.4](#). By [Theorem 0.9.7](#) (7),  $\mathbf{V}\mathbf{c} = \mathbf{y}$  has exactly one solution  $\mathbf{c} = \mathbf{V}^{-1}\mathbf{y}$ . The polynomial with these coefficients is the unique  $p$  required. This proves the theorem. ■

The proof is a model of how this chapter turns questions into systems. Existence for every right-hand side and uniqueness are both properties of the square matrix  $\mathbf{V}$ , and for a square matrix they come together. So we only had to prove uniqueness for the zero data, and that was a statement about roots. Chapter 5 gives an explicit formula for  $p$ , the Lagrange interpolation formula. Here is the computation by elimination.

#### Example. (A cubic through four points)

Find the polynomial  $p \in \mathbb{Q}[x]_{\leq 3}$  whose graph passes through  $(-1, 2)$ ,  $(0, 1)$ ,  $(1, 2)$  and  $(2, 11)$ .

*Solution.* The  $x$ -values  $-1, 0, 1, 2$  are distinct, so by [Theorem 2.7.3](#) there is exactly one such  $p = c_0 + c_1x + c_2x^2 + c_3x^3$ . The conditions  $p(x_i) = y_i$  give the augmented matrix below. We first swap the rows for  $x = -1$  and  $x = 0$ , so that the simplest row leads:

$$\begin{aligned} \left( \begin{array}{cccc|c} 1 & -1 & 1 & -1 & 2 \\ 1 & 0 & 0 & 0 & 1 \\ 1 & 1 & 1 & 1 & 2 \\ 1 & 2 & 4 & 8 & 11 \end{array} \right) & \xrightarrow{R_1 \leftrightarrow R_2} \left( \begin{array}{cccc|c} 1 & 0 & 0 & 0 & 1 \\ 1 & -1 & 1 & -1 & 2 \\ 1 & 1 & 1 & 1 & 2 \\ 1 & 2 & 4 & 8 & 11 \end{array} \right) \\ & \xrightarrow[\substack{R_2 \rightarrow R_2 - R_1, \quad R_3 \rightarrow R_3 - R_1 \\ R_4 \rightarrow R_4 - R_1}]{R_2 \rightarrow R_2 - R_1, \quad R_3 \rightarrow R_3 - R_1} \left( \begin{array}{cccc|c} 1 & 0 & 0 & 0 & 1 \\ 0 & -1 & 1 & -1 & 1 \\ 0 & 1 & 1 & 1 & 1 \\ 0 & 2 & 4 & 8 & 10 \end{array} \right). \end{aligned}$$

Next clear column 2 below the pivot  $-1$ :

$$\begin{array}{l} R_3 \rightarrow R_3 + R_2 \\ R_4 \rightarrow R_4 + 2R_2 \end{array} \rightarrow \left( \begin{array}{cccc|c} 1 & 0 & 0 & 0 & 1 \\ 0 & -1 & 1 & -1 & 1 \\ 0 & 0 & 2 & 0 & 2 \\ 0 & 0 & 6 & 6 & 12 \end{array} \right) \xrightarrow{R_4 \rightarrow R_4 - 3R_3} \left( \begin{array}{cccc|c} 1 & 0 & 0 & 0 & 1 \\ 0 & -1 & 1 & -1 & 1 \\ 0 & 0 & 2 & 0 & 2 \\ 0 & 0 & 0 & 6 & 6 \end{array} \right).$$

Back substitution:  $6c_3 = 6$ , so  $c_3 = 1$ ;  $2c_2 = 2$ , so  $c_2 = 1$ ;  $-c_1 + c_2 - c_3 = 1$ , so  $c_1 = -1$ ; and  $c_0 = 1$ . Hence

$$p = 1 - x + x^2 + x^3.$$

Check:  $p(-1) = 1+1+1-1 = 2$ ,  $p(0) = 1$ ,  $p(1) = 1-1+1+1 = 2$ ,  $p(2) = 1-2+4+8 = 11$ .

**Warning.** The  $x$ -values must be distinct as elements of  $F$ . If  $x_0 = x_1$ , the first two equations of  $\mathbf{V}\mathbf{c} = \mathbf{y}$  have the same left-hand side, so the system is inconsistent for any data with  $y_0 \neq y_1$ . This depends on the field: over  $\mathbb{F}_3$  the numbers  $-1$  and  $2$  are the **same** element, so the four points of [Example \(A cubic through four points\)](#) do not have distinct  $x$ -values there. Note also that the theorem promises degree **at most**  $n$ : the three points  $(0, 0)$ ,  $(1, 1)$ ,  $(2, 2)$  are interpolated by  $p = x$ , of degree 1.

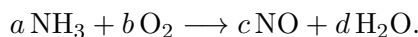
**Quick check.** Find the polynomial  $p \in \mathbb{F}_3[x]_{\leq 2}$  with  $p(0) = 1$ ,  $p(1) = 0$  and  $p(2) = 0$ .

**Solution.** The  $x$ -values  $0, 1, 2$  are distinct in  $\mathbb{F}_3$ , so there is exactly one such  $p$ . Since  $1$  and  $2$  are roots, try  $p = a(x-1)(x-2)$ , which has degree at most 2 and vanishes at  $1$  and  $2$ . Then  $p(0) = a \cdot (-1)(-2) = 2a$ , and  $2a = 1$  gives  $a = 2$ , since  $2 \cdot 2 = 4 = 1$  in  $\mathbb{F}_3$ . Expanding,  $p = 2(x^2 - 3x + 2) = 2(x^2 + 2) = 2x^2 + 1$ . Check:  $p(0) = 1$ ,  $p(1) = 3 = 0$ ,  $p(2) = 9 = 0$ . By uniqueness, this is the answer.

**What the model leaves out.** Interpolation fits the data **exactly**. When the  $y_i$  are measurements with errors, an exact fit of high degree follows the noise and can swing wildly between the data points. The honest tool there is a low-degree polynomial that fits approximately, found by least squares in Chapter 10.

## 2.7.2 Balancing chemical equations

When ammonia burns in oxygen over a catalyst, it produces nitric oxide and water:



The coefficients  $a, b, c, d$  count molecules. The model rests on one physical law: **atoms are neither created nor destroyed**, so each element appears equally often on both sides. For each element this gives one linear equation:

$$\begin{array}{ll} \text{N:} & a = c, \\ \text{H:} & 3a = 2d, \\ \text{O:} & 2b = c + d. \end{array}$$

Moving everything to the left gives a homogeneous system, so the balanced coefficient vectors form the null space of its coefficient matrix. We want positive integers, but we compute over  $\mathbb{Q}$ . Elimination uses only the four field operations on integer data, so it never leaves  $\mathbb{Q}$ , and a rational solution can be scaled to an integer one at the end.

**Example. (Burning ammonia)**

Find all solutions of the system above over  $\mathbb{Q}$ , and hence the balanced equation with the smallest positive integer coefficients.

*Solution.* In the order  $(a, b, c, d)$ , with rows N, H, O, the coefficient matrix is reduced as follows:

$$\begin{pmatrix} 1 & 0 & -1 & 0 \\ 3 & 0 & 0 & -2 \\ 0 & 2 & -1 & -1 \end{pmatrix} \xrightarrow{R_2 \rightarrow R_2 - 3R_1} \begin{pmatrix} 1 & 0 & -1 & 0 \\ 0 & 0 & 3 & -2 \\ 0 & 2 & -1 & -1 \end{pmatrix} \xrightarrow{R_2 \leftrightarrow R_3} \begin{pmatrix} 1 & 0 & -1 & 0 \\ 0 & 2 & -1 & -1 \\ 0 & 0 & 3 & -2 \end{pmatrix}.$$

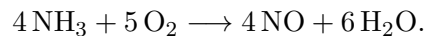
Scale rows 2 and 3 by  $\frac{1}{2}$  and  $\frac{1}{3}$ , then clear column 3 above its pivot:

$$\begin{array}{c} R_2 \rightarrow \frac{1}{2} R_2 \\ R_3 \rightarrow \frac{1}{3} R_3 \end{array} \rightarrow \begin{pmatrix} 1 & 0 & -1 & 0 \\ 0 & 1 & -\frac{1}{2} & -\frac{1}{2} \\ 0 & 0 & 1 & -\frac{2}{3} \end{pmatrix} \xrightarrow{\begin{array}{c} R_1 \rightarrow R_1 + R_3 \\ R_2 \rightarrow R_2 + \frac{1}{2} R_3 \end{array}} \begin{pmatrix} 1 & 0 & 0 & -\frac{2}{3} \\ 0 & 1 & 0 & -\frac{5}{6} \\ 0 & 0 & 1 & -\frac{2}{3} \end{pmatrix}.$$

This is the RREF. The pivot columns are 1, 2, 3, and  $d$  is the only free variable. By [Theorem 2.2.4](#), with  $d = t$ , the solutions are

$$(a, b, c, d) = t \left( \frac{2}{3}, \frac{5}{6}, \frac{2}{3}, 1 \right), \quad t \in \mathbb{Q}.$$

So the null space is a line: rank 3 and nullity  $4 - 3 = 1$ , as [Theorem 2.5.7](#) predicts. The coefficients are integers exactly when  $\frac{2}{3}t$ ,  $\frac{5}{6}t$  and  $t$  are integers, which forces  $t$  to be a multiple of 6. The smallest positive choice  $t = 6$  gives  $(4, 5, 4, 6)$ :



Check: N:  $4 = 4$ . H:  $12 = 12$ . O:  $10 = 4 + 6$ .

Nullity 1 is what makes “the balanced equation” meaningful: every balancing is a multiple of one. Positivity was luck of the data here. Nothing in the algebra guarantees that the null space contains a vector with all entries positive, and when it does not, the proposed reaction cannot be balanced at all.

**Warning.** A null space of dimension 2 or more does not give one balanced equation. For  $a \text{H}_2 + b \text{O}_2 \rightarrow c \text{H}_2\text{O} + d \text{H}_2\text{O}_2$ , the equations  $2a = 2c + 2d$  (hydrogen) and  $2b = c + 2d$  (oxygen) have rank 2 and nullity 2. Both  $(2, 1, 2, 0)$  and  $(1, 1, 0, 1)$  balance, and so does every positive combination of them. The “reaction” is really two reactions,  $2\text{H}_2 + \text{O}_2 \rightarrow 2\text{H}_2\text{O}$  and  $\text{H}_2 + \text{O}_2 \rightarrow \text{H}_2\text{O}_2$ , and the proportions depend on chemistry, not on algebra.

**What the model leaves out.** Conservation of atoms is necessary, not sufficient. The equations cannot tell whether a reaction actually happens, and reactions involving ions also conserve electric charge, which adds one more linear equation.

### 2.7.3 Flows in a network

A small town center has four intersections and five one-way streets. Cars enter and leave the center at some intersections. We measure the external traffic and want to know the traffic on

each street.

The model has one assumption: at every intersection, as many cars leave per minute as arrive, since cars do not pile up or vanish. In electrical circuits the same rule for currents is **Kirchhoff's current law**. To write it down, number the intersections (**vertices**)  $1, \dots, m$  and the streets (**edges**)  $1, \dots, k$ . Each edge  $e$  runs from a vertex  $\text{tail}(e)$  to a **different** vertex  $\text{head}(e)$ . The unknown  $f_e$  is the flow along edge  $e$ . The data  $s_v$  is the net external supply at  $v$ : cars entering the network there per minute, with a negative sign for cars leaving. Conservation at  $v$  says

$$(\text{flow out of } v) - (\text{flow into } v) = s_v.$$

### Definition 2.7.1 — Incidence matrix

The **incidence matrix** of a network with vertices  $1, \dots, m$  and edges  $1, \dots, k$ , where  $k \geq 1$ , is  $\mathbf{N} \in M_{m \times k}(\mathbb{R})$  with

$$\mathbf{N}_{ve} = \begin{cases} 1 & \text{if } v = \text{tail}(e), \\ -1 & \text{if } v = \text{head}(e), \\ 0 & \text{otherwise.} \end{cases}$$

Row  $v$  of  $\mathbf{N}\mathbf{f}$  is  $\sum_e \mathbf{N}_{ve}f_e$ , the flow out of  $v$  minus the flow into  $v$ . So conservation at every vertex is the single matrix equation  $\mathbf{N}\mathbf{f} = \mathbf{s}$ . Column  $e$  of  $\mathbf{N}$  has exactly one 1 and one  $-1$ , so its entries sum to 0. Adding up all the conservation equations therefore gives  $0 = s_1 + \dots + s_m$ : **total inflow equals total outflow**. This is necessary for a solution. The next result shows that, for a network in one piece, it is also sufficient.

A network is **connected** if for any two vertices  $v, w$  there is a sequence of vertices  $v = w_0, w_1, \dots, w_t = w$  in which each consecutive pair is joined by an edge, in either direction.

### Proposition 2.7.4 — Rank of an incidence matrix

Let  $\mathbf{N} \in M_{m \times k}(\mathbb{R})$  be the incidence matrix of a **connected** network with  $m$  vertices. Then  $\text{rank } \mathbf{N} = m - 1$ .

*Idea.* Rank is easier to see through the transpose. A vector  $\mathbf{y} \in \mathbb{R}^m$  assigns a number to each vertex, and  $\mathbf{N}^\top \mathbf{y}$  records, for each edge, the difference of the numbers at its two ends. So  $\mathbf{N}^\top \mathbf{y} = \mathbf{0}$  says that the numbers agree across every edge. Along a chain of edges they then agree everywhere, so the null space of  $\mathbf{N}^\top$  is the line of constant vectors. Rank-nullity and “row rank equals column rank” finish.

*Proof.* Let  $\mathbf{y} \in \mathbb{R}^m$ . For each edge  $e$ , entry  $e$  of  $\mathbf{N}^\top \mathbf{y}$  is  $\sum_v \mathbf{N}_{ve}y_v = y_{\text{tail}(e)} - y_{\text{head}(e)}$ , by Definition 0.9.10 and Definition 2.7.1. Hence  $\mathbf{N}^\top \mathbf{y} = \mathbf{0}$  if and only if  $y_u = y_w$  whenever  $u$  and  $w$  are joined by an edge.

Suppose  $\mathbf{N}^\top \mathbf{y} = \mathbf{0}$ . Given any vertex  $w$ , connectedness gives a sequence  $1 = w_0, w_1, \dots, w_t = w$  with consecutive vertices joined by edges, so  $y_1 = y_{w_1} = \dots = y_w$ . Hence  $\mathbf{y} = y_1 \mathbf{1}$ , where  $\mathbf{1} = (1, \dots, 1) \in \mathbb{R}^m$ . Conversely  $\mathbf{N}^\top \mathbf{1} = \mathbf{0}$ , since all differences vanish. So  $\text{nul}(\mathbf{N}^\top) = \text{span}(\mathbf{1})$ , and since  $\mathbf{1} \neq \mathbf{0}$ , its dimension is 1.

The matrix  $\mathbf{N}^\top$  has  $m$  columns, so Theorem 2.5.7 gives  $\text{rank } \mathbf{N}^\top = m - 1$ . By Theo-



rem 2.5.5,  $\text{rank } \mathbf{N} = \text{rank } \mathbf{N}^\top = m - 1$ . This proves the proposition. ■

### Corollary 2.7.5 — Solvable flow problems

Let  $\mathbf{N} \in M_{m \times k}(\mathbb{R})$  be the incidence matrix of a connected network, and let  $\mathbf{s} \in \mathbb{R}^m$ . Then  $\mathbf{N}\mathbf{f} = \mathbf{s}$  has a solution if and only if  $s_1 + \cdots + s_m = 0$ . In that case the solutions form  $\mathbf{f}_0 + \text{nul}(\mathbf{N})$  for any one solution  $\mathbf{f}_0$ , and  $\dim \text{nul}(\mathbf{N}) = k - m + 1$ .

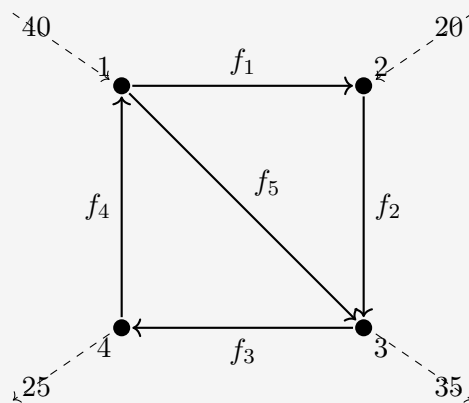
*Proof.* Let  $W = \{\mathbf{z} \in \mathbb{R}^m : z_1 + \cdots + z_m = 0\}$ . It is the solution set of the homogeneous system with the  $1 \times m$  coefficient matrix  $(1 \ \cdots \ 1)$ , so it is a subspace by Theorem 2.1.2. That matrix has rank 1, since its column space is spanned by the non-zero column (1) and so equals  $\mathbb{R}^1$ . By Theorem 2.5.7,  $\dim W = m - 1$ .

Each column of  $\mathbf{N}$  has entries summing to 0, so it lies in  $W$ , and since  $W$  is a subspace,  $\text{col}(\mathbf{N}) \subseteq W$  by Theorem 1.3.1. Also  $\dim \text{col}(\mathbf{N}) = \text{rank } \mathbf{N} = m - 1 = \dim W$ , by Definition 2.5.4 and Proposition 2.7.4. Hence  $\text{col}(\mathbf{N}) = W$  by Theorem 1.6.8. By Theorem 2.1.1,  $\mathbf{N}\mathbf{f} = \mathbf{s}$  is consistent if and only if  $\mathbf{s} \in \text{col}(\mathbf{N}) = W$ , which is the first claim. The description of all solutions is Theorem 2.1.3, and  $\dim \text{nul}(\mathbf{N}) = k - (m - 1)$  by Theorem 2.5.7, since  $\mathbf{N}$  has  $k$  columns. ■

The  $k - m + 1$  free parameters have a clear meaning: they are flows around **loops**. Adding the same amount of traffic all the way around a closed circuit of streets changes no intersection's balance.

#### Example. (Traffic in a town center)

Four intersections are joined by one-way streets  $1 \rightarrow 2$ ,  $2 \rightarrow 3$ ,  $3 \rightarrow 4$ ,  $4 \rightarrow 1$  and  $1 \rightarrow 3$ , carrying unknown flows  $f_1, \dots, f_5$  (cars per minute) in that order. From outside, 40 cars per minute enter at intersection 1 and 20 at intersection 2, while 35 leave at intersection 3 and 25 at intersection 4.



Find all flows that satisfy conservation. If street  $4 \rightarrow 1$  is closed for repairs and flows cannot be negative, what is the largest flow the diagonal street  $1 \rightarrow 3$  may have to

carry?

*Solution.* The network is connected, and  $\mathbf{s} = (40, 20, -35, -25)$  sums to 0, so by [Corollary 2.7.5](#) a solution exists and the solutions have  $5 - 4 + 1 = 2$  free parameters. We row reduce  $[\mathbf{N} \mid \mathbf{s}]$ , with rows for vertices 1, 2, 3, 4 and columns for  $f_1, \dots, f_5$ :

$$\begin{aligned} & \left( \begin{array}{ccccc|c} 1 & 0 & 0 & -1 & 1 & 40 \\ -1 & 1 & 0 & 0 & 0 & 20 \\ 0 & -1 & 1 & 0 & -1 & -35 \\ 0 & 0 & -1 & 1 & 0 & -25 \end{array} \right) \xrightarrow{R_2 \rightarrow R_2 + R_1} \left( \begin{array}{ccccc|c} 1 & 0 & 0 & -1 & 1 & 40 \\ 0 & 1 & 0 & -1 & 1 & 60 \\ 0 & -1 & 1 & 0 & -1 & -35 \\ 0 & 0 & -1 & 1 & 0 & -25 \end{array} \right) \\ & \xrightarrow{R_3 \rightarrow R_3 + R_2} \left( \begin{array}{ccccc|c} 1 & 0 & 0 & -1 & 1 & 40 \\ 0 & 1 & 0 & -1 & 1 & 60 \\ 0 & 0 & 1 & -1 & 0 & 25 \\ 0 & 0 & -1 & 1 & 0 & -25 \end{array} \right) \xrightarrow{R_4 \rightarrow R_4 + R_3} \left( \begin{array}{ccccc|c} 1 & 0 & 0 & -1 & 1 & 40 \\ 0 & 1 & 0 & -1 & 1 & 60 \\ 0 & 0 & 1 & -1 & 0 & 25 \\ 0 & 0 & 0 & 0 & 0 & 0 \end{array} \right). \end{aligned}$$

This is the RREF, with pivots in columns 1, 2, 3, so  $\text{rank } \mathbf{N} = 3 = m - 1$ , as [Proposition 2.7.4](#) predicts. The free variables are  $f_4 = \alpha$  and  $f_5 = \beta$ , and by [Theorem 2.2.4](#),

$$\mathbf{f} = \begin{pmatrix} 40 \\ 60 \\ 25 \\ 0 \\ 0 \end{pmatrix} + \alpha \begin{pmatrix} 1 \\ 1 \\ 1 \\ 1 \\ 0 \end{pmatrix} + \beta \begin{pmatrix} -1 \\ -1 \\ 0 \\ 0 \\ 1 \end{pmatrix}, \quad \alpha, \beta \in \mathbb{R}.$$

The two null space vectors are the loops:  $\alpha$  sends extra cars around  $1 \rightarrow 2 \rightarrow 3 \rightarrow 4 \rightarrow 1$ , and  $\beta$  moves cars from the route  $1 \rightarrow 2 \rightarrow 3$  onto the diagonal  $1 \rightarrow 3$ .

With street  $4 \rightarrow 1$  closed,  $\alpha = f_4 = 0$  and  $\mathbf{f} = (40 - \beta, 60 - \beta, 25, 0, \beta)$ . Non-negativity requires  $\beta \geq 0$ ,  $40 - \beta \geq 0$  and  $60 - \beta \geq 0$ , that is,  $0 \leq \beta \leq 40$ . So the diagonal may carry up to 40 cars per minute, which happens when street  $1 \rightarrow 2$  is empty. Check, for  $\mathbf{f} = (0, 20, 25, 0, 40)$ : at vertex 1, out 0 + 40 minus in 0 is 40; at 2, out 20 minus in 0 is 20; at 3, out 25 minus in 20 + 40 is  $-35$ ; at 4, out 0 minus in 25 is  $-25$ .

**What the model leaves out.** Linear algebra finds every flow that conserves cars, including physically meaningless ones with negative entries, and it cannot say which of the many valid flows drivers actually choose. Non-negativity and street capacities turn the problem into one about inequalities, which is the subject of linear programming. The matrix  $\mathbf{N}\mathbf{N}^\top$  of this model, the graph Laplacian, returns in Chapter 23.

## 2.7.4 Lights out over $\mathbb{F}_2$

A puzzle has a  $3 \times 3$  grid of lights, each on or off. Pressing a light **toggles** it and its horizontal and vertical neighbors (a corner has two neighbors, an edge light three, the center four). Starting from some pattern, the goal is to switch every light off. Can every starting pattern be solved?

The model rests on two observations about the puzzle, and together they say that the right field is  $\mathbb{F}_2 = \{0, 1\}$ .

1. Encode “on” as 1 and “off” as 0. Toggling a light is adding 1 in  $\mathbb{F}_2$ , because  $0 + 1 = 1$  and  $1 + 1 = 0$ . The effect of several presses on a light is the sum of their effects, and since addition is commutative, **the order of presses does not matter**.

2. Pressing a button twice toggles each affected light twice, which is  $1 + 1 = 0$ : nothing. So a strategy is just the **set** of buttons pressed once, a vector  $\mathbf{x} \in \mathbb{F}_2^9$ .

Number the cells  $(i, j)$ , row  $i$ , column  $j$ , and list them row by row. Pressing button  $(i, j)$  adds to the pattern a fixed vector  $\mathbf{a}_{ij} \in \mathbb{F}_2^9$ , with 1's at  $(i, j)$  and its neighbors. Let  $\mathbf{A} \in M_9(\mathbb{F}_2)$  be the matrix with these columns. Starting from pattern  $\mathbf{b}$  and pressing the buttons in  $\mathbf{x}$ , the final pattern is  $\mathbf{b} + \mathbf{A}\mathbf{x}$  by [Theorem 0.9.3](#), and we want it to be  $\mathbf{0}$ . Since  $-\mathbf{b} = \mathbf{b}$  in  $\mathbb{F}_2^9$ , the puzzle is the linear system

$$\mathbf{A}\mathbf{x} = \mathbf{b} \quad \text{over } \mathbb{F}_2, \quad \mathbf{A} = \begin{pmatrix} 1 & 1 & 0 & 1 & 0 & 0 & 0 & 0 & 0 \\ 1 & 1 & 1 & 0 & 1 & 0 & 0 & 0 & 0 \\ 0 & 1 & 1 & 0 & 0 & 1 & 0 & 0 & 0 \\ 1 & 0 & 0 & 1 & 1 & 0 & 1 & 0 & 0 \\ 0 & 1 & 0 & 1 & 1 & 1 & 0 & 1 & 0 \\ 0 & 0 & 1 & 0 & 1 & 1 & 0 & 0 & 1 \\ 0 & 0 & 0 & 1 & 0 & 0 & 1 & 1 & 0 \\ 0 & 0 & 0 & 0 & 1 & 0 & 1 & 1 & 1 \\ 0 & 0 & 0 & 0 & 0 & 1 & 0 & 1 & 1 \end{pmatrix}.$$

By [Theorem 2.1.1](#), pattern  $\mathbf{b}$  is solvable exactly when  $\mathbf{b} \in \text{col}(\mathbf{A})$ . So “is every pattern solvable?” asks whether  $\text{rank } \mathbf{A} = 9$ .

**Example. (Every pattern of the  $3 \times 3$  puzzle is solvable)**

Show that  $\text{rank } \mathbf{A} = 9$ . Deduce that every one of the  $2^9 = 512$  starting patterns can be switched off, by exactly one set of presses.

*Solution.* We show that  $\mathbf{A}\mathbf{x} = \mathbf{0}$  has only the trivial solution. Write the presses as a grid  $x_{ij}$ , and suppose  $\mathbf{A}\mathbf{x} = \mathbf{0}$ : the presses change no light. The equation for light  $(i, j)$  says that the presses at  $(i, j)$  and its neighbors add up to 0. We solve these equations row by row, remembering that  $-u = u$  in  $\mathbb{F}_2$ . Name the top row  $x_{11} = p$ ,  $x_{12} = q$ ,  $x_{13} = r$ .

*Lights of row 1 determine the presses of row 2.* Light  $(1, 1)$  gives  $x_{11} + x_{12} + x_{21} = 0$ , so  $x_{21} = p + q$ . Likewise lights  $(1, 2)$  and  $(1, 3)$  give

$$x_{22} = x_{11} + x_{12} + x_{13} = p + q + r, \quad x_{23} = x_{12} + x_{13} = q + r.$$

*Lights of row 2 determine the presses of row 3.* Light  $(2, 1)$  gives  $x_{31} = x_{11} + x_{21} + x_{22} = p + (p + q) + (p + q + r) = p + r$ , using  $3p = p$  and  $2q = 0$ . Light  $(2, 2)$  gives  $x_{32} = x_{12} + x_{21} + x_{22} + x_{23} = q + (p + q) + (p + q + r) + (q + r) = 0$ . Light  $(2, 3)$  gives  $x_{33} = x_{13} + x_{22} + x_{23} = r + (p + q + r) + (q + r) = p + r$ .

*Lights of row 3 give three conditions on  $p, q, r$ .*

$$\begin{aligned} (3, 1): \quad & x_{21} + x_{31} + x_{32} = (p + q) + (p + r) + 0 = q + r = 0, \\ (3, 2): \quad & x_{22} + x_{31} + x_{32} + x_{33} = (p + q + r) + (p + r) + 0 + (p + r) \\ & \quad \quad \quad = p + q + r = 0, \\ (3, 3): \quad & x_{23} + x_{32} + x_{33} = (q + r) + 0 + (p + r) = p + q = 0. \end{aligned}$$

The first and third give  $r = q$  and  $p = q$ , and then the second gives  $3q = q = 0$ . So  $p = q = r = 0$ , and the formulas above make every  $x_{ij} = 0$ .

Hence the null space of  $\mathbf{A}$  is  $\{\mathbf{0}\}$ , and  $\text{rank } \mathbf{A} = 9 - 0 = 9$  by [Theorem 2.5.7](#). So  $\text{col}(\mathbf{A})$  is a subspace of  $\mathbb{F}_2^9$  of dimension 9, and  $\text{col}(\mathbf{A}) = \mathbb{F}_2^9$  by [Theorem 1.6.8](#). Every pattern

$\mathbf{b}$  lies in  $\text{col}(\mathbf{A})$ , so every pattern is solvable. The solution is unique by [Corollary 2.1.4](#), since  $\mathbf{Ax} = \mathbf{0}$  has only the trivial solution. In particular exactly one of the 512 press sets solves each of the 512 patterns.

The same chase finds an actual solution: the constants from  $\mathbf{b}$  simply ride along.

**Example. (Switching off the center)**

Only the center light is on. Which buttons should be pressed?

*Solution.* Now  $\mathbf{b}$  has a 1 at  $(2, 2)$  only, so the equation for light  $(2, 2)$  has right-hand side 1 and all others 0. Rows 1 and 2 of lights give, as before,  $x_{21} = p + q$ ,  $x_{22} = p + q + r$ ,  $x_{23} = q + r$ ,  $x_{31} = p + r$ ,  $x_{33} = p + r$ , and now

$$x_{32} = 1 + x_{12} + x_{21} + x_{22} + x_{23} = 1.$$

The lights of row 3 give  $(p + q) + (p + r) + 1 = 0$ ,  $(p + q + r) + (p + r) + 1 + (p + r) = 0$  and  $(q + r) + 1 + (p + r) = 0$ , that is,

$$q + r = 1, \quad p + q + r = 1, \quad p + q = 1.$$

Adding the first two gives  $p = 0$ ; then  $q = 1$  and  $r = 0$ . Substituting,  $x_{21} = x_{22} = x_{23} = 1$ ,  $x_{31} = x_{33} = 0$  and  $x_{32} = 1$ . So we press the center and its four neighbors, a plus sign:

|   |   |   |
|---|---|---|
|   | × |   |
| × | × | × |
|   | × |   |

Check, counting how often each light is toggled. The center is toggled by all five presses: odd, so it goes from on to off. A corner, say  $(1, 1)$ , is toggled by presses at  $(1, 2)$  and  $(2, 1)$ : even, so it stays off. An edge light, say  $(1, 2)$ , is toggled by presses at  $(1, 2)$  and  $(2, 2)$ : even, so it stays off. By the symmetry of the plus sign the other corners and edges behave the same way.

*Quick check.* Starting from all lights **off**, which set of presses produces the pattern with only the center light **on**?

*Solution.* The same plus sign. Starting from  $\mathbf{0}$ , presses  $\mathbf{x}$  produce the pattern  $\mathbf{Ax}$ , so we need  $\mathbf{Ax} = \mathbf{b}$  with  $\mathbf{b}$  the center pattern, which is exactly the system just solved. Over  $\mathbb{F}_2$ , turning a pattern on and turning it off are the same problem.

*Remark.* Larger boards behave differently. A computation over  $\mathbb{F}_2$ , by the same elimination, gives  $\text{rank } \mathbf{A} = 12$  for the  $4 \times 4$  board and  $\text{rank } \mathbf{A} = 23$  for the  $5 \times 5$  board. On the  $5 \times 5$  board,  $\text{col}(\mathbf{A})$  has dimension 23, so it contains  $2^{23}$  patterns (coordinates in a basis match its elements with  $\mathbb{F}_2^{23}$ ), and only one pattern in four can be switched off.

**What the model leaves out.** Nothing, which is rare. The puzzle is linear over  $\mathbb{F}_2$  by its rules, so the model is exact. It even predicts features that are not obvious when playing, such as the irrelevance of the order of presses.

## 2.7.5 Exercises

### 2.7.5.1 A. Check your understanding

#### Exercise A1.

- (a) State the polynomial interpolation theorem, including its hypotheses.
- (b) True or false: there is a polynomial  $p \in \mathbb{R}[x]_{\leq 1}$  with  $p(1) = 2$  and  $p(1) = 3$ . Which hypothesis of the theorem fails?
- (c) In balancing a chemical equation, what does it mean if the null space of the coefficient matrix has dimension 1? Dimension 2?
- (d) Why is lights out modeled over  $\mathbb{F}_2$  and not over  $\mathbb{R}$ ?
- (e) A connected network has 6 vertices and 9 edges. What is the rank of its incidence matrix, and how many free parameters does the solution set of a consistent flow problem have?

*Solution.*

- (a) See [Theorem 2.7.3](#): for **distinct**  $x_0, \dots, x_n \in F$  and any  $y_0, \dots, y_n \in F$ , there is exactly one  $p \in F[x]_{\leq n}$  with  $p(x_i) = y_i$  for all  $i$ .
- (b) False:  $p(1)$  is one number, and  $2 \neq 3$ . The theorem needs distinct  $x$ -values, and here  $x_0 = x_1 = 1$ .
- (c) Dimension 1: all balancings are multiples of one vector, so there is one balanced equation up to scaling (provided a positive multiple exists). Dimension 2 or more: there are independent balancings, and the equation is a mixture of several reactions whose proportions algebra cannot decide.
- (d) A light has two states, and toggling twice restores it, so toggling is adding 1 with  $1 + 1 = 0$ , which is arithmetic in  $\mathbb{F}_2$ . Over  $\mathbb{R}$ , pressing a button twice would count as 2, not 0.
- (e) By [Proposition 2.7.4](#) the rank is  $6 - 1 = 5$ . By [Corollary 2.7.5](#) a consistent problem has  $9 - 6 + 1 = 4$  free parameters.

### 2.7.5.2 B. Practice

#### Exercise B1: Interpolating four points.

Find the polynomial  $p \in \mathbb{Q}[x]_{\leq 3}$  with  $p(0) = -1$ ,  $p(1) = 1$ ,  $p(2) = 3$  and  $p(3) = 11$ . Justify that there is only one.

*Solution.* The  $x$ -values 0, 1, 2, 3 are distinct, so there is exactly one such  $p = c_0 + c_1x +$

$c_2x^2 + c_3x^3$ , by Theorem 2.7.3. Its coefficients solve

$$\begin{pmatrix} 1 & 0 & 0 & 0 & -1 \\ 1 & 1 & 1 & 1 & 1 \\ 1 & 2 & 4 & 8 & 3 \\ 1 & 3 & 9 & 27 & 11 \end{pmatrix} \xrightarrow[R_4 \rightarrow R_4 - R_1]{R_2 \rightarrow R_2 - R_1, R_3 \rightarrow R_3 - R_1} \begin{pmatrix} 1 & 0 & 0 & 0 & -1 \\ 0 & 1 & 1 & 1 & 2 \\ 0 & 2 & 4 & 8 & 4 \\ 0 & 3 & 9 & 27 & 12 \end{pmatrix}$$

$$\xrightarrow[R_4 \rightarrow R_4 - 3R_2]{R_3 \rightarrow R_3 - 2R_2} \begin{pmatrix} 1 & 0 & 0 & 0 & -1 \\ 0 & 1 & 1 & 1 & 2 \\ 0 & 0 & 2 & 6 & 0 \\ 0 & 0 & 6 & 24 & 6 \end{pmatrix}$$

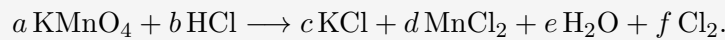
and then  $R_4 - 3R_3$  turns the last row into  $(0, 0, 0, 6 \mid 6)$ . Back substitution:  $c_3 = 1$ ;  $2c_2 + 6 = 0$ , so  $c_2 = -3$ ;  $c_1 - 3 + 1 = 2$ , so  $c_1 = 4$ ;  $c_0 = -1$ . Hence

$$p = -1 + 4x - 3x^2 + x^3.$$

Check:  $p(0) = -1$ ,  $p(1) = -1 + 4 - 3 + 1 = 1$ ,  $p(2) = -1 + 8 - 12 + 8 = 3$ ,  $p(3) = -1 + 12 - 27 + 27 = 11$ .

### Exercise B2: Balancing a redox reaction.

Potassium permanganate reacts with hydrochloric acid:



Write the conservation equations for K, Mn, O, H and Cl, find all rational solutions, and hence the balanced equation with the smallest positive integer coefficients.

*Solution.* Counting atoms of each element on both sides:

$$\text{K: } a = c, \quad \text{Mn: } a = d, \quad \text{O: } 4a = e, \quad \text{H: } b = 2e, \quad \text{Cl: } b = c + 2d + 2f.$$

In the order  $(a, b, c, d, e, f)$ , the coefficient matrix has rows  $(1, 0, -1, 0, 0, 0)$ ,  $(1, 0, 0, -1, 0, 0)$ ,  $(4, 0, 0, 0, -1, 0)$ ,  $(0, 1, 0, 0, -2, 0)$  and  $(0, 1, -1, -2, 0, -2)$ . The equations are nearly solved already, so we eliminate by substitution, which is row reduction written in words. The K, Mn and O equations give  $c = a$ ,  $d = a$ ,  $e = 4a$ . The H equation gives  $b = 2e = 8a$ . The Cl equation gives  $2f = b - c - 2d = 8a - a - 2a = 5a$ , so  $a = \frac{2}{5}f$ . Taking  $f = t$  as the free variable, the RREF of the coefficient matrix is

$$\begin{pmatrix} 1 & 0 & 0 & 0 & 0 & -\frac{2}{5} \\ 0 & 1 & 0 & 0 & 0 & -\frac{16}{5} \\ 0 & 0 & 1 & 0 & 0 & -\frac{3}{5} \\ 0 & 0 & 0 & 1 & 0 & -\frac{2}{5} \\ 0 & 0 & 0 & 0 & 1 & -\frac{3}{5} \end{pmatrix},$$

and the solutions are  $(a, b, c, d, e, f) = t(\frac{2}{5}, \frac{16}{5}, \frac{2}{5}, \frac{2}{5}, \frac{8}{5}, 1)$ ,  $t \in \mathbb{Q}$ . The rank is 5 and the nullity  $6 - 5 = 1$ , so the balancing is unique up to scaling. All entries are integers exactly when  $t$  is a multiple of 5, and  $t = 5$  gives



Check: K  $2 = 2$ ; Mn  $2 = 2$ ; O  $8 = 8$ ; H  $16 = 16$ ; Cl  $16 = 2 + 4 + 10$ .

**Exercise B3: A ring road.**

Four intersections lie on a ring of one-way streets  $1 \rightarrow 2$ ,  $2 \rightarrow 3$ ,  $3 \rightarrow 4$ ,  $4 \rightarrow 1$ , with flows  $f_1, f_2, f_3, f_4$  in that order. From outside, 30 cars per minute enter at intersection 1 and 10 at intersection 3, while 15 leave at intersection 2 and 25 at intersection 4.

- Write down the incidence matrix  $\mathbf{N}$  and the supply vector  $\mathbf{s}$ , and explain without computation why the system  $\mathbf{N}\mathbf{f} = \mathbf{s}$  is consistent.
- Find all solutions.
- If all flows must be non-negative, what is the smallest possible flow on each street?

*Solution.*

- With rows for the vertices and columns for the edges,

$$\mathbf{N} = \begin{pmatrix} 1 & 0 & 0 & -1 \\ -1 & 1 & 0 & 0 \\ 0 & -1 & 1 & 0 \\ 0 & 0 & -1 & 1 \end{pmatrix}, \quad \mathbf{s} = \begin{pmatrix} 30 \\ -15 \\ 10 \\ -25 \end{pmatrix}.$$

The ring is connected, and  $30 - 15 + 10 - 25 = 0$ , so the system is consistent by [Corollary 2.7.5](#).

- Row reducing  $[\mathbf{N} \mid \mathbf{s}]$ :  $R_2 + R_1$  gives  $(0, 1, 0, -1 \mid 15)$ ; then  $R_3 + R_2$  gives  $(0, 0, 1, -1 \mid 25)$ ; then  $R_4 + R_3$  gives the zero row. The pivots are in columns 1, 2, 3, and  $f_4 = t$  is free, so

$$\mathbf{f} = (30 + t, 15 + t, 25 + t, t), \quad t \in \mathbb{R}.$$

There is  $4 - 4 + 1 = 1$  parameter, the flow around the ring. Check at  $t = 0$ : vertex 1,  $30 - 0 = 30$ ; vertex 2,  $15 - 30 = -15$ ; vertex 3,  $25 - 15 = 10$ ; vertex 4,  $0 - 25 = -25$ .

- All flows are non-negative exactly when  $t \geq 0$ , since then  $30 + t, 15 + t, 25 + t \geq 0$  too. Each flow increases with  $t$ , so all are smallest at  $t = 0$ :  $\mathbf{f} = (30, 15, 25, 0)$ . No car needs to use street  $4 \rightarrow 1$ .

**2.7.5.3 C. Going deeper****Exercise C1: The  $2 \times 2$  puzzle.**

Consider lights out on a  $2 \times 2$  grid, where pressing a light toggles it and its two horizontal and vertical neighbors. Number the cells  $1 = (1, 1)$ ,  $2 = (1, 2)$ ,  $3 = (2, 1)$ ,  $4 = (2, 2)$ .

- Write down the matrix  $\mathbf{A} \in M_4(\mathbb{F}_2)$  of the puzzle.
- Show that  $\mathbf{A}^2 = \mathbf{I}$ .
- Deduce that every pattern is solvable by exactly one set of presses, and find the presses that switch off the pattern with only light 1 on.

*Solution.*

- (a) Button  $j$  toggles every light except the one diagonally opposite. So

$$\mathbf{A} = \begin{pmatrix} 1 & 1 & 1 & 0 \\ 1 & 1 & 0 & 1 \\ 1 & 0 & 1 & 1 \\ 0 & 1 & 1 & 1 \end{pmatrix}.$$

- (b) Let  $\mathbf{J}$  be the  $4 \times 4$  all-ones matrix and  $\mathbf{Q}$  the permutation matrix exchanging  $1 \leftrightarrow 4$  and  $2 \leftrightarrow 3$ , which has 1's on the anti-diagonal. Then  $\mathbf{A} = \mathbf{J} - \mathbf{Q} = \mathbf{J} + \mathbf{Q}$  over  $\mathbb{F}_2$ . Every entry of  $\mathbf{J}^2$  is  $1 + 1 + 1 + 1 = 0$ , so  $\mathbf{J}^2 = 0$ . Each row and each column of  $\mathbf{Q}$  has a single 1, so  $\mathbf{JQ} = \mathbf{J}$  and  $\mathbf{QJ} = \mathbf{J}$ . And  $\mathbf{Q} = \mathbf{P}_\sigma$  for  $\sigma = (1\ 4)(2\ 3)$ , with  $\sigma \circ \sigma = \text{id}$ , so  $\mathbf{Q}^2 = \mathbf{P}_{\text{id}} = \mathbf{I}$  by Lemma 2.6.6 (a). By Theorem 0.9.4,

$$\mathbf{A}^2 = \mathbf{J}^2 + \mathbf{JQ} + \mathbf{QJ} + \mathbf{Q}^2 = 0 + \mathbf{J} + \mathbf{J} + \mathbf{I} = \mathbf{I},$$

because  $\mathbf{J} + \mathbf{J} = 2\mathbf{J} = 0$  over  $\mathbb{F}_2$ .

- (c) By (b),  $\mathbf{A}$  is invertible with  $\mathbf{A}^{-1} = \mathbf{A}$ . By Theorem 0.9.7 (7),  $\mathbf{Ax} = \mathbf{b}$  has exactly one solution  $\mathbf{x} = \mathbf{Ab}$  for every  $\mathbf{b} \in \mathbb{F}_2^4$ . For  $\mathbf{b} = \mathbf{e}_1$ ,  $\mathbf{x} = \mathbf{Ae}_1 = (1, 1, 1, 0)$ : press lights 1, 2, 3. Check: light 1 is toggled by presses 1, 2, 3, three times, so it goes off; light 2 by presses 1 and 2; light 3 by presses 1 and 3; light 4 by presses 2 and 3; each of these is toggled twice and stays off.

### Exercise C2: Interpolating values and slopes.

For  $p = c_0 + c_1x + c_2x^2 + c_3x^3 \in F[x]_{\leq 3}$ , define its (formal) derivative  $p' = c_1 + 2c_2x + 3c_3x^2$ .

- (a) Prove that for **every** field  $F$  and all  $a, b, c, d \in F$  there is exactly one  $p \in F[x]_{\leq 3}$  with

$$p(0) = a, \quad p'(0) = b, \quad p(1) = c, \quad p'(1) = d.$$

- (b) Find this  $p$  over  $\mathbb{Q}$  for  $(a, b, c, d) = (0, 1, 1, 0)$ .

*Hint: in (a), check that elimination never divides by an element that could be zero in some field.*

*Solution.*

- (a) The four conditions are linear in  $c_0, \dots, c_3$ :  $p(0) = c_0$ ,  $p'(0) = c_1$ ,  $p(1) = c_0 + c_1 + c_2 + c_3$ ,  $p'(1) = c_1 + 2c_2 + 3c_3$ . Row reducing the augmented matrix, using only additions of multiples of rows,

$$\begin{aligned} \left( \begin{array}{cccc|c} 1 & 0 & 0 & 0 & a \\ 0 & 1 & 0 & 0 & b \\ 1 & 1 & 1 & 1 & c \\ 0 & 1 & 2 & 3 & d \end{array} \right) & \xrightarrow[R_4 \rightarrow R_4 - R_2]{R_3 \rightarrow R_3 - R_1 - R_2} \left( \begin{array}{cccc|c} 1 & 0 & 0 & 0 & a \\ 0 & 1 & 0 & 0 & b \\ 0 & 0 & 1 & 1 & c - a - b \\ 0 & 0 & 2 & 3 & d - b \end{array} \right) \\ & \xrightarrow{R_4 \rightarrow R_4 - 2R_3} \left( \begin{array}{cccc|c} 1 & 0 & 0 & 0 & a \\ 0 & 1 & 0 & 0 & b \\ 0 & 0 & 1 & 1 & c - a - b \\ 0 & 0 & 0 & 1 & 2a + b - 2c + d \end{array} \right). \end{aligned}$$

Here  $3 - 2 = 1$  holds in every field. The coefficient matrix is now upper triangular



with every diagonal entry equal to 1, so it is invertible by [Lemma 2.6.2](#), and back substitution gives the unique solution without any division:

$$\begin{aligned}c_3 &= 2a + b - 2c + d, \\c_2 &= (c - a - b) - c_3 = -3a - 2b + 3c - d, \\c_1 &= b, \quad c_0 = a.\end{aligned}$$

Row operations preserve the solution set ([Theorem 2.2.2](#)), so the original system also has exactly this one solution, for every field  $F$ .

- (b) With  $(a, b, c, d) = (0, 1, 1, 0)$ :  $c_3 = 0 + 1 - 2 + 0 = -1$ ,  $c_2 = 0 - 2 + 3 - 0 = 1$ ,  $c_1 = 1$ ,  $c_0 = 0$ . So  $p = x + x^2 - x^3$ . Check:  $p(0) = 0$ ;  $p' = 1 + 2x - 3x^2$ , so  $p'(0) = 1$ ;  $p(1) = 1 + 1 - 1 = 1$ ;  $p'(1) = 1 + 2 - 3 = 0$ .

### Exercise C3: Networks in several pieces.

Let  $\mathbf{N} \in M_{m \times k}(\mathbb{R})$  be the incidence matrix of a network with  $m$  vertices. Say that vertices  $v$  and  $w$  are **linked** if they are joined by a sequence of vertices in which consecutive ones share an edge, in either direction (every vertex is linked to itself by the sequence of length zero).

- (a) Show that “linked” is an equivalence relation. Its equivalence classes are called the **components** of the network; let  $c$  be their number.
- (b) Prove that  $\text{rank } \mathbf{N} = m - c$ .

*Solution.*

- (a) *Reflexive*: by the sequence of length zero. *Symmetric*: reversing a sequence from  $v$  to  $w$  gives one from  $w$  to  $v$ , and consecutive vertices still share an edge. *Transitive*: following a sequence from  $u$  to  $v$  by one from  $v$  to  $w$  gives a sequence from  $u$  to  $w$ . So “linked” is an equivalence relation ([Definition 0.5.2](#)).
- (b) Let  $C_1, \dots, C_c$  be the components, and let  $\mathbf{1}_{C_r} \in \mathbb{R}^m$  have entry 1 at the vertices of  $C_r$  and 0 elsewhere. As in the proof of [Proposition 2.7.4](#),  $\mathbf{N}^\top \mathbf{y} = \mathbf{0}$  if and only if  $y_u = y_w$  for every edge joining  $u$  and  $w$ . Following sequences of edges, this holds if and only if  $\mathbf{y}$  is constant on each component, that is,  $\mathbf{y} = \sum_r \lambda_r \mathbf{1}_{C_r}$  with  $\lambda_r$  the common value on  $C_r$ . (Conversely each  $\mathbf{1}_{C_r}$  is in the null space, since both ends of an edge lie in the same component.) So  $(\mathbf{1}_{C_1}, \dots, \mathbf{1}_{C_c})$  spans  $\text{nul}(\mathbf{N}^\top)$ . It is independent: if  $\sum_r \lambda_r \mathbf{1}_{C_r} = \mathbf{0}$ , then reading the entry at any vertex of  $C_r$ , where only the  $r$ -th vector is non-zero because the components are disjoint ([Theorem 0.5.2](#)), gives  $\lambda_r = 0$ . Hence  $\dim \text{nul}(\mathbf{N}^\top) = c$ . Since  $\mathbf{N}^\top$  has  $m$  columns, [Theorem 2.5.7](#) gives  $\text{rank } \mathbf{N}^\top = m - c$ , and [Theorem 2.5.5](#) gives  $\text{rank } \mathbf{N} = m - c$ , as claimed.

# Chapter 3

## Linear Maps

Chapter 1 studied vector spaces one at a time. This chapter studies the functions between them that respect the structure: **linear maps**, the functions that send sums to sums and scalar multiples to scalar multiples. Differentiation of polynomials, the trace of a matrix, a rotation of the plane and multiplication by a matrix are all linear maps, and one theory covers them all.

Two subspaces measure how far a linear map is from being injective and surjective: its kernel and its image. The Rank–Nullity Theorem ties their dimensions together, and turns many questions about maps into counting. Then we choose bases and write every linear map as a matrix. This explains why matrix multiplication is defined the way it is, and it raises the central question of the rest of the book: how does the matrix change when the bases change, and how simple can we make it?

**What you need.** From Chapter 1, bases and dimension ([Definition 1.5.1](#), [Definition 1.6.1](#)), basis extension ([Theorem 1.6.5](#)) and counting instead of checking ([Theorem 1.6.4](#)). From Chapter 2, the null space and column space of a matrix ([Definition 2.5.3](#), [Definition 2.5.1](#)), its rank ([Definition 2.5.4](#), [Theorem 2.5.7](#)) and the invertible matrix theorem ([Theorem 2.4.4](#)). From Chapter 0, quotient sets and well-defined functions on them ([Theorem 0.5.3](#)), for the section on quotient spaces.

### 3.0.1 Roadmap

- **Linear maps.** The definition, a gallery of examples and non-examples, and the fact that a linear map is determined by its values on a basis.
- **Kernel and image.** The two subspaces attached to every linear map, and the kernel test for injectivity.
- **The Rank–Nullity Theorem.**  $\dim V = \text{nullity } T + \text{rank } T$ , and what counting can decide.
- **The algebra of linear maps.** Adding, scaling and composing maps; polynomials in an operator.
- **Isomorphisms.** When two spaces are “the same”, and why dimension alone decides it.
- **The matrix of a linear map.** Coordinates turn maps into matrices and composition into matrix multiplication.
- **Change of basis and similarity.** How the matrix of a map depends on the chosen bases.
- **Rank, equivalence and factorizations.** With free choice of bases on both sides, every

matrix becomes  $\begin{pmatrix} \mathbf{I}_r & 0 \\ 0 & 0 \end{pmatrix}$ .

- **Products and quotient spaces.** Building new spaces from old ones, and the isomorphism theorems.
- **Projections and the trace.** Direct sums seen through operators, and a basis-free trace.

### 3.0.2 Named moves

- **Injective via the kernel.** To show  $T$  is injective, show  $T\mathbf{v} = \mathbf{0}$  forces  $\mathbf{v} = \mathbf{0}$ .
- **Count with Rank–Nullity.** When  $\dim V = \dim W$ , injective, surjective and bijective are the same; check the easiest one.
- **Basis of the kernel, then extend.** The proof pattern behind Rank–Nullity, the same one that proved the dimension formula for sums.
- **Abstract space  $\rightarrow$  write down a matrix.** Choose a basis that makes the map simple, then compute with its matrix.
- **The change-of-basis square.** Go around the square instead of across it:  $[T]_{\mathcal{B}'} = \mathbf{P}^{-1}[T]_{\mathcal{B}}\mathbf{P}$ .
- **Check well-definedness on cosets.** Every operation on a quotient space, and every map out of one, starts with this check.

## 3.1 Linear Maps

Chapter 1 studied vector spaces one at a time, and Chapter 2 studied one kind of question inside  $F^n$ : does  $\mathbf{Ax} = \mathbf{b}$  have a solution? This chapter studies the functions **between** vector spaces that respect addition and scaling. This section defines them, collects examples from geometry, matrices, calculus and algebra, and proves the fact that makes them manageable: a linear map is completely determined by what it does to a basis.

### 3.1.1 Functions that behave like matrices

Objects come first, then the maps between them that respect their structure. For sets the maps are functions; for groups they are homomorphisms (Definition 0.10.3). For vector spaces the structure is addition and scalar multiplication, so we want functions that respect those two operations.

The standard example is a matrix. For  $\mathbf{A} \in M_{m \times n}(F)$ , the rules of matrix multiplication (Theorem 0.9.4) give

$$\mathbf{A}(\mathbf{x} + \mathbf{y}) = \mathbf{Ax} + \mathbf{Ay}, \quad \mathbf{A}(c\mathbf{x}) = c(\mathbf{Ax}) \quad \text{for all } \mathbf{x}, \mathbf{y} \in F^n, c \in F.$$

Now look at three functions that are not given by matrices at all.

- **Differentiation** on polynomials:  $(p + q)' = p' + q'$  and  $(cp)' = cp'$ .
- **Trace** on square matrices:  $\text{tr}(\mathbf{A} + \mathbf{B}) = \text{tr } \mathbf{A} + \text{tr } \mathbf{B}$  and  $\text{tr}(c\mathbf{A}) = c \text{tr } \mathbf{A}$  (Theorem 0.9.6).
- **Rotation** of the plane about the origin: rotating a parallelogram gives a parallelogram, so the rotation of a sum is the sum of the rotations, and rotating a stretched arrow gives the stretched rotated arrow.

The inputs are polynomials, matrices and arrows, not columns. Yet all three obey exactly the two rules that matrices obey. Whatever we can prove from those two rules alone will hold for all of them at once. So we give the two rules a name.

*A linear map is a function that preserves linear combinations.*

### 3.1.2 The definition

#### Definition 3.1.1 — Linear Map

Let  $V$  and  $W$  be vector spaces **over the same field**  $F$ . A function  $T: V \rightarrow W$  is a **linear map** (or **linear transformation**) **over**  $F$  if

(LT1)  $T(\mathbf{u} + \mathbf{v}) = T(\mathbf{u}) + T(\mathbf{v})$  **for all**  $\mathbf{u}, \mathbf{v} \in V$ , and

(LT2)  $T(c\mathbf{v}) = cT(\mathbf{v})$  **for all**  $c \in F$  and **all**  $\mathbf{v} \in V$ .

A linear map  $T: V \rightarrow V$  from a space to itself is also called a **linear operator** on  $V$ . We often write  $T\mathbf{v}$  for  $T(\mathbf{v})$ .

In words: (LT1), **additivity**, says that adding first and then applying  $T$  gives the same result as applying  $T$  first and then adding. The sum on the left is taken in  $V$ ; the sum on the right is taken in  $W$ . (LT2), **homogeneity**, says the same for scaling: the scalar  $c$  can be pulled out of  $T$ . Both conditions must hold for **every** choice of vectors and scalars, so one counterexample is enough to show that a function is not linear.

The phrase “over the same field” matters. Scalars from  $F$  act on both sides of (LT2), so both spaces must accept them. We will see below that one function can be linear over  $\mathbb{R}$  and not linear over  $\mathbb{C}$ . A general linear map is always named  $T$ ,  $S$  or  $R$  in this book, never  $f$ ; a specific map may get a letter that recalls what it does, such as  $D$  for differentiation.

### 3.1.3 Examples

We check the examples against (LT1) and (LT2), grouped by where they come from.

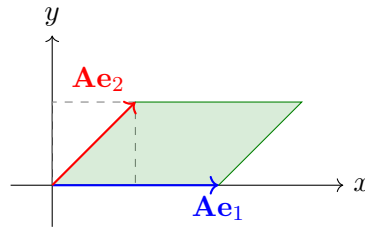
**Matrices.**

#### Example. (Multiplication by a Matrix)

Let  $\mathbf{A} \in M_{m \times n}(F)$ . Show that  $T_{\mathbf{A}}: F^n \rightarrow F^m$ ,  $T_{\mathbf{A}}(\mathbf{x}) = \mathbf{A}\mathbf{x}$ , is a linear map.

*Solution.* Let  $\mathbf{x}, \mathbf{y} \in F^n$  and  $c \in F$ . By the distributive law in [Theorem 0.9.4](#),  $T_{\mathbf{A}}(\mathbf{x} + \mathbf{y}) = \mathbf{A}(\mathbf{x} + \mathbf{y}) = \mathbf{A}\mathbf{x} + \mathbf{A}\mathbf{y} = T_{\mathbf{A}}(\mathbf{x}) + T_{\mathbf{A}}(\mathbf{y})$ , which is (LT1). By the scalar rule in the same theorem,  $T_{\mathbf{A}}(c\mathbf{x}) = \mathbf{A}(c\mathbf{x}) = c(\mathbf{A}\mathbf{x}) = cT_{\mathbf{A}}(\mathbf{x})$ , which is (LT2). Hence  $T_{\mathbf{A}}$  is linear.

We will use the name  $T_{\mathbf{A}}$  for this map throughout the chapter. For  $F = \mathbb{R}$  and  $n = m = 2$ , the map  $T_{\mathbf{A}}$  sends the unit square to the parallelogram spanned by  $\mathbf{A}\mathbf{e}_1$  and  $\mathbf{A}\mathbf{e}_2$ , the columns of  $\mathbf{A}$  ([Theorem 0.9.3](#)).



The unit square (dashed) and its image under  $\mathbf{A} = \begin{pmatrix} 2 & 1 \\ 0 & 1 \end{pmatrix}$ .

**Example. (Maps Given by Formulas in Coordinates)**

- (a) Show that  $T: F^3 \rightarrow F^2$ ,  $T(a_1, a_2, a_3) = (2a_1 + a_3, a_2 - a_1)$ , is linear.
- (b) Let  $\mathcal{B} = (\mathbf{v}_1, \dots, \mathbf{v}_n)$  be a basis of a vector space  $V$  over  $F$ . Show that the **coordinate map**  $V \rightarrow F^n$ ,  $\mathbf{v} \mapsto [\mathbf{v}]_{\mathcal{B}}$ , is linear.

*Solution.*

- (a) The formula is a matrix product:

$$\begin{pmatrix} 2 & 0 & 1 \\ -1 & 1 & 0 \end{pmatrix} \begin{pmatrix} a_1 \\ a_2 \\ a_3 \end{pmatrix} = \begin{pmatrix} 2a_1 + a_3 \\ -a_1 + a_2 \end{pmatrix}.$$

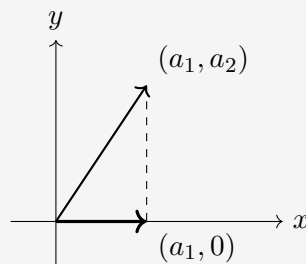
So  $T = T_{\mathbf{A}}$  for this  $2 \times 3$  matrix  $\mathbf{A}$ , and  $T$  is linear by [Example \(Multiplication by a Matrix\)](#). The same argument shows that every map  $F^n \rightarrow F^m$  whose output entries are combinations of the input entries, with constant coefficients and no constant terms, is linear.

- (b) By [Theorem 1.5.2](#),  $[\mathbf{u} + \mathbf{v}]_{\mathcal{B}} = [\mathbf{u}]_{\mathcal{B}} + [\mathbf{v}]_{\mathcal{B}}$  and  $[c\mathbf{v}]_{\mathcal{B}} = c[\mathbf{v}]_{\mathcal{B}}$ . These are (LT1) and (LT2).

**Geometry.**

**Example. (Projection onto an Axis)**

Show that  $T: \mathbb{R}^2 \rightarrow \mathbb{R}^2$ ,  $T(a_1, a_2) = (a_1, 0)$ , is linear.



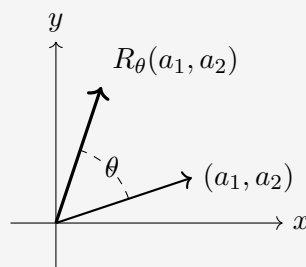
*Solution.* Let  $\mathbf{u} = (a_1, a_2)$ ,  $\mathbf{v} = (b_1, b_2)$  and  $c \in \mathbb{R}$ . Then

$$\begin{aligned} T(\mathbf{u} + \mathbf{v}) &= (a_1 + b_1, 0) = (a_1, 0) + (b_1, 0) = T(\mathbf{u}) + T(\mathbf{v}), \\ T(c\mathbf{u}) &= (ca_1, 0) = c(a_1, 0) = cT(\mathbf{u}). \end{aligned}$$

So (LT1) and (LT2) hold. The map drops each point vertically onto the  $x$ -axis, like a shadow cast by light from straight above.

### Example. (Rotation of the Plane)

Let  $\theta \in \mathbb{R}$ , and let  $R_\theta: \mathbb{R}^2 \rightarrow \mathbb{R}^2$  rotate each vector counterclockwise about the origin by the angle  $\theta$ . Find a formula for  $R_\theta$ , and deduce that it is linear.



*Solution.* Write  $(a_1, a_2) = (r \cos \varphi, r \sin \varphi)$  in polar coordinates. Rotating adds  $\theta$  to the angle and keeps the length  $r$ , so by the addition formulas for sine and cosine,

$$\begin{aligned} R_\theta(a_1, a_2) &= (r \cos(\varphi + \theta), r \sin(\varphi + \theta)) \\ &= (a_1 \cos \theta - a_2 \sin \theta, a_1 \sin \theta + a_2 \cos \theta). \end{aligned}$$

This is  $T_{\mathbf{A}}$  for  $\mathbf{A} = \begin{pmatrix} \cos \theta & -\sin \theta \\ \sin \theta & \cos \theta \end{pmatrix}$ , so  $R_\theta$  is linear by [Example \(Multiplication by a Matrix\)](#).

**Calculus.** For a polynomial  $p = a_0 + a_1x + \cdots + a_nx^n \in F[x]$ , define its **(formal) derivative** as  $p' = a_1 + 2a_2x + \cdots + na_nx^{n-1}$ , where  $ka_k$  means  $a_k$  added to itself  $k$  times. Over  $\mathbb{R}$  this is the derivative of calculus.

### Example. (Differentiation)

Show that  $D: F[x] \rightarrow F[x]$ ,  $D(p) = p'$ , is linear. Show also that  $D$  maps  $F[x]_{\leq n}$  into  $F[x]_{\leq n-1}$  for  $n \geq 1$ .

*Solution.* Let  $p = \sum_k a_k x^k$  and  $q = \sum_k b_k x^k$ , and  $c \in F$ . The coefficient of  $x^{k-1}$  in  $D(p+q)$  is  $k(a_k + b_k) = ka_k + kb_k$ , which is the coefficient of  $x^{k-1}$  in  $D(p) + D(q)$ . The coefficient of  $x^{k-1}$  in  $D(cp)$  is  $k(ca_k) = c(ka_k)$ , which is the coefficient in  $cD(p)$ . Both steps use only the field axioms. Two polynomials with the same coefficients are equal ([Definition 0.8.1](#)), so (LT1) and (LT2) hold. If  $\deg p \leq n$ , the highest power in  $p'$  is at most  $x^{n-1}$ , so  $D$  restricts to a linear map  $F[x]_{\leq n} \rightarrow F[x]_{\leq n-1}$ .

**Example. (Integration)**

Show that  $J: \mathbb{R}[x] \rightarrow \mathbb{R}[x]$ ,  $J(p)(x) = \int_0^x p(t) dt$ , is linear.

*Solution.* On coefficients,  $J$  sends  $a_0 + a_1x + \cdots + a_nx^n$  to  $a_0x + \frac{a_1}{2}x^2 + \cdots + \frac{a_n}{n+1}x^{n+1}$ , so it does map polynomials to polynomials. By linearity of the integral from calculus,

$$\begin{aligned} J(p+q)(x) &= \int_0^x (p(t) + q(t)) dt = J(p)(x) + J(q)(x), \\ J(cp)(x) &= \int_0^x cp(t) dt = cJ(p)(x) \end{aligned}$$

for every  $x \in \mathbb{R}$ . Hence  $J$  is linear. (The coefficient formula divides by  $k+1$ ; over a field such as  $\mathbb{F}_2$ , where  $1+1=0$ , it breaks down.)

**Algebra and sequences.****Example. (Multiplication by  $x$ )**

Show that  $M: F[x] \rightarrow F[x]$ ,  $M(p) = xp$ , is linear.

*Solution.* By the distributive laws in [Theorem 0.8.1](#),  $x(p+q) = xp+xq$ , and  $x(cp) = c(xp)$  because the product of polynomials is associative and commutative and the constant  $c$  is a polynomial. These are (LT1) and (LT2).

**Example. (Trace and Transpose)**

Show that the trace  $\text{tr}: M_n(F) \rightarrow F$  and the transpose  $M_{m \times n}(F) \rightarrow M_{n \times m}(F)$ ,  $\mathbf{A} \mapsto \mathbf{A}^\top$ , are linear.

*Solution.* For the trace, [Theorem 0.9.6](#) (1) says  $\text{tr}(\mathbf{A} + \mathbf{B}) = \text{tr} \mathbf{A} + \text{tr} \mathbf{B}$  and  $\text{tr}(c\mathbf{A}) = c \text{tr} \mathbf{A}$ , which are (LT1) and (LT2); the target is  $F$ , regarded as a vector space over itself. For the transpose, the  $(i, j)$  entry of  $(\mathbf{A} + \mathbf{B})^\top$  is  $a_{ji} + b_{ji}$ , the  $(i, j)$  entry of  $\mathbf{A}^\top + \mathbf{B}^\top$ ; and the  $(i, j)$  entry of  $(c\mathbf{A})^\top$  is  $ca_{ji}$ , the  $(i, j)$  entry of  $c\mathbf{A}^\top$ .

**Example. (The Shift Maps on Sequences)**

On the space  $F^\mathbb{N}$  of sequences ([Example \(The Standard Vector Spaces\)](#)), define the **right shift** and the **left shift**

$$R(s_0, s_1, s_2, \dots) = (0, s_0, s_1, \dots), \quad L(s_0, s_1, s_2, \dots) = (s_1, s_2, s_3, \dots).$$

Show that both are linear.

*Solution.* The operations on  $F^\mathbb{N}$  are termwise. The entry in position  $k \geq 1$  of  $R(\mathbf{s} + \mathbf{t})$  is  $s_{k-1} + t_{k-1}$ , which is the entry of  $R(\mathbf{s}) + R(\mathbf{t})$ ; in position 0 both have  $0 = 0 + 0$ . Similarly,  $R(c\mathbf{s})$  has entries  $0 = c \cdot 0$  and  $cs_{k-1}$ , the entries of  $cR(\mathbf{s})$ . For  $L$ , the entry in position  $k$  of  $L(\mathbf{s} + \mathbf{t})$  is  $s_{k+1} + t_{k+1}$ , and of  $L(c\mathbf{s})$  is  $cs_{k+1}$ , as required.

The right shift is the map promised in the warning of Chapter 1, §8, on what changes in infinite dimension: it matches  $F^{\mathbb{N}}$  one-to-one with the sequences whose first entry is 0, respecting addition and scaling. In the next section we measure exactly how  $R$  and  $L$  fail to be bijective.

**Example. (Composition with a Fixed Function)**

Let  $X$  and  $Y$  be sets and  $g: X \rightarrow Y$  a function. Show that  $C_g: F^Y \rightarrow F^X$ ,  $C_g(h) = h \circ g$ , is linear. Deduce that  $\mathbb{R}[x] \rightarrow \mathbb{R}[x]$ ,  $p(x) \mapsto p(x+1)$ , is linear.

*Solution.* The operations on function spaces are pointwise (Example (The Standard Vector Spaces)). For  $t \in X$ ,

$$C_g(h+k)(t) = (h+k)(g(t)) = h(g(t)) + k(g(t)) = (C_g(h) + C_g(k))(t),$$

and  $C_g(ch)(t) = ch(g(t)) = (cC_g(h))(t)$ . So (LT1) and (LT2) hold. The output is again a function, not a number; linearity is about the function  $h$  varying, while  $g$  stays fixed.

For the second claim, take  $X = Y = \mathbb{R}$  and  $g(t) = t+1$ , and regard each real polynomial as a function  $\mathbb{R} \rightarrow \mathbb{R}$ . Then  $p(x+1) = C_g(p)$ , which is again a polynomial of the same degree. The identities  $C_g(p+q) = C_g(p) + C_g(q)$  and  $C_g(cp) = cC_g(p)$  hold as functions, hence as polynomials, because two real polynomials with the same values everywhere are equal (a consequence of Lemma 2.7.2). So the map is linear. (It translates the graph of  $p$  one unit to the left.)

**The degenerate maps.** Three maps are always available: the identity on any space, the inclusion of any subspace, and the zero map between any two spaces. They look too simple to matter. They matter because every general statement about linear maps must be true for them, so they are the first test of any conjecture.

**Example. (Identity Map)**

The identity  $\text{id}_V: V \rightarrow V$ ,  $\mathbf{v} \mapsto \mathbf{v}$ , is linear:  $\text{id}_V(\mathbf{u} + \mathbf{v}) = \mathbf{u} + \mathbf{v} = \text{id}_V(\mathbf{u}) + \text{id}_V(\mathbf{v})$  and  $\text{id}_V(c\mathbf{v}) = c\mathbf{v} = c \text{id}_V(\mathbf{v})$ .

**Example. (Inclusion Map)**

If  $U$  is a subspace of  $V$ , the inclusion  $\iota: U \rightarrow V$ ,  $\iota(\mathbf{u}) = \mathbf{u}$ , is linear, because the operations of  $U$  are those of  $V$ . Its formula is the identity's, but its codomain is bigger. For instance  $\mathbb{R}[x]_{\leq 2} \rightarrow \mathbb{R}[x]$  is an inclusion.

**Example. (Zero Map)**

For any  $V$  and  $W$  over  $F$ , the zero map  $0: V \rightarrow W$ ,  $\mathbf{v} \mapsto \mathbf{0}_W$ , is linear:  $\mathbf{0}_W = \mathbf{0}_W + \mathbf{0}_W$  and  $\mathbf{0}_W = c\mathbf{0}_W$  (Theorem 1.1.6).

## 3.1.4 Three consequences of linearity

The two axioms already force several other rules. The first is the quickest test for non-linearity.



**Theorem 3.1.1 — Linear Maps Send Zero to Zero**

Let  $T: V \rightarrow W$  be linear. Then  $T(\mathbf{0}_V) = \mathbf{0}_W$ .

*Proof.* By (LT1),  $T(\mathbf{0}_V) = T(\mathbf{0}_V + \mathbf{0}_V) = T(\mathbf{0}_V) + T(\mathbf{0}_V)$ . Adding  $-T(\mathbf{0}_V)$  to both sides gives  $\mathbf{0}_W = T(\mathbf{0}_V)$ . ■

The proof used only (LT1). Homogeneity with  $c = 0$  would also give it:  $T(\mathbf{0}_V) = T(0\mathbf{v}) = 0T(\mathbf{v}) = \mathbf{0}_W$ .

**Theorem 3.1.2 — Linear Maps Preserve Negatives**

Let  $T: V \rightarrow W$  be linear. Then  $T(-\mathbf{v}) = -T(\mathbf{v})$  and  $T(\mathbf{u} - \mathbf{v}) = T(\mathbf{u}) - T(\mathbf{v})$  for all  $\mathbf{u}, \mathbf{v} \in V$ .

*Proof.* By Theorem 1.1.7,  $-\mathbf{v} = (-1)\mathbf{v}$ . So (LT2) gives  $T(-\mathbf{v}) = (-1)T(\mathbf{v}) = -T(\mathbf{v})$ , using Theorem 1.1.7 again in  $W$ . Then (LT1) gives  $T(\mathbf{u} - \mathbf{v}) = T(\mathbf{u} + (-\mathbf{v})) = T(\mathbf{u}) + T(-\mathbf{v}) = T(\mathbf{u}) - T(\mathbf{v})$ . ■

Checking two conditions separately is often longer than necessary. One combined condition suffices.

**Theorem 3.1.3 — One-Line Test for Linearity**

Let  $V$  and  $W$  be vector spaces over  $F$ , and let  $T: V \rightarrow W$  be a function. Then  $T$  is linear **if and only if**

$$T(c\mathbf{u} + \mathbf{v}) = cT(\mathbf{u}) + T(\mathbf{v}) \quad \text{for all } c \in F \text{ and all } \mathbf{u}, \mathbf{v} \in V.$$

*Proof.* ( $\Rightarrow$ ) Suppose  $T$  is linear. By (LT1) and then (LT2),  $T(c\mathbf{u} + \mathbf{v}) = T(c\mathbf{u}) + T(\mathbf{v}) = cT(\mathbf{u}) + T(\mathbf{v})$ .

( $\Leftarrow$ ) Suppose the condition holds. Taking  $c = 1$  gives  $T(\mathbf{u} + \mathbf{v}) = T(\mathbf{u}) + T(\mathbf{v})$ , which is (LT1). Taking  $\mathbf{u} = \mathbf{v} = \mathbf{0}$  and  $c = 1$  gives  $T(\mathbf{0}) = T(\mathbf{0}) + T(\mathbf{0})$ , so  $T(\mathbf{0}) = \mathbf{0}$ . Now taking  $\mathbf{v} = \mathbf{0}$  gives  $T(c\mathbf{u}) = cT(\mathbf{u}) + T(\mathbf{0}) = cT(\mathbf{u})$ , which is (LT2). ■

Finally, the slogan. Two applications of the axioms handle a combination of two vectors; induction handles any finite combination.

**Theorem 3.1.4 — Linear Maps Preserve Linear Combinations**

Let  $T: V \rightarrow W$  be linear, let  $\mathbf{v}_1, \dots, \mathbf{v}_k \in V$  and  $a_1, \dots, a_k \in F$ , where  $k \geq 0$ . Then

$$T(a_1\mathbf{v}_1 + \dots + a_k\mathbf{v}_k) = a_1T(\mathbf{v}_1) + \dots + a_kT(\mathbf{v}_k).$$

*Proof.* We use induction on  $k$ . For  $k = 0$  both sides are empty sums, so the claim is  $T(\mathbf{0}) = \mathbf{0}$ , which is [Theorem 3.1.1](#). Let  $k \geq 1$  and suppose the claim holds for  $k - 1$  vectors. By (LT1), then the induction hypothesis and (LT2),

$$T\left(\sum_{i=1}^{k-1} a_i\mathbf{v}_i + a_k\mathbf{v}_k\right) = T\left(\sum_{i=1}^{k-1} a_i\mathbf{v}_i\right) + T(a_k\mathbf{v}_k) = \sum_{i=1}^{k-1} a_iT(\mathbf{v}_i) + a_kT(\mathbf{v}_k).$$

This completes the induction. ■

In words: to apply  $T$  to a combination, apply it to each vector and keep the same coefficients. This is the property the slogan promised, and it is the property every later theorem about linear maps uses.

### 3.1.5 Non-examples

A function fails to be linear as soon as **one** instance of (LT1) or (LT2) fails. The cheapest instance to try is  $\mathbf{0}$ , by [Theorem 3.1.1](#).

#### Example. (A Constant Function)

Is  $T: \mathbb{R} \rightarrow \mathbb{R}$ ,  $T(x) = 5$ , linear?

*Solution.* No. A linear map sends  $0$  to  $0$  by [Theorem 3.1.1](#), but  $T(0) = 5 \neq 0$ .

#### Example. (Translation)

Let  $\mathbf{b} \in \mathbb{R}^2$  be **non-zero**. Is the translation  $T: \mathbb{R}^2 \rightarrow \mathbb{R}^2$ ,  $T(\mathbf{v}) = \mathbf{v} + \mathbf{b}$ , linear?

*Solution.* No:  $T(\mathbf{0}) = \mathbf{b} \neq \mathbf{0}$ , which is impossible for a linear map by [Theorem 3.1.1](#). This is a minimal change of the identity map ([Example \(Identity Map\)](#)): the formula differs only by the constant  $\mathbf{b}$ , and that constant is exactly what breaks linearity. For the record, (LT1) fails too:  $T(\mathbf{u} + \mathbf{v}) = \mathbf{u} + \mathbf{v} + \mathbf{b}$ , while  $T(\mathbf{u}) + T(\mathbf{v}) = \mathbf{u} + \mathbf{v} + 2\mathbf{b}$ . Maps of the form  $\mathbf{v} \mapsto \mathbf{A}\mathbf{v} + \mathbf{b}$  are called **affine**; they are linear exactly when  $\mathbf{b} = \mathbf{0}$ .

Passing the zero test does not make a function linear.

#### Example. (Squaring)

Is  $T: \mathbb{R}^2 \rightarrow \mathbb{R}^2$ ,  $T(a_1, a_2) = (a_1^2, 0)$ , linear?

*Solution.* No. Here  $T(\mathbf{0}) = \mathbf{0}$ , so the quick test says nothing. But (LT2) fails for  $\mathbf{v} = (1, 0)$  and  $c = 2$ :  $T(2\mathbf{v}) = T(2, 0) = (4, 0)$ , while  $2T(\mathbf{v}) = 2(1, 0) = (2, 0)$ . Compare with the projection  $(a_1, a_2) \mapsto (a_1, 0)$  of [Example \(Projection onto an Axis\)](#): squaring the first entry is the one change, and it destroys homogeneity.

### Example. (Length)

Is  $T: \mathbb{R}^2 \rightarrow \mathbb{R}$ ,  $T(x, y) = \sqrt{x^2 + y^2}$ , linear?

*Solution.* No. Take  $\mathbf{v} = (1, 0)$  and  $c = -1$ . Then  $T(-\mathbf{v}) = 1$ , but  $-T(\mathbf{v}) = -1$ , so (LT2) fails. Length does satisfy  $T(c\mathbf{v}) = |c|T(\mathbf{v})$ , which agrees with  $cT(\mathbf{v})$  only for  $c \geq 0$ . Additivity fails too:  $T((1, 0) + (0, 1)) = \sqrt{2}$ , but  $T(1, 0) + T(0, 1) = 2$ .

The last non-example shows that linearity depends on the field, not only on the function.

### Example. (Complex Conjugation: the Field Matters)

Let  $T: \mathbb{C} \rightarrow \mathbb{C}$ ,  $T(z) = \bar{z}$ . Is  $T$  linear when  $\mathbb{C}$  is regarded as a vector space **over**  $\mathbb{C}$ ? Over  $\mathbb{R}$ ?

*Solution.* (LT1) holds in both cases, since  $\overline{z + w} = \bar{z} + \bar{w}$  ([Theorem 0.6.2](#)).

**Over  $\mathbb{C}$ : not linear.** The scalars are all complex numbers. Take  $c = i$  and  $z = 1$ : then  $T(i \cdot 1) = \bar{i} = -i$ , but  $iT(1) = i \cdot 1 = i$ . So (LT2) fails.

**Over  $\mathbb{R}$ : linear.** Now (LT2) is only required for real  $c$ . For  $c \in \mathbb{R}$ ,  $T(cz) = \overline{cz} = c\bar{z} = cT(z)$ , by [Theorem 0.6.2](#) and  $\bar{c} = c$ .

*Warning.* “Linear” here is not the “linear function” of school. A function  $f(x) = mx + b$  has a straight-line graph, but it is a linear map  $\mathbb{R} \rightarrow \mathbb{R}$  only when  $b = 0$ :  $f(0) = b$ , and [Theorem 3.1.1](#) demands  $f(0) = 0$ . In fact every linear map  $T: \mathbb{R} \rightarrow \mathbb{R}$  has the form  $T(x) = mx$ : by (LT2),  $T(x) = T(x \cdot 1) = xT(1)$ , so  $m = T(1)$ . And  $T(\mathbf{0}) = \mathbf{0}$  is **necessary but not sufficient** for linearity, as [Example \(Squaring\)](#) shows.

*Quick check.* Which of these maps  $\mathbb{R}^2 \rightarrow \mathbb{R}^2$  are linear? (i)  $(x, y) \mapsto (y, x)$ ; (ii)  $(x, y) \mapsto (x + y, 1)$ ; (iii)  $(x, y) \mapsto (xy, 0)$ .

*Solution.*

- (i) Linear: it is  $T_{\mathbf{A}}$  with  $\mathbf{A} = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}$ . (ii) Not linear:  $(0, 0) \mapsto (0, 1) \neq \mathbf{0}$ . (iii) Not linear, although  $\mathbf{0} \mapsto \mathbf{0}$ :  $(2, 2) \mapsto (4, 0)$ , but  $2 \cdot T(1, 1) = 2(1, 0) = (2, 0)$ , so (LT2) fails.

**Why this definition.** Could one condition be dropped? Complex conjugation on  $\mathbb{C}$  over  $\mathbb{C}$  satisfies (LT1) but not (LT2), so additivity alone is not enough. (Over  $\mathbb{Q}$ , surprisingly, additivity does imply homogeneity; see [Exercise \(C1: Over  \$\mathbb{Q}\$ , Additivity Is Enough\)](#).) Conversely,  $T: \mathbb{R}^2 \rightarrow \mathbb{R}$ ,  $T(x, y) = \sqrt[3]{x^3 + y^3}$  (the real cube root), satisfies (LT2), since  $\sqrt[3]{c^3x^3 + c^3y^3} = c\sqrt[3]{x^3 + y^3}$  for every real  $c$ . But  $T(1, 0) + T(0, 1) = 2$ , while  $T(1, 1) = \sqrt[3]{2}$ , so (LT1) fails. Neither condition implies the other, and we need both to get [Theorem 3.1.4](#). The name comes from

the case  $\mathbb{R} \rightarrow \mathbb{R}$  just described: the graph of a linear map is a line through the origin.

### 3.1.6 A linear map is determined by a basis

A function on an infinite set normally needs infinitely many values to describe it. A linear map needs far fewer. If  $\mathcal{B} = (\mathbf{v}_1, \dots, \mathbf{v}_n)$  is a basis of  $V$ , every  $\mathbf{v} \in V$  is a combination  $a_1\mathbf{v}_1 + \dots + a_n\mathbf{v}_n$ , and [Theorem 3.1.4](#) forces

$$T(\mathbf{v}) = a_1T(\mathbf{v}_1) + \dots + a_nT(\mathbf{v}_n).$$

So the  $n$  vectors  $T(\mathbf{v}_1), \dots, T(\mathbf{v}_n)$  already fix every value of  $T$ . The converse question is whether **any** choice of these  $n$  vectors comes from a linear map. It does, and that is the theorem.

#### Theorem 3.1.5 — Linear Maps Are Determined by a Basis

Let  $V$  and  $W$  be vector spaces over  $F$ , let  $\mathcal{B} = (\mathbf{v}_1, \dots, \mathbf{v}_n)$  be a basis of  $V$ , and let  $\mathbf{w}_1, \dots, \mathbf{w}_n \in W$  be **any** vectors. Then there **exists exactly one** linear map  $T: V \rightarrow W$  such that

$$T(\mathbf{v}_i) = \mathbf{w}_i \quad \text{for } i = 1, \dots, n.$$

It is given by  $T(a_1\mathbf{v}_1 + \dots + a_n\mathbf{v}_n) = a_1\mathbf{w}_1 + \dots + a_n\mathbf{w}_n$  for all  $a_1, \dots, a_n \in F$ .

*Idea.* **Uniqueness** is the computation above: any linear  $T$  with these values must be given by the formula. **Existence** asks us to *use* the formula as a definition. The only danger is that a vector might have two expressions  $\sum a_i\mathbf{v}_i$ , giving two candidate values; the basis rules that out, because coordinates are unique. So: ① define  $T(\mathbf{v})$  from the coordinates of  $\mathbf{v}$ ; ② check (LT1), (LT2), using that coordinates respect the operations; ③ check  $T(\mathbf{v}_i) = \mathbf{w}_i$ ; ④ prove uniqueness by [Theorem 3.1.4](#).

*Proof. Existence.* Let  $\mathbf{v} \in V$ . By [Theorem 1.5.1](#), there are **unique** scalars  $a_1, \dots, a_n \in F$  with  $\mathbf{v} = a_1\mathbf{v}_1 + \dots + a_n\mathbf{v}_n$ ; they are the entries of  $[\mathbf{v}]_{\mathcal{B}}$  ([Definition 1.5.2](#)). Define

$$T(\mathbf{v}) := a_1\mathbf{w}_1 + \dots + a_n\mathbf{w}_n.$$

Since the scalars are unique, this assigns exactly one vector of  $W$  to each  $\mathbf{v} \in V$ , so  $T: V \rightarrow W$  is a well-defined function.

Let  $\mathbf{u}, \mathbf{v} \in V$  and  $c \in F$ , with  $[\mathbf{u}]_{\mathcal{B}} = (b_1, \dots, b_n)$  and  $[\mathbf{v}]_{\mathcal{B}} = (a_1, \dots, a_n)$ . By [Theorem 1.5.2](#),  $[c\mathbf{u} + \mathbf{v}]_{\mathcal{B}} = c[\mathbf{u}]_{\mathcal{B}} + [\mathbf{v}]_{\mathcal{B}} = (cb_1 + a_1, \dots, cb_n + a_n)$ . Hence, by the definition of  $T$  and the vector space axioms in  $W$ ,

$$T(c\mathbf{u} + \mathbf{v}) = \sum_{i=1}^n (cb_i + a_i)\mathbf{w}_i = c \sum_{i=1}^n b_i\mathbf{w}_i + \sum_{i=1}^n a_i\mathbf{w}_i = cT(\mathbf{u}) + T(\mathbf{v}).$$

By [Theorem 3.1.3](#),  $T$  is linear. Moreover,  $\mathbf{v}_i = 0\mathbf{v}_1 + \dots + 1\mathbf{v}_i + \dots + 0\mathbf{v}_n$ , so  $[\mathbf{v}_i]_{\mathcal{B}} = \mathbf{e}_i$  and  $T(\mathbf{v}_i) = \mathbf{w}_i$ .

*Uniqueness.* Let  $S: V \rightarrow W$  be any linear map with  $S(\mathbf{v}_i) = \mathbf{w}_i$  for all  $i$ . For  $\mathbf{v} = a_1\mathbf{v}_1 + \dots + a_n\mathbf{v}_n \in V$ , [Theorem 3.1.4](#) gives

$$S(\mathbf{v}) = a_1S(\mathbf{v}_1) + \dots + a_nS(\mathbf{v}_n) = a_1\mathbf{w}_1 + \dots + a_n\mathbf{w}_n = T(\mathbf{v}).$$

Since  $S(\mathbf{v}) = T(\mathbf{v})$  for every  $\mathbf{v} \in V$ , the functions are equal:  $S = T$ . This proves the theorem. ■

The theorem has two halves, and they are used differently. Uniqueness says that **to check two linear maps are equal, check them on a basis**. Existence says that **to build a linear map, choose the images of a basis freely**. The same holds for the infinite bases of Chapter 1, §8, because every vector is still a unique **finite** combination of basis vectors; we record this as [Theorem 3.1.6](#) below. Later in this chapter this theorem turns into the matrix of a linear map: once bases are fixed, the  $n$  vectors  $T(\mathbf{v}_i)$ , written in coordinates, are the columns.

### Theorem 3.1.6 — Linear Maps on an Arbitrary Basis

Let  $V$  and  $W$  be vector spaces over  $F$ , let  $B$  be a basis of  $V$  (possibly infinite, in the sense of [Theorem 1.8.2](#)), and for each  $\mathbf{b} \in B$  let  $\mathbf{w}_{\mathbf{b}} \in W$  be any vector. Then there is exactly one linear map  $T: V \rightarrow W$  with  $T\mathbf{b} = \mathbf{w}_{\mathbf{b}}$  for every  $\mathbf{b} \in B$ .

*Proof.* Let  $\mathbf{v} \in V$ . Since  $B$  spans  $V$  and is independent,  $\mathbf{v}$  is a finite combination  $\mathbf{v} = \sum_{\mathbf{b} \in B} a_{\mathbf{b}} \mathbf{b}$  in which only finitely many  $a_{\mathbf{b}}$  are non-zero, and these coefficients are unique: two such expressions differ by a finite combination of distinct elements of  $B$  equal to  $\mathbf{0}$ , whose coefficients all vanish by independence. Define  $T\mathbf{v} = \sum_{\mathbf{b} \in B} a_{\mathbf{b}} \mathbf{w}_{\mathbf{b}}$ , a finite sum. The coefficients of  $\mathbf{v} + \mathbf{u}$  and of  $c\mathbf{v}$  are  $a_{\mathbf{b}} + a'_{\mathbf{b}}$  and  $ca_{\mathbf{b}}$  by uniqueness, so  $T$  is linear, and  $T\mathbf{b} = \mathbf{w}_{\mathbf{b}}$  because the coefficients of  $\mathbf{b}$  are 1 at  $\mathbf{b}$  and 0 elsewhere. If  $S$  is another linear map with  $S\mathbf{b} = \mathbf{w}_{\mathbf{b}}$  for all  $\mathbf{b}$ , then  $S\mathbf{v} = \sum a_{\mathbf{b}} S\mathbf{b} = T\mathbf{v}$  by [Theorem 3.1.4](#). This proves the theorem. ■

#### Example. (Finding a Formula from Values on a Basis)

Show that there is exactly one linear map  $T: \mathbb{R}^2 \rightarrow \mathbb{R}^3$  with

$$T(1, 1) = (1, 2, 0), \quad T(1, -1) = (3, 0, 2),$$

and find  $T(x, y)$ .

*Solution.* First,  $((1, 1), (1, -1))$  is a basis of  $\mathbb{R}^2$ . The matrix with these columns is  $\begin{pmatrix} 1 & 1 \\ 1 & -1 \end{pmatrix}$ , and  $ad - bc = 1 \cdot (-1) - 1 \cdot 1 = -2 \neq 0$ . So the matrix is invertible ([Theorem 0.9.8](#)), and its columns form a basis of  $\mathbb{R}^2$  ([Theorem 2.4.4](#)). By [Theorem 3.1.5](#), exactly one such  $T$  exists.

To find it, write  $(x, y)$  in the basis. Solving  $a(1, 1) + b(1, -1) = (x, y)$ , that is  $a + b = x$  and  $a - b = y$ , gives  $a = \frac{x+y}{2}$  and  $b = \frac{x-y}{2}$ . By the formula in the theorem,

$$T(x, y) = \frac{x+y}{2}(1, 2, 0) + \frac{x-y}{2}(3, 0, 2) = (2x - y, x + y, x - y).$$

Check:  $T(1, 1) = (1, 2, 0)$  and  $T(1, -1) = (3, 0, 2)$ .

**Example. (Too Many Conditions)**

Is there a linear map  $T: \mathbb{R}^2 \rightarrow \mathbb{R}^3$  with

$$T(1, 0) = (2, 3, 4), \quad T(1, 1) = (3, 4, 5), \quad T(2, 3) = (4, 5, 6)?$$

*Solution.* No. The matrix  $\begin{pmatrix} 1 & 1 \\ 0 & 1 \end{pmatrix}$  with columns  $(1, 0)$  and  $(1, 1)$  has  $ad - bc = 1 \cdot 1 - 1 \cdot 0 = 1 \neq 0$ , so as in [Example \(Finding a Formula from Values on a Basis\)](#) these two vectors form a basis of  $\mathbb{R}^2$ . By [Theorem 3.1.5](#), the first two conditions already determine  $T$ , and the third condition must be checked against them, not imposed. Since  $(2, 3) = -1 \cdot (1, 0) + 3 \cdot (1, 1)$ , any linear  $T$  with the first two values satisfies

$$T(2, 3) = -T(1, 0) + 3T(1, 1) = -(2, 3, 4) + 3(3, 4, 5) = (7, 9, 11)$$

by [Theorem 3.1.4](#). As  $(7, 9, 11) \neq (4, 5, 6)$ , no such  $T$  exists. Had the third value been  $(7, 9, 11)$ , the answer would be yes, with exactly one such map.

**Example. (Recognizing Differentiation)**

Show that the differentiation map  $D: \mathbb{R}[x]_{\leq 3} \rightarrow \mathbb{R}[x]_{\leq 2}$  is the **only** linear map  $T: \mathbb{R}[x]_{\leq 3} \rightarrow \mathbb{R}[x]_{\leq 2}$  with

$$T(1) = 0, \quad T(x) = 1, \quad T(x^2) = 2x, \quad T(x^3) = 3x^2.$$

*Solution.* The list  $(1, x, x^2, x^3)$  is a basis of  $\mathbb{R}[x]_{\leq 3}$  ([Example \(Standard Bases\)](#)). The map  $D$  is linear ([Example \(Differentiation\)](#)), and  $D(1) = 0$ ,  $D(x) = 1$ ,  $D(x^2) = 2x$ ,  $D(x^3) = 3x^2$ , so  $D$  has the four prescribed values. By the uniqueness half of [Theorem 3.1.5](#), every linear  $T$  with these values equals  $D$ . Four values, the derivatives of the monomials, pin down the derivative of every cubic.

**Warning.** The values must be prescribed on a basis, or at least on an independent list. On a dependent list, the values may contradict each other. There is **no** linear  $T: \mathbb{R}^2 \rightarrow \mathbb{R}^2$  with  $T(1, 0) = (1, 0)$  and  $T(2, 0) = (0, 1)$ : since  $(2, 0) = 2(1, 0)$ , (LT2) forces  $T(2, 0) = 2T(1, 0) = (2, 0) \neq (0, 1)$ . On the other hand, a list that does not span leaves  $T$  undetermined: many linear maps satisfy  $T(1, 0) = (1, 0)$  alone.

*Quick check.* Let  $T: \mathbb{R}^2 \rightarrow \mathbb{R}$  be linear with  $T(1, 2) = 3$  and  $T(2, 5) = 1$ . Find  $T(x, y)$ .

*Solution.* Here  $ad - bc = 1 \cdot 5 - 2 \cdot 2 = 1 \neq 0$ , so  $((1, 2), (2, 5))$  is a basis. Solving  $a(1, 2) + b(2, 5) = (x, y)$ , that is  $a + 2b = x$  and  $2a + 5b = y$ , gives  $b = y - 2x$  and  $a = 5x - 2y$ . Hence  $T(x, y) = 3(5x - 2y) + 1 \cdot (y - 2x) = 13x - 5y$ . Check:  $13 - 10 = 3$  and  $26 - 25 = 1$ .

We now have a supply of linear maps and a way to build new ones from a basis. The next section attaches to every linear map two subspaces that measure how far it is from being injective and from being surjective.

## 3.1.7 Exercises

### 3.1.7.1 A. Check your understanding

#### Exercise A1.

- (a) Define a linear map  $T: V \rightarrow W$  over  $F$ .
- (b) State the theorem that says a linear map is determined by a basis, including both existence and uniqueness.
- (c) True or false: if  $T: V \rightarrow W$  satisfies  $T(\mathbf{0}) = \mathbf{0}$ , then  $T$  is linear. Justify your answer.
- (d) True or false: there is a linear map  $T: \mathbb{R}^2 \rightarrow \mathbb{R}^2$  with  $T(1, 1) = (1, 0)$  and  $T(2, 2) = (3, 0)$ . Justify your answer.
- (e) True or false: complex conjugation  $\mathbb{C} \rightarrow \mathbb{C}$  is linear over  $\mathbb{R}$ . Justify your answer.

*Solution.*

- (a) See [Definition 3.1.1](#):  $V, W$  are vector spaces over the same field  $F$ , and  $T(\mathbf{u} + \mathbf{v}) = T(\mathbf{u}) + T(\mathbf{v})$  and  $T(c\mathbf{v}) = cT(\mathbf{v})$  for all  $\mathbf{u}, \mathbf{v} \in V$  and  $c \in F$ .
- (b) See [Theorem 3.1.5](#): if  $(\mathbf{v}_1, \dots, \mathbf{v}_n)$  is a basis of  $V$  and  $\mathbf{w}_1, \dots, \mathbf{w}_n \in W$  are arbitrary, there is one and only one linear  $T: V \rightarrow W$  with  $T(\mathbf{v}_i) = \mathbf{w}_i$  for all  $i$ .
- (c) False.  $T: \mathbb{R}^2 \rightarrow \mathbb{R}^2$ ,  $T(a_1, a_2) = (a_1^2, 0)$ , sends  $\mathbf{0}$  to  $\mathbf{0}$  but is not linear ([Example \(Squaring\)](#)).
- (d) False. By (LT2),  $T(2, 2) = 2T(1, 1) = (2, 0) \neq (3, 0)$ .
- (e) True. See [Example \(Complex Conjugation: the Field Matters\)](#): for real  $c$ ,  $\overline{cz} = c\overline{z}$ , and  $\overline{z + w} = \overline{z} + \overline{w}$ .

### 3.1.7.2 B. Practice

#### Exercise B1: Which Maps Are Linear?.

Determine which of the following functions are linear maps over  $\mathbb{R}$ . Justify your answer.

- (a)  $T: \mathbb{R}^3 \rightarrow \mathbb{R}^2$ ,  $T(x, y, z) = (x - 2z, 3y)$ .
- (b)  $T: \mathbb{R}^2 \rightarrow \mathbb{R}^2$ ,  $T(x, y) = (x + 1, y)$ .
- (c)  $T: M_2(\mathbb{R}) \rightarrow \mathbb{R}$ ,  $T\begin{pmatrix} a & b \\ c & d \end{pmatrix} = ad - bc$ .
- (d)  $T: \mathbb{R}[x]_{\leq 2} \rightarrow \mathbb{R}[x]_{\leq 3}$ ,  $T(p) = x^2 p'$ .
- (e)  $T: \mathbb{R}^2 \rightarrow \mathbb{R}$ ,  $T(x, y) = xy$ .
- (f)  $T: \mathbb{R}^{\mathbb{N}} \rightarrow \mathbb{R}^{\mathbb{N}}$ ,  $T(s_0, s_1, s_2, \dots) = (s_1 - s_0, s_2 - s_1, s_3 - s_2, \dots)$ .

*Solution.*

- (a) Linear.  $T = T_{\mathbf{A}}$  with  $\mathbf{A} = \begin{pmatrix} 1 & 0 & -2 \\ 0 & 3 & 0 \end{pmatrix}$ , which is linear by [Example \(Multiplication by a Matrix\)](#).
- (b) Not linear.  $T(0, 0) = (1, 0) \neq \mathbf{0}$ , contradicting [Theorem 3.1.1](#).

- (c) Not linear. Take  $\mathbf{I}_2$  and  $c = 2$ :  $T(2\mathbf{I}_2) = 2 \cdot 2 - 0 = 4$ , but  $2T(\mathbf{I}_2) = 2 \cdot 1 = 2$ . So (LT2) fails. (Additivity fails too:  $T(\mathbf{E}_{11}) + T(\mathbf{E}_{22}) = 0 + 0 = 0$ , but  $T(\mathbf{E}_{11} + \mathbf{E}_{22}) = T(\mathbf{I}_2) = 1$ .)
- (d) Linear. For  $p \in \mathbb{R}[x]_{\leq 2}$ ,  $p'$  has degree at most 1, so  $x^2 p' \in \mathbb{R}[x]_{\leq 3}$ . For  $p, q$  and  $c \in \mathbb{R}$ , by [Example \(Differentiation\)](#) and the distributive law,  $T(cp + q) = x^2(cp' + q') = cx^2 p' + x^2 q' = cT(p) + T(q)$ . By [Theorem 3.1.3](#),  $T$  is linear.
- (e) Not linear, although  $T(0, 0) = 0$ .  $T(2(1, 1)) = T(2, 2) = 4$ , but  $2T(1, 1) = 2$ .
- (f) Linear. The entry in position  $k$  of  $T(cs + \mathbf{t})$  is  $(cs_{k+1} + t_{k+1}) - (cs_k + t_k) = c(s_{k+1} - s_k) + (t_{k+1} - t_k)$ , which is the entry in position  $k$  of  $cT(\mathbf{s}) + T(\mathbf{t})$ . By [Theorem 3.1.3](#),  $T$  is linear.

### Exercise B2: A Formula from Values on a Basis.

- (a) Show that there is exactly one linear map  $T: \mathbb{R}^2 \rightarrow \mathbb{R}^2$  with  $T(2, 1) = (1, 4)$  and  $T(1, 1) = (0, 3)$ . Hence find  $T(x, y)$  and a matrix  $\mathbf{A}$  with  $T = T_{\mathbf{A}}$ .
- (b) Let  $S: \mathbb{R}[x]_{\leq 2} \rightarrow \mathbb{R}$  be linear with  $S(1) = 1$ ,  $S(1 + x) = 3$  and  $S(1 + x + x^2) = 6$ . Find  $S(a + bx + cx^2)$ .

*Solution.*

- (a) The matrix with columns  $(2, 1)$  and  $(1, 1)$  has  $ad - bc = 2 \cdot 1 - 1 \cdot 1 = 1 \neq 0$ , so it is invertible ([Theorem 0.9.8](#)) and its columns form a basis of  $\mathbb{R}^2$  ([Theorem 2.4.4](#)). By [Theorem 3.1.5](#), exactly one such  $T$  exists. Solving  $a(2, 1) + b(1, 1) = (x, y)$ , that is  $2a + b = x$  and  $a + b = y$ , gives  $a = x - y$  and  $b = 2y - x$ . Hence

$$T(x, y) = (x - y)(1, 4) + (2y - x)(0, 3) = (x - y, x + 2y).$$

Check:  $T(2, 1) = (1, 4)$  and  $T(1, 1) = (0, 3)$ . Hence  $T = T_{\mathbf{A}}$  with  $\mathbf{A} = \begin{pmatrix} 1 & -1 \\ 1 & 2 \end{pmatrix}$ .

- (b) The polynomials  $1, 1 + x, 1 + x + x^2$  are non-zero of distinct degrees, so they are independent ([Theorem 1.4.4](#)); there are  $3 = \dim \mathbb{R}[x]_{\leq 2}$  of them, so they form a basis ([Theorem 1.6.4](#)). By (LT1) and [Theorem 3.1.2](#),  $S(x) = S(1 + x) - S(1) = 2$  and  $S(x^2) = S(1 + x + x^2) - S(1 + x) = 3$ . By [Theorem 3.1.4](#),

$$S(a + bx + cx^2) = aS(1) + bS(x) + cS(x^2) = a + 2b + 3c.$$

### Exercise B3: Evaluation and Integration.

- (a) Prove that  $\varphi: \mathbb{R}[x] \rightarrow \mathbb{R}$ ,  $\varphi(p) = p(3)$ , is linear.
- (b) Prove that  $\psi: \mathbb{R}[x] \rightarrow \mathbb{R}$ ,  $\psi(p) = \int_0^1 p(t) dt$ , is linear.
- (c) Hence find  $\varphi(a + bx + cx^2)$  and  $\psi(a + bx + cx^2)$ .

*Solution.*

- (a) Let  $p, q \in \mathbb{R}[x]$  and  $c \in \mathbb{R}$ . By [Theorem 0.8.5](#),  $(cp + q)(3) = cp(3) + q(3)$ , where  $cp$  is the product of the constant polynomial  $c$  with  $p$ , whose value at 3 is  $cp(3)$ . So  $\varphi(cp + q) = c\varphi(p) + \varphi(q)$ , and  $\varphi$  is linear by [Theorem 3.1.3](#).
- (b) By linearity of the integral,  $\int_0^1 (cp(t) + q(t)) dt = c \int_0^1 p(t) dt + \int_0^1 q(t) dt$ . So  $\psi(cp +$



$q) = c\psi(p) + \psi(q)$ , and  $\psi$  is linear by [Theorem 3.1.3](#).

- (c) On the basis:  $\varphi(1) = 1$ ,  $\varphi(x) = 3$ ,  $\varphi(x^2) = 9$ , and  $\psi(1) = 1$ ,  $\psi(x) = \frac{1}{2}$ ,  $\psi(x^2) = \frac{1}{3}$ .  
By [Theorem 3.1.4](#),

$$\varphi(a + bx + cx^2) = a + 3b + 9c, \quad \psi(a + bx + cx^2) = a + \frac{1}{2}b + \frac{1}{3}c.$$

### 3.1.7.3 C. Going deeper

#### Exercise C1: Over $\mathbb{Q}$ , Additivity Is Enough.

Let  $V$  and  $W$  be vector spaces over  $\mathbb{Q}$ , and let  $T: V \rightarrow W$  be **additive**:  $T(\mathbf{u} + \mathbf{v}) = T(\mathbf{u}) + T(\mathbf{v})$  for all  $\mathbf{u}, \mathbf{v} \in V$ .

- (a) Prove that  $T(n\mathbf{v}) = nT(\mathbf{v})$  for every  $n \in \mathbb{N}$  and  $\mathbf{v} \in V$ .
- (b) Prove that  $T(n\mathbf{v}) = nT(\mathbf{v})$  for every  $n \in \mathbb{Z}$ .
- (c) Prove that  $T(q\mathbf{v}) = qT(\mathbf{v})$  for every  $q \in \mathbb{Q}$ . Deduce that  $T$  is linear.

*Hint: for (c), apply (b) to the vector  $q\mathbf{v}$ .*

*Solution.*

- (a) We use induction on  $n$ . For  $n = 0$ : the proof of [Theorem 3.1.1](#) used only additivity, so  $T(\mathbf{0}) = \mathbf{0}$ , that is,  $T(0\mathbf{v}) = 0T(\mathbf{v})$ . If  $T(n\mathbf{v}) = nT(\mathbf{v})$ , then by additivity  $T((n+1)\mathbf{v}) = T(n\mathbf{v} + \mathbf{v}) = T(n\mathbf{v}) + T(\mathbf{v}) = nT(\mathbf{v}) + T(\mathbf{v}) = (n+1)T(\mathbf{v})$ .
- (b) Let  $n \in \mathbb{N}$ . By additivity and  $T(\mathbf{0}) = \mathbf{0}$ ,  $\mathbf{0} = T(n\mathbf{v} + (-n)\mathbf{v}) = T(n\mathbf{v}) + T((-n)\mathbf{v})$ , so  $T((-n)\mathbf{v}) = -T(n\mathbf{v}) = -nT(\mathbf{v}) = (-n)T(\mathbf{v})$ , using (a). Every integer is  $n$  or  $-n$  for some  $n \in \mathbb{N}$ .
- (c) Let  $q = m/n$  with  $m \in \mathbb{Z}$  and  $n \geq 1$ . By (b) applied to the vector  $q\mathbf{v}$ , and then to  $\mathbf{v}$ ,

$$nT(q\mathbf{v}) = T(nq\mathbf{v}) = T(m\mathbf{v}) = mT(\mathbf{v}).$$

Multiplying by  $\frac{1}{n} \in \mathbb{Q}$  gives  $T(q\mathbf{v}) = \frac{m}{n}T(\mathbf{v}) = qT(\mathbf{v})$ . So (LT2) holds for every scalar  $q \in \mathbb{Q}$ , and with additivity, which is (LT1),  $T$  is linear.

#### Exercise C2: Over $\mathbb{C}$ , Additivity Is Not Enough.

- (a) Give an additive map  $\mathbb{C} \rightarrow \mathbb{C}$  that is **not** linear over  $\mathbb{C}$ , and explain which step of [Exercise \(C1: Over  \$\mathbb{Q}\$ , Additivity Is Enough\)](#) fails for it.
- (b) Let  $V$  and  $W$  be vector spaces over  $\mathbb{C}$ . Regarding them also as spaces over  $\mathbb{R}$ , let  $T: V \rightarrow W$  be linear **over**  $\mathbb{R}$ , and suppose  $T(i\mathbf{v}) = iT(\mathbf{v})$  for all  $\mathbf{v} \in V$ . Prove that  $T$  is linear over  $\mathbb{C}$ .

*Solution.*

- (a) Complex conjugation  $T(z) = \bar{z}$  is additive and not  $\mathbb{C}$ -linear ([Example \(Complex Conjugation: the Field Matters\)](#)). The argument of [Exercise \(C1: Over  \$\mathbb{Q}\$ , Additivity Is Enough\)](#) still shows  $T(qz) = qT(z)$  for rational  $q$ , and that is true for conjugation. What fails is that the rational numbers are not all the scalars: nothing in that argument reaches a scalar like  $i$ , which is not built from 1 by adding,

subtracting, multiplying and dividing.

- (b) (LT1) holds because  $T$  is linear over  $\mathbb{R}$ . For (LT2), let  $c = a + bi \in \mathbb{C}$  with  $a, b \in \mathbb{R}$ , and  $\mathbf{v} \in V$ . By the vector space axioms,  $c\mathbf{v} = a\mathbf{v} + b(i\mathbf{v})$ . Using (LT1), then  $\mathbb{R}$ -homogeneity, then the hypothesis,

$$\begin{aligned} T(c\mathbf{v}) &= T(a\mathbf{v}) + T(b(i\mathbf{v})) = aT(\mathbf{v}) + bT(i\mathbf{v}) \\ &= aT(\mathbf{v}) + b iT(\mathbf{v}) = (a + bi)T(\mathbf{v}) = cT(\mathbf{v}). \end{aligned}$$

Hence  $T$  is linear over  $\mathbb{C}$ .

## 3.2 Kernel and Image

For any function, Chapter 0 asked two questions: is it injective, and is it surjective? For a function between sets, answering them means comparing outputs pair by pair and hunting for preimages point by point. For a linear map both questions reduce to a single subspace each. This section defines the two subspaces, the kernel and the image, shows how to compute them, and proves the test that makes injectivity cheap: a linear map is injective exactly when its kernel is zero.

### 3.2.1 Two questions, two subspaces

Let  $T: V \rightarrow W$  be linear. Start with injectivity. Suppose two inputs collide,  $T(\mathbf{u}) = T(\mathbf{v})$ . By [Theorem 3.1.2](#) this is the same as

$$T(\mathbf{u} - \mathbf{v}) = T(\mathbf{u}) - T(\mathbf{v}) = \mathbf{0}.$$

So a collision between  $\mathbf{u}$  and  $\mathbf{v}$  is the same thing as their difference being sent to  $\mathbf{0}$ . To detect every collision at once, it is enough to know **which vectors are sent to  $\mathbf{0}$** . For a general function this trick is unavailable, because there is no subtraction; linearity turns a question about all pairs into a question about one set.

Surjectivity asks whether every  $\mathbf{w} \in W$  is an output, so the set to know is **the set of all outputs**. Chapter 2 met both sets for  $T_{\mathbf{A}}(\mathbf{x}) = \mathbf{Ax}$ : the system  $\mathbf{Ax} = \mathbf{b}$  is consistent exactly when  $\mathbf{b}$  is in the column space ([Theorem 2.1.1](#)), and a consistent system has a unique solution exactly when  $\mathbf{Ax} = \mathbf{0}$  has only the trivial solution ([Corollary 2.1.4](#)). The definitions below give these two sets a name for every linear map. They are defined for what they measure.

*The kernel is everything  $T$  sends to zero; the image is everything  $T$  reaches.*

#### Definition 3.2.1 — Kernel

Let  $T: V \rightarrow W$  be a linear map over  $F$ . The **kernel** of  $T$  is the subset of the **domain**

$$\ker T := \{\mathbf{v} \in V : T(\mathbf{v}) = \mathbf{0}_W\} \subseteq V.$$

#### Definition 3.2.2 — Image

Let  $T: V \rightarrow W$  be a linear map over  $F$ . The **image** of  $T$  is the subset of the **codomain**

$$\operatorname{im} T := \{T(\mathbf{v}) : \mathbf{v} \in V\} = \{\mathbf{w} \in W : \mathbf{w} = T(\mathbf{v}) \text{ for some } \mathbf{v} \in V\} \subseteq W.$$

In words: a vector  $\mathbf{v}$  is in  $\ker T$  when  $\mathbf{v}$  lives in  $V$  **and**  $T$  sends it to the zero vector of  $W$ . A vector  $\mathbf{w}$  is in  $\operatorname{im} T$  when  $\mathbf{w}$  lives in  $W$  **and** there **exists** at least one input with output  $\mathbf{w}$ ; the input need not be unique. In the language of Definition 0.4.2,  $\ker T = T^{-1}(\{\mathbf{0}\})$  is the preimage of the zero vector, and  $\operatorname{im} T = T(V)$  is the image of the function  $T$ . The kernel is also called the **null space** of  $T$ , and the image its **range**. For group homomorphisms, the kernel was defined the same way (Definition 0.10.3), with  $\mathbf{0}$  playing the role of the identity element.

There is nothing to check for well-definedness: both sets are described by conditions that either hold or fail. What does need checking is that they are **subspaces**, not just subsets.

### Theorem 3.2.1 — The Kernel Is a Subspace

Let  $T: V \rightarrow W$  be linear. Then  $\ker T$  is a subspace of  $V$ .

*Proof.* We use the Subspace Test (Theorem 1.2.1).

- (1) By Theorem 3.1.1,  $T(\mathbf{0}_V) = \mathbf{0}_W$ , so  $\mathbf{0}_V \in \ker T$ .
- (2) Let  $\mathbf{u}, \mathbf{v} \in \ker T$ , so  $T(\mathbf{u}) = T(\mathbf{v}) = \mathbf{0}_W$ . By (LT1),  $T(\mathbf{u} + \mathbf{v}) = T(\mathbf{u}) + T(\mathbf{v}) = \mathbf{0}_W + \mathbf{0}_W = \mathbf{0}_W$ . Therefore  $\mathbf{u} + \mathbf{v} \in \ker T$ .
- (3) Let  $\mathbf{v} \in \ker T$  and  $c \in F$ . By (LT2),  $T(c\mathbf{v}) = cT(\mathbf{v}) = c\mathbf{0}_W = \mathbf{0}_W$ , using Theorem 1.1.6. Therefore  $c\mathbf{v} \in \ker T$ .

■

### Theorem 3.2.2 — The Image Is a Subspace

Let  $T: V \rightarrow W$  be linear. Then:

- (a)  $\operatorname{im} T$  is a subspace of  $W$ ;
- (b)  $T$  is surjective if and only if  $\operatorname{im} T = W$ .

*Proof.*

- (a) We use the Subspace Test (Theorem 1.2.1). (1)  $\mathbf{0}_W = T(\mathbf{0}_V)$  by Theorem 3.1.1, so  $\mathbf{0}_W \in \operatorname{im} T$ . (2) Let  $\mathbf{w}_1, \mathbf{w}_2 \in \operatorname{im} T$ , say  $\mathbf{w}_1 = T(\mathbf{v}_1)$  and  $\mathbf{w}_2 = T(\mathbf{v}_2)$  with  $\mathbf{v}_1, \mathbf{v}_2 \in V$ . By (LT1),  $\mathbf{w}_1 + \mathbf{w}_2 = T(\mathbf{v}_1 + \mathbf{v}_2)$ , and  $\mathbf{v}_1 + \mathbf{v}_2 \in V$ , so  $\mathbf{w}_1 + \mathbf{w}_2 \in \operatorname{im} T$ . (3) Let  $\mathbf{w} = T(\mathbf{v}) \in \operatorname{im} T$  and  $c \in F$ . By (LT2),  $c\mathbf{w} = T(c\mathbf{v})$  with  $c\mathbf{v} \in V$ , so  $c\mathbf{w} \in \operatorname{im} T$ .
- (b) The inclusion  $\operatorname{im} T \subseteq W$  always holds. By Definition 0.4.3,  $T$  is surjective exactly when every  $\mathbf{w} \in W$  equals  $T(\mathbf{v})$  for some  $\mathbf{v}$ , that is, when  $W \subseteq \operatorname{im} T$ . So  $T$  is surjective if and only if  $\operatorname{im} T = W$ .

■

This pays off a pattern from Chapter 1, §2. The null space of a matrix (Example (Solutions of a Homogeneous System)), the symmetric matrices (Example (Symmetric Matrices)) and the trace-free matrices (Example (Trace-Free Matrices)) were each shown to be subspaces by a separate check, and that section promised a single theorem. Each is the kernel of a linear map ( $T_{\mathbf{A}}, \mathbf{A} \mapsto \mathbf{A} - \mathbf{A}^T$ , and the trace), so Theorem 3.2.1 covers them all at once.

**First examples.** Start with the maps that have the most extreme kernels and images.

**Example. (The Zero Map and the Identity)**

Find the kernel and image of the zero map  $0: V \rightarrow W$  and of the identity  $\text{id}_V: V \rightarrow V$ .

*Solution.* The zero map sends every vector to  $\mathbf{0}_W$ , so  $\ker 0 = V$  and  $\text{im } 0 = \{\mathbf{0}_W\}$ . The identity sends  $\mathbf{v}$  to  $\mathbf{v}$ , which is  $\mathbf{0}$  only when  $\mathbf{v} = \mathbf{0}$ , so  $\ker \text{id}_V = \{\mathbf{0}\}$ . Every  $\mathbf{v}$  is its own image, so  $\text{im } \text{id}_V = V$ .

These two are the ends of the scale. The kernel is always between  $\{\mathbf{0}\}$  and  $V$ , and the image between  $\{\mathbf{0}\}$  and  $W$ . The zero map kills everything and reaches nothing but  $\mathbf{0}$ ; the identity kills nothing and reaches everything.

**Example. (Projection onto an Axis)**

Find the kernel and image of the projection  $T: \mathbb{R}^2 \rightarrow \mathbb{R}^2$ ,  $T(a_1, a_2) = (a_1, 0)$ , of Example (Projection onto an Axis).

*Solution.*  $T(a_1, a_2) = (0, 0)$  exactly when  $a_1 = 0$ . So  $\ker T = \{(0, a_2) : a_2 \in \mathbb{R}\}$ , the  $y$ -axis. Every output has second entry 0, and every  $(a_1, 0)$  is an output, namely  $T(a_1, 0)$ . So  $\text{im } T$  is the  $x$ -axis. The kernel is the direction the light shines along; the image is the screen the shadow falls on.

**A non-example by minimal change.** Replace  $\mathbf{0}$  by another vector: for the projection,  $\{\mathbf{v} : T(\mathbf{v}) = (1, 0)\} = \{(1, a_2) : a_2 \in \mathbb{R}\}$ . This is the vertical line through  $(1, 0)$ . It does not contain  $\mathbf{0}$ , and it is not closed under addition, since  $(1, 0) + (1, 0) = (2, 0)$  lies off the line. So it is **not** a subspace. It is the kernel shifted by  $(1, 0)$ . The same thing happened for systems in Chapter 2: the solutions of  $\mathbf{Ax} = \mathbf{b}$  are  $\mathbf{p} + \text{nul}(\mathbf{A})$  (Theorem 2.1.3).

**Why the preimage of  $\mathbf{0}$ ?** Because  $\mathbf{0}$  is the one output that every linear map is guaranteed to hit, so its preimage is never empty, and because all other preimages are translates of it. If  $T(\mathbf{p}) = \mathbf{w}$ , then  $T(\mathbf{v}) = \mathbf{w}$  exactly when  $T(\mathbf{v} - \mathbf{p}) = \mathbf{0}$ , that is, when  $\mathbf{v} \in \mathbf{p} + \ker T$ . The kernel describes the shape of every non-empty preimage at once.

**Warning. Keep track of where each subspace lives.** The kernel sits inside the **domain**  $V$ ; the image sits inside the **codomain**  $W$ . For  $T_{\mathbf{A}}: F^n \rightarrow F^m$ , the kernel consists of columns with  $n$  entries and the image of columns with  $m$  entries, so “ $\ker T = \text{im } T$ ” does not even make sense unless  $V = W$ . And the image is **not** the codomain in general: the differentiation map  $D: \mathbb{R}[x]_{\leq 3} \rightarrow \mathbb{R}[x]_{\leq 3}$  has codomain  $\mathbb{R}[x]_{\leq 3}$ , but  $x^3$  is not the derivative of any cubic, so  $\text{im } D \neq \mathbb{R}[x]_{\leq 3}$ .

### 3.2.2 Injectivity via the kernel

The computation that opened the section now becomes a theorem. It is the reason to prefer the kernel over the definition of injectivity whenever a map is linear.

#### Theorem 3.2.3 — Kernel Test for Injectivity

Let  $T: V \rightarrow W$  be linear. Then  $T$  is injective **if and only if**  $\ker T = \{0\}$ .

*Proof.* ( $\Rightarrow$ ) Suppose  $T$  is injective, and let  $\mathbf{v} \in \ker T$ . Then  $T(\mathbf{v}) = \mathbf{0}_W = T(\mathbf{0}_V)$  by Theorem 3.1.1. Injectivity gives  $\mathbf{v} = \mathbf{0}_V$ . Hence  $\ker T \subseteq \{0\}$ , and the reverse inclusion holds because  $\ker T$  is a subspace (Theorem 3.2.1).

( $\Leftarrow$ ) Suppose  $\ker T = \{0\}$ , and let  $T(\mathbf{u}) = T(\mathbf{v})$ . By Theorem 3.1.2,  $T(\mathbf{u} - \mathbf{v}) = T(\mathbf{u}) - T(\mathbf{v}) = \mathbf{0}$ , so  $\mathbf{u} - \mathbf{v} \in \ker T = \{0\}$ . Hence  $\mathbf{u} = \mathbf{v}$ , and  $T$  is injective. ■

Compare the work. The definition asks: for **all pairs**  $\mathbf{u}, \mathbf{v}$ , does  $T(\mathbf{u}) = T(\mathbf{v})$  force  $\mathbf{u} = \mathbf{v}$ ? The theorem asks only: for **each**  $\mathbf{v}$ , does  $T(\mathbf{v}) = \mathbf{0}$  force  $\mathbf{v} = \mathbf{0}$ ? The second question is usually a homogeneous linear system or a coefficient comparison. For  $T_{\mathbf{A}}$ , the theorem says that  $\mathbf{x} \mapsto \mathbf{Ax}$  is injective exactly when  $\mathbf{Ax} = \mathbf{0}$  has only the trivial solution, which is the content of Corollary 2.1.4.

### 3.2.3 Computing the image

The image is defined by an existence condition, which is hard to check directly. When  $V$  has a spanning list, applying  $T$  to it gives a spanning list of  $\operatorname{im} T$ .

#### Theorem 3.2.4 — The Image Is Spanned by the Images of a Spanning List

Let  $T: V \rightarrow W$  be linear, and suppose  $(\mathbf{v}_1, \dots, \mathbf{v}_k)$  spans  $V$ . Then

$$\operatorname{im} T = \operatorname{span}(T(\mathbf{v}_1), \dots, T(\mathbf{v}_k)).$$

In particular, if  $V$  is finite-dimensional, then so is  $\operatorname{im} T$ , even when  $W$  is not.

*Proof.* ( $\supseteq$ ) Each  $T(\mathbf{v}_i)$  lies in  $\operatorname{im} T$ , which is a subspace by Theorem 3.2.2. By Theorem 1.3.1 (3), the span of these vectors is contained in  $\operatorname{im} T$ .

( $\subseteq$ ) Let  $\mathbf{w} \in \operatorname{im} T$ , say  $\mathbf{w} = T(\mathbf{v})$  with  $\mathbf{v} \in V$ . Since  $(\mathbf{v}_1, \dots, \mathbf{v}_k)$  spans  $V$ ,  $\mathbf{v} = a_1\mathbf{v}_1 + \dots + a_k\mathbf{v}_k$  for some  $a_i \in F$ . By Theorem 3.1.4,  $\mathbf{w} = a_1T(\mathbf{v}_1) + \dots + a_kT(\mathbf{v}_k) \in \operatorname{span}(T(\mathbf{v}_1), \dots, T(\mathbf{v}_k))$ .

If  $V$  is finite-dimensional, some finite list spans  $V$  (Definition 1.3.4), and its images form a finite list spanning  $\operatorname{im} T$ . ■

So the recipe for the image is: apply  $T$  to a basis (or any spanning list) of  $V$ , then sift the results to a basis if one is wanted ([Theorem 1.5.4](#)). The images of a basis need not be independent; the projection sends the basis  $(\mathbf{e}_1, \mathbf{e}_2)$  to  $(\mathbf{e}_1, \mathbf{0})$ .

### 3.2.4 Examples

We now compute kernels and images across the families. In each case we then read off injectivity from the kernel and surjectivity from the image.

#### Example. (A Matrix Map)

Let  $\mathbf{A} = \begin{pmatrix} 1 & 2 & 0 & 3 \\ 2 & 4 & 1 & 8 \\ -1 & -2 & 1 & -1 \end{pmatrix} \in M_{3 \times 4}(\mathbb{R})$ . Find bases of  $\ker T_{\mathbf{A}}$  and  $\operatorname{im} T_{\mathbf{A}}$  for  $T_{\mathbf{A}}: \mathbb{R}^4 \rightarrow \mathbb{R}^3$ , and decide whether  $T_{\mathbf{A}}$  is injective or surjective.

*Solution.* By definition,  $\ker T_{\mathbf{A}} = \{\mathbf{x} : \mathbf{A}\mathbf{x} = \mathbf{0}\} = \operatorname{nul}(\mathbf{A})$  ([Definition 2.5.3](#)) and  $\operatorname{im} T_{\mathbf{A}} = \{\mathbf{A}\mathbf{x} : \mathbf{x} \in \mathbb{R}^4\} = \operatorname{col}(\mathbf{A})$  ([Definition 2.5.1](#)). So Chapter 2's methods apply. Row reduce:

$$\mathbf{A} \xrightarrow[R_3+R_1]{R_2-2R_1} \begin{pmatrix} 1 & 2 & 0 & 3 \\ 0 & 0 & 1 & 2 \\ 0 & 0 & 1 & 2 \end{pmatrix} \xrightarrow{R_3-R_2} \begin{pmatrix} 1 & 2 & 0 & 3 \\ 0 & 0 & 1 & 2 \\ 0 & 0 & 0 & 0 \end{pmatrix}.$$

This is the RREF, with pivot columns 1 and 3 and free columns 2 and 4. By [Theorem 2.5.4](#), a basis of  $\ker T_{\mathbf{A}}$  is

$$((-2, 1, 0, 0), (-3, 0, -2, 1)).$$

By [Theorem 2.5.2](#), the pivot columns of  $\mathbf{A}$  form a basis of  $\operatorname{im} T_{\mathbf{A}}$ :  $((1, 2, -1), (0, 1, 1))$ .

Since  $\ker T_{\mathbf{A}} \neq \{\mathbf{0}\}$ ,  $T_{\mathbf{A}}$  is not injective ([Theorem 3.2.3](#)); for instance  $T_{\mathbf{A}}(-2, 1, 0, 0) = \mathbf{0} = T_{\mathbf{A}}(\mathbf{0})$ . Since  $\dim \operatorname{im} T_{\mathbf{A}} = 2 < 3 = \dim \mathbb{R}^3$ ,  $\operatorname{im} T_{\mathbf{A}} \neq \mathbb{R}^3$ , so  $T_{\mathbf{A}}$  is not surjective ([Theorem 3.2.2](#)).

#### Example. (Differentiation)

Find the kernel and image of  $D: \mathbb{R}[x]_{\leq 3} \rightarrow \mathbb{R}[x]_{\leq 2}$ ,  $D(p) = p'$ .

*Solution. Kernel.* Let  $p = a_0 + a_1x + a_2x^2 + a_3x^3$ . Then  $D(p) = a_1 + 2a_2x + 3a_3x^2$ , which is the zero polynomial exactly when  $a_1 = 2a_2 = 3a_3 = 0$ . Since  $2 \neq 0$  and  $3 \neq 0$  in  $\mathbb{R}$ , this means  $a_1 = a_2 = a_3 = 0$ . So  $\ker D = \{a_0 : a_0 \in \mathbb{R}\} = \operatorname{span}(1)$ , the constants.

*Image.* By [Theorem 3.2.4](#) applied to the basis  $(1, x, x^2, x^3)$ ,

$$\begin{aligned} \operatorname{im} D &= \operatorname{span}(D(1), D(x), D(x^2), D(x^3)) = \operatorname{span}(0, 1, 2x, 3x^2) \\ &= \operatorname{span}(1, x, x^2) = \mathbb{R}[x]_{\leq 2}. \end{aligned}$$

So  $D$  is surjective but not injective:  $p$  and  $p + 5$  have the same derivative. Note  $\dim \ker D + \dim \operatorname{im} D = 1 + 3 = 4 = \dim \mathbb{R}[x]_{\leq 3}$ .

**Example. (Trace)**

Find the kernel and image of  $\text{tr}: M_2(\mathbb{R}) \rightarrow \mathbb{R}$ .

*Solution. Kernel.* A matrix  $\begin{pmatrix} a & b \\ c & d \end{pmatrix}$  has trace 0 exactly when  $d = -a$ , that is, when it equals

$$\begin{pmatrix} a & b \\ c & -a \end{pmatrix} = a \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix} + b\mathbf{E}_{12} + c\mathbf{E}_{21}.$$

So  $\mathbf{E}_{11} - \mathbf{E}_{22}, \mathbf{E}_{12}, \mathbf{E}_{21}$  span  $\ker \text{tr}$ . They are independent: the combination above is the zero matrix only if  $a = b = c = 0$ . Hence they form a basis, and  $\dim \ker \text{tr} = 3$ .

*Image.*  $\text{im tr}$  is a subspace of  $\mathbb{R}$ , and it contains  $\text{tr } \mathbf{E}_{11} = 1$ . The lazy preimage  $r\mathbf{E}_{11}$  of any  $r \in \mathbb{R}$  shows  $\text{im tr} = \mathbb{R}$ . So the trace is surjective but not injective. Again  $3 + 1 = 4 = \dim M_2(\mathbb{R})$ .

The next two examples live in infinite-dimensional spaces, and the pattern “one property but not the other” goes the opposite way.

**Example. (Integration)**

Find the kernel and image of  $J: \mathbb{R}[x] \rightarrow \mathbb{R}[x]$ ,  $J(p)(x) = \int_0^x p(t) dt$ , of [Example \(Integration\)](#).

*Solution. Kernel.* For  $p = a_0 + a_1x + \cdots + a_nx^n$ ,  $J(p) = a_0x + \frac{a_1}{2}x^2 + \cdots + \frac{a_n}{n+1}x^{n+1}$ . This is the zero polynomial only if every  $\frac{a_k}{k+1} = 0$ , that is, every  $a_k = 0$ . So  $\ker J = \{0\}$ , and  $J$  is injective by [Theorem 3.2.3](#).

*Image.* Let  $U = \{q \in \mathbb{R}[x] : q(0) = 0\}$ , the polynomials with constant coefficient 0. Every  $J(p)$  has constant coefficient 0, so  $\text{im } J \subseteq U$ . Conversely, if  $q = b_1x + b_2x^2 + \cdots + b_mx^m \in U$ , then  $p = b_1 + 2b_2x + \cdots + mb_mx^{m-1}$  satisfies  $J(p) = q$ . Hence  $\text{im } J = U$ . Since  $1 \notin U$ ,  $J$  is **not** surjective.

**Example. (The Shifts: Injective but Not Surjective, and Conversely)**

Find the kernels and images of the right and left shifts  $R, L: F^{\mathbb{N}} \rightarrow F^{\mathbb{N}}$  of [Example \(The Shift Maps on Sequences\)](#).

*Solution. Right shift.*  $R(s_0, s_1, \dots) = (0, s_0, s_1, \dots)$  is the zero sequence only if every  $s_k = 0$ . So  $\ker R = \{\mathbf{0}\}$ , and  $R$  is injective. Every output has first entry 0; conversely,  $(0, t_1, t_2, \dots) = R(t_1, t_2, \dots)$ . So  $\text{im } R = \{\mathbf{t} \in F^{\mathbb{N}} : t_0 = 0\} \neq F^{\mathbb{N}}$ , and  $R$  is not surjective:  $\mathbf{e}_0 = (1, 0, 0, \dots)$  is never an output.

*Left shift.*  $L(s_0, s_1, s_2, \dots) = (s_1, s_2, \dots)$  is zero exactly when  $s_1 = s_2 = \cdots = 0$ , with  $s_0$  arbitrary. So  $\ker L = \text{span}(\mathbf{e}_0) \neq \{\mathbf{0}\}$ , and  $L$  is not injective. Every sequence is an output, since  $L(R(\mathbf{t})) = \mathbf{t}$ . So  $\text{im } L = F^{\mathbb{N}}$ , and  $L$  is surjective.

This pays off the pointer from Chapter 1, §8:  $R$  matches  $F^{\mathbb{N}}$  one-to-one with its proper subspace  $\text{im } R$ . For a map  $V \rightarrow V$  on a **finite** set, injective and surjective are equivalent ([Theorem 0.4.9](#)).

The shifts, and the pair  $J$  and  $D$  on  $\mathbb{R}[x]$  (where  $D(J(p)) = p$  makes  $D$  surjective, while  $D(1) = 0$  makes it not injective), show that for linear maps on an infinite-dimensional space this equivalence fails. The next section proves that it holds again when  $V$  is finite-dimensional.

*Quick check.* Find  $\ker T$  and  $\operatorname{im} T$  for the map  $T: \mathbb{R}^2 \rightarrow \mathbb{R}^3$ ,  $T(x, y) = (2x - y, x + y, x - y)$ , of [Example \(Finding a Formula from Values on a Basis\)](#). Is  $T$  injective? Surjective?

*Solution.* If  $T(x, y) = \mathbf{0}$ , then  $x + y = 0$  and  $x - y = 0$ , so  $x = y = 0$ . Hence  $\ker T = \{\mathbf{0}\}$ , and  $T$  is injective. By [Theorem 3.2.4](#),  $\operatorname{im} T = \operatorname{span}(T(\mathbf{e}_1), T(\mathbf{e}_2)) = \operatorname{span}((2, 1, 1), (-1, 1, -1))$ , a plane in  $\mathbb{R}^3$ . A span of two vectors has dimension at most  $2 < 3$ , so  $\operatorname{im} T \neq \mathbb{R}^3$  and  $T$  is not surjective.

### 3.2.5 What injective and surjective maps do to lists

Injectivity and surjectivity are statements about vectors. They translate into statements about lists: injective maps keep independent lists independent, and surjective maps keep spanning lists spanning. On a basis, the translation goes both ways.

#### Theorem 3.2.5 — Injective Maps Preserve Independence, Surjective Maps Preserve Spanning

Let  $T: V \rightarrow W$  be linear.

- (a) If  $T$  is injective and  $(\mathbf{v}_1, \dots, \mathbf{v}_k)$  is linearly independent in  $V$ , then  $(T(\mathbf{v}_1), \dots, T(\mathbf{v}_k))$  is linearly independent in  $W$ .
- (b) If  $T$  is surjective and  $(\mathbf{v}_1, \dots, \mathbf{v}_k)$  spans  $V$ , then  $(T(\mathbf{v}_1), \dots, T(\mathbf{v}_k))$  spans  $W$ .
- (c) If  $(\mathbf{v}_1, \dots, \mathbf{v}_n)$  is a basis of  $V$ , then  $T$  is injective if and only if  $(T(\mathbf{v}_1), \dots, T(\mathbf{v}_n))$  is linearly independent, and  $T$  is surjective if and only if  $(T(\mathbf{v}_1), \dots, T(\mathbf{v}_n))$  spans  $W$ .

*Proof.*

- (a) Let  $a_1T(\mathbf{v}_1) + \dots + a_kT(\mathbf{v}_k) = \mathbf{0}$ . By [Theorem 3.1.4](#),  $T(a_1\mathbf{v}_1 + \dots + a_k\mathbf{v}_k) = \mathbf{0}$ , so  $a_1\mathbf{v}_1 + \dots + a_k\mathbf{v}_k \in \ker T$ . Since  $T$  is injective,  $\ker T = \{\mathbf{0}\}$  ([Theorem 3.2.3](#)), so  $a_1\mathbf{v}_1 + \dots + a_k\mathbf{v}_k = \mathbf{0}$ . The independence of  $(\mathbf{v}_1, \dots, \mathbf{v}_k)$  gives  $a_1 = \dots = a_k = 0$ .
- (b) By [Theorem 3.2.4](#),  $\operatorname{span}(T(\mathbf{v}_1), \dots, T(\mathbf{v}_k)) = \operatorname{im} T$ , and  $\operatorname{im} T = W$  because  $T$  is surjective ([Theorem 3.2.2](#)).
- (c) *Injectivity.*  $(\Rightarrow)$  is (a).  $(\Leftarrow)$  Suppose the images are independent, and let  $\mathbf{v} \in \ker T$ . Write  $\mathbf{v} = a_1\mathbf{v}_1 + \dots + a_n\mathbf{v}_n$ . By [Theorem 3.1.4](#),  $\mathbf{0} = T(\mathbf{v}) = a_1T(\mathbf{v}_1) + \dots + a_nT(\mathbf{v}_n)$ , so all  $a_i = 0$  and  $\mathbf{v} = \mathbf{0}$ . Hence  $\ker T = \{\mathbf{0}\}$ , and  $T$  is injective by [Theorem 3.2.3](#).

*Surjectivity.* By [Theorem 3.2.4](#),  $\operatorname{im} T = \operatorname{span}(T(\mathbf{v}_1), \dots, T(\mathbf{v}_n))$ . By [Theorem 3.2.2](#),  $T$  is surjective exactly when this span is  $W$ . ■



**Warning. Without injectivity, independence can be lost.** The projection  $T(a_1, a_2) = (a_1, 0)$  sends the independent list  $((1, 0), (1, 1))$  to  $((1, 0), (1, 0))$ , which is dependent. The kernel is to blame:  $(1, 1) - (1, 0) = (0, 1)$  lies in  $\ker T$ . Likewise, a map that is not surjective sends **no** list to a spanning list of  $W$ , since all the images lie in  $\operatorname{im} T \neq W$ : the inclusion  $\mathbb{R}[x]_{\leq 2} \rightarrow \mathbb{R}[x]_{\leq 3}$  sends the basis  $(1, x, x^2)$  to a list that misses  $x^3$ .

Part (c) turns a question about a map into a question about  $n$  vectors in  $W$ , which is a question we can settle by elimination. In the next section, counting dimensions will let us check only **one** of injectivity and surjectivity when  $\dim V = \dim W$ .

## 3.2.6 Exercises

### 3.2.6.1 A. Check your understanding

#### Exercise A1.

- Define the kernel and the image of a linear map  $T: V \rightarrow W$ . Of which spaces are they subspaces?
- State the kernel test for injectivity.
- True or false: if  $T(\mathbf{u}) = T(\mathbf{v})$ , then  $\mathbf{u} - \mathbf{v} \in \ker T$ . Justify your answer.
- True or false: for  $D: \mathbb{R}[x]_{\leq 3} \rightarrow \mathbb{R}[x]_{\leq 3}$ ,  $D(p) = p'$ , we have  $\operatorname{im} D = \mathbb{R}[x]_{\leq 3}$ . Justify your answer.
- True or false: for a linear  $T: V \rightarrow W$  and **any**  $\mathbf{w} \in W$ , the set  $\{\mathbf{v} \in V : T(\mathbf{v}) = \mathbf{w}\}$  is a subspace of  $V$ . Justify your answer.

*Solution.*

- See [Definition 3.2.1](#) and [Definition 3.2.2](#):  $\ker T = \{\mathbf{v} \in V : T(\mathbf{v}) = \mathbf{0}\}$ , a subspace of  $V$  ([Theorem 3.2.1](#));  $\operatorname{im} T = \{T(\mathbf{v}) : \mathbf{v} \in V\}$ , a subspace of  $W$  ([Theorem 3.2.2](#)).
- See [Theorem 3.2.3](#): a linear map is injective if and only if its kernel is  $\{\mathbf{0}\}$ .
- True. By [Theorem 3.1.2](#),  $T(\mathbf{u} - \mathbf{v}) = T(\mathbf{u}) - T(\mathbf{v}) = \mathbf{0}$ .
- False. Every derivative of a polynomial of degree at most 3 has degree at most 2, so  $x^3 \notin \operatorname{im} D$ .
- False. If  $\mathbf{w} \neq \mathbf{0}$ , the set does not contain  $\mathbf{0}$ , since  $T(\mathbf{0}) = \mathbf{0} \neq \mathbf{w}$ . For example, for the projection of [Example \(Projection onto an Axis\)](#) and  $\mathbf{w} = (1, 0)$ , the set is the line  $\{(1, a_2)\}$ . It is a subspace exactly when  $\mathbf{w} = \mathbf{0}$ , in which case it is  $\ker T$ .

### 3.2.6.2 B. Practice

#### Exercise B1: Kernel and Image of a Matrix Map.

Let  $\mathbf{B} = \begin{pmatrix} 1 & -1 & 2 & 0 \\ 2 & -2 & 5 & 1 \\ 0 & 0 & 1 & 1 \end{pmatrix} \in M_{3 \times 4}(\mathbb{R})$ . Find bases of  $\ker T_{\mathbf{B}}$  and  $\operatorname{im} T_{\mathbf{B}}$ . Hence decide whether  $T_{\mathbf{B}}: \mathbb{R}^4 \rightarrow \mathbb{R}^3$  is injective, and whether it is surjective.

*Solution.* As in [Example \(A Matrix Map\)](#),  $\ker T_{\mathbf{B}} = \text{nul}(\mathbf{B})$  and  $\text{im } T_{\mathbf{B}} = \text{col}(\mathbf{B})$ . Row reduce:

$$\mathbf{B} \xrightarrow{R_2-2R_1} \begin{pmatrix} 1 & -1 & 2 & 0 \\ 0 & 0 & 1 & 1 \\ 0 & 0 & 1 & 1 \end{pmatrix} \xrightarrow[R_1-2R_2]{R_3-R_2} \begin{pmatrix} 1 & -1 & 0 & -2 \\ 0 & 0 & 1 & 1 \\ 0 & 0 & 0 & 0 \end{pmatrix}.$$

The pivot columns are 1 and 3; the free columns are 2 and 4. By [Theorem 2.5.4](#),  $\ker T_{\mathbf{B}}$  has basis  $((1, 1, 0, 0), (2, 0, -1, 1))$ . Check:  $\mathbf{B}(2, 0, -1, 1) = (2-2, 4-5+1, -1+1) = \mathbf{0}$ . By [Theorem 2.5.2](#),  $\text{im } T_{\mathbf{B}}$  has basis  $((1, 2, 0), (2, 5, 1))$ .

Hence  $\ker T_{\mathbf{B}} \neq \{\mathbf{0}\}$ , so  $T_{\mathbf{B}}$  is not injective ([Theorem 3.2.3](#)). And  $\dim \text{im } T_{\mathbf{B}} = 2 < 3$ , so  $\text{im } T_{\mathbf{B}} \neq \mathbb{R}^3$  and  $T_{\mathbf{B}}$  is not surjective ([Theorem 3.2.2](#)).

### Exercise B2: Kernels and Images Beyond $F^n$ .

For each map, find a basis of the kernel and a basis of the image, and decide whether the map is injective and whether it is surjective.

- (a)  $T: \mathbb{R}[x]_{\leq 2} \rightarrow \mathbb{R}[x]_{\leq 2}$ ,  $T(p) = p' + p$ .
- (b)  $S: M_2(\mathbb{R}) \rightarrow M_2(\mathbb{R})$ ,  $S(\mathbf{A}) = \mathbf{A} - \mathbf{A}^\top$ .

*Solution.* Both maps are linear. For  $c \in \mathbb{R}$ , by linearity of differentiation ([Example \(Differentiation\)](#)),  $T(cp + q) = (cp' + q') + (cp + q) = cT(p) + T(q)$ . By linearity of the transpose ([Example \(Trace and Transpose\)](#)),  $S(c\mathbf{A} + \mathbf{B}) = c\mathbf{A} + \mathbf{B} - c\mathbf{A}^\top - \mathbf{B}^\top = cS(\mathbf{A}) + S(\mathbf{B})$ . Both are linear by [Theorem 3.1.3](#).

- (a) *Kernel.* Let  $p = a + bx + cx^2$ . Then  $T(p) = (b + 2cx) + (a + bx + cx^2) = (a + b) + (b + 2c)x + cx^2$ . This is zero exactly when  $c = 0$ ,  $b + 2c = 0$  and  $a + b = 0$ , that is,  $c = b = a = 0$ . So  $\ker T = \{0\}$ , with the empty list as basis, and  $T$  is injective.

*Image.* We show  $T$  is surjective. Given  $q \in \mathbb{R}[x]_{\leq 2}$ , let  $p = q - q' + q''$ . Then  $p \in \mathbb{R}[x]_{\leq 2}$ , and since  $q''' = 0$ ,

$$T(p) = (q' - q'' + q''') + (q - q' + q'') = q.$$

So  $\text{im } T = \mathbb{R}[x]_{\leq 2}$ , with basis  $(1, x, x^2)$ , and  $T$  is surjective.

- (b) *Kernel.*  $S(\mathbf{A}) = 0$  exactly when  $\mathbf{A}^\top = \mathbf{A}$ . A matrix  $\begin{pmatrix} a & b \\ c & d \end{pmatrix}$  is symmetric exactly when  $b = c$ , that is, when it equals  $a\mathbf{E}_{11} + d\mathbf{E}_{22} + b(\mathbf{E}_{12} + \mathbf{E}_{21})$ . These three matrices are independent (compare entries), so  $(\mathbf{E}_{11}, \mathbf{E}_{22}, \mathbf{E}_{12} + \mathbf{E}_{21})$  is a basis of  $\ker S$ .  $S$  is not injective.

*Image.* By [Theorem 3.2.4](#) with the basis  $(\mathbf{E}_{11}, \mathbf{E}_{12}, \mathbf{E}_{21}, \mathbf{E}_{22})$ ,  $\text{im } S$  is spanned by  $S(\mathbf{E}_{11}) = 0$ ,  $S(\mathbf{E}_{12}) = \mathbf{E}_{12} - \mathbf{E}_{21}$ ,  $S(\mathbf{E}_{21}) = \mathbf{E}_{21} - \mathbf{E}_{12}$  and  $S(\mathbf{E}_{22}) = 0$ . So  $\text{im } S = \text{span}(\mathbf{E}_{12} - \mathbf{E}_{21})$ , with basis  $(\mathbf{E}_{12} - \mathbf{E}_{21})$ , the skew-symmetric matrices. Since  $\mathbf{I}_2 \notin \text{im } S$ ,  $S$  is not surjective.

### 3.2.6.3 C. Going deeper

In these exercises, for linear maps  $T: U \rightarrow V$  and  $S: V \rightarrow W$  we write  $ST$  for the composite  $S \circ T: U \rightarrow W$ , which applies  $T$  first. It is linear:  $ST(c\mathbf{u} + \mathbf{u}') = S(cT(\mathbf{u}) + T(\mathbf{u}')) = cST(\mathbf{u}) + ST(\mathbf{u}')$ , by [Theorem 3.1.3](#) for  $T$  and then for  $S$ .

**Exercise C1: Kernels and Images of a Composite.**

Let  $T: U \rightarrow V$  and  $S: V \rightarrow W$  be linear.

- (a) Prove that  $\ker T \subseteq \ker(ST)$  and  $\operatorname{im}(ST) \subseteq \operatorname{im} S$ .
- (b) Give an example, with  $U = V = W = \mathbb{R}^2$ , where  $\ker T \neq \ker(ST)$ .
- (c) Deduce that if  $ST$  is injective, then  $T$  is injective, and if  $ST$  is surjective, then  $S$  is surjective.

*Solution.*

- (a) Let  $\mathbf{u} \in \ker T$ . Then  $ST(\mathbf{u}) = S(\mathbf{0}) = \mathbf{0}$  by [Theorem 3.1.1](#), so  $\mathbf{u} \in \ker(ST)$ . Let  $\mathbf{w} \in \operatorname{im}(ST)$ , say  $\mathbf{w} = S(T(\mathbf{u}))$ . Then  $\mathbf{w} = S(\mathbf{v})$  with  $\mathbf{v} = T(\mathbf{u}) \in V$ , so  $\mathbf{w} \in \operatorname{im} S$ .
- (b) Let  $T = \operatorname{id}_{\mathbb{R}^2}$  and let  $S$  be the projection  $S(a_1, a_2) = (a_1, 0)$ . Then  $\ker T = \{\mathbf{0}\}$ , while  $\ker(ST) = \ker S$  is the  $y$ -axis ([Example \(Projection onto an Axis\)](#)).
- (c) If  $ST$  is injective, then  $\ker(ST) = \{\mathbf{0}\}$  ([Theorem 3.2.3](#)), so  $\ker T = \{\mathbf{0}\}$  by (a), and  $T$  is injective. If  $ST$  is surjective, then  $W = \operatorname{im}(ST) \subseteq \operatorname{im} S \subseteq W$  by (a), so  $\operatorname{im} S = W$  and  $S$  is surjective ([Theorem 3.2.2](#)).

**Exercise C2: When Is a Composite Injective?.**

Let  $T: U \rightarrow V$  and  $S: V \rightarrow W$  be linear, and suppose  $T$  is injective. Prove that  $ST$  is injective if and only if  $\ker S \cap \operatorname{im} T = \{\mathbf{0}\}$ , that is, if and only if  $S$  is injective on  $\operatorname{im} T$ . Give an example where  $S$  is not injective but  $ST$  is.

*Solution.* ( $\Rightarrow$ ) Suppose  $ST$  is injective, and let  $\mathbf{v} \in \ker S \cap \operatorname{im} T$ . Then  $\mathbf{v} = T(\mathbf{u})$  for some  $\mathbf{u} \in U$ , and  $ST(\mathbf{u}) = S(\mathbf{v}) = \mathbf{0}$ . So  $\mathbf{u} \in \ker(ST) = \{\mathbf{0}\}$  ([Theorem 3.2.3](#)), and  $\mathbf{v} = T(\mathbf{0}) = \mathbf{0}$ .

( $\Leftarrow$ ) Suppose  $\ker S \cap \operatorname{im} T = \{\mathbf{0}\}$ , and let  $\mathbf{u} \in \ker(ST)$ . Then  $T(\mathbf{u}) \in \operatorname{im} T$  and  $S(T(\mathbf{u})) = \mathbf{0}$ , so  $T(\mathbf{u}) \in \ker S \cap \operatorname{im} T = \{\mathbf{0}\}$ . Since  $T$  is injective,  $\mathbf{u} = \mathbf{0}$ . Hence  $\ker(ST) = \{\mathbf{0}\}$ , and  $ST$  is injective.

Example: let  $T: \mathbb{R} \rightarrow \mathbb{R}^2$ ,  $T(a) = (a, 0)$ , which is injective with  $\operatorname{im} T$  the  $x$ -axis, and let  $S: \mathbb{R}^2 \rightarrow \mathbb{R}$ ,  $S(a_1, a_2) = a_1$ . Then  $\ker S$  is the  $y$ -axis, so  $S$  is not injective, but  $\ker S \cap \operatorname{im} T = \{\mathbf{0}\}$ , and indeed  $ST(a) = a$  is injective.

**Exercise C3: Differentiation over  $\mathbb{F}_2$ .**

Let  $D: \mathbb{F}_2[x]_{\leq 3} \rightarrow \mathbb{F}_2[x]_{\leq 3}$  be the formal derivative ([Example \(Differentiation\)](#)). Find bases of  $\ker D$  and  $\operatorname{im} D$ . Which step of [Example \(Differentiation\)](#) fails over  $\mathbb{F}_2$ ?

*Solution.* For  $p = a_0 + a_1x + a_2x^2 + a_3x^3$ ,  $D(p) = a_1 + 2a_2x + 3a_3x^2$ . In  $\mathbb{F}_2$ ,  $2 = 0$  and  $3 = 1$ , so  $D(p) = a_1 + a_3x^2$ .

*Kernel.*  $D(p) = 0$  exactly when  $a_1 = a_3 = 0$ , so  $\ker D = \{a_0 + a_2x^2\}$ , with basis  $(1, x^2)$ . The polynomials  $1$  and  $x^2$  are independent by [Theorem 1.4.4](#).

*Image.* By [Theorem 3.2.4](#),  $\operatorname{im} D = \operatorname{span}(D(1), D(x), D(x^2), D(x^3)) = \operatorname{span}(0, 1, 0, x^2)$ , with basis  $(1, x^2)$ .

The failing step is “ $2a_2 = 0$  forces  $a_2 = 0$ ”, which divides by 2; over  $\mathbb{F}_2$ ,  $2 = 0$ , and  $D(x^2) = 2x = 0$  although  $x^2$  is not constant. So over  $\mathbb{F}_2$  the kernel of differentiation is bigger than the constants, and here  $\ker D = \operatorname{im} D$ .

## 3.3 The Rank–Nullity Theorem

The examples of the last section kept producing the same coincidence. Differentiation  $\mathbb{R}[x]_{\leq 3} \rightarrow \mathbb{R}[x]_{\leq 2}$  has a kernel of dimension 1 and an image of dimension 3, and  $1+3 = 4 = \dim \mathbb{R}[x]_{\leq 3}$ . The trace on  $M_2(\mathbb{R})$  gives  $3+1 = 4$  again. For matrices, Chapter 2 proved that  $\operatorname{rank} \mathbf{A} + \operatorname{nullity} \mathbf{A}$  is the number of columns (Theorem 2.5.7). This section proves the same equation for every linear map on a finite-dimensional space, with no matrices at all, and then turns it into the most useful counting tool of the chapter.

### 3.3.1 Rank and nullity

The two numbers in the equation are dimensions of the two subspaces of Definition 3.2.1 and Definition 3.2.2, so they are defined whenever those subspaces are finite-dimensional.

#### Definition 3.3.1 — Nullity

Let  $T: V \rightarrow W$  be a linear map with  $\ker T$  finite-dimensional. The **nullity** of  $T$  is  $\operatorname{nullity} T := \dim \ker T$ .

#### Definition 3.3.2 — Rank

Let  $T: V \rightarrow W$  be a linear map with  $\operatorname{im} T$  finite-dimensional. The **rank** of  $T$  is  $\operatorname{rank} T := \dim \operatorname{im} T$ .

If  $V$  is finite-dimensional, both numbers are defined:  $\ker T$  is a subspace of  $V$ , hence finite-dimensional (Theorem 1.6.7), and  $\operatorname{im} T$  is finite-dimensional by Theorem 3.2.4. The nullity counts the independent directions that  $T$  destroys; the rank counts the independent directions that survive in the output.

#### Example. (Rank and Nullity in Examples)

Find the nullity and rank of (a) differentiation  $D: \mathbb{R}[x]_{\leq 3} \rightarrow \mathbb{R}[x]_{\leq 2}$ ; (b) the matrix map  $T_{\mathbf{A}}: F^n \rightarrow F^m$  for  $\mathbf{A} \in M_{m \times n}(F)$ ; (c) the zero map and the identity on a space of dimension  $n$ .

*Solution.*

- (a) By Example (Differentiation),  $\ker D = \operatorname{span}(1)$  and  $\operatorname{im} D = \mathbb{R}[x]_{\leq 2}$ . So  $\operatorname{nullity} D = 1$  and  $\operatorname{rank} D = 3$ .
- (b) By Example (A Matrix Map),  $\ker T_{\mathbf{A}} = \operatorname{nul}(\mathbf{A})$  and  $\operatorname{im} T_{\mathbf{A}} = \operatorname{col}(\mathbf{A})$ . So  $\operatorname{nullity} T_{\mathbf{A}} = \dim \operatorname{nul}(\mathbf{A}) = \operatorname{nullity} \mathbf{A}$  and  $\operatorname{rank} T_{\mathbf{A}} = \dim \operatorname{col}(\mathbf{A}) = \operatorname{rank} \mathbf{A}$  (Definition 2.5.5, Def-

inition 2.5.4). The new names agree with the old ones.

- (c) By [Example \(The Zero Map and the Identity\)](#), the zero map  $V \rightarrow V$  has nullity  $n$  and rank 0, and the identity has nullity 0 and rank  $n$ .

**A non-example.** The integration map  $J: \mathbb{R}[x] \rightarrow \mathbb{R}[x]$  of [Example \(Integration\)](#) has nullity 0, but its image contains the independent list  $(x, x^2, \dots, x^k)$  for every  $k$ , so the image is not finite-dimensional ([Theorem 1.6.3](#)) and  $\text{rank } J$  is not defined. What separates  $J$  from the examples above is that its domain is infinite-dimensional.

### 3.3.2 The theorem

We want to prove that for every linear  $T: V \rightarrow W$  with  $V$  finite-dimensional,

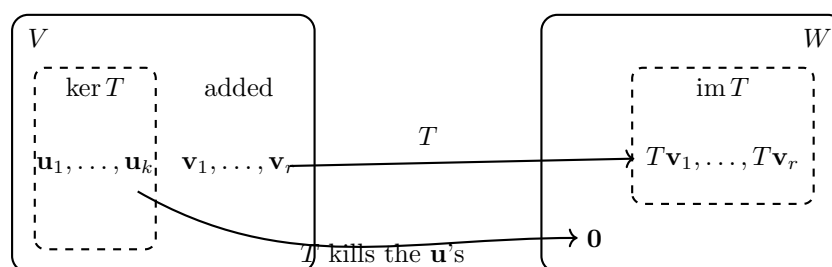
$$\dim \ker T + \dim \text{im } T = \dim V.$$

This is a dimension formula, so we look for the proof pattern that proved the last one, the Dimension Formula for Sums ([Theorem 1.7.3](#)): take a basis of the **smallest** space, extend it, make a Big Claim about the vectors that were added, and count. Which space is the smallest? The kernel lives in  $V$  and the image lives in  $W$ , different spaces that cannot be compared. The only inclusion available is  $\ker T \subseteq V$ . So we start with a basis of  $\ker T$  and extend it to a basis of  $V$ . The image has to be reached by applying  $T$ , and the kernel vectors contribute nothing when we do. So the natural candidate for a basis of  $\text{im } T$  is the images of the **added** vectors.

#### Theorem 3.3.1 — Rank–Nullity Theorem

Let  $V$  be a **finite-dimensional** vector space, let  $W$  be any vector space over the same field, and let  $T: V \rightarrow W$  be linear. Then  $\ker T$  and  $\text{im } T$  are finite-dimensional, and

$$\text{nullity } T + \text{rank } T = \dim V.$$



*Idea.* The picture is the whole proof. The  $\mathbf{u}$ 's all go to  $\mathbf{0}$ ; the  $\mathbf{v}$ 's carry everything else.

① Take a basis  $(\mathbf{u}_1, \dots, \mathbf{u}_k)$  of  $\ker T$ , the smallest space in sight. ② Extend it to a basis  $(\mathbf{u}_1, \dots, \mathbf{u}_k, \mathbf{v}_1, \dots, \mathbf{v}_r)$  of  $V$  (Basis Extension). ③ **Big Claim:**  $(T\mathbf{v}_1, \dots, T\mathbf{v}_r)$  is a basis of  $\text{im } T$ . *Spanning:* apply  $T$  to a general vector of  $V$ ; the  $\mathbf{u}$ -terms die because they are in the kernel. *Independence:* the only way to use linearity is backwards, turning  $\sum b_j T\mathbf{v}_j = \mathbf{0}$  into  $T(\sum b_j \mathbf{v}_j) = \mathbf{0}$ . Then  $\sum b_j \mathbf{v}_j$  is in the kernel, where the  $\mathbf{u}$ 's are a basis, and the big basis of  $V$  finishes it. ④ Count:  $\dim V = k + r = \text{nullity } T + \text{rank } T$ .

This is the move of [Theorem 1.7.3](#): basis of the smallest space, extend, Big Claim, count.

The new ingredient is that the Big Claim is about the **images** of the added vectors.

*Proof. Step 1: bases.* By Theorem 3.2.1,  $\ker T$  is a subspace of  $V$ . Since  $V$  is finite-dimensional, so is  $\ker T$  (Theorem 1.6.7), and it has a basis  $(\mathbf{u}_1, \dots, \mathbf{u}_k)$  with  $k = \text{nullity } T$  (Corollary 1.5.5). This list is linearly independent in  $V$ , so by the Basis Extension Theorem (Theorem 1.6.5) there are  $\mathbf{v}_1, \dots, \mathbf{v}_r \in V$  such that  $(\mathbf{u}_1, \dots, \mathbf{u}_k, \mathbf{v}_1, \dots, \mathbf{v}_r)$  is a basis of  $V$ . Thus  $\dim V = k + r$ .

### Step 2: the Big Claim.

*Claim.* The list  $(T\mathbf{v}_1, \dots, T\mathbf{v}_r)$  is a basis of  $\text{im } T$ .

*Proof. Spanning.* By Theorem 3.2.4 applied to the basis of Step 1,  $\text{im } T = \text{span}(T\mathbf{u}_1, \dots, T\mathbf{u}_k, T\mathbf{v}_1, \dots, T\mathbf{v}_r)$ . Each  $T\mathbf{u}_i = \mathbf{0}$ , since  $\mathbf{u}_i \in \ker T$ , and deleting a zero vector from a list does not change its span, because  $\mathbf{0}$  lies in every span (Theorem 1.3.5). Hence  $\text{im } T = \text{span}(T\mathbf{v}_1, \dots, T\mathbf{v}_r)$ .

*Independence.* Let  $b_1, \dots, b_r \in F$  satisfy  $b_1 T\mathbf{v}_1 + \dots + b_r T\mathbf{v}_r = \mathbf{0}$ . By Theorem 3.1.4,  $T(b_1 \mathbf{v}_1 + \dots + b_r \mathbf{v}_r) = \mathbf{0}$ , so  $b_1 \mathbf{v}_1 + \dots + b_r \mathbf{v}_r \in \ker T$ . Since  $(\mathbf{u}_1, \dots, \mathbf{u}_k)$  spans  $\ker T$ , there are  $a_1, \dots, a_k \in F$  with

$$b_1 \mathbf{v}_1 + \dots + b_r \mathbf{v}_r = a_1 \mathbf{u}_1 + \dots + a_k \mathbf{u}_k.$$

Rearranging,  $(-a_1)\mathbf{u}_1 + \dots + (-a_k)\mathbf{u}_k + b_1 \mathbf{v}_1 + \dots + b_r \mathbf{v}_r = \mathbf{0}$ . This is a combination of the basis  $(\mathbf{u}_1, \dots, \mathbf{u}_k, \mathbf{v}_1, \dots, \mathbf{v}_r)$  of  $V$ , which is linearly independent, so all its coefficients are 0. In particular  $b_1 = \dots = b_r = 0$ . Hence  $(T\mathbf{v}_1, \dots, T\mathbf{v}_r)$  is linearly independent. ■

**Step 3: count.** By the claim,  $\text{im } T$  has a basis of length  $r$ , so it is finite-dimensional with  $\text{rank } T = r$  (Definition 1.6.1). Therefore

$$\text{nullity } T + \text{rank } T = k + r = \dim V,$$

as claimed. ■

The proof reads correctly when  $k = 0$  (injective  $T$ : the  $\mathbf{v}$ 's are a basis of  $V$ ) and when  $r = 0$  ( $\ker T = V$ : the claim says  $\text{im } T$  has the empty basis). Notice what was **not** used: nothing about  $W$  beyond its being a vector space. The theorem counts only in the domain.

Applied to  $T_{\mathbf{A}}: F^n \rightarrow F^m$ , where  $\text{nullity } T_{\mathbf{A}} = \text{nullity } \mathbf{A}$  and  $\text{rank } T_{\mathbf{A}} = \text{rank } \mathbf{A}$  (Example (Rank and Nullity in Examples)), the theorem says  $\text{nullity } \mathbf{A} + \text{rank } \mathbf{A} = n$ . This is Theorem 2.5.7 again, now with a second proof. Chapter 2 counted free and pivot columns of a reduced matrix; here we counted basis vectors, and no row operation appeared. In particular, **the solution space of a homogeneous system  $\mathbf{A}\mathbf{x} = \mathbf{0}$  in  $n$  unknowns has dimension  $n - \text{rank } \mathbf{A}$** , whatever method is used to find the rank.

In the online version you can test the count on any matrix map.

```
from sympy import Matrix
```

```

A = Matrix([
    [1, 2, 0, 3],
    [2, 4, 1, 8],
    [-1, -2, 1, -1],
])
rank = A.rank()                # dim im T_A
nullity = len(A.nullspace())    # dim ker T_A: length of a basis of the kernel
print("rank =", rank, " nullity =", nullity, " dim of domain =", A.cols)
rank + nullity == A.cols

```

### 3.3.3 Count instead of check

Rank–Nullity links the two questions of the last section. Injectivity is about the kernel, surjectivity about the image, and the theorem says the two dimensions add up to a fixed number. So one controls the other.

#### Corollary 3.3.2 — Consequences of Rank–Nullity

Let  $V$  and  $W$  be finite-dimensional vector spaces over  $F$ , and let  $T: V \rightarrow W$  be linear.

- (a)  $T$  is injective if and only if  $\text{rank } T = \dim V$ .
- (b)  $T$  is surjective if and only if  $\text{rank } T = \dim W$ .
- (c) If  $\dim V > \dim W$ , then  $T$  is **not** injective.
- (d) If  $\dim V < \dim W$ , then  $T$  is **not** surjective.
- (e) If  $\dim V = \dim W$ , then  $T$  is injective  $\iff T$  is surjective  $\iff T$  is bijective.

*Proof.*

- (a) By [Theorem 3.2.3](#),  $T$  is injective if and only if  $\ker T = \{\mathbf{0}\}$ . A finite-dimensional space is  $\{\mathbf{0}\}$  exactly when its dimension is 0, because the empty list is a basis of  $\{\mathbf{0}\}$  and a basis of length 0 spans only  $\{\mathbf{0}\}$ . So  $T$  is injective if and only if  $\text{nullity } T = 0$ , which by [Theorem 3.3.1](#) means  $\text{rank } T = \dim V$ .
- (b) If  $T$  is surjective, then  $\text{im } T = W$  ([Theorem 3.2.2](#)), so  $\text{rank } T = \dim W$ . Conversely, if  $\text{rank } T = \dim W$ , then  $\text{im } T$  is a subspace of  $W$  of the same dimension, so  $\text{im } T = W$  by [Theorem 1.6.8](#), and  $T$  is surjective.
- (c) Since  $\text{im } T$  is a subspace of  $W$ ,  $\text{rank } T \leq \dim W < \dim V$  by [Theorem 1.6.7](#). By (a),  $T$  is not injective.
- (d) By [Theorem 3.3.1](#),  $\text{rank } T = \dim V - \text{nullity } T \leq \dim V < \dim W$ . By (b),  $T$  is not surjective.
- (e) If  $\dim V = \dim W$ , then (a) and (b) have the same right-hand side, so  $T$  is injective if and only if it is surjective. Hence either property gives both, which is bijectivity; and bijective maps are injective by definition.

■

Part (e) is the linear version of the pigeonhole principle for finite sets ([Theorem 0.4.9](#)), with dimension in place of the number of elements. It is a **count instead of check**: to show that a map between spaces of the same finite dimension is bijective, check **only** the kernel, usually the

easier half, and let the count supply surjectivity. For  $T_{\mathbf{A}}$  with  $\mathbf{A}$  square, it is the equivalence of (b) and (e) in the Invertible Matrix Theorem (Theorem 2.4.4). Parts (c) and (d) are the linear version of “no injection from a bigger finite set into a smaller one, and no surjection the other way”.

**Warning. Both finiteness and equal dimension are needed.** Rank–Nullity requires  $V$  to be finite-dimensional;  $W$  may be anything, as for the inclusion  $\mathbb{R}[x]_{\leq 2} \rightarrow \mathbb{R}[x]$ , with nullity 0 and rank 3. Part (e) fails when the spaces are infinite-dimensional, even for an operator  $V \rightarrow V$ : the right shift  $R$  on  $F^{\mathbb{N}}$  is injective but not surjective, and the left shift  $L$  is surjective but not injective (Example (The Shifts: Injective but Not Surjective, and Conversely)). And the hypothesis  $\dim V = \dim W$  cannot be dropped:  $T(x, y) = (2x - y, x + y, x - y)$  from  $\mathbb{R}^2$  to  $\mathbb{R}^3$  is injective but not surjective.

*Quick check.* Answer each question with a count.

- (a) Can a linear map  $\mathbb{R}^5 \rightarrow \mathbb{R}^2$  be injective?
- (b) Is there a linear map  $T: \mathbb{R}^6 \rightarrow \mathbb{R}^4$  with nullity  $T = 1$ ?
- (c) A linear operator  $T$  on  $\mathbb{R}[x]_{\leq 3}$  satisfies  $\ker T = \{0\}$ . Must  $T$  be surjective?

*Solution.*

- (a) No, by Corollary 3.3.2 (c):  $5 > 2$ . Concretely, the rank is at most 2, so the nullity is at least 3.
- (b) No. By Theorem 3.3.1 the rank would be  $6 - 1 = 5$ , but  $\operatorname{im} T \subseteq \mathbb{R}^4$  has dimension at most 4.
- (c) Yes. The domain and codomain are both  $\mathbb{R}[x]_{\leq 3}$ , of dimension 4, and  $T$  is injective, so it is surjective by Corollary 3.3.2 (e).

### 3.3.4 Polynomial interpolation by counting

Chapter 2 proved that through  $n + 1$  points with distinct  $x$ -values there passes exactly one polynomial of degree at most  $n$  (Theorem 2.7.3). Existence for **all** data and uniqueness came together there because a square matrix was involved. Rank–Nullity gives the same result as a count, with no matrix written down.

#### Example. (Interpolation as a Count)

Let  $x_0, \dots, x_n \in F$  be distinct, and define  $E: F[x]_{\leq n} \rightarrow F^{n+1}$  by  $E(p) = (p(x_0), \dots, p(x_n))$ . Show that  $E$  is bijective, and deduce Theorem 2.7.3.

*Solution. Linear.* For  $p, q \in F[x]_{\leq n}$  and  $c \in F$ , Theorem 0.8.5 gives  $(cp + q)(x_i) = cp(x_i) + q(x_i)$  in each entry, so  $E(cp + q) = cE(p) + E(q)$ , and  $E$  is linear by Theorem 3.1.3.

*Check the kernel.* If  $E(p) = \mathbf{0}$ , then  $p$  has the  $n + 1$  distinct roots  $x_0, \dots, x_n$  while  $\deg p \leq n$ . A non-zero polynomial of degree at most  $n$  has at most  $n$  distinct roots (Lemma 2.7.2), so  $p = 0$ . Hence  $\ker E = \{0\}$ , and  $E$  is injective (Theorem 3.2.3).

*Count.*  $\dim F[x]_{\leq n} = n + 1$ , since  $(1, x, \dots, x^n)$  is a basis (Example (Standard Bases)), and  $\dim F^{n+1} = n + 1$ . By Corollary 3.3.2 (e),  $E$  is bijective.



Surjectivity of  $E$  says that for every  $(y_0, \dots, y_n) \in F^{n+1}$  some  $p \in F[x]_{\leq n}$  has  $p(x_i) = y_i$  for all  $i$ ; injectivity says there is only one. This is [Theorem 2.7.3](#).

Compare the two arguments. Both prove uniqueness for the zero data from the root bound. Chapter 2 then invoked the Invertible Matrix Theorem for the Vandermonde matrix; here the count  $n+1 = n+1$  does the same job. The proof of existence never constructs a polynomial. That is the price and the power of counting: it tells us a solution exists without saying what it is.

### 3.3.5 Existence with conditions on derivatives

The same argument works when some conditions are on derivatives, where no Vandermonde matrix is at hand.

#### Example. (Prescribing Values and Slopes)

Show that for all  $a, b, c, d \in \mathbb{R}$  there is **exactly one**  $p \in \mathbb{R}[x]_{\leq 3}$  with

$$p(0) = a, \quad p(1) = b, \quad p'(0) = c, \quad p'(1) = d.$$

Hence find the cubic that rises from 0 at  $x = 0$  to 1 at  $x = 1$  with slope 0 at both ends.

*Solution.* Define  $T: \mathbb{R}[x]_{\leq 3} \rightarrow \mathbb{R}^4$  by  $T(p) = (p(0), p(1), p'(0), p'(1))$ . Each entry is linear in  $p$ : evaluation respects sums and scalar multiples ([Theorem 0.8.5](#)), and so does differentiation ([Example \(Differentiation\)](#)). Hence  $T$  is linear by [Theorem 3.1.3](#).

*Kernel.* Let  $p = c_0 + c_1x + c_2x^2 + c_3x^3$  with  $T(p) = \mathbf{0}$ . Then  $p(0) = c_0 = 0$  and  $p'(0) = c_1 = 0$ . So  $p = c_2x^2 + c_3x^3$ , and the remaining conditions read

$$p(1) = c_2 + c_3 = 0, \quad p'(1) = 2c_2 + 3c_3 = 0.$$

Subtracting twice the first from the second gives  $c_3 = 0$ , and then  $c_2 = 0$ . Hence  $\ker T = \{0\}$ .

*Count.*  $\dim \mathbb{R}[x]_{\leq 3} = 4 = \dim \mathbb{R}^4$ , so  $T$  is bijective by [Corollary 3.3.2](#) (e). Surjectivity is existence for every  $(a, b, c, d)$ , and injectivity is uniqueness.

For the data  $(0, 1, 0, 0)$ , try  $p = \alpha x^2 + \beta x^3$ , which already has  $p(0) = p'(0) = 0$ . Then  $p(1) = \alpha + \beta = 1$  and  $p'(1) = 2\alpha + 3\beta = 0$  give  $\beta = -2$  and  $\alpha = 3$ . So  $p = 3x^2 - 2x^3$ ; by uniqueness it is the only such cubic.

### 3.3.6 Dimensions of subspaces cut out by conditions

Many subspaces are defined as “all vectors satisfying some linear conditions”, which is to say as the kernel of a linear map. If the map is **surjective**, Rank–Nullity gives the dimension of the kernel at once, and surjectivity is often easy to see.

**Example. (Dimension of a Kernel from a Surjective Map)**

Find the dimension of each subspace.

- (a) The trace-free matrices  $U = \{\mathbf{A} \in M_n(F) : \operatorname{tr} \mathbf{A} = 0\}$  (Example (Trace-Free Matrices)).
- (b)  $U' = \{p \in \mathbb{R}[x]_{\leq n} : \int_0^1 p(t) dt = 0\}$ .

*Solution.*

- (a)  $U = \ker \operatorname{tr}$ , where  $\operatorname{tr}: M_n(F) \rightarrow F$  is linear (Example (Trace and Transpose)). Its image is a subspace of  $F$  containing  $\operatorname{tr} \mathbf{E}_{11} = 1$ , so it contains  $c \cdot 1 = c$  for every  $c \in F$ ; it is all of  $F$ , and  $\operatorname{rank} \operatorname{tr} = 1$ . Since  $\dim M_n(F) = n^2$  (Example (Standard Bases)), Theorem 3.3.1 gives  $\dim U = n^2 - 1$ .
- (b)  $U' = \ker \psi$ , where  $\psi: \mathbb{R}[x]_{\leq n} \rightarrow \mathbb{R}$ ,  $\psi(p) = \int_0^1 p(t) dt$ , is linear by Exercise (B3: Evaluation and Integration) (b). Since  $\psi(1) = 1$ , the image is a subspace of  $\mathbb{R}$  containing 1, hence all of  $\mathbb{R}$ , so  $\operatorname{rank} \psi = 1$ . By Theorem 3.3.1,  $\dim U' = (n+1) - 1 = n$ .

Without the theorem, (a) needs an explicit basis of  $n^2 - 1$  matrices together with a proof of independence and spanning. With it, one matrix of trace 1 is enough.

## 3.3.7 Exercises

### 3.3.7.1 A. Check your understanding

#### Exercise A1.

- (a) Define the rank and the nullity of a linear map.
- (b) State the Rank–Nullity Theorem, with its hypotheses.
- (c) True or false: there is an injective linear map  $\mathbb{R}^4 \rightarrow \mathbb{R}^3$ . Justify your answer.
- (d) True or false: if  $V$  is finite-dimensional and  $T: V \rightarrow V$  is linear with  $\operatorname{im} T = V$ , then  $T$  is injective. Justify your answer.
- (e) True or false: every injective linear map  $F^{\mathbb{N}} \rightarrow F^{\mathbb{N}}$  is surjective. Justify your answer.
- (f) Name the four steps of the proof of Rank–Nullity.

*Solution.*

- (a) See Definition 3.3.1 and Definition 3.3.2:  $\operatorname{nullity} T = \dim \ker T$  and  $\operatorname{rank} T = \dim \operatorname{im} T$ , when these are finite-dimensional.
- (b) See Theorem 3.3.1: if  $V$  is finite-dimensional and  $T: V \rightarrow W$  is linear, then  $\operatorname{nullity} T + \operatorname{rank} T = \dim V$ .
- (c) False.  $4 > 3$ , so no linear map  $\mathbb{R}^4 \rightarrow \mathbb{R}^3$  is injective (Corollary 3.3.2 (c)).
- (d) True.  $T$  is surjective, and its domain and codomain are the same finite-dimensional space, so  $T$  is injective by Corollary 3.3.2 (e).

- (e) False. The right shift  $R$  is injective but not surjective ([Example \(The Shifts: Injective but Not Surjective, and Conversely\)](#)). The count fails because  $F^{\mathbb{N}}$  is not finite-dimensional.
- (f) Take a basis of  $\ker T$ ; extend it to a basis of  $V$ ; prove the Big Claim that the images of the added vectors form a basis of  $\operatorname{im} T$  (spanning, then independence); count.

### 3.3.7.2 B. Practice

#### Exercise B1: Verifying Rank–Nullity.

For each map, find a basis of the kernel and a basis of the image, and verify that  $\operatorname{nullity} T + \operatorname{rank} T = \dim V$ .

- (a)  $T_{\mathbf{C}}: \mathbb{R}^4 \rightarrow \mathbb{R}^3$ , where  $\mathbf{C} = \begin{pmatrix} 1 & 0 & 1 & 2 \\ 0 & 1 & 1 & -1 \\ 1 & 1 & 2 & 1 \end{pmatrix}$ .
- (b)  $T: \mathbb{R}[x]_{\leq 3} \rightarrow \mathbb{R}^2$ ,  $T(p) = (p(0), p(1))$ .
- (c)  $T: M_2(\mathbb{R}) \rightarrow M_2(\mathbb{R})$ ,  $T(\mathbf{A}) = \mathbf{A} + \mathbf{A}^{\top}$ .

*Solution.*

- (a) Row reduce:  $R_3 - R_1$  gives  $(0, 1, 1, -1)$ , and then  $R_3 - R_2$  gives a zero row, so

$$\mathbf{C} \longrightarrow \begin{pmatrix} 1 & 0 & 1 & 2 \\ 0 & 1 & 1 & -1 \\ 0 & 0 & 0 & 0 \end{pmatrix},$$

which is the RREF. The free columns are 3 and 4. By [Theorem 2.5.4](#),  $\ker T_{\mathbf{C}}$  has basis  $((-1, -1, 1, 0), (-2, 1, 0, 1))$ . By [Theorem 2.5.2](#),  $\operatorname{im} T_{\mathbf{C}}$  has basis  $((1, 0, 1), (0, 1, 1))$ . So  $2 + 2 = 4 = \dim \mathbb{R}^4$ .

- (b)  $T$  is linear by [Theorem 0.8.5](#). *Kernel:* for  $p = a_0 + a_1x + a_2x^2 + a_3x^3$ ,  $T(p) = \mathbf{0}$  means  $a_0 = 0$  and  $a_1 + a_2 + a_3 = 0$ , that is,  $p = a_2(x^2 - x) + a_3(x^3 - x)$ . The list  $(x^2 - x, x^3 - x)$  spans  $\ker T$ , and it is independent by [Theorem 1.4.4](#), so it is a basis. *Image:*  $T(1) = (1, 1)$  and  $T(x) = (0, 1)$  are independent, so they span a 2-dimensional subspace of  $\mathbb{R}^2$ , which is  $\mathbb{R}^2$  ([Theorem 1.6.8](#)). So  $\operatorname{im} T = \mathbb{R}^2$  with basis  $((1, 1), (0, 1))$ . Hence  $2 + 2 = 4 = \dim \mathbb{R}[x]_{\leq 3}$ .
- (c)  $T$  is linear because the transpose is ([Example \(Trace and Transpose\)](#)). *Kernel:*  $\mathbf{A} + \mathbf{A}^{\top} = \mathbf{0}$  means  $a_{11} = a_{22} = 0$  and  $a_{21} = -a_{12}$ , so  $\ker T = \operatorname{span}(\mathbf{E}_{12} - \mathbf{E}_{21})$ , with basis  $(\mathbf{E}_{12} - \mathbf{E}_{21})$ . *Image:* each  $\mathbf{A} + \mathbf{A}^{\top}$  is symmetric, since  $(\mathbf{A} + \mathbf{A}^{\top})^{\top} = \mathbf{A}^{\top} + \mathbf{A}$ , and each symmetric  $\mathbf{B}$  equals  $T(\frac{1}{2}\mathbf{B})$ . So  $\operatorname{im} T$  is the space of symmetric matrices, with basis  $(\mathbf{E}_{11}, \mathbf{E}_{22}, \mathbf{E}_{12} + \mathbf{E}_{21})$  (as in [Exercise \(B2: Kernels and Images Beyond  \$F^n\$ \)](#)). Hence  $1 + 3 = 4 = \dim M_2(\mathbb{R})$ .

#### Exercise B2: Bijective by the Kernel Alone.

Let  $n \geq 0$  and  $T: \mathbb{R}[x]_{\leq n} \rightarrow \mathbb{R}[x]_{\leq n}$ ,  $T(p) = p + xp'$ .

- (a) Show that  $T$  is a linear map into  $\mathbb{R}[x]_{\leq n}$  and that  $\ker T = \{0\}$ . Deduce that  $T$  is bijective.

- (b) Hence show that there is exactly one  $p \in \mathbb{R}[x]_{\leq 2}$  with  $p + xp' = 1 + x + x^2$ , and find it.

*Solution.*

- (a) For  $p = \sum_{k=0}^n a_k x^k$ ,  $xp' = \sum_{k=1}^n k a_k x^k$ , so

$$T(p) = \sum_{k=0}^n (1+k)a_k x^k \in \mathbb{R}[x]_{\leq n}.$$

Linearity:  $T(cp+q) = (cp+q) + x(cp'+q') = c(p+xp') + (q+xq') = cT(p) + T(q)$ , using [Example \(Differentiation\)](#). If  $T(p) = 0$ , then  $(1+k)a_k = 0$  for each  $k$ ; since  $1+k \neq 0$  in  $\mathbb{R}$ , every  $a_k = 0$ . So  $\ker T = \{0\}$ , and  $T$  is injective ([Theorem 3.2.3](#)). Domain and codomain have the same finite dimension  $n+1$ , so  $T$  is bijective by [Corollary 3.3.2 \(e\)](#).

- (b) With  $n = 2$ , surjectivity gives a solution and injectivity makes it unique. By the formula in (a),  $T(a_0 + a_1x + a_2x^2) = a_0 + 2a_1x + 3a_2x^2$ , so we need  $a_0 = 1$ ,  $2a_1 = 1$ ,  $3a_2 = 1$ . Hence  $p = 1 + \frac{1}{2}x + \frac{1}{3}x^2$ . Check:  $xp' = \frac{1}{2}x + \frac{2}{3}x^2$ , and  $p + xp' = 1 + x + x^2$ .

### Exercise B3: A Dimension via a Surjective Map.

Let  $U = \{p \in \mathbb{R}[x]_{\leq 4} : p(1) = p(2) = 0\}$ . Use a surjective linear map to show that  $\dim U = 3$ . Hence find a basis of  $U$ .

*Solution.* Define  $T: \mathbb{R}[x]_{\leq 4} \rightarrow \mathbb{R}^2$ ,  $T(p) = (p(1), p(2))$ . It is linear by [Theorem 0.8.5](#), and  $U = \ker T$ . Its image contains  $T(1) = (1, 1)$  and  $T(x) = (1, 2)$ , which are independent, so  $\text{im } T$  is a 2-dimensional subspace of  $\mathbb{R}^2$  and equals  $\mathbb{R}^2$  ([Theorem 1.6.8](#)). Thus  $\text{rank } T = 2$ , and by [Theorem 3.3.1](#),  $\dim U = 5 - 2 = 3$ .

The polynomials  $(x-1)(x-2)$ ,  $x(x-1)(x-2)$  and  $x^2(x-1)(x-2)$  vanish at 1 and 2 and have degrees  $2, 3, 4 \leq 4$ , so they lie in  $U$ . They are non-zero of distinct degrees, hence independent ([Theorem 1.4.4](#)). There are  $3 = \dim U$  of them, so they form a basis of  $U$  ([Theorem 1.6.4](#)).

### 3.3.7.3 C. Going deeper

#### Exercise C1: A Lower Bound for the Rank of a Composite.

Let  $U$  and  $V$  be finite-dimensional, let  $W$  be a vector space, all over  $F$ , and let  $T: U \rightarrow V$  and  $S: V \rightarrow W$  be linear. Write  $ST$  for the composite  $S \circ T$ , which is linear (see the note before [Exercise \(C1: Kernels and Images of a Composite\)](#)). Let  $S': \text{im } T \rightarrow W$  be the restriction of  $S$  to  $\text{im } T$ , that is,  $S'(\mathbf{v}) = S(\mathbf{v})$  for  $\mathbf{v} \in \text{im } T$ .

- Show that  $S'$  is linear, that  $\text{im } S' = \text{im}(ST)$  and that  $\ker S' = \ker S \cap \text{im } T$ .
- Deduce that  $\text{rank}(ST) = \text{rank } T - \dim(\ker S \cap \text{im } T)$ .
- Hence prove **Sylvester's inequality**:  $\text{rank}(ST) \geq \text{rank } S + \text{rank } T - \dim V$ .
- Deduce that  $\text{rank}(\mathbf{AB}) \geq \text{rank } \mathbf{A} + \text{rank } \mathbf{B} - n$  for  $\mathbf{A} \in M_{m \times n}(F)$  and  $\mathbf{B} \in M_{n \times p}(F)$ .

*Hint: for (c), compare  $\ker S \cap \operatorname{im} T$  with  $\ker S$ .*

*Solution.*

- (a)  $S'$  satisfies (LT1) and (LT2) because  $S$  does and  $\operatorname{im} T$  is a subspace (Theorem 3.2.2) with the operations of  $V$ . A vector  $\mathbf{w}$  lies in  $\operatorname{im} S'$  exactly when  $\mathbf{w} = S(\mathbf{v})$  with  $\mathbf{v} = T(\mathbf{u})$  for some  $\mathbf{u} \in U$ , that is, when  $\mathbf{w} = ST(\mathbf{u})$ ; so  $\operatorname{im} S' = \operatorname{im}(ST)$ . A vector  $\mathbf{v} \in \operatorname{im} T$  lies in  $\ker S'$  exactly when  $S(\mathbf{v}) = \mathbf{0}$ ; so  $\ker S' = \ker S \cap \operatorname{im} T$ .
- (b)  $\operatorname{im} T$  is finite-dimensional with  $\dim \operatorname{im} T = \operatorname{rank} T$ , since  $U$  is finite-dimensional (Theorem 3.2.4). By Theorem 3.3.1 applied to  $S': \operatorname{im} T \rightarrow W$  and by (a),

$$\operatorname{rank} T = \dim \ker S' + \dim \operatorname{im} S' = \dim(\ker S \cap \operatorname{im} T) + \operatorname{rank}(ST).$$

- (c)  $\ker S \cap \operatorname{im} T$  is a subspace of  $\ker S$ , so  $\dim(\ker S \cap \operatorname{im} T) \leq \operatorname{nullity} S = \dim V - \operatorname{rank} S$ , by Theorem 1.6.7 and Theorem 3.3.1 for  $S$ . Substituting into (b),

$$\operatorname{rank}(ST) \geq \operatorname{rank} T - (\dim V - \operatorname{rank} S) = \operatorname{rank} S + \operatorname{rank} T - \dim V.$$

- (d) Take  $T = T_{\mathbf{B}}: F^p \rightarrow F^n$  and  $S = T_{\mathbf{A}}: F^n \rightarrow F^m$ . By associativity (Theorem 0.9.4),  $T_{\mathbf{A}}T_{\mathbf{B}}(\mathbf{x}) = \mathbf{A}(\mathbf{B}\mathbf{x}) = (\mathbf{A}\mathbf{B})\mathbf{x}$ , so  $ST = T_{\mathbf{A}\mathbf{B}}$ . By Example (Rank and Nullity in Examples) (b), ranks of matrix maps are ranks of matrices, and  $\dim F^n = n$ . So (c) gives  $\operatorname{rank}(\mathbf{A}\mathbf{B}) \geq \operatorname{rank} \mathbf{A} + \operatorname{rank} \mathbf{B} - n$ .

### Exercise C2: When the Kernel Equals the Image.

Let  $V$  be finite-dimensional and  $T: V \rightarrow V$  linear with  $\ker T = \operatorname{im} T$ .

- (a) Prove that  $\dim V$  is even.  
 (b) Give an example of such a  $T$  on  $\mathbb{R}^2$ , and one on  $\mathbb{R}^4$ .  
 (c) Is there such a  $T$  on  $\mathbb{R}^3$ ? Justify your answer.

*Solution.*

- (a) By Theorem 3.3.1,  $\dim V = \operatorname{nullity} T + \operatorname{rank} T = \dim \ker T + \dim \operatorname{im} T = 2 \dim \operatorname{im} T$ , which is even.
- (b) On  $\mathbb{R}^2$ , let  $T(x, y) = (y, 0)$ . Then  $T(x, y) = \mathbf{0}$  exactly when  $y = 0$ , so  $\ker T$  is the  $x$ -axis; and the outputs  $(y, 0)$  fill the  $x$ -axis, so  $\operatorname{im} T$  is the  $x$ -axis too. On  $\mathbb{R}^4$ , let  $T(x_1, x_2, x_3, x_4) = (x_2, 0, x_4, 0)$ . Its kernel is  $\{x_2 = x_4 = 0\} = \operatorname{span}(\mathbf{e}_1, \mathbf{e}_3)$ , and its image is  $\{(a, 0, b, 0)\} = \operatorname{span}(\mathbf{e}_1, \mathbf{e}_3)$ . Both maps are matrix maps, hence linear.
- (c) No. By (a),  $\dim V$  would be even, but  $\dim \mathbb{R}^3 = 3$ .

## 3.4 The Algebra of Linear Maps

So far we have studied one linear map at a time: its kernel, its image, and the count that ties them together. But linear maps rarely come alone. The map  $p \mapsto p + p'$  is the identity plus differentiation, and rotating then projecting is one map followed by another. This section shows that linear maps can be added, scaled and composed, that these operations obey almost all the

rules of arithmetic, and that an operator can be plugged into a polynomial just as a square matrix can.

### 3.4.1 Adding and scaling linear maps

Look at three maps on  $\mathbb{R}[x]_{\leq 2}$  that have already appeared in this chapter or will soon:

$$p \mapsto p + p', \quad p \mapsto 3p', \quad p \mapsto p' - 2p.$$

Each is built from two simpler maps, the identity  $\text{id}$  and differentiation  $D$ , by the recipe “apply both to  $p$ , then combine the outputs with the operations of the target space”. The first is “ $\text{id} + D$ ”, the second “ $3D$ ”, the third “ $D - 2\text{id}$ ”. Instead of checking linearity for each such combination separately, we would like to check it once for the recipe. The recipe itself deserves a name.

*Linear maps into the same space can be added and scaled output by output, and the results are again linear maps.*

#### Definition 3.4.1 — The space of linear maps

Let  $V$  and  $W$  be vector spaces over **the same field**  $F$ . We write  $\mathcal{L}(V, W)$  for the set of **all** linear maps  $V \rightarrow W$ , and  $\mathcal{L}(V) := \mathcal{L}(V, V)$  for the set of linear operators on  $V$  (Definition 3.1.1). For  $S, T \in \mathcal{L}(V, W)$  and  $c \in F$ , the **sum**  $S + T$  and the **scalar multiple**  $cT$  are the functions  $V \rightarrow W$  defined by

$$(S + T)(\mathbf{v}) := S\mathbf{v} + T\mathbf{v}, \quad (cT)(\mathbf{v}) := c(T\mathbf{v}) \quad \text{for every } \mathbf{v} \in V.$$

As in Definition 3.1.1, we write  $T\mathbf{v}$  for  $T(\mathbf{v})$  when no brackets are needed. In words: to evaluate  $S + T$  at  $\mathbf{v}$ , evaluate both maps at  $\mathbf{v}$  and add the results **in**  $W$ ; to evaluate  $cT$ , evaluate  $T$  and scale the result in  $W$ . The operations of  $V$  play no role in the definition. That is why  $S$  and  $T$  must have the **same codomain**: the sum  $S\mathbf{v} + T\mathbf{v}$  has to be computed in one space. Both maps must also be defined on the same  $V$ .

We have used these operations before without naming them: the operations on  $F^X$  in Example (The Standard Vector Spaces) (d) are defined by the same pointwise recipe, with values added in  $F$  instead of  $W$ .

It turns out that the recipe never leaves the world of linear maps, and that  $\mathcal{L}(V, W)$  is itself a vector space. This is the first sign that maps between vector spaces are as good as vectors.

#### Theorem 3.4.1 — $\mathcal{L}(V, W)$ Is a Vector Space

Let  $V$  and  $W$  be vector spaces over  $F$ . If  $S, T \in \mathcal{L}(V, W)$  and  $c \in F$ , then  $S + T$  and  $cT$  are linear. With the operations of Definition 3.4.1,  $\mathcal{L}(V, W)$  is a vector space over  $F$ . Its zero vector is the zero map  $\mathbf{v} \mapsto \mathbf{0}_W$ , and the negative of  $T$  is  $(-1)T: \mathbf{v} \mapsto -T\mathbf{v}$ .

*Idea.* There are two layers. The set  $W^V$  of **all** functions  $V \rightarrow W$ , with the pointwise operations, is a vector space, for the same reason  $F^X$  is: each axiom is checked one input at a time, where it becomes an axiom of  $W$ . Then  $\mathcal{L}(V, W)$  sits inside  $W^V$ , so the Subspace Test finishes the job, and its three conditions are exactly “the zero map is

linear”, “a sum of linear maps is linear” and “a multiple of a linear map is linear”.

*Proof.* Let  $W^V$  be the set of all functions  $V \rightarrow W$ , with the operations of [Definition 3.4.1](#) applied to arbitrary functions. The verification in [Example \(The Standard Vector Spaces\)](#) (d) goes through word for word with  $W$  in place of  $F$ : two functions  $V \rightarrow W$  are equal when they agree at every  $\mathbf{v} \in V$  ([Definition 0.4.1](#)), and at each  $\mathbf{v}$  every axiom for  $W^V$  becomes the same axiom in  $W$ , with zero function  $\mathbf{v} \mapsto \mathbf{0}_W$  and negative  $\mathbf{v} \mapsto -f(\mathbf{v})$ . Hence  $W^V$  is a vector space over  $F$ , and  $\mathcal{L}(V, W) \subseteq W^V$ . We apply [Theorem 1.2.1](#).

- (1) The zero map is linear:  $\mathbf{0}_W = \mathbf{0}_W + \mathbf{0}_W$  and  $\mathbf{0}_W = c\mathbf{0}_W$  by [Theorem 1.1.6](#). So  $\mathbf{0} \in \mathcal{L}(V, W)$ .
- (2) Let  $S, T \in \mathcal{L}(V, W)$ ,  $\mathbf{u}, \mathbf{v} \in V$  and  $a \in F$ . By the definition of  $S + T$ , the linearity of  $S$  and  $T$ , and the axioms of  $W$ ,

$$(S + T)(\mathbf{u} + \mathbf{v}) = S\mathbf{u} + S\mathbf{v} + T\mathbf{u} + T\mathbf{v} = (S + T)\mathbf{u} + (S + T)\mathbf{v},$$

$$(S + T)(a\mathbf{v}) = aS\mathbf{v} + aT\mathbf{v} = a(S\mathbf{v} + T\mathbf{v}) = a(S + T)\mathbf{v}.$$

So  $S + T \in \mathcal{L}(V, W)$ .

- (3) Let  $c \in F$  and  $T \in \mathcal{L}(V, W)$ . Then  $(cT)(\mathbf{u} + \mathbf{v}) = c(T\mathbf{u} + T\mathbf{v}) = (cT)\mathbf{u} + (cT)\mathbf{v}$ , and  $(cT)(a\mathbf{v}) = c(aT\mathbf{v}) = (ca)T\mathbf{v} = (ac)T\mathbf{v} = a(cT)\mathbf{v}$ , using the axioms of  $W$  and commutativity of multiplication in  $F$ . So  $cT \in \mathcal{L}(V, W)$ .

By [Theorem 1.2.1](#),  $\mathcal{L}(V, W)$  is a subspace of  $W^V$ , hence a vector space over  $F$  with the same zero and negatives. The negative of  $T$  in  $W^V$  is  $\mathbf{v} \mapsto -T\mathbf{v} = (-1)T\mathbf{v}$ , by [Theorem 1.1.7](#). This proves the theorem. ■

The step that used commutativity of  $F$  is worth a second look:  $cT$  is linear because  $c$  can be moved past the scalar  $a$ . This is one place where “ $F$  is a field” is used silently.

### Example. (Sums of Matrix Maps and Other First Examples)

For  $\mathbf{A} \in M_{m \times n}(F)$ , write  $T_{\mathbf{A}}: F^n \rightarrow F^m$ ,  $\mathbf{x} \mapsto \mathbf{A}\mathbf{x}$ , for the matrix map of [Example \(Multiplication by a Matrix\)](#).

- (a) Show that  $T_{\mathbf{A}} + T_{\mathbf{B}} = T_{\mathbf{A} + \mathbf{B}}$  and  $cT_{\mathbf{A}} = T_{c\mathbf{A}}$  for  $\mathbf{A}, \mathbf{B} \in M_{m \times n}(F)$  and  $c \in F$ .
- (b) Describe  $\mathcal{L}(V, W)$  when  $V = \{\mathbf{0}\}$ , and when  $W = \{\mathbf{0}\}$ .
- (c) On  $\mathbb{R}[x]_{\leq 2}$ , write  $T(p) = p' - 2p$  in terms of  $\text{id}$  and  $D$ , and compute  $T(1 + x^2)$ .

*Solution.*

- (a) Let  $\mathbf{x} \in F^n$ . By [Definition 3.4.1](#) and distributivity ([Theorem 0.9.4](#)),

$$(T_{\mathbf{A}} + T_{\mathbf{B}})(\mathbf{x}) = \mathbf{A}\mathbf{x} + \mathbf{B}\mathbf{x} = (\mathbf{A} + \mathbf{B})\mathbf{x} = T_{\mathbf{A} + \mathbf{B}}(\mathbf{x}),$$

$$(cT_{\mathbf{A}})(\mathbf{x}) = c(\mathbf{A}\mathbf{x}) = (c\mathbf{A})\mathbf{x} = T_{c\mathbf{A}}(\mathbf{x}).$$

The maps agree at every  $\mathbf{x}$ , so they are equal. So adding and scaling matrix maps is adding and scaling matrices.

- (b) If  $V = \{\mathbf{0}\}$ , a linear map must send  $\mathbf{0}$  to  $\mathbf{0}_W$ , so the zero map is the **only** element:  $\mathcal{L}(\{\mathbf{0}\}, W) = \{\mathbf{0}\}$ . If  $W = \{\mathbf{0}\}$ , every function  $V \rightarrow \{\mathbf{0}\}$  is the zero map, and it is linear, so again  $\mathcal{L}(V, \{\mathbf{0}\}) = \{\mathbf{0}\}$ . These degenerate cases matter because formulas like “ $\dim \mathcal{L}(V, W) = \dim V \cdot \dim W$ ”, which we prove later in this chapter, must give 0 here, and they do.
- (c)  $T = D - 2\text{id}$ , which is linear by [Theorem 3.4.1](#). With  $p = 1 + x^2$ ,  $p' = 2x$ , so  $T(p) = 2x - 2 - 2x^2$ .

Here is a non-example by minimal change. Keep  $D$  on  $\mathbb{R}[x]_{\leq 2}$ , but replace  $\text{id}$  by the evaluation map  $E(p) = p(0) \in \mathbb{R}$ . Both maps are linear and both are defined on  $\mathbb{R}[x]_{\leq 2}$ . But  $D$  lands in  $\mathbb{R}[x]_{\leq 2}$  and  $E$  lands in  $\mathbb{R}$ , so  $D + E$  would require adding a polynomial to a number. The clause that fails is “ $S, T \in \mathcal{L}(V, W)$ ” with the **same**  $W$ : the sum is not defined.

## 3.4.2 Composition

Addition makes  $\mathcal{L}(V, W)$  a vector space, but the operation that really matters for maps is doing one after another. For functions, Chapter 0 wrote this as  $S \circ T$  ([Definition 0.4.4](#)). For linear maps we drop the circle.

*Remark. Notation.* For  $T \in \mathcal{L}(U, V)$  and  $S \in \mathcal{L}(V, W)$ , we write  $ST := S \circ T: U \rightarrow W$ , so  $(ST)\mathbf{u} = S(T\mathbf{u})$ . **The map written on the right acts first.** For an operator  $T \in \mathcal{L}(V)$ ,  $T^2 = TT$  means “apply  $T$  twice”, never “square the output”.

Is  $ST$  linear, and how does composition interact with the sum? It turns out that composition behaves like a multiplication: associative, with identities, and distributive over addition on both sides.

### Theorem 3.4.2 — Composition of Linear Maps

Let  $U, V, W, X$  be vector spaces over  $F$ .

- (a) **(Linearity)** If  $T \in \mathcal{L}(U, V)$  and  $S \in \mathcal{L}(V, W)$ , then  $ST \in \mathcal{L}(U, W)$ .
- (b) **(Associativity)** If  $T \in \mathcal{L}(U, V)$ ,  $S \in \mathcal{L}(V, W)$  and  $R \in \mathcal{L}(W, X)$ , then  $R(ST) = (RS)T$ .
- (c) **(Identities)** If  $T \in \mathcal{L}(U, V)$ , then  $T\text{id}_U = T = \text{id}_V T$ .
- (d) **(Distributivity)** If  $T, T_1, T_2 \in \mathcal{L}(U, V)$ ,  $S, S_1, S_2 \in \mathcal{L}(V, W)$  and  $c \in F$ , then

$$\begin{aligned}(S_1 + S_2)T &= S_1T + S_2T, \\ S(T_1 + T_2) &= ST_1 + ST_2, \\ (cS)T &= c(ST) = S(cT).\end{aligned}$$

*Idea.* Parts (b) and (c) hold for all functions, so there is nothing new to do. Part (a) is a two-line check. The interesting point is in (d): the three identities are not equally cheap. In  $(S_1 + S_2)T$  the map  $T$  acts first and the sum is taken afterwards, so the identity is just the definition of the sum. In  $S(T_1 + T_2)$  the sum is taken **before**  $S$  acts, and pulling it out of  $S$  needs  $S$  to be linear. So we should watch where the linearity of the **left** factor is spent.



*Proof.*

- (a) Let  $\mathbf{u}, \mathbf{u}' \in U$  and  $a \in F$ . Since  $T$  and then  $S$  are linear,

$$ST(\mathbf{u} + \mathbf{u}') = S(T\mathbf{u} + T\mathbf{u}') = ST\mathbf{u} + ST\mathbf{u}', \quad ST(a\mathbf{u}) = S(aT\mathbf{u}) = aST\mathbf{u}.$$

- (b) and (c) hold for arbitrary functions, by [Theorem 0.4.2](#) and the observation after [Definition 0.4.5](#).

- (c) Each side is a function  $U \rightarrow W$ , so we compare values at  $\mathbf{u} \in U$ . By [Definition 3.4.1](#),

$$(S_1 + S_2)T\mathbf{u} = S_1(T\mathbf{u}) + S_2(T\mathbf{u}) = (S_1T + S_2T)\mathbf{u},$$

with no linearity used. Next, since  $S$  is linear,

$$S(T_1 + T_2)\mathbf{u} = S(T_1\mathbf{u} + T_2\mathbf{u}) = ST_1\mathbf{u} + ST_2\mathbf{u} = (ST_1 + ST_2)\mathbf{u}.$$

Finally  $(cS)(T\mathbf{u}) = c(ST\mathbf{u})$  by definition, and  $S(cT\mathbf{u}) = c(ST\mathbf{u})$  because  $S$  is linear. This proves the theorem. ■

In short,  $\mathcal{L}(V)$  carries a vector space structure and a multiplication (composition) that is associative, has the unit  $\text{id}_V$ , distributes over addition, and lets scalars move freely through products. A vector space with such a multiplication is called an **algebra** over  $F$ ;  $M_n(F)$  is the other algebra we know ([Theorem 0.9.4](#)). The two will turn out to be the same algebra in disguise when we write operators as matrices later in this chapter.

### Example. (Rotate, Then Project)

Let  $R: \mathbb{R}^2 \rightarrow \mathbb{R}^2$  be the rotation by  $90^\circ$  counterclockwise,  $R(x, y) = (-y, x)$ , and let  $P: \mathbb{R}^2 \rightarrow \mathbb{R}^2$ ,  $P(x, y) = (x, 0)$ , be the projection onto the  $x$ -axis. Compute  $PR$  and  $RP$ .

*Solution.* In  $PR$  the rotation acts first:

$$PR(x, y) = P(-y, x) = (-y, 0), \quad RP(x, y) = R(x, 0) = (0, x).$$

At  $(1, 0)$ :  $PR(1, 0) = (0, 0)$ , while  $RP(1, 0) = (0, 1)$ . So  $PR \neq RP$ . Geometrically,  $PR$  rotates  $\mathbf{e}_1$  onto the  $y$ -axis and then squashes it to  $\mathbf{0}$ ;  $RP$  leaves  $\mathbf{e}_1$  alone under the projection and then rotates it to  $\mathbf{e}_2$ . In matrix form,  $P = T_{\mathbf{A}}$  and  $R = T_{\mathbf{B}}$  with  $\mathbf{A} = \begin{pmatrix} 1 & 0 \\ 0 & 0 \end{pmatrix}$  and  $\mathbf{B} = \begin{pmatrix} 0 & -1 \\ 1 & 0 \end{pmatrix}$ , and indeed  $\mathbf{AB} = \begin{pmatrix} 0 & -1 \\ 0 & 0 \end{pmatrix} \neq \begin{pmatrix} 0 & 0 \\ 1 & 0 \end{pmatrix} = \mathbf{BA}$ .

**Warning. Composition is not commutative, and the order is read right to left.** In [Example \(Rotate, Then Project\)](#),  $PR \neq RP$ . It can be worse: for  $T \in \mathcal{L}(\mathbb{R}^2, \mathbb{R}^3)$  and  $S \in \mathcal{L}(\mathbb{R}^3, \mathbb{R}^2)$ , both  $ST \in \mathcal{L}(\mathbb{R}^2)$  and  $TS \in \mathcal{L}(\mathbb{R}^3)$  are defined, but they live on different spaces, so they cannot even be compared. And for  $T \in \mathcal{L}(\mathbb{R}^2, \mathbb{R}^3)$ ,  $S \in \mathcal{L}(\mathbb{R}^3, \mathbb{R}^4)$ ,  $ST$  is defined but  $TS$  is not. The thing you do first goes on the **right**.

*Quick check.* Let  $T: \mathbb{R}^3 \rightarrow \mathbb{R}^2$  and  $S: \mathbb{R}^2 \rightarrow \mathbb{R}^4$  be linear. Which of  $ST$  and  $TS$  is defined, and between which spaces does it go?

*Solution.*  $ST = S \circ T$  first applies  $T: \mathbb{R}^3 \rightarrow \mathbb{R}^2$ , then  $S: \mathbb{R}^2 \rightarrow \mathbb{R}^4$ , so  $ST \in \mathcal{L}(\mathbb{R}^3, \mathbb{R}^4)$ .  $TS$  would apply  $S$  first, landing in  $\mathbb{R}^4$ , where  $T$  is not defined; so  $TS$  is not defined.

Because of the missing commutativity, school identities about products must be rechecked. By distributivity (Theorem 3.4.2 (d)),

$$(S + T)^2 = (S + T)(S + T) = S^2 + ST + TS + T^2,$$

which equals  $S^2 + 2ST + T^2$  only if  $ST = TS$ . With  $P$  and  $R$  as above,  $PR + RP: (x, y) \mapsto (-y, x)$ , while  $2PR: (x, y) \mapsto (-2y, 0)$ ; so  $(P + R)^2 \neq P^2 + 2PR + R^2$ . This is the operator version of the matrix warning in Chapter 0, §9.

### 3.4.3 Operators, powers and polynomials

Composition of operators on one space  $V$  never leaves  $\mathcal{L}(V)$ , so we may compose an operator with itself as often as we like. Repeated composition leads to powers, and combinations of powers to polynomials, exactly as for square matrices (Definition 0.9.14).

#### Definition 3.4.2 — Powers and Polynomials of an Operator

Let  $V$  be a vector space over  $F$  and  $T \in \mathcal{L}(V)$ . The **powers** of  $T$  are defined recursively by

$$T^0 := \text{id}_V, \quad T^{k+1} := T^k T \quad (k \in \mathbb{N}).$$

For  $p = a_0 + a_1x + \cdots + a_Nx^N \in F[x]$ , the operator  $p(T) \in \mathcal{L}(V)$  is

$$p(T) := a_0 \text{id}_V + a_1T + a_2T^2 + \cdots + a_NT^N.$$

In words:  $T^k$  applies  $T$  exactly  $k$  times, and  $T^0$  does nothing. In  $p(T)$ , the **constant term becomes**  $a_0 \text{id}_V$ , because an operator cannot be added to a scalar. Each  $T^k$  is linear by Theorem 3.4.2 (a) and induction, and  $p(T)$  is linear by Theorem 3.4.1. As for matrices,  $p(T)$  does not depend on how many zero coefficients we write, since  $0 \cdot T^k$  is the zero map.

Plugging in an operator respects sums and products of polynomials, just as plugging in a matrix does.

#### Theorem 3.4.3 — Polynomials in One Operator

Let  $T \in \mathcal{L}(V)$  and  $p, q \in F[x]$ . Then:

- (a)  $T^j T^k = T^{j+k}$  for all  $j, k \in \mathbb{N}$ ;
- (b)  $(p + q)(T) = p(T) + q(T)$  and  $(pq)(T) = p(T)q(T)$ ;
- (c)  $p(T)q(T) = q(T)p(T)$ .

*Idea.* The proof of Theorem 0.9.9 used four matrix rules: associativity, identity, distributivity and “scalars move through products”. Theorem 3.4.2 supplies exactly these four rules for operators. So the old proof transfers line by line. Part (c) is the surprise:  $T$  need not commute with other operators, but it commutes with every polynomial in

itself.

*Proof.*

- (a) Fix  $j$  and use induction on  $k$  ([Theorem 0.2.7](#)). For  $k = 0$ ,  $T^j T^0 = T^j \text{id}_V = T^j$  by [Theorem 3.4.2](#) (c). If  $T^j T^k = T^{j+k}$ , then by [Definition 3.4.2](#) and associativity ([Theorem 3.4.2](#) (b)),

$$T^j T^{k+1} = T^j (T^k T) = (T^j T^k) T = T^{j+k} T = T^{j+k+1}.$$

- (b) Let  $p = \sum_{i=0}^m a_i x^i$  and  $q = \sum_{j=0}^l b_j x^j$ . The statement for sums holds because  $(a_i + b_i)T^i = a_i T^i + b_i T^i$  in the vector space  $\mathcal{L}(V)$  ([Theorem 3.4.1](#)). For products, the coefficient of  $x^k$  in  $pq$  is  $\sum_{i+j=k} a_i b_j$  ([Definition 0.8.2](#)), so

$$\begin{aligned} (pq)(T) &= \sum_k \left( \sum_{i+j=k} a_i b_j \right) T^k = \sum_{i=0}^m \sum_{j=0}^l (a_i b_j) T^i T^j \\ &= \left( \sum_{i=0}^m a_i T^i \right) \left( \sum_{j=0}^l b_j T^j \right) = p(T) q(T), \end{aligned}$$

where the second equality uses (a), and the third uses both distributive laws and  $(a_i T^i)(b_j T^j) = (a_i b_j) T^i T^j$ , all from [Theorem 3.4.2](#) (d).

- (c) Since  $pq = qp$  in  $F[x]$  ([Theorem 0.8.1](#)), part (b) gives  $p(T)q(T) = (pq)(T) = (qp)(T) = q(T)p(T)$ . This proves the theorem. ■

In practice, (b) says that **a factorization of  $p$  is a factorization of  $p(T)$** : for instance  $T^2 - 3T + 2\text{id}_V = (T - \text{id}_V)(T - 2\text{id}_V) = (T - 2\text{id}_V)(T - \text{id}_V)$ .

### Example. (Differentiation Is Nilpotent)

Let  $n \geq 0$  and let  $D \in \mathcal{L}(F[x]_{\leq n})$  be (formal) differentiation ([Example \(Differentiation\)](#)),  $D(a_0 + a_1x + \cdots + a_nx^n) = a_1 + 2a_2x + \cdots + na_nx^{n-1}$ , where  $ka_k$  means  $a_k$  added to itself  $k$  times. Show that  $D^{n+1} = 0$ . Show also that  $D^n \neq 0$  when  $F = \mathbb{R}$ .

*Solution.* For  $0 \leq k \leq n$ ,  $D(x^k) = kx^{k-1}$  if  $k \geq 1$  and  $D(1) = 0$ . So  $D$  sends each  $x^k$  to a scalar multiple of  $x^{k-1}$ , or to 0 when  $k = 0$ . Applying this  $k + 1$  times,  $D^{k+1}(x^k)$  is a scalar multiple of  $D(1) = 0$ , so  $D^{k+1}(x^k) = 0$ . Since  $k \leq n$ , also  $D^{n+1}(x^k) = D^{n-k}(D^{k+1}(x^k)) = D^{n-k}(0) = 0$ , by [Theorem 3.4.3](#) (a).

Now let  $p = a_0 + a_1x + \cdots + a_nx^n$ . Since  $D^{n+1}$  is linear, [Theorem 3.1.4](#) gives

$$D^{n+1}(p) = a_0 D^{n+1}(1) + a_1 D^{n+1}(x) + \cdots + a_n D^{n+1}(x^n) = 0.$$

Hence  $D^{n+1} = 0$ .

Over  $\mathbb{R}$ ,  $D^n(x^n) = n(n-1)\cdots 1 = n! \neq 0$ , so  $D^n \neq 0$ . Over  $\mathbb{F}_2$  with  $n = 2$ , however,  $D(x^2) = 2x = 0$ , and  $D^2 = 0$  already; the field matters for the exact power, but never for  $D^{n+1} = 0$ .

An operator  $T$  with  $T^k = 0$  for some  $k \geq 1$  is called **nilpotent**. The example shows that a non-zero operator can have a power equal to zero, just as the matrix  $\begin{pmatrix} 0 & 1 \\ 0 & 0 \end{pmatrix}$  can. Differentiation is the model nilpotent operator, and it returns when we study canonical forms.

### Example. (Operators That Satisfy a Polynomial)

With  $P$  and  $R$  as in [Example \(Rotate, Then Project\)](#), find non-zero polynomials  $p$  and  $q$  with  $p(P) = 0$  and  $q(R) = 0$ . Also, for an arbitrary  $p \in F[x]$ , compute  $p(\text{id}_V)$  and  $p(Z)$ , where  $Z \in \mathcal{L}(V)$  is the zero operator.

*Solution.* For the projection,  $P^2(x, y) = P(x, 0) = (x, 0) = P(x, y)$ , so  $P^2 = P$  and  $p = x^2 - x$  satisfies  $p(P) = P^2 - P = 0$ . By [Theorem 3.4.3 \(b\)](#), this says  $P(P - \text{id}) = 0$ : whatever  $P - \text{id}$  produces is killed by  $P$ , and indeed  $(P - \text{id})(x, y) = (0, -y)$  lies on the  $y$ -axis.

For the rotation,  $R^2(x, y) = R(-y, x) = (-x, -y)$ , so  $R^2 = -\text{id}$ , and  $q = x^2 + 1$  satisfies  $q(R) = R^2 + \text{id} = 0$ . Rotating twice by  $90^\circ$  is rotating by  $180^\circ$ . Since  $q$  has no real roots,  $q$  has no real linear factor, so the trick “factor  $q$ , then factor  $q(R)$ ” gives nothing over  $\mathbb{R}$ .

For  $p = a_0 + a_1x + \cdots + a_Nx^N$ : every power of  $\text{id}_V$  is  $\text{id}_V$ , so  $p(\text{id}_V) = (a_0 + a_1 + \cdots + a_N)\text{id}_V = p(1)\text{id}_V$ . Every power  $Z^k$  with  $k \geq 1$  is the zero map, but  $Z^0 = \text{id}_V$ , so  $p(Z) = a_0\text{id}_V$ , the scalar  $p(0)$  times  $\text{id}_V$ . The second computation is the reason for the convention  $T^0 = \text{id}_V$ : with it, “the constant term” of  $p(T)$  is what survives when  $T$  is the zero operator.

**Warning.** The constant term is  $a_0\text{id}_V$ , **not**  $a_0$ . Writing “ $P^2 - P + 3$ ” for an operator is a type error:  $P^2 - P$  is a map and 3 is a number. For  $p = x^2 - x + 3$ ,  $p(P) = P^2 - P + 3\text{id}$ , which is  $3\text{id}$  since  $P^2 = P$ . Forgetting the  $\text{id}$  leads to “ $p(P) = 3$ ”, which is not an operator at all.

*Quick check.* With  $P$  the projection above, compute  $p(P)$  for  $p = x^3 - 2x + 1$ .

*Solution.* Since  $P^2 = P$ , also  $P^3 = P^2P = PP = P$ . So  $p(P) = P^3 - 2P + \text{id} = P - 2P + \text{id} = \text{id} - P$ , which is the map  $(x, y) \mapsto (0, y)$ : the projection onto the  $y$ -axis.

Polynomial relations such as  $P^2 = P$ ,  $R^2 = -\text{id}$  and  $D^{n+1} = 0$  carry real information about an operator. In the next section we use one to write down an inverse without solving any equations, and later in the book the polynomials that kill an operator lead to the minimal polynomial and to canonical forms.

## 3.4.4 Exercises

### 3.4.4.1 A. Check your understanding

#### Exercise A1.

- (a) For  $S, T \in \mathcal{L}(V, W)$  and  $c \in F$ , define  $S + T$  and  $cT$ . Why must  $S$  and  $T$  have the

same codomain?

- (b) True or false: for all  $S, T \in \mathcal{L}(V)$ ,  $ST = TS$ . Justify your answer.
- (c) Let  $T \in \mathcal{L}(\mathbb{R}^5, \mathbb{R}^2)$  and  $S \in \mathcal{L}(\mathbb{R}^2, \mathbb{R}^5)$ . Which of  $ST, TS$  are defined, and in which spaces do they lie?
- (d) For  $T \in \mathcal{L}(V)$  and  $p = 4$  (a constant polynomial), what is  $p(T)$ ?
- (e) True or false: for all  $S, T \in \mathcal{L}(V)$ ,  $(S+T)^2 = S^2 + 2ST + T^2$ . Justify your answer.

*Solution.*

- (a)  $(S+T)(\mathbf{v}) = S\mathbf{v} + T\mathbf{v}$  and  $(cT)(\mathbf{v}) = cT\mathbf{v}$  for every  $\mathbf{v} \in V$  (Definition 3.4.1). The sum  $S\mathbf{v} + T\mathbf{v}$  must be computed in a single vector space, so both values must lie in the same  $W$ .
- (b) False. In Example (Rotate, Then Project),  $PR(1, 0) = (0, 0)$  but  $RP(1, 0) = (0, 1)$ .
- (c) Both are defined.  $ST$  applies  $T$  first:  $ST \in \mathcal{L}(\mathbb{R}^5)$ .  $TS$  applies  $S$  first:  $TS \in \mathcal{L}(\mathbb{R}^2)$ .
- (d)  $p(T) = 4\text{id}_V$ , by Definition 3.4.2.
- (e) False. By Theorem 3.4.2 (d),  $(S+T)^2 = S^2 + ST + TS + T^2$ , which differs from  $S^2 + 2ST + T^2$  by  $TS - ST$ . With  $S = P, T = R$  of Example (Rotate, Then Project),  $RP - PR: (x, y) \mapsto (y, x)$  is not the zero map, so the identity fails.

### 3.4.4.2 B. Practice

#### Exercise B1: Computing compositions.

Let  $S, T \in \mathcal{L}(\mathbb{R}^2)$  be given by  $S(x, y) = (x + y, y)$  and  $T(x, y) = (y, x)$ .

- (a) Compute  $ST, TS$  and  $2S - T$ .
- (b) Compute  $S^2$  and  $S^k$  for every  $k \in \mathbb{N}$ .
- (c) Hence show that  $(S - \text{id})^2 = 0$ .

*Solution.*

- (a) Applying the right-hand map first,

$$ST(x, y) = S(y, x) = (x + y, x), \quad TS(x, y) = T(x + y, y) = (y, x + y).$$

So  $ST \neq TS$ ; for instance  $ST(1, 0) = (1, 1)$  and  $TS(1, 0) = (0, 1)$ . By Definition 3.4.1,  $(2S - T)(x, y) = (2x + 2y, 2y) - (y, x) = (2x + y, 2y - x)$ .

- (b)  $S^2(x, y) = S(x + y, y) = (x + 2y, y)$ . We claim  $S^k(x, y) = (x + ky, y)$  for all  $k \in \mathbb{N}$ , by induction on  $k$ . For  $k = 0$ ,  $S^0 = \text{id}$  sends  $(x, y)$  to  $(x + 0y, y)$ . If the claim holds for  $k$ , then

$$S^{k+1}(x, y) = S^k(S(x, y)) = S^k(x + y, y) = (x + y + ky, y) = (x + (k + 1)y, y),$$

where the first equality is Definition 3.4.2,  $S^{k+1} = S^k S$ .

- (c)  $(S - \text{id})(x, y) = (y, 0)$ , so  $(S - \text{id})^2(x, y) = (S - \text{id})(y, 0) = (0, 0)$ . Hence  $(S - \text{id})^2 = 0$ , that is,  $S - \text{id}$  is nilpotent. Expanding with Theorem 3.4.3 (b),  $S^2 - 2S + \text{id} = 0$ , which matches (b):  $(x + 2y, y) - 2(x + y, y) + (x, y) = (0, 0)$ .

**Exercise B2: Operators that preserve a subspace.**

Let  $U$  be a subspace of  $V$ , and let

$$\mathcal{A}_U := \{T \in \mathcal{L}(V) : T\mathbf{u} \in U \text{ for every } \mathbf{u} \in U\}.$$

Prove that  $\mathcal{A}_U$  is a subspace of  $\mathcal{L}(V)$ , that  $\text{id}_V \in \mathcal{A}_U$ , and that  $ST \in \mathcal{A}_U$  whenever  $S, T \in \mathcal{A}_U$ .

*Solution.* We use [Theorem 1.2.1](#) in the vector space  $\mathcal{L}(V)$  ([Theorem 3.4.1](#)).

- (1) The zero map sends every  $\mathbf{u} \in U$  to  $\mathbf{0}$ , and  $\mathbf{0} \in U$  because  $U$  is a subspace. So  $0 \in \mathcal{A}_U$ .
- (2) Let  $S, T \in \mathcal{A}_U$  and  $\mathbf{u} \in U$ . Then  $S\mathbf{u} \in U$  and  $T\mathbf{u} \in U$ , so  $(S+T)\mathbf{u} = S\mathbf{u} + T\mathbf{u} \in U$ , since  $U$  is closed under addition. Therefore  $S + T \in \mathcal{A}_U$ .
- (3) Let  $c \in F$  and  $T \in \mathcal{A}_U$ . For  $\mathbf{u} \in U$ ,  $(cT)\mathbf{u} = cT\mathbf{u} \in U$ , since  $U$  is closed under scalar multiplication. Therefore  $cT \in \mathcal{A}_U$ .

Hence  $\mathcal{A}_U$  is a subspace of  $\mathcal{L}(V)$ . Next,  $\text{id}_V \mathbf{u} = \mathbf{u} \in U$  for every  $\mathbf{u} \in U$ , so  $\text{id}_V \in \mathcal{A}_U$ . Finally let  $S, T \in \mathcal{A}_U$  and  $\mathbf{u} \in U$ . Then  $T\mathbf{u} \in U$ , and applying the hypothesis on  $S$  to the vector  $T\mathbf{u} \in U$  gives  $ST\mathbf{u} = S(T\mathbf{u}) \in U$ . This shows  $ST \in \mathcal{A}_U$ .

**Exercise B3: A polynomial that kills  $D$ .**

Let  $D \in \mathcal{L}(\mathbb{R}[x]_{\leq 2})$  be differentiation.

- (a) Find a non-zero polynomial  $p$  with  $p(D) = 0$ .
- (b) Show that no non-zero polynomial  $q$  of degree at most 2 satisfies  $q(D) = 0$ .
- (c) Hence, or otherwise, compute  $(\text{id} + D)^3$  as a combination of  $\text{id}$ ,  $D$ ,  $D^2$ , and evaluate  $(\text{id} + D)^3(x^2)$ .

*Solution.*

- (a) By [Example \(Differentiation Is Nilpotent\)](#) with  $n = 2$ ,  $D^3 = 0$ . So  $p = x^3$  works.
- (b) Let  $q = a + bx + cx^2$  with  $q(D) = a\text{id} + bD + cD^2 = 0$ . Applying this operator to  $x^2$ , with  $D(x^2) = 2x$  and  $D^2(x^2) = 2$ ,

$$0 = ax^2 + 2bx + 2c.$$

A polynomial is zero exactly when all its coefficients are, so  $a = 0$ ,  $2b = 0$  and  $2c = 0$ . Hence  $a = b = c = 0$  and  $q = 0$ .

- (c) The operator  $\text{id} + D$  is  $q(D)$  for  $q = 1 + x$ . Applying the product rule of [Theorem 3.4.3](#) (b) twice,  $(\text{id} + D)^3 = q(D)q(D)q(D) = (q^3)(D)$ , and  $q^3 = 1 + 3x + 3x^2 + x^3$ . So

$$(\text{id} + D)^3 = \text{id} + 3D + 3D^2 + D^3 = \text{id} + 3D + 3D^2,$$

using  $D^3 = 0$  from (a). Therefore  $(\text{id} + D)^3(x^2) = x^2 + 3 \cdot 2x + 3 \cdot 2 = x^2 + 6x + 6$ .

**3.4.4.3 C. Going deeper**

**Exercise C1: The Heisenberg relation.**

Let  $F$  be a field, let  $D \in \mathcal{L}(F[x])$  be formal differentiation,  $D(\sum_k a_k x^k) = \sum_{k \geq 1} k a_k x^{k-1}$  ([Example \(Differentiation\)](#)), and let  $S \in \mathcal{L}(F[x])$  be multiplication by  $x$ ,  $S(p) = xp$  (the map  $M$  of [Example \(Multiplication by  \$x\$ \)](#)).

- (a) Prove that  $DS - SD = \text{id}$ .
- (b) Let  $T \in \mathcal{L}(F^n, F^n)$ . Prove that  $T = T_{\mathbf{A}}$ , where  $\mathbf{A} \in M_{n \times n}(F)$  is the matrix with columns  $T\mathbf{e}_1, \dots, T\mathbf{e}_n$ .
- (c) Now suppose that  $F$  has **characteristic 0** ([Definition 0.7.2](#)), and let  $n \geq 1$ . Deduce that there are no operators  $D', S' \in \mathcal{L}(F^n)$  with  $D'S' - S'D' = \text{id}$ .
- (d) Show that the assumption on the characteristic in (c) cannot be dropped.

*Hint: for (c), use (b) and consider traces.*

*Solution.*

- (a) Both sides are linear ([Theorem 3.4.1](#), [Theorem 3.4.2](#)). We first compare them on  $x^k$ ,  $k \in \mathbb{N}$ . We have  $DS(x^k) = D(x^{k+1}) = (k+1)x^k$ . If  $k \geq 1$ ,  $SD(x^k) = x \cdot kx^{k-1} = kx^k$ ; if  $k = 0$ ,  $SD(1) = x \cdot 0 = 0 = 0 \cdot x^0$ . In both cases

$$(DS - SD)(x^k) = (k+1)x^k - kx^k = x^k = \text{id}(x^k).$$

Every  $p \in F[x]$  is a finite combination  $p = \sum_k a_k x^k$ , so by [Theorem 3.1.4](#),  $(DS - SD)(p) = \sum_k a_k (DS - SD)(x^k) = \sum_k a_k x^k = p$ . Hence  $DS - SD = \text{id}$ .

- (b) Let  $\mathbf{x} = (x_1, \dots, x_n) = x_1 \mathbf{e}_1 + \dots + x_n \mathbf{e}_n \in F^n$ . By [Theorem 3.1.4](#) and [Theorem 0.9.3](#),

$$T\mathbf{x} = x_1 T\mathbf{e}_1 + \dots + x_n T\mathbf{e}_n = \mathbf{A}\mathbf{x} = T_{\mathbf{A}}(\mathbf{x}).$$

Since  $\mathbf{x}$  was arbitrary,  $T = T_{\mathbf{A}}$ .

- (c) Suppose, for a contradiction, that  $D', S' \in \mathcal{L}(F^n)$  satisfy  $D'S' - S'D' = \text{id}$ . By (b),  $D' = T_{\mathbf{A}}$  and  $S' = T_{\mathbf{B}}$  for some  $\mathbf{A}, \mathbf{B} \in M_n(F)$ , and  $\text{id} = T_{\mathbf{I}_n}$ . By associativity of matrix multiplication,  $T_{\mathbf{A}}T_{\mathbf{B}}(\mathbf{x}) = \mathbf{A}(\mathbf{B}\mathbf{x}) = (\mathbf{A}\mathbf{B})\mathbf{x}$ , so  $T_{\mathbf{A}}T_{\mathbf{B}} = T_{\mathbf{A}\mathbf{B}}$ , and similarly  $T_{\mathbf{B}}T_{\mathbf{A}} = T_{\mathbf{B}\mathbf{A}}$ . With [Example \(Sums of Matrix Maps and Other First Examples\)](#) (a),

$$T_{\mathbf{A}\mathbf{B} - \mathbf{B}\mathbf{A}} = T_{\mathbf{A}}T_{\mathbf{B}} - T_{\mathbf{B}}T_{\mathbf{A}} = \text{id} = T_{\mathbf{I}_n}.$$

Evaluating at  $\mathbf{e}_j$  gives equal  $j$ -th columns ([Theorem 0.9.3](#)), so  $\mathbf{A}\mathbf{B} - \mathbf{B}\mathbf{A} = \mathbf{I}_n$ . Taking traces and using [Theorem 0.9.6](#),

$$n \cdot 1 = \text{tr } \mathbf{I}_n = \text{tr}(\mathbf{A}\mathbf{B}) - \text{tr}(\mathbf{B}\mathbf{A}) = 0.$$

Since  $n \geq 1$  and  $F$  has characteristic 0,  $n \cdot 1 \neq 0$  by [Definition 0.7.2](#). This is a contradiction, so no such  $D', S'$  exist. In words: over  $\mathbb{Q}$ ,  $\mathbb{R}$ ,  $\mathbb{C}$  or any field of characteristic 0, the relation  $DS - SD = \text{id}$  of (a) has no analogue on  $F^n$ ,  $n \geq 1$ . Since every  $n$ -dimensional space is isomorphic to  $F^n$ , as we show in the next section, the same holds on every non-zero finite-dimensional space. (On the zero space  $\text{id} = 0$ , so the relation holds trivially there; that is why  $n \geq 1$ .)

- (d) Over  $\mathbb{F}_2$  (characteristic 2), take  $\mathbf{A} = \begin{pmatrix} 0 & 0 \\ 1 & 0 \end{pmatrix}$  and  $\mathbf{B} = \begin{pmatrix} 0 & 1 \\ 0 & 0 \end{pmatrix}$  as in [Exercise \(C1:  \$\mathbf{AB} - \mathbf{BA} = \mathbf{I}\$  is impossible, or is it?\)](#). There  $\mathbf{AB} - \mathbf{BA} = \begin{pmatrix} -1 & 0 \\ 0 & 1 \end{pmatrix} = \mathbf{I}_2$ , because  $-1 = 1$  in  $\mathbb{F}_2$ . By the computation in (c),  $D' = T_{\mathbf{A}}$  and  $S' = T_{\mathbf{B}}$  satisfy  $D'S' - S'D' = T_{\mathbf{AB}-\mathbf{BA}} = \text{id}$  on  $\mathbb{F}_2^2$ . So the conclusion of (c) fails without the characteristic assumption; the proof breaks exactly at  $n \cdot 1 \neq 0$ , since  $2 \cdot 1 = 0$  in  $\mathbb{F}_2$ .

### Exercise C2: Commuting nilpotent operators.

Let  $S, T \in \mathcal{L}(V)$ .

- Suppose  $ST = TS$ . Prove that  $(S + T)^n = \sum_{k=0}^n \binom{n}{k} S^k T^{n-k}$  for every  $n \in \mathbb{N}$ , where  $\binom{n}{k} S^k T^{n-k}$  means  $S^k T^{n-k}$  added to itself  $\binom{n}{k}$  times.
- Deduce that if  $ST = TS$ ,  $S^a = 0$  and  $T^b = 0$  for some  $a, b \geq 1$ , then  $(S + T)^{a+b-1} = 0$ .
- Show that (b) fails without  $ST = TS$ : give nilpotent  $S, T \in \mathcal{L}(\mathbb{R}^2)$  such that  $S + T$  is not nilpotent.

*Hint: for (c), try maps that send one standard basis vector to the other and the other to 0.*

*Solution.*

- First,  $T^j S = ST^j$  for all  $j \in \mathbb{N}$ , by induction on  $j$ : it holds for  $j = 0$ , and if  $T^j S = ST^j$ , then  $T^{j+1} S = T^j(TS) = T^j(ST) = (T^j S)T = (ST^j)T = ST^{j+1}$ , using associativity ([Theorem 3.4.2](#) (b)) and [Theorem 3.4.3](#) (a) in the form  $T^{j+1} = T^j T$ .

Now induct on  $n$ . For  $n = 0$  both sides are  $\text{id}_V$ . Suppose the formula holds for  $n$ . By [Theorem 3.4.2](#) (d),

$$(S + T)^{n+1} = (S + T)^n(S + T) = \sum_{k=0}^n \binom{n}{k} S^k T^{n-k} S + \sum_{k=0}^n \binom{n}{k} S^k T^{n-k+1}.$$

By the first paragraph,  $S^k T^{n-k} S = S^k (T^{n-k} S) = S^k S T^{n-k} = S^{k+1} T^{n-k}$ . Shifting the index in the first sum, the coefficient of  $S^k T^{n+1-k}$  becomes  $\binom{n}{k-1} + \binom{n}{k} = \binom{n+1}{k}$  (Pascal's rule, with  $\binom{n}{-1} = \binom{n}{n+1} = 0$ ). This is the formula for  $n + 1$ .

- By (a) with  $n = a + b - 1$ ,  $(S + T)^{a+b-1}$  is a sum of multiples of  $S^k T^{a+b-1-k}$ ,  $0 \leq k \leq a + b - 1$ . If  $k \geq a$ , then  $S^k = S^{k-a} S^a = 0$  by [Theorem 3.4.3](#) (a), so the term is 0 by [Theorem 3.4.2](#) (d). If  $k \leq a - 1$ , then  $a + b - 1 - k \geq b$ , so  $T^{a+b-1-k} = 0$  likewise. Every term vanishes, so  $(S + T)^{a+b-1} = 0$ .
- Let  $S(x, y) = (y, 0)$  and  $T(x, y) = (0, x)$ . Then  $S^2(x, y) = S(y, 0) = (0, 0)$  and  $T^2(x, y) = T(0, x) = (0, 0)$ , so both are nilpotent. But  $(S + T)(x, y) = (y, x)$ , so  $(S + T)^2(x, y) = (x, y)$ , that is,  $(S + T)^2 = \text{id}$ . Then  $(S + T)^{2k} = \text{id} \neq 0$  and  $(S + T)^{2k+1} = S + T \neq 0$  for every  $k$ , so  $S + T$  is not nilpotent. Here  $ST(x, y) = (x, 0) \neq (0, y) = TS(x, y)$ , so the hypothesis of (b) indeed fails.



## 3.5 Invertible Maps and Isomorphisms

In Chapter 1 we noticed that a basis of length  $n$  turns every vector of  $V$  into a column in  $F^n$ , and that sums and multiples survive the translation. Then  $\mathbb{R}[x]_{\leq 2}$  and  $\mathbb{R}^3$  behave “the same”, although they are different sets. This section makes “the same” precise. We define invertible linear maps and isomorphisms, prove that finite-dimensional spaces are classified by one number, their dimension, and collect everything we know about invertibility of an operator into one theorem.

### 3.5.1 When are two spaces the same?

The naive notion of sameness is equality of sets, and it fails at once. The polynomial  $3 - x + 2x^2$  is not a column vector, so  $\mathbb{R}[x]_{\leq 2} \neq \mathbb{R}^3$ . Yet every computation in one space has a twin in the other. Match  $a_0 + a_1x + a_2x^2$  with  $(a_0, a_1, a_2)$ :

$$(1 + 2x) + (x - x^2) = 1 + 3x - x^2 \quad \longleftrightarrow \quad (1, 2, 0) + (0, 1, -1) = (1, 3, -1).$$

A reader who only sees one side cannot tell which space the computation happened in. Three features make this work. The matching is **one-to-one** and **onto**, so nothing is lost or left over. It **respects sums and multiples** in the direction polynomial  $\rightarrow$  column. And it respects them in the **reverse** direction too, so a conclusion drawn among columns can be carried back. In the language of this chapter: a linear map with a linear map going back.

*An isomorphism is a perfect dictionary between two vector spaces: every word has exactly one translation, and translating respects sums and scalar multiples in both directions.*

#### Definition 3.5.1 — Invertible Linear Map

Let  $V$  and  $W$  be vector spaces over  $F$ , and let  $T \in \mathcal{L}(V, W)$ . Then  $T$  is **invertible** if there **exists a linear** map  $S \in \mathcal{L}(W, V)$  such that

$$ST = \text{id}_V \quad \text{and} \quad TS = \text{id}_W.$$

In words:  $S$  undoes  $T$  on  $V$  (first  $T$ , then  $S$ , and we are back where we started),  $T$  undoes  $S$  on  $W$ , and the undoing map is itself linear. The definition asks for **one** map  $S$  doing **both** jobs.

**Well-definedness.** If  $S$  exists, it is an inverse **function** of  $T$  in the sense of [Definition 0.4.7](#), and inverse functions are unique by [Theorem 0.4.7](#). So we may speak of **the** inverse and write  $T^{-1} := S$ . The same theorem says an invertible  $T$  is a bijection.

Is the word “linear” in the definition doing any work? One could imagine a linear bijection whose inverse function is not linear, which would then fail the definition. It turns out this never happens:

#### Theorem 3.5.1 — The Inverse of a Linear Bijection Is Linear

Let  $T \in \mathcal{L}(V, W)$  be bijective. Then its inverse function  $T^{-1}: W \rightarrow V$  is linear. Consequently, a linear map  $T$  is invertible if and only if it is bijective.

*Idea.* To show  $T^{-1}(\mathbf{w}_1 + \mathbf{w}_2) = T^{-1}\mathbf{w}_1 + T^{-1}\mathbf{w}_2$ , we cannot compute  $T^{-1}$  directly. But we can apply  $T$ , which we know is linear, to the right-hand side, and see that it lands on  $\mathbf{w}_1 + \mathbf{w}_2$ . Since  $T$  is injective, only one vector does that, namely  $T^{-1}(\mathbf{w}_1 + \mathbf{w}_2)$ .

*Proof.* Let  $\mathbf{w}_1, \mathbf{w}_2 \in W$  and  $c \in F$ , and put  $\mathbf{v}_i := T^{-1}\mathbf{w}_i$ , so that  $T\mathbf{v}_i = \mathbf{w}_i$ . Since  $T$  is linear,

$$T(\mathbf{v}_1 + \mathbf{v}_2) = \mathbf{w}_1 + \mathbf{w}_2, \quad T(c\mathbf{v}_1) = c\mathbf{w}_1.$$

Applying  $T^{-1}$  to both equations gives  $T^{-1}(\mathbf{w}_1 + \mathbf{w}_2) = \mathbf{v}_1 + \mathbf{v}_2 = T^{-1}\mathbf{w}_1 + T^{-1}\mathbf{w}_2$  and  $T^{-1}(c\mathbf{w}_1) = c\mathbf{v}_1 = cT^{-1}\mathbf{w}_1$ . Hence  $T^{-1}$  is linear.

For the last statement: if  $T$  is invertible, it is bijective by [Theorem 0.4.7](#). Conversely, if  $T$  is bijective, its inverse function  $S = T^{-1}$  satisfies  $ST = \text{id}_V$  and  $TS = \text{id}_W$  by [Definition 0.4.7](#), and  $S$  is linear by the first part. So  $T$  is invertible. This proves the theorem. ■

So in practice we check invertibility by checking **bijectivity**, and never need to verify linearity of the inverse. Invertible linear maps are important enough to have a second name.

### Definition 3.5.2 — Isomorphism

Let  $V$  and  $W$  be vector spaces over the same field  $F$ . An **isomorphism** from  $V$  to  $W$  is an invertible linear map  $T \in \mathcal{L}(V, W)$ ; equivalently, by [Theorem 3.5.1](#), a **bijective** linear map.

### Definition 3.5.3 — Isomorphic Spaces

Vector spaces  $V$  and  $W$  over  $F$  are **isomorphic**, written  $V \cong W$ , if there **exists an** isomorphism  $V \rightarrow W$ .

In words:  $V \cong W$  says that **some** perfect dictionary exists; it does not name one. The name comes from Greek *isos* (equal) and *morphe* (form): the two spaces have the same form, though not the same elements. An isomorphism  $V \rightarrow V$  is also called an **invertible operator**.

### Example. (The Standard Isomorphisms)

Show that each map is an isomorphism, and give its inverse.

- (a)  $\Phi: \mathbb{R}[x]_{\leq n} \rightarrow \mathbb{R}^{n+1}$ ,  $a_0 + a_1x + \cdots + a_nx^n \mapsto (a_0, a_1, \dots, a_n)$ .
- (b)  $\Psi: M_{m \times n}(F) \rightarrow F^{mn}$ , which lists the entries row by row:  $(a_{ij}) \mapsto (a_{11}, \dots, a_{1n}, a_{21}, \dots, a_{mn})$ .
- (c)  $\Theta: \mathbb{C} \rightarrow \mathbb{R}^2$ ,  $a + bi \mapsto (a, b)$ , where  $\mathbb{C}$  is a vector space **over**  $\mathbb{R}$ .
- (d)  $T_{\mathbf{A}}: F^n \rightarrow F^n$ ,  $\mathbf{x} \mapsto \mathbf{A}\mathbf{x}$ , for an invertible  $\mathbf{A} \in M_n(F)$ ; and  $\text{id}_V$  for any  $V$ .

*Solution.*

- (a) *Linear.* Polynomials are added and scaled coefficient by coefficient, and so are

- columns, so  $\Phi(p + q) = \Phi(p) + \Phi(q)$  and  $\Phi(cp) = c\Phi(p)$ . *Bijective.* The map  $(a_0, \dots, a_n) \mapsto a_0 + a_1x + \dots + a_nx^n$  is a two-sided inverse function, because two polynomials are equal exactly when all their coefficients agree ([Definition 0.8.1](#)). By [Theorem 3.5.1](#),  $\Phi$  is an isomorphism, and  $\Phi^{-1}$  is that map. Hence  $\mathbb{R}[x]_{\leq n} \cong \mathbb{R}^{n+1}$ .
- (b) Matrices are added and scaled entrywise, and so are vectors of  $F^{mn}$ , so  $\Psi$  is linear. The map that writes the first  $n$  entries of a vector into row 1, the next  $n$  into row 2, and so on, undoes  $\Psi$  on both sides. Hence  $M_{m \times n}(F) \cong F^{mn}$ .
- (c) For  $a, b, a', b', c \in \mathbb{R}$ :  $(a + bi) + (a' + b'i) = (a + a') + (b + b')i$  and  $c(a + bi) = ca + cbi$ , so  $\Theta$  is linear over  $\mathbb{R}$ . Every complex number has unique real and imaginary parts ([Definition 0.6.1](#)), so  $(a, b) \mapsto a + bi$  is a two-sided inverse. Hence  $\mathbb{C} \cong \mathbb{R}^2$  **as real vector spaces**. In the same way [Example \(The Complexification of  \$\mathbb{R}^n\$  Is  \$\mathbb{C}^n\$ \)](#) showed that  $(\mathbf{x}, \mathbf{y}) \mapsto \mathbf{x} + i\mathbf{y}$  is a bijection  $(\mathbb{R}^n)_{\mathbb{C}} \rightarrow \mathbb{C}^n$  respecting complex scalars; in our new language,  $(\mathbb{R}^n)_{\mathbb{C}} \cong \mathbb{C}^n$  over  $\mathbb{C}$ .
- (d)  $T_{\mathbf{A}}$  is linear ([Example \(Multiplication by a Matrix\)](#)). Since  $\mathbf{A}(\mathbf{A}^{-1}\mathbf{x}) = (\mathbf{A}\mathbf{A}^{-1})\mathbf{x} = \mathbf{x}$  and  $\mathbf{A}^{-1}(\mathbf{A}\mathbf{x}) = \mathbf{x}$  by [Theorem 0.9.4](#), the linear map  $T_{\mathbf{A}^{-1}}$  is a two-sided inverse:  $T_{\mathbf{A}}^{-1} = T_{\mathbf{A}^{-1}}$ . The identity  $\text{id}_V$  is its own inverse. This degenerate example matters: it shows  $V \cong V$ , and even the zero space is isomorphic to itself, via the zero map  $\{\mathbf{0}\} \rightarrow \{\mathbf{0}\}$ , which is  $\text{id}_{\{\mathbf{0}\}}$ .

Here is a non-example by minimal change. Take (a) with  $n = 3$ , but keep only three coefficients:  $\Phi': \mathbb{R}[x]_{\leq 3} \rightarrow \mathbb{R}^3$ ,  $a_0 + a_1x + a_2x^2 + a_3x^3 \mapsto (a_0, a_1, a_2)$ . It is still linear, and still onto, since  $(a_0, a_1, a_2) = \Phi'(a_0 + a_1x + a_2x^2)$ . But  $\Phi'(x^3) = \mathbf{0} = \Phi'(0)$ , so  $\Phi'$  is **not injective**. No  $S$  can satisfy  $S\Phi' = \text{id}$ , because  $S\Phi'(x^3) = S(\mathbf{0}) = 0 \neq x^3$ . The clause  $ST = \text{id}_V$  is the one that fails.

**Why this definition.** Could we drop one of the two equations? The next example says no.

#### Example. (One Equation Is Not Enough)

On  $\mathbb{R}[x]$ , let  $D$  be differentiation and  $J(p) := \int_0^x p(t) dt$ , so that  $J(x^k) = \frac{x^{k+1}}{k+1}$  ([Example \(Differentiation\)](#), [Example \(Integration\)](#)). On the sequence space  $F^{\mathbb{N}}$ , let  $L(a_0, a_1, a_2, \dots) := (a_1, a_2, a_3, \dots)$  and  $R(a_0, a_1, a_2, \dots) := (0, a_0, a_1, \dots)$  be the left and right shifts of [Example \(The Shift Maps on Sequences\)](#). Show that  $DJ = \text{id}$  and  $LR = \text{id}$ , but that none of  $D, J, L, R$  is invertible.

*Solution.* All four maps are linear, by [Example \(Differentiation\)](#), [Example \(Integration\)](#) and [Example \(The Shift Maps on Sequences\)](#). For each  $k \in \mathbb{N}$ ,  $DJ(x^k) = D(\frac{x^{k+1}}{k+1}) = x^k$ , so  $DJ$  and  $\text{id}$  agree on every  $x^k$ . Every polynomial is a finite combination of the  $x^k$ , so  $DJ = \text{id}$  by [Theorem 3.1.4](#). Directly,  $LR(a_0, a_1, \dots) = L(0, a_0, a_1, \dots) = (a_0, a_1, \dots)$ , so  $LR = \text{id}$ .

But  $JD(1) = J(0) = 0 \neq 1$ , and  $RL(1, 0, 0, \dots) = R(0, 0, \dots) = \mathbf{0}$ . In fact  $D(1) = 0 = D(0)$  and  $L(1, 0, 0, \dots) = \mathbf{0} = L(\mathbf{0})$ , so  $D$  and  $L$  are not injective; and  $1 \notin \text{im } J$  (every  $J(p)$  has constant coefficient 0) and  $(1, 0, 0, \dots) \notin \text{im } R$ , so  $J$  and  $R$  are not surjective. None of the four is bijective, so none is invertible ([Theorem 3.5.1](#)).

So a one-sided inverse can exist without invertibility. The spaces here are infinite-dimensional, and that is not an accident: later in this section we prove that in finite dimension one equation does imply the other.

*Warning.* **Isomorphic is not equal, and an isomorphism is a choice.**  $\mathbb{R}[x]_{\leq 2} \cong \mathbb{R}^3$  does not make a polynomial a column. Worse, there are many isomorphisms, and none is singled out. The coefficient map of [Example \(The Standard Isomorphisms\)](#) sends  $x$  to  $(0, 1, 0)$ . The coordinate map for the basis  $(1, 1+x, 1+x+x^2)$  of [Example \(Coordinates of a Polynomial\)](#) is also an isomorphism (we prove this below), and it sends  $x = -1 \cdot 1 + 1 \cdot (1+x)$  to  $(-1, 1, 0)$ . A general  $n$ -dimensional space has no “standard” basis, so there is no natural isomorphism  $F^n \rightarrow V$ ; every one depends on a choice. Chapter 4 returns to this point with dual spaces.

The first payoff of the definition is that an isomorphism carries all linear structure across, in both directions. We record the two facts we need.

### Theorem 3.5.2 — Isomorphisms Carry Bases to Bases

Let  $T: V \rightarrow W$  be an isomorphism, and let  $(\mathbf{v}_1, \dots, \mathbf{v}_n)$  be a basis of  $V$ . Then  $(T\mathbf{v}_1, \dots, T\mathbf{v}_n)$  is a basis of  $W$ .

*Proof.* Since  $T$  is injective, the list  $(T\mathbf{v}_1, \dots, T\mathbf{v}_n)$  is linearly independent by [Theorem 3.2.5](#). By [Theorem 3.2.4](#), it spans  $\text{im } T$ , and  $\text{im } T = W$  because  $T$  is surjective. Hence it is a basis of  $W$ . ■

### Theorem 3.5.3 — Isomorphism Is an Equivalence Relation

For vector spaces over  $F$ :  $V \cong V$ ; if  $V \cong W$  then  $W \cong V$ ; and if  $U \cong V$  and  $V \cong W$  then  $U \cong W$ .

*Proof.*  $\text{id}_V$  is an isomorphism ([Example \(The Standard Isomorphisms\)](#) (d)). If  $T: V \rightarrow W$  is an isomorphism, then  $T^{-1}$  is linear ([Theorem 3.5.1](#)) and has inverse  $T$ , so it is an isomorphism  $W \rightarrow V$ . If  $T: U \rightarrow V$  and  $S: V \rightarrow W$  are isomorphisms, then  $ST$  is linear by [Theorem 3.4.2](#) and bijective by [Theorem 0.4.3](#), hence an isomorphism by [Theorem 3.5.1](#), with  $(ST)^{-1} = T^{-1}S^{-1}$  by [Theorem 0.4.8](#). This proves the theorem. ■

So  $\cong$  sorts vector spaces over  $F$  into classes ([Definition 0.5.2](#)). The next question is which spaces land in the same class.

## 3.5.2 Classification by dimension

By [Theorem 3.5.2](#), isomorphic spaces have bases of the same length. Is that the **only** obstruction? Two spaces of dimension 6, say  $M_{2 \times 3}(\mathbb{R})$  and  $\mathbb{R}[x]_{\leq 5}$ , look nothing alike. The following theorem says that this does not matter: for finite-dimensional spaces, dimension is a **complete invariant**.

### Theorem 3.5.4 — Classification of Finite-Dimensional Spaces

Let  $V$  and  $W$  be finite-dimensional vector spaces over the same field  $F$ . Then

$$V \cong W \iff \dim V = \dim W.$$

*Idea.*  $(\Rightarrow)$  is the payoff we just proved: an isomorphism moves a basis to a basis of the same length. For  $(\Leftarrow)$  we must **build** an isomorphism, and the only material we have is a basis on each side,  $(\mathbf{v}_1, \dots, \mathbf{v}_n)$  and  $(\mathbf{w}_1, \dots, \mathbf{w}_n)$ . The obvious dictionary sends  $\mathbf{v}_i$  to  $\mathbf{w}_i$ . Linear maps can be prescribed freely on a basis, so this defines  $T$ , and the reverse prescription defines  $S$ . To see  $ST = \text{id}_V$  we do not compute: both sides are linear and agree on the basis, so they are equal. The hypothesis “same dimension” is exactly what makes the two bases the same length, so that “ $\mathbf{v}_i \mapsto \mathbf{w}_i$ ” makes sense.

*Proof.*  $(\Rightarrow)$  Suppose  $T: V \rightarrow W$  is an isomorphism, and let  $(\mathbf{v}_1, \dots, \mathbf{v}_n)$  be a basis of  $V$ , with  $n = \dim V$  (Corollary 1.5.5). By Theorem 3.5.2,  $(T\mathbf{v}_1, \dots, T\mathbf{v}_n)$  is a basis of  $W$ , so  $\dim W = n = \dim V$ .

$(\Leftarrow)$  Suppose  $\dim V = \dim W = n$ , and choose bases  $(\mathbf{v}_1, \dots, \mathbf{v}_n)$  of  $V$  and  $(\mathbf{w}_1, \dots, \mathbf{w}_n)$  of  $W$  (Corollary 1.5.5). By Theorem 3.1.5 there are linear maps  $T \in \mathcal{L}(V, W)$  with  $T\mathbf{v}_i = \mathbf{w}_i$  and  $S \in \mathcal{L}(W, V)$  with  $S\mathbf{w}_i = \mathbf{v}_i$ , for  $i = 1, \dots, n$ . Then  $ST \in \mathcal{L}(V)$  is linear (Theorem 3.4.2) and  $ST\mathbf{v}_i = S\mathbf{w}_i = \mathbf{v}_i = \text{id}_V \mathbf{v}_i$  for every  $i$ . Two linear maps that agree on a basis are equal, by the uniqueness part of Theorem 3.1.5, so  $ST = \text{id}_V$ . Swapping the roles of  $V$  and  $W$  gives  $TS = \text{id}_W$ . Hence  $T$  is invertible, and  $V \cong W$ . This proves the theorem. ■

The theorem needs both spaces to be over the **same** field; this is part of Definition 3.5.3.  $\mathbb{C}$  over  $\mathbb{C}$  has dimension 1 and  $\mathbb{R}$  over  $\mathbb{R}$  has dimension 1, but the question “is  $\mathbb{C} \cong \mathbb{R}$ ?” makes no sense, since a linear map must respect one common set of scalars.

Taking  $W = F^n$  with its standard basis, the proof builds a specific isomorphism  $V \rightarrow F^n$ : the one sending  $\mathbf{v}_i$  to  $\mathbf{e}_i$ . It is an old friend.

### Corollary 3.5.5 — The Coordinate Isomorphism

Let  $\mathcal{B} = (\mathbf{v}_1, \dots, \mathbf{v}_n)$  be a basis of a vector space  $V$  over  $F$ . Then the coordinate map

$$V \rightarrow F^n, \quad \mathbf{v} \mapsto [\mathbf{v}]_{\mathcal{B}}$$

is an isomorphism, with inverse  $(a_1, \dots, a_n) \mapsto a_1\mathbf{v}_1 + \dots + a_n\mathbf{v}_n$ . In particular, every  $n$ -dimensional vector space over  $F$  is isomorphic to  $F^n$ .

*Proof.* By Theorem 1.5.2, the coordinate map is linear, and every column  $(a_1, \dots, a_n) \in F^n$  is the coordinate vector of exactly one vector of  $V$ , namely  $a_1\mathbf{v}_1 + \dots + a_n\mathbf{v}_n$ . So the coordinate map is bijective, with the stated inverse function, and it is an isomorphism by Theorem 3.5.1. ■

This is the promise of Chapter 1 kept: the correspondence  $\mathbf{v} \mapsto [\mathbf{v}]_{\mathcal{B}}$  from [Theorem 1.5.2](#) is an isomorphism. Its practical meaning is the slogan behind the rest of this chapter: **any computation with sums and multiples in an  $n$ -dimensional space can be done with columns in  $F^n$** , at the price of choosing a basis.

**Example. (Sorting Spaces by Dimension)**

Which of the following real vector spaces are isomorphic to each other?

- (i)  $\mathbb{R}[x]_{\leq 5}$ ;
- (ii)  $M_{2 \times 3}(\mathbb{R})$ ;
- (iii) the upper triangular matrices in  $M_3(\mathbb{R})$ ;
- (iv)  $\mathbb{C}^3$ , regarded as a vector space over  $\mathbb{R}$ ;
- (v)  $U = \{p \in \mathbb{R}[x]_{\leq 6} : p(1) = 0\}$ ;
- (vi)  $\mathbb{R}^5$ .

*Solution.* By [Theorem 3.5.4](#) it suffices to compute dimensions over  $\mathbb{R}$ .

- (i) The basis  $(1, x, \dots, x^5)$  gives dimension 6. (ii)  $\dim M_{2 \times 3}(\mathbb{R}) = 2 \cdot 3 = 6$  ([Example \(Dimensions of the Running Examples\)](#)). (iii) An upper triangular matrix is  $\sum_{i \leq j} a_{ij} \mathbf{E}_{ij}$ , and this expression is unique because the coefficient of  $\mathbf{E}_{ij}$  is the  $(i, j)$ -entry. By [Theorem 1.5.1](#) the six matrix units  $\mathbf{E}_{ij}$  with  $1 \leq i \leq j \leq 3$  form a basis, so the dimension is 6. (iv)  $\dim_{\mathbb{C}} \mathbb{C}^3 = 3$ , so  $\dim_{\mathbb{R}} \mathbb{C}^3 = 6$  by [Theorem 1.9.2](#).
- (ii) The evaluation map  $E: \mathbb{R}[x]_{\leq 6} \rightarrow \mathbb{R}$ ,  $p \mapsto p(1)$ , is linear, and surjective because  $E(c) = c$  for each constant  $c$ . Its kernel is  $U$ . By [Theorem 3.3.1](#),  $\dim U = \dim \mathbb{R}[x]_{\leq 6} - \text{rank } E = 7 - 1 = 6$ .
- (iii)  $\dim \mathbb{R}^5 = 5$ .

Hence the spaces (i)–(v) are all isomorphic to each other (and to  $\mathbb{R}^6$ ), while  $\mathbb{R}^5$  is isomorphic to none of them. We never wrote down an isomorphism between, say, (iii) and (v); the theorem guarantees one.

*Quick check.* Is  $M_2(\mathbb{R})$  isomorphic to  $\mathbb{R}[x]_{\leq 3}$ ? Is it isomorphic to  $\mathbb{R}[x]_{\leq 4}$ ?

*Solution.*  $\dim M_2(\mathbb{R}) = 4 = \dim \mathbb{R}[x]_{\leq 3}$ , so  $M_2(\mathbb{R}) \cong \mathbb{R}[x]_{\leq 3}$  by [Theorem 3.5.4](#). But  $\dim \mathbb{R}[x]_{\leq 4} = 5 \neq 4$  (not 4: count  $1, x, x^2, x^3, x^4$ ), so  $M_2(\mathbb{R}) \not\cong \mathbb{R}[x]_{\leq 4}$ .

### 3.5.3 One-sided inverses

For square matrices, Chapter 2 proved that one equation is enough: if  $\mathbf{A}, \mathbf{B} \in M_n(F)$  and  $\mathbf{AB} = \mathbf{I}_n$ , then  $\mathbf{BA} = \mathbf{I}_n$  ([Theorem 2.4.5](#)). [Example \(One Equation Is Not Enough\)](#) shows that this fails for operators on infinite-dimensional spaces. And it fails between spaces of different dimension: for  $T: \mathbb{R}^2 \rightarrow \mathbb{R}^3$ ,  $(x, y) \mapsto (x, y, 0)$ , and  $S: \mathbb{R}^3 \rightarrow \mathbb{R}^2$ ,  $(x, y, z) \mapsto (x, y)$ , we have  $ST = \text{id}_{\mathbb{R}^2}$  but  $TS(0, 0, 1) = \mathbf{0}$ , so  $TS \neq \text{id}_{\mathbb{R}^3}$ . What survives is exactly the case that looks like “square”:

**Theorem 3.5.6 — One-Sided Inverses of Linear Maps**

Let  $V$  and  $W$  be finite-dimensional vector spaces over  $F$  with  $\dim V = \dim W$ . Let  $T \in \mathcal{L}(V, W)$  and  $S \in \mathcal{L}(W, V)$ . If  $ST = \text{id}_V$ , then  $TS = \text{id}_W$ . In that case  $T$  and  $S$  are both isomorphisms,  $S = T^{-1}$  and  $T = S^{-1}$ .

*Idea.* Without any dimension hypothesis,  $ST = \text{id}_V$  already gives half of what we need:  $T$  must be injective, since  $S$  recovers  $\mathbf{v}$  from  $T\mathbf{v}$ . The missing half, surjectivity, is exactly what the Rank–Nullity Theorem supplies for free when  $\dim V = \dim W$  is finite. Once  $T$  is known to be invertible, we cancel it:  $S = (ST)T^{-1} = T^{-1}$ . So the one hypothesis the infinite-dimensional counterexamples violate, finite equal dimension, is spent on a single line: injective  $\Rightarrow$  surjective.

*Proof.* Suppose  $ST = \text{id}_V$ , and let  $\mathbf{v} \in \ker T$ . Then  $\mathbf{v} = \text{id}_V \mathbf{v} = S(T\mathbf{v}) = S\mathbf{0} = \mathbf{0}$ , since  $S$  is linear. So  $\ker T = \{\mathbf{0}\}$ , and  $T$  is injective by Theorem 3.2.3. Since  $V$  and  $W$  are finite-dimensional with  $\dim V = \dim W$ , Corollary 3.3.2 shows that  $T$  is also surjective. Hence  $T$  is bijective, and by Theorem 3.5.1 it has a linear inverse  $T^{-1} \in \mathcal{L}(W, V)$ . By Theorem 3.4.2 (b) and (c),

$$S = S \text{id}_W = S(TT^{-1}) = (ST)T^{-1} = \text{id}_V T^{-1} = T^{-1}.$$

Therefore  $TS = TT^{-1} = \text{id}_W$ . Finally  $S = T^{-1}$  is an isomorphism with inverse  $T$  (Theorem 3.5.3), so  $T = S^{-1}$ . This proves the theorem. ■

*Remark.* Compare the two proofs. For matrices (Theorem 2.4.5), the step “ $\mathbf{B}\mathbf{x} = \mathbf{0}$  has only the trivial solution, hence  $\mathbf{B}$  is invertible” came from row reduction, through the Invertible Matrix Theorem. Here the same step, “injective, hence bijective”, comes from Rank–Nullity, with no matrices at all. Once we can write maps as matrices (next section), the two theorems become the same theorem seen from two sides.

In practice this means: **to verify that  $S$  is the inverse of  $T$  between spaces of the same finite dimension, one composition suffices.** By symmetry,  $TS = \text{id}_W$  also implies  $ST = \text{id}_V$ : apply the theorem with the roles of  $T$  and  $S$  (and of  $V$  and  $W$ ) swapped.

**3.5.4 The Invertible Operator Theorem**

For an operator on a finite-dimensional space, invertibility can now be detected in many different ways. Chapter 2 collected the matrix versions in the Invertible Matrix Theorem (Theorem 2.4.4), and said the list would grow. Here is its first growth, stated for operators.

**Theorem 3.5.7 — Invertible Operator Theorem**

Let  $V$  be a **finite-dimensional** vector space over  $F$ , and let  $T \in \mathcal{L}(V)$ . The following are equivalent.

- (a)  $T$  is invertible.

- (b)  $T$  is injective.
- (c)  $\ker T = \{\mathbf{0}\}$ , that is, nullity  $T = 0$ .
- (d)  $T$  is surjective.
- (e)  $\text{rank } T = \dim V$ .
- (f) There is  $S \in \mathcal{L}(V)$  with  $ST = \text{id}_V$ .
- (g) There is  $S \in \mathcal{L}(V)$  with  $TS = \text{id}_V$ .
- (h)  $T$  maps **some** basis of  $V$  to a basis of  $V$ .
- (i)  $T$  maps **every** basis of  $V$  to a basis of  $V$ .

*Idea.* Nine conditions, but almost every arrow is a citation. The plan:

- ① the core, (a)  $\Leftrightarrow$  (b)  $\Leftrightarrow$  (c)  $\Leftrightarrow$  (d)  $\Leftrightarrow$  (e): kernel test for injectivity, Rank–Nullity for “injective  $\Leftrightarrow$  surjective”, equal dimension for “image is everything”;
- ② the one-sided inverses (f), (g): trivially implied by (a), and they imply (a) by [Theorem 3.5.6](#) with  $W = V$ ;
- ③ the bases, (a)  $\Rightarrow$  (i)  $\Rightarrow$  (h)  $\Rightarrow$  (d): isomorphisms carry bases to bases, some basis exists, and an image containing a basis is everything.

*Proof.* (a)  $\Rightarrow$  (b). An invertible map is bijective ([Theorem 3.5.1](#)).

(b)  $\Leftrightarrow$  (c). This is [Theorem 3.2.3](#), and  $\ker T = \{\mathbf{0}\}$  is the same as  $\dim \ker T = 0$ .

(b)  $\Leftrightarrow$  (d). Since  $T$  maps the finite-dimensional space  $V$  to a space of the same dimension, this is [Corollary 3.3.2](#).

(d)  $\Leftrightarrow$  (e). If  $\text{im } T = V$ , then  $\text{rank } T = \dim V$ . Conversely, if  $\text{rank } T = \dim V$ , then  $\text{im } T$  is a subspace of  $V$  of dimension  $\dim V$ , so  $\text{im } T = V$  by [Theorem 1.6.8](#).

(b) and (d)  $\Rightarrow$  (a). If  $T$  is injective, it is also surjective by (b)  $\Rightarrow$  (d), hence bijective, hence invertible by [Theorem 3.5.1](#). Together with the arrows above, (a)–(e) are equivalent.

(a)  $\Rightarrow$  (f) and (a)  $\Rightarrow$  (g). Take  $S = T^{-1}$ .

(f)  $\Rightarrow$  (a) and (g)  $\Rightarrow$  (a). Apply [Theorem 3.5.6](#) with  $W = V$ : if  $ST = \text{id}_V$ , then  $T$  is invertible; if  $TS = \text{id}_V$ , the same theorem with the roles of  $S$  and  $T$  swapped shows that  $S$  is invertible with  $S^{-1} = T$ , so  $T$  is invertible.

(a)  $\Rightarrow$  (i). This is [Theorem 3.5.2](#).

(i)  $\Rightarrow$  (h).  $V$  has a basis by [Corollary 1.5.5](#), and by (i) its image is a basis.

(h)  $\Rightarrow$  (d). Let  $(\mathbf{v}_1, \dots, \mathbf{v}_n)$  be a basis whose image  $(T\mathbf{v}_1, \dots, T\mathbf{v}_n)$  is a basis of  $V$ . Each  $T\mathbf{v}_i$  lies in the subspace  $\text{im } T$ , so  $V = \text{span}(T\mathbf{v}_1, \dots, T\mathbf{v}_n) \subseteq \text{im } T$  by [Theorem 1.3.1](#). Hence  $\text{im } T = V$ .

All nine conditions are equivalent. This proves the theorem. ■

As with the matrix theorem, the list is a menu. To prove that an operator **is** invertible, pick the cheapest item, usually (c): take  $\mathbf{v}$  with  $T\mathbf{v} = \mathbf{0}$  and show  $\mathbf{v} = \mathbf{0}$ . To prove that it is **not**,



exhibit one non-zero vector in the kernel.

**Example. (Invertibility Without Solving Equations)**

- (a) Let  $T \in \mathcal{L}(\mathbb{R}[x]_{\leq 2})$ ,  $T(p) = p + xp'$ . Show that  $T$  is invertible and find  $T^{-1}$ .
- (b) Let  $V$  be **any** vector space and  $T \in \mathcal{L}(V)$  with  $T^2 = T + \text{id}_V$ . Show that  $T$  is invertible, with  $T^{-1} = T - \text{id}_V$ .

*Solution.*

- (a) Let  $p = a + bx + cx^2$ . Then  $xp' = bx + 2cx^2$ , so

$$T(a + bx + cx^2) = a + 2bx + 3cx^2.$$

If  $T(p) = 0$ , then  $a = 2b = 3c = 0$ , so  $a = b = c = 0$  and  $p = 0$ . Hence  $\ker T = \{0\}$ , and  $T$  is invertible by [Theorem 3.5.7](#) ((c)  $\Rightarrow$  (a)). We did not have to check surjectivity: the count did it for us. For the inverse, let  $S(a + bx + cx^2) := a + \frac{b}{2}x + \frac{c}{3}x^2$ , which is linear. Then  $TS(a + bx + cx^2) = a + 2 \cdot \frac{b}{2}x + 3 \cdot \frac{c}{3}x^2 = a + bx + cx^2$ , so  $TS = \text{id}$ , and by [Theorem 3.5.6](#),  $S = T^{-1}$ .

- (b) The operators  $T$  and  $T - \text{id}_V$  are polynomials in  $T$ , so they commute ([Theorem 3.4.3](#) (c)). By [Theorem 3.4.2](#) (d) and the hypothesis,

$$T(T - \text{id}_V) = T^2 - T = \text{id}_V = (T - \text{id}_V)T.$$

So  $T - \text{id}_V$  is a two-sided inverse, and  $T$  is invertible by [Definition 3.5.1](#). Here  $V$  may be infinite-dimensional, since we verified **both** equations.

**Warning. The Invertible Operator Theorem needs finite dimension, and it is about operators.** On  $\mathbb{R}[x]$ , differentiation  $D$  is surjective (every polynomial has an antiderivative  $J(p)$ ) but not injective, and  $S(p) = xp$  is injective but not surjective ( $1 \notin \text{im } S$ ). And for a map  $T: V \rightarrow W$  between **different** spaces, “injective” does not give “surjective” unless  $\dim V = \dim W$ : the inclusion  $(x, y) \mapsto (x, y, 0)$  of  $\mathbb{R}^2$  into  $\mathbb{R}^3$  is injective and not surjective.

*Quick check.* Is  $T \in \mathcal{L}(M_2(\mathbb{R}))$ ,  $T(\mathbf{A}) = \mathbf{A} + \mathbf{A}^\top$ , invertible? Use the cheapest item of [Theorem 3.5.7](#).

*Solution.* No. Let  $\mathbf{A} = \begin{pmatrix} 0 & 1 \\ -1 & 0 \end{pmatrix} \neq 0$ . Then  $\mathbf{A}^\top = -\mathbf{A}$ , so  $T(\mathbf{A}) = 0$ , and  $\ker T \neq \{0\}$ . Item (c) fails, so  $T$  is not invertible.

Isomorphisms let us move a whole space into  $F^n$ . The next section moves **maps** as well: once bases are chosen, every linear map between finite-dimensional spaces becomes a matrix, and composition becomes matrix multiplication.

## 3.5.5 Exercises

### 3.5.5.1 A. Check your understanding

**Exercise A1.**

- (a) Define what it means for  $T \in \mathcal{L}(V, W)$  to be an isomorphism, and what  $V \cong W$  means.
- (b) True or false:  $\mathbb{R}[x]_{\leq 3} \cong M_2(\mathbb{R})$ . Justify your answer.
- (c) True or false: if  $\dim V = \dim W = 5$  and  $T \in \mathcal{L}(V, W)$  is injective, then  $T$  is an isomorphism. Justify your answer.
- (d) True or false: for every vector space  $V$  and all  $S, T \in \mathcal{L}(V)$ ,  $ST = \text{id}_V$  implies  $TS = \text{id}_V$ . Justify your answer.
- (e) True or false: if  $V \cong W$ , then every linear map  $V \rightarrow W$  is an isomorphism. Justify your answer.
- (f) Which theorem would you use to show that a given operator on a finite-dimensional space is invertible by computing only its kernel?

*Solution.*

- (a)  $T$  is an isomorphism if it is invertible, that is, there is  $S \in \mathcal{L}(W, V)$  with  $ST = \text{id}_V$  and  $TS = \text{id}_W$ ; equivalently,  $T$  is a bijective linear map ([Theorem 3.5.1](#)).  $V \cong W$  means that some isomorphism  $V \rightarrow W$  exists.
- (b) True. Both have dimension 4, so they are isomorphic by [Theorem 3.5.4](#).
- (c) True. By [Corollary 3.3.2](#) an injective linear map between spaces of the same finite dimension is surjective, so  $T$  is bijective, hence an isomorphism.
- (d) False. On  $F^{\mathbb{N}}$ , the shifts of [Example \(One Equation Is Not Enough\)](#) satisfy  $LR = \text{id}$  but  $RL \neq \text{id}$ . The statement is true when  $V$  is finite-dimensional ([Theorem 3.5.6](#)).
- (e) False.  $\mathbb{R}^2 \cong \mathbb{R}^2$ , but the zero map  $\mathbb{R}^2 \rightarrow \mathbb{R}^2$  is not injective.
- (f) The Invertible Operator Theorem, [Theorem 3.5.7](#), (c)  $\Rightarrow$  (a).

**3.5.5.2 B. Practice****Exercise B1: An explicit isomorphism.**

Let  $U = \{\mathbf{A} \in M_2(\mathbb{R}) : \mathbf{A}^\top = \mathbf{A}\}$  be the space of symmetric  $2 \times 2$  real matrices. Find an explicit isomorphism  $U \rightarrow \mathbb{R}[x]_{\leq 2}$ , and verify that it is one.

*Solution.* Every  $\mathbf{A} \in U$  has the form  $\begin{pmatrix} a & b \\ b & c \end{pmatrix}$  with  $a, b, c \in \mathbb{R}$ . Define

$$T \begin{pmatrix} a & b \\ b & c \end{pmatrix} := a + bx + cx^2, \quad S(a + bx + cx^2) := \begin{pmatrix} a & b \\ b & c \end{pmatrix}.$$

*Linearity of  $T$ .* Adding two symmetric matrices adds the entries  $a, b, c$ , and adding polynomials adds coefficients, so  $T(\mathbf{A} + \mathbf{A}') = T(\mathbf{A}) + T(\mathbf{A}')$ ; similarly  $T(\lambda \mathbf{A}) = \lambda T(\mathbf{A})$ .

*Inverse.*  $S$  takes values in  $U$ , and  $ST(\mathbf{A}) = \mathbf{A}$  for every  $\mathbf{A} \in U$ , since  $\mathbf{A}$  is determined by its entries  $a, b, c$ . Also  $TS(p) = p$  for every  $p$ , since  $p$  is determined by its coefficients. So  $T$  has a two-sided inverse and is bijective, hence an isomorphism by [Theorem 3.5.1](#). (Alternatively: both spaces have dimension 3 by [Example \(Dimensions of the Running Examples\)](#), so by [Theorem 3.5.6](#) the single equation  $ST = \text{id}$  would already suffice.)

**Exercise B2: Inverting  $\text{id} + D$ .**

Let  $n \geq 0$ , and let  $T \in \mathcal{L}(\mathbb{R}[x]_{\leq n})$ ,  $T(p) = p + p'$ .

- (a) Show that  $T$  is an isomorphism by computing its kernel.
- (b) Show that  $T^{-1} = \sum_{k=0}^n (-1)^k D^k$ , where  $D$  is differentiation.
- (c) For  $n = 2$ , find the polynomial  $p$  with  $p + p' = x^2$ .

*Hint: for (b), use the polynomial identity  $(1+x)\sum_{k=0}^n (-x)^k = 1 + (-1)^n x^{n+1}$ .*

*Solution.*

- (a) Let  $p \in \ker T$ , so  $p' = -p$ . If  $p \neq 0$ , let  $d = \deg p \geq 0$ . Differentiation lowers the degree of a non-zero polynomial by at least one (it kills constants), so  $\deg p' < d = \deg(-p)$ , contradicting  $p' = -p$ . Hence  $p = 0$  and  $\ker T = \{0\}$ . By [Theorem 3.5.7](#) ((c)  $\Rightarrow$  (a)),  $T$  is an isomorphism.
- (b)  $T = \text{id} + D = q(D)$  with  $q = 1 + x$ , and  $S := \sum_{k=0}^n (-1)^k D^k = r(D)$  with  $r = \sum_{k=0}^n (-x)^k$ . The identity in the hint is the telescoping sum  $\sum_{k=0}^n (-x)^k - \sum_{k=0}^n (-x)^{k+1} = 1 - (-x)^{n+1}$ . By [Theorem 3.4.3](#) (b),

$$TS = (qr)(D) = \text{id} + (-1)^n D^{n+1} = \text{id},$$

since  $D^{n+1} = 0$  on  $\mathbb{R}[x]_{\leq n}$  ([Example \(Differentiation Is Nilpotent\)](#)). By [Theorem 3.5.6](#),  $S = T^{-1}$ .

- (c)  $p = T^{-1}(x^2) = x^2 - D(x^2) + D^2(x^2) = x^2 - 2x + 2$ . Check:  $p + p' = (x^2 - 2x + 2) + (2x - 2) = x^2$ .

**Exercise B3: Deciding isomorphism.**

Determine which of the following pairs of real vector spaces are isomorphic. Justify your answer.

- (a)  $\mathbb{R}^4$  and  $M_2(\mathbb{R})$ .
- (b)  $\mathbb{R}[x]_{\leq 3}$  and  $\{\mathbf{A} \in M_3(\mathbb{R}) : \mathbf{A}^\top = -\mathbf{A}\}$ .
- (c)  $\{p \in \mathbb{R}[x]_{\leq 4} : p(2) = 0\}$  and  $\mathbb{R}^4$ .
- (d)  $\mathbb{C}^2$  (over  $\mathbb{R}$ ) and  $\mathbb{R}[x]_{\leq 3}$ .
- (e) The plane  $\{(x, y, z) \in \mathbb{R}^3 : x + y + z = 0\}$  and  $\mathbb{C}$  (over  $\mathbb{R}$ ).

*Solution.* By [Theorem 3.5.4](#), in each case we compare dimensions over  $\mathbb{R}$ .

- (a) Isomorphic:  $\dim \mathbb{R}^4 = 4 = \dim M_2(\mathbb{R})$ .
- (b) Not isomorphic.  $\dim \mathbb{R}[x]_{\leq 3} = 4$ . If  $\mathbf{A}^\top = -\mathbf{A}$ , then  $a_{ii} = -a_{ii}$ , so  $2a_{ii} = 0$  and  $a_{ii} = 0$ , and  $a_{ji} = -a_{ij}$ . Hence  $\mathbf{A} = a_{12}(\mathbf{E}_{12} - \mathbf{E}_{21}) + a_{13}(\mathbf{E}_{13} - \mathbf{E}_{31}) + a_{23}(\mathbf{E}_{23} - \mathbf{E}_{32})$ , and the coefficients are entries of  $\mathbf{A}$ , so this expression is unique. By [Theorem 1.5.1](#) these three matrices form a basis, and the dimension is  $3 \neq 4$ .
- (c) Isomorphic. The evaluation  $p \mapsto p(2)$ ,  $\mathbb{R}[x]_{\leq 4} \rightarrow \mathbb{R}$ , is linear and surjective (constants), with kernel the given space, so by [Theorem 3.3.1](#) its dimension is  $5 - 1 = 4$ .
- (d) Isomorphic:  $\dim_{\mathbb{R}} \mathbb{C}^2 = 2 \dim_{\mathbb{C}} \mathbb{C}^2 = 4$  by [Theorem 1.9.2](#), and  $\dim \mathbb{R}[x]_{\leq 3} = 4$ .

- (e) Isomorphic. The plane is the kernel of the surjective linear map  $(x, y, z) \mapsto x + y + z$  onto  $\mathbb{R}$ , so it has dimension  $3 - 1 = 2$  by [Theorem 3.3.1](#); and  $\dim_{\mathbb{R}} \mathbb{C} = 2$  ([Example \(Dimensions of the Running Examples\)](#)).

### 3.5.5.3 C. Going deeper

#### Exercise C1: Invertible products.

- (a) Let  $V$  be finite-dimensional and  $S, T \in \mathcal{L}(V)$ . Prove that if  $ST$  is invertible, then  $S$  and  $T$  are both invertible. Deduce that  $TS$  is then invertible too.
- (b) Show that (a) fails for  $V = F^{\mathbb{N}}$ : give  $S, T \in \mathcal{L}(F^{\mathbb{N}})$  with  $ST$  invertible but neither  $S$  nor  $T$  invertible, and  $TS$  not invertible.

*Hint: for (a), regroup  $(ST)(ST)^{-1}$  and  $(ST)^{-1}(ST)$ .*

*Solution.*

- (a) The inverse  $(ST)^{-1}$  lies in  $\mathcal{L}(V)$  ([Theorem 3.5.1](#)), and  $(ST)(ST)^{-1} = \text{id}_V = (ST)^{-1}(ST)$ . By associativity ([Theorem 3.4.2 \(b\)](#)),  $S(T(ST)^{-1}) = \text{id}_V$  and  $((ST)^{-1}S)T = \text{id}_V$ . By [Theorem 3.5.7](#), (g)  $\Rightarrow$  (a) applied to  $S$  with right inverse  $T(ST)^{-1}$  shows that  $S$  is invertible, and (f)  $\Rightarrow$  (a) applied to  $T$  with left inverse  $(ST)^{-1}S$  shows that  $T$  is invertible. Then  $TS$  is a composition of isomorphisms, hence an isomorphism by [Theorem 3.5.3](#).
- (b) Let  $S = L$  and  $T = R$  be the left and right shifts of [Example \(One Equation Is Not Enough\)](#). Then  $ST = LR = \text{id}$ , which is invertible. But  $L$  is not injective and  $R$  is not surjective, so neither is invertible, and  $TS = RL$  sends  $(1, 0, 0, \dots)$  to  $\mathbf{0}$ , so it is not injective and not invertible. The proof of (a) breaks at the appeal to [Theorem 3.5.7](#), which needs finite dimension.

#### Exercise C2: A space isomorphic to a proper subspace.

- (a) Let  $V$  be finite-dimensional and  $U$  a subspace of  $V$  with  $U \cong V$ . Prove that  $U = V$ .
- (b) Let  $U = \{p \in F[x] : p(0) = 0\}$ , the polynomials with constant coefficient 0. Show that  $T: F[x] \rightarrow U$ ,  $T(p) = xp$ , is an isomorphism. Hence (a) fails without finite dimension.

*Solution.*

- (a)  $U$  is finite-dimensional by [Theorem 1.6.7](#). Since  $U \cong V$ , [Theorem 3.5.4](#) gives  $\dim U = \dim V$ , and then  $U = V$  by [Theorem 1.6.8](#).
- (b) For  $p = \sum_k a_k x^k$ ,  $xp = \sum_k a_k x^{k+1}$  has constant coefficient 0, so  $T$  maps into  $U$ , and  $T$  is linear because multiplication in  $F[x]$  distributes and commutes with scalars ([Theorem 0.8.1](#)). *Injective:* if  $xp = 0$ , then  $p = 0$  by [Corollary 0.8.4](#), since  $x \neq 0$ . *Surjective:* if  $q = \sum_{k \geq 1} b_k x^k \in U$ , then  $q = x \sum_{k \geq 1} b_k x^{k-1} = T(\sum_{k \geq 1} b_k x^{k-1})$ . So  $T$  is a bijective linear map, an isomorphism by [Theorem 3.5.1](#), and  $U \cong F[x]$ . But  $1 \notin U$ , so  $U \neq F[x]$ . This is the phenomenon noted in Chapter 1, §8, now as an isomorphism.

## 3.6 The Matrix of a Linear Map

An isomorphism  $V \cong F^n$  turns the vectors of an abstract space into columns. This section does the same for **maps**. Once a basis is fixed on each side, every linear map between finite-dimensional spaces becomes a matrix, applying the map becomes multiplying by that matrix, and composing maps becomes multiplying matrices. This last fact finally explains the strange multiplication rule of Chapter 0, and it gives us the working method of the rest of the book: to study an abstract map, write down its matrix, compute there, and translate back.

### 3.6.1 Writing a map as a matrix

We have seen one family of linear maps that is completely described by a table of numbers: the matrix maps  $T_{\mathbf{A}}: F^n \rightarrow F^m$ ,  $\mathbf{x} \mapsto \mathbf{A}\mathbf{x}$ . In fact **every** linear map  $T: F^n \rightarrow F^m$  is one of them. By [Theorem 3.1.4](#) and [Theorem 0.9.3](#), for  $\mathbf{x} = x_1\mathbf{e}_1 + \cdots + x_n\mathbf{e}_n$ ,

$$T\mathbf{x} = x_1T\mathbf{e}_1 + \cdots + x_nT\mathbf{e}_n = \mathbf{A}\mathbf{x}, \quad \text{where } \mathbf{A} = (T\mathbf{e}_1 \ \cdots \ T\mathbf{e}_n).$$

The  $j$ -th column of  $\mathbf{A}$  is the image of the  $j$ -th basis vector. We want the same description for maps between arbitrary finite-dimensional spaces, such as differentiation on polynomials.

Take  $D: \mathbb{R}[x]_{\leq 3} \rightarrow \mathbb{R}[x]_{\leq 2}$ . Then

$$D(a_0 + a_1x + a_2x^2 + a_3x^3) = a_1 + 2a_2x + 3a_3x^2.$$

A polynomial is not a column, but its coordinates are. In the bases  $(1, x, x^2, x^3)$  and  $(1, x, x^2)$ , the input has coordinates  $(a_0, a_1, a_2, a_3)$  and the output has coordinates  $(a_1, 2a_2, 3a_3)$ , and

$$\begin{pmatrix} a_1 \\ 2a_2 \\ 3a_3 \end{pmatrix} = \begin{pmatrix} 0 & 1 & 0 & 0 \\ 0 & 0 & 2 & 0 \\ 0 & 0 & 0 & 3 \end{pmatrix} \begin{pmatrix} a_0 \\ a_1 \\ a_2 \\ a_3 \end{pmatrix}.$$

Look at the columns of this matrix:  $(0, 0, 0)$ ,  $(1, 0, 0)$ ,  $(0, 2, 0)$ ,  $(0, 0, 3)$ . They are the coordinates of  $D(1) = 0$ ,  $D(x) = 1$ ,  $D(x^2) = 2x$ ,  $D(x^3) = 3x^2$ . The standard example generalizes: record the images of the input basis, in coordinates of the output basis, as columns.

*The matrix of a linear map lists, column by column, where the input basis vectors go, written in coordinates of the output basis.*

#### Definition 3.6.1 — Matrix of a Linear Map

Let  $V$  and  $W$  be finite-dimensional vector spaces over  $F$ , with **ordered** bases  $\mathcal{B} = (\mathbf{v}_1, \dots, \mathbf{v}_n)$  of  $V$  and  $\mathcal{C} = (\mathbf{w}_1, \dots, \mathbf{w}_m)$  of  $W$ , and let  $T \in \mathcal{L}(V, W)$ . The **matrix of  $T$  with respect to  $\mathcal{B}$  and  $\mathcal{C}$**  is the  $m \times n$  matrix

$$[T]_{\mathcal{B}}^{\mathcal{C}} := ([T\mathbf{v}_1]_{\mathcal{C}} \ [T\mathbf{v}_2]_{\mathcal{C}} \ \cdots \ [T\mathbf{v}_n]_{\mathcal{C}}) \in M_{m \times n}(F),$$

whose  $j$ -th column is the coordinate vector of  $T\mathbf{v}_j$  with respect to  $\mathcal{C}$ . Equivalently, its entries  $a_{ij}$  are the **unique** scalars with

$$T\mathbf{v}_j = a_{1j}\mathbf{w}_1 + a_{2j}\mathbf{w}_2 + \cdots + a_{mj}\mathbf{w}_m \quad (j = 1, \dots, n).$$

When  $V = W$  and  $\mathcal{B} = \mathcal{C}$ , we write  $[T]_{\mathcal{B}} := [T]_{\mathcal{B}}^{\mathcal{B}}$  and call it the matrix of the operator  $T$  with respect to  $\mathcal{B}$ .

In words, clause by clause:

- **The input basis  $\mathcal{B}$  is the subscript, the output basis  $\mathcal{C}$  is the superscript.** The subscript sits next to where the input goes, as in  $[T]_{\mathcal{B}}^{\mathcal{C}}[\mathbf{v}]_{\mathcal{B}}$  below.
- **Columns correspond to input basis vectors**, so there are  $n = \dim V$  of them; **rows correspond to output basis vectors**, so there are  $m = \dim W$ . The size is “ $\dim(\text{codomain}) \times \dim(\text{domain})$ ”, the same order as  $M_{m \times n}$  for  $T_{\mathbf{A}}: F^n \rightarrow F^m$ .
- Column  $j$  answers one question: **where does  $\mathbf{v}_j$  go?**

**Well-definedness.** Each  $T\mathbf{v}_j$  lies in  $W$ , and  $\mathcal{C}$  is a basis, so  $[T\mathbf{v}_j]_{\mathcal{C}}$  exists and is unique by Theorem 1.5.1. So the matrix is determined by  $T$ ,  $\mathcal{B}$  and  $\mathcal{C}$ . It is **not** determined by  $T$  alone: change either basis, or even the **order** of the vectors in one of them, and the matrix changes. That is why the bases are ordered.

### Example. (Differentiation on Cubic Polynomials)

Let  $\mathcal{E}_3 = (1, x, x^2, x^3)$  and  $\mathcal{E}_2 = (1, x, x^2)$ . Find  $[D]_{\mathcal{E}_3}^{\mathcal{E}_2}$  for  $D: \mathbb{R}[x]_{\leq 3} \rightarrow \mathbb{R}[x]_{\leq 2}$ , and  $[D]_{\mathcal{E}_3}$  for  $D$  regarded as an operator on  $\mathbb{R}[x]_{\leq 3}$ .

*Solution.* The images of the input basis are  $D(1) = 0$ ,  $D(x) = 1$ ,  $D(x^2) = 2x$ ,  $D(x^3) = 3x^2$ . In  $\mathcal{E}_2$  their coordinates are  $(0, 0, 0)$ ,  $(1, 0, 0)$ ,  $(0, 2, 0)$ ,  $(0, 0, 3)$ ; in  $\mathcal{E}_3$  the same vectors have a fourth coordinate 0. So

$$[D]_{\mathcal{E}_3}^{\mathcal{E}_2} = \begin{pmatrix} 0 & 1 & 0 & 0 \\ 0 & 0 & 2 & 0 \\ 0 & 0 & 0 & 3 \end{pmatrix}, \quad [D]_{\mathcal{E}_3} = \begin{pmatrix} 0 & 1 & 0 & 0 \\ 0 & 0 & 2 & 0 \\ 0 & 0 & 0 & 3 \\ 0 & 0 & 0 & 0 \end{pmatrix}.$$

The same rule  $p \mapsto p'$  gives a  $3 \times 4$  or a  $4 \times 4$  matrix, depending on the codomain we declare. The superdiagonal 1, 2, 3 records “ $x^k \mapsto kx^{k-1}$ ”.

### Example. (Matrix Maps and the Identity)

- Let  $\mathbf{A} \in M_{m \times n}(F)$ , and let  $\mathcal{E}_n, \mathcal{E}_m$  be the standard bases of  $F^n, F^m$ . Show that  $[T_{\mathbf{A}}]_{\mathcal{E}_n}^{\mathcal{E}_m} = \mathbf{A}$ .
- Let  $\mathcal{B}$  be any basis of an  $n$ -dimensional space  $V$ . Find  $[\text{id}_V]_{\mathcal{B}}$  and the matrix of the zero map  $V \rightarrow W$  for any bases.

*Solution.*

- The  $j$ -th column of  $[T_{\mathbf{A}}]_{\mathcal{E}_n}^{\mathcal{E}_m}$  is  $[\mathbf{A}\mathbf{e}_j]_{\mathcal{E}_m} = \mathbf{A}\mathbf{e}_j$ , since coordinates in the standard basis of  $F^m$  are the entries themselves; and  $\mathbf{A}\mathbf{e}_j$  is the  $j$ -th column of  $\mathbf{A}$  (Theorem 0.9.3). So the matrix of a matrix map in the standard bases is the matrix we started with.
- $\text{id}_V \mathbf{v}_j = \mathbf{v}_j = 0\mathbf{v}_1 + \cdots + 1\mathbf{v}_j + \cdots + 0\mathbf{v}_n$ , so the  $j$ -th column of  $[\text{id}_V]_{\mathcal{B}}$  is  $\mathbf{e}_j$ , and  $[\text{id}_V]_{\mathcal{B}} = \mathbf{I}_n$ . The zero map sends every  $\mathbf{v}_j$  to  $\mathbf{0}$ , whose coordinates are all 0, so its matrix is the zero matrix of size  $\dim W \times \dim V$ , whatever the bases. These

degenerate cases fix the expectations “identity map, identity matrix” and “zero map, zero matrix”. The first needs the **same** basis on both sides, as the warning below shows.

### Example. (The Transpose Map)

Let  $T: M_2(F) \rightarrow M_2(F)$ ,  $T(\mathbf{X}) = \mathbf{X}^\top$ , and let  $\mathcal{E} = (\mathbf{E}_{11}, \mathbf{E}_{12}, \mathbf{E}_{21}, \mathbf{E}_{22})$ . Find  $[T]_{\mathcal{E}}$ .

*Solution.*  $T$  is linear by [Theorem 0.9.5](#). Its values on the basis are  $\mathbf{E}_{11}^\top = \mathbf{E}_{11}$ ,  $\mathbf{E}_{12}^\top = \mathbf{E}_{21}$ ,  $\mathbf{E}_{21}^\top = \mathbf{E}_{12}$ ,  $\mathbf{E}_{22}^\top = \mathbf{E}_{22}$ . In  $\mathcal{E}$ , these have coordinates  $\mathbf{e}_1, \mathbf{e}_3, \mathbf{e}_2, \mathbf{e}_4$ . So

$$[T]_{\mathcal{E}} = \begin{pmatrix} 1 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 1 \end{pmatrix}.$$

A map on a space of  $2 \times 2$  matrices is described by a  $4 \times 4$  matrix, because  $\dim M_2(F) = 4$ . Do not confuse the matrix  $\mathbf{X}$  that  $T$  acts on with the matrix  $[T]_{\mathcal{E}}$  that describes  $T$ .

### Example. (A Reflection in Two Bases)

Let  $T: \mathbb{R}^2 \rightarrow \mathbb{R}^2$  be the reflection across the line  $y = x$ ,  $T(x, y) = (y, x)$ . Find  $[T]_{\mathcal{E}}$  for the standard basis  $\mathcal{E} = (\mathbf{e}_1, \mathbf{e}_2)$ , and  $[T]_{\mathcal{B}}$  for  $\mathcal{B} = (\mathbf{u}_1, \mathbf{u}_2)$  with  $\mathbf{u}_1 = (1, 1)$  and  $\mathbf{u}_2 = (1, -1)$ .

*Solution.*  $T\mathbf{e}_1 = (0, 1) = \mathbf{e}_2$  and  $T\mathbf{e}_2 = \mathbf{e}_1$ , so  $[T]_{\mathcal{E}} = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}$ .

The vectors  $\mathbf{u}_1, \mathbf{u}_2$  are independent (neither is a multiple of the other), hence a basis of  $\mathbb{R}^2$  by [Theorem 1.6.4](#). The first lies **on** the mirror line and the second is **perpendicular** to it:  $T\mathbf{u}_1 = (1, 1) = 1\mathbf{u}_1 + 0\mathbf{u}_2$  and  $T\mathbf{u}_2 = (-1, 1) = 0\mathbf{u}_1 + (-1)\mathbf{u}_2$ . So

$$[T]_{\mathcal{B}} = \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix}.$$

In the basis adapted to the geometry, the reflection is visibly “keep one direction, flip the other”. The same map has a diagonal matrix in one basis and a non-diagonal one in another. How the two matrices are related is the subject of the next section.

The matrix also works in reverse: from  $[T]_{\mathcal{B}}$  alone we can recover  $T$ . Take  $\mathbf{v} = (3, 1)$ . Its coordinates in  $\mathcal{B}$  solve  $a\mathbf{u}_1 + b\mathbf{u}_2 = (3, 1)$ , that is,  $a + b = 3$  and  $a - b = 1$ , so  $[\mathbf{v}]_{\mathcal{B}} = (2, 1)$ , that is,  $\mathbf{v} = 2\mathbf{u}_1 + \mathbf{u}_2$ . By linearity,  $T\mathbf{v} = 2T\mathbf{u}_1 + T\mathbf{u}_2$ . The columns of  $[T]_{\mathcal{B}}$  say  $T\mathbf{u}_1 = 1\mathbf{u}_1 + 0\mathbf{u}_2$  and  $T\mathbf{u}_2 = 0\mathbf{u}_1 + (-1)\mathbf{u}_2$ , so  $T\mathbf{v} = 2\mathbf{u}_1 - \mathbf{u}_2 = (1, 3)$ , which is indeed  $(y, x)$  at  $(3, 1)$ . In coordinates, we turned  $(2, 1)$  into  $(2, -1) = [T]_{\mathcal{B}} \begin{pmatrix} 2 \\ 1 \end{pmatrix}$ ; the next subsection proves that this “coordinates in, multiply, combine out” recipe always works.

Here is a non-example by minimal change. In [Example \(Differentiation on Cubic Polynomials\)](#), write the coordinate vectors  $[D(x^j)]_{\mathcal{E}_2}$  as **rows** instead of columns. The numbers are the same,

but the result is the  $4 \times 3$  matrix  $\begin{pmatrix} 0 & 0 & 0 \\ 1 & 0 & 0 \\ 0 & 2 & 0 \\ 0 & 0 & 3 \end{pmatrix}$ , and it cannot even be multiplied by the coordinate vector  $(a_0, a_1, a_2, a_3)$  of an input polynomial. The clause that fails is “the  $j$ -th **column** is  $[T\mathbf{v}_j]_{\mathcal{C}}$ ”.

**Why this definition.** Why columns, and why coordinates in  $\mathcal{C}$ ? Because these are the choices under which “apply  $T$ ” becomes “multiply by the matrix”, as the  $D$  computation above suggested; we prove this next. With rows, we would get the transpose, and the natural formula  $[T\mathbf{v}]_{\mathcal{C}} = \mathbf{A}[\mathbf{v}]_{\mathcal{B}}$  would fail. The convention that the input basis is the subscript is chosen to make that formula easy to read: the  $\mathcal{B}$  of the matrix sits next to the  $\mathcal{B}$  of the vector.

*Warning. Three ways to get a matrix of a map wrong.* (1) **Columns, not rows.**

The  $j$ -th **column** is  $[T\mathbf{v}_j]_{\mathcal{C}}$ , as in the non-example above. (2) **Order matters.** In

[Example \(A Reflection in Two Bases\)](#), reordering the standard basis as  $(\mathbf{e}_2, \mathbf{e}_1)$  leaves

$[T] = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}$  unchanged, but reordering  $\mathcal{B}$  as  $(\mathbf{u}_2, \mathbf{u}_1)$  gives  $\begin{pmatrix} -1 & 0 \\ 0 & 1 \end{pmatrix} \neq [T]_{\mathcal{B}}$ . (3)

**Input subscript, output superscript.**  $[T]_{\mathcal{B}}^{\mathcal{C}}$  and  $[T]_{\mathcal{C}}^{\mathcal{B}}$  are different objects, and for  $T: V \rightarrow W$  with  $V \neq W$  only one of them makes sense. Also  $[\text{id}_V]_{\mathcal{B}}^{\mathcal{C}} \neq \mathbf{I}_n$  when  $\mathcal{B} \neq \mathcal{C}$ :

for  $\mathcal{B} = (\mathbf{e}_1, \mathbf{e}_2)$  and  $\mathcal{C} = (\mathbf{e}_2, \mathbf{e}_1)$  in  $\mathbb{R}^2$ ,  $[\text{id}]_{\mathcal{B}}^{\mathcal{C}} = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}$ .

*Quick check.* What is the size of  $[T]_{\mathcal{B}}^{\mathcal{C}}$  for a linear map  $T: M_{2 \times 3}(\mathbb{R}) \rightarrow \mathbb{R}[x]_{\leq 2}$ ? And what is the first column of  $[T]_{\mathcal{E}'}^{\mathcal{E}}$  for  $T: \mathbb{R}[x]_{\leq 2} \rightarrow \mathbb{R}^2$ ,  $T(p) = (p(0), p(1))$ , with  $\mathcal{E} = (1, x, x^2)$  and  $\mathcal{E}'$  the standard basis of  $\mathbb{R}^2$ ?

*Solution.*  $\dim \mathbb{R}[x]_{\leq 2} = 3$  and  $\dim M_{2 \times 3}(\mathbb{R}) = 6$ , so the matrix is  $3 \times 6$ : rows for the codomain, columns for the domain. For the second map, the first column is the coordinate vector of the image of the first basis vector 1:  $T(1) = (1, 1)$ , so the column is  $(1, 1)$ .

## 3.6.2 The fundamental square

Here is the promise behind the definition: once coordinates are taken on both sides, applying  $T$  is multiplying by its matrix.

### Theorem 3.6.1 — Coordinates of an Image

Let  $V$  and  $W$  be finite-dimensional vector spaces over  $F$  with ordered bases  $\mathcal{B} = (\mathbf{v}_1, \dots, \mathbf{v}_n)$  and  $\mathcal{C} = (\mathbf{w}_1, \dots, \mathbf{w}_m)$ , and let  $T \in \mathcal{L}(V, W)$ . Then for every  $\mathbf{v} \in V$ ,

$$[T\mathbf{v}]_{\mathcal{C}} = [T]_{\mathcal{B}}^{\mathcal{C}}[\mathbf{v}]_{\mathcal{B}}.$$

Moreover,  $[T]_{\mathcal{B}}^{\mathcal{C}}$  is the **only** matrix  $\mathbf{A} \in M_{m \times n}(F)$  with  $[T\mathbf{v}]_{\mathcal{C}} = \mathbf{A}[\mathbf{v}]_{\mathcal{B}}$  for all  $\mathbf{v} \in V$ .

The statement says that the following square **commutes**: starting from  $\mathbf{v}$  in the top-left corner, the two routes to the bottom-right corner give the same column.



$$\begin{array}{ccc}
\mathbf{v} & & T\mathbf{v} \\
V & \xrightarrow{T} & W \\
\downarrow \mathbf{v} \mapsto [\mathbf{v}]_{\mathcal{B}} & & \downarrow \mathbf{w} \mapsto [\mathbf{w}]_{\mathcal{C}} \\
F^n & \xrightarrow{\text{multiply by } \begin{bmatrix} [T]_{\mathcal{B}}^{\mathcal{C}} \\ [T\mathbf{v}]_{\mathcal{C}} \end{bmatrix}} & F^m \\
[\mathbf{v}]_{\mathcal{B}} & & [T\mathbf{v}]_{\mathcal{C}} = [T]_{\mathcal{B}}^{\mathcal{C}}[\mathbf{v}]_{\mathcal{B}}
\end{array}$$

The top row is the abstract world, where  $T$  acts on vectors of  $V$ . The bottom row is the matrix world, where a matrix acts on columns. The vertical arrows are the coordinate isomorphisms of [Corollary 3.5.5](#), so nothing is lost going down or coming back up. **Right then down** (apply  $T$ , then take coordinates) equals **down then right** (take coordinates, then multiply by the matrix).

*Idea.* Both sides are linear in  $\mathbf{v}$ , so it is enough to check the formula on the basis vectors  $\mathbf{v}_j$ , and there it is the definition:  $[\mathbf{v}_j]_{\mathcal{B}} = \mathbf{e}_j$ , and multiplying a matrix by  $\mathbf{e}_j$  picks out its  $j$ -th column, which was defined to be  $[T\mathbf{v}_j]_{\mathcal{C}}$ . Written out for a general  $\mathbf{v}$ , this becomes a one-line chain. Uniqueness uses the same observation backwards: any  $\mathbf{A}$  that works must have  $[T\mathbf{v}_j]_{\mathcal{C}}$  as its  $j$ -th column.

*Proof.* Let  $\mathbf{v} \in V$  and  $[\mathbf{v}]_{\mathcal{B}} = (x_1, \dots, x_n)$ , so that  $\mathbf{v} = x_1\mathbf{v}_1 + \dots + x_n\mathbf{v}_n$ . Since  $T$  is linear, [Theorem 3.1.4](#) gives  $T\mathbf{v} = x_1T\mathbf{v}_1 + \dots + x_nT\mathbf{v}_n$ . Since taking coordinates respects sums and scalar multiples ([Theorem 1.5.2](#)),

$$[T\mathbf{v}]_{\mathcal{C}} = x_1[T\mathbf{v}_1]_{\mathcal{C}} + \dots + x_n[T\mathbf{v}_n]_{\mathcal{C}} = [T]_{\mathcal{B}}^{\mathcal{C}} \begin{pmatrix} x_1 \\ \vdots \\ x_n \end{pmatrix} = [T]_{\mathcal{B}}^{\mathcal{C}}[\mathbf{v}]_{\mathcal{B}},$$

where the middle equality is [Theorem 0.9.3](#) applied to the columns  $[T\mathbf{v}_j]_{\mathcal{C}}$  of  $[T]_{\mathcal{B}}^{\mathcal{C}}$ .

For uniqueness, suppose  $\mathbf{A} \in M_{m \times n}(F)$  satisfies  $[T\mathbf{v}]_{\mathcal{C}} = \mathbf{A}[\mathbf{v}]_{\mathcal{B}}$  for all  $\mathbf{v}$ . Taking  $\mathbf{v} = \mathbf{v}_j$ , whose coordinate vector is  $\mathbf{e}_j$ , the  $j$ -th column of  $\mathbf{A}$  is  $\mathbf{A}\mathbf{e}_j = [T\mathbf{v}_j]_{\mathcal{C}}$  ([Theorem 0.9.3](#)), which is the  $j$ -th column of  $[T]_{\mathcal{B}}^{\mathcal{C}}$ . Hence  $\mathbf{A} = [T]_{\mathcal{B}}^{\mathcal{C}}$ . This proves the theorem. ■

This is the dictionary that lets us “write down a matrix first”. A question about  $T$  (its kernel, its image, a preimage) becomes a question about a matrix, which Chapter 2 taught us to answer by row reduction; the coordinate isomorphisms carry the answer back.

### Example. (Kernel and Image Through a Matrix)

Let  $T: \mathbb{R}[x]_{\leq 2} \rightarrow \mathbb{R}^2$ ,  $T(p) = (p(1), p(-1))$ . Find  $\ker T$  and decide whether  $T$  is surjective, by working with the matrix of  $T$  in the bases  $\mathcal{E} = (1, x, x^2)$  and the standard basis  $\mathcal{E}'$  of  $\mathbb{R}^2$ .

*Solution.* Write down the matrix.  $T(1) = (1, 1)$ ,  $T(x) = (1, -1)$ ,  $T(x^2) = (1, 1)$ , so

$$\mathbf{A} = [T]_{\mathcal{E}}^{\mathcal{E}'} = \begin{pmatrix} 1 & 1 & 1 \\ 1 & -1 & 1 \end{pmatrix}.$$

*Solve in the matrix world.* Subtracting the first row from the second gives  $\begin{pmatrix} 1 & 1 & 1 \\ 0 & -2 & 0 \end{pmatrix}$ , so  $\mathbf{A}\mathbf{x} = \mathbf{0}$  means  $x_2 = 0$  and  $x_1 = -x_3$ :  $\text{nul}(\mathbf{A}) = \text{span}((-1, 0, 1))$ . Both rows are pivot rows, so  $\text{rank } \mathbf{A} = 2$ .

*Translate back.* By [Theorem 3.6.1](#),  $T(p) = \mathbf{0}$  exactly when  $\mathbf{A}[p]_{\mathcal{E}} = \mathbf{0}$  (the only vector with zero coordinates is  $\mathbf{0}$ ). The coordinate vector  $(-1, 0, 1)$  belongs to  $p = -1 + x^2$ . So

$$\ker T = \text{span}(x^2 - 1),$$

as a check confirms:  $x^2 - 1$  vanishes at 1 and  $-1$ . By [Theorem 3.3.1](#),  $\text{rank } T = 3 - 1 = 2 = \dim \mathbb{R}^2$ , so  $\text{im } T = \mathbb{R}^2$  by [Theorem 1.6.8](#), and  $T$  is surjective: any two values  $p(1)$ ,  $p(-1)$  can be prescribed.

### 3.6.3 Composition is matrix multiplication

In Chapter 0, §9 we asked why matrices are multiplied row-against-column. The answer given there was **substitution**. If  $\mathbf{y} = \mathbf{B}\mathbf{x}$  and  $\mathbf{z} = \mathbf{A}\mathbf{y}$ , with

$$\mathbf{B} = \begin{pmatrix} 2 & 1 \\ 1 & 4 \end{pmatrix}, \quad \mathbf{A} = \begin{pmatrix} 1 & 2 \\ 3 & -1 \end{pmatrix},$$

then substituting gives  $\mathbf{z} = \begin{pmatrix} 4 & 9 \\ 5 & -1 \end{pmatrix} \mathbf{x}$ , and the table of the combined substitution turned out to be  $\mathbf{AB}$ . At the time we promised that this fact would become a statement about linear maps. We can now say what it is. Each substitution is a linear map,  $T_{\mathbf{B}}: \mathbf{x} \mapsto \mathbf{y}$  and  $T_{\mathbf{A}}: \mathbf{y} \mapsto \mathbf{z}$ , and substituting one into the other is **composing** them:  $\mathbf{z} = T_{\mathbf{A}}(T_{\mathbf{B}}\mathbf{x}) = (T_{\mathbf{A}}T_{\mathbf{B}})\mathbf{x}$ . So the Chapter 0 computation says  $T_{\mathbf{A}}T_{\mathbf{B}} = T_{\mathbf{AB}}$ . Here is the general statement, for arbitrary spaces and bases.

#### Theorem 3.6.2 — Matrix of a Composition

Let  $U, V, W$  be finite-dimensional vector spaces over  $F$  with ordered bases  $\mathcal{B}$ ,  $\mathcal{C}$ ,  $\mathcal{D}$  respectively. Let  $T \in \mathcal{L}(U, V)$  and  $S \in \mathcal{L}(V, W)$ . Then

$$[ST]_{\mathcal{B}}^{\mathcal{D}} = [S]_{\mathcal{C}}^{\mathcal{D}} [T]_{\mathcal{B}}^{\mathcal{C}}.$$

Notice how the notation checks itself: the middle basis  $\mathcal{C}$  appears as the output of  $T$  and the input of  $S$ , and it “cancels”, leaving  $\mathcal{B}$  below and  $\mathcal{D}$  above. The order of the factors is the order of the maps: the map applied first is on the right, in both  $ST$  and the product.

*Idea.* Glue two commutative squares side by side. The left square is  $T$  with its matrix, the right square is  $S$  with its matrix, and they share the middle column  $V \rightarrow F^{\dim V}$ . Going along the top from  $U$  to  $W$  is  $ST$ ; going along the bottom is multiplication by  $[T]_{\mathcal{B}}^{\mathcal{C}}$  followed by multiplication by  $[S]_{\mathcal{C}}^{\mathcal{D}}$ , which is multiplication by their product. The big rectangle commutes because both small squares do, and the uniqueness part of [Theorem 3.6.1](#) identifies the product as the matrix of  $ST$ .

$$\begin{array}{ccccc}
 U & \xrightarrow{T} & V & \xrightarrow{S} & W \\
 \downarrow [\cdot]_{\mathcal{B}} & & \downarrow [\cdot]_{\mathcal{C}} & & \downarrow [\cdot]_{\mathcal{D}} \\
 F^p & \xrightarrow{[T]_{\mathcal{B}}^{\mathcal{C}}} & F^m & \xrightarrow{[S]_{\mathcal{C}}^{\mathcal{D}}} & F^m
 \end{array}$$

*Proof.* Let  $p = \dim U$  and  $m = \dim W$ , and let  $\mathbf{u} \in U$ . Applying [Theorem 3.6.1](#) first to  $S$  and the vector  $T\mathbf{u}$ , then to  $T$  and  $\mathbf{u}$ ,

$$[ST\mathbf{u}]_{\mathcal{D}} = [S(T\mathbf{u})]_{\mathcal{D}} = [S]_{\mathcal{C}}^{\mathcal{D}}[T\mathbf{u}]_{\mathcal{C}} = [S]_{\mathcal{C}}^{\mathcal{D}}([T]_{\mathcal{B}}^{\mathcal{C}}[\mathbf{u}]_{\mathcal{B}}) = ([S]_{\mathcal{C}}^{\mathcal{D}}[T]_{\mathcal{B}}^{\mathcal{C}})[\mathbf{u}]_{\mathcal{B}},$$

where the last equality is associativity of matrix multiplication ([Theorem 0.9.4](#)). The map  $ST$  is linear ([Theorem 3.4.2](#)), and the  $m \times p$  matrix  $[S]_{\mathcal{C}}^{\mathcal{D}}[T]_{\mathcal{B}}^{\mathcal{C}}$  satisfies  $[ST\mathbf{u}]_{\mathcal{D}} = ([S]_{\mathcal{C}}^{\mathcal{D}}[T]_{\mathcal{B}}^{\mathcal{C}})[\mathbf{u}]_{\mathcal{B}}$  for every  $\mathbf{u} \in U$ . By the uniqueness part of [Theorem 3.6.1](#), it equals  $[ST]_{\mathcal{B}}^{\mathcal{D}}$ . This proves the theorem. ■

This is the payoff of the hook in Chapter 0, §9 ([Definition 0.9.9](#)). The row-times-column rule was not an arbitrary convention: it is the **only** rule under which the matrix of a composition is the product of the matrices, because composing linear maps is substituting one coordinate formula into another. Everything Chapter 0 proved about matrix multiplication now has a meaning for maps, and conversely:

- **associativity** of matrix multiplication is associativity of composition ([Theorem 0.4.2](#));
- $\mathbf{AB} \neq \mathbf{BA}$  in general is  $ST \neq TS$  ([Example \(Rotate, Then Project\)](#));
- the size condition “columns of  $\mathbf{A}$  = rows of  $\mathbf{B}$ ” is “the codomain of the first map is the domain of the second”;
- in particular  $T_{\mathbf{A}}T_{\mathbf{B}} = T_{\mathbf{AB}}$ : apply the theorem with standard bases and [Example \(Matrix Maps and the Identity\)](#) (a).

### Example. (Multiply by $x$ , Then Differentiate)

Let  $S: \mathbb{R}[x]_{\leq 2} \rightarrow \mathbb{R}[x]_{\leq 3}$ ,  $S(p) = xp$ , and  $D: \mathbb{R}[x]_{\leq 3} \rightarrow \mathbb{R}[x]_{\leq 2}$ ,  $D(p) = p'$ , with the monomial bases  $\mathcal{E}_2 = (1, x, x^2)$  and  $\mathcal{E}_3 = (1, x, x^2, x^3)$ . Compute  $[DS]_{\mathcal{E}_2}^{\mathcal{E}_3}$  and  $[SD]_{\mathcal{E}_3}^{\mathcal{E}_2}$ , both directly and as products.

*Solution. Matrices of the factors.*  $S(1) = x$ ,  $S(x) = x^2$ ,  $S(x^2) = x^3$ , and  $[D]_{\mathcal{E}_3}^{\mathcal{E}_2}$  was found in [Example \(Differentiation on Cubic Polynomials\)](#):

$$[S]_{\mathcal{E}_3}^{\mathcal{E}_2} = \begin{pmatrix} 0 & 0 & 0 \\ 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{pmatrix}, \quad [D]_{\mathcal{E}_3}^{\mathcal{E}_2} = \begin{pmatrix} 0 & 1 & 0 & 0 \\ 0 & 0 & 2 & 0 \\ 0 & 0 & 0 & 3 \end{pmatrix}.$$

*Products.* By [Theorem 3.6.2](#),

$$[DS]_{\mathcal{E}_2}^{\mathcal{E}_2} = [D]_{\mathcal{E}_3}^{\mathcal{E}_2} [S]_{\mathcal{E}_2}^{\mathcal{E}_3} = \begin{pmatrix} 1 & 0 & 0 \\ 0 & 2 & 0 \\ 0 & 0 & 3 \end{pmatrix}, \quad [SD]_{\mathcal{E}_3}^{\mathcal{E}_3} = [S]_{\mathcal{E}_2}^{\mathcal{E}_3} [D]_{\mathcal{E}_3}^{\mathcal{E}_2} = \begin{pmatrix} 0 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 2 & 0 \\ 0 & 0 & 0 & 3 \end{pmatrix}.$$

*Directly.*  $DS(x^k) = (x^{k+1})' = (k+1)x^k$  for  $k = 0, 1, 2$ , giving the diagonal  $1, 2, 3$ . And  $SD(x^k) = x \cdot kx^{k-1} = kx^k$  for  $k \geq 1$ , with  $SD(1) = 0$ , giving the diagonal  $0, 1, 2, 3$ . Both computations agree. The two composites are not even the same size:  $DS$  is an operator on  $\mathbb{R}[x]_{\leq 2}$  and  $SD$  an operator on  $\mathbb{R}[x]_{\leq 3}$ .

For operators, the theorem applies with  $\mathcal{B} = \mathcal{C} = \mathcal{D}$ , and gives powers and polynomials.

### Corollary 3.6.3 — Matrices of Powers and Polynomials

Let  $\mathcal{B} = (\mathbf{v}_1, \dots, \mathbf{v}_n)$  be a basis of a finite-dimensional space  $V$ , let  $T, S \in \mathcal{L}(V)$ ,  $c \in F$  and  $p \in F[x]$ . Then  $[ST]_{\mathcal{B}} = [S]_{\mathcal{B}}[T]_{\mathcal{B}}$ ,  $[S+T]_{\mathcal{B}} = [S]_{\mathcal{B}} + [T]_{\mathcal{B}}$ ,  $[cT]_{\mathcal{B}} = c[T]_{\mathcal{B}}$ ,  $[T^k]_{\mathcal{B}} = ([T]_{\mathcal{B}})^k$  for  $k \in \mathbb{N}$ , and

$$[p(T)]_{\mathcal{B}} = p([T]_{\mathcal{B}}).$$

*Proof.* The first formula is [Theorem 3.6.2](#) with  $\mathcal{B} = \mathcal{C} = \mathcal{D}$ . For sums and scalar multiples, the  $j$ -th column of  $[S+T]_{\mathcal{B}}$  is  $[S\mathbf{v}_j + T\mathbf{v}_j]_{\mathcal{B}} = [S\mathbf{v}_j]_{\mathcal{B}} + [T\mathbf{v}_j]_{\mathcal{B}}$  by [Theorem 1.5.2](#), and similarly  $[cT\mathbf{v}_j]_{\mathcal{B}} = c[T\mathbf{v}_j]_{\mathcal{B}}$ . For powers we induct on  $k$ :  $[T^0]_{\mathcal{B}} = [\text{id}_V]_{\mathcal{B}} = \mathbf{I}_n$  by [Example \(Matrix Maps and the Identity\)](#) (b), and if  $[T^k]_{\mathcal{B}} = ([T]_{\mathcal{B}})^k$ , then  $[T^{k+1}]_{\mathcal{B}} = [T^k T]_{\mathcal{B}} = ([T]_{\mathcal{B}})^k [T]_{\mathcal{B}} = ([T]_{\mathcal{B}})^{k+1}$ , by the first formula and [Definition 0.9.14](#). Finally, for  $p = \sum_k a_k x^k$ , combining the three rules gives  $[p(T)]_{\mathcal{B}} = \sum_k a_k [T^k]_{\mathcal{B}} = \sum_k a_k ([T]_{\mathcal{B}})^k = p([T]_{\mathcal{B}})$ , using [Definition 3.4.2](#) and [Definition 0.9.14](#). ■

For example, with  $[D]_{\mathcal{E}_3}$  from [Example \(Differentiation on Cubic Polynomials\)](#),  $([D]_{\mathcal{E}_3})^4 = [D^4]_{\mathcal{E}_3} = 0$ , recovering [Example \(Differentiation Is Nilpotent\)](#) in matrix form.

## 3.6.4 Linear maps are matrices

The sums in [Corollary 3.6.3](#) work between different spaces too, and together with the construction of maps from values on a basis they show that, after choosing bases, there is no difference at all between linear maps and matrices.

### Theorem 3.6.4 — $\mathcal{L}(V, W) \cong M_{m \times n}(F)$

Let  $V$  and  $W$  be finite-dimensional vector spaces over  $F$  with ordered bases  $\mathcal{B} = (\mathbf{v}_1, \dots, \mathbf{v}_n)$  and  $\mathcal{C} = (\mathbf{w}_1, \dots, \mathbf{w}_m)$ . Then the map

$$\Phi: \mathcal{L}(V, W) \rightarrow M_{m \times n}(F), \quad T \mapsto [T]_{\mathcal{C}}^{\mathcal{B}},$$

is an isomorphism. In particular,  $\dim \mathcal{L}(V, W) = mn = (\dim V)(\dim W)$ .

*Idea.* Three things to check. *Linear:* columns of  $[S + T]_{\mathcal{B}}^{\mathcal{C}}$  are coordinates of  $S\mathbf{v}_j + T\mathbf{v}_j$ , and coordinates respect sums. *Injective:* a map whose matrix is zero kills every basis vector, so it is the zero map. *Surjective:* a matrix prescribes the value of each  $\mathbf{v}_j$ , and a linear map with prescribed values on a basis exists. The last two are the uniqueness and existence halves of the same theorem about bases, which is why a linear map is “nothing but” its values on a basis.

*Proof. Linear.* Let  $S, T \in \mathcal{L}(V, W)$  and  $c \in F$ . By Definition 3.4.1 and Theorem 1.5.2, the  $j$ -th column of  $[S + T]_{\mathcal{B}}^{\mathcal{C}}$  is  $[S\mathbf{v}_j + T\mathbf{v}_j]_{\mathcal{C}} = [S\mathbf{v}_j]_{\mathcal{C}} + [T\mathbf{v}_j]_{\mathcal{C}}$ , and that of  $[cT]_{\mathcal{B}}^{\mathcal{C}}$  is  $[cT\mathbf{v}_j]_{\mathcal{C}} = c[T\mathbf{v}_j]_{\mathcal{C}}$ . Hence  $\Phi(S + T) = \Phi(S) + \Phi(T)$  and  $\Phi(cT) = c\Phi(T)$ .

*Injective.* Suppose  $\Phi(T) = 0$ . Then  $[T\mathbf{v}_j]_{\mathcal{C}} = \mathbf{0}$ , so  $T\mathbf{v}_j = \mathbf{0}$  for every  $j$ . The zero map also sends every  $\mathbf{v}_j$  to  $\mathbf{0}$ , so  $T = 0$  by the uniqueness part of Theorem 3.1.5. Hence  $\ker \Phi = \{0\}$ , and  $\Phi$  is injective by Theorem 3.2.3.

*Surjective.* Let  $\mathbf{A} = (a_{ij}) \in M_{m \times n}(F)$ . By the existence part of Theorem 3.1.5 there is  $T \in \mathcal{L}(V, W)$  with  $T\mathbf{v}_j = a_{1j}\mathbf{w}_1 + \cdots + a_{mj}\mathbf{w}_m$  for each  $j$ . By Definition 3.6.1,  $[T]_{\mathcal{B}}^{\mathcal{C}} = \mathbf{A}$ .

So  $\Phi$  is a bijective linear map, an isomorphism by Theorem 3.5.1. For the dimension, we cannot quote Theorem 3.5.4 directly, since we do not yet know that  $\mathcal{L}(V, W)$  is finite-dimensional. Instead,  $M_{m \times n}(F)$  has a basis of  $mn$  matrix units (Example (Dimensions of the Running Examples)), and the inverse isomorphism  $\Phi^{-1}$  carries it to a basis of  $\mathcal{L}(V, W)$  of length  $mn$  (Theorem 3.5.2, applied to  $\Phi^{-1}$ , which is an isomorphism by Theorem 3.5.3). Hence  $\dim \mathcal{L}(V, W) = mn$ . This proves the theorem. ■

With Theorem 3.6.2, the isomorphism  $\Phi$  also turns composition into multiplication. So, for  $V = W$  and  $\mathcal{B} = \mathcal{C}$ , the algebra  $\mathcal{L}(V)$  of the previous sections and the algebra  $M_n(F)$  of Chapter 0 are the same algebra, written in two languages. The dictionary depends on the basis, as every isomorphism to  $F^n$  does.

## 3.6.5 Reading rank and invertibility from a matrix

Chapter 2 developed tools for the rank and the invertibility of matrices. The square of Theorem 3.6.1 transfers all of them to linear maps.

### Theorem 3.6.5 — Rank and Invertibility from the Matrix

Let  $V$  and  $W$  be finite-dimensional vector spaces over  $F$  with ordered bases  $\mathcal{B}$  and  $\mathcal{C}$ , let  $T \in \mathcal{L}(V, W)$ , and let  $\mathbf{A} = [T]_{\mathcal{B}}^{\mathcal{C}}$ .

- (a) The coordinate map  $\mathbf{v} \mapsto [\mathbf{v}]_{\mathcal{B}}$  restricts to an isomorphism  $\ker T \rightarrow \text{nul}(\mathbf{A})$ , and the coordinate map  $\mathbf{w} \mapsto [\mathbf{w}]_{\mathcal{C}}$  restricts to an isomorphism  $\text{im } T \rightarrow \text{col}(\mathbf{A})$ . Hence

$$\text{rank } T = \text{rank } \mathbf{A} \quad \text{and} \quad \text{nullity } T = \text{nullity } \mathbf{A}.$$

- (b) Suppose  $\dim V = \dim W$ . Then  $T$  is invertible if and only if  $\mathbf{A}$  is invertible, and

in that case  $[T^{-1}]_{\mathcal{C}}^{\mathcal{B}} = \mathbf{A}^{-1}$ .

*Idea.* For (a), the commutative square says a vector is killed by  $T$  exactly when its coordinate column is killed by  $\mathbf{A}$ , and the outputs of  $T$ , in coordinates, are exactly the outputs of  $\mathbf{A}$ . Isomorphic subspaces have the same dimension. For (b),  $T^{-1}T = \text{id}_V$  translates, by the composition theorem, into a matrix equation with  $\mathbf{I}_n$  on the right; conversely an inverse matrix is the matrix of **some** map, by the isomorphism  $\mathcal{L}(W, V) \cong M_n(F)$ , and that map is the inverse of  $T$ .

*Proof.* Let  $n = \dim V$  and  $m = \dim W$ , and write  $C_{\mathcal{B}}: V \rightarrow F^n$  and  $C_{\mathcal{C}}: W \rightarrow F^m$  for the coordinate isomorphisms (Corollary 3.5.5).

- (a) *Kernel.* For  $\mathbf{v} \in V$ :  $\mathbf{v} \in \ker T$  iff  $T\mathbf{v} = \mathbf{0}$  iff  $[T\mathbf{v}]_{\mathcal{C}} = \mathbf{0}$  (since  $C_{\mathcal{C}}$  is injective) iff  $\mathbf{A}[\mathbf{v}]_{\mathcal{B}} = \mathbf{0}$  (Theorem 3.6.1) iff  $[\mathbf{v}]_{\mathcal{B}} \in \text{nul}(\mathbf{A})$ . So  $C_{\mathcal{B}}$  maps  $\ker T$  into  $\text{nul}(\mathbf{A})$ , and every  $\mathbf{x} \in \text{nul}(\mathbf{A})$  is  $[\mathbf{v}]_{\mathcal{B}}$  for some  $\mathbf{v} \in V$  (surjectivity of  $C_{\mathcal{B}}$ ), which then lies in  $\ker T$ . Thus the restriction of  $C_{\mathcal{B}}$  is a linear map  $\ker T \rightarrow \text{nul}(\mathbf{A})$ , injective because  $C_{\mathcal{B}}$  is, and surjective. By Theorem 3.5.1 it is an isomorphism.

*Image.* By Theorem 3.6.1,  $C_{\mathcal{C}}(T\mathbf{v}) = \mathbf{A}C_{\mathcal{B}}(\mathbf{v})$  for every  $\mathbf{v} \in V$ . As  $\mathbf{v}$  runs over  $V$ ,  $C_{\mathcal{B}}(\mathbf{v})$  runs over all of  $F^n$ , so  $\{C_{\mathcal{C}}(\mathbf{w}) : \mathbf{w} \in \text{im } T\} = \{\mathbf{A}\mathbf{x} : \mathbf{x} \in F^n\} = \text{col}(\mathbf{A})$  (Definition 2.5.1). So the restriction of  $C_{\mathcal{C}}$  is a linear map  $\text{im } T \rightarrow \text{col}(\mathbf{A})$ , surjective by this computation and injective because  $C_{\mathcal{C}}$  is; hence an isomorphism.

The subspaces involved are finite-dimensional (Theorem 1.6.7), so Theorem 3.5.4 gives  $\dim \ker T = \dim \text{nul}(\mathbf{A})$  and  $\dim \text{im } T = \dim \text{col}(\mathbf{A})$ , which are the stated equalities by Definition 3.3.1, Definition 3.3.2, Definition 2.5.5 and Definition 2.5.4.

- (b) ( $\Rightarrow$ ) Suppose  $T$  is invertible. By Theorem 3.6.2 and Example (Matrix Maps and the Identity) (b),

$$[T^{-1}]_{\mathcal{C}}^{\mathcal{B}} \mathbf{A} = [T^{-1}T]_{\mathcal{B}}^{\mathcal{B}} = [\text{id}_V]_{\mathcal{B}} = \mathbf{I}_n, \quad \mathbf{A} [T^{-1}]_{\mathcal{C}}^{\mathcal{B}} = [TT^{-1}]_{\mathcal{C}}^{\mathcal{C}} = [\text{id}_W]_{\mathcal{C}} = \mathbf{I}_n.$$

So  $\mathbf{A}$  is invertible with  $\mathbf{A}^{-1} = [T^{-1}]_{\mathcal{C}}^{\mathcal{B}}$  (Definition 0.9.13).

( $\Leftarrow$ ) Suppose  $\mathbf{A}$  is invertible. By Theorem 3.6.4 (for  $\mathcal{L}(W, V)$  with bases  $\mathcal{C}$  and  $\mathcal{B}$ ), there is  $S \in \mathcal{L}(W, V)$  with  $[S]_{\mathcal{C}}^{\mathcal{B}} = \mathbf{A}^{-1}$ . Then by Theorem 3.6.2,  $[ST]_{\mathcal{B}} = \mathbf{A}^{-1}\mathbf{A} = \mathbf{I}_n = [\text{id}_V]_{\mathcal{B}}$ . Since  $T' \mapsto [T']_{\mathcal{B}}$  is injective on  $\mathcal{L}(V)$  (Theorem 3.6.4),  $ST = \text{id}_V$ . By Theorem 3.5.6,  $T$  is invertible with  $T^{-1} = S$ . This proves the theorem. ■

So we may add one more item to the Invertible Operator Theorem (Theorem 3.5.7): for  $T \in \mathcal{L}(V)$  and **any** basis  $\mathcal{B}$  of  $V$ ,  $T$  is invertible if and only if  $[T]_{\mathcal{B}}$  is invertible. Every item of the Invertible Matrix Theorem (Theorem 2.4.4) now becomes a test for operators.

**Example. (Rank of  $\mathbf{X} \mapsto \mathbf{X} + \mathbf{X}^{\top}$ )**

Let  $T: M_2(\mathbb{R}) \rightarrow M_2(\mathbb{R})$ ,  $T(\mathbf{X}) = \mathbf{X} + \mathbf{X}^{\top}$ , and  $\mathcal{E} = (\mathbf{E}_{11}, \mathbf{E}_{12}, \mathbf{E}_{21}, \mathbf{E}_{22})$ . Find  $[T]_{\mathcal{E}}$ , use it to find  $\text{rank } T$ , a basis of  $\ker T$  and a basis of  $\text{im } T$ , and decide whether  $T$  is invertible. Then do the same for the transpose map of Example (The Transpose Map).

*Solution. Matrix.*  $T(\mathbf{E}_{11}) = 2\mathbf{E}_{11}$ ,  $T(\mathbf{E}_{12}) = \mathbf{E}_{12} + \mathbf{E}_{21} = T(\mathbf{E}_{21})$ ,  $T(\mathbf{E}_{22}) = 2\mathbf{E}_{22}$ . So

$$\mathbf{A} = [T]_{\mathcal{E}} = \begin{pmatrix} 2 & 0 & 0 & 0 \\ 0 & 1 & 1 & 0 \\ 0 & 1 & 1 & 0 \\ 0 & 0 & 0 & 2 \end{pmatrix}.$$

*Matrix world.* Subtracting row 2 from row 3 turns row 3 into a zero row; moving it to the bottom gives an echelon form with three non-zero rows and pivots in columns 1, 2, 4. So  $\text{rank } \mathbf{A} = 3$  and  $\text{nullity } \mathbf{A} = 1$ . The free column is column 3, and  $\mathbf{A}\mathbf{x} = \mathbf{0}$  gives  $x_1 = x_4 = 0$ ,  $x_2 = -x_3$ , so  $\text{nul}(\mathbf{A}) = \text{span}((0, -1, 1, 0))$ . By Theorem 2.5.2, the pivot columns  $(2, 0, 0, 0)$ ,  $(0, 1, 1, 0)$ ,  $(0, 0, 0, 2)$  form a basis of  $\text{col}(\mathbf{A})$ .

*Translate back.* By Theorem 3.6.5,  $\text{rank } T = 3$ . The column  $(0, -1, 1, 0)$  is the coordinate vector of  $\mathbf{E}_{21} - \mathbf{E}_{12}$ , so  $\ker T = \text{span}\left(\begin{pmatrix} 0 & -1 \\ 1 & 0 \end{pmatrix}\right)$ , the skew-symmetric matrices. The basis of  $\text{col}(\mathbf{A})$  translates to  $2\mathbf{E}_{11}$ ,  $\mathbf{E}_{12} + \mathbf{E}_{21}$ ,  $2\mathbf{E}_{22}$ , a basis of  $\text{im } T$ , which is therefore the space of symmetric matrices. Since  $\text{rank } \mathbf{A} = 3 < 4$ ,  $\mathbf{A}$  is not invertible (Theorem 2.4.4), and neither is  $T$ .

*Transpose.* The matrix  $\mathbf{P} = [\mathbf{X} \mapsto \mathbf{X}^{\top}]_{\mathcal{E}}$  from Example (The Transpose Map) satisfies  $\mathbf{P}^2 = \mathbf{I}_4$  (swapping two coordinates twice does nothing). So  $\mathbf{P}$  is invertible with  $\mathbf{P}^{-1} = \mathbf{P}$ , and by Theorem 3.6.5 (b) the transpose map is invertible and is its own inverse, which is  $(\mathbf{X}^{\top})^{\top} = \mathbf{X}$  again.

We now have both halves of the working method: a basis turns spaces into  $F^n$  and maps into matrices, and every computation can be done with Chapter 2's tools. What we have not yet controlled is the choice. Example (A Reflection in Two Bases) shows that one map can have a complicated matrix in one basis and a diagonal one in another; the next section finds the formula relating the two.

## 3.6.6 Exercises

### 3.6.6.1 A. Check your understanding

#### Exercise A1.

- Define  $[T]_{\mathcal{B}}^{\mathcal{C}}$  for  $T \in \mathcal{L}(V, W)$  and ordered bases  $\mathcal{B}$  of  $V$ ,  $\mathcal{C}$  of  $W$ .
- State the formula relating  $[T\mathbf{v}]_{\mathcal{C}}$  and  $[\mathbf{v}]_{\mathcal{B}}$ , and the formula for the matrix of a composition.
- True or false: the  $j$ -th **row** of  $[T]_{\mathcal{B}}^{\mathcal{C}}$  is  $[T\mathbf{v}_j]_{\mathcal{C}}$ . Justify your answer.
- True or false:  $[\text{id}_V]_{\mathcal{B}}^{\mathcal{C}} = \mathbf{I}_n$  for any two bases  $\mathcal{B}, \mathcal{C}$  of an  $n$ -dimensional space  $V$ . Justify your answer.
- What is  $\dim \mathcal{L}(\mathbb{R}^3, \mathbb{R}[x]_{\leq 2})$ ?

*Solution.*

- If  $\mathcal{B} = (\mathbf{v}_1, \dots, \mathbf{v}_n)$ , then  $[T]_{\mathcal{B}}^{\mathcal{C}}$  is the  $\dim W \times n$  matrix whose  $j$ -th column is  $[T\mathbf{v}_j]_{\mathcal{C}}$  (Definition 3.6.1).

- (b)  $[T\mathbf{v}]_{\mathcal{C}} = [T]_{\mathcal{B}}^{\mathcal{C}}[\mathbf{v}]_{\mathcal{B}}$  ([Theorem 3.6.1](#)), and  $[ST]_{\mathcal{B}}^{\mathcal{D}} = [S]_{\mathcal{C}}^{\mathcal{D}}[T]_{\mathcal{B}}^{\mathcal{C}}$  ([Theorem 3.6.2](#)).
- (c) False. It is the  $j$ -th **column**. For  $D: \mathbb{R}[x]_{\leq 3} \rightarrow \mathbb{R}[x]_{\leq 2}$  in [Example \(Differentiation on Cubic Polynomials\)](#),  $[D(x)]_{\mathcal{E}_2} = (1, 0, 0)$  is the second column, while the second row is  $(0, 0, 2, 0)$ .
- (d) False. In  $\mathbb{R}^2$  with  $\mathcal{B} = (\mathbf{e}_1, \mathbf{e}_2)$  and  $\mathcal{C} = (\mathbf{e}_2, \mathbf{e}_1)$ ,  $[\mathbf{e}_1]_{\mathcal{C}} = (0, 1)$ , so the first column of  $[\text{id}]_{\mathcal{B}}^{\mathcal{C}}$  is  $(0, 1) \neq \mathbf{e}_1$ . It is true when  $\mathcal{B} = \mathcal{C}$  ([Example \(Matrix Maps and the Identity\)](#) (b)).
- (e)  $3 \cdot 3 = 9$ , by [Theorem 3.6.4](#).

### 3.6.6.2 B. Practice

#### Exercise B1: Matrices in given bases.

Find the matrix of each operator in the given basis. (Both maps are linear; you may take this as given.)

- (a)  $T: \mathbb{R}[x]_{\leq 2} \rightarrow \mathbb{R}[x]_{\leq 2}$ ,  $T(p) = p(x+1)$ , in  $\mathcal{E} = (1, x, x^2)$ .
- (b)  $T: M_2(\mathbb{R}) \rightarrow M_2(\mathbb{R})$ ,  $T(\mathbf{X}) = \mathbf{B}\mathbf{X}$  with  $\mathbf{B} = \begin{pmatrix} 1 & 2 \\ 3 & 4 \end{pmatrix}$ , in  $\mathcal{E} = (\mathbf{E}_{11}, \mathbf{E}_{12}, \mathbf{E}_{21}, \mathbf{E}_{22})$ .

*Solution.*

- (a)  $T(1) = 1$ ,  $T(x) = x+1 = 1+x$ ,  $T(x^2) = (x+1)^2 = 1+2x+x^2$ . Their coordinates are  $(1, 0, 0)$ ,  $(1, 1, 0)$ ,  $(1, 2, 1)$ , so

$$[T]_{\mathcal{E}} = \begin{pmatrix} 1 & 1 & 1 \\ 0 & 1 & 2 \\ 0 & 0 & 1 \end{pmatrix}.$$

- (b) Multiplying  $\mathbf{B}$  by a matrix unit  $\mathbf{E}_{kj}$  on the right keeps column  $k$  of  $\mathbf{B}$  and moves it to column  $j$ :

$$\mathbf{B}\mathbf{E}_{11} = \begin{pmatrix} 1 & 0 \\ 3 & 0 \end{pmatrix}, \quad \mathbf{B}\mathbf{E}_{12} = \begin{pmatrix} 0 & 1 \\ 0 & 3 \end{pmatrix}, \quad \mathbf{B}\mathbf{E}_{21} = \begin{pmatrix} 2 & 0 \\ 4 & 0 \end{pmatrix}, \quad \mathbf{B}\mathbf{E}_{22} = \begin{pmatrix} 0 & 2 \\ 0 & 4 \end{pmatrix}.$$

Reading entries in the order  $(1, 1), (1, 2), (2, 1), (2, 2)$  gives the columns  $(1, 0, 3, 0)$ ,  $(0, 1, 0, 3)$ ,  $(2, 0, 4, 0)$ ,  $(0, 2, 0, 4)$ , so

$$[T]_{\mathcal{E}} = \begin{pmatrix} 1 & 0 & 2 & 0 \\ 0 & 1 & 0 & 2 \\ 3 & 0 & 4 & 0 \\ 0 & 3 & 0 & 4 \end{pmatrix}.$$

#### Exercise B2: Recovering a map from its matrix.

Let  $\mathcal{B} = (1, 1+x, x^2)$ , a basis of  $\mathbb{R}[x]_{\leq 2}$ , and let  $\mathcal{E}'$  be the standard basis of  $\mathbb{R}^2$ . A linear



map  $T: \mathbb{R}[x]_{\leq 2} \rightarrow \mathbb{R}^2$  has

$$[T]_{\mathcal{B}}^{\mathcal{E}'} = \begin{pmatrix} 1 & 0 & 2 \\ 0 & 1 & -1 \end{pmatrix}.$$

Find  $T(a + bx + cx^2)$ . Hence find a basis of  $\ker T$ .

*Solution.* First find the coordinates of  $p = a + bx + cx^2$  in  $\mathcal{B}$ :  $p = (a-b) \cdot 1 + b(1+x) + cx^2$ , so  $[p]_{\mathcal{B}} = (a-b, b, c)$ , unique by [Theorem 1.5.1](#). By [Theorem 3.6.1](#), and since coordinates in  $\mathcal{E}'$  are the entries,

$$T(p) = \begin{pmatrix} 1 & 0 & 2 \\ 0 & 1 & -1 \end{pmatrix} \begin{pmatrix} a-b \\ b \\ c \end{pmatrix} = (a-b+2c, b-c).$$

Hence  $T(p) = \mathbf{0}$  iff  $b = c$  and  $a = b - 2c = -c$ , that is,  $p = c(-1 + x + x^2)$ . So  $(x^2 + x - 1)$  is a basis of  $\ker T$ . Check:  $[x^2 + x - 1]_{\mathcal{B}} = (-2, 1, 1)$ , and  $\begin{pmatrix} 1 & 0 & 2 \\ 0 & 1 & -1 \end{pmatrix} (-2, 1, 1) = (0, 0)$ .

### Exercise B3: A composition computed two ways.

Let  $T: \mathbb{R}[x]_{\leq 1} \rightarrow \mathbb{R}[x]_{\leq 2}$ ,  $T(p) = xp$ , and  $S: \mathbb{R}[x]_{\leq 2} \rightarrow \mathbb{R}^2$ ,  $S(p) = (p(0), p(1))$ . Use the monomial bases and the standard basis of  $\mathbb{R}^2$ . Compute the matrix of  $ST$  directly, and again as a product of the matrices of  $S$  and  $T$ .

*Solution.* Let  $\mathcal{E}_1 = (1, x)$ ,  $\mathcal{E}_2 = (1, x, x^2)$  and  $\mathcal{E}'$  the standard basis of  $\mathbb{R}^2$ .

*Factors.*  $T(1) = x$ ,  $T(x) = x^2$ , so  $[T]_{\mathcal{E}_2}^{\mathcal{E}_1} = \begin{pmatrix} 0 & 0 \\ 1 & 0 \\ 0 & 1 \end{pmatrix}$ .  $S(1) = (1, 1)$ ,  $S(x) = (0, 1)$ ,

$S(x^2) = (0, 1)$ , so  $[S]_{\mathcal{E}'}^{\mathcal{E}_2} = \begin{pmatrix} 1 & 0 & 0 \\ 1 & 1 & 1 \end{pmatrix}$ .

*Product.* By [Theorem 3.6.2](#),

$$[ST]_{\mathcal{E}'}^{\mathcal{E}_1} = \begin{pmatrix} 1 & 0 & 0 \\ 1 & 1 & 1 \end{pmatrix} \begin{pmatrix} 0 & 0 \\ 1 & 0 \\ 0 & 1 \end{pmatrix} = \begin{pmatrix} 0 & 0 \\ 1 & 1 \end{pmatrix}.$$

*Directly.*  $ST(1) = S(x) = (0, 1)$  and  $ST(x) = S(x^2) = (0, 1)$ , so the columns are  $(0, 1)$  and  $(0, 1)$ , in agreement. In words,  $ST(a + bx) = (0, a + b)$ .

### 3.6.6.3 C. Going deeper

#### Exercise C1: Upper triangular matrices and invariant subspaces.

Let  $V$  be finite-dimensional and  $T \in \mathcal{L}(V)$ . A subspace  $U$  of  $V$  is  **$T$ -invariant** if  $T\mathbf{u} \in U$  for every  $\mathbf{u} \in U$ . Let  $\mathcal{B} = (\mathbf{v}_1, \dots, \mathbf{v}_n)$  be a basis of  $V$ , and for  $1 \leq k \leq n$  let  $U_k = \text{span}(\mathbf{v}_1, \dots, \mathbf{v}_k)$ . Prove that  $[T]_{\mathcal{B}}$  is upper triangular if and only if every  $U_k$  is  $T$ -invariant.

*Solution.* Let  $[T]_{\mathcal{B}} = (a_{ij})$ , so  $T\mathbf{v}_j = \sum_{i=1}^n a_{ij}\mathbf{v}_i$  with unique coefficients (Definition 3.6.1). By Definition 0.9.5,  $[T]_{\mathcal{B}}$  is upper triangular iff  $a_{ij} = 0$  whenever  $i > j$ , that is, iff

$$T\mathbf{v}_j \in U_j \quad \text{for every } j = 1, \dots, n. \quad (*)$$

Indeed, if  $a_{ij} = 0$  for  $i > j$ , then  $T\mathbf{v}_j = \sum_{i \leq j} a_{ij}\mathbf{v}_i \in U_j$ ; conversely, if  $T\mathbf{v}_j = \sum_{i \leq j} b_i\mathbf{v}_i$ , then by uniqueness of coordinates  $a_{ij} = 0$  for  $i > j$ .

( $\Rightarrow$ ) Assume (\*), and fix  $k$ . For  $j \leq k$ ,  $T\mathbf{v}_j \in U_j \subseteq U_k$ . Let  $\mathbf{u} = \sum_{j \leq k} c_j\mathbf{v}_j \in U_k$ . Then  $T\mathbf{u} = \sum_{j \leq k} c_j T\mathbf{v}_j$  by Theorem 3.1.4, a combination of vectors of the subspace  $U_k$ , so  $T\mathbf{u} \in U_k$  by Theorem 1.3.1. Hence  $U_k$  is  $T$ -invariant.

( $\Leftarrow$ ) Assume every  $U_k$  is  $T$ -invariant. For each  $j$ ,  $\mathbf{v}_j \in U_j$ , so  $T\mathbf{v}_j \in U_j$  by invariance of  $U_j$ . This is (\*), so  $[T]_{\mathcal{B}}$  is upper triangular.

### Exercise C2: Integrating with a matrix.

Let  $f_1(x) = e^x$ ,  $f_2(x) = xe^x$ ,  $f_3(x) = x^2e^x$ , and let  $V = \text{span}(f_1, f_2, f_3)$  inside the real vector space of all functions  $\mathbb{R} \rightarrow \mathbb{R}$  (Example (The Standard Vector Spaces) (d)).

- Prove that  $\mathcal{B} = (f_1, f_2, f_3)$  is a basis of  $V$ .
- Show that differentiation maps  $V$  into  $V$ , and find  $[D]_{\mathcal{B}}$  for  $D \in \mathcal{L}(V)$ ,  $D(f) = f'$ .
- Hence show that  $D$  is invertible on  $V$ , and use  $([D]_{\mathcal{B}})^{-1}$  to find a function  $g \in V$  with  $g' = x^2e^x$ .

*Hint:* for (a), divide a vanishing combination by  $e^x$  and count roots.

*Solution.*

- $\mathcal{B}$  spans  $V$  by definition. Suppose  $af_1 + bf_2 + cf_3 = 0$ , the zero function. Then  $(a + bx + cx^2)e^x = 0$  for every  $x \in \mathbb{R}$ , and  $e^x \neq 0$ , so the polynomial  $a + bx + cx^2$  vanishes at every real number. It has infinitely many roots, so it is the zero polynomial by Lemma 2.7.2, and  $a = b = c = 0$ . Hence  $\mathcal{B}$  is independent, and a basis of  $V$ .
- By the product rule,

$$f'_1 = e^x = f_1, \quad f'_2 = e^x + xe^x = f_1 + f_2, \quad f'_3 = 2xe^x + x^2e^x = 2f_2 + f_3.$$

Each lies in  $V$ , and  $D$  is linear, so  $D$  maps every combination of  $f_1, f_2, f_3$  into  $V$  (Theorem 3.1.4). Reading off coordinates as columns,

$$[D]_{\mathcal{B}} = \begin{pmatrix} 1 & 1 & 0 \\ 0 & 1 & 2 \\ 0 & 0 & 1 \end{pmatrix}.$$

- The matrix is upper triangular with non-zero diagonal entries, and one checks directly that

$$\begin{pmatrix} 1 & -1 & 2 \\ 0 & 1 & -2 \\ 0 & 0 & 1 \end{pmatrix} \begin{pmatrix} 1 & 1 & 0 \\ 0 & 1 & 2 \\ 0 & 0 & 1 \end{pmatrix} = \mathbf{I}_3,$$

so  $[D]_{\mathcal{B}}$  is invertible with this inverse (Theorem 2.4.5). By Theorem 3.6.5 (b),  $D$  is invertible on  $V$  and  $[D^{-1}]_{\mathcal{B}} = ([D]_{\mathcal{B}})^{-1}$ . We want  $g = D^{-1}(f_3)$ . Since  $[f_3]_{\mathcal{B}} = (0, 0, 1)$ , Theorem 3.6.1 gives

$$[g]_{\mathcal{B}} = \begin{pmatrix} 1 & -1 & 2 \\ 0 & 1 & -2 \\ 0 & 0 & 1 \end{pmatrix} \begin{pmatrix} 0 \\ 0 \\ 1 \end{pmatrix} = \begin{pmatrix} 2 \\ -2 \\ 1 \end{pmatrix},$$

so  $g = 2f_1 - 2f_2 + f_3 = (x^2 - 2x + 2)e^x$ . Check:  $g' = (2x - 2)e^x + (x^2 - 2x + 2)e^x = x^2e^x$ . No integration by parts was needed.

## 3.7 Change of Basis and Similarity

The matrix  $[T]_{\mathcal{B}}^{\mathcal{C}}$  of a linear map depends on two choices of basis, and different choices can give very different matrices for the same map. This section answers two questions left open by that fact. How are two matrices of the same map related? And which properties of a matrix belong to the map itself, so that no choice of basis can change them? The answers are the change-of-basis formula, drawn as a commutative square, and the relation of similarity.

### 3.7.1 Change-of-coordinates matrices

Recall the definition of the matrix of a map (Definition 3.6.1). If  $\mathcal{B} = (\mathbf{v}_1, \dots, \mathbf{v}_n)$  is a basis of  $V$  and  $\mathcal{C}$  is a basis of  $W$ , then  $[T]_{\mathcal{B}}^{\mathcal{C}}$  is the matrix whose  $j$ -th column is  $[T\mathbf{v}_j]_{\mathcal{C}}$ , and it turns coordinates into coordinates:  $[T\mathbf{v}]_{\mathcal{C}} = [T]_{\mathcal{B}}^{\mathcal{C}}[\mathbf{v}]_{\mathcal{B}}$  (Theorem 3.6.1).

Recall the reflection from the previous section (Example (A Reflection in Two Bases)), which motivates this one. Let  $R: \mathbb{R}^2 \rightarrow \mathbb{R}^2$  be the reflection across the line  $y = x$ , so  $R(x, y) = (y, x)$ . In the standard basis  $\mathcal{E} = (\mathbf{e}_1, \mathbf{e}_2)$ , we have  $R\mathbf{e}_1 = \mathbf{e}_2$  and  $R\mathbf{e}_2 = \mathbf{e}_1$ , so

$$[R]_{\mathcal{E}} = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}.$$

Now use the basis  $\mathcal{B} = ((1, 1), (1, -1))$ , one vector on the mirror line and one perpendicular to it. Then  $R(1, 1) = (1, 1)$  and  $R(1, -1) = (-1, 1) = -(1, -1)$ , so

$$[R]_{\mathcal{B}} = \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix}.$$

The second matrix shows at a glance what  $R$  does: it keeps one direction and flips the other. The two matrices describe the same map, so there must be a rule converting one into the other. The first step is to convert the **coordinates** of a single vector.

Take  $\mathbf{v} = (3, 1)$ . Its standard coordinates are  $[\mathbf{v}]_{\mathcal{E}} = (3, 1)$ . In the basis  $\mathcal{B}$ , we solve  $a(1, 1) + b(1, -1) = (3, 1)$ , which gives  $a = 2$ ,  $b = 1$ , so  $[\mathbf{v}]_{\mathcal{B}} = (2, 1)$ . The same vector has two different coordinate columns, and we want a matrix that translates between them. Since the identity map  $\text{id}_V$  sends every vector to itself, the matrix of  $\text{id}_V$ , with one basis on the input side and the other on the output side, should do exactly this job.

*A change-of-coordinates matrix is the matrix of “do nothing”, read with one basis on the way in and another on the way out.*

**Definition 3.7.1 — Change-of-coordinates matrix**

Let  $V$  be a finite-dimensional vector space over  $F$ , and let  $\mathcal{B} = (\mathbf{v}_1, \dots, \mathbf{v}_n)$  and  $\mathcal{C}$  be **ordered** bases of  $V$ . The **change-of-coordinates matrix from  $\mathcal{B}$  to  $\mathcal{C}$**  is the  $n \times n$  matrix

$$[\text{id}]_{\mathcal{B}}^{\mathcal{C}} := [\text{id}_V]_{\mathcal{B}}^{\mathcal{C}},$$

the matrix of the identity map  $\text{id}_V: V \rightarrow V$  with respect to the input basis  $\mathcal{B}$  and the output basis  $\mathcal{C}$ . Its  $j$ -th column is  $[\mathbf{v}_j]_{\mathcal{C}}$ .

In words: to build  $[\text{id}]_{\mathcal{B}}^{\mathcal{C}}$ , take the vectors of the **old** basis  $\mathcal{B}$  one at a time, write each in the **new** basis  $\mathcal{C}$ , and put the coordinate columns side by side. The column description is [Definition 3.6.1](#) with  $T = \text{id}_V$ , since  $\text{id}_V(\mathbf{v}_j) = \mathbf{v}_j$ . Both bases must be bases of the **same** space  $V$ , and both are ordered: reordering either one permutes the columns or the rows.

**Examples.**

- **Into the standard basis of  $F^n$ .** If  $\mathcal{B} = (\mathbf{b}_1, \dots, \mathbf{b}_n)$  is a basis of  $F^n$  and  $\mathcal{E}$  is the standard basis, then  $[\mathbf{b}_j]_{\mathcal{E}} = \mathbf{b}_j$ . So  $[\text{id}]_{\mathcal{B}}^{\mathcal{E}}$  is the matrix whose columns are the vectors of  $\mathcal{B}$  themselves. For  $\mathcal{B} = ((1, 1), (1, -1))$  in  $\mathbb{R}^2$ ,  $[\text{id}]_{\mathcal{B}}^{\mathcal{E}} = \begin{pmatrix} 1 & 1 \\ 1 & -1 \end{pmatrix}$ , and indeed  $\begin{pmatrix} 1 & 1 \\ 1 & -1 \end{pmatrix} \begin{pmatrix} 2 \\ 1 \end{pmatrix} = \begin{pmatrix} 3 \\ 1 \end{pmatrix}$  turns  $[\mathbf{v}]_{\mathcal{B}}$  into  $[\mathbf{v}]_{\mathcal{E}}$  for  $\mathbf{v} = (3, 1)$ .
- **Out of the standard basis.** In the other direction,  $[\text{id}]_{\mathcal{E}}^{\mathcal{B}}$  has columns  $[\mathbf{e}_1]_{\mathcal{B}}$  and  $[\mathbf{e}_2]_{\mathcal{B}}$ . Solving  $a(1, 1) + b(1, -1) = (1, 0)$  gives  $a = b = \frac{1}{2}$ , and solving for  $(0, 1)$  gives  $a = \frac{1}{2}$ ,  $b = -\frac{1}{2}$ . So  $[\text{id}]_{\mathcal{E}}^{\mathcal{B}} = \frac{1}{2} \begin{pmatrix} 1 & 1 \\ 1 & -1 \end{pmatrix}$ . This is the inverse of the previous matrix, which the next theorem explains.
- **Polynomials.** In  $F[x]_{\leq 2}$ , let  $\mathcal{B} = (1, x, x^2)$  and  $\mathcal{C} = (1, x - 1, (x - 1)^2)$ . Expanding  $(x - 1)^2 = 1 - 2x + x^2$ , the vectors of  $\mathcal{C}$  have  $\mathcal{B}$ -coordinates  $(1, 0, 0)$ ,  $(-1, 1, 0)$  and  $(1, -2, 1)$ , so

$$[\text{id}]_{\mathcal{C}}^{\mathcal{B}} = \begin{pmatrix} 1 & -1 & 1 \\ 0 & 1 & -2 \\ 0 & 0 & 1 \end{pmatrix}.$$

- **Degenerate case.** If  $\mathcal{C} = \mathcal{B}$ , then  $[\mathbf{v}_j]_{\mathcal{B}} = \mathbf{e}_j$ , so  $[\text{id}]_{\mathcal{B}} = \mathbf{I}_n$ . Changing nothing is recorded by the identity matrix, as it should be.

**Non-example by minimal change.** Replace the basis  $((1, 1), (1, -1))$  by the list  $\mathcal{L} = ((1, 1), (2, 2))$ . We can still write the matrix with these vectors as columns, but  $\mathcal{L}$  is not a basis, so “coordinates with respect to  $\mathcal{L}$ ” are not defined:  $(2, 2) = 2(1, 1) + 0(2, 2) = 0(1, 1) + 1(2, 2)$  has two coordinate columns, and  $(1, 0)$  has none. The clause that fails is “ $\mathcal{B}$  is a basis”, which [Definition 1.5.2](#) needs for coordinates to exist and be unique.

**Warning. Watch the direction.** The columns of  $[\text{id}]_{\mathcal{B}}^{\mathcal{C}}$  are the  **$\mathcal{B}$ -vectors written in  $\mathcal{C}$** , and the matrix turns  $\mathcal{B}$ -coordinates into  $\mathcal{C}$ -coordinates. Using it the other way gives nonsense. With  $\mathcal{B} = ((1, 1), (1, -1))$ , the matrix  $\begin{pmatrix} 1 & 1 \\ 1 & -1 \end{pmatrix} = [\text{id}]_{\mathcal{B}}^{\mathcal{E}}$  applied to  $[\mathbf{v}]_{\mathcal{E}} = (3, 1)$  gives  $(4, 2)$ , which is **not**  $[\mathbf{v}]_{\mathcal{B}} = (2, 1)$ . A reliable test: the first column must be the first **input** basis vector written in the **output** basis. Here  $(1, 1) = 1 \cdot \mathbf{e}_1 + 1 \cdot \mathbf{e}_2$ , so the column  $(1, 1)$  belongs to  $[\text{id}]_{\mathcal{B}}^{\mathcal{E}}$ , which accepts  $\mathcal{B}$ -coordinates.

The examples suggest three facts: the matrix converts coordinates, converting twice multiplies

the matrices, and converting back inverts the matrix. All three follow from results about matrices of maps, applied to the identity.

### Theorem 3.7.1 — Change of Coordinates

Let  $V$  be a vector space of dimension  $n$ , and let  $\mathcal{B}, \mathcal{C}, \mathcal{D}$  be bases of  $V$ .

- (a) For every  $\mathbf{v} \in V$ ,  $[\mathbf{v}]_{\mathcal{C}} = [\text{id}]_{\mathcal{B}}^{\mathcal{C}}[\mathbf{v}]_{\mathcal{B}}$ . Moreover,  $[\text{id}]_{\mathcal{B}}^{\mathcal{C}}$  is the **only** matrix  $M \in M_n(F)$  with  $[\mathbf{v}]_{\mathcal{C}} = M[\mathbf{v}]_{\mathcal{B}}$  for every  $\mathbf{v} \in V$ .
- (b)  $[\text{id}]_{\mathcal{B}}^{\mathcal{D}} = [\text{id}]_{\mathcal{C}}^{\mathcal{D}}[\text{id}]_{\mathcal{B}}^{\mathcal{C}}$ .
- (c)  $[\text{id}]_{\mathcal{B}}^{\mathcal{C}}$  is invertible, and  $([\text{id}]_{\mathcal{B}}^{\mathcal{C}})^{-1} = [\text{id}]_{\mathcal{C}}^{\mathcal{B}}$ .

*Proof.*

- (a) Apply [Theorem 3.6.1](#), including its uniqueness clause, to  $\text{id}_V$  with input basis  $\mathcal{B}$  and output basis  $\mathcal{C}$ : since  $\text{id}_V(\mathbf{v}) = \mathbf{v}$ , it says  $[\mathbf{v}]_{\mathcal{C}} = [\text{id}]_{\mathcal{B}}^{\mathcal{C}}[\mathbf{v}]_{\mathcal{B}}$  for every  $\mathbf{v}$ , and that no other matrix has this property.
- (b) Since  $\text{id}_V = \text{id}_V \circ \text{id}_V$ , [Theorem 3.6.2](#), with  $\mathcal{B}$  on the input,  $\mathcal{C}$  in the middle and  $\mathcal{D}$  on the output, gives  $[\text{id}]_{\mathcal{B}}^{\mathcal{D}} = [\text{id}]_{\mathcal{C}}^{\mathcal{D}}[\text{id}]_{\mathcal{B}}^{\mathcal{C}}$ .
- (c) Taking  $\mathcal{D} = \mathcal{B}$  in (b) gives  $[\text{id}]_{\mathcal{C}}^{\mathcal{B}}[\text{id}]_{\mathcal{B}}^{\mathcal{C}} = [\text{id}]_{\mathcal{B}}^{\mathcal{B}} = \mathbf{I}_n$ , where the last equality holds because, writing  $\mathcal{B} = (\mathbf{v}_1, \dots, \mathbf{v}_n)$ , the  $j$ -th column of  $[\text{id}]_{\mathcal{B}}^{\mathcal{B}}$  is  $[\mathbf{v}_j]_{\mathcal{B}} = \mathbf{e}_j$ . Swapping the roles of  $\mathcal{B}$  and  $\mathcal{C}$  gives  $[\text{id}]_{\mathcal{B}}^{\mathcal{C}}[\text{id}]_{\mathcal{C}}^{\mathcal{B}} = \mathbf{I}_n$ . By [Definition 0.9.13](#),  $[\text{id}]_{\mathcal{B}}^{\mathcal{C}}$  is invertible with inverse  $[\text{id}]_{\mathcal{C}}^{\mathcal{B}}$ . ■

So every change-of-coordinates matrix is invertible. The converse is also true, and it will matter when we compare matrices: every invertible matrix is a change-of-coordinates matrix, for a suitable new basis.

### Proposition 3.7.2 — Invertible matrices are changes of basis

Let  $\mathcal{B} = (\mathbf{v}_1, \dots, \mathbf{v}_n)$  be a basis of  $V$ , and let  $\mathbf{P} = (p_{ij}) \in M_n(F)$  be invertible. Put

$$\mathbf{v}'_j := p_{1j}\mathbf{v}_1 + p_{2j}\mathbf{v}_2 + \dots + p_{nj}\mathbf{v}_n \quad (j = 1, \dots, n).$$

Then  $\mathcal{B}' = (\mathbf{v}'_1, \dots, \mathbf{v}'_n)$  is a basis of  $V$ , and  $[\text{id}]_{\mathcal{B}'}^{\mathcal{B}} = \mathbf{P}$ .

*Proof.* By [Definition 1.5.2](#),  $[\mathbf{v}'_j]_{\mathcal{B}}$  is the  $j$ -th column  $\mathbf{p}_j$  of  $\mathbf{P}$ . Let  $c_1, \dots, c_n \in F$  satisfy  $c_1\mathbf{v}'_1 + \dots + c_n\mathbf{v}'_n = \mathbf{0}$ . Taking  $\mathcal{B}$ -coordinates and using [Theorem 1.5.2](#),

$$c_1\mathbf{p}_1 + \dots + c_n\mathbf{p}_n = [\mathbf{0}]_{\mathcal{B}} = \mathbf{0},$$

that is,  $\mathbf{P}\mathbf{c} = \mathbf{0}$  with  $\mathbf{c} = (c_1, \dots, c_n)$ , by [Theorem 0.9.3](#). Since  $\mathbf{P}$  is invertible, [Theorem 2.4.4](#) ((a)  $\Rightarrow$  (b)) gives  $\mathbf{c} = \mathbf{0}$ . So  $\mathcal{B}'$  is linearly independent. It has length  $n = \dim V$ , hence it is a basis by [Theorem 1.6.4](#) (a). Finally, the  $j$ -th column of  $[\text{id}]_{\mathcal{B}'}^{\mathcal{B}}$  is  $[\mathbf{v}'_j]_{\mathcal{B}} = \mathbf{p}_j$ , so  $[\text{id}]_{\mathcal{B}'}^{\mathcal{B}} = \mathbf{P}$ . ■

**Example. (Coordinates around the point 1)**

In  $\mathbb{R}[x]_{\leq 2}$ , let  $\mathcal{B} = (1, x, x^2)$  and  $\mathcal{C} = (1, x - 1, (x - 1)^2)$ . Find  $[\text{id}]_{\mathcal{B}}^{\mathcal{C}}$ , and use it to write  $p = 2 + 3x + x^2$  in powers of  $x - 1$ .

*Solution.* From the examples above,  $[\text{id}]_{\mathcal{C}}^{\mathcal{B}} = \begin{pmatrix} 1 & -1 & 1 \\ 0 & 1 & -2 \\ 0 & 0 & 1 \end{pmatrix}$ . By Theorem 3.7.1 (c),  $[\text{id}]_{\mathcal{B}}^{\mathcal{C}}$  is its inverse. We can also read it off directly:  $1 = 1$ ,  $x = 1 + (x - 1)$ , and  $x^2 = (1 + (x - 1))^2 = 1 + 2(x - 1) + (x - 1)^2$ . So

$$[\text{id}]_{\mathcal{B}}^{\mathcal{C}} = \begin{pmatrix} 1 & 1 & 1 \\ 0 & 1 & 2 \\ 0 & 0 & 1 \end{pmatrix},$$

and multiplying the two matrices gives  $\mathbf{I}_3$ , as the theorem predicts. By Theorem 3.7.1 (a),

$$[p]_{\mathcal{C}} = \begin{pmatrix} 1 & 1 & 1 \\ 0 & 1 & 2 \\ 0 & 0 & 1 \end{pmatrix} \begin{pmatrix} 2 \\ 3 \\ 1 \end{pmatrix} = \begin{pmatrix} 6 \\ 5 \\ 1 \end{pmatrix}.$$

Hence  $2 + 3x + x^2 = 6 + 5(x - 1) + (x - 1)^2$ . Check at  $x = 1$ : both sides equal 6. The  $\mathcal{C}$ -coordinates are  $p(1)$ ,  $p'(1)$  and  $p''(1)/2$ , the Taylor coefficients at 1.

## 3.7.2 Change of basis for linear maps

Now return to a linear map  $T: V \rightarrow W$  between finite-dimensional spaces. Suppose we know its matrix  $[T]_{\mathcal{B}}^{\mathcal{C}}$  and want the matrix  $[T]_{\mathcal{B}'}^{\mathcal{C}'}$  for new bases  $\mathcal{B}'$  of  $V$  and  $\mathcal{C}'$  of  $W$ . The trick is to write  $T$  in a slightly more elaborate way that does nothing:

$$T = \text{id}_W \circ T \circ \text{id}_V.$$

Each factor can now carry its own pair of bases, and the matrix of the composite is the product of the matrices. The picture keeps track of which basis sits where.

$$\begin{array}{ccc}
 V \text{ with } \mathcal{B} & \xrightarrow{\begin{matrix} T \\ [T]_{\mathcal{B}}^{\mathcal{C}} \end{matrix}} & W \text{ with } \mathcal{C} \\
 \begin{matrix} \uparrow \text{id}_V \\ [\text{id}]_{\mathcal{B}'}^{\mathcal{B}} \end{matrix} & & \begin{matrix} \downarrow \text{id}_W \\ [\text{id}]_{\mathcal{C}}^{\mathcal{C}'} \end{matrix} \\
 V \text{ with } \mathcal{B}' & \xrightarrow{\begin{matrix} T \\ [T]_{\mathcal{B}'}^{\mathcal{C}'} \text{ (wanted)} \end{matrix}} & W \text{ with } \mathcal{C}'
 \end{array}$$

Going along the bottom edge applies  $T$ . Going up, across the top and down applies  $\text{id}_W \circ T \circ \text{id}_V$ , which is the same map. So the two routes must have the same matrix. Reading the long route, the step done **first** (up the left side) is written on the **right**.

**Theorem 3.7.3 — Change of Basis for Linear Maps**

Let  $V$  and  $W$  be finite-dimensional vector spaces over  $F$ , let  $\mathcal{B}, \mathcal{B}'$  be bases of  $V$  and  $\mathcal{C}, \mathcal{C}'$  bases of  $W$ , and let  $T: V \rightarrow W$  be linear. Then

$$[T]_{\mathcal{B}'}^{\mathcal{C}'} = [\text{id}]_{\mathcal{C}}^{\mathcal{C}'} [T]_{\mathcal{B}}^{\mathcal{C}} [\text{id}]_{\mathcal{B}'}^{\mathcal{B}}.$$

In particular, for an operator  $T \in \mathcal{L}(V)$  and bases  $\mathcal{B}, \mathcal{B}'$  of  $V$ , with  $\mathbf{P} = [\text{id}]_{\mathcal{B}'}^{\mathcal{B}}$ ,

$$[T]_{\mathcal{B}'} = \mathbf{P}^{-1} [T]_{\mathcal{B}} \mathbf{P}.$$

*Idea.* Nothing is computed: the formula is the change-of-basis square read along its long route. ① Write  $T = \text{id}_W \circ T \circ \text{id}_V$ . ② Give the middle factor the bases whose matrix we know. ③ Apply the composition rule twice. Do not memorize the formula; redraw the square and read it off, right to left.

*Proof.* Since  $T = \text{id}_W \circ (T \circ \text{id}_V)$ , Theorem 3.6.2 with bases  $\mathcal{B}', \mathcal{C}, \mathcal{C}'$  gives  $[T]_{\mathcal{B}'}^{\mathcal{C}'} = [\text{id}]_{\mathcal{C}}^{\mathcal{C}'} [T \circ \text{id}_V]_{\mathcal{B}'}^{\mathcal{C}}$ . Applying Theorem 3.6.2 again, with bases  $\mathcal{B}', \mathcal{B}, \mathcal{C}$ , gives  $[T \circ \text{id}_V]_{\mathcal{B}'}^{\mathcal{C}} = [T]_{\mathcal{B}}^{\mathcal{C}} [\text{id}]_{\mathcal{B}'}^{\mathcal{B}}$ . Combining the two equations proves the first formula.

For an operator, take  $W = V$ ,  $\mathcal{C} = \mathcal{B}$  and  $\mathcal{C}' = \mathcal{B}'$ . The first formula becomes  $[T]_{\mathcal{B}'} = [\text{id}]_{\mathcal{B}}^{\mathcal{B}'} [T]_{\mathcal{B}} \mathbf{P}$ , and  $[\text{id}]_{\mathcal{B}}^{\mathcal{B}'} = \mathbf{P}^{-1}$  by Theorem 3.7.1 (c). This proves the theorem. ■

We will call this move **the change-of-basis square**: whenever a question involves the same map in two bases, draw the square, label the arrows with bases, and read the formula off the long route. It also explains the word “coordinates”: a matrix is a map seen through a choice of coordinates, and the side arrows re-express those coordinates.

**Example. (The reflection in a mirror-adapted basis)**

Let  $R(x, y) = (y, x)$  on  $\mathbb{R}^2$ , and  $\mathcal{B} = ((1, 1), (1, -1))$ . Compute  $[R]_{\mathcal{B}}$  from  $[R]_{\mathcal{E}}$  using Theorem 3.7.3, and compare with the direct computation at the start of the section.

*Solution.* Here  $\mathbf{P} = [\text{id}]_{\mathcal{B}}^{\mathcal{E}} = \begin{pmatrix} 1 & 1 \\ 1 & -1 \end{pmatrix}$ , with inverse  $\mathbf{P}^{-1} = [\text{id}]_{\mathcal{E}}^{\mathcal{B}} = \frac{1}{2} \begin{pmatrix} 1 & 1 \\ 1 & -1 \end{pmatrix}$ , found in the examples after Definition 3.7.1. Then

$$\begin{aligned} \mathbf{P}^{-1} [R]_{\mathcal{E}} \mathbf{P} &= \frac{1}{2} \begin{pmatrix} 1 & 1 \\ 1 & -1 \end{pmatrix} \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix} \begin{pmatrix} 1 & 1 \\ 1 & -1 \end{pmatrix} \\ &= \frac{1}{2} \begin{pmatrix} 1 & 1 \\ 1 & -1 \end{pmatrix} \begin{pmatrix} 1 & -1 \\ 1 & 1 \end{pmatrix} = \frac{1}{2} \begin{pmatrix} 2 & 0 \\ 0 & -2 \end{pmatrix} = \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix}, \end{aligned}$$

which agrees with the direct computation. The direct route was shorter here, because we chose  $\mathcal{B}$  so that  $R$  acts simply on it. The formula earns its keep when the new matrix is not visible by inspection, or when we want to go back from the simple matrix to the standard one:  $[R]_{\mathcal{E}} = \mathbf{P} [R]_{\mathcal{B}} \mathbf{P}^{-1}$ .

The next example changes basis in a polynomial space, where “standard” does not mean “best”.

**Example. (Differentiation with a rescaled basis)**

Let  $D: \mathbb{R}[x]_{\leq 2} \rightarrow \mathbb{R}[x]_{\leq 2}$  be differentiation,  $\mathcal{B} = (1, x, x^2)$  and  $\mathcal{B}' = (1, x, \frac{1}{2}x^2)$ . Find  $[D]_{\mathcal{B}'}$ .

*Solution.* In  $\mathcal{B}$ ,  $D(1) = 0$ ,  $D(x) = 1$  and  $D(x^2) = 2x$ , so  $\mathbf{A} = [D]_{\mathcal{B}} = \begin{pmatrix} 0 & 1 & 0 \\ 0 & 0 & 2 \\ 0 & 0 & 0 \end{pmatrix}$ .

The vectors of  $\mathcal{B}'$  have  $\mathcal{B}$ -coordinates  $\mathbf{e}_1$ ,  $\mathbf{e}_2$ ,  $\frac{1}{2}\mathbf{e}_3$ , so  $\mathbf{P} = [\text{id}]_{\mathcal{B}'}^{\mathcal{B}} = \text{diag}(1, 1, \frac{1}{2})$  and  $\mathbf{P}^{-1} = \text{diag}(1, 1, 2)$ . Multiplying a matrix on the left by a diagonal matrix scales its rows, and on the right scales its columns, so

$$\mathbf{P}^{-1}\mathbf{A}\mathbf{P} = \begin{pmatrix} 0 & 1 \cdot 1 \cdot 1 & 0 \\ 0 & 0 & 1 \cdot 2 \cdot \frac{1}{2} \\ 0 & 0 & 0 \end{pmatrix} = \begin{pmatrix} 0 & 1 & 0 \\ 0 & 0 & 1 \\ 0 & 0 & 0 \end{pmatrix}.$$

Directly:  $D(1) = 0$ ,  $D(x) = 1$ ,  $D(\frac{1}{2}x^2) = x$ , so  $D$  shifts each vector of  $\mathcal{B}'$  to the previous one, and the matrix has 1's just above the diagonal. The factor 2 has disappeared into the basis.

*Quick check.* Let  $T: V \rightarrow W$ , and suppose we change only the basis of  $W$ , from  $\mathcal{C}$  to  $\mathcal{C}'$ , keeping  $\mathcal{B}$ . Which of the two side matrices in [Theorem 3.7.3](#) becomes the identity, and what is the formula?

*Solution.* With  $\mathcal{B}' = \mathcal{B}$ , the left side of the square is  $[\text{id}]_{\mathcal{B}} = \mathbf{I}_n$ . The formula becomes  $[T]_{\mathcal{B}}^{\mathcal{C}'} = [\text{id}]_{\mathcal{C}'}^{\mathcal{C}} [T]_{\mathcal{B}}^{\mathcal{C}}$ : only a multiplication on the left. Changing the output basis acts on the rows; changing the input basis acts on the columns.

### 3.7.3 Similar matrices

For an operator, the two bases on the two sides of the square are the same, and the formula takes the special shape  $\mathbf{P}^{-1}\mathbf{A}\mathbf{P}$ . Recall from [Example \(Similarity\)](#) that we already gave this shape a name in Chapter 0 and checked that it defines an equivalence relation. We now record it as a definition, because it is exactly the relation “same operator, different basis”.

*Two square matrices are similar when one is obtained from the other by a change of basis applied on both sides at once.*

**Definition 3.7.2 — Similar matrices**

Let  $\mathbf{A}, \mathbf{B} \in M_n(F)$ . We say  $\mathbf{A}$  is **similar** to  $\mathbf{B}$ , written  $\mathbf{A} \sim \mathbf{B}$ , if there **exists** an **invertible** matrix  $\mathbf{P} \in M_n(F)$  such that

$$\mathbf{B} = \mathbf{P}^{-1}\mathbf{A}\mathbf{P}.$$

In words: the **same**  $\mathbf{P}$  appears on both sides, once inverted. By [Example \(Similarity\)](#),  $\sim$  is an equivalence relation, so we may say “ $\mathbf{A}$  and  $\mathbf{B}$  are similar” without worrying about the order. The matrix  $\mathbf{P}$  is not unique: for instance  $\mathbf{A} = \mathbf{P}^{-1}\mathbf{A}\mathbf{P}$  holds for every invertible  $\mathbf{P}$  that commutes with  $\mathbf{A}$ , such as any non-zero multiple of  $\mathbf{I}_n$ . Writing  $\mathbf{B} = \mathbf{Q}\mathbf{A}\mathbf{Q}^{-1}$  defines the same



relation, with  $\mathbf{Q} = \mathbf{P}^{-1}$ .

**Examples.**

- **The reflection.**  $\begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix} \sim \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix}$ , by [Example \(The reflection in a mirror-adapted basis\)](#).
- **A triangular matrix.**  $\begin{pmatrix} 1 & 2 \\ 0 & 3 \end{pmatrix} \sim \text{diag}(1, 3)$ , as computed in [Example \(Similarity\)](#) with  $\mathbf{P} = \begin{pmatrix} 1 & 1 \\ 0 & 1 \end{pmatrix}$ .
- **Reordering a basis.**  $\text{diag}(1, 2) \sim \text{diag}(2, 1)$ : with  $\mathbf{P} = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix} = \mathbf{P}^{-1}$ , we get  $\mathbf{P}^{-1} \text{diag}(1, 2) \mathbf{P} = \text{diag}(2, 1)$ . This is the operator with matrix  $\text{diag}(1, 2)$  in  $(\mathbf{v}_1, \mathbf{v}_2)$ , read in the reordered basis  $(\mathbf{v}_2, \mathbf{v}_1)$ .
- **Degenerate case: scalar matrices.** For  $c \in F$ ,  $\mathbf{P}^{-1}(c\mathbf{I}_n)\mathbf{P} = c\mathbf{P}^{-1}\mathbf{P} = c\mathbf{I}_n$  for every invertible  $\mathbf{P}$ . So  $c\mathbf{I}_n$  is similar only to itself. In particular  $\mathbf{I}_n$  and the zero matrix are alone in their classes: the identity operator looks like  $\mathbf{I}_n$  in **every** basis.

**Non-example by minimal change.** Allow two different invertible matrices,  $\mathbf{B} = \mathbf{QAP}$ . With  $\mathbf{A} = \mathbf{I}_2$ ,  $\mathbf{Q} = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}$  and  $\mathbf{P} = \mathbf{I}_2$ , we get  $\mathbf{B} = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix} \neq \mathbf{I}_2$ . So  $\mathbf{B} = \mathbf{QAP}$  holds, but  $\mathbf{B}$  is not similar to  $\mathbf{I}_2$ , which is similar only to itself. The clause that failed is “the **same**  $\mathbf{P}$  on both sides, once inverted”. The relaxed relation is also useful; it is the subject of the next section.

The definition was designed to capture one idea, and the next theorem says it captures exactly that idea.

### Theorem 3.7.4 — Similar Matrices Represent the Same Operator

Let  $\mathbf{A}, \mathbf{B} \in M_n(F)$ .

- If  $V$  is a vector space of dimension  $n$ ,  $T \in \mathcal{L}(V)$ , and  $\mathcal{B}, \mathcal{C}$  are bases of  $V$ , then  $[T]_{\mathcal{B}} \sim [T]_{\mathcal{C}}$ .
- Conversely, suppose  $\mathbf{A} \sim \mathbf{B}$ . Let  $V$  be any vector space of dimension  $n$  with a basis  $\mathcal{B}$ , and let  $T \in \mathcal{L}(V)$  be an operator with  $[T]_{\mathcal{B}} = \mathbf{A}$ . Then there is a basis  $\mathcal{C}$  of  $V$  with  $[T]_{\mathcal{C}} = \mathbf{B}$ .
- Such an operator  $T$  in (b) always exists. In particular,  $\mathbf{A} \sim \mathbf{B}$  if and only if  $\mathbf{A}$  and  $\mathbf{B}$  are the matrices of the operator  $\mathbf{x} \mapsto \mathbf{Ax}$  on  $F^n$  in the standard basis and in some basis of  $F^n$ , respectively.

*Proof.*

- By [Theorem 3.7.3](#),  $[T]_{\mathcal{C}} = \mathbf{P}^{-1}[T]_{\mathcal{B}}\mathbf{P}$  with  $\mathbf{P} = [\text{id}]_{\mathcal{C}}^{\mathcal{B}}$ , which is invertible by [Theorem 3.7.1](#) (c).
- Suppose  $\mathbf{B} = \mathbf{P}^{-1}\mathbf{AP}$  with  $\mathbf{P}$  invertible. By [Proposition 3.7.2](#), there is a basis  $\mathcal{C}$  of  $V$  with  $[\text{id}]_{\mathcal{C}}^{\mathcal{B}} = \mathbf{P}$ . By [Theorem 3.7.3](#),  $[T]_{\mathcal{C}} = \mathbf{P}^{-1}[T]_{\mathcal{B}}\mathbf{P} = \mathbf{P}^{-1}\mathbf{AP} = \mathbf{B}$ .
- Write  $\mathcal{B} = (\mathbf{v}_1, \dots, \mathbf{v}_n)$ . By [Theorem 3.1.5](#), there is a linear map  $T: V \rightarrow V$  with  $T\mathbf{v}_j = a_{1j}\mathbf{v}_1 + \dots + a_{nj}\mathbf{v}_n$  for each  $j$ . The  $j$ -th column of  $[T]_{\mathcal{B}}$  is  $[T\mathbf{v}_j]_{\mathcal{B}}$ , which is the  $j$ -th column of  $\mathbf{A}$ ; so  $[T]_{\mathcal{B}} = \mathbf{A}$ . For  $V = F^n$  with the standard basis  $\mathcal{E}$ , the operator  $\mathbf{x} \mapsto \mathbf{Ax}$  works, since  $\mathbf{Ae}_j$  is the  $j$ -th column of  $\mathbf{A}$  ([Theorem 0.9.3](#)). The

last statement now follows from (a) and (b). ■

This is the promise of Chapter 0 kept: similar matrices describe the same operator in two coordinate systems. So a question of the form “are  $\mathbf{A}$  and  $\mathbf{B}$  similar?” is really the question “is there a basis in which the operator  $\mathbf{x} \mapsto \mathbf{A}\mathbf{x}$  has matrix  $\mathbf{B}$ ?”. Deciding it is one of the central problems of the book, and it will take until Chapter 9 to answer it in general.

### 3.7.4 Similarity invariants

If a quantity computed from a matrix does not change under  $\mathbf{A} \mapsto \mathbf{P}^{-1}\mathbf{A}\mathbf{P}$ , then it is really a property of the operator, and we call it a **similarity invariant**. Invariants are the cheap way to prove that two matrices are **not** similar: find one invariant on which they differ. The trace is the first example, and it is the reason the trace will make sense for operators in this chapter’s last section.

#### Theorem 3.7.5 — Trace Is a Similarity Invariant

If  $\mathbf{A}, \mathbf{B} \in M_n(F)$  are similar, then  $\text{tr } \mathbf{A} = \text{tr } \mathbf{B}$ .

*Proof.* Let  $\mathbf{B} = \mathbf{P}^{-1}\mathbf{A}\mathbf{P}$  with  $\mathbf{P}$  invertible. By Theorem 0.9.6 (3), applied to the matrices  $\mathbf{P}^{-1}$  and  $\mathbf{A}\mathbf{P}$ ,

$$\text{tr } \mathbf{B} = \text{tr } (\mathbf{P}^{-1}(\mathbf{A}\mathbf{P})) = \text{tr } ((\mathbf{A}\mathbf{P})\mathbf{P}^{-1}) = \text{tr } (\mathbf{A}(\mathbf{P}\mathbf{P}^{-1})) = \text{tr } \mathbf{A},$$

where the third equality is associativity of matrix multiplication. This proves the theorem. ■

The proof never used that  $\mathbf{B}$  is “similar” in any geometric sense, only that the trace ignores the order of two factors. Several other quantities survive a change of basis, for simpler reasons.

#### Proposition 3.7.6 — More similarity invariants

Let  $\mathbf{A}, \mathbf{B} \in M_n(F)$  with  $\mathbf{B} = \mathbf{P}^{-1}\mathbf{A}\mathbf{P}$  for an invertible  $\mathbf{P}$ .

- (a)  $\text{rank } \mathbf{A} = \text{rank } \mathbf{B}$ .
- (b)  $\mathbf{A}$  is invertible if and only if  $\mathbf{B}$  is invertible.
- (c) For every polynomial  $p \in F[x]$ ,  $p(\mathbf{B}) = \mathbf{P}^{-1}p(\mathbf{A})\mathbf{P}$ . In particular,  $p(\mathbf{A}) = 0$  if and only if  $p(\mathbf{B}) = 0$ ; for example,  $\mathbf{A}^2 = 0$  if and only if  $\mathbf{B}^2 = 0$ .

*Proof.*

- (a) By Theorem 2.5.9, multiplying by invertible matrices on either side does not change the rank, so  $\text{rank}(\mathbf{P}^{-1}\mathbf{A}\mathbf{P}) = \text{rank}(\mathbf{A}\mathbf{P}) = \text{rank } \mathbf{A}$ .
- (b) If  $\mathbf{A}$  is invertible, then  $\mathbf{B} = \mathbf{P}^{-1}\mathbf{A}\mathbf{P}$  is a product of invertible matrices, hence

invertible by [Theorem 0.9.7](#) (part 3). Since  $\mathbf{A} = \mathbf{PBP}^{-1} = (\mathbf{P}^{-1})^{-1}\mathbf{BP}^{-1}$ , the same argument with the roles swapped gives the converse.

- (c) First,  $\mathbf{B}^k = \mathbf{P}^{-1}\mathbf{A}^k\mathbf{P}$  for every  $k \geq 0$ , by induction on  $k$ : for  $k = 0$  both sides are  $\mathbf{I}_n$ , and if it holds for  $k$ , then  $\mathbf{B}^{k+1} = \mathbf{B}^k\mathbf{B} = (\mathbf{P}^{-1}\mathbf{A}^k\mathbf{P})(\mathbf{P}^{-1}\mathbf{A}\mathbf{P}) = \mathbf{P}^{-1}\mathbf{A}^k(\mathbf{P}\mathbf{P}^{-1})\mathbf{A}\mathbf{P} = \mathbf{P}^{-1}\mathbf{A}^{k+1}\mathbf{P}$ . Now let  $p = c_0 + c_1x + \cdots + c_dx^d$ . By [Definition 0.9.14](#) and the distributive and scalar laws of [Theorem 0.9.4](#),

$$p(\mathbf{B}) = \sum_{k=0}^d c_k \mathbf{B}^k = \sum_{k=0}^d c_k \mathbf{P}^{-1} \mathbf{A}^k \mathbf{P} = \mathbf{P}^{-1} \left( \sum_{k=0}^d c_k \mathbf{A}^k \right) \mathbf{P} = \mathbf{P}^{-1} p(\mathbf{A}) \mathbf{P}.$$

If  $p(\mathbf{A}) = 0$ , then  $p(\mathbf{B}) = \mathbf{P}^{-1}0\mathbf{P} = 0$ . Conversely,  $\mathbf{A} = \mathbf{PBP}^{-1}$  is of the same form with  $\mathbf{P}^{-1}$  in place of  $\mathbf{P}$ , so  $p(\mathbf{B}) = 0$  implies  $p(\mathbf{A}) = 0$ . ■

For example,  $\begin{pmatrix} 1 & 2 \\ 0 & 3 \end{pmatrix}$  and  $\begin{pmatrix} 1 & 0 \\ 0 & 2 \end{pmatrix}$  are **not** similar, because their traces are 4 and 3. And  $\begin{pmatrix} 0 & 1 \\ 0 & 0 \end{pmatrix}$  is not similar to the zero matrix, because their ranks are 1 and 0 (or because the zero matrix is similar only to itself).

It is tempting to turn this around and decide similarity by comparing a list of invariants. That fails.

*Warning. Shared invariants do not imply similarity.* Let  $\mathbf{J} = \begin{pmatrix} 1 & 1 \\ 0 & 1 \end{pmatrix}$ . Then  $\mathbf{I}_2$  and  $\mathbf{J}$  have the same trace 2, the same rank 2, both are invertible, and both satisfy  $(\mathbf{X} - \mathbf{I}_2)^2 = 0$ . But they are **not** similar: for every invertible  $\mathbf{P}$ ,  $\mathbf{P}^{-1}\mathbf{I}_2\mathbf{P} = \mathbf{I}_2 \neq \mathbf{J}$ . A sharper invariant does separate them: combining (a) and (c) of [Proposition 3.7.6](#), similar matrices  $\mathbf{A}$  and  $\mathbf{B}$  have  $\text{rank } p(\mathbf{A}) = \text{rank } p(\mathbf{B})$  for every polynomial  $p$ , and  $\text{rank}(\mathbf{I}_2 - \mathbf{I}_2) = 0$  while  $\text{rank}(\mathbf{J} - \mathbf{I}_2) = 1$ . Which invariants suffice in general is the question Chapter 9 answers.

*Quick check.* Are  $\text{diag}(2, 5)$  and  $\text{diag}(5, 2)$  similar? Are  $\begin{pmatrix} 2 & 0 \\ 0 & 2 \end{pmatrix}$  and  $\begin{pmatrix} 2 & 1 \\ 0 & 2 \end{pmatrix}$  similar?

*Solution.* The first pair is similar: with  $\mathbf{P} = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}$ , which is its own inverse,  $\mathbf{P}^{-1} \text{diag}(2, 5) \mathbf{P} = \text{diag}(5, 2)$ . The second pair is not:  $2\mathbf{I}_2$  is a scalar matrix, so it is similar only to itself, although both matrices have trace 4 and rank 2.

### 3.7.5 A basis adapted to the operator

The reflection became diagonal once we chose one basis vector the map keeps and one it flips. The same idea works whenever a map is simple on two complementary pieces of the space. For a map of rank 1 on  $\mathbb{R}^2$ , the natural pieces are the image and the kernel, provided they are two different lines. (They need not be: for  $\mathbf{x} \mapsto \mathbf{N}\mathbf{x}$  with  $\mathbf{N} = \begin{pmatrix} 0 & 1 \\ 0 & 0 \end{pmatrix}$ , both are  $\text{span}(\mathbf{e}_1)$ , and [Exercise \(C3: Every non-zero  \$2 \times 2\$  matrix with square zero\)](#) shows what to do instead.)

**Example. (Making a matrix diagonal with a well-chosen basis)**

Let  $T: \mathbb{R}^2 \rightarrow \mathbb{R}^2$ ,  $T(x, y) = (2x - y, 2x - y)$ . Find a basis  $\mathcal{B}$  of  $\mathbb{R}^2$  such that  $[T]_{\mathcal{B}}$  is diagonal, and check the answer with [Theorem 3.7.3](#).

*Solution.* In the standard basis  $\mathbf{A} = [T]_{\mathcal{E}} = \begin{pmatrix} 2 & -1 \\ 2 & -1 \end{pmatrix}$ .

Both entries of  $T(x, y)$  are equal, so  $\text{im } T \subseteq \text{span}((1, 1))$ , and  $T(1, 1) = (1, 1)$  shows  $\text{im } T = \text{span}((1, 1))$ . The kernel is  $\{(x, y) : y = 2x\} = \text{span}((1, 2))$ . So  $T$  fixes  $(1, 1)$  and kills  $(1, 2)$ . These two vectors are not multiples of each other, so  $\mathcal{B} = ((1, 1), (1, 2))$  is independent, and it is a basis of  $\mathbb{R}^2$  by [Theorem 1.6.4](#). Since  $T(1, 1) = 1 \cdot (1, 1) + 0 \cdot (1, 2)$  and  $T(1, 2) = \mathbf{0}$ ,

$$[T]_{\mathcal{B}} = \begin{pmatrix} 1 & 0 \\ 0 & 0 \end{pmatrix}.$$

Check:  $\mathbf{P} = [\text{id}]_{\mathcal{B}}^{\mathcal{E}} = \begin{pmatrix} 1 & 1 \\ 1 & 2 \end{pmatrix}$  has  $ad - bc = 1$ , so  $\mathbf{P}^{-1} = \begin{pmatrix} 2 & -1 \\ -1 & 1 \end{pmatrix}$  by [Theorem 0.9.8](#).

Then

$$\mathbf{P}^{-1} \mathbf{A} \mathbf{P} = \begin{pmatrix} 2 & -1 \\ -1 & 1 \end{pmatrix} \begin{pmatrix} 2 & -1 \\ 2 & -1 \end{pmatrix} \begin{pmatrix} 1 & 1 \\ 1 & 2 \end{pmatrix} = \begin{pmatrix} 2 & -1 \\ 0 & 0 \end{pmatrix} \begin{pmatrix} 1 & 1 \\ 1 & 2 \end{pmatrix} = \begin{pmatrix} 1 & 0 \\ 0 & 0 \end{pmatrix}.$$

As [Theorem 3.7.5](#) and [Proposition 3.7.6](#) require,  $\text{tr } \mathbf{A} = 1$  and  $\text{rank } \mathbf{A} = 1$  match the diagonal matrix. Its form also explains a property of  $\mathbf{A}$  that is not obvious from its entries:  $\text{diag}(1, 0)^2 = \text{diag}(1, 0)$ , so  $\mathbf{A}^2 = \mathbf{A}$  by [Proposition 3.7.6](#) (c) applied to  $p = x^2 - x$ .

The basis in this example consisted of vectors that  $T$  sends to multiples of themselves. Chapter 8 turns this into a method: such vectors are called eigenvectors, and an operator has a diagonal matrix exactly when there is a basis of them. Operators with  $T^2 = T$ , like this one, are the projections of the last section of this chapter.

## 3.7.6 Exercises

### 3.7.6.1 A. Check your understanding

#### Exercise A1.

- Define the change-of-coordinates matrix  $[\text{id}]_{\mathcal{B}}^{\mathcal{C}}$ , and say what its columns are.
- State the change-of-basis formula for an operator  $T \in \mathcal{L}(V)$  and bases  $\mathcal{B}, \mathcal{B}'$ , saying exactly which matrix  $\mathbf{P}$  is.
- True or false:  $[\text{id}]_{\mathcal{B}}^{\mathcal{C}} = [\text{id}]_{\mathcal{C}}^{\mathcal{B}}$  for all bases  $\mathcal{B}, \mathcal{C}$  of  $V$ . Justify your answer.
- True or false: if  $\mathbf{A} \sim \mathbf{B}$ , then  $\text{tr } \mathbf{A} = \text{tr } \mathbf{B}$  and  $\text{rank } \mathbf{A} = \text{rank } \mathbf{B}$ . Justify your answer.
- True or false: if  $\text{tr } \mathbf{A} = \text{tr } \mathbf{B}$  and  $\text{rank } \mathbf{A} = \text{rank } \mathbf{B}$ , then  $\mathbf{A} \sim \mathbf{B}$ . Justify your answer.
- Describe the method for recovering the change-of-basis formula without memorizing it.

*Solution.*

- (a) For bases  $\mathcal{B} = (\mathbf{v}_1, \dots, \mathbf{v}_n)$  and  $\mathcal{C}$  of  $V$ ,  $[\text{id}]_{\mathcal{B}}^{\mathcal{C}}$  is the matrix of  $\text{id}_V$  with input basis  $\mathcal{B}$  and output basis  $\mathcal{C}$  (Definition 3.7.1). Its  $j$ -th column is  $[\mathbf{v}_j]_{\mathcal{C}}$ .
- (b)  $[T]_{\mathcal{B}'} = \mathbf{P}^{-1}[T]_{\mathcal{B}}\mathbf{P}$  with  $\mathbf{P} = [\text{id}]_{\mathcal{B}'}^{\mathcal{B}}$ , whose columns are the  $\mathcal{B}$ -coordinates of the vectors of  $\mathcal{B}'$  (Theorem 3.7.3).
- (c) False. By Theorem 3.7.1 (c) the two matrices are inverse to each other, and an invertible matrix usually differs from its inverse: for  $\mathcal{B} = ((1, 1), (1, -1))$  and the standard basis  $\mathcal{E}$  of  $\mathbb{R}^2$ ,  $[\text{id}]_{\mathcal{B}}^{\mathcal{E}} = \begin{pmatrix} 1 & 1 \\ 1 & -1 \end{pmatrix}$  but  $[\text{id}]_{\mathcal{E}}^{\mathcal{B}} = \frac{1}{2} \begin{pmatrix} 1 & 1 \\ 1 & -1 \end{pmatrix}$ .
- (d) True, by Theorem 3.7.5 and Proposition 3.7.6 (a).
- (e) False.  $\mathbf{I}_2$  and  $\begin{pmatrix} 1 & 1 \\ 0 & 1 \end{pmatrix}$  both have trace 2 and rank 2, but  $\mathbf{I}_2$  is similar only to itself.
- (f) Draw the change-of-basis square:  $T$  along the top and bottom with the old and new bases, identity maps up the left side and down the right side. Since  $T = \text{id}_W \circ T \circ \text{id}_V$ , the matrix of the bottom arrow is the product of the matrices along the long route, with the first step on the right.

### 3.7.6.2 B. Practice

#### Exercise B1: Change-of-coordinates matrices.

- (a) In  $\mathbb{R}^2$ , let  $\mathcal{B} = ((1, 2), (3, 5))$  and  $\mathcal{C} = ((1, 1), (1, 2))$ . Find  $[\text{id}]_{\mathcal{B}}^{\mathcal{C}}$ . If  $[\mathbf{v}]_{\mathcal{B}} = (1, -1)$ , find  $[\mathbf{v}]_{\mathcal{C}}$ , and check your answer by computing  $\mathbf{v}$  both ways.
- (b) In  $\mathbb{R}[x]_{\leq 2}$ , let  $\mathcal{B} = (1, x, x^2)$  and  $\mathcal{C} = (1, 1+x, 1+x+x^2)$ . Find  $[\text{id}]_{\mathcal{C}}^{\mathcal{B}}$  and  $[\text{id}]_{\mathcal{B}}^{\mathcal{C}}$ . Hence find the  $\mathcal{C}$ -coordinates of  $3 - x + 2x^2$ .

*Solution.*

- (a) The columns are the  $\mathcal{C}$ -coordinates of  $(1, 2)$  and  $(3, 5)$ . Solving  $a(1, 1) + b(1, 2) = (1, 2)$  gives  $a + b = 1$ ,  $a + 2b = 2$ , so  $b = 1$ ,  $a = 0$ . Solving  $a(1, 1) + b(1, 2) = (3, 5)$  gives  $b = 2$ ,  $a = 1$ . Hence

$$[\text{id}]_{\mathcal{B}}^{\mathcal{C}} = \begin{pmatrix} 0 & 1 \\ 1 & 2 \end{pmatrix}, \quad [\mathbf{v}]_{\mathcal{C}} = \begin{pmatrix} 0 & 1 \\ 1 & 2 \end{pmatrix} \begin{pmatrix} 1 \\ -1 \end{pmatrix} = \begin{pmatrix} -1 \\ -1 \end{pmatrix}$$

by Theorem 3.7.1 (a). Check: from  $\mathcal{B}$ ,  $\mathbf{v} = (1, 2) - (3, 5) = (-2, -3)$ ; from  $\mathcal{C}$ ,  $\mathbf{v} = -(1, 1) - (1, 2) = (-2, -3)$ .

- (b) The vectors of  $\mathcal{C}$  have  $\mathcal{B}$ -coordinates  $(1, 0, 0)$ ,  $(1, 1, 0)$ ,  $(1, 1, 1)$ , so  $[\text{id}]_{\mathcal{C}}^{\mathcal{B}} = \begin{pmatrix} 1 & 1 & 1 \\ 0 & 1 & 1 \\ 0 & 0 & 1 \end{pmatrix}$ .

For the other direction, write the vectors of  $\mathcal{B}$  in  $\mathcal{C}$ :  $1 = 1$ ,  $x = (1 + x) - 1$ ,  $x^2 = (1 + x + x^2) - (1 + x)$ . So

$$[\text{id}]_{\mathcal{B}}^{\mathcal{C}} = \begin{pmatrix} 1 & -1 & 0 \\ 0 & 1 & -1 \\ 0 & 0 & 1 \end{pmatrix},$$

and multiplying the two matrices gives  $\mathbf{I}_3$ , as [Theorem 3.7.1](#) (c) requires. Hence

$$[3 - x + 2x^2]_{\mathcal{C}} = \begin{pmatrix} 1 & -1 & 0 \\ 0 & 1 & -1 \\ 0 & 0 & 1 \end{pmatrix} \begin{pmatrix} 3 \\ -1 \\ 2 \end{pmatrix} = \begin{pmatrix} 4 \\ -3 \\ 2 \end{pmatrix}.$$

Check:  $4 - 3(1 + x) + 2(1 + x + x^2) = 3 - x + 2x^2$ .

### Exercise B2: A matrix in a new basis, two ways.

Let  $T: \mathbb{R}^2 \rightarrow \mathbb{R}^2$ ,  $T(x, y) = (4x - 2y, x + y)$ , and  $\mathcal{B} = ((1, 1), (2, 1))$ .

- (a) Compute  $[T]_{\mathcal{B}}$  directly from [Definition 3.6.1](#).
- (b) Compute it again with [Theorem 3.7.3](#), starting from the standard matrix.

*Solution.*

- (a)  $T(1, 1) = (2, 2) = 2(1, 1) + 0(2, 1)$  and  $T(2, 1) = (6, 3) = 0(1, 1) + 3(2, 1)$ . Hence  $[T]_{\mathcal{B}} = \text{diag}(2, 3)$ .

- (b)  $\mathbf{A} = [T]_{\mathcal{E}} = \begin{pmatrix} 4 & -2 \\ 1 & 1 \end{pmatrix}$  and  $\mathbf{P} = [\text{id}]_{\mathcal{B}}^{\mathcal{E}} = \begin{pmatrix} 1 & 2 \\ 1 & 1 \end{pmatrix}$ . Here  $ad - bc = -1$ , so by [Theorem 0.9.8](#)  $\mathbf{P}^{-1} = \begin{pmatrix} -1 & 2 \\ 1 & -1 \end{pmatrix}$ . Then

$$\mathbf{P}^{-1}\mathbf{A}\mathbf{P} = \begin{pmatrix} -1 & 2 \\ 1 & -1 \end{pmatrix} \begin{pmatrix} 4 & -2 \\ 1 & 1 \end{pmatrix} \begin{pmatrix} 1 & 2 \\ 1 & 1 \end{pmatrix} = \begin{pmatrix} -2 & 4 \\ 3 & -3 \end{pmatrix} \begin{pmatrix} 1 & 2 \\ 1 & 1 \end{pmatrix} = \begin{pmatrix} 2 & 0 \\ 0 & 3 \end{pmatrix},$$

in agreement with (a).

### Exercise B3: Similar or not?

Determine which of the following pairs of real matrices are similar. Justify your answer, either with an invariant or with an explicit invertible  $\mathbf{P}$ .

- (a)  $\begin{pmatrix} 1 & 2 \\ 0 & 3 \end{pmatrix}$  and  $\begin{pmatrix} 1 & 0 \\ 0 & 2 \end{pmatrix}$ .
- (b)  $\begin{pmatrix} 0 & 1 \\ 0 & 0 \end{pmatrix}$  and  $\begin{pmatrix} 0 & 0 \\ 0 & 0 \end{pmatrix}$ .
- (c)  $\begin{pmatrix} 1 & 1 \\ 0 & 2 \end{pmatrix}$  and  $\begin{pmatrix} 1 & 0 \\ 0 & 2 \end{pmatrix}$ .
- (d)  $\begin{pmatrix} 2 & 0 \\ 0 & 2 \end{pmatrix}$  and  $\begin{pmatrix} 2 & 1 \\ 0 & 2 \end{pmatrix}$ .
- (e)  $\begin{pmatrix} 1 & 1 \\ 1 & 1 \end{pmatrix}$  and  $\begin{pmatrix} 2 & 0 \\ 0 & 0 \end{pmatrix}$ .

*Solution.*

- (a) Not similar: the traces are 4 and 3, contradicting [Theorem 3.7.5](#).

- (b) Not similar: the ranks are 1 and 0, contradicting [Proposition 3.7.6](#) (a).  
 (c) Similar. Let  $\mathbf{A}$  be the first matrix. We look for a basis in which  $\mathbf{x} \mapsto \mathbf{A}\mathbf{x}$  is diagonal:

$\mathbf{A}\mathbf{e}_1 = \mathbf{e}_1$ , and  $\mathbf{A}(1, 1) = (2, 2) = 2(1, 1)$ . With  $\mathbf{P} = \begin{pmatrix} 1 & 1 \\ 0 & 1 \end{pmatrix}$ ,  $\mathbf{P}^{-1} = \begin{pmatrix} 1 & -1 \\ 0 & 1 \end{pmatrix}$  and

$$\mathbf{P}^{-1}\mathbf{A}\mathbf{P} = \begin{pmatrix} 1 & -1 \\ 0 & 1 \end{pmatrix} \begin{pmatrix} 1 & 1 \\ 0 & 2 \end{pmatrix} \begin{pmatrix} 1 & 1 \\ 0 & 1 \end{pmatrix} = \begin{pmatrix} 1 & -1 \\ 0 & 2 \end{pmatrix} \begin{pmatrix} 1 & 1 \\ 0 & 1 \end{pmatrix} = \begin{pmatrix} 1 & 0 \\ 0 & 2 \end{pmatrix}.$$

- (d) Not similar:  $2\mathbf{I}_2$  is a scalar matrix, so  $\mathbf{P}^{-1}(2\mathbf{I}_2)\mathbf{P} = 2\mathbf{I}_2$  for every invertible  $\mathbf{P}$ , and the second matrix is not  $2\mathbf{I}_2$ . The traces and ranks agree, so invariants alone would not have decided this.  
 (e) Similar. Let  $\mathbf{A}$  be the first matrix.  $\mathbf{A}(1, 1) = (2, 2)$  and  $\mathbf{A}(1, -1) = (0, 0)$ . With  $\mathbf{P} = \begin{pmatrix} 1 & 1 \\ 1 & -1 \end{pmatrix}$  and  $\mathbf{P}^{-1} = \frac{1}{2} \begin{pmatrix} 1 & 1 \\ 1 & -1 \end{pmatrix}$ ,

$$\mathbf{P}^{-1}\mathbf{A}\mathbf{P} = \frac{1}{2} \begin{pmatrix} 1 & 1 \\ 1 & -1 \end{pmatrix} \begin{pmatrix} 2 & 0 \\ 2 & 0 \end{pmatrix} = \frac{1}{2} \begin{pmatrix} 4 & 0 \\ 0 & 0 \end{pmatrix} = \begin{pmatrix} 2 & 0 \\ 0 & 0 \end{pmatrix},$$

where  $\mathbf{A}\mathbf{P} = \begin{pmatrix} 2 & 0 \\ 2 & 0 \end{pmatrix}$  has columns  $\mathbf{A}(1, 1)$  and  $\mathbf{A}(1, -1)$ .

### 3.7.6.3 C. Going deeper

#### Exercise C1: Matrices similar only to themselves.

Let  $n \geq 1$  and  $\mathbf{A} \in M_n(F)$ .

- (a) Prove that  $\mathbf{A}$  is similar only to itself if and only if  $\mathbf{A} = c\mathbf{I}_n$  for some  $c \in F$ .  
 (b) Show that  $\mathbf{N} = \begin{pmatrix} 0 & 1 \\ 0 & 0 \end{pmatrix}$  and  $\mathbf{N}^\top = \begin{pmatrix} 0 & 0 \\ 1 & 0 \end{pmatrix}$  are similar, and explain the answer in terms of the operator  $\mathbf{x} \mapsto \mathbf{N}\mathbf{x}$  and a reordered basis.

*Hint: for (a), use the invertible matrices  $\mathbf{I}_n + \mathbf{E}_{ij}$  with  $i \neq j$ .*

*Solution.*

- (a) ( $\Leftarrow$ ) If  $\mathbf{A} = c\mathbf{I}_n$ , then  $\mathbf{P}^{-1}\mathbf{A}\mathbf{P} = c\mathbf{P}^{-1}\mathbf{P} = c\mathbf{I}_n = \mathbf{A}$  for every invertible  $\mathbf{P}$ , so the only matrix similar to  $\mathbf{A}$  is  $\mathbf{A}$ .  
 ( $\Rightarrow$ ) Suppose  $\mathbf{P}^{-1}\mathbf{A}\mathbf{P} = \mathbf{A}$  for every invertible  $\mathbf{P}$ , that is,  $\mathbf{A}\mathbf{P} = \mathbf{P}\mathbf{A}$ . If  $n = 1$ ,  $\mathbf{A} = (a_{11}) = a_{11}\mathbf{I}_1$ . Let  $n \geq 2$ , and fix  $i \neq j$ . The matrix  $\mathbf{I}_n + \mathbf{E}_{ij}$  is invertible, with inverse  $\mathbf{I}_n - \mathbf{E}_{ij}$ , since  $\mathbf{E}_{ij}\mathbf{E}_{ij} = 0$  for  $i \neq j$ . So  $\mathbf{A}(\mathbf{I}_n + \mathbf{E}_{ij}) = (\mathbf{I}_n + \mathbf{E}_{ij})\mathbf{A}$ , which gives  $\mathbf{A}\mathbf{E}_{ij} = \mathbf{E}_{ij}\mathbf{A}$ . Compare the entries in position  $(k, l)$ : by [Definition 0.9.9](#),

$$(\mathbf{A}\mathbf{E}_{ij})_{kl} = \begin{cases} a_{ki} & l = j \\ 0 & l \neq j \end{cases}, \quad (\mathbf{E}_{ij}\mathbf{A})_{kl} = \begin{cases} a_{jl} & k = i \\ 0 & k \neq i. \end{cases}$$

Taking  $l = j$  and  $k \neq i$  gives  $a_{ki} = 0$ . Taking  $k = i$  and  $l = j$  gives  $a_{ii} = a_{jj}$ . Since  $i \neq j$  were arbitrary (and every index  $i$  has some  $j \neq i$  because  $n \geq 2$ ), all off-diagonal entries of  $\mathbf{A}$  vanish and all diagonal entries are equal to  $c = a_{11}$ . Hence  $\mathbf{A} = c\mathbf{I}_n$ .

- (b) Let  $\mathbf{P} = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}$ , so  $\mathbf{P}^{-1} = \mathbf{P}$ . Then  $\mathbf{NP} = \begin{pmatrix} 1 & 0 \\ 0 & 0 \end{pmatrix}$  and  $\mathbf{P}^{-1}\mathbf{NP} = \begin{pmatrix} 0 & 0 \\ 1 & 0 \end{pmatrix} = \mathbf{N}^\top$ . In terms of operators:  $T\mathbf{x} = \mathbf{N}\mathbf{x}$  has  $T\mathbf{e}_1 = \mathbf{0}$  and  $T\mathbf{e}_2 = \mathbf{e}_1$ . In the reordered basis  $\mathcal{B} = (\mathbf{e}_2, \mathbf{e}_1)$ , the first basis vector goes to the second and the second goes to  $\mathbf{0}$ , so  $[T]_{\mathcal{B}} = \mathbf{N}^\top$ . The same operator has both matrices, as [Theorem 3.7.4](#) predicts.

### Exercise C2: $\mathbf{AB}$ and $\mathbf{BA}$ .

Let  $\mathbf{A}, \mathbf{B} \in M_n(F)$ .

- (a) Prove that if  $\mathbf{A}$  is invertible, then  $\mathbf{AB} \sim \mathbf{BA}$ .  
 (b) Show that (a) fails without the invertibility hypothesis, even though  $\text{tr}(\mathbf{AB}) = \text{tr}(\mathbf{BA})$  always holds.

*Solution.*

- (a) Since  $\mathbf{A}$  is invertible,  $\mathbf{A}^{-1}(\mathbf{AB})\mathbf{A} = (\mathbf{A}^{-1}\mathbf{A})(\mathbf{BA}) = \mathbf{BA}$  by associativity. With  $\mathbf{P} = \mathbf{A}$  this is  $\mathbf{BA} = \mathbf{P}^{-1}(\mathbf{AB})\mathbf{P}$ , so  $\mathbf{AB} \sim \mathbf{BA}$ .  
 (b) Take  $\mathbf{A} = \begin{pmatrix} 1 & 0 \\ 0 & 0 \end{pmatrix}$  and  $\mathbf{B} = \begin{pmatrix} 0 & 1 \\ 0 & 0 \end{pmatrix}$ . Then  $\mathbf{AB} = \begin{pmatrix} 0 & 1 \\ 0 & 0 \end{pmatrix}$ , of rank 1, and  $\mathbf{BA} = \mathbf{0}$ , of rank 0. By [Proposition 3.7.6](#) (a),  $\mathbf{AB} \not\sim \mathbf{BA}$ . Both traces are 0, in agreement with [Theorem 0.9.6](#); the trace simply cannot detect the difference.

### Exercise C3: Every non-zero $2 \times 2$ matrix with square zero.

Let  $\mathbf{N} \in M_2(F)$  with  $\mathbf{N} \neq \mathbf{0}$  and  $\mathbf{N}^2 = \mathbf{0}$ , and let  $T: F^2 \rightarrow F^2$ ,  $T\mathbf{x} = \mathbf{N}\mathbf{x}$ .

- (a) Explain why there is  $\mathbf{v} \in F^2$  with  $\mathbf{N}\mathbf{v} \neq \mathbf{0}$ , and show that  $\mathcal{B} = (\mathbf{N}\mathbf{v}, \mathbf{v})$  is a basis of  $F^2$ .  
 (b) Compute  $[T]_{\mathcal{B}}$ , and deduce that  $\mathbf{N} \sim \begin{pmatrix} 0 & 1 \\ 0 & 0 \end{pmatrix}$ .  
 (c) Hence show that  $\begin{pmatrix} 2 & -4 \\ 1 & -2 \end{pmatrix} \sim \begin{pmatrix} 0 & 1 \\ 0 & 0 \end{pmatrix}$  in  $M_2(\mathbb{R})$ , with an explicit  $\mathbf{P}$ .

*Solution.*

- (a) Since  $\mathbf{N} \neq \mathbf{0}$ , some column  $\mathbf{N}\mathbf{e}_j$  is non-zero ([Theorem 0.9.3](#)); take  $\mathbf{v} = \mathbf{e}_j$ . Let  $a\mathbf{N}\mathbf{v} + b\mathbf{v} = \mathbf{0}$ . Multiplying by  $\mathbf{N}$  and using  $\mathbf{N}^2 = \mathbf{0}$  gives  $b\mathbf{N}\mathbf{v} = \mathbf{0}$ , so  $b = 0$  because  $\mathbf{N}\mathbf{v} \neq \mathbf{0}$  ([Theorem 1.1.9](#)). Then  $a\mathbf{N}\mathbf{v} = \mathbf{0}$  forces  $a = 0$  for the same reason. So  $\mathcal{B}$  is linearly independent of length  $2 = \dim F^2$ , hence a basis by [Theorem 1.6.4](#).  
 (b)  $T(\mathbf{N}\mathbf{v}) = \mathbf{N}^2\mathbf{v} = \mathbf{0}$  and  $T\mathbf{v} = \mathbf{N}\mathbf{v} = 1 \cdot \mathbf{N}\mathbf{v} + 0 \cdot \mathbf{v}$ . The columns of  $[T]_{\mathcal{B}}$  are therefore  $(0, 0)$  and  $(1, 0)$ , so  $[T]_{\mathcal{B}} = \begin{pmatrix} 0 & 1 \\ 0 & 0 \end{pmatrix}$ . Also  $[T]_{\mathcal{E}} = \mathbf{N}$ . By [Theorem 3.7.4](#)  
 (a),  $\mathbf{N} \sim \begin{pmatrix} 0 & 1 \\ 0 & 0 \end{pmatrix}$ .  
 (c) Let  $\mathbf{N} = \begin{pmatrix} 2 & -4 \\ 1 & -2 \end{pmatrix}$ . Then  $\mathbf{N}^2 = \begin{pmatrix} 4-4 & -8+8 \\ 2-2 & -4+4 \end{pmatrix} = \mathbf{0}$  and  $\mathbf{N} \neq \mathbf{0}$ . Take  $\mathbf{v} = \mathbf{e}_1$ ,



so  $\mathbf{N}\mathbf{v} = (2, 1)$ , and  $\mathcal{B} = ((2, 1), (1, 0))$ . By [Theorem 3.7.3](#),  $\mathbf{P} = [\text{id}]_{\mathcal{B}}^{\mathcal{E}} = \begin{pmatrix} 2 & 1 \\ 1 & 0 \end{pmatrix}$  satisfies  $\mathbf{P}^{-1}\mathbf{N}\mathbf{P} = \begin{pmatrix} 0 & 1 \\ 0 & 0 \end{pmatrix}$ . Check:  $\mathbf{P}^{-1} = \begin{pmatrix} 0 & 1 \\ 1 & -2 \end{pmatrix}$  by [Theorem 0.9.8](#) (here  $ad - bc = -1$ ),  $\mathbf{N}\mathbf{P} = \begin{pmatrix} 0 & 2 \\ 0 & 1 \end{pmatrix}$ , and  $\mathbf{P}^{-1}\mathbf{N}\mathbf{P} = \begin{pmatrix} 0 & 1 \\ 0 & 0 \end{pmatrix}$ .

## 3.8 Rank, Equivalence and Factorizations

Similarity compares two matrices of one operator, where the same basis is used for input and output, and the last section showed that deciding it is hard. This section asks what happens when the two bases may be chosen independently, as they may for any map  $T: V \rightarrow W$ . The answer is strikingly simple: the rank is then the only thing that matters. On the way we meet the column–row factorization, which writes every matrix as a product through a space of dimension equal to its rank, and we collect the basic inequalities for the rank of a composition.

### 3.8.1 The column–row factorization

In Chapter 2 we computed the rank of a matrix by row reduction, and the reduction gave more than a number. It produced a basis of the column space, the pivot columns of  $\mathbf{A}$  ([Theorem 2.5.2](#)), and it expressed every other column in terms of them, with coefficients read off the reduced row echelon form. Recording both pieces of information as matrices gives a factorization.

#### Theorem 3.8.1 — Column–Row Factorization

Let  $\mathbf{A} \in M_{m \times n}(F)$  have rank  $r \geq 1$ , with pivot columns  $j_1 < \cdots < j_r$ . Let  $\mathbf{C} \in M_{m \times r}(F)$  be the matrix whose columns are the pivot columns  $\mathbf{a}_{j_1}, \dots, \mathbf{a}_{j_r}$  of  $\mathbf{A}$ , and let  $\mathbf{R} \in M_{r \times n}(F)$  consist of the first  $r$  rows of the reduced row echelon form of  $\mathbf{A}$ , that is, its non-zero rows. Then

$$\mathbf{A} = \mathbf{C}\mathbf{R}.$$

Moreover, the columns of  $\mathbf{C}$  are linearly independent and the rows of  $\mathbf{R}$  are linearly independent.

*Idea.* Column  $l$  of  $\mathbf{C}\mathbf{R}$  is  $\mathbf{C}$  times column  $l$  of  $\mathbf{R}$ , that is, a combination of the pivot columns of  $\mathbf{A}$  with the entries of column  $l$  of the RREF as coefficients. For a pivot column the coefficients are a standard basis vector, and for a free column they are exactly the coefficients that [Theorem 2.5.2](#) (b) provides. So the factorization is that theorem written as a matrix identity.

*Proof.* Let  $\mathbf{E}$  be the RREF of  $\mathbf{A}$ , with entries  $e_{kl}$ . By [Theorem 2.5.5](#),  $\mathbf{E}$  has exactly  $r$  pivots, so its non-zero rows are rows  $1, \dots, r$ . Fix a column index  $l$ . By [Theorem 0.9.3](#), column  $l$  of  $\mathbf{C}\mathbf{R}$  is

$$\mathbf{C} \begin{pmatrix} e_{1l} \\ \vdots \\ e_{rl} \end{pmatrix} = e_{1l}\mathbf{a}_{j_1} + \cdots + e_{rl}\mathbf{a}_{j_r}.$$

*Case 1:  $l = j_i$  is a pivot column.* By Lemma 2.3.3 (a), column  $j_i$  of  $\mathbf{E}$  is  $\mathbf{e}_i$ , so the combination is  $\mathbf{a}_{j_i} = \mathbf{a}_i$ .

*Case 2:  $l$  is a free column.* Let  $t$  be the number of pivot columns to the left of  $l$ . For  $k > t$ , the leading entry of row  $k$  of  $\mathbf{E}$  lies in column  $j_k > l$ , so  $e_{kl} = 0$ . The combination is therefore  $e_{1l}\mathbf{a}_{j_1} + \cdots + e_{tl}\mathbf{a}_{j_t}$ , which equals  $\mathbf{a}_l$  by Theorem 2.5.2 (b).

In both cases column  $l$  of  $\mathbf{CR}$  is column  $l$  of  $\mathbf{A}$ ; hence  $\mathbf{A} = \mathbf{CR}$ . The columns of  $\mathbf{C}$  form a basis of  $\text{col}(\mathbf{A})$  by Theorem 2.5.2 (a), so they are independent. The rows of  $\mathbf{R}$  are the non-zero rows of  $\mathbf{E}$ , which form a basis of  $\text{row}(\mathbf{A})$  by Theorem 2.5.3, so they are independent. This proves the theorem. ■

In words: every matrix of rank  $r$  is a “tall” matrix with independent columns times a “wide” matrix with independent rows, and the middle size is exactly  $r$ . The factorization cannot be squeezed further.

### Proposition 3.8.2 — Rank as the size of the thinnest factorization

Let  $\mathbf{A} \in M_{m \times n}(F)$  be non-zero. Then  $\text{rank } \mathbf{A}$  is the **smallest** integer  $s \geq 1$  such that  $\mathbf{A} = \mathbf{BD}$  for some  $\mathbf{B} \in M_{m \times s}(F)$  and  $\mathbf{D} \in M_{s \times n}(F)$ .

*Proof.* Let  $r = \text{rank } \mathbf{A}$ ; then  $r \geq 1$  since  $\mathbf{A} \neq 0$ . By Theorem 3.8.1, a factorization with  $s = r$  exists. Conversely, suppose  $\mathbf{A} = \mathbf{BD}$  with  $\mathbf{B} \in M_{m \times s}(F)$ . By Theorem 2.5.9,  $r = \text{rank}(\mathbf{BD}) \leq \text{rank } \mathbf{B}$ , and  $\text{rank } \mathbf{B} \leq s$  because  $\text{col}(\mathbf{B})$  is spanned by  $s$  columns (Theorem 1.6.3). Hence  $r \leq s$ . ■

This gives a second view of “row rank equals column rank”. Let  $\mathbf{A} \neq 0$ , and let  $\mathbf{B}$  be the matrix whose  $s = \dim \text{col}(\mathbf{A})$  columns form a basis of  $\text{col}(\mathbf{A})$ . Each column  $\mathbf{a}_j$  of  $\mathbf{A}$  is  $\mathbf{B}\mathbf{d}_j$  for its coordinate column  $\mathbf{d}_j$  (Theorem 0.9.3), so  $\mathbf{A} = \mathbf{BD}$  with  $\mathbf{D} = (\mathbf{d}_1 \ \cdots \ \mathbf{d}_n)$ . Then every row of  $\mathbf{A}$  is a row of  $\mathbf{B}$  times  $\mathbf{D}$ , a combination of the  $s$  rows of  $\mathbf{D}$  (Theorem 0.9.2 (3) and the row form of Theorem 0.9.3), so  $\dim \text{row}(\mathbf{A}) \leq s = \dim \text{col}(\mathbf{A})$ . Applying the same argument to  $\mathbf{A}^\top$  gives the reverse inequality. In the language of maps, the factorization says that  $\mathbf{x} \mapsto \mathbf{Ax}$  can be done in two steps,  $F^n \rightarrow F^r \rightarrow F^m$ : first a surjection onto a space of dimension  $r$ , then an injection.

#### Example. (A column–row factorization)

Find the column–row factorization of  $\mathbf{A} = \begin{pmatrix} 1 & 2 & 0 & 3 \\ 2 & 4 & 1 & 4 \\ 3 & 6 & 1 & 7 \end{pmatrix} \in M_{3 \times 4}(\mathbb{R})$ .

*Solution.* Row reduce. Subtracting 2 times row 1 from row 2 and 3 times row 1 from row 3 gives rows  $(1, 2, 0, 3)$ ,  $(0, 0, 1, -2)$ ,  $(0, 0, 1, -2)$ . Subtracting row 2 from row 3 gives the RREF

$$\begin{pmatrix} 1 & 2 & 0 & 3 \\ 0 & 0 & 1 & -2 \\ 0 & 0 & 0 & 0 \end{pmatrix}.$$

The pivot columns are columns 1 and 3, so  $r = 2$ , and by [Theorem 3.8.1](#)

$$\mathbf{A} = \mathbf{C}\mathbf{R} = \begin{pmatrix} 1 & 0 \\ 2 & 1 \\ 3 & 1 \end{pmatrix} \begin{pmatrix} 1 & 2 & 0 & 3 \\ 0 & 0 & 1 & -2 \end{pmatrix}.$$

Check by rows: row 2 of  $\mathbf{C}\mathbf{R}$  is  $2(1, 2, 0, 3) + (0, 0, 1, -2) = (2, 4, 1, 4)$ , and row 3 is  $3(1, 2, 0, 3) + (0, 0, 1, -2) = (3, 6, 1, 7)$ . The last column of  $\mathbf{R}$  says  $\mathbf{a}_4 = 3\mathbf{a}_1 - 2\mathbf{a}_3$ , and indeed  $3(1, 2, 3) - 2(0, 1, 1) = (3, 4, 7)$ .

## 3.8.2 Equivalent matrices

Go back to the change-of-basis square for a map  $T: V \rightarrow W$  ([Theorem 3.7.3](#)):

$$[T]_{\mathcal{B}'}^{\mathcal{C}'} = [\text{id}]_{\mathcal{C}'}^{\mathcal{C}'} [T]_{\mathcal{B}}^{\mathcal{C}} [\text{id}]_{\mathcal{B}'}^{\mathcal{B}}.$$

When  $V \neq W$ , nothing ties the basis of  $V$  to the basis of  $W$ . So the two outer matrices are **unrelated** invertible matrices, of sizes  $m \times m$  and  $n \times n$ . Any two matrices of the same map are related in this way, and this recurring shape deserves a name.

*Two matrices are equivalent when they describe the same linear map, with the input basis and the output basis each changed freely.*

### Definition 3.8.1 — Equivalent matrices

Let  $\mathbf{A}, \mathbf{B} \in M_{m \times n}(F)$ . We say  $\mathbf{A}$  is **equivalent** to  $\mathbf{B}$  if there **exist** invertible matrices  $\mathbf{Q} \in M_m(F)$  and  $\mathbf{P} \in M_n(F)$  such that

$$\mathbf{B} = \mathbf{Q}\mathbf{A}\mathbf{P}.$$

In words: we may multiply on the left by **any** invertible matrix and on the right by **any** invertible matrix, and the two need not be related. Both sizes are forced:  $\mathbf{Q}$  acts on the  $m$  rows and  $\mathbf{P}$  on the  $n$  columns. This is an equivalence relation on  $M_{m \times n}(F)$ :  $\mathbf{A} = \mathbf{I}_m \mathbf{A} \mathbf{I}_n$ ; if  $\mathbf{B} = \mathbf{Q}\mathbf{A}\mathbf{P}$  then  $\mathbf{A} = \mathbf{Q}^{-1} \mathbf{B} \mathbf{P}^{-1}$ ; and if also  $\mathbf{C} = \mathbf{Q}' \mathbf{B} \mathbf{P}'$ , then  $\mathbf{C} = (\mathbf{Q}' \mathbf{Q}) \mathbf{A} (\mathbf{P} \mathbf{P}')$ , where products of invertible matrices are invertible by [Theorem 0.9.7](#) (part 3).

**Examples.**

- **Row equivalence.** If  $\mathbf{B}$  is obtained from  $\mathbf{A}$  by row operations, then  $\mathbf{B} = \mathbf{E}\mathbf{A}$  with  $\mathbf{E}$  invertible ([Theorem 2.4.7](#)), so  $\mathbf{A}$  and  $\mathbf{B}$  are equivalent with  $\mathbf{Q} = \mathbf{E}$  and  $\mathbf{P} = \mathbf{I}_n$ . Column operations correspond to multiplying on the right, and they are allowed too.
- **Similarity.** If  $\mathbf{B} = \mathbf{P}^{-1} \mathbf{A} \mathbf{P}$ , then  $\mathbf{A}$  and  $\mathbf{B}$  are equivalent, with  $\mathbf{Q} = \mathbf{P}^{-1}$ .
- **The identity and a swap.**  $\mathbf{I}_2$  and  $S = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}$  are equivalent, since  $S = S \mathbf{I}_2$ . They are not similar, because  $\mathbf{I}_2$  is similar only to itself.
- **Degenerate case.** The zero matrix  $0_{m \times n}$  is equivalent only to itself, since  $\mathbf{Q}0\mathbf{P} = 0$ . It is the matrix of the zero map in every pair of bases.

**Non-example by minimal change.** Keep  $\mathbf{B} = \mathbf{Q}\mathbf{A}\mathbf{P}$  but drop the requirement that  $\mathbf{Q}$  be invertible. Then every matrix could be sent to the zero matrix with  $\mathbf{Q} = 0$ , and the relation would not even be symmetric:  $0 = 0 \cdot \mathbf{I}_2 \cdot \mathbf{I}_2$ , but  $\mathbf{I}_2 \neq \mathbf{Q}0\mathbf{P}$  for any  $\mathbf{Q}, \mathbf{P}$ . The clause “invertible” is what makes  $\mathbf{Q}\mathbf{A}\mathbf{P}$  a change of basis rather than an arbitrary composition.

**Warning. Equivalent is much weaker than similar.** The matrices  $\mathbf{I}_2$  and  $J = \begin{pmatrix} 1 & 1 \\ 0 & 1 \end{pmatrix}$  are equivalent, since  $J = J\mathbf{I}_2\mathbf{I}_2$  and  $J$  is invertible, but they are not similar (see the warning in the previous section). For square matrices, “equivalent” allows a different change of basis on each side, and that freedom destroys almost all the structure an operator has.

How much structure survives? The rank does, since multiplying by invertible matrices does not change it ([Theorem 2.5.9](#)). The theorem below says that nothing else survives. The target is the simplest matrix of each rank.

For  $0 \leq r \leq \min(m, n)$ , let  $\mathbf{N}_r \in M_{m \times n}(F)$  be the block matrix

$$\mathbf{N}_r := \begin{pmatrix} \mathbf{I}_r & 0 \\ 0 & 0 \end{pmatrix},$$

with 1 in positions  $(1, 1), \dots, (r, r)$  and 0 everywhere else. Its first  $r$  columns are  $\mathbf{e}_1, \dots, \mathbf{e}_r \in F^m$ , and its remaining columns are zero.

**Theorem 3.8.3 — Rank Normal Form** (a) Let  $V$  and  $W$  be finite-dimensional with  $\dim V = n$  and  $\dim W = m$ , and let  $T: V \rightarrow W$  be linear of rank  $r$ . Then there are bases  $\mathcal{B}$  of  $V$  and  $\mathcal{C}$  of  $W$  such that  $[T]_{\mathcal{B}}^{\mathcal{C}} = \mathbf{N}_r$ .  
 (b) Every  $\mathbf{A} \in M_{m \times n}(F)$  is equivalent to  $\mathbf{N}_r$ , where  $r = \text{rank } \mathbf{A}$ .  
 (c) Two matrices  $\mathbf{A}, \mathbf{B} \in M_{m \times n}(F)$  are equivalent if and only if  $\text{rank } \mathbf{A} = \text{rank } \mathbf{B}$ .

*Idea.* Work backwards from the goal.  $[T]_{\mathcal{B}}^{\mathcal{C}} = \mathbf{N}_r$  says:  $T$  sends the first  $r$  vectors of  $\mathcal{B}$  to the first  $r$  vectors of  $\mathcal{C}$ , and sends the last  $n - r$  vectors of  $\mathcal{B}$  to  $\mathbf{0}$ . So ① the last  $n - r$  vectors of  $\mathcal{B}$  should be a basis of  $\ker T$ , and by Rank–Nullity that is exactly the right number; ② the first  $r$  vectors of  $\mathcal{B}$  extend it to a basis of  $V$ ; ③ their images are forced to be the first  $r$  vectors of  $\mathcal{C}$ , and we must show they are independent, which follows by counting; ④ extend them to a basis of  $W$ . These are the bases of the proof of Rank–Nullity, one on each side.

*Proof.*

- (a) By the Rank–Nullity Theorem ([Theorem 3.3.1](#)),  $\dim \ker T = n - r$ . Let  $(\mathbf{u}_{r+1}, \dots, \mathbf{u}_n)$  be a basis of  $\ker T$ , a subspace of  $V$  by [Theorem 3.2.1](#). By the Basis Extension Theorem ([Theorem 1.6.5](#)) there are  $\mathbf{u}_1, \dots, \mathbf{u}_r \in V$  such that  $\mathcal{B} = (\mathbf{u}_1, \dots, \mathbf{u}_r, \mathbf{u}_{r+1}, \dots, \mathbf{u}_n)$  is a basis of  $V$ ; the new vectors may be listed first, since reordering a basis gives a basis. Put  $\mathbf{w}_i = T\mathbf{u}_i$  for  $i = 1, \dots, r$ .

By [Theorem 3.2.4](#),  $\text{im } T = \text{span}(T\mathbf{u}_1, \dots, T\mathbf{u}_n)$ . Since  $T\mathbf{u}_j = \mathbf{0}$  for  $j > r$ , removing these vectors does not change the span, so  $(\mathbf{w}_1, \dots, \mathbf{w}_r)$  spans  $\text{im } T$ . This list has length  $r = \dim \text{im } T$ , so it is a basis of  $\text{im } T$  by [Theorem 1.6.4](#) (b); in particular it is linearly independent in  $W$ . By the Basis Extension Theorem there are  $\mathbf{w}_{r+1}, \dots, \mathbf{w}_m \in W$  such that  $\mathcal{C} = (\mathbf{w}_1, \dots, \mathbf{w}_m)$  is a basis of  $W$ .

Now read off the columns of  $[T]_{\mathcal{B}}^{\mathcal{C}}$  ([Definition 3.6.1](#)). For  $i \leq r$ ,  $T\mathbf{u}_i = \mathbf{w}_i$ , whose  $\mathcal{C}$ -coordinate vector is  $\mathbf{e}_i$ . For  $j > r$ ,  $T\mathbf{u}_j = \mathbf{0}$ , whose coordinate vector is  $\mathbf{0}$ . These are the columns of  $\mathbf{N}_r$ .

- (b) Let  $T: F^n \rightarrow F^m$ ,  $T\mathbf{x} = \mathbf{A}\mathbf{x}$ , and let  $\mathcal{E}_n, \mathcal{E}_m$  be the standard bases. The  $j$ -th column of  $[T]_{\mathcal{E}_n}^{\mathcal{E}_m}$  is  $[\mathbf{A}\mathbf{e}_j]_{\mathcal{E}_m} = \mathbf{A}\mathbf{e}_j$ , the  $j$ -th column of  $\mathbf{A}$  ([Theorem 0.9.3](#)), so

$[T]_{\mathcal{E}_n}^{\mathcal{E}_m} = \mathbf{A}$ . By [Theorem 3.6.5](#),  $\text{rank } T = \text{rank } \mathbf{A} = r$ . Take  $\mathcal{B}, \mathcal{C}$  as in (a). By [Theorem 3.7.3](#),

$$\mathbf{N}_r = [T]_{\mathcal{B}}^{\mathcal{C}} = [\text{id}]_{\mathcal{E}_m}^{\mathcal{C}} \mathbf{A} [\text{id}]_{\mathcal{B}}^{\mathcal{E}_n},$$

and both outer matrices are invertible by [Theorem 3.7.1](#) (c). Hence  $\mathbf{A}$  is equivalent to  $\mathbf{N}_r$ .

- (c) ( $\Rightarrow$ ) If  $\mathbf{B} = \mathbf{QAP}$  with  $\mathbf{Q}, \mathbf{P}$  invertible, then  $\text{rank } \mathbf{B} = \text{rank } \mathbf{A}$  by [Theorem 2.5.9](#). ( $\Leftarrow$ ) If  $\text{rank } \mathbf{A} = \text{rank } \mathbf{B} = r$ , then by (b) both  $\mathbf{A}$  and  $\mathbf{B}$  are equivalent to  $\mathbf{N}_r$ . Since equivalence is symmetric and transitive,  $\mathbf{A}$  is equivalent to  $\mathbf{B}$ . This proves the theorem. ■

So the equivalence classes of  $M_{m \times n}(F)$  are exactly the sets of matrices of rank  $r$ , for  $r = 0, 1, \dots, \min(m, n)$ : there are  $\min(m, n) + 1$  of them, whatever the field. Every linear map between finite-dimensional spaces looks like “keep  $r$  coordinates and delete the rest” once the bases on both sides are chosen well. **The right pair of bases makes any single map trivial.**

*Remark.* Contrast this with operators. For  $T \in \mathcal{L}(V)$  we must use **one** basis for input and output, and the relation becomes similarity. There the rank is far from the whole story:  $\mathbf{I}_2$  and  $\begin{pmatrix} 1 & 1 \\ 0 & 1 \end{pmatrix}$  have the same rank but are not similar. Classifying operators up to a change of one basis is a much harder problem, answered in Chapter 9 by the Jordan and rational canonical forms.

The proof is also an algorithm. To find  $\mathbf{Q}$  and  $\mathbf{P}$  for a concrete matrix, compute a basis of  $\text{nul}(\mathbf{A})$ , extend it, and take images.

**Example. (Bringing a  $2 \times 3$  matrix to normal form)**

Let  $\mathbf{A} = \begin{pmatrix} 1 & 2 & 1 \\ 2 & 4 & 3 \end{pmatrix} \in M_{2 \times 3}(\mathbb{R})$ . Find invertible  $\mathbf{Q} \in M_2(\mathbb{R})$  and  $\mathbf{P} \in M_3(\mathbb{R})$  with  $\mathbf{QAP} = \mathbf{N}_r$ .

*Solution.* Subtracting 2 times row 1 from row 2 gives  $\begin{pmatrix} 1 & 2 & 1 \\ 0 & 0 & 1 \end{pmatrix}$ , so  $r = \text{rank } \mathbf{A} = 2$ , and the solutions of  $\mathbf{Ax} = \mathbf{0}$  satisfy  $x_3 = 0$ ,  $x_1 = -2x_2$ . Hence  $\text{nul}(\mathbf{A}) = \text{span}((-2, 1, 0))$ , as Rank–Nullity predicts ( $3 - 2 = 1$ ).

Following the proof, let  $\mathbf{u}_3 = (-2, 1, 0)$ , and extend with  $\mathbf{u}_1 = \mathbf{e}_1$ ,  $\mathbf{u}_2 = \mathbf{e}_3$ . The list  $(\mathbf{e}_1, \mathbf{e}_3, \mathbf{u}_3)$  is independent: in  $a\mathbf{e}_1 + b\mathbf{e}_3 + c(-2, 1, 0) = (a - 2c, c, b) = \mathbf{0}$ , the second and third entries give  $c = b = 0$ , and then  $a = 0$ . So it is a basis  $\mathcal{B}$  of  $\mathbb{R}^3$ . The images are  $\mathbf{w}_1 = \mathbf{Ae}_1 = (1, 2)$  and  $\mathbf{w}_2 = \mathbf{Ae}_3 = (1, 3)$ , which already form a basis  $\mathcal{C}$  of  $\mathbb{R}^2$  since  $r = m = 2$ . Therefore

$$\mathbf{P} = [\text{id}]_{\mathcal{B}}^{\mathcal{E}_3} = \begin{pmatrix} 1 & 0 & -2 \\ 0 & 0 & 1 \\ 0 & 1 & 0 \end{pmatrix}, \quad \mathbf{Q} = [\text{id}]_{\mathcal{E}_2}^{\mathcal{C}} = \begin{pmatrix} 1 & 1 \\ 2 & 3 \end{pmatrix}^{-1} = \begin{pmatrix} 3 & -1 \\ -2 & 1 \end{pmatrix},$$

using [Theorem 3.7.1](#) (c) and [Theorem 0.9.8](#). Check:  $\mathbf{AP} = \begin{pmatrix} 1 & 1 & 0 \\ 2 & 3 & 0 \end{pmatrix}$  (the columns are

$\mathbf{w}_1, \mathbf{w}_2, \mathbf{0}$ ), and

$$\mathbf{QAP} = \begin{pmatrix} 3 & -1 \\ -2 & 1 \end{pmatrix} \begin{pmatrix} 1 & 1 & 0 \\ 2 & 3 & 0 \end{pmatrix} = \begin{pmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \end{pmatrix} = \mathbf{N}_2.$$

The pair  $(\mathbf{Q}, \mathbf{P})$  is far from unique; any choice of extension gives another.

*Quick check.* How many equivalence classes does  $M_{3 \times 2}(F)$  have, and which is the class of  $\begin{pmatrix} 1 & 2 \\ 2 & 4 \\ 0 & 0 \end{pmatrix}$ ?

*Solution.* Ranks in  $M_{3 \times 2}(F)$  range over 0, 1, 2, so there are  $\min(3, 2) + 1 = 3$  classes by [Theorem 3.8.3](#) (c). The given matrix is non-zero and its second column is twice the first, so its rank is 1, and it lies in the class of  $\mathbf{N}_1$ : all matrices of rank 1.

### 3.8.3 Rank inequalities for compositions

The rank of a composition cannot exceed the rank of either factor, and it cannot fall too far either. Both facts come from the same observation:  $ST$  is  $S$  applied to the subspace  $\text{im } T$ . For a subspace  $X \subseteq W$  and a linear map  $S: W \rightarrow U$ , the **restriction**  $S|_X: X \rightarrow U$ ,  $\mathbf{x} \mapsto S\mathbf{x}$ , is linear, since its values are those of  $S$ . Its kernel is  $\ker S \cap X$ .

#### Theorem 3.8.4 — Rank of a Composition

Let  $T: V \rightarrow W$  and  $S: W \rightarrow U$  be linear, with  $V$  and  $W$  finite-dimensional. Then

$$\text{rank}(ST) \leq \min(\text{rank } S, \text{rank } T).$$

*Proof.* Consider the restriction  $S' = S|_{\text{im } T}: \text{im } T \rightarrow U$ . For  $\mathbf{v} \in V$ ,  $(ST)(\mathbf{v}) = S(T\mathbf{v}) = S'(T\mathbf{v})$ , so  $\text{im}(ST) \subseteq \text{im } S'$ ; conversely every  $S'\mathbf{w}$  with  $\mathbf{w} = T\mathbf{v} \in \text{im } T$  equals  $(ST)(\mathbf{v})$ . Hence  $\text{im}(ST) = \text{im } S'$ . Since  $\text{im } T$  is finite-dimensional (a subspace of  $W$ , [Theorem 1.6.7](#)), Rank–Nullity ([Theorem 3.3.1](#)) applied to  $S'$  gives

$$\text{rank}(ST) = \dim \text{im } S' = \dim \text{im } T - \dim \ker S' \leq \text{rank } T.$$

Also  $\text{im}(ST) = \text{im } S' \subseteq \text{im } S$ , and  $\text{im } S$  is finite-dimensional by Rank–Nullity for  $S$ , so  $\text{rank}(ST) \leq \text{rank } S$  by [Theorem 1.6.7](#). This proves the theorem. ■

The lower bound uses the same restriction, but now we control its kernel instead of ignoring it.

#### Theorem 3.8.5 — Sylvester’s Rank Inequality

Let  $T: V \rightarrow W$  and  $S: W \rightarrow U$  be linear, with  $V$  and  $W$  finite-dimensional. Then

$$\text{rank}(ST) \geq \text{rank } S + \text{rank } T - \dim W.$$

In matrix form: for  $\mathbf{A} \in M_{m \times n}(F)$  and  $\mathbf{B} \in M_{n \times p}(F)$ ,  $\text{rank}(\mathbf{AB}) \geq \text{rank } \mathbf{A} + \text{rank } \mathbf{B} - n$ .

*Idea.* Of the  $\text{rank } T$  dimensions that reach  $W$ , the map  $S$  can destroy only those lying in  $\ker S$ , and  $\ker S$  has dimension  $\dim W - \text{rank } S$ . Rank–Nullity for the restriction  $S|_{\text{im } T}$  turns this sentence into an inequality.

*Proof.* Let  $S' = S|_{\text{im } T}$ , so that  $\text{im } S' = \text{im}(ST)$  as in the proof of [Theorem 3.8.4](#), and  $\ker S' = \ker S \cap \text{im } T \subseteq \ker S$ . By Rank–Nullity for  $S'$  and then for  $S$ ,

$$\begin{aligned} \text{rank } T &= \text{rank}(ST) + \dim(\ker S \cap \text{im } T) \\ &\leq \text{rank}(ST) + \dim \ker S = \text{rank}(ST) + \dim W - \text{rank } S, \end{aligned}$$

where the inequality is [Theorem 1.6.7](#). Rearranging gives the claim. For matrices, apply it to  $S\mathbf{y} = \mathbf{A}\mathbf{y}$  on  $W = F^n$  and  $T\mathbf{x} = \mathbf{B}\mathbf{x}$  on  $V = F^p$ : the matrix of  $ST$  in standard bases is  $\mathbf{AB}$  by [Theorem 3.6.2](#), and ranks of maps and matrices agree by [Theorem 3.6.5](#). ■

If you worked [Exercise \(C1: A Lower Bound for the Rank of a Composite\)](#), you have seen this proof already; we record the result here as a theorem so that later arguments can cite it. A typical use: if  $\mathbf{AB} = 0$  with  $\mathbf{A} \in M_{m \times n}(F)$  and  $\mathbf{B} \in M_{n \times p}(F)$ , then  $0 \geq \text{rank } \mathbf{A} + \text{rank } \mathbf{B} - n$ , that is,

$$\text{rank } \mathbf{A} + \text{rank } \mathbf{B} \leq n.$$

For instance, there are no  $\mathbf{A}, \mathbf{B} \in M_3(\mathbb{R})$  of rank 2 with  $\mathbf{AB} = 0$ .

**Warning.** The rank of a product depends on the order of the factors. For  $\mathbf{A} = \begin{pmatrix} 1 & 0 \\ 0 & 0 \end{pmatrix}$  and  $\mathbf{B} = \begin{pmatrix} 0 & 1 \\ 0 & 0 \end{pmatrix}$ ,  $\mathbf{AB} = \mathbf{B}$  has rank 1 but  $\mathbf{BA} = 0$  has rank 0. [Theorem 2.5.9](#) and Sylvester’s inequality bound  $\text{rank}(\mathbf{AB})$  from both sides; they do not determine it, and they say nothing that would make  $\text{rank}(\mathbf{AB}) = \text{rank}(\mathbf{BA})$ .

## 3.8.4 Exercises

### 3.8.4.1 A. Check your understanding

#### Exercise A1.

- Define what it means for  $\mathbf{A}, \mathbf{B} \in M_{m \times n}(F)$  to be equivalent.
- State the Rank Normal Form theorem for a matrix  $\mathbf{A} \in M_{m \times n}(F)$ .
- True or false: equivalent square matrices are similar. Justify your answer.
- True or false:  $\text{rank}(\mathbf{AB}) = \text{rank}(\mathbf{BA})$  for all  $\mathbf{A}, \mathbf{B} \in M_n(F)$ . Justify your answer.
- How many equivalence classes does  $M_{2 \times 5}(\mathbb{F}_7)$  have?
- In the proof of the normal form for  $T: V \rightarrow W$ , which vectors form the end of the basis of  $V$ , and which vectors form the start of the basis of  $W$ ?

*Solution.*

- There are invertible  $\mathbf{Q} \in M_m(F)$  and  $\mathbf{P} \in M_n(F)$  with  $\mathbf{B} = \mathbf{QAP}$  ([Defini-](#)

tion 3.8.1).

- (b) If  $r = \text{rank } \mathbf{A}$ , there are invertible  $\mathbf{Q}, \mathbf{P}$  with  $\mathbf{QAP} = \begin{pmatrix} \mathbf{I}_r & 0 \\ 0 & 0 \end{pmatrix}$ ; and two  $m \times n$  matrices are equivalent if and only if they have the same rank ([Theorem 3.8.3](#)).
- (c) False.  $\mathbf{I}_2$  and  $\begin{pmatrix} 1 & 1 \\ 0 & 1 \end{pmatrix}$  both have rank 2, so they are equivalent, but  $\mathbf{I}_2$  is similar only to itself.
- (d) False. With  $\mathbf{A} = \begin{pmatrix} 1 & 0 \\ 0 & 0 \end{pmatrix}$  and  $\mathbf{B} = \begin{pmatrix} 0 & 1 \\ 0 & 0 \end{pmatrix}$ ,  $\text{rank}(\mathbf{AB}) = 1$  and  $\text{rank}(\mathbf{BA}) = 0$ .
- (e) The possible ranks are 0, 1, 2, so there are 3 classes. The field plays no role.
- (f) The last  $n - r$  vectors of the basis of  $V$  form a basis of  $\ker T$ . The first  $r$  vectors of the basis of  $W$  are the images of the first  $r$  basis vectors of  $V$ , and they form a basis of  $\text{im } T$ .

### 3.8.4.2 B. Practice

#### Exercise B1: A column–row factorization.

Find the column–row factorization of

$$\mathbf{A} = \begin{pmatrix} 1 & -1 & 2 & 0 \\ 2 & -2 & 5 & 1 \\ 1 & -1 & 3 & 1 \end{pmatrix} \in M_{3 \times 4}(\mathbb{R}).$$

Hence write the second and fourth columns of  $\mathbf{A}$  as combinations of its pivot columns.

*Solution.* Subtract 2 times row 1 from row 2 and row 1 from row 3: the rows become  $(1, -1, 2, 0)$ ,  $(0, 0, 1, 1)$ ,  $(0, 0, 1, 1)$ . Subtract row 2 from row 3, and then 2 times row 2 from row 1:

$$\text{RREF} = \begin{pmatrix} 1 & -1 & 0 & -2 \\ 0 & 0 & 1 & 1 \\ 0 & 0 & 0 & 0 \end{pmatrix}.$$

The pivot columns are 1 and 3, so  $\text{rank } \mathbf{A} = 2$ , and by [Theorem 3.8.1](#)

$$\mathbf{A} = \begin{pmatrix} 1 & 2 \\ 2 & 5 \\ 1 & 3 \end{pmatrix} \begin{pmatrix} 1 & -1 & 0 & -2 \\ 0 & 0 & 1 & 1 \end{pmatrix}.$$

Check by rows:  $2(1, -1, 0, -2) + 5(0, 0, 1, 1) = (2, -2, 5, 1)$  and  $(1, -1, 0, -2) + 3(0, 0, 1, 1) = (1, -1, 3, 1)$ ; row 1 is  $(1, -1, 0, -2) + 2(0, 0, 1, 1)$ . Reading columns 2 and 4 of  $\mathbf{R}$ :  $\mathbf{a}_2 = -\mathbf{a}_1$  and  $\mathbf{a}_4 = -2\mathbf{a}_1 + \mathbf{a}_3$ . Indeed  $-2(1, 2, 1) + (2, 5, 3) = (0, 1, 1) = \mathbf{a}_4$ .

#### Exercise B2: Normal form of a rank-one matrix.

Let  $\mathbf{A} = \begin{pmatrix} 1 & -1 & 2 \\ -2 & 2 & -4 \end{pmatrix} \in M_{2 \times 3}(\mathbb{R})$ . Find invertible  $\mathbf{Q} \in M_2(\mathbb{R})$  and  $\mathbf{P} \in M_3(\mathbb{R})$  such that  $\mathbf{QAP} = \begin{pmatrix} 1 & 0 & 0 \\ 0 & 0 & 0 \end{pmatrix}$ , following the proof of [Theorem 3.8.3](#).



*Solution.* Row 2 is  $-2$  times row 1, and  $\mathbf{A} \neq \mathbf{0}$ , so  $\text{rank } \mathbf{A} = 1$ . The null space is  $\{x_1 - x_2 + 2x_3 = 0\}$ ; setting the free variables  $x_2, x_3$  to  $(1, 0)$  and  $(0, 1)$  gives the basis  $\mathbf{u}_2 = (1, 1, 0)$ ,  $\mathbf{u}_3 = (-2, 0, 1)$ . Extend with  $\mathbf{u}_1 = \mathbf{e}_1$ : in  $a\mathbf{e}_1 + b\mathbf{u}_2 + c\mathbf{u}_3 = (a+b-2c, b, c) = \mathbf{0}$ , the last two entries give  $b = c = 0$  and then  $a = 0$ , so  $\mathcal{B} = (\mathbf{e}_1, \mathbf{u}_2, \mathbf{u}_3)$  is a basis of  $\mathbb{R}^3$ . Its first image is  $\mathbf{w}_1 = \mathbf{A}\mathbf{e}_1 = (1, -2)$ , which we extend to the basis  $\mathcal{C} = ((1, -2), \mathbf{e}_2)$  of  $\mathbb{R}^2$  (independent, since  $a(1, -2) + b(0, 1) = (a, -2a + b) = \mathbf{0}$  forces  $a = b = 0$ ). Then

$$\mathbf{P} = \begin{pmatrix} 1 & 1 & -2 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{pmatrix}, \quad \mathbf{Q} = \begin{pmatrix} 1 & 0 \\ -2 & 1 \end{pmatrix}^{-1} = \begin{pmatrix} 1 & 0 \\ 2 & 1 \end{pmatrix}.$$

Check:  $\mathbf{A}\mathbf{P}$  has columns  $\mathbf{A}\mathbf{e}_1 = (1, -2)$ ,  $\mathbf{A}\mathbf{u}_2 = \mathbf{0}$ ,  $\mathbf{A}\mathbf{u}_3 = \mathbf{0}$ , and  $\mathbf{Q}(1, -2) = (1, 0)$ . So  $\mathbf{QAP} = \begin{pmatrix} 1 & 0 & 0 \\ 0 & 0 & 0 \end{pmatrix}$ . Both  $\mathbf{Q}$  and  $\mathbf{P}$  are invertible by [Theorem 3.7.1 \(c\)](#).

### Exercise B3: Rank-one pieces.

Let  $\mathbf{A} \in M_{m \times n}(F)$ .

- Use [Theorem 3.8.1](#) to show that if  $\text{rank } \mathbf{A} = 1$ , then  $\mathbf{A} = \mathbf{xy}^\top$  for some non-zero  $\mathbf{x} \in F^m$  and  $\mathbf{y} \in F^n$ .
- Show that if  $\text{rank } \mathbf{A} = r \geq 1$ , then  $\mathbf{A}$  is a sum of  $r$  matrices of rank 1, and that it is not a sum of fewer than  $r$  matrices of rank 1.
- Write  $\begin{pmatrix} 1 & 2 & 0 & 3 \\ 2 & 4 & 1 & 4 \\ 3 & 6 & 1 & 7 \end{pmatrix}$  as a sum of two rank-one matrices.

*Solution.*

- With  $r = 1$ ,  $\mathbf{C} \in M_{m \times 1}(F)$  is a single column  $\mathbf{x}$ , and  $\mathbf{R} \in M_{1 \times n}(F)$  is a single row, which we write as  $\mathbf{y}^\top$  with  $\mathbf{y} \in F^n$ . So  $\mathbf{A} = \mathbf{xy}^\top$ . Both are non-zero: the columns of  $\mathbf{C}$  and the rows of  $\mathbf{R}$  are linearly independent by [Theorem 3.8.1](#), and a list containing a zero vector is dependent. (This recovers one direction of [Exercise \(C2: Matrices of rank one\)](#).)
- Write  $\mathbf{A} = \mathbf{CR}$  with columns  $\mathbf{c}_1, \dots, \mathbf{c}_r$  of  $\mathbf{C}$  and rows  $\mathbf{r}_1, \dots, \mathbf{r}_r$  of  $\mathbf{R}$ , and let  $c_{ik}$ ,  $r_{kl}$  be the entries of  $\mathbf{C}$ ,  $\mathbf{R}$ . The  $m \times n$  matrix  $\mathbf{c}_k \mathbf{r}_k$  has  $(i, l)$ -entry  $c_{ik} r_{kl}$ , so by [Theorem 0.9.2 \(1\)](#),

$$(\mathbf{CR})_{il} = \sum_{k=1}^r c_{ik} r_{kl} = \sum_{k=1}^r (\mathbf{c}_k \mathbf{r}_k)_{il}, \quad \text{that is,} \quad \mathbf{A} = \mathbf{c}_1 \mathbf{r}_1 + \dots + \mathbf{c}_r \mathbf{r}_r.$$

Each  $\mathbf{c}_k \mathbf{r}_k$  has rank 1 by the  $(\Leftarrow)$  direction of [Exercise \(C2: Matrices of rank one\)](#), since  $\mathbf{c}_k \neq \mathbf{0}$  (a column of  $\mathbf{C}$ , whose columns are independent) and  $\mathbf{r}_k \neq \mathbf{0}$  (a row of  $\mathbf{R}$ , whose rows are independent). If  $\mathbf{A} = \mathbf{A}_1 + \dots + \mathbf{A}_s$  with each  $\text{rank } \mathbf{A}_i = 1$ , then by [Exercise \(C1: Rank of a sum\)](#) applied repeatedly,  $r = \text{rank } \mathbf{A} \leq \text{rank } \mathbf{A}_1 + \dots + \text{rank } \mathbf{A}_s = s$ .

- By [Example \(A column-row factorization\)](#),  $\mathbf{A} = \mathbf{CR}$  with  $\mathbf{C} = \begin{pmatrix} 1 & 0 \\ 2 & 1 \\ 3 & 1 \end{pmatrix}$  and

$$\mathbf{R} = \begin{pmatrix} 1 & 2 & 0 & 3 \\ 0 & 0 & 1 & -2 \end{pmatrix}, \text{ so}$$

$$\begin{aligned} \mathbf{A} &= \begin{pmatrix} 1 \\ 2 \\ 3 \end{pmatrix} \begin{pmatrix} 1 & 2 & 0 & 3 \end{pmatrix} + \begin{pmatrix} 0 \\ 1 \\ 1 \end{pmatrix} \begin{pmatrix} 0 & 0 & 1 & -2 \end{pmatrix} \\ &= \begin{pmatrix} 1 & 2 & 0 & 3 \\ 2 & 4 & 0 & 6 \\ 3 & 6 & 0 & 9 \end{pmatrix} + \begin{pmatrix} 0 & 0 & 0 & 0 \\ 0 & 0 & 1 & -2 \\ 0 & 0 & 1 & -2 \end{pmatrix}. \end{aligned}$$

### 3.8.4.3 C. Going deeper

#### Exercise C1: Frobenius's rank inequality.

Let  $\mathbf{A} \in M_{m \times n}(F)$ ,  $\mathbf{B} \in M_{n \times p}(F)$  and  $\mathbf{C} \in M_{p \times q}(F)$ . Write  $N = \text{nul}(\mathbf{A}) \subseteq F^n$ , and let  $S: F^n \rightarrow F^m$  be  $S\mathbf{y} = \mathbf{A}\mathbf{y}$ .

- (a) Show that  $\text{col}(\mathbf{AB}) = \{\mathbf{A}\mathbf{y} : \mathbf{y} \in \text{col}(\mathbf{B})\}$ , and deduce from Rank–Nullity for  $S|_{\text{col}(\mathbf{B})}$  that  $\text{rank}(\mathbf{AB}) = \text{rank} \mathbf{B} - \dim(N \cap \text{col}(\mathbf{B}))$ .
- (b) Show similarly that  $\text{rank}(\mathbf{ABC}) = \text{rank}(\mathbf{BC}) - \dim(N \cap \text{col}(\mathbf{BC}))$ .
- (c) Prove that  $\text{col}(\mathbf{BC}) \subseteq \text{col}(\mathbf{B})$ , and deduce **Frobenius's inequality**

$$\text{rank}(\mathbf{ABC}) + \text{rank} \mathbf{B} \geq \text{rank}(\mathbf{AB}) + \text{rank}(\mathbf{BC}).$$

- (d) Deduce Sylvester's inequality for matrices from (c).

*Hint: for (d), look for a special choice of the middle factor.*

*Solution.*

- (a) By [Theorem 0.9.2](#), every vector  $\mathbf{ABx}$  equals  $\mathbf{Ay}$  with  $\mathbf{y} = \mathbf{Bx} \in \text{col}(\mathbf{B})$ ; conversely every  $\mathbf{y} \in \text{col}(\mathbf{B})$  is  $\mathbf{Bx}$  for some  $\mathbf{x}$ , and then  $\mathbf{Ay} = (\mathbf{AB})\mathbf{x} \in \text{col}(\mathbf{AB})$ . So  $\text{col}(\mathbf{AB}) = \text{im}(S|_{\text{col}(\mathbf{B})})$ . The kernel of  $S|_{\text{col}(\mathbf{B})}$  is  $N \cap \text{col}(\mathbf{B})$ , and  $\text{col}(\mathbf{B})$  is finite-dimensional, so by [Theorem 3.3.1](#),

$$\text{rank} \mathbf{B} = \dim \text{col}(\mathbf{B}) = \text{rank}(\mathbf{AB}) + \dim(N \cap \text{col}(\mathbf{B})).$$

- (b) The same argument with  $\mathbf{BC}$  in place of  $\mathbf{B}$  gives  $\text{col}(\mathbf{ABC}) = \text{im}(S|_{\text{col}(\mathbf{BC})})$ , with kernel  $N \cap \text{col}(\mathbf{BC})$ , so  $\text{rank}(\mathbf{BC}) = \text{rank}(\mathbf{ABC}) + \dim(N \cap \text{col}(\mathbf{BC}))$ .
- (c) Every vector  $\mathbf{BCx} = \mathbf{B}(\mathbf{Cx})$  lies in  $\text{col}(\mathbf{B})$ , so  $\text{col}(\mathbf{BC}) \subseteq \text{col}(\mathbf{B})$ . Hence  $N \cap \text{col}(\mathbf{BC}) \subseteq N \cap \text{col}(\mathbf{B})$ , and by [Theorem 1.6.7](#),  $\dim(N \cap \text{col}(\mathbf{BC})) \leq \dim(N \cap \text{col}(\mathbf{B}))$ . By (b) and then (a),

$$\begin{aligned} \text{rank}(\mathbf{ABC}) &= \text{rank}(\mathbf{BC}) - \dim(N \cap \text{col}(\mathbf{BC})) \\ &\geq \text{rank}(\mathbf{BC}) - \dim(N \cap \text{col}(\mathbf{B})) \\ &= \text{rank}(\mathbf{BC}) - \text{rank} \mathbf{B} + \text{rank}(\mathbf{AB}). \end{aligned}$$

Rearranging gives Frobenius's inequality.

- (d) Let  $\mathbf{A} \in M_{m \times n}(F)$  and  $\mathbf{C} \in M_{n \times q}(F)$ , and apply (c) with  $p = n$  and  $\mathbf{B} = \mathbf{I}_n$ .

Then  $\text{rank}(\mathbf{AC}) + n \geq \text{rank } \mathbf{A} + \text{rank } \mathbf{C}$ , since  $\text{rank } \mathbf{I}_n = n$ . This is [Theorem 3.8.5](#) for matrices.

### Exercise C2: Counting over $\mathbb{F}_2$ .

- (a) Show that  $M_2(\mathbb{F}_2)$  contains exactly 1 matrix of rank 0, 6 of rank 2 and 9 of rank 1.
- (b) Deduce the sizes of the equivalence classes of  $M_2(\mathbb{F}_2)$ .
- (c) Compare with similarity: how many matrices of  $M_2(\mathbb{F}_2)$  are similar to  $\mathbf{I}_2$ ? Is the class of  $\mathbf{I}_2$  under equivalence a union of several similarity classes?

*Solution.*

- (a) The only matrix of rank 0 is 0. A matrix has rank 2 exactly when its two columns span  $\mathbb{F}_2^2$ , that is ([Theorem 1.6.4](#)), when they form a basis of  $\mathbb{F}_2^2$ . The first column can be any of the 3 non-zero vectors of  $\mathbb{F}_2^2$ . The second must lie outside the span of the first, which has 2 elements, so there are  $4 - 2 = 2$  choices. That gives  $3 \cdot 2 = 6$  matrices of rank 2. Since  $|M_2(\mathbb{F}_2)| = 2^4 = 16$ , the remaining  $16 - 1 - 6 = 9$  matrices have rank 1.
- (b) By [Theorem 3.8.3](#) (c), the classes are the sets of matrices of a fixed rank, of sizes 1, 9 and 6.
- (c)  $\mathbf{I}_2$  is similar only to itself, so its similarity class has 1 element, while its equivalence class, the invertible matrices, has 6. Since similar matrices are equivalent, each equivalence class is a union of similarity classes, and here the class of the 6 invertible matrices contains the similarity class  $\{\mathbf{I}_2\}$  together with others. For example  $\begin{pmatrix} 1 & 1 \\ 0 & 1 \end{pmatrix}$  is invertible but not similar to  $\mathbf{I}_2$ , so there are at least two similarity classes inside.

## 3.9 Products and Quotient Spaces

So far every new vector space in this book has been found **inside** an old one, as a subspace. This section builds new spaces in two other ways. The first glues spaces together side by side, making a product; it is the “external” direct sum promised in Chapter 1. The second, the quotient space  $V/U$ , is subtler and more important: it collapses a subspace  $U$  to a single point and keeps only what  $V$  knows beyond  $U$ . Quotients let us state precisely what a linear map remembers and what it forgets, in the three Isomorphism Theorems.

### 3.9.1 Products of vector spaces

In Chapter 1 we wrote  $V = U \oplus W$  for subspaces  $U$  and  $W$  of one space  $V$  ([Definition 1.7.2](#)). The two pieces already lived in a common space, and “direct” was a property of how they sat there. But often we want to combine two spaces that have nothing to do with each other, such as  $\mathbb{R}^2$  and  $\mathbb{R}[x]_{\leq 1}$ , into one space in which each keeps its identity. There is no common space to add inside, so we build one.

The model is  $F^2$ . A vector of  $F^2$  is a pair  $(a, b)$  of scalars, and we add and scale pairs entry by entry. Nothing in that recipe needs the entries to be scalars: the entries could be vectors, taken from any vector spaces over  $F$ .

*The product of vector spaces is the space of lists with one entry from each space, added and scaled entry by entry.*

### Definition 3.9.1 — Product of vector spaces

Let  $V_1, \dots, V_k$  be vector spaces over the **same** field  $F$ . Their **product** (or **external direct sum**) is the Cartesian product (Definition 0.3.10)

$$V_1 \times \cdots \times V_k = \{(\mathbf{v}_1, \dots, \mathbf{v}_k) : \mathbf{v}_i \in V_i \text{ for each } i\},$$

with addition and scalar multiplication defined **entrywise**: for  $a \in F$ ,

$$\begin{aligned} (\mathbf{v}_1, \dots, \mathbf{v}_k) + (\mathbf{w}_1, \dots, \mathbf{w}_k) &:= (\mathbf{v}_1 + \mathbf{w}_1, \dots, \mathbf{v}_k + \mathbf{w}_k), \\ a(\mathbf{v}_1, \dots, \mathbf{v}_k) &:= (a\mathbf{v}_1, \dots, a\mathbf{v}_k). \end{aligned}$$

In words: the  $i$ -th entry of a sum is the sum of the  $i$ -th entries, computed **in**  $V_i$ , and similarly for scalar multiples. Two elements are equal exactly when all their entries are equal. Every operation happens inside one  $V_i$  at a time, which is why all the  $V_i$  must share the field  $F$ : the same scalar  $a$  has to act on every entry.

**It is a vector space.** Both operations land in  $V_1 \times \cdots \times V_k$ , since  $\mathbf{v}_i + \mathbf{w}_i \in V_i$  and  $a\mathbf{v}_i \in V_i$ . Each axiom of Definition 1.1.1 is an equation between lists, so it holds when it holds in each entry, and in each entry it is the same axiom in  $V_i$ . For example, (VS6) reads  $(a + b)(\mathbf{v}_1, \dots, \mathbf{v}_k) = ((a + b)\mathbf{v}_1, \dots, (a + b)\mathbf{v}_k) = (a\mathbf{v}_1 + b\mathbf{v}_1, \dots, a\mathbf{v}_k + b\mathbf{v}_k) = a(\mathbf{v}_1, \dots, \mathbf{v}_k) + b(\mathbf{v}_1, \dots, \mathbf{v}_k)$ , using (VS6) in each  $V_i$ . The zero vector is  $(\mathbf{0}, \dots, \mathbf{0})$ , with the zero of  $V_i$  in the  $i$ -th entry, and the additive inverse of  $(\mathbf{v}_1, \dots, \mathbf{v}_k)$  is  $(-\mathbf{v}_1, \dots, -\mathbf{v}_k)$ . The other axioms are checked in exactly the same way.

#### Examples.

- **The standard example.**  $F \times \cdots \times F$  ( $n$  factors) is  $F^n$  with its usual operations.
- **Unrelated spaces.** An element of  $\mathbb{R}^2 \times \mathbb{R}[x]_{\leq 1}$  is a pair  $((a, b), c + dx)$ , and  $((1, 0), x) + ((2, 5), 3 - x) = ((3, 5), 3)$ .
- **Augmented data.** A linear system  $\mathbf{Ax} = \mathbf{b}$  with  $m$  equations in  $n$  unknowns is a single element  $(\mathbf{A}, \mathbf{b})$  of  $M_{m \times n}(F) \times F^m$ .
- **Degenerate cases.** For  $k = 1$  the product is  $V_1$  itself, with entries written in parentheses. And  $V \times \{\mathbf{0}\}$  consists of the pairs  $(\mathbf{v}, \mathbf{0})$ ; the map  $\mathbf{v} \mapsto (\mathbf{v}, \mathbf{0})$  is a linear bijection  $V \rightarrow V \times \{\mathbf{0}\}$ , so a zero factor adds nothing.

**Non-example by minimal change.** Take  $V_1 = \mathbb{C}$  as a vector space over  $\mathbb{C}$  and  $V_2 = \mathbb{R}$  over  $\mathbb{R}$ . The set  $\mathbb{C} \times \mathbb{R}$  is fine, but the entrywise rule  $i \cdot (z, t) = (iz, it)$  fails:  $it \notin \mathbb{R}$  for  $t \neq 0$ . The clause that fails is “over the **same** field”. Restricting the scalars of  $\mathbb{C}$  to  $\mathbb{R}$  (Definition 1.9.1) repairs it, and  $\mathbb{C} \times \mathbb{R}$  over  $\mathbb{R}$  is a vector space of dimension 3.

Bases of the factors combine into a basis of the product. For  $\mathbf{v} \in V_i$ , write  $\iota_i(\mathbf{v})$  for the list with  $\mathbf{v}$  in the  $i$ -th entry and  $\mathbf{0}$  in every other entry. The map  $\iota_i: V_i \rightarrow V_1 \times \cdots \times V_k$  is linear, because the operations are entrywise.

**Theorem 3.9.1 — Dimension of a Product**

Let  $V_1, \dots, V_k$  be finite-dimensional vector spaces over  $F$ , and let  $(\mathbf{b}_{i1}, \dots, \mathbf{b}_{in_i})$  be a basis of  $V_i$  for each  $i$ . Then the list of all vectors  $\iota_i(\mathbf{b}_{ij})$ , for  $i = 1, \dots, k$  and  $j = 1, \dots, n_i$ , is a basis of  $V_1 \times \dots \times V_k$ . In particular,

$$\dim(V_1 \times \dots \times V_k) = \dim V_1 + \dots + \dim V_k.$$

*Proof. Spanning.* Let  $(\mathbf{v}_1, \dots, \mathbf{v}_k)$  be in the product. Since the operations are entrywise,  $(\mathbf{v}_1, \dots, \mathbf{v}_k) = \iota_1(\mathbf{v}_1) + \dots + \iota_k(\mathbf{v}_k)$ . Each  $\mathbf{v}_i$  is a combination  $\sum_j c_{ij} \mathbf{b}_{ij}$ , and  $\iota_i$  is linear, so  $\iota_i(\mathbf{v}_i) = \sum_j c_{ij} \iota_i(\mathbf{b}_{ij})$  (Theorem 3.1.4). Hence  $(\mathbf{v}_1, \dots, \mathbf{v}_k)$  is a combination of the vectors  $\iota_i(\mathbf{b}_{ij})$ .

*Independence.* Let  $\sum_{i=1}^k \sum_{j=1}^{n_i} c_{ij} \iota_i(\mathbf{b}_{ij}) = \mathbf{0}$ . The  $i$ -th entry of the left side is  $\sum_j c_{ij} \mathbf{b}_{ij}$ , because every  $\iota_{i'}(\mathbf{b}_{i'j})$  with  $i' \neq i$  has  $\mathbf{0}$  there. So  $\sum_j c_{ij} \mathbf{b}_{ij} = \mathbf{0}$  in  $V_i$  for each  $i$ , and the independence of the basis of  $V_i$  gives  $c_{ij} = 0$  for all  $j$ .

The list has  $n_1 + \dots + n_k$  vectors, which gives the dimension (Definition 1.6.1). ■

For instance,  $\dim(\mathbb{R}^2 \times \mathbb{R}[x]_{\leq 1}) = 2+2 = 4$ , so this product is isomorphic to  $\mathbb{R}^4$  by Theorem 3.5.4.

How does the external product relate to the internal direct sum of Chapter 1? If  $U_1, \dots, U_k$  are subspaces of  $V$ , there is a natural linear map from the product to  $V$ : add up the entries. The sum is direct exactly when this map loses no information.

**Theorem 3.9.2 — Internal and External Direct Sums** (a) Let  $U_1, \dots, U_k$  be subspaces of a vector space  $V$ , and let

$$S: U_1 \times \dots \times U_k \rightarrow V, \quad S(\mathbf{u}_1, \dots, \mathbf{u}_k) = \mathbf{u}_1 + \dots + \mathbf{u}_k.$$

Then  $S$  is linear with  $\text{im } S = U_1 + \dots + U_k$ . It is injective if and only if the sum  $U_1 + \dots + U_k$  is direct. In that case  $S$  is an isomorphism from  $U_1 \times \dots \times U_k$  onto  $U_1 \oplus \dots \oplus U_k$ .

- (b) Conversely, let  $V_1, \dots, V_k$  be vector spaces over  $F$ . Then the images  $\iota_i(V_i)$  are subspaces of  $V_1 \times \dots \times V_k$ , each isomorphic to  $V_i$  via  $\iota_i$ , and

$$V_1 \times \dots \times V_k = \iota_1(V_1) \oplus \dots \oplus \iota_k(V_k).$$

*Proof.*

- (a) For lists  $(\mathbf{u}_i)$ ,  $(\mathbf{u}'_i)$  and  $a \in F$ , regrouping the finite sums gives  $S((\mathbf{u}_i) + (\mathbf{u}'_i)) = \sum_i (\mathbf{u}_i + \mathbf{u}'_i) = S((\mathbf{u}_i)) + S((\mathbf{u}'_i))$  and  $S(a(\mathbf{u}_i)) = \sum_i a\mathbf{u}_i = aS((\mathbf{u}_i))$ , so  $S$  is linear. Its image is  $\{\mathbf{u}_1 + \dots + \mathbf{u}_k : \mathbf{u}_i \in U_i\}$ , which is  $U_1 + \dots + U_k$  by Definition 1.7.1. Its kernel consists of the lists with  $\mathbf{u}_1 + \dots + \mathbf{u}_k = \mathbf{0}$ . By Theorem 3.2.3,  $S$  is injective if and only if the only such list is  $(\mathbf{0}, \dots, \mathbf{0})$ , which is condition (b) of Theorem 1.7.5, equivalent to the sum being direct. In that case  $S$ , viewed as a map onto its image, is a linear bijection, hence an isomorphism (Definition 3.5.2).
- (b) Each  $\iota_i$  is linear and injective (its kernel is  $\{\mathbf{0}\}$ , since  $\iota_i(\mathbf{v})$  has  $\mathbf{v}$  as its  $i$ -th entry),

so  $\iota_i(V_i)$  is a subspace (Theorem 3.2.2) and  $\iota_i$  is a bijection onto it. Every list satisfies  $(\mathbf{v}_1, \dots, \mathbf{v}_k) = \iota_1(\mathbf{v}_1) + \dots + \iota_k(\mathbf{v}_k)$ , so the product is the sum of the subspaces  $\iota_i(V_i)$ . If  $\iota_1(\mathbf{v}_1) + \dots + \iota_k(\mathbf{v}_k) = \mathbf{0}$ , then  $(\mathbf{v}_1, \dots, \mathbf{v}_k) = (\mathbf{0}, \dots, \mathbf{0})$ , so every piece  $\iota_i(\mathbf{v}_i)$  is  $\mathbf{0}$ . By Theorem 1.7.5 ((b)  $\Rightarrow$  (a)) the sum is direct. ■

So internal and external direct sums are the same thing seen from two sides. An internal direct sum is a product that happens to live inside  $V$ ; a product is an internal direct sum of copies of its factors. Many books write  $\oplus$  for both. This book keeps  $\times$  for the external construction, so that  $U \oplus W$  always refers to subspaces of one space.

*Quick check.* Use Theorem 3.9.2 (a) and Rank–Nullity to explain why, for **finite-dimensional** subspaces  $U, W$  of  $V$ , the sum  $U + W$  is direct if and only if  $\dim(U + W) = \dim U + \dim W$ .

*Solution.* The addition map  $S: U \times W \rightarrow V$  has image  $U + W$ , and  $\dim(U \times W) = \dim U + \dim W$  by Theorem 3.9.1. By Theorem 3.3.1,  $\dim U + \dim W = \dim(U + W) + \dim \ker S$ . So  $\dim(U + W) = \dim U + \dim W$  exactly when  $\ker S = \{\mathbf{0}\}$ , that is, when  $S$  is injective, which by the theorem means the sum is direct. This recovers Theorem 1.7.4 (a)  $\Leftrightarrow$  (c).

## 3.9.2 Cosets

The second construction starts from a picture you have already seen. In Chapter 1 we compared the line  $U = \{x + 2y = 0\}$  through the origin with the parallel line  $L = \{x + 2y = 3\}$ , which is not a subspace. In Chapter 2 we found that the solutions of a consistent system  $\mathbf{Ax} = \mathbf{b}$  form a set  $\mathbf{p} + \text{nul}(\mathbf{A})$ , a subspace shifted by one particular solution (Theorem 2.1.3). Both are translates of a subspace. They deserve a name.

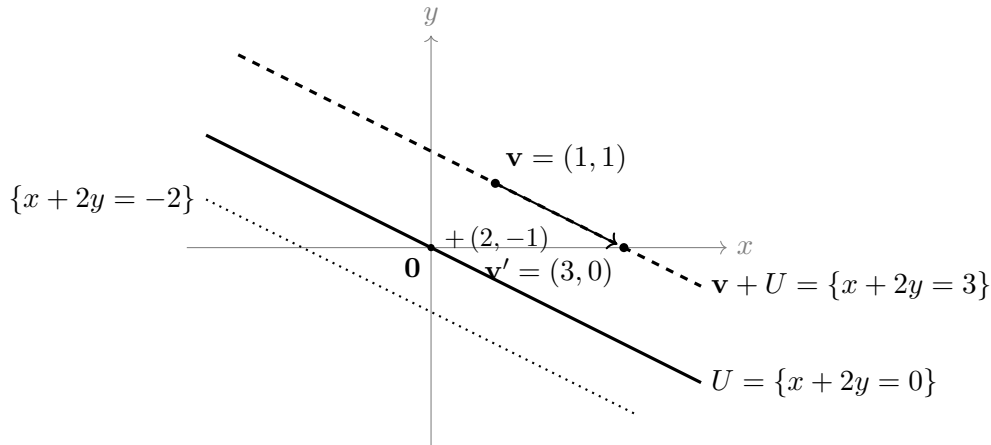
### Definition 3.9.2 — Coset

Let  $U$  be a subspace of a vector space  $V$ , and let  $\mathbf{v} \in V$ . The **coset** of  $U$  through  $\mathbf{v}$  is the subset

$$\mathbf{v} + U := \{\mathbf{v} + \mathbf{u} : \mathbf{u} \in U\} \subseteq V.$$

Any  $\mathbf{w} \in \mathbf{v} + U$  is called a **representative** of the coset.

In  $\mathbb{R}^2$ , with  $U = \{x + 2y = 0\} = \text{span}((2, -1))$ , the coset through  $(3, 0)$  is the line  $\{x + 2y = 3\}$ : a point  $(3, 0) + t(2, -1) = (3 + 2t, -t)$  satisfies  $x + 2y = 3$ , and every solution of  $x + 2y = 3$  is of this form with  $t = -y$ . In general  $(a, b) + U = \{x + 2y = a + 2b\}$ , so the cosets of  $U$  are exactly the lines parallel to  $U$ .



The picture shows three cosets of  $U$ :  $U$  itself (solid), the line through  $\mathbf{v} = (1, 1)$  (dashed), and another parallel line (dotted). The points  $\mathbf{v} = (1, 1)$  and  $\mathbf{v}' = (3, 0)$  lie on the same dashed line, because  $\mathbf{v}' - \mathbf{v} = (2, -1) \in U$ ; so they are two representatives of **one** coset. That observation is the whole theory of cosets.

### Lemma 3.9.3 — When two cosets are equal

Let  $U$  be a subspace of  $V$ . For  $\mathbf{v}, \mathbf{w} \in V$ , write  $\mathbf{v} \sim \mathbf{w}$  if  $\mathbf{v} - \mathbf{w} \in U$ .

- (a)  $\sim$  is an equivalence relation on  $V$ , and the equivalence class of  $\mathbf{v}$  is the coset  $\mathbf{v} + U$ .
- (b)  $\mathbf{v} + U = \mathbf{w} + U$  if and only if  $\mathbf{v} - \mathbf{w} \in U$ .
- (c) Two cosets of  $U$  are either equal or disjoint, and every vector of  $V$  lies in exactly one coset.

*Proof.*

- (a) We check Definition 0.5.2. (E1)  $\mathbf{v} - \mathbf{v} = \mathbf{0} \in U$ . (E2) If  $\mathbf{v} - \mathbf{w} \in U$ , then  $\mathbf{w} - \mathbf{v} = (-1)(\mathbf{v} - \mathbf{w}) \in U$  by Theorem 1.1.7 and closure under scalar multiplication. (E3) If  $\mathbf{v} - \mathbf{w} \in U$  and  $\mathbf{w} - \mathbf{x} \in U$ , then  $\mathbf{v} - \mathbf{x} = (\mathbf{v} - \mathbf{w}) + (\mathbf{w} - \mathbf{x}) \in U$  by closure under addition. For the class,  $\mathbf{w} \sim \mathbf{v}$  means  $\mathbf{w} - \mathbf{v} = \mathbf{u}$  for some  $\mathbf{u} \in U$ , that is,  $\mathbf{w} = \mathbf{v} + \mathbf{u} \in \mathbf{v} + U$ ; so the class  $[\mathbf{v}]$  of Definition 0.5.3 is  $\mathbf{v} + U$ .
- (b) By (a) and Lemma 0.5.1 (a),  $\mathbf{v} + U = \mathbf{w} + U$  if and only if  $\mathbf{v} \sim \mathbf{w}$ , that is,  $\mathbf{v} - \mathbf{w} \in U$ .
- (c) This is Lemma 0.5.1 (b), together with  $\mathbf{v} \in \mathbf{v} + U$  (take  $\mathbf{u} = \mathbf{0}$ ).

■

#### More examples.

- **Vertical lines.** In  $\mathbb{R}^3$ , let  $U = \text{span}(\mathbf{e}_3)$ , the  $z$ -axis. The coset  $(a, b, c) + U$  is the vertical line through  $(a, b, 0)$ , and  $(a, b, c) + U = (a', b', c') + U$  exactly when  $a = a'$  and  $b = b'$ . A coset is determined by the shadow of any of its points on the  $xy$ -plane.
- **Polynomials with a prescribed value.** In  $F[x]_{\leq n}$ , let  $U = \{p : p(0) = 0\}$ , a subspace (the kernel of evaluation at 0). Then  $p + U = q + U$  exactly when  $(p - q)(0) = 0$ , that is,  $p(0) = q(0)$ . So  $p + U$  is the set of all polynomials with the same constant term as  $p$ .
- **Degenerate cases.** If  $U = \{\mathbf{0}\}$ , each coset  $\mathbf{v} + U = \{\mathbf{v}\}$  is a single vector. If  $U = V$ , there is only one coset,  $\mathbf{v} + V = V$ , because  $\mathbf{v} - \mathbf{w} \in V$  always.

**Non-example by minimal change.** Try to use the line  $L = \{x + 2y = 3\}$ , which is not a subspace, in place of  $U$ . The rule “ $\mathbf{v} \sim \mathbf{w}$  if  $\mathbf{v} - \mathbf{w} \in L$ ” is not even reflexive:  $\mathbf{v} - \mathbf{v} = \mathbf{0} \notin L$ . What fails is the hypothesis that  $U$  is a subspace, which (E1), (E2) and (E3) each used.

*Warning.* **A coset has many names, and it is usually not a subspace.** In the example above,  $(1, 1) + U$  and  $(3, 0) + U$  are the **same** set, although  $(1, 1) \neq (3, 0)$ . Never conclude  $\mathbf{v} = \mathbf{w}$  from  $\mathbf{v} + U = \mathbf{w} + U$ ; conclude only  $\mathbf{v} - \mathbf{w} \in U$ . And  $\mathbf{v} + U$  is a subspace only when it contains  $\mathbf{0}$ , that is, when  $\mathbf{v} \in U$  and so  $\mathbf{v} + U = U$ . The dashed line above is not a subspace.

### 3.9.3 The quotient space

Now we want to treat each coset as a single point. The motivation is a kind of deliberate forgetting. Suppose we only care about vectors of  $\mathbb{R}^2$  “up to” adding something from  $U$ , for instance because we only ever measure the quantity  $x + 2y$ . Then  $(1, 1)$  and  $(3, 0)$  carry the same information, and the space of all the information we keep is the set of lines parallel to  $U$ . We would like this set to be a vector space, so that we can add and scale lines.

The recipe is the one used for  $\mathbb{Z}/n\mathbb{Z}$  in Chapter 0 ([Example \(Addition modulo  \$n\$  is well defined\)](#)): define the operations through representatives, then check that the choice of representatives does not matter.

*The quotient space  $V/U$  is  $V$  with everything in  $U$  declared to be zero.*

#### Definition 3.9.3 — Quotient space

Let  $U$  be a subspace of a vector space  $V$  over  $F$ . The **quotient space**  $V/U$  (read “ $V$  modulo  $U$ ”) is the set of all cosets of  $U$ ,

$$V/U := \{\mathbf{v} + U : \mathbf{v} \in V\},$$

with the operations

$$\begin{aligned} (\mathbf{v} + U) + (\mathbf{w} + U) &:= (\mathbf{v} + \mathbf{w}) + U, \\ a(\mathbf{v} + U) &:= a\mathbf{v} + U \quad (\mathbf{v}, \mathbf{w} \in V, a \in F). \end{aligned}$$

The **quotient map** is  $\pi: V \rightarrow V/U$ ,  $\pi(\mathbf{v}) = \mathbf{v} + U$ .

In words: an element of  $V/U$  is a whole coset, a subset of  $V$ . By [Lemma 3.9.3](#) (a),  $V/U$  is the quotient set of  $V$  by the relation  $\mathbf{v} \sim \mathbf{w} \Leftrightarrow \mathbf{v} - \mathbf{w} \in U$  ([Definition 0.5.5](#)), and  $\pi$  is its quotient map. To add two cosets, pick a representative of each, add them in  $V$ , and take the coset of the result. To scale a coset, scale a representative. The operations are written with representatives, so before anything else we must check that they are **well defined**.

The danger is real. By [Lemma 3.9.3](#),  $(1, 1) + U$  and  $(3, 0) + U$  are one element of  $\mathbb{R}^2/U$ . If the sum  $((1, 1) + U) + (\mathbf{w} + U)$  depended on which of the two names we used, the “definition” would assign two values to one pair of cosets, and it would not define a function at all ([Definition 0.4.1](#)). The theorem below rules this out, and then shows that the operations make  $V/U$  a vector space.



### Theorem 3.9.4 — The Quotient Space Is a Vector Space

Let  $U$  be a subspace of a vector space  $V$  over  $F$ .

- (a) The addition and scalar multiplication in [Definition 3.9.3](#) are well defined: the right-hand sides depend only on the cosets  $\mathbf{v} + U$  and  $\mathbf{w} + U$ , not on the representatives  $\mathbf{v}$  and  $\mathbf{w}$ .
- (b) With these operations  $V/U$  is a vector space over  $F$ . Its zero vector is  $\mathbf{0} + U = U$ , and the additive inverse of  $\mathbf{v} + U$  is  $(-\mathbf{v}) + U$ .
- (c) The quotient map  $\pi: V \rightarrow V/U$  is linear and surjective, and  $\ker \pi = U$ .

*Idea.* Chapter 0's [Theorem 0.5.3](#) turns a function on  $V$  that is constant on classes into a function on  $V/U$ . It is stated for functions of one variable, so we use it one variable at a time. ① For scalar multiplication by a fixed  $a$ : if  $\mathbf{v} - \mathbf{v}' \in U$ , then  $a\mathbf{v} - a\mathbf{v}' = a(\mathbf{v} - \mathbf{v}') \in U$ . ② For addition, first freeze the second summand  $\mathbf{w}$  and vary the first; then show that the resulting function on cosets does not change when  $\mathbf{w}$  is replaced by an equivalent  $\mathbf{w}'$ . Each check is one line and uses exactly one closure property of  $U$ . Once the operations exist, every axiom for cosets is the same axiom in  $V$ , passed through  $\pi$ .

*Proof.* Throughout,  $\mathbf{v} \sim \mathbf{v}'$  means  $\mathbf{v} - \mathbf{v}' \in U$ , and by [Lemma 3.9.3](#) (b),  $\mathbf{v} \sim \mathbf{v}'$  exactly when  $\mathbf{v} + U = \mathbf{v}' + U$ .

- (a) *Scalar multiplication.* Fix  $a \in F$ , and let  $g_a: V \rightarrow V/U$ ,  $g_a(\mathbf{v}) = a\mathbf{v} + U$ . If  $\mathbf{v} \sim \mathbf{v}'$ , then  $a\mathbf{v} - a\mathbf{v}' = a(\mathbf{v} - \mathbf{v}') \in U$  by (VS5) and closure of  $U$  under scalar multiplication, so  $g_a(\mathbf{v}) = g_a(\mathbf{v}')$ . By [Theorem 0.5.3](#) (a), there is exactly one function  $\bar{g}_a: V/U \rightarrow V/U$  with  $\bar{g}_a(\mathbf{v} + U) = a\mathbf{v} + U$  for every  $\mathbf{v} \in V$ . We define  $a(\mathbf{v} + U) := \bar{g}_a(\mathbf{v} + U)$ .

*Addition, first variable.* Fix  $\mathbf{w} \in V$ , and let  $f_{\mathbf{w}}: V \rightarrow V/U$ ,  $f_{\mathbf{w}}(\mathbf{v}) = (\mathbf{v} + \mathbf{w}) + U$ . If  $\mathbf{v} \sim \mathbf{v}'$ , then  $(\mathbf{v} + \mathbf{w}) - (\mathbf{v}' + \mathbf{w}) = \mathbf{v} - \mathbf{v}' \in U$ , so  $f_{\mathbf{w}}(\mathbf{v}) = f_{\mathbf{w}}(\mathbf{v}')$ . By [Theorem 0.5.3](#) (a), there is exactly one function  $\bar{f}_{\mathbf{w}}: V/U \rightarrow V/U$  with  $\bar{f}_{\mathbf{w}}(\mathbf{v} + U) = (\mathbf{v} + \mathbf{w}) + U$  for every  $\mathbf{v}$ .

*Addition, second variable.* Let  $\mathcal{F}$  be the set of all functions  $V/U \rightarrow V/U$ , and let  $h: V \rightarrow \mathcal{F}$ ,  $h(\mathbf{w}) = \bar{f}_{\mathbf{w}}$ . Suppose  $\mathbf{w} \sim \mathbf{w}'$ . For every  $\mathbf{v} \in V$ ,  $(\mathbf{v} + \mathbf{w}) - (\mathbf{v} + \mathbf{w}') = \mathbf{w} - \mathbf{w}' \in U$ , so

$$\bar{f}_{\mathbf{w}}(\mathbf{v} + U) = (\mathbf{v} + \mathbf{w}) + U = (\mathbf{v} + \mathbf{w}') + U = \bar{f}_{\mathbf{w}'}(\mathbf{v} + U).$$

Since every element of  $V/U$  has the form  $\mathbf{v} + U$ , the functions  $\bar{f}_{\mathbf{w}}$  and  $\bar{f}_{\mathbf{w}'}$  are equal, that is,  $h(\mathbf{w}) = h(\mathbf{w}')$ . By [Theorem 0.5.3](#) (a) again, there is exactly one function  $\bar{h}: V/U \rightarrow \mathcal{F}$  with  $\bar{h}(\mathbf{w} + U) = \bar{f}_{\mathbf{w}}$ . We define  $X + Y := \bar{h}(Y)(X)$  for  $X, Y \in V/U$ . Then

$$(\mathbf{v} + U) + (\mathbf{w} + U) = \bar{h}(\mathbf{w} + U)(\mathbf{v} + U) = \bar{f}_{\mathbf{w}}(\mathbf{v} + U) = (\mathbf{v} + \mathbf{w}) + U,$$

which is the formula of [Definition 3.9.3](#), and its value depends only on the two cosets.

- (b) Every element of  $V/U$  is  $\pi(\mathbf{v}) = \mathbf{v} + U$  for some  $\mathbf{v}$ , and by (a) we have  $\pi(\mathbf{v}) + \pi(\mathbf{w}) = \pi(\mathbf{v} + \mathbf{w})$  and  $a\pi(\mathbf{v}) = \pi(a\mathbf{v})$ . Let  $X = \pi(\mathbf{x})$ ,  $Y = \pi(\mathbf{y})$ ,  $Z = \pi(\mathbf{z})$  be in  $V/U$ ,

and  $a, b \in F$ . Each axiom follows by moving  $\pi$  outside, using the axiom in  $V$ , and moving  $\pi$  back:

- (VS1)  $X + Y = \pi(\mathbf{x} + \mathbf{y}) = \pi(\mathbf{y} + \mathbf{x}) = Y + X$ .
- (VS2)  $(X + Y) + Z = \pi((\mathbf{x} + \mathbf{y}) + \mathbf{z}) = \pi(\mathbf{x} + (\mathbf{y} + \mathbf{z})) = X + (Y + Z)$ .
- (VS3)  $X + \pi(\mathbf{0}) = \pi(\mathbf{x} + \mathbf{0}) = \pi(\mathbf{x}) = X$ , so  $\pi(\mathbf{0}) = \mathbf{0} + U = U$  is a zero vector.
- (VS4)  $X + \pi(-\mathbf{x}) = \pi(\mathbf{x} + (-\mathbf{x})) = \pi(\mathbf{0})$ , so  $\pi(-\mathbf{x}) = (-\mathbf{x}) + U$  is an additive inverse of  $X$ .
- (VS5)  $a(X + Y) = \pi(a(\mathbf{x} + \mathbf{y})) = \pi(a\mathbf{x} + a\mathbf{y}) = aX + aY$ .
- (VS6)  $(a + b)X = \pi((a + b)\mathbf{x}) = \pi(a\mathbf{x} + b\mathbf{x}) = aX + bX$ .
- (VS7)  $a(bX) = \pi(a(b\mathbf{x})) = \pi((ab)\mathbf{x}) = (ab)X$ .
- (VS8)  $1X = \pi(1\mathbf{x}) = \pi(\mathbf{x}) = X$ .

Hence  $V/U$  is a vector space over  $F$ .

- (c) The identities  $\pi(\mathbf{v} + \mathbf{w}) = \pi(\mathbf{v}) + \pi(\mathbf{w})$  and  $\pi(a\mathbf{v}) = a\pi(\mathbf{v})$  from (b) say that  $\pi$  is linear. It is surjective by the definition of  $V/U$ . Finally,  $\mathbf{v} \in \ker \pi$  means  $\mathbf{v} + U = \mathbf{0} + U$ , which by [Lemma 3.9.3](#) (b) means  $\mathbf{v} = \mathbf{v} - \mathbf{0} \in U$ . So  $\ker \pi = U$ . ■

Part (c) has a pleasant reading: **every subspace is a kernel**. Given  $U \subseteq V$ , the quotient map is a linear map whose kernel is exactly  $U$ .

### Examples.

- **Parallel lines.** In  $\mathbb{R}^2/U$  with  $U = \{x + 2y = 0\}$ , write  $\ell_c$  for the line  $\{x + 2y = c\}$ . Since  $(a, b) \in \ell_{a+2b}$ , the definition gives  $\ell_c + \ell_d = \ell_{c+d}$  and  $t\ell_c = \ell_{tc}$ : to add two parallel lines, add their “heights”  $c$  and  $d$ . The zero vector is  $\ell_0 = U$ .
- **Constant terms.** In  $F[x]_{\leq n}/U$  with  $U = \{p : p(0) = 0\}$ , the coset  $p + U$  is determined by  $p(0)$ , and  $(p + U) + (q + U) = (p + q) + U$  is determined by  $p(0) + q(0)$ . All that survives of a polynomial is its constant term.
- **Degenerate cases.**  $V/\{\mathbf{0}\}$  has the cosets  $\{\mathbf{v}\}$ , added exactly as in  $V$ : nothing is forgotten.  $V/V$  has the single element  $V$ , so it is the zero space: everything is forgotten.

**Non-example by minimal change.** Replace the line  $U$  by the union of the two axes,  $S = \text{span}(\mathbf{e}_1) \cup \text{span}(\mathbf{e}_2) \subseteq \mathbb{R}^2$ . It still contains  $\mathbf{0}$  and is still closed under scalar multiplication, but it is not closed under addition. Then “ $\mathbf{v} - \mathbf{w} \in S$ ” is not transitive:  $(1, 1) - (0, 1) = (1, 0) \in S$  and  $(0, 1) - (0, 0) = (0, 1) \in S$ , but  $(1, 1) - (0, 0) = (1, 1) \notin S$ . Concretely, the “cosets”  $(1, 1) + S$  and  $(0, 0) + S = S$  both contain  $(0, 1) = (1, 1) + (-1, 0)$ , yet they are different sets, since  $\mathbf{0} \in S$  but  $(1, 1) + \mathbf{s} = \mathbf{0}$  would need  $\mathbf{s} = (-1, -1) \notin S$ . The sets  $\mathbf{v} + S$  do not split  $\mathbb{R}^2$  into disjoint pieces, so there is no set “ $\mathbb{R}^2/S$ ” of the kind the definition needs. The clause that fails is “ $U$  is a subspace”, specifically closure under addition.

**Why this definition.** Each part of the hypothesis that  $U$  is a subspace has a job:  $\mathbf{0} \in U$  makes  $\sim$  reflexive; closure under scalar multiplication makes  $\sim$  symmetric and scaling of cosets well defined; closure under addition makes  $\sim$  transitive and addition of cosets well defined. The non-example shows what goes wrong when one of them is missing. The notation  $V/U$  and the phrase “modulo  $U$ ” come from  $\mathbb{Z}/n\mathbb{Z}$ , where integers are identified when their difference lies in  $n\mathbb{Z}$ .

*Warning.*  $V/U$  is not a subspace of  $V$ . Its elements are subsets of  $V$ , not vectors of  $V$ , and its addition is a new operation. For example,  $\mathbb{R}^3/\text{span}(\mathbf{e}_3)$  is not the  $xy$ -plane: its elements are vertical lines. The  $xy$ -plane is a **complement** of the  $z$ -axis, one of many

(Chapter 1), whereas  $V/U$  involves no choice at all. The two are isomorphic, as we show below, but they are different objects.

In finite dimension the size of a quotient is exactly what one would guess from the picture: we lose  $\dim U$  dimensions.

### Theorem 3.9.5 — Dimension of a Quotient

Let  $V$  be finite-dimensional and  $U$  a subspace of  $V$ . Then  $V/U$  is finite-dimensional and

$$\dim(V/U) = \dim V - \dim U.$$

More precisely, if  $(\mathbf{u}_1, \dots, \mathbf{u}_m, \mathbf{w}_1, \dots, \mathbf{w}_s)$  is a basis of  $V$  such that  $(\mathbf{u}_1, \dots, \mathbf{u}_m)$  is a basis of  $U$ , then  $(\mathbf{w}_1 + U, \dots, \mathbf{w}_s + U)$  is a basis of  $V/U$ .

*Proof.* By Theorem 3.9.4 (c),  $\pi: V \rightarrow V/U$  is linear, surjective and has kernel  $U$ . By the Rank–Nullity Theorem (Theorem 3.3.1),  $\dim \pi = \dim V/U$  is finite-dimensional and  $\dim V = \dim U + \dim(V/U)$ , which is the formula.

For the basis, let  $\mathbf{v} + U \in V/U$  and write  $\mathbf{v} = \sum_i a_i \mathbf{u}_i + \sum_j b_j \mathbf{w}_j$ . Since  $\pi$  is linear and  $\pi(\mathbf{u}_i) = U$  is the zero vector, Theorem 3.1.4 gives  $\mathbf{v} + U = \sum_j b_j(\mathbf{w}_j + U)$ . So the list  $(\mathbf{w}_1 + U, \dots, \mathbf{w}_s + U)$  spans  $V/U$ . Its length is  $s = \dim V - \dim U = \dim(V/U)$ , so it is a basis by Theorem 1.6.4 (b). ■

Such a basis always exists: take a basis of  $U$  and extend it (Theorem 1.6.5). So to find a basis of a quotient, **extend a basis of  $U$  to one of  $V$  and keep the cosets of the new vectors.**

#### Example. (A basis of a quotient of $\mathbb{R}^3$ )

Let  $U = \text{span}((1, 1, 1)) \subseteq \mathbb{R}^3$ . Find a basis of  $\mathbb{R}^3/U$ , and write  $(2, 5, -1) + U$  in it.

*Solution.* Extend  $(1, 1, 1)$  by  $\mathbf{e}_1$  and  $\mathbf{e}_2$ . The list  $((1, 1, 1), \mathbf{e}_1, \mathbf{e}_2)$  is independent: in  $a(1, 1, 1) + b\mathbf{e}_1 + c\mathbf{e}_2 = (a + b, a + c, a) = \mathbf{0}$ , the third entry gives  $a = 0$ , and then  $b = c = 0$ . It has length 3, so it is a basis of  $\mathbb{R}^3$  (Theorem 1.6.4). By Theorem 3.9.5,  $(\mathbf{e}_1 + U, \mathbf{e}_2 + U)$  is a basis of  $\mathbb{R}^3/U$ , which has dimension  $3 - 1 = 2$ .

To write  $(2, 5, -1) + U$ , solve  $(2, 5, -1) = a(1, 1, 1) + b\mathbf{e}_1 + c\mathbf{e}_2$ : the third entry gives  $a = -1$ , then  $b = 3$  and  $c = 6$ . Hence

$$(2, 5, -1) + U = 3(\mathbf{e}_1 + U) + 6(\mathbf{e}_2 + U),$$

and indeed  $(2, 5, -1) - (3, 6, 0) = (-1, -1, -1) \in U$ . The coordinates  $(3, 6)$  are  $(x - z, y - z)$  evaluated at  $(2, 5, -1)$ : they are exactly the quantities that do not change when a multiple of  $(1, 1, 1)$  is added.

The warning above said that a quotient is not a complement. The next result says how they are related: any complement is a concrete model of the quotient.

**Proposition 3.9.6 — A complement is a model of the quotient**

Let  $U$  be a subspace of  $V$ , and let  $W$  be a subspace with  $V = U \oplus W$ . Then the restriction of the quotient map,  $W \rightarrow V/U$ ,  $\mathbf{w} \mapsto \mathbf{w} + U$ , is an isomorphism.

*Proof.* The restriction of the linear map  $\pi$  to  $W$  is linear. *Injective:* if  $\mathbf{w} + U = U$  with  $\mathbf{w} \in W$ , then  $\mathbf{w} \in U$  by Lemma 3.9.3 (b), so  $\mathbf{w} \in U \cap W = \{\mathbf{0}\}$  (Theorem 1.7.4); by Theorem 3.2.3 the restriction is injective. *Surjective:* given  $\mathbf{v} + U$ , write  $\mathbf{v} = \mathbf{u} + \mathbf{w}$  with  $\mathbf{u} \in U$ ,  $\mathbf{w} \in W$ ; then  $\mathbf{v} - \mathbf{w} = \mathbf{u} \in U$ , so  $\mathbf{v} + U = \mathbf{w} + U$ . A linear bijection is an isomorphism (Definition 3.5.2). ■

So every complement of  $U$  is isomorphic to  $V/U$ , which gives a second proof that all complements have the same dimension. The quotient is the choice-free object that all the complements imitate.

*Quick check.* In  $\mathbb{R}^2/U$  with  $U = \text{span}((1, 1))$ , is  $(2, 3) + U = (0, 1) + U$ ? Is  $(1, 0) + U = (0, 1) + U$ ? What is  $\dim(\mathbb{R}^2/U)$ ?

*Solution.* By Lemma 3.9.3 (b):  $(2, 3) - (0, 1) = (2, 2) \in U$ , so the first two cosets are equal.  $(1, 0) - (0, 1) = (1, -1) \notin U$ , so the second pair are different cosets. By Theorem 3.9.5,  $\dim(\mathbb{R}^2/U) = 2 - 1 = 1$ .

**3.9.4 The First Isomorphism Theorem**

A linear map  $T: V \rightarrow W$  forgets exactly the information in its kernel:  $T\mathbf{v} = T\mathbf{v}'$  if and only if  $T(\mathbf{v} - \mathbf{v}') = \mathbf{0}$ , that is,  $\mathbf{v} - \mathbf{v}' \in \ker T$ . In the language of this section,  $T$  takes the same value at two vectors exactly when they lie in the same coset of  $\ker T$ . So  $T$  is really a function of the coset, and on cosets it forgets nothing.

**Theorem 3.9.7 — First Isomorphism Theorem**

Let  $T: V \rightarrow W$  be a linear map, and let  $K = \ker T$ . Then the rule

$$\bar{T}: V/K \rightarrow W, \quad \bar{T}(\mathbf{v} + K) = T\mathbf{v}$$

defines a linear map, which is injective with  $\text{im } \bar{T} = \text{im } T$ , and satisfies  $\bar{T} \circ \pi = T$ . Hence

$$V/\ker T \cong \text{im } T.$$

*Proof.* *Well defined.*  $K$  is a subspace by Theorem 3.2.1, so  $V/K$  is a vector space. If  $\mathbf{v} + K = \mathbf{v}' + K$ , then  $\mathbf{v} - \mathbf{v}' \in K$  by Lemma 3.9.3 (b), so  $T\mathbf{v} - T\mathbf{v}' = T(\mathbf{v} - \mathbf{v}') = \mathbf{0}$ . Hence  $T$  is constant on each coset of  $K$ , and by Theorem 0.5.3 (a) there is exactly one function  $\bar{T}: V/K \rightarrow W$  with  $\bar{T}(\mathbf{v} + K) = T\mathbf{v}$ , that is,  $\bar{T} \circ \pi = T$ .

*Linear.* For  $\mathbf{v}, \mathbf{w} \in V$  and  $a \in F$ , by [Definition 3.9.3](#) and the linearity of  $T$ ,

$$\begin{aligned}\bar{T}((\mathbf{v} + K) + (\mathbf{w} + K)) &= \bar{T}((\mathbf{v} + \mathbf{w}) + K) = T(\mathbf{v} + \mathbf{w}) \\ &= T\mathbf{v} + T\mathbf{w} = \bar{T}(\mathbf{v} + K) + \bar{T}(\mathbf{w} + K),\end{aligned}$$

and similarly  $\bar{T}(a(\mathbf{v} + K)) = \bar{T}(a\mathbf{v} + K) = T(a\mathbf{v}) = a\bar{T}(\mathbf{v} + K)$ .

*Injective.* If  $\bar{T}(\mathbf{v} + K) = \mathbf{0}$ , then  $T\mathbf{v} = \mathbf{0}$ , so  $\mathbf{v} \in K$  and  $\mathbf{v} + K = K$ , the zero of  $V/K$ . By [Theorem 3.2.3](#),  $\bar{T}$  is injective.

*Image.* The values of  $\bar{T}$  are the vectors  $T\mathbf{v}$ ,  $\mathbf{v} \in V$ , so  $\text{im } \bar{T} = \text{im } T$ . Regarded as a map  $V/K \rightarrow \text{im } T$ ,  $\bar{T}$  is a linear bijection, hence an isomorphism ([Definition 3.5.2](#)).

■

So every linear map factors as “collapse the kernel, then embed”:  $T = \bar{T} \circ \pi$ , with  $\pi$  surjective and  $\bar{T}$  injective. When  $V$  is finite-dimensional, taking dimensions with [Theorem 3.9.5](#) gives  $\dim V - \dim \ker T = \dim \text{im } T$ , which is Rank–Nullity again. The theorem is stronger than the count, because it names the isomorphism, and it holds in infinite dimension too.

The practical use runs the other way: to identify a quotient  $V/U$ , find a surjective linear map out of  $V$  whose kernel is exactly  $U$ .

#### Example. (Identifying quotients)

Use the First Isomorphism Theorem to show:

- (a)  $\mathbb{R}^3/L \cong \mathbb{R}^2$ , where  $L = \text{span}((1, 1, 1))$ ;
- (b)  $\mathbb{R}[x]_{\leq n}/U \cong \mathbb{R}$ , where  $U = \{p : p(0) = 0\}$ .

*Solution.*

- (a) Let  $T: \mathbb{R}^3 \rightarrow \mathbb{R}^2$ ,  $T(x, y, z) = (x - z, y - z)$ , which is linear because each entry is a linear expression. It is surjective, since  $T(a, b, 0) = (a, b)$ . Its kernel is  $\{x = z, y = z\} = \{(t, t, t)\} = L$ . By [Theorem 3.9.7](#),  $(x, y, z) + L \mapsto (x - z, y - z)$  is an isomorphism  $\mathbb{R}^3/L \rightarrow \mathbb{R}^2$ . This explains the coordinates found in [Example \(A basis of a quotient of  \$\mathbb{R}^3\$ \)](#). The check that it is well defined is visible: adding  $(t, t, t)$  to  $(x, y, z)$  does not change  $x - z$  or  $y - z$ .
- (b) Let  $\text{ev}_0: \mathbb{R}[x]_{\leq n} \rightarrow \mathbb{R}$ ,  $p \mapsto p(0)$ . It is linear by [Theorem 0.8.5](#), surjective because the constant polynomial  $c$  maps to  $c$ , and its kernel is  $U$  by definition. By [Theorem 3.9.7](#),  $p + U \mapsto p(0)$  is an isomorphism  $\mathbb{R}[x]_{\leq n}/U \rightarrow \mathbb{R}$ . In particular  $\dim(\mathbb{R}[x]_{\leq n}/U) = 1$  by [Theorem 3.5.4](#), so  $\dim U = (n + 1) - 1 = n$  by [Theorem 3.9.5](#).

## 3.9.5 The Second and Third Isomorphism Theorems

Two more identifications come from the same method: write down a surjective map, compute its kernel, and apply the First Isomorphism Theorem. The first compares a sum of subspaces

with an intersection.

### Theorem 3.9.8 — Second Isomorphism Theorem

Let  $U$  and  $W$  be subspaces of a vector space  $V$ . Then  $U \cap W$  is a subspace of  $U$ ,  $W$  is a subspace of  $U + W$ , and

$$U/(U \cap W) \rightarrow (U + W)/W, \quad \mathbf{u} + (U \cap W) \mapsto \mathbf{u} + W$$

is a well-defined isomorphism. Hence  $(U + W)/W \cong U/(U \cap W)$ .

*Idea.* Picture two planes  $U$  and  $W$  in  $\mathbb{R}^3$  meeting in a line. Modulo  $W$ , a vector of  $U + W$  is represented by its  $U$ -part alone, since the  $W$ -part is forgotten. The only vectors of  $U$  that become zero are those already in  $W$ . So the map “ $\mathbf{u} \mapsto \mathbf{u} + W$ ” from  $U$  is onto, with kernel  $U \cap W$ .

*Proof.*  $U \cap W$  is a subspace (Theorem 1.2.2) contained in  $U$ , and  $W$  is a subspace contained in  $U + W$  (Theorem 1.7.1), so both quotients exist. Let  $\pi_W: U + W \rightarrow (U + W)/W$  be the quotient map, and let  $S: U \rightarrow (U + W)/W$ ,  $S\mathbf{u} = \mathbf{u} + W$ , be its restriction to  $U$ ;  $S$  is linear because  $\pi_W$  is (Theorem 3.9.4).

*$S$  is surjective.* An element of  $(U + W)/W$  is  $(\mathbf{u} + \mathbf{w}) + W$  with  $\mathbf{u} \in U$ ,  $\mathbf{w} \in W$ . Since  $(\mathbf{u} + \mathbf{w}) - \mathbf{u} = \mathbf{w} \in W$ , Lemma 3.9.3 (b) gives  $(\mathbf{u} + \mathbf{w}) + W = \mathbf{u} + W = S\mathbf{u}$ .

$\ker S = U \cap W$ . For  $\mathbf{u} \in U$ ,  $S\mathbf{u} = W$  means  $\mathbf{u} \in W$  (Lemma 3.9.3 (b)), that is,  $\mathbf{u} \in U \cap W$ .

By Theorem 3.9.7 applied to  $S$ , the rule  $\mathbf{u} + (U \cap W) \mapsto S\mathbf{u} = \mathbf{u} + W$  is a well-defined injective linear map with image  $\text{im } S = (U + W)/W$ , hence an isomorphism. (Directly: if  $\mathbf{u} - \mathbf{u}' \in U \cap W$ , then  $\mathbf{u} - \mathbf{u}' \in W$ , so  $\mathbf{u} + W = \mathbf{u}' + W$ , and the value does not depend on the representative.)

■

If  $U$  and  $W$  are finite-dimensional, then so is  $U + W$  (it is spanned by the union of bases, Proposition 1.7.2), and Theorem 3.9.5 turns the theorem into  $\dim(U + W) - \dim W = \dim U - \dim(U \cap W)$ . That is the dimension formula for sums (Theorem 1.7.3), now as a consequence of an isomorphism rather than of a basis computation. An exercise below asks for the details.

The third theorem says that quotients can be taken in stages.

### Theorem 3.9.9 — Third Isomorphism Theorem

Let  $U \subseteq W \subseteq V$  be subspaces. Then  $W/U = \{\mathbf{w} + U : \mathbf{w} \in W\}$  is a subspace of  $V/U$ , and

$$(V/U)/(W/U) \rightarrow V/W, \quad (\mathbf{v} + U) + W/U \mapsto \mathbf{v} + W$$

is a well-defined isomorphism. Hence  $(V/U)/(W/U) \cong V/W$ .

*Proof.* Let  $S: V/U \rightarrow V/W$ ,  $S(\mathbf{v} + U) = \mathbf{v} + W$ .

*Well defined.* If  $\mathbf{v} + U = \mathbf{v}' + U$ , then  $\mathbf{v} - \mathbf{v}' \in U \subseteq W$ , so  $\mathbf{v} + W = \mathbf{v}' + W$  by

**Lemma 3.9.3** (b). By [Theorem 0.5.3](#) (a), applied to  $V \rightarrow V/W$ ,  $\mathbf{v} \mapsto \mathbf{v} + W$ , the rule defines a function on  $V/U$ .

*Linear and surjective.*  $S((\mathbf{v} + U) + (\mathbf{w} + U)) = S((\mathbf{v} + \mathbf{w}) + U) = (\mathbf{v} + \mathbf{w}) + W = S(\mathbf{v} + U) + S(\mathbf{w} + U)$ , and similarly for scalars, by [Definition 3.9.3](#) in both quotients. Every  $\mathbf{v} + W$  equals  $S(\mathbf{v} + U)$ .

*Kernel.*  $S(\mathbf{v} + U) = W$  means  $\mathbf{v} \in W$ . If  $\mathbf{v} \in W$ , then  $\mathbf{v} + U \in W/U$ . Conversely, if  $\mathbf{v} + U = \mathbf{w} + U$  with  $\mathbf{w} \in W$ , then  $\mathbf{v} - \mathbf{w} \in U \subseteq W$ , so  $\mathbf{v} \in W$ . Hence  $\ker S = W/U$ . In particular  $W/U$  is a subspace of  $V/U$ , by [Theorem 3.2.1](#).

By [Theorem 3.9.7](#) applied to  $S$ ,  $(V/U)/(W/U) \cong \text{im } S = V/W$ , via  $(\mathbf{v} + U) + W/U \mapsto S(\mathbf{v} + U) = \mathbf{v} + W$ .

■

The notation  $W/U$  is used with two meanings, and they agree: the quotient space of  $W$  by  $U$ , and the set of cosets  $\mathbf{w} + U$  with  $\mathbf{w} \in W$ , sitting inside  $V/U$ . The coset  $\mathbf{w} + U$  is the same subset of  $V$  in both readings, and the operations agree because both are computed with representatives. For example, with  $V = \mathbb{R}^3$ ,  $U$  the  $z$ -axis and  $W$  the  $xz$ -plane,  $V/W \cong \mathbb{R}$  through the  $y$ -coordinate, and the theorem says that first forgetting  $z$  and then forgetting  $x$  leaves the same thing as forgetting both at once.

## 3.9.6 The universal property of the quotient

The proofs of the three theorems repeated one move: a linear map that vanishes on  $U$  passes to  $V/U$ . We record that move once, in its general form. It is the standard way to **define** linear maps out of a quotient.

### Theorem 3.9.10 — Universal Property of the Quotient

Let  $U$  be a subspace of  $V$ ,  $\pi: V \rightarrow V/U$  the quotient map, and  $T: V \rightarrow W$  a linear map.

- (a) If  $U \subseteq \ker T$ , then there is **exactly one** linear map  $\bar{T}: V/U \rightarrow W$  with  $\bar{T} \circ \pi = T$ , namely  $\bar{T}(\mathbf{v} + U) = T\mathbf{v}$ . Its image is  $\text{im } T$ , and it is injective if and only if  $U = \ker T$ .
- (b) Conversely, if some function  $\bar{T}: V/U \rightarrow W$  satisfies  $\bar{T} \circ \pi = T$ , then  $U \subseteq \ker T$ .

*Proof.*

- (a) If  $\mathbf{v} - \mathbf{v}' \in U \subseteq \ker T$ , then  $T\mathbf{v} = T\mathbf{v}'$ . By [Theorem 0.5.3](#) (a) there is exactly one function  $\bar{T}$  with  $\bar{T}(\mathbf{v} + U) = T\mathbf{v}$ , and it is linear by the computation in the proof of [Theorem 3.9.7](#), which used only this formula. Any linear  $\bar{T}$  with  $\bar{T} \circ \pi = T$  satisfies this formula, so it is this one. The image is  $\{T\mathbf{v}\} = \text{im } T$ . The kernel of  $\bar{T}$  is  $\{\mathbf{v} + U : \mathbf{v} \in \ker T\}$ ; it is  $\{U\}$ , the zero subspace, exactly when every  $\mathbf{v} \in \ker T$  lies in  $U$ , that is, when  $\ker T \subseteq U$ , which together with  $U \subseteq \ker T$  means  $U = \ker T$ . Apply [Theorem 3.2.3](#).
- (b) Let  $\mathbf{u} \in U$ . Then  $\pi(\mathbf{u}) = U = \pi(\mathbf{0})$ , so  $T\mathbf{u} = \bar{T}(\pi(\mathbf{u})) = \bar{T}(\pi(\mathbf{0})) = T(\mathbf{0}) = \mathbf{0}$  by [Theorem 3.1.1](#). Hence  $U \subseteq \ker T$ .

In words: **linear maps out of  $V/U$  are the same as linear maps out of  $V$  that kill  $U$ .** The First Isomorphism Theorem is the case  $U = \ker T$ . For example, the map  $T: \mathbb{R}^3 \rightarrow \mathbb{R}$ ,  $T(x, y, z) = x$ , kills the  $z$ -axis  $U$ , so it induces  $\bar{T}: \mathbb{R}^3/U \rightarrow \mathbb{R}$ ,  $(x, y, z) + U \mapsto x$ . Since  $\ker T$  is the  $yz$ -plane, strictly larger than  $U$ , this  $\bar{T}$  is not injective: it also kills the coset  $\mathbf{e}_2 + U$ .

Quotients will return in Chapter 8, where an operator  $T$  with  $T(U) \subseteq U$  induces an operator on  $V/U$ , and this is the engine of induction arguments on the dimension.

## 3.9.7 Exercises

### 3.9.7.1 A. Check your understanding

#### Exercise A1.

- Define the quotient space  $V/U$  and its two operations.
- State the condition, in terms of  $\mathbf{v} - \mathbf{w}$ , for  $\mathbf{v} + U = \mathbf{w} + U$ .
- True or false:  $V/U$  is a subspace of  $V$ . Justify your answer.
- State the First Isomorphism Theorem.
- If  $U$  is a 2-dimensional subspace of  $\mathbb{R}^5$ , what is  $\dim(\mathbb{R}^5/U)$ ? What is  $\dim(\mathbb{R}^5 \times U)$ ?
- What must be checked before the formula  $(\mathbf{v} + U) + (\mathbf{w} + U) = (\mathbf{v} + \mathbf{w}) + U$  can be used as a definition, and which property of  $U$  makes the check work?

*Solution.*

- For a subspace  $U$  of  $V$ ,  $V/U = \{\mathbf{v} + U : \mathbf{v} \in V\}$ , with  $(\mathbf{v} + U) + (\mathbf{w} + U) = (\mathbf{v} + \mathbf{w}) + U$  and  $a(\mathbf{v} + U) = a\mathbf{v} + U$  (Definition 3.9.3).
- $\mathbf{v} + U = \mathbf{w} + U$  if and only if  $\mathbf{v} - \mathbf{w} \in U$  (Lemma 3.9.3).
- False. The elements of  $V/U$  are cosets, which are subsets of  $V$ , not vectors of  $V$ .
- If  $T: V \rightarrow W$  is linear, then  $\mathbf{v} + \ker T \mapsto T\mathbf{v}$  is a well-defined isomorphism  $V/\ker T \rightarrow \operatorname{im} T$  (Theorem 3.9.7).
- $5 - 2 = 3$  by Theorem 3.9.5, and  $5 + 2 = 7$  by Theorem 3.9.1.
- That the right-hand side does not depend on the representatives: if  $\mathbf{v} - \mathbf{v}' \in U$  and  $\mathbf{w} - \mathbf{w}' \in U$ , then  $(\mathbf{v} + \mathbf{w}) + U = (\mathbf{v}' + \mathbf{w}') + U$ . This works because  $(\mathbf{v} + \mathbf{w}) - (\mathbf{v}' + \mathbf{w}') = (\mathbf{v} - \mathbf{v}') + (\mathbf{w} - \mathbf{w}')$  and  $U$  is closed under addition.

### 3.9.7.2 B. Practice

#### Exercise B1: A basis of a quotient of $\mathbb{R}^4$ .

Let  $U = \operatorname{span}((1, 0, 1, 0), (0, 1, 0, 1)) \subseteq \mathbb{R}^4$ .

- Find  $\dim(\mathbb{R}^4/U)$  and a basis of  $\mathbb{R}^4/U$ .
- Write  $(3, 1, 4, 1) + U$  in your basis.
- Decide whether  $(1, 2, 3, 4) + U = (3, 4, 1, 2) + U$ .



*Solution.*

- (a) The two spanning vectors are independent (look at the first two entries), so  $\dim U = 2$  and  $\dim(\mathbb{R}^4/U) = 4 - 2 = 2$  by [Theorem 3.9.5](#). Extend by  $\mathbf{e}_1, \mathbf{e}_2$ . If  $a(1, 0, 1, 0) + b(0, 1, 0, 1) + c\mathbf{e}_1 + d\mathbf{e}_2 = (a + c, b + d, a, b) = \mathbf{0}$ , then  $a = b = 0$  from the last two entries and  $c = d = 0$  from the first two. So the four vectors are independent, hence a basis of  $\mathbb{R}^4$  ([Theorem 1.6.4](#)), and by [Theorem 3.9.5](#)  $(\mathbf{e}_1 + U, \mathbf{e}_2 + U)$  is a basis of  $\mathbb{R}^4/U$ .
- (b) Solve  $(3, 1, 4, 1) = (a + c, b + d, a, b)$ :  $a = 4, b = 1, c = -1, d = 0$ . Since the  $U$ -part becomes zero in the quotient,  $(3, 1, 4, 1) + U = -1(\mathbf{e}_1 + U) + 0(\mathbf{e}_2 + U)$ . Check:  $(3, 1, 4, 1) + \mathbf{e}_1 = (4, 1, 4, 1) = 4(1, 0, 1, 0) + (0, 1, 0, 1) \in U$ .
- (c) The difference is  $(1, 2, 3, 4) - (3, 4, 1, 2) = (-2, -2, 2, 2)$ . A vector  $(a, b, a, b)$  of  $U$  has equal first and third entries, but  $-2 \neq 2$ . So the difference is not in  $U$ , and the cosets are different by [Lemma 3.9.3](#).

### Exercise B2: A map that factors through a quotient.

Let  $T: \mathbb{R}[x]_{\leq 2} \rightarrow \mathbb{R}^2$ ,  $T(p) = (p(1), p'(1))$ , and  $U = \text{span}((x - 1)^2)$ .

- (a) Show that  $U \subseteq \ker T$ , so that  $T$  induces a linear map  $\bar{T}: \mathbb{R}[x]_{\leq 2}/U \rightarrow \mathbb{R}^2$  with  $\bar{T}(p + U) = T(p)$ .
- (b) Prove that  $\bar{T}$  is an isomorphism.
- (c) Compute  $\bar{T}(x^2 + U)$ , and find the coset  $\bar{T}^{-1}(0, 1)$ .

*Solution.*

- (a)  $T$  is linear, since evaluation and differentiation are. For  $q = (x - 1)^2$ ,  $q(1) = 0$  and  $q'(1) = 2(1 - 1) = 0$ , so  $T(q) = \mathbf{0}$ , and hence  $T(cq) = \mathbf{0}$  for all  $c$ . So  $U \subseteq \ker T$ , and [Theorem 3.9.10](#) (a) gives the linear map  $\bar{T}$ .
- (b)  $T(1) = (1, 0)$  and  $T(x - 1) = (0, 1)$ , so  $T$  is surjective and  $\text{rank } T = 2$ . By [Theorem 3.3.1](#),  $\dim \ker T = 3 - 2 = 1 = \dim U$ , so  $\ker T = U$  by [Theorem 1.6.8](#). By [Theorem 3.9.10](#) (a),  $\bar{T}$  is injective with image  $\text{im } T = \mathbb{R}^2$ . Hence  $\bar{T}$  is a linear bijection, an isomorphism.
- (c)  $\bar{T}(x^2 + U) = T(x^2) = (1, 2)$ . Since  $T(x - 1) = (0, 1)$ ,  $\bar{T}^{-1}(0, 1) = (x - 1) + U$ , the set of all  $x - 1 + c(x - 1)^2$ .

### Exercise B3: Matrices modulo scalars.

Let  $n \geq 1$  and  $S = \{c\mathbf{I}_n : c \in F\} \subseteq M_n(F)$ .

- (a) Show that  $S$  is a subspace and find  $\dim(M_n(F)/S)$ .
- (b) For  $n = 2$ , find a basis of  $M_2(F)/S$ .
- (c) Show that  $\mathbf{A} + S = \mathbf{B} + S$  if and only if  $\mathbf{A} - \mathbf{B}$  is a scalar matrix, and decide whether  $\begin{pmatrix} 3 & 1 \\ 0 & 5 \end{pmatrix} + S = \begin{pmatrix} 1 & 1 \\ 0 & 3 \end{pmatrix} + S$ .

*Solution.*

- (a)  $S = \text{span}(\mathbf{I}_n)$  is a subspace by [Theorem 1.3.1](#), of dimension 1 since  $\mathbf{I}_n \neq 0$ . By [Theorem 3.9.5](#),  $\dim(M_n(F)/S) = n^2 - 1$ .
- (b) Extend  $(\mathbf{I}_2)$  by  $\mathbf{E}_{11}, \mathbf{E}_{12}, \mathbf{E}_{21}$ . If  $a\mathbf{I}_2 + b\mathbf{E}_{11} + c\mathbf{E}_{12} + d\mathbf{E}_{21} = \begin{pmatrix} a+b & c \\ d & a \end{pmatrix} = 0$ , then  $a = c = d = 0$  and then  $b = 0$ . So the four matrices are independent, hence a basis of  $M_2(F)$  ([Theorem 1.6.4](#)), and  $(\mathbf{E}_{11} + S, \mathbf{E}_{12} + S, \mathbf{E}_{21} + S)$  is a basis of  $M_2(F)/S$  by [Theorem 3.9.5](#).
- (c) This is [Lemma 3.9.3](#) (b) for the subspace  $S$ . Here the difference is  $\begin{pmatrix} 2 & 0 \\ 0 & 2 \end{pmatrix} = 2\mathbf{I}_2 \in S$ , so the two cosets are equal.

### 3.9.7.3 C. Going deeper

#### Exercise C1: The dimension formula from isomorphisms.

Let  $U$  and  $W$  be finite-dimensional subspaces of a vector space  $V$ .

- (a) Using [Theorem 3.9.8](#) and [Theorem 3.9.5](#), prove that  $\dim(U + W) = \dim U + \dim W - \dim(U \cap W)$ .
- (b) Give a second proof: apply Rank–Nullity to the addition map  $S: U \times W \rightarrow V$ ,  $S(\mathbf{u}, \mathbf{w}) = \mathbf{u} + \mathbf{w}$ , after showing that  $\mathbf{u} \mapsto (\mathbf{u}, -\mathbf{u})$  is an isomorphism from  $U \cap W$  onto  $\ker S$ .

*Solution.*

- (a)  $U + W$  is finite-dimensional, being spanned by a basis of  $U$  followed by a basis of  $W$  ([Proposition 1.7.2](#)), and  $U \cap W \subseteq U$  is finite-dimensional ([Theorem 1.6.7](#)). By [Theorem 3.9.8](#),  $(U + W)/W \cong U/(U \cap W)$ , and isomorphic finite-dimensional spaces have equal dimension ([Theorem 3.5.4](#)). Applying [Theorem 3.9.5](#) to both sides,

$$\dim(U + W) - \dim W = \dim U - \dim(U \cap W),$$

which rearranges to the formula.

- (b) The map  $S$  is linear with image  $U + W$  by [Theorem 3.9.2](#) (a). Let  $J: U \cap W \rightarrow U \times W$ ,  $J\mathbf{u} = (\mathbf{u}, -\mathbf{u})$ ; it is well defined because  $\mathbf{u} \in U$  and  $-\mathbf{u} \in W$ , and linear because the operations are entrywise. It is injective, since  $(\mathbf{u}, -\mathbf{u}) = (\mathbf{0}, \mathbf{0})$  forces  $\mathbf{u} = \mathbf{0}$ . Its image lies in  $\ker S$ , since  $\mathbf{u} + (-\mathbf{u}) = \mathbf{0}$ . Conversely, if  $(\mathbf{u}, \mathbf{w}) \in \ker S$ , then  $\mathbf{w} = -\mathbf{u}$ ; so  $\mathbf{u} = -\mathbf{w} \in W$ , hence  $\mathbf{u} \in U \cap W$  and  $(\mathbf{u}, \mathbf{w}) = J\mathbf{u}$ . Thus  $J$  is an isomorphism onto  $\ker S$ , and  $\dim \ker S = \dim(U \cap W)$ . By [Theorem 3.9.1](#) and [Theorem 3.3.1](#),

$$\dim U + \dim W = \dim(U \times W) = \dim(U + W) + \dim(U \cap W).$$

#### Exercise C2: The operator induced on a quotient.

Let  $V$  be finite-dimensional,  $T \in \mathcal{L}(V)$ , and  $U$  a subspace with  $T\mathbf{u} \in U$  for every  $\mathbf{u} \in U$ .

- (a) Prove that  $\bar{T}(\mathbf{v} + U) := T\mathbf{v} + U$  is a well-defined linear operator on  $V/U$ .
- (b) Let  $(\mathbf{u}_1, \dots, \mathbf{u}_m)$  be a basis of  $U$ , extended to a basis  $\mathcal{B} = (\mathbf{u}_1, \dots, \mathbf{u}_m, \mathbf{w}_1, \dots, \mathbf{w}_s)$

of  $V$ . Show that

$$[T]_{\mathcal{B}} = \begin{pmatrix} \mathbf{A} & \mathbf{B} \\ 0 & D \end{pmatrix},$$

where  $\mathbf{A} \in M_m(F)$  is the matrix of the restriction  $T|_U: U \rightarrow U$  in  $(\mathbf{u}_1, \dots, \mathbf{u}_m)$ ,  $D \in M_s(F)$  is the matrix of  $\bar{T}$  in  $(\mathbf{w}_1 + U, \dots, \mathbf{w}_s + U)$ , and  $0$  is the  $s \times m$  zero matrix.

- (c) Check (b) when  $T$  is differentiation on  $\mathbb{R}[x]_{\leq 2}$ , with  $U = \mathbb{R}[x]_{\leq 1}$  and  $\mathcal{B} = (1, x, x^2)$ .

*Hint: for (a), apply Theorem 3.9.10 to  $\pi \circ T$ .*

*Solution.*

- (a) The map  $\pi \circ T: V \rightarrow V/U$ ,  $\mathbf{v} \mapsto T\mathbf{v} + U$ , is linear as a composition of linear maps (Theorem 3.4.2). For  $\mathbf{u} \in U$ ,  $T\mathbf{u} \in U$ , so  $(\pi \circ T)(\mathbf{u}) = U$ , the zero of  $V/U$ ; thus  $U \subseteq \ker(\pi \circ T)$ . By Theorem 3.9.10 (a) there is exactly one linear map  $\bar{T}: V/U \rightarrow V/U$  with  $\bar{T}(\mathbf{v} + U) = T\mathbf{v} + U$ .
- (b) For  $j \leq m$ ,  $T\mathbf{u}_j \in U$ , so it is a combination of  $\mathbf{u}_1, \dots, \mathbf{u}_m$  only, and its  $\mathcal{B}$ -coordinates are 0 in the last  $s$  places. These first  $m$  columns, restricted to their first  $m$  entries, are the columns of the matrix  $\mathbf{A}$  of  $T|_U$  (which maps  $U$  to  $U$  by hypothesis and is linear). For the remaining columns, write  $T\mathbf{w}_k = \sum_i b_{ik}\mathbf{u}_i + \sum_l d_{lk}\mathbf{w}_l$ . Passing to  $V/U$ , where  $\mathbf{u}_i + U = U$  is zero, gives  $\bar{T}(\mathbf{w}_k + U) = \sum_l d_{lk}(\mathbf{w}_l + U)$ . So the  $s \times s$  matrix  $(d_{lk})$  is the matrix  $D$  of  $\bar{T}$  in the basis  $(\mathbf{w}_1 + U, \dots, \mathbf{w}_s + U)$  of Theorem 3.9.5, and the bottom-right block of  $[T]_{\mathcal{B}}$  is  $D$ .
- (c)  $T(1) = 0$ ,  $T(x) = 1$ ,  $T(x^2) = 2x$ , so  $[T]_{\mathcal{B}} = \begin{pmatrix} 0 & 1 & 0 \\ 0 & 0 & 2 \\ 0 & 0 & 0 \end{pmatrix}$ . Differentiation maps  $U = \mathbb{R}[x]_{\leq 1}$  into itself. The block  $\mathbf{A} = \begin{pmatrix} 0 & 1 \\ 0 & 0 \end{pmatrix}$  is the matrix of  $T$  on  $U$  in  $(1, x)$ , the lower-left block is  $\begin{pmatrix} 0 & 0 \end{pmatrix}$ , and the block  $D$  is the  $1 \times 1$  matrix  $(0)$ : indeed  $\bar{T}(x^2 + U) = 2x + U = U$ , the zero coset, since  $2x \in U$ .

### Exercise C3: Counting cosets over a finite field.

Let  $F = \mathbb{F}_q$  be a finite field with  $q$  elements,  $V$  a vector space over  $F$  of dimension  $n$ , and  $U$  a subspace of dimension  $m$ .

- (a) Show that  $|V| = q^n$ .
- (b) Show that every coset  $\mathbf{v} + U$  has exactly  $|U|$  elements, and deduce  $|V/U| = q^{n-m}$  by counting.
- (c) Check that (b) agrees with Theorem 3.9.5.

*Solution.*

- (a) Choose a basis  $(\mathbf{v}_1, \dots, \mathbf{v}_n)$ . By Theorem 1.5.1,  $(a_1, \dots, a_n) \mapsto a_1\mathbf{v}_1 + \dots + a_n\mathbf{v}_n$  is a bijection  $F^n \rightarrow V$ , and  $F^n$  has  $q^n$  elements.
- (b) The map  $U \rightarrow \mathbf{v} + U$ ,  $\mathbf{u} \mapsto \mathbf{v} + \mathbf{u}$ , is surjective by Definition 3.9.2 and injective because  $\mathbf{v} + \mathbf{u} = \mathbf{v} + \mathbf{u}'$  implies  $\mathbf{u} = \mathbf{u}'$ . So  $|\mathbf{v} + U| = |U| = q^m$  by (a) applied to  $U$ . By Lemma 3.9.3 (c), the cosets partition  $V$  into  $|V/U|$  disjoint sets of size  $q^m$ , so  $q^n = |V/U| \cdot q^m$ , and  $|V/U| = q^{n-m}$ .

- (c) By [Theorem 3.9.5](#),  $V/U$  is a vector space over  $\mathbb{F}_q$  of dimension  $n - m$ , so by (a) it has  $q^{n-m}$  elements, in agreement with (b). For example,  $\mathbb{F}_2^3$  modulo a line has  $8/2 = 4$  cosets.

## 3.10 Projections and the Trace of an Operator

Chapter 1 ended its discussion of complements with a promise: a splitting  $V = U \oplus W$  should give a projection of  $V$  onto  $U$  along  $W$ , and different complements should give different projections. This section keeps that promise. Projections are the operators that are simplest in a well-chosen basis, and computing their matrices there leads to a second topic: the trace of an operator, a number attached to  $T \in \mathcal{L}(V)$  that no choice of basis can change. We end with involutions, which split a space into two pieces in the same way.

### 3.10.1 Projections

Suppose  $V = U \oplus W$ . By [Definition 1.7.2](#), every  $\mathbf{v} \in V$  can be written as  $\mathbf{v} = \mathbf{u} + \mathbf{w}$  with  $\mathbf{u} \in U$  and  $\mathbf{w} \in W$  in **exactly one** way. So the rule “ $\mathbf{v} \mapsto \mathbf{u}$ ”, keep the  $U$ -piece and throw away the  $W$ -piece, is a well-defined function  $V \rightarrow V$ . In  $\mathbb{R}^3 = \{\text{the } xy\text{-plane}\} \oplus \{\text{the } z\text{-axis}\}$ , it sends  $(a, b, c)$  to  $(a, b, 0)$ : the shadow of a point on the floor when light falls straight down.

What property does such a map have that singles it out? Apply it twice. The output  $\mathbf{u}$  already lies in  $U$ , and its splitting is  $\mathbf{u} = \mathbf{u} + \mathbf{0}$ , so applying the rule again returns  $\mathbf{u}$ . Doing it twice is the same as doing it once. We take this as the definition, because it can be checked without knowing  $U$  and  $W$  in advance.

*A projection is an operator that leaves alone everything it has already produced.*

#### Definition 3.10.1 — Projection

Let  $V$  be a vector space over  $F$ . An operator  $P \in \mathcal{L}(V)$  is a **projection** if

$$P^2 = P,$$

that is,  $P(P\mathbf{v}) = P\mathbf{v}$  for every  $\mathbf{v} \in V$ .

In words:  $P^2$  is the composite  $PP$  ([Definition 3.4.2](#)), so the condition says that  $P$  fixes each of its own outputs. Equivalently,  $P\mathbf{x} = \mathbf{x}$  for every  $\mathbf{x} \in \text{im } P$ : if  $\mathbf{x} = P\mathbf{v}$ , then  $P\mathbf{x} = P^2(\mathbf{v}) = P\mathbf{v} = \mathbf{x}$ , and conversely this for  $\mathbf{x} = P\mathbf{v}$  is  $P^2(\mathbf{v}) = P\mathbf{v}$ . No inner product and no notion of angle is involved. The projections of this section are often called **oblique** projections, to distinguish them from the orthogonal projections met later, in the chapters on inner product spaces.

**Examples.**

- **Onto an axis.**  $P(x, y) = (x, 0)$  on  $F^2$ . Then  $P^2(x, y) = P(x, 0) = (x, 0)$ , so  $P^2 = P$ .
- **A rank-one projection.**  $T(x, y) = (2x - y, 2x - y)$  on  $\mathbb{R}^2$  from [Example \(Making a matrix diagonal with a well-chosen basis\)](#). We saw there that its matrix  $\mathbf{A}$  satisfies

$\mathbf{A}^2 = \mathbf{A}$ ; directly,  $T(2x - y, 2x - y) = (4x - 2y - 2x + y, \dots) = (2x - y, 2x - y)$ .

- **Constant term.** On  $F[x]_{\leq n}$ ,  $P(p) = p(0)$ , the constant polynomial with value  $p(0)$ . The constant polynomial  $c$  has value  $c$  at 0, so  $P(P(p)) = P(p)$ .
- **Degenerate cases.** The identity  $\text{id}_V$  and the zero operator are projections, since  $\text{id}_V^2 = \text{id}_V$  and  $0^2 = 0$ . They correspond to keeping everything and keeping nothing.

**Non-example by minimal change.** Change the first example to  $T(x, y) = (2x, 0)$  on  $\mathbb{R}^2$ . Its image is still the  $x$ -axis and its kernel is still the  $y$ -axis. But  $T^2(x, y) = (4x, 0) \neq T(x, y)$  when  $x \neq 0$ :  $T$  does not fix the vectors of its own image, since  $T(1, 0) = (2, 0)$ . So having “the right image and kernel” is not enough; the clause  $P^2 = P$  is what forces  $P$  to be the identity on its image. Likewise the reflection  $R(x, y) = (y, x)$  is not a projection:  $R^2 = \text{id} \neq R$ .

Here is the promised correspondence. It says that projections and direct sum decompositions are two descriptions of the same thing.

### Theorem 3.10.1 — Projections and Direct Sums

Let  $V$  be a vector space.

- (a) If  $P \in \mathcal{L}(V)$  is a projection, then

$$V = \text{im } P \oplus \ker P,$$

$P$  is the identity on  $\text{im } P$ , and  $P$  is zero on  $\ker P$ .

- (b) Conversely, suppose  $V = U \oplus W$ . Then there is **exactly one** operator  $P \in \mathcal{L}(V)$  with  $P\mathbf{u} = \mathbf{u}$  for all  $\mathbf{u} \in U$  and  $P\mathbf{w} = \mathbf{0}$  for all  $\mathbf{w} \in W$ , namely  $P(\mathbf{u} + \mathbf{w}) = \mathbf{u}$ . It is a projection, with  $\text{im } P = U$  and  $\ker P = W$ . It is called the **projection onto  $U$  along  $W$** .

*Idea.* For (a), every vector splits as “what  $P$  keeps” plus “what  $P$  removes”:  $\mathbf{v} = P\mathbf{v} + (\mathbf{v} - P\mathbf{v})$ . The first piece is in the image by definition, and  $P^2 = P$  is exactly what puts the second piece in the kernel. For (b), uniqueness of the splitting makes the rule a function, and uniqueness again makes it linear.

*Proof.*

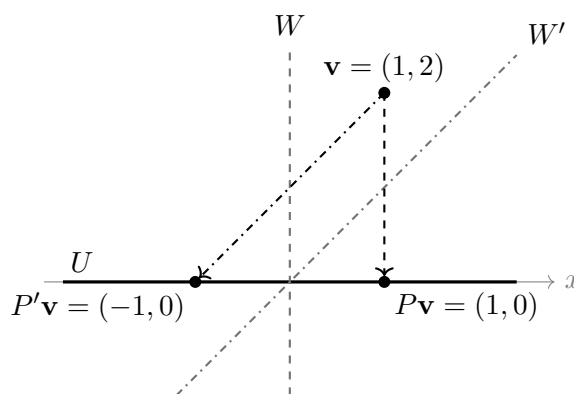
- (a) Let  $\mathbf{v} \in V$ . Then  $\mathbf{v} = P\mathbf{v} + (\mathbf{v} - P\mathbf{v})$ , where  $P\mathbf{v} \in \text{im } P$  and  $P(\mathbf{v} - P\mathbf{v}) = P\mathbf{v} - P^2(\mathbf{v}) = \mathbf{0}$  because  $P^2 = P$ . Hence  $V = \text{im } P + \ker P$ . If  $\mathbf{x} \in \text{im } P \cap \ker P$ , then  $\mathbf{x} = P\mathbf{x}$ , as shown after [Definition 3.10.1](#), and  $P\mathbf{x} = \mathbf{0}$ ; so  $\mathbf{x} = \mathbf{0}$ . By [Theorem 1.7.4](#),  $V = \text{im } P \oplus \ker P$ . We have also seen that  $P$  fixes  $\text{im } P$ , and  $P$  is zero on  $\ker P$  by definition.
- (b) *Existence.* By [Definition 1.7.2](#), each  $\mathbf{v} \in V$  has exactly one splitting  $\mathbf{v} = \mathbf{u} + \mathbf{w}$ , so  $P\mathbf{v} := \mathbf{u}$  defines a function  $V \rightarrow V$ . It is linear: if  $\mathbf{v} = \mathbf{u} + \mathbf{w}$  and  $\mathbf{v}' = \mathbf{u}' + \mathbf{w}'$ , then  $\mathbf{v} + \mathbf{v}' = (\mathbf{u} + \mathbf{u}') + (\mathbf{w} + \mathbf{w}')$  and  $a\mathbf{v} = a\mathbf{u} + a\mathbf{w}$  are splittings, because  $U$  and  $W$  are subspaces, so by uniqueness  $P(\mathbf{v} + \mathbf{v}') = \mathbf{u} + \mathbf{u}' = P\mathbf{v} + P\mathbf{v}'$  and  $P(a\mathbf{v}) = a\mathbf{u} = aP\mathbf{v}$ . For  $\mathbf{u} \in U$  the splitting is  $\mathbf{u} = \mathbf{u} + \mathbf{0}$ , so  $P\mathbf{u} = \mathbf{u}$ ; for  $\mathbf{w} \in W$  it is  $\mathbf{w} = \mathbf{0} + \mathbf{w}$ , so  $P\mathbf{w} = \mathbf{0}$ .

*Properties.* Every value  $P\mathbf{v}$  lies in  $U$ , and every  $\mathbf{u} \in U$  is  $P\mathbf{u}$ ; so  $\text{im } P = U$ . Then  $P(P\mathbf{v}) = P\mathbf{v}$  because  $P$  fixes  $U$ ; so  $P^2 = P$ . Finally  $P(\mathbf{u} + \mathbf{w}) = \mathbf{u}$  is  $\mathbf{0}$  exactly when  $\mathbf{u} = \mathbf{0}$ , that is, when  $\mathbf{v} = \mathbf{w} \in W$ ; so  $\ker P = W$ .

*Uniqueness.* If  $P' \in \mathcal{L}(V)$  is identity on  $U$  and zero on  $W$ , then  $P'(\mathbf{u} + \mathbf{w}) = P'\mathbf{u} + P'\mathbf{w} = \mathbf{u}$  by linearity, so  $P' = P$ . ■

So a projection **is** a direct sum decomposition  $V = U \oplus W$ , recorded as an operator:  $U$  is its image and  $W$  its kernel. Given  $P$ , the decomposition is  $\text{im } P \oplus \ker P$ ; given the decomposition,  $P$  is the projection onto  $U$  along  $W$ . The two constructions undo each other, by the “exactly one” in (b).

Now the promise from Chapter 1. There, the  $x$ -axis  $U$  in  $\mathbb{R}^2$  had many complements, for instance the  $y$ -axis  $W$  and the line  $W'$  spanned by  $(1, 1)$ . Each gives a projection onto the same line  $U$ .



To project  $\mathbf{v}$  onto  $U$  along a complement, slide  $\mathbf{v}$  parallel to that complement until it hits  $U$ . Along the  $y$ -axis,  $(1, 2) = (1, 0) + (0, 2)$  gives  $P\mathbf{v} = (1, 0)$  (dashed arrow). Along  $W'$ ,  $(1, 2) = (-1, 0) + (2, 2)$  gives  $P'\mathbf{v} = (-1, 0)$  (dash-dotted arrow). In general  $P(x, y) = (x, 0)$  and  $P'(x, y) = (x - y, 0)$ , with standard matrices  $\begin{pmatrix} 1 & 0 \\ 0 & 0 \end{pmatrix}$  and  $\begin{pmatrix} 1 & -1 \\ 0 & 0 \end{pmatrix}$ .

**Warning.** “**Projection onto  $U$** ” is incomplete: the direction “**along  $W$** ” **matters**. The two operators above are both projections, both have image  $U$ , and they are different:  $P(1, 2) = (1, 0)$  but  $P'(1, 2) = (-1, 0)$ . Since  $U$  has infinitely many complements in  $\mathbb{R}^2$ , it is the image of infinitely many projections. A projection is determined by its image **and** its kernel together.

In finite dimension, the direct sum  $V = \text{im } P \oplus \ker P$  hands us a basis in which  $P$  is as simple as possible.

### Corollary 3.10.2 — The matrix of a projection

Let  $V$  be finite-dimensional and  $P \in \mathcal{L}(V)$  a projection with  $\text{rank } P = r$ . Let  $(\mathbf{u}_1, \dots, \mathbf{u}_r)$  be a basis of  $\text{im } P$  and  $(\mathbf{w}_1, \dots, \mathbf{w}_s)$  a basis of  $\ker P$ . Then  $\mathcal{B} = (\mathbf{u}_1, \dots, \mathbf{u}_r, \mathbf{w}_1, \dots, \mathbf{w}_s)$  is a basis of  $V$ , and

$$[P]_{\mathcal{B}} = \begin{pmatrix} \mathbf{I}_r & 0 \\ 0 & 0 \end{pmatrix} = \text{diag}(1, \dots, 1, 0, \dots, 0),$$

with  $r$  ones.

*Proof.* By [Theorem 3.10.1](#) (a),  $V = \operatorname{im} P \oplus \ker P$ , so the concatenated list  $\mathcal{B}$  is a basis of  $V$  by [Theorem 1.7.5](#) ((a)  $\Rightarrow$  (d)). Since  $P\mathbf{u}_i = \mathbf{u}_i$  and  $P\mathbf{w}_j = \mathbf{0}$ , the columns of  $[P]_{\mathcal{B}}$  are  $\mathbf{e}_1, \dots, \mathbf{e}_r$  followed by  $s$  zero columns ([Definition 3.6.1](#)). ■

Conversely, an operator whose matrix in some basis is  $\operatorname{diag}(1, \dots, 1, 0, \dots, 0)$  is a projection, since that matrix squares to itself. With the change-of-basis square, this gives a way to write down any projection in standard coordinates.

### Example. (Projecting onto a plane along a line)

In  $\mathbb{R}^3$ , let  $U = \{(x, y, z) : x + y + z = 0\}$  and  $L = \operatorname{span}((1, 1, 1))$ , so that  $\mathbb{R}^3 = U \oplus L$  by [Example \(A plane and a line in  \$\mathbb{R}^3\$ \)](#). Find the standard matrix of the projection  $P$  onto  $U$  along  $L$ , and check that it squares to itself.

*Solution. Direct route.* By [Example \(A plane and a line in  \$\mathbb{R}^3\$ \)](#),  $(a, b, c)$  splits as  $(a - m, b - m, c - m) + m(1, 1, 1)$  with  $m = \frac{a+b+c}{3}$ , and the first piece lies in  $U$ . So

$$P(a, b, c) = \left( \frac{2a-b-c}{3}, \frac{-a+2b-c}{3}, \frac{-a-b+2c}{3} \right), \quad [P]_{\mathcal{E}} = \frac{1}{3} \begin{pmatrix} 2 & -1 & -1 \\ -1 & 2 & -1 \\ -1 & -1 & 2 \end{pmatrix}.$$

*Adapted basis.*  $(1, -1, 0)$  and  $(0, -1, 1)$  lie in  $U$  and are independent (look at the first and third entries), so they form a basis of the 2-dimensional plane  $U$ . With  $\mathcal{B} = ((1, -1, 0), (0, -1, 1), (1, 1, 1))$ , [Corollary 3.10.2](#) gives  $[P]_{\mathcal{B}} = \operatorname{diag}(1, 1, 0)$ , and by [Theorem 3.7.3](#)  $[P]_{\mathcal{E}} = M \operatorname{diag}(1, 1, 0) M^{-1}$  with  $M = [\operatorname{id}]_{\mathcal{B}}^{\mathcal{E}}$ , whose columns are the vectors of  $\mathcal{B}$ . Multiplying out gives the same matrix as the direct route.

*Check.* Let  $\mathbf{A}$  be the matrix found. The first column of  $\mathbf{A}^2$  is  $\mathbf{A}$  applied to  $\frac{1}{3}(2, -1, -1)$ , which is  $\frac{1}{9}(4 + 1 + 1, -2 - 2 + 1, -2 + 1 - 2) = \frac{1}{3}(2, -1, -1)$ , the first column of  $\mathbf{A}$ ; the other columns work the same way by symmetry of the entries. Also  $\mathbf{A}(1, 1, 1) = \mathbf{0}$  and  $\mathbf{A}(1, -1, 0) = (1, -1, 0)$ , as a projection onto  $U$  along  $L$  must satisfy. The division by 3 needs  $3 \neq 0$  in the field, which holds over  $\mathbb{R}$ .

*Quick check.* Is  $P = \begin{pmatrix} 1 & 1 \\ 0 & 0 \end{pmatrix}$  (acting on  $\mathbb{R}^2$ ) a projection? If so, onto which line, and along which line?

*Solution.*  $P^2 = \begin{pmatrix} 1 & 1 \\ 0 & 0 \end{pmatrix} \begin{pmatrix} 1 & 1 \\ 0 & 0 \end{pmatrix} = \begin{pmatrix} 1 & 1 \\ 0 & 0 \end{pmatrix} = P$ , so yes. Its image is spanned by its columns,  $\operatorname{span}((1, 0))$ , the  $x$ -axis. Its kernel is  $\{x + y = 0\} = \operatorname{span}((1, -1))$ . By [Theorem 3.10.1](#), it is the projection onto the  $x$ -axis along the line  $y = -x$ .

## 3.10.2 The trace of an operator

For a projection of rank  $r$ , the diagonal matrix of [Corollary 3.10.2](#) has trace  $1 + \dots + 1 = r \cdot 1$ . But traces are defined for matrices, and the same projection has many other matrices, such as the  $3 \times 3$  matrix of [Example \(Projecting onto a plane along a line\)](#), whose trace is  $\frac{2+2+2}{3} = 2$ .

The agreement is no accident. In [Theorem 3.7.5](#) we proved that similar matrices have the same trace, and by [Theorem 3.7.4](#) all matrices of one operator are similar. So the trace belongs to the operator.

*The trace of an operator is the trace of its matrix in any basis; the basis does not matter.*

### Definition 3.10.2 — Trace of an operator

Let  $V$  be a finite-dimensional vector space over  $F$  with  $\dim V \geq 1$ , and let  $T \in \mathcal{L}(V)$ . The **trace** of  $T$  is

$$\operatorname{tr} T := \operatorname{tr}[T]_{\mathcal{B}},$$

where  $\mathcal{B}$  is **any** basis of  $V$ . (For  $V = \{\mathbf{0}\}$  we set  $\operatorname{tr} T = 0$ .)

In words: to find  $\operatorname{tr} T$ , choose a basis, write down the matrix of  $T$  with that **same** basis on input and output, and add the diagonal entries.

**Well-definedness.** The definition names a basis but claims the answer does not depend on it. Let  $\mathcal{B}$  and  $\mathcal{C}$  be bases of  $V$ . By [Theorem 3.7.3](#),  $[T]_{\mathcal{C}} = P^{-1}[T]_{\mathcal{B}}P$  with  $P = [\operatorname{id}]_{\mathcal{C}}^{\mathcal{B}}$  invertible, so the two matrices are similar, and by [Theorem 3.7.5](#) they have the same trace. Hence  $\operatorname{tr} T$  depends only on  $T$ .

**Examples.**

- **Identity.** In every basis,  $[\operatorname{id}_V]_{\mathcal{B}} = \mathbf{I}_n$ , so  $\operatorname{tr} \operatorname{id}_V = n \cdot 1$ , where  $n = \dim V$ . Over  $\mathbb{R}$  this is  $n$ ; over  $\mathbb{F}_2$  it is 0 when  $n$  is even.
- **Differentiation.** On  $F[x]_{\leq n}$  with the basis  $(1, x, \dots, x^n)$ ,  $D(x^k) = kx^{k-1}$  has no  $x^k$ -component, so every diagonal entry of the matrix is 0 and  $\operatorname{tr} D = 0$ .
- **A shift of the variable.** On  $\mathbb{R}[x]_{\leq 2}$ , let  $T(p) = p(x+1)$ . In  $(1, x, x^2)$ ,  $T(1) = 1$ ,  $T(x) = 1 + x$ ,  $T(x^2) = 1 + 2x + x^2$ , so the matrix is  $\begin{pmatrix} 1 & 1 & 1 \\ 0 & 1 & 2 \\ 0 & 0 & 1 \end{pmatrix}$  and  $\operatorname{tr} T = 3$ .
- **The reflection.**  $R(x, y) = (y, x)$  has trace 0, whether computed from  $\begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}$  or from  $\operatorname{diag}(1, -1)$  ([Example \(The reflection in a mirror-adapted basis\)](#)).

*Warning.* **The trace needs the same basis on both sides.** For an operator  $T \in \mathcal{L}(V)$ , the matrix  $[T]_{\mathcal{B}}^{\mathcal{C}}$  with  $\mathcal{B} \neq \mathcal{C}$  is square, but its trace depends on the bases. For  $\operatorname{id}$  on  $\mathbb{R}^2$ , with  $\mathcal{E}$  the standard basis and  $\mathcal{C} = ((2, 0), (0, 1))$ , we get  $[\operatorname{id}]_{\mathcal{C}}^{\mathcal{E}} = \operatorname{diag}(\frac{1}{2}, 1)$ , of trace  $\frac{3}{2}$ , while  $\operatorname{tr} \operatorname{id} = 2$ . Different bases on input and output give equivalent matrices, not similar ones, and equivalence does not preserve the trace. For the same reason there is no trace of a map  $V \rightarrow W$  between different spaces.

The rules for the trace of a matrix carry over at once.

### Theorem 3.10.3 — Properties of the Trace of an Operator



Let  $V$  be finite-dimensional with  $n = \dim V$ , let  $S, T \in \mathcal{L}(V)$  and  $c \in F$ . Then

$$\begin{aligned}\operatorname{tr}(S + T) &= \operatorname{tr} S + \operatorname{tr} T, \\ \operatorname{tr}(cT) &= c \operatorname{tr} T, \\ \operatorname{tr}(ST) &= \operatorname{tr}(TS), \\ \operatorname{tr} \operatorname{id}_V &= n \cdot 1.\end{aligned}$$

*Proof.* For  $n = 0$  all traces are 0. Otherwise fix a basis  $\mathcal{B}$  of  $V$ . The map  $T \mapsto [T]_{\mathcal{B}}$  is linear by [Theorem 3.6.4](#), so  $[S + T]_{\mathcal{B}} = [S]_{\mathcal{B}} + [T]_{\mathcal{B}}$  and  $[cT]_{\mathcal{B}} = c[T]_{\mathcal{B}}$ ; and  $[ST]_{\mathcal{B}} = [S]_{\mathcal{B}}[T]_{\mathcal{B}}$  by [Theorem 3.6.2](#), and likewise for  $TS$ . The first three identities now follow from [Theorem 0.9.6](#), since  $\operatorname{tr} T$  may be computed in  $\mathcal{B}$ . The last was computed in the examples. ■

A first consequence: on a finite-dimensional space  $V$  of dimension  $n \geq 1$  over a field in which  $n \cdot 1 \neq 0$ , such as  $\mathbb{R}$ , there are no operators with  $ST - TS = \operatorname{id}_V$ , since the left side has trace 0 and the right side has trace  $n \cdot 1$ .

Now the trace of a projection. Over  $\mathbb{R}$  it is simply the rank, but the precise statement is an equation in the field, and reading it as an equation of integers needs a hypothesis.

### Theorem 3.10.4 — Trace of a Projection

Let  $V$  be a finite-dimensional vector space over  $F$  and  $P \in \mathcal{L}(V)$  a projection. Then

$$\operatorname{tr} P = (\operatorname{rank} P) \cdot 1 \quad \text{in } F.$$

If  $F$  has characteristic 0, or characteristic  $p > \dim V$ , then  $\operatorname{rank} P$  is determined by  $\operatorname{tr} P$ : it is the unique integer  $r$  with  $0 \leq r \leq \dim V$  and  $r \cdot 1 = \operatorname{tr} P$ . In particular  $\operatorname{rank} P = \operatorname{tr} P$  for projections on real or complex spaces.

*Proof.* Let  $r = \operatorname{rank} P$ . If  $V = \{0\}$ , both sides are 0. Otherwise, by [Corollary 3.10.2](#) there is a basis  $\mathcal{B}$  with  $[P]_{\mathcal{B}} = \operatorname{diag}(1, \dots, 1, 0, \dots, 0)$  with  $r$  ones, so  $\operatorname{tr} P = r \cdot 1$  by [Definition 3.10.2](#).

For the second statement, let  $n = \dim V$ . Since  $0 \leq r \leq n$  and  $r \cdot 1 = \operatorname{tr} P$ , it remains to show uniqueness. Suppose  $k \cdot 1 = l \cdot 1$  for integers  $0 \leq l \leq k \leq n$ . By [Lemma 0.7.4](#),  $(k - l) \cdot 1 + l \cdot 1 = k \cdot 1 = l \cdot 1$ , and adding  $-(l \cdot 1)$  to both sides gives  $(k - l) \cdot 1 = 0$ . If  $k - l > 0$ , then by [Definition 0.7.2](#) the characteristic of  $F$  is non-zero and at most  $k - l \leq n$ , which contradicts the hypothesis. Hence  $k = l$ . In  $\mathbb{R}$  and  $\mathbb{C}$ ,  $r \cdot 1$  is the number  $r$ . ■

The hypothesis cannot be dropped. Over  $\mathbb{F}_2$ , the identity of  $\mathbb{F}_2^2$  is a projection of rank 2, and its trace is  $1 + 1 = 0$ , the same as the trace of the zero projection, of rank 0. The equation  $\operatorname{tr} P = (\operatorname{rank} P) \cdot 1$  still holds there; what fails is recovering the integer  $\operatorname{rank} P$  from the field element  $\operatorname{tr} P$ .

### 3.10.3 Involutions and reflections

A projection satisfies  $P^2 = P$ . The other simple polynomial relation is  $T^2 = \text{id}_V$ : doing  $T$  twice changes nothing. Such an operator is called an **involution**. Examples are the reflection  $R(x, y) = (y, x)$ , the transpose  $\mathbf{A} \mapsto \mathbf{A}^\top$  on  $M_n(F)$ , and  $f(x) \mapsto f(-x)$  on functions  $\mathbb{R} \rightarrow \mathbb{R}$ . In Chapter 1 the last two produced direct sums: symmetric plus skew-symmetric matrices ([Example \(Symmetric and skew-symmetric matrices\)](#)), and even plus odd functions ([Example \(Even and odd functions\)](#)). The next theorem explains both at once. For an involution  $T$ , write

$$E_+ := \{\mathbf{v} \in V : T\mathbf{v} = \mathbf{v}\}, \quad E_- := \{\mathbf{v} \in V : T\mathbf{v} = -\mathbf{v}\}.$$

These are subspaces, since  $E_+ = \ker(T - \text{id}_V)$  and  $E_- = \ker(T + \text{id}_V)$  ([Theorem 3.2.1](#)).

#### Theorem 3.10.5 — Involutions Split the Space

Let  $F$  be a field of characteristic not 2,  $V$  a vector space over  $F$ , and  $T \in \mathcal{L}(V)$  with  $T^2 = \text{id}_V$ . Then

$$V = E_+ \oplus E_-,$$

and the projection onto  $E_+$  along  $E_-$  is  $\frac{1}{2}(\text{id}_V + T)$ .

*Idea.* Copy the even/odd split:  $\mathbf{v} = \frac{1}{2}(\mathbf{v} + T\mathbf{v}) + \frac{1}{2}(\mathbf{v} - T\mathbf{v})$ . Applying  $T$  swaps  $\mathbf{v}$  and  $T\mathbf{v}$ , so the first piece is fixed and the second changes sign. Both halves of the argument divide by 2, which is where the characteristic enters.

*Proof.* Since the characteristic is not 2,  $2 = 1 + 1 \neq 0$  in  $F$  ([Definition 0.7.2](#)), so  $\frac{1}{2}$  exists. Let  $\mathbf{v} \in V$ , and put  $\mathbf{v}_+ = \frac{1}{2}(\mathbf{v} + T\mathbf{v})$  and  $\mathbf{v}_- = \frac{1}{2}(\mathbf{v} - T\mathbf{v})$ . Then  $\mathbf{v} = \mathbf{v}_+ + \mathbf{v}_-$ , and by linearity and  $T^2 = \text{id}_V$ ,

$$T\mathbf{v}_+ = \frac{1}{2}(T\mathbf{v} + \mathbf{v}) = \mathbf{v}_+, \quad T\mathbf{v}_- = \frac{1}{2}(T\mathbf{v} - \mathbf{v}) = -\mathbf{v}_-.$$

So  $\mathbf{v}_+ \in E_+$ ,  $\mathbf{v}_- \in E_-$ , and  $V = E_+ + E_-$ . If  $\mathbf{v} \in E_+ \cap E_-$ , then  $\mathbf{v} = T\mathbf{v} = -\mathbf{v}$ , so  $2\mathbf{v} = \mathbf{0}$  and  $\mathbf{v} = \frac{1}{2}(2\mathbf{v}) = \mathbf{0}$ . By [Theorem 1.7.4](#),  $V = E_+ \oplus E_-$ .

The operator  $\frac{1}{2}(\text{id}_V + T)$  is linear by [Theorem 3.4.1](#), sends  $\mathbf{v}$  to  $\mathbf{v}_+$ , fixes  $E_+$  (where  $T\mathbf{v} = \mathbf{v}$ ) and kills  $E_-$  (where  $T\mathbf{v} = -\mathbf{v}$ ). By the uniqueness in [Theorem 3.10.1](#) (b), it is the projection onto  $E_+$  along  $E_-$ . ■

For the transpose,  $E_+$  is the symmetric and  $E_-$  the skew-symmetric matrices; for  $f(x) \mapsto f(-x)$ , they are the even and odd functions; for the reflection  $R(x, y) = (y, x)$ , they are the mirror line  $y = x$  and the perpendicular line  $y = -x$ , and the adapted basis  $((1, 1), (1, -1))$  is the one of [Example \(The reflection in a mirror-adapted basis\)](#). In finite dimension, a basis of  $E_+$  followed by a basis of  $E_-$  gives  $[T]_{\mathcal{B}} = \text{diag}(1, \dots, 1, -1, \dots, -1)$ , and so  $\text{tr } T = (\dim E_+) \cdot 1 - (\dim E_-) \cdot 1$ . In Chapter 8 the non-zero vectors of  $E_+$  and  $E_-$  will be called eigenvectors with eigenvalues 1 and  $-1$ .

**Warning.** In characteristic 2 an involution need not split the space. Over  $\mathbb{F}_2$ ,

let  $T\mathbf{x} = \mathbf{J}\mathbf{x}$  with  $\mathbf{J} = \begin{pmatrix} 1 & 1 \\ 0 & 1 \end{pmatrix}$ . Then  $\mathbf{J}^2 = \begin{pmatrix} 1 & 2 \\ 0 & 1 \end{pmatrix} = \mathbf{I}_2$ , since  $2 = 0$ . But  $-1 = 1$ , so  $E_- = E_+ = \ker(\mathbf{J} - \mathbf{I}_2) = \text{span}(\mathbf{e}_1)$ , and  $E_+ + E_- = \text{span}(\mathbf{e}_1) \neq \mathbb{F}_2^2$ .

### 3.10.4 Exercises

#### 3.10.4.1 A. Check your understanding

##### Exercise A1.

- Define a projection  $P \in \mathcal{L}(V)$ , and state the correspondence between projections and direct sum decompositions.
- True or false: a projection is determined by its image. Justify your answer.
- True or false: an operator  $T$  on  $\mathbb{R}^2$  with  $\text{im } T$  the  $x$ -axis and  $\ker T$  the  $y$ -axis is a projection. Justify your answer.
- Explain why  $\text{tr } T$  does not depend on the basis used to compute it.
- True or false: for every field  $F$  and every projection  $P$  on a finite-dimensional space over  $F$ , the integer  $\text{rank } P$  can be recovered from  $\text{tr } P \in F$ . Justify your answer.
- Where does the hypothesis “characteristic not 2” enter the proof of [Theorem 3.10.5](#)?

*Solution.*

- $P$  is a projection if  $P^2 = P$  ([Definition 3.10.1](#)). By [Theorem 3.10.1](#), a projection gives  $V = \text{im } P \oplus \ker P$ , and each decomposition  $V = U \oplus W$  gives exactly one projection with image  $U$  and kernel  $W$ .
- False. On  $\mathbb{R}^2$ ,  $P(x, y) = (x, 0)$  and  $P'(x, y) = (x - y, 0)$  are different projections with the same image, the  $x$ -axis. A projection is determined by its image and its kernel together.
- False.  $T(x, y) = (2x, 0)$  has this image and kernel, but  $T^2 \neq T$ .
- Matrices of  $T$  in two bases are similar by [Theorem 3.7.3](#), and similar matrices have equal traces by [Theorem 3.7.5](#).
- False. Over  $\mathbb{F}_2$ ,  $\text{id}$  on  $\mathbb{F}_2^2$  has rank 2 and the zero operator has rank 0, but both have trace 0. It is true when the characteristic is 0 or larger than  $\dim V$  ([Theorem 3.10.4](#)).
- Twice: to define  $\mathbf{v}_\pm = \frac{1}{2}(\mathbf{v} \pm T\mathbf{v})$ , and to pass from  $2\mathbf{v} = \mathbf{0}$  to  $\mathbf{v} = \mathbf{0}$ . Both need  $2 \neq 0$  in  $F$ .

#### 3.10.4.2 B. Practice

##### Exercise B1: A projection in $\mathbb{R}^3$ .

Let  $U = \{(x, y, z) \in \mathbb{R}^3 : x + y - z = 0\}$  and  $L = \text{span}((1, 1, 1))$ .

- Show that  $\mathbb{R}^3 = U \oplus L$ .
- Find the standard matrix  $\mathbf{A}$  of the projection  $P$  onto  $U$  along  $L$ .

- (c) Verify that  $\mathbf{A}^2 = \mathbf{A}$ , and that  $\text{tr } \mathbf{A} = \text{rank } \mathbf{A}$ .

*Solution.*

- (a)  $U$  is the kernel of  $(x, y, z) \mapsto x + y - z$ , which is non-zero; its image is then a non-zero subspace of  $\mathbb{R}$ , hence all of  $\mathbb{R}$  (Theorem 1.6.8), so  $\dim U = 3 - 1 = 2$  by Theorem 3.3.1. If  $t(1, 1, 1) \in U$ , then  $t + t - t = t = 0$ , so  $U \cap L = \{\mathbf{0}\}$ . By Theorem 1.7.4,  $\dim(U + L) = 2 + 1 = 3$ , so  $U + L = \mathbb{R}^3$  by Theorem 1.6.8, and  $\mathbb{R}^3 = U \oplus L$ .
- (b) Split  $(a, b, c) = \mathbf{u} + t(1, 1, 1)$  with  $\mathbf{u} = (a - t, b - t, c - t) \in U$ : the condition  $(a - t) + (b - t) - (c - t) = 0$  gives  $t = a + b - c$ . So

$$P(a, b, c) = (-b + c, -a + c, -a - b + 2c), \quad \mathbf{A} = \begin{pmatrix} 0 & -1 & 1 \\ -1 & 0 & 1 \\ -1 & -1 & 2 \end{pmatrix}.$$

Check:  $\mathbf{A}(1, 1, 1) = \mathbf{0}$ , and for the vectors  $(1, 0, 1), (0, 1, 1) \in U$ ,  $\mathbf{A}(1, 0, 1) = (1, 0, 1)$  and  $\mathbf{A}(0, 1, 1) = (0, 1, 1)$ .

- (c) Row by row,

$$\mathbf{A}^2 = \begin{pmatrix} 0+1-1 & 0+0-1 & 0-1+2 \\ 0+0-1 & 1+0-1 & -1+0+2 \\ 0+1-2 & 1+0-2 & -1-1+4 \end{pmatrix} = \begin{pmatrix} 0 & -1 & 1 \\ -1 & 0 & 1 \\ -1 & -1 & 2 \end{pmatrix} = \mathbf{A}.$$

$\text{tr } \mathbf{A} = 0 + 0 + 2 = 2$ , and  $\text{rank } \mathbf{A} = \text{rank } P = \dim U = 2$  by Theorem 3.10.1 (b), in agreement with Theorem 3.10.4.

### Exercise B2: The complementary projection.

Let  $P \in \mathcal{L}(V)$  be a projection.

- (a) Prove that  $\text{id}_V - P$  is a projection.
- (b) Prove that  $\text{im}(\text{id}_V - P) = \ker P$  and  $\ker(\text{id}_V - P) = \text{im } P$ .
- (c) Deduce that if  $P$  is the projection onto  $U$  along  $W$ , then  $\text{id}_V - P$  is the projection onto  $W$  along  $U$ .

*Solution.*

- (a) By Theorem 3.4.2 (composition distributes over sums of maps),  $(\text{id}_V - P)^2 = \text{id}_V - 2P + P^2 = \text{id}_V - 2P + P = \text{id}_V - P$ .
- (b)  $(\subseteq)$   $P((\text{id}_V - P)(\mathbf{v})) = P\mathbf{v} - P^2(\mathbf{v}) = \mathbf{0}$ , so  $\text{im}(\text{id}_V - P) \subseteq \ker P$ .  $(\supseteq)$  If  $P\mathbf{v} = \mathbf{0}$ , then  $\mathbf{v} = (\text{id}_V - P)(\mathbf{v}) \in \text{im}(\text{id}_V - P)$ . For the kernel:  $(\text{id}_V - P)(\mathbf{v}) = \mathbf{0}$  means  $\mathbf{v} = P\mathbf{v}$ . Such  $\mathbf{v}$  lies in  $\text{im } P$ , and conversely every  $\mathbf{x} \in \text{im } P$  satisfies  $P\mathbf{x} = \mathbf{x}$ , as shown after Definition 3.10.1. So  $\ker(\text{id}_V - P) = \text{im } P$ .
- (c) By Theorem 3.10.1 (b),  $\text{im } P = U$  and  $\ker P = W$ . By (a) and (b),  $\text{id}_V - P$  is a projection with image  $W$  and kernel  $U$ . By the uniqueness in Theorem 3.10.1 (b), applied to  $V = W \oplus U$ , it is the projection onto  $W$  along  $U$ .

**Exercise B3: Traces of operators on matrices.**

- (a) Let  $\tau: M_n(F) \rightarrow M_n(F)$ ,  $\tau(\mathbf{A}) = \mathbf{A}^\top$ . Show that  $\text{tr } \tau = n \cdot 1$ .
- (b) Fix  $\mathbf{B} \in M_2(F)$ , and let  $L_{\mathbf{B}}: M_2(F) \rightarrow M_2(F)$ ,  $L_{\mathbf{B}}(\mathbf{X}) = \mathbf{B}\mathbf{X}$ . Show that  $\text{tr } L_{\mathbf{B}} = 2 \text{tr } \mathbf{B}$ .

*Hint: use the basis of matrix units  $\mathbf{E}_{ij}$ .*

*Solution.*

- (a) Order the basis  $(\mathbf{E}_{ij})$  of  $M_n(F)$  in any way. Since  $\tau(\mathbf{E}_{ij}) = \mathbf{E}_{ji}$ , the coordinate column of  $\tau(\mathbf{E}_{ij})$  has a 1 in the position of  $\mathbf{E}_{ji}$  and 0 elsewhere. The diagonal entry of the matrix in the column of  $\mathbf{E}_{ij}$  is therefore 1 if  $\mathbf{E}_{ji} = \mathbf{E}_{ij}$ , that is  $i = j$ , and 0 otherwise. There are  $n$  basis vectors  $\mathbf{E}_{ii}$ , so  $\text{tr } \tau = n \cdot 1$  by [Definition 3.10.2](#). (Consistently with [Theorem 3.10.5](#) over  $\mathbb{R}$ : symmetric matrices have dimension  $\frac{n(n+1)}{2}$ , skew-symmetric ones  $\frac{n(n-1)}{2}$ , and the difference is  $n$ .)
- (b) Use the basis  $(\mathbf{E}_{11}, \mathbf{E}_{12}, \mathbf{E}_{21}, \mathbf{E}_{22})$  and write  $\mathbf{B} = (b_{ij})$ . The product  $\mathbf{B}\mathbf{E}_{ij}$  has the  $i$ -th column of  $\mathbf{B}$  in column  $j$  and zeros elsewhere, so  $\mathbf{B}\mathbf{E}_{ij} = b_{1i}\mathbf{E}_{1j} + b_{2i}\mathbf{E}_{2j}$ . The diagonal entry in the column of  $\mathbf{E}_{ij}$  is the coefficient of  $\mathbf{E}_{ij}$  itself, which is  $b_{ii}$ . Adding over the four basis vectors,  $\text{tr } L_{\mathbf{B}} = b_{11} + b_{11} + b_{22} + b_{22} = 2 \text{tr } \mathbf{B}$ .

**3.10.4.3 C. Going deeper****Exercise C1: When is a sum of projections a projection?**

Let  $F$  have characteristic not 2, and let  $P, Q \in \mathcal{L}(V)$  be projections.

- (a) Prove that  $P + Q$  is a projection if and only if  $PQ = QP = 0$ .
- (b) Show that (a) fails over  $\mathbb{F}_2$ .
- (c) Suppose  $V$  is finite-dimensional over  $\mathbb{R}$  and  $P + Q$  is a projection. Deduce that  $\text{rank}(P + Q) = \text{rank } P + \text{rank } Q$ .

*Hint: for  $(\Rightarrow)$  in (a), use the relation  $PQ + QP = 0$  together with  $P^2 = P$ .*

*Solution.*

- (a) By [Theorem 3.4.2](#),  $(P + Q)^2 = P^2 + PQ + QP + Q^2 = P + Q + PQ + QP$ .  
 $(\Leftarrow)$  If  $PQ = QP = 0$ , this is  $P + Q$ .  
 $(\Rightarrow)$  If  $(P + Q)^2 = P + Q$ , then  $PQ + QP = 0$ . Multiplying on the left by  $P$  and using  $P^2 = P$  gives  $PQ + PQP = 0$ ; multiplying on the right by  $P$  gives  $PQP + QP = 0$ . Subtracting,  $PQ - QP = 0$ , so  $PQ = QP$ . Then  $2PQ = PQ + QP = 0$ , and since  $2 \neq 0$  in  $F$ ,  $PQ = 0$ . Hence also  $QP = 0$ .
- (b) Over  $\mathbb{F}_2$ , let  $P = Q = \text{id}_V$  with  $V \neq \{\mathbf{0}\}$ . Then  $P + Q = 2 \text{id}_V = 0$ , which is a projection, but  $PQ = \text{id}_V \neq 0$ . The step from  $2PQ = 0$  to  $PQ = 0$  is the one that fails.
- (c) By [Theorem 3.10.4](#) over  $\mathbb{R}$ , and [Theorem 3.10.3](#),

$$\text{rank}(P + Q) = \text{tr}(P + Q) = \text{tr } P + \text{tr } Q = \text{rank } P + \text{rank } Q.$$

**Exercise C2: The trace is the only trace-like functional.**

Let  $n \geq 1$ , and let  $\varphi: M_n(F) \rightarrow F$  be linear with  $\varphi(\mathbf{AB}) = \varphi(\mathbf{BA})$  for all  $\mathbf{A}, \mathbf{B} \in M_n(F)$ .

- (a) Using  $\mathbf{E}_{ij}\mathbf{E}_{jk} = \mathbf{E}_{ik}$  and  $\mathbf{E}_{jk}\mathbf{E}_{ij} = 0$  for  $k \neq i$ , show that  $\varphi(\mathbf{E}_{ik}) = 0$  whenever  $i \neq k$ .
- (b) Show that  $\varphi(\mathbf{E}_{ii}) = \varphi(\mathbf{E}_{jj})$  for all  $i, j$ .
- (c) Deduce that  $\varphi = c \operatorname{tr}$  for some  $c \in F$ , and conversely that every  $c \operatorname{tr}$  has the property.

*Solution.* Recall that  $\mathbf{E}_{ij}\mathbf{E}_{kl}$  is  $\mathbf{E}_{il}$  if  $j = k$  and 0 otherwise, by [Definition 0.9.9](#).

- (a) Let  $i \neq k$ . Then  $\mathbf{E}_{ik} = \mathbf{E}_{ii}\mathbf{E}_{ik}$  and  $\mathbf{E}_{ik}\mathbf{E}_{ii} = 0$ , because  $k \neq i$ . By the hypothesis and linearity,  $\varphi(\mathbf{E}_{ik}) = \varphi(\mathbf{E}_{ii}\mathbf{E}_{ik}) = \varphi(\mathbf{E}_{ik}\mathbf{E}_{ii}) = \varphi(0) = 0$ .
- (b)  $\mathbf{E}_{ii} = \mathbf{E}_{ij}\mathbf{E}_{ji}$  and  $\mathbf{E}_{jj} = \mathbf{E}_{ji}\mathbf{E}_{ij}$ . Hence  $\varphi(\mathbf{E}_{ii}) = \varphi(\mathbf{E}_{ij}\mathbf{E}_{ji}) = \varphi(\mathbf{E}_{ji}\mathbf{E}_{ij}) = \varphi(\mathbf{E}_{jj})$ .
- (c) Let  $c = \varphi(\mathbf{E}_{11})$ . For  $\mathbf{A} = (a_{ij}) = \sum_{i,j} a_{ij}\mathbf{E}_{ij}$ , linearity with (a) and (b) gives

$$\varphi(\mathbf{A}) = \sum_{i,j} a_{ij}\varphi(\mathbf{E}_{ij}) = \sum_i a_{ii}\varphi(\mathbf{E}_{ii}) = c \sum_i a_{ii} = c \operatorname{tr} \mathbf{A}.$$

Conversely,  $c \operatorname{tr}$  is linear and  $c \operatorname{tr}(\mathbf{AB}) = c \operatorname{tr}(\mathbf{BA})$  by [Theorem 0.9.6](#). So the linear functionals with  $\varphi(\mathbf{AB}) = \varphi(\mathbf{BA})$  are exactly the multiples of the trace.

# Chapter 4

## Duality

A linear map  $V \rightarrow F$  turns each vector into a single number. The  $i$ -th coordinate in a basis does this, and so do evaluating a polynomial at a point, integrating a function, and taking the trace of a matrix. These **linear functionals** are the measurements we can make on a vector space, and together they form a vector space of their own, the dual space  $V^*$ .

The dual space mirrors  $V$  with every inclusion reversed. A subspace corresponds to the functionals that vanish on it. An injective map corresponds to a surjective one. The kernel of a map corresponds to the image of its dual map, whose matrix is the transpose. Passing to the dual turns a hard question into an easy one surprisingly often. It also explains facts we already know from a new angle: why a subspace of  $F^n$  is always the solution set of a homogeneous system, and why row rank equals column rank.

**What you need.** From Chapter 1, bases, dimension and complements ([Definition 1.5.1](#), [Definition 1.6.1](#), [Theorem 1.7.6](#)), and bases of infinite-dimensional spaces ([Theorem 1.8.2](#)) for the remarks on infinite dimension. From Chapter 2, the row space and null space of a matrix ([Definition 2.5.2](#), [Definition 2.5.3](#)) and inverses by row reduction ([Theorem 2.4.6](#)). From Chapter 3, linear maps and the space  $\mathcal{L}(V, W)$  ([Definition 3.1.1](#), [Theorem 3.4.1](#)), the Rank–Nullity Theorem ([Theorem 3.3.1](#)), the matrix of a map and change of basis ([Definition 3.6.1](#), [Theorem 3.7.1](#)), and quotient spaces with the First Isomorphism Theorem ([Definition 3.9.3](#), [Theorem 3.9.7](#)).

### 4.0.1 Roadmap

- **Linear functionals and the dual space.** Measurements as vectors; dual bases; separating vectors by functionals.
- **Annihilators.** The functionals that vanish on a subspace, and the formula  $\dim U + \dim U^\circ = \dim V$ .
- **The double dual.** A natural isomorphism  $V \rightarrow V^{**}$  that needs no basis, and what “natural” means.
- **The dual map and the transpose.** Pulling functionals back along a map; its matrix is the transpose; kernels and images trade places.
- **Duality for subspaces and quotients.** The dual of a subspace is a quotient of the dual, and the dual of a quotient is an annihilator; hyperplanes and codimension.
- **Hyperplanes and linear systems revisited.** Every subspace is cut out by equations; converting between parametric and implicit descriptions.

## 4.0.2 Named moves

- **Test against functionals.** To show a vector is zero, show that every functional (or every vector of a dual basis) sends it to zero.
- **Pass to the dual and count.** Replace a subspace by its annihilator and use  $\dim U + \dim U^0 = \dim V$ .
- **Natural or chosen?** Ask whether an isomorphism needs a basis.  $V \cong V^*$  does;  $V \cong V^{**}$  does not.
- **Dualize a statement.** Injective becomes surjective, kernel becomes image, a matrix becomes its transpose, and inclusions reverse.

# 4.1 Linear Functionals and the Dual Space

Chapter 3 studied linear maps  $V \rightarrow W$  into arbitrary targets. The smallest non-zero target is the field  $F$  itself, and maps into it are everywhere: a coordinate, the value of a polynomial at a point, an integral, the trace. This section names these maps and shows that they form a vector space  $V^*$ , of the same dimension as  $V$  when  $V$  is finite-dimensional. It also proves the fact that makes them useful: every non-zero vector is detected by at least one of them.

## 4.1.1 Measuring a vector by a number

After studying vector spaces and the maps between them, we look at the simplest possible target. Here are four maps we have already met, from four different families.

- **A coordinate.** Fix a basis  $\mathcal{B} = (\mathbf{v}_1, \dots, \mathbf{v}_n)$  of  $V$ . The map  $V \rightarrow F$  sending  $\mathbf{v}$  to the first entry of  $[\mathbf{v}]_{\mathcal{B}}$  respects sums and multiples, by [Theorem 1.5.2](#).
- **Evaluation.** The map  $F[x] \rightarrow F$ ,  $p \mapsto p(3)$ , satisfies  $(p + q)(3) = p(3) + q(3)$  and  $(cp)(3) = cp(3)$ , by [Theorem 0.8.5](#) (a constant  $c$  is a polynomial with value  $c$ ).
- **Integration.** The map  $\mathbb{R}[x] \rightarrow \mathbb{R}$ ,  $p \mapsto \int_0^1 p(t) dt$ , is linear by the linearity of the integral from calculus.
- **Trace.** The map  $\text{tr}: M_n(F) \rightarrow F$  is linear ([Example \(Trace and Transpose\)](#)).

The inputs are columns, polynomials and matrices. The outputs are always single scalars. Each map is a way of **measuring** a vector by one number, and each measurement respects sums and scalar multiples. The same expression keeps recurring, “a linear map whose target is  $F$ ”, and we will see that these maps form a space with a structure of its own. So we give them a name.

*A linear functional is a linear measurement: it assigns a scalar to each vector, and the measurement of a linear combination is the same combination of the measurements.*

### Definition 4.1.1 — Linear Functional

Let  $V$  be a vector space over  $F$ , and regard  $F$  as a vector space over itself. A **linear functional** on  $V$  is a linear map  $\varphi: V \rightarrow F$ . That is, for all  $\mathbf{u}, \mathbf{v} \in V$  and all  $c \in F$ ,

$$\varphi(\mathbf{u} + \mathbf{v}) = \varphi(\mathbf{u}) + \varphi(\mathbf{v}), \quad \varphi(c\mathbf{v}) = c\varphi(\mathbf{v}).$$



**Definition 4.1.2 — Dual Space**

Let  $V$  be a vector space over  $F$ . The **dual space** of  $V$  is the set of **all** linear functionals on  $V$ ,

$$V^* := \mathcal{L}(V, F),$$

with the pointwise operations  $(\varphi + \psi)(\mathbf{v}) := \varphi(\mathbf{v}) + \psi(\mathbf{v})$  and  $(c\varphi)(\mathbf{v}) := c\varphi(\mathbf{v})$  for  $\varphi, \psi \in V^*$ ,  $c \in F$  and  $\mathbf{v} \in V$ .

In words: a linear functional is a linear map (Definition 3.1.1) whose codomain is the one-dimensional space  $F$ . Its values lie in the **same** field  $F$  over which  $V$  is a vector space. The first condition says that measuring a sum gives the sum of the measurements, and the second says that scaling a vector scales its measurement. The dual space collects **every** such measurement, and it adds them the natural way: the measurement  $\varphi + \psi$  reports the sum of the two readings. We write functionals with Greek letters  $\varphi, \psi$ , which keeps them apart from vectors of  $V$ .

**Well-definedness.** Two things need checking: that  $\varphi + \psi$  and  $c\varphi$  are again linear, and that the vector space axioms hold. Both are the case  $W = F$  of Theorem 3.4.1. So  $V^*$  is a vector space over  $F$ . Its zero vector is the **zero functional**  $\mathbf{v} \mapsto 0$ , and the negative of  $\varphi$  is  $\mathbf{v} \mapsto -\varphi(\mathbf{v})$ . Note that  $V^*$  is a set of **functions on**  $V$ , not a subset of  $V$ .

**Example. (Four Families of Functionals)**

Show that each map is a linear functional.

- (a)  $\varepsilon_c: F[x] \rightarrow F$ ,  $p \mapsto p(c)$ , for a fixed  $c \in F$  (**evaluation at**  $c$ ).
- (b)  $\iota: \mathbb{R}[x]_{\leq 2} \rightarrow \mathbb{R}$ ,  $p \mapsto \int_0^1 p(t) dt$ .
- (c)  $\text{tr}: M_n(F) \rightarrow F$ .
- (d) The zero functional on any  $V$ . Also find the dual space of the zero space  $\{\mathbf{0}\}$ .

*Solution.*

- (a) By Theorem 0.8.5,  $(p+q)(c) = p(c) + q(c)$ , and, regarding  $ap$  as the product of the constant polynomial  $a$  with  $p$ ,  $(ap)(c) = a(c)p(c) = ap(c)$ , because the constant polynomial  $a$  has value  $a$  at  $c$ . These are the two conditions of Definition 4.1.1.
- (b) By linearity of the integral,  $\int_0^1 (p+q) = \int_0^1 p + \int_0^1 q$  and  $\int_0^1 ap = a \int_0^1 p$ . On coefficients,  $\iota(a_0 + a_1x + a_2x^2) = a_0 + \frac{a_1}{2} + \frac{a_2}{3}$ .
- (c) This is Theorem 0.9.6 (1):  $\text{tr}(\mathbf{A} + \mathbf{B}) = \text{tr} \mathbf{A} + \text{tr} \mathbf{B}$  and  $\text{tr}(c\mathbf{A}) = c \text{tr} \mathbf{A}$ .
- (d)  $0 = 0 + 0$  and  $0 = c \cdot 0$  in  $F$ , so the zero functional is linear. On  $\{\mathbf{0}\}$ , a linear functional must send  $\mathbf{0}$  to 0 (Theorem 3.1.1), and there is only one vector. So the zero functional is the only element, and  $\{\mathbf{0}\}^* = \{0\}$ . This degenerate case matters: it is the case  $n = 0$  of the count  $\dim V^* = \dim V$  proved below.

The family  $F^n$  can be described completely. A row vector times a column is a scalar, and every functional on  $F^n$  arises this way.

**Theorem 4.1.1 — Functionals on  $F^n$  Are Row Vectors**

For  $\mathbf{a} = (a_1, \dots, a_n) \in F^n$ , define  $\varphi_{\mathbf{a}}: F^n \rightarrow F$  by

$$\varphi_{\mathbf{a}}(\mathbf{x}) := \mathbf{a}^\top \mathbf{x} = a_1x_1 + \dots + a_nx_n.$$

Then  $\varphi_{\mathbf{a}} \in (F^n)^*$ . Every  $\varphi \in (F^n)^*$  equals  $\varphi_{\mathbf{a}}$  for **exactly one**  $\mathbf{a}$ , namely  $\mathbf{a} = (\varphi(\mathbf{e}_1), \dots, \varphi(\mathbf{e}_n))$ . The map  $F^n \rightarrow (F^n)^*$ ,  $\mathbf{a} \mapsto \varphi_{\mathbf{a}}$ , is an isomorphism.

*Proof.* The map  $\varphi_{\mathbf{a}}$  is  $T_{\mathbf{a}^\top}$  for the  $1 \times n$  matrix  $\mathbf{a}^\top$ , with a  $1 \times 1$  matrix read as a scalar, so it is linear by [Example \(Multiplication by a Matrix\)](#). Let  $\varphi \in (F^n)^*$  and put  $a_i := \varphi(\mathbf{e}_i)$ . Then  $\varphi_{\mathbf{a}}(\mathbf{e}_i) = a_i = \varphi(\mathbf{e}_i)$  for each  $i$ . Two linear maps that agree on the basis  $(\mathbf{e}_1, \dots, \mathbf{e}_n)$  are equal by the uniqueness part of [Theorem 3.1.5](#), so  $\varphi = \varphi_{\mathbf{a}}$ . If  $\varphi_{\mathbf{a}} = \varphi_{\mathbf{b}}$ , then  $a_i = \varphi_{\mathbf{a}}(\mathbf{e}_i) = \varphi_{\mathbf{b}}(\mathbf{e}_i) = b_i$  for every  $i$ , so  $\mathbf{a} = \mathbf{b}$ .

Hence  $\mathbf{a} \mapsto \varphi_{\mathbf{a}}$  is a bijection  $F^n \rightarrow (F^n)^*$ . It is linear, since  $\varphi_{\mathbf{a}+c\mathbf{b}}(\mathbf{x}) = \sum_i (a_i + cb_i)x_i = \varphi_{\mathbf{a}}(\mathbf{x}) + c\varphi_{\mathbf{b}}(\mathbf{x})$  for every  $\mathbf{x}$ . By [Theorem 3.5.1](#) it is an isomorphism. ■

So in  $F^n$  a functional **is** a row vector acting by matrix multiplication, and its  $1 \times n$  matrix in the standard bases is  $(\varphi(\mathbf{e}_1) \ \dots \ \varphi(\mathbf{e}_n))$ . This is how we compute with functionals: rows act on columns.

**Non-examples by minimal change.** Each of the following changes one of the examples slightly. In each case something still works, and we name a condition that fails.

- On  $\mathbb{R}^2$ , the length  $(x, y) \mapsto \sqrt{x^2 + y^2}$  ([Example \(Length\)](#)). It sends  $\mathbf{0}$  to 0 and respects **positive** scalars. But with  $c = -1$  and  $\mathbf{v} = (1, 0)$  it gives 1, not  $-1$ , so the scaling condition fails.
- On  $\mathbb{R}[x]$ , squaring the evaluation at 0:  $p \mapsto p(0)^2$ . The outputs are still scalars. But for  $p = q = 1$  we get  $(1 + 1)^2 = 4 \neq 1 + 1$ , so additivity fails.
- On  $M_2(F)$ , replace the trace  $a + d$  of  $\mathbf{A} = \begin{pmatrix} a & b \\ c & d \end{pmatrix}$  by  $ad - bc$ . Then  $\mathbf{E}_{11} \mapsto 0$  and  $\mathbf{E}_{22} \mapsto 0$ , but  $\mathbf{E}_{11} + \mathbf{E}_{22} = \mathbf{I}_2 \mapsto 1$ , so additivity fails. The expression  $ad - bc$  is linear in each **row** separately, which is a different property.

**Why this definition.** Why single out the target  $F$ ? Because nothing is lost. A linear map  $T: V \rightarrow F^m$  is the same thing as  $m$  functionals: if  $\varphi_i(\mathbf{v})$  denotes the  $i$ -th entry of  $T\mathbf{v}$ , then each  $\varphi_i$  is linear and  $T\mathbf{v} = (\varphi_1(\mathbf{v}), \dots, \varphi_m(\mathbf{v}))$ ; conversely any  $m$  functionals stacked this way give a linear map. So functionals are the “rows” of linear maps, and understanding them one at a time is a way of understanding all maps into  $F^m$ . The name comes from analysis, where the vectors were themselves functions and a map such as  $f \mapsto \int_0^1 f$  was called a “functional”, a function of a function. The word “dual” will explain itself in this chapter: every statement about  $V$  has a mirror statement about  $V^*$ .

**Warning.** The values must lie in the field of scalars of  $V$ . Regard  $\mathbb{C}$  as a vector space over  $\mathbb{C}$ . The real part  $z \mapsto \operatorname{Re} z$  is additive, and it respects **real** scalars, but it is **not** a linear functional on  $\mathbb{C}$  over  $\mathbb{C}$ : with  $c = i$  and  $z = 1$ ,  $\operatorname{Re}(i \cdot 1) = 0 \neq i \cdot \operatorname{Re}(1) = i$ . It is a functional on  $\mathbb{C}$  regarded as a vector space over  $\mathbb{R}$ . The dual space depends on the field.

The first payoff is a count that holds for **every** non-zero functional: it hits every scalar, and in finite dimension it kills exactly one dimension.

**Proposition 4.1.2 — A Non-Zero Functional Is Surjective**

Let  $V$  be a vector space over  $F$ , and let  $\varphi \in V^*$  be **non-zero**. Then  $\varphi$  is surjective. If  $V$  is finite-dimensional, then  $\dim \ker \varphi = \dim V - 1$ .

*Proof.* Since  $\varphi \neq 0$ , there is  $\mathbf{u} \in V$  with  $c := \varphi(\mathbf{u}) \neq 0$ . For every  $a \in F$ , linearity gives  $\varphi(ac^{-1}\mathbf{u}) = ac^{-1}c = a$ , so  $\varphi$  is surjective. Hence  $\text{im } \varphi = F$  and  $\text{rank } \varphi = 1$ . If  $V$  is finite-dimensional, the Rank–Nullity Theorem (Theorem 3.3.1) gives  $\dim \ker \varphi = \dim V - 1$ . ■

For example,  $\ker \text{tr}$  in  $M_n(F)$  has dimension  $n^2 - 1$ , and the polynomials in  $F[x]_{\leq n}$  with  $p(c) = 0$  form a subspace of dimension  $n$ . Every non-zero functional cuts out a subspace one dimension smaller than the whole space. Later in this chapter these kernels, the hyperplanes, become the building blocks of every subspace.

*Quick check.* Which of the following are linear functionals? (a)  $p \mapsto p(0) + 1$  on  $\mathbb{R}[x]$ . (b)  $p \mapsto p(1) - 2p'(0)$  on  $\mathbb{R}[x]$ . (c)  $\mathbf{A} \mapsto a_{12}$  on  $M_2(F)$ .

*Solution.*

- (a) No. It sends the zero polynomial to 1, but a linear map sends 0 to 0 (Theorem 3.1.1). (b) Yes. Evaluation at 1, differentiation and evaluation at 0 are linear, and  $V^*$  is closed under linear combinations and composition with linear maps (Theorem 3.4.1, Theorem 3.4.2). (c) Yes. It is the coordinate of  $\mathbf{A}$  at  $\mathbf{E}_{12}$  in the standard basis of Example (Standard Bases), and coordinates are linear (Theorem 1.5.2).

**4.1.2 The dual basis**

Return to the coordinate functionals. Let  $\mathcal{B} = (\mathbf{v}_1, \dots, \mathbf{v}_n)$  be a basis of  $V$ , and let  $\varphi_i(\mathbf{v})$  be the  $i$ -th entry of  $[\mathbf{v}]_{\mathcal{B}}$ . The basis vector  $\mathbf{v}_j$  has coordinate vector  $\mathbf{e}_j$ , so

$$\varphi_i(\mathbf{v}_j) = \delta_{ij} = \begin{cases} 1 & \text{if } i = j, \\ 0 & \text{if } i \neq j, \end{cases}$$

with the Kronecker delta of Definition 0.9.3. Each  $\varphi_i$  reads off one coordinate and ignores the others. These  $n$  functionals are the natural candidates for a basis of  $V^*$ , and the theorem says they are one.

**Theorem 4.1.3 — The Dual Basis**

Let  $V$  be a finite-dimensional vector space over  $F$ , and let  $\mathcal{B} = (\mathbf{v}_1, \dots, \mathbf{v}_n)$  be a basis of  $V$ .

- (a) There are **unique**  $\varphi_1, \dots, \varphi_n \in V^*$  with  $\varphi_i(\mathbf{v}_j) = \delta_{ij}$  for all  $i, j$ . For every  $\mathbf{v} \in V$ ,  $\varphi_i(\mathbf{v})$  is the  $i$ -th entry of  $[\mathbf{v}]_{\mathcal{B}}$ .  
 (b) The list  $(\varphi_1, \dots, \varphi_n)$  is a basis of  $V^*$ .

(c) For every  $\mathbf{v} \in V$  and every  $\varphi \in V^*$ ,

$$\mathbf{v} = \sum_{i=1}^n \varphi_i(\mathbf{v}) \mathbf{v}_i, \quad \varphi = \sum_{i=1}^n \varphi(\mathbf{v}_i) \varphi_i.$$

*Idea.* ① A functional is a linear map into  $F$ , and a linear map may be prescribed freely on a basis, so existence and uniqueness in (a) are one citation. ② Independence: apply a vanishing combination  $\sum c_i \varphi_i$  to  $\mathbf{v}_j$ ; the delta kills every term but  $c_j$ . ③ Spanning: we cannot count, because  $\dim V^*$  is exactly what we do not know yet. So we guess the coefficients. If  $\varphi = \sum c_i \varphi_i$ , then applying both sides to  $\mathbf{v}_j$  forces  $c_j = \varphi(\mathbf{v}_j)$ . With this guess, the two sides are linear maps that agree on a basis, hence equal.

*Proof.*

- (a) For each  $i$ , apply [Theorem 3.1.5](#) with  $W = F$  and the prescribed values  $\delta_{i1}, \dots, \delta_{in} \in F$ . It gives exactly one linear map  $\varphi_i: V \rightarrow F$  with  $\varphi_i(\mathbf{v}_j) = \delta_{ij}$  for all  $j$ . If  $\mathbf{v} = a_1 \mathbf{v}_1 + \dots + a_n \mathbf{v}_n$ , then by [Theorem 3.1.4](#)

$$\varphi_i(\mathbf{v}) = \sum_{j=1}^n a_j \varphi_i(\mathbf{v}_j) = \sum_{j=1}^n a_j \delta_{ij} = a_i,$$

the  $i$ -th entry of  $[\mathbf{v}]_{\mathcal{B}}$  ([Definition 1.5.2](#)).

- (b) *Independence.* Let  $c_1 \varphi_1 + \dots + c_n \varphi_n = 0$  in  $V^*$ . Applying both sides to  $\mathbf{v}_j$  and using the pointwise operations of [Definition 4.1.2](#),

$$0 = \sum_{i=1}^n c_i \varphi_i(\mathbf{v}_j) = \sum_{i=1}^n c_i \delta_{ij} = c_j$$

for each  $j$ . Hence  $(\varphi_1, \dots, \varphi_n)$  is linearly independent.

*Spanning.* Let  $\varphi \in V^*$ , and put  $\psi := \sum_{i=1}^n \varphi(\mathbf{v}_i) \varphi_i \in V^*$ . For each  $j$ ,  $\psi(\mathbf{v}_j) = \sum_i \varphi(\mathbf{v}_i) \delta_{ij} = \varphi(\mathbf{v}_j)$ . So  $\psi$  and  $\varphi$  are linear maps  $V \rightarrow F$  that agree on the basis  $\mathcal{B}$ , and  $\psi = \varphi$  by the uniqueness part of [Theorem 3.1.5](#). Hence  $\varphi \in \text{span}(\varphi_1, \dots, \varphi_n)$ , and  $(\varphi_1, \dots, \varphi_n)$  is a basis of  $V^*$ .

- (c) The second formula was proved in the spanning step. The first is the definition of coordinates, since  $\varphi_i(\mathbf{v})$  is the  $i$ -th coordinate of  $\mathbf{v}$  by (a). This proves the theorem. ■

The two formulas in (c) are mirror images. A vector is recovered from its measurements by the dual basis; a functional is recovered from its values on the basis.

### Definition 4.1.3 — Dual Basis

Let  $\mathcal{B} = (\mathbf{v}_1, \dots, \mathbf{v}_n)$  be a basis of a finite-dimensional vector space  $V$  over  $F$ . The basis  $(\varphi_1, \dots, \varphi_n)$  of  $V^*$  given by [Theorem 4.1.3](#), characterized by  $\varphi_i(\mathbf{v}_j) = \delta_{ij}$ , is **the dual basis** of  $\mathcal{B}$ . We write it  $\mathcal{B}^*$ .

Counting the dual basis gives the dimension of the dual space at once.

### Corollary 4.1.4 — Dimension of the Dual Space

Let  $V$  be a finite-dimensional vector space over  $F$ . Then  $V^*$  is finite-dimensional and  $\dim V^* = \dim V$ . In particular  $V^* \cong V$ .

*Proof.* Let  $n = \dim V$  and choose a basis of  $V$  (Corollary 1.5.5). By Theorem 4.1.3 (b), its dual basis is a basis of  $V^*$  of length  $n$ . Hence  $\dim V^* = n$ , and  $V^* \cong V$  by Theorem 3.5.4. ■

For instance  $\dim M_{m \times n}(F)^* = mn$  and  $\dim F[x]_{\leq n}^* = n + 1$ .

*Remark.* Many books write  $\mathbf{v}_i^*$  for the dual basis functional  $\varphi_i$ . We avoid that notation for two reasons. It suggests that  $\varphi_i$  is determined by  $\mathbf{v}_i$  alone, which is false (see the warning below). And in this book the star on a map is reserved for adjoints, which come much later. We write  $\varphi_i$ , and speak of “the dual basis of  $\mathcal{B}$ ”.

How do we find a dual basis in practice? In  $F^n$ , functionals are rows, and the conditions  $\varphi_i(\mathbf{v}_j) = \delta_{ij}$  turn into a single matrix equation.

#### Example. (The Dual Basis Is Read Off the Inverse Matrix)

- (a) Find the dual basis of  $\mathcal{B} = ((1, 1), (1, -1))$  in  $\mathbb{R}^2$ .
- (b) More generally, let  $\mathcal{B} = (\mathbf{v}_1, \dots, \mathbf{v}_n)$  be a basis of  $F^n$ , and let  $\mathbf{P} = (\mathbf{v}_1 \ \cdots \ \mathbf{v}_n)$ . Show that  $\mathbf{P}$  is invertible and that the dual basis of  $\mathcal{B}$  is  $\varphi_i(\mathbf{x}) = \mathbf{r}_i \mathbf{x}$ , where  $\mathbf{r}_i$  is the  $i$ -th row of  $\mathbf{P}^{-1}$ .

*Solution.*

- (a) By Theorem 4.1.1,  $\varphi_1(x, y) = px + qy$  for some  $p, q \in \mathbb{R}$ , that is,  $\varphi_1(\mathbf{x}) = (p \ q) \mathbf{x}$ . The conditions  $\varphi_1(1, 1) = 1$  and  $\varphi_1(1, -1) = 0$  say  $(p \ q) \mathbf{P} = (1 \ 0)$  with  $\mathbf{P} = \begin{pmatrix} 1 & 1 \\ 1 & -1 \end{pmatrix}$ . Likewise the row  $\mathbf{r}_2$  of  $\varphi_2$  satisfies  $\mathbf{r}_2 \mathbf{P} = (0 \ 1)$ . Stacking the two rows into a matrix  $\mathbf{R}$ , the four conditions become  $\mathbf{R} \mathbf{P} = \mathbf{I}_2$ , so  $\mathbf{R} = \mathbf{P}^{-1}$ . By Theorem 0.9.8, with  $ad - bc = -2$ ,

$$\mathbf{P}^{-1} = \frac{1}{-2} \begin{pmatrix} -1 & -1 \\ -1 & 1 \end{pmatrix} = \begin{pmatrix} \frac{1}{2} & \frac{1}{2} \\ \frac{1}{2} & -\frac{1}{2} \end{pmatrix}.$$

Hence

$$\varphi_1(x, y) = \frac{x+y}{2}, \quad \varphi_2(x, y) = \frac{x-y}{2}.$$

Check:  $\varphi_1(1, 1) = 1$ ,  $\varphi_1(1, -1) = 0$ ,  $\varphi_2(1, 1) = 0$ ,  $\varphi_2(1, -1) = 1$ . And the first formula of Theorem 4.1.3 (c) for  $\mathbf{v} = (3, 1)$  reads  $(3, 1) = 2(1, 1) + 1(1, -1)$ , which is true. The second formula expresses a functional in the dual basis through its values on  $\mathcal{B}$ : for  $\psi(x, y) = x$ , the values are  $\psi(1, 1) = 1$  and  $\psi(1, -1) = 1$ , so  $\psi = \varphi_1 + \varphi_2$ , and indeed  $\frac{x+y}{2} + \frac{x-y}{2} = x$ .

- (b) The columns of  $\mathbf{P}$  form a basis of  $F^n$ , so  $\mathbf{P}$  is invertible by Theorem 2.4.4. Let  $\mathbf{R} \in M_n(F)$  have rows  $\mathbf{r}_1, \dots, \mathbf{r}_n$ , and let  $\psi_i(\mathbf{x}) := \mathbf{r}_i \mathbf{x}$ . By the entry formula for a

product (Theorem 0.9.2),  $\psi_i(\mathbf{v}_j) = \mathbf{r}_i \mathbf{v}_j = (\mathbf{R}\mathbf{P})_{ij}$ . Hence  $\psi_i(\mathbf{v}_j) = \delta_{ij}$  for all  $i, j$  if and only if  $\mathbf{R}\mathbf{P} = \mathbf{I}_n$ , if and only if  $\mathbf{R} = \mathbf{P}^{-1}$  (multiply by  $\mathbf{P}^{-1}$  on the right). Now take  $\mathbf{R} = \mathbf{P}^{-1}$ : the functionals  $\psi_i(\mathbf{x}) = \mathbf{r}_i \mathbf{x}$  are linear (Theorem 4.1.1) and satisfy  $\psi_i(\mathbf{v}_j) = \delta_{ij}$ , so by the uniqueness in Theorem 4.1.3 (a) they are the dual basis.

This agrees with Chapter 3:  $\mathbf{P} = [\text{id}]_{\mathcal{B}}^{\mathcal{E}}$  and  $\mathbf{P}^{-1} = [\text{id}]_{\mathcal{E}}^{\mathcal{B}}$  (Theorem 3.7.1), so  $\mathbf{P}^{-1}\mathbf{x} = [\mathbf{x}]_{\mathcal{B}}$ , and row  $i$  of  $\mathbf{P}^{-1}$  computes the  $i$ -th coordinate, as Theorem 4.1.3 (a) predicts. The columns of  $\mathbf{P}$  are the basis; the rows of  $\mathbf{P}^{-1}$  are the dual basis.

**Warning.** A dual basis functional depends on the whole basis, not only on its own vector. In  $\mathbb{R}^2$ , the basis  $((1, 0), (0, 1))$  has first dual functional  $\varphi_1(x, y) = x$ . The basis  $((1, 0), (1, 1))$  has the same first vector, but  $\mathbf{P}^{-1} = \begin{pmatrix} 1 & -1 \\ 0 & 1 \end{pmatrix}$ , so its first dual functional is  $\varphi_1(x, y) = x - y$ . Changing  $\mathbf{v}_2$  changed  $\varphi_1$ , because  $\varphi_1$  must vanish on  $\mathbf{v}_2$ .

The same method works away from  $F^n$ , once we recognize the coordinates.

**Example. (Dual Bases in  $\mathbb{R}[x]_{\leq 2}$ )**

- (a) Show that the dual basis of  $(1, x, x^2)$  in  $\mathbb{R}[x]_{\leq 2}$  is  $(\varphi_0, \varphi_1, \varphi_2)$  with  $\varphi_0(p) = p(0)$ ,  $\varphi_1(p) = p'(0)$  and  $\varphi_2(p) = \frac{1}{2}p''(0)$ . (We index from 0 to match the exponents.)
- (b) Let  $\ell_{-1} = \frac{1}{2}x(x-1)$ ,  $\ell_0 = 1 - x^2$  and  $\ell_1 = \frac{1}{2}x(x+1)$ . Show that  $(\ell_{-1}, \ell_0, \ell_1)$  is a basis of  $\mathbb{R}[x]_{\leq 2}$  whose dual basis is  $(\varepsilon_{-1}, \varepsilon_0, \varepsilon_1)$ , the evaluations at  $-1, 0, 1$ . Deduce that  $p = p(-1)\ell_{-1} + p(0)\ell_0 + p(1)\ell_1$  for every  $p \in \mathbb{R}[x]_{\leq 2}$ .

*Solution.*

- (a) Each  $\varphi_k$  is linear, as a composition of differentiation (Example (Differentiation)) with evaluation at 0, scaled. For  $p = a_0 + a_1x + a_2x^2$ , we have  $p(0) = a_0$ ,  $p'(0) = a_1$  and  $\frac{1}{2}p''(0) = \frac{1}{2} \cdot 2a_2 = a_2$ . So  $\varphi_k(x^j) = \delta_{kj}$  for  $j, k \in \{0, 1, 2\}$ , and by the uniqueness in Theorem 4.1.3 (a) these are the dual basis. Over a general field  $F$ , the dual basis of  $(1, x, x^2)$  is still “the coefficient of  $x^k$ ”, but the formula  $\frac{1}{2}p''(0)$  needs  $2 \neq 0$  in  $F$ .
- (b) The values of the  $\ell$ ’s at the three points are

$$\begin{aligned}\ell_{-1}(-1) &= 1, \ell_{-1}(0) = 0, \ell_{-1}(1) = 0; \\ \ell_0(-1) &= 0, \ell_0(0) = 1, \ell_0(1) = 0; \\ \ell_1(-1) &= 0, \ell_1(0) = 0, \ell_1(1) = 1.\end{aligned}$$

That is,  $\varepsilon_i(\ell_j) = \delta_{ij}$  for  $i, j \in \{-1, 0, 1\}$ . *Basis.* Let  $c_{-1}\ell_{-1} + c_0\ell_0 + c_1\ell_1 = 0$ . Evaluating at  $j \in \{-1, 0, 1\}$  gives  $c_j = 0$ . So the list is independent, of length  $3 = \dim \mathbb{R}[x]_{\leq 2}$ , hence a basis by Theorem 1.6.4. *Dual basis.* The evaluations are linear (Example (Four Families of Functionals)) and satisfy  $\varepsilon_i(\ell_j) = \delta_{ij}$ , so they are the dual basis by the uniqueness in Theorem 4.1.3 (a). The first formula of Theorem 4.1.3 (c) now reads  $p = \sum_i \varepsilon_i(p) \ell_i = p(-1)\ell_{-1} + p(0)\ell_0 + p(1)\ell_1$ .

The last formula rebuilds a polynomial from three of its values. It is a preview of the Lagrange interpolation formula of Chapter 5, and it explains where the  $\ell$ ’s come from: they are the basis dual to the evaluations.

*Quick check.* Let  $(\mathbf{v}_1, \mathbf{v}_2, \mathbf{v}_3)$  be a basis of  $V$  with dual basis  $(\varphi_1, \varphi_2, \varphi_3)$ . Compute  $\varphi_2(3\mathbf{v}_1 - \mathbf{v}_2 + 5\mathbf{v}_3)$  and  $(2\varphi_1 + \varphi_3)(\mathbf{v}_3)$ .

*Solution.*  $\varphi_2(3\mathbf{v}_1 - \mathbf{v}_2 + 5\mathbf{v}_3) = 3 \cdot 0 - 1 \cdot 1 + 5 \cdot 0 = -1$ , the second coordinate.  $(2\varphi_1 + \varphi_3)(\mathbf{v}_3) = 2 \cdot 0 + 1 = 1$ .

*Warning.*  $V \cong V^*$  is true in finite dimension, but no isomorphism is singled out. Corollary 4.1.4 gives an isomorphism by sending  $\mathbf{v}_i \mapsto \varphi_i$ , and that depends on the basis. In  $\mathbb{R}^2$  with the standard basis, it sends  $(1, 0)$  to the functional  $(x, y) \mapsto x$ . With the basis  $((1, 0), (1, 1))$ , it sends the **same** vector  $(1, 0)$  to  $(x, y) \mapsto x - y$ , by the warning above. A vector does not come with “its” functional. Later in this chapter we make precise the sense in which  $V^*$  needs a choice to be identified with  $V$ , while the double dual  $V^{**}$  does not.

### 4.1.3 Testing against functionals

In  $F^n$  a vector is zero exactly when all its entries are zero, and the entries are the values of the coordinate functionals. The same holds in any finite-dimensional space, and it gives a flexible way to prove that a vector is zero.

#### Theorem 4.1.5 — Functionals Separate Vectors

Let  $V$  be a finite-dimensional vector space over  $F$ .

- (a) If  $\mathbf{v} \in V$  and  $\mathbf{v} \neq \mathbf{0}$ , then there is  $\varphi \in V^*$  with  $\varphi(\mathbf{v}) = 1$ .
- (b) If  $\mathbf{v} \in V$  satisfies  $\varphi(\mathbf{v}) = 0$  for **every**  $\varphi \in V^*$ , then  $\mathbf{v} = \mathbf{0}$ . It suffices that  $\varphi_i(\mathbf{v}) = 0$  for the functionals  $\varphi_i$  of **one** dual basis.
- (c) For  $\mathbf{u}, \mathbf{w} \in V$ :  $\mathbf{u} = \mathbf{w}$  if and only if  $\varphi(\mathbf{u}) = \varphi(\mathbf{w})$  for every  $\varphi \in V^*$ .

*Idea.* We need a functional that does not vanish at  $\mathbf{v}$ , and the only functionals we can build at will are dual basis functionals. The first one takes the value 1 on the first basis vector. So make  $\mathbf{v}$  the first vector of a basis: this is possible exactly because  $\mathbf{v} \neq \mathbf{0}$ .

*Proof.*

- (a) Since  $\mathbf{v} \neq \mathbf{0}$ , the list  $(\mathbf{v})$  is linearly independent: if  $a\mathbf{v} = \mathbf{0}$ , then  $a = 0$  by Theorem 1.1.9. By the Basis Extension Theorem (Theorem 1.6.5), there is a basis  $(\mathbf{v}, \mathbf{v}_2, \dots, \mathbf{v}_n)$  of  $V$ . Let  $(\varphi_1, \dots, \varphi_n)$  be its dual basis (Theorem 4.1.3). Then  $\varphi_1(\mathbf{v}) = 1$ .
- (b) The first statement is the contrapositive of (a). For the second, if  $(\varphi_1, \dots, \varphi_n)$  is the dual basis of a basis  $(\mathbf{v}_1, \dots, \mathbf{v}_n)$  and every  $\varphi_i(\mathbf{v}) = 0$ , then  $\mathbf{v} = \sum_i \varphi_i(\mathbf{v})\mathbf{v}_i = \mathbf{0}$  by Theorem 4.1.3 (c).
- (c) If  $\mathbf{u} = \mathbf{w}$ , the values agree. Conversely, if  $\varphi(\mathbf{u}) = \varphi(\mathbf{w})$  for every  $\varphi$ , then  $\varphi(\mathbf{u} - \mathbf{w}) = \varphi(\mathbf{u}) - \varphi(\mathbf{w}) = 0$  for every  $\varphi$ , so  $\mathbf{u} - \mathbf{w} = \mathbf{0}$  by (b).

■

This is the signature move of the chapter: **to show a vector is zero, test it against every functional, and a dual basis suffices.** It turns one equation in  $V$  into scalar equations, which are often easier to reach.

*Remark.* Parts (a), (b) (first sentence) and (c) hold for **every** vector space  $V$ ; only (a) needs a new construction. By [Theorem 1.8.2](#),  $\{\mathbf{v}\}$  extends to a basis  $B$  of  $V$ . Every  $\mathbf{w} \in V$  is a finite combination of distinct vectors of  $B$ , with coefficients that are unique because  $B$  is independent. Let  $\varphi(\mathbf{w})$  be the coefficient of  $\mathbf{v}$  in this expansion (and 0 if  $\mathbf{v}$  does not occur). Adding or scaling expansions gives expansions, so by uniqueness coefficients add and scale, and  $\varphi$  is linear with  $\varphi(\mathbf{v}) = 1$ . The price is Zorn's Lemma ([Theorem 1.8.1](#)): for  $\mathbb{R}$  over  $\mathbb{Q}$ , no one has an explicit formula for such a functional.

### Example. (A Matrix Killed by Every Trace)

Let  $\mathbf{A} \in M_n(F)$  satisfy  $\text{tr}(\mathbf{A}\mathbf{X}) = 0$  for every  $\mathbf{X} \in M_n(F)$ . Show that  $\mathbf{A} = 0$ . Deduce that every linear functional on  $M_n(F)$  is  $\mathbf{X} \mapsto \text{tr}(\mathbf{A}\mathbf{X})$  for exactly one  $\mathbf{A} \in M_n(F)$ .

*Solution.* The matrix units  $\mathbf{E}_{ij}$  form a basis of  $M_n(F)$  ([Example \(Standard Bases\)](#)), and  $\mathbf{A} = \sum_{i,j} a_{ij} \mathbf{E}_{ij}$ . By [Theorem 4.1.3](#) (a), its dual basis consists of the coordinate functionals  $\varepsilon_{ij}(\mathbf{A}) = a_{ij}$ . We compute  $\varepsilon_{ij}(\mathbf{A})$  as a trace. By [Definition 0.9.9](#),  $(\mathbf{A}\mathbf{E}_{ji})_{kk} = \sum_l a_{kl}(\mathbf{E}_{ji})_{lk}$ , and  $(\mathbf{E}_{ji})_{lk} = 1$  only for  $l = j$  and  $k = i$ . So the only non-zero diagonal entry of  $\mathbf{A}\mathbf{E}_{ji}$  is  $(\mathbf{A}\mathbf{E}_{ji})_{ii} = a_{ij}$ , and

$$\text{tr}(\mathbf{A}\mathbf{E}_{ji}) = a_{ij} = \varepsilon_{ij}(\mathbf{A}).$$

Taking  $\mathbf{X} = \mathbf{E}_{ji}$  in the hypothesis gives  $\varepsilon_{ij}(\mathbf{A}) = 0$  for all  $i, j$ . By [Theorem 4.1.5](#) (b),  $\mathbf{A} = 0$ .

For the deduction, let  $\Theta: M_n(F) \rightarrow M_n(F)^*$ ,  $\Theta(\mathbf{A})(\mathbf{X}) := \text{tr}(\mathbf{A}\mathbf{X})$ . Each  $\Theta(\mathbf{A})$  is linear in  $\mathbf{X}$ , and  $\Theta$  is linear in  $\mathbf{A}$ , both by [Theorem 0.9.6](#) (1) and the distributive laws of [Theorem 0.9.4](#). We just showed  $\ker \Theta = \{0\}$ , so  $\Theta$  is injective ([Theorem 3.2.3](#)). Since  $\dim M_n(F)^* = n^2 = \dim M_n(F)$  ([Corollary 4.1.4](#)),  $\Theta$  is bijective by [Corollary 3.3.2](#) (e). This is the matrix analogue of [Theorem 4.1.1](#).

## 4.1.4 When $V$ is infinite-dimensional

The main results of the last two subsections assumed finite dimension, or used a finite basis. That hypothesis does real work. Consider  $F[x]$ , with its basis  $\{1, x, x^2, \dots\}$  ([Example \(A Basis of  \$F\[x\]\$ \)](#)). For  $k \in \mathbb{N}$  let

$$\varphi_k(p) := \text{the coefficient of } x^k \text{ in } p.$$

These are the coordinate functionals, and  $\varphi_k(x^j) = \delta_{kj}$ , exactly as in [Theorem 4.1.3](#). But they are not a basis of the dual space.

### Proposition 4.1.6 — The Coordinate Functionals Do Not Span $F[x]^*$

The functionals  $\varphi_k$  ( $k \in \mathbb{N}$ ) lie in  $F[x]^*$  and form a linearly independent set. But they



do **not** span  $F[x]^*$ : the evaluation  $\varepsilon_1: p \mapsto p(1)$  is **not** a linear combination of finitely many  $\varphi_k$ .

*Proof.* Polynomials are added and scaled coefficient by coefficient (Definition 0.8.1), so each  $\varphi_k$  is linear. *Independence.* Let  $k_1, \dots, k_m$  be distinct and  $c_1\varphi_{k_1} + \dots + c_m\varphi_{k_m} = 0$ . Applying both sides to  $x^{k_j}$  gives  $c_j = 0$ , since  $\varphi_{k_i}(x^{k_j}) = \delta_{k_i k_j}$ .

*Not spanning.* The evaluation  $\varepsilon_1$  is linear by Example (Four Families of Functionals) (a), and  $\varepsilon_1(x^N) = 1^N = 1$  for every  $N \in \mathbb{N}$ . Suppose, for a contradiction, that  $\varepsilon_1 = c_1\varphi_{k_1} + \dots + c_m\varphi_{k_m}$  for some  $m \in \mathbb{N}$ , distinct  $k_j$  and  $c_j \in F$ . Choose  $N \in \mathbb{N}$  larger than every  $k_j$ . Then  $\varphi_{k_j}(x^N) = 0$  for every  $j$ , so the right-hand side sends  $x^N$  to 0, while  $\varepsilon_1(x^N) = 1$ . This contradiction shows  $\varepsilon_1 \notin \text{span}\{\varphi_k : k \in \mathbb{N}\}$ . ■

Where did the proof of Theorem 4.1.3 break? In the spanning step. The candidate  $\sum_i \varphi(\mathbf{v}_i)\varphi_i$  for  $\varphi = \varepsilon_1$  would be  $\varphi_0 + \varphi_1 + \varphi_2 + \dots$ , an infinite sum, and infinite sums mean nothing in a vector space (the warning in Chapter 1, §8). A functional on  $F[x]$  may take non-zero values on infinitely many basis vectors, while a finite combination of the  $\varphi_k$  cannot.

*Remark.* More is true, and we state it without proof. For every infinite-dimensional  $V$ , a basis of  $V^*$  is **strictly larger**, as an infinite set, than a basis of  $V$ ; in particular  $V^* \not\cong V$ . So “ $\dim V^* = \dim V$ ” is a genuinely finite-dimensional fact, and this is why finite dimension appears in the hypotheses of most theorems in this chapter. A first step towards seeing how large  $F[x]^*$  is appears in Exercise C4 below.

## 4.1.5 Exercises

### 4.1.5.1 A. Check your understanding

#### Exercise A1.

- Define a linear functional on a vector space  $V$  over  $F$ , and the dual space  $V^*$ .
- True or false:  $p \mapsto p(1)^2$  is a linear functional on  $\mathbb{R}[x]$ . Justify your answer.
- True or false: if  $(\mathbf{v}_1, \mathbf{v}_2)$  and  $(\mathbf{v}_1, \mathbf{w}_2)$  are bases of  $V$ , then the first functionals of their dual bases are equal. Justify your answer.
- What is  $\dim M_{2 \times 3}(F)^*$ ?
- State the formula that expresses  $\varphi \in V^*$  in the dual basis of  $(\mathbf{v}_1, \dots, \mathbf{v}_n)$ .
- Name the move you would use to prove that a vector  $\mathbf{v}$  of a finite-dimensional space is  $\mathbf{0}$  without computing  $\mathbf{v}$  itself.

*Solution.*

- See Definition 4.1.1 and Definition 4.1.2: a linear functional is a linear map  $V \rightarrow F$ , and  $V^* = \mathcal{L}(V, F)$  with pointwise addition and scaling.
- False. For  $p = 1$  and  $c = 2$ , the polynomial  $cp = 2$  is sent to  $2^2 = 4$ , while  $c$  times the image of  $p$  is  $2 \cdot 1^2 = 2$ . So the scaling condition fails.
- False. By the warning after Example (The Dual Basis Is Read Off the Inverse

Matrix), in  $\mathbb{R}^2$  the bases  $((1, 0), (0, 1))$  and  $((1, 0), (1, 1))$  have first dual functionals  $x$  and  $x - y$ .

- (d) 6, by [Corollary 4.1.4](#), since  $\dim M_{2 \times 3}(F) = 6$ .
- (e)  $\varphi = \sum_{i=1}^n \varphi(\mathbf{v}_i) \varphi_i$ , by [Theorem 4.1.3](#) (c).
- (f) Test against functionals: show  $\varphi(\mathbf{v}) = 0$  for every  $\varphi \in V^*$ , or only for the functionals of one dual basis, and apply [Theorem 4.1.5](#) (b).

#### 4.1.5.2 B. Practice

##### Exercise B1: Finding dual bases.

- (a) Find the dual basis of  $((1, 0, 1), (0, 1, 1), (1, 1, 1))$  in  $\mathbb{R}^3$ .
- (b) Show that the dual basis of  $(1, x - 1, (x - 1)^2)$  in  $\mathbb{R}[x]_{\leq 2}$  is  $p \mapsto p(1)$ ,  $p \mapsto p'(1)$ ,  $p \mapsto \frac{1}{2}p''(1)$ . Hence write  $p = 2 + 3x - x^2$  in this basis.

*Solution.*

- (a) Let  $\mathbf{P}$  have the three vectors as columns. Row reduction of  $[\mathbf{P} \mid \mathbf{I}_3]$  ([Theorem 2.4.6](#)) gives

$$\mathbf{P} = \begin{pmatrix} 1 & 0 & 1 \\ 0 & 1 & 1 \\ 1 & 1 & 1 \end{pmatrix}, \quad \mathbf{P}^{-1} = \begin{pmatrix} 0 & -1 & 1 \\ -1 & 0 & 1 \\ 1 & 1 & -1 \end{pmatrix},$$

and one checks  $\mathbf{P}^{-1}\mathbf{P} = \mathbf{I}_3$ . By [Example \(The Dual Basis Is Read Off the Inverse Matrix\)](#) (b), the dual basis is

$$\varphi_1(x, y, z) = -y + z, \quad \varphi_2(x, y, z) = -x + z, \quad \varphi_3(x, y, z) = x + y - z.$$

For instance  $\varphi_3(1, 0, 1) = 0$ ,  $\varphi_3(0, 1, 1) = 0$  and  $\varphi_3(1, 1, 1) = 1$ .

- (b) The three maps are linear, as combinations of differentiation and evaluation at 1. Call them  $\psi_1, \psi_2, \psi_3$ . On the basis:

$$\begin{aligned} \psi_1(1) &= 1, \quad \psi_1(x - 1) = 0, \quad \psi_1((x - 1)^2) = 0; \\ \psi_2(1) &= 0, \quad \psi_2(x - 1) = 1, \quad \psi_2((x - 1)^2) = 2(1 - 1) = 0; \end{aligned}$$

$$\psi_3(1) = 0, \quad \psi_3(x - 1) = 0, \quad \psi_3((x - 1)^2) = \frac{1}{2} \cdot 2 = 1.$$

So  $\psi_i$  takes the value  $\delta_{ij}$  on the  $j$ -th basis vector, and by the uniqueness in [Theorem 4.1.3](#) (a) they are the dual basis. (The list  $(1, x - 1, (x - 1)^2)$  is a basis: it has distinct degrees 0, 1, 2, so it is independent by [Theorem 1.4.4](#), and it has length  $3 = \dim \mathbb{R}[x]_{\leq 2}$ , so [Theorem 1.6.4](#) applies.) For  $p = 2 + 3x - x^2$ :  $p(1) = 4$ ,  $p'(1) = 3 - 2 = 1$ ,  $\frac{1}{2}p''(1) = -1$ . By [Theorem 4.1.3](#) (c),

$$p = 4 + (x - 1) - (x - 1)^2.$$

Check:  $4 + x - 1 - (x^2 - 2x + 1) = 2 + 3x - x^2$ .

##### Exercise B2: Which maps are functionals?.

Determine which of the following are linear functionals. Justify your answer.

- (a)  $\mathbb{R}^3 \rightarrow \mathbb{R}, (x, y, z) \mapsto 2x - z$ .
- (b)  $\mathbb{R}^3 \rightarrow \mathbb{R}, (x, y, z) \mapsto xy$ .
- (c)  $M_2(\mathbb{R}) \rightarrow \mathbb{R}, \mathbf{A} \mapsto a_{12} - 3a_{21}$ .
- (d)  $\mathbb{R}[x] \rightarrow \mathbb{R}, p \mapsto \int_0^1 t p(t) dt$ .
- (e)  $\mathbb{R}[x]_{\leq 2} \rightarrow \mathbb{R}, p \mapsto p(0) + p'(0) + 1$ .
- (f)  $\mathbb{C}^2 \rightarrow \mathbb{C}, (z, w) \mapsto z + \bar{w}$ , where  $\mathbb{C}^2$  is a vector space over  $\mathbb{C}$ .

*Solution.*

- (a) Yes: it is  $\varphi_{\mathbf{a}}$  with  $\mathbf{a} = (2, 0, -1)$ , linear by [Theorem 4.1.1](#).
- (b) No.  $(1, 1, 0) \mapsto 1$ , but  $2(1, 1, 0) = (2, 2, 0) \mapsto 4 \neq 2$ , so the scaling condition fails.
- (c) Yes: it is  $\varepsilon_{12} - 3\varepsilon_{21}$ , a combination of coordinate functionals for the standard basis, and  $V^*$  is a vector space ([Definition 4.1.2](#)).
- (d) Yes. By linearity of the integral,  $\int_0^1 t(p + q)(t) dt = \int_0^1 tp(t) dt + \int_0^1 tq(t) dt$  and  $\int_0^1 t(cp)(t) dt = c \int_0^1 tp(t) dt$ .
- (e) No. The zero polynomial is sent to  $1 \neq 0$ , contradicting [Theorem 3.1.1](#).
- (f) No. With  $c = i$  and  $(z, w) = (0, 1)$ :  $(0, i) \mapsto \bar{i} = -i$ , while  $i \cdot (0 + \bar{1}) = i$ . The map is additive and respects real scalars, but it is not linear over  $\mathbb{C}$ .

### Exercise B3: Coordinates of a functional.

- (a) Let  $(\varphi_1, \varphi_2)$  be the dual basis of  $((1, 1), (1, -1))$  from [Example \(The Dual Basis Is Read Off the Inverse Matrix\)](#). Write  $\psi(x, y) = 3x + 5y$  as a combination of  $\varphi_1$  and  $\varphi_2$ .
- (b) Let  $(\varphi_0, \varphi_1, \varphi_2)$  be the dual basis of  $(1, x, x^2)$  from [Example \(Dual Bases in  \$\mathbb{R}\[x\]\_{\leq 2}\$ \)](#). Write  $\iota(p) = \int_0^1 p(t) dt$  as a combination of them.

*Solution.*

- (a) By [Theorem 4.1.3](#) (c),  $\psi = \psi(1, 1)\varphi_1 + \psi(1, -1)\varphi_2 = 8\varphi_1 - 2\varphi_2$ . Check:  $8 \cdot \frac{x+y}{2} - 2 \cdot \frac{x-y}{2} = 4x + 4y - x + y = 3x + 5y$ .
- (b) By [Theorem 4.1.3](#) (c),  $\iota = \iota(1)\varphi_0 + \iota(x)\varphi_1 + \iota(x^2)\varphi_2 = \varphi_0 + \frac{1}{2}\varphi_1 + \frac{1}{3}\varphi_2$ . Check: for  $p = a_0 + a_1x + a_2x^2$ , the right-hand side is  $a_0 + \frac{a_1}{2} + \frac{a_2}{3} = \int_0^1 p$ .

### 4.1.5.3 C. Going deeper

#### Exercise C1: Functionals with the same kernel.

Let  $V$  be **any** vector space over  $F$ , and let  $\varphi, \psi \in V^*$  with  $\ker \varphi = \ker \psi$ . Prove that  $\psi = c\varphi$  for some  $c \in F$ . Is the converse true?

*Hint: if  $\varphi \neq 0$ , fix  $\mathbf{u}$  with  $\varphi(\mathbf{u}) = 1$  and consider  $\mathbf{v} - \varphi(\mathbf{v})\mathbf{u}$ .*

*Solution.* If  $\varphi = 0$ , then  $\ker \psi = \ker \varphi = V$ , so  $\psi = 0 = 1 \cdot \varphi$ . Suppose  $\varphi \neq 0$ . By

**Proposition 4.1.2** there is  $\mathbf{u} \in V$  with  $\varphi(\mathbf{u}) = 1$ . Let  $\mathbf{v} \in V$ . Then

$$\varphi(\mathbf{v} - \varphi(\mathbf{v})\mathbf{u}) = \varphi(\mathbf{v}) - \varphi(\mathbf{v})\varphi(\mathbf{u}) = 0,$$

so  $\mathbf{v} - \varphi(\mathbf{v})\mathbf{u} \in \ker \varphi = \ker \psi$ . Therefore  $0 = \psi(\mathbf{v} - \varphi(\mathbf{v})\mathbf{u})$ , that is,  $\psi(\mathbf{v}) = \psi(\mathbf{u})\varphi(\mathbf{v})$ . Since  $\mathbf{v}$  was arbitrary,  $\psi = c\varphi$  with  $c = \psi(\mathbf{u})$ .

The converse holds only for  $c \neq 0$ : if  $\psi = c\varphi$  with  $c \neq 0$ , then  $\psi(\mathbf{v}) = 0 \iff \varphi(\mathbf{v}) = 0$ . For  $c = 0$  and  $\varphi \neq 0$ ,  $\ker \psi = V \neq \ker \varphi$ .

### Exercise C2: Functionals vanishing where others vanish.

Let  $V$  be a vector space over  $F$ , and let  $\varphi_1, \dots, \varphi_k, \psi \in V^*$ .

- (a) Prove that if  $\ker \varphi_1 \cap \dots \cap \ker \varphi_k \subseteq \ker \psi$ , then  $\psi \in \text{span}(\varphi_1, \dots, \varphi_k)$ .
- (b) Deduce that if  $V$  is finite-dimensional, then  $(\varphi_1, \dots, \varphi_k)$  spans  $V^*$  if and only if  $\ker \varphi_1 \cap \dots \cap \ker \varphi_k = \{\mathbf{0}\}$ .

*Hint: for (a), consider the linear map  $T: V \rightarrow F^k$ ,  $\mathbf{v} \mapsto (\varphi_1(\mathbf{v}), \dots, \varphi_k(\mathbf{v}))$ .*

*Solution.*

- (a) Let  $T$  be as in the hint; it is linear because each  $\varphi_i$  is. Its kernel is  $\ker \varphi_1 \cap \dots \cap \ker \varphi_k$ . Define  $S: \text{im } T \rightarrow F$  by  $S(T\mathbf{v}) := \psi(\mathbf{v})$ . This is well defined: if  $T\mathbf{v} = T\mathbf{v}'$ , then  $\mathbf{v} - \mathbf{v}' \in \ker T \subseteq \ker \psi$ , so  $\psi(\mathbf{v}) = \psi(\mathbf{v}')$ . It is linear, since  $S(cT\mathbf{v} + T\mathbf{v}') = S(T(c\mathbf{v} + \mathbf{v}')) = \psi(c\mathbf{v} + \mathbf{v}') = c\psi(\mathbf{v}) + \psi(\mathbf{v}') = cS(T\mathbf{v}) + S(T\mathbf{v}')$ . Now extend  $S$  to  $F^k$ . The subspace  $\text{im } T \subseteq F^k$  has a basis  $(\mathbf{y}_1, \dots, \mathbf{y}_r)$  ([Theorem 1.6.7](#), [Corollary 1.5.5](#)), which extends to a basis  $(\mathbf{y}_1, \dots, \mathbf{y}_k)$  of  $F^k$  by [Theorem 1.6.5](#). By [Theorem 3.1.5](#) there is  $\tilde{S} \in (F^k)^*$  with  $\tilde{S}(\mathbf{y}_j) = S(\mathbf{y}_j)$  for  $j \leq r$  and  $\tilde{S}(\mathbf{y}_j) = 0$  for  $j > r$ . Then  $\tilde{S}$  and  $S$  agree on a basis of  $\text{im } T$ , so  $\tilde{S} = S$  on  $\text{im } T$ . By [Theorem 4.1.1](#),  $\tilde{S}(z_1, \dots, z_k) = a_1 z_1 + \dots + a_k z_k$  for some  $a_i \in F$ . Hence for every  $\mathbf{v} \in V$ ,

$$\psi(\mathbf{v}) = S(T\mathbf{v}) = \tilde{S}(\varphi_1(\mathbf{v}), \dots, \varphi_k(\mathbf{v})) = a_1 \varphi_1(\mathbf{v}) + \dots + a_k \varphi_k(\mathbf{v}),$$

so  $\psi = a_1 \varphi_1 + \dots + a_k \varphi_k$ . Note that  $V$  was arbitrary; only  $F^k$  needed to be finite-dimensional.

- (b) ( $\Leftarrow$ ) If the intersection is  $\{\mathbf{0}\}$ , it lies in  $\ker \psi$  for every  $\psi \in V^*$ , so every  $\psi$  is in the span by (a). ( $\Rightarrow$ ) Suppose the  $\varphi_i$  span  $V^*$ , and let  $\mathbf{v}$  lie in every  $\ker \varphi_i$ . Every  $\psi \in V^*$  is a combination  $\sum c_i \varphi_i$ , so  $\psi(\mathbf{v}) = \sum c_i \varphi_i(\mathbf{v}) = 0$ . By [Theorem 4.1.5](#) (b),  $\mathbf{v} = \mathbf{0}$ .

### Exercise C3: Evaluations as a dual basis.

Let  $n \in \mathbb{N}$ .

- (a) Let  $x_0, \dots, x_n \in F$  be **distinct**. Prove that the evaluations  $(\varepsilon_{x_0}, \dots, \varepsilon_{x_n})$  form a basis of  $F[x]_{\leq n}^*$ , and that they are the dual basis of some basis of  $F[x]_{\leq n}$ .
- (b) Over  $\mathbb{F}_2$  with  $n = 2$ , show that the evaluations at the elements of  $\mathbb{F}_2$  do **not** span  $\mathbb{F}_2[x]_{\leq 2}^*$ .

*Hint: for (a), use Theorem 2.7.3 to build the basis first.*

*Solution.*

- (a) For each  $j$ , Theorem 2.7.3 gives  $\ell_j \in F[x]_{\leq n}$  with  $\ell_j(x_i) = \delta_{ij}$  for  $i = 0, \dots, n$ , because the  $x_i$  are distinct. If  $\sum_j c_j \ell_j = 0$ , evaluating at  $x_i$  gives  $c_i = 0$ . So  $(\ell_0, \dots, \ell_n)$  is independent of length  $n+1 = \dim F[x]_{\leq n}$ , hence a basis by Theorem 1.6.4. The evaluations are linear (Example (Four Families of Functionals)) and  $\varepsilon_{x_i}(\ell_j) = \delta_{ij}$ , so by the uniqueness in Theorem 4.1.3 (a) they are the dual basis of  $(\ell_0, \dots, \ell_n)$ , and by Theorem 4.1.3 (b) a basis of  $F[x]_{\leq n}^*$ .
- (b)  $\mathbb{F}_2 = \{0, 1\}$  has only two elements, so only  $\varepsilon_0$  and  $\varepsilon_1$  are available, and  $\dim \mathbb{F}_2[x]_{\leq 2}^* = 3$  by Corollary 4.1.4. Two functionals cannot span a 3-dimensional space (Theorem 1.6.3). Concretely,  $p = x^2 + x$  is non-zero but  $p(0) = 0$  and  $p(1) = 1 + 1 = 0$ ; the coefficient functional of  $x^2$  takes the value 1 on  $p$ , so it is not a combination of  $\varepsilon_0, \varepsilon_1$ . The hypothesis “distinct points” in (a) needs  $n+1$  distinct elements of  $F$ , which a small field may not have.

#### Exercise C4: How large is $F[x]^*$ ?

Let  $F^{\mathbb{N}}$  be the space of sequences with entries in  $F$ .

- (a) Prove that  $\Phi: F[x]^* \rightarrow F^{\mathbb{N}}$ ,  $\varphi \mapsto (\varphi(1), \varphi(x), \varphi(x^2), \dots)$ , is an isomorphism.
- (b) Identify  $\Phi(\varphi_k)$  for the coordinate functionals  $\varphi_k$  of Proposition 4.1.6, and  $\Phi(\varepsilon_1)$ . Explain how Proposition 4.1.6 matches Example (The Vectors  $\mathbf{e}_i$  Do Not Span  $F^{\mathbb{N}}$ ).

*Hint: for surjectivity, remember that a polynomial has only finitely many non-zero coefficients.*

*Solution.*

- (a) *Linear.* The  $k$ -th entry of  $\Phi(c\varphi + \psi)$  is  $(c\varphi + \psi)(x^k) = c\varphi(x^k) + \psi(x^k)$ , by the point-wise operations of Definition 4.1.2. *Injective.* If  $\Phi(\varphi) = \mathbf{0}$ , then  $\varphi$  vanishes on the basis  $\{x^k\}$  of Example (A Basis of  $F[x]$ ). Every polynomial is a finite combination of the  $x^k$ , so  $\varphi = 0$  by Theorem 3.1.4; hence  $\Phi$  is injective by Theorem 3.2.3. *Surjective.* Given  $(a_0, a_1, \dots) \in F^{\mathbb{N}}$ , define  $\varphi(b_0 + b_1x + \dots + b_Nx^N) := a_0b_0 + \dots + a_Nb_N$ . This is a finite sum, it is well defined because coefficients are unique, and it is linear because coefficients add and scale; it is the linear map with  $\varphi(x^k) = a_k$  given by Theorem 3.1.6. Then  $\Phi(\varphi) = (a_0, a_1, \dots)$ . By Theorem 3.5.1,  $\Phi$  is an isomorphism.
- (b)  $\Phi(\varphi_k) = (\delta_{k0}, \delta_{k1}, \dots) = \mathbf{e}_k$ , and  $\Phi(\varepsilon_1) = (1, 1, 1, \dots)$ . An isomorphism carries spans to spans, so  $\varepsilon_1 \in \text{span}\{\varphi_k\}$  would give  $(1, 1, 1, \dots) \in \text{span}\{\mathbf{e}_k\}$ . Example (The Vectors  $\mathbf{e}_i$  Do Not Span  $F^{\mathbb{N}}$ ) showed this is false, because finite combinations of the  $\mathbf{e}_k$  have only finitely many non-zero entries. So  $F[x]$ , with its basis  $\{x^k\}$  indexed by  $\mathbb{N}$ , has a dual as large as the whole sequence space  $F^{\mathbb{N}}$ .

## 4.2 Annihilators

Section 1 built the dual space  $V^*$ , the space of linear measurements on  $V$ . Now we connect subspaces of  $V$  with subspaces of  $V^*$ . A plane in  $\mathbb{R}^3$  can be described by a spanning list or by an equation, and an equation is a functional. This section collects all the equations that a set of vectors satisfies into one subspace of  $V^*$ , the annihilator, counts its dimension, and shows that it turns sums into intersections and intersections into sums.

### 4.2.1 The equations a subspace satisfies

Every object comes with its subobjects. The dual space  $V^*$  has subspaces of its own, and the question is which of them are attached to subspaces of  $V$ . The standard example shows the way.

Let  $U = \text{span}((1, 0, 1), (0, 1, 1))$ , a plane in  $\mathbb{R}^3$ . The functional  $\varphi(x, y, z) = x + y - z$  vanishes on both spanning vectors, so by linearity it vanishes on all of  $U$ : the plane satisfies the equation  $x + y - z = 0$ . So do  $2x + 2y - 2z = 0$  and the trivial equation  $0 = 0$ . Which functionals vanish on  $U$ ? By [Theorem 4.1.1](#) a functional is  $\psi(x, y, z) = ax + by + cz$ . It vanishes on the two spanning vectors if and only if  $a + c = 0$  and  $b + c = 0$ , that is,  $\psi = -c(x + y - z) = -c\varphi$ . So the equations of the plane form the **line**  $\text{span}(\varphi)$  in  $(\mathbb{R}^3)^*$ .

For the  $z$ -axis  $\text{span}((0, 0, 1))$ , the same computation gives  $c = 0$ : the equations are  $ax + by = 0$ , a **plane** in  $(\mathbb{R}^3)^*$ . A bigger subspace satisfies fewer equations, and in both cases the dimensions add up to 3. The set of equations deserves a name, and the pattern deserves a theorem.

*The annihilator of a set of vectors is the space of all linear equations that every one of those vectors satisfies.*

#### Definition 4.2.1 — Annihilator

Let  $V$  be a vector space over  $F$ , and let  $S \subseteq V$  be **any subset**. The **annihilator** of  $S$  is

$$S^0 := \{\varphi \in V^* : \varphi(s) = 0 \text{ for all } s \in S\}.$$

In words:  $S^0$  is a subset of the **dual space**  $V^*$ , not of  $V$ . A functional belongs to it when it vanishes at **every** vector of  $S$ ; a single non-zero value disqualifies it. The set  $S$  needs no structure: it may be a subspace, a finite list of vectors, or even empty. The superscript 0 recalls the value that the functionals take on  $S$ .

**Well-definedness.** The definition makes sense for every subset. What needs checking is that the result is a subspace, which is what makes its dimension meaningful.

#### Proposition 4.2.1 — The Annihilator Is a Subspace

Let  $V$  be a vector space over  $F$ , and let  $S \subseteq V$ . Then  $S^0$  is a subspace of  $V^*$ .

*Proof.* We use the subspace test ([Theorem 1.2.1](#)) in  $V^*$ . The zero functional sends every  $s \in S$  to 0, so it lies in  $S^0$ . Let  $\varphi, \psi \in S^0$  and  $c \in F$ . For every  $s \in S$ , the pointwise

operations of Definition 4.1.2 give

$$(\varphi + \psi)(\mathbf{s}) = \varphi(\mathbf{s}) + \psi(\mathbf{s}) = 0 + 0 = 0, \quad (c\varphi)(\mathbf{s}) = c\varphi(\mathbf{s}) = c \cdot 0 = 0.$$

Hence  $\varphi + \psi \in S^0$  and  $c\varphi \in S^0$ , and  $S^0$  is a subspace of  $V^*$ . ■

### Example. (First Annihilators)

- (a) For any vector space  $V$ , find  $\{\mathbf{0}\}^0$ ,  $\emptyset^0$  and  $V^0$ .
- (b) In  $M_2(F)$ , find the annihilator of the single matrix  $\mathbf{E}_{11}$ .
- (c) In  $\mathbb{R}[x]_{\leq 2}$ , find the annihilator of  $S = \{1, x\}$ .

*Solution.*

- (a) Every functional sends  $\mathbf{0}$  to 0 (Theorem 3.1.1), so  $\{\mathbf{0}\}^0 = V^*$ . The condition “for all  $\mathbf{s} \in \emptyset$ ” holds vacuously, so  $\emptyset^0 = V^*$ . A functional in  $V^0$  takes the value 0 at every vector, so it is the zero functional, and  $V^0 = \{0\}$ . These degenerate cases fix the direction of everything that follows: the smallest subspace of  $V$  has the largest annihilator, and the largest has the smallest.
- (b) Let  $\varepsilon_{ij}(\mathbf{A}) = a_{ij}$  be the dual basis of the standard basis  $(\mathbf{E}_{11}, \mathbf{E}_{12}, \mathbf{E}_{21}, \mathbf{E}_{22})$ . By Theorem 4.1.3 (c), every  $\psi \in M_2(F)^*$  is  $\psi = \sum_{i,j} \psi(\mathbf{E}_{ij}) \varepsilon_{ij}$ . Such a  $\psi$  lies in  $\{\mathbf{E}_{11}\}^0$  exactly when  $\psi(\mathbf{E}_{11}) = 0$ . Hence

$$\{\mathbf{E}_{11}\}^0 = \text{span}(\varepsilon_{12}, \varepsilon_{21}, \varepsilon_{22}),$$

the functionals  $\mathbf{A} \mapsto c_{12}a_{12} + c_{21}a_{21} + c_{22}a_{22}$ . It has dimension 3, and  $3 + 1 = 4 = \dim M_2(F)$ .

- (c) Let  $(\varphi_0, \varphi_1, \varphi_2)$  be the dual basis of  $(1, x, x^2)$ , so  $\varphi_k$  reads off the coefficient of  $x^k$ . A functional  $\psi = \psi(1)\varphi_0 + \psi(x)\varphi_1 + \psi(x^2)\varphi_2$  lies in  $S^0$  exactly when  $\psi(1) = \psi(x) = 0$ , that is, when  $\psi = \psi(x^2)\varphi_2$ . So  $S^0 = \text{span}(\varphi_2)$ : the only equations satisfied by both 1 and  $x$  are the multiples of “the coefficient of  $x^2$  is 0”.

**Non-example by minimal change.** Shift the plane of the opening example off the origin:  $A = \{(x, y, z) \in \mathbb{R}^3 : x + y - z = 1\}$ . The set is still cut out by the same left-hand side, but  $\varphi(x, y, z) = x + y - z$  takes the value 1, not 0, on  $A$ , so the defining clause  $\varphi(\mathbf{s}) = 0$  fails and  $\varphi \notin A^0$ . In fact  $A^0 = \{0\}$ : the set  $A$  contains  $(1, 0, 0)$ ,  $(0, 1, 0)$  and  $(0, 0, -1)$ , and a functional  $ax + by + cz$  vanishing at these three vectors has  $a = b = c = 0$ . Annihilators record only **homogeneous** equations.

**Why this definition.** We allow any subset  $S$ , not only subspaces, because subspaces are usually handed to us by spanning lists, and we want to test a functional on the list alone. The theorem at the end of this section shows that this loses nothing:  $S^0 = (\text{span } S)^0$ . The name is literal. To annihilate is to reduce to nothing, and the functionals in  $S^0$  reduce every vector of  $S$  to 0.

**Warning.**  $U^0$  lives in  $V^*$ , and it is not a complement of  $U$ . It is tempting to turn the functional  $ax + by$  into the vector  $(a, b)$  and to picture  $U^0$  as a subspace of  $F^2$  “perpendicular” to  $U$ . Over  $\mathbb{F}_2$  this picture fails completely. Let  $U = \text{span}((1, 1)) \subseteq \mathbb{F}_2^2$ .

A functional  $ax + by$  vanishes on  $(1, 1)$  exactly when  $a + b = 0$ , that is,  $a = -b$ , so  $U^0 = \{0, x + y\}$ . The functional  $x + y$  corresponds to the vector  $(1, 1)$ , which lies in  $U$ . So the “picture” of  $U^0$  is  $U$  itself, not a complement. Perpendicularity needs an inner product, which is the orthogonal complement of Chapter 10; the annihilator needs nothing.

The payoff of the definition is the pattern we saw in the opening example. The equations of a subspace take up exactly the dimensions that the subspace leaves free.

## 4.2.2 Counting equations

### Theorem 4.2.2 — Dimension of the Annihilator

Let  $V$  be a finite-dimensional vector space over  $F$ , and let  $U$  be a subspace of  $V$ . Then

$$\dim U + \dim U^0 = \dim V.$$

More precisely, if  $(\mathbf{u}_1, \dots, \mathbf{u}_m, \mathbf{v}_{m+1}, \dots, \mathbf{v}_n)$  is a basis of  $V$  extending a basis  $(\mathbf{u}_1, \dots, \mathbf{u}_m)$  of  $U$ , with dual basis  $(\varphi_1, \dots, \varphi_n)$ , then  $(\varphi_{m+1}, \dots, \varphi_n)$  is a basis of  $U^0$ .

*Idea.* This is a dimension formula, so we start from the smallest subspace,  $U$ , take a basis and extend it to  $V$ . The dual basis now splits in two. The last  $n - m$  functionals vanish on every  $\mathbf{u}_i$ , by the delta, so they vanish on  $U$ . The first  $m$  do not. To see that nothing else is in  $U^0$ , expand a functional  $\psi \in U^0$  in the dual basis: its coefficients are its values on the basis, and the first  $m$  values are  $\psi(\mathbf{u}_i) = 0$ .

*Proof.* Since  $U$  is a subspace of the finite-dimensional space  $V$ , it has a basis  $(\mathbf{u}_1, \dots, \mathbf{u}_m)$  (Theorem 1.6.7, Corollary 1.5.5), and by the Basis Extension Theorem (Theorem 1.6.5) it extends to a basis  $\mathcal{B} = (\mathbf{u}_1, \dots, \mathbf{u}_m, \mathbf{v}_{m+1}, \dots, \mathbf{v}_n)$  of  $V$ , where  $n = \dim V$ . Let  $(\varphi_1, \dots, \varphi_n)$  be the dual basis of  $\mathcal{B}$  (Theorem 4.1.3).

The  $\varphi_j$  with  $j > m$  lie in  $U^0$ . Let  $j > m$  and  $\mathbf{u} \in U$ , say  $\mathbf{u} = a_1\mathbf{u}_1 + \dots + a_m\mathbf{u}_m$ . By Theorem 3.1.4,  $\varphi_j(\mathbf{u}) = \sum_{i=1}^m a_i\varphi_j(\mathbf{u}_i) = 0$ , since  $\varphi_j(\mathbf{u}_i) = \delta_{ji} = 0$  for  $i \leq m < j$ .

They span  $U^0$ . Let  $\psi \in U^0$ . By Theorem 4.1.3 (c),

$$\psi = \sum_{i=1}^m \psi(\mathbf{u}_i) \varphi_i + \sum_{j=m+1}^n \psi(\mathbf{v}_j) \varphi_j = \sum_{j=m+1}^n \psi(\mathbf{v}_j) \varphi_j,$$

because  $\psi(\mathbf{u}_i) = 0$  for  $\mathbf{u}_i \in U$ . Hence  $U^0 \subseteq \text{span}(\varphi_{m+1}, \dots, \varphi_n)$ , and the reverse inclusion holds because  $U^0$  is a subspace (Proposition 4.2.1, Theorem 1.3.1).

They are independent, being part of the basis  $(\varphi_1, \dots, \varphi_n)$  of  $V^*$ . Therefore  $(\varphi_{m+1}, \dots, \varphi_n)$  is a basis of  $U^0$ , and  $\dim U^0 = n - m = \dim V - \dim U$ . This proves the theorem. ■

Alternatively, restrict functionals to  $U$ . The map  $R: V^* \rightarrow U^*$ ,  $\varphi \mapsto \varphi|_U$ , is linear, and its kernel is  $U^0$  by definition. It is surjective: given  $\xi \in U^*$ , Theorem 3.1.5 gives  $\varphi \in V^*$  with  $\varphi(\mathbf{u}_i) = \xi(\mathbf{u}_i)$  and  $\varphi(\mathbf{v}_j) = 0$ , and then  $\varphi|_U = \xi$  because both agree on a basis of  $U$ . By the Rank–



Nullity Theorem ([Theorem 3.3.1](#)) and [Corollary 4.1.4](#),  $\dim V = \dim V^* = \dim U^0 + \dim U^* = \dim U^0 + \dim U$ . This route shows **why** the formula holds: a functional on  $V$  is a functional on  $U$  plus a choice of values off  $U$ , and  $U^0$  measures that choice.

The theorem is the second signature move of the chapter: **pass to the dual and count**. One immediate consequence: if  $U$  is a **proper** subspace of a finite-dimensional  $V$ , then  $\dim U^0 > 0$ , so some **non-zero** functional vanishes on  $U$ . Equivalently, a list spans a finite-dimensional  $V$  if and only if the only functional that kills every vector of the list is 0: apply the previous sentence to the span  $U$  of the list, and note that a functional kills every vector of the list exactly when it kills  $U$ , by [Theorem 3.1.4](#).

In  $F^n$  the whole computation becomes elimination, because functionals are rows.

### Example. (An Annihilator as a Null Space)

Find a basis of  $U^0$  for  $U = \text{span}((1, 2, 0), (0, 1, 1)) \subseteq \mathbb{R}^3$ , and check the dimension formula.

*Solution.* By [Theorem 4.1.1](#), every functional is  $\varphi_{\mathbf{a}}(\mathbf{x}) = \mathbf{a}^\top \mathbf{x}$  for a unique  $\mathbf{a} = (a_1, a_2, a_3)$ . By linearity,  $\varphi_{\mathbf{a}}$  vanishes on  $U$  if and only if it vanishes on the two spanning vectors (a combination of vectors killed by  $\varphi_{\mathbf{a}}$  is killed, by [Theorem 3.1.4](#)). These two conditions are

$$\begin{aligned} a_1 + 2a_2 &= 0, & a_2 + a_3 &= 0, \\ \text{that is, } \mathbf{A}\mathbf{a} &= \mathbf{0} \quad \text{with} \quad \mathbf{A} = \begin{pmatrix} 1 & 2 & 0 \\ 0 & 1 & 1 \end{pmatrix}, \end{aligned}$$

the matrix whose **rows** are the spanning vectors. So  $\varphi_{\mathbf{a}} \in U^0$  if and only if  $\mathbf{a} \in \text{nul}(\mathbf{A})$ . Subtracting twice row 2 from row 1 gives the RREF

$$\begin{pmatrix} 1 & 0 & -2 \\ 0 & 1 & 1 \end{pmatrix},$$

with  $a_3$  free. By [Theorem 2.5.4](#),  $\text{nul}(\mathbf{A}) = \text{span}((2, -1, 1))$ . Hence  $U^0 = \text{span}(\psi)$  with

$$\psi(x, y, z) = 2x - y + z.$$

Check:  $\psi(1, 2, 0) = 2 - 2 + 0 = 0$  and  $\psi(0, 1, 1) = -1 + 1 = 0$ . The two spanning vectors are independent (neither is a multiple of the other), so  $\dim U = 2$ , and  $\dim U + \dim U^0 = 2 + 1 = 3$ , as [Theorem 4.2.2](#) predicts. As a by-product,  $U$  is the plane  $2x - y + z = 0$ :  $U \subseteq \ker \psi$ , both have dimension 2 ([Proposition 4.1.2](#)), and [Theorem 1.6.8](#) gives equality.

The same recipe works in general. For  $U = \text{span}(\mathbf{u}_1, \dots, \mathbf{u}_m) \subseteq F^n$ , let  $\mathbf{A}$  have rows  $\mathbf{u}_1^\top, \dots, \mathbf{u}_m^\top$ . Then  $U^0 = \{\varphi_{\mathbf{a}} : \mathbf{a} \in \text{nul}(\mathbf{A})\}$ , and the isomorphism of [Theorem 4.1.1](#) carries a basis of  $\text{nul}(\mathbf{A})$  to a basis of  $U^0$ .

Away from  $F^n$ , counting often replaces the computation entirely.

### Example. (The Equations of a Root Condition)

Let  $U = \{p \in \mathbb{R}[x]_{\leq 2} : p(1) = 0\}$ . Find  $U^0$ .

*Solution.* Let  $\varepsilon_1(p) = p(1)$ , a linear functional (Example (Four Families of Functionals)). By definition  $U = \ker \varepsilon_1$ , so  $\varepsilon_1 \in U^0$ . Since  $\varepsilon_1(1) = 1$ ,  $\varepsilon_1 \neq 0$ , and Proposition 4.1.2 gives  $\dim U = 3 - 1 = 2$ . By Theorem 4.2.2,  $\dim U^0 = 3 - 2 = 1$ . Now  $\text{span}(\varepsilon_1)$  is a subspace of  $U^0$  (Proposition 4.2.1) of dimension 1, so  $U^0 = \text{span}(\varepsilon_1)$  by Theorem 1.6.8.

We never found a basis of  $U$  and never solved an equation. The conclusion is concrete: a functional on  $\mathbb{R}[x]_{\leq 2}$  vanishes on every polynomial with a root at 1 **only if** it is a multiple of  $p \mapsto p(1)$ .

*Quick check.* Let  $U = \text{span}((1, 1, 1)) \subseteq \mathbb{R}^3$ . What is  $\dim U^0$ ? Name a basis of  $U^0$  without solving a system.

*Solution.*  $\dim U^0 = 3 - 1 = 2$  by Theorem 4.2.2. The functionals  $x - y$  and  $y - z$  vanish at  $(1, 1, 1)$ , hence on  $U$ , and they are independent: in  $a(x - y) + b(y - z) = 0$ , the coefficient of  $x$  gives  $a = 0$ , and then  $b = 0$ . Two independent vectors in a 2-dimensional space form a basis (Theorem 1.6.4).

### 4.2.3 Sums, intersections and spans

Taking annihilators reverses inclusions: more vectors satisfy fewer common equations. It also exchanges sums with intersections. In  $\mathbb{R}^3$ , let  $U$  be the  $xy$ -plane and  $W$  the  $xz$ -plane. Then  $U^0 = \text{span}(z)$  and  $W^0 = \text{span}(y)$ , writing  $x, y, z$  for the coordinate functionals. The intersection  $U \cap W$  is the  $x$ -axis, with annihilator  $\text{span}(y, z) = U^0 + W^0$ . The sum  $U + W = \mathbb{R}^3$  has annihilator  $\{0\} = U^0 \cap W^0$ . The theorem says this always happens.

#### Theorem 4.2.3 — Properties of Annihilators

Let  $V$  be a vector space over  $F$ .

- (a) If  $S \subseteq T \subseteq V$ , then  $T^0 \subseteq S^0$ .
- (b) For every subset  $S \subseteq V$ ,  $S^0 = (\text{span } S)^0$ .
- (c) For subspaces  $U, W$  of  $V$ ,  $(U + W)^0 = U^0 \cap W^0$ .
- (d) If  $V$  is **finite-dimensional**, then for subspaces  $U, W$  of  $V$ ,  $(U \cap W)^0 = U^0 + W^0$ .

*Idea.* Parts (a)–(c) are unwinding definitions: fix a functional in one side and test it on the vectors of the other. In (d), the inclusion  $U^0 + W^0 \subseteq (U \cap W)^0$  is also direct. The reverse inclusion is the hard half. Given  $\varphi$  vanishing on  $U \cap W$ , we would have to **split** it as  $\psi + \chi$  with  $\psi$  killing  $U$  and  $\chi$  killing  $W$ , and nothing in sight tells us how. So we count instead: one inclusion plus equal dimensions gives equality. The plan is ① compute  $\dim(U^0 + W^0)$  by the dimension formula for sums inside  $V^*$ ; ② replace  $U^0 \cap W^0$  by  $(U + W)^0$  using (c); ③ turn every annihilator dimension into a dimension in  $V$  with Theorem 4.2.2; ④ recognize the dimension formula in  $V$ . Finite dimension is spent in step ③.

*Proof.*

- (a) Let  $\varphi \in T^0$ . Every  $\mathbf{s} \in S$  lies in  $T$ , so  $\varphi(\mathbf{s}) = 0$ . Hence  $\varphi \in S^0$ .

- (b) Since  $S \subseteq \text{span } S$  (Theorem 1.3.1), part (a) gives  $(\text{span } S)^0 \subseteq S^0$ . Conversely, let  $\varphi \in S^0$  and  $\mathbf{v} \in \text{span } S$ , say  $\mathbf{v} = a_1 \mathbf{s}_1 + \cdots + a_k \mathbf{s}_k$  with  $\mathbf{s}_i \in S$  and  $k \geq 0$ . By Theorem 3.1.4,  $\varphi(\mathbf{v}) = \sum_i a_i \varphi(\mathbf{s}_i) = 0$ . Hence  $\varphi \in (\text{span } S)^0$ .
- (c) Since  $U \subseteq U+W$  and  $W \subseteq U+W$  (Theorem 1.7.1), part (a) gives  $(U+W)^0 \subseteq U^0$  and  $(U+W)^0 \subseteq W^0$ , so  $(U+W)^0 \subseteq U^0 \cap W^0$ . Conversely, let  $\varphi \in U^0 \cap W^0$ , and let  $\mathbf{u} + \mathbf{w} \in U+W$  with  $\mathbf{u} \in U$ ,  $\mathbf{w} \in W$ . Then  $\varphi(\mathbf{u} + \mathbf{w}) = \varphi(\mathbf{u}) + \varphi(\mathbf{w}) = 0 + 0 = 0$ . Hence  $\varphi \in (U+W)^0$ .
- (d)  $(\supseteq)$  Let  $\varphi = \psi + \chi$  with  $\psi \in U^0$  and  $\chi \in W^0$ , and let  $\mathbf{v} \in U \cap W$ . Since  $\mathbf{v} \in U$  and  $\mathbf{v} \in W$ ,  $\varphi(\mathbf{v}) = \psi(\mathbf{v}) + \chi(\mathbf{v}) = 0 + 0 = 0$ . Hence  $U^0 + W^0 \subseteq (U \cap W)^0$ .

$(\subseteq)$  Let  $n = \dim V$ . The subspaces  $U$ ,  $W$ ,  $U \cap W$  (Theorem 1.2.2) and  $U+W$  (Theorem 1.7.1) of  $V$  are finite-dimensional by Theorem 1.6.7. The space  $V^*$  has dimension  $n$  by Corollary 4.1.4, so its subspaces  $U^0$ ,  $W^0$  are finite-dimensional too. By the Dimension Formula for Sums (Theorem 1.7.3) in  $V^*$ , then (c),

$$\begin{aligned} \dim(U^0 + W^0) &= \dim U^0 + \dim W^0 - \dim(U^0 \cap W^0) \\ &= \dim U^0 + \dim W^0 - \dim(U+W)^0. \end{aligned}$$

By Theorem 4.2.2 applied to  $U$ ,  $W$  and  $U+W$ , the right-hand side is

$$\begin{aligned} (n - \dim U) + (n - \dim W) - (n - \dim(U+W)) \\ = n - (\dim U + \dim W - \dim(U+W)) \\ = n - \dim(U \cap W), \end{aligned}$$

where the last equality is Theorem 1.7.3 in  $V$ . By Theorem 4.2.2 once more,  $n - \dim(U \cap W) = \dim(U \cap W)^0$ . So  $U^0 + W^0$  is a subspace of  $(U \cap W)^0$  (by  $(\supseteq)$  and Theorem 1.7.1) with the same finite dimension, and Theorem 1.6.8 gives  $U^0 + W^0 = (U \cap W)^0$ . This proves the theorem. ■

Part (b) is what made the computations of this section legitimate: to find the annihilator of a subspace, test functionals on a spanning list only. Parts (c) and (d) form a dictionary. A subspace given as a **sum** has equations obtained by **intersecting** equation spaces; a subspace given as an **intersection** has equations obtained by **adding** them. Later in this chapter this dictionary becomes a practical method for computing sums and intersections of subspaces.

*Remark.* Parts (a)–(c) and the inclusion  $(\supseteq)$  in (d) hold in every vector space; only the counting step used finite dimension. The equality (d) is in fact true for **every**  $V$ , but the proof must then construct the splitting  $\varphi = \psi + \chi$  by hand; Exercise C1 below does this. In finite dimension, (a) also becomes strict: if  $U \subsetneq W$ , then  $\dim U < \dim W$  (Theorem 1.6.8), so  $\dim W^0 < \dim U^0$  and  $W^0 \subsetneq U^0$ .

## 4.2.4 Exercises

### 4.2.4.1 A. Check your understanding

#### Exercise A1.

- (a) Define the annihilator  $S^0$  of a subset  $S$  of a vector space  $V$ .
- (b) True or false:  $U^0$  is a subspace of  $V$ . Justify your answer.
- (c) Let  $U$  be a 2-dimensional subspace of  $\mathbb{R}^5$ . What is  $\dim U^0$ ?
- (d) For a finite-dimensional  $V$ , what are  $\{\mathbf{0}\}^0$  and  $V^0$ ?
- (e) True or false: if  $U \subsetneq W$  are subspaces of a finite-dimensional  $V$ , then  $W^0 \subsetneq U^0$ . Justify your answer.
- (f) To compute  $U^0$  for  $U = \text{span}(\mathbf{u}_1, \dots, \mathbf{u}_m) \subseteq F^n$ , which matrix do you row reduce, and why?

*Solution.*

- (a)  $S^0 = \{\varphi \in V^* : \varphi(\mathbf{s}) = 0 \text{ for all } \mathbf{s} \in S\}$  (Definition 4.2.1).
- (b) False.  $U^0$  is a subspace of the dual space  $V^*$  (Proposition 4.2.1); its elements are functionals, not vectors of  $V$ .
- (c)  $\dim U^0 = 5 - 2 = 3$ , by Theorem 4.2.2.
- (d)  $\{\mathbf{0}\}^0 = V^*$  and  $V^0 = \{0\}$ , by Example (First Annihilators) (a).
- (e) True. By Theorem 4.2.3 (a),  $W^0 \subseteq U^0$ . Since  $U \subsetneq W$ ,  $\dim U < \dim W$  by Theorem 1.6.8, so  $\dim W^0 = \dim V - \dim W < \dim V - \dim U = \dim U^0$ . Hence the inclusion is strict.
- (f) The matrix  $\mathbf{A}$  whose rows are  $\mathbf{u}_1^\top, \dots, \mathbf{u}_m^\top$ . By Theorem 4.1.1 a functional is  $\mathbf{x} \mapsto \mathbf{a}^\top \mathbf{x}$ , and it kills every  $\mathbf{u}_i$  exactly when  $\mathbf{A}\mathbf{a} = \mathbf{0}$ ; so  $U^0$  corresponds to  $\text{nul}(\mathbf{A})$  (Example (An Annihilator as a Null Space)).

#### 4.2.4.2 B. Practice

##### Exercise B1: Computing annihilators.

- (a) Let  $U = \text{span}((1, 2, 1, 0), (2, 4, 3, 1), (1, 2, 2, 1)) \subseteq \mathbb{R}^4$ . Find a basis of  $U^0$ , and verify that  $\dim U + \dim U^0 = 4$ .
- (b) Let  $U = \{p \in \mathbb{R}[x]_{\leq 3} : p(1) = p(-1) = 0\}$ . Show that  $U^0 = \text{span}(\varepsilon_1, \varepsilon_{-1})$ , where  $\varepsilon_c(p) = p(c)$ .

*Solution.*

- (a) Let  $\mathbf{A}$  have the three spanning vectors as rows. As in Example (An Annihilator as a Null Space),  $\varphi_{\mathbf{a}} \in U^0$  if and only if  $\mathbf{A}\mathbf{a} = \mathbf{0}$ . Subtracting 2 times row 1 from row 2 and row 1 from row 3 gives rows  $(1, 2, 1, 0), (0, 0, 1, 1), (0, 0, 1, 1)$ ; subtracting row 2 from row 3 and then row 2 from row 1 gives the RREF

$$\begin{pmatrix} 1 & 2 & 0 & -1 \\ 0 & 0 & 1 & 1 \\ 0 & 0 & 0 & 0 \end{pmatrix}.$$

The free variables are  $a_2$  and  $a_4$ , and Theorem 2.5.4 gives  $\text{nul}(\mathbf{A}) = \text{span}((-2, 1, 0, 0), (1, 0, -1, 1))$ . Hence  $U^0$  has the basis

$$\psi_1(x_1, x_2, x_3, x_4) = -2x_1 + x_2, \quad \psi_2(x_1, x_2, x_3, x_4) = x_1 - x_3 + x_4.$$

Check on the first two spanning vectors:  $\psi_1(1, 2, 1, 0) = 0$ ,  $\psi_2(1, 2, 1, 0) = 0$ ,  $\psi_1(2, 4, 3, 1) = 0$ ,  $\psi_2(2, 4, 3, 1) = 2 - 3 + 1 = 0$ ; the third vector is their difference,  $(2, 4, 3, 1) - (1, 2, 1, 0)$ . The RREF has two pivots, so  $\dim U = \dim \text{row}(\mathbf{A}) = 2$  (Theorem 2.5.5), and  $2 + 2 = 4$ .

- (b) By definition  $\varepsilon_1, \varepsilon_{-1} \in U^0$ . *Independent:* if  $a\varepsilon_1 + b\varepsilon_{-1} = 0$ , applying it to  $x + 1$  gives  $2a = 0$  and applying it to  $x - 1$  gives  $-2b = 0$ . *Dimension of  $U$ :* the map  $T: \mathbb{R}[x]_{\leq 3} \rightarrow \mathbb{R}^2$ ,  $p \mapsto (p(1), p(-1))$ , is linear with kernel  $U$ , and its image contains  $T(x + 1) = (2, 0)$  and  $T(x - 1) = (0, -2)$ , which span  $\mathbb{R}^2$ . So  $\text{rank } T = 2$  and  $\dim U = 4 - 2 = 2$  by Theorem 3.3.1. By Theorem 4.2.2,  $\dim U^0 = 2$ . Hence the 2-dimensional subspace  $\text{span}(\varepsilon_1, \varepsilon_{-1})$  of  $U^0$  is all of  $U^0$  (Theorem 1.6.8).

### Exercise B2: Equations of the symmetric matrices.

Let  $U = \{\mathbf{A} \in M_2(\mathbb{R}) : \mathbf{A}^\top = \mathbf{A}\}$ . Find  $U^0$ , and verify the dimension formula.

*Solution.* Every symmetric matrix is  $a\mathbf{E}_{11} + b(\mathbf{E}_{12} + \mathbf{E}_{21}) + c\mathbf{E}_{22}$ , so  $U = \text{span}(\mathbf{E}_{11}, \mathbf{E}_{12} + \mathbf{E}_{21}, \mathbf{E}_{22})$ . Let  $(\varepsilon_{ij})$  be the dual basis of the standard basis, so that every  $\psi \in M_2(\mathbb{R})^*$  is  $\psi = \sum_{i,j} c_{ij}\varepsilon_{ij}$  with  $c_{ij} = \psi(\mathbf{E}_{ij})$  (Theorem 4.1.3 (c)). By Theorem 4.2.3 (b),  $\psi \in U^0$  if and only if  $\psi$  kills the three spanning matrices:

$$c_{11} = 0, \quad c_{12} + c_{21} = 0, \quad c_{22} = 0.$$

So  $\psi = c_{12}(\varepsilon_{12} - \varepsilon_{21})$ , and  $U^0 = \text{span}(\varepsilon_{12} - \varepsilon_{21})$ : the only equation of the symmetric matrices, up to scaling, is  $a_{12} - a_{21} = 0$ . The three spanning matrices are independent (they involve disjoint sets of entries), so  $\dim U = 3$ , and  $\dim U + \dim U^0 = 3 + 1 = 4 = \dim M_2(\mathbb{R})$ .

### Exercise B3: The equations of a hyperplane.

Let  $V$  be a finite-dimensional vector space over  $F$ , and let  $\varphi \in V^*$  be non-zero. Prove that  $(\ker \varphi)^0 = \text{span}(\varphi)$ .

*Solution.* By definition of the kernel,  $\varphi$  vanishes on  $\ker \varphi$ , so  $\varphi \in (\ker \varphi)^0$ , and  $\text{span}(\varphi) \subseteq (\ker \varphi)^0$  by Proposition 4.2.1. Since  $\varphi \neq 0$ ,  $\dim \text{span}(\varphi) = 1$ , and Proposition 4.1.2 gives  $\dim \ker \varphi = \dim V - 1$ . By Theorem 4.2.2,  $\dim(\ker \varphi)^0 = \dim V - (\dim V - 1) = 1$ . A 1-dimensional subspace of a 1-dimensional space is the whole space (Theorem 1.6.8), so  $(\ker \varphi)^0 = \text{span}(\varphi)$ .

#### 4.2.4.3 C. Going deeper

### Exercise C1: The intersection formula without counting.

Let  $U, W$  be subspaces of a vector space  $V$  over  $F$ , and let  $\varphi \in (U \cap W)^0$ .

- (a) Suppose  $V$  is finite-dimensional. Construct  $\psi \in U^0$  and  $\chi \in W^0$  with  $\varphi = \psi + \chi$ , without using any dimension formula.
- (b) Explain how to modify the construction when  $V$  is infinite-dimensional. Deduce that  $(U \cap W)^0 = U^0 + W^0$  holds in every vector space.

*Hint: extend a basis of  $U \cap W$  to a basis of  $U$  and, separately, to a basis of  $W$ .*

*Solution.*

- (a) Let  $(\mathbf{x}_1, \dots, \mathbf{x}_k)$  be a basis of  $U \cap W$  (Theorem 1.2.2, Theorem 1.6.7). Extend it to a basis  $(\mathbf{x}_1, \dots, \mathbf{x}_k, \mathbf{u}_1, \dots, \mathbf{u}_p)$  of  $U$  and to a basis  $(\mathbf{x}_1, \dots, \mathbf{x}_k, \mathbf{w}_1, \dots, \mathbf{w}_q)$  of  $W$  (Theorem 1.6.5).

The combined list  $(\mathbf{x}_1, \dots, \mathbf{x}_k, \mathbf{u}_1, \dots, \mathbf{u}_p, \mathbf{w}_1, \dots, \mathbf{w}_q)$  is independent. Let  $\sum a_i \mathbf{x}_i + \sum b_i \mathbf{u}_i + \sum c_j \mathbf{w}_j = \mathbf{0}$ . Then  $\sum c_j \mathbf{w}_j = -\sum a_i \mathbf{x}_i - \sum b_i \mathbf{u}_i$  lies in  $W$  and in  $U$ , hence in  $U \cap W$ , so  $\sum c_j \mathbf{w}_j = \sum d_i \mathbf{x}_i$  for some  $d_i \in F$ . Since  $(\mathbf{x}, \mathbf{w})$  is a basis of  $W$ , all  $c_j$  and  $d_i$  are 0. Then  $\sum a_i \mathbf{x}_i + \sum b_i \mathbf{u}_i = \mathbf{0}$ , and since  $(\mathbf{x}, \mathbf{u})$  is a basis of  $U$ , all  $a_i$  and  $b_i$  are 0.

Extend the combined list to a basis of  $V$  by vectors  $\mathbf{y}_1, \dots, \mathbf{y}_r$  (Theorem 1.6.5). By Theorem 3.1.5 there is  $\psi \in V^*$  with

$$\psi(\mathbf{x}_i) = 0, \quad \psi(\mathbf{u}_i) = 0, \quad \psi(\mathbf{w}_j) = \varphi(\mathbf{w}_j), \quad \psi(\mathbf{y}_l) = 0.$$

Then  $\psi$  kills a basis of  $U$ , so  $\psi \in U^0$  (Theorem 4.2.3 (b)). Put  $\chi := \varphi - \psi$ . Then  $\chi(\mathbf{x}_i) = \varphi(\mathbf{x}_i) - 0 = 0$ , because  $\mathbf{x}_i \in U \cap W$  and  $\varphi \in (U \cap W)^0$ ; and  $\chi(\mathbf{w}_j) = \varphi(\mathbf{w}_j) - \varphi(\mathbf{w}_j) = 0$ . So  $\chi$  kills a basis of  $W$ , and  $\chi \in W^0$ . Hence  $\varphi = \psi + \chi \in U^0 + W^0$ .

- (b) Replace the finite lists by sets. By Theorem 1.8.3,  $U \cap W$  has a basis  $X$ , and by Theorem 1.8.2 it extends to a basis  $X \cup P$  of  $U$  and to a basis  $X \cup Q$  of  $W$ , with  $P \cap X = Q \cap X = \emptyset$ . First,  $P \cap Q = \emptyset$ : a vector of  $P \cap Q$  lies in  $U \cap W = \text{span } X$ , which contradicts the independence of  $X \cup P$ . So every vector of  $X \cup P \cup Q$  lies in exactly one of  $X$ ,  $P$ ,  $Q$ . A linear relation among finitely many distinct vectors of  $X \cup P \cup Q$  splits into an  $X$ -part, a  $P$ -part and a  $Q$ -part, and the independence argument of (a) applies word for word; so  $X \cup P \cup Q$  is independent. By Theorem 1.8.2 it extends to a basis  $B$  of  $V$ . Every  $\mathbf{v} \in V$  is a finite combination  $\mathbf{v} = \sum_{\mathbf{b}} c_{\mathbf{b}} \mathbf{b}$  of distinct vectors of  $B$ , with unique coefficients because  $B$  is independent. Define

$$\psi(\mathbf{v}) := \sum_{\mathbf{b} \in Q} c_{\mathbf{b}} \varphi(\mathbf{b}),$$

a finite sum over the vectors of  $Q$  that occur. Adding or scaling expansions gives expansions, so by uniqueness the coefficients add and scale, and  $\psi$  is linear. It vanishes on  $X \cup P$  and agrees with  $\varphi$  on  $Q$ , so the rest of (a) is unchanged:  $\psi \in U^0$  and  $\chi = \varphi - \psi$  kills the basis  $X \cup Q$  of  $W$ . Together with the inclusion  $(\supseteq)$  of Theorem 4.2.3 (d), which used no finiteness, this gives  $(U \cap W)^0 = U^0 + W^0$  in every vector space.

### Exercise C2: Equations of a double root.

Let  $U = \{p \in \mathbb{R}[x]_{\leq 3} : p(1) = 0 \text{ and } p'(1) = 0\}$ , and let  $\varepsilon_1(p) = p(1)$  and  $\eta_1(p) = p'(1)$ . Prove that  $U^0 = \text{span}(\varepsilon_1, \eta_1)$ . Deduce that  $\psi(p) = p(0) - p(2) + 2p'(1)$  does **not** vanish on  $U$ .

*Solution.* Both  $\varepsilon_1$  and  $\eta_1$  are linear (evaluation, and differentiation followed by evaluation), and they vanish on  $U$  by definition. *Independent:* if  $a\varepsilon_1 + b\eta_1 = 0$ , applying it to 1 gives  $a = 0$ , and applying it to  $x - 1$  gives  $b = 0$ . *Dimension of  $U$ :* the linear map

$T: \mathbb{R}[x]_{\leq 3} \rightarrow \mathbb{R}^2$ ,  $p \mapsto (p(1), p'(1))$ , has kernel  $U$ , and  $T(1) = (1, 0)$ ,  $T(x-1) = (0, 1)$ , so  $T$  is surjective. By [Theorem 3.3.1](#),  $\dim U = 4 - 2 = 2$ , and by [Theorem 4.2.2](#),  $\dim U^0 = 2$ . The 2-dimensional subspace  $\text{span}(\varepsilon_1, \eta_1)$  of  $U^0$  is therefore all of  $U^0$  ([Theorem 1.6.8](#)).

For the deduction, suppose  $\psi \in U^0$ . Then  $\psi = a\varepsilon_1 + b\eta_1$ . Applying both sides to 1 gives  $1 - 1 + 0 = a$ , so  $a = 0$ ; applying them to  $x - 1$  gives  $-1 - 1 + 2 = b$ , so  $b = 0$ . Then  $\psi = 0$ . But  $\psi(x^2) = 0 - 4 + 4 = 0$  and  $\psi(x^3) = 0 - 8 + 6 = -2 \neq 0$ , a contradiction. Hence  $\psi \notin U^0$ . Check with an explicit witness:  $p = x(x-1)^2 \in U$  has  $\psi(p) = 0 - 2 + 0 = -2 \neq 0$ .

### Exercise C3: Dual of a direct sum decomposition.

- (a) Let  $V$  be finite-dimensional with  $V = U \oplus W$ . Prove that  $V^* = U^0 \oplus W^0$ .
- (b) In  $\mathbb{R}^2$ , let  $U = \text{span}((1, 0))$  and  $W = \text{span}((1, 1))$ . Write  $\psi(x, y) = x$  as  $\psi_U + \psi_W$  with  $\psi_U \in U^0$  and  $\psi_W \in W^0$ .

*Solution.*

- (a) Since the sum is direct,  $U \cap W = \{0\}$  ([Theorem 1.7.4](#)), and  $U + W = V$ . By [Theorem 4.2.3](#) (c) and [Example \(First Annihilators\)](#) (a),  $U^0 \cap W^0 = (U + W)^0 = V^0 = \{0\}$ . By [Theorem 4.2.3](#) (d),  $U^0 + W^0 = (U \cap W)^0 = \{0\}^0 = V^*$ . So  $V^* = U^0 + W^0$ , and the sum is direct by [Theorem 1.7.4](#).
- (b) A functional  $ax + by$  vanishes on  $(1, 0)$  if and only if  $a = 0$ , so  $U^0 = \text{span}(y)$ . It vanishes on  $(1, 1)$  if and only if  $b = -a$ , so  $W^0 = \text{span}(x - y)$ . Solving  $x = sy + t(x - y)$  gives  $t = 1$  and  $s - t = 0$ , so  $s = 1$ :

$$\psi = \underbrace{y}_{\in U^0} + \underbrace{(x - y)}_{\in W^0}.$$

By (a) this decomposition is unique.

## 4.3 The Double Dual

In a finite-dimensional space, the dual basis gives an isomorphism  $V \cong V^*$ , sending  $\mathbf{v}_i$  to  $\varphi_i$ . But that isomorphism depends on the basis: change the basis and the same vector goes to a different functional. Nothing stops us from taking duals twice. The dual  $V^*$  is a vector space, so it has a dual of its own,  $V^{**}$ , and two chosen bases give  $V \cong V^* \cong V^{**}$ . This section shows that the composite isomorphism  $V \rightarrow V^{**}$  needs **no** choice at all: it is built from evaluation alone. We use it to make one meaning of the word “natural” precise, and to prove that a subspace can be recovered from its annihilator. At the end we see that in infinite dimension the construction still makes sense, but it is no longer an isomorphism.

### 4.3.1 Measuring the measurements

We have studied a space  $V$  and its space of measurements  $V^* = \mathcal{L}(V, F)$  ([Definition 4.1.2](#)). The next object is the space of measurements **of**  $V^*$ .

Look at the expression  $\varphi(\mathbf{v})$ , with  $\varphi \in V^*$  and  $\mathbf{v} \in V$ . It has two inputs. If we fix  $\varphi$  and let  $\mathbf{v}$  vary, we get the functional  $\varphi$  itself. If we fix  $\mathbf{v}$  and let  $\varphi$  vary, we get a rule that takes a functional and returns a number. For example, take  $\mathbf{v} = (3, -1) \in \mathbb{R}^2$ . A functional on  $\mathbb{R}^2$  is  $\varphi(x, y) = ax + by$  for unique  $a, b$  (Theorem 4.1.1), and the rule returns  $\varphi(3, -1) = 3a - b$ . So the fixed vector  $(3, -1)$  measures each measurement. This rule appears every time we test a vector against functionals, so it deserves a name.

*A vector measures a functional by letting the functional act on it.*

### Definition 4.3.1 — Double Dual and Evaluation Map

Let  $V$  be a vector space over  $F$ . The **double dual** of  $V$  is  $V^{**} := (V^*)^*$ , the space of linear functionals on  $V^*$ .

For  $\mathbf{v} \in V$ , **evaluation at  $\mathbf{v}$**  is the function

$$\text{ev}_{\mathbf{v}}: V^* \rightarrow F, \quad \text{ev}_{\mathbf{v}}(\varphi) := \varphi(\mathbf{v}).$$

The **evaluation map** of  $V$  is

$$\text{ev}_V: V \rightarrow V^{**}, \quad \mathbf{v} \mapsto \text{ev}_{\mathbf{v}}.$$

When only one space is in play we write  $\text{ev}$  for  $\text{ev}_V$ .

In words, clause by clause:

- $V^{**}$  consists of **linear** maps  $V^* \rightarrow F$ . Its elements take a whole functional as input.
- $\text{ev}_{\mathbf{v}}$  is **defined on**  $V^*$ , not on  $V$ . The vector  $\mathbf{v}$  is fixed once and for all; the variable is  $\varphi$ .
- The evaluation map sends each vector  $\mathbf{v}$  to the function  $\text{ev}_{\mathbf{v}}$ . A bold subscript  $\text{ev}_{\mathbf{v}}$  is a single element of  $V^{**}$ ; an italic capital  $\text{ev}_V$  is the whole map.

**Well-definedness.** The definition claims that  $\text{ev}_{\mathbf{v}} \in V^{**}$ , so we must check that  $\text{ev}_{\mathbf{v}}$  is linear. The operations on  $V^* = \mathcal{L}(V, F)$  are pointwise (Definition 3.4.1). So for  $\varphi, \psi \in V^*$  and  $a, b \in F$ ,

$$\text{ev}_{\mathbf{v}}(a\varphi + b\psi) = (a\varphi + b\psi)(\mathbf{v}) = a\varphi(\mathbf{v}) + b\psi(\mathbf{v}) = a\text{ev}_{\mathbf{v}}(\varphi) + b\text{ev}_{\mathbf{v}}(\psi).$$

Notice what the definition did **not** use: no basis, no coordinates, no choice of any kind. Only  $\mathbf{v}$  and the meaning of “functional” appear.

### Example. (Evaluation in Three Spaces)

- Let  $V = F^n$ . Describe  $\text{ev}_{\mathbf{x}}$  for  $\mathbf{x} \in F^n$  in terms of row vectors.
- Let  $V = \mathbb{R}[x]_{\leq 2}$ , and let  $(\varphi_0, \varphi_1, \varphi_2)$  be the dual basis of  $(1, x, x^2)$ , so that  $\varphi_k(p)$  is the coefficient of  $x^k$  in  $p$  (Example (Dual Bases in  $\mathbb{R}[x]_{\leq 2}$ )). Compute  $\text{ev}_{x^2}$  on  $\varphi_0, \varphi_1, \varphi_2$  and on the functional  $\varepsilon_3(p) = p(3)$ . Which element of  $V^{**}$  is  $\text{ev}_{x^2}$ ?
- Describe  $\text{ev}_{\mathbf{0}}$ , and the evaluation map of  $V = \{\mathbf{0}\}$ .

*Solution.*

- By Theorem 4.1.1, every  $\varphi \in (F^n)^*$  has the form  $\varphi(\mathbf{y}) = \mathbf{a}^\top \mathbf{y}$  for a unique  $\mathbf{a} \in F^n$ . So  $\text{ev}_{\mathbf{x}}(\varphi) = \mathbf{a}^\top \mathbf{x}$ : the vector  $\mathbf{x}$  measures the row  $\mathbf{a}^\top$  by standing to its right in the product. The same product  $\mathbf{a}^\top \mathbf{x}$  is read as “ $\mathbf{a}^\top$  measures  $\mathbf{x}$ ” or “ $\mathbf{x}$  measures



$\mathbf{a}^\top$ ”, depending on which factor we hold fixed.

- (b)  $\text{ev}_{x^2}(\varphi_k) = \varphi_k(x^2)$ , the coefficient of  $x^k$  in  $x^2$ . So the values on  $\varphi_0, \varphi_1, \varphi_2$  are  $0, 0, 1$ . Also  $\text{ev}_{x^2}(\varepsilon_3) = \varepsilon_3(x^2) = 9$ . A linear functional on  $V^*$  is determined by its values on the basis  $(\varphi_0, \varphi_1, \varphi_2)$  of  $V^*$  (Theorem 3.1.5). The values  $0, 0, 1$  say that  $\text{ev}_{x^2}$  is the third vector of the dual basis of  $(\varphi_0, \varphi_1, \varphi_2)$  (Theorem 4.1.3). In the same way,  $\text{ev}_1$  and  $\text{ev}_x$  are the first and second vectors of that dual basis. We will see that this is no accident.
- (c)  $\text{ev}_0(\varphi) = \varphi(0) = 0$  for every  $\varphi$ , because a linear map sends  $0$  to  $0$  (Theorem 3.1.1). So  $\text{ev}_0$  is the zero functional on  $V^*$ . If  $V = \{0\}$ , the only functional on  $V$  is the zero functional, so  $V^* = \{0\}$  and  $V^{**} = \{0\}$ , and  $\text{ev}_V$  is the only map  $\{0\} \rightarrow \{0\}$ . This degenerate case is a first sign that  $\text{ev}$  loses nothing: the only vector it sends to zero is  $0$ .

Here is a non-example by minimal change. On  $V = \mathbb{R}^2$ , replace  $\text{ev}_v$  by  $q_v(\varphi) := \varphi(v)^2$ . It is still a function  $V^* \rightarrow \mathbb{R}$ , and it is still built without any choice. But take  $v = (1, 0)$  and  $\varphi(x, y) = x$ . Then  $q_v(2\varphi) = 2^2 = 4$ , while  $2q_v(\varphi) = 2$ . The clause that fails is **linearity** on  $V^*$ , so  $q_v \notin V^{**}$ .

**Why this definition.** Given only a vector  $v$  and a functional  $\varphi$ , the one number we can form without choosing anything is  $\varphi(v)$ ; anything else, such as  $\varphi(v)^2$ , is built from it and usually breaks linearity. The pointwise operations of Definition 3.4.1 are exactly what make  $\varphi \mapsto \varphi(v)$  linear. The name “double dual” is literal: it is the dual of the dual.

*Warning. Two different “evaluations”.* On  $V = \mathbb{R}[x]_{\leq 2}$ , the functional  $\varepsilon_3: p \mapsto p(3)$  lies in  $V^*$ : it evaluates **polynomials at a number**. The element  $\text{ev}_p: \varphi \mapsto \varphi(p)$  lies in  $V^{**}$ : it evaluates **functionals at a polynomial**. They meet in one number,  $\text{ev}_p(\varepsilon_3) = p(3) = \varepsilon_3(p)$ , read from opposite sides. Also,  $\text{ev}_v$  is not the vector  $v$ : it is a function on  $V^*$ .

*Quick check.* Let  $v = (1, 2) \in \mathbb{R}^2$ . Compute  $\text{ev}_v(\varphi)$  for  $\varphi(x, y) = 3x - y$ . Is there a **non-zero**  $\psi \in (\mathbb{R}^2)^*$  with  $\text{ev}_v(\psi) = 0$ ?

*Solution.*  $\text{ev}_v(\varphi) = \varphi(1, 2) = 3 - 2 = 1$ . Yes:  $\psi(x, y) = 2x - y$  is non-zero and  $\psi(1, 2) = 0$ . A single  $\text{ev}_v$  kills many functionals; what the next theorem says is that for  $v \neq 0$  it does not kill **all** of them.

## 4.3.2 The evaluation map never loses information

The first result says that  $\text{ev}$  is a linear map, and that it is injective in every vector space, of any dimension.

### Theorem 4.3.1 — The Evaluation Map Is Linear and Injective

Let  $V$  be any vector space over  $F$ . Then the evaluation map  $\text{ev}_V: V \rightarrow V^{**}$  is linear and injective.

*Idea.* Linearity is an identity between two elements of  $V^{**}$ , that is, between two functions on  $V^*$ ; compare them at an arbitrary  $\varphi$ . For injectivity, use the kernel test. If  $\text{ev}_v = 0$ , then **every** functional vanishes at  $v$ . This is the signature move of the chapter: to show a vector is zero, test it against every functional. In finite dimension Theorem 4.1.5 does

the job. In general we repeat its proof with the Basis Extension Theorem for arbitrary spaces.

*Proof. Linear.* Let  $\mathbf{u}, \mathbf{v} \in V$  and  $a, b \in F$ . For every  $\varphi \in V^*$ , since  $\varphi$  is linear and the operations on  $V^{**}$  are pointwise (Definition 3.4.1),

$$\text{ev}_{a\mathbf{u}+b\mathbf{v}}(\varphi) = \varphi(a\mathbf{u} + b\mathbf{v}) = a\varphi(\mathbf{u}) + b\varphi(\mathbf{v}) = (a\text{ev}_{\mathbf{u}} + b\text{ev}_{\mathbf{v}})(\varphi).$$

Two functions on  $V^*$  that agree at every  $\varphi$  are equal, so  $\text{ev}_{a\mathbf{u}+b\mathbf{v}} = a\text{ev}_{\mathbf{u}} + b\text{ev}_{\mathbf{v}}$ .

*Injective.* By Theorem 3.2.3 it suffices to show that  $\text{ev}_{\mathbf{v}} = 0$  implies  $\mathbf{v} = \mathbf{0}$ . We prove the contrapositive. Let  $\mathbf{v} \neq \mathbf{0}$ . If  $V$  is finite-dimensional, Theorem 4.1.5 (a) gives  $\varphi \in V^*$  with  $\varphi(\mathbf{v}) = 1$ . In general,  $\{\mathbf{v}\}$  is linearly independent, so by Theorem 1.8.2 there is a basis  $B$  of  $V$  with  $\mathbf{v} \in B$ . By Theorem 3.1.6, which allows infinite bases, there is a linear map  $\varphi: V \rightarrow F$  with  $\varphi(\mathbf{v}) = 1$  and  $\varphi(\mathbf{b}) = 0$  for every  $\mathbf{b} \in B \setminus \{\mathbf{v}\}$ . In either case  $\varphi \in V^*$  and  $\text{ev}_{\mathbf{v}}(\varphi) = \varphi(\mathbf{v}) = 1 \neq 0$ , so  $\text{ev}_{\mathbf{v}} \neq 0$ . This proves that  $\text{ev}_V$  is injective. ■

In infinite dimension the proof repeats the remark after Theorem 4.1.5, and Zorn's Lemma enters through Theorem 1.8.2. In finite dimension no choice principle is needed.

So every vector  $\mathbf{v}$  is completely remembered by the list of all its measurements  $\varphi(\mathbf{v})$ . In finite dimension a count turns “injective” into “bijective”.

### Theorem 4.3.2 — The Double Dual Isomorphism

Let  $V$  be a finite-dimensional vector space over  $F$ . Then the evaluation map  $\text{ev}_V: V \rightarrow V^{**}$  is an isomorphism. Moreover, if  $\mathcal{B} = (\mathbf{v}_1, \dots, \mathbf{v}_n)$  is a basis of  $V$  with dual basis  $\mathcal{B}^* = (\varphi_1, \dots, \varphi_n)$ , then  $(\text{ev}_{\mathbf{v}_1}, \dots, \text{ev}_{\mathbf{v}_n})$  is the dual basis of  $\mathcal{B}^*$ .

*Idea.* Count instead of check. The dual basis has as many vectors as the basis, so  $\dim V^* = \dim V$ , and the same holds one level up. An injective linear map between spaces of the same finite dimension is bijective. For the second statement, “dual basis” is the condition  $\theta_i(\varphi_j) = \delta_{ij}$ , and for  $\theta_i = \text{ev}_{\mathbf{v}_i}$  that condition is the definition of  $\varphi_j$  read backwards.

*Proof.* Let  $n = \dim V$ . By Corollary 4.1.4,  $V^*$  is finite-dimensional with  $\dim V^* = n$ , and applying it again to  $V^*$  gives  $\dim V^{**} = n$ . By Theorem 4.3.1,  $\text{ev}_V$  is linear and injective. Since  $\dim V = \dim V^{**}$  is finite, Corollary 3.3.2 (e) shows that  $\text{ev}_V$  is bijective, hence an isomorphism (Definition 3.5.2).

For the second statement,  $\text{ev}_{\mathbf{v}_i}(\varphi_j) = \varphi_j(\mathbf{v}_i) = \delta_{ji} = \delta_{ij}$  for all  $i, j$ , by the definition of the dual basis. By the uniqueness in Theorem 4.1.3 (a), applied to the basis  $\mathcal{B}^*$  of  $V^*$ , the dual basis of  $\mathcal{B}^*$  (Definition 4.1.3) is the only list  $(\theta_1, \dots, \theta_n)$  in  $V^{**}$  with  $\theta_i(\varphi_j) = \delta_{ij}$ . Hence it is  $(\text{ev}_{\mathbf{v}_1}, \dots, \text{ev}_{\mathbf{v}_n})$ , as claimed. ■

The theorem says: **every** linear functional on  $V^*$  is evaluation at exactly one vector. To find the vector, read off its coordinates from the values on a dual basis. For instance, on

$V = \mathbb{R}[x]_{\leq 2}$  with the coefficient functionals  $\varphi_0, \varphi_1, \varphi_2$  of [Example \(Evaluation in Three Spaces\)](#), let  $\theta \in V^{**}$  be the functional with  $\theta(\varphi_0) = 2$ ,  $\theta(\varphi_1) = 0$ ,  $\theta(\varphi_2) = -1$ . By the second statement,  $\theta = 2\text{ev}_1 - \text{ev}_{x^2} = \text{ev}_{2-x^2}$ , since both sides agree on the basis  $(\varphi_0, \varphi_1, \varphi_2)$  and  $\text{ev}$  is linear. So, for example,  $\theta(\varepsilon_2) = \varepsilon_2(2 - x^2) = -2$ , where  $\varepsilon_2(p) = p(2)$ . Part (b) of [Example \(Evaluation in Three Spaces\)](#) was the case  $\theta = \text{ev}_{x^2}$ .

### 4.3.3 What “natural” means

Both  $V \cong V^*$  and  $V \cong V^{**}$  hold in finite dimension, and both follow from counting dimensions. Yet they are very different isomorphisms. The difference is whether a choice is needed.

**The isomorphism  $V \rightarrow V^*$  depends on the basis.** For a basis  $\mathcal{B} = (\mathbf{v}_1, \dots, \mathbf{v}_n)$  of  $V$ , let  $\Theta_{\mathcal{B}}: V \rightarrow V^*$  be the linear map with  $\Theta_{\mathcal{B}}(\mathbf{v}_i) = \varphi_i$  ([Theorem 3.1.5](#)). It sends a basis to a basis, so it is an isomorphism, by the argument in the proof of [Theorem 3.5.4](#). Now take  $V = \mathbb{R}$ , a space of dimension 1. With the basis (1), the dual basis is  $\varphi(t) = t$ , and  $\Theta_{(1)}(1)$  is  $t \mapsto t$ . With the basis (2), the dual basis is  $\varphi(t) = t/2$ , since that is the functional sending 2 to 1. So  $\Theta_{(2)}(2) = (t \mapsto t/2)$ , and by linearity  $\Theta_{(2)}(1) = \frac{1}{2}\Theta_{(2)}(2)$  is  $t \mapsto t/4$ . The same vector 1 goes to two different functionals. Every basis is as good as any other, so none of these isomorphisms is preferred.

**The isomorphism  $V \rightarrow V^{**}$  does not.** Build it the same way, choosing bases twice. First  $\Theta_{\mathcal{B}}: V \rightarrow V^*$  sends  $\mathbf{v}_i \mapsto \varphi_i$ . Then, using the basis  $\mathcal{B}^*$  of  $V^*$ , the map  $\Theta_{\mathcal{B}^*}: V^* \rightarrow V^{**}$  sends each  $\varphi_i$  to the  $i$ -th vector of the dual basis of  $\mathcal{B}^*$ . By [Theorem 4.3.2](#), that vector is  $\text{ev}_{\mathbf{v}_i}$ . So the composite  $\Theta_{\mathcal{B}^*}\Theta_{\mathcal{B}}$  sends  $\mathbf{v}_i \mapsto \text{ev}_{\mathbf{v}_i}$  for every  $i$ . The linear map  $\text{ev}_V$  does the same, so by the uniqueness part of [Theorem 3.1.5](#),

$$\Theta_{\mathcal{B}^*}\Theta_{\mathcal{B}} = \text{ev}_V \quad \text{for every basis } \mathcal{B} \text{ of } V.$$

Each factor depends on  $\mathcal{B}$ , but the composite does not: the two choices cancel. This is the first meaning of **natural**: an isomorphism is natural if it can be defined without choosing anything, and  $\text{ev}_V$  is defined by the formula  $\text{ev}_{\mathbf{v}}(\varphi) = \varphi(\mathbf{v})$  alone.

There is a second, sharper meaning. A choice-free construction should be compatible with every linear map: if  $T: V \rightarrow W$ , then first applying  $T$  and then identifying  $W$  with  $W^{**}$  should agree with first identifying  $V$  with  $V^{**}$  and then applying “the map that  $T$  induces on double duals”. To state this we need to know how a linear map acts on functionals. That is the **dual map**, the subject of the next section, where we prove that  $\text{ev}$  is compatible with every linear map, and see that no family of isomorphisms  $V \rightarrow V^*$  can be.

*Quick check.* In  $V = \mathbb{R}$  with the basis  $\mathcal{B} = (2)$ , compute  $\Theta_{\mathcal{B}^*}\Theta_{\mathcal{B}}(1)$  as an element of  $V^{**}$ , and evaluate it on  $\psi(t) = 5t$ . Compare with  $\text{ev}_1(\psi)$ .

*Solution.* The dual basis of  $\mathcal{B}$  is  $\varphi_1(t) = t/2$ , so  $\Theta_{\mathcal{B}}(1) = \frac{1}{2}\Theta_{\mathcal{B}}(2) = \frac{1}{2}\varphi_1$ . The dual basis of  $\mathcal{B}^* = (\varphi_1)$  is the  $\theta \in V^{**}$  with  $\theta(\varphi_1) = 1$ , and  $\text{ev}_2(\varphi_1) = \varphi_1(2) = 1$ , so  $\theta = \text{ev}_2$  and  $\Theta_{\mathcal{B}^*}(\varphi_1) = \text{ev}_2$ . Hence  $\Theta_{\mathcal{B}^*}\Theta_{\mathcal{B}}(1) = \frac{1}{2}\text{ev}_2 = \text{ev}_1$ , using linearity of  $\text{ev}$ . On  $\psi$  it gives  $\text{ev}_1(\psi) = \psi(1) = 5$ . The factor  $\frac{1}{2}$  from the first step and the factor 2 hidden in  $\text{ev}_2$  cancel, exactly as the general computation predicts.

*Remark.* Because  $\text{ev}_V$  is natural, it is common to **identify** a finite-dimensional  $V$  with  $V^{**}$ , and to regard a vector  $\mathbf{v}$  as the functional  $\varphi \mapsto \varphi(\mathbf{v})$  on  $V^*$ . We will do this only with an explicit “under the identification  $\mathbf{v} \leftrightarrow \text{ev}_{\mathbf{v}}$ ”, as in the next corollary.

### 4.3.4 Consequences in finite dimension

Throughout this subsection,  $V$  is finite-dimensional. The isomorphism  $\text{ev}_V$  lets us run arguments about  $V^*$  “one level up” and bring the answer back to  $V$ .

The dual basis construction goes from a basis of  $V$  to a basis of  $V^*$ . Does every basis of  $V^*$  arise this way? A list of  $n$  independent measurements should be the coordinate functions of some basis, and the double dual shows it is.

#### Corollary 4.3.3 — Every Basis of the Dual Is a Dual Basis

Let  $V$  be a finite-dimensional vector space over  $F$ , and let  $(\psi_1, \dots, \psi_n)$  be a basis of  $V^*$ . Then there is exactly one basis  $(\mathbf{v}_1, \dots, \mathbf{v}_n)$  of  $V$  whose dual basis is  $(\psi_1, \dots, \psi_n)$ , that is, with

$$\psi_i(\mathbf{v}_j) = \delta_{ij} \quad \text{for all } i, j.$$

*Idea.* Go up one level, where the problem is already solved. The basis  $(\psi_1, \dots, \psi_n)$  of  $V^*$  has a dual basis  $(\theta_1, \dots, \theta_n)$  in  $V^{**}$ . Every  $\theta_j$  is evaluation at some vector, because  $\text{ev}$  is onto. Those vectors are the answer, since  $\theta_j(\psi_i) = \delta_{ij}$  says  $\psi_i(\mathbf{v}_j) = \delta_{ij}$ . Uniqueness is a “test against functionals” argument.

*Proof. Existence.* Here  $V^*$  is finite-dimensional (Corollary 4.1.4). By Theorem 4.1.3 applied to  $V^*$ , there is a basis  $(\theta_1, \dots, \theta_n)$  of  $V^{**}$  with  $\theta_j(\psi_i) = \delta_{ij}$ . By Theorem 4.3.2,  $\text{ev}_V$  is an isomorphism, so there are unique  $\mathbf{v}_j \in V$  with  $\text{ev}_{\mathbf{v}_j} = \theta_j$ . Its inverse is an isomorphism (Theorem 3.5.3), so  $(\mathbf{v}_1, \dots, \mathbf{v}_n)$  is a basis of  $V$  by Theorem 3.5.2. For all  $i, j$ ,

$$\psi_i(\mathbf{v}_j) = \text{ev}_{\mathbf{v}_j}(\psi_i) = \theta_j(\psi_i) = \delta_{ij}.$$

By the uniqueness in Theorem 4.1.3 (a), the dual basis of  $(\mathbf{v}_1, \dots, \mathbf{v}_n)$  is  $(\psi_1, \dots, \psi_n)$ .

*Uniqueness.* Suppose  $(\mathbf{u}_1, \dots, \mathbf{u}_n)$  is also a basis with  $\psi_i(\mathbf{u}_j) = \delta_{ij}$ . Fix  $j$ . Then  $\psi_i(\mathbf{v}_j - \mathbf{u}_j) = \delta_{ij} - \delta_{ij} = 0$  for every  $i$ . Every  $\varphi \in V^*$  is a combination of the  $\psi_i$ , so  $\varphi(\mathbf{v}_j - \mathbf{u}_j) = 0$  for every  $\varphi \in V^*$ . By Theorem 4.1.5 (b),  $\mathbf{v}_j - \mathbf{u}_j = \mathbf{0}$ . Hence  $\mathbf{u}_j = \mathbf{v}_j$  for every  $j$ . ■

In coordinates this is a matrix inversion, the mirror image of Example (The Dual Basis Is Read Off the Inverse Matrix). There, a basis was given and its dual basis appeared as the rows of an inverse matrix. Here the functionals are given, and the basis appears as the columns of an inverse matrix.

#### Example. (A Basis With Prescribed Dual Basis)

On  $V = \mathbb{R}[x]_{\leq 2}$ , consider the functionals

$$\psi_1(p) = p(0), \quad \psi_2(p) = p(1), \quad \psi_3(p) = p'(1).$$

Show that  $(\psi_1, \psi_2, \psi_3)$  is a basis of  $V^*$ , and find the basis  $(p_1, p_2, p_3)$  of  $V$  whose dual basis it is.

*Solution.* Write down a matrix. For  $p = a + bx + cx^2$ , with  $[p]_{\mathcal{E}} = (a, b, c)$  in  $\mathcal{E} = (1, x, x^2)$ ,

$$\psi_1(p) = a, \quad \psi_2(p) = a + b + c, \quad \psi_3(p) = b + 2c.$$

So  $\psi_i(p) = \mathbf{r}_i[p]_{\mathcal{E}}$ , where  $\mathbf{r}_i$  is row  $i$  of

$$\mathbf{A} = \begin{pmatrix} 1 & 0 & 0 \\ 1 & 1 & 1 \\ 0 & 1 & 2 \end{pmatrix}.$$

*Translate the condition.* Let  $\mathbf{P}$  be the  $3 \times 3$  matrix whose  $j$ -th column is  $[p_j]_{\mathcal{E}}$ . By [Theorem 0.9.2](#), the  $(i, j)$  entry of  $\mathbf{AP}$  is  $\mathbf{r}_i[p_j]_{\mathcal{E}} = \psi_i(p_j)$ . So the condition  $\psi_i(p_j) = \delta_{ij}$  says exactly  $\mathbf{AP} = \mathbf{I}_3$ .

*Solve.* Row reduction ([Theorem 2.4.6](#)) gives

$$\mathbf{A}^{-1} = \begin{pmatrix} 1 & 0 & 0 \\ -2 & 2 & -1 \\ 1 & -1 & 1 \end{pmatrix},$$

and one checks  $\mathbf{AA}^{-1} = \mathbf{I}_3$ , so  $\mathbf{A}$  is invertible by [Theorem 2.4.5](#). Multiplying  $\mathbf{AP} = \mathbf{I}_3$  on the left by  $\mathbf{A}^{-1}$  forces  $\mathbf{P} = \mathbf{A}^{-1}$ . Since the columns of the invertible matrix  $\mathbf{P}$  are independent,  $(p_1, p_2, p_3)$  is a basis of  $V$  (via the coordinate isomorphism, [Corollary 3.5.5](#)). By the uniqueness in [Theorem 4.1.3](#) (a), its dual basis is  $(\psi_1, \psi_2, \psi_3)$ , which in particular is a basis of  $V^*$ . Reading the columns,

$$p_1 = 1 - 2x + x^2 = (1 - x)^2, \quad p_2 = 2x - x^2, \quad p_3 = x^2 - x.$$

*Check.*  $p_1(0) = 1, p_1(1) = 0, p_1'(1) = 0; p_2(0) = 0, p_2(1) = 1, p_2'(1) = 2 - 2 = 0; p_3(0) = 0, p_3(1) = 0, p_3'(1) = 1.$

So every  $p \in \mathbb{R}[x]_{\leq 2}$  is  $p = p(0)p_1 + p(1)p_2 + p'(1)p_3$ : the coordinates of  $p$  in this basis are the three measurements.

The next consequence is the one we will use most. A subspace  $U$  determines the set  $U^0$  of functionals vanishing on it. Going back, the functionals in  $U^0$  vanish together on some set of vectors, which contains  $U$ . Is it exactly  $U$ ?

### Corollary 4.3.4 — The Annihilator of the Annihilator

Let  $V$  be a finite-dimensional vector space over  $F$ , and let  $U$  be a subspace of  $V$ . Write  $U^{00} := (U^0)^0 \subseteq V^{**}$ . Then

$$U^{00} = \text{ev}_V(U) = \{\text{ev}_{\mathbf{u}} : \mathbf{u} \in U\}.$$

So, under the identification  $\mathbf{v} \leftrightarrow \text{ev}_{\mathbf{v}}$ , we have  $U^{00} = U$ . Equivalently,

$$U = \{\mathbf{v} \in V : \varphi(\mathbf{v}) = 0 \text{ for every } \varphi \in U^0\}.$$

*Idea.* One inclusion is a tautology: if  $\mathbf{u} \in U$  and  $\varphi$  kills  $U$ , then  $\text{ev}_{\mathbf{u}}(\varphi) = \varphi(\mathbf{u}) = 0$ . For the other, count. The dimension formula applied twice, once in  $V$  and once in  $V^*$ , gives

$\dim U^{00} = \dim U$ , and  $\text{ev}$  is injective, so  $\dim \text{ev}_V(U) = \dim U$  too. One inclusion plus equal dimension gives equality.

*Proof.* Let  $n = \dim V$ , so  $\dim V^* = n$  by [Corollary 4.1.4](#).

*Inclusion.* Let  $\mathbf{u} \in U$ . For every  $\varphi \in U^0$ ,  $\text{ev}_{\mathbf{u}}(\varphi) = \varphi(\mathbf{u}) = 0$  by [Definition 4.2.1](#). Hence  $\text{ev}_{\mathbf{u}} \in (U^0)^0$ , and  $\text{ev}_V(U) \subseteq U^{00}$ .

*Dimensions.* By [Theorem 4.2.2](#) in  $V$ ,  $\dim U^0 = n - \dim U$ . By [Theorem 4.2.2](#) in the  $n$ -dimensional space  $V^*$ , applied to its subspace  $U^0$ ,

$$\dim U^{00} = n - \dim U^0 = n - (n - \dim U) = \dim U.$$

The restriction of  $\text{ev}_V$  to  $U$  is linear and injective ([Theorem 4.3.1](#)), so it is an isomorphism onto its image  $\text{ev}_V(U)$ , and  $\dim \text{ev}_V(U) = \dim U$  ([Theorem 3.5.4](#)). So  $\text{ev}_V(U)$  is a subspace of  $U^{00}$  of the same finite dimension, and [Theorem 1.6.8](#) gives  $\text{ev}_V(U) = U^{00}$ .

*The equivalent form.* Let  $\mathbf{v} \in V$ . Then  $\varphi(\mathbf{v}) = 0$  for every  $\varphi \in U^0$  if and only if  $\text{ev}_{\mathbf{v}} \in U^{00} = \text{ev}_V(U)$ , if and only if  $\text{ev}_{\mathbf{v}} = \text{ev}_{\mathbf{u}}$  for some  $\mathbf{u} \in U$ . Since  $\text{ev}_V$  is injective, this happens if and only if  $\mathbf{v} \in U$ .

■

The last form says: **a subspace is exactly the common zero set of the functionals that vanish on it.** No vector outside  $U$  sneaks through all the equations. In section 6 of this chapter this becomes the statement that every subspace of  $F^n$  is the solution set of a homogeneous linear system. An immediate consequence is that the annihilator loses no information.

### Corollary 4.3.5 — Subspaces Are Determined by Their Annihilators

Let  $V$  be a finite-dimensional vector space, and let  $U$  and  $W$  be subspaces of  $V$ . Then  $U = W$  if and only if  $U^0 = W^0$ .

*Proof.*  $(\Rightarrow)$  is immediate.  $(\Leftarrow)$  Suppose  $U^0 = W^0$ . By [Corollary 4.3.4](#),  $U = \{\mathbf{v} : \varphi(\mathbf{v}) = 0 \text{ for all } \varphi \in U^0\} = \{\mathbf{v} : \varphi(\mathbf{v}) = 0 \text{ for all } \varphi \in W^0\} = W$ .

■

*Quick check.* Is [Corollary 4.3.5](#) still true if  $U$  and  $W$  are allowed to be arbitrary subsets of  $V$ ?

*Solution.* No. In  $\mathbb{R}^2$ , take  $U = \{(1, 0)\}$  and  $W = \{(2, 0)\}$ . A functional  $\varphi(x, y) = ax + by$  vanishes at  $(1, 0)$  exactly when  $a = 0$ , and at  $(2, 0)$  exactly when  $2a = 0$ , that is, again when  $a = 0$ . So  $U^0 = W^0$  but  $U \neq W$ . An annihilator only sees the span,  $U^0 = (\text{span } U)^0$  ([Theorem 4.2.3](#) (b)).

### 4.3.5 Infinite dimension: the evaluation map is not onto

Nothing in [Definition 4.3.1](#) or [Theorem 4.3.1](#) needed finite dimension. The count in [Theorem 4.3.2](#) did.

*Warning.*  $V \cong V^{**}$  **through**  $\text{ev}$  **is a finite-dimensional fact.** For  $V = F[x]$ , the evaluation map is injective but **not** surjective: there is a linear functional on  $F[x]^*$  that is not evaluation at any polynomial. The next example constructs one.

#### Example. (A Functional on $F[x]^*$ That Is Not an Evaluation)

Let  $\varphi_k \in F[x]^*$  ( $k \in \mathbb{N}$ ) be the coordinate functionals,  $\varphi_k(p)$  = the coefficient of  $x^k$  in  $p$ , and let  $\varepsilon_1(p) = p(1)$ , as in [Proposition 4.1.6](#). Show that there is  $\theta \in F[x]**$  with  $\theta \neq \text{ev}_p$  for every  $p \in F[x]$ .

*Solution. Step 1:*  $S = \{\varphi_k : k \in \mathbb{N}\} \cup \{\varepsilon_1\}$  is linearly independent. By [Proposition 4.1.6](#), the  $\varphi_k$  form an independent set and  $\varepsilon_1 \notin \text{span}\{\varphi_k : k \in \mathbb{N}\}$ . Let  $L$  be a finite list of distinct elements of  $S$ . If  $\varepsilon_1$  is not in  $L$ , then  $L$  is independent. Otherwise, list  $\varepsilon_1$  last; the others are independent and  $\varepsilon_1$  is not in their span, so  $L$  is independent by [Lemma 1.5.3](#).

*Step 2: build  $\theta$ .* By [Theorem 1.8.2](#) there is a basis  $B$  of  $F[x]^*$  containing  $S$ . By [Theorem 3.1.6](#) for the (infinite) basis  $B$ , there is a linear  $\theta: F[x]^* \rightarrow F$  with  $\theta(\varepsilon_1) = 1$  and  $\theta(\beta) = 0$  for every  $\beta \in B \setminus \{\varepsilon_1\}$ . In particular  $\theta(\varphi_k) = 0$  for all  $k$ .

*Step 3:  $\theta$  is not an evaluation.* Suppose  $\theta = \text{ev}_p$  for some  $p \in F[x]$ . Then for every  $k$ , the coefficient of  $x^k$  in  $p$  is  $\varphi_k(p) = \text{ev}_p(\varphi_k) = \theta(\varphi_k) = 0$ . So  $p = 0$ , and  $\theta = \text{ev}_0 = 0$ . This contradicts  $\theta(\varepsilon_1) = 1$ . Hence  $\theta \notin \text{ev}(F[x])$ , and  $\text{ev}$  is not surjective.

The functional  $\theta$  exists, but nobody can write it down: it comes from Zorn's Lemma through [Theorem 1.8.2](#). The deeper reason for the failure is size.  $F[x]^*$  is so much bigger than  $F[x]$  ([Exercise \(C4: How large is  \$F\[x\]^\*\$ ?\)](#) identifies it with the whole sequence space  $F^{\mathbb{N}}$ ) that  $F[x]**$  is bigger still, and in fact no isomorphism  $F[x] \cong F[x]**$  exists at all, a statement about the sizes of bases that we do not prove. In finite dimension, sizes cannot grow, and the evaluation map is the perfect dictionary of [Theorem 4.3.2](#).

### 4.3.6 Exercises

#### 4.3.6.1 A. Check your understanding

##### Exercise A1.

- Define  $\text{ev}_{\mathbf{v}}$  for  $\mathbf{v} \in V$ , and the evaluation map  $\text{ev}_V$ . In which space does  $\text{ev}_{\mathbf{v}}$  live?
- State a hypothesis on  $V$  under which  $\text{ev}_V$  is an isomorphism, and a space for which it is not.
- True or false: if  $V$  is finite-dimensional and  $\mathbf{v} \neq \mathbf{0}$ , then  $\text{ev}_{\mathbf{v}}(\varphi) \neq 0$  for **every** non-zero  $\varphi \in V^*$ . Justify your answer.
- True or false: if  $U$  and  $W$  are subspaces of  $\mathbb{R}^5$  with  $U^0 = W^0$ , then  $U = W$ .

Justify your answer.

- (e) Name the method used to prove  $U^{00} = \text{ev}_V(U)$  in [Corollary 4.3.4](#).

*Solution.*

- (a)  $\text{ev}_{\mathbf{v}}: V^* \rightarrow F$ ,  $\text{ev}_{\mathbf{v}}(\varphi) = \varphi(\mathbf{v})$ , and  $\text{ev}_V: V \rightarrow V^{**}$ ,  $\mathbf{v} \mapsto \text{ev}_{\mathbf{v}}$  ([Definition 4.3.1](#)).  $\text{ev}_{\mathbf{v}}$  lives in  $V^{**}$ .
- (b) If  $V$  is finite-dimensional,  $\text{ev}_V$  is an isomorphism ([Theorem 4.3.2](#)). For  $V = F[x]$  it is not surjective ([Example \(A Functional on  \$F\[x\]^\*\$  That Is Not an Evaluation\)](#)).
- (c) False. It says only that **some**  $\varphi$  has  $\varphi(\mathbf{v}) \neq 0$  ([Theorem 4.3.1](#)). In  $\mathbb{R}^2$ ,  $\mathbf{v} = (1, 2)$  and  $\varphi(x, y) = 2x - y \neq 0$  give  $\text{ev}_{\mathbf{v}}(\varphi) = 0$ .
- (d) True, by [Corollary 4.3.5](#), since  $\mathbb{R}^5$  is finite-dimensional.
- (e) One inclusion plus equal dimension:  $\text{ev}_V(U) \subseteq U^{00}$  directly, then  $\dim U^{00} = \dim U = \dim \text{ev}_V(U)$  from the dimension formula for annihilators (twice) and injectivity of  $\text{ev}$ , and [Theorem 1.6.8](#).

### 4.3.6.2 B. Practice

#### Exercise B1: A basis with a given dual basis.

- (a) Let  $\psi_1(x, y) = 2x + y$  and  $\psi_2(x, y) = x + y$  on  $\mathbb{R}^2$ . Find the basis  $(\mathbf{v}_1, \mathbf{v}_2)$  of  $\mathbb{R}^2$  whose dual basis is  $(\psi_1, \psi_2)$ .
- (b) Let  $\psi_1(p) = p(0)$  and  $\psi_2(p) = \int_0^1 p(x) dx$  on  $\mathbb{R}[x]_{\leq 1}$ . Find the basis  $(p_1, p_2)$  of  $\mathbb{R}[x]_{\leq 1}$  whose dual basis is  $(\psi_1, \psi_2)$ .

*Solution.*

- (a) As in [Example \(A Basis With Prescribed Dual Basis\)](#),  $\psi_i(\mathbf{v}) = \mathbf{r}_i \mathbf{v}$  with  $\mathbf{r}_i$  the rows of  $\mathbf{A} = \begin{pmatrix} 2 & 1 \\ 1 & 1 \end{pmatrix}$ , and the condition  $\psi_i(\mathbf{v}_j) = \delta_{ij}$  says  $\mathbf{A}\mathbf{P} = \mathbf{I}_2$  for  $\mathbf{P} = (\mathbf{v}_1 \ \mathbf{v}_2)$ . Here  $2 \cdot 1 - 1 \cdot 1 = 1 \neq 0$ , so by [Theorem 0.9.8](#)

$$\mathbf{A}^{-1} = \begin{pmatrix} 1 & -1 \\ -1 & 2 \end{pmatrix},$$

$\mathbf{P} = \mathbf{A}^{-1}$ , so  $\mathbf{v}_1 = (1, -1)$  and  $\mathbf{v}_2 = (-1, 2)$ . These are independent (the columns of an invertible matrix), hence a basis. Check:  $\psi_1(\mathbf{v}_1) = 1$ ,  $\psi_2(\mathbf{v}_1) = 0$ ,  $\psi_1(\mathbf{v}_2) = 0$ ,  $\psi_2(\mathbf{v}_2) = 1$ . By the uniqueness part of [Theorem 4.1.3](#), the dual basis of  $(\mathbf{v}_1, \mathbf{v}_2)$  is  $(\psi_1, \psi_2)$ .

- (b) For  $p = a + bx$ ,  $\psi_1(p) = a$  and  $\psi_2(p) = a + \frac{1}{2}b$ . So  $\mathbf{A} = \begin{pmatrix} 1 & 0 \\ 1 & 1/2 \end{pmatrix}$  and

$$\mathbf{P} = \mathbf{A}^{-1} = \begin{pmatrix} 1 & 0 \\ -2 & 2 \end{pmatrix},$$

whose columns are the coordinates of  $p_1 = 1 - 2x$  and  $p_2 = 2x$  in  $(1, x)$ . Check:  $p_1(0) = 1$ ,  $\int_0^1 (1 - 2x) dx = 1 - 1 = 0$ ,  $p_2(0) = 0$ ,  $\int_0^1 2x dx = 1$ . As in (a),  $(p_1, p_2)$  is a basis with dual basis  $(\psi_1, \psi_2)$ .



**Exercise B2:**  $U^{00} = U$  for a line.

Let  $U = \text{span}((1, 2, -1)) \subseteq \mathbb{R}^3$ .

- (a) Find a basis of  $U^0$ .
- (b) Find all  $\mathbf{v} \in \mathbb{R}^3$  with  $\varphi(\mathbf{v}) = 0$  for every  $\varphi \in U^0$ , and check that the answer is  $U$ .
- (c) Hence verify that  $U^{00} = \text{ev}(U)$ , and confirm the dimension count.

*Solution.*

- (a) A functional  $\varphi(x, y, z) = ax + by + cz$  (Theorem 4.1.1) lies in  $U^0$  iff it vanishes at  $(1, 2, -1)$  (Theorem 4.2.3 (b)), that is,  $a + 2b - c = 0$ . The solutions are  $(a, b, c) = s(-2, 1, 0) + t(1, 0, 1)$ , so  $\varphi_1(x, y, z) = 2x - y$  and  $\varphi_2(x, y, z) = x + z$  span  $U^0$  (we replaced  $(-2, 1, 0)$  by its negative). They are independent, since neither is a multiple of the other, so  $(\varphi_1, \varphi_2)$  is a basis of  $U^0$ . This matches  $\dim U^0 = 3 - 1 = 2$  (Theorem 4.2.2).
- (b) Every  $\varphi \in U^0$  is a combination of  $\varphi_1, \varphi_2$ , so  $\varphi(\mathbf{v}) = 0$  for all  $\varphi \in U^0$  iff  $\varphi_1(\mathbf{v}) = \varphi_2(\mathbf{v}) = 0$ , that is,  $2x - y = 0$  and  $x + z = 0$ . With  $x = t$ :  $y = 2t$ ,  $z = -t$ , so  $\mathbf{v} = t(1, 2, -1)$ . The set is exactly  $U$ .
- (c) Let  $\theta \in U^{00} \subseteq (\mathbb{R}^3)^{**}$ . By Theorem 4.3.2,  $\theta = \text{ev}_{\mathbf{v}}$  for some  $\mathbf{v}$ , and  $\theta \in (U^0)^0$  means  $\varphi(\mathbf{v}) = 0$  for all  $\varphi \in U^0$ . By (b),  $\mathbf{v} \in U$ , so  $U^{00} \subseteq \text{ev}(U)$ ; the reverse inclusion is immediate from Definition 4.2.1. Dimensions:  $\dim U^{00} = 3 - \dim U^0 = 1 = \dim U$ .

**Exercise B3: Recognizing an evaluation.**

Let  $V = \mathbb{R}[x]_{\leq 2}$ , with the coefficient functionals  $\varphi_0, \varphi_1, \varphi_2$  of Example (Evaluation in Three Spaces). Let  $\theta \in V^{**}$  be the linear functional with  $\theta(\varphi_0) = 1$ ,  $\theta(\varphi_1) = -3$ ,  $\theta(\varphi_2) = 2$ . Find  $p \in V$  with  $\theta = \text{ev}_p$ . Hence compute  $\theta(\varepsilon_1)$  and  $\theta(\psi)$ , where  $\varepsilon_1(q) = q(1)$  and  $\psi(q) = q'(0)$ .

*Solution.* By Theorem 4.3.2,  $(\text{ev}_1, \text{ev}_x, \text{ev}_{x^2})$  is the dual basis of  $(\varphi_0, \varphi_1, \varphi_2)$ . Hence  $\text{ev}_1 - 3\text{ev}_x + 2\text{ev}_{x^2}$  takes the values  $1, -3, 2$  on  $\varphi_0, \varphi_1, \varphi_2$ . So does  $\theta$ , and a linear functional is determined by its values on a basis (Theorem 3.1.5). Since  $\text{ev}$  is linear (Theorem 4.3.1),

$$\theta = \text{ev}_1 - 3\text{ev}_x + 2\text{ev}_{x^2} = \text{ev}_p, \quad p = 1 - 3x + 2x^2.$$

Hence  $\theta(\varepsilon_1) = \varepsilon_1(p) = 1 - 3 + 2 = 0$  and  $\theta(\psi) = \psi(p) = p'(0) = -3$ .

**4.3.6.3 C. Going deeper****Exercise C1: Subspaces of  $V$  and of  $V^*$ .**

Let  $V$  be a finite-dimensional vector space over  $F$ . Prove that  $U \mapsto U^0$  is a bijection from the set of subspaces of  $V$  to the set of subspaces of  $V^*$ , and that it reverses inclusions in both directions:  $U \subseteq W$  if and only if  $W^0 \subseteq U^0$ .

*Hint: for surjectivity, given a subspace  $X \subseteq V^*$ , consider  $U = \{\mathbf{v} \in V : \varphi(\mathbf{v}) = 0 \text{ for all } \varphi \in X\}$ .*

*Solution.* Let  $n = \dim V = \dim V^*$  (Corollary 4.1.4). By Proposition 4.2.1,  $U^0$  is a subspace of  $V^*$ , so the map is well defined.

*Injective.* This is Corollary 4.3.5.

*Inclusions.* If  $U \subseteq W$ , then  $W^0 \subseteq U^0$  by Theorem 4.2.3 (a). Conversely, suppose  $W^0 \subseteq U^0$ , and let  $\mathbf{u} \in U$ . Every  $\varphi \in W^0$  lies in  $U^0$ , so  $\varphi(\mathbf{u}) = 0$ . By Corollary 4.3.4 (the equivalent form, for  $W$ ),  $\mathbf{u} \in W$ . Hence  $U \subseteq W$ .

*Surjective.* Let  $X$  be a subspace of  $V^*$ , and let  $U = \{\mathbf{v} \in V : \varphi(\mathbf{v}) = 0 \text{ for all } \varphi \in X\}$ . For  $\mathbf{v} \in V$ ,  $\mathbf{v} \in U$  iff  $\text{ev}_{\mathbf{v}}(\varphi) = 0$  for all  $\varphi \in X$  iff  $\text{ev}_{\mathbf{v}} \in X^0$ , where  $X^0 \subseteq V^{**}$ . So  $U = \text{ev}_V^{-1}(X^0)$ , and since  $\text{ev}_V$  is an isomorphism (Theorem 4.3.2),  $U$  is a subspace and  $\text{ev}_V$  restricts to an isomorphism  $U \rightarrow X^0$ . By Theorem 4.2.2 in  $V^*$ ,  $\dim U = \dim X^0 = n - \dim X$ . By definition of  $U$ , every  $\varphi \in X$  vanishes on  $U$ , so  $X \subseteq U^0$ . By Theorem 4.2.2 in  $V$ ,  $\dim U^0 = n - \dim U = \dim X$ . By Theorem 1.6.8,  $X = U^0$ . This proves the claim.

### Exercise C2: Spans cut out by equations.

Let  $V$  be finite-dimensional and  $S \subseteq V$  any subset.

- Prove that  $\{\mathbf{v} \in V : \varphi(\mathbf{v}) = 0 \text{ for every } \varphi \in S^0\} = \text{span}(S)$ .
- Deduce: a vector  $\mathbf{v}$  lies in  $\text{span}(S)$  if and only if every functional that vanishes on  $S$  vanishes at  $\mathbf{v}$ .
- Show by an example that (a) can fail if “ $\text{span}(S)$ ” is replaced by “ $S$ ”.

*Solution.*

- $\text{span}(S)$  is a subspace (Theorem 1.3.1), and  $S^0 = (\text{span } S)^0$  by Theorem 4.2.3 (b). Applying Corollary 4.3.4 to  $U = \text{span}(S)$ ,

$$\{\mathbf{v} : \varphi(\mathbf{v}) = 0 \ \forall \varphi \in S^0\} = \{\mathbf{v} : \varphi(\mathbf{v}) = 0 \ \forall \varphi \in (\text{span } S)^0\} = \text{span}(S).$$

- This is (a) read for a single vector  $\mathbf{v}$ .
- Take  $S = \{(1, 0)\} \subseteq \mathbb{R}^2$ . Then  $S^0$  consists of the functionals  $(x, y) \mapsto by$ , and their common zero set is the whole  $x$ -axis, which contains  $(2, 0) \notin S$ .

### Exercise C3: Where finite dimension is needed.

Let  $V = F[x]$ , let  $\varphi_k$  and  $\varepsilon_1$  be as in Example (A Functional on  $F[x]^*$  That Is Not an Evaluation), and let  $X = \text{span}\{\varphi_k : k \in \mathbb{N}\} \subseteq V^*$ .

- Show that  $\{p \in F[x] : \varphi(p) = 0 \text{ for all } \varphi \in X\} = \{0\}$ .
- Deduce that  $X \neq U^0$  for **every** subspace  $U$  of  $F[x]$ . So the map  $U \mapsto U^0$  of Exercise (C1: Subspaces of  $V$  and of  $V^*$ ) is not surjective for  $F[x]$ .
- Which step of the solution of Exercise (C1: Subspaces of  $V$  and of  $V^*$ ) fails here?

*Solution.*

- If  $\varphi(p) = 0$  for all  $\varphi \in X$ , then in particular  $\varphi_k(p) = 0$  for every  $k$ , so every coefficient of  $p$  is 0 and  $p = 0$ . Conversely  $\varphi(0) = 0$  for every  $\varphi$ .

- (b) Suppose  $X = U^0$ . Every  $\varphi \in X = U^0$  vanishes on  $U$ , so  $U \subseteq \{p : \varphi(p) = 0 \ \forall \varphi \in X\} = \{0\}$  by (a). Hence  $U = \{0\}$  and  $U^0 = V^*$  (every functional vanishes at 0). But  $\varepsilon_1 \in V^*$  and  $\varepsilon_1 \notin X$ , by [Proposition 4.1.6](#). So  $X \neq V^* = U^0$ , a contradiction.
- (c) The dimension count. In [Exercise \(C1: Subspaces of  \$V\$  and of  \$V^\*\$ \)](#) we used that  $\text{ev}_V$  is onto and that  $\dim U + \dim U^0 = \dim V$  to force  $X = U^0$  from  $X \subseteq U^0$ . Here  $U = \{0\}$  and  $X \subsetneq U^0 = V^*$ : the inclusion holds, but there is no finite dimension to compare, and a proper subspace of an infinite-dimensional space can be “as large as” the whole space.

## 4.4 The Dual Map and the Transpose

So far each space has had its own dual, but the duals of different spaces have not talked to each other. Linear algebra is about maps, so the question is: what does a linear map  $T: V \rightarrow W$  do to functionals? This section shows that  $T$  moves vectors forward but moves measurements **backward**, giving the dual map  $T': W^* \rightarrow V^*$ . In dual bases its matrix is the transpose of the matrix of  $T$ , which finally explains what the transpose of Chapter 0 means. The kernel and image of  $T'$  are annihilators of the image and kernel of  $T$ , and counting dimensions gives a third proof that row rank equals column rank. At the end we return to the double dual and prove that the evaluation map is compatible with every linear map.

### 4.4.1 Pulling functionals back

After spaces come the maps between them. We have a linear map  $T: V \rightarrow W$  and the two spaces of measurements  $V^*$  and  $W^*$ . We want to know how  $T$  relates them.

Start with the standard example, a matrix map  $T_{\mathbf{A}}: F^n \rightarrow F^m$ ,  $\mathbf{x} \mapsto \mathbf{A}\mathbf{x}$ , with  $\mathbf{A} \in M_{m \times n}(F)$ . A functional on  $F^m$  is  $\psi(\mathbf{y}) = \mathbf{b}^\top \mathbf{y}$  for a unique  $\mathbf{b} \in F^m$  ([Theorem 4.1.1](#)). Given  $\psi$ , there is one obvious way to measure a vector  $\mathbf{x} \in F^n$ : move it to  $F^m$  with  $T_{\mathbf{A}}$ , then measure. The result is

$$\psi(T_{\mathbf{A}}\mathbf{x}) = \mathbf{b}^\top \mathbf{A}\mathbf{x} = (\mathbf{A}^\top \mathbf{b})^\top \mathbf{x},$$

using  $(\mathbf{A}^\top \mathbf{b})^\top = \mathbf{b}^\top \mathbf{A}$  ([Theorem 0.9.5](#)). So “move, then measure” is again a functional, now on  $F^n$ , and it is given by the vector  $\mathbf{A}^\top \mathbf{b}$ . The transpose appears by itself. The recipe “compose with  $T$ ” makes sense for any linear map, and it deserves a name.

*The dual map pulls a measurement of  $W$  back to a measurement of  $V$ : first move the vector with  $T$ , then measure it in  $W$ .*

#### Definition 4.4.1 — Dual Map

Let  $V$  and  $W$  be vector spaces over  $F$ , and let  $T \in \mathcal{L}(V, W)$ . The **dual map** of  $T$  is the map

$$T': W^* \rightarrow V^*, \quad T'(\psi) := \psi T = \psi \circ T.$$

That is,  $(T'\psi)(\mathbf{v}) = \psi(T\mathbf{v})$  for every  $\psi \in W^*$  and every  $\mathbf{v} \in V$ .

In words, clause by clause:

- The input of  $T'$  is a functional on the **codomain**  $W$ , and the output is a functional on the **domain**  $V$ . The arrow is reversed.
- The formula  $(T'\psi)(\mathbf{v}) = \psi(T\mathbf{v})$  has two steps: to measure  $\mathbf{v}$  with  $T'\psi$ , send  $\mathbf{v}$  to  $W$  with  $T$ , then apply  $\psi$  there.
- No basis and no finite dimension are involved.

**Well-definedness.** Two things must be checked. First,  $T'\psi$  must lie in  $V^*$ : it is the composite of the linear maps  $T: V \rightarrow W$  and  $\psi: W \rightarrow F$ , so it is linear by [Theorem 3.4.2](#) (a). Second,  $T'$  should be linear, so that  $T' \in \mathcal{L}(W^*, V^*)$ . By the distributive laws in [Theorem 3.4.2](#) (d), for  $\psi_1, \psi_2 \in W^*$  and  $c \in F$ ,

$$\begin{aligned} T'(\psi_1 + \psi_2) &= (\psi_1 + \psi_2)T = \psi_1 T + \psi_2 T = T'\psi_1 + T'\psi_2, \\ T'(c\psi) &= (c\psi)T = c(\psi T) = cT'\psi. \end{aligned}$$

### Example. (Dual Maps in Four Settings)

- For  $\mathbf{A} \in M_{m \times n}(F)$ , describe  $T'_{\mathbf{A}}$  in terms of the vectors  $\mathbf{b}$  and  $\mathbf{A}^{\top} \mathbf{b}$  that represent functionals.
- Let  $D: \mathbb{R}[x]_{\leq 3} \rightarrow \mathbb{R}[x]_{\leq 2}$ ,  $D(p) = p'$ , and let  $\varepsilon_2 \in (\mathbb{R}[x]_{\leq 2})^*$  be  $\varepsilon_2(q) = q(2)$ . Compute  $D'(\varepsilon_2)$ .
- Let  $S: F[x] \rightarrow F[x]$ ,  $S(p) = xp$ , and let  $\varphi_k \in F[x]^*$  send a polynomial to its coefficient of  $x^k$ . Compute  $S'(\varphi_k)$  for every  $k \in \mathbb{N}$ .
- Find  $(\text{id}_V)'$ , the dual map of the zero map  $V \rightarrow W$ , and the dual map of the inclusion  $\iota: U \rightarrow V$ ,  $\iota(\mathbf{u}) = \mathbf{u}$ , of a subspace  $U \subseteq V$ .

*Solution.*

- If  $\psi(\mathbf{y}) = \mathbf{b}^{\top} \mathbf{y}$ , the computation before the definition shows  $(T'_{\mathbf{A}}\psi)(\mathbf{x}) = (\mathbf{A}^{\top} \mathbf{b})^{\top} \mathbf{x}$ . So, identifying a functional with its vector,  $T'_{\mathbf{A}}$  is  $\mathbf{b} \mapsto \mathbf{A}^{\top} \mathbf{b}$ . In row language:  $T'_{\mathbf{A}}$  sends the row  $\mathbf{b}^{\top}$  to the row  $\mathbf{b}^{\top} \mathbf{A}$ .
- For  $p \in \mathbb{R}[x]_{\leq 3}$ ,  $(D'\varepsilon_2)(p) = \varepsilon_2(Dp) = p'(2)$ . So  $D'$  turns “the value at 2” into “the slope at 2”.
- $(S'\varphi_k)(p) = \varphi_k(xp)$ . Multiplying by  $x$  shifts every coefficient up one place, so the coefficient of  $x^k$  in  $xp$  is the coefficient of  $x^{k-1}$  in  $p$  when  $k \geq 1$ , and 0 when  $k = 0$ . Hence  $S'\varphi_k = \varphi_{k-1}$  for  $k \geq 1$ , and  $S'\varphi_0 = 0$ . Here  $F[x]$  is infinite-dimensional, and the definition works all the same.
- $(\text{id}_V)'\varphi = \varphi \circ \text{id}_V = \varphi$ , so  $(\text{id}_V)' = \text{id}_{V^*}$ . For the zero map  $0: V \rightarrow W$ ,  $(0'\psi)(\mathbf{v}) = \psi(0) = 0$  ([Theorem 3.1.1](#)), so  $0'$  is the zero map  $W^* \rightarrow V^*$ . For the inclusion,  $(\iota'\varphi)(\mathbf{u}) = \varphi(\mathbf{u})$  for  $\mathbf{u} \in U$ , so  $\iota'\varphi = \varphi|_U$  is the **restriction** of  $\varphi$  to  $U$ . The restriction map  $V^* \rightarrow U^*$  is thus a dual map; we study it further in the next section.

Here is a non-example by minimal change. Try to push functionals **forward** instead: for  $T: V \rightarrow W$  and  $\varphi \in V^*$ , look for a  $\psi \in W^*$  with  $\psi T = \varphi$ , and call it “ $T_*\varphi$ ”. Take  $T: \mathbb{R} \rightarrow \mathbb{R}^2$ ,  $T(t) = (t, 0)$ , and  $\varphi(t) = t$ . Then  $\psi(x, y) = x + cy$  satisfies  $\psi(T(t)) = t$  for **every**  $c \in \mathbb{R}$ , so there is no single answer. And for the zero map  $\mathbb{R} \rightarrow \mathbb{R}^2$ , no  $\psi$  at all satisfies  $\psi \circ 0 = \varphi$ . What fails is well-definedness: a function must assign **exactly one** value. Composition with  $T$  goes only one way, on the right, and that forces the direction  $W^* \rightarrow V^*$ .

**Why this definition.** From  $T: V \rightarrow W$  and  $\psi: W \rightarrow F$ , the only composite we can form is

$\psi T$ ; the non-example shows that the other direction is not a function. Linearity of  $T'$  comes for free from the distributive laws. The name “dual map” says that  $T'$  is the map  $T$  induces between the duals. It is also called the **transpose** of  $T$ , for a reason that the matrix theorem below makes exact. We write  $T'$  rather than  $T^*$ , because  $T^*$  is reserved for the adjoint of Chapter 10, which is related to  $T'$  but needs an inner product.

**Warning. The dual map goes backwards.** For  $T: V \rightarrow W$ , the dual map is  $T': W^* \rightarrow V^*$ , **not**  $V^* \rightarrow W^*$ . As a result, composition reverses order: if  $T: U \rightarrow V$  and  $S: V \rightarrow W$ , then  $(ST)' = T'S'$ , just as  $(\mathbf{AB})^\top = \mathbf{B}^\top \mathbf{A}^\top$ . The other order  $S'T'$  does not even make sense:  $T'$  lands in  $U^*$ , where  $S'$  is not defined.

*Quick check.* Let  $T: \mathbb{R}^3 \rightarrow \mathbb{R}^2$ ,  $T(x, y, z) = (x - z, 2y)$ , and  $\psi(u, v) = u + v$ . Compute  $T'\psi$ . If  $T'$  is written as a matrix in some bases, what is its size?

*Solution.*  $(T'\psi)(x, y, z) = \psi(x - z, 2y) = x + 2y - z$ . The map  $T'$  goes from  $(\mathbb{R}^2)^*$ , of dimension 2, to  $(\mathbb{R}^3)^*$ , of dimension 3 (Corollary 4.1.4), so its matrix is  $3 \times 2$ , the size of the transpose of the  $2 \times 3$  matrix of  $T$ .

The first payoff is that duals behave well under the operations on maps, with one reversal.

### Theorem 4.4.1 — Properties of the Dual Map

Let  $U, V, W$  be vector spaces over  $F$ .

- (a) For  $S, T \in \mathcal{L}(V, W)$  and  $a, b \in F$ ,  $(aS + bT)' = aS' + bT'$ .
- (b) For  $T \in \mathcal{L}(U, V)$  and  $S \in \mathcal{L}(V, W)$ ,  $(ST)' = T'S'$ .
- (c)  $(\text{id}_V)' = \text{id}_{V^*}$ .
- (d) If  $T \in \mathcal{L}(V, W)$  is invertible, then  $T'$  is invertible and  $(T')^{-1} = (T^{-1})'$ .

*Idea.* Each identity is an equality of maps into a dual space, so we evaluate both sides at a functional, and then evaluate the resulting functionals at a vector. After that, each line is the definition of  $T'$  read twice. Part (d) needs no computation: apply (b) and (c) to  $T^{-1}T = \text{id}_V$  and  $TT^{-1} = \text{id}_W$ .

*Proof.*

- (a) Let  $\psi \in W^*$  and  $\mathbf{v} \in V$ . Using the linearity of  $\psi$  and the pointwise operations (Definition 3.4.1),

$$((aS + bT)'\psi)(\mathbf{v}) = \psi(aS\mathbf{v} + bT\mathbf{v}) = a\psi(S\mathbf{v}) + b\psi(T\mathbf{v}) = (aS'\psi + bT'\psi)(\mathbf{v}).$$

Hence  $(aS + bT)'\psi = (aS' + bT')\psi$  for every  $\psi$ , and  $(aS + bT)' = aS' + bT'$ .

- (b) Let  $\psi \in W^*$ . By Definition 4.4.1 and associativity of composition (Theorem 3.4.2 (b)),

$$(ST)'\psi = \psi(ST) = (\psi S)T = T'(\psi S) = T'(S'\psi) = (T'S')\psi.$$

Hence  $(ST)' = T'S'$ .

- (c) This is Example (Dual Maps in Four Settings) (d).

(d) Suppose  $T$  is invertible with inverse  $T^{-1} \in \mathcal{L}(W, V)$ . By (b) and (c),

$$T'(T^{-1})' = (T^{-1}T)' = (\text{id}_V)' = \text{id}_{V^*}, \quad (T^{-1})'T' = (TT^{-1})' = (\text{id}_W)' = \text{id}_{W^*}.$$

The map  $(T^{-1})'$  is linear, so by [Definition 3.5.1](#)  $T'$  is invertible with inverse  $(T^{-1})'$ . This proves the theorem. ■

Part (a) says that  $T \mapsto T'$  is itself a linear map  $\mathcal{L}(V, W) \rightarrow \mathcal{L}(W^*, V^*)$ . Part (b) is the reversal of the warning. Parts (c) and (d) say that dualizing turns isomorphisms into isomorphisms, so isomorphic spaces have isomorphic duals.

## 4.4.2 The matrix of the dual map

Now we fix bases and compute. Let  $\mathcal{B} = (\mathbf{v}_1, \dots, \mathbf{v}_n)$  be a basis of  $V$  with dual basis  $\mathcal{B}^* = (\varphi_1, \dots, \varphi_n)$  ([Definition 4.1.3](#)), and  $\mathcal{C} = (\mathbf{w}_1, \dots, \mathbf{w}_m)$  a basis of  $W$  with dual basis  $\mathcal{C}^* = (\psi_1, \dots, \psi_m)$ . The map  $T$  has the  $m \times n$  matrix  $[T]_{\mathcal{B}}^{\mathcal{C}}$  ([Definition 3.6.1](#)). The dual map  $T': W^* \rightarrow V^*$  has input basis  $\mathcal{C}^*$  and output basis  $\mathcal{B}^*$ , so its matrix is  $[T']_{\mathcal{C}^*}^{\mathcal{B}^*}$ , of size  $n \times m$ . The size already suggests the answer, and [Example \(Dual Maps in Four Settings\)](#) (a) suggests it for matrix maps.

### Theorem 4.4.2 — The Matrix of the Dual Map Is the Transpose

Let  $V$  and  $W$  be finite-dimensional vector spaces over  $F$ , with ordered bases  $\mathcal{B}$  of  $V$  and  $\mathcal{C}$  of  $W$ , and let  $\mathcal{B}^*$  and  $\mathcal{C}^*$  be their dual bases. For every  $T \in \mathcal{L}(V, W)$ ,

$$[T']_{\mathcal{C}^*}^{\mathcal{B}^*} = ([T]_{\mathcal{B}}^{\mathcal{C}})^{\top}.$$

*Idea.* Both matrices are tables of the same numbers  $\psi_i(T\mathbf{v}_j)$ . In  $[T]_{\mathcal{B}}^{\mathcal{C}}$ , the  $(i, j)$  entry is the  $\mathbf{w}_i$ -coordinate of  $T\mathbf{v}_j$ , and a dual basis vector **reads off** a coordinate: that coordinate is  $\psi_i(T\mathbf{v}_j)$ . In  $[T']_{\mathcal{C}^*}^{\mathcal{B}^*}$ , column  $i$  holds the  $\mathcal{B}^*$ -coordinates of  $T'\psi_i$ , and the  $\varphi_j$ -coordinate of a functional is its value at  $\mathbf{v}_j$ : that is  $(T'\psi_i)(\mathbf{v}_j) = \psi_i(T\mathbf{v}_j)$ . The same number sits at  $(i, j)$  in one matrix and at  $(j, i)$  in the other.

*Proof.* Write  $\mathcal{B} = (\mathbf{v}_1, \dots, \mathbf{v}_n)$ ,  $\mathcal{B}^* = (\varphi_1, \dots, \varphi_n)$ ,  $\mathcal{C} = (\mathbf{w}_1, \dots, \mathbf{w}_m)$ ,  $\mathcal{C}^* = (\psi_1, \dots, \psi_m)$ , and  $\mathbf{A} = (a_{kj}) = [T]_{\mathcal{B}}^{\mathcal{C}}$ . By [Definition 3.6.1](#),  $a_{ij}$  is the  $i$ -th entry of  $[T\mathbf{v}_j]_{\mathcal{C}}$ , and by [Theorem 4.1.3](#) (a) that entry is read off by  $\psi_i$ :

$$\psi_i(T\mathbf{v}_j) = a_{ij}. \quad (*)$$

Next, by [Theorem 4.1.3](#) (c), every  $\varphi \in V^*$  satisfies  $\varphi = \sum_{j=1}^n \varphi(\mathbf{v}_j)\varphi_j$ . Since  $\mathcal{B}^*$  is a basis, coordinates in it are unique ([Theorem 1.5.1](#)), so  $[\varphi]_{\mathcal{B}^*} = (\varphi(\mathbf{v}_1), \dots, \varphi(\mathbf{v}_n))$ .

By [Definition 3.6.1](#), column  $i$  of  $[T']_{\mathcal{C}^*}^{\mathcal{B}^*}$  is  $[T'\psi_i]_{\mathcal{B}^*}$ . By the previous paragraph and (\*), its  $j$ -th entry is

$$(T'\psi_i)(\mathbf{v}_j) = \psi_i(T\mathbf{v}_j) = a_{ij}.$$

So the  $(j, i)$  entry of  $[T']_{\mathcal{C}^*}^{\mathcal{B}^*}$  is  $a_{ij}$ , which is the  $(j, i)$  entry of  $\mathbf{A}^{\top}$  ([Definition 0.9.10](#)). Both matrices are  $n \times m$ , so  $[T']_{\mathcal{C}^*}^{\mathcal{B}^*} = \mathbf{A}^{\top}$ . This proves the theorem.



This is the meaning of the transpose. In Chapter 0, [Definition 0.9.10](#) was a rule for rearranging entries. Now we see it as the matrix shadow of pulling functionals back: **transposing a matrix is dualizing the map**, provided we use dual bases on both sides. Every property of the transpose becomes a property of dual maps, and conversely; for instance, [Theorem 4.4.1](#) (b) and this theorem together give  $(\mathbf{A}\mathbf{B})^\top = \mathbf{B}^\top \mathbf{A}^\top$  a second time (Exercise B3 below).

**Example. (A  $2 \times 3$  Example)**

Let  $T: \mathbb{R}[x]_{\leq 2} \rightarrow \mathbb{R}^2$ ,  $T(p) = (p(1), p'(1))$ . Use the basis  $\mathcal{B} = (1, x, x^2)$ , whose dual basis  $\mathcal{B}^* = (\varphi_0, \varphi_1, \varphi_2)$  consists of  $\varphi_0(p) = p(0)$ ,  $\varphi_1(p) = p'(0)$ ,  $\varphi_2(p) = p''(0)/2$  ([Example \(Dual Bases in  \$\mathbb{R}\[x\]\_{\leq 2}\$ \)](#)), and the standard basis  $\mathcal{C} = (\mathbf{e}_1, \mathbf{e}_2)$  of  $\mathbb{R}^2$ , with dual basis  $\mathcal{C}^* = (\eta_1, \eta_2)$ ,  $\eta_1(u, v) = u$ ,  $\eta_2(u, v) = v$ .

- Find  $\mathbf{A} = [T]_{\mathcal{B}}^{\mathcal{C}}$ .
- Compute  $T'\eta_1$  and  $T'\eta_2$  directly as combinations of  $\varphi_0, \varphi_1, \varphi_2$ , write down  $[T']_{\mathcal{C}^*}^{\mathcal{B}^*}$ , and compare with  $\mathbf{A}^\top$ .
- Use the matrix to compute  $T'\psi$  for  $\psi = -2\eta_1 + \eta_2$ , and check the answer directly.

*Solution.*

- $T(1) = (1, 0)$ ,  $T(x) = (1, 1)$ ,  $T(x^2) = (1, 2)$ . These are the columns:

$$\mathbf{A} = [T]_{\mathcal{B}}^{\mathcal{C}} = \begin{pmatrix} 1 & 1 & 1 \\ 0 & 1 & 2 \end{pmatrix}.$$

- Let  $p = a + bx + cx^2$ , so  $\varphi_0(p) = a$ ,  $\varphi_1(p) = b$ ,  $\varphi_2(p) = c$ . Then

$$\begin{aligned} (T'\eta_1)(p) &= \eta_1(Tp) = p(1) = a + b + c, \\ (T'\eta_2)(p) &= \eta_2(Tp) = p'(1) = b + 2c. \end{aligned}$$

So  $T'\eta_1 = \varphi_0 + \varphi_1 + \varphi_2$  and  $T'\eta_2 = \varphi_1 + 2\varphi_2$ . Their coordinate vectors in  $\mathcal{B}^*$  are the columns of

$$[T']_{\mathcal{C}^*}^{\mathcal{B}^*} = \begin{pmatrix} 1 & 0 \\ 1 & 1 \\ 1 & 2 \end{pmatrix} = \mathbf{A}^\top,$$

as [Theorem 4.4.2](#) predicts. Read in words, the first column is Taylor's formula  $p(1) = p(0) + p'(0) + \frac{1}{2}p''(0)$  for quadratics, and the second is  $p'(1) = p'(0) + p''(0)$ .

- $[\psi]_{\mathcal{C}^*} = (-2, 1)$ . By [Theorem 3.6.1](#),

$$[T'\psi]_{\mathcal{B}^*} = \mathbf{A}^\top \begin{pmatrix} -2 \\ 1 \end{pmatrix} = \begin{pmatrix} -2 \\ -1 \\ 0 \end{pmatrix},$$

so  $T'\psi = -2\varphi_0 - \varphi_1$ . Directly:  $(T'\psi)(p) = -2p(1) + p'(1) = -2(a+b+c) + (b+2c) = -2a - b$ , which is  $(-2\varphi_0 - \varphi_1)(p)$ .

**Warning. Dual bases on both sides.** The transpose rule needs the **dual** bases  $\mathcal{C}^*$  and  $\mathcal{B}^*$ . With other bases of  $W^*$  and  $V^*$  the matrix of  $T'$  is usually not  $\mathbf{A}^\top$ . In [Example \(A  \$2 \times 3\$  Example\)](#), keep  $\mathcal{B}^*$  but use the basis  $(\eta_1 + \eta_2, \eta_2)$  of  $(\mathbb{R}^2)^*$ : then

$T'(\eta_1 + \eta_2) = \varphi_0 + 2\varphi_1 + 3\varphi_2$ , so the first column is  $(1, 2, 3)$ , not the first column  $(1, 1, 1)$  of  $\mathbf{A}^\top$ .

### 4.4.3 Kernel and image of the dual map

The kernel and image of  $T'$  are subspaces of  $W^*$  and  $V^*$ . The natural candidates for them are annihilators of the subspaces that  $T$  defines, and the dictionary swaps kernel and image.

#### Theorem 4.4.3 — Kernel and Image of the Dual Map

Let  $V$  and  $W$  be vector spaces over  $F$ , and let  $T \in \mathcal{L}(V, W)$ .

- (a)  $\ker T' = (\operatorname{im} T)^0$ .
- (b) If  $V$  and  $W$  are finite-dimensional, then  $\operatorname{im} T' = (\ker T)^0$ .

*Idea.*

- (a) is a chain of definitions:  $\psi T = 0$  says that  $\psi$  vanishes on every  $T\mathbf{v}$ , that is, on  $\operatorname{im} T$ . For (b), one inclusion is just as easy:  $\psi T$  kills every vector that  $T$  kills. The reverse inclusion asks us to **extend** a functional given on  $V$  to one of the form  $\psi T$ , which is a construction. Instead we count. Rank–Nullity for  $T'$ , together with (a) and the dimension formula for annihilators, gives  $\dim \operatorname{im} T' = \operatorname{rank} T$ ; the dimension formula in  $V$  and Rank–Nullity for  $T$  give  $\dim(\ker T)^0 = \operatorname{rank} T$ . Finite dimension is spent exactly on these counts.

*Proof.*

- (a) Let  $\psi \in W^*$ . Then  $\psi \in \ker T'$  iff  $\psi T = 0$  iff  $\psi(T\mathbf{v}) = 0$  for every  $\mathbf{v} \in V$  iff  $\psi(\mathbf{w}) = 0$  for every  $\mathbf{w} \in \operatorname{im} T$  (Definition 3.2.2) iff  $\psi \in (\operatorname{im} T)^0$  (Definition 4.2.1).
- (b) Let  $n = \dim V$  and  $m = \dim W$ , so  $\dim V^* = n$  and  $\dim W^* = m$  (Corollary 4.1.4).

*Inclusion.* Let  $\varphi \in \operatorname{im} T'$ , say  $\varphi = \psi T$  with  $\psi \in W^*$ . For  $\mathbf{u} \in \ker T$ ,  $\varphi(\mathbf{u}) = \psi(T\mathbf{u}) = \psi(\mathbf{0}) = 0$ . Hence  $\varphi \in (\ker T)^0$ , and  $\operatorname{im} T' \subseteq (\ker T)^0$ .

*Dimensions.* By the Rank–Nullity Theorem (Theorem 3.3.1) for  $T': W^* \rightarrow V^*$ , then (a), then Theorem 4.2.2 for the subspace  $\operatorname{im} T$  of  $W$ ,

$$\begin{aligned} \dim \operatorname{im} T' &= m - \dim \ker T' = m - \dim(\operatorname{im} T)^0 \\ &= m - (m - \dim \operatorname{im} T) = \operatorname{rank} T. \end{aligned}$$

By Theorem 4.2.2 for the subspace  $\ker T$  of  $V$ , and Rank–Nullity for  $T$ ,

$$\dim(\ker T)^0 = n - \dim \ker T = \operatorname{rank} T.$$

So  $\operatorname{im} T'$  is a subspace of the finite-dimensional space  $(\ker T)^0$  with the same dimension, and Theorem 1.6.8 gives  $\operatorname{im} T' = (\ker T)^0$ . This proves the theorem. ■

For a matrix map, this is a statement about rows. By Example (Dual Maps in Four Settings)



(a),  $T'_\mathbf{A}$  sends the row  $\mathbf{b}^\top$  to  $\mathbf{b}^\top \mathbf{A}$ , so  $\text{im } T'_\mathbf{A}$  is the set of combinations of the rows of  $\mathbf{A}$ : it is the row space, read as functionals. Part (b) then says that **the functionals vanishing on  $\text{nul}(\mathbf{A})$  are exactly the combinations of the rows of  $\mathbf{A}$** . Part (a) says that the functionals vanishing on  $\text{col}(\mathbf{A})$  are the rows  $\mathbf{b}^\top$  with  $\mathbf{b}^\top \mathbf{A} = 0$ , that is,  $\mathbf{A}^\top \mathbf{b} = \mathbf{0}$ .

**Example. (Kernel and Image of a Dual Map)**

Let  $T = T_\mathbf{A}: \mathbb{R}^3 \rightarrow \mathbb{R}^3$  with

$$\mathbf{A} = \begin{pmatrix} 1 & 0 & 2 \\ 0 & 1 & -1 \\ 1 & 1 & 1 \end{pmatrix}.$$

Find bases of  $\ker T'$  and  $\text{im } T'$ , written as explicit functionals, and check [Theorem 4.4.3](#).

*Solution. The matrix world.* The third row of  $\mathbf{A}$  is the sum of the first two, and subtracting them leaves the RREF  $\begin{pmatrix} 1 & 0 & 2 \\ 0 & 1 & -1 \\ 0 & 0 & 0 \end{pmatrix}$ . So  $\text{rank } \mathbf{A} = 2$ , and  $\mathbf{A}\mathbf{x} = \mathbf{0}$  gives  $x = -2z$ ,  $y = z$ :  $\ker T = \text{span}((-2, 1, 1))$ . The pivot columns give  $\text{im } T = \text{span}((1, 0, 1), (0, 1, 1))$  ([Theorem 2.5.2](#)).

*Kernel of  $T'$ .* By [Theorem 4.4.2](#) with standard bases and [Example \(Matrix Maps and the Identity\)](#) (a), a functional  $\psi(u, v, w) = b_1u + b_2v + b_3w$  lies in  $\ker T'$  iff  $\mathbf{A}^\top \mathbf{b} = \mathbf{0}$ :

$$b_1 + b_3 = 0, \quad b_2 + b_3 = 0, \quad 2b_1 - b_2 + b_3 = 0.$$

The first two give  $\mathbf{b} = t(1, 1, -1)$ , and these satisfy the third. So  $\ker T' = \text{span}(\psi_0)$  with  $\psi_0(u, v, w) = u + v - w$ . *Check:*  $\psi_0(1, 0, 1) = 0$  and  $\psi_0(0, 1, 1) = 0$ , so  $\psi_0$  vanishes on  $\text{im } T$ ; and  $\dim(\text{im } T)^0 = 3 - 2 = 1$  ([Theorem 4.2.2](#)). So  $\ker T' = (\text{im } T)^0$ .

*Image of  $T'$ .* By [Example \(Dual Maps in Four Settings\)](#) (a),  $\text{im } T'$  is spanned by the rows of  $\mathbf{A}$  read as functionals, and the first two rows already span (the third is their sum):

$$\varphi_1(x, y, z) = x + 2z, \quad \varphi_2(x, y, z) = y - z.$$

They are independent (neither is a multiple of the other), so  $(\varphi_1, \varphi_2)$  is a basis of  $\text{im } T'$ . *Check:*  $\varphi_1(-2, 1, 1) = 0$  and  $\varphi_2(-2, 1, 1) = 0$ , so both vanish on  $\ker T$ , and  $\dim(\ker T)^0 = 3 - 1 = 2$ . So  $\text{im } T' = (\ker T)^0$ .

Counting dimensions in the proof of [Theorem 4.4.3](#) already gave  $\dim \text{im } T' = \text{rank } T$ . We record it, together with its most famous consequence.

**Corollary 4.4.4 — Rank of the Dual Map** (a) Let  $V$  and  $W$  be finite-dimensional vector spaces over  $F$ , and  $T \in \mathcal{L}(V, W)$ . Then  $\text{rank } T' = \text{rank } T$ .  
 (b) For every  $\mathbf{A} \in M_{m \times n}(F)$ ,  $\text{rank } \mathbf{A}^\top = \text{rank } \mathbf{A}$ . Equivalently,  $\dim \text{row}(\mathbf{A}) = \dim \text{col}(\mathbf{A})$ : **row rank equals column rank**.

*Proof.*

(a) This is the first display in the proof of [Theorem 4.4.3](#) (b).

(b) Let  $\mathcal{E}_n, \mathcal{E}_m$  be the standard bases. By [Example \(Matrix Maps and the Identity\)](#)

(a),  $[T_{\mathbf{A}}]_{\mathcal{E}_n^m}^{\mathcal{E}_m^n} = \mathbf{A}$ , so by [Theorem 4.4.2](#),  $[T'_{\mathbf{A}}]_{\mathcal{E}_m^n}^{\mathcal{E}_n^m} = \mathbf{A}^\top$ . By [Theorem 3.6.5](#) (a), applied to  $T_{\mathbf{A}}$  and to  $T'_{\mathbf{A}}$ , and by (a),

$$\text{rank } \mathbf{A} = \text{rank } T_{\mathbf{A}} = \text{rank } T'_{\mathbf{A}} = \text{rank } \mathbf{A}^\top.$$

Finally, the transpose  $M_{1 \times n}(F) \rightarrow F^n$  is linear ([Theorem 0.9.5](#)) and bijective (it is its own inverse), and it sends row  $i$  of  $\mathbf{A}$  to column  $i$  of  $\mathbf{A}^\top$ . By [Theorem 3.1.4](#) it maps  $\text{row}(\mathbf{A})$  onto  $\text{col}(\mathbf{A}^\top)$ , so these spaces are isomorphic and  $\dim \text{row}(\mathbf{A}) = \dim \text{col}(\mathbf{A}^\top) = \text{rank } \mathbf{A}^\top = \text{rank } \mathbf{A} = \dim \text{col}(\mathbf{A})$  ([Theorem 3.5.4](#), [Definition 2.5.4](#)). ■

This is the third proof of this fact in the book, and each one explains it differently.

- **Chapter 2** ([Theorem 2.5.5](#)) counted pivots. Row reduction preserves the row space and the relations among the columns, and in the RREF both dimensions visibly equal the number of pivots. The proof is an algorithm.
- **Chapter 3, §8** factored  $\mathbf{A} = \mathbf{B}\mathbf{D}$  through a space of dimension  $\dim \text{col}(\mathbf{A})$ , after [Proposition 3.8.2](#), and saw that the rows of  $\mathbf{A}$  are combinations of the few rows of  $\mathbf{D}$ . The proof is a factorization.
- **Here** there are no row operations and no factorizations. The equality is the dimension formula  $\dim U + \dim U^0 = \dim V$ , applied once to  $\text{im } T$  in  $W$  and once to  $\ker T$  in  $V$ , glued together by Rank–Nullity. The rows of  $\mathbf{A}$  are the functionals that  $T'_{\mathbf{A}}$  produces, and the columns span the image of  $T_{\mathbf{A}}$ ; the two counts agree because pulling back measurements loses exactly as much as pushing forward vectors.

(The dimension formula for annihilators in section 2 was proved by extending a basis of  $U$ , without using ranks of matrices, so this proof does not go in a circle.)

The dictionary also swaps injective and surjective.

### Corollary 4.4.5 — Injective and Surjective Swap Under Duality

Let  $V$  and  $W$  be finite-dimensional vector spaces over  $F$ , and  $T \in \mathcal{L}(V, W)$ . Then:

- (a)  $T$  is surjective if and only if  $T'$  is injective;
- (b)  $T$  is injective if and only if  $T'$  is surjective.

*Proof.* Let  $n = \dim V$  and  $m = \dim W$ . A finite-dimensional space is  $\{0\}$  exactly when its dimension is 0.

- (a) By [Theorem 3.2.3](#) and [Theorem 4.4.3](#) (a),  $T'$  is injective iff  $(\text{im } T)^0 = \{0\}$ . By [Theorem 4.2.2](#),  $\dim(\text{im } T)^0 = m - \text{rank } T$ , so this holds iff  $\text{rank } T = m$ , that is, iff  $T$  is surjective ([Corollary 3.3.2](#) (b)).
- (b)  $T'$  is surjective iff  $\text{im } T' = V^*$ , iff  $\dim \text{im } T' = n$  (by [Theorem 1.6.8](#) for one direction, since  $\dim V^* = n$ ). By [Corollary 4.4.4](#) (a),  $\dim \text{im } T' = \text{rank } T$ , so this holds iff  $\text{rank } T = n$ , that is, iff  $T$  is injective ([Corollary 3.3.2](#) (a)). ■

For  $T_{\mathbf{A}}$ , part (a) reads:  $\mathbf{A}\mathbf{x} = \mathbf{b}$  is solvable for every  $\mathbf{b}$  exactly when  $\mathbf{A}^\top \mathbf{y} = \mathbf{0}$  has only the

trivial solution. So a question about the columns of  $\mathbf{A}$  turns into a question about the columns of  $\mathbf{A}^\top$ , the rows of  $\mathbf{A}$ . Exercise C1 below sharpens this into a test for a single  $\mathbf{b}$ .

### 4.4.4 Naturality of the double dual

In the previous section we called  $\text{ev}_V: V \rightarrow V^{**}$  natural because it needs no choice, and promised a sharper sense: compatibility with every linear map. With dual maps available, we can state it. A linear map  $T: V \rightarrow W$  has a dual  $T': W^* \rightarrow V^*$ , and that has a dual in turn,

$$T'' := (T')': V^{**} \rightarrow W^{**},$$

which points **forward** again, because we reversed twice.

#### Theorem 4.4.6 — The Evaluation Map Is Natural

Let  $V$  and  $W$  be vector spaces over  $F$ , and let  $T \in \mathcal{L}(V, W)$ . Then

$$T'' \text{ ev}_V = \text{ev}_W T.$$

That is,  $T''(\text{ev}_{\mathbf{v}}) = \text{ev}_{T\mathbf{v}}$  for every  $\mathbf{v} \in V$ . If  $V$  and  $W$  are finite-dimensional, then  $T'' = \text{ev}_W T \text{ ev}_V^{-1}$ .

The statement says that the square below commutes, for **every** linear map  $T$ : going right then down (apply  $T$ , then evaluate) equals going down then right (evaluate, then apply  $T''$ ).

$$\begin{array}{ccc} V & \xrightarrow{T} & W \\ \text{ev}_V \downarrow & & \downarrow \text{ev}_W \\ V^{**} & \xrightarrow{T''} & W^{**} \end{array}$$

*Proof.* Let  $\mathbf{v} \in V$ . Both  $T''(\text{ev}_{\mathbf{v}})$  and  $\text{ev}_{T\mathbf{v}}$  are functionals on  $W^*$ , so let  $\psi \in W^*$ . By Definition 4.4.1 applied to  $T'$ , then to  $T$ ,

$$(T''(\text{ev}_{\mathbf{v}}))(\psi) = \text{ev}_{\mathbf{v}}(T'\psi) = (T'\psi)(\mathbf{v}) = \psi(T\mathbf{v}) = \text{ev}_{T\mathbf{v}}(\psi).$$

Hence  $T''(\text{ev}_{\mathbf{v}}) = \text{ev}_{T\mathbf{v}}$  for every  $\mathbf{v}$ , which is  $T'' \text{ ev}_V = \text{ev}_W T$ . If  $V$  is finite-dimensional,  $\text{ev}_V$  is an isomorphism (Theorem 4.3.2); composing on the right with  $\text{ev}_V^{-1}$  gives  $T'' = \text{ev}_W T \text{ ev}_V^{-1}$ . ■

So, once  $V$  is identified with  $V^{**}$  and  $W$  with  $W^{**}$  through evaluation, the double dual map  $T''$  is  $T$ . The matrices agree: by Theorem 4.3.2,  $\text{ev}$  carries  $\mathcal{B}$  and  $\mathcal{C}$  to the dual bases  $\mathcal{B}^{**}$  and  $\mathcal{C}^{**}$  of  $\mathcal{B}^*$  and  $\mathcal{C}^*$ , and two applications of Theorem 4.4.2 give  $[T'']_{\mathcal{B}^{**}}^{\mathcal{C}^{**}} = (([T]_{\mathcal{B}}^{\mathcal{C}})^\top)^\top = [T]_{\mathcal{B}}^{\mathcal{C}}$  (Theorem 0.9.5). The rule  $(\mathbf{A}^\top)^\top = \mathbf{A}$  is the matrix form of naturality.

*Remark.* No such statement holds for  $V \cong V^*$ . Suppose that for every finite-dimensional real space  $V$  we had an isomorphism  $\theta_V: V \rightarrow V^*$ , compatible with isomorphisms in the only way the directions allow:  $\theta_V = T' \theta_W T$  for every isomorphism  $T: V \rightarrow W$ . Take  $V = W = \mathbb{R}$  and  $T = 2 \text{ id}$ . Then  $T' = 2 \text{ id}$  by [Theorem 4.4.1](#) (a) and (c), so  $\theta_{\mathbb{R}} = 2 \cdot 2 \cdot \theta_{\mathbb{R}} = 4\theta_{\mathbb{R}}$ , which forces  $\theta_{\mathbb{R}} = 0$ , not an isomorphism. The dual space is isomorphic to  $V$ , but not naturally.

## 4.4.5 Exercises

### 4.4.5.1 A. Check your understanding

#### Exercise A1.

- Define the dual map  $T'$  of  $T \in \mathcal{L}(V, W)$ , including its domain and codomain.
- State the relation between  $[T']_{\mathcal{C}^*}^{\mathcal{B}^*}$  and  $[T]_{\mathcal{B}}^{\mathcal{C}}$ , with its hypotheses.
- True or false:  $(ST)' = S'T'$  whenever  $ST$  is defined. Justify your answer.
- True or false:  $\ker T' = (\ker T)^0$  for every linear map between finite-dimensional spaces. Justify your answer.
- True or false: for  $T \in \mathcal{L}(\mathbb{R}^4, \mathbb{R}^3)$ , the dual map  $T'$  is never surjective. Justify your answer.
- Name the two tools combined to prove  $\text{rank } T' = \text{rank } T$ .

*Solution.*

- For  $T: V \rightarrow W$ ,  $T': W^* \rightarrow V^*$ ,  $T'(\psi) = \psi T$  ([Definition 4.4.1](#)).
- If  $V, W$  are finite-dimensional with bases  $\mathcal{B}, \mathcal{C}$  and dual bases  $\mathcal{B}^*, \mathcal{C}^*$ , then  $[T']_{\mathcal{C}^*}^{\mathcal{B}^*} = ([T]_{\mathcal{B}}^{\mathcal{C}})^{\top}$  ([Theorem 4.4.2](#)).
- False. The correct rule is  $(ST)' = T'S'$  ([Theorem 4.4.1](#) (b)); for  $T: U \rightarrow V$ ,  $S: V \rightarrow W$  with  $U \neq W$ , the composite  $S'T'$  is not even defined.
- False.  $\ker T' = (\text{im } T)^0$ . For the zero map  $T: \mathbb{R} \rightarrow \mathbb{R}$ ,  $\ker T = \mathbb{R}$ , so  $(\ker T)^0 = \{0\}$ , while  $T' = 0$  has kernel  $\mathbb{R}^* \neq \{0\}$ .
- True.  $T'$  maps  $(\mathbb{R}^3)^*$ , of dimension 3, to  $(\mathbb{R}^4)^*$ , of dimension 4, so it cannot be surjective ([Corollary 3.3.2](#) (d)). This matches [Corollary 4.4.5](#):  $T$  cannot be injective.
- The Rank–Nullity Theorem and the dimension formula for annihilators,  $\dim U + \dim U^0 = \dim V$  (together with  $\ker T' = (\text{im } T)^0$ ).

### 4.4.5.2 B. Practice

#### Exercise B1: Differentiation in dual bases.

Let  $D: \mathbb{R}[x]_{\leq 2} \rightarrow \mathbb{R}[x]_{\leq 2}$ ,  $D(p) = p'$ , and let  $\mathcal{B} = (1, x, x^2)$  with dual basis  $\mathcal{B}^* = (\varphi_0, \varphi_1, \varphi_2)$ ,  $\varphi_k(p) =$  the coefficient of  $x^k$  in  $p$ . Compute  $D'\varphi_0$ ,  $D'\varphi_1$ ,  $D'\varphi_2$  directly, write down  $[D']_{\mathcal{B}^*}$ , and check that it is  $([D]_{\mathcal{B}})^{\top}$ . Hence describe  $\ker D'$  and  $\text{im } D'$ , and verify [Theorem 4.4.3](#).

*Solution.* Let  $p = a + bx + cx^2$ , so  $Dp = b + 2cx$ . Then  $(D'\varphi_0)(p) = \varphi_0(Dp) = b$ ,  $(D'\varphi_1)(p) = 2c$ ,  $(D'\varphi_2)(p) = 0$ . Hence

$$D'\varphi_0 = \varphi_1, \quad D'\varphi_1 = 2\varphi_2, \quad D'\varphi_2 = 0.$$

The coordinate vectors  $(0, 1, 0)$ ,  $(0, 0, 2)$ ,  $(0, 0, 0)$  are the columns of  $[D']_{\mathcal{B}^*}$ . On the other side,  $D(1) = 0$ ,  $D(x) = 1$ ,  $D(x^2) = 2x$ , so

$$[D]_{\mathcal{B}} = \begin{pmatrix} 0 & 1 & 0 \\ 0 & 0 & 2 \\ 0 & 0 & 0 \end{pmatrix}, \quad [D']_{\mathcal{B}^*} = \begin{pmatrix} 0 & 0 & 0 \\ 1 & 0 & 0 \\ 0 & 2 & 0 \end{pmatrix} = ([D]_{\mathcal{B}})^{\top},$$

as [Theorem 4.4.2](#) predicts.

From the formulas,  $D'(c_0\varphi_0 + c_1\varphi_1 + c_2\varphi_2) = c_0\varphi_1 + 2c_1\varphi_2$ , which is 0 iff  $c_0 = c_1 = 0$ . So  $\ker D' = \text{span}(\varphi_2)$  and  $\text{im } D' = \text{span}(\varphi_1, \varphi_2)$ . On the other hand  $\text{im } D = \mathbb{R}[x]_{\leq 1} = \text{span}(1, x)$ , and  $\varphi_2$  is exactly the functional (up to scalars) vanishing on it, since  $c_0\varphi_0 + c_1\varphi_1 + c_2\varphi_2$  kills 1 and  $x$  iff  $c_0 = c_1 = 0$ . So  $\ker D' = (\text{im } D)^0$ . Also  $\ker D = \text{span}(1)$ , and a functional kills 1 iff  $c_0 = 0$ , so  $(\ker D)^0 = \text{span}(\varphi_1, \varphi_2) = \text{im } D'$ .

### Exercise B2: Kernel and image of a dual map.

Let  $\mathbf{A} = \begin{pmatrix} 1 & 2 & -1 \\ -2 & -4 & 2 \end{pmatrix}$  and  $T = T_{\mathbf{A}}: \mathbb{R}^3 \rightarrow \mathbb{R}^2$ . Find bases of  $\ker T'$  and  $\text{im } T'$  as explicit functionals, and verify  $\dim \ker T' + \dim \text{im } T' = 2$  and  $\text{rank } T' = \text{rank } T$ .

*Solution.* The second row is  $-2$  times the first, so  $\text{rank } \mathbf{A} = 1$ ,  $\text{im } T = \text{col}(\mathbf{A}) = \text{span}((1, -2))$ , and  $\ker T = \{(x, y, z) : x + 2y - z = 0\}$ , of dimension 2.

*Kernel of  $T'$ .* As in [Example \(Kernel and Image of a Dual Map\)](#),  $\psi(u, v) = b_1u + b_2v$  lies in  $\ker T'$  iff  $\mathbf{A}^{\top} \mathbf{b} = \mathbf{0}$ , that is,  $b_1 - 2b_2 = 0$ ,  $2b_1 - 4b_2 = 0$ ,  $-b_1 + 2b_2 = 0$ . So  $\mathbf{b} = t(2, 1)$ , and  $\ker T' = \text{span}(\psi_0)$  with  $\psi_0(u, v) = 2u + v$ . Check:  $\psi_0(1, -2) = 0$ , so  $\psi_0 \in (\text{im } T)^0$ , which has dimension  $2 - 1 = 1$ .

*Image of  $T'$ .* By [Example \(Dual Maps in Four Settings\)](#) (a),  $\text{im } T'$  is spanned by the rows of  $\mathbf{A}$  read as functionals,  $x + 2y - z$  and  $-2x - 4y + 2z$ . The second is  $-2$  times the first, so  $\text{im } T' = \text{span}(\varphi_0)$  with  $\varphi_0(x, y, z) = x + 2y - z$ . Check:  $\varphi_0$  vanishes on  $\ker T$  by the description of  $\ker T$ , and  $\dim(\ker T)^0 = 3 - 2 = 1$ .

Hence  $\dim \ker T' + \dim \text{im } T' = 1 + 1 = 2 = \dim(\mathbb{R}^2)^*$ , and  $\text{rank } T' = 1 = \text{rank } T$ .

### Exercise B3: The transpose of a product, again.

Let  $\mathbf{A} \in M_{m \times n}(F)$  and  $\mathbf{B} \in M_{n \times p}(F)$ . Using  $T_{\mathbf{A}}T_{\mathbf{B}} = T_{\mathbf{AB}}$  (Chapter 3, §6), [Theorem 4.4.1](#) (b) and [Theorem 4.4.2](#), give a new proof that  $(\mathbf{AB})^{\top} = \mathbf{B}^{\top} \mathbf{A}^{\top}$ .

*Solution.* Let  $\mathcal{E}_p, \mathcal{E}_n, \mathcal{E}_m$  be the standard bases, with dual bases  $\mathcal{E}_p^*, \mathcal{E}_n^*, \mathcal{E}_m^*$ . By [Example \(Matrix Maps and the Identity\)](#) (a),  $[T_{\mathbf{A}}]_{\mathcal{E}_n^*}^{\mathcal{E}_m^*} = \mathbf{A}$ ,  $[T_{\mathbf{B}}]_{\mathcal{E}_p^*}^{\mathcal{E}_n^*} = \mathbf{B}$ , and  $[T_{\mathbf{AB}}]_{\mathcal{E}_p^*}^{\mathcal{E}_m^*} = \mathbf{AB}$ . Since  $T_{\mathbf{AB}} = T_{\mathbf{A}}T_{\mathbf{B}}$ , [Theorem 4.4.1](#) (b) gives  $T'_{\mathbf{AB}} = T'_{\mathbf{B}}T'_{\mathbf{A}}$ . Taking matrices, with

[Theorem 3.6.2](#) in the middle and [Theorem 4.4.2](#) on the outside,

$$(\mathbf{AB})^\top = [T'_{\mathbf{AB}}]_{\mathcal{E}_m^*}^{\mathcal{E}_p^*} = [T'_\mathbf{B} T'_\mathbf{A}]_{\mathcal{E}_m^*}^{\mathcal{E}_p^*} = [T'_\mathbf{B}]_{\mathcal{E}_n^*}^{\mathcal{E}_p^*} [T'_\mathbf{A}]_{\mathcal{E}_m^*}^{\mathcal{E}_n^*} = \mathbf{B}^\top \mathbf{A}^\top.$$

The reversal of order is now explained: it is the reversal of arrows under duality.

#### 4.4.5.3 C. Going deeper

##### Exercise C1: The Fredholm alternative.

- (a) Let  $V, W$  be finite-dimensional,  $T \in \mathcal{L}(V, W)$  and  $\mathbf{w} \in W$ . Prove that  $T\mathbf{v} = \mathbf{w}$  has a solution  $\mathbf{v} \in V$  if and only if  $\psi(\mathbf{w}) = 0$  for every  $\psi \in \ker T'$ .
- (b) Deduce: for  $\mathbf{A} \in M_{m \times n}(F)$  and  $\mathbf{b} \in F^m$ , the system  $\mathbf{Ax} = \mathbf{b}$  is consistent if and only if  $\mathbf{y}^\top \mathbf{b} = 0$  for every  $\mathbf{y} \in F^m$  with  $\mathbf{A}^\top \mathbf{y} = \mathbf{0}$ .
- (c) Let  $\mathbf{A} = \begin{pmatrix} 1 & 1 \\ 1 & 2 \\ 1 & 3 \end{pmatrix}$ . Find a single linear condition on  $\mathbf{b} = (b_1, b_2, b_3) \in \mathbb{R}^3$  that is equivalent to consistency of  $\mathbf{Ax} = \mathbf{b}$ .

*Hint: for (a), combine [Theorem 4.4.3](#) with the description of a subspace by its annihilator in the previous section.*

*Solution.*

- (a)  $T\mathbf{v} = \mathbf{w}$  has a solution iff  $\mathbf{w} \in \text{im } T$ . By [Theorem 4.4.3](#) (a),  $\ker T' = (\text{im } T)^0$ . By [Corollary 4.3.4](#) applied to the subspace  $\text{im } T$  of the finite-dimensional space  $W$ ,

$$\begin{aligned} \text{im } T &= \{\mathbf{w} \in W : \psi(\mathbf{w}) = 0 \text{ for every } \psi \in (\text{im } T)^0\} \\ &= \{\mathbf{w} \in W : \psi(\mathbf{w}) = 0 \text{ for every } \psi \in \ker T'\}. \end{aligned}$$

This is the claim.

- (b) Apply (a) to  $T_\mathbf{A}: F^n \rightarrow F^m$ . By [Theorem 4.1.1](#), every  $\psi \in (F^m)^*$  is  $\psi(\mathbf{u}) = \mathbf{y}^\top \mathbf{u}$  for a unique  $\mathbf{y} \in F^m$ . By [Example \(Dual Maps in Four Settings\)](#) (a),  $T'_\mathbf{A} \psi$  is represented by  $\mathbf{A}^\top \mathbf{y}$ , and the zero functional by  $\mathbf{0}$ ; by uniqueness of the representing vector,  $\psi \in \ker T'_\mathbf{A}$  iff  $\mathbf{A}^\top \mathbf{y} = \mathbf{0}$ . For such  $\psi$ ,  $\psi(\mathbf{b}) = \mathbf{y}^\top \mathbf{b}$ . So (a) says:  $\mathbf{Ax} = \mathbf{b}$  is consistent iff  $\mathbf{y}^\top \mathbf{b} = 0$  for every  $\mathbf{y}$  with  $\mathbf{A}^\top \mathbf{y} = \mathbf{0}$ .
- (c)  $\mathbf{A}^\top \mathbf{y} = \mathbf{0}$  reads  $y_1 + y_2 + y_3 = 0$  and  $y_1 + 2y_2 + 3y_3 = 0$ . Subtracting,  $y_2 + 2y_3 = 0$ , so  $\mathbf{y} = t(1, -2, 1)$ . By (b),  $\mathbf{Ax} = \mathbf{b}$  is consistent iff  $\mathbf{y}^\top \mathbf{b} = 0$  for all these  $\mathbf{y}$ , that is, iff

$$b_1 - 2b_2 + b_3 = 0.$$

In words, the system  $x_1 + kx_2 = b_k$  ( $k = 1, 2, 3$ ) asks for a line through the points  $(1, b_1), (2, b_2), (3, b_3)$ , and it exists exactly when the second difference  $b_3 - 2b_2 + b_1$  vanishes. No inner product was used:  $\mathbf{y}^\top \mathbf{b}$  is just a functional applied to  $\mathbf{b}$ .

##### Exercise C2: Invariant subspaces and annihilators.

Let  $V$  be finite-dimensional,  $T \in \mathcal{L}(V)$ , and  $U$  a subspace of  $V$ . Recall that  $U$  is *T-invariant* if  $T\mathbf{u} \in U$  for every  $\mathbf{u} \in U$ . Prove that  $U$  is  $T$ -invariant if and only if  $U^0$  is  $T'$ -invariant.

*Solution.* ( $\Rightarrow$ ) Suppose  $U$  is  $T$ -invariant, and let  $\varphi \in U^0$ . For every  $\mathbf{u} \in U$ ,  $(T'\varphi)(\mathbf{u}) = \varphi(T\mathbf{u}) = 0$ , because  $T\mathbf{u} \in U$ . Hence  $T'\varphi \in U^0$ , and  $U^0$  is  $T'$ -invariant.

( $\Leftarrow$ ) Suppose  $U^0$  is  $T'$ -invariant, and let  $\mathbf{u} \in U$ . For every  $\varphi \in U^0$ ,  $T'\varphi \in U^0$ , so  $\varphi(T\mathbf{u}) = (T'\varphi)(\mathbf{u}) = 0$ . Thus  $T\mathbf{u}$  is killed by every functional in  $U^0$ . By [Corollary 4.3.4](#),  $T\mathbf{u} \in U$ . Hence  $U$  is  $T$ -invariant.

### Exercise C3: Duality without finite dimension.

Let  $V$  and  $W$  be arbitrary vector spaces over  $F$ , and  $T \in \mathcal{L}(V, W)$ .

- (a) Prove that if  $T$  is surjective, then  $T'$  is injective.
- (b) Prove that if  $T$  is injective, then  $T'$  is surjective.
- (c) Check (b) on  $S: F[x] \rightarrow F[x]$ ,  $S(p) = xp$ , from [Example \(Dual Maps in Four Settings\)](#) (c): given  $\varphi \in F[x]^*$ , find  $\psi$  with  $S'\psi = \varphi$ . Is  $S'$  injective?

*Hint:* for (b), use [Theorem 1.8.2](#) in  $W$ .

*Solution.*

- (a) By [Theorem 4.4.3](#) (a), which needs no finite dimension,  $\ker T' = (\operatorname{im} T)^0 = W^0$ . A functional vanishing on all of  $W$  is the zero functional, so  $\ker T' = \{0\}$  and  $T'$  is injective ([Theorem 3.2.3](#)).
- (b) Let  $\varphi \in V^*$ . By [Theorem 1.8.3](#),  $V$  has a basis  $B$ . Since  $T$  is injective,  $T$  is one-to-one on  $B$ , and  $T(B)$  is linearly independent: a finite combination  $\sum c_i T\mathbf{b}_i = T(\sum c_i \mathbf{b}_i) = \mathbf{0}$  with distinct  $\mathbf{b}_i$  forces  $\sum c_i \mathbf{b}_i = \mathbf{0}$  ([Theorem 3.2.3](#)), hence all  $c_i = 0$ . By [Theorem 1.8.2](#) there is a basis  $C \supseteq T(B)$  of  $W$ . By [Theorem 3.1.6](#), there is  $\psi \in W^*$  with  $\psi(T\mathbf{b}) = \varphi(\mathbf{b})$  for  $\mathbf{b} \in B$ , and  $\psi(\mathbf{c}) = 0$  for  $\mathbf{c} \in C \setminus T(B)$ ; this is well defined because each element of  $T(B)$  is  $T\mathbf{b}$  for exactly one  $\mathbf{b}$ . Then  $\psi T$  and  $\varphi$  are linear and agree on the basis  $B$ , so  $T'\psi = \psi T = \varphi$  by the uniqueness part of [Theorem 3.1.6](#). Hence  $T'$  is surjective.
- (c) Here  $B = \{1, x, x^2, \dots\}$  and  $S(B) = \{x, x^2, \dots\}$ , extended to a basis by adding 1. The recipe gives  $\psi(x^{k+1}) = \varphi(x^k)$  for  $k \geq 0$  and  $\psi(1) = 0$ . Then  $(S'\psi)(x^k) = \psi(x^{k+1}) = \varphi(x^k)$ , so  $S'\psi = \varphi$ . The dual map  $S'$  is not injective:  $S'\varphi_0 = 0$  by [Example \(Dual Maps in Four Settings\)](#) (c). This matches the contrapositive of (a):  $S$  is not surjective, since  $1 \notin \operatorname{im} S$ , and indeed  $\varphi_0$  is a non-zero functional vanishing on  $\operatorname{im} S$ .

## 4.5 Duality for Subspaces and Quotients

A subspace  $U$  of  $V$  gives two new vector spaces:  $U$  itself, and the quotient  $V/U$  of Chapter 3. Each has a dual space, and so far  $U^*$  and  $(V/U)^*$  have nothing to do with  $V^*$ . This section connects them. The answer is a swap: the dual of the quotient is a **subspace** of  $V^*$ , namely the annihilator  $U^0$ , and the dual of the subspace is a **quotient** of  $V^*$ , namely  $V^*/U^0$ . Both identifications come from the First Isomorphism Theorem, and neither needs a basis. At the end we use them to describe hyperplanes, the subspaces cut out by a single equation, in any dimension.

### 4.5.1 Restricting functionals to a subspace

Let  $U$  be a subspace of  $V$ . A functional  $\varphi \in V^*$  measures every vector of  $V$ , so in particular it measures the vectors of  $U$ . Forgetting the rest gives the **restriction**  $\varphi|_U: U \rightarrow F$ ,  $\mathbf{u} \mapsto \varphi(\mathbf{u})$ , which is a functional on  $U$ . This defines the restriction map

$$\rho: V^* \rightarrow U^*, \quad \rho(\varphi) = \varphi|_U.$$

It is not a new kind of map. Let  $\iota: U \rightarrow V$ ,  $\iota(\mathbf{u}) = \mathbf{u}$ , be the inclusion. Its dual map (Definition 4.4.1) sends  $\varphi$  to  $\varphi \circ \iota$ , and  $(\varphi \circ \iota)(\mathbf{u}) = \varphi(\mathbf{u})$ . So  $\rho = \iota'$ , and in particular  $\rho$  is linear.

Two questions decide what  $\rho$  does. Which functionals restrict to zero? By Definition 4.2.1, exactly those in  $U^0$ . And does every functional on  $U$  come from one on  $V$ ? That is an extension problem, and it has a positive answer in every dimension.

#### Theorem 4.5.1 — Dual of a Subspace

Let  $V$  be a vector space over  $F$ , of any dimension, and let  $U$  be a subspace of  $V$ . The restriction map  $\rho: V^* \rightarrow U^*$ ,  $\rho(\varphi) = \varphi|_U$ , is linear and surjective, and  $\ker \rho = U^0$ . Hence

$$V^*/U^0 \rightarrow U^*, \quad \varphi + U^0 \mapsto \varphi|_U$$

is a well-defined isomorphism, and  $U^* \cong V^*/U^0$ .

*Idea.* The kernel is the definition of  $U^0$  read aloud. The real content is surjectivity: given  $\psi$  on  $U$ , we must extend it to all of  $V$ . A basis tells us how. Extend a basis of  $U$  to a basis of  $V$ ; a linear map may be prescribed freely on a basis, so declare the extension to agree with  $\psi$  on the basis of  $U$  and to be 0 on the new basis vectors. Then the First Isomorphism Theorem turns “surjective with kernel  $U^0$ ” into the isomorphism.

*Proof.* We saw above that  $\rho = \iota'$  is linear. For  $\varphi \in V^*$ ,  $\rho(\varphi) = 0$  means  $\varphi(\mathbf{u}) = 0$  for every  $\mathbf{u} \in U$ , which is  $\varphi \in U^0$  by Definition 4.2.1. So  $\ker \rho = U^0$ .

*Surjective.* Let  $\psi \in U^*$ . By Theorem 1.8.3,  $U$  has a basis  $B_U$ . It is a linearly independent subset of  $V$ , so by Theorem 1.8.2 there is a basis  $B$  of  $V$  with  $B_U \subseteq B$ . By Theorem 3.1.6 there is a linear map  $\varphi: V \rightarrow F$  with

$$\varphi(\mathbf{b}) = \psi(\mathbf{b}) \quad \text{for } \mathbf{b} \in B_U, \quad \varphi(\mathbf{b}) = 0 \quad \text{for } \mathbf{b} \in B \setminus B_U.$$

Then  $\varphi|_U$  and  $\psi$  are linear maps  $U \rightarrow F$  that agree on the basis  $B_U$  of  $U$ , so  $\varphi|_U = \psi$  by the uniqueness part of Theorem 3.1.6. Thus  $\rho(\varphi) = \psi$ , and  $\rho$  is surjective. (If  $V$  is finite-dimensional, the finite versions Corollary 1.5.5, Theorem 1.6.5 and Theorem 3.1.5 suffice, and no choice principle is needed.)

By the First Isomorphism Theorem (Theorem 3.9.7),  $\varphi + \ker \rho \mapsto \rho(\varphi)$  is a well-defined isomorphism  $V^*/\ker \rho \rightarrow \text{im } \rho$ . Since  $\ker \rho = U^0$  and  $\text{im } \rho = U^*$ , this is the claimed isomorphism. ■

In words: **a functional on  $U$  is a functional on  $V$ , up to adding something that vanishes on  $U$ .** Two extensions of the same  $\psi$  differ by an element of  $U^0$ , and every element of  $U^0$  can be added.



When  $V$  is finite-dimensional, taking dimensions recovers the count of this chapter. By [Corollary 4.1.4](#),  $\dim U^* = \dim U$  and  $\dim V^* = \dim V$ , and by [Theorem 3.9.5](#),  $\dim(V^*/U^0) = \dim V - \dim U^0$ . So the isomorphism says  $\dim U = \dim V - \dim U^0$ , which is [Theorem 4.2.2](#) again.

## 4.5.2 Functionals on a quotient

Now the other space. A functional  $\psi$  on  $V/U$  measures cosets. Composing with the quotient map  $\pi: V \rightarrow V/U$ ,  $\pi(\mathbf{v}) = \mathbf{v} + U$ , turns it into a measurement of vectors,  $\mathbf{v} \mapsto \psi(\mathbf{v} + U)$ . This functional on  $V$  cannot tell apart two vectors in the same coset, so in particular it is zero on  $U$ . The theorem says that nothing is lost and nothing else appears: the functionals on  $V/U$  are **exactly** the functionals on  $V$  that vanish on  $U$ .

### Theorem 4.5.2 — Dual of a Quotient

Let  $V$  be a vector space over  $F$ , of any dimension, let  $U$  be a subspace, and let  $\pi: V \rightarrow V/U$  be the quotient map. Then the dual map

$$\pi': (V/U)^* \rightarrow V^*, \quad \pi'(\psi) = \psi \circ \pi,$$

is injective, and its image is  $U^0$ . Hence  $\pi'$  is an isomorphism from  $(V/U)^*$  onto  $U^0$ , and  $(V/U)^* \cong U^0$ .

*Idea.* Injectivity uses only that  $\pi$  is surjective: if  $\psi \circ \pi$  is zero, then  $\psi$  is zero on every coset. For the image, one inclusion is “ $\pi$  kills  $U$ ”. The other inclusion asks us to build a functional on  $V/U$  from a functional on  $V$  that vanishes on  $U$ , and that is exactly what the universal property of the quotient does.

*Proof.* The map  $\pi'$  is linear by [Definition 4.4.1](#).

*Injective.* Suppose  $\pi'(\psi) = \psi \circ \pi = 0$ . Every element of  $V/U$  is  $\pi(\mathbf{v})$  for some  $\mathbf{v} \in V$ , because  $\pi$  is surjective ([Theorem 3.9.4](#) (c)). Then  $\psi(\pi(\mathbf{v})) = 0$  for every  $\mathbf{v}$ , so  $\psi = 0$ . By [Theorem 3.2.3](#),  $\pi'$  is injective.

$\text{im } \pi' \subseteq U^0$ . For  $\mathbf{u} \in U$ ,  $\pi(\mathbf{u}) = U$  is the zero vector of  $V/U$ , so  $(\psi \circ \pi)(\mathbf{u}) = \psi(\mathbf{0}) = 0$ . Hence  $\psi \circ \pi \in U^0$ .

$U^0 \subseteq \text{im } \pi'$ . Let  $\varphi \in U^0$ . Then  $U \subseteq \ker \varphi$ , so by [Theorem 3.9.10](#) (a) there is a linear map  $\psi: V/U \rightarrow F$  with  $\psi \circ \pi = \varphi$ . This  $\psi$  lies in  $(V/U)^*$ , and  $\pi'(\psi) = \varphi$ .

So  $\pi'$  is an injective linear map with image  $U^0$ ; regarded as a map onto  $U^0$  it is a linear bijection, hence an isomorphism. ■

Concretely, the inverse isomorphism  $U^0 \rightarrow (V/U)^*$  is  $\varphi \mapsto \bar{\varphi}$ , where  $\bar{\varphi}(\mathbf{v} + U) = \varphi(\mathbf{v})$ . The formula is well defined precisely because  $\varphi$  vanishes on  $U$ : if  $\mathbf{v} - \mathbf{v}' \in U$ , then  $\varphi(\mathbf{v}) - \varphi(\mathbf{v}') = \varphi(\mathbf{v} - \mathbf{v}') = 0$ .

The two theorems fit together as a mirror:

|             | subspace $U \subseteq V$                            | quotient $V/U$                                      |
|-------------|-----------------------------------------------------|-----------------------------------------------------|
| its dual is | a <b>quotient</b> of $V^*$ :<br>$U^* \cong V^*/U^0$ | a <b>subspace</b> of $V^*$ :<br>$(V/U)^* \cong U^0$ |
| the map     | $\iota'$ : restrict                                 | $\pi'$ : compose with $\pi$                         |
| the map is  | surjective                                          | injective                                           |

The last row is no accident:  $\iota$  is injective and  $\pi$  is surjective, and dualizing swaps the two properties.

**Alternatively**, in finite dimension both theorems follow from [Theorem 4.4.3](#), which says  $\ker T' = (\operatorname{im} T)^0$  and  $\operatorname{im} T' = (\ker T)^0$ . For  $T = \pi$ :  $\operatorname{im} \pi = V/U$ , whose annihilator is  $\{0\}$ , and  $\ker \pi = U$ , so  $\pi'$  is injective with image  $U^0$ . For  $T = \iota$ :  $\operatorname{im} \iota = U$ , so  $\ker \iota' = U^0$ , and  $\ker \iota = \{0\}$ , whose annihilator is all of  $U^*$ , so  $\iota'$  is surjective. The proofs above avoid dimension counts, which is why they work in every dimension.

*Quick check.* Let  $V$  be a vector space of dimension 6 and  $U$  a subspace of dimension 2. Find  $\dim U^0$ ,  $\dim(V/U)^*$ ,  $\dim U^*$  and  $\dim(V^*/U^0)$ , and check that the two theorems are consistent with these numbers.

*Solution.* By [Theorem 4.2.2](#),  $\dim U^0 = 6 - 2 = 4$ . By [Theorem 3.9.5](#) and [Corollary 4.1.4](#),  $\dim(V/U)^* = \dim(V/U) = 4$ , matching  $(V/U)^* \cong U^0$ . Next,  $\dim U^* = \dim U = 2$ , and  $\dim(V^*/U^0) = 6 - 4 = 2$ , matching  $U^* \cong V^*/U^0$ .

### 4.5.3 A line in $\mathbb{R}^3$ , in coordinates

The theorems name their isomorphisms without choosing anything. To see them act, we now choose bases and compute. Recall that a functional on  $\mathbb{R}^3$  is determined by its values on  $\mathbf{e}_1, \mathbf{e}_2, \mathbf{e}_3$  ([Theorem 4.1.1](#)). So every  $\varphi \in (\mathbb{R}^3)^*$  has the form  $\varphi(x, y, z) = ax + by + cz$ , with  $(a, b, c) = (\varphi(\mathbf{e}_1), \varphi(\mathbf{e}_2), \varphi(\mathbf{e}_3))$ . We write such a functional simply as  $ax + by + cz$ .

#### Example. (Both isomorphisms for a line)

Let  $\mathbf{u} = (1, 2, -1)$  and  $U = \operatorname{span}(\mathbf{u}) \subseteq \mathbb{R}^3$ .

- Find a basis of  $U^0$ .
- Find a basis of  $\mathbb{R}^3/U$ , its dual basis  $(\psi_1, \psi_2)$  of  $(\mathbb{R}^3/U)^*$ , and the images  $\pi'(\psi_1), \pi'(\psi_2)$ .
- Describe the restriction map  $\rho: (\mathbb{R}^3)^* \rightarrow U^*$  and the isomorphism  $(\mathbb{R}^3)^*/U^0 \rightarrow U^*$ .

*Solution.*

- By [Theorem 4.2.3](#) (b),  $U^0 = \{\mathbf{u}\}^0$ , so  $ax + by + cz \in U^0$  exactly when  $a + 2b - c = 0$ , that is,  $c = a + 2b$ . Taking  $(a, b) = (1, 0)$  and  $(0, 1)$  gives

$$\varphi_1 = x + z, \quad \varphi_2 = y + 2z.$$

They are independent (look at the coefficients of  $x$  and  $y$ ), and  $\dim U^0 = 3 - 1 = 2$  by [Theorem 4.2.2](#), so  $(\varphi_1, \varphi_2)$  is a basis of  $U^0$  ([Theorem 1.6.4](#)).

- The list  $(\mathbf{u}, \mathbf{e}_1, \mathbf{e}_2)$  is independent: in  $a\mathbf{u} + b\mathbf{e}_1 + c\mathbf{e}_2 = (a + b, 2a + c, -a) = \mathbf{0}$ , the

third entry gives  $a = 0$ , and then  $b = c = 0$ . It has length 3, so it is a basis of  $\mathbb{R}^3$ , and by [Theorem 3.9.5](#)  $(\mathbf{e}_1 + U, \mathbf{e}_2 + U)$  is a basis of  $\mathbb{R}^3/U$ . Its dual basis consists of the  $\psi_i$  with  $\psi_i(\mathbf{e}_j + U) = \delta_{ij}$  ([Theorem 4.1.3](#)).

The functional  $\pi'(\psi_1) = \psi_1 \circ \pi$  on  $\mathbb{R}^3$  takes the values  $\psi_1(\mathbf{e}_1 + U) = 1$  at  $\mathbf{e}_1$ ,  $\psi_1(\mathbf{e}_2 + U) = 0$  at  $\mathbf{e}_2$ , and  $\psi_1(U) = 0$  at  $\mathbf{u}$ . Writing it as  $ax + by + cz$ , this says  $a = 1$ ,  $b = 0$  and  $a + 2b - c = 0$ , so  $c = 1$ . Similarly  $\pi'(\psi_2)$  has  $a = 0$ ,  $b = 1$ ,  $c = 2$ . Hence

$$\pi'(\psi_1) = x + z = \varphi_1, \quad \pi'(\psi_2) = y + 2z = \varphi_2.$$

So  $\pi'$  carries the dual basis of  $(\mathbb{R}^3/U)^*$  onto the basis of  $U^0$  found in (a), as [Theorem 4.5.2](#) predicts. Read backwards:  $\psi_1((x, y, z) + U) = x + z$  and  $\psi_2((x, y, z) + U) = y + 2z$ . These are well defined, since adding  $t\mathbf{u} = (t, 2t, -t)$  changes  $x + z$  by  $t - t = 0$  and  $y + 2z$  by  $2t - 2t = 0$ .

- (c) The space  $U^*$  is one-dimensional, with basis the functional  $\xi$  determined by  $\xi(\mathbf{u}) = 1$  (the dual basis of  $(\mathbf{u})$ ). A functional  $\varphi = ax + by + cz$  restricts to  $\varphi|_U = \varphi(\mathbf{u})\xi = (a + 2b - c)\xi$ , since both sides take the value  $a + 2b - c$  at  $\mathbf{u}$ . So  $\rho$  is “evaluate at  $\mathbf{u}$ ”. Its kernel is  $\{a + 2b - c = 0\} = U^0$ , and it is surjective, since  $\rho(x) = \xi$ . The isomorphism of [Theorem 4.5.1](#) is

$$(ax + by + cz) + U^0 \mapsto (a + 2b - c)\xi.$$

For instance  $x + U^0 = -z + U^0$ , because  $x - (-z) = \varphi_1 \in U^0$ ; both restrict to  $\xi$ .

## 4.5.4 Why “natural” matters here

In the example, bases entered only when we wanted numbers. The maps  $\rho$  and  $\pi'$  were defined before any basis was chosen: restrict a functional, or compose it with  $\pi$ . That is what makes the two isomorphisms **natural**, in the sense of the discussion of the double dual. They are part of the structure of  $V$  and  $U$ , not of a coordinate system.

There is a tempting shortcut that is **not** natural. Since  $U^*$  is a quotient of  $V^*$ , one might try to find a copy of  $U^*$  **inside**  $V^*$ , as a subspace. This can be done, but only by making a choice. Pick a complement  $W$  of  $U$ , so  $V = U \oplus W$ . A functional that vanishes on  $W$  is determined by its values on  $U$ , so the restriction  $\rho$  maps  $W^0$  isomorphically onto  $U^*$  ([Exercise \(C2: Complements and a choice of copy of  \$U^\*\$ \)](#)). But different complements give different subspaces  $W^0$ .

*Warning.*  $U^*$  is **not** a subspace of  $V^*$ . Its elements are functionals on  $U$ , which cannot be evaluated at vectors outside  $U$ . To place  $U^*$  inside  $V^*$  one must extend each functional, and extensions are not unique. In the example above, the complement  $W_1 = \text{span}(\mathbf{e}_1, \mathbf{e}_2)$  of  $U$  gives  $W_1^0 = \text{span}(z)$ , while  $W_2 = \text{span}(\mathbf{e}_1, \mathbf{e}_3)$  gives  $W_2^0 = \text{span}(y)$ . These are different lines in  $(\mathbb{R}^3)^*$ , and both restrict isomorphically onto  $U^*$ :  $z|_U = -\xi$  and  $y|_U = 2\xi$ . The choice-free statement is  $U^* \cong V^*/U^0$ , a quotient.

The same caution applies on the other side, in reverse.  $(V/U)^*$  is naturally a subspace of  $V^*$ , namely  $U^0$ , but  $V/U$  itself is not a subspace of  $V$ . Duality turns the quotient, which has no natural home inside  $V$ , into a subspace, which does.

### 4.5.5 Codimension and hyperplanes

The quotient measures how far a subspace is from filling the whole space. In  $\mathbb{R}^3$  a plane through the origin misses “one direction” and a line misses “two directions”. For finite-dimensional  $V$  the number of missing directions is  $\dim V - \dim U$ . But this formula is useless in  $F[x]$ , where  $\dim V$  and  $\dim U$  may both be infinite, although  $U = \{p : p(0) = 0\}$  misses only one direction: every polynomial is a constant plus an element of  $U$ . The quotient gives the right number in every case.

*The codimension of  $U$  counts the directions in  $V$  that  $U$  is missing.*

#### Definition 4.5.1 — Codimension

Let  $U$  be a subspace of a vector space  $V$  over  $F$ . If  $V/U$  is finite-dimensional, the **codimension** of  $U$  in  $V$  is

$$\operatorname{codim} U := \dim(V/U).$$

If  $V/U$  is infinite-dimensional, we say that  $U$  has **infinite codimension**.

In words: we do not compare the sizes of  $U$  and  $V$ . We collapse  $U$  to zero and measure what is left. The codimension depends on the ambient space, so when that is unclear we write “codimension of  $U$  in  $V$ ”.

#### Definition 4.5.2 — Hyperplane

A **hyperplane** in  $V$  is a subspace  $H$  of  $V$  of codimension **exactly** 1, that is, with  $\dim(V/H) = 1$ .

**Examples.**

- **Finite dimension.** If  $V$  is finite-dimensional, then  $\operatorname{codim} U = \dim V - \dim U$  by [Theorem 3.9.5](#). The hyperplanes of  $F^n$  are its subspaces of dimension  $n - 1$ : the lines through  $\mathbf{0}$  in  $F^2$  and the planes through  $\mathbf{0}$  in  $F^3$ .
- **Trace.** In  $M_n(F)$  with  $n \geq 1$ , the trace-zero matrices are the kernel of  $\operatorname{tr}: M_n(F) \rightarrow F$ . This map is linear and onto, since  $\operatorname{tr}(c\mathbf{E}_{11}) = c$ , so  $M_n(F)/\ker \operatorname{tr} \cong F$  by [Theorem 3.9.7](#), and the trace-zero matrices form a hyperplane.
- **Polynomials.** In  $F[x]$ , let  $U = \{p : p(0) = 0\}$ . Evaluation at 0 is a linear map  $F[x] \rightarrow F$  with kernel  $U$ , and it is surjective because the constant  $c$  maps to  $c$ . By [Theorem 3.9.7](#),  $F[x]/U \cong F$ , so  $U$  is a hyperplane in the infinite-dimensional space  $F[x]$ . In the same way  $\{p : p(0) = p(1) = 0\}$  has codimension 2 ([Exercise \(B3: Dimension computations\)](#)).
- **Degenerate cases.**  $\operatorname{codim} V = 0$ , since  $V/V$  is the zero space, and  $\operatorname{codim}\{\mathbf{0}\} = \dim V$  when  $V$  is finite-dimensional. In particular  $V$  itself is **not** a hyperplane, and the only hyperplane of a one-dimensional space is  $\{\mathbf{0}\}$ . The zero space  $\{\mathbf{0}\}$  has no hyperplanes at all.

**Non-example by minimal change.** In  $\mathbb{R}^3$ , the plane  $\{x + y + z = 0\}$  is a hyperplane. Replace it by the line  $\{x + y + z = 0, x - y = 0\} = \operatorname{span}((1, 1, -2))$ . It is still a subspace, but its codimension is  $3 - 1 = 2$ , so the clause “codimension **exactly** 1” fails. Replace it instead by  $\{x + y + z = 1\}$ : now it is not a subspace at all, so it has no codimension. It is a coset of a hyperplane, an object we return to in the next section.

**Why this definition.** Codimension is the dual notion of dimension, and the duality theorems make this precise. If  $U$  has finite codimension, then by [Theorem 4.5.2](#), [Corollary 4.1.4](#) and [Theorem 3.5.4](#),

$$\operatorname{codim} U = \dim(V/U) = \dim(V/U)^* = \dim U^0,$$

and this holds even when  $V$  is infinite-dimensional. So **dimension counts vectors, and codimension counts independent equations**. For finite-dimensional  $V$  this is [Theorem 4.2.2](#) again:  $\dim U^0 = \dim V - \dim U$ .

*Warning.* In infinite dimension a hyperplane need not be “smaller” than the whole space. In  $F[x]$ , the hyperplane  $U = \{p : p(0) = 0\} = \{xq : q \in F[x]\}$  is isomorphic to  $F[x]$  itself, via  $q \mapsto xq$ : this map is linear, injective (if  $xq = 0$  then  $q = 0$ ) and onto  $U$ . So you cannot detect codimension by comparing  $U$  with  $V$  up to isomorphism, and the formula  $\operatorname{codim} U = \dim V - \dim U$  must never be used outside finite dimension. Always compute  $\dim(V/U)$ , or count equations.

Hyperplanes are exactly the subspaces cut out by **one** non-trivial linear equation. In  $F^n$  this is familiar:  $a_1x_1 + \cdots + a_nx_n = 0$  with the  $a_i$  not all zero defines a subspace of dimension  $n - 1$ . The quotient makes the statement true in every dimension.

### Theorem 4.5.3 — Hyperplanes Are Kernels of Functionals

Let  $V$  be a vector space over  $F$ , of any dimension, and let  $H$  be a subspace of  $V$ . Then  $H$  is a hyperplane if and only if  $H = \ker \varphi$  for some **non-zero**  $\varphi \in V^*$ .

*Idea.* A non-zero functional maps onto the one-dimensional space  $F$ , so the First Isomorphism Theorem makes  $V/\ker \varphi$  one-dimensional. Conversely, if  $V/H$  is one-dimensional, it “is”  $F$  after choosing an isomorphism, and composing that isomorphism with  $\pi$  gives a functional whose kernel is  $H$ .

*Proof.* ( $\Leftarrow$ ) Suppose  $\varphi \in V^*$  is non-zero. Then  $\operatorname{im} \varphi$  is a subspace of  $F$  ([Theorem 3.2.2](#)) containing a non-zero scalar, so  $\dim \operatorname{im} \varphi = 1 = \dim F$ , and  $\operatorname{im} \varphi = F$  by [Theorem 1.6.8](#). By [Theorem 3.9.7](#),  $V/\ker \varphi \cong F$ , so  $\dim(V/\ker \varphi) = 1$ , and  $\ker \varphi$  is a hyperplane.

( $\Rightarrow$ ) Suppose  $\dim(V/H) = 1$ . By [Theorem 3.5.4](#) there is an isomorphism  $\theta: V/H \rightarrow F$ . Let  $\varphi = \theta \circ \pi$ , where  $\pi: V \rightarrow V/H$  is the quotient map. It is linear as a composition of linear maps ([Theorem 3.4.2](#)), so  $\varphi \in V^*$ . For  $\mathbf{v} \in V$ ,  $\varphi(\mathbf{v}) = 0$  means  $\theta(\mathbf{v} + H) = 0$ , which holds exactly when  $\mathbf{v} + H$  is the zero coset, since  $\theta$  is injective; and  $\mathbf{v} + H = H$  means  $\mathbf{v} \in H$  by [Lemma 3.9.3](#) (b). So  $\ker \varphi = H$ . Finally  $\varphi \neq 0$ : since  $V/H \neq \{0\}$ , we have  $H \neq V$ , so some  $\mathbf{v} \notin H = \ker \varphi$ . ■

The functional is not unique, but almost: two functionals with the same kernel are scalar multiples of each other ([Exercise \(C1: Functionals with the same kernel\)](#); [Exercise \(B2: Functionals with the same kernel\)](#) gives a second proof). So hyperplanes of  $V$  correspond to non-zero functionals **up to a non-zero scalar**, just as the plane  $x + y + z = 0$  is also  $2x + 2y + 2z = 0$ .

A hyperplane together with any one vector outside it fills the space.

*Remark.* Let  $H = \ker \varphi$  be a hyperplane and  $\mathbf{v}_0 \notin H$ . Then  $V = H \oplus \operatorname{span}(\mathbf{v}_0)$ . Indeed

$\varphi(\mathbf{v}_0) \neq 0$ , and every  $\mathbf{v} \in V$  splits as

$$\mathbf{v} = \left( \mathbf{v} - \frac{\varphi(\mathbf{v})}{\varphi(\mathbf{v}_0)} \mathbf{v}_0 \right) + \frac{\varphi(\mathbf{v})}{\varphi(\mathbf{v}_0)} \mathbf{v}_0,$$

where the first bracket lies in  $\ker \varphi$  because  $\varphi$  of it is  $\varphi(\mathbf{v}) - \varphi(\mathbf{v}) = 0$ . And  $H \cap \text{span}(\mathbf{v}_0) = \{\mathbf{0}\}$ , since  $\varphi(c\mathbf{v}_0) = c\varphi(\mathbf{v}_0) = 0$  forces  $c = 0$ .

*Quick check.* Decide which of the following are hyperplanes, and for each hyperplane give a non-zero functional with that kernel.

- (a) The symmetric matrices in  $M_2(F)$ .
- (b) The symmetric matrices in  $M_3(F)$ .
- (c)  $\{p \in F[x] : p(1) = 0\}$  in  $F[x]$ .

*Solution.*

- (a) Yes.  $\mathbf{A}$  is symmetric exactly when  $a_{12} - a_{21} = 0$ , so the symmetric matrices are the kernel of  $\mathbf{A} \mapsto a_{12} - a_{21}$ , which is linear and non-zero (it takes the value 1 at  $\mathbf{E}_{12}$ ). By [Theorem 4.5.3](#) this is a hyperplane.
- (b) No. The symmetric  $3 \times 3$  matrices form a subspace of dimension 6 (choose the entries on and above the diagonal freely), so the codimension is  $9 - 6 = 3 \neq 1$ .
- (c) Yes. It is the kernel of  $p \mapsto p(1)$ , which is linear ([Theorem 0.8.5](#)) and non-zero (it sends the constant 1 to 1).

## 4.5.6 Exercises

### 4.5.6.1 A. Check your understanding

#### Exercise A1.

- (a) State the theorem identifying  $(V/U)^*$  with a subspace of  $V^*$ , including the map.
- (b) Define the codimension of a subspace, and define a hyperplane.
- (c) True or false:  $U^*$  is a subspace of  $V^*$ . Justify your answer.
- (d) Let  $U$  be a subspace of dimension 2 in  $F^5$ . Find  $\dim(F^5/U)^*$  and  $\text{codim } U$ .
- (e) Is  $\{(x_1, x_2, x_3, x_4) \in \mathbb{R}^4 : x_1 = x_2 = 0\}$  a hyperplane in  $\mathbb{R}^4$ ? Justify your answer.
- (f) True or false: every non-zero linear functional  $\varphi: V \rightarrow F$  is surjective. Justify your answer.

*Solution.*

- (a) For a subspace  $U$  of  $V$  with quotient map  $\pi$ , the dual map  $\pi': (V/U)^* \rightarrow V^*$ ,  $\psi \mapsto \psi \circ \pi$ , is injective with image  $U^0$ ; so  $(V/U)^* \cong U^0$  ([Theorem 4.5.2](#)).
- (b) If  $V/U$  is finite-dimensional,  $\text{codim } U = \dim(V/U)$ ; otherwise  $U$  has infinite codimension. A hyperplane is a subspace of codimension exactly 1.
- (c) False. Elements of  $U^*$  are defined only on  $U$ , so they are not elements of  $V^*$ . What is true is  $U^* \cong V^*/U^0$ , a quotient ([Theorem 4.5.1](#)).

- (d)  $\dim(F^5/U)^* = \dim(F^5/U) = 5 - 2 = 3$ , by [Corollary 4.1.4](#) and [Theorem 3.9.5](#). So  $\operatorname{codim} U = 3$ .
- (e) No. The subspace is  $\operatorname{span}(\mathbf{e}_3, \mathbf{e}_4)$ , of dimension 2, so its codimension is  $4 - 2 = 2$ .
- (f) True, in any dimension. This is [Proposition 4.1.2](#):  $\operatorname{im} \varphi$  is a subspace of the one-dimensional space  $F$  containing a non-zero element, so it is  $F$ .

#### 4.5.6.2 B. Practice

##### Exercise B1: The dual of a quotient of $\mathbb{R}^4$ .

Let  $U = \operatorname{span}((1, -1, 0, 2), (0, 1, 1, -1)) \subseteq \mathbb{R}^4$ , and write functionals on  $\mathbb{R}^4$  as  $a_1x_1 + a_2x_2 + a_3x_3 + a_4x_4$ .

- (a) Find a basis  $(\varphi_1, \varphi_2)$  of  $U^0$ .
- (b) Let  $\psi_1, \psi_2 \in (\mathbb{R}^4/U)^*$  be the functionals with  $\pi'(\psi_i) = \varphi_i$ . Compute  $\psi_1$  and  $\psi_2$  at the coset  $(2, 0, 1, 3) + U$ .
- (c) Find a basis of  $\mathbb{R}^4/U$  whose dual basis is  $(\psi_1, \psi_2)$ .

*Solution.*

- (a) A functional lies in  $U^0$  exactly when it vanishes at both spanning vectors ([Theorem 4.2.3](#) (b)), that is,

$$a_1 - a_2 + 2a_4 = 0, \quad a_2 + a_3 - a_4 = 0.$$

The second gives  $a_3 = a_4 - a_2$ , and the first gives  $a_1 = a_2 - 2a_4$ . Taking  $(a_2, a_4) = (1, 0)$  and  $(-1, -1)$  gives  $(1, 1, -1, 0)$  and  $(1, -1, 0, -1)$ , that is,

$$\varphi_1 = x_1 + x_2 - x_3, \quad \varphi_2 = x_1 - x_2 - x_4.$$

Check:  $\varphi_1(1, -1, 0, 2) = 0$ ,  $\varphi_1(0, 1, 1, -1) = 0$ ,  $\varphi_2(1, -1, 0, 2) = 1 + 1 - 2 = 0$ ,  $\varphi_2(0, 1, 1, -1) = -1 + 1 = 0$ . They are independent (compare the coefficients of  $x_3$  and  $x_4$ ). The spanning vectors of  $U$  are independent (compare first entries, then second), so  $\dim U^0 = 4 - 2 = 2$  by [Theorem 4.2.2](#), and  $(\varphi_1, \varphi_2)$  is a basis by [Theorem 1.6.4](#).

- (b) By [Theorem 4.5.2](#),  $\psi_i(\mathbf{v} + U) = \varphi_i(\mathbf{v})$ . So  $\psi_1((2, 0, 1, 3) + U) = 2 + 0 - 1 = 1$  and  $\psi_2((2, 0, 1, 3) + U) = 2 - 0 - 3 = -1$ .
- (c) We need cosets  $\mathbf{c}_1, \mathbf{c}_2$  with  $\psi_i(\mathbf{c}_j) = \delta_{ij}$ . The vector  $-\mathbf{e}_3$  has  $\varphi_1(-\mathbf{e}_3) = 1$  and  $\varphi_2(-\mathbf{e}_3) = 0$ , and  $-\mathbf{e}_4$  has  $\varphi_1(-\mathbf{e}_4) = 0$ ,  $\varphi_2(-\mathbf{e}_4) = 1$ . Put  $\mathbf{c}_1 = -\mathbf{e}_3 + U$  and  $\mathbf{c}_2 = -\mathbf{e}_4 + U$ . If  $a\mathbf{c}_1 + b\mathbf{c}_2 = \mathbf{0}$ , applying  $\psi_1$  and  $\psi_2$  gives  $a = 0$  and  $b = 0$ ; so  $(\mathbf{c}_1, \mathbf{c}_2)$  is independent, and since  $\dim(\mathbb{R}^4/U) = 2$  it is a basis ([Theorem 1.6.4](#)). By construction  $\psi_i(\mathbf{c}_j) = \delta_{ij}$ , so  $(\psi_1, \psi_2)$  is its dual basis ([Theorem 4.1.3](#)).

##### Exercise B2: Functionals with the same kernel.

Let  $V$  be a vector space over  $F$ , of any dimension, and let  $\varphi, \psi \in V^*$  be non-zero. Prove that  $\ker \varphi = \ker \psi$  if and only if  $\psi = c\varphi$  for some non-zero  $c \in F$ . ([Exercise \(C1: Functionals with the same kernel\)](#) proved  $(\Rightarrow)$  by a different route; here use the hyperplane decomposition.)

*Hint: the remark after Theorem 4.5.3.*

*Solution.* ( $\Leftarrow$ ) If  $\psi = c\varphi$  with  $c \neq 0$ , then  $\psi(\mathbf{v}) = 0$  if and only if  $c\varphi(\mathbf{v}) = 0$ , if and only if  $\varphi(\mathbf{v}) = 0$ , since  $c$  is invertible. So  $\ker \psi = \ker \varphi$ .

( $\Rightarrow$ ) Suppose  $\ker \varphi = \ker \psi = H$ . Since  $\varphi \neq 0$ , choose  $\mathbf{v}_0$  with  $\varphi(\mathbf{v}_0) \neq 0$ ; then  $\mathbf{v}_0 \notin H$ , so also  $\psi(\mathbf{v}_0) \neq 0$ . Put  $c = \psi(\mathbf{v}_0)/\varphi(\mathbf{v}_0)$ , which is non-zero, and  $\chi = \psi - c\varphi \in V^*$ . Then  $\chi$  vanishes on  $H$ , because both  $\psi$  and  $\varphi$  do, and  $\chi(\mathbf{v}_0) = \psi(\mathbf{v}_0) - c\varphi(\mathbf{v}_0) = 0$ . By Theorem 4.5.3  $H$  is a hyperplane, and by the remark after it  $V = H \oplus \text{span}(\mathbf{v}_0)$ . A linear map vanishing on  $H$  and on  $\mathbf{v}_0$  vanishes on every  $\mathbf{h} + a\mathbf{v}_0$ , hence on  $V$ . So  $\chi = 0$ , that is,  $\psi = c\varphi$ .

### Exercise B3: Dimension computations.

- (a) Let  $\dim V = 7$  and  $\dim U = 3$ . Find  $\dim U^*$ ,  $\dim(V^*/U^0)$ ,  $\dim(V/U)^*$  and  $\text{codim } U$ .
- (b) In  $V = F[x]_{\leq 3}$ , let  $U = \{p : p(0) = p(1) = 0\}$ . Using the map  $T: V \rightarrow F^2$ ,  $T(p) = (p(0), p(1))$ , find  $\text{codim } U$  and  $\dim U$ , and write  $U$  as an intersection of two hyperplanes.
- (c) Hence find a basis of  $U^0$  consisting of evaluation functionals.

*Solution.*

- (a)  $\dim U^* = 3$  by Corollary 4.1.4.  $\dim U^0 = 7 - 3 = 4$  by Theorem 4.2.2, so  $\dim(V^*/U^0) = 7 - 4 = 3$  (Theorem 3.9.5), consistent with Theorem 4.5.1.  $\dim(V/U)^* = \dim(V/U) = 4$ , consistent with Theorem 4.5.2, and  $\text{codim } U = 4$ .
- (b)  $T$  is linear because evaluation is (Theorem 0.8.5), and  $\ker T = U$ . It is surjective:  $T(1) = (1, 1)$  and  $T(x) = (0, 1)$  span  $F^2$ . By Theorem 3.9.7,  $V/U \cong F^2$ , so  $\text{codim } U = 2$ , and  $\dim U = 4 - 2 = 2$  by Theorem 3.9.5. Writing  $\varepsilon_0(p) = p(0)$  and  $\varepsilon_1(p) = p(1)$ , we have  $U = \ker \varepsilon_0 \cap \ker \varepsilon_1$ , and both kernels are hyperplanes by Theorem 4.5.3, since both functionals send 1 to 1.
- (c) Both  $\varepsilon_0$  and  $\varepsilon_1$  lie in  $U^0$ . They are independent: if  $a\varepsilon_0 + b\varepsilon_1 = 0$ , evaluating at  $x$  gives  $b = 0$ , and then evaluating at 1 gives  $a = 0$ . Since  $\dim U^0 = 4 - 2 = 2$ ,  $(\varepsilon_0, \varepsilon_1)$  is a basis of  $U^0$  (Theorem 1.6.4).

### 4.5.6.3 C. Going deeper

#### Exercise C1: Subspaces of finite codimension are intersections of hyperplanes.

Let  $V$  be a vector space over  $F$ , of any dimension, and let  $U$  be a subspace of finite codimension  $k \geq 1$ .

- (a) Prove that there are non-zero  $\varphi_1, \dots, \varphi_k \in U^0$  with  $U = \ker \varphi_1 \cap \dots \cap \ker \varphi_k$ . Deduce that  $U$  is an intersection of  $k$  hyperplanes.
- (b) Prove that  $U$  is not an intersection of fewer than  $k$  hyperplanes.



*Hint: for (a), take a dual basis of  $(V/U)^*$ ; for (b), consider the map  $V \rightarrow F^m$ ,  $\mathbf{v} \mapsto (\varphi_1(\mathbf{v}), \dots, \varphi_m(\mathbf{v}))$ .*

*Solution.*

- (a) Let  $\pi: V \rightarrow V/U$  be the quotient map. Since  $\dim(V/U) = k$ , choose a basis  $(\mathbf{c}_1, \dots, \mathbf{c}_k)$  of  $V/U$  and let  $(\psi_1, \dots, \psi_k)$  be its dual basis (Theorem 4.1.3). Put  $\varphi_i = \psi_i \circ \pi$ . By Theorem 4.5.2,  $\varphi_i \in U^0$ , and  $\varphi_i \neq 0$  because  $\pi'$  is injective and  $\psi_i \neq 0$ .  
 Since each  $\varphi_i$  lies in  $U^0$ ,  $U \subseteq \ker \varphi_i$  for each  $i$ . Conversely, let  $\mathbf{v} \in \ker \varphi_1 \cap \dots \cap \ker \varphi_k$ , and write  $\mathbf{v} + U = \sum_j a_j \mathbf{c}_j$ . Applying  $\psi_i$  gives  $a_i = \psi_i(\mathbf{v} + U) = \varphi_i(\mathbf{v}) = 0$  for every  $i$ . So  $\mathbf{v} + U$  is the zero coset, and  $\mathbf{v} \in U$  by Lemma 3.9.3 (b). This proves  $U = \bigcap_i \ker \varphi_i$ . Each  $\ker \varphi_i$  is a hyperplane by Theorem 4.5.3.
- (b) Suppose  $U = H_1 \cap \dots \cap H_m$  with hyperplanes  $H_i$ . By Theorem 4.5.3,  $H_i = \ker \varphi_i$  for non-zero  $\varphi_i \in V^*$ . The map  $S: V \rightarrow F^m$ ,  $S(\mathbf{v}) = (\varphi_1(\mathbf{v}), \dots, \varphi_m(\mathbf{v}))$ , is linear, since each entry is, and  $\ker S = \bigcap_i \ker \varphi_i = U$ . By Theorem 3.9.7,  $V/U \cong \text{im } S$ , a subspace of  $F^m$ . Hence  $k = \dim(V/U) = \dim \text{im } S \leq m$  by Theorem 1.6.7. So at least  $k$  hyperplanes are needed.

### Exercise C2: Complements and a choice of copy of $U^*$ .

Let  $V$  be finite-dimensional, and let  $U, W$  be subspaces with  $V = U \oplus W$ .

- (a) Prove that  $V^* = U^0 \oplus W^0$  without using Theorem 4.2.3 (d). (This is Exercise (C3: Dual of a direct sum decomposition) (a), which used it.)
- (b) Deduce that the restriction map  $\rho: V^* \rightarrow U^*$  restricts to an isomorphism  $W^0 \rightarrow U^*$ .
- (c) In  $\mathbb{R}^2$ , let  $U = \text{span}(\mathbf{e}_1)$ ,  $W_1 = \text{span}(\mathbf{e}_2)$  and  $W_2 = \text{span}((1, 1))$ . Find  $W_1^0$  and  $W_2^0$ , and find the element of each that restricts to the functional  $\xi \in U^*$  with  $\xi(\mathbf{e}_1) = 1$ . Conclude that the copy of  $U^*$  inside  $V^*$  depends on the complement.

*Solution.*

- (a) Let  $n = \dim V$ . If  $\varphi \in U^0 \cap W^0$ , then  $\varphi$  vanishes on  $U$  and on  $W$ , hence on every  $\mathbf{u} + \mathbf{w}$ , that is, on  $U + W = V$ ; so  $\varphi = 0$ . By Theorem 4.2.2,  $\dim U^0 + \dim W^0 = (n - \dim U) + (n - \dim W) = 2n - n = n$ , using  $\dim U + \dim W = n$  for a direct sum (Theorem 1.7.4). By Theorem 1.7.3,  $\dim(U^0 + W^0) = n - 0 = n = \dim V^*$ , so  $U^0 + W^0 = V^*$  (Theorem 1.6.8), and the sum is direct because the intersection is  $\{0\}$ .
- (b) The restriction of  $\rho$  to  $W^0$  is linear. Its kernel is  $W^0 \cap \ker \rho = W^0 \cap U^0 = \{0\}$  by Theorem 4.5.1 and (a), so it is injective. Its domain has dimension  $n - \dim W = \dim U = \dim U^*$ . An injective linear map between spaces of the same finite dimension is an isomorphism (Corollary 3.3.2 (e)).
- (c) Write functionals on  $\mathbb{R}^2$  as  $ax + by$ .  $W_1^0$  consists of those with  $b = 0$ , so  $W_1^0 = \text{span}(x)$ .  $W_2^0$  consists of those with  $a + b = 0$ , so  $W_2^0 = \text{span}(x - y)$ . Restricting to  $U$ :  $x|_U$  and  $(x - y)|_U$  both take the value 1 at  $\mathbf{e}_1$ , so both equal  $\xi$ . The elements  $x$  and  $x - y$  are different, so the subspaces  $W_1^0 \neq W_2^0$  of  $V^*$  are two different copies of  $U^*$ , one for each complement.

## 4.6 Hyperplanes and Linear Systems Revisited

Chapter 2 met subspaces of  $F^n$  in two disguises. A **parametric** description lists spanning vectors, as in  $\text{col}(\mathbf{A})$ . An **implicit** description lists equations, as in  $\text{nul}(\mathbf{A})$ . We moved from equations to spanning vectors by elimination, but we never asked whether the reverse is always possible, or how many equations it takes. Duality answers both questions at once: the equations of a subspace  $U$  are the elements of  $U^0$ . This section turns that sentence into an algorithm, proves once more that row rank and nullity add up to  $n$ , and ends the chapter with a dictionary between a space and its dual.

### 4.6.1 Row vectors are functionals

Recall from [Theorem 4.1.1](#) how functionals on  $F^n$  look. For  $\mathbf{a} = (a_1, \dots, a_n) \in F^n$ , put

$$\varphi_{\mathbf{a}}: F^n \rightarrow F, \quad \varphi_{\mathbf{a}}(\mathbf{x}) = \mathbf{a}^\top \mathbf{x} = a_1x_1 + \dots + a_nx_n.$$

By that theorem, every  $\varphi \in (F^n)^*$  is  $\varphi_{\mathbf{a}}$  for exactly one  $\mathbf{a}$ , namely  $a_j = \varphi(\mathbf{e}_j)$ , and  $\mathbf{a} \mapsto \varphi_{\mathbf{a}}$  is an isomorphism  $F^n \rightarrow (F^n)^*$ .

The functional  $\varphi_{\mathbf{a}}$  is “multiply by the **row** vector  $\mathbf{a}^\top$ ”. So the identification is really between functionals and row vectors, and that is what makes a system of equations a dual object. Write the rows of  $\mathbf{A} \in M_{m \times n}(F)$  as  $\mathbf{a}_1^\top, \dots, \mathbf{a}_m^\top$  with  $\mathbf{a}_i \in F^n$ . The  $i$ -th entry of  $\mathbf{A}\mathbf{x}$  is  $\mathbf{a}_i^\top \mathbf{x} = \varphi_{\mathbf{a}_i}(\mathbf{x})$ , so the homogeneous system  $\mathbf{A}\mathbf{x} = \mathbf{0}$  says exactly that  $\varphi_{\mathbf{a}_1}(\mathbf{x}) = \dots = \varphi_{\mathbf{a}_m}(\mathbf{x}) = 0$ . Hence

$$\text{nul}(\mathbf{A}) = \ker \varphi_{\mathbf{a}_1} \cap \dots \cap \ker \varphi_{\mathbf{a}_m}.$$

Each non-zero row contributes a hyperplane ([Theorem 4.5.3](#)), and the null space is their intersection. **An equation is a functional, and a system is a list of functionals.**

The identification uses the standard basis of  $F^n$ , and nothing else. Under it, the coordinate functionals  $\mathbf{x} \mapsto x_i$ , which form the dual basis of  $(\mathbf{e}_1, \dots, \mathbf{e}_n)$ , correspond to  $\mathbf{e}_1, \dots, \mathbf{e}_n$ , that is, to the rows  $\mathbf{e}_1^\top, \dots, \mathbf{e}_n^\top$ .

### 4.6.2 Every subspace is a solution set

To describe  $U$  by equations, we need equations that hold on  $U$ , that is, functionals in  $U^0$ , and enough of them that they hold **only** on  $U$ . The first requirement is the definition of  $U^0$ . The second follows from the separation of points, applied in the quotient.

#### Lemma 4.6.1 — A subspace is cut out by its annihilator

Let  $V$  be a finite-dimensional vector space and  $U$  a subspace. For  $\mathbf{v} \in V$ ,

$$\mathbf{v} \in U \iff \varphi(\mathbf{v}) = 0 \text{ for every } \varphi \in U^0.$$

If  $(\varphi_1, \dots, \varphi_k)$  is a basis of  $U^0$ , then  $U = \ker \varphi_1 \cap \dots \cap \ker \varphi_k$ .

*Proof.* ( $\Rightarrow$ ) This is [Definition 4.2.1](#).

( $\Leftarrow$ ) Suppose  $\mathbf{v} \notin U$ . Then  $\mathbf{v} + U$  is a non-zero vector of  $V/U$  by [Lemma 3.9.3](#) (b), and  $V/U$  is finite-dimensional ([Theorem 3.9.5](#)). By [Theorem 4.1.5](#) there is  $\psi \in (V/U)^*$  with  $\psi(\mathbf{v} + U) \neq 0$ . By [Theorem 4.5.2](#),  $\varphi = \psi \circ \pi$  lies in  $U^0$ , and  $\varphi(\mathbf{v}) = \psi(\mathbf{v} + U) \neq 0$ . This proves the contrapositive.

For the last statement,  $U \subseteq \ker \varphi_i$  for each  $i$  since  $\varphi_i \in U^0$ . Conversely, if  $\varphi_i(\mathbf{v}) = 0$  for all  $i$ , then every  $\varphi = \sum_i c_i \varphi_i \in U^0$  has  $\varphi(\mathbf{v}) = 0$ , so  $\mathbf{v} \in U$  by ( $\Leftarrow$ ).

■

The lemma is [Corollary 4.3.4](#) in disguise. That corollary says  $U^{00} = U$  under  $V \cong V^{**}$ ; the lemma unpacks it into a statement about vectors. In  $F^n$ , with rows in place of functionals, it becomes the main theorem of this section.

### Theorem 4.6.2 — Every Subspace Is a Solution Set

Let  $U$  be a subspace of  $F^n$  with  $\dim U = k$ .

- (a) There is a matrix  $\mathbf{A} \in M_{(n-k) \times n}(F)$  with  $U = \text{nul}(\mathbf{A})$ . One may take as rows of  $\mathbf{A}$  the vectors  $\mathbf{a}_1^\top, \dots, \mathbf{a}_{n-k}^\top$  for any  $\mathbf{a}_i \in F^n$  such that  $(\varphi_{\mathbf{a}_1}, \dots, \varphi_{\mathbf{a}_{n-k}})$  is a basis of  $U^0$ .
- (b) If  $\mathbf{B} \in M_{m \times n}(F)$  and  $\text{nul}(\mathbf{B}) = U$ , then  $m \geq n - k$ .

So  $U$  is the solution set of a homogeneous system of exactly  $n - k = \text{codim } U$  equations, and of no system with fewer. (When  $k = n$ , that is  $U = F^n$ , this is the empty system, with no equations.)

*Proof.*

- (a) By [Theorem 4.2.2](#),  $\dim U^0 = n - k$ . Choose a basis of  $U^0$ ; by the identification above it is  $(\varphi_{\mathbf{a}_1}, \dots, \varphi_{\mathbf{a}_{n-k}})$  for some  $\mathbf{a}_i \in F^n$ . Let  $\mathbf{A}$  have rows  $\mathbf{a}_1^\top, \dots, \mathbf{a}_{n-k}^\top$ . Then

$$\text{nul}(\mathbf{A}) = \ker \varphi_{\mathbf{a}_1} \cap \dots \cap \ker \varphi_{\mathbf{a}_{n-k}} = U,$$

where the second equality is [Lemma 4.6.1](#).

- (b) By [Theorem 2.5.7](#),  $\text{rank } \mathbf{B} = n - \text{nullity } \mathbf{B} = n - k$ . By [Theorem 2.5.5](#),  $\text{rank } \mathbf{B} = \dim \text{row}(\mathbf{B})$ . The row space is spanned by the  $m$  rows of  $\mathbf{B}$ , so  $m \geq \dim \text{row}(\mathbf{B})$  by [Theorem 1.6.3](#) (b). Hence  $m \geq n - k$ .

■

The count is the one promised by codimension: **dimension counts free parameters, codimension counts independent equations**, and in  $F^n$  the two add up to  $n$ .

#### Example. (Equations for a plane in $\mathbb{R}^4$ )

Let  $U = \text{span}(\mathbf{u}_1, \mathbf{u}_2) \subseteq \mathbb{R}^4$ , where  $\mathbf{u}_1 = (1, 1, 0, 1)$  and  $\mathbf{u}_2 = (0, 1, 1, 1)$ . Find a system of linear equations whose solution set is  $U$ .

*Solution.* The two vectors are independent (compare first entries, then third), so  $\dim U = 2$ , and we expect  $4 - 2 = 2$  equations. For  $\mathbf{a} = (a_1, a_2, a_3, a_4)$ , the functional  $\varphi_{\mathbf{a}}$  lies in  $U^0$  exactly when it vanishes at both spanning vectors (Theorem 4.2.3 (b)):

$$a_1 + a_2 + a_4 = 0, \quad a_2 + a_3 + a_4 = 0.$$

This is the system  $\mathbf{M}\mathbf{a} = \mathbf{0}$  for the matrix  $\mathbf{M}$  whose **rows** are  $\mathbf{u}_1^\top$  and  $\mathbf{u}_2^\top$ . Subtracting the second row from the first gives the RREF

$$\mathbf{M} = \begin{pmatrix} 1 & 1 & 0 & 1 \\ 0 & 1 & 1 & 1 \end{pmatrix} \rightarrow \begin{pmatrix} 1 & 0 & -1 & 0 \\ 0 & 1 & 1 & 1 \end{pmatrix},$$

with free variables  $a_3, a_4$ . By Theorem 2.5.4, a basis of the solutions is  $(1, -1, 1, 0)$  (from  $a_3 = 1, a_4 = 0$ ) and  $(0, -1, 0, 1)$  (from  $a_3 = 0, a_4 = 1$ ). The corresponding functionals form a basis of  $U^0$ , and by Theorem 4.6.2,

$$U = \{\mathbf{x} \in \mathbb{R}^4 : x_1 - x_2 + x_3 = 0, \quad x_4 - x_2 = 0\}.$$

**Check.** Both equations hold at  $\mathbf{u}_1$ :  $1 - 1 + 0 = 0$  and  $1 - 1 = 0$ ; and at  $\mathbf{u}_2$ :  $0 - 1 + 1 = 0$  and  $1 - 1 = 0$ . The coefficient matrix  $\begin{pmatrix} 1 & -1 & 1 & 0 \\ 0 & -1 & 0 & 1 \end{pmatrix}$  has rank 2, so its null space has dimension  $4 - 2 = 2$ , and it contains the independent vectors  $\mathbf{u}_1, \mathbf{u}_2$ ; hence it equals  $U$  (Theorem 1.6.8). The implicit description says:  $U$  consists of the vectors with  $x_4 = x_2$  and  $x_1 + x_3 = x_2$ .

### 4.6.3 From spans to equations and back

The example is an instance of a general procedure. Both directions of the conversion are “compute a null space”, applied to transposed data.

*Algorithm. Parametric to implicit.* Input: vectors  $\mathbf{v}_1, \dots, \mathbf{v}_p \in F^n$  with  $U = \text{span}(\mathbf{v}_1, \dots, \mathbf{v}_p)$ .

1. Form  $\mathbf{M} \in M_{p \times n}(F)$  whose  $i$ -th **row** is  $\mathbf{v}_i^\top$ .
2. Row reduce  $\mathbf{M}$  and find a basis  $\mathbf{n}_1, \dots, \mathbf{n}_d$  of  $\text{nul}(\mathbf{M})$  as in Theorem 2.5.4.
3. Output the matrix  $\mathbf{A} \in M_{d \times n}(F)$  whose  $j$ -th row is  $\mathbf{n}_j^\top$ , that is, the equations  $\mathbf{n}_j^\top \mathbf{x} = 0$  for  $j = 1, \dots, d$ .

**Implicit to parametric.** Input:  $\mathbf{A} \in M_{m \times n}(F)$ . Output: a basis of  $\text{nul}(\mathbf{A})$ , found by row reduction as in Theorem 2.5.4.

The second half was proved correct in Chapter 2. The first half is new.

#### Theorem 4.6.3 — Correctness of the Conversion

Let  $\mathbf{v}_1, \dots, \mathbf{v}_p \in F^n$ ,  $U = \text{span}(\mathbf{v}_1, \dots, \mathbf{v}_p)$ , and let  $\mathbf{M}$  and  $\mathbf{A}$  be as in the algorithm. Then  $\text{nul}(\mathbf{A}) = U$ , and  $\mathbf{A}$  has exactly  $d = n - \dim U$  rows, which are linearly independent.

*Idea.* The only thing to see is what step 2 computes. The  $i$ -th entry of  $\mathbf{M}\mathbf{n}$  is  $\mathbf{v}_i^\top \mathbf{n} = \varphi_{\mathbf{n}}(\mathbf{v}_i)$ . So  $\mathbf{n} \in \text{nul}(\mathbf{M})$  says that the functional  $\varphi_{\mathbf{n}}$  vanishes at every  $\mathbf{v}_i$ , that is,  $\varphi_{\mathbf{n}} \in U^0$ . Step 2 therefore computes a basis of  $U^0$ , and the theorem of the previous subsection does the rest.

*Proof.* For  $\mathbf{n} \in F^n$ , the  $i$ -th entry of  $\mathbf{M}\mathbf{n}$  is  $\mathbf{v}_i^\top \mathbf{n} = \mathbf{n}^\top \mathbf{v}_i = \varphi_{\mathbf{n}}(\mathbf{v}_i)$ , where the middle equality holds because both sides equal  $\sum_j (\mathbf{v}_i)_j n_j$ . Hence

$$\mathbf{n} \in \text{nul}(\mathbf{M}) \iff \varphi_{\mathbf{n}}(\mathbf{v}_i) = 0 \text{ for all } i \iff \varphi_{\mathbf{n}} \in \{\mathbf{v}_1, \dots, \mathbf{v}_p\}^0 = U^0,$$

where the last equality is Theorem 4.2.3 (b). So the isomorphism  $\mathbf{n} \mapsto \varphi_{\mathbf{n}}$  maps  $\text{nul}(\mathbf{M})$  onto  $U^0$ . An isomorphism carries a basis to a basis (Theorem 3.5.2), so  $(\varphi_{\mathbf{n}_1}, \dots, \varphi_{\mathbf{n}_d})$  is a basis of  $U^0$ . In particular  $d = \dim U^0 = n - \dim U$  by Theorem 4.2.2, and the  $\mathbf{n}_j$ , hence the rows  $\mathbf{n}_j^\top$ , are independent. By Theorem 4.6.2 (a), the matrix  $\mathbf{A}$  with rows  $\mathbf{n}_1^\top, \dots, \mathbf{n}_d^\top$  satisfies  $\text{nul}(\mathbf{A}) = U$ . ■

The algorithm does not need the  $\mathbf{v}_i$  to be independent. Redundant spanning vectors only produce zero rows in the RREF of  $\mathbf{M}$ , and the count  $d = n - \text{rank } \mathbf{M}$  automatically uses  $\dim U = \text{rank } \mathbf{M}$ .

Running both halves in turn is a useful self-check. Starting from  $\mathbf{A}$  in Example (Equations for a plane in  $\mathbb{R}^4$ ) and computing  $\text{nul}(\mathbf{A})$  must return a basis of  $U$ ; starting from that basis and running the first half must return equations with the same solution set, though not necessarily the same equations.

**Warning.** Equations for a subspace are far from unique, and more equations do not mean a smaller solution set. In Example (Equations for a plane in  $\mathbb{R}^4$ ), the system  $x_1 - x_2 + x_3 = 0$ ,  $x_4 - x_2 = 0$ ,  $x_1 + x_3 - x_4 = 0$  has the same solution set  $U$ : the third equation is the sum of the first and the negative of the second, so it adds a functional already in  $U^0$ . Three equations in  $\mathbb{R}^4$  cut out a two-dimensional subspace here, not a one-dimensional one. Each new equation lowers the dimension by one **only if** its functional is not a combination of the previous ones. A theorem later in this section makes this precise.

## 4.6.4 Affine solution sets

An inhomogeneous equation  $\mathbf{a}^\top \mathbf{x} = b$  with  $\mathbf{a} \neq \mathbf{0}$  defines a set that is not a subspace when  $b \neq 0$ . It is a translate of the hyperplane  $H = \ker \varphi_{\mathbf{a}}$ : if  $\mathbf{p}$  is one solution, then by Theorem 2.1.3 the solutions are  $\mathbf{p} + H$ . We call a coset  $\mathbf{p} + H$  of a hyperplane  $H$  (Definition 3.9.2) an **affine hyperplane**. In  $\mathbb{R}^3$ , the plane  $x + y + z = 1$  is an affine hyperplane, parallel to the hyperplane  $x + y + z = 0$ .

A system  $\mathbf{A}\mathbf{x} = \mathbf{b}$  with rows  $\mathbf{a}_i^\top$  and right-hand sides  $b_i$  asks for the vectors lying on all the sets  $\{\mathbf{x} : \mathbf{a}_i^\top \mathbf{x} = b_i\}$  at once. Each such set is an affine hyperplane if  $\mathbf{a}_i \neq \mathbf{0}$ ; if  $\mathbf{a}_i = \mathbf{0}$ , it is all of  $F^n$  (when  $b_i = 0$ ) or empty (when  $b_i \neq 0$ ). So **the solution set of a linear system is an intersection of affine hyperplanes**, apart from these trivial rows. The duality theorems give the converse.

**Proposition 4.6.4 — Cosets as intersections of affine hyperplanes**

Let  $U$  be a subspace of  $F^n$  of dimension  $k < n$ , and let  $\mathbf{p} \in F^n$ . Let  $\mathbf{A} \in M_{(n-k) \times n}(F)$  be as in [Theorem 4.6.2](#) (a), with rows  $\mathbf{a}_1^\top, \dots, \mathbf{a}_{n-k}^\top$ . Then

$$\mathbf{p} + U = \{\mathbf{x} \in F^n : \mathbf{A}\mathbf{x} = \mathbf{A}\mathbf{p}\} = \bigcap_{i=1}^{n-k} \{\mathbf{x} : \mathbf{a}_i^\top \mathbf{x} = \mathbf{a}_i^\top \mathbf{p}\},$$

an intersection of  $n - k$  affine hyperplanes.

*Proof.* By [Theorem 4.6.2](#),  $\text{nul}(\mathbf{A}) = U$ . The vector  $\mathbf{p}$  is one solution of  $\mathbf{A}\mathbf{x} = \mathbf{A}\mathbf{p}$ , so by [Theorem 2.1.3](#) its solution set is  $\mathbf{p} + \text{nul}(\mathbf{A}) = \mathbf{p} + U$ . The  $i$ -th equation reads  $\mathbf{a}_i^\top \mathbf{x} = \mathbf{a}_i^\top \mathbf{p}$ . Each  $\varphi_{\mathbf{a}_i}$  is non-zero, because it belongs to a basis of  $U^0$ ; so  $\ker \varphi_{\mathbf{a}_i}$  is a hyperplane ([Theorem 4.5.3](#)), and  $\{\mathbf{x} : \mathbf{a}_i^\top \mathbf{x} = \mathbf{a}_i^\top \mathbf{p}\} = \mathbf{p} + \ker \varphi_{\mathbf{a}_i}$  is an affine hyperplane, again by [Theorem 2.1.3](#) applied to this one equation. ■

Unlike hyperplanes, which all contain  $\mathbf{0}$ , affine hyperplanes can be disjoint:  $x_1 + x_2 = 0$  and  $x_1 + x_2 = 1$  are parallel lines in  $\mathbb{R}^2$ . That is the geometric picture of an inconsistent system.

**4.6.5 The row space annihilates the null space**

In Chapter 2, [Theorem 2.5.7](#) said  $\text{rank } \mathbf{A} + \text{nullity } \mathbf{A} = n$ , and its proof counted pivot and free columns in the RREF. The duality of this chapter explains the same equation without elimination: the rows of  $\mathbf{A}$  are exactly the equations satisfied by the null space.

**Theorem 4.6.5 — The Row Space Is the Annihilator of the Null Space**

Let  $\mathbf{A} \in M_{m \times n}(F)$ . Identifying a row vector  $\mathbf{a}^\top \in M_{1 \times n}(F)$  with the functional  $\varphi_{\mathbf{a}} \in (F^n)^*$ ,

$$\{\varphi_{\mathbf{a}} : \mathbf{a}^\top \in \text{row}(\mathbf{A})\} = \text{nul}(\mathbf{A})^0.$$

Consequently  $\dim \text{row}(\mathbf{A}) + \dim \text{nul}(\mathbf{A}) = n$ .

*Idea.* Let  $T = T_{\mathbf{A}} : F^n \rightarrow F^m$ . Its kernel is  $\text{nul}(\mathbf{A})$ , and [Theorem 4.4.3](#) says  $\text{im } T' = (\ker T)^0$ . So it is enough to see that  $\text{im } T'$  is the row space. Pull back a functional  $\varphi_{\mathbf{y}}$  on  $F^m$ : the result is  $\mathbf{x} \mapsto \mathbf{y}^\top \mathbf{A}\mathbf{x}$ , which is multiplication by the row  $\mathbf{y}^\top \mathbf{A}$ , the combination of the rows of  $\mathbf{A}$  with coefficients  $y_1, \dots, y_m$ . The transpose of Chapter 0 appears exactly here:  $\mathbf{y}^\top \mathbf{A} = (\mathbf{A}^\top \mathbf{y})^\top$ .

*Proof.* Let  $T = T_{\mathbf{A}} : F^n \rightarrow F^m$ ,  $T(\mathbf{x}) = \mathbf{A}\mathbf{x}$ , so that  $\ker T = \text{nul}(\mathbf{A})$ . Every functional on  $F^m$  is  $\varphi_{\mathbf{y}}$  for a unique  $\mathbf{y} \in F^m$ , as recalled at the start of this section. By [Definition 4.4.1](#), associativity of matrix multiplication, and [Theorem 0.9.5](#),

$$\begin{aligned} T'(\varphi_{\mathbf{y}})(\mathbf{x}) &= \varphi_{\mathbf{y}}(\mathbf{A}\mathbf{x}) = \mathbf{y}^\top (\mathbf{A}\mathbf{x}) = (\mathbf{y}^\top \mathbf{A})\mathbf{x} \\ &= (\mathbf{A}^\top \mathbf{y})^\top \mathbf{x} = \varphi_{\mathbf{A}^\top \mathbf{y}}(\mathbf{x}) \quad (\mathbf{x} \in F^n), \end{aligned}$$

so  $T'(\varphi_{\mathbf{y}}) = \varphi_{\mathbf{A}^\top \mathbf{y}}$ . The columns of  $\mathbf{A}^\top$  are  $\mathbf{a}_1, \dots, \mathbf{a}_m$ , where  $\mathbf{a}_i^\top$  are the rows of  $\mathbf{A}$ , so  $\mathbf{A}^\top \mathbf{y} = y_1 \mathbf{a}_1 + \dots + y_m \mathbf{a}_m$  by Theorem 0.9.3. As  $\mathbf{y}$  runs over  $F^m$ , these are all the combinations of  $\mathbf{a}_1, \dots, \mathbf{a}_m$ , and since transposing respects combinations (Theorem 0.9.5), they are exactly the vectors  $\mathbf{a}$  with  $\mathbf{a}^\top \in \text{row}(\mathbf{A})$  (Definition 2.5.2). Hence

$$\text{im } T' = \{\varphi_{\mathbf{a}} : \mathbf{a}^\top \in \text{row}(\mathbf{A})\}.$$

By Theorem 4.4.3,  $\text{im } T' = (\ker T)^0 = \text{nul}(\mathbf{A})^0$ , which is the first claim.

For the count,  $\mathbf{a}^\top \mapsto \varphi_{\mathbf{a}}$  is an isomorphism  $M_{1 \times n}(F) \rightarrow (F^n)^*$  (transposition followed by  $\mathbf{a} \mapsto \varphi_{\mathbf{a}}$ ), so  $\dim \text{row}(\mathbf{A}) = \dim \text{nul}(\mathbf{A})^0 = n - \dim \text{nul}(\mathbf{A})$  by Theorem 4.2.2. ■

The count is Rank–Nullity for matrices once more, now as a statement about annihilators:  $\text{rank } \mathbf{A} = \dim \text{row}(\mathbf{A})$  counts independent equations and nullity  $\mathbf{A}$  counts free parameters. It also says that the given equations generate **all** the equations satisfied by  $\text{nul}(\mathbf{A})$ : any linear equation that holds on the solution set of  $\mathbf{A}\mathbf{x} = \mathbf{0}$  is a combination of the rows of  $\mathbf{A}$ .

### Corollary 4.6.6 — Same solutions, same row space

Let  $\mathbf{A} \in M_{m \times n}(F)$  and  $\mathbf{B} \in M_{p \times n}(F)$ . Then  $\text{nul}(\mathbf{A}) = \text{nul}(\mathbf{B})$  if and only if  $\text{row}(\mathbf{A}) = \text{row}(\mathbf{B})$ .

*Proof.* ( $\Leftarrow$ ) If  $\text{row}(\mathbf{A}) = \text{row}(\mathbf{B})$ , then  $\text{nul}(\mathbf{A})^0 = \text{nul}(\mathbf{B})^0$  by Theorem 4.6.5, so  $\text{nul}(\mathbf{A}) = \text{nul}(\mathbf{B})$  by Corollary 4.3.5.

( $\Rightarrow$ ) If  $\text{nul}(\mathbf{A}) = \text{nul}(\mathbf{B})$ , then  $\{\varphi_{\mathbf{a}} : \mathbf{a}^\top \in \text{row}(\mathbf{A})\} = \text{nul}(\mathbf{A})^0 = \{\varphi_{\mathbf{a}} : \mathbf{a}^\top \in \text{row}(\mathbf{B})\}$  by Theorem 4.6.5. Since  $\mathbf{a}^\top \mapsto \varphi_{\mathbf{a}}$  is injective,  $\text{row}(\mathbf{A}) = \text{row}(\mathbf{B})$ . ■

Chapter 2 showed that row operations preserve the null space (Theorem 2.5.1). The corollary gives a converse in terms of row spaces: two systems in the same unknowns have the same solutions **exactly** when each equation of one is a combination of the equations of the other.

## 4.6.6 Sums and intersections by duality

Parametric and implicit descriptions are good at different things. A **sum** is easy parametrically:  $U + W$  is spanned by a spanning list of  $U$  followed by one of  $W$  (Proposition 1.7.2). An **intersection** is easy implicitly: if  $U = \text{nul}(\mathbf{A})$  and  $W = \text{nul}(\mathbf{B})$ , then  $\mathbf{x} \in U \cap W$  exactly when  $\mathbf{x}$  satisfies both systems, so

$$U \cap W = \text{nul} \begin{pmatrix} \mathbf{A} \\ \mathbf{B} \end{pmatrix},$$

the null space of the matrix with the rows of  $\mathbf{A}$  above the rows of  $\mathbf{B}$ . Duality explains why each is easy where it is. The rows of the stacked matrix span  $\text{row}(\mathbf{A}) + \text{row}(\mathbf{B})$ , which corresponds to  $U^0 + W^0 = (U \cap W)^0$  by Theorem 4.6.5 and Theorem 4.2.3 (d); and  $(U + W)^0 = U^0 \cap W^0$  says that the equations of a sum are the equations common to both pieces.

**Example. (An intersection computed from equations)**

In  $\mathbb{R}^4$ , let  $U = \text{span}((1, 0, 1, 2), (0, 1, 1, 1))$  and  $W = \text{span}((1, 1, 0, 1), (1, 3, 0, 1))$ , as in [Example \(A basis of an intersection\)](#). Find  $U \cap W$  by first finding equations for  $U$  and for  $W$ .

*Solution. Equations for  $U$ .* A row  $(a_1, a_2, a_3, a_4)$  annihilates both spanning vectors when  $a_1 + a_3 + 2a_4 = 0$  and  $a_2 + a_3 + a_4 = 0$ . This system is already in RREF, with free  $a_3, a_4$ ; the basic solutions are  $(-1, -1, 1, 0)$  and  $(-2, -1, 0, 1)$ . Changing signs, which does not change the kernels,

$$U = \{x_1 + x_2 - x_3 = 0, \quad 2x_1 + x_2 - x_4 = 0\}.$$

*Equations for  $W$ .* Here  $a_1 + a_2 + a_4 = 0$  and  $a_1 + 3a_2 + a_4 = 0$ . Subtracting gives  $a_2 = 0$ , and then  $a_1 = -a_4$ , with  $a_3$  free. The basic solutions are  $(0, 0, 1, 0)$  and  $(-1, 0, 0, 1)$ , so

$$W = \{x_3 = 0, \quad x_4 - x_1 = 0\}.$$

*Intersection.* Stack the four equations. From  $W$ :  $x_3 = 0$  and  $x_4 = x_1$ . Then  $x_1 + x_2 - x_3 = 0$  gives  $x_2 = -x_1$ , and the last equation  $2x_1 + x_2 - x_4 = 2x_1 - x_1 - x_1 = 0$  holds automatically. So

$$U \cap W = \{(t, -t, 0, t) : t \in \mathbb{R}\} = \text{span}((1, -1, 0, 1)),$$

in agreement with Chapter 2.

The redundancy of the fourth equation is not a coincidence. The four rows span  $U^0 + W^0 = (U \cap W)^0$ , which has dimension  $4 - 1 = 3$ , so exactly one of the four equations must be a consequence of the others.

**Contrast with Chapter 2.** There we put the four spanning vectors as **columns** of one matrix and read the intersection off a null vector, a relation  $\mathbf{u}_1 - \mathbf{u}_2 = 2\mathbf{w}_1 - \mathbf{w}_2$ . Here we computed two small null spaces (the equations) and then one more (the intersection). The equation method pays off when the subspaces are given by equations in the first place, or when several intersections with the same  $U$  are needed: the equations of  $U$  are computed once.

## 4.6.7 Independent functionals

The warning above said that  $k$  equations cut out codimension  $k$  only when they are independent. We now make that precise, in any dimension. Given  $\varphi_1, \dots, \varphi_k \in V^*$ , bundle them into one linear map

$$\Phi: V \rightarrow F^k, \quad \Phi(\mathbf{v}) = (\varphi_1(\mathbf{v}), \dots, \varphi_k(\mathbf{v})),$$

which is linear because each entry is. Its kernel is  $\ker \varphi_1 \cap \dots \cap \ker \varphi_k$ , the common solution set of the  $k$  equations.

### Theorem 4.6.7 — Independent Functionals

Let  $V$  be a vector space over  $F$ , of any dimension, and let  $\varphi_1, \dots, \varphi_k \in V^*$ . The following are equivalent:



- (a)  $(\varphi_1, \dots, \varphi_k)$  is linearly independent in  $V^*$ ;
- (b)  $\Phi: V \rightarrow F^k$  is surjective;
- (c)  $\ker \varphi_1 \cap \dots \cap \ker \varphi_k$  has codimension exactly  $k$ .

*Idea.* A dependence  $\sum c_i \varphi_i = 0$  says that  $\sum c_i y_i = 0$  for every  $\mathbf{y} = \Phi(\mathbf{v})$  in the image: the image lies in the kernel of a non-zero functional on  $F^k$ . Conversely, if the image is a proper subspace of  $F^k$ , some non-zero functional on  $F^k$  kills it, and its coefficients give a dependence. The First Isomorphism Theorem turns “surjective” into “codimension  $k$ ”.

*Proof.*

- (a)  $\Rightarrow$  (b). Suppose  $\Phi$  is not surjective. Then  $\text{im } \Phi$  is a proper subspace of  $F^k$ , so  $\dim \text{im } \Phi < k$  and, by Theorem 4.2.2,  $(\text{im } \Phi)^0 \neq \{0\}$ . Take a non-zero functional in it; as at the start of this section it is  $\varphi_{\mathbf{c}}, \mathbf{y} \mapsto c_1 y_1 + \dots + c_k y_k$ , for a non-zero  $\mathbf{c} \in F^k$ . For every  $\mathbf{v} \in V$ ,  $\Phi(\mathbf{v}) \in \text{im } \Phi$ , so

$$(c_1 \varphi_1 + \dots + c_k \varphi_k)(\mathbf{v}) = \varphi_{\mathbf{c}}(\Phi(\mathbf{v})) = 0.$$

Thus  $c_1 \varphi_1 + \dots + c_k \varphi_k = 0$  with  $\mathbf{c} \neq \mathbf{0}$ , and the list is dependent. This proves the contrapositive.

- (b)  $\Rightarrow$  (a). Suppose  $\Phi$  is surjective and  $c_1 \varphi_1 + \dots + c_k \varphi_k = 0$ . For each  $j$ , choose  $\mathbf{v}_j$  with  $\Phi(\mathbf{v}_j) = \mathbf{e}_j$ , that is,  $\varphi_i(\mathbf{v}_j) = \delta_{ij}$ . Evaluating the relation at  $\mathbf{v}_j$  gives  $c_j = 0$ . So the list is independent.
- (c)  $\Leftrightarrow$  (c). By Theorem 3.9.7,  $V/\ker \Phi \cong \text{im } \Phi$ , and  $\ker \Phi = \bigcap_i \ker \varphi_i$ . So the codimension of  $\bigcap_i \ker \varphi_i$  is  $\dim \text{im } \Phi$ , which equals  $k$  exactly when  $\text{im } \Phi = F^k$  (Theorem 1.6.8). ■

For  $V = F^n$  and  $\varphi_i = \varphi_{\mathbf{a}_i}$ , the map  $\Phi$  is  $T_{\mathbf{A}}$ , where  $\mathbf{A}$  has rows  $\mathbf{a}_i^{\top}$ . The theorem then says: the rows of  $\mathbf{A}$  are independent if and only if  $\mathbf{A}\mathbf{x} = \mathbf{b}$  is solvable for **every**  $\mathbf{b} \in F^k$ . Independent equations never contradict each other.

*Quick check.* In  $\mathbb{R}^3$ , consider  $\varphi_1 = x+y$ ,  $\varphi_2 = y+z$ ,  $\varphi_3 = x-z$ . What is the codimension of  $\ker \varphi_1 \cap \ker \varphi_2 \cap \ker \varphi_3$ ? Is the system  $x+y=1$ ,  $y+z=1$ ,  $x-z=1$  solvable?

*Solution.*  $\varphi_3 = \varphi_1 - \varphi_2$ , so the list is dependent, while  $(\varphi_1, \varphi_2)$  is independent (look at the coefficients of  $x$  and  $z$ ). The three kernels intersect in  $\ker \varphi_1 \cap \ker \varphi_2$ , of codimension 2 by Theorem 4.6.7, so the intersection is a line, not  $\{0\}$ . The system is not solvable: any solution would give  $1 = x - z = (x + y) - (y + z) = 1 - 1 = 0$ . This is (b) failing:  $(1, 1, 1) \notin \text{im } \Phi$ .

## 4.6.8 Summary and transfer

The chapter has built a dictionary. On the left is a finite-dimensional space  $V$  of dimension  $n$ , or a linear map  $T: V \rightarrow W$  between finite-dimensional spaces; on the right is its mirror image in the dual. Every entry has been proved in this chapter, and every arrow reverses inclusions or directions.

| In $V$ (vectors)                                          | In $V^*$ (measurements)                                                  |
|-----------------------------------------------------------|--------------------------------------------------------------------------|
| a subspace $U$ , $\dim U = k$                             | its annihilator $U^0$ , $\dim U^0 = n - k$                               |
| $U \subseteq W$                                           | $W^0 \subseteq U^0$                                                      |
| $U + W$                                                   | $U^0 \cap W^0$                                                           |
| $U \cap W$                                                | $U^0 + W^0$                                                              |
| $U$ itself, recovered as $U^{00}$                         | $U^0$ determines $U$                                                     |
| the quotient $V/U$                                        | the subspace $U^0 \cong (V/U)^*$                                         |
| the subspace $U$                                          | the quotient $V^*/U^0 \cong U^*$                                         |
| a hyperplane $H$                                          | a non-zero functional, up to scale                                       |
| a spanning list (parametric form)                         | a list of equations (implicit form)                                      |
| dimension (free parameters)                               | codimension (independent equations)                                      |
| a basis $(\mathbf{v}_1, \dots, \mathbf{v}_n)$             | its dual basis $(\varphi_1, \dots, \varphi_n)$                           |
| $T: V \rightarrow W$                                      | $T': W^* \rightarrow V^*$ , backwards                                    |
| $T$ injective                                             | $T'$ surjective                                                          |
| $T$ surjective                                            | $T'$ injective                                                           |
| $\ker T$                                                  | $\operatorname{im} T' = (\ker T)^0$                                      |
| $\operatorname{im} T$                                     | $\ker T' = (\operatorname{im} T)^0$                                      |
| $\operatorname{rank} T$                                   | $\operatorname{rank} T' = \operatorname{rank} T$                         |
| the matrix $\mathbf{A} = [T]_{\mathcal{B}}^{\mathcal{C}}$ | the transpose $\mathbf{A}^\top = [T']_{\mathcal{C}^*}^{\mathcal{B}^*}$   |
| $\operatorname{nul}(\mathbf{A})$                          | $\operatorname{row}(\mathbf{A})$ , as $\operatorname{nul}(\mathbf{A})^0$ |
| $V$                                                       | $V^{**}$ , naturally isomorphic via $\operatorname{ev}_V$                |

Three moves make the dictionary usable, and they transfer to any situation with a pairing between two spaces.

- **Test against functionals.** To show a vector is zero, or lies in a subspace, check that enough functionals vanish on it ([Theorem 4.1.5](#), [Lemma 4.6.1](#)).
- **Pass to the dual and count.** A question about a subspace can be translated into one about its annihilator, where  $\dim U + \dim U^0 = n$  often settles it. The third proof of row rank = column rank ([Corollary 4.4.4](#)) and the count in [Theorem 4.6.5](#) both work this way.
- **Prefer the natural map.** Isomorphisms such as  $V \cong V^{**}$ ,  $(V/U)^* \cong U^0$  and  $U^* \cong V^*/U^0$  are defined without a basis, so they can be used in any coordinates and survive changes of basis. Isomorphisms such as  $V \cong V^*$  depend on a choice, and a proof that uses one must say which.

In Chapter 10, an inner product will give a **chosen** isomorphism  $V \cong V^*$ , and the annihilator  $U^0$  will reappear inside  $V$  itself as the orthogonal complement. The dictionary above is the part of that story that needs no inner product.

## 4.6.9 Exercises

### 4.6.9.1 A. Check your understanding

#### Exercise A1.

- Explain how a row vector  $\mathbf{a}^\top$  (with  $\mathbf{a} \in F^n$ ) defines a linear functional on  $F^n$ , and why every functional arises this way.
- What is the least number of homogeneous linear equations whose solution set is a given 3-dimensional subspace of  $F^7$ ?

- (c) True or false: every system of three homogeneous equations in  $\mathbb{R}^5$  has a solution set of dimension 2. Justify your answer.
- (d) State the relation between  $\text{row}(\mathbf{A})$  and  $\text{nul}(\mathbf{A})$  in terms of annihilators.
- (e) True or false: if  $\mathbf{A}$  and  $\mathbf{B}$  have  $n$  columns and  $\text{nul}(\mathbf{A}) = \text{nul}(\mathbf{B})$ , then  $\mathbf{A}$  and  $\mathbf{B}$  have the same number of rows. Justify your answer.
- (f) Name the dual counterparts of: a sum of subspaces; an injective linear map; the matrix of a linear map.

*Solution.*

- (a)  $\varphi_{\mathbf{a}}(\mathbf{x}) = \mathbf{a}^\top \mathbf{x}$  is linear by the rules of matrix multiplication (Theorem 0.9.4). A functional  $\varphi$  equals  $\varphi_{\mathbf{a}}$  with  $a_j = \varphi(\mathbf{e}_j)$ , because both agree on the standard basis (Theorem 3.1.5); this is Theorem 4.1.1.
- (b)  $7 - 3 = 4$ , by Theorem 4.6.2.
- (c) False. The equations may be dependent:  $x_1 = 0$  written three times has a solution set of dimension 4. The dimension is  $5 - \text{rank } \mathbf{A}$ , which is 2 only when the three rows are independent.
- (d) Identifying rows with functionals,  $\text{row}(\mathbf{A}) = \text{nul}(\mathbf{A})^0$  (Theorem 4.6.5).
- (e) False. By Corollary 4.6.6 they have the same row space, but not necessarily the same number of rows:  $\mathbf{A} = \begin{pmatrix} 1 & 0 \end{pmatrix}$  and  $\mathbf{B} = \begin{pmatrix} 1 & 0 \\ 2 & 0 \end{pmatrix}$  both have null space  $\text{span}(\mathbf{e}_2)$ .
- (f) An intersection of annihilators,  $(U + W)^0 = U^0 \cap W^0$ ; a surjective dual map; the transpose matrix.

#### 4.6.9.2 B. Practice

##### Exercise B1: Equations for a span.

Let  $U = \text{span}((1, 2, 0, -1), (2, 1, 1, 0), (3, 3, 1, -1)) \subseteq \mathbb{R}^4$ .

- (a) Find  $\dim U$ .
- (b) Find a system of homogeneous linear equations with solution set  $U$ , using as few equations as possible.
- (c) Hence write the affine plane  $(1, 0, 0, 0) + U$  as the solution set of a linear system.

*Solution.*

- (a) The third vector is the sum of the first two, so  $U = \text{span}((1, 2, 0, -1), (2, 1, 1, 0))$  by Theorem 1.3.5. These two vectors are independent, since neither is a multiple of the other, so they form a basis of  $U$  and  $\dim U = 2$ .
- (b) By Theorem 4.6.3, we need a basis of  $\text{nul}(\mathbf{M})$ , where  $\mathbf{M}$  has the three vectors as rows. Row reduction gives

$$\mathbf{M} = \begin{pmatrix} 1 & 2 & 0 & -1 \\ 2 & 1 & 1 & 0 \\ 3 & 3 & 1 & -1 \end{pmatrix} \longrightarrow \begin{pmatrix} 1 & 0 & \frac{2}{3} & \frac{1}{3} \\ 0 & 1 & -\frac{1}{3} & -\frac{2}{3} \\ 0 & 0 & 0 & 0 \end{pmatrix}.$$

The free variables are the third and fourth. Setting them to  $(3, 0)$  and  $(0, 3)$  to

clear denominators gives the null vectors  $(-2, 1, 3, 0)$  and  $(-1, 2, 0, 3)$ . So

$$U = \{\mathbf{x} \in \mathbb{R}^4 : -2x_1 + x_2 + 3x_3 = 0, \quad -x_1 + 2x_2 + 3x_4 = 0\}.$$

Check on the first two spanning vectors, which suffice since they span  $U$ :  $-2 + 2 + 0 = 0$ ,  $-1 + 4 - 3 = 0$ ;  $-4 + 1 + 3 = 0$ ,  $-2 + 2 + 0 = 0$ . Two equations is the minimum, by [Theorem 4.6.2](#) (b).

- (c) By [Proposition 4.6.4](#), replace each right-hand side by the value of the left-hand side at  $(1, 0, 0, 0)$ :

$$(1, 0, 0, 0) + U = \{\mathbf{x} : -2x_1 + x_2 + 3x_3 = -2, \quad -x_1 + 2x_2 + 3x_4 = -1\}.$$

### Exercise B2: A spanning list for a system.

Let  $U = \{\mathbf{x} \in \mathbb{R}^4 : x_1 + x_2 - x_3 + 2x_4 = 0, \quad 2x_1 + 2x_2 + x_3 + x_4 = 0\}$ .

- Find a basis of  $U$ .
- Run the parametric-to-implicit algorithm on your basis, and check that the resulting equations have the same solution set as the given ones, by comparing row spaces.

*Solution.*

- (a) Row reduce the coefficient matrix: subtract twice row 1 from row 2 to get  $(0, 0, 3, -3)$ , divide by 3, then add row 2 to row 1:

$$\begin{pmatrix} 1 & 1 & -1 & 2 \\ 2 & 2 & 1 & 1 \end{pmatrix} \longrightarrow \begin{pmatrix} 1 & 1 & 0 & 1 \\ 0 & 0 & 1 & -1 \end{pmatrix}.$$

The pivots are in columns 1 and 3, and  $x_2, x_4$  are free. By [Theorem 2.5.4](#), a basis is  $\mathbf{n}_1 = (-1, 1, 0, 0)$  and  $\mathbf{n}_2 = (-1, 0, 1, 1)$ .

- (b) Put  $\mathbf{n}_1^\top, \mathbf{n}_2^\top$  as the rows of  $\mathbf{M}$  and solve  $\mathbf{M}\mathbf{a} = \mathbf{0}$ :  $-a_1 + a_2 = 0$  and  $-a_1 + a_3 + a_4 = 0$ . So  $a_2 = a_1$  and  $a_3 = a_1 - a_4$ , with  $a_1, a_4$  free; the basic solutions are  $(1, 1, 1, 0)$  and  $(0, 0, -1, 1)$ . The new system is  $x_1 + x_2 + x_3 = 0$ ,  $-x_3 + x_4 = 0$ . Its rows lie in the row space of the original matrix, which equals the row space of the RREF ([Theorem 2.5.1](#)), since  $(1, 1, 1, 0) = (1, 1, 0, 1) + (0, 0, 1, -1)$  and  $(0, 0, -1, 1) = -(0, 0, 1, -1)$ . Both row spaces are two-dimensional, so they are equal ([Theorem 1.6.8](#)), and the two systems have the same solution set by [Corollary 4.6.6](#).

### Exercise B3: Two planes in $\mathbb{R}^4$ .

Let  $U = \text{span}((1, 0, 0, 1), (0, 1, 1, 0))$  and  $W = \text{span}((1, 1, 0, 0), (0, 0, 1, 1))$  in  $\mathbb{R}^4$ .

- Find two equations for  $U$  and two for  $W$ .
- Use them to find  $U \cap W$ .
- Find a single equation for  $U + W$ , using  $(U + W)^0 = U^0 \cap W^0$ .

*Solution.*

- (a) Each pair is independent, so each plane has dimension 2 and needs two equations. The functionals  $x_1 - x_4$  and  $x_2 - x_3$  vanish at  $(1, 0, 0, 1)$  and  $(0, 1, 1, 0)$ , and they are independent; since  $\dim U^0 = 2$  (Theorem 4.2.2), they form a basis of  $U^0$ . So  $U = \{x_1 = x_4, x_2 = x_3\}$  by Theorem 4.6.2. In the same way  $x_1 - x_2$  and  $x_3 - x_4$  form a basis of  $W^0$ , and  $W = \{x_1 = x_2, x_3 = x_4\}$ .
- (b)  $U \cap W$  is the solution set of all four equations:  $x_1 = x_4, x_2 = x_3, x_1 = x_2, x_3 = x_4$ . They force  $x_1 = x_2 = x_3 = x_4$ , so  $U \cap W = \text{span}((1, 1, 1, 1))$ .
- (c) A functional in  $U^0 \cap W^0$  has the form  $a(x_1 - x_4) + b(x_2 - x_3) = c(x_1 - x_2) + d(x_3 - x_4)$ . Comparing coefficients of  $x_1, x_2, x_3, x_4$ :  $a = c, b = -c, -b = d, -a = -d$ . So  $a = c = d$  and  $b = -a$ , and  $U^0 \cap W^0 = \text{span}(x_1 - x_2 + x_3 - x_4)$ . By Theorem 4.2.3 (c) and Lemma 4.6.1,

$$U + W = \{\mathbf{x} : x_1 - x_2 + x_3 - x_4 = 0\},$$

a hyperplane. Consistently,  $\dim(U + W) = 2 + 2 - 1 = 3$  by Theorem 1.7.3.

#### 4.6.9.3 C. Going deeper

##### Exercise C1: Avoiding finitely many subspaces.

Let  $F$  be an **infinite** field,  $V$  a vector space over  $F$  with  $1 \leq \dim V = n < \infty$ , and  $U_1, \dots, U_k$  proper subspaces of  $V$ .

- (a) Prove that for each  $i$  there is a non-zero  $\varphi_i \in V^*$  with  $U_i \subseteq \ker \varphi_i$ .
- (b) Let  $(\mathbf{b}_1, \dots, \mathbf{b}_n)$  be a basis of  $V$ , and for  $t \in F$  put  $\mathbf{v}(t) = \mathbf{b}_1 + t\mathbf{b}_2 + t^2\mathbf{b}_3 + \dots + t^{n-1}\mathbf{b}_n$ . Show that  $t \mapsto \varphi_i(\mathbf{v}(t))$  is given by a non-zero polynomial of degree at most  $n - 1$ .
- (c) Deduce that some vector of  $V$  lies in none of  $U_1, \dots, U_k$ . In other words,  $V$  is not a union of finitely many proper subspaces.
- (d) Show that the hypothesis that  $F$  is infinite cannot be dropped: over  $\mathbb{F}_q$ , the space  $\mathbb{F}_q^2$  is the union of  $q + 1$  proper subspaces.

*Hint: for (c), use Lemma 2.7.2.*

*Solution.*

- (a) Since  $U_i \neq V$ , pick  $\mathbf{v} \notin U_i$ . By Lemma 4.6.1 there is  $\varphi_i \in U_i^0$  with  $\varphi_i(\mathbf{v}) \neq 0$ . Then  $\varphi_i \neq 0$  and  $U_i \subseteq \ker \varphi_i$ .
- (b) By linearity,  $\varphi_i(\mathbf{v}(t)) = \sum_{j=1}^n \varphi_i(\mathbf{b}_j) t^{j-1}$ , which is the value at  $t$  of the polynomial  $p_i(x) = \sum_{j=1}^n \varphi_i(\mathbf{b}_j) x^{j-1}$  of degree at most  $n - 1$ . Its coefficients are  $\varphi_i(\mathbf{b}_1), \dots, \varphi_i(\mathbf{b}_n)$ , which are not all zero, since otherwise  $\varphi_i$  would vanish on a basis and be zero (Theorem 3.1.5). So  $p_i \neq 0$ .
- (c) By Lemma 2.7.2, each  $p_i$  has at most  $n - 1$  roots in  $F$ , so together they have at most  $k(n - 1)$  roots. Since  $F$  is infinite, there is  $t_0 \in F$  that is a root of none of them. Then  $\varphi_i(\mathbf{v}(t_0)) = p_i(t_0) \neq 0$ , so  $\mathbf{v}(t_0) \notin \ker \varphi_i \supseteq U_i$ , for every  $i$ .
- (d)  $\mathbb{F}_q^2$  has  $q^2 - 1$  non-zero vectors, and each line through  $\mathbf{0}$  contains  $q - 1$  of them (the non-zero multiples of one vector). Two distinct lines meet only in  $\mathbf{0}$ , so there

are  $(q^2 - 1)/(q - 1) = q + 1$  lines, and every non-zero vector lies on one of them. Hence  $\mathbb{F}_q^2$  is the union of its  $q + 1$  lines, each a proper subspace. For  $q = 2$  these are  $\text{span}((1, 0))$ ,  $\text{span}((0, 1))$  and  $\text{span}((1, 1))$ , as in [Exercise \(C1: Covering a Space by Subspaces\)](#). The proof of (c) breaks exactly where it used infinitely many  $t$ : here  $k(n - 1) = q + 1$  exceeds  $|F| = q$ .

### Exercise C2: The least number of equations for a system.

Let  $\mathbf{A} \in M_{m \times n}(F)$  and  $\mathbf{b} \in F^m$ , and suppose the solution set  $S$  of  $\mathbf{A}\mathbf{x} = \mathbf{b}$  is non-empty.

- (a) Prove that if  $\mathbf{B} \in M_{p \times n}(F)$ ,  $\mathbf{c} \in F^p$  and  $\mathbf{B}\mathbf{x} = \mathbf{c}$  has the same solution set  $S$ , then  $\text{nul}(\mathbf{B}) = \text{nul}(\mathbf{A})$ .
- (b) Deduce that every system with solution set  $S$  has at least  $\text{rank } \mathbf{A}$  equations, and that some system with solution set  $S$  has exactly  $\text{rank } \mathbf{A}$  equations.
- (c) Show by an example that (b) fails without the hypothesis  $S \neq \emptyset$ .

*Solution.*

- (a) Fix  $\mathbf{p} \in S$ . By [Theorem 2.1.3](#),  $S = \mathbf{p} + \text{nul}(\mathbf{A})$  and also  $S = \mathbf{p} + \text{nul}(\mathbf{B})$ . For  $\mathbf{h} \in \text{nul}(\mathbf{A})$ ,  $\mathbf{p} + \mathbf{h} \in S = \mathbf{p} + \text{nul}(\mathbf{B})$ , so  $\mathbf{h} \in \text{nul}(\mathbf{B})$ . Swapping the roles of  $\mathbf{A}$  and  $\mathbf{B}$  gives the reverse inclusion, so  $\text{nul}(\mathbf{A}) = \text{nul}(\mathbf{B})$ .
- (b) Let  $k = \dim \text{nul}(\mathbf{A}) = n - \text{rank } \mathbf{A}$  ([Theorem 2.5.7](#)). If  $\mathbf{B}\mathbf{x} = \mathbf{c}$  with  $p$  equations has solution set  $S$ , then  $\text{nul}(\mathbf{B}) = \text{nul}(\mathbf{A})$  by (a), so  $p \geq n - k = \text{rank } \mathbf{A}$  by [Theorem 4.6.2](#) (b). Conversely, take  $\mathbf{A}_0 \in M_{(n-k) \times n}(F)$  with  $\text{nul}(\mathbf{A}_0) = \text{nul}(\mathbf{A})$ , which exists by [Theorem 4.6.2](#) (a) (if  $k = n$ , take the empty system, whose solution set is  $F^n = S$ ). By [Proposition 4.6.4](#), or directly by [Theorem 2.1.3](#), the system  $\mathbf{A}_0\mathbf{x} = \mathbf{A}_0\mathbf{p}$  has solution set  $\mathbf{p} + \text{nul}(\mathbf{A}) = S$ , and it has  $n - k = \text{rank } \mathbf{A}$  equations.
- (c) Over any field,  $0x_1 = 1$  (in  $F^1$ ) has empty solution set and coefficient matrix of rank 0, yet no system with 0 equations has empty solution set, since the empty system is solved by every vector. So at least one equation is needed, which is more than  $\text{rank } \mathbf{A} = 0$ .

# Chapter 5

## Polynomials

Integers and polynomials share one arithmetic. Both can be divided with a remainder, and from that single tool everything else follows: greatest common divisors, Bézout identities, unique factorization into primes or irreducibles, and the Chinese Remainder Theorem. This chapter develops that arithmetic for  $F[x]$  in full, because from Chapter 8 on the structure of an operator is read off from polynomials: which polynomials kill it, how they factor, and which of their factors are coprime.

Along the way we keep two promises made in Chapter 0: we prove division with remainder, stated there only for orientation, and the Fundamental Theorem of Algebra, that every non-constant complex polynomial has a root. We also prove that polynomials factor uniquely into irreducibles, just as integers factor uniquely into primes. The chapter ends by plugging an operator into a polynomial, where the arithmetic of  $F[x]$  turns into a splitting of the space into pieces.

**What you need.** From Chapter 0, induction and well-ordering ([Theorem 0.2.10](#), [Theorem 0.2.11](#)), divisibility of integers and Bézout's lemma ([Definition 0.2.2](#), [Lemma 0.7.2](#)), the fields  $\mathbb{F}_p$  ([Theorem 0.7.3](#)), polynomials, degree and evaluation ([Definition 0.8.1](#), [Definition 0.8.3](#), [Theorem 0.8.3](#), [Theorem 0.8.5](#)) and complex numbers ([Theorem 0.6.6](#)). From Chapter 2, the factor theorem and root bound for linear factors ([Lemma 2.7.1](#), [Lemma 2.7.2](#)) and interpolation ([Theorem 2.7.3](#)). From Chapter 3, polynomials of an operator ([Definition 3.4.2](#)), quotient spaces and the First Isomorphism Theorem ([Definition 3.9.3](#), [Theorem 3.9.7](#)). From Chapter 4, dual bases ([Theorem 4.1.3](#)).

### 5.0.1 Roadmap

- **Division with remainder.** Long division always works over a field, with a unique quotient and remainder; the remainder and factor theorems in general.
- **Ideals and greatest common divisors.** Every ideal of  $F[x]$  is generated by one polynomial; gcds, Bézout identities and the Euclidean algorithm.
- **Irreducible polynomials and unique factorization.** The primes of  $F[x]$ , and why factorization into them is unique.
- **Roots, multiplicity and derivatives.** Repeated roots detected by the formal derivative, and Taylor expansion of a polynomial.
- **The Fundamental Theorem of Algebra.** A complete proof, and what it says about real polynomials.

- **Interpolation and the Chinese Remainder Theorem.** The Lagrange formula, solving simultaneous congruences, and the quotient-space picture behind both.
- **Polynomials of an operator.** Every operator on a finite-dimensional space satisfies a polynomial, and coprime factors split the space.

## 5.0.2 Named moves

- **Divide and look at the remainder.** Most statements about divisibility reduce to one division.
- **The smallest-degree element generates.** In an ideal, a non-zero element of least degree divides every other element.
- **Evaluate at a root.** Plugging a root of the divisor into  $f = qg + r$  kills  $qg$  and reads off information about the remainder.
- **Compare degrees.** Uniqueness of quotients, factorizations and interpolants usually comes from a degree count.
- **Bézout gives projections.** When  $\gcd(p, q) = 1$ , write  $1 = ap + bq$ ; evaluated at an operator, the two terms split the space.

# 5.1 Division with Remainder

Integers can be divided with a remainder:  $17 = 3 \cdot 5 + 2$ , and the remainder 2 is smaller than the divisor 5. Polynomials over a field can be divided in the same way, with “smaller” measured by degree. Chapter 0 stated this fact without proof, and Chapter 2 proved the special case of dividing by  $x - c$ . This section proves division with remainder in general, over any field, and draws the first consequences: the language of divisibility, the Remainder and Factor Theorems for every divisor  $x - c$ , a sharper root bound, and the fact that over an infinite field a polynomial is determined by its values. Throughout,  $F$  is a field unless we say otherwise, and it matters: the warning in the second subsection shows division failing when the coefficients only form  $\mathbb{Z}$ .

## 5.1.1 Long division by hand

At school one divides  $x^4 + 2x^3 - x + 3$  by  $x^2 + x - 1$  with a layout that imitates long division of numbers. Let us do it in words, so that we can see what each step does.

Write  $f = x^4 + 2x^3 - x + 3$  and  $g = x^2 + x - 1$ , in  $\mathbb{Q}[x]$ .

1. The leading term of  $f$  is  $x^4$ , and the leading term of  $g$  is  $x^2$ . The quotient  $x^4/x^2 = x^2$  is the first term of the answer. Subtract  $x^2g = x^4 + x^3 - x^2$ :

$$f - x^2g = x^3 + x^2 - x + 3.$$

2. The leading term is now  $x^3$ , and  $x^3/x^2 = x$ . Subtract  $xg = x^3 + x^2 - x$ :

$$f - x^2g - xg = 3.$$

3. What is left, 3, has degree 0, smaller than  $\deg g = 2$ . No multiple  $cx^k g$  with  $k \geq 0$  can remove it without creating terms of degree at least 2. We stop.



So  $f = (x^2 + x)g + 3$ : the **quotient** is  $x^2 + x$  and the **remainder** is 3. Expanding  $(x^2 + x)(x^2 + x - 1) = x^4 + 2x^3 - x$  confirms it.

Two questions are hidden in this routine. **Why does it always stop?** Each step removes the current leading term, so the degree of what is left drops by at least one. Degrees are natural numbers, and a strictly decreasing sequence of natural numbers cannot go on forever. **Why is the answer unique?** Someone else might hope to find a different quotient and a different remainder of degree less than 2. We will see that they cannot, by comparing degrees.

There is also a third, quieter question. In step 1 we divided the leading coefficient of the current polynomial by the leading coefficient of  $g$ . Here that coefficient was 1. In general it is some  $b \neq 0$ , and we need  $b^{-1}$ . Over a field this always exists. That is exactly where the field enters.

## 5.1.2 The Division Theorem

The routine above works for every pair of polynomials with non-zero divisor. Here is the precise statement.

### Theorem 5.1.1 — Polynomial Division with Remainder

Let  $F$  be a field, and let  $f, g \in F[x]$  with  $g \neq 0$ . Then there exist **unique** polynomials  $q, r \in F[x]$  such that

$$f = qg + r \quad \text{and} \quad \deg r < \deg g.$$

The polynomial  $q$  is the **quotient** and  $r$  is the **remainder** of  $f$  on division by  $g$ . Because  $\deg 0 = -\infty$ , the condition  $\deg r < \deg g$  allows  $r = 0$ , which is the case “ $g$  goes into  $f$  exactly”.

*Idea.* Existence is the hand routine turned into an induction on  $\deg f$ . If  $\deg f < \deg g$ , there is nothing to do:  $q = 0$ ,  $r = f$ . Otherwise subtract  $(a/b)x^{n-m}g$ , where  $ax^n$  and  $bx^m$  are the leading terms of  $f$  and  $g$ . This kills the leading term of  $f$ , so the induction hypothesis applies to what is left. Uniqueness is “take the difference”: if  $qg + r = q'g + r'$ , then  $(q - q')g = r' - r$ . The right side has degree less than  $\deg g$ , while a non-zero multiple of  $g$  has degree at least  $\deg g$ .

*Proof.* Let  $m = \deg g$ , which lies in  $\mathbb{N}$  since  $g \neq 0$ , and let  $b \neq 0$  be the leading coefficient of  $g$ .

*Existence.* If  $f = 0$ , then  $q = 0$  and  $r = 0$  work. For non-zero  $f$  we prove, by strong induction on  $n \in \mathbb{N}$  (Theorem 0.2.10 with  $n_0 = 0$ ), the statement

$P(n)$ : for every  $f \in F[x]$  with  $\deg f = n$ , there are  $q, r \in F[x]$  with  $f = qg + r$  and  $\deg r < m$ .

Let  $n \in \mathbb{N}$ , assume  $P(k)$  for all  $k < n$ , and let  $\deg f = n$  with leading coefficient  $a$ .

*Case 1:*  $n < m$ . Take  $q = 0$  and  $r = f$ . Then  $f = 0 \cdot g + f$  and  $\deg r = n < m$ .

*Case 2:*  $n \geq m$ . Since  $F$  is a field and  $b \neq 0$ , the inverse  $b^{-1}$  exists. Put

$$f_1 = f - ab^{-1}x^{n-m}g.$$

By [Theorem 0.8.3](#), the polynomial  $ab^{-1}x^{n-m}g$  has degree  $(n-m) + m = n$  and leading coefficient  $ab^{-1} \cdot b = a$ . So the coefficient of  $x^n$  in  $f_1$  is  $a - a = 0$ , and all coefficients of higher powers are 0 as well. Hence  $f_1 = 0$  or  $\deg f_1 < n$ . If  $f_1 = 0$ , then  $f = ab^{-1}x^{n-m}g + 0$ , and we take  $q = ab^{-1}x^{n-m}$ ,  $r = 0$ . Otherwise  $k = \deg f_1 < n$ , and the induction hypothesis  $P(k)$  gives  $q_1, r_1$  with  $f_1 = q_1g + r_1$  and  $\deg r_1 < m$ . Then

$$f = f_1 + ab^{-1}x^{n-m}g = (ab^{-1}x^{n-m} + q_1)g + r_1,$$

so  $q = ab^{-1}x^{n-m} + q_1$  and  $r = r_1$  work. This proves  $P(n)$ , and by strong induction  $P(n)$  holds for every  $n \in \mathbb{N}$ .

*Uniqueness.* Suppose  $f = qg + r = q'g + r'$  with  $\deg r < m$  and  $\deg r' < m$ . Then  $(q - q')g = r' - r$ . By [Theorem 0.8.2](#),  $\deg(r' - r) \leq \max(\deg r', \deg r) < m$ , since  $-r$  has the same degree as  $r$ . Suppose  $q \neq q'$ . Then  $\deg(q - q') \geq 0$ , and [Theorem 0.8.3](#) gives  $\deg((q - q')g) = \deg(q - q') + m \geq m$ , a contradiction. Hence  $q = q'$ . Then  $r' - r = 0 \cdot g = 0$ , so  $r = r'$ . This proves the theorem. ■

The field axioms did essential work in two places: the inverse  $b^{-1}$  in Case 2, and [Theorem 0.8.3](#) in the uniqueness step, which needs a product of non-zero coefficients to be non-zero. The proof is also an algorithm. Case 2, repeated, is the long division of the previous subsection, and the induction guarantees that it stops after at most  $\deg f - \deg g + 1$  subtractions.

The hypothesis “over a field” is not decoration. **The leading coefficient of  $g$  must be invertible.**

*Warning.* **Division with remainder fails over  $\mathbb{Z}$ .** Write  $\mathbb{Z}[x]$  for the polynomials in  $\mathbb{Q}[x]$  whose coefficients are all integers. Try to divide  $f = x^2$  by  $g = 2x$  inside  $\mathbb{Z}[x]$ : we want  $q, r \in \mathbb{Z}[x]$  with  $x^2 = 2x \cdot q + r$  and  $\deg r < 1$ , so  $r$  is a constant. Comparing coefficients of  $x^2$  gives  $2q_1 = 1$ , where  $q_1$  is the coefficient of  $x$  in  $q$ . No integer  $q_1$  satisfies this, so no such  $q, r$  exist. Over  $\mathbb{Q}$  there is no problem:  $x^2 = 2x \cdot \frac{1}{2}x + 0$ . The obstacle is precisely that 2 has no inverse in  $\mathbb{Z}$ .

Division by a divisor whose leading coefficient **is** invertible still works over  $\mathbb{Z}$ , for example by a monic polynomial; see Exercise C1.

*Quick check.* Why must the inequality  $\deg r < \deg g$  be strict? Find two different pairs  $(q, r)$  with  $x^2 = q \cdot (x^2 + 1) + r$  and  $\deg r \leq 2$ .

*Solution.* Both  $x^2 = 1 \cdot (x^2 + 1) + (-1)$  and  $x^2 = 0 \cdot (x^2 + 1) + x^2$  work, with remainders of degree 0 and 2. With  $\deg r \leq \deg g$  allowed, uniqueness fails, because we may move one copy of  $g$  between  $qg$  and  $r$ . The strict inequality is exactly what rules this out: in the uniqueness proof,  $\deg(r' - r) < \deg g$  is the step that forces  $q = q'$ .

Now a division in which the leading coefficient of the divisor is not 1, so fractions appear.

### Example. (Long Division with a Non-Monic Divisor)

In  $\mathbb{Q}[x]$ , divide  $f = 2x^4 - 3x^3 + x^2 + 4x - 1$  by  $g = 2x^2 - x + 3$  with remainder.

*Solution.* We follow Case 2 of the proof, where  $b = 2$  and  $b^{-1} = \frac{1}{2}$ .

*Step 1.* The leading term of  $f$  is  $2x^4$ , and  $2x^4/2x^2 = x^2$ . Subtracting  $x^2g = 2x^4 - x^3 + 3x^2$  leaves

$$f - x^2g = -2x^3 - 2x^2 + 4x - 1.$$

*Step 2.* The leading term is  $-2x^3$ , and  $-2x^3/2x^2 = -x$ . Subtracting  $-xg = -2x^3 + x^2 - 3x$  leaves

$$-3x^2 + 7x - 1.$$

*Step 3.* The leading term is  $-3x^2$ , and  $-3x^2/2x^2 = -\frac{3}{2}$ . Subtracting  $-\frac{3}{2}g = -3x^2 + \frac{3}{2}x - \frac{9}{2}$  leaves

$$(7 - \frac{3}{2})x + (-1 + \frac{9}{2}) = \frac{11}{2}x + \frac{7}{2}.$$

This has degree  $1 < 2 = \deg g$ , so we stop. Hence

$$2x^4 - 3x^3 + x^2 + 4x - 1 = (x^2 - x - \frac{3}{2})(2x^2 - x + 3) + \frac{11}{2}x + \frac{7}{2},$$

with quotient  $q = x^2 - x - \frac{3}{2}$  and remainder  $r = \frac{11}{2}x + \frac{7}{2}$ . By the uniqueness in [Theorem 5.1.1](#), any other correct method must produce the same  $q$  and  $r$ .

Notice that the quotient has a non-integer coefficient although  $f$  and  $g$  have integer coefficients. This is the warning above in action: over  $\mathbb{Z}$  this division would be impossible.

### 5.1.3 Divisibility

With integers, “3 divides 12” means that 12 is an integer multiple of 3 ([Definition 0.2.2](#)), and it is the case “remainder zero” of integer division. Polynomials get the same word. The field is part of the meaning, so we put it in the definition.

*$g$  divides  $f$  when  $f$  is a polynomial multiple of  $g$ , with the multiplier taken over the same field.*

#### Definition 5.1.1 — Divisibility, Associates

Let  $F$  be a field and  $f, g \in F[x]$ .

- We say  $g$  **divides**  $f$  **over**  $F$ , and write  $g \mid f$ , if there **exists**  $h \in F[x]$  with  $f = gh$ . Then  $g$  is a **divisor** (or **factor**) of  $f$ , and  $f$  is a **multiple** of  $g$ . We write  $g \nmid f$  if  $g$  does not divide  $f$ .
- We say  $f$  and  $g$  are **associates** if  $f = cg$  for some **non-zero** constant  $c \in F$ .

In words:  $g \mid f$  asks for a multiplier  $h$  **in**  $F[x]$ , so the same pair can behave differently over different fields. Associates differ only by a non-zero scalar factor. Since  $c^{-1}$  exists, “associates” is symmetric:  $f = cg$  gives  $g = c^{-1}f$ .

Examples, simplest first.

- **Degenerate cases.** Every  $g$  divides 0, since  $0 = g \cdot 0$ . On the other hand  $0 \mid f$  only for  $f = 0$ , since  $0 \cdot h = 0$ . Every non-zero constant  $c$  divides every  $f$ , since  $f = c \cdot (c^{-1}f)$ . These cases matter: they say that non-zero constants are invisible to divisibility, just as  $\pm 1$  are for integers.
- In  $\mathbb{Q}[x]$ ,  $x - 1 \mid x^2 - 1$ , since  $x^2 - 1 = (x - 1)(x + 1)$ . Also  $2x - 2 \mid x^2 - 1$ , since

$x^2 - 1 = (2x - 2) \cdot \frac{1}{2}(x + 1)$ . The polynomials  $x - 1$  and  $2x - 2$  are associates.

- In  $\mathbb{F}_2[x]$ ,  $x + 1 \mid x^2 + 1$ , since  $(x + 1)^2 = x^2 + 2x + 1 = x^2 + 1$  there. In  $\mathbb{Q}[x]$ , however,  $x + 1 \nmid x^2 + 1$ , as the next paragraph shows.
- In  $\mathbb{C}[x]$ ,  $x - i \mid x^2 + 1 = (x - i)(x + i)$ .

For a non-example by minimal change, keep  $x^2 + 1$  and  $x + 1$  but move from  $\mathbb{F}_2$  to  $\mathbb{Q}$ . Both are still polynomials, and division with remainder still applies. What fails is the clause “there **exists**  $h \in F[x]$ ”. If  $x^2 + 1 = (x + 1)h$  with  $h \in \mathbb{Q}[x]$ , then evaluating at  $-1$  ([Theorem 0.8.5](#)) gives  $2 = 0 \cdot h(-1) = 0$ , which is false in  $\mathbb{Q}$ . Over  $\mathbb{F}_2$  the same computation reads  $0 = 0$ , and there is no contradiction.

*Warning.* Divisibility depends on where the multiplier may live. In  $\mathbb{Q}[x]$ ,  $2x + 2 \mid x + 1$ , because  $x + 1 = (2x + 2) \cdot \frac{1}{2}$ . If we insisted on integer coefficients,  $2x + 2$  would **not** divide  $x + 1$ : any integer multiple  $(2x + 2)h$  has all coefficients even. Over a field, a non-zero scalar factor never affects divisibility, so it is pointless to ask whether  $x - 1$  or  $3x - 3$  is “the” factor of  $x^2 - 1$ .

Why this definition? Because it makes “divides” the case of remainder zero, which is the link to division we need. The basic rules follow quickly.

### Proposition 5.1.2 — Rules for Divisibility

Let  $F$  be a field and  $f, g, h \in F[x]$ .

- If  $f \mid g$  and  $g \mid h$ , then  $f \mid h$ .
- If  $h \mid f$  and  $h \mid g$ , then  $h \mid af + bg$  **for all**  $a, b \in F[x]$ .
- If  $g \mid f$  and  $f \neq 0$ , then  $\deg g \leq \deg f$ .
- $f \mid g$  and  $g \mid f$  if and only if  $f$  and  $g$  are associates or both are 0. In particular, if  $f \mid g$ ,  $g \mid f$ , and  $f, g$  are both monic, then  $f = g$ .
- If  $g \neq 0$ , then  $g \mid f$  if and only if the remainder of  $f$  on division by  $g$  is 0.

*Proof.*

- Write  $g = fu$  and  $h = gv$  with  $u, v \in F[x]$ . Then  $h = f(uv)$  by [Theorem 0.8.1](#), so  $f \mid h$ .
- Write  $f = hu$  and  $g = hv$ . Then  $af + bg = h(au + bv)$  by [Theorem 0.8.1](#), so  $h \mid af + bg$ .
- Write  $f = gu$ . Since  $f \neq 0$ , both  $g$  and  $u$  are non-zero, so  $\deg u \geq 0$ . By [Theorem 0.8.3](#),  $\deg f = \deg g + \deg u \geq \deg g$ .
- ( $\Leftarrow$ ) If  $f = cg$  with  $c \neq 0$ , then  $g \mid f$  and  $g = c^{-1}f$  gives  $f \mid g$ . If  $f = g = 0$ , both divide each other. ( $\Rightarrow$ ) Suppose  $g = fu$  and  $f = gv$ . If  $f = 0$ , then  $g = 0 \cdot u = 0$ , so both are 0. If  $f \neq 0$ , then  $f = f(uv)$ , so  $f(1 - uv) = 0$ , and [Corollary 0.8.4](#) gives  $uv = 1$ . By [Theorem 0.8.3](#),  $\deg u + \deg v = \deg 1 = 0$  with both degrees in  $\mathbb{N}$ , so  $u$  and  $v$  are non-zero constants. Hence  $f = vg$  with  $v \neq 0$ , and  $f, g$  are associates. If both are monic, comparing leading coefficients in  $f = vg$  gives  $1 = v \cdot 1$ , so  $f = g$ .
- If the remainder is 0, then  $f = qg$ , so  $g \mid f$ . Conversely, if  $f = hg$ , then  $f = hg + 0$  with  $\deg 0 = -\infty < \deg g$ . This is a division with remainder, so by the uniqueness in [Theorem 5.1.1](#), the remainder of  $f$  on division by  $g$  is 0. This proves the proposition.

Part (d) is the polynomial form of a move we will use again and again: **to show two monic polynomials are equal, show that each divides the other**. It also gives each non-zero polynomial a canonical representative. If  $f \neq 0$  has leading coefficient  $a$ , then  $a^{-1}f$  is monic and an associate of  $f$ , and it is the **only** monic associate: two monic associates divide each other, so they are equal by (d). For example, the monic associate of  $3x^2 - 6$  in  $\mathbb{Q}[x]$  is  $x^2 - 2$ . The greatest common divisor in the next section will be defined as a monic polynomial for exactly this reason.

## 5.1.4 The Remainder and Factor Theorems

Division by a polynomial of degree 1 leaves a remainder of degree less than 1, that is, a constant. Which constant? Chapter 2 already answered this. The Factor Theorem for a Linear Divisor (Lemma 2.7.1) showed, by an induction that was long division in disguise, that  $p = (x - c)q + p(c)$  for some  $q$ . We now read that lemma through Theorem 5.1.1. What is new is that  $q$  and the constant are **unique**, so  $p(c)$  is **the** remainder in the sense of division.

### Theorem 5.1.3 — Remainder Theorem and Factor Theorem

Let  $F$  be a field,  $f \in F[x]$  and  $c \in F$ .

- (a) The remainder of  $f$  on division by  $x - c$  is the constant  $f(c)$ .
- (b)  $c$  is a root of  $f$  if and only if  $x - c \mid f$ .

*Proof.*

- (a) By Lemma 2.7.1,  $f = (x - c)q + f(c)$  for some  $q \in F[x]$ . The constant  $f(c)$  has degree 0 or  $-\infty$ , which is less than  $\deg(x - c) = 1$ . So this is a division with remainder, and by the uniqueness in Theorem 5.1.1,  $f(c)$  is the remainder.
- (b) By Proposition 5.1.2 (e),  $x - c \mid f$  if and only if the remainder is 0, which by (a) means  $f(c) = 0$ . This proves the theorem.

In other words, the Factor Theorem of Chapter 2 is really a statement about division. Division alone would also give it: dividing  $f$  by  $x - c$  leaves a constant remainder  $r$ , and evaluating  $f = (x - c)q + r$  at  $c$  gives  $r = f(c)$ . Since Chapter 2 already proved the lemma, we cite it rather than prove it again, and we will use the name “Remainder Theorem” for both parts.

### Example. (Remainders Without Dividing)

- (a) Find the remainder of  $f = x^5 - 3x^3 + 2x + 7$  on division by  $x - 2$  in  $\mathbb{Q}[x]$ . (b) Show that  $x + 1 \nmid x^2 + 1$  in  $\mathbb{Q}[x]$ , but  $x + 1 \mid x^2 + 1$  in  $\mathbb{F}_2[x]$ . (c) Find the remainder of  $x^5$  on division by  $x^2 - x$  in  $\mathbb{Q}[x]$ , again without long division.

*Solution.*

- (a) By [Theorem 5.1.3](#) (a), the remainder is  $f(2) = 32 - 24 + 4 + 7 = 19$ . No long division is needed.
- (b) Here  $x+1 = x-(-1)$ . In  $\mathbb{Q}$ ,  $(-1)^2 + 1 = 2 \neq 0$ , so  $-1$  is not a root and  $x+1 \nmid x^2+1$  by part (b) of the theorem; the remainder is 2. In  $\mathbb{F}_2$ ,  $-1 = 1$  and  $1^2 + 1 = 0$ , so  $x+1 \mid x^2+1$ . Indeed  $x^2+1 = (x+1)^2$  in  $\mathbb{F}_2[x]$ . The same polynomial symbols, over two fields, give different answers.
- (c) The divisor has degree 2, so by [Theorem 5.1.1](#),  $x^5 = q(x^2 - x) + r$  with  $\deg r \leq 1$ , say  $r = ax + b$ . The divisor  $x^2 - x = x(x-1)$  vanishes at 0 and at 1, so evaluating both sides there ([Theorem 0.8.5](#)) kills the term  $q(x^2 - x)$ : at 0,  $0 = b$ ; at 1,  $1 = a + b$ . Hence  $a = 1$ ,  $b = 0$ , and the remainder is  $x$ . As a check,  $x^5 - x = x(x^4 - 1) = (x^2 - x)(x^3 + x^2 + x + 1)$ . The move is: **evaluate at the roots of the divisor**, one equation per root.

The Factor Theorem peels off one root at a time. Doing it for several distinct roots at once strengthens the Root bound of Chapter 2 ([Lemma 2.7.2](#)), which said that a non-zero polynomial of degree at most  $n$  has at most  $n$  distinct roots. The stronger version says **why** there are so few: every root contributes its own linear factor.

### Corollary 5.1.4 — Distinct Roots Give a Product of Linear Factors

Let  $F$  be a field,  $f \in F[x]$ , and let  $c_1, \dots, c_k \in F$  be **distinct** roots of  $f$ , with  $k \geq 1$ . Then

$$(x - c_1)(x - c_2) \cdots (x - c_k) \mid f.$$

Consequently, if  $f \neq 0$ , then  $k \leq \deg f$ : a non-zero polynomial of degree  $n$  has at most  $n$  distinct roots in  $F$ .

*Idea.* Peel off  $x - c_1$  by the Factor Theorem. What remains,  $h$ , must still vanish at  $c_2$ , because  $f(c_2) = (c_2 - c_1)h(c_2)$  and  $c_2 - c_1 \neq 0$ . So peel off  $x - c_2$  from  $h$ , and so on. Distinctness is what makes the factors  $c_k - c_i$  non-zero.

*Proof.* We use induction on  $k$ . For  $k = 1$ , this is [Theorem 5.1.3](#) (b).

Let  $k \geq 2$ , and assume the statement for  $k-1$  distinct roots. By the induction hypothesis,  $f = (x - c_1) \cdots (x - c_{k-1}) h$  for some  $h \in F[x]$ . Evaluating at  $c_k$  with [Theorem 0.8.5](#),

$$0 = f(c_k) = (c_k - c_1) \cdots (c_k - c_{k-1}) h(c_k).$$

Since the  $c_i$  are distinct, each  $c_k - c_i$  with  $i < k$  is non-zero, and a product of non-zero elements of a field is non-zero ([Theorem 0.7.1](#)). Hence  $h(c_k) = 0$ , and [Theorem 5.1.3](#) (b) gives  $h = (x - c_k) h_1$  with  $h_1 \in F[x]$ . Therefore  $f = (x - c_1) \cdots (x - c_k) h_1$ , which completes the induction.

For the count, suppose  $f \neq 0$  and write  $f = (x - c_1) \cdots (x - c_k) h_1$  as above. Then  $h_1 \neq 0$ , and by [Theorem 0.8.3](#),  $\deg f = k + \deg h_1 \geq k$ . This proves the corollary. ■

The count is [Lemma 2.7.2](#) again, now as a consequence of a divisibility statement. The divisibility

statement is the more useful one. For instance, it tells us at once that a polynomial vanishing at 1, 2 and 3 is a multiple of  $(x-1)(x-2)(x-3)$ .

## 5.1.5 Polynomials and polynomial functions

Chapter 0 warned that a polynomial is not the function it defines: over  $\mathbb{F}_2$ , the non-zero polynomial  $x^2 + x$  has the zero function (Example (Different polynomials, same function)). It also promised that over an infinite field the distinction disappears. Chapter 2 kept that promise in a paragraph right after Lemma 2.7.2. We record the fact here as a theorem, in a slightly sharper form that counts how many values are needed.

### Theorem 5.1.5 — Values Determine a Polynomial

Let  $F$  be a field and  $p, q \in F[x]$ .

- (a) Let  $n \in \mathbb{N}$  with  $\deg p \leq n$  and  $\deg q \leq n$ . If  $p(c) = q(c)$  for at least  $n+1$  **distinct** elements  $c \in F$ , then  $p = q$ .
- (b) If  $F$  is **infinite** and  $p(c) = q(c)$  for **every**  $c \in F$ , then  $p = q$ .

*Proof.*

- (a) Let  $h = p - q$ . By Theorem 0.8.2,  $\deg h \leq n$ . By Theorem 0.8.5,  $h(c) = p(c) - q(c) = 0$  for each of the given  $n+1$  distinct elements, so  $h$  has at least  $n+1$  distinct roots. By Lemma 2.7.2, a non-zero polynomial of degree at most  $n$  has at most  $n$  distinct roots, so  $h$  cannot be non-zero. Hence  $h = 0$  and  $p = q$ .
- (b) Let  $n = \max(\deg p, \deg q, 0) \in \mathbb{N}$ . Since  $F$  is infinite, it contains  $n+1$  distinct elements, and  $p$  and  $q$  agree at each of them. Part (a) gives  $p = q$ . This proves the theorem. ■

So over  $\mathbb{Q}$ ,  $\mathbb{R}$  and  $\mathbb{C}$ , we may compare polynomials by comparing their functions, and we may read off coefficients from values. Part (a) holds over **every** field, including finite ones: a quadratic over  $\mathbb{F}_5$  is determined by its values at any three distinct points.

*Warning.* Part (b) is false over every finite field. Over  $\mathbb{F}_p$ , the polynomial  $x^p - x$  is non-zero but vanishes at every element (Exercise (C2: A non-zero polynomial vanishing everywhere)). So  $x^p$  and  $x$  define the same function  $\mathbb{F}_p \rightarrow \mathbb{F}_p$  although they are different polynomials. There is no contradiction with (a): the degrees are at most  $p$ , and (a) would need agreement at  $p+1$  distinct points, but  $\mathbb{F}_p$  has only  $p$  elements.

*Quick check.* Over  $\mathbb{F}_3$ , do the polynomials  $x^3 + 2x$  and 0 define the same function? Does this contradict Theorem 5.1.5?

*Solution.* Yes. The values of  $x^3 + 2x$  at 0, 1, 2 are 0,  $1 + 2 = 3 = 0$ , and  $8 + 4 = 12 = 0$ . So the function is zero, although the polynomial is not. There is no contradiction:  $\mathbb{F}_3$  is finite, so part (b) does not apply, and part (a) with  $n = 3$  would need 4 distinct points. In fact  $x^3 + 2x = x^3 - x$ , the polynomial of the warning with  $p = 3$ .

Division with remainder is the one tool this chapter keeps using. In the next section it shows that every set of polynomials closed under the right operations consists of the multiples of a single polynomial, and that gives greatest common divisors.

## 5.1.6 Exercises

### 5.1.6.1 A. Check your understanding

#### Exercise A1.

- (a) State the theorem on division with remainder in  $F[x]$ , including every hypothesis.
- (b) Determine whether the following statement is true: “for every field  $F$  and all  $f, g \in F[x]$ , if  $f \mid g$  and  $g \mid f$ , then  $f = g$ .” Justify your answer. Does your answer change if  $F = \mathbb{F}_2$ ?
- (c) Let  $c \in F$  be non-zero. What are the quotient and the remainder of  $f$  on division by the constant polynomial  $c$ ?
- (d) Name the method for finding the remainder of  $f$  on division by  $x - c$  without dividing.
- (e) Determine whether the following statement is true: “if  $F$  is a field and  $p \in F[x]$  satisfies  $p(c) = 0$  for every  $c \in F$ , then  $p = 0$ .” Justify your answer.

*Solution.*

- (a) Let  $F$  be a field and  $f, g \in F[x]$  with  $g \neq 0$ . Then there are unique  $q, r \in F[x]$  with  $f = qg + r$  and  $\deg r < \deg g$  ([Theorem 5.1.1](#)).
- (b) False in general. In  $\mathbb{Q}[x]$ , take  $f = x$  and  $g = 2x$ . Then  $g = 2f$  and  $f = \frac{1}{2}g$ , so each divides the other, but  $f \neq g$ . By [Proposition 5.1.2](#) (d), mutual divisibility only gives associates. Over  $\mathbb{F}_2$ , the statement is **true**: the only non-zero constant in  $\mathbb{F}_2$  is 1, so associates are equal, and if  $f \mid g$  and  $g \mid f$ , then either both are 0 or  $f = 1 \cdot g$ .
- (c) The remainder must have degree less than  $\deg c = 0$ , so  $r = 0$ . Then  $f = qc$ , so  $q = c^{-1}f$ . In particular every polynomial is divisible by every non-zero constant.
- (d) The Remainder Theorem ([Theorem 5.1.3](#)): the remainder is the constant  $f(c)$ .
- (e) False. Over  $\mathbb{F}_2$ ,  $p = x^2 + x$  is non-zero but  $p(0) = p(1) = 0$  ([Example \(Different polynomials, same function\)](#)). The statement is true when  $F$  is infinite, by [Theorem 5.1.5](#) (b).

### 5.1.6.2 B. Practice

#### Exercise B1: Dividing Over Two Fields.

- (a) In  $\mathbb{Q}[x]$ , divide  $f = x^4 - 3x^3 + 2x - 5$  by  $g = 3x^2 - 1$  with remainder.
- (b) In  $\mathbb{F}_5[x]$ , divide  $f = x^4 + 2x^3 + 3x + 1$  by  $g = 2x^2 + 3$  with remainder. Hence decide whether  $g \mid f$  in  $\mathbb{F}_5[x]$ .

*Hint for (b): first find the inverse of 2 in  $\mathbb{F}_5$ .*



*Solution.*

- (a) The leading coefficient of  $g$  is 3, with inverse  $\frac{1}{3}$ . First,  $x^4/3x^2 = \frac{1}{3}x^2$ , and  $f - \frac{1}{3}x^2g = -3x^3 + \frac{1}{3}x^2 + 2x - 5$ . Next,  $-3x^3/3x^2 = -x$ , and subtracting  $-xg = -3x^3 + x$  leaves  $\frac{1}{3}x^2 + x - 5$ . Next,  $\frac{1}{3}x^2/3x^2 = \frac{1}{9}$ , and subtracting  $\frac{1}{9}g = \frac{1}{3}x^2 - \frac{1}{9}$  leaves  $x - \frac{44}{9}$ , of degree  $1 < 2$ . Hence

$$x^4 - 3x^3 + 2x - 5 = \left(\frac{1}{3}x^2 - x + \frac{1}{9}\right)(3x^2 - 1) + x - \frac{44}{9},$$

with quotient  $\frac{1}{3}x^2 - x + \frac{1}{9}$  and remainder  $x - \frac{44}{9}$ , which are the only ones by [Theorem 5.1.1](#).

- (b) In  $\mathbb{F}_5$ ,  $2 \cdot 3 = 6 = 1$ , so  $2^{-1} = 3$ . First, the leading term  $x^4$  divided by  $2x^2$  is  $3x^2$ , and  $3x^2g = 6x^4 + 9x^2 = x^4 + 4x^2$ . Subtracting leaves  $2x^3 - 4x^2 + 3x + 1 = 2x^3 + x^2 + 3x + 1$ . Next,  $2x^3/2x^2 = x$ , and subtracting  $xg = 2x^3 + 3x$  leaves  $x^2 + 1$ . Next,  $x^2/2x^2 = 3$ , and subtracting  $3g = 6x^2 + 9 = x^2 + 4$  leaves  $1 - 4 = -3 = 2$ . Hence

$$x^4 + 2x^3 + 3x + 1 = (3x^2 + x + 3)(2x^2 + 3) + 2 \quad \text{in } \mathbb{F}_5[x].$$

The remainder 2 is non-zero, so  $g \nmid f$  in  $\mathbb{F}_5[x]$ , by [Proposition 5.1.2](#) (e).

### Exercise B2: A Factor of $x^n - c^n$ .

Let  $F$  be a field,  $c \in F$  and  $n \geq 1$ . Prove that  $x - c \mid x^n - c^n$  in  $F[x]$ , and verify that the quotient is  $x^{n-1} + cx^{n-2} + \cdots + c^{n-2}x + c^{n-1}$ .

*Solution.* The value of  $f = x^n - c^n$  at  $c$  is  $c^n - c^n = 0$ , so  $c$  is a root of  $f$ , and [Theorem 5.1.3](#) (b) gives  $x - c \mid f$ . For the quotient, let  $s = \sum_{j=0}^{n-1} c^j x^{n-1-j}$ . Then

$$(x - c)s = \sum_{j=0}^{n-1} c^j x^{n-j} - \sum_{j=0}^{n-1} c^{j+1} x^{n-1-j} = \sum_{j=0}^{n-1} c^j x^{n-j} - \sum_{j=1}^n c^j x^{n-j} = x^n - c^n,$$

since the two sums share the terms with  $1 \leq j \leq n-1$ , leaving  $c^0 x^n$  from the first and  $c^n x^0$  from the second. This shows  $x^n - c^n = (x - c)s$ .

### Exercise B3: A Remainder Without Long Division.

- (a) In  $\mathbb{Q}[x]$ , find the remainder of  $x^{100}$  on division by  $x^2 - 1$  without long division.  
 (b) Show that the answer to (a) is the remainder of  $x^{100}$  on division by  $x^2 - 1$  over **every** field  $F$ , including  $\mathbb{F}_2$ . Explain why the method of (a) alone does not settle the case  $F = \mathbb{F}_2$ .

*Hint for (a):* part (c) of [Example \(Remainders Without Dividing\)](#). *Hint for (b):* imitate the identity of B2.

*Solution.*

- (a) By [Theorem 5.1.1](#),  $x^{100} = q(x^2 - 1) + r$  with  $\deg r \leq 1$ , say  $r = ax + b$ . Evaluating at 1 and  $-1$ , where  $x^2 - 1$  vanishes ([Theorem 0.8.5](#)), gives  $1 = a + b$  and  $1 = (-1)^{100} = -a + b$ . Adding and subtracting,  $2b = 2$  and  $2a = 0$ , so  $b = 1$  and  $a = 0$ . The remainder is 1.

- (b) Over any field, let  $s = \sum_{j=0}^{49} x^{2j}$ . Then

$$(x^2 - 1)s = \sum_{j=0}^{49} x^{2j+2} - \sum_{j=0}^{49} x^{2j} = x^{100} - 1,$$

because the terms  $x^2, x^4, \dots, x^{98}$  occur in both sums. Hence  $x^{100} = s \cdot (x^2 - 1) + 1$ , and  $\deg 1 = 0 < 2$ , so by the uniqueness in [Theorem 5.1.1](#) the remainder is 1 over every field. The method of (a) divides by 2, and over  $\mathbb{F}_2$  we have  $2 = 0$ ; also  $1 = -1$  there, so evaluating at  $\pm 1$  gives only the single equation  $a + b = 1$ , which does not determine  $a$  and  $b$ .

### 5.1.6.3 C. Going deeper

#### Exercise C1: Division by a Monic Polynomial Over $\mathbb{Z}$ .

As in the warning after [Theorem 5.1.1](#),  $\mathbb{Z}[x]$  denotes the set of polynomials in  $\mathbb{Q}[x]$  all of whose coefficients are integers. It is closed under addition and multiplication.

- (a) Let  $f, g \in \mathbb{Z}[x]$  with  $g$  **monic**. Prove that there exist  $q, r \in \mathbb{Z}[x]$  with  $f = qg + r$  and  $\deg r < \deg g$ .
- (b) Prove that  $q$  and  $r$  in (a) are unique, even among  $q, r \in \mathbb{Q}[x]$ .
- (c) Does (a) remain true if the leading coefficient of  $g$  is  $-1$  instead of 1? Justify your answer.

*Hint for (a): find the one step in the proof of [Theorem 5.1.1](#) that leaves  $\mathbb{Z}[x]$ .*

*Solution.*

- (a) We repeat the existence proof of [Theorem 5.1.1](#), checking that every polynomial stays in  $\mathbb{Z}[x]$ . Let  $m = \deg g$ . The only step that could leave  $\mathbb{Z}[x]$  is forming  $ab^{-1}x^{n-m}g$ , where  $b$  is the leading coefficient of  $g$ . Here  $b = 1$ , so  $ab^{-1} = a \in \mathbb{Z}$  and  $f_1 = f - ax^{n-m}g \in \mathbb{Z}[x]$ . The strong induction on  $\deg f$  ([Theorem 0.2.10](#)) then runs word for word inside  $\mathbb{Z}[x]$ : in Case 1,  $q = 0$  and  $r = f$  are in  $\mathbb{Z}[x]$ ; in Case 2,  $q = ax^{n-m} + q_1$  and  $r = r_1$  are in  $\mathbb{Z}[x]$  because  $q_1, r_1$  are, by the induction hypothesis applied to  $f_1 \in \mathbb{Z}[x]$ . This proves existence in  $\mathbb{Z}[x]$ .
- (b) Suppose  $f = qg + r = q'g + r'$  with  $q, q', r, r' \in \mathbb{Q}[x]$  and  $\deg r, \deg r' < \deg g$ . Since  $\mathbb{Q}$  is a field and  $g \neq 0$ , the uniqueness part of [Theorem 5.1.1](#) in  $\mathbb{Q}[x]$  gives  $q = q'$  and  $r = r'$ . In particular the pair found in (a) is the unique one.
- (c) Yes. If the leading coefficient is  $b = -1$ , then  $b^{-1} = -1 \in \mathbb{Z}$ , so  $ab^{-1} = -a \in \mathbb{Z}$ , and the argument of (a) goes through unchanged. What matters is that the leading coefficient has an inverse in  $\mathbb{Z}$ , which is true exactly for  $\pm 1$ .

#### Exercise C2: Remainders on Division by a Quadratic.

Let  $F$  be a field,  $f \in F[x]$  and  $a, b \in F$ .

- (a) Suppose  $a \neq b$ . Prove that the remainder of  $f$  on division by  $(x - a)(x - b)$  is

$$r = f(a) \frac{x - b}{a - b} + f(b) \frac{x - a}{b - a},$$

where  $\frac{x-b}{a-b}$  means  $(a-b)^{-1}(x-b)$ .

- (b) Now let  $a = b$ . Write  $f = (x-a)h + f(a)$  by [Theorem 5.1.3](#). Prove that the remainder of  $f$  on division by  $(x-a)^2$  is  $h(a)(x-a) + f(a)$ .
- (c) Hence find the remainder of  $x^{100}$  on division by  $x^2 - 3x + 2$  in  $\mathbb{Q}[x]$ .

*Solution.*

- (a) Since  $a \neq b$ ,  $a-b$  is invertible, so  $r$  is a well-defined polynomial of degree at most 1. By [Theorem 5.1.1](#),  $f = q(x-a)(x-b) + s$  with  $\deg s \leq 1$ . Evaluating at  $a$  and at  $b$  ([Theorem 0.8.5](#)) gives  $s(a) = f(a)$  and  $s(b) = f(b)$ . Also  $r(a) = f(a) \cdot 1 + f(b) \cdot 0 = f(a)$  and  $r(b) = f(b)$ . So  $s$  and  $r$  have degree at most 1 and agree at the 2 distinct points  $a, b$ . By [Theorem 5.1.5](#) (a) with  $n = 1$ ,  $s = r$ . This proves the formula.
- (b) By [Theorem 5.1.3](#) (a) applied to  $h$ ,  $h = (x-a)k + h(a)$  for some  $k \in F[x]$ . Substituting,

$$f = (x-a)((x-a)k + h(a)) + f(a) = k(x-a)^2 + (h(a)(x-a) + f(a)).$$

The last bracket has degree at most  $1 < 2 = \deg(x-a)^2$ , so by the uniqueness in [Theorem 5.1.1](#) it is the remainder.

- (c) Here  $x^2 - 3x + 2 = (x-1)(x-2)$ , with  $a = 1 \neq b = 2$ ,  $f(1) = 1$  and  $f(2) = 2^{100}$ . By (a),

$$r = 1 \cdot \frac{x-2}{1-2} + 2^{100} \cdot \frac{x-1}{2-1} = (2^{100} - 1)x + (2 - 2^{100}).$$

As a check,  $r(1) = 1$  and  $r(2) = 2^{101} - 2 + 2 - 2^{100} = 2^{100}$ .

### Exercise C3: Equal Functions Over $\mathbb{F}_p$ .

Let  $p$  be a prime and  $f, g \in \mathbb{F}_p[x]$ .

- (a) Prove that  $f(c) = g(c)$  for every  $c \in \mathbb{F}_p$  if and only if  $x^p - x \mid f - g$  in  $\mathbb{F}_p[x]$ .
- (b) Deduce that  $f$  and its remainder on division by  $x^p - x$  define the same function  $\mathbb{F}_p \rightarrow \mathbb{F}_p$ . Use this to find a polynomial of degree less than 5 that defines the same function on  $\mathbb{F}_5$  as  $x^7 + x$ .

*Hint for (a):* [Exercise \(C2: A non-zero polynomial vanishing everywhere\)](#) and [Corollary 5.1.4](#).

*Solution.*

- (a) ( $\Leftarrow$ ) Suppose  $f - g = (x^p - x)h$ . By [Exercise \(C2: A non-zero polynomial vanishing everywhere\)](#),  $c^p - c = 0$  for every  $c \in \mathbb{F}_p$ . By [Theorem 0.8.5](#),  $f(c) - g(c) = (c^p - c)h(c) = 0$  for every  $c$ . ( $\Rightarrow$ ) Suppose  $f(c) = g(c)$  for every  $c \in \mathbb{F}_p$ . Then the  $p$  distinct elements  $0, 1, \dots, p-1$  of  $\mathbb{F}_p$  are roots of  $f - g$ . Let  $P = \prod_{c \in \mathbb{F}_p} (x - c)$ . By [Corollary 5.1.4](#),  $P \mid f - g$ . It remains to show  $P = x^p - x$ . By [Exercise \(C2: A non-zero polynomial vanishing everywhere\)](#), every  $c \in \mathbb{F}_p$  is a root of  $x^p - x$ , so [Corollary 5.1.4](#) gives  $x^p - x = Pu$  for some  $u$ . Both  $P$  and  $x^p - x$  are monic of degree  $p$  ([Theorem 0.8.3](#)), so  $\deg u = 0$ , and comparing leading coefficients gives  $u = 1$ . Hence  $P = x^p - x$ , and  $x^p - x \mid f - g$ .
- (b) Write  $f = q(x^p - x) + r$  by [Theorem 5.1.1](#). Then  $x^p - x \mid f - r$ , so  $f$  and  $r$  define

the same function by (a). For  $f = x^7 + x$  over  $\mathbb{F}_5$ :  $x^7 = x^2 \cdot x^5 = x^2(x^5 - x) + x^3$ , so  $x^7 + x = x^2(x^5 - x) + (x^3 + x)$  with  $\deg(x^3 + x) = 3 < 5$ . Hence  $x^3 + x$  defines the same function as  $x^7 + x$  on  $\mathbb{F}_5$ .

## 5.2 Ideals and Greatest Common Divisors

Division with remainder is now available over every field. This section uses it to answer a question about two polynomials  $f$  and  $g$ : what do all the combinations  $af + bg$ , with **polynomial** coefficients  $a$  and  $b$ , look like? The answer is that they are exactly the multiples of one polynomial, and that polynomial is the greatest common divisor of  $f$  and  $g$ . Along the way we meet ideals, the subsets of  $F[x]$  adapted to multiplication, and we prove that each of them is generated by a single polynomial. The Euclidean algorithm then computes the greatest common divisor and the coefficients  $a, b$  by repeated division. Throughout,  $F$  is a field.

### 5.2.1 All the combinations of two polynomials

Take  $f = x^2 - 1$  and  $g = x^2 + x - 2$  in  $\mathbb{Q}[x]$ , and look at the set

$$\{af + bg : a, b \in \mathbb{Q}[x]\}$$

of all their combinations with polynomial coefficients. It contains  $f$  (take  $a = 1, b = 0$ ),  $g$ , and  $g - f = x - 1$ .

Since  $f = (x - 1)(x + 1)$  and  $g = (x - 1)(x + 2)$ , every combination is

$$af + bg = (x - 1)(a(x + 1) + b(x + 2)),$$

a multiple of  $x - 1$ . Conversely, every multiple  $h(x - 1)$  is a combination, namely  $(-h)f + hg$ . So the set of combinations is **exactly** the set of multiples of  $x - 1$ . Its non-zero members of smallest degree are the non-zero scalar multiples of  $x - 1$ . By [Proposition 5.1.2](#) (c), no non-zero constant is a multiple of  $x - 1$ , so there is nothing of degree 0.

Two features of this set will carry the whole section. It is closed under addition, and it absorbs multiplication by **any** polynomial: if  $af + bg$  is in the set, so is  $h(af + bg) = (ha)f + (hb)g$ . Sets with these two features appear everywhere, so they get a name.

### 5.2.2 Ideals

In Chapter 1 we studied  $F[x]$  as a vector space and its subspaces: subsets closed under addition and under multiplication by **scalars**. But  $F[x]$  also has a multiplication of its own. The subsets that respect it are closed under multiplication by **polynomials**. Three recurring examples share this feature:

- the multiples of a fixed polynomial  $p$ ;
- the combinations  $af + bg$  of two fixed polynomials;
- the polynomials that vanish at a fixed point  $c$ : if  $f(c) = 0$ , then  $(hf)(c) = h(c) \cdot 0 = 0$  for every  $h$ .

An ideal is a set of polynomials that is closed under addition and swallows multiplication by anything.

### Definition 5.2.1 — Ideal of $F[x]$

Let  $F$  be a field. A subset  $I \subseteq F[x]$  is an **ideal** of  $F[x]$  if

- (I1)  $I$  is **non-empty**;
- (I2)  $f + g \in I$  for all  $f, g \in I$ ;
- (I3)  $hf \in I$  for all  $f \in I$  and every  $h \in F[x]$ .

For  $f_1, \dots, f_k \in F[x]$ , write

$$\langle f_1, \dots, f_k \rangle := \{a_1 f_1 + \dots + a_k f_k : a_1, \dots, a_k \in F[x]\}.$$

In particular  $\langle p \rangle = \{hp : h \in F[x]\}$  is the set of all multiples of  $p$ .

In words: (I1) rules out the empty set. (I2) says sums stay inside. (I3) is the new clause: a member of  $I$  multiplied by **any** polynomial, not just by a constant, is still a member. Taking  $h = 0$  in (I3) shows  $0 \in I$ , and taking  $h = -1$  shows  $-f \in I$ . Taking  $h = c$  constant shows  $I$  is closed under scalar multiplication, so **every ideal is a subspace** of  $F[x]$  by Theorem 1.2.1.

The angle brackets here have nothing to do with the inner products of later chapters:  $\langle f, g \rangle$  is a set of polynomials, not a number.

**Well-definedness.** We have named the set  $\langle f_1, \dots, f_k \rangle$ , and must check that it is an ideal. It is in fact the smallest ideal containing  $f_1, \dots, f_k$ .

### Proposition 5.2.1 — The Ideal Generated by Finitely Many Polynomials

Let  $F$  be a field and  $f_1, \dots, f_k \in F[x]$ , with  $k \geq 1$ . Then  $\langle f_1, \dots, f_k \rangle$  is an ideal of  $F[x]$  containing  $f_1, \dots, f_k$ . Moreover, every ideal  $J$  of  $F[x]$  with  $f_1, \dots, f_k \in J$  satisfies  $\langle f_1, \dots, f_k \rangle \subseteq J$ .

*Proof.* Let  $I = \langle f_1, \dots, f_k \rangle$ . Taking  $a_i = 1$  and  $a_j = 0$  for  $j \neq i$  shows  $f_i \in I$ ; in particular (I1) holds. For (I2),  $\sum a_i f_i + \sum b_i f_i = \sum (a_i + b_i) f_i \in I$ . For (I3),  $h \sum a_i f_i = \sum (ha_i) f_i \in I$ , using Theorem 0.8.1 in both cases. So  $I$  is an ideal.

Now let  $J$  be an ideal containing every  $f_i$ . By (I3) for  $J$ , each  $a_i f_i \in J$ , and by (I2) applied  $k - 1$  times,  $a_1 f_1 + \dots + a_k f_k \in J$ . Hence  $I \subseteq J$ . This proves the proposition. ■

The ideal  $\langle f_1, \dots, f_k \rangle$  is the **ideal generated by**  $f_1, \dots, f_k$ . An ideal of the form  $\langle p \rangle$ , generated by one polynomial, is called **principal**.

### Example. (First Ideals)

Check each of the following against Definition 5.2.1.

- (a)  $\{0\}$  and  $F[x]$ ; and any ideal containing a non-zero constant.
- (b)  $I_c = \{f \in F[x] : f(c) = 0\}$  for a fixed  $c \in F$ .
- (c) For  $\mathbf{A} = \begin{pmatrix} 0 & 1 \\ 0 & 0 \end{pmatrix} \in M_2(F)$ , the set  $\{p \in F[x] : p(\mathbf{A}) = 0\}$ .

*Solution.*

- (a)  $\{0\} = \langle 0 \rangle$  and  $F[x] = \langle 1 \rangle$  are ideals by [Proposition 5.2.1](#). They are the degenerate cases, and the second one is the one to recognize: if an ideal  $I$  contains a non-zero constant  $c$ , then  $1 = c^{-1}c \in I$  by (I3), and then  $h = h \cdot 1 \in I$  for every  $h$ , so  $I = F[x]$ . An ideal that contains a non-zero constant is everything.
- (b)  $0 \in I_c$ . If  $f(c) = g(c) = 0$ , then  $(f+g)(c) = 0$  and  $(hf)(c) = h(c)f(c) = 0$  for every  $h \in F[x]$ , by [Theorem 0.8.5](#). So  $I_c$  is an ideal. By [Theorem 5.1.3](#) (b),  $f(c) = 0$  exactly when  $x - c \mid f$ , so  $I_c = \langle x - c \rangle$ .
- (c) Call the set  $I$ . The zero polynomial is in  $I$ . If  $p(\mathbf{A}) = q(\mathbf{A}) = 0$ , then by [Theorem 0.9.9](#),  $(p+q)(\mathbf{A}) = p(\mathbf{A}) + q(\mathbf{A}) = 0$  and  $(hp)(\mathbf{A}) = h(\mathbf{A})p(\mathbf{A}) = h(\mathbf{A}) \cdot 0 = 0$  for every  $h$ . So  $I$  is an ideal. It contains  $x^2$ , since  $\mathbf{A}^2 = 0$ . It contains no non-zero polynomial of degree at most 1:  $(a+bx)(\mathbf{A}) = \begin{pmatrix} a & b \\ 0 & a \end{pmatrix} = 0$  forces  $a = b = 0$ . The same argument works for any linear operator  $T$  in place of  $\mathbf{A}$ , with [Theorem 3.4.3](#). In Chapter 8 the ideal of all polynomials killing an operator is where the minimal polynomial comes from.

Now the non-example, by a minimal change. Take  $F[x]_{\leq n}$  for some  $n \in \mathbb{N}$ . It is non-empty, closed under addition, and closed under multiplication by **constants**: it is a subspace of  $F[x]$  ([Example \(Polynomials of Bounded Degree\)](#)). But it fails (I3):  $x^n \in F[x]_{\leq n}$  and  $x \in F[x]$ , while  $x \cdot x^n = x^{n+1} \notin F[x]_{\leq n}$ . The exact clause that fails is “for **every**  $h \in F[x]$ ”.

Why insist on every  $h$ , and not just constants? Because the main theorem below divides a member  $f$  of  $I$  by another member  $d$  and needs the remainder  $f - qd$  to lie in  $I$  again. The quotient  $q$  is an arbitrary polynomial, so we need  $qd \in I$ . With only closure under scalars, the argument breaks, and indeed  $F[x]_{\leq n}$  is not the set of multiples of any single polynomial. The name “ideal” is historical and says nothing about the definition.

*Warning.* Do not confuse  $\langle f, g \rangle$  with  $\text{span}(f, g)$ . The span uses **scalar** coefficients, the ideal uses **polynomial** coefficients. For example,  $\text{span}(x) = \{cx : c \in F\}$  is a line in  $F[x]$ , while  $\langle x \rangle$  contains  $x^2, x^3, \dots$  and is infinite-dimensional. Every ideal is a subspace, but most subspaces, such as  $F[x]_{\leq n}$  or  $\text{span}(x)$ , are not ideals.

*Quick check.* Is  $J = \{f \in \mathbb{R}[x] : f(1) = 0 \text{ or } f(2) = 0\}$  an ideal of  $\mathbb{R}[x]$ ?

*Solution.* No.  $J$  is non-empty and satisfies (I3), since  $(hf)(c) = h(c)f(c)$ . But it fails (I2):  $x - 1 \in J$  and  $x - 2 \in J$ , while their sum  $2x - 3$  takes the values  $-1$  at 1 and 1 at 2, so  $2x - 3 \notin J$ . The set with “and” in place of “or” is an ideal.

### 5.2.3 Every ideal is generated by one polynomial

The payoff of the definition is that ideals of  $F[x]$  are very simple: each one is the set of multiples of a single polynomial. The proof is the argument of the opening example, made general. We have met it before for integers, in [Lemma 0.7.2](#), where the smallest positive combination  $ax + ny$  turned out to divide  $a$  and  $n$ .

#### Theorem 5.2.2 — Every Ideal of $F[x]$ Is Principal

Let  $F$  be a field and  $I$  an ideal of  $F[x]$ . If  $I \neq \{0\}$ , there is **exactly one monic** polynomial  $d$  with  $I = \langle d \rangle$ . Moreover,  $d$  is the unique monic polynomial of least degree in  $I$ .

*Idea.* Pick a non-zero  $d \in I$  of smallest degree, made monic. Every multiple of  $d$  is in  $I$  by (I3). For the other inclusion, divide any  $f \in I$  by  $d$ : the remainder  $r = f - qd$  lies in  $I$  and has degree smaller than  $\deg d$ . Minimality leaves only one option,  $r = 0$ . Uniqueness is mutual divisibility of monic polynomials.

*Proof.* Suppose  $I \neq \{0\}$ . The degrees of the non-zero elements of  $I$  form a non-empty set of natural numbers, so by [Theorem 0.2.11](#) it has a least element  $m$ . Choose  $g \in I$  with  $\deg g = m$ , and let  $c$  be its leading coefficient. Put  $d = c^{-1}g$ . Then  $d \in I$  by (I3), and  $d$  is monic of degree  $m$ .

( $\supseteq$ ) For every  $h \in F[x]$ ,  $hd \in I$  by (I3). Hence  $\langle d \rangle \subseteq I$ .

( $\subseteq$ ) Let  $f \in I$ . By [Theorem 5.1.1](#),  $f = qd + r$  with  $\deg r < m$ . By (I3),  $-qd \in I$ , and by (I2),  $r = f + (-qd) \in I$ . If  $r \neq 0$ , then  $r$  is a non-zero element of  $I$  of degree less than  $m$ , contradicting the choice of  $m$ . Hence  $r = 0$ , so  $f = qd \in \langle d \rangle$ .

Therefore  $I = \langle d \rangle$ . For uniqueness, let  $d'$  be monic with  $I = \langle d' \rangle$ . Then  $d \in \langle d' \rangle$  and  $d' \in \langle d \rangle$ , so  $d' \mid d$  and  $d \mid d'$ . Both are monic, so  $d = d'$  by [Proposition 5.1.2](#) (d).

Finally, let  $e \in I$  be monic of least degree  $m$ . Then  $d \mid e$ , so  $e = ud$  with  $\deg u = 0$  by [Theorem 0.8.3](#), and comparing leading coefficients gives  $u = 1$ . Hence  $e = d$ . This proves the theorem. ■

For  $I = \{0\}$  the only generator is 0, since  $\langle p \rangle$  contains  $p$ . So every ideal of  $F[x]$  is principal, with a canonical generator: 0 for the zero ideal, and the monic polynomial of least degree otherwise. The named move is: **the smallest-degree element of an ideal generates it**. In [Example \(First Ideals\)](#) (c), the ideal contains  $x^2$  and nothing non-zero of smaller degree, so it equals  $\langle x^2 \rangle$ .

The theorem also turns divisibility into containment. For  $p, q \in F[x]$ ,

$$\langle p \rangle \subseteq \langle q \rangle \iff q \mid p,$$

because  $\langle p \rangle \subseteq \langle q \rangle$  holds exactly when  $p \in \langle q \rangle$ , by [Proposition 5.2.1](#). The bigger the ideal, the smaller the generator.

*Quick check.* Find the monic generator of  $\langle x^2 - 1, x^3 - 1 \rangle \subseteq \mathbb{Q}[x]$  by finding an element of small degree in it.

*Solution.* The ideal contains  $(x^3 - 1) - x(x^2 - 1) = x - 1$ . Both generators are multiples of  $x - 1$ , since  $x^2 - 1 = (x - 1)(x + 1)$  and  $x^3 - 1 = (x - 1)(x^2 + x + 1)$ , so every combination is a multiple of  $x - 1$  by Proposition 5.1.2 (b). A non-zero constant is not a multiple of  $x - 1$ , so  $x - 1$  has least degree in the ideal. By Theorem 5.2.2, the ideal is  $\langle x - 1 \rangle$ .

## 5.2.4 The greatest common divisor

Apply Theorem 5.2.2 to the ideal  $\langle f, g \rangle$  of all combinations. It has a canonical generator, and in the opening example that generator,  $x - 1$ , was a common divisor of  $f$  and  $g$ . A naive attempt to define “the greatest common divisor” as “a common divisor of largest degree” runs into a problem at once:  $x - 1$ ,  $2x - 2$  and  $-\frac{1}{3}x + \frac{1}{3}$  are all common divisors of  $x^2 - 1$  and  $x^2 + x - 2$  of the largest possible degree. Degree cannot tell associates apart, and it also does not say how the common divisors relate to each other. The generator of the ideal fixes both problems.

*The gcd of  $f$  and  $g$  is the monic polynomial whose multiples are exactly the combinations of  $f$  and  $g$ .*

### Definition 5.2.2 — Greatest Common Divisor

Let  $F$  be a field and  $f, g \in F[x]$ . If  $f$  and  $g$  are **not both zero**, the **greatest common divisor**  $\gcd(f, g)$  is the **unique monic** polynomial  $d \in F[x]$  with

$$\langle f, g \rangle = \langle d \rangle.$$

We set  $\gcd(0, 0) = 0$ .

In words: form the ideal of all combinations  $af + bg$  with  $a, b \in F[x]$ ; it is generated by one polynomial; take the monic one. The definition refers to  $F[x]$  twice, in the coefficients  $a, b$  and in the generator  $d$ , so at first sight the gcd depends on the field. We show below that it does not change when the field is enlarged.

**Well-definedness.** By Proposition 5.2.1,  $\langle f, g \rangle$  is an ideal. If  $f$  and  $g$  are not both zero, it contains a non-zero element, so by Theorem 5.2.2 there is exactly one monic  $d$  with  $\langle f, g \rangle = \langle d \rangle$ . The convention  $\gcd(0, 0) = 0$  keeps the equation  $\langle f, g \rangle = \langle \gcd(f, g) \rangle$  true in the remaining case. The definition is symmetric:  $\gcd(f, g) = \gcd(g, f)$ .

### Example. (First Greatest Common Divisors)

- (a) Let  $f \neq 0$  have leading coefficient  $a$ . Find  $\gcd(f, 0)$  and  $\gcd(f, c)$  for a non-zero constant  $c$ .
- (b) Find  $\gcd(x^2 - 1, x^2 + x - 2)$  in  $\mathbb{Q}[x]$ .
- (c) Find  $\gcd(x^2 + 1, x + 1)$  in  $\mathbb{Q}[x]$  and in  $\mathbb{F}_2[x]$ .

*Solution.*

- (a)  $\langle f, 0 \rangle = \{af : a \in F[x]\} = \langle f \rangle = \langle a^{-1}f \rangle$ , and  $a^{-1}f$  is monic, so  $\gcd(f, 0) = a^{-1}f$ , the monic associate of  $f$ . This degenerate case shows why the gcd must be made



monic:  $f$  itself generates the ideal too, but is not the canonical choice. Next,  $\langle f, c \rangle$  contains the non-zero constant  $c$ , so it is  $F[x] = \langle 1 \rangle$  by [Example \(First Ideals\)](#) (a). Hence  $\gcd(f, c) = 1$ .

- (b) The opening example showed  $\langle x^2 - 1, x^2 + x - 2 \rangle = \langle x - 1 \rangle$ , and  $x - 1$  is monic. So the gcd is  $x - 1$ .
- (c) In  $\mathbb{Q}[x]$ : dividing,  $x^2 + 1 = (x - 1)(x + 1) + 2$ . So  $2 = (x^2 + 1) - (x - 1)(x + 1)$  lies in the ideal, which is therefore  $\mathbb{Q}[x]$  by [Example \(First Ideals\)](#) (a). Hence  $\gcd = 1$ . In  $\mathbb{F}_2[x]$ :  $x^2 + 1 = (x + 1)^2$ , so  $x^2 + 1 \in \langle x + 1 \rangle$ , and  $\langle x^2 + 1, x + 1 \rangle = \langle x + 1 \rangle$  by [Proposition 5.2.1](#). Hence  $\gcd = x + 1$ . The symbols are the same, but  $\mathbb{F}_2$  is not a field containing  $\mathbb{Q}$ , so the two answers need not agree.

For a non-example by minimal change, keep (b) but replace  $x - 1$  by the associate  $3x - 3$ . It is still a common divisor of largest degree and still generates the ideal, but it is **not**  $\gcd(x^2 - 1, x^2 + x - 2)$ : the clause “monic” fails. Or keep “monic” but lower the degree: 1 is a monic common divisor, and it does not generate the ideal, since  $1 \notin \langle x - 1 \rangle$ .

Why define the gcd through the ideal, rather than as “the largest common divisor”? Because the ideal definition delivers three things at once: existence and uniqueness (from [Theorem 5.2.2](#)), the fact that the gcd is a combination of  $f$  and  $g$ , and the fact that it is “greatest” in the strongest sense, namely that every common divisor **divides** it. Here is the last point.

### Theorem 5.2.3 — Characterization of the gcd

Let  $F$  be a field,  $f, g \in F[x]$ , and  $d = \gcd(f, g)$ .

- (a)  $d \mid f$  and  $d \mid g$ .
- (b) If  $h \in F[x]$  satisfies  $h \mid f$  and  $h \mid g$ , then  $h \mid d$ .
- (c) If  $f, g$  are not both zero, then  $d$  is the **only** monic polynomial with properties (a) and (b).

*Proof.*

- (a)  $f = 1 \cdot f + 0 \cdot g \in \langle f, g \rangle = \langle d \rangle$ , so  $d \mid f$ . Similarly  $d \mid g$ .
- (b) Since  $d \in \langle d \rangle = \langle f, g \rangle$ , there are  $a, b \in F[x]$  with  $d = af + bg$ . If  $h \mid f$  and  $h \mid g$ , then  $h \mid af + bg = d$  by [Proposition 5.1.2](#) (b).
- (c) Let  $e$  be monic with properties (a) and (b). Since  $d$  is a common divisor of  $f$  and  $g$ , property (b) for  $e$  gives  $d \mid e$ . Since  $e$  is a common divisor, property (b) for  $d$  gives  $e \mid d$ . Both are monic ( $e$  by assumption, and  $d$  by [Definition 5.2.2](#), since  $f, g$  are not both zero), so  $d = e$  by [Proposition 5.1.2](#) (d). This proves the theorem. ■

So  $\gcd(f, g)$  is a common divisor that every common divisor divides. If  $f, g$  are not both zero, then  $d \neq 0$ , and by [Proposition 5.1.2](#) (c) every common divisor has degree at most  $\deg d$ ; “greatest” is true for degrees as well. The proof of (b) used one fact that deserves its own statement, because it is the most useful property of the gcd.

**Corollary 5.2.4 — Bézout's Identity for Polynomials**

Let  $F$  be a field and  $f, g \in F[x]$ . Then there exist  $a, b \in F[x]$  with

$$\gcd(f, g) = af + bg.$$

*Proof.* By [Definition 5.2.2](#),  $\gcd(f, g) \in \langle \gcd(f, g) \rangle = \langle f, g \rangle$ , and the elements of  $\langle f, g \rangle$  are exactly the combinations  $af + bg$ . ■

The coefficients are not unique: if  $d = af + bg$ , then also  $d = (a + g)f + (b - f)g$ . Bézout's identity is the polynomial version of [Lemma 0.7.2](#), and it will be the tool behind the Chinese Remainder Theorem and the splitting of kernels later in this chapter.

**5.2.5 The Euclidean algorithm**

The definition tells us what the gcd is, but not how to find it. Factoring  $f$  and  $g$  is usually hard. Division is easy, and one observation turns division into an algorithm: dividing does not change the ideal.

**Lemma 5.2.5 — Dividing Does Not Change the Ideal**

Let  $F$  be a field and  $f, g, q, r \in F[x]$  with  $f = qg + r$ . Then  $\langle f, g \rangle = \langle g, r \rangle$ . Consequently  $\gcd(f, g) = \gcd(g, r)$ .

*Proof.* ( $\supseteq$ ) The ideal  $\langle f, g \rangle$  contains  $g$ , and  $r = f + (-q)g$  by (I2) and (I3). By [Proposition 5.2.1](#),  $\langle g, r \rangle \subseteq \langle f, g \rangle$ .

( $\subseteq$ ) The ideal  $\langle g, r \rangle$  contains  $g$  and  $f = qg + r$ . By [Proposition 5.2.1](#),  $\langle f, g \rangle \subseteq \langle g, r \rangle$ .

Hence the two ideals are equal. The gcd is the canonical generator of the ideal ([Definition 5.2.2](#)), and  $f, g$  are both zero exactly when  $g, r$  are, so  $\gcd(f, g) = \gcd(g, r)$ . ■

Now divide repeatedly. Each division replaces the pair  $(f, g)$  by  $(g, r)$ , a pair with the same gcd and a smaller second entry.

**Theorem 5.2.6 — Euclidean Algorithm**

Let  $F$  be a field and  $f, g \in F[x]$  with  $g \neq 0$ . Put  $r_{-1} = f$  and  $r_0 = g$ . As long as  $r_k \neq 0$ , divide  $r_{k-1}$  by  $r_k$  ([Theorem 5.1.1](#)):

$$r_{k-1} = q_{k+1}r_k + r_{k+1}, \quad \deg r_{k+1} < \deg r_k.$$

Then:

- (a) the process stops: there is an index  $n$  with  $0 \leq n \leq \deg g$  such that  $r_n \neq 0$  and  $r_{n+1} = 0$ ;
- (b) if  $c$  is the leading coefficient of this last non-zero remainder  $r_n$ , then  $\gcd(f, g) = c^{-1}r_n$ .

*Idea.* Two things to see. The degrees  $\deg r_0 > \deg r_1 > \deg r_2 > \dots$  strictly decrease, and natural numbers cannot decrease forever, so some remainder is 0. By [Lemma 5.2.5](#), the ideal  $\langle r_{k-1}, r_k \rangle$  never changes along the way; at the end it is  $\langle r_n, 0 \rangle = \langle r_n \rangle$ , and making  $r_n$  monic gives the gcd.

*Proof.*

- (a) As long as  $r_1, \dots, r_j$  are all non-zero, their degrees are natural numbers with  $\deg g = \deg r_0 > \deg r_1 > \dots > \deg r_j \geq 0$ . So  $r_1, \dots, r_j$  are  $j$  distinct values in  $\{0, 1, \dots, \deg g - 1\}$ , and  $j \leq \deg g$ . Hence the remainders cannot all be non-zero for  $j = \deg g + 1$ : there is a first index  $n + 1$  with  $r_{n+1} = 0$ , and  $n \leq \deg g$ . Also  $n \geq 0$  since  $r_0 = g \neq 0$ .
- (b) We show by induction on  $k$  that  $\langle r_{k-1}, r_k \rangle = \langle f, g \rangle$  for  $k = 0, 1, \dots, n+1$ . For  $k = 0$  this is the definition of  $r_{-1}, r_0$ . If it holds for some  $k \leq n$ , then  $r_{k-1} = q_{k+1}r_k + r_{k+1}$ , and [Lemma 5.2.5](#) gives  $\langle r_k, r_{k+1} \rangle = \langle r_{k-1}, r_k \rangle = \langle f, g \rangle$ .

For  $k = n + 1$  this reads  $\langle f, g \rangle = \langle r_n, 0 \rangle = \langle r_n \rangle$ . Since  $c \neq 0$ ,  $\langle r_n \rangle = \langle c^{-1}r_n \rangle$ , and  $c^{-1}r_n$  is monic. Because  $g \neq 0$ , [Definition 5.2.2](#) gives  $\gcd(f, g) = c^{-1}r_n$ . This proves the theorem. ■

If  $g = 0$ , no division is needed:  $\gcd(f, 0)$  is the monic associate of  $f$  (or 0), as in [Example \(First Greatest Common Divisors\)](#) (a).

The algorithm also produces Bézout coefficients. Each remainder is a combination of the previous two,  $r_{k+1} = r_{k-1} - q_{k+1}r_k$ , so by induction each remainder is a combination of  $f$  and  $g$ . Keeping track of the coefficients gives the following procedure.

*Algorithm. Extended Euclidean algorithm.* Input:  $f, g \in F[x]$  with  $g \neq 0$ .

1. Set  $(r_{-1}, a_{-1}, b_{-1}) = (f, 1, 0)$  and  $(r_0, a_0, b_0) = (g, 0, 1)$ .
2. For  $k = 0, 1, 2, \dots$ , as long as  $r_k \neq 0$ : divide  $r_{k-1} = q_{k+1}r_k + r_{k+1}$ , and set

$$a_{k+1} = a_{k-1} - q_{k+1}a_k, \quad b_{k+1} = b_{k-1} - q_{k+1}b_k.$$

3. When  $r_{n+1} = 0$ , let  $c$  be the leading coefficient of  $r_n$ . Output  $\gcd(f, g) = c^{-1}r_n$  and the coefficients  $a = c^{-1}a_n$ ,  $b = c^{-1}b_n$ .

The invariant  $r_k = a_k f + b_k g$  holds for  $k = -1, 0$  by the choice in step 1. If it holds for  $k - 1$  and  $k$ , then

$$r_{k+1} = r_{k-1} - q_{k+1}r_k = (a_{k-1} - q_{k+1}a_k)f + (b_{k-1} - q_{k+1}b_k)g = a_{k+1}f + b_{k+1}g.$$

So at the end  $\gcd(f, g) = c^{-1}r_n = af + bg$ . By hand, it is often quicker to write each division as “remainder = ...” and substitute backwards, which is the same computation.

**Example. (The Euclidean Algorithm with Back-Substitution)**

In  $\mathbb{Q}[x]$ , let  $f = x^4 - x^3 - 3x^2 + x + 2$  and  $g = x^3 - 2x - 1$ . Find  $\gcd(f, g)$  and polynomials  $a, b$  with  $\gcd(f, g) = af + bg$ .

*Solution. Divisions.* First divide  $f$  by  $g$ . The leading terms give  $x$ , and  $f - xg = -x^3 - x^2 + 2x + 2$ ; then  $-1$ , and adding  $g$  leaves  $-x^2 + 1$ . So

$$f = (x - 1)g + r_1, \quad r_1 = 1 - x^2.$$

Next divide  $g$  by  $r_1$ . The leading terms give  $x^3/(-x^2) = -x$ , and  $g - (-x)r_1 = x^3 - 2x - 1 + x - x^3 = -x - 1$ . So

$$g = (-x)r_1 + r_2, \quad r_2 = -x - 1.$$

Next divide  $r_1$  by  $r_2$ . Since  $(x - 1)(-x - 1) = -x^2 + 1$ ,

$$r_1 = (x - 1)r_2 + 0.$$

The last non-zero remainder is  $r_2 = -x - 1$ , with leading coefficient  $-1$ . By [Theorem 5.2.6](#),  $\gcd(f, g) = x + 1$ .

*Back-substitution.* From the first two divisions,

$$r_1 = f - (x - 1)g, \quad r_2 = g + xr_1.$$

Substituting the first into the second,

$$r_2 = g + x(f - (x - 1)g) = xf + (1 + x - x^2)g.$$

Multiplying by  $c^{-1} = -1$ ,

$$x + 1 = (-x)f + (x^2 - x - 1)g.$$

In the notation of the algorithm,  $(a_1, b_1) = (1, 1 - x)$  and  $(a_2, b_2) = (x, 1 + x - x^2)$ .

*Checks.* Expanding,  $-x(x^4 - x^3 - 3x^2 + x + 2) = -x^5 + x^4 + 3x^3 - x^2 - 2x$  and  $(x^2 - x - 1)(x^3 - 2x - 1) = x^5 - x^4 - 3x^3 + x^2 + 3x + 1$ , which sum to  $x + 1$ . As an independent check that  $x + 1$  divides both,  $f(-1) = 1 + 1 - 3 - 1 + 2 = 0$  and  $g(-1) = -1 + 2 - 1 = 0$ , so [Theorem 5.1.3](#) applies.

## 5.2.6 Coprime polynomials

The most important value a gcd can take is 1. Then  $f$  and  $g$  share no non-constant factor, and Bézout's identity writes 1 itself as a combination.

### Definition 5.2.3 — Coprime Polynomials

Let  $F$  be a field. Polynomials  $f, g \in F[x]$  are **coprime** (or **relatively prime**) over  $F$  if  $\gcd(f, g) = 1$ .

### Proposition 5.2.7 — Coprime Means 1 Is a Combination

Let  $F$  be a field and  $f, g \in F[x]$ . Then  $f$  and  $g$  are coprime if and only if there exist  $a, b \in F[x]$  with  $af + bg = 1$ .

*Proof.* ( $\Rightarrow$ ) This is [Corollary 5.2.4](#) with  $\gcd(f, g) = 1$ .

( $\Leftarrow$ ) If  $af + bg = 1$ , then  $1 \in \langle f, g \rangle$ , so  $\langle f, g \rangle = F[x] = \langle 1 \rangle$  by [Example \(First Ideals\)](#) (a). In particular  $f, g$  are not both zero, and the monic generator is 1. Hence  $\gcd(f, g) = 1$ . ■

Examples: for **distinct**  $c, c' \in F$ , the polynomials  $x - c$  and  $x - c'$  are coprime, since

$$(c' - c)^{-1}(x - c) - (c' - c)^{-1}(x - c') = (c' - c)^{-1}(c' - c) = 1.$$

A non-zero constant is coprime to every polynomial ([Example \(First Greatest Common Divisors\)](#) (a)). The polynomial 0 is coprime only to the non-zero constants. Non-example:  $x^2 - 1$  and  $x + 1$  are not coprime, since  $x + 1$  divides both, so  $x + 1 \mid \gcd$  by [Theorem 5.2.3](#) (b), and the gcd has degree at least 1.

Coprimality is what lets a divisor pass through a product.

### Theorem 5.2.8 — A Coprime Factor Can Be Canceled

Let  $F$  be a field and  $p, q, r \in F[x]$ . If  $p \mid qr$  and  $\gcd(p, q) = 1$ , then  $p \mid r$ .

*Proof.* By [Proposition 5.2.7](#), there are  $a, b \in F[x]$  with  $ap + bq = 1$ . Multiplying by  $r$ ,  $r = (ar)p + b(qr)$ . Here  $p \mid p$  and  $p \mid qr$  by hypothesis, so  $p \mid r$  by [Proposition 5.1.2](#) (b). This proves the theorem. ■

*Warning.* The coprimality hypothesis cannot be dropped. Take  $p = x^2$  and  $q = r = x$ . Then  $p \mid qr = x^2$ , but  $p \nmid r$ , since  $\deg x^2 > \deg x$  ([Proposition 5.1.2](#) (c)). Here  $\gcd(p, q) = x \neq 1$ : part of  $p$  is absorbed by  $q$  and part by  $r$ .

In the next section, [Theorem 5.2.8](#) becomes Euclid's lemma for irreducible polynomials, which is the heart of unique factorization.

**Least common multiples**, briefly. For  $f, g \in F[x]$ , the intersection  $\langle f \rangle \cap \langle g \rangle$  is again an ideal: it contains 0, so (I1) holds, and (I2) and (I3) hold in each of  $\langle f \rangle$  and  $\langle g \rangle$ , hence in the intersection. Its members are the common multiples of  $f$  and  $g$ .

### Definition 5.2.4 — Least Common Multiple

Let  $F$  be a field and  $f, g \in F[x]$ . The **least common multiple**  $\text{lcm}(f, g)$  is the monic generator of the ideal  $\langle f \rangle \cap \langle g \rangle$  if this ideal is non-zero, and 0 otherwise.

Exactly as for the gcd,  $\text{lcm}(f, g)$  is a common multiple of  $f$  and  $g$  that divides every common

multiple, by [Theorem 5.2.2](#). If  $f = 0$  or  $g = 0$ , the only common multiple is 0; otherwise  $fg$  is a non-zero common multiple. For monic  $f$  and  $g$ ,  $\gcd(f, g) \cdot \text{lcm}(f, g) = fg$ ; this is proved in Exercise C3. For example, in  $\mathbb{Q}[x]$ ,  $\gcd(x^2 - 1, x^2 + x - 2) = x - 1$ , so  $\text{lcm}(x^2 - 1, x^2 + x - 2) = (x - 1)(x + 1)(x + 2)$ .

## 5.2.7 The gcd does not depend on the field

A polynomial with rational coefficients is also a polynomial with real and with complex coefficients. Its gcd with another rational polynomial can be computed in  $\mathbb{Q}[x]$ ,  $\mathbb{R}[x]$  or  $\mathbb{C}[x]$ , and the definition does not obviously give the same answer, because the coefficients  $a, b$  in  $\langle f, g \rangle$  range over a larger set. The Euclidean algorithm shows the answers agree. Recall that a subfield  $K$  of a field  $L$  is a subset closed under the field operations ([Definition 1.9.1](#)); then  $K[x] \subseteq L[x]$ , with the same addition and multiplication.

### Proposition 5.2.9 — The gcd Is the Same Over a Larger Field

Let  $K$  be a subfield of a field  $L$ , and let  $f, g \in K[x]$ . Then the greatest common divisor of  $f$  and  $g$  computed in  $K[x]$  equals the one computed in  $L[x]$ . In particular,  $f$  and  $g$  are coprime over  $K$  if and only if they are coprime over  $L$ .

*Proof.* If  $g = 0$ , both gcds are the monic associate of  $f$  (or 0), by [Example \(First Greatest Common Divisors\)](#) (a), and this does not depend on the field. Let  $g \neq 0$ , and run the Euclidean algorithm in  $K[x]$ :  $r_{k-1} = q_{k+1}r_k + r_{k+1}$  with  $\deg r_{k+1} < \deg r_k$ , all in  $K[x]$ . Each of these equations also holds in  $L[x]$ , with the same degree condition. By the uniqueness in [Theorem 5.1.1](#) over  $L$ , they are exactly the divisions that the algorithm performs in  $L[x]$ . So both runs produce the same last non-zero remainder  $r_n$ , and [Theorem 5.2.6](#) gives the same gcd  $c^{-1}r_n$  in both. This proves the proposition. ■

*Warning.* The gcd does not see the field, but **factorization does**. Over  $\mathbb{Q}$ , the polynomial  $x^2 - 2$  has no root, by [Theorem 0.2.3](#), so it has no factor  $x - c$  with  $c \in \mathbb{Q}$  ([Theorem 5.1.3](#)). Over  $\mathbb{R}$  it factors as  $(x - \sqrt{2})(x + \sqrt{2})$ . Yet  $\gcd(x^2 - 2, x^2 - 2x)$ , say, is the same over  $\mathbb{Q}$  and over  $\mathbb{R}$ . So “coprime” is a robust notion, while “cannot be factored” depends on the field; the next section makes the second notion precise.

A useful consequence: if  $f, g \in \mathbb{Q}[x]$  are coprime over  $\mathbb{Q}$ , then they have **no common root in  $\mathbb{C}$** . Indeed,  $af + bg = 1$  with  $a, b \in \mathbb{Q}[x] \subseteq \mathbb{C}[x]$ , and evaluating at a common root  $z$  would give  $0 = 1$  by [Theorem 0.8.5](#). So common complex roots can be detected by a computation that never leaves  $\mathbb{Q}$ .

## 5.2.8 Exercises

### 5.2.8.1 A. Check your understanding

#### Exercise A1.

- (a) Define an ideal of  $F[x]$ .
- (b) State the theorem describing all ideals of  $F[x]$ .
- (c) Determine whether the following statement is true: “for all  $f, g \in F[x]$ ,  $\text{span}(f, g) = \langle f, g \rangle$ .” Justify your answer.
- (d) Is  $\text{gcd}(2x - 2, 4x^2 - 4) = 2x - 2$  in  $\mathbb{Q}[x]$ ? Justify your answer.
- (e) Determine whether the following statement is true: “if  $af + bg = x - 1$  for some  $a, b \in \mathbb{Q}[x]$ , then  $\text{gcd}(f, g) = x - 1$ .” Justify your answer.
- (f) Name an algorithm that computes  $\text{gcd}(f, g)$  using only division with remainder (Theorem 5.1.1).

*Solution.*

- (a) A subset  $I \subseteq F[x]$  is an ideal if it is non-empty,  $f + g \in I$  for all  $f, g \in I$ , and  $hf \in I$  for all  $f \in I$  and every  $h \in F[x]$  (Definition 5.2.1).
- (b) If  $I \neq \{0\}$  is an ideal of  $F[x]$ , there is exactly one monic  $d$  with  $I = \langle d \rangle$ , namely the monic polynomial of least degree in  $I$  (Theorem 5.2.2).
- (c) False. Take  $f = g = x$ . Then  $\text{span}(x, x) = \{cx : c \in F\}$  does not contain  $x^2$ , while  $x^2 = x \cdot x \in \langle x, x \rangle$ . The span allows only scalar coefficients.
- (d) No. The gcd is monic by definition. Since  $4x^2 - 4 = (2x - 2)(2x + 2)$ , we have  $\langle 2x - 2, 4x^2 - 4 \rangle = \langle 2x - 2 \rangle = \langle x - 1 \rangle$ , so the gcd is  $x - 1$ .
- (e) False. Take  $f = x$  and  $g = x - 1$ , with  $a = 0$ ,  $b = 1$ . Then  $af + bg = x - 1$ , but  $\text{gcd}(x, x - 1) = 1$ , because  $x - (x - 1) = 1$  (Proposition 5.2.7). What is true is that  $\text{gcd}(f, g)$  divides every combination  $af + bg$ , by Theorem 5.2.3 (a) and Proposition 5.1.2 (b).
- (f) The Euclidean algorithm (Theorem 5.2.6).

### 5.2.8.2 B. Practice

#### Exercise B1: gcd and Bézout Coefficients of Two Cubics.

In  $\mathbb{Q}[x]$ , let  $f = x^3 + x^2 + x + 1$  and  $g = x^3 + x + 2$ . Use the Euclidean algorithm to find  $d = \text{gcd}(f, g)$  and polynomials  $a, b \in \mathbb{Q}[x]$  with  $d = af + bg$ . Hence find all complex numbers that are roots of both  $f$  and  $g$ .

*Solution.* Divide  $f$  by  $g$ :  $f = 1 \cdot g + r_1$  with  $r_1 = f - g = x^2 - 1$ . Divide  $g$  by  $r_1$ :  $x \cdot r_1 = x^3 - x$ , and  $g - xr_1 = 2x + 2$ , so  $g = xr_1 + r_2$  with  $r_2 = 2x + 2$ . Divide  $r_1$  by  $r_2$ :  $x^2 - 1 = \frac{1}{2}(x - 1)(2x + 2) + 0$ . The last non-zero remainder is  $r_2 = 2x + 2$ , with leading coefficient 2, so  $d = x + 1$  by Theorem 5.2.6.

Back-substituting,  $r_2 = g - xr_1 = g - x(f - g) = -xf + (1 + x)g$ . Dividing by 2,

$$x + 1 = \left(-\frac{x}{2}\right)f + \frac{x+1}{2}g.$$

Check:  $-\frac{x}{2}(x^3 + x^2 + x + 1) + \frac{x+1}{2}(x^3 + x + 2) = \frac{1}{2}(-x^4 - x^3 - x^2 - x + x^4 + x^3 + x^2 + 3x + 2) = x + 1$ .

Hence: let  $z \in \mathbb{C}$ . If  $f(z) = g(z) = 0$ , then evaluating the identity above in  $\mathbb{C}$  (Theorem 0.8.5) gives  $z + 1 = 0$ . Conversely,  $x + 1$  divides  $f$  and  $g$  (Theorem 5.2.3 (a)), so  $-1$

is a root of both. Therefore  $-1$  is the only common complex root.

### Exercise B2: Ideal or Not?

Determine which of the following subsets of  $\mathbb{R}[x]$  are ideals. For each ideal, find its monic generator. Justify your answers.

- (a)  $S_1 = \{f \in \mathbb{R}[x] : f(1) = f(-1)\}$ .
- (b)  $S_2 = \{f \in \mathbb{R}[x] : f(0) = 0 \text{ and } f(1) = 0\}$ .
- (c)  $S_3 = \{f \in \mathbb{R}[x] : \deg f \geq 3\} \cup \{0\}$ .
- (d)  $S_4 = \{f \in \mathbb{R}[x] : f(i) = 0\}$ , where  $f(i)$  means the value at  $i \in \mathbb{C}$  of  $f$  regarded as a complex polynomial.

*Solution.*

- (a) Not an ideal.  $S_1$  is a subspace, and  $1 \in S_1$ . But  $x \cdot 1 = x$  has  $x(1) = 1 \neq -1 = x(-1)$ , so  $x \notin S_1$ , and (I3) fails.
- (b) An ideal. The argument of [Example \(First Ideals\)](#) (b), applied at both points, gives (I1)–(I3). If  $f \in S_2$ , then 0 and 1 are distinct roots of  $f$ , so  $x(x-1) \mid f$  by [Corollary 5.1.4](#). Conversely every multiple of  $x^2 - x$  vanishes at 0 and 1. Hence  $S_2 = \langle x^2 - x \rangle$ , with monic generator  $x^2 - x$ .
- (c) Not an ideal.  $x^3 \in S_3$  and  $-x^3 + x \in S_3$ , but their sum  $x$  has degree 1 and is non-zero, so (I2) fails.
- (d) An ideal: evaluation at  $i$  respects sums and products ([Theorem 0.8.5](#) over  $\mathbb{C}$ ), so the argument of [Example \(First Ideals\)](#) (b) applies. It contains  $x^2 + 1$ . A non-zero element of degree at most 1 would be  $a + bx$  with  $a, b \in \mathbb{R}$  not both zero and  $a + bi = 0$ , which is impossible since the real and imaginary parts give  $a = b = 0$ . So the least degree of a non-zero element is 2. If  $d$  is the monic generator ([Theorem 5.2.2](#)), then  $\deg d = 2$  and  $d \mid x^2 + 1$ , so  $x^2 + 1 = ud$  with  $u$  constant, and comparing leading coefficients gives  $u = 1$ . Hence  $S_4 = \langle x^2 + 1 \rangle$ .

### Exercise B3: The gcd of $x^m - 1$ and $x^n - 1$ .

Let  $F$  be a field. For integers  $m, n \geq 0$ , not both zero, let  $\gcd(m, n)$  denote the largest positive integer dividing both.

- (a) For  $m \geq n \geq 1$ , verify that  $x^m - 1 = x^{m-n}(x^n - 1) + (x^{m-n} - 1)$ .
- (b) Prove that  $\gcd(x^m - 1, x^n - 1) = x^{\gcd(m, n)} - 1$  in  $F[x]$  for all  $m, n \geq 0$  not both zero, where  $x^0 - 1 = 0$ .
- (c) Hence compute  $\gcd(x^{12} - 1, x^{18} - 1)$ .

*Hint for (b):* [Theorem 5.2.6](#), run on the exponents.

*Solution.*

- (a) Expanding,  $x^{m-n}(x^n - 1) + x^{m-n} - 1 = x^m - x^{m-n} + x^{m-n} - 1 = x^m - 1$ .
- (b) We use strong induction on  $s = m + n \geq 1$  ([Theorem 0.2.10](#)). If  $n = 0$ , then  $m \geq 1$ ,  $\gcd(m, 0) = m$ , and  $\gcd(x^m - 1, 0)$  is the monic associate of  $x^m - 1$ , which



is  $x^m - 1$  itself ([Example \(First Greatest Common Divisors\)](#) (a)). The case  $m = 0$  is the same, since both gcds are symmetric. Now let  $m, n \geq 1$ ; by symmetry we may assume  $m \geq n$ . By (a) and [Lemma 5.2.5](#),

$$\gcd(x^m - 1, x^n - 1) = \gcd(x^n - 1, x^{m-n} - 1).$$

The pair  $(n, m - n)$  is not both zero and has sum  $m < m + n$ , so by the induction hypothesis the right side is  $x^{\gcd(n, m-n)} - 1$ . Finally,  $\gcd(n, m - n) = \gcd(m, n)$ : if  $k \mid m$  and  $k \mid n$ , then  $k \mid m - n$ ; and if  $k \mid n$  and  $k \mid m - n$ , then  $k \mid (m - n) + n = m$ . So the two pairs have the same positive common divisors, and the same largest one. This completes the induction.

(c) By (b),  $\gcd(x^{12} - 1, x^{18} - 1) = x^{\gcd(12, 18)} - 1 = x^6 - 1$ .

### 5.2.8.3 C. Going deeper

#### Exercise C1: Polynomials Vanishing on a Set.

Let  $F$  be a field and  $S \subseteq F$  non-empty. Put  $I(S) = \{f \in F[x] : f(s) = 0 \text{ for all } s \in S\}$ .

- (a) Prove that  $I(S)$  is an ideal of  $F[x]$ .
- (b) Suppose  $S = \{s_1, \dots, s_k\}$  is finite, with the  $s_i$  distinct. Prove that  $I(S) = \langle (x - s_1) \cdots (x - s_k) \rangle$ .
- (c) Suppose  $S$  is infinite. Prove that  $I(S) = \{0\}$ .
- (d) Deduce from (b) and [Exercise \(C3: Equal Functions Over  \$\mathbb{F}\_p\$ \)](#) that  $I(\mathbb{F}_p) = \langle x^p - x \rangle$  for every prime  $p$ .

*Solution.*

- (a) The zero polynomial vanishes everywhere, so  $I(S) \neq \emptyset$ . If  $f, g \in I(S)$  and  $h \in F[x]$ , then for every  $s \in S$ , [Theorem 0.8.5](#) gives  $(f + g)(s) = 0 + 0 = 0$  and  $(hf)(s) = h(s) \cdot 0 = 0$ . So (I2) and (I3) hold.
- (b) Let  $P = (x - s_1) \cdots (x - s_k)$ . Every multiple  $hP$  vanishes at each  $s_i$ , since  $P(s_i) = 0$ ; so  $\langle P \rangle \subseteq I(S)$ . Conversely, if  $f \in I(S)$ , then  $s_1, \dots, s_k$  are distinct roots of  $f$ , and [Corollary 5.1.4](#) gives  $P \mid f$ . Hence  $I(S) = \langle P \rangle$ , and since  $P$  is monic, it is the monic generator.
- (c) Let  $f \in I(S)$ , and suppose  $f \neq 0$ . Since  $S$  is infinite, it contains  $\deg f + 1$  distinct elements, all roots of  $f$ . This contradicts [Corollary 5.1.4](#). Hence  $f = 0$ .
- (d) By (b),  $I(\mathbb{F}_p) = \langle P \rangle$  with  $P = \prod_{c \in \mathbb{F}_p} (x - c)$ . The solution of [Exercise \(C3: Equal Functions Over  \$\mathbb{F}\_p\$ \)](#) (a) showed  $P = x^p - x$ . Hence  $I(\mathbb{F}_p) = \langle x^p - x \rangle$ .

#### Exercise C2: A Non-Principal Ideal in Two Variables.

A **polynomial in two variables** over  $F$  is a finite sum  $P = \sum_{i,j} a_{ij} x^i y^j$  with  $a_{ij} \in F$ , only finitely many non-zero; two are equal when all coefficients agree. They are added coefficient by coefficient and multiplied by expanding and collecting, and the set  $F[x, y]$  of all of them satisfies the laws of [Theorem 0.8.1](#), which you may use. Ideals of  $F[x, y]$  and  $\langle P_1, \dots, P_k \rangle$  are defined exactly as in [Definition 5.2.1](#), and [Proposition 5.2.1](#) holds with the same proof. For  $P \neq 0$ , let  $\deg_y P$  be the largest  $j$  with  $a_{ij} \neq 0$  for some  $i$ , and define  $\deg_x P$  similarly.

- (a) Prove that  $\deg_y(PQ) = \deg_y P + \deg_y Q$  for non-zero  $P, Q \in F[x, y]$ . By symmetry the same holds for  $\deg_x$ .
- (b) Show that  $\langle x, y \rangle$  consists of the polynomials whose constant coefficient  $a_{00}$  is 0.
- (c) Prove that  $\langle x, y \rangle \neq \langle D \rangle$  for every  $D \in F[x, y]$ .
- (d) Explain precisely which step of the proof of [Theorem 5.2.2](#) has no analogue in  $F[x, y]$ .

*Hint for (a):* write  $P = \sum_j p_j(x)y^j$  with  $p_j \in F[x]$ , and use [Corollary 0.8.4](#).

*Solution.*

- (a) Group terms by the power of  $y$ :  $P = \sum_{j=0}^s p_j y^j$  and  $Q = \sum_{j=0}^t q_j y^j$  with  $p_j, q_j \in F[x]$ ,  $s = \deg_y P$ ,  $t = \deg_y Q$ , so  $p_s \neq 0$  and  $q_t \neq 0$ . Expanding, the terms of  $PQ$  with  $y$ -power greater than  $s+t$  are all zero, and the terms with  $y^{s+t}$  come only from  $p_s y^s \cdot q_t y^t$ , giving  $(p_s q_t) y^{s+t}$ . By [Corollary 0.8.4](#),  $p_s q_t \neq 0$  in  $F[x]$ , so some coefficient of  $x^i y^{s+t}$  in  $PQ$  is non-zero. Hence  $\deg_y(PQ) = s+t$ .
- (b) Every  $Ax + By$  has constant coefficient 0, since each term of  $Ax$  and  $By$  involves  $x$  or  $y$  to a positive power. Conversely, if  $a_{00} = 0$ , then every term  $a_{ij}x^i y^j$  with  $i \geq 1$  is a multiple of  $x$ , and every term with  $i = 0, j \geq 1$  is a multiple of  $y$ . Grouping them,  $P = Ax + By$  for some  $A, B \in F[x, y]$ .
- (c) Suppose  $\langle x, y \rangle = \langle D \rangle$ . Since  $x \in \langle D \rangle$ ,  $x = HD$  for some  $H$ , so  $D \neq 0$  and  $H \neq 0$ . By (a),  $0 = \deg_y x = \deg_y H + \deg_y D$ , so  $\deg_y D = 0$ :  $D$  involves no  $y$ . Similarly  $y = KD$  gives  $\deg_x D = 0$ . So  $D$  is a non-zero constant. But  $D \in \langle x, y \rangle$ , so its constant coefficient is 0 by (b), that is,  $D = 0$ . This contradiction shows that  $\langle x, y \rangle$  is not principal.
- (d) The proof of [Theorem 5.2.2](#) picks a non-zero  $d \in I$  of least degree, writes each  $f \in I$  as  $f = qd + r$  with  $r$  smaller than  $d$  by [Theorem 5.1.1](#), notes  $r = f - qd \in I$ , and concludes  $r = 0$  from minimality. Measure size in  $F[x, y]$  by the **total degree**, the largest  $i+j$  with  $a_{ij} \neq 0$ . Choosing  $d$  still works: by (b), every non-zero element of  $\langle x, y \rangle$  has total degree at least 1, and those of total degree 1 are the  $ax + by$  with  $(a, b) \neq (0, 0)$ . Let  $d$  be any of them. Then  $\langle d \rangle \subseteq \langle x, y \rangle$  by (I3), and the inclusion is strict by (c), so some  $f \in \langle x, y \rangle$  is not a multiple of  $d$ . For this  $f$  and **every**  $Q \in F[x, y]$ , the element  $R = f - Qd$  lies in  $\langle x, y \rangle$  and is non-zero, so its total degree is at least 1, the total degree of  $d$ . So no remainder smaller than  $d$  exists: the step with no analogue is **division with remainder**, the appeal to [Theorem 5.1.1](#). Concretely, for  $d = x$  and  $f = y$ , every term of  $Qx$  contains  $x$ , so the term  $y$  survives in  $y - Qx$ . The underlying reason is that, viewed as a polynomial in  $y$  with coefficients in  $F[x]$ , the leading coefficient  $x$  of  $d$  is not invertible in  $F[x]$ , which is exactly the obstacle of the warning after [Theorem 5.1.1](#).

### Exercise C3: gcd Times lcm.

Let  $F$  be a field and  $f, g \in F[x]$  **monic**. Put  $d = \gcd(f, g)$  and write  $f = df_1$ ,  $g = dg_1$  with  $f_1, g_1 \in F[x]$ .

- (a) Prove that  $\gcd(f_1, g_1) = 1$ .
- (b) Prove that  $m = df_1 g_1$  is a monic common multiple of  $f$  and  $g$  that divides every common multiple of  $f$  and  $g$ .

- (c) Deduce that  $\text{lcm}(f, g) = df_1g_1$ , and hence  $\text{gcd}(f, g) \cdot \text{lcm}(f, g) = fg$ .  
 (d) Show by an example in  $\mathbb{Q}[x]$  that the formula in (c) fails if  $f$  and  $g$  are not monic.

*Hint for (b): use Theorem 5.2.8.*

*Solution.*

- (a) The quotients exist by Theorem 5.2.3 (a), and  $d \neq 0$  since  $f \neq 0$ . By Corollary 5.2.4,  $d = af + bg = d(af_1 + bg_1)$ . Canceling  $d \neq 0$  (Corollary 0.8.4 (2)) gives  $af_1 + bg_1 = 1$ , so  $\text{gcd}(f_1, g_1) = 1$  by Proposition 5.2.7.  
 (b) Comparing leading coefficients in  $f = df_1$  with  $f, d$  monic shows  $f_1$  is monic; likewise  $g_1$ . So  $m$  is monic, and  $m = fg_1 = gf_1$  is a common multiple. Let  $M$  be any common multiple, say  $M = fu$ . Since  $g \mid M$ , write  $M = gv$ . Then  $df_1u = dg_1v$ , and canceling  $d$  gives  $f_1u = g_1v$ , so  $g_1 \mid f_1u$ . By (a) and Theorem 5.2.8,  $g_1 \mid u$ , say  $u = g_1w$ . Hence  $M = fg_1w = mw$ , and  $m \mid M$ .  
 (c) By (b),  $m \in \langle f \rangle \cap \langle g \rangle$ , and every element of this ideal is a multiple of  $m$ . So  $\langle f \rangle \cap \langle g \rangle = \langle m \rangle$ , and  $m$  is monic, hence  $\text{lcm}(f, g) = m$  by Definition 5.2.4 and Theorem 5.2.2. Then  $\text{gcd}(f, g) \cdot \text{lcm}(f, g) = d \cdot df_1g_1 = (df_1)(dg_1) = fg$ .  
 (d) Take  $f = g = 2x$ . Then  $\text{gcd}(f, g) = x$  and  $\text{lcm}(f, g) = x$ , both monic, so their product is  $x^2$ , while  $fg = 4x^2$ . In general the two sides differ by the product of the leading coefficients.

## 5.3 Irreducible Polynomials and Unique Factorization

Every integer  $n \geq 2$  is a product of primes, and Chapter 0 remarked that the convention “1 is not prime” is what makes such factorizations unique. Polynomials over a field behave the same way, with irreducible polynomials in the role of primes and non-zero constants in the role of  $\pm 1$ . This section defines irreducible polynomials, proves Euclid’s lemma from the gcd machinery of the previous section, and proves that every non-constant polynomial factors into irreducibles in exactly one way. The catch is that “irreducible” depends on the field: the same polynomial may factor over  $\mathbb{R}$  but not over  $\mathbb{Q}$ , or over  $\mathbb{F}_5$  but not over  $\mathbb{F}_3$ . So we also collect tools for deciding irreducibility over specific fields: roots in low degree, the Rational Root Theorem, and Eisenstein’s criterion. Throughout,  $F$  is a field.

### 5.3.1 Irreducible polynomials

For integers, the building blocks are the primes: integers  $p \geq 2$  whose only positive divisors are 1 and  $p$  (Definition 0.2.2). To transplant this to  $F[x]$ , recall from the first section that every non-zero constant divides every polynomial, and that multiplying by a non-zero constant changes nothing for divisibility. So the trivial divisors of  $p$  are the non-zero constants and the associates  $cp$ . A building block should have no others.

*An irreducible polynomial is a non-constant polynomial that cannot be split into two factors of smaller degree.*

**Definition 5.3.1 — Irreducible Polynomial**

Let  $F$  be a field. A polynomial  $p \in F[x]$  is **irreducible over  $F$**  if

- (1)  $p$  is **non-constant**, that is,  $\deg p \geq 1$ ; and
- (2) whenever  $p = gh$  with  $g, h \in F[x]$ , **one of**  $g, h$  is a constant.

A non-constant polynomial that is not irreducible is **reducible over  $F$** .

In words: condition (1) excludes 0 and the non-zero constants. Condition (2) says every factorization of  $p$  is trivial: one factor is a constant, which is non-zero because  $p \neq 0$ , and the other is then an associate of  $p$ . The phrase **over  $F$**  is part of the notion, since  $g$  and  $h$  must lie in  $F[x]$ .

Since degrees add ([Theorem 0.8.3](#)), a factorization  $p = gh$  with  $p \neq 0$  has  $\deg g + \deg h = \deg p$ . So  $g$  is constant exactly when  $\deg h = \deg p$ . This gives a working form of the definition: **a non-constant  $p$  is reducible if and only if  $p = gh$  with  $1 \leq \deg g < \deg p$  and  $1 \leq \deg h < \deg p$ .**

**Example. (First Irreducible and Reducible Polynomials)**

- (a) Show that every polynomial of degree 1 is irreducible over every field.
- (b) Show that the constants 0 and 2 are not irreducible over  $\mathbb{Q}$ .
- (c) Show that  $x^2 + x$  is reducible over every field, and that  $x^3$  is reducible over every field.

*Solution.*

- (a) If  $\deg p = 1$  and  $p = gh$ , then  $\deg g + \deg h = 1$  with both degrees in  $\mathbb{N}$ , so one of them is 0: one factor is a constant. Condition (1) holds since  $\deg p = 1$ . Hence  $p$  is irreducible, whatever the field. For example  $x$ ,  $2x + 4$  and  $x - \sqrt{2}$  (over  $\mathbb{R}$ ) are irreducible.
- (b) Both are constants, so condition (1) fails. They are the degenerate cases, and they matter: 2 divides everything in  $\mathbb{Q}[x]$ , like 1 among the integers, and it is not a building block.
- (c)  $x^2 + x = x(x + 1)$  with both factors of degree 1, and  $x^3 = x \cdot x^2$  with factors of degrees 1 and 2. In each case both factors are non-constant, so condition (2) fails.

Why insist on “non-constant”? For the same reason 1 is not a prime. If non-zero constants were allowed as building blocks, factorizations could never be unique:  $x = 2 \cdot \frac{1}{2}x = 2 \cdot 3 \cdot \frac{1}{6}x = \dots$  over  $\mathbb{Q}$ , with any number of constant factors. The factorization theorem below pulls out one constant in front and uses only non-constant, monic irreducibles.

For polynomials of degree 2 and 3, irreducibility is decided by roots alone.

**Theorem 5.3.1 — Irreducibility in Degree 2 and 3**

Let  $F$  be a field and  $p \in F[x]$ .

- (a)  $p$  has a factor of degree 1 in  $F[x]$  if and only if  $p$  has a root in  $F$ . In particular, if  $\deg p \geq 2$  and  $p$  has a root in  $F$ , then  $p$  is reducible over  $F$ .

(b) If  $\deg p \in \{2, 3\}$ , then  $p$  is irreducible over  $F$  if and only if  $p$  has **no** root in  $F$ .

*Proof.*

- (a) ( $\Leftarrow$ ) If  $c \in F$  is a root, then  $p = (x - c)q$  by [Theorem 5.1.3](#) (b). ( $\Rightarrow$ ) If  $p = (ax + b)q$  with  $a \neq 0$ , then  $c = -a^{-1}b$  satisfies  $ac + b = 0$ , so  $p(c) = 0 \cdot q(c) = 0$  by [Theorem 0.8.5](#). For the last sentence, if  $\deg p \geq 2$  and  $p = (x - c)q$ , then  $\deg q = \deg p - 1 \geq 1$ , so both factors are non-constant of degree less than  $\deg p$ , and  $p$  is reducible.
- (b) We prove both directions by contraposition. ( $\Rightarrow$ ) If  $p$  has a root, it is reducible by (a), since  $\deg p \geq 2$ . ( $\Leftarrow$ ) Suppose  $p$  is reducible. By the working form of the definition,  $p = gh$  with  $\deg g, \deg h \geq 1$  and  $\deg g + \deg h = \deg p \in \{2, 3\}$ . Two integers that are at least 1 and sum to 2 or 3 cannot both be at least 2, so one of  $g, h$  has degree 1. By (a),  $p$  has a root in  $F$ . This proves the theorem. ■

Now the standard examples, each checked with the theorem.

- $x^2 + 1$  **over  $\mathbb{R}$  and over  $\mathbb{C}$** . For real  $c$ ,  $c^2 + 1 \geq 1 > 0$ , so there is no real root, and  $x^2 + 1$  is irreducible over  $\mathbb{R}$  (and over  $\mathbb{Q}$ ). Over  $\mathbb{C}$ ,  $x^2 + 1 = (x - i)(x + i)$  is reducible.
- $x^2 - 2$  **over  $\mathbb{Q}$  and over  $\mathbb{R}$** . A rational root would be a rational number with square 2, which does not exist ([Theorem 0.2.3](#)). So  $x^2 - 2$  is irreducible over  $\mathbb{Q}$ . Over  $\mathbb{R}$ ,  $x^2 - 2 = (x - \sqrt{2})(x + \sqrt{2})$ .
- $x^2 + x + 1$  **over  $\mathbb{F}_2$** . The values at 0 and 1 are 1 and  $1 + 1 + 1 = 1$ . No root, so it is irreducible over  $\mathbb{F}_2$ . It is the only irreducible polynomial of degree 2 over  $\mathbb{F}_2$  (Exercise B3).

The non-example by minimal change is  $x^2 + x$  over  $\mathbb{F}_2$ : change the constant term of  $x^2 + x + 1$  from 1 to 0, and 0 becomes a root, so  $x^2 + x = x(x + 1)$  is reducible. What fails is condition (2).

*Warning.* “No roots” implies “irreducible” **only in degrees 2 and 3**. The polynomial  $(x^2 + 1)^2 = x^4 + 2x^2 + 1$  has no real root, since its values are at least 1, but it is reducible over  $\mathbb{R}$ : it is a product of two factors of degree 2. In degree 4 and higher, a polynomial can split into non-linear factors without having any root.

*Quick check.* Over which of the fields  $\mathbb{Q}$ ,  $\mathbb{Q}(\sqrt{2})$  ([Example \( \$\mathbb{Q}\(\sqrt{2}\)\$  Is a Field\)](#)),  $\mathbb{R}$  is  $x^2 - 2$  irreducible? Is  $x^2 + 1$  irreducible over  $\mathbb{F}_5$ ?

*Solution.* Over  $\mathbb{Q}$  only. The field  $\mathbb{Q}(\sqrt{2})$  contains  $\sqrt{2}$ , a root of  $x^2 - 2$ , so  $x^2 - 2 = (x - \sqrt{2})(x + \sqrt{2})$  is reducible there, and likewise over  $\mathbb{R}$ ; over  $\mathbb{Q}$  it is irreducible, as shown above. Over  $\mathbb{F}_5$ ,  $2^2 + 1 = 5 = 0$ , so 2 is a root and  $x^2 + 1$  is reducible by [Theorem 5.3.1](#): indeed  $x^2 + 1 = (x - 2)(x + 2)$  in  $\mathbb{F}_5[x]$ .

## 5.3.2 Euclid’s lemma

What makes primes useful is not their definition but a property the definition implies: if a prime divides a product, it divides one of the factors. For polynomials this follows quickly from the previous section. The first step is that an irreducible polynomial has only two possible gcds with anything.

**Lemma 5.3.2 — The gcd with an Irreducible Polynomial**

Let  $F$  be a field,  $p \in F[x]$  irreducible over  $F$ , and  $f \in F[x]$ . Then either  $p \mid f$  or  $\gcd(p, f) = 1$ .

*Proof.* Let  $d = \gcd(p, f)$ . Since  $p \neq 0$ ,  $d$  is monic, and  $d \mid p$  by Theorem 5.2.3 (a), say  $p = du$ . Since  $p$  is irreducible,  $d$  or  $u$  is a constant. If  $d$  is a constant, then  $d = 1$ , since  $d$  is monic. If  $u$  is a constant, it is non-zero, and  $p = ud$  divides every multiple of  $d$ ; since  $d \mid f$  by Theorem 5.2.3 (a), Proposition 5.1.2 (a) gives  $p \mid f$ . This proves the lemma. ■

**Theorem 5.3.3 — Euclid's Lemma for Polynomials**

Let  $F$  be a field and  $p \in F[x]$  irreducible over  $F$ . If  $q, r \in F[x]$  and  $p \mid qr$ , then  $p \mid q$  or  $p \mid r$ .

*Proof.* Suppose  $p \nmid q$ . By Lemma 5.3.2,  $\gcd(p, q) = 1$ . Since  $p \mid qr$ , Theorem 5.2.8 gives  $p \mid r$ . This proves the theorem. ■

The hypothesis “irreducible” is essential. The polynomial  $x^2$  divides  $x \cdot x$  but divides neither factor; the lemma does not apply because  $x^2 = x \cdot x$  is reducible. Over  $\mathbb{F}_2$ ,  $x^2 + 1 = (x + 1)^2$  divides  $(x + 1)(x + 1)$  but not  $x + 1$ , for the same reason.

For unique factorization we need the lemma for products of any length, and we need to know what “divides” means between two monic irreducibles.

**Corollary 5.3.4 — Euclid's Lemma for Several Factors**

Let  $F$  be a field and  $p \in F[x]$  irreducible over  $F$ .

- (a) If  $q_1, \dots, q_k \in F[x]$  with  $k \geq 1$  and  $p \mid q_1 q_2 \cdots q_k$ , then  $p \mid q_i$  for some  $i$ .
- (b) If  $q$  is irreducible over  $F$ ,  $p \mid q$ , and  $p, q$  are both monic, then  $p = q$ .

*Proof.*

- (a) We use induction on  $k$ . For  $k = 1$  there is nothing to prove. Let  $k \geq 2$  and assume the statement for  $k - 1$  factors. Since  $p \mid (q_1 \cdots q_{k-1}) q_k$ , Theorem 5.3.3 gives  $p \mid q_1 \cdots q_{k-1}$  or  $p \mid q_k$ . In the first case the induction hypothesis gives  $p \mid q_i$  for some  $i \leq k - 1$ . Either way the statement holds for  $k$ .
- (b) Write  $q = pu$ . Since  $q$  is irreducible,  $p$  or  $u$  is constant. But  $p$  is not constant, being irreducible, so  $u$  is a non-zero constant. Comparing leading coefficients of the monic polynomials  $q$  and  $p$  gives  $u = 1$ , so  $p = q$ . ■

### 5.3.3 Unique factorization

Here is the theorem that Chapter 0's remark about primes was pointing towards, now for polynomials. The constant in front is forced to be the leading coefficient; the rest is a product of monic irreducibles, unique up to order.

#### Theorem 5.3.5 — Unique Factorization in $F[x]$

Let  $F$  be a field and  $f \in F[x]$  non-constant.

- (a) **Existence.** There are a non-zero constant  $c \in F$ , an integer  $k \geq 1$ , and monic polynomials  $p_1, \dots, p_k \in F[x]$ , each irreducible over  $F$ , with

$$f = c p_1 p_2 \cdots p_k.$$

- (b) **Uniqueness.** If also  $f = c' q_1 q_2 \cdots q_l$  with  $c' \in F$  non-zero,  $l \geq 1$ , and  $q_1, \dots, q_l$  monic and irreducible over  $F$ , then  $c = c'$ ,  $k = l$ , and after renumbering the  $q_j$  we have  $p_i = q_i$  for every  $i$ .

In both factorizations, the constant is the leading coefficient of  $f$ .

*Idea. Existence:* if  $f$  is irreducible, stop; otherwise split it into two factors of smaller degree and factor each of them, which is strong induction on the degree. **Uniqueness:** compare leading coefficients to get  $c = c'$ . Then  $p_k$  divides  $q_1 \cdots q_l$ , so by Euclid's lemma it equals some  $q_j$ . Cancel it from both sides; the two shorter factorizations are equal by induction on the number of factors  $k$ . The one thing to watch is that after canceling, the right-hand side must still have at least one factor.

*Proof.* First a remark used in both parts. A product of monic polynomials is monic, by Theorem 0.8.3 and induction on the number of factors. So if  $f = c p_1 \cdots p_k$  with the  $p_i$  monic, then the leading coefficient of  $f$  is  $c \cdot 1 = c$ . This proves the final sentence of the theorem.

- (a) We prove by strong induction on  $n \geq 1$  (Theorem 0.2.10 with  $n_0 = 1$ ) that every  $f$  of degree  $n$  has a factorization as in (a). Let  $\deg f = n$ , assume the statement for all degrees  $1 \leq m < n$ , and let  $c$  be the leading coefficient of  $f$ .

*Case 1:  $f$  is irreducible over  $F$ .* Then  $f = c(c^{-1}f)$ . The polynomial  $c^{-1}f$  is monic, and it is irreducible: it is non-constant, and if  $c^{-1}f = gh$ , then  $f = (cg)h$ , so  $cg$  or  $h$  is constant, hence  $g$  or  $h$  is constant. So  $k = 1$ ,  $p_1 = c^{-1}f$  works.

*Case 2:  $f$  is reducible over  $F$ .* By the working form of the definition,  $f = gh$  with  $1 \leq \deg g < n$  and  $1 \leq \deg h < n$ . By the induction hypothesis,  $g = a p_1 \cdots p_s$  and  $h = b p_{s+1} \cdots p_{s+t}$  with  $a, b$  non-zero constants,  $s, t \geq 1$ , and all  $p_i$  monic irreducible. Then  $f = (ab) p_1 \cdots p_{s+t}$ , with  $ab \neq 0$  by Theorem 0.7.1. This is a factorization as in (a), with  $k = s + t$ .

- (b) By the first remark,  $c$  and  $c'$  are both the leading coefficient of  $f$ , so  $c = c'$ . Multiplying by  $c^{-1}$ , we get  $p_1 \cdots p_k = q_1 \cdots q_l$ . So it suffices to prove, by induction on  $k \geq 1$ , the statement

$U(k)$ : whenever  $p_1 \cdots p_k = q_1 \cdots q_l$  with  $l \geq 1$  and all  $p_i, q_j$  monic and irreducible over  $F$ , we have  $k = l$ , and after renumbering the  $q_j$ ,  $p_i = q_i$  for



every  $i$ .

We first note: if a monic irreducible  $q$  equals a product  $r_1 \cdots r_m$  of  $m \geq 2$  monic irreducibles, then  $q = r_1 \cdot (r_2 \cdots r_m)$  with both factors non-constant, since each  $r_i$  has degree at least 1; this contradicts the irreducibility of  $q$ . So **a monic irreducible is never a product of two or more monic irreducibles.** (\*)

*Base case*  $k = 1$ . Then  $p_1 = q_1 \cdots q_l$ . By (\*),  $l = 1$ , and so  $p_1 = q_1$ .

*Inductive step.* Let  $k \geq 2$  and assume  $U(k-1)$ . Suppose  $p_1 \cdots p_k = q_1 \cdots q_l$ . If  $l = 1$ , then  $q_1 = p_1 \cdots p_k$  with  $k \geq 2$ , which is impossible by (\*). So  $l \geq 2$ . Now  $p_k \mid q_1 \cdots q_l$ , so by Corollary 5.3.4 (a),  $p_k \mid q_j$  for some  $j$ , and by Corollary 5.3.4 (b),  $p_k = q_j$ . Renumber the  $q$ 's so that  $j = l$ . Then

$$(p_1 \cdots p_{k-1}) p_k = (q_1 \cdots q_{l-1}) p_k,$$

and since  $p_k \neq 0$ , Corollary 0.8.4 (2) lets us cancel it:  $p_1 \cdots p_{k-1} = q_1 \cdots q_{l-1}$ , with  $k-1 \geq 1$  and  $l-1 \geq 1$  factors. By  $U(k-1)$ ,  $k-1 = l-1$ , and after renumbering  $q_1, \dots, q_{l-1}$ ,  $p_i = q_i$  for  $i \leq k-1$ . Together with  $p_k = q_k$ , this proves  $U(k)$ .

By induction,  $U(k)$  holds for every  $k \geq 1$ , which proves (b). ■

Grouping equal factors, every non-constant  $f \in F[x]$  can be written uniquely as

$$f = c p_1^{e_1} p_2^{e_2} \cdots p_s^{e_s}$$

with  $c$  the leading coefficient,  $p_1, \dots, p_s$  **distinct** monic irreducibles over  $F$ , and exponents  $e_i \geq 1$ . The monic irreducible factors of the characteristic and minimal polynomials of an operator will organize the structure theory of Chapters 8 and 9.

*Warning.* Uniqueness needs the normalization “monic”. Without it, constants can move between factors: over  $\mathbb{Q}$ ,  $2x^2 - 2 = (2x - 2)(x + 1) = (x - 1)(2x + 2) = (\frac{1}{2}x - \frac{1}{2})(4x + 4)$ . All of these are “factorizations into irreducibles”, and they differ only by associates. The theorem removes this freedom by collecting all constants into  $c$ . Uniqueness also refers to a **fixed** field: over  $\mathbb{Q}$  the factorization of  $x^2 - 2$  is  $x^2 - 2$  itself, and over  $\mathbb{R}$  it is  $(x - \sqrt{2})(x + \sqrt{2})$ .

The next example shows the field dependence in one table.

### Example. (Factoring $x^4 - 1$ Over Six Fields)

Factor  $x^4 - 1$  into a constant times monic irreducibles over  $\mathbb{Q}$ ,  $\mathbb{R}$ ,  $\mathbb{C}$ ,  $\mathbb{F}_2$ ,  $\mathbb{F}_3$ , and, for contrast,  $\mathbb{F}_5$ .

*Solution.* Over every field,  $x^4 - 1 = (x^2 - 1)(x^2 + 1) = (x - 1)(x + 1)(x^2 + 1)$ , by expanding. The factors  $x - 1$  and  $x + 1$  are irreducible by Example (First Irreducible and Reducible Polynomials) (a). Everything depends on  $x^2 + 1$ , which by Theorem 5.3.1 is irreducible exactly when it has no root in the field. The constant in front is always the leading coefficient 1.



- $\mathbb{Q}$  and  $\mathbb{R}$ . For real  $c$ ,  $c^2 + 1 \geq 1$ , so there is no root, and  $x^2 + 1$  is irreducible.
- $\mathbb{C}$ .  $x^2 + 1 = (x - i)(x + i)$ .
- $\mathbb{F}_2$ . Here  $-1 = 1$ , so  $x - 1 = x + 1$ , and  $x^2 + 1 = (x + 1)^2$  because  $2x = 0$ . All four factors coincide.
- $\mathbb{F}_3$ . The values of  $x^2 + 1$  at  $0, 1, 2$  are  $1, 2, 5 = 2$ . No root, so  $x^2 + 1$  is irreducible. Also  $x - 1 = x + 2$ .
- $\mathbb{F}_5$ .  $2^2 + 1 = 5 = 0$  and  $3^2 + 1 = 10 = 0$ , so  $x^2 + 1 = (x - 2)(x - 3) = (x + 3)(x + 2)$ .

| field          | factorization of $x^4 - 1$ into monic irreducibles |
|----------------|----------------------------------------------------|
| $\mathbb{Q}$   | $(x - 1)(x + 1)(x^2 + 1)$                          |
| $\mathbb{R}$   | $(x - 1)(x + 1)(x^2 + 1)$                          |
| $\mathbb{C}$   | $(x - 1)(x + 1)(x - i)(x + i)$                     |
| $\mathbb{F}_2$ | $(x + 1)^4$                                        |
| $\mathbb{F}_3$ | $(x + 1)(x + 2)(x^2 + 1)$                          |
| $\mathbb{F}_5$ | $(x + 1)(x + 2)(x + 3)(x + 4)$                     |

By Theorem 5.3.5, each row is **the** factorization over that field, up to order. Notice that the numbers of irreducible factors are 3, 3, 4, 4, 3, 4, and that over  $\mathbb{F}_2$  a repeated factor appears although  $x^4 - 1$  has no repeated factor over the other fields.

### 5.3.4 Tools over $\mathbb{Q}$ : rational roots and Eisenstein's criterion

Over  $\mathbb{Q}$ , Theorem 5.3.1 reduces irreducibility of a quadratic or cubic to the question of rational roots. There are infinitely many rational numbers to try, but for integer coefficients only finitely many candidates can work. We write  $\mathbb{Z}[x]$  for the polynomials in  $\mathbb{Q}[x]$  whose coefficients are all integers, as in the first section. Every rational number can be written as  $u/v$  with  $u \in \mathbb{Z}$ ,  $v \geq 1$ , and the only positive integer dividing both  $u$  and  $v$  equal to 1: take  $v \geq 1$  as small as possible, since a common divisor  $e > 1$  would give the representation  $(u/e)/(v/e)$  with a smaller denominator. We say  $u/v$  is **in lowest terms**.

#### Theorem 5.3.6 — Rational Root Theorem

Let  $f = a_n x^n + a_{n-1} x^{n-1} + \cdots + a_0 \in \mathbb{Z}[x]$  with  $n \geq 1$  and  $a_n \neq 0$ . If  $u/v \in \mathbb{Q}$  is in lowest terms and  $f(u/v) = 0$ , then  $u \mid a_0$  and  $v \mid a_n$ .

*Idea.* Clear denominators:  $v^n f(u/v) = a_n u^n + a_{n-1} u^{n-1} v + \cdots + a_0 v^n = 0$ . Every term except the last contains  $u$ , so  $u$  divides  $a_0 v^n$ ; every term except the first contains  $v$ , so  $v$  divides  $a_n u^n$ . Since  $u$  and  $v$  share no factor, the powers of  $v$  can be stripped off one at a time, using Bézout for integers.

*Proof.* Multiplying  $f(u/v) = 0$  by  $v^n$  gives

$$a_n u^n + a_{n-1} u^{n-1} v + \cdots + a_1 u v^{n-1} + a_0 v^n = 0.$$

Hence  $a_0 v^n = -u(a_n u^{n-1} + a_{n-1} u^{n-2} v + \cdots + a_1 v^{n-1})$ , so  $u \mid a_0 v^n$ ; and  $a_n u^n = -v(a_{n-1} u^{n-1} + \cdots + a_0 v^{n-1})$ , so  $v \mid a_n u^n$ .

Since  $v \geq 1$  and the only positive common divisor of  $u$  and  $v$  is 1, [Lemma 0.7.2](#) gives integers  $s, t$  with  $us + vt = 1$ . If  $b \in \mathbb{Z}$  and  $k \geq 1$  with  $u \mid bv^k$ , then

$$bv^{k-1} = bv^{k-1}(us + vt) = u(bv^{k-1}s) + t(bv^k)$$

is a sum of two multiples of  $u$ , so  $u \mid bv^{k-1}$ . Applying this  $n$  times to  $b = a_0$  gives  $u \mid a_0$ . Similarly, if  $v \mid cu^k$  with  $k \geq 1$ , then  $cu^{k-1} = s(cu^k) + v(cu^{k-1}t)$ , so  $v \mid cu^{k-1}$ ; applying this  $n$  times to  $c = a_n$  gives  $v \mid a_n$ . This proves the theorem. ■

A polynomial in  $\mathbb{Q}[x]$  with fractional coefficients can first be multiplied by a common denominator; this does not change its roots or its irreducibility.

### Example. (A Cubic That Is Irreducible Over $\mathbb{Q}$ )

Decide whether  $f = 2x^3 - 3x^2 + 4x - 1$  is irreducible over  $\mathbb{Q}$ .

*Solution.* By [Theorem 5.3.6](#), a rational root  $u/v$  in lowest terms has  $u \mid -1$  and  $v \mid 2$  with  $v \geq 1$ . So  $u = \pm 1$  and  $v \in \{1, 2\}$ , and the candidates are  $1, -1, \frac{1}{2}, -\frac{1}{2}$ . Evaluating,

$$\begin{aligned} f(1) &= 2, & f(-1) &= -10, \\ f\left(\frac{1}{2}\right) &= \frac{1}{4} - \frac{3}{4} + 2 - 1 = \frac{1}{2}, \\ f\left(-\frac{1}{2}\right) &= -\frac{1}{4} - \frac{3}{4} - 2 - 1 = -4. \end{aligned}$$

None is 0, so  $f$  has no rational root. Since  $\deg f = 3$ , [Theorem 5.3.1](#) shows that  $f$  is irreducible over  $\mathbb{Q}$ .

Beyond degree 3, roots are not enough, as the warning after [Theorem 5.3.1](#) showed. One classical criterion handles many polynomials of any degree at once. It looks at divisibility of the coefficients by a prime.

### Theorem 5.3.7 — Eisenstein's Criterion

Let  $f = a_n x^n + a_{n-1} x^{n-1} + \cdots + a_0 \in \mathbb{Z}[x]$  with  $n \geq 1$ , and let  $p$  be a prime such that

- (1)  $p \nmid a_n$ ;
- (2)  $p \mid a_i$  for every  $i = 0, 1, \dots, n-1$ ;
- (3)  $p^2 \nmid a_0$ .

Then  $f$  is irreducible over  $\mathbb{Q}$ .

The proof needs one more idea, **Gauss's lemma**: a polynomial with integer coefficients that factors over  $\mathbb{Q}$  already factors over  $\mathbb{Z}$ , with factors of the same degrees. With it, one reduces the coefficients modulo  $p$  and compares. Exercise C2 of this section proves Gauss's lemma and then Eisenstein's criterion, step by step; no result of this chapter depends on the criterion. Here we only use it in examples.

**Example. (Irreducible Polynomials of Every Degree Over  $\mathbb{Q}$ )**

- (a) Show that  $x^n - 2$  is irreducible over  $\mathbb{Q}$  for every  $n \geq 1$ . (b) Show that  $x^4 + 10x + 5$  is irreducible over  $\mathbb{Q}$ .

*Solution.*

- (a) Take  $p = 2$ . The leading coefficient 1 is not divisible by 2; the coefficients  $a_1, \dots, a_{n-1}$  are 0 and  $a_0 = -2$ , all divisible by 2; and  $4 \nmid -2$ . By [Theorem 5.3.7](#),  $x^n - 2$  is irreducible over  $\mathbb{Q}$ . So over  $\mathbb{Q}$  there are irreducible polynomials of every degree. (Over  $\mathbb{R}$ , by contrast,  $x^n - 2$  has the root  $\sqrt[n]{2}$  and is reducible for  $n \geq 2$ .)
- (b) Take  $p = 5$ :  $5 \nmid 1$ ; 5 divides 0, 0, 10, 5; and  $25 \nmid 5$ . So  $x^4 + 10x + 5$  is irreducible over  $\mathbb{Q}$ . Roots alone could not have shown this, since a quartic may factor into two quadratics.

Eisenstein's criterion is sufficient, not necessary. The polynomial  $x^2 + 1$  is irreducible over  $\mathbb{Q}$ , but no prime divides its constant term 1, so condition (2) fails for every  $p$ .

## 5.3.5 Exercises

### 5.3.5.1 A. Check your understanding

#### Exercise A1.

- (a) Define what it means for  $p \in F[x]$  to be irreducible over  $F$ .
- (b) State the Unique Factorization Theorem in  $F[x]$ .
- (c) Determine whether the following statement is true: “a polynomial in  $\mathbb{R}[x]$  of degree at least 1 with no real root is irreducible over  $\mathbb{R}$ .” Justify your answer.
- (d) Is the constant 2 irreducible over  $\mathbb{Q}$ ? Is  $2x + 4$ ?
- (e) Is  $x^2 + 1$  irreducible over  $\mathbb{F}_5$ ? Justify your answer.
- (f) Name a tool that reduces the irreducibility of a cubic in  $\mathbb{Z}[x]$  over  $\mathbb{Q}$  to finitely many evaluations.

*Solution.*

- (a)  $p$  is irreducible over  $F$  if  $\deg p \geq 1$  and, whenever  $p = gh$  with  $g, h \in F[x]$ , one of  $g, h$  is a constant ([Definition 5.3.1](#)).
- (b) Every non-constant  $f \in F[x]$  can be written as  $f = cp_1 \cdots p_k$  with  $c \in F$  non-zero and  $p_1, \dots, p_k$  monic and irreducible over  $F$ ; the constant  $c$  is the leading coefficient of  $f$ , and the  $p_i$  are unique up to order ([Theorem 5.3.5](#)).
- (c) False.  $(x^2 + 1)^2$  has no real root, but it is the product of two non-constant factors, so it is reducible over  $\mathbb{R}$ . The statement is true in degrees 2 and 3 only ([Theorem 5.3.1](#)).
- (d) No: 2 is constant, and irreducible polynomials are non-constant by definition. Yes:  $2x + 4$  has degree 1, so it is irreducible ([Example \(First Irreducible and Reducible Polynomials\)](#) (a)).

- (e) No. In  $\mathbb{F}_5$ ,  $2^2 + 1 = 0$ , so  $x^2 + 1 = (x - 2)(x + 2)$  is reducible by [Theorem 5.3.1](#).  
 (f) The Rational Root Theorem ([Theorem 5.3.6](#)), combined with [Theorem 5.3.1](#).

### 5.3.5.2 B. Practice

#### Exercise B1: Irreducible or Not?.

Determine whether each polynomial is irreducible over each of the given fields. Justify your answers.

- (a)  $x^2 + 2$  over  $\mathbb{Q}$ ,  $\mathbb{R}$ ,  $\mathbb{F}_2$  and  $\mathbb{F}_3$ .  
 (b)  $x^3 + x + 1$  over  $\mathbb{F}_2$  and over  $\mathbb{F}_3$ .  
 (c)  $x^3 - 3x - 1$  over  $\mathbb{Q}$ .  
 (d)  $x^4 + 4$  over  $\mathbb{Q}$ .  
 (e)  $x^5 - 6x + 3$  over  $\mathbb{Q}$ .

*Solution.*

- (a) Over  $\mathbb{Q}$  and  $\mathbb{R}$ : for real  $c$ ,  $c^2 + 2 \geq 2 > 0$ , so there is no root, and  $x^2 + 2$  is irreducible by [Theorem 5.3.1](#). Over  $\mathbb{F}_2$ :  $2 = 0$ , so  $x^2 + 2 = x \cdot x$  is reducible. Over  $\mathbb{F}_3$ :  $2 = -1$ , so  $x^2 + 2 = x^2 - 1 = (x - 1)(x + 1)$  is reducible.  
 (b) Over  $\mathbb{F}_2$ : the values at 0 and 1 are 1 and  $3 = 1$ , so there is no root, and the cubic is irreducible by [Theorem 5.3.1](#). Over  $\mathbb{F}_3$ : the value at 1 is  $3 = 0$ , so it is reducible; indeed  $x^3 + x + 1 = (x - 1)(x^2 + x + 2)$  in  $\mathbb{F}_3[x]$ , as expanding shows:  $x^3 + x^2 + 2x - x^2 - x - 2 = x^3 + x - 2 = x^3 + x + 1$ .  
 (c) By [Theorem 5.3.6](#), a rational root  $u/v$  in lowest terms has  $u \mid -1$  and  $v \mid 1$ , so it is 1 or  $-1$ . The values are  $1 - 3 - 1 = -3$  and  $-1 + 3 - 1 = 1$ . No rational root, so the cubic is irreducible over  $\mathbb{Q}$  by [Theorem 5.3.1](#).  
 (d) Reducible, although it has no rational root (its values are at least 4): expanding,  $(x^2 + 2x + 2)(x^2 - 2x + 2) = x^4 - 4x^2 + 4 + 4x^2 = x^4 + 4$ . This is the warning after [Theorem 5.3.1](#) in action.  
 (e) Irreducible by [Theorem 5.3.7](#) with  $p = 3$ :  $3 \nmid 1$ ; 3 divides the coefficients 0, 0, 0,  $-6$ , 3 of  $x^4, x^3, x^2, x, 1$ ; and  $9 \nmid 3$ .

#### Exercise B2: Factoring $x^6 - 1$ Over $\mathbb{Q}$ .

Factor  $x^6 - 1$  into monic irreducible polynomials over  $\mathbb{Q}$ , and justify that each factor is irreducible. Hence find its factorization into monic irreducibles over  $\mathbb{F}_2$  as well. Is it obtained by reducing the coefficients of each rational factor modulo 2? Explain why, in general, this has to be checked rather than assumed.

*Solution.* Using  $x^6 - 1 = (x^3 - 1)(x^3 + 1)$  and [Exercise \(B2: A Factor of  \$x^n - c^n\$ \)](#) (with  $c = 1$  and  $c = -1$ ),

$$x^6 - 1 = (x - 1)(x^2 + x + 1)(x + 1)(x^2 - x + 1),$$

which one checks by expanding:  $(x - 1)(x^2 + x + 1) = x^3 - 1$  and  $(x + 1)(x^2 - x + 1) = x^3 + 1$ . The linear factors are irreducible. For the quadratics,  $c^2 \pm c + 1 = (c \pm \frac{1}{2})^2 + \frac{3}{4} > 0$  for real

$c$ , so they have no rational root and are irreducible by [Theorem 5.3.1](#). By [Theorem 5.3.5](#) this is the factorization over  $\mathbb{Q}$ .

Over  $\mathbb{F}_2$ , reducing coefficients gives  $(x+1)(x^2+x+1)(x+1)(x^2+x+1)$ , since  $-1 = 1$ . Here  $x+1$  is irreducible and  $x^2+x+1$  is irreducible over  $\mathbb{F}_2$  (it has no root), so the factorization over  $\mathbb{F}_2$  is  $(x+1)^2(x^2+x+1)^2$ . So yes, in this case the factorization is obtained by reduction, but only because the reduced factors happen to stay irreducible, and that has to be checked: two distinct rational factors,  $x-1$  and  $x+1$ , became equal, and in general a factor that is irreducible over  $\mathbb{Q}$  may become reducible after reduction, as  $x^2+1$  does over  $\mathbb{F}_2$  in [Example \(Factoring  \$x^4-1\$  Over Six Fields\)](#).

### Exercise B3: Small Irreducibles Over $\mathbb{F}_2$ .

List all irreducible polynomials over  $\mathbb{F}_2$  of degree at most 3. Justify that your list is complete.

*Hint: a polynomial over  $\mathbb{F}_2$  has the root 0 exactly when its constant term is 0, and the root 1 exactly when its number of non-zero terms is even.*

*Solution.* Every non-zero polynomial over  $\mathbb{F}_2$  is monic, since its leading coefficient is 1. For  $f = \sum a_k x^k$ ,  $f(0) = a_0$  and  $f(1) = \sum a_k$ , which is 0 exactly when the number of non-zero coefficients is even.

*Degree 1:*  $x$  and  $x+1$ , both irreducible ([Example \(First Irreducible and Reducible Polynomials\)](#) (a)).

*Degrees 2 and 3:* by [Theorem 5.3.1](#),  $f$  is irreducible exactly when  $f(0) = 1$  and  $f(1) = 1$ , that is, the constant term is 1 and the number of non-zero terms is odd. In degree 2, the polynomials with constant term 1 are  $x^2+1$  (two terms) and  $x^2+x+1$  (three terms), so only  $x^2+x+1$  is irreducible. In degree 3, they are  $x^3+1$  (two terms),  $x^3+x+1$  and  $x^3+x^2+1$  (three terms), and  $x^3+x^2+x+1$  (four terms). So the irreducible ones are  $x^3+x+1$  and  $x^3+x^2+1$ .

Hence the complete list is  $x$ ,  $x+1$ ,  $x^2+x+1$ ,  $x^3+x+1$ ,  $x^3+x^2+1$ .

### 5.3.5.3 C. Going deeper

#### Exercise C1: Infinitely Many Irreducibles.

Let  $F$  be any field.

- Prove that there are infinitely many monic irreducible polynomials over  $F$ .
- Deduce that for every prime  $p$  and every  $N \in \mathbb{N}$ , there is an irreducible polynomial over  $\mathbb{F}_p$  of degree greater than  $N$ .
- Explain why (a) is immediate when  $F$  is infinite, and use [Exercise \(B3: Small Irreducibles Over  \$\mathbb{F}\_2\$ \)](#) to name an irreducible polynomial of degree 3 over  $\mathbb{F}_2$ .

*Hint for (a): imitate the proof that there are infinitely many primes ([Theorem 0.2.13](#)).*

*Solution.*

- (a) Suppose, for a contradiction, that  $p_1, \dots, p_k$  are all the monic irreducible polynomials over  $F$ . The list is non-empty, since  $x$  is monic and irreducible. Let  $f = p_1 p_2 \cdots p_k + 1$ . Each  $p_i$  has degree at least 1, so  $\deg(p_1 \cdots p_k) \geq 1$  by [Theorem 0.8.3](#), and  $f$  is non-constant by [Theorem 0.8.2](#). By [Theorem 5.3.5](#) (a),  $f$  has a monic irreducible factor, which must be some  $p_j$ . Then  $p_j \mid f$  and  $p_j \mid p_1 \cdots p_k$ , so  $p_j \mid f - p_1 \cdots p_k = 1$  by [Proposition 5.1.2](#) (b). By [Proposition 5.1.2](#) (c),  $\deg p_j \leq \deg 1 = 0$ , contradicting  $\deg p_j \geq 1$ . Hence there are infinitely many monic irreducibles.
- (b) A monic polynomial of degree  $n$  over  $\mathbb{F}_p$  is determined by its  $n$  lower coefficients, so there are exactly  $p^n$  of them, and there are  $1 + p + \cdots + p^N$  monic polynomials of degree at most  $N$ , a finite number. By (a), infinitely many monic irreducibles exist, so some of them have degree greater than  $N$ .
- (c) If  $F$  is infinite, the polynomials  $x - c$  for  $c \in F$  are infinitely many distinct monic polynomials of degree 1, all irreducible by [Example \(First Irreducible and Reducible Polynomials\)](#) (a). Over a finite field there are only finitely many polynomials of degree 1, so the content of (a) is that irreducibles of higher and higher degree keep appearing. Over  $\mathbb{F}_2$ , for example,  $x^3 + x + 1$  is irreducible of degree 3 by [Exercise \(B3: Small Irreducibles Over  \$\mathbb{F}\_2\$ \)](#).

### Exercise C2: Gauss's Lemma and Eisenstein's Criterion.

Let  $\mathbb{Z}[x]$  be as in [Theorem 5.3.6](#), and let  $p$  be a prime. For  $g = \sum b_i x^i \in \mathbb{Z}[x]$ , define its **reduction modulo  $p$**  as  $\bar{g} = \sum [b_i] x^i \in \mathbb{F}_p[x]$ , where  $[b]$  is the class of  $b$  in  $\mathbb{F}_p = \mathbb{Z}/p\mathbb{Z}$  ([Example \(Arithmetic on  \$\mathbb{Z}/n\mathbb{Z}\$  and the Field  \$\mathbb{F}\_2\$ \)](#)).

- (a) Prove that  $\overline{g+h} = \bar{g} + \bar{h}$  and  $\overline{gh} = \bar{g}\bar{h}$  for all  $g, h \in \mathbb{Z}[x]$ , and that  $\deg \bar{g} \leq \deg g$ .
- (b) Let  $g, h \in \mathbb{Z}[x]$ , and suppose  $p$  divides every coefficient of  $gh$ . Prove that  $p$  divides every coefficient of  $g$  or every coefficient of  $h$ .
- (c) (**Gauss's lemma.**) Let  $f \in \mathbb{Z}[x]$ , and suppose  $f = gh$  with  $g, h \in \mathbb{Q}[x]$ . Prove that there is a non-zero  $\lambda \in \mathbb{Q}$  with  $\lambda g \in \mathbb{Z}[x]$  and  $\lambda^{-1}h \in \mathbb{Z}[x]$ .
- (d) Let  $F$  be a field,  $c \in F$  non-zero,  $n \in \mathbb{N}$ , and  $u, v \in F[x]$  with  $uv = cx^n$ . Prove that  $u = \beta x^s$  and  $v = \gamma x^t$  for some non-zero  $\beta, \gamma \in F$  and  $s, t \in \mathbb{N}$ .
- (e) Prove [Theorem 5.3.7](#).

*Hint for (c): among all ways of writing  $kf = (\alpha g)(\beta h)$  with  $\alpha, \beta \in \mathbb{Q}$ ,  $\alpha\beta = k$  a positive integer and  $\alpha g, \beta h \in \mathbb{Z}[x]$ , take  $k$  as small as possible. Hint for (e): suppose  $f = gh$  with  $g, h$  non-constant, use (c), and reduce modulo  $p$ .*

*Solution.*

- (a) By [Example \(Arithmetic on  \$\mathbb{Z}/n\mathbb{Z}\$  and the Field  \$\mathbb{F}\_2\$ \)](#),  $[a] + [b] = [a+b]$  and  $[a][b] = [ab]$ . Let  $g = \sum b_i x^i$  and  $h = \sum c_i x^i$ . The coefficient of  $x^k$  in  $g+h$  is  $b_k + c_k$ , whose class is  $[b_k] + [c_k]$ , the coefficient of  $x^k$  in  $\bar{g} + \bar{h}$ . The coefficient of  $x^k$  in  $gh$  is  $\sum_{i+j=k} b_i c_j$ , whose class is  $\sum_{i+j=k} [b_i][c_j]$ , the coefficient of  $x^k$  in  $\bar{g}\bar{h}$  ([Definition 0.8.2](#)). Finally, if  $b_i = 0$  then  $[b_i] = 0$ , so  $\bar{g}$  has no non-zero coefficient beyond position  $\deg g$ .
- (b) The hypothesis says  $\overline{gh} = 0$ . By (a),  $\bar{g}\bar{h} = 0$  in  $\mathbb{F}_p[x]$ . Since  $p$  is prime,  $\mathbb{F}_p$  is a field ([Theorem 0.7.3](#)), so [Corollary 0.8.4](#) gives  $\bar{g} = 0$  or  $\bar{h} = 0$ . That is,  $p$  divides every coefficient of  $g$  or every coefficient of  $h$ .

- (c) Choose positive integers  $m, n$  with  $mg \in \mathbb{Z}[x]$  and  $nh \in \mathbb{Z}[x]$ , for instance the products of the denominators of the coefficients. Let  $K$  be the set of positive integers  $k$  for which there are  $\alpha, \beta \in \mathbb{Q}$  with  $\alpha\beta = k$ ,  $\alpha g \in \mathbb{Z}[x]$  and  $\beta h \in \mathbb{Z}[x]$ . Then  $mn \in K$ , so by [Theorem 0.2.11](#)  $K$  has a least element  $k$ , with corresponding  $\alpha, \beta$ . Suppose  $k > 1$ . By [Proposition 0.2.12](#), some prime  $q$  divides  $k$ . Now  $(\alpha g)(\beta h) = \alpha\beta gh = kf$ , and every coefficient of  $kf$  is divisible by  $q$ . By (b) with the prime  $q$ ,  $q$  divides every coefficient of  $\alpha g$  or every coefficient of  $\beta h$ ; say the first (the other case is symmetric). Then  $(\alpha/q)g \in \mathbb{Z}[x]$ ,  $\beta h \in \mathbb{Z}[x]$ , and  $(\alpha/q)\beta = k/q$  is a positive integer smaller than  $k$ , which lies in  $K$ . This contradicts the minimality of  $k$ . Hence  $k = 1$ , so  $\beta = \alpha^{-1}$ , and  $\lambda = \alpha$  works.
- (d) Since  $uv \neq 0$ , both  $u, v$  are non-zero. Write  $u = \sum u_i x^i$ ,  $v = \sum v_j x^j$ , let  $i_0$  and  $j_0$  be the smallest indices with  $u_{i_0} \neq 0$  and  $v_{j_0} \neq 0$ , and let  $s = \deg u$ ,  $t = \deg v$ . The coefficient of  $x^{i_0+j_0}$  in  $uv$  is  $\sum_{i+j=i_0+j_0} u_i v_j$ . A term with  $i < i_0$  has  $u_i = 0$ , and a term with  $i > i_0$  has  $j < j_0$ , so  $v_j = 0$ . So this coefficient is  $u_{i_0} v_{j_0} \neq 0$ . The only non-zero coefficient of  $cx^n$  is at  $x^n$ , so  $i_0 + j_0 = n$ . By [Theorem 0.8.3](#), also  $s + t = n$ . Since  $i_0 \leq s$  and  $j_0 \leq t$ , this forces  $i_0 = s$  and  $j_0 = t$ . So  $u$  has exactly one non-zero coefficient,  $u = u_s x^s$ , and likewise  $v = v_t x^t$ .
- (e) Suppose  $f$  is reducible over  $\mathbb{Q}$ :  $f = gh$  with  $g, h \in \mathbb{Q}[x]$ ,  $\deg g = s \geq 1$ ,  $\deg h = t \geq 1$ , and  $s + t = n$ . By (c), replacing  $g, h$  by  $\lambda g, \lambda^{-1} h$  (which have the same degrees), we may assume  $g, h \in \mathbb{Z}[x]$ . Write  $g_0, h_0$  for their constant coefficients, so  $a_0 = g_0 h_0$ . Reduce modulo  $p$ . By conditions (1) and (2),  $\bar{f} = [a_n]x^n$  with  $[a_n] \neq 0$ . By (a),  $\bar{g}\bar{h} = [a_n]x^n$ , so by (d),  $\bar{g} = \beta x^{s'}$  and  $\bar{h} = \gamma x^{t'}$  with  $\beta, \gamma \neq 0$ . By [Theorem 0.8.3](#),  $s' + t' = n = s + t$ , while  $s' \leq s$  and  $t' \leq t$  by (a). Hence  $s' = s \geq 1$  and  $t' = t \geq 1$ . So the constant coefficients of  $\bar{g}$  and  $\bar{h}$  are 0, that is,  $p \mid g_0$  and  $p \mid h_0$ . Then  $p^2 \mid g_0 h_0 = a_0$ , contradicting condition (3). Therefore  $f$  is irreducible over  $\mathbb{Q}$ ; it is non-constant since  $n \geq 1$ .

## 5.4 Roots, Multiplicity and Derivatives

Chapter 2 bounded the number of roots of a polynomial, and the first section of this chapter restated the bound as [Corollary 5.1.4](#): a non-zero polynomial of degree  $n$  has at most  $n$  roots. The bound counts each root once, and so it misses something. The polynomial  $(x - 1)^2$  has degree 2 but only one root, and the root 1 is somehow “used twice”. This section makes “used twice” precise with the multiplicity of a root, sharpens the root bound, and then gives a test for repeated roots that needs no factoring: the formal derivative. The derivative behaves differently in characteristic  $p$ , and we say exactly where. The section ends with Taylor expansion of a polynomial, which turns out to be a dual basis from Chapter 4.

### 5.4.1 Multiplicity of a root

By the Factor Theorem ([Lemma 2.7.1](#), restated in general as [Theorem 5.1.3](#)),  $c$  is a root of  $f$  exactly when  $x - c$  divides  $f$ . A repeated root should mean that a higher power of  $x - c$  divides  $f$ . So we measure how many factors  $x - c$  can be pulled out.

*The multiplicity of  $c$  counts how many times  $x - c$  divides  $f$ .*



**Definition 5.4.1 — Multiplicity of a Root**

Let  $F$  be a field,  $f \in F[x]$  **non-zero** and  $c \in F$ . The **multiplicity of  $c$  as a root of  $f$** , written  $\text{mult}_c(f)$ , is the **largest**  $m \in \mathbb{N}$  such that  $(x - c)^m \mid f$ .

The root  $c$  is **simple** if  $\text{mult}_c(f) = 1$ , and **multiple** (or **repeated**) if  $\text{mult}_c(f) \geq 2$ .

In words: keep dividing  $f$  by  $x - c$  as long as the division leaves no remainder, and count the successful divisions. Divisibility is meant in the sense of [Definition 5.1.1](#), so  $(x - c)^m \mid f$  says  $f = (x - c)^m g$  for some  $g \in F[x]$ .

There is something to check: a largest such  $m$  must exist. The value  $m = 0$  always works, since  $(x - c)^0 = 1$  divides every polynomial. And if  $f = (x - c)^m g$ , then  $g \neq 0$  because  $f \neq 0$ , so [Theorem 0.8.3](#) gives  $\deg f = m + \deg g \geq m$ . The admissible  $m$  form a non-empty set of natural numbers bounded by  $\deg f$ , so it has a largest element. Moreover  $\text{mult}_c(f) \geq 1$  exactly when  $x - c \mid f$ , that is, exactly when  $c$  is a root. So multiplicity 0 means “not a root”.

The hypothesis  $f \neq 0$  is not decoration. Every power  $(x - c)^m$  divides the zero polynomial, since  $0 = (x - c)^m \cdot 0$ . There is no largest  $m$ , and the zero polynomial has no multiplicities.

A definition by “largest  $m$ ” is awkward to use directly, because to confirm  $m$  we must rule out all larger powers. The next lemma replaces it with a single test at  $c$ .

**Lemma 5.4.1 — Multiplicity via the Cofactor**

Let  $f \in F[x]$  be non-zero,  $c \in F$  and  $m \in \mathbb{N}$ . Then  $\text{mult}_c(f) = m$  if and only if

$$f = (x - c)^m g \quad \text{for some } g \in F[x] \text{ with } g(c) \neq 0.$$

*Proof.* ( $\Rightarrow$ ) Suppose  $\text{mult}_c(f) = m$ . Then  $f = (x - c)^m g$  for some  $g \in F[x]$ . If  $g(c) = 0$ , then  $g = (x - c)h$  by [Lemma 2.7.1](#), so  $f = (x - c)^{m+1}h$ , contradicting the maximality of  $m$ . Hence  $g(c) \neq 0$ .

( $\Leftarrow$ ) Suppose  $f = (x - c)^m g$  with  $g(c) \neq 0$ . Then  $(x - c)^m \mid f$ , so  $\text{mult}_c(f) \geq m$ . Suppose, for a contradiction, that  $(x - c)^{m+1} \mid f$ , say  $f = (x - c)^{m+1}h$ . Then  $(x - c)^m g = (x - c)^{m+1}h$ . Since  $(x - c)^m \neq 0$ , cancellation ([Corollary 0.8.4](#)) gives  $g = (x - c)h$ , and evaluating at  $c$  gives  $g(c) = 0$  by [Theorem 0.8.5](#), a contradiction. Every higher power  $(x - c)^k$  with  $k > m + 1$  would also give  $(x - c)^{m+1} \mid f$ . Hence  $\text{mult}_c(f) = m$ . ■

The first payoff is that multiplicities add when polynomials multiply, just as exponents do.

**Theorem 5.4.2 — Multiplicity of a Product**

Let  $f, g \in F[x]$  be non-zero and  $c \in F$ . Then  $\text{mult}_c(fg) = \text{mult}_c(f) + \text{mult}_c(g)$ .

*Proof.* Let  $m = \text{mult}_c(f)$  and  $n = \text{mult}_c(g)$ . By [Lemma 5.4.1](#),  $f = (x - c)^m u$  and  $g = (x - c)^n v$  with  $u(c) \neq 0$  and  $v(c) \neq 0$ . Then  $fg = (x - c)^{m+n} uv$ , and  $(uv)(c) = u(c)v(c) \neq 0$  by [Theorem 0.8.5](#) and [Theorem 0.7.1](#). By [Lemma 5.4.1](#) again,  $\text{mult}_c(fg) = m + n$ .



**Example. (Finding a Multiplicity by Repeated Division)**

Let  $f = x^4 - 2x^3 + 2x - 1 \in \mathbb{Q}[x]$ . Find the multiplicity of 1 and of  $-1$  as roots of  $f$ .

*Solution.* We have  $f(1) = 1 - 2 + 2 - 1 = 0$ , so we divide by  $x - 1$ . Long division gives

$$f = (x - 1)(x^3 - x^2 - x + 1).$$

The cofactor  $g_1 = x^3 - x^2 - x + 1$  has  $g_1(1) = 0$ , so we divide again:  $g_1 = (x - 1)(x^2 - 1)$ . The new cofactor  $g_2 = x^2 - 1$  has  $g_2(1) = 0$ , and  $g_2 = (x - 1)(x + 1)$ . Now  $g_3 = x + 1$  has  $g_3(1) = 2 \neq 0$ . Hence

$$f = (x - 1)^3(x + 1), \quad g_3(1) \neq 0,$$

and [Lemma 5.4.1](#) gives  $\text{mult}_1(f) = 3$ . For  $-1$ , write  $f = (x + 1) \cdot (x - 1)^3$ , whose cofactor  $(x - 1)^3$  takes the value  $(-2)^3 = -8 \neq 0$  at  $-1$ . So  $\text{mult}_{-1}(f) = 1$ , a simple root. The multiplicities add up to  $3 + 1 = 4 = \deg f$ .

*Warning. Multiplicity is not visible from the set of roots.* The polynomials  $(x - 1)^3(x + 1)$  and  $(x - 1)(x + 1)^3$  have the same roots  $\{1, -1\}$ , the same degree and the same leading coefficient, but they are different polynomials. “The roots of  $f$ ” as a set loses information; the roots **with their multiplicities** do not, as the next theorem shows for polynomials that split.

*Quick check.* Over  $\mathbb{R}$ , what is the multiplicity of 0 as a root of  $x^3(x^2 + 1)$ ? Of 2 as a root of  $(x^2 - 4)^2$ ?

*Solution.* For  $x^3(x^2 + 1)$ , the cofactor  $x^2 + 1$  is  $1 \neq 0$  at 0, so the multiplicity is 3 by [Lemma 5.4.1](#). For  $(x^2 - 4)^2 = (x - 2)^2(x + 2)^2$ , the cofactor  $(x + 2)^2$  is  $16 \neq 0$  at 2, so the multiplicity is 2.

## 5.4.2 Counting roots with multiplicity

In [Example \(Finding a Multiplicity by Repeated Division\)](#) the multiplicities added up to exactly the degree. That cannot always happen:  $x^2 + 1 \in \mathbb{R}[x]$  has degree 2 and no real roots at all. The difference is whether  $f$  breaks into linear factors, so we give that a name.

### Definition 5.4.2 — Splitting into Linear Factors

Let  $F$  be a field. A non-zero  $f \in F[x]$  of degree  $n$  **splits over**  $F$  if there are  $a \in F$  and  $a_1, \dots, a_n \in F$ , not necessarily distinct, with

$$f = a(x - a_1)(x - a_2) \cdots (x - a_n).$$

A non-zero constant splits (take  $n = 0$  and the empty product 1). The polynomial  $x^2 - 1 =$

$(x-1)(x+1)$  splits over  $\mathbb{Q}$ . The polynomial  $x^2-2$  does not split over  $\mathbb{Q}$ , since a linear factor  $x-a_1$  with  $a_1 \in \mathbb{Q}$  would give a rational root, but it splits over  $\mathbb{R}$  as  $(x-\sqrt{2})(x+\sqrt{2})$ . The phrase **over**  $F$  carries weight: the  $a_i$  must lie in  $F$ .

It turns out that the multiplicities always fit inside the degree, and they fill it exactly when  $f$  splits:

### Theorem 5.4.3 — Roots Counted with Multiplicity

Let  $F$  be a field and  $f \in F[x]$  non-zero. Let  $c_1, \dots, c_k$  be the distinct roots of  $f$  in  $F$ , with multiplicities  $m_1, \dots, m_k$ . Then

$$f = (x - c_1)^{m_1} \cdots (x - c_k)^{m_k} g$$

for some  $g \in F[x]$  with **no** root in  $F$ . Consequently

$$m_1 + \cdots + m_k \leq \deg f,$$

with equality if and only if  $f$  splits over  $F$ .

*Idea.* Peel off the roots one at a time. After removing  $(x - c_1)^{m_1}$ , the other roots  $c_j$  are still roots of the cofactor, because  $c_j - c_1 \neq 0$ , and by Theorem 5.4.2 they keep their multiplicities. Whatever is left at the end has no roots, and its degree is exactly the gap between  $\deg f$  and  $\sum m_i$ . The gap is zero exactly when the leftover is a constant.

*Proof.* The set of roots is finite by Corollary 5.1.4, so the list  $c_1, \dots, c_k$  makes sense. We prove the following claim by induction on  $j \in \{0, 1, \dots, k\}$ :  $f = (x - c_1)^{m_1} \cdots (x - c_j)^{m_j} g_j$  for some non-zero  $g_j \in F[x]$  with

$$\text{mult}_{c_i}(g_j) = 0 \text{ for } i \leq j, \quad \text{mult}_{c_i}(g_j) = m_i \text{ for } i > j.$$

For  $j = 0$  take  $g_0 = f$ . Suppose the claim holds for some  $j < k$ . By Lemma 5.4.1 applied to  $g_j$ , we have  $g_j = (x - c_{j+1})^{m_{j+1}} g_{j+1}$  with  $g_{j+1}(c_{j+1}) \neq 0$ , which gives the product formula for  $j+1$  and  $\text{mult}_{c_{j+1}}(g_{j+1}) = 0$ . For  $i \neq j+1$ , the polynomial  $(x - c_{j+1})^{m_{j+1}}$  takes the value  $(c_i - c_{j+1})^{m_{j+1}} \neq 0$  at  $c_i$ , since the roots are distinct. So its multiplicity at  $c_i$  is 0 by Lemma 5.4.1 (with cofactor the polynomial itself), and Theorem 5.4.2 gives

$$\text{mult}_{c_i}(g_j) = 0 + \text{mult}_{c_i}(g_{j+1}).$$

This is 0 for  $i \leq j$  and  $m_i$  for  $i > j+1$ , which completes the induction.

Put  $g = g_k$ . By the claim,  $\text{mult}_{c_i}(g) = 0$ , that is,  $g(c_i) \neq 0$ , for every  $i$ . If  $d \in F$  were a root of  $g$ , then  $f(d) = (d - c_1)^{m_1} \cdots (d - c_k)^{m_k} g(d) = 0$ , so  $d$  would be one of the  $c_i$ , which is impossible. Hence  $g$  has no root in  $F$ .

Since  $f \neq 0$ , also  $g \neq 0$ , and Theorem 0.8.3 gives  $\deg f = m_1 + \cdots + m_k + \deg g \geq m_1 + \cdots + m_k$ .

( $\Rightarrow$ ) If equality holds, then  $\deg g = 0$ , so  $g = a$  is a non-zero constant and  $f = a(x - c_1)^{m_1} \cdots (x - c_k)^{m_k}$  splits over  $F$ .

( $\Leftarrow$ ) Suppose  $f = a(x - a_1) \cdots (x - a_n)$  with  $n = \deg f$ . Group equal factors:  $f = a(x - d_1)^{n_1} \cdots (x - d_l)^{n_l}$  with  $d_1, \dots, d_l$  distinct and  $n_1 + \cdots + n_l = n$ . By the computation

above, the cofactor  $a \prod_{s \neq r} (x - d_s)^{n_s}$  is non-zero at  $d_r$ , so  $\text{mult}_{d_r}(f) = n_r$  by Lemma 5.4.1. Every root  $d$  of  $f$  satisfies  $a \prod_s (d - d_s)^{n_s} = 0$ , so  $d$  is one of the  $d_r$ . Hence the roots of  $f$  are exactly  $d_1, \dots, d_l$ , and their multiplicities add up to  $n = \deg f$ . This proves the theorem. ■

The theorem sharpens Corollary 5.1.4, because the number  $k$  of distinct roots is at most  $m_1 + \dots + m_k$ . It also shows that a split polynomial is determined by its leading coefficient and its roots with multiplicities. Whether a given polynomial splits depends on the field: the fifth-degree polynomial  $(x-1)^3(x^2+1)$  has root multiplicities adding up to  $3 < 5$  over  $\mathbb{R}$ , but it splits over  $\mathbb{C}$  as  $(x-1)^3(x-i)(x+i)$ . In the next section we prove that over  $\mathbb{C}$  **every** polynomial splits.

### 5.4.3 The formal derivative

How do we recognize a repeated root without factoring? Over  $\mathbb{R}$  calculus has an answer. At a simple root the graph of  $f$  crosses the axis with non-zero slope, while at a double root, as for  $(x-1)^2$  at 1, it touches the axis and the slope is 0. We would like to use this over every field, including  $\mathbb{F}_p$ , where there are no limits and no slopes. So we keep the formula of the derivative and throw away the limit.

*The formal derivative is the power rule, applied coefficient by coefficient.*

#### Definition 5.4.3 — Formal Derivative

Let  $F$  be a field and  $f = a_0 + a_1x + \dots + a_nx^n \in F[x]$ . The **formal derivative** of  $f$  is

$$f' = a_1 + 2a_2x + 3a_3x^2 + \dots + na_nx^{n-1} = \sum_{k=1}^n ka_kx^{k-1},$$

where  $ka_k$  is the integer multiple of  $a_k$  (the sum of  $k$  copies of  $a_k$ ). Higher derivatives are defined by  $f^{(0)} = f$  and  $f^{(k+1)} = (f^{(k)})'$ ; we also write  $f'' = f^{(2)}$ .

In words: each monomial  $a_kx^k$  becomes  $ka_kx^{k-1}$ , and constants disappear. This is the map  $D$  of Example (Differentiation), where we already saw that  $D: F[x] \rightarrow F[x]$  is linear. Over  $\mathbb{R}$  it agrees with the derivative of calculus on polynomial functions, since calculus gives the same power rule. By Lemma 0.7.4,  $ka_k = (k \cdot 1)a_k$ , so the integer  $k$  acts through the element  $k \cdot 1$  of  $F$ . **That element can be 0**, and this is the source of every difference between characteristic 0 and characteristic  $p$ .

Some examples. Over  $\mathbb{Q}$ ,  $(x^4 - 2x^3 + 2x - 1)' = 4x^3 - 6x^2 + 2$ . Over any field, the derivative of a constant is 0 and  $(x - c)' = 1$ . Over  $\mathbb{F}_3$ ,  $(x^3 + x)' = 3x^2 + 1 = 1$ , because  $3 \cdot 1 = 0$  there. Over  $\mathbb{F}_2$ ,  $(x^2)' = 2x = 0$ : a non-constant polynomial with derivative zero.

In characteristic 0 the last example cannot happen. If  $\deg f = n \geq 1$ , the coefficient of  $x^{n-1}$  in  $f'$  is  $(n \cdot 1)a_n$ , a product of two non-zero elements, so  $\deg f' = n - 1$  and in particular  $f' \neq 0$ . In characteristic  $p$  the same coefficient vanishes whenever  $p \mid n$ , because then  $n \cdot 1 = (n/p) \cdot (p \cdot 1) = 0$  by Lemma 0.7.4.

The derivative obeys the rules we know from calculus. The proofs cannot use limits, so they go through monomials.

### Theorem 5.4.4 — Rules for the Formal Derivative

Let  $F$  be a field,  $f, g \in F[x]$ ,  $a, b, c \in F$  and  $m \geq 1$  an integer. Then:

- (a)  $(af + bg)' = af' + bg'$ , and  $a' = 0$  for the constant polynomial  $a$ ;
- (b)  $(fg)' = f'g + fg'$  (product rule);
- (c)  $(g^m)' = mg^{m-1}g'$ ; in particular  $((x - c)^m)' = m(x - c)^{m-1}$ .

*Idea.* Both sides of the product rule are linear in  $f$  for fixed  $g$ , and linear in  $g$  for fixed  $f$ . So it is enough to check the rule when  $f$  and  $g$  are monomials, where it is the identity  $i + j = i + j$  for exponents. Checking on monomials, and extending by linearity, is the standard way to prove any identity about  $D$ .

*Proof.*

- (a) Linearity is [Example \(Differentiation\)](#). A constant has no coefficients in positions  $k \geq 1$ , so its derivative is 0.
- (b) First let  $f = x^i$  and  $g = x^j$  with  $i, j \in \mathbb{N}$ . If  $i, j \geq 1$ , then

$$(x^{i+j})' = (i+j)x^{i+j-1} = ix^{i-1}x^j + x^i jx^{j-1} = f'g + fg',$$

using  $(i+j) \cdot 1 = i \cdot 1 + j \cdot 1$  from [Lemma 0.7.4](#). If  $i = 0$ , then  $f = 1$ ,  $f' = 0$  and both sides equal  $jx^{j-1}$  (or 0 when  $j = 0$ ); the case  $j = 0$  is symmetric.

Now let  $f = \sum_i a_i x^i$  and  $g = \sum_j b_j x^j$ . By [Theorem 0.8.1](#),  $fg = \sum_{i,j} a_i b_j x^i x^j$ , a finite sum. Applying (a) and the monomial case,

$$\begin{aligned} (fg)' &= \sum_{i,j} a_i b_j ((x^i)'x^j + x^i(x^j)') \\ &= \left( \sum_i a_i (x^i)' \right) \left( \sum_j b_j x^j \right) + \left( \sum_i a_i x^i \right) \left( \sum_j b_j (x^j)' \right) \\ &= f'g + fg', \end{aligned}$$

where the middle equality regroups with the distributive laws, and the last uses (a) again.

- (c) We use induction on  $m$ . For  $m = 1$  both sides are  $g'$ . If  $(g^m)' = mg^{m-1}g'$ , then by (b)

$$(g^{m+1})' = (g^m g)' = mg^{m-1}g'g + g^m g' = (m+1)g^m g'.$$

For  $g = x - c$  we have  $g' = 1$  by (a). This proves the theorem. ■

## 5.4.4 Repeated roots and the derivative

Now we can turn the calculus picture into a theorem valid over every field.

**Theorem 5.4.5 — Repeated Root Test**

Let  $F$  be a field,  $f \in F[x]$  non-zero and  $c \in F$ .

- (a)  $c$  is a multiple root of  $f$  if and only if  $f(c) = 0$  and  $f'(c) = 0$ .
- (b) Suppose  $\text{mult}_c(f) = m \geq 1$  and  $m \cdot 1 \neq 0$  in  $F$  (this holds automatically if  $F$  has characteristic 0). Then  $f' \neq 0$  and  $\text{mult}_c(f') = m - 1$ .

*Idea.* Write  $f = (x - c)g$  and differentiate:  $f' = g + (x - c)g'$ , so  $f'(c) = g(c)$ . The derivative at  $c$  reads off exactly the cofactor's value, which is what [Lemma 5.4.1](#) needs. For (b), differentiate  $(x - c)^m g$  instead; the factor  $m$  appears, and it must not be 0 in  $F$ .

*Proof.*

- (a) ( $\Rightarrow$ ) Suppose  $f = (x - c)^2 g$ . By [Theorem 5.4.4](#) (b) and (c),  $f' = 2(x - c)g + (x - c)^2 g'$ , and evaluating at  $c$  gives  $f(c) = 0$  and  $f'(c) = 0$ .  
 ( $\Leftarrow$ ) Suppose  $f(c) = 0$  and  $f'(c) = 0$ . By [Lemma 2.7.1](#),  $f = (x - c)g$  for some  $g \in F[x]$ . By [Theorem 5.4.4](#),  $f' = g + (x - c)g'$ , so  $0 = f'(c) = g(c)$ . By [Lemma 2.7.1](#) again,  $g = (x - c)h$ , so  $f = (x - c)^2 h$  and  $\text{mult}_c(f) \geq 2$ .
- (b) By [Lemma 5.4.1](#),  $f = (x - c)^m g$  with  $g(c) \neq 0$ . By [Theorem 5.4.4](#),

$$f' = m(x - c)^{m-1}g + (x - c)^m g' = (x - c)^{m-1} h, \quad h := mg + (x - c)g'.$$

Then  $h(c) = (m \cdot 1)g(c)$ , a product of two non-zero elements of  $F$ , so  $h(c) \neq 0$ . In particular  $h \neq 0$ , so  $f' \neq 0$  by [Corollary 0.8.4](#), and  $\text{mult}_c(f') = m - 1$  by [Lemma 5.4.1](#). ■

Part (a) holds over **every** field. Part (b) needs its hypothesis. Over  $\mathbb{F}_2$ , let  $f = x^2(x + 1) = x^3 + x^2$ . Then  $\text{mult}_0(f) = 2$ , and  $f' = 3x^2 + 2x = x^2$ , so  $\text{mult}_0(f') = 2$ , not 1. Here  $m \cdot 1 = 2 \cdot 1 = 0$ , and the factor  $m$  killed the term that should have lowered the multiplicity.

Part (a) finds a repeated root at a known point  $c$ . To decide whether **any** repeated root exists, without knowing the roots, we use the gcd of  $f$  and  $f'$ , which the Euclidean algorithm ([Theorem 5.2.6](#)) computes by division alone. Repeated roots are a special case of repeated irreducible factors:  $c$  is a multiple root exactly when the irreducible polynomial  $x - c$  appears squared.

**Corollary 5.4.6 — Repeated Factors and  $\gcd(f, f')$** 

Let  $F$  be a field and  $f \in F[x]$  non-constant.

- (a) (**Any field.**) If  $p \in F[x]$  is irreducible and  $p^2 \mid f$ , then  $p \mid \gcd(f, f')$ . Hence, if  $\gcd(f, f') = 1$ , then no irreducible polynomial divides  $f$  twice; in particular  $f$  has no multiple root in  $F$ .
- (b) (**Characteristic 0.**) If  $F$  has characteristic 0 and  $\gcd(f, f') \neq 1$ , then  $p^2 \mid f$  for some irreducible  $p \in F[x]$ .

*Idea.* For (a), differentiate  $p^2h$ : every term still contains  $p$ . For (b), run it backwards. An irreducible  $p$  dividing both  $f = pg$  and  $f' = p'g + pg'$  must divide  $p'g$ . If  $p$  does not divide  $p'$ , then Euclid's lemma pushes  $p$  into  $g$ , and  $p^2 \mid f$ . The only question is whether  $p \mid p'$  can happen, and that is exactly where the characteristic enters.

*Proof.*

- (a) Write  $f = p^2h$ . By Theorem 5.4.4,  $f' = 2pp'h + p^2h' = p(2p'h + ph')$ . So  $p$  is a common divisor of  $f$  and  $f'$ , and  $p \mid \gcd(f, f')$  by Theorem 5.2.3. If  $\gcd(f, f') = 1$ , this would give  $p \mid 1$ , which is impossible because  $p$  is non-constant (Definition 5.3.1) and a non-zero multiple of  $p$  has degree at least  $\deg p \geq 1$  by Theorem 0.8.3. If  $c$  were a multiple root, then  $(x - c)^2 \mid f$ . But  $x - c$  is irreducible (Definition 5.3.1), because a product of two non-constant polynomials has degree at least 2. This is again impossible.
- (b) Let  $d = \gcd(f, f') \neq 1$ . Since  $d \mid f$  and  $f \neq 0$ ,  $d \neq 0$ ; as  $d$  is monic and  $d \neq 1$ ,  $\deg d \geq 1$ . By Theorem 5.3.5,  $d$  has a monic irreducible factor  $p$ . Then  $p \mid f$  and  $p \mid f'$  (Theorem 5.2.3). Write  $f = pg$ . By Theorem 5.4.4,  $f' = p'g + pg'$ , so  $p$  divides  $f' - pg' = p'g$ .

Since  $F$  has characteristic 0 and  $\deg p \geq 1$ , we showed after Definition 5.4.3 that  $p' \neq 0$  and  $\deg p' = \deg p - 1$ . A non-zero multiple of  $p$  has degree at least  $\deg p$ , so  $p \nmid p'$ . By Euclid's lemma (Theorem 5.3.3),  $p \mid g$ . Hence  $p^2 \mid pg = f$ . This proves the corollary. ■

**Warning.** In characteristic  $p$ , a non-constant polynomial can have derivative 0. Over  $\mathbb{F}_p$ ,  $(x^p)' = (p \cdot 1)x^{p-1} = 0$ . The proof of (b) used that an irreducible factor  $q$  of  $f$  has  $q' \neq 0$  of smaller degree, so that  $q \nmid q'$ ; in characteristic  $p$  that step is no longer automatic, because  $q'$  may be 0. So the correct statements are:  $\gcd(f, f') = 1$  implies no repeated irreducible factor, over **any** field; the converse is proved here **only in characteristic 0**. Do not use “repeated factor if and only if  $\gcd(f, f') \neq 1$ ” over a field of characteristic  $p$  without checking that no irreducible factor of  $f$  has derivative 0.

*Remark.* The converse really can fail in characteristic  $p$ , though not over the fields  $\mathbb{F}_p$ . Over the field of rational functions in a variable  $t$  with coefficients in  $\mathbb{F}_p$ , which this book does not construct, the polynomial  $x^p - t$  is irreducible, so it has no repeated factor, yet its derivative is 0.

### Example. (Detecting a Repeated Root with a gcd)

Decide whether  $f = x^4 - 2x^3 + 2x - 1 \in \mathbb{Q}[x]$  has a repeated irreducible factor, using only  $\gcd(f, f')$ . Does  $f$  have a multiple root in  $\mathbb{C}$ ?

*Solution.* Here  $f' = 4x^3 - 6x^2 + 2$ . Run the Euclidean algorithm (Theorem 5.2.6). Dividing  $f$  by  $f'$ ,

$$f = \left(\frac{1}{4}x - \frac{1}{8}\right)f' + \left(-\frac{3}{4}x^2 + \frac{3}{2}x - \frac{3}{4}\right),$$

and the remainder is  $-\frac{3}{4}(x^2 - 2x + 1)$ . Dividing  $f'$  by  $x^2 - 2x + 1$ ,

$$f' = (4x + 2)(x^2 - 2x + 1) + 0.$$

(We divided by the monic multiple of the remainder; scaling a divisor by a non-zero constant changes only the quotient, so the next remainder is still 0.) The last non-zero remainder, made monic, is the gcd:  $\gcd(f, f') = x^2 - 2x + 1 = (x - 1)^2 \neq 1$ . By [Corollary 5.4.6](#) (b), since  $\mathbb{Q}$  has characteristic 0,  $f$  has a repeated irreducible factor. Indeed  $x - 1$  divides the gcd, so 1 is a common root of  $f$  and  $f'$ , and by [Theorem 5.4.5](#) (a) it is a multiple root. This matches [Example \(Finding a Multiplicity by Repeated Division\)](#), where  $\text{mult}_1(f) = 3$ , and [Theorem 5.4.5](#) (b) gives  $\text{mult}_1(f') = 2$ , consistent with the factor  $(x - 1)^2$  of the gcd.

So yes,  $f$  has a multiple root in  $\mathbb{C}$ , namely  $1 \in \mathbb{Q}$ . The gcd test is useful over  $\mathbb{C}$  even for polynomials with rational coefficients: the Euclidean algorithm uses only divisions in  $\mathbb{Q}[x]$ , and division with remainder is unique ([Theorem 5.1.1](#)), so running it in  $\mathbb{C}[x]$  produces the same remainders and the same gcd. Had the gcd been 1, part (a) over  $\mathbb{C}$  would have ruled out every multiple complex root, without our finding a single root.

*Quick check.* Over  $\mathbb{R}$ ,  $f = x^3$  satisfies  $f(0) = f'(0) = 0$ . Does [Theorem 5.4.5](#) say that 0 has multiplicity 2?

*Solution.* No. Part (a) says only that the multiplicity is **at least** 2. The actual multiplicity is 3. Part (b) applies, since  $\mathbb{R}$  has characteristic 0:  $\text{mult}_0(f') = \text{mult}_0(3x^2) = 2 = 3 - 1$ . To pin the multiplicity down with derivatives, find the first  $k$  with  $f^{(k)}(0) \neq 0$ : here  $f''(0) = 0$  and  $f'''(0) = 6 \neq 0$ , so  $k = 3$ .

## 5.4.5 Taylor expansion of a polynomial

The answer to the check suggests that, in characteristic 0, the derivatives of  $f$  at  $c$  know everything about  $f$  near  $c$ . Calculus makes this precise with Taylor's formula. For polynomials the formula is exact, and it has a clean linear-algebra meaning: the numbers  $f^{(k)}(c)/k!$  are the coordinates of  $f$  in the basis of powers of  $x - c$ , so the maps  $f \mapsto f^{(k)}(c)/k!$  form a dual basis in the sense of [Definition 4.1.3](#). In [Example \(Dual Bases in  \$\mathbb{R}\[x\]\_{\leq 2}\$ \)](#) we saw this for  $c = 0$  and  $n = 2$ .

We need one derivative computation. For integers  $0 \leq j \leq k$ , repeated use of [Theorem 5.4.4](#) (c) and (a) gives

$$((x - c)^k)^{(j)} = k(k - 1) \cdots (k - j + 1) (x - c)^{k-j},$$

by induction on  $j$ , and  $((x - c)^k)^{(j)} = 0$  for  $j > k$ , since the  $k$ -th derivative is the constant  $k!$ . Evaluating at  $c$ , every term with  $j < k$  vanishes because of the factor  $(x - c)^{k-j}$ . Hence

$$((x - c)^k)^{(j)}(c) = \begin{cases} k! \cdot 1 & \text{if } j = k, \\ 0 & \text{if } j \neq k. \end{cases}$$

### Theorem 5.4.7 — Taylor Expansion of a Polynomial

Let  $F$  be a field,  $n \in \mathbb{N}$  and  $c \in F$ .

- (a) The list  $\mathcal{B}_c = (1, x - c, (x - c)^2, \dots, (x - c)^n)$  is a basis of  $F[x]_{\leq n}$ .
- (b) Suppose  $F$  has characteristic 0, or characteristic  $p > n$ . For  $k = 0, \dots, n$  define

$\varphi_k: F[x]_{\leq n} \rightarrow F$  by  $\varphi_k(f) = \frac{f^{(k)}(c)}{k!}$ . Then  $(\varphi_0, \dots, \varphi_n)$  is the dual basis of  $\mathcal{B}_c$ , and every  $f \in F[x]_{\leq n}$  satisfies

$$f = \sum_{k=0}^n \frac{f^{(k)}(c)}{k!} (x-c)^k.$$

*Idea.* ① The powers of  $x-c$  have degrees  $0, 1, \dots, n$ , so no combination of them can cancel down to zero; there are  $n+1$  of them, so counting finishes (a). ② The computation above says exactly  $\varphi_j((x-c)^k) = \delta_{jk}$ , once we may divide by  $k!$ . ③ A list of functionals with the  $\delta$  property is the dual basis, and the dual basis expansion  $\mathbf{v} = \sum \varphi_k(\mathbf{v})\mathbf{v}_k$  of Theorem 4.1.3 is Taylor's formula.

*Proof.*

- (a) Let  $b_0 + b_1(x-c) + \dots + b_n(x-c)^n = 0$ , and suppose not all  $b_k$  are zero. Let  $k$  be the largest index with  $b_k \neq 0$ . By Theorem 0.8.3,  $b_k(x-c)^k$  has degree  $k$ , and each earlier term has degree less than  $k$ , so by Theorem 0.8.2 the sum has degree  $k \geq 0$ . This contradicts the sum being 0, of degree  $-\infty$ . Hence  $\mathcal{B}_c$  is linearly independent. It has  $n+1$  elements and  $\dim F[x]_{\leq n} = n+1$  by Example (Standard Bases), so  $\mathcal{B}_c$  is a basis by Theorem 1.6.4 (a).
- (b) First,  $k! \cdot 1 \neq 0$  in  $F$  for  $0 \leq k \leq n$ . By Lemma 0.7.4,  $k! \cdot 1 = (1 \cdot 1)(2 \cdot 1) \cdots (k \cdot 1)$ . In characteristic 0 each factor is non-zero by Definition 0.7.2; in characteristic  $p > n$ , each factor  $j \cdot 1$  with  $1 \leq j \leq k < p$  is non-zero by the minimality of  $p$ . A product of non-zero elements is non-zero (Theorem 0.7.1). So  $\varphi_k$  is defined.

Each  $\varphi_k$  is linear:  $f \mapsto f^{(k)}$  is linear as a composite of copies of  $D$  (Example (Differentiation)), evaluation at  $c$  is linear by Theorem 0.8.5, and so is multiplication by  $(k!)^{-1}$ . The derivatives of a polynomial of degree at most  $n$  have degree at most  $n$ , so  $\varphi_k \in F[x]_{\leq n}^*$ . By the computation before the theorem,

$$\varphi_j((x-c)^k) = \frac{1}{j!} ((x-c)^k)^{(j)}(c) = \delta_{jk} \quad (0 \leq j, k \leq n).$$

By the uniqueness in Theorem 4.1.3 (a),  $(\varphi_0, \dots, \varphi_n)$  is the dual basis of  $\mathcal{B}_c$ . The first formula of Theorem 4.1.3 (c),  $f = \sum_k \varphi_k(f)(x-c)^k$ , is the Taylor expansion. This proves the theorem. ■

### Example. (Re-centering a Cubic)

Expand  $f = x^3 - 2x + 5 \in \mathbb{Q}[x]$  in powers of  $x-1$ .

*Solution.* The derivatives are  $f' = 3x^2 - 2$ ,  $f'' = 6x$ ,  $f''' = 6$ . At  $c = 1$ :  $f(1) = 4$ ,  $f'(1) = 1$ ,  $f''(1) = 6$ ,  $f'''(1) = 6$ . By Theorem 5.4.7 (b), since  $\mathbb{Q}$  has characteristic 0,

$$\begin{aligned} f &= 4 + 1 \cdot (x-1) + \frac{6}{2}(x-1)^2 + \frac{6}{6}(x-1)^3 \\ &= 4 + (x-1) + 3(x-1)^2 + (x-1)^3. \end{aligned}$$



Check by expanding:  $(x-1)^3 + 3(x-1)^2 + (x-1) + 4 = (x^3 - 3x^2 + 3x - 1) + (3x^2 - 6x + 3) + (x-1) + 4 = x^3 - 2x + 5$ .

The expansion also reads off root information at 1: the constant term  $f(1) = 4 \neq 0$ , so 1 is not a root. In general,  $\text{mult}_c(f)$  is the index of the first non-zero coordinate of  $f$  in the basis  $\mathcal{B}_c$  (over any field, by part (a); in characteristic 0 these coordinates are the Taylor coefficients), because  $f = (x-c)^m(b_m + b_{m+1}(x-c) + \cdots)$  and the bracket is  $b_m$  at  $c$ .

**Warning.** **Taylor's formula needs  $k! \neq 0$ .** Over  $\mathbb{F}_2$ , take  $f = x^2$  and  $c = 0$ . Then  $f' = 2x = 0$ , so **every** derivative of  $f$ , at every point, is 0, even though  $f \neq 0$ . No formula in the values  $f^{(k)}(0)$  can recover the coefficient 1 of  $x^2$ . Part (a) of [Theorem 5.4.7](#) still holds over  $\mathbb{F}_2$ :  $x^2$  has coordinates  $(0, 0, 1)$  in  $(1, x, x^2)$ . What fails is that these coordinates are derivatives divided by  $k!$ .

The two halves of this section meet in Chapter 8, where a matrix is diagonalizable exactly when its minimal polynomial splits and has no repeated roots; the gcd test above is how such a condition is checked in practice. Before that, the next section proves that over  $\mathbb{C}$  the sum of the multiplicities always equals the degree.

## 5.4.6 Exercises

### 5.4.6.1 A. Check your understanding

#### Exercise A1.

- Define the multiplicity of  $c$  as a root of a non-zero  $f \in F[x]$ . Why is  $f \neq 0$  required?
- True or false: if  $f(c) = f'(c) = 0$ , then  $c$  has multiplicity exactly 2. Justify your answer.
- True or false: over every field,  $f' = 0$  implies that  $f$  is constant. Justify your answer.
- True or false: if  $f \in \mathbb{R}[x]$  has degree 4 and the multiplicities of its real roots add up to 4, then  $f$  splits over  $\mathbb{R}$ . Justify your answer.
- True or false: if  $f \in \mathbb{Q}[x]$  is non-constant and  $\gcd(f, f') \neq 1$ , then  $f$  has a repeated irreducible factor. Justify your answer.
- Name the dual basis ([Theorem 4.1.3](#)) of  $(1, x - c, \dots, (x - c)^n)$  in  $\mathbb{R}[x]_{\leq n}$ .

*Solution.*

- It is the largest  $m \in \mathbb{N}$  with  $(x-c)^m \mid f$  ([Definition 5.4.1](#)). For  $f = 0$ , every power divides  $f$ , so no largest  $m$  exists.
- False. For  $f = x^3$  and  $c = 0$ ,  $f(0) = f'(0) = 0$ , but  $\text{mult}_0(f) = 3$ . [Theorem 5.4.5](#) (a) only gives multiplicity at least 2.
- False. Over  $\mathbb{F}_2$ ,  $(x^2)' = 2x = 0$ . (In characteristic 0 it is true:  $\deg f' = \deg f - 1$  whenever  $\deg f \geq 1$ .)
- True. By [Theorem 5.4.3](#), the sum of multiplicities equals the degree if and only if

$f$  splits over  $\mathbb{R}$ .

- (e) True.  $\mathbb{Q}$  has characteristic 0, so this is [Corollary 5.4.6](#) (b).  
 (f)  $\varphi_k(f) = f^{(k)}(c)/k!$  for  $k = 0, \dots, n$ , by [Theorem 5.4.7](#) (b);  $\mathbb{R}$  has characteristic 0.

### 5.4.6.2 B. Practice

#### Exercise B1: Multiplicities over different fields.

- (a) Let  $f = x^5 - 3x^4 + 4x^3 - 4x^2 + 3x - 1$ . Find  $\text{mult}_1(f)$  in  $\mathbb{Q}[x]$ . Hence decide whether  $f$  splits over  $\mathbb{R}$ , and whether it splits over  $\mathbb{C}$ .  
 (b) In  $\mathbb{F}_2[x]$ , find the roots of  $g = x^4 + x^2$  and their multiplicities. Does  $g$  split over  $\mathbb{F}_2$ ?

*Solution.*

- (a) Here  $f(1) = 1 - 3 + 4 - 4 + 3 - 1 = 0$ . Dividing by  $x - 1$  repeatedly,

$$f = (x - 1)(x^4 - 2x^3 + 2x^2 - 2x + 1),$$

$$x^4 - 2x^3 + 2x^2 - 2x + 1 = (x - 1)(x^3 - x^2 + x - 1),$$

and  $x^3 - x^2 + x - 1 = (x - 1)(x^2 + 1)$ . Since  $x^2 + 1$  takes the value  $2 \neq 0$  at 1, [Lemma 5.4.1](#) gives  $f = (x - 1)^3(x^2 + 1)$  and  $\text{mult}_1(f) = 3$ . Over  $\mathbb{R}$ ,  $x^2 + 1$  has no root ( $x^2 + 1 \geq 1$ ), so the only real root of  $f$  is 1, and  $3 < 5 = \deg f$ . By [Theorem 5.4.3](#),  $f$  does not split over  $\mathbb{R}$ . Over  $\mathbb{C}$ ,  $x^2 + 1 = (x - i)(x + i)$ , so  $f = (x - 1)^3(x - i)(x + i)$  splits.

- (b) In  $\mathbb{F}_2[x]$ ,  $x^4 + x^2 = x^2(x^2 + 1)$ , and  $x^2 + 1 = (x + 1)^2$  because  $(x + 1)^2 = x^2 + 2x + 1 = x^2 + 1$ . Hence  $g = x^2(x + 1)^2 = (x - 0)^2(x - 1)^2$ , using  $-1 = 1$ . The cofactor  $(x + 1)^2$  is 1 at 0, and  $x^2$  is 1 at 1, so  $\text{mult}_0(g) = \text{mult}_1(g) = 2$  by [Lemma 5.4.1](#). These are the only elements of  $\mathbb{F}_2$ , the multiplicities add to  $4 = \deg g$ , and  $g$  splits over  $\mathbb{F}_2$ .

#### Exercise B2: A gcd test.

Decide whether  $f = x^4 - 2x^2 + 1 \in \mathbb{Q}[x]$  has a repeated root, by computing  $\gcd(f, f')$  with the Euclidean algorithm. Hence factor  $f$  and give all multiplicities.

*Solution.* We have  $f' = 4x^3 - 4x$ . Dividing,  $f = \frac{1}{4}x \cdot f' + (-x^2 + 1)$ , since  $\frac{1}{4}x(4x^3 - 4x) = x^4 - x^2$ . Next,  $f' = (-4x)(-x^2 + 1) + 0$ . So the last non-zero remainder is  $-x^2 + 1$ , and by [Theorem 5.2.6](#)  $\gcd(f, f') = x^2 - 1 \neq 1$ . Its roots 1 and  $-1$  are common roots of  $f$  and  $f'$ , since  $x^2 - 1 = (x - 1)(x + 1)$  divides both. By [Theorem 5.4.5](#) (a), 1 and  $-1$  are multiple roots of  $f$ .

To factor, note that  $(x^2 - 1)^2 = x^4 - 2x^2 + 1 = f$ , so  $f = (x - 1)^2(x + 1)^2$ . By [Lemma 5.4.1](#),  $\text{mult}_1(f) = 2$  (cofactor  $(x + 1)^2$ , value 4 at 1) and  $\text{mult}_{-1}(f) = 2$ .

#### Exercise B3: The multiplicity of 1.

Let  $n \geq 2$  and  $f = x^n - nx + n - 1 \in \mathbb{Q}[x]$ . Prove that 1 is a root of  $f$  of multiplicity

exactly 2.

*Solution.* We have  $f(1) = 1 - n + n - 1 = 0$ ,  $f' = nx^{n-1} - n$ , so  $f'(1) = 0$ , and  $f'' = n(n-1)x^{n-2}$ , so  $f''(1) = n(n-1) \neq 0$  since  $n \geq 2$ . In particular  $f \neq 0$ . By [Theorem 5.4.5](#) (a),  $m := \text{mult}_1(f) \geq 2$ . Since  $\mathbb{Q}$  has characteristic 0, part (b) gives  $\text{mult}_1(f') = m - 1 \geq 1$ . Apply (a) to  $f' \neq 0$ : since  $f''(1) \neq 0$ , 1 is not a multiple root of  $f'$ , so  $m - 1 = 1$ . Hence  $m = 2$ .

### 5.4.6.3 C. Going deeper

#### Exercise C1: Real roots of the derivative.

Let  $f \in \mathbb{R}[x]$  have degree  $n \geq 2$  and split over  $\mathbb{R}$ . Prove that  $f'$  splits over  $\mathbb{R}$ . You may use **Rolle's theorem** from calculus: if  $a < b$  are real and  $f(a) = f(b) = 0$ , then  $f'(\xi) = 0$  for some  $\xi$  with  $a < \xi < b$ .

*Hint: count the roots of  $f'$  with multiplicity, and compare with  $\deg f'$ .*

*Solution.* Let  $c_1 < c_2 < \dots < c_k$  be the distinct real roots of  $f$ , with multiplicities  $m_1, \dots, m_k$ . Since  $f$  splits over  $\mathbb{R}$ , [Theorem 5.4.3](#) gives  $m_1 + \dots + m_k = n$ .

By Rolle's theorem, for each  $i = 1, \dots, k-1$  there is  $\xi_i$  with  $c_i < \xi_i < c_{i+1}$  and  $f'(\xi_i) = 0$ . These  $k-1$  numbers are distinct, since they lie in disjoint open intervals, and none of them is a  $c_j$ . Moreover, each  $c_i$  with  $m_i \geq 2$  is a root of  $f'$  of multiplicity  $m_i - 1$ , by [Theorem 5.4.5](#) (b) (characteristic 0); for  $m_i = 1$  the count  $m_i - 1 = 0$  is also correct.

Since  $\deg f = n \geq 1$  and  $\mathbb{R}$  has characteristic 0,  $\deg f' = n - 1$ , so  $f' \neq 0$ . Adding the multiplicities of the distinct roots  $\xi_1, \dots, \xi_{k-1}, c_1, \dots, c_k$  of  $f'$  gives at least

$$(k-1) + \sum_{i=1}^k (m_i - 1) = (k-1) + n - k = n - 1 = \deg f'.$$

By [Theorem 5.4.3](#) the sum of the multiplicities of all roots of  $f'$  is at most  $\deg f'$ , so it equals  $\deg f'$ , and  $f'$  splits over  $\mathbb{R}$ .

#### Exercise C2: $x^p - x$ over $\mathbb{F}_p$ .

Let  $p$  be a prime and  $f = x^p - x \in \mathbb{F}_p[x]$ .

- Compute  $f'$  and deduce that  $f$  has no repeated irreducible factor.
- Using [Exercise \(C2: A non-zero polynomial vanishing everywhere\)](#), deduce that  $x^p - x = \prod_{a \in \mathbb{F}_p} (x - a)$ .
- Explain why the argument of (a) gives nothing for  $g = x^p - 1$ . Then prove that  $g = (x - 1)^p$  in  $\mathbb{F}_p[x]$ .

*Hint for (c): for  $0 < k < p$ , use the integer identity  $k \binom{p}{k} = p \binom{p-1}{k-1}$ .*

*Solution.*

- By [Definition 5.4.3](#),  $f' = px^{p-1} - 1 = -1$ , since  $p \cdot 1 = 0$  in  $\mathbb{F}_p$ . Every common

divisor of  $f$  and  $-1$  divides 1, so  $\gcd(f, f') = 1$ , and [Corollary 5.4.6](#) (a), which holds over every field, shows  $f$  has no repeated irreducible factor.

- (b) By [Exercise \(C2: A non-zero polynomial vanishing everywhere\)](#) (c), every  $a \in \mathbb{F}_p$  is a root of  $f$ , so each of the  $p$  elements has multiplicity at least 1. The multiplicities add up to at least  $p = \deg f$ , and at most  $\deg f$  by [Theorem 5.4.3](#). So they add up to exactly  $p$ , and each is 1. By [Theorem 5.4.3](#),  $f = \prod_{b \in \mathbb{F}_p} (x - b)g$  with  $\deg g = p - p = 0$ , so  $g$  is a constant; comparing leading coefficients ([Theorem 0.8.3](#)),  $g = 1$ .
- (c) Here  $g' = px^{p-1} = 0$ , so  $\gcd(g, g') = g \neq 1$  (the monic generator of the ideal generated by  $g$  and 0). Part (a) of [Corollary 5.4.6](#) needs  $\gcd = 1$ , so it says nothing, and part (b) is only available in characteristic 0. Still,  $g(1) = 0$  and  $g'(1) = 0$ , so 1 is a multiple root by [Theorem 5.4.5](#) (a).

For the factorization, expand with the binomial theorem (valid in the commutative ring  $\mathbb{F}_p[x]$ , [Theorem 0.8.1](#)):  $(x - 1)^p = \sum_{k=0}^p \binom{p}{k} (-1)^{p-k} x^k$ , with the integers  $\binom{p}{k}$  acting as integer multiples. For  $0 < k < p$ , the identity  $k \binom{p}{k} = p \binom{p-1}{k-1}$  and [Lemma 0.7.4](#) give  $(k \cdot 1) \left( \binom{p}{k} \cdot 1 \right) = (p \cdot 1) \left( \binom{p-1}{k-1} \cdot 1 \right) = 0$  in  $\mathbb{F}_p$ . Since  $0 < k < p$ ,  $k \cdot 1 \neq 0$  by [Definition 0.7.2](#), so  $\binom{p}{k} \cdot 1 = 0$  ([Theorem 0.7.1](#)). Hence  $(x - 1)^p = x^p + (-1)^p$ . For odd  $p$  this is  $x^p - 1$ ; for  $p = 2$  it is  $x^2 + 1 = x^2 - 1$ . So  $g = (x - 1)^p$ , and  $\text{mult}_1(g) = p$  by [Lemma 5.4.1](#).

### Exercise C3: Removing repeated factors.

Let  $F$  be a field and  $f = ap_1^{e_1} \cdots p_r^{e_r} \in F[x]$ , where  $a \in F \setminus \{0\}$ , the  $p_i$  are **distinct** monic irreducible polynomials and  $e_i \geq 1$ .

- (a) Suppose  $F$  has characteristic 0. Prove that  $\gcd(f, f') = p_1^{e_1-1} \cdots p_r^{e_r-1}$ , and deduce that  $f / \gcd(f, f') = ap_1 \cdots p_r$ .
- (b) Over  $\mathbb{F}_2$ , let  $f = x^2(x + 1)$ . Compute  $f / \gcd(f, f')$  and show that the conclusion of (a) fails.

*Hint for (a): write  $D = p_1^{e_1-1} \cdots p_r^{e_r-1}$ . Show  $D \mid \gcd(f, f')$ ; for the reverse, show  $p_i^{e_i} \nmid f'$  using Euclid's lemma.*

*Solution.*

- (a) Let  $D = \prod_i p_i^{e_i-1}$  and  $d = \gcd(f, f')$ . By [Theorem 5.4.4](#) (b) and (c), applied repeatedly,

$$f' = a \sum_{i=1}^r e_i p_i^{e_i-1} p_i' \prod_{j \neq i} p_j^{e_j}.$$

Every term is divisible by  $D$ , and so is  $f$ . Hence  $D \mid d$  by [Theorem 5.2.3](#).

Conversely,  $d \mid f$ , so  $f = dh$ . Factoring  $d$  and  $h$  into monic irreducibles and comparing with the factorization of  $f$  by the uniqueness in [Theorem 5.3.5](#),  $d = \prod_i p_i^{k_i}$  with  $0 \leq k_i \leq e_i$  (as  $d$  is monic). Suppose  $k_i = e_i$  for some  $i$ . Then  $p_i^{e_i} \mid f'$ . In the formula for  $f'$  every term with index  $j \neq i$  contains  $p_i^{e_i}$ , so  $p_i^{e_i}$  divides the remaining term  $ae_i p_i^{e_i-1} p_i' \prod_{j \neq i} p_j^{e_j}$ . Canceling  $p_i^{e_i-1}$  ([Corollary 0.8.4](#)),  $p_i \mid (ae_i) p_i' \prod_{j \neq i} p_j^{e_j}$ . Here  $ae_i = a(e_i \cdot 1)$  is a non-zero constant (characteristic 0),  $p_i \nmid p_i'$  because  $p_i' \neq 0$  has smaller degree, and  $p_i \nmid p_j$  for  $j \neq i$  because both are monic irreducible and distinct. Applying [Theorem 5.3.3](#) repeatedly gives a

contradiction. Hence  $k_i \leq e_i - 1$  for all  $i$ , so  $d \mid D$ . Both are monic and divide each other, so they have the same degree and  $d = D$ . Finally  $f = D \cdot a p_1 \cdots p_r$ , so  $f/d = a p_1 \cdots p_r$ .

- (b) In  $\mathbb{F}_2[x]$ ,  $f = x^3 + x^2$  and  $f' = 3x^2 + 2x = x^2$ . Since  $x^2 \mid f$ ,  $\gcd(f, f') = x^2$ , and  $f/\gcd(f, f') = x + 1$ . The formula of (a) would predict  $\gcd(f, f') = x$  and  $f/\gcd(f, f') = x(x+1)$ . The irreducible factor  $x$  has been lost. The step that fails is “ $ae_i \neq 0$ ”: here  $e_1 = 2$  and  $2 \cdot 1 = 0$  in  $\mathbb{F}_2$ .

## 5.5 The Fundamental Theorem of Algebra

Chapter 0 stated the Fundamental Theorem of Algebra ([Theorem 0.6.7](#)) and promised a proof. Everything algebraic that the proof needs is now in place: division, the Factor Theorem, multiplicities, and the polar form of complex numbers with De Moivre’s formula. What algebra alone cannot supply is a property of the real numbers that goes beyond the field axioms, and some such property must enter, because  $\mathbb{Q}$  is a field too and there  $x^2 - 2$  has no root. We use completeness, in the form of the extreme value theorem. This section isolates the analytic input in two clearly labeled facts, proves everything else in full, and then draws the consequences for real polynomials: conjugate pairs, irreducible factors of degree at most 2, and real roots of odd-degree polynomials.

### 5.5.1 The goal and the plan

Here is the theorem we want, in the language of this chapter. A polynomial  $p \in \mathbb{C}[x]$  of degree  $n \geq 1$  is exactly the expression  $a_n x^n + \cdots + a_0$  with  $a_n \neq 0$  of [Theorem 0.6.7](#).

#### Theorem 5.5.1 — Fundamental Theorem of Algebra

Every non-constant polynomial  $p \in \mathbb{C}[x]$  has a root in  $\mathbb{C}$ .

We prove it after three lemmas. The proof is a minimum argument, and it runs in three steps.

① **Growth.** Far from the origin,  $|p(z)|$  is large, because the leading term  $a_n z^n$  outweighs all the others. We will prove this with an explicit radius.

② **A minimum exists.** On a large closed disc, the continuous function  $z \mapsto |p(z)|$  attains a smallest value. By ①, nothing outside the disc does better, so this is the smallest value of  $|p|$  on all of  $\mathbb{C}$ .

③ **A non-zero value is never the minimum.** If  $p(z_0) \neq 0$ , we can move from  $z_0$  a little in a well-chosen direction and make  $|p|$  strictly smaller. This is d’Alembert’s lemma, and it is pure algebra of  $\mathbb{C}$ .

Putting ② and ③ together, the minimum value must be 0, and the point where it is attained is a root.

Steps ① and ③ are proved from scratch. Step ② is where analysis enters, and we say exactly how.

*Remark. What we assume from analysis.* The proof uses exactly two facts from real analysis, and nothing else: no limits, no derivatives, no intermediate value theorem. We identify  $\mathbb{C}$  with  $\mathbb{R}^2$  as in [Definition 0.6.1](#), so that  $|z - w|$  is the usual distance in the plane.

- (A1) **Continuity of polynomial functions.** For every  $p \in \mathbb{C}[x]$ , the function  $\mathbb{C} \rightarrow \mathbb{R}$ ,  $z \mapsto |p(z)|$ , is continuous.
- (A2) **Extreme value theorem on a closed disc.** Let  $\rho \geq 0$  and  $D = \{z \in \mathbb{C} : |z| \leq \rho\}$ . If  $g: D \rightarrow \mathbb{R}$  is continuous, then there is  $w \in D$  with  $g(w) \leq g(z)$  for **every**  $z \in D$ .

Fact (A2) is where the completeness of  $\mathbb{R}$  is used. Everything else below is the arithmetic of  $\mathbb{C}$  from Chapter 0, including polar form and [Theorem 0.6.6](#).

## 5.5.2 Step 1: growth

We use three small facts about the modulus repeatedly. For  $z, w, z_1, \dots, z_m \in \mathbb{C}$  and  $j \in \mathbb{N}$ :

$$|z^j| = |z|^j, \quad |z_1 + \dots + z_m| \leq |z_1| + \dots + |z_m|, \quad |z + w| \geq |z| - |w|.$$

The first follows from [Theorem 0.6.2](#) (d) by induction on  $j$ , and the second from [Theorem 0.6.4](#) by induction on  $m$ . For the third, apply the triangle inequality to  $z = (z + w) + (-w)$  and use  $|-w| = |w|$ .

The leading term wins as soon as  $|z|$  exceeds a radius computed from the coefficients.

### Lemma 5.5.2 — Growth of a Polynomial

Let  $p = a_n x^n + \dots + a_1 x + a_0 \in \mathbb{C}[x]$  with  $n \geq 1$  and  $a_n \neq 0$ . Put

$$A = |a_0| + |a_1| + \dots + |a_{n-1}|, \quad R = \max\left\{1, \frac{2A}{|a_n|}\right\}.$$

Then for every  $z \in \mathbb{C}$  with  $|z| \geq R$ ,

$$|p(z)| \geq \frac{1}{2}|a_n||z|^n \geq \frac{1}{2}|a_n||z|.$$

In particular, for every real  $M \geq 0$ , we have  $|p(z)| > M$  whenever  $|z| \geq R$  and  $|z| > 2M/|a_n|$ .

*Idea.* Compare the leading term with the rest. When  $|z| \geq 1$ , every lower power  $|z|^j$  is at most  $|z|^{n-1}$ , so the rest is at most  $A|z|^{n-1}$ , while the leading term is  $|a_n||z|^n$ . The rest is at most half the leading term once  $A \leq \frac{1}{2}|a_n||z|$ , which is how  $R$  was found.

*Proof.* Let  $|z| \geq R$ . Since  $|z| \geq 1$ , we have  $|z|^j \leq |z|^{n-1}$  for  $0 \leq j \leq n-1$ . By the triangle inequality ([Theorem 0.6.4](#)),

$$\left| \sum_{j=0}^{n-1} a_j z^j \right| \leq \sum_{j=0}^{n-1} |a_j| |z|^j \leq A |z|^{n-1}.$$

Since  $|z| \geq 2A/|a_n|$ , we have  $A \leq \frac{1}{2}|a_n||z|$ . By the reverse triangle inequality,

$$|p(z)| \geq |a_n||z|^n - A|z|^{n-1} \geq |a_n||z|^n - \frac{1}{2}|a_n||z|^n = \frac{1}{2}|a_n||z|^n.$$

Since  $|z| \geq 1$  and  $n \geq 1$ ,  $|z|^n \geq |z|$ , which gives the second inequality. Finally, if also  $|z| > 2M/|a_n|$ , then  $|p(z)| \geq \frac{1}{2}|a_n||z| > M$ . This proves the lemma. ■

For example, for  $p = x^3 - 5x + 1$  we get  $A = 6$  and  $R = 12$ , so  $|p(z)| \geq \frac{1}{2}|z|^3$  whenever  $|z| \geq 12$ . The radius is far from optimal, and it does not need to be. We only need **some** radius beyond which  $|p|$  is large.

### 5.5.3 Step 2: the minimum is attained

#### Lemma 5.5.3 — The Minimum of $|p|$ Is Attained

Let  $p \in \mathbb{C}[x]$ . Then there is  $z_0 \in \mathbb{C}$  with  $|p(z_0)| \leq |p(z)|$  for **every**  $z \in \mathbb{C}$ .

*Idea.* Use (A2) on a disc so big that outside it,  $|p|$  exceeds  $|p(0)|$ . The disc contains 0, so its minimum is at most  $|p(0)|$ , and the points outside cannot compete. [Lemma 5.5.2](#) tells us how big “so big” is.

*Proof.* If  $p$  is constant, every  $z_0$  works. Suppose  $p = a_n x^n + \cdots + a_0$  with  $n \geq 1$  and  $a_n \neq 0$ , and let  $R$  be the radius of [Lemma 5.5.2](#). Put  $M = |p(0)|$  and

$$\rho = R + \frac{2M}{|a_n|}, \quad D = \{z \in \mathbb{C} : |z| \leq \rho\}.$$

By (A1),  $z \mapsto |p(z)|$  is continuous on  $D$ , so by (A2) there is  $z_0 \in D$  with  $|p(z_0)| \leq |p(z)|$  for every  $z \in D$ .

Now let  $z \notin D$ , so  $|z| > \rho$ . Then  $|z| > R$ , and  $|z| > 2M/|a_n|$  because  $R \geq 1 > 0$ . By [Lemma 5.5.2](#),  $|p(z)| > M = |p(0)|$ . Since  $0 \in D$ ,  $|p(0)| \geq |p(z_0)|$ . Hence  $|p(z)| > |p(z_0)|$  for  $z \notin D$ , and  $|p(z)| \geq |p(z_0)|$  for  $z \in D$ . This proves the lemma. ■

### 5.5.4 Step 3: d’Alembert’s lemma

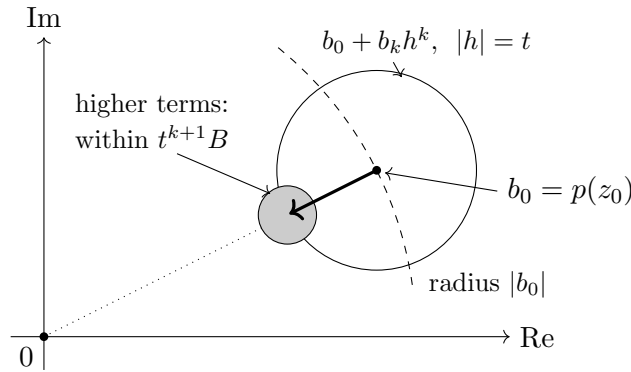
It remains to show that a non-zero value of  $|p|$  is never the smallest one. Near  $z_0$ , write  $p(z_0 + h) = b_0 + b_k h^k + (\text{terms of higher degree in } h)$ , where  $b_0 = p(z_0)$  and  $b_k h^k$  is the first non-constant term that is actually present. For small  $h$ , the term  $b_k h^k$  dominates the later ones, so we choose the direction of  $h$  to make  $b_k h^k$  point from  $b_0$  straight toward 0.

#### Lemma 5.5.4 — d’Alembert’s Lemma

Let  $p \in \mathbb{C}[x]$  be non-constant, and let  $z_0 \in \mathbb{C}$  with  $p(z_0) \neq 0$ . Then there is  $z \in \mathbb{C}$  with

$$|p(z)| < |p(z_0)|.$$

*Idea.* Picture the values  $p(z_0 + h)$  in the plane. As  $h = tu$  runs over a small circle  $|h| = t$ , the term  $b_0 + b_k h^k$  runs  $k$  times around the circle of radius  $t^k |b_k|$  centered at  $b_0 = p(z_0)$ . One point of that circle lies on the segment from  $b_0$  to 0, at distance  $|b_0| - t^k |b_k|$  from 0. Reaching it needs a  $k$ -th root of a unit complex number, which De Moivre supplies. The higher terms move the value by at most a multiple of  $t^{k+1}$ , a blur much smaller than the gain  $t^k |b_k|$  when  $t$  is small.



The dashed arc is part of the circle of radius  $|b_0|$  about 0, the old value of  $|p|$ . The solid circle shows where  $b_0 + b_k h^k$  goes for  $|h| = t$ . The arrow is the chosen direction, and the shaded disc is the uncertainty caused by the higher terms. The whole shaded disc lies strictly inside the dashed circle, which is the inequality  $|p(z_0 + h)| < |p(z_0)|$ .

*Proof.* Let  $n = \deg p \geq 1$ .

**Step 1: re-center at  $z_0$ .** By Theorem 5.4.7 (a),  $(1, x - z_0, \dots, (x - z_0)^n)$  is a basis of  $\mathbb{C}[x]_{\leq n}$ , so  $p = \sum_{j=0}^n b_j (x - z_0)^j$  for some  $b_0, \dots, b_n \in \mathbb{C}$ . Evaluating at  $z_0$  gives  $b_0 = p(z_0) \neq 0$ . The coefficient of  $x^n$  on the right is  $b_n$ , since  $(x - z_0)^j$  has degree  $j$  and  $(x - z_0)^n$  is monic; so  $b_n$  is the leading coefficient of  $p$ , and  $b_n \neq 0$ . By Theorem 0.8.5, for every  $h \in \mathbb{C}$ ,

$$p(z_0 + h) = \sum_{j=0}^n b_j h^j.$$

Since  $b_n \neq 0$  and  $n \geq 1$ , there is a **smallest**  $k \geq 1$  with  $b_k \neq 0$ . Then  $b_1 = \dots = b_{k-1} = 0$ , and

$$p(z_0 + h) = b_0 + b_k h^k + \sum_{j=k+1}^n b_j h^j. \quad (*)$$

**Step 2: choose the direction.** Since  $b_0 \neq 0$  and  $b_k \neq 0$ , the number  $w = -b_0/b_k$  is non-zero. Write it in polar form,  $w = |w|e^{i\alpha}$  with  $\alpha \in \mathbb{R}$ , and put  $u = e^{i\alpha/k}$ . Then  $|u| = 1$ , and  $u^k = e^{i\alpha}$  by Theorem 0.6.6. Using  $b_k w = -b_0$  and  $|w| = |b_0|/|b_k|$  (Theorem 0.6.2 (d)),

$$b_k u^k = \frac{b_k w}{|w|} = -\frac{|b_k|}{|b_0|} b_0.$$



Hence, for every real  $t > 0$ ,

$$b_0 + b_k(tu)^k = b_0 \left( 1 - t^k \frac{|b_k|}{|b_0|} \right).$$

**Step 3: choose the size.** Put  $B = |b_{k+1}| + \cdots + |b_n|$  (so  $B = 0$  if  $k = n$ ), and

$$t = \min \left\{ 1, \frac{|b_0|}{|b_k|}, \frac{|b_k|}{2B} \right\},$$

where the last entry is omitted if  $B = 0$ . Then  $t > 0$ , and  $t$  has three properties.

- (i)  $0 < t^k |b_k| / |b_0| \leq 1$ . Indeed, if  $|b_0| / |b_k| \geq 1$ , then  $t^k \leq 1 \leq |b_0| / |b_k|$ ; otherwise  $t \leq |b_0| / |b_k| < 1$ , and  $t^k \leq t \leq |b_0| / |b_k|$ .
- (ii)  $tB \leq \frac{1}{2}|b_k|$ . This is trivial if  $B = 0$ , and follows from  $t \leq |b_k| / (2B)$  otherwise.
- (iii)  $t^j \leq t^{k+1}$  for every  $j \geq k+1$ , since  $0 < t \leq 1$ .

**Step 4: estimate.** Let  $z = z_0 + tu$ . By Step 2 and (i), the number  $1 - t^k |b_k| / |b_0|$  is real and lies in  $[0, 1)$ , so

$$|b_0 + b_k(tu)^k| = |b_0| \left( 1 - t^k \frac{|b_k|}{|b_0|} \right) = |b_0| - t^k |b_k|.$$

By the triangle inequality ([Theorem 0.6.4](#)),  $|u| = 1$  and (iii),

$$\left| \sum_{j=k+1}^n b_j(tu)^j \right| \leq \sum_{j=k+1}^n |b_j| t^j \leq t^{k+1} B.$$

Combining these with (\*), the triangle inequality and (ii),

$$|p(z)| \leq |b_0| - t^k |b_k| + t^{k+1} B = |b_0| - t^k (|b_k| - tB) \leq |b_0| - \frac{1}{2} t^k |b_k| < |b_0|,$$

where the last step uses  $t > 0$  and  $b_k \neq 0$ . Since  $|b_0| = |p(z_0)|$ , this proves the lemma. ■

Every hypothesis did a job. Non-constant gave the index  $k$ . The condition  $p(z_0) \neq 0$  let us divide by  $|b_0|$  and fix a direction toward 0. And the field  $\mathbb{C}$  supplied the  $k$ -th root  $u$ , the one step that has no analogue over  $\mathbb{R}$ .

### Example. (One Step Downhill)

Let  $p = x^3 + x^2 + 1 \in \mathbb{C}[x]$  and  $z_0 = 0$ . Follow the proof of [Lemma 5.5.4](#) to find  $z$  with  $|p(z)| < |p(0)|$ .

*Solution.* Here  $p(0) = 1$  and the expansion at  $z_0 = 0$  is  $p$  itself:  $b_0 = 1$ ,  $b_1 = 0$ ,  $b_2 = 1$ ,  $b_3 = 1$ . The first non-zero  $b_j$  with  $j \geq 1$  is  $b_2$ , so  $k = 2$  and  $B = |b_3| = 1$ .

Direction:  $w = -b_0/b_2 = -1 = e^{i\pi}$ , so  $\alpha = \pi$  and  $u = e^{i\pi/2} = i$ . Indeed  $b_2 u^2 = -1$  points from  $b_0 = 1$  toward 0.

Size:  $t = \min\{1, 1/1, 1/(2 \cdot 1)\} = \frac{1}{2}$ . So  $z = \frac{i}{2}$ , and

$$p\left(\frac{i}{2}\right) = -\frac{i}{8} - \frac{1}{4} + 1 = \frac{3}{4} - \frac{i}{8}, \quad |p\left(\frac{i}{2}\right)| = \sqrt{\frac{9}{16} + \frac{1}{64}} = \frac{\sqrt{37}}{8} \approx 0.76 < 1.$$

The proof guarantees  $|p(z)| \leq 1 - \frac{1}{2} \cdot \frac{1}{4} \cdot 1 = \frac{7}{8}$ , and  $\sqrt{37}/8 \leq \frac{7}{8}$  since  $37 \leq 49$ . The direction matters: along the real axis,  $p(\frac{1}{2}) = \frac{1}{8} + \frac{1}{4} + 1 = \frac{11}{8} > 1$ , because there  $b_2 h^2 = \frac{1}{4}$  points away from 0.

### 5.5.5 The proof of the theorem

The three steps now combine into a proof of [Theorem 5.5.1](#).

*Proof.* Let  $p \in \mathbb{C}[x]$  be non-constant. By [Lemma 5.5.3](#) there is  $z_0 \in \mathbb{C}$  with  $|p(z_0)| \leq |p(z)|$  for every  $z \in \mathbb{C}$ . Suppose, for a contradiction, that  $p(z_0) \neq 0$ . By [Lemma 5.5.4](#) there is  $z$  with  $|p(z)| < |p(z_0)|$ , contradicting the choice of  $z_0$ . Hence  $p(z_0) = 0$ , and  $z_0$  is a root of  $p$ . ■

This proves Chapter 0's [Theorem 0.6.7](#): a polynomial  $a_n z^n + \cdots + a_0$  with  $n \geq 1$  and  $a_n \neq 0$  is a non-constant element of  $\mathbb{C}[x]$ , and we have just shown that it has a root.

*Quick check.* The polynomial  $p = x^2 + 1 \in \mathbb{R}[x]$  has no real root, yet the function  $c \mapsto |p(c)|$  on  $\mathbb{R}$  has a minimum, at  $c = 0$ . Which step of the argument breaks if we try to run it over  $\mathbb{R}$ ?

*Solution.* Growth and the attained minimum both hold on  $\mathbb{R}$ : the minimum value is  $|p(0)| = 1$ . The step that fails is d'Alembert's lemma. At  $z_0 = 0$  we have  $b_0 = 1$ ,  $k = 2$ ,  $b_2 = 1$ , so we need a direction  $u$  with  $u^2 = w = -1$ . There is no such real  $u$ , and indeed  $p(h) = 1 + h^2 \geq 1$  for every real  $h$ . Over  $\mathbb{C}$ ,  $u = i$  works and  $p(ti) = 1 - t^2 < 1$  for  $0 < t \leq 1$ .

**Warning. The theorem is an existence theorem.** It says a root exists; it does not say how to find one, and the proof does not produce one. There is no formula for the roots of a general polynomial of degree 5 or more built from the coefficients by  $+$ ,  $-$ ,  $\times$ ,  $\div$  and  $k$ -th roots (a theorem from Galois theory, which we do not prove). Numerical root-finding algorithms approximate roots; that is a different subject, and the Fundamental Theorem of Algebra is not one of those algorithms.

### 5.5.6 Complex polynomials split

Combining the theorem with the root count of the previous section gives the form in which it is used most.

**Corollary 5.5.5 — Complex Polynomials Split** (a) Every non-zero  $p \in \mathbb{C}[x]$  splits over  $\mathbb{C}$ : if  $\deg p = n$  and  $p$  has leading coefficient  $a$ , then  $p = a(x - z_1)(x - z_2) \cdots (x - z_n)$  for some  $z_1, \dots, z_n \in \mathbb{C}$ . In particular the multiplicities of

the roots of  $p$  add up to  $n$ .

- (b) The irreducible polynomials in  $\mathbb{C}[x]$  are exactly the polynomials of degree 1.

*Proof.*

- (a) By [Theorem 5.4.3](#),  $p = (x - c_1)^{m_1} \cdots (x - c_k)^{m_k} g$ , where  $c_1, \dots, c_k$  are the distinct roots of  $p$  in  $\mathbb{C}$  and  $g \in \mathbb{C}[x]$  has no root in  $\mathbb{C}$ . By [Theorem 5.5.1](#),  $g$  is constant, and  $g \neq 0$  because  $p \neq 0$ . So  $p$  splits over  $\mathbb{C}$ , and the multiplicities add up to  $n$  by the same theorem. Comparing leading coefficients ([Theorem 0.8.3](#)), the constant is  $a$ .
- (b) A polynomial of degree 1 is irreducible, since a factorization into two non-constant factors would have degree at least 2. Conversely, let  $p \in \mathbb{C}[x]$  be irreducible. It is non-constant ([Definition 5.3.1](#)), so by (a) it has a factor  $x - z_1$ , say  $p = (x - z_1)q$ . Since  $p$  is irreducible and  $x - z_1$  is not constant,  $q$  is a non-zero constant, and  $\deg p = 1$ .

■

So in  $\mathbb{C}[x]$ , unique factorization ([Theorem 5.3.5](#)) is completely explicit: the monic irreducible factors are the  $x - z$ , and the exponent of  $x - z$  is the multiplicity of  $z$ . In Chapter 8 this is what gives every operator on a non-zero finite-dimensional complex vector space an eigenvalue. (That fact also has a proof using only linear algebra, and one can run the logic in the other direction; we will not need it.)

**Example. (Factoring  $x^4 + 1$  over  $\mathbb{C}$  and over  $\mathbb{R}$ )**

Factor  $f = x^4 + 1$  into irreducible polynomials in  $\mathbb{C}[x]$  and in  $\mathbb{R}[x]$ .

*Solution. Roots.* We solve  $z^4 = -1 = e^{i\pi}$  as in [Example \(Roots of Unity\)](#). Write  $z = re^{i\theta}$  with  $r > 0$ ; by [Theorem 0.6.6](#),  $z^4 = r^4 e^{4i\theta}$ . Comparing polar forms,  $r^4 = 1$ , so  $r = 1$ , and  $4\theta = \pi + 2\pi j$  for an integer  $j$ . The values  $j = 0, 1, 2, 3$  give four distinct roots

$$z_0 = e^{i\pi/4} = \frac{\sqrt{2}}{2} + \frac{\sqrt{2}}{2}i, \quad z_1 = e^{3i\pi/4} = -\frac{\sqrt{2}}{2} + \frac{\sqrt{2}}{2}i, \quad z_2 = \overline{z_1}, \quad z_3 = \overline{z_0}.$$

*Over  $\mathbb{C}$ .* Four distinct roots, each of multiplicity at least 1, and  $\deg f = 4$ , so by [Theorem 5.4.3](#) each multiplicity is 1 and

$$x^4 + 1 = (x - z_0)(x - z_1)(x - z_2)(x - z_3),$$

the leading coefficient being 1. The factors are irreducible by [Corollary 5.5.5](#) (b).

*Over  $\mathbb{R}$ .* Pair each root with its conjugate. For any  $z \in \mathbb{C}$ ,

$$(x - z)(x - \overline{z}) = x^2 - (z + \overline{z})x + z\overline{z} = x^2 - 2\operatorname{Re}(z)x + |z|^2 \in \mathbb{R}[x],$$

by [Theorem 0.6.2](#) (c). With  $\operatorname{Re} z_0 = \frac{\sqrt{2}}{2}$ ,  $\operatorname{Re} z_1 = -\frac{\sqrt{2}}{2}$  and  $|z_0| = |z_1| = 1$ ,

$$x^4 + 1 = (x^2 - \sqrt{2}x + 1)(x^2 + \sqrt{2}x + 1).$$

Check: the product is  $(x^2 + 1)^2 - 2x^2 = x^4 + 1$ . Each quadratic has non-real roots only, hence no real root, so it is irreducible over  $\mathbb{R}$  by [Theorem 5.3.1](#). Note that  $x^4 + 1$  has no real root but is **reducible** over  $\mathbb{R}$ .

## 5.5.7 Real polynomials

A real polynomial is also a complex polynomial, so the theorem applies to it; the roots it provides may be non-real. They are not scattered arbitrarily. Chapter 0 observed that non-real roots of a real polynomial come in conjugate pairs; we now show that the pairs also share their multiplicity. For  $g = \sum_j c_j x^j \in \mathbb{C}[x]$  write  $\bar{g} = \sum_j \bar{c}_j x^j$ , the polynomial with conjugated coefficients.

### Theorem 5.5.6 — Conjugate Roots Have Equal Multiplicity

Let  $f \in \mathbb{R}[x]$  be non-zero, regarded as an element of  $\mathbb{C}[x]$ , and let  $z \in \mathbb{C}$ . Then  $f(\bar{z}) = \overline{f(z)}$ . In particular  $z$  is a root of  $f$  if and only if  $\bar{z}$  is, and

$$\text{mult}_{\bar{z}}(f) = \text{mult}_z(f).$$

*Idea.* Conjugation respects sums and products of complex numbers, so applying it to all coefficients respects sums and products of polynomials. A real polynomial is unchanged by it. So conjugating a factorization  $f = (x - z)^m g$  gives a factorization  $f = (x - \bar{z})^m \bar{g}$ , and the cofactor test transfers.

*Proof.* Let  $g = \sum_j c_j x^j$  and  $h = \sum_j d_j x^j$  be in  $\mathbb{C}[x]$ , and  $z \in \mathbb{C}$ . By [Theorem 0.6.2](#) (a) and (b), the coefficient of  $x^k$  in  $\overline{gh}$  is  $\overline{\sum_{i+j=k} c_i d_j} = \sum_{i+j=k} \bar{c}_i \bar{d}_j$ , which is the coefficient of  $x^k$  in  $\bar{g}\bar{h}$ . So  $\overline{gh} = \bar{g}\bar{h}$ . By the same parts,  $\overline{g(z)} = \bar{g}(\bar{z}) = \overline{\sum_j c_j z^j} = \sum_j \bar{c}_j \bar{z}^j = \bar{g}(\bar{z})$ .

Since  $f$  has real coefficients,  $\bar{f} = f$ . Hence  $f(\bar{z}) = \bar{f}(\bar{z}) = \overline{f(z)}$ , and  $f(z) = 0$  if and only if  $f(\bar{z}) = 0$ .

Let  $m = \text{mult}_z(f)$ . By [Lemma 5.4.1](#),  $f = (x - z)^m g$  with  $g \in \mathbb{C}[x]$  and  $g(z) \neq 0$ . Conjugating, and using  $\overline{x - z} = x - \bar{z}$ ,

$$f = \bar{f} = (x - \bar{z})^m \bar{g}, \quad \bar{g}(\bar{z}) = \overline{g(z)} \neq 0.$$

By [Lemma 5.4.1](#) again,  $\text{mult}_{\bar{z}}(f) = m$ . This proves the theorem. ■

For instance, a real polynomial with  $2 + i$  as a root of multiplicity 3 also has  $2 - i$  as a root of multiplicity 3, so by [Theorem 5.4.3](#) its degree is at least 6. Now the irreducible real polynomials can be listed.

### Corollary 5.5.7 — Irreducible Real Polynomials

The irreducible polynomials in  $\mathbb{R}[x]$  are exactly the polynomials of degree 1, and the polynomials of degree 2 with no real root. In particular, every irreducible polynomial in  $\mathbb{R}[x]$  has degree 1 or 2.

*Idea.* An irreducible real  $p$  has a complex root  $z$  by the theorem. If  $z$  is real,  $x - z$  is a real factor and  $p$  is linear. If not,  $\bar{z}$  is another root, and  $(x - z)(x - \bar{z})$  is a **real** quadratic dividing  $p$ . The only subtle point is that it divides  $p$  in  $\mathbb{R}[x]$  and not merely in  $\mathbb{C}[x]$ ; uniqueness of division settles that.

*Proof.* ( $\Leftarrow$ ) A polynomial of degree 1 is irreducible, since a product of two non-constant polynomials has degree at least 2 ([Theorem 0.8.3](#)). A polynomial of degree 2 with no real root is irreducible by [Theorem 5.3.1](#).

( $\Rightarrow$ ) Let  $p \in \mathbb{R}[x]$  be irreducible. It is non-constant, so by [Theorem 5.5.1](#) it has a root  $z \in \mathbb{C}$ .

*Case 1:*  $z \in \mathbb{R}$ . By [Lemma 2.7.1](#) in  $\mathbb{R}[x]$ ,  $p = (x - z)q$  with  $q \in \mathbb{R}[x]$ . Since  $p$  is irreducible and  $x - z$  is not constant,  $q$  is a non-zero constant, so  $\deg p = 1$ .

*Case 2:*  $z \notin \mathbb{R}$ . Then  $\bar{z} \neq z$ , and  $\bar{z}$  is also a root by [Theorem 5.5.6](#). Applying [Corollary 5.1.4](#) in  $\mathbb{C}[x]$  to the two distinct roots  $z, \bar{z}$  gives  $p = (x - z)(x - \bar{z})h$  for some  $h \in \mathbb{C}[x]$ . Put  $q = (x - z)(x - \bar{z}) = x^2 - 2\operatorname{Re}(z)x + |z|^2$ , which lies in  $\mathbb{R}[x]$ . Divide in  $\mathbb{R}[x]$  ([Theorem 5.1.1](#)):  $p = qh_1 + r$  with  $h_1, r \in \mathbb{R}[x]$  and  $\deg r < 2$ . This is also a division with remainder in  $\mathbb{C}[x]$ , and so is  $p = qh + 0$ . By the uniqueness in [Theorem 5.1.1](#) over  $\mathbb{C}$ ,  $r = 0$  and  $h = h_1 \in \mathbb{R}[x]$ . So  $p = qh_1$  in  $\mathbb{R}[x]$ . Since  $p$  is irreducible and  $q$  is not constant,  $h_1$  is a non-zero constant, and  $\deg p = 2$ . Finally,  $p$  has no real root, because a real root  $c$  would give a factor  $x - c$  and  $p = (x - c)q'$  with  $\deg q' = 1$ , contradicting irreducibility. This proves the corollary. ■

*Remark.* Completing the square decides which quadratics occur:  $x^2 + \beta x + \gamma = (x + \frac{\beta}{2})^2 - \frac{\beta^2 - 4\gamma}{4}$  has a real root if and only if  $\beta^2 - 4\gamma \geq 0$ . So the monic irreducible real quadratics are those with  $\beta^2 < 4\gamma$ .

Unique factorization in  $\mathbb{R}[x]$  now takes a concrete form.

### Corollary 5.5.8 — Factorization over $\mathbb{R}$

Every non-constant  $f \in \mathbb{R}[x]$  can be written as

$$f = a(x - c_1) \cdots (x - c_r) q_1 \cdots q_s,$$

where  $a \in \mathbb{R}$  is the leading coefficient of  $f$ ,  $c_1, \dots, c_r \in \mathbb{R}$  (not necessarily distinct), and  $q_1, \dots, q_s \in \mathbb{R}[x]$  are monic quadratics with no real root. Then  $r + 2s = \deg f$ , and  $r$  is the sum of the multiplicities of the real roots of  $f$ . The factorization is unique up to the order of the factors.

*Proof.* By [Theorem 5.3.5](#),  $f = ap_1 \cdots p_k$  with  $p_i \in \mathbb{R}[x]$  monic irreducible, unique up to order. By [Corollary 5.5.7](#) each  $p_i$  is either  $x - c$  with  $c \in \mathbb{R}$  or a monic quadratic with no real root; list the first kind as  $x - c_1, \dots, x - c_r$  and the second as  $q_1, \dots, q_s$ . Then  $\deg f = r + 2s$  by [Theorem 0.8.3](#). For a real number  $c$ , each  $q_j$  has multiplicity 0 at  $c$ , and  $x - c_i$  has multiplicity 1 or 0 according as  $c_i = c$  or not; by [Theorem 5.4.2](#),  $\operatorname{mult}_c(f)$

is the number of indices  $i$  with  $c_i = c$ . Adding over the real roots of  $f$  gives  $r$ .

■

The last consequence is one that calculus proves with the intermediate value theorem. Here it falls out of counting degrees.

### Corollary 5.5.9 — Odd Degree Forces a Real Root

Every  $f \in \mathbb{R}[x]$  of odd degree has a root in  $\mathbb{R}$ .

*Proof.* By Corollary 5.5.8,  $\deg f = r + 2s$ . Since  $\deg f$  is odd,  $r$  is odd, so  $r \geq 1$ , and  $c_1$  is a real root of  $f$ .

■

*Warning.* No real root does not mean irreducible over  $\mathbb{R}$ . By Corollary 5.5.7, an irreducible real polynomial has degree at most 2; so  $x^4 + 1$ , which has no real root, is still reducible over  $\mathbb{R}$  (Example (Factoring  $x^4 + 1$  over  $\mathbb{C}$  and over  $\mathbb{R}$ )). The test “reducible if and only if it has a root” is only valid in degrees 2 and 3 (Theorem 5.3.1).

## 5.5.8 Exercises

### 5.5.8.1 A. Check your understanding

#### Exercise A1.

- State the Fundamental Theorem of Algebra, and list the two facts from analysis used in its proof in this section.
- True or false: every polynomial in  $\mathbb{R}[x]$  of degree 4 has a real root. Justify your answer.
- True or false: if  $f \in \mathbb{R}[x]$  has  $2 + i$  as a root of multiplicity 3, then  $\deg f \geq 6$ . Justify your answer.
- True or false: every polynomial in  $\mathbb{R}[x]$  of degree 5 has a real root. Justify your answer.
- True or false: every irreducible polynomial in  $\mathbb{F}_2[x]$  has degree at most 2. Justify your answer.
- In the proof of d’Alembert’s lemma (Lemma 5.5.4), which property of  $\mathbb{C}$  is used that  $\mathbb{R}$  lacks?

*Solution.*

- Every non-constant  $p \in \mathbb{C}[x]$  has a root in  $\mathbb{C}$  (Theorem 5.5.1). The analytic facts are (A1) continuity of  $z \mapsto |p(z)|$  on  $\mathbb{C}$ , and (A2) the extreme value theorem for continuous real functions on a closed disc.
- False.  $x^4 + 1 \geq 1$  at every real number, so it has no real root.
- True. By Theorem 5.5.6,  $2 - i$  is also a root of multiplicity 3, and  $2 - i \neq 2 + i$ .

By [Theorem 5.4.3](#),  $\deg f \geq 3 + 3 = 6$ .

- (d) True, by [Corollary 5.5.9](#).
- (e) False.  $x^3 + x + 1$  takes the value 1 at both 0 and 1, so it has no root in  $\mathbb{F}_2$ , and a cubic without a root is irreducible by [Theorem 5.3.1](#). [Corollary 5.5.7](#) is special to  $\mathbb{R}$ .
- (f) Every non-zero complex number has a  $k$ -th root; the proof takes a  $k$ -th root of a complex number of modulus 1 by [Theorem 0.6.6](#). Over  $\mathbb{R}$ ,  $-1$  has no square root.

### 5.5.8.2 B. Practice

#### Exercise B1: $x^4 + 4$ .

Find all complex roots of  $x^4 + 4$ , and factor it into irreducible polynomials in  $\mathbb{C}[x]$  and in  $\mathbb{R}[x]$ .

*Solution.* Write  $z = re^{i\theta}$  with  $r > 0$ . By [Theorem 0.6.6](#),  $z^4 = r^4 e^{4i\theta}$ , and  $-4 = 4e^{i\pi}$ . Comparing polar forms,  $r^4 = 4$ , so  $r = \sqrt{2}$  (the only positive real number with fourth power 4, since  $t \mapsto t^4$  is strictly increasing on positive reals), and  $4\theta = \pi + 2\pi j$ ,  $j \in \mathbb{Z}$ . The values  $j = 0, 1, 2, 3$  give the four distinct roots

$$\sqrt{2}e^{i\pi/4} = 1 + i, \quad \sqrt{2}e^{3i\pi/4} = -1 + i, \quad \sqrt{2}e^{5i\pi/4} = -1 - i, \quad \sqrt{2}e^{7i\pi/4} = 1 - i.$$

By [Theorem 5.4.3](#), four distinct roots of a degree-4 monic polynomial each have multiplicity 1, and

$$x^4 + 4 = (x - 1 - i)(x - 1 + i)(x + 1 - i)(x + 1 + i) \quad \text{in } \mathbb{C}[x].$$

Pairing conjugates with  $(x - z)(x - \bar{z}) = x^2 - 2\operatorname{Re}(z)x + |z|^2$ , and  $|1 \pm i|^2 = |-1 \pm i|^2 = 2$ ,

$$x^4 + 4 = (x^2 - 2x + 2)(x^2 + 2x + 2) \quad \text{in } \mathbb{R}[x].$$

Check:  $(x^2 + 2)^2 - (2x)^2 = x^4 + 4x^2 + 4 - 4x^2 = x^4 + 4$ . Both quadratics have  $\beta^2 - 4\gamma = 4 - 8 < 0$ , so they have no real roots and are irreducible by [Corollary 5.5.7](#).

#### Exercise B2: $x^6 + 1$ .

Find the factorization of  $x^6 + 1$  into irreducible polynomials in  $\mathbb{R}[x]$ .

*Solution.* The roots solve  $z^6 = -1 = e^{i\pi}$ . As in B1,  $z = e^{i\theta}$  with  $6\theta = \pi + 2\pi j$ , and  $j = 0, \dots, 5$  give six distinct roots at the angles  $30^\circ, 90^\circ, 150^\circ, 210^\circ, 270^\circ, 330^\circ$ :

$$\frac{\sqrt{3}}{2} \pm \frac{1}{2}i, \quad \pm i, \quad -\frac{\sqrt{3}}{2} \pm \frac{1}{2}i.$$

Each has multiplicity 1 by [Theorem 5.4.3](#), and none is real. They form three conjugate pairs, each of modulus 1, with real parts  $\frac{\sqrt{3}}{2}, 0, -\frac{\sqrt{3}}{2}$ . Pairing them as in [Example \(Factoring  \$x^4 + 1\$  over  \$\mathbb{C}\$  and over  \$\mathbb{R}\$ \)](#),

$$x^6 + 1 = (x^2 + 1)(x^2 - \sqrt{3}x + 1)(x^2 + \sqrt{3}x + 1).$$

Check:  $(x^2 - \sqrt{3}x + 1)(x^2 + \sqrt{3}x + 1) = (x^2 + 1)^2 - 3x^2 = x^4 - x^2 + 1$ , and  $(x^2 + 1)(x^4 -$

$x^2 + 1) = x^6 + 1$ . Each quadratic factor has only non-real roots, so it is irreducible over  $\mathbb{R}$  by [Corollary 5.5.7](#). By the uniqueness in [Corollary 5.5.8](#), this is the factorization.

### Exercise B3: Real cubics.

Let  $f \in \mathbb{R}[x]$  have degree 3, and suppose the multiplicities of the real roots of  $f$  add up to 1.

- Prove that  $f = a(x - c)(x - z)(x - \bar{z})$  for some  $a, c \in \mathbb{R}$  and some  $z \in \mathbb{C} \setminus \mathbb{R}$ , and that  $z$  and  $\bar{z}$  are simple roots.
- Show that the hypothesis cannot be weakened to “ $f$  has exactly one real root” by considering  $(x - 1)^3$ .

*Solution.*

- By [Corollary 5.5.8](#),  $f = a(x - c_1) \cdots (x - c_r) q_1 \cdots q_s$  with  $r + 2s = 3$  and  $r$  equal to the sum of the real multiplicities, which is 1. So  $r = 1$ ,  $s = 1$ , and  $f = a(x - c)q$  with  $c \in \mathbb{R}$  and  $q$  a monic real quadratic without real roots. By [Theorem 5.5.1](#),  $q$  has a root  $z \in \mathbb{C}$ , which is not real. By [Theorem 5.5.6](#),  $\bar{z} \neq z$  is also a root of  $q$ . The two multiplicities in  $q$  are each at least 1 and add up to at most 2 ([Theorem 5.4.3](#)), so they equal 1 and  $q = (x - z)(x - \bar{z})$ , the leading coefficient being 1. Finally, by [Theorem 5.4.2](#),  $\text{mult}_z(f) = \text{mult}_z(a(x - c)) + \text{mult}_z(q) = 0 + 1 = 1$ , since  $a(z - c) \neq 0$ ; similarly for  $\bar{z}$ .
- $(x - 1)^3$  has exactly one real root, 1, but no non-real roots at all: it already splits over  $\mathbb{R}$ . Here the real multiplicities add up to 3, not 1.

### Exercise B4: Running the proof on an example.

Let  $p = x^4 - x^2 + 2 \in \mathbb{C}[x]$ .

- Compute the radius  $R$  of [Lemma 5.5.2](#) for  $p$ .
- Follow the proof of [Lemma 5.5.4](#) at  $z_0 = 0$ : find  $k$ ,  $B$ ,  $u$  and  $t$ , and verify that  $|p(tu)| < |p(0)|$ .
- Compute  $p(i/2)$ , and explain why the direction  $i$  does not work here, although it did in [Example \(One Step Downhill\)](#).

*Solution.*

- Here  $a_4 = 1$  and  $A = |2| + |0| + |-1| + |0| = 3$ , so  $R = \max\{1, 6\} = 6$ . Hence  $|p(z)| \geq \frac{1}{2}|z|^4$  for  $|z| \geq 6$ .
- At  $z_0 = 0$  the coefficients are  $b_0 = 2$ ,  $b_1 = 0$ ,  $b_2 = -1$ ,  $b_3 = 0$ ,  $b_4 = 1$ . So  $k = 2$  and  $B = |b_3| + |b_4| = 1$ . Then  $w = -b_0/b_2 = 2 = 2e^{i \cdot 0}$ , so  $\alpha = 0$  and  $u = 1$ . Next  $t = \min\{1, 2/1, 1/2\} = \frac{1}{2}$ . So  $z = \frac{1}{2}$ , and  $p(\frac{1}{2}) = \frac{1}{16} - \frac{1}{4} + 2 = \frac{29}{16} < 2 = |p(0)|$ . The proof guarantees at most  $2 - \frac{1}{2} \cdot \frac{1}{4} \cdot 1 = \frac{15}{8} = \frac{30}{16}$ , consistent with  $\frac{29}{16}$ .
- $p(i/2) = \frac{1}{16} + \frac{1}{4} + 2 = \frac{37}{16} > 2$ . The right direction depends on the sign of  $b_k$  relative to  $b_0$ : we need  $b_2 u^2$  to point from  $b_0$  toward 0, that is,  $b_2 u^2$  a negative multiple of  $b_0 = 2$ . With  $b_2 = -1$  this means  $u^2 > 0$ , so  $u = \pm 1$ ; for  $u = i$ ,  $b_2 u^2 = 1$  points



away from 0. In [Example \(One Step Downhill\)](#),  $b_2 = +1$  and  $u = i$  was the right choice.

### 5.5.8.3 C. Going deeper

#### Exercise C1: Fields where the theorem fails.

Call a field  $F$  **algebraically closed** if every non-constant polynomial in  $F[x]$  has a root in  $F$ .

- (a) Let  $p$  be a prime. Show that  $x^p - x + 1$  has no root in  $\mathbb{F}_p$ . More generally, show that no finite field is algebraically closed.
- (b) Give explicit witnesses that  $\mathbb{R}$  and  $\mathbb{Q}$  are not algebraically closed.
- (c) For your witness over  $\mathbb{R}$ , explain which step of the proof of the Fundamental Theorem of Algebra ([Theorem 5.5.1](#)) breaks.
- (d) For  $f = x^2 - 2$  over  $\mathbb{Q}$ , prove that  $r \mapsto |f(r)|$  has **no** minimum on  $\mathbb{Q}$ . Which step of the proof breaks now?

*Hint for (d): if  $r_0$  were a minimum point with  $r_0^2 > \frac{2}{5}$ , compare  $r_0$  with  $r_1 = \frac{r_0^2 + 2}{2r_0}$ .*

*Solution.*

- (a) By [Exercise \(C2: A non-zero polynomial vanishing everywhere\)](#) (c),  $a^p - a = 0$  for every  $a \in \mathbb{F}_p$ , so  $a^p - a + 1 = 1 \neq 0$ . For a finite field  $F = \{c_1, \dots, c_q\}$ , let  $g = (x - c_1) \cdots (x - c_q) + 1$ . It has degree  $q \geq 2$  ([Theorem 0.8.3](#)), and  $g(c_i) = 0 + 1 = 1 \neq 0$  for every  $i$  by [Theorem 0.8.5](#). So  $g$  is a non-constant polynomial without a root in  $F$ .
- (b) Over  $\mathbb{R}$ ,  $x^2 + 1$  has no root, since  $c^2 + 1 \geq 1$ . Over  $\mathbb{Q}$ ,  $x^2 - 2$  has no root, since  $\sqrt{2}$  is irrational ([Theorem 0.2.3](#)) and the only real roots are  $\pm\sqrt{2}$ .
- (c) For  $x^2 + 1$  over  $\mathbb{R}$ , growth holds, and  $c \mapsto |c^2 + 1|$  attains its minimum 1 at  $c = 0$ . The step that breaks is d'Alembert's lemma ([Lemma 5.5.4](#)): at  $z_0 = 0$  we have  $b_0 = 1$ ,  $k = 2$ ,  $b_2 = 1$ , and we need  $u$  with  $u^2 = -1$ , which does not exist in  $\mathbb{R}$ .
- (d) Suppose  $r_0 \in \mathbb{Q}$  satisfies  $|r_0^2 - 2| \leq |r^2 - 2|$  for every  $r \in \mathbb{Q}$ . Put  $m = |r_0^2 - 2|$ ; then  $m > 0$  by (b).  
*Case 1:*  $r_0^2 \leq \frac{2}{5}$ . Then  $m = 2 - r_0^2 \geq \frac{8}{5} > 1 = |1^2 - 2|$ , contradicting minimality at  $r = 1$ .  
*Case 2:*  $r_0^2 > \frac{2}{5}$ . Then  $r_0 \neq 0$ , and  $r_1 = (r_0^2 + 2)/(2r_0) \in \mathbb{Q}$ . Expanding,

$$r_1^2 - 2 = \frac{(r_0^2 + 2)^2 - 8r_0^2}{4r_0^2} = \frac{(r_0^2 - 2)^2}{4r_0^2}, \quad \text{so} \quad |r_1^2 - 2| = m \cdot \frac{m}{4r_0^2}.$$

We claim  $m < 4r_0^2$ . If  $r_0^2 \geq 2$ , then  $m = r_0^2 - 2 < 4r_0^2$ . If  $\frac{2}{5} < r_0^2 < 2$ , then  $m = 2 - r_0^2 < \frac{8}{5} < 4r_0^2$ . Hence  $|r_1^2 - 2| < m$ , contradicting minimality.

So there is no minimum. The step of the proof that breaks is [Lemma 5.5.3](#), which rests on the extreme value theorem (A2). That fact is about the completeness of  $\mathbb{R}$ , and  $\mathbb{Q}$  is not complete: the values  $|r^2 - 2|$  get arbitrarily close to 0 without reaching it.

**Exercise C2: Sums of two squares.**

Let  $f \in \mathbb{R}[x]$  be monic with **no** real root.

- (a) Prove that  $\deg f$  is even and that  $f(c) > 0$  for every  $c \in \mathbb{R}$ .
- (b) Prove that  $f = g^2 + h^2$  for some  $g, h \in \mathbb{R}[x]$ .

*Hint for (b): each irreducible factor of  $f$  is  $(x - z)(x - \bar{z})$ . Collect one root from each pair into a complex polynomial  $G$ , and split  $G$  into real and imaginary parts.*

*Solution.*

- (a) By [Corollary 5.5.8](#),  $f = (x - c_1) \cdots (x - c_r) q_1 \cdots q_s$ , with leading coefficient 1 since  $f$  is monic. A factor  $x - c_1$  would give the real root  $c_1$ , so  $r = 0$  and  $\deg f = 2s$  is even. (If  $s = 0$ , then  $f = 1$  and the claims hold.) Each  $q_j = x^2 + \beta x + \gamma$  has no real root, so by the remark after [Corollary 5.5.7](#),  $\beta^2 < 4\gamma$ . For real  $c$ ,

$$q_j(c) = \left(c + \frac{\beta}{2}\right)^2 + \frac{4\gamma - \beta^2}{4} > 0.$$

By [Theorem 0.8.5](#),  $f(c) = q_1(c) \cdots q_s(c)$  is a product of positive reals, so  $f(c) > 0$ .

- (b) For each  $j$ ,  $q_j$  is non-constant, so by [Theorem 5.5.1](#) it has a root  $z_j \in \mathbb{C}$ , not real. As in [Exercise \(B3: Real cubics\)](#) (a),  $q_j = (x - z_j)(x - \bar{z}_j)$ . Let  $G = (x - z_1) \cdots (x - z_s) \in \mathbb{C}[x]$ . By the product rule for conjugate polynomials proved in [Theorem 5.5.6](#),  $\bar{G} = (x - \bar{z}_1) \cdots (x - \bar{z}_s)$ , so  $f = G\bar{G}$  (commutativity, [Theorem 0.8.1](#)).

Write each coefficient of  $G$  as  $\alpha_k + i\beta_k$  with  $\alpha_k, \beta_k \in \mathbb{R}$ , and put  $g = \sum_k \alpha_k x^k$  and  $h = \sum_k \beta_k x^k$  in  $\mathbb{R}[x]$ . Then  $G = g + ih$  and  $\bar{G} = g - ih$ , so

$$f = (g + ih)(g - ih) = g^2 - i^2 h^2 + (ihg - igh) = g^2 + h^2.$$

For example, for  $x^4 + 1$  take  $z_1 = e^{i\pi/4}$  and  $z_2 = e^{3i\pi/4}$  from [Example \(Factoring  \$x^4 + 1\$  over  \$\mathbb{C}\$  and over  \$\mathbb{R}\$ \)](#). Then  $z_1 + z_2 = i\sqrt{2}$  and  $z_1 z_2 = e^{i\pi} = -1$ , so  $G = x^2 - i\sqrt{2}x - 1$ ,  $g = x^2 - 1$ ,  $h = -\sqrt{2}x$ , and  $x^4 + 1 = (x^2 - 1)^2 + 2x^2$ .

## 5.6 Interpolation and the Chinese Remainder Theorem

Chapter 2 proved that through  $n + 1$  points with distinct  $x$ -values there passes exactly one polynomial of degree at most  $n$  ([Theorem 2.7.3](#)). The proof went through an invertible matrix, and it never wrote the polynomial down. This section gives the formula. Then it reads interpolation in the arithmetic of this chapter: “ $p(c) = y$ ” says that  $p$  leaves remainder  $y$  on division by  $x - c$ . Prescribing remainders modulo several pairwise coprime polynomials at once is the Chinese Remainder Theorem. We prove it twice, once with Bézout’s identity, which gives a formula, and once by counting dimensions in the quotient spaces  $F[x]/\langle m \rangle$ .

## 5.6.1 The Lagrange basis

Suppose we want the polynomial of degree at most  $n$  taking prescribed values  $y_0, \dots, y_n$  at distinct points  $c_0, \dots, c_n$ . The problem is linear in the data: if  $p$  fits the values  $(y_i)$  and  $q$  fits  $(y'_i)$ , then  $p + q$  fits  $(y_i + y'_i)$ , by [Theorem 0.8.5](#). So it is enough to solve the simplest data sets, a single 1 and zeros everywhere else, and add up.

The polynomial that vanishes at every  $c_j$  with  $j \neq i$  is easy to write:  $\prod_{j \neq i} (x - c_j)$ . At  $c_i$  it takes the value  $\prod_{j \neq i} (c_i - c_j)$ , which is a product of non-zero scalars, so we can divide by it. In [Example \(Dual Bases in  \$\mathbb{R}\[x\]\_{\leq 2}\$ \)](#) we met these polynomials for the three points  $-1, 0, 1$ . Here is the general case.

*The  $i$ -th Lagrange polynomial is the polynomial of degree  $n$  that is 1 at the  $i$ -th node and 0 at every other node.*

### Definition 5.6.1 — Lagrange Basis

Let  $n \in \mathbb{N}$ , and let  $c_0, c_1, \dots, c_n \in F$  be **distinct**. For  $i = 0, \dots, n$ , the  $i$ -th **Lagrange polynomial** for these nodes is

$$\ell_i(x) := \prod_{\substack{j=0 \\ j \neq i}}^n \frac{x - c_j}{c_i - c_j} \in F[x].$$

The list  $(\ell_0, \dots, \ell_n)$  is the **Lagrange basis** for the nodes  $c_0, \dots, c_n$ .

In words: the numerator makes  $\ell_i$  vanish at every node except  $c_i$ , and the denominator rescales so that the value at  $c_i$  is exactly 1. The points  $c_i$  are called **nodes**.

**Well-definedness.** Each denominator  $c_i - c_j$  with  $j \neq i$  is non-zero because the nodes are **distinct**, so the quotients exist in  $F$ . The product has  $n$  linear factors, each with leading coefficient  $(c_i - c_j)^{-1} \neq 0$ , so  $\deg \ell_i = n$  by [Theorem 0.8.3](#). The name “basis” is justified by the theorem below.

### Examples.

- **One node.** For  $n = 0$  the product is empty, so  $\ell_0 = 1$ . The degenerate case is consistent: a polynomial of degree at most 0 with value  $y_0$  at  $c_0$  is the constant  $y_0 = y_0 \ell_0$ .
- **Two nodes.** For  $c_0 = 0, c_1 = 1$  we get  $\ell_0 = \frac{x-1}{0-1} = 1 - x$  and  $\ell_1 = \frac{x-0}{1-0} = x$ . The line through  $(0, y_0)$  and  $(1, y_1)$  is  $y_0(1 - x) + y_1 x$ .
- **Three nodes.** For  $-1, 0, 1$ ,  $\ell_{-1} = \frac{x(x-1)}{(-1)(-2)} = \frac{1}{2}x(x-1)$ ,  $\ell_0 = \frac{(x+1)(x-1)}{(1)(-1)} = 1 - x^2$  and  $\ell_1 = \frac{1}{2}x(x+1)$ , the polynomials of [Example \(Dual Bases in  \$\mathbb{R}\[x\]\_{\leq 2}\$ \)](#). (Here we index the nodes by their values.)

**Non-example by minimal change.** Take the nodes  $0, 1, 1$ . The numerator of  $\ell_1$  is still defined,  $(x - 0)(x - 1)$ , but the denominator contains  $c_1 - c_2 = 0$ , which has no inverse. No rescaling helps: a polynomial that vanishes at  $c_2 = 1$  cannot take the value 1 at  $c_1 = 1$ . The clause that fails is “distinct”.

Everything about the Lagrange polynomials follows from one table of values, the Kronecker delta  $\ell_i(c_j) = \delta_{ij}$ .

### Theorem 5.6.1 — Lagrange Interpolation

Let  $n \in \mathbb{N}$ , let  $c_0, \dots, c_n \in F$  be distinct, and let  $(\ell_0, \dots, \ell_n)$  be the Lagrange basis for these nodes.

- (a)  $\ell_i(c_j) = \delta_{ij}$  for all  $i, j \in \{0, \dots, n\}$ .
- (b)  $(\ell_0, \dots, \ell_n)$  is a basis of  $F[x]_{\leq n}$ , and every  $p \in F[x]_{\leq n}$  satisfies

$$p = \sum_{i=0}^n p(c_i) \ell_i.$$

- (c) For all  $y_0, \dots, y_n \in F$ , the unique  $p \in F[x]_{\leq n}$  with  $p(c_i) = y_i$  for every  $i$  is

$$p = \sum_{i=0}^n y_i \ell_i.$$

*Idea.* ① The delta table is a direct evaluation. ② Evaluating a vanishing combination  $\sum a_i \ell_i$  at  $c_j$  kills every term but  $a_j$ , so the list is independent, and it has the right length, so we count instead of checking spanning. ③ The expansion of  $p$  and the interpolation formula are the same computation: both sides of a candidate identity are polynomials of degree at most  $n$  with the same values at  $n+1$  nodes.

*Proof.*

- (a) By Theorem 0.8.5,  $\ell_i(c_j) = \prod_{k \neq i} \frac{c_j - c_k}{c_i - c_k}$ . If  $j = i$ , every factor is 1, so  $\ell_i(c_i) = 1$ . If  $j \neq i$ , the factor with  $k = j$  is 0, so  $\ell_i(c_j) = 0$ .
- (b) Each  $\ell_i$  has degree  $n$ , so it lies in  $F[x]_{\leq n}$ . Let  $a_0 \ell_0 + \dots + a_n \ell_n = 0$ . Evaluating at  $c_j$  and using (a) gives  $0 = \sum_i a_i \delta_{ij} = a_j$  for each  $j$ . Hence the list is linearly independent. It has length  $n+1 = \dim F[x]_{\leq n}$  (Example (Standard Bases)), so it is a basis by Theorem 1.6.4 (a).  
Now let  $p \in F[x]_{\leq n}$ , and put  $q = p - \sum_i p(c_i) \ell_i \in F[x]_{\leq n}$ . By (a),  $q(c_j) = p(c_j) - p(c_j) = 0$  for every  $j$ . So  $q$  has the  $n+1$  distinct roots  $c_0, \dots, c_n$  and degree at most  $n$ . By Corollary 5.1.4,  $q = 0$ , which is the expansion of  $p$ .
- (c) By (a),  $(\sum_i y_i \ell_i)(c_j) = \sum_i y_i \delta_{ij} = y_j$ , so  $\sum_i y_i \ell_i$  has the required values, and it lies in  $F[x]_{\leq n}$ . By Theorem 2.7.3, it is the only such polynomial. This proves the theorem.

■

So Chapter 2's existence statement now comes with a formula, and the uniqueness half of (c) is the expansion in (b) read backwards: a polynomial of degree at most  $n$  is determined by its values at the nodes, because those values are its coordinates in the Lagrange basis.

#### Example. (The Chapter 2 Cubic by Lagrange's Formula)

Use Theorem 5.6.1 to find the  $p \in \mathbb{Q}[x]_{\leq 3}$  with  $p(-1) = 2$ ,  $p(0) = 1$ ,  $p(1) = 2$  and  $p(2) = 11$ , and compare with Example (A cubic through four points).

*Solution.* The nodes  $-1, 0, 1, 2$  are distinct. Write  $\ell_{-1}, \ell_0, \ell_1, \ell_2$  for the Lagrange polynomials, indexed by the node. The denominators are  $\prod_{j \neq i} (c_i - c_j)$ :

$$\begin{aligned}\ell_{-1} &= \frac{x(x-1)(x-2)}{(-1)(-2)(-3)} = -\frac{1}{6}x(x-1)(x-2), \\ \ell_0 &= \frac{(x+1)(x-1)(x-2)}{(1)(-1)(-2)} = \frac{1}{2}(x+1)(x-1)(x-2), \\ \ell_1 &= \frac{(x+1)x(x-2)}{(2)(1)(-1)} = -\frac{1}{2}(x+1)x(x-2), \\ \ell_2 &= \frac{(x+1)x(x-1)}{(3)(2)(1)} = \frac{1}{6}(x+1)x(x-1).\end{aligned}$$

Expanding,

$$\begin{aligned}\ell_{-1} &= -\frac{1}{6}x^3 + \frac{1}{2}x^2 - \frac{1}{3}x, \\ \ell_0 &= \frac{1}{2}x^3 - x^2 - \frac{1}{2}x + 1, \\ \ell_1 &= -\frac{1}{2}x^3 + \frac{1}{2}x^2 + x, \\ \ell_2 &= \frac{1}{6}x^3 - \frac{1}{6}x.\end{aligned}$$

By [Theorem 5.6.1](#) (c),  $p = 2\ell_{-1} + \ell_0 + 2\ell_1 + 11\ell_2$ . Collecting coefficients:

$$\begin{aligned}x^3: & -\frac{1}{3} + \frac{1}{2} - 1 + \frac{11}{6} = 1, \\ x^2: & 1 - 1 + 1 + 0 = 1, \\ x: & -\frac{2}{3} - \frac{1}{2} + 2 - \frac{11}{6} = -1, \\ 1: & 1.\end{aligned}$$

Hence  $p = x^3 + x^2 - x + 1$ , the polynomial found by elimination in [Example \(A cubic through four points\)](#), as uniqueness demands. The elimination solved a  $4 \times 4$  system for these particular values. The Lagrange polynomials were computed once, and would serve for any other values at the same nodes.

**Warning.** The interpolant has degree at most  $n$ , not exactly  $n$ . Through the three points  $(0, 0), (1, 1), (2, 2)$  the formula gives  $0 \cdot \ell_0 + 1 \cdot \ell_1 + 2\ell_2$ , where  $\ell_1 = -x(x-2)$  and  $\ell_2 = \frac{1}{2}x(x-1)$ ; this is  $-x^2 + 2x + x^2 - x = x$ . The degree-2 terms cancel, because the points lie on a line. Each  $\ell_i$  has degree exactly  $n$ , but a combination of them need not.

**Quick check.** Let  $n \geq 1$  and let  $(\ell_0, \dots, \ell_n)$  be the Lagrange basis for distinct nodes  $c_0, \dots, c_n$ . What is  $\sum_{i=0}^n c_i \ell_i$ ?

*Solution.* It is the polynomial  $x$ . Indeed  $x \in F[x]_{\leq n}$  since  $n \geq 1$ , and its values at the nodes are  $c_0, \dots, c_n$ , so [Theorem 5.6.1](#) (b) gives  $x = \sum_i c_i \ell_i$ .

## 5.6.2 Evaluations are the dual basis

The delta table  $\ell_i(c_j) = \delta_{ij}$  can be read the other way round. For  $c \in F$ , evaluation at  $c$  is the linear functional  $\varepsilon_c: p \mapsto p(c)$  ([Example \(Four Families of Functionals\)](#)), which we restrict to  $F[x]_{\leq n}$ . Then the table says  $\varepsilon_{c_j}(\ell_i) = \delta_{ji}$ : the evaluations at the nodes behave on the Lagrange basis exactly as a dual basis must ([Theorem 4.1.3](#)). [Exercise \(C3: Evaluations as a dual basis\)](#)

proved from [Theorem 2.7.3](#) that the evaluations form the dual basis of **some** basis. We can now name that basis.

### Corollary 5.6.2 — Evaluations Are Dual to the Lagrange Basis

Let  $c_0, \dots, c_n \in F$  be distinct, with Lagrange basis  $(\ell_0, \dots, \ell_n)$ . Then  $(\varepsilon_{c_0}, \dots, \varepsilon_{c_n})$  is the dual basis of  $(\ell_0, \dots, \ell_n)$  in  $F[x]_{\leq n}^*$ . In particular, every linear functional  $\varphi$  on  $F[x]_{\leq n}$  is a combination of evaluations at the nodes:

$$\varphi = \sum_{i=0}^n \varphi(\ell_i) \varepsilon_{c_i}.$$

*Proof.* The evaluations are linear ([Example \(Four Families of Functionals\)](#)), and  $\varepsilon_{c_i}(\ell_j) = \ell_j(c_i) = \delta_{ij}$  by [Theorem 5.6.1](#) (a). Since  $(\ell_0, \dots, \ell_n)$  is a basis ([Theorem 5.6.1](#) (b)), the uniqueness in [Theorem 4.1.3](#) (a) shows that the evaluations are its dual basis. The formula for  $\varphi$  is the second formula of [Theorem 4.1.3](#) (c). ■

The first formula of [Theorem 4.1.3](#) (c),  $p = \sum_i \varepsilon_{c_i}(p) \ell_i$ , is the expansion in [Theorem 5.6.1](#) (b). So Lagrange's formula is the statement “a vector is recovered from its dual-basis coordinates”, for the space  $F[x]_{\leq n}$  and the measurements “value at  $c_i$ ”.

The formula for  $\varphi$  has a practical reading: **any** linear measurement of a polynomial of degree at most  $n$  can be computed from  $n + 1$  of its values, with weights  $\varphi(\ell_i)$  that do not depend on the polynomial.

#### Example. (Integrals and Slopes from Values)

- (a) Find  $w_0, w_1, w_2 \in \mathbb{R}$  such that  $\int_0^1 p(t) dt = w_0 p(0) + w_1 p(\frac{1}{2}) + w_2 p(1)$  for every  $p \in \mathbb{R}[x]_{\leq 2}$ .
- (b) Express  $p \mapsto p'(0)$  on  $\mathbb{R}[x]_{\leq 2}$  as a combination of  $\varepsilon_{-1}, \varepsilon_0, \varepsilon_1$ .

*Solution.*

- (a) Integration over  $[0, 1]$  is a linear functional on  $\mathbb{R}[x]_{\leq 2}$  ([Example \(Four Families of Functionals\)](#)). For the nodes  $0, \frac{1}{2}, 1$ ,

$$\begin{aligned} \ell_0 &= \frac{(x - \frac{1}{2})(x - 1)}{(-\frac{1}{2})(-1)} = 2x^2 - 3x + 1, \\ \ell_{1/2} &= \frac{x(x - 1)}{(\frac{1}{2})(-\frac{1}{2})} = -4x^2 + 4x, \\ \ell_1 &= \frac{x(x - \frac{1}{2})}{(1)(\frac{1}{2})} = 2x^2 - x. \end{aligned}$$

Their integrals are  $\frac{2}{3} - \frac{3}{2} + 1 = \frac{1}{6}$ ,  $-\frac{4}{3} + 2 = \frac{2}{3}$  and  $\frac{2}{3} - \frac{1}{2} = \frac{1}{6}$ . By [Corollary 5.6.2](#),

$$\int_0^1 p(t) dt = \frac{1}{6} p(0) + \frac{2}{3} p(\frac{1}{2}) + \frac{1}{6} p(1) \quad \text{for all } p \in \mathbb{R}[x]_{\leq 2}.$$

Check with  $p = x^2$ : the right side is  $\frac{2}{3} \cdot \frac{1}{4} + \frac{1}{6} = \frac{1}{3} = \int_0^1 t^2 dt$ . This is Simpson's rule, which here is exact rather than approximate because  $p$  has degree at most 2.

- (b) With  $\ell_{-1} = \frac{1}{2}x(x-1)$ ,  $\ell_0 = 1-x^2$  and  $\ell_1 = \frac{1}{2}x(x+1)$ , the derivatives at 0 are  $-\frac{1}{2}$ , 0 and  $\frac{1}{2}$ . Hence  $p'(0) = \frac{1}{2}(p(1) - p(-1))$  for every  $p \in \mathbb{R}[x]_{\leq 2}$ , the symmetric difference quotient, exact on quadratics.

### 5.6.3 Congruences and the Chinese Remainder Theorem

Now we change the point of view. By [Theorem 5.1.3](#), the remainder of  $p$  on division by  $x-c$  is  $p(c)$ . So the interpolation conditions  $p(c_i) = y_i$  say that  $p$  leaves the remainder  $y_i$  on division by  $x-c_i$ , for each  $i$ . Nothing in this reformulation needs the divisors to be linear. What happens if we prescribe the remainders modulo arbitrary polynomials  $m_1, \dots, m_k$ ? The integers faced the same question long ago, which explains the name of the answer. We first need the language of congruences, copied from  $\mathbb{Z}$ .

#### Definition 5.6.2 — Congruence Modulo a Polynomial

Let  $m \in F[x]$ . Two polynomials  $f, g \in F[x]$  are **congruent modulo  $m$** , written  $f \equiv g \pmod{m}$ , if  $m \mid f - g$ .

For example,  $x^2 \equiv -1 \pmod{x^2+1}$ , and  $f \equiv f(c) \pmod{x-c}$  for every  $f$ , by [Theorem 5.1.3](#). Congruence modulo  $m$  is an equivalence relation, and it respects sums and products: if  $f \equiv f'$  and  $g \equiv g' \pmod{m}$ , then  $f+g \equiv (f'+g') \pmod{m}$  and  $fg \equiv f'g' \pmod{m}$ . By [Theorem 5.1.1](#), if  $m \neq 0$ , every  $f$  is congruent modulo  $m$  to **exactly one** polynomial of degree less than  $\deg m$ , its remainder:  $f = qm + r$  gives  $f \equiv r$ , and if  $f \equiv r \equiv r'$  with both degrees below  $\deg m$ , then  $m \mid r - r'$  with  $\deg(r - r') < \deg m$ , which forces  $r - r' = 0$ .

Interpolation needed distinct nodes. The general condition is that the moduli share no factor, and the proof needs a way to pass from “coprime in pairs” to “coprime to the product”.

#### Lemma 5.6.3 — Coprime to Each, Coprime to the Product

- (a) Let  $p, q_1, \dots, q_s \in F[x]$ . If  $p$  is coprime to each  $q_j$ , then there are  $a, b \in F[x]$  with  $ap + bq_1 \cdots q_s = 1$ , and  $p$  is coprime to  $q_1 \cdots q_s$ .
- (b) Let  $m_1, \dots, m_k \in F[x]$  be non-zero and **pairwise coprime**, that is,  $\gcd(m_i, m_j) = 1$  whenever  $i \neq j$ . If  $m_i \mid h$  for every  $i$ , then  $m_1 \cdots m_k \mid h$ .

*Proof.*

- (a) By [Corollary 5.2.4](#), for each  $j$  there are  $a_j, b_j$  with  $a_j p + b_j q_j = 1$ . Multiplying these  $s$  identities,

$$1 = \prod_{j=1}^s (a_j p + b_j q_j) = ap + (b_1 \cdots b_s) q_1 \cdots q_s$$

for some  $a \in F[x]$ , because every term of the expanded product except  $(b_1 q_1) \cdots (b_s q_s)$  contains a factor  $a_j p$ . Let  $d = \gcd(p, q_1 \cdots q_s)$ . By [Theorem 5.2.3](#),  $d$  divides  $p$  and

$q_1 \cdots q_s$ , hence divides  $ap + bq_1 \cdots q_s = 1$ , where  $b = b_1 \cdots b_s$ . So  $d$  is a non-zero constant by [Theorem 0.8.3](#), and being monic,  $d = 1$ . Hence  $p$  and  $q_1 \cdots q_s$  are coprime ([Definition 5.2.3](#)).

- (b) We use induction on  $k$ . For  $k = 1$  there is nothing to prove. Let  $k \geq 2$ . The moduli  $m_1, \dots, m_{k-1}$  are pairwise coprime and each divides  $h$ , so by the induction hypothesis  $M := m_1 \cdots m_{k-1}$  divides  $h$ , say  $h = Mg$ . Since  $m_k \mid h = Mg$ , and  $m_k$  is coprime to  $M$  by (a), [Theorem 5.2.8](#) gives  $m_k \mid g$ , say  $g = m_k g'$ . Then  $h = m_1 \cdots m_k g'$ , as claimed. ■

### Theorem 5.6.4 — Chinese Remainder Theorem for Polynomials

Let  $m_1, \dots, m_k \in F[x]$  be non-zero and pairwise coprime, and put  $m = m_1 \cdots m_k$ . For all  $r_1, \dots, r_k \in F[x]$ :

- (a) there is  $f \in F[x]$  with  $f \equiv r_i \pmod{m_i}$  for every  $i$ ;
- (b) if  $f$  and  $g$  both satisfy (a), then  $f \equiv g \pmod{m}$ ;
- (c) there is **exactly one** such  $f$  with  $\deg f < \deg m$ .

*Idea.* Copy Lagrange. There,  $\ell_i$  was 1 at  $c_i$  and 0 at the other nodes; in the new language,  $\ell_i \equiv 1 \pmod{x - c_i}$  and  $\ell_i \equiv 0 \pmod{x - c_j}$ . So we look for  $e_i$  with  $e_i \equiv 1 \pmod{m_i}$  and  $e_i \equiv 0 \pmod{m_j}$  for  $j \neq i$ , and then  $f = \sum_i r_i e_i$  works. To be 0 modulo every other  $m_j$ ,  $e_i$  should be a multiple of  $M_i = \prod_{j \neq i} m_j$ ; to be 1 modulo  $m_i$ , it should be  $1 - a_i m_i$ . Bézout's identity  $a_i m_i + b_i M_i = 1$  provides exactly such an element:  $e_i = b_i M_i$ .

*Proof.*

- (a) Fix  $i$ , and let  $M_i = \prod_{j \neq i} m_j$ . Since  $m_i$  is coprime to each  $m_j$  with  $j \neq i$ , [Lemma 5.6.3](#) (a) gives  $a_i, b_i \in F[x]$  with  $a_i m_i + b_i M_i = 1$ . Put  $e_i = b_i M_i$ . Then  $e_i - 1 = -a_i m_i$ , so  $e_i \equiv 1 \pmod{m_i}$ , and  $m_j \mid M_i \mid e_i$  for  $j \neq i$ , so  $e_i \equiv 0 \pmod{m_j}$ . Let  $f = r_1 e_1 + \cdots + r_k e_k$ . Since congruence modulo  $m_j$  respects sums and products,

$$f \equiv r_1 \cdot 0 + \cdots + r_j \cdot 1 + \cdots + r_k \cdot 0 = r_j \pmod{m_j}$$

for each  $j$ .

- (b) If  $f$  and  $g$  satisfy (a), then  $f - g \equiv r_i - r_i = 0 \pmod{m_i}$ , that is,  $m_i \mid f - g$ , for every  $i$ . By [Lemma 5.6.3](#) (b),  $m \mid f - g$ .
- (c) Since each  $m_i \neq 0$ ,  $m \neq 0$  ([Theorem 0.8.3](#)). Let  $f$  be as in (a), and let  $f_0$  be its remainder on division by  $m$  ([Theorem 5.1.1](#)). Then  $m_i \mid m \mid f - f_0$ , so  $f_0 \equiv f \equiv r_i \pmod{m_i}$  for each  $i$ , and  $\deg f_0 < \deg m$ . If  $g$  is another solution with  $\deg g < \deg m$ , then  $m \mid f_0 - g$  by (b), and  $\deg(f_0 - g) < \deg m$ , so  $f_0 = g$ . This proves the theorem. ■

**Lagrange is the case**  $m_i = x - c_i$ . For distinct nodes,  $(x - c_j) - (x - c_i) = c_i - c_j$  is a non-zero constant, so  $\frac{1}{c_i - c_j}(x - c_j) - \frac{1}{c_i - c_j}(x - c_i) = 1$ , and any common divisor of  $x - c_i$  and  $x - c_j$  divides



1: the moduli are pairwise coprime. The congruence  $f \equiv y_i \pmod{x - c_i}$  means  $f(c_i) = y_i$  by [Theorem 5.1.3](#), and  $\deg m = n + 1$ . So part (c) is [Theorem 2.7.3](#). The element  $e_i$  has degree less than  $n + 1$  after reduction modulo  $m$ , is 1 at  $c_i$  and 0 at the other nodes, so after reduction it is  $\ell_i$ .

**Example. (A Linear and a Quadratic Modulus)**

Find the polynomial  $f \in \mathbb{Q}[x]$  of degree less than 3 with

$$f \equiv 1 \pmod{x - 2}, \quad f \equiv 2x - 1 \pmod{x^2 + 1}.$$

*Solution. Bézout.* Divide  $x^2 + 1$  by  $x - 2$ :  $x^2 + 1 = (x + 2)(x - 2) + 5$ . Hence

$$\frac{1}{5}(x^2 + 1) - \frac{1}{5}(x + 2)(x - 2) = 1,$$

and the moduli are coprime, since any common divisor divides 1.

*The elements  $e_i$ .* Put  $e_1 = \frac{1}{5}(x^2 + 1)$  and  $e_2 = -\frac{1}{5}(x + 2)(x - 2) = 1 - e_1$ . Then  $e_1$  is a multiple of  $x^2 + 1$  and  $e_1 \equiv 1 + \frac{1}{5}(x + 2)(x - 2) \equiv 1 \pmod{x - 2}$ ; symmetrically,  $e_2 \equiv 0 \pmod{x - 2}$  and  $e_2 \equiv 1 \pmod{x^2 + 1}$ .

*Combine and reduce.* By the proof of [Theorem 5.6.4](#),  $f \equiv 1 \cdot e_1 + (2x - 1)e_2 \pmod{(x - 2)(x^2 + 1)}$ . Expanding,

$$e_1 + (2x - 1)e_2 = \frac{1}{5}(x^2 + 1) - \frac{1}{5}(2x - 1)(x^2 - 4) = \frac{1}{5}(-2x^3 + 2x^2 + 8x - 3).$$

Dividing by  $m = (x - 2)(x^2 + 1) = x^3 - 2x^2 + x - 2$  gives quotient  $-\frac{2}{5}$  and remainder

$$f = \frac{1}{5}(-2x^2 + 10x - 7).$$

*Check.*  $f(2) = \frac{1}{5}(-8 + 20 - 7) = 1$ , so  $f \equiv 1 \pmod{x - 2}$ . Modulo  $x^2 + 1$ ,  $x^2 \equiv -1$ , so  $f \equiv \frac{1}{5}(2 + 10x - 7) = 2x - 1$ .

**Warning. The moduli must be coprime in pairs; coprime as a whole family is not enough.** Without coprimality a system can be unsolvable: with  $m_1 = x$  and  $m_2 = x^2$ , the system  $f \equiv 0 \pmod{x}$ ,  $f \equiv 1 \pmod{x^2}$  has no solution: the first condition says  $f(0) = 0$ , the second says  $f = 1 + x^2q$ , so  $f(0) = 1$ . For three moduli,  $x$ ,  $x - 1$  and  $x(x - 1)$  have no common factor, yet the first and third share  $x$ , and  $f \equiv 0 \pmod{x}$ ,  $f \equiv 1 \pmod{x(x - 1)}$  is again impossible.

**Example. (Values and Slopes by CRT)**

- (a) Let  $c, a, b \in F$  and  $f \in F[x]$ . Show that  $f \equiv a + b(x - c) \pmod{(x - c)^2}$  if and only if  $f(c) = a$  and  $f'(c) = b$ .
- (b) Find the  $p \in \mathbb{Q}[x]_{\leq 3}$  with  $p(0) = 2$ ,  $p'(0) = -1$ ,  $p(1) = 1$  and  $p'(1) = 3$ .

*Solution.*

- (a) Divide by  $(x - c)^2$ :  $f = q(x - c)^2 + r$  with  $\deg r \leq 1$  ([Theorem 5.1.1](#)). Writing  $r = r_0 + r_1(x - c)$ , which is possible because  $(1, x - c)$  is a basis of  $F[x]_{\leq 1}$  by [Theorem 1.4.4](#) and [Theorem 1.6.4](#), the uniqueness of the remainder says  $f \equiv a + b(x - c)$  exactly when  $r_0 = a$  and  $r_1 = b$ . Now  $f(c) = r_0$ . By [Theorem 5.4.4](#),

$f' = q'(x - c)^2 + 2q(x - c) + r_1$ , so  $f'(c) = r_1$ . Hence the congruence holds if and only if  $f(c) = a$  and  $f'(c) = b$ .

- (b) By (a), the conditions are  $p \equiv 2 - x \pmod{x^2}$  and  $p \equiv 1 + 3(x - 1) \pmod{(x - 1)^2}$ . The moduli are coprime, since

$$(2x + 1)(x - 1)^2 - (2x - 3)x^2 = (2x^3 - 3x^2 + 1) - (2x^3 - 3x^2) = 1.$$

So  $e_0 = (2x + 1)(x - 1)^2$  is  $\equiv 0 \pmod{(x - 1)^2}$  and  $\equiv 1 \pmod{x^2}$ , while  $e_1 = -(2x - 3)x^2 = 3x^2 - 2x^3$  is  $\equiv 0 \pmod{x^2}$  and  $\equiv 1 \pmod{(x - 1)^2}$ . The polynomial  $e_1$  is the smooth step  $3x^2 - 2x^3$  of [Example \(Prescribing Values and Slopes\)](#): value 0 and slope 0 at 0, value 1 and slope 0 at 1. By [Theorem 5.6.4](#),  $p$  is the remainder of

$$(2 - x)e_0 + (3x - 2)e_1 = -8x^4 + 20x^3 - 12x^2 - x + 2$$

on division by  $x^2(x - 1)^2 = x^4 - 2x^3 + x^2$ . Adding  $8(x^4 - 2x^3 + x^2)$  gives

$$p = 4x^3 - 4x^2 - x + 2.$$

*Check.*  $p(0) = 2$ ;  $p' = 12x^2 - 8x - 1$ , so  $p'(0) = -1$ ;  $p(1) = 4 - 4 - 1 + 2 = 1$ ;  $p'(1) = 12 - 8 - 1 = 3$ .

The same recipe handles any finite set of nodes  $c_i$  with multiplicities  $k_i$ : the moduli  $(x - c_i)^{k_i}$  are pairwise coprime for distinct nodes, by [Lemma 5.6.3](#) (a) applied twice: first  $x - c_i$  is coprime to  $(x - c_j)^{k_j}$ , and then  $(x - c_j)^{k_j}$  is coprime to  $(x - c_i)^{k_i}$ . So for any prescribed remainders modulo  $(x - c_i)^{k_i}$ , there is exactly one polynomial of degree less than  $k_1 + \cdots + k_s$  with those remainders. In characteristic 0, the Taylor expansion of [Theorem 5.4.7](#) translates “the remainder modulo  $(x - c)^k$ ” into the values  $f(c), f'(c), \dots, f^{(k-1)}(c)$ , and this is **Hermite interpolation**.

## 5.6.4 The quotient space $F[x]/\langle m \rangle$

Congruence modulo  $m$  is equality of cosets. The ideal  $\langle m \rangle$  is a subspace of  $F[x]$ : it is non-empty and closed under addition and under multiplication by constant polynomials ([Definition 5.2.1](#)). So the quotient space  $F[x]/\langle m \rangle$  of [Definition 3.9.3](#) exists, and by [Lemma 3.9.3](#) (b),

$$f + \langle m \rangle = g + \langle m \rangle \iff f - g \in \langle m \rangle \iff f \equiv g \pmod{m}.$$

An element of  $F[x]/\langle m \rangle$  is a polynomial “known only modulo  $m$ ”. We cannot quote [Theorem 3.9.5](#) for its dimension, because  $F[x]$  is infinite-dimensional ([Theorem 1.3.3](#)). The remainder does the job instead.

### Theorem 5.6.5 — Dimension of $F[x]/\langle m \rangle$

Let  $m \in F[x]$  be non-zero of degree  $d$ . Then  $F[x]/\langle m \rangle$  is finite-dimensional with

$$\dim F[x]/\langle m \rangle = \deg m = d.$$

If  $d \geq 1$ , the cosets  $(1 + \langle m \rangle, x + \langle m \rangle, \dots, x^{d-1} + \langle m \rangle)$  form a basis.

*Proof.* If  $d = 0$ , then  $m$  is a non-zero constant, and every  $f = (m^{-1}f)m$  lies in  $\langle m \rangle$ . So  $\langle m \rangle = F[x]$  and the quotient is the zero space, of dimension 0.

Let  $d \geq 1$ , and define  $\rho: F[x] \rightarrow F[x]_{\leq d-1}$  by letting  $\rho(f)$  be the remainder of  $f$  on division by  $m$  (Theorem 5.1.1). *Linear.* If  $f = q_1m + \rho(f)$  and  $g = q_2m + \rho(g)$ , then for  $c \in F$ ,

$$cf + g = (cq_1 + q_2)m + (c\rho(f) + \rho(g)),$$

and  $\deg(c\rho(f) + \rho(g)) < d$  by Theorem 0.8.2. By the uniqueness in Theorem 5.1.1,  $\rho(cf + g) = c\rho(f) + \rho(g)$ . *Surjective.* If  $\deg r < d$ , then  $r = 0 \cdot m + r$ , so  $\rho(r) = r$ . *Kernel.*  $\rho(f) = 0$  exactly when  $f = qm$  for some  $q$ , that is,  $f \in \langle m \rangle$ .

By the First Isomorphism Theorem (Theorem 3.9.7),  $\bar{\rho}: F[x]/\langle m \rangle \rightarrow F[x]_{\leq d-1}$ ,  $f + \langle m \rangle \mapsto \rho(f)$ , is a well-defined injective linear map with image  $\text{im } \bar{\rho} = F[x]_{\leq d-1}$ , so it is an isomorphism. Since  $\bar{\rho}(x^j + \langle m \rangle) = x^j$  for  $j < d$ , and  $(1, x, \dots, x^{d-1})$  is a basis of  $F[x]_{\leq d-1}$  (Example (Standard Bases)), the inverse isomorphism  $\bar{\rho}^{-1}$  carries it to the basis  $(x^j + \langle m \rangle)_{j < d}$  (Theorem 3.5.2). Hence  $\dim F[x]/\langle m \rangle = d$ . ■

For example,  $\mathbb{R}[x]/\langle x^2 + 1 \rangle$  has basis  $(1 + \langle x^2 + 1 \rangle, x + \langle x^2 + 1 \rangle)$ , so each element is  $a + bx + \langle x^2 + 1 \rangle$  for unique  $a, b \in \mathbb{R}$ .

*Remark.* Cosets can also be **multiplied**:  $(f + \langle m \rangle)(g + \langle m \rangle) := fg + \langle m \rangle$  is well defined, because congruence modulo  $m$  respects products. In  $\mathbb{R}[x]/\langle x^2 + 1 \rangle$ ,  $x \cdot x = x^2 \equiv -1$ , so the coset of  $x$  squares to minus the coset of 1, and  $(a + bx)(a' + b'x) \equiv (aa' - bb') + (ab' + a'b)x$ : this is multiplication in  $\mathbb{C}$ , with the coset of  $x$  playing the role of  $i$ .

The Chinese Remainder Theorem now becomes a statement about linear maps. Reducing a polynomial modulo each  $m_i$  records its  $k$  remainders; the theorem says that this record, taken modulo  $m$ , is a perfect dictionary. Recall from Definition 3.9.1 that  $V_1 \times \dots \times V_k$  is the space of lists with entrywise operations.

### Theorem 5.6.6 — CRT as an Isomorphism

Let  $m_1, \dots, m_k \in F[x]$  be non-zero and pairwise coprime, and put  $m = m_1 \cdots m_k$ . Then

$$\begin{aligned} \Phi: F[x]/\langle m \rangle &\rightarrow F[x]/\langle m_1 \rangle \times \cdots \times F[x]/\langle m_k \rangle, \\ f + \langle m \rangle &\mapsto (f + \langle m_1 \rangle, \dots, f + \langle m_k \rangle), \end{aligned}$$

is a well-defined isomorphism of vector spaces.

*Idea.* The map that does not care about well-definedness is  $\Psi: f \mapsto (f + \langle m_i \rangle)_i$  on  $F[x]$  itself. Its kernel is the set of polynomials divisible by every  $m_i$ , which is  $\langle m \rangle$  by Lemma 5.6.3. So the First Isomorphism Theorem makes  $\Phi$  well defined and injective in one stroke. For surjectivity we do **not** construct a preimage. We count: both sides have dimension  $\deg m$ , because degrees add under products.

*Proof.* Let  $\Psi: F[x] \rightarrow F[x]/\langle m_1 \rangle \times \cdots \times F[x]/\langle m_k \rangle$ ,  $\Psi(f) = (f + \langle m_1 \rangle, \dots, f + \langle m_k \rangle)$ . Each entry is a quotient map, which is linear by Theorem 3.9.4 (c), and the operations in the product are entrywise, so  $\Psi$  is linear. By Lemma 3.9.3 (b),  $\Psi(f) = \mathbf{0}$  if and only if  $m_i \mid f$  for every  $i$ . If  $m \mid f$ , then  $m_i \mid f$  for all  $i$ , since  $m_i \mid m$ ; conversely, if every  $m_i \mid f$ , then  $m \mid f$  by Lemma 5.6.3 (b). Hence  $\ker \Psi = \langle m \rangle$ .

By the First Isomorphism Theorem (Theorem 3.9.7), the rule  $f + \langle m \rangle \mapsto \Psi(f)$  is a

well-defined injective linear map; this is  $\Phi$ . By [Theorem 5.6.5](#), [Theorem 0.8.3](#) and [Theorem 3.9.1](#),

$$\begin{aligned}\dim F[x]/\langle m \rangle &= \deg m = \sum_{i=1}^k \deg m_i = \sum_{i=1}^k \dim F[x]/\langle m_i \rangle \\ &= \dim \left( F[x]/\langle m_1 \rangle \times \cdots \times F[x]/\langle m_k \rangle \right).\end{aligned}$$

Since  $\Phi$  is injective between spaces of the same finite dimension, it is bijective by [Corollary 3.3.2](#) (e), hence an isomorphism ([Definition 3.5.2](#)).

■

Surjectivity of  $\Phi$  is the existence part of [Theorem 5.6.4](#), and injectivity is the uniqueness part. So the count has given a second proof of the Chinese Remainder Theorem, one that never uses Bézout's identity for existence. It is the same trade as in [Example \(Interpolation as a Count\)](#): counting proves that a solution exists, and the Bézout construction says what it is. By the remark above,  $\Phi$  also respects products of cosets, since reduction modulo each  $m_i$  does. In Chapter 8 this splitting of  $F[x]/\langle m \rangle$  into pieces, one for each coprime factor of  $m$ , reappears as a splitting of a space on which an operator acts. The next section begins that story.

*Quick check.* What is  $\dim \mathbb{Q}[x]/\langle x^3 - x \rangle$ ? Into which three spaces does [Theorem 5.6.6](#) split it, and what does  $\Phi$  do to the coset of  $x^2$ ?

*Solution.* By [Theorem 5.6.5](#), the dimension is 3. Since  $x^3 - x = x(x-1)(x+1)$  with distinct linear factors, which are pairwise coprime,  $\Phi$  maps it isomorphically onto  $\mathbb{Q}[x]/\langle x \rangle \times \mathbb{Q}[x]/\langle x-1 \rangle \times \mathbb{Q}[x]/\langle x+1 \rangle$ , a product of three one-dimensional spaces. Modulo  $x - c$ , a polynomial is congruent to its value at  $c$ , so  $\Phi(x^2 + \langle x^3 - x \rangle)$  is the list of cosets of the constants 0, 1, 1: the values of  $x^2$  at 0, 1, -1.

## 5.6.5 Exercises

### 5.6.5.1 A. Check your understanding

#### Exercise A1.

- Define the Lagrange basis for distinct nodes  $c_0, \dots, c_n$ , and state the value of  $\ell_i$  at  $c_j$ .
- True or false: for four distinct nodes, the polynomial of degree at most 3 through the four points always has degree exactly 3. Justify your answer.
- State the Chinese Remainder Theorem for polynomials, with all hypotheses.
- True or false: if  $\gcd(m_1, m_2, m_3) = 1$ , then every system  $f \equiv r_i \pmod{m_i}$ ,  $i = 1, 2, 3$ , has a solution. Justify your answer.
- What is  $\dim_{\mathbb{R}} \mathbb{R}[x]/\langle x^4 + 1 \rangle$ ? Give a basis.
- Which functional on  $F[x]_{\leq n}$  is the  $i$ -th vector of the dual basis ([Theorem 4.1.3](#)) of the Lagrange basis?

*Solution.*

- (a) For distinct  $c_0, \dots, c_n \in F$ ,  $\ell_i = \prod_{j \neq i} \frac{x - c_j}{c_i - c_j}$  (Definition 5.6.1), and  $\ell_i(c_j) = \delta_{ij}$  (Theorem 5.6.1 (a)).
- (b) False. The points  $(0, 0), (1, 0), (2, 0), (3, 0)$  are fitted by the zero polynomial, of degree  $-\infty$ ; by uniqueness (Theorem 2.7.3), that is the interpolant.
- (c) If  $m_1, \dots, m_k \in F[x]$  are non-zero and pairwise coprime and  $m = m_1 \cdots m_k$ , then for all  $r_1, \dots, r_k \in F[x]$  there is  $f$  with  $f \equiv r_i \pmod{m_i}$  for each  $i$ , unique modulo  $m$ , and exactly one such  $f$  has  $\deg f < \deg m$  (Theorem 5.6.4).
- (d) False.  $x, x - 1, x(x - 1)$  have gcd 1 as a family, but  $f \equiv 0 \pmod{x}$ ,  $f \equiv 0 \pmod{x - 1}$ ,  $f \equiv 1 \pmod{x(x - 1)}$  has no solution: the last condition gives  $f(0) = 1$ , contradicting the first.
- (e) 4, with basis  $(1, x, x^2, x^3)$  of cosets modulo  $\langle x^4 + 1 \rangle$ , by Theorem 5.6.5.
- (f) Evaluation at the  $i$ -th node,  $\varepsilon_{c_i} : p \mapsto p(c_i)$  (Corollary 5.6.2).

### 5.6.5.2 B. Practice

#### Exercise B1: A Lagrange interpolant.

Let  $c_0, c_1, c_2, c_3 = 0, 1, 2, 3$ . Write down the Lagrange basis for these nodes in factored form, and use it to find the  $p \in \mathbb{Q}[x]_{\leq 3}$  with  $p(0) = 1$ ,  $p(1) = 0$ ,  $p(2) = 1$  and  $p(3) = 10$ . Hence compute  $p(4)$ .

*Solution.* The denominators  $\prod_{j \neq i} (c_i - c_j)$  are  $(-1)(-2)(-3) = -6$ ,  $(1)(-1)(-2) = 2$ ,  $(2)(1)(-1) = -2$  and  $(3)(2)(1) = 6$ . So

$$\begin{aligned}\ell_0 &= -\frac{1}{6}(x-1)(x-2)(x-3), \\ \ell_1 &= \frac{1}{2}x(x-2)(x-3), \\ \ell_2 &= -\frac{1}{2}x(x-1)(x-3), \\ \ell_3 &= \frac{1}{6}x(x-1)(x-2).\end{aligned}$$

By Theorem 5.6.1 (c),  $p = \ell_0 + 0 \cdot \ell_1 + \ell_2 + 10\ell_3$ . Expanding each needed term,

$$\begin{aligned}\ell_0 &= -\frac{1}{6}(x^3 - 6x^2 + 11x - 6), \\ \ell_2 &= -\frac{1}{2}(x^3 - 4x^2 + 3x), \\ 10\ell_3 &= \frac{5}{3}(x^3 - 3x^2 + 2x).\end{aligned}$$

The coefficient of  $x^3$  is  $-\frac{1}{6} - \frac{1}{2} + \frac{5}{3} = 1$ ; of  $x^2$ ,  $1 + 2 - 5 = -2$ ; of  $x$ ,  $-\frac{11}{6} - \frac{3}{2} + \frac{10}{3} = 0$ ; the constant is 1. Hence

$$p = x^3 - 2x^2 + 1.$$

Check:  $p(0) = 1$ ,  $p(1) = 0$ ,  $p(2) = 8 - 8 + 1 = 1$ ,  $p(3) = 27 - 18 + 1 = 10$ . Hence  $p(4) = 64 - 32 + 1 = 33$ .

#### Exercise B2: A CRT system.

Find the polynomial  $f \in \mathbb{Q}[x]$  of degree less than 3 with  $f \equiv 2 \pmod{x-1}$  and  $f \equiv x \pmod{x^2+1}$ . Hence describe all  $g \in \mathbb{Q}[x]$  satisfying both congruences.

*Solution. Bézout.*  $x^2 + 1 = (x + 1)(x - 1) + 2$ , so  $\frac{1}{2}(x^2 + 1) - \frac{1}{2}(x + 1)(x - 1) = 1$ , and the moduli are coprime.

*The elements  $e_i$ .*  $e_1 = \frac{1}{2}(x^2 + 1)$  is a multiple of  $x^2 + 1$  and  $e_1 = 1 + \frac{1}{2}(x + 1)(x - 1) \equiv 1 \pmod{x - 1}$ . And  $e_2 = 1 - e_1 = -\frac{1}{2}(x + 1)(x - 1)$  is a multiple of  $x - 1$  with  $e_2 \equiv 1 \pmod{x^2 + 1}$ .

*Combine and reduce.* By the proof of [Theorem 5.6.4](#),  $f \equiv 2e_1 + xe_2 = (x^2 + 1) - \frac{1}{2}x(x^2 - 1) = -\frac{1}{2}x^3 + x^2 + \frac{1}{2}x + 1$ . Modulo  $m = (x - 1)(x^2 + 1) = x^3 - x^2 + x - 1$ , add  $\frac{1}{2}m$ :

$$f = -\frac{1}{2}x^3 + x^2 + \frac{1}{2}x + 1 + \frac{1}{2}(x^3 - x^2 + x - 1) = \frac{1}{2}x^2 + x + \frac{1}{2} = \frac{1}{2}(x + 1)^2.$$

*Check.*  $f(1) = 2$ . Modulo  $x^2 + 1$ ,  $f \equiv \frac{1}{2}(-1) + x + \frac{1}{2} = x$ .

By [Theorem 5.6.4](#) (b), and since every  $g \equiv f \pmod{m}$  satisfies both congruences (because  $x - 1$  and  $x^2 + 1$  divide  $m$ ), the solutions are exactly  $g = \frac{1}{2}(x + 1)^2 + h(x - 1)(x^2 + 1)$  with  $h \in \mathbb{Q}[x]$ .

### Exercise B3: The Lagrange basis sums to one.

Let  $c_0, \dots, c_n \in F$  be distinct with Lagrange basis  $(\ell_0, \dots, \ell_n)$ .

- (a) Prove that  $\ell_0 + \ell_1 + \dots + \ell_n = 1$ .
- (b) Prove more generally that  $\sum_{i=0}^n c_i^k \ell_i = x^k$  for  $k = 0, 1, \dots, n$  (with  $c^0 = 1$ ).
- (c) For  $n = 2$  and nodes  $0, 1, 2$ , verify (a) by expanding.

*Solution.*

- (a) This is (b) with  $k = 0$ .
- (b) Let  $0 \leq k \leq n$ . Then  $x^k \in F[x]_{\leq n}$ , and its value at  $c_i$  is  $c_i^k$  ([Definition 0.8.6](#)). By [Theorem 5.6.1](#) (b),  $x^k = \sum_i c_i^k \ell_i$ .
- (c)  $\ell_0 = \frac{(x-1)(x-2)}{2} = \frac{1}{2}x^2 - \frac{3}{2}x + 1$ ,  $\ell_1 = \frac{x(x-2)}{-1} = -x^2 + 2x$ ,  $\ell_2 = \frac{x(x-1)}{2} = \frac{1}{2}x^2 - \frac{1}{2}x$ . Adding:  $x^2$  has coefficient  $\frac{1}{2} - 1 + \frac{1}{2} = 0$ ,  $x$  has  $-\frac{3}{2} + 2 - \frac{1}{2} = 0$ , and the constant is 1. So the sum is 1.

### 5.6.5.3 C. Going deeper

#### Exercise C1: Interpolation over a finite field.

Let  $p$  be a prime and  $F = \mathbb{F}_p$ .

- (a) Explain why [Theorem 5.6.1](#) cannot be applied with more than  $p$  nodes.
- (b) Let  $E: \mathbb{F}_p[x]_{\leq p} \rightarrow \mathbb{F}_p^p$ ,  $E(f) = (f(0), f(1), \dots, f(p-1))$ . Prove that  $E$  is linear and surjective but not injective, and find a non-zero polynomial in  $\ker E$ .
- (c) Deduce that every function  $\mathbb{F}_p \rightarrow \mathbb{F}_p$  is the polynomial function of exactly one polynomial of degree at most  $p-1$ , but of more than one polynomial of degree at most  $p$ .

*Solution.*

- (a) The theorem needs distinct nodes, and  $\mathbb{F}_p$  has only  $p$  elements, so there are no  $p + 1$  distinct nodes.
- (b) *Linear.* Each entry  $f \mapsto f(c)$  is linear by [Theorem 0.8.5](#). *Surjective.* The  $p$  elements  $0, 1, \dots, p - 1$  of  $\mathbb{F}_p$  are distinct nodes, so for every  $(y_0, \dots, y_{p-1})$ , [Theorem 5.6.1](#) (c) with  $n = p - 1$  gives  $f = \sum_i y_i \ell_i \in \mathbb{F}_p[x]_{\leq p-1} \subseteq \mathbb{F}_p[x]_{\leq p}$  with  $E(f) = (y_0, \dots, y_{p-1})$ . *Not injective.*  $\dim \mathbb{F}_p[x]_{\leq p} = p + 1 > p = \dim \mathbb{F}_p^p$ , so  $E$  is not injective by [Corollary 3.3.2](#) (c). Concretely,  $g = \prod_{c \in \mathbb{F}_p} (x - c)$  is monic of degree  $p$ , hence non-zero, and  $g(c) = 0$  for every  $c \in \mathbb{F}_p$ , because one factor vanishes ([Theorem 0.8.5](#)).
- (c) A function  $\mathbb{F}_p \rightarrow \mathbb{F}_p$  is the same as its list of values  $(y_0, \dots, y_{p-1})$ . By (b), the restriction of  $E$  to  $\mathbb{F}_p[x]_{\leq p-1}$  is surjective, and it is injective because a polynomial of degree at most  $p - 1$  vanishing at  $p$  distinct points is 0 ([Corollary 5.1.4](#)). So exactly one polynomial of degree at most  $p - 1$  has the given values. In degree at most  $p$ , if  $f$  has the given values, so does  $f + g \neq f$ , with  $g$  from (b). Unlike over  $\mathbb{R}$ , a polynomial over  $\mathbb{F}_p$  is not determined by its function.

### Exercise C2: Newton's form of the interpolant.

Let  $c_0, \dots, c_n \in F$  be distinct. Put  $N_0 = 1$  and  $N_k = (x - c_0)(x - c_1) \cdots (x - c_{k-1})$  for  $1 \leq k \leq n + 1$ .

- (a) Prove that  $(N_0, \dots, N_n)$  is a basis of  $F[x]_{\leq n}$ .
- (b) Let  $p = \sum_{k=0}^n a_k N_k$ . Show that  $p(c_i) = \sum_{k=0}^i a_k N_k(c_i)$  and  $N_i(c_i) \neq 0$ . Explain why the coefficients  $a_0, a_1, \dots$  of the interpolant can be found one at a time, and show that  $a_0 = y_0$  and  $a_1 = \frac{y_1 - y_0}{c_1 - c_0}$ .
- (c) Let  $p_n \in F[x]_{\leq n}$  interpolate  $y_0, \dots, y_n$  at  $c_0, \dots, c_n$ , and let  $c_{n+1}$  be a further node, distinct from the others, with a value  $y_{n+1}$ . Prove that the interpolant  $p_{n+1} \in F[x]_{\leq n+1}$  is

$$p_{n+1} = p_n + \frac{y_{n+1} - p_n(c_{n+1})}{N_{n+1}(c_{n+1})} N_{n+1}.$$

- (d) Find the Newton form of the interpolant of [Exercise \(B1: A Lagrange interpolant\)](#), and then the interpolant after adding the point  $(4, 0)$ . Why is this more economical than recomputing with Lagrange's formula?

*Hint: for (c), check the values of the right-hand side at all  $n + 2$  nodes.*

*Solution.*

- (a)  $N_k$  is a product of  $k$  monic linear factors, so it is monic of degree  $k$  ([Theorem 0.8.3](#)). The polynomials  $N_0, \dots, N_n$  are non-zero with pairwise distinct degrees, hence linearly independent by [Theorem 1.4.4](#). The list has length  $n + 1 = \dim F[x]_{\leq n}$ , so it is a basis by [Theorem 1.6.4](#) (a).
- (b) For  $k > i$ ,  $N_k$  contains the factor  $x - c_i$ , so  $N_k(c_i) = 0$ ; this gives the formula for  $p(c_i)$ . And  $N_i(c_i) = \prod_{j < i} (c_i - c_j)$  is a product of non-zero scalars, since the nodes are distinct, so it is non-zero. The conditions  $p(c_i) = y_i$  therefore form a lower triangular system: the  $i$ -th equation involves only  $a_0, \dots, a_i$ , with coefficient  $N_i(c_i) \neq 0$  on  $a_i$ . Solving in the order  $i = 0, 1, \dots, n$  determines each  $a_i$  from the earlier ones. For  $i = 0$ :  $p(c_0) = a_0$ , so  $a_0 = y_0$ . For  $i = 1$ :  $y_1 = a_0 + a_1(c_1 - c_0)$ , so

$a_1 = \frac{y_1 - y_0}{c_1 - c_0}$ , the slope of the chord, the first “divided difference”.

- (c) Let  $q$  be the right-hand side; the division is legal because  $N_{n+1}(c_{n+1}) \neq 0$  by (b). Then  $\deg q \leq n + 1$ . For  $i \leq n$ ,  $N_{n+1}(c_i) = 0$ , so  $q(c_i) = p_n(c_i) = y_i$ . At  $c_{n+1}$ ,  $q(c_{n+1}) = p_n(c_{n+1}) + y_{n+1} - p_n(c_{n+1}) = y_{n+1}$ . By [Theorem 2.7.3](#),  $q = p_{n+1}$ .
- (d) With nodes  $0, 1, 2, 3$ :  $N_1 = x$ ,  $N_2 = x(x - 1)$ ,  $N_3 = x(x - 1)(x - 2)$ . Forward substitution:  $a_0 = p(0) = 1$ ;  $0 = p(1) = 1 + a_1$ , so  $a_1 = -1$ ;  $1 = p(2) = 1 - 2 + 2a_2$ , so  $a_2 = 1$ ;  $10 = p(3) = 1 - 3 + 6 + 6a_3$ , so  $a_3 = 1$ . Thus

$$p_3 = 1 - x + x(x - 1) + x(x - 1)(x - 2) = x^3 - 2x^2 + 1,$$

agreeing with [Exercise \(B1: A Lagrange interpolant\)](#). Adding  $(4, 0)$ :  $p_3(4) = 33$  and  $N_4(4) = 4 \cdot 3 \cdot 2 \cdot 1 = 24$ , so by (c)

$$\begin{aligned} p_4 &= x^3 - 2x^2 + 1 - \frac{33}{24}x(x - 1)(x - 2)(x - 3) \\ &= x^3 - 2x^2 + 1 - \frac{11}{8}x(x - 1)(x - 2)(x - 3). \end{aligned}$$

One checks  $p_4(4) = 33 - \frac{11}{8} \cdot 24 = 0$ . Adding a node changes **every** Lagrange polynomial, since each  $\ell_i$  acquires a new factor; in Newton’s form the old coefficients  $a_0, \dots, a_n$  survive and only one new coefficient is computed.

## 5.7 Polynomials of an Operator

Chapter 3 taught us to substitute an operator into a polynomial: for  $T \in \mathcal{L}(V)$  and  $p \in F[x]$ , the operator  $p(T)$  replaces each  $x^k$  by  $T^k$  ([Definition 3.4.2](#)). That section used the idea for a few examples. With the arithmetic of this chapter in hand, the idea becomes a method. Divisibility, ideals and Bézout’s identity in  $F[x]$  turn into statements about  $T$ : every operator on a finite-dimensional space satisfies a polynomial equation, the polynomials it satisfies form an ideal, and a factorization of such a polynomial into coprime pieces splits the space. The last statement, the kernel splitting lemma, is the engine that Chapter 8 will run to take operators apart.

### 5.7.1 Evaluating at an operator

Fix  $T \in \mathcal{L}(V)$ , where  $V$  is any vector space over  $F$ . [Theorem 3.4.3](#) already says that substituting  $T$  respects sums and products of polynomials, and that any two polynomials in  $T$  commute. The new point of view is to treat “substitute  $T$ ” as a single map from  $F[x]$  to  $\mathcal{L}(V)$ , and to ask what it preserves. Both sides are vector spaces, and both have a multiplication. The map respects all of the structure.

#### Theorem 5.7.1 — Evaluation at an Operator

Let  $V$  be a vector space over  $F$  and  $T \in \mathcal{L}(V)$ . Then the map

$$F[x] \rightarrow \mathcal{L}(V), \quad p \mapsto p(T),$$

has the following properties.

- (a) It is **linear**:  $(cp + q)(T) = cp(T) + q(T)$  for all  $p, q \in F[x]$  and  $c \in F$ .



- (b) It is **multiplicative** and sends 1 to  $\text{id}_V$ :  $(pq)(T) = p(T)q(T)$  and  $1(T) = \text{id}_V$ .
- (c) Its image  $\{p(T) : p \in F[x]\}$  is a subspace of  $\mathcal{L}(V)$ , closed under composition, in which any two operators commute.
- (d) If  $S \in \mathcal{L}(V)$  commutes with  $T$ , then  $S$  commutes with  $p(T)$  for every  $p \in F[x]$ .

*Proof.*

- (a) Let  $p = \sum_k a_k x^k$  and  $q = \sum_k b_k x^k$ . The coefficient of  $x^k$  in  $cp + q$  is  $ca_k + b_k$ , so by [Definition 3.4.2](#) and the vector space laws in  $\mathcal{L}(V)$  ([Theorem 3.4.1](#)),

$$(cp + q)(T) = \sum_k (ca_k + b_k)T^k = c \sum_k a_k T^k + \sum_k b_k T^k = cp(T) + q(T).$$

- (b) The product rule is [Theorem 3.4.3](#) (b). The constant polynomial 1 gives  $1 \cdot T^0 = \text{id}_V$ .
- (c) The image of a linear map is a subspace ([Theorem 3.2.2](#)). For polynomials  $p, q$ , the composite  $p(T)q(T)$  equals  $(pq)(T)$  by (b), so it lies in the image, and  $p(T)q(T) = q(T)p(T)$  by [Theorem 3.4.3](#) (c).
- (d) Suppose  $ST = TS$ . We show  $ST^k = T^k S$  by induction on  $k$ . For  $k = 0$ , both sides are  $S$ . If  $ST^k = T^k S$ , then by associativity ([Theorem 3.4.2](#))

$$ST^{k+1} = (ST^k)T = T^k(ST) = T^k(TS) = T^{k+1}S.$$

For  $p = \sum_k a_k x^k$ , distributivity ([Theorem 3.4.2](#)) gives  $Sp(T) = \sum_k a_k ST^k = \sum_k a_k T^k S = p(T)S$ .

■

Parts (a) and (b) together say that  $p \mapsto p(T)$  is a **homomorphism of algebras**: it translates every identity built from sums, scalar multiples and products of polynomials into the same identity for operators. (This map should not be confused with the evaluation map  $V \rightarrow V^{**}$  of Chapter 4; here we evaluate polynomials, not functionals.) Part (d) is what we will use most. It says that anything commuting with  $T$  commutes with the whole family of polynomials in  $T$ .

For matrices nothing new is needed. If  $V$  is finite-dimensional with basis  $\mathcal{B}$ , then  $[p(T)]_{\mathcal{B}} = p([T]_{\mathcal{B}})$  by [Corollary 3.6.3](#). So every statement in this section holds verbatim for  $\mathbf{A} \in M_n(F)$ , applied to the operator  $T_{\mathbf{A}}$ .

#### Example. (Two Ways to Evaluate)

Let  $\mathbf{A} = \begin{pmatrix} 3 & 1 \\ -1 & 1 \end{pmatrix} \in M_2(\mathbb{Q})$ . Compute  $p(\mathbf{A})$  for  $p = x^2 - 4x + 4$  and  $q = x^2 - 3x$ , once by expanding and once by factoring.

*Solution. Expanding.*  $\mathbf{A}^2 = \begin{pmatrix} 9-1 & 3+1 \\ -3-1 & -1+1 \end{pmatrix} = \begin{pmatrix} 8 & 4 \\ -4 & 0 \end{pmatrix}$ . Hence

$$p(\mathbf{A}) = \begin{pmatrix} 8-12+4 & 4-4 \\ -4+4 & 0-4+4 \end{pmatrix} = 0,$$

$$q(\mathbf{A}) = \begin{pmatrix} 8-9 & 4-3 \\ -4+3 & 0-3 \end{pmatrix} = \begin{pmatrix} -1 & 1 \\ -1 & -3 \end{pmatrix}.$$

*Factoring.*  $p = (x-2)^2$ , so  $p(\mathbf{A}) = (\mathbf{A} - 2\mathbf{I})^2$  by [Theorem 5.7.1](#) (b). Here  $\mathbf{A} - 2\mathbf{I} = \begin{pmatrix} 1 & 1 \\ -1 & -1 \end{pmatrix}$ , whose square is  $\begin{pmatrix} 1-1 & 1-1 \\ -1+1 & -1+1 \end{pmatrix} = 0$ . Likewise  $q = x(x-3)$ , so

$$q(\mathbf{A}) = \mathbf{A}(\mathbf{A} - 3\mathbf{I}) = \begin{pmatrix} 3 & 1 \\ -1 & 1 \end{pmatrix} \begin{pmatrix} 0 & 1 \\ -1 & -2 \end{pmatrix} = \begin{pmatrix} -1 & 1 \\ -1 & -3 \end{pmatrix},$$

in agreement with the expansion. The factored form showed at a glance why  $p(\mathbf{A}) = 0$ : the matrix  $\mathbf{A} - 2\mathbf{I}$  squares to zero.

*Warning.* **A product of polynomials in  $T$  can vanish without either factor vanishing.** In  $F[x]$ ,  $pq = 0$  forces  $p = 0$  or  $q = 0$  ([Corollary 0.8.4](#)). Evaluation at  $T$  does not carry this over. For the projection  $P(x, y) = (x, 0)$  on  $F^2$ , the polynomials  $p = x$  and  $q = x - 1$  give  $p(P)q(P) = P^2 - P = 0$ , while  $p(P) = P \neq 0$  and  $q(P) = P - \text{id} \neq 0$ . The map  $p \mapsto p(T)$  respects products, but it is not injective, and it is exactly its kernel that makes this possible.

## 5.7.2 Annihilating polynomials

In Chapter 3 we saw operators satisfying polynomial equations: a projection satisfies  $P^2 - P = 0$ , the rotation of  $\mathbb{R}^2$  by a right angle satisfies  $R^2 + \text{id} = 0$  ([Example \(Operators That Satisfy a Polynomial\)](#)), and differentiation on  $F[x]_{\leq n}$  satisfies  $D^{n+1} = 0$  ([Example \(Differentiation Is Nilpotent\)](#)). Each such equation is a non-zero polynomial in the kernel of  $p \mapsto p(T)$ . Does every operator have one? We need a word for these polynomials.

### Definition 5.7.1 — Annihilating Polynomial

Let  $T \in \mathcal{L}(V)$ . A polynomial  $p \in F[x]$  **annihilates**  $T$  if  $p(T) = 0$ , the zero operator on  $V$ .

For example,  $x^2 - x$  annihilates every projection,  $x^2 + 1$  annihilates the rotation  $R$ , and the zero polynomial annihilates every operator. On the zero space  $V = \{\mathbf{0}\}$ , even the constant 1 annihilates  $T$ , because  $1(T) = \text{id}_V$  is the zero map there. On  $V \neq \{\mathbf{0}\}$ , a non-zero constant  $c$  never annihilates, since  $c \text{id}_V \neq 0$ . (The word “annihilate” is also used for the annihilator  $U^0$  of Chapter 4; the two uses share the idea “sends to zero” and nothing more.)

In finite dimension a counting argument settles existence. The space  $\mathcal{L}(V)$  is finite-dimensional, so the powers of  $T$  cannot all be independent.

### Theorem 5.7.2 — Every Operator Satisfies a Polynomial

Let  $V$  be a finite-dimensional vector space over  $F$  with  $\dim V = n$ , and let  $T \in \mathcal{L}(V)$ . Then there is a **non-zero**  $p \in F[x]$  with  $\deg p \leq n^2$  and  $p(T) = 0$ . The same holds for every  $\mathbf{A} \in M_n(F)$ .

*Proof.* If  $n = 0$ , the constant 1 works, as noted above. Let  $n \geq 1$ . By Theorem 3.6.4,  $\dim \mathcal{L}(V) = n \cdot n = n^2$ . The list  $(\text{id}_V, T, T^2, \dots, T^{n^2})$  in  $\mathcal{L}(V)$  has length  $n^2 + 1 > n^2$ , so it is linearly dependent by Theorem 1.6.3 (a). Hence there are  $a_0, \dots, a_{n^2} \in F$ , **not all zero**, with  $a_0 \text{id}_V + a_1 T + \dots + a_{n^2} T^{n^2} = 0$ . The polynomial  $p = a_0 + a_1 x + \dots + a_{n^2} x^{n^2}$  is non-zero, has degree at most  $n^2$ , and satisfies  $p(T) = 0$  by Definition 3.4.2. For  $\mathbf{A} \in M_n(F)$ , apply the same argument in  $M_n(F)$ , which has dimension  $n^2$  (Example (Dimensions of the Running Examples)).

■

The bound  $n^2$  is crude. In Chapter 8, the Cayley–Hamilton Theorem will produce an annihilating polynomial of degree exactly  $n$ . The finite dimension is essential: on  $F[x]$ , let  $S$  be multiplication by  $x$ ,  $S(f) = xf$ . Then  $S^k(1) = x^k$ , so  $p(S)(1) = p$  for every  $p \in F[x]$ , and  $p(S) \neq 0$  whenever  $p \neq 0$ .

The proof is also an algorithm: search for the first power of  $\mathbf{A}$  that is a combination of the earlier ones. This is a question about linear dependence, which elimination answers (Theorem 2.5.10).

#### Example. (Finding an Annihilating Polynomial)

Find a non-zero polynomial of least possible degree that annihilates  $\mathbf{B} = \begin{pmatrix} 2 & 1 & 1 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{pmatrix} \in M_3(\mathbb{Q})$ .

*Solution.* *Degree 1.*  $\mathbf{B}$  is not a scalar multiple of  $\mathbf{I}_3$  (its  $(1, 2)$ -entry is non-zero), so  $(\mathbf{I}_3, \mathbf{B})$  is independent, and no non-zero polynomial of degree at most 1 annihilates  $\mathbf{B}$ .

*Degree 2.* Compute

$$\mathbf{B}^2 = \begin{pmatrix} 4 & 3 & 3 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{pmatrix}.$$

We look for  $a, b, c$  with  $a\mathbf{I}_3 + b\mathbf{B} + c\mathbf{B}^2 = 0$ . Reading entries: position  $(1, 2)$  gives  $b + 3c = 0$ ; position  $(1, 1)$  gives  $a + 2b + 4c = 0$ ; position  $(2, 2)$  gives  $a + b + c = 0$ . The positions  $(1, 3)$  and  $(3, 3)$  repeat the  $(1, 2)$  and  $(2, 2)$  equations, and all other entries are  $0 = 0$ . With  $c = 1$ :  $b = -3$  and  $a = 2$ , and indeed  $2 - 6 + 4 = 0$ . So

$$\mathbf{B}^2 - 3\mathbf{B} + 2\mathbf{I}_3 = 0,$$

and  $p = x^2 - 3x + 2 = (x-1)(x-2)$  annihilates  $\mathbf{B}$ . Check with the factored form:  $\mathbf{B} - \mathbf{I}_3 =$

$$\begin{pmatrix} 1 & 1 & 1 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{pmatrix} \text{ and } \mathbf{B} - 2\mathbf{I}_3 = \begin{pmatrix} 0 & 1 & 1 \\ 0 & -1 & 0 \\ 0 & 0 & -1 \end{pmatrix}, \text{ whose product is } \begin{pmatrix} 0 & 1-1 & 1-1 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{pmatrix} = 0.$$

The bound of [Theorem 5.7.2](#) allowed degree 9; the search stopped at degree 2.

Annihilating polynomials are far from unique: if  $p$  annihilates  $T$ , so does every multiple  $qp$ , since  $(qp)(T) = q(T)p(T) = q(T) \cdot 0 = 0$ . In [Example \(Finding an Annihilating Polynomial\)](#),  $x^3 - 3x^2 + 2x$  annihilates  $\mathbf{B}$  too. This closure under multiplication is the defining property of an ideal, and ideals of  $F[x]$  are completely understood ([Theorem 5.2.2](#)).

### Theorem 5.7.3 — The Annihilating Polynomials Form an Ideal

Let  $V$  be a vector space over  $F$  and  $T \in \mathcal{L}(V)$ , and let

$$I_T := \{p \in F[x] : p(T) = 0\}.$$

- (a)  $I_T$  is an ideal of  $F[x]$ .
- (b) Either  $I_T = \{0\}$ , or there is a **unique monic**  $m \in F[x]$  with  $I_T = \langle m \rangle$ . In the second case, for every  $p \in F[x]$ ,

$$p(T) = 0 \iff m \mid p,$$

and  $m$  has the least degree among the non-zero polynomials annihilating  $T$ .

- (c) If  $V$  is finite-dimensional, then  $I_T \neq \{0\}$ , and  $\deg m \leq (\dim V)^2$ .

*Proof.*

- (a) By [Theorem 5.7.1](#) (a),  $I_T$  is the kernel of a linear map, so it is a subspace ([Theorem 3.2.1](#)); in particular it contains 0 and is closed under addition. If  $p \in I_T$  and  $q \in F[x]$ , then  $(qp)(T) = q(T)p(T) = 0$  by [Theorem 5.7.1](#) (b), so  $qp \in I_T$ . By [Definition 5.2.1](#),  $I_T$  is an ideal.
- (b) By [Theorem 5.2.2](#), either  $I_T = \{0\}$ , or  $I_T = \langle m \rangle$  for a unique monic  $m$ , which has least degree among the non-zero elements of  $I_T$ . In the second case,  $p(T) = 0$  means  $p \in \langle m \rangle$ , that is,  $p = qm$  for some  $q$ , that is,  $m \mid p$ .
- (c) By [Theorem 5.7.2](#),  $I_T$  contains a non-zero polynomial of degree at most  $(\dim V)^2$ . So  $I_T \neq \{0\}$ , and  $\deg m \leq (\dim V)^2$  because  $m$  has the least degree. ■

This monic generator is one of the most important invariants of an operator. **Chapter 8 names it the minimal polynomial of  $T$**  and relates its roots to eigenvalues; here it is simply “the monic generator of  $I_T$ ”. Part (b) turns every polynomial identity satisfied by  $T$  into a divisibility statement about one fixed polynomial.

#### Examples.

- **Projections.** Let  $P$  be a projection with  $P \neq 0$  and  $P \neq \text{id}_V$ . Then  $x^2 - x = x(x-1) \in I_P$ , so the monic generator divides  $x(x-1)$ . By unique factorization ([Theorem 5.3.5](#)), its monic divisors are 1,  $x$ ,  $x-1$  and  $x(x-1)$ . The first three do not annihilate  $P$ :  $1(P) = \text{id}_V \neq 0$ ,  $x(P) = P \neq 0$  and  $(x-1)(P) = P - \text{id}_V \neq 0$ . So the generator is  $x^2 - x$ .

- **Degenerate cases.** For the identity on  $V \neq \{0\}$ , the generator is  $x - 1$ ; for the zero operator on  $V \neq \{0\}$ , it is  $x$ ; on  $V = \{0\}$ , it is 1.
- **The matrix  $\mathbf{B}$ .** In [Example \(Finding an Annihilating Polynomial\)](#), no non-zero polynomial of degree at most 1 annihilates  $\mathbf{B}$ , and  $x^2 - 3x + 2$  does, so the generator is  $x^2 - 3x + 2$ .
- **Infinite dimension.** For multiplication by  $x$  on  $F[x]$ , we saw that  $I_S = \{0\}$ .

*Quick check.* Let  $R(x, y) = (-y, x)$  be the rotation of  $\mathbb{R}^2$  by a right angle. What is the monic generator of  $I_R \subseteq \mathbb{R}[x]$ ?

*Solution.*  $R^2 = -\text{id}$ , so  $x^2 + 1 \in I_R$ , and the generator divides  $x^2 + 1$ . Over  $\mathbb{R}$ ,  $x^2 + 1$  has no root, so its only monic divisors are 1 and  $x^2 + 1$  (a monic divisor of degree 1 would be  $x - c$  with  $c$  a root). Since  $1(R) = \text{id} \neq 0$ , the generator is  $x^2 + 1$ .

### 5.7.3 Kernels of polynomials in $T$

The subspaces we can build from  $T$  and a polynomial  $p$  are  $\ker p(T)$  and  $\text{im } p(T)$ . For  $p = x - \lambda$ , the kernel  $\ker(T - \lambda \text{id}_V)$  consists of the vectors that  $T$  merely scales by  $\lambda$ ; for a projection,  $\ker P$  and  $\ker(P - \text{id}_V)$  are the two pieces of the direct sum. These subspaces have a property that general subspaces lack:  $T$  does not move vectors out of them.

#### Theorem 5.7.4 — $T$ Preserves Kernels and Images of Polynomials in $T$ (a)

Let  $R, S \in \mathcal{L}(V)$  with  $SR = RS$ . Then  $S\mathbf{v} \in \ker R$  for every  $\mathbf{v} \in \ker R$ , and  $S\mathbf{w} \in \text{im } R$  for every  $\mathbf{w} \in \text{im } R$ .

- (b) In particular, for  $T \in \mathcal{L}(V)$  and  $p, q \in F[x]$ , the operator  $q(T)$  maps  $\ker p(T)$  into  $\ker p(T)$  and  $\text{im } p(T)$  into  $\text{im } p(T)$ . Taking  $q = x$ :  $T$  maps  $\ker p(T)$  and  $\text{im } p(T)$  into themselves.

*Proof.*

- (a) Let  $\mathbf{v} \in \ker R$ . Then  $R(S\mathbf{v}) = S(R\mathbf{v}) = S\mathbf{0} = \mathbf{0}$ , so  $S\mathbf{v} \in \ker R$ . Let  $\mathbf{w} = R\mathbf{u} \in \text{im } R$ . Then  $S\mathbf{w} = S(R\mathbf{u}) = R(S\mathbf{u}) \in \text{im } R$ .
- (b) By [Theorem 5.7.1](#) (c),  $q(T)$  commutes with  $p(T)$ , so (a) applies with  $R = p(T)$  and  $S = q(T)$ . For  $q = x$ ,  $q(T) = T$ .

■

In Chapter 8, a subspace that an operator maps into itself will be called an invariant subspace, and the notion will be developed properly there. For now the theorem is what matters: whenever  $U = \ker p(T)$ , the rule  $\mathbf{u} \mapsto T\mathbf{u}$  is an operator on  $U$ , and so is  $\mathbf{u} \mapsto q(T)\mathbf{u}$ .

The inclusion can fail for subspaces not built from  $T$ . For the rotation  $R$  of the check above,  $R$  moves the  $x$ -axis to the  $y$ -axis, so the  $x$ -axis is not of the form  $\ker p(R)$  or  $\text{im } p(R)$  for any  $p$ .

### 5.7.4 The kernel splitting lemma

Chapter 3 contains two theorems with the same shape. A projection satisfies  $P(P - \text{id}_V) = 0$ , and [Theorem 3.10.1](#) splits  $V$  as  $\text{im } P \oplus \ker P$ . An involution in characteristic not 2 satisfies

$(T - \text{id}_V)(T + \text{id}_V) = 0$ , and [Theorem 3.10.5](#) splits  $V$  as  $\ker(T - \text{id}_V) \oplus \ker(T + \text{id}_V)$ . In both, a polynomial annihilating  $T$  factors into two pieces with no common factor, and  $V$  is the direct sum of the kernels of the pieces. In both, the projections turned out to be polynomials in  $T$ :  $P$  itself, and  $\frac{1}{2}(\text{id}_V + T)$ . The general statement needs only the arithmetic of  $F[x]$ .

### Theorem 5.7.5 — Kernel Splitting Lemma

Let  $V$  be a vector space over  $F$ , let  $T \in \mathcal{L}(V)$ , and let  $p, q \in F[x]$  be **coprime**. Choose  $a, b \in F[x]$  with  $ap + bq = 1$  ([Corollary 5.2.4](#)), and put

$$W = \ker(pq)(T), \quad E_1 = b(T)q(T), \quad E_2 = a(T)p(T).$$

Then:

- (a)  $W = \ker p(T) \oplus \ker q(T)$ ;
- (b)  $E_1 + E_2 = \text{id}_V$ , and for every  $\mathbf{w} \in W$ , the splitting of  $\mathbf{w}$  in (a) is

$$\mathbf{w} = E_1\mathbf{w} + E_2\mathbf{w}, \quad E_1\mathbf{w} \in \ker p(T), \quad E_2\mathbf{w} \in \ker q(T);$$

so  $E_1$  maps  $W$  into  $W$  and acts on  $W$  as the projection onto  $\ker p(T)$  along  $\ker q(T)$ , and  $E_2$  acts on  $W$  as the projection onto  $\ker q(T)$  along  $\ker p(T)$ ;

- (c) in particular, if  $(pq)(T) = 0$ , then

$$V = \ker p(T) \oplus \ker q(T),$$

and the projections of  $V$  onto the two summands along each other are the polynomials  $(bq)(T)$  and  $(ap)(T)$  in  $T$ .

*Idea.* Substitute  $T$  into Bézout's identity:  $\text{id}_V = a(T)p(T) + b(T)q(T)$ . Applied to a vector  $\mathbf{w}$ , this writes  $\mathbf{w} = E_1\mathbf{w} + E_2\mathbf{w}$ , and it does so for **every** vector of  $V$ . Now look at the pieces. ①  $p(T)$  kills  $E_1\mathbf{w} = b(T)q(T)\mathbf{w}$ , because  $p(T)b(T)q(T) = b(T)(pq)(T)$  and  $(pq)(T)\mathbf{w} = \mathbf{0}$  for  $\mathbf{w} \in W$ : the factor  $q$  in  $E_1$  combines with  $p$  to the whole product. Symmetrically  $q(T)$  kills  $E_2\mathbf{w}$ . ② If  $\mathbf{w}$  is killed by both  $p(T)$  and  $q(T)$ , then both pieces are  $\mathbf{0}$ , so  $\mathbf{w} = \mathbf{0}$ . ③ Direct sum by the intersection criterion; the projections are read off from uniqueness. Coprimality enters only through the identity  $ap + bq = 1$ , and without it step ② has nothing to stand on.

*Proof.* Since  $ap + bq = 1$ , [Theorem 5.7.1](#) gives

$$\text{id}_V = a(T)p(T) + b(T)q(T) = E_2 + E_1.$$

Throughout, polynomials in  $T$  commute with each other ([Theorem 5.7.1](#) (c)).

*The kernels lie in  $W$ .* If  $p(T)\mathbf{v} = \mathbf{0}$ , then  $(pq)(T)\mathbf{v} = q(T)p(T)\mathbf{v} = \mathbf{0}$ ; so  $\ker p(T) \subseteq W$ , and likewise  $\ker q(T) \subseteq W$ .

*The pieces.* Let  $\mathbf{w} \in W$ . Then  $\mathbf{w} = \text{id}_V \mathbf{w} = E_1\mathbf{w} + E_2\mathbf{w}$ . Moreover

$$p(T)(E_1\mathbf{w}) = p(T)b(T)q(T)\mathbf{w} = b(T)(pq)(T)\mathbf{w} = b(T)\mathbf{0} = \mathbf{0},$$

so  $E_1\mathbf{w} \in \ker p(T)$ , and similarly  $q(T)(E_2\mathbf{w}) = a(T)(pq)(T)\mathbf{w} = \mathbf{0}$ , so  $E_2\mathbf{w} \in \ker q(T)$ . Hence  $W = \ker p(T) + \ker q(T)$ .

*The intersection.* Let  $\mathbf{v} \in \ker p(T) \cap \ker q(T)$ . Then  $E_1\mathbf{v} = b(T)q(T)\mathbf{v} = \mathbf{0}$  and  $E_2\mathbf{v} = a(T)p(T)\mathbf{v} = \mathbf{0}$ , so  $\mathbf{v} = E_1\mathbf{v} + E_2\mathbf{v} = \mathbf{0}$ . By [Theorem 1.7.4](#), the sum is direct, which proves (a).

*The projections.* The splitting of  $\mathbf{w} \in W$  in (a) is unique ([Definition 1.7.2](#)), and we have exhibited it as  $\mathbf{w} = E_1\mathbf{w} + E_2\mathbf{w}$ ; this proves the formula in (b). In particular  $E_1\mathbf{w} \in \ker p(T) \subseteq W$ , so  $E_1$  maps  $W$  into  $W$ . As an operator on  $W$ ,  $E_1$  is linear, fixes every  $\mathbf{u} \in \ker p(T)$  (whose splitting is  $\mathbf{u} = \mathbf{u} + \mathbf{0}$ ) and kills every vector of  $\ker q(T)$  (whose splitting is  $\mathbf{0} + \mathbf{v}$ ). By the uniqueness in [Theorem 3.10.1](#) (b), applied to  $W = \ker p(T) \oplus \ker q(T)$ , it is the projection onto  $\ker p(T)$  along  $\ker q(T)$ . Swapping the roles of  $p$  and  $q$  gives the statement for  $E_2$ .

- (c) If  $(pq)(T) = 0$ , then  $W = V$ , and (a) and (b) become the statements in (c), with  $E_1 = (bq)(T)$  and  $E_2 = (ap)(T)$  by [Theorem 5.7.1](#) (b). This proves the theorem. ■

Bézout's coefficients  $a, b$  are not unique, but the projections on  $W$  are, by [Theorem 3.10.1](#) (b). So different choices of  $a, b$  give polynomials  $bq$  that may differ, while  $(bq)(T)$  always does the same thing on  $W$ . When  $(pq)(T) = 0$ , the lemma turns a factorization of an annihilating polynomial into a direct sum decomposition of the whole space, with explicit projections, and each summand is carried into itself by  $T$  ([Theorem 5.7.4](#)).

### Example. (Recovering Projections and Direct Sums)

Let  $P \in \mathcal{L}(V)$  be a projection. Apply [Theorem 5.7.5](#) with  $p = x$  and  $q = x - 1$ , and compare with [Theorem 3.10.1](#) (a).

*Solution. Coprime.*  $1 \cdot x + (-1)(x - 1) = 1$ , so we may take  $a = 1$ ,  $b = -1$ , and any common divisor of  $x$  and  $x - 1$  divides 1. *Annihilating.*  $(pq)(P) = P^2 - P = 0$  because  $P$  is a projection.

By [Theorem 5.7.5](#) (c),

$$V = \ker P \oplus \ker(P - \text{id}_V),$$

with projection  $E_1 = b(P)q(P) = -(P - \text{id}_V) = \text{id}_V - P$  onto  $\ker P$  along  $\ker(P - \text{id}_V)$ , and  $E_2 = a(P)p(P) = P$  onto  $\ker(P - \text{id}_V)$  along  $\ker P$ .

To compare, note that  $\ker(P - \text{id}_V) = \text{im } P$ . ( $\subseteq$ ) If  $P\mathbf{v} = \mathbf{v}$ , then  $\mathbf{v} \in \text{im } P$ . ( $\supseteq$ ) If  $\mathbf{v} = P\mathbf{u}$ , then  $P\mathbf{v} = P^2\mathbf{u} = P\mathbf{u} = \mathbf{v}$ . So the lemma gives  $V = \text{im } P \oplus \ker P$ , with  $P$  the projection onto  $\text{im } P$  along  $\ker P$ : this is [Theorem 3.10.1](#) (a). It also reproves [Exercise \(B2: The complementary projection\)](#), that  $\text{id}_V - P$  is the complementary projection.

### Example. (Recovering the Involution Decomposition)

Let  $F$  have characteristic not 2, and let  $T \in \mathcal{L}(V)$  with  $T^2 = \text{id}_V$ . Apply [Theorem 5.7.5](#) with  $p = x - 1$  and  $q = x + 1$ , and compare with [Theorem 3.10.5](#). What goes wrong in characteristic 2?

*Solution. Coprime.* Since  $2 \neq 0$  in  $F$  ([Definition 0.7.2](#)),

$$-\frac{1}{2}(x - 1) + \frac{1}{2}(x + 1) = 1,$$

so  $a = -\frac{1}{2}$  and  $b = \frac{1}{2}$  work, and a common divisor of  $p$  and  $q$  divides 1. *Annihilating.*  $(pq)(T) = T^2 - \text{id}_V = 0$ .

By [Theorem 5.7.5](#) (c),

$$V = \ker(T - \text{id}_V) \oplus \ker(T + \text{id}_V) = E_+ \oplus E_-,$$

and the projection onto  $E_+$  along  $E_-$  is  $b(T)q(T) = \frac{1}{2}(T + \text{id}_V)$ . These are exactly the conclusions of [Theorem 3.10.5](#). The other projection is  $a(T)p(T) = \frac{1}{2}(\text{id}_V - T)$ .

In characteristic 2,  $x + 1 = x - 1$ , so  $p = q$  and  $\gcd(p, q) = x - 1 \neq 1$ . The lemma does not apply, and the warning after [Theorem 3.10.5](#) shows that the conclusion really fails there.

The lemma applies just as well when neither factor is linear, and then it finds subspaces that are not visible from single vectors.

### Example. (A Splitting with a Repeated Factor)

Let  $\mathbf{A} = \begin{pmatrix} 0 & 1 & -1 \\ 0 & 0 & 1 \\ 0 & 0 & 1 \end{pmatrix} \in M_3(\mathbb{Q})$ . Verify that  $\mathbf{A}^2(\mathbf{A} - \mathbf{I}_3) = 0$ , and use [Theorem 5.7.5](#) with  $p = x^2$ ,  $q = x - 1$  to split  $\mathbb{Q}^3$ , giving both projections as polynomials in  $\mathbf{A}$ .

*Solution. The relation.*  $\mathbf{A}^2 = \begin{pmatrix} 0 & 0 & 0 \\ 0 & 0 & 1 \\ 0 & 0 & 1 \end{pmatrix}$ , and  $\mathbf{A}^3 = \mathbf{A} \cdot \mathbf{A}^2 = \begin{pmatrix} 0 & 0 & 0 \\ 0 & 0 & 1 \\ 0 & 0 & 1 \end{pmatrix} = \mathbf{A}^2$ . So  $\mathbf{A}^2(\mathbf{A} - \mathbf{I}_3) = \mathbf{A}^3 - \mathbf{A}^2 = 0$ .

*Bézout.*  $x^2 = (x+1)(x-1) + 1$ , so  $1 \cdot x^2 + (-(x+1))(x-1) = 1$ : take  $a = 1$ ,  $b = -(x+1)$ . Then

$$\begin{aligned} \mathbf{E}_1 &= b(\mathbf{A})q(\mathbf{A}) = -(\mathbf{A} + \mathbf{I}_3)(\mathbf{A} - \mathbf{I}_3) = \mathbf{I}_3 - \mathbf{A}^2 \\ &= \begin{pmatrix} 1 & 0 & 0 \\ 0 & 1 & -1 \\ 0 & 0 & 0 \end{pmatrix}, \\ \mathbf{E}_2 &= a(\mathbf{A})p(\mathbf{A}) = \mathbf{A}^2. \end{aligned}$$

*The summands.*  $\ker \mathbf{A}^2 = \{(x, y, z) : z = 0\} = \text{span}(\mathbf{e}_1, \mathbf{e}_2)$ , and  $\ker(\mathbf{A} - \mathbf{I}_3)$  is cut out by  $-x + y - z = 0$ ,  $-y + z = 0$ ,  $0 = 0$ , so it is  $\text{span}((0, 1, 1))$ . By [Theorem 5.7.5](#) (c),

$$\mathbb{Q}^3 = \text{span}(\mathbf{e}_1, \mathbf{e}_2) \oplus \text{span}((0, 1, 1)),$$

and for  $\mathbf{v} = (x, y, z)$ , the splitting is  $\mathbf{E}_1\mathbf{v} + \mathbf{E}_2\mathbf{v} = (x, y - z, 0) + (0, z, z)$ . One checks directly that  $(x, y - z, 0)$  has third entry 0 and  $(0, z, z)$  is a multiple of  $(0, 1, 1)$ .

Note that  $\ker \mathbf{A} = \text{span}(\mathbf{e}_1)$  is strictly smaller than  $\ker \mathbf{A}^2$ . The factor  $p = x^2$  could not be replaced by  $x$ : the polynomial  $x(x - 1)$  does not annihilate  $\mathbf{A}$ , since  $\mathbf{A}^2 - \mathbf{A} \neq 0$ .

**Warning. Without coprimality the kernels need not fill  $\ker(pq)(T)$ , or even form a direct sum.** Let  $T = T_{\mathbf{N}}$  with  $\mathbf{N} = \begin{pmatrix} 0 & 1 \\ 0 & 0 \end{pmatrix}$  on  $F^2$ , and  $p = q = x$ . Then  $(pq)(T) = \mathbf{N}^2 = 0$ , so  $\ker(pq)(T) = F^2$ . But  $\ker p(T) = \ker q(T) = \ker \mathbf{N} = \text{span}(\mathbf{e}_1)$ . The two kernels are equal, so their sum is not direct, and  $\text{span}(\mathbf{e}_1) + \text{span}(\mathbf{e}_1) = \text{span}(\mathbf{e}_1) \neq F^2$ .



Here  $\gcd(x, x) = x$ , and no identity  $ax + bx = 1$  exists, since its left side vanishes at 0.

*Quick check.* Let  $T \in \mathcal{L}(V)$  satisfy  $T^2 = 3T$ , over a field with  $3 \neq 0$ . Split  $V$  with the lemma, and write the projection onto  $\ker T$  as a polynomial in  $T$ .

*Solution.*  $x(x - 3)$  annihilates  $T$ . Since  $\frac{1}{3}x - \frac{1}{3}(x - 3) = 1$ ,  $p = x$  and  $q = x - 3$  are coprime with  $a = \frac{1}{3}$ ,  $b = -\frac{1}{3}$ . By [Theorem 5.7.5](#) (c),  $V = \ker T \oplus \ker(T - 3\text{id}_V)$ , and the projection onto  $\ker T$  along  $\ker(T - 3\text{id}_V)$  is  $b(T)q(T) = -\frac{1}{3}(T - 3\text{id}_V) = \text{id}_V - \frac{1}{3}T$ .

In Chapter 8 the lemma is used with an annihilating polynomial factored into powers of distinct irreducibles,  $m = p_1^{e_1} \cdots p_k^{e_k}$ . Distinct monic irreducibles are coprime, and so are their powers, so the lemma with  $k$  factors ([Exercise C2: Splitting with  \$k\$  coprime factors](#)) splits  $V$  into the pieces  $\ker p_i(T)^{e_i}$ . That is the primary decomposition, and the canonical forms of Chapter 9 are built on it.

## 5.7.5 Exercises

### 5.7.5.1 A. Check your understanding

#### Exercise A1.

- State what it means that  $p \mapsto p(T)$  is linear and multiplicative.
- True or false: if  $p(T)q(T) = 0$ , then  $p(T) = 0$  or  $q(T) = 0$ . Justify your answer.
- Why does every operator on a finite-dimensional space have a non-zero annihilating polynomial? Is the same true on  $F[x]$ ?
- True or false: if  $p$  annihilates  $T$  and  $p \mid q$ , then  $q$  annihilates  $T$ . Justify your answer.
- State the kernel splitting lemma for  $(pq)(T) = 0$ , and say where coprimality is used in its proof.
- True or false: for every subspace  $U$  of  $V$  and every  $T \in \mathcal{L}(V)$ ,  $T$  maps  $U$  into  $U$ . Justify your answer.

*Solution.*

- $(cp + q)(T) = cp(T) + q(T)$  and  $(pq)(T) = p(T)q(T)$  for all  $p, q \in F[x]$  and  $c \in F$  ([Theorem 5.7.1](#)).
- False. For the projection  $P(x, y) = (x, 0)$ ,  $P(P - \text{id}) = 0$  while  $P \neq 0$  and  $P - \text{id} \neq 0$ .
- $\mathcal{L}(V)$  has dimension  $n^2$ , so the  $n^2 + 1$  operators  $\text{id}_V, T, \dots, T^{n^2}$  are linearly dependent, and a dependence is a non-zero annihilating polynomial ([Theorem 5.7.2](#)). On  $F[x]$ , multiplication by  $x$  has no non-zero annihilating polynomial, since  $p(S)(1) = p$ .
- True. If  $q = hp$ , then  $q(T) = h(T)p(T) = 0$ . This is the ideal property ([Theorem 5.7.3](#)).
- If  $p, q$  are coprime and  $(pq)(T) = 0$ , then  $V = \ker p(T) \oplus \ker q(T)$ , and the projections are polynomials in  $T$  ([Theorem 5.7.5](#)). Coprimality gives  $ap + bq = 1$ , hence

$\text{id}_V = a(T)p(T) + b(T)q(T)$ , which both produces the splitting of each vector and shows the intersection is  $\{\mathbf{0}\}$ .

- (f) False. The rotation  $R(x, y) = (-y, x)$  maps the  $x$ -axis to the  $y$ -axis. The statement is true for  $U = \ker p(T)$  and  $U = \text{im } p(T)$  (Theorem 5.7.4).

### 5.7.5.2 B. Practice

#### Exercise B1: A polynomial of a $2 \times 2$ matrix.

Let  $\mathbf{A} = \begin{pmatrix} 1 & 2 \\ 3 & -1 \end{pmatrix} \in M_2(\mathbb{Q})$ .

- (a) Compute  $p(\mathbf{A})$  for  $p = x^2 - 7$  and for  $q = x^2 + x - 5$ .  
 (b) Hence show that  $\mathbf{A}$  is invertible and find  $\mathbf{A}^{-1}$  as a polynomial in  $\mathbf{A}$ .

*Solution.*

- (a)  $\mathbf{A}^2 = \begin{pmatrix} 1+6 & 2-2 \\ 3-3 & 6+1 \end{pmatrix} = 7\mathbf{I}_2$ . Hence  $p(\mathbf{A}) = \mathbf{A}^2 - 7\mathbf{I}_2 = 0$ . For  $q$ , divide:  $q = (x^2 - 7) + (x + 2)$ , so by Theorem 5.7.1  $q(\mathbf{A}) = p(\mathbf{A}) + \mathbf{A} + 2\mathbf{I}_2 = \begin{pmatrix} 3 & 2 \\ 3 & 1 \end{pmatrix}$ . (Directly:  $7\mathbf{I}_2 + \mathbf{A} - 5\mathbf{I}_2 = \mathbf{A} + 2\mathbf{I}_2$ .)  
 (b) From  $\mathbf{A}^2 = 7\mathbf{I}_2$ ,  $\mathbf{A}(\frac{1}{7}\mathbf{A}) = (\frac{1}{7}\mathbf{A})\mathbf{A} = \mathbf{I}_2$ . So  $\mathbf{A}$  is invertible with  $\mathbf{A}^{-1} = \frac{1}{7}\mathbf{A} = \begin{pmatrix} \frac{1}{7} & \frac{2}{7} \\ \frac{3}{7} & -\frac{1}{7} \end{pmatrix}$ .

#### Exercise B2: An annihilating polynomial by elimination.

Let  $\mathbf{C} = \begin{pmatrix} 0 & 0 & 2 \\ 1 & 0 & -1 \\ 0 & 1 & 2 \end{pmatrix} \in M_3(\mathbb{Q})$ .

- (a) Compute  $\mathbf{C}^2$  and  $\mathbf{C}^3$ , and show that  $(\mathbf{I}_3, \mathbf{C}, \mathbf{C}^2)$  is linearly independent.  
 (b) Find a linear dependence among  $\mathbf{I}_3, \mathbf{C}, \mathbf{C}^2, \mathbf{C}^3$ , and hence a monic polynomial  $p$  of degree 3 with  $p(\mathbf{C}) = 0$ .  
 (c) Hence show that  $\mathbf{C}$  is invertible and write  $\mathbf{C}^{-1}$  as a polynomial in  $\mathbf{C}$ .

*Solution.*

- (a) Multiplying out,

$$\mathbf{C}^2 = \begin{pmatrix} 0 & 2 & 4 \\ 0 & -1 & 0 \\ 1 & 2 & 3 \end{pmatrix}, \quad \mathbf{C}^3 = \mathbf{C} \cdot \mathbf{C}^2 = \begin{pmatrix} 2 & 4 & 6 \\ -1 & 0 & 1 \\ 2 & 3 & 6 \end{pmatrix}.$$

Let  $a\mathbf{I}_3 + b\mathbf{C} + c\mathbf{C}^2 = 0$ . The  $(2, 1)$ -entry gives  $b = 0$ ; the  $(3, 1)$ -entry gives  $c = 0$ ; then the  $(1, 1)$ -entry gives  $a = 0$ . So the list is independent.

- (b) Let  $a\mathbf{I}_3 + b\mathbf{C} + c\mathbf{C}^2 + d\mathbf{C}^3 = 0$ . Reading entries:  $(2, 1)$ :  $b - d = 0$ ;  $(3, 1)$ :  $c + 2d = 0$ ;  $(1, 1)$ :  $a + 2d = 0$ . With  $d = 1$ :  $b = 1$ ,  $c = -2$ ,  $a = -2$ . Check all entries

of  $-2\mathbf{I}_3 + \mathbf{C} - 2\mathbf{C}^2 + \mathbf{C}^3$ : row 1 is  $(-2 + 0 - 0 + 2, 0 - 4 + 4, 2 - 8 + 6) = (0, 0, 0)$ ; row 2 is  $(1 - 0 - 1, -2 + 0 + 2 + 0, -1 - 0 + 1) = (0, 0, 0)$ ; row 3 is  $(0 - 2 + 2, 1 - 4 + 3, -2 + 2 - 6 + 6) = (0, 0, 0)$ . Hence

$$p = x^3 - 2x^2 + x - 2 = (x - 2)(x^2 + 1)$$

satisfies  $p(\mathbf{C}) = 0$ . By (a) no non-zero polynomial of degree at most 2 annihilates  $\mathbf{C}$ , so  $p$  is the monic generator of  $I_{\mathbf{C}}$  (Theorem 5.7.3).

- (c)  $\mathbf{C}^3 - 2\mathbf{C}^2 + \mathbf{C} = 2\mathbf{I}_3$ , that is,  $\mathbf{C}(\mathbf{C}^2 - 2\mathbf{C} + \mathbf{I}_3) = (\mathbf{C}^2 - 2\mathbf{C} + \mathbf{I}_3)\mathbf{C} = 2\mathbf{I}_3$ , using Theorem 5.7.1 (c) for the second equality. Since  $2 \neq 0$  in  $\mathbb{Q}$ ,  $\mathbf{C}$  is invertible with

$$\mathbf{C}^{-1} = \frac{1}{2}(\mathbf{C}^2 - 2\mathbf{C} + \mathbf{I}_3) = \begin{pmatrix} \frac{1}{2} & 1 & 0 \\ -1 & 0 & 1 \\ \frac{1}{2} & 0 & 0 \end{pmatrix}.$$

### Exercise B3: Operators with $T^3 = T$ .

Let  $F$  have characteristic not 2, and let  $T \in \mathcal{L}(V)$  with  $T^3 = T$ .

- Use Theorem 5.7.5 with  $p = x$  and  $q = x^2 - 1$  to show  $V = \ker T \oplus \ker(T^2 - \text{id}_V)$ , with projection  $\text{id}_V - T^2$  onto  $\ker T$ .
- Show that  $V = \ker T \oplus \ker(T - \text{id}_V) \oplus \ker(T + \text{id}_V)$ .
- Verify that the projections onto the three summands in (b) are  $\text{id}_V - T^2$ ,  $\frac{1}{2}(T^2 + T)$  and  $\frac{1}{2}(T^2 - T)$ , in the sense that these three operators sum to  $\text{id}_V$  and each maps  $V$  into the corresponding kernel.

*Solution.*

- $x \cdot x + (-1)(x^2 - 1) = 1$ : take  $a = x$ ,  $b = -1$ . Then  $p$  and  $q$  are coprime, and  $(pq)(T) = T^3 - T = 0$ . By Theorem 5.7.5 (c),  $V = \ker T \oplus \ker(T^2 - \text{id}_V)$ , and the projection onto  $\ker T$  is  $b(T)q(T) = -(T^2 - \text{id}_V) = \text{id}_V - T^2$ .
- Let  $W = \ker(T^2 - \text{id}_V)$ . Apply Theorem 5.7.5 (a) with  $p_1 = x - 1$ ,  $q_1 = x + 1$ , which are coprime as in Example (Recovering the Involution Decomposition):  $W = \ker(p_1 q_1)(T) = \ker(T - \text{id}_V) \oplus \ker(T + \text{id}_V)$ . Write  $K_0 = \ker T$ ,  $K_+ = \ker(T - \text{id}_V)$ ,  $K_- = \ker(T + \text{id}_V)$ . Then  $V = K_0 + K_+ + K_-$  by (a). If  $\mathbf{u}_0 + \mathbf{u}_+ + \mathbf{u}_- = \mathbf{0}$  with  $\mathbf{u}_0 \in K_0$ ,  $\mathbf{u}_{\pm} \in K_{\pm}$ , then  $\mathbf{u}_+ + \mathbf{u}_- \in W$ , so directness in (a) gives  $\mathbf{u}_0 = \mathbf{0}$  and  $\mathbf{u}_+ + \mathbf{u}_- = \mathbf{0}$ , and directness of  $W = K_+ \oplus K_-$  gives  $\mathbf{u}_+ = \mathbf{u}_- = \mathbf{0}$ . By Theorem 1.7.5 ((b)  $\Rightarrow$  (a)), the sum is direct.
- The sum is  $\text{id}_V - T^2 + \frac{1}{2}T^2 + \frac{1}{2}T + \frac{1}{2}T^2 - \frac{1}{2}T = \text{id}_V$ . Using  $T^3 = T$ :  $T(\text{id}_V - T^2) = T - T^3 = 0$ ;  $(T - \text{id}_V)\frac{1}{2}(T^2 + T) = \frac{1}{2}(T^3 - T) = 0$ ;  $(T + \text{id}_V)\frac{1}{2}(T^2 - T) = \frac{1}{2}(T^3 - T) = 0$ . So the three operators map  $V$  into  $K_0$ ,  $K_+$ ,  $K_-$  respectively. For  $\mathbf{v} \in V$ ,  $\mathbf{v}$  is the sum of its three images, and by the uniqueness of the splitting in (b) these are the components of  $\mathbf{v}$ . Hence each operator is the projection onto its summand along the sum of the other two.

#### 5.7.5.3 C. Going deeper

**Exercise C1: Inverses are polynomials.**

Let  $V$  be a vector space over  $F$  and  $T \in \mathcal{L}(V)$ .

- (a) Suppose  $p(T) = 0$  for some  $p \in F[x]$  with  $p(0) \neq 0$ . Prove that  $T$  is invertible and that  $T^{-1} = g(T)$  for some  $g \in F[x]$ .
- (b) Deduce that if  $V$  is finite-dimensional and  $T$  is invertible, then  $T^{-1}$  is a polynomial in  $T$ .
- (c) Show that the monic generator  $m$  of  $I_T$ , for  $V$  finite-dimensional and non-zero, satisfies  $m(0) \neq 0$  if and only if  $T$  is invertible.

*Hint: for (b), write a non-zero annihilating polynomial as  $x^k p_1$  with  $p_1(0) \neq 0$ .*

*Solution.*

- (a) Write  $p = c + xg_0$ , where  $c = p(0) \neq 0$  and  $g_0 \in F[x]$ . Then  $0 = p(T) = c \operatorname{id}_V + Tg_0(T)$ , so  $Tg_0(T) = -c \operatorname{id}_V$ . Put  $g = -c^{-1}g_0$ . Then  $Tg(T) = \operatorname{id}_V$ , and  $g(T)T = Tg(T) = \operatorname{id}_V$  by [Theorem 5.7.1](#) (c). Hence  $T$  is invertible with  $T^{-1} = g(T)$ .
- (b) If  $V = \{0\}$ , then  $T^{-1} = \operatorname{id}_V = 1(T)$ . Otherwise, by [Theorem 5.7.2](#) there is a non-zero  $p$  with  $p(T) = 0$ . Let  $k$  be the least index with a non-zero coefficient in  $p$ ; then  $p = x^k p_1$  with  $p_1(0) \neq 0$ . Now  $0 = p(T) = T^k p_1(T)$ . Since  $T$  is invertible, so is  $T^k$ , with inverse  $(T^{-1})^k$ , and composing on the left with it gives  $p_1(T) = 0$ . By (a),  $T^{-1}$  is a polynomial in  $T$ .
- (c) ( $\Rightarrow$ ) If  $m(0) \neq 0$ , apply (a) to  $m$ . ( $\Leftarrow$ ) Suppose  $T$  is invertible and  $m(0) = 0$ . Then  $m = xm_1$  with  $m_1$  monic of degree  $\deg m - 1$ , and  $0 = m(T) = Tm_1(T)$ . Composing with  $T^{-1}$  gives  $m_1(T) = 0$ , so  $m \mid m_1$  by [Theorem 5.7.3](#) (b). Since  $V \neq \{0\}$ ,  $m$  is not constant, so  $m_1 \neq 0$  and  $\deg m \leq \deg m_1 = \deg m - 1$  by [Theorem 0.8.3](#), a contradiction. Hence  $m(0) \neq 0$ .

**Exercise C2: Splitting with  $k$  coprime factors.**

Let  $T \in \mathcal{L}(V)$ , and let  $p_1, \dots, p_k \in F[x]$  be non-zero and pairwise coprime, with  $k \geq 2$ . Put  $p = p_1 \cdots p_k$  and  $W = \ker p(T)$ .

- (a) Use [Theorem 5.6.4](#) to find  $e_1, \dots, e_k \in F[x]$  with  $e_i \equiv 1 \pmod{p_i}$  and  $e_i \equiv 0 \pmod{p_j}$  for  $j \neq i$ . Show that  $p \mid e_1 + \cdots + e_k - 1$  and that  $p \mid p_i e_i$  for each  $i$ .
- (b) Prove that for every  $\mathbf{w} \in W$ ,  $\mathbf{w} = e_1(T)\mathbf{w} + \cdots + e_k(T)\mathbf{w}$  with  $e_i(T)\mathbf{w} \in \ker p_i(T)$ .
- (c) Prove that  $W = \ker p_1(T) \oplus \cdots \oplus \ker p_k(T)$ , and that  $e_i(T)$  is the identity on  $\ker p_i(T)$  and zero on  $\ker p_j(T)$  for  $j \neq i$ .
- (d) Deduce: if  $p(T) = 0$ , then  $V = \ker p_1(T) \oplus \cdots \oplus \ker p_k(T)$ , and the projection onto the  $i$ -th summand along the others is the polynomial  $e_i(T)$ .

*Hint: for the second divisibility in (a), use [Lemma 5.6.3](#) (b).*

*Solution.*

- (a) By [Theorem 5.6.4](#) (a), applied to the pairwise coprime moduli  $p_1, \dots, p_k$  with  $r_i = 1$  and  $r_j = 0$  for  $j \neq i$ , such an  $e_i$  exists. The sum  $s = e_1 + \cdots + e_k$  satisfies  $s \equiv 1$

(mod  $p_j$ ) for each  $j$ , since exactly one summand is  $\equiv 1$  and the others are  $\equiv 0$ . The constant 1 satisfies the same congruences, so  $s \equiv 1 \pmod{p}$  by [Theorem 5.6.4](#) (b). For the second claim,  $p_j \mid e_i$  for every  $j \neq i$ , and these  $p_j$  are pairwise coprime, so  $\prod_{j \neq i} p_j \mid e_i$  by [Lemma 5.6.3](#) (b). Multiplying by  $p_i$ ,  $p \mid p_i e_i$ .

- (b) Write  $e_1 + \cdots + e_k = 1 + hp$  and  $p_i e_i = h_i p$ . By [Theorem 5.7.1](#), for  $\mathbf{w} \in W$ ,

$$\begin{aligned} e_1(T)\mathbf{w} + \cdots + e_k(T)\mathbf{w} &= \mathbf{w} + h(T)p(T)\mathbf{w} = \mathbf{w}, \\ p_i(T)(e_i(T)\mathbf{w}) &= h_i(T)p(T)\mathbf{w} = \mathbf{0}. \end{aligned}$$

- (c) Each  $\ker p_i(T) \subseteq W$ , since  $p(T) = (\prod_{j \neq i} p_j)(T)p_i(T)$ . With (b),  $W = \ker p_1(T) + \cdots + \ker p_k(T)$ . Let  $\mathbf{u}_j \in \ker p_j(T)$ . For  $j \neq i$ ,  $p_j \mid e_i$ , say  $e_i = g p_j$ , so  $e_i(T)\mathbf{u}_j = g(T)p_j(T)\mathbf{u}_j = \mathbf{0}$ . And  $e_i = 1 + g'p_i$ , so  $e_i(T)\mathbf{u}_i = \mathbf{u}_i + g'(T)p_i(T)\mathbf{u}_i = \mathbf{u}_i$ . This proves the second statement. Now suppose  $\mathbf{u}_1 + \cdots + \mathbf{u}_k = \mathbf{0}$  with  $\mathbf{u}_j \in \ker p_j(T)$ . Applying  $e_i(T)$  gives  $\mathbf{u}_i = \mathbf{0}$ , for each  $i$ . By [Theorem 1.7.5](#) ((b)  $\Rightarrow$  (a)), the sum is direct.
- (d) If  $p(T) = 0$ , then  $W = V$ . By (c),  $V$  is the direct sum, and  $e_i(T) \in \mathcal{L}(V)$  is the identity on the  $i$ -th summand and zero on the others. For  $\mathbf{v} = \mathbf{u}_1 + \cdots + \mathbf{u}_k$ ,  $e_i(T)\mathbf{v} = \mathbf{u}_i$ , which is the projection onto  $\ker p_i(T)$  along the sum of the other summands. For  $k = 2$  this is [Theorem 5.7.5](#) (c).

### Exercise C3: The same rotation over $\mathbb{R}$ and over $\mathbb{C}$ .

Let  $\mathbf{A} = \begin{pmatrix} 0 & -1 \\ 1 & 0 \end{pmatrix}$ , so that  $\mathbf{A}^2 + \mathbf{I}_2 = \mathbf{0}$ .

- (a) Over  $\mathbb{C}$ , apply [Theorem 5.7.5](#) to  $T_{\mathbf{A}}$  on  $\mathbb{C}^2$  with  $p = x - i$  and  $q = x + i$ . Find both kernels and write the projection onto  $\ker(\mathbf{A} - i\mathbf{I}_2)$  as a polynomial in  $\mathbf{A}$ .
- (b) Over  $\mathbb{R}$ , show that the only subspaces of  $\mathbb{R}^2$  that  $T_{\mathbf{A}}$  maps into themselves are  $\{\mathbf{0}\}$  and  $\mathbb{R}^2$ . Explain why this matches the fact that  $x^2 + 1$  admits no factorization into coprime non-constant factors in  $\mathbb{R}[x]$ .

*Solution.*

- (a)  $(x + i) - (x - i) = 2i$ , so  $\frac{i}{2}(x - i) - \frac{i}{2}(x + i) = \frac{i}{2}(-2i) = 1$ : take  $a = \frac{i}{2}$  and  $b = -\frac{i}{2}$ . So  $p, q$  are coprime, and  $(pq)(\mathbf{A}) = \mathbf{A}^2 + \mathbf{I}_2 = \mathbf{0}$ . By [Theorem 5.7.5](#) (c),  $\mathbb{C}^2 = \ker(\mathbf{A} - i\mathbf{I}_2) \oplus \ker(\mathbf{A} + i\mathbf{I}_2)$ . Solving,  $(\mathbf{A} - i\mathbf{I}_2)(z, w) = (-iz - w, z - iw) = \mathbf{0}$  gives  $w = -iz$ , so  $\ker(\mathbf{A} - i\mathbf{I}_2) = \text{span}((1, -i))$ ; similarly  $\ker(\mathbf{A} + i\mathbf{I}_2) = \text{span}((1, i))$ . The projection onto  $\ker(\mathbf{A} - i\mathbf{I}_2)$  is

$$b(\mathbf{A})q(\mathbf{A}) = -\frac{i}{2}(\mathbf{A} + i\mathbf{I}_2) = \frac{1}{2}(\mathbf{I}_2 - i\mathbf{A}) = \begin{pmatrix} \frac{1}{2} & \frac{i}{2} \\ -\frac{i}{2} & \frac{1}{2} \end{pmatrix}.$$

Check: it sends  $(1, -i)$  to  $(\frac{1}{2} + \frac{1}{2}, -\frac{i}{2} - \frac{i}{2}) = (1, -i)$  and  $(1, i)$  to  $(\frac{1}{2} - \frac{1}{2}, -\frac{i}{2} + \frac{i}{2}) = \mathbf{0}$ .

- (b) A subspace of  $\mathbb{R}^2$  other than  $\{\mathbf{0}\}$  and  $\mathbb{R}^2$  has dimension 1 ([Theorem 1.6.7](#)), so it is  $\text{span}(\mathbf{v})$  with  $\mathbf{v} = (s, t) \neq \mathbf{0}$ . If  $\mathbf{A}\mathbf{v} \in \text{span}(\mathbf{v})$ , then  $(-t, s) = c(s, t)$  for some  $c \in \mathbb{R}$ , so  $-t = cs$  and  $s = ct$ , giving  $s = -c^2s$  and  $t = -c^2t$ . Since  $1 + c^2 > 0$ ,  $s = t = 0$ , a contradiction. So only  $\{\mathbf{0}\}$  and  $\mathbb{R}^2$  are mapped into themselves.

Over  $\mathbb{R}$ ,  $x^2 + 1$  has no root, so its only monic divisors are 1 and  $x^2 + 1$ , and the only factorizations  $x^2 + 1 = pq$  into monic factors have  $p = 1$  or  $q = 1$ . The lemma then gives

only  $\ker p(T) \oplus \ker q(T) = \{\mathbf{0}\} \oplus \mathbb{R}^2$ , and by (b) there is no finer splitting into subspaces preserved by  $T_{\mathbf{A}}$  at all. Enlarging the field to  $\mathbb{C}$  creates the coprime factors  $x \mp i$ , and with them the splitting of (a).

# Chapter 6

## Determinants

A square matrix is invertible or it is not, and so far deciding which has meant row reducing it. The determinant packages that decision into a single number. The same number measures how the linear map scales areas and volumes, and its sign records whether the map preserves or reverses orientation. It also produces the characteristic polynomial, the gateway to eigenvalues in Chapter 8.

This chapter does not start from a formula. It starts from what a signed volume must do. Scaling one edge scales the volume. Sliding one edge along another does not change it. The unit cube has volume 1. We prove that exactly one function has these properties, write it down, and then derive every property of the determinant from that uniqueness: multiplicativity, the transpose rule, cofactor expansion, the inverse formula and Cramer's rule. Nothing is assumed, and no formula is used before it is proved.

**What you need.** From Chapter 0, permutations and transpositions ([Theorem 0.10.4](#)) and the characteristic of a field ([Definition 0.7.2](#)). From Chapter 1, linear independence and the Linear Dependence Lemma ([Definition 1.4.1](#), [Theorem 1.4.1](#)). From Chapter 2, elementary matrices and the invertible matrix theorem ([Definition 2.4.1](#), [Theorem 2.4.4](#)). From Chapter 3, the matrix of a linear map and similarity ([Definition 3.6.1](#), [Theorem 3.7.4](#)). From Chapter 4, linear functionals ([Definition 4.1.1](#)). From Chapter 5, the root bound for polynomials ([Corollary 5.1.4](#)).

### 6.0.1 Roadmap

- **Signed area and volume.** Three rules that any reasonable notion of signed volume obeys, and the formula  $ad - bc$  forced by them.
- **Multilinear alternating forms.** The rules made precise for functions of  $n$  vectors, and their first consequences.
- **Permutations and sign.** Counting inversions, and why the parity of a permutation is well defined.
- **Existence and uniqueness of the determinant.** Exactly one alternating form takes the value 1 on the identity; the Leibniz formula.
- **Row operations and multiplicativity.** Computing determinants by elimination,  $\det(\mathbf{AB}) = \det \mathbf{A} \det \mathbf{B}$ , and the determinant of an operator.
- **Cofactor expansion, the adjugate and Cramer's rule.** Expanding along a row or column, and a formula for the inverse.
- **Minors, rank and Cauchy–Binet.** Rank read off from minors, and the determinant of

a product of rectangular matrices.

- **Special determinants.** Vandermonde, tridiagonal and block triangular matrices, and rank-one updates.
- **Orientation and volume.** The determinant as a volume scaling factor, and the two orientations of  $\mathbb{R}^n$ .
- **The characteristic polynomial.**  $\det(x\mathbf{I} - \mathbf{A})$ , its coefficients, and why similar matrices share it.

## 6.0.2 Named moves

- **Derive from the rules, not the formula.** Prove determinant facts from multilinearity, alternation and normalization, so no property is assumed before it is proved.
- **Uniqueness does the work.** To prove two expressions agree, show both are alternating multilinear in the columns and agree on the identity.
- **Build from elementary matrices.** Check a claim for elementary matrices, then extend it to products.
- **Look-alike matrices.** A matrix with two equal rows has determinant zero; expanding such a matrix proves the adjugate identity.
- **Induct on size.** Expand along a row or column to reduce an  $n \times n$  determinant to smaller ones.

# 6.1 Signed Area and Volume

In Chapter 0 a single number,  $ad - bc$ , decided whether a  $2 \times 2$  matrix is invertible ([Theorem 0.9.8](#)). For an  $n \times n$  matrix, Chapter 2 answered the same question by row reduction, which is a procedure, not a number. Is there one number for every square matrix that does the job of  $ad - bc$ ? This section does not define it yet. Instead it asks what such a number should **mean**, finds the answer in area and volume, and extracts a short list of rules. From the rules alone,  $ad - bc$  comes out, with no other choice possible.

## 6.1.1 Area as a function of two vectors

Take two vectors  $\mathbf{u}, \mathbf{v} \in \mathbb{R}^2$  and the parallelogram they span,

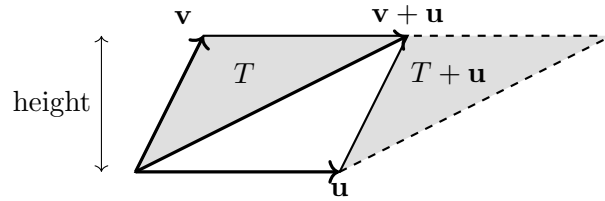
$$P(\mathbf{u}, \mathbf{v}) = \{s\mathbf{u} + t\mathbf{v} : 0 \leq s \leq 1, 0 \leq t \leq 1\}.$$

Its corners are  $\mathbf{0}$ ,  $\mathbf{u}$ ,  $\mathbf{v}$  and  $\mathbf{u} + \mathbf{v}$ . We want to understand its area as a **function of the two vectors**, without drawing a coordinate grid and counting squares. Four features are visible without any formula.

**Scaling a side scales the area.** Replacing  $\mathbf{v}$  by  $2\mathbf{v}$  stacks two copies of  $P(\mathbf{u}, \mathbf{v})$  on top of each other, so the area doubles. Replacing  $\mathbf{v}$  by  $3\mathbf{v}$  triples it, and replacing  $\mathbf{v}$  by  $\frac{1}{2}\mathbf{v}$  halves it.

**Shearing does not change the area.** Replace  $\mathbf{v}$  by  $\mathbf{v} + c\mathbf{u}$ . The top edge slides along its own line, parallel to the base  $\mathbf{u}$ . The base stays the same, and so does the height, so the area stays the same. The picture shows  $c = 1$ : the triangle cut off on the left of  $P(\mathbf{u}, \mathbf{v})$  reappears, moved by  $\mathbf{u}$ , on the right of  $P(\mathbf{u}, \mathbf{v} + \mathbf{u})$ .





The solid outline is  $P(\mathbf{u}, \mathbf{v})$  and the dashed one completes  $P(\mathbf{u}, \mathbf{v} + \mathbf{u})$ . Both consist of the unshaded middle triangle plus a shaded triangle, and the two shaded triangles  $T$  and  $T + \mathbf{u}$  are translates of each other.

**Equal sides give area zero.**  $P(\mathbf{u}, \mathbf{u})$  is a segment, which has no area. More generally, if  $\mathbf{v}$  is a multiple of  $\mathbf{u}$ , the parallelogram is flat.

**The unit square has area one.**  $P(\mathbf{e}_1, \mathbf{e}_2)$  is the unit square. This fixes the unit of measurement.

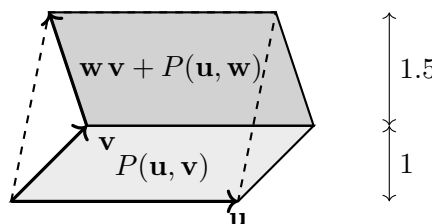
## 6.1.2 Why the area needs a sign

The first feature is not quite right as stated. Replacing  $\mathbf{v}$  by  $-\mathbf{v}$  moves the parallelogram to the other side of the line of  $\mathbf{u}$  (the point reflection  $\mathbf{x} \mapsto \mathbf{u} - \mathbf{x}$  carries  $P(\mathbf{u}, \mathbf{v})$  onto  $P(\mathbf{u}, -\mathbf{v})$ ) and does **not** change its area, so ordinary area satisfies  $\text{area}(\mathbf{u}, c\mathbf{v}) = |c| \text{area}(\mathbf{u}, \mathbf{v})$ , with an absolute value. That absolute value spoils a second, very useful property.

Fix the base  $\mathbf{u}$  and let the second vector vary. The area is base times height, and heights along a fixed direction add up: if  $\mathbf{v}$  reaches height 1 above the line of  $\mathbf{u}$  and  $\mathbf{w}$  reaches height 1.5, then  $\mathbf{v} + \mathbf{w}$  reaches height 2.5. So we expect

$$\text{area}(\mathbf{u}, \mathbf{v} + \mathbf{w}) = \text{area}(\mathbf{u}, \mathbf{v}) + \text{area}(\mathbf{u}, \mathbf{w}).$$

The picture shows it:  $P(\mathbf{u}, \mathbf{w})$ , moved up by  $\mathbf{v}$ , sits exactly on top of  $P(\mathbf{u}, \mathbf{v})$ , and the two together have the same base and total height as  $P(\mathbf{u}, \mathbf{v} + \mathbf{w})$  (dashed).



Now take  $\mathbf{w} = -\mathbf{v}$ . Then  $\mathbf{v} + \mathbf{w} = \mathbf{0}$  and the left side is  $\text{area}(\mathbf{u}, \mathbf{0}) = 0$ , while the right side is  $2 \text{area}(\mathbf{u}, \mathbf{v})$ , which is positive. Additivity fails, because  $-\mathbf{v}$  points **below** the line of  $\mathbf{u}$  and its height should count as negative. The repair is to give area a sign.

*Signed area is ordinary area, counted positive when the second vector lies on the counterclockwise side of the first, and negative when it lies on the clockwise side.*

So  $P(\mathbf{e}_1, \mathbf{e}_2)$  has signed area  $+1$ , because  $\mathbf{e}_2$  is a quarter turn counterclockwise from  $\mathbf{e}_1$ , and  $P(\mathbf{e}_2, \mathbf{e}_1)$  has signed area  $-1$ . With the sign in place, scaling holds with  $c$  instead of  $|c|$ , and heights add with their signs, so additivity holds for all  $\mathbf{v}, \mathbf{w}$ .

Write  $D(\mathbf{u}, \mathbf{v})$  for the signed area. The pictures suggest that  $D: \mathbb{R}^2 \times \mathbb{R}^2 \rightarrow \mathbb{R}$  obeys the following rules, **for all**  $\mathbf{u}, \mathbf{v}, \mathbf{w} \in \mathbb{R}^2$  and **all**  $c \in \mathbb{R}$ :

- (R1) (**Scaling**)  $D(c\mathbf{u}, \mathbf{v}) = cD(\mathbf{u}, \mathbf{v})$  and  $D(\mathbf{u}, c\mathbf{v}) = cD(\mathbf{u}, \mathbf{v})$ .
- (R2) (**Shear**)  $D(\mathbf{u} + c\mathbf{v}, \mathbf{v}) = D(\mathbf{u}, \mathbf{v})$  and  $D(\mathbf{u}, \mathbf{v} + c\mathbf{u}) = D(\mathbf{u}, \mathbf{v})$ .
- (R3) (**Equal sides**)  $D(\mathbf{u}, \mathbf{u}) = 0$ .
- (R4) (**Normalization**)  $D(\mathbf{e}_1, \mathbf{e}_2) = 1$ .
- (R5) (**Additivity**)  $D(\mathbf{u} + \mathbf{w}, \mathbf{v}) = D(\mathbf{u}, \mathbf{v}) + D(\mathbf{w}, \mathbf{v})$  and  $D(\mathbf{u}, \mathbf{v} + \mathbf{w}) = D(\mathbf{u}, \mathbf{v}) + D(\mathbf{u}, \mathbf{w})$ .

These are rules read off from pictures, not yet a definition. The question is what they force.

### 6.1.3 The rules force $ad - bc$

A  $2 \times 2$  matrix  $\mathbf{A} = \begin{pmatrix} a & b \\ c & d \end{pmatrix}$  has columns  $\mathbf{u} = (a, c)$  and  $\mathbf{v} = (b, d)$ , and  $P(\mathbf{u}, \mathbf{v})$  is the image of the unit square under  $\mathbf{x} \mapsto \mathbf{A}\mathbf{x}$ . We now show that the first four rules leave no freedom at all. The method is the column version of elimination: use shears to clear entries until the parallelogram becomes a rectangle.

#### Example. ( $ad - bc$ from the Rules)

Let  $D: \mathbb{R}^2 \times \mathbb{R}^2 \rightarrow \mathbb{R}$  be **any** function satisfying (R1)–(R4). Show that  $D((a, c), (b, d)) = ad - bc$  for all  $a, b, c, d \in \mathbb{R}$ .

*Solution.* Write  $\mathbf{u} = (a, c)$  and  $\mathbf{v} = (b, d)$ . Two consequences of (R1) come first. Taking the scalar 0 in (R1) gives  $D(\mathbf{u}, \mathbf{0}) = D(\mathbf{u}, 0\mathbf{v}) = 0 \cdot D(\mathbf{u}, \mathbf{v}) = 0$ , and likewise  $D(\mathbf{0}, \mathbf{v}) = 0$ . Using (R1) twice and then (R4),

$$\begin{aligned} D((p, 0), (0, q)) &= D(p\mathbf{e}_1, q\mathbf{e}_2) = pq D(\mathbf{e}_1, \mathbf{e}_2) \\ &= pq \quad \text{for all } p, q \in \mathbb{R}. \end{aligned} \tag{*}$$

*Case 1:*  $a \neq 0$ . By the second shear rule with scalar  $-b/a$ ,

$$D(\mathbf{u}, \mathbf{v}) = D\left(\mathbf{u}, \mathbf{v} - \frac{b}{a}\mathbf{u}\right) = D(\mathbf{u}, (0, d')), \quad \text{where } d' = d - \frac{bc}{a}.$$

If  $d' = 0$ , this is  $D(\mathbf{u}, \mathbf{0}) = 0 = ad'$ . If  $d' \neq 0$ , the first shear rule with scalar  $-c/d'$  clears the second entry of  $\mathbf{u}$ :

$$D(\mathbf{u}, (0, d')) = D\left(\mathbf{u} - \frac{c}{d'}(0, d'), (0, d')\right) = D((a, 0), (0, d')) = ad'$$

by (\*). In both sub-cases  $D(\mathbf{u}, \mathbf{v}) = ad' = ad - bc$ .

*Case 2:*  $a = 0$  and  $b \neq 0$ . By the first shear rule with scalar 1,  $D(\mathbf{u}, \mathbf{v}) = D(\mathbf{u} + \mathbf{v}, \mathbf{v})$ , and  $\mathbf{u} + \mathbf{v} = (b, c + d)$  has first entry  $b \neq 0$ . Case 1, applied to the columns  $(b, c + d)$  and  $(b, d)$ , gives

$$D(\mathbf{u}, \mathbf{v}) = bd - b(c + d) = -bc = ad - bc,$$

since  $a = 0$ .

*Case 3:*  $a = b = 0$ . If  $c = 0$ , then  $\mathbf{u} = \mathbf{0}$  and  $D(\mathbf{u}, \mathbf{v}) = 0$ . If  $c \neq 0$ , the second shear rule with scalar  $-d/c$  gives  $D(\mathbf{u}, \mathbf{v}) = D(\mathbf{u}, \mathbf{v} - \frac{d}{c}\mathbf{u}) = D(\mathbf{u}, \mathbf{0}) = 0$ . In both sub-cases  $D(\mathbf{u}, \mathbf{v}) = 0 = ad - bc$ .

The three cases cover all  $a, b, c, d$ , so  $D((a, c), (b, d)) = ad - bc$ , as claimed.

So **at most one** function obeys (R1)–(R4). Conversely,  $(a, c), (b, d) \mapsto ad - bc$  obeys all five rules; for instance, for the second shear rule, replacing  $(b, d)$  by  $(b + ta, d + tc)$  gives  $a(d + tc) - (b + ta)c = ad - bc$ . The other checks are just as short. Hence **exactly one** function satisfies (R1)–(R4), and it satisfies (R5) as well.

Two things in this computation are worth noticing. First, (R3) was never used. It follows from the others:  $D(\mathbf{u}, \mathbf{u}) = D(\mathbf{u}, \mathbf{u} - \mathbf{u}) = D(\mathbf{u}, \mathbf{0}) = 0$ , by a shear and then scaling by 0. Second, the argument used only addition, subtraction, multiplication and division by non-zero numbers. So over **any** field  $F$ , exactly one function  $F^2 \times F^2 \rightarrow F$  satisfies (R1)–(R4), namely  $ad - bc$ . The picture needs  $\mathbb{R}$ ; the algebra does not.

There is also a shorter route, if we allow additivity. Expanding  $\mathbf{u} = a\mathbf{e}_1 + c\mathbf{e}_2$  and  $\mathbf{v} = b\mathbf{e}_1 + d\mathbf{e}_2$  with (R5) and (R1) gives four terms:

$$D(\mathbf{u}, \mathbf{v}) = ab D(\mathbf{e}_1, \mathbf{e}_1) + ad D(\mathbf{e}_1, \mathbf{e}_2) + cb D(\mathbf{e}_2, \mathbf{e}_1) + cd D(\mathbf{e}_2, \mathbf{e}_2).$$

The first and last vanish by (R3),  $D(\mathbf{e}_1, \mathbf{e}_2) = 1$  by (R4), and  $D(\mathbf{e}_2, \mathbf{e}_1) = -1$  (see the Quick check below). This expansion is the route that generalizes to  $n$  vectors later in the chapter, while the elimination route of [Example \( \$ad - bc\$  from the Rules\)](#) becomes the practical way to compute.

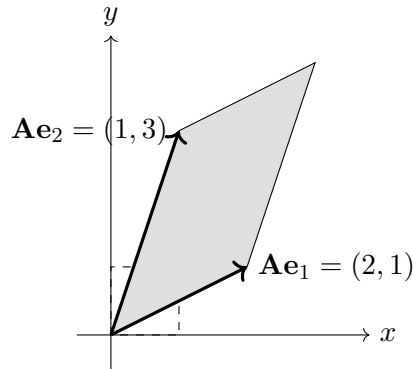
*Quick check.* Using only (R1) and (R2), show that  $D(\mathbf{u}, \mathbf{v}) = -D(\mathbf{v}, \mathbf{u})$  for all  $\mathbf{u}, \mathbf{v}$ . Deduce  $D(\mathbf{e}_2, \mathbf{e}_1)$ .

*Solution.* Apply shears and one scaling:

$$\begin{aligned} D(\mathbf{u}, \mathbf{v}) &\stackrel{(R2)}{=} D(\mathbf{u} + \mathbf{v}, \mathbf{v}) \\ &\stackrel{(R2)}{=} D(\mathbf{u} + \mathbf{v}, \mathbf{v} - (\mathbf{u} + \mathbf{v})) = D(\mathbf{u} + \mathbf{v}, -\mathbf{u}) \\ &\stackrel{(R2)}{=} D((\mathbf{u} + \mathbf{v}) + (-\mathbf{u}), -\mathbf{u}) = D(\mathbf{v}, -\mathbf{u}) \\ &\stackrel{(R1)}{=} -D(\mathbf{v}, \mathbf{u}). \end{aligned}$$

With  $\mathbf{u} = \mathbf{e}_1$ ,  $\mathbf{v} = \mathbf{e}_2$  and (R4),  $D(\mathbf{e}_2, \mathbf{e}_1) = -D(\mathbf{e}_1, \mathbf{e}_2) = -1$ . This agrees with  $ad - bc = 0 \cdot 0 - 1 \cdot 1$  for the columns  $(0, 1), (1, 0)$ .

Now compare with Chapter 0. By [Theorem 0.9.8](#),  $\mathbf{A}$  is invertible **if and only if**  $ad - bc \neq 0$ . In the language of this section:  $\mathbf{A}$  is invertible exactly when the parallelogram spanned by its columns is **not flat**, that is, when the columns do not lie on a common line through the origin. The number that decides invertibility and the signed area of the image of the unit square are the same number. Drag the images of  $\mathbf{e}_1$  and  $\mathbf{e}_2$  below and watch the signed area pass through 0 exactly when the two columns line up.



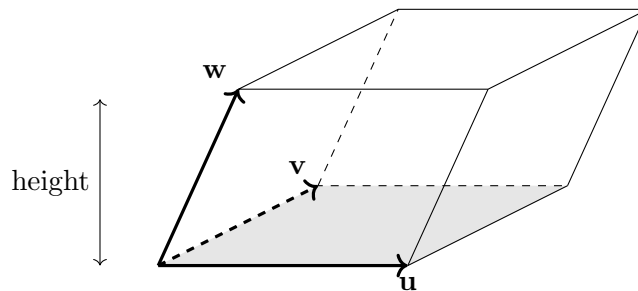
The unit square (dashed) and its image under  $\mathbf{A} = \begin{pmatrix} 2 & 1 \\ 1 & 3 \end{pmatrix}$ , a parallelogram of signed area  $2 \cdot 3 - 1 \cdot 1 = 5$ . The second column lies counterclockwise from the first, so the sign is positive.

### 6.1.4 Volume in three dimensions

The same story plays out one dimension up. Three vectors  $\mathbf{u}, \mathbf{v}, \mathbf{w} \in \mathbb{R}^3$  span a **parallelepiped**

$$P(\mathbf{u}, \mathbf{v}, \mathbf{w}) = \{r\mathbf{u} + s\mathbf{v} + t\mathbf{w} : 0 \leq r, s, t \leq 1\},$$

a slanted box whose faces are parallelograms.



Its volume is the area of the base  $P(\mathbf{u}, \mathbf{v})$  (shaded) times the height of  $\mathbf{w}$  above the plane of the base. Each feature from the plane has a counterpart:

- scaling one edge scales the volume;
- replacing  $\mathbf{w}$  by  $\mathbf{w} + a\mathbf{u} + b\mathbf{v}$  slides the top face within its own plane, so the base and height, hence the volume, are unchanged, and similarly for shears of  $\mathbf{u}$  or  $\mathbf{v}$ ;
- if two edges are equal, the box is flat and the volume is 0;
- the unit cube  $P(\mathbf{e}_1, \mathbf{e}_2, \mathbf{e}_3)$  has volume 1;
- with a sign attached, heights add, so the signed volume is additive in each edge.

The sign in three dimensions records whether  $(\mathbf{u}, \mathbf{v}, \mathbf{w})$  is arranged like  $(\mathbf{e}_1, \mathbf{e}_2, \mathbf{e}_3)$ , the right-hand rule. We make this precise as orientation later in the chapter. For now the point is that the rules are **the same rules**, now for a function  $D(\mathbf{u}, \mathbf{v}, \mathbf{w})$  of three vectors. The Quick check above also carries over word for word: shearing and scaling two of the arguments while the third stays fixed shows that swapping two arguments changes the sign.

## 6.1.5 What we will prove

Here is the goal for the next three sections. Let  $F$  be a field and  $n \geq 1$ .

There is **exactly one** function  $D$  of  $n$  vectors in  $F^n$  that is linear in each argument separately, is 0 whenever two arguments are equal, and takes the value 1 at  $(\mathbf{e}_1, \dots, \mathbf{e}_n)$ .

Its value on the columns of a matrix  $\mathbf{A}$  will be the determinant  $\det \mathbf{A}$ . The plan follows the shorter route for  $n = 2$ . Expanding every argument in the standard basis gives  $n^n$  terms. The rule for equal arguments kills every term with a repeated basis vector, which leaves one term for each rearrangement  $(\mathbf{e}_{\sigma(1)}, \dots, \mathbf{e}_{\sigma(n)})$  of the standard basis. Each surviving value is  $\pm 1$ , reached from  $D(\mathbf{e}_1, \dots, \mathbf{e}_n) = 1$  by swaps, with one sign change per swap. That creates two tasks:

1. Name and study functions of several vectors with these rules. This is the next section, on multilinear alternating forms.
2. Make sure the sign of a rearrangement is well defined, even though it can be reached by different sequences of swaps. This is the section on permutations and sign.

The existence and uniqueness theorem itself comes after both.

*Warning. Sarrus's rule is a  $3 \times 3$  coincidence.* For a  $3 \times 3$  matrix, the “diagonals” shortcut (add the three products along the down-right diagonals of the array with its first two columns copied to the right, subtract the three along the up-right diagonals) gives the correct six terms, because there are exactly  $3! = 6$  rearrangements. The same picture for  $4 \times 4$  gives  $4 + 4 = 8$  products, while the expansion above has  $4! = 24$  terms. It fails already for the matrix with columns  $\mathbf{e}_2, \mathbf{e}_1, \mathbf{e}_3, \mathbf{e}_4$ : every one of the eight wrapped diagonals passes through a zero entry, so the shortcut returns 0, but the swap rule gives  $D(\mathbf{e}_2, \mathbf{e}_1, \mathbf{e}_3, \mathbf{e}_4) = -D(\mathbf{e}_1, \mathbf{e}_2, \mathbf{e}_3, \mathbf{e}_4) = -1$ . Never use Sarrus's rule beyond  $3 \times 3$ , and never use it as a proof tool.

## 6.1.6 Exercises

### 6.1.6.1 A. Check your understanding

#### Exercise A1.

- (a) State the four rules (R1)–(R4) for signed area.
- (b) True or false: ordinary (unsigned) area satisfies  $\text{area}(\mathbf{u}, \mathbf{v} + \mathbf{w}) = \text{area}(\mathbf{u}, \mathbf{v}) + \text{area}(\mathbf{u}, \mathbf{w})$  for all  $\mathbf{u}, \mathbf{v}, \mathbf{w} \in \mathbb{R}^2$ . Justify your answer.
- (c) Find the signed area of  $P((1, 3), (2, 4))$ . Is  $\begin{pmatrix} 1 & 2 \\ 3 & 4 \end{pmatrix}$  invertible?
- (d) Explain why any  $D$  satisfying (R1) and (R2) has  $D(\mathbf{u}, 5\mathbf{u}) = 0$ .
- (e) How many terms does the full expansion for  $4 \times 4$  matrices have, and how many products does the  $4 \times 4$  analogue of Sarrus's rule use?

*Solution.*

- (a) For all  $\mathbf{u}, \mathbf{v} \in \mathbb{R}^2$  and  $c \in \mathbb{R}$ :  $D(c\mathbf{u}, \mathbf{v}) = cD(\mathbf{u}, \mathbf{v}) = D(\mathbf{u}, c\mathbf{v})$ ;  $D(\mathbf{u} + c\mathbf{v}, \mathbf{v}) = D(\mathbf{u}, \mathbf{v}) = D(\mathbf{u}, \mathbf{v} + c\mathbf{u})$ ;  $D(\mathbf{u}, \mathbf{u}) = 0$ ;  $D(\mathbf{e}_1, \mathbf{e}_2) = 1$ .
- (b) False. Take  $\mathbf{u} = \mathbf{e}_1$ ,  $\mathbf{v} = \mathbf{e}_2$ ,  $\mathbf{w} = -\mathbf{e}_2$ . The left side is the area of  $P(\mathbf{e}_1, \mathbf{0})$ , which is 0; the right side is  $1 + 1 = 2$ .
- (c) With  $a = 1$ ,  $b = 2$ ,  $c = 3$ ,  $d = 4$ , the signed area is  $ad - bc = 4 - 6 = -2$ . Since  $-2 \neq 0$ , the matrix is invertible by [Theorem 0.9.8](#).
- (d) By the second shear rule with scalar  $-5$ ,  $D(\mathbf{u}, 5\mathbf{u}) = D(\mathbf{u}, 5\mathbf{u} - 5\mathbf{u}) = D(\mathbf{u}, \mathbf{0})$ , and  $D(\mathbf{u}, \mathbf{0}) = D(\mathbf{u}, 0 \cdot \mathbf{u}) = 0 \cdot D(\mathbf{u}, \mathbf{u}) = 0$  by (R1).
- (e) The expansion has  $4! = 24$  terms, one for each rearrangement of  $\mathbf{e}_1, \dots, \mathbf{e}_4$ . The Sarrus-type diagram has only 8 products, so it must miss 16 of them.

### 6.1.6.2 B. Practice

#### Exercise B1: Signed Areas from the Rules.

Using only (R1)–(R4), in the manner of [Example \( \$ad - bc\$  from the Rules\)](#), compute the following. Then check each answer against  $ad - bc$ .

- (a)  $D((2, 1), (4, 5))$ .
- (b)  $D((1, 2), (3, 6))$ .
- (c)  $D((0, 3), (2, 0))$ .

*Solution.*

- (a) By the second shear rule with scalar  $-2$ ,  $D((2, 1), (4, 5)) = D((2, 1), (4, 5) - 2(2, 1)) = D((2, 1), (0, 3))$ . By the first shear rule with scalar  $-\frac{1}{3}$ , this equals  $D((2, 1) - \frac{1}{3}(0, 3), (0, 3)) = D((2, 0), (0, 3))$ . By (R1) twice and (R4),  $D(2\mathbf{e}_1, 3\mathbf{e}_2) = 6D(\mathbf{e}_1, \mathbf{e}_2) = 6$ . Check:  $2 \cdot 5 - 4 \cdot 1 = 6$ .
- (b) By the second shear rule with scalar  $-3$ ,  $D((1, 2), (3, 6)) = D((1, 2), (0, 0))$ , and  $D(\mathbf{u}, \mathbf{0}) = D(\mathbf{u}, 0 \cdot \mathbf{u}) = 0$  by (R1). So the value is 0. Check:  $1 \cdot 6 - 3 \cdot 2 = 0$ . The columns lie on one line, and the parallelogram is flat.
- (c) By (R1) twice,  $D(3\mathbf{e}_2, 2\mathbf{e}_1) = 6D(\mathbf{e}_2, \mathbf{e}_1)$ . By the Quick check (which uses only (R1) and (R2)),  $D(\mathbf{e}_2, \mathbf{e}_1) = -D(\mathbf{e}_1, \mathbf{e}_2) = -1$  by (R4). So the value is  $-6$ . Check: with columns  $(0, 3)$  and  $(2, 0)$ ,  $a = 0$ ,  $b = 2$ ,  $c = 3$ ,  $d = 0$ , and  $ad - bc = -6$ . The sign is negative because  $(2, 0)$  lies clockwise from  $(0, 3)$ .

#### Exercise B2: Equal Sides Force the Swap Rule.

Let  $D: \mathbb{R}^2 \times \mathbb{R}^2 \rightarrow \mathbb{R}$  satisfy (R1), (R3) and (R5), but not necessarily (R2).

- (a) Prove that  $D(\mathbf{u}, \mathbf{v}) = -D(\mathbf{v}, \mathbf{u})$  for all  $\mathbf{u}, \mathbf{v} \in \mathbb{R}^2$ .
- (b) Suppose moreover that (R4) holds. Deduce that  $D((a, c), (b, d)) = ad - bc$ .

*Hint: expand  $D(\mathbf{u} + \mathbf{v}, \mathbf{u} + \mathbf{v})$ .*

*Solution.*

- (a) Let  $\mathbf{u}, \mathbf{v} \in \mathbb{R}^2$ . By (R3),  $D(\mathbf{u} + \mathbf{v}, \mathbf{u} + \mathbf{v}) = 0$ . Expanding with (R5) in the first argument and then in the second,

$$0 = D(\mathbf{u}, \mathbf{u}) + D(\mathbf{u}, \mathbf{v}) + D(\mathbf{v}, \mathbf{u}) + D(\mathbf{v}, \mathbf{v}) = D(\mathbf{u}, \mathbf{v}) + D(\mathbf{v}, \mathbf{u}),$$

where the second equality uses (R3) twice. Hence  $D(\mathbf{u}, \mathbf{v}) = -D(\mathbf{v}, \mathbf{u})$ .

- (b) Write  $(a, c) = a\mathbf{e}_1 + c\mathbf{e}_2$  and  $(b, d) = b\mathbf{e}_1 + d\mathbf{e}_2$ . By (R5) and (R1) in each argument,

$$D((a, c), (b, d)) = ab D(\mathbf{e}_1, \mathbf{e}_1) + ad D(\mathbf{e}_1, \mathbf{e}_2) + cb D(\mathbf{e}_2, \mathbf{e}_1) + cd D(\mathbf{e}_2, \mathbf{e}_2).$$

By (R3) the first and last terms are 0. By (R4),  $D(\mathbf{e}_1, \mathbf{e}_2) = 1$ , and by (a),  $D(\mathbf{e}_2, \mathbf{e}_1) = -1$ . Hence  $D((a, c), (b, d)) = ad - bc$ .

### 6.1.6.3 C. Going deeper

#### Exercise C1: Area of a Triangle.

Let  $\mathbf{p}, \mathbf{q}, \mathbf{r} \in \mathbb{R}^2$ .

- Explain why the triangle with vertices  $\mathbf{p}, \mathbf{q}, \mathbf{r}$  has area  $\frac{1}{2}|D(\mathbf{q} - \mathbf{p}, \mathbf{r} - \mathbf{p})|$ , where  $D(\mathbf{u}, \mathbf{v})$  is the signed area.
- Find the area of the triangle with vertices  $(1, 2)$ ,  $(4, 3)$ ,  $(2, 6)$ .
- Hence decide whether the points  $(0, 1)$ ,  $(2, 4)$ ,  $(6, 10)$  lie on a line. Justify your answer.

*Solution.*

- Translating by  $-\mathbf{p}$  does not change areas, so the triangle has the same area as the triangle with vertices  $\mathbf{0}, \mathbf{u} = \mathbf{q} - \mathbf{p}, \mathbf{v} = \mathbf{r} - \mathbf{p}$ . That triangle is half of the parallelogram  $P(\mathbf{u}, \mathbf{v})$ : the diagonal from  $\mathbf{u}$  to  $\mathbf{v}$  cuts  $P(\mathbf{u}, \mathbf{v})$  into the triangle with vertices  $\mathbf{0}, \mathbf{u}, \mathbf{v}$  and the triangle with vertices  $\mathbf{u} + \mathbf{v}, \mathbf{v}, \mathbf{u}$ , and the point reflection  $\mathbf{x} \mapsto \mathbf{u} + \mathbf{v} - \mathbf{x}$  maps the first onto the second, so they have equal area. The area of  $P(\mathbf{u}, \mathbf{v})$  is  $|D(\mathbf{u}, \mathbf{v})|$ , since the signed area differs from the area only by its sign. Hence the triangle has area  $\frac{1}{2}|D(\mathbf{q} - \mathbf{p}, \mathbf{r} - \mathbf{p})|$ .
- Here  $\mathbf{q} - \mathbf{p} = (3, 1)$  and  $\mathbf{r} - \mathbf{p} = (1, 4)$ , so  $D = 3 \cdot 4 - 1 \cdot 1 = 11$  and the area is  $\frac{11}{2}$ .
- With  $\mathbf{p} = (0, 1)$ ,  $\mathbf{q} - \mathbf{p} = (2, 3)$  and  $\mathbf{r} - \mathbf{p} = (6, 9)$ , so  $D = 2 \cdot 9 - 6 \cdot 3 = 0$  and the “triangle” has area 0. Indeed  $(6, 9) = 3(2, 3)$ , so  $\mathbf{r} - \mathbf{p}$  is a multiple of  $\mathbf{q} - \mathbf{p}$ , and all three points lie on the line through  $(0, 1)$  with direction  $(2, 3)$ . Yes, they are collinear.

#### Exercise C2: The $1 \times 1$ Case.

For  $n = 1$ , the rules ask for a function  $f: F \rightarrow F$  of one vector in  $F^1 = F$ .

- Explain why the rule “zero whenever two arguments are equal” imposes no condition when  $n = 1$ .

- (b) Prove that the only function  $f: F \rightarrow F$  that is linear with  $f(1) = 1$  is  $f(a) = a$ . Hence the determinant of a  $1 \times 1$  matrix  $(a)$  must be  $a$ .

*Solution.*

- (a) The rule speaks about two **different** positions  $i \neq j$  holding equal vectors. With one argument there is no pair of positions, so the condition is vacuously true (Example (A vacuously true implication)).
- (b) Let  $f$  be linear with  $f(1) = 1$ , and let  $a \in F$ . Since  $a = a \cdot 1$ , homogeneity gives  $f(a) = f(a \cdot 1) = af(1) = a$ . Conversely,  $a \mapsto a$  is linear and sends 1 to 1. Hence  $f(a) = a$  is the only such function.

## 6.2 Multilinear Alternating Forms

The previous section found that signed area obeys a few rules: it is linear in each side separately, it vanishes when two sides are equal, and swapping the sides changes its sign. This section gives these rules names for functions of  $k$  vectors in an arbitrary vector space, and derives everything that follows from them before any formula exists. The main result is that “zero on equal arguments” already implies the swap rule, the shear rule and vanishing on dependent lists. The swap rule, in turn, gives back “zero on equal arguments” only when  $1 + 1 \neq 0$  in the field.

Throughout,  $V$  is a vector space over a field  $F$ ,  $k \geq 1$  is an integer, and  $V^k = V \times \cdots \times V$  is the set of  $k$ -tuples of vectors (Definition 0.3.11).

### 6.2.1 Multilinear forms

Signed area is not a linear function of the pair  $(\mathbf{u}, \mathbf{v})$ : doubling both sides multiplies the area by four, not by two. What it does have is linearity in  $\mathbf{u}$  when  $\mathbf{v}$  is held fixed, and linearity in  $\mathbf{v}$  when  $\mathbf{u}$  is held fixed. This “one slot at a time” linearity recurs throughout the book: in the dot product, later in inner products and bilinear forms, and in the determinant. So we give it a name.

*A multilinear form is a function of several vectors that is linear in each vector while the others are held fixed.*

#### Definition 6.2.1 — Multilinear Form

A function  $f: V^k \rightarrow F$  is  $k$ -**linear**, or a **multilinear form** on  $V$ , if **for every** position  $i \in \{1, \dots, k\}$  and **every** choice of  $\mathbf{v}_1, \dots, \mathbf{v}_{i-1}, \mathbf{v}_{i+1}, \dots, \mathbf{v}_k \in V$ , the function

$$V \rightarrow F, \quad \mathbf{x} \mapsto f(\mathbf{v}_1, \dots, \mathbf{v}_{i-1}, \mathbf{x}, \mathbf{v}_{i+1}, \dots, \mathbf{v}_k)$$

is linear. A 2-linear form is called **bilinear**.

In words: fix all arguments except the one in position  $i$ . What remains is a function of one vector, and it must be a linear functional (Definition 4.1.1). Written out, for all  $\mathbf{x}, \mathbf{y} \in V$  and



$c \in F$ ,

$$\begin{aligned} f(\dots, \mathbf{x} + \mathbf{y}, \dots) &= f(\dots, \mathbf{x}, \dots) + f(\dots, \mathbf{y}, \dots), \\ f(\dots, c\mathbf{x}, \dots) &= c f(\dots, \mathbf{x}, \dots), \end{aligned}$$

where the dots stand for the same fixed vectors on both sides. The condition is imposed for **every** position  $i$ , one at a time.

Some examples:

- **Signed area.** On  $F^2$ ,  $f(\mathbf{u}, \mathbf{v}) = u_1v_2 - u_2v_1$  is bilinear. For fixed  $\mathbf{v}$ , the map  $\mathbf{u} \mapsto v_2u_1 - v_1u_2$  is a linear functional on  $F^2$ , and for fixed  $\mathbf{u}$ , so is  $\mathbf{v} \mapsto u_1v_2 - u_2v_1$ .
- **The dot product.** On  $F^n$ ,  $(\mathbf{u}, \mathbf{v}) \mapsto u_1v_1 + \dots + u_nv_n$  is bilinear, by the same one-slot check.
- **Products of functionals.** If  $\varphi_1, \dots, \varphi_k \in V^*$ , then  $f(\mathbf{v}_1, \dots, \mathbf{v}_k) = \varphi_1(\mathbf{v}_1)\varphi_2(\mathbf{v}_2) \cdots \varphi_k(\mathbf{v}_k)$  is  $k$ -linear: in position  $i$  it is the linear functional  $\varphi_i$  multiplied by the fixed scalar  $\prod_{j \neq i} \varphi_j(\mathbf{v}_j)$ . For instance, on  $F[x]$ ,  $(p, q) \mapsto p(0)q(1)$  is bilinear, since evaluation at a point is linear ([Theorem 0.8.5](#)).
- **The degenerate case  $k = 1$ .** A 1-linear form is exactly a linear functional on  $V$ . So multilinear forms generalize the dual space of Chapter 4.

*Warning. Multilinear is not linear.* A bilinear form is **not** a linear function of the pair. For the signed area  $f(\mathbf{u}, \mathbf{v}) = u_1v_2 - u_2v_1$  on  $\mathbb{R}^2$ , scaling both arguments by 2 gives  $f(2\mathbf{e}_1, 2\mathbf{e}_2) = 4$ , not  $2f(\mathbf{e}_1, \mathbf{e}_2) = 2$ . Adding in both slots at once is not allowed either:  $f(\mathbf{e}_1 + \mathbf{0}, \mathbf{e}_1 + \mathbf{e}_2) = f(\mathbf{e}_1, \mathbf{e}_1 + \mathbf{e}_2) = 1$ , while  $f(\mathbf{e}_1, \mathbf{e}_1) + f(\mathbf{0}, \mathbf{e}_2) = 0$ . Pull out scalars and split sums **one slot at a time**.

## 6.2.2 Alternating and skew-symmetric forms

The second rule of signed area was geometric: if two edges coincide, the parallelogram or box is flat. There are two natural ways to put this into algebra. One says the value is **zero** when two arguments are equal. The other says the value **changes sign** when two arguments are swapped. They look like two phrasings of one idea, but they are not the same, and we must choose.

*An alternating form vanishes as soon as two of its arguments coincide.*

### Definition 6.2.2 — Alternating Form

A  $k$ -linear form  $f: V^k \rightarrow F$  is **alternating** if  $f(\mathbf{v}_1, \dots, \mathbf{v}_k) = 0$  **whenever**  $\mathbf{v}_i = \mathbf{v}_j$  for **some** pair of positions  $i \neq j$ .

### Definition 6.2.3 — Skew-Symmetric Form

A  $k$ -linear form  $f: V^k \rightarrow F$  is **skew-symmetric** if, **for all**  $\mathbf{v}_1, \dots, \mathbf{v}_k \in V$  and **all** positions  $i < j$ , swapping the arguments in positions  $i$  and  $j$  multiplies the value by  $-1$ :

$$f(\mathbf{v}_1, \dots, \mathbf{v}_j, \dots, \mathbf{v}_i, \dots, \mathbf{v}_k) = -f(\mathbf{v}_1, \dots, \mathbf{v}_i, \dots, \mathbf{v}_j, \dots, \mathbf{v}_k),$$

where on the left  $\mathbf{v}_j$  sits in position  $i$ ,  $\mathbf{v}_i$  sits in position  $j$ , and all other positions are unchanged.

In words: “alternating” asks for the value 0 as soon as **any two** positions, adjacent or not, hold the same vector, whatever the remaining arguments are. It is a condition on the inputs where two entries coincide, and says nothing directly about other inputs. “Skew-symmetric” is a condition on **every** input: it compares the value at a list with the value at the list with two entries exchanged.

### Example. (Alternating Forms)

Check that each of the following is an alternating form.

- (a) On  $F^2$ , for a fixed  $c \in F$ :  $f_c(\mathbf{u}, \mathbf{v}) = c(u_1v_2 - u_2v_1)$ .
- (b) On  $F^3$ , for fixed coordinate indices  $1 \leq r < s \leq 3$ :  $g_{rs}(\mathbf{u}, \mathbf{v}) = u_rv_s - u_sv_r$ .
- (c) On  $F[x]$ :  $h(p, q) = p(0)q(1) - p(1)q(0)$ .
- (d) On any  $V$ , with  $k = 1$ : any linear functional  $\varphi \in V^*$ .

*Solution.*

- (a) Bilinearity is the one-slot check above, multiplied by the constant  $c$ . With  $k = 2$  the only pair of positions is  $(1, 2)$ , and  $f_c(\mathbf{u}, \mathbf{u}) = c(u_1u_2 - u_2u_1) = 0$  by commutativity in  $F$ . So  $f_c$  is alternating. For  $c = 0$  it is the zero form, which is alternating for the silly reason that it is zero everywhere; for  $c = 1$  it is the signed area of the previous section.
- (b) In each slot  $g_{rs}$  is a linear functional, as in (a), and  $g_{rs}(\mathbf{u}, \mathbf{u}) = u_ru_s - u_su_r = 0$ . Over  $\mathbb{R}$  these three forms are, up to sign and order, the components of the cross product:  $\mathbf{u} \times \mathbf{v} = (g_{23}(\mathbf{u}, \mathbf{v}), -g_{13}(\mathbf{u}, \mathbf{v}), g_{12}(\mathbf{u}, \mathbf{v}))$ . This example shows that the number of arguments need not equal the dimension: here  $k = 2$  and  $\dim F^3 = 3$ .
- (c) Evaluation at 0 and at 1 are linear in  $p$  ([Theorem 0.8.5](#)), so for fixed  $q$  the map  $p \mapsto q(1)p(0) - q(0)p(1)$  is linear, and similarly in  $q$ . Also  $h(p, p) = p(0)p(1) - p(1)p(0) = 0$ . It is not the zero form:  $h(1, x) = 1 \cdot 1 - 1 \cdot 0 = 1$ .
- (d) A linear functional is 1-linear. With a single position there is **no** pair  $i \neq j$ , so the alternating condition holds vacuously ([Example \(A vacuously true implication\)](#)). This degenerate case matters because it shows that “alternating” is purely a condition about **pairs** of positions; the interesting content starts at  $k = 2$ .

Now a non-example, by a minimal change of (a). Over  $\mathbb{R}$ , replace the minus sign by a plus:  $s(\mathbf{u}, \mathbf{v}) = u_1v_2 + u_2v_1$ . Bilinearity survives, by the same one-slot check. But  $s(\mathbf{e}_1 + \mathbf{e}_2, \mathbf{e}_1 + \mathbf{e}_2) = 1 + 1 = 2 \neq 0$ , so the alternating clause fails. The dot product fails in the same way,  $\mathbf{e}_1 \cdot \mathbf{e}_1 = 1$ . A different minimal change keeps the alternating clause and breaks the other one. The unsigned area  $|u_1v_2 - u_2v_1|$  on  $\mathbb{R}^2$  is still 0 whenever  $\mathbf{u} = \mathbf{v}$ , but it is not linear in  $\mathbf{u}$ : at  $\mathbf{v} = \mathbf{e}_2$  it sends  $\mathbf{e}_1 \mapsto 1$  and  $-\mathbf{e}_1 \mapsto 1 \neq -1$ . This is the failure that forced a sign onto area in the previous section.

*Quick check.* Is  $f(\mathbf{u}, \mathbf{v}) = u_1v_2$  on  $F^2$  alternating? Is it skew-symmetric?

*Solution.* It is bilinear, but neither. With  $\mathbf{u} = \mathbf{v} = \mathbf{e}_1 + \mathbf{e}_2$ ,  $f(\mathbf{u}, \mathbf{u}) = 1 \cdot 1 = 1 \neq 0$ , so it is not alternating. And  $f(\mathbf{e}_1, \mathbf{e}_2) = 1$  while  $f(\mathbf{e}_2, \mathbf{e}_1) = 0$ ; skew-symmetry would need  $0 = -1$ , which fails in every field.

**Why this definition.** We chose “zero when two arguments are equal” as the basic notion, and in the form “**any two** positions”. There are two reasons. First, it is what the picture says

directly: two equal edges make a flat box. Second, as we prove next, it implies the swap rule over **every** field, while the swap rule implies it only when  $1 + 1 \neq 0$ . Allowing any pair of positions, not just adjacent ones, also spares us a separate argument moving equal entries next to each other.

### 6.2.3 Consequences of the alternating condition

Everything that elimination does to a determinant will come from the following theorem. Its four parts are the column operations of Chapter 2 seen through a form: swap two arguments, add a multiple of one to another, scale one, and the extreme case of a dependent list.

#### Theorem 6.2.1 — Properties of Alternating Forms

Let  $f: V^k \rightarrow F$  be an alternating  $k$ -linear form, let  $\mathbf{v}_1, \dots, \mathbf{v}_k \in V$ , and let  $i \neq j$  be positions in  $\{1, \dots, k\}$ .

- (a) **(Swap)** Exchanging the arguments in positions  $i$  and  $j$  multiplies  $f(\mathbf{v}_1, \dots, \mathbf{v}_k)$  by  $-1$ . In particular,  $f$  is skew-symmetric.
- (b) **(Shear)** For every  $c \in F$ , replacing  $\mathbf{v}_i$  by  $\mathbf{v}_i + c\mathbf{v}_j$  does not change  $f(\mathbf{v}_1, \dots, \mathbf{v}_k)$ .
- (c) **(Scaling)** For every  $c \in F$ , replacing  $\mathbf{v}_i$  by  $c\mathbf{v}_i$  multiplies  $f(\mathbf{v}_1, \dots, \mathbf{v}_k)$  by  $c$ .
- (d) **(Dependence)** If the list  $(\mathbf{v}_1, \dots, \mathbf{v}_k)$  is linearly dependent, then  $f(\mathbf{v}_1, \dots, \mathbf{v}_k) = 0$ . In particular,  $f(\mathbf{v}_1, \dots, \mathbf{v}_k) = 0$  if some  $\mathbf{v}_i = \mathbf{0}$ .

*Idea.* Parts (a)–(c) involve at most the two slots  $i$  and  $j$ , so freeze the other  $k - 2$  arguments. What is left is a bilinear form  $g(\mathbf{x}, \mathbf{y})$  with  $g(\mathbf{x}, \mathbf{x}) = 0$ , and the  $k$ -argument statements become two-argument statements about  $g$ . For the swap, the move is to feed  $g$  a sum in both slots,  $g(\mathbf{x} + \mathbf{y}, \mathbf{x} + \mathbf{y}) = 0$ , and expand: the two “square” terms die and the two cross terms must cancel. For (d), write one vector as a combination of the others and expand in its slot: every resulting term has a repeated argument.

*Proof.* Suppose first  $i < j$ ; the case  $i > j$  is the same with the roles of  $i$  and  $j$  exchanged. For  $\mathbf{x}, \mathbf{y} \in V$ , let

$$g(\mathbf{x}, \mathbf{y}) = f(\mathbf{v}_1, \dots, \mathbf{v}_{i-1}, \mathbf{x}, \mathbf{v}_{i+1}, \dots, \mathbf{v}_{j-1}, \mathbf{y}, \mathbf{v}_{j+1}, \dots, \mathbf{v}_k),$$

the value of  $f$  with  $\mathbf{x}$  in position  $i$ ,  $\mathbf{y}$  in position  $j$ , and  $\mathbf{v}_m$  in every other position  $m$ . Since  $f$  is  $k$ -linear,  $g$  is bilinear, and since  $f$  is alternating,  $g(\mathbf{x}, \mathbf{x}) = 0$  for every  $\mathbf{x} \in V$ . In this notation  $f(\mathbf{v}_1, \dots, \mathbf{v}_k) = g(\mathbf{v}_i, \mathbf{v}_j)$ .

- (c) This is linearity of  $f$  in position  $i$ :  $g(c\mathbf{v}_i, \mathbf{v}_j) = c g(\mathbf{v}_i, \mathbf{v}_j)$ .
- (d) By linearity in the first slot of  $g$  and  $g(\mathbf{v}_j, \mathbf{v}_j) = 0$ ,

$$g(\mathbf{v}_i + c\mathbf{v}_j, \mathbf{v}_j) = g(\mathbf{v}_i, \mathbf{v}_j) + c g(\mathbf{v}_j, \mathbf{v}_j) = g(\mathbf{v}_i, \mathbf{v}_j).$$

- (e) By bilinearity of  $g$ ,

$$\begin{aligned} 0 &= g(\mathbf{v}_i + \mathbf{v}_j, \mathbf{v}_i + \mathbf{v}_j) \\ &= g(\mathbf{v}_i, \mathbf{v}_i) + g(\mathbf{v}_i, \mathbf{v}_j) + g(\mathbf{v}_j, \mathbf{v}_i) + g(\mathbf{v}_j, \mathbf{v}_j) \\ &= g(\mathbf{v}_i, \mathbf{v}_j) + g(\mathbf{v}_j, \mathbf{v}_i), \end{aligned}$$

where the first and last equalities use  $g(\mathbf{x}, \mathbf{x}) = 0$ . Hence  $g(\mathbf{v}_j, \mathbf{v}_i) = -g(\mathbf{v}_i, \mathbf{v}_j)$ , which is the swap statement. Since  $\mathbf{v}_1, \dots, \mathbf{v}_k$  and  $i < j$  were arbitrary,  $f$  is skew-symmetric (Definition 6.2.3).

- (f) Suppose  $(\mathbf{v}_1, \dots, \mathbf{v}_k)$  is linearly dependent. By the Linear Dependence Lemma (Theorem 1.4.1), there is an index  $j$  with  $\mathbf{v}_j \in \text{span}(\mathbf{v}_1, \dots, \mathbf{v}_{j-1})$ . If  $j = 1$ , then  $\mathbf{v}_1 = \mathbf{0} = 0 \cdot \mathbf{v}_1$ , and by (c) with  $c = 0$ ,  $f(\mathbf{v}_1, \dots, \mathbf{v}_k) = 0 \cdot f(\mathbf{v}_1, \dots, \mathbf{v}_k) = 0$ . If  $j \geq 2$ , write  $\mathbf{v}_j = a_1 \mathbf{v}_1 + \dots + a_{j-1} \mathbf{v}_{j-1}$  with  $a_m \in F$ . By linearity of  $f$  in position  $j$ ,

$$f(\mathbf{v}_1, \dots, \mathbf{v}_k) = \sum_{m=1}^{j-1} a_m f(\mathbf{v}_1, \dots, \mathbf{v}_{j-1}, \mathbf{v}_m, \mathbf{v}_{j+1}, \dots, \mathbf{v}_k).$$

In the  $m$ -th term, the vector  $\mathbf{v}_m$  sits both in position  $m$  and in position  $j \neq m$ , so the term is 0 because  $f$  is alternating. Hence  $f(\mathbf{v}_1, \dots, \mathbf{v}_k) = 0$ . The last sentence of (d) holds because a list containing  $\mathbf{0}$  is dependent. This proves the theorem. ■

For the determinant, (a)–(c) will be the rules for column operations, and (d) explains at once why a matrix with dependent columns must have determinant 0. Part (d) also bounds the number of arguments: on a space of dimension  $n$ , any  $k > n$  vectors are dependent, so a non-zero alternating form needs  $k \leq n$  (Exercise C1).

## 6.2.4 Alternating versus skew-symmetric

Part (a) says alternating forms are skew-symmetric. The converse depends on the field.

### Theorem 6.2.2 — Skew-Symmetric and Alternating Forms

Let  $f: V^k \rightarrow F$  be a  $k$ -linear form.

- (a) If  $f$  is alternating, then  $f$  is skew-symmetric.
- (b) If  $f$  is skew-symmetric and the characteristic of  $F$  is **not** 2, then  $f$  is alternating.

*Proof.*

- (a) is Theorem 6.2.1 (a).
- (b) Suppose  $\mathbf{v}_i = \mathbf{v}_j$  for some  $i < j$ . Exchanging the arguments in positions  $i$  and  $j$  does not change the list, so skew-symmetry gives  $f(\mathbf{v}_1, \dots, \mathbf{v}_k) = -f(\mathbf{v}_1, \dots, \mathbf{v}_k)$ , that is,  $(1+1)f(\mathbf{v}_1, \dots, \mathbf{v}_k) = 0$ . If  $1+1 = 0$  in  $F$ , then  $2 \cdot 1 = 0$  while  $1 \cdot 1 = 1 \neq 0$ , so the smallest positive  $n$  with  $n \cdot 1 = 0$  would be 2, and the characteristic of  $F$  would be 2 (Definition 0.7.2). So  $1+1 \neq 0$ , and multiplying by  $(1+1)^{-1}$  gives  $f(\mathbf{v}_1, \dots, \mathbf{v}_k) = 0$ . Hence  $f$  is alternating. ■

**Warning.** Skew-symmetric does not imply alternating over  $\mathbb{F}_2$ . In  $\mathbb{F}_2$  we have  $-1 = 1$ , so “skew-symmetric” just means “symmetric”. Take  $f(\mathbf{u}, \mathbf{v}) = u_1 v_1$  on  $\mathbb{F}_2^2$ . It is bilinear, and  $f(\mathbf{v}, \mathbf{u}) = v_1 u_1 = u_1 v_1 = -f(\mathbf{u}, \mathbf{v})$ , so it is skew-symmetric. But

$f(\mathbf{e}_1, \mathbf{e}_1) = 1 \neq 0$ , so it is **not** alternating. Checking all 16 pairs  $(\mathbf{u}, \mathbf{v})$  confirms both claims:  $f(\mathbf{u}, \mathbf{v}) = f(\mathbf{v}, \mathbf{u})$  always, and  $f(\mathbf{u}, \mathbf{u}) = u_1$  is 1 for  $\mathbf{u} = (1, 0)$  and  $\mathbf{u} = (1, 1)$ . This is why the determinant is built on alternating forms, not skew-symmetric ones: the construction must work over  $\mathbb{F}_2$  too.

## 6.2.5 The space of alternating forms

Alternating forms can be added and scaled, and the result is again alternating. This is what makes the uniqueness question later in the chapter a question about dimension.

### Theorem 6.2.3 — Alternating Forms Form a Vector Space

The  $k$ -linear forms on  $V$  form a subspace of the vector space  $F^{V^k}$  of all functions  $V^k \rightarrow F$ , with the pointwise operations of [Example \(The Standard Vector Spaces\)](#) (d). The alternating  $k$ -linear forms on  $V$  form a subspace of it. In particular, both are vector spaces over  $F$ .

*Proof.* We use the Subspace Test ([Theorem 1.2.1](#)) twice.

The zero function is  $k$ -linear, since in each position it is the zero functional, and it is alternating, since all its values are 0.

Let  $f$  and  $g$  be  $k$ -linear and  $a \in F$ . Fix a position  $i$  and vectors in the other positions, and write  $f(\dots, \mathbf{x}, \dots)$  for the value with  $\mathbf{x}$  in position  $i$ . For  $\mathbf{x}, \mathbf{y} \in V$  and  $c \in F$ , the pointwise definition and linearity of  $f$  and  $g$  in position  $i$  give

$$\begin{aligned} (f + g)(\dots, \mathbf{x} + \mathbf{y}, \dots) &= f(\dots, \mathbf{x}, \dots) + f(\dots, \mathbf{y}, \dots) \\ &\quad + g(\dots, \mathbf{x}, \dots) + g(\dots, \mathbf{y}, \dots) \\ &= (f + g)(\dots, \mathbf{x}, \dots) + (f + g)(\dots, \mathbf{y}, \dots) \end{aligned}$$

and

$$(f + g)(\dots, c\mathbf{x}, \dots) = cf(\dots, \mathbf{x}, \dots) + cg(\dots, \mathbf{x}, \dots) = c(f + g)(\dots, \mathbf{x}, \dots).$$

So  $f + g$  is  $k$ -linear. In the same way  $(af)(\dots, \mathbf{x} + \mathbf{y}, \dots) = af(\dots, \mathbf{x}, \dots) + af(\dots, \mathbf{y}, \dots)$  and  $(af)(\dots, c\mathbf{x}, \dots) = c(af)(\dots, \mathbf{x}, \dots)$ , using commutativity in  $F$ , so  $af$  is  $k$ -linear.

If moreover  $f$  and  $g$  are alternating and  $\mathbf{v}_i = \mathbf{v}_j$  for some  $i \neq j$ , then  $(f + g)(\mathbf{v}_1, \dots, \mathbf{v}_k) = 0 + 0 = 0$  and  $(af)(\mathbf{v}_1, \dots, \mathbf{v}_k) = a \cdot 0 = 0$ . So the alternating forms are closed under both operations. This proves the theorem. ■

For  $V = F^2$  and  $k = 2$ , Exercise B2 shows that every alternating bilinear form is one of the forms  $f_c$  of [Example \(Alternating Forms\)](#) (a), with  $c = f(\mathbf{e}_1, \mathbf{e}_2)$ . So that space is spanned by the single non-zero form  $f_1$ , and has dimension 1. Later in this chapter we prove the same for all  $n$ : the alternating  $n$ -linear forms on  $F^n$  form a space of dimension exactly 1, spanned by the determinant. The missing ingredient is how the value of an alternating form changes when its arguments are rearranged by more than one swap, which is the subject of the next section.

## 6.2.6 Exercises

### 6.2.6.1 A. Check your understanding

#### Exercise A1.

- (a) Define an alternating  $k$ -linear form on a vector space  $V$  over  $F$ .
- (b) True or false: a bilinear form  $f$  on  $\mathbb{R}^2$  with  $f(\mathbf{e}_1, \mathbf{e}_1) = f(\mathbf{e}_2, \mathbf{e}_2) = 0$  is alternating. Justify your answer.
- (c) True or false: every skew-symmetric bilinear form on  $\mathbb{R}^5$  is alternating. Justify your answer.
- (d) Let  $f$  be an alternating 3-linear form on  $\mathbb{R}^3$  and  $\mathbf{u}, \mathbf{v} \in \mathbb{R}^3$ . What is  $f(\mathbf{u}, \mathbf{v}, 2\mathbf{u} - \mathbf{v})$ ?
- (e) Name the move that proves an alternating form changes sign under a swap.

*Solution.*

- (a) A function  $f: V^k \rightarrow F$  that is linear in each argument when the others are fixed (Definition 6.2.1), and satisfies  $f(\mathbf{v}_1, \dots, \mathbf{v}_k) = 0$  whenever  $\mathbf{v}_i = \mathbf{v}_j$  for some  $i \neq j$  (Definition 6.2.2).
- (b) False. The form  $s(\mathbf{u}, \mathbf{v}) = u_1v_2 + u_2v_1$  is bilinear with  $s(\mathbf{e}_1, \mathbf{e}_1) = s(\mathbf{e}_2, \mathbf{e}_2) = 0$ , but  $s(\mathbf{e}_1 + \mathbf{e}_2, \mathbf{e}_1 + \mathbf{e}_2) = 2 \neq 0$ . Checking the basis vectors is not enough, because the alternating condition is about **all** vectors.
- (c) True.  $\mathbb{R}$  has characteristic 0, so Theorem 6.2.2 (b) applies.
- (d) 0. The list  $(\mathbf{u}, \mathbf{v}, 2\mathbf{u} - \mathbf{v})$  is linearly dependent, since  $2\mathbf{u} - \mathbf{v} - (2\mathbf{u} - \mathbf{v}) = \mathbf{0}$  with coefficients  $2, -1, -1$  not all zero. By Theorem 6.2.1 (d) the value is 0.
- (e) Put the sum of the two arguments into both positions, use  $f(\dots, \mathbf{x} + \mathbf{y}, \dots, \mathbf{x} + \mathbf{y}, \dots) = 0$ , and expand by linearity in both positions; the terms with equal arguments vanish and the two cross terms must cancel.

### 6.2.6.2 B. Practice

#### Exercise B1: Multilinear or Alternating?.

Determine which of the following functions are multilinear, and which of those are alternating. Justify your answer.

- (a)  $\mathbb{R}^2 \times \mathbb{R}^2 \rightarrow \mathbb{R}$ ,  $(\mathbf{u}, \mathbf{v}) \mapsto 3u_1v_2 - 3u_2v_1$ .
- (b)  $\mathbb{R}^2 \times \mathbb{R}^2 \rightarrow \mathbb{R}$ ,  $(\mathbf{u}, \mathbf{v}) \mapsto u_1v_2 - u_2v_1 + 1$ .
- (c)  $\mathbb{R}^3 \times \mathbb{R}^3 \times \mathbb{R}^3 \rightarrow \mathbb{R}$ ,  $(\mathbf{u}, \mathbf{v}, \mathbf{w}) \mapsto u_1v_2w_3$ .
- (d)  $M_2(\mathbb{R}) \times M_2(\mathbb{R}) \rightarrow \mathbb{R}$ ,  $(\mathbf{A}, \mathbf{B}) \mapsto \text{tr}(\mathbf{AB}) - \text{tr}(\mathbf{BA})$ .
- (e)  $\mathbb{R}[x] \times \mathbb{R}[x] \rightarrow \mathbb{R}$ ,  $(p, q) \mapsto p(1)q'(1) - p'(1)q(1)$ .

*Solution.*

- (a) Multilinear and alternating: it is  $f_3$  from Example (Alternating Forms) (a) with  $c = 3$ .
- (b) Not multilinear. A linear functional sends  $\mathbf{0}$  to 0, but at  $\mathbf{u} = \mathbf{0}$  the map  $\mathbf{v} \mapsto$

- $0 - 0 + 1 = 1$  does not. So it is not an alternating form either; in any case its value at  $(\mathbf{u}, \mathbf{u})$  is  $1 \neq 0$ .
- (c) Multilinear: in each position it is a coordinate functional times a fixed scalar, as for products of functionals. Not alternating: with  $\mathbf{u} = \mathbf{v} = \mathbf{e}_1 + \mathbf{e}_2$  and  $\mathbf{w} = \mathbf{e}_3$ , the value is  $1 \cdot 1 \cdot 1 = 1 \neq 0$ .
- (d) By [Theorem 0.9.6](#),  $\text{tr}(\mathbf{AB}) = \text{tr}(\mathbf{BA})$ , so this is the zero function. The zero function is multilinear and alternating ([Theorem 6.2.3](#)). This is the degenerate case.
- (e) Multilinear:  $p \mapsto p(1)$  and  $p \mapsto p'(1)$  are linear, the second as a composition of differentiation ([Example \(Differentiation\)](#)) with evaluation ([Theorem 0.8.5](#)). So for fixed  $q$ ,  $p \mapsto q'(1)p(1) - q(1)p'(1)$  is linear, and similarly in  $q$ . Alternating:  $p(1)p'(1) - p'(1)p(1) = 0$ .

### Exercise B2: Alternating Bilinear Forms on $F^2$ .

Let  $f$  be an alternating bilinear form on  $F^2$ . Prove that

$$f(\mathbf{u}, \mathbf{v}) = f(\mathbf{e}_1, \mathbf{e}_2)(u_1v_2 - u_2v_1) \quad \text{for all } \mathbf{u}, \mathbf{v} \in F^2.$$

Hence show that the alternating bilinear forms on  $F^2$  form a space of dimension 1.

*Solution.* Write  $\mathbf{u} = u_1\mathbf{e}_1 + u_2\mathbf{e}_2$  and  $\mathbf{v} = v_1\mathbf{e}_1 + v_2\mathbf{e}_2$ . By linearity in the first and then in the second argument,

$$f(\mathbf{u}, \mathbf{v}) = u_1v_1f(\mathbf{e}_1, \mathbf{e}_1) + u_1v_2f(\mathbf{e}_1, \mathbf{e}_2) + u_2v_1f(\mathbf{e}_2, \mathbf{e}_1) + u_2v_2f(\mathbf{e}_2, \mathbf{e}_2).$$

Since  $f$  is alternating,  $f(\mathbf{e}_1, \mathbf{e}_1) = f(\mathbf{e}_2, \mathbf{e}_2) = 0$ , and by [Theorem 6.2.1](#) (a),  $f(\mathbf{e}_2, \mathbf{e}_1) = -f(\mathbf{e}_1, \mathbf{e}_2)$ . Hence  $f(\mathbf{u}, \mathbf{v}) = f(\mathbf{e}_1, \mathbf{e}_2)(u_1v_2 - u_2v_1)$ .

By [Theorem 6.2.3](#) these forms form a vector space, and we have just shown  $f = f(\mathbf{e}_1, \mathbf{e}_2)f_1$ , where  $f_1(\mathbf{u}, \mathbf{v}) = u_1v_2 - u_2v_1$  is alternating by [Example \(Alternating Forms\)](#) (a). So  $(f_1)$  spans the space. It is non-zero, since  $f_1(\mathbf{e}_1, \mathbf{e}_2) = 1$ , so the list  $(f_1)$  is linearly independent and hence a basis. The dimension is 1.

### Exercise B3: The Triple Product.

For  $\mathbf{v}, \mathbf{w} \in \mathbb{R}^3$  let  $\mathbf{v} \times \mathbf{w} = (v_2w_3 - v_3w_2, v_3w_1 - v_1w_3, v_1w_2 - v_2w_1)$ , and define  $f(\mathbf{u}, \mathbf{v}, \mathbf{w}) = u_1(\mathbf{v} \times \mathbf{w})_1 + u_2(\mathbf{v} \times \mathbf{w})_2 + u_3(\mathbf{v} \times \mathbf{w})_3$ . Show that  $f$  is an alternating 3-linear form on  $\mathbb{R}^3$  with  $f(\mathbf{e}_1, \mathbf{e}_2, \mathbf{e}_3) = 1$ .

*Solution. Multilinear.* Each component of  $\mathbf{v} \times \mathbf{w}$  is  $g_{rs}(\mathbf{v}, \mathbf{w})$  or  $-g_{rs}(\mathbf{v}, \mathbf{w})$  for a form of [Example \(Alternating Forms\)](#) (b), hence bilinear in  $(\mathbf{v}, \mathbf{w})$ . For fixed  $\mathbf{v}, \mathbf{w}$ ,  $f$  is a linear functional of  $\mathbf{u}$  with coefficients  $(\mathbf{v} \times \mathbf{w})_m$ . For fixed  $\mathbf{u}$  and  $\mathbf{w}$ ,  $f = \sum_m u_m(\mathbf{v} \times \mathbf{w})_m$  is a combination of linear functionals of  $\mathbf{v}$ , hence linear in  $\mathbf{v}$ ; likewise in  $\mathbf{w}$ .

*Alternating.* There are three pairs of positions. If  $\mathbf{v} = \mathbf{w}$ , every component  $v_rv_s - v_sv_r$  is 0, so  $\mathbf{v} \times \mathbf{v} = \mathbf{0}$  and  $f = 0$ . If  $\mathbf{u} = \mathbf{v}$ ,

$$f(\mathbf{u}, \mathbf{u}, \mathbf{w}) = u_1(u_2w_3 - u_3w_2) + u_2(u_3w_1 - u_1w_3) + u_3(u_1w_2 - u_2w_1) = 0,$$

since the six terms cancel in pairs:  $u_1u_2w_3$  with  $-u_2u_1w_3$ ,  $-u_1u_3w_2$  with  $u_3u_1w_2$ , and  $u_2u_3w_1$  with  $-u_3u_2w_1$ . If  $\mathbf{u} = \mathbf{w}$ , the same expansion  $u_1(v_2u_3 - v_3u_2) + u_2(v_3u_1 - v_1u_3) + u_3(v_1u_2 - v_2u_1)$  cancels in pairs in the same way.

*Value.*  $\mathbf{e}_2 \times \mathbf{e}_3 = (1 \cdot 1 - 0 \cdot 0, 0 \cdot 0 - 0 \cdot 1, 0 \cdot 0 - 1 \cdot 0) = (1, 0, 0)$ , so  $f(\mathbf{e}_1, \mathbf{e}_2, \mathbf{e}_3) = 1$ .

### 6.2.6.3 C. Going deeper

#### Exercise C1: Too Many Arguments.

Let  $V$  be finite-dimensional with  $\dim V = n$ .

- (a) Prove that every alternating  $k$ -linear form on  $V$  with  $k > n$  is the zero form.
- (b) Deduce that the only alternating 3-linear form on  $F^2$  is zero, and give a non-zero alternating 2-linear form on  $F^2$  to show that the bound in (a) cannot be lowered to  $k \geq n$ .

*Solution.*

- (a) Let  $f$  be an alternating  $k$ -linear form with  $k > n$ , and let  $\mathbf{v}_1, \dots, \mathbf{v}_k \in V$ . By [Theorem 1.6.3](#) (a), a list of length  $k > \dim V$  is linearly dependent. By [Theorem 6.2.1](#) (d),  $f(\mathbf{v}_1, \dots, \mathbf{v}_k) = 0$ . Since the vectors were arbitrary,  $f$  is the zero form.
- (b) Here  $\dim F^2 = 2 < 3$ , so (a) applies. For  $k = n = 2$ , the form  $f_1(\mathbf{u}, \mathbf{v}) = u_1v_2 - u_2v_1$  of [Example \(Alternating Forms\)](#) (a) is alternating with  $f_1(\mathbf{e}_1, \mathbf{e}_2) = 1 \neq 0$ .

#### Exercise C2: Bilinear Forms over $\mathbb{F}_2$ .

- (a) Let  $f$  be a bilinear form on  $F^2$  and write  $b_{rs} = f(\mathbf{e}_r, \mathbf{e}_s)$ . Show that  $f(\mathbf{u}, \mathbf{v}) = \sum_{r,s=1}^2 b_{rs}u_rv_s$ , and that conversely every choice of four scalars  $b_{rs}$  defines a bilinear form this way. Hence there are exactly 16 bilinear forms on  $\mathbb{F}_2^2$ .
- (b) Prove that a bilinear form on  $\mathbb{F}_2^2$  is skew-symmetric if and only if  $b_{12} = b_{21}$ . How many are there?
- (c) Prove that a bilinear form on  $\mathbb{F}_2^2$  is alternating if and only if  $b_{11} = b_{22} = 0$  and  $b_{12} = b_{21}$ . How many are there?
- (d) Over  $\mathbb{R}$ , how do the analogous two sets of bilinear forms on  $\mathbb{R}^2$  compare? Justify your answer.

*Solution.*

- (a) Expanding  $\mathbf{u} = u_1\mathbf{e}_1 + u_2\mathbf{e}_2$  and  $\mathbf{v} = v_1\mathbf{e}_1 + v_2\mathbf{e}_2$  by linearity in each argument gives  $f(\mathbf{u}, \mathbf{v}) = \sum_{r,s} u_rv_sf(\mathbf{e}_r, \mathbf{e}_s) = \sum_{r,s} b_{rs}u_rv_s$ . Conversely, for any scalars  $b_{rs}$ , the function  $\sum_{r,s} b_{rs}u_rv_s$  is linear in  $\mathbf{u}$  for fixed  $\mathbf{v}$  (a combination of the coordinates  $u_r$  with coefficients  $\sum_s b_{rs}v_s$ ) and similarly in  $\mathbf{v}$ , and it takes the value  $b_{rs}$  at  $(\mathbf{e}_r, \mathbf{e}_s)$ . So bilinear forms on  $F^2$  correspond exactly to quadruples  $(b_{11}, b_{12}, b_{21}, b_{22})$ , and over  $\mathbb{F}_2$  there are  $2^4 = 16$ .
- (b) In  $\mathbb{F}_2$ ,  $-a = a$  for every  $a$ , so  $f$  is skew-symmetric exactly when  $f(\mathbf{v}, \mathbf{u}) = f(\mathbf{u}, \mathbf{v})$  for all  $\mathbf{u}, \mathbf{v}$ . ( $\Rightarrow$ ) Taking  $\mathbf{u} = \mathbf{e}_1$ ,  $\mathbf{v} = \mathbf{e}_2$  gives  $b_{21} = b_{12}$ . ( $\Leftarrow$ ) If  $b_{12} = b_{21}$ , then  $b_{rs} = b_{sr}$  for all  $r, s$ , and  $f(\mathbf{v}, \mathbf{u}) = \sum_{r,s} b_{rs}v_ru_s = \sum_{r,s} b_{sr}u_sv_r = f(\mathbf{u}, \mathbf{v})$ . The entries  $b_{11}, b_{22}$  and the common value  $b_{12} = b_{21}$  are free, so there are  $2^3 = 8$



skew-symmetric forms.

- (c) ( $\Rightarrow$ ) If  $f$  is alternating, then  $b_{11} = f(\mathbf{e}_1, \mathbf{e}_1) = 0$  and  $b_{22} = 0$ , and  $0 = f(\mathbf{e}_1 + \mathbf{e}_2, \mathbf{e}_1 + \mathbf{e}_2) = b_{11} + b_{12} + b_{21} + b_{22} = b_{12} + b_{21}$ , so  $b_{21} = -b_{12} = b_{12}$ . ( $\Leftarrow$ ) With  $b_{11} = b_{22} = 0$  and  $b_{12} = b_{21} = b$ ,  $f(\mathbf{u}, \mathbf{u}) = b(u_1u_2 + u_2u_1) = (1+1)b u_1u_2 = 0$ . Only  $b \in \{0, 1\}$  is free, so there are 2 alternating forms. Hence  $8 - 2 = 6$  skew-symmetric forms over  $\mathbb{F}_2$  are not alternating;  $u_1v_1$  (with  $b_{11} = 1$ , all others 0) is one of them, as in the warning. A brute-force check of all 16 forms against all pairs in  $\mathbb{F}_2^2$  gives the same counts, 8 and 2.
- (d) Over  $\mathbb{R}$  the two sets are **equal**. Alternating forms are skew-symmetric by [Theorem 6.2.2](#) (a), and since  $\mathbb{R}$  has characteristic 0, skew-symmetric forms are alternating by part (b) of that theorem. Concretely, both conditions say  $b_{11} = b_{22} = 0$  and  $b_{21} = -b_{12}$ .

## 6.3 Permutations and Sign

An alternating form changes sign when two of its arguments are swapped ([Theorem 6.2.1](#)). To compare  $f(\mathbf{v}_3, \mathbf{v}_1, \mathbf{v}_2)$  with  $f(\mathbf{v}_1, \mathbf{v}_2, \mathbf{v}_3)$ , we reach one list from the other by swaps and collect a factor  $-1$  per swap. But the same rearrangement can be reached by different sequences of swaps, and Chapter 0 showed that even their number is not unique. If one route used an even number of swaps and another an odd number, we would get  $f(\mathbf{v}_1, \mathbf{v}_2, \mathbf{v}_3) = -f(\mathbf{v}_1, \mathbf{v}_2, \mathbf{v}_3)$  for every alternating form, which is absurd for signed volume. This section rules that out. It attaches to each permutation a sign defined without any choices, proves that the sign is multiplicative, and deduces that the parity of the number of transpositions is well defined, as Chapter 0 promised.

### 6.3.1 Permutations, recalled

Recall from Chapter 0 that  $S_n$  is the group of bijections  $\{1, \dots, n\} \rightarrow \{1, \dots, n\}$ , with  $n!$  elements. A permutation can be written in two-line notation, with  $\sigma(i)$  below  $i$ , or as a product of cycles. The product  $\sigma\tau$  is the composition  $\sigma \circ \tau$ : **first**  $\tau$ , **then**  $\sigma$ , so products of cycles are read from right to left ([Example \(Computing with cycles\)](#)). A transposition  $(i\ j)$  swaps  $i$  and  $j$  and fixes everything else, and for  $n \geq 2$  every permutation is a product of transpositions ([Theorem 0.10.4](#)). The product is far from unique, since

$$(1\ 2\ 3) = (1\ 3)(1\ 2) = (1\ 2)(2\ 3) = (1\ 3)(1\ 2)(1\ 2)(1\ 2).$$

Here the numbers of transpositions are 2, 2, 4, all even. The goal is to show this is no accident.

Why can we not simply **define** the sign of  $\sigma$  as  $(-1)^r$ , where  $\sigma$  is a product of  $r$  transpositions? Because that definition depends on a choice of product, and we would have to prove that every choice gives the same answer. That is the statement we are trying to prove. We need a definition that involves no choice at all.

### 6.3.2 Inversions and the sign

To measure how much a permutation scrambles  $1, 2, \dots, n$ , look at the bottom row  $\sigma(1), \sigma(2), \dots, \sigma(n)$  of its two-line notation and count the pairs of entries that are in the wrong order. A sorted row

has none, and each swap of two neighboring entries changes the count by exactly one. So the count is a natural candidate for tracking the parity of swaps.

*The sign of a permutation records whether it puts an even or an odd number of pairs out of order.*

### Definition 6.3.1 — Inversion

Let  $\sigma \in S_n$ . An **inversion** of  $\sigma$  is a pair  $(i, j)$  of positions with  $1 \leq i < j \leq n$  and  $\sigma(i) > \sigma(j)$ . We write  $\text{inv}(\sigma)$  for the number of inversions of  $\sigma$ .

### Definition 6.3.2 — Sign of a Permutation

The **sign** of  $\sigma \in S_n$  is

$$\text{sgn } \sigma = (-1)^{\text{inv}(\sigma)} \in \{1, -1\}.$$

The permutation  $\sigma$  is **even** if  $\text{sgn } \sigma = 1$  and **odd** if  $\text{sgn } \sigma = -1$ .

In words: an inversion is a pair of positions whose order  $\sigma$  **reverses**. The condition  $i < j$  makes each pair of positions count once, and  $\sigma(i) > \sigma(j)$  says the values come out in the opposite order. Since  $\text{inv}(\sigma)$  is the size of a subset of the finite set of pairs  $i < j$ , it is a well-defined integer between 0 and  $n(n-1)/2$ , and there is nothing to choose. That is the whole reason for this definition.

**The identity.** Its bottom row is sorted, so  $\text{inv}(\text{id}) = 0$  and  $\text{id}$  is even. In  $S_1 = \{\text{id}\}$  this is the only permutation, with sign 1. The degenerate case matters: the identity is the empty product of transpositions, and its sign  $1 = (-1)^0$  is what a product of 0 swaps should give.

**All of  $S_3$ .** Here is each permutation with its bottom row and inversions.

| $\sigma$   | id    | (1 2)  | (2 3)  | (1 3)                  | (1 2 3) | (1 3 2)        |
|------------|-------|--------|--------|------------------------|---------|----------------|
| bottom row | 1 2 3 | 2 1 3  | 1 3 2  | 3 2 1                  | 2 3 1   | 3 1 2          |
| inversions | none  | (1, 2) | (2, 3) | (1, 2), (1, 3), (2, 3) | (2, 3)  | (1, 2), (1, 3) |
| sgn        | 1     | -1     | -1     | -1                     | 1       | 1              |

For (1 2 3), the bottom row is  $\sigma(1), \sigma(2), \sigma(3) = 2, 3, 1$ , and the out-of-order pairs of positions are (1, 3), since  $2 > 1$ , and (2, 3), since  $3 > 1$ .

Two things stand out. The three transpositions are odd, although (1 3) has three inversions, not one. And the two 3-cycles are even, consistent with  $(1\ 2\ 3) = (1\ 3)(1\ 2)$  being a product of two transpositions.

**Why all pairs, not just neighbors.** A tempting shortcut counts only **adjacent** positions  $(i, i+1)$  with  $\sigma(i) > \sigma(i+1)$ . For (1 3), with bottom row 3, 2, 1, this count is 2, which would make (1 3) “even”. But  $(1\ 3) = (1\ 2)(2\ 3)(1\ 2)$ , as one checks by following  $1 \mapsto 2 \mapsto 3 \mapsto 3$ ,  $2 \mapsto 1 \mapsto 1 \mapsto 2$ ,  $3 \mapsto 3 \mapsto 2 \mapsto 1$ , and each factor has adjacent count 1. A count that is to be multiplicative must give  $(-1)^3 = -1$  here, so the adjacent count is the wrong one.

*Quick check.* Find  $\text{sgn}(1\ 4)$  in  $S_4$  by counting inversions.

*Solution.* The bottom row is 4, 2, 3, 1. The pairs of positions in the wrong order are (1, 2), (1, 3), (1, 4), since 4 is larger than everything after it, and (2, 4), (3, 4), since  $2 > 1$  and  $3 > 1$ . The pair (2, 3) is in order. So  $\text{inv}(1\ 4) = 5$  and  $\text{sgn}(1\ 4) = -1$ .

### 6.3.3 The sign is multiplicative

The key property is that signs multiply when permutations are composed.

#### Theorem 6.3.1 — The Sign Is Multiplicative

For all  $\sigma, \tau \in S_n$ ,

$$\text{sgn}(\sigma\tau) = \text{sgn } \sigma \cdot \text{sgn } \tau.$$

*Idea.* Count inversions one pair at a time, and record each pair as  $+1$  (order kept) or  $-1$  (order reversed). The sign is the product of these  $\pm 1$ 's. Applying  $\tau$  and then  $\sigma$  moves a pair of positions  $\{i, j\}$  to  $\{\tau(i), \tau(j)\}$  and then to  $\{\sigma\tau(i), \sigma\tau(j)\}$ . The order is reversed overall exactly when it is reversed at one of the two stages but not both, which is multiplication of  $\pm 1$ 's. Finally, as  $\{i, j\}$  runs over all pairs, so does  $\{\tau(i), \tau(j)\}$ , so the contributions of  $\sigma$  multiply to  $\text{sgn } \sigma$ , just in a different order.

*Proof.* For a non-zero integer  $m$  let  $s(m) = 1$  if  $m > 0$  and  $s(m) = -1$  if  $m < 0$ . Then  $s(mm') = s(m)s(m')$  for non-zero integers  $m, m'$ , by the sign rules for products of integers.

Let  $\mathcal{P}$  be the set of 2-element subsets  $\{i, j\}$  of  $\{1, \dots, n\}$ . For  $\rho \in S_n$  and  $\{i, j\} \in \mathcal{P}$ , define

$$\varepsilon_\rho(\{i, j\}) = s((\rho(i) - \rho(j))(i - j)).$$

This is defined:  $i \neq j$  and  $\rho(i) \neq \rho(j)$  since  $\rho$  is injective, so the product is non-zero; and exchanging the names  $i$  and  $j$  negates both factors, so the product does not depend on the order in which the subset is written. If  $i < j$ , then  $i - j < 0$ , so  $\varepsilon_\rho(\{i, j\}) = -1$  exactly when  $\rho(i) > \rho(j)$ , that is, when  $(i, j)$  is an inversion of  $\rho$ . Hence, by [Definition 6.3.2](#),

$$\text{sgn } \rho = \prod_{P \in \mathcal{P}} \varepsilon_\rho(P). \quad (1)$$

Now let  $\{i, j\} \in \mathcal{P}$ . Since  $\tau(i) \neq \tau(j)$ , we have  $(\tau(i) - \tau(j))^2 > 0$ . Multiplying inside  $s$  by this positive integer does not change the value, so

$$\begin{aligned} \varepsilon_{\sigma\tau}(\{i, j\}) &= s((\sigma(\tau(i)) - \sigma(\tau(j)))(\tau(i) - \tau(j)) \cdot (\tau(i) - \tau(j))(i - j)) \\ &= \varepsilon_\sigma(\{\tau(i), \tau(j)\}) \varepsilon_\tau(\{i, j\}). \end{aligned}$$

Let  $\hat{\tau}: \mathcal{P} \rightarrow \mathcal{P}$ ,  $\{i, j\} \mapsto \{\tau(i), \tau(j)\}$ ; the image has two elements because  $\tau$  is injective. The same construction with  $\tau^{-1}$  gives an inverse function, so  $\hat{\tau}$  is a bijection ([Theorem 0.4.7](#)). Taking the product over all  $P \in \mathcal{P}$  and using (1) three times,

$$\text{sgn}(\sigma\tau) = \prod_{P \in \mathcal{P}} \varepsilon_\sigma(\hat{\tau}(P)) \cdot \prod_{P \in \mathcal{P}} \varepsilon_\tau(P) = \prod_{Q \in \mathcal{P}} \varepsilon_\sigma(Q) \cdot \text{sgn } \tau = \text{sgn } \sigma \cdot \text{sgn } \tau,$$

where the second equality substitutes  $Q = \hat{\tau}(P)$ , which runs over  $\mathcal{P}$  exactly once because  $\hat{\tau}$  is a bijection, and reorders the finite product of integers. This proves the theorem. ■

By induction on the number of factors,  $\text{sgn}(\sigma_1\sigma_2\cdots\sigma_r) = \text{sgn}\sigma_1\cdots\text{sgn}\sigma_r$ . In the language of Chapter 0, the set  $\{1, -1\}$  is a group under multiplication (it contains 1, and it is closed under products and inverses, since  $(-1)^{-1} = -1$ ), and the theorem says exactly that

$$\text{sgn}: S_n \rightarrow \{1, -1\}$$

is a group homomorphism (Definition 0.10.3).

## 6.3.4 Transpositions and parity

Now we can compute the sign of the building blocks.

### Corollary 6.3.2 — Sign of a Transposition

Every transposition is odd:  $\text{sgn}(i\ j) = -1$  for all  $i \neq j$  in  $\{1, \dots, n\}$ .

*Proof.* Since  $(i\ j) = (j\ i)$ , we may assume  $i < j$ . Let  $\tau = (i\ j)$  and let  $p < q$  be positions. We decide when  $\tau(p) > \tau(q)$ .

- If neither  $p$  nor  $q$  lies in  $\{i, j\}$ , then  $\tau(p) = p < q = \tau(q)$ : no inversion.
- If  $(p, q) = (i, j)$ , then  $\tau(i) = j > i = \tau(j)$ : one inversion.
- If  $p = i$  and  $q \neq j$ , then  $\tau(p) = j$  and  $\tau(q) = q$ , an inversion exactly when  $i < q < j$ . This gives  $j - i - 1$  inversions.
- If  $q = j$  and  $p \neq i$ , then  $\tau(p) = p$  and  $\tau(q) = i$ , an inversion exactly when  $i < p < j$ . This gives  $j - i - 1$  inversions.
- If  $p = j$  (so  $q > j$ ), then  $\tau(p) = i < q = \tau(q)$ ; if  $q = i$  (so  $p < i$ ), then  $\tau(p) = p < j = \tau(q)$ : no inversion.

These cases cover every pair, so  $\text{inv}(\tau) = 1 + 2(j - i - 1)$ , an odd number. Hence  $\text{sgn}\tau = -1$ . ■

The Quick check above is the case  $(i, j) = (1, 4)$ :  $1 + 2 \cdot 2 = 5$  inversions. Combining the corollary with multiplicativity gives the fact that Chapter 0 left open.

### Corollary 6.3.3 — Parity Is Well Defined

Let  $n \geq 2$  and  $\sigma \in S_n$ . If  $\sigma = \tau_1 \cdots \tau_r$  is a product of  $r$  transpositions, then  $\text{sgn}\sigma = (-1)^r$ . Consequently, if also  $\sigma = \tau'_1 \cdots \tau'_s$  with transpositions  $\tau'_m$ , then  $r$  and  $s$  are both even or both odd.

*Proof.* By Theorem 6.3.1, extended to  $r$  factors, and Corollary 6.3.2,  $\text{sgn}\sigma = \text{sgn}\tau_1 \cdots \text{sgn}\tau_r = (-1)^r$ . In the same way  $\text{sgn}\sigma = (-1)^s$ . Hence  $(-1)^r = (-1)^s$ , so  $r - s$  is even.

So a permutation is even exactly when it is a product of an even number of transpositions, and the “number of swaps” has a well-defined parity even though the number itself is not defined.

### Proposition 6.3.4 — Sign of a Cycle

A  $k$ -cycle has sign  $(-1)^{k-1}$ . In particular, cycles of even length are odd and cycles of odd length are even.

*Proof.* For  $k = 1$  the cycle is id, with sign  $1 = (-1)^0$ . Let  $k \geq 2$  and let  $\gamma = (i_1 \ i_2 \ \cdots \ i_k)$ . We claim

$$\gamma = (i_1 \ i_k)(i_1 \ i_{k-1}) \cdots (i_1 \ i_3)(i_1 \ i_2).$$

Apply the factors from right to left. The element  $i_1$  is sent to  $i_2$  by  $(i_1 \ i_2)$ , and no later factor involves  $i_2$ . For  $2 \leq m \leq k-1$ , the element  $i_m$  is fixed by the factors before  $(i_1 \ i_m)$ , sent to  $i_1$  by it, then to  $i_{m+1}$  by  $(i_1 \ i_{m+1})$ , and no later factor involves  $i_{m+1}$ . The element  $i_k$  is fixed until the last factor  $(i_1 \ i_k)$  sends it to  $i_1$ . Elements outside the cycle are fixed by every factor. This agrees with  $\gamma$  everywhere, proving the claim. The product has  $k-1$  transpositions, so  $\text{sgn } \gamma = (-1)^{k-1}$  by Corollary 6.3.3.

**Warning.** Only the parity of the number of transpositions is determined, and it is not the parity of the cycle length. The identity in  $S_3$  equals  $(1 \ 2)(1 \ 2)$  and also  $(1 \ 2)(1 \ 3)(1 \ 2)(1 \ 3)(1 \ 2)(1 \ 3)$ , since  $((1 \ 2)(1 \ 3))^3 = (1 \ 3 \ 2)^3 = \text{id}$ ; the numbers 2 and 6 differ, only their parity agrees. And a 3-cycle, “of length 3”, is **even**:  $(1 \ 2 \ 3) = (1 \ 3)(1 \ 2)$ . One more trap concerns order:  $(1 \ 3)(1 \ 2) = (1 \ 2 \ 3)$  but  $(1 \ 2)(1 \ 3) = (1 \ 3 \ 2)$ . Multiplying in the wrong order gives the wrong permutation and the wrong bottom row, even though, by Theorem 6.3.1, it happens to give the right sign.

## 6.3.5 The alternating group

Since  $\text{sgn}$  is a homomorphism, its kernel is a subgroup of  $S_n$  (Theorem 0.10.3).

### Definition 6.3.3 — Alternating Group

The **alternating group**  $A_n$  is the kernel of  $\text{sgn}: S_n \rightarrow \{1, -1\}$ , that is, the subgroup of **even** permutations in  $S_n$ .

For example,  $A_3 = \{\text{id}, (1 \ 2 \ 3), (1 \ 3 \ 2)\}$  by the table above, and  $A_2 = \{\text{id}\}$ . For  $n \geq 2$ , exactly half of the permutations are even (Exercise C1). The odd permutations do **not** form a subgroup: they do not contain id. The name is no coincidence: as we see below and in the next section, rearranging the arguments of an alternating form by an even permutation leaves its value unchanged.

## 6.3.6 Back to alternating forms

Here is why this section exists. Let  $f$  be an alternating 3-linear form and compare  $f(\mathbf{v}_3, \mathbf{v}_1, \mathbf{v}_2)$  with  $f(\mathbf{v}_1, \mathbf{v}_2, \mathbf{v}_3)$ . Swapping the first two arguments of  $(\mathbf{v}_3, \mathbf{v}_1, \mathbf{v}_2)$  gives  $(\mathbf{v}_1, \mathbf{v}_3, \mathbf{v}_2)$ , and swapping the last two then gives  $(\mathbf{v}_1, \mathbf{v}_2, \mathbf{v}_3)$ . By [Theorem 6.2.1](#) (a) each swap costs a factor  $-1$ , so  $f(\mathbf{v}_3, \mathbf{v}_1, \mathbf{v}_2) = (-1)^2 f(\mathbf{v}_1, \mathbf{v}_2, \mathbf{v}_3) = f(\mathbf{v}_1, \mathbf{v}_2, \mathbf{v}_3)$ . The list  $(\mathbf{v}_3, \mathbf{v}_1, \mathbf{v}_2)$  is  $(\mathbf{v}_{\sigma(1)}, \mathbf{v}_{\sigma(2)}, \mathbf{v}_{\sigma(3)})$  for  $\sigma = (1\ 3\ 2)$ , with bottom row 3, 1, 2, and the factor 1 we found is  $\text{sgn } \sigma$ .

A different sequence of swaps could have been used, possibly a longer one. By [Corollary 6.3.3](#), every such sequence has an even number of swaps, so every route gives the same factor. In general, the next section proves that  $f(\mathbf{v}_{\sigma(1)}, \dots, \mathbf{v}_{\sigma(k)}) = \text{sgn } \sigma \cdot f(\mathbf{v}_1, \dots, \mathbf{v}_k)$  for every alternating  $k$ -linear form, and uses it to turn each of the  $n!$  surviving terms of the expansion sketched in the first section of this chapter into  $\pm D(\mathbf{e}_1, \dots, \mathbf{e}_n)$ . Without the corollary, the factor computed swap by swap could depend on the route, and could not be identified with  $\text{sgn } \sigma$ .

## 6.3.7 Computing signs in practice

There are now two independent ways to find a sign: count inversions, or factor into cycles and use [Proposition 6.3.4](#) with multiplicativity. Doing both is a good check.

### Example. (A Sign Computed Two Ways)

Find the sign of the permutation  $\sigma \in S_6$  with two-line notation  $\begin{pmatrix} 1 & 2 & 3 & 4 & 5 & 6 \\ 4 & 5 & 3 & 6 & 2 & 1 \end{pmatrix}$ , first by counting inversions and then from its cycles.

*Solution. Inversions.* Go through the bottom row 4, 5, 3, 6, 2, 1 from left to right, and for each entry count the **later** entries that are smaller:

| position $i$      | 1       | 2       | 3    | 4    | 5 | 6    |
|-------------------|---------|---------|------|------|---|------|
| $\sigma(i)$       | 4       | 5       | 3    | 6    | 2 | 1    |
| later and smaller | 3, 2, 1 | 3, 2, 1 | 2, 1 | 2, 1 | 1 | none |
| count             | 3       | 3       | 2    | 2    | 1 | 0    |

So  $\text{inv}(\sigma) = 3 + 3 + 2 + 2 + 1 + 0 = 11$ , which is odd, and  $\text{sgn } \sigma = -1$ .

*Cycles.* Following elements as in [Example \(Computing with cycles\)](#):  $1 \mapsto 4 \mapsto 6 \mapsto 1$ , then  $2 \mapsto 5 \mapsto 2$ , and  $3 \mapsto 3$ . So  $\sigma = (1\ 4\ 6)(2\ 5)$ . By [Proposition 6.3.4](#), the 3-cycle has sign 1 and the transposition has sign  $-1$ , so by [Theorem 6.3.1](#),  $\text{sgn } \sigma = 1 \cdot (-1) = -1$ . The two methods agree.

The following cell checks the example, and also tests [Theorem 6.3.1](#) and [Proposition 6.3.4](#) on all of  $S_5$  by brute force. Permutations are stored as tuples of values  $(\sigma(1), \dots, \sigma(n))$ , shifted to start at 0.

```
from itertools import permutations, combinations

def inv(s):
    return sum(1 for i, j in combinations(range(len(s)), 2) if s[i] > s[j])
```

```

def sgn(s):
    return (-1) ** inv(s)

def compose(s, t):          # (s t)(i) = s(t(i)): first t, then s
    return tuple(s[t[i]] for i in range(len(t)))

def cycle(entries, n):      # the cycle (entries[0] entries[1] ...), 0-based
    s = list(range(n))
    for a, b in zip(entries, entries[1:] + entries[:1]):
        s[a] = b
    return tuple(s)

sigma = (4, 5, 3, 6, 2, 1)
sigma0 = tuple(x - 1 for x in sigma)
print("inversions:", inv(sigma0), " sign:", sgn(sigma0))
print("(1 4 6)(2 5) equals sigma:", compose(cycle([0, 3, 5], 6), cycle([1, 4], 6))
      == sigma0)

S5 = list(permutations(range(5)))
print("multiplicative on S_5:", all(sgn(compose(s, t)) == sgn(s) * sgn(t) for s in
    S5 for t in S5))
print("k-cycles have sign (-1)^(k-1):", all(sgn(cycle(list(range(k))), 5)) == (-1) **
    (k - 1) for k in range(1, 6)))

```

## 6.3.8 Exercises

### 6.3.8.1 A. Check your understanding

#### Exercise A1.

- Define an inversion of  $\sigma \in S_n$  and the sign  $\text{sgn } \sigma$ .
- True or false: every 3-cycle is odd. Justify your answer.
- True or false: some permutation is a product of 5 transpositions and also a product of 8 transpositions. Justify your answer.
- What is the kernel of  $\text{sgn}: S_n \rightarrow \{1, -1\}$ ? Is the set of odd permutations a subgroup?
- True or false:  $\text{sgn}(\sigma\tau) = \text{sgn}(\tau\sigma)$  for all  $\sigma, \tau \in S_n$ , even when  $\sigma\tau \neq \tau\sigma$ . Justify your answer.
- Let  $f$  be an alternating 3-linear form. Express  $f(\mathbf{v}_2, \mathbf{v}_3, \mathbf{v}_1)$  in terms of  $f(\mathbf{v}_1, \mathbf{v}_2, \mathbf{v}_3)$ .

*Solution.*

- An inversion of  $\sigma$  is a pair  $(i, j)$  with  $1 \leq i < j \leq n$  and  $\sigma(i) > \sigma(j)$  ([Definition 6.3.1](#));  $\text{sgn } \sigma = (-1)^{\text{inv}(\sigma)}$  ([Definition 6.3.2](#)).
- False. By [Proposition 6.3.4](#) a 3-cycle has sign  $(-1)^2 = 1$ , so it is even; for instance  $(1\ 2\ 3) = (1\ 3)(1\ 2)$ .
- False. By [Corollary 6.3.3](#) the numbers of transpositions in two such products have the same parity, and 5 and 8 do not.
- The kernel is the alternating group  $A_n$  of even permutations ([Definition 6.3.3](#)).

The odd permutations do not form a subgroup:  $\text{id}$  is even, so it is not among them.

- (e) True. By [Theorem 6.3.1](#) both sides equal  $\text{sgn } \sigma \cdot \text{sgn } \tau$ , and multiplication of integers is commutative. For example  $(1\ 3)(1\ 2) \neq (1\ 2)(1\ 3)$ , but both are 3-cycles, with sign 1.
- (f) Swapping the first two arguments of  $(\mathbf{v}_2, \mathbf{v}_3, \mathbf{v}_1)$  gives  $(\mathbf{v}_3, \mathbf{v}_2, \mathbf{v}_1)$ , and swapping the first and third then gives  $(\mathbf{v}_1, \mathbf{v}_2, \mathbf{v}_3)$ . Each swap costs a factor  $-1$  by [Theorem 6.2.1](#) (a), so  $f(\mathbf{v}_2, \mathbf{v}_3, \mathbf{v}_1) = f(\mathbf{v}_1, \mathbf{v}_2, \mathbf{v}_3)$ . This matches the sign of the rearrangement  $\sigma$  with bottom row 2, 3, 1, namely  $\sigma = (1\ 2\ 3)$ , which is even.

### 6.3.8.2 B. Practice

#### Exercise B1: Computing Signs.

Find the sign of each permutation, both by counting inversions and by using cycles.

- (a)  $\sigma \in S_4$  with two-line notation  $\begin{pmatrix} 1 & 2 & 3 & 4 \\ 2 & 4 & 1 & 3 \end{pmatrix}$ .
- (b)  $(1\ 2)(3\ 4\ 5) \in S_5$ .
- (c)  $(1\ 2\ 3)(2\ 3\ 4) \in S_4$ .

*Solution.*

- (a) *Inversions.* In the bottom row 2, 4, 1, 3, the entry 2 is followed by the smaller entry 1, the entry 4 by 1 and 3, and the entry 1 by nothing smaller. So  $\text{inv}(\sigma) = 1 + 2 + 0 = 3$  and  $\text{sgn } \sigma = -1$ . *Cycles.*  $1 \mapsto 2 \mapsto 4 \mapsto 3 \mapsto 1$ , so  $\sigma = (1\ 2\ 4\ 3)$ , a 4-cycle, with sign  $(-1)^3 = -1$  by [Proposition 6.3.4](#).
- (b) *Cycles.* By [Proposition 6.3.4](#) and [Theorem 6.3.1](#), the sign is  $(-1) \cdot (-1)^2 = -1$ . *Inversions.* The permutation sends  $1 \mapsto 2$ ,  $2 \mapsto 1$ ,  $3 \mapsto 4$ ,  $4 \mapsto 5$ ,  $5 \mapsto 3$ , so the bottom row is 2, 1, 4, 5, 3. The inversions are (1, 2), (3, 5) and (4, 5), so there are 3, and the sign is  $-1$ .
- (c) *Cycles.* Both factors are 3-cycles, so the sign is  $1 \cdot 1 = 1$ . *Inversions.* Apply (2 3 4) first:  $1 \mapsto 1 \mapsto 2$ ,  $2 \mapsto 3 \mapsto 1$ ,  $3 \mapsto 4 \mapsto 4$ ,  $4 \mapsto 2 \mapsto 3$ . The bottom row is 2, 1, 4, 3, with inversions (1, 2) and (3, 4). So the sign is  $(-1)^2 = 1$ , and in fact the product is  $(1\ 2)(3\ 4)$ .

#### Exercise B2: Sign of the Inverse.

Prove that  $\text{sgn}(\sigma^{-1}) = \text{sgn } \sigma$  for every  $\sigma \in S_n$ .

*Solution.* Let  $\sigma \in S_n$ . By [Theorem 6.3.1](#),  $\text{sgn } \sigma \cdot \text{sgn}(\sigma^{-1}) = \text{sgn}(\sigma\sigma^{-1}) = \text{sgn}(\text{id}) = 1$ , since  $\text{id}$  has no inversions. Since  $\text{sgn } \sigma \in \{1, -1\}$  and  $(\pm 1)^{-1} = \pm 1$ , multiplying both sides by  $\text{sgn } \sigma$  gives  $\text{sgn}(\sigma^{-1}) = \text{sgn } \sigma$ .

#### Exercise B3: The Reversal Permutation.



Let  $\rho \in S_n$  be the reversal,  $\rho(i) = n + 1 - i$ . Show that  $\text{inv}(\rho) = n(n-1)/2$ . Hence find the  $n$  for which  $\rho$  is even, and decide the sign of  $\rho$  for  $n = 6$  and  $n = 9$ .

*Solution.* For  $1 \leq i < j \leq n$ ,  $\rho(i) = n + 1 - i > n + 1 - j = \rho(j)$ . So **every** pair  $i < j$  is an inversion, and  $\text{inv}(\rho)$  is the number of 2-element subsets of  $\{1, \dots, n\}$ , which is  $n(n-1)/2$ . Hence  $\text{sgn } \rho = (-1)^{n(n-1)/2}$ .

Write  $n = 4q + t$  with  $t \in \{0, 1, 2, 3\}$ . Then  $n(n-1)/2$  is  $2q(4q-1)$  for  $t = 0$ ,  $2q(4q+1)$  for  $t = 1$ ,  $(4q+1)(2q+1)$  for  $t = 2$ , and  $(2q+1)(4q+3)$  for  $t = 3$ . The first two are even and the last two are odd. So  $\rho$  is even exactly when  $n$  leaves remainder 0 or 1 on division by 4. For  $n = 6$ ,  $\text{inv}(\rho) = 15$  and  $\rho$  is odd; for  $n = 9$ ,  $\text{inv}(\rho) = 36$  and  $\rho$  is even.

### 6.3.8.3 C. Going deeper

#### Exercise C1: Half the Permutations Are Even.

Let  $n \geq 2$ . Prove that exactly  $n!/2$  permutations in  $S_n$  are even.

*Hint:* consider  $\sigma \mapsto (1\ 2)\sigma$ .

*Solution.* Let  $E$  and  $O$  be the sets of even and odd permutations; they are disjoint and  $E \cup O = S_n$ , so  $|E| + |O| = n!$ . Let  $\tau = (1\ 2)$ , which exists since  $n \geq 2$ , and define  $L: S_n \rightarrow S_n$ ,  $L(\sigma) = \tau\sigma$ . By [Theorem 6.3.1](#) and [Corollary 6.3.2](#),  $\text{sgn } L(\sigma) = -\text{sgn } \sigma$ , so  $L$  maps  $E$  into  $O$  and  $O$  into  $E$ . Since  $\tau\tau = \text{id}$ ,  $L(L(\sigma)) = \sigma$ , so  $L$  restricted to  $E$  is a bijection  $E \rightarrow O$  with inverse  $L$  restricted to  $O$  ([Theorem 0.4.7](#)). Hence  $|E| = |O| = n!/2$ .

#### Exercise C2: Sign from the Number of Cycles.

Suppose  $\sigma \in S_n$  is written as a product  $\sigma = \gamma_1\gamma_2 \cdots \gamma_c$  of **disjoint** cycles (no element appears in two of them) whose lengths  $k_1, \dots, k_c$  add up to  $n$ , where fixed points are included as cycles of length 1. (Such a product always exists: follow each element around its loop, as in [Example \(Computing with cycles\)](#).)

- Prove that  $\text{sgn } \sigma = (-1)^{n-c}$ .
- Use (a) to find the sign of  $\sigma = (1\ 4\ 6)(2\ 5) \in S_6$ , and compare with [Example \(A Sign Computed Two Ways\)](#).
- Deduce that an  $n$ -cycle in  $S_n$  and the identity in  $S_n$  have the same sign exactly when  $n$  is odd.

*Solution.*

- By [Theorem 6.3.1](#) and [Proposition 6.3.4](#),

$$\text{sgn } \sigma = \prod_{m=1}^c \text{sgn } \gamma_m = \prod_{m=1}^c (-1)^{k_m-1} = (-1)^{(k_1+\cdots+k_c)-c} = (-1)^{n-c}.$$

- Including the fixed point 3 as the cycle (3),  $\sigma = (1\ 4\ 6)(2\ 5)(3)$ , with lengths  $3 + 2 + 1 = 6$  and  $c = 3$ . So  $\text{sgn } \sigma = (-1)^{6-3} = -1$ , as in [Example \(A Sign Computed Two Ways\)](#).
- An  $n$ -cycle has  $c = 1$  and sign  $(-1)^{n-1}$ ; the identity consists of  $n$  cycles of length

1, so  $c = n$  and its sign is  $(-1)^0 = 1$ . These agree exactly when  $n - 1$  is even, that is, when  $n$  is odd.

## 6.4 Existence and Uniqueness of the Determinant

In the first section of this chapter, signed area obeyed three rules: it is linear in each side, it vanishes when two sides are equal, and the unit square has area 1. In the second section we named functions obeying the first two rules, alternating multilinear forms, and in the third we built the sign of a permutation. This section puts the pieces together. We prove that on  $F^n$  there is **exactly one** alternating  $n$ -linear form that takes the value 1 on the standard basis, and we find a formula for it. That form is the determinant.

### 6.4.1 The goal and the plan

Here is the statement we are after.

There is exactly one function  $D: F^n \times \cdots \times F^n \rightarrow F$  (with  $n$  factors) that is  $n$ -linear (Definition 6.2.1), alternating (Definition 6.2.2), and satisfies  $D(\mathbf{e}_1, \dots, \mathbf{e}_n) = 1$ .

It has two halves of a very different nature. **Uniqueness** says: if such a  $D$  exists, its values are forced. **Existence** says: some function really has all three properties. We prove uniqueness first, because its proof is how we *find* the formula. Suppose  $D$  exists, expand everything using the rules, and see what is left. The formula that comes out is then the only candidate, and existence reduces to checking that the candidate obeys the rules.

Throughout, a list of  $n$  columns  $\mathbf{a}_1, \dots, \mathbf{a}_n \in F^n$  and the matrix  $\mathbf{A} = (\mathbf{a}_1 \ \cdots \ \mathbf{a}_n) \in M_n(F)$  carry the same information, and we pass between them freely. We write  $a_{ij}$  for the  $i$ -th entry of  $\mathbf{a}_j$ , so that

$$\mathbf{a}_j = a_{1j}\mathbf{e}_1 + a_{2j}\mathbf{e}_2 + \cdots + a_{nj}\mathbf{e}_n = \sum_{i=1}^n a_{ij}\mathbf{e}_i.$$

### 6.4.2 Rearranging the arguments

Expanding a form on basis vectors will produce values like  $f(\mathbf{e}_3, \mathbf{e}_1, \mathbf{e}_2)$ : the standard basis in a scrambled order. Alternating forms handle scrambling by a sign, and the sign is exactly the one built from inversions.

#### Lemma 6.4.1 — Permuting the arguments of an alternating form

Let  $f$  be an alternating  $n$ -linear form on a vector space  $V$  over  $F$ , let  $\mathbf{v}_1, \dots, \mathbf{v}_n \in V$ ,

and let  $\sigma \in S_n$ . Then

$$f(\mathbf{v}_{\sigma(1)}, \mathbf{v}_{\sigma(2)}, \dots, \mathbf{v}_{\sigma(n)}) = \text{sgn}(\sigma) f(\mathbf{v}_1, \mathbf{v}_2, \dots, \mathbf{v}_n).$$

*Idea.* A transposition swaps two arguments, and each swap costs a factor  $-1$  by [Theorem 6.2.1](#). Every permutation is a product of transpositions, so the value picks up  $(-1)^k$  for a product of  $k$  of them. Multiplicativity of the sign says that  $(-1)^k$  is  $\text{sgn}(\sigma)$ , whichever product we chose.

*Proof.* If  $n = 1$ , then  $S_1 = \{\text{id}\}$  and  $\text{sgn}(\text{id}) = 1$ , since the identity has no inversions ([Definition 6.3.2](#)); there is nothing to prove. Let  $n \geq 2$ . For  $\pi \in S_n$ , write  $f_\pi = f(\mathbf{v}_{\pi(1)}, \dots, \mathbf{v}_{\pi(n)})$ .

First let  $\pi \in S_n$  and let  $\tau = (p \ q)$  be a transposition. The  $j$ -th argument of  $f_{\pi\tau}$  is  $\mathbf{v}_{\pi(\tau(j))}$ . For  $j \neq p, q$  this is  $\mathbf{v}_{\pi(j)}$ ; for  $j = p$  it is  $\mathbf{v}_{\pi(q)}$ ; for  $j = q$  it is  $\mathbf{v}_{\pi(p)}$ . So the arguments of  $f_{\pi\tau}$  are those of  $f_\pi$  with positions  $p$  and  $q$  exchanged. By [Theorem 6.2.1](#), swapping two arguments changes the sign, so

$$f_{\pi\tau} = -f_\pi. \quad (*)$$

Now let  $\sigma \in S_n$ . By [Theorem 0.10.4](#),  $\sigma = \tau_1 \tau_2 \cdots \tau_k$  for some transpositions  $\tau_1, \dots, \tau_k$  with  $k \geq 1$ . Put  $\pi_0 = \text{id}$  and  $\pi_m = \tau_1 \cdots \tau_m$ , so that  $\pi_m = \pi_{m-1} \tau_m$ . Applying  $(*)$   $k$  times,

$$f_\sigma = f_{\pi_k} = -f_{\pi_{k-1}} = \cdots = (-1)^k f_{\pi_0} = (-1)^k f(\mathbf{v}_1, \dots, \mathbf{v}_n).$$

On the other hand, by [Theorem 6.3.1](#) and [Corollary 6.3.2](#),  $\text{sgn}(\sigma) = \text{sgn}(\tau_1) \cdots \text{sgn}(\tau_k) = (-1)^k$ . This proves the lemma. ■

The proof chose a factorization of  $\sigma$  into transpositions, and different choices can have different lengths. The answer does not depend on the choice, because only the parity of  $k$  enters, and [Corollary 6.3.3](#) says the parity is determined by  $\sigma$ . In the proof this is absorbed into the equation  $(-1)^k = \text{sgn}(\sigma)$ .

### 6.4.3 Uniqueness

We now expand an alternating form completely. It helps to see the case  $n = 2$  first.

#### Theorem 6.4.2 — Uniqueness of Alternating Forms

Let  $f$  be an alternating  $n$ -linear form on  $F^n$ , and let  $\mathbf{a}_1, \dots, \mathbf{a}_n \in F^n$  have entries  $a_{ij}$  as above. Then

$$f(\mathbf{a}_1, \dots, \mathbf{a}_n) = f(\mathbf{e}_1, \dots, \mathbf{e}_n) \sum_{\sigma \in S_n} \text{sgn}(\sigma) a_{\sigma(1)1} a_{\sigma(2)2} \cdots a_{\sigma(n)n}.$$

In particular, an alternating  $n$ -linear form on  $F^n$  is determined by the single value  $f(\mathbf{e}_1, \dots, \mathbf{e}_n)$ .

*Idea.* For  $n = 2$ , write  $\mathbf{a}_1 = a_{11}\mathbf{e}_1 + a_{21}\mathbf{e}_2$  and  $\mathbf{a}_2 = a_{12}\mathbf{e}_1 + a_{22}\mathbf{e}_2$ . Linearity in the first argument, then in the second, gives four terms:

$$\begin{aligned} f(\mathbf{a}_1, \mathbf{a}_2) &= a_{11}f(\mathbf{e}_1, \mathbf{a}_2) + a_{21}f(\mathbf{e}_2, \mathbf{a}_2) \\ &= a_{11}a_{12}f(\mathbf{e}_1, \mathbf{e}_1) + a_{11}a_{22}f(\mathbf{e}_1, \mathbf{e}_2) \\ &\quad + a_{21}a_{12}f(\mathbf{e}_2, \mathbf{e}_1) + a_{21}a_{22}f(\mathbf{e}_2, \mathbf{e}_2). \end{aligned}$$

The first and last terms vanish because two arguments are equal, and  $f(\mathbf{e}_2, \mathbf{e}_1) = -f(\mathbf{e}_1, \mathbf{e}_2)$ . What is left is  $(a_{11}a_{22} - a_{21}a_{12})f(\mathbf{e}_1, \mathbf{e}_2)$ , the formula for  $n = 2$ .

In general the four terms become one term for each way of choosing, for every argument  $j$ , which basis vector  $\mathbf{e}_{\varphi(j)}$  to take from  $\mathbf{a}_j$ . A choice is a function  $\varphi: \{1, \dots, n\} \rightarrow \{1, \dots, n\}$ , and there are  $n^n$  of them. The ones that pick some basis vector twice die. The survivors pick every basis vector once, so they are permutations, and [Lemma 6.4.1](#) converts each into a sign.

*Proof.* Write  $[k] = \{1, \dots, k\}$  for  $k \geq 1$ . We first expand the arguments one at a time.

*Claim.* For each  $k \in \{0, 1, \dots, n\}$ ,

$$f(\mathbf{a}_1, \dots, \mathbf{a}_n) = \sum_{\varphi: [k] \rightarrow [n]} a_{\varphi(1)1} \cdots a_{\varphi(k)k} f(\mathbf{e}_{\varphi(1)}, \dots, \mathbf{e}_{\varphi(k)}, \mathbf{a}_{k+1}, \dots, \mathbf{a}_n),$$

where the sum runs over **all** functions  $\varphi: [k] \rightarrow [n]$ . For  $k = 0$  we read the right side as the single term  $f(\mathbf{a}_1, \dots, \mathbf{a}_n)$ .

*Proof.* We use induction on  $k$ . For  $k = 0$  both sides are  $f(\mathbf{a}_1, \dots, \mathbf{a}_n)$ . Suppose the claim holds for some  $k < n$ . In each term, the argument in position  $k + 1$  is  $\mathbf{a}_{k+1} = \sum_{i=1}^n a_{i,k+1}\mathbf{e}_i$ . Since  $f$  is linear in argument  $k + 1$ ,

$$f(\mathbf{e}_{\varphi(1)}, \dots, \mathbf{e}_{\varphi(k)}, \mathbf{a}_{k+1}, \dots, \mathbf{a}_n) = \sum_{i=1}^n a_{i,k+1} f(\mathbf{e}_{\varphi(1)}, \dots, \mathbf{e}_{\varphi(k)}, \mathbf{e}_i, \mathbf{a}_{k+2}, \dots, \mathbf{a}_n).$$

Substituting this into the induction hypothesis gives a sum over all pairs  $(\varphi, i)$  with  $\varphi: [k] \rightarrow [n]$  and  $i \in [n]$ . Such a pair is the same thing as a function  $\psi: [k+1] \rightarrow [n]$ , namely  $\psi(j) = \varphi(j)$  for  $j \leq k$  and  $\psi(k+1) = i$ ; and each  $\psi$  arises from exactly one pair. The term for the pair  $(\varphi, i)$  is  $a_{\psi(1)1} \cdots a_{\psi(k+1),k+1} f(\mathbf{e}_{\psi(1)}, \dots, \mathbf{e}_{\psi(k+1)}, \mathbf{a}_{k+2}, \dots, \mathbf{a}_n)$ . This is the claim for  $k + 1$ . ■

Taking  $k = n$  in the claim,

$$f(\mathbf{a}_1, \dots, \mathbf{a}_n) = \sum_{\varphi: [n] \rightarrow [n]} a_{\varphi(1)1} \cdots a_{\varphi(n)n} f(\mathbf{e}_{\varphi(1)}, \dots, \mathbf{e}_{\varphi(n)}). \quad (**)$$

We sort the functions  $\varphi$  into two kinds.

*Case 1:  $\varphi$  is not injective.* Then  $\varphi(p) = \varphi(q)$  for some  $p \neq q$ , so the arguments in positions  $p$  and  $q$  of  $f(\mathbf{e}_{\varphi(1)}, \dots, \mathbf{e}_{\varphi(n)})$  are equal. Since  $f$  is alternating, this value is 0, and the whole term is 0.

*Case 2:  $\varphi$  is injective.* Since  $[n]$  is finite,  $\varphi$  is bijective by [Theorem 0.4.9](#), so  $\varphi = \sigma$  for some  $\sigma \in S_n$ . By [Lemma 6.4.1](#) with  $\mathbf{v}_i = \mathbf{e}_i$ ,

$$f(\mathbf{e}_{\sigma(1)}, \dots, \mathbf{e}_{\sigma(n)}) = \operatorname{sgn}(\sigma) f(\mathbf{e}_1, \dots, \mathbf{e}_n).$$

Hence only the terms of Case 2 survive in  $(**)$ , there is exactly one for each  $\sigma \in S_n$ , and

$$f(\mathbf{a}_1, \dots, \mathbf{a}_n) = \sum_{\sigma \in S_n} a_{\sigma(1)1} \cdots a_{\sigma(n)n} \operatorname{sgn}(\sigma) f(\mathbf{e}_1, \dots, \mathbf{e}_n),$$

which is the stated formula after moving the scalar  $f(\mathbf{e}_1, \dots, \mathbf{e}_n)$  to the front. This proves the theorem. ■

So the three rules leave no freedom at all. If a normalized alternating form exists, it **must** be

$$\mathbf{a}_1, \dots, \mathbf{a}_n \longmapsto \sum_{\sigma \in S_n} \operatorname{sgn}(\sigma) a_{\sigma(1)1} \cdots a_{\sigma(n)n}.$$

Out of  $n^n$  terms, the rules killed all but  $n!$ . For  $n = 3$  that is 6 out of 27.

## 6.4.4 Existence

The candidate has been found. Now we check it against the rules, forgetting where it came from.

### Theorem 6.4.3 — Existence: the Leibniz Formula

Define  $D: F^n \times \cdots \times F^n \rightarrow F$  by

$$D(\mathbf{a}_1, \dots, \mathbf{a}_n) = \sum_{\sigma \in S_n} \operatorname{sgn}(\sigma) a_{\sigma(1)1} a_{\sigma(2)2} \cdots a_{\sigma(n)n},$$

where  $a_{ij}$  is the  $i$ -th entry of  $\mathbf{a}_j$ . Then  $D$  is an alternating  $n$ -linear form on  $F^n$ , and  $D(\mathbf{e}_1, \dots, \mathbf{e}_n) = 1$ .

*Idea.* Linearity is visible: each product takes exactly one entry from each column. For alternation, suppose columns  $p$  and  $q$  are equal. Pair each  $\sigma$  with  $\sigma(p \ q)$ . The two permutations have opposite signs. Their products use the same entries, except that the entries taken from columns  $p$  and  $q$  trade places, and those two columns are equal. So each pair contributes  $x - x = 0$ .

Why pair terms instead of arguing “swapping the equal columns changes  $D$  to  $-D$ , so  $D = -D$  and  $D = 0$ ”? Because  $D = -D$  gives  $2D = 0$ , and over a field with  $1 + 1 = 0$ , such as  $\mathbb{F}_2$ , that says nothing. The pairing argument never divides by 2.

*Proof. Multilinearity.* Fix  $k \in [n]$  and fix all columns except the  $k$ -th. For a fixed  $\sigma$ , the product  $a_{\sigma(1)1} \cdots a_{\sigma(n)n}$  contains exactly one entry of  $\mathbf{a}_k$ , namely  $a_{\sigma(k)k}$ , the  $\sigma(k)$ -th coordinate of  $\mathbf{a}_k$ . So the term for  $\sigma$  equals  $c_\sigma a_{\sigma(k)k}$ , where  $c_\sigma = \operatorname{sgn}(\sigma) \prod_{j \neq k} a_{\sigma(j)j}$  does not depend on  $\mathbf{a}_k$ . The map  $\mathbf{a}_k \mapsto a_{\sigma(k)k}$  is linear, since coordinates of a sum are sums of coordinates and coordinates of  $c\mathbf{a}_k$  are  $c$  times the coordinates. So each term is linear

in  $\mathbf{a}_k$ , and hence so is their sum  $D$ .

*Alternation.* Suppose  $\mathbf{a}_p = \mathbf{a}_q$  with  $p < q$ , that is,  $a_{ip} = a_{iq}$  for all  $i$ . Let  $\tau = (p \ q)$ , let  $A_n = \{\sigma \in S_n : \text{sgn}(\sigma) = 1\}$  be the even permutations (Definition 6.3.3), so that  $S_n \setminus A_n = \{\sigma \in S_n : \text{sgn}(\sigma) = -1\}$  consists of the odd ones. Since  $\text{sgn}(\sigma\tau) = \text{sgn}(\sigma)\text{sgn}(\tau) = -\text{sgn}(\sigma)$  by Theorem 6.3.1 and Corollary 6.3.2, the map  $\sigma \mapsto \sigma\tau$  sends  $A_n$  to  $S_n \setminus A_n$ . It is a bijection  $A_n \rightarrow S_n \setminus A_n$ , because  $\rho \mapsto \rho\tau$  sends  $S_n \setminus A_n$  back to  $A_n$  and  $\sigma\tau\tau = \sigma$ . Every  $\sigma \in S_n$  has sign 1 or  $-1$ , so

$$D(\mathbf{a}_1, \dots, \mathbf{a}_n) = \sum_{\sigma \in A_n} \left( \prod_{j=1}^n a_{\sigma(j)j} - \prod_{j=1}^n a_{\sigma\tau(j)j} \right).$$

Fix  $\sigma \in A_n$  and compare the two products factor by factor. For  $j \neq p, q$ ,  $\sigma\tau(j) = \sigma(j)$ , so the factors agree. For  $j = p$ , the second product has  $a_{\sigma(q)p} = a_{\sigma(q)q}$ , which is the factor of the first product at  $j = q$ . For  $j = q$ , the second product has  $a_{\sigma(p)q} = a_{\sigma(p)p}$ , the factor of the first product at  $j = p$ . So the two products consist of the same  $n$  factors in a different order, and they are equal because multiplication in  $F$  is commutative. Every bracket is 0, and  $D(\mathbf{a}_1, \dots, \mathbf{a}_n) = 0$ .

*Normalization.* For  $\mathbf{a}_j = \mathbf{e}_j$ ,  $a_{ij}$  is 1 if  $i = j$  and 0 otherwise. The product  $a_{\sigma(1)1} \cdots a_{\sigma(n)n}$  is therefore 1 if  $\sigma(j) = j$  for all  $j$ , and 0 otherwise. Only  $\sigma = \text{id}$  contributes, and  $\text{sgn}(\text{id}) = 1$ . Hence  $D(\mathbf{e}_1, \dots, \mathbf{e}_n) = 1$ . This proves the theorem. ■

Uniqueness and existence together answer the question of the first section: the rules of signed area single out exactly one function.

## 6.4.5 The determinant

We now have a function of square matrices that is forced on us by three reasonable rules, and it deserves a name. It is the central notion of this chapter.

*The determinant is the only way to attach a number to  $n$  vectors in  $F^n$  that is linear in each vector, vanishes when two vectors coincide, and gives 1 on the standard basis.*

### Definition 6.4.1 — Determinant

Let  $n \geq 1$  and  $\mathbf{A} = (a_{ij}) \in M_n(F)$ . The **determinant** of  $\mathbf{A}$  is

$$\det \mathbf{A} := \sum_{\sigma \in S_n} \text{sgn}(\sigma) a_{\sigma(1)1} a_{\sigma(2)2} \cdots a_{\sigma(n)n} \in F.$$

For columns  $\mathbf{a}_1, \dots, \mathbf{a}_n \in F^n$  we also write  $\det(\mathbf{a}_1, \dots, \mathbf{a}_n)$  for the determinant of the matrix  $(\mathbf{a}_1 \ \cdots \ \mathbf{a}_n)$ .

In words: run over **all**  $n!$  permutations  $\sigma$ . For each, pick from column  $j$  the entry in row  $\sigma(j)$ ; this picks one entry in every column and, because  $\sigma$  is a bijection, one entry in every row. Multiply the  $n$  picked entries, attach the sign of  $\sigma$ , and add up. The determinant is defined **only for square matrices**, and its value lies in the field  $F$  of the entries.

**Well-definedness.** The formula is a finite sum, so it always gives a number. What the name “**the** determinant” really claims is that nothing else satisfies the three rules, and that is [Theorem 6.4.2](#); [Theorem 6.4.3](#) shows  $\det$  does satisfy them. The sign  $\text{sgn}(\sigma)$  is fixed by counting inversions ([Definition 6.3.2](#)), and by [Corollary 6.3.3](#) any factorization of  $\sigma$  into transpositions gives the same sign, so either method may be used to evaluate the formula.

### Examples.

- $n = 1$ .  $S_1 = \{\text{id}\}$ , so  $\det(a) = a$ . A degenerate case, but it is the base of every induction on size.
- $n = 2$ .  $S_2 = \{\text{id}, (1\ 2)\}$ , with signs 1 and  $-1$ . The identity picks  $a_{11}a_{22}$ , and  $(1\ 2)$  picks  $a_{21}a_{12}$ . So

$$\det \begin{pmatrix} a & b \\ c & d \end{pmatrix} = ad - cb,$$

the value derived from the rules in [Example \( \$ad - bc\$  from the Rules\)](#), and the number whose non-vanishing decides invertibility in [Theorem 0.9.8](#).

- $n = 3$ . The six permutations of  $S_3$  and their signs (a 3-cycle has sign 1, a transposition  $-1$ ) give

$$\begin{aligned} \det \mathbf{A} = & a_{11}a_{22}a_{33} - a_{21}a_{12}a_{33} - a_{31}a_{22}a_{13} - a_{11}a_{32}a_{23} \\ & + a_{21}a_{32}a_{13} + a_{31}a_{12}a_{23}, \end{aligned}$$

the terms coming from  $\text{id}, (1\ 2), (1\ 3), (2\ 3), (1\ 2\ 3), (1\ 3\ 2)$  in this order. For instance  $\sigma = (1\ 2\ 3)$  has  $\sigma(1) = 2, \sigma(2) = 3, \sigma(3) = 1$ , and picks  $a_{21}a_{32}a_{13}$ .

- **A numerical  $3 \times 3$ .** For  $\mathbf{A} = \begin{pmatrix} 2 & 1 & 3 \\ 0 & 4 & 1 \\ 5 & 2 & 1 \end{pmatrix}$ , the six terms in the same order are 8,  $-0$ ,  $-60$ ,  $-4$ , 0 and 5, so  $\det \mathbf{A} = 8 - 60 - 4 + 5 = -51$ .
- **Identity and zero.**  $\det \mathbf{I}_n = 1$  by the normalization in [Theorem 6.4.3](#), and  $\det 0 = 0$ , since every product contains a zero factor.

**Non-example by minimal change.** Delete the sign. The **permanent**  $\text{per } \mathbf{A} = \sum_{\sigma} a_{\sigma(1)1} \cdots a_{\sigma(n)n}$  is still  $n$ -linear in the columns, by the same argument as in [Theorem 6.4.3](#), and still gives 1 at  $\mathbf{I}_n$ . But it is not alternating:  $\text{per} \begin{pmatrix} 1 & 1 \\ 1 & 1 \end{pmatrix} = 1 + 1 = 2$ , which is non-zero over  $\mathbb{R}$  although the two columns are equal. The sign is exactly what makes the pairing argument cancel.

**Why this definition.** Each rule has a job. Without normalization, the uniqueness theorem only pins the form down up to the scalar  $f(\mathbf{e}_1, \dots, \mathbf{e}_n)$ . Without alternation, the permanent and many other functions qualify. Multilinearity is what let us expand at all. The choice of **columns** rather than rows is a convention: we show below that the transpose has the same determinant, so rows would give the same function. We have avoided “Sarrus’s rule” on purpose. It is a picture of the six-term formula for  $n = 3$  and nothing more, and for  $n = 4$  its diagonal pattern would produce 8 products while the Leibniz formula has 24.

*Warning.* **The determinant is not linear.** It is linear in **each column separately**, not in the matrix. For  $n = 2$ ,  $\det \mathbf{I}_2 = 1$  but  $\det(\mathbf{I}_2 + \mathbf{I}_2) = \det(2\mathbf{I}_2) = 4 \neq 1 + 1$ . In general each product in the Leibniz formula has  $n$  factors, and scaling  $\mathbf{A}$  scales every factor, so

$$\det(c\mathbf{A}) = c^n \det \mathbf{A}, \quad \text{in particular} \quad \det(-\mathbf{A}) = (-1)^n \det \mathbf{A}.$$

So  $\det(-\mathbf{A}) = \det \mathbf{A}$  for  $2 \times 2$  matrices, not  $-\det \mathbf{A}$ .

*Quick check.* Which permutations give a non-zero term in the determinant of  $\begin{pmatrix} 0 & 0 & 1 \\ 0 & 2 & 0 \\ 3 & 0 & 0 \end{pmatrix}$ , and what is the determinant?

*Solution.* Column 1 has its only non-zero entry in row 3, column 2 in row 2, and column 3 in row 1. A non-zero term needs  $a_{\sigma(j)j} \neq 0$  for every  $j$ , so  $\sigma(1) = 3$ ,  $\sigma(2) = 2$ ,  $\sigma(3) = 1$ , that is,  $\sigma = (1\ 3)$ , of sign  $-1$ . The determinant is  $-3 \cdot 2 \cdot 1 = -6$ .

The payoff for the forms of the second section is immediate. There we saw that the alternating  $n$ -linear forms on  $F^n$  form a vector space (Theorem 6.2.3) and promised its dimension.

### Corollary 6.4.4 — Alternating $n$ -Forms on $F^n$ Form a Line

The vector space of alternating  $n$ -linear forms on  $F^n$  has dimension 1, with basis  $(\det)$ . Explicitly, every alternating  $n$ -linear form  $f$  on  $F^n$  satisfies

$$f(\mathbf{a}_1, \dots, \mathbf{a}_n) = f(\mathbf{e}_1, \dots, \mathbf{e}_n) \det(\mathbf{a}_1, \dots, \mathbf{a}_n).$$

*Proof.* By Theorem 6.4.3,  $\det$  is an element of this space, and it is not the zero form since  $\det(\mathbf{e}_1, \dots, \mathbf{e}_n) = 1$ . So the list  $(\det)$  is linearly independent. By Theorem 6.4.2 and Definition 6.4.1, every  $f$  in the space satisfies the displayed formula, so  $f = f(\mathbf{e}_1, \dots, \mathbf{e}_n) \det$  lies in the span of  $\det$ . Hence  $(\det)$  is a basis, and the dimension is 1. ■

This corollary is the most useful form of uniqueness. To prove that some function  $g$  of  $n$  columns is a multiple of the determinant, check that  $g$  is multilinear and alternating, and compute  $g(\mathbf{e}_1, \dots, \mathbf{e}_n)$ . The multiplicativity of the determinant in the next section is proved exactly this way.

## 6.4.6 Transposes and triangular matrices

The definition read the matrix column by column. Reading it row by row gives nothing new.

### Theorem 6.4.5 — Determinant of the Transpose

For every  $\mathbf{A} \in M_n(F)$ ,  $\det \mathbf{A}^\top = \det \mathbf{A}$ . Consequently,  $\det \mathbf{A}$  is also an alternating  $n$ -linear function of the **rows** of  $\mathbf{A}$ , with value 1 at  $\mathbf{I}_n$ .

*Idea.* A term of  $\det \mathbf{A}^\top$  picks the entries  $a_{j\sigma(j)}$ , one in each row. The same  $n$  positions are also “one in each column”: position  $(j, \sigma(j))$  is  $(\sigma^{-1}(i), i)$  with  $i = \sigma(j)$ . So the term is the term of  $\det \mathbf{A}$  for  $\sigma^{-1}$ , and  $\sigma^{-1}$  has the same sign as  $\sigma$ .

*Proof.* The  $(i, j)$ -entry of  $\mathbf{A}^\top$  is  $a_{ji}$  (Definition 0.9.10), so by Definition 6.4.1

$$\det \mathbf{A}^\top = \sum_{\sigma \in S_n} \text{sgn}(\sigma) a_{1\sigma(1)} a_{2\sigma(2)} \cdots a_{n\sigma(n)}.$$



Fix  $\sigma$ . As  $j$  runs through  $[n]$ ,  $i = \sigma(j)$  runs through  $[n]$  exactly once, and  $j = \sigma^{-1}(i)$ . Since multiplication in  $F$  is commutative, we may reorder the factors by  $i$ :

$$\prod_{j=1}^n a_{j\sigma(j)} = \prod_{i=1}^n a_{\sigma^{-1}(i)i}.$$

Next,  $\operatorname{sgn}(\sigma) \operatorname{sgn}(\sigma^{-1}) = \operatorname{sgn}(\sigma\sigma^{-1}) = \operatorname{sgn}(\operatorname{id}) = 1$  by [Theorem 6.3.1](#), and both signs lie in  $\{1, -1\}$ , so  $\operatorname{sgn}(\sigma^{-1}) = \operatorname{sgn}(\sigma)$ . Therefore

$$\det \mathbf{A}^\top = \sum_{\sigma \in S_n} \operatorname{sgn}(\sigma^{-1}) \prod_{i=1}^n a_{\sigma^{-1}(i)i} = \sum_{\rho \in S_n} \operatorname{sgn}(\rho) \prod_{i=1}^n a_{\rho(i)i} = \det \mathbf{A},$$

where the second equality substitutes  $\rho = \sigma^{-1}$ ; this is allowed because  $\sigma \mapsto \sigma^{-1}$  is a bijection of  $S_n$  onto itself, being its own inverse.

For the second statement, let  $\mathbf{r}_1, \dots, \mathbf{r}_n$  be the rows of  $\mathbf{A}$ . The columns of  $\mathbf{A}^\top$  are  $\mathbf{r}_1^\top, \dots, \mathbf{r}_n^\top$ , so  $\det \mathbf{A} = \det \mathbf{A}^\top = \det(\mathbf{r}_1^\top, \dots, \mathbf{r}_n^\top)$ . The map  $\mathbf{r} \mapsto \mathbf{r}^\top$  is linear ([Theorem 0.9.5](#)), so by [Theorem 6.4.3](#) the function  $(\mathbf{r}_1, \dots, \mathbf{r}_n) \mapsto \det(\mathbf{r}_1^\top, \dots, \mathbf{r}_n^\top)$  is linear in each  $\mathbf{r}_i$ . If  $\mathbf{r}_p = \mathbf{r}_q$  for some  $p \neq q$ , then  $\mathbf{r}_p^\top = \mathbf{r}_q^\top$ , and the value is 0. Finally  $\mathbf{I}_n^\top = \mathbf{I}_n$ . This proves the theorem. ■

So every statement about columns has a twin about rows. In particular [Theorem 6.2.1](#) applies to rows: swapping two rows changes the sign of  $\det$ , and adding a multiple of one row to another does not change it. The next section turns this into a method of computation.

Before that, one class of matrices whose determinant can be read off directly. Recall that  $\mathbf{A}$  is upper triangular if  $a_{ij} = 0$  whenever  $i > j$  ([Definition 0.9.5](#)).

### Theorem 6.4.6 — Determinant of a Triangular Matrix

If  $\mathbf{A} \in M_n(F)$  is upper triangular or lower triangular, then

$$\det \mathbf{A} = a_{11} a_{22} \cdots a_{nn}.$$

*Idea.* Which permutations can pick only entries on or above the diagonal? Picking  $a_{\sigma(j)j}$  with  $\sigma(j) \leq j$  for every  $j$  forces  $\sigma = \operatorname{id}$ : the numbers  $\sigma(1), \dots, \sigma(n)$  are  $1, \dots, n$  in some order, so they have the same sum, and none of them is allowed to exceed its partner.

*Proof.* Suppose  $\mathbf{A}$  is upper triangular, and let  $\sigma \in S_n$  with  $\sigma \neq \operatorname{id}$ . We claim  $\sigma(j) > j$  for some  $j$ . Otherwise  $\sigma(j) \leq j$  for all  $j$ . Since  $\sigma$  is a bijection of  $[n]$ ,  $\sum_{j=1}^n \sigma(j) = \sum_{j=1}^n j$ , so  $\sum_{j=1}^n (j - \sigma(j)) = 0$  is a sum of non-negative integers equal to 0. Hence  $\sigma(j) = j$  for every  $j$ , contradicting  $\sigma \neq \operatorname{id}$ . For such  $j$ ,  $a_{\sigma(j)j} = 0$  by upper triangularity, so the term of  $\sigma$  is 0. Only the identity remains, and  $\det \mathbf{A} = \operatorname{sgn}(\operatorname{id}) a_{11} \cdots a_{nn} = a_{11} \cdots a_{nn}$ .

If  $\mathbf{A}$  is lower triangular, then  $\mathbf{A}^\top$  is upper triangular with the same diagonal, and  $\det \mathbf{A} = \det \mathbf{A}^\top = a_{11} \cdots a_{nn}$  by [Theorem 6.4.5](#). This proves the theorem. ■

For example,  $\det \begin{pmatrix} 2 & 7 & -1 \\ 0 & 3 & 5 \\ 0 & 0 & -4 \end{pmatrix} = 2 \cdot 3 \cdot (-4) = -24$ : six terms, five of which vanish. Diagonal matrices are a special case,  $\det \text{diag}(d_1, \dots, d_n) = d_1 \cdots d_n$ . Together with Chapter 2's result that an upper triangular matrix is invertible exactly when its diagonal entries are non-zero (Lemma 2.6.2), this already shows that for upper triangular matrices,  $\mathbf{A}$  is invertible if and only if  $\det \mathbf{A} \neq 0$ . The next section proves this for all square matrices.

## 6.4.7 Determinants over a commutative ring

The characteristic polynomial at the end of this chapter is the determinant of a matrix whose entries are **polynomials**, such as  $\begin{pmatrix} x-1 & -2 \\ -3 & x-4 \end{pmatrix}$ . Polynomials form no field, since  $x$  has no inverse in  $F[x]$ . So we record exactly what this section used.

*Remark. Commutative rings.* A **commutative ring** is a set  $R$  with an addition and a multiplication satisfying all the axioms of a field (Definition 0.7.1) except, possibly, the existence of multiplicative inverses (F8). The integers  $\mathbb{Z}$  and the polynomials  $F[x]$  are examples (Theorem 0.8.1). For  $\mathbf{A} \in M_n(R)$ , the Leibniz formula of Definition 6.4.1 still defines  $\det \mathbf{A} \in R$ , reading  $\text{sgn}(\sigma)r$  as  $r$  or  $-r$ .

**Every proof in this section uses only addition, subtraction and multiplication**, together with commutativity, and never a multiplicative inverse. So the following hold verbatim over any commutative ring  $R$ , with “linear in each argument” meaning additive and compatible with multiplication by elements of  $R$ : Lemma 6.4.1, Theorem 6.4.2, Theorem 6.4.3, Theorem 6.4.5 and Theorem 6.4.6. The only outside input is the swap rule of Theorem 6.2.1, and it too is division-free: expanding  $0 = f(\dots, \mathbf{u} + \mathbf{v}, \dots, \mathbf{u} + \mathbf{v}, \dots)$  by additivity leaves  $f(\dots, \mathbf{u}, \dots, \mathbf{v}, \dots) + f(\dots, \mathbf{v}, \dots, \mathbf{u}, \dots) = 0$ . The parts of Theorem 6.2.1 about dependent arguments are not used; they involve solving for one vector in terms of others, which needs division.

In the characteristic-polynomial section we will apply these results with  $R = F[x]$ . The next two sections flag, in the same way, which of their proofs survive the passage to a ring.

## 6.4.8 Exercises

### 6.4.8.1 A. Check your understanding

#### Exercise A1.

- State the three properties that characterize  $\det: M_n(F) \rightarrow F$ , and write down the Leibniz formula.
- How many terms does the Leibniz formula have for  $n = 4$ ? How many of the corresponding permutations have sign 1?
- True or false:  $\det(\mathbf{A} + \mathbf{B}) = \det \mathbf{A} + \det \mathbf{B}$  for all  $\mathbf{A}, \mathbf{B} \in M_2(\mathbb{R})$ . Justify your answer.
- True or false:  $\det(-\mathbf{A}) = -\det \mathbf{A}$  for all  $\mathbf{A} \in M_3(\mathbb{R})$ . Justify your answer.
- In the proof of Theorem 6.4.2, why does a term with a non-injective  $\varphi$  vanish?

- (f) Explain why the existence proof pairs  $\sigma$  with  $\sigma(p\ q)$  instead of arguing “ $D = -D$ , so  $D = 0$ ”.

*Solution.*

- (a)  $\det$  is  $n$ -linear in the columns, alternating (zero whenever two columns are equal), and  $\det \mathbf{I}_n = 1$ . By [Theorem 6.4.2](#) and [Theorem 6.4.3](#) it is the only such function, and  $\det \mathbf{A} = \sum_{\sigma \in S_n} \operatorname{sgn}(\sigma) a_{\sigma(1)1} \cdots a_{\sigma(n)n}$ .
- (b) One term per permutation:  $4! = 24$ . In the proof of [Theorem 6.4.3](#),  $\sigma \mapsto \sigma(1\ 2)$  is a bijection from the permutations of sign 1 to those of sign  $-1$ , so each set has 12 elements.
- (c) False. For  $\mathbf{A} = \mathbf{B} = \mathbf{I}_2$ ,  $\det(\mathbf{A} + \mathbf{B}) = \det(2\mathbf{I}_2) = 4$  but  $\det \mathbf{A} + \det \mathbf{B} = 2$ .
- (d) True. Each Leibniz product has 3 factors, each multiplied by  $-1$ , so  $\det(-\mathbf{A}) = (-1)^3 \det \mathbf{A} = -\det \mathbf{A}$ . (For  $n = 2$  it is false.)
- (e) If  $\varphi(p) = \varphi(q)$  with  $p \neq q$ , the form is evaluated with the equal arguments  $\mathbf{e}_{\varphi(p)}$  in positions  $p$  and  $q$ , and an alternating form vanishes there.
- (f)  $D = -D$  only gives  $2D = 0$ . Over a field in which  $2 = 0$ , such as  $\mathbb{F}_2$ , this does not imply  $D = 0$ . The pairing shows that the terms cancel in pairs, which works over every field.

### 6.4.8.2 B. Practice

#### Exercise B1: Six-term determinants.

Using the Leibniz formula, listing all six terms, compute the determinants of

$$\mathbf{A} = \begin{pmatrix} 3 & 1 & 2 \\ 1 & 0 & 4 \\ 2 & 5 & 1 \end{pmatrix}, \quad \mathbf{B} = \begin{pmatrix} 1 & -1 & 2 \\ 0 & 3 & 1 \\ 4 & 2 & -2 \end{pmatrix} \in M_3(\mathbb{R}).$$

Then compute  $\det \mathbf{A}^\top$  in the same way, and hence verify [Theorem 6.4.5](#) for this  $\mathbf{A}$ .

*Solution.* We use the order  $\text{id}, (1\ 2), (1\ 3), (2\ 3), (1\ 2\ 3), (1\ 3\ 2)$  and the six-term formula from the examples after [Definition 6.4.1](#):

$$a_{11}a_{22}a_{33}, -a_{21}a_{12}a_{33}, -a_{31}a_{22}a_{13}, -a_{11}a_{32}a_{23}, a_{21}a_{32}a_{13}, a_{31}a_{12}a_{23}.$$

For  $\mathbf{A}$ :  $3 \cdot 0 \cdot 1 = 0$ ;  $-1 \cdot 1 \cdot 1 = -1$ ;  $-2 \cdot 0 \cdot 2 = 0$ ;  $-3 \cdot 5 \cdot 4 = -60$ ;  $1 \cdot 5 \cdot 2 = 10$ ;  $2 \cdot 1 \cdot 4 = 8$ . Hence  $\det \mathbf{A} = 0 - 1 + 0 - 60 + 10 + 8 = -43$ .

For  $\mathbf{B}$ :  $1 \cdot 3 \cdot (-2) = -6$ ;  $-0 \cdot (-1) \cdot (-2) = 0$ ;  $-4 \cdot 3 \cdot 2 = -24$ ;  $-1 \cdot 2 \cdot 1 = -2$ ;  $0 \cdot 2 \cdot 2 = 0$ ;  $4 \cdot (-1) \cdot 1 = -4$ . Hence  $\det \mathbf{B} = -6 + 0 - 24 - 2 + 0 - 4 = -36$ .

For  $\mathbf{A}^\top = \begin{pmatrix} 3 & 1 & 2 \\ 1 & 0 & 5 \\ 2 & 4 & 1 \end{pmatrix}$ :  $3 \cdot 0 \cdot 1 = 0$ ;  $-1 \cdot 1 \cdot 1 = -1$ ;  $-2 \cdot 0 \cdot 2 = 0$ ;  $-3 \cdot 4 \cdot 5 = -60$ ;  $1 \cdot 4 \cdot 2 = 8$ ;  $2 \cdot 1 \cdot 5 = 10$ . The sum is  $-43 = \det \mathbf{A}$ , as [Theorem 6.4.5](#) predicts; the six terms are the same numbers, with the two 3-cycles exchanged, because  $(1\ 2\ 3)^{-1} = (1\ 3\ 2)$ .

**Exercise B2: The determinant of  $\mathbf{P}_\sigma$ .**

Let  $\sigma \in S_n$  and let  $\mathbf{P}_\sigma = (\mathbf{e}_{\sigma(1)} \cdots \mathbf{e}_{\sigma(n)})$  be its permutation matrix (Definition 2.6.2).

(a) Prove that  $\det \mathbf{P}_\sigma = \operatorname{sgn}(\sigma)$ .

(b) Hence compute  $\det \begin{pmatrix} 0 & 0 & 1 \\ 1 & 0 & 0 \\ 0 & 1 & 0 \end{pmatrix}$  and  $\det \begin{pmatrix} 0 & 1 & 0 & 0 \\ 1 & 0 & 0 & 0 \\ 0 & 0 & 0 & 1 \\ 0 & 0 & 1 & 0 \end{pmatrix}$ .

*Solution.*

(a) By Theorem 6.4.3,  $\det$  is an alternating  $n$ -linear form on  $F^n$ . By Lemma 6.4.1 with  $\mathbf{v}_i = \mathbf{e}_i$ ,

$$\det \mathbf{P}_\sigma = \det(\mathbf{e}_{\sigma(1)}, \dots, \mathbf{e}_{\sigma(n)}) = \operatorname{sgn}(\sigma) \det(\mathbf{e}_1, \dots, \mathbf{e}_n) = \operatorname{sgn}(\sigma) \det \mathbf{I}_n = \operatorname{sgn}(\sigma).$$

(b) The first matrix has columns  $\mathbf{e}_2, \mathbf{e}_3, \mathbf{e}_1$ , so it is  $\mathbf{P}_\sigma$  with  $\sigma(1) = 2$ ,  $\sigma(2) = 3$ ,  $\sigma(3) = 1$ , that is,  $\sigma = (1\ 2\ 3)$ . A 3-cycle is  $(1\ 3)(1\ 2)$  (Example (Computing with cycles)), a product of two transpositions, so its sign is 1 and the determinant is 1. The second matrix has columns  $\mathbf{e}_2, \mathbf{e}_1, \mathbf{e}_4, \mathbf{e}_3$ , so it is  $\mathbf{P}_\sigma$  with  $\sigma = (1\ 2)(3\ 4)$ , of sign  $(-1)^2 = 1$  by Theorem 6.3.1 and Corollary 6.3.2. Its determinant is 1.

**Exercise B3: A zero row.**

Let  $\mathbf{A} \in M_n(F)$  have a zero row. Prove directly from Definition 6.4.1, without using Theorem 6.4.5, that  $\det \mathbf{A} = 0$ . Then prove the same for a zero column.

*Solution.* Suppose row  $i$  of  $\mathbf{A}$  is zero, and let  $\sigma \in S_n$ . Since  $\sigma$  is surjective, there is  $j$  with  $\sigma(j) = i$ . The factor  $a_{\sigma(j)j} = a_{ij}$  of the term for  $\sigma$  is 0, so the term is 0. This holds for every  $\sigma$ , so  $\det \mathbf{A} = 0$ .

Suppose column  $j$  is zero. For every  $\sigma$ , the term for  $\sigma$  contains the factor  $a_{\sigma(j)j} = 0$ , so again every term vanishes and  $\det \mathbf{A} = 0$ .

**6.4.8.3 C. Going deeper****Exercise C1: Too many zeros.**

Let  $\mathbf{A} \in M_n(F)$ .

- (a) Prove that if more than  $n^2 - n$  entries of  $\mathbf{A}$  are zero, then  $\det \mathbf{A} = 0$ .
- (b) Show that the bound is sharp: give, for each  $n$ , a matrix with exactly  $n^2 - n$  zero entries and non-zero determinant.

*Solution.*

- (a) If more than  $n^2 - n$  entries are zero, then fewer than  $n$  entries are non-zero. For  $\sigma \in S_n$ , the term for  $\sigma$  is a product of the  $n$  entries in positions  $(\sigma(1), 1), \dots, (\sigma(n), n)$ . These positions are distinct, since their column indices are distinct. They cannot

all hold non-zero entries, because there are fewer than  $n$  such entries. So at least one factor is 0, and the term is 0. Hence  $\det \mathbf{A} = 0$ .

- (b)  $\mathbf{I}_n$  has exactly  $n^2 - n$  zero entries, and  $\det \mathbf{I}_n = 1 \neq 0$ .

### Exercise C2: Counting the cost.

A computer performs about  $10^9$  multiplications per second.

- (a) Show that evaluating the Leibniz formula term by term for an  $n \times n$  matrix takes  $n!(n-1)$  multiplications. Estimate the time this takes for  $n = 20$ .
- (b) Gaussian elimination (Chapter 2) brings  $\mathbf{A}$  to upper triangular form. Suppose no row swaps are needed. At step  $k$  (for  $k = 1, \dots, n-1$ ), each of the  $n-k$  rows below the pivot requires one division to find its multiplier and  $n-k$  multiplications to update its entries in columns  $k+1, \dots, n$ . Show that the total number of multiplications and divisions is  $\frac{(n-1)n(n+1)}{3}$ , and evaluate it for  $n = 20$ .

In the next section we will see that the determinant can be read off from this triangular form with  $n-1$  further multiplications.

*Solution.*

- (a) There is one term for each  $\sigma \in S_n$ , that is,  $n!$  terms, and each is a product of  $n$  entries, which takes  $n-1$  multiplications. For  $n = 20$ ,  $20! \cdot 19 = 46\,225\,138\,155\,356\,160\,000 \approx 4.6 \times 10^{19}$ . At  $10^9$  per second this takes about  $4.6 \times 10^{10}$  seconds, roughly 1 460 years.
- (b) Step  $k$  costs  $(n-k)(1+(n-k))$  operations. With  $m = n-k$  running from 1 to  $n-1$ , the total is

$$\begin{aligned} \sum_{m=1}^{n-1} m(m+1) &= \sum_{m=1}^{n-1} m^2 + \sum_{m=1}^{n-1} m \\ &= \frac{(n-1)n(2n-1)}{6} + \frac{(n-1)n}{2} \\ &= \frac{(n-1)n((2n-1)+3)}{6} = \frac{(n-1)n(n+1)}{3}. \end{aligned}$$

For  $n = 20$  this is  $\frac{19 \cdot 20 \cdot 21}{3} = 2\,660$ , a few microseconds. The Leibniz formula is a definition and a proof tool, not an algorithm.

### Exercise C3: The permanent, and a change of field.

For  $\mathbf{A} \in M_n(F)$  define  $\text{per } \mathbf{A} = \sum_{\sigma \in S_n} a_{\sigma(1)1} \cdots a_{\sigma(n)n}$ .

- (a) Prove that  $\text{per}$  is  $n$ -linear in the columns and  $\text{per } \mathbf{I}_n = 1$ .
- (b) Let  $n \geq 2$ . Prove that over  $\mathbb{R}$ ,  $\text{per}$  is **not** alternating.
- (c) Prove that over  $\mathbb{F}_2$ ,  $\text{per } \mathbf{A} = \det \mathbf{A}$  for every  $\mathbf{A} \in M_n(\mathbb{F}_2)$ . Explain why this does not contradict (b) and [Theorem 6.4.2](#).

*Solution.*

- (a) The proof of multilinearity in [Theorem 6.4.3](#) never used the signs: for fixed  $k$ , the term for  $\sigma$  is a constant times  $a_{\sigma(k)k}$ , which is linear in the  $k$ -th column. The normalization argument there also applies: only  $\sigma = \text{id}$  contributes at  $\mathbf{I}_n$ , with product 1.
- (b) Let  $\mathbf{J} \in M_n(\mathbb{R})$  have every entry equal to 1. Its first two columns are equal, but every term of  $\text{per } \mathbf{J}$  is 1, so  $\text{per } \mathbf{J} = n! \neq 0$  in  $\mathbb{R}$ .
- (c) In  $\mathbb{F}_2$ ,  $-1 = 1$ , so  $\text{sgn}(\sigma) \cdot r = r$  for every  $r$ , and the two formulas agree term by term. There is no contradiction: (b) is a statement over  $\mathbb{R}$ , where  $n! \neq 0$ . Over  $\mathbb{F}_2$ ,  $\text{per } \mathbf{J} = n! \cdot 1 = 0$  for  $n \geq 2$ , and  $\text{per} = \det$  is alternating, as [Theorem 6.4.3](#) guarantees. Uniqueness is about a fixed field, and over each field there is exactly one normalized alternating form.

## 6.5 Row Operations and Multiplicativity

The previous section defined the determinant by the Leibniz formula and proved it is the only normalized alternating form. The formula settles every theoretical question in principle, but with  $n!$  terms it is useless for computing, and it hides the two facts that make determinants worth having. The first is that  $\det \mathbf{A} \neq 0$  exactly when  $\mathbf{A}$  is invertible. The second is that  $\det(\mathbf{AB}) = \det \mathbf{A} \det \mathbf{B}$ . This section proves both. Along the way we get a fast algorithm, and we extend the determinant from matrices to operators.

### 6.5.1 Row operations

Row reduction is the workhorse of Chapter 2, and the three rules of a determinant say precisely how it reacts to each elementary row operation. Everything follows from [Theorem 6.4.5](#), which lets us treat  $\det$  as a function of the rows.

#### Theorem 6.5.1 — Row Operations and the Determinant

Let  $\mathbf{A} \in M_n(F)$  and let  $\mathbf{B}$  be obtained from  $\mathbf{A}$  by one operation.

- (a) **(Swap)** If  $\mathbf{B}$  is obtained by  $R_i \leftrightarrow R_j$  with  $i \neq j$ , then  $\det \mathbf{B} = -\det \mathbf{A}$ .
- (b) **(Scale)** If  $\mathbf{B}$  is obtained by  $R_i \rightarrow cR_i$  with  $c \in F$ , then  $\det \mathbf{B} = c \det \mathbf{A}$ .
- (c) **(Replace)** If  $\mathbf{B}$  is obtained by  $R_i \rightarrow R_i + cR_j$  with  $i \neq j$  and  $c \in F$ , then  $\det \mathbf{B} = \det \mathbf{A}$ .
- (d) The same three statements hold for the corresponding operations on **columns**.
- (e) For the elementary matrices of [Definition 2.4.1](#),

$$\det \mathbf{P}_{ij} = -1, \quad \det \mathbf{D}_i(c) = c, \quad \det(\mathbf{I}_n + c\mathbf{E}_{ij}) = 1 \quad (i \neq j),$$

and  $\det(\mathbf{EA}) = \det \mathbf{E} \det \mathbf{A}$  for every elementary matrix  $\mathbf{E} \in M_n(F)$ .

*Proof.* By [Theorem 6.4.5](#),  $\det$  is an alternating  $n$ -linear function of the rows  $\mathbf{r}_1, \dots, \mathbf{r}_n$  of  $\mathbf{A}$ . We write  $\det(\mathbf{r}_1, \dots, \mathbf{r}_n)$  for it in this proof.

- (b) This is linearity in row  $i$ .
- (c) By linearity in row  $i$ ,

$$\begin{aligned}\det \mathbf{B} &= \det(\dots, \mathbf{r}_i + c\mathbf{r}_j, \dots, \mathbf{r}_j, \dots) \\ &= \det(\dots, \mathbf{r}_i, \dots, \mathbf{r}_j, \dots) + c \det(\dots, \mathbf{r}_j, \dots, \mathbf{r}_j, \dots),\end{aligned}$$

where the displayed arguments sit in positions  $i$  and  $j$ . The last determinant has two equal rows, so it is 0. Hence  $\det \mathbf{B} = \det \mathbf{A}$ .

- (d) This is the swap rule of [Theorem 6.2.1](#), applied to the rows.
- (e) By [Theorem 6.4.3](#),  $\det$  is an alternating  $n$ -linear function of the columns, and the arguments for (a), (b), (c) apply word for word.
- (f) The matrices  $\mathbf{P}_{ij}$ ,  $\mathbf{D}_i(c)$  and  $\mathbf{I}_n + c\mathbf{E}_{ij}$  are obtained from  $\mathbf{I}_n$  by a swap, a scaling and a replacement respectively. By (a), (b), (c) and  $\det \mathbf{I}_n = 1$ , their determinants are  $-1$ ,  $c$  and  $1$ . Now let  $\mathbf{E} = \rho(\mathbf{I}_n)$  be elementary. By [Theorem 2.4.1](#),  $\mathbf{EA} = \rho(\mathbf{A})$ . By (a), (b) or (c),  $\det \rho(\mathbf{A})$  is  $\det \mathbf{A}$  times  $-1$ ,  $c$  or  $1$  respectively, and this factor is  $\det \mathbf{E}$ . Hence  $\det(\mathbf{EA}) = \det \mathbf{E} \det \mathbf{A}$ . This proves the theorem. ■

Read part (e) as a bookkeeping rule. A swap flips the sign, a scaling by  $c$  multiplies by  $c$ , and a replacement changes nothing. Replacement, the operation used most in elimination, is free.

**Warning. A combined move is not a replacement.** The move  $R_2 \rightarrow 3R_2 - R_1$  looks harmless but multiplies the determinant by 3: it is the scaling  $R_2 \rightarrow 3R_2$  followed by the replacement  $R_2 \rightarrow R_2 - R_1$ . For  $\mathbf{A} = \begin{pmatrix} 1 & 2 \\ 1 & 3 \end{pmatrix}$ , with  $\det \mathbf{A} = 1$ , it produces  $\begin{pmatrix} 1 & 2 \\ 2 & 7 \end{pmatrix}$ , whose determinant is 3. Keep replacements in the form “row  $i$  plus a multiple of another row”, with coefficient 1 on row  $i$ .

Since elimination brings any square matrix to an upper triangular one, and triangular determinants are products of diagonal entries ([Theorem 6.4.6](#)), we have an algorithm.

#### Example. ( $\mathbf{A}$ $4 \times 4$ Determinant by Elimination)

Compute  $\det \mathbf{A}$  for

$$\mathbf{A} = \begin{pmatrix} 0 & 2 & 4 & 2 \\ 1 & 1 & 0 & 3 \\ 2 & 3 & 1 & 5 \\ -1 & 2 & 3 & 1 \end{pmatrix} \in M_4(\mathbb{R}).$$

*Solution.* We record each operation and its effect, using [Theorem 6.5.1](#).

**Step 1.** The  $(1,1)$ -entry is 0, so swap:  $R_1 \leftrightarrow R_2$ . This multiplies the determinant by

−1:

$$\det \mathbf{A} = -\det \begin{pmatrix} 1 & 1 & 0 & 3 \\ 0 & 2 & 4 & 2 \\ 2 & 3 & 1 & 5 \\ -1 & 2 & 3 & 1 \end{pmatrix}.$$

**Step 2.** Clear column 1 with the replacements  $R_3 \rightarrow R_3 - 2R_1$  and  $R_4 \rightarrow R_4 + R_1$ , which change nothing. Row 2 has a common factor 2; pulling it out is linearity in row 2:

$$\det \mathbf{A} = -\det \begin{pmatrix} 1 & 1 & 0 & 3 \\ 0 & 2 & 4 & 2 \\ 0 & 1 & 1 & -1 \\ 0 & 3 & 3 & 4 \end{pmatrix} = -2 \det \begin{pmatrix} 1 & 1 & 0 & 3 \\ 0 & 1 & 2 & 1 \\ 0 & 1 & 1 & -1 \\ 0 & 3 & 3 & 4 \end{pmatrix}.$$

**Step 3.** Clear column 2:  $R_3 \rightarrow R_3 - R_2$  and  $R_4 \rightarrow R_4 - 3R_2$ . Then clear column 3:  $R_4 \rightarrow R_4 - 3R_3$ . None of these changes the determinant:

$$\det \mathbf{A} = -2 \det \begin{pmatrix} 1 & 1 & 0 & 3 \\ 0 & 1 & 2 & 1 \\ 0 & 0 & -1 & -2 \\ 0 & 0 & -3 & 1 \end{pmatrix} = -2 \det \begin{pmatrix} 1 & 1 & 0 & 3 \\ 0 & 1 & 2 & 1 \\ 0 & 0 & -1 & -2 \\ 0 & 0 & 0 & 7 \end{pmatrix}.$$

**Step 4.** The last matrix is upper triangular, so by [Theorem 6.4.6](#) its determinant is  $1 \cdot 1 \cdot (-1) \cdot 7 = -7$ . Hence  $\det \mathbf{A} = -2 \cdot (-7) = 14$ .

Elimination costs roughly  $n^3/3$  arithmetic operations, and reading off the triangular determinant takes only  $n - 1$  more multiplications, against  $n!(n - 1)$  multiplications for the Leibniz formula ([Exercise \(C2: Counting the cost\)](#)). For  $n = 20$  that is a few thousand operations against about  $4.6 \times 10^{19}$ . Every practical determinant computation is some form of elimination.

## 6.5.2 Determinant and invertibility

The same bookkeeping answers the invertibility question. Row reduction multiplies the determinant by non-zero factors only, so it cannot turn a non-zero determinant into zero or back. And the end of the reduction is either  $\mathbf{I}_n$  or a matrix with a zero row.

### Theorem 6.5.2 — Invertible if and only if Non-Zero Determinant

A matrix  $\mathbf{A} \in M_n(F)$  is invertible if and only if  $\det \mathbf{A} \neq 0$ .

*Proof.* Let  $\mathbf{R}$  be the reduced row echelon form of  $\mathbf{A}$  ([Theorem 2.2.3](#), [Theorem 2.3.4](#)). By [Definition 2.2.2](#) and [Theorem 2.4.1](#), there are elementary matrices  $\mathbf{E}_1, \dots, \mathbf{E}_k$  (possibly  $k = 0$ ) with  $\mathbf{R} = \mathbf{E}_k \cdots \mathbf{E}_1 \mathbf{A}$ . Applying [Theorem 6.5.1](#) (e)  $k$  times,

$$\det \mathbf{R} = \det \mathbf{E}_k \cdots \det \mathbf{E}_1 \det \mathbf{A}.$$

Each  $\det \mathbf{E}_m$  is  $-1$ ,  $1$ , or some  $c \neq 0$ , since a scaling operation requires  $c \neq 0$ . A product of non-zero elements of a field is non-zero ([Theorem 0.7.1](#)), so  $\det \mathbf{R} = 0$  if and only if  $\det \mathbf{A} = 0$ .



By [Lemma 2.4.3](#), either  $\mathbf{R} = \mathbf{I}_n$ , and then  $\det \mathbf{R} = 1 \neq 0$ ; or the last row of  $\mathbf{R}$  is zero. In the second case, scaling that row by 0 leaves  $\mathbf{R}$  unchanged, so  $\det \mathbf{R} = 0 \cdot \det \mathbf{R} = 0$  by [Theorem 6.5.1](#) (b). Hence  $\det \mathbf{A} \neq 0$  if and only if  $\mathbf{R} = \mathbf{I}_n$ , which by [Theorem 2.4.4](#) holds if and only if  $\mathbf{A}$  is invertible. ■

This is the promised addition to the list of equivalent conditions. We collect it with the conditions from Chapter 3 about the map  $T_{\mathbf{A}}: \mathbf{x} \mapsto \mathbf{A}\mathbf{x}$ , and with two conditions about rows that the determinant makes easy.

### Theorem 6.5.3 — Invertible Matrix Theorem, with Determinants

Let  $\mathbf{A} \in M_n(F)$  and  $T_{\mathbf{A}}: F^n \rightarrow F^n$ ,  $\mathbf{x} \mapsto \mathbf{A}\mathbf{x}$ . The following are equivalent, and each is equivalent to each of the eight conditions of [Theorem 2.4.4](#).

- (a)  $\mathbf{A}$  is invertible.
- (b)  $T_{\mathbf{A}}$  is injective.
- (c)  $T_{\mathbf{A}}$  is surjective.
- (d)  $\text{rank } \mathbf{A} = n$ .
- (e)  $\mathbf{A}^{\top}$  is invertible.
- (f) The rows of  $\mathbf{A}$  are linearly independent.
- (g)  $\det \mathbf{A} \neq 0$ .

*Proof.* Item (a) is item (a) of [Theorem 2.4.4](#), so it suffices to connect each new item with that theorem.

(a)  $\Leftrightarrow$  (g). This is [Theorem 6.5.2](#).

(a)  $\Leftrightarrow$  (b). By [Theorem 3.5.7](#) ((b)  $\Leftrightarrow$  (c)) with  $V = F^n$ ,  $T_{\mathbf{A}}$  is injective if and only if  $\ker T_{\mathbf{A}} = \{\mathbf{x} : \mathbf{A}\mathbf{x} = \mathbf{0}\}$  is  $\{\mathbf{0}\}$ . That is item (b) of [Theorem 2.4.4](#).

(b)  $\Leftrightarrow$  (c). This is [Theorem 3.5.7](#) ((b)  $\Leftrightarrow$  (d)), since  $F^n$  is finite-dimensional.

(c)  $\Leftrightarrow$  (d). By [Theorem 0.9.3](#),  $\text{im } T_{\mathbf{A}}$  is the set of combinations of the columns of  $\mathbf{A}$ , that is,  $\text{col}(\mathbf{A})$ . So  $T_{\mathbf{A}}$  is surjective if and only if  $\text{col}(\mathbf{A}) = F^n$ . Since  $\text{col}(\mathbf{A})$  is a subspace of  $F^n$ , this holds if and only if  $\text{rank } \mathbf{A} = \dim \text{col}(\mathbf{A}) = n$  ([Definition 2.5.4](#), [Theorem 1.6.8](#)).

(a)  $\Leftrightarrow$  (e). By [Theorem 6.4.5](#),  $\det \mathbf{A}^{\top} = \det \mathbf{A}$ . Applying (a)  $\Leftrightarrow$  (g) to  $\mathbf{A}$  and to  $\mathbf{A}^{\top}$ , both “ $\mathbf{A}$  invertible” and “ $\mathbf{A}^{\top}$  invertible” are equivalent to  $\det \mathbf{A} \neq 0$ .

(e)  $\Leftrightarrow$  (f). Let  $\mathbf{r}_1, \dots, \mathbf{r}_n$  be the rows of  $\mathbf{A}$ . The columns of  $\mathbf{A}^{\top}$  are  $\mathbf{r}_1^{\top}, \dots, \mathbf{r}_n^{\top}$ . By [Theorem 0.9.5](#),  $c_1\mathbf{r}_1 + \dots + c_n\mathbf{r}_n = \mathbf{0}$  if and only if  $c_1\mathbf{r}_1^{\top} + \dots + c_n\mathbf{r}_n^{\top} = \mathbf{0}$ , so the rows of  $\mathbf{A}$  are independent exactly when the columns of  $\mathbf{A}^{\top}$  are. By [Theorem 2.4.4](#) ((a)  $\Leftrightarrow$  (f)) applied to  $\mathbf{A}^{\top}$ , this holds exactly when  $\mathbf{A}^{\top}$  is invertible. This proves the theorem. ■

The determinant is now one more item on the menu, and for the right problems it is the best one. For a concrete matrix with a parameter, one polynomial in the parameter decides invertibility at once. For an abstract matrix, such as  $\mathbf{A} + \mathbf{I}$  when  $\mathbf{A}^k = \mathbf{0}$ , the determinant is usually hopeless,

and the kernel is the item to try first.

*Quick check.* For which  $t \in \mathbb{R}$  is  $\begin{pmatrix} 1 & t \\ t & 4 \end{pmatrix}$  invertible? Over  $\mathbb{F}_3$ , for which  $t$ ?

*Solution.* The determinant is  $4 - t^2 = (2 - t)(2 + t)$ . Over  $\mathbb{R}$  the matrix is invertible exactly for  $t \neq \pm 2$ . Over  $\mathbb{F}_3$ ,  $4 = 1$ , so the determinant is  $1 - t^2$ , which is 1 for  $t = 0$  and 0 for  $t = 1, 2$ ; the matrix is invertible only for  $t = 0$ .

### 6.5.3 Multiplicativity

Now the second fact. A natural first attempt is to multiply out the Leibniz formula for  $\mathbf{AB}$ , whose entries are sums, and regroup. That is legitimate, but the bookkeeping is horrible. The uniqueness theorem does the work instead: we exhibit  $\mathbf{B} \mapsto \det(\mathbf{AB})$  as an alternating form in the columns of  $\mathbf{B}$ , and then it has no choice.

#### Theorem 6.5.4 — Multiplicativity of the Determinant

For all  $\mathbf{A}, \mathbf{B} \in M_n(F)$ ,

$$\det(\mathbf{AB}) = \det \mathbf{A} \det \mathbf{B}.$$

*Idea.* Fix  $\mathbf{A}$ . The  $j$ -th column of  $\mathbf{AB}$  is  $\mathbf{Ab}_j$ , so  $g(\mathbf{b}_1, \dots, \mathbf{b}_n) = \det(\mathbf{Ab}_1, \dots, \mathbf{Ab}_n)$  is  $\det(\mathbf{AB})$  viewed as a function of the columns of  $\mathbf{B}$ . It inherits linearity in each  $\mathbf{b}_j$  from  $\det$  and from matrix multiplication, and equal columns  $\mathbf{b}_p = \mathbf{b}_q$  give equal columns  $\mathbf{Ab}_p = \mathbf{Ab}_q$ . So  $g$  is an alternating form, hence a multiple of  $\det$ , and the multiple is  $g(\mathbf{e}_1, \dots, \mathbf{e}_n) = \det \mathbf{A}$ .

*Proof.* Fix  $\mathbf{A} \in M_n(F)$ , and define  $g: F^n \times \dots \times F^n \rightarrow F$  by

$$g(\mathbf{b}_1, \dots, \mathbf{b}_n) = \det(\mathbf{Ab}_1, \dots, \mathbf{Ab}_n).$$

By Theorem 0.9.2 (columns), if  $\mathbf{B}$  has columns  $\mathbf{b}_1, \dots, \mathbf{b}_n$ , then  $\mathbf{AB}$  has columns  $\mathbf{Ab}_1, \dots, \mathbf{Ab}_n$ , so  $g(\mathbf{b}_1, \dots, \mathbf{b}_n) = \det(\mathbf{AB})$ .

*$g$  is  $n$ -linear.* Fix  $k$  and all  $\mathbf{b}_j$  with  $j \neq k$ . The map  $\mathbf{b}_k \mapsto \mathbf{Ab}_k$  is linear by Theorem 0.9.4, and  $\det$  is linear in its  $k$ -th argument by Theorem 6.4.3. A composite of linear maps is linear, so  $g$  is linear in  $\mathbf{b}_k$ .

*$g$  is alternating.* If  $\mathbf{b}_p = \mathbf{b}_q$  with  $p \neq q$ , then  $\mathbf{Ab}_p = \mathbf{Ab}_q$ , so  $g(\mathbf{b}_1, \dots, \mathbf{b}_n) = 0$  since  $\det$  is alternating.

By Corollary 6.4.4,  $g(\mathbf{b}_1, \dots, \mathbf{b}_n) = g(\mathbf{e}_1, \dots, \mathbf{e}_n) \det(\mathbf{b}_1, \dots, \mathbf{b}_n)$ . Finally  $\mathbf{Ae}_j$  is the  $j$ -th column of  $\mathbf{A}$  (Theorem 0.9.3), so  $g(\mathbf{e}_1, \dots, \mathbf{e}_n) = \det \mathbf{A}$ . Hence  $\det(\mathbf{AB}) = \det \mathbf{A} \det \mathbf{B}$ . ■

*Remark.* **Over a commutative ring.** This proof uses only Corollary 6.4.4 (in the explicit form of Theorem 6.4.2), linearity of matrix multiplication, and the Leibniz for-

mula. None of these divides. So  $\det(\mathbf{AB}) = \det \mathbf{A} \det \mathbf{B}$  holds for  $\mathbf{A}, \mathbf{B} \in M_n(R)$  over any commutative ring  $R$ , such as  $F[x]$ .

Alternatively, the building blocks of Chapter 2 give a proof. If  $\mathbf{A}$  is invertible, then  $\mathbf{A} = \mathbf{E}_1 \cdots \mathbf{E}_k$  with elementary  $\mathbf{E}_m$  (Theorem 2.4.4 (d)), and peeling off one factor at a time with Theorem 6.5.1 (e) gives  $\det(\mathbf{AB}) = \det \mathbf{E}_1 \cdots \det \mathbf{E}_k \det \mathbf{B}$ ; the case  $\mathbf{B} = \mathbf{I}_n$  of the same computation shows  $\det \mathbf{A} = \det \mathbf{E}_1 \cdots \det \mathbf{E}_k$ . If  $\mathbf{A}$  is not invertible, neither is  $\mathbf{AB}$ : otherwise  $(\mathbf{AB})\mathbf{C} = \mathbf{I}_n$  for some  $\mathbf{C}$ , so  $\mathbf{A}(\mathbf{BC}) = \mathbf{I}_n$  and  $\mathbf{A}$  would be invertible by Theorem 2.4.5. Then both sides are 0 by Theorem 6.5.2. This route leans on row reduction, which divides, so unlike the first proof it needs a field.

For reference, here are the two identities used most often, now both proved.

### Theorem 6.5.5 — Properties of Determinants

Let  $\mathbf{A}, \mathbf{B} \in M_n(F)$ . Then:

1.  $\det(\mathbf{AB}) = \det \mathbf{A} \det \mathbf{B}$ ;
2.  $\det \mathbf{A}^\top = \det \mathbf{A}$ .

*Proof.* Part 1 is Theorem 6.5.4 and part 2 is Theorem 6.4.5. ■

The first consequences are immediate.

### Corollary 6.5.6 — Determinant of the Inverse

If  $\mathbf{A} \in M_n(F)$  is invertible, then  $\det \mathbf{A} \neq 0$  and  $\det(\mathbf{A}^{-1}) = (\det \mathbf{A})^{-1}$ .

*Proof.* By Theorem 6.5.4,  $\det \mathbf{A} \det(\mathbf{A}^{-1}) = \det(\mathbf{AA}^{-1}) = \det \mathbf{I}_n = 1$ . Hence  $\det \mathbf{A} \neq 0$ , and  $\det(\mathbf{A}^{-1})$  is the inverse of  $\det \mathbf{A}$  in  $F$ . ■

**Warning.** Multiplicativity does not make  $\det$  additive, and it does not make products commute. For  $\mathbf{A} = \begin{pmatrix} 1 & 2 \\ 3 & 4 \end{pmatrix}$  and  $\mathbf{B} = \begin{pmatrix} 0 & 1 \\ 1 & 5 \end{pmatrix}$ ,  $\mathbf{AB} = \begin{pmatrix} 2 & 11 \\ 4 & 23 \end{pmatrix} \neq \begin{pmatrix} 3 & 4 \\ 16 & 22 \end{pmatrix} = \mathbf{BA}$ , yet  $\det(\mathbf{AB}) = \det(\mathbf{BA}) = 2$ , since both equal  $\det \mathbf{A} \det \mathbf{B} = (-2)(-1)$ . On the other hand,  $\det(\mathbf{A} + \mathbf{B})$  is not determined by  $\det \mathbf{A}$  and  $\det \mathbf{B}$  at all:  $\mathbf{I}_2$  and  $\mathbf{I}_2$  have determinants 1, 1 and  $\det(\mathbf{I}_2 + \mathbf{I}_2) = 4$ , while  $\mathbf{I}_2$  and  $\mathbf{C} = \begin{pmatrix} 0 & 1 \\ -1 & 0 \end{pmatrix}$  also have determinants 1, 1 but  $\det(\mathbf{I}_2 + \mathbf{C}) = \det \begin{pmatrix} 1 & 1 \\ -1 & 1 \end{pmatrix} = 2$ .

**Quick check.** Let  $\mathbf{A} = \begin{pmatrix} 2 & 1 \\ 7 & 4 \end{pmatrix}$ . Without multiplying any matrices, find  $\det(\mathbf{A}^5)$ ,  $\det(\mathbf{A}^{-1})$  and  $\det(\mathbf{A}^\top \mathbf{A})$ .

*Solution.*  $\det \mathbf{A} = 8 - 7 = 1$ . By [Theorem 6.5.4](#),  $\det(\mathbf{A}^5) = (\det \mathbf{A})^5 = 1$ . By [Corollary 6.5.6](#),  $\det(\mathbf{A}^{-1}) = 1^{-1} = 1$ . By [Theorem 6.5.5](#),  $\det(\mathbf{A}^\top \mathbf{A}) = \det \mathbf{A}^\top \det \mathbf{A} = (\det \mathbf{A})^2 = 1$ .

## 6.5.4 The determinant of an operator

In Chapter 3 the trace passed from matrices to operators because similar matrices have the same trace ([Definition 3.10.2](#)). Multiplicativity gives the same for determinants.

### Corollary 6.5.7 — Determinant Is a Similarity Invariant

If  $\mathbf{A}, \mathbf{B} \in M_n(F)$  are similar, then  $\det \mathbf{A} = \det \mathbf{B}$ .

*Proof.* Let  $\mathbf{B} = \mathbf{P}^{-1}\mathbf{A}\mathbf{P}$  with  $\mathbf{P}$  invertible. By [Theorem 6.5.4](#) and [Corollary 6.5.6](#),

$$\det \mathbf{B} = \det(\mathbf{P}^{-1}) \det \mathbf{A} \det \mathbf{P} = (\det \mathbf{P})^{-1} \det \mathbf{P} \det \mathbf{A} = \det \mathbf{A},$$

where the middle step uses commutativity of multiplication in  $F$ . ■

The converse fails:  $\mathbf{I}_2$  and  $\begin{pmatrix} 1 & 1 \\ 0 & 1 \end{pmatrix}$  both have determinant 1, but the only matrix similar to  $\mathbf{I}_2$  is  $\mathbf{P}^{-1}\mathbf{I}_2\mathbf{P} = \mathbf{I}_2$  itself.

By [Theorem 3.7.4](#), all matrices of one operator in the various bases are similar to each other. So the corollary lets us define:

### Definition 6.5.1 — Determinant of an operator

Let  $V$  be a finite-dimensional vector space over  $F$  with  $\dim V \geq 1$ , and let  $T \in \mathcal{L}(V)$ . The **determinant** of  $T$  is

$$\det T := \det[T]_{\mathcal{B}},$$

where  $\mathcal{B}$  is **any** basis of  $V$ . For  $V = \{\mathbf{0}\}$  we set  $\det T = 1$ .

**Well-definedness.** If  $\mathcal{B}$  and  $\mathcal{C}$  are bases of  $V$ , then  $[T]_{\mathcal{C}} = \mathbf{P}^{-1}[T]_{\mathcal{B}}\mathbf{P}$  with  $\mathbf{P} = [\text{id}]_{\mathcal{C}}^{\mathcal{B}}$  invertible, by [Theorem 3.7.3](#) and [Theorem 3.7.4](#) (a). By [Corollary 6.5.7](#) the two determinants agree. The convention for  $V = \{\mathbf{0}\}$  is the empty product, and it keeps the next theorem true in that case.

**Examples.**

- **Scalars.** For  $c \in F$  and  $\dim V = n \geq 1$ ,  $[c \text{id}_V]_{\mathcal{B}} = c\mathbf{I}_n$  in every basis, so  $\det(c \text{id}_V) = c^n$  by [Theorem 6.4.6](#).
- **A reflection.**  $R(x, y) = (y, x)$  on  $\mathbb{R}^2$  has standard matrix  $\begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}$ , of determinant  $-1$ . In the basis  $((1, 1), (1, -1))$  its matrix is  $\text{diag}(1, -1)$  ([Example \(The reflection in a mirror-adapted basis\)](#)), again of determinant  $-1$ , as it must be.
- **Differentiation.** On  $\mathbb{R}[x]_{\leq 3}$  with basis  $(1, x, x^2, x^3)$ , the matrix of  $\mathbf{D}$  is upper triangular

with zero diagonal ([Example \(Differentiation on Cubic Polynomials\)](#)), so  $\det \mathbf{D} = 0$ .

The determinant of a map  $T: V \rightarrow W$  between **different** spaces is not defined, even when  $\dim V = \dim W$ . A matrix  $[T]_{\mathcal{B}}^{\mathcal{C}}$  exists, but changing  $\mathcal{C}$  alone multiplies it on the left by an arbitrary invertible matrix, which can change its determinant to any non-zero value. For  $\text{id}: \mathbb{R}^2 \rightarrow \mathbb{R}^2$  with input basis  $\mathcal{E}$  and output basis  $((2, 0), (0, 1))$ , the matrix is  $\text{diag}(\frac{1}{2}, 1)$ , of determinant  $\frac{1}{2}$ , not 1.

### Theorem 6.5.8 — Properties of the Determinant of an Operator

Let  $V$  be a finite-dimensional vector space over  $F$  and  $S, T \in \mathcal{L}(V)$ . Then

$$\det(ST) = \det S \det T, \quad \det \text{id}_V = 1,$$

and  $T$  is invertible if and only if  $\det T \neq 0$ .

*Proof.* If  $V = \{0\}$ , all determinants are 1 and the only operator is  $\text{id}_V$ , which is invertible. Otherwise fix a basis  $\mathcal{B}$  of  $V$ . By [Theorem 3.6.2](#),  $[ST]_{\mathcal{B}} = [S]_{\mathcal{B}}[T]_{\mathcal{B}}$ , so  $\det(ST) = \det S \det T$  by [Theorem 6.5.4](#). Next  $[\text{id}_V]_{\mathcal{B}} = \mathbf{I}_n$ , of determinant 1. Finally, by [Theorem 3.6.5](#) (b),  $T$  is invertible if and only if  $[T]_{\mathcal{B}}$  is, which by [Theorem 6.5.2](#) holds if and only if  $\det T = \det [T]_{\mathcal{B}} \neq 0$ . ■

So the determinant is a property of the operator, not of the coordinates. Differentiation on  $\mathbb{R}[x]_{\leq 3}$  is not invertible because its determinant is 0, whichever basis we compute in. In the section on orientation and volume we will see what the number  $\det T$  measures geometrically.

## 6.5.5 The determinant as a homomorphism

In Chapter 0 we met groups and homomorphisms, and promised that the determinant would be one. The general linear group  $\text{GL}_n(F)$  is the group of invertible matrices in  $M_n(F)$  under multiplication, and  $F \setminus \{0\}$  is a group under multiplication ([Example \(Groups\)](#)).

### Theorem 6.5.9 — The Determinant Is a Group Homomorphism

Let  $n \geq 1$ . The determinant restricts to a **surjective** group homomorphism

$$\det: \text{GL}_n(F) \rightarrow F \setminus \{0\}.$$

Its kernel is the **special linear group**  $\text{SL}_n(F) := \{\mathbf{A} \in M_n(F) : \det \mathbf{A} = 1\}$ , which is therefore a subgroup of  $\text{GL}_n(F)$ .

*Proof.* By [Theorem 6.5.2](#),  $\det \mathbf{A} \neq 0$  for  $\mathbf{A} \in \text{GL}_n(F)$ , so  $\det$  maps  $\text{GL}_n(F)$  into  $F \setminus \{0\}$ , and [Theorem 6.5.4](#) says it is a homomorphism ([Definition 0.10.3](#)). For  $c \neq 0$ , the matrix  $\text{diag}(c, 1, \dots, 1)$  has determinant  $c$  by [Theorem 6.4.6](#), which is non-zero, so it lies in  $\text{GL}_n(F)$ ; hence  $\det$  is surjective. A matrix with determinant 1 is invertible by [Theorem 6.5.2](#), so the kernel  $\{\mathbf{A} \in \text{GL}_n(F) : \det \mathbf{A} = 1\}$  is  $\text{SL}_n(F)$ . It is a subgroup by [Theorem 0.10.3](#).

For example,  $\begin{pmatrix} 1 & 1 \\ 0 & 1 \end{pmatrix}$  and the elementary matrices  $\mathbf{I}_n + c\mathbf{E}_{ij}$  lie in  $\mathrm{SL}_n(F)$ , while  $\mathbf{P}_{12}$ , of determinant  $-1$ , does not. Over  $\mathbb{R}$ , the rotation matrices  $\begin{pmatrix} \cos \theta & -\sin \theta \\ \sin \theta & \cos \theta \end{pmatrix}$  have determinant  $\cos^2 \theta + \sin^2 \theta = 1$  and lie in  $\mathrm{SL}_2(\mathbb{R})$ . The sign  $\mathrm{sgn}: S_n \rightarrow \{1, -1\}$  of the third section is the other homomorphism promised in Chapter 0, and the two are linked:  $\det \mathbf{P}_\sigma = \mathrm{sgn}(\sigma)$  ([Exercise \(B2: The determinant of  \$\mathbf{P}\_\sigma\$ \)](#)), and  $\mathbf{P}_\sigma \mathbf{P}_\tau = \mathbf{P}_{\sigma\tau}$  ([Lemma 2.6.6](#)).

## 6.5.6 Which proofs survive over a ring

*Remark. Commutative rings.* Over a commutative ring  $R$  (as in the remark at the end of the previous section), the following proofs of this section use no division and remain valid for matrices in  $M_n(R)$ : [Theorem 6.5.1](#), with  $c \in R$  arbitrary in (b) and (c); [Theorem 6.5.4](#), by its first proof; and [Theorem 6.5.5](#). The proofs of [Corollary 6.5.6](#) and [Corollary 6.5.7](#) also survive for matrices that have an inverse in  $M_n(R)$  (then  $\det \mathbf{A} \det(\mathbf{A}^{-1}) = 1$ , so  $\det \mathbf{A}$  has an inverse in  $R$ ), for instance an invertible  $\mathbf{P} \in M_n(F)$  regarded inside  $M_n(F[x])$ ; this is the case needed for characteristic polynomials. What fails over a ring is the converse direction: a matrix whose determinant is non-zero need not be invertible, since  $\det$  may be a non-zero element with no inverse, such as  $2 \in \mathbb{Z}$  or  $x \in F[x]$ . By contrast, [Theorem 6.5.2](#), [Theorem 6.5.3](#) and the “Alternatively” proof of multiplicativity run through row reduction, which divides by pivots, and they are proved here only over a field.

## 6.5.7 Exercises

### 6.5.7.1 A. Check your understanding

#### Exercise A1.

- State how each of the three elementary row operations changes the determinant.
- True or false: the move  $R_2 \rightarrow 3R_2 - R_1$  does not change the determinant. Justify your answer.
- True or false: if  $\mathbf{A}, \mathbf{B} \in M_n(F)$  have  $\det \mathbf{A} = \det \mathbf{B}$ , then  $\mathbf{A}$  and  $\mathbf{B}$  are similar. Justify your answer.
- True or false:  $\det(\mathbf{AB}) = \det(\mathbf{BA})$  for all  $\mathbf{A}, \mathbf{B} \in M_n(F)$ . Justify your answer.
- Explain why  $\det T$  does not depend on the basis used to compute it.
- Is  $\det: M_n(F) \rightarrow F$  a group homomorphism? Explain.

*Solution.*

- By [Theorem 6.5.1](#): a swap multiplies it by  $-1$ , scaling a row by  $c$  multiplies it by  $c$ , and adding a multiple of one row to a different row leaves it unchanged.
- False. The move is  $R_2 \rightarrow 3R_2$  followed by  $R_2 \rightarrow R_2 - R_1$ , so it multiplies the

determinant by 3. For  $\mathbf{I}_2$  it gives  $\begin{pmatrix} 1 & 0 \\ -1 & 3 \end{pmatrix}$ , of determinant  $3 \neq 1$ .

- (c) False.  $\mathbf{I}_2$  and  $\begin{pmatrix} 1 & 1 \\ 0 & 1 \end{pmatrix}$  both have determinant 1, but  $\mathbf{P}^{-1}\mathbf{I}_2\mathbf{P} = \mathbf{I}_2$  for every invertible  $\mathbf{P}$ , so  $\mathbf{I}_2$  is similar only to itself.
- (d) True. Both equal  $\det \mathbf{A} \det \mathbf{B}$  by [Theorem 6.5.4](#), since multiplication in  $F$  is commutative.
- (e) Matrices of  $T$  in two bases are similar ([Theorem 3.7.3](#)), and similar matrices have equal determinants ([Corollary 6.5.7](#)).
- (f) No:  $M_n(F)$  is not a group under multiplication, since the zero matrix has no inverse. The restriction  $\det: \mathrm{GL}_n(F) \rightarrow F \setminus \{0\}$  is a homomorphism ([Theorem 6.5.9](#)).

### 6.5.7.2 B. Practice

#### Exercise B1: Determinants by elimination.

Compute the determinants of the following real matrices by row reduction, recording the effect of each operation.

$$\mathbf{A} = \begin{pmatrix} 1 & 2 & 0 & 1 \\ 2 & 3 & 1 & 4 \\ 0 & 1 & 3 & 2 \\ -1 & 1 & 2 & 5 \end{pmatrix}, \quad \mathbf{B} = \begin{pmatrix} 2 & -1 & 3 & 1 \\ 4 & -2 & 7 & 3 \\ -2 & 2 & -1 & 4 \\ 6 & -3 & 9 & 5 \end{pmatrix}.$$

*Solution.* For  $\mathbf{A}$ : the replacements  $R_2 \rightarrow R_2 - 2R_1$  and  $R_4 \rightarrow R_4 + R_1$  give rows  $(1, 2, 0, 1)$ ,  $(0, -1, 1, 2)$ ,  $(0, 1, 3, 2)$ ,  $(0, 3, 2, 6)$ . Then  $R_3 \rightarrow R_3 + R_2$  and  $R_4 \rightarrow R_4 + 3R_2$  give  $(0, 0, 4, 4)$  and  $(0, 0, 5, 12)$ . Pulling the factor 4 out of row 3 leaves  $(0, 0, 1, 1)$ , and  $R_4 \rightarrow R_4 - 5R_3$  gives  $(0, 0, 0, 7)$ . None of the replacements changes the determinant ([Theorem 6.5.1](#)), so

$$\det \mathbf{A} = 4 \det \begin{pmatrix} 1 & 2 & 0 & 1 \\ 0 & -1 & 1 & 2 \\ 0 & 0 & 1 & 1 \\ 0 & 0 & 0 & 7 \end{pmatrix} = 4 \cdot (1 \cdot (-1) \cdot 1 \cdot 7) = -28,$$

by [Theorem 6.4.6](#).

For  $\mathbf{B}$ :  $R_2 \rightarrow R_2 - 2R_1$ ,  $R_3 \rightarrow R_3 + R_1$ ,  $R_4 \rightarrow R_4 - 3R_1$  give rows  $(2, -1, 3, 1)$ ,  $(0, 0, 1, 1)$ ,  $(0, 1, 2, 5)$ ,  $(0, 0, 0, 2)$ , without changing the determinant. The  $(2, 2)$ -entry is 0, so swap  $R_2 \leftrightarrow R_3$ , which multiplies the determinant by  $-1$ . The result is upper triangular with diagonal  $2, 1, 1, 2$ , so

$$\det \mathbf{B} = -(2 \cdot 1 \cdot 1 \cdot 2) = -4.$$

#### Exercise B2: A parameter.

Let  $t \in \mathbb{R}$  and  $\mathbf{A}_t = \begin{pmatrix} t & 1 & 1 \\ 1 & t & 1 \\ 1 & 1 & t \end{pmatrix}$ . Find  $\det \mathbf{A}_t$  as a factored polynomial in  $t$ . Hence determine for which  $t$  the matrix  $\mathbf{A}_t$  is invertible.

*Solution.* The replacements  $R_1 \rightarrow R_1 + R_2$  and  $R_1 \rightarrow R_1 + R_3$  do not change the determinant and make row 1 equal to  $(t+2, t+2, t+2)$ . By linearity in row 1 (this holds for every value of  $t+2$ , including 0),

$$\begin{aligned}\det \mathbf{A}_t &= (t+2) \det \begin{pmatrix} 1 & 1 & 1 \\ 1 & t & 1 \\ 1 & 1 & t \end{pmatrix} \\ &= (t+2) \det \begin{pmatrix} 1 & 1 & 1 \\ 0 & t-1 & 0 \\ 0 & 0 & t-1 \end{pmatrix} = (t+2)(t-1)^2,\end{aligned}$$

where the second step uses  $R_2 \rightarrow R_2 - R_1$ ,  $R_3 \rightarrow R_3 - R_1$ , and the last uses [Theorem 6.4.6](#). By [Theorem 6.5.2](#),  $\mathbf{A}_t$  is invertible if and only if  $t \neq 1$  and  $t \neq -2$ .

### Exercise B3: The shift operator.

Let  $T: \mathbb{R}[x]_{\leq 3} \rightarrow \mathbb{R}[x]_{\leq 3}$ ,  $T(p) = p(x+1)$ . Find  $\det T$ . Hence show that  $T$  is invertible.

*Solution.* In the basis  $\mathcal{B} = (1, x, x^2, x^3)$ ,  $T(1) = 1$ ,  $T(x) = 1 + x$ ,  $T(x^2) = 1 + 2x + x^2$  and  $T(x^3) = 1 + 3x + 3x^2 + x^3$ . So

$$[T]_{\mathcal{B}} = \begin{pmatrix} 1 & 1 & 1 & 1 \\ 0 & 1 & 2 & 3 \\ 0 & 0 & 1 & 3 \\ 0 & 0 & 0 & 1 \end{pmatrix},$$

which is upper triangular with ones on the diagonal. By [Definition 6.5.1](#) and [Theorem 6.4.6](#),  $\det T = 1$ . Since  $\det T \neq 0$ ,  $T$  is invertible by [Theorem 6.5.8](#). (Its inverse is  $p \mapsto p(x-1)$ .)

### 6.5.7.3 C. Going deeper

#### Exercise C1: Orthogonal-type matrices.

- (a) Let  $\mathbf{A} \in M_n(\mathbb{R})$  satisfy  $\mathbf{A}^\top \mathbf{A} = \mathbf{I}_n$ . Prove that  $\det \mathbf{A} = 1$  or  $\det \mathbf{A} = -1$ .
- (b) Show that both values occur, for every  $n \geq 1$ .
- (c) Over  $\mathbb{F}_2$ , what can  $\det \mathbf{A}$  be if  $\mathbf{A}^\top \mathbf{A} = \mathbf{I}_n$ ?

*Solution.*

- (a) By [Theorem 6.5.5](#),  $1 = \det \mathbf{I}_n = \det(\mathbf{A}^\top \mathbf{A}) = \det \mathbf{A}^\top \det \mathbf{A} = (\det \mathbf{A})^2$ . A real number with square 1 is 1 or  $-1$ .
- (b)  $\mathbf{I}_n^\top \mathbf{I}_n = \mathbf{I}_n$  and  $\det \mathbf{I}_n = 1$ . For  $\mathbf{D} = \text{diag}(-1, 1, \dots, 1)$ ,  $\mathbf{D}^\top \mathbf{D} = \mathbf{D}^2 = \mathbf{I}_n$  and  $\det \mathbf{D} = -1$  by [Theorem 6.4.6](#).
- (c) The same computation gives  $(\det \mathbf{A})^2 = 1$ . In  $\mathbb{F}_2$ ,  $\det \mathbf{A} \in \{0, 1\}$  and  $0^2 = 0$ , so  $\det \mathbf{A} = 1$ . This agrees with (a), since  $-1 = 1$  in  $\mathbb{F}_2$ .



**Exercise C2: Left multiplication on matrices.**

Let  $\mathbf{A} \in M_n(F)$ , and let  $L_{\mathbf{A}}: M_n(F) \rightarrow M_n(F)$ ,  $L_{\mathbf{A}}(\mathbf{X}) = \mathbf{A}\mathbf{X}$ . This is a linear operator on a space of dimension  $n^2$ . Prove that

$$\det L_{\mathbf{A}} = (\det \mathbf{A})^n.$$

- (a) Order the matrix units column by column,  $\mathcal{B} = (\mathbf{E}_{11}, \mathbf{E}_{21}, \dots, \mathbf{E}_{n1}, \mathbf{E}_{12}, \dots, \mathbf{E}_{n2}, \dots, \mathbf{E}_{1n}, \dots, \mathbf{E}_{nn})$ . Show that  $[L_{\mathbf{A}}]_{\mathcal{B}}$  consists of  $n$  copies of  $\mathbf{A}$  placed along the diagonal, with zeros elsewhere.
- (b) Show that  $L_{\mathbf{A}\mathbf{B}} = L_{\mathbf{A}}L_{\mathbf{B}}$ , and prove the formula when  $\mathbf{A}$  is an elementary matrix.
- (c) Prove the formula for invertible  $\mathbf{A}$ , and then for non-invertible  $\mathbf{A}$ .

*Hint: for a swap matrix, identify  $[L_{\mathbf{P}_{ij}}]_{\mathcal{B}}$  as a permutation matrix and use [Exercise \(B2: The determinant of  \$\mathbf{P}\_{\sigma}\$ \)](#).*

*Solution.*

- (a) By [Theorem 0.9.2](#), the  $l$ -th column of  $\mathbf{A}\mathbf{E}_{kl}$  is  $\mathbf{A}\mathbf{e}_k$ , the  $k$ -th column of  $\mathbf{A}$ , and the other columns are zero. So  $\mathbf{A}\mathbf{E}_{kl} = \sum_i a_{ik}\mathbf{E}_{il}$ . In  $\mathcal{B}$ , the coordinates of this vector are zero outside the  $l$ -th block of  $n$  positions, and inside that block they are the  $k$ -th column of  $\mathbf{A}$ . Hence the columns of  $[L_{\mathbf{A}}]_{\mathcal{B}}$  belonging to the  $l$ -th block are those of  $\mathbf{A}$ , placed in the rows of the  $l$ -th block:  $[L_{\mathbf{A}}]_{\mathcal{B}}$  has  $n$  copies of  $\mathbf{A}$  along the diagonal.
- (b)  $L_{\mathbf{A}\mathbf{B}}(\mathbf{X}) = \mathbf{A}\mathbf{B}\mathbf{X} = L_{\mathbf{A}}(L_{\mathbf{B}}(\mathbf{X}))$  by associativity. Now let  $\mathbf{E}$  be elementary. If  $\mathbf{E} = \mathbf{I}_n + c\mathbf{E}_{ij}$  with  $i \neq j$ , then by (a)  $[L_{\mathbf{E}}]_{\mathcal{B}}$  is triangular (upper if  $i < j$ , lower if  $i > j$ ) with all diagonal entries 1, so  $\det L_{\mathbf{E}} = 1 = (\det \mathbf{E})^n$  by [Theorem 6.4.6](#) and [Theorem 6.5.1](#) (e). If  $\mathbf{E} = \mathbf{D}_i(c)$ , then  $[L_{\mathbf{E}}]_{\mathcal{B}}$  is diagonal with  $c$  appearing  $n$  times and 1 elsewhere, so  $\det L_{\mathbf{E}} = c^n = (\det \mathbf{E})^n$ . If  $\mathbf{E} = \mathbf{P}_{ij}$ , then  $[L_{\mathbf{E}}]_{\mathcal{B}}$  has exactly one 1 in each row and column, so it is a permutation matrix  $\mathbf{P}_{\sigma}$  ([Definition 2.6.2](#)), and  $\sigma$  exchanges the positions of  $\mathbf{E}_{il}$  and  $\mathbf{E}_{jl}$  for each  $l$ :  $\sigma$  is a product of  $n$  disjoint transpositions. By [Exercise \(B2: The determinant of  \$\mathbf{P}\_{\sigma}\$ \)](#), [Theorem 6.3.1](#) and [Corollary 6.3.2](#),  $\det L_{\mathbf{E}} = \operatorname{sgn}(\sigma) = (-1)^n = (\det \mathbf{E})^n$ .
- (c) If  $\mathbf{A}$  is invertible, then  $\mathbf{A} = \mathbf{E}_1 \cdots \mathbf{E}_k$  with elementary  $\mathbf{E}_m$  ([Theorem 2.4.4](#) (d)). By (b) and [Theorem 6.5.8](#),  $\det L_{\mathbf{A}} = \det L_{\mathbf{E}_1} \cdots \det L_{\mathbf{E}_k} = (\det \mathbf{E}_1)^n \cdots (\det \mathbf{E}_k)^n = (\det \mathbf{E}_1 \cdots \det \mathbf{E}_k)^n = (\det \mathbf{A})^n$ , where the last step is [Theorem 6.5.4](#). If  $\mathbf{A}$  is not invertible, there is  $\mathbf{v} \neq \mathbf{0}$  with  $\mathbf{A}\mathbf{v} = \mathbf{0}$  ([Theorem 2.4.4](#) (b)). The matrix  $\mathbf{X} = (\mathbf{v} \ \mathbf{0} \ \cdots \ \mathbf{0})$  is non-zero and  $\mathbf{A}\mathbf{X} = \mathbf{0}$ , so  $L_{\mathbf{A}}$  is not injective, hence not invertible, and  $\det L_{\mathbf{A}} = 0$  by [Theorem 6.5.8](#). Also  $\det \mathbf{A} = 0$  by [Theorem 6.5.2](#), so  $(\det \mathbf{A})^n = 0$  as  $n \geq 1$ .

**Exercise C3: Non-square factors.**

- (a) Give  $\mathbf{A} \in M_{2 \times 1}(\mathbb{R})$  and  $\mathbf{B} \in M_{1 \times 2}(\mathbb{R})$  with  $\det(\mathbf{B}\mathbf{A}) \neq \det(\mathbf{A}\mathbf{B})$ .
- (b) Let  $m > n$ ,  $\mathbf{A} \in M_{m \times n}(F)$  and  $\mathbf{B} \in M_{n \times m}(F)$ . Prove that  $\det(\mathbf{A}\mathbf{B}) = 0$ .
- (c) Explain why (a) does not contradict [Theorem 6.5.4](#).

*Solution.*

- (a) Let  $\mathbf{A} = \begin{pmatrix} 1 \\ 0 \end{pmatrix}$  and  $\mathbf{B} = \begin{pmatrix} 1 & 0 \end{pmatrix}$ . Then  $\mathbf{BA} = (1)$ , of determinant 1, and  $\mathbf{AB} = \begin{pmatrix} 1 & 0 \\ 0 & 0 \end{pmatrix}$ , of determinant 0.
- (b) By [Theorem 2.5.9](#),  $\text{rank}(\mathbf{AB}) \leq \text{rank } \mathbf{A} \leq n$ , since  $\text{col}(\mathbf{A})$  is spanned by  $n$  columns. So  $\text{rank}(\mathbf{AB}) < m$ , and the  $m \times m$  matrix  $\mathbf{AB}$  is not invertible by [Theorem 6.5.3](#) (d). Hence  $\det(\mathbf{AB}) = 0$  by [Theorem 6.5.2](#).
- (c) [Theorem 6.5.4](#) concerns two **square** matrices of the same size, and then  $\det(\mathbf{AB}) = \det(\mathbf{BA})$ . Here  $\mathbf{A}$  and  $\mathbf{B}$  are not square,  $\det \mathbf{A}$  and  $\det \mathbf{B}$  are not defined, and  $\mathbf{AB}$ ,  $\mathbf{BA}$  have different sizes.

## 6.6 Cofactor Expansion, the Adjugate and Cramer's Rule

We can now compute determinants by elimination and we know what they decide. This section adds a third view: a determinant of size  $n$  is a combination of determinants of size  $n - 1$ . This recursive formula, the Laplace expansion, is rarely the fastest way to compute, but it has two important consequences. It gives an explicit formula for the inverse of a matrix, through the adjugate, and an explicit formula for the solution of a square linear system, Cramer's rule. Both rest on one identity built from the entries alone, with no division, and that identity holds for matrices of polynomials too, which is what later chapters need.

### 6.6.1 Minors and cofactors

Group the six terms of a  $3 \times 3$  determinant (from the examples after [Definition 6.4.1](#)) by the entry they take from column 1:

$$\det \mathbf{A} = a_{11}(a_{22}a_{33} - a_{32}a_{23}) - a_{21}(a_{12}a_{33} - a_{32}a_{13}) + a_{31}(a_{12}a_{23} - a_{22}a_{13}).$$

Each bracket is a  $2 \times 2$  determinant. The bracket beside  $a_{11}$  is the determinant of what remains of  $\mathbf{A}$  after deleting row 1 and column 1; the bracket beside  $a_{21}$  is what remains after deleting row 2 and column 1; and similarly for  $a_{31}$ . The signs alternate. These smaller determinants recur constantly, so they get names.

*A minor is the determinant of what is left after crossing out one row and one column; a cofactor is that minor with a checkerboard sign.*

#### Definition 6.6.1 — Minor and cofactor

Let  $n \geq 2$ ,  $\mathbf{A} \in M_n(F)$  and  $i, j \in \{1, \dots, n\}$ . The  $(i, j)$ -**minor** of  $\mathbf{A}$  is

$$M_{ij} := \det(\text{the } (n-1) \times (n-1) \text{ matrix obtained from } \mathbf{A} \text{ by deleting row } i \text{ and column } j).$$

The  $(i, j)$ -**cofactor** of  $\mathbf{A}$  is  $C_{ij} := (-1)^{i+j} M_{ij}$ .

The rows and columns that remain keep their original order. The sign  $(-1)^{i+j}$  depends only on the position, and follows a checkerboard starting with  $+$  in the top-left corner:

$$\begin{pmatrix} + & - & + & \cdots \\ - & + & - & \cdots \\ + & - & + & \cdots \\ \vdots & \vdots & \vdots & \ddots \end{pmatrix}.$$

When the matrix matters we write  $M_{ij}(\mathbf{A})$  and  $C_{ij}(\mathbf{A})$ .

**Examples.** For  $\mathbf{A} = \begin{pmatrix} a & b \\ c & d \end{pmatrix}$ , the minors are  $1 \times 1$  determinants:  $M_{11} = d$ ,  $M_{12} = c$ ,  $M_{21} = b$ ,  $M_{22} = a$ , and the cofactors are  $C_{11} = d$ ,  $C_{12} = -c$ ,  $C_{21} = -b$ ,  $C_{22} = a$ . For the matrix  $\mathbf{A} = \begin{pmatrix} 2 & 1 & 3 \\ 0 & 4 & 1 \\ 5 & 2 & 1 \end{pmatrix}$  of the previous sections, the cofactors along column 1 are

$$\begin{aligned} C_{11} &= + \det \begin{pmatrix} 4 & 1 \\ 2 & 1 \end{pmatrix} = 2, \\ C_{21} &= - \det \begin{pmatrix} 1 & 3 \\ 2 & 1 \end{pmatrix} = 5, \\ C_{31} &= + \det \begin{pmatrix} 1 & 3 \\ 4 & 1 \end{pmatrix} = -11, \end{aligned}$$

and  $a_{11}C_{11} + a_{21}C_{21} + a_{31}C_{31} = 4 + 0 - 55 = -51$ , which is  $\det \mathbf{A}$  as computed before. For the identity  $\mathbf{I}_n$ ,  $C_{ii} = \det \mathbf{I}_{n-1} = 1$ , and  $C_{ij} = 0$  for  $i \neq j$ , because deleting row  $i$  and a different column  $j$  leaves a matrix with a zero column.

**Non-example by minimal change.** A minor and a cofactor differ only by a sign, and forgetting the sign is the commonest slip. For the  $2 \times 2$  matrix above,  $M_{12} = c$  but  $C_{12} = -c$ ; using minors throughout would turn  $a_{11}C_{11} + a_{21}C_{21} = ad - cb$  into  $ad + cb$ , which is the permanent, not the determinant.

## 6.6.2 The Laplace expansion

The  $3 \times 3$  grouping above is a special case of a general identity, valid along **every** row and **every** column.

### Theorem 6.6.1 — Laplace Expansion

Let  $n \geq 2$  and  $\mathbf{A} \in M_n(F)$ , with cofactors  $C_{ij}$ .

(a) **(Along column  $j$ )** For each  $j \in \{1, \dots, n\}$ ,

$$\det \mathbf{A} = a_{1j}C_{1j} + a_{2j}C_{2j} + \cdots + a_{nj}C_{nj} = \sum_{i=1}^n a_{ij}C_{ij}.$$

(b) (**Along row  $i$** ) For each  $i \in \{1, \dots, n\}$ ,

$$\det \mathbf{A} = a_{i1}C_{i1} + a_{i2}C_{i2} + \cdots + a_{in}C_{in} = \sum_{j=1}^n a_{ij}C_{ij}.$$

*Idea.* The plan for (a):

- ① Write column  $j$  as  $\sum_i a_{ij}\mathbf{e}_i$  and use linearity in column  $j$ . This leaves  $n$  determinants, the  $i$ -th with column  $j$  replaced by  $\mathbf{e}_i$ .
- ② In such a determinant, move the lone 1 to the top-left corner. To keep the other rows and columns in their original order, use **adjacent** swaps:  $j - 1$  column swaps and  $i - 1$  row swaps, costing  $(-1)^{(i-1)+(j-1)} = (-1)^{i+j}$ .
- ③ A matrix whose first column is  $\mathbf{e}_1$  has the same determinant as its lower-right  $(n - 1) \times (n - 1)$  block: only permutations fixing 1 contribute to the Leibniz formula. After ②, that block is exactly the matrix defining  $M_{ij}$ .

Part (b) is part (a) for the transpose.

*Proof.*

- (a) Fix  $j$ . For  $i \in \{1, \dots, n\}$ , let  $\mathbf{B}^{(i)}$  be the matrix obtained from  $\mathbf{A}$  by replacing column  $j$  with  $\mathbf{e}_i$ .

*Claim.*  $\det \mathbf{B}^{(i)} = C_{ij}$ .

*Proof.* Write  $\mathbf{B} = \mathbf{B}^{(i)}$ , with columns  $\mathbf{b}_1, \dots, \mathbf{b}_n$ , so  $\mathbf{b}_j = \mathbf{e}_i$  and  $\mathbf{b}_k$  is the  $k$ -th column of  $\mathbf{A}$  for  $k \neq j$ .

*Moving the column.* Swap columns  $j-1$  and  $j$ , then columns  $j-2$  and  $j-1$ , and so on, ending with columns 1 and 2. After these  $j-1$  adjacent swaps the columns are  $\mathbf{b}_j, \mathbf{b}_1, \dots, \mathbf{b}_{j-1}, \mathbf{b}_{j+1}, \dots, \mathbf{b}_n$ : the column  $\mathbf{b}_j$  is first and the others are in their original order. Call this matrix  $\mathbf{B}_1$ . By [Theorem 6.5.1](#) (d), each swap multiplies the determinant by  $-1$ , so  $\det \mathbf{B}_1 = (-1)^{j-1} \det \mathbf{B}$ .

*Moving the row.* In the same way, move row  $i$  of  $\mathbf{B}_1$  to the top by  $i-1$  adjacent row swaps, keeping the other rows in their original order. Call the result  $\mathbf{B}_2$ . By [Theorem 6.5.1](#) (a),  $\det \mathbf{B}_2 = (-1)^{i-1} \det \mathbf{B}_1$ .

*The shape of  $\mathbf{B}_2$ .* The first column of  $\mathbf{B}_1$  is  $\mathbf{e}_i$ , and moving row  $i$  to the top turns it into  $\mathbf{e}_1$ . For  $p, q \geq 2$ , the  $(p, q)$ -entry of  $\mathbf{B}_2$  comes from the rows of  $\mathbf{A}$  other than  $i$  and the columns of  $\mathbf{A}$  other than  $j$ , both in their original order. So the lower-right  $(n-1) \times (n-1)$  block of  $\mathbf{B}_2$  is the matrix  $\mathbf{A}'$  obtained from  $\mathbf{A}$  by deleting row  $i$  and column  $j$ . Writing  $b'_{pq}$  for the entries of  $\mathbf{B}_2$ , we have  $b'_{11} = 1$ ,  $b'_{p1} = 0$  for  $p \geq 2$ , and  $b'_{pq} = a'_{p-1, q-1}$  for  $p, q \geq 2$ .

*The block determinant.* In the Leibniz formula for  $\det \mathbf{B}_2$ , the factor  $b'_{\sigma(1)1}$  is 0 unless  $\sigma(1) = 1$ . Let  $G = \{\sigma \in S_n : \sigma(1) = 1\}$ . For  $\sigma \in G$ , define  $\rho \in S_{n-1}$  by  $\rho(k) = \sigma(k+1) - 1$ ; this is a bijection of  $\{1, \dots, n-1\}$ , because  $\sigma$  maps  $\{2, \dots, n\}$  bijectively onto itself. The map  $\sigma \mapsto \rho$  is a bijection  $G \rightarrow S_{n-1}$ , with inverse  $\rho \mapsto \sigma$ ,  $\sigma(1) = 1$ ,  $\sigma(k+1) = \rho(k) + 1$ . An inversion of  $\sigma$  ([Definition 6.3.1](#)) is a pair  $p < q$  with  $\sigma(p) > \sigma(q)$ ; it never has  $p = 1$ , since  $\sigma(1) = 1$  is smallest, and for  $p, q \geq 2$  it is exactly an inversion  $(p-1, q-1)$  of  $\rho$ . So  $\text{sgn}(\sigma) = \text{sgn}(\rho)$  by [Definition 6.3.2](#). Therefore

$$\det \mathbf{B}_2 = \sum_{\sigma \in G} \text{sgn}(\sigma) \cdot 1 \cdot \prod_{q=2}^n b'_{\sigma(q)q} = \sum_{\rho \in S_{n-1}} \text{sgn}(\rho) \prod_{k=1}^{n-1} a'_{\rho(k)k} = \det \mathbf{A}' = M_{ij}.$$

Combining,  $\det \mathbf{B} = (-1)^{j-1} (-1)^{i-1} \det \mathbf{B}_2 = (-1)^{i+j} M_{ij} = C_{ij}$ . ■

Now the  $j$ -th column of  $\mathbf{A}$  is  $\sum_{i=1}^n a_{ij} \mathbf{e}_i$ , and the other columns of  $\mathbf{A}$  and of each  $\mathbf{B}^{(i)}$  agree. By linearity of  $\det$  in column  $j$  ([Theorem 6.4.3](#)) and the claim,

$$\det \mathbf{A} = \sum_{i=1}^n a_{ij} \det \mathbf{B}^{(i)} = \sum_{i=1}^n a_{ij} C_{ij}.$$

- (b) Fix  $i$ . The  $(k, i)$ -entry of  $\mathbf{A}^\top$  is  $a_{ik}$ . Deleting row  $k$  and column  $i$  from  $\mathbf{A}^\top$  gives the transpose of the matrix obtained from  $\mathbf{A}$  by deleting row  $i$  and column  $k$ , so  $M_{ki}(\mathbf{A}^\top) = M_{ik}(\mathbf{A})$  by [Theorem 6.4.5](#), and hence  $C_{ki}(\mathbf{A}^\top) = C_{ik}(\mathbf{A})$ . Applying (a) to  $\mathbf{A}^\top$  along column  $i$ , and [Theorem 6.4.5](#) again,

$$\det \mathbf{A} = \det \mathbf{A}^\top = \sum_{k=1}^n a_{ik} C_{ki}(\mathbf{A}^\top) = \sum_{k=1}^n a_{ik} C_{ik}(\mathbf{A}).$$

This proves the theorem. ■

*Quick check.* In step ② of the Idea, why not move column  $j$  to the front with the single swap of columns 1 and  $j$ ?

*Solution.* A single swap costs only one factor  $-1$ , but it puts the old first column in position  $j$ , so the lower-right block is no longer the matrix defining  $M_{ij}$ : its columns are out of order. For  $j \geq 3$ , putting that column back in place within the block takes  $j - 2$  further adjacent swaps, and the total sign is again  $(-1)^{1+(j-2)} = (-1)^{j-1}$ . Adjacent swaps keep the count honest from the start.

*Remark. Over a commutative ring.* The proof uses multilinearity and the swap rule for columns and rows, the Leibniz formula, and [Theorem 6.4.5](#). All of these are division-free (see the remarks at the ends of the previous two sections). So the Laplace expansion holds for matrices with entries in any commutative ring  $R$ , for instance  $F[x]$ .

The expansion is most useful along a row or column with many zeros, since each zero entry kills its term without computing the minor.

**Example. (A  $4 \times 4$  Expansion)**

Compute  $\det \mathbf{A}$  for

$$\mathbf{A} = \begin{pmatrix} 3 & 1 & 0 & 2 \\ 1 & 0 & 2 & 0 \\ 2 & 4 & 1 & 3 \\ 0 & 1 & 5 & 2 \end{pmatrix} \in M_4(\mathbb{R})$$

by expanding along a suitable row.

*Solution.* Row 2 has two zeros. Its signs in the checkerboard are  $-$ ,  $+$ ,  $-$ ,  $+$ . By [Theorem 6.6.1](#) (b) along row 2,

$$\det \mathbf{A} = 1 \cdot C_{21} + 0 \cdot C_{22} + 2 \cdot C_{23} + 0 \cdot C_{24} = -M_{21} - 2M_{23}.$$

Deleting row 2 and column 1, and then row 2 and column 3,

$$M_{21} = \det \begin{pmatrix} 1 & 0 & 2 \\ 4 & 1 & 3 \\ 1 & 5 & 2 \end{pmatrix}, \quad M_{23} = \det \begin{pmatrix} 3 & 1 & 2 \\ 2 & 4 & 3 \\ 0 & 1 & 2 \end{pmatrix}.$$

Expand  $M_{21}$  along its first row, which has a zero:

$$M_{21} = 1 \cdot \det \begin{pmatrix} 1 & 3 \\ 5 & 2 \end{pmatrix} - 0 + 2 \cdot \det \begin{pmatrix} 4 & 1 \\ 1 & 5 \end{pmatrix} = (2 - 15) + 2(20 - 1) = 25.$$

Expand  $M_{23}$  along its third row  $(0, 1, 2)$ , with signs  $+, -, +$ :

$$M_{23} = -1 \cdot \det \begin{pmatrix} 3 & 2 \\ 2 & 3 \end{pmatrix} + 2 \cdot \det \begin{pmatrix} 3 & 1 \\ 2 & 4 \end{pmatrix} = -(9 - 4) + 2(12 - 2) = 15.$$

Hence  $\det \mathbf{A} = -25 - 2 \cdot 15 = -55$ .

*Warning.* **The sign belongs to the position, not to the place in the row.** Along row 2 the signs start with  $-$ . Using the first-row pattern  $+, -, +, -$  in the example above would give  $1 \cdot 25 + 2 \cdot 15 = 55$ , the wrong sign. Always read the sign as  $(-1)^{i+j}$ .

Expansion alone is slow: expanding fully down to  $1 \times 1$  determinants reproduces all  $n!$  Leibniz terms. In practice one first creates zeros with replacements, which do not change the determinant, and then expands along the line of zeros.

### 6.6.3 The adjugate

In Chapter 0 we inverted a  $2 \times 2$  matrix by multiplying it by  $\begin{pmatrix} d & -b \\ -c & a \end{pmatrix}$  (Theorem 0.9.8). Compare with the cofactors listed after Definition 6.6.1:  $C_{11} = d$ ,  $C_{12} = -c$ ,  $C_{21} = -b$ ,  $C_{22} = a$ . The matrix of Chapter 0 is the matrix of cofactors, **transposed**. That recurring matrix deserves a name in every size.

*The adjugate is the transposed matrix of cofactors.*

#### Definition 6.6.2 — Adjugate

Let  $n \geq 2$  and  $\mathbf{A} \in M_n(F)$ . The **adjugate** of  $\mathbf{A}$  is the matrix  $\text{adj } \mathbf{A} \in M_n(F)$  with entries

$$(\text{adj } \mathbf{A})_{ij} := C_{ji},$$

that is, the **transpose** of the matrix  $(C_{ij})$  of cofactors. For  $n = 1$  we set  $\text{adj } \mathbf{A} := (1)$ .

In words: the entry in row  $i$ , column  $j$  of  $\text{adj } \mathbf{A}$  is the cofactor from row  $j$ , column  $i$  of  $\mathbf{A}$ . The adjugate is defined for **every** square matrix, invertible or not.

**Examples.** For  $n = 2$ ,  $\text{adj} \begin{pmatrix} a & b \\ c & d \end{pmatrix} = \begin{pmatrix} d & -b \\ -c & a \end{pmatrix}$ . For  $\mathbf{I}_n$ , the cofactors computed above give  $\text{adj } \mathbf{I}_n = \mathbf{I}_n$ . For

$$\mathbf{B} = \begin{pmatrix} 1 & 2 & 0 \\ 0 & 1 & 1 \\ 1 & 0 & 3 \end{pmatrix},$$

the nine cofactors, row by row, are  $C_{11} = 3$ ,  $C_{12} = 1$ ,  $C_{13} = -1$ ;  $C_{21} = -6$ ,  $C_{22} = 3$ ,  $C_{23} = 2$ ;  $C_{31} = 2$ ,  $C_{32} = -1$ ,  $C_{33} = 1$ . Transposing,

$$\text{adj } \mathbf{B} = \begin{pmatrix} 3 & -6 & 2 \\ 1 & 3 & -1 \\ -1 & 2 & 1 \end{pmatrix}, \quad \mathbf{B} \text{adj } \mathbf{B} = \begin{pmatrix} 5 & 0 & 0 \\ 0 & 5 & 0 \\ 0 & 0 & 5 \end{pmatrix},$$

and  $\det \mathbf{B} = 1 \cdot 3 + 2 \cdot 1 + 0 \cdot (-1) = 5$ , by expansion along row 1.

**Non-example by minimal change.** Skip the transpose. The cofactor matrix  $\begin{pmatrix} 3 & 1 & -1 \\ -6 & 3 & 2 \\ 2 & -1 & 1 \end{pmatrix}$  has the same entries in other places, and  $\mathbf{B}$  times it is  $\begin{pmatrix} -9 & 7 & 3 \\ -4 & 2 & 3 \\ 9 & -2 & 2 \end{pmatrix}$ , which is not a multiple of  $\mathbf{I}_3$ . The transpose is what lines up row  $i$  of  $\mathbf{B}$  with the cofactors of row  $i$ .

The product  $\mathbf{B} \operatorname{adj} \mathbf{B} = (\det \mathbf{B}) \mathbf{I}_3$  is no accident. Its diagonal entries are row expansions of  $\det \mathbf{B}$ . Its off-diagonal entries are expansions too, of a matrix that looks like  $\mathbf{B}$  but has a repeated row.

### Theorem 6.6.2 — The Adjugate Identity

For every  $\mathbf{A} \in M_n(F)$ ,

$$\mathbf{A} \operatorname{adj} \mathbf{A} = \operatorname{adj} \mathbf{A} \mathbf{A} = (\det \mathbf{A}) \mathbf{I}_n.$$

*Idea.* The  $(i, k)$ -entry of  $\mathbf{A} \operatorname{adj} \mathbf{A}$  is  $\sum_j a_{ij} C_{kj}$ : the entries of row  $i$  against the cofactors of row  $k$ . If  $i = k$ , this is the expansion of  $\det \mathbf{A}$  along row  $i$ . If  $i \neq k$ , build a look-alike matrix  $\tilde{\mathbf{A}}$ : copy  $\mathbf{A}$ , but overwrite row  $k$  with row  $i$ . The cofactors along row  $k$  never see row  $k$ , because computing them deletes it, so  $\tilde{\mathbf{A}}$  has the same cofactors along row  $k$  as  $\mathbf{A}$ . Expanding  $\det \tilde{\mathbf{A}}$  along row  $k$  gives exactly  $\sum_j a_{ij} C_{kj}$ , and  $\det \tilde{\mathbf{A}} = 0$  because two rows are equal.

*Proof.* For  $n = 1$ ,  $\mathbf{A} = (a)$ ,  $\operatorname{adj} \mathbf{A} = (1)$ , and both products are  $(a) = (\det \mathbf{A}) \mathbf{I}_1$ . Let  $n \geq 2$ .

*The product  $\mathbf{A} \operatorname{adj} \mathbf{A}$ .* By Theorem 0.9.2 and Definition 6.6.2, its  $(i, k)$ -entry is

$$\sum_{j=1}^n a_{ij} (\operatorname{adj} \mathbf{A})_{jk} = \sum_{j=1}^n a_{ij} C_{kj}.$$

If  $i = k$ , this is  $\det \mathbf{A}$  by Theorem 6.6.1 (b) along row  $i$ . Suppose  $i \neq k$ , and let  $\tilde{\mathbf{A}}$  be the matrix obtained from  $\mathbf{A}$  by replacing row  $k$  with a copy of row  $i$ . Rows  $i$  and  $k$  of  $\tilde{\mathbf{A}}$  are equal, so  $\det \tilde{\mathbf{A}} = 0$  by Theorem 6.4.5. Expanding along row  $k$  by Theorem 6.6.1 (b),

$$0 = \det \tilde{\mathbf{A}} = \sum_{j=1}^n \tilde{a}_{kj} C_{kj}(\tilde{\mathbf{A}}).$$

Here  $\tilde{a}_{kj} = a_{ij}$ . The minor  $M_{kj}(\tilde{\mathbf{A}})$  is computed from  $\tilde{\mathbf{A}}$  with row  $k$  deleted, and outside row  $k$  the matrices  $\tilde{\mathbf{A}}$  and  $\mathbf{A}$  agree; so  $C_{kj}(\tilde{\mathbf{A}}) = C_{kj}(\mathbf{A})$ . Hence  $\sum_j a_{ij} C_{kj} = 0$ . Therefore  $\mathbf{A} \operatorname{adj} \mathbf{A} = (\det \mathbf{A}) \mathbf{I}_n$ .

*The product  $\operatorname{adj} \mathbf{A} \mathbf{A}$ .* Its  $(i, k)$ -entry is  $\sum_j C_{ji} a_{jk}$ . If  $i = k$ , this is  $\det \mathbf{A}$  by Theorem 6.6.1 (a) along column  $i$ . If  $i \neq k$ , let  $\hat{\mathbf{A}}$  be  $\mathbf{A}$  with column  $i$  replaced by a copy of column  $k$ . It has two equal columns, so  $\det \hat{\mathbf{A}} = 0$  by Theorem 6.4.3. Expanding along column  $i$ ,  $0 = \sum_j \hat{a}_{ji} C_{ji}(\hat{\mathbf{A}}) = \sum_j a_{jk} C_{ji}(\mathbf{A})$ , since deleting column  $i$  removes the only column where  $\hat{\mathbf{A}}$  and  $\mathbf{A}$  differ. Therefore  $\operatorname{adj} \mathbf{A} \mathbf{A} = (\det \mathbf{A}) \mathbf{I}_n$ . This proves the theorem. ■



*Remark. Over a commutative ring.* This proof uses only [Theorem 6.6.1](#) and the alternating property, so the adjugate identity holds for  $\mathbf{A} \in M_n(R)$  over any commutative ring  $R$ . For  $R = F[x]$ , the matrix  $\text{adj}(x\mathbf{I} - \mathbf{A})$  has polynomial entries and  $(x\mathbf{I} - \mathbf{A}) \text{adj}(x\mathbf{I} - \mathbf{A}) = \det(x\mathbf{I} - \mathbf{A})\mathbf{I}$ . This identity reappears in Chapter 8. Over  $R$  the identity also shows that  $\mathbf{A}$  has an inverse in  $M_n(R)$  as soon as  $\det \mathbf{A}$  has an inverse in  $R$ , namely  $(\det \mathbf{A})^{-1} \text{adj } \mathbf{A}$ .

Dividing by the determinant, when we may, gives the inverse.

### Corollary 6.6.3 — Inverse via the Adjugate

Let  $\mathbf{A} \in M_n(F)$  with  $\det \mathbf{A} \neq 0$ . Then  $\mathbf{A}$  is invertible and

$$\mathbf{A}^{-1} = \frac{1}{\det \mathbf{A}} \text{adj } \mathbf{A}.$$

*Proof.* By [Theorem 6.6.2](#),  $\mathbf{A}((\det \mathbf{A})^{-1} \text{adj } \mathbf{A}) = (\det \mathbf{A})^{-1}(\det \mathbf{A})\mathbf{I}_n = \mathbf{I}_n$ , using that scalars commute with matrices ([Theorem 0.9.4](#)). By [Theorem 2.4.5](#),  $\mathbf{A}$  is invertible with inverse  $(\det \mathbf{A})^{-1} \text{adj } \mathbf{A}$ . ■

For  $n = 2$  this is [Theorem 0.9.8](#) again. For the matrix  $\mathbf{B}$  above,  $\mathbf{B}^{-1} = \frac{1}{5} \begin{pmatrix} 3 & -6 & 2 \\ 1 & 3 & -1 \\ -1 & 2 & 1 \end{pmatrix}$ .

*Warning. The adjugate exists for singular matrices too.* Nothing in [Definition 6.6.2](#) needs  $\det \mathbf{A} \neq 0$ . For  $S = \begin{pmatrix} 1 & 2 & 3 \\ 2 & 4 & 6 \\ 1 & 0 & 1 \end{pmatrix}$ , with  $\det S = 0$ , one computes

$\text{adj } S = \begin{pmatrix} 4 & -2 & 0 \\ 4 & -2 & 0 \\ -4 & 2 & 0 \end{pmatrix} \neq 0$ , and  $S \text{adj } S = 0$ , as [Theorem 6.6.2](#) predicts. So “adj  $\mathbf{A}$  exists” says nothing about invertibility, and the formula  $\mathbf{A}^{-1} = \text{adj } \mathbf{A} / \det \mathbf{A}$  must never be used before checking  $\det \mathbf{A} \neq 0$ .

*Quick check.* What are  $\text{adj}(c\mathbf{I}_n)$  for  $c \in F$  and  $n \geq 2$ , and  $\text{adj } 0$  for the zero matrix in  $M_n(F)$ ,  $n \geq 2$ ? What is  $\text{adj } 0$  for  $n = 1$ ?

*Solution.* Deleting row  $i$  and column  $i$  from  $c\mathbf{I}_n$  leaves  $c\mathbf{I}_{n-1}$ , of determinant  $c^{n-1}$ ; deleting row  $i$  and a different column leaves a zero column, of determinant 0. So  $\text{adj}(c\mathbf{I}_n) = c^{n-1}\mathbf{I}_n$ ; in particular  $\text{adj } 0 = 0$  for  $n \geq 2$ , since  $0^{n-1} = 0$ . For  $n = 1$ ,  $\text{adj}(0) = (1)$  by definition, and indeed  $(0)(1) = (0) = (\det 0)\mathbf{I}_1$ .

## 6.6.4 Cramer's rule

For an invertible  $\mathbf{A}$ , the system  $\mathbf{A}\mathbf{x} = \mathbf{b}$  has exactly one solution. The adjugate expresses it in closed form, and the closed form has a direct proof by linearity that shows where it comes from.

For  $k \in \{1, \dots, n\}$ , write  $\mathbf{A}_k(\mathbf{b})$  for the matrix obtained from  $\mathbf{A}$  by replacing column  $k$  with  $\mathbf{b}$ .

### Theorem 6.6.4 — Cramer's Rule

Let  $\mathbf{A} \in M_n(F)$  be invertible and  $\mathbf{b} \in F^n$ . The unique solution  $\mathbf{x}$  of  $\mathbf{A}\mathbf{x} = \mathbf{b}$  is given by

$$x_k = \frac{\det \mathbf{A}_k(\mathbf{b})}{\det \mathbf{A}}, \quad k = 1, \dots, n.$$

*Idea.* If  $\mathbf{A}\mathbf{x} = \mathbf{b}$ , then  $\mathbf{b} = x_1\mathbf{a}_1 + \dots + x_n\mathbf{a}_n$ . Put this combination into column  $k$  and expand by linearity. Every term except the  $k$ -th has the column  $\mathbf{a}_j$  twice, so it vanishes, and what remains is  $x_k \det \mathbf{A}$ .

*Proof.* By Theorem 0.9.7 (7),  $\mathbf{A}\mathbf{x} = \mathbf{b}$  has exactly one solution  $\mathbf{x}$ . By Theorem 0.9.3,  $\mathbf{b} = \sum_{j=1}^n x_j \mathbf{a}_j$ , where  $\mathbf{a}_j$  is the  $j$ -th column of  $\mathbf{A}$ . Since  $\det$  is linear in column  $k$  (Theorem 6.4.3),

$$\det \mathbf{A}_k(\mathbf{b}) = \sum_{j=1}^n x_j \det \mathbf{A}_k(\mathbf{a}_j).$$

For  $j \neq k$ , the matrix  $\mathbf{A}_k(\mathbf{a}_j)$  has the column  $\mathbf{a}_j$  in positions  $j$  and  $k$ , so its determinant is 0. For  $j = k$ ,  $\mathbf{A}_k(\mathbf{a}_k) = \mathbf{A}$ . Hence  $\det \mathbf{A}_k(\mathbf{b}) = x_k \det \mathbf{A}$ . Since  $\mathbf{A}$  is invertible,  $\det \mathbf{A} \neq 0$  by Theorem 6.5.2, and dividing gives the formula. ■

Alternatively,  $\mathbf{x} = \mathbf{A}^{-1}\mathbf{b} = (\det \mathbf{A})^{-1} \operatorname{adj} \mathbf{A} \mathbf{b}$  by Corollary 6.6.3, and the  $k$ -th entry of  $\operatorname{adj} \mathbf{A} \mathbf{b}$  is  $\sum_j C_{jk} b_j$ , which is the expansion of  $\det \mathbf{A}_k(\mathbf{b})$  along column  $k$  (Theorem 6.6.1 (a)); the cofactors along column  $k$  of  $\mathbf{A}_k(\mathbf{b})$  and of  $\mathbf{A}$  agree, because computing them deletes column  $k$ .

For example, the system  $2x + y = 5$ ,  $7x + 4y = 18$  has coefficient matrix of determinant  $8 - 7 = 1$ , so

$$x = \det \begin{pmatrix} 5 & 1 \\ 18 & 4 \end{pmatrix} = 2, \quad y = \det \begin{pmatrix} 2 & 5 \\ 7 & 18 \end{pmatrix} = 1,$$

and indeed  $2 \cdot 2 + 1 = 5$  and  $7 \cdot 2 + 4 = 18$ .

**Warning. Cramer's rule is a theoretical tool, not an algorithm.** It needs  $n + 1$  determinants of size  $n$ . Evaluated by the Leibniz formula this is about  $(n + 1)!$  products; even with each determinant computed by elimination it costs about  $n^4/3$  operations, while Gaussian elimination solves the system once in about  $n^3/3$ . For  $n = 20$ , term-by-term Leibniz evaluation at  $10^9$  multiplications per second would take tens of thousands of years. Use Cramer's rule to understand **how** the solution depends on  $\mathbf{A}$  and  $\mathbf{b}$ , and elimination to find it.

## 6.6.5 Integer matrices

The adjugate formula shows that each entry of  $\mathbf{A}^{-1}$  is a polynomial in the entries of  $\mathbf{A}$  divided by the single polynomial  $\det \mathbf{A}$ . In particular, the entries of  $\mathbf{A}^{-1}$  change continuously with those of  $\mathbf{A}$  over  $\mathbb{R}$ , as long as  $\det \mathbf{A} \neq 0$ . Here is an arithmetic consequence. A matrix with integer entries lies in  $M_n(\mathbb{Q})$ , and we may ask when its inverse again has integer entries.

**Corollary 6.6.5 — Integer Matrices with Integer Inverses**

Let  $\mathbf{A} \in M_n(\mathbb{Q})$  have all entries in  $\mathbb{Z}$ .

- (a)  $\det \mathbf{A} \in \mathbb{Z}$ , and  $\operatorname{adj} \mathbf{A}$  has all entries in  $\mathbb{Z}$ .
- (b)  $\mathbf{A}$  is invertible and  $\mathbf{A}^{-1}$  has all entries in  $\mathbb{Z}$  if and only if  $\det \mathbf{A} = 1$  or  $\det \mathbf{A} = -1$ .

*Proof.*

- (a) The Leibniz formula (Definition 6.4.1) is a sum of products of entries with signs  $\pm 1$ , so the determinant of a matrix with integer entries is an integer. Each cofactor is  $\pm$  such a determinant, so  $\operatorname{adj} \mathbf{A}$  has integer entries (for  $n = 1$ ,  $\operatorname{adj} \mathbf{A} = (1)$ ).
- (b)  $(\Leftarrow)$  If  $\det \mathbf{A} = \pm 1$ , then  $\mathbf{A}$  is invertible and  $\mathbf{A}^{-1} = \pm \operatorname{adj} \mathbf{A}$  by Corollary 6.6.3, which has integer entries by (a).  $(\Rightarrow)$  Suppose  $\mathbf{A}^{-1}$  has integer entries. By Theorem 6.5.4,  $\det \mathbf{A} \det(\mathbf{A}^{-1}) = \det \mathbf{I}_n = 1$ , and both factors are integers by (a). The only integers whose product is 1 are  $1 \cdot 1$  and  $(-1)(-1)$ , so  $\det \mathbf{A} = \pm 1$ . ■

For example,  $\begin{pmatrix} 2 & 1 \\ 5 & 3 \end{pmatrix}$  has determinant 1 and inverse  $\begin{pmatrix} 3 & -1 \\ -5 & 2 \end{pmatrix}$ , while  $\begin{pmatrix} 2 & 1 \\ 0 & 1 \end{pmatrix}$  has determinant 2 and inverse  $\begin{pmatrix} 1/2 & -1/2 \\ 0 & 1 \end{pmatrix}$ . In the language of the remark after Theorem 6.6.2, part (b) says that a matrix over the ring  $\mathbb{Z}$  is invertible over  $\mathbb{Z}$  exactly when its determinant is invertible in  $\mathbb{Z}$ .

**6.6.6 Exercises****6.6.6.1 A. Check your understanding****Exercise A1.**

- (a) Define the  $(i, j)$ -minor, the  $(i, j)$ -cofactor and the adjugate of  $\mathbf{A} \in M_n(F)$ ,  $n \geq 2$ .
- (b) What are the signs  $(-1)^{i+j}$  of the cofactors in positions  $(2, 3)$  and  $(4, 4)$ ?
- (c) True or false:  $\operatorname{adj} \mathbf{A}$  is defined only when  $\mathbf{A}$  is invertible. Justify your answer.
- (d) True or false: Cramer's rule (Theorem 6.6.4) gives the solutions of every square linear system  $\mathbf{A}\mathbf{x} = \mathbf{b}$ . Justify your answer.
- (e) In the proof of Theorem 6.6.2, which matrix is expanded to show that  $\sum_j a_{ij}C_{kj} = 0$  for  $i \neq k$ , and why is its determinant 0?
- (f) True or false: if  $\mathbf{A}$  has integer entries and  $\det \mathbf{A} = 2$ , then  $\mathbf{A}^{-1}$  has integer entries. Justify your answer.

*Solution.*

- (a)  $M_{ij}$  is the determinant of the  $(n-1) \times (n-1)$  matrix obtained by deleting row  $i$  and column  $j$ ;  $C_{ij} = (-1)^{i+j}M_{ij}$ ;  $\operatorname{adj} \mathbf{A}$  is the matrix with  $(i, j)$ -entry  $C_{ji}$  (Definition 6.6.1, Definition 6.6.2).

- (b)  $(-1)^5 = -1$  and  $(-1)^8 = 1$ .
- (c) False. It is defined for every square matrix; for instance  $\text{adj} \begin{pmatrix} 1 & 1 \\ 1 & 1 \end{pmatrix} = \begin{pmatrix} 1 & -1 \\ -1 & 1 \end{pmatrix}$ , although the matrix is singular.
- (d) False. It requires  $\det \mathbf{A} \neq 0$ . For  $\mathbf{A} = \begin{pmatrix} 1 & 1 \\ 1 & 1 \end{pmatrix}$  and  $\mathbf{b} = (1, 1)$ , the system has infinitely many solutions, and the formula would divide by 0.
- (e) The matrix  $\tilde{\mathbf{A}}$  obtained from  $\mathbf{A}$  by replacing row  $k$  with row  $i$ , expanded along row  $k$ . It has two equal rows,  $i$  and  $k$ , so its determinant is 0.
- (f) False. By [Corollary 6.6.5](#), an integer inverse forces  $\det \mathbf{A} = \pm 1$ . For instance  $\text{diag}(2, 1)$  has inverse  $\text{diag}(\frac{1}{2}, 1)$ .

### 6.6.6.2 B. Practice

#### Exercise B1: Adjugate and inverse.

Let  $\mathbf{A} = \begin{pmatrix} 1 & 2 & -1 \\ 0 & 1 & 3 \\ 2 & 1 & 1 \end{pmatrix} \in M_3(\mathbb{R})$ . Compute all nine cofactors,  $\text{adj } \mathbf{A}$  and  $\det \mathbf{A}$ . Verify that  $\mathbf{A} \text{adj } \mathbf{A} = (\det \mathbf{A}) \mathbf{I}_3$ , and hence write down  $\mathbf{A}^{-1}$ .

*Solution.* The cofactors are

$$\begin{aligned} C_{11} &= +(1 \cdot 1 - 3 \cdot 1) = -2, & C_{12} &= -(0 \cdot 1 - 3 \cdot 2) = 6, \\ C_{13} &= +(0 \cdot 1 - 1 \cdot 2) = -2, & C_{21} &= -(2 \cdot 1 - (-1) \cdot 1) = -3, \\ C_{22} &= +(1 \cdot 1 - (-1) \cdot 2) = 3, & C_{23} &= -(1 \cdot 1 - 2 \cdot 2) = 3, \\ C_{31} &= +(2 \cdot 3 - (-1) \cdot 1) = 7, & C_{32} &= -(1 \cdot 3 - (-1) \cdot 0) = -3, \\ C_{33} &= +(1 \cdot 1 - 2 \cdot 0) = 1. \end{aligned}$$

Transposing the cofactor matrix ([Definition 6.6.2](#)),

$$\text{adj } \mathbf{A} = \begin{pmatrix} -2 & -3 & 7 \\ 6 & 3 & -3 \\ -2 & 3 & 1 \end{pmatrix}.$$

Expanding along row 1 ([Theorem 6.6.1](#)),  $\det \mathbf{A} = 1 \cdot (-2) + 2 \cdot 6 + (-1)(-2) = 12$ . Row 1 of  $\mathbf{A}$  against the columns of  $\text{adj } \mathbf{A}$  gives  $-2 + 12 + 2 = 12$ ,  $-3 + 6 - 3 = 0$ ,  $7 - 6 - 1 = 0$ ; row 2 gives  $0 + 6 - 6 = 0$ ,  $0 + 3 + 9 = 12$ ,  $0 - 3 + 3 = 0$ ; row 3 gives  $-4 + 6 - 2 = 0$ ,  $-6 + 3 + 3 = 0$ ,  $14 - 3 + 1 = 12$ . So  $\mathbf{A} \text{adj } \mathbf{A} = 12 \mathbf{I}_3$ . By [Corollary 6.6.3](#),

$$\mathbf{A}^{-1} = \frac{1}{12} \begin{pmatrix} -2 & -3 & 7 \\ 6 & 3 & -3 \\ -2 & 3 & 1 \end{pmatrix}.$$

#### Exercise B2: Cramer's rule.

Use Cramer's rule to solve

$$x_1 + 2x_2 - x_3 = -3, \quad x_2 + 3x_3 = 5, \quad 2x_1 + x_2 + x_3 = 3$$

over  $\mathbb{R}$ . Check your answer by substitution.

*Solution.* The coefficient matrix is the matrix  $\mathbf{A}$  of [Exercise \(B1: Adjugate and inverse\)](#), with  $\det \mathbf{A} = 12 \neq 0$ , and  $\mathbf{b} = (-3, 5, 3)$ . Expanding each  $\mathbf{A}_k(\mathbf{b})$  along its first row,

$$\begin{aligned}\det \mathbf{A}_1(\mathbf{b}) &= \det \begin{pmatrix} -3 & 2 & -1 \\ 5 & 1 & 3 \\ 3 & 1 & 1 \end{pmatrix} \\ &= -3(1 - 3) - 2(5 - 9) - 1(5 - 3) = 6 + 8 - 2 = 12, \\ \det \mathbf{A}_2(\mathbf{b}) &= \det \begin{pmatrix} 1 & -3 & -1 \\ 0 & 5 & 3 \\ 2 & 3 & 1 \end{pmatrix} \\ &= 1(5 - 9) + 3(0 - 6) - 1(0 - 10) = -4 - 18 + 10 = -12, \\ \det \mathbf{A}_3(\mathbf{b}) &= \det \begin{pmatrix} 1 & 2 & -3 \\ 0 & 1 & 5 \\ 2 & 1 & 3 \end{pmatrix} \\ &= 1(3 - 5) - 2(0 - 10) - 3(0 - 2) = -2 + 20 + 6 = 24.\end{aligned}$$

By [Theorem 6.6.4](#),  $x_1 = 12/12 = 1$ ,  $x_2 = -12/12 = -1$ ,  $x_3 = 24/12 = 2$ . Check:  $1 - 2 - 2 = -3$ ,  $-1 + 6 = 5$ ,  $2 - 1 + 2 = 3$ .

### Exercise B3: Choosing the best line.

Compute  $\det \mathbf{N}$  for

$$\mathbf{N} = \begin{pmatrix} 2 & 0 & 0 & 3 \\ 1 & 4 & 0 & -1 \\ 0 & 5 & 0 & 2 \\ 3 & 1 & 2 & 0 \end{pmatrix} \in M_4(\mathbb{R}),$$

expanding along the row or column that needs the least work.

*Solution.* Column 3 has a single non-zero entry,  $a_{43} = 2$ , whose sign is  $(-1)^{4+3} = -1$ . By [Theorem 6.6.1](#) (a) along column 3,  $\det \mathbf{N} = -2M_{43}$ , where

$$\begin{aligned}M_{43} &= \det \begin{pmatrix} 2 & 0 & 3 \\ 1 & 4 & -1 \\ 0 & 5 & 2 \end{pmatrix} \\ &= 2(4 \cdot 2 - (-1) \cdot 5) - 0 + 3(1 \cdot 5 - 4 \cdot 0) = 26 + 15 = 41,\end{aligned}$$

expanding along the first row. Hence  $\det \mathbf{N} = -82$ .

### 6.6.6.3 C. Going deeper

#### Exercise C1: The determinant of the adjugate.

Let  $n \geq 2$  and  $\mathbf{A} \in M_n(F)$ . Prove that  $\det(\operatorname{adj} \mathbf{A}) = (\det \mathbf{A})^{n-1}$ .

*Hint: treat  $\det \mathbf{A} \neq 0$  and  $\det \mathbf{A} = 0$  separately; in the second case, ask whether  $\operatorname{adj} \mathbf{A}$  can be invertible.*

*Solution. Case 1:*  $\det \mathbf{A} \neq 0$ . By [Theorem 6.6.2](#) and [Theorem 6.5.4](#),  $\det \mathbf{A} \det(\operatorname{adj} \mathbf{A}) = \det((\det \mathbf{A})\mathbf{I}_n) = (\det \mathbf{A})^n$ , where the last step is [Theorem 6.4.6](#). Dividing by  $\det \mathbf{A} \neq 0$  gives  $\det(\operatorname{adj} \mathbf{A}) = (\det \mathbf{A})^{n-1}$ .

*Case 2:*  $\det \mathbf{A} = 0$ . Then  $\mathbf{A} \operatorname{adj} \mathbf{A} = \mathbf{0}$  by [Theorem 6.6.2](#). Suppose, for a contradiction, that  $\operatorname{adj} \mathbf{A}$  is invertible. Then  $\mathbf{A} = (\mathbf{A} \operatorname{adj} \mathbf{A})(\operatorname{adj} \mathbf{A})^{-1} = \mathbf{0}$ . But every minor of the zero matrix is the determinant of a zero matrix of size  $n-1 \geq 1$ , which is 0, so  $\operatorname{adj} \mathbf{0} = \mathbf{0}$ , which is not invertible. This contradiction shows that  $\operatorname{adj} \mathbf{A}$  is not invertible, so  $\det(\operatorname{adj} \mathbf{A}) = 0 = (\det \mathbf{A})^{n-1}$  by [Theorem 6.5.2](#), since  $n-1 \geq 1$ .

### Exercise C2: The rank of the adjugate.

Let  $n \geq 2$  and  $\mathbf{A} \in M_n(F)$ . Prove that

$$\operatorname{rank}(\operatorname{adj} \mathbf{A}) = \begin{cases} n & \text{if } \operatorname{rank} \mathbf{A} = n, \\ 1 & \text{if } \operatorname{rank} \mathbf{A} = n-1, \\ 0 & \text{if } \operatorname{rank} \mathbf{A} \leq n-2. \end{cases}$$

- (a) Prove the case  $\operatorname{rank} \mathbf{A} = n$ .
- (b) Suppose  $\operatorname{rank} \mathbf{A} \leq n-2$ . Prove that every minor  $M_{ij}$  is 0.
- (c) Suppose  $\operatorname{rank} \mathbf{A} = n-1$ . Prove that every column of  $\operatorname{adj} \mathbf{A}$  lies in  $\operatorname{nul}(\mathbf{A})$ , and deduce  $\operatorname{rank}(\operatorname{adj} \mathbf{A}) \leq 1$ .
- (d) Suppose  $\operatorname{rank} \mathbf{A} = n-1$ . Prove that some minor  $M_{ij}$  is non-zero, and conclude.

*Hint for (d):* choose  $n-1$  independent columns of  $\mathbf{A}$ , then  $n-1$  independent rows of the resulting  $n \times (n-1)$  matrix.

*Solution.*

- (a) If  $\operatorname{rank} \mathbf{A} = n$ , then  $\mathbf{A}$  is invertible by [Theorem 6.5.3](#), so  $\det \mathbf{A} \neq 0$  and  $\operatorname{adj} \mathbf{A} = (\det \mathbf{A})\mathbf{A}^{-1}$  by [Corollary 6.6.3](#). This is invertible, since  $\mathbf{A}^{-1}$  is and  $\det \mathbf{A} \neq 0$  ([Theorem 0.9.7](#)), so  $\operatorname{rank}(\operatorname{adj} \mathbf{A}) = n$ .
- (b) Fix  $i, j$ , and let  $\mathbf{A}'$  be  $\mathbf{A}$  with row  $i$  and column  $j$  deleted. The  $n-1$  columns of  $\mathbf{A}$  other than column  $j$  lie in  $\operatorname{col}(\mathbf{A})$ , of dimension  $\operatorname{rank} \mathbf{A} \leq n-2$ , so they are linearly dependent by [Theorem 1.6.3](#):  $\sum_{k \neq j} c_k \mathbf{a}_k = \mathbf{0}$  with the  $c_k$  not all zero. Deleting the  $i$ -th entry of every vector preserves this relation, so the columns of  $\mathbf{A}'$  are dependent. By [Theorem 2.4.4](#),  $\mathbf{A}'$  is not invertible, and  $M_{ij} = \det \mathbf{A}' = 0$  by [Theorem 6.5.2](#). Hence  $\operatorname{adj} \mathbf{A} = \mathbf{0}$  and  $\operatorname{rank}(\operatorname{adj} \mathbf{A}) = 0$ .
- (c) Since  $\operatorname{rank} \mathbf{A} < n$ ,  $\det \mathbf{A} = 0$  by [Theorem 6.5.3](#), so  $\mathbf{A} \operatorname{adj} \mathbf{A} = \mathbf{0}$  by [Theorem 6.6.2](#). By [Theorem 0.9.2](#),  $\mathbf{A}$  kills each column of  $\operatorname{adj} \mathbf{A}$ , so  $\operatorname{col}(\operatorname{adj} \mathbf{A}) \subseteq \operatorname{nul}(\mathbf{A})$ . By [Theorem 2.5.7](#),  $\dim \operatorname{nul}(\mathbf{A}) = n - (n-1) = 1$ . Hence  $\operatorname{rank}(\operatorname{adj} \mathbf{A}) \leq 1$ .
- (d) By [Theorem 2.5.2](#), the pivot columns of  $\mathbf{A}$  form a basis of  $\operatorname{col}(\mathbf{A})$ ; there are  $n-1$  of them, so exactly one column index  $j$  is not a pivot column. Let  $\mathbf{A}_0 \in M_{n \times (n-1)}(F)$  consist of the other  $n-1$  columns. They are independent, so  $\operatorname{rank} \mathbf{A}_0 = n-1$ , and  $\dim \operatorname{row}(\mathbf{A}_0) = n-1$  by [Theorem 2.5.5](#). The  $n$  rows of  $\mathbf{A}_0$  span  $\operatorname{row}(\mathbf{A}_0)$ , so by [Theorem 1.5.4](#) some  $n-1$  of them are independent; since  $n-1$  independent vectors in a space of dimension  $n-1$  form a basis, sifting leaves out exactly one row, say row  $i$ . Deleting row  $i$  from  $\mathbf{A}_0$  gives an  $(n-1) \times (n-1)$  matrix  $\mathbf{A}'$  with independent rows, so  $\det \mathbf{A}' \neq 0$  by [Theorem 6.5.3](#). But  $\mathbf{A}'$  is  $\mathbf{A}$  with row

$i$  and column  $j$  deleted, so  $M_{ij} \neq 0$ , hence  $C_{ij} \neq 0$  and  $\text{adj } \mathbf{A} \neq 0$ . With (c),  $\text{rank}(\text{adj } \mathbf{A}) = 1$ .

**Exercise C3: An adjugate over  $F[x]$ .**

Work in  $M_2(\mathbb{R}[x])$ , using the remarks after [Theorem 6.6.1](#) and [Theorem 6.6.2](#).

- (a) For  $\mathbf{A} = \begin{pmatrix} 1 & 2 \\ 3 & 4 \end{pmatrix}$ , compute  $\det(x\mathbf{I} - \mathbf{A})$  and  $\text{adj}(x\mathbf{I} - \mathbf{A})$ , and verify  $(x\mathbf{I} - \mathbf{A}) \text{adj}(x\mathbf{I} - \mathbf{A}) = \det(x\mathbf{I} - \mathbf{A}) \mathbf{I}$  by multiplying out.
- (b) For a general  $\mathbf{A} = \begin{pmatrix} a & b \\ c & d \end{pmatrix} \in M_2(\mathbb{R})$ , show that  $\text{adj}(x\mathbf{I} - \mathbf{A}) = x\mathbf{I} - \text{adj } \mathbf{A}$  and  $\mathbf{A} + \text{adj } \mathbf{A} = (\text{tr } \mathbf{A})\mathbf{I}$ .
- (c) Deduce from (b), without using [Theorem 6.6.2](#) over  $\mathbb{R}[x]$ , that  $(x\mathbf{I} - \mathbf{A}) \text{adj}(x\mathbf{I} - \mathbf{A}) = (x^2 - (\text{tr } \mathbf{A})x + \det \mathbf{A})\mathbf{I}$ , and check that  $x^2 - (\text{tr } \mathbf{A})x + \det \mathbf{A} = \det(x\mathbf{I} - \mathbf{A})$ .

*Solution.*

- (a)  $x\mathbf{I} - \mathbf{A} = \begin{pmatrix} x-1 & -2 \\ -3 & x-4 \end{pmatrix}$ , so  $\det(x\mathbf{I} - \mathbf{A}) = (x-1)(x-4) - 6 = x^2 - 5x - 2$ ,  
and by [Definition 6.6.2](#),  $\text{adj}(x\mathbf{I} - \mathbf{A}) = \begin{pmatrix} x-4 & 2 \\ 3 & x-1 \end{pmatrix}$ . The product has entries  
 $(x-1)(x-4) - 6 = x^2 - 5x - 2$ ,  $2(x-1) - 2(x-1) = 0$ ,  $-3(x-4) + 3(x-4) = 0$   
and  $-6 + (x-4)(x-1) = x^2 - 5x - 2$ , so it is  $(x^2 - 5x - 2)\mathbf{I}$ .
- (b)  $x\mathbf{I} - \mathbf{A} = \begin{pmatrix} x-a & -b \\ -c & x-d \end{pmatrix}$ , so  $\text{adj}(x\mathbf{I} - \mathbf{A}) = \begin{pmatrix} x-d & b \\ c & x-a \end{pmatrix} = x\mathbf{I} - \begin{pmatrix} d & -b \\ -c & a \end{pmatrix} =$   
 $x\mathbf{I} - \text{adj } \mathbf{A}$ . Also  $\mathbf{A} + \text{adj } \mathbf{A} = \begin{pmatrix} a+d & 0 \\ 0 & a+d \end{pmatrix} = (\text{tr } \mathbf{A})\mathbf{I}$ .
- (c) Since  $x\mathbf{I}$  commutes with every matrix, expanding with (b) gives

$$\begin{aligned} (x\mathbf{I} - \mathbf{A})(x\mathbf{I} - \text{adj } \mathbf{A}) &= x^2\mathbf{I} - x(\mathbf{A} + \text{adj } \mathbf{A}) + \mathbf{A} \text{adj } \mathbf{A} \\ &= x^2\mathbf{I} - (\text{tr } \mathbf{A})x\mathbf{I} + (ad - bc)\mathbf{I}, \end{aligned}$$

where  $\mathbf{A} \text{adj } \mathbf{A} = (ad - bc)\mathbf{I}$  is the computation of [Theorem 0.9.8](#) over  $\mathbb{R}$ . Finally  $\det(x\mathbf{I} - \mathbf{A}) = (x-a)(x-d) - bc = x^2 - (a+d)x + (ad - bc)$ , which is  $x^2 - (\text{tr } \mathbf{A})x + \det \mathbf{A}$ .

## 6.7 Minors, Rank and Cauchy–Binet

The determinant of a square matrix answers a yes-or-no question: is the matrix invertible? A rectangular matrix has no determinant, and a singular square matrix has determinant 0, which says nothing about *how* singular it is. Yet every matrix is full of square pieces, and each piece has a determinant. This section shows that these small determinants measure the rank exactly. It then answers a second question left open by multiplicativity: if  $\mathbf{A}$  is  $m \times n$  and  $\mathbf{B}$  is  $n \times m$ , the product  $\mathbf{AB}$  is square, so what is  $\det(\mathbf{AB})$ ? The answer, the Cauchy–Binet formula, is proved by the same multilinear expansion that gave uniqueness of the determinant in [Theorem 6.4.2](#).

## 6.7.1 Submatrices and minors

In [Definition 6.6.1](#) we deleted one row and one column of a square matrix and took the determinant of what was left. That is a special case of a more flexible operation: keep **any**  $k$  rows and **any**  $k$  columns. The recurring object is the square block sitting at the crossings of the chosen rows and columns, and it deserves a name.

*A minor is the determinant of a square block cut out of a matrix by choosing some rows and the same number of columns.*

For a positive integer  $m$ , write  $[m] = \{1, 2, \dots, m\}$ .

### Definition 6.7.1 — Submatrix and Minor

Let  $\mathbf{A} = (a_{ij}) \in M_{m \times n}(F)$ , and let  $1 \leq k \leq \min(m, n)$ . Let  $I = \{i_1 < i_2 < \dots < i_k\} \subseteq [m]$  and  $J = \{j_1 < j_2 < \dots < j_k\} \subseteq [n]$  be sets with **exactly**  $k$  elements each, listed in **increasing** order. The **submatrix** of  $\mathbf{A}$  on rows  $I$  and columns  $J$  is

$$\mathbf{A}_{I,J} := (a_{i_p j_q})_{p,q=1}^k \in M_k(F),$$

the  $k \times k$  matrix whose  $(p, q)$ -entry is  $a_{i_p j_q}$ . Its determinant  $\det \mathbf{A}_{I,J}$  is a  $k \times k$  **minor** of  $\mathbf{A}$ .

In words: cross out every row not in  $I$  and every column not in  $J$ , and push the surviving entries together **without changing their order**. The chosen rows need not be adjacent, and neither need the columns. The sets  $I$  and  $J$  have the **same** size, so the block is square and has a determinant.

A  $1 \times 1$  minor is a single entry,  $\det \mathbf{A}_{\{i\},\{j\}} = a_{ij}$ . For a square  $\mathbf{A} \in M_n(F)$  with  $n \geq 2$ , the minor  $M_{ij}$  of [Definition 6.6.1](#) is the  $(n-1) \times (n-1)$  minor with  $I = [n] \setminus \{i\}$  and  $J = [n] \setminus \{j\}$ . The only  $n \times n$  minor of a square matrix is  $\det \mathbf{A}$  itself. An  $m \times n$  matrix has  $\binom{m}{k} \binom{n}{k}$  minors of size  $k$ , one for each choice of  $I$  and  $J$ .

**Example.** Let

$$\mathbf{A} = \begin{pmatrix} 1 & 2 & 0 \\ 0 & 1 & 3 \end{pmatrix} \in M_{2 \times 3}(\mathbb{Q}).$$

Its  $2 \times 2$  minors all use  $I = \{1, 2\}$ , and there are three choices of  $J$ :

$$\begin{aligned} \det \mathbf{A}_{I,\{1,2\}} &= \det \begin{pmatrix} 1 & 2 \\ 0 & 1 \end{pmatrix} = 1, \\ \det \mathbf{A}_{I,\{1,3\}} &= \det \begin{pmatrix} 1 & 0 \\ 0 & 3 \end{pmatrix} = 3, \\ \det \mathbf{A}_{I,\{2,3\}} &= \det \begin{pmatrix} 2 & 0 \\ 1 & 3 \end{pmatrix} = 6. \end{aligned}$$

Its six  $1 \times 1$  minors are its entries. It has no  $3 \times 3$  minors, since it has only two rows.

**A degenerate example.** Every minor of the zero matrix  $0_{m \times n}$  is 0: each block is a zero matrix, whose determinant is 0 by [Theorem 6.4.6](#). This case matters because it is exactly the matrix of rank 0, and the theorem below has to treat it separately.

**Non-example by minimal change.** Take the same rows and columns but list the columns of



$\mathbf{A}$  above in the order 3, 1: the block  $\begin{pmatrix} 0 & 1 \\ 3 & 0 \end{pmatrix}$  has determinant  $-3$ , not 3. Every entry is still an entry of  $\mathbf{A}$  from rows  $\{1, 2\}$  and columns  $\{1, 3\}$ ; what fails is the clause **listed in increasing order**. Reordering columns changes the determinant by a sign (Theorem 6.2.1), so the order convention is what makes “the” minor on  $(I, J)$  a single well-defined number.

**Why this definition.** Insisting on increasing order costs nothing and removes all sign ambiguity. Insisting on  $|I| = |J|$  is forced: only square blocks have determinants.

*Warning. Rank is not read off principal minors.* A **principal** minor uses the same index set for rows and columns ( $I = J$ ). It is tempting to test rank using only these, but they can all vanish for a non-zero matrix. For  $\mathbf{N} = \begin{pmatrix} 0 & 1 \\ 0 & 0 \end{pmatrix}$ , the principal minors are 0, 0 and  $\det \mathbf{N} = 0$ , yet  $\text{rank } \mathbf{N} = 1$ . The non-zero minor is the off-diagonal entry  $\det \mathbf{N}_{\{1\}, \{2\}} = 1$ . The theorem below uses **all** minors.

*Quick check.* How many  $2 \times 2$  minors does a  $3 \times 4$  matrix have? How many  $3 \times 3$  minors?

*Solution.* A  $2 \times 2$  minor needs 2 of the 3 rows and 2 of the 4 columns:  $\binom{3}{2} \binom{4}{2} = 3 \cdot 6 = 18$ . A  $3 \times 3$  minor uses all three rows and 3 of the 4 columns:  $\binom{3}{3} \binom{4}{3} = 4$ .

## 6.7.2 Rank via minors

A non-zero determinant of a square block means the block is invertible, so its columns are independent. Independent columns inside a block come from independent columns of the whole matrix. So a large non-zero minor forces large rank. The converse, that large rank produces a large non-zero minor, needs us to find the right rows as well as the right columns. Both directions together give a description of rank using only determinants:

### Theorem 6.7.1 — Rank via Minors

Let  $\mathbf{A} \in M_{m \times n}(F)$  be **non-zero**.

- (a)  $\text{rank } \mathbf{A}$  is the **largest** integer  $k$  such that  $\mathbf{A}$  has a **non-zero**  $k \times k$  minor.
- (b) Equivalently, for each  $1 \leq k \leq \min(m, n)$ :  $\text{rank } \mathbf{A} \geq k$  if and only if some  $k \times k$  minor of  $\mathbf{A}$  is non-zero; and  $\text{rank } \mathbf{A} < k$  if and only if **every**  $k \times k$  minor of  $\mathbf{A}$  is zero.

*Idea.* Two inequalities. ① A non-zero  $k \times k$  minor on rows  $I$  and columns  $J$  makes  $\mathbf{A}_{I,J}$  invertible; a dependence among the columns of  $\mathbf{A}$  indexed by  $J$  would restrict to a dependence among the columns of  $\mathbf{A}_{I,J}$ . So  $\text{rank } \mathbf{A} \geq k$ . ② If  $\text{rank } \mathbf{A} = r$ , choose  $r$  independent columns (pivot columns). The  $m \times r$  matrix they form has rank  $r$ , so by row rank = column rank it has  $r$  independent rows. Those rows cut out an  $r \times r$  block with independent rows, hence a non-zero determinant.

*Proof. Step 1: a non-zero  $k \times k$  minor forces  $\text{rank } \mathbf{A} \geq k$ .* Let  $I = \{i_1 < \dots < i_k\}$  and  $J = \{j_1 < \dots < j_k\}$  with  $\det \mathbf{A}_{I,J} \neq 0$ . Write  $\mathbf{a}_1, \dots, \mathbf{a}_n$  for the columns of  $\mathbf{A}$ . Suppose  $c_1 \mathbf{a}_{j_1} + \dots + c_k \mathbf{a}_{j_k} = \mathbf{0}$  with  $c_1, \dots, c_k \in F$ . Reading only the entries in rows  $i_1, \dots, i_k$  of this vector equation gives  $\sum_q c_q a_{i_p j_q} = 0$  for  $p = 1, \dots, k$ , that is,  $\mathbf{A}_{I,J} \mathbf{c} = \mathbf{0}$

with  $\mathbf{c} = (c_1, \dots, c_k)$ . By [Theorem 6.5.2](#),  $\mathbf{A}_{I,J}$  is invertible, so  $\mathbf{c} = \mathbf{0}$  by [Theorem 2.4.4](#). Hence  $(\mathbf{a}_{j_1}, \dots, \mathbf{a}_{j_k})$  is a linearly independent list in  $\text{col}(\mathbf{A})$ , and by [Theorem 1.6.3](#) (a),  $k \leq \dim \text{col}(\mathbf{A}) = \text{rank } \mathbf{A}$ .

**Step 2:  $\mathbf{A}$  has a non-zero  $r \times r$  minor, where  $r = \text{rank } \mathbf{A}$ .** Since  $\mathbf{A} \neq \mathbf{0}$ , some column is non-zero, so  $r \geq 1$ . By [Theorem 2.5.2](#) (a), the pivot columns  $\mathbf{a}_{j_1}, \dots, \mathbf{a}_{j_r}$  ( $j_1 < \dots < j_r$ ) are linearly independent. Let  $J = \{j_1, \dots, j_r\}$  and let  $\mathbf{A}' = \mathbf{A}_{[m],J} \in M_{m \times r}(F)$  be the matrix with these columns. By [Theorem 2.5.10](#) (a),  $\text{rank } \mathbf{A}' = r$ , so by [Theorem 2.5.5](#) the row space of  $\mathbf{A}'$  has dimension  $r$ . The  $m$  rows of  $\mathbf{A}'$  span its row space, so by [Theorem 1.5.4](#) some  $r$  of them, in rows  $i_1 < \dots < i_r$ , form a basis of it; in particular they are linearly independent. These are exactly the rows of  $\mathbf{B} = \mathbf{A}_{I,J} \in M_r(F)$ , where  $I = \{i_1, \dots, i_r\}$ . Transposition is a linear bijection from  $1 \times r$  rows to columns in  $F^r$ , so the columns of  $\mathbf{B}^\top$  are linearly independent. By [Theorem 2.4.4](#),  $\mathbf{B}^\top$  is invertible, so  $\det \mathbf{B}^\top \neq 0$  by [Theorem 6.5.2](#), and  $\det \mathbf{B} = \det \mathbf{B}^\top \neq 0$  by [Theorem 6.4.5](#).

**Conclusion.** By Step 2 there is a non-zero minor of size  $r$ , and by Step 1 there is none of size larger than  $r$ . This proves (a). For (b): if  $\text{rank } \mathbf{A} \geq k$ , then Step 2 gives a non-zero  $r \times r$  minor with  $r \geq k$ , and we still need one of size exactly  $k$ . Expanding that  $r \times r$  determinant along any row ([Theorem 6.6.1](#)) writes it as a combination of  $(r-1) \times (r-1)$  minors of the block, each of which is also an  $(r-1) \times (r-1)$  minor of  $\mathbf{A}$ ; since the determinant is non-zero, one of these minors is non-zero. Repeating  $r-k$  times gives a non-zero  $k \times k$  minor. Conversely, a non-zero  $k \times k$  minor gives  $\text{rank } \mathbf{A} \geq k$  by Step 1. The second equivalence in (b) is the negation of the first. ■

The theorem turns “rank at most  $k-1$ ” into a finite list of polynomial equations in the entries: all  $k \times k$  minors vanish. That is why rank can only **drop** when the entries satisfy special equations, as the next example shows.

### Example. (Rank of a Family of Matrices)

For  $t \in \mathbb{Q}$ , let

$$\mathbf{A}(t) = \begin{pmatrix} 1 & 2 & 1 \\ 2 & t & 3 \\ 3 & 6 & t \end{pmatrix}.$$

Find  $\text{rank } \mathbf{A}(t)$  for every  $t$ .

*Solution.* The only  $3 \times 3$  minor is  $\det \mathbf{A}(t)$ . By cofactor expansion along the first row ([Theorem 6.6.1](#)),

$$\begin{aligned} \det \mathbf{A}(t) &= 1 \cdot (t^2 - 18) - 2 \cdot (2t - 9) + 1 \cdot (12 - 3t) \\ &= t^2 - 7t + 12 = (t-3)(t-4). \end{aligned}$$

So for  $t \neq 3, 4$ ,  $\mathbf{A}(t)$  has a non-zero  $3 \times 3$  minor and  $\text{rank } \mathbf{A}(t) = 3$  by [Theorem 6.7.1](#).

For  $t \in \{3, 4\}$  every  $3 \times 3$  minor is zero, so  $\text{rank } \mathbf{A}(t) \leq 2$ . Look for a  $2 \times 2$  minor that does not involve  $t$ : rows  $\{1, 2\}$  and columns  $\{1, 3\}$  give

$$\det \begin{pmatrix} 1 & 1 \\ 2 & 3 \end{pmatrix} = 1 \neq 0.$$

Hence  $\text{rank } \mathbf{A}(3) = \text{rank } \mathbf{A}(4) = 2$ . The same minor shows  $\text{rank } \mathbf{A}(t) \geq 2$  for every  $t$ , so rank 1 never occurs.

*Remark.* To **compute** the rank of one given matrix, elimination is far cheaper: a  $10 \times 10$  matrix has  $\binom{10}{5}^2 = 63504$  minors of size 5. Minors earn their keep in theory. They show that rank is unchanged by transposing (the minors of  $\mathbf{A}^\top$  are the minors of  $\mathbf{A}$ , by [Theorem 6.4.5](#)) and by enlarging the field ([Exercise \(C2: Rank does not depend on the field\)](#)), and they describe the matrices of rank below  $k$  by polynomial equations.

### 6.7.3 The Cauchy–Binet formula

Multiplicativity, [Theorem 6.5.4](#), handles  $\det(\mathbf{AB})$  when both factors are square. Now let  $\mathbf{A} \in M_{m \times n}(F)$  and  $\mathbf{B} \in M_{n \times m}(F)$ . The product  $\mathbf{AB}$  is  $m \times m$ , so it has a determinant, but  $\det \mathbf{A}$  and  $\det \mathbf{B}$  do not exist. The natural candidates to replace them are the largest minors: the  $m \times m$  minors of  $\mathbf{A}$ , which keep all rows and choose  $m$  columns  $S$ , and the  $m \times m$  minors of  $\mathbf{B}$ , which keep all columns and choose the **same**  $m$  rows  $S$ .

#### Theorem 6.7.2 — Cauchy–Binet Formula

Let  $\mathbf{A} \in M_{m \times n}(F)$  and  $\mathbf{B} \in M_{n \times m}(F)$ . Then

$$\det(\mathbf{AB}) = \sum_{\substack{S \subseteq [n] \\ |S|=m}} \det \mathbf{A}_{[m],S} \det \mathbf{B}_{S,[m]},$$

the sum running over all  $m$ -element subsets  $S$  of  $[n]$ . In particular:

- (a) if  $m > n$ , the sum is empty and  $\det(\mathbf{AB}) = 0$ ;
- (b) if  $m = n$ , the only subset is  $S = [n]$  and the formula is  $\det(\mathbf{AB}) = \det \mathbf{A} \det \mathbf{B}$ .

*Idea.* Treat  $\det$  as a function of the  $m$  columns of  $\mathbf{AB}$ . Column  $k$  of  $\mathbf{AB}$  is  $\mathbf{A}\mathbf{b}_k = \sum_j b_{jk} \mathbf{a}_j$ , a combination of the columns of  $\mathbf{A}$ . Expand by multilinearity in all  $m$  columns at once, exactly as in the uniqueness proof: one term for each way of picking a column  $\mathbf{a}_{\varphi(k)}$  of  $\mathbf{A}$  in slot  $k$ . ① Terms that pick the same column twice vanish, because the determinant is alternating. ② The surviving picks use  $m$  **distinct** columns, forming a set  $S$ , in some order; putting them in increasing order costs the sign of the reordering ([Lemma 6.4.1](#)). ③ Collecting all orders of the same  $S$  produces a Leibniz sum, which is  $\det \mathbf{B}_{S,[m]}$ .

*Proof.* Write  $\mathbf{a}_1, \dots, \mathbf{a}_n \in F^m$  for the columns of  $\mathbf{A}$  and  $b_{jk}$  for the entries of  $\mathbf{B}$ . For  $\mathbf{x}_1, \dots, \mathbf{x}_m \in F^m$ , the function  $D(\mathbf{x}_1, \dots, \mathbf{x}_m) = \det(\mathbf{x}_1, \dots, \mathbf{x}_m)$  is an alternating  $m$ -linear form by [Theorem 6.4.3](#).

By [Theorem 0.9.2](#), column  $k$  of  $\mathbf{AB}$  is  $\mathbf{A}\mathbf{b}_k = \sum_{j=1}^n b_{jk} \mathbf{a}_j$ . Expanding  $D$  by linearity in each of its  $m$  arguments in turn,

$$\det(\mathbf{AB}) = D\left(\sum_{j=1}^n b_{j1} \mathbf{a}_j, \dots, \sum_{j=1}^n b_{jm} \mathbf{a}_j\right) = \sum_{\varphi} \left(\prod_{k=1}^m b_{\varphi(k)k}\right) D(\mathbf{a}_{\varphi(1)}, \dots, \mathbf{a}_{\varphi(m)}),$$

where  $\varphi$  runs over **all** functions  $[m] \rightarrow [n]$ . If  $\varphi$  is not injective, two arguments of  $D$  are equal and the term is 0 because  $D$  is alternating (Definition 6.2.2). So only injective  $\varphi$  contribute. In particular, if  $m > n$  there is no injective  $\varphi$ , and  $\det(\mathbf{AB}) = 0$ , which proves (a).

An injective  $\varphi$  has an image  $S = \{s_1 < \cdots < s_m\} \subseteq [n]$  with  $m$  elements, and then  $\varphi(k) = s_{\pi(k)}$  for exactly one  $\pi \in S_m$ . Conversely each pair  $(S, \pi)$  gives an injective  $\varphi$ . By Lemma 6.4.1 with  $\mathbf{v}_p = \mathbf{a}_{s_p}$ ,

$$D(\mathbf{a}_{\varphi(1)}, \dots, \mathbf{a}_{\varphi(m)}) = \operatorname{sgn}(\pi) D(\mathbf{a}_{s_1}, \dots, \mathbf{a}_{s_m}) = \operatorname{sgn}(\pi) \det \mathbf{A}_{[m], S},$$

since  $\mathbf{A}_{[m], S}$  has columns  $\mathbf{a}_{s_1}, \dots, \mathbf{a}_{s_m}$ . Hence

$$\det(\mathbf{AB}) = \sum_{|S|=m} \det \mathbf{A}_{[m], S} \sum_{\pi \in S_m} \operatorname{sgn}(\pi) \prod_{k=1}^m b_{s_{\pi(k)} k}.$$

The matrix  $\mathbf{B}_{S, [m]}$  has  $(p, k)$ -entry  $b_{s_p k}$ , so its  $(\pi(k), k)$ -entry is  $b_{s_{\pi(k)} k}$ , and the inner sum is  $\det \mathbf{B}_{S, [m]}$  by Definition 6.4.1. This proves the formula. For  $m = n$ , the only  $S$  is  $[n]$ , with  $\mathbf{A}_{[n], [n]} = \mathbf{A}$  and  $\mathbf{B}_{[n], [n]} = \mathbf{B}$ , which gives (b). ■

Part (b) is a second proof of multiplicativity, and part (a) matches what rank predicts: if  $m > n$ , then  $\operatorname{rank}(\mathbf{AB}) \leq \operatorname{rank} \mathbf{A} \leq n < m$  by Theorem 2.5.9, so  $\mathbf{AB}$  is not invertible. The formula also passes a sanity check at  $m = 1$ : a  $1 \times n$  row times an  $n \times 1$  column is the  $1 \times 1$  matrix  $\sum_j a_{1j} b_{j1}$ , and the  $1 \times 1$  minors are single entries.

*Remark.* The proof uses only sums, differences and products of entries, together with Lemma 6.4.1, which is division-free by the remark in §4. So, like the results flagged in §4–§6, the Cauchy–Binet formula holds for matrices with entries in any commutative ring.

#### Example. (Cauchy–Binet for a $2 \times 3$ times $3 \times 2$ Product)

Let

$$\mathbf{A} = \begin{pmatrix} 1 & 2 & 0 \\ 0 & 1 & 3 \end{pmatrix}, \quad \mathbf{B} = \begin{pmatrix} 1 & 0 \\ 1 & 1 \\ 2 & -1 \end{pmatrix}$$

over  $\mathbb{Q}$ . Compute  $\det(\mathbf{AB})$  directly and by the Cauchy–Binet formula. Then compute  $\det(\mathbf{BA})$ .

*Solution. Directly.*  $\mathbf{AB} = \begin{pmatrix} 1+2+0 & 0+2+0 \\ 0+1+6 & 0+1-3 \end{pmatrix} = \begin{pmatrix} 3 & 2 \\ 7 & -2 \end{pmatrix}$ , so  $\det(\mathbf{AB}) = -6 - 14 = -20$ .

*By Cauchy–Binet.* Here  $m = 2$ ,  $n = 3$ , and there are three subsets  $S$ . The minors of  $\mathbf{A}$  were computed after Definition 6.7.1. The minors of  $\mathbf{B}$  keep both columns and rows  $S$ :

| $S$        | $\det \mathbf{A}_{[2], S}$ | $\det \mathbf{B}_{S, [2]}$                              | product |
|------------|----------------------------|---------------------------------------------------------|---------|
| $\{1, 2\}$ | 1                          | $\det \begin{pmatrix} 1 & 0 \\ 1 & 1 \end{pmatrix} = 1$ | 1       |

|            |   |                                                                     |
|------------|---|---------------------------------------------------------------------|
| $\{1, 3\}$ | 3 | $\det \begin{pmatrix} 1 & 0 \\ 2 & -1 \end{pmatrix} = -1 \quad -3$  |
| $\{2, 3\}$ | 6 | $\det \begin{pmatrix} 1 & 1 \\ 2 & -1 \end{pmatrix} = -3 \quad -18$ |

The sum is  $1 - 3 - 18 = -20$ , in agreement.

*The other product.*  $\mathbf{BA}$  is  $3 \times 3$ , and here the roles are  $m = 3 > n = 2$ , so  $\det(\mathbf{BA}) = 0$  by Theorem 6.7.2 (a). Indeed  $\mathbf{BA} = \begin{pmatrix} 1 & 2 & 0 \\ 1 & 3 & 3 \\ 2 & 3 & -3 \end{pmatrix}$ , whose third row is 3 times the first minus the second, so  $\det(\mathbf{BA}) = 0$ .

**Warning.** For non-square factors,  $\det(\mathbf{AB})$  and  $\det(\mathbf{BA})$  differ. For square matrices  $\det(\mathbf{AB}) = \det \mathbf{A} \det \mathbf{B} = \det(\mathbf{BA})$ . In the example above,  $\det(\mathbf{AB}) = -20$  while  $\det(\mathbf{BA}) = 0$ . An even smaller case:  $\mathbf{A} = \begin{pmatrix} 1 & 1 \end{pmatrix}$  and  $\mathbf{B} = \begin{pmatrix} 1 \\ 1 \end{pmatrix}$  give  $\mathbf{AB} = (2)$  and  $\mathbf{BA} = \begin{pmatrix} 1 & 1 \\ 1 & 1 \end{pmatrix}$ , with determinants 2 and 0. Also, do not write “ $\det \mathbf{A} \det \mathbf{B}$ ” for non-square  $\mathbf{A}$ ,  $\mathbf{B}$ : those symbols are undefined, and the sum over  $S$  is what replaces them.

*Quick check.* Let  $\mathbf{u}, \mathbf{v} \in F^n$  with  $n \geq 2$ . Without computing, what is  $\det(\mathbf{uv}^\top)$ ?

*Solution.*  $\mathbf{uv}^\top$  is the product of the  $n \times 1$  matrix  $\mathbf{u}$  and the  $1 \times n$  matrix  $\mathbf{v}^\top$ . Here the product is  $n \times n$  and the middle size is  $1 < n$ , so  $\det(\mathbf{uv}^\top) = 0$  by Theorem 6.7.2 (a).

## 6.7.4 Sums of squares over the reals

Over  $\mathbb{R}$ , take  $\mathbf{B} = \mathbf{A}^\top$ . The minors of  $\mathbf{A}^\top$  are the minors of  $\mathbf{A}$  with rows and columns exchanged, so Cauchy–Binet writes  $\det(\mathbf{AA}^\top)$  as a sum of squares.

### Corollary 6.7.3 — Gram Determinants Are Non-negative

Let  $\mathbf{A} \in M_{m \times n}(\mathbb{R})$ . Then

$$\det(\mathbf{AA}^\top) = \sum_{\substack{S \subseteq [n] \\ |S|=m}} (\det \mathbf{A}_{[m],S})^2 \geq 0,$$

with equality if and only if  $\text{rank } \mathbf{A} < m$ .

*Proof.* By Definition 0.9.10, the  $(p, k)$ -entry of  $(\mathbf{A}^\top)_{S,[m]}$  is the  $(k, s_p)$ -entry of  $\mathbf{A}$ , so  $(\mathbf{A}^\top)_{S,[m]} = (\mathbf{A}_{[m],S})^\top$ . By Theorem 6.4.5,  $\det(\mathbf{A}^\top)_{S,[m]} = \det \mathbf{A}_{[m],S}$ . Substituting into Theorem 6.7.2 with  $\mathbf{B} = \mathbf{A}^\top$  gives the formula. A sum of squares of real numbers is  $\geq 0$ , and it is 0 if and only if every term is 0. Every  $m \times m$  minor of  $\mathbf{A}$  has the form  $\det \mathbf{A}_{[m],S}$ , since an  $m \times m$  minor must use all  $m$  rows. If  $m > n$ , the sum is empty and  $\text{rank } \mathbf{A} \leq n < m$ . Otherwise, by Theorem 6.7.1 (b) (or directly if  $\mathbf{A} = 0$ ), all these

minors vanish if and only if  $\text{rank } \mathbf{A} < m$ .

■

The matrix  $\mathbf{A}\mathbf{A}^\top$  collects the dot products of the rows of  $\mathbf{A}$  with each other. Such matrices of pairwise inner products are **Gram matrices**; Chapters 10 and 12 study them, and there this corollary becomes the statement that a Gram matrix is positive semidefinite. For  $m = 2$  the corollary is a classical identity about two vectors, [Exercise \(B3: The Lagrange identity\)](#). Over  $\mathbb{C}$  or  $\mathbb{F}_p$  the conclusion “ $\geq 0$ ” has no meaning, and the equality case can fail: over  $\mathbb{C}$ ,  $\mathbf{A} = \begin{pmatrix} 1 & i \end{pmatrix}$  has rank 1 but  $\mathbf{A}\mathbf{A}^\top = (1 + i^2) = (0)$ .

## 6.7.5 Exercises

### 6.7.5.1 A. Check your understanding

#### Exercise A1.

- (a) Define the  $k \times k$  minor of  $\mathbf{A} \in M_{m \times n}(F)$  on rows  $I$  and columns  $J$ .
- (b) True or false: if every principal minor of a square matrix is zero, the matrix is zero. Justify your answer.
- (c) True or false: if  $\mathbf{A} \in M_{4 \times 5}(F)$  has a non-zero  $3 \times 3$  minor, then  $\text{rank } \mathbf{A} \geq 3$ . Justify your answer.
- (d) True or false: if  $\mathbf{A} \in M_3(F)$  has a zero  $2 \times 2$  minor, then  $\text{rank } \mathbf{A} \leq 1$ . Justify your answer.
- (e) State the Cauchy–Binet formula, and say what it gives when  $m > n$ .
- (f) Let  $\mathbf{A} \in M_{2 \times 3}(\mathbb{R})$ . Why is  $\det(\mathbf{A}^\top \mathbf{A}) = 0$ ?

*Solution.*

- (a) For  $I = \{i_1 < \dots < i_k\} \subseteq [m]$  and  $J = \{j_1 < \dots < j_k\} \subseteq [n]$ , it is  $\det \mathbf{A}_{I,J}$ , where  $\mathbf{A}_{I,J} \in M_k(F)$  has  $(p, q)$ -entry  $a_{i_p j_q}$  ([Definition 6.7.1](#)).
- (b) False.  $\mathbf{N} = \begin{pmatrix} 0 & 1 \\ 0 & 0 \end{pmatrix}$  has principal minors  $0, 0, 0$  but  $\mathbf{N} \neq 0$ .
- (c) True, by Step 1 of the proof of [Theorem 6.7.1](#), or by part (b) of that theorem.
- (d) False. The rank is decided by whether **some** minor is non-zero, not by one zero minor.  $\mathbf{I}_3$  has the zero  $2 \times 2$  minor on rows  $\{1, 2\}$  and columns  $\{2, 3\}$ , namely  $\det \begin{pmatrix} 0 & 0 \\ 1 & 0 \end{pmatrix} = 0$ , but  $\text{rank } \mathbf{I}_3 = 3$ .
- (e) For  $\mathbf{A} \in M_{m \times n}(F)$  and  $\mathbf{B} \in M_{n \times m}(F)$ ,  $\det(\mathbf{A}\mathbf{B}) = \sum_{|S|=m} \det \mathbf{A}_{[m],S} \det \mathbf{B}_{S,[m]}$ , summed over  $m$ -subsets  $S \subseteq [n]$ . If  $m > n$  there are no such subsets and  $\det(\mathbf{A}\mathbf{B}) = 0$ .
- (f)  $\mathbf{A}^\top \mathbf{A}$  is  $3 \times 3$ , the product of a  $3 \times 2$  and a  $2 \times 3$  matrix. By [Theorem 6.7.2](#) (a) with  $m = 3 > n = 2$ ,  $\det(\mathbf{A}^\top \mathbf{A}) = 0$ .

### 6.7.5.2 B. Practice

**Exercise B1: Rank with a parameter.**

For  $s \in \mathbb{Q}$ , let  $\mathbf{A}(s) = \begin{pmatrix} 1 & s & 1 \\ s & 1 & 1 \\ 1 & 1 & s \end{pmatrix}$ . Use minors to find  $\text{rank } \mathbf{A}(s)$  for every  $s$ .

*Solution.* By cofactor expansion along the first row ([Theorem 6.6.1](#)),

$$\begin{aligned} \det \mathbf{A}(s) &= 1 \cdot (s-1) - s(s^2-1) + 1 \cdot (s-1) \\ &= -s^3 + 3s - 2 = -(s-1)^2(s+2). \end{aligned}$$

If  $s \neq 1, -2$ , this  $3 \times 3$  minor is non-zero, so  $\text{rank } \mathbf{A}(s) = 3$  by [Theorem 6.7.1](#).

If  $s = -2$ , then  $\det \mathbf{A}(-2) = 0$ , so  $\text{rank } \mathbf{A}(-2) \leq 2$ . The minor on rows  $\{1, 2\}$  and columns  $\{1, 2\}$  is  $\det \begin{pmatrix} 1 & -2 \\ -2 & 1 \end{pmatrix} = 1 - 4 = -3 \neq 0$ . Hence  $\text{rank } \mathbf{A}(-2) = 2$ .

If  $s = 1$ , every entry is 1. Every  $2 \times 2$  minor is  $\det \begin{pmatrix} 1 & 1 \\ 1 & 1 \end{pmatrix} = 0$ , and the  $1 \times 1$  minors are non-zero. Hence  $\text{rank } \mathbf{A}(1) = 1$ .

**Exercise B2: Checking Cauchy–Binet.**

Let  $\mathbf{A} = \begin{pmatrix} 1 & 0 & 2 \\ -1 & 1 & 1 \end{pmatrix}$  and  $\mathbf{B} = \begin{pmatrix} 2 & 1 \\ 0 & 1 \\ 1 & -1 \end{pmatrix}$  over  $\mathbb{Q}$ .

- Compute  $\det(\mathbf{AB})$  directly.
- Compute  $\det(\mathbf{AB})$  by the Cauchy–Binet formula.
- Hence, without multiplying, find  $\det(\mathbf{BA})$ .

*Solution.*

(a)  $\mathbf{AB} = \begin{pmatrix} 2+0+2 & 1+0-2 \\ -2+0+1 & -1+1-1 \end{pmatrix} = \begin{pmatrix} 4 & -1 \\ -1 & -1 \end{pmatrix}$ , so  $\det(\mathbf{AB}) = -4 - 1 = -5$ .

(b) The three subsets of  $[3]$  of size 2 give

$$\begin{aligned} S = \{1, 2\} : \det \begin{pmatrix} 1 & 0 \\ -1 & 1 \end{pmatrix} \det \begin{pmatrix} 2 & 1 \\ 0 & 1 \end{pmatrix} &= 1 \cdot 2 = 2, \\ S = \{1, 3\} : \det \begin{pmatrix} 1 & 2 \\ -1 & 1 \end{pmatrix} \det \begin{pmatrix} 2 & 1 \\ 1 & -1 \end{pmatrix} &= 3 \cdot (-3) = -9, \\ S = \{2, 3\} : \det \begin{pmatrix} 0 & 2 \\ 1 & 1 \end{pmatrix} \det \begin{pmatrix} 0 & 1 \\ 1 & -1 \end{pmatrix} &= (-2)(-1) = 2. \end{aligned}$$

The sum is  $2 - 9 + 2 = -5$ , agreeing with (a).

(c)  $\mathbf{BA} \in M_3(\mathbb{Q})$  is a product through  $F^2$ , so  $\det(\mathbf{BA}) = 0$  by [Theorem 6.7.2](#) (a).

**Exercise B3: The Lagrange identity.**

Let  $\mathbf{a} = (a_1, \dots, a_n)$  and  $\mathbf{b} = (b_1, \dots, b_n)$  be vectors in  $\mathbb{R}^n$ ,  $n \geq 2$ , and write  $\mathbf{a} \cdot \mathbf{b} = \sum_i a_i b_i$ .

- (a) Let  $\mathbf{M} \in M_{2 \times n}(\mathbb{R})$  have rows  $\mathbf{a}^\top$  and  $\mathbf{b}^\top$ . Compute  $\mathbf{M}\mathbf{M}^\top$ .  
 (b) Hence prove the **Lagrange identity**

$$(\mathbf{a} \cdot \mathbf{a})(\mathbf{b} \cdot \mathbf{b}) - (\mathbf{a} \cdot \mathbf{b})^2 = \sum_{1 \leq i < j \leq n} (a_i b_j - a_j b_i)^2.$$

- (c) Check it for  $\mathbf{a} = (1, 2, 3)$  and  $\mathbf{b} = (4, 5, 6)$ , and deduce  $(\mathbf{a} \cdot \mathbf{b})^2 \leq (\mathbf{a} \cdot \mathbf{a})(\mathbf{b} \cdot \mathbf{b})$  for all  $\mathbf{a}, \mathbf{b} \in \mathbb{R}^n$ .

*Solution.*

- (a) By [Theorem 0.9.2](#), the  $(p, q)$ -entry of  $\mathbf{M}\mathbf{M}^\top$  is row  $p$  of  $\mathbf{M}$  times column  $q$  of  $\mathbf{M}^\top$ , which is row  $q$  of  $\mathbf{M}$  transposed. Hence

$$\mathbf{M}\mathbf{M}^\top = \begin{pmatrix} \mathbf{a} \cdot \mathbf{a} & \mathbf{a} \cdot \mathbf{b} \\ \mathbf{b} \cdot \mathbf{a} & \mathbf{b} \cdot \mathbf{b} \end{pmatrix}, \quad \mathbf{b} \cdot \mathbf{a} = \mathbf{a} \cdot \mathbf{b}.$$

- (b) The left side of the identity is  $\det(\mathbf{M}\mathbf{M}^\top)$ . By [Corollary 6.7.3](#) with  $m = 2$ , it equals the sum over  $S = \{i < j\}$  of  $(\det \mathbf{M}_{[2], S})^2$ , and  $\mathbf{M}_{[2], \{i, j\}} = \begin{pmatrix} a_i & a_j \\ b_i & b_j \end{pmatrix}$  has determinant  $a_i b_j - a_j b_i$ . This proves the identity.  
 (c) Left side:  $\mathbf{a} \cdot \mathbf{a} = 14$ ,  $\mathbf{b} \cdot \mathbf{b} = 77$ ,  $\mathbf{a} \cdot \mathbf{b} = 32$ , so  $14 \cdot 77 - 32^2 = 1078 - 1024 = 54$ . Right side: the pairs  $(1, 2), (1, 3), (2, 3)$  give  $5 - 8 = -3, 6 - 12 = -6, 12 - 15 = -3$ , and  $9 + 36 + 9 = 54$ . In general the right side is a sum of squares of reals, hence  $\geq 0$ , so  $(\mathbf{a} \cdot \mathbf{b})^2 \leq (\mathbf{a} \cdot \mathbf{a})(\mathbf{b} \cdot \mathbf{b})$ .

### 6.7.5.3 C. Going deeper

#### Exercise C1: Counting spanning trees.

Consider the network with vertices  $1, 2, 3, 4$  and five edges, oriented as  $e_1 = 1 \rightarrow 2$ ,  $e_2 = 2 \rightarrow 3$ ,  $e_3 = 3 \rightarrow 4$ ,  $e_4 = 4 \rightarrow 1$ ,  $e_5 = 1 \rightarrow 3$ . A **spanning tree** is a set of 3 edges that connects all four vertices (ignoring orientation). Let  $\mathbf{N} \in M_{4 \times 5}(\mathbb{R})$  be its incidence matrix ([Definition 2.7.1](#)), let  $\mathbf{L} = \mathbf{N}\mathbf{N}^\top$ , and let  $\mathbf{N}_0$  and  $\mathbf{L}_0$  be  $\mathbf{N}$  and  $\mathbf{L}$  with the row (and, for  $\mathbf{L}$ , the column) of vertex 4 deleted.

- (a) Compute  $\mathbf{L}$  and  $\det \mathbf{L}_0$ .  
 (b) List the spanning trees and count them.  
 (c) Show that  $\mathbf{L}_0 = \mathbf{N}_0 \mathbf{N}_0^\top$ , and use [Corollary 6.7.3](#) to write  $\det \mathbf{L}_0$  as a sum over 3-element sets of edges.  
 (d) Compute the minor of  $\mathbf{N}_0$  for the tree  $\{e_1, e_2, e_3\}$  and for the non-tree  $\{e_1, e_2, e_5\}$ . Explain, for these two sets, why the answers are  $\pm 1$  and  $0$ .

The **matrix-tree theorem** says that for every connected network,  $\det \mathbf{L}_0$  equals the number of spanning trees; parts (c) and (d) are the heart of its proof, which Chapter 23 completes.



*Hint: for (d), a set of edges containing a cycle gives dependent columns.*

*Solution.*

- (a) Column  $e$  of  $\mathbf{N}$  has 1 at the tail and  $-1$  at the head:

$$\mathbf{N} = \begin{pmatrix} 1 & 0 & 0 & -1 & 1 \\ -1 & 1 & 0 & 0 & 0 \\ 0 & -1 & 1 & 0 & -1 \\ 0 & 0 & -1 & 1 & 0 \end{pmatrix}, \quad \mathbf{L} = \mathbf{N}\mathbf{N}^\top = \begin{pmatrix} 3 & -1 & -1 & -1 \\ -1 & 2 & -1 & 0 \\ -1 & -1 & 3 & -1 \\ -1 & 0 & -1 & 2 \end{pmatrix}.$$

(Diagonal entries count the edges at a vertex; off-diagonal entries are  $-1$  for each edge joining two vertices.) Expanding along the first row,

$$\begin{aligned} \det \mathbf{L}_0 &= \det \begin{pmatrix} 3 & -1 & -1 \\ -1 & 2 & -1 \\ -1 & -1 & 3 \end{pmatrix} \\ &= 3(6 - 1) + 1 \cdot (-3 - 1) - 1 \cdot (1 + 2) = 15 - 4 - 3 = 8. \end{aligned}$$

- (b) There are  $\binom{5}{3} = 10$  sets of three edges. Three edges on four vertices connect everything exactly when they contain no cycle, and the only 3-edge cycles are the triangles  $\{e_1, e_2, e_5\}$  (vertices 1, 2, 3) and  $\{e_3, e_4, e_5\}$  (vertices 1, 3, 4). So there are  $10 - 2 = 8$  spanning trees, matching  $\det \mathbf{L}_0$ .
- (c) Deleting row 4 of  $\mathbf{N}$  deletes row 4 of  $\mathbf{N}\mathbf{N}^\top$  (row view) and column 4 of  $\mathbf{N}\mathbf{N}^\top$  (column view of  $\mathbf{N}^\top$ ), by [Theorem 0.9.2](#). So  $\mathbf{L}_0 = \mathbf{N}_0\mathbf{N}_0^\top$ , with  $\mathbf{N}_0 \in M_{3 \times 5}(\mathbb{R})$ , and [Corollary 6.7.3](#) gives  $\det \mathbf{L}_0 = \sum_{|S|=3} (\det(\mathbf{N}_0)_{[3],S})^2$ , one term for each set  $S$  of three edges.

- (d) For  $S = \{e_1, e_2, e_3\}$ ,  $(\mathbf{N}_0)_{[3],S} = \begin{pmatrix} 1 & 0 & 0 \\ -1 & 1 & 0 \\ 0 & -1 & 1 \end{pmatrix}$ , which is lower triangular with determinant 1 ([Theorem 6.4.6](#)). For  $S = \{e_1, e_2, e_5\}$ , the columns are  $(1, -1, 0)$ ,  $(0, 1, -1)$ ,  $(1, 0, -1)$ , and the third is the sum of the first two: going around the cycle  $1 \rightarrow 2 \rightarrow 3$  equals going directly  $1 \rightarrow 3$ . So the columns are dependent and the minor is 0 by [Theorem 6.2.1](#). Thus the non-tree contributes  $0^2$  and the tree contributes  $1^2$ .

### Exercise C2: Rank does not depend on the field.

Let  $K$  be a subfield of a field  $L$  ([Definition 1.9.1](#)), and let  $\mathbf{A} \in M_{m \times n}(K)$ . We may also regard  $\mathbf{A}$  as a matrix in  $M_{m \times n}(L)$ .

- (a) Explain why each minor  $\det \mathbf{A}_{I,J}$  is the same element whether computed in  $K$  or in  $L$ .
- (b) Prove that the rank of  $\mathbf{A}$  over  $K$  equals its rank over  $L$ .
- (c) Deduce: if a system  $\mathbf{A}\mathbf{x} = \mathbf{0}$  with rational coefficients has a non-zero complex solution, it has a non-zero rational solution.

*Solution.*

- (a) By [Definition 6.4.1](#),  $\det \mathbf{A}_{I,J}$  is a finite sum of  $\pm$  products of entries of  $\mathbf{A}$ . The

entries lie in  $K$ , and  $K$  has the same addition and multiplication as  $L$ , so the sum is the same element of  $K \subseteq L$ .

- (b) If  $\mathbf{A} = 0$ , both ranks are 0. Otherwise, by [Theorem 6.7.1](#) (a) applied over  $K$  and over  $L$ , each rank is the largest  $k$  for which some  $k \times k$  minor is non-zero. By (a) the list of minors, and which of them are non-zero, is the same in both cases. Hence the two ranks agree.
- (c) Take  $K = \mathbb{Q}$ ,  $L = \mathbb{C}$ . A non-zero complex solution means nullity  $\mathbf{A} \geq 1$  over  $\mathbb{C}$ , so  $\text{rank}_{\mathbb{C}} \mathbf{A} \leq n - 1$  by [Theorem 2.5.7](#). By (b),  $\text{rank}_{\mathbb{Q}} \mathbf{A} \leq n - 1$ , so nullity  $\mathbf{A} \geq 1$  over  $\mathbb{Q}$ , and a non-zero rational solution exists.

## 6.8 Special Determinants

A determinant of size  $n$  has  $n!$  terms, so a general formula for a large matrix is hopeless. Some matrices, though, have so much structure that their determinants collapse to a short product or a recurrence. This section computes the most useful of them: the Vandermonde determinant, which governs polynomial interpolation; tridiagonal determinants, which satisfy a three-term recurrence; block triangular determinants, which split into smaller ones; and the determinant of a rank-one update of the identity. The proofs share one idea: find the structure first (a polynomial with known roots, a row with two non-zero entries, a pattern of zeros) and let the defining properties of the determinant do the rest.

### 6.8.1 The Vandermonde determinant

Chapter 2 proved that there is exactly one polynomial of degree at most  $n$  through  $n + 1$  points with distinct first coordinates ([Theorem 2.7.3](#)). Behind it sat a square matrix of powers of the nodes. Its determinant turns out to be a product of all the differences of the nodes, and that product explains at a glance why distinct nodes are exactly what is needed.

For  $x_1, \dots, x_n \in F$ , the **Vandermonde matrix** is

$$\mathbf{V}(x_1, \dots, x_n) := \begin{pmatrix} 1 & x_1 & x_1^2 & \cdots & x_1^{n-1} \\ 1 & x_2 & x_2^2 & \cdots & x_2^{n-1} \\ \vdots & \vdots & \vdots & & \vdots \\ 1 & x_n & x_n^2 & \cdots & x_n^{n-1} \end{pmatrix} \in M_n(F),$$

whose  $(i, j)$ -entry is  $x_i^{j-1}$ . Row  $i$  lists the powers  $1, x_i, \dots, x_i^{n-1}$  of the  $i$ -th node. For  $n = 1$  it is the  $1 \times 1$  matrix  $(1)$ , and for  $n = 2$  its determinant is  $x_2 - x_1$ .

#### Theorem 6.8.1 — Vandermonde Determinant

Let  $n \geq 1$  and  $x_1, \dots, x_n \in F$ . Then

$$\det \mathbf{V}(x_1, \dots, x_n) = \prod_{1 \leq i < j \leq n} (x_j - x_i),$$

where the empty product (for  $n = 1$ ) is 1.

*Idea.* Induction on  $n$ . Replace the last node  $x_n$  by a variable  $t$ . Expanding along the last row shows the determinant is a polynomial  $f(t)$  of degree at most  $n - 1$ , with leading coefficient the smaller Vandermonde determinant. Setting  $t = x_i$  for  $i < n$  makes two rows equal, so  $x_1, \dots, x_{n-1}$  are roots of  $f$ . A polynomial of degree at most  $n - 1$  with these  $n - 1$  roots is its leading coefficient times  $\prod_{i < n} (t - x_i)$ . Now put  $t = x_n$  back. The roots argument needs the  $x_i$  distinct; if they are not, both sides are 0 anyway.

*Proof.* We use induction on  $n$  (Theorem 0.2.7). For  $n = 1$ ,  $\det(1) = 1$ , the empty product.

Let  $n \geq 2$ , assume the formula for  $n - 1$  nodes, and write  $\mathbf{V} = \mathbf{V}(x_1, \dots, x_n)$ . Let  $C_{n1}, \dots, C_{nn}$  be the cofactors of  $\mathbf{V}$  along row  $n$  (Definition 6.6.1). Each  $C_{nj}$  is computed after deleting row  $n$ , so it does not involve  $x_n$ . Define

$$f(t) := \sum_{j=1}^n C_{nj} t^{j-1} \in F[t].$$

For  $c \in F$ , let  $\mathbf{V}_c$  be  $\mathbf{V}$  with row  $n$  replaced by  $(1, c, \dots, c^{n-1})$ . Then  $\mathbf{V}_c$  has the same cofactors along row  $n$  as  $\mathbf{V}$ , so by Theorem 6.6.1 along row  $n$ ,

$$\det \mathbf{V}_c = \sum_{j=1}^n c^{j-1} C_{nj} = f(c). \quad (*)$$

In particular  $\det \mathbf{V} = f(x_n)$ . The coefficient of  $t^{n-1}$  in  $f$  is  $C_{nn} = (-1)^{2n} \det \mathbf{V}(x_1, \dots, x_{n-1})$ , since deleting row  $n$  and column  $n$  of  $\mathbf{V}$  leaves the Vandermonde matrix of the first  $n - 1$  nodes.

*Case 1: two of  $x_1, \dots, x_{n-1}$  are equal*, say  $x_p = x_q$  with  $p < q < n$ . Then rows  $p$  and  $q$  of  $\mathbf{V}$  are equal. By Theorem 6.4.5,  $\det \mathbf{V} = \det \mathbf{V}^\top$ , and  $\mathbf{V}^\top$  has two equal columns, so  $\det \mathbf{V} = 0$  because the determinant is alternating (Definition 6.2.2). The product also vanishes, since it contains the factor  $x_q - x_p = 0$ .

*Case 2:  $x_1, \dots, x_{n-1}$  are distinct.* For  $i < n$ , the matrix  $\mathbf{V}_{x_i}$  has rows  $i$  and  $n$  equal, so  $f(x_i) = \det \mathbf{V}_{x_i} = 0$  by (\*) and the argument of Case 1. Hence  $x_1, \dots, x_{n-1}$  are  $n - 1$  distinct roots of  $f$ , and by Corollary 5.1.4,

$$g(t) := (t - x_1)(t - x_2) \cdots (t - x_{n-1}) \text{ divides } f.$$

Write  $f = hg$  with  $h \in F[t]$ . If  $h \neq 0$ , then  $\deg h = \deg f - (n - 1) \leq 0$  by Theorem 0.8.3, since  $\deg f \leq n - 1$ ; so in every case  $h$  is a constant  $h \in F$ . As  $g$  is monic of degree  $n - 1$ , the coefficient of  $t^{n-1}$  in  $hg$  is  $h$ , so  $h = C_{nn} = \det \mathbf{V}(x_1, \dots, x_{n-1})$ . Evaluating at  $x_n$  (Theorem 0.8.5) and using the induction hypothesis,

$$\begin{aligned} \det \mathbf{V} &= f(x_n) = \det \mathbf{V}(x_1, \dots, x_{n-1}) \prod_{i=1}^{n-1} (x_n - x_i) \\ &= \prod_{1 \leq i < j \leq n-1} (x_j - x_i) \prod_{i=1}^{n-1} (x_n - x_i). \end{aligned}$$

The two products together run over all pairs  $i < j \leq n$ , the second one collecting exactly the pairs with  $j = n$ . This proves the formula for  $n$ , and the theorem follows by induction.

**Example. (A  $3 \times 3$  Vandermonde Determinant)**

Compute  $\det \mathbf{V}(1, 2, 4)$  from the formula and by cofactor expansion.

*Solution.* The formula gives  $(2 - 1)(4 - 1)(4 - 2) = 1 \cdot 3 \cdot 2 = 6$ . Directly, expanding

$$\mathbf{V}(1, 2, 4) = \begin{pmatrix} 1 & 1 & 1 \\ 1 & 2 & 4 \\ 1 & 4 & 16 \end{pmatrix}$$

along the first row gives  $1 \cdot (32 - 16) - 1 \cdot (16 - 4) + 1 \cdot (4 - 2) = 16 - 12 + 2 = 6$ .

The product vanishes exactly when a factor does, because a field has no zero divisors. So:

**Corollary 6.8.2 — Distinct Nodes Give an Invertible Vandermonde Matrix**

$\mathbf{V}(x_1, \dots, x_n)$  is invertible if and only if  $x_1, \dots, x_n$  are **distinct**.

*Proof.* In a field, a product is non-zero if and only if every factor is non-zero (Theorem 0.7.1 (f), applied repeatedly). So by Theorem 6.8.1,  $\det \mathbf{V}(x_1, \dots, x_n) \neq 0$  if and only if  $x_j \neq x_i$  for all  $i < j$ . By Theorem 6.5.2, this is equivalent to invertibility.

Now the promised two-line proof of Theorem 2.7.3. Given distinct  $x_0, \dots, x_n$  and values  $y_0, \dots, y_n$ , a polynomial  $p = c_0 + c_1x + \dots + c_nx^n$  satisfies  $p(x_i) = y_i$  for all  $i$  if and only if  $\mathbf{V}(x_0, \dots, x_n) \mathbf{c} = \mathbf{y}$ , since row  $i$  of this system reads  $c_0 + c_1x_i + \dots + c_nx_i^n = y_i$ . By Corollary 6.8.2 the matrix is invertible, so the system has exactly one solution  $\mathbf{c} = \mathbf{V}(x_0, \dots, x_n)^{-1} \mathbf{y}$  (Theorem 2.4.4), and the interpolating polynomial exists and is unique.

*Remark.* This is not a proof from nothing. The root bound of Chapter 5 did the real work inside Theorem 6.8.1, exactly as the root bound did in Chapter 2. What the determinant adds is an explicit number: it says how close to singular the interpolation problem is, since nodes that nearly coincide make the product small.

*Warning. Watch the shape and the order.* The formula is for rows  $(1, x_i, \dots, x_i^{n-1})$  starting at the power 0, and for the factors  $x_j - x_i$  with  $i < j$ , “later node minus earlier node”. Dropping the column of ones changes the answer:  $\det \begin{pmatrix} x_1 & x_1^2 \\ x_2 & x_2^2 \end{pmatrix} = x_1x_2(x_2 - x_1)$ , not  $x_2 - x_1$ . Writing the factors as  $x_i - x_j$  instead flips the sign of each of the  $\binom{n}{2}$  factors, so it multiplies the answer by  $(-1)^{\binom{n}{2}}$ : already for  $n = 2$  and  $n = 3$  this is  $-1$ , a wrong sign.

*Quick check.* Without expanding, find  $\det \mathbf{V}(1, 2, 3, 5)$ .

*Solution.* The pairs  $i < j$  give  $(2-1)(3-1)(5-1)(3-2)(5-2)(5-3) = 1 \cdot 2 \cdot 4 \cdot 1 \cdot 3 \cdot 2 = 48$ .

## 6.8.2 Tridiagonal determinants

Many matrices in applications have non-zero entries only on the diagonal and next to it: discretized differential equations, chains of springs, the recurrences of orthogonal polynomials. Such a matrix is **tridiagonal**. Its determinant is not a closed formula in general, but it obeys a recurrence in the size, because the last row has only two non-zero entries.

Let  $\mathbf{T}_n \in M_n(F)$  have diagonal entries  $a_1, \dots, a_n$ , entries  $b_1, \dots, b_{n-1}$  just above the diagonal ( $(i, i+1)$ -entry  $b_i$ ), entries  $c_1, \dots, c_{n-1}$  just below it ( $((i+1, i)$ -entry  $c_i$ ), and zeros elsewhere:

$$\mathbf{T}_n = \begin{pmatrix} a_1 & b_1 & & & \\ c_1 & a_2 & b_2 & & \\ & c_2 & \ddots & \ddots & \\ & & \ddots & a_{n-1} & b_{n-1} \\ & & & c_{n-1} & a_n \end{pmatrix}.$$

For  $1 \leq k \leq n$ , the top-left  $k \times k$  block of  $\mathbf{T}_n$  is the tridiagonal matrix  $\mathbf{T}_k$  built from  $a_1, \dots, a_k, b_1, \dots, b_{k-1}, c_1, \dots, c_{k-1}$ . Write  $D_k = \det \mathbf{T}_k$ , and set  $D_0 = 1$ .

### Theorem 6.8.3 — Tridiagonal Recurrence

With this notation,  $D_1 = a_1$ , and for  $2 \leq k \leq n$ ,

$$D_k = a_k D_{k-1} - b_{k-1} c_{k-1} D_{k-2}.$$

*Proof.*  $D_1 = \det(a_1) = a_1$ . For  $k = 2$ ,  $D_2 = a_1 a_2 - b_1 c_1 = a_2 D_1 - b_1 c_1 D_0$ . Let  $k \geq 3$ . Row  $k$  of  $\mathbf{T}_k$  has only two possibly non-zero entries,  $c_{k-1}$  in column  $k-1$  and  $a_k$  in column  $k$ . By Theorem 6.6.1 along row  $k$ ,

$$D_k = c_{k-1}(-1)^{2k-1} M_{k,k-1} + a_k(-1)^{2k} M_{kk} = a_k M_{kk} - c_{k-1} M_{k,k-1},$$

where  $M_{kk} = \det \mathbf{T}_{k-1} = D_{k-1}$ . The minor  $M_{k,k-1}$  is the determinant of the  $(k-1) \times (k-1)$  matrix obtained by deleting row  $k$  and column  $k-1$ . Its last column consists of rows  $1, \dots, k-1$  of column  $k$  of  $\mathbf{T}_k$ , which is  $b_{k-1}$  in the last position and 0 elsewhere. Expanding along this last column (Theorem 6.6.1), with sign  $(-1)^{(k-1)+(k-1)} = 1$ , gives  $M_{k,k-1} = b_{k-1} \det \mathbf{T}_{k-2} = b_{k-1} D_{k-2}$ , because deleting the last row and column leaves rows and columns  $1, \dots, k-2$  of  $\mathbf{T}_k$ . Substituting proves the recurrence. ■

### Example. (Fibonacci Numbers as Determinants)

Let  $\mathbf{T}_n$  have 1 on the diagonal, 1 above it and  $-1$  below it. Find  $D_n = \det \mathbf{T}_n$ .

*Solution.* Here  $a_k = 1$  and  $b_{k-1} c_{k-1} = 1 \cdot (-1) = -1$ , so Theorem 6.8.3 gives  $D_k =$

$D_{k-1} + D_{k-2}$  for  $k \geq 2$ , with  $D_0 = 1$  and  $D_1 = 1$ . Hence

$$D_1, D_2, D_3, \dots = 1, 2, 3, 5, 8, 13, 21, 34, \dots,$$

the Fibonacci numbers:  $D_n = F_{n+1}$ , where  $F_1 = F_2 = 1$  and  $F_k = F_{k-1} + F_{k-2}$ . For instance

$$D_3 = \det \begin{pmatrix} 1 & 1 & 0 \\ -1 & 1 & 1 \\ 0 & -1 & 1 \end{pmatrix} = 1 \cdot (1 + 1) - 1 \cdot (-1 - 0) + 0 = 3.$$

Only the products  $b_{k-1}c_{k-1}$  enter the recurrence, never  $b_{k-1}$  and  $c_{k-1}$  separately. For example, for a real number  $t$ , putting  $2t$  on the diagonal and 1 on both neighboring lines gives  $D_k = 2t D_{k-1} - D_{k-2}$ , so  $D_1 = 2t$ ,  $D_2 = 4t^2 - 1$ ,  $D_3 = 8t^3 - 4t$ . These are the Chebyshev polynomials of the second kind.

### 6.8.3 Block triangular matrices

A matrix can be cut into blocks by a horizontal and a vertical line. When the block in the lower left is zero, the determinant should see only the two square blocks on the diagonal, just as a triangular determinant sees only diagonal entries. This result is used constantly, for instance in Chapter 7 for Schur complements.

#### Theorem 6.8.4 — Block Triangular Determinant

Let  $\mathbf{A} \in M_k(F)$ ,  $\mathbf{D} \in M_l(F)$  and  $\mathbf{B} \in M_{k \times l}(F)$ , with  $k, l \geq 1$ . Then

$$\det \begin{pmatrix} \mathbf{A} & \mathbf{B} \\ 0 & \mathbf{D} \end{pmatrix} = \det \mathbf{A} \cdot \det \mathbf{D},$$

where 0 is the  $l \times k$  zero matrix. The same holds for  $\begin{pmatrix} \mathbf{A} & 0 \\ \mathbf{C} & \mathbf{D} \end{pmatrix}$  with  $\mathbf{C} \in M_{l \times k}(F)$ .

*Idea.* In the Leibniz formula for the whole matrix, a term picks one entry from each column. The first  $k$  columns have zeros below row  $k$ , so a non-zero term must pick rows  $1, \dots, k$  for the first  $k$  columns, and is then forced to pick rows  $k+1, \dots, n$  for the last  $l$  columns. Such a permutation is a permutation of each block, its inversions are the inversions inside the blocks, and the Leibniz sum factors into two Leibniz sums.

*Proof.* Let  $n = k + l$  and  $\mathbf{M} = \begin{pmatrix} \mathbf{A} & \mathbf{B} \\ 0 & \mathbf{D} \end{pmatrix} = (m_{ij})$ . By Definition 6.4.1,  $\det \mathbf{M} = \sum_{\sigma \in S_n} \text{sgn}(\sigma) \prod_{j=1}^n m_{\sigma(j)j}$ . Since  $m_{ij} = 0$  for  $i > k \geq j$ , the term of  $\sigma$  is 0 unless  $\sigma(j) \leq k$  for every  $j \leq k$ . For such  $\sigma$ , the injective map  $\sigma$  sends  $\{1, \dots, k\}$  into itself, hence onto itself (a finite set), and so it sends  $\{k+1, \dots, n\}$  onto itself. Therefore  $\sigma$  determines  $\alpha \in S_k$  and  $\beta \in S_l$  by

$$\alpha(j) = \sigma(j) \quad (j \leq k), \quad \beta(q) = \sigma(k+q) - k \quad (q \leq l),$$

and every pair  $(\alpha, \beta)$  arises from exactly one such  $\sigma$ .

We compare signs through inversions (Definition 6.3.1). A pair  $i < j$  with  $i, j \leq k$  is

an inversion of  $\sigma$  exactly when it is one of  $\alpha$ ; a pair  $k + p < k + q$  is an inversion of  $\sigma$  exactly when  $p < q$  is one of  $\beta$ ; and a pair  $i \leq k < j$  is never an inversion, since  $\sigma(i) \leq k < \sigma(j)$ . By Definition 6.3.2,  $\text{sgn}(\sigma) = \text{sgn}(\alpha) \text{sgn}(\beta)$ . Also  $m_{\sigma(j)j} = a_{\alpha(j)j}$  for  $j \leq k$  and  $m_{\sigma(k+q),k+q} = d_{\beta(q)q}$ . Hence

$$\det \mathbf{M} = \sum_{\alpha \in S_k} \sum_{\beta \in S_l} \text{sgn}(\alpha) \text{sgn}(\beta) \prod_{j=1}^k a_{\alpha(j)j} \prod_{q=1}^l d_{\beta(q)q} = \det \mathbf{A} \cdot \det \mathbf{D},$$

using Definition 6.4.1 twice. For the lower block triangular case, the transpose of  $\begin{pmatrix} \mathbf{A} & 0 \\ \mathbf{C} & \mathbf{D} \end{pmatrix}$  is  $\begin{pmatrix} \mathbf{A}^\top & \mathbf{C}^\top \\ 0 & \mathbf{D}^\top \end{pmatrix}$ , so by Theorem 6.4.5 and the first case its determinant is  $\det \mathbf{A}^\top \det \mathbf{D}^\top = \det \mathbf{A} \det \mathbf{D}$ . ■

The proof uses only sums and products of entries, so, like the results flagged in §4–§6, it holds for matrices over any commutative ring; §10 uses this over  $F[x]$ . By induction on the number of blocks, a matrix with square blocks  $\mathbf{A}_1, \dots, \mathbf{A}_r$  on the diagonal and zeros below them has determinant  $\det \mathbf{A}_1 \cdots \det \mathbf{A}_r$ ; Theorem 6.4.6 is the case where every block is  $1 \times 1$ .

For example,

$$\det \begin{pmatrix} 2 & 1 & 5 & -3 \\ 1 & 1 & 0 & 7 \\ 0 & 0 & 3 & 1 \\ 0 & 0 & 4 & 2 \end{pmatrix} = \det \begin{pmatrix} 2 & 1 \\ 1 & 1 \end{pmatrix} \det \begin{pmatrix} 3 & 1 \\ 4 & 2 \end{pmatrix} = 1 \cdot 2 = 2,$$

and the entries 5,  $-3$ , 0, 7 of the upper right block play no role.

**Warning.** There is no “block  $ad-bc$ ” rule. For a general block matrix,  $\det \begin{pmatrix} \mathbf{A} & \mathbf{B} \\ \mathbf{C} & \mathbf{D} \end{pmatrix} \neq \det \mathbf{A} \det \mathbf{D} - \det \mathbf{B} \det \mathbf{C}$ . Take  $\mathbf{A} = \mathbf{D} = 0$  and  $\mathbf{B} = \mathbf{C} = \mathbf{I}_2$ : the  $4 \times 4$  matrix  $\begin{pmatrix} 0 & \mathbf{I}_2 \\ \mathbf{I}_2 & 0 \end{pmatrix}$  is the permutation matrix of  $(1\ 3)(2\ 4)$ , a product of two transpositions, so its determinant is  $+1$ ; the rule would give  $0 \cdot 0 - 1 \cdot 1 = -1$ . The zero block in the corner is what the theorem needs. The correct formulas for general blocks use Schur complements, in Chapter 7.

**Quick check.** Is  $\det \begin{pmatrix} \mathbf{A} & \mathbf{B} \\ 0 & \mathbf{D} \end{pmatrix} = \det \mathbf{A} \det \mathbf{D}$  meaningful when  $\mathbf{A}$  is  $2 \times 3$  and  $\mathbf{D}$  is  $3 \times 2$  (so the whole matrix is  $5 \times 5$ )?

**Solution.** No. The theorem requires  $\mathbf{A}$  and  $\mathbf{D}$  square; here  $\det \mathbf{A}$  and  $\det \mathbf{D}$  are undefined. In fact the whole determinant is 0 whatever the entries: the first 3 columns have their non-zero entries only in the first 2 rows, so they are linearly dependent in a 2-dimensional space, and a determinant with dependent columns is 0 (Theorem 6.2.1).

## 6.8.4 Rank-one updates

Changing a matrix by a rank-one matrix  $\mathbf{u}\mathbf{v}^\top$  is the smallest possible change in the sense of rank, and it happens often: updating one observation in statistics, modifying one row of a system, or

adding the same number to every entry. The determinant of the updated identity turns out to be a single dot product.

### Theorem 6.8.5 — Matrix Determinant Lemma

Let  $\mathbf{u}, \mathbf{v} \in F^n$ . Then

$$\det(\mathbf{I}_n + \mathbf{u}\mathbf{v}^\top) = 1 + \mathbf{v}^\top \mathbf{u}.$$

*Idea.* Border  $\mathbf{I}_n + \mathbf{u}\mathbf{v}^\top$  by an extra column  $\mathbf{u}$  and an extra row  $(\mathbf{0}^\top, 1)$ , giving a block triangular matrix  $\mathbf{M}$  with the same determinant. Then clear the update by elimination with blocks. Rows: add  $\mathbf{v}^\top$  times the top  $n$  rows to the new bottom row (multiply by  $\mathbf{L}$  on the left). Columns: subtract the last column times  $\mathbf{v}^\top$  from the first  $n$  columns (multiply by  $\mathbf{L}'$  on the right). The factors  $\mathbf{L}, \mathbf{L}'$  have determinant 1, and the result is block triangular with  $\mathbf{I}_n$  and the  $1 \times 1$  block  $1 + \mathbf{v}^\top \mathbf{u}$  on the diagonal.

*Proof.* Consider the  $(n+1) \times (n+1)$  block matrices

$$\mathbf{L} = \begin{pmatrix} \mathbf{I}_n & \mathbf{0} \\ \mathbf{v}^\top & 1 \end{pmatrix}, \quad \mathbf{M} = \begin{pmatrix} \mathbf{I}_n + \mathbf{u}\mathbf{v}^\top & \mathbf{u} \\ \mathbf{0}^\top & 1 \end{pmatrix}, \quad \mathbf{L}' = \begin{pmatrix} \mathbf{I}_n & \mathbf{0} \\ -\mathbf{v}^\top & 1 \end{pmatrix}.$$

Splitting the sum  $\sum_j$  in Theorem 0.9.2 (1) into the indices  $j \leq n$  and  $j = n+1$  shows that matrices partitioned in this way multiply block by block, like  $2 \times 2$  matrices whose entries are blocks. Hence

$$\mathbf{LM} = \begin{pmatrix} \mathbf{I}_n + \mathbf{u}\mathbf{v}^\top & \mathbf{u} \\ \mathbf{v}^\top + (\mathbf{v}^\top \mathbf{u})\mathbf{v}^\top & \mathbf{v}^\top \mathbf{u} + 1 \end{pmatrix},$$

where we used  $\mathbf{v}^\top(\mathbf{u}\mathbf{v}^\top) = (\mathbf{v}^\top \mathbf{u})\mathbf{v}^\top$  by associativity,  $\mathbf{v}^\top \mathbf{u}$  being a  $1 \times 1$  matrix, that is, a scalar. Multiplying on the right by  $\mathbf{L}'$ ,

$$\mathbf{LML}' = \begin{pmatrix} \mathbf{I}_n + \mathbf{u}\mathbf{v}^\top - \mathbf{u}\mathbf{v}^\top & \mathbf{u} \\ \mathbf{v}^\top + (\mathbf{v}^\top \mathbf{u})\mathbf{v}^\top - (1 + \mathbf{v}^\top \mathbf{u})\mathbf{v}^\top & 1 + \mathbf{v}^\top \mathbf{u} \end{pmatrix} = \begin{pmatrix} \mathbf{I}_n & \mathbf{u} \\ \mathbf{0}^\top & 1 + \mathbf{v}^\top \mathbf{u} \end{pmatrix}.$$

By Theorem 6.8.4,  $\det \mathbf{L} = \det \mathbf{I}_n \cdot 1 = 1$ ,  $\det \mathbf{L}' = 1$ ,  $\det \mathbf{M} = \det(\mathbf{I}_n + \mathbf{u}\mathbf{v}^\top) \cdot 1$ , and  $\det(\mathbf{LML}') = \det \mathbf{I}_n (1 + \mathbf{v}^\top \mathbf{u}) = 1 + \mathbf{v}^\top \mathbf{u}$ . By Theorem 6.5.4,  $\det(\mathbf{LML}') = \det \mathbf{L} \det \mathbf{M} \det \mathbf{L}' = \det(\mathbf{I}_n + \mathbf{u}\mathbf{v}^\top)$ . This proves the lemma. ■

### Example. ( $a$ on the Diagonal, 1 Elsewhere)

Let  $a \in \mathbb{R}$  and  $n \geq 1$ , and let  $\mathbf{A}_n \in M_n(\mathbb{R})$  have every diagonal entry  $a$  and every other entry 1. Find  $\det \mathbf{A}_n$ .

*Solution.* Let  $\mathbf{1} = (1, \dots, 1) \in \mathbb{R}^n$ . Every entry of  $\mathbf{1}\mathbf{1}^\top$  is 1, so  $\mathbf{A}_n = (a-1)\mathbf{I}_n + \mathbf{1}\mathbf{1}^\top$ .

*Case  $a \neq 1$ .* Then  $\mathbf{A}_n = (a-1)(\mathbf{I}_n + \mathbf{u}\mathbf{u}^\top)$  with  $\mathbf{u} = \frac{1}{a-1}\mathbf{1}$ . Scaling an  $n \times n$  matrix by  $c$  multiplies its determinant by  $c^n$  (multilinearity in each of the  $n$  columns), so by Theorem 6.8.5

$$\det \mathbf{A}_n = (a-1)^n \left(1 + \frac{n}{a-1}\right) = (a-1)^{n-1}(a-1+n),$$



since  $\mathbf{1}^\top \mathbf{u} = \frac{n}{a-1}$ .

*Case  $a = 1$ .* For  $n \geq 2$ , all columns are equal, so  $\det \mathbf{A}_n = 0$ , which the formula also gives. For  $n = 1$ ,  $\det \mathbf{A}_1 = 1$ , and the formula gives  $(a-1)^0 \cdot 1 = 1$ .

So  $\det \mathbf{A}_n = (a-1)^{n-1}(a+n-1)$  for every  $a$ . For  $n = 3$  this is  $(a-1)^2(a+2)$ ; with  $a = 3$  it gives  $4 \cdot 5 = 20$ .

*Warning.* The lemma is about  $\mathbf{I}_n + \mathbf{u}\mathbf{v}^\top$ , not  $\mathbf{u}\mathbf{v}^\top$  or  $\mathbf{u}^\top \mathbf{v}$ . The rank-one matrix  $\mathbf{u}\mathbf{v}^\top$  is  $n \times n$  and, for  $n \geq 2$ , has determinant 0; the scalar  $\mathbf{v}^\top \mathbf{u}$  is its trace. Do not confuse  $\det(\mathbf{I}_n + \mathbf{u}\mathbf{v}^\top) = 1 + \mathbf{v}^\top \mathbf{u}$  with  $\det(\mathbf{I}_n) + \det(\mathbf{u}\mathbf{v}^\top) = 1$  (for  $n \geq 2$ ), which is false whenever  $\mathbf{v}^\top \mathbf{u} \neq 0$ : for  $\mathbf{u} = \mathbf{v} = \mathbf{e}_1$ ,  $\mathbf{I}_n + \mathbf{e}_1\mathbf{e}_1^\top = \text{diag}(2, 1, \dots, 1)$  has determinant 2.

## 6.8.5 Exercises

### 6.8.5.1 A. Check your understanding

#### Exercise A1.

- State the Vandermonde determinant formula, including the order of the factors.
- True or false:  $\mathbf{V}(x_1, \dots, x_n)$  is invertible whenever  $x_1, \dots, x_n$  are non-zero. Justify your answer.
- Write down the recurrence for the determinant of a tridiagonal matrix, and say which row the proof expands along.
- True or false:  $\det \begin{pmatrix} \mathbf{A} & \mathbf{B} \\ \mathbf{C} & \mathbf{D} \end{pmatrix} = \det \mathbf{A} \det \mathbf{D}$  whenever  $\mathbf{A}$  and  $\mathbf{D}$  are square and  $\mathbf{B} = \mathbf{0}$ . Justify your answer.
- True or false:  $\det(\mathbf{I}_n + \mathbf{u}\mathbf{v}^\top) = \det(\mathbf{I}_n + \mathbf{v}\mathbf{u}^\top)$  for all  $\mathbf{u}, \mathbf{v} \in F^n$ . Justify your answer.

*Solution.*

- $\det \mathbf{V}(x_1, \dots, x_n) = \prod_{1 \leq i < j \leq n} (x_j - x_i)$ , where  $\mathbf{V}$  has  $(i, j)$ -entry  $x_i^{j-1}$  (Theorem 6.8.1).
- False.  $\mathbf{V}(1, 1) = \begin{pmatrix} 1 & 1 \\ 1 & 1 \end{pmatrix}$  has non-zero nodes and determinant 0. The condition is that the nodes are **distinct** (Corollary 6.8.2); 0 is allowed as a node.
- $D_k = a_k D_{k-1} - b_{k-1} c_{k-1} D_{k-2}$ , with  $D_0 = 1$ ,  $D_1 = a_1$  (Theorem 6.8.3). The proof expands along the last row, which has only two non-zero entries.
- True. With  $\mathbf{B} = \mathbf{0}$  the matrix is lower block triangular, and Theorem 6.8.4 applies.
- True. Both equal  $1 + \mathbf{v}^\top \mathbf{u} = 1 + \mathbf{u}^\top \mathbf{v}$  by Theorem 6.8.5, since  $\mathbf{v}^\top \mathbf{u} = \sum_i v_i u_i = \mathbf{u}^\top \mathbf{v}$ .

### 6.8.5.2 B. Practice

#### Exercise B1: A $4 \times 4$ Vandermonde determinant.

Compute  $\det \begin{pmatrix} 1 & -1 & 1 & -1 \\ 1 & 0 & 0 & 0 \\ 1 & 1 & 1 & 1 \\ 1 & 2 & 4 & 8 \end{pmatrix}$ . Hence decide whether there is exactly one polynomial  $p \in \mathbb{R}[x]_{\leq 3}$  with  $p(-1) = 3$ ,  $p(0) = 1$ ,  $p(1) = 0$ ,  $p(2) = 5$ .

*Solution.* Row  $i$  is  $(1, x_i, x_i^2, x_i^3)$  with nodes  $-1, 0, 1, 2$ , so this is  $\mathbf{V}(-1, 0, 1, 2)$ . By [Theorem 6.8.1](#) its determinant is

$$(0 - (-1))(1 - (-1))(2 - (-1))(1 - 0)(2 - 0)(2 - 1) = 1 \cdot 2 \cdot 3 \cdot 1 \cdot 2 \cdot 1 = 12.$$

(Check by expanding along row 2, which is  $(1, 0, 0, 0)$ : the determinant is  $(-1)^{2+1} \det \begin{pmatrix} -1 & 1 & -1 \\ 1 & 1 & 1 \\ 2 & 4 & 8 \end{pmatrix} = -(-1(8 - 4) - 1(8 - 2) - 1(4 - 2)) = -(-4 - 6 - 2) = 12$ .) Hence the coefficient matrix of the interpolation system is invertible, and there is exactly one  $p \in \mathbb{R}[x]_{\leq 3}$  with the given values.

### Exercise B2: The second-difference matrix.

Let  $\mathbf{K}_n \in M_n(\mathbb{R})$  have 2 on the diagonal,  $-1$  directly above and below it, and 0 elsewhere. Prove that  $\det \mathbf{K}_n = n + 1$ . Hence show that  $\mathbf{K}_n$  is invertible for every  $n$ .

*Solution.* Here  $a_k = 2$  and  $b_{k-1}c_{k-1} = (-1)(-1) = 1$ . By [Theorem 6.8.3](#),  $D_k = 2D_{k-1} - D_{k-2}$  for  $k \geq 2$ , with  $D_0 = 1$ ,  $D_1 = 2$ . We show  $D_k = k + 1$  by strong induction on  $k$  ([Theorem 0.2.10](#)). It holds for  $k = 0, 1$ . If it holds for  $k - 1$  and  $k - 2$ , then  $D_k = 2k - (k - 1) = k + 1$ . Hence  $\det \mathbf{K}_n = D_n = n + 1 \neq 0$ , and  $\mathbf{K}_n$  is invertible by [Theorem 6.5.2](#).

### Exercise B3: A rank-one update.

Let  $\mathbf{u} = (1, 2, 3)$  and  $\mathbf{v} = (1, -1, 2)$  in  $\mathbb{Q}^3$ .

- Use [Theorem 6.8.5](#) to find  $\det(\mathbf{I}_3 + \mathbf{u}\mathbf{v}^\top)$ .
- Write out  $\mathbf{I}_3 + \mathbf{u}\mathbf{v}^\top$  and check your answer by elimination.
- Find all  $t \in \mathbb{Q}$  for which  $\mathbf{I}_3 + t\mathbf{u}\mathbf{v}^\top$  is **not** invertible.

*Solution.*

- $\mathbf{v}^\top \mathbf{u} = 1 - 2 + 6 = 5$ , so  $\det(\mathbf{I}_3 + \mathbf{u}\mathbf{v}^\top) = 6$ .

- $\mathbf{u}\mathbf{v}^\top = \begin{pmatrix} 1 & -1 & 2 \\ 2 & -2 & 4 \\ 3 & -3 & 6 \end{pmatrix}$ , so  $\mathbf{I}_3 + \mathbf{u}\mathbf{v}^\top = \begin{pmatrix} 2 & -1 & 2 \\ 2 & -1 & 4 \\ 3 & -3 & 7 \end{pmatrix}$ . Subtract row 1 from row 2, which does not change the determinant ([Theorem 6.5.1](#)), to get row 2 equal to  $(0, 0, 2)$ . Expanding along that row,

$$\det = 2 \cdot (-1)^{2+3} \det \begin{pmatrix} 2 & -1 \\ 3 & -3 \end{pmatrix} = -2(-6 + 3) = 6.$$

- $\mathbf{I}_3 + t\mathbf{u}\mathbf{v}^\top = \mathbf{I}_3 + (t\mathbf{u})\mathbf{v}^\top$ , whose determinant is  $1 + t\mathbf{v}^\top \mathbf{u} = 1 + 5t$ . It is zero

exactly for  $t = -\frac{1}{5}$ , and by [Theorem 6.5.2](#) that is the only  $t$  for which the matrix is not invertible.

### 6.8.5.3 C. Going deeper

#### Exercise C1: The Cauchy determinant.

Let  $x_1, \dots, x_n, y_1, \dots, y_n \in F$  with  $x_i + y_j \neq 0$  for all  $i, j$ , and let  $\mathbf{Q}_n \in M_n(F)$  have  $(i, j)$ -entry  $\frac{1}{x_i + y_j}$ . The goal is

$$\det \mathbf{Q}_n = \frac{\prod_{1 \leq i < j \leq n} (x_j - x_i)(y_j - y_i)}{\prod_{i=1}^n \prod_{j=1}^n (x_i + y_j)}.$$

- (a) For  $i < n$ , subtract row  $n$  from row  $i$ . Show that the new  $(i, j)$ -entry is  $\frac{x_n - x_i}{(x_i + y_j)(x_n + y_j)}$ , and deduce that  $\det \mathbf{Q}_n = \frac{\prod_{i < n} (x_n - x_i)}{\prod_j (x_n + y_j)} \det \mathbf{Q}'$ , where  $\mathbf{Q}'$  agrees with  $\mathbf{Q}_n$  in rows  $1, \dots, n-1$  and has last row  $(1, \dots, 1)$ .
- (b) In  $\mathbf{Q}'$ , for  $j < n$ , subtract column  $n$  from column  $j$ . Deduce that  $\det \mathbf{Q}' = \frac{\prod_{j < n} (y_n - y_j)}{\prod_{i < n} (x_i + y_n)} \det \mathbf{Q}_{n-1}$ , where  $\mathbf{Q}_{n-1}$  is built from  $x_1, \dots, x_{n-1}$  and  $y_1, \dots, y_{n-1}$ .
- (c) Prove the formula by induction on  $n$ .
- (d) The **Hilbert matrix**  $\mathbf{H}_n$  has  $(i, j)$ -entry  $\frac{1}{i+j-1}$ . Show that it is a Cauchy matrix, and use (c) to compute  $\det \mathbf{H}_3$ .

*Hint: in (a) and (b), factor scalars out of whole rows and columns using multilinearity.*

*Solution.*

- (a) For  $i < n$ ,  $\frac{1}{x_i + y_j} - \frac{1}{x_n + y_j} = \frac{(x_n + y_j) - (x_i + y_j)}{(x_i + y_j)(x_n + y_j)} = \frac{x_n - x_i}{(x_i + y_j)(x_n + y_j)}$ . These row operations do not change the determinant ([Theorem 6.5.1](#)). In the new matrix, row  $i < n$  has the factor  $x_n - x_i$  in every entry, and column  $j$  has the factor  $\frac{1}{x_n + y_j}$  in every entry (in row  $n$ ,  $\frac{1}{x_n + y_j} = \frac{1}{x_n + y_j} \cdot 1$ ). Pulling these scalars out of each row and column by multilinearity ([Theorem 6.5.1](#), in its row and column forms) leaves the matrix  $\mathbf{Q}'$  with entries  $\frac{1}{x_i + y_j}$  in rows  $i < n$  and 1 in row  $n$ . This gives the stated formula.
- (b) For  $j < n$ , subtracting column  $n$  from column  $j$  leaves the determinant unchanged, makes the last row  $(0, \dots, 0, 1)$ , and for  $i < n$  gives  $\frac{1}{x_i + y_j} - \frac{1}{x_i + y_n} = \frac{y_n - y_j}{(x_i + y_j)(x_i + y_n)}$ . Pull  $y_n - y_j$  out of column  $j$  ( $j < n$ ) and  $\frac{1}{x_i + y_n}$  out of row  $i$  ( $i < n$ ); the last row  $(0, \dots, 0, 1)$  is unaffected by the column factors because its first  $n-1$  entries are 0, and row factors do not touch row  $n$ . What remains has  $\frac{1}{x_i + y_j}$  in positions  $i, j < n$ , entries  $(x_i + y_n) \cdot \frac{1}{x_i + y_n} = 1$  in column  $n$  above the corner, and last row  $(0, \dots, 0, 1)$ . Expanding along the last row ([Theorem 6.6.1](#)) gives  $\det \mathbf{Q}_{n-1}$ , with sign  $(-1)^{2n} = 1$ .
- (c) For  $n = 1$ ,  $\det \mathbf{Q}_1 = \frac{1}{x_1 + y_1}$ , which is the formula with empty products upstairs. For  $n \geq 2$ , combining (a) and (b),

$$\det \mathbf{Q}_n = \frac{\prod_{i < n} (x_n - x_i)(y_n - y_i)}{\prod_{j \leq n} (x_n + y_j) \prod_{i < n} (x_i + y_n)} \det \mathbf{Q}_{n-1}.$$

The numerator supplies exactly the factors of  $\prod_{i < j} (x_j - x_i)(y_j - y_i)$  with  $j = n$ , and the denominator supplies exactly the factors  $x_i + y_j$  with  $\max(i, j) = n$ , each once. By the induction hypothesis,  $\det \mathbf{Q}_{n-1}$  supplies all the remaining factors. This proves the formula.

- (d) With  $x_i = i$  and  $y_j = j - 1$ ,  $x_i + y_j = i + j - 1 \geq 1$ , so  $\mathbf{H}_n$  is a Cauchy matrix over  $\mathbb{Q}$ . For  $n = 3$ , the numerator is  $((2-1)(3-1)(3-2))^2 = 4$ , since  $y_j - y_i = j - i = x_j - x_i$ . The denominator is the product of all  $i + j - 1$ : the rows give  $1 \cdot 2 \cdot 3 = 6$ ,  $2 \cdot 3 \cdot 4 = 24$ ,  $3 \cdot 4 \cdot 5 = 60$ , in total 8640. Hence  $\det \mathbf{H}_3 = \frac{4}{8640} = \frac{1}{2160}$ . (In general  $\det \mathbf{H}_n = c_n^4 / c_{2n}$  with  $c_n = \prod_{k=1}^{n-1} k!$ ; we state this without proof. These determinants shrink extremely fast, which is why Hilbert matrices are a standard test of numerical methods.)

### Exercise C2: A circulant determinant over $\mathbb{C}$ .

Let  $a, b, c \in \mathbb{C}$ , let  $\omega = e^{2\pi i/3}$ , and let

$$\mathbf{Z} = \begin{pmatrix} a & b & c \\ c & a & b \\ b & c & a \end{pmatrix}.$$

- (a) For  $k = 0, 1, 2$ , let  $\mathbf{w}_k = (1, \omega^k, \omega^{2k})$  and  $\lambda_k = a + b\omega^k + c\omega^{2k}$ . Show that  $\mathbf{Z}\mathbf{w}_k = \lambda_k \mathbf{w}_k$ .
- (b) Let  $\mathbf{W}$  be the matrix with columns  $\mathbf{w}_0, \mathbf{w}_1, \mathbf{w}_2$ . Show that  $\mathbf{Z}\mathbf{W} = \mathbf{W} \operatorname{diag}(\lambda_0, \lambda_1, \lambda_2)$  and that  $\det \mathbf{W} \neq 0$ . Deduce  $\det \mathbf{Z} = \lambda_0 \lambda_1 \lambda_2$ .
- (c) Using  $\omega^3 = 1$  and  $1 + \omega + \omega^2 = 0$ , deduce  $\det \mathbf{Z} = a^3 + b^3 + c^3 - 3abc$ , and check this against the Leibniz formula.

Circulant matrices of every size are treated in Chapter 11.

*Solution.*

- (a) By [Example \(Roots of Unity\)](#),  $\omega^3 = 1$ , so  $\omega^{-k} = \omega^{2k}$  and  $\omega^{-2k} = \omega^k$ . Row 1 of  $\mathbf{Z}\mathbf{w}_k$  is  $a + b\omega^k + c\omega^{2k} = \lambda_k$ . Row 2 is  $c + a\omega^k + b\omega^{2k} = \omega^k(c\omega^{-k} + a + b\omega^k) = \omega^k(a + b\omega^k + c\omega^{2k}) = \omega^k \lambda_k$ . Row 3 is  $b + c\omega^k + a\omega^{2k} = \omega^{2k}(b\omega^{-2k} + c\omega^{-k} + a) = \omega^{2k} \lambda_k$ . Hence  $\mathbf{Z}\mathbf{w}_k = \lambda_k \mathbf{w}_k$ .
- (b) By [Theorem 0.9.2](#), column  $k$  of  $\mathbf{Z}\mathbf{W}$  is  $\mathbf{Z}\mathbf{w}_k = \lambda_k \mathbf{w}_k$ , which is column  $k$  of  $\mathbf{W} \operatorname{diag}(\lambda_0, \lambda_1, \lambda_2)$ . The  $(i, k)$ -entry of  $\mathbf{W}$  is  $(\omega^k)^{i-1}$ , so  $\mathbf{W} = \mathbf{V}(1, \omega, \omega^2)^\top$ . The nodes  $1, \omega, \omega^2$  are distinct (they are  $e^{2\pi i k/3}$  for  $k = 0, 1, 2$ ), so  $\det \mathbf{W} = \det \mathbf{V}(1, \omega, \omega^2) \neq 0$  by [Theorem 6.4.5](#) and [Corollary 6.8.2](#). By [Theorem 6.5.4](#) and [Theorem 6.4.6](#),  $\det \mathbf{Z} \det \mathbf{W} = \det \mathbf{W} \lambda_0 \lambda_1 \lambda_2$ , and canceling  $\det \mathbf{W} \neq 0$  gives  $\det \mathbf{Z} = \lambda_0 \lambda_1 \lambda_2$ .
- (c) Multiply out  $\lambda_1 \lambda_2 = (a + b\omega + c\omega^2)(a + b\omega^2 + c\omega)$ . The nine terms are  $a^2, ab\omega^2 + ab\omega, ac\omega + ac\omega^2, b^2\omega^3, bc\omega^2, bc\omega^4 = bc\omega, c^2\omega^3$ . Using  $\omega^3 = 1$  and  $\omega + \omega^2 = -1$ ,

$$\lambda_1 \lambda_2 = a^2 + b^2 + c^2 - ab - ac - bc.$$

Then  $\det \mathbf{Z} = (a + b + c)(a^2 + b^2 + c^2 - ab - ac - bc) = a^3 + b^3 + c^3 - 3abc$ , expanding and canceling. The Leibniz formula for  $\mathbf{Z}$  gives the three even terms  $a \cdot a \cdot a, c \cdot c \cdot c$  (from  $\sigma = (1 \ 2 \ 3)$ ) and  $b \cdot b \cdot b$ , and the three odd terms, each equal to  $abc$  with sign  $-$ : the same answer.

**Exercise C3: Rank-one updates of an invertible matrix.**

Let  $\mathbf{A} \in M_n(F)$  be invertible and  $\mathbf{u}, \mathbf{v} \in F^n$ .

- (a) Prove that  $\det(\mathbf{A} + \mathbf{u}\mathbf{v}^\top) = \det \mathbf{A} (1 + \mathbf{v}^\top \mathbf{A}^{-1} \mathbf{u})$ .
- (b) Deduce that  $\mathbf{A} + \mathbf{u}\mathbf{v}^\top$  is invertible if and only if  $\mathbf{v}^\top \mathbf{A}^{-1} \mathbf{u} \neq -1$ .
- (c) Give an example over  $\mathbb{Q}$  with  $n = 2$  where  $\mathbf{A}$  is invertible,  $\mathbf{u}\mathbf{v}^\top \neq 0$ , and  $\mathbf{A} + \mathbf{u}\mathbf{v}^\top$  is **not** invertible.

*Solution.*

- (a) Since  $\mathbf{A}$  is invertible,  $\mathbf{A} + \mathbf{u}\mathbf{v}^\top = \mathbf{A}(\mathbf{I}_n + \mathbf{A}^{-1}\mathbf{u}\mathbf{v}^\top) = \mathbf{A}(\mathbf{I}_n + (\mathbf{A}^{-1}\mathbf{u})\mathbf{v}^\top)$  by Theorem 0.9.4. By Theorem 6.5.4 and Theorem 6.8.5 applied to the vectors  $\mathbf{A}^{-1}\mathbf{u}$  and  $\mathbf{v}$ ,  $\det(\mathbf{A} + \mathbf{u}\mathbf{v}^\top) = \det \mathbf{A} (1 + \mathbf{v}^\top \mathbf{A}^{-1} \mathbf{u})$ .
- (b)  $\det \mathbf{A} \neq 0$  by Theorem 6.5.2, so  $\det(\mathbf{A} + \mathbf{u}\mathbf{v}^\top) \neq 0$  exactly when  $1 + \mathbf{v}^\top \mathbf{A}^{-1} \mathbf{u} \neq 0$ . Apply Theorem 6.5.2 again.
- (c) Take  $\mathbf{A} = \mathbf{I}_2$ ,  $\mathbf{u} = (-1, 0)$ ,  $\mathbf{v} = (1, 0)$ . Then  $\mathbf{v}^\top \mathbf{u} = -1$ , and indeed  $\mathbf{A} + \mathbf{u}\mathbf{v}^\top = \begin{pmatrix} 0 & 0 \\ 0 & 1 \end{pmatrix}$  is not invertible.

## 6.9 Orientation and Volume

The chapter began in §1 with a picture: the area of a parallelogram, with a sign attached, obeys a few simple rules, and those rules pinned down  $ad - bc$ . We have since built the determinant in every size from the same rules. It is time to return to geometry and say precisely what the determinant measures. There are two halves to the answer. Its absolute value measures volume, and its sign measures orientation, the difference between a left hand and a right hand. Both halves need care. “Volume” in  $\mathbb{R}^7$  has no meaning until someone defines it, and “orientation” has no meaning at all over  $\mathbb{C}$ .

Throughout this section the field is  $\mathbb{R}$ .

### 6.9.1 What should the volume of a parallelepiped be?

For vectors  $\mathbf{a}_1, \dots, \mathbf{a}_n \in \mathbb{R}^n$ , the **parallelepiped** they span is the set

$$P(\mathbf{a}_1, \dots, \mathbf{a}_n) := \{t_1\mathbf{a}_1 + \dots + t_n\mathbf{a}_n : 0 \leq t_i \leq 1 \text{ for every } i\}.$$

For  $n = 2$  it is the parallelogram with sides  $\mathbf{a}_1, \mathbf{a}_2$  at the origin, and for  $n = 3$  a slanted box. For  $n = 1$  it is the segment from 0 to  $\mathbf{a}_1$ .

We would like to say “its volume is  $|\det(\mathbf{a}_1 \ \dots \ \mathbf{a}_n)|$ ”. But a theorem needs two sides, and one side is missing: we have no definition of the volume of a subset of  $\mathbb{R}^n$ . School geometry supplies one for  $n = 2$  and  $n = 3$  through “base times height”, and the proper general notion, Lebesgue measure, belongs to a course in measure theory. So we proceed honestly in three steps. We **define** the volume of a parallelepiped by the determinant. We **prove** that this is the only

function obeying the rules that any reasonable notion of volume must obey. And we **state**, without proof, that it agrees with Lebesgue measure.

*The volume of a parallelepiped is the size of the determinant of its edge vectors.*

### Definition 6.9.1 — Volume of a Parallelepiped

Let  $n \geq 1$  and  $\mathbf{a}_1, \dots, \mathbf{a}_n \in \mathbb{R}^n$ . The ( $n$ -dimensional) **volume** of the parallelepiped spanned by  $\mathbf{a}_1, \dots, \mathbf{a}_n$  is

$$\text{vol}(\mathbf{a}_1, \dots, \mathbf{a}_n) := |\det(\mathbf{a}_1 \ \mathbf{a}_2 \ \cdots \ \mathbf{a}_n)| \geq 0.$$

In words: stack the  $n$  edge vectors as the columns of a square matrix, take its determinant, and forget the sign. There must be **exactly**  $n$  vectors in  $\mathbb{R}^n$ , so that the matrix is square.

#### Examples.

- The unit cube  $P(\mathbf{e}_1, \dots, \mathbf{e}_n)$  has volume  $|\det \mathbf{I}_n| = 1$ .
- A box with edges  $d_1\mathbf{e}_1, \dots, d_n\mathbf{e}_n$  has volume  $|\det \text{diag}(d_1, \dots, d_n)| = |d_1 \cdots d_n|$  by [Theorem 6.4.6](#): the product of its side lengths, as it should be.
- The parallelepiped spanned by  $(1, 2, 0)$ ,  $(0, 1, 1)$ ,  $(1, 0, 3)$  in  $\mathbb{R}^3$  has volume

$$\left| \det \begin{pmatrix} 1 & 0 & 1 \\ 2 & 1 & 0 \\ 0 & 1 & 3 \end{pmatrix} \right| = |1 \cdot (3 - 0) - 0 + 1 \cdot (2 - 0)| = 5,$$

expanding along the first row.

- **Degenerate case.** If  $\mathbf{a}_1, \dots, \mathbf{a}_n$  are linearly dependent, the parallelepiped is flattened into a proper subspace, and its volume is 0 by [Theorem 6.2.1](#). For  $n = 2$ ,  $P((1, 2), (2, 4))$  is the segment from  $(0, 0)$  to  $(3, 6)$ , with area  $|4 - 4| = 0$ . This case matters: “volume zero” is the geometric face of “not invertible”.

**Non-example by minimal change.** Drop one vector:  $(1, 2, 0)$  and  $(0, 1, 1)$  span a genuine parallelogram in  $\mathbb{R}^3$ , but the matrix  $\begin{pmatrix} 1 & 0 \\ 2 & 1 \\ 0 & 1 \end{pmatrix}$  is  $3 \times 2$  and has no determinant. The clause that fails is **exactly  $n$  vectors in  $\mathbb{R}^n$** . The 3-dimensional volume of that flat parallelogram would be 0, and its area needs a different formula ([Exercise \(C1: Area of a triangle in space\)](#)).

**Why this definition.** Reordering the vectors multiplies the determinant by a sign ([Theorem 6.2.1](#)), which the absolute value removes, so the volume does not depend on the order in which the edges are listed. The absolute value is also what makes volume non-negative. The sign we threw away is not garbage: it is the orientation, the subject of the second half of this section.

**Warning. Volume is not linear.** Scaling **one** edge by 2 doubles the volume, but scaling the whole parallelepiped by 2 multiplies it by  $2^n$ :  $\text{vol}(2\mathbf{a}_1, \dots, 2\mathbf{a}_n) = |\det(2\mathbf{A})| = 2^n \text{vol}(\mathbf{a}_1, \dots, \mathbf{a}_n)$ , where  $\mathbf{A}$  has columns  $\mathbf{a}_1, \dots, \mathbf{a}_n$ . A cube of side 2 in  $\mathbb{R}^3$  has volume 8, not 2; this failure persists even with the sign kept, since  $\det$  is linear in each edge separately, not in all edges at once. The absolute value also destroys additivity in a single edge:  $\text{vol}(\mathbf{a}_1 + \mathbf{b}_1, \mathbf{a}_2, \dots)$  is in general not  $\text{vol}(\mathbf{a}_1, \mathbf{a}_2, \dots) + \text{vol}(\mathbf{b}_1, \mathbf{a}_2, \dots)$ : with  $\mathbf{a}_1 = \mathbf{e}_1$ ,  $\mathbf{b}_1 = -\mathbf{e}_1$  in  $\mathbb{R}^2$  and  $\mathbf{a}_2 = \mathbf{e}_2$ , the left side is 0 and the right side is 2.

## 6.9.2 Volume is forced by its rules

Why should anyone accept [Definition 6.9.1](#)? Because of what any notion of volume of parallelepipeds must satisfy. The pictures in §1 showed three facts about area, and they have obvious  $n$ -dimensional versions: stretching one edge by a factor  $c$  multiplies the volume by  $|c|$ ; sliding one edge parallel to another (a shear) does not change the volume; and the unit cube has volume 1. The next theorem says that these three rules alone determine the volume completely. So any definition that obeys them, whether from school geometry or from measure theory, must agree with ours.

### Theorem 6.9.1 — Volume Is Determined by Its Rules

Let  $n \geq 1$ , and let  $g$  be a function assigning a real number  $g(\mathbf{a}_1, \dots, \mathbf{a}_n)$  to every list of  $n$  vectors in  $\mathbb{R}^n$ . Suppose that for every list, every  $i \neq j$  and every  $c \in \mathbb{R}$ :

- (V1)  $g(\mathbf{a}_1, \dots, c\mathbf{a}_i, \dots, \mathbf{a}_n) = |c|g(\mathbf{a}_1, \dots, \mathbf{a}_i, \dots, \mathbf{a}_n)$  (only slot  $i$  changes);
- (V2)  $g(\mathbf{a}_1, \dots, \mathbf{a}_i + c\mathbf{a}_j, \dots, \mathbf{a}_n) = g(\mathbf{a}_1, \dots, \mathbf{a}_i, \dots, \mathbf{a}_n)$  (only slot  $i$  changes);
- (V3)  $g(\mathbf{e}_1, \dots, \mathbf{e}_n) = 1$ .

Then  $g(\mathbf{a}_1, \dots, \mathbf{a}_n) = \text{vol}(\mathbf{a}_1, \dots, \mathbf{a}_n)$  for every list. Moreover,  $\text{vol}$  itself satisfies (V1)–(V3).

*Idea.* Think of  $g$  as a function of the matrix  $\mathbf{M}$  with columns  $\mathbf{a}_1, \dots, \mathbf{a}_n$ . ① If the columns are dependent, shear one column to  $\mathbf{0}$  and scale by 0:  $g = 0$ . ② Each elementary column operation changes  $g$  by the factor  $|\det \mathbf{E}|$ . Scaling and shearing are (V1) and (V2); a swap is not among the rules, but three shears and a sign change perform it. ③ An invertible  $\mathbf{M}$  is a product of elementary matrices, so peel them off one at a time until only  $\mathbf{I}$  is left.

*Proof.* For  $\mathbf{M} \in M_n(\mathbb{R})$  with columns  $\mathbf{a}_1, \dots, \mathbf{a}_n$ , write  $h(\mathbf{M}) = g(\mathbf{a}_1, \dots, \mathbf{a}_n)$ .

**Step 1: dependent columns give  $h(\mathbf{M}) = 0$ .** Suppose the columns are linearly dependent. By [Theorem 1.4.1](#), some  $\mathbf{a}_i = c_1\mathbf{a}_1 + \dots + c_{i-1}\mathbf{a}_{i-1}$  (for  $i = 1$ ,  $\mathbf{a}_1 = \mathbf{0}$ ). Applying (V2) to slot  $i$  with the scalars  $-c_1, \dots, -c_{i-1}$ , one at a time, does not change  $g$ , and it replaces  $\mathbf{a}_i$  by  $\mathbf{0}$  while the other slots stay fixed. Then (V1) with  $c = 0$  gives  $g(\dots, \mathbf{0}, \dots) = g(\dots, 0 \cdot \mathbf{0}, \dots) = 0 \cdot g(\dots, \mathbf{0}, \dots) = 0$ . Hence  $h(\mathbf{M}) = 0$ .

**Step 2:  $h(\mathbf{ME}) = |\det \mathbf{E}| h(\mathbf{M})$  for every elementary matrix  $\mathbf{E}$ .** By [Theorem 2.4.1](#) and [Theorem 0.9.5](#),  $\mathbf{ME} = (\mathbf{E}^\top \mathbf{M}^\top)^\top$ , and  $\mathbf{E}^\top$  is the elementary matrix  $\mathbf{P}_{ij}$ ,  $\mathbf{D}_i(c)$  or  $\mathbf{I} + c\mathbf{E}_{ji}$  when  $\mathbf{E}$  is  $\mathbf{P}_{ij}$ ,  $\mathbf{D}_i(c)$  or  $\mathbf{I} + c\mathbf{E}_{ij}$ . So right multiplication by  $\mathbf{E}$  performs a column operation: swap columns  $i$  and  $j$ ; multiply column  $i$  by  $c \neq 0$ ; or add  $c$  times column  $i$  to column  $j$ . The determinants are  $\det \mathbf{P}_{ij} = -1$ ,  $\det \mathbf{D}_i(c) = c$  and  $\det(\mathbf{I} + c\mathbf{E}_{ij}) = 1$  by [Theorem 6.5.1](#).

- For  $\mathbf{E} = \mathbf{I} + c\mathbf{E}_{ij}$ , (V2) gives  $h(\mathbf{ME}) = h(\mathbf{M}) = |1| h(\mathbf{M})$ .
- For  $\mathbf{E} = \mathbf{D}_i(c)$ , (V1) gives  $h(\mathbf{ME}) = |c| h(\mathbf{M})$ .
- For  $\mathbf{E} = \mathbf{P}_{ij}$ , watch only slots  $i$  and  $j$ , holding  $\mathbf{x}$  and  $\mathbf{y}$ . By (V2) three times,  $(\mathbf{x}, \mathbf{y}) \rightarrow (\mathbf{x} + \mathbf{y}, \mathbf{y}) \rightarrow (\mathbf{x} + \mathbf{y}, -\mathbf{x}) \rightarrow (\mathbf{y}, -\mathbf{x})$  leaves  $g$  unchanged, where the second step subtracts slot  $i$  from slot  $j$  and the third adds slot  $j$  to slot  $i$ . By (V1) with  $c = -1$  on slot  $j$ ,  $g$  at  $(\mathbf{y}, \mathbf{x})$  equals  $|-1|$  times  $g$  at  $(\mathbf{y}, -\mathbf{x})$ . Hence  $h(\mathbf{MP}_{ij}) = h(\mathbf{M}) = |-1| h(\mathbf{M})$ .

**Step 3: conclusion.** If the columns of  $\mathbf{M}$  are dependent, then  $h(\mathbf{M}) = 0$  by Step 1 and  $\det \mathbf{M} = 0$  by [Theorem 6.2.1](#), so  $h(\mathbf{M}) = |\det \mathbf{M}|$ . Otherwise  $\mathbf{M}$  is invertible by [Theorem 2.4.4](#), hence  $\mathbf{M} = \mathbf{E}_1 \mathbf{E}_2 \cdots \mathbf{E}_k$  is a product of elementary matrices by the same theorem. Applying Step 2  $k$  times, starting from  $\mathbf{I}$ , and then (V3),

$$h(\mathbf{M}) = h(\mathbf{I} \mathbf{E}_1 \cdots \mathbf{E}_k) = |\det \mathbf{E}_1| \cdots |\det \mathbf{E}_k| h(\mathbf{I}) = |\det(\mathbf{E}_1 \cdots \mathbf{E}_k)| = |\det \mathbf{M}|,$$

where the third equality uses [Theorem 6.5.4](#) and  $|xy| = |x||y|$ . This proves  $g = \text{vol}$ .

Finally,  $\text{vol}$  satisfies (V1) because the determinant is linear in each column ([Theorem 6.4.3](#)) and  $|cd| = |c||d|$ ; (V2) by [Theorem 6.2.1](#); and (V3) because  $\det \mathbf{I}_n = 1$ . ■

**Agreement with area and volume from school.** For  $n = 2$ , let  $g(\mathbf{a}_1, \mathbf{a}_2)$  be the area of the parallelogram  $P(\mathbf{a}_1, \mathbf{a}_2)$  in the sense of school geometry, base times height. The pictures of §1 are exactly (V1)–(V3) for this  $g$ : stretching a side by  $c$  stretches the base or the height by  $|c|$ ; a shear  $\mathbf{a}_1 \mapsto \mathbf{a}_1 + c\mathbf{a}_2$  slides the opposite side along a line parallel to the base  $\mathbf{a}_2$ , keeping base and height; and the unit square has area 1. Hence, by [Theorem 6.9.1](#), school area equals  $|\det(\mathbf{a}_1 \ \mathbf{a}_2)|$ . For  $n = 3$  the same argument works with “area of base times height”. What the theorem cannot supply is the geometric input itself: it takes the three rules as given.

**Agreement with Lebesgue measure.** In measure theory, every reasonable subset  $S \subseteq \mathbb{R}^n$  has an  $n$ -dimensional volume, its Lebesgue measure. One can prove that the Lebesgue measure of  $P(\mathbf{a}_1, \dots, \mathbf{a}_n)$  satisfies (V1)–(V3), and hence equals  $\text{vol}(\mathbf{a}_1, \dots, \mathbf{a}_n)$ . We state this without proof; nothing in this book depends on it.

*Quick check.* Which of the rules (V1)–(V3) does the function  $g(\mathbf{a}_1, \mathbf{a}_2) = |\det(\mathbf{a}_1 \ \mathbf{a}_2)|^2$  on  $\mathbb{R}^2$  break?

*Solution.* It satisfies (V2), since shears do not change the determinant, and (V3), since  $1^2 = 1$ . It breaks (V1):  $g(2\mathbf{e}_1, \mathbf{e}_2) = 4$ , while  $2|g(\mathbf{e}_1, \mathbf{e}_2)| = 2$ . So the theorem does not apply, and indeed  $g \neq \text{vol}$ .

### 6.9.3 Linear maps scale volume

Applying a linear map to a parallelepiped gives another parallelepiped: by linearity,  $T(\sum t_i \mathbf{a}_i) = \sum t_i T\mathbf{a}_i$ , so  $T(P(\mathbf{a}_1, \dots, \mathbf{a}_n)) = P(T\mathbf{a}_1, \dots, T\mathbf{a}_n)$ . How does the volume change? Multiplicativity answers this in one line, and the answer does not depend on the parallelepiped.

#### Theorem 6.9.2 — Linear Maps Scale Volume by $|\det|$

Let  $T \in \mathcal{L}(\mathbb{R}^n)$  and  $\mathbf{a}_1, \dots, \mathbf{a}_n \in \mathbb{R}^n$ . Then

$$\text{vol}(T\mathbf{a}_1, \dots, T\mathbf{a}_n) = |\det T| \cdot \text{vol}(\mathbf{a}_1, \dots, \mathbf{a}_n).$$

In particular, the image of the unit cube under  $T$  has volume  $|\det T|$ .

*Proof.* Let  $\mathbf{B}$  be the matrix of  $T$  in the standard basis, so  $T\mathbf{x} = \mathbf{B}\mathbf{x}$  for all  $\mathbf{x}$  ([Theorem 3.6.1](#)) and  $\det T = \det \mathbf{B}$  ([Definition 6.5.1](#)). Let  $\mathbf{A} = (\mathbf{a}_1 \ \cdots \ \mathbf{a}_n)$ . By [Theo-](#)



rem 0.9.2,  $\mathbf{BA} = (\mathbf{Ba}_1 \ \cdots \ \mathbf{Ba}_n)$ . By Theorem 6.5.4,

$$\text{vol}(T\mathbf{a}_1, \dots, T\mathbf{a}_n) = |\det(\mathbf{BA})| = |\det \mathbf{B}| |\det \mathbf{A}| = |\det T| \text{vol}(\mathbf{a}_1, \dots, \mathbf{a}_n).$$

For the unit cube take  $\mathbf{a}_i = \mathbf{e}_i$ , with  $\text{vol}(\mathbf{e}_1, \dots, \mathbf{e}_n) = 1$ .

■

This is the meaning of the determinant of a real operator: **the factor by which it scales every volume**. An operator with  $\det T = 0$  squashes  $\mathbb{R}^n$  into a lower-dimensional subspace, where  $n$ -dimensional volume is 0. In multivariable calculus the same fact, applied to the linear approximation of a smooth map, becomes the Jacobian factor  $|\det \mathbf{J}|$  in the change-of-variables formula for integrals.

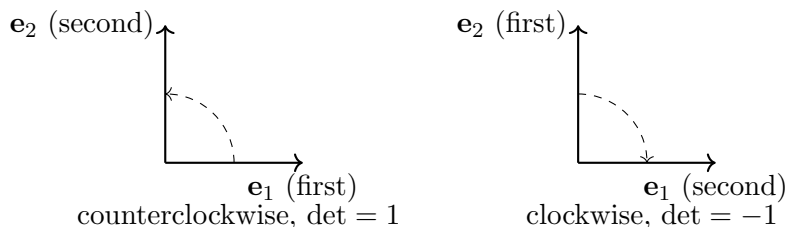
### Example. (Area of an Image)

Let  $T: \mathbb{R}^2 \rightarrow \mathbb{R}^2$  be  $T(x, y) = (3x + y, x + 2y)$ . Find the area of the image of the unit square, and of the image of the parallelogram  $P((1, 1), (-1, 2))$ .

*Solution.* The standard matrix is  $\mathbf{B} = \begin{pmatrix} 3 & 1 \\ 1 & 2 \end{pmatrix}$ , with  $\det T = 6 - 1 = 5$ . By Theorem 6.9.2 the image of the unit square, the parallelogram spanned by  $(3, 1)$  and  $(1, 2)$ , has area 5. The parallelogram  $P((1, 1), (-1, 2))$  has area  $|2 + 1| = 3$ , so its image has area  $5 \cdot 3 = 15$ . Check directly:  $T(1, 1) = (4, 3)$  and  $T(-1, 2) = (-1, 3)$ , and  $|\det \begin{pmatrix} 4 & -1 \\ 3 & 3 \end{pmatrix}| = |12 + 3| = 15$ .

## 6.9.4 Orientation

Now the sign. In  $\mathbb{R}^2$ , the ordered basis  $(\mathbf{e}_1, \mathbf{e}_2)$  turns **counterclockwise** from its first vector to its second, while  $(\mathbf{e}_2, \mathbf{e}_1)$  turns clockwise. No rotation carries one arrangement to the other: a rotation of the plane keeps counterclockwise turns counterclockwise. The determinant sees the difference:  $\det(\mathbf{e}_1 \ \mathbf{e}_2) = 1 > 0$  and  $\det(\mathbf{e}_2 \ \mathbf{e}_1) = -1 < 0$ .



To compare two bases of an arbitrary real space, we compare them with each other rather than with a fixed standard basis, using the change-of-coordinates matrix of Chapter 3.

*Two bases have the same orientation when the matrix converting one into the other has positive determinant.*

**Definition 6.9.2 — Orientation**

Let  $V$  be a vector space **over**  $\mathbb{R}$  with  $\dim V = n \geq 1$ , and let  $\mathcal{B}$  and  $\mathcal{C}$  be **ordered** bases of  $V$ . We say  $\mathcal{B}$  and  $\mathcal{C}$  have the **same orientation** if

$$\det[\text{id}]_{\mathcal{B}}^{\mathcal{C}} > 0,$$

and **opposite orientations** if  $\det[\text{id}]_{\mathcal{B}}^{\mathcal{C}} < 0$ .

In words: write each vector of  $\mathcal{B}$  in coordinates with respect to  $\mathcal{C}$ , put the coordinate columns side by side (Definition 3.7.1), and look at the **sign** of the determinant. Exactly one of the two cases occurs, because  $[\text{id}]_{\mathcal{B}}^{\mathcal{C}}$  is invertible (Theorem 3.7.1 (c)), so its determinant is non-zero (Theorem 6.5.2), and a non-zero real number is either positive or negative. The bases must be **ordered**: reordering a basis can change the answer.

For  $V = \mathbb{R}^n$  there is a shortcut. If  $\mathbf{B}$  and  $\mathbf{C}$  are the matrices whose columns are the vectors of  $\mathcal{B}$  and  $\mathcal{C}$ , then  $[\text{id}]_{\mathcal{B}}^{\mathcal{E}} = \mathbf{B}$  and  $[\text{id}]_{\mathcal{C}}^{\mathcal{E}} = \mathbf{C}$  for the standard basis  $\mathcal{E}$ . By Theorem 3.7.1 (b) and (c),  $[\text{id}]_{\mathcal{B}}^{\mathcal{C}} = \mathbf{C}^{-1}\mathbf{B}$ . Since  $\det \mathbf{C} \det \mathbf{C}^{-1} = \det \mathbf{I} = 1$  (Theorem 6.5.4),  $\det[\text{id}]_{\mathcal{B}}^{\mathcal{C}} = \det \mathbf{B} / \det \mathbf{C}$ . So  $\mathcal{B}$  and  $\mathcal{C}$  have the same orientation if and only if  $\det \mathbf{B}$  and  $\det \mathbf{C}$  have the same sign.

**Examples.**

- In  $\mathbb{R}^2$ ,  $(\mathbf{e}_1, \mathbf{e}_2)$  and  $(\mathbf{e}_2, \mathbf{e}_1)$  have opposite orientations:  $\det \mathbf{I}_2 = 1$  and  $\det \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix} = -1$ .
- In  $\mathbb{R}^3$ ,  $(\mathbf{e}_1, \mathbf{e}_2, \mathbf{e}_3)$  and  $(\mathbf{e}_2, \mathbf{e}_3, \mathbf{e}_1)$  have the **same** orientation. The second matrix is the permutation matrix of a 3-cycle, whose sign is  $(-1)^{3-1} = 1$ . A cyclic shift of three vectors is two swaps.
- In  $V = \mathbb{R}[x]_{\leq 1}$ , the bases  $\mathcal{B} = (1, x)$  and  $\mathcal{C} = (x, 1+x)$  have  $1 = -x + (1+x)$  and  $x = x + 0 \cdot (1+x)$ , so  $[\text{id}]_{\mathcal{B}}^{\mathcal{C}} = \begin{pmatrix} -1 & 1 \\ 1 & 0 \end{pmatrix}$ , with determinant  $-1$ . They have opposite orientations. Orientation makes sense in any real space, with no picture at all.
- **Degenerate case,  $n = 1$ .** Bases of  $\mathbb{R}$  are single non-zero numbers  $(a)$ , and  $[\text{id}]_{(a)}^{(b)} = (a/b)$ . So  $(a)$  and  $(b)$  have the same orientation exactly when  $a$  and  $b$  have the same sign. The two orientations of a line are its two directions.

**Non-example by minimal change.** In  $\mathbb{R}^2$ ,  $(-\mathbf{e}_1, -\mathbf{e}_2)$  has the same orientation as  $(\mathbf{e}_1, \mathbf{e}_2)$ :  $\det(-\mathbf{I}_2) = 1$ , and indeed it is the rotation of  $(\mathbf{e}_1, \mathbf{e}_2)$  by a half-turn. Change only **one** sign, to  $(\mathbf{e}_1, -\mathbf{e}_2)$ : the determinant becomes  $-1$ , and the orientation flips. What changed is the sign of the determinant, which is the whole content of the definition. In  $\mathbb{R}^3$ , by contrast,  $-\mathbf{I}_3$  has determinant  $-1$ : negating all three vectors reverses orientation.

**Why this definition.** Using the change-of-coordinates matrix, rather than the matrices  $\mathbf{B}$  and  $\mathbf{C}$  above, makes the definition work in every real space, including spaces like  $\mathbb{R}[x]_{\leq 1}$  that have no standard picture. Requiring  $\det > 0$  rather than  $\det = 1$  is what gives a coarse, two-valued answer: rescaling a basis vector by a positive number should not change its handedness.

The payoff is that “same orientation” really sorts all bases into two families, as the pictures suggest.

**Theorem 6.9.3 — There Are Exactly Two Orientations**

Let  $V$  be a vector space over  $\mathbb{R}$  with  $\dim V = n \geq 1$ .

- (a) “Having the same orientation” is an equivalence relation on the set of ordered bases of  $V$ .
- (b) It has **exactly two** equivalence classes. If  $\mathcal{B} = (\mathbf{v}_1, \dots, \mathbf{v}_n)$ , then  $\mathcal{B}^- = (-\mathbf{v}_1, \mathbf{v}_2, \dots, \mathbf{v}_n)$  lies in the class not containing  $\mathcal{B}$ .

*Idea.* The three axioms are three facts about change-of-coordinates matrices:  $[\text{id}]_{\mathcal{B}}^{\mathcal{B}} = \mathbf{I}$ ; the matrix back is the inverse; and matrices compose along a chain  $\mathcal{B} \rightarrow \mathcal{C} \rightarrow \mathcal{D}$ . Determinants turn these into  $1 > 0$ ,  $1/\text{positive} > 0$  and  $\text{positive} \cdot \text{positive} > 0$ . For “exactly two”, flipping one vector gives a second class, and “negative times negative is positive” shows there is no third.

*Proof.* Write  $d(\mathcal{B}, \mathcal{C}) = \det[\text{id}]_{\mathcal{B}}^{\mathcal{C}}$ . By Theorem 3.7.1 (b) and Theorem 6.5.4, for bases  $\mathcal{B}, \mathcal{C}, \mathcal{D}$ ,

$$d(\mathcal{B}, \mathcal{D}) = d(\mathcal{C}, \mathcal{D}) d(\mathcal{B}, \mathcal{C}). \quad (*)$$

- (a) *Reflexive:*  $[\text{id}]_{\mathcal{B}}^{\mathcal{B}} = \mathbf{I}_n$ , so  $d(\mathcal{B}, \mathcal{B}) = 1 > 0$ . *Symmetric:* by (\*) with  $\mathcal{D} = \mathcal{B}$ ,  $d(\mathcal{C}, \mathcal{B}) d(\mathcal{B}, \mathcal{C}) = d(\mathcal{B}, \mathcal{B}) = 1$ , so  $d(\mathcal{C}, \mathcal{B}) = 1/d(\mathcal{B}, \mathcal{C})$ , which is positive when  $d(\mathcal{B}, \mathcal{C})$  is. *Transitive:* if  $d(\mathcal{B}, \mathcal{C}) > 0$  and  $d(\mathcal{C}, \mathcal{D}) > 0$ , then  $d(\mathcal{B}, \mathcal{D}) > 0$  by (\*), since a product of positive reals is positive. By Definition 0.5.2, this is an equivalence relation.
- (b) *At least two classes.*  $\mathcal{B}^-$  is a basis, and  $[\text{id}]_{\mathcal{B}^-}^{\mathcal{B}^-} = \text{diag}(-1, 1, \dots, 1)$ , because  $-\mathbf{v}_1 = (-1)\mathbf{v}_1$  and  $\mathbf{v}_j = 1 \cdot \mathbf{v}_j$ . Its determinant is  $-1$  by Theorem 6.4.6. So  $d(\mathcal{B}^-, \mathcal{B}) < 0$ , and  $\mathcal{B}^-$  and  $\mathcal{B}$  lie in different classes.

*At most two classes.* Let  $\mathcal{C}$  and  $\mathcal{D}$  be bases, neither of which has the same orientation as  $\mathcal{B}$ . Then  $d(\mathcal{C}, \mathcal{B}) < 0$  and  $d(\mathcal{D}, \mathcal{B}) < 0$ , and by symmetry  $d(\mathcal{B}, \mathcal{D}) < 0$  as well. By (\*),  $d(\mathcal{C}, \mathcal{D}) = d(\mathcal{B}, \mathcal{D}) d(\mathcal{C}, \mathcal{B})$  is a product of two negative reals, hence positive. So all bases outside the class of  $\mathcal{B}$  lie in one class, and there are exactly two classes. ■

Choosing one of the two classes is called **orienting**  $V$ ; its bases are then called **positively oriented**, and the others **negatively oriented**. On  $\mathbb{R}^n$  the standard choice is the class of  $(\mathbf{e}_1, \dots, \mathbf{e}_n)$ . By the shortcut above,  $(\mathbf{a}_1, \dots, \mathbf{a}_n)$  is positively oriented if and only if  $\det(\mathbf{a}_1 \ \cdots \ \mathbf{a}_n) > 0$ . So the determinant of a real square matrix carries two independent pieces of information: its absolute value is a volume, and its sign is an orientation.

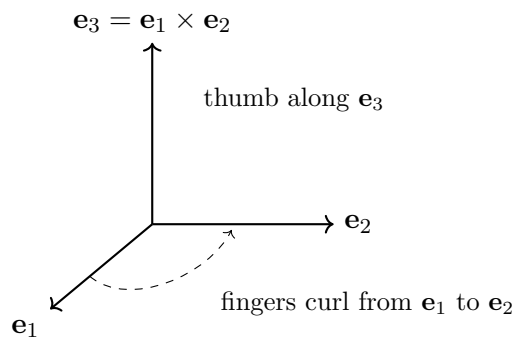
*Warning. Orientation needs  $\mathbb{R}$ .* The proof used two facts about the reals: every non-zero number is positive or negative, and products and inverses of positive numbers are positive. Over  $\mathbb{C}$  there is no such splitting. Suppose we tried “ $\mathcal{B}$  and  $\mathcal{C}$  have the same orientation if  $d(\mathcal{B}, \mathcal{C})$  has positive real part”. In  $\mathbb{C}^1$ , with  $z = e^{i\pi/3}$ , the bases  $(\mathbf{e}_1)$ ,  $(z\mathbf{e}_1)$ ,  $(z^2\mathbf{e}_1)$  give  $d((\mathbf{e}_1), (z\mathbf{e}_1)) = z^{-1}$  and  $d((z\mathbf{e}_1), (z^2\mathbf{e}_1)) = z^{-1}$ , both with real part  $\frac{1}{2} > 0$ , but  $d((\mathbf{e}_1), (z^2\mathbf{e}_1)) = z^{-2}$  has real part  $-\frac{1}{2}$ : transitivity fails. Over  $\mathbb{F}_2$ ,  $-1 = 1$ , so the second basis  $\mathcal{B}^-$  of the proof is  $\mathcal{B}$  itself. The theorem holds verbatim over any field with a compatible order, such as  $\mathbb{Q}$ , but not over  $\mathbb{C}$  or a finite field.

*Quick check.* Do  $((1, 2), (3, 4))$  and  $((3, 4), (1, 2))$  have the same orientation? Does  $((1, 2), (3, 4))$  have the standard orientation of  $\mathbb{R}^2$ ?

*Solution.*  $\det \begin{pmatrix} 1 & 3 \\ 2 & 4 \end{pmatrix} = 4 - 6 = -2$  and  $\det \begin{pmatrix} 3 & 1 \\ 4 & 2 \end{pmatrix} = 6 - 4 = 2$ . The signs differ, so the two bases have opposite orientations. The first has negative determinant, so it is negatively oriented: it is **not** in the class of  $(\mathbf{e}_1, \mathbf{e}_2)$ .

### 6.9.5 The right-hand rule

In  $\mathbb{R}^3$ , the standard orientation is the one described by the right hand: curl the fingers of the right hand from the first basis vector toward the second, and the thumb points along the third. For  $(\mathbf{e}_1, \mathbf{e}_2, \mathbf{e}_3)$  with the usual picture of the axes, this is the familiar arrangement below.



The **cross product** of  $\mathbf{a} = (a_1, a_2, a_3)$  and  $\mathbf{b} = (b_1, b_2, b_3)$  in  $\mathbb{R}^3$  is

$$\mathbf{a} \times \mathbf{b} := (a_2b_3 - a_3b_2, a_3b_1 - a_1b_3, a_1b_2 - a_2b_1).$$

Physics defines it by the right-hand rule. Here is the linear-algebra reason the two descriptions match:  $(\mathbf{a}, \mathbf{b}, \mathbf{a} \times \mathbf{b})$  is always positively oriented.

Expand  $\det(\mathbf{a} \ \mathbf{b} \ \mathbf{a} \times \mathbf{b})$  along its third column (Theorem 6.6.1). The cofactors of that column are  $C_{13} = a_2b_3 - a_3b_2$ ,  $C_{23} = -(a_1b_3 - a_3b_1)$  and  $C_{33} = a_1b_2 - a_2b_1$ , which are exactly the three components of  $\mathbf{a} \times \mathbf{b}$ . Hence

$$\det(\mathbf{a} \ \mathbf{b} \ \mathbf{a} \times \mathbf{b}) = (a_2b_3 - a_3b_2)^2 + (a_3b_1 - a_1b_3)^2 + (a_1b_2 - a_2b_1)^2 \geq 0.$$

If  $\mathbf{a}$  and  $\mathbf{b}$  are linearly independent, the  $3 \times 2$  matrix  $(\mathbf{a} \ \mathbf{b})$  has rank 2, so one of its  $2 \times 2$  minors, which are  $\pm$  the components of  $\mathbf{a} \times \mathbf{b}$ , is non-zero (Theorem 6.7.1). Then the determinant is strictly positive, so  $(\mathbf{a}, \mathbf{b}, \mathbf{a} \times \mathbf{b})$  is a basis (Theorem 6.5.2) with the standard orientation. For example,  $\mathbf{e}_1 \times \mathbf{e}_2 = (0, 0, 1) = \mathbf{e}_3$ , matching the picture.

### 6.9.6 Orientation-preserving maps

An invertible operator carries each basis to a basis. Does it keep the orientation? The answer is the same for every basis and depends only on the sign of  $\det T$ .

#### Proposition 6.9.4 — Determinant Sign and Orientation

Let  $V$  be a vector space over  $\mathbb{R}$  with  $\dim V = n \geq 1$ , let  $T \in \mathcal{L}(V)$  be invertible, and let

$\mathcal{B} = (\mathbf{v}_1, \dots, \mathbf{v}_n)$  be any basis of  $V$ . Then  $T\mathcal{B} = (T\mathbf{v}_1, \dots, T\mathbf{v}_n)$  is a basis of  $V$ , and

$$[\text{id}]_{T\mathcal{B}}^{\mathcal{B}} = [T]_{\mathcal{B}}.$$

Hence  $\mathcal{B}$  and  $T\mathcal{B}$  have the same orientation if and only if  $\det T > 0$ , and this does not depend on  $\mathcal{B}$ .

*Proof.* Since  $T$  is invertible, it is an isomorphism, so  $T\mathcal{B}$  is a basis by [Theorem 3.5.2](#). By [Definition 3.7.1](#), column  $j$  of  $[\text{id}]_{T\mathcal{B}}^{\mathcal{B}}$  is  $[T\mathbf{v}_j]_{\mathcal{B}}$ , which is column  $j$  of  $[T]_{\mathcal{B}}$  by [Definition 3.6.1](#). Hence the two matrices are equal, and by [Definition 6.5.1](#),  $\det[\text{id}]_{T\mathcal{B}}^{\mathcal{B}} = \det T$ . By [Definition 6.9.2](#) and the symmetry in [Theorem 6.9.3 \(a\)](#),  $\mathcal{B}$  and  $T\mathcal{B}$  have the same orientation exactly when  $\det T > 0$ . ■

An invertible operator with  $\det T > 0$  is **orientation-preserving**, and one with  $\det T < 0$  is **orientation-reversing**. Rotations of the plane preserve orientation and reflections reverse it ([Exercise \(C2: Rotations and reflections\)](#)). In  $\mathbb{R}^3$ , the map  $\mathbf{x} \mapsto -\mathbf{x}$  reverses orientation, since  $\det(-\mathbf{I}_3) = -1$ : a right hand, pushed through the origin, becomes a left hand.

## 6.9.7 Exercises

### 6.9.7.1 A. Check your understanding

#### Exercise A1.

- State the definition of the volume of the parallelepiped spanned by  $\mathbf{a}_1, \dots, \mathbf{a}_n \in \mathbb{R}^n$ , and say what it is when the vectors are dependent.
- True or false:  $\text{vol}(\mathbf{a}_1 + \mathbf{b}, \mathbf{a}_2, \dots, \mathbf{a}_n) = \text{vol}(\mathbf{a}_1, \dots, \mathbf{a}_n) + \text{vol}(\mathbf{b}, \mathbf{a}_2, \dots, \mathbf{a}_n)$ . Justify your answer.
- Which three rules characterize volume in [Theorem 6.9.1](#)?
- True or false: if  $\det T = -2$ , then  $T$  doubles every volume. Justify your answer.
- Define “same orientation” for two ordered bases of a real vector space.
- True or false: for every  $n \geq 1$ , the map  $\mathbf{x} \mapsto -\mathbf{x}$  on  $\mathbb{R}^n$  reverses orientation. Justify your answer.

*Solution.*

- $\text{vol}(\mathbf{a}_1, \dots, \mathbf{a}_n) = |\det(\mathbf{a}_1 \ \cdots \ \mathbf{a}_n)|$  ([Definition 6.9.1](#)). For dependent vectors it is 0, by [Theorem 6.2.1](#).
- False. With  $n = 2$ ,  $\mathbf{a}_1 = \mathbf{e}_1$ ,  $\mathbf{b} = -\mathbf{e}_1$ ,  $\mathbf{a}_2 = \mathbf{e}_2$ : the left side is  $\text{vol}(\mathbf{0}, \mathbf{e}_2) = 0$ , the right side is  $1 + 1 = 2$ . The absolute value destroys additivity.
- (V1) scaling one vector by  $c$  scales the volume by  $|c|$ ; (V2) adding a multiple of one vector to another does not change it; (V3) the unit cube has volume 1.
- True. By [Theorem 6.9.2](#),  $\text{vol}(T\mathbf{a}_1, \dots, T\mathbf{a}_n) = |-2| \text{vol}(\mathbf{a}_1, \dots, \mathbf{a}_n)$ . The sign only affects orientation.
- $\mathcal{B}$  and  $\mathcal{C}$  have the same orientation if  $\det[\text{id}]_{\mathcal{B}}^{\mathcal{C}} > 0$  ([Definition 6.9.2](#)).

- (f) False.  $\det(-\mathbf{I}_n) = (-1)^n$ , so by [Proposition 6.9.4](#) the map reverses orientation for odd  $n$  and preserves it for even  $n$ . In  $\mathbb{R}^2$  it is the rotation by a half-turn.

### 6.9.7.2 B. Practice

#### Exercise B1: Volume of a parallelepiped.

Find the volume of the parallelepiped in  $\mathbb{R}^3$  spanned by  $(2, 1, 0)$ ,  $(1, 3, 1)$  and  $(0, 1, 1)$ . Hence find the volume of its image under the map  $T(x, y, z) = (x + y, y + z, 2z)$ .

*Solution.* Expanding along the first row,

$$\det \begin{pmatrix} 2 & 1 & 0 \\ 1 & 3 & 1 \\ 0 & 1 & 1 \end{pmatrix} = 2(3 - 1) - 1 \cdot (1 - 0) + 0 = 3,$$

so the volume is 3. The standard matrix of  $T$  is  $\begin{pmatrix} 1 & 1 & 0 \\ 0 & 1 & 1 \\ 0 & 0 & 2 \end{pmatrix}$ , upper triangular with determinant 2 ([Theorem 6.4.6](#)). By [Theorem 6.9.2](#) the image has volume  $2 \cdot 3 = 6$ .

#### Exercise B2: Comparing orientations.

In  $\mathbb{R}^3$ , let  $\mathcal{B} = ((1, 1, 0), (0, 1, 1), (1, 0, 1))$ ,  $\mathcal{C} = ((0, 1, 0), (1, 0, 0), (1, 1, 1))$  and  $\mathcal{D} = ((1, 0, 1), (1, 1, 0), (0, 1, 1))$ . Determine which pairs have the same orientation, and which are positively oriented. Justify your answer.

*Solution.* By the shortcut after [Definition 6.9.2](#), compare signs of the determinants of the matrices with these columns. Expanding along the first row:

$$\det \begin{pmatrix} 1 & 0 & 1 \\ 1 & 1 & 0 \\ 0 & 1 & 1 \end{pmatrix} = 1 \cdot (1 - 0) - 0 + 1 \cdot (1 - 0) = 2, \quad \det \begin{pmatrix} 0 & 1 & 1 \\ 1 & 0 & 1 \\ 0 & 0 & 1 \end{pmatrix} = -1,$$

the second by expanding along the last row:  $1 \cdot \det \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix} = -1$ . The list  $\mathcal{D}$  is  $\mathcal{B}$  shifted cyclically,  $(\mathbf{b}_3, \mathbf{b}_1, \mathbf{b}_2)$ ; moving the last column to the front takes two column swaps, so by [Theorem 6.2.1](#) its determinant is  $(-1)^2 \cdot 2 = 2$ . Hence  $\mathcal{B}$  and  $\mathcal{D}$  are positively oriented and have the same orientation;  $\mathcal{C}$  is negatively oriented, opposite to both.

#### Exercise B3: Areas of images.

Let  $T(x, y) = (2x + y, -x + 3y)$  on  $\mathbb{R}^2$ .

- Find the area of the image of the unit square under  $T$ .
- Find the area of the image of the parallelogram  $P((1, 0), (1, 2))$ , first by [Theorem 6.9.2](#) and then directly.
- Is  $T$  orientation-preserving?

*Solution.*

- (a) The standard matrix  $\begin{pmatrix} 2 & 1 \\ -1 & 3 \end{pmatrix}$  has determinant  $6 + 1 = 7$ , so the image of the unit square has area 7.
- (b)  $\text{vol}((1, 0), (1, 2)) = |\det \begin{pmatrix} 1 & 1 \\ 0 & 2 \end{pmatrix}| = 2$ , so the image has area  $7 \cdot 2 = 14$ . Directly,  $T(1, 0) = (2, -1)$  and  $T(1, 2) = (4, 5)$ , and  $|\det \begin{pmatrix} 2 & 4 \\ -1 & 5 \end{pmatrix}| = |10 + 4| = 14$ .
- (c) Yes:  $\det T = 7 > 0$ , so by [Proposition 6.9.4](#)  $T$  preserves orientation.

### 6.9.7.3 C. Going deeper

#### Exercise C1: Area of a triangle in space.

For  $\mathbf{u}, \mathbf{v} \in \mathbb{R}^3$ , let  $\mathbf{A} = (\mathbf{u} \ \mathbf{v}) \in M_{3 \times 2}(\mathbb{R})$ . We **define** the area of the parallelogram  $P(\mathbf{u}, \mathbf{v}) \subseteq \mathbb{R}^3$  to be  $\sqrt{\det(\mathbf{A}^\top \mathbf{A})}$ ; Chapter 12 justifies this with inner products, where  $\det(\mathbf{A}^\top \mathbf{A})$  is recognized as a Gram determinant.

- (a) Show that  $\det(\mathbf{A}^\top \mathbf{A}) \geq 0$ , so the square root exists, and that  $\det(\mathbf{A}^\top \mathbf{A}) = (u_2v_3 - u_3v_2)^2 + (u_3v_1 - u_1v_3)^2 + (u_1v_2 - u_2v_1)^2$ .
- (b) Show that if  $\mathbf{u}$  and  $\mathbf{v}$  lie in the plane  $z = 0$ , this area equals the area  $\text{vol}((u_1, u_2), (v_1, v_2))$  of [Definition 6.9.1](#) in  $\mathbb{R}^2$ .
- (c) Find the area of the triangle with vertices  $(1, 0, 0)$ ,  $(0, 2, 0)$ ,  $(0, 0, 3)$ .

*Hint:* apply [Corollary 6.7.3](#) to  $\mathbf{A}^\top$ .

*Solution.*

- (a)  $\mathbf{A}^\top \in M_{2 \times 3}(\mathbb{R})$  and  $(\mathbf{A}^\top)^\top = \mathbf{A}$ , so by [Corollary 6.7.3](#) applied to  $\mathbf{A}^\top$ ,  $\det(\mathbf{A}^\top \mathbf{A}) = \sum_{|S|=2} (\det(\mathbf{A}^\top)_{[2], S})^2 \geq 0$ . The  $2 \times 2$  minors of  $\mathbf{A}^\top = \begin{pmatrix} u_1 & u_2 & u_3 \\ v_1 & v_2 & v_3 \end{pmatrix}$  are  $u_1v_2 - u_2v_1$ ,  $u_1v_3 - u_3v_1$ ,  $u_2v_3 - u_3v_2$  for  $S = \{1, 2\}, \{1, 3\}, \{2, 3\}$ ; squaring removes the signs, giving the stated sum.
- (b) If  $u_3 = v_3 = 0$ , two of the three squares vanish, and  $\det(\mathbf{A}^\top \mathbf{A}) = (u_1v_2 - u_2v_1)^2$ . Hence  $\sqrt{\det(\mathbf{A}^\top \mathbf{A})} = |u_1v_2 - u_2v_1| = |\det \begin{pmatrix} u_1 & v_1 \\ u_2 & v_2 \end{pmatrix}|$ , which is  $\text{vol}((u_1, u_2), (v_1, v_2))$ .
- (c) The triangle is half of the parallelogram spanned by  $\mathbf{u} = (0, 2, 0) - (1, 0, 0) = (-1, 2, 0)$  and  $\mathbf{v} = (0, 0, 3) - (1, 0, 0) = (-1, 0, 3)$ . By (a),  $\det(\mathbf{A}^\top \mathbf{A}) = (6 - 0)^2 + (0 + 3)^2 + (0 + 2)^2 = 36 + 9 + 4 = 49$ . As a check,  $\mathbf{A}^\top \mathbf{A} = \begin{pmatrix} 5 & 1 \\ 1 & 10 \end{pmatrix}$  has determinant  $50 - 1 = 49$ . So the parallelogram has area 7, and the triangle has area  $\frac{7}{2}$ .

#### Exercise C2: Rotations and reflections.

For  $\theta \in \mathbb{R}$ , let  $\mathbf{R}_\theta = \begin{pmatrix} \cos \theta & -\sin \theta \\ \sin \theta & \cos \theta \end{pmatrix}$  (rotation by  $\theta$ ) and  $\mathbf{S}_\theta = \begin{pmatrix} \cos 2\theta & \sin 2\theta \\ \sin 2\theta & -\cos 2\theta \end{pmatrix}$  (reflection in the line through 0 at angle  $\theta$ ).

- (a) Show that every rotation preserves orientation and area, and every reflection reverses orientation and preserves area.
- (b) Show that  $\mathbf{S}_\theta \mathbf{S}_\varphi$  is orientation-preserving for all  $\theta, \varphi$ . (It is in fact a rotation, but you need not show this.)
- (c) Prove that no  $\mathbf{M} \in M_2(\mathbb{R})$  satisfies  $\mathbf{M}^2 = \mathbf{S}_0 = \text{diag}(1, -1)$ . Does the same argument rule out  $\mathbf{M}^2 = -\mathbf{I}_2$ ?

*Solution.*

- (a)  $\det \mathbf{R}_\theta = \cos^2 \theta + \sin^2 \theta = 1 > 0$  and  $\det \mathbf{S}_\theta = -\cos^2 2\theta - \sin^2 2\theta = -1 < 0$ . By [Proposition 6.9.4](#), rotations preserve and reflections reverse orientation. Both have  $|\det| = 1$ , so both preserve area by [Theorem 6.9.2](#).
- (b) By [Theorem 6.5.4](#),  $\det(\mathbf{S}_\theta \mathbf{S}_\varphi) = (-1)(-1) = 1 > 0$ .
- (c) If  $\mathbf{M}^2 = \mathbf{S}_0$ , then  $(\det \mathbf{M})^2 = \det \mathbf{S}_0 = -1$  by [Theorem 6.5.4](#), impossible for a real number  $\det \mathbf{M}$ . The argument says nothing about  $-\mathbf{I}_2$ , since  $\det(-\mathbf{I}_2) = 1$ ; and indeed  $\mathbf{R}_{\pi/2}^2 = \mathbf{R}_{\pi/2} \mathbf{R}_{\pi/2} = \begin{pmatrix} 0 & -1 \\ 1 & 0 \end{pmatrix}^2 = -\mathbf{I}_2$ . An orientation-reversing map can never be the square of a real map, since the square of anything preserves orientation.

## 6.10 The Characteristic Polynomial

For a square matrix  $\mathbf{A}$  over  $F$  and a scalar  $c \in F$ , when is  $c\mathbf{I} - \mathbf{A}$  invertible? The question matters:  $c\mathbf{I} - \mathbf{A}$  fails to be invertible exactly when  $\mathbf{A}\mathbf{v} = c\mathbf{v}$  for some non-zero vector  $\mathbf{v}$ , the situation Chapter 8 is built around. The determinant answers the question for each single  $c$ . This section lets  $c$  vary. Replacing the number  $c$  by a variable  $x$  turns  $\det(c\mathbf{I} - \mathbf{A})$  into a polynomial, the characteristic polynomial, and the scalars we are looking for become its roots. The price is that  $x\mathbf{I} - \mathbf{A}$  is a matrix of **polynomials**, not of field elements, so we must check which determinant facts survive there. The remarks at the ends of §4, §5 and §6 did exactly that bookkeeping, and here we cash them in.

### 6.10.1 From a number to a variable

Take  $\mathbf{A} = \begin{pmatrix} 1 & 2 \\ 3 & 4 \end{pmatrix} \in M_2(\mathbb{Q})$ . For each  $c \in \mathbb{Q}$ ,

$$\det(c\mathbf{I} - \mathbf{A}) = \det \begin{pmatrix} c-1 & -2 \\ -3 & c-4 \end{pmatrix} = (c-1)(c-4) - 6 = c^2 - 5c - 2.$$

The same expression  $c^2 - 5c - 2$  appears whatever  $c$  is. It is the value at  $c$  of one polynomial, and that polynomial deserves a name. To define it directly, without first choosing  $c$ , we put the variable  $x$  into the matrix and take the determinant of

$$x\mathbf{I} - \mathbf{A} = \begin{pmatrix} x-1 & -2 \\ -3 & x-4 \end{pmatrix},$$

a matrix whose entries lie in  $F[x]$ .



Is that allowed?  $F[x]$  is not a field, but it is a **commutative ring** in the sense of the remark in §4: all the field axioms hold except the existence of inverses ([Theorem 0.8.1](#)). The Leibniz formula of [Definition 6.4.1](#) uses only sums, differences and products of entries, so it defines  $\det \mathbf{M} \in F[x]$  for every  $\mathbf{M} \in M_n(F[x])$ .

*The characteristic polynomial of  $\mathbf{A}$  is the determinant of  $x\mathbf{I} - \mathbf{A}$ , computed with polynomial entries.*

### Definition 6.10.1 — Characteristic Polynomial

Let  $n \geq 1$  and  $\mathbf{A} = (a_{ij}) \in M_n(F)$ . The **characteristic polynomial** of  $\mathbf{A}$  is

$$p_{\mathbf{A}}(x) := \det(x\mathbf{I}_n - \mathbf{A}) = \sum_{\sigma \in S_n} \operatorname{sgn}(\sigma) \prod_{j=1}^n (x \delta_{\sigma(j)j} - a_{\sigma(j)j}) \in F[x],$$

where  $\delta_{ij} = 1$  if  $i = j$  and 0 otherwise, so that  $x\delta_{ij} - a_{ij}$  is the  $(i, j)$ -entry of  $x\mathbf{I}_n - \mathbf{A}$ . The determinant is the Leibniz sum of [Definition 6.4.1](#), computed in the commutative ring  $F[x]$ .

In words: subtract  $\mathbf{A}$  from  $x$  times the identity, so that  $x$  appears only on the diagonal; then take the determinant exactly as for a matrix of numbers, multiplying and adding polynomials instead of scalars. The result is **one** polynomial, not a number.

**Well-definedness.** Each term of the Leibniz sum is a product of  $n$  polynomials, and the sum is finite, so  $p_{\mathbf{A}}$  is a polynomial. It depends only on  $\mathbf{A}$ .

**How to compute it.** By the remarks in §4, §5 and §6, the following results were proved without dividing, and so hold for matrices over  $F[x]$ : [Theorem 6.4.5](#) and [Theorem 6.4.6](#) (§4), [Theorem 6.5.4](#) by its first proof (§5), [Theorem 6.6.1](#) (§6), and [Theorem 6.8.4](#) (§8). So we may expand  $\det(x\mathbf{I} - \mathbf{A})$  along any row or column, and read it off from triangular or block triangular shape. What we may **not** do is row reduce  $x\mathbf{I} - \mathbf{A}$  by dividing by a polynomial entry, as the warning below explains.

**Examples.**

- $1 \times 1$ . For  $\mathbf{A} = (a)$ ,  $p_{\mathbf{A}}(x) = x - a$ .
- $2 \times 2$ . For  $\mathbf{A} = \begin{pmatrix} a & b \\ c & d \end{pmatrix}$ ,

$$\begin{aligned} p_{\mathbf{A}}(x) &= (x - a)(x - d) - bc = x^2 - (a + d)x + (ad - bc) \\ &= x^2 - (\operatorname{tr} \mathbf{A})x + \det \mathbf{A}. \end{aligned}$$

For  $\mathbf{A} = \begin{pmatrix} 1 & 2 \\ 3 & 4 \end{pmatrix}$  this is  $x^2 - 5x - 2$ , as above.

- **Triangular.** If  $\mathbf{A}$  is upper or lower triangular, so is  $x\mathbf{I} - \mathbf{A}$ , with diagonal entries  $x - a_{ii}$ . By [Theorem 6.4.6](#) over  $F[x]$ ,  $p_{\mathbf{A}}(x) = (x - a_{11})(x - a_{22}) \cdots (x - a_{nn})$ .
- **Degenerate cases.** The zero matrix has  $p_0(x) = x^n$ , and the identity has  $p_{\mathbf{I}_n}(x) = (x - 1)^n$ ; both are triangular. These two polynomials are different, although both matrices are as simple as possible, which already shows that  $p_{\mathbf{A}}$  records something about  $\mathbf{A}$ .

**Example. ( $\mathbf{A}$   $3 \times 3$  Characteristic Polynomial)**

Find  $p_{\mathbf{A}}$  for  $\mathbf{A} = \begin{pmatrix} 2 & 0 & 1 \\ 1 & 3 & 1 \\ 1 & 0 & 2 \end{pmatrix} \in M_3(\mathbb{Q})$ .

*Solution.* We have

$$x\mathbf{I} - \mathbf{A} = \begin{pmatrix} x-2 & 0 & -1 \\ -1 & x-3 & -1 \\ -1 & 0 & x-2 \end{pmatrix}.$$

Column 2 has a single non-zero entry,  $x-3$  in position  $(2,2)$ . Expanding along it ([Theorem 6.6.1](#) over  $F[x]$ ),

$$\begin{aligned} p_{\mathbf{A}}(x) &= (x-3)(-1)^{2+2} \det \begin{pmatrix} x-2 & -1 \\ -1 & x-2 \end{pmatrix} \\ &= (x-3)((x-2)^2 - 1) = (x-1)(x-3)^2. \end{aligned}$$

Multiplied out,  $p_{\mathbf{A}}(x) = x^3 - 7x^2 + 15x - 9$ . Compare with  $\text{tr } \mathbf{A} = 2 + 3 + 2 = 7$  and  $\det \mathbf{A} = 9$  (expand  $\mathbf{A}$  along column 2:  $3(4-1) = 9$ ).

**A companion-type example.** Let  $\mathbf{A} = \begin{pmatrix} 0 & 0 & 6 \\ 1 & 0 & -11 \\ 0 & 1 & 6 \end{pmatrix}$ , with ones just below the diagonal, a

last column  $(6, -11, 6)$ , and zeros elsewhere. Expanding  $x\mathbf{I} - \mathbf{A} = \begin{pmatrix} x & 0 & -6 \\ -1 & x & 11 \\ 0 & -1 & x-6 \end{pmatrix}$  along the first row ([Theorem 6.6.1](#) over  $F[x]$ ),

$$\begin{aligned} p_{\mathbf{A}}(x) &= x \det \begin{pmatrix} x & 11 \\ -1 & x-6 \end{pmatrix} + (-6) \det \begin{pmatrix} -1 & x \\ 0 & -1 \end{pmatrix} \\ &= x(x^2 - 6x + 11) - 6 = x^3 - 6x^2 + 11x - 6, \end{aligned}$$

which factors as  $(x-1)(x-2)(x-3)$ . The last column of  $\mathbf{A}$  lists the coefficients  $-6, 11, -6$  of  $1, x, x^2$  in  $p_{\mathbf{A}}$ , with their signs changed. This is no accident: [Exercise \(C1: Companion matrices\)](#) shows that the same pattern works in every size.

**Non-example by minimal change.** Swap the order of subtraction:  $\det(\mathbf{A} - x\mathbf{I})$ . Each of the  $n$  columns of  $\mathbf{A} - x\mathbf{I}$  is  $-1$  times the corresponding column of  $x\mathbf{I} - \mathbf{A}$ , so by multilinearity  $\det(\mathbf{A} - x\mathbf{I}) = (-1)^n p_{\mathbf{A}}(x)$ . For  $\mathbf{A} = (a)$  this is  $a - x$ , whose leading coefficient is  $-1$ . What still works: the roots are the same. What fails: the result is **not monic** when  $n$  is odd.

**Why this definition.** Some books use  $\det(\mathbf{A} - x\mathbf{I})$ . We use  $x\mathbf{I} - \mathbf{A}$  so that the characteristic polynomial is always **monic** ([Theorem 6.10.2](#) below), which makes statements like “ $p_{\mathbf{A}} = (x-1)(x-3)^2$ ” true without sign bookkeeping. The name comes from the fact that the roots of this polynomial are the “characteristic values” of  $\mathbf{A}$ , which Chapter 8 calls eigenvalues.

*Warning.* Do not row reduce  $x\mathbf{I} - \mathbf{A}$  by dividing by polynomials. Over a field, a matrix with non-zero determinant is invertible ([Theorem 6.5.2](#)), but that proof runs through elimination, which divides by pivots, and it fails over  $F[x]$ . For  $\mathbf{A} = (0) \in M_1(F)$ , the matrix  $x\mathbf{I} - \mathbf{A} = (x)$  has determinant  $x \neq 0$ , yet it has no inverse in  $M_1(F[x])$ : if  $xq(x) = 1$  for a polynomial  $q$ , then  $q \neq 0$ , and  $1 + \deg q = \deg 1 = 0$  by [Theorem 0.8.3](#), which is impossible since  $\deg q \geq 0$ . Likewise, “divide row 1 of  $x\mathbf{I} - \mathbf{A}$

by  $x - 1$  produces entries that are not polynomials and that make no sense at  $x = 1$ . Compute  $p_{\mathbf{A}}$  by expansion, by triangular or block shape, or by the Leibniz formula.

Before looking at the coefficients, we record the fact that makes  $p_{\mathbf{A}}$  useful: evaluating it at a scalar  $c$  gives the same result as putting  $c$  in place of  $x$  before taking the determinant.

### Lemma 6.10.1 — Evaluating the Characteristic Polynomial

Let  $\mathbf{A} \in M_n(F)$  and  $c \in F$ . Then  $p_{\mathbf{A}}(c) = \det(c\mathbf{I}_n - \mathbf{A})$ .

*Proof.* By Theorem 0.8.5, evaluation at  $c$  turns sums of polynomials into sums and products into products, and sends a constant  $a$  to  $a$ . Applying it to the Leibniz sum in Definition 6.10.1 term by term,

$$p_{\mathbf{A}}(c) = \sum_{\sigma \in S_n} \operatorname{sgn}(\sigma) \prod_{j=1}^n (c\delta_{\sigma(j)j} - a_{\sigma(j)j}),$$

and  $c\delta_{ij} - a_{ij}$  is the  $(i, j)$ -entry of  $c\mathbf{I}_n - \mathbf{A}$ . By Definition 6.4.1, the right side is  $\det(c\mathbf{I}_n - \mathbf{A})$ . ■

## 6.10.2 The shape of the characteristic polynomial

The  $2 \times 2$  example suggests a pattern: the leading coefficient is 1, the next coefficient is minus the trace, and the constant term is the determinant up to sign. The same holds in every size.

### Theorem 6.10.2 — Coefficients of the Characteristic Polynomial

Let  $n \geq 1$  and  $\mathbf{A} \in M_n(F)$ . Then  $p_{\mathbf{A}}$  is **monic of degree  $n$** , and

$$p_{\mathbf{A}}(x) = x^n - (\operatorname{tr} \mathbf{A})x^{n-1} + \cdots + (-1)^n \det \mathbf{A},$$

that is, the coefficient of  $x^{n-1}$  is  $-\operatorname{tr} \mathbf{A}$  and the constant term is  $(-1)^n \det \mathbf{A}$ . (For  $n = 1$  these two statements concern the same coefficient, and both say it is  $-a_{11}$ .)

*Idea.* Look at the Leibniz sum term by term and count powers of  $x$ . A factor  $x\delta_{\sigma(j)j} - a_{\sigma(j)j}$  contains  $x$  only if  $\sigma(j) = j$ . A permutation other than the identity moves at least **two** indices, so its term has degree at most  $n - 2$  and cannot affect the top two coefficients. The identity contributes  $\prod_j (x - a_{jj})$ , whose top two coefficients are 1 and  $-\sum_j a_{jj}$ . For the constant term, set  $x = 0$ .

*Proof.* Write  $m_{ij} = x\delta_{ij} - a_{ij}$  for the entries of  $x\mathbf{I} - \mathbf{A}$ . By Definition 6.10.1,  $p_{\mathbf{A}} = \sum_{\sigma} t_{\sigma}$  with  $t_{\sigma} = \operatorname{sgn}(\sigma) \prod_j m_{\sigma(j)j}$ .

*Terms with  $\sigma \neq \operatorname{id}$ .* If  $\sigma(j) \neq j$  for some  $j$ , then  $\sigma(\sigma(j)) \neq \sigma(j)$ , because  $\sigma$  is injective. So  $\sigma$  moves at least two indices, and at most  $n - 2$  indices are fixed. For a moved index  $j$ ,  $m_{\sigma(j)j} = -a_{\sigma(j)j}$  has degree at most 0; for a fixed index it has degree at most 1. By

**Theorem 0.8.3**,  $\deg t_\sigma \leq n - 2$ .

*The term  $\sigma = \text{id}$ .* We show by induction on  $k \geq 1$  that  $\prod_{j=1}^k (x - a_{jj}) = x^k - s_k x^{k-1} + r_k$  with  $s_k = a_{11} + \cdots + a_{kk}$  and  $\deg r_k \leq k - 2$ . For  $k = 1$ , take  $r_1 = 0$ . If it holds for  $k$ , then with  $a = a_{k+1,k+1}$ ,

$$(x^k - s_k x^{k-1} + r_k)(x - a) = x^{k+1} - (s_k + a)x^k + (s_k a x^{k-1} + r_k x - a r_k),$$

and the bracket has degree at most  $k - 1$  by [Theorem 0.8.2](#) and [Theorem 0.8.3](#). This completes the induction. Since  $\text{sgn}(\text{id}) = 1$ ,  $t_{\text{id}} = x^n - (\text{tr } \mathbf{A}) x^{n-1} + r_n$  with  $\deg r_n \leq n - 2$ .

Adding,  $p_{\mathbf{A}} = x^n - (\text{tr } \mathbf{A}) x^{n-1} + (r_n + \sum_{\sigma \neq \text{id}} t_\sigma)$ , and the bracket has degree at most  $n - 2$  by [Theorem 0.8.2](#). Hence  $p_{\mathbf{A}}$  is monic of degree  $n$ , with coefficient  $-\text{tr } \mathbf{A}$  at  $x^{n-1}$  ([Definition 0.9.12](#)).

*Constant term.* The constant term of a polynomial is its value at 0 ([Definition 0.8.6](#)). By [Lemma 6.10.1](#) with  $c = 0$ ,  $p_{\mathbf{A}}(0) = \det(-\mathbf{A})$ . The matrix  $-\mathbf{A}$  is obtained from  $\mathbf{A}$  by multiplying each of its  $n$  columns by  $-1$ , so  $\det(-\mathbf{A}) = (-1)^n \det \mathbf{A}$  by [Theorem 6.5.1](#) (d). This proves the theorem. ■

So the characteristic polynomial packages two invariants we already know, the trace and the determinant, and adds  $n - 2$  more coefficients in between. For  $n = 2$  nothing is in between, which is why  $p_{\mathbf{A}}(x) = x^2 - (\text{tr } \mathbf{A})x + \det \mathbf{A}$  for every  $2 \times 2$  matrix.

*Quick check.* Without expanding any determinant, find the coefficient of  $x^2$  and the constant term of  $p_{\mathbf{A}}$  for  $\mathbf{A} = \begin{pmatrix} 0 & 1 & 1 \\ 1 & 0 & 1 \\ 1 & 1 & 0 \end{pmatrix}$ , given that  $\det \mathbf{A} = 2$ .

*Solution.* By [Theorem 6.10.2](#), the coefficient of  $x^2$  is  $-\text{tr } \mathbf{A} = 0$ , and the constant term is  $(-1)^3 \det \mathbf{A} = -2$ . (In full,  $p_{\mathbf{A}}(x) = x^3 - 3x - 2 = (x - 2)(x + 1)^2$ .)

### 6.10.3 Similar matrices, same polynomial

Similar matrices describe the same operator in different bases ([Theorem 3.7.4](#)), so everything intrinsic to the operator should agree for them. The trace and determinant do ([Theorem 3.7.5](#), [Corollary 6.5.7](#)). The characteristic polynomial, which contains both, does too, and the proof is the proof for the determinant, run over  $F[x]$ .

#### **Theorem 6.10.3 — Similar Matrices Have the Same Characteristic Polynomial**

If  $\mathbf{A}, \mathbf{B} \in M_n(F)$  are similar, then  $p_{\mathbf{A}} = p_{\mathbf{B}}$ .

*Proof.* Let  $\mathbf{B} = \mathbf{P}^{-1}\mathbf{A}\mathbf{P}$  with  $\mathbf{P} \in M_n(F)$  invertible. Regard  $\mathbf{P}$ ,  $\mathbf{P}^{-1}$ ,  $\mathbf{A}$  and  $\mathbf{I}$  as matrices over  $F[x]$  whose entries happen to be constants. The laws of matrix arithmetic in [Theorem 0.9.4](#) are proved from the ring axioms alone, so they hold in  $M_n(F[x])$ .

Hence

$$\mathbf{P}^{-1}(x\mathbf{I} - \mathbf{A})\mathbf{P} = x\mathbf{P}^{-1}\mathbf{P} - \mathbf{P}^{-1}\mathbf{A}\mathbf{P} = x\mathbf{I} - \mathbf{B}.$$

By [Theorem 6.5.4](#), valid over  $F[x]$  by the remark in §5,

$$\begin{aligned} p_{\mathbf{B}} &= \det(\mathbf{P}^{-1}(x\mathbf{I} - \mathbf{A})\mathbf{P}) = \det(\mathbf{P}^{-1}) \det(x\mathbf{I} - \mathbf{A}) \det \mathbf{P} \\ &= \det(\mathbf{P}^{-1}) \det(\mathbf{P}) p_{\mathbf{A}}, \end{aligned}$$

using commutativity of  $F[x]$  in the last step. Finally  $\det(\mathbf{P}^{-1}) \det \mathbf{P} = \det(\mathbf{P}^{-1}\mathbf{P}) = \det \mathbf{I} = 1$  by [Theorem 6.5.4](#) over  $F$ . Hence  $p_{\mathbf{B}} = p_{\mathbf{A}}$ . ■

This is exactly the case singled out in the remark in §5: the proof of [Corollary 6.5.7](#) survives over  $F[x]$  because  $\mathbf{P}$  is invertible already over  $F$ . The theorem lets us attach a characteristic polynomial to an operator.

### Definition 6.10.2 — Characteristic Polynomial of an Operator

Let  $V$  be a vector space over  $F$  with  $\dim V = n \geq 1$ , and let  $T \in \mathcal{L}(V)$ . The **characteristic polynomial** of  $T$  is

$$p_T(x) := p_{[T]_{\mathcal{B}}}(x) = \det(x\mathbf{I}_n - [T]_{\mathcal{B}}),$$

where  $\mathcal{B}$  is **any** basis of  $V$ .

This does not depend on  $\mathcal{B}$ : for bases  $\mathcal{B}$  and  $\mathcal{C}$ , the matrices  $[T]_{\mathcal{B}}$  and  $[T]_{\mathcal{C}}$  are similar by [Theorem 3.7.4](#) (a), so they have the same characteristic polynomial by [Theorem 6.10.3](#). By [Theorem 6.10.2](#),  $p_T$  is monic of degree  $\dim V$ , with coefficient  $-\operatorname{tr} T$  at  $x^{n-1}$  and constant term  $(-1)^n \det T$  ([Definition 3.10.2](#), [Definition 6.5.1](#)).

### Example. (Characteristic Polynomials of Two Operators)

On  $V = \mathbb{R}[x]_{\leq 2}$ , find the characteristic polynomials of differentiation  $D(q) = q'$  and of the shift  $S(q)(x) = q(x+1)$ .

*Solution.* Use the basis  $\mathcal{B} = (1, x, x^2)$ . Since  $D(1) = 0$ ,  $D(x) = 1$ ,  $D(x^2) = 2x$ , and  $S(1) = 1$ ,  $S(x) = 1 + x$ ,  $S(x^2) = 1 + 2x + x^2$ ,

$$[D]_{\mathcal{B}} = \begin{pmatrix} 0 & 1 & 0 \\ 0 & 0 & 2 \\ 0 & 0 & 0 \end{pmatrix}, \quad [S]_{\mathcal{B}} = \begin{pmatrix} 1 & 1 & 1 \\ 0 & 1 & 2 \\ 0 & 0 & 1 \end{pmatrix}.$$

Both are upper triangular, so  $p_D(t) = t^3$  and  $p_S(t) = (t-1)^3$ . (We name the variable  $t$  here, since  $x$  is already the variable of the polynomials in  $V$ .) As a check,  $\operatorname{tr} S = 3$  and  $\det S = 1$ , matching the coefficient  $-3$  of  $t^2$  and the constant term  $(-1)^3 \cdot 1$ .

*Warning.* **Equal characteristic polynomials do not make matrices similar.** The converse of [Theorem 6.10.3](#) is false. Recall the pair from the warning in Chapter 3, §7:  $\mathbf{I}_2$  and  $\mathbf{J} = \begin{pmatrix} 1 & 1 \\ 0 & 1 \end{pmatrix}$  both have characteristic polynomial  $(x-1)^2$ , since both are

triangular with diagonal 1, 1. But  $\mathbf{I}_2$  is similar only to itself, so  $\mathbf{J} \not\sim \mathbf{I}_2$ . In particular the characteristic polynomial does not determine the matrix, not even up to similarity. Deciding similarity needs finer invariants; Chapter 9 finds a complete set.

### 6.10.4 Roots and singular matrices

We can now answer the opening question. The determinant tests invertibility of  $c\mathbf{I} - \mathbf{A}$  for one  $c$  at a time; the characteristic polynomial tests all  $c$  at once, and the answer is a finite set.

#### Theorem 6.10.4 — Roots of the Characteristic Polynomial

Let  $\mathbf{A} \in M_n(F)$  with  $n \geq 1$ , and let  $c \in F$ .

- (a) The following are equivalent: (i)  $p_{\mathbf{A}}(c) = 0$ ; (ii)  $c\mathbf{I} - \mathbf{A}$  is **not** invertible; (iii) there is a **non-zero**  $\mathbf{v} \in F^n$  with  $\mathbf{A}\mathbf{v} = c\mathbf{v}$ .
- (b) There are **at most**  $n$  scalars  $c \in F$  with these properties.

*Proof.*

- (a) By Lemma 6.10.1,  $p_{\mathbf{A}}(c) = \det(c\mathbf{I} - \mathbf{A})$ , which is a determinant of a matrix over the field  $F$ . By Theorem 6.5.2, it is zero if and only if  $c\mathbf{I} - \mathbf{A}$  is not invertible; this is (i)  $\Leftrightarrow$  (ii). By Theorem 2.4.4,  $c\mathbf{I} - \mathbf{A}$  is not invertible if and only if  $(c\mathbf{I} - \mathbf{A})\mathbf{v} = \mathbf{0}$  for some  $\mathbf{v} \neq \mathbf{0}$ , and  $(c\mathbf{I} - \mathbf{A})\mathbf{v} = c\mathbf{v} - \mathbf{A}\mathbf{v}$  by Theorem 0.9.4; this is (ii)  $\Leftrightarrow$  (iii).
- (b) By Theorem 6.10.2,  $p_{\mathbf{A}}$  is monic of degree  $n$ , in particular non-zero. By Corollary 5.1.4, it has at most  $n$  distinct roots in  $F$ , and by (a) these are exactly the scalars in question. ■

Notice where the evaluation happens: **first** substitute  $c$ , getting a matrix over the field  $F$ , **then** use the invertibility test, which is valid over fields. That order is what the warning about  $F[x]$  above requires.

The same holds for an operator  $T$  on  $V$  with  $\dim V = n \geq 1$ :  $p_T(c) = 0$  if and only if  $c \operatorname{id}_V - T$  is not invertible, by (a) applied to  $[T]_{\mathcal{B}}$  together with Theorem 3.6.5 (b), since  $[c \operatorname{id}_V - T]_{\mathcal{B}} = c\mathbf{I} - [T]_{\mathcal{B}}$ .

The scalars  $c$  of this theorem are the **eigenvalues** of  $\mathbf{A}$ , and the vectors  $\mathbf{v}$  in (iii) its eigenvectors. They are the subject of Chapter 8, which starts from this theorem.

**Examples.**

- For  $\mathbf{A}$  in Example (A  $3 \times 3$  Characteristic Polynomial),  $p_{\mathbf{A}} = (x - 1)(x - 3)^2$ , so  $c\mathbf{I} - \mathbf{A}$  is singular exactly for  $c = 1$  and  $c = 3$ . For instance  $3\mathbf{I} - \mathbf{A} = \begin{pmatrix} 1 & 0 & -1 \\ -1 & 0 & -1 \\ -1 & 0 & 1 \end{pmatrix}$  has a zero column, and  $\mathbf{v} = \mathbf{e}_2$  satisfies  $\mathbf{A}\mathbf{e}_2 = 3\mathbf{e}_2$ .
- The rotation  $\mathbf{R} = \begin{pmatrix} 0 & -1 \\ 1 & 0 \end{pmatrix}$  has  $p_{\mathbf{R}} = x^2 + 1$ . Over  $\mathbb{R}$  this has no root, so  $c\mathbf{I} - \mathbf{R}$  is invertible for **every** real  $c$ : no non-zero real vector is merely stretched by a quarter-turn.

Over  $\mathbb{C}$ , the roots  $\pm i$  appear, and for example  $\mathbf{R}(1, -i) = (i, 1) = i(1, -i)$ . The field matters.

*Quick check.* For which  $c \in \mathbb{Q}$  is  $c\mathbf{I} - \mathbf{A}$  invertible, where  $\mathbf{A} = \begin{pmatrix} 5 & -2 \\ 3 & 0 \end{pmatrix}$ ?

*Solution.*  $p_{\mathbf{A}}(x) = x^2 - (\text{tr } \mathbf{A})x + \det \mathbf{A} = x^2 - 5x + 6 = (x-2)(x-3)$ . By Theorem 6.10.4,  $c\mathbf{I} - \mathbf{A}$  is invertible exactly for  $c \neq 2, 3$ .

## 6.10.5 A tempting false proof

Here is a statement that is true, and important: **every square matrix satisfies its own characteristic polynomial**,  $p_{\mathbf{A}}(\mathbf{A}) = 0$ . This is the Cayley–Hamilton Theorem, proved in Chapter 8. Here  $p_{\mathbf{A}}(\mathbf{A})$  means substituting the matrix  $\mathbf{A}$  into the polynomial  $p_{\mathbf{A}}$ , replacing each  $x^k$  by  $\mathbf{A}^k$  and the constant term  $a_0$  by  $a_0\mathbf{I}$  (Definition 0.9.14). For a  $2 \times 2$  matrix it says  $\mathbf{A}^2 - (\text{tr } \mathbf{A})\mathbf{A} + (\det \mathbf{A})\mathbf{I} = 0$ , which you can check by multiplying out.

A one-line “proof” suggests itself, and it is wrong.

*Warning. The fake proof of Cayley–Hamilton.* “Since  $p_{\mathbf{A}}(x) = \det(x\mathbf{I} - \mathbf{A})$ , put  $x = \mathbf{A}$ :  $p_{\mathbf{A}}(\mathbf{A}) = \det(\mathbf{A}\mathbf{I} - \mathbf{A}) = \det(0) = 0$ .” This is invalid for two reasons. First, the two sides are different kinds of object:  $p_{\mathbf{A}}(\mathbf{A})$  is an  $n \times n$  **matrix**, while  $\det(\cdots)$  is a **scalar**, so “ $= 0$ ” compares a matrix with a number. Second, Lemma 6.10.1 lets us substitute only a **scalar**  $c \in F$  into the entries of  $x\mathbf{I} - \mathbf{A}$ . Substituting a matrix for  $x$  inside the entries would produce a matrix whose entries are matrices, and “ $\mathbf{A}\mathbf{I} - \mathbf{A}$ ” silently treats the  $x$  in the entry  $x - a_{11}$  as if the entry became  $\mathbf{A} - a_{11}$ , which is not what substitution into a polynomial means.

The same “argument” proves false statements, which shows it is not a proof. Replace  $\det$  by  $\text{tr}$ : let  $q(x) = \text{tr}(x\mathbf{I} - \mathbf{A}) = nx - \text{tr } \mathbf{A}$ . The fake reasoning gives “ $q(\mathbf{A}) = \text{tr}(\mathbf{A}\mathbf{I} - \mathbf{A}) = \text{tr } 0 = 0$ ”. But for  $\mathbf{A} = \text{diag}(1, 0)$ ,  $q(x) = 2x - 1$  and  $q(\mathbf{A}) = 2\mathbf{A} - \mathbf{I} = \text{diag}(1, -1) \neq 0$ .

## 6.10.6 Exercises

### 6.10.6.1 A. Check your understanding

#### Exercise A1.

- Define the characteristic polynomial of  $\mathbf{A} \in M_n(F)$ , and say in which ring its determinant is computed.
- State the degree, leading coefficient, coefficient of  $x^{n-1}$  and constant term of  $p_{\mathbf{A}}$ .
- True or false:  $\det(\mathbf{A} - x\mathbf{I}) = p_{\mathbf{A}}(x)$  for every  $\mathbf{A} \in M_3(F)$ . Justify your answer.
- True or false: if  $p_{\mathbf{A}} = p_{\mathbf{B}}$ , then  $\mathbf{A} \sim \mathbf{B}$ . Justify your answer.
- What does  $p_{\mathbf{A}}(c) = 0$  say about the matrix  $c\mathbf{I} - \mathbf{A}$ ? About vectors?
- Explain in one sentence why “ $p_{\mathbf{A}}(\mathbf{A}) = \det(\mathbf{A}\mathbf{I} - \mathbf{A}) = 0$ ” is not a proof.

*Solution.*

- (a)  $p_{\mathbf{A}}(x) = \det(x\mathbf{I}_n - \mathbf{A})$ , the Leibniz sum computed in the commutative ring  $F[x]$  (Definition 6.10.1).
- (b) Degree  $n$ , leading coefficient 1, coefficient  $-\operatorname{tr} \mathbf{A}$  at  $x^{n-1}$ , constant term  $(-1)^n \det \mathbf{A}$  (Theorem 6.10.2).
- (c) False.  $\det(\mathbf{A} - x\mathbf{I}) = (-1)^3 p_{\mathbf{A}}(x) = -p_{\mathbf{A}}(x)$ , since each of the three columns changes sign. For  $\mathbf{A} = 0$ ,  $\det(-x\mathbf{I}_3) = -x^3 \neq x^3 = p_{\mathbf{A}}(x)$ .
- (d) False.  $\mathbf{I}_2$  and  $\begin{pmatrix} 1 & 1 \\ 0 & 1 \end{pmatrix}$  both have  $p = (x-1)^2$  but are not similar.
- (e)  $c\mathbf{I} - \mathbf{A}$  is not invertible; equivalently there is a non-zero  $\mathbf{v}$  with  $\mathbf{A}\mathbf{v} = c\mathbf{v}$  (Theorem 6.10.4).
- (f)  $p_{\mathbf{A}}(\mathbf{A})$  is a matrix obtained by substituting  $\mathbf{A}$  into a polynomial, while  $\det(\mathbf{A}\mathbf{I} - \mathbf{A})$  is a scalar, and substitution into the entries of  $x\mathbf{I} - \mathbf{A}$  is only justified for scalars (Lemma 6.10.1).

### 6.10.6.2 B. Practice

#### Exercise B1: Computing characteristic polynomials.

Find the characteristic polynomial of each matrix over  $\mathbb{Q}$ , factor it as far as possible over  $\mathbb{Q}$ , and check the trace and determinant against Theorem 6.10.2.

$$\mathbf{A} = \begin{pmatrix} 1 & 1 & 0 \\ 0 & 1 & 1 \\ 1 & 0 & 1 \end{pmatrix}, \quad \mathbf{B} = \begin{pmatrix} 2 & 1 & 0 & 0 \\ 0 & 2 & 0 & 0 \\ 0 & 0 & 1 & 3 \\ 0 & 0 & 1 & -1 \end{pmatrix}.$$

Hence list the  $c \in \mathbb{Q}$  for which  $c\mathbf{I} - \mathbf{A}$ , respectively  $c\mathbf{I} - \mathbf{B}$ , is not invertible.

*Solution.* For  $\mathbf{A}$ . Expanding  $x\mathbf{I} - \mathbf{A} = \begin{pmatrix} x-1 & -1 & 0 \\ 0 & x-1 & -1 \\ -1 & 0 & x-1 \end{pmatrix}$  along the first row (Theorem 6.6.1 over  $F[x]$ ),

$$\begin{aligned} p_{\mathbf{A}}(x) &= (x-1)((x-1)^2 - 0) - (-1)(0-1) + 0 \\ &= (x-1)^3 - 1 = x^3 - 3x^2 + 3x - 2. \end{aligned}$$

Check:  $-\operatorname{tr} \mathbf{A} = -3$ ;  $\det \mathbf{A} = 1 \cdot (1-0) - 1 \cdot (0-1) + 0 = 2$ , and  $(-1)^3 \cdot 2 = -2$ . Since  $p_{\mathbf{A}}(2) = 1 - 1 = 0$ , divide by  $x-2$ :  $p_{\mathbf{A}} = (x-2)(x^2 - x + 1)$ . The quadratic has discriminant  $1 - 4 = -3 < 0$ , so no rational (indeed no real) root. By Theorem 6.10.4,  $c\mathbf{I} - \mathbf{A}$  is singular only for  $c = 2$ .

For  $\mathbf{B}$ .  $x\mathbf{I} - \mathbf{B}$  is block upper triangular with diagonal blocks  $x\mathbf{I} - \begin{pmatrix} 2 & 1 \\ 0 & 2 \end{pmatrix}$  and  $x\mathbf{I} - \begin{pmatrix} 1 & 3 \\ 1 & -1 \end{pmatrix}$ . By Theorem 6.8.4 over  $F[x]$  and the  $2 \times 2$  formula,

$$p_{\mathbf{B}}(x) = (x-2)^2 (x^2 - 0 \cdot x + (-1-3)) = (x-2)^2 (x^2 - 4) = (x-2)^3 (x+2).$$



Multiplied out,  $p_{\mathbf{B}} = x^4 - 4x^3 + 16x - 16$ . Check:  $-\operatorname{tr} \mathbf{B} = -(2 + 2 + 1 - 1) = -4$ ;  $\det \mathbf{B} = 4 \cdot (-4) = -16$  by [Theorem 6.8.4](#), and  $(-1)^4(-16) = -16$ . So  $c\mathbf{I} - \mathbf{B}$  is singular exactly for  $c = 2$  and  $c = -2$ .

### Exercise B2: Transpose.

Prove that  $p_{\mathbf{A}^\top} = p_{\mathbf{A}}$  for every  $\mathbf{A} \in M_n(F)$ . Hence show that  $c\mathbf{I} - \mathbf{A}$  is invertible if and only if  $c\mathbf{I} - \mathbf{A}^\top$  is.

*Solution.* By [Theorem 0.9.5](#),  $(x\mathbf{I} - \mathbf{A})^\top = x\mathbf{I}^\top - \mathbf{A}^\top = x\mathbf{I} - \mathbf{A}^\top$ ; these rules are proved entrywise from the ring axioms, so they hold over  $F[x]$ . By [Theorem 6.4.5](#), which holds over  $F[x]$  by the remark in §4,

$$p_{\mathbf{A}^\top} = \det(x\mathbf{I} - \mathbf{A}^\top) = \det((x\mathbf{I} - \mathbf{A})^\top) = \det(x\mathbf{I} - \mathbf{A}) = p_{\mathbf{A}}.$$

By [Theorem 6.10.4](#),  $c\mathbf{I} - \mathbf{A}$  is not invertible exactly when  $p_{\mathbf{A}}(c) = 0$ , which is exactly when  $p_{\mathbf{A}^\top}(c) = 0$ , that is, when  $c\mathbf{I} - \mathbf{A}^\top$  is not invertible.

### Exercise B3: $\mathbf{AB}$ and $\mathbf{BA}$ .

Let  $\mathbf{A}, \mathbf{B} \in M_n(F)$  with  $\mathbf{A}$  invertible.

- Show that  $\mathbf{BA} = \mathbf{A}^{-1}(\mathbf{AB})\mathbf{A}$ , and deduce  $p_{\mathbf{AB}} = p_{\mathbf{BA}}$ .
- Deduce that  $\operatorname{tr}(\mathbf{AB}) = \operatorname{tr}(\mathbf{BA})$  and  $\det(\mathbf{AB}) = \det(\mathbf{BA})$  in this case.
- Check (a) for  $\mathbf{A} = \begin{pmatrix} 1 & 1 \\ 0 & 1 \end{pmatrix}$ ,  $\mathbf{B} = \begin{pmatrix} 0 & 0 \\ 1 & 0 \end{pmatrix}$ , and note that  $\mathbf{AB} \neq \mathbf{BA}$ .

(Chapter 8 removes the hypothesis that  $\mathbf{A}$  is invertible.)

*Solution.*

- By associativity,  $\mathbf{A}^{-1}(\mathbf{AB})\mathbf{A} = (\mathbf{A}^{-1}\mathbf{A})(\mathbf{BA}) = \mathbf{BA}$ . So  $\mathbf{BA} \sim \mathbf{AB}$  ([Definition 3.7.2](#), with  $\mathbf{P} = \mathbf{A}$ ), and  $p_{\mathbf{AB}} = p_{\mathbf{BA}}$  by [Theorem 6.10.3](#).
- By [Theorem 6.10.2](#), the coefficient of  $x^{n-1}$  is  $-\operatorname{tr}(\mathbf{AB})$  in  $p_{\mathbf{AB}}$  and  $-\operatorname{tr}(\mathbf{BA})$  in  $p_{\mathbf{BA}}$ ; the constant terms are  $(-1)^n \det(\mathbf{AB})$  and  $(-1)^n \det(\mathbf{BA})$ . Equal polynomials have equal coefficients.
- $\mathbf{AB} = \begin{pmatrix} 1 & 0 \\ 1 & 0 \end{pmatrix}$  and  $\mathbf{BA} = \begin{pmatrix} 0 & 0 \\ 1 & 1 \end{pmatrix}$ , so  $\mathbf{AB} \neq \mathbf{BA}$ . Both have trace 1 and determinant 0, so both characteristic polynomials are  $x^2 - x = x(x - 1)$ .

#### 6.10.6.3 C. Going deeper

### Exercise C1: Companion matrices.

Let  $n \geq 2$  and  $p(x) = x^n + a_{n-1}x^{n-1} + \cdots + a_1x + a_0 \in F[x]$ . The **companion matrix** of  $p$  is the matrix  $\mathbf{C}(p) \in M_n(F)$  with 1 in each position  $(i+1, i)$  just below the diagonal,

last column  $(-a_0, -a_1, \dots, -a_{n-1})$ , and 0 elsewhere:

$$\mathbf{C}(p) = \begin{pmatrix} 0 & 0 & \cdots & 0 & -a_0 \\ 1 & 0 & \cdots & 0 & -a_1 \\ 0 & 1 & \cdots & 0 & -a_2 \\ \vdots & & \ddots & & \vdots \\ 0 & 0 & \cdots & 1 & -a_{n-1} \end{pmatrix}.$$

(The matrix  $\mathbf{C}$  of [Exercise \(B2: An annihilating polynomial by elimination\)](#) is  $\mathbf{C}(x^3 - 2x^2 + x - 2)$ .)

- (a) Let  $\mathbf{M} = x\mathbf{I} - \mathbf{C}(p)$ . For each  $i$ , show that deleting row  $i$  and column  $n$  of  $\mathbf{M}$  leaves a block diagonal matrix whose blocks are an  $(i-1) \times (i-1)$  lower triangular matrix with diagonal  $x, \dots, x$  and an  $(n-i) \times (n-i)$  upper triangular matrix with diagonal  $-1, \dots, -1$  (either block may be absent when  $i = 1$  or  $i = n$ ). Deduce that the cofactor  $C_{in}$  of  $\mathbf{M}$  equals  $x^{i-1}$ .
- (b) Deduce, by expanding along the last column, that  $p_{\mathbf{C}(p)} = p$ .
- (c) Verify (b) for  $p = x^4 + x^3 - 2x + 3$  by writing  $\mathbf{C}(p)$  and checking the coefficients predicted by [Theorem 6.10.2](#).
- (d) Deduce that every monic polynomial of degree  $n \geq 1$  over  $F$  is the characteristic polynomial of some  $n \times n$  matrix over  $F$ .

*Solution.*

- (a) The entries of  $\mathbf{M}$  are:  $x$  at  $(j, j)$  for  $j < n$ ;  $-1$  at  $(j+1, j)$ ;  $a_{j-1}$  at  $(j, n)$  for  $j < n$ ;  $x + a_{n-1}$  at  $(n, n)$ ; 0 elsewhere. Delete row  $i$  and column  $n$ . In the remaining matrix, rows  $1, \dots, i-1$  have non-zero entries only in columns  $j-1$  and  $j$  for row  $j$ , all at most  $i-1$ ; and rows  $i+1, \dots, n$  have non-zero entries only in columns  $j-1$  and  $j \leq n-1$ , all at least  $i$ . So the matrix is block diagonal with blocks on rows and columns  $\{1, \dots, i-1\}$  and on rows  $\{i+1, \dots, n\}$ , columns  $\{i, \dots, n-1\}$ . The first block has  $x$  on its diagonal and  $-1$  just below it, so it is lower triangular. In the second block, row  $j$  of  $\mathbf{M}$  (for  $j > i$ ) has  $-1$  in column  $j-1$ , which is the diagonal position of the block, and  $x$  in column  $j$ , one step to the right; so it is upper triangular with diagonal  $-1, \dots, -1$ . By [Theorem 6.8.4](#) and [Theorem 6.4.6](#) over  $F[x]$  (both division-free), the minor is  $M_{in} = x^{i-1}(-1)^{n-i}$ , where an absent block contributes the factor 1 (for  $i = 1$  or  $i = n$  the matrix is a single triangular block). Hence  $C_{in} = (-1)^{i+n}(-1)^{n-i}x^{i-1} = (-1)^{2n}x^{i-1} = x^{i-1}$ .
- (b) By [Theorem 6.6.1](#) along column  $n$ , valid over  $F[x]$ ,

$$\begin{aligned} p_{\mathbf{C}(p)} = \det \mathbf{M} &= \sum_{i=1}^{n-1} a_{i-1} x^{i-1} + (x + a_{n-1}) x^{n-1} \\ &= a_0 + a_1 x + \cdots + a_{n-1} x^{n-1} + x^n = p. \end{aligned}$$

- (c) Here  $a_3 = 1$ ,  $a_2 = 0$ ,  $a_1 = -2$ ,  $a_0 = 3$ , so

$$\mathbf{C}(p) = \begin{pmatrix} 0 & 0 & 0 & -3 \\ 1 & 0 & 0 & 2 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & -1 \end{pmatrix}.$$

Its trace is  $-1$ , so [Theorem 6.10.2](#) predicts the coefficient  $1 = a_3$  at  $x^3$ , as in  $p$ . Its determinant, expanding along the first row, whose only non-zero entry is  $-3$  in position  $(1, 4)$ , is  $(-3)(-1)^{1+4} \det \mathbf{I}_3 = 3$ , since deleting row 1 and column 4 leaves  $\mathbf{I}_3$ . So the predicted constant term is  $(-1)^4 \cdot 3 = 3 = a_0$ .

- (d) For  $n \geq 2$ , take  $\mathbf{C}(p)$  and use (b). For  $n = 1$ ,  $p = x + a_0$  is  $p_{(-a_0)}$ .

### Exercise C2: Nilpotent $2 \times 2$ matrices.

Let  $\mathbf{A} \in M_2(F)$  satisfy  $\mathbf{A}^k = 0$  for some  $k \geq 1$ .

- Show that the constant term of  $p_{\mathbf{A}}$  is 0.
- By multiplying out, verify that  $\mathbf{A}^2 - (\operatorname{tr} \mathbf{A})\mathbf{A} + (\det \mathbf{A})\mathbf{I} = 0$  for every  $\mathbf{A} \in M_2(F)$ .
- Deduce that  $\operatorname{tr} \mathbf{A} = 0$ , and hence  $p_{\mathbf{A}} = x^2$  and  $\mathbf{A}^2 = 0$ .
- Give an example showing that  $\mathbf{A}^2 = 0$  does not force  $\mathbf{A} = 0$ .

*Solution.*

- By [Theorem 6.5.4](#),  $(\det \mathbf{A})^k = \det(\mathbf{A}^k) = \det 0 = 0$ , so  $\det \mathbf{A} = 0$  because a field has no zero divisors ([Theorem 0.7.1](#)). The constant term is  $(-1)^2 \det \mathbf{A} = 0$  by [Theorem 6.10.2](#).
- Let  $\mathbf{A} = \begin{pmatrix} a & b \\ c & d \end{pmatrix}$ . Then  $\mathbf{A}^2 = \begin{pmatrix} a^2 + bc & ab + bd \\ ca + dc & bc + d^2 \end{pmatrix}$  and  $(a+d)\mathbf{A} = \begin{pmatrix} a^2 + ad & ab + bd \\ ac + dc & ad + d^2 \end{pmatrix}$ . The difference is  $\begin{pmatrix} bc - ad & 0 \\ 0 & bc - ad \end{pmatrix} = -(\det \mathbf{A})\mathbf{I}$ , as claimed.
- By (a) and (b),  $\mathbf{A}^2 = (\operatorname{tr} \mathbf{A})\mathbf{A}$ . By induction,  $\mathbf{A}^m = (\operatorname{tr} \mathbf{A})^{m-1}\mathbf{A}$  for all  $m \geq 1$ : if it holds for  $m$ , then  $\mathbf{A}^{m+1} = (\operatorname{tr} \mathbf{A})^{m-1}\mathbf{A}^2 = (\operatorname{tr} \mathbf{A})^m\mathbf{A}$ . If  $\mathbf{A} = 0$ , then  $\operatorname{tr} \mathbf{A} = 0$ . If  $\mathbf{A} \neq 0$ , then  $0 = \mathbf{A}^k = (\operatorname{tr} \mathbf{A})^{k-1}\mathbf{A}$  forces  $(\operatorname{tr} \mathbf{A})^{k-1} = 0$ , since a non-zero matrix times a scalar is zero only if the scalar is zero; here  $k \geq 2$ , because  $k = 1$  would give  $\mathbf{A} = 0$ . Hence  $\operatorname{tr} \mathbf{A} = 0$  by [Theorem 0.7.1](#) (f), applied repeatedly. So  $p_{\mathbf{A}} = x^2 - 0 \cdot x + 0 = x^2$ , and  $\mathbf{A}^2 = (\operatorname{tr} \mathbf{A})\mathbf{A} = 0$ .
- $\mathbf{A} = \begin{pmatrix} 0 & 1 \\ 0 & 0 \end{pmatrix}$  has  $\mathbf{A}^2 = 0$  and  $\mathbf{A} \neq 0$ .

### Exercise C3: Odd size over the reals.

Let  $n$  be odd and  $\mathbf{A} \in M_n(\mathbb{R})$ .

- Prove that there is  $c \in \mathbb{R}$  and a non-zero  $\mathbf{v} \in \mathbb{R}^n$  with  $\mathbf{A}\mathbf{v} = c\mathbf{v}$ .
- Show by an example that (a) fails for  $n = 2$ .
- Show that (a) holds for every  $\mathbf{A} \in M_n(\mathbb{C})$  and every  $n \geq 1$ .

*Solution.*

- By [Theorem 6.10.2](#),  $p_{\mathbf{A}} \in \mathbb{R}[x]$  has odd degree  $n$ . By [Corollary 5.5.9](#), it has a real root  $c$ . By [Theorem 6.10.4](#), there is a non-zero  $\mathbf{v} \in \mathbb{R}^n$  with  $\mathbf{A}\mathbf{v} = c\mathbf{v}$ .
- The rotation  $\mathbf{R} = \begin{pmatrix} 0 & -1 \\ 1 & 0 \end{pmatrix}$  has  $p_{\mathbf{R}} = x^2 + 1$ , with no real root; by [Theorem 6.10.4](#)

no real  $c$  and non-zero real  $\mathbf{v}$  satisfy  $\mathbf{R}\mathbf{v} = c\mathbf{v}$ .

- (c)  $p_{\mathbf{A}} \in \mathbb{C}[x]$  has degree  $n \geq 1$ , so it has a root  $c \in \mathbb{C}$  by [Theorem 5.5.1](#). Apply [Theorem 6.10.4](#) over  $F = \mathbb{C}$ .

# Chapter 7

## Block Matrices

A large matrix often has structure that its entries hide. Split it into blocks, smaller matrices arranged in a grid, and the structure shows. A direct sum of subspaces becomes a block-diagonal matrix. A subspace that an operator maps into itself becomes a block of zeros in the corner. A system of equations that splits into two groups becomes a two-by-two system whose “entries” are matrices.

This chapter makes block arithmetic rigorous and then puts it to work. Multiplying block by block, eliminating a whole block at once, and taking the Schur complement  $\mathbf{D} - \mathbf{CA}^{-1}\mathbf{B}$  turn statements about big matrices into statements about small ones: ranks, determinants and inverses. The Kronecker product rewrites matrix equations such as  $\mathbf{AX} + \mathbf{XB} = \mathbf{C}$  as ordinary linear systems. The chapter ends with a question that looks like it needs heavy machinery but does not: which matrices can be written as a commutator  $\mathbf{XY} - \mathbf{YX}$ ?

**What you need.** From Chapter 0, matrix multiplication ([Definition 0.9.9](#)) and polynomials without zero divisors ([Corollary 0.8.4](#)). From Chapter 2, rank ([Definition 2.5.4](#)). From Chapter 3, the matrix of a linear map and change of basis ([Definition 3.6.1](#), [Theorem 3.7.3](#)), quotient spaces ([Definition 3.9.3](#)), the rank inequalities ([Theorem 3.8.5](#)) and the trace ([Definition 3.10.2](#)). From Chapter 6, determinants of block triangular matrices and the characteristic polynomial ([Theorem 6.8.4](#), [Theorem 6.10.2](#)).

### 7.0.1 Roadmap

- **Partitions and block multiplication.** Multiplying matrices block by block, and what block-diagonal and block-triangular shapes say about a linear map.
- **Rank and block matrices.** Rank inequalities proved by clearing blocks, including Frobenius’s inequality.
- **Schur complements.** The Schur complement, the determinant, rank and inverse of a block matrix, the Woodbury formula, and determinants with commuting blocks over any field.
- **The Kronecker product.** A product of matrices of different sizes, and matrix equations turned into linear systems.
- **Commutators and trace-zero matrices.** Every trace-zero matrix over a field of characteristic zero is a commutator.

## 7.0.2 Named moves

- **Partition to match the structure.** Choose the blocks so that the zeros you know about sit in whole blocks.
- **Eliminate a block.** Multiply by an invertible block triangular matrix to clear a block, just as a row operation clears an entry.
- **Pass to the Schur complement.** Reduce a question about  $\begin{pmatrix} \mathbf{A} & \mathbf{B} \\ \mathbf{C} & \mathbf{D} \end{pmatrix}$  with  $\mathbf{A}$  invertible to a question about  $\mathbf{A}$  and  $\mathbf{D} - \mathbf{CA}^{-1}\mathbf{B}$ .
- **Vectorize.** Turn a linear equation in an unknown matrix  $\mathbf{X}$  into a linear system in the entries of  $\mathbf{X}$ .
- **Adjoin an indeterminate.** To remove an invertibility hypothesis, replace  $\mathbf{D}$  by  $\mathbf{D} + x\mathbf{I}$ , work over  $F[x]$ , and set  $x = 0$  at the end.

## 7.1 Partitions and Block Multiplication

A large matrix often has structure that its individual entries hide: a corner of zeros, a copy of the identity, a smaller matrix repeated. Chapter 6 already met one such shape, the block triangular matrix, and computed its determinant from two smaller determinants. This section makes the idea systematic. We cut matrices into rectangular pieces, prove that products can be computed piece by piece, and use the pieces to read off structure: direct sums become block diagonal matrices, and a subspace that an operator maps into itself becomes a zero block in the corner.

### 7.1.1 Cutting a matrix into blocks

Take two  $4 \times 4$  matrices over  $\mathbb{Q}$  whose top-right  $2 \times 2$  corners are zero:

$$\mathbf{M} = \left( \begin{array}{cc|cc} 1 & 2 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ \hline 1 & 0 & 2 & 1 \\ 2 & 1 & 1 & 1 \end{array} \right), \quad \mathbf{N} = \left( \begin{array}{cc|cc} 1 & -1 & 0 & 0 \\ 1 & 0 & 0 & 0 \\ \hline 0 & 1 & 1 & 0 \\ 1 & 1 & 3 & 1 \end{array} \right).$$

The lines cut each matrix into four  $2 \times 2$  pieces. Name them:  $\mathbf{M} = \begin{pmatrix} \mathbf{A} & \mathbf{0} \\ \mathbf{C} & \mathbf{D} \end{pmatrix}$  and  $\mathbf{N} = \begin{pmatrix} \mathbf{E} & \mathbf{0} \\ \mathbf{G} & \mathbf{H} \end{pmatrix}$ , with  $\mathbf{A} = \begin{pmatrix} 1 & 2 \\ 0 & 1 \end{pmatrix}$ ,  $\mathbf{C} = \begin{pmatrix} 1 & 0 \\ 2 & 1 \end{pmatrix}$ , and so on. If we pretend that the pieces are numbers and multiply as for  $2 \times 2$  matrices, keeping the order of every product, we get

$$\mathbf{MN} \stackrel{?}{=} \begin{pmatrix} \mathbf{AE} + \mathbf{0G} & \mathbf{A0} + \mathbf{0H} \\ \mathbf{CE} + \mathbf{DG} & \mathbf{C0} + \mathbf{DH} \end{pmatrix} = \begin{pmatrix} \mathbf{AE} & \mathbf{0} \\ \mathbf{CE} + \mathbf{DG} & \mathbf{DH} \end{pmatrix}.$$

With  $\mathbf{AE} = \begin{pmatrix} 3 & -1 \\ 1 & 0 \end{pmatrix}$ ,  $\mathbf{CE} + \mathbf{DG} = \begin{pmatrix} 2 & 2 \\ 4 & 0 \end{pmatrix}$  and  $\mathbf{DH} = \begin{pmatrix} 5 & 1 \\ 4 & 1 \end{pmatrix}$ , this predicts

$$\mathbf{MN} = \begin{pmatrix} 3 & -1 & 0 & 0 \\ 1 & 0 & 0 & 0 \\ 2 & 2 & 5 & 1 \\ 4 & 0 & 4 & 1 \end{pmatrix},$$

and multiplying out the sixteen entries directly gives exactly this matrix. Two things happened. The zero corner of the product came for free, with no arithmetic. And the product of two  $4 \times 4$  matrices became four products of  $2 \times 2$  matrices, **AE**, **CE**, **DG** and **DH**; the other four products in the block formula have a zero factor and cost nothing. To use this safely we need to say precisely what “cutting” means and prove that the pretend multiplication is always correct.

*Draw horizontal lines between rows and vertical lines between columns; the pieces are the blocks.*

### Definition 7.1.1 — Block Partition

Let  $\mathbf{A} = (a_{ij}) \in M_{m \times n}(F)$ . A **row partition** of  $\mathbf{A}$  is a list of **positive** integers  $m_1, \dots, m_a$  with  $m_1 + \dots + m_a = m$ , and a **column partition** is a list of positive integers  $n_1, \dots, n_b$  with  $n_1 + \dots + n_b = n$ . Put  $\mu_r = m_1 + \dots + m_r$  and  $\nu_s = n_1 + \dots + n_s$ , with  $\mu_0 = \nu_0 = 0$ . For  $1 \leq r \leq a$  and  $1 \leq s \leq b$ , the  $(r, s)$  **block** of  $\mathbf{A}$  is the matrix  $\mathbf{A}_{rs} \in M_{m_r \times n_s}(F)$  with entries

$$(\mathbf{A}_{rs})_{ij} = a_{\mu_{r-1}+i, \nu_{s-1}+j} \quad (1 \leq i \leq m_r, 1 \leq j \leq n_s).$$

We then write  $\mathbf{A} = (\mathbf{A}_{rs})$ , displayed as an  $a \times b$  array of blocks, and call  $\mathbf{A}$  a **block matrix**.

Let  $\mathbf{B} \in M_{n \times p}(F)$ . Partitions of  $\mathbf{A}$  and  $\mathbf{B}$  are **conformable** if the column partition of  $\mathbf{A}$  and the row partition of  $\mathbf{B}$  are **the same list**  $n_1, \dots, n_b$ , in the same order.

In words: the row partition says where the horizontal lines go. Block row  $r$  consists of the rows with indices in  $I_r = \{\mu_{r-1}+1, \dots, \mu_r\}$ , a run of  $m_r$  consecutive rows, and similarly block column  $s$  consists of the columns in  $J_s = \{\nu_{s-1}+1, \dots, \nu_s\}$ . The block  $\mathbf{A}_{rs}$  is what survives when we keep only the rows in  $I_r$  and the columns in  $J_s$ , renumbered from 1. Conformability says that the vertical lines through  $\mathbf{A}$  sit at the same places as the horizontal lines through  $\mathbf{B}$ . That is exactly what will make each product “block of  $\mathbf{A}$  times block of  $\mathbf{B}$ ” defined.

**Well-definedness.** The runs  $I_1, \dots, I_a$  are disjoint and cover  $\{1, \dots, m\}$ , and the same holds for  $J_1, \dots, J_b$ . So every entry of  $\mathbf{A}$  lies in exactly one block. Conversely, if we are given matrices  $\mathbf{A}_{rs} \in M_{m_r \times n_s}(F)$ , there is exactly one  $\mathbf{A} \in M_{m \times n}(F)$  with these blocks. Block sizes must fit: all blocks in block row  $r$  have  $m_r$  rows, and all blocks in block column  $s$  have  $n_s$  columns. We allow only positive parts, so no block is empty.

#### Examples.

- **Every entry a block.** With  $m_r = 1$  and  $n_s = 1$  for all  $r, s$ , the block  $\mathbf{A}_{rs}$  is the  $1 \times 1$  matrix  $(a_{rs})$ . A matrix is its own finest block matrix.
- **One block.** With  $a = b = 1$ , the single block is  $\mathbf{A}$  itself. This degenerate partition matters because a statement about block matrices then says something about ordinary matrices, and the statement must not break.
- **Columns as blocks.** Row partition  $m$ , column partition  $1, 1, \dots, 1$ . The blocks are the  $n$  columns of  $\mathbf{A}$ :  $\mathbf{A} = (\mathbf{a}_1 \ \dots \ \mathbf{a}_n)$ , as in Chapter 0.
- **The hook.** For  $\mathbf{M}$  and  $\mathbf{N}$  above, both partitions are  $2, 2$ , so the column partition of  $\mathbf{M}$  equals the row partition of  $\mathbf{N}$ : they are conformable.

**Non-example by minimal change.** Keep  $\mathbf{M}$  with column partition  $2, 2$ , but cut  $\mathbf{N}$  with row partition  $1, 3$ . The product  $\mathbf{MN}$  is still defined, and both partitions are legitimate. What fails is conformability: the first block column of  $\mathbf{M}$  has 2 columns, while the first block row of  $\mathbf{N}$  has 1 row, so the “product”  $\mathbf{A} \cdot \mathbf{N}_{11}$  of a  $2 \times 2$  matrix with a  $1 \times 2$  matrix is undefined.

*Quick check.* Let  $\mathbf{A} \in M_{3 \times 5}(F)$  have row partition 1, 2 and column partition 3, 2, and let  $\mathbf{B} \in M_{5 \times 4}(F)$  have row partition 2, 3. Are the partitions conformable? What are the sizes of the blocks  $\mathbf{A}_{21}$  and  $\mathbf{B}_{2t}$ ?

*Solution.* No. The column partition of  $\mathbf{A}$  is 3, 2 and the row partition of  $\mathbf{B}$  is 2, 3. Both lists sum to 5, but they differ in order, and conformability asks for the same list. The block  $\mathbf{A}_{21}$  is  $2 \times 3$  (block row 2 has 2 rows, block column 1 has 3 columns), and every  $\mathbf{B}_{2t}$  has 3 rows.

## 7.1.2 Block multiplication

Here is the theorem that justifies the computation in the hook. The proof is nothing but the entry formula for a product, with the sum over the middle index cut into the runs  $J_1, \dots, J_b$ .

### Theorem 7.1.1 — Block Multiplication

Let  $\mathbf{A} \in M_{m \times n}(F)$  and  $\mathbf{B} \in M_{n \times p}(F)$  be conformably partitioned:  $\mathbf{A}$  with row partition  $m_1, \dots, m_a$  and column partition  $n_1, \dots, n_b$ , and  $\mathbf{B}$  with row partition  $n_1, \dots, n_b$  and column partition  $p_1, \dots, p_c$ . Partition  $\mathbf{AB} \in M_{m \times p}(F)$  with row partition  $m_1, \dots, m_a$  and column partition  $p_1, \dots, p_c$ . Then for all  $1 \leq r \leq a$  and  $1 \leq t \leq c$ ,

$$(\mathbf{AB})_{rt} = \sum_{s=1}^b \mathbf{A}_{rs} \mathbf{B}_{st} = \mathbf{A}_{r1} \mathbf{B}_{1t} + \mathbf{A}_{r2} \mathbf{B}_{2t} + \cdots + \mathbf{A}_{rb} \mathbf{B}_{bt}.$$

Each product  $\mathbf{A}_{rs} \mathbf{B}_{st}$  is defined and lies in  $M_{m_r \times p_t}(F)$ .

*Idea.* Look at one entry of  $\mathbf{AB}$ , in row  $i^*$  and column  $k^*$ . It is a sum over the middle index  $j = 1, \dots, n$ . Cut that sum into the runs  $J_1, \dots, J_b$  given by the common partition. The part over  $J_s$  pairs row  $i^*$  of  $\mathbf{A}$ , restricted to the columns of block column  $s$ , with column  $k^*$  of  $\mathbf{B}$ , restricted to the rows of block row  $s$ . That is one entry of  $\mathbf{A}_{rs} \mathbf{B}_{st}$ . The only real work is translating indices between the big matrix and the blocks.

*Proof.* Let  $\mu_r, \nu_s, \rho_t$  be the partial sums of the three partitions, as in Definition 7.1.1, and  $J_s = \{\nu_{s-1} + 1, \dots, \nu_s\}$ . The block  $\mathbf{A}_{rs}$  is  $m_r \times n_s$  and  $\mathbf{B}_{st}$  is  $n_s \times p_t$ , so their product is defined and is  $m_r \times p_t$ , as is the block  $(\mathbf{AB})_{rt}$ .

Fix  $r, t$ , and let  $1 \leq i \leq m_r$  and  $1 \leq k \leq p_t$ . Put  $i^* = \mu_{r-1} + i$  and  $k^* = \rho_{t-1} + k$ . By Definition 7.1.1 and Definition 0.9.9,

$$((\mathbf{AB})_{rt})_{ik} = (\mathbf{AB})_{i^*k^*} = \sum_{j=1}^n a_{i^*j} b_{jk^*}.$$

The sets  $J_1, \dots, J_b$  are disjoint with union  $\{1, \dots, n\}$ , so the finite sum may be grouped according to them. In the group for  $J_s$ , substitute  $j = \nu_{s-1} + l$ ; as  $l$  runs through



$1, \dots, n_s$ ,  $j$  runs through  $J_s$  exactly once. Hence

$$\begin{aligned} ((\mathbf{AB})_{rt})_{ik} &= \sum_{s=1}^b \sum_{j \in J_s} a_{i^*j} b_{jk^*} \\ &= \sum_{s=1}^b \sum_{l=1}^{n_s} a_{\mu_{r-1}+i, \nu_{s-1}+l} b_{\nu_{s-1}+l, \rho_{t-1}+k} \\ &= \sum_{s=1}^b \sum_{l=1}^{n_s} (\mathbf{A}_{rs})_{il} (\mathbf{B}_{st})_{lk}, \end{aligned}$$

where the last step is the definition of the blocks  $\mathbf{A}_{rs}$  and  $\mathbf{B}_{st}$ . By [Definition 0.9.9](#) the inner sum is  $(\mathbf{A}_{rs}\mathbf{B}_{st})_{ik}$ , and by [Definition 0.9.6](#) the outer sum is the  $(i, k)$ -entry of  $\sum_s \mathbf{A}_{rs}\mathbf{B}_{st}$ . Since  $i$  and  $k$  were arbitrary,  $(\mathbf{AB})_{rt} = \sum_s \mathbf{A}_{rs}\mathbf{B}_{st}$ . This proves the theorem. ■

So conformably partitioned matrices multiply like matrices whose entries are matrices. The commonest case is a  $2 \times 2$  block matrix:

$$\begin{pmatrix} \mathbf{A} & \mathbf{B} \\ \mathbf{C} & \mathbf{D} \end{pmatrix} \begin{pmatrix} \mathbf{A}' & \mathbf{B}' \\ \mathbf{C}' & \mathbf{D}' \end{pmatrix} = \begin{pmatrix} \mathbf{AA}' + \mathbf{BC}' & \mathbf{AB}' + \mathbf{BD}' \\ \mathbf{CA}' + \mathbf{DC}' & \mathbf{CB}' + \mathbf{DD}' \end{pmatrix},$$

whenever the partitions are conformable. Sums and scalar multiples of matrices with the same partition are also computed blockwise, straight from [Definition 0.9.6](#) and [Definition 0.9.7](#), since each entry lies in exactly one block.

**Over a commutative ring.** The proof uses only sums and products of entries and the re-grouping of a finite sum. So block multiplication holds verbatim for matrices with entries in any commutative ring  $R$  in the sense of Chapter 6 §4, for instance  $R = F[x]$ . Section 3 uses this.

*Warning. Keep the order inside every block product.* The top-left block of the product above is  $\mathbf{AA}' + \mathbf{BC}'$ , not  $\mathbf{A}'\mathbf{A} + \mathbf{C}'\mathbf{B}$ : blocks are matrices, and matrices do not commute. In the hook,  $\mathbf{CE} + \mathbf{DG} = \begin{pmatrix} 2 & 2 \\ 4 & 0 \end{pmatrix}$ , while  $\mathbf{EC} + \mathbf{GD} = \begin{pmatrix} 0 & 0 \\ 4 & 2 \end{pmatrix}$ . For the same reason, the formula  $ad - bc$  for a  $2 \times 2$  determinant does **not** survive with blocks: in general  $\det \begin{pmatrix} \mathbf{A} & \mathbf{B} \\ \mathbf{C} & \mathbf{D} \end{pmatrix} \neq \det(\mathbf{AD} - \mathbf{BC})$ . Take  $\mathbf{A} = \mathbf{I}_2$ ,  $\mathbf{B} = \mathbf{E}_{12}$ ,  $\mathbf{C} = \mathbf{E}_{21}$ ,  $\mathbf{D} = \mathbf{E}_{11}$  in  $M_2(\mathbb{Q})$ . Then  $\mathbf{AD} - \mathbf{BC} = \mathbf{E}_{11} - \mathbf{E}_{12}\mathbf{E}_{21} = \mathbf{E}_{11} - \mathbf{E}_{11} = 0$ , with determinant 0. But

$$\mathbf{M} = \begin{pmatrix} 1 & 0 & 0 & 1 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 1 & 0 & 0 & 0 \end{pmatrix},$$

and subtracting row 4 from row 1, which does not change the determinant ([Theorem 6.5.1](#)), gives the permutation matrix that swaps  $\mathbf{e}_1$  and  $\mathbf{e}_4$ , whose determinant is  $-1$ . So  $\det \mathbf{M} = -1 \neq 0$ . Section 3 shows that the formula does hold when  $\mathbf{C}$  and  $\mathbf{D}$  commute; here  $\mathbf{CD} = \mathbf{E}_{21} \neq 0 = \mathbf{DC}$ .

Transposing a block matrix transposes each block and reflects the array of blocks.

### Proposition 7.1.2 — Transpose of a Block Matrix

Let  $\mathbf{A} \in M_{m \times n}(F)$  have row partition  $m_1, \dots, m_a$ , column partition  $n_1, \dots, n_b$  and blocks  $\mathbf{A}_{rs}$ . Give  $\mathbf{A}^\top$  the row partition  $n_1, \dots, n_b$  and the column partition  $m_1, \dots, m_a$ .

Then  $(\mathbf{A}^\top)_{sr} = (\mathbf{A}_{rs})^\top$ . In particular  $\begin{pmatrix} \mathbf{A} & \mathbf{B} \\ \mathbf{C} & \mathbf{D} \end{pmatrix}^\top = \begin{pmatrix} \mathbf{A}^\top & \mathbf{C}^\top \\ \mathbf{B}^\top & \mathbf{D}^\top \end{pmatrix}$ .

*Proof.* For  $1 \leq j \leq n_s$  and  $1 \leq i \leq m_r$ , by Definition 7.1.1 and Definition 0.9.10,  $((\mathbf{A}^\top)_{sr})_{ji} = (\mathbf{A}^\top)_{\nu_{s-1}+j, \mu_{r-1}+i} = a_{\mu_{r-1}+i, \nu_{s-1}+j} = (\mathbf{A}_{rs})_{ij} = ((\mathbf{A}_{rs})^\top)_{ji}$ . ■

## 7.1.3 Rows, columns and outer products

The finest and coarsest partitions turn Theorem 7.1.1 into facts we already know, and one mixed choice gives something new.

Partition  $\mathbf{A} \in M_{m \times n}(F)$  trivially (one block) and  $\mathbf{B} \in M_{n \times p}(F)$  into columns (row partition  $n$ , column partition  $1, \dots, 1$ ). These are conformable, and the theorem says that the  $k$ -th block of  $\mathbf{AB}$ , its  $k$ -th column, is  $\mathbf{A}\mathbf{b}_k$ . That is the column view of Theorem 0.9.2. Partitioning  $\mathbf{A}$  into rows and  $\mathbf{B}$  trivially gives the row view in the same way.

Now cut  $\mathbf{A}$  into columns and  $\mathbf{B}$  into rows. Both middle partitions are  $1, 1, \dots, 1$  (with  $n$  ones), so they are conformable, and the product has a single block.

### Corollary 7.1.3 — Outer Product Expansion

Let  $\mathbf{A} \in M_{m \times n}(F)$  have columns  $\mathbf{a}_1, \dots, \mathbf{a}_n \in F^m$ , and let  $\mathbf{B} \in M_{n \times p}(F)$  have rows  $\mathbf{r}_1, \dots, \mathbf{r}_n \in M_{1 \times p}(F)$ . Then

$$\mathbf{AB} = \mathbf{a}_1\mathbf{r}_1 + \mathbf{a}_2\mathbf{r}_2 + \cdots + \mathbf{a}_n\mathbf{r}_n = \sum_{k=1}^n \mathbf{a}_k\mathbf{r}_k,$$

where each  $\mathbf{a}_k\mathbf{r}_k \in M_{m \times p}(F)$  is column  $k$  of  $\mathbf{A}$  times row  $k$  of  $\mathbf{B}$ .

*Proof.* Give  $\mathbf{A}$  the row partition  $m$  and the column partition  $1, \dots, 1$ , so that its blocks are  $\mathbf{A}_{1k} = \mathbf{a}_k$ ; give  $\mathbf{B}$  the row partition  $1, \dots, 1$  and the column partition  $p$ , so that  $\mathbf{B}_{k1} = \mathbf{r}_k$ . The partitions are conformable, and by Theorem 7.1.1 the single block of  $\mathbf{AB}$  is  $\sum_k \mathbf{A}_{1k}\mathbf{B}_{k1} = \sum_k \mathbf{a}_k\mathbf{r}_k$ . ■

Each term  $\mathbf{a}_k\mathbf{r}_k$  is a column times a row, an **outer product**, and it has rank at most 1: its columns are the multiples  $r_{kj}\mathbf{a}_k$  of the single vector  $\mathbf{a}_k$ . So the corollary writes every product of an  $m \times n$  and an  $n \times p$  matrix as a sum of  $n$  matrices of rank at most one, in the spirit of Exercise (B3: Rank-one pieces), which writes a matrix of rank  $r$  as a sum of  $r$  rank-one matrices. For

example,

$$\begin{aligned}
 & \begin{pmatrix} 1 & 2 \\ 3 & 4 \\ 0 & 1 \end{pmatrix} \begin{pmatrix} 1 & 0 & 2 \\ -1 & 1 & 1 \end{pmatrix} \\
 &= \begin{pmatrix} 1 \\ 3 \\ 0 \end{pmatrix} \begin{pmatrix} 1 & 0 & 2 \end{pmatrix} + \begin{pmatrix} 2 \\ 4 \\ 1 \end{pmatrix} \begin{pmatrix} -1 & 1 & 1 \end{pmatrix} \\
 &= \begin{pmatrix} 1 & 0 & 2 \\ 3 & 0 & 6 \\ 0 & 0 & 0 \end{pmatrix} + \begin{pmatrix} -2 & 2 & 2 \\ -4 & 4 & 4 \\ -1 & 1 & 1 \end{pmatrix} \\
 &= \begin{pmatrix} -1 & 2 & 4 \\ -1 & 4 & 10 \\ -1 & 1 & 1 \end{pmatrix},
 \end{aligned}$$

which is the product computed entry by entry.

## 7.1.4 Block diagonal matrices

Diagonal matrices are the easiest matrices to compute with: they multiply, power and invert entry by entry along the diagonal. Replace the diagonal entries by square blocks and all of this survives, one block at a time.

*A block diagonal matrix is several square matrices placed corner to corner along the diagonal, with zeros everywhere else.*

### Definition 7.1.2 — Block Diagonal Matrix

Let  $k_1, \dots, k_r$  be positive integers and  $\mathbf{A}_i \in M_{k_i}(F)$  **square** for each  $i$ . Their **direct sum**  $\mathbf{A}_1 \oplus \dots \oplus \mathbf{A}_r$  is the matrix in  $M_{k_1+\dots+k_r}(F)$  which, with row partition and column partition both equal to  $k_1, \dots, k_r$ , has  $(i, i)$  block  $\mathbf{A}_i$  and  $(i, j)$  block 0 for  $i \neq j$ . A square matrix is **block diagonal with respect to** the partition  $k_1, \dots, k_r$  if it has this form. For two blocks,

$$\mathbf{A} \oplus \mathbf{B} = \begin{pmatrix} \mathbf{A} & 0 \\ 0 & \mathbf{B} \end{pmatrix}.$$

In words: the same partition is used for rows and columns, so the diagonal blocks are square; the blocks on the diagonal may be anything; every block off the diagonal is zero. Whether a matrix is block diagonal depends on the partition, which is part of the statement.

**Examples.**

- **Diagonal matrices.**  $\text{diag}(d_1, \dots, d_n) = (d_1) \oplus \dots \oplus (d_n)$ , with all  $k_i = 1$ .
- **The identity.**  $\mathbf{I}_k \oplus \mathbf{I}_l = \mathbf{I}_{k+l}$ , and more generally  $\mathbf{I}_k \oplus 0_l$  is the matrix of a projection (Corollary 3.10.2).
- **A genuine block.**  $\begin{pmatrix} 1 & 1 \\ 0 & 1 \end{pmatrix} \oplus (2) = \begin{pmatrix} 1 & 1 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 2 \end{pmatrix}$ , with partition 2, 1.
- **One block.** With  $r = 1$ ,  $\mathbf{A}_1$  itself. Every square matrix is block diagonal with respect to the trivial partition, which is why the partition must always be named.

**Non-example by minimal change.** Change the  $(1, 3)$ -entry of the third example from 0 to 1. The partition  $2, 1$  still has square diagonal blocks  $\begin{pmatrix} 1 & 1 \\ 0 & 1 \end{pmatrix}$  and  $(2)$ , and the  $(2, 1)$  block  $\begin{pmatrix} 0 & 0 \end{pmatrix}$  is still zero. What fails is the clause that the  $(1, 2)$  block is zero: it is now  $\begin{pmatrix} 1 \\ 0 \end{pmatrix}$ . The matrix is block upper triangular, which is treated below, but not block diagonal for this partition.

**Why the symbol  $\oplus$ .** The matrix  $\mathbf{A} \oplus \mathbf{B}$  acts on  $F^{k+l} = \text{span}(\mathbf{e}_1, \dots, \mathbf{e}_k) \oplus \text{span}(\mathbf{e}_{k+1}, \dots, \mathbf{e}_{k+l})$  by sending each of the two summands into itself, acting as  $\mathbf{A}$  on the first and as  $\mathbf{B}$  on the second. The theorem on adapted bases below makes this precise for any operator and any direct sum, and it is the reason the same symbol is used.

*Warning.*  $\mathbf{A} \oplus \mathbf{B}$  is a matrix, not a subspace, and the order matters. For subspaces,  $U \oplus W$  and  $W \oplus U$  are the same set. For matrices,  $(1) \oplus (2) = \text{diag}(1, 2)$  and  $(2) \oplus (1) = \text{diag}(2, 1)$  are different matrices. They are similar (swap the two basis vectors), but they are not equal.

### Theorem 7.1.4 — Arithmetic of Block Diagonal Matrices

Let  $\mathbf{A}, \mathbf{A}' \in M_k(F)$ ,  $\mathbf{B}, \mathbf{B}' \in M_l(F)$  and  $c \in F$ . Then:

- (a)  $(\mathbf{A} \oplus \mathbf{B}) + (\mathbf{A}' \oplus \mathbf{B}') = (\mathbf{A} + \mathbf{A}') \oplus (\mathbf{B} + \mathbf{B}')$ ,  $c(\mathbf{A} \oplus \mathbf{B}) = c\mathbf{A} \oplus c\mathbf{B}$ , and  $(\mathbf{A} \oplus \mathbf{B})(\mathbf{A}' \oplus \mathbf{B}') = \mathbf{A}\mathbf{A}' \oplus \mathbf{B}\mathbf{B}'$ ;
- (b)  $(\mathbf{A} \oplus \mathbf{B})^j = \mathbf{A}^j \oplus \mathbf{B}^j$  for every  $j \in \mathbb{N}$ , and  $p(\mathbf{A} \oplus \mathbf{B}) = p(\mathbf{A}) \oplus p(\mathbf{B})$  for every  $p \in F[x]$ ;
- (c)  $\mathbf{A} \oplus \mathbf{B}$  is invertible if and only if  $\mathbf{A}$  and  $\mathbf{B}$  are both invertible, and then  $(\mathbf{A} \oplus \mathbf{B})^{-1} = \mathbf{A}^{-1} \oplus \mathbf{B}^{-1}$ ;
- (d)  $\det(\mathbf{A} \oplus \mathbf{B}) = \det \mathbf{A} \det \mathbf{B}$  and  $\text{tr}(\mathbf{A} \oplus \mathbf{B}) = \text{tr} \mathbf{A} + \text{tr} \mathbf{B}$ .

*Proof.*

- (a) Sums and scalar multiples are computed blockwise, and by [Theorem 7.1.1](#) the product has blocks  $\mathbf{A}\mathbf{A}' + 0 \cdot 0 = \mathbf{A}\mathbf{A}'$ ,  $\mathbf{A}0 + 0\mathbf{B}' = 0$ ,  $0\mathbf{A}' + \mathbf{B}0 = 0$  and  $0 \cdot 0 + \mathbf{B}\mathbf{B}' = \mathbf{B}\mathbf{B}'$ .
- (b) By induction on  $j$ . For  $j = 0$ ,  $(\mathbf{A} \oplus \mathbf{B})^0 = \mathbf{I}_{k+l} = \mathbf{I}_k \oplus \mathbf{I}_l$  by [Definition 0.9.14](#). If  $(\mathbf{A} \oplus \mathbf{B})^j = \mathbf{A}^j \oplus \mathbf{B}^j$ , then  $(\mathbf{A} \oplus \mathbf{B})^{j+1} = (\mathbf{A}^j \oplus \mathbf{B}^j)(\mathbf{A} \oplus \mathbf{B}) = \mathbf{A}^{j+1} \oplus \mathbf{B}^{j+1}$  by (a). For  $p = \sum_j c_j x^j$ , (a) gives  $p(\mathbf{A} \oplus \mathbf{B}) = \sum_j c_j (\mathbf{A}^j \oplus \mathbf{B}^j) = (\sum_j c_j \mathbf{A}^j) \oplus (\sum_j c_j \mathbf{B}^j) = p(\mathbf{A}) \oplus p(\mathbf{B})$ .
- (c) The matrix  $\mathbf{A} \oplus \mathbf{B}$  is block upper triangular with square diagonal blocks, so  $\det(\mathbf{A} \oplus \mathbf{B}) = \det \mathbf{A} \det \mathbf{B}$  by [Theorem 6.8.4](#). Its diagonal entries are those of  $\mathbf{A}$  followed by those of  $\mathbf{B}$ , so the traces add ([Definition 0.9.12](#)).
- (d)  $(\Leftarrow)$  By (a),  $(\mathbf{A} \oplus \mathbf{B})(\mathbf{A}^{-1} \oplus \mathbf{B}^{-1}) = \mathbf{I}_k \oplus \mathbf{I}_l = \mathbf{I}_{k+l}$ , so by [Theorem 2.4.5](#)  $\mathbf{A} \oplus \mathbf{B}$  is invertible with inverse  $\mathbf{A}^{-1} \oplus \mathbf{B}^{-1}$ .  $(\Rightarrow)$  If  $\mathbf{A} \oplus \mathbf{B}$  is invertible, then  $\det \mathbf{A} \det \mathbf{B} \neq 0$  by (d) and [Theorem 6.5.2](#). A product in a field is non-zero only if both factors are ([Theorem 0.7.1](#) (d)), so  $\det \mathbf{A} \neq 0 \neq \det \mathbf{B}$ , and both are invertible by [Theorem 6.5.2](#).

■

By induction on the number of blocks, the same rules hold for  $\mathbf{A}_1 \oplus \cdots \oplus \mathbf{A}_r$ : everything is done block by block. For instance  $\left(\begin{pmatrix} 1 & 1 \\ 0 & 1 \end{pmatrix} \oplus (2)\right)^5 = \begin{pmatrix} 1 & 5 \\ 0 & 1 \end{pmatrix} \oplus (32)$ , since  $\begin{pmatrix} 1 & 1 \\ 0 & 1 \end{pmatrix}^j = \begin{pmatrix} 1 & j \\ 0 & 1 \end{pmatrix}$  by induction.

## 7.1.5 Block triangular matrices

Allow one more non-zero block:  $\mathbf{M} = \begin{pmatrix} \mathbf{A} & \mathbf{B} \\ 0 & \mathbf{D} \end{pmatrix}$  with  $\mathbf{A} \in M_k(F)$ ,  $\mathbf{D} \in M_l(F)$  square and  $\mathbf{B} \in M_{k \times l}(F)$ . Such a matrix is **block upper triangular**; with the zero block above the diagonal instead, **block lower triangular**. Chapter 6 computed  $\det \mathbf{M} = \det \mathbf{A} \det \mathbf{D}$  (Theorem 6.8.4). Products and inverses keep the shape.

### Theorem 7.1.5 — Block Triangular Products and Inverses

Let  $k, l \geq 1$ , and let  $\mathbf{M} = \begin{pmatrix} \mathbf{A} & \mathbf{B} \\ 0 & \mathbf{D} \end{pmatrix}$  and  $\mathbf{M}' = \begin{pmatrix} \mathbf{A}' & \mathbf{B}' \\ 0 & \mathbf{D}' \end{pmatrix}$ , with  $\mathbf{A}, \mathbf{A}' \in M_k(F)$ ,  $\mathbf{D}, \mathbf{D}' \in M_l(F)$  and  $\mathbf{B}, \mathbf{B}' \in M_{k \times l}(F)$ .

- (a)  $\mathbf{M}\mathbf{M}' = \begin{pmatrix} \mathbf{A}\mathbf{A}' & \mathbf{A}\mathbf{B}' + \mathbf{B}\mathbf{D}' \\ 0 & \mathbf{D}\mathbf{D}' \end{pmatrix}$ ; in particular it is block upper triangular.
- (b)  $\mathbf{M}$  is invertible if and only if  $\mathbf{A}$  and  $\mathbf{D}$  are both invertible.
- (c) In that case

$$\mathbf{M}^{-1} = \begin{pmatrix} \mathbf{A}^{-1} & -\mathbf{A}^{-1}\mathbf{B}\mathbf{D}^{-1} \\ 0 & \mathbf{D}^{-1} \end{pmatrix}.$$

*Idea.* Where does the formula come from? Solve  $\mathbf{M} \begin{pmatrix} \mathbf{x} \\ \mathbf{y} \end{pmatrix} = \begin{pmatrix} \mathbf{f} \\ \mathbf{g} \end{pmatrix}$  by back substitution with blocks. The second block row says  $\mathbf{D}\mathbf{y} = \mathbf{g}$ , so  $\mathbf{y} = \mathbf{D}^{-1}\mathbf{g}$ . The first says  $\mathbf{A}\mathbf{x} + \mathbf{B}\mathbf{y} = \mathbf{f}$ , so  $\mathbf{x} = \mathbf{A}^{-1}\mathbf{f} - \mathbf{A}^{-1}\mathbf{B}\mathbf{D}^{-1}\mathbf{g}$ . Reading off the coefficients of  $\mathbf{f}$  and  $\mathbf{g}$  gives the four blocks of  $\mathbf{M}^{-1}$ . The proof only has to check the answer.

*Proof.*

- (a) By Theorem 7.1.1, the blocks of  $\mathbf{M}\mathbf{M}'$  are  $\mathbf{A}\mathbf{A}' + \mathbf{B}0 = \mathbf{A}\mathbf{A}'$ ,  $\mathbf{A}\mathbf{B}' + \mathbf{B}\mathbf{D}'$ ,  $0\mathbf{A}' + \mathbf{D}0 = 0$  and  $0\mathbf{B}' + \mathbf{D}\mathbf{D}' = \mathbf{D}\mathbf{D}'$ .
- (b) By Theorem 6.8.4,  $\det \mathbf{M} = \det \mathbf{A} \det \mathbf{D}$ . By Theorem 6.5.2,  $\mathbf{M}$  is invertible if and only if  $\det \mathbf{A} \det \mathbf{D} \neq 0$ , which in a field happens exactly when  $\det \mathbf{A} \neq 0$  and  $\det \mathbf{D} \neq 0$  (Theorem 0.7.1 (f) and (d)), that is, when  $\mathbf{A}$  and  $\mathbf{D}$  are both invertible.
- (c) Let  $\mathbf{X}$  be the displayed matrix. By (a) with  $\mathbf{A}' = \mathbf{A}^{-1}$ ,  $\mathbf{B}' = -\mathbf{A}^{-1}\mathbf{B}\mathbf{D}^{-1}$  and  $\mathbf{D}' = \mathbf{D}^{-1}$ ,

$$\mathbf{M}\mathbf{X} = \begin{pmatrix} \mathbf{A}\mathbf{A}^{-1} & -\mathbf{A}\mathbf{A}^{-1}\mathbf{B}\mathbf{D}^{-1} + \mathbf{B}\mathbf{D}^{-1} \\ 0 & \mathbf{D}\mathbf{D}^{-1} \end{pmatrix} = \begin{pmatrix} \mathbf{I}_k & 0 \\ 0 & \mathbf{I}_l \end{pmatrix} = \mathbf{I}_{k+l}.$$

By Theorem 2.4.5,  $\mathbf{X} = \mathbf{M}^{-1}$ .

■

Transposing with [Proposition 7.1.2](#) turns this into the lower triangular version: if  $\mathbf{A}$  and  $\mathbf{D}$  are invertible, then  $\begin{pmatrix} \mathbf{A} & 0 \\ \mathbf{C} & \mathbf{D} \end{pmatrix}^{-1} = \begin{pmatrix} \mathbf{A}^{-1} & 0 \\ -\mathbf{D}^{-1}\mathbf{C}\mathbf{A}^{-1} & \mathbf{D}^{-1} \end{pmatrix}$ . A special case is used constantly:  $\begin{pmatrix} \mathbf{I}_k & \mathbf{X} \\ 0 & \mathbf{I}_l \end{pmatrix}^{-1} = \begin{pmatrix} \mathbf{I}_k & -\mathbf{X} \\ 0 & \mathbf{I}_l \end{pmatrix}$  for every  $\mathbf{X} \in M_{k \times l}(F)$ . Adding  $\mathbf{X}$  times the second block of coordinates to the first is undone by subtracting it.

*Quick check.* The matrix  $\mathbf{M} = \begin{pmatrix} 1 & 0 & 0 \\ 0 & 0 & 1 \\ 0 & 1 & 0 \end{pmatrix}$ , with row partition 2, 1 and column partition 1, 2, has a zero lower-left block and is invertible. Does this contradict [Theorem 7.1.5](#)?

*Solution.* No. The theorem needs the diagonal blocks square, which means equal row and column partitions. Here the top-left block  $\begin{pmatrix} 1 \\ 0 \end{pmatrix}$  is  $2 \times 1$  and the bottom-right block  $\begin{pmatrix} 1 & 0 \end{pmatrix}$  is  $1 \times 2$ , so “ $\mathbf{A}$  is invertible” does not even make sense. (The lower-left block is the  $1 \times 1$  entry in position  $(3, 1)$ , which is 0.)

## 7.1.6 Matrices of operators in adapted bases

Now the shapes pay off. For an operator  $T \in \mathcal{L}(V)$  and a subspace  $U \subseteq V$ , we say that  $T$  **maps  $U$  into  $U$**  if  $T\mathbf{u} \in U$  for every  $\mathbf{u} \in U$ . Then the **restriction**  $T|_U: U \rightarrow U$ ,  $\mathbf{u} \mapsto T\mathbf{u}$ , is an operator on  $U$ , linear because its values are those of  $T$ . Such subspaces are the subject of Chapter 8; here we only ask what they do to the matrix of  $T$ .

When  $V = U \oplus W$  and a basis of  $U$  is followed by a basis of  $W$ , the combined list is a basis of  $V$  ([Theorem 1.7.5](#)), which we call **adapted** to the decomposition. If  $T$  maps both pieces into themselves, the matrix splits.

### Theorem 7.1.6 — Direct Sums Give Block Diagonal Matrices

Let  $V = U \oplus W$  be finite-dimensional with  $U \neq \{0\} \neq W$ , let  $\mathcal{B}_U = (\mathbf{u}_1, \dots, \mathbf{u}_k)$  and  $\mathcal{B}_W = (\mathbf{w}_1, \dots, \mathbf{w}_l)$  be bases of  $U$  and  $W$ , and let  $\mathcal{B} = (\mathbf{u}_1, \dots, \mathbf{u}_k, \mathbf{w}_1, \dots, \mathbf{w}_l)$ . For  $T \in \mathcal{L}(V)$ , the following are equivalent:

- (a)  $T$  maps  $U$  into  $U$  and  $W$  into  $W$ ;
- (b)  $[T]_{\mathcal{B}}$  is block diagonal with respect to the partition  $k, l$ .

In that case  $[T]_{\mathcal{B}} = [T|_U]_{\mathcal{B}_U} \oplus [T|_W]_{\mathcal{B}_W}$ .

*Proof.* By [Theorem 1.7.5](#),  $\mathcal{B}$  is a basis of  $V$ . By [Definition 3.6.1](#), column  $j \leq k$  of  $[T]_{\mathcal{B}}$  holds the coordinates of  $T\mathbf{u}_j$ , and column  $k + q$  those of  $T\mathbf{w}_q$ .

(a)  $\Rightarrow$  (b). Since  $T\mathbf{u}_j \in U = \text{span}(\mathcal{B}_U)$ , write  $T\mathbf{u}_j = \sum_i a_{ij}\mathbf{u}_i$ . By uniqueness of coordinates ([Theorem 1.5.1](#)), the last  $l$  entries of column  $j$  are 0 and the first  $k$  are  $a_{1j}, \dots, a_{kj}$ , which is column  $j$  of  $[T|_U]_{\mathcal{B}_U}$ . In the same way  $T\mathbf{w}_q = \sum_p d_{pq}\mathbf{w}_p$ , so column  $k + q$  is 0 in its first  $k$  entries and equals column  $q$  of  $[T|_W]_{\mathcal{B}_W}$  in its last  $l$ . This shows  $[T]_{\mathcal{B}} = [T|_U]_{\mathcal{B}_U} \oplus [T|_W]_{\mathcal{B}_W}$ .

(b)  $\Rightarrow$  (a). If the  $(2, 1)$  block is zero, then the coordinates of each  $T\mathbf{u}_j$  vanish in the

last  $l$  places, so  $T\mathbf{u}_j \in \text{span}(\mathcal{B}_U) = U$ . For  $\mathbf{u} = \sum_j c_j \mathbf{u}_j \in U$ , linearity gives  $T\mathbf{u} = \sum_j c_j T\mathbf{u}_j \in U$ , since  $U$  is a subspace. The  $(1, 2)$  block being zero gives  $T\mathbf{w}_q \in W$  in the same way, and hence  $T$  maps  $W$  into  $W$ . ■

For example, let  $T = \frac{d^2}{dx^2}$  on  $\mathbb{R}[x]_{\leq 3} = U \oplus W$  with  $U = \text{span}(1, x^2)$  (even polynomials) and  $W = \text{span}(x, x^3)$  (odd polynomials). Since  $T(1) = 0$ ,  $T(x^2) = 2$ ,  $T(x) = 0$  and  $T(x^3) = 6x$ ,  $T$  maps each piece into itself, and in the basis  $\mathcal{B} = (1, x^2, x, x^3)$

$$[T]_{\mathcal{B}} = \begin{pmatrix} 0 & 2 \\ 0 & 0 \end{pmatrix} \oplus \begin{pmatrix} 0 & 6 \\ 0 & 0 \end{pmatrix}.$$

In the standard order  $(1, x, x^2, x^3)$  the same operator has the matrix with 2 in position  $(1, 3)$  and 6 in position  $(2, 4)$ , which is not block diagonal for the partition 2, 2. **Ordering the basis to match the decomposition is part of the method.**

If  $T$  maps only one subspace  $U$  into itself, we can still take a basis of  $U$  first, but the remaining basis vectors need not be sent anywhere special. What survives is a zero block in the lower left, and the lower-right block turns out to describe the operator that  $T$  induces on the quotient  $V/U$ . This was [Exercise \(C2: The operator induced on a quotient\)](#) in Chapter 3; we now record it as a theorem, with its converse.

### Theorem 7.1.7 — Subspaces Mapped into Themselves Give Block Triangular Matrices

Let  $V$  be finite-dimensional,  $T \in \mathcal{L}(V)$ , and  $U$  a subspace with  $\{0\} \neq U \neq V$ . Let  $\mathcal{B}_U = (\mathbf{u}_1, \dots, \mathbf{u}_k)$  be a basis of  $U$ , extended to a basis  $\mathcal{B} = (\mathbf{u}_1, \dots, \mathbf{u}_k, \mathbf{w}_1, \dots, \mathbf{w}_l)$  of  $V$ .

- (a)  $T$  maps  $U$  into  $U$  if and only if the lower-left  $l \times k$  block of  $[T]_{\mathcal{B}}$  is zero.
- (b) In that case the rule  $\bar{T}(\mathbf{v} + U) = T\mathbf{v} + U$  defines an operator  $\bar{T}$  on  $V/U$ , and

$$[T]_{\mathcal{B}} = \begin{pmatrix} [T|_U]_{\mathcal{B}_U} & \mathbf{B} \\ 0 & [\bar{T}]_{\bar{\mathcal{B}}} \end{pmatrix}$$

for some  $\mathbf{B} \in M_{k \times l}(F)$ , where  $\bar{\mathcal{B}} = (\mathbf{w}_1 + U, \dots, \mathbf{w}_l + U)$  is a basis of  $V/U$ .

*Idea.* Read the matrix one column at a time. The first  $k$  columns describe  $T\mathbf{u}_j$ , which lie in  $U$  exactly when their  $\mathbf{w}$ -coordinates vanish. The last  $l$  columns describe  $T\mathbf{w}_q$ , which may have any  $\mathbf{u}$ -part; passing to  $V/U$  erases that part and leaves precisely the  $\mathbf{w}$ -coordinates, which are the lower-right block.

*Proof.*

- (a) Column  $j \leq k$  of  $[T]_{\mathcal{B}}$  is the coordinate vector of  $T\mathbf{u}_j$  ([Definition 3.6.1](#)), and by [Theorem 1.5.1](#) its last  $l$  entries vanish if and only if  $T\mathbf{u}_j \in \text{span}(\mathbf{u}_1, \dots, \mathbf{u}_k) = U$ . If  $T$  maps  $U$  into  $U$ , this holds for every  $j$ . Conversely, if it holds for every  $j$ , then for  $\mathbf{u} = \sum_j c_j \mathbf{u}_j \in U$  we get  $T\mathbf{u} = \sum_j c_j T\mathbf{u}_j \in U$ , because  $U$  is a subspace.
- (b) Let  $\pi: V \rightarrow V/U$  be the quotient map, linear with kernel  $U$  ([Theorem 3.9.4](#)). The composite  $\pi T$  is linear ([Theorem 3.4.2](#)), and for  $\mathbf{u} \in U$ ,  $\pi(T\mathbf{u}) = \mathbf{0}$  because

$T\mathbf{u} \in U$ . So  $U \subseteq \ker(\pi T)$ , and by [Theorem 3.9.10](#) (a) there is exactly one linear  $\bar{T}: V/U \rightarrow V/U$  with  $\bar{T}(\mathbf{v} + U) = T\mathbf{v} + U$ . By [Theorem 3.9.5](#),  $\bar{\mathcal{B}}$  is a basis of  $V/U$ .

By (a), the lower-left block is zero, and as in the proof of [Theorem 7.1.6](#), the upper-left block is  $[T|_U]_{\mathcal{B}_U}$ . For  $1 \leq q \leq l$ , write  $T\mathbf{w}_q = \sum_{i=1}^k b_{iq}\mathbf{u}_i + \sum_{p=1}^l d_{pq}\mathbf{w}_p$ , so that column  $k+q$  of  $[T]_{\mathcal{B}}$  is  $(b_{1q}, \dots, b_{kq}, d_{1q}, \dots, d_{lq})$ . Applying the linear map  $\pi$ , and using  $\pi(\mathbf{u}_i) = \mathbf{0}$ ,

$$\bar{T}(\mathbf{w}_q + U) = \pi(T\mathbf{w}_q) = \sum_{p=1}^l d_{pq}(\mathbf{w}_p + U).$$

By [Definition 3.6.1](#), column  $q$  of  $[\bar{T}]_{\bar{\mathcal{B}}}$  is  $(d_{1q}, \dots, d_{lq})$ , which is the lower-right part of column  $k+q$  of  $[T]_{\mathcal{B}}$ . This proves (b), with  $\mathbf{B} = (b_{iq})$ . ■

The upper-left block depends only on the basis  $\mathcal{B}_U$ , and the lower-right block only on the cosets  $\mathbf{w}_1 + U, \dots, \mathbf{w}_l + U$ . So replacing each  $\mathbf{w}_q$  by  $\mathbf{w}_q + \mathbf{u}'_q$  with  $\mathbf{u}'_q \in U$  can change the top-right block  $\mathbf{B}$ , but leaves both diagonal blocks unchanged. The theorem splits  $T$  into two smaller operators, and every quantity that sees only diagonal blocks splits with it. By [Theorem 6.8.4](#),  $\det T = \det(T|_U) \det \bar{T}$ . Applied over  $F[x]$  (Chapter 6 §8 notes that the block triangular determinant holds over any commutative ring),  $x\mathbf{I} - [T]_{\mathcal{B}} = \begin{pmatrix} x\mathbf{I}_k - [T|_U]_{\mathcal{B}_U} & -\mathbf{B} \\ 0 & x\mathbf{I}_l - [\bar{T}]_{\bar{\mathcal{B}}} \end{pmatrix}$  gives  $p_T = p_{T|_U} p_{\bar{T}}$ . Chapter 8 builds on exactly this factorization.

### Example. (A Plane Mapped into Itself)

Let  $T: \mathbb{R}^3 \rightarrow \mathbb{R}^3$ ,  $T(x, y, z) = (y, z, x)$ , and  $U = \{(x, y, z) : x + y + z = 0\}$ . Show that  $T$  maps  $U$  into  $U$ , find  $[T]_{\mathcal{B}}$  for the basis  $\mathcal{B} = (\mathbf{u}_1, \mathbf{u}_2, \mathbf{e}_3)$  with  $\mathbf{u}_1 = (1, -1, 0)$ ,  $\mathbf{u}_2 = (0, 1, -1)$ , and interpret the lower-right block.

*Solution.* If  $x + y + z = 0$ , then the entries of  $T(x, y, z) = (y, z, x)$  also sum to 0, so  $T$  maps  $U$  into  $U$ . The vectors  $\mathbf{u}_1, \mathbf{u}_2$  lie in  $U$  and are independent (look at the first entries, then the last), and  $\dim U = 2$  since  $U$  is the kernel of the non-zero functional  $(x, y, z) \mapsto x + y + z$  ([Theorem 3.3.1](#)). So  $(\mathbf{u}_1, \mathbf{u}_2)$  is a basis of  $U$  ([Theorem 1.6.4](#)), and  $\mathcal{B}$  is a basis of  $\mathbb{R}^3$ : in a relation  $a\mathbf{u}_1 + b\mathbf{u}_2 + c\mathbf{e}_3 = \mathbf{0}$ , the entries sum to  $c = 0$ , and then  $a = b = 0$ , so  $\mathcal{B}$  is an independent list of 3 vectors ([Theorem 1.6.4](#)). Now

$$\begin{aligned} T\mathbf{u}_1 &= (-1, 0, 1) = -\mathbf{u}_1 - \mathbf{u}_2, \\ T\mathbf{u}_2 &= (1, -1, 0) = \mathbf{u}_1, \\ T\mathbf{e}_3 &= (0, 1, 0) = \mathbf{u}_2 + \mathbf{e}_3, \end{aligned}$$

so

$$[T]_{\mathcal{B}} = \left( \begin{array}{cc|c} -1 & 1 & 0 \\ -1 & 0 & 1 \\ 0 & 0 & 1 \end{array} \right).$$

The lower-left block is zero, as [Theorem 7.1.7](#) predicts. The lower-right block (1) says  $\bar{T}(\mathbf{e}_3 + U) = \mathbf{e}_3 + U$ , so  $\bar{T}$  is the identity on the line  $\mathbb{R}^3/U$ . Indeed  $T\mathbf{v} - \mathbf{v} = (y - x, z - y, x - z)$  has entries summing to 0, so  $T\mathbf{v} + U = \mathbf{v} + U$  for every  $\mathbf{v}$ :  $T$  moves vectors, but never changes the sum of their coordinates, and that sum is all that  $\mathbb{R}^3/U$  remembers.



## 7.1.7 Exercises

### 7.1.7.1 A. Check your understanding

#### Exercise A1.

- Define what it means for partitions of  $\mathbf{A} \in M_{m \times n}(F)$  and  $\mathbf{B} \in M_{n \times p}(F)$  to be conformable.
- State the block multiplication formula for  $\begin{pmatrix} \mathbf{A} & \mathbf{B} \\ \mathbf{C} & \mathbf{D} \end{pmatrix} \begin{pmatrix} \mathbf{A}' & \mathbf{B}' \\ \mathbf{C}' & \mathbf{D}' \end{pmatrix}$ .
- True or false: the top-left block of that product is  $\mathbf{A}'\mathbf{A} + \mathbf{C}'\mathbf{B}$ . Justify your answer.
- True or false: if  $\mathbf{A} \in M_k(F)$  and  $\mathbf{B} \in M_l(F)$ , then  $\mathbf{A} \oplus \mathbf{B} = \mathbf{B} \oplus \mathbf{A}$ . Justify your answer.
- True or false: if  $\mathbf{M} = \begin{pmatrix} \mathbf{A} & \mathbf{B} \\ 0 & \mathbf{D} \end{pmatrix}$  with  $\mathbf{A}$ ,  $\mathbf{D}$  square is invertible, then  $\mathbf{M}^{-1}$  is block upper triangular. Justify your answer.
- In a basis whose first  $k$  vectors form a basis of  $U$ , what does a zero lower-left block of  $[T]_{\mathcal{B}}$  say about  $T$ ?

*Solution.*

- The column partition of  $\mathbf{A}$  and the row partition of  $\mathbf{B}$  are the same list  $n_1, \dots, n_b$ , in the same order (Definition 7.1.1).
- It is  $\begin{pmatrix} \mathbf{AA}' + \mathbf{BC}' & \mathbf{AB}' + \mathbf{BD}' \\ \mathbf{CA}' + \mathbf{DC}' & \mathbf{CB}' + \mathbf{DD}' \end{pmatrix}$ , provided the partitions are conformable (Theorem 7.1.1).
- False in general. The block is  $\mathbf{AA}' + \mathbf{BC}'$ , with each block of the left factor on the left. For  $4 \times 4$  matrices with partition 2, 2, take  $\mathbf{A} = \mathbf{E}_{12}$ ,  $\mathbf{A}' = \mathbf{E}_{21}$  in  $M_2(F)$  and all other blocks zero. Then the top-left block of the product is  $\mathbf{AA}' = \mathbf{E}_{11}$ , while  $\mathbf{A}'\mathbf{A} + \mathbf{C}'\mathbf{B} = \mathbf{E}_{22}$ .
- False. For  $k = l = 1$ ,  $(1) \oplus (2) = \text{diag}(1, 2) \neq \text{diag}(2, 1) = (2) \oplus (1)$ .
- True. By Theorem 7.1.5 (b),  $\mathbf{A}$  and  $\mathbf{D}$  are invertible, and (c) gives  $\mathbf{M}^{-1}$  with lower-left block 0.
- That  $T$  maps  $U$  into  $U$  (Theorem 7.1.7 (a)).

### 7.1.7.2 B. Practice

#### Exercise B1: Multiplying block matrices.

Let

$$\mathbf{M} = \left( \begin{array}{cc|c} 1 & 2 & 1 \\ 0 & 1 & -1 \\ \hline 2 & 0 & 3 \end{array} \right), \quad \mathbf{N} = \left( \begin{array}{cc|c} 1 & 0 & 0 \\ 1 & 1 & 2 \\ \hline 1 & -1 & 1 \end{array} \right) \in M_3(\mathbb{Q}),$$

both with row and column partition 2, 1.

- Compute  $\mathbf{MN}$  blockwise, and check one block against the entry formula.

- (b) Explain why the partitions would **not** be conformable if  $\mathbf{N}$  were given the row partition 1, 2 instead.

*Solution.*

- (a) Write  $\mathbf{M} = \begin{pmatrix} \mathbf{A} & \mathbf{b} \\ \mathbf{c} & d \end{pmatrix}$  and  $\mathbf{N} = \begin{pmatrix} \mathbf{E} & \mathbf{f} \\ \mathbf{g} & h \end{pmatrix}$  with  $\mathbf{A} = \begin{pmatrix} 1 & 2 \\ 0 & 1 \end{pmatrix}$ ,  $\mathbf{b} = \begin{pmatrix} 1 \\ -1 \end{pmatrix}$ ,  $\mathbf{c} = (2 \ 0)$ ,  $d = (3)$ ,  $\mathbf{E} = \begin{pmatrix} 1 & 0 \\ 1 & 1 \end{pmatrix}$ ,  $\mathbf{f} = \begin{pmatrix} 0 \\ 2 \end{pmatrix}$ ,  $\mathbf{g} = (1 \ -1)$ ,  $h = (1)$ . The partitions are conformable (both middle partitions are 2, 1), so by [Theorem 7.1.1](#)

$$\mathbf{AE} + \mathbf{bg} = \begin{pmatrix} 3 & 2 \\ 1 & 1 \end{pmatrix} + \begin{pmatrix} 1 & -1 \\ -1 & 1 \end{pmatrix} = \begin{pmatrix} 4 & 1 \\ 0 & 2 \end{pmatrix},$$

$$\mathbf{Af} + \mathbf{bh} = \begin{pmatrix} 4 \\ 2 \end{pmatrix} + \begin{pmatrix} 1 \\ -1 \end{pmatrix} = \begin{pmatrix} 5 \\ 1 \end{pmatrix},$$

$$\mathbf{cE} + d\mathbf{g} = (2 \ 0) + (3 \ -3) = (5 \ -3), \quad \mathbf{cf} + dh = (0) + (3) = (3).$$

Hence

$$\mathbf{MN} = \begin{pmatrix} 4 & 1 & 5 \\ 0 & 2 & 1 \\ 5 & -3 & 3 \end{pmatrix}.$$

Check the (3, 3)-entry directly: row 3 of  $\mathbf{M}$  is (2, 0, 3) and column 3 of  $\mathbf{N}$  is (0, 2, 1), giving  $0 + 0 + 3 = 3$ .

- (b) The column partition of  $\mathbf{M}$  is 2, 1, and the row partition 1, 2 is a different list. Concretely, the top-left block  $\mathbf{A}$  of  $\mathbf{M}$  is  $2 \times 2$ , while the top-left block of  $\mathbf{N}$  would be  $1 \times 2$ , and a  $2 \times 2$  matrix cannot multiply a  $1 \times 2$  matrix.

### Exercise B2: Inverting a block triangular matrix.

Use [Theorem 7.1.5](#) to find the inverse of

$$\mathbf{M} = \begin{pmatrix} 2 & 1 & 1 & 0 \\ 1 & 1 & 2 & 1 \\ 0 & 0 & 1 & 2 \\ 0 & 0 & 0 & 1 \end{pmatrix} \in M_4(\mathbb{Q}),$$

and check one block of  $\mathbf{MM}^{-1}$ .

*Solution.* With the partition 2, 2,  $\mathbf{M} = \begin{pmatrix} \mathbf{A} & \mathbf{B} \\ \mathbf{0} & \mathbf{D} \end{pmatrix}$  with  $\mathbf{A} = \begin{pmatrix} 2 & 1 \\ 1 & 1 \end{pmatrix}$ ,  $\mathbf{B} = \begin{pmatrix} 1 & 0 \\ 2 & 1 \end{pmatrix}$ ,  $\mathbf{D} = \begin{pmatrix} 1 & 2 \\ 0 & 1 \end{pmatrix}$ . Both  $\det \mathbf{A} = 1$  and  $\det \mathbf{D} = 1$  are non-zero, so by [Theorem 0.9.8](#)

$$\mathbf{A}^{-1} = \begin{pmatrix} 1 & -1 \\ -1 & 2 \end{pmatrix}, \quad \mathbf{D}^{-1} = \begin{pmatrix} 1 & -2 \\ 0 & 1 \end{pmatrix}.$$

Then  $\mathbf{A}^{-1}\mathbf{B} = \begin{pmatrix} -1 & -1 \\ 3 & 2 \end{pmatrix}$  and  $-\mathbf{A}^{-1}\mathbf{B}\mathbf{D}^{-1} = -\begin{pmatrix} -1 & 1 \\ 3 & -4 \end{pmatrix} = \begin{pmatrix} 1 & -1 \\ -3 & 4 \end{pmatrix}$ . By [Theo-](#)

rem 7.1.5 (c),

$$\mathbf{M}^{-1} = \begin{pmatrix} 1 & -1 & 1 & -1 \\ -1 & 2 & -3 & 4 \\ 0 & 0 & 1 & -2 \\ 0 & 0 & 0 & 1 \end{pmatrix}.$$

Check the top-right block of  $\mathbf{M}\mathbf{M}^{-1}$ :  $\mathbf{A}(-\mathbf{A}^{-1}\mathbf{B}\mathbf{D}^{-1}) + \mathbf{B}\mathbf{D}^{-1} = \begin{pmatrix} 2 & 1 \\ 1 & 1 \end{pmatrix} \begin{pmatrix} 1 & -1 \\ -3 & 4 \end{pmatrix} + \begin{pmatrix} 1 & 0 \\ 2 & 1 \end{pmatrix} \begin{pmatrix} 1 & -2 \\ 0 & 1 \end{pmatrix} = \begin{pmatrix} -1 & 2 \\ -2 & 3 \end{pmatrix} + \begin{pmatrix} 1 & -2 \\ 2 & -3 \end{pmatrix} = \mathbf{0}$ , as it must be.

### Exercise B3: A projection in an adapted basis.

Let  $\mathbf{P} = \begin{pmatrix} 0 & 1 & -1 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{pmatrix} \in M_3(\mathbb{R})$ ,  $U = \text{span}((1, 1, 0), (0, 1, 1))$  and  $W = \text{span}(\mathbf{e}_1)$ .

- Show that  $\mathbf{P}^2 = \mathbf{P}$ , that  $\mathbf{P}\mathbf{u} = \mathbf{u}$  for  $\mathbf{u} \in U$  and  $\mathbf{P}\mathbf{e}_1 = \mathbf{0}$ , and that  $\mathbb{R}^3 = U \oplus W$ .
- Let  $\mathcal{B} = ((1, 1, 0), (0, 1, 1), \mathbf{e}_1)$ . Use Theorem 7.1.6 to write down  $[T_{\mathbf{P}}]_{\mathcal{B}}$  without computing, and confirm it as  $\mathbf{S}^{-1}\mathbf{P}\mathbf{S}$ , where  $\mathbf{S}$  has the vectors of  $\mathcal{B}$  as columns.
- Hence find  $\mathbf{P}^{100}$  and  $\text{tr } \mathbf{P}$ .

*Solution.*

- Multiplying,  $\mathbf{P}^2$  has rows  $(0, 1, -1)$ ,  $(0, 1, 0)$ ,  $(0, 0, 1)$ , so  $\mathbf{P}^2 = \mathbf{P}$ . Also  $\mathbf{P}(1, 1, 0) = (1, 1, 0)$ ,  $\mathbf{P}(0, 1, 1) = (0, 1, 1)$  and  $\mathbf{P}\mathbf{e}_1 = \mathbf{0}$  (the first column is zero), so  $\mathbf{P}$  fixes every vector of  $U$  by linearity. The three vectors of  $\mathcal{B}$  are the columns of  $\mathbf{S} = \begin{pmatrix} 1 & 0 & 1 \\ 1 & 1 & 0 \\ 0 & 1 & 0 \end{pmatrix}$ , and expanding along the last column,  $\det \mathbf{S} = 1 \cdot (1 - 0) = 1 \neq 0$ . So  $\mathbf{S}$  is invertible (Theorem 6.5.2), its columns form a basis of  $\mathbb{R}^3$  (Theorem 2.4.4), and  $\mathbb{R}^3 = U \oplus W$  by Theorem 1.7.5.
- $T_{\mathbf{P}}$  maps  $U$  into  $U$  (as the identity) and  $W$  into  $W$  (as zero). By Theorem 7.1.6,  $[T_{\mathbf{P}}]_{\mathcal{B}} = \mathbf{I}_2 \oplus (0) = \text{diag}(1, 1, 0)$ . By Theorem 3.7.3,  $[T_{\mathbf{P}}]_{\mathcal{B}} = \mathbf{S}^{-1}\mathbf{P}\mathbf{S}$ . Here  $\mathbf{S}^{-1} = \begin{pmatrix} 0 & 1 & -1 \\ 0 & 0 & 1 \\ 1 & -1 & 1 \end{pmatrix}$  (check  $\mathbf{S}\mathbf{S}^{-1} = \mathbf{I}_3$ ), and  $\mathbf{P}\mathbf{S}$  has columns  $\mathbf{P}(1, 1, 0) = (1, 1, 0)$ ,  $\mathbf{P}(0, 1, 1) = (0, 1, 1)$ ,  $\mathbf{P}\mathbf{e}_1 = \mathbf{0}$ , so  $\mathbf{S}^{-1}\mathbf{P}\mathbf{S} = \mathbf{S}^{-1} \begin{pmatrix} 1 & 0 & 0 \\ 1 & 1 & 0 \\ 0 & 1 & 0 \end{pmatrix} = \text{diag}(1, 1, 0)$ .
- $\mathbf{P} = \mathbf{S} \text{diag}(1, 1, 0) \mathbf{S}^{-1}$ , so  $\mathbf{P}^{100} = \mathbf{S} \text{diag}(1, 1, 0)^{100} \mathbf{S}^{-1} = \mathbf{S} \text{diag}(1, 1, 0) \mathbf{S}^{-1} = \mathbf{P}$  by Theorem 7.1.4 (b) (or directly from  $\mathbf{P}^2 = \mathbf{P}$ ). By Theorem 3.7.5,  $\text{tr } \mathbf{P} = \text{tr } \text{diag}(1, 1, 0) = 2$ , which matches  $0 + 1 + 1$ .

### 7.1.7.3 C. Going deeper

#### Exercise C1: Square-zero matrices of half rank.

Let  $n \geq 1$ .

- (a) For  $\mathbf{X} \in M_n(F)$ , let  $\mathbf{N}_{\mathbf{X}} = \begin{pmatrix} 0 & \mathbf{X} \\ 0 & 0 \end{pmatrix} \in M_{2n}(F)$ . Show that  $\mathbf{N}_{\mathbf{X}}^2 = 0$  and that  $\text{rank } \mathbf{N}_{\mathbf{X}} = \text{rank } \mathbf{X}$ .
- (b) Let  $V$  have dimension  $2n$ , and let  $T \in \mathcal{L}(V)$  satisfy  $T^2 = 0$  and  $\text{rank } T = n$ . Show that  $\text{im } T = \ker T$ . Choose  $\mathbf{w}_1, \dots, \mathbf{w}_n$  with  $(T\mathbf{w}_1, \dots, T\mathbf{w}_n)$  a basis of  $\text{im } T$ , and put  $\mathbf{u}_i = T\mathbf{w}_i$ . Prove that  $\mathcal{B} = (\mathbf{u}_1, \dots, \mathbf{u}_n, \mathbf{w}_1, \dots, \mathbf{w}_n)$  is a basis of  $V$  and that  $[T]_{\mathcal{B}} = \mathbf{N}_{\mathbf{I}_n}$ .
- (c) Deduce that for every invertible  $\mathbf{X} \in M_n(F)$ , the matrix  $\mathbf{N}_{\mathbf{X}}$  is similar to  $\mathbf{N}_{\mathbf{I}_n}$ .

*Hint: for the independence in (b), apply  $T$ .*

*Solution.*

- (a) By [Theorem 7.1.1](#),  $\mathbf{N}_{\mathbf{X}}^2 = \begin{pmatrix} 0 \cdot 0 + \mathbf{X} \cdot 0 & 0 \cdot \mathbf{X} + \mathbf{X} \cdot 0 \\ 0 & 0 \end{pmatrix} = 0$ . The first  $n$  columns of  $\mathbf{N}_{\mathbf{X}}$  are zero, and column  $n + j$  is  $\begin{pmatrix} \mathbf{x}_j \\ 0 \end{pmatrix}$ , where  $\mathbf{x}_j$  is column  $j$  of  $\mathbf{X}$ . A combination  $\sum_j c_j \begin{pmatrix} \mathbf{x}_j \\ 0 \end{pmatrix}$  is zero exactly when  $\sum_j c_j \mathbf{x}_j = \mathbf{0}$ , so a list of these columns is independent exactly when the corresponding columns of  $\mathbf{X}$  are. Hence  $\dim \text{col}(\mathbf{N}_{\mathbf{X}}) = \dim \text{col}(\mathbf{X})$ , that is,  $\text{rank } \mathbf{N}_{\mathbf{X}} = \text{rank } \mathbf{X}$ .
- (b) If  $\mathbf{v} = T\mathbf{x} \in \text{im } T$ , then  $T\mathbf{v} = T^2\mathbf{x} = \mathbf{0}$ , so  $\text{im } T \subseteq \ker T$ . By [Theorem 3.3.1](#),  $\dim \ker T = 2n - n = n = \dim \text{im } T$ , so  $\text{im } T = \ker T$  by [Theorem 1.6.8](#). The vectors  $\mathbf{w}_i$  exist because  $(T\mathbf{w}_1, \dots, T\mathbf{w}_n)$  can be any basis of  $\text{im } T$ , each of whose vectors is  $T$  of something. Let  $\sum_i a_i \mathbf{u}_i + \sum_i b_i \mathbf{w}_i = \mathbf{0}$ . Applying  $T$ , and using  $T\mathbf{u}_i = T^2\mathbf{w}_i = \mathbf{0}$ , gives  $\sum_i b_i T\mathbf{w}_i = \mathbf{0}$ . The  $T\mathbf{w}_i$  form a basis of  $\text{im } T$ , so all  $b_i = 0$ , and then  $\sum_i a_i \mathbf{u}_i = \mathbf{0}$  gives all  $a_i = 0$ . So  $\mathcal{B}$  is a linearly independent list of  $2n = \dim V$  vectors, hence a basis ([Theorem 1.6.4](#)). Since  $T\mathbf{u}_i = \mathbf{0}$  and  $T\mathbf{w}_i = \mathbf{u}_i$ , column  $i \leq n$  of  $[T]_{\mathcal{B}}$  is zero and column  $n + i$  is  $\mathbf{e}_i$ . So  $[T]_{\mathcal{B}} = \begin{pmatrix} 0 & \mathbf{I}_n \\ 0 & 0 \end{pmatrix} = \mathbf{N}_{\mathbf{I}_n}$ .
- (c) If  $\mathbf{X}$  is invertible, then by (a) the operator  $T = T_{\mathbf{N}_{\mathbf{X}}}$  on  $F^{2n}$  satisfies  $T^2 = 0$  and  $\text{rank } T = \text{rank } \mathbf{N}_{\mathbf{X}} = \text{rank } \mathbf{X} = n$  ([Theorem 3.6.5](#), and [Theorem 2.4.4](#) for  $\text{rank } \mathbf{X} = n$ ). By (b) there is a basis  $\mathcal{B}$  with  $[T]_{\mathcal{B}} = \mathbf{N}_{\mathbf{I}_n}$ . The matrix of  $T$  in the standard basis is  $\mathbf{N}_{\mathbf{X}}$ , so  $\mathbf{N}_{\mathbf{X}}$  and  $\mathbf{N}_{\mathbf{I}_n}$  are similar by [Theorem 3.7.4](#).

### Exercise C2: Powers of block triangular matrices.

Let  $\mathbf{A} \in M_k(F)$ ,  $\mathbf{D} \in M_l(F)$  and  $\mathbf{B} \in M_{k \times l}(F)$ , and let  $\mathbf{M} = \begin{pmatrix} \mathbf{A} & \mathbf{B} \\ 0 & \mathbf{D} \end{pmatrix}$ .

- (a) Prove that for every  $j \geq 1$ ,

$$\mathbf{M}^j = \begin{pmatrix} \mathbf{A}^j & \mathbf{S}_j \\ 0 & \mathbf{D}^j \end{pmatrix}, \quad \mathbf{S}_j = \sum_{i=0}^{j-1} \mathbf{A}^i \mathbf{B} \mathbf{D}^{j-1-i}.$$

- (b) Deduce that if  $\mathbf{D} = 0$ , then  $\mathbf{M}^j = \begin{pmatrix} \mathbf{A}^j & \mathbf{A}^{j-1} \mathbf{B} \\ 0 & 0 \end{pmatrix}$  for  $j \geq 1$ , and that if moreover  $\mathbf{A}^r = 0$ , then  $\mathbf{M}^{r+1} = 0$ .

- (c) For  $\mathbf{A} = \begin{pmatrix} 0 & 1 \\ 0 & 0 \end{pmatrix}$ ,  $\mathbf{B} = \begin{pmatrix} 1 \\ 1 \end{pmatrix}$  and  $\mathbf{D} = (0)$ , find the least  $j$  with  $\mathbf{M}^j = 0$ .

*Solution.*

- (a) By induction on  $j$ . For  $j = 1$ ,  $\mathbf{S}_1 = \mathbf{A}^0 \mathbf{B} \mathbf{D}^0 = \mathbf{B}$ . Suppose the formula holds for  $j$ . By Theorem 7.1.5 (a),

$$\mathbf{M}^{j+1} = \mathbf{M}^j \mathbf{M} = \begin{pmatrix} \mathbf{A}^j \mathbf{A} & \mathbf{A}^j \mathbf{B} + \mathbf{S}_j \mathbf{D} \\ 0 & \mathbf{D}^j \mathbf{D} \end{pmatrix},$$

and  $\mathbf{A}^j \mathbf{B} + \mathbf{S}_j \mathbf{D} = \mathbf{A}^j \mathbf{B} \mathbf{D}^0 + \sum_{i=0}^{j-1} \mathbf{A}^i \mathbf{B} \mathbf{D}^{j-i} = \sum_{i=0}^j \mathbf{A}^i \mathbf{B} \mathbf{D}^{j-i} = \mathbf{S}_{j+1}$ . This completes the induction.

- (b) If  $\mathbf{D} = 0$ , every term of  $\mathbf{S}_j$  with  $j - 1 - i \geq 1$  contains a factor  $\mathbf{D} = 0$ , and only  $i = j - 1$  survives, with  $\mathbf{D}^0 = \mathbf{I}_l$ . So  $\mathbf{S}_j = \mathbf{A}^{j-1} \mathbf{B}$  and  $\mathbf{D}^j = 0$ . If also  $\mathbf{A}^r = 0$ , then  $\mathbf{M}^{r+1}$  has blocks  $\mathbf{A}^{r+1} = 0$  and  $\mathbf{A}^r \mathbf{B} = 0$ , so  $\mathbf{M}^{r+1} = 0$ .

- (c) Here  $\mathbf{A}^2 = 0$ , so  $\mathbf{M}^3 = 0$  by (b). But  $\mathbf{M}^2 = \begin{pmatrix} \mathbf{A}^2 & \mathbf{A} \mathbf{B} \\ 0 & 0 \end{pmatrix}$  with  $\mathbf{A} \mathbf{B} = \begin{pmatrix} 1 \\ 1 \end{pmatrix} \neq 0$ ,

so  $\mathbf{M}^2 \neq 0$ ; and  $\mathbf{M} \neq 0$ . The least such  $j$  is 3. Explicitly,  $\mathbf{M} = \begin{pmatrix} 0 & 1 & 1 \\ 0 & 0 & 1 \\ 0 & 0 & 0 \end{pmatrix}$  and

$$\mathbf{M}^2 = \mathbf{E}_{13}.$$

## 7.2 Rank and Block Matrices

Rank is the most robust number attached to a matrix, and Chapters 2 and 3 found several inequalities for it: the rank of a product is at most the rank of each factor, and Sylvester's inequality bounds it from below. Those proofs worked with subspaces. This section proves rank inequalities by a different and very mechanical method. We place the matrices we care about as blocks of one bigger matrix, and then change that matrix by block row and column operations, which never change the rank, until its rank can be read off. The method gives short proofs of the Frobenius inequality and of identities such as  $\text{rank}(\mathbf{I}_m - \mathbf{A} \mathbf{B}) + n = \text{rank}(\mathbf{I}_n - \mathbf{B} \mathbf{A}) + m$ , which are awkward to reach with subspaces alone.

### 7.2.1 Three basic inequalities

We start with what the shape of a block matrix alone says about its rank. Write  $\begin{pmatrix} \mathbf{A} & \mathbf{B} \end{pmatrix}$  for the matrix obtained by placing  $\mathbf{B}$  to the right of  $\mathbf{A}$  (same number of rows), and  $\begin{pmatrix} \mathbf{A} \\ \mathbf{C} \end{pmatrix}$  for  $\mathbf{C}$  below  $\mathbf{A}$  (same number of columns). In these results the blocks need **not** be square.

#### Theorem 7.2.1 — Rank Inequalities for Block Matrices

Let  $\mathbf{A} \in M_{m \times k}(F)$ ,  $\mathbf{B} \in M_{m \times l}(F)$ ,  $\mathbf{C} \in M_{p \times k}(F)$  and  $\mathbf{D} \in M_{p \times l}(F)$ .

- (a)  $\max(\text{rank } \mathbf{A}, \text{rank } \mathbf{B}) \leq \text{rank} \begin{pmatrix} \mathbf{A} & \mathbf{B} \end{pmatrix} \leq \text{rank } \mathbf{A} + \text{rank } \mathbf{B}$ , and likewise  $\max(\text{rank } \mathbf{A}, \text{rank } \mathbf{C}) \leq$

- $$\text{rank} \begin{pmatrix} \mathbf{A} \\ \mathbf{C} \end{pmatrix} \leq \text{rank } \mathbf{A} + \text{rank } \mathbf{C}.$$
- (b)  $\text{rank} \begin{pmatrix} \mathbf{A} & \mathbf{0} \\ \mathbf{0} & \mathbf{D} \end{pmatrix} = \text{rank } \mathbf{A} + \text{rank } \mathbf{D}.$
- (c)  $\text{rank} \begin{pmatrix} \mathbf{A} & \mathbf{B} \\ \mathbf{0} & \mathbf{D} \end{pmatrix} \geq \text{rank } \mathbf{A} + \text{rank } \mathbf{D}$  and  $\text{rank} \begin{pmatrix} \mathbf{A} & \mathbf{0} \\ \mathbf{C} & \mathbf{D} \end{pmatrix} \geq \text{rank } \mathbf{A} + \text{rank } \mathbf{D}.$

*Idea.* Rank is the dimension of the column space, so look at columns. In (a) the columns of  $\begin{pmatrix} \mathbf{A} & \mathbf{B} \end{pmatrix}$  are those of  $\mathbf{A}$  together with those of  $\mathbf{B}$ , and dimensions of sums are at most sums of dimensions. In (b) the columns coming from  $\mathbf{A}$  and from  $\mathbf{D}$  live in two subspaces that meet only in  $\mathbf{0}$ . In (c) we cannot use the column space of  $\mathbf{B}$ , which is unknown, so instead we pick independent columns of  $\mathbf{A}$  and of  $\mathbf{D}$  and show that the columns of the big matrix standing above them are still independent: the zero block lets us peel off the  $\mathbf{D}$ -part first.

*Proof.* For  $\mathbf{y} \in F^m$  and  $\mathbf{z} \in F^p$ , write  $\begin{pmatrix} \mathbf{y} \\ \mathbf{z} \end{pmatrix} \in F^{m+p}$  for the stacked vector.

- (a) The columns of  $\begin{pmatrix} \mathbf{A} & \mathbf{B} \end{pmatrix}$  are the columns of  $\mathbf{A}$  followed by the columns of  $\mathbf{B}$ , so its column space is  $\text{col}(\mathbf{A}) + \text{col}(\mathbf{B})$  by [Proposition 1.7.2](#). It contains  $\text{col}(\mathbf{A})$  and  $\text{col}(\mathbf{B})$ , which gives the lower bound by [Theorem 1.6.7](#), and by [Theorem 1.7.3](#)

$$\text{rank} \begin{pmatrix} \mathbf{A} & \mathbf{B} \end{pmatrix} = \text{rank } \mathbf{A} + \text{rank } \mathbf{B} - \dim(\text{col } \mathbf{A} \cap \text{col } \mathbf{B}) \leq \text{rank } \mathbf{A} + \text{rank } \mathbf{B}.$$

For the stacked matrix, [Proposition 7.1.2](#) gives  $\begin{pmatrix} \mathbf{A} \\ \mathbf{C} \end{pmatrix}^\top = (\mathbf{A}^\top \quad \mathbf{C}^\top)$ , and a matrix and its transpose have the same rank ([Theorem 2.5.5](#)). Apply the first part to  $\mathbf{A}^\top$  and  $\mathbf{C}^\top$ .

- (b) Let  $\mathbf{M} = \begin{pmatrix} \mathbf{A} & \mathbf{0} \\ \mathbf{0} & \mathbf{D} \end{pmatrix}$ . Its first  $k$  columns are  $\begin{pmatrix} \mathbf{a}_j \\ \mathbf{0} \end{pmatrix}$  and its last  $l$  columns are  $\begin{pmatrix} \mathbf{0} \\ \mathbf{d}_q \end{pmatrix}$ , where  $\mathbf{a}_j, \mathbf{d}_q$  are the columns of  $\mathbf{A}, \mathbf{D}$ . So  $\text{col}(\mathbf{M}) = Y + Z$  with  $Y = \left\{ \begin{pmatrix} \mathbf{y} \\ \mathbf{0} \end{pmatrix} : \mathbf{y} \in \text{col } \mathbf{A} \right\}$  and  $Z = \left\{ \begin{pmatrix} \mathbf{0} \\ \mathbf{z} \end{pmatrix} : \mathbf{z} \in \text{col } \mathbf{D} \right\}$ . A vector in  $Y \cap Z$  has both parts zero, so  $Y \cap Z = \{\mathbf{0}\}$ . The map  $\mathbf{y} \mapsto \begin{pmatrix} \mathbf{y} \\ \mathbf{0} \end{pmatrix}$  is linear and injective from  $\text{col } \mathbf{A}$  onto  $Y$ , so  $\dim Y = \text{rank } \mathbf{A}$  ([Theorem 3.5.4](#)); likewise  $\dim Z = \text{rank } \mathbf{D}$ . By [Theorem 1.7.3](#),  $\text{rank } \mathbf{M} = \text{rank } \mathbf{A} + \text{rank } \mathbf{D} - 0$ .
- (c) Let  $\mathbf{M} = \begin{pmatrix} \mathbf{A} & \mathbf{B} \\ \mathbf{0} & \mathbf{D} \end{pmatrix}$ ,  $r = \text{rank } \mathbf{A}$  and  $s = \text{rank } \mathbf{D}$ . By [Theorem 2.5.2](#) there are columns  $\mathbf{a}_{j_1}, \dots, \mathbf{a}_{j_r}$  of  $\mathbf{A}$  that are linearly independent, and columns  $\mathbf{d}_{q_1}, \dots, \mathbf{d}_{q_s}$  of  $\mathbf{D}$  that are linearly independent. The corresponding columns of  $\mathbf{M}$  are  $\begin{pmatrix} \mathbf{a}_{j_i} \\ \mathbf{0} \end{pmatrix}$  and  $\begin{pmatrix} \mathbf{b}_{q_i} \\ \mathbf{d}_{q_i} \end{pmatrix}$ , where  $\mathbf{b}_q$  is column  $q$  of  $\mathbf{B}$ . Let

$$\sum_{i=1}^r \alpha_i \begin{pmatrix} \mathbf{a}_{j_i} \\ \mathbf{0} \end{pmatrix} + \sum_{i=1}^s \beta_i \begin{pmatrix} \mathbf{b}_{q_i} \\ \mathbf{d}_{q_i} \end{pmatrix} = \mathbf{0}.$$

The last  $p$  entries give  $\sum_i \beta_i \mathbf{d}_{q_i} = \mathbf{0}$ , so all  $\beta_i = 0$ . The first  $m$  entries then

give  $\sum_i \alpha_i \mathbf{a}_{j_i} = \mathbf{0}$ , so all  $\alpha_i = 0$ . Hence  $\text{col}(\mathbf{M})$  contains  $r + s$  linearly independent vectors, and  $\text{rank } \mathbf{M} \geq r + s$  by [Theorem 1.6.3](#). For the lower triangular shape,  $\begin{pmatrix} \mathbf{A} & \mathbf{0} \\ \mathbf{C} & \mathbf{D} \end{pmatrix}^\top = \begin{pmatrix} \mathbf{A}^\top & \mathbf{C}^\top \\ \mathbf{0} & \mathbf{D}^\top \end{pmatrix}$  by [Proposition 7.1.2](#); apply the first case and [Theorem 2.5.5](#). ■

All three inequalities, the two in (a) and the one in (c), can be strict, and small witnesses show it. For  $\mathbf{A} = \mathbf{B} = (1)$ ,  $\text{rank} \begin{pmatrix} 1 & 1 \end{pmatrix} = 1 < 2$ : the two column spaces overlap. For  $\mathbf{A} = \mathbf{e}_1$  and  $\mathbf{B} = \mathbf{e}_2$  in  $F^2$ ,  $\text{rank} \begin{pmatrix} \mathbf{e}_1 & \mathbf{e}_2 \end{pmatrix} = \text{rank } \mathbf{I}_2 = 2 > \max(1, 1)$ : each block adds something the other lacks. For  $\mathbf{A} = \mathbf{D} = (0)$  and  $\mathbf{B} = (1)$ ,  $\text{rank} \begin{pmatrix} 0 & 1 \\ 0 & 0 \end{pmatrix} = 1 > 0 + 0$ : the off-diagonal block contributes rank of its own. The proof of (a) says exactly when the upper bound is attained:  $\text{rank} \begin{pmatrix} \mathbf{A} & \mathbf{B} \end{pmatrix} = \text{rank } \mathbf{A} + \text{rank } \mathbf{B}$  if and only if  $\text{col } \mathbf{A} \cap \text{col } \mathbf{B} = \{\mathbf{0}\}$ .

**Warning.** The ranks of the four blocks do not determine the rank of the whole.

Over  $\mathbb{Q}$ ,  $\begin{pmatrix} 1 & 1 \\ 1 & 1 \end{pmatrix}$  and  $\begin{pmatrix} 1 & 1 \\ 1 & -1 \end{pmatrix}$  have all four  $1 \times 1$  blocks of rank 1, but ranks 1 and 2. Only special shapes, like the zero blocks in (b) and (c), turn block ranks into information about the whole matrix.

*Quick check.* Without computing, find  $\text{rank} \begin{pmatrix} 1 & 2 & 0 & 0 & 0 \\ 2 & 4 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 & 3 \\ 0 & 0 & 0 & 1 & 5 \end{pmatrix}$ .

*Solution.* With row partition 2, 2 and column partition 2, 3, the matrix is  $\begin{pmatrix} \mathbf{A} & \mathbf{0} \\ \mathbf{0} & \mathbf{D} \end{pmatrix}$  with  $\mathbf{A} = \begin{pmatrix} 1 & 2 \\ 2 & 4 \end{pmatrix}$  of rank 1 (the second row is twice the first) and  $\mathbf{D} = \begin{pmatrix} 1 & 0 & 3 \\ 0 & 1 & 5 \end{pmatrix}$  of rank 2 (the first two columns are  $\mathbf{e}_1, \mathbf{e}_2$ ). By [Theorem 7.2.1](#) (b), the rank is  $1 + 2 = 3$ .

## 7.2.2 Block elimination

Gaussian elimination changes a matrix by row operations, which are multiplications by invertible matrices on the left, and the rank never changes. The same is true when the “rows” are whole block rows and the multipliers are matrices. The matrices that do this are the block triangular matrices with identity blocks on the diagonal from [Theorem 7.1.5](#), together with block swaps.

Fix a partition of  $\mathbf{M} = \begin{pmatrix} \mathbf{A} & \mathbf{B} \\ \mathbf{C} & \mathbf{D} \end{pmatrix} \in M_{(m+p) \times (k+l)}(F)$  with  $\mathbf{A} \in M_{m \times k}(F)$ . For matrices  $\mathbf{X} \in M_{m \times p}(F)$  and  $\mathbf{Y} \in M_{k \times l}(F)$ , [Theorem 7.1.1](#) gives

$$\begin{pmatrix} \mathbf{I}_m & \mathbf{X} \\ \mathbf{0} & \mathbf{I}_p \end{pmatrix} \begin{pmatrix} \mathbf{A} & \mathbf{B} \\ \mathbf{C} & \mathbf{D} \end{pmatrix} = \begin{pmatrix} \mathbf{A} + \mathbf{XC} & \mathbf{B} + \mathbf{XD} \\ \mathbf{C} & \mathbf{D} \end{pmatrix},$$

$$\begin{pmatrix} \mathbf{A} & \mathbf{B} \\ \mathbf{C} & \mathbf{D} \end{pmatrix} \begin{pmatrix} \mathbf{I}_k & \mathbf{Y} \\ \mathbf{0} & \mathbf{I}_l \end{pmatrix} = \begin{pmatrix} \mathbf{A} & \mathbf{AY} + \mathbf{B} \\ \mathbf{C} & \mathbf{CY} + \mathbf{D} \end{pmatrix}.$$

In words: multiplying on the **left** adds  $\mathbf{X}$  times block row 2 to block row 1, with  $\mathbf{X}$  on the left

of the row; multiplying on the **right** adds block column 1 times  $\mathbf{Y}$  to block column 2, with  $\mathbf{Y}$  on the right of the column. The lower triangular multipliers  $\begin{pmatrix} \mathbf{I}_m & 0 \\ \mathbf{X}' & \mathbf{I}_p \end{pmatrix}$  and  $\begin{pmatrix} \mathbf{I}_k & 0 \\ \mathbf{Y}' & \mathbf{I}_l \end{pmatrix}$  act the same way in the other direction. Sides matter: the multiplier sits on the side of the product where the matrix operation happens.

### Theorem 7.2.2 — Block Elimination Preserves Rank

Let  $\mathbf{M} = \begin{pmatrix} \mathbf{A} & \mathbf{B} \\ \mathbf{C} & \mathbf{D} \end{pmatrix}$  be as above.

- (a) For all  $\mathbf{X} \in M_{m \times p}(F)$ ,  $\mathbf{X}' \in M_{p \times m}(F)$ ,  $\mathbf{Y} \in M_{k \times l}(F)$  and  $\mathbf{Y}' \in M_{l \times k}(F)$ , the matrices

$$\begin{pmatrix} \mathbf{I}_m & \mathbf{X} \\ 0 & \mathbf{I}_p \end{pmatrix}, \quad \begin{pmatrix} \mathbf{I}_m & 0 \\ \mathbf{X}' & \mathbf{I}_p \end{pmatrix}, \quad \begin{pmatrix} \mathbf{I}_k & \mathbf{Y} \\ 0 & \mathbf{I}_l \end{pmatrix}, \quad \begin{pmatrix} \mathbf{I}_k & 0 \\ \mathbf{Y}' & \mathbf{I}_l \end{pmatrix}$$

are invertible, and so are the block swaps  $\begin{pmatrix} 0 & \mathbf{I}_p \\ \mathbf{I}_m & 0 \end{pmatrix} \in M_{m+p}(F)$  and  $\begin{pmatrix} 0 & \mathbf{I}_k \\ \mathbf{I}_l & 0 \end{pmatrix} \in M_{k+l}(F)$ . Hence multiplying  $\mathbf{M}$  on the left or right by any product of such matrices does not change its rank. The swaps give  $\begin{pmatrix} 0 & \mathbf{I}_p \\ \mathbf{I}_m & 0 \end{pmatrix} \mathbf{M} = \begin{pmatrix} \mathbf{C} & \mathbf{D} \\ \mathbf{A} & \mathbf{B} \end{pmatrix}$  and  $\mathbf{M} \begin{pmatrix} 0 & \mathbf{I}_k \\ \mathbf{I}_l & 0 \end{pmatrix} = \begin{pmatrix} \mathbf{B} & \mathbf{A} \\ \mathbf{D} & \mathbf{C} \end{pmatrix}$ .

- (b) If  $\text{col}(\mathbf{B}) \subseteq \text{col}(\mathbf{A})$ , or if  $\text{row}(\mathbf{B}) \subseteq \text{row}(\mathbf{D})$ , then  $\text{rank} \begin{pmatrix} \mathbf{A} & \mathbf{B} \\ 0 & \mathbf{D} \end{pmatrix} = \text{rank } \mathbf{A} + \text{rank } \mathbf{D}$ .

*Proof.*

- (a) By Theorem 7.1.5 (c), with identity blocks on the diagonal,  $\begin{pmatrix} \mathbf{I}_m & \mathbf{X} \\ 0 & \mathbf{I}_p \end{pmatrix}$  has inverse  $\begin{pmatrix} \mathbf{I}_m & -\mathbf{X} \\ 0 & \mathbf{I}_p \end{pmatrix}$ ; the lower triangular case follows by transposing (Proposition 7.1.2, Theorem 0.9.7). In the swap  $\mathbf{S} = \begin{pmatrix} 0 & \mathbf{I}_p \\ \mathbf{I}_m & 0 \end{pmatrix}$  the row partition is  $p, m$  and the column partition is  $m, p$ . Let  $\mathbf{S}' = \begin{pmatrix} 0 & \mathbf{I}_m \\ \mathbf{I}_p & 0 \end{pmatrix}$ , with row partition  $m, p$  and column partition  $p, m$ . These are conformable, and Theorem 7.1.1 gives  $\mathbf{S}'\mathbf{S} = \begin{pmatrix} \mathbf{I}_m & 0 \\ 0 & \mathbf{I}_p \end{pmatrix} = \mathbf{I}_{m+p}$ , so  $\mathbf{S}$  is invertible by Theorem 2.4.5; the other swap is the same with  $k, l$  in place of  $m, p$ . The two displayed products are Theorem 7.1.1. Rank is unchanged by invertible factors on either side (Theorem 2.5.9).
- (b) Suppose  $\text{col}(\mathbf{B}) \subseteq \text{col}(\mathbf{A})$ . Each column  $\mathbf{b}_q$  of  $\mathbf{B}$  is then  $\mathbf{A}\mathbf{y}_q$  for some  $\mathbf{y}_q \in F^k$  (Definition 2.5.1, Theorem 0.9.3). Let  $\mathbf{Y} \in M_{k \times l}(F)$  have columns  $\mathbf{y}_q$ , so that  $\mathbf{A}\mathbf{Y} = \mathbf{B}$  by Theorem 0.9.2. By the computation before the theorem,

$$\begin{pmatrix} \mathbf{A} & \mathbf{B} \\ 0 & \mathbf{D} \end{pmatrix} \begin{pmatrix} \mathbf{I}_k & -\mathbf{Y} \\ 0 & \mathbf{I}_l \end{pmatrix} = \begin{pmatrix} \mathbf{A} & -\mathbf{A}\mathbf{Y} + \mathbf{B} \\ 0 & \mathbf{D} \end{pmatrix} = \begin{pmatrix} \mathbf{A} & 0 \\ 0 & \mathbf{D} \end{pmatrix},$$

and (a) together with Theorem 7.2.1 (b) gives the claim. If  $\text{row}(\mathbf{B}) \subseteq \text{row}(\mathbf{D})$ , the same argument applied to rows gives  $\mathbf{B} = \mathbf{X}\mathbf{D}$  for some  $\mathbf{X} \in M_{m \times p}(F)$ , and



$$\begin{pmatrix} \mathbf{I}_m & -\mathbf{X} \\ 0 & \mathbf{I}_p \end{pmatrix} \begin{pmatrix} \mathbf{A} & \mathbf{B} \\ 0 & \mathbf{D} \end{pmatrix} = \begin{pmatrix} \mathbf{A} & 0 \\ 0 & \mathbf{D} \end{pmatrix}.$$

■

Part (b) is the typical use of (a): a block that is a multiple of a neighboring block can be cleared, exactly as an entry below a pivot is cleared in Gaussian elimination. The zero block of [Theorem 7.2.1](#) (c) is what lets the cleared matrix fall apart into a direct sum.

### 7.2.3 Frobenius's inequality by blocks

Recall Sylvester's rank inequality ([Theorem 3.8.5](#)): for  $\mathbf{A} \in M_{m \times n}(F)$  and  $\mathbf{B} \in M_{n \times p}(F)$ ,  $\text{rank}(\mathbf{AB}) \geq \text{rank } \mathbf{A} + \text{rank } \mathbf{B} - n$ . In [Exercise \(C1: Frobenius's rank inequality\)](#) we met a stronger statement, Frobenius's inequality, and proved it by comparing intersections of subspaces with a null space. Here is a proof in which nothing is chosen: we write down one block matrix whose rank we know, and eliminate.

#### Theorem 7.2.3 — Frobenius's Rank Inequality

Let  $\mathbf{A} \in M_{m \times n}(F)$ ,  $\mathbf{B} \in M_{n \times p}(F)$  and  $\mathbf{C} \in M_{p \times q}(F)$ . Then

$$\text{rank}(\mathbf{ABC}) + \text{rank } \mathbf{B} \geq \text{rank}(\mathbf{AB}) + \text{rank}(\mathbf{BC}).$$

*Idea.* The left side is the rank of  $\begin{pmatrix} \mathbf{ABC} & 0 \\ 0 & \mathbf{B} \end{pmatrix}$ , by [Theorem 7.2.1](#) (b). The right side is a lower bound for any block triangular matrix with diagonal blocks  $\mathbf{AB}$  and  $\mathbf{BC}$ , by part (c). So we want to turn the first matrix into the second by block elimination. Found backwards:  $\mathbf{A}$  times the second block row puts  $\mathbf{AB}$  into the top-right corner; then the second block column times  $\mathbf{C}$  cancels  $\mathbf{ABC}$  and puts  $\mathbf{BC}$  into the bottom-left corner. A swap of the two block columns finishes the job.

*Proof.* Let  $\mathbf{P} = \begin{pmatrix} \mathbf{ABC} & 0 \\ 0 & \mathbf{B} \end{pmatrix}$ , with row partition  $m, n$  and column partition  $q, p$ . By [Theorem 7.2.1](#) (b),  $\text{rank } \mathbf{P} = \text{rank}(\mathbf{ABC}) + \text{rank } \mathbf{B}$ . By [Theorem 7.1.1](#),

$$\begin{aligned} \begin{pmatrix} \mathbf{I}_m & \mathbf{A} \\ 0 & \mathbf{I}_n \end{pmatrix} \mathbf{P} &= \begin{pmatrix} \mathbf{ABC} & \mathbf{AB} \\ 0 & \mathbf{B} \end{pmatrix}, \\ \begin{pmatrix} \mathbf{ABC} & \mathbf{AB} \\ 0 & \mathbf{B} \end{pmatrix} \begin{pmatrix} \mathbf{I}_q & 0 \\ -\mathbf{C} & \mathbf{I}_p \end{pmatrix} &= \begin{pmatrix} \mathbf{ABC} - \mathbf{ABC} & \mathbf{AB} \\ -\mathbf{BC} & \mathbf{B} \end{pmatrix} \\ &= \begin{pmatrix} 0 & \mathbf{AB} \\ -\mathbf{BC} & \mathbf{B} \end{pmatrix}. \end{aligned}$$

Multiplying on the right by the block swap  $\begin{pmatrix} 0 & \mathbf{I}_q \\ \mathbf{I}_p & 0 \end{pmatrix}$  exchanges the two block columns

and gives  $\mathbf{Q} = \begin{pmatrix} \mathbf{AB} & 0 \\ \mathbf{B} & -\mathbf{BC} \end{pmatrix}$ . By [Theorem 7.2.2](#) (a),  $\text{rank } \mathbf{Q} = \text{rank } \mathbf{P}$ . By [Theorem 7.2.1](#) (c),  $\text{rank } \mathbf{Q} \geq \text{rank}(\mathbf{AB}) + \text{rank}(-\mathbf{BC}) = \text{rank}(\mathbf{AB}) + \text{rank}(\mathbf{BC})$ , where  $\text{rank}(-\mathbf{BC}) = \text{rank}(\mathbf{BC})$  because  $-\mathbf{BC}$  and  $\mathbf{BC}$  have the same column space. Hence  $\text{rank}(\mathbf{ABC}) + \text{rank } \mathbf{B} \geq \text{rank}(\mathbf{AB}) + \text{rank}(\mathbf{BC})$ .

Taking  $\mathbf{B} = \mathbf{I}_n$  gives  $\text{rank}(\mathbf{AC}) + n \geq \text{rank } \mathbf{A} + \text{rank } \mathbf{C}$  for  $\mathbf{A} \in M_{m \times n}(F)$  and  $\mathbf{C} \in M_{n \times q}(F)$ , which is Sylvester's inequality again. Taking  $\mathbf{A} = \mathbf{I}_n$  or  $\mathbf{C} = \mathbf{I}_p$  gives an equality and no information, as it should. The inequality says that multiplying by  $\mathbf{A}$  on the left loses at least as much rank from  $\mathbf{BC}$  as from  $\mathbf{B}$ : the rank lost is  $\text{rank}(\mathbf{BC}) - \text{rank}(\mathbf{ABC}) \leq \text{rank } \mathbf{B} - \text{rank}(\mathbf{AB})$ . A smaller column space has less to lose.

## 7.2.4 A worked example: $\mathbf{I} - \mathbf{AB}$ and $\mathbf{I} - \mathbf{BA}$

Products in the two orders,  $\mathbf{AB}$  and  $\mathbf{BA}$ , are different matrices of possibly different sizes, and Chapter 3 showed that even their ranks can differ. Yet  $\mathbf{I} - \mathbf{AB}$  and  $\mathbf{I} - \mathbf{BA}$  are tightly linked. The block matrix that links them contains  $\mathbf{I}_m$ ,  $\mathbf{A}$ ,  $\mathbf{B}$  and  $\mathbf{I}_n$ , and each identity block can be used as a pivot to clear its neighbors.

### Example. (Rank of $\mathbf{I} - \mathbf{AB}$ versus $\mathbf{I} - \mathbf{BA}$ )

Let  $\mathbf{A} \in M_{m \times n}(F)$  and  $\mathbf{B} \in M_{n \times m}(F)$ .

- (a) Prove that  $\text{rank}(\mathbf{I}_m - \mathbf{AB}) + n = \text{rank}(\mathbf{I}_n - \mathbf{BA}) + m$ .
- (b) Deduce that for  $m = n$ ,  $\text{rank}(\mathbf{I}_n - \mathbf{AB}) = \text{rank}(\mathbf{I}_n - \mathbf{BA})$ , and  $\mathbf{I}_n - \mathbf{AB}$  is invertible if and only if  $\mathbf{I}_n - \mathbf{BA}$  is.
- (c) Check (a) for  $\mathbf{A} = \begin{pmatrix} 1 & 1 & 0 \\ 0 & 1 & 1 \end{pmatrix}$  and  $\mathbf{B} = \begin{pmatrix} 1 & 0 \\ 0 & 0 \\ 0 & 1 \end{pmatrix}$  over  $\mathbb{Q}$ .

*Solution.*

- (a) Let  $\mathbf{M} = \begin{pmatrix} \mathbf{I}_m & \mathbf{A} \\ \mathbf{B} & \mathbf{I}_n \end{pmatrix} \in M_{m+n}(F)$ , with partition  $m, n$  for rows and columns. We eliminate in two different ways.

*Using  $\mathbf{I}_m$  as the pivot.* Subtract  $\mathbf{B}$  times block row 1 from block row 2, then subtract block column 1 times  $\mathbf{A}$  from block column 2. By [Theorem 7.1.1](#),

$$\begin{pmatrix} \mathbf{I}_m & 0 \\ -\mathbf{B} & \mathbf{I}_n \end{pmatrix} \mathbf{M} = \begin{pmatrix} \mathbf{I}_m & \mathbf{A} \\ 0 & \mathbf{I}_n - \mathbf{BA} \end{pmatrix},$$

$$\begin{pmatrix} \mathbf{I}_m & \mathbf{A} \\ 0 & \mathbf{I}_n - \mathbf{BA} \end{pmatrix} \begin{pmatrix} \mathbf{I}_m & -\mathbf{A} \\ 0 & \mathbf{I}_n \end{pmatrix} = \begin{pmatrix} \mathbf{I}_m & 0 \\ 0 & \mathbf{I}_n - \mathbf{BA} \end{pmatrix}.$$

By [Theorem 7.2.2](#) (a) and [Theorem 7.2.1](#) (b),  $\text{rank } \mathbf{M} = m + \text{rank}(\mathbf{I}_n - \mathbf{BA})$ .

*Using  $\mathbf{I}_n$  as the pivot.* Subtract  $\mathbf{A}$  times block row 2 from block row 1, then subtract block column 2 times  $\mathbf{B}$  from block column 1:

$$\begin{pmatrix} \mathbf{I}_m & -\mathbf{A} \\ 0 & \mathbf{I}_n \end{pmatrix} \mathbf{M} = \begin{pmatrix} \mathbf{I}_m - \mathbf{AB} & 0 \\ \mathbf{B} & \mathbf{I}_n \end{pmatrix},$$

$$\begin{pmatrix} \mathbf{I}_m - \mathbf{AB} & 0 \\ \mathbf{B} & \mathbf{I}_n \end{pmatrix} \begin{pmatrix} \mathbf{I}_m & 0 \\ -\mathbf{B} & \mathbf{I}_n \end{pmatrix} = \begin{pmatrix} \mathbf{I}_m - \mathbf{AB} & 0 \\ 0 & \mathbf{I}_n \end{pmatrix}.$$

So  $\text{rank } \mathbf{M} = \text{rank}(\mathbf{I}_m - \mathbf{AB}) + n$  in the same way. Comparing the two expressions for  $\text{rank } \mathbf{M}$  proves (a).

- (b) For  $m = n$  the two added terms are equal and cancel. A square matrix is invertible exactly when its rank is  $n$  ([Theorem 2.4.4](#)), so  $\mathbf{I}_n - \mathbf{AB}$  is invertible if and only if  $\mathbf{I}_n - \mathbf{BA}$  is.

- (c) Here  $\mathbf{AB} = \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix} = \mathbf{I}_2$ , so  $\text{rank}(\mathbf{I}_2 - \mathbf{AB}) = 0$ , and

$$\mathbf{BA} = \begin{pmatrix} 1 & 1 & 0 \\ 0 & 0 & 0 \\ 0 & 1 & 1 \end{pmatrix}, \quad \mathbf{I}_3 - \mathbf{BA} = \begin{pmatrix} 0 & -1 & 0 \\ 0 & 1 & 0 \\ 0 & -1 & 0 \end{pmatrix},$$

which has rank 1. Indeed  $0 + 3 = 1 + 2$ . So  $\mathbf{AB} = \mathbf{I}_2$  while  $\mathbf{BA} \neq \mathbf{I}_3$ , and the identity accounts for the difference exactly:  $\mathbf{I}_3 - \mathbf{BA}$  has rank  $3 - 2 = 1$ .

The trick is worth naming. **To compare two expressions, find one block matrix from which both can be obtained by elimination, and compute its rank twice.** Section 3 does the same with determinants and inverses, and the pair  $\mathbf{I} - \mathbf{AB}$ ,  $\mathbf{I} - \mathbf{BA}$  returns there as Sylvester's determinant identity.

## 7.2.5 Exercises

### 7.2.5.1 A. Check your understanding

#### Exercise A1.

- State the three inequalities of [Theorem 7.2.1](#).
- True or false:  $\text{rank} \begin{pmatrix} \mathbf{A} & \mathbf{B} \\ 0 & \mathbf{D} \end{pmatrix} = \text{rank } \mathbf{A} + \text{rank } \mathbf{D}$  for all blocks of compatible sizes. Justify your answer.
- Which matrix do you multiply  $\begin{pmatrix} \mathbf{A} & \mathbf{B} \\ \mathbf{C} & \mathbf{D} \end{pmatrix}$  by, and on which side, to replace  $\mathbf{C}$  by  $\mathbf{C} - \mathbf{XA}$ ?
- True or false:  $\text{rank} \begin{pmatrix} \mathbf{A} & \mathbf{B} \end{pmatrix} = \text{rank } \mathbf{A} + \text{rank } \mathbf{B}$  whenever  $\mathbf{A}$  and  $\mathbf{B}$  have the same number of rows. Justify your answer.
- State Frobenius's inequality, and explain how Sylvester's inequality follows from it.

*Solution.*

- $\max(\text{rank } \mathbf{A}, \text{rank } \mathbf{B}) \leq \text{rank} \begin{pmatrix} \mathbf{A} & \mathbf{B} \end{pmatrix} \leq \text{rank } \mathbf{A} + \text{rank } \mathbf{B}$  (and the same for stacked blocks);  $\text{rank} \begin{pmatrix} \mathbf{A} & 0 \\ 0 & \mathbf{D} \end{pmatrix} = \text{rank } \mathbf{A} + \text{rank } \mathbf{D}$ ; and both block triangular shapes have rank at least  $\text{rank } \mathbf{A} + \text{rank } \mathbf{D}$ .
- False.  $\begin{pmatrix} 0 & 1 \\ 0 & 0 \end{pmatrix}$  with  $1 \times 1$  blocks has rank 1, but  $\text{rank}(0) + \text{rank}(0) = 0$ . Only  $\geq$  holds in general.
- Multiply on the left by  $\begin{pmatrix} \mathbf{I} & 0 \\ -\mathbf{X} & \mathbf{I} \end{pmatrix}$ : block row 2 becomes  $(\mathbf{C} - \mathbf{XA} \quad \mathbf{D} - \mathbf{XB})$  by [Theorem 7.1.1](#). The rank does not change ([Theorem 7.2.2](#)).
- False.  $\begin{pmatrix} 1 & 1 \end{pmatrix}$  has rank 1, not  $1+1$ . Equality holds exactly when  $\text{col } \mathbf{A} \cap \text{col } \mathbf{B} = \{\mathbf{0}\}$ .

- (e)  $\text{rank}(\mathbf{ABC}) + \text{rank} \mathbf{B} \geq \text{rank}(\mathbf{AB}) + \text{rank}(\mathbf{BC})$  for matrices of compatible sizes. With  $\mathbf{B} = \mathbf{I}_n$  it reads  $\text{rank}(\mathbf{AC}) + n \geq \text{rank} \mathbf{A} + \text{rank} \mathbf{C}$ , which is Sylvester's inequality.

### 7.2.5.2 B. Practice

#### Exercise B1: Ranks of block matrices.

Let  $\mathbf{A} = \begin{pmatrix} 1 & 2 \\ 2 & 4 \end{pmatrix}$  and  $\mathbf{D} = \begin{pmatrix} 1 & 1 \\ 0 & 1 \end{pmatrix}$  in  $M_2(\mathbb{Q})$ . Find the rank of each  $4 \times 4$  matrix, citing the result you use. In which case is the lower bound of [Theorem 7.2.1](#) (c) strict?

- (a)  $\mathbf{M}_1 = \begin{pmatrix} \mathbf{A} & \mathbf{0} \\ \mathbf{0} & \mathbf{D} \end{pmatrix}$ .  
 (b)  $\mathbf{M}_2 = \begin{pmatrix} \mathbf{A} & \mathbf{B} \\ \mathbf{0} & \mathbf{D} \end{pmatrix}$  with  $\mathbf{B} = \begin{pmatrix} 1 & 3 \\ 2 & 6 \end{pmatrix}$ .  
 (c)  $\mathbf{M}_3 = \begin{pmatrix} \mathbf{A} & \mathbf{I}_2 \\ \mathbf{0} & \mathbf{0} \end{pmatrix}$ .

*Solution.*  $\text{rank} \mathbf{A} = 1$ , since the second row is twice the first and  $\mathbf{A} \neq \mathbf{0}$ ;  $\text{rank} \mathbf{D} = 2$ , since  $\det \mathbf{D} = 1 \neq 0$ .

- (a) By [Theorem 7.2.1](#) (b),  $\text{rank} \mathbf{M}_1 = 1 + 2 = 3$ .  
 (b) The columns of  $\mathbf{B}$  are  $(1, 2)$  and  $(3, 6) = 3(1, 2)$ , and  $(1, 2)$  is the first column of  $\mathbf{A}$ . So  $\text{col } \mathbf{B} \subseteq \text{col } \mathbf{A}$ , and by [Theorem 7.2.2](#) (b),  $\text{rank} \mathbf{M}_2 = 1 + 2 = 3$ . (Explicitly  $\mathbf{B} = \mathbf{AY}$  with  $\mathbf{Y} = \begin{pmatrix} 1 & 3 \\ 0 & 0 \end{pmatrix}$ .)  
 (c) The lower bound is  $\text{rank} \mathbf{A} + \text{rank} \mathbf{0} = 1$ . But the last two columns of  $\mathbf{M}_3$  are  $\mathbf{e}_1, \mathbf{e}_2$ , and every column lies in  $\text{span}(\mathbf{e}_1, \mathbf{e}_2)$ , so  $\text{rank} \mathbf{M}_3 = 2$ . This is the strict case: the block  $\mathbf{I}_2$  is not a multiple of  $\mathbf{A}$  or of the zero block.

#### Exercise B2: Shifted products.

Let  $\mathbf{A} \in M_{m \times n}(F)$ ,  $\mathbf{B} \in M_{n \times m}(F)$  and  $c \in F$  with  $c \neq 0$ .

- (a) By adapting [Example \(Rank of  \$\mathbf{I} - \mathbf{AB}\$  versus  \$\mathbf{I} - \mathbf{BA}\$ \)](#), prove that  $\text{rank}(c\mathbf{I}_m - \mathbf{AB}) + n = \text{rank}(c\mathbf{I}_n - \mathbf{BA}) + m$ .  
 (b) Check (a) over  $\mathbb{Q}$  for  $c = 2$ ,  $\mathbf{A} = \begin{pmatrix} 1 & 1 \\ 0 & 1 \\ 1 & 0 \end{pmatrix}$  and  $\mathbf{B} = \begin{pmatrix} 1 & 0 & 1 \\ 0 & 1 & 0 \end{pmatrix}$ .  
 (c) Show by an example with  $m = 1$ ,  $n = 2$  that (a) can fail for  $c = 0$ .

*Solution.*

- (a) Since  $c \neq 0$ , put  $\mathbf{A}' = c^{-1}\mathbf{A}$ . Then  $c\mathbf{I}_m - \mathbf{AB} = c(\mathbf{I}_m - \mathbf{A}'\mathbf{B})$  and  $c\mathbf{I}_n - \mathbf{BA} = c(\mathbf{I}_n - \mathbf{BA}')$ . Multiplying a matrix by the non-zero scalar  $c$  does not change its column space, hence not its rank. By [Example \(Rank of  \$\mathbf{I} - \mathbf{AB}\$  versus  \$\mathbf{I} - \mathbf{BA}\$ \)](#)

(a) applied to  $\mathbf{A}'$  and  $\mathbf{B}$ ,

$$\begin{aligned}\operatorname{rank}(c\mathbf{I}_m - \mathbf{AB}) + n &= \operatorname{rank}(\mathbf{I}_m - \mathbf{A}'\mathbf{B}) + n \\ &= \operatorname{rank}(\mathbf{I}_n - \mathbf{BA}') + m \\ &= \operatorname{rank}(c\mathbf{I}_n - \mathbf{BA}) + m.\end{aligned}$$

(b) Here  $m = 3$ ,  $n = 2$ , and

$$2\mathbf{I}_3 - \mathbf{AB} = \begin{pmatrix} 1 & -1 & -1 \\ 0 & 1 & 0 \\ -1 & 0 & 1 \end{pmatrix}, \quad 2\mathbf{I}_2 - \mathbf{BA} = \begin{pmatrix} 0 & -1 \\ 0 & 1 \end{pmatrix},$$

since  $\mathbf{AB} = \begin{pmatrix} 1 & 1 & 1 \\ 0 & 1 & 0 \\ 1 & 0 & 1 \end{pmatrix}$  and  $\mathbf{BA} = \begin{pmatrix} 2 & 1 \\ 0 & 1 \end{pmatrix}$ . In the first matrix, row 1 equals  $-(\text{row } 2) - (\text{row } 3)$ , and rows 2 and 3 are not multiples of each other, so its rank is 2 ([Theorem 2.5.5](#)). The second has rank 1. Indeed  $2 + 2 = 1 + 3$ .

(c) For  $c = 0$  the identity would say  $\operatorname{rank}(\mathbf{AB}) + n = \operatorname{rank}(\mathbf{BA}) + m$ . Take  $\mathbf{A} = \begin{pmatrix} 1 & 0 \end{pmatrix}$  and  $\mathbf{B} = \begin{pmatrix} 1 \\ 0 \end{pmatrix}$ :  $\mathbf{AB} = (1)$  has rank 1, and  $\mathbf{BA} = \mathbf{E}_{11} \in M_2(F)$  has rank 1, but  $1 + 2 \neq 1 + 1$ .

### Exercise B3: Idempotents by blocks.

Let  $\mathbf{A} \in M_n(F)$ .

(a) Starting from  $\begin{pmatrix} \mathbf{A} & 0 \\ 0 & \mathbf{I}_n - \mathbf{A} \end{pmatrix}$ , use block elimination to prove

$$\operatorname{rank} \mathbf{A} + \operatorname{rank}(\mathbf{I}_n - \mathbf{A}) = n + \operatorname{rank}(\mathbf{A} - \mathbf{A}^2).$$

(b) Deduce that  $\mathbf{A}^2 = \mathbf{A}$  if and only if  $\operatorname{rank} \mathbf{A} + \operatorname{rank}(\mathbf{I}_n - \mathbf{A}) = n$ .

(c) Compare (b) with what Chapter 3 says about projections.

*Hint: first add block row 1 to block row 2, then add block column 2 to block column 1.*

*Solution.*

(a) Write  $\mathbf{I} = \mathbf{I}_n$ . By [Theorem 7.1.1](#), each step being multiplication by one of the matrices of [Theorem 7.2.2](#) (a):

$$\begin{aligned}\begin{pmatrix} \mathbf{A} & 0 \\ 0 & \mathbf{I} - \mathbf{A} \end{pmatrix} &\xrightarrow{\text{row}_2 + \text{row}_1} \begin{pmatrix} \mathbf{A} & 0 \\ \mathbf{A} & \mathbf{I} - \mathbf{A} \end{pmatrix} \\ &\xrightarrow{\text{col}_1 + \text{col}_2} \begin{pmatrix} \mathbf{A} & 0 \\ \mathbf{I} & \mathbf{I} - \mathbf{A} \end{pmatrix} \\ &\xrightarrow{\text{row}_1 - \mathbf{A} \text{ row}_2} \begin{pmatrix} 0 & \mathbf{A}^2 - \mathbf{A} \\ \mathbf{I} & \mathbf{I} - \mathbf{A} \end{pmatrix} \\ &\xrightarrow{\text{col}_2 - \text{col}_1(\mathbf{I} - \mathbf{A})} \begin{pmatrix} 0 & \mathbf{A}^2 - \mathbf{A} \\ \mathbf{I} & 0 \end{pmatrix}.\end{aligned}$$

The multipliers are, in order,  $\begin{pmatrix} \mathbf{I} & 0 \\ \mathbf{I} & \mathbf{I} \end{pmatrix}$  on the left,  $\begin{pmatrix} \mathbf{I} & 0 \\ \mathbf{I} & \mathbf{I} \end{pmatrix}$  on the right,  $\begin{pmatrix} \mathbf{I} & -\mathbf{A} \\ 0 & \mathbf{I} \end{pmatrix}$  on the left and  $\begin{pmatrix} \mathbf{I} & -(\mathbf{I} - \mathbf{A}) \\ 0 & \mathbf{I} \end{pmatrix}$  on the right. In the third step the top row becomes  $(\mathbf{A} - \mathbf{A}\mathbf{I} \quad 0 - \mathbf{A}(\mathbf{I} - \mathbf{A})) = (0 \quad \mathbf{A}^2 - \mathbf{A})$ , and in the fourth the bottom-right block becomes  $(\mathbf{I} - \mathbf{A}) - \mathbf{I}(\mathbf{I} - \mathbf{A}) = 0$ . Swapping the block columns (Theorem 7.2.2 (a)) gives  $\begin{pmatrix} \mathbf{A}^2 - \mathbf{A} & 0 \\ 0 & \mathbf{I} \end{pmatrix}$ , of rank  $\text{rank}(\mathbf{A}^2 - \mathbf{A}) + n$  by Theorem 7.2.1 (b). The first matrix has rank  $\text{rank } \mathbf{A} + \text{rank}(\mathbf{I} - \mathbf{A})$ , and rank is unchanged throughout. Since  $\mathbf{A}^2 - \mathbf{A} = -(\mathbf{A} - \mathbf{A}^2)$  has the same rank as  $\mathbf{A} - \mathbf{A}^2$ , (a) follows.

- (b) By (a),  $\text{rank } \mathbf{A} + \text{rank}(\mathbf{I} - \mathbf{A}) = n$  if and only if  $\text{rank}(\mathbf{A} - \mathbf{A}^2) = 0$ , that is,  $\mathbf{A} - \mathbf{A}^2 = 0$ .
- (c) If  $\mathbf{A}^2 = \mathbf{A}$ , then  $T_{\mathbf{A}}$  is a projection (Definition 3.10.1), so  $F^n = \text{im } T_{\mathbf{A}} \oplus \ker T_{\mathbf{A}}$  by Theorem 3.10.1, with  $\ker T_{\mathbf{A}} = \text{col}(\mathbf{I} - \mathbf{A})$  (a vector  $\mathbf{v}$  with  $\mathbf{A}\mathbf{v} = \mathbf{0}$  equals  $(\mathbf{I} - \mathbf{A})\mathbf{v}$ , and conversely  $\mathbf{A}(\mathbf{I} - \mathbf{A}) = \mathbf{A} - \mathbf{A}^2 = 0$ ). So  $\text{rank } \mathbf{A} + \text{rank}(\mathbf{I} - \mathbf{A}) = \dim \text{im } T_{\mathbf{A}} + \dim \ker T_{\mathbf{A}} = n$ , which is the forward direction of (b). The block argument also gives the converse, and it never mentions subspaces.

### 7.2.5.3 C. Going deeper

#### Exercise C1: The rank of $\mathbf{A} - \mathbf{ABA}$ .

Let  $\mathbf{A} \in M_{m \times n}(F)$  and  $\mathbf{B} \in M_{n \times m}(F)$ .

- (a) Verify, by block multiplication, the following chain, in which each arrow is multiplication by an invertible block triangular matrix:

$$\begin{pmatrix} \mathbf{A} & 0 \\ 0 & \mathbf{I}_n - \mathbf{BA} \end{pmatrix} \rightarrow \begin{pmatrix} \mathbf{A} & \mathbf{A} \\ 0 & \mathbf{I}_n - \mathbf{BA} \end{pmatrix} \rightarrow \begin{pmatrix} \mathbf{A} & \mathbf{A} \\ \mathbf{BA} & \mathbf{I}_n \end{pmatrix} \\ \rightarrow \begin{pmatrix} \mathbf{A} - \mathbf{ABA} & \mathbf{A} \\ 0 & \mathbf{I}_n \end{pmatrix} \rightarrow \begin{pmatrix} \mathbf{A} - \mathbf{ABA} & 0 \\ 0 & \mathbf{I}_n \end{pmatrix}.$$

Name each multiplier and the side it acts on.

- (b) Deduce that  $\text{rank}(\mathbf{A} - \mathbf{ABA}) = \text{rank } \mathbf{A} + \text{rank}(\mathbf{I}_n - \mathbf{BA}) - n$ .
- (c) Suppose  $\mathbf{ABA} = \mathbf{A}$ . Show that  $\mathbf{BA}$  is idempotent and that  $\text{rank}(\mathbf{BA}) = \text{rank } \mathbf{A}$ .

*Hint for (c): use (b), then Exercise (B3: Idempotents by blocks).*

*Solution.*

- (a) The partitions are  $m, n$  for rows and  $n, n$  for columns. By Theorem 7.1.1:

1. Right multiplication by  $\begin{pmatrix} \mathbf{I}_n & \mathbf{I}_n \\ 0 & \mathbf{I}_n \end{pmatrix}$  adds block column 1 to block column 2:  $\begin{pmatrix} \mathbf{A} & \mathbf{A} + 0 \\ 0 & 0 + (\mathbf{I}_n - \mathbf{BA}) \end{pmatrix}$ .
2. Left multiplication by  $\begin{pmatrix} \mathbf{I}_m & 0 \\ \mathbf{B} & \mathbf{I}_n \end{pmatrix}$  adds  $\mathbf{B}$  times block row 1 to block row 2: the new row is  $(\mathbf{BA} \quad \mathbf{BA} + \mathbf{I}_n - \mathbf{BA}) = (\mathbf{BA} \quad \mathbf{I}_n)$ .
3. Right multiplication by  $\begin{pmatrix} \mathbf{I}_n & 0 \\ -\mathbf{BA} & \mathbf{I}_n \end{pmatrix}$  subtracts block column 2 times  $\mathbf{BA}$  from

block column 1: the new column is  $\begin{pmatrix} \mathbf{A} - \mathbf{ABA} \\ \mathbf{BA} - \mathbf{I}_n \mathbf{BA} \end{pmatrix} = \begin{pmatrix} \mathbf{A} - \mathbf{ABA} \\ 0 \end{pmatrix}$ .

4. Left multiplication by  $\begin{pmatrix} \mathbf{I}_m & -\mathbf{A} \\ 0 & \mathbf{I}_n \end{pmatrix}$  subtracts  $\mathbf{A}$  times block row 2 from block row 1: the new row is  $(\mathbf{A} - \mathbf{ABA} - \mathbf{A} \cdot 0 \quad \mathbf{A} - \mathbf{AI}_n) = (\mathbf{A} - \mathbf{ABA} \quad 0)$ .

All four multipliers are invertible by [Theorem 7.2.2](#) (a).

- (b) By [Theorem 7.2.2](#) (a) the first and last matrices have the same rank, and by [Theorem 7.2.1](#) (b) these ranks are  $\text{rank } \mathbf{A} + \text{rank}(\mathbf{I}_n - \mathbf{BA})$  and  $\text{rank}(\mathbf{A} - \mathbf{ABA}) + n$ . Rearranging gives the formula.
- (c)  $(\mathbf{BA})^2 = \mathbf{B}(\mathbf{ABA}) = \mathbf{BA}$ , so  $\mathbf{BA}$  is idempotent. By (b),  $0 = \text{rank}(\mathbf{A} - \mathbf{ABA}) = \text{rank } \mathbf{A} + \text{rank}(\mathbf{I}_n - \mathbf{BA}) - n$ , so  $\text{rank}(\mathbf{I}_n - \mathbf{BA}) = n - \text{rank } \mathbf{A}$ . By [Exercise \(B3: Idempotents by blocks\)](#) (b) applied to the idempotent  $\mathbf{BA}$ ,  $\text{rank}(\mathbf{BA}) = n - \text{rank}(\mathbf{I}_n - \mathbf{BA}) = \text{rank } \mathbf{A}$ .

### Exercise C2: Involutions and the field.

Let  $\mathbf{A} \in M_n(F)$ .

- (a) Suppose the characteristic of  $F$  is not 2. Starting from  $\begin{pmatrix} \mathbf{A} - \mathbf{I}_n & 0 \\ 0 & \mathbf{A} + \mathbf{I}_n \end{pmatrix}$ , prove that

$$\text{rank}(\mathbf{A} - \mathbf{I}_n) + \text{rank}(\mathbf{A} + \mathbf{I}_n) = n + \text{rank}(\mathbf{A}^2 - \mathbf{I}_n).$$

- (b) Deduce that if  $\mathbf{A}^2 = \mathbf{I}_n$  (and  $\text{char } F \neq 2$ ), then  $\text{rank}(\mathbf{A} - \mathbf{I}_n) + \text{rank}(\mathbf{A} + \mathbf{I}_n) = n$ .
- (c) Show that (b) fails over  $\mathbb{F}_2$ , and point to the step of (a) that breaks.

*Hint for (a): add block column 2 to block column 1, and subtract block row 2 from block row 1; then  $-2\mathbf{I}_n$  is an invertible pivot.*

*Solution.*

- (a) Write  $\mathbf{I} = \mathbf{I}_n$  and  $\mathbf{P} = \mathbf{A} + \mathbf{I}$ . Since  $\text{char } F \neq 2$ ,  $2 \neq 0$  in  $F$ , so  $\frac{1}{2}$  exists and  $-2\mathbf{I}$  is invertible. By [Theorem 7.1.1](#), with each multiplier from [Theorem 7.2.2](#) (a):

$$\begin{aligned} \begin{pmatrix} \mathbf{A} - \mathbf{I} & 0 \\ 0 & \mathbf{P} \end{pmatrix} &\xrightarrow{\text{col}_1 + \text{col}_2} \begin{pmatrix} \mathbf{A} - \mathbf{I} & 0 \\ \mathbf{P} & \mathbf{P} \end{pmatrix} \xrightarrow{\text{row}_1 - \text{row}_2} \begin{pmatrix} -2\mathbf{I} & -\mathbf{P} \\ \mathbf{P} & \mathbf{P} \end{pmatrix} \\ &\xrightarrow{\text{col}_2 - \text{col}_1 \cdot \frac{1}{2}\mathbf{P}} \begin{pmatrix} -2\mathbf{I} & 0 \\ \mathbf{P} & \mathbf{P} - \frac{1}{2}\mathbf{P}^2 \end{pmatrix} \\ &\xrightarrow{\text{row}_2 + \frac{1}{2}\mathbf{P} \text{ row}_1} \begin{pmatrix} -2\mathbf{I} & 0 \\ 0 & \mathbf{P} - \frac{1}{2}\mathbf{P}^2 \end{pmatrix}. \end{aligned}$$

In the second step  $(\mathbf{A} - \mathbf{I}) - \mathbf{P} = -2\mathbf{I}$ ; in the third,  $-\mathbf{P} - (-2\mathbf{I})\frac{1}{2}\mathbf{P} = 0$  and  $\mathbf{P} - \mathbf{P} \cdot \frac{1}{2}\mathbf{P} = \mathbf{P} - \frac{1}{2}\mathbf{P}^2$ ; in the fourth,  $\mathbf{P} + \frac{1}{2}\mathbf{P}(-2\mathbf{I}) = 0$ . Finally

$$\mathbf{P} - \frac{1}{2}\mathbf{P}^2 = \frac{1}{2}\mathbf{P}(2\mathbf{I} - \mathbf{P}) = \frac{1}{2}(\mathbf{A} + \mathbf{I})(\mathbf{I} - \mathbf{A}) = \frac{1}{2}(\mathbf{I} - \mathbf{A}^2),$$

where the product  $(\mathbf{A} + \mathbf{I})(\mathbf{I} - \mathbf{A}) = \mathbf{I} - \mathbf{A} + \mathbf{A} - \mathbf{A}^2$  expands by distributivity. The first matrix has rank  $\text{rank}(\mathbf{A} - \mathbf{I}) + \text{rank}(\mathbf{A} + \mathbf{I})$  and the last has rank  $n +$

$\text{rank}(\frac{1}{2}(\mathbf{I} - \mathbf{A}^2)) = n + \text{rank}(\mathbf{A}^2 - \mathbf{I})$  by [Theorem 7.2.1](#) (b), non-zero scalar multiples having the same column space. Rank is unchanged throughout.

(b) If  $\mathbf{A}^2 = \mathbf{I}_n$ , then  $\text{rank}(\mathbf{A}^2 - \mathbf{I}_n) = 0$ .

(c) Over  $\mathbb{F}_2$ , take  $\mathbf{A} = \mathbf{I}_n$ . Then  $\mathbf{A}^2 = \mathbf{I}_n$ , but  $\mathbf{A} - \mathbf{I}_n = 0$  and  $\mathbf{A} + \mathbf{I}_n = 2\mathbf{I}_n = 0$ , so the left side is  $0 \neq n$ . The step that breaks is the pivot:  $-2\mathbf{I}_n = 0$  is not invertible, and  $\frac{1}{2}$  does not exist. (Chapter 3's decomposition of an involution into  $\pm 1$  parts, [Theorem 3.10.5](#), needs the same hypothesis.)

## 7.3 Schur Complements

Section 1 inverted block triangular matrices, and Section 2 cleared blocks without changing the rank. A general block matrix  $\begin{pmatrix} \mathbf{A} & \mathbf{B} \\ \mathbf{C} & \mathbf{D} \end{pmatrix}$  has no zero block to help, but when one diagonal block is invertible, a single step of block elimination creates one. What is left in the opposite corner is a new matrix, the Schur complement, and it carries everything that the invertible block does not: the determinant, the rank and the inverse of the whole matrix can all be read off from the block and its complement. This section introduces it, derives these formulas, uses the two complements of one matrix to prove the Sherman–Morrison–Woodbury formula, and ends with the determinant of a block matrix whose lower blocks commute, over every field.

### 7.3.1 Eliminating a block

Suppose we must solve a linear system whose unknowns come in two groups,  $\mathbf{x} \in F^k$  and  $\mathbf{y} \in F^l$ :

$$\begin{pmatrix} \mathbf{A} & \mathbf{B} \\ \mathbf{C} & \mathbf{D} \end{pmatrix} \begin{pmatrix} \mathbf{x} \\ \mathbf{y} \end{pmatrix} = \begin{pmatrix} \mathbf{f} \\ \mathbf{g} \end{pmatrix}, \quad \text{that is,} \quad \begin{aligned} \mathbf{Ax} + \mathbf{By} &= \mathbf{f}, \\ \mathbf{Cx} + \mathbf{Dy} &= \mathbf{g}, \end{aligned}$$

with  $\mathbf{A} \in M_k(F)$  invertible. Do what we would do for two equations in two numbers: solve the first equation for  $\mathbf{x}$ , and substitute. The first equation gives  $\mathbf{x} = \mathbf{A}^{-1}(\mathbf{f} - \mathbf{By})$ . Substituting into the second,

$$\mathbf{CA}^{-1}\mathbf{f} - \mathbf{CA}^{-1}\mathbf{By} + \mathbf{Dy} = \mathbf{g}, \quad \text{so} \quad (\mathbf{D} - \mathbf{CA}^{-1}\mathbf{B})\mathbf{y} = \mathbf{g} - \mathbf{CA}^{-1}\mathbf{f}.$$

The unknown  $\mathbf{x}$  has disappeared, and  $\mathbf{y}$  solves a smaller system whose matrix is  $\mathbf{D} - \mathbf{CA}^{-1}\mathbf{B}$ . Once  $\mathbf{y}$  is known,  $\mathbf{x} = \mathbf{A}^{-1}(\mathbf{f} - \mathbf{By})$ . So the big system is solvable for every right-hand side exactly when the small matrix  $\mathbf{D} - \mathbf{CA}^{-1}\mathbf{B}$  allows it, and this matrix will keep appearing.

For  $k = l = 1$  we have met it before:  $d - cb/a = (ad - bc)/a$  is the determinant divided by the pivot  $a$ , which is what Gaussian elimination leaves in the corner after clearing the entry  $c$ .

*The Schur complement of an invertible block is what remains in the opposite corner after that block has been used to eliminate its row and column.*

#### Definition 7.3.1 — Schur Complement

Let  $\mathbf{M} = \begin{pmatrix} \mathbf{A} & \mathbf{B} \\ \mathbf{C} & \mathbf{D} \end{pmatrix} \in M_{(k+p) \times (k+l)}(F)$  be a block matrix with  $\mathbf{A} \in M_k(F)$ ,  $\mathbf{B} \in M_{k \times l}(F)$ ,  $\mathbf{C} \in M_{p \times k}(F)$  and  $\mathbf{D} \in M_{p \times l}(F)$ , where  $k, p, l \geq 1$ .



- (a) If  $\mathbf{A}$  is **invertible**, the **Schur complement of  $\mathbf{A}$  in  $\mathbf{M}$**  is

$$\mathbf{M}/\mathbf{A} := \mathbf{D} - \mathbf{C}\mathbf{A}^{-1}\mathbf{B} \in M_{p \times l}(F).$$

- (b) Symmetrically, if  $\mathbf{M} = \begin{pmatrix} \mathbf{A} & \mathbf{B} \\ \mathbf{C} & \mathbf{D} \end{pmatrix}$  with  $\mathbf{D} \in M_l(F)$  **invertible**,  $\mathbf{A} \in M_{m \times k}(F)$ ,  $\mathbf{B} \in M_{m \times l}(F)$  and  $\mathbf{C} \in M_{l \times k}(F)$ , the **Schur complement of  $\mathbf{D}$  in  $\mathbf{M}$**  is  $\mathbf{M}/\mathbf{D} := \mathbf{A} - \mathbf{B}\mathbf{D}^{-1}\mathbf{C} \in M_{m \times k}(F)$ .

In words, clause by clause. The block whose complement we take must be **square and invertible**, since its inverse appears in the formula; the other blocks may be rectangular. The complement has the size of the **opposite corner**:  $\mathbf{M}/\mathbf{A}$  has the shape of  $\mathbf{D}$ , and  $\mathbf{M}/\mathbf{D}$  has the shape of  $\mathbf{A}$ . In  $\mathbf{C}\mathbf{A}^{-1}\mathbf{B}$  the factors come in the only order the sizes allow:  $\mathbf{C}$  is  $p \times k$ ,  $\mathbf{A}^{-1}$  is  $k \times k$ ,  $\mathbf{B}$  is  $k \times l$ . A way to remember it: go from the corner  $\mathbf{D}$  to  $\mathbf{A}$  along its row ( $\mathbf{C}$ ), invert, and come back along its column ( $\mathbf{B}$ ).

**Well-definedness.**  $\mathbf{M}/\mathbf{A}$  depends on  $\mathbf{M}$  and on the chosen partition, which says which corner is  $\mathbf{A}$ . The inverse  $\mathbf{A}^{-1}$  is unique (Theorem 0.9.7), so there is no other choice to make.

**Examples.**

- **Scalars.** For  $\mathbf{M} = \begin{pmatrix} a & b \\ c & d \end{pmatrix}$  with  $a \neq 0$ ,  $\mathbf{M}/a = d - ca^{-1}b = (ad - bc)/a$ . In particular  $a \cdot (\mathbf{M}/a) = \det \mathbf{M}$ .
- **An identity corner.** If  $\mathbf{A} = \mathbf{I}_k$ , then  $\mathbf{M}/\mathbf{I}_k = \mathbf{D} - \mathbf{C}\mathbf{B}$ . No inversion is needed at all.
- **Block triangular.** If  $\mathbf{C} = 0$ , then  $\mathbf{M}/\mathbf{A} = \mathbf{D}$ : nothing needs to be eliminated.
- **A  $3 \times 3$  example.** For  $\mathbf{M} = \left( \begin{array}{cc|c} 2 & 1 & 1 \\ 1 & 1 & 0 \\ 3 & 1 & 4 \end{array} \right)$  over  $\mathbb{Q}$ ,  $\mathbf{A} = \begin{pmatrix} 2 & 1 \\ 1 & 1 \end{pmatrix}$  has  $\mathbf{A}^{-1} = \begin{pmatrix} 1 & -1 \\ -1 & 2 \end{pmatrix}$ ,

and  $\mathbf{C}\mathbf{A}^{-1} = (3 \ 1) \mathbf{A}^{-1} = (2 \ -1)$ , so  $\mathbf{M}/\mathbf{A} = 4 - (2 \ -1) \begin{pmatrix} 1 \\ 0 \end{pmatrix} = (2)$ . This  $1 \times 1$  complement is the last pivot of Gaussian elimination on  $\mathbf{M}$ : subtracting  $\frac{1}{2}$  row 1 from row 2 and  $\frac{3}{2}$  row 1 from row 3 gives the rows  $(0, \frac{1}{2}, -\frac{1}{2})$  and  $(0, -\frac{1}{2}, \frac{5}{2})$ , and adding the first of these to the second gives  $(0, 0, 2)$ .

**Non-example by minimal change.** For  $\mathbf{M} = \begin{pmatrix} 1 & 1 \\ 1 & 0 \end{pmatrix}$ ,  $\mathbf{M}/(1) = 0 - 1 \cdot 1 \cdot 1 = (-1)$ . Change the corner entry 1 to 0:  $\mathbf{M}' = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}$ . The blocks  $\mathbf{B}$ ,  $\mathbf{C}$ ,  $\mathbf{D}$  and the partition are unchanged, and  $\mathbf{M}'$  is still invertible, but  $\mathbf{M}'/(0)$  is **not defined**, because the clause “ $\mathbf{A}$  is invertible” fails. Here  $\mathbf{M}'/\mathbf{D}$  is not defined either, since  $\mathbf{D} = (0)$ .

**Why this definition.** The formula is read off from the elimination in the hook, so the question is only about the name and the hypotheses. “Complement” refers to position:  $\mathbf{M}/\mathbf{A}$  sits in the corner complementary to  $\mathbf{A}$ . The quotient notation  $\mathbf{M}/\mathbf{A}$  is suggested by the first example,  $\det(\mathbf{M}/a) = \det \mathbf{M} / \det a$ , and the determinant formula below shows that this holds for blocks of every size. Invertibility of  $\mathbf{A}$  cannot be dropped, since  $\mathbf{A}^{-1}$  is in the formula; the price is that some invertible matrices, like  $\mathbf{M}'$ , have no Schur complement for a given partition.

*Warning.*  $\mathbf{M}$  can be invertible while  $\mathbf{A}$  is not. For  $\mathbf{M}' = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}$ , neither corner is invertible, so neither Schur complement exists for the partition 1, 1. Before using a formula below, check that the block you invert is invertible. If it is not, try the other

corner  $(\mathbf{M}/\mathbf{D})$ , a different partition, or first permute rows:  $\begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix} \mathbf{M}' = \mathbf{I}_2$ . The same issue appears in Gaussian elimination as a zero pivot.

*Quick check.* Let  $\mathbf{M} = \left( \begin{array}{c|cc} 2 & 4 & 2 \\ \hline 1 & 3 & 5 \\ 3 & 1 & 0 \end{array} \right)$  with partition 1, 2. Compute  $\mathbf{M}/(2)$ .

*Solution.* Here  $\mathbf{A} = (2)$ ,  $\mathbf{B} = (4 \ 2)$ ,  $\mathbf{C} = \begin{pmatrix} 1 \\ 3 \end{pmatrix}$ ,  $\mathbf{D} = \begin{pmatrix} 3 & 5 \\ 1 & 0 \end{pmatrix}$ . Then  $\mathbf{CA}^{-1}\mathbf{B} = \frac{1}{2} \begin{pmatrix} 4 & 2 \\ 12 & 6 \end{pmatrix} = \begin{pmatrix} 2 & 1 \\ 6 & 3 \end{pmatrix}$ , so  $\mathbf{M}/(2) = \begin{pmatrix} 1 & 4 \\ -5 & -3 \end{pmatrix}$ . These are exactly the entries left in rows 2, 3 after clearing column 1 with the pivot 2.

## 7.3.2 The block LDU factorization

The elimination of the hook, written with matrices, factors  $\mathbf{M}$  into a lower block triangular matrix, a block diagonal matrix and an upper block triangular matrix. Everything else in this section follows from this factorization.

### Theorem 7.3.1 — Block LDU Factorization

Let  $\mathbf{M} = \begin{pmatrix} \mathbf{A} & \mathbf{B} \\ \mathbf{C} & \mathbf{D} \end{pmatrix}$  be partitioned as in Definition 7.3.1.

(a) If  $\mathbf{A} \in M_k(F)$  is invertible, then

$$\mathbf{M} = \begin{pmatrix} \mathbf{I}_k & 0 \\ \mathbf{CA}^{-1} & \mathbf{I}_p \end{pmatrix} \begin{pmatrix} \mathbf{A} & 0 \\ 0 & \mathbf{M}/\mathbf{A} \end{pmatrix} \begin{pmatrix} \mathbf{I}_k & \mathbf{A}^{-1}\mathbf{B} \\ 0 & \mathbf{I}_l \end{pmatrix}.$$

(b) If  $\mathbf{D} \in M_l(F)$  is invertible, then

$$\mathbf{M} = \begin{pmatrix} \mathbf{I}_m & \mathbf{BD}^{-1} \\ 0 & \mathbf{I}_l \end{pmatrix} \begin{pmatrix} \mathbf{M}/\mathbf{D} & 0 \\ 0 & \mathbf{D} \end{pmatrix} \begin{pmatrix} \mathbf{I}_k & 0 \\ \mathbf{D}^{-1}\mathbf{C} & \mathbf{I}_l \end{pmatrix}.$$

*Idea.* The left factor in (a) records the row operation “subtract  $\mathbf{CA}^{-1}$  times block row 1 from block row 2”, which turns  $\mathbf{C}$  into 0 and  $\mathbf{D}$  into  $\mathbf{M}/\mathbf{A}$ . The right factor records the column operation “subtract block column 1 times  $\mathbf{A}^{-1}\mathbf{B}$  from block column 2”, which turns  $\mathbf{B}$  into 0. Undoing both operations gives the factorization. To prove it we need not retrace the elimination: multiplying out is enough.

*Proof.*

(a) Write  $\mathbf{S} = \mathbf{M}/\mathbf{A} = \mathbf{D} - \mathbf{CA}^{-1}\mathbf{B}$ . The sizes are conformable throughout. By Theorem 7.1.1,

$$\begin{pmatrix} \mathbf{A} & 0 \\ 0 & \mathbf{S} \end{pmatrix} \begin{pmatrix} \mathbf{I}_k & \mathbf{A}^{-1}\mathbf{B} \\ 0 & \mathbf{I}_l \end{pmatrix} = \begin{pmatrix} \mathbf{A} & \mathbf{AA}^{-1}\mathbf{B} \\ 0 & \mathbf{S} \end{pmatrix} = \begin{pmatrix} \mathbf{A} & \mathbf{B} \\ 0 & \mathbf{S} \end{pmatrix},$$

and then

$$\begin{pmatrix} \mathbf{I}_k & 0 \\ \mathbf{CA}^{-1} & \mathbf{I}_p \end{pmatrix} \begin{pmatrix} \mathbf{A} & \mathbf{B} \\ 0 & \mathbf{S} \end{pmatrix} = \begin{pmatrix} \mathbf{A} & \mathbf{B} \\ \mathbf{CA}^{-1}\mathbf{A} & \mathbf{CA}^{-1}\mathbf{B} + \mathbf{S} \end{pmatrix} = \begin{pmatrix} \mathbf{A} & \mathbf{B} \\ \mathbf{C} & \mathbf{D} \end{pmatrix},$$

since  $\mathbf{CA}^{-1}\mathbf{A} = \mathbf{C}$  and  $\mathbf{CA}^{-1}\mathbf{B} + (\mathbf{D} - \mathbf{CA}^{-1}\mathbf{B}) = \mathbf{D}$ . By associativity ([Theorem 0.9.4](#)) the triple product equals  $\mathbf{M}$ .

(b) Write  $\mathbf{S}' = \mathbf{M}/\mathbf{D} = \mathbf{A} - \mathbf{BD}^{-1}\mathbf{C}$ . By [Theorem 7.1.1](#),

$$\begin{pmatrix} \mathbf{S}' & 0 \\ 0 & \mathbf{D} \end{pmatrix} \begin{pmatrix} \mathbf{I}_k & 0 \\ \mathbf{D}^{-1}\mathbf{C} & \mathbf{I}_l \end{pmatrix} = \begin{pmatrix} \mathbf{S}' & 0 \\ \mathbf{C} & \mathbf{D} \end{pmatrix},$$

$$\begin{pmatrix} \mathbf{I}_m & \mathbf{BD}^{-1} \\ 0 & \mathbf{I}_l \end{pmatrix} \begin{pmatrix} \mathbf{S}' & 0 \\ \mathbf{C} & \mathbf{D} \end{pmatrix} = \begin{pmatrix} \mathbf{S}' + \mathbf{BD}^{-1}\mathbf{C} & \mathbf{BD}^{-1}\mathbf{D} \\ \mathbf{C} & \mathbf{D} \end{pmatrix} = \mathbf{M}.$$

■

The outer factors are block triangular with identity blocks on the diagonal. They are invertible, with inverses obtained by changing the sign of the off-diagonal block ([Theorem 7.1.5](#)), and they have determinant 1. So, up to these harmless factors,  $\mathbf{M}$  is the block diagonal matrix  $\mathbf{A} \oplus (\mathbf{M}/\mathbf{A})$  when  $\mathbf{A}$  and  $\mathbf{D}$  are square. Determinant, rank and invertibility cannot tell the difference.

### 7.3.3 Determinant and rank

The outer factors of [Theorem 7.3.1](#) do not change the determinant or the rank, so both can be read off from the middle factor.

#### Theorem 7.3.2 — Schur Determinant Formula

Let  $\mathbf{M} = \begin{pmatrix} \mathbf{A} & \mathbf{B} \\ \mathbf{C} & \mathbf{D} \end{pmatrix} \in M_{k+l}(F)$  with  $\mathbf{A} \in M_k(F)$ ,  $\mathbf{D} \in M_l(F)$  and  $k, l \geq 1$ .

- (a) If  $\mathbf{A}$  is invertible, then  $\det \mathbf{M} = \det \mathbf{A} \cdot \det(\mathbf{M}/\mathbf{A})$ .
- (b) If  $\mathbf{D}$  is invertible, then  $\det \mathbf{M} = \det \mathbf{D} \cdot \det(\mathbf{M}/\mathbf{D})$ .

*Proof.*

- (a) By [Theorem 7.3.1](#) (a),  $\mathbf{M} = \mathbf{L}(\mathbf{A} \oplus (\mathbf{M}/\mathbf{A}))\mathbf{U}$  with  $\mathbf{L} = \begin{pmatrix} \mathbf{I}_k & 0 \\ \mathbf{CA}^{-1} & \mathbf{I}_l \end{pmatrix}$  and  $\mathbf{U} = \begin{pmatrix} \mathbf{I}_k & \mathbf{A}^{-1}\mathbf{B} \\ 0 & \mathbf{I}_l \end{pmatrix}$ . Here  $p = l$  because  $\mathbf{M}$  is square, so  $\mathbf{M}/\mathbf{A} \in M_l(F)$ . By [Theorem 6.8.4](#),  $\det \mathbf{L} = \det \mathbf{I}_k \det \mathbf{I}_l = 1$ ,  $\det \mathbf{U} = 1$ , and  $\det(\mathbf{A} \oplus (\mathbf{M}/\mathbf{A})) = \det \mathbf{A} \det(\mathbf{M}/\mathbf{A})$ . By [Theorem 6.5.4](#),

$$\det \mathbf{M} = \det \mathbf{L} \cdot \det \mathbf{A} \det(\mathbf{M}/\mathbf{A}) \cdot \det \mathbf{U} = \det \mathbf{A} \det(\mathbf{M}/\mathbf{A}).$$

- (b) The same argument with [Theorem 7.3.1](#) (b), where  $\mathbf{M}/\mathbf{D} \in M_k(F)$  because  $\mathbf{M}$  is square.

■

This is the correct replacement for the false rule  $\det \mathbf{M} = \det(\mathbf{AD} - \mathbf{BC})$  of Section 1: the determinant of  $\mathbf{M}$  is that of  $\mathbf{A}$  times that of the complement. For  $k = l = 1$  it is  $a(d - ca^{-1}b) = ad - bc$ . Chapter 6 promised this, and it relies on [Theorem 6.8.4](#) in exactly the form proved there, with square diagonal blocks.

### Theorem 7.3.3 — Schur Rank Formula

Let  $\mathbf{M} = \begin{pmatrix} \mathbf{A} & \mathbf{B} \\ \mathbf{C} & \mathbf{D} \end{pmatrix}$  be partitioned as in [Definition 7.3.1](#). If  $\mathbf{A} \in M_k(F)$  is invertible, then

$$\text{rank } \mathbf{M} = k + \text{rank}(\mathbf{M}/\mathbf{A}).$$

Symmetrically, if  $\mathbf{D} \in M_l(F)$  is invertible, then  $\text{rank } \mathbf{M} = l + \text{rank}(\mathbf{M}/\mathbf{D})$ .

*Proof.* By [Theorem 7.3.1](#) (a),  $\mathbf{M} = \mathbf{L} \begin{pmatrix} \mathbf{A} & 0 \\ 0 & \mathbf{M}/\mathbf{A} \end{pmatrix} \mathbf{U}$ , where  $\mathbf{L}$  and  $\mathbf{U}$  are invertible by [Theorem 7.2.2](#) (a). By [Theorem 2.5.9](#),  $\text{rank } \mathbf{M}$  equals the rank of the middle factor, which is  $\text{rank } \mathbf{A} + \text{rank}(\mathbf{M}/\mathbf{A}) = k + \text{rank}(\mathbf{M}/\mathbf{A})$  by [Theorem 7.2.1](#) (b) and [Theorem 2.4.4](#). The second statement follows in the same way from [Theorem 7.3.1](#) (b). ■

In words: an invertible block of size  $k$  contributes exactly  $k$  to the rank, and all the remaining rank sits in its complement. For example, in [Example \(Rank of  \$\mathbf{I} - \mathbf{AB}\$  versus  \$\mathbf{I} - \mathbf{BA}\$ \)](#) the matrix  $\begin{pmatrix} \mathbf{I}_m & \mathbf{A} \\ \mathbf{B} & \mathbf{I}_n \end{pmatrix}$  has the two complements  $\mathbf{I}_n - \mathbf{BA}$  and  $\mathbf{I}_m - \mathbf{AB}$ , and the theorem gives both expressions for its rank at once.

## 7.3.4 Inverting a block matrix

Since  $\mathbf{A} \oplus (\mathbf{M}/\mathbf{A})$  is invertible exactly when  $\mathbf{M}/\mathbf{A}$  is, the factorization also inverts  $\mathbf{M}$ : invert the three factors and multiply them in the reverse order.

### Theorem 7.3.4 — Block Inverse Formula

Let  $\mathbf{M} = \begin{pmatrix} \mathbf{A} & \mathbf{B} \\ \mathbf{C} & \mathbf{D} \end{pmatrix} \in M_{k+l}(F)$  with  $\mathbf{A} \in M_k(F)$  and  $\mathbf{D} \in M_l(F)$ .

- (a) Suppose  $\mathbf{A}$  is invertible, and let  $\mathbf{S} = \mathbf{M}/\mathbf{A}$ . Then  $\mathbf{M}$  is invertible if and only if  $\mathbf{S}$  is invertible, and in that case

$$\mathbf{M}^{-1} = \begin{pmatrix} \mathbf{A}^{-1} + \mathbf{A}^{-1}\mathbf{B}\mathbf{S}^{-1}\mathbf{C}\mathbf{A}^{-1} & -\mathbf{A}^{-1}\mathbf{B}\mathbf{S}^{-1} \\ -\mathbf{S}^{-1}\mathbf{C}\mathbf{A}^{-1} & \mathbf{S}^{-1} \end{pmatrix}.$$

In particular the lower-right  $l \times l$  block of  $\mathbf{M}^{-1}$  is  $(\mathbf{M}/\mathbf{A})^{-1}$ .

- (b) Suppose  $\mathbf{D}$  is invertible, and let  $\mathbf{S}' = \mathbf{M}/\mathbf{D}$ . Then  $\mathbf{M}$  is invertible if and only if  $\mathbf{S}'$  is invertible, and in that case

$$\mathbf{M}^{-1} = \begin{pmatrix} \mathbf{S}'^{-1} & -\mathbf{S}'^{-1}\mathbf{B}\mathbf{D}^{-1} \\ -\mathbf{D}^{-1}\mathbf{C}\mathbf{S}'^{-1} & \mathbf{D}^{-1} + \mathbf{D}^{-1}\mathbf{C}\mathbf{S}'^{-1}\mathbf{B}\mathbf{D}^{-1} \end{pmatrix}.$$

In particular the upper-left  $k \times k$  block of  $\mathbf{M}^{-1}$  is  $(\mathbf{M}/\mathbf{D})^{-1}$ .

*Proof.*

- (a) By [Theorem 7.3.3](#),  $\text{rank } \mathbf{M} = k + \text{rank } \mathbf{S}$ . So  $\text{rank } \mathbf{M} = k + l$  if and only if  $\text{rank } \mathbf{S} = l$ , and by [Theorem 2.4.4](#) this says that  $\mathbf{M}$  is invertible if and only if  $\mathbf{S}$  is.

Suppose  $\mathbf{S}$  is invertible. In [Theorem 7.3.1](#) (a),  $\mathbf{M} = \mathbf{L}(\mathbf{A} \oplus \mathbf{S})\mathbf{U}$ , where  $\mathbf{L}^{-1} = \begin{pmatrix} \mathbf{I}_k & 0 \\ -\mathbf{C}\mathbf{A}^{-1} & \mathbf{I}_l \end{pmatrix}$  and  $\mathbf{U}^{-1} = \begin{pmatrix} \mathbf{I}_k & -\mathbf{A}^{-1}\mathbf{B} \\ 0 & \mathbf{I}_l \end{pmatrix}$  by [Theorem 7.1.5](#), and  $(\mathbf{A} \oplus \mathbf{S})^{-1} = \mathbf{A}^{-1} \oplus \mathbf{S}^{-1}$  by [Theorem 7.1.4](#) (c). By [Theorem 0.9.7](#) (3),  $\mathbf{M}^{-1} = \mathbf{U}^{-1}(\mathbf{A}^{-1} \oplus \mathbf{S}^{-1})\mathbf{L}^{-1}$ . By [Theorem 7.1.1](#),

$$\begin{aligned} \mathbf{U}^{-1}(\mathbf{A}^{-1} \oplus \mathbf{S}^{-1}) &= \begin{pmatrix} \mathbf{A}^{-1} & -\mathbf{A}^{-1}\mathbf{B}\mathbf{S}^{-1} \\ 0 & \mathbf{S}^{-1} \end{pmatrix}, \\ \begin{pmatrix} \mathbf{A}^{-1} & -\mathbf{A}^{-1}\mathbf{B}\mathbf{S}^{-1} \\ 0 & \mathbf{S}^{-1} \end{pmatrix} \begin{pmatrix} \mathbf{I}_k & 0 \\ -\mathbf{C}\mathbf{A}^{-1} & \mathbf{I}_l \end{pmatrix} \\ &= \begin{pmatrix} \mathbf{A}^{-1} + \mathbf{A}^{-1}\mathbf{B}\mathbf{S}^{-1}\mathbf{C}\mathbf{A}^{-1} & -\mathbf{A}^{-1}\mathbf{B}\mathbf{S}^{-1} \\ -\mathbf{S}^{-1}\mathbf{C}\mathbf{A}^{-1} & \mathbf{S}^{-1} \end{pmatrix}. \end{aligned}$$

- (b) The equivalence follows from the second statement of [Theorem 7.3.3](#) in the same way. By [Theorem 7.3.1](#) (b),  $\mathbf{M}^{-1} = \begin{pmatrix} \mathbf{I}_k & 0 \\ -\mathbf{D}^{-1}\mathbf{C} & \mathbf{I}_l \end{pmatrix} (\mathbf{S}'^{-1} \oplus \mathbf{D}^{-1}) \begin{pmatrix} \mathbf{I}_k & -\mathbf{B}\mathbf{D}^{-1} \\ 0 & \mathbf{I}_l \end{pmatrix}$ , and

$$\begin{pmatrix} \mathbf{I}_k & 0 \\ -\mathbf{D}^{-1}\mathbf{C} & \mathbf{I}_l \end{pmatrix} \begin{pmatrix} \mathbf{S}'^{-1} & 0 \\ 0 & \mathbf{D}^{-1} \end{pmatrix} = \begin{pmatrix} \mathbf{S}'^{-1} & 0 \\ -\mathbf{D}^{-1}\mathbf{C}\mathbf{S}'^{-1} & \mathbf{D}^{-1} \end{pmatrix},$$

whose product with  $\begin{pmatrix} \mathbf{I}_k & -\mathbf{B}\mathbf{D}^{-1} \\ 0 & \mathbf{I}_l \end{pmatrix}$  is the displayed matrix by [Theorem 7.1.1](#).

■

Nobody should memorize these blocks. Remember instead that  $\mathbf{M}^{-1}$  is the product of three easy inverses in reverse order, and that the corner opposite  $\mathbf{A}$  in  $\mathbf{M}^{-1}$  is the inverse of the complement of  $\mathbf{A}$ .

**Example. ( $\mathbf{A}$   $4 \times 4$  Inverse through a  $2 \times 2$  Schur Complement)**

Invert

$$\mathbf{M} = \left( \begin{array}{cc|cc} 1 & 1 & 0 & -1 \\ 1 & 2 & -1 & -1 \\ \hline -1 & -1 & 1 & 1 \\ 0 & 1 & 0 & -1 \end{array} \right) \in M_4(\mathbb{Q})$$

using [Theorem 7.3.4](#), and find  $\det \mathbf{M}$ .

*Solution.* The blocks are  $\mathbf{A} = \begin{pmatrix} 1 & 1 \\ 1 & 2 \end{pmatrix}$ ,  $\mathbf{B} = \begin{pmatrix} 0 & -1 \\ -1 & -1 \end{pmatrix}$ ,  $\mathbf{C} = \begin{pmatrix} -1 & -1 \\ 0 & 1 \end{pmatrix}$ ,  $\mathbf{D} = \begin{pmatrix} 1 & 1 \\ 0 & -1 \end{pmatrix}$ .

*Step 1: invert  $\mathbf{A}$ .*  $\det \mathbf{A} = 1$ , so  $\mathbf{A}^{-1} = \begin{pmatrix} 2 & -1 \\ -1 & 1 \end{pmatrix}$  by [Theorem 0.9.8](#).

Step 2: the two “half” products.

$$\mathbf{CA}^{-1} = \begin{pmatrix} -1 & 0 \\ -1 & 1 \end{pmatrix}, \quad \mathbf{A}^{-1}\mathbf{B} = \begin{pmatrix} 1 & -1 \\ -1 & 0 \end{pmatrix}, \quad \mathbf{CA}^{-1}\mathbf{B} = \begin{pmatrix} 0 & 1 \\ -1 & 0 \end{pmatrix}.$$

Step 3: the complement and its inverse.  $\mathbf{S} = \mathbf{M}/\mathbf{A} = \mathbf{D} - \mathbf{CA}^{-1}\mathbf{B} = \begin{pmatrix} 1 & 0 \\ 1 & -1 \end{pmatrix}$ . Its determinant is  $-1 \neq 0$ , so  $\mathbf{M}$  is invertible by [Theorem 7.3.4](#) (a), and  $\mathbf{S}^{-1} = \begin{pmatrix} 1 & 0 \\ 1 & -1 \end{pmatrix} = \mathbf{S}$  by [Theorem 0.9.8](#). By [Theorem 7.3.2](#),  $\det \mathbf{M} = \det \mathbf{A} \det \mathbf{S} = -1$ .

Step 4: the blocks of  $\mathbf{M}^{-1}$ . Using Step 2,

$$\begin{aligned} -\mathbf{A}^{-1}\mathbf{BS}^{-1} &= -\begin{pmatrix} 1 & -1 \\ -1 & 0 \end{pmatrix} \begin{pmatrix} 1 & 0 \\ 1 & -1 \end{pmatrix} = \begin{pmatrix} 0 & -1 \\ 1 & 0 \end{pmatrix}, \\ -\mathbf{S}^{-1}\mathbf{CA}^{-1} &= -\begin{pmatrix} 1 & 0 \\ 1 & -1 \end{pmatrix} \begin{pmatrix} -1 & 0 \\ -1 & 1 \end{pmatrix} = \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix}, \end{aligned}$$

and  $\mathbf{A}^{-1} + \mathbf{A}^{-1}\mathbf{BS}^{-1}\mathbf{CA}^{-1} = \mathbf{A}^{-1} + (\mathbf{A}^{-1}\mathbf{BS}^{-1})(\mathbf{CA}^{-1}) = \begin{pmatrix} 2 & -1 \\ -1 & 1 \end{pmatrix} + \begin{pmatrix} 0 & 1 \\ -1 & 0 \end{pmatrix} \begin{pmatrix} -1 & 0 \\ -1 & 1 \end{pmatrix} = \begin{pmatrix} 2 & -1 \\ -1 & 1 \end{pmatrix} + \begin{pmatrix} -1 & 1 \\ 1 & 0 \end{pmatrix} = \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix}$ . Hence

$$\mathbf{M}^{-1} = \left( \begin{array}{cc|cc} 1 & 0 & 0 & -1 \\ 0 & 1 & 1 & 0 \\ \hline 1 & 0 & 1 & 0 \\ 0 & 1 & 1 & -1 \end{array} \right).$$

Check one row: row 1 of  $\mathbf{M}$  times column 1 of  $\mathbf{M}^{-1}$  is  $1 + 0 + 0 + 0 = 1$ , and times column 3 is  $0 + 1 + 0 - 1 = 0$ . All the work was with  $2 \times 2$  matrices, and only two of them had to be inverted.

*Remark.* When  $\mathbf{M}$  is real symmetric,  $\mathbf{M}/\mathbf{A}$  is again symmetric, and Chapter 12 uses Schur complements to test whether  $\mathbf{M}$  is positive definite one block at a time. Chapter 23 reads Gaussian elimination itself as a sequence of Schur complements of  $1 \times 1$  pivots, which is the point of view behind block algorithms in numerical linear algebra.

### 7.3.5 The Sherman–Morrison–Woodbury formula

Chapter 6 proved the Matrix Determinant Lemma,  $\det(\mathbf{I}_n + \mathbf{u}\mathbf{v}^\top) = 1 + \mathbf{v}^\top\mathbf{u}$  ([Theorem 6.8.5](#)), by bordering  $\mathbf{I}_n + \mathbf{u}\mathbf{v}^\top$  into a larger block matrix. There is a matching statement about inverses: if we already know  $\mathbf{A}^{-1}$  and change  $\mathbf{A}$  by a low-rank term  $\mathbf{UCV}$ , the new inverse should cost only a small inversion. The trick is to find one block matrix that has  $\mathbf{A}$  in one corner and  $\mathbf{A} + \mathbf{UCV}$  as a Schur complement, and then to compute its inverse twice.

#### Theorem 7.3.5 — Sherman–Morrison–Woodbury Formula

Let  $n, k \geq 1$ , let  $\mathbf{A} \in M_n(F)$  and  $\mathbf{C} \in M_k(F)$  be invertible, and let  $\mathbf{U} \in M_{n \times k}(F)$  and  $\mathbf{V} \in M_{k \times n}(F)$ . Put  $\mathbf{W} = \mathbf{C}^{-1} + \mathbf{VA}^{-1}\mathbf{U} \in M_k(F)$ .

- (a)  $\mathbf{A} + \mathbf{UCV}$  is invertible if and only if  $\mathbf{W}$  is invertible.  
 (b) In that case

$$\begin{aligned} (\mathbf{A} + \mathbf{UCV})^{-1} &= \mathbf{A}^{-1} - \mathbf{A}^{-1}\mathbf{U}\mathbf{W}^{-1}\mathbf{VA}^{-1} \\ &= \mathbf{A}^{-1} - \mathbf{A}^{-1}\mathbf{U}(\mathbf{C}^{-1} + \mathbf{VA}^{-1}\mathbf{U})^{-1}\mathbf{VA}^{-1}. \end{aligned}$$

- (c)  $\det(\mathbf{A} + \mathbf{UCV}) = \det \mathbf{A} \cdot \det \mathbf{C} \cdot \det \mathbf{W}$ .

*Idea.* Look for a block matrix whose two Schur complements are  $\mathbf{A} + \mathbf{UCV}$  and something involving  $\mathbf{W}$ . The complement of the lower-right corner of  $\mathbf{M} = \begin{pmatrix} \mathbf{A} & \mathbf{U} \\ \mathbf{V} & \mathbf{D} \end{pmatrix}$  is  $\mathbf{A} - \mathbf{UD}^{-1}\mathbf{V}$ , which is  $\mathbf{A} + \mathbf{UCV}$  when  $\mathbf{D}^{-1} = -\mathbf{C}$ , that is,  $\mathbf{D} = -\mathbf{C}^{-1}$ . The complement of the upper-left corner is then  $-\mathbf{C}^{-1} - \mathbf{VA}^{-1}\mathbf{U} = -\mathbf{W}$ . Now [Theorem 7.3.4](#) computes the upper-left block of  $\mathbf{M}^{-1}$  in two ways: as  $(\mathbf{M}/\mathbf{D})^{-1} = (\mathbf{A} + \mathbf{UCV})^{-1}$ , and from the formula in terms of  $\mathbf{A}$  and  $\mathbf{M}/\mathbf{A}$ . Setting them equal is the Woodbury formula. The determinant formula of the previous subsection, used twice, gives (c).

*Proof.* Let  $\mathbf{D} = -\mathbf{C}^{-1}$ , which is invertible with  $\mathbf{D}^{-1} = -\mathbf{C}$  by [Theorem 0.9.7](#) (2) and (5), and let  $\mathbf{M} = \begin{pmatrix} \mathbf{A} & \mathbf{U} \\ \mathbf{V} & \mathbf{D} \end{pmatrix} \in M_{n+k}(F)$ . By [Definition 7.3.1](#),

$$\begin{aligned} \mathbf{M}/\mathbf{A} &= \mathbf{D} - \mathbf{VA}^{-1}\mathbf{U} = -\mathbf{C}^{-1} - \mathbf{VA}^{-1}\mathbf{U} = -\mathbf{W}, \\ \mathbf{M}/\mathbf{D} &= \mathbf{A} - \mathbf{UD}^{-1}\mathbf{V} = \mathbf{A} + \mathbf{UCV}. \end{aligned}$$

- (a) By [Theorem 7.3.4](#) (a) and (b),  $\mathbf{M}$  is invertible if and only if  $-\mathbf{W}$  is, and if and only if  $\mathbf{A} + \mathbf{UCV}$  is. Since  $-\mathbf{W}$  is invertible exactly when  $\mathbf{W}$  is ([Theorem 0.9.7](#) (5)), this proves (a).  
 (b) Suppose  $\mathbf{W}$  is invertible, so  $\mathbf{M}$  is invertible, and  $(-\mathbf{W})^{-1} = -\mathbf{W}^{-1}$ . The upper-left  $n \times n$  block of  $\mathbf{M}^{-1}$  is  $(\mathbf{M}/\mathbf{D})^{-1} = (\mathbf{A} + \mathbf{UCV})^{-1}$  by [Theorem 7.3.4](#) (b). By [Theorem 7.3.4](#) (a), applied to  $\mathbf{M}$ , whose blocks are  $\mathbf{A}, \mathbf{U}, \mathbf{V}, \mathbf{D}$ , and with  $\mathbf{S} = \mathbf{M}/\mathbf{A} = -\mathbf{W}$ , the same block is

$$\mathbf{A}^{-1} + \mathbf{A}^{-1}\mathbf{U}(-\mathbf{W})^{-1}\mathbf{VA}^{-1} = \mathbf{A}^{-1} - \mathbf{A}^{-1}\mathbf{U}\mathbf{W}^{-1}\mathbf{VA}^{-1}.$$

The two expressions are the same block of the same matrix, so they are equal.

- (c) By [Theorem 7.3.2](#) (a) and (b),  $\det \mathbf{A} \det(-\mathbf{W}) = \det \mathbf{M} = \det \mathbf{D} \det(\mathbf{A} + \mathbf{UCV})$ . Multiplying a  $k \times k$  matrix by  $-1$  multiplies each of its  $k$  rows by  $-1$ , so by [Theorem 6.5.1](#)  $\det(-\mathbf{W}) = (-1)^k \det \mathbf{W}$  and  $\det \mathbf{D} = (-1)^k \det(\mathbf{C}^{-1}) = (-1)^k (\det \mathbf{C})^{-1}$ , using [Corollary 6.5.6](#). Hence  $\det \mathbf{A} (-1)^k \det \mathbf{W} = (-1)^k (\det \mathbf{C})^{-1} \det(\mathbf{A} + \mathbf{UCV})$ . Multiplying both sides by  $(-1)^k \det \mathbf{C}$  gives (c). ■

The formula is useful when  $k$  is much smaller than  $n$  and  $\mathbf{A}^{-1}$  is already known or easy (diagonal, say): the only new inversion is of the  $k \times k$  matrix  $\mathbf{W}$ . The most common case is a rank-one update,  $k = 1$  and  $\mathbf{C} = (1)$ .

### Corollary 7.3.6 — Sherman–Morrison Formula

Let  $\mathbf{A} \in M_n(F)$  be invertible and  $\mathbf{u}, \mathbf{v} \in F^n$ . Then  $\mathbf{A} + \mathbf{u}\mathbf{v}^\top$  is invertible if and only if  $1 + \mathbf{v}^\top \mathbf{A}^{-1} \mathbf{u} \neq 0$ , and in that case

$$(\mathbf{A} + \mathbf{u}\mathbf{v}^\top)^{-1} = \mathbf{A}^{-1} - \frac{\mathbf{A}^{-1} \mathbf{u} \mathbf{v}^\top \mathbf{A}^{-1}}{1 + \mathbf{v}^\top \mathbf{A}^{-1} \mathbf{u}}.$$

Moreover  $\det(\mathbf{A} + \mathbf{u}\mathbf{v}^\top) = \det \mathbf{A} (1 + \mathbf{v}^\top \mathbf{A}^{-1} \mathbf{u})$ .

*Proof.* Apply [Theorem 7.3.5](#) with  $k = 1$ ,  $\mathbf{C} = (1)$ ,  $\mathbf{U} = \mathbf{u}$  and  $\mathbf{V} = \mathbf{v}^\top$ . Then  $\mathbf{W} = (1 + \mathbf{v}^\top \mathbf{A}^{-1} \mathbf{u})$  is a  $1 \times 1$  matrix, invertible exactly when its entry is non-zero, with inverse the reciprocal. Multiplying by a  $1 \times 1$  matrix is multiplying by its entry, which gives the displayed formula; and  $\det \mathbf{W}$  is the entry itself. ■

For example, over  $\mathbb{Q}$  let  $\mathbf{A} = \text{diag}(1, -1)$ , so  $\mathbf{A}^{-1} = \mathbf{A}$ , and  $\mathbf{u} = (1, 1)$ . For  $\mathbf{v} = (2, 1)$ ,  $\mathbf{A}^{-1} \mathbf{u} = (1, -1)$ ,  $\mathbf{v}^\top \mathbf{A}^{-1} = (2 \ -1)$  and  $1 + \mathbf{v}^\top \mathbf{A}^{-1} \mathbf{u} = 1 + (2 - 1) = 2 \neq 0$ . So  $\mathbf{A} + \mathbf{u}\mathbf{v}^\top = \begin{pmatrix} 3 & 1 \\ 2 & 0 \end{pmatrix}$  has determinant  $\det \mathbf{A} \cdot 2 = -2$ , and

$$\begin{aligned} (\mathbf{A} + \mathbf{u}\mathbf{v}^\top)^{-1} &= \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix} - \frac{1}{2} \begin{pmatrix} 1 \\ -1 \end{pmatrix} (2 \ -1) \\ &= \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix} - \begin{pmatrix} 1 & -\frac{1}{2} \\ -1 & \frac{1}{2} \end{pmatrix} = \begin{pmatrix} 0 & \frac{1}{2} \\ 1 & -\frac{3}{2} \end{pmatrix}, \end{aligned}$$

which agrees with [Theorem 0.9.8](#). For  $\mathbf{v} = (1, 2)$  instead,  $1 + \mathbf{v}^\top \mathbf{A}^{-1} \mathbf{u} = 1 + (1 - 2) = 0$ , and indeed  $\mathbf{A} + \mathbf{u}\mathbf{v}^\top = \begin{pmatrix} 2 & 2 \\ 1 & 1 \end{pmatrix}$  is not invertible.

With  $\mathbf{A} = \mathbf{I}_n$ , the determinant statement is the Matrix Determinant Lemma of Chapter 6, now obtained as a special case of a formula about general low-rank updates, and it matches [Exercise \(C3: Rank-one updates of an invertible matrix\)](#).

### 7.3.6 Commuting blocks

Section 1 warned that  $\det \begin{pmatrix} \mathbf{A} & \mathbf{B} \\ \mathbf{C} & \mathbf{D} \end{pmatrix} \neq \det(\mathbf{AD} - \mathbf{BC})$  in general. With the Schur complement we can see what goes wrong and what extra hypothesis repairs it. Take all four blocks in  $M_n(F)$  and suppose  $\mathbf{D}$  is invertible. By [Theorem 7.3.2](#) (b),

$$\begin{aligned} \det \mathbf{M} &= \det \mathbf{D} \det(\mathbf{A} - \mathbf{BD}^{-1} \mathbf{C}) \\ &= \det((\mathbf{A} - \mathbf{BD}^{-1} \mathbf{C}) \mathbf{D}) = \det(\mathbf{AD} - \mathbf{BD}^{-1} \mathbf{CD}), \end{aligned}$$

using [Theorem 6.5.4](#). If  $\mathbf{C}$  and  $\mathbf{D}$  commute, then  $\mathbf{D}^{-1} \mathbf{CD} = \mathbf{D}^{-1} \mathbf{DC} = \mathbf{C}$ , and the right side is  $\det(\mathbf{AD} - \mathbf{BC})$ . So commuting is exactly what lets  $\mathbf{D}^{-1}$  cancel. The real question is how to remove the hypothesis that  $\mathbf{D}$  is invertible, which is not needed for the conclusion to make sense.

Over  $\mathbb{R}$  one could perturb:  $\mathbf{D} + t\mathbf{I}_n$  is invertible for all but finitely many  $t$ , both sides are polynomials in  $t$ , and polynomials that agree at infinitely many points are equal. Over a finite field such as  $\mathbb{F}_2$  there may be no invertible perturbation at all. We use instead a perturbation



by an **indeterminate**: replace  $\mathbf{D}$  by  $\mathbf{D} + x\mathbf{I}_n$ , a matrix with entries in  $F[x]$ . Its determinant is a monic polynomial, hence not the zero polynomial, and that is all the cancellation needs. First we record that evaluating polynomial entries commutes with matrix operations.

For  $\mathbf{N} = (n_{ij})$  with entries in  $F[x]$  and  $c \in F$ , write  $\mathbf{N}(c) = (n_{ij}(c))$  for the matrix over  $F$  obtained by evaluating every entry at  $c$ . A matrix over  $F$  is a matrix over  $F[x]$  with constant entries, and then  $\mathbf{N}(c) = \mathbf{N}$ .

### Lemma 7.3.7 — Evaluating a Matrix of Polynomials

Let  $c \in F$ , and let  $\mathbf{N}, \mathbf{N}'$  be matrices with entries in  $F[x]$  of sizes for which the operations below are defined. Then  $(\mathbf{N} + \mathbf{N}')(c) = \mathbf{N}(c) + \mathbf{N}'(c)$ ,  $(\mathbf{N}\mathbf{N}')(c) = \mathbf{N}(c)\mathbf{N}'(c)$ , and, for square  $\mathbf{N}$ ,  $(\det \mathbf{N})(c) = \det(\mathbf{N}(c))$ .

*Proof.* By [Theorem 0.8.5](#), evaluation at  $c$  sends sums of polynomials to sums, products to products, and constants to themselves. The  $(i, j)$ -entry of  $\mathbf{N} + \mathbf{N}'$  is  $n_{ij} + n'_{ij}$  and that of  $\mathbf{N}\mathbf{N}'$  is  $\sum_l n_{il}n'_{lj}$ , so evaluating gives the corresponding entries of  $\mathbf{N}(c) + \mathbf{N}'(c)$  and  $\mathbf{N}(c)\mathbf{N}'(c)$ . The determinant  $\det \mathbf{N}$  is the Leibniz sum  $\sum_{\sigma} \operatorname{sgn}(\sigma) \prod_j n_{\sigma(j)j}$  of [Definition 6.4.1](#), computed in  $F[x]$ , and evaluating it term by term gives the Leibniz sum for  $\mathbf{N}(c)$ . This is the argument of [Lemma 6.10.1](#). ■

### Theorem 7.3.8 — Determinant with Commuting Blocks

Let  $n \geq 1$  and  $\mathbf{A}, \mathbf{B}, \mathbf{C}, \mathbf{D} \in M_n(F)$  over **any** field  $F$ . If  $\mathbf{CD} = \mathbf{DC}$ , then

$$\det \begin{pmatrix} \mathbf{A} & \mathbf{B} \\ \mathbf{C} & \mathbf{D} \end{pmatrix} = \det(\mathbf{AD} - \mathbf{BC}).$$

*Idea.* ① Work over the commutative ring  $F[x]$ , replacing  $\mathbf{D}$  by  $\mathbf{D}_x = \mathbf{D} + x\mathbf{I}_n$ , which still commutes with  $\mathbf{C}$ . ② Multiply  $\mathbf{M}_x = \begin{pmatrix} \mathbf{A} & \mathbf{B} \\ \mathbf{C} & \mathbf{D}_x \end{pmatrix}$  on the right by  $\begin{pmatrix} \mathbf{D}_x & 0 \\ -\mathbf{C} & \mathbf{I}_n \end{pmatrix}$ : the lower-left block becomes  $\mathbf{CD}_x - \mathbf{D}_x\mathbf{C} = 0$  and the upper-left block becomes  $\mathbf{AD}_x - \mathbf{BC}$ . This is the Schur computation above with  $\mathbf{D}^{-1}$  cleared from the denominators. ③ Take determinants:  $\det \mathbf{M}_x \det \mathbf{D}_x = \det(\mathbf{AD}_x - \mathbf{BC}) \det \mathbf{D}_x$ . ④ Cancel  $\det \mathbf{D}_x$ , which is legal because it is a non-zero polynomial and  $F[x]$  has no zero divisors. ⑤ Set  $x = 0$ . Each step uses only determinant facts that Chapter 6 proved over commutative rings.

*Proof.* Regard  $\mathbf{A}, \mathbf{B}, \mathbf{C}, \mathbf{D}$  as matrices over  $F[x]$  with constant entries, and let  $\mathbf{D}_x = \mathbf{D} + x\mathbf{I}_n$ . The proofs of [Theorem 0.9.4](#) use only the ring axioms, so they hold for matrices over  $F[x]$ ; in particular

$$\mathbf{CD}_x = \mathbf{CD} + x\mathbf{C} = \mathbf{DC} + x\mathbf{C} = \mathbf{D}_x\mathbf{C},$$

using the hypothesis  $\mathbf{CD} = \mathbf{DC}$ . Let

$$\mathbf{M}_x = \begin{pmatrix} \mathbf{A} & \mathbf{B} \\ \mathbf{C} & \mathbf{D}_x \end{pmatrix}, \quad \mathbf{R} = \begin{pmatrix} \mathbf{D}_x & 0 \\ -\mathbf{C} & \mathbf{I}_n \end{pmatrix},$$

both in  $M_{2n}(F[x])$ . By [Theorem 7.1.1](#), which holds over  $F[x]$  by the remark following it,

$$\mathbf{M}_x \mathbf{R} = \begin{pmatrix} \mathbf{A}\mathbf{D}_x - \mathbf{B}\mathbf{C} & \mathbf{B} \\ \mathbf{C}\mathbf{D}_x - \mathbf{D}_x\mathbf{C} & \mathbf{D}_x \end{pmatrix} = \begin{pmatrix} \mathbf{A}\mathbf{D}_x - \mathbf{B}\mathbf{C} & \mathbf{B} \\ 0 & \mathbf{D}_x \end{pmatrix}.$$

By the remarks in Chapter 6 §5 and §8, [Theorem 6.5.4](#) and [Theorem 6.8.4](#) hold over the commutative ring  $F[x]$ . Hence  $\det(\mathbf{M}_x \mathbf{R}) = \det \mathbf{M}_x \det \mathbf{R}$ , with  $\det \mathbf{R} = \det \mathbf{D}_x \det \mathbf{I}_n = \det \mathbf{D}_x$ , and  $\det(\mathbf{M}_x \mathbf{R}) = \det(\mathbf{A}\mathbf{D}_x - \mathbf{B}\mathbf{C}) \det \mathbf{D}_x$ . So, in  $F[x]$ ,

$$\det \mathbf{M}_x \cdot \det \mathbf{D}_x = \det(\mathbf{A}\mathbf{D}_x - \mathbf{B}\mathbf{C}) \cdot \det \mathbf{D}_x. \quad (*)$$

Now  $\mathbf{D}_x = x\mathbf{I}_n - (-\mathbf{D})$ , so  $\det \mathbf{D}_x = p_{-\mathbf{D}}(x)$  is the characteristic polynomial of  $-\mathbf{D} \in M_n(F)$  ([Definition 6.10.1](#)). By [Theorem 6.10.2](#) it is monic of degree  $n$ , so  $\det \mathbf{D}_x \neq 0$  in  $F[x]$ . By [Corollary 0.8.4](#) (2), we may cancel it from (\*):

$$\det \mathbf{M}_x = \det(\mathbf{A}\mathbf{D}_x - \mathbf{B}\mathbf{C}) \quad \text{in } F[x].$$

Finally evaluate both sides at 0. By [Lemma 7.3.7](#),  $(\det \mathbf{M}_x)(0) = \det(\mathbf{M}_x(0)) = \det \mathbf{M}$ , since  $\mathbf{D}_x(0) = \mathbf{D}$ ; and  $(\mathbf{A}\mathbf{D}_x - \mathbf{B}\mathbf{C})(0) = \mathbf{A}\mathbf{D} - \mathbf{B}\mathbf{C}$ , so  $(\det(\mathbf{A}\mathbf{D}_x - \mathbf{B}\mathbf{C}))(0) = \det(\mathbf{A}\mathbf{D} - \mathbf{B}\mathbf{C})$ . Hence  $\det \mathbf{M} = \det(\mathbf{A}\mathbf{D} - \mathbf{B}\mathbf{C})$ . ■

The cancellation in  $F[x]$  is the heart of the argument, so it is worth being precise about why it is legal. Over  $F$  we cannot divide by  $\det \mathbf{D}$  when it is 0. Over  $F[x]$  we still cannot divide, since  $\det \mathbf{D}_x$  usually has no inverse, but we do not need to: cancellation only needs “ $pq = pr$  and  $p \neq 0$  imply  $q = r$ ”, and this holds in  $F[x]$  because a product of non-zero polynomials is non-zero ([Corollary 0.8.4](#)). The perturbation  $x\mathbf{I}_n$  makes  $\det \mathbf{D}_x$  non-zero **as a polynomial** even when every value  $\det(\mathbf{D} + c\mathbf{I}_n)$  with  $c \in F$  is zero. For example, over  $\mathbb{F}_2$ , let  $\mathbf{D} = \mathbf{E}_{22} = \text{diag}(0, 1)$ . Then  $\det(\mathbf{D} + 0\mathbf{I}_2) = 0$  and  $\det(\mathbf{D} + \mathbf{I}_2) = \det \text{diag}(1, 0) = 0$ , because  $1 + 1 = 0$  in  $\mathbb{F}_2$ ; no scalar shift makes  $\mathbf{D}$  invertible. But  $\det(\mathbf{D} + x\mathbf{I}_2) = x(x + 1) = x^2 + x$  is not the zero polynomial.

For example, over  $\mathbb{Q}$ , let  $\mathbf{A} = \begin{pmatrix} 1 & 2 \\ 0 & 1 \end{pmatrix}$ ,  $\mathbf{B} = \begin{pmatrix} 1 & 0 \\ 1 & 1 \end{pmatrix}$ ,  $\mathbf{D} = \begin{pmatrix} 0 & 1 \\ 0 & 0 \end{pmatrix}$  and  $\mathbf{C} = 2\mathbf{I}_2 + \mathbf{D} = \begin{pmatrix} 2 & 1 \\ 0 & 2 \end{pmatrix}$ , which commutes with  $\mathbf{D}$ . Here  $\mathbf{D}$  is not invertible, so the Schur route is closed. The theorem gives  $\det \mathbf{M} = \det(\mathbf{A}\mathbf{D} - \mathbf{B}\mathbf{C}) = \det \begin{pmatrix} -2 & 0 \\ -2 & -3 \end{pmatrix} = 6$ .

**Warning.** Which blocks commute decides which product appears. The hypothesis  $\mathbf{C}\mathbf{D} = \mathbf{D}\mathbf{C}$  gives  $\det(\mathbf{A}\mathbf{D} - \mathbf{B}\mathbf{C})$ . If instead  $\mathbf{A}$  and  $\mathbf{C}$  commute, the correct formula is  $\det(\mathbf{A}\mathbf{D} - \mathbf{C}\mathbf{B})$  ([Exercise \(C3: When A and C commute\)](#)), with  $\mathbf{C}$  and  $\mathbf{B}$  in the other order. The two can differ: for the blocks  $\mathbf{A} = \mathbf{I}_2$ ,  $\mathbf{B} = \mathbf{E}_{12}$ ,  $\mathbf{C} = \mathbf{E}_{21}$ ,  $\mathbf{D} = \mathbf{E}_{11}$  of the warning in Section 1,  $\mathbf{A}$  commutes with  $\mathbf{C}$  but  $\mathbf{C}$  does not commute with  $\mathbf{D}$ ; indeed  $\det \mathbf{M} = -1 = \det(\mathbf{E}_{11} - \mathbf{E}_{22}) = \det(\mathbf{A}\mathbf{D} - \mathbf{C}\mathbf{B})$ , while  $\det(\mathbf{A}\mathbf{D} - \mathbf{B}\mathbf{C}) = 0$ .

## 7.3.7 Exercises

### 7.3.7.1 A. Check your understanding

#### Exercise A1.

- (a) Define the Schur complement  $\mathbf{M}/\mathbf{A}$ , including the hypothesis on  $\mathbf{A}$  and the size of  $\mathbf{M}/\mathbf{A}$ .
- (b) State the block LDU factorization of  $\mathbf{M}$  in terms of  $\mathbf{A}$  and  $\mathbf{M}/\mathbf{A}$ .
- (c) True or false: if  $\mathbf{M} = \begin{pmatrix} \mathbf{A} & \mathbf{B} \\ \mathbf{C} & \mathbf{D} \end{pmatrix}$  is invertible and  $\mathbf{A}$  is square, then  $\mathbf{A}$  is invertible. Justify your answer.
- (d) If  $\mathbf{A}$  is invertible and  $\mathbf{M}$  is invertible, which block of  $\mathbf{M}^{-1}$  is  $(\mathbf{M}/\mathbf{A})^{-1}$ ?
- (e) True or false: for all  $\mathbf{A}, \mathbf{B}, \mathbf{C}, \mathbf{D} \in M_n(F)$  with  $\mathbf{CD} = \mathbf{DC}$ ,  $\det \begin{pmatrix} \mathbf{A} & \mathbf{B} \\ \mathbf{C} & \mathbf{D} \end{pmatrix} = \det(\mathbf{AD} - \mathbf{BC})$ , even if  $\mathbf{D}$  is not invertible and  $F = \mathbb{F}_2$ . Justify your answer.

*Solution.*

- (a) For  $\mathbf{M} = \begin{pmatrix} \mathbf{A} & \mathbf{B} \\ \mathbf{C} & \mathbf{D} \end{pmatrix}$  with  $\mathbf{A} \in M_k(F)$  invertible,  $\mathbf{B} \in M_{k \times l}(F)$ ,  $\mathbf{C} \in M_{p \times k}(F)$ ,  $\mathbf{D} \in M_{p \times l}(F)$ :  $\mathbf{M}/\mathbf{A} = \mathbf{D} - \mathbf{CA}^{-1}\mathbf{B} \in M_{p \times l}(F)$ , the size of  $\mathbf{D}$  (Definition 7.3.1).
- (b)  $\mathbf{M} = \begin{pmatrix} \mathbf{I}_k & 0 \\ \mathbf{CA}^{-1} & \mathbf{I}_p \end{pmatrix} \begin{pmatrix} \mathbf{A} & 0 \\ 0 & \mathbf{M}/\mathbf{A} \end{pmatrix} \begin{pmatrix} \mathbf{I}_k & \mathbf{A}^{-1}\mathbf{B} \\ 0 & \mathbf{I}_l \end{pmatrix}$  (Theorem 7.3.1).
- (c) False.  $\begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}$  is invertible, but  $\mathbf{A} = (0)$  is not.
- (d) The lower-right block, in the position of  $\mathbf{D}$  (Theorem 7.3.4 (a)).
- (e) True. Theorem 7.3.8 holds over every field and needs no invertibility; its proof cancels the non-zero polynomial  $\det(\mathbf{D} + x\mathbf{I}_n)$  in  $F[x]$ .

### 7.3.7.2 B. Practice

#### Exercise B1: A Schur complement, a determinant and a rank.

Let

$$\mathbf{M} = \left( \begin{array}{cc|cc} 2 & 1 & 1 & 0 \\ 1 & 1 & 3 & 1 \\ \hline 1 & 2 & 9 & 5 \\ 0 & 1 & 6 & 4 \end{array} \right) \in M_4(\mathbb{Q}).$$

Compute  $\mathbf{M}/\mathbf{A}$  for the upper-left block  $\mathbf{A}$ . Hence find  $\det \mathbf{M}$  and  $\text{rank } \mathbf{M}$ , and decide whether  $\mathbf{M}$  is invertible.

*Solution.*  $\mathbf{A} = \begin{pmatrix} 2 & 1 \\ 1 & 1 \end{pmatrix}$  has  $\det \mathbf{A} = 1$ , so  $\mathbf{A}^{-1} = \begin{pmatrix} 1 & -1 \\ -1 & 2 \end{pmatrix}$  (Theorem 0.9.8). With  $\mathbf{B} = \begin{pmatrix} 1 & 0 \\ 3 & 1 \end{pmatrix}$ ,  $\mathbf{C} = \begin{pmatrix} 1 & 2 \\ 0 & 1 \end{pmatrix}$ ,  $\mathbf{D} = \begin{pmatrix} 9 & 5 \\ 6 & 4 \end{pmatrix}$ :

$$\mathbf{CA}^{-1} = \begin{pmatrix} -1 & 3 \\ -1 & 2 \end{pmatrix}, \quad \mathbf{CA}^{-1}\mathbf{B} = \begin{pmatrix} 8 & 3 \\ 5 & 2 \end{pmatrix}, \quad \mathbf{M}/\mathbf{A} = \begin{pmatrix} 1 & 2 \\ 1 & 2 \end{pmatrix}.$$

By Theorem 7.3.2,  $\det \mathbf{M} = \det \mathbf{A} \det(\mathbf{M}/\mathbf{A}) = 1 \cdot 0 = 0$ . By Theorem 7.3.3,  $\text{rank } \mathbf{M} = 2 + \text{rank}(\mathbf{M}/\mathbf{A}) = 2 + 1 = 3$ , since  $\mathbf{M}/\mathbf{A}$  is non-zero with equal rows. So  $\mathbf{M}$  is not invertible (Theorem 2.4.4).

**Exercise B2: A rank-one update.**

Let  $\mathbf{A} = \text{diag}(1, 2, 4) \in M_3(\mathbb{Q})$ ,  $\mathbf{u} = (1, 1, 1)$  and  $\mathbf{v} = (1, -1, 2)$ . Use [Corollary 7.3.6](#) to find  $\det(\mathbf{A} + \mathbf{u}\mathbf{v}^\top)$  and  $(\mathbf{A} + \mathbf{u}\mathbf{v}^\top)^{-1}$ , and check one entry.

*Solution.*  $\mathbf{A}^{-1} = \text{diag}(1, \frac{1}{2}, \frac{1}{4})$ , so  $\mathbf{A}^{-1}\mathbf{u} = (1, \frac{1}{2}, \frac{1}{4})$  and  $\mathbf{v}^\top \mathbf{A}^{-1} = (1 \quad -\frac{1}{2} \quad \frac{1}{2})$ . Then  $\mathbf{v}^\top \mathbf{A}^{-1}\mathbf{u} = 1 - \frac{1}{2} + \frac{1}{2} = 1$ , so  $1 + \mathbf{v}^\top \mathbf{A}^{-1}\mathbf{u} = 2 \neq 0$ . By [Corollary 7.3.6](#),  $\mathbf{A} + \mathbf{u}\mathbf{v}^\top$  is invertible,  $\det(\mathbf{A} + \mathbf{u}\mathbf{v}^\top) = 8 \cdot 2 = 16$ , and

$$\begin{aligned} (\mathbf{A} + \mathbf{u}\mathbf{v}^\top)^{-1} &= \text{diag}\left(1, \frac{1}{2}, \frac{1}{4}\right) - \frac{1}{2} \begin{pmatrix} 1 \\ \frac{1}{2} \\ \frac{1}{4} \end{pmatrix} \begin{pmatrix} 1 & -\frac{1}{2} & \frac{1}{2} \end{pmatrix} \\ &= \begin{pmatrix} 1 & 0 & 0 \\ 0 & \frac{1}{2} & 0 \\ 0 & 0 & \frac{1}{4} \end{pmatrix} - \begin{pmatrix} \frac{1}{2} & -\frac{1}{4} & \frac{1}{4} \\ \frac{1}{4} & -\frac{1}{8} & \frac{1}{8} \\ \frac{1}{8} & -\frac{1}{16} & \frac{1}{16} \end{pmatrix} \\ &= \begin{pmatrix} \frac{1}{2} & \frac{1}{4} & -\frac{1}{4} \\ -\frac{1}{4} & \frac{5}{8} & -\frac{1}{8} \\ -\frac{1}{8} & \frac{1}{16} & \frac{3}{16} \end{pmatrix}. \end{aligned}$$

Check:  $\mathbf{A} + \mathbf{u}\mathbf{v}^\top = \begin{pmatrix} 2 & -1 & 2 \\ 1 & 1 & 2 \\ 1 & -1 & 6 \end{pmatrix}$ , and its row 1 times column 1 of the answer is  $1 + \frac{1}{4} - \frac{1}{4} = 1$ , times column 2 is  $\frac{1}{2} - \frac{5}{8} + \frac{1}{8} = 0$ .

**Exercise B3: Identity corners.**

Let  $\mathbf{B} \in M_{k \times l}(F)$ ,  $\mathbf{C} \in M_{l \times k}(F)$  and  $\mathbf{M} = \begin{pmatrix} \mathbf{I}_k & \mathbf{B} \\ \mathbf{C} & \mathbf{I}_l \end{pmatrix}$ .

- Show that  $\mathbf{M}$  is invertible if and only if  $\mathbf{S} = \mathbf{I}_l - \mathbf{CB}$  is, and write down  $\mathbf{M}^{-1}$  in terms of  $\mathbf{S}^{-1}$ .
- Deduce the inverse of  $\begin{pmatrix} \mathbf{I}_k & \mathbf{B} \\ 0 & \mathbf{I}_l \end{pmatrix}$ .
- For  $\mathbf{B} = \begin{pmatrix} 1 \\ 2 \end{pmatrix}$  and  $\mathbf{C} = (1 \quad 1)$  over  $\mathbb{Q}$ , compute  $\mathbf{M}^{-1}$ .

*Solution.*

- The block  $\mathbf{I}_k$  is invertible and  $\mathbf{M}/\mathbf{I}_k = \mathbf{I}_l - \mathbf{C}\mathbf{I}_k^{-1}\mathbf{B} = \mathbf{I}_l - \mathbf{CB}$ . By [Theorem 7.3.4](#) (a),  $\mathbf{M}$  is invertible if and only if  $\mathbf{I}_l - \mathbf{CB}$  is, and then, with  $\mathbf{A}^{-1} = \mathbf{I}_k$ ,

$$\mathbf{M}^{-1} = \begin{pmatrix} \mathbf{I}_k + \mathbf{BS}^{-1}\mathbf{C} & -\mathbf{BS}^{-1} \\ -\mathbf{S}^{-1}\mathbf{C} & \mathbf{S}^{-1} \end{pmatrix}.$$

- With  $\mathbf{C} = 0$ ,  $\mathbf{S} = \mathbf{I}_l - \mathbf{CB} = \mathbf{I}_l$  and  $\mathbf{S}^{-1} = \mathbf{I}_l$ , so the inverse is  $\begin{pmatrix} \mathbf{I}_k & -\mathbf{B} \\ 0 & \mathbf{I}_l \end{pmatrix}$ , in agreement with [Theorem 7.1.5](#).

- $\mathbf{CB} = (3)$ , so  $\mathbf{S} = \mathbf{I}_1 - \mathbf{CB} = (-2)$  and  $\mathbf{S}^{-1} = (-\frac{1}{2})$ . Then  $\mathbf{BS}^{-1}\mathbf{C} = -\frac{1}{2} \begin{pmatrix} 1 & 1 \\ 2 & 2 \end{pmatrix}$ ,

$-\mathbf{B}\mathbf{S}^{-1} = \begin{pmatrix} \frac{1}{2} \\ 1 \end{pmatrix}$  and  $-\mathbf{S}^{-1}\mathbf{C} = \begin{pmatrix} \frac{1}{2} & \frac{1}{2} \end{pmatrix}$ , so

$$\mathbf{M}^{-1} = \begin{pmatrix} \frac{1}{2} & -\frac{1}{2} & \frac{1}{2} \\ -1 & 0 & 1 \\ \frac{1}{2} & \frac{1}{2} & -\frac{1}{2} \end{pmatrix}, \quad \mathbf{M} = \begin{pmatrix} 1 & 0 & 1 \\ 0 & 1 & 2 \\ 1 & 1 & 1 \end{pmatrix}.$$

Check: row 3 of  $\mathbf{M}$  times column 3 of  $\mathbf{M}^{-1}$  is  $\frac{1}{2} + 1 - \frac{1}{2} = 1$ , and times column 1 is  $\frac{1}{2} - 1 + \frac{1}{2} = 0$ .

### 7.3.7.3 C. Going deeper

#### Exercise C1: Sylvester's determinant identity.

Let  $\mathbf{A} \in M_{m \times n}(F)$  and  $\mathbf{B} \in M_{n \times m}(F)$  with  $m, n \geq 1$ .

- (a) Compute both Schur complements of the identity blocks in  $\mathbf{M} = \begin{pmatrix} \mathbf{I}_m & -\mathbf{A} \\ \mathbf{B} & \mathbf{I}_n \end{pmatrix}$ , and deduce

$$\det(\mathbf{I}_m + \mathbf{A}\mathbf{B}) = \det(\mathbf{I}_n + \mathbf{B}\mathbf{A}).$$

- (b) Deduce that for every non-zero  $c \in F$ ,  $\det(c\mathbf{I}_m - \mathbf{A}\mathbf{B}) = c^{m-n} \det(c\mathbf{I}_n - \mathbf{B}\mathbf{A})$ .  
(c) Deduce [Theorem 6.8.5](#) from (a).

*Solution.*

- (a) Both identity blocks are invertible. By [Definition 7.3.1](#),

$$\begin{aligned} \mathbf{M}/\mathbf{I}_m &= \mathbf{I}_n - \mathbf{B}\mathbf{I}_m^{-1}(-\mathbf{A}) = \mathbf{I}_n + \mathbf{B}\mathbf{A}, \\ \mathbf{M}/\mathbf{I}_n &= \mathbf{I}_m - (-\mathbf{A})\mathbf{I}_n^{-1}\mathbf{B} = \mathbf{I}_m + \mathbf{A}\mathbf{B}. \end{aligned}$$

By [Theorem 7.3.2](#) (a) and (b),  $\det \mathbf{M} = \det \mathbf{I}_m \det(\mathbf{I}_n + \mathbf{B}\mathbf{A})$  and  $\det \mathbf{M} = \det \mathbf{I}_n \det(\mathbf{I}_m + \mathbf{A}\mathbf{B})$ . Hence  $\det(\mathbf{I}_m + \mathbf{A}\mathbf{B}) = \det(\mathbf{I}_n + \mathbf{B}\mathbf{A})$ .

- (b) Let  $\mathbf{A}' = -c^{-1}\mathbf{A}$ . Then  $c\mathbf{I}_m - \mathbf{A}\mathbf{B} = c(\mathbf{I}_m + \mathbf{A}'\mathbf{B})$  and  $c\mathbf{I}_n - \mathbf{B}\mathbf{A} = c(\mathbf{I}_n + \mathbf{B}\mathbf{A}')$ . Multiplying an  $m \times m$  matrix by  $c$  multiplies each of its  $m$  rows by  $c$ , so it multiplies the determinant by  $c^m$  ([Theorem 6.5.1](#)). By (a) applied to  $\mathbf{A}'$  and  $\mathbf{B}$ ,

$$\begin{aligned} \det(c\mathbf{I}_m - \mathbf{A}\mathbf{B}) &= c^m \det(\mathbf{I}_m + \mathbf{A}'\mathbf{B}) = c^m \det(\mathbf{I}_n + \mathbf{B}\mathbf{A}') \\ &= c^m c^{-n} \det(c\mathbf{I}_n - \mathbf{B}\mathbf{A}) = c^{m-n} \det(c\mathbf{I}_n - \mathbf{B}\mathbf{A}). \end{aligned}$$

- (c) Let  $\mathbf{u}, \mathbf{v} \in F^n$ , and apply (a) with the sizes  $m, n$  there replaced by  $n, 1$ :  $\mathbf{A} = \mathbf{u} \in M_{n \times 1}(F)$  and  $\mathbf{B} = \mathbf{v}^\top \in M_{1 \times n}(F)$ . Then  $\det(\mathbf{I}_n + \mathbf{u}\mathbf{v}^\top) = \det(\mathbf{I}_1 + \mathbf{v}^\top \mathbf{u}) = 1 + \mathbf{v}^\top \mathbf{u}$ , because the determinant of a  $1 \times 1$  matrix is its entry.

#### Exercise C2: Pivots as Schur complements.

Let  $\mathbf{A} \in M_n(F)$ , and for  $1 \leq k \leq n$  let  $\mathbf{A}_k$  be its leading principal submatrix (the top-left  $k \times k$  corner), as in [Exercise \(C1: Leading principal submatrices\)](#).

- (a) Suppose  $\mathbf{A}_{k-1}$  is invertible, where  $2 \leq k \leq n$ , and write  $\mathbf{A}_k = \begin{pmatrix} \mathbf{A}_{k-1} & \mathbf{b} \\ \mathbf{c}^\top & d \end{pmatrix}$ . Show

- that  $s_k := \mathbf{A}_k / \mathbf{A}_{k-1}$  is the scalar  $d - \mathbf{c}^\top \mathbf{A}_{k-1}^{-1} \mathbf{b}$ , and that  $\det \mathbf{A}_k = \det \mathbf{A}_{k-1} \cdot s_k$ .
- (b) Suppose  $\mathbf{A}_1, \dots, \mathbf{A}_n$  are all invertible, and put  $s_1 = a_{11}$ . Prove that  $\det \mathbf{A}_k = s_1 s_2 \cdots s_k$  for every  $k$ , and that every  $s_k$  is non-zero.
- (c) Keep the hypothesis of (b). In the proof of [Exercise \(C1: Leading principal submatrices\)](#) (c), the last pivot is  $z = d - \mathbf{x}^\top \mathbf{y}$  with  $\mathbf{y} = \mathbf{L}'^{-1} \mathbf{b}$  and  $\mathbf{x}^\top = \mathbf{c}^\top \mathbf{U}'^{-1}$ . Show that  $z = \mathbf{A}_n / \mathbf{A}_{n-1}$ . Deduce that the diagonal entries of  $\mathbf{U}$  in the LU factorization are  $u_{kk} = \det \mathbf{A}_k / \det \mathbf{A}_{k-1}$ , with  $\det \mathbf{A}_0 := 1$ .
- (d) Check (c) for  $\mathbf{A} = \begin{pmatrix} 2 & 1 & 1 \\ 4 & 3 & 3 \\ 8 & 7 & 9 \end{pmatrix}$  over  $\mathbb{Q}$ .

*Solution.*

- (a) With the partition  $k-1, 1$ , [Definition 7.3.1](#) gives the  $1 \times 1$  matrix  $\mathbf{A}_k / \mathbf{A}_{k-1} = d - \mathbf{c}^\top \mathbf{A}_{k-1}^{-1} \mathbf{b}$ , which we identify with its entry. By [Theorem 7.3.2](#) (a),  $\det \mathbf{A}_k = \det \mathbf{A}_{k-1} \det(s_k) = \det \mathbf{A}_{k-1} \cdot s_k$ .
- (b) By induction on  $k$ . For  $k = 1$ ,  $\det \mathbf{A}_1 = a_{11} = s_1$ . If  $\det \mathbf{A}_{k-1} = s_1 \cdots s_{k-1}$ , then by (a), which applies since  $\mathbf{A}_{k-1}$  is invertible,  $\det \mathbf{A}_k = s_1 \cdots s_{k-1} s_k$ . Since  $\mathbf{A}_k$  is invertible,  $\det \mathbf{A}_k \neq 0$  ([Theorem 6.5.2](#)), so no factor  $s_k$  is zero ([Theorem 0.7.1](#) (d)).
- (c)  $\mathbf{A}_{n-1} = \mathbf{L}' \mathbf{U}'$  with both factors invertible, so  $\mathbf{A}_{n-1}^{-1} = \mathbf{U}'^{-1} \mathbf{L}'^{-1}$  ([Theorem 0.9.7](#) (3)) and  $\mathbf{x}^\top \mathbf{y} = \mathbf{c}^\top \mathbf{U}'^{-1} \mathbf{L}'^{-1} \mathbf{b} = \mathbf{c}^\top \mathbf{A}_{n-1}^{-1} \mathbf{b}$ . Hence  $z = d - \mathbf{c}^\top \mathbf{A}_{n-1}^{-1} \mathbf{b} = \mathbf{A}_n / \mathbf{A}_{n-1} = s_n$  by (a). The same construction applied to  $\mathbf{A}_k$  (whose LU factors are the top-left corners  $\mathbf{L}_k, \mathbf{U}_k$  of  $\mathbf{L}, \mathbf{U}$ , by [Exercise \(C1: Leading principal submatrices\)](#) (a) and [Theorem 2.6.5](#)) shows  $u_{kk} = s_k$  for every  $k$ . By (a) and (b),  $s_k = \det \mathbf{A}_k / \det \mathbf{A}_{k-1}$ , with  $s_1 = \det \mathbf{A}_1 / 1$ .
- (d)  $\det \mathbf{A}_1 = 2$ ,  $\det \mathbf{A}_2 = 6 - 4 = 2$ , and expanding,  $\det \mathbf{A}_3 = 2(27 - 21) - 1(36 - 24) + 1(28 - 24) = 12 - 12 + 4 = 4$ . The predicted pivots are 2,  $2/2 = 1$ ,  $4/2 = 2$ . Elimination gives  $\mathbf{L} = \begin{pmatrix} 1 & 0 & 0 \\ 2 & 1 & 0 \\ 4 & 3 & 1 \end{pmatrix}$  and  $\mathbf{U} = \begin{pmatrix} 2 & 1 & 1 \\ 0 & 1 & 1 \\ 0 & 0 & 2 \end{pmatrix}$ : subtract 2 and 4 times row 1 to get rows  $(0, 1, 1)$  and  $(0, 3, 5)$ , then 3 times the new row 2 to get  $(0, 0, 2)$ . The diagonal of  $\mathbf{U}$  is 2, 1, 2, as predicted.

### Exercise C3: When $\mathbf{A}$ and $\mathbf{C}$ commute.

Let  $n \geq 1$  and  $\mathbf{A}, \mathbf{B}, \mathbf{C}, \mathbf{D} \in M_n(F)$  with  $\mathbf{AC} = \mathbf{CA}$ , and  $\mathbf{M} = \begin{pmatrix} \mathbf{A} & \mathbf{B} \\ \mathbf{C} & \mathbf{D} \end{pmatrix}$ .

- (a) Assume  $\mathbf{A}$  is invertible. Use [Theorem 7.3.2](#) to prove  $\det \mathbf{M} = \det(\mathbf{AD} - \mathbf{CB})$ .
- (b) Remove the assumption that  $\mathbf{A}$  is invertible, over any field, by adapting the proof of [Theorem 7.3.8](#) with  $\mathbf{A}_x = \mathbf{A} + x\mathbf{I}_n$ .
- (c) For  $\mathbf{A} = \mathbf{I}_2$ ,  $\mathbf{B} = \mathbf{E}_{12}$ ,  $\mathbf{C} = \mathbf{E}_{21}$ ,  $\mathbf{D} = \mathbf{E}_{11}$  in  $M_2(\mathbb{Q})$ , check that  $\mathbf{AC} = \mathbf{CA}$ , and compare  $\det \mathbf{M}$ ,  $\det(\mathbf{AD} - \mathbf{CB})$  and  $\det(\mathbf{AD} - \mathbf{BC})$ .

*Hint for (b): multiply  $\mathbf{M}_x$  on the left by a block matrix that clears the lower-left block.*

*Solution.*

- (a) By [Theorem 7.3.2](#) (a) and [Theorem 6.5.4](#),

$$\begin{aligned}\det \mathbf{M} &= \det \mathbf{A} \det(\mathbf{D} - \mathbf{CA}^{-1}\mathbf{B}) \\ &= \det(\mathbf{A}(\mathbf{D} - \mathbf{CA}^{-1}\mathbf{B})) = \det(\mathbf{AD} - \mathbf{ACA}^{-1}\mathbf{B}).\end{aligned}$$

Since  $\mathbf{AC} = \mathbf{CA}$ ,  $\mathbf{ACA}^{-1} = \mathbf{CAA}^{-1} = \mathbf{C}$ , so  $\det \mathbf{M} = \det(\mathbf{AD} - \mathbf{CB})$ .

- (b) Let  $\mathbf{A}_x = \mathbf{A} + x\mathbf{I}_n$  over  $F[x]$ . Then  $\mathbf{A}_x\mathbf{C} = \mathbf{AC} + x\mathbf{C} = \mathbf{CA} + x\mathbf{C} = \mathbf{CA}_x$ . Let  $\mathbf{M}_x = \begin{pmatrix} \mathbf{A}_x & \mathbf{B} \\ \mathbf{C} & \mathbf{D} \end{pmatrix}$  and  $\mathbf{L} = \begin{pmatrix} \mathbf{I}_n & 0 \\ -\mathbf{C} & \mathbf{A}_x \end{pmatrix}$ . By [Theorem 7.1.1](#) over  $F[x]$ ,

$$\mathbf{LM}_x = \begin{pmatrix} \mathbf{A}_x & \mathbf{B} \\ -\mathbf{CA}_x + \mathbf{A}_x\mathbf{C} & -\mathbf{CB} + \mathbf{A}_x\mathbf{D} \end{pmatrix} = \begin{pmatrix} \mathbf{A}_x & \mathbf{B} \\ 0 & \mathbf{A}_x\mathbf{D} - \mathbf{CB} \end{pmatrix}.$$

By [Theorem 6.5.4](#) and [Theorem 6.8.4](#) over  $F[x]$  (Chapter 6 §5 and §8),  $\det \mathbf{A}_x \det \mathbf{M}_x = \det \mathbf{A}_x \det(\mathbf{A}_x\mathbf{D} - \mathbf{CB})$ , using  $\det \mathbf{L} = \det \mathbf{I}_n \det \mathbf{A}_x$ . As in the proof of [Theorem 7.3.8](#),  $\det \mathbf{A}_x = p_{-\mathbf{A}}(x)$  is monic, hence non-zero, and [Corollary 0.8.4](#) (2) gives  $\det \mathbf{M}_x = \det(\mathbf{A}_x\mathbf{D} - \mathbf{CB})$ . Evaluating at  $x = 0$  with [Lemma 7.3.7](#) gives  $\det \mathbf{M} = \det(\mathbf{AD} - \mathbf{CB})$ .

- (c)  $\mathbf{AC} = \mathbf{C} = \mathbf{CA}$ , since  $\mathbf{A} = \mathbf{I}_2$ . As computed in Section 1,  $\det \mathbf{M} = -1$ . Also  $\mathbf{AD} - \mathbf{CB} = \mathbf{E}_{11} - \mathbf{E}_{21}\mathbf{E}_{12} = \mathbf{E}_{11} - \mathbf{E}_{22} = \text{diag}(1, -1)$ , with determinant  $-1$ , in agreement with (b). But  $\mathbf{AD} - \mathbf{BC} = \mathbf{E}_{11} - \mathbf{E}_{12}\mathbf{E}_{21} = \mathbf{E}_{11} - \mathbf{E}_{11} = 0$ , with determinant 0. So with  $\mathbf{AC} = \mathbf{CA}$  the product must be taken as  $\mathbf{CB}$ , not  $\mathbf{BC}$ ; here  $\mathbf{CD} = \mathbf{E}_{21} \neq 0 = \mathbf{DC}$ , so [Theorem 7.3.8](#) does not apply.

## 7.4 The Kronecker Product

So far the unknown in a matrix equation has been a vector:  $\mathbf{Ax} = \mathbf{b}$ . Many natural questions have a matrix as the unknown instead. Which matrices commute with a given  $\mathbf{A}$ ? Which  $\mathbf{X}$  solve  $\mathbf{AX} + \mathbf{XB} = \mathbf{C}$ ? Such an equation is still linear in the entries of  $\mathbf{X}$ , so in principle it is an ordinary linear system. The trouble is bookkeeping: we need to know what the coefficient matrix of that system is. This section builds it from  $\mathbf{A}$  and  $\mathbf{B}$  with one operation, the Kronecker product, and turns matrix equations into systems we already know how to solve.

### 7.4.1 From a matrix equation to a linear system

Take  $\mathbf{A}, \mathbf{B} \in M_2(F)$  and an unknown  $\mathbf{X} = (x_{ij}) \in M_2(F)$ , and look at the equation  $\mathbf{AXB} = \mathbf{C}$ . By [Definition 0.9.9](#), applied twice,

$$(\mathbf{AXB})_{rs} = \sum_{i=1}^2 \sum_{j=1}^2 a_{ri} x_{ij} b_{js}.$$

So each entry of  $\mathbf{AXB}$  is a linear combination of the four unknowns  $x_{11}, x_{21}, x_{12}, x_{22}$ , and the coefficient of  $x_{ij}$  in entry  $(r, s)$  is the product  $b_{js}a_{ri}$ : **an entry of  $\mathbf{B}$  times an entry of  $\mathbf{A}$** . Writing the four equations as a  $4 \times 4$  system, every coefficient is such a product. For instance, if we list the unknowns column by column as  $(x_{11}, x_{21}, x_{12}, x_{22})$  and the equations in the same order, the first row of the coefficient matrix is

$$(b_{11}a_{11} \quad b_{11}a_{12} \quad b_{21}a_{11} \quad b_{21}a_{12}).$$

The left half of this row is  $b_{11}$  times row 1 of  $\mathbf{A}$ , and the right half is  $b_{21}$  times row 1 of  $\mathbf{A}$ .

That is a recurring pattern: a big matrix whose entries are all the products of the entries of two small matrices, arranged in blocks. It deserves a name.

## 7.4.2 The Kronecker product

The Kronecker product of  $\mathbf{A}$  and  $\mathbf{B}$  is the block matrix obtained by replacing each entry  $a_{ij}$  of  $\mathbf{A}$  by the block  $a_{ij}\mathbf{B}$ .

### Definition 7.4.1 — Kronecker product

Let  $\mathbf{A} = (a_{ij}) \in M_{m \times n}(F)$  and  $\mathbf{B} = (b_{kl}) \in M_{p \times q}(F)$ , of **any** sizes. The **Kronecker product** of  $\mathbf{A}$  and  $\mathbf{B}$  is the  $mp \times nq$  block matrix

$$\mathbf{A} \otimes \mathbf{B} := \begin{pmatrix} a_{11}\mathbf{B} & a_{12}\mathbf{B} & \cdots & a_{1n}\mathbf{B} \\ a_{21}\mathbf{B} & a_{22}\mathbf{B} & \cdots & a_{2n}\mathbf{B} \\ \vdots & \vdots & & \vdots \\ a_{m1}\mathbf{B} & a_{m2}\mathbf{B} & \cdots & a_{mn}\mathbf{B} \end{pmatrix} \in M_{mp \times nq}(F),$$

with  $m$  rows and  $n$  columns of blocks, each block of size  $p \times q$  (Definition 7.1.1).

In words: the outer layout is the layout of  $\mathbf{A}$ , and the inner pattern of every block is  $\mathbf{B}$ . The sizes multiply, rows with rows and columns with columns. No condition on the sizes is needed, unlike for an ordinary product.

To work with single entries we need to know where each product  $a_{ij}b_{kl}$  sits. Block  $(i, j)$  occupies rows  $(i-1)p+1, \dots, ip$  and columns  $(j-1)q+1, \dots, jq$ . Inside it, entry  $(k, l)$  is  $a_{ij}b_{kl}$ . So

$$\begin{aligned} (\mathbf{A} \otimes \mathbf{B})_{(i-1)p+k, (j-1)q+l} &= a_{ij}b_{kl} \\ (1 \leq i \leq m, 1 \leq j \leq n, 1 \leq k \leq p, 1 \leq l \leq q). \end{aligned} \quad (*)$$

**Every entry is reached exactly once.** Division with remainder writes each row index  $r \in \{1, \dots, mp\}$  uniquely as  $r = (i-1)p + k$  with  $1 \leq i \leq m$  and  $1 \leq k \leq p$  (divide  $r-1$  by  $p$ ), and likewise for columns. So  $(*)$  is a complete description of  $\mathbf{A} \otimes \mathbf{B}$ , and it is the formula we use in proofs. In words, the rows of  $\mathbf{A} \otimes \mathbf{B}$  are labeled by pairs (row of  $\mathbf{A}$ , row of  $\mathbf{B}$ ) in dictionary order, and the columns by pairs (column of  $\mathbf{A}$ , column of  $\mathbf{B}$ ) in dictionary order.

**Examples.**

1. **A  $2 \times 2$  example.** With  $\mathbf{A} = \begin{pmatrix} 1 & 2 \\ 3 & 4 \end{pmatrix}$  and  $\mathbf{B} = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}$ ,

$$\mathbf{A} \otimes \mathbf{B} = \begin{pmatrix} 1\mathbf{B} & 2\mathbf{B} \\ 3\mathbf{B} & 4\mathbf{B} \end{pmatrix} = \begin{pmatrix} 0 & 1 & 0 & 2 \\ 1 & 0 & 2 & 0 \\ 0 & 3 & 0 & 4 \\ 3 & 0 & 4 & 0 \end{pmatrix}.$$

Check one entry against  $(*)$ : row 3 =  $(2-1) \cdot 2 + 1$  and column 2 =  $(1-1) \cdot 2 + 2$  give  $a_{21}b_{12} = 3 \cdot 1 = 3$ , as displayed.

2. **Vectors.** For column vectors  $\mathbf{u} \in F^m$  and  $\mathbf{v} \in F^p$  (so  $n = q = 1$ ),  $\mathbf{u} \otimes \mathbf{v} \in F^{mp}$  lists all products  $u_i v_k$  in dictionary order. In particular, for standard basis vectors,  $\mathbf{e}_i \otimes \mathbf{e}_k =$



- $\mathbf{e}_{(i-1)p+k} \in F^{mp}$ : the only non-zero product is  $1 \cdot 1$  in position  $(i-1)p+k$ . For a column  $\mathbf{u} \in F^m$  and a row  $\mathbf{v}^\top$ ,  $\mathbf{u} \otimes \mathbf{v}^\top$  is the  $m \times p$  matrix with entries  $u_i v_k$ , which is the outer product  $\mathbf{u}\mathbf{v}^\top$  of [Corollary 7.1.3](#).
3. **Identity blocks.**  $\mathbf{I}_n \otimes \mathbf{B}$  has  $\mathbf{B}$  in each diagonal block and zero blocks elsewhere, so it is the block diagonal matrix  $\mathbf{B} \oplus \cdots \oplus \mathbf{B}$  with  $n$  copies of  $\mathbf{B}$  ([Definition 7.1.2](#)). By contrast,  $\mathbf{A} \otimes \mathbf{I}_m$  replaces each entry  $a_{ij}$  by the  $m \times m$  block  $a_{ij}\mathbf{I}_m$ . Taking both,  $\mathbf{I}_n \otimes \mathbf{I}_m = \mathbf{I}_{nm}$ : by  $(*)$  its entries are  $\delta_{ij}\delta_{kl}$ , which is 1 exactly when the row and column labels  $(i, k)$  and  $(j, l)$  agree.
4. **The degenerate case**  $1 \times 1$ . If  $\mathbf{A} = (a)$  is  $1 \times 1$ , then  $\mathbf{A} \otimes \mathbf{B} = a\mathbf{B}$ ; if  $\mathbf{B} = (b)$ , then  $\mathbf{A} \otimes \mathbf{B} = b\mathbf{A}$ . So a  $1 \times 1$  factor behaves like a scalar, and  $(1) \otimes \mathbf{B} = \mathbf{B}$ . This case is worth keeping in mind: it shows that the Kronecker product extends scalar multiplication, which is why it is bilinear below.

**Non-example by minimal change.** Swap the two factors of Example 1:

$$\mathbf{B} \otimes \mathbf{A} = \begin{pmatrix} 0\mathbf{A} & 1\mathbf{A} \\ 1\mathbf{A} & 0\mathbf{A} \end{pmatrix} = \begin{pmatrix} 0 & 0 & 1 & 2 \\ 0 & 0 & 3 & 4 \\ 1 & 2 & 0 & 0 \\ 3 & 4 & 0 & 0 \end{pmatrix}.$$

It has the same size and the same multiset of entries, since both list all products  $a_{ij}b_{kl}$ . What fails is the position formula  $(*)$ : in  $\mathbf{B} \otimes \mathbf{A}$  the **outer** layout comes from  $\mathbf{B}$ , so the products sit in different places. For instance, the  $(1, 2)$ -entry is  $a_{11}b_{12} = 1$  in  $\mathbf{A} \otimes \mathbf{B}$  but  $b_{11}a_{12} = 0$  in  $\mathbf{B} \otimes \mathbf{A}$ .

**Why this definition.** The dictionary order in  $(*)$  is a convention, and the other order would simply give  $\mathbf{B} \otimes \mathbf{A}$ . We fix it because it makes the block picture true: the Kronecker product is literally “ $\mathbf{A}$  with entries replaced by blocks”, so block multiplication ([Theorem 7.1.1](#)) applies to it directly. The symbol  $\otimes$  anticipates the tensor product of Chapter 14, of which this is the matrix version.

*Warning.* **The Kronecker product is not commutative, and it is not the ordinary product.** The two matrices in Example 1 and the non-example differ, so  $\mathbf{A} \otimes \mathbf{B} \neq \mathbf{B} \otimes \mathbf{A}$  in general, even for square matrices of the same size. Also do not confuse  $\mathbf{A} \otimes \mathbf{B}$  with  $\mathbf{AB}$ : for  $\mathbf{A}, \mathbf{B} \in M_2(F)$ ,  $\mathbf{AB}$  is  $2 \times 2$  while  $\mathbf{A} \otimes \mathbf{B}$  is  $4 \times 4$ , and  $\mathbf{A} \otimes \mathbf{B}$  is defined for all sizes. We will see below that  $\mathbf{A} \otimes \mathbf{B}$  and  $\mathbf{B} \otimes \mathbf{A}$  are always **similar** when  $\mathbf{A}$  and  $\mathbf{B}$  are square, so they share rank, trace and determinant, but they are not equal.

### 7.4.3 Algebraic properties

The Kronecker product interacts well with every operation we have. The most useful rule is the one connecting it to ordinary multiplication: multiply the two left factors together and the two right factors together.

#### Theorem 7.4.1 — Properties of the Kronecker Product

Let  $c \in F$ . Whenever the sizes make each side defined:

- (a) **(Bilinearity)**  $(\mathbf{A} + \mathbf{A}') \otimes \mathbf{B} = \mathbf{A} \otimes \mathbf{B} + \mathbf{A}' \otimes \mathbf{B}$ ,  $\mathbf{A} \otimes (\mathbf{B} + \mathbf{B}') = \mathbf{A} \otimes \mathbf{B} + \mathbf{A} \otimes \mathbf{B}'$ ,  
and  $(c\mathbf{A}) \otimes \mathbf{B} = c(\mathbf{A} \otimes \mathbf{B}) = \mathbf{A} \otimes (c\mathbf{B})$ .
- (b) **(Associativity)**  $(\mathbf{A} \otimes \mathbf{B}) \otimes \mathbf{C} = \mathbf{A} \otimes (\mathbf{B} \otimes \mathbf{C})$  for all  $\mathbf{A} \in M_{m \times n}(F)$ ,  $\mathbf{B} \in M_{p \times q}(F)$ ,

$$\mathbf{C} \in M_{r \times s}(F).$$

- (c) **(Mixed product rule)** For  $\mathbf{A} \in M_{m \times n}(F)$ ,  $\mathbf{C} \in M_{n \times t}(F)$ ,  $\mathbf{B} \in M_{p \times q}(F)$  and  $\mathbf{D} \in M_{q \times u}(F)$ ,

$$(\mathbf{A} \otimes \mathbf{B})(\mathbf{C} \otimes \mathbf{D}) = (\mathbf{AC}) \otimes (\mathbf{BD}).$$

- (d) **(Transpose)**  $(\mathbf{A} \otimes \mathbf{B})^\top = \mathbf{A}^\top \otimes \mathbf{B}^\top$ .  
 (e) **(Inverse)** If  $\mathbf{A} \in M_n(F)$  and  $\mathbf{B} \in M_m(F)$  are invertible, then  $\mathbf{A} \otimes \mathbf{B}$  is invertible and  $(\mathbf{A} \otimes \mathbf{B})^{-1} = \mathbf{A}^{-1} \otimes \mathbf{B}^{-1}$ .

*Idea.* Parts (a), (b) and (d) are entry checks with (\*). For (c), both factors on the left are block matrices, and the columns of blocks of  $\mathbf{A} \otimes \mathbf{B}$  ( $n$  of them, each  $q$  wide) match the rows of blocks of  $\mathbf{C} \otimes \mathbf{D}$  ( $n$  of them, each  $q$  tall). So we may multiply blockwise, and each block of the product is a sum  $\sum_j (a_{ij}\mathbf{B})(c_{jk}\mathbf{D})$  in which the scalars come out. Part (e) is (c) with  $\mathbf{C} = \mathbf{A}^{-1}$ ,  $\mathbf{D} = \mathbf{B}^{-1}$ .

*Proof.*

- (a) By (\*), the entry of  $(\mathbf{A} + \mathbf{A}') \otimes \mathbf{B}$  in position  $((i-1)p+k, (j-1)q+l)$  is  $(a_{ij} + a'_{ij})b_{kl} = a_{ij}b_{kl} + a'_{ij}b_{kl}$ , which is the entry of  $\mathbf{A} \otimes \mathbf{B} + \mathbf{A}' \otimes \mathbf{B}$  in the same position. The other identities are checked the same way, using  $(ca_{ij})b_{kl} = c(a_{ij}b_{kl}) = a_{ij}(cb_{kl})$ .  
 (b) Both sides are  $mpr \times nqs$ . Let  $1 \leq i \leq m$ ,  $1 \leq k \leq p$ ,  $1 \leq t \leq r$ , and similarly  $j, l, u$  for columns. Applying (\*) twice, the entry of  $(\mathbf{A} \otimes \mathbf{B}) \otimes \mathbf{C}$  in row  $((i-1)p+k-1)r+t$  and column  $((j-1)q+l-1)s+u$  is  $(\mathbf{A} \otimes \mathbf{B})_{(i-1)p+k, (j-1)q+l} c_{tu} = a_{ij}b_{kl}c_{tu}$ . Likewise the entry of  $\mathbf{A} \otimes (\mathbf{B} \otimes \mathbf{C})$  in row  $(i-1)pr + (k-1)r+t$  and column  $(j-1)qs + (l-1)s+u$  is  $a_{ij}(\mathbf{B} \otimes \mathbf{C})_{(k-1)r+t, (l-1)s+u} = a_{ij}b_{kl}c_{tu}$ . Since  $((i-1)p+k-1)r+t = (i-1)pr + (k-1)r+t$ , and similarly for columns, the two matrices have the same entry in every position; every position arises from exactly one choice of indices, by division with remainder. Hence they are equal.  
 (c) Partition  $\mathbf{A} \otimes \mathbf{B}$  into  $m$  rows and  $n$  columns of blocks as in Definition 7.4.1, each block  $p \times q$ , and  $\mathbf{C} \otimes \mathbf{D}$  into  $n$  rows and  $t$  columns of blocks, each  $q \times u$ . The column partition of the first matrix ( $n$  parts of size  $q$ ) equals the row partition of the second, so the partitions are conformable, and by Theorem 7.1.1 the  $(i, k)$  block of the product is

$$\sum_{j=1}^n (a_{ij}\mathbf{B})(c_{jk}\mathbf{D}) = \sum_{j=1}^n a_{ij}c_{jk} \mathbf{BD} = (\mathbf{AC})_{ik} \mathbf{BD},$$

where the first equality uses Theorem 0.9.4 (4) and the second uses Definition 0.9.9. This is the  $(i, k)$  block of  $(\mathbf{AC}) \otimes (\mathbf{BD})$ , a matrix partitioned into  $m$  rows and  $t$  columns of  $p \times u$  blocks. So the two matrices agree block by block, hence entry by entry.

- (d)  $(\mathbf{A} \otimes \mathbf{B})^\top$  is  $nq \times mp$ . By Definition 0.9.10 and (\*), its entry in row  $(j-1)q+l$  and column  $(i-1)p+k$  is  $(\mathbf{A} \otimes \mathbf{B})_{(i-1)p+k, (j-1)q+l} = a_{ij}b_{kl} = (\mathbf{A}^\top)_{ji}(\mathbf{B}^\top)_{lk}$ , which by (\*) is the entry of  $\mathbf{A}^\top \otimes \mathbf{B}^\top$  in the same position.  
 (e) By (c) and Example 3,  $(\mathbf{A} \otimes \mathbf{B})(\mathbf{A}^{-1} \otimes \mathbf{B}^{-1}) = (\mathbf{AA}^{-1}) \otimes (\mathbf{BB}^{-1}) = \mathbf{I}_n \otimes \mathbf{I}_m = \mathbf{I}_{nm}$ , and similarly  $(\mathbf{A}^{-1} \otimes \mathbf{B}^{-1})(\mathbf{A} \otimes \mathbf{B}) = \mathbf{I}_{nm}$ . By Definition 0.9.13,  $\mathbf{A} \otimes \mathbf{B}$  is invertible with inverse  $\mathbf{A}^{-1} \otimes \mathbf{B}^{-1}$ . ■

The mixed product rule is the engine of everything that follows. One immediate consequence splits a Kronecker product of square matrices into two simpler factors: for  $\mathbf{A} \in M_n(F)$  and  $\mathbf{B} \in M_m(F)$ ,

$$\mathbf{A} \otimes \mathbf{B} = (\mathbf{A}\mathbf{I}_n) \otimes (\mathbf{I}_m\mathbf{B}) = (\mathbf{A} \otimes \mathbf{I}_m)(\mathbf{I}_n \otimes \mathbf{B}), \quad (**)$$

and in the same way  $\mathbf{A} \otimes \mathbf{B} = (\mathbf{I}_n \otimes \mathbf{B})(\mathbf{A} \otimes \mathbf{I}_m)$ . So  $\mathbf{A} \otimes \mathbf{I}_m$  and  $\mathbf{I}_n \otimes \mathbf{B}$  commute, even though  $\mathbf{A}$  and  $\mathbf{B}$  have nothing to do with each other.

For example, with  $\mathbf{A} = \begin{pmatrix} 1 & -1 \\ 0 & 2 \end{pmatrix}$ ,  $\mathbf{B} = \begin{pmatrix} 2 & 1 \\ 1 & 1 \end{pmatrix}$ ,  $\mathbf{C} = \begin{pmatrix} 1 & 0 \\ 1 & 1 \end{pmatrix}$  and  $\mathbf{D} = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}$ , multiplying two  $4 \times 4$  matrices directly gives

$$(\mathbf{A} \otimes \mathbf{B})(\mathbf{C} \otimes \mathbf{D}) = \begin{pmatrix} 2 & 1 & -2 & -1 \\ 1 & 1 & -1 & -1 \\ 0 & 0 & 4 & 2 \\ 0 & 0 & 2 & 2 \end{pmatrix} \begin{pmatrix} 0 & 1 & 0 & 0 \\ 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 1 \\ 1 & 0 & 1 & 0 \end{pmatrix} = \begin{pmatrix} 0 & 0 & -1 & -2 \\ 0 & 0 & -1 & -1 \\ 2 & 4 & 2 & 4 \\ 2 & 2 & 2 & 2 \end{pmatrix},$$

while the rule needs only two  $2 \times 2$  products,  $\mathbf{AC} = \begin{pmatrix} 0 & -1 \\ 2 & 2 \end{pmatrix}$  and  $\mathbf{BD} = \begin{pmatrix} 1 & 2 \\ 1 & 1 \end{pmatrix}$ , whose Kronecker product is the same matrix.

## 7.4.4 Swapping the factors

The warning said that  $\mathbf{A} \otimes \mathbf{B}$  and  $\mathbf{B} \otimes \mathbf{A}$  differ. They differ only in the order in which rows and columns are labeled:  $\mathbf{A} \otimes \mathbf{B}$  uses (index of  $\mathbf{A}$ , index of  $\mathbf{B}$ ) in dictionary order,  $\mathbf{B} \otimes \mathbf{A}$  the reverse order. Relabeling rows and columns by the same permutation is conjugation by a permutation matrix, and that is what the next result makes precise.

First, a fact about permutation matrices. Recall that for  $\sigma \in S_N$ , the permutation matrix  $\mathbf{P}_\sigma$  has  $j$ -th column  $\mathbf{e}_{\sigma(j)}$ , and  $\mathbf{P}_\sigma^{-1} = \mathbf{P}_\sigma^\top$  (Definition 2.6.2, Lemma 2.6.6). For every  $\mathbf{M} \in M_N(F)$ ,

$$(\mathbf{P}_\sigma^\top \mathbf{M} \mathbf{P}_\sigma)_{rs} = \mathbf{M}_{\sigma(r)\sigma(s)} \quad (1 \leq r, s \leq N). \quad (\dagger)$$

Indeed, by Theorem 0.9.2 column  $s$  of  $\mathbf{M} \mathbf{P}_\sigma$  is  $\mathbf{M} \mathbf{e}_{\sigma(s)}$ , which is column  $\sigma(s)$  of  $\mathbf{M}$ . Row  $r$  of  $\mathbf{P}_\sigma^\top$  is  $\mathbf{e}_{\sigma(r)}^\top$ , so by the same theorem row  $r$  of  $\mathbf{P}_\sigma^\top \mathbf{L}$  is row  $\sigma(r)$  of  $\mathbf{L}$ , for any  $\mathbf{L} \in M_N(F)$ . Combining the two gives  $(\dagger)$ .

### Proposition 7.4.2 — Swapping the Factors Is a Permutation Similarity

Let  $n, m \geq 1$ . There is a permutation matrix  $\mathbf{P} \in M_{nm}(F)$ , depending only on  $n$  and  $m$ , such that

$$\mathbf{B} \otimes \mathbf{A} = \mathbf{P}^{-1}(\mathbf{A} \otimes \mathbf{B})\mathbf{P} \quad \text{for all } \mathbf{A} \in M_n(F), \mathbf{B} \in M_m(F).$$

In particular  $\mathbf{A} \otimes \mathbf{B} \sim \mathbf{B} \otimes \mathbf{A}$ , and  $\mathbf{I}_m \otimes \mathbf{A} = \mathbf{P}^{-1}(\mathbf{A} \otimes \mathbf{I}_m)\mathbf{P}$ .

*Proof.* Every  $r \in \{1, \dots, nm\}$  can be written uniquely as  $r = (k-1)n + i$  with  $1 \leq k \leq m$  and  $1 \leq i \leq n$ . Define  $\sigma(r) := (i-1)m + k$ . As  $(k, i)$  runs over all pairs, so does  $(i, k)$ , and each value  $(i-1)m + k$  is taken by exactly one pair; hence  $\sigma$  is a bijection of  $\{1, \dots, nm\}$ , that is,  $\sigma \in S_{nm}$ . Put  $\mathbf{P} = \mathbf{P}_\sigma$ .

Let  $r = (k-1)n + i$  and  $s = (l-1)n + j$ . By  $(*)$  applied to  $\mathbf{B} \otimes \mathbf{A}$ ,  $(\mathbf{B} \otimes \mathbf{A})_{rs} = b_{kl}a_{ij}$ .

By (\*) applied to  $\mathbf{A} \otimes \mathbf{B}$ ,

$$(\mathbf{A} \otimes \mathbf{B})_{\sigma(r)\sigma(s)} = (\mathbf{A} \otimes \mathbf{B})_{(i-1)m+k, (j-1)m+l} = a_{ij}b_{kl}.$$

So by (†),  $\mathbf{P}_\sigma^\top (\mathbf{A} \otimes \mathbf{B}) \mathbf{P}_\sigma$  and  $\mathbf{B} \otimes \mathbf{A}$  agree in every entry, and  $\mathbf{P}_\sigma^\top = \mathbf{P}_\sigma^{-1}$  by Lemma 2.6.6 (b). The last claim is the case  $\mathbf{B} = \mathbf{I}_m$ . ■

For example, when  $n = m = 2$ ,  $\sigma$  fixes 1 and 4 and swaps 2 and 3, so  $\mathbf{P}$  is the elementary matrix  $\mathbf{P}_{23}$ : conjugating by it swaps rows 2, 3 and columns 2, 3. Doing this to  $\mathbf{A} \otimes \mathbf{B}$  in Example 1 does give  $\mathbf{B} \otimes \mathbf{A}$ , as a direct check shows. For rectangular  $\mathbf{A} \in M_{m \times n}(F)$  and  $\mathbf{B} \in M_{p \times q}(F)$ , the same argument gives  $\mathbf{B} \otimes \mathbf{A} = \mathbf{P}_\sigma^\top (\mathbf{A} \otimes \mathbf{B}) \mathbf{P}_\tau$  with one permutation for the rows and another for the columns.

## 7.4.5 Rank, trace and determinant

Each of the three basic numbers attached to a matrix behaves multiplicatively. The formulas for trace and rank are what one would guess. The determinant is subtler, because the sizes enter as exponents.

### Theorem 7.4.3 — Rank, Trace and Determinant of a Kronecker Product (a)

For  $\mathbf{A} \in M_{m \times n}(F)$  and  $\mathbf{B} \in M_{p \times q}(F)$ ,  $\text{rank}(\mathbf{A} \otimes \mathbf{B}) = \text{rank } \mathbf{A} \cdot \text{rank } \mathbf{B}$ .

(b) For  $\mathbf{A} \in M_n(F)$  and  $\mathbf{B} \in M_m(F)$ ,  $\text{tr}(\mathbf{A} \otimes \mathbf{B}) = \text{tr } \mathbf{A} \cdot \text{tr } \mathbf{B}$ .

(c) For  $\mathbf{A} \in M_n(F)$  and  $\mathbf{B} \in M_m(F)$ ,  $\det(\mathbf{A} \otimes \mathbf{B}) = (\det \mathbf{A})^m (\det \mathbf{B})^n$ .

*Idea.* For rank, bring both factors to rank normal form,  $\mathbf{A} = \mathbf{Q}\mathbf{N}_r\mathbf{P}$  and  $\mathbf{B} = \mathbf{Q}'\mathbf{N}_s\mathbf{P}'$ . The mixed product rule moves the invertible factors outside, where they do not change rank, and  $\mathbf{N}_r \otimes \mathbf{N}_s$  visibly has rank  $rs$ . Trace is a sum over diagonal entries, and the diagonal entries of  $\mathbf{A} \otimes \mathbf{B}$  are the products  $a_{ii}b_{kk}$ . For the determinant, split  $\mathbf{A} \otimes \mathbf{B} = (\mathbf{A} \otimes \mathbf{I}_m)(\mathbf{I}_n \otimes \mathbf{B})$  by (\*\*). The second factor is block diagonal with  $n$  copies of  $\mathbf{B}$ . The first is not, but by Proposition 7.4.2 it is similar to  $\mathbf{I}_m \otimes \mathbf{A}$ , which is block diagonal with  $m$  copies of  $\mathbf{A}$ .

*Proof.*

- (a) Let  $r = \text{rank } \mathbf{A}$  and  $s = \text{rank } \mathbf{B}$ . By Theorem 3.8.3 (b) and Definition 3.8.1, there are invertible  $\mathbf{Q}_1, \mathbf{P}_1, \mathbf{Q}_2, \mathbf{P}_2$  with  $\mathbf{N}_r = \mathbf{Q}_1\mathbf{A}\mathbf{P}_1$  and  $\mathbf{N}_s = \mathbf{Q}_2\mathbf{B}\mathbf{P}_2$ , where  $\mathbf{N}_r \in M_{m \times n}(F)$  and  $\mathbf{N}_s \in M_{p \times q}(F)$  have  $\mathbf{I}_r, \mathbf{I}_s$  in the top left corner and zeros elsewhere. Hence  $\mathbf{A} = \mathbf{Q}_1^{-1}\mathbf{N}_r\mathbf{P}_1^{-1}$  and  $\mathbf{B} = \mathbf{Q}_2^{-1}\mathbf{N}_s\mathbf{P}_2^{-1}$ , and by Theorem 7.4.1 (c), used twice,

$$\mathbf{A} \otimes \mathbf{B} = (\mathbf{Q}_1^{-1} \otimes \mathbf{Q}_2^{-1})(\mathbf{N}_r \otimes \mathbf{N}_s)(\mathbf{P}_1^{-1} \otimes \mathbf{P}_2^{-1}).$$

The outer factors are invertible by Theorem 7.4.1 (e), so  $\text{rank}(\mathbf{A} \otimes \mathbf{B}) = \text{rank}(\mathbf{N}_r \otimes \mathbf{N}_s)$  by Theorem 2.5.9.

By (\*), the entry of  $\mathbf{N}_r \otimes \mathbf{N}_s$  in position  $((i-1)p+k, (j-1)q+l)$  is 1 if  $i = j \leq r$  and  $k = l \leq s$ , and 0 otherwise. So for each of the  $rs$  pairs  $(i, k)$  with  $i \leq r, k \leq s$ , column  $(i-1)q+k$  equals  $\mathbf{e}_{(i-1)p+k}$ , and every other column is zero. Distinct pairs

give distinct standard basis vectors, so these  $rs$  columns are linearly independent and span the column space. Hence  $\text{rank}(\mathbf{N}_r \otimes \mathbf{N}_s) = rs$  by [Definition 2.5.4](#), which proves (a). (If  $r = 0$  or  $s = 0$ , the matrix is zero and both sides are 0.)

- (b) By  $(*)$  with  $p = q = m$ , the diagonal entries of  $\mathbf{A} \otimes \mathbf{B}$  are those with  $(i-1)m + k = (j-1)m + l$ , that is  $i = j$  and  $k = l$  by uniqueness of division with remainder, and they equal  $a_{ii}b_{kk}$ . By [Definition 0.9.12](#),

$$\text{tr}(\mathbf{A} \otimes \mathbf{B}) = \sum_{i=1}^n \sum_{k=1}^m a_{ii}b_{kk} = \left( \sum_{i=1}^n a_{ii} \right) \left( \sum_{k=1}^m b_{kk} \right) = \text{tr } \mathbf{A} \cdot \text{tr } \mathbf{B}.$$

- (c) First, a block diagonal matrix  $\mathbf{M}_1 \oplus \cdots \oplus \mathbf{M}_t$  with square blocks  $\mathbf{M}_1, \dots, \mathbf{M}_t$  has determinant  $\det \mathbf{M}_1 \cdots \det \mathbf{M}_t$ . For  $t = 1$  there is nothing to prove, and for  $t \geq 2$  the matrix is  $\mathbf{M}_1 \oplus (\mathbf{M}_2 \oplus \cdots \oplus \mathbf{M}_t)$ , so [Theorem 7.1.4](#) (d) (or [Theorem 6.8.4](#)) and induction on  $t$  give the claim.

By Example 3,  $\mathbf{I}_n \otimes \mathbf{B} = \mathbf{B} \oplus \cdots \oplus \mathbf{B}$  with  $n$  blocks, so  $\det(\mathbf{I}_n \otimes \mathbf{B}) = (\det \mathbf{B})^n$ . Likewise  $\mathbf{I}_m \otimes \mathbf{A} = \mathbf{A} \oplus \cdots \oplus \mathbf{A}$  with  $m$  blocks, so  $\det(\mathbf{I}_m \otimes \mathbf{A}) = (\det \mathbf{A})^m$ . By [Proposition 7.4.2](#),  $\mathbf{A} \otimes \mathbf{I}_m$  is similar to  $\mathbf{I}_m \otimes \mathbf{A}$ , so  $\det(\mathbf{A} \otimes \mathbf{I}_m) = (\det \mathbf{A})^m$  by [Corollary 6.5.7](#). Finally, by  $(**)$  and [Theorem 6.5.4](#),

$$\det(\mathbf{A} \otimes \mathbf{B}) = \det(\mathbf{A} \otimes \mathbf{I}_m) \det(\mathbf{I}_n \otimes \mathbf{B}) = (\det \mathbf{A})^m (\det \mathbf{B})^n.$$

■

The exponents are easy to get backwards, and a size count fixes them. In  $\mathbf{A} \otimes \mathbf{B}$  with  $\mathbf{A} \in M_n(F)$ ,  $\mathbf{B} \in M_m(F)$ , the matrix  $\mathbf{A}$  is “used”  $m$  times (once for each index of  $\mathbf{B}$ ), so  $\det \mathbf{A}$  appears to the power  $m$ , the size of the **other** factor. For  $\mathbf{A} = c\mathbf{I}_n$  and  $\mathbf{B} = \mathbf{I}_m$ ,  $\mathbf{A} \otimes \mathbf{B} = c\mathbf{I}_{nm}$  has determinant  $c^{nm} = (c^n)^m = (\det \mathbf{A})^m$ , which confirms the exponent.

For the matrices of the example after [Theorem 7.4.1](#),  $\det \mathbf{A} = 2$ ,  $\det \mathbf{B} = 1$ ,  $\text{tr } \mathbf{A} = 3$  and  $\text{tr } \mathbf{B} = 3$ , so  $\det(\mathbf{A} \otimes \mathbf{B}) = 2^2 \cdot 1^2 = 4$ ,  $\text{tr}(\mathbf{A} \otimes \mathbf{B}) = 9$ , and  $\text{rank}(\mathbf{A} \otimes \mathbf{B}) = 2 \cdot 2 = 4$ . Directly, the diagonal of  $\mathbf{A} \otimes \mathbf{B}$  is 2, 1, 4, 2, with sum 9.

*Quick check.* Let  $\mathbf{A} \in M_2(\mathbb{R})$  have  $\text{tr } \mathbf{A} = 4$ ,  $\det \mathbf{A} = 3$ , and let  $\mathbf{B} \in M_3(\mathbb{R})$  have  $\text{tr } \mathbf{B} = 2$ ,  $\det \mathbf{B} = -1$ . Find  $\text{tr}(\mathbf{A} \otimes \mathbf{B})$ ,  $\det(\mathbf{A} \otimes \mathbf{B})$  and  $\det(\mathbf{B} \otimes \mathbf{A})$ .

*Solution.* By [Theorem 7.4.3](#),  $\text{tr}(\mathbf{A} \otimes \mathbf{B}) = 4 \cdot 2 = 8$  and  $\det(\mathbf{A} \otimes \mathbf{B}) = (\det \mathbf{A})^3 (\det \mathbf{B})^2 = 27 \cdot 1 = 27$ . With the roles swapped,  $\det(\mathbf{B} \otimes \mathbf{A}) = (\det \mathbf{B})^2 (\det \mathbf{A})^3 = 27$ , as it must be since  $\mathbf{B} \otimes \mathbf{A} \sim \mathbf{A} \otimes \mathbf{B}$  ([Proposition 7.4.2](#)). The common wrong answer  $(\det \mathbf{A})^2 (\det \mathbf{B})^3 = -9$  puts each matrix’s own size in its exponent.

## 7.4.6 Vectorization

We return to the question of the opening: turning a matrix equation into a linear system. For that we need to flatten the unknown matrix into a column, in an order that matches  $(*)$ .

*To vectorize a matrix is to stack its columns into one long column.*

**Definition 7.4.2 — Vectorization**

Let  $\mathbf{X} = (x_{ij}) \in M_{m \times n}(F)$  have columns  $\mathbf{x}_1, \dots, \mathbf{x}_n \in F^m$ . Its **vectorization** is the column vector

$$\text{vec } \mathbf{X} := \begin{pmatrix} \mathbf{x}_1 \\ \mathbf{x}_2 \\ \vdots \\ \mathbf{x}_n \end{pmatrix} \in F^{mn},$$

whose entry in position  $(j-1)m+i$  is  $x_{ij}$ , for  $1 \leq i \leq m$ ,  $1 \leq j \leq n$ .

For example,  $\text{vec} \begin{pmatrix} 1 & 2 \\ 3 & 4 \end{pmatrix} = (1, 3, 2, 4)$ , read down the first column and then down the second.

For a column  $\mathbf{x} \in F^m$  (the case  $n = 1$ ),  $\text{vec } \mathbf{x} = \mathbf{x}$ . The map  $\text{vec}: M_{m \times n}(F) \rightarrow F^{mn}$  is linear, since each entry of  $\text{vec } \mathbf{X}$  is an entry of  $\mathbf{X}$ , and bijective, since it only rearranges entries; so it is an isomorphism. It sends the matrix unit  $\mathbf{E}_{ij}$  to  $\mathbf{e}_{(j-1)m+i} = \mathbf{e}_j \otimes \mathbf{e}_i$  (Example 2, with the index of the column first). More generally,  $\text{vec}(\mathbf{u}\mathbf{v}^\top) = \mathbf{v} \otimes \mathbf{u}$  for  $\mathbf{u} \in F^m$  and  $\mathbf{v} \in F^n$ : column  $j$  of  $\mathbf{u}\mathbf{v}^\top$  is  $v_j\mathbf{u}$ .

Two conventions are in use, and they differ. Some books stack the **rows** instead, which exchanges the roles of the two factors in the next theorem. We always stack columns.

**Theorem 7.4.4 — The vec Identity**

Let  $\mathbf{A} \in M_{k \times m}(F)$ ,  $\mathbf{X} \in M_{m \times n}(F)$  and  $\mathbf{B} \in M_{n \times l}(F)$ . Then

$$\text{vec}(\mathbf{A}\mathbf{X}\mathbf{B}) = (\mathbf{B}^\top \otimes \mathbf{A}) \text{vec } \mathbf{X}.$$

*Idea.* Both sides are columns in  $F^{kl}$ . We compare one entry, using the formula for a triple product on the left and  $(*)$  on the right. The factor  $\mathbf{A}$  acts on the row index of  $\mathbf{X}$ , which is the inner index in  $\text{vec}$ ; the factor  $\mathbf{B}$  acts on the column index, which is the outer index. That is why  $\mathbf{B}$  comes first, and it enters transposed because it multiplies  $\mathbf{X}$  from the right.

*Proof.* Both sides lie in  $F^{kl}$ :  $\mathbf{A}\mathbf{X}\mathbf{B} \in M_{k \times l}(F)$ , and  $\mathbf{B}^\top \otimes \mathbf{A} \in M_{lk \times nm}(F)$  acts on  $\text{vec } \mathbf{X} \in F^{mn}$ . Fix  $1 \leq r \leq k$  and  $1 \leq s \leq l$ , and consider position  $(s-1)k+r$ . By Definition 7.4.2 and Definition 0.9.9 applied twice, the left side has entry

$$(\mathbf{A}\mathbf{X}\mathbf{B})_{rs} = \sum_{i=1}^m \sum_{j=1}^n a_{ri} x_{ij} b_{js}.$$

On the right, write the column index  $c \in \{1, \dots, nm\}$  of  $\mathbf{B}^\top \otimes \mathbf{A}$  uniquely as  $c = (j-1)m+i$  with  $1 \leq j \leq n$ ,  $1 \leq i \leq m$ . By  $(*)$ ,  $(\mathbf{B}^\top \otimes \mathbf{A})_{(s-1)k+r, (j-1)m+i} = (\mathbf{B}^\top)_{sj} a_{ri} = b_{js} a_{ri}$ , and by Definition 7.4.2 the  $c$ -th entry of  $\text{vec } \mathbf{X}$  is  $x_{ij}$ . Hence the entry of the right side in position  $(s-1)k+r$  is

$$\sum_{c=1}^{nm} (\mathbf{B}^\top \otimes \mathbf{A})_{(s-1)k+r, c} (\text{vec } \mathbf{X})_c = \sum_{j=1}^n \sum_{i=1}^m b_{js} a_{ri} x_{ij},$$

which equals the left side. As  $(r, s)$  ranges over all pairs,  $(s-1)k + r$  ranges over all positions, so the two vectors are equal. ■

Taking  $\mathbf{B} = \mathbf{I}_n$  or  $\mathbf{A} = \mathbf{I}_m$  gives the two one-sided forms, which are used most often:

$$\text{vec}(\mathbf{AX}) = (\mathbf{I}_n \otimes \mathbf{A}) \text{vec } \mathbf{X}, \quad \text{vec}(\mathbf{XB}) = (\mathbf{B}^\top \otimes \mathbf{I}_m) \text{vec } \mathbf{X}.$$

The first says that multiplying  $\mathbf{X}$  by  $\mathbf{A}$  on the left multiplies each column of  $\mathbf{X}$  by  $\mathbf{A}$ :  $\mathbf{I}_n \otimes \mathbf{A} = \mathbf{A} \oplus \cdots \oplus \mathbf{A}$  acts on the stacked columns one at a time.

Now every linear matrix equation built from products becomes a linear system. The most important one is the **Sylvester equation**

$$\mathbf{AX} + \mathbf{XB} = \mathbf{C}, \quad \mathbf{A} \in M_m(F), \mathbf{B} \in M_n(F), \mathbf{C} \in M_{m \times n}(F),$$

for an unknown  $\mathbf{X} \in M_{m \times n}(F)$ . Applying  $\text{vec}$ , which is linear, and the two one-sided forms,

$$(\mathbf{I}_n \otimes \mathbf{A} + \mathbf{B}^\top \otimes \mathbf{I}_m) \text{vec } \mathbf{X} = \text{vec } \mathbf{C}.$$

So the Sylvester equation has **exactly one** solution for every  $\mathbf{C}$  if and only if the  $mn \times mn$  matrix  $\mathbf{I}_n \otimes \mathbf{A} + \mathbf{B}^\top \otimes \mathbf{I}_m$  is invertible ([Theorem 2.4.4](#), transported through the isomorphism  $\text{vec}$ ). The equation  $\mathbf{AX} = \mathbf{XA}$  of the opening, which describes the matrices commuting with  $\mathbf{A}$ , is the case  $\mathbf{B} = -\mathbf{A}$ ,  $\mathbf{C} = 0$ : its solutions form the null space of  $\mathbf{I}_m \otimes \mathbf{A} - \mathbf{A}^\top \otimes \mathbf{I}_m$ .

#### Example. (A Sylvester Equation as a Linear System)

Solve  $\mathbf{AX} + \mathbf{XB} = \mathbf{C}$  over  $\mathbb{Q}$ , where

$$\mathbf{A} = \begin{pmatrix} 1 & 1 \\ 0 & 2 \end{pmatrix}, \quad \mathbf{B} = \begin{pmatrix} 1 & 0 \\ 1 & 1 \end{pmatrix}, \quad \mathbf{C} = \begin{pmatrix} 1 & -1 \\ 1 & 3 \end{pmatrix}.$$

*Solution.* Here  $m = n = 2$ . Compute the two Kronecker products from [Definition 7.4.1](#):

$$\mathbf{I}_2 \otimes \mathbf{A} = \begin{pmatrix} 1 & 1 & 0 & 0 \\ 0 & 2 & 0 & 0 \\ 0 & 0 & 1 & 1 \\ 0 & 0 & 0 & 2 \end{pmatrix}, \quad \mathbf{B}^\top \otimes \mathbf{I}_2 = \begin{pmatrix} 1\mathbf{I}_2 & 1\mathbf{I}_2 \\ 0\mathbf{I}_2 & 1\mathbf{I}_2 \end{pmatrix} = \begin{pmatrix} 1 & 0 & 1 & 0 \\ 0 & 1 & 0 & 1 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{pmatrix}.$$

With  $\text{vec } \mathbf{X} = (x_{11}, x_{21}, x_{12}, x_{22})$  and  $\text{vec } \mathbf{C} = (1, 1, -1, 3)$ , the system  $(\mathbf{I}_2 \otimes \mathbf{A} + \mathbf{B}^\top \otimes \mathbf{I}_2) \text{vec } \mathbf{X} = \text{vec } \mathbf{C}$  is

$$\begin{pmatrix} 2 & 1 & 1 & 0 \\ 0 & 3 & 0 & 1 \\ 0 & 0 & 2 & 1 \\ 0 & 0 & 0 & 3 \end{pmatrix} \begin{pmatrix} x_{11} \\ x_{21} \\ x_{12} \\ x_{22} \end{pmatrix} = \begin{pmatrix} 1 \\ 1 \\ -1 \\ 3 \end{pmatrix}.$$

The coefficient matrix is upper triangular with non-zero diagonal entries 2, 3, 2, 3, so its determinant is  $36 \neq 0$  ([Theorem 6.4.6](#)) and the solution is unique. Back substitution gives  $x_{22} = 1$ ; then  $2x_{12} + 1 = -1$ , so  $x_{12} = -1$ ; then  $3x_{21} + 1 = 1$ , so  $x_{21} = 0$ ; then  $2x_{11} + 0 - 1 = 1$ , so  $x_{11} = 1$ . Hence

$$\mathbf{X} = \begin{pmatrix} 1 & -1 \\ 0 & 1 \end{pmatrix}.$$



*Check:*  $\mathbf{A}\mathbf{X} = \begin{pmatrix} 1 & 0 \\ 0 & 2 \end{pmatrix}$  and  $\mathbf{X}\mathbf{B} = \begin{pmatrix} 0 & -1 \\ 1 & 1 \end{pmatrix}$ , whose sum is  $\mathbf{C}$ . As a check on the bookkeeping, row 1 of the system is the  $(1, 1)$  entry of  $\mathbf{A}\mathbf{X} + \mathbf{X}\mathbf{B}$ , namely  $(x_{11} + x_{21}) + (x_{11} + x_{12})$ .

In this example the coefficient matrix came out triangular, with diagonal entries  $a_{ii} + b_{jj}$ . That is no accident when  $\mathbf{A}$  is upper triangular and  $\mathbf{B}$  lower triangular, as [Exercise \(C2: Kronecker sums\)](#) below shows. In general, whether  $\mathbf{I}_n \otimes \mathbf{A} + \mathbf{B}^\top \otimes \mathbf{I}_m$  is invertible depends on the eigenvalues of  $\mathbf{A}$  and  $\mathbf{B}$ : over  $\mathbb{C}$ , the equation  $\mathbf{A}\mathbf{X} - \mathbf{X}\mathbf{B} = \mathbf{C}$  is uniquely solvable for every  $\mathbf{C}$  exactly when  $\mathbf{A}$  and  $\mathbf{B}$  have no common eigenvalue. That criterion needs eigenvalues and is proved in Chapter 11, and its quantitative form appears in Chapter 19.

**Warning.** Keep the order and the transpose in  $\text{vec}(\mathbf{A}\mathbf{X}\mathbf{B}) = (\mathbf{B}^\top \otimes \mathbf{A}) \text{vec } \mathbf{X}$ . Both  $\mathbf{A} \otimes \mathbf{B}^\top$  and  $\mathbf{B} \otimes \mathbf{A}$  look plausible, and both are wrong. With  $\mathbf{X} = \mathbf{E}_{11} \in M_2(F)$ ,  $\mathbf{A} = \mathbf{I}_2$  and  $\mathbf{B} = \mathbf{E}_{21}$ , we get  $\mathbf{A}\mathbf{X}\mathbf{B} = \mathbf{E}_{11}\mathbf{E}_{21} = 0$ , and indeed  $(\mathbf{B}^\top \otimes \mathbf{I}_2)\mathbf{e}_1 = (\mathbf{E}_{12} \otimes \mathbf{I}_2)\mathbf{e}_1 = 0$ , because the first column of  $\mathbf{E}_{12} \otimes \mathbf{I}_2$  is zero. But  $(\mathbf{B} \otimes \mathbf{I}_2)\mathbf{e}_1 = (\mathbf{E}_{21} \otimes \mathbf{I}_2)\mathbf{e}_1 = \mathbf{e}_3 \neq 0$ . So  $\mathbf{B} \otimes \mathbf{A}$  fails. A similar test with  $\mathbf{A} = \mathbf{E}_{12}$ ,  $\mathbf{X} = \mathbf{E}_{21}$ ,  $\mathbf{B} = \mathbf{I}_2$  refutes  $\mathbf{A} \otimes \mathbf{B}^\top$ . If a formula is in doubt, test it on matrix units.

The Kronecker product is the coordinate shadow of a basis-free construction: the tensor product of linear maps, in Chapter 14, whose matrix in suitable bases is exactly  $\mathbf{A} \otimes \mathbf{B}$ . The dictionary order in  $(*)$  will reappear there as an ordering of a basis  $\mathbf{v}_i \otimes \mathbf{w}_k$ .

## 7.4.7 Exercises

### 7.4.7.1 A. Check your understanding

#### Exercise A1.

- Define  $\mathbf{A} \otimes \mathbf{B}$  for  $\mathbf{A} \in M_{m \times n}(F)$  and  $\mathbf{B} \in M_{p \times q}(F)$ , and state its size.
- True or false: for all  $\mathbf{A}, \mathbf{B} \in M_2(F)$ ,  $\mathbf{A} \otimes \mathbf{B} = \mathbf{B} \otimes \mathbf{A}$ . Justify your answer.
- State the mixed product rule, including the conditions on the sizes.
- True or false:  $\det(\mathbf{A} \otimes \mathbf{B}) = \det \mathbf{A} \det \mathbf{B}$  for all  $\mathbf{A}, \mathbf{B} \in M_2(\mathbb{R})$ . Justify your answer.
- State the vec identity, and name the matrix  $\mathbf{M}$  with  $\text{vec}(\mathbf{X}\mathbf{B}) = \mathbf{M} \text{vec } \mathbf{X}$  for  $\mathbf{X} \in M_{m \times n}(F)$ ,  $\mathbf{B} \in M_n(F)$ .

*Solution.*

- $\mathbf{A} \otimes \mathbf{B}$  is the  $mp \times nq$  block matrix with  $(i, j)$  block  $a_{ij}\mathbf{B}$  ([Definition 7.4.1](#)); equivalently, its entry in position  $((i-1)p+k, (j-1)q+l)$  is  $a_{ij}b_{kl}$ .
- False. For  $\mathbf{A} = \begin{pmatrix} 1 & 2 \\ 3 & 4 \end{pmatrix}$  and  $\mathbf{B} = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}$ , the  $(1, 2)$  entry of  $\mathbf{A} \otimes \mathbf{B}$  is  $a_{11}b_{12} = 1$  and that of  $\mathbf{B} \otimes \mathbf{A}$  is  $b_{11}a_{12} = 0$ . The two are similar ([Proposition 7.4.2](#)), not equal.
- If  $\mathbf{A} \in M_{m \times n}(F)$ ,  $\mathbf{C} \in M_{n \times t}(F)$ ,  $\mathbf{B} \in M_{p \times q}(F)$  and  $\mathbf{D} \in M_{q \times u}(F)$ , then  $(\mathbf{A} \otimes \mathbf{B})(\mathbf{C} \otimes \mathbf{D}) = (\mathbf{A}\mathbf{C}) \otimes (\mathbf{B}\mathbf{D})$  ([Theorem 7.4.1](#) (c)). The conditions are exactly that  $\mathbf{A}\mathbf{C}$  and  $\mathbf{B}\mathbf{D}$  are defined.
- False. By [Theorem 7.4.3](#),  $\det(\mathbf{A} \otimes \mathbf{B}) = (\det \mathbf{A})^2(\det \mathbf{B})^2$ . For  $\mathbf{A} = \mathbf{B} = 2\mathbf{I}_2$ ,



$\mathbf{A} \otimes \mathbf{B} = 4\mathbf{I}_4$  has determinant 256, while  $\det \mathbf{A} \det \mathbf{B} = 16$ .

(e)  $\text{vec}(\mathbf{AXB}) = (\mathbf{B}^\top \otimes \mathbf{A}) \text{vec } \mathbf{X}$  (Theorem 7.4.4). With  $\mathbf{A} = \mathbf{I}_m$ ,  $\mathbf{M} = \mathbf{B}^\top \otimes \mathbf{I}_m$ .

### 7.4.7.2 B. Practice

#### Exercise B1: Computing with Kronecker products.

Let  $\mathbf{A} = \begin{pmatrix} 2 & 0 \\ 1 & -1 \end{pmatrix}$  and  $\mathbf{B} = \begin{pmatrix} 1 & 1 \\ 0 & 3 \end{pmatrix}$  over  $\mathbb{Q}$ .

- Write out  $\mathbf{A} \otimes \mathbf{B}$  and  $\mathbf{B} \otimes \mathbf{A}$ .
- Compute  $\text{tr}$ ,  $\det$  and  $\text{rank}$  of  $\mathbf{A} \otimes \mathbf{B}$  from Theorem 7.4.3, and check the trace directly.
- Compute  $\mathbf{A}^2 \otimes \mathbf{B}^2$  and use it to write down  $(\mathbf{A} \otimes \mathbf{B})^2$  without multiplying  $4 \times 4$  matrices. Justify the step.
- Hence find  $(\mathbf{A} \otimes \mathbf{B})^{-1}$ .

*Solution.*

- By Definition 7.4.1,

$$\mathbf{A} \otimes \mathbf{B} = \begin{pmatrix} 2\mathbf{B} & 0\mathbf{B} \\ 1\mathbf{B} & -1\mathbf{B} \end{pmatrix} = \begin{pmatrix} 2 & 2 & 0 & 0 \\ 0 & 6 & 0 & 0 \\ 1 & 1 & -1 & -1 \\ 0 & 3 & 0 & -3 \end{pmatrix},$$

$$\mathbf{B} \otimes \mathbf{A} = \begin{pmatrix} 1\mathbf{A} & 1\mathbf{A} \\ 0\mathbf{A} & 3\mathbf{A} \end{pmatrix} = \begin{pmatrix} 2 & 0 & 2 & 0 \\ 1 & -1 & 1 & -1 \\ 0 & 0 & 6 & 0 \\ 0 & 0 & 3 & -3 \end{pmatrix}.$$

- $\text{tr } \mathbf{A} = 1$ ,  $\text{tr } \mathbf{B} = 4$ ,  $\det \mathbf{A} = -2$ ,  $\det \mathbf{B} = 3$ , and both have rank 2. By Theorem 7.4.3,  $\text{tr}(\mathbf{A} \otimes \mathbf{B}) = 4$ ,  $\det(\mathbf{A} \otimes \mathbf{B}) = (-2)^2 \cdot 3^2 = 36$  and  $\text{rank}(\mathbf{A} \otimes \mathbf{B}) = 4$ . Directly, the diagonal of  $\mathbf{A} \otimes \mathbf{B}$  is 2, 6, -1, -3, with sum 4.
- By the mixed product rule (Theorem 7.4.1 (c)) with  $\mathbf{C} = \mathbf{A}$  and  $\mathbf{D} = \mathbf{B}$ ,  $(\mathbf{A} \otimes \mathbf{B})^2 = \mathbf{A}^2 \otimes \mathbf{B}^2$ . Here  $\mathbf{A}^2 = \begin{pmatrix} 4 & 0 \\ 1 & 1 \end{pmatrix}$  and  $\mathbf{B}^2 = \begin{pmatrix} 1 & 4 \\ 0 & 9 \end{pmatrix}$ , so

$$(\mathbf{A} \otimes \mathbf{B})^2 = \begin{pmatrix} 4\mathbf{B}^2 & 0 \\ \mathbf{B}^2 & \mathbf{B}^2 \end{pmatrix} = \begin{pmatrix} 4 & 16 & 0 & 0 \\ 0 & 36 & 0 & 0 \\ 1 & 4 & 1 & 4 \\ 0 & 9 & 0 & 9 \end{pmatrix}.$$

- $\mathbf{A}^{-1} = \frac{1}{-2} \begin{pmatrix} -1 & 0 \\ -1 & 2 \end{pmatrix} = \begin{pmatrix} \frac{1}{2} & 0 \\ \frac{1}{2} & -1 \end{pmatrix}$  and  $\mathbf{B}^{-1} = \frac{1}{3} \begin{pmatrix} 3 & -1 \\ 0 & 1 \end{pmatrix}$  by Theorem 0.9.8. By Theorem 7.4.1 (e),

$$(\mathbf{A} \otimes \mathbf{B})^{-1} = \mathbf{A}^{-1} \otimes \mathbf{B}^{-1} = \begin{pmatrix} \frac{1}{2}\mathbf{B}^{-1} & 0 \\ \frac{1}{2}\mathbf{B}^{-1} & -\mathbf{B}^{-1} \end{pmatrix} = \begin{pmatrix} \frac{1}{2} & -\frac{1}{6} & 0 & 0 \\ 0 & \frac{1}{6} & 0 & 0 \\ \frac{1}{2} & -\frac{1}{6} & -1 & \frac{1}{3} \\ 0 & \frac{1}{6} & 0 & -\frac{1}{3} \end{pmatrix}.$$

Multiplying by  $\mathbf{A} \otimes \mathbf{B}$  gives  $\mathbf{I}_4$ .

### Exercise B2: One-sided equations.

Let  $\mathbf{A} \in M_m(F)$  and  $\mathbf{C} \in M_{m \times n}(F)$ , and consider  $\mathbf{AX} = \mathbf{C}$  for  $\mathbf{X} \in M_{m \times n}(F)$ .

- Write the equation in the form  $\mathbf{M} \operatorname{vec} \mathbf{X} = \operatorname{vec} \mathbf{C}$ , and describe  $\mathbf{M}$  as a block matrix.
- Explain why the system splits into  $n$  separate systems, one for each column of  $\mathbf{X}$ .
- Hence show that  $\mathbf{AX} = \mathbf{C}$  has exactly one solution for every  $\mathbf{C}$  if and only if  $\mathbf{A}$  is invertible.
- Solve  $\mathbf{XA} = \mathbf{C}$  over  $\mathbb{Q}$  by the same method, where  $\mathbf{A} = \begin{pmatrix} 1 & 2 \\ 1 & 3 \end{pmatrix}$  and  $\mathbf{C} = \begin{pmatrix} 3 & 7 \\ 0 & 1 \end{pmatrix}$ .

*Solution.*

- By Theorem 7.4.4 with  $\mathbf{B} = \mathbf{I}_n$ ,  $\operatorname{vec}(\mathbf{AX}) = (\mathbf{I}_n \otimes \mathbf{A}) \operatorname{vec} \mathbf{X}$ , so  $\mathbf{M} = \mathbf{I}_n \otimes \mathbf{A} = \mathbf{A} \oplus \cdots \oplus \mathbf{A}$ , with  $n$  blocks.
- Write  $\operatorname{vec} \mathbf{X}$  and  $\operatorname{vec} \mathbf{C}$  as stacks of the columns  $\mathbf{x}_j$  and  $\mathbf{c}_j$ . By Theorem 7.1.1,  $(\mathbf{A} \oplus \cdots \oplus \mathbf{A}) \operatorname{vec} \mathbf{X}$  is the stack of the  $\mathbf{Ax}_j$ . So the system says  $\mathbf{Ax}_j = \mathbf{c}_j$  for  $j = 1, \dots, n$ , and the unknowns of different columns never meet.
- ( $\Leftarrow$ ) If  $\mathbf{A}$  is invertible, then  $\mathbf{AX} = \mathbf{C}$  forces  $\mathbf{X} = \mathbf{A}^{-1}\mathbf{C}$ , and this  $\mathbf{X}$  is a solution. ( $\Rightarrow$ ) Suppose  $\mathbf{AX} = \mathbf{C}$  is solvable for every  $\mathbf{C}$ . Given  $\mathbf{c} \in F^m$ , take  $\mathbf{C}$  with every column equal to  $\mathbf{c}$ ; by (b), a solution gives  $\mathbf{x}$  with  $\mathbf{Ax} = \mathbf{c}$ . So  $\mathbf{Ax} = \mathbf{c}$  is solvable for every  $\mathbf{c}$ , and  $\mathbf{A}$  is invertible by Theorem 2.4.4 ((e)  $\Rightarrow$  (a)).
- By Theorem 7.4.4 with the left factor  $\mathbf{I}_2$ ,  $\operatorname{vec}(\mathbf{XA}) = (\mathbf{A}^\top \otimes \mathbf{I}_2) \operatorname{vec} \mathbf{X}$ . With  $\mathbf{A}^\top = \begin{pmatrix} 1 & 1 \\ 2 & 3 \end{pmatrix}$ ,  $\operatorname{vec} \mathbf{X} = (x_{11}, x_{21}, x_{12}, x_{22})$  and  $\operatorname{vec} \mathbf{C} = (3, 0, 7, 1)$ , the system is

$$\begin{pmatrix} 1 & 0 & 1 & 0 \\ 0 & 1 & 0 & 1 \\ 2 & 0 & 3 & 0 \\ 0 & 2 & 0 & 3 \end{pmatrix} \begin{pmatrix} x_{11} \\ x_{21} \\ x_{12} \\ x_{22} \end{pmatrix} = \begin{pmatrix} 3 \\ 0 \\ 7 \\ 1 \end{pmatrix}.$$

This time the system splits by the **rows** of  $\mathbf{X}$ : equations 1, 3 involve only  $x_{11}, x_{12}$ , and equations 2, 4 only  $x_{21}, x_{22}$ . From  $x_{11} + x_{12} = 3$  and  $2x_{11} + 3x_{12} = 7$  we get  $x_{12} = 1$ ,  $x_{11} = 2$ . From  $x_{21} + x_{22} = 0$  and  $2x_{21} + 3x_{22} = 1$  we get  $x_{22} = 1$ ,  $x_{21} = -1$ . Hence

$$\mathbf{X} = \begin{pmatrix} 2 & 1 \\ -1 & 1 \end{pmatrix}.$$

$$\text{Check: } \mathbf{XA} = \begin{pmatrix} 2+1 & 4+3 \\ -1+1 & -2+3 \end{pmatrix} = \begin{pmatrix} 3 & 7 \\ 0 & 1 \end{pmatrix} = \mathbf{C}.$$

### Exercise B3: Which identities hold?.

Determine which of the following statements hold for **all**  $\mathbf{A}, \mathbf{B}, \mathbf{C} \in M_2(\mathbb{R})$ . Justify your answer.

- (a)  $(\mathbf{A} \otimes \mathbf{B})(\mathbf{A} \otimes \mathbf{B}) = \mathbf{A}^2 \otimes \mathbf{B}^2$ .
- (b)  $(\mathbf{A} + \mathbf{B}) \otimes \mathbf{C} = \mathbf{A} \otimes \mathbf{C} + \mathbf{B} \otimes \mathbf{C}$ .
- (c)  $(\mathbf{A} \otimes \mathbf{B})^\top = \mathbf{B}^\top \otimes \mathbf{A}^\top$ .
- (d)  $\mathbf{A} \otimes \mathbf{B} = 0$  implies  $\mathbf{A} = 0$  or  $\mathbf{B} = 0$ .
- (e)  $\text{tr}(\mathbf{A} \otimes \mathbf{B}) = \text{tr}(\mathbf{B} \otimes \mathbf{A})$ .
- (f)  $\mathbf{A} \otimes \mathbf{B} + \mathbf{B} \otimes \mathbf{A} = 2(\mathbf{A} \otimes \mathbf{B})$ .

*Solution.*

- (a) Holds, by the mixed product rule (Theorem 7.4.1 (c)).
- (b) Holds, by bilinearity (Theorem 7.4.1 (a)).
- (c) Fails. By Theorem 7.4.1 (d),  $(\mathbf{A} \otimes \mathbf{B})^\top = \mathbf{A}^\top \otimes \mathbf{B}^\top$ , and this differs from  $\mathbf{B}^\top \otimes \mathbf{A}^\top$  in general. Take  $\mathbf{A} = \begin{pmatrix} 1 & 3 \\ 2 & 4 \end{pmatrix}$  and  $\mathbf{B} = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix} = \mathbf{B}^\top$ . By (\*), the  $(1, 2)$  entry of  $\mathbf{A}^\top \otimes \mathbf{B}^\top$  is  $(\mathbf{A}^\top)_{11}b_{12} = 1$ , while that of  $\mathbf{B}^\top \otimes \mathbf{A}^\top$  is  $b_{11}(\mathbf{A}^\top)_{12} = 0$ . Unlike  $(\mathbf{AB})^\top = \mathbf{B}^\top \mathbf{A}^\top$ , the order of the factors does not reverse.
- (d) Holds. If  $\mathbf{A} \neq 0$  and  $\mathbf{B} \neq 0$ , then  $\text{rank } \mathbf{A} \geq 1$  and  $\text{rank } \mathbf{B} \geq 1$ , so  $\text{rank}(\mathbf{A} \otimes \mathbf{B}) \geq 1$  by Theorem 7.4.3 (a), and  $\mathbf{A} \otimes \mathbf{B} \neq 0$ . (Directly: if  $a_{ij} \neq 0$  and  $b_{kl} \neq 0$ , the entry  $a_{ij}b_{kl}$  of  $\mathbf{A} \otimes \mathbf{B}$  is non-zero.)
- (e) Holds: both equal  $\text{tr } \mathbf{A} \text{tr } \mathbf{B}$  by Theorem 7.4.3 (b).
- (f) Fails, since it is equivalent to  $\mathbf{B} \otimes \mathbf{A} = \mathbf{A} \otimes \mathbf{B}$ , which is false for the matrices of Example 1 after Definition 7.4.1 (see the non-example there).

### 7.4.7.3 C. Going deeper

**Exercise C1: When is a Kronecker product invertible?**

- (a) Let  $\mathbf{A} \in M_n(F)$  and  $\mathbf{B} \in M_m(F)$ . Prove that  $\mathbf{A} \otimes \mathbf{B}$  is invertible if and only if both  $\mathbf{A}$  and  $\mathbf{B}$  are invertible.
- (b) Let  $\mathbf{A} \in M_{m \times n}(F)$  and  $\mathbf{B} \in M_{p \times q}(F)$  with  $mp = nq$ , so that  $\mathbf{A} \otimes \mathbf{B}$  is square. Prove that if  $m \neq n$ , then  $\mathbf{A} \otimes \mathbf{B}$  is **not** invertible.
- (c) Give an example of a  $2 \times 1$  matrix  $\mathbf{A}$  and a  $1 \times 2$  matrix  $\mathbf{B}$  for which  $\mathbf{A} \otimes \mathbf{B}$  is a non-zero  $2 \times 2$  matrix, and compute its rank.

*Solution.*

- (a) By Theorem 7.4.3 (c),  $\det(\mathbf{A} \otimes \mathbf{B}) = (\det \mathbf{A})^m (\det \mathbf{B})^n$ . In a field a product is non-zero if and only if each factor is, so this is non-zero if and only if  $\det \mathbf{A} \neq 0$  and  $\det \mathbf{B} \neq 0$  (here  $n, m \geq 1$ ). By Theorem 6.5.2, applied to all three matrices,  $\mathbf{A} \otimes \mathbf{B}$  is invertible if and only if  $\mathbf{A}$  and  $\mathbf{B}$  are.
- (b) Suppose  $m < n$ . Since  $mp = nq$  and  $m < n$ , we get  $p > q$ . By Theorem 7.4.3 (a) and the bound  $\text{rank } \mathbf{M} \leq (\text{number of rows})$  and  $\leq (\text{number of columns})$ ,

$$\text{rank}(\mathbf{A} \otimes \mathbf{B}) = \text{rank } \mathbf{A} \text{rank } \mathbf{B} \leq m \cdot q < n \cdot q = nq,$$

so the  $nq \times nq$  matrix  $\mathbf{A} \otimes \mathbf{B}$  is not invertible (Theorem 2.4.4). If  $m > n$ , then  $p < q$ , and  $\text{rank } \mathbf{A} \text{rank } \mathbf{B} \leq n \cdot p < m \cdot p = mp = nq$  in the same way.

- (c) Take  $\mathbf{A} = \begin{pmatrix} 1 \\ 2 \end{pmatrix}$  and  $\mathbf{B} = \begin{pmatrix} 3 & 1 \end{pmatrix}$ . Then  $\mathbf{A} \otimes \mathbf{B} = \begin{pmatrix} 1\mathbf{B} \\ 2\mathbf{B} \end{pmatrix} = \begin{pmatrix} 3 & 1 \\ 6 & 2 \end{pmatrix}$ , the outer product  $\mathbf{AB}$ , of rank  $1 \cdot 1 = 1$ : its second row is twice its first. So it is square and non-zero but singular, as (b) predicts.

### Exercise C2: Kronecker sums.

Let  $\mathbf{A} \in M_n(F)$  and  $\mathbf{B} \in M_m(F)$ , and put  $\mathbf{K} := \mathbf{A} \otimes \mathbf{I}_m + \mathbf{I}_n \otimes \mathbf{B} \in M_{nm}(F)$ .

- (a) Prove that  $(\mathbf{A} \otimes \mathbf{I}_m)(\mathbf{I}_n \otimes \mathbf{B}) = (\mathbf{I}_n \otimes \mathbf{B})(\mathbf{A} \otimes \mathbf{I}_m)$ , and deduce that  $\mathbf{K}^2 = \mathbf{A}^2 \otimes \mathbf{I}_m + 2(\mathbf{A} \otimes \mathbf{B}) + \mathbf{I}_n \otimes \mathbf{B}^2$ .
- (b) Show that for  $\mathbf{X} \in M_{m \times n}(F)$ ,  $\text{vec}(\mathbf{BX} + \mathbf{XA}^\top) = \mathbf{K} \text{vec } \mathbf{X}$ .
- (c) Suppose  $\mathbf{A}$  and  $\mathbf{B}$  are upper triangular. Prove that  $\mathbf{K}$  is upper triangular with diagonal entries  $a_{ii} + b_{kk}$ , and deduce that  $\mathbf{BX} + \mathbf{XA}^\top = \mathbf{C}$  has exactly one solution for every  $\mathbf{C} \in M_{m \times n}(F)$  if and only if  $a_{ii} + b_{kk} \neq 0$  for all  $i, k$ .
- (d) For  $\mathbf{A} = \begin{pmatrix} 1 & 2 \\ 0 & -1 \end{pmatrix}$  and  $\mathbf{B} = \begin{pmatrix} 1 & 0 \\ 0 & 3 \end{pmatrix}$  over  $\mathbb{Q}$ , decide whether  $\mathbf{BX} + \mathbf{XA}^\top = \mathbf{C}$  is solvable for every  $\mathbf{C}$ . If not, find a  $\mathbf{C}$  with no solution.

*Solution.*

- (a) By the mixed product rule, both products equal  $(\mathbf{A}\mathbf{I}_n) \otimes (\mathbf{I}_m\mathbf{B}) = (\mathbf{I}_n\mathbf{A}) \otimes (\mathbf{B}\mathbf{I}_m) = \mathbf{A} \otimes \mathbf{B}$ , as in (\*\*). Expanding  $\mathbf{K}^2$  by distributivity (Theorem 0.9.4) gives  $(\mathbf{A} \otimes \mathbf{I}_m)^2 + (\mathbf{A} \otimes \mathbf{I}_m)(\mathbf{I}_n \otimes \mathbf{B}) + (\mathbf{I}_n \otimes \mathbf{B})(\mathbf{A} \otimes \mathbf{I}_m) + (\mathbf{I}_n \otimes \mathbf{B})^2$ . The middle terms are both  $\mathbf{A} \otimes \mathbf{B}$ , and the outer ones are  $\mathbf{A}^2 \otimes \mathbf{I}_m$  and  $\mathbf{I}_n \otimes \mathbf{B}^2$  by the mixed product rule.
- (b) By Theorem 7.4.4,  $\text{vec}(\mathbf{BXI}_n) = (\mathbf{I}_n \otimes \mathbf{B}) \text{vec } \mathbf{X}$  and  $\text{vec}(\mathbf{I}_m\mathbf{XA}^\top) = ((\mathbf{A}^\top)^\top \otimes \mathbf{I}_m) \text{vec } \mathbf{X} = (\mathbf{A} \otimes \mathbf{I}_m) \text{vec } \mathbf{X}$ . Add, using linearity of  $\text{vec}$ .
- (c) By (\*), the entry of  $\mathbf{A} \otimes \mathbf{I}_m$  in position  $((i-1)m+k, (j-1)m+l)$  is  $a_{ij}\delta_{kl}$ , which is zero unless  $i \leq j$  and  $k = l$ ; in that case  $(i-1)m+k \leq (j-1)m+l$ . The entry of  $\mathbf{I}_n \otimes \mathbf{B}$  there is  $\delta_{ij}b_{kl}$ , zero unless  $i = j$  and  $k \leq l$ , and again the row index is at most the column index. So both are upper triangular, hence so is  $\mathbf{K}$ . On the diagonal,  $i = j$  and  $k = l$ , and the entry is  $a_{ii} + b_{kk}$ . By Theorem 6.4.6,  $\det \mathbf{K} = \prod_{i,k} (a_{ii} + b_{kk})$ . By (b), the equation is  $\mathbf{K} \text{vec } \mathbf{X} = \text{vec } \mathbf{C}$ , and since  $\text{vec}$  is an isomorphism, it has exactly one solution for every  $\mathbf{C}$  if and only if  $\mathbf{K}$  is invertible (Theorem 2.4.4), that is  $\det \mathbf{K} \neq 0$  (Theorem 6.5.2), that is every  $a_{ii} + b_{kk} \neq 0$ .
- (d) The diagonal entries  $a_{ii} + b_{kk}$  are  $1 + 1 = 2$ ,  $1 + 3 = 4$ ,  $-1 + 1 = 0$ ,  $-1 + 3 = 2$ , so by (c) the equation is not solvable for every  $\mathbf{C}$ . Explicitly,

$$\mathbf{K} = \begin{pmatrix} 1\mathbf{I}_2 & 2\mathbf{I}_2 \\ 0\mathbf{I}_2 & -1\mathbf{I}_2 \end{pmatrix} + \begin{pmatrix} \mathbf{B} & 0 \\ 0 & \mathbf{B} \end{pmatrix} = \begin{pmatrix} 2 & 0 & 2 & 0 \\ 0 & 4 & 0 & 2 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 2 \end{pmatrix},$$

whose third row is zero. So  $\mathbf{K} \text{vec } \mathbf{X}$  always has third entry 0, and any  $\mathbf{C}$  with  $(\text{vec } \mathbf{C})_3 = c_{12} \neq 0$  has no solution, for example  $\mathbf{C} = \mathbf{E}_{12}$ . (Directly: the  $(1, 2)$  entry of  $\mathbf{BX} + \mathbf{XA}^\top$  is  $x_{12} + (x_{11}a_{21} + x_{12}a_{22}) = x_{12} - x_{12} = 0$ .)

## 7.5 Commutators and Trace-Zero Matrices

The expression  $\mathbf{AB} - \mathbf{BA}$  has come up several times already. In Chapter 0 its trace was 0, which showed that  $\mathbf{AB} - \mathbf{BA} = \mathbf{I}_n$  is impossible over  $\mathbb{R}$  (Exercise (C1:  $\mathbf{AB} - \mathbf{BA} = \mathbf{I}$  is impossible, or is it?)). In Chapter 3 the same trace argument showed that the relation  $DS - SD = \text{id}$  between differentiation and multiplication by  $x$  on  $F[x]$  has no finite-dimensional analogue in characteristic 0 (Exercise (C1: The Heisenberg relation)). Both arguments use one direction only: every matrix of the form  $\mathbf{AB} - \mathbf{BA}$  has trace zero. This section asks whether the converse holds. Is every trace-zero matrix of the form  $\mathbf{AB} - \mathbf{BA}$ ? The answer is yes over a field of characteristic 0, and the proof is a good example of the block techniques of this chapter: move the matrix into a convenient shape by a similarity, one row and column at a time, and then solve an equation that has become entrywise.

**Which fields.** Commutators, their basic rules, the first similarity lemma and the final theorem on spans work over **every** field  $F$ . The lemma on zero diagonals and Shoda's theorem assume that  $F$  has **characteristic** 0. We point out exactly where that hypothesis is used, and what happens in characteristic  $p$ .

### 7.5.1 Commutators

Two matrices  $\mathbf{A}$  and  $\mathbf{B}$  commute when  $\mathbf{AB} = \mathbf{BA}$ , and most pairs do not. The difference  $\mathbf{AB} - \mathbf{BA}$  measures the failure, and it keeps recurring: in the trace arguments above, in the relation between differentiation and multiplication by  $x$ , and in the description of trace-like functionals (Exercise (C2: The trace is the only trace-like functional)). So we give it a name.

*The commutator of two matrices is the amount by which they fail to commute.*

#### Definition 7.5.1 — Commutator

Let  $n \geq 1$  and  $\mathbf{A}, \mathbf{B} \in M_n(F)$ . The **commutator** of  $\mathbf{A}$  and  $\mathbf{B}$  is

$$[\mathbf{A}, \mathbf{B}] := \mathbf{AB} - \mathbf{BA} \in M_n(F).$$

A matrix  $\mathbf{C} \in M_n(F)$  is a **commutator** if  $\mathbf{C} = [\mathbf{X}, \mathbf{Y}]$  for **some**  $\mathbf{X}, \mathbf{Y} \in M_n(F)$ . For operators  $S, T \in \mathcal{L}(V)$  on a vector space  $V$ , we write likewise  $[S, T] := ST - TS \in \mathcal{L}(V)$ .

In words:  $[\mathbf{A}, \mathbf{B}]$  is “ $\mathbf{A}$  then  $\mathbf{B}$ ” minus “ $\mathbf{B}$  then  $\mathbf{A}$ ”, in that order, and it is the zero matrix exactly when  $\mathbf{A}$  and  $\mathbf{B}$  commute. The second sentence of the definition is about a single matrix: being a commutator means that **there exist** two matrices whose commutator it is, and it says nothing about which ones.

**Examples.**

1. **Degenerate cases.**  $[\mathbf{A}, \mathbf{A}] = 0$  and  $[\mathbf{A}, c\mathbf{I}_n] = c\mathbf{A} - c\mathbf{A} = 0$  for every  $c \in F$ . So the zero matrix is a commutator, in many ways. In  $M_1(F) = F$  every commutator is  $ab - ba = 0$ , so for  $n = 1$  the only commutator is 0; this is the smallest case, and it already matches the trace condition, since the only  $1 \times 1$  matrix of trace 0 is (0).
2. **Matrix units.** Recall that  $\mathbf{E}_{ij}\mathbf{E}_{kl}$  is  $\mathbf{E}_{il}$  if  $j = k$  and 0 otherwise (Definition 0.9.9). So

for  $i \neq k$ ,

$$[\mathbf{E}_{ij}, \mathbf{E}_{jk}] = \mathbf{E}_{ik} - 0 = \mathbf{E}_{ik}, \quad \text{and} \quad [\mathbf{E}_{ij}, \mathbf{E}_{ji}] = \mathbf{E}_{ii} - \mathbf{E}_{jj} \quad (i \neq j).$$

For instance  $[\mathbf{E}_{12}, \mathbf{E}_{21}] = \text{diag}(1, -1)$  in  $M_2(F)$ , and  $[\mathbf{E}_{11}, \mathbf{E}_{12}] = \mathbf{E}_{12}$ . Both have trace 0.

3. **Operators on an infinite-dimensional space.** On  $F[x]$ , with  $D$  differentiation and  $S$  multiplication by  $x$ , [Exercise \(C1: The Heisenberg relation\)](#) (a) shows  $[D, S] = \text{id}$ . So the identity is a commutator of operators on  $F[x]$ . This does not contradict anything, because there is no trace on  $\mathcal{L}(F[x])$ : the trace was defined only on finite-dimensional spaces ([Definition 3.10.2](#)).

**Non-example by minimal change.** Change the minus sign to a plus:  $\mathbf{AB} + \mathbf{BA}$ . It is still linear in  $\mathbf{A}$  and in  $\mathbf{B}$ , and it is still built from the two products  $\mathbf{AB}$  and  $\mathbf{BA}$ . What is lost is the trace:  $\text{tr}(\mathbf{AB} + \mathbf{BA}) = 2 \text{tr}(\mathbf{AB})$ , which has no reason to vanish. For example  $\mathbf{E}_{12}\mathbf{E}_{21} + \mathbf{E}_{21}\mathbf{E}_{12} = \mathbf{E}_{11} + \mathbf{E}_{22} = \mathbf{I}_2$ , with trace 2. It also no longer detects commuting:  $\mathbf{AA} + \mathbf{AA} = 2\mathbf{A}^2$  is non-zero for  $\mathbf{A} = \mathbf{I}_n$  over  $\mathbb{Q}$ , although  $\mathbf{A}$  commutes with itself.

**Why this definition.** The order in  $\mathbf{AB} - \mathbf{BA}$  is a convention; the other order gives  $[\mathbf{B}, \mathbf{A}] = -[\mathbf{A}, \mathbf{B}]$ , which is a commutator too. The notion is invariant under a change of basis: a similarity applied to  $\mathbf{X}$  and  $\mathbf{Y}$  applies to  $[\mathbf{X}, \mathbf{Y}]$ . So “being a commutator” is a property of the operator  $\mathbf{x} \mapsto \mathbf{C}\mathbf{x}$ , not of the basis used to write  $\mathbf{C}$ , and we are free to choose a convenient basis. This is what the proof of the main theorem exploits.

The basic rules are collected here.

### Proposition 7.5.1 — Rules for Commutators

Let  $\mathbf{A}, \mathbf{A}', \mathbf{B} \in M_n(F)$ ,  $c \in F$ , and let  $\mathbf{P} \in M_n(F)$  be invertible. Then:

- (a)  $[\mathbf{A} + \mathbf{A}', \mathbf{B}] = [\mathbf{A}, \mathbf{B}] + [\mathbf{A}', \mathbf{B}]$ ,  $[c\mathbf{A}, \mathbf{B}] = c[\mathbf{A}, \mathbf{B}]$ , and  $[\mathbf{B}, \mathbf{A}] = -[\mathbf{A}, \mathbf{B}]$ ; in particular the commutator is linear in each slot.
- (b)  $\mathbf{P}^{-1}[\mathbf{A}, \mathbf{B}]\mathbf{P} = [\mathbf{P}^{-1}\mathbf{A}\mathbf{P}, \mathbf{P}^{-1}\mathbf{B}\mathbf{P}]$ .
- (c)  $\text{tr}[\mathbf{A}, \mathbf{B}] = 0$ .

*Proof.*

- (a) By [Theorem 0.9.4](#),  $(\mathbf{A} + \mathbf{A}')\mathbf{B} - \mathbf{B}(\mathbf{A} + \mathbf{A}') = (\mathbf{AB} - \mathbf{BA}) + (\mathbf{A}'\mathbf{B} - \mathbf{BA}')$  and  $(c\mathbf{A})\mathbf{B} - \mathbf{B}(c\mathbf{A}) = c(\mathbf{AB} - \mathbf{BA})$ . The last identity is  $\mathbf{BA} - \mathbf{AB} = -(\mathbf{AB} - \mathbf{BA})$ . Linearity in the second slot follows from the first slot and antisymmetry.
- (b) Inserting  $\mathbf{PP}^{-1} = \mathbf{I}_n$ ,  $\mathbf{P}^{-1}\mathbf{A}\mathbf{B}\mathbf{P} = (\mathbf{P}^{-1}\mathbf{A}\mathbf{P})(\mathbf{P}^{-1}\mathbf{B}\mathbf{P})$ , and likewise for  $\mathbf{BA}$ . Subtract, using distributivity.
- (c) By [Theorem 0.9.6](#) (1) and (3),  $\text{tr}(\mathbf{AB} - \mathbf{BA}) = \text{tr}(\mathbf{AB}) - \text{tr}(\mathbf{BA}) = 0$ .

■

So every commutator lies in the subspace of trace-zero matrices of [Example \(Trace-Free Matrices\)](#). The question of this section is whether it fills that subspace.

## 7.5.2 The identity in characteristic $p$

Rule (c) is the whole content of the Heisenberg exercises: if  $[\mathbf{X}, \mathbf{Y}] = \mathbf{I}_n$ , then  $n \cdot 1 = \text{tr } \mathbf{I}_n = 0$ , which is impossible when  $F$  has characteristic 0 (Definition 0.7.2). The obstruction is the number  $n \cdot 1$ , not the identity matrix itself. In characteristic  $p$ , the  $p \times p$  identity has trace  $p \cdot 1 = 0$ , and the obstruction disappears. In fact the pair  $D, S$  of Example 3 survives in a finite form.

### Example. (The Identity as a Commutator in Characteristic $p$ )

Let  $F$  have prime characteristic  $p$  (for instance  $F = \mathbb{F}_p$ ). Let  $\mathbf{D}_p \in M_p(F)$  have  $(k, k+1)$ -entry  $k \cdot 1$  for  $1 \leq k \leq p-1$ , and let  $\mathbf{S}_p \in M_p(F)$  have  $(k+1, k)$ -entry 1 for  $1 \leq k \leq p-1$ , all other entries 0. Show that  $[\mathbf{D}_p, \mathbf{S}_p] = \mathbf{I}_p$ .

*Solution.* The columns of these matrices describe differentiation and multiplication by  $x$  on the list  $1, x, \dots, x^{p-1}$ , where  $x^p$  is replaced by 0. By Theorem 0.9.2,  $\mathbf{S}_p \mathbf{e}_k = \mathbf{e}_{k+1}$  for  $k < p$  and  $\mathbf{S}_p \mathbf{e}_p = \mathbf{0}$ , while  $\mathbf{D}_p \mathbf{e}_1 = \mathbf{0}$  and  $\mathbf{D}_p \mathbf{e}_k = (k-1)\mathbf{e}_{k-1}$  for  $k \geq 2$ . Hence

$$\mathbf{D}_p \mathbf{S}_p \mathbf{e}_k = k \mathbf{e}_k \quad (k < p), \quad \mathbf{D}_p \mathbf{S}_p \mathbf{e}_p = \mathbf{0}, \quad \mathbf{S}_p \mathbf{D}_p \mathbf{e}_k = (k-1) \mathbf{e}_k \quad (1 \leq k \leq p),$$

where for  $k=1$  we used  $\mathbf{S}_p \mathbf{D}_p \mathbf{e}_1 = \mathbf{0} = 0 \mathbf{e}_1$ . Subtracting,  $[\mathbf{D}_p, \mathbf{S}_p] \mathbf{e}_k = \mathbf{e}_k$  for  $k < p$ , and  $[\mathbf{D}_p, \mathbf{S}_p] \mathbf{e}_p = -(p-1)\mathbf{e}_p = (1-p)\mathbf{e}_p = \mathbf{e}_p$ , because  $p \cdot 1 = 0$  in  $F$  (Definition 0.7.2, Lemma 0.7.4). So every column of  $[\mathbf{D}_p, \mathbf{S}_p]$  is the corresponding column of  $\mathbf{I}_p$ . For  $p=2$  this is the pair of Exercise (C1: The Heisenberg relation) (d), transposed.

*Warning.* “The identity is never a commutator” is a statement about characteristic 0. The trace test says that a commutator has trace 0, and  $\text{tr } \mathbf{I}_n = n \cdot 1$ . Over  $\mathbb{R}$  this is  $n \neq 0$ ; over  $\mathbb{F}_p$  it is 0 whenever  $p$  divides  $n$ , and then the test is silent. The example above shows that  $\mathbf{I}_p$  really is a commutator over  $\mathbb{F}_p$ . Whenever an argument ends with “ $n \cdot 1 = 0$ , contradiction”, check which fields it covers.

## 7.5.3 Trace-zero matrices with zero diagonal

We want to write a given trace-zero  $\mathbf{A}$  as  $[\mathbf{X}, \mathbf{Y}]$ . Look first at the easiest choice of  $\mathbf{X}$ , a diagonal matrix  $\mathbf{X} = \text{diag}(x_1, \dots, x_n)$ . For any  $\mathbf{Y} = (y_{ij})$ , Definition 0.9.9 gives  $(\mathbf{XY})_{ij} = x_i y_{ij}$  and  $(\mathbf{YX})_{ij} = y_{ij} x_j$ , so

$$[\mathbf{X}, \mathbf{Y}]_{ij} = (x_i - x_j) y_{ij}. \quad (*)$$

The diagonal entries of  $[\mathbf{X}, \mathbf{Y}]$  are all 0, and each off-diagonal entry can be anything, provided  $x_i \neq x_j$ . So a diagonal  $\mathbf{X}$  with distinct entries produces **exactly** the matrices with zero diagonal. A trace-zero matrix need not have zero diagonal, but by Proposition 7.5.1 (b) we may replace  $\mathbf{A}$  by any matrix similar to it. The plan is therefore:

- ① find a similarity that makes the diagonal of  $\mathbf{A}$  zero; ② solve  $(*)$  entrywise.

Step ① is the heart of the matter. We do it one diagonal entry at a time, starting with the  $(1, 1)$ -entry.

A matrix is **scalar** if it equals  $\lambda \mathbf{I}_n$  for some  $\lambda \in F$ . A scalar matrix is similar only to itself, since  $\mathbf{P}^{-1}(\lambda \mathbf{I}_n)\mathbf{P} = \lambda \mathbf{I}_n$ , so its diagonal cannot be changed. Every other matrix can at least be given a zero in the corner. This first lemma holds over every field.

### Lemma 7.5.2 — A Non-Scalar Matrix Has a Similar Matrix with Zero Corner

Let  $F$  be any field, let  $n \geq 2$ , and let  $\mathbf{A} \in M_n(F)$  be **not** scalar. Then there is an invertible  $\mathbf{P} \in M_n(F)$  such that the first column of  $\mathbf{P}^{-1}\mathbf{A}\mathbf{P}$  is  $\mathbf{e}_2$ . In particular

$$\mathbf{P}^{-1}\mathbf{A}\mathbf{P} = \begin{pmatrix} 0 & \mathbf{r}^\top \\ \mathbf{e}_1 & \mathbf{A}_1 \end{pmatrix}$$

for some  $\mathbf{r} \in F^{n-1}$  and  $\mathbf{A}_1 \in M_{n-1}(F)$  with  $\text{tr } \mathbf{A}_1 = \text{tr } \mathbf{A}$ , where  $\mathbf{e}_1 \in F^{n-1}$ .

*Idea.* Work backwards from the goal. The first column of  $\mathbf{P}^{-1}\mathbf{A}\mathbf{P}$  is the coordinate vector of  $\mathbf{A}\mathbf{p}_1$  in the basis formed by the columns  $\mathbf{p}_1, \dots, \mathbf{p}_n$  of  $\mathbf{P}$ . For it to be  $\mathbf{e}_2$ , we need  $\mathbf{A}\mathbf{p}_1 = \mathbf{p}_2$ . So we want a vector  $\mathbf{v}$  such that  $\mathbf{v}$  and  $\mathbf{A}\mathbf{v}$  are independent, and then we build a basis starting  $\mathbf{v}, \mathbf{A}\mathbf{v}$ . Such a  $\mathbf{v}$  exists unless  $\mathbf{A}$  sends every vector to a multiple of itself, and a short argument shows that this forces  $\mathbf{A}$  to be scalar.

*Proof.* We first show that there is  $\mathbf{v} \in F^n$  with  $(\mathbf{v}, \mathbf{A}\mathbf{v})$  linearly independent. Suppose not. Then for every  $\mathbf{v} \neq \mathbf{0}$  there are  $a, b \in F$ , not both zero, with  $a\mathbf{v} + b\mathbf{A}\mathbf{v} = \mathbf{0}$ . If  $b = 0$ , then  $a \neq 0$  and  $a\mathbf{v} = \mathbf{0}$ , which forces  $\mathbf{v} = \mathbf{0}$  (Theorem 1.1.9), a contradiction; so  $b \neq 0$  and  $\mathbf{A}\mathbf{v} = \lambda_{\mathbf{v}}\mathbf{v}$  with  $\lambda_{\mathbf{v}} = -a/b$ . Apply this to  $\mathbf{e}_i, \mathbf{e}_j$  and  $\mathbf{e}_i + \mathbf{e}_j$  for  $i \neq j$ :

$$\lambda_{\mathbf{e}_i+\mathbf{e}_j}\mathbf{e}_i + \lambda_{\mathbf{e}_i+\mathbf{e}_j}\mathbf{e}_j = \mathbf{A}(\mathbf{e}_i + \mathbf{e}_j) = \mathbf{A}\mathbf{e}_i + \mathbf{A}\mathbf{e}_j = \lambda_{\mathbf{e}_i}\mathbf{e}_i + \lambda_{\mathbf{e}_j}\mathbf{e}_j.$$

Since  $\mathbf{e}_i, \mathbf{e}_j$  are linearly independent, comparing coefficients (Theorem 1.4.3) gives  $\lambda_{\mathbf{e}_i} = \lambda_{\mathbf{e}_i+\mathbf{e}_j} = \lambda_{\mathbf{e}_j}$ . Hence all  $\lambda_{\mathbf{e}_i}$  equal a common  $\lambda$ , so  $\mathbf{A}\mathbf{e}_i = \lambda\mathbf{e}_i$  for every  $i$ , and by Theorem 0.9.2 the columns of  $\mathbf{A}$  are those of  $\lambda\mathbf{I}_n$ . So  $\mathbf{A} = \lambda\mathbf{I}_n$  is scalar, contrary to the hypothesis.

Take such a  $\mathbf{v}$ . By Theorem 1.6.5, the independent list  $(\mathbf{v}, \mathbf{A}\mathbf{v})$  extends to a basis  $(\mathbf{v}, \mathbf{A}\mathbf{v}, \mathbf{w}_3, \dots, \mathbf{w}_n)$  of  $F^n$ . Let  $\mathbf{P}$  be the matrix with these columns; it is invertible by Theorem 2.4.4. Then  $\mathbf{P}\mathbf{e}_1 = \mathbf{v}$  and  $\mathbf{P}\mathbf{e}_2 = \mathbf{A}\mathbf{v}$  (Theorem 0.9.3), so the first column of  $\mathbf{P}^{-1}\mathbf{A}\mathbf{P}$  is

$$\mathbf{P}^{-1}\mathbf{A}\mathbf{P}\mathbf{e}_1 = \mathbf{P}^{-1}\mathbf{A}\mathbf{v} = \mathbf{P}^{-1}\mathbf{P}\mathbf{e}_2 = \mathbf{e}_2.$$

Partitioning  $\mathbf{P}^{-1}\mathbf{A}\mathbf{P}$  after the first row and column gives the displayed form. Finally,  $\text{tr}(\mathbf{P}^{-1}\mathbf{A}\mathbf{P}) = \text{tr } \mathbf{A}$  by Theorem 3.7.5, and  $\text{tr}(\mathbf{P}^{-1}\mathbf{A}\mathbf{P}) = 0 + \text{tr } \mathbf{A}_1$  by Definition 0.9.12, so  $\text{tr } \mathbf{A}_1 = \text{tr } \mathbf{A}$ . ■

*Quick check.* Over  $\mathbb{Q}$ , let  $\mathbf{A} = \begin{pmatrix} 2 & 1 \\ 0 & -2 \end{pmatrix}$ . Using  $\mathbf{v} = \mathbf{e}_2$  in the proof of Lemma 7.5.2, find  $\mathbf{P}$  and  $\mathbf{P}^{-1}\mathbf{A}\mathbf{P}$ , and check that its diagonal is zero.

*Solution.*  $\mathbf{A}\mathbf{e}_2 = (1, -2)$ , which is not a multiple of  $\mathbf{e}_2$ , so  $(\mathbf{e}_2, \mathbf{A}\mathbf{e}_2)$  is independent and already a basis of  $\mathbb{Q}^2$ . Then  $\mathbf{P} = \begin{pmatrix} 0 & 1 \\ 1 & -2 \end{pmatrix}$ , with  $\det \mathbf{P} = -1$  and  $\mathbf{P}^{-1} = \begin{pmatrix} 2 & 1 \\ 1 & 0 \end{pmatrix}$ . We



get  $\mathbf{AP} = \begin{pmatrix} 1 & 0 \\ -2 & 4 \end{pmatrix}$  and  $\mathbf{P}^{-1}\mathbf{AP} = \begin{pmatrix} 0 & 4 \\ 1 & 0 \end{pmatrix}$ . The first column is  $\mathbf{e}_2$ , as the lemma promises; the  $(2, 2)$ -entry is 0 as well, because the trace is 0 and trace is a similarity invariant.

The lemma controls one diagonal entry and hands us a smaller matrix  $\mathbf{A}_1$  with the same trace. That is exactly the setting for induction on the size. The only thing that can stop the induction is a scalar  $\mathbf{A}_1$ , and here the trace saves us, provided the field has characteristic 0.

### Lemma 7.5.3 — Trace Zero Means Similar to Zero Diagonal

Let  $F$  be a field of **characteristic** 0, let  $n \geq 1$ , and let  $\mathbf{A} \in M_n(F)$  with  $\text{tr } \mathbf{A} = 0$ . Then  $\mathbf{A}$  is similar to a matrix all of whose diagonal entries are 0.

*Idea.* Induction on  $n$ . If  $\mathbf{A}$  is scalar,  $\mathbf{A} = \lambda \mathbf{I}_n$  with  $n\lambda = 0$ , and characteristic 0 forces  $\lambda = 0$ : then  $\mathbf{A} = 0$  already has zero diagonal. Otherwise, Lemma 7.5.2 puts a 0 in the corner and leaves a trace-zero block  $\mathbf{A}_1$  of size  $n-1$ . The induction hypothesis conjugates  $\mathbf{A}_1$  by some  $\mathbf{Q}$  into zero diagonal. Conjugating the whole matrix by the block diagonal matrix  $(1) \oplus \mathbf{Q}$  does this to the lower block without disturbing the corner.

*Proof.* We use induction on  $n$  (Theorem 0.2.7). For  $n = 1$ ,  $\mathbf{A} = (\text{tr } \mathbf{A}) = (0)$ , which has zero diagonal.

Let  $n \geq 2$ , and assume the statement for trace-zero matrices of size  $n-1$ .

*Case 1:  $\mathbf{A}$  is scalar,* say  $\mathbf{A} = \lambda \mathbf{I}_n$ . Then  $0 = \text{tr } \mathbf{A} = n \cdot \lambda = (n \cdot 1)\lambda$  by Definition 0.9.12 and Lemma 0.7.4. Since  $F$  has characteristic 0 and  $n \geq 1$ ,  $n \cdot 1 \neq 0$  (Definition 0.7.2), so  $\lambda = 0$ . Hence  $\mathbf{A} = 0$ , which has zero diagonal and is similar to itself.

*Case 2:  $\mathbf{A}$  is not scalar.* By Lemma 7.5.2 there is an invertible  $\mathbf{P}$  with

$$\mathbf{M} := \mathbf{P}^{-1}\mathbf{AP} = \begin{pmatrix} 0 & \mathbf{r}^\top \\ \mathbf{e}_1 & \mathbf{A}_1 \end{pmatrix}, \quad \text{tr } \mathbf{A}_1 = \text{tr } \mathbf{A} = 0.$$

By the induction hypothesis there is an invertible  $\mathbf{Q} \in M_{n-1}(F)$  such that  $\mathbf{Z}_1 := \mathbf{Q}^{-1}\mathbf{A}_1\mathbf{Q}$  has zero diagonal. Let  $\mathbf{R} = (1) \oplus \mathbf{Q} = \begin{pmatrix} 1 & \mathbf{0}^\top \\ \mathbf{0} & \mathbf{Q} \end{pmatrix}$ . By Theorem 7.1.4,  $\mathbf{R}$  is invertible with  $\mathbf{R}^{-1} = (1) \oplus \mathbf{Q}^{-1}$ . By Theorem 7.1.1,

$$\begin{aligned} \mathbf{R}^{-1}\mathbf{MR} &= \begin{pmatrix} 1 & \mathbf{0}^\top \\ \mathbf{0} & \mathbf{Q}^{-1} \end{pmatrix} \begin{pmatrix} 0 & \mathbf{r}^\top \\ \mathbf{e}_1 & \mathbf{A}_1 \end{pmatrix} \begin{pmatrix} 1 & \mathbf{0}^\top \\ \mathbf{0} & \mathbf{Q} \end{pmatrix} \\ &= \begin{pmatrix} 0 & \mathbf{r}^\top \mathbf{Q} \\ \mathbf{Q}^{-1}\mathbf{e}_1 & \mathbf{Q}^{-1}\mathbf{A}_1\mathbf{Q} \end{pmatrix} = \begin{pmatrix} 0 & \mathbf{r}^\top \mathbf{Q} \\ \mathbf{Q}^{-1}\mathbf{e}_1 & \mathbf{Z}_1 \end{pmatrix}. \end{aligned}$$

Its  $(1, 1)$ -entry is 0, and its other diagonal entries are those of  $\mathbf{Z}_1$ , which are 0. Since  $\mathbf{R}^{-1}\mathbf{MR} = (\mathbf{PR})^{-1}\mathbf{A}(\mathbf{PR})$  by Theorem 0.9.7 (3), and  $\mathbf{PR}$  is invertible,  $\mathbf{A}$  is similar to a matrix with zero diagonal. This completes the induction. ■

*Remark.* The characteristic was used only in Case 1, and only through  $k \cdot 1 \neq 0$  for the sizes  $k = 2, \dots, n$  that the induction passes through. So the lemma holds, with the same

proof, whenever  $F$  has characteristic 0 or characteristic  $p > n$ . Some hypothesis on the characteristic is needed: for  $n = p$ , over  $\mathbb{F}_p$ , the matrix  $\mathbf{I}_p$  has trace  $p \cdot 1 = 0$ , but it is scalar, so it is similar only to itself, and its diagonal entries are 1. The same happens for  $\lambda \mathbf{I}_n$  with  $\lambda \neq 0$  whenever  $p$  divides  $n$ .

## 7.5.4 Shoda's theorem

With a zero diagonal available, the diagonal choice of  $\mathbf{X}$  in (\*) finishes the job. All we need are  $n$  distinct scalars for the diagonal of  $\mathbf{X}$ , and characteristic 0 provides them.

### Theorem 7.5.4 — Shoda's Theorem

Let  $F$  be a field of **characteristic** 0, let  $n \geq 1$ , and let  $\mathbf{A} \in M_n(F)$ . Then  $\mathbf{A}$  is a commutator, that is  $\mathbf{A} = [\mathbf{X}, \mathbf{Y}]$  for some  $\mathbf{X}, \mathbf{Y} \in M_n(F)$ , if and only if  $\text{tr } \mathbf{A} = 0$ .

*Idea.* One direction is Proposition 7.5.1 (c). For the other: ① by Lemma 7.5.3,  $\mathbf{A} = \mathbf{PZP}^{-1}$  with  $\mathbf{Z}$  of zero diagonal; ② take  $\mathbf{X}_0 = \text{diag}(0, 1, \dots, n-1)$ , whose entries are distinct in characteristic 0; ③ solve  $[\mathbf{X}_0, \mathbf{Y}_0] = \mathbf{Z}$  entrywise with (\*), dividing by  $x_i - x_j \neq 0$ ; ④ conjugate back with Proposition 7.5.1 (b).

*Proof.* ( $\Rightarrow$ ) If  $\mathbf{A} = [\mathbf{X}, \mathbf{Y}]$ , then  $\text{tr } \mathbf{A} = 0$  by Proposition 7.5.1 (c).

( $\Leftarrow$ ) Suppose  $\text{tr } \mathbf{A} = 0$ . By Lemma 7.5.3 there is an invertible  $\mathbf{P}$  such that  $\mathbf{Z} = (z_{ij}) := \mathbf{P}^{-1}\mathbf{A}\mathbf{P}$  has  $z_{ii} = 0$  for all  $i$ .

For  $1 \leq i \leq n$  let  $x_i := (i-1) \cdot 1 \in F$ , and  $\mathbf{X}_0 := \text{diag}(x_1, \dots, x_n)$ . For  $i < j$ , Lemma 0.7.4 gives  $x_j - x_i = (j-i) \cdot 1$ , which is non-zero because  $j-i \geq 1$  and  $F$  has characteristic 0 (Definition 0.7.2). So the  $x_i$  are distinct, and we may define  $\mathbf{Y}_0 = (y_{ij}) \in M_n(F)$  by

$$y_{ij} := \frac{z_{ij}}{x_i - x_j} \quad (i \neq j), \quad y_{ii} := 0.$$

By (\*),  $[\mathbf{X}_0, \mathbf{Y}_0]_{ij} = (x_i - x_j)y_{ij} = z_{ij}$  for  $i \neq j$ , and  $[\mathbf{X}_0, \mathbf{Y}_0]_{ii} = 0 = z_{ii}$ . Hence  $[\mathbf{X}_0, \mathbf{Y}_0] = \mathbf{Z}$ .

Let  $\mathbf{X} = \mathbf{P}\mathbf{X}_0\mathbf{P}^{-1}$  and  $\mathbf{Y} = \mathbf{P}\mathbf{Y}_0\mathbf{P}^{-1}$ . Applying Proposition 7.5.1 (b) with the invertible matrix  $\mathbf{P}^{-1}$ ,

$$[\mathbf{X}, \mathbf{Y}] = [\mathbf{P}\mathbf{X}_0\mathbf{P}^{-1}, \mathbf{P}\mathbf{Y}_0\mathbf{P}^{-1}] = \mathbf{P}[\mathbf{X}_0, \mathbf{Y}_0]\mathbf{P}^{-1} = \mathbf{PZP}^{-1} = \mathbf{A}.$$

This proves that  $\mathbf{A}$  is a commutator. ■

The proof used characteristic 0 twice: in the lemma, to rule out non-zero scalar blocks of trace zero, and here, to find  $n$  distinct diagonal entries. The second use needs only  $(j-i) \cdot 1 \neq 0$  for  $1 \leq j-i \leq n-1$ , and with the remark after Lemma 7.5.3, the same proof works whenever the characteristic of  $F$  is 0 or larger than  $n$ . Neither use is essential for the **theorem**. In fact the statement “a square matrix over a field is a commutator if and only if its trace is 0” is true over **every** field. The proof in positive characteristic needs a different normal form and more work, and we do not give it. Example (The Identity as a Commutator in Characteristic  $p$ ) is a small

instance:  $\mathbf{I}_p$  over  $\mathbb{F}_p$  is a commutator, although [Lemma 7.5.3](#) fails for it.

The proof is constructive, and it is worth running once in full.

**Example. (Writing a Trace-Zero  $3 \times 3$  Matrix as a Commutator)**

Over  $\mathbb{Q}$ , write  $\mathbf{A} = \begin{pmatrix} 1 & 2 & 0 \\ 0 & 1 & 1 \\ 1 & 0 & -2 \end{pmatrix}$  as a commutator  $[\mathbf{X}, \mathbf{Y}]$ .

*Solution.* The trace is  $1 + 1 - 2 = 0$ . We follow the proofs step by step.

*First corner.*  $\mathbf{A}$  is not scalar. Take  $\mathbf{v} = \mathbf{e}_1$ ; then  $\mathbf{A}\mathbf{v} = (1, 0, 1)$  is not a multiple of  $\mathbf{e}_1$ . Extend  $(\mathbf{e}_1, \mathbf{A}\mathbf{e}_1)$  by  $\mathbf{e}_2$ , which is not in their span, to a basis, and let

$$\mathbf{P}_1 = \begin{pmatrix} 1 & 1 & 0 \\ 0 & 0 & 1 \\ 0 & 1 & 0 \end{pmatrix}, \quad \mathbf{M} = \mathbf{P}_1^{-1} \mathbf{A} \mathbf{P}_1 = \begin{pmatrix} 0 & 2 & 2 \\ 1 & -1 & 0 \\ 0 & 1 & 1 \end{pmatrix}.$$

(To check  $\mathbf{M}$  without inverting, verify  $\mathbf{P}_1 \mathbf{M} = \mathbf{A} \mathbf{P}_1$ : both equal  $\begin{pmatrix} 1 & 1 & 2 \\ 0 & 1 & 1 \\ 1 & -1 & 0 \end{pmatrix}$ .) The

first column is  $\mathbf{e}_2$ , and the lower block is  $\mathbf{A}_1 = \begin{pmatrix} -1 & 0 \\ 1 & 1 \end{pmatrix}$ , with trace 0.

*Second corner.*  $\mathbf{A}_1$  is not scalar. With  $\mathbf{w} = \mathbf{e}_1 \in \mathbb{Q}^2$ ,  $\mathbf{A}_1 \mathbf{w} = (-1, 1)$  is independent of  $\mathbf{e}_1$ , so

$$\mathbf{Q} = \begin{pmatrix} 1 & -1 \\ 0 & 1 \end{pmatrix}, \quad \mathbf{Q}^{-1} = \begin{pmatrix} 1 & 1 \\ 0 & 1 \end{pmatrix}, \quad \mathbf{Z}_1 = \mathbf{Q}^{-1} \mathbf{A}_1 \mathbf{Q} = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}.$$

*Assemble.* With  $\mathbf{r}^\top = (2, 2)$ , the proof of [Lemma 7.5.3](#) gives  $\mathbf{r}^\top \mathbf{Q} = (2, 0)$  and  $\mathbf{Q}^{-1} \mathbf{e}_1 = \mathbf{e}_1$ , so for  $\mathbf{P} = \mathbf{P}_1((1) \oplus \mathbf{Q})$ ,

$$\mathbf{P} = \begin{pmatrix} 1 & 1 & -1 \\ 0 & 0 & 1 \\ 0 & 1 & -1 \end{pmatrix}, \quad \mathbf{Z} = \mathbf{P}^{-1} \mathbf{A} \mathbf{P} = \begin{pmatrix} 0 & 2 & 0 \\ 1 & 0 & 1 \\ 0 & 1 & 0 \end{pmatrix}, \quad \mathbf{P}^{-1} = \begin{pmatrix} 1 & 0 & -1 \\ 0 & 1 & 1 \\ 0 & 1 & 0 \end{pmatrix}.$$

*Solve entrywise.* With  $\mathbf{X}_0 = \text{diag}(0, 1, 2)$ , the rule  $y_{ij} = z_{ij}/(x_i - x_j)$  gives  $y_{12} = \frac{2}{-1} = -2$ ,  $y_{21} = \frac{1}{1} = 1$ ,  $y_{23} = \frac{1}{-1} = -1$ ,  $y_{32} = \frac{1}{1} = 1$ , and  $y_{13} = y_{31} = 0$ :

$$\mathbf{Y}_0 = \begin{pmatrix} 0 & -2 & 0 \\ 1 & 0 & -1 \\ 0 & 1 & 0 \end{pmatrix}.$$

*Conjugate back.*  $\mathbf{X} = \mathbf{P} \mathbf{X}_0 \mathbf{P}^{-1}$  and  $\mathbf{Y} = \mathbf{P} \mathbf{Y}_0 \mathbf{P}^{-1}$  are

$$\mathbf{X} = \begin{pmatrix} 0 & -1 & 1 \\ 0 & 2 & 0 \\ 0 & -1 & 1 \end{pmatrix}, \quad \mathbf{Y} = \begin{pmatrix} 1 & -4 & -4 \\ 0 & 1 & 1 \\ 1 & -2 & -2 \end{pmatrix}.$$

$$\text{Check: } \mathbf{XY} = \begin{pmatrix} 1 & -3 & -3 \\ 0 & 2 & 2 \\ 1 & -3 & -3 \end{pmatrix} \text{ and } \mathbf{YX} = \begin{pmatrix} 0 & -5 & -3 \\ 0 & 1 & 1 \\ 0 & -3 & -1 \end{pmatrix}, \text{ and } \mathbf{XY} - \mathbf{YX} = \mathbf{A}.$$

Nothing about these  $\mathbf{X}$  and  $\mathbf{Y}$  is unique. Adding to  $\mathbf{Y}$  any matrix that commutes with  $\mathbf{X}$ , for instance a polynomial in  $\mathbf{X}$ , gives another solution, by [Proposition 7.5.1](#) (a).

## 7.5.5 The span of the commutators

Shoda's theorem is about single commutators. A weaker question, with an easier answer that holds over **every** field, is which matrices are **sums** of commutators. Since the trace-zero matrices form a subspace and contain all commutators, the span of the commutators lies inside it. Matrix units show that it is all of it.

### Theorem 7.5.5 — Commutators Span the Trace-Zero Matrices

Let  $F$  be any field and  $n \geq 1$ . The span of all commutators  $[\mathbf{X}, \mathbf{Y}]$  with  $\mathbf{X}, \mathbf{Y} \in M_n(F)$  is the subspace  $\{\mathbf{A} \in M_n(F) : \text{tr } \mathbf{A} = 0\}$ . More precisely, every trace-zero matrix is a linear combination of the commutators

$$\mathbf{E}_{ij} = [\mathbf{E}_{ii}, \mathbf{E}_{ij}] \quad (i \neq j) \quad \text{and} \quad \mathbf{E}_{ii} - \mathbf{E}_{nn} = [\mathbf{E}_{in}, \mathbf{E}_{ni}] \quad (1 \leq i \leq n-1).$$

*Proof.* Let  $W = \{\mathbf{A} \in M_n(F) : \text{tr } \mathbf{A} = 0\}$ , a subspace by [Example \(Trace-Free Matrices\)](#). By [Proposition 7.5.1](#) (c), every commutator lies in  $W$ , so their span is contained in  $W$  ([Theorem 1.3.1](#), the span being the smallest subspace containing them).

For the reverse inclusion, the displayed identities hold by [Example 2](#) after [Definition 7.5.1](#): for  $i \neq j$ ,  $[\mathbf{E}_{ii}, \mathbf{E}_{ij}] = \mathbf{E}_{ij}$ , and for  $i < n$ ,  $[\mathbf{E}_{in}, \mathbf{E}_{ni}] = \mathbf{E}_{ii} - \mathbf{E}_{nn}$ . Now let  $\mathbf{A} = (a_{ij}) \in W$ . Writing  $\mathbf{A}$  in the basis of matrix units ([Example \(Standard Bases\)](#)) and using  $a_{nn} = -\sum_{i < n} a_{ii}$ ,

$$\mathbf{A} = \sum_{i \neq j} a_{ij} \mathbf{E}_{ij} + \sum_{i=1}^{n-1} a_{ii} \mathbf{E}_{ii} + a_{nn} \mathbf{E}_{nn} = \sum_{i \neq j} a_{ij} \mathbf{E}_{ij} + \sum_{i=1}^{n-1} a_{ii} (\mathbf{E}_{ii} - \mathbf{E}_{nn}).$$

So  $\mathbf{A}$  is a linear combination of commutators, hence lies in their span. (For  $n = 1$ ,  $W = \{0\}$  and both sides are  $\{0\}$ .)

■

The theorem lists  $n(n-1) + (n-1) = n^2 - 1$  commutators spanning the trace-zero matrices. Since  $\text{tr}: M_n(F) \rightarrow F$  is onto ( $\text{tr}(c\mathbf{E}_{11}) = c$ ), the Rank–Nullity Theorem ([Theorem 3.3.1](#)) gives that its kernel has dimension  $n^2 - 1$ , so by [Theorem 1.6.4](#) (b) these commutators form a basis of it. Over a field of characteristic 0, Shoda's theorem says much more: the sum of commutators produced here can always be collapsed into a single commutator. In general the set of commutators is not obviously closed under addition, and that is exactly what makes Shoda's theorem a theorem.

**Warning.** The span of a set is not the set. [Theorem 7.5.5](#) shows that every trace-zero matrix is a **sum** of commutators, over any field, by a two-line argument. It does not show that every trace-zero matrix **is** a commutator. For comparison, in  $M_2(F)$  the

matrices of rank at most 1 span everything (the four matrix units have rank 1), yet  $\mathbf{I}_2$  has rank 2: a set can span a subspace without filling it. For commutators it happens that the set does fill the subspace, but the proof needs the similarity argument of [Lemma 7.5.3](#), not just matrix units.

The span theorem gives a clean proof of a fact from Chapter 3: a linear functional  $\varphi$  on  $M_n(F)$  with  $\varphi(\mathbf{AB}) = \varphi(\mathbf{BA})$  is a multiple of the trace. Such a  $\varphi$  kills every commutator, hence their span, which is the kernel of the trace; [Exercise \(C1: Trace-like functionals, again\)](#) below completes the argument.

## 7.5.6 Exercises

### 7.5.6.1 A. Check your understanding

#### Exercise A1.

- Define the commutator  $[\mathbf{A}, \mathbf{B}]$ , and state what it means for a matrix to be a commutator.
- State Shoda's theorem, including the hypothesis on the field.
- True or false:  $[\mathbf{A}, \mathbf{B}] = [\mathbf{B}, \mathbf{A}]$  for all  $\mathbf{A}, \mathbf{B} \in M_n(F)$ . Justify your answer.
- True or false: over every field, every trace-zero matrix is similar to a matrix with zero diagonal. Justify your answer.
- Name the two places where the proof of Shoda's theorem uses characteristic 0.
- Explain why [Theorem 7.5.5](#) does not by itself prove Shoda's theorem.

*Solution.*

- $[\mathbf{A}, \mathbf{B}] = \mathbf{AB} - \mathbf{BA}$  ([Definition 7.5.1](#)). A matrix  $\mathbf{C} \in M_n(F)$  is a commutator if  $\mathbf{C} = [\mathbf{X}, \mathbf{Y}]$  for some  $\mathbf{X}, \mathbf{Y} \in M_n(F)$ .
- If  $F$  has characteristic 0 and  $\mathbf{A} \in M_n(F)$ , then  $\mathbf{A} = [\mathbf{X}, \mathbf{Y}]$  for some  $\mathbf{X}, \mathbf{Y} \in M_n(F)$  if and only if  $\text{tr } \mathbf{A} = 0$  ([Theorem 7.5.4](#)).
- False.  $[\mathbf{B}, \mathbf{A}] = -[\mathbf{A}, \mathbf{B}]$  ([Proposition 7.5.1](#) (a)), so equality holds only when  $[\mathbf{A}, \mathbf{B}] = 0$ . For example  $[\mathbf{E}_{12}, \mathbf{E}_{21}] = \text{diag}(1, -1)$  and  $[\mathbf{E}_{21}, \mathbf{E}_{12}] = \text{diag}(-1, 1)$  over  $\mathbb{Q}$ .
- False. Over  $\mathbb{F}_2$ ,  $\mathbf{I}_2$  has trace  $1 + 1 = 0$ , but it is scalar, so every matrix similar to it equals  $\mathbf{P}^{-1}\mathbf{I}_2\mathbf{P} = \mathbf{I}_2$ , whose diagonal entries are 1.
- In [Lemma 7.5.3](#), to conclude from  $(n \cdot 1)\lambda = 0$  that a trace-zero scalar matrix  $\lambda\mathbf{I}_n$  is 0; and in [Theorem 7.5.4](#), to find  $n$  distinct scalars  $x_i = (i - 1) \cdot 1$  so that we can divide by  $x_i - x_j$ .
- It says every trace-zero matrix is a **linear combination** of commutators. Shoda's theorem says it is a **single** commutator. A sum of commutators need not visibly be a commutator, since  $[\mathbf{X}, \mathbf{Y}] + [\mathbf{X}', \mathbf{Y}']$  has no evident expression as  $[\mathbf{X}'', \mathbf{Y}']$ .

### 7.5.6.2 B. Practice

**Exercise B1: A  $2 \times 2$  commutator.**

Over  $\mathbb{Q}$ , let  $\mathbf{A} = \begin{pmatrix} 3 & -2 \\ 1 & -3 \end{pmatrix}$ .

- (a) Using  $\mathbf{v} = \mathbf{e}_1$  as in [Lemma 7.5.2](#), find an invertible  $\mathbf{P}$  such that  $\mathbf{Z} = \mathbf{P}^{-1}\mathbf{A}\mathbf{P}$  has zero diagonal.
- (b) With  $\mathbf{X}_0 = \text{diag}(0, 1)$ , find  $\mathbf{Y}_0$  with  $[\mathbf{X}_0, \mathbf{Y}_0] = \mathbf{Z}$ .
- (c) Hence write  $\mathbf{A} = [\mathbf{X}, \mathbf{Y}]$ , and check your answer by multiplying out.

*Solution.*

- (a)  $\text{tr } \mathbf{A} = 0$ , and  $\mathbf{A}\mathbf{e}_1 = (3, 1)$  is not a multiple of  $\mathbf{e}_1$ , so  $(\mathbf{e}_1, \mathbf{A}\mathbf{e}_1)$  is a basis of  $\mathbb{Q}^2$ .

Let  $\mathbf{P} = \begin{pmatrix} 1 & 3 \\ 0 & 1 \end{pmatrix}$ , so  $\mathbf{P}^{-1} = \begin{pmatrix} 1 & -3 \\ 0 & 1 \end{pmatrix}$ . Then  $\mathbf{A}\mathbf{P} = \begin{pmatrix} 3 & 7 \\ 1 & 0 \end{pmatrix}$  and

$$\mathbf{Z} = \mathbf{P}^{-1}\mathbf{A}\mathbf{P} = \begin{pmatrix} 0 & 7 \\ 1 & 0 \end{pmatrix}.$$

The first column is  $\mathbf{e}_2$  by [Lemma 7.5.2](#), and the  $(2, 2)$ -entry is 0 because the trace is 0.

- (b) By (\*),  $[\mathbf{X}_0, \mathbf{Y}]_{12} = (0 - 1)y_{12}$  and  $[\mathbf{X}_0, \mathbf{Y}]_{21} = (1 - 0)y_{21}$ , while the diagonal of  $[\mathbf{X}_0, \mathbf{Y}]$  is zero. So  $y_{12} = -7$ ,  $y_{21} = 1$ , and we may take  $y_{11} = y_{22} = 0$ :

$$\mathbf{Y}_0 = \begin{pmatrix} 0 & -7 \\ 1 & 0 \end{pmatrix}.$$

- (c) By the proof of [Theorem 7.5.4](#),  $\mathbf{X} = \mathbf{P}\mathbf{X}_0\mathbf{P}^{-1}$  and  $\mathbf{Y} = \mathbf{P}\mathbf{Y}_0\mathbf{P}^{-1}$ :

$$\mathbf{X} = \begin{pmatrix} 0 & 3 \\ 0 & 1 \end{pmatrix} \begin{pmatrix} 1 & -3 \\ 0 & 1 \end{pmatrix} = \begin{pmatrix} 0 & 3 \\ 0 & 1 \end{pmatrix}, \quad \mathbf{Y} = \begin{pmatrix} 3 & -7 \\ 1 & 0 \end{pmatrix} \begin{pmatrix} 1 & -3 \\ 0 & 1 \end{pmatrix} = \begin{pmatrix} 3 & -16 \\ 1 & -3 \end{pmatrix},$$

using  $\mathbf{P}\mathbf{X}_0 = \begin{pmatrix} 0 & 3 \\ 0 & 1 \end{pmatrix}$  and  $\mathbf{P}\mathbf{Y}_0 = \begin{pmatrix} 3 & -7 \\ 1 & 0 \end{pmatrix}$ . Check:  $\mathbf{X}\mathbf{Y} = \begin{pmatrix} 3 & -9 \\ 1 & -3 \end{pmatrix}$ ,  $\mathbf{Y}\mathbf{X} = \begin{pmatrix} 0 & -7 \\ 0 & 0 \end{pmatrix}$ , and  $\mathbf{X}\mathbf{Y} - \mathbf{Y}\mathbf{X} = \begin{pmatrix} 3 & -2 \\ 1 & -3 \end{pmatrix} = \mathbf{A}$ .

**Exercise B2: A commutator over  $\mathbb{F}_2$ .**

Work over  $\mathbb{F}_2$ , and let  $\mathbf{A} = \begin{pmatrix} 1 & 1 \\ 0 & 1 \end{pmatrix}$ .

- (a) Show that  $\text{tr } \mathbf{A} = 0$  and that  $\mathbf{A}$  is not scalar.
- (b) Explain why  $\mathbf{v} = \mathbf{e}_1$  cannot be used in [Lemma 7.5.2](#), and use  $\mathbf{v} = \mathbf{e}_2$  instead to find  $\mathbf{P}$  with  $\mathbf{P}^{-1}\mathbf{A}\mathbf{P}$  of zero diagonal.
- (c) The field  $\mathbb{F}_2$  has exactly two elements. Use them as the diagonal of  $\mathbf{X}_0$  to write  $\mathbf{A}$  as a commutator, by hand.
- (d) Explain why the same method cannot write  $\mathbf{I}_2 \in M_2(\mathbb{F}_2)$  as a commutator, and recall a pair  $\mathbf{X}, \mathbf{Y}$  that does.

*Solution.*

- (a)  $\text{tr } \mathbf{A} = 1 + 1 = 0$  in  $\mathbb{F}_2$ .  $\mathbf{A}$  has a non-zero off-diagonal entry, while every scalar matrix has zero off-diagonal entries.
- (b)  $\mathbf{A}\mathbf{e}_1 = \mathbf{e}_1$ , so  $(\mathbf{e}_1, \mathbf{A}\mathbf{e}_1)$  is dependent. With  $\mathbf{v} = \mathbf{e}_2$ ,  $\mathbf{A}\mathbf{e}_2 = (1, 1)$ , which is not a multiple of  $\mathbf{e}_2$ , so  $(\mathbf{e}_2, \mathbf{A}\mathbf{e}_2)$  is a basis. Let  $\mathbf{P} = \begin{pmatrix} 0 & 1 \\ 1 & 1 \end{pmatrix}$ ; since  $\det \mathbf{P} = -1 = 1$ ,  $\mathbf{P}^{-1} = \begin{pmatrix} 1 & 1 \\ 1 & 0 \end{pmatrix}$  (check:  $\mathbf{P}\mathbf{P}^{-1} = \begin{pmatrix} 1 & 0 \\ 2 & 1 \end{pmatrix} = \mathbf{I}_2$  in  $\mathbb{F}_2$ ). Then  $\mathbf{A}\mathbf{P} = \begin{pmatrix} 1 & 0 \\ 1 & 1 \end{pmatrix}$  and  $\mathbf{Z} = \mathbf{P}^{-1}\mathbf{A}\mathbf{P} = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}$ .
- (c) Take  $\mathbf{X}_0 = \text{diag}(0, 1)$ . By (\*),  $[\mathbf{X}_0, \mathbf{Y}]_{12} = (0 - 1)y_{12} = y_{12}$  (as  $-1 = 1$ ) and  $[\mathbf{X}_0, \mathbf{Y}]_{21} = y_{21}$ . So  $\mathbf{Y}_0 = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}$  gives  $[\mathbf{X}_0, \mathbf{Y}_0] = \mathbf{Z}$ . Conjugating back,  $\mathbf{P}\mathbf{X}_0 = \begin{pmatrix} 0 & 1 \\ 0 & 1 \end{pmatrix}$  and  $\mathbf{P}\mathbf{Y}_0 = \begin{pmatrix} 1 & 0 \\ 1 & 1 \end{pmatrix}$ , so

$$\mathbf{X} = \mathbf{P}\mathbf{X}_0\mathbf{P}^{-1} = \begin{pmatrix} 1 & 0 \\ 1 & 0 \end{pmatrix}, \quad \mathbf{Y} = \mathbf{P}\mathbf{Y}_0\mathbf{P}^{-1} = \begin{pmatrix} 1 & 1 \\ 0 & 1 \end{pmatrix}.$$

Check:  $\mathbf{X}\mathbf{Y} = \begin{pmatrix} 1 & 1 \\ 1 & 1 \end{pmatrix}$ ,  $\mathbf{Y}\mathbf{X} = \begin{pmatrix} 2 & 0 \\ 1 & 0 \end{pmatrix} = \begin{pmatrix} 0 & 0 \\ 1 & 0 \end{pmatrix}$ , so  $[\mathbf{X}, \mathbf{Y}] = \begin{pmatrix} 1 & 1 \\ 0 & 1 \end{pmatrix} = \mathbf{A}$ . (Here  $\mathbf{Y} = \mathbf{A}$ : indeed  $[\mathbf{X}, \mathbf{A}] = \mathbf{A}$ .)

- (d) The method writes  $\mathbf{Z} = \mathbf{P}^{-1}\mathbf{I}_2\mathbf{P}$  as  $[\mathbf{X}_0, \mathbf{Y}_0]$  with  $\mathbf{X}_0$  diagonal, and by (\*) such a commutator has zero diagonal. But  $\mathbf{I}_2$  is scalar, so  $\mathbf{P}^{-1}\mathbf{I}_2\mathbf{P} = \mathbf{I}_2$  for every invertible  $\mathbf{P}$ , and its diagonal entries are 1. So the method cannot start. Nevertheless  $\mathbf{I}_2$  is a commutator: with  $\mathbf{X} = \begin{pmatrix} 0 & 0 \\ 1 & 0 \end{pmatrix}$  and  $\mathbf{Y} = \begin{pmatrix} 0 & 1 \\ 0 & 0 \end{pmatrix}$ ,  $[\mathbf{X}, \mathbf{Y}] = \text{diag}(-1, 1) = \mathbf{I}_2$  in  $\mathbb{F}_2$  (Exercise (C1: The Heisenberg relation) (d), or Example (The Identity as a Commutator in Characteristic  $p$ ) with  $p = 2$ ).

### Exercise B3: Commuting with a diagonal matrix.

Let  $F$  be any field, let  $d_1, \dots, d_n \in F$  be **distinct**, let  $\mathbf{D} = \text{diag}(d_1, \dots, d_n)$ , and define  $L: M_n(F) \rightarrow M_n(F)$  by  $L(\mathbf{X}) = [\mathbf{D}, \mathbf{X}]$ .

- (a) Show that  $L$  is linear.
- (b) Prove that  $\ker L$  is the set of diagonal matrices and  $\text{im } L$  is the set of matrices with zero diagonal.
- (c) Check that your answer to (b) agrees with the Rank–Nullity Theorem.
- (d) Hence decide whether  $\begin{pmatrix} 0 & 1 & 2 \\ 3 & 0 & 4 \\ 5 & 6 & 0 \end{pmatrix} \in M_3(\mathbb{Q})$  is a commutator with  $\mathbf{X} = \text{diag}(1, 2, 3)$ , and if so find  $\mathbf{Y}$ .

*Solution.*

- (a) By Proposition 7.5.1 (a),  $[\mathbf{D}, \mathbf{X} + \mathbf{X}'] = [\mathbf{D}, \mathbf{X}] + [\mathbf{D}, \mathbf{X}']$  and  $[\mathbf{D}, c\mathbf{X}] = c[\mathbf{D}, \mathbf{X}]$ .
- (b) By (\*),  $L(\mathbf{X})_{ij} = (d_i - d_j)x_{ij}$ . Since  $d_i - d_j \neq 0$  for  $i \neq j$ ,  $L(\mathbf{X}) = 0$  if and only

if  $x_{ij} = 0$  for all  $i \neq j$ , that is,  $\mathbf{X}$  is diagonal. Every  $L(\mathbf{X})$  has zero diagonal. Conversely, if  $\mathbf{Z}$  has zero diagonal, the matrix  $\mathbf{Y}$  with  $y_{ij} = z_{ij}/(d_i - d_j)$  for  $i \neq j$  and  $y_{ii} = 0$  satisfies  $L(\mathbf{Y}) = \mathbf{Z}$ . So  $\text{im } L$  is exactly the set of zero-diagonal matrices.

- (c) The diagonal matrices have basis  $\mathbf{E}_{11}, \dots, \mathbf{E}_{nn}$ , so nullity  $L = n$ . The zero-diagonal matrices have basis  $\mathbf{E}_{ij}$  ( $i \neq j$ ), so  $\text{rank } L = n^2 - n$ . Their sum is  $n^2 = \dim M_n(F)$ , as [Theorem 3.3.1](#) requires.
- (d) Yes, since the matrix has zero diagonal and 1, 2, 3 are distinct. With  $d = (1, 2, 3)$ ,  $y_{ij} = z_{ij}/(d_i - d_j)$  gives

$$\mathbf{Y} = \begin{pmatrix} 0 & \frac{1}{-1} & \frac{2}{-2} \\ \frac{3}{\frac{1}{2}} & 0 & \frac{4}{-1} \\ \frac{5}{2} & \frac{6}{1} & 0 \end{pmatrix} = \begin{pmatrix} 0 & -1 & -1 \\ 3 & 0 & -4 \\ \frac{5}{2} & 6 & 0 \end{pmatrix},$$

and by (b),  $[\text{diag}(1, 2, 3), \mathbf{Y}]$  is the given matrix.

### 7.5.6.3 C. Going deeper

#### Exercise C1: Trace-like functionals, again.

Let  $F$  be any field,  $n \geq 1$ , and let  $\varphi: M_n(F) \rightarrow F$  be linear with  $\varphi(\mathbf{AB}) = \varphi(\mathbf{BA})$  for all  $\mathbf{A}, \mathbf{B} \in M_n(F)$ .

- (a) Prove that  $\varphi$  vanishes on every trace-zero matrix.
- (b) Deduce that  $\varphi = c \text{ tr}$ , where  $c = \varphi(\mathbf{E}_{11})$ .
- (c) Suppose  $F$  has characteristic 0 and, in addition,  $\varphi(\mathbf{I}_n) = n \cdot 1$ . Deduce that  $\varphi = \text{tr}$ .
- (d) Show that the conclusion of (c) fails over  $\mathbb{F}_p$  when  $n = p$ .

*Hint: for (a), use [Theorem 7.5.5](#).*

*Solution.*

- (a) For all  $\mathbf{X}, \mathbf{Y}$ ,  $\varphi([\mathbf{X}, \mathbf{Y}]) = \varphi(\mathbf{XY}) - \varphi(\mathbf{YX}) = 0$  by linearity and the hypothesis. So the kernel of  $\varphi$ , a subspace, contains every commutator, hence their span ([Theorem 1.3.1](#)), which is the set of trace-zero matrices by [Theorem 7.5.5](#).
- (b) Let  $\mathbf{A} \in M_n(F)$ . Since  $\text{tr } \mathbf{E}_{11} = 1$ , the matrix  $\mathbf{A} - (\text{tr } \mathbf{A})\mathbf{E}_{11}$  has trace 0 ([Theorem 0.9.6](#) (1)), so by (a) and linearity,

$$0 = \varphi(\mathbf{A} - (\text{tr } \mathbf{A})\mathbf{E}_{11}) = \varphi(\mathbf{A}) - (\text{tr } \mathbf{A})\varphi(\mathbf{E}_{11}).$$

Hence  $\varphi(\mathbf{A}) = c \text{ tr } \mathbf{A}$  with  $c = \varphi(\mathbf{E}_{11})$ . This recovers [Exercise \(C2: The trace is the only trace-like functional\)](#) by a different route: the matrix-unit computations are now packaged in [Theorem 7.5.5](#).

- (c) By (b),  $n \cdot 1 = \varphi(\mathbf{I}_n) = c \text{ tr } \mathbf{I}_n = c(n \cdot 1)$ . Since  $F$  has characteristic 0,  $n \cdot 1 \neq 0$ , so  $c = 1$  and  $\varphi = \text{tr}$ .
- (d) Over  $\mathbb{F}_p$  with  $n = p$ , take  $\varphi = 0$ . It is linear, satisfies  $\varphi(\mathbf{AB}) = 0 = \varphi(\mathbf{BA})$ , and  $\varphi(\mathbf{I}_p) = 0 = p \cdot 1$ . But  $\varphi \neq \text{tr}$ , since  $\text{tr } \mathbf{E}_{11} = 1$ . The step that fails is canceling  $n \cdot 1 = 0$  in (c).



**Exercise C2: When  $\mathbf{A}$  commutes with  $[\mathbf{A}, \mathbf{B}]$ .**

Let  $F$  be a field of characteristic 0, let  $\mathbf{A}, \mathbf{B} \in M_n(F)$ , put  $\mathbf{C} = [\mathbf{A}, \mathbf{B}]$ , and suppose that  $\mathbf{A}\mathbf{C} = \mathbf{C}\mathbf{A}$ . The goal is to prove that  $\mathbf{C}$  is **nilpotent**, that is,  $\mathbf{C}^s = 0$  for some  $s \geq 1$ .

- (a) Prove that  $\mathbf{C}^k = [\mathbf{A}, \mathbf{B}\mathbf{C}^{k-1}]$  for every  $k \geq 1$ . Deduce that  $\text{tr } \mathbf{C}^k = 0$  for every  $k \geq 1$ .
- (b) Show that there are an integer  $s \geq 1$  and  $g \in F[x]$  with  $g(0) \neq 0$  such that  $\mathbf{C}^s g(\mathbf{C}) = 0$ .
- (c) Let  $U = \text{col}(\mathbf{C}^s)$  and  $r = \dim U$ , and suppose  $r \geq 1$ . Extend a basis of  $U$  to a basis of  $F^n$ , and let  $\mathbf{P}$  be the matrix with these columns. Show that

$$\mathbf{P}^{-1}\mathbf{C}\mathbf{P} = \begin{pmatrix} \mathbf{D} & * \\ 0 & \mathbf{N} \end{pmatrix}, \quad \mathbf{D} \in M_r(F),$$

with  $\mathbf{N}^s = 0$  and  $g(\mathbf{D}) = 0$ .

- (d) Write  $g = g_0 + xh$  with  $g_0 = g(0) \neq 0$ . Show that  $\mathbf{I}_r$  is a linear combination of  $\mathbf{D}^s, \mathbf{D}^{s+1}, \dots, \mathbf{D}^d$  for some  $d \geq s$ , and use (a) to reach a contradiction.
- (e) Conclude that  $\mathbf{C}^s = 0$ .

*Hint: in (c), compare the columns of  $\mathbf{P}^{-1}\mathbf{C}^k\mathbf{P}$  for  $k = 1$  and  $k = s$  with the coordinates of vectors of  $U$ ; in (d), take traces.*

*Solution.*

- (a) First,  $\mathbf{C}^{k-1}\mathbf{A} = \mathbf{A}\mathbf{C}^{k-1}$  for all  $k \geq 1$ , by induction on  $k$  from  $\mathbf{C}\mathbf{A} = \mathbf{A}\mathbf{C}$ . Hence

$$\begin{aligned} [\mathbf{A}, \mathbf{B}\mathbf{C}^{k-1}] &= \mathbf{A}\mathbf{B}\mathbf{C}^{k-1} - \mathbf{B}\mathbf{C}^{k-1}\mathbf{A} = \mathbf{A}\mathbf{B}\mathbf{C}^{k-1} - \mathbf{B}\mathbf{A}\mathbf{C}^{k-1} \\ &= (\mathbf{A}\mathbf{B} - \mathbf{B}\mathbf{A})\mathbf{C}^{k-1} = \mathbf{C}^k. \end{aligned}$$

By [Proposition 7.5.1](#) (c),  $\text{tr } \mathbf{C}^k = 0$ .

- (b) By [Theorem 5.7.2](#) there is a non-zero  $f \in F[x]$  with  $f(\mathbf{C}) = 0$ . Let  $t \geq 0$  be the largest integer with  $x^t$  dividing  $f$ , and write  $f = x^t g$ ; then  $x \nmid g$ , so  $g(0) \neq 0$  by [Theorem 5.1.3](#). Put  $s = t + 1 \geq 1$ . By [Theorem 0.9.9](#),  $\mathbf{C}^s g(\mathbf{C}) = \mathbf{C} f(\mathbf{C}) = 0$ .
- (c) Let  $\mathbf{P}$  have columns  $\mathbf{p}_1, \dots, \mathbf{p}_n$ , where  $(\mathbf{p}_1, \dots, \mathbf{p}_r)$  is a basis of  $U$ ;  $\mathbf{P}$  is invertible by [Theorem 2.4.4](#). For any  $k \geq 1$ , column  $j$  of  $\mathbf{P}^{-1}\mathbf{C}^k\mathbf{P}$  is  $\mathbf{P}^{-1}(\mathbf{C}^k\mathbf{p}_j)$ , the vector  $\mathbf{c}$  with  $\mathbf{P}\mathbf{c} = \mathbf{C}^k\mathbf{p}_j$ , that is, the coefficients of  $\mathbf{C}^k\mathbf{p}_j$  in terms of  $\mathbf{p}_1, \dots, \mathbf{p}_n$  ([Theorem 0.9.3](#)). If  $\mathbf{y} \in U$ , then  $\mathbf{y} = \mathbf{C}^s\mathbf{z}$  and  $\mathbf{C}\mathbf{y} = \mathbf{C}^s(\mathbf{C}\mathbf{z}) \in U$ , so  $\mathbf{C}$  maps  $U$  into  $U$ ; hence for  $j \leq r$  the vector  $\mathbf{C}\mathbf{p}_j$  is a combination of  $\mathbf{p}_1, \dots, \mathbf{p}_r$ , and column  $j$  of  $\mathbf{P}^{-1}\mathbf{C}\mathbf{P}$  has zeros below row  $r$ . This gives the block form. By [Theorem 7.1.1](#) and induction on  $k$ ,  $(\mathbf{P}^{-1}\mathbf{C}\mathbf{P})^k = \begin{pmatrix} \mathbf{D}^k & * \\ 0 & \mathbf{N}^k \end{pmatrix}$ , and by [Proposition 3.7.6](#) (c) this is

$$\mathbf{P}^{-1}\mathbf{C}^k\mathbf{P}; \text{ more generally } \mathbf{P}^{-1}q(\mathbf{C})\mathbf{P} = \begin{pmatrix} q(\mathbf{D}) & * \\ 0 & q(\mathbf{N}) \end{pmatrix} \text{ for every } q \in F[x].$$

For  $k = s$ , every vector  $\mathbf{C}^s\mathbf{p}_j$  lies in  $U$ , so every column of  $\mathbf{P}^{-1}\mathbf{C}^s\mathbf{P}$  is zero below row  $r$ , and  $\mathbf{N}^s = 0$ . For  $q = g$ : for  $j \leq r$ ,  $\mathbf{p}_j \in U$  is  $\mathbf{C}^s\mathbf{z}$  for some  $\mathbf{z}$ , so  $g(\mathbf{C})\mathbf{p}_j = g(\mathbf{C})\mathbf{C}^s\mathbf{z} = \mathbf{0}$  by (b) and [Theorem 0.9.9](#). So the first  $r$  columns of  $\mathbf{P}^{-1}g(\mathbf{C})\mathbf{P}$  are zero, and in particular  $g(\mathbf{D}) = 0$ .

- (d) From  $g(\mathbf{D}) = g_0\mathbf{I}_r + \mathbf{D}h(\mathbf{D}) = 0$  we get  $\mathbf{I}_r = \mathbf{D}h_1(\mathbf{D})$  with  $h_1 = -g_0^{-1}h$ . Since

polynomials in  $\mathbf{D}$  commute ([Theorem 0.9.9](#)),

$$\mathbf{I}_r = (\mathbf{D} h_1(\mathbf{D}))^s = \mathbf{D}^s h_1(\mathbf{D})^s = \sum_{k=s}^d c_k \mathbf{D}^k$$

for some  $c_k \in F$ , as every term of  $x^s h_1(x)^s$  has degree at least  $s$ . For  $k \geq s$ ,  $\mathbf{N}^k = \mathbf{N}^s \mathbf{N}^{k-s} = 0$ , so by (c), [Theorem 3.7.5](#) and [Definition 0.9.12](#),  $\text{tr } \mathbf{C}^k = \text{tr}(\mathbf{P}^{-1} \mathbf{C}^k \mathbf{P}) = \text{tr } \mathbf{D}^k + \text{tr } \mathbf{N}^k = \text{tr } \mathbf{D}^k$ . Taking traces,

$$r \cdot 1 = \text{tr } \mathbf{I}_r = \sum_{k=s}^d c_k \text{tr } \mathbf{D}^k = \sum_{k=s}^d c_k \text{tr } \mathbf{C}^k = 0$$

by (a). But  $r \geq 1$  and  $F$  has characteristic 0, so  $r \cdot 1 \neq 0$ , a contradiction.

- (e) By (d),  $r = 0$ , so  $\text{col}(\mathbf{C}^s) = \{\mathbf{0}\}$  and  $\mathbf{C}^s = 0$ . For example, with  $\mathbf{A} = \mathbf{E}_{12} + \mathbf{E}_{23}$  and  $\mathbf{B} = \text{diag}(0, 1, 2)$  in  $M_3(\mathbb{Q})$ ,  $\mathbf{C} = [\mathbf{A}, \mathbf{B}] = \mathbf{E}_{12} + \mathbf{E}_{23} = \mathbf{A}$ , which commutes with  $\mathbf{A}$ , and indeed  $\mathbf{C}^3 = 0$ . In Chapter 8 the statement “ $\text{tr } \mathbf{C}^k = 0$  for all  $k \geq 1$ ” forces nilpotency in characteristic 0” becomes a statement about eigenvalues.

# Chapter 8

## Eigenvalues and Diagonalization

Chapter 3 showed that one operator has many matrices, one for each basis, and asked how simple the matrix can be made. This chapter answers the first half of that question. The simplest possible matrix is diagonal, and a matrix of an operator is diagonal exactly when every basis vector is only stretched, never turned. Such vectors are **eigenvectors**, and the stretching factors are **eigenvalues**.

Not every operator has enough eigenvectors, and the chapter explains precisely when it does. Two polynomials carry the information: the characteristic polynomial from Chapter 6 and the **minimal polynomial**, the smallest polynomial that kills the operator. The Cayley–Hamilton Theorem ties them together. When eigenvectors run short, operators over  $\mathbb{C}$  can still be made triangular, and the space splits into pieces on which the operator is easier to understand. The chapter ends with what diagonalization is for: powers of matrices, linear recurrences such as the Fibonacci numbers, and the long-run behavior of Markov chains.

**What you need.** From Chapter 3, the matrix of a linear map, change of basis and similarity (Definition 3.6.1, Theorem 3.7.3, Theorem 3.7.4), and quotient spaces (Definition 3.9.3). From Chapter 5, polynomials of an operator, the ideal of polynomials that kill it, and the kernel splitting lemma (Theorem 5.7.1, Theorem 5.7.3, Theorem 5.7.5), and the Fundamental Theorem of Algebra (Theorem 5.5.1). From Chapter 6, the characteristic polynomial and the adjugate (Definition 6.10.1, Theorem 6.6.2). From Chapter 7, block triangular matrices (Theorem 7.1.7).

### 8.0.1 Roadmap

- **Invariant subspaces.** Subspaces an operator maps into themselves, and the block triangular matrices they produce.
- **Eigenvalues and eigenvectors.** Directions that are only stretched; eigenvalues as roots of the characteristic polynomial; every complex operator has one.
- **Algebraic and geometric multiplicity.** Two ways to count a repeated eigenvalue, and why the second never exceeds the first.
- **Diagonalization.** Exactly when an operator has a basis of eigenvectors, and how to compute with one.
- **The minimal polynomial.** The monic polynomial of least degree that kills an operator, and why its roots are the eigenvalues.
- **The Cayley–Hamilton Theorem.** Every square matrix satisfies its own characteristic polynomial.

- **Triangularization.** An operator has a triangular matrix exactly when its characteristic polynomial splits.
- **Primary decomposition.** The factorization of the minimal polynomial splits the space, and decides diagonalizability.
- **Commuting operators.** Operators that commute share eigenvectors, and can be triangularized or diagonalized together.
- **Spectral mapping,  $AB$  and  $BA$ .** Eigenvalues of polynomials in an operator, and why  $AB$  and  $BA$  share their non-zero eigenvalues.
- **Powers, recurrences and Markov chains.** Diagonalization at work.

## 8.0.2 Named moves

- **Ask when the matrix is diagonal.** Unwinding the columns of a diagonal matrix gives the eigenvector equation  $T\mathbf{v} = \lambda\mathbf{v}$ .
- **Eigenvalue means a kernel.**  $\lambda$  is an eigenvalue exactly when  $T - \lambda I$  is not injective.
- **Apply  $T - \lambda I$  to a relation.** Kills one eigenvector and shortens a dependence, the heart of “distinct eigenvalues give independent eigenvectors”.
- **Turn a relation into a divisibility.** If  $p(T) = 0$ , then the minimal polynomial divides  $p$ .
- **One eigenvector, extend, recurse.** Split off an eigenvector, pass to the quotient, and use induction on dimension.
- **Search inside an eigenspace.** A commuting operator maps each eigenspace into itself, so look for a common eigenvector there.

# 8.1 Invariant Subspaces

Chapter 3 raised a question it could not answer: given an operator  $T$  on a finite-dimensional space, how simple can its matrix  $[T]_{\mathcal{B}}$  be made by choosing the basis  $\mathcal{B}$  well? Chapter 7 found the first piece of an answer. A zero block in the lower-left corner of  $[T]_{\mathcal{B}}$  appears exactly when the first few basis vectors span a subspace that  $T$  maps into itself. This section gives that property a name, collects the standard examples, and shows that it can fail badly: some operators map almost no subspace into itself. It ends by recording how such subspaces cut an operator, and its characteristic polynomial, into smaller pieces. The next section looks at the smallest non-zero case, a line, and finds eigenvectors there.

## 8.1.1 Subspaces an operator maps into itself

After a vector space we studied its subspaces, and after linear maps their kernels and images. For an operator  $T \in \mathcal{L}(V)$ , the domain and the codomain are the same space, so a new question makes sense: which subspaces of  $V$  are compatible with  $T$ ?

Here is the result that makes the question worth asking. Let  $V$  be finite-dimensional,  $T \in \mathcal{L}(V)$ , and  $U$  a subspace with  $\{\mathbf{0}\} \neq U \neq V$ . Take a basis  $(\mathbf{u}_1, \dots, \mathbf{u}_k)$  of  $U$  and extend it to a basis  $\mathcal{B} = (\mathbf{u}_1, \dots, \mathbf{u}_k, \mathbf{w}_1, \dots, \mathbf{w}_l)$  of  $V$ . By [Theorem 7.1.7](#), the lower-left  $l \times k$  block of  $[T]_{\mathcal{B}}$  is zero **if and only if**  $T\mathbf{u} \in U$  for every  $\mathbf{u} \in U$ . For example, the cyclic shift  $T(x, y, z) = (y, z, x)$  on  $\mathbb{R}^3$  keeps the plane  $x + y + z = 0$  in place, and in [Example \(A Plane Mapped into Itself\)](#) this produced a matrix with a zero corner. The property “ $T$  maps  $U$  into  $U$ ” is what we need to

control, so it gets a name.

*A subspace is invariant under  $T$  when  $T$  never moves a vector of the subspace out of it.*

### Definition 8.1.1 — Invariant Subspace

Let  $V$  be a vector space over  $F$  and  $T \in \mathcal{L}(V)$ . A subspace  $U$  of  $V$  is **invariant under  $T$** , or  **$T$ -invariant**, if

$$T\mathbf{u} \in U \quad \text{for every } \mathbf{u} \in U,$$

that is, if  $T(U) \subseteq U$ .

In words: we start from a **subspace**  $U$ , not an arbitrary subset. We apply  $T$  to **every** vector of  $U$ , and each image must land back **in**  $U$ . Nothing is required of vectors outside  $U$ , and nothing says that a vector of  $U$  stays where it is:  $T\mathbf{u}$  may be a different vector of  $U$ . The condition is an inclusion  $T(U) \subseteq U$ , not an equality.

Checking “for every  $\mathbf{u} \in U$ ” looks like infinitely many checks. Linearity reduces it to finitely many when  $U$  has a finite spanning list.

### Lemma 8.1.1 — Invariance Can Be Checked on a Spanning List

Let  $T \in \mathcal{L}(V)$  and  $U = \text{span}(\mathbf{u}_1, \dots, \mathbf{u}_k)$ . Then  $U$  is  $T$ -invariant if and only if  $T\mathbf{u}_i \in U$  for each  $i = 1, \dots, k$ .

*Proof.*  $(\Rightarrow)$  Each  $\mathbf{u}_i$  lies in  $U$ .  $(\Leftarrow)$  Let  $\mathbf{u} = c_1\mathbf{u}_1 + \dots + c_k\mathbf{u}_k \in U$ . By [Theorem 3.1.4](#),  $T\mathbf{u} = c_1T\mathbf{u}_1 + \dots + c_kT\mathbf{u}_k$ , a linear combination of vectors of  $U$ , which lies in  $U$  because  $U$  is a subspace ([Theorem 1.2.1](#)). ■

### Examples.

- **The trivial subspaces.** For every  $T \in \mathcal{L}(V)$ , both  $\{\mathbf{0}\}$  and  $V$  are  $T$ -invariant:  $T\mathbf{0} = \mathbf{0}$  ([Theorem 3.1.1](#)), and  $T\mathbf{v} \in V$  always. These are the degenerate cases, and they matter as a warning about what the word does **not** tell us: invariance carries information only for subspaces strictly between  $\{\mathbf{0}\}$  and  $V$ . At the other extreme, for  $T = c \text{id}_V$  **every** subspace is invariant, since  $c\mathbf{u} \in U$  whenever  $\mathbf{u} \in U$ .
- **Kernel and image.**  $\ker T$  is invariant: if  $\mathbf{u} \in \ker T$ , then  $T\mathbf{u} = \mathbf{0} \in \ker T$ . So is  $\text{im } T$ : if  $\mathbf{w} \in \text{im } T$ , then  $T\mathbf{w}$  is  $T$  of something, so  $T\mathbf{w} \in \text{im } T$ . More generally, for every polynomial  $p \in F[x]$ , the subspaces  $\ker p(T)$  and  $\text{im } p(T)$  are  $T$ -invariant, by [Theorem 5.7.4](#) (b). For instance  $\ker(T - c \text{id}_V)$  is invariant for every scalar  $c$ .
- **A line in the plane.** Let  $\mathbf{A} = \begin{pmatrix} 2 & 1 \\ 0 & 3 \end{pmatrix}$  act on  $\mathbb{R}^2$ . The line  $\text{span}(\mathbf{e}_1)$  is invariant, by [Lemma 8.1.1](#), because  $\mathbf{A}\mathbf{e}_1 = (2, 0) = 2\mathbf{e}_1 \in \text{span}(\mathbf{e}_1)$ . The line  $\text{span}((1, 1))$  is invariant as well, since  $\mathbf{A}(1, 1) = (3, 3) = 3(1, 1)$ .
- **Polynomials of bounded degree.** Let  $D: F[x] \rightarrow F[x]$  be differentiation ([Example \(Differentiation\)](#)). For each  $k \in \mathbb{N}$ , the subspace  $F[x]_{\leq k}$  is  $D$ -invariant, because differentiating does not raise the degree. It is spanned by  $1, x, \dots, x^k$ , and  $D(x^j) = jx^{j-1} \in F[x]_{\leq k}$ .

**Non-example by minimal change.** Keep  $\mathbf{A} = \begin{pmatrix} 2 & 1 \\ 0 & 3 \end{pmatrix}$  and replace the line  $\text{span}(\mathbf{e}_1)$  by the line  $\text{span}(\mathbf{e}_2)$ . It is still a subspace, and  $\mathbf{A}\mathbf{e}_2 = (1, 3)$  is still some vector of  $\mathbb{R}^2$ . What fails is the clause “ $T\mathbf{u} \in U$  for every  $\mathbf{u} \in U$ ” at  $\mathbf{u} = \mathbf{e}_2$ : the vector  $(1, 3)$  is not a multiple of  $\mathbf{e}_2$ . In the same way,  $\text{span}(x, x^2)$  inside  $\mathbb{R}[x]_{\leq 3}$  is not  $D$ -invariant, because  $D(x) = 1 \notin \text{span}(x, x^2)$ .

The examples with lines all have the same shape, which is worth recording now, because the whole next section grows out of it.

### Proposition 8.1.2 — Invariant Lines

Let  $T \in \mathcal{L}(V)$  and  $\mathbf{v} \in V$  with  $\mathbf{v} \neq \mathbf{0}$ . Then  $\text{span}(\mathbf{v})$  is  $T$ -invariant if and only if  $T\mathbf{v} = \lambda\mathbf{v}$  for some  $\lambda \in F$ .

*Proof.* By Lemma 8.1.1,  $\text{span}(\mathbf{v})$  is invariant if and only if  $T\mathbf{v} \in \text{span}(\mathbf{v})$ , and  $\text{span}(\mathbf{v}) = \{\lambda\mathbf{v} : \lambda \in F\}$  by Definition 1.3.2. ■

So an invariant line is a line on which  $T$  acts by stretching with a single factor  $\lambda$ . Does every operator have one? Not over  $\mathbb{R}$ .

### Example. (A Rotation Has No Invariant Lines)

Let  $R: \mathbb{R}^2 \rightarrow \mathbb{R}^2$ ,  $R(x, y) = (-y, x)$ , be the rotation by a right angle. Find all  $R$ -invariant subspaces of  $\mathbb{R}^2$ .

*Solution.* A subspace  $U$  of  $\mathbb{R}^2$  has  $\dim U \in \{0, 1, 2\}$  by Theorem 1.6.7. If  $\dim U = 0$ , then  $U = \{\mathbf{0}\}$ ; if  $\dim U = 2$ , then  $U = \mathbb{R}^2$  by Theorem 1.6.8. Both are invariant.

Suppose  $\dim U = 1$ . A basis of  $U$  is a single vector  $\mathbf{v} = (a, b) \neq \mathbf{0}$ , and  $U = \text{span}(\mathbf{v})$ . If  $U$  were invariant, then by Proposition 8.1.2 there would be  $\lambda \in \mathbb{R}$  with  $R\mathbf{v} = \lambda\mathbf{v}$ , that is,

$$-b = \lambda a, \quad a = \lambda b.$$

Substituting the first equation into the second gives  $a = \lambda b = \lambda(-\lambda a) = -\lambda^2 a$ , so  $(1 + \lambda^2)a = 0$ . Since  $\lambda$  is real,  $1 + \lambda^2 \geq 1 > 0$ , so  $a = 0$ . Then  $b = -\lambda a = 0$  by the first equation, and  $\mathbf{v} = \mathbf{0}$ , a contradiction.

Hence the only  $R$ -invariant subspaces of  $\mathbb{R}^2$  are  $\{\mathbf{0}\}$  and  $\mathbb{R}^2$ . Geometrically, a quarter-turn moves every line through the origin onto the perpendicular line.

The same computation over  $\mathbb{C}$  does not end in a contradiction:  $1 + \lambda^2 = 0$  has the solutions  $\lambda = \pm i$ , and we will see in the next section that  $\mathbb{C}^2$  does contain  $R$ -invariant lines. Whether invariant lines exist depends on the field.

**Why this definition.** Two variations suggest themselves, and both are worse.

- Demand  $T(U) = U$ . Then  $\ker T$  would fail whenever  $T$  is not injective, since then  $T(\ker T) = \{\mathbf{0}\} \neq \ker T$ . But the kernel is exactly the kind of subspace that produces a zero block, and Theorem 7.1.7 needs only the inclusion.
- Demand  $T\mathbf{u} = \mathbf{u}$  for every  $\mathbf{u} \in U$ . That says  $T$  acts as the identity on  $U$ , a far stronger

condition. The line  $\text{span}((1, 1))$  above is invariant under  $\mathbf{A}$ , but  $\mathbf{A}$  triples its vectors.

The name records that the subspace, as a whole, is unchanged in the sense that it is carried into itself; its individual vectors may move.

**Warning. Invariant does not mean fixed.** The vectors of a  $T$ -invariant subspace may move, as long as they stay inside it. For the swap  $T(x, y) = (y, x)$  on  $\mathbb{R}^2$ , the whole plane is invariant, yet  $T(1, 0) = (0, 1) \neq (1, 0)$ . Conversely, containing a fixed vector does not make a subspace invariant. For  $T(x, y, z) = (x, z, y + z)$  on  $\mathbb{R}^3$ , the plane  $z = 0$  contains  $\mathbf{e}_1$ , and  $T\mathbf{e}_1 = \mathbf{e}_1$ ; but it also contains  $\mathbf{e}_2$ , and  $T\mathbf{e}_2 = \mathbf{e}_3$  leaves the plane.

**Quick check.** Let  $T(x, y) = (x + y, y)$  on  $\mathbb{R}^2$ . Is the  $x$ -axis  $T$ -invariant? Is the  $y$ -axis?

**Solution.** By Lemma 8.1.1 it suffices to test one spanning vector.  $T\mathbf{e}_1 = (1, 0) = \mathbf{e}_1$ , which lies on the  $x$ -axis, so the  $x$ -axis is invariant.  $T\mathbf{e}_2 = (1, 1)$ , which is not a multiple of  $\mathbf{e}_2$ , so the  $y$ -axis is **not** invariant.

## 8.1.2 Restriction and the induced operator

An invariant subspace  $U$  lets us split  $T$  into two smaller operators. One lives on  $U$ : since  $T$  sends  $U$  into  $U$ , we may simply forget the rest of  $V$ . The other lives on the quotient  $V/U$ , which records what  $U$  cannot see. Chapter 7 met both in the proof of Theorem 7.1.7, and Chapter 3 constructed the second in Exercise (C2: The operator induced on a quotient). We now give them names.

### Definition 8.1.2 — Restriction and Induced Operator

Let  $T \in \mathcal{L}(V)$  and let  $U$  be a  $T$ -invariant subspace of  $V$ .

- (a) The **restriction** of  $T$  to  $U$  is the operator  $T|_U: U \rightarrow U$ ,  $\mathbf{u} \mapsto T\mathbf{u}$ .
- (b) The **induced operator** (or quotient operator) on  $V/U$  is  $\bar{T}: V/U \rightarrow V/U$ ,  $\mathbf{v} + U \mapsto T\mathbf{v} + U$ .

**Well-definedness.** For (a), the rule makes sense as a map **into**  $U$  only because  $U$  is invariant; it is linear because  $T$  is, and  $U$  carries the operations of  $V$ . For (b), the rule is phrased in terms of a representative  $\mathbf{v}$  of the coset, so we must check that it does not depend on that choice. Suppose  $\mathbf{v} + U = \mathbf{v}' + U$ . Then  $\mathbf{v} - \mathbf{v}' \in U$  (Lemma 3.9.3), so  $T\mathbf{v} - T\mathbf{v}' = T(\mathbf{v} - \mathbf{v}') \in U$  by invariance, and hence  $T\mathbf{v} + U = T\mathbf{v}' + U$ . Linearity follows from the coset operations (Theorem 3.9.4): for instance  $\bar{T}((\mathbf{v} + U) + (\mathbf{w} + U)) = \bar{T}(\mathbf{v} + \mathbf{w} + U) = T\mathbf{v} + T\mathbf{w} + U = \bar{T}(\mathbf{v} + U) + \bar{T}(\mathbf{w} + U)$ , and scalars work the same way. Invariance was used exactly once in each part, which is where the hypothesis earns its place.

**Examples.**

- **Differentiation.** Let  $D$  act on  $V = \mathbb{R}[x]_{\leq 2}$  and  $U = \mathbb{R}[x]_{\leq 1}$ , which is  $D$ -invariant. The restriction  $D|_U$  is differentiation on  $\mathbb{R}[x]_{\leq 1}$ . The quotient  $V/U$  has dimension 1 (Theorem 3.9.5), with basis  $x^2 + U$ , and  $\bar{D}(x^2 + U) = 2x + U = \mathbf{0} + U$ , because  $2x \in U$ . So  $\bar{D} = 0$ .
- **The cyclic shift.** For  $T(x, y, z) = (y, z, x)$  and the plane  $U$  where  $x + y + z = 0$ , Example (A Plane Mapped into Itself) showed that  $\bar{T}$  is the identity of the line  $\mathbb{R}^3/U$ : the shift

permutes coordinates and so never changes their sum.

- **Degenerate cases.** For  $U = V$ , the restriction is  $T$  itself and  $V/V$  is the zero space. For  $U = \{0\}$ , the restriction is the operator on the zero space, and  $\bar{T}$  is  $T$  in disguise, since  $\mathbf{v} + \{0\}$  is just  $\mathbf{v}$ .

With these names, the matrix theorem of Chapter 7 reads as follows. We add the consequence for determinants and characteristic polynomials, which is what the rest of this chapter uses.

### Theorem 8.1.3 — Invariant Subspaces and Block Triangular Matrices

Let  $V$  be finite-dimensional,  $T \in \mathcal{L}(V)$ , and  $U$  a subspace with  $\{0\} \neq U \neq V$ . Let  $\mathcal{B}_U = (\mathbf{u}_1, \dots, \mathbf{u}_k)$  be a basis of  $U$ , extended to a basis  $\mathcal{B} = (\mathbf{u}_1, \dots, \mathbf{u}_k, \mathbf{w}_1, \dots, \mathbf{w}_l)$  of  $V$ , and put  $\bar{\mathcal{B}} = (\mathbf{w}_1 + U, \dots, \mathbf{w}_l + U)$ .

- (a)  $U$  is  $T$ -invariant if and only if  $[T]_{\mathcal{B}}$  has zero lower-left  $l \times k$  block.
- (b) In that case

$$[T]_{\mathcal{B}} = \begin{pmatrix} [T|_U]_{\mathcal{B}_U} & \mathbf{B} \\ 0 & [\bar{T}]_{\bar{\mathcal{B}}} \end{pmatrix}$$

for some  $\mathbf{B} \in M_{k \times l}(F)$ , and

$$\det T = \det(T|_U) \det \bar{T}, \quad p_T = p_{T|_U} p_{\bar{T}}.$$

*Proof.* Part (a), and the block form in (b), are Theorem 7.1.7 in the language of Definition 8.1.1 and Definition 8.1.2;  $\bar{\mathcal{B}}$  is a basis of  $V/U$  by Theorem 3.9.5.

Write  $\mathbf{A} = [T|_U]_{\mathcal{B}_U} \in M_k(F)$  and  $\mathbf{D} = [\bar{T}]_{\bar{\mathcal{B}}} \in M_l(F)$ . By Definition 6.5.1 and Theorem 6.8.4,  $\det T = \det [T]_{\mathcal{B}} = \det \mathbf{A} \det \mathbf{D} = \det(T|_U) \det \bar{T}$ . For the characteristic polynomials,

$$x\mathbf{I}_{k+l} - [T]_{\mathcal{B}} = \begin{pmatrix} x\mathbf{I}_k - \mathbf{A} & -\mathbf{B} \\ 0 & x\mathbf{I}_l - \mathbf{D} \end{pmatrix}$$

is block upper triangular with square diagonal blocks, as a matrix over  $F[x]$ . The block triangular determinant holds over the commutative ring  $F[x]$  (the remark after Theorem 6.8.4), so  $\det(x\mathbf{I} - [T]_{\mathcal{B}}) = \det(x\mathbf{I}_k - \mathbf{A}) \det(x\mathbf{I}_l - \mathbf{D})$ . By Definition 6.10.2, this says  $p_T = p_{T|_U} p_{\bar{T}}$ . ■

The theorem is the reason invariant subspaces matter. It splits the study of  $T$  into the study of two operators on smaller spaces, and every quantity that sees only the diagonal blocks splits with it. For differentiation on  $\mathbb{R}[x]_{\leq 2}$ , in the basis  $(1, x, x^2)$ ,

$$[D]_{\mathcal{B}} = \left( \begin{array}{cc|c} 0 & 1 & 0 \\ 0 & 0 & 2 \\ 0 & 0 & 0 \end{array} \right),$$

with upper-left block  $[D|_U] = \begin{pmatrix} 0 & 1 \\ 0 & 0 \end{pmatrix}$ , lower-right block  $[\bar{D}] = (0)$ , and  $p_D = x^3 = x^2 \cdot x$ . Section 7 of this chapter will run this splitting repeatedly, by induction on the dimension.



### 8.1.3 Sums, intersections and direct sums

Invariant subspaces can be combined. Sums and intersections of invariant subspaces are invariant, in contrast to many other properties, and the proof is one line each.

#### Proposition 8.1.4 — Sums and Intersections of Invariant Subspaces

Let  $T \in \mathcal{L}(V)$  and let  $U_1, \dots, U_k$  be  $T$ -invariant subspaces of  $V$ . Then  $U_1 + \dots + U_k$  and  $U_1 \cap \dots \cap U_k$  are  $T$ -invariant.

*Proof.* Both are subspaces, by Theorem 1.7.1 and Theorem 1.2.2. Let  $\mathbf{u} = \mathbf{u}_1 + \dots + \mathbf{u}_k$  with  $\mathbf{u}_i \in U_i$ . Then  $T\mathbf{u} = T\mathbf{u}_1 + \dots + T\mathbf{u}_k$  with  $T\mathbf{u}_i \in U_i$  by invariance, so  $T\mathbf{u} \in U_1 + \dots + U_k$ . If  $\mathbf{u} \in U_1 \cap \dots \cap U_k$ , then  $T\mathbf{u} \in U_i$  for each  $i$ , by invariance of each  $U_i$ , so  $T\mathbf{u} \in U_1 \cap \dots \cap U_k$ . ■

When  $V$  is a **direct** sum of invariant subspaces, all the off-diagonal blocks vanish, not just one corner. Chapter 7 proved this for two summands (Theorem 7.1.6). The diagonalization theorem of Section 4 needs any number of summands.

#### Theorem 8.1.5 — Invariant Direct Sums and Block Diagonal Matrices

Let  $V$  be finite-dimensional with  $V = U_1 \oplus \dots \oplus U_k$ , where each  $U_i \neq \{\mathbf{0}\}$  has a basis  $\mathcal{B}_i$  of length  $n_i$ . Let  $\mathcal{B}$  be the list  $\mathcal{B}_1, \dots, \mathcal{B}_k$  written one after another, and  $T \in \mathcal{L}(V)$ . Then  $\mathcal{B}$  is a basis of  $V$ , and the following are equivalent:

- (a) each  $U_i$  is  $T$ -invariant;
- (b)  $[T]_{\mathcal{B}}$  is block diagonal with respect to the partition  $n_1, \dots, n_k$ .

In that case

$$[T]_{\mathcal{B}} = [T|_{U_1}]_{\mathcal{B}_1} \oplus \dots \oplus [T|_{U_k}]_{\mathcal{B}_k}, \quad p_T = p_{T|_{U_1}} \cdots p_{T|_{U_k}}.$$

*Idea.* Read the matrix one block of columns at a time. The columns of block  $j$  are the coordinates of  $T\mathbf{b}$  for the vectors  $\mathbf{b}$  of  $\mathcal{B}_j$ . Such a  $T\mathbf{b}$  lies in  $U_j$  exactly when all its coordinates outside block  $j$  vanish, because a vector of  $U_j$  is already a combination of  $\mathcal{B}_j$ , and coordinates are unique.

*Proof.* By Theorem 1.7.5 ((a)  $\Rightarrow$  (d)),  $\mathcal{B}$  is a basis of  $V$ . Fix  $j$  and a vector  $\mathbf{b}$  of  $\mathcal{B}_j$ ; the column of  $[T]_{\mathcal{B}}$  belonging to  $\mathbf{b}$  is  $[T\mathbf{b}]_{\mathcal{B}}$  (Definition 3.6.1). We claim:  $T\mathbf{b} \in U_j$  if and only if the entries of this column outside the rows of block  $j$  are all 0. If  $T\mathbf{b} \in U_j = \text{span}(\mathcal{B}_j)$ , then  $T\mathbf{b}$  is a combination of  $\mathcal{B}_j$  alone, which is a representation in the basis  $\mathcal{B}$  with zero coefficients outside block  $j$ ; by Theorem 1.5.1, it is **the** coordinate vector. Conversely, if those entries vanish, then  $T\mathbf{b}$  is a combination of  $\mathcal{B}_j$ , so it lies in  $U_j$ .

(a)  $\Rightarrow$  (b). If  $U_j$  is invariant, then  $T\mathbf{b} \in U_j$  for each  $\mathbf{b}$  in  $\mathcal{B}_j$ , so by the claim every

block  $(i, j)$  with  $i \neq j$  is zero. As  $j$  was arbitrary,  $[T]_{\mathcal{B}}$  is block diagonal. Moreover the entries of column  $\mathbf{b}$  in the rows of block  $j$  are the coordinates of  $T\mathbf{b} = T|_{U_j}\mathbf{b}$  in  $\mathcal{B}_j$ , so the  $(j, j)$  block is  $[T|_{U_j}]_{\mathcal{B}_j}$ .

- (b)  $\Rightarrow$  (a). If the blocks  $(i, j)$  with  $i \neq j$  vanish, the claim gives  $T\mathbf{b} \in U_j$  for every  $\mathbf{b}$  in  $\mathcal{B}_j$ , and  $U_j$  is invariant by [Lemma 8.1.1](#).

Finally,  $x\mathbf{I} - [T]_{\mathcal{B}}$  is block diagonal with diagonal blocks  $x\mathbf{I}_{n_j} - [T|_{U_j}]_{\mathcal{B}_j}$ . Its determinant over  $F[x]$  is the product of the determinants of these blocks (the remark after [Theorem 6.8.4](#), applied by induction on the number of blocks), which is  $p_T = \prod_j p_{T|_{U_j}}$  by [Definition 6.10.2](#). ■

So an operator that preserves each piece of a direct sum decomposition is, in a suitable basis, a list of independent smaller operators. The best possible case is a decomposition into invariant **lines**: then every block is  $1 \times 1$ , and  $[T]_{\mathcal{B}}$  is diagonal. By [Proposition 8.1.2](#), that happens exactly when  $V$  has a basis of vectors  $\mathbf{v}$  with  $T\mathbf{v} = \lambda\mathbf{v}$ . This observation is the starting point of the next section.

A tempting shortcut is to assume that once one invariant subspace is found, a complement of it can be chosen invariant too, and the matrix becomes block diagonal. This fails.

*Warning.* **The complement of an invariant subspace need not be invariant.**

Let  $\mathbf{N} = \begin{pmatrix} 0 & 1 \\ 0 & 0 \end{pmatrix}$  act on  $F^2$ . The line  $U = \text{span}(\mathbf{e}_1)$  is invariant, since  $\mathbf{N}\mathbf{e}_1 = \mathbf{0}$ . Every complement of  $U$  is a line  $W = \text{span}((a, 1))$  for some  $a \in F$  (a line not equal to  $U$  contains a vector with second entry 1). But  $\mathbf{N}(a, 1) = (1, 0)$ , which is not a multiple of  $(a, 1)$ . So **no** complement of  $U$  is  $\mathbf{N}$ -invariant, and no basis makes  $[\mathbf{N}]$  block diagonal with two  $1 \times 1$  blocks.

*Quick check.* Let  $U_1, U_2$  be  $T$ -invariant. Must the union  $U_1 \cup U_2$  be  $T$ -invariant? Must it even be a subspace?

*Solution.* The question has a trap: “invariant subspace” requires a subspace, and a union of two subspaces is usually not one ([Proposition 1.2.3](#)). For  $T = \text{id}_{\mathbb{R}^2}$ , the two axes are invariant, but their union is not a subspace, so the word does not apply. The subspace that plays the role of the union is the sum  $U_1 + U_2$ , which is invariant by [Proposition 8.1.4](#).

## 8.1.4 Exercises

### 8.1.4.1 A. Check your understanding

#### Exercise A1.

- (a) Define what it means for a subspace  $U$  of  $V$  to be invariant under  $T \in \mathcal{L}(V)$ .
- (b) True or false: for every  $T \in \mathcal{L}(V)$ , the subspaces  $\ker T$  and  $\text{im } T$  are  $T$ -invariant. Justify your answer.
- (c) True or false: if  $U$  is  $T$ -invariant, then  $T\mathbf{u} = \mathbf{u}$  for every  $\mathbf{u} \in U$ . Justify your

answer.

- (d) True or false: if  $U$  is invariant under  $S$  and under  $T$ , then  $U$  is invariant under  $S + T$  and under  $ST$ . Justify your answer.
- (e) For  $\mathbf{v} \neq \mathbf{0}$ , what condition on  $T\mathbf{v}$  makes  $\text{span}(\mathbf{v})$  invariant?
- (f) In a basis whose first  $k$  vectors form a basis of a  $T$ -invariant subspace  $U$ , which block of  $[T]_{\mathcal{B}}$  is zero, and what are the two diagonal blocks?

*Solution.*

- (a)  $U$  is  $T$ -invariant if  $T\mathbf{u} \in U$  for every  $\mathbf{u} \in U$  (Definition 8.1.1).
- (b) True. If  $\mathbf{u} \in \ker T$ , then  $T\mathbf{u} = \mathbf{0} \in \ker T$ . If  $\mathbf{w} \in \text{im } T$ , then  $T\mathbf{w} \in \text{im } T$  by definition of the image.
- (c) False. For  $T = 2 \text{ id}_{\mathbb{R}^2}$ , the subspace  $\mathbb{R}^2$  is invariant, but  $T\mathbf{e}_1 = 2\mathbf{e}_1 \neq \mathbf{e}_1$ .
- (d) True. For  $\mathbf{u} \in U$ , both  $S\mathbf{u}$  and  $T\mathbf{u}$  lie in  $U$ , so  $(S + T)\mathbf{u} = S\mathbf{u} + T\mathbf{u} \in U$  since  $U$  is a subspace. Also  $T\mathbf{u} \in U$ , so  $(ST)\mathbf{u} = S(T\mathbf{u}) \in U$  by invariance under  $S$ .
- (e)  $T\mathbf{v} = \lambda\mathbf{v}$  for some scalar  $\lambda$  (Proposition 8.1.2).
- (f) The lower-left block is zero. The upper-left block is  $[T|_U]_{\mathcal{B}_U}$  and the lower-right block is the matrix of the induced operator  $\bar{T}$  on  $V/U$  (Theorem 8.1.3).

#### 8.1.4.2 B. Practice

##### Exercise B1: All invariant subspaces of a diagonal matrix.

Let  $\mathbf{A} = \text{diag}(1, 2)$  act on  $\mathbb{R}^2$ . Find all  $\mathbf{A}$ -invariant subspaces of  $\mathbb{R}^2$ .

*Solution.* As in Example (A Rotation Has No Invariant Lines), a subspace of  $\mathbb{R}^2$  is  $\{\mathbf{0}\}$ ,  $\mathbb{R}^2$ , or a line  $\text{span}(\mathbf{v})$  with  $\mathbf{v} = (a, b) \neq \mathbf{0}$ . The first two are invariant. By Proposition 8.1.2, the line is invariant if and only if  $\mathbf{A}\mathbf{v} = \lambda\mathbf{v}$  for some  $\lambda \in \mathbb{R}$ , that is,

$$a = \lambda a, \quad 2b = \lambda b.$$

If  $a \neq 0$ , the first equation gives  $\lambda = 1$ , and then  $2b = b$  gives  $b = 0$ ; so  $\mathbf{v}$  is a multiple of  $\mathbf{e}_1$ . If  $a = 0$ , then  $b \neq 0$ , the second equation gives  $\lambda = 2$ , and  $\mathbf{v}$  is a multiple of  $\mathbf{e}_2$ . Conversely  $\mathbf{A}\mathbf{e}_1 = \mathbf{e}_1$  and  $\mathbf{A}\mathbf{e}_2 = 2\mathbf{e}_2$ . Hence the invariant subspaces are exactly

$$\{\mathbf{0}\}, \quad \text{span}(\mathbf{e}_1), \quad \text{span}(\mathbf{e}_2), \quad \mathbb{R}^2.$$

Compare with the identity  $\mathbf{I}_2$ , for which every line is invariant: changing one diagonal entry from 1 to 2 destroys all lines except the two axes.

##### Exercise B2: Kernels of $T - c \text{ id}$ .

Let  $T \in \mathcal{L}(V)$  and  $c \in F$ .

- (a) Prove directly from the definition that  $\ker(T - c \text{ id}_V)$  is  $T$ -invariant.
- (b) Hence show that **every** subspace of  $\ker(T - c \text{ id}_V)$  is  $T$ -invariant.

*Solution.*

- (a) Let  $K = \ker(T - c \operatorname{id}_V)$ , a subspace by [Theorem 3.2.1](#). For  $\mathbf{u} \in K$  we have  $T\mathbf{u} - c\mathbf{u} = \mathbf{0}$ , that is,  $T\mathbf{u} = c\mathbf{u}$ . Since  $K$  is a subspace,  $c\mathbf{u} \in K$ . Hence  $T\mathbf{u} \in K$ , and  $K$  is  $T$ -invariant.
- (b) Let  $W$  be a subspace of  $K$  and  $\mathbf{w} \in W$ . Since  $\mathbf{w} \in K$ , part (a) gives  $T\mathbf{w} = c\mathbf{w}$ , and  $c\mathbf{w} \in W$  because  $W$  is a subspace. Hence  $W$  is  $T$ -invariant. (On  $K$ ,  $T$  acts as  $c \operatorname{id}$ , and for a scalar operator every subspace is invariant.)

### Exercise B3: Invariance under differentiation.

Let  $D$  be differentiation on  $V = \mathbb{R}[x]_{\leq 3}$ . Determine which of the following subspaces are  $D$ -invariant. Justify your answer.

- (a)  $U_1 = \operatorname{span}(1, x)$ .
- (b)  $U_2 = \operatorname{span}(x, x^2)$ .
- (c)  $U_3 = \{p \in V : p(0) = 0\}$ .
- (d)  $U_4 = \operatorname{span}(x^2 + 2x, 2x + 2, 2)$ .
- (e)  $U_5 = \operatorname{span}(1, x^2)$ , the even polynomials in  $V$ .

*Solution.* We use [Lemma 8.1.1](#) throughout.

- (a) Invariant:  $D(1) = 0 \in U_1$  and  $D(x) = 1 \in U_1$ .
- (b) Not invariant:  $D(x) = 1$ , and  $1 \notin \operatorname{span}(x, x^2)$  since every element of  $U_2$  has constant term 0.
- (c) Not invariant:  $x \in U_3$ , but  $D(x) = 1$  and 1 evaluated at 0 is  $1 \neq 0$ .
- (d) Invariant:  $D(x^2 + 2x) = 2x + 2$ ,  $D(2x + 2) = 2$  and  $D(2) = 0$ , all in  $U_4$ . (In fact  $U_4 = \mathbb{R}[x]_{\leq 2}$ , since the three spanning polynomials have degrees 2, 1, 0 and so are independent by [Theorem 1.4.4](#).)
- (e) Not invariant:  $D(x^2) = 2x$ , which is odd and non-zero, so it is not in  $\operatorname{span}(1, x^2)$ .

#### 8.1.4.3 C. Going deeper

### Exercise C1: A chain of invariant subspaces.

Let  $F$  be any field and

$$\mathbf{N} = \begin{pmatrix} 0 & 1 & 0 \\ 0 & 0 & 1 \\ 0 & 0 & 0 \end{pmatrix} \in M_3(F),$$

acting on  $F^3$ , so  $\mathbf{N}\mathbf{e}_1 = \mathbf{0}$ ,  $\mathbf{N}\mathbf{e}_2 = \mathbf{e}_1$ ,  $\mathbf{N}\mathbf{e}_3 = \mathbf{e}_2$ . Put  $W_0 = \{\mathbf{0}\}$ ,  $W_1 = \operatorname{span}(\mathbf{e}_1)$ ,  $W_2 = \operatorname{span}(\mathbf{e}_1, \mathbf{e}_2)$ ,  $W_3 = F^3$ .

- (a) Show that each  $W_j$  is  $\mathbf{N}$ -invariant.
- (b) Let  $U$  be an  $\mathbf{N}$ -invariant subspace, and let  $j$  be the largest index such that some vector of  $U$  has a non-zero  $j$ -th entry (put  $j = 0$  if  $U = \{\mathbf{0}\}$ ). Prove that  $U = W_j$ .
- (c) Deduce that the  $\mathbf{N}$ -invariant subspaces are exactly  $W_0 \subset W_1 \subset W_2 \subset W_3$ , and that any two of them are comparable under inclusion.

*Hint: for (b), apply  $\mathbf{N}$  and  $\mathbf{N}^2$  to a vector with non-zero  $j$ -th entry.*

*Solution.*

- (a)  $W_0$  and  $W_3$  are invariant for every operator. By [Lemma 8.1.1](#),  $W_1$  is invariant since  $\mathbf{N}\mathbf{e}_1 = \mathbf{0}$ , and  $W_2$  is invariant since  $\mathbf{N}\mathbf{e}_1 = \mathbf{0}$  and  $\mathbf{N}\mathbf{e}_2 = \mathbf{e}_1$  lie in  $W_2$ .
- (b) If  $U = \{\mathbf{0}\}$ , then  $j = 0$  and  $U = W_0$ . Otherwise  $j \geq 1$ . By the choice of  $j$ , every vector of  $U$  has entries 0 beyond position  $j$ , so  $U \subseteq W_j$ . Pick  $\mathbf{v} = (a, b, c) \in U$  with non-zero  $j$ -th entry. We have  $\mathbf{N}\mathbf{v} = (b, c, 0)$  and  $\mathbf{N}^2\mathbf{v} = (c, 0, 0)$ , and all of these lie in  $U$  by invariance.
- Case  $j = 3$ . Here  $c \neq 0$ . Then  $\mathbf{e}_1 = c^{-1}\mathbf{N}^2\mathbf{v} \in U$ ,  $\mathbf{e}_2 = c^{-1}(\mathbf{N}\mathbf{v} - b\mathbf{e}_1) \in U$  and  $\mathbf{e}_3 = c^{-1}(\mathbf{v} - a\mathbf{e}_1 - b\mathbf{e}_2) \in U$ . So  $W_3 \subseteq U$ .
- Case  $j = 2$ . Here  $c = 0$  and  $b \neq 0$ . Then  $\mathbf{e}_1 = b^{-1}\mathbf{N}\mathbf{v} \in U$  and  $\mathbf{e}_2 = b^{-1}(\mathbf{v} - a\mathbf{e}_1) \in U$ , so  $W_2 \subseteq U$ .
- Case  $j = 1$ . Here  $b = c = 0$  and  $a \neq 0$ , so  $\mathbf{e}_1 = a^{-1}\mathbf{v} \in U$  and  $W_1 \subseteq U$ .
- In each case  $W_j \subseteq U \subseteq W_j$ , so  $U = W_j$ .
- (c) By (a) each  $W_j$  is invariant, and by (b) every invariant subspace is one of them. Since  $W_0 \subset W_1 \subset W_2 \subset W_3$ , any two are comparable. In particular  $\mathbf{N}$  has exactly one invariant line,  $W_1$ , and exactly one invariant plane,  $W_2$ . Contrast [Exercise \(B1: All invariant subspaces of a diagonal matrix\)](#), where  $\text{diag}(1, 2)$  has two invariant lines (in the sense of [Proposition 8.1.2](#)), neither containing the other.

### Exercise C2: Invariance under the inverse.

- (a) Let  $V$  be finite-dimensional,  $T \in \mathcal{L}(V)$  invertible, and  $U$  a  $T$ -invariant subspace. Prove that  $T(U) = U$ , and deduce that  $U$  is  $T^{-1}$ -invariant.
- (b) Show that (a) fails without finite dimension. On  $V = F^{\mathbb{Z}}$ , the space of all functions  $\mathbb{Z} \rightarrow F$  written as two-sided sequences  $(s_n)_{n \in \mathbb{Z}}$ , let  $(Ss)_n = s_{n-1}$ . Show that  $S$  is invertible, that  $U = \{s : s_n = 0 \text{ for all } n < 0\}$  is  $S$ -invariant, and that  $U$  is **not**  $S^{-1}$ -invariant.

*Hint: for (a), consider the restriction  $T|_U$  and its kernel.*

*Solution.*

- (a) The restriction  $T|_U \in \mathcal{L}(U)$  exists by invariance. Its kernel is  $\ker T \cap U = \{\mathbf{0}\}$ , since  $T$  is injective, so  $T|_U$  is injective ([Theorem 3.2.3](#)). The space  $U$  is finite-dimensional ([Theorem 1.6.7](#)), so by [Theorem 3.5.7](#),  $T|_U$  is surjective onto  $U$ . That is,  $T(U) = U$ .
- Now let  $\mathbf{u} \in U$ . By surjectivity of  $T|_U$ , there is  $\mathbf{u}' \in U$  with  $T\mathbf{u}' = \mathbf{u}$ . Applying  $T^{-1}$  gives  $T^{-1}\mathbf{u} = \mathbf{u}' \in U$ . Hence  $U$  is  $T^{-1}$ -invariant.
- (b)  $V = F^{\mathbb{Z}}$  is a vector space with pointwise operations ([Example \(The Standard Vector Spaces\)](#)), and  $S$  is linear because it only re-indexes entries. Define  $(S's)_n = s_{n+1}$ . Then  $(S'Ss)_n = (Ss)_{n+1} = s_n$  and  $(SS's)_n = (S's)_{n-1} = s_n$ , so  $S$  is invertible with  $S^{-1} = S'$ .

$U$  is a subspace (the conditions  $s_n = 0$  are preserved by sums and scalar multiples). If  $s \in U$  and  $n < 0$ , then  $n - 1 < 0$ , so  $(Ss)_n = s_{n-1} = 0$ ; thus  $Ss \in U$ , and  $U$  is  $S$ -invariant.

Let  $\delta \in U$  be the sequence with  $\delta_0 = 1$  and  $\delta_n = 0$  for  $n \neq 0$ . Then  $(S^{-1}\delta)_{-1} = \delta_0 = 1 \neq 0$ , so  $S^{-1}\delta \notin U$ . Hence  $U$  is not  $S^{-1}$ -invariant. In (a), the step that breaks is “ $T|_U$  injective, hence surjective”, which needs finite dimension: here  $S|_U$  is injective but its image misses  $\delta$ .

## 8.2 Eigenvalues and Eigenvectors

The previous section ended with an observation: an operator has a diagonal matrix exactly when the space splits into invariant lines, and a line  $\text{span}(\mathbf{v})$  is invariant exactly when  $T\mathbf{v}$  is a multiple of  $\mathbf{v}$ . This section turns that observation into the central notion of the chapter. We name the vectors that an operator only stretches, and the stretching factors. We find the factors as roots of the characteristic polynomial of Chapter 6, and we prove the two facts that everything later rests on: eigenvectors for different factors are independent, and over  $\mathbb{C}$  every operator has at least one eigenvector.

### 8.2.1 When is a matrix diagonal?

Chapter 3 asked how simple  $[T]_{\mathcal{B}}$  can be made by choosing  $\mathcal{B}$ . The simplest square matrices are the diagonal ones, so start with the wish and unwind it.

Let  $V$  be finite-dimensional with basis  $\mathcal{B} = (\mathbf{v}_1, \dots, \mathbf{v}_n)$ , and  $T \in \mathcal{L}(V)$ . By Definition 3.6.1, column  $j$  of  $[T]_{\mathcal{B}}$  holds the coordinates of  $T\mathbf{v}_j$ . The matrix is  $\text{diag}(\lambda_1, \dots, \lambda_n)$  exactly when column  $j$  is  $\lambda_j \mathbf{e}_j$  for each  $j$ , that is,

$$T\mathbf{v}_j = \lambda_j \mathbf{v}_j \quad (j = 1, \dots, n).$$

So the wish forces an equation: every basis vector must be sent to a multiple of itself. We met this in Chapter 3, where the reflection  $R(x, y) = (y, x)$  became  $\text{diag}(1, -1)$  in the basis  $((1, 1), (1, -1))$  (Example (The reflection in a mirror-adapted basis)), precisely because  $R(1, 1) = (1, 1)$  and  $R(1, -1) = -(1, -1)$ .

Where should we look for vectors with  $T\mathbf{v} = \lambda\mathbf{v}$ ? Rewrite  $\lambda\mathbf{v}$  as  $\lambda \text{id}_V(\mathbf{v})$ . Then the equation says  $(T - \lambda \text{id}_V)\mathbf{v} = \mathbf{0}$ : the vector lies in a **kernel**. Kernels are subspaces we know how to compute, so this rewriting is a plan, not just a restatement.

*An eigenvector of  $T$  is a non-zero vector that  $T$  only stretches, and the eigenvalue is the stretching factor.*

#### Definition 8.2.1 — Eigenvalue and Eigenvector

Let  $V$  be a vector space over  $F$  and  $T \in \mathcal{L}(V)$ . A scalar  $\lambda \in F$  is an **eigenvalue** of  $T$  if there **exists** a **non-zero** vector  $\mathbf{v} \in V$  with

$$T\mathbf{v} = \lambda\mathbf{v}.$$

Every **non-zero**  $\mathbf{v} \in V$  with  $T\mathbf{v} = \lambda\mathbf{v}$  is called an **eigenvector** of  $T$  for the eigenvalue  $\lambda$ .

For a matrix  $\mathbf{A} \in M_n(F)$ , the eigenvalues and eigenvectors of  $\mathbf{A}$  are those of  $T_{\mathbf{A}}: F^n \rightarrow F^n$ : a scalar  $\lambda \in F$  and a non-zero  $\mathbf{v} \in F^n$  with  $\mathbf{A}\mathbf{v} = \lambda\mathbf{v}$ .

In words: an eigenvalue is a scalar **in the field**  $F$  over which  $V$  is a vector space. It qualifies if **some** non-zero vector is sent to  $\lambda$  times itself. The vector must be non-zero; the scalar may be anything, including 0. An eigenvector is a witness for its eigenvalue, and there are always many witnesses: if  $\mathbf{v}$  is one, so is  $c\mathbf{v}$  for every  $c \neq 0$ , since  $T(c\mathbf{v}) = cT\mathbf{v} = \lambda(c\mathbf{v})$ .

**Well-definedness.** An eigenvector determines its eigenvalue. If  $T\mathbf{v} = \lambda\mathbf{v}$  and  $T\mathbf{v} = \mu\mathbf{v}$  with  $\mathbf{v} \neq \mathbf{0}$ , then  $(\lambda - \mu)\mathbf{v} = \mathbf{0}$ , and [Theorem 1.1.9](#) gives  $\lambda = \mu$ . So “the eigenvalue of  $\mathbf{v}$ ” makes sense, while “the eigenvector of  $\lambda$ ” does not.

The eigenvectors for one eigenvalue, together with  $\mathbf{0}$ , are exactly the kernel found above. That set is a subspace, and it deserves its own name.

### Definition 8.2.2 — Eigenspace

Let  $T \in \mathcal{L}(V)$  and  $\lambda \in F$ . The **eigenspace** of  $T$  for  $\lambda$  is

$$E_{\lambda}(T) := \ker(T - \lambda \operatorname{id}_V) = \{\mathbf{v} \in V : T\mathbf{v} = \lambda\mathbf{v}\}.$$

For  $\mathbf{A} \in M_n(F)$  we write  $E_{\lambda}(\mathbf{A}) = E_{\lambda}(T_{\mathbf{A}}) = \operatorname{nul}(\mathbf{A} - \lambda\mathbf{I}_n)$ .

$E_{\lambda}(T)$  is a subspace by [Theorem 3.2.1](#), and it is  $T$ -invariant by [Theorem 5.7.4](#) (b) with  $p = x - \lambda$ . It is defined for **every**  $\lambda \in F$ , and  $\lambda$  is an eigenvalue exactly when  $E_{\lambda}(T) \neq \{\mathbf{0}\}$ ; in that case the eigenvectors for  $\lambda$  are the non-zero vectors of  $E_{\lambda}(T)$ . The eigenspace is the home of all eigenvectors for  $\lambda$  at once, and it is one of the invariant subspaces of the previous section: on it,  $T$  acts as  $\lambda \operatorname{id}$ .

#### Examples.

- **Diagonal matrices.** Let  $\mathbf{A} = \operatorname{diag}(d_1, \dots, d_n)$ . Then  $\mathbf{A}\mathbf{e}_j = d_j\mathbf{e}_j$  and  $\mathbf{e}_j \neq \mathbf{0}$ , so each  $d_j$  is an eigenvalue with eigenvector  $\mathbf{e}_j$ . The standard basis is a basis of eigenvectors, which is the observation we started from, read backwards.
- **Degenerate cases.** On  $V \neq \{\mathbf{0}\}$ , the identity has the eigenvalue 1 with  $E_1(\operatorname{id}_V) = V$ : **every** non-zero vector is an eigenvector. The zero operator has the eigenvalue 0 with  $E_0 = V$ . On  $V = \{\mathbf{0}\}$ , no operator has any eigenvalue, because there is no non-zero vector to be a witness. That last case matters: it is why the existence theorem later in this section must assume  $V \neq \{\mathbf{0}\}$ .
- **Projections.** Let  $P \in \mathcal{L}(V)$  be a projection,  $P^2 = P$  ([Definition 3.10.1](#)). If  $P\mathbf{v} = \lambda\mathbf{v}$  with  $\mathbf{v} \neq \mathbf{0}$ , then  $\lambda\mathbf{v} = P\mathbf{v} = P^2\mathbf{v} = P(\lambda\mathbf{v}) = \lambda^2\mathbf{v}$ , so  $(\lambda^2 - \lambda)\mathbf{v} = \mathbf{0}$  and  $\lambda(\lambda - 1) = 0$  by [Theorem 1.1.9](#). So the only possible eigenvalues are 0 and 1. By definition  $E_0(P) = \ker P$ . Also  $E_1(P) = \operatorname{im} P$ : a vector with  $P\mathbf{v} = \mathbf{v}$  is in the image, and conversely  $P$  is the identity on  $\operatorname{im} P$  by [Theorem 3.10.1](#) (a). So a vector is fixed by  $P$  exactly when it is in the image, and killed exactly when it is in the kernel.
- **Differentiation of polynomials.** Let  $D$  act on  $V = \mathbb{R}[x]_{\leq n}$  with  $n \geq 1$ . Suppose  $Dp = \lambda p$  with  $p \neq 0$ . If  $\lambda \neq 0$ , then  $\deg(\lambda p) = \deg p$ , while  $Dp$  is either 0 or of degree  $\deg p - 1$ ; neither equals  $\deg p$  for  $p \neq 0$ . So  $\lambda = 0$ , and  $E_0(D) = \ker D$  is the space of constant polynomials. Differentiation on  $\mathbb{R}[x]_{\leq n}$  has exactly one eigenvalue, and its eigenvectors span only a line.

**Non-example by minimal change.** For  $\mathbf{A} = \begin{pmatrix} 2 & 1 \\ 0 & 3 \end{pmatrix}$ , the vector  $\mathbf{e}_1$  is an eigenvector:  $\mathbf{A}\mathbf{e}_1 = 2\mathbf{e}_1$ . Move to the neighboring basis vector  $\mathbf{e}_2$ . It is still non-zero, and  $\mathbf{A}\mathbf{e}_2 = (1, 3)$  is still a vector of  $\mathbb{R}^2$ . What fails is the equation  $\mathbf{A}\mathbf{v} = \lambda\mathbf{v}$ : no  $\lambda$  makes  $(1, 3) = (0, \lambda)$ . So  $\mathbf{e}_2$  is not an eigenvector, and the line  $\text{span}(\mathbf{e}_2)$  is not invariant, as we saw in the previous section.

**Why this definition.** The clause “**non-zero**” is what gives the notion content. The zero vector satisfies  $T\mathbf{0} = \lambda\mathbf{0}$  for **every**  $\lambda \in F$ , so if we allowed  $\mathbf{v} = \mathbf{0}$ , every scalar would be an eigenvalue of every operator. On the other hand we do **not** exclude  $\lambda = 0$ : the equation  $T\mathbf{v} = 0\mathbf{v}$  says  $\mathbf{v} \in \ker T$ , and a non-zero kernel vector is exactly the kind of information we want to record. Requiring  $\lambda \in F$ , rather than in some larger field, makes the notion depend on the field, and this is deliberate: a real matrix may have no real eigenvalues, and then no real basis diagonalizes it. The prefix *eigen* is German for “own”: an eigenvector is a direction that belongs to  $T$ , which  $T$  maps into itself.

*Warning.* The eigenvalue may be 0; the eigenvector may not. For  $\mathbf{A} = \begin{pmatrix} 1 & 2 \\ 2 & 4 \end{pmatrix}$ , the vector  $(2, -1)$  satisfies  $\mathbf{A}(2, -1) = (0, 0) = 0 \cdot (2, -1)$ , so 0 is an eigenvalue. By contrast,  $\mathbf{A}\mathbf{0} = 7 \cdot \mathbf{0}$  does **not** make 7 an eigenvalue. And say “**an** eigenvector for 0”, never “the”:  $(2, -1)$ ,  $(-4, 2)$  and  $(200, -100)$  are all eigenvectors for 0.

*Quick check.* Let  $T(x, y, z) = (x, z, y)$  on  $\mathbb{R}^3$ . Which of  $\mathbf{e}_1$ ,  $(0, 1, 1)$ ,  $(0, 1, -1)$ ,  $\mathbf{e}_2$  are eigenvectors of  $T$ , and for which eigenvalues?

*Solution.*  $T\mathbf{e}_1 = \mathbf{e}_1$ , so  $\mathbf{e}_1$  is an eigenvector for 1.  $T(0, 1, 1) = (0, 1, 1)$ , an eigenvector for 1 as well.  $T(0, 1, -1) = (0, -1, 1) = -(0, 1, -1)$ , an eigenvector for  $-1$ .  $T\mathbf{e}_2 = \mathbf{e}_3$ , which is not a multiple of  $\mathbf{e}_2$ , so  $\mathbf{e}_2$  is **not** an eigenvector. The three eigenvectors form a basis of  $\mathbb{R}^3$ , and in that basis  $[T] = \text{diag}(1, 1, -1)$ .

## 8.2.2 Finding eigenvalues

The definition asks for a non-zero vector in  $\ker(T - \lambda \text{id}_V)$ , a question about one  $\lambda$  at a time. In finite dimension, Chapters 3 and 6 turn it into a test that handles all  $\lambda$  at once. The last item is the one we compute with.

### Theorem 8.2.1 — Characterizations of Eigenvalues

Let  $V$  be a finite-dimensional vector space over  $F$  with  $\dim V = n \geq 1$ , let  $T \in \mathcal{L}(V)$  and  $\lambda \in F$ . The following are equivalent:

- (a)  $\lambda$  is an eigenvalue of  $T$ ;
- (b)  $E_\lambda(T) \neq \{\mathbf{0}\}$ , that is,  $T - \lambda \text{id}_V$  is **not injective**;
- (c)  $T - \lambda \text{id}_V$  is **not invertible**;
- (d)  $\det(\lambda \text{id}_V - T) = 0$ ;
- (e)  $p_T(\lambda) = 0$ .

In particular  $T$  has at most  $n$  eigenvalues. For  $\mathbf{A} \in M_n(F)$ :  $\lambda$  is an eigenvalue of  $\mathbf{A}$  if and only if  $p_{\mathbf{A}}(\lambda) = 0$ , if and only if  $\lambda\mathbf{I} - \mathbf{A}$  is not invertible.



*Proof.*

- (a)  $\Leftrightarrow$  (b). By [Definition 8.2.1](#) and [Definition 8.2.2](#),  $\lambda$  is an eigenvalue exactly when  $E_\lambda(T) = \ker(T - \lambda \operatorname{id}_V)$  contains a non-zero vector. By [Theorem 3.2.3](#), this happens exactly when  $T - \lambda \operatorname{id}_V$  is not injective.
- (b)  $\Leftrightarrow$  (c).  $V$  is finite-dimensional, so by [Theorem 3.5.7](#) ((a)  $\Leftrightarrow$  (b)) an operator on  $V$  is invertible if and only if it is injective.
- (c)  $\Leftrightarrow$  (d). The operators  $\lambda \operatorname{id}_V - T$  and  $T - \lambda \operatorname{id}_V$  differ by the factor  $-1$ , so one is invertible exactly when the other is. By [Theorem 6.5.8](#),  $\lambda \operatorname{id}_V - T$  is invertible if and only if  $\det(\lambda \operatorname{id}_V - T) \neq 0$ .
- (d)  $\Leftrightarrow$  (e). Fix a basis  $\mathcal{B}$  of  $V$  and put  $\mathbf{A} = [T]_{\mathcal{B}}$ . By [Corollary 3.6.3](#),  $[\lambda \operatorname{id}_V - T]_{\mathcal{B}} = \lambda \mathbf{I}_n - \mathbf{A}$ , and by [Theorem 3.6.5](#) (b),  $\lambda \operatorname{id}_V - T$  is invertible if and only if  $\lambda \mathbf{I}_n - \mathbf{A}$  is. By [Theorem 6.10.4](#) ((a), (i)  $\Leftrightarrow$  (ii)),  $\lambda \mathbf{I}_n - \mathbf{A}$  is not invertible if and only if  $p_{\mathbf{A}}(\lambda) = 0$ , and  $p_{\mathbf{A}} = p_T$  by [Definition 6.10.2](#).

The bound on the number of eigenvalues is [Theorem 6.10.4](#) (b) applied to  $\mathbf{A}$ , since by (e) the eigenvalues of  $T$  are the roots of  $p_{\mathbf{A}}$  in  $F$ . The statement for matrices is the case  $T = T_{\mathbf{A}}$ ,  $\mathcal{B}$  standard,  $[T_{\mathbf{A}}]_{\mathcal{B}} = \mathbf{A}$ . ■

For matrices, this is [Theorem 6.10.4](#) read in the new language: Chapter 6 found the scalars  $c$  with  $c\mathbf{I} - \mathbf{A}$  singular, and those are the eigenvalues. It gives a procedure with two steps.

1. Compute  $p_{\mathbf{A}}(x) = \det(x\mathbf{I} - \mathbf{A})$  and find its roots in  $F$ . These are the eigenvalues.
2. For each eigenvalue  $\lambda$ , solve the homogeneous system  $(\mathbf{A} - \lambda\mathbf{I})\mathbf{x} = \mathbf{0}$  by elimination. Its solution set is  $E_\lambda(\mathbf{A})$ , and a basis of it lists “all” eigenvectors for  $\lambda$  ([Theorem 2.5.4](#)).

### Example. (Eigenvalues and Eigenspaces of a $2 \times 2$ Matrix)

Find the eigenvalues and eigenspaces of  $\mathbf{A} = \begin{pmatrix} 1 & 2 \\ 3 & 2 \end{pmatrix} \in M_2(\mathbb{R})$ .

*Solution. Step 1.* For a  $2 \times 2$  matrix,  $p_{\mathbf{A}}(x) = x^2 - (\operatorname{tr} \mathbf{A})x + \det \mathbf{A}$  (the examples after [Definition 6.10.1](#)). Here  $\operatorname{tr} \mathbf{A} = 3$  and  $\det \mathbf{A} = 2 - 6 = -4$ , so

$$p_{\mathbf{A}}(x) = x^2 - 3x - 4 = (x - 4)(x + 1).$$

By [Theorem 8.2.1](#), the eigenvalues are 4 and  $-1$ .

*Step 2.* For  $\lambda = 4$ ,

$$\mathbf{A} - 4\mathbf{I} = \begin{pmatrix} -3 & 2 \\ 3 & -2 \end{pmatrix}.$$

The second row is  $-1$  times the first, so the system reduces to  $-3x + 2y = 0$ , with solutions  $(x, y) = t(2, 3)$ . Hence  $E_4(\mathbf{A}) = \operatorname{span}((2, 3))$ . For  $\lambda = -1$ ,

$$\mathbf{A} + \mathbf{I} = \begin{pmatrix} 2 & 2 \\ 3 & 3 \end{pmatrix},$$

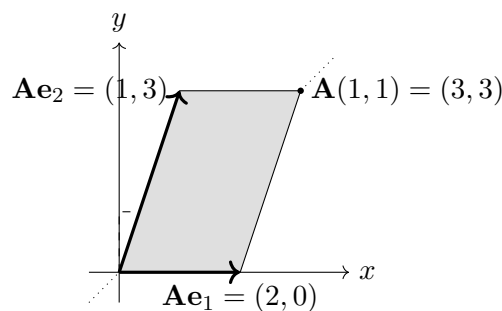
which reduces to  $x + y = 0$ , so  $E_{-1}(\mathbf{A}) = \operatorname{span}((1, -1))$ .

*Check.*  $\mathbf{A}(2, 3) = (2+6, 6+6) = (8, 12) = 4(2, 3)$  and  $\mathbf{A}(1, -1) = (1-2, 3-2) = (-1, 1) = -1(1, -1)$ . In the basis  $((2, 3), (1, -1))$ , the operator  $T_{\mathbf{A}}$  has matrix  $\text{diag}(4, -1)$ , by the computation at the start of this section.

In Step 2, the matrix  $\mathbf{A} - \lambda\mathbf{I}$  **must** be singular; if elimination produces only the trivial solution, then  $\lambda$  was not a root of  $p_{\mathbf{A}}$ , and Step 1 contains an error. This is a built-in check.

**Triangular matrices.** If  $\mathbf{A}$  is upper or lower triangular, then  $p_{\mathbf{A}}(x) = (x - a_{11}) \cdots (x - a_{nn})$  (the examples after Definition 6.10.1), so the eigenvalues of a triangular matrix are its diagonal entries. For  $\begin{pmatrix} 2 & 1 \\ 0 & 3 \end{pmatrix}$  they are 2 and 3, with eigenvectors  $\mathbf{e}_1$  and  $(1, 1)$ , found in the previous section.

You can watch eigenvectors in the widget below. It shows the unit square and its image under  $\mathbf{A} = \begin{pmatrix} 2 & 1 \\ 0 & 3 \end{pmatrix}$ . The image  $\mathbf{A}\mathbf{e}_1$  lies on the  $x$ -axis, which says that  $\mathbf{e}_1$  is an eigenvector. The far corner of the image parallelogram is  $\mathbf{A}(1, 1) = \mathbf{A}\mathbf{e}_1 + \mathbf{A}\mathbf{e}_2 = (3, 3)$ , and it lies on the diagonal line through  $(1, 1)$ : so  $(1, 1)$  is an eigenvector too, for the eigenvalue 3. Drag  $\mathbf{A}\mathbf{e}_2$  to  $(1, 2)$ , which changes the matrix to  $\begin{pmatrix} 2 & 1 \\ 0 & 2 \end{pmatrix}$ . Now the far corner  $(3, 2)$  leaves the diagonal, and only the  $x$ -axis is still mapped into itself. The next section explains what was lost.



The unit square (dashed) and its image under  $\mathbf{A} = \begin{pmatrix} 2 & 1 \\ 0 & 3 \end{pmatrix}$ . The image of  $\mathbf{e}_1$  stays on the  $x$ -axis (eigenvalue 2), and the image of  $(1, 1)$  stays on the dotted diagonal (eigenvalue 3). The image of  $\mathbf{e}_2$  leaves the  $y$ -axis, so  $\mathbf{e}_2$  is not an eigenvector.

The set of all eigenvalues is used so often that it gets a symbol.

### Definition 8.2.3 — Spectrum

Let  $T \in \mathcal{L}(V)$ . The **spectrum** of  $T$  is the set  $\text{spec}(T) \subseteq F$  of all eigenvalues of  $T$ . For  $\mathbf{A} \in M_n(F)$ ,  $\text{spec}(\mathbf{A}) := \text{spec}(T_{\mathbf{A}})$ .

For instance  $\text{spec} \begin{pmatrix} 1 & 2 \\ 3 & 2 \end{pmatrix} = \{4, -1\}$ ,  $\text{spec}(\text{id}_V) = \{1\}$  for  $V \neq \{0\}$ , and the spectrum of a projection is contained in  $\{0, 1\}$ . By Theorem 8.2.1, if  $\dim V = n \geq 1$ , then  $\text{spec}(T)$  is the set of roots of  $p_T$  in  $F$ , and it has at most  $n$  elements. It may be empty: the characteristic polynomial of the rotation  $\mathbf{R} = \begin{pmatrix} 0 & -1 \\ 1 & 0 \end{pmatrix}$  is  $x^2 + 1$ , which has no real root, so  $\text{spec}(\mathbf{R}) = \emptyset$  over  $\mathbb{R}$ . This matches Example (A Rotation Has No Invariant Lines): no invariant lines, no eigenvectors.

**Infinite dimension.** Without a characteristic polynomial, we go back to the definition, and anything can happen.

- **Differentiation of smooth functions.** Let  $V$  be the space of infinitely differentiable functions  $\mathbb{R} \rightarrow \mathbb{R}$  (a subspace of  $\mathbb{R}^{\mathbb{R}}$ , by the argument of [Example \(Differentiable Functions\)](#) applied to all derivatives), and  $Df = f'$ . Here we take three facts from calculus as known:  $e^{\lambda x}$  is smooth with derivative  $\lambda e^{\lambda x}$ ; the product rule; and a function on  $\mathbb{R}$  with derivative 0 everywhere is constant. For **every**  $\lambda \in \mathbb{R}$ , the function  $f(x) = e^{\lambda x}$  is non-zero and satisfies  $Df = \lambda f$ . So  $\text{spec}(D) = \mathbb{R}$ , an infinite set. In fact  $E_{\lambda}(D) = \text{span}(e^{\lambda x})$ : if  $f' = \lambda f$ , then by the product rule  $g(x) = e^{-\lambda x} f(x)$  has  $g' = e^{-\lambda x}(f' - \lambda f) = 0$ , so  $g$  is a constant  $c$  and  $f = ce^{\lambda x}$ .
- **The right shift.** Let  $R(s_0, s_1, s_2, \dots) = (0, s_0, s_1, \dots)$  on  $F^{\mathbb{N}}$  ([Example \(The Shift Maps on Sequences\)](#)). If  $R\mathbf{s} = \lambda\mathbf{s}$ , compare entries:  $0 = \lambda s_0$ , and  $s_k = \lambda s_{k+1}$  for all  $k \geq 0$ . If  $\lambda = 0$ , the second equation gives  $s_k = 0$  for all  $k$ . If  $\lambda \neq 0$ , the first gives  $s_0 = 0$ , and then  $s_{k+1} = \lambda^{-1}s_k$  gives  $s_1 = 0$ ,  $s_2 = 0$ , and so on by induction. Either way  $\mathbf{s} = \mathbf{0}$ . So  $\text{spec}(R) = \emptyset$ , over **every** field.

*Warning.* In infinite dimension, “not invertible” does not produce an eigenvalue. The step (b)  $\Leftrightarrow$  (c) of [Theorem 8.2.1](#) used finite dimension. The right shift  $R$  is injective but not surjective ([Example \(The Shifts: Injective but Not Surjective, and Conversely\)](#)), so  $R - 0 \cdot \text{id}$  is not invertible, yet 0 is not an eigenvalue of  $R$ : nothing non-zero is sent to  $\mathbf{0}$ .

### 8.2.3 Distinct eigenvalues give independent eigenvectors

In [Example \(Eigenvalues and Eigenspaces of a  \$2 \times 2\$  Matrix\)](#) the eigenvectors  $(2, 3)$  and  $(1, -1)$  for the eigenvalues 4 and  $-1$  formed a basis. That is no accident. Eigenvectors for different eigenvalues can never be linearly dependent.

#### **Theorem 8.2.2 — Distinct Eigenvalues Give Independent Eigenvectors**

Let  $T \in \mathcal{L}(V)$ , let  $\lambda_1, \dots, \lambda_k \in F$  be **distinct** eigenvalues of  $T$ , and for each  $i$  let  $\mathbf{v}_i$  be an eigenvector of  $T$  for  $\lambda_i$ . Then  $(\mathbf{v}_1, \dots, \mathbf{v}_k)$  is linearly independent.

*Idea.* Try two vectors first. From  $a\mathbf{v}_1 + b\mathbf{v}_2 = \mathbf{0}$ , we want to kill one term, and the operator  $T - \lambda_2 \text{id}_V$  does exactly that: it sends  $\mathbf{v}_2$  to  $\mathbf{0}$  and  $\mathbf{v}_1$  to  $(\lambda_1 - \lambda_2)\mathbf{v}_1$ . So  $a(\lambda_1 - \lambda_2)\mathbf{v}_1 = \mathbf{0}$ , and since  $\lambda_1 \neq \lambda_2$  and  $\mathbf{v}_1 \neq \mathbf{0}$ ,  $a = 0$ ; then  $b = 0$ . In general, applying  $T - \lambda_j \text{id}_V$  to a dependence kills the  $j$ -th vector and rescales the others by non-zero factors, producing a **shorter** dependence. So a shortest dependence cannot exist. The Linear Dependence Lemma lets us pick that shortest one cleanly.

*Proof.* Suppose, for a contradiction, that  $(\mathbf{v}_1, \dots, \mathbf{v}_k)$  is linearly dependent. By [Theorem 1.4.1](#) (1), there is an index  $j$  with  $\mathbf{v}_j \in \text{span}(\mathbf{v}_1, \dots, \mathbf{v}_{j-1})$ ; choose the **smallest** such  $j$ . Since  $\mathbf{v}_1 \neq \mathbf{0}$ ,  $j \geq 2$ . The list  $(\mathbf{v}_1, \dots, \mathbf{v}_{j-1})$  is linearly independent: otherwise [Theorem 1.4.1](#) (1), applied to it, would give a smaller index with the same property.

Write

$$\mathbf{v}_j = a_1 \mathbf{v}_1 + \cdots + a_{j-1} \mathbf{v}_{j-1} \quad (a_i \in F).$$

Apply  $T - \lambda_j \text{id}_V$  to both sides. Since  $(T - \lambda_j \text{id}_V)\mathbf{v}_i = (\lambda_i - \lambda_j)\mathbf{v}_i$  for each  $i$ , linearity gives

$$\mathbf{0} = a_1(\lambda_1 - \lambda_j)\mathbf{v}_1 + \cdots + a_{j-1}(\lambda_{j-1} - \lambda_j)\mathbf{v}_{j-1}.$$

By independence of  $(\mathbf{v}_1, \dots, \mathbf{v}_{j-1})$ , every coefficient  $a_i(\lambda_i - \lambda_j)$  is 0. The eigenvalues are distinct, so  $\lambda_i - \lambda_j \neq 0$  for  $i < j$ , and hence  $a_i = 0$  for all  $i < j$ . Then  $\mathbf{v}_j = \mathbf{0}$ , which contradicts the fact that an eigenvector is non-zero. This proves that  $(\mathbf{v}_1, \dots, \mathbf{v}_k)$  is linearly independent. ■

The theorem works in any vector space, finite-dimensional or not. For instance, for distinct real numbers  $\lambda_1, \dots, \lambda_k$ , the functions  $e^{\lambda_1 x}, \dots, e^{\lambda_k x}$  are eigenvectors of differentiation for distinct eigenvalues, so they are linearly independent, with no computation at all. Two consequences follow at once, and the second is the first half of a promise made in Chapter 1, §7: eigenspaces sit inside  $V$  as a direct sum.

### Corollary 8.2.3 — Eigenspaces Form a Direct Sum

Let  $T \in \mathcal{L}(V)$  and let  $\lambda_1, \dots, \lambda_k \in F$  be distinct. Then the sum  $E_{\lambda_1}(T) + \cdots + E_{\lambda_k}(T)$  is direct. If  $V$  is finite-dimensional, then

$$\dim E_{\lambda_1}(T) + \cdots + \dim E_{\lambda_k}(T) \leq \dim V,$$

and  $T$  has at most  $\dim V$  distinct eigenvalues.

*Proof.* By Theorem 1.7.5 ((b)  $\Rightarrow$  (a)), it suffices to show: if  $\mathbf{u}_1 + \cdots + \mathbf{u}_k = \mathbf{0}$  with  $\mathbf{u}_i \in E_{\lambda_i}(T)$ , then every  $\mathbf{u}_i = \mathbf{0}$ . Suppose not, and let  $S$  be the non-empty set of indices  $i$  with  $\mathbf{u}_i \neq \mathbf{0}$ . For  $i \in S$ ,  $\mathbf{u}_i$  is an eigenvector of  $T$  for  $\lambda_i$ , and these eigenvalues are distinct. By Theorem 8.2.2, the list  $(\mathbf{u}_i)_{i \in S}$  is linearly independent. But  $\sum_{i \in S} 1 \cdot \mathbf{u}_i = \sum_{i=1}^k \mathbf{u}_i = \mathbf{0}$  is a relation with all coefficients  $1 \neq 0$ , a contradiction. So the sum is direct.

If  $V$  is finite-dimensional, Theorem 1.7.5 ((a)  $\Rightarrow$  (e)) gives  $\dim(E_{\lambda_1}(T) \oplus \cdots \oplus E_{\lambda_k}(T)) = \sum_i \dim E_{\lambda_i}(T)$ , and this is at most  $\dim V$  by Theorem 1.6.7. If the  $\lambda_i$  are eigenvalues, each  $\dim E_{\lambda_i}(T) \geq 1$ , so  $k \leq \dim V$ . ■

The corollary gives a second, polynomial-free proof that an operator on an  $n$ -dimensional space has at most  $n$  eigenvalues. Section 4 shows that  $T$  has a diagonal matrix exactly when the inequality is an equality.

## 8.2.4 Existence over the complex numbers

We have seen operators without eigenvalues: the rotation of  $\mathbb{R}^2$ , and the right shift. The rotation fails because  $\mathbb{R}$  is missing the roots of  $x^2 + 1$ ; the shift fails because its space is infinite-dimensional. Remove both obstacles and eigenvalues always exist.

### Theorem 8.2.4 — Complex Operators Have Eigenvalues

Let  $V$  be a finite-dimensional vector space over  $\mathbb{C}$  with  $V \neq \{0\}$ , and let  $T \in \mathcal{L}(V)$ . Then  $T$  has an eigenvalue. The same holds for every  $\mathbf{A} \in M_n(\mathbb{C})$  with  $n \geq 1$ .

*Proof.* Let  $n = \dim V \geq 1$ . By Theorem 6.10.2 and Definition 6.10.2,  $p_T \in \mathbb{C}[x]$  is monic of degree  $n \geq 1$ , so it is not constant. By the Fundamental Theorem of Algebra (Theorem 5.5.1),  $p_T(\lambda) = 0$  for some  $\lambda \in \mathbb{C}$ . By Theorem 8.2.1 ((e)  $\Rightarrow$  (a)),  $\lambda$  is an eigenvalue of  $T$ . For  $\mathbf{A} \in M_n(\mathbb{C})$ , apply this to  $T_{\mathbf{A}}$ . ■

Alternatively, determinants can be avoided altogether, and the proof then shows where the eigenvector comes from. By Theorem 5.7.2, there is a non-zero  $p \in \mathbb{C}[x]$  with  $p(T) = 0$ . It is not a non-zero constant  $c$ , since  $c \operatorname{id}_V \neq 0$  on  $V \neq \{0\}$ . By Corollary 5.5.5,  $p = c(x - z_1) \cdots (x - z_m)$  with  $c \neq 0$  and  $m \geq 1$ , so by Theorem 5.7.1 (a) and (b),

$$0 = p(T) = c(T - z_1 \operatorname{id}_V)(T - z_2 \operatorname{id}_V) \cdots (T - z_m \operatorname{id}_V).$$

Since  $c \neq 0$ , multiplying by  $c^{-1}$  shows that the composite  $(T - z_1 \operatorname{id}_V) \cdots (T - z_m \operatorname{id}_V)$  is the zero operator, which is not injective on  $V \neq \{0\}$ . A composition of injective maps is injective (Theorem 0.4.3 (a), applied  $m - 1$  times), so at least one factor  $T - z_j \operatorname{id}_V$  is not injective, and  $z_j$  is an eigenvalue by Theorem 8.2.1 ((b)  $\Rightarrow$  (a)).

Both routes use the Fundamental Theorem of Algebra, and both need all three hypotheses. Over  $\mathbb{R}$  the rotation has no eigenvalue. In infinite dimension the right shift on  $\mathbb{C}^{\mathbb{N}}$  has none, and indeed no non-zero polynomial annihilates it: for  $p = a_0 + a_1x + \cdots + a_Nx^N \neq 0$ , the sequence  $p(R)(1, 0, 0, \dots) = (a_0, a_1, \dots, a_N, 0, \dots)$  is not zero. On  $V = \{0\}$  there is nothing to be an eigenvector. This theorem pays off the promise made in Chapter 5, and it is the base step of every structure theorem over  $\mathbb{C}$  in this book: find one eigenvector, then look at what is left.

*Quick check.* Does every operator on  $\mathbb{C}^3$  have an invariant line? Does every operator on  $\mathbb{R}^2$ ?

*Solution.* Every operator on  $\mathbb{C}^3$  has one: by Theorem 8.2.4 it has an eigenvector  $\mathbf{v}$ , and  $\operatorname{span}(\mathbf{v})$  is invariant by Proposition 8.1.2. Not every operator on  $\mathbb{R}^2$ : the rotation by a right angle has no invariant line (Example (A Rotation Has No Invariant Lines)).

## 8.2.5 Real matrices and complex eigenvalues

A real matrix may have no real eigenvalues, but it is also a complex matrix, and as such it has eigenvalues by Theorem 8.2.4. This is the situation Chapter 1, §9 prepared for. Let  $\mathbf{A} \in M_n(\mathbb{R})$ . The complexification of  $\mathbb{R}^n$  is  $\mathbb{C}^n$ : the pair  $(\mathbf{x}, \mathbf{y})$  is the vector  $\mathbf{x} + i\mathbf{y}$  (Example (The Complexification of  $\mathbb{R}^n$  Is  $\mathbb{C}^n$ )). The real operator  $T_{\mathbf{A}}$  extends to the complexification by the only rule compatible with complex scalars,

$$\mathbf{x} + i\mathbf{y} \mapsto \mathbf{A}\mathbf{x} + i\mathbf{A}\mathbf{y} = \mathbf{A}(\mathbf{x} + i\mathbf{y}),$$

which is just  $T_{\mathbf{A}}$  for  $\mathbf{A}$  regarded as a matrix in  $M_n(\mathbb{C})$ . By Theorem 1.9.3, the real standard basis is also a complex basis of  $\mathbb{C}^n$ , and in it the extended operator still has matrix  $\mathbf{A}$ ; in particular

it has the same characteristic polynomial  $p_{\mathbf{A}} \in \mathbb{R}[x] \subseteq \mathbb{C}[x]$ . So the **complex eigenvalues** of a real matrix are the roots of  $p_{\mathbf{A}}$  in  $\mathbb{C}$ , with eigenvectors in  $\mathbb{C}^n$ , and the real eigenvalues are among them. The complex ones cannot appear alone.

### Theorem 8.2.5 — Complex Eigenvalues of Real Matrices Come in Conjugate Pairs

Let  $\mathbf{A} \in M_n(\mathbb{R})$ , regarded as an element of  $M_n(\mathbb{C})$ . If  $\lambda \in \mathbb{C}$  is an eigenvalue of  $\mathbf{A}$  with eigenvector  $\mathbf{v} \in \mathbb{C}^n$ , then  $\bar{\lambda}$  is an eigenvalue of  $\mathbf{A}$  with eigenvector  $\bar{\mathbf{v}}$ , the vector of conjugated entries. Moreover:

- (a)  $\text{mult}_{\lambda}(p_{\mathbf{A}}) = \text{mult}_{\bar{\lambda}}(p_{\mathbf{A}})$ ;
- (b) if  $\lambda \notin \mathbb{R}$ , then  $(\mathbf{v}, \bar{\mathbf{v}})$  is linearly independent over  $\mathbb{C}$ .

*Proof.* Let  $\mathbf{A}\mathbf{v} = \lambda\mathbf{v}$  with  $\mathbf{v} = (v_1, \dots, v_n) \neq \mathbf{0}$ . For each  $k$ , the  $k$ -th entry of  $\mathbf{A}\mathbf{v}$  is  $\sum_j a_{kj}v_j$  (Definition 0.9.9). By Theorem 0.6.2 (a) and (b), and since  $\overline{a_{kj}} = a_{kj}$  for the real entries of  $\mathbf{A}$ ,

$$\overline{\sum_j a_{kj}v_j} = \sum_j a_{kj}\bar{v}_j, \quad \overline{\lambda v_k} = \bar{\lambda} \bar{v}_k.$$

Conjugating the equation  $\mathbf{A}\mathbf{v} = \lambda\mathbf{v}$  entry by entry therefore gives  $\mathbf{A}\bar{\mathbf{v}} = \bar{\lambda}\bar{\mathbf{v}}$ . Since  $\overline{\bar{v}_k} = v_k$ ,  $\bar{\mathbf{v}} = \mathbf{0}$  would force  $\mathbf{v} = \mathbf{0}$ ; so  $\bar{\mathbf{v}} \neq \mathbf{0}$ , and  $\bar{\mathbf{v}}$  is an eigenvector for  $\bar{\lambda}$ .

- (a) The entries of  $x\mathbf{I} - \mathbf{A}$  lie in  $\mathbb{R}[x]$ , so its determinant  $p_{\mathbf{A}}$  lies in  $\mathbb{R}[x]$  (Definition 6.10.1). By Theorem 5.5.6,  $\text{mult}_{\bar{\lambda}}(p_{\mathbf{A}}) = \text{mult}_{\lambda}(p_{\mathbf{A}})$ .
- (b) If  $\lambda \notin \mathbb{R}$ , then  $\bar{\lambda} \neq \lambda$ , so  $\mathbf{v}$  and  $\bar{\mathbf{v}}$  are eigenvectors for distinct eigenvalues, and they are independent by Theorem 8.2.2.

■

For the rotation  $\mathbf{R} = \begin{pmatrix} 0 & -1 \\ 1 & 0 \end{pmatrix}$ , the vector  $(1, -i)$  satisfies  $\mathbf{R}(1, -i) = (i, 1) = i(1, -i)$ , so  $i$  is an eigenvalue. Conjugating,  $\mathbf{R}(1, i) = (-i, 1) = -i(1, i)$ . Over  $\mathbb{C}$ , the two eigenvectors form a basis of  $\mathbb{C}^2$ , and the rotation has a diagonal matrix  $\text{diag}(i, -i)$  in it, although no real basis diagonalizes it. So the field matters for eigenvalues, and it will matter for diagonalization.

#### Example. (A Real Matrix with Non-Real Eigenvalues)

Find the eigenvalues of  $\mathbf{A} = \begin{pmatrix} 3 & -2 \\ 4 & -1 \end{pmatrix}$  over  $\mathbb{C}$ , with an eigenvector for each.

*Solution.* Here  $\text{tr } \mathbf{A} = 2$  and  $\det \mathbf{A} = -3 + 8 = 5$ , so  $p_{\mathbf{A}}(x) = x^2 - 2x + 5 = (x - 1)^2 + 4$ . Over  $\mathbb{R}$  this has no root, so  $\text{spec}(\mathbf{A}) = \emptyset$  over  $\mathbb{R}$ . Over  $\mathbb{C}$ ,  $(x - 1)^2 = -4$  gives the roots  $1 + 2i$  and  $1 - 2i$ .

For  $\lambda = 1 + 2i$ ,

$$\mathbf{A} - \lambda\mathbf{I} = \begin{pmatrix} 2 - 2i & -2 \\ 4 & -2 - 2i \end{pmatrix}.$$

The first row gives  $(2 - 2i)x = 2y$ , that is,  $y = (1 - i)x$ ; take  $\mathbf{v} = (1, 1 - i)$ . Check the second row:  $4 \cdot 1 + (-2 - 2i)(1 - i) = 4 + (-2 + 2i - 2i + 2i^2) = 4 - 4 = 0$ . So  $\mathbf{v}$  is an

eigenvector for  $1 + 2i$ . By [Theorem 8.2.5](#),  $\bar{\mathbf{v}} = (1, 1 + i)$  is an eigenvector for  $1 - 2i$ , with no further computation.

## 8.2.6 The invertible matrix theorem grows

Chapter 2 listed eight conditions equivalent to invertibility ([Theorem 2.4.4](#)), and Chapter 6 added the determinant ([Theorem 6.5.3](#)). Eigenvalues give one more: a matrix is singular exactly when it kills a non-zero vector, which is to say, when it stretches some non-zero vector by the factor 0.

### Theorem 8.2.6 — Invertible Matrix Theorem, with Eigenvalues

Let  $\mathbf{A} \in M_n(F)$  with  $n \geq 1$ . The following are equivalent, and each is equivalent to each of the conditions of [Theorem 6.5.3](#) and [Theorem 2.4.4](#).

- (a)  $\mathbf{A}$  is invertible.
- (b) 0 is **not** an eigenvalue of  $\mathbf{A}$ , that is,  $0 \notin \text{spec}(\mathbf{A})$ .

Likewise, an operator  $T$  on a finite-dimensional space  $V \neq \{0\}$  is invertible if and only if  $0 \notin \text{spec}(T)$ .

*Proof.* By [Definition 8.2.2](#),  $E_0(\mathbf{A}) = \ker(T_{\mathbf{A}} - 0 \text{ id}) = \ker T_{\mathbf{A}}$ . So  $0 \notin \text{spec}(\mathbf{A})$  if and only if  $\ker T_{\mathbf{A}} = \{0\}$ , if and only if  $T_{\mathbf{A}}$  is injective ([Theorem 3.2.3](#)), if and only if  $\mathbf{A}$  is invertible ([Theorem 6.5.3](#), (a)  $\Leftrightarrow$  (b)). For an operator,  $E_0(T) = \ker T$ , and [Theorem 3.5.7](#) ((a)  $\Leftrightarrow$  (c)) finishes the same way. ■

For example,  $\text{spec} \begin{pmatrix} 1 & 2 \\ 3 & 2 \end{pmatrix} = \{4, -1\}$  does not contain 0, so that matrix is invertible, while  $\begin{pmatrix} 1 & 2 \\ 2 & 4 \end{pmatrix}$  has the eigenvalue 0 and is singular.

Eigenvalues are easy to compute for one matrix, and it is tempting to assume they behave well under the operations we use on matrices. Two such assumptions are false.

**Warning. Eigenvalues do not add or multiply.** Let  $\mathbf{A} = \begin{pmatrix} 0 & 1 \\ 0 & 0 \end{pmatrix}$  and  $\mathbf{B} = \begin{pmatrix} 0 & 0 \\ 1 & 0 \end{pmatrix}$ . Both are triangular with zero diagonal, so  $\text{spec}(\mathbf{A}) = \text{spec}(\mathbf{B}) = \{0\}$ . But  $\mathbf{A} + \mathbf{B} = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}$  has  $p(x) = x^2 - 1$  and eigenvalues  $\pm 1$ , which are not sums of eigenvalues of  $\mathbf{A}$  and  $\mathbf{B}$ ; and  $\mathbf{AB} = \begin{pmatrix} 1 & 0 \\ 0 & 0 \end{pmatrix}$  has the eigenvalue 1, which is not a product of them.

**Warning. Row operations change eigenvalues.** Elimination preserves the solution set of  $\mathbf{Ax} = \mathbf{0}$  ([Theorem 2.2.2](#)), not the solution sets of  $\mathbf{Ax} = \lambda \mathbf{x}$ . For  $\mathbf{A} = \begin{pmatrix} 1 & 2 \\ 3 & 2 \end{pmatrix}$ , with eigenvalues 4 and  $-1$ , subtracting 3 times row 1 from row 2 gives the triangular matrix  $\begin{pmatrix} 1 & 2 \\ 0 & -4 \end{pmatrix}$ , whose eigenvalues are 1 and  $-4$ . So never row reduce  $\mathbf{A}$  to find its

eigenvalues; row reduce  $\mathbf{A} - \lambda\mathbf{I}$  to find eigenvectors, once  $\lambda$  is known.

## 8.2.7 Exercises

### 8.2.7.1 A. Check your understanding

#### Exercise A1.

- Define eigenvalue, eigenvector and eigenspace of an operator  $T \in \mathcal{L}(V)$ .
- True or false:  $\mathbf{0}$  is an eigenvector of every operator. Justify your answer.
- True or false: if 0 is an eigenvalue of  $\mathbf{A} \in M_n(F)$ , then  $\mathbf{A}$  is not invertible. Justify your answer.
- State four conditions equivalent to “ $\lambda$  is an eigenvalue of  $T$ ” for an operator on a finite-dimensional space  $V \neq \{\mathbf{0}\}$ .
- True or false: every operator on  $\mathbb{R}^2$  has an eigenvalue. Justify your answer.
- True or false: the eigenvalues of  $\mathbf{A}$  are the diagonal entries of a row echelon form of  $\mathbf{A}$ . Justify your answer.

*Solution.*

- $\lambda \in F$  is an eigenvalue of  $T$  if  $T\mathbf{v} = \lambda\mathbf{v}$  for some **non-zero**  $\mathbf{v} \in V$ ; each such non-zero  $\mathbf{v}$  is an eigenvector for  $\lambda$  (Definition 8.2.1). The eigenspace is  $E_\lambda(T) = \ker(T - \lambda \text{id}_V)$  (Definition 8.2.2).
- False. Eigenvectors are non-zero by definition. (The equation  $T\mathbf{0} = \lambda\mathbf{0}$  holds for every  $\lambda$ , which is why  $\mathbf{0}$  is excluded.)
- True, by Theorem 8.2.6.
- By Theorem 8.2.1:  $T - \lambda \text{id}_V$  is not injective;  $T - \lambda \text{id}_V$  is not invertible;  $\det(\lambda \text{id}_V - T) = 0$ ;  $p_T(\lambda) = 0$ .
- False. The rotation  $R(x, y) = (-y, x)$  has  $p_R = x^2 + 1$ , with no real root, so it has no real eigenvalue.
- False. Row operations change eigenvalues:  $\begin{pmatrix} 1 & 2 \\ 3 & 2 \end{pmatrix}$  has eigenvalues 4,  $-1$ , but the echelon form  $\begin{pmatrix} 1 & 2 \\ 0 & -4 \end{pmatrix}$  has diagonal entries 1,  $-4$ .

### 8.2.7.2 B. Practice

#### Exercise B1: Computing eigenvalues and eigenspaces.

Find the eigenvalues and a basis of each eigenspace of the following real matrices.

- $\mathbf{A} = \begin{pmatrix} 5 & 4 \\ -2 & -1 \end{pmatrix}$ .
- $\mathbf{B} = \begin{pmatrix} 2 & 0 & 1 \\ -1 & 3 & 1 \\ 2 & -2 & 1 \end{pmatrix}$ .



Hence write down a basis of  $\mathbb{R}^2$  consisting of eigenvectors of  $\mathbf{A}$ , and a basis of  $\mathbb{R}^3$  consisting of eigenvectors of  $\mathbf{B}$ .

*Solution.*

- (a)  $\operatorname{tr} \mathbf{A} = 4$  and  $\det \mathbf{A} = -5 + 8 = 3$ , so  $p_{\mathbf{A}}(x) = x^2 - 4x + 3 = (x - 1)(x - 3)$ , and  $\operatorname{spec}(\mathbf{A}) = \{1, 3\}$  by [Theorem 8.2.1](#). For  $\lambda = 1$ ,  $\mathbf{A} - \mathbf{I} = \begin{pmatrix} 4 & 4 \\ -2 & -2 \end{pmatrix}$  reduces to  $x + y = 0$ , so  $E_1(\mathbf{A}) = \operatorname{span}((1, -1))$ . For  $\lambda = 3$ ,  $\mathbf{A} - 3\mathbf{I} = \begin{pmatrix} 2 & 4 \\ -2 & -4 \end{pmatrix}$  reduces to  $x + 2y = 0$ , so  $E_3(\mathbf{A}) = \operatorname{span}((2, -1))$ . Check:  $\mathbf{A}(1, -1) = (1, -1)$  and  $\mathbf{A}(2, -1) = (6, -3) = 3(2, -1)$ .

- (b) Expanding  $\det(x\mathbf{I} - \mathbf{B})$  along the first row,

$$\begin{aligned} p_{\mathbf{B}}(x) &= \det \begin{pmatrix} x-2 & 0 & -1 \\ 1 & x-3 & -1 \\ -2 & 2 & x-1 \end{pmatrix} \\ &= (x-2)[(x-3)(x-1) + 2] + (-1)[1 \cdot 2 - (x-3)(-2)]. \end{aligned}$$

The first bracket is  $x^2 - 4x + 5$  and the second is  $2x - 4 = 2(x - 2)$ , so

$$\begin{aligned} p_{\mathbf{B}}(x) &= (x-2)(x^2 - 4x + 5) - 2(x-2) \\ &= (x-2)(x^2 - 4x + 3) = (x-1)(x-2)(x-3). \end{aligned}$$

So  $\operatorname{spec}(\mathbf{B}) = \{1, 2, 3\}$ . Solving  $(\mathbf{B} - \lambda\mathbf{I})\mathbf{x} = \mathbf{0}$ :

- $\lambda = 1$ :  $\mathbf{B} - \mathbf{I} = \begin{pmatrix} 1 & 0 & 1 \\ -1 & 2 & 1 \\ 2 & -2 & 0 \end{pmatrix}$ . The first row gives  $z = -x$ , the third gives  $y = x$ , and the second is then  $-x + 2x - x = 0$ . So  $E_1(\mathbf{B}) = \operatorname{span}((1, 1, -1))$ .
- $\lambda = 2$ :  $\mathbf{B} - 2\mathbf{I} = \begin{pmatrix} 0 & 0 & 1 \\ -1 & 1 & 1 \\ 2 & -2 & -1 \end{pmatrix}$ . The first row gives  $z = 0$ , and then the second gives  $y = x$ . So  $E_2(\mathbf{B}) = \operatorname{span}((1, 1, 0))$ .
- $\lambda = 3$ :  $\mathbf{B} - 3\mathbf{I} = \begin{pmatrix} -1 & 0 & 1 \\ -1 & 0 & 1 \\ 2 & -2 & -2 \end{pmatrix}$ . The first row gives  $z = x$ , and the third gives  $2x - 2y - 2x = 0$ , so  $y = 0$ . So  $E_3(\mathbf{B}) = \operatorname{span}((1, 0, 1))$ .

Check:  $\mathbf{B}(1, 1, -1) = (1, 1, -1)$ ,  $\mathbf{B}(1, 1, 0) = (2, 2, 0)$ ,  $\mathbf{B}(1, 0, 1) = (3, 0, 3)$ .

Hence: the eigenvalues of  $\mathbf{A}$  are distinct, so by [Theorem 8.2.2](#) the eigenvectors  $((1, -1), (2, -1))$  are independent, and two independent vectors in  $\mathbb{R}^2$  form a basis ([Theorem 1.6.4](#)). In the same way  $((1, 1, -1), (1, 1, 0), (1, 0, 1))$  is a basis of  $\mathbb{R}^3$  consisting of eigenvectors of  $\mathbf{B}$ .

### Exercise B2: The transpose map.

Let  $F$  be a field of characteristic not 2, and let  $\tau: M_2(F) \rightarrow M_2(F)$ ,  $\tau(\mathbf{X}) = \mathbf{X}^\top$ .

- (a) Show that every eigenvalue  $\lambda$  of  $\tau$  satisfies  $\lambda^2 = 1$ .

- (b) Find  $\text{spec}(\tau)$ , and describe  $E_1(\tau)$  and  $E_{-1}(\tau)$  with a basis of each.

*Solution.*

- (a) Let  $\tau(\mathbf{X}) = \lambda\mathbf{X}$  with  $\mathbf{X} \neq 0$ . Since  $(\mathbf{X}^\top)^\top = \mathbf{X}$  (Theorem 0.9.5),  $\mathbf{X} = \tau(\tau(\mathbf{X})) = \tau(\lambda\mathbf{X}) = \lambda\tau(\mathbf{X}) = \lambda^2\mathbf{X}$ . So  $(\lambda^2 - 1)\mathbf{X} = 0$  with  $\mathbf{X} \neq 0$ , and  $\lambda^2 = 1$  by Theorem 1.1.9.
- (b) From  $\lambda^2 - 1 = (\lambda - 1)(\lambda + 1) = 0$ , the only candidates are 1 and  $-1$ , which are different because the characteristic is not 2.  $E_1(\tau) = \{\mathbf{X} : \mathbf{X}^\top = \mathbf{X}\}$  is the space of symmetric matrices. Writing  $\mathbf{X} = \begin{pmatrix} a & b \\ c & d \end{pmatrix}$ , the condition is  $b = c$ , so  $E_1(\tau) = \text{span}(\mathbf{E}_{11}, \mathbf{E}_{22}, \mathbf{E}_{12} + \mathbf{E}_{21})$ , and these three matrices are independent (compare entries).  $E_{-1}(\tau) = \{\mathbf{X} : \mathbf{X}^\top = -\mathbf{X}\}$  is the space of skew-symmetric matrices. The condition  $\mathbf{X}^\top = -\mathbf{X}$  says  $a = -a$ ,  $d = -d$  and  $c = -b$ ; since  $2 \neq 0$ ,  $a = d = 0$ , so  $E_{-1}(\tau) = \text{span}(\mathbf{E}_{12} - \mathbf{E}_{21})$ . Both eigenspaces are non-zero, so  $\text{spec}(\tau) = \{1, -1\}$ . This agrees with Theorem 3.10.5:  $\tau^2 = \text{id}$ , and  $M_2(F) = E_1(\tau) \oplus E_{-1}(\tau)$ , with dimensions  $3 + 1 = 4$ .

### Exercise B3: Eigenvalues of powers and inverses.

Let  $T \in \mathcal{L}(V)$  and let  $\mathbf{v}$  be an eigenvector of  $T$  for  $\lambda$ .

- (a) Prove that  $\mathbf{v}$  is an eigenvector of  $T^2$  for  $\lambda^2$ , and more generally of  $p(T)$  for  $p(\lambda)$ , for every  $p \in F[x]$ .
- (b) Suppose  $T$  is invertible. Prove that  $\lambda \neq 0$  and that  $\mathbf{v}$  is an eigenvector of  $T^{-1}$  for  $\lambda^{-1}$ .
- (c) Show that the converse of the first statement in (a) fails: give  $T \in \mathcal{L}(\mathbb{R}^2)$  such that  $-1$  is an eigenvalue of  $T^2$  but  $T$  has no eigenvalue.

*Solution.*

- (a)  $T^2\mathbf{v} = T(T\mathbf{v}) = T(\lambda\mathbf{v}) = \lambda T\mathbf{v} = \lambda^2\mathbf{v}$ , and  $\mathbf{v} \neq \mathbf{0}$ . By induction on  $k$ ,  $T^k\mathbf{v} = \lambda^k\mathbf{v}$ : if  $T^k\mathbf{v} = \lambda^k\mathbf{v}$ , then  $T^{k+1}\mathbf{v} = T(\lambda^k\mathbf{v}) = \lambda^{k+1}\mathbf{v}$ . For  $p = \sum_k a_k x^k$ , Definition 3.4.2 gives  $p(T)\mathbf{v} = \sum_k a_k T^k\mathbf{v} = \sum_k a_k \lambda^k\mathbf{v} = p(\lambda)\mathbf{v}$ .
- (b) An invertible operator is injective, so  $\ker T = \{\mathbf{0}\}$  and  $T\mathbf{v} \neq \mathbf{0}$ ; as  $T\mathbf{v} = \lambda\mathbf{v}$ ,  $\lambda \neq 0$  by Theorem 1.1.9. Applying  $T^{-1}$  to  $T\mathbf{v} = \lambda\mathbf{v}$  gives  $\mathbf{v} = \lambda T^{-1}\mathbf{v}$ , so  $T^{-1}\mathbf{v} = \lambda^{-1}\mathbf{v}$ .
- (c) Take the rotation  $R(x, y) = (-y, x)$ . It has no real eigenvalue (Example (A Rotation Has No Invariant Lines)). But  $R^2(x, y) = R(-y, x) = (-x, -y)$ , so  $R^2 = -\text{id}$ , and every non-zero vector is an eigenvector of  $R^2$  for  $-1$ . Part (a) would need a real  $\lambda$  with  $\lambda^2 = -1$ , and there is none.

### 8.2.7.3 C. Going deeper

#### Exercise C1: Every vector an eigenvector.

Let  $V$  be a vector space over  $F$  and  $T \in \mathcal{L}(V)$ . Suppose every non-zero vector of  $V$  is an eigenvector of  $T$ . Prove that  $T = c \text{id}_V$  for some  $c \in F$ .

*Hint: compare the eigenvalues of  $\mathbf{v}$ ,  $\mathbf{w}$  and  $\mathbf{v} + \mathbf{w}$ .*

*Solution.* If  $V = \{\mathbf{0}\}$ , then  $T = 0 = 1 \cdot \text{id}_V$ . Otherwise, for each  $\mathbf{v} \neq \mathbf{0}$  let  $\lambda_{\mathbf{v}}$  be its eigenvalue, which is unique by the well-definedness remark after [Definition 8.2.1](#). We show that  $\lambda_{\mathbf{v}} = \lambda_{\mathbf{w}}$  for all non-zero  $\mathbf{v}, \mathbf{w}$ .

*Case 1:*  $(\mathbf{v}, \mathbf{w})$  is linearly dependent. Then  $\mathbf{w} = a\mathbf{v}$  for some  $a \neq 0$ , and  $T\mathbf{w} = aT\mathbf{v} = \lambda_{\mathbf{v}}(a\mathbf{v}) = \lambda_{\mathbf{v}}\mathbf{w}$ . By uniqueness of the eigenvalue of  $\mathbf{w}$ ,  $\lambda_{\mathbf{w}} = \lambda_{\mathbf{v}}$ .

*Case 2:*  $(\mathbf{v}, \mathbf{w})$  is linearly independent. Then  $\mathbf{v} + \mathbf{w} \neq \mathbf{0}$ , and

$$\lambda_{\mathbf{v}+\mathbf{w}}\mathbf{v} + \lambda_{\mathbf{v}+\mathbf{w}}\mathbf{w} = T(\mathbf{v} + \mathbf{w}) = T\mathbf{v} + T\mathbf{w} = \lambda_{\mathbf{v}}\mathbf{v} + \lambda_{\mathbf{w}}\mathbf{w}.$$

So  $(\lambda_{\mathbf{v}+\mathbf{w}} - \lambda_{\mathbf{v}})\mathbf{v} + (\lambda_{\mathbf{v}+\mathbf{w}} - \lambda_{\mathbf{w}})\mathbf{w} = \mathbf{0}$ , and by independence  $\lambda_{\mathbf{v}} = \lambda_{\mathbf{v}+\mathbf{w}} = \lambda_{\mathbf{w}}$ .

Fix  $\mathbf{v}_0 \neq \mathbf{0}$  and put  $c = \lambda_{\mathbf{v}_0}$ . Then  $T\mathbf{v} = c\mathbf{v}$  for every  $\mathbf{v} \neq \mathbf{0}$ , and also for  $\mathbf{v} = \mathbf{0}$ . Hence  $T = c \text{id}_V$ .

### Exercise C2: Nilpotent operators.

An operator  $T \in \mathcal{L}(V)$  is **nilpotent** if  $T^k = 0$  for some integer  $k \geq 1$ .

- (a) Let  $V \neq \{\mathbf{0}\}$  and  $T$  nilpotent. Prove that  $\text{spec}(T) = \{0\}$ . (No finite dimension is needed.)
- (b) Let  $\mathbf{A} \in M_n(\mathbb{C})$ ,  $n \geq 1$ , be nilpotent. Deduce that  $p_{\mathbf{A}}(x) = x^n$ .
- (c) Let  $\mathbf{A} \in M_n(F)$ ,  $n \geq 1$ , be nilpotent. Prove that  $\mathbf{I} + \mathbf{A}$  is invertible.

*Solution.*

- (a) Let  $T\mathbf{v} = \lambda\mathbf{v}$  with  $\mathbf{v} \neq \mathbf{0}$ . By [Exercise \(B3: Eigenvalues of powers and inverses\)](#) (a),  $\mathbf{0} = T^k\mathbf{v} = \lambda^k\mathbf{v}$ , so  $\lambda^k = 0$  by [Theorem 1.1.9](#), and hence  $\lambda = 0$  (a product of non-zero elements of a field is non-zero, [Theorem 0.7.1](#)). So  $\text{spec}(T) \subseteq \{0\}$ . Conversely,  $T^k = 0$  is not injective on  $V \neq \{\mathbf{0}\}$ ; if  $T$  were injective, so would be the composite  $T^k$  ([Theorem 0.4.3](#)). So  $\ker T \neq \{\mathbf{0}\}$ , and  $0 \in \text{spec}(T)$ .
- (b) By [Corollary 5.5.5](#) and [Theorem 6.10.2](#),  $p_{\mathbf{A}} = (x - z_1) \cdots (x - z_n)$  for some  $z_i \in \mathbb{C}$ . Each  $z_i$  is a root of  $p_{\mathbf{A}}$ , hence an eigenvalue of  $\mathbf{A}$  by [Theorem 8.2.1](#), hence  $z_i = 0$  by (a). So  $p_{\mathbf{A}} = x^n$ .
- (c) By [Theorem 8.2.6](#), it suffices to show  $0 \notin \text{spec}(\mathbf{I} + \mathbf{A})$ . Suppose  $(\mathbf{I} + \mathbf{A})\mathbf{v} = \mathbf{0}$  with  $\mathbf{v} \neq \mathbf{0}$ . Then  $\mathbf{A}\mathbf{v} = -\mathbf{v}$ , so  $-1 \in \text{spec}(\mathbf{A})$ . By (a),  $\text{spec}(\mathbf{A}) = \{0\}$ , but  $-1 \neq 0$  in every field. This contradiction shows that  $\mathbf{I} + \mathbf{A}$  is invertible.

### Exercise C3: Real matrices of odd size.

- (a) Prove that every  $\mathbf{A} \in M_n(\mathbb{R})$  with  $n$  odd has a real eigenvalue.
- (b) Deduce that every operator on a real vector space of dimension 3 has an invariant line.
- (c) Show that (a) fails for  $n = 4$ : find  $\mathbf{A} \in M_4(\mathbb{R})$  with no real eigenvalue.

*Solution.*

- (a) By [Theorem 6.10.2](#),  $p_{\mathbf{A}} \in \mathbb{R}[x]$  is monic of degree  $n$ , which is odd. By [Corollary 5.5.9](#),  $p_{\mathbf{A}}(\lambda) = 0$  for some  $\lambda \in \mathbb{R}$ , and  $\lambda$  is an eigenvalue of  $\mathbf{A}$  by [Theorem 8.2.1](#).

- (b) Let  $\dim V = 3$  over  $\mathbb{R}$  and  $T \in \mathcal{L}(V)$ . Choose a basis  $\mathcal{B}$ . By (a), the real matrix  $[T]_{\mathcal{B}}$  has a real eigenvalue  $\lambda$ , which is a root of  $p_{[T]_{\mathcal{B}}} = p_T$  (Theorem 8.2.1, Definition 6.10.2). So  $\lambda$  is an eigenvalue of  $T$ , again by Theorem 8.2.1, and  $T$  has an eigenvector  $\mathbf{v}$  for it. By Proposition 8.1.2,  $\text{span}(\mathbf{v})$  is invariant.
- (c) Let  $\mathbf{R} = \begin{pmatrix} 0 & -1 \\ 1 & 0 \end{pmatrix}$  and  $\mathbf{A} = \mathbf{R} \oplus \mathbf{R} \in M_4(\mathbb{R})$ . Then  $x\mathbf{I}_4 - \mathbf{A} = (x\mathbf{I}_2 - \mathbf{R}) \oplus (x\mathbf{I}_2 - \mathbf{R})$  is block diagonal, so by the block triangular determinant over  $\mathbb{R}[x]$  (the remark after Theorem 6.8.4),  $p_{\mathbf{A}} = p_{\mathbf{R}}^2 = (x^2 + 1)^2$ . For real  $\lambda$ ,  $(\lambda^2 + 1)^2 \geq 1 > 0$ , so  $p_{\mathbf{A}}$  has no real root, and  $\mathbf{A}$  has no real eigenvalue. The argument of (a) needs odd degree.

## 8.3 Algebraic and Geometric Multiplicity

An eigenvalue can be counted in two ways. It is a root of the characteristic polynomial, and a root has a multiplicity. It also has an eigenspace, and a subspace has a dimension. The two numbers need not agree, and the gap between them is exactly what can stop an operator from having a diagonal matrix. This section defines both counts, proves that the dimension never exceeds the root multiplicity, and shows that the root multiplicities package the trace and the determinant. The next section turns the comparison into a test for diagonalizability.

### 8.3.1 Two ways to count an eigenvalue

Compare two matrices over any field  $F$ :

$$2\mathbf{I}_2 = \begin{pmatrix} 2 & 0 \\ 0 & 2 \end{pmatrix}, \quad \mathbf{J} = \begin{pmatrix} 2 & 1 \\ 0 & 2 \end{pmatrix}.$$

Both are triangular with diagonal 2, 2, so both have characteristic polynomial  $(x - 2)^2$  and spectrum  $\{2\}$ . By the only measure we have so far, the eigenvalue data agree. Yet the eigenvectors are very different. For  $2\mathbf{I}_2$ , every non-zero vector is an eigenvector and  $E_2 = F^2$ . For  $\mathbf{J}$ , the matrix  $\mathbf{J} - 2\mathbf{I} = \begin{pmatrix} 0 & 1 \\ 0 & 0 \end{pmatrix}$  kills exactly the multiples of  $\mathbf{e}_1$ , so  $E_2(\mathbf{J}) = \text{span}(\mathbf{e}_1)$  is a line. The root 2 “counts twice” in both characteristic polynomials, but it supplies two independent eigenvectors for  $2\mathbf{I}_2$  and only one for  $\mathbf{J}$ . Both numbers deserve names.

*The algebraic multiplicity counts an eigenvalue as a root; the geometric multiplicity counts the independent eigenvectors it supplies.*

#### Definition 8.3.1 — Algebraic Multiplicity

Let  $V$  be a finite-dimensional vector space over  $F$  with  $\dim V \geq 1$ , let  $T \in \mathcal{L}(V)$  and  $\lambda \in F$ . The **algebraic multiplicity** of  $\lambda$  for  $T$  is

$$a_T(\lambda) := \text{mult}_{\lambda}(p_T),$$

the multiplicity of  $\lambda$  as a root of the characteristic polynomial (Definition 5.4.1). For  $\mathbf{A} \in M_n(F)$ ,  $a_{\mathbf{A}}(\lambda) := \text{mult}_{\lambda}(p_{\mathbf{A}})$ .

**Definition 8.3.2 — Geometric Multiplicity**

Let  $V$  be a finite-dimensional vector space over  $F$ ,  $T \in \mathcal{L}(V)$  and  $\lambda \in F$ . The **geometric multiplicity** of  $\lambda$  for  $T$  is

$$g_T(\lambda) := \dim E_\lambda(T) = \dim \ker(T - \lambda \operatorname{id}_V).$$

For  $\mathbf{A} \in M_n(F)$ ,  $g_{\mathbf{A}}(\lambda) := \dim E_\lambda(\mathbf{A}) = n - \operatorname{rank}(\mathbf{A} - \lambda \mathbf{I}_n)$ .

When  $T$  or  $\mathbf{A}$  is clear from context we write simply  $a(\lambda)$  and  $g(\lambda)$ .

In words:  $a_T(\lambda)$  is the **largest**  $m$  such that  $(x - \lambda)^m$  divides  $p_T$ . It is a statement about a polynomial and can be computed by factoring.  $g_T(\lambda)$  is the dimension of the subspace of solutions of  $T\mathbf{v} = \lambda\mathbf{v}$ , including  $\mathbf{0}$ . It is the largest number of **linearly independent** eigenvectors for  $\lambda$ , not the number of eigenvectors, which is infinite over an infinite field. For matrices the second formula comes from [Theorem 2.5.7](#) applied to  $\mathbf{A} - \lambda\mathbf{I}_n$ , and it is the one we compute with: find the rank of  $\mathbf{A} - \lambda\mathbf{I}$  by elimination and subtract from  $n$ .

**Well-definedness.** Both numbers are defined for every  $\lambda \in F$ , eigenvalue or not. By [Theorem 8.2.1](#),  $\lambda$  is an eigenvalue if and only if  $p_T(\lambda) = 0$ , if and only if  $a_T(\lambda) \geq 1$ ; and by [Definition 8.2.1](#), if and only if  $E_\lambda(T) \neq \{\mathbf{0}\}$ , that is,  $g_T(\lambda) \geq 1$ . So for a scalar that is not an eigenvalue both multiplicities are 0, and for an eigenvalue both are at least 1. Neither depends on a basis:  $p_T$  does not ([Definition 6.10.2](#)), and  $E_\lambda(T)$  is defined without one. If  $\mathcal{B}$  is a basis of  $V$ , then  $a_T(\lambda) = a_{[T]_{\mathcal{B}}}(\lambda)$  because  $p_T = p_{[T]_{\mathcal{B}}}$ , and  $g_T(\lambda) = g_{[T]_{\mathcal{B}}}(\lambda)$  because the coordinate map is an isomorphism from  $\ker(T - \lambda \operatorname{id}_V)$  onto  $\operatorname{nul}([T]_{\mathcal{B}} - \lambda \mathbf{I})$  ([Theorem 3.6.5](#) (a) and [Corollary 3.6.3](#)).

**Examples.**

- **The hook.** For  $2\mathbf{I}_2$ :  $a(2) = 2$  and  $g(2) = 2 - \operatorname{rank} 0 = 2$ . For  $\mathbf{J}$ :  $a(2) = 2$  and  $g(2) = 2 - \operatorname{rank} \begin{pmatrix} 0 & 1 \\ 0 & 0 \end{pmatrix} = 1$ .
- **A  $1 \times 1$  matrix.** For  $\mathbf{A} = (c)$ ,  $p_{\mathbf{A}} = x - c$ , so  $a(c) = 1$ , and  $\mathbf{A} - c\mathbf{I} = (0)$  has rank 0, so  $g(c) = 1$ . This degenerate case already shows the pattern of the theorem below: the two counts can only differ when there is room, a repeated root.
- **Differentiation.** On  $V = \mathbb{R}[x]_{\leq n}$ , the matrix of  $D$  in the basis  $(1, x, \dots, x^n)$  is strictly upper triangular, so  $p_D = x^{n+1}$  and  $a_D(0) = n + 1$ . But  $E_0(D)$  is the line of constant polynomials (the examples after [Definition 8.2.2](#)), so  $g_D(0) = 1$ . The gap can be as large as  $n$ .

- **A matrix from Chapter 6.** For  $\mathbf{A} = \begin{pmatrix} 2 & 0 & 1 \\ 1 & 3 & 1 \\ 1 & 0 & 2 \end{pmatrix}$ , [Example \(A  \$3 \times 3\$  Characteristic](#)

[Polynomial](#)) found  $p_{\mathbf{A}} = (x - 1)(x - 3)^2$ , so  $a(1) = 1$  and  $a(3) = 2$ . The matrix  $\mathbf{A} - 3\mathbf{I} = \begin{pmatrix} -1 & 0 & 1 \\ 1 & 0 & 1 \\ 1 & 0 & -1 \end{pmatrix}$  has a zero column, and its first two rows are independent while the third

is minus the first, so its rank is 2 and  $g(3) = 3 - 2 = 1$ . Similarly  $\mathbf{A} - \mathbf{I} = \begin{pmatrix} 1 & 0 & 1 \\ 1 & 2 & 1 \\ 1 & 0 & 1 \end{pmatrix}$  has third row equal to the first and first two rows independent, so its rank is 2 and  $g(1) = 1$ .

**Minimal change.** Take the triangular matrices

$$\mathbf{B} = \begin{pmatrix} 2 & 0 & 1 \\ 0 & 2 & 0 \\ 0 & 0 & 1 \end{pmatrix}, \quad \mathbf{B}' = \begin{pmatrix} 2 & 1 & 1 \\ 0 & 2 & 0 \\ 0 & 0 & 1 \end{pmatrix},$$

which differ in one entry. Both have  $p = (x - 2)^2(x - 1)$ , so  $a(2) = 2$  for both: the algebraic multiplicity does not notice the change. But  $\mathbf{B} - 2\mathbf{I}$  has non-zero rows  $(0, 0, 1)$  and  $(0, 0, -1)$ , rank 1, so  $g_{\mathbf{B}}(2) = 2$ ; while  $\mathbf{B}' - 2\mathbf{I}$  has rows  $(0, 1, 1)$  and  $(0, 0, -1)$ , rank 2, so  $g_{\mathbf{B}'}(2) = 1$ . One off-diagonal entry, invisible to  $p$ , removed an eigenvector.

**Why these definitions.** The algebraic multiplicity is the natural count for a polynomial: with it, the roots of  $p_T$  add up to  $\deg p_T = \dim V$  whenever  $p_T$  splits (Theorem 5.4.3), and the trace and determinant come out right (Theorem 8.3.3 below). The geometric multiplicity is the natural count for eigenvectors: it is the number of basis vectors we can take from  $E_{\lambda}(T)$  when we try to build a basis of eigenvectors. The names record where each number lives, in algebra (a polynomial) or in geometry (a subspace).

*Warning.* The geometric multiplicity is not read off from  $p_T$ . Two operators with the same characteristic polynomial can have different geometric multiplicities:  $2\mathbf{I}_2$  and  $\mathbf{J}$  above, or  $\mathbf{B}$  and  $\mathbf{B}'$ . To find  $g(\lambda)$  you must compute  $\text{rank}(\mathbf{A} - \lambda\mathbf{I})$ . Also,  $g(\lambda)$  is **not** “how many eigenvectors”  $\lambda$  has: over  $\mathbb{R}$ ,  $E_{\lambda} \neq \{\mathbf{0}\}$  always contains infinitely many eigenvectors.

*Quick check.* Find  $a(0)$  and  $g(0)$  for  $\mathbf{N} = \begin{pmatrix} 0 & 1 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{pmatrix} \in M_3(F)$ .

*Solution.*  $\mathbf{N}$  is triangular with zero diagonal, so  $p_{\mathbf{N}} = x^3$  and  $a(0) = 3$ . The rank of  $\mathbf{N} - 0\mathbf{I} = \mathbf{N}$  is 1, so  $g(0) = 3 - 1 = 2$ , with  $E_0(\mathbf{N}) = \text{span}(\mathbf{e}_1, \mathbf{e}_3)$ .

## 8.3.2 The geometric multiplicity never exceeds the algebraic

In every example so far,  $g(\lambda) \leq a(\lambda)$ . That is always the case.

### Theorem 8.3.1 — Geometric Multiplicity Is at Most Algebraic Multiplicity

Let  $V$  be a finite-dimensional vector space over  $F$  with  $\dim V = n \geq 1$ , let  $T \in \mathcal{L}(V)$ , and let  $\lambda \in F$  be an eigenvalue of  $T$ . Then

$$1 \leq g_T(\lambda) \leq a_T(\lambda).$$

*Idea.* Start from the smallest subspace in sight,  $E_{\lambda}(T)$ . Take a basis of it and extend it to a basis of  $V$ . In that basis, the first  $g$  columns of the matrix are  $\lambda\mathbf{e}_1, \dots, \lambda\mathbf{e}_g$ , so the matrix is block upper triangular with top-left block  $\lambda\mathbf{I}_g$ . The block triangular determinant then factors  $(x - \lambda)^g$  out of  $p_T$ , and a root with a factor  $(x - \lambda)^g$  has multiplicity at least  $g$ . The inequality can be strict because the other block may contribute further

factors  $x - \lambda$ , as it does for  $\mathbf{J}$ .

*Proof.* Since  $\lambda$  is an eigenvalue,  $E_\lambda(T) \neq \{\mathbf{0}\}$ , so  $g = g_T(\lambda) \geq 1$ . Let  $(\mathbf{v}_1, \dots, \mathbf{v}_g)$  be a basis of  $E_\lambda(T)$ . By Theorem 1.6.5, it extends to a basis  $\mathcal{B} = (\mathbf{v}_1, \dots, \mathbf{v}_g, \mathbf{w}_1, \dots, \mathbf{w}_{n-g})$  of  $V$ .

*Case  $g = n$ .* Then  $T\mathbf{v}_i = \lambda\mathbf{v}_i$  for every basis vector, so  $[T]_{\mathcal{B}} = \lambda\mathbf{I}_n$  by Definition 3.6.1, and  $p_T = (x - \lambda)^n$  by the triangular case of Definition 6.10.1. Hence  $a_T(\lambda) = n = g$ .

*Case  $g < n$ .* For  $i \leq g$ ,  $T\mathbf{v}_i = \lambda\mathbf{v}_i$ , so column  $i$  of  $[T]_{\mathcal{B}}$  is  $\lambda\mathbf{e}_i$ . Hence

$$[T]_{\mathcal{B}} = \begin{pmatrix} \lambda\mathbf{I}_g & \mathbf{B} \\ 0 & \mathbf{D} \end{pmatrix}, \quad x\mathbf{I}_n - [T]_{\mathcal{B}} = \begin{pmatrix} (x - \lambda)\mathbf{I}_g & -\mathbf{B} \\ 0 & x\mathbf{I}_{n-g} - \mathbf{D} \end{pmatrix},$$

for some  $\mathbf{B} \in M_{g \times (n-g)}(F)$  and  $\mathbf{D} \in M_{n-g}(F)$ . The second matrix is block upper triangular over  $F[x]$  with square diagonal blocks. By Theorem 6.8.4, which holds over the commutative ring  $F[x]$  (the remark after it), and Definition 6.10.2,

$$p_T = \det((x - \lambda)\mathbf{I}_g) \det(x\mathbf{I}_{n-g} - \mathbf{D}) = (x - \lambda)^g p_{\mathbf{D}},$$

where  $\det((x - \lambda)\mathbf{I}_g) = (x - \lambda)^g$  because the matrix is diagonal. So  $(x - \lambda)^g \mid p_T$ . By Definition 5.4.1,  $a_T(\lambda)$ , the **largest**  $m$  with  $(x - \lambda)^m \mid p_T$ , satisfies  $a_T(\lambda) \geq g$ . This proves the theorem. ■

The inequality is strict exactly when  $\lambda$  is also a root of  $p_{\mathbf{D}}$ , the characteristic polynomial of the “rest” of the operator. For  $\mathbf{J}$ , with  $\mathcal{B} = (\mathbf{e}_1, \mathbf{e}_2)$ , the block  $\mathbf{D}$  is  $(2)$ , and  $p_{\mathbf{D}} = x - 2$  contributes the second factor.

A useful consequence: if  $a(\lambda) = 1$ , then  $g(\lambda) = 1$ . A **simple** root of  $p_T$  always has a one-dimensional eigenspace, and so the only eigenvalues whose eigenspaces need checking are the repeated ones.

Both multiplicities are properties of the operator, not of the basis, so they cannot distinguish similar matrices. That makes them tools for proving that two matrices are **not** similar.

### Proposition 8.3.2 — Similar Matrices Have the Same Multiplicities

Let  $\mathbf{A}, \mathbf{B} \in M_n(F)$  with  $\mathbf{A} \sim \mathbf{B}$ , and let  $\lambda \in F$ . Then  $a_{\mathbf{A}}(\lambda) = a_{\mathbf{B}}(\lambda)$  and  $g_{\mathbf{A}}(\lambda) = g_{\mathbf{B}}(\lambda)$ .

*Proof.* Let  $\mathbf{B} = \mathbf{P}^{-1}\mathbf{A}\mathbf{P}$ . By Theorem 6.10.3,  $p_{\mathbf{A}} = p_{\mathbf{B}}$ , so the algebraic multiplicities agree. Also  $\mathbf{B} - \lambda\mathbf{I} = \mathbf{P}^{-1}\mathbf{A}\mathbf{P} - \lambda\mathbf{P}^{-1}\mathbf{P} = \mathbf{P}^{-1}(\mathbf{A} - \lambda\mathbf{I})\mathbf{P}$ , so  $\mathbf{B} - \lambda\mathbf{I} \sim \mathbf{A} - \lambda\mathbf{I}$ , and these have the same rank by Proposition 3.7.6 (a). Hence  $g_{\mathbf{B}}(\lambda) = n - \text{rank}(\mathbf{B} - \lambda\mathbf{I}) = n - \text{rank}(\mathbf{A} - \lambda\mathbf{I}) = g_{\mathbf{A}}(\lambda)$ . ■

For example,  $2\mathbf{I}_2$  and  $\mathbf{J}$  have the same characteristic polynomial but  $g(2) = 2$  versus  $g(2) = 1$ , so they are not similar. The same test shows  $\mathbf{B} \not\sim \mathbf{B}'$  in the minimal change above.

**Warning.** A repeated eigenvalue does not force a shortage of eigenvectors, and equal eigenvalues do not force similarity. The identity  $\mathbf{I}_2$  has the repeated eigenvalue 1, yet  $g(1) = a(1) = 2$  and its matrix is as simple as possible. In the other direction,  $\mathbf{I}_2$  and  $\begin{pmatrix} 1 & 1 \\ 0 & 1 \end{pmatrix}$  have the same eigenvalues with the same algebraic multiplicities, and they are not similar. Even agreement of **both** multiplicities for every eigenvalue does not force similarity; [Exercise \(C2: Multiplicities do not determine similarity\)](#) gives two  $4 \times 4$  matrices showing this.

### 8.3.3 Multiplicities, trace and determinant

By [Theorem 5.4.3](#), the algebraic multiplicities of the eigenvalues of  $T$  add up to at most  $\deg p_T = \dim V$ , with equality exactly when  $p_T$  splits over  $F$ . Over  $\mathbb{C}$  this always happens ([Corollary 5.5.5](#)). When it does,  $p_T$  is determined by the eigenvalues and their algebraic multiplicities,

$$p_T = (x - \lambda_1)^{a(\lambda_1)} \cdots (x - \lambda_k)^{a(\lambda_k)},$$

and two of its coefficients are familiar: by [Theorem 6.10.2](#), the coefficient of  $x^{n-1}$  is  $-\operatorname{tr} T$  and the constant term is  $(-1)^n \det T$ . Comparing the two descriptions expresses trace and determinant through eigenvalues.

#### Theorem 8.3.3 — Trace and Determinant from Eigenvalues

Let  $V$  be a finite-dimensional vector space over  $F$  with  $\dim V = n \geq 1$ , and let  $T \in \mathcal{L}(V)$ . Suppose  $p_T$  splits over  $F$ , and let  $\lambda_1, \dots, \lambda_k$  be the distinct eigenvalues of  $T$ . Then

$$a(\lambda_1) + \cdots + a(\lambda_k) = n, \quad \operatorname{tr} T = \sum_{i=1}^k a(\lambda_i) \lambda_i, \quad \det T = \prod_{i=1}^k \lambda_i^{a(\lambda_i)}.$$

In words: listing each eigenvalue as often as its algebraic multiplicity, the trace is the sum of the list and the determinant is its product. In particular this holds for every operator on a non-zero finite-dimensional complex space, and for every  $\mathbf{A} \in M_n(\mathbb{C})$ .

*Idea.* Rather than expanding a product of linear factors by hand, notice that the product is itself a characteristic polynomial: that of the diagonal matrix with the eigenvalues on its diagonal. Chapter 6 already knows the  $x^{n-1}$  coefficient and the constant term of any characteristic polynomial, so we read them off twice and compare.

*Proof.* Since  $p_T$  splits and is monic of degree  $n$  ([Theorem 6.10.2](#)), [Definition 5.4.2](#) gives  $p_T = (x - c_1) \cdots (x - c_n)$  for some  $c_1, \dots, c_n \in F$ . For  $c, \lambda \in F$ , the polynomial  $x - c$  has  $\operatorname{mult}_\lambda(x - c) = 1$  if  $c = \lambda$  and 0 otherwise, since  $\lambda$  is a root of  $x - c$  only when  $c = \lambda$  ([Theorem 5.1.3](#)). So by [Theorem 5.4.2](#),  $\operatorname{mult}_\lambda(p_T) = \sum_j \operatorname{mult}_\lambda(x - c_j)$  is the number of indices  $j$  with  $c_j = \lambda$ . In particular the values occurring among  $c_1, \dots, c_n$  are exactly the roots of  $p_T$  in  $F$ , which are  $\lambda_1, \dots, \lambda_k$  by [Theorem 8.2.1](#), and  $\lambda_i$  occurs exactly  $a(\lambda_i)$  times. Counting the  $n$  factors gives  $\sum_i a(\lambda_i) = n$ , and

$$c_1 + \cdots + c_n = \sum_i a(\lambda_i) \lambda_i, \quad c_1 \cdots c_n = \prod_i \lambda_i^{a(\lambda_i)}.$$

Let  $\mathbf{C} = \operatorname{diag}(c_1, \dots, c_n)$ . It is triangular, so  $p_{\mathbf{C}} = (x - c_1) \cdots (x - c_n) = p_T$  (the examples



after Definition 6.10.1). Apply Theorem 6.10.2 to both  $\mathbf{C}$  and  $T$ : the coefficient of  $x^{n-1}$  in this one polynomial is  $-\operatorname{tr} \mathbf{C} = -(c_1 + \cdots + c_n)$  and also  $-\operatorname{tr} T$ ; the constant term is  $(-1)^n \det \mathbf{C} = (-1)^n c_1 \cdots c_n$  (Theorem 6.4.6) and also  $(-1)^n \det T$ . Hence  $\operatorname{tr} T = c_1 + \cdots + c_n$  and  $\det T = c_1 \cdots c_n$ , which are the displayed formulas. The complex case follows from Corollary 5.5.5. ■

### Examples.

- For  $\mathbf{A} = \begin{pmatrix} 2 & 0 & 1 \\ 1 & 3 & 1 \\ 1 & 0 & 2 \end{pmatrix}$ , with  $a(1) = 1$  and  $a(3) = 2$ :  $\operatorname{tr} \mathbf{A} = 2 + 3 + 2 = 7 = 1 + 3 + 3$ , and  $\det \mathbf{A} = 9 = 1 \cdot 3 \cdot 3$ .
- For the real matrix  $\begin{pmatrix} 3 & -2 \\ 4 & -1 \end{pmatrix}$  of Example (A Real Matrix with Non-Real Eigenvalues),  $p$  does not split over  $\mathbb{R}$ , but it splits over  $\mathbb{C}$  with roots  $1 \pm 2i$ . The theorem over  $\mathbb{C}$  gives  $(1 + 2i) + (1 - 2i) = 2 = \operatorname{tr} \mathbf{A}$  and  $(1 + 2i)(1 - 2i) = 1 + 4 = 5 = \det \mathbf{A}$ . The trace and determinant are real, while the eigenvalues are not.
- For a  $2 \times 2$  matrix whose characteristic polynomial splits, the theorem says the eigenvalues  $\lambda, \mu$  (listed with multiplicity) satisfy  $\lambda + \mu = \operatorname{tr} \mathbf{A}$  and  $\lambda\mu = \det \mathbf{A}$ . This is a quick way to guess eigenvalues: for  $\begin{pmatrix} 1 & 2 \\ 3 & 2 \end{pmatrix}$ , we want two numbers with sum 3 and product  $-4$ , namely 4 and  $-1$ .

**Without splitting, the theorem says nothing.** If  $p_T$  does not split, the eigenvalues in  $F$ , counted with algebraic multiplicity, do not account for all of  $p_T$ , and summing or multiplying them gives no information. For  $\mathbf{A} = \begin{pmatrix} 3 & -2 \\ 4 & -1 \end{pmatrix}$  over  $\mathbb{R}$  there are no eigenvalues: the multiplicities add up to  $0 \neq 2$ , the empty sum  $0$  is not  $\operatorname{tr} \mathbf{A} = 2$ , and the empty product  $1$  is not  $\det \mathbf{A} = 5$ . For the real rotation  $\begin{pmatrix} 0 & -1 \\ 1 & 0 \end{pmatrix}$ , with trace  $0$  and determinant  $1$ , the empty sum and product happen to give the right values, but only by coincidence. For real matrices the remedy is to pass to  $\mathbb{C}$ , where  $p_{\mathbf{A}}$  always splits, as in the second example above.

## 8.3.4 Exercises

### 8.3.4.1 A. Check your understanding

#### Exercise A1.

- Define the algebraic and the geometric multiplicity of  $\lambda \in F$  for  $\mathbf{A} \in M_n(F)$ .
- State the inequality between them for an eigenvalue  $\lambda$ , and name the tool its proof uses.
- True or false: if  $a(\lambda) = 1$ , then  $g(\lambda) = 1$ . Justify your answer.
- True or false: for every  $\mathbf{A} \in M_n(\mathbb{C})$ , the geometric multiplicities of the eigenvalues add up to  $n$ . Justify your answer.
- True or false: if  $\mathbf{A}, \mathbf{B} \in M_n(F)$  have the same characteristic polynomial, then  $g_{\mathbf{A}}(\lambda) = g_{\mathbf{B}}(\lambda)$  for every  $\lambda$ . Justify your answer.
- For  $\mathbf{A} \in M_3(\mathbb{C})$  with eigenvalues  $2$  (algebraic multiplicity  $2$ ) and  $-1$ , what are

$\text{tr } \mathbf{A}$  and  $\det \mathbf{A}$ ?

*Solution.*

- (a)  $a_{\mathbf{A}}(\lambda) = \text{mult}_{\lambda}(p_{\mathbf{A}})$ , the largest  $m$  with  $(x - \lambda)^m \mid p_{\mathbf{A}}$ ;  $g_{\mathbf{A}}(\lambda) = \dim E_{\lambda}(\mathbf{A}) = n - \text{rank}(\mathbf{A} - \lambda \mathbf{I})$  (Definition 8.3.1, Definition 8.3.2).
- (b)  $1 \leq g(\lambda) \leq a(\lambda)$  (Theorem 8.3.1). The proof extends a basis of  $E_{\lambda}$  to a basis of the space and uses the block triangular determinant (Theorem 6.8.4) over  $F[x]$ .
- (c) True. An eigenvalue has  $g(\lambda) \geq 1$ , and  $g(\lambda) \leq a(\lambda) = 1$ .
- (d) False. For  $\mathbf{J} = \begin{pmatrix} 2 & 1 \\ 0 & 2 \end{pmatrix}$ , the only eigenvalue is 2, with  $g(2) = 1 \neq 2$ . (The algebraic multiplicities do add up to  $n$  over  $\mathbb{C}$ .)
- (e) False.  $2\mathbf{I}_2$  and  $\mathbf{J}$  both have  $p = (x - 2)^2$ , but  $g(2) = 2$  and  $g(2) = 1$ .
- (f) By Theorem 8.3.3, listing eigenvalues with algebraic multiplicity as 2, 2,  $-1$ :  $\text{tr } \mathbf{A} = 3$  and  $\det \mathbf{A} = -4$ .

### 8.3.4.2 B. Practice

#### Exercise B1: Computing multiplicities.

For each real matrix, find every eigenvalue together with its algebraic and geometric multiplicity.

- (a)  $\mathbf{A}_1 = \begin{pmatrix} 4 & -1 \\ 1 & 2 \end{pmatrix}$ .
- (b)  $\mathbf{A}_2 = \begin{pmatrix} 2 & 1 & 1 \\ 0 & 1 & -1 \\ -1 & -1 & 1 \end{pmatrix}$ .
- (c)  $\mathbf{A}_3 = \begin{pmatrix} 2 & 1 & 1 \\ -1 & 0 & -1 \\ 1 & 1 & 2 \end{pmatrix}$ , given that  $p_{\mathbf{A}_3} = p_{\mathbf{A}_2}$ .
- (d) The matrix of the operator  $S(q)(x) = q(x + 1)$  on  $\mathbb{R}[x]_{\leq 2}$ , in the basis  $(1, x, x^2)$ .

*Solution.*

- (a)  $\text{tr } \mathbf{A}_1 = 6$ ,  $\det \mathbf{A}_1 = 8 + 1 = 9$ , so  $p = x^2 - 6x + 9 = (x - 3)^2$ : the only eigenvalue is 3, with  $a(3) = 2$ .  $\mathbf{A}_1 - 3\mathbf{I} = \begin{pmatrix} 1 & -1 \\ 1 & -1 \end{pmatrix}$  has rank 1, so  $g(3) = 1$ .
- (b) Expand  $\det(x\mathbf{I} - \mathbf{A}_2)$  along the first column:

$$\det \begin{pmatrix} x-2 & -1 & -1 \\ 0 & x-1 & 1 \\ 1 & 1 & x-1 \end{pmatrix} = (x-2)[(x-1)^2 - 1] + 1 \cdot \det \begin{pmatrix} -1 & -1 \\ x-1 & 1 \end{pmatrix}.$$

The first bracket is  $x^2 - 2x = x(x-2)$  and the last determinant is  $-1 + (x-1) = x-2$ , so  $p_{\mathbf{A}_2} = (x-2)(x^2 - 2x + 1) = (x-1)^2(x-2)$ . Thus  $a(1) = 2$  and  $a(2) = 1$ . By

**Theorem 8.3.1**,  $g(2) = 1$ . For  $\lambda = 1$ ,

$$\mathbf{A}_2 - \mathbf{I} = \begin{pmatrix} 1 & 1 & 1 \\ 0 & 0 & -1 \\ -1 & -1 & 0 \end{pmatrix}.$$

Row 3 is  $-(\text{row } 1) - (\text{row } 2)$ , and rows 1 and 2 are independent, so the rank is 2 and  $g(1) = 1$ .

(c) Here  $a(1) = 2$  and  $a(2) = 1$ , as for  $\mathbf{A}_2$ , so  $g(2) = 1$ . Now

$$\mathbf{A}_3 - \mathbf{I} = \begin{pmatrix} 1 & 1 & 1 \\ -1 & -1 & -1 \\ 1 & 1 & 1 \end{pmatrix}$$

has all rows multiples of  $(1, 1, 1)$ , so its rank is 1 and  $g(1) = 2$ . By [Proposition 8.3.2](#),  $\mathbf{A}_2$  and  $\mathbf{A}_3$  are not similar, although they have the same characteristic polynomial.

(d)  $S(1) = 1$ ,  $S(x) = 1 + x$ ,  $S(x^2) = 1 + 2x + x^2$ , so the matrix is  $\begin{pmatrix} 1 & 1 & 1 \\ 0 & 1 & 2 \\ 0 & 0 & 1 \end{pmatrix}$  (as in [Example \(Characteristic Polynomials of Two Operators\)](#)). It is triangular, so  $p = (x - 1)^3$  and  $a(1) = 3$ . The matrix minus  $\mathbf{I}$  is  $\begin{pmatrix} 0 & 1 & 1 \\ 0 & 0 & 2 \\ 0 & 0 & 0 \end{pmatrix}$ , of rank 2, so  $g(1) = 1$ : the only polynomials with  $q(x + 1) = q(x)$  in  $\mathbb{R}[x]_{\leq 2}$  are the constants.

### Exercise B2: A parameter that merges eigenvalues.

For  $t \in \mathbb{R}$ , let  $\mathbf{A}_t = \begin{pmatrix} t & 1 \\ -1 & 3 \end{pmatrix}$ . Find all  $t$  for which  $\mathbf{A}_t$  has a real eigenvalue  $\lambda$  with  $g(\lambda) < a(\lambda)$ .

*Solution.*  $\text{tr } \mathbf{A}_t = t + 3$  and  $\det \mathbf{A}_t = 3t + 1$ , so  $p(x) = x^2 - (t + 3)x + (3t + 1)$ . For a  $2 \times 2$  matrix,  $g(\lambda) < a(\lambda)$  is possible only if  $a(\lambda) = 2$ , because  $a(\lambda) = 1$  forces  $g(\lambda) = 1$  ([Theorem 8.3.1](#)). So  $p$  must have a repeated real root, which happens exactly when its discriminant vanishes:

$$(t + 3)^2 - 4(3t + 1) = t^2 - 6t + 5 = (t - 1)(t - 5) = 0.$$

If  $t = 1$ , then  $p = x^2 - 4x + 4 = (x - 2)^2$  and  $\mathbf{A}_1 - 2\mathbf{I} = \begin{pmatrix} -1 & 1 \\ -1 & 1 \end{pmatrix}$ , of rank 1, so  $g(2) = 1 < 2 = a(2)$ . If  $t = 5$ , then  $p = (x - 4)^2$  and  $\mathbf{A}_5 - 4\mathbf{I} = \begin{pmatrix} 1 & 1 \\ -1 & -1 \end{pmatrix}$ , of rank 1, so  $g(4) = 1 < 2 = a(4)$ . Hence the answer is  $t \in \{1, 5\}$ . (In both cases  $g = a$  was impossible anyway:  $g(\lambda) = 2$  would mean  $\mathbf{A}_t - \lambda\mathbf{I} = 0$ , but the off-diagonal entries of  $\mathbf{A}_t$  are non-zero.)

### Exercise B3: Eigenvalues from trace and determinant.

(a) Let  $\mathbf{A} \in M_3(\mathbb{R})$  have the eigenvalue  $1 + i$  (as a complex matrix) and  $\det \mathbf{A} = 6$ . Find all eigenvalues of  $\mathbf{A}$  over  $\mathbb{C}$ , and  $\text{tr } \mathbf{A}$ .

- (b) Let  $\mathbf{B} \in M_2(\mathbb{R})$  with  $\operatorname{tr} \mathbf{B} = 6$  and  $\det \mathbf{B} = 9$ . Show that  $\operatorname{spec}(\mathbf{B}) = \{3\}$  with  $a(3) = 2$ . Give one such  $\mathbf{B}$  with  $g(3) = 2$  and one with  $g(3) = 1$ .

*Solution.*

- (a) By [Theorem 8.2.5](#),  $1 - i$  is also an eigenvalue. By [Corollary 5.5.5](#),  $p_{\mathbf{A}}$  splits over  $\mathbb{C}$  into three linear factors, and two of them are  $x - (1 + i)$  and  $x - (1 - i)$ ; let  $\mu$  be the third root. By [Theorem 8.3.3](#),  $6 = \det \mathbf{A} = (1 + i)(1 - i)\mu = 2\mu$ , so  $\mu = 3$ . Hence the eigenvalues are  $1 + i, 1 - i, 3$ , and  $\operatorname{tr} \mathbf{A} = (1 + i) + (1 - i) + 3 = 5$ .
- (b) By [Theorem 6.10.2](#),  $p_{\mathbf{B}} = x^2 - 6x + 9 = (x - 3)^2$ . So  $\operatorname{spec}(\mathbf{B}) = \{3\}$  and  $a(3) = 2$ . For  $\mathbf{B} = 3\mathbf{I}_2$ ,  $g(3) = 2$ . For  $\mathbf{B} = \begin{pmatrix} 4 & -1 \\ 1 & 2 \end{pmatrix}$  of [Exercise \(B1: Computing multiplicities\)](#) (a),  $\operatorname{tr} \mathbf{B} = 6$ ,  $\det \mathbf{B} = 9$  and  $g(3) = 1$ .

### 8.3.4.3 C. Going deeper

#### Exercise C1: Nilpotent matrices.

Let  $\mathbf{A} \in M_n(\mathbb{C})$ ,  $n \geq 1$ , be nilpotent ( $\mathbf{A}^k = 0$  for some  $k \geq 1$ ).

- (a) Prove that  $a(0) = n$  and  $g(0) = n - \operatorname{rank} \mathbf{A}$ .
- (b) Deduce that  $g(0) = a(0)$  if and only if  $\mathbf{A} = 0$ .
- (c) For  $\mathbf{N} = \begin{pmatrix} 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \\ 0 & 0 & 0 & 0 \end{pmatrix}$ , check that  $\mathbf{N}^4 = 0$  and compute  $g(0)$ . How large can  $a(0) - g(0)$  be for a nilpotent  $n \times n$  matrix?

*Solution.*

- (a) By [Exercise \(C2: Nilpotent operators\)](#) (b),  $p_{\mathbf{A}} = x^n$ , so  $a(0) = \operatorname{mult}_0(x^n) = n$ . By [Definition 8.3.2](#),  $g(0) = n - \operatorname{rank}(\mathbf{A} - 0\mathbf{I}) = n - \operatorname{rank} \mathbf{A}$ .
- (b) By (a),  $g(0) = a(0)$  if and only if  $n - \operatorname{rank} \mathbf{A} = n$ , that is,  $\operatorname{rank} \mathbf{A} = 0$ , that is,  $\mathbf{A} = 0$  (a matrix of rank 0 has no non-zero column).
- (c)  $\mathbf{N}\mathbf{e}_1 = \mathbf{0}$  and  $\mathbf{N}\mathbf{e}_j = \mathbf{e}_{j-1}$  for  $j \geq 2$ . So  $\mathbf{N}^j\mathbf{e}_i = \mathbf{0}$  once  $j \geq i$ , and  $\mathbf{N}^4\mathbf{e}_i = \mathbf{0}$  for  $i \leq 4$ ; hence  $\mathbf{N}^4 = 0$ . The columns of  $\mathbf{N}$  are  $\mathbf{0}, \mathbf{e}_1, \mathbf{e}_2, \mathbf{e}_3$ , so  $\operatorname{rank} \mathbf{N} = 3$  and  $g(0) = 1$ , while  $a(0) = 4$ . In general, a nilpotent  $\mathbf{A}$  has  $0 \in \operatorname{spec}(\mathbf{A})$  ([Exercise \(C2: Nilpotent operators\)](#) (a)), so  $g(0) \geq 1$  and  $a(0) - g(0) \leq n - 1$ ; the  $n \times n$  analogue of  $\mathbf{N}$ , with ones just above the diagonal, has rank  $n - 1$  and attains  $a(0) - g(0) = n - 1$ .

#### Exercise C2: Multiplicities do not determine similarity.

Let

$$\mathbf{M}_1 = \begin{pmatrix} 1 & 1 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 1 \\ 0 & 0 & 0 & 1 \end{pmatrix}, \quad \mathbf{M}_2 = \begin{pmatrix} 1 & 1 & 0 & 0 \\ 0 & 1 & 1 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{pmatrix} \in M_4(\mathbb{R}).$$

- (a) Show that  $\mathbf{M}_1$  and  $\mathbf{M}_2$  have the same eigenvalues with the same algebraic **and** geometric multiplicities.
- (b) Compute  $\text{rank}(\mathbf{M}_1 - \mathbf{I})^2$  and  $\text{rank}(\mathbf{M}_2 - \mathbf{I})^2$ . Deduce that  $\mathbf{M}_1 \not\sim \mathbf{M}_2$ .
- (c) Construct a matrix  $\mathbf{M} \in M_4(\mathbb{R})$  with  $\text{spec}(\mathbf{M}) = \{2, 5\}$ ,  $a(2) = 3$ ,  $g(2) = 1$ , and  $a(5) = g(5) = 1$ .

*Hint: for (b), if  $\mathbf{B} = \mathbf{P}^{-1}\mathbf{A}\mathbf{P}$ , relate  $(\mathbf{B} - \mathbf{I})^2$  to  $(\mathbf{A} - \mathbf{I})^2$ .*

*Solution.*

- (a) Both matrices are upper triangular with diagonal 1, 1, 1, 1, so both have  $p = (x - 1)^4$ , spectrum  $\{1\}$  and  $a(1) = 4$ . The matrix  $\mathbf{M}_1 - \mathbf{I}$  has columns  $\mathbf{0}, \mathbf{e}_1, \mathbf{0}, \mathbf{e}_3$ , of rank 2;  $\mathbf{M}_2 - \mathbf{I}$  has columns  $\mathbf{0}, \mathbf{e}_1, \mathbf{e}_2, \mathbf{0}$ , also of rank 2. So  $g(1) = 4 - 2 = 2$  for both.
- (b) Let  $\mathbf{N}_1 = \mathbf{M}_1 - \mathbf{I}$ , so  $\mathbf{N}_1\mathbf{e}_2 = \mathbf{e}_1$ ,  $\mathbf{N}_1\mathbf{e}_4 = \mathbf{e}_3$  and  $\mathbf{N}_1\mathbf{e}_1 = \mathbf{N}_1\mathbf{e}_3 = \mathbf{0}$ . Then  $\mathbf{N}_1^2$  kills every  $\mathbf{e}_j$ , so  $\mathbf{N}_1^2 = \mathbf{0}$  and  $\text{rank } \mathbf{N}_1^2 = 0$ . Let  $\mathbf{N}_2 = \mathbf{M}_2 - \mathbf{I}$ , so  $\mathbf{N}_2\mathbf{e}_2 = \mathbf{e}_1$ ,  $\mathbf{N}_2\mathbf{e}_3 = \mathbf{e}_2$ , and  $\mathbf{N}_2\mathbf{e}_1 = \mathbf{N}_2\mathbf{e}_4 = \mathbf{0}$ . Then  $\mathbf{N}_2^2\mathbf{e}_3 = \mathbf{e}_1 \neq \mathbf{0}$  and  $\mathbf{N}_2^2$  kills  $\mathbf{e}_1, \mathbf{e}_2, \mathbf{e}_4$ , so  $\mathbf{N}_2^2 = \mathbf{E}_{13}$  has rank 1.

Suppose  $\mathbf{M}_2 = \mathbf{P}^{-1}\mathbf{M}_1\mathbf{P}$ . With  $p(x) = (x - 1)^2$ , [Proposition 3.7.6](#) (c) gives  $(\mathbf{M}_2 - \mathbf{I})^2 = \mathbf{P}^{-1}(\mathbf{M}_1 - \mathbf{I})^2\mathbf{P}$ , and then (a) of the same proposition gives equal ranks,  $1 = 0$ . This contradiction shows  $\mathbf{M}_1 \not\sim \mathbf{M}_2$ . So eigenvalues together with both multiplicities do not determine a matrix up to similarity; Chapter 9 finds invariants that do.

- (c) Take

$$\mathbf{M} = \begin{pmatrix} 2 & 1 & 0 & 0 \\ 0 & 2 & 1 & 0 \\ 0 & 0 & 2 & 0 \\ 0 & 0 & 0 & 5 \end{pmatrix}.$$

It is triangular, so  $p_{\mathbf{M}} = (x - 2)^3(x - 5)$ ,  $a(2) = 3$  and  $a(5) = 1$ , hence  $g(5) = 1$  by [Theorem 8.3.1](#). The matrix  $\mathbf{M} - 2\mathbf{I}$  has columns  $\mathbf{0}, \mathbf{e}_1, \mathbf{e}_2, 3\mathbf{e}_4$ , which has rank 3, so  $g(2) = 4 - 3 = 1$ .

## 8.4 Diagonalization

We can now answer the question of Chapter 3 in the best case. An operator has a diagonal matrix in some basis exactly when there are enough eigenvectors to form a basis, and the previous two sections give precise ways to count “enough”: by dimensions of eigenspaces, or by comparing geometric with algebraic multiplicities. This section proves that these tests agree, shows how to find the diagonalizing basis for a concrete matrix, and uses it to compute powers. The payoff is a closed formula for the Fibonacci numbers. We also meet the operators for which the method fails, and see that the field is part of the answer.

### 8.4.1 Diagonalizable operators

Chapter 3 introduced similarity to compare the matrices of one operator in different bases ([Theorem 3.7.4](#)), and asked for the simplest representative. Section 2 unwound the wish “[ $T$ ] $_{\mathcal{B}}$

is diagonal” and found that it says exactly that every vector of  $\mathcal{B}$  is an eigenvector. Diagonal matrices are the matrices we can compute with most easily: they add, multiply, power and invert entry by entry. So the operators that admit such a basis deserve a name.

*An operator is diagonalizable when some basis consists of vectors that the operator only stretches.*

### Definition 8.4.1 — Diagonalizable

Let  $V$  be a finite-dimensional vector space over  $F$ . An operator  $T \in \mathcal{L}(V)$  is **diagonalizable** if there **exists** a basis  $\mathcal{B}$  of  $V$  such that  $[T]_{\mathcal{B}}$  is a diagonal matrix.

A matrix  $\mathbf{A} \in M_n(F)$  is **diagonalizable over  $F$**  if it is similar to a diagonal matrix: there **exist** an **invertible**  $\mathbf{P} \in M_n(F)$  and a diagonal  $\mathbf{D} \in M_n(F)$  with  $\mathbf{P}^{-1}\mathbf{A}\mathbf{P} = \mathbf{D}$ , equivalently  $\mathbf{A} = \mathbf{P}\mathbf{D}\mathbf{P}^{-1}$ .

In words: for an operator, we are free to choose **any** basis, and we ask whether **one** of them makes the matrix diagonal; most bases will not. For a matrix, the change-of-basis matrix  $\mathbf{P}$  and the diagonal entries of  $\mathbf{D}$  must both have entries in  $F$ . The two notions match:  $\mathbf{A}$  is diagonalizable over  $F$  if and only if  $T_{\mathbf{A}} \in \mathcal{L}(F^n)$  is diagonalizable, by Theorem 3.7.4 (c), because the matrices of  $T_{\mathbf{A}}$  in the bases of  $F^n$  are exactly the matrices similar to  $\mathbf{A}$ . Likewise,  $T$  is diagonalizable if and only if  $[T]_{\mathcal{C}}$  is diagonalizable for one, equivalently every, basis  $\mathcal{C}$  of  $V$ , by Theorem 3.7.4 (a) and (b).

#### Examples.

- **Diagonal matrices** are diagonalizable, with  $\mathbf{P} = \mathbf{I}$ . So is every scalar operator  $c \operatorname{id}_V$ , whose matrix is  $c\mathbf{I}$  in every basis.
- **The degenerate cases.** Every  $1 \times 1$  matrix is diagonal. On  $V = \{0\}$ , the empty basis gives the empty matrix, which counts as diagonal. These cases are harmless, but they remind us that diagonalizability is a question about repeated eigenvalues and missing eigenvectors, which need room to occur.
- **A  $2 \times 2$  matrix.** In Example (Eigenvalues and Eigenspaces of a  $2 \times 2$  Matrix),  $\mathbf{A} = \begin{pmatrix} 1 & 2 \\ 3 & 2 \end{pmatrix}$  has eigenvectors  $(2, 3)$  for 4 and  $(1, -1)$  for  $-1$ . They form a basis of  $\mathbb{R}^2$ , so  $\mathbf{A}$  is diagonalizable, with  $\mathbf{P} = \begin{pmatrix} 2 & 1 \\ 3 & -1 \end{pmatrix}$  and  $\mathbf{D} = \operatorname{diag}(4, -1)$ ; we check this below.
- **Projections and involutions.** A projection  $\mathbf{P}$  satisfies  $V = \operatorname{im} \mathbf{P} \oplus \ker \mathbf{P}$  with  $\mathbf{P}$  the identity on  $\operatorname{im} \mathbf{P}$  and zero on  $\ker \mathbf{P}$  (Theorem 3.10.1). A basis of  $\operatorname{im} \mathbf{P}$  followed by a basis of  $\ker \mathbf{P}$  gives  $[\mathbf{P}] = \operatorname{diag}(1, \dots, 1, 0, \dots, 0)$ . In characteristic not 2, an involution splits  $V = E_+ \oplus E_-$  (Theorem 3.10.5), and gives  $\operatorname{diag}(1, \dots, 1, -1, \dots, -1)$  in the same way.

**Non-example by minimal change.** Change the  $(1, 2)$ -entry of  $\mathbf{I}_2$  from 0 to 1, giving  $\mathbf{J} = \begin{pmatrix} 1 & 1 \\ 0 & 1 \end{pmatrix}$  over any field  $F$ . It is still triangular with the single eigenvalue 1, and still invertible.

What fails is the existence of a basis of eigenvectors:  $E_1(\mathbf{J}) = \ker \begin{pmatrix} 0 & 1 \\ 0 & 0 \end{pmatrix} = \operatorname{span}(\mathbf{e}_1)$  is a line, so every eigenvector of  $\mathbf{J}$  is a multiple of  $\mathbf{e}_1$ , and no two eigenvectors are independent. By the unwinding in Section 2, no basis makes  $[T_{\mathbf{J}}]$  diagonal. (Directly: if  $\mathbf{P}^{-1}\mathbf{J}\mathbf{P} = \mathbf{D}$  were diagonal, its diagonal entries would be eigenvalues of  $\mathbf{J}$ , so  $\mathbf{D} = \mathbf{I}$ , and then  $\mathbf{J} = \mathbf{P}\mathbf{I}\mathbf{P}^{-1} = \mathbf{I}$ , which is false.)

**Why this definition.** We ask for similarity to a diagonal matrix, not equality, because the

basis is ours to choose; a matrix like  $\begin{pmatrix} 1 & 2 \\ 3 & 2 \end{pmatrix}$  has nothing diagonal about it until the right basis is found. We insist that  $\mathbf{P}$  and  $\mathbf{D}$  have entries in  $F$ , because otherwise the word would not depend on the field and would hide a real phenomenon: the rotation  $\mathbf{R} = \begin{pmatrix} 0 & -1 \\ 1 & 0 \end{pmatrix}$  has no real eigenvector, so it is not diagonalizable over  $\mathbb{R}$ , while over  $\mathbb{C}$  it becomes  $\text{diag}(i, -i)$  (Section 2). The name says what the definition promises: the operator can be **made** diagonal.

**Warning. Diagonalizability depends on the field, and even on its characteristic.** The rotation  $\mathbf{R}$  is diagonalizable over  $\mathbb{C}$  but not over  $\mathbb{R}$ . A subtler case:  $\mathbf{S} = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}$  is diagonalizable over  $\mathbb{Q}$ , with eigenvectors  $(1, 1)$  for 1 and  $(1, -1)$  for  $-1$ . Over  $\mathbb{F}_2$  the same matrix has  $p_{\mathbf{S}} = x^2 - 1 = (x - 1)^2$ , since  $-1 = 1$ , and  $\mathbf{S} - \mathbf{I} = \begin{pmatrix} 1 & 1 \\ 1 & 1 \end{pmatrix}$  has rank 1, so its only eigenvectors are the multiples of  $(1, 1)$ , and  $\mathbf{S}$  is **not** diagonalizable over  $\mathbb{F}_2$ . Always say over which field.

## 8.4.2 The Diagonalization Theorem

How do we decide whether a basis of eigenvectors exists without searching for one? Eigenvectors for different eigenvalues are automatically independent ([Theorem 8.2.2](#)), so the only question is whether each eigenspace supplies enough of them. The theorem below lists five equivalent ways to say “enough”.

### Theorem 8.4.1 — Diagonalization Theorem

Let  $V$  be a finite-dimensional vector space over  $F$  with  $\dim V = n \geq 1$ , let  $T \in \mathcal{L}(V)$ , and let  $\lambda_1, \dots, \lambda_k$  be the distinct eigenvalues of  $T$  (possibly  $k = 0$ ). The following are equivalent:

- (a)  $T$  is diagonalizable;
- (b)  $V$  has a basis consisting of eigenvectors of  $T$ ;
- (c)  $V = E_{\lambda_1}(T) \oplus \dots \oplus E_{\lambda_k}(T)$ ;
- (d)  $\dim E_{\lambda_1}(T) + \dots + \dim E_{\lambda_k}(T) = n$ ;
- (e)  $p_T$  splits over  $F$ , and  $g_T(\lambda_i) = a_T(\lambda_i)$  for every  $i = 1, \dots, k$ .

In that case, if  $\mathcal{B}$  is a basis of eigenvectors, then  $[T]_{\mathcal{B}}$  is diagonal, its diagonal entries are the eigenvalues of the corresponding basis vectors, and each  $\lambda_i$  appears on the diagonal exactly  $a_T(\lambda_i) = g_T(\lambda_i)$  times.

(For  $k = 0$  the sum in (c) is  $\{\mathbf{0}\}$  and the sum in (d) is 0; all five conditions are then false, since  $n \geq 1$ .)

**Idea. Plan:** prove (a)  $\Rightarrow$  (b)  $\Rightarrow$  (c)  $\Rightarrow$  (d)  $\Rightarrow$  (e)  $\Rightarrow$  (b)  $\Rightarrow$  (a).

① (a)  $\Leftrightarrow$  (b) is the unwinding of columns from Section 2. ② From a basis of eigenvectors, the eigenspaces add up to  $V$ , and [Corollary 8.2.3](#) makes the sum direct. ③ Dimensions add in a direct sum. ④ (d)  $\Rightarrow$  (e) is a sandwich:  $n = \sum g \leq \sum a \leq n$ , so every inequality is an equality. ⑤ (e)  $\Rightarrow$  (b) runs the sandwich backwards and glues bases of

the eigenspaces together.

The hard work was done in Sections 2 and 3; this proof only assembles it.

*Proof.*

- (a)  $\Leftrightarrow$  (b). Let  $\mathcal{B} = (\mathbf{v}_1, \dots, \mathbf{v}_n)$  be a basis of  $V$ . By [Definition 3.6.1](#), column  $j$  of  $[T]_{\mathcal{B}}$  is  $[T\mathbf{v}_j]_{\mathcal{B}}$ , and this column is  $d_j \mathbf{e}_j$  if and only if  $T\mathbf{v}_j = d_j \mathbf{v}_j$ . Hence  $[T]_{\mathcal{B}} = \text{diag}(d_1, \dots, d_n)$  if and only if  $T\mathbf{v}_j = d_j \mathbf{v}_j$  for every  $j$ . Since basis vectors are non-zero, this says that every  $\mathbf{v}_j$  is an eigenvector of  $T$ , for the eigenvalue  $d_j$ . So a basis makes  $[T]_{\mathcal{B}}$  diagonal exactly when it consists of eigenvectors, which also proves the final statement about the diagonal entries, apart from the count.
- (b)  $\Rightarrow$  (c). Let  $\mathcal{B}$  be a basis of eigenvectors. Each vector of  $\mathcal{B}$  lies in  $E_{\lambda_i}(T)$  for some  $i$ , so  $\mathcal{B} \subseteq E_{\lambda_1}(T) + \dots + E_{\lambda_k}(T)$ . This sum is a subspace ([Theorem 1.7.1](#)) containing a spanning list of  $V$ , so it equals  $V$ . By [Corollary 8.2.3](#) the sum is direct.
- (c)  $\Rightarrow$  (d). By [Theorem 1.7.5](#) ((a)  $\Rightarrow$  (e)),  $n = \dim V = \sum_i \dim E_{\lambda_i}(T)$ .
- (d)  $\Rightarrow$  (e). By [Theorem 8.3.1](#),  $g_T(\lambda_i) \leq a_T(\lambda_i)$  for each  $i$ . The eigenvalues are exactly the roots of  $p_T$  in  $F$  ([Theorem 8.2.1](#)), so by [Theorem 5.4.3](#),  $\sum_i a_T(\lambda_i) \leq \deg p_T = n$  ([Theorem 6.10.2](#)). With (d),

$$n = \sum_{i=1}^k g_T(\lambda_i) \leq \sum_{i=1}^k a_T(\lambda_i) \leq n.$$

Hence both inequalities are equalities. The second equality,  $\sum_i a_T(\lambda_i) = \deg p_T$ , means that  $p_T$  splits over  $F$  ([Theorem 5.4.3](#)). The first says  $\sum_i (a_T(\lambda_i) - g_T(\lambda_i)) = 0$ , a sum of non-negative integers, so every term is 0 and  $g_T(\lambda_i) = a_T(\lambda_i)$  for each  $i$ .

- (e)  $\Rightarrow$  (b). Since  $p_T$  splits, [Theorem 5.4.3](#) gives  $\sum_i a_T(\lambda_i) = n$ , and so  $\sum_i g_T(\lambda_i) = n$  by (e). For each  $i$  let  $\mathcal{B}_i$  be a basis of  $E_{\lambda_i}(T)$ , of length  $g_T(\lambda_i)$ , and let  $\mathcal{B}$  be the list  $\mathcal{B}_1, \dots, \mathcal{B}_k$ . The sum  $W = E_{\lambda_1}(T) + \dots + E_{\lambda_k}(T)$  is direct ([Corollary 8.2.3](#)), so by [Theorem 1.7.5](#) ((a)  $\Rightarrow$  (d)),  $\mathcal{B}$  is a basis of  $W$ , and  $\dim W = n = \dim V$ . By [Theorem 1.6.8](#),  $W = V$ . So  $\mathcal{B}$  is a basis of  $V$ , and its vectors are eigenvectors, being non-zero vectors of eigenspaces.

Finally, suppose  $\mathcal{B}$  is a basis of eigenvectors, and let  $m_i$  be the number of its vectors with eigenvalue  $\lambda_i$ . By (a)  $\Leftrightarrow$  (b),  $[T]_{\mathcal{B}}$  is diagonal with  $\lambda_i$  appearing  $m_i$  times, so  $p_T = \prod_i (x - \lambda_i)^{m_i}$  (the triangular case after [Definition 6.10.1](#)). Since the  $\lambda_i$  are distinct,  $(x - \lambda_i)^{m_i}$  divides  $p_T$  and the remaining factors do not vanish at  $\lambda_i$ , so  $m_i = a_T(\lambda_i)$  by [Lemma 5.4.1](#); and  $a_T(\lambda_i) = g_T(\lambda_i)$  by (e). This proves the theorem. ■

This is the theorem promised in Chapter 1, §7: **an operator is diagonalizable exactly when its eigenspaces split  $V$  as a direct sum**, and then  $T$  acts on the  $i$ -th piece as the scalar  $\lambda_i$ . In the language of [Theorem 8.1.5](#), the pieces are invariant, the blocks are  $\lambda_i \mathbf{I}_{g(\lambda_i)}$ , and the matrix is block diagonal with scalar blocks.

Condition (e) is the practical test. It separates two different ways to fail, and they have different remedies. If  $p_T$  does not split, the field is too small, and enlarging it (from  $\mathbb{R}$  to  $\mathbb{C}$ ) may help. If  $p_T$  splits but some  $g(\lambda) < a(\lambda)$ , there are genuinely too few eigenvectors, and no change of field helps; Chapter 9 explains what the best matrix is then.



The easiest case of (e) needs no rank computation at all.

### Corollary 8.4.2 — Distinct Eigenvalues Imply Diagonalizable

Let  $V$  be a finite-dimensional vector space over  $F$  with  $\dim V = n \geq 1$ , and let  $T \in \mathcal{L}(V)$  have  $n$  distinct eigenvalues. Then  $T$  is diagonalizable. The same holds for  $\mathbf{A} \in M_n(F)$  with  $n$  distinct eigenvalues in  $F$ .

*Proof.* Let  $\lambda_1, \dots, \lambda_n$  be the eigenvalues. Each  $\lambda_i$  is an eigenvalue, so  $E_{\lambda_i}(T) \neq \{\mathbf{0}\}$  (Definition 8.2.2) and  $\dim E_{\lambda_i}(T) \geq 1$ , so  $\sum_i \dim E_{\lambda_i}(T) \geq n$ , and by Corollary 8.2.3 the sum is at most  $n$ . Hence it equals  $n$ , and  $T$  is diagonalizable by Theorem 8.4.1 ((d)  $\Rightarrow$  (a)).

■

So a matrix whose characteristic polynomial has  $n$  distinct roots in  $F$  is diagonalizable over  $F$ ; for example  $\begin{pmatrix} 2 & 1 \\ 0 & 3 \end{pmatrix}$  is, because its eigenvalues 2 and 3 are distinct. This is the generic situation, but it is only a sufficient condition.

**Warning. The converse fails: diagonalizable matrices may have repeated eigenvalues.**  $\mathbf{I}_n$  is diagonal, and its only eigenvalue is 1. Less trivially, the matrix of Example (Diagonalizing a  $3 \times 3$  Matrix with a Repeated Eigenvalue) below has eigenvalues 1, 1, 3 and is diagonalizable. A repeated eigenvalue is a reason to **check**  $g(\lambda) = a(\lambda)$ , not a verdict.

*Quick check.* Which of  $\begin{pmatrix} 3 & 1 \\ 0 & 5 \end{pmatrix}$ ,  $\begin{pmatrix} 3 & 1 \\ 0 & 3 \end{pmatrix}$ ,  $\begin{pmatrix} 3 & 0 \\ 0 & 3 \end{pmatrix}$  are diagonalizable over  $\mathbb{R}$ ?

*Solution.* The first has distinct eigenvalues 3, 5, so it is diagonalizable by Corollary 8.4.2. The second has  $a(3) = 2$  but  $\text{rank} \begin{pmatrix} 0 & 1 \\ 0 & 0 \end{pmatrix} = 1$ , so  $g(3) = 1 < a(3)$ , and it is not diagonalizable by Theorem 8.4.1 (e). The third is already diagonal.

## 8.4.3 Computing $\mathbf{P}$ and $\mathbf{D}$

For a matrix  $\mathbf{A} \in M_n(F)$ , a basis of eigenvectors turns into the matrices  $\mathbf{P}$  and  $\mathbf{D}$  of Definition 8.4.1 directly. Let  $(\mathbf{v}_1, \dots, \mathbf{v}_n)$  be a basis of  $F^n$  with  $\mathbf{A}\mathbf{v}_j = d_j\mathbf{v}_j$ , let  $\mathbf{P}$  be the matrix with columns  $\mathbf{v}_1, \dots, \mathbf{v}_n$ , and  $\mathbf{D} = \text{diag}(d_1, \dots, d_n)$ . By the column view of matrix multiplication (Theorem 0.9.2), column  $j$  of  $\mathbf{AP}$  is  $\mathbf{A}\mathbf{v}_j = d_j\mathbf{v}_j$ , and column  $j$  of  $\mathbf{PD}$  is  $\mathbf{P}(d_j\mathbf{e}_j) = d_j\mathbf{v}_j$ . So

$$\mathbf{AP} = \mathbf{PD}.$$

The columns of  $\mathbf{P}$  form a basis, so  $\mathbf{P}$  is invertible (Theorem 2.4.4), and multiplying on the left by  $\mathbf{P}^{-1}$  gives  $\mathbf{P}^{-1}\mathbf{AP} = \mathbf{D}$ . The **order** matters in one way only: the  $j$ -th column of  $\mathbf{P}$  must be an eigenvector for the  $j$ -th diagonal entry of  $\mathbf{D}$ . Reordering the eigenvectors reorders the diagonal of  $\mathbf{D}$  in the same way, and any order works.

For  $\mathbf{A} = \begin{pmatrix} 1 & 2 \\ 3 & 2 \end{pmatrix}$ , with  $\mathbf{P} = \begin{pmatrix} 2 & 1 \\ 3 & -1 \end{pmatrix}$  and  $\mathbf{D} = \text{diag}(4, -1)$ :  $\mathbf{AP} = \begin{pmatrix} 8 & -1 \\ 12 & 1 \end{pmatrix}$  and  $\mathbf{PD} =$

$\begin{pmatrix} 8 & -1 \\ 12 & 1 \end{pmatrix}$ . Checking  $\mathbf{AP} = \mathbf{PD}$  avoids computing  $\mathbf{P}^{-1}$ , and together with  $\det \mathbf{P} = -5 \neq 0$  it proves  $\mathbf{P}^{-1}\mathbf{AP} = \mathbf{D}$ .

The full procedure, then: compute  $p_{\mathbf{A}}$  and its roots; if it does not split over  $F$ , stop (not diagonalizable over  $F$ ); for each eigenvalue compute a basis of  $E_{\lambda}(\mathbf{A})$ ; if some  $g(\lambda) < a(\lambda)$ , stop; otherwise put the basis vectors as columns of  $\mathbf{P}$ .

**Example. (Diagonalizing a  $3 \times 3$  Matrix with a Repeated Eigenvalue)**

Let

$$\mathbf{A} = \begin{pmatrix} 3 & 2 & -2 \\ -2 & -1 & 2 \\ -2 & -2 & 3 \end{pmatrix} \in M_3(\mathbb{R}).$$

Decide whether  $\mathbf{A}$  is diagonalizable over  $\mathbb{R}$ . If it is, find an invertible  $\mathbf{P}$  and a diagonal  $\mathbf{D}$  with  $\mathbf{P}^{-1}\mathbf{AP} = \mathbf{D}$ .

*Solution. Characteristic polynomial.* Expanding along the first row ([Theorem 6.6.1](#) over  $\mathbb{R}[x]$ ),

$$\begin{aligned} p_{\mathbf{A}}(x) &= \det \begin{pmatrix} x-3 & -2 & 2 \\ 2 & x+1 & -2 \\ 2 & 2 & x-3 \end{pmatrix} \\ &= (x-3)[(x+1)(x-3)+4] + 2[2(x-3)+4] \\ &\quad + 2[4-2(x+1)]. \end{aligned}$$

The brackets are  $x^2 - 2x + 1 = (x-1)^2$ ,  $2(x-1)$  and  $-2(x-1)$ , so the last two terms cancel and

$$p_{\mathbf{A}}(x) = (x-1)^2(x-3).$$

It splits over  $\mathbb{R}$ , with  $a(1) = 2$  and  $a(3) = 1$ . As a check,  $\text{tr } \mathbf{A} = 5 = 1 + 1 + 3$  ([Theorem 8.3.3](#)).

*Eigenspace for 1.*

$$\mathbf{A} - \mathbf{I} = \begin{pmatrix} 2 & 2 & -2 \\ -2 & -2 & 2 \\ -2 & -2 & 2 \end{pmatrix}.$$

Every row is a multiple of  $(1, 1, -1)$ , so the rank is 1 and  $g(1) = 3 - 1 = 2 = a(1)$ . The system reduces to  $x + y - z = 0$ ; with the free variables  $y$  and  $z$ , a basis of  $E_1(\mathbf{A})$  is  $(-1, 1, 0)$ ,  $(1, 0, 1)$  ([Theorem 2.5.4](#)).

*Eigenspace for 3.* Here  $g(3) = 1 = a(3)$  by [Theorem 8.3.1](#), so condition (e) of [Theorem 8.4.1](#) holds and  $\mathbf{A}$  is diagonalizable. To find the eigenvector,

$$\mathbf{A} - 3\mathbf{I} = \begin{pmatrix} 0 & 2 & -2 \\ -2 & -4 & 2 \\ -2 & -2 & 0 \end{pmatrix}.$$

Row 1 gives  $y = z$ , and row 3 gives  $x = -y$ ; row 2 then reads  $2y - 4y + 2y = 0$ . So  $E_3(\mathbf{A}) = \text{span}((1, -1, -1))$ .

*The matrices.* Put the eigenvectors as columns, in the order of the eigenvalues we want

on the diagonal:

$$\mathbf{P} = \begin{pmatrix} -1 & 1 & 1 \\ 1 & 0 & -1 \\ 0 & 1 & -1 \end{pmatrix}, \quad \mathbf{D} = \text{diag}(1, 1, 3).$$

*Check.* The columns of  $\mathbf{AP}$  are  $\mathbf{A}(-1, 1, 0) = (-1, 1, 0)$ ,  $\mathbf{A}(1, 0, 1) = (1, 0, 1)$  and  $\mathbf{A}(1, -1, -1) = (3, -3, -3)$ , which are the columns of  $\mathbf{PD}$ . Expanding along the first column,  $\det \mathbf{P} = -1 \cdot (0 + 1) - 1 \cdot (-1 - 1) = 1 \neq 0$ , so  $\mathbf{P}$  is invertible, and in fact

$$\mathbf{P}^{-1} = \begin{pmatrix} 1 & 2 & -1 \\ 1 & 1 & 0 \\ 1 & 1 & -1 \end{pmatrix},$$

as multiplying out  $\mathbf{PP}^{-1} = \mathbf{I}$  confirms. Hence  $\mathbf{P}^{-1}\mathbf{AP} = \mathbf{D}$ .

Had we listed the eigenvectors in the order  $(1, -1, -1), (-1, 1, 0), (1, 0, 1)$ , we would get  $\mathbf{D} = \text{diag}(3, 1, 1)$  instead. Both are correct.

## 8.4.4 Powers and polynomials

Why diagonalize? Because powers of diagonal matrices are free:  $\text{diag}(d_1, \dots, d_n)^k = \text{diag}(d_1^k, \dots, d_n^k)$ . Similarity carries this over.

### Theorem 8.4.3 — Powers and Polynomials of a Diagonalizable Matrix

Let  $\mathbf{A} = \mathbf{PDP}^{-1}$  with  $\mathbf{P} \in M_n(F)$  invertible and  $\mathbf{D} = \text{diag}(d_1, \dots, d_n)$ . Then for every  $k \in \mathbb{N}$  and every  $p \in F[x]$ ,

$$\mathbf{A}^k = \mathbf{P} \text{diag}(d_1^k, \dots, d_n^k) \mathbf{P}^{-1}, \quad p(\mathbf{A}) = \mathbf{P} \text{diag}(p(d_1), \dots, p(d_n)) \mathbf{P}^{-1}.$$

If moreover every  $d_i \neq 0$ , then  $\mathbf{A}$  is invertible and  $\mathbf{A}^{-1} = \mathbf{P} \text{diag}(d_1^{-1}, \dots, d_n^{-1}) \mathbf{P}^{-1}$ .

*Proof.* Since  $\mathbf{D} = \mathbf{P}^{-1}\mathbf{AP}$ , Proposition 3.7.6 (c) gives  $p(\mathbf{D}) = \mathbf{P}^{-1}p(\mathbf{A})\mathbf{P}$ , so  $p(\mathbf{A}) = \mathbf{P}p(\mathbf{D})\mathbf{P}^{-1}$ . A diagonal matrix is the direct sum  $(d_1) \oplus \dots \oplus (d_n)$  of  $1 \times 1$  blocks, so by Theorem 7.1.4 (b), extended to  $n$  blocks by induction on the number of blocks,  $p(\mathbf{D}) = (p(d_1)) \oplus \dots \oplus (p(d_n)) = \text{diag}(p(d_1), \dots, p(d_n))$ . Taking  $p = x^k$  gives the formula for  $\mathbf{A}^k$ . If every  $d_i \neq 0$ , put  $\mathbf{B} = \mathbf{P} \text{diag}(d_1^{-1}, \dots, d_n^{-1}) \mathbf{P}^{-1}$ ; then  $\mathbf{AB} = \mathbf{PDP}^{-1}\mathbf{P} \text{diag}(d_i^{-1}) \mathbf{P}^{-1} = \mathbf{P} \text{diag}(d_i d_i^{-1}) \mathbf{P}^{-1} = \mathbf{PP}^{-1} = \mathbf{I}$ , and  $\mathbf{B} = \mathbf{A}^{-1}$  by Theorem 2.4.5. ■

In words: to apply a polynomial to a diagonalizable matrix, apply it to the eigenvalues and change basis back. The work is done once, when  $\mathbf{P}$  is found; every power after that costs a single product of three matrices. For Example (Diagonalizing a  $3 \times 3$  Matrix with a Repeated Eigenvalue),

$$\mathbf{A}^k = \mathbf{P} \text{diag}(1, 1, 3^k) \mathbf{P}^{-1} = \begin{pmatrix} 3^k & 3^k - 1 & 1 - 3^k \\ 1 - 3^k & 2 - 3^k & 3^k - 1 \\ 1 - 3^k & 1 - 3^k & 3^k \end{pmatrix},$$

which is  $\mathbf{I}$  for  $k = 0$  and  $\mathbf{A}$  for  $k = 1$ , as it must be.

The most famous application is to a recurrence. A linear recurrence of order two becomes a first-order recurrence for vectors, whose solution is a matrix power.

**Example. (A Closed Formula for the Fibonacci Numbers)**

The Fibonacci numbers are defined by  $F_0 = 0$ ,  $F_1 = 1$  and  $F_{k+2} = F_{k+1} + F_k$  for  $k \geq 0$ . (Here  $F_k$  is a number, not a field; we work over  $\mathbb{R}$ .) Find a formula for  $F_k$  by diagonalizing a  $2 \times 2$  matrix.

*Solution. A vector recurrence.* Put  $\mathbf{x}_k = (F_{k+1}, F_k) \in \mathbb{R}^2$ . The recurrence says

$$\mathbf{x}_{k+1} = \begin{pmatrix} F_{k+2} \\ F_{k+1} \end{pmatrix} = \begin{pmatrix} F_{k+1} + F_k \\ F_{k+1} \end{pmatrix} = \mathbf{A}\mathbf{x}_k, \quad \mathbf{A} = \begin{pmatrix} 1 & 1 \\ 1 & 0 \end{pmatrix}.$$

By induction on  $k$ ,  $\mathbf{x}_k = \mathbf{A}^k \mathbf{x}_0$  with  $\mathbf{x}_0 = (1, 0)$ .

*Eigenvalues.*  $\text{tr } \mathbf{A} = 1$  and  $\det \mathbf{A} = -1$ , so  $p_{\mathbf{A}}(x) = x^2 - x - 1$ , with the two real roots

$$\varphi = \frac{1 + \sqrt{5}}{2}, \quad \psi = \frac{1 - \sqrt{5}}{2}.$$

They are distinct, so  $\mathbf{A}$  is diagonalizable over  $\mathbb{R}$  (Corollary 8.4.2). Each root satisfies  $\lambda^2 = \lambda + 1$ , and we will use  $\varphi + \psi = 1$ ,  $\varphi\psi = -1$  and  $\varphi - \psi = \sqrt{5}$ .

*Eigenvectors.* For a root  $\lambda$ , the vector  $(\lambda, 1)$  is an eigenvector:  $\mathbf{A}(\lambda, 1) = (\lambda + 1, \lambda) = (\lambda^2, \lambda) = \lambda(\lambda, 1)$ . So

$$\mathbf{P} = \begin{pmatrix} \varphi & \psi \\ 1 & 1 \end{pmatrix}, \quad \mathbf{D} = \text{diag}(\varphi, \psi), \quad \mathbf{P}^{-1} = \frac{1}{\sqrt{5}} \begin{pmatrix} 1 & -\psi \\ -1 & \varphi \end{pmatrix},$$

where  $\det \mathbf{P} = \varphi - \psi = \sqrt{5} \neq 0$  and  $\mathbf{P}^{-1}$  comes from Theorem 0.9.8.

*The power.* By Theorem 8.4.3,  $\mathbf{x}_k = \mathbf{P}\mathbf{D}^k\mathbf{P}^{-1}\mathbf{x}_0$ . Working from the right:

$$\begin{aligned} \mathbf{P}^{-1}\mathbf{x}_0 &= \frac{1}{\sqrt{5}} \begin{pmatrix} 1 \\ -1 \end{pmatrix}, \\ \mathbf{D}^k\mathbf{P}^{-1}\mathbf{x}_0 &= \frac{1}{\sqrt{5}} \begin{pmatrix} \varphi^k \\ -\psi^k \end{pmatrix}, \\ \mathbf{P}\mathbf{D}^k\mathbf{P}^{-1}\mathbf{x}_0 &= \frac{1}{\sqrt{5}} \begin{pmatrix} \varphi^{k+1} - \psi^{k+1} \\ \varphi^k - \psi^k \end{pmatrix}. \end{aligned}$$

The second entry of  $\mathbf{x}_k$  is  $F_k$ , so

$$F_k = \frac{\varphi^k - \psi^k}{\sqrt{5}} = \frac{1}{\sqrt{5}} \left[ \left( \frac{1 + \sqrt{5}}{2} \right)^k - \left( \frac{1 - \sqrt{5}}{2} \right)^k \right].$$

*Check.*  $k = 0$ : 0.  $k = 1$ :  $(\varphi - \psi)/\sqrt{5} = 1$ .  $k = 2$ :  $(\varphi^2 - \psi^2)/\sqrt{5} = (\varphi - \psi)(\varphi + \psi)/\sqrt{5} = 1$ . And the right side satisfies the recurrence, because  $\varphi^{k+2} = \varphi^{k+1} + \varphi^k$  (multiply  $\varphi^2 = \varphi + 1$  by  $\varphi^k$ ), and likewise for  $\psi$ .

Since  $|\psi| < 1$ , the term  $\psi^k/\sqrt{5}$  has absolute value less than  $1/2$ , so  $F_k$  is the integer nearest to  $\varphi^k/\sqrt{5}$ ; the Fibonacci numbers grow like powers of  $\varphi \approx 1.618$ , the larger eigenvalue.

The irrational numbers in the formula are forced. Over  $\mathbb{Q}$ , the polynomial  $x^2 - x - 1$  has no root (by [Theorem 5.3.6](#), the only candidates are  $\pm 1$ , and neither works), so  $\mathbf{A}$  has no eigenvalue in  $\mathbb{Q}$  and is **not** diagonalizable over  $\mathbb{Q}$ , although all its entries and powers are integers. Chapter 6 met the same numbers as determinants of tridiagonal matrices ([Example \(Fibonacci Numbers as Determinants\)](#)); here the recurrence is solved rather than just recognized. Section 11 of this chapter treats general linear recurrences and Markov chains the same way.

## 8.4.5 When diagonalization fails

Two examples collect the ways [Theorem 8.4.1](#) (e) can fail.

**Too few eigenvectors.**  $\mathbf{J} = \begin{pmatrix} 1 & 1 \\ 0 & 1 \end{pmatrix}$  is not diagonalizable over **any** field, because  $p_{\mathbf{J}} = (x-1)^2$  splits but  $g(1) = 1 < 2 = a(1)$ . Enlarging the field changes nothing: the rank of  $\mathbf{J} - \mathbf{I}$  is 1 over every field. Its powers are still easy,  $\mathbf{J}^k = \begin{pmatrix} 1 & k \\ 0 & 1 \end{pmatrix}$  by induction, but they are not of the form  $\mathbf{P} \operatorname{diag}(1, 1) \mathbf{P}^{-1} = \mathbf{I}$ ; the entry  $k$ , growing linearly rather than like a power, is the footprint of the missing eigenvector. Chapter 9 shows that such blocks are the only additional ingredient.

**Too small a field.** The rotation  $\mathbf{R} = \begin{pmatrix} 0 & -1 \\ 1 & 0 \end{pmatrix}$  has  $p_{\mathbf{R}} = x^2 + 1$ , which does not split over  $\mathbb{R}$ , so  $\mathbf{R}$  is not diagonalizable over  $\mathbb{R}$ . Over  $\mathbb{C}$  it has the distinct eigenvalues  $\pm i$  with eigenvectors  $(1, -i)$  and  $(1, i)$ , so with  $\mathbf{P} = \begin{pmatrix} 1 & 1 \\ -i & i \end{pmatrix}$ ,

$$\mathbf{P}^{-1} \mathbf{R} \mathbf{P} = \begin{pmatrix} i & 0 \\ 0 & -i \end{pmatrix}.$$

Then  $\mathbf{R}^k = \mathbf{P} \operatorname{diag}(i^k, (-i)^k) \mathbf{P}^{-1}$ , and since  $i^4 = 1$ ,  $\mathbf{R}^4 = \mathbf{I}$ : four quarter-turns make a full turn, as they should.

When a matrix depends on a parameter, the same test decides diagonalizability for every value at once.

### Example. (Diagonalizability with a Parameter)

For which  $t \in \mathbb{R}$  is  $\mathbf{A}_t = \begin{pmatrix} 2 & t & 0 \\ 0 & 2 & 0 \\ 0 & 0 & 3 \end{pmatrix}$  diagonalizable over  $\mathbb{R}$ ?

*Solution.* The matrix is upper triangular, so  $p_{\mathbf{A}_t}(x) = (x-2)^2(x-3)$  for every  $t$ , by [Theorem 6.4.6](#). This splits, with  $a(2) = 2$  and  $a(3) = 1$ . Since  $1 \leq g(3) \leq a(3) = 1$  by [Theorem 8.3.1](#), only  $\lambda = 2$  needs checking. Here

$$\mathbf{A}_t - 2\mathbf{I} = \begin{pmatrix} 0 & t & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 1 \end{pmatrix},$$

whose rank is 1 if  $t = 0$  and 2 if  $t \neq 0$ . By rank-nullity for matrices (Theorem 2.5.7),  $g(2) = 3 - \text{rank}(\mathbf{A}_t - 2\mathbf{I})$  is 2 when  $t = 0$  and 1 when  $t \neq 0$ . By Theorem 8.4.1 (e),  $\mathbf{A}_t$  is diagonalizable exactly when  $t = 0$ .

Notice what did not change: the eigenvalues and their algebraic multiplicities are the same for every  $t$ . Only the geometric multiplicity, a rank, detects the difference.

## 8.4.6 Exercises

### 8.4.6.1 A. Check your understanding

#### Exercise A1.

- Define what it means for  $\mathbf{A} \in M_n(F)$  to be diagonalizable over  $F$ .
- State the Diagonalization Theorem, with at least four equivalent conditions.
- True or false: a diagonalizable  $n \times n$  matrix has  $n$  distinct eigenvalues. Justify your answer.
- True or false: every invertible matrix is diagonalizable. Justify your answer.
- True or false: every diagonalizable matrix is invertible. Justify your answer.
- If  $\mathbf{P}^{-1}\mathbf{A}\mathbf{P} = \mathbf{D}$  is diagonal, what are the columns of  $\mathbf{P}$ , and how are they matched with the diagonal of  $\mathbf{D}$ ?

*Solution.*

- There are an invertible  $\mathbf{P} \in M_n(F)$  and a diagonal  $\mathbf{D} \in M_n(F)$  with  $\mathbf{P}^{-1}\mathbf{A}\mathbf{P} = \mathbf{D}$  (Definition 8.4.1).
- See Theorem 8.4.1: for  $T \in \mathcal{L}(V)$ ,  $\dim V = n \geq 1$ , with distinct eigenvalues  $\lambda_1, \dots, \lambda_k$ , the following are equivalent:  $T$  is diagonalizable;  $V$  has a basis of eigenvectors;  $V = \bigoplus_i E_{\lambda_i}(T)$ ;  $\sum_i \dim E_{\lambda_i}(T) = n$ ;  $p_T$  splits and  $g(\lambda_i) = a(\lambda_i)$  for all  $i$ .
- False.  $\mathbf{I}_n$  is diagonal, with the single eigenvalue 1.
- False.  $\begin{pmatrix} 1 & 1 \\ 0 & 1 \end{pmatrix}$  has determinant 1 but is not diagonalizable.
- False. The zero matrix is diagonal and not invertible.
- The columns of  $\mathbf{P}$  form a basis of eigenvectors of  $\mathbf{A}$ , and column  $j$  is an eigenvector for the  $j$ -th diagonal entry of  $\mathbf{D}$ , since  $\mathbf{A}\mathbf{P} = \mathbf{P}\mathbf{D}$ .

### 8.4.6.2 B. Practice

#### Exercise B1: Diagonalize or show impossible.

For each real matrix, either find an invertible  $\mathbf{P}$  and a diagonal  $\mathbf{D}$  with  $\mathbf{P}^{-1}\mathbf{A}\mathbf{P} = \mathbf{D}$ , or show that none exist.

- $\mathbf{A}_1 = \begin{pmatrix} 0 & 1 \\ -6 & 5 \end{pmatrix}$ .

$$(b) \mathbf{A}_2 = \begin{pmatrix} 1 & 1 & 1 \\ 2 & 0 & 1 \\ -3 & 3 & 2 \end{pmatrix}.$$

$$(c) \mathbf{A}_3 = \begin{pmatrix} -1 & 2 & -2 \\ -2 & 3 & -2 \\ 2 & -2 & 3 \end{pmatrix}.$$

*Solution.*

(a)  $\text{tr } \mathbf{A}_1 = 5$ ,  $\det \mathbf{A}_1 = 6$ , so  $p = x^2 - 5x + 6 = (x - 2)(x - 3)$ . The eigenvalues are distinct, so  $\mathbf{A}_1$  is diagonalizable (Corollary 8.4.2).  $\mathbf{A}_1 - 2\mathbf{I} = \begin{pmatrix} -2 & 1 \\ -6 & 3 \end{pmatrix}$  gives  $y = 2x$ , eigenvector  $(1, 2)$ ;  $\mathbf{A}_1 - 3\mathbf{I} = \begin{pmatrix} -3 & 1 \\ -6 & 2 \end{pmatrix}$  gives  $y = 3x$ , eigenvector  $(1, 3)$ . So

$$\mathbf{P} = \begin{pmatrix} 1 & 1 \\ 2 & 3 \end{pmatrix}, \quad \mathbf{D} = \text{diag}(2, 3), \quad \mathbf{P}^{-1} = \begin{pmatrix} 3 & -1 \\ -2 & 1 \end{pmatrix},$$

$$\text{and indeed } \mathbf{A}_1 \mathbf{P} = \begin{pmatrix} 2 & 3 \\ 4 & 9 \end{pmatrix} = \mathbf{P} \mathbf{D}.$$

(b) Expanding along the first row,

$$\begin{aligned} p(x) &= \det \begin{pmatrix} x-1 & -1 & -1 \\ -2 & x & -1 \\ 3 & -3 & x-2 \end{pmatrix} \\ &= (x-1)[x(x-2)-3] + [-2(x-2)+3] - [6-3x]. \end{aligned}$$

The first bracket is  $(x-3)(x+1)$ , and the last two terms add up to  $-2x+7-6+3x = x+1$ . So  $p(x) = (x+1)[(x-1)(x-3)+1] = (x+1)(x-2)^2$ , and  $a(2) = 2$ . Now

$$\mathbf{A}_2 - 2\mathbf{I} = \begin{pmatrix} -1 & 1 & 1 \\ 2 & -2 & 1 \\ -3 & 3 & 0 \end{pmatrix}.$$

Row 3 equals row 1 minus row 2, and rows 1 and 2 are not proportional, so the rank is 2 and  $g(2) = 1 < 2 = a(2)$ . By Theorem 8.4.1 (e),  $\mathbf{A}_2$  is **not** diagonalizable, over  $\mathbb{R}$  or over  $\mathbb{C}$ .

(c) Expanding along the first row,

$$\begin{aligned} p(x) &= \det \begin{pmatrix} x+1 & -2 & 2 \\ 2 & x-3 & 2 \\ -2 & 2 & x-3 \end{pmatrix} \\ &= (x+1)[(x-3)^2-4] + 2[2(x-3)+4] \\ &\quad + 2[4+2(x-3)]. \end{aligned}$$

The first bracket is  $(x-5)(x-1)$  and the other two are  $2(x-1)$  each, so  $p(x) = (x-1)[(x+1)(x-5)+8] = (x-1)(x^2-4x+3) = (x-1)^2(x-3)$ . For  $\lambda = 1$ ,

$\mathbf{A}_3 - \mathbf{I} = \begin{pmatrix} -2 & 2 & -2 \\ -2 & 2 & -2 \\ 2 & -2 & 2 \end{pmatrix}$  has rank 1, so  $g(1) = 2 = a(1)$ , and  $E_1$  is the plane  $x - y + z = 0$ , with basis  $(1, 1, 0), (-1, 0, 1)$ . Also  $g(3) = 1 = a(3)$ , so  $\mathbf{A}_3$  is

diagonalizable. From  $\mathbf{A}_3 - 3\mathbf{I} = \begin{pmatrix} -4 & 2 & -2 \\ -2 & 0 & -2 \\ 2 & -2 & 0 \end{pmatrix}$ , row 3 gives  $y = x$  and row 2 gives  $z = -x$ , so  $E_3 = \text{span}((1, 1, -1))$ . Hence

$$\mathbf{P} = \begin{pmatrix} 1 & -1 & 1 \\ 1 & 0 & 1 \\ 0 & 1 & -1 \end{pmatrix}, \quad \mathbf{D} = \text{diag}(1, 1, 3).$$

Check:  $\mathbf{A}_3(1, 1, 0) = (1, 1, 0)$ ,  $\mathbf{A}_3(-1, 0, 1) = (-1, 0, 1)$ ,  $\mathbf{A}_3(1, 1, -1) = (3, 3, -3)$ , so  $\mathbf{A}_3\mathbf{P} = \mathbf{PD}$ ; and  $\det \mathbf{P} = 1(0 - 1) + 1(-1 - 0) + 1(1 - 0) = -1 \neq 0$ .

### Exercise B2: A tenth power.

Let  $\mathbf{A} = \begin{pmatrix} 1 & 2 \\ 2 & 1 \end{pmatrix} \in M_2(\mathbb{R})$ .

- (a) Diagonalize  $\mathbf{A}$ .
- (b) Hence find a formula for  $\mathbf{A}^k$ ,  $k \in \mathbb{N}$ , and compute  $\mathbf{A}^{10}$ .

*Solution.*

- (a)  $\text{tr } \mathbf{A} = 2$ ,  $\det \mathbf{A} = -3$ , so  $p = x^2 - 2x - 3 = (x - 3)(x + 1)$ .  $\mathbf{A} - 3\mathbf{I} = \begin{pmatrix} -2 & 2 \\ 2 & -2 \end{pmatrix}$  gives the eigenvector  $(1, 1)$ , and  $\mathbf{A} + \mathbf{I} = \begin{pmatrix} 2 & 2 \\ 2 & 2 \end{pmatrix}$  gives  $(1, -1)$ . So  $\mathbf{P} = \begin{pmatrix} 1 & 1 \\ 1 & -1 \end{pmatrix}$ ,  $\mathbf{D} = \text{diag}(3, -1)$ , and  $\mathbf{P}^{-1} = \frac{1}{2} \begin{pmatrix} 1 & 1 \\ 1 & -1 \end{pmatrix}$  by [Theorem 0.9.8](#).
- (b) By [Theorem 8.4.3](#),

$$\mathbf{A}^k = \frac{1}{2} \begin{pmatrix} 1 & 1 \\ 1 & -1 \end{pmatrix} \begin{pmatrix} 3^k & 0 \\ 0 & (-1)^k \end{pmatrix} \begin{pmatrix} 1 & 1 \\ 1 & -1 \end{pmatrix} = \frac{1}{2} \begin{pmatrix} 3^k + (-1)^k & 3^k - (-1)^k \\ 3^k - (-1)^k & 3^k + (-1)^k \end{pmatrix}.$$

For  $k = 1$  this gives  $\frac{1}{2} \begin{pmatrix} 2 & 4 \\ 4 & 2 \end{pmatrix} = \mathbf{A}$ . With  $3^{10} = 59049$  and  $(-1)^{10} = 1$ ,

$$\mathbf{A}^{10} = \begin{pmatrix} 29525 & 29524 \\ 29524 & 29525 \end{pmatrix}.$$

### Exercise B3: A matrix with a parameter.

For  $t \in \mathbb{R}$ , let  $\mathbf{A}_t = \begin{pmatrix} 1 & 1 & 0 \\ 0 & t & 0 \\ 0 & 0 & 2 \end{pmatrix}$ . Determine all  $t$  for which  $\mathbf{A}_t$  is diagonalizable over  $\mathbb{R}$ . Justify your answer.

*Solution.*  $\mathbf{A}_t$  is upper triangular, so  $p(x) = (x - 1)(x - t)(x - 2)$ , which splits over  $\mathbb{R}$ .

*Case  $t \notin \{1, 2\}$ .* The eigenvalues  $1, t, 2$  are distinct, so  $\mathbf{A}_t$  is diagonalizable ([Corollary 8.4.2](#)).



Case  $t = 1$ . Then  $a(1) = 2$ , and  $\mathbf{A}_1 - \mathbf{I} = \begin{pmatrix} 0 & 1 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 1 \end{pmatrix}$  has rank 2, so  $g(1) = 1 < 2$ . Not diagonalizable.

Case  $t = 2$ . Then  $a(2) = 2$ , and  $\mathbf{A}_2 - 2\mathbf{I} = \begin{pmatrix} -1 & 1 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{pmatrix}$  has rank 1, so  $g(2) = 2 = a(2)$ ; also  $g(1) = a(1) = 1$ . Diagonalizable, with eigenvectors  $\mathbf{e}_1$  for 1 and  $(1, 1, 0), \mathbf{e}_3$  for 2.

Hence  $\mathbf{A}_t$  is diagonalizable over  $\mathbb{R}$  if and only if  $t \neq 1$ . In both exceptional cases a diagonal entry is repeated, but only for  $t = 1$  is an eigenvector lost: the entry 1 in position  $(1, 2)$  couples  $\mathbf{e}_1$  and  $\mathbf{e}_2$ , and it matters only when their diagonal entries agree.

### 8.4.6.3 C. Going deeper

#### Exercise C1: Matrices with $\mathbf{A}^k = \mathbf{I}$ .

Let  $\mathbf{A} \in M_n(\mathbb{C})$ ,  $n \geq 1$ , and  $k \geq 1$ .

- (a) Prove that if  $\mathbf{A}^k = \mathbf{I}$ , then every eigenvalue  $\lambda$  of  $\mathbf{A}$  satisfies  $\lambda^k = 1$ .
- (b) Prove that if  $\mathbf{A}$  is diagonalizable and every eigenvalue satisfies  $\lambda^k = 1$ , then  $\mathbf{A}^k = \mathbf{I}$ .
- (c) Show that the hypothesis “diagonalizable” in (b) cannot be dropped.

*Solution.*

- (a) Let  $\mathbf{A}\mathbf{v} = \lambda\mathbf{v}$  with  $\mathbf{v} \neq \mathbf{0}$ . By [Exercise \(B3: Eigenvalues of powers and inverses\)](#) (a),  $\mathbf{v} = \mathbf{I}\mathbf{v} = \mathbf{A}^k\mathbf{v} = \lambda^k\mathbf{v}$ , so  $(\lambda^k - 1)\mathbf{v} = \mathbf{0}$  and  $\lambda^k = 1$  by [Theorem 1.1.9](#).
- (b) Write  $\mathbf{A} = \mathbf{P}\mathbf{D}\mathbf{P}^{-1}$  with  $\mathbf{D} = \text{diag}(d_1, \dots, d_n)$  ([Definition 8.4.1](#)). The  $d_i$  are eigenvalues of  $\mathbf{A}$ :  $\mathbf{D}$  and  $\mathbf{A}$  are similar, so  $p_{\mathbf{D}} = p_{\mathbf{A}}$  ([Theorem 6.10.3](#)), and each  $d_i$  is a root of  $p_{\mathbf{D}} = \prod_j (x - d_j)$ . So  $d_i^k = 1$  for all  $i$ , and by [Theorem 8.4.3](#),  $\mathbf{A}^k = \mathbf{P} \text{diag}(d_1^k, \dots, d_n^k) \mathbf{P}^{-1} = \mathbf{P}\mathbf{I}\mathbf{P}^{-1} = \mathbf{I}$ .
- (c) Let  $\mathbf{J} = \begin{pmatrix} 1 & 1 \\ 0 & 1 \end{pmatrix}$ . Its only eigenvalue is 1, and  $1^k = 1$ . But  $\mathbf{J}^k = \begin{pmatrix} 1 & k \\ 0 & 1 \end{pmatrix}$  by induction ( $\mathbf{J}^{k+1} = \mathbf{J}^k\mathbf{J}$ ), which is not  $\mathbf{I}$  since  $k \neq 0$  in  $\mathbb{C}$ . (Section 8 proves the converse of (b) in a strong form: over  $\mathbb{C}$ ,  $\mathbf{A}^k = \mathbf{I}$  forces  $\mathbf{A}$  to be diagonalizable.)

#### Exercise C2: Polynomials of diagonalizable matrices.

- (a) Prove that if  $\mathbf{A} \in M_n(F)$  is diagonalizable over  $F$ , then so is  $p(\mathbf{A})$  for every  $p \in F[x]$ , and that every eigenvalue of  $p(\mathbf{A})$  has the form  $p(\lambda)$  with  $\lambda \in \text{spec}(\mathbf{A})$ .
- (b) Show that the converse of the first statement fails: find a non-diagonalizable  $\mathbf{A}$  and a polynomial  $p$  with  $p(\mathbf{A})$  diagonalizable.

*Solution.*

- (a) Let  $\mathbf{A} = \mathbf{P}\mathbf{D}\mathbf{P}^{-1}$ ,  $\mathbf{D} = \text{diag}(d_1, \dots, d_n)$ . By [Theorem 8.4.3](#),  $p(\mathbf{A}) = \mathbf{P} \text{diag}(p(d_1), \dots, p(d_n)) \mathbf{P}^{-1}$ , so  $p(\mathbf{A})$  is similar to a diagonal matrix, that is, diagonalizable. By [Theorem 6.10.3](#),

$p_p(\mathbf{A}) = \prod_i (x - p(d_i))$ , so by [Theorem 8.2.1](#) every eigenvalue of  $p(\mathbf{A})$  is some  $p(d_i)$ . As in [Exercise \(C1: Matrices with  \$\mathbf{A}^k = \mathbf{I}\$ \)](#) (b), each  $d_i$  is an eigenvalue of  $\mathbf{A}$ .

- (b) Let  $\mathbf{A} = \begin{pmatrix} 0 & 1 \\ 0 & 0 \end{pmatrix}$  and  $p = x^2$ . Then  $\mathbf{A}$  is not diagonalizable:  $p_{\mathbf{A}} = x^2$  gives  $a(0) = 2$ , while  $\text{rank } \mathbf{A} = 1$  gives  $g(0) = 1$  ([Theorem 8.4.1](#) (e)). But  $p(\mathbf{A}) = \mathbf{A}^2 = 0$  is diagonal.

### Exercise C3: The transpose map on $M_n(F)$ .

Let  $n \geq 1$  and  $\tau: M_n(F) \rightarrow M_n(F)$ ,  $\tau(\mathbf{X}) = \mathbf{X}^\top$ .

- Suppose the characteristic of  $F$  is not 2. Prove that  $\tau$  is diagonalizable, and find  $\dim E_1(\tau)$  and  $\dim E_{-1}(\tau)$ .
- Under the same hypothesis, deduce  $p_\tau$ ,  $\text{tr } \tau$  and  $\det \tau$ .
- Now let  $F = \mathbb{F}_2$  and  $n \geq 2$ . Prove that  $\tau$  is **not** diagonalizable.

*Hint: for the dimensions, use the matrices  $\mathbf{E}_{ij} + \mathbf{E}_{ji}$  and  $\mathbf{E}_{ij} - \mathbf{E}_{ji}$ .*

*Solution.*

- (a) Since  $(\mathbf{X}^\top)^\top = \mathbf{X}$  ([Theorem 0.9.5](#)),  $\tau^2 = \text{id}$ . By [Theorem 3.10.5](#),  $M_n(F) = E_+ \oplus E_-$ , where  $E_+ = E_1(\tau)$  is the space of symmetric matrices and  $E_- = E_{-1}(\tau)$  the space of skew-symmetric matrices.

A symmetric  $\mathbf{X} = (x_{ij})$  is determined by its entries on and above the diagonal, and  $\mathbf{X} = \sum_i x_{ii} \mathbf{E}_{ii} + \sum_{i < j} x_{ij} (\mathbf{E}_{ij} + \mathbf{E}_{ji})$ . These  $n + \binom{n}{2}$  matrices are symmetric and independent, since their non-zero entries sit in disjoint positions. So  $\dim E_1(\tau) = n(n+1)/2$ . A skew-symmetric  $\mathbf{X}$  has  $x_{ii} = -x_{ii}$ , so  $2x_{ii} = 0$  and  $x_{ii} = 0$  (as  $2 \neq 0$ ); and  $\mathbf{X} = \sum_{i < j} x_{ij} (\mathbf{E}_{ij} - \mathbf{E}_{ji})$ , with these  $\binom{n}{2}$  matrices independent for the same reason. So  $\dim E_{-1}(\tau) = n(n-1)/2$ .

The eigenvalues of  $\tau$  lie among  $\pm 1$  (as in [Exercise \(B2: The transpose map\)](#)), and  $n(n+1)/2 + n(n-1)/2 = n^2 = \dim M_n(F)$ . If  $n \geq 2$ , both eigenspaces are non-zero, and the dimensions of the eigenspaces add up to  $n^2$ ; if  $n = 1$ ,  $\tau = \text{id}$  and  $E_1(\tau)$  is everything. Either way [Theorem 8.4.1](#) ((d)  $\Rightarrow$  (a)) shows that  $\tau$  is diagonalizable.

- (b) In a basis of eigenvectors,  $[\tau] = \text{diag}(1, \dots, 1, -1, \dots, -1)$  with  $n(n+1)/2$  ones and  $n(n-1)/2$  entries  $-1$  ([Theorem 8.4.1](#)). This matrix is triangular, so

$$\begin{aligned} p_\tau &= (x-1)^{n(n+1)/2} (x+1)^{n(n-1)/2}, \\ \text{tr } \tau &= \frac{n(n+1)}{2} - \frac{n(n-1)}{2} = n, \\ \det \tau &= (-1)^{n(n-1)/2}. \end{aligned}$$

As a check on the trace: in the basis  $(\mathbf{E}_{ij})$ ,  $\tau(\mathbf{E}_{ij}) = \mathbf{E}_{ji}$ , so the diagonal entry of  $[\tau]$  at  $\mathbf{E}_{ij}$  is 1 if  $i = j$  and 0 otherwise, and the trace is  $n$ .

- (c) Over  $\mathbb{F}_2$ ,  $-1 = 1$ . If  $\tau(\mathbf{X}) = \lambda \mathbf{X}$  with  $\mathbf{X} \neq 0$ , then as in [Exercise \(B2: The transpose map\)](#) (a),  $\lambda^2 = 1$ , so  $(\lambda - 1)^2 = \lambda^2 - 2\lambda + 1 = \lambda^2 + 1 = 0$  and  $\lambda = 1$ . So  $\text{spec}(\tau) = \{1\}$  and  $E_1(\tau)$  is the space of symmetric matrices. The spanning argument in (a) did not divide by 2, so  $\dim E_1(\tau) = n(n+1)/2$ . For  $n \geq 2$ ,  $n(n+1)/2 < n^2$ , so condition (d) of [Theorem 8.4.1](#) fails, and  $\tau$  is not diagonalizable.

For instance,  $\mathbf{E}_{12}$  is not a sum of eigenvectors.

## 8.5 The Minimal Polynomial

Section 4 decided diagonalizability by counting eigenvectors, one eigenspace at a time. There is a second source of information about an operator, which Chapter 5 set up: the polynomials that kill it. On a finite-dimensional space they form an ideal of  $F[x]$ , and that ideal has a single monic generator. This section gives the generator its name, the minimal polynomial, and shows that its roots are exactly the eigenvalues. We learn two ways to compute it and check that similar matrices share it. The next section compares it with the characteristic polynomial, and Section 8 reads diagonalizability off its factorization.

### 8.5.1 The polynomial that divides all the others

Chapter 5 turned every polynomial identity satisfied by an operator into a statement about one ideal. For  $T \in \mathcal{L}(V)$ , the set

$$I_T = \{p \in F[x] : p(T) = 0\}$$

is an ideal of  $F[x]$  ([Theorem 5.7.3 \(a\)](#)). If  $V$  is finite-dimensional, it contains a non-zero polynomial ([Theorem 5.7.3 \(c\)](#)). Then it consists of the multiples of a unique monic polynomial ([Theorem 5.7.3 \(b\)](#)).

Look at what this says for one operator. A projection  $P$  with  $P \neq 0, \text{id}_V$  satisfies  $P^2 = P$ , so  $x^2 - x$  kills it. So do  $x^3 - x^2$ ,  $(x^2 - x)(x + 5)$  and  $x^{10} - x^9$ , and a short calculation shows each one is a multiple of  $x^2 - x$ . This is no accident. Every annihilating polynomial of  $P$  is a multiple of  $x^2 - x$ , which Chapter 5 found to be the monic generator of  $I_P$ . One polynomial controls the whole list, and whenever something in mathematics is unique we give it a name.

*The minimal polynomial of  $T$  is the one monic polynomial that kills  $T$  and divides every other polynomial that kills  $T$ .*

#### Definition 8.5.1 — Minimal Polynomial

Let  $V$  be a **finite-dimensional** vector space over  $F$  and  $T \in \mathcal{L}(V)$ . The **minimal polynomial** of  $T$  is the unique **monic** polynomial  $m_T \in F[x]$  with

$$I_T = \{p \in F[x] : p(T) = 0\} = \langle m_T \rangle.$$

Equivalently,  $m_T$  is the monic polynomial of **least degree** with  $m_T(T) = 0$ .

For  $\mathbf{A} \in M_n(F)$ , the minimal polynomial is  $m_{\mathbf{A}} := m_{T_{\mathbf{A}}}$ : the monic polynomial of least degree with  $m_{\mathbf{A}}(\mathbf{A}) = 0$ .

In words: we look at **all** polynomials  $p$  with  $p(T) = 0$ , where  $p(T)$  replaces  $x^k$  by  $T^k$  and the constant term  $c$  by  $c \text{id}_V$  ([Definition 3.4.2](#)). They are exactly the multiples of one polynomial  $m_T$ . Among the generators of this ideal, which differ by non-zero constant factors, we pick **the monic** one. The second sentence of the definition gives the same polynomial described by size: it is the monic polynomial of **least degree** that kills  $T$ .

**Well-definedness.** Three things need checking, and Chapter 5 did all three. *Existence:*  $I_T \neq \{0\}$  because  $V$  is finite-dimensional (Theorem 5.7.3 (c)). *Uniqueness:* a non-zero ideal of  $F[x]$  has exactly one monic generator (Theorem 5.2.2). *The two descriptions agree:* the same theorem says that the monic generator is the unique monic polynomial of least degree in the ideal. For a matrix,  $p(T_{\mathbf{A}}) = 0$  if and only if  $p(\mathbf{A}) = 0$ , because  $[p(T_{\mathbf{A}})]_{\mathcal{E}} = p(\mathbf{A})$  in the standard basis  $\mathcal{E}$  (Corollary 3.6.3). So  $I_{T_{\mathbf{A}}} = \{p : p(\mathbf{A}) = 0\}$ , and the matrix version is the operator version for  $T_{\mathbf{A}}$ .

The name records the second description:  $m_T$  is the annihilating polynomial of minimal degree. The first description is the one we use in proofs, because it hands us a divisibility for free.

### Theorem 8.5.1 — Annihilating Polynomials Are Multiples of the Minimal Polynomial

Let  $V$  be finite-dimensional and  $T \in \mathcal{L}(V)$ . For every  $p \in F[x]$ ,

$$p(T) = 0 \iff m_T \mid p.$$

In particular, a non-zero polynomial that annihilates  $T$  has degree at least  $\deg m_T$ . If  $V \neq \{0\}$ , then  $\deg m_T \geq 1$ .

*Proof.* By Definition 8.5.1,  $p(T) = 0$  means  $p \in \langle m_T \rangle$ , that is,  $p = q m_T$  for some  $q \in F[x]$ , that is,  $m_T \mid p$ . If  $p \neq 0$  and  $m_T \mid p$ , then  $\deg m_T \leq \deg p$  by Proposition 5.1.2 (c). Finally, the only monic polynomial of degree 0 is the constant 1, and  $1(T) = \text{id}_V \neq 0$  when  $V \neq \{0\}$ ; so  $m_T \neq 1$  and  $\deg m_T \geq 1$ . ■

This is the move we will use again and again: **a polynomial relation becomes a divisibility**. If we learn that  $T^3 = T$ , we do not need to know anything else about  $T$  to conclude that  $m_T \mid x^3 - x = x(x-1)(x+1)$ . That leaves only finitely many candidates for  $m_T$ . To list candidates like these we need to know the monic divisors of a factored polynomial, and unique factorization describes them exactly.

### Lemma 8.5.2 — Monic Divisors of a Factored Polynomial

Let  $f = p_1^{e_1} \cdots p_k^{e_k} \in F[x]$ , where  $k \geq 1$ ,  $p_1, \dots, p_k$  are **distinct** monic irreducible polynomials and  $e_i \geq 1$ . Then the monic divisors of  $f$  are exactly the polynomials

$$p_1^{f_1} \cdots p_k^{f_k} \quad \text{with } 0 \leq f_i \leq e_i \text{ for each } i.$$

*Idea.* One direction is a product: the missing factors form the cofactor. For the other, factor a monic divisor  $d$  into irreducibles, multiply that factorization by one of the cofactors, and compare the result with the factorization of  $f$ . Uniqueness says the two lists of irreducibles agree, so the irreducibles in  $d$  are among the  $p_i$ , and none can occur more often in  $d$  than in  $f$ .

*Proof.* Each such product divides  $f$ , with cofactor  $p_1^{e_1-f_1} \cdots p_k^{e_k-f_k}$ . Conversely, let  $d$  be monic with  $f = dh$ . If  $d = 1$ , take every  $f_i = 0$ . Otherwise  $d$  is non-constant, and by Theorem 5.3.5 (a) it is a product  $q_1 \cdots q_s$  of monic irreducibles; the constant in front is

the leading coefficient of  $d$ , which is 1. The cofactor  $h$  is non-zero (as  $f \neq 0$ ), so it is either a non-zero constant  $c$  or, by the same theorem,  $c r_1 \cdots r_t$  with  $r_j$  monic irreducible. Multiplying,  $f = c q_1 \cdots q_s r_1 \cdots r_t$  (with no  $r$ 's in the first case) is a factorization of  $f$  into a constant and monic irreducibles. By [Theorem 5.3.5](#) (b), after renumbering it is the factorization  $f = 1 \cdot p_1^{e_1} \cdots p_k^{e_k}$ , in which  $p_i$  occurs exactly  $e_i$  times. So each  $q_j$  is one of the  $p_i$ , and  $p_i$  occurs at most  $e_i$  times among  $q_1, \dots, q_s$ . Hence  $d = p_1^{f_1} \cdots p_k^{f_k}$  with  $0 \leq f_i \leq e_i$ . ■

**Examples.** Each computation follows the same plan: find one annihilating polynomial, list its monic divisors with [Lemma 8.5.2](#), and discard those that do not annihilate.

- **A diagonal matrix with a repeated entry.** Let  $\mathbf{A} = \text{diag}(1, 1, 2)$ . Diagonal matrices multiply entry by entry, so

$$(\mathbf{A} - \mathbf{I})(\mathbf{A} - 2\mathbf{I}) = \text{diag}(0, 0, 1) \text{diag}(-1, -1, 0) = 0.$$

By [Theorem 8.5.1](#),  $m_{\mathbf{A}}$  is a monic divisor of  $(x-1)(x-2)$ : one of 1,  $x-1$ ,  $x-2$ ,  $(x-1)(x-2)$ . None of the first three annihilates  $\mathbf{A}$ , since  $\mathbf{I}$ ,  $\mathbf{A} - \mathbf{I} = \text{diag}(0, 0, 1)$  and  $\mathbf{A} - 2\mathbf{I} = \text{diag}(-1, -1, 0)$  are non-zero. So  $m_{\mathbf{A}} = (x-1)(x-2)$ . The repeated diagonal entry 1 appears only **once** in  $m_{\mathbf{A}}$ , while the characteristic polynomial is  $p_{\mathbf{A}} = (x-1)^2(x-2)$ .

- **A matrix with too few eigenvectors.** Let  $\mathbf{J} = \begin{pmatrix} 1 & 1 \\ 0 & 1 \end{pmatrix}$  over any field. Then  $\mathbf{J} - \mathbf{I} = \begin{pmatrix} 0 & 1 \\ 0 & 0 \end{pmatrix} \neq 0$ , and  $(\mathbf{J} - \mathbf{I})^2 = 0$ . So  $m_{\mathbf{J}}$  is a monic divisor of  $(x-1)^2$ , namely 1,  $x-1$  or  $(x-1)^2$ , and the first two do not annihilate  $\mathbf{J}$ . Hence  $m_{\mathbf{J}} = (x-1)^2$ , which equals  $p_{\mathbf{J}}$ .
- **Projections.** Chapter 5 found, by this plan, that a projection  $P$  with  $P \neq 0$  and  $P \neq \text{id}_V$  has  $m_P = x^2 - x$  (the examples after [Theorem 5.7.3](#)).
- **Differentiation.** Let  $D$  be differentiation on  $\mathbb{R}[x]_{\leq n}$ . Then  $D^{n+1} = 0$  and  $D^n \neq 0$  ([Example \(Differentiation Is Nilpotent\)](#)). By [Lemma 8.5.2](#), the monic divisors of  $x^{n+1}$  are the powers  $x^j$  with  $0 \leq j \leq n+1$ . For  $j \leq n$ ,  $D^j \neq 0$ , since  $D^n = D^{n-j}D^j$  would otherwise be 0. Hence  $m_D = x^{n+1}$ , of degree  $n+1 = \dim \mathbb{R}[x]_{\leq n}$ .
- **Degenerate cases.** On  $V \neq \{0\}$ , the identity has  $m = x-1$  and the zero operator has  $m = x$ : each is killed by a polynomial of degree 1, which is the least possible degree. More generally  $m_{c \text{id}_V} = x-c$ , and every  $1 \times 1$  matrix  $(c)$  has  $m = x-c$ . On  $V = \{0\}$ , even the constant 1 annihilates, so  $m_T = 1$ . This last case is the reason for the hypothesis  $V \neq \{0\}$  in [Theorem 8.5.1](#).

**Non-example by minimal change.** For  $\mathbf{J}$  above, take  $q = x-1$  instead of  $(x-1)^2$ . It is still monic, it has the least degree a minimal polynomial can have on a non-zero space, and its only root is the eigenvalue 1 of  $\mathbf{J}$ . What fails is the clause  $q(\mathbf{J}) = 0$ :  $q(\mathbf{J}) = \mathbf{J} - \mathbf{I} \neq 0$ . In the other direction,  $(x-1)^3$  is monic and does annihilate  $\mathbf{J}$ , but it fails the clause  $I_{\mathbf{J}} = \langle q \rangle$ : the annihilating polynomial  $(x-1)^2$  is not a multiple of  $(x-1)^3$ .

*Quick check.* Find the minimal polynomial of  $2\mathbf{I}_3$  and of  $\mathbf{N} = \begin{pmatrix} 0 & 1 & 0 \\ 0 & 0 & 1 \\ 0 & 0 & 0 \end{pmatrix} \in M_3(F)$ .

*Solution.*  $2\mathbf{I}_3 - 2\mathbf{I}_3 = 0$ , so  $m_{2\mathbf{I}_3}$  divides  $x-2$ ; it is not 1, since  $\mathbf{I}_3 \neq 0$ . Hence  $m_{2\mathbf{I}_3} = x-2$ . For  $\mathbf{N}$ :  $\mathbf{N}\mathbf{e}_1 = 0$ ,  $\mathbf{N}\mathbf{e}_2 = \mathbf{e}_1$ ,  $\mathbf{N}\mathbf{e}_3 = \mathbf{e}_2$ , so  $\mathbf{N}^2\mathbf{e}_3 = \mathbf{e}_1 \neq 0$  and  $\mathbf{N}^3$  kills  $\mathbf{e}_1, \mathbf{e}_2, \mathbf{e}_3$ . Thus

$\mathbf{N}^3 = 0 \neq \mathbf{N}^2$ . By Lemma 8.5.2,  $m_{\mathbf{N}} \in \{1, x, x^2, x^3\}$ , and only  $x^3$  annihilates  $\mathbf{N}$  (the others give  $\mathbf{I}_3, \mathbf{N}, \mathbf{N}^2 \neq 0$ ). So  $m_{\mathbf{N}} = x^3$ .

**Why this definition.** Each clause has a job.

- *Monic.* The polynomials  $m$  and  $5m$  generate the same ideal, so “a generator” is determined only up to a non-zero constant. Insisting on leading coefficient 1 picks one, and makes statements like “ $m_{\mathbf{A}} = (x-1)(x-2)$ ” meaningful.
- *Generator, not only least degree.* The two descriptions give the same polynomial. Stating the definition through the ideal puts Theorem 8.5.1 in front of us, and divisibility is the property we use.
- *Finite dimension.* Without it,  $I_T$  can be  $\{0\}$ , and then there is nothing to name. For multiplication by  $x$  on  $F[x]$ ,  $p(S)(1) = p$ , so  $p(S) \neq 0$  for every  $p \neq 0$  (the remark after Theorem 5.7.2). The same happens for the right shift on  $F^{\mathbb{N}}$ .

*Warning.* **The minimal polynomial is not the characteristic polynomial.** For  $\mathbf{I}_2$ ,  $m = x - 1$  while  $p = (x - 1)^2$ ; for  $\text{diag}(1, 1, 2)$  above,  $m = (x - 1)(x - 2)$  while  $p = (x - 1)^2(x - 2)$ . The two can differ in their exponents, and  $m$  can have smaller degree than the size of the matrix. How they are related is the subject of the next section. Its main theorem is one that Chapter 6 warned against “proving” by substituting  $x = \mathbf{A}$  into  $\det(x\mathbf{I} - \mathbf{A})$ ; that shortcut is not a proof and must not be used here either.

## 8.5.2 The roots are the eigenvalues

The minimal polynomial is defined by an ideal, and eigenvalues by vectors. They are closely linked.

### Theorem 8.5.3 — Roots of the Minimal Polynomial

Let  $V$  be finite-dimensional,  $T \in \mathcal{L}(V)$  and  $\lambda \in F$ . Then  $m_T(\lambda) = 0$  if and only if  $\lambda$  is an eigenvalue of  $T$ . Consequently, if  $\dim V \geq 1$ , then  $m_T$  and  $p_T$  have the same roots in  $F$ .

*Idea.* ( $\Leftarrow$ ) is a one-liner: apply  $m_T(T) = 0$  to an eigenvector  $\mathbf{v}$ . Since  $T$  scales  $\mathbf{v}$  by  $\lambda$ , every polynomial in  $T$  scales  $\mathbf{v}$  by the value of the polynomial at  $\lambda$ . For ( $\Rightarrow$ ), a root gives a factor:  $m_T = (x - \lambda)q$ . The cofactor  $q$  has smaller degree, so by minimality it does **not** kill  $T$ , and some vector  $\mathbf{w}$  survives it. Then  $q(T)\mathbf{w}$  is non-zero and is killed by  $T - \lambda \text{id}_V$ . The eigenvector is manufactured by the part of  $m_T$  we removed.

*Proof.* ( $\Leftarrow$ ) Let  $\mathbf{v} \neq \mathbf{0}$  with  $T\mathbf{v} = \lambda\mathbf{v}$ . By Exercise (B3: Eigenvalues of powers and inverses) (a),  $p(T)\mathbf{v} = p(\lambda)\mathbf{v}$  for every  $p \in F[x]$ . With  $p = m_T$ ,

$$\mathbf{0} = m_T(T)\mathbf{v} = m_T(\lambda)\mathbf{v}.$$

Since  $\mathbf{v} \neq \mathbf{0}$ , Theorem 1.1.9 gives  $m_T(\lambda) = 0$ .

( $\Rightarrow$ ) Suppose  $m_T(\lambda) = 0$ . By Theorem 5.1.3 (b),  $m_T = (x - \lambda)q$  for some  $q \in F[x]$ , and  $q$  is monic with  $\deg q = \deg m_T - 1$  by Theorem 0.8.3. By Theorem 8.5.1, a non-zero polynomial of degree less than  $\deg m_T$  does not annihilate  $T$ ; so  $q(T) \neq 0$ , and there is

$\mathbf{w} \in V$  with  $\mathbf{v} := q(T)\mathbf{w} \neq \mathbf{0}$ . By [Theorem 5.7.1](#) (b),

$$(T - \lambda \operatorname{id}_V)\mathbf{v} = (T - \lambda \operatorname{id}_V)q(T)\mathbf{w} = m_T(T)\mathbf{w} = \mathbf{0}.$$

So  $T\mathbf{v} = \lambda\mathbf{v}$  with  $\mathbf{v} \neq \mathbf{0}$ , and  $\lambda$  is an eigenvalue of  $T$ .

If  $\dim V \geq 1$ , the eigenvalues of  $T$  are exactly the roots of  $p_T$  in  $F$  ([Theorem 8.2.1](#), (a)  $\Leftrightarrow$  (e)), so  $m_T$  and  $p_T$  have the same roots in  $F$ . This proves the theorem. ■

So the minimal polynomial knows the spectrum, just as the characteristic polynomial does, although the multiplicities of the roots may differ. The examples bear this out.  $\operatorname{diag}(1, 1, 2)$  has spectrum  $\{1, 2\}$ , the roots of  $(x - 1)(x - 2)$ .  $\mathbf{J}$  has spectrum  $\{1\}$ , the root of  $(x - 1)^2$ . The rotation  $R$  of  $\mathbb{R}^2$  by a right angle has  $m_R = x^2 + 1$  (the check after [Theorem 5.7.3](#)), with no real root, and indeed  $\operatorname{spec}(R) = \emptyset$  over  $\mathbb{R}$ . Its minimal polynomial has no linear factor at all.

Two consequences are worth stating.

- **Invertibility.** Taking  $\lambda = 0$ : for  $V \neq \{\mathbf{0}\}$ ,  $T$  is invertible if and only if  $m_T(0) \neq 0$ . Indeed  $m_T(0) = 0$  if and only if  $0 \in \operatorname{spec}(T)$ , if and only if  $T = T - 0 \operatorname{id}_V$  is not invertible ([Theorem 8.2.1](#), (a)  $\Leftrightarrow$  (c) with  $\lambda = 0$ ; for matrices this is [Theorem 8.2.6](#)). Chapter 5 proved the same statement without eigenvalues ([Exercise \(C1: Inverses are polynomials\)](#) (c)).
- **Over  $\mathbb{C}$ .** If  $F = \mathbb{C}$  and  $V \neq \{\mathbf{0}\}$ , then  $m_T$  splits ([Corollary 5.5.5](#)), and its roots are the distinct eigenvalues  $\lambda_1, \dots, \lambda_k$ . So

$$m_T = (x - \lambda_1)^{s_1} \cdots (x - \lambda_k)^{s_k} \quad \text{with every } s_i \geq 1.$$

Only the exponents remain to be found.

### 8.5.3 Computing the minimal polynomial

There are two practical methods. The first needs nothing but elimination. The second uses the eigenvalues, when we know them, to cut the search down to a few candidates.

**Method 1: the first dependence among powers.** The proof of [Theorem 5.7.2](#) searched for a dependence among  $\operatorname{id}_V, T, T^2, \dots$ . The **first** dependence that appears gives  $m_T$  exactly.

#### Proposition 8.5.4 — The First Dependence Among the Powers

Let  $V$  be finite-dimensional and  $T \in \mathcal{L}(V)$ . Let  $d \geq 0$  be the **least** integer such that the list  $(\operatorname{id}_V, T, \dots, T^d)$  in  $\mathcal{L}(V)$  is linearly dependent. Then there are unique  $c_0, \dots, c_{d-1} \in F$  with

$$T^d = c_0 \operatorname{id}_V + c_1 T + \cdots + c_{d-1} T^{d-1},$$

and  $m_T = x^d - c_{d-1}x^{d-1} - \cdots - c_1x - c_0$ . In particular  $\deg m_T = d$ . The same holds for  $\mathbf{A} \in M_n(F)$ , with powers of  $\mathbf{A}$  in  $M_n(F)$ .



*Proof.* Such a  $d$  exists: with  $n = \dim V$ , the space  $\mathcal{L}(V)$  has dimension  $n^2$  (Theorem 3.6.4), so the list  $(\text{id}_V, T, \dots, T^{n^2})$ , of length  $n^2 + 1$ , is linearly dependent (Theorem 1.6.3 (a)). By minimality of  $d$ , the list  $(\text{id}_V, \dots, T^{d-1})$  is independent (for  $d = 0$  it is empty). By Lemma 1.5.3, since appending  $T^d$  makes it dependent,  $T^d \in \text{span}(\text{id}_V, \dots, T^{d-1})$ . The coefficients are unique by Theorem 1.4.3.

Let  $q = x^d - c_{d-1}x^{d-1} - \dots - c_0$ . It is monic, and  $q(T) = 0$  by the displayed relation. Let  $p \neq 0$  with  $\deg p < d$ . Then  $p(T) = \sum_{k < d} a_k T^k$  with the  $a_k$  not all zero, and this is not 0 because  $(\text{id}_V, \dots, T^{d-1})$  is independent. So no non-zero polynomial of degree less than  $d$  annihilates  $T$ , and  $q$  is the monic annihilating polynomial of least degree. By Definition 8.5.1,  $q = m_T$ . ■

### Example. (The Minimal Polynomial from the First Dependence)

Find the minimal polynomial of

$$\mathbf{A} = \begin{pmatrix} 3 & 2 & -2 \\ -2 & -1 & 2 \\ -2 & -2 & 3 \end{pmatrix} \in M_3(\mathbb{R}),$$

the matrix of Example (Diagonalizing a  $3 \times 3$  Matrix with a Repeated Eigenvalue), without using its eigenvalues.

*Solution.* *Degree 1.*  $\mathbf{A}$  is not a scalar multiple of  $\mathbf{I}_3$ , since its  $(1, 2)$ -entry is non-zero. So  $(\mathbf{I}_3, \mathbf{A})$  is independent.

*Degree 2.* Multiplying out,

$$\mathbf{A}^2 = \begin{pmatrix} 9 & 8 & -8 \\ -8 & -7 & 8 \\ -8 & -8 & 9 \end{pmatrix}.$$

We look for  $a, b$  with  $\mathbf{A}^2 = a\mathbf{A} + b\mathbf{I}_3$ . The  $(1, 2)$ -entries give  $8 = 2a$ , so  $a = 4$ . The  $(1, 1)$ -entries give  $9 = 3a + b$ , so  $b = -3$ . Now check every entry of  $4\mathbf{A} - 3\mathbf{I}_3$ :

$$4\mathbf{A} - 3\mathbf{I}_3 = \begin{pmatrix} 12-3 & 8 & -8 \\ -8 & -4-3 & 8 \\ -8 & -8 & 12-3 \end{pmatrix} = \begin{pmatrix} 9 & 8 & -8 \\ -8 & -7 & 8 \\ -8 & -8 & 9 \end{pmatrix} = \mathbf{A}^2.$$

So  $(\mathbf{I}_3, \mathbf{A}, \mathbf{A}^2)$  is dependent, and  $d = 2$ . By Proposition 8.5.4,

$$m_{\mathbf{A}} = x^2 - 4x + 3 = (x - 1)(x - 3).$$

This fits Theorem 8.5.3: Example (Diagonalizing a  $3 \times 3$  Matrix with a Repeated Eigenvalue) found the eigenvalues 1 and 3, and  $p_{\mathbf{A}} = (x - 1)^2(x - 3)$ . Once again the repeated eigenvalue 1 appears only once in  $m_{\mathbf{A}}$ . Section 8 explains that this is exactly why  $\mathbf{A}$  was diagonalizable.

**Method 2: eigenvalues and trial exponents.** Suppose  $p_T$  splits,  $p_T = (x - \lambda_1)^{a_1} \dots (x - \lambda_k)^{a_k}$  with distinct  $\lambda_i$ . By Theorem 8.5.3, every  $x - \lambda_i$  divides  $m_T$ . If we find **any** annihilating polynomial of the form  $q = (x - \lambda_1)^{t_1} \dots (x - \lambda_k)^{t_k}$ , then  $m_T \mid q$ . By Lemma 8.5.2,  $m_T =$



$(x - \lambda_1)^{s_1} \cdots (x - \lambda_k)^{s_k}$  with  $1 \leq s_i \leq t_i$ , so only finitely many candidates remain, and we test them from the smallest degree up. The next section proves that  $q = p_T$  always annihilates  $T$ , so the search always succeeds with  $t_i \leq a_i$ . Here we verify that directly.

**Example. (The Minimal Polynomial from the Eigenvalues)**

Let  $\mathbf{A} = \begin{pmatrix} 2 & 0 & 1 \\ 1 & 3 & 1 \\ 1 & 0 & 2 \end{pmatrix} \in M_3(\mathbb{R})$ , with  $p_{\mathbf{A}} = (x - 1)(x - 3)^2$  (Example (A  $3 \times 3$  Characteristic Polynomial)). Find  $m_{\mathbf{A}}$ .

*Solution.* By Theorem 8.5.3, both  $x - 1$  and  $x - 3$  divide  $m_{\mathbf{A}}$ , so the candidate of least degree is  $(x - 1)(x - 3)$ . With

$$\mathbf{A} - \mathbf{I} = \begin{pmatrix} 1 & 0 & 1 \\ 1 & 2 & 1 \\ 1 & 0 & 1 \end{pmatrix}, \quad \mathbf{A} - 3\mathbf{I} = \begin{pmatrix} -1 & 0 & 1 \\ 1 & 0 & 1 \\ 1 & 0 & -1 \end{pmatrix},$$

the product is

$$(\mathbf{A} - \mathbf{I})(\mathbf{A} - 3\mathbf{I}) = \begin{pmatrix} 0 & 0 & 0 \\ 2 & 0 & 2 \\ 0 & 0 & 0 \end{pmatrix} \neq 0.$$

So  $m_{\mathbf{A}} \neq (x - 1)(x - 3)$ . Raise one exponent and try  $(x - 1)(x - 3)^2$  (had this failed, the other candidate of degree 3, namely  $(x - 1)^2(x - 3)$ , would come next). Polynomials in  $\mathbf{A}$  commute (Theorem 5.7.1 (c)), so we can multiply the matrix just found by  $\mathbf{A} - 3\mathbf{I}$  on the right:

$$(\mathbf{A} - \mathbf{I})(\mathbf{A} - 3\mathbf{I})^2 = \begin{pmatrix} 0 & 0 & 0 \\ 2 & 0 & 2 \\ 0 & 0 & 0 \end{pmatrix} \begin{pmatrix} -1 & 0 & 1 \\ 1 & 0 & 1 \\ 1 & 0 & -1 \end{pmatrix} = \begin{pmatrix} 0 & 0 & 0 \\ -2 + 2 & 0 & 2 - 2 \\ 0 & 0 & 0 \end{pmatrix} = 0.$$

So  $m_{\mathbf{A}} \mid (x - 1)(x - 3)^2$ , and by Lemma 8.5.2,  $m_{\mathbf{A}} = (x - 1)^s(x - 3)^t$  with  $1 \leq s \leq 1$  and  $1 \leq t \leq 2$ . The case  $t = 1$  was excluded, so

$$m_{\mathbf{A}} = (x - 1)(x - 3)^2 = p_{\mathbf{A}}.$$

In Section 3 we found  $g(3) = 1 < 2 = a(3)$  for this matrix. The exponent 2 of  $x - 3$  in  $m_{\mathbf{A}}$  is the polynomial trace of the same shortage of eigenvectors.

The product in Method 2 need not be multiplied out completely: to rule out a candidate it is enough to find **one** non-zero entry, and to confirm one, the factored form usually makes the zero visible.

## 8.5.4 Similar matrices and block diagonal matrices

The minimal polynomial belongs to the operator, not to a basis. That is what makes it an invariant of similarity.

### Theorem 8.5.5 — Similar Matrices Have the Same Minimal Polynomial

Let  $V$  be finite-dimensional with a basis  $\mathcal{B}$ , and let  $T \in \mathcal{L}(V)$ . Then  $m_T = m_{[T]_{\mathcal{B}}}$ . If  $\mathbf{A}, \mathbf{B} \in M_n(F)$  are similar, then  $m_{\mathbf{A}} = m_{\mathbf{B}}$ .

*Proof.* By Corollary 3.6.3,  $[p(T)]_{\mathcal{B}} = p([T]_{\mathcal{B}})$  for every  $p \in F[x]$ . An operator is zero exactly when its matrix is zero (Theorem 3.6.4), so  $p(T) = 0$  if and only if  $p([T]_{\mathcal{B}}) = 0$ . Thus  $T$  and  $[T]_{\mathcal{B}}$  have the same ideal of annihilating polynomials, and hence the same monic generator. If  $\mathbf{B} = \mathbf{P}^{-1}\mathbf{A}\mathbf{P}$ , then  $p(\mathbf{B}) = \mathbf{P}^{-1}p(\mathbf{A})\mathbf{P}$  by Proposition 3.7.6 (c), so again  $p(\mathbf{A}) = 0$  if and only if  $p(\mathbf{B}) = 0$ , and  $m_{\mathbf{A}} = m_{\mathbf{B}}$ . ■

The theorem gives a quick non-similarity test. For instance  $\mathbf{I}_2$  and  $\mathbf{J} = \begin{pmatrix} 1 & 1 \\ 0 & 1 \end{pmatrix}$  have the same characteristic polynomial, but  $m_{\mathbf{I}_2} = x - 1 \neq (x - 1)^2 = m_{\mathbf{J}}$ , so they are not similar.

Chapter 7 showed that a block diagonal matrix computes polynomials blockwise:  $p(\mathbf{A} \oplus \mathbf{B}) = p(\mathbf{A}) \oplus p(\mathbf{B})$  (Theorem 7.1.4 (b)). So a polynomial kills  $\mathbf{A} \oplus \mathbf{B}$  exactly when it kills both blocks. In the language of ideals, the answer is the least common multiple.

### Proposition 8.5.6 — Minimal Polynomial of a Block Diagonal Matrix

Let  $\mathbf{A} \in M_k(F)$  and  $\mathbf{B} \in M_l(F)$  with  $k, l \geq 1$ . Then  $m_{\mathbf{A} \oplus \mathbf{B}} = \text{lcm}(m_{\mathbf{A}}, m_{\mathbf{B}})$ .

*Proof.* Let  $p \in F[x]$ . By Theorem 7.1.4 (b),  $p(\mathbf{A} \oplus \mathbf{B}) = p(\mathbf{A}) \oplus p(\mathbf{B})$ , which is zero if and only if  $p(\mathbf{A}) = 0$  and  $p(\mathbf{B}) = 0$ . By Theorem 8.5.1, this holds if and only if  $p \in \langle m_{\mathbf{A}} \rangle \cap \langle m_{\mathbf{B}} \rangle$ . So  $I_{\mathbf{A} \oplus \mathbf{B}} = \langle m_{\mathbf{A}} \rangle \cap \langle m_{\mathbf{B}} \rangle$ . This ideal contains  $m_{\mathbf{A} \oplus \mathbf{B}} \neq 0$ , and by Definition 5.2.4 its monic generator is  $\text{lcm}(m_{\mathbf{A}}, m_{\mathbf{B}})$ . By Definition 8.5.1 its monic generator is also  $m_{\mathbf{A} \oplus \mathbf{B}}$ , and the monic generator is unique (Theorem 5.2.2). ■

For example,  $\text{diag}(1, 1, 2) = \mathbf{I}_2 \oplus (2)$  has  $m = \text{lcm}(x - 1, x - 2) = (x - 1)(x - 2)$ , as computed above. And  $\mathbf{J} \oplus (1)$  has  $m = \text{lcm}((x - 1)^2, x - 1) = (x - 1)^2$ : a block can contribute a factor that the other block already contains, and then the factor is not repeated. This is the difference from the characteristic polynomial, which multiplies:  $p_{\mathbf{A} \oplus \mathbf{B}} = p_{\mathbf{A}} p_{\mathbf{B}}$ .

**Warning.** Equal minimal polynomials do not make matrices similar.  $\text{diag}(1, 1, 2)$  and  $\text{diag}(1, 2, 2)$  both have minimal polynomial  $(x - 1)(x - 2)$ , by Proposition 8.5.6. But their characteristic polynomials  $(x - 1)^2(x - 2)$  and  $(x - 1)(x - 2)^2$  differ, so they are not similar (Theorem 6.10.3).

**Remark.** Even the minimal and characteristic polynomials **together** do not decide similarity. With  $\mathbf{J} = \begin{pmatrix} 1 & 1 \\ 0 & 1 \end{pmatrix}$ , the matrices  $\mathbf{J} \oplus \mathbf{J}$  and  $\mathbf{J} \oplus \mathbf{I}_2$  in  $M_4(F)$  both have characteristic polynomial  $(x - 1)^4$  and, by Proposition 8.5.6, minimal polynomial  $(x - 1)^2$ , as  $\text{lcm}((x - 1)^2, (x - 1)^2)$  and  $\text{lcm}((x - 1)^2, x - 1)$  are both  $(x - 1)^2$ . Yet  $(\mathbf{J} \oplus \mathbf{J}) - \mathbf{I}_4$

has rank 2 and  $(\mathbf{J} \oplus \mathbf{I}_2) - \mathbf{I}_4$  has rank 1; if  $\mathbf{B} = \mathbf{P}^{-1}\mathbf{A}\mathbf{P}$ , then  $\mathbf{B} - \mathbf{I} = \mathbf{P}^{-1}(\mathbf{A} - \mathbf{I})\mathbf{P}$ , so these ranks would have to agree (Proposition 3.7.6 (a)), and the two matrices are not similar. The invariant that does decide similarity is the subject of Chapter 9.

## 8.5.5 Exercises

### 8.5.5.1 A. Check your understanding

#### Exercise A1.

- (a) Define the minimal polynomial of an operator  $T$  on a finite-dimensional space, and give its two equivalent descriptions.
- (b) True or false:  $m_{\mathbf{A}} = p_{\mathbf{A}}$  for every  $\mathbf{A} \in M_n(F)$ . Justify your answer.
- (c) State the relation between the roots of  $m_T$  and the eigenvalues of  $T$ .
- (d) Let  $\mathbf{A} \neq 0$  satisfy  $\mathbf{A}^3 = 0$ . Which polynomials can  $m_{\mathbf{A}}$  be?
- (e) True or false: if  $m_{\mathbf{A}} = m_{\mathbf{B}}$ , then  $\mathbf{A} \sim \mathbf{B}$ . Justify your answer.
- (f) Explain why multiplication by  $x$  on  $F[x]$  has no minimal polynomial.

*Solution.*

- (a)  $m_T$  is the unique monic polynomial with  $\{p : p(T) = 0\} = \langle m_T \rangle$ ; equivalently, the monic polynomial of least degree with  $m_T(T) = 0$  (Definition 8.5.1).
- (b) False. For  $\mathbf{I}_2$ ,  $m = x - 1$  and  $p = (x - 1)^2$ .
- (c) They coincide:  $m_T(\lambda) = 0$  if and only if  $\lambda$  is an eigenvalue of  $T$  (Theorem 8.5.3).
- (d) By Theorem 8.5.1,  $m_{\mathbf{A}} \mid x^3$ , so  $m_{\mathbf{A}} \in \{1, x, x^2, x^3\}$  by Lemma 8.5.2. It is not 1 (as  $\mathbf{I} \neq 0$ ) and not  $x$  (as  $\mathbf{A} \neq 0$ ). So  $m_{\mathbf{A}} = x^2$  or  $m_{\mathbf{A}} = x^3$ , and both occur:  $\begin{pmatrix} 0 & 1 \\ 0 & 0 \end{pmatrix}$  and the matrix  $\mathbf{N}$  of the Quick check.
- (e) False.  $\text{diag}(1, 1, 2)$  and  $\text{diag}(1, 2, 2)$  have the same minimal polynomial but different characteristic polynomials.
- (f) For  $S(f) = xf$ , we have  $p(S)(1) = p$ , so  $p(S) \neq 0$  for all  $p \neq 0$ . The ideal  $I_S$  is  $\{0\}$  and has no monic generator. The space  $F[x]$  is infinite-dimensional, so the definition does not apply.

### 8.5.5.2 B. Practice

#### Exercise B1: Computing minimal polynomials.

Find the minimal polynomial of each real matrix.

- (a)  $\mathbf{A}_1 = \begin{pmatrix} 4 & -1 \\ 1 & 2 \end{pmatrix}$ .
- (b)  $\mathbf{A}_2 = \begin{pmatrix} 1 & 1 & 1 \\ 1 & 1 & 1 \\ 1 & 1 & 1 \end{pmatrix}$ .

- (c)  $\mathbf{A}_3 = \begin{pmatrix} 2 & 1 & 1 \\ 0 & 1 & -1 \\ -1 & -1 & 1 \end{pmatrix}$ , given that  $p_{\mathbf{A}_3} = (x-1)^2(x-2)$ .
- (d)  $\mathbf{A}_4 = \begin{pmatrix} 2 & 1 & 1 \\ -1 & 0 & -1 \\ 1 & 1 & 2 \end{pmatrix}$ , given that  $p_{\mathbf{A}_4} = (x-1)^2(x-2)$ .

Hence decide whether  $\mathbf{A}_3$  and  $\mathbf{A}_4$  are similar.

*Solution.*

- (a)  $\mathbf{A}_1$  is not scalar, so  $\deg m_{\mathbf{A}_1} \geq 2$ . Here  $\text{tr } \mathbf{A}_1 = 6$  and  $\det \mathbf{A}_1 = 9$ , and  $\mathbf{A}_1^2 = \begin{pmatrix} 15 & -6 \\ 6 & 3 \end{pmatrix} = 6\mathbf{A}_1 - 9\mathbf{I}_2$ , as one checks entry by entry. By [Proposition 8.5.4](#),  $m_{\mathbf{A}_1} = x^2 - 6x + 9 = (x-3)^2$ .
- (b)  $\mathbf{A}_2$  is not scalar. Every entry of  $\mathbf{A}_2^2$  is  $1 + 1 + 1 = 3$ , so  $\mathbf{A}_2^2 = 3\mathbf{A}_2$ . By [Proposition 8.5.4](#),  $m_{\mathbf{A}_2} = x^2 - 3x = x(x-3)$ .
- (c) By [Theorem 8.5.3](#),  $(x-1)(x-2) \mid m_{\mathbf{A}_3}$ . First candidate:

$$\begin{aligned} (\mathbf{A}_3 - \mathbf{I})(\mathbf{A}_3 - 2\mathbf{I}) &= \begin{pmatrix} 1 & 1 & 1 \\ 0 & 0 & -1 \\ -1 & -1 & 0 \end{pmatrix} \begin{pmatrix} 0 & 1 & 1 \\ 0 & -1 & -1 \\ -1 & -1 & -1 \end{pmatrix} \\ &= \begin{pmatrix} -1 & -1 & -1 \\ 1 & 1 & 1 \\ 0 & 0 & 0 \end{pmatrix} \neq \mathbf{0}. \end{aligned}$$

Next candidate: multiplying this on the left by  $\mathbf{A}_3 - \mathbf{I}$ , the rows of the result are  $(1, 1, 1)\mathbf{M}$ ,  $(0, 0, -1)\mathbf{M}$  and  $(-1, -1, 0)\mathbf{M}$ , where  $\mathbf{M}$  is the matrix just found. Since the rows of  $\mathbf{M}$  sum to zero and its third row is zero, all three products are  $\mathbf{0}$ . So  $(\mathbf{A}_3 - \mathbf{I})^2(\mathbf{A}_3 - 2\mathbf{I}) = \mathbf{0}$ , and by [Lemma 8.5.2](#),  $m_{\mathbf{A}_3} = (x-1)^2(x-2)$ .

- (d) Again  $(x-1)(x-2) \mid m_{\mathbf{A}_4}$ , and

$$(\mathbf{A}_4 - \mathbf{I})(\mathbf{A}_4 - 2\mathbf{I}) = \begin{pmatrix} 1 & 1 & 1 \\ -1 & -1 & -1 \\ 1 & 1 & 1 \end{pmatrix} \begin{pmatrix} 0 & 1 & 1 \\ -1 & -2 & -1 \\ 1 & 1 & 0 \end{pmatrix} = \mathbf{0},$$

since each column of the second matrix has entries summing to 0 and each row of the first is a multiple of  $(1, 1, 1)$ . So  $m_{\mathbf{A}_4} = (x-1)(x-2)$ .

Hence  $m_{\mathbf{A}_3} \neq m_{\mathbf{A}_4}$ , and by [Theorem 8.5.5](#)  $\mathbf{A}_3$  and  $\mathbf{A}_4$  are **not** similar, although their characteristic polynomials agree. (This matches [Exercise \(B1: Computing multiplicities\)](#), where the same two matrices appear as  $\mathbf{A}_2$  and  $\mathbf{A}_3$ , with  $g(1) = 1$  for the first and  $g(1) = 2$  for the second.)

### Exercise B2: Operators with $T^2 = T$ .

Let  $V \neq \{0\}$  be finite-dimensional and  $T \in \mathcal{L}(V)$  with  $T^2 = T$ .

- (a) Show that  $m_T \in \{x, x-1, x(x-1)\}$ .
- (b) Show that  $m_T = x$  exactly when  $T = 0$ , and  $m_T = x-1$  exactly when  $T = \text{id}_V$ .

(c) What is  $m_T$  on  $V = \{0\}$ ?

*Solution.*

- (a)  $x^2 - x = x(x-1)$  annihilates  $T$ , so  $m_T \mid x(x-1)$  by [Theorem 8.5.1](#). By [Lemma 8.5.2](#), the monic divisors are  $1, x, x-1, x(x-1)$ . Since  $V \neq \{0\}$ ,  $\deg m_T \geq 1$ , so  $m_T \neq 1$ .
- (b) If  $m_T = x$ , then  $T = m_T(T) = 0$ . Conversely, if  $T = 0$ , then  $x$  annihilates  $T$ , so  $m_T \mid x$ , and  $m_T = x$  by (a). Likewise  $m_T = x-1$  if and only if  $T - \text{id}_V = 0$ . In all other cases  $m_T = x(x-1)$ .
- (c) On  $V = \{0\}$ ,  $1(T) = \text{id}_V = 0$ , so  $m_T = 1$ .

### Exercise B3: Companion matrices.

Let  $n \geq 2$ ,  $p = x^n + a_{n-1}x^{n-1} + \cdots + a_1x + a_0 \in F[x]$ , and let  $\mathbf{C} = \mathbf{C}(p)$  be its companion matrix ([Exercise \(C1: Companion matrices\)](#)).

- (a) Show that  $\mathbf{C}\mathbf{e}_k = \mathbf{e}_{k+1}$  for  $1 \leq k \leq n-1$ , so  $\mathbf{C}^k\mathbf{e}_1 = \mathbf{e}_{k+1}$  for  $0 \leq k \leq n-1$ .
- (b) Deduce that no non-zero polynomial of degree less than  $n$  annihilates  $\mathbf{C}$ .
- (c) Show that  $p(\mathbf{C})\mathbf{e}_1 = \mathbf{0}$ , and hence  $p(\mathbf{C})\mathbf{e}_{k+1} = \mathbf{0}$  for all  $k \leq n-1$ .
- (d) Deduce that  $m_{\mathbf{C}} = p = p_{\mathbf{C}}$ .

*Solution.*

- (a) Column  $k$  of  $\mathbf{C}$ , for  $k \leq n-1$ , has a single 1 in row  $k+1$ , so  $\mathbf{C}\mathbf{e}_k = \mathbf{e}_{k+1}$ . By induction on  $k$ ,  $\mathbf{C}^k\mathbf{e}_1 = \mathbf{C}(\mathbf{C}^{k-1}\mathbf{e}_1) = \mathbf{C}\mathbf{e}_k = \mathbf{e}_{k+1}$ .
- (b) Let  $q = \sum_{k=0}^d b_k x^k \neq 0$  with  $d \leq n-1$ . By (a),  $q(\mathbf{C})\mathbf{e}_1 = \sum_k b_k \mathbf{C}^k \mathbf{e}_1 = \sum_k b_k \mathbf{e}_{k+1}$ , which is not  $\mathbf{0}$  because some  $b_k \neq 0$ . So  $q(\mathbf{C}) \neq 0$ .
- (c) By (a),  $\mathbf{C}^n \mathbf{e}_1 = \mathbf{C}\mathbf{e}_n$ , the last column  $(-a_0, \dots, -a_{n-1}) = -\sum_{k=0}^{n-1} a_k \mathbf{e}_{k+1}$ . Hence

$$p(\mathbf{C})\mathbf{e}_1 = \mathbf{C}^n \mathbf{e}_1 + \sum_{k=0}^{n-1} a_k \mathbf{C}^k \mathbf{e}_1 = -\sum_k a_k \mathbf{e}_{k+1} + \sum_k a_k \mathbf{e}_{k+1} = \mathbf{0}.$$

For  $k \leq n-1$ , polynomials in  $\mathbf{C}$  commute ([Theorem 5.7.1 \(c\)](#)), so  $p(\mathbf{C})\mathbf{e}_{k+1} = p(\mathbf{C})\mathbf{C}^k \mathbf{e}_1 = \mathbf{C}^k p(\mathbf{C})\mathbf{e}_1 = \mathbf{0}$ .

- (d) By (c),  $p(\mathbf{C})$  kills the standard basis, so  $p(\mathbf{C}) = 0$  and  $m_{\mathbf{C}} \mid p$ . By (b),  $\deg m_{\mathbf{C}} \geq n = \deg p$ . Two monic polynomials, one dividing the other, with  $\deg m_{\mathbf{C}} \geq \deg p$ , are equal ([Proposition 5.1.2 \(c\), \(d\)](#)). So  $m_{\mathbf{C}} = p$ , and  $p = p_{\mathbf{C}}$  by [Exercise \(C1: Companion matrices\) \(b\)](#). Every monic polynomial of degree  $n \geq 2$  is therefore the minimal polynomial of some  $n \times n$  matrix.

### 8.5.5.3 C. Going deeper

#### Exercise C1: Cyclic vectors.

Let  $\dim V = n \geq 1$  and  $T \in \mathcal{L}(V)$ . A vector  $\mathbf{v}$  is a **cyclic vector** for  $T$  if  $(\mathbf{v}, T\mathbf{v}, \dots, T^{n-1}\mathbf{v})$  is a basis of  $V$ .

- (a) Prove that if  $T$  has a cyclic vector, then  $\deg m_T \geq n$ .

- (b) Suppose  $\mathbf{v}$  is cyclic and  $T^n \mathbf{v} = -(a_0 \mathbf{v} + a_1 T \mathbf{v} + \cdots + a_{n-1} T^{n-1} \mathbf{v})$ . Show that the matrix of  $T$  in the basis  $(\mathbf{v}, \dots, T^{n-1} \mathbf{v})$  is the companion matrix of  $p = x^n + a_{n-1}x^{n-1} + \cdots + a_0$  (for  $n \geq 2$ ), and deduce  $m_T = p_T = p$ .
- (c) Find a cyclic vector for  $\mathbf{J} = \begin{pmatrix} 1 & 1 \\ 0 & 1 \end{pmatrix}$ , and show that  $\mathbf{I}_2$  has none.

(The converse of (a), that  $\deg m_T = n$  forces a cyclic vector, is proved in Chapter 9.)

*Solution.*

- (a) Let  $q = \sum_{k=0}^d b_k x^k \neq 0$  with  $d \leq n-1$ . Then  $q(T)\mathbf{v} = \sum_k b_k T^k \mathbf{v} \neq \mathbf{0}$ , because the vectors  $T^k \mathbf{v}$ ,  $k \leq n-1$ , are independent and some  $b_k \neq 0$ . So  $q(T) \neq 0$ , and by [Theorem 8.5.1](#),  $\deg m_T \geq n$ .
- (b) Let  $\mathcal{B} = (\mathbf{v}, T\mathbf{v}, \dots, T^{n-1}\mathbf{v})$ . For  $k \leq n-2$ ,  $T(T^k \mathbf{v}) = T^{k+1} \mathbf{v}$ , the  $(k+2)$ -th basis vector, so column  $k+1$  of  $[T]_{\mathcal{B}}$  is  $\mathbf{e}_{k+2}$ . The last column is  $[T^n \mathbf{v}]_{\mathcal{B}} = (-a_0, \dots, -a_{n-1})$ . This is  $\mathbf{C}(p)$ . By [Theorem 8.5.5](#) and [Exercise \(B3: Companion matrices\)](#) (d),  $m_T = m_{\mathbf{C}(p)} = p$ . By [Definition 6.10.2](#) and [Exercise \(C1: Companion matrices\)](#) (b),  $p_T = p_{\mathbf{C}(p)} = p$ . (For  $n=1$ ,  $T\mathbf{v} = -a_0 \mathbf{v}$  and  $T = -a_0 \text{id}_V$ , so  $m_T = p_T = x + a_0$  directly.)
- (c)  $\mathbf{J}\mathbf{e}_2 = (1, 1)$ , and  $(\mathbf{e}_2, (1, 1))$  is independent, so  $\mathbf{e}_2$  is cyclic. For  $\mathbf{I}_2$ , every  $\mathbf{v}$  has  $\mathbf{I}_2 \mathbf{v} = \mathbf{v}$ , so  $(\mathbf{v}, \mathbf{I}_2 \mathbf{v})$  is dependent. This matches (a):  $\deg m_{\mathbf{I}_2} = 1 < 2$ .

### Exercise C2: Minimal polynomials of the pieces.

Let  $V$  be finite-dimensional,  $T \in \mathcal{L}(V)$ , and  $U$  a  $T$ -invariant subspace, with restriction  $T|_U$  and induced operator  $\bar{T}$  on  $V/U$  ([Definition 8.1.2](#)).

- (a) Prove that  $p(T|_U) = p(T)|_U$  and  $p(\bar{T})(\mathbf{v} + U) = p(T)\mathbf{v} + U$  for every  $p \in F[x]$ .
- (b) Deduce that  $m_{T|_U} \mid m_T$  and  $m_{\bar{T}} \mid m_T$ .
- (c) Prove that  $m_T \mid m_{T|_U} m_{\bar{T}}$ .
- (d) For  $\mathbf{J} = \begin{pmatrix} 1 & 1 \\ 0 & 1 \end{pmatrix}$  and  $U = \text{span}(\mathbf{e}_1)$ , compute all three minimal polynomials, and show that  $m_T$  need not equal  $\text{lcm}(m_{T|_U}, m_{\bar{T}})$ .

*Hint: for (c), apply  $m_{\bar{T}}(T)$  to  $\mathbf{v}$  and ask where the result lies.*

*Solution.*

- (a) First,  $(T|_U)^k = T^k|_U$  by induction on  $k$ : for  $\mathbf{u} \in U$ ,  $(T|_U)^{k+1} \mathbf{u} = T|_U((T|_U)^k \mathbf{u}) = T(T^k \mathbf{u}) = T^{k+1} \mathbf{u}$ . Taking linear combinations,  $p(T|_U)\mathbf{u} = p(T)\mathbf{u}$ . For the quotient,  $\bar{T}^k(\mathbf{v} + U) = T^k \mathbf{v} + U$  by the same induction, using [Definition 8.1.2](#) (b) at each step, and  $p(\bar{T})(\mathbf{v} + U) = \sum_k c_k (T^k \mathbf{v} + U) = p(T)\mathbf{v} + U$  by the coset operations ([Theorem 3.9.4](#)).
- (b) By (a),  $m_T(T|_U) = m_T(T)|_U = 0$  and  $m_T(\bar{T})(\mathbf{v} + U) = \mathbf{0} + U$  for all  $\mathbf{v}$ . By [Theorem 8.5.1](#), applied to  $T|_U \in \mathcal{L}(U)$  and to  $\bar{T} \in \mathcal{L}(V/U)$ , both finite-dimensional,  $m_{T|_U} \mid m_T$  and  $m_{\bar{T}} \mid m_T$ .
- (c) Let  $\mathbf{v} \in V$ . By (a),  $m_{\bar{T}}(T)\mathbf{v} + U = m_{\bar{T}}(\bar{T})(\mathbf{v} + U) = \mathbf{0} + U$ , so  $\mathbf{u} := m_{\bar{T}}(T)\mathbf{v} \in U$ . By (a) again,  $m_{T|_U}(T)\mathbf{u} = m_{T|_U}(T|_U)\mathbf{u} = \mathbf{0}$ . Hence  $(m_{T|_U} m_{\bar{T}})(T)\mathbf{v} = m_{T|_U}(T) m_{\bar{T}}(T)\mathbf{v} = \mathbf{0}$  ([Theorem 5.7.1](#) (b)). As  $\mathbf{v}$  was arbitrary,  $m_{T|_U} m_{\bar{T}}$  anni-

hilates  $T$ , and  $m_T \mid m_{T|_U} m_{\bar{T}}$  by [Theorem 8.5.1](#).

- (d)  $\mathbf{J}\mathbf{e}_1 = \mathbf{e}_1$ , so  $T|_U = \text{id}_U$  and  $m_{T|_U} = x - 1$ . The quotient  $F^2/U$  has basis  $\mathbf{e}_2 + U$ , and  $\bar{T}(\mathbf{e}_2 + U) = (1, 1) + U = \mathbf{e}_2 + U$ , so  $\bar{T} = \text{id}$  and  $m_{\bar{T}} = x - 1$ . But  $m_{\mathbf{J}} = (x - 1)^2$ , computed earlier. So  $m_{\mathbf{J}} = m_{T|_U} m_{\bar{T}}$ , while  $\text{lcm}(m_{T|_U}, m_{\bar{T}}) = x - 1 \neq m_{\mathbf{J}}$ . Compare [Proposition 8.5.6](#): when  $U$  has an invariant complement the lcm is right, but in general it is not.

## 8.6 The Cayley–Hamilton Theorem

An operator on an  $n$ -dimensional space now carries two monic polynomials. The characteristic polynomial  $p_T$  comes from determinants and has degree exactly  $n$ . The minimal polynomial  $m_T$  comes from the ideal of polynomials that kill  $T$ , and the previous section showed that it has the same roots as  $p_T$ . Chapter 6 stated, without proof, the fact that ties the two together: every square matrix satisfies its own characteristic polynomial. Chapter 6 also warned against a one-line fake proof of that statement. This section gives a real proof, built on the adjugate identity of Chapter 6 read over  $F[x]$ . It then draws the consequences:  $m_T$  divides  $p_T$ , inverses are polynomials, and high powers of a matrix reduce to low ones.

### 8.6.1 The statement, and a proof that is not one

Chapter 5 proved that every operator on an  $n$ -dimensional space is killed by some non-zero polynomial of degree at most  $n^2$  ([Theorem 5.7.2](#)). The bound came from counting:  $\mathcal{L}(V)$  has dimension  $n^2$ . Counting cannot do better, yet in every example so far a polynomial of degree at most  $n$  sufficed. The characteristic polynomial always has degree  $n$ , and it is the natural candidate.

#### Theorem 8.6.1 — Cayley–Hamilton Theorem

Let  $n \geq 1$  and  $\mathbf{A} \in M_n(F)$ . Then

$$p_{\mathbf{A}}(\mathbf{A}) = 0.$$

Likewise, if  $V$  is a finite-dimensional vector space over  $F$  with  $\dim V \geq 1$  and  $T \in \mathcal{L}(V)$ , then  $p_T(T) = 0$ .

Here  $p_{\mathbf{A}}(\mathbf{A})$  means what it always means ([Definition 0.9.14](#)): if  $p_{\mathbf{A}} = x^n + c_{n-1}x^{n-1} + \cdots + c_1x + c_0$ , then

$$p_{\mathbf{A}}(\mathbf{A}) = \mathbf{A}^n + c_{n-1}\mathbf{A}^{n-1} + \cdots + c_1\mathbf{A} + c_0\mathbf{I}_n,$$

an  $n \times n$  matrix, with the constant term becoming  $c_0\mathbf{I}_n$ .

For  $n = 2$  the theorem can be checked by hand. With  $\mathbf{A} = \begin{pmatrix} a & b \\ c & d \end{pmatrix}$ ,  $p_{\mathbf{A}} = x^2 - (\text{tr } \mathbf{A})x + \det \mathbf{A}$ , and [Exercise \(C2: Nilpotent  \$2 \times 2\$  matrices\)](#) (b) multiplied out

$$\mathbf{A}^2 - (a + d)\mathbf{A} + (ad - bc)\mathbf{I}_2 = 0.$$

For  $n = 3$ , the same brute force already involves cubic expressions in nine entries, and for general

$n$  it is hopeless. We need an argument. There is a tempting one, and we name it before it can do any harm.

**Warning. The fake proof of Cayley–Hamilton, again.** “Since  $p_{\mathbf{A}}(x) = \det(x\mathbf{I} - \mathbf{A})$ , substitute  $x = \mathbf{A}$  to get  $p_{\mathbf{A}}(\mathbf{A}) = \det(\mathbf{A}\mathbf{I} - \mathbf{A}) = \det 0 = 0$ .” Chapter 6 (the warning at the end of its §10) explained why this is not a proof. The left side  $p_{\mathbf{A}}(\mathbf{A})$  is a **matrix**, while  $\det(\cdots)$  is a **scalar**, so the equation compares objects of different kinds. And substituting  $\mathbf{A}$  for  $x$  **inside the entries** of  $x\mathbf{I} - \mathbf{A}$  is not what substituting into a polynomial means: the entry  $x - a_{11}$  would have to become the matrix  $\mathbf{A} - a_{11}\mathbf{I}$ , and the determinant of a matrix whose entries are matrices is undefined. The same “argument” applied to  $q(x) = \operatorname{tr}(x\mathbf{I} - \mathbf{A}) = nx - \operatorname{tr} \mathbf{A}$  would prove  $q(\mathbf{A}) = 0$ , which is false for  $\mathbf{A} = \operatorname{diag}(1, 0)$ : there  $q(\mathbf{A}) = 2\mathbf{A} - \mathbf{I} = \operatorname{diag}(1, -1)$ . Any correct proof must keep  $x$  a polynomial variable until the very end, and only then put a matrix in its place.

## 8.6.2 A proof through the adjugate

The adjugate identity of Chapter 6 says  $\mathbf{M} \operatorname{adj} \mathbf{M} = (\det \mathbf{M}) \mathbf{I}$  for every square matrix  $\mathbf{M}$ , and the remark after [Theorem 6.6.2](#) notes that its proof works over any commutative ring. Over  $F[x]$ , with  $\mathbf{M} = x\mathbf{I} - \mathbf{A}$ , it reads

$$(x\mathbf{I} - \mathbf{A}) \operatorname{adj}(x\mathbf{I} - \mathbf{A}) = p_{\mathbf{A}}(x) \mathbf{I}. \quad (1)$$

This is an identity between matrices of **polynomials**, and it contains  $p_{\mathbf{A}}$  on the right. The proof extracts  $p_{\mathbf{A}}(\mathbf{A})$  from it legally.

*Idea.* Try  $n = 2$  first. By [Exercise \(C3: An adjugate over  \$F\[x\]\$ \)](#) (b) (done there over  $\mathbb{R}$ , but the computation works over any field),  $\operatorname{adj}(x\mathbf{I} - \mathbf{A}) = x\mathbf{I} - \operatorname{adj} \mathbf{A}$ , a polynomial in  $x$  with **matrix coefficients**  $\mathbf{B}_1 = \mathbf{I}$  and  $\mathbf{B}_0 = -\operatorname{adj} \mathbf{A}$ . Expanding the left side of (1),

$$(x\mathbf{I} - \mathbf{A})(x\mathbf{I} - \operatorname{adj} \mathbf{A}) = x^2 \mathbf{I} - x(\mathbf{A} + \operatorname{adj} \mathbf{A}) + \mathbf{A} \operatorname{adj} \mathbf{A}.$$

Comparing the coefficients of  $x^2$ ,  $x$  and 1 with those of  $(x^2 - (\operatorname{tr} \mathbf{A})x + \det \mathbf{A})\mathbf{I}$  gives three matrix equations:

$$\mathbf{I} = \mathbf{I}, \quad -(\mathbf{A} + \operatorname{adj} \mathbf{A}) = -(\operatorname{tr} \mathbf{A}) \mathbf{I}, \quad \mathbf{A} \operatorname{adj} \mathbf{A} = (\det \mathbf{A}) \mathbf{I}.$$

Now multiply the first by  $\mathbf{A}^2$ , the second by  $\mathbf{A}$ , the third by  $\mathbf{I}$ , and add. The right sides add up to  $\mathbf{A}^2 - (\operatorname{tr} \mathbf{A})\mathbf{A} + (\det \mathbf{A})\mathbf{I} = p_{\mathbf{A}}(\mathbf{A})$ . The left sides add up to  $\mathbf{A}^2 - \mathbf{A}^2 - \mathbf{A} \operatorname{adj} \mathbf{A} + \mathbf{A} \operatorname{adj} \mathbf{A}$ , in which every term cancels against its neighbor. So  $p_{\mathbf{A}}(\mathbf{A}) = 0$ . This is the whole proof; for general  $n$  there are  $n + 1$  equations and the cancellation is a telescoping sum. The variable  $x$  is never replaced by a matrix: we compare coefficients first, and only then multiply the resulting **matrix** equations by powers of  $\mathbf{A}$ .

*Proof.* Let  $p_{\mathbf{A}} = c_n x^n + c_{n-1} x^{n-1} + \cdots + c_1 x + c_0$  with  $c_n = 1$  ([Theorem 6.10.2](#)).

**Step 1: the adjugate as a polynomial with matrix coefficients.** If  $n = 1$ , then  $\operatorname{adj}(x\mathbf{I} - \mathbf{A}) = (1)$  by [Definition 6.6.2](#). If  $n \geq 2$ , each entry of  $\operatorname{adj}(x\mathbf{I} - \mathbf{A})$  is a cofactor  $\pm \det \mathbf{N}$ , where  $\mathbf{N}$  is an  $(n - 1) \times (n - 1)$  submatrix of  $x\mathbf{I} - \mathbf{A}$  ([Definition 6.6.2](#), [Definition 6.6.1](#)). Every entry of  $\mathbf{N}$  lies in  $F[x]_{\leq 1}$ . A product of  $n - 1$  such polynomials lies in  $F[x]_{\leq n-1}$  ([Theorem 0.8.3](#), the zero polynomial being harmless), and so does a sum of such products ([Theorem 0.8.2](#)); in particular the Leibniz sum  $\det \mathbf{N}$  does ([Definition 6.4.1](#)). In both cases every entry of  $\operatorname{adj}(x\mathbf{I} - \mathbf{A})$  lies in  $F[x]_{\leq n-1}$ . For



$0 \leq k \leq n-1$ , let  $\mathbf{B}_k \in M_n(F)$  be the matrix whose  $(i, j)$ -entry is the coefficient of  $x^k$  in the  $(i, j)$ -entry of  $\text{adj}(x\mathbf{I} - \mathbf{A})$ . Then, entry by entry,

$$\text{adj}(x\mathbf{I} - \mathbf{A}) = \sum_{k=0}^{n-1} x^k \mathbf{B}_k.$$

Put  $\mathbf{B}_{-1} = \mathbf{B}_n = 0$  for convenience.

**Step 2: compare coefficients.** By (1), which holds by [Theorem 6.6.2](#) over the commutative ring  $F[x]$  (the remark after it), the  $(i, j)$ -entries of both sides of (1) agree. On the left, by [Definition 0.9.9](#),

$$\sum_{l=1}^n (x\delta_{il} - a_{il}) \sum_{k=0}^{n-1} (\mathbf{B}_k)_{lj} x^k = \sum_{k=0}^{n-1} (\mathbf{B}_k)_{ij} x^{k+1} - \sum_{k=0}^{n-1} (\mathbf{A}\mathbf{B}_k)_{ij} x^k,$$

where  $\delta_{il} = 1$  if  $i = l$  and 0 otherwise. So for  $0 \leq k \leq n$ , the coefficient of  $x^k$  is  $(\mathbf{B}_{k-1})_{ij} - (\mathbf{A}\mathbf{B}_k)_{ij}$ . On the right the  $(i, j)$ -entry is  $p_{\mathbf{A}} \delta_{ij}$ , whose coefficient of  $x^k$  is  $c_k \delta_{ij}$ . Two polynomials are equal exactly when their coefficients are equal ([Definition 0.8.1](#)). Since this holds for every  $(i, j)$ ,

$$\mathbf{B}_{k-1} - \mathbf{A}\mathbf{B}_k = c_k \mathbf{I} \quad (0 \leq k \leq n). \quad (2)$$

**Step 3: telescope.** Multiply equation (2) for  $k$  on the left by  $\mathbf{A}^k$ , and add the  $n+1$  equations. On the right we get  $\sum_{k=0}^n c_k \mathbf{A}^k = p_{\mathbf{A}}(\mathbf{A})$  by [Definition 0.9.14](#). On the left, writing  $\mathbf{S}_k = \mathbf{A}^k \mathbf{B}_{k-1}$  for  $0 \leq k \leq n+1$ ,

$$\sum_{k=0}^n (\mathbf{A}^k \mathbf{B}_{k-1} - \mathbf{A}^{k+1} \mathbf{B}_k) = \sum_{k=0}^n (\mathbf{S}_k - \mathbf{S}_{k+1}) = \mathbf{S}_0 - \mathbf{S}_{n+1} = \mathbf{B}_{-1} - \mathbf{A}^{n+1} \mathbf{B}_n = 0,$$

since  $\mathbf{B}_{-1} = \mathbf{B}_n = 0$ . Hence  $p_{\mathbf{A}}(\mathbf{A}) = 0$ .

**The operator version.** Let  $\mathcal{B}$  be a basis of  $V$  and  $\mathbf{A} = [T]_{\mathcal{B}}$ . By [Definition 6.10.2](#),  $p_T = p_{\mathbf{A}}$ . By [Corollary 3.6.3](#),  $[p_T(T)]_{\mathcal{B}} = p_T(\mathbf{A}) = p_{\mathbf{A}}(\mathbf{A}) = 0$ , so  $p_T(T) = 0$  ([Theorem 3.6.4](#)). This proves the theorem. ■

Compare with the fake proof. There,  $\mathbf{A}$  was pushed into the entries of  $x\mathbf{I} - \mathbf{A}$ . Here, (1) is first unpacked into the  $n+1$  matrix equations (2), which contain no  $x$  at all, and  $\mathbf{A}$  enters only when we multiply those equations by  $\mathbf{A}^k$ . Every multiplication is on the **left**, so no commutativity of matrices was used; the only commutativity used is inside  $F[x]$ , in the adjugate identity.

Nothing in the proof divides. The determinant, the adjugate identity, coefficient comparison and matrix multiplication all make sense over a commutative ring, and [Exercise \(C1: Cayley–Hamilton over a commutative ring\)](#) asks you to check that the theorem holds, with the same proof, for matrices with entries in a commutative ring such as  $\mathbb{Z}$  or  $\mathbb{R}[t]$ .

The proof also produced more than it was asked for. Equation (2) with  $k = n$  says  $\mathbf{B}_{n-1} = \mathbf{I}$ , and the others say  $\mathbf{B}_{k-1} = \mathbf{A}\mathbf{B}_k + c_k \mathbf{I}$ . Working down from  $\mathbf{B}_{n-1}$ , every  $\mathbf{B}_k$  is a polynomial in  $\mathbf{A}$ ; [Exercise \(C2: The adjugate is a polynomial in the matrix\)](#) turns this into a formula for  $\text{adj } \mathbf{A}$ .

### 8.6.3 Consequences

The first consequence is the promised comparison of the two polynomials.

#### Corollary 8.6.2 — The Minimal Polynomial Divides the Characteristic Polynomial

Let  $V$  be finite-dimensional with  $\dim V = n \geq 1$ , and let  $T \in \mathcal{L}(V)$ .

- (a)  $m_T \mid p_T$ ; in particular  $1 \leq \deg m_T \leq n$ .
- (b) If  $p_T$  splits over  $F$ , say  $p_T = (x - \lambda_1)^{a_1} \cdots (x - \lambda_k)^{a_k}$  with distinct  $\lambda_i$ , then

$$m_T = (x - \lambda_1)^{s_1} \cdots (x - \lambda_k)^{s_k} \quad \text{with } 1 \leq s_i \leq a_i = a_T(\lambda_i).$$

The same holds for  $\mathbf{A} \in M_n(F)$  with  $m_{\mathbf{A}}$  and  $p_{\mathbf{A}}$ . In particular, over  $\mathbb{C}$ , (b) applies to every  $\mathbf{A} \in M_n(\mathbb{C})$ .

*Proof.*

- (a) By Theorem 8.6.1,  $p_T(T) = 0$ , so  $m_T \mid p_T$  by Theorem 8.5.1. Since  $p_T$  is monic of degree  $n$  (Theorem 6.10.2),  $\deg m_T \leq n$  by Proposition 5.1.2 (c), and  $\deg m_T \geq 1$  by Theorem 8.5.1, as  $V \neq \{0\}$ .
- (b) A polynomial  $x - \lambda$  of degree 1 is irreducible: if  $x - \lambda = gh$ , then  $\deg g + \deg h = 1$  (Theorem 0.8.3), so one factor is constant. The  $x - \lambda_i$  are distinct and monic. By (a) and Lemma 8.5.2,  $m_T = \prod_i (x - \lambda_i)^{s_i}$  with  $0 \leq s_i \leq a_i$ . Each  $\lambda_i$  is a root of  $p_T$ , hence an eigenvalue (Theorem 8.2.1), hence a root of  $m_T$  (Theorem 8.5.3), so  $s_i \geq 1$ . Finally  $a_i = \text{mult}_{\lambda_i}(p_T) = a_T(\lambda_i)$  by Lemma 5.4.1, since the other factors do not vanish at  $\lambda_i$ . The matrix case is the operator  $T_{\mathbf{A}}$ , and over  $\mathbb{C}$  every  $p_{\mathbf{A}}$  splits (Corollary 5.5.5). ■

This pays off a promise from Chapter 5: the bound  $n^2$  of Theorem 5.7.2 improves to  $n$ , and  $p_T$  is an explicit annihilating polynomial of degree exactly  $n$ . It also makes Method 2 of the previous section a finite search that always succeeds. For a matrix with  $p_{\mathbf{A}} = (x - 2)^2(x - 3)$ , the corollary leaves exactly two candidates,  $(x - 2)(x - 3)$  and  $(x - 2)^2(x - 3)$ , and one matrix product decides between them.

**Warning. Cayley–Hamilton gives an annihilating polynomial, not the minimal one.** From  $p_{\mathbf{A}}(\mathbf{A}) = 0$  we may **not** conclude  $m_{\mathbf{A}} = p_{\mathbf{A}}$ ; only  $m_{\mathbf{A}} \mid p_{\mathbf{A}}$ . For  $\mathbf{A} = \text{diag}(1, 1, 2)$ ,  $p_{\mathbf{A}} = (x - 1)^2(x - 2)$  kills  $\mathbf{A}$ , and so does the proper divisor  $(x - 1)(x - 2) = m_{\mathbf{A}}$ . Nor may we conclude that every divisor of  $p_{\mathbf{A}}$  with the right roots kills  $\mathbf{A}$ : for  $\mathbf{J} = \begin{pmatrix} 1 & 1 \\ 0 & 1 \end{pmatrix}$ ,  $x - 1$  has the right root and does not.

The second consequence expresses inverses through powers.

#### Corollary 8.6.3 — The Inverse Is a Polynomial in the Matrix

Let  $n \geq 1$ ,  $\mathbf{A} \in M_n(F)$ , and  $p_{\mathbf{A}} = x^n + c_{n-1}x^{n-1} + \cdots + c_1x + c_0$ . Then  $c_0 = (-1)^n \det \mathbf{A}$ . If  $\mathbf{A}$  is invertible, then  $c_0 \neq 0$  and

$$\mathbf{A}^{-1} = -\frac{1}{c_0}(\mathbf{A}^{n-1} + c_{n-1}\mathbf{A}^{n-2} + \cdots + c_2\mathbf{A} + c_1\mathbf{I}).$$

(For  $n = 1$  the bracket is  $\mathbf{I}$  alone, and the formula reads  $\mathbf{A}^{-1} = -c_0^{-1}\mathbf{I}$ .) The same holds for an invertible operator  $T$  on a space of dimension  $n \geq 1$ , with  $\text{id}_V$  in place of  $\mathbf{I}$ .

*Proof.* By Theorem 6.10.2,  $c_0 = (-1)^n \det \mathbf{A}$ , which is non-zero when  $\mathbf{A}$  is invertible (Theorem 6.5.3). By Theorem 8.6.1,

$$\mathbf{A}(\mathbf{A}^{n-1} + c_{n-1}\mathbf{A}^{n-2} + \cdots + c_1\mathbf{I}) = \mathbf{A}^n + c_{n-1}\mathbf{A}^{n-1} + \cdots + c_1\mathbf{A} = -c_0\mathbf{I}.$$

Multiplying by  $-c_0^{-1}$ , the matrix in parentheses times  $-c_0^{-1}$  is a right inverse of  $\mathbf{A}$ , hence the inverse by Theorem 2.4.5. For an operator, apply this to  $[T]_{\mathcal{B}}$  and use Corollary 3.6.3. ■

So inverting an  $n \times n$  matrix never requires more than its first  $n - 1$  powers and the coefficients of  $p_{\mathbf{A}}$ . The theorem also reduces **every** power of  $\mathbf{A}$  to the first  $n - 1$ . Divide  $x^N$  by  $p_{\mathbf{A}}$  (Theorem 5.1.1):  $x^N = qp_{\mathbf{A}} + r$  with  $\deg r < n$ . Then by Theorem 5.7.1 and Theorem 8.6.1,

$$\mathbf{A}^N = q(\mathbf{A})p_{\mathbf{A}}(\mathbf{A}) + r(\mathbf{A}) = r(\mathbf{A}).$$

When  $p_{\mathbf{A}}$  has  $n$  distinct roots  $\lambda_1, \dots, \lambda_n$ , the remainder can be found without long division. Substituting  $x = \lambda_i$  in  $x^N = qp_{\mathbf{A}} + r$  gives  $r(\lambda_i) = \lambda_i^N$  for each  $i$ , and these  $n$  conditions determine  $r \in F[x]_{\leq n-1}$  by Lagrange interpolation (Theorem 5.6.1). This substitution is legal:  $\lambda_i$  is a **scalar** plugged into an identity of polynomials.

### Example. (Cayley–Hamilton for a $3 \times 3$ Matrix)

Let

$$\mathbf{A} = \begin{pmatrix} 1 & 1 & 0 \\ 0 & 0 & 1 \\ 1 & 0 & 1 \end{pmatrix} \in M_3(\mathbb{Q}).$$

Compute  $p_{\mathbf{A}}$  and verify  $p_{\mathbf{A}}(\mathbf{A}) = 0$ . Then use it to find  $\mathbf{A}^{-1}$  and  $\mathbf{A}^5$ , and write down the matrices  $\mathbf{B}_k$  of the proof.

*Solution. Characteristic polynomial.* Expand  $\det(x\mathbf{I} - \mathbf{A})$  along the first row (Theorem 6.6.1 over  $\mathbb{Q}[x]$ ):

$$\begin{aligned} p_{\mathbf{A}} &= \det \begin{pmatrix} x-1 & -1 & 0 \\ 0 & x & -1 \\ -1 & 0 & x-1 \end{pmatrix} \\ &= (x-1) \det \begin{pmatrix} x & -1 \\ 0 & x-1 \end{pmatrix} + (-1)(-1)^{1+2} \det \begin{pmatrix} 0 & -1 \\ -1 & x-1 \end{pmatrix}. \end{aligned}$$

The two minors are  $x(x-1)$  and  $-1$ , so  $p_{\mathbf{A}} = x(x-1)^2 - 1 = x^3 - 2x^2 + x - 1$ . As a check,  $\text{tr } \mathbf{A} = 2$  matches the coefficient  $-2$  of  $x^2$ , and  $c_0 = -1 = (-1)^3 \det \mathbf{A}$  says  $\det \mathbf{A} = 1$ .

*Verification.* Multiplying out,

$$\mathbf{A}^2 = \begin{pmatrix} 1 & 1 & 1 \\ 1 & 0 & 1 \\ 2 & 1 & 1 \end{pmatrix}, \quad \mathbf{A}^3 = \mathbf{A} \mathbf{A}^2 = \begin{pmatrix} 2 & 1 & 2 \\ 2 & 1 & 1 \\ 3 & 2 & 2 \end{pmatrix}.$$

Then  $\mathbf{A}^3 - 2\mathbf{A}^2 + \mathbf{A} - \mathbf{I}$  has first row  $(2-2+1-1, 1-2+1, 2-2+0) = (0, 0, 0)$ , second row  $(2-2+0, 1-0+0-1, 1-2+1) = (0, 0, 0)$ , and third row  $(3-4+1, 2-2+0, 2-2+1-1) = (0, 0, 0)$ . So  $p_{\mathbf{A}}(\mathbf{A}) = 0$ , as the theorem predicts.

*Inverse.* Here  $c_0 = -1$ ,  $c_1 = 1$ ,  $c_2 = -2$ . By [Corollary 8.6.3](#),

$$\mathbf{A}^{-1} = \mathbf{A}^2 - 2\mathbf{A} + \mathbf{I} = \begin{pmatrix} 0 & -1 & 1 \\ 1 & 1 & -1 \\ 0 & 1 & 0 \end{pmatrix}.$$

Check: the first row of  $\mathbf{A} \mathbf{A}^{-1}$  is  $(0+1, -1+1, 1-1) = (1, 0, 0)$ , and the other rows work the same way. The entries are integers, as they must be for an integer matrix of determinant 1 ([Corollary 6.6.5](#)).

*Fifth power.* The relation  $\mathbf{A}^3 = 2\mathbf{A}^2 - \mathbf{A} + \mathbf{I}$  lets us lower degrees step by step:

$$\mathbf{A}^4 = 2\mathbf{A}^3 - \mathbf{A}^2 + \mathbf{A} = 2(2\mathbf{A}^2 - \mathbf{A} + \mathbf{I}) - \mathbf{A}^2 + \mathbf{A} = 3\mathbf{A}^2 - \mathbf{A} + 2\mathbf{I},$$

$$\mathbf{A}^5 = 3\mathbf{A}^3 - \mathbf{A}^2 + 2\mathbf{A} = 3(2\mathbf{A}^2 - \mathbf{A} + \mathbf{I}) - \mathbf{A}^2 + 2\mathbf{A} = 5\mathbf{A}^2 - \mathbf{A} + 3\mathbf{I}.$$

(Equivalently,  $x^5 = (x^2 + 2x + 3)p_{\mathbf{A}} + (5x^2 - x + 3)$ .) Hence

$$\mathbf{A}^5 = 5 \begin{pmatrix} 1 & 1 & 1 \\ 1 & 0 & 1 \\ 2 & 1 & 1 \end{pmatrix} - \begin{pmatrix} 1 & 1 & 0 \\ 0 & 0 & 1 \\ 1 & 0 & 1 \end{pmatrix} + 3\mathbf{I} = \begin{pmatrix} 7 & 4 & 5 \\ 5 & 3 & 4 \\ 9 & 5 & 7 \end{pmatrix}.$$

*The matrices of the proof.* By (2),  $\mathbf{B}_2 = \mathbf{I}$ ,  $\mathbf{B}_1 = \mathbf{A}\mathbf{B}_2 + c_2\mathbf{I} = \mathbf{A} - 2\mathbf{I}$  and  $\mathbf{B}_0 = \mathbf{A}\mathbf{B}_1 + c_1\mathbf{I} = \mathbf{A}^2 - 2\mathbf{A} + \mathbf{I}$ . So

$$\begin{aligned} \text{adj}(x\mathbf{I} - \mathbf{A}) &= x^2\mathbf{I} + x(\mathbf{A} - 2\mathbf{I}) + (\mathbf{A}^2 - 2\mathbf{A} + \mathbf{I}) \\ &= \begin{pmatrix} x^2 - x & x - 1 & 1 \\ 1 & x^2 - 2x + 1 & x - 1 \\ x & 1 & x^2 - x \end{pmatrix}, \end{aligned}$$

which one can confirm entry by entry from cofactors of  $x\mathbf{I} - \mathbf{A}$ ; for instance the  $(1, 3)$ -entry is the cofactor  $C_{31} = \det \begin{pmatrix} -1 & 0 \\ x & -1 \end{pmatrix} = 1$ . The last equation of (2),  $-\mathbf{A}\mathbf{B}_0 = c_0\mathbf{I} = -\mathbf{I}$ , is the statement  $\mathbf{A}^{-1} = \mathbf{B}_0$  found above.

*Quick check.* Let  $\mathbf{A} \in M_2(\mathbb{R})$  with  $\text{tr } \mathbf{A} = 5$  and  $\det \mathbf{A} = 6$ . Write  $\mathbf{A}^2$  and  $\mathbf{A}^3$  as combinations of  $\mathbf{A}$  and  $\mathbf{I}$ , and write  $\mathbf{A}^{-1}$  the same way.

*Solution.*  $p_{\mathbf{A}} = x^2 - 5x + 6$ , so by [Theorem 8.6.1](#)  $\mathbf{A}^2 = 5\mathbf{A} - 6\mathbf{I}$ . Then  $\mathbf{A}^3 = 5\mathbf{A}^2 - 6\mathbf{A} = 5(5\mathbf{A} - 6\mathbf{I}) - 6\mathbf{A} = 19\mathbf{A} - 30\mathbf{I}$ . By [Corollary 8.6.3](#) with  $c_0 = 6$  and  $c_1 = -5$ ,  $\mathbf{A}^{-1} = -\frac{1}{6}(\mathbf{A} - 5\mathbf{I}) = \frac{1}{6}(5\mathbf{I} - \mathbf{A})$ .

**Alternatively.** When  $p_T$  splits, there is a second proof that avoids determinants of polynomial

matrices entirely. The next section shows that such an operator has an upper triangular matrix, and a triangular matrix reveals directly why the product  $(T - \lambda_1 \text{id}_V) \cdots (T - \lambda_n \text{id}_V)$  is zero. That proof is set out as an exercise there. The adjugate proof has the advantage of working over every field at once, with no splitting hypothesis.

## 8.6.4 Exercises

### 8.6.4.1 A. Check your understanding

#### Exercise A1.

- State the Cayley–Hamilton Theorem for  $\mathbf{A} \in M_n(F)$ .
- True or false:  $\deg m_{\mathbf{A}} \leq n$  for every  $\mathbf{A} \in M_n(F)$ ,  $n \geq 1$ . Justify your answer.
- In one sentence, why is “ $p_{\mathbf{A}}(\mathbf{A}) = \det(\mathbf{A}\mathbf{I} - \mathbf{A}) = 0$ ” not a proof?
- Suppose  $p_{\mathbf{A}} = (x - 1)^2(x + 2)$ . Which polynomials can  $m_{\mathbf{A}}$  be?
- True or false: if  $p_{\mathbf{A}}(\mathbf{A}) = 0$ , then  $p_{\mathbf{A}}$  is the minimal polynomial of  $\mathbf{A}$ . Justify your answer.
- How do you write  $\mathbf{A}^{-1}$  as a polynomial in  $\mathbf{A}$  when  $\mathbf{A}$  is invertible?

*Solution.*

- If  $n \geq 1$  and  $\mathbf{A} \in M_n(F)$ , then  $p_{\mathbf{A}}(\mathbf{A}) = 0$  ([Theorem 8.6.1](#)).
- True. By [Corollary 8.6.2](#),  $m_{\mathbf{A}} \mid p_{\mathbf{A}}$ , and  $\deg p_{\mathbf{A}} = n$ .
- It substitutes a matrix into the entries of  $x\mathbf{I} - \mathbf{A}$  and then equates the matrix  $p_{\mathbf{A}}(\mathbf{A})$  with the scalar  $\det 0$ , neither of which is legitimate.
- By [Corollary 8.6.2](#) (b),  $m_{\mathbf{A}} = (x - 1)^s(x + 2)^t$  with  $1 \leq s \leq 2$  and  $t = 1$ : so  $(x - 1)(x + 2)$  or  $(x - 1)^2(x + 2)$ . Both occur, for  $\text{diag}(1, 1, -2)$  and for  $\begin{pmatrix} 1 & 1 \\ 0 & 1 \end{pmatrix} \oplus (-2)$  ([Proposition 8.5.6](#)).
- False.  $p_{\mathbf{I}_2} = (x - 1)^2$  kills  $\mathbf{I}_2$ , but  $m_{\mathbf{I}_2} = x - 1$ .
- If  $p_{\mathbf{A}} = x^n + c_{n-1}x^{n-1} + \cdots + c_0$ , then  $c_0 = (-1)^n \det \mathbf{A} \neq 0$  and  $\mathbf{A}^{-1} = -c_0^{-1}(\mathbf{A}^{n-1} + c_{n-1}\mathbf{A}^{n-2} + \cdots + c_1\mathbf{I})$  ([Corollary 8.6.3](#)).

### 8.6.4.2 B. Practice

#### Exercise B1: Inverse and a hundredth power.

Let  $\mathbf{A} = \begin{pmatrix} 3 & -2 \\ 1 & 0 \end{pmatrix} \in M_2(\mathbb{R})$ .

- Find  $p_{\mathbf{A}}$ , and use the Cayley–Hamilton Theorem to compute  $\mathbf{A}^{-1}$ .
- Hence find  $\mathbf{A}^{100}$  as a combination  $a\mathbf{A} + b\mathbf{I}$ , and write it as a matrix.

*Solution.*

- $\text{tr } \mathbf{A} = 3$  and  $\det \mathbf{A} = 2$ , so  $p_{\mathbf{A}} = x^2 - 3x + 2 = (x - 1)(x - 2)$ . By [Theorem 8.6.1](#),

$\mathbf{A}^2 - 3\mathbf{A} + 2\mathbf{I} = 0$ , that is,  $\mathbf{A}(3\mathbf{I} - \mathbf{A}) = 2\mathbf{I}$ . By [Corollary 8.6.3](#),

$$\mathbf{A}^{-1} = \frac{1}{2}(3\mathbf{I} - \mathbf{A}) = \begin{pmatrix} 0 & 1 \\ -\frac{1}{2} & \frac{3}{2} \end{pmatrix}.$$

Check:  $\mathbf{A}\mathbf{A}^{-1} = \begin{pmatrix} 0+1 & 3-3 \\ 0 & 1 \end{pmatrix} = \mathbf{I}$ .

- (b) By [Theorem 5.1.1](#),  $x^{100} = qp_{\mathbf{A}} + (ax + b)$  for some  $q$  and scalars  $a, b$ . Substituting the roots  $x = 1$  and  $x = 2$  of  $p_{\mathbf{A}}$  gives  $1 = a + b$  and  $2^{100} = 2a + b$ . So  $a = 2^{100} - 1$  and  $b = 2 - 2^{100}$ . By [Theorem 8.6.1](#),  $\mathbf{A}^{100} = q(\mathbf{A})p_{\mathbf{A}}(\mathbf{A}) + a\mathbf{A} + b\mathbf{I} = a\mathbf{A} + b\mathbf{I}$ , that is,

$$\mathbf{A}^{100} = (2^{100} - 1)\mathbf{A} + (2 - 2^{100})\mathbf{I} = \begin{pmatrix} 2^{101} - 1 & 2 - 2^{101} \\ 2^{100} - 1 & 2 - 2^{100} \end{pmatrix}.$$

For  $k = 1$  in place of 100, the same recipe gives  $a = 1$ ,  $b = 0$ , and returns  $\mathbf{A}$ , as it should.

### Exercise B2: Trace zero in size two.

- Let  $\mathbf{A} \in M_2(F)$  with  $\text{tr } \mathbf{A} = 0$ . Prove that  $\mathbf{A}^2 = -(\det \mathbf{A})\mathbf{I}$ .
- Deduce that for all  $\mathbf{X}, \mathbf{Y} \in M_2(F)$ , the matrix  $(\mathbf{XY} - \mathbf{YX})^2$  is a scalar multiple of  $\mathbf{I}$ .
- Show that the converse of (a) fails: give  $\mathbf{A} \in M_2(\mathbb{R})$  with  $\mathbf{A}^2$  scalar and  $\text{tr } \mathbf{A} \neq 0$ .

*Solution.*

- By [Theorem 8.6.1](#),  $\mathbf{A}^2 - (\text{tr } \mathbf{A})\mathbf{A} + (\det \mathbf{A})\mathbf{I} = 0$ . With  $\text{tr } \mathbf{A} = 0$ , this is  $\mathbf{A}^2 = -(\det \mathbf{A})\mathbf{I}$ .
- By [Theorem 3.10.3](#) applied to  $T_{\mathbf{X}}, T_{\mathbf{Y}}$ , or directly by  $\text{tr}(\mathbf{XY}) = \text{tr}(\mathbf{YX})$ , we have  $\text{tr}(\mathbf{XY} - \mathbf{YX}) = 0$ . By (a),  $(\mathbf{XY} - \mathbf{YX})^2 = -\det(\mathbf{XY} - \mathbf{YX})\mathbf{I}$ .
- $\mathbf{A} = \mathbf{I}_2$  has  $\mathbf{A}^2 = \mathbf{I}_2$  and  $\text{tr } \mathbf{A} = 2 \neq 0$ .

### Exercise B3: Minimal polynomial from one product.

Each of the real matrices

$$\mathbf{M}_1 = \begin{pmatrix} 2 & 1 & 0 \\ 0 & 2 & 0 \\ -1 & 1 & 3 \end{pmatrix}, \quad \mathbf{M}_2 = \begin{pmatrix} 2 & 0 & 0 \\ 0 & 2 & 0 \\ -1 & 0 & 3 \end{pmatrix}$$

has characteristic polynomial  $(x - 2)^2(x - 3)$ . Find  $m_{\mathbf{M}_1}$  and  $m_{\mathbf{M}_2}$ . Are  $\mathbf{M}_1$  and  $\mathbf{M}_2$  similar?

*Solution.* By [Corollary 8.6.2](#) (b), each minimal polynomial is  $(x - 2)(x - 3)$  or  $(x - 2)^2(x - 3)$ , and only the first needs testing.

For  $\mathbf{M}_1$ :  $\mathbf{M}_1 - 2\mathbf{I} = \begin{pmatrix} 0 & 1 & 0 \\ 0 & 0 & 0 \\ -1 & 1 & 1 \end{pmatrix}$  and  $\mathbf{M}_1 - 3\mathbf{I} = \begin{pmatrix} -1 & 1 & 0 \\ 0 & -1 & 0 \\ -1 & 1 & 0 \end{pmatrix}$ . The first row of the product  $(\mathbf{M}_1 - 2\mathbf{I})(\mathbf{M}_1 - 3\mathbf{I})$  is row 2 of  $\mathbf{M}_1 - 3\mathbf{I}$ , namely  $(0, -1, 0) \neq \mathbf{0}$ . So

$$m_{\mathbf{M}_1} = (x - 2)^2(x - 3).$$

For  $\mathbf{M}_2$ :  $\mathbf{M}_2 - 2\mathbf{I} = \begin{pmatrix} 0 & 0 & 0 \\ 0 & 0 & 0 \\ -1 & 0 & 1 \end{pmatrix}$  and  $\mathbf{M}_2 - 3\mathbf{I} = \begin{pmatrix} -1 & 0 & 0 \\ 0 & -1 & 0 \\ -1 & 0 & 0 \end{pmatrix}$ . The only non-zero row of the product is  $-(\text{row } 1) + (\text{row } 3)$  of  $\mathbf{M}_2 - 3\mathbf{I}$ , which is  $(1, 0, 0) + (-1, 0, 0) = \mathbf{0}$ . So the product is 0, and  $m_{\mathbf{M}_2} = (x - 2)(x - 3)$ .

The minimal polynomials differ, so  $\mathbf{M}_1 \not\sim \mathbf{M}_2$  by [Theorem 8.5.5](#).

### 8.6.4.3 C. Going deeper

#### Exercise C1: Cayley–Hamilton over a commutative ring.

Let  $R$  be a commutative ring (the remark in Chapter 6, §4), and suppose, as for  $F[x]$ , that the polynomials in  $x$  with coefficients in  $R$  form a commutative ring  $R[x]$  in which two polynomials are equal exactly when all their coefficients are equal. For  $\mathbf{A} \in M_n(R)$ , define  $p_{\mathbf{A}} = \det(x\mathbf{I} - \mathbf{A}) \in R[x]$ .

- Go through the proof of [Theorem 8.6.1](#) and name, for each step, the fact it uses. Explain why each fact holds over  $R$  and  $R[x]$ , and conclude that  $p_{\mathbf{A}}(\mathbf{A}) = 0$  for every  $\mathbf{A} \in M_n(R)$ .
- Let  $R = \mathbb{R}[t]$  and  $\mathbf{A} = \begin{pmatrix} t & 1 \\ t^2 & 0 \end{pmatrix}$ . Compute  $p_{\mathbf{A}}$  and verify  $p_{\mathbf{A}}(\mathbf{A}) = 0$  directly.
- Let  $\mathbf{A} \in M_n(\mathbb{Z})$  with  $\det \mathbf{A} = \pm 1$ . Using (a) with  $R = \mathbb{Z}$ , show that  $\mathbf{A}^{-1}$  has integer entries.

*Solution.*

- The shape of  $p_{\mathbf{A}}$ .* In the Leibniz sum for  $\det(x\mathbf{I} - \mathbf{A})$ , the identity permutation contributes  $(x - a_{11}) \cdots (x - a_{nn})$ , of degree  $n$  with leading coefficient 1, and every other term omits at least two diagonal entries, so it has degree at most  $n - 2$ . Hence  $p_{\mathbf{A}} = x^n + c_{n-1}x^{n-1} + \cdots + c_0$  with  $c_k \in R$ , which is what the proof uses in place of [Theorem 6.10.2](#); only sums and products of entries appear. *Step 1* uses the Leibniz formula over  $R[x]$  (which only multiplies and adds entries) and the degree bounds for sums and products, which hold because the degree of a product of polynomials is at most the sum of the degrees; no inverse is needed for the upper bound. *Step 2* uses the adjugate identity over the commutative ring  $R[x]$ , which holds by the remark after [Theorem 6.6.2](#), and coefficient comparison in  $R[x]$ , which is the assumption on  $R[x]$ . *Step 3* uses only associativity and distributivity of matrix multiplication over  $R$  and a telescoping sum. No step divides, so each works verbatim over  $R$ , and  $p_{\mathbf{A}}(\mathbf{A}) = 0$ .

- $x\mathbf{I} - \mathbf{A} = \begin{pmatrix} x - t & -1 \\ -t^2 & x \end{pmatrix}$ , so  $p_{\mathbf{A}} = x(x - t) - t^2 = x^2 - tx - t^2$ . Then

$$\mathbf{A}^2 = \begin{pmatrix} t^2 + t^2 & t \\ t^3 & t^2 \end{pmatrix}, \quad \mathbf{A}^2 - t\mathbf{A} - t^2\mathbf{I} = \begin{pmatrix} 2t^2 - t^2 - t^2 & t - t \\ t^3 - t^3 & t^2 - 0 - t^2 \end{pmatrix} = \mathbf{0}.$$

- The Leibniz formula for  $\det(x\mathbf{I} - \mathbf{A})$  only adds and multiplies integers and  $x$ , so the coefficients  $c_k$  of  $p_{\mathbf{A}}$  are integers. Regarding  $\mathbf{A}$  as a rational matrix, [Theorem 6.10.2](#) gives  $c_0 = (-1)^n \det \mathbf{A} = \pm 1$ . By (a),  $\mathbf{A}(\mathbf{A}^{n-1} + c_{n-1}\mathbf{A}^{n-2} + \cdots + c_1\mathbf{I}) = -c_0\mathbf{I}$ .

Since  $c_0^2 = 1$ , multiplying by  $-c_0$  gives  $\mathbf{A}\mathbf{B} = \mathbf{I}$  with  $\mathbf{B} = -c_0(\mathbf{A}^{n-1} + \cdots + c_1\mathbf{I}) \in M_n(\mathbb{Z})$ . Regarding  $\mathbf{A}$  and  $\mathbf{B}$  as rational matrices, [Theorem 2.4.5](#) gives  $\mathbf{B} = \mathbf{A}^{-1}$ , which therefore has integer entries. (This reproves part of [Corollary 6.6.5](#).)

**Exercise C2: The adjugate is a polynomial in the matrix.**

Let  $n \geq 2$ ,  $\mathbf{A} \in M_n(F)$ ,  $p_{\mathbf{A}} = x^n + c_{n-1}x^{n-1} + \cdots + c_0$ , and let  $\mathbf{B}_0, \dots, \mathbf{B}_{n-1}$  be as in the proof of [Theorem 8.6.1](#).

- (a) Using equation (2) of the proof, show that  $\mathbf{B}_0 = \mathbf{A}^{n-1} + c_{n-1}\mathbf{A}^{n-2} + \cdots + c_2\mathbf{A} + c_1\mathbf{I}$ .
- (b) Show that the constant matrix  $\mathbf{B}_0$  equals  $\text{adj}(-\mathbf{A})$ , and that  $\text{adj}(-\mathbf{A}) = (-1)^{n-1} \text{adj } \mathbf{A}$ .
- (c) Deduce that for **every**  $\mathbf{A} \in M_n(F)$ , invertible or not,

$$\text{adj } \mathbf{A} = (-1)^{n-1} (\mathbf{A}^{n-1} + c_{n-1}\mathbf{A}^{n-2} + \cdots + c_2\mathbf{A} + c_1\mathbf{I}).$$

- (d) Check (c) for  $\mathbf{S} = \begin{pmatrix} 1 & 2 & 3 \\ 2 & 4 & 6 \\ 1 & 0 & 1 \end{pmatrix}$ , whose adjugate Chapter 6 computed (the warning after [Corollary 6.6.3](#)), given that  $p_{\mathbf{S}} = x^3 - 6x^2 + 2x$ .

*Solution.*

- (a) Equation (2) with  $k = n$  gives  $\mathbf{B}_{n-1} = c_n\mathbf{I} = \mathbf{I}$ , and for  $1 \leq k \leq n-1$  it gives  $\mathbf{B}_{k-1} = \mathbf{A}\mathbf{B}_k + c_k\mathbf{I}$ . By downward induction on  $k$ ,  $\mathbf{B}_{k-1} = \mathbf{A}^{n-k} + c_{n-1}\mathbf{A}^{n-k-1} + \cdots + c_k\mathbf{I}$ : this holds for  $k = n$ , and if it holds for  $k$ , then  $\mathbf{B}_{k-2} = \mathbf{A}\mathbf{B}_{k-1} + c_{k-1}\mathbf{I} = \mathbf{A}^{n-k+1} + c_{n-1}\mathbf{A}^{n-k} + \cdots + c_k\mathbf{A} + c_{k-1}\mathbf{I}$ . The case  $k = 1$  is the claim.
- (b) Each entry of  $\text{adj}(x\mathbf{I} - \mathbf{A})$  is a cofactor  $\pm \det \mathbf{N}$  of  $x\mathbf{I} - \mathbf{A}$ , and its constant coefficient is its value at 0. By [Lemma 7.3.7](#),  $(\det \mathbf{N})(0) = \det \mathbf{N}(0)$ , and  $\mathbf{N}(0)$  is the corresponding submatrix of  $-\mathbf{A}$ . So the matrix of constant coefficients,  $\mathbf{B}_0$ , is  $\text{adj}(-\mathbf{A})$ . Each cofactor of  $-\mathbf{A}$  is  $\pm$  the determinant of an  $(n-1) \times (n-1)$  submatrix of  $-\mathbf{A}$ , which is  $-1$  times the corresponding submatrix of  $\mathbf{A}$ . Multiplying each of the  $n-1$  rows by  $-1$  multiplies the determinant by  $(-1)^{n-1}$  ([Theorem 6.5.1](#)). Hence  $\text{adj}(-\mathbf{A}) = (-1)^{n-1} \text{adj } \mathbf{A}$ .
- (c) By (a) and (b),  $(-1)^{n-1} \text{adj } \mathbf{A} = \mathbf{B}_0 = \mathbf{A}^{n-1} + c_{n-1}\mathbf{A}^{n-2} + \cdots + c_1\mathbf{I}$ ; multiply by  $(-1)^{n-1}$ . Nothing assumed  $\det \mathbf{A} \neq 0$ .
- (d) Here  $n = 3$ ,  $c_2 = -6$ ,  $c_1 = 2$ , so the formula predicts  $\text{adj } \mathbf{S} = \mathbf{S}^2 - 6\mathbf{S} + 2\mathbf{I}$ . Now

$$\begin{aligned} \mathbf{S}^2 &= \begin{pmatrix} 8 & 10 & 18 \\ 16 & 20 & 36 \\ 2 & 2 & 4 \end{pmatrix}, \\ \mathbf{S}^2 - 6\mathbf{S} + 2\mathbf{I} &= \begin{pmatrix} 8-6+2 & 10-12 & 18-18 \\ 16-12 & 20-24+2 & 36-36 \\ 2-6 & 2 & 4-6+2 \end{pmatrix} \\ &= \begin{pmatrix} 4 & -2 & 0 \\ 4 & -2 & 0 \\ -4 & 2 & 0 \end{pmatrix}, \end{aligned}$$

which is the adjugate found in Chapter 6.



**Exercise C3: Nilpotent matrices of size  $n$ .**

Let  $F$  be any field,  $n \geq 1$ , and  $\mathbf{A} \in M_n(F)$  with  $\mathbf{A}^k = 0$  for some  $k \geq 1$ .

- (a) Prove that  $m_{\mathbf{A}} = x^s$  for some  $s$  with  $1 \leq s \leq n$ .
- (b) Deduce that  $\mathbf{A}^n = 0$ .
- (c) Give a  $3 \times 3$  example with  $\mathbf{A}^2 \neq 0$ , and explain why no  $3 \times 3$  matrix has  $\mathbf{A}^3 \neq 0 = \mathbf{A}^4$ .

*Solution.*

- (a)  $x^k$  annihilates  $\mathbf{A}$ , so  $m_{\mathbf{A}} \mid x^k$  (Theorem 8.5.1). The polynomial  $x$  is monic and irreducible (degree 1), so by Lemma 8.5.2,  $m_{\mathbf{A}} = x^s$  with  $0 \leq s \leq k$ . By Corollary 8.6.2 (a),  $1 \leq s = \deg m_{\mathbf{A}} \leq n$ .
- (b)  $x^n = x^{n-s} x^s$ , so  $m_{\mathbf{A}} \mid x^n$  and  $\mathbf{A}^n = 0$  by Theorem 8.5.1.
- (c) The matrix  $\mathbf{N}$  of the Quick check in Section 5 has  $\mathbf{N}^2 \neq 0 = \mathbf{N}^3$ . If  $\mathbf{A} \in M_3(F)$  satisfies  $\mathbf{A}^4 = 0$ , then  $\mathbf{A}$  is nilpotent and  $\mathbf{A}^3 = 0$  by (b), so  $\mathbf{A}^3 \neq 0 = \mathbf{A}^4$  is impossible. Note that no splitting or complex numbers were needed; compare Exercise (C2: Nilpotent operators), which worked over  $\mathbb{C}$ .

## 8.7 Triangularization

Section 4 found two ways for diagonalization to fail. The field can be too small, as for a rotation of  $\mathbb{R}^2$ . Or there can be too few eigenvectors, as for  $\begin{pmatrix} 1 & 1 \\ 0 & 1 \end{pmatrix}$ , and no change of field helps. In the second case we should ask for less. The next simplest shape after a diagonal matrix is an upper triangular one, and triangular matrices are almost as good for computation: their eigenvalues, trace and determinant are visible on the diagonal. This section proves that an operator has a triangular matrix exactly when its characteristic polynomial splits, so every complex matrix is similar to a triangular one. The proof is the induction on dimension that Section 1 promised: take one eigenvector, pass to the quotient, and repeat.

### 8.7.1 Triangular matrices and chains of invariant subspaces

As in Section 2, start with the wish and unwind it. Let  $\mathcal{B} = (\mathbf{v}_1, \dots, \mathbf{v}_n)$  be a basis of  $V$  and  $T \in \mathcal{L}(V)$ . Column  $j$  of  $[T]_{\mathcal{B}}$  holds the coordinates of  $T\mathbf{v}_j$  (Definition 3.6.1). The matrix is upper triangular when every entry below the diagonal is 0 (Definition 0.9.5), that is, when  $T\mathbf{v}_j$  is a combination of  $\mathbf{v}_1, \dots, \mathbf{v}_j$  only. So the first basis vector must be sent to a multiple of itself, the second into the plane of the first two, and so on. In the language of Section 1, each of the growing spans must be invariant. Chapter 3 proved this as an exercise (Exercise (C1: Upper triangular matrices and invariant subspaces)), where the word *invariant* was introduced just for that problem; here it is again, with one more piece of information about the diagonal.

### Theorem 8.7.1 — Triangular Matrices and Chains of Invariant Subspaces

Let  $V$  be finite-dimensional with basis  $\mathcal{B} = (\mathbf{v}_1, \dots, \mathbf{v}_n)$ , let  $T \in \mathcal{L}(V)$ , and put  $U_0 = \{0\}$  and  $U_k = \text{span}(\mathbf{v}_1, \dots, \mathbf{v}_k)$  for  $1 \leq k \leq n$ . The following are equivalent:

- (a)  $[T]_{\mathcal{B}}$  is upper triangular;
- (b)  $T\mathbf{v}_k \in U_k$  for every  $k = 1, \dots, n$ ;
- (c) each  $U_k$  is  $T$ -invariant.

In that case, if  $d_1, \dots, d_n$  are the diagonal entries of  $[T]_{\mathcal{B}}$ , then  $(T - d_k \text{id}_V)(U_k) \subseteq U_{k-1}$  for every  $k$ .

*Proof.* Write  $[T]_{\mathcal{B}} = (a_{ij})$ , so that  $T\mathbf{v}_j = \sum_i a_{ij}\mathbf{v}_i$  with unique coefficients (Definition 3.6.1, Theorem 1.5.1).

- (a)  $\Leftrightarrow$  (b). If  $a_{ij} = 0$  for  $i > j$ , then  $T\mathbf{v}_j = \sum_{i \leq j} a_{ij}\mathbf{v}_i \in U_j$ . Conversely, if  $T\mathbf{v}_j \in U_j$ , it has a representation using only  $\mathbf{v}_1, \dots, \mathbf{v}_j$ ; by uniqueness,  $a_{ij} = 0$  for  $i > j$ .
- (b)  $\Rightarrow$  (c). For  $j \leq k$ ,  $T\mathbf{v}_j \in U_j \subseteq U_k$ , so  $U_k$  is invariant by Lemma 8.1.1.
- (c)  $\Rightarrow$  (b).  $\mathbf{v}_k \in U_k$ , so  $T\mathbf{v}_k \in U_k$  by invariance.

Finally, suppose (a) holds and let  $\mathbf{u} = \sum_{j \leq k} b_j \mathbf{v}_j \in U_k$ . For  $j < k$ ,  $(T - d_k \text{id}_V)\mathbf{v}_j = T\mathbf{v}_j - d_k \mathbf{v}_j \in U_j \subseteq U_{k-1}$ . For  $j = k$ ,  $(T - d_k \text{id}_V)\mathbf{v}_k = \sum_{i \leq k} a_{ik}\mathbf{v}_i - a_{kk}\mathbf{v}_k = \sum_{i < k} a_{ik}\mathbf{v}_i \in U_{k-1}$ , since  $d_k = a_{kk}$ . By linearity,  $(T - d_k \text{id}_V)\mathbf{u} \in U_{k-1}$ . ■

A chain  $U_1 \subset U_2 \subset \dots \subset U_n = V$  of subspaces with  $\dim U_k = k$  is often called a **flag**, after the picture of a point on a line in a plane in space. The theorem says: an upper triangular matrix is the same thing as a flag of invariant subspaces, together with a basis adapted to it. The last statement says what the diagonal entry  $d_k$  measures. Modulo the smaller space  $U_{k-1}$ , the operator  $T$  acts on the new basis vector  $\mathbf{v}_k$  as multiplication by  $d_k$ .

The operators for which such a basis exists deserve a name.

### Definition 8.7.1 — Triangularizable

Let  $V$  be a finite-dimensional vector space over  $F$ . An operator  $T \in \mathcal{L}(V)$  is **triangularizable over  $F$**  if there **exists** a basis  $\mathcal{B}$  of  $V$  such that  $[T]_{\mathcal{B}}$  is upper triangular. A matrix  $\mathbf{A} \in M_n(F)$  is **triangularizable over  $F$**  if there **exist** an **invertible**  $\mathbf{P} \in M_n(F)$  and an upper triangular  $\mathbf{U} \in M_n(F)$  with  $\mathbf{P}^{-1}\mathbf{A}\mathbf{P} = \mathbf{U}$ .

As for diagonalizability,  $\mathbf{A}$  is triangularizable over  $F$  exactly when  $T_{\mathbf{A}}$  is, by Theorem 3.7.4.

#### Examples.

- **Diagonalizable operators** are triangularizable, since a diagonal matrix is upper triangular. So are all operators on a space of dimension 1, and the zero and identity operators.
- **Differentiation.** For  $D$  on  $\mathbb{R}[x]_{\leq n}$  in the basis  $(1, x, \dots, x^n)$ , the matrix is upper triangular.

lar with zero diagonal:  $D(x^k) = kx^{k-1} \in \text{span}(1, \dots, x^k)$ . The flag is  $\mathbb{R}[x]_{\leq 0} \subset \mathbb{R}[x]_{\leq 1} \subset \dots \subset \mathbb{R}[x]_{\leq n}$ , a chain of invariant subspaces we met in Section 1.  $D$  is not diagonalizable (its only eigenvectors are the constants), but it is triangularizable.

- **Too few eigenvectors.**  $\mathbf{J} = \begin{pmatrix} 1 & 1 \\ 0 & 1 \end{pmatrix}$  is already triangular, with flag  $\text{span}(\mathbf{e}_1) \subset F^2$ .

**Non-example by minimal change.** Reverse the basis for  $D$  on  $\mathbb{R}[x]_{\leq 2}$ : take  $(x^2, x, 1)$ . The spans are  $\text{span}(x^2) \subset \text{span}(x^2, x) \subset \mathbb{R}[x]_{\leq 2}$ , still a chain of subspaces of dimensions 1, 2, 3. What fails is invariance of the first one:  $D(x^2) = 2x \notin \text{span}(x^2)$ . Accordingly, the matrix in this basis is

$$\begin{pmatrix} 0 & 0 & 0 \\ 2 & 0 & 0 \\ 0 & 1 & 0 \end{pmatrix},$$

which is lower triangular, not upper. Being triangularizable is a property of  $T$ ; being triangular is a property of  $T$  **and** a basis.

A triangularizable operator on  $V \neq \{\mathbf{0}\}$  must have an invariant line  $U_1 = \text{span}(\mathbf{v}_1)$ , hence an eigenvector (Proposition 8.1.2). So the rotation  $R$  of  $\mathbb{R}^2$  by a right angle, which has no invariant line (Example (A Rotation Has No Invariant Lines)), is **not** triangularizable over  $\mathbb{R}$ . An eigenvector to start the flag is exactly what is needed, and the next theorem shows that it is also enough, provided eigenvectors keep existing at every stage.

*Quick check.* Is  $\begin{pmatrix} 2 & 0 \\ 1 & 2 \end{pmatrix} \in M_2(\mathbb{R})$  triangularizable over  $\mathbb{R}$ ? If so, find a basis in which its matrix is upper triangular.

*Solution.* Yes. We need a first basis vector that is an eigenvector. The matrix sends  $\mathbf{e}_2$  to  $(0, 2) = 2\mathbf{e}_2$ , so start with  $\mathbf{e}_2$ , and complete with  $\mathbf{e}_1$ . In the basis  $(\mathbf{e}_2, \mathbf{e}_1)$ , the image of  $\mathbf{e}_2$  is  $2\mathbf{e}_2$  and the image of  $\mathbf{e}_1$  is  $(2, 1) = \mathbf{e}_2 + 2\mathbf{e}_1$ , so the matrix is  $\begin{pmatrix} 2 & 1 \\ 0 & 2 \end{pmatrix}$ , which is upper triangular. (The standard basis gives a lower triangular matrix:  $\text{span}(\mathbf{e}_1)$  is not invariant.)

## 8.7.2 The Triangularization Theorem

If  $T$  has a triangular matrix, its characteristic polynomial is a product of linear factors, one for each diagonal entry. The converse is the real content: splitting of  $p_T$  is enough to build the flag.

### Theorem 8.7.2 — Triangularization Theorem

Let  $V$  be a finite-dimensional vector space over  $F$  with  $\dim V = n \geq 1$ , and let  $T \in \mathcal{L}(V)$ . Then  $T$  is triangularizable over  $F$  if and only if  $p_T$  splits over  $F$ . The same holds for  $\mathbf{A} \in M_n(F)$  and  $p_{\mathbf{A}}$ .

*Idea.* Take one eigenvector, extend, recurse. Since  $p_T$  splits, it has a root in  $F$ , so  $T$  has an eigenvector  $\mathbf{v}_1$ . Put  $U = \text{span}(\mathbf{v}_1)$  and extend  $\mathbf{v}_1$  to a basis. By Theorem 8.1.3 the matrix looks like

$$\begin{pmatrix} \lambda & * \\ 0 & [\bar{T}] \end{pmatrix},$$

where  $\bar{T}$  is the induced operator on the smaller space  $V/U$ . The first column is already right. If  $[\bar{T}]$  could be made triangular, the whole matrix would be. Now  $\bar{T}$  lives on a space of dimension  $n - 1$ , and  $p_T = (x - \lambda)p_{\bar{T}}$ , so  $p_{\bar{T}}$  splits too: this is the induction on dimension. The only technical point is that a basis of  $V/U$  lifts to vectors that, together with  $\mathbf{v}_1$ , form a basis of  $V$ .

*Proof.* ( $\Rightarrow$ ) Let  $[T]_{\mathcal{B}}$  be upper triangular with diagonal entries  $d_1, \dots, d_n$ . The matrix  $x\mathbf{I}_n - [T]_{\mathcal{B}}$  is upper triangular with diagonal entries  $x - d_i$ , and [Theorem 6.4.6](#) holds over  $F[x]$  (the remark at the end of Chapter 6, §4). So  $p_T = (x - d_1) \cdots (x - d_n)$  by [Definition 6.10.2](#), which splits ([Definition 5.4.2](#)).

( $\Leftarrow$ ) We prove, by induction on  $n \geq 1$ , that every operator on an  $n$ -dimensional space whose characteristic polynomial splits is triangularizable. For  $n = 1$ , every  $1 \times 1$  matrix is upper triangular.

Let  $n \geq 2$ , and assume the statement for spaces of dimension  $n - 1$ . Since  $p_T$  is monic of degree  $n$  ([Theorem 6.10.2](#)) and splits,  $p_T = (x - c_1)(x - c_2) \cdots (x - c_n)$  with  $c_i \in F$ . As  $p_T(c_1) = 0$ , [Theorem 8.2.1](#) gives an eigenvector  $\mathbf{v}_1$  with  $T\mathbf{v}_1 = c_1\mathbf{v}_1$ . Let  $U = \text{span}(\mathbf{v}_1)$ . It is  $T$ -invariant ([Proposition 8.1.2](#)),  $\dim U = 1$ , and  $U \neq V$  since  $n \geq 2$ . Let  $\bar{T}$  be the induced operator on  $V/U$  ([Definition 8.1.2](#)), a space of dimension  $n - 1 \geq 1$  ([Theorem 3.9.5](#)).

$p_{\bar{T}}$  splits. The restriction  $T|_U$  is  $c_1 \text{id}_U$ , whose matrix in the basis  $(\mathbf{v}_1)$  is  $(c_1)$ , so  $p_{T|_U} = x - c_1$ . By [Theorem 8.1.3](#) (b),

$$(x - c_1)(x - c_2) \cdots (x - c_n) = p_T = (x - c_1)p_{\bar{T}}.$$

Canceling the non-zero factor  $x - c_1$  ([Corollary 0.8.4](#) (2)) gives  $p_{\bar{T}} = (x - c_2) \cdots (x - c_n)$ , which splits.

*Apply the induction hypothesis.* There is a basis  $\bar{\mathcal{C}} = (\mathbf{w}_2 + U, \dots, \mathbf{w}_n + U)$  of  $V/U$  with  $[\bar{T}]_{\bar{\mathcal{C}}}$  upper triangular, for some  $\mathbf{w}_2, \dots, \mathbf{w}_n \in V$ .

*Lift the basis.* Let  $\mathcal{B} = (\mathbf{v}_1, \mathbf{w}_2, \dots, \mathbf{w}_n)$ . Suppose  $a\mathbf{v}_1 + b_2\mathbf{w}_2 + \cdots + b_n\mathbf{w}_n = \mathbf{0}$ . Passing to cosets, and using  $\mathbf{v}_1 \in U$ , gives  $b_2(\mathbf{w}_2 + U) + \cdots + b_n(\mathbf{w}_n + U) = \mathbf{0} + U$  ([Theorem 3.9.4](#)). Since  $\bar{\mathcal{C}}$  is a basis, all  $b_i = 0$ . Then  $a\mathbf{v}_1 = \mathbf{0}$  with  $\mathbf{v}_1 \neq \mathbf{0}$ , so  $a = 0$  ([Theorem 1.1.9](#)). Thus  $\mathcal{B}$  is a linearly independent list of  $n$  vectors in  $V$ , hence a basis ([Theorem 1.6.4](#)).

*The matrix.*  $\mathcal{B}$  extends the basis  $(\mathbf{v}_1)$  of  $U$ , and the cosets of the added vectors form  $\bar{\mathcal{C}}$ . By [Theorem 8.1.3](#) (b),

$$[T]_{\mathcal{B}} = \begin{pmatrix} c_1 & * \\ 0 & [\bar{T}]_{\bar{\mathcal{C}}} \end{pmatrix},$$

where the 0 is a column of  $n - 1$  zeros. The block  $[\bar{T}]_{\bar{\mathcal{C}}}$  is upper triangular, so  $[T]_{\mathcal{B}}$  is upper triangular. This completes the induction.

For  $\mathbf{A} \in M_n(F)$ , apply the result to  $T_{\mathbf{A}}$ , using [Theorem 3.7.4](#). ■

The proof is an algorithm, and its output is a flag. The first invariant subspace is the eigenline  $U$ ; the  $k$ -th is spanned by  $\mathbf{v}_1$  and the lifts  $\mathbf{w}_2, \dots, \mathbf{w}_k$  of the first  $k - 1$  vectors of the flag for  $\bar{T}$ . The eigenvalue found at each stage is the next diagonal entry.

Over the complex numbers the hypothesis is automatic.

### Corollary 8.7.3 — Complex Operators Are Triangularizable

Every operator on a finite-dimensional complex vector space  $V \neq \{0\}$  is triangularizable over  $\mathbb{C}$ . Every  $\mathbf{A} \in M_n(\mathbb{C})$ ,  $n \geq 1$ , is similar to an upper triangular matrix in  $M_n(\mathbb{C})$ .

*Proof.* By Theorem 6.10.2,  $p_T$  is non-zero, and by Corollary 5.5.5 (a) it splits over  $\mathbb{C}$ . Apply Theorem 8.7.2. ■

So over  $\mathbb{C}$  the question of Chapter 3 always has at least this answer: a basis exists in which the matrix has zeros below the diagonal. A real matrix can always be triangularized as a complex matrix, but perhaps only with complex entries.

**Warning.** Triangularizing may require leaving the field, and the triangular form is far from unique. The rotation  $\mathbf{R} = \begin{pmatrix} 0 & -1 \\ 1 & 0 \end{pmatrix}$  has  $p_{\mathbf{R}} = x^2 + 1$ , which does not split over  $\mathbb{R}$ , so no **real** invertible  $\mathbf{P}$  makes  $\mathbf{P}^{-1}\mathbf{R}\mathbf{P}$  triangular; over  $\mathbb{C}$  it becomes  $\text{diag}(i, -i)$ . And a triangularizable matrix has many triangular forms.  $\begin{pmatrix} 1 & 1 \\ 0 & 2 \end{pmatrix}$  and  $\begin{pmatrix} 2 & 0 \\ 0 & 1 \end{pmatrix}$  are both triangular forms of the same operator (the first has distinct eigenvalues 1, 2, so it is diagonalizable, and similar to the second). The off-diagonal entries change with the flag and the basis, and even the order of the diagonal entries is not fixed. Only the diagonal entries, as a list with repetitions, are determined, as the next theorem shows.

### 8.7.3 What the diagonal records

A diagonalization displays the eigenvalues. A triangularization does too, with their algebraic multiplicities.

#### Theorem 8.7.4 — The Diagonal of a Triangular Form

Let  $V$  be finite-dimensional with  $\dim V = n \geq 1$ ,  $T \in \mathcal{L}(V)$ , and let  $\mathcal{B}$  be a basis with  $[T]_{\mathcal{B}}$  upper triangular, with diagonal entries  $d_1, \dots, d_n$ . Then:

- (a)  $p_T = (x - d_1)(x - d_2) \cdots (x - d_n)$ ;
- (b)  $\text{spec}(T) = \{d_1, \dots, d_n\}$ , and each eigenvalue  $\lambda$  occurs among  $d_1, \dots, d_n$  exactly  $a_T(\lambda)$  times;
- (c)  $\text{tr } T = d_1 + \cdots + d_n$  and  $\det T = d_1 \cdots d_n$ .

In particular, any two triangular forms of  $T$  have the same diagonal entries, up to order.

*Proof.*

- (a) was shown in the proof of [Theorem 8.7.2](#), ( $\Rightarrow$ ).
- (b) By [Theorem 8.2.1](#), the eigenvalues are the roots of  $p_T$  in  $F$ , and  $\lambda$  is a root of  $\prod_i (x - d_i)$  exactly when  $\prod_i (\lambda - d_i) = 0$ , that is, when  $\lambda = d_i$  for some  $i$  ([Theorem 0.7.1](#)). By [Theorem 5.4.2](#),  $a_T(\lambda) = \text{mult}_\lambda(p_T) = \sum_i \text{mult}_\lambda(x - d_i)$ . Each term is 1 if  $d_i = \lambda$  and 0 otherwise, by [Theorem 5.1.3](#) (b), since  $x - d_i$  is not divisible by  $(x - \lambda)^2$  for degree reasons. So the sum counts the indices with  $d_i = \lambda$ .
- (c) By [Definition 3.10.2](#),  $\text{tr } T = \text{tr}[T]_{\mathcal{B}} = \sum_i d_i$ . By [Definition 6.5.1](#) and [Theorem 6.4.6](#),  $\det T = \det[T]_{\mathcal{B}} = \prod_i d_i$ .

The final statement follows from (b): the number of times  $\lambda$  occurs on the diagonal is  $a_T(\lambda)$ , which does not depend on the basis. ■

Part (c) recovers [Theorem 8.3.3](#). In Section 3 it came from comparing coefficients of  $p_T$ ; here it is read off a matrix. For instance, a complex  $3 \times 3$  matrix with  $p_{\mathbf{A}} = (x - 1)^2(x + 4)$  is similar to a triangular matrix with diagonal 1, 1,  $-4$  in some order, so  $\text{tr } \mathbf{A} = -2$  and  $\det \mathbf{A} = -4$ , however complicated  $\mathbf{A}$  looks.

The theorem does **not** say that a triangular form with **distinct** diagonal entries is itself diagonal:  $\begin{pmatrix} 1 & 1 \\ 0 & 2 \end{pmatrix}$  is a counterexample. What does follow is that  $T$  is then diagonalizable ([Corollary 8.4.2](#)), so some **other** basis makes its matrix diagonal. And an operator with repeated diagonal entries may or may not be diagonalizable: compare  $\mathbf{I}_2$  with  $\mathbf{J}$ .

## 8.7.4 Triangularizing a matrix in practice

For a matrix  $\mathbf{A} \in M_n(F)$  whose characteristic polynomial splits, the proof of [Theorem 8.7.2](#) becomes a procedure.

1. Find an eigenvalue  $\lambda_1$  and an eigenvector  $\mathbf{v}_1$ .
2. Extend  $\mathbf{v}_1$  to a basis  $(\mathbf{v}_1, \mathbf{w}_2, \dots, \mathbf{w}_n)$  of  $F^n$ , choosing the  $\mathbf{w}_i$  as simple as possible, for example standard basis vectors.
3. Compute the matrix of the induced operator on  $F^n / \text{span}(\mathbf{v}_1)$  in the basis  $(\mathbf{w}_2 + U, \dots, \mathbf{w}_n + U)$ . Its entries are the coordinates of  $\mathbf{A}\mathbf{w}_j$  along  $\mathbf{w}_2, \dots, \mathbf{w}_n$ , ignoring the  $\mathbf{v}_1$ -coordinate.
4. Triangularize this  $(n - 1) \times (n - 1)$  matrix in the same way, and lift each coset back to its representative.

### Example. (Triangularizing a Matrix with a Single Eigenvector Line)

Let

$$\mathbf{A} = \begin{pmatrix} 2 & 0 & 1 \\ 0 & 2 & 1 \\ 1 & -1 & 2 \end{pmatrix} \in M_3(\mathbb{R}).$$

Show that  $\mathbf{A}$  is not diagonalizable but is triangularizable over  $\mathbb{R}$ , and find an invertible  $\mathbf{P}$  with  $\mathbf{P}^{-1}\mathbf{A}\mathbf{P}$  upper triangular.

*Solution.* The characteristic polynomial. Expanding along the first row ([Theorem 6.6.1](#)

over  $\mathbb{R}[x]$ ),

$$\begin{aligned} p_{\mathbf{A}} &= \det \begin{pmatrix} x-2 & 0 & -1 \\ 0 & x-2 & -1 \\ -1 & 1 & x-2 \end{pmatrix} \\ &= (x-2)[(x-2)^2 + 1] + (-1) \det \begin{pmatrix} 0 & x-2 \\ -1 & 1 \end{pmatrix}. \end{aligned}$$

The last determinant is  $0 \cdot 1 - (x-2)(-1) = x-2$ , so  $p_{\mathbf{A}} = (x-2)[(x-2)^2 + 1 - 1] = (x-2)^3$ . It splits, so  $\mathbf{A}$  is triangularizable by [Theorem 8.7.2](#).

*Eigenvectors.*  $\mathbf{A} - 2\mathbf{I} = \begin{pmatrix} 0 & 0 & 1 \\ 0 & 0 & 1 \\ 1 & -1 & 0 \end{pmatrix}$  has rank 2, so  $g(2) = 1 < 3 = a(2)$ , and  $\mathbf{A}$  is not diagonalizable ([Theorem 8.4.1](#) (e)). The system gives  $z = 0$  and  $x = y$ , so  $E_2(\mathbf{A}) = \text{span}(\mathbf{v}_1)$  with  $\mathbf{v}_1 = (1, 1, 0)$ : all eigenvectors lie on one line.

*Step 1.* Let  $U = \text{span}(\mathbf{v}_1)$ , and extend to the basis  $(\mathbf{v}_1, \mathbf{e}_2, \mathbf{e}_3)$  of  $\mathbb{R}^3$  (the matrix with these columns has determinant 1).

*Step 2: the induced operator.* Express the images of  $\mathbf{e}_2$  and  $\mathbf{e}_3$  in the basis:

$$\mathbf{A}\mathbf{e}_2 = (0, 2, -1) = 0\mathbf{v}_1 + 2\mathbf{e}_2 - \mathbf{e}_3, \quad \mathbf{A}\mathbf{e}_3 = (1, 1, 2) = 1\mathbf{v}_1 + 0\mathbf{e}_2 + 2\mathbf{e}_3.$$

Dropping the  $\mathbf{v}_1$ -coordinates, the induced operator  $\bar{\mathbf{A}}$  on  $\mathbb{R}^3/U$  has, in the basis  $(\mathbf{e}_2 + U, \mathbf{e}_3 + U)$ , the matrix

$$[\bar{\mathbf{A}}] = \begin{pmatrix} 2 & 0 \\ -1 & 2 \end{pmatrix},$$

with  $p_{\bar{\mathbf{A}}} = (x-2)^2$ , as the proof predicts.

*Step 3: triangularize  $\bar{\mathbf{A}}$ .* Its second column says  $\bar{\mathbf{A}}(\mathbf{e}_3 + U) = 2(\mathbf{e}_3 + U)$ , so  $\mathbf{e}_3 + U$  is an eigenvector. In the basis  $(\mathbf{e}_3 + U, \mathbf{e}_2 + U)$ ,  $\bar{\mathbf{A}}(\mathbf{e}_2 + U) = -(\mathbf{e}_3 + U) + 2(\mathbf{e}_2 + U)$ , and the matrix becomes  $\begin{pmatrix} 2 & -1 \\ 0 & 2 \end{pmatrix}$ .

*Step 4: lift.* The basis  $\mathcal{B} = (\mathbf{v}_1, \mathbf{e}_3, \mathbf{e}_2)$  of  $\mathbb{R}^3$  gives

$$\mathbf{P} = \begin{pmatrix} 1 & 0 & 0 \\ 1 & 0 & 1 \\ 0 & 1 & 0 \end{pmatrix}, \quad \mathbf{P}^{-1}\mathbf{A}\mathbf{P} = \begin{pmatrix} 2 & 1 & 0 \\ 0 & 2 & -1 \\ 0 & 0 & 2 \end{pmatrix}.$$

*Check.*  $\mathbf{A}\mathbf{v}_1 = (2, 2, 0) = 2\mathbf{v}_1$ ;  $\mathbf{A}\mathbf{e}_3 = (1, 1, 2) = \mathbf{v}_1 + 2\mathbf{e}_3$ ;  $\mathbf{A}\mathbf{e}_2 = (0, 2, -1) = -\mathbf{e}_3 + 2\mathbf{e}_2$ . These are the columns  $(2, 0, 0)$ ,  $(1, 2, 0)$ ,  $(0, -1, 2)$  of the triangular matrix, read in the basis  $\mathcal{B}$ . Also  $\det \mathbf{P} = -1 \neq 0$ .

The flag is  $\text{span}((1, 1, 0)) \subset \{(a, a, b) : a, b \in \mathbb{R}\} \subset \mathbb{R}^3$ . One can check the plane directly:  $\mathbf{A}(a, a, b) = (2a + b, 2a + b, 2b)$  lies in it again. The diagonal is  $2, 2, 2$ , as [Theorem 8.7.4](#) requires.

Different choices in Step 2 give different triangular matrices, with the same diagonal. Chapter 9 shows how to choose the basis so that the entries above the diagonal are as simple as possible (the Jordan form). Chapter 11 shows that over  $\mathbb{C}$  with an inner product, the basis can even be chosen orthonormal (Schur's theorem).

## 8.7.5 Exercises

### 8.7.5.1 A. Check your understanding

#### Exercise A1.

- (a) Define what it means for  $\mathbf{A} \in M_n(F)$  to be triangularizable over  $F$ .
- (b) State the Triangularization Theorem.
- (c) For a basis  $(\mathbf{v}_1, \dots, \mathbf{v}_n)$ , which subspaces must be invariant for  $[T]_{\mathcal{B}}$  to be upper triangular?
- (d) True or false: every  $\mathbf{A} \in M_n(\mathbb{R})$  is triangularizable over  $\mathbb{R}$ . Justify your answer.
- (e) True or false: every triangularizable matrix is diagonalizable. Justify your answer.
- (f) Let  $\mathbf{A} \in M_4(\mathbb{C})$  have  $p_{\mathbf{A}} = (x - 3)^3(x + 1)$ . What are the diagonal entries of any upper triangular matrix similar to  $\mathbf{A}$ , and what are  $\text{tr } \mathbf{A}$  and  $\det \mathbf{A}$ ?

*Solution.*

- (a) There are an invertible  $\mathbf{P} \in M_n(F)$  and an upper triangular  $\mathbf{U} \in M_n(F)$  with  $\mathbf{P}^{-1}\mathbf{A}\mathbf{P} = \mathbf{U}$  (Definition 8.7.1).
- (b) An operator on a space of dimension  $n \geq 1$  over  $F$  is triangularizable over  $F$  if and only if its characteristic polynomial splits over  $F$  (Theorem 8.7.2).
- (c) All the spans  $\text{span}(\mathbf{v}_1, \dots, \mathbf{v}_k)$ ,  $1 \leq k \leq n$  (Theorem 8.7.1).
- (d) False. The rotation  $\begin{pmatrix} 0 & -1 \\ 1 & 0 \end{pmatrix}$  has  $p = x^2 + 1$ , which does not split over  $\mathbb{R}$ .
- (e) False.  $\begin{pmatrix} 1 & 1 \\ 0 & 1 \end{pmatrix}$  is triangular but not diagonalizable.
- (f) By Theorem 8.7.4, the diagonal is  $3, 3, 3, -1$  in some order, so  $\text{tr } \mathbf{A} = 8$  and  $\det \mathbf{A} = -27$ .

### 8.7.5.2 B. Practice

#### Exercise B1: Triangularizing matrices.

For each real matrix, find an invertible  $\mathbf{P}$  such that  $\mathbf{P}^{-1}\mathbf{A}\mathbf{P}$  is upper triangular, and write down  $\mathbf{P}^{-1}\mathbf{A}\mathbf{P}$ .

- (a)  $\mathbf{A} = \begin{pmatrix} 3 & 1 \\ -1 & 1 \end{pmatrix}$ .
- (b)  $\mathbf{A} = \begin{pmatrix} 2 & 0 & 1 \\ 1 & 3 & 1 \\ 1 & 0 & 2 \end{pmatrix}$ , with  $p_{\mathbf{A}} = (x - 1)(x - 3)^2$  (Example (A  $3 \times 3$  Characteristic Polynomial)).

*Solution.*

- (a)  $\text{tr } \mathbf{A} = 4$ ,  $\det \mathbf{A} = 4$ , so  $p_{\mathbf{A}} = (x - 2)^2$ , which splits.  $\mathbf{A} - 2\mathbf{I} = \begin{pmatrix} 1 & 1 \\ -1 & -1 \end{pmatrix}$  gives



the eigenvector  $\mathbf{v}_1 = (1, -1)$ . Extend with  $\mathbf{e}_2$ :  $\mathbf{A}\mathbf{e}_2 = (1, 1) = \mathbf{v}_1 + 2\mathbf{e}_2$ . So with

$$\mathbf{P} = \begin{pmatrix} 1 & 0 \\ -1 & 1 \end{pmatrix}, \quad \mathbf{P}^{-1}\mathbf{A}\mathbf{P} = \begin{pmatrix} 2 & 1 \\ 0 & 2 \end{pmatrix}.$$

(In size 2, one eigenvector already fixes the flag.)

- (b)  $\mathbf{A}\mathbf{e}_2 = (0, 3, 0) = 3\mathbf{e}_2$ , so  $\mathbf{v}_1 = \mathbf{e}_2$  is an eigenvector for 3. Let  $U = \text{span}(\mathbf{e}_2)$  and extend to  $(\mathbf{e}_2, \mathbf{e}_1, \mathbf{e}_3)$ . Then  $\mathbf{A}\mathbf{e}_1 = (2, 1, 1) = \mathbf{e}_2 + 2\mathbf{e}_1 + \mathbf{e}_3$  and  $\mathbf{A}\mathbf{e}_3 = (1, 1, 2) = \mathbf{e}_2 + \mathbf{e}_1 + 2\mathbf{e}_3$ , so the induced operator has matrix  $\begin{pmatrix} 2 & 1 \\ 1 & 2 \end{pmatrix}$  in the basis  $(\mathbf{e}_1 + U, \mathbf{e}_3 + U)$ . Its characteristic polynomial is  $x^2 - 4x + 3 = (x - 3)(x - 1)$ , with eigenvectors  $(1, 1)$  for 3 and  $(1, -1)$  for 1, that is,  $\mathbf{e}_1 + \mathbf{e}_3 + U$  and  $\mathbf{e}_1 - \mathbf{e}_3 + U$ . Lifting gives the basis  $(\mathbf{e}_2, \mathbf{e}_1 + \mathbf{e}_3, \mathbf{e}_1 - \mathbf{e}_3)$ . Now

$$\begin{aligned} \mathbf{A}(\mathbf{e}_1 + \mathbf{e}_3) &= (3, 2, 3) = 2\mathbf{e}_2 + 3(\mathbf{e}_1 + \mathbf{e}_3), \\ \mathbf{A}(\mathbf{e}_1 - \mathbf{e}_3) &= (1, 0, -1) = \mathbf{e}_1 - \mathbf{e}_3, \end{aligned}$$

so

$$\mathbf{P} = \begin{pmatrix} 0 & 1 & 1 \\ 1 & 0 & 0 \\ 0 & 1 & -1 \end{pmatrix}, \quad \mathbf{P}^{-1}\mathbf{A}\mathbf{P} = \begin{pmatrix} 3 & 2 & 0 \\ 0 & 3 & 0 \\ 0 & 0 & 1 \end{pmatrix},$$

with  $\det \mathbf{P} = 2 \neq 0$  (expand along the second row). The diagonal 3, 3, 1 agrees with  $p_{\mathbf{A}}$ .

### Exercise B2: Nilpotent matrices are strictly triangular.

Let  $\mathbf{A} \in M_n(\mathbb{C})$ ,  $n \geq 1$ , with  $\mathbf{A}^k = 0$  for some  $k \geq 1$ . Prove that  $\mathbf{A}$  is similar to a **strictly** upper triangular matrix, that is, an upper triangular matrix with zero diagonal.

*Solution.* By Corollary 8.7.3,  $\mathbf{P}^{-1}\mathbf{A}\mathbf{P} = \mathbf{U}$  is upper triangular for some invertible  $\mathbf{P}$ . By Theorem 8.7.4 (b), each diagonal entry of  $\mathbf{U}$  is an eigenvalue of  $\mathbf{A}$ . By Exercise (C2: Nilpotent operators) (a), the only eigenvalue of a nilpotent operator on a non-zero space is 0. So every diagonal entry of  $\mathbf{U}$  is 0, and  $\mathbf{U}$  is strictly upper triangular. (The statement holds over every field, with a different proof: use the chain  $\ker \mathbf{A} \subseteq \ker \mathbf{A}^2 \subseteq \cdots \subseteq \ker \mathbf{A}^k = F^n$  to build the basis.)

### Exercise B3: Eigenvalues of powers from a triangular form.

Let  $\mathbf{A} \in M_n(\mathbb{C})$ ,  $n \geq 1$ , and let  $\mathbf{P}^{-1}\mathbf{A}\mathbf{P} = \mathbf{U}$  be upper triangular with diagonal entries  $d_1, \dots, d_n$ .

- Show that the product of two upper triangular matrices is upper triangular, with diagonal entries the products of the corresponding diagonal entries.
- Deduce that  $p_{\mathbf{A}^k} = (x - d_1^k) \cdots (x - d_n^k)$  and  $\text{tr}(\mathbf{A}^k) = d_1^k + \cdots + d_n^k$  for every  $k \geq 1$ .
- Suppose  $p_{\mathbf{A}} = (x - 1)(x + 1)(x - 2)$ . Find  $p_{\mathbf{A}^2}$ ,  $\text{tr} \mathbf{A}^2$  and  $\det \mathbf{A}^2$ .

*Solution.*

- Let  $\mathbf{U}, \mathbf{U}'$  be upper triangular. The  $(i, j)$ -entry of  $\mathbf{U}\mathbf{U}'$  is  $\sum_l u_{il}u'_{lj}$ . A term can

be non-zero only if  $u_{il} \neq 0$  and  $u'_{lj} \neq 0$ , which forces  $i \leq l \leq j$ . So the entry is 0 when  $i > j$ , and for  $i = j$  only  $l = i$  survives, giving  $u_{ii}u'_{ii}$ .

- (b)  $\mathbf{P}^{-1}\mathbf{A}^k\mathbf{P} = (\mathbf{P}^{-1}\mathbf{A}\mathbf{P})^k = \mathbf{U}^k$ , by induction on  $k$ . By (a) and induction,  $\mathbf{U}^k$  is upper triangular with diagonal entries  $d_i^k$ . By [Theorem 8.7.4](#) (a) and (c), applied to  $T_{\mathbf{A}^k}$  in the basis given by the columns of  $\mathbf{P}$ ,  $p_{\mathbf{A}^k} = \prod_i (x - d_i^k)$  and  $\text{tr } \mathbf{A}^k = \sum_i d_i^k$ .
- (c) By [Theorem 8.7.4](#), the diagonal of  $\mathbf{U}$  is  $1, -1, 2$  in some order. By (b),  $p_{\mathbf{A}^2} = (x-1)(x-1)(x-4) = (x-1)^2(x-4)$ ,  $\text{tr } \mathbf{A}^2 = 1+1+4 = 6$ , and  $\det \mathbf{A}^2 = 1 \cdot 1 \cdot 4 = 4$ . The last agrees with  $\det \mathbf{A}^2 = (\det \mathbf{A})^2 = (-2)^2$ .

### 8.7.5.3 C. Going deeper

#### Exercise C1: A second proof of Cayley–Hamilton.

Let  $V$  be finite-dimensional with  $\dim V = n \geq 1$ , and let  $T \in \mathcal{L}(V)$  have a basis  $\mathcal{B} = (\mathbf{v}_1, \dots, \mathbf{v}_n)$  with  $[T]_{\mathcal{B}}$  upper triangular, with diagonal entries  $d_1, \dots, d_n$ . Let  $U_0 = \{\mathbf{0}\}$  and  $U_k = \text{span}(\mathbf{v}_1, \dots, \mathbf{v}_k)$ .

- (a) Let  $S_k = (T - d_1 \text{id}_V)(T - d_2 \text{id}_V) \cdots (T - d_k \text{id}_V)$ . Prove by induction on  $k$  that  $S_k(U_k) = \{\mathbf{0}\}$  for  $1 \leq k \leq n$ .
- (b) Deduce that  $p_T(T) = 0$ , without using the adjugate.
- (c) Deduce the Cayley–Hamilton Theorem for every  $\mathbf{A} \in M_n(\mathbb{C})$ , and then for every  $\mathbf{A} \in M_n(\mathbb{R})$ .
- (d) Why does this argument, on its own, not prove the theorem for  $\mathbf{A} \in M_2(\mathbb{F}_2)$  with  $p_{\mathbf{A}} = x^2 + x + 1$ ?

*Hint: for (a), use the last statement of [Theorem 8.7.1](#).*

*Solution.*

- (a) By [Theorem 8.7.1](#), each  $U_k$  is invariant and  $(T - d_k \text{id}_V)(U_k) \subseteq U_{k-1}$ . For  $k = 1$ ,  $S_1(U_1) = (T - d_1 \text{id}_V)(U_1) \subseteq U_0 = \{\mathbf{0}\}$ . Suppose  $S_{k-1}(U_{k-1}) = \{\mathbf{0}\}$  for some  $k \geq 2$ . Since  $S_k = S_{k-1}(T - d_k \text{id}_V)$ , for  $\mathbf{u} \in U_k$  we get  $S_k \mathbf{u} = S_{k-1}((T - d_k \text{id}_V)\mathbf{u}) \in S_{k-1}(U_{k-1}) = \{\mathbf{0}\}$ .
- (b) By [Theorem 8.7.4](#) (a),  $p_T = \prod_{i=1}^n (x - d_i)$ , so  $p_T(T) = S_n$  by [Theorem 5.7.1](#) (b). By (a),  $S_n$  kills  $U_n = V$ , so  $p_T(T) = 0$ .
- (c) Let  $\mathbf{A} \in M_n(\mathbb{C})$ . By [Corollary 8.7.3](#),  $T_{\mathbf{A}}$  has a basis in which its matrix is upper triangular, and (b) gives  $p_{\mathbf{A}}(T_{\mathbf{A}}) = 0$ , so  $p_{\mathbf{A}}(\mathbf{A}) = 0$  by [Corollary 3.6.3](#). For  $\mathbf{A} \in M_n(\mathbb{R})$ , regard  $\mathbf{A}$  as an element of  $M_n(\mathbb{C})$ . The Leibniz formula gives the same polynomial  $p_{\mathbf{A}}$  with real coefficients, and  $p_{\mathbf{A}}(\mathbf{A})$  is computed from the same entries by the same sums and products; it is 0 by the complex case.
- (d) Over  $\mathbb{F}_2$ ,  $x^2 + x + 1$  has no root (its values at 0 and 1 are both 1), so it does not split, and the matrix is not triangularizable over  $\mathbb{F}_2$  by [Theorem 8.7.2](#). To run the argument we would need a larger field containing  $\mathbb{F}_2$  in which the polynomial splits, and we have not constructed one. The adjugate proof of [Theorem 8.6.1](#) needs no such field. (For instance  $\mathbf{A} = \begin{pmatrix} 0 & 1 \\ 1 & 1 \end{pmatrix}$  has this characteristic polynomial, and indeed  $\mathbf{A}^2 + \mathbf{A} + \mathbf{I} = 0$  over  $\mathbb{F}_2$ .)

**Exercise C2: Real operators have small invariant subspaces.**

Let  $\mathbf{A} \in M_n(\mathbb{R})$ ,  $n \geq 1$ , regarded also as a complex matrix.

- (a) Let  $\lambda = a + bi$  ( $a, b \in \mathbb{R}$ ) be a complex eigenvalue of  $\mathbf{A}$  with eigenvector  $\mathbf{z} = \mathbf{x} + i\mathbf{y}$ , where  $\mathbf{x}, \mathbf{y} \in \mathbb{R}^n$ . Show that  $\mathbf{Ax} = a\mathbf{x} - b\mathbf{y}$  and  $\mathbf{Ay} = b\mathbf{x} + a\mathbf{y}$ .
- (b) Deduce that  $\text{span}(\mathbf{x}, \mathbf{y}) \subseteq \mathbb{R}^n$  is  $\mathbf{A}$ -invariant, and has dimension 1 or 2.
- (c) Deduce that every operator on a real vector space  $V$  with  $1 \leq \dim V < \infty$  has an invariant subspace of dimension 1 or 2.
- (d) For  $\mathbf{C} = \begin{pmatrix} 0 & 0 & 2 \\ 1 & 0 & -1 \\ 0 & 1 & 2 \end{pmatrix}$ , check that  $\mathbf{z} = (0, -2, 1) + i(-2, 1, 0)$  is an eigenvector for  $i$ , and find the invariant plane of (b).

*Solution.*

- (a) The eigenvalue exists by [Theorem 8.2.4](#). Since  $\mathbf{A}$  has real entries,  $\mathbf{Az} = \mathbf{Ax} + i\mathbf{Ay}$  with  $\mathbf{Ax}, \mathbf{Ay} \in \mathbb{R}^n$ . On the other hand,  $\lambda\mathbf{z} = (a + bi)(\mathbf{x} + i\mathbf{y}) = (a\mathbf{x} - b\mathbf{y}) + i(b\mathbf{x} + a\mathbf{y})$ . A vector of  $\mathbb{C}^n$  determines its real and imaginary parts entry by entry, so  $\mathbf{Ax} = a\mathbf{x} - b\mathbf{y}$  and  $\mathbf{Ay} = b\mathbf{x} + a\mathbf{y}$ .
- (b) By (a),  $\mathbf{Ax}$  and  $\mathbf{Ay}$  lie in  $W = \text{span}(\mathbf{x}, \mathbf{y})$ , so  $W$  is invariant by [Lemma 8.1.1](#). Since  $\mathbf{z} \neq \mathbf{0}$ , at least one of  $\mathbf{x}, \mathbf{y}$  is non-zero, so  $\dim W \geq 1$ , and  $\dim W \leq 2$  since  $W$  is spanned by two vectors.
- (c) Choose a basis  $\mathcal{B}$  of  $V$ , let  $\mathbf{A} = [\mathbf{A}]_{\mathcal{B}} \in M_n(\mathbb{R})$ , and let  $W$  be as in (b). Let  $\Phi: V \rightarrow \mathbb{R}^n$  be the coordinate isomorphism  $\mathbf{v} \mapsto [\mathbf{v}]_{\mathcal{B}}$  ([Corollary 3.5.5](#)). Then  $\Phi^{-1}(W)$  is a subspace of the same dimension as  $W$ , and for  $\mathbf{v} \in \Phi^{-1}(W)$ ,  $\Phi(T\mathbf{v}) = \mathbf{A}\Phi(\mathbf{v}) \in W$  ([Theorem 3.6.1](#)), so  $T\mathbf{v} \in \Phi^{-1}(W)$ .
- (d) With  $\mathbf{x} = (0, -2, 1)$  and  $\mathbf{y} = (-2, 1, 0)$ :  $\mathbf{Cx} = (2, -1, 0) = -\mathbf{y}$  and  $\mathbf{Cy} = (0, -2, 1) = \mathbf{x}$ . So  $\mathbf{Cz} = \mathbf{Cx} + i\mathbf{Cy} = -\mathbf{y} + i\mathbf{x} = i(\mathbf{x} + i\mathbf{y}) = i\mathbf{z}$ , and  $\mathbf{z}$  is an eigenvector for  $i$  (here  $a = 0$ ,  $b = 1$ , matching (a)). The invariant plane is  $\text{span}((0, -2, 1), (-2, 1, 0))$ , which is the plane  $x + 2y + 4z = 0$ . Over  $\mathbb{R}$ ,  $\mathbf{C}$  also has the eigenvalue 2 with eigenvector  $(1, 0, 1)$ , an invariant line.

**Exercise C3: Traces of powers detect nilpotency.**

Let  $\mathbf{C} \in M_n(\mathbb{C})$ ,  $n \geq 1$ , with  $\text{tr}(\mathbf{C}^k) = 0$  for  $k = 1, \dots, n$ . The goal is to prove that  $\mathbf{C}^n = \mathbf{0}$ .

- (a) Let  $d_1, \dots, d_n$  be the diagonal entries of a triangular form of  $\mathbf{C}$ . Show that  $d_1^k + \dots + d_n^k = 0$  for  $k = 1, \dots, n$ .
- (b) Suppose some  $d_i \neq 0$ . Let  $\mu_1, \dots, \mu_r$  ( $r \geq 1$ ) be the distinct non-zero values among the  $d_i$ , and let  $m_j \geq 1$  be the number of indices  $i$  with  $d_i = \mu_j$ . Show that  $\sum_{j=1}^r m_j \mu_j^k = 0$  for  $k = 1, \dots, r$ .
- (c) Show that the  $r \times r$  matrix with  $(k, j)$ -entry  $\mu_j^k$  is invertible, and derive a contradiction from (b).
- (d) Conclude that every  $d_i = 0$ , that  $p_{\mathbf{C}} = x^n$ , and that  $\mathbf{C}^n = \mathbf{0}$ .
- (e) Show that the statement fails over  $\mathbb{F}_2$ .

*Hint: for (c), factor the matrix as a transposed Vandermonde matrix times a diagonal matrix.*

*Solution.*

- (a) By [Corollary 8.7.3](#) a triangular form exists, and by [Exercise \(B3: Eigenvalues of powers from a triangular form\)](#) (b),  $\text{tr } \mathbf{C}^k = \sum_i d_i^k$ , which is 0 by hypothesis.
- (b) Indices with  $d_i = 0$  contribute nothing to  $\sum_i d_i^k$  for  $k \geq 1$ . Grouping the other indices by their value,  $\sum_i d_i^k = \sum_j m_j \mu_j^k$ . Since  $r \leq n$ , (a) applies for  $k = 1, \dots, r$ .
- (c) Let  $\mathbf{M}$  be the matrix with  $(k, j)$ -entry  $\mu_j^k$ . The Vandermonde matrix  $\mathbf{V} = \mathbf{V}(\mu_1, \dots, \mu_r)$  has  $(j, k)$ -entry  $\mu_j^{k-1}$ , so  $\mathbf{V}^\top$  has  $(k, j)$ -entry  $\mu_j^{k-1}$ , and multiplying column  $j$  by  $\mu_j$  gives  $\mathbf{M} = \mathbf{V}^\top \text{diag}(\mu_1, \dots, \mu_r)$ . Since the  $\mu_j$  are distinct,  $\mathbf{V}$  is invertible ([Corollary 6.8.2](#)), hence so is  $\mathbf{V}^\top$  ([Theorem 6.4.5](#)); since the  $\mu_j$  are non-zero, the diagonal matrix is invertible. So  $\mathbf{M}$  is invertible. By (b),  $\mathbf{M}(m_1, \dots, m_r) = \mathbf{0}$ , so  $(m_1, \dots, m_r) = \mathbf{0}$  in  $\mathbb{C}^r$ . But each  $m_j$  is a positive integer, which is non-zero in  $\mathbb{C}$ . This contradiction shows that no  $d_i$  is non-zero.
- (d) By (c), all  $d_i = 0$ , so  $p_{\mathbf{C}} = x^n$  by [Theorem 8.7.4](#) (a). By [Theorem 8.6.1](#),  $\mathbf{C}^n = p_{\mathbf{C}}(\mathbf{C}) = \mathbf{0}$ .
- (e) Over  $\mathbb{F}_2$ ,  $\mathbf{C} = \mathbf{I}_2$  has  $\text{tr } \mathbf{C}^k = \text{tr } \mathbf{I}_2 = 1 + 1 = 0$  for every  $k$ , but  $\mathbf{C}^k = \mathbf{I}_2 \neq \mathbf{0}$ . The step that fails is (c): the positive integer  $m_1 = 2$  is 0 in  $\mathbb{F}_2$ . Compare [Exercise \(C2: When  \$\mathbf{A}\$  commutes with  \$\[\mathbf{A}, \mathbf{B}\]\$ \)](#), where in characteristic 0 the same conclusion was reached for a commutator by a route that avoids eigenvalues.

## 8.8 Primary Decomposition

Chapter 5 ended with a lemma about any polynomial that kills an operator: if it factors into coprime pieces, the space splits into the kernels of the pieces. The previous sections produced the best polynomial to feed it, the minimal polynomial, which factors into powers of distinct irreducibles. This section runs the lemma on that factorization. The result, the primary decomposition, cuts  $V$  into invariant pieces, one for each irreducible factor of  $m_T$ , and on each piece the operator has a minimal polynomial that is a single prime power. Two answers follow at once. An operator is diagonalizable exactly when its minimal polynomial is a product of distinct linear factors. And every complex matrix with  $\mathbf{A}^k = \mathbf{I}$  is diagonalizable, a promise from Section 4. Finally, when  $m_T$  splits, the pieces are the spaces on which Chapter 9 builds the Jordan form.

### 8.8.1 Splitting the space along the minimal polynomial

Here is the question. Diagonalization splits  $V$  into eigenspaces, and it can fail. What splitting survives for **every** operator? The minimal polynomial suggests one. By [Theorem 5.3.5](#), if  $V \neq \{\mathbf{0}\}$ , then  $m_T$  (monic, of degree at least 1 by [Theorem 8.5.1](#)) factors as

$$m_T = p_1^{e_1} p_2^{e_2} \cdots p_k^{e_k},$$

with  $p_1, \dots, p_k$  **distinct** monic irreducible polynomials and exponents  $e_i \geq 1$ . Powers of distinct irreducibles have no common factor, so the kernel splitting lemma of Chapter 5, in its form with  $k$  factors ([Exercise \(C2: Splitting with  \$k\$  coprime factors\)](#)), applies to them. It gives a direct sum. The new information is what happens on the pieces.

*The primary decomposition splits  $V$  into one invariant piece for each irreducible factor of  $m_T$ , and on that piece the operator is killed by exactly that factor's power.*

### Theorem 8.8.1 — Primary Decomposition Theorem

Let  $V \neq \{0\}$  be a finite-dimensional vector space over  $F$ , let  $T \in \mathcal{L}(V)$ , and let  $m_T = p_1^{e_1} \cdots p_k^{e_k}$ , where  $p_1, \dots, p_k$  are distinct monic irreducible polynomials and  $e_i \geq 1$ . Put

$$V_i := \ker p_i(T)^{e_i} \quad (i = 1, \dots, k).$$

Then:

- (a)  $V = V_1 \oplus V_2 \oplus \cdots \oplus V_k$ ;
- (b) each  $V_i$  is  $T$ -invariant and non-zero;
- (c) the restriction  $T|_{V_i}$  has minimal polynomial  $m_{T|_{V_i}} = p_i^{e_i}$ ;
- (d) for each  $i$  there is  $h_i \in F[x]$  such that  $h_i(T)$  is the projection of  $V$  onto  $V_i$  along  $\bigoplus_{j \neq i} V_j$ .

In words: (a) every vector is **uniquely** a sum of pieces, one from each  $V_i$ . (b)  $T$  maps each piece into itself, so  $T$  is determined by the  $k$  restrictions, and in a basis made of bases of the  $V_i$  its matrix is block diagonal ([Theorem 8.1.5](#)). (c) Nothing is wasted: on  $V_i$  the operator is killed by  $p_i^{e_i}$ , and by no proper divisor of it. (d) The pieces can be computed with polynomials in  $T$  alone. The word *primary* refers to the prime powers  $p_i^{e_i}$ .

*Idea.* ① The splitting is Chapter 5's lemma: distinct irreducible powers are pairwise coprime, and their product  $m_T$  kills  $T$ . ② Invariance holds for every kernel of a polynomial in  $T$ . ③ For (c), each restriction is killed by  $p_i^{e_i}$ , so its minimal polynomial is some  $p_i^{f_i}$  with  $f_i \leq e_i$ . Conversely, the product of these minimal polynomials kills  $T$  piece by piece, hence on all of  $V$ , so it is a multiple of  $m_T$ . A degree count then forces  $f_i = e_i$ .

*Proof. Coprime.* Let  $i \neq j$  and  $d = \gcd(p_i^{e_i}, p_j^{e_j})$ , which is monic. By [Lemma 8.5.2](#),  $d = p_i^f$  and  $d = p_j^g$  for some  $f, g \geq 0$ . By the uniqueness in [Theorem 5.3.5](#), and since  $p_i \neq p_j$ , this forces  $f = g = 0$ , so  $d = 1$ .

- (a) and (d). If  $k = 1$ , then  $V_1 = \ker m_T(T) = V$  and  $h_1 = 1$  works. If  $k \geq 2$ , apply [Exercise \(C2: Splitting with  \$k\$  coprime factors\)](#) (d) to the pairwise coprime polynomials  $p_1^{e_1}, \dots, p_k^{e_k}$ , whose product  $m_T$  annihilates  $T$ . It gives  $V = V_1 \oplus \cdots \oplus V_k$ , with the projections given by polynomials  $h_i(T)$ .
- (b) Each  $V_i$  is the kernel of a polynomial in  $T$ , so it is  $T$ -invariant by [Theorem 5.7.4](#) (b). That  $V_i \neq \{0\}$  will follow from (c): the minimal polynomial of an operator on the zero space is 1, and  $p_i^{e_i} \neq 1$ .
- (c) Let  $q_i = m_{T|_{V_i}}$ . By [Exercise \(C2: Minimal polynomials of the pieces\)](#) (a),  $p_i^{e_i}(T|_{V_i}) = p_i^{e_i}(T)|_{V_i} = 0$ , so  $q_i \mid p_i^{e_i}$  ([Theorem 8.5.1](#)), and  $q_i = p_i^{f_i}$  with  $0 \leq f_i \leq e_i$  by [Lemma 8.5.2](#). Put  $q = q_1 \cdots q_k$ . For  $\mathbf{v}_i \in V_i$ , polynomials in  $T$  commute ([Theo-](#)

rem 5.7.1 (c)), so

$$q(T)\mathbf{v}_i = \left(\prod_{j \neq i} q_j\right)(T) q_i(T)\mathbf{v}_i = \left(\prod_{j \neq i} q_j\right)(T) q_i(T|_{V_i})\mathbf{v}_i = \mathbf{0},$$

using [Exercise \(C2: Minimal polynomials of the pieces\)](#) (a) again. By (a), every  $\mathbf{v} \in V$  is a sum of such  $\mathbf{v}_i$ , so  $q(T) = 0$ , and  $m_T \mid q$  by [Theorem 8.5.1](#). Comparing degrees ([Proposition 5.1.2](#) (c), [Theorem 0.8.3](#)),

$$\sum_{i=1}^k e_i \deg p_i = \deg m_T \leq \deg q = \sum_{i=1}^k f_i \deg p_i.$$

Since  $f_i \leq e_i$  and  $\deg p_i \geq 1$  for every  $i$ , each term on the right is at most the matching term on the left, so equality holds term by term, and  $f_i = e_i$ . Hence  $m_{T|_{V_i}} = p_i^{e_i}$ . This proves the theorem. ■

The theorem reduces the study of  $T$  to the study of operators whose minimal polynomial is a power of **one** irreducible. For a matrix, choose a basis of each  $V_i$  and put them together:  $\mathbf{A}$  is similar to a block diagonal matrix  $\mathbf{A}_1 \oplus \cdots \oplus \mathbf{A}_k$  with  $m_{\mathbf{A}_i} = p_i^{e_i}$ , and then  $m_{\mathbf{A}} = \text{lcm}(m_{\mathbf{A}_1}, \dots, m_{\mathbf{A}_k})$  by [Proposition 8.5.6](#), applied  $k-1$  times, consistent with (c).

**Example. (A Primary Decomposition with an Irreducible Quadratic)**

Let  $\mathbf{C} = \begin{pmatrix} 0 & 0 & 2 \\ 1 & 0 & -1 \\ 0 & 1 & 2 \end{pmatrix} \in M_3(\mathbb{R})$ . Chapter 5 showed that its minimal polynomial is  $m_{\mathbf{C}} = (x-2)(x^2+1)$  ([Exercise \(B2: An annihilating polynomial by elimination\)](#)). Find the primary decomposition of  $\mathbb{R}^3$ , the projections as polynomials in  $\mathbf{C}$ , and the matrix of  $\mathbf{C}$  in an adapted basis.

*Solution. The factorization.*  $x-2$  is irreducible (degree 1), and  $x^2+1$  is irreducible over  $\mathbb{R}$  because it has no real root ([Theorem 5.3.1](#) (b)). So  $k=2$ ,  $p_1 = x-2$ ,  $p_2 = x^2+1$ ,  $e_1 = e_2 = 1$ .

*The pieces.*  $V_1 = \ker(\mathbf{C} - 2\mathbf{I})$ : the matrix  $\mathbf{C} - 2\mathbf{I} = \begin{pmatrix} -2 & 0 & 2 \\ 1 & -2 & -1 \\ 0 & 1 & 0 \end{pmatrix}$  gives  $y=0$  and  $x=z$ , so  $V_1 = \text{span}((1, 0, 1))$ . For  $V_2 = \ker(\mathbf{C}^2 + \mathbf{I})$ , multiply out:

$$\mathbf{C}^2 = \begin{pmatrix} 0 & 2 & 4 \\ 0 & -1 & 0 \\ 1 & 2 & 3 \end{pmatrix}, \quad \mathbf{C}^2 + \mathbf{I} = \begin{pmatrix} 1 & 2 & 4 \\ 0 & 0 & 0 \\ 1 & 2 & 4 \end{pmatrix}.$$

So  $V_2$  is the plane  $x + 2y + 4z = 0$ , with basis  $(-2, 1, 0)$ ,  $(-4, 0, 1)$ . As the theorem predicts,  $\dim V_1 + \dim V_2 = 3$ , and  $(1, 0, 1) \notin V_2$  since  $1 + 0 + 4 \neq 0$ .

*The projections.* Bézout's identity for  $p_1, p_2$ : since  $x^2+1 = (x+2)(x-2) + 5$ ,

$$\frac{1}{5}(x^2+1) - \frac{1}{5}(x+2)(x-2) = 1.$$

Following [Theorem 5.7.5](#) (c) with  $p = x - 2$ ,  $q = x^2 + 1$ ,  $a = -\frac{1}{5}(x + 2)$  and  $b = \frac{1}{5}$ , the projection onto  $V_1$  along  $V_2$  is  $h_1(\mathbf{C}) = b(\mathbf{C})q(\mathbf{C}) = \frac{1}{5}(\mathbf{C}^2 + \mathbf{I})$ , and the projection onto  $V_2$  along  $V_1$  is  $h_2(\mathbf{C}) = \mathbf{I} - h_1(\mathbf{C})$ . Concretely,

$$h_1(\mathbf{C}) \begin{pmatrix} x \\ y \\ z \end{pmatrix} = \frac{x + 2y + 4z}{5} \begin{pmatrix} 1 \\ 0 \\ 1 \end{pmatrix},$$

which lies in  $V_1$ , is  $\mathbf{0}$  on  $V_2$ , and fixes  $(1, 0, 1)$ .

*The adapted basis.*  $\mathbf{C}(1, 0, 1) = (2, 0, 2)$ . On  $V_2$ ,  $\mathbf{C}(-2, 1, 0) = (0, -2, 1) = -2(-2, 1, 0) + (-4, 0, 1)$  and  $\mathbf{C}(-4, 0, 1) = (2, -5, 2) = -5(-2, 1, 0) + 2(-4, 0, 1)$ . So in the basis  $((1, 0, 1), (-2, 1, 0), (-4, 0, 1))$ ,

$$[\mathbf{C}] = (2) \oplus \begin{pmatrix} -2 & -5 \\ 1 & 2 \end{pmatrix}.$$

The second block has trace 0 and determinant  $-4 + 5 = 1$ , so its characteristic polynomial is  $x^2 + 1$ ; it is not a scalar matrix, so its minimal polynomial is also  $x^2 + 1$ , as (c) says. Over  $\mathbb{R}$ , no further splitting of  $V_2$  is possible, since  $\mathbf{C}$  has no eigenvector in  $V_2$ . Over  $\mathbb{C}$ , the factor  $x^2 + 1 = (x - i)(x + i)$  splits, and so does the plane.

**Warning.** The pieces are kernels of the powers  $p_i^{e_i}$ , not of the  $p_i$ . Replacing  $\ker p_i(T)^{e_i}$  by  $\ker p_i(T)$  can lose most of the space. For  $\mathbf{J} = \begin{pmatrix} 1 & 1 \\ 0 & 1 \end{pmatrix}$ ,  $m_{\mathbf{J}} = (x - 1)^2$ , and the primary decomposition has the single piece  $\ker(\mathbf{J} - \mathbf{I})^2 = F^2$ ; but  $\ker(\mathbf{J} - \mathbf{I}) = \text{span}(\mathbf{e}_1)$  is only a line. The sum of the kernels  $\ker p_i(T)$  is still direct, but it equals  $V$  only when every  $e_i = 1$ , which is the subject of the next subsection.

**Quick check.** Let  $T \in \mathcal{L}(V)$ ,  $V \neq \{\mathbf{0}\}$  finite-dimensional over  $\mathbb{Q}$ , satisfy  $T^3 = T^2$  but  $T^2 \neq T$ . Which polynomials can  $m_T$  be? What are the pieces of the primary decomposition?

**Solution.**  $x^3 - x^2 = x^2(x - 1)$  annihilates  $T$ , so by [Lemma 8.5.2](#)  $m_T = x^a(x - 1)^b$  with  $a \leq 2$ ,  $b \leq 1$ , not both 0. Since  $x(x - 1)$  does not annihilate  $T$ , and neither do its divisors  $x$  and  $x - 1$  (they would give  $T = 0$  or  $T = \text{id}_V$ , both satisfying  $T^2 = T$ ), we need  $a = 2$ . So  $m_T = x^2$  or  $m_T = x^2(x - 1)$ . In the first case the only piece is  $\ker T^2 = V$ ; in the second,  $V = \ker T^2 \oplus \ker(T - \text{id}_V)$ .

## 8.8.2 Diagonalizability and the minimal polynomial

When every  $p_i$  is linear and every  $e_i = 1$ , the pieces are kernels of  $T - \lambda_i \text{id}_V$ , that is, eigenspaces. That is exactly diagonalizability, and it gives the criterion Chapter 5 promised.

### Theorem 8.8.2 — Diagonalizability via the Minimal Polynomial

Let  $V$  be a finite-dimensional vector space over  $F$  with  $\dim V \geq 1$ , and let  $T \in \mathcal{L}(V)$ . The following are equivalent:



- (a)  $T$  is diagonalizable;
- (b)  $m_T = (x - \lambda_1)(x - \lambda_2) \cdots (x - \lambda_k)$  for some **distinct**  $\lambda_1, \dots, \lambda_k \in F$ , that is,  $m_T$  splits over  $F$  with no repeated root;
- (c) some polynomial  $(x - \mu_1) \cdots (x - \mu_r)$  with **distinct**  $\mu_1, \dots, \mu_r \in F$  annihilates  $T$ .

In that case  $\lambda_1, \dots, \lambda_k$  are the distinct eigenvalues of  $T$ . The same holds for  $\mathbf{A} \in M_n(F)$ .

*Idea.*

- (a)  $\Rightarrow$  (b): the polynomial  $\prod_i (x - \lambda_i)$  over the distinct eigenvalues kills each vector of an eigenbasis, hence kills  $T$ ; and  $m_T$  must contain every factor  $x - \lambda_i$ , because its roots are the eigenvalues. (b)  $\Rightarrow$  (c) is free. (c)  $\Rightarrow$  (a):  $m_T$  divides a product of distinct linear factors, so it is one, and the primary decomposition with every  $e_i = 1$  splits  $V$  into eigenspaces.

*Proof.*

- (a)  $\Rightarrow$  (b). Let  $\lambda_1, \dots, \lambda_k$  be the distinct eigenvalues of  $T$ , let  $\mathcal{B}$  be a basis of eigenvectors (Theorem 8.4.1 (b)), and put  $q = (x - \lambda_1) \cdots (x - \lambda_k)$ . For  $\mathbf{v} \in \mathcal{B}$  with eigenvalue  $\lambda_j$ , Exercise (B3: Eigenvalues of powers and inverses) (a) gives  $q(T)\mathbf{v} = q(\lambda_j)\mathbf{v} = \mathbf{0}$ , as  $x - \lambda_j$  is a factor of  $q$ . So  $q(T)$  and the zero operator agree on a basis, and  $q(T) = 0$  by the uniqueness in Theorem 3.1.5. Hence  $m_T \mid q$ , and by Lemma 8.5.2  $m_T$  is a product of some of the factors  $x - \lambda_i$ . Each  $\lambda_i$  is a root of  $m_T$  by Theorem 8.5.3, so no factor is missing, and  $m_T = q$ .
- (b)  $\Rightarrow$  (c). Take  $q = m_T$ .
- (c)  $\Rightarrow$  (a). Let  $q = (x - \mu_1) \cdots (x - \mu_r)$  with distinct  $\mu_j$  annihilate  $T$ . Each factor  $x - \mu_j$  is monic irreducible, being of degree 1 (Theorem 0.8.3: a factorization would leave one factor constant). By Theorem 8.5.1 and Lemma 8.5.2,  $m_T = (x - \lambda_1) \cdots (x - \lambda_k)$  for some distinct  $\lambda_i$  among the  $\mu_j$ , and  $k \geq 1$  since  $V \neq \{\mathbf{0}\}$ . By Theorem 8.5.3,  $\lambda_1, \dots, \lambda_k$  are exactly the eigenvalues of  $T$ . Apply Theorem 8.8.1 with  $p_i = x - \lambda_i$  and  $e_i = 1$ :

$$V = \ker(T - \lambda_1 \text{id}_V) \oplus \cdots \oplus \ker(T - \lambda_k \text{id}_V) = E_{\lambda_1}(T) \oplus \cdots \oplus E_{\lambda_k}(T).$$

This is condition (c) of Theorem 8.4.1, so  $T$  is diagonalizable. The statement for matrices follows by applying this to  $T_{\mathbf{A}}$ , since  $m_{\mathbf{A}} = m_{T_{\mathbf{A}}}$ . ■

This is the promise of Chapter 5 kept: **diagonalizable means the minimal polynomial splits with distinct roots**. The criterion is often the fastest test, because it needs no eigenvectors, only one polynomial identity. Chapter 5 also supplied a way to detect repeated factors without finding roots: if  $\gcd(m_T, m'_T) = 1$ , then no irreducible factor of  $m_T$  is repeated (Corollary 5.4.6 (a)). So  $T$  is diagonalizable as soon as  $m_T$  splits and is coprime to its derivative.

Condition (c) is especially useful because it does not require knowing  $m_T$  at all.

- **Projections.** If  $P^2 = P$ , then  $x(x - 1)$  annihilates  $P$ , and  $0 \neq 1$  in every field. So every projection on a finite-dimensional space is diagonalizable, over every field.
- **Involutions.** If  $T^2 = \text{id}_V$  and the characteristic of  $F$  is not 2, then  $(x - 1)(x + 1)$  annihilates  $T$ , and  $1 \neq -1$  because  $2 \neq 0$ . So  $T$  is diagonalizable. In characteristic 2 this



fails: over  $\mathbb{F}_2$ ,  $\mathbf{J} = \begin{pmatrix} 1 & 1 \\ 0 & 1 \end{pmatrix}$  has  $\mathbf{J}^2 = \begin{pmatrix} 1 & 2 \\ 0 & 1 \end{pmatrix} = \mathbf{I}$ , but  $m_{\mathbf{J}} = (x-1)^2$ .

- **Too few eigenvectors.**  $\mathbf{J}$  over any field has  $m_{\mathbf{J}} = (x-1)^2$ , with a repeated root, so it is not diagonalizable, confirming Section 4 without any rank computation. Likewise the matrix of [Example \(The Minimal Polynomial from the Eigenvalues\)](#), with  $m = (x-1)(x-3)^2$ , is not diagonalizable, while that of [Example \(The Minimal Polynomial from the First Dependence\)](#), with  $m = (x-1)(x-3)$ , is.

The most striking use settles the question left open in Section 4 ([Exercise \(C1: Matrices with  \$\mathbf{A}^k = \mathbf{I}\$ \)](#)).

### Example. (Complex Matrices of Finite Order Are Diagonalizable)

Let  $\mathbf{A} \in M_n(\mathbb{C})$ ,  $n \geq 1$ , satisfy  $\mathbf{A}^k = \mathbf{I}$  for some  $k \geq 1$ . Show that  $\mathbf{A}$  is diagonalizable, with eigenvalues among the  $k$ -th roots of unity. Show that both “over  $\mathbb{C}$ ” and “characteristic 0” matter.

*Solution. Diagonalizable.* The polynomial  $q = x^k - 1$  annihilates  $\mathbf{A}$ . By [Corollary 5.5.5](#),  $q = (x - \zeta_1) \cdots (x - \zeta_k)$  for some  $\zeta_j \in \mathbb{C}$ . The  $\zeta_j$  are distinct: if  $c$  were a multiple root, then  $q(c) = 0$  and  $q'(c) = kc^{k-1} = 0$  by [Theorem 5.4.5](#) (a). But  $q(c) = 0$  means  $c^k = 1$ , so  $c \neq 0$ ; and  $k \neq 0$  in  $\mathbb{C}$ ; so  $kc^{k-1} \neq 0$ , a contradiction. By [Theorem 8.8.2](#) ((c)  $\Rightarrow$  (a)),  $\mathbf{A}$  is diagonalizable. Its eigenvalues are roots of  $m_{\mathbf{A}}$ , which divides  $q$ , so each eigenvalue  $\lambda$  satisfies  $\lambda^k = 1$ . (Explicitly, the  $\zeta_j$  are  $e^{2\pi i j/k}$ ,  $j = 0, \dots, k-1$ .)

*Over  $\mathbb{R}$  it fails.* The rotation  $\mathbf{R} = \begin{pmatrix} 0 & -1 \\ 1 & 0 \end{pmatrix}$  has  $\mathbf{R}^4 = \mathbf{I}$  but is not diagonalizable over  $\mathbb{R}$ , since  $m_{\mathbf{R}} = x^2 + 1$  does not split. The step that breaks is the splitting of  $x^4 - 1 = (x-1)(x+1)(x^2+1)$ .

*In characteristic  $p$  it fails.* Over  $\mathbb{F}_p$ ,  $\mathbf{J} = \begin{pmatrix} 1 & 1 \\ 0 & 1 \end{pmatrix}$  has  $\mathbf{J}^p = \begin{pmatrix} 1 & p \\ 0 & 1 \end{pmatrix} = \mathbf{I}$  (the powers of  $\mathbf{J}$  computed in Section 4), but  $m_{\mathbf{J}} = (x-1)^2$ . The step that breaks is distinctness: the derivative of  $x^p - 1$  is  $px^{p-1} = 0$  in  $\mathbb{F}_p$ , and indeed 1 is a multiple root of  $x^p - 1$ .

## 8.8.3 Restrictions and polynomials of diagonalizable operators

The criterion in terms of a single annihilating polynomial passes easily to smaller spaces.

### Corollary 8.8.3 — Restrictions of Diagonalizable Operators

Let  $V$  be finite-dimensional,  $T \in \mathcal{L}(V)$  diagonalizable, and  $U$  a  $T$ -invariant subspace. Then  $T|_U$  is diagonalizable. Moreover  $U = \bigoplus_{\lambda \in \text{spec}(T)} (U \cap E_{\lambda}(T))$ .

*Proof.* If  $U = \{\mathbf{0}\}$ , the empty basis gives the empty (diagonal) matrix, and both statements are trivial. Let  $U \neq \{\mathbf{0}\}$ . By [Theorem 8.8.2](#) ((a)  $\Rightarrow$  (b)),  $m_T = (x - \lambda_1) \cdots (x - \lambda_k)$  with distinct  $\lambda_i$ , the eigenvalues of  $T$ . By [Exercise \(C2: Minimal polynomials of the pieces\)](#) (a),  $m_T(T|_U) = m_T(T)|_U = 0$ , so  $m_T$  is a product of distinct linear factors

annihilating  $T|_U$ . By [Theorem 8.8.2](#) ((c)  $\Rightarrow$  (a)),  $T|_U$  is diagonalizable.

For the decomposition, [Theorem 8.4.1](#) (c) applied to  $T|_U$  gives  $U = \bigoplus_{\mu} E_{\mu}(T|_U)$ , over the eigenvalues  $\mu$  of  $T|_U$ . An eigenvector of  $T|_U$  is an eigenvector of  $T$  lying in  $U$ , so  $E_{\mu}(T|_U) = U \cap E_{\mu}(T)$ , and every such  $\mu$  is in  $\text{spec}(T)$ . For  $\lambda \in \text{spec}(T)$  that is not an eigenvalue of  $T|_U$ ,  $U \cap E_{\lambda}(T) = \{\mathbf{0}\}$ , and adding zero summands does not change a direct sum. ■

The second statement says that an invariant subspace of a diagonalizable operator is spanned by the eigenvectors it contains. It will be the key step in Section 9, where two commuting diagonalizable operators are diagonalized by a single basis.

Diagonalizability is also preserved by polynomials: if  $T$  is diagonalizable and  $p \in F[x]$ , then every eigenvector of  $T$  for  $\lambda$  is an eigenvector of  $p(T)$  for  $p(\lambda)$  ([Exercise \(B3: Eigenvalues of powers and inverses\)](#) (a)), so a basis of eigenvectors of  $T$  is also one for  $p(T)$ . This is the operator form of [Exercise \(C2: Polynomials of diagonalizable matrices\)](#). The converse fails, as that exercise shows.

## 8.8.4 When the minimal polynomial splits: a first look at Chapter 9

Suppose  $p_T$  splits,  $p_T = (x - \lambda_1)^{a_1} \cdots (x - \lambda_k)^{a_k}$  with distinct  $\lambda_i$ ; this always happens over  $\mathbb{C}$ . By [Corollary 8.6.2](#),  $m_T = (x - \lambda_1)^{s_1} \cdots (x - \lambda_k)^{s_k}$  with  $1 \leq s_i \leq a_i$ , and the primary decomposition reads

$$V = \ker(T - \lambda_1 \text{id}_V)^{s_1} \oplus \cdots \oplus \ker(T - \lambda_k \text{id}_V)^{s_k}.$$

On the  $i$ -th piece,  $T|_{V_i} = \lambda_i \text{id}_{V_i} + N_i$ , where  $N_i = (T - \lambda_i \text{id}_V)|_{V_i}$  satisfies  $N_i^{s_i} = 0$ , because  $V_i = \ker(T - \lambda_i \text{id}_V)^{s_i}$ . So every operator whose characteristic polynomial splits is, piece by piece, a scalar plus a nilpotent operator. Chapter 9 calls the pieces **generalized eigenspaces** and finds the best basis for each nilpotent part, which leads to the Jordan form. Here we record only their dimensions.

### Corollary 8.8.4 — Dimensions of the Primary Components

Let  $V$  be finite-dimensional with  $\dim V \geq 1$ ,  $T \in \mathcal{L}(V)$ , and suppose  $p_T$  splits over  $F$ , with distinct eigenvalues  $\lambda_1, \dots, \lambda_k$  and  $m_T = \prod_i (x - \lambda_i)^{s_i}$ . Let  $V_i = \ker(T - \lambda_i \text{id}_V)^{s_i}$ . Then

$$p_{T|_{V_i}} = (x - \lambda_i)^{\dim V_i} \quad \text{and} \quad \dim V_i = a_T(\lambda_i).$$

Moreover  $E_{\lambda_i}(T) \subseteq V_i$ , with equality if and only if  $s_i = 1$ .

*Idea.* The decomposition makes  $p_T$  the product of the characteristic polynomials of the restrictions. Each of those factors is a monic divisor of  $p_T$ , and its roots are the eigenvalues of the restriction, which the minimal polynomial  $(x - \lambda_i)^{s_i}$  pins down to the single value  $\lambda_i$ . So the  $i$ -th factor is  $(x - \lambda_i)^{\dim V_i}$ , and multiplying the factors back together identifies  $\dim V_i$  as the multiplicity of  $\lambda_i$  in  $p_T$ .

*Proof.* By Theorem 8.8.1,  $V = V_1 \oplus \cdots \oplus V_k$  with each  $V_i$  invariant, non-zero, and  $m_{T|_{V_i}} = (x - \lambda_i)^{s_i}$ . By Theorem 8.1.5,  $p_T = p_{T|_{V_1}} \cdots p_{T|_{V_k}}$ . So each  $p_{T|_{V_i}}$  is a monic divisor of  $p_T$ , and by Lemma 8.5.2 it is a product of powers of the  $x - \lambda_j$ . Its roots are the eigenvalues of  $T|_{V_i}$  (Theorem 8.2.1), which are the roots of  $m_{T|_{V_i}} = (x - \lambda_i)^{s_i}$  (Theorem 8.5.3), that is, only  $\lambda_i$ . So  $p_{T|_{V_i}} = (x - \lambda_i)^{d_i}$ , with  $d_i = \deg p_{T|_{V_i}} = \dim V_i$  (Theorem 6.10.2). Then  $p_T = \prod_i (x - \lambda_i)^{d_i}$ , and since the  $\lambda_i$  are distinct, Lemma 5.4.1 gives  $a_T(\lambda_i) = d_i$ .

If  $T\mathbf{v} = \lambda_i \mathbf{v}$ , then  $(T - \lambda_i \text{id}_V)^{s_i} \mathbf{v} = \mathbf{0}$  since  $s_i \geq 1$ , so  $E_{\lambda_i}(T) \subseteq V_i$ . If  $s_i = 1$ , the two spaces are equal by definition. Conversely, if  $E_{\lambda_i}(T) = V_i$ , then  $x - \lambda_i$  annihilates  $T|_{V_i}$ , so  $m_{T|_{V_i}} = (x - \lambda_i)^{s_i}$  divides  $x - \lambda_i$ , and  $s_i = 1$ . ■

So, when  $p_T$  splits, the dimension of the  $i$ -th piece is the algebraic multiplicity  $a(\lambda_i)$ , and inside it the eigenspace has dimension  $g(\lambda_i)$ . The operator is diagonalizable exactly when all these pieces are eigenspaces, which ties together the three criteria of this chapter:  $g(\lambda_i) = a(\lambda_i)$  for all  $i$  (Section 4), every  $s_i = 1$  (this section), and every piece  $\ker(T - \lambda_i \text{id}_V)^{s_i}$  equal to  $\ker(T - \lambda_i \text{id}_V)$ .

**Warning.** A single eigenvalue gives a single piece, whatever the operator looks like. If  $p_T = (x - \lambda)^n$ , then the primary decomposition is  $V = \ker(T - \lambda \text{id}_V)^s$ , all of  $V$ , and says nothing new. For instance  $2\mathbf{I}_3$ ,  $\begin{pmatrix} 2 & 1 \\ 0 & 2 \end{pmatrix} \oplus (2)$  and the matrix  $\mathbf{A}$  of Example (Triangularizing a Matrix with a Single Eigenvector Line) all have  $p = (x - 2)^3$  and a primary decomposition with one piece, yet they are pairwise non-similar: their minimal polynomials are  $x - 2$ ,  $(x - 2)^2$  (Proposition 8.5.6) and  $(x - 2)^3$ , the last because  $(\mathbf{A} - 2\mathbf{I})^2 = \begin{pmatrix} 1 & -1 & 0 \\ 1 & -1 & 0 \\ 0 & 0 & 0 \end{pmatrix} \neq 0$ , and equal minimal polynomials are necessary for similarity (Theorem 8.5.5). Distinguishing operators within one piece is the work of Chapter 9.

## 8.8.5 Exercises

### 8.8.5.1 A. Check your understanding

#### Exercise A1.

- State the Primary Decomposition Theorem.
- State the criterion for diagonalizability in terms of the minimal polynomial.
- True or false: if  $m_{\mathbf{A}} = x(x - 1)(x + 1)$  over  $\mathbb{Q}$ , then  $\mathbf{A}$  is diagonalizable over  $\mathbb{Q}$ . Justify your answer.
- True or false: if  $\mathbf{A} \in M_n(\mathbb{R})$  and  $\mathbf{A}^k = \mathbf{I}$  for some  $k \geq 1$ , then  $\mathbf{A}$  is diagonalizable over  $\mathbb{R}$ . Justify your answer.
- Let  $m_T = (x - 2)^2(x + 1)$ . What are the summands of the primary decomposition, and is  $T$  diagonalizable?
- True or false: the restriction of a diagonalizable operator to an invariant subspace is diagonalizable. Justify your answer.

*Solution.*

- (a) If  $V \neq \{\mathbf{0}\}$  is finite-dimensional and  $m_T = p_1^{e_1} \cdots p_k^{e_k}$  with distinct monic irreducible  $p_i$ , then  $V = \bigoplus_i \ker p_i(T)^{e_i}$ ; the summands are invariant and non-zero,  $T$  restricted to the  $i$ -th has minimal polynomial  $p_i^{e_i}$ , and the projections are polynomials in  $T$  ([Theorem 8.8.1](#)).
- (b)  $T$  is diagonalizable if and only if  $m_T$  is a product of distinct monic linear factors over  $F$  ([Theorem 8.8.2](#)).
- (c) True:  $m_{\mathbf{A}}$  splits over  $\mathbb{Q}$  with distinct roots  $0, 1, -1$ .
- (d) False. The rotation by a right angle satisfies  $\mathbf{R}^4 = \mathbf{I}$  but has no real eigenvalue ([Example \(Complex Matrices of Finite Order Are Diagonalizable\)](#)).
- (e)  $\ker(T - 2\text{id}_V)^2$  and  $\ker(T + \text{id}_V)$ .  $T$  is not diagonalizable, since  $m_T$  has the repeated root 2.
- (f) True, by [Corollary 8.8.3](#).

### 8.8.5.2 B. Practice

#### Exercise B1: Diagonalizable or not?

Determine whether each matrix is diagonalizable over the given field. Justify your answer.

- (a)  $\mathbf{A}_1 = \begin{pmatrix} 2 & 1 & 1 \\ -1 & 0 & -1 \\ 1 & 1 & 2 \end{pmatrix}$  over  $\mathbb{R}$  (see [Exercise \(B1: Computing minimal polynomials\)](#)).
- (b)  $\mathbf{M}_1 = \begin{pmatrix} 2 & 1 & 0 \\ 0 & 2 & 0 \\ -1 & 1 & 3 \end{pmatrix}$  over  $\mathbb{R}$  (see [Exercise \(B3: Minimal polynomial from one product\)](#)).
- (c) Any  $\mathbf{A} \in M_5(\mathbb{Q})$  with  $\mathbf{A}^3 = 4\mathbf{A}$ .
- (d) Any  $\mathbf{A} \in M_2(\mathbb{R})$  with  $\mathbf{A}^2 + \mathbf{I} = 0$ , over  $\mathbb{R}$  and over  $\mathbb{C}$ .

*Solution.*

- (a) By [Exercise \(B1: Computing minimal polynomials\)](#) (d),  $m_{\mathbf{A}_1} = (x - 1)(x - 2)$ , which has distinct linear factors. Diagonalizable, by [Theorem 8.8.2](#).
- (b) By [Exercise \(B3: Minimal polynomial from one product\)](#),  $m_{\mathbf{M}_1} = (x - 2)^2(x - 3)$ , with a repeated root. Not diagonalizable.
- (c)  $x^3 - 4x = x(x - 2)(x + 2)$  annihilates  $\mathbf{A}$ , and  $0, 2, -2$  are distinct in  $\mathbb{Q}$ . Diagonalizable, by condition (c) of [Theorem 8.8.2](#).
- (d)  $m_{\mathbf{A}} \mid x^2 + 1$ . Over  $\mathbb{R}$ ,  $x^2 + 1$  has no root, so it is irreducible ([Theorem 5.3.1](#)), and  $m_{\mathbf{A}} = x^2 + 1$  (it is not 1 since  $\mathbf{A}$  acts on  $\mathbb{R}^2 \neq \{\mathbf{0}\}$ ). It does not split over  $\mathbb{R}$ , so  $\mathbf{A}$  is not diagonalizable over  $\mathbb{R}$ . Over  $\mathbb{C}$ ,  $x^2 + 1 = (x - i)(x + i)$  with  $i \neq -i$ , so condition (c) holds and  $\mathbf{A}$  is diagonalizable over  $\mathbb{C}$ .

**Exercise B2: Decomposing  $\mathbb{R}^3$ .**

Let  $\mathbf{A} = \begin{pmatrix} 2 & 1 & 1 \\ 0 & 1 & -1 \\ -1 & -1 & 1 \end{pmatrix} \in M_3(\mathbb{R})$ , which has  $m_{\mathbf{A}} = (x-1)^2(x-2)$  (Exercise B1: Computing minimal polynomials) (c)).

- (a) Find bases of  $V_1 = \ker(\mathbf{A} - \mathbf{I})^2$  and  $V_2 = \ker(\mathbf{A} - 2\mathbf{I})$ , and check that  $\mathbb{R}^3 = V_1 \oplus V_2$ .
- (b) Compare  $V_1$  with  $E_1(\mathbf{A})$ .
- (c) Using  $(x-1)^2 - x(x-2) = 1$ , write the two projections as polynomials in  $\mathbf{A}$ , and compute them.
- (d) Write the matrix of  $T_{\mathbf{A}}$  in the basis formed by your bases of  $V_1$  and  $V_2$ .

*Solution.*

$$(a) \quad (\mathbf{A} - \mathbf{I})^2 = \begin{pmatrix} 1 & 1 & 1 \\ 0 & 0 & -1 \\ -1 & -1 & 0 \end{pmatrix}^2 = \begin{pmatrix} 0 & 0 & 0 \\ 1 & 1 & 0 \\ -1 & -1 & 0 \end{pmatrix}, \text{ whose kernel is } x + y = 0, \text{ with}$$

basis  $(-1, 1, 0)$ ,  $(0, 0, 1)$ .  $\mathbf{A} - 2\mathbf{I} = \begin{pmatrix} 0 & 1 & 1 \\ 0 & -1 & -1 \\ -1 & -1 & -1 \end{pmatrix}$  gives  $y + z = 0$  and  $x = -y - z = 0$ , so  $V_2 = \text{span}((0, -1, 1))$ . The three vectors form a basis of  $\mathbb{R}^3$ : the matrix with these columns has determinant  $-1 \cdot (0 \cdot 1 - 1 \cdot (-1)) - 0 + 0 = -1 \neq 0$ , expanding along the first row. So  $\mathbb{R}^3 = V_1 \oplus V_2$ , as Theorem 8.8.1 guarantees.

- (b)  $\mathbf{A} - \mathbf{I}$  has rank 2 (its third row is minus the sum of the first two, and the first two are independent), so  $E_1(\mathbf{A})$  is a line,  $\text{span}((-1, 1, 0))$ , strictly inside the plane  $V_1$ . This matches Corollary 8.8.4:  $\dim V_1 = a(1) = 2$ ,  $g(1) = 1$ ,  $s_1 = 2$ .
- (c) With  $p = (x-1)^2$ ,  $q = x-2$ ,  $a = 1$ ,  $b = -x$  in Theorem 5.7.5 (c), the projection onto  $V_1$  along  $V_2$  is  $b(\mathbf{A})q(\mathbf{A}) = -\mathbf{A}(\mathbf{A} - 2\mathbf{I})$ , and onto  $V_2$  along  $V_1$  it is  $a(\mathbf{A})p(\mathbf{A}) = (\mathbf{A} - \mathbf{I})^2$ . Computing,

$$-\mathbf{A}(\mathbf{A} - 2\mathbf{I}) = \begin{pmatrix} 1 & 0 & 0 \\ -1 & 0 & 0 \\ 1 & 1 & 1 \end{pmatrix}, \quad (\mathbf{A} - \mathbf{I})^2 = \begin{pmatrix} 0 & 0 & 0 \\ 1 & 1 & 0 \\ -1 & -1 & 0 \end{pmatrix}.$$

They add up to  $\mathbf{I}$ . The columns of the first satisfy  $x + y = 0$ , so they lie in  $V_1$ ; the columns of the second are multiples of  $(0, -1, 1)$ , so they lie in  $V_2$ .

- (d)  $\mathbf{A}(-1, 1, 0) = (-1, 1, 0)$ ,  $\mathbf{A}(0, 0, 1) = (1, -1, 1) = -(-1, 1, 0) + (0, 0, 1)$ , and  $\mathbf{A}(0, -1, 1) = (0, -2, 2) = 2(0, -1, 1)$ . So the matrix is

$$\begin{pmatrix} 1 & -1 \\ 0 & 1 \end{pmatrix} \oplus (2),$$

block diagonal as Theorem 8.1.5 predicts, with first block of minimal polynomial  $(x-1)^2$ .

**Exercise B3: Matrices with  $\mathbf{A}^3 = \mathbf{A}$ .**

Let  $F$  have characteristic not 2, and let  $\mathbf{A} \in M_n(F)$ ,  $n \geq 1$ , satisfy  $\mathbf{A}^3 = \mathbf{A}$ .

- (a) Prove that  $\mathbf{A}$  is diagonalizable, with  $\text{spec}(\mathbf{A}) \subseteq \{0, 1, -1\}$ .
- (b) Now let  $F = \mathbb{R}$ . Prove that  $\text{rank } \mathbf{A} = \text{tr}(\mathbf{A}^2)$ .

*Solution.*

- (a)  $x^3 - x = x(x-1)(x+1)$  annihilates  $\mathbf{A}$ . The roots  $0, 1, -1$  are distinct:  $1 \neq 0$ ,  $-1 \neq 0$ , and  $1 \neq -1$  because  $2 \neq 0$ . By [Theorem 8.8.2](#) ((c)  $\Rightarrow$  (a)),  $\mathbf{A}$  is diagonalizable, and its eigenvalues are roots of  $m_{\mathbf{A}} \mid x^3 - x$ , so they lie in  $\{0, 1, -1\}$ .
- (b) Write  $\mathbf{A} = \mathbf{P}\mathbf{D}\mathbf{P}^{-1}$  with  $\mathbf{D}$  diagonal, entries in  $\{0, 1, -1\}$ . Then  $\text{rank } \mathbf{A} = \text{rank } \mathbf{D}$  ([Proposition 3.7.6](#) (a)), which is the number  $r$  of non-zero diagonal entries. Also  $\mathbf{A}^2 = \mathbf{P}\mathbf{D}^2\mathbf{P}^{-1}$  ([Theorem 8.4.3](#)), and  $\mathbf{D}^2$  is diagonal with  $r$  entries 1 and the rest 0. By [Theorem 3.7.5](#),  $\text{tr } \mathbf{A}^2 = \text{tr } \mathbf{D}^2 = r$ .

### 8.8.5.3 C. Going deeper

**Exercise C1:**  $\mathbf{A}^p = \mathbf{A}$  over  $\mathbb{F}_p$ .

Let  $p$  be a prime and  $\mathbf{A} \in M_n(\mathbb{F}_p)$ ,  $n \geq 1$ , with  $\mathbf{A}^p = \mathbf{A}$ .

- (a) Let  $a \in \mathbb{F}_p$  be non-zero. Show that  $b \mapsto ab$  is a bijection of  $\mathbb{F}_p \setminus \{0\}$ , and deduce  $a^{p-1} = 1$ . Conclude that  $a^p = a$  for **every**  $a \in \mathbb{F}_p$ .
- (b) Deduce that  $x^p - x = \prod_{a \in \mathbb{F}_p} (x - a)$  in  $\mathbb{F}_p[x]$ .
- (c) Prove that  $\mathbf{A}$  is diagonalizable over  $\mathbb{F}_p$ .
- (d) Show that  $\mathbf{J} = \begin{pmatrix} 1 & 1 \\ 0 & 1 \end{pmatrix}$  satisfies  $\mathbf{J}^p = \mathbf{I}$  over  $\mathbb{F}_p$  but is not diagonalizable, and explain why this does not contradict (c).

*Hint: for (a), compare the product of all non-zero elements with the product of all their multiples by  $a$ .*

*Solution.*

- (a) The map  $b \mapsto ab$  sends non-zero elements to non-zero elements ([Theorem 0.7.1](#)), and it is injective, since  $ab = ab'$  gives  $b = b'$  after multiplying by  $a^{-1}$ . An injective map from the finite set  $\mathbb{F}_p \setminus \{0\}$  to itself is bijective ([Theorem 0.4.9](#)). So the elements  $ab$ , for  $b \neq 0$ , are the non-zero elements in another order. Let  $c$  be the product of all non-zero elements; it is non-zero. Multiplying the  $p-1$  elements  $ab$  gives  $a^{p-1}c = c$ , by commutativity, so  $a^{p-1} = 1$ . Multiplying by  $a$  gives  $a^p = a$ , which also holds for  $a = 0$ .
- (b) By (a), each of the  $p$  distinct elements of  $\mathbb{F}_p$  is a root of  $f = x^p - x$ . By [Theorem 5.4.3](#), the multiplicities of these roots, each at least 1, add up to at most  $\deg f = p$ . So each multiplicity is 1, the sum equals  $p$ , and  $f$  splits; as  $f$  is monic,  $f = \prod_{a \in \mathbb{F}_p} (x - a)$ .
- (c) By (b),  $x^p - x$  is a product of distinct monic linear factors, and it annihilates  $\mathbf{A}$ . By [Theorem 8.8.2](#) ((c)  $\Rightarrow$  (a)),  $\mathbf{A}$  is diagonalizable, with eigenvalues in  $\mathbb{F}_p$ .
- (d) As in [Example \(Complex Matrices of Finite Order Are Diagonalizable\)](#),  $\mathbf{J}^p = \begin{pmatrix} 1 & p \\ 0 & 1 \end{pmatrix} = \mathbf{I}$  in  $\mathbb{F}_p$ , while  $m_{\mathbf{J}} = (x-1)^2$ , so  $\mathbf{J}$  is not diagonalizable. There is no contradiction:  $\mathbf{J}$  satisfies  $\mathbf{J}^p = \mathbf{I}$ , not  $\mathbf{J}^p = \mathbf{J}$  (for  $p \geq 2$ ,  $\mathbf{J}^p = \mathbf{I} \neq \mathbf{J}$ ). The

polynomial  $x^p - 1$  has the multiple root 1, because its derivative  $px^{p-1}$  is 0 in  $\mathbb{F}_p$  (Theorem 5.4.5 (a)).

### Exercise C2: Irreducible minimal polynomials.

Let  $V \neq \{0\}$  be finite-dimensional,  $T \in \mathcal{L}(V)$ , and suppose  $m_T$  is **irreducible**, of degree  $d$ . For  $\mathbf{v} \in V$ , let  $W_{\mathbf{v}} = \{q(T)\mathbf{v} : q \in F[x]\}$ .

- Show that  $W_{\mathbf{v}}$  is the smallest  $T$ -invariant subspace of  $V$  containing  $\mathbf{v}$ .
- Let  $\mathbf{v} \neq 0$ . Prove that  $(\mathbf{v}, T\mathbf{v}, \dots, T^{d-1}\mathbf{v})$  is a basis of  $W_{\mathbf{v}}$ . In particular  $\dim W_{\mathbf{v}} = d$ .
- Deduce that the only  $T$ -invariant subspaces of  $W_{\mathbf{v}}$  are  $\{0\}$  and  $W_{\mathbf{v}}$ .
- Prove that  $V$  is a direct sum of subspaces of the form  $W_{\mathbf{v}}$ , and hence that  $d$  divides  $\dim V$ .
- Illustrate with the rotation  $\mathbf{R}$  of  $\mathbb{R}^2$  and with  $\mathbf{R} \oplus \mathbf{R}$  on  $\mathbb{R}^4$ .

*Hint: for (b), use the gcd of  $m_T$  with a polynomial of degree less than  $d$ . For (d), take a direct sum  $W_{\mathbf{v}_1} \oplus \dots \oplus W_{\mathbf{v}_s}$  that is as large as possible.*

*Solution.*

- $W_{\mathbf{v}}$  is the image of the linear map  $F[x] \rightarrow V$ ,  $q \mapsto q(T)\mathbf{v}$  (linear by Theorem 5.7.1 (a)), so it is a subspace (Theorem 3.2.2). It contains  $\mathbf{v} = 1(T)\mathbf{v}$ , and  $T(q(T)\mathbf{v}) = (xq)(T)\mathbf{v} \in W_{\mathbf{v}}$ , so it is invariant. If  $W$  is any invariant subspace containing  $\mathbf{v}$ , then  $T^k\mathbf{v} \in W$  for all  $k$  by induction, so every  $q(T)\mathbf{v} \in W$ , and  $W_{\mathbf{v}} \subseteq W$ .
- Independence.* Suppose  $q(T)\mathbf{v} = 0$  for some non-zero  $q$  of degree less than  $d$ . Let  $g = \gcd(q, m_T)$ . It is a monic divisor of the irreducible  $m_T$ , so  $g = 1$  or  $g = m_T$  (Definition 5.3.1). It is not  $m_T$ , since  $m_T \nmid q$  for degree reasons (Proposition 5.1.2 (c)). So  $g = 1$ , and  $aq + bm_T = 1$  for some  $a, b$  (Corollary 5.2.4). Then  $\mathbf{v} = a(T)q(T)\mathbf{v} + b(T)m_T(T)\mathbf{v} = 0$ , a contradiction. So no non-trivial combination of  $\mathbf{v}, \dots, T^{d-1}\mathbf{v}$  is 0.  
*Spanning.* For  $q \in F[x]$ , divide:  $q = sm_T + r$  with  $\deg r < d$  (Theorem 5.1.1). Then  $q(T)\mathbf{v} = r(T)\mathbf{v}$ , a combination of  $\mathbf{v}, \dots, T^{d-1}\mathbf{v}$ .
- Let  $W \subseteq W_{\mathbf{v}}$  be invariant and non-zero, and take  $\mathbf{w} \in W$  with  $\mathbf{w} \neq 0$ . By (a),  $W_{\mathbf{w}} \subseteq W$ . By (b),  $\dim W_{\mathbf{w}} = d = \dim W_{\mathbf{v}}$ , so  $W_{\mathbf{w}} = W_{\mathbf{v}}$  (Theorem 1.6.8), and hence  $W = W_{\mathbf{v}}$ .
- Among all families  $\mathbf{v}_1, \dots, \mathbf{v}_s$  of non-zero vectors for which the sum  $S = W_{\mathbf{v}_1} + \dots + W_{\mathbf{v}_s}$  is direct, choose one with  $s$  as large as possible; this is possible since  $\dim S = sd \leq \dim V$  by (b) and Theorem 1.7.5. Suppose  $S \neq V$ , and take  $\mathbf{v} \in V \setminus S$ . The subspace  $W_{\mathbf{v}} \cap S$  is invariant (Proposition 8.1.4) and lies in  $W_{\mathbf{v}}$ , but does not contain  $\mathbf{v}$ ; by (c) it is  $\{0\}$ . Then the sum  $S + W_{\mathbf{v}}$  is direct: if  $\mathbf{u}_1 + \dots + \mathbf{u}_s + \mathbf{w} = 0$  with  $\mathbf{u}_j \in W_{\mathbf{v}_j}$  and  $\mathbf{w} \in W_{\mathbf{v}}$ , then  $\mathbf{w} \in W_{\mathbf{v}} \cap S = \{0\}$ , and directness of  $S$  gives all  $\mathbf{u}_j = 0$ . This contradicts the maximality of  $s$ . So  $V = W_{\mathbf{v}_1} \oplus \dots \oplus W_{\mathbf{v}_s}$ , and  $\dim V = sd$ .
- For the rotation  $\mathbf{R}$ ,  $m_{\mathbf{R}} = x^2 + 1$  is irreducible over  $\mathbb{R}$ ,  $d = 2$ , and  $W_{\mathbf{v}} = \mathbb{R}^2$  for every  $\mathbf{v} \neq 0$ : no line is invariant (Example (A Rotation Has No Invariant Lines)). For  $\mathbf{R} \oplus \mathbf{R}$ ,  $m = \text{lcm}(x^2 + 1, x^2 + 1) = x^2 + 1$  (Proposition 8.5.6), and  $\mathbb{R}^4 = W_{\mathbf{e}_1} \oplus W_{\mathbf{e}_3}$  with  $W_{\mathbf{e}_1} = \text{span}(\mathbf{e}_1, \mathbf{e}_2)$  and  $W_{\mathbf{e}_3} = \text{span}(\mathbf{e}_3, \mathbf{e}_4)$ ; indeed  $2 \mid 4$ .



## 8.9 Commuting Operators

So far we have diagonalized or triangularized one operator at a time. Many problems hand us several at once: two matrices that describe the same system, or a matrix together with all its powers. Can a single basis make **all** of them diagonal, or all of them triangular? A necessary condition is easy to spot, because diagonal matrices commute. This section shows that commuting is also enough, as long as each operator can be handled on its own. The whole argument rests on one observation: an operator that commutes with  $T$  maps each eigenspace of  $T$  into itself, so we can search for common eigenvectors inside an eigenspace.

### 8.9.1 When can two operators share a basis?

Let  $V$  be finite-dimensional and  $S, T \in \mathcal{L}(V)$ . Suppose one basis  $\mathcal{B}$  makes both  $[S]_{\mathcal{B}}$  and  $[T]_{\mathcal{B}}$  diagonal. Diagonal matrices multiply entry by entry on the diagonal, so  $[S]_{\mathcal{B}}[T]_{\mathcal{B}} = [T]_{\mathcal{B}}[S]_{\mathcal{B}}$ . By [Corollary 3.6.3](#) this says  $[ST]_{\mathcal{B}} = [TS]_{\mathcal{B}}$ , and since  $T \mapsto [T]_{\mathcal{B}}$  is injective ([Theorem 3.6.4](#)),  $ST = TS$ . So **sharing a diagonalizing basis forces commuting**. The same is not true for triangular matrices:  $\begin{pmatrix} 1 & 1 \\ 0 & 2 \end{pmatrix}$  and  $\begin{pmatrix} 1 & 0 \\ 0 & 2 \end{pmatrix}$  are both upper triangular, and their products in the two orders are  $\begin{pmatrix} 1 & 2 \\ 0 & 4 \end{pmatrix}$  and  $\begin{pmatrix} 1 & 1 \\ 0 & 4 \end{pmatrix}$ . We will see, though, that commuting is a convenient sufficient condition in the triangular case as well.

The question for this section is the converse: if  $ST = TS$ , can we find a common basis? Here are the commuting pairs to keep in mind.

- **Polynomials in one operator.** For every  $p, q \in F[x]$ ,  $p(T)q(T) = q(T)p(T)$  ([Theorem 5.7.1](#) (c)). In particular  $T$  commutes with  $T^2$ , with  $T^{-1}$  when it exists (both products are  $\text{id}_V$ ), and with every scalar operator  $c \text{id}_V$ .
- **Diagonal matrices** commute with each other, and so do block diagonal matrices whose corresponding blocks commute ([Theorem 7.1.4](#)).
- **A non-commuting pair.**  $\mathbf{E}_{12} = \begin{pmatrix} 0 & 1 \\ 0 & 0 \end{pmatrix}$  and  $\mathbf{E}_{21} = \begin{pmatrix} 0 & 0 \\ 1 & 0 \end{pmatrix}$  give  $\mathbf{E}_{12}\mathbf{E}_{21} = \mathbf{E}_{11} \neq \mathbf{E}_{22} = \mathbf{E}_{21}\mathbf{E}_{12}$ .

The last pair already shows that something must be assumed. Over any field, the eigenvectors of  $\mathbf{E}_{12}$  are the non-zero multiples of  $\mathbf{e}_1$ , and those of  $\mathbf{E}_{21}$  are the non-zero multiples of  $\mathbf{e}_2$ . They share no eigenvector at all, let alone a basis of them.

### 8.9.2 Commuting operators preserve eigenspaces

How does the hypothesis  $ST = TS$  interact with eigenvectors? Take  $T\mathbf{v} = \lambda\mathbf{v}$  and ask what  $T$  does to  $S\mathbf{v}$ . There is no  $S$  in the eigenvector equation yet, so we apply  $S$  to both sides.

#### Theorem 8.9.1 — Commuting Operators Preserve Eigenspaces

Let  $V$  be a vector space over  $F$ , and let  $S, T \in \mathcal{L}(V)$  with  $ST = TS$ . Then for every  $\lambda \in F$ , the eigenspace  $E_{\lambda}(T)$  is  $S$ -invariant.



*Proof.* Let  $\mathbf{v} \in E_\lambda(T)$ , so  $T\mathbf{v} = \lambda\mathbf{v}$ . Since  $ST = TS$  and  $S$  is linear,

$$T(S\mathbf{v}) = (TS)\mathbf{v} = (ST)\mathbf{v} = S(\lambda\mathbf{v}) = \lambda S\mathbf{v}.$$

Hence  $S\mathbf{v} \in E_\lambda(T)$  by Definition 8.2.2. This shows that  $S$  maps  $E_\lambda(T)$  into itself. ■

No finite dimension is needed, and  $\lambda$  need not be an eigenvalue (then  $E_\lambda(T) = \{\mathbf{0}\}$  and there is nothing to prove). By symmetry, every eigenspace of  $S$  is  $T$ -invariant too.

In matrix language the theorem is a statement about block shapes. Take  $T = \text{diag}(1, 1, 2)$  on  $F^3$ , with  $E_1(T) = \text{span}(\mathbf{e}_1, \mathbf{e}_2)$  and  $E_2(T) = \text{span}(\mathbf{e}_3)$ , and let  $S$  be any matrix with  $ST = TS$ . Both eigenspaces are  $S$ -invariant, and  $F^3 = E_1(T) \oplus E_2(T)$ , so by Theorem 8.1.5 the matrix  $S$  is block diagonal:

$$S = \begin{pmatrix} s_{11} & s_{12} & 0 \\ s_{21} & s_{22} & 0 \\ 0 & 0 & s_{33} \end{pmatrix}.$$

Conversely, every matrix of this shape commutes with  $T$ , since  $T = \mathbf{I}_2 \oplus (2)$  and  $\mathbf{I}_2$  commutes with every  $2 \times 2$  block (Theorem 7.1.4). Inside the repeated eigenspace  $S$  can be anything; this freedom is what the rest of the section has to deal with.

**Non-example.** Take  $T = \text{diag}(1, 2)$  and the swap  $S = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}$ . Then  $S\mathbf{e}_1 = \mathbf{e}_2 \notin E_1(T) = \text{span}(\mathbf{e}_1)$ , so  $E_1(T)$  is not  $S$ -invariant. The theorem is not contradicted:  $ST = \begin{pmatrix} 0 & 2 \\ 1 & 0 \end{pmatrix}$  and  $TS = \begin{pmatrix} 0 & 1 \\ 2 & 0 \end{pmatrix}$  differ.

*Quick check.* Let  $T = \mathbf{I}_2$  and  $S = \begin{pmatrix} 1 & 1 \\ 0 & 1 \end{pmatrix}$  over  $\mathbb{R}$ . Do  $S$  and  $T$  commute? Is every eigenvector of  $T$  an eigenvector of  $S$ ? Do  $S$  and  $T$  have a common eigenvector?

*Solution.* Yes,  $\mathbf{I}_2$  commutes with every  $2 \times 2$  matrix. No: every non-zero vector is an eigenvector of  $\mathbf{I}_2$ , but  $S\mathbf{e}_2 = (1, 1)$  is not a multiple of  $\mathbf{e}_2$ . Yes:  $\mathbf{e}_1$  is an eigenvector of both, for the eigenvalues 1 and 1. The theorem says  $S$  maps  $E_1(T) = \mathbb{R}^2$  into itself, which is true but empty of information; the common eigenvector must be **searched for** inside that eigenspace, and here it is an eigenvector of  $S$ .

### 8.9.3 A common eigenvector

Here is the plan for finding a common eigenvector of commuting  $S$  and  $T$ . Pick an eigenvalue  $\lambda$  of  $T$  and put  $U = E_\lambda(T) \neq \{\mathbf{0}\}$ . By Theorem 8.9.1,  $S$  restricts to an operator  $S|_U$  on  $U$ . If  $S|_U$  has an eigenvector  $\mathbf{u}$ , then  $S\mathbf{u} = \mu\mathbf{u}$  and, since  $\mathbf{u} \in U$ , also  $T\mathbf{u} = \lambda\mathbf{u}$ . So everything reduces to whether  $S|_U$  has an eigenvalue. Over  $\mathbb{C}$  it does (Theorem 8.2.4). Over a general field we need to know that the characteristic polynomial of a restriction splits when that of  $S$  does. We record this, together with the same fact for the induced operator on the quotient, since triangularization will need both.

### Lemma 8.9.2 — Restrictions and Quotients Inherit Splitting

Let  $V$  be a finite-dimensional vector space over  $F$ , let  $T \in \mathcal{L}(V)$ , and let  $U$  be a  $T$ -invariant subspace with  $\{0\} \neq U \neq V$ . If  $p_T$  splits over  $F$ , then  $p_{T|_U}$  and  $p_{\bar{T}}$  split over  $F$ , where  $\bar{T}$  is the induced operator on  $V/U$ .

*Idea.* Section 1 factors  $p_T$  as  $p_{T|_U} p_{\bar{T}}$ , so the two factors can only have roots that  $p_T$  has. A polynomial splits exactly when its root multiplicities add up to its degree (Theorem 5.4.3). Count: the multiplicities of the two factors add up to those of  $p_T$ , which use up the whole degree; so neither factor can be short of its own degree.

*Proof.* By Theorem 8.1.3,  $p_T = fg$  with  $f = p_{T|_U}$  and  $g = p_{\bar{T}}$ , both monic (Theorem 6.10.2). Let  $c_1, \dots, c_k$  be the distinct roots of  $p_T$  in  $F$ . Every root of  $f$  or of  $g$  in  $F$  is a root of  $p_T$  (Theorem 0.8.5), so it is among the  $c_i$ , and a  $c_i$  that is not a root of  $f$  has  $\text{mult}_{c_i}(f) = 0$ . Hence, by Theorem 5.4.3 applied to  $f$  and to  $g$ ,

$$\sum_{i=1}^k \text{mult}_{c_i}(f) \leq \deg f, \quad \sum_{i=1}^k \text{mult}_{c_i}(g) \leq \deg g.$$

Adding, and using Theorem 5.4.2 and then Theorem 5.4.3 for the split polynomial  $p_T$ ,

$$\begin{aligned} \deg f + \deg g &= \deg p_T = \sum_{i=1}^k \text{mult}_{c_i}(p_T) \\ &= \sum_{i=1}^k \text{mult}_{c_i}(f) + \sum_{i=1}^k \text{mult}_{c_i}(g) \\ &\leq \deg f + \deg g. \end{aligned}$$

So both inequalities above are equalities, and  $f$  and  $g$  split over  $F$  by Theorem 5.4.3. ■

Now the common eigenvector. We prove it for a whole family of commuting operators at once, possibly infinite, since the proof costs no more and the next theorems need it. A family  $\mathcal{F} \subseteq \mathcal{L}(V)$  is **commuting** if  $ST = TS$  for all  $S, T \in \mathcal{F}$ .

### Theorem 8.9.3 — Commuting Operators Have a Common Eigenvector

Let  $V \neq \{0\}$  be a finite-dimensional vector space over  $F$ , and let  $\mathcal{F} \subseteq \mathcal{L}(V)$  be a non-empty commuting family such that  $p_T$  splits over  $F$  for every  $T \in \mathcal{F}$ . Then there is a non-zero  $\mathbf{v} \in V$  that is an eigenvector of **every**  $T \in \mathcal{F}$ . This applies to every commuting family when  $F = \mathbb{C}$ .

*Idea.* For two operators, the plan above works: an eigenvector of  $S|_U$  with  $U = E_\lambda(T)$ . For many operators we would like to repeat it, shrinking  $U$  each time, and a family may be infinite. So we jump straight to the end of the shrinking: take a non-zero subspace  $W$ , invariant under the whole family, that is as small as possible. If some  $T \in \mathcal{F}$  were not a scalar on  $W$ , an eigenspace of  $T|_W$  would be a smaller such subspace. So every

operator in the family is a scalar on  $W$ , and any non-zero vector of  $W$  is a common eigenvector.

*Proof.* Call a subspace  $\mathcal{F}$ -**invariant** if it is  $T$ -invariant for every  $T \in \mathcal{F}$ . The space  $V$  is a non-zero  $\mathcal{F}$ -invariant subspace, so among all non-zero  $\mathcal{F}$ -invariant subspaces there is one,  $W$ , of smallest dimension.

Let  $T \in \mathcal{F}$ . Then  $T|_W$  is an operator on  $W \neq \{0\}$ , and  $p_{T|_W}$  splits: if  $W = V$  this is the hypothesis, and otherwise it is [Lemma 8.9.2](#). Since  $p_{T|_W}$  is monic of degree  $\dim W \geq 1$ , it has a root  $\lambda \in F$ , which is an eigenvalue of  $T|_W$  by [Theorem 8.2.1](#). Put

$$W' = \{\mathbf{w} \in W : T\mathbf{w} = \lambda\mathbf{w}\} = W \cap E_\lambda(T),$$

which is non-zero because it contains an eigenvector of  $T|_W$ . For every  $S \in \mathcal{F}$ , both  $W$  and  $E_\lambda(T)$  are  $S$ -invariant, the first by the choice of  $W$  and the second by [Theorem 8.9.1](#); so  $W'$  is  $S$ -invariant by [Proposition 8.1.4](#). Hence  $W'$  is a non-zero  $\mathcal{F}$ -invariant subspace contained in  $W$ , and by the minimality of  $\dim W$ ,  $\dim W' = \dim W$ . By [Theorem 1.6.8](#),  $W' = W$ , that is,  $T\mathbf{w} = \lambda\mathbf{w}$  for every  $\mathbf{w} \in W$ .

So every  $T \in \mathcal{F}$  acts on  $W$  as multiplication by a scalar  $\lambda_T$ . Any non-zero  $\mathbf{v} \in W$  then satisfies  $T\mathbf{v} = \lambda_T\mathbf{v}$  for every  $T \in \mathcal{F}$ , as claimed. Over  $\mathbb{C}$  every characteristic polynomial splits ([Corollary 5.5.5](#)), so the hypothesis holds automatically. ■

The proof never had to name the eigenvalues. It found a subspace on which every operator of the family is a scalar, and the scalars came for free.

*Warning.* The **splitting hypothesis cannot be dropped, and neither can commuting**. Over  $\mathbb{R}$ , the rotation  $\mathbf{R} = \begin{pmatrix} 0 & -1 \\ 1 & 0 \end{pmatrix}$  commutes with itself, yet the family  $\{\mathbf{R}\}$  has no real eigenvector at all, since  $p_{\mathbf{R}} = x^2 + 1$  has no real root. Over  $\mathbb{C}$ , the non-commuting pair  $\mathbf{E}_{12}, \mathbf{E}_{21}$  has no common eigenvector, as we saw above.

## 8.9.4 Simultaneous triangularization

Section 7 triangularized one operator by splitting off an eigenvector and passing to the quotient. For a commuting family the same recursion works, because the eigenvector we split off can be chosen common to all.

### Theorem 8.9.4 — Simultaneous Triangularization

Let  $V$  be a finite-dimensional vector space over  $F$  with  $\dim V = n \geq 1$ , and let  $\mathcal{F} \subseteq \mathcal{L}(V)$  be a non-empty commuting family such that  $p_T$  splits over  $F$  for every  $T \in \mathcal{F}$ . Then there is **one** basis  $\mathcal{B}$  of  $V$  such that  $[T]_{\mathcal{B}}$  is upper triangular for **every**  $T \in \mathcal{F}$ .

In particular, if  $\mathbf{A}_1, \dots, \mathbf{A}_r \in M_n(\mathbb{C})$  commute pairwise, there is an invertible  $\mathbf{P} \in M_n(\mathbb{C})$  such that  $\mathbf{P}^{-1}\mathbf{A}_i\mathbf{P}$  is upper triangular for every  $i$ .

*Idea.* **Take one common eigenvector, extend, recurse.** ① [Theorem 8.9.3](#) gives  $\mathbf{v}_1$ , an eigenvector of every  $T \in \mathcal{F}$ , so  $U = \text{span}(\mathbf{v}_1)$  is invariant under the family. ② On  $V/U$ , the induced operators still commute, and their characteristic polynomials still split. ③ By induction on the dimension, one basis of  $V/U$  triangularizes all of them. ④ Lift it back and put  $\mathbf{v}_1$  in front; the block triangular form of Section 1 does the rest.

*Proof.* We use induction on  $n$ . If  $n = 1$ , every  $1 \times 1$  matrix is upper triangular, so any basis works.

Let  $n \geq 2$ , and assume the theorem for spaces of dimension  $n - 1$ . By [Theorem 8.9.3](#) there is  $\mathbf{v}_1 \neq \mathbf{0}$  with  $T\mathbf{v}_1 = \lambda_T \mathbf{v}_1$  for every  $T \in \mathcal{F}$ . Let  $U = \text{span}(\mathbf{v}_1)$ ; it is  $T$ -invariant for every  $T \in \mathcal{F}$  ([Proposition 8.1.2](#)), and  $\{\mathbf{0}\} \neq U \neq V$  since  $\dim U = 1 < n$ . For  $T \in \mathcal{F}$  let  $\bar{T}$  be the induced operator on  $V/U$  ([Definition 8.1.2](#)), and put  $\bar{\mathcal{F}} = \{\bar{T} : T \in \mathcal{F}\}$ .

The family  $\bar{\mathcal{F}}$  is commuting: for  $S, T \in \mathcal{F}$  and  $\mathbf{v} \in V$ ,

$$\bar{S}\bar{T}(\mathbf{v} + U) = \bar{S}(T\mathbf{v} + U) = ST\mathbf{v} + U = TS\mathbf{v} + U = \bar{T}\bar{S}(\mathbf{v} + U).$$

Each  $p_{\bar{T}}$  splits by [Lemma 8.9.2](#), and  $\dim V/U = n - 1$  by [Theorem 3.9.5](#). By the induction hypothesis there are  $\mathbf{w}_2, \dots, \mathbf{w}_n \in V$  such that  $\bar{\mathcal{B}} = (\mathbf{w}_2 + U, \dots, \mathbf{w}_n + U)$  is a basis of  $V/U$  in which every  $[\bar{T}]_{\bar{\mathcal{B}}}$  is upper triangular.

Let  $\mathcal{B} = (\mathbf{v}_1, \mathbf{w}_2, \dots, \mathbf{w}_n)$ . It is linearly independent: if  $a_1\mathbf{v}_1 + a_2\mathbf{w}_2 + \dots + a_n\mathbf{w}_n = \mathbf{0}$ , then passing to  $V/U$ , where  $\mathbf{v}_1 + U$  is the zero coset, gives  $a_2(\mathbf{w}_2 + U) + \dots + a_n(\mathbf{w}_n + U) = \mathbf{0}$ , so  $a_2 = \dots = a_n = 0$  by the independence of  $\bar{\mathcal{B}}$ ; then  $a_1\mathbf{v}_1 = \mathbf{0}$  with  $\mathbf{v}_1 \neq \mathbf{0}$  gives  $a_1 = 0$ . A linearly independent list of length  $n = \dim V$  is a basis ([Theorem 1.6.4](#)). Now  $\mathcal{B}$  is a basis of  $U$  extended to a basis of  $V$ , and  $\bar{\mathcal{B}}$  is exactly the list of cosets in [Theorem 8.1.3](#). Hence for every  $T \in \mathcal{F}$ ,

$$[T]_{\mathcal{B}} = \begin{pmatrix} \lambda_T & * \\ 0 & [\bar{T}]_{\bar{\mathcal{B}}} \end{pmatrix},$$

with a  $1 \times 1$  upper-left block, a zero column below it, and an upper triangular lower-right block. Such a matrix is upper triangular. This completes the induction.

For the matrix statement, apply this to the family  $\{T_{\mathbf{A}_1}, \dots, T_{\mathbf{A}_r}\}$  on  $\mathbb{C}^n$ , whose characteristic polynomials split by [Corollary 5.5.5](#), and let  $\mathbf{P}$  have the vectors of  $\mathcal{B}$  as columns; then  $\mathbf{P}^{-1}\mathbf{A}_i\mathbf{P} = [T_{\mathbf{A}_i}]_{\mathcal{B}}$  ([Theorem 3.7.3](#)).

■

A first use: if  $S$  and  $T$  commute and both characteristic polynomials split, the diagonal entries of  $[S + T]_{\mathcal{B}}$  and  $[ST]_{\mathcal{B}}$  are the sums and products of the diagonal entries of  $[S]_{\mathcal{B}}$  and  $[T]_{\mathcal{B}}$ , position by position. For the sum this is entrywise; for the product, the  $(i, i)$ -entry of  $\mathbf{XY}$  is  $\sum_k x_{ik}y_{ki}$  ([Definition 0.9.9](#)), and when  $\mathbf{X}$  and  $\mathbf{Y}$  are upper triangular only  $k = i$  can contribute, since  $x_{ik} = 0$  for  $k < i$  and  $y_{ki} = 0$  for  $k > i$ . By [Theorem 8.7.4](#), those diagonal entries are the eigenvalues. So the eigenvalues of  $S + T$  are of the form  $\lambda + \mu$ , with  $\lambda$  an eigenvalue of  $S$  and  $\mu$  one of  $T$ , **paired by position in a common triangular form**. Section 2 warned that nothing like this holds without commuting.

## 8.9.5 Simultaneous diagonalization

Diagonalization is the stronger wish, and it needs a stronger hypothesis: each operator must be diagonalizable on its own. The name for the goal:

### Definition 8.9.1 — Simultaneously Diagonalizable

Let  $V$  be finite-dimensional. A family  $\mathcal{F} \subseteq \mathcal{L}(V)$  is **simultaneously diagonalizable** if there is **one** basis  $\mathcal{B}$  of  $V$  such that  $[T]_{\mathcal{B}}$  is diagonal for **every**  $T \in \mathcal{F}$ . Matrices  $\mathbf{A}_1, \dots, \mathbf{A}_r \in M_n(F)$  are simultaneously diagonalizable over  $F$  if there is **one** invertible  $\mathbf{P} \in M_n(F)$  with every  $\mathbf{P}^{-1}\mathbf{A}_i\mathbf{P}$  diagonal.

Equivalently, by the unwinding in Section 4,  $V$  has a basis whose vectors are eigenvectors of every  $T \in \mathcal{F}$ . The opening of this section showed that a simultaneously diagonalizable family is commuting. For diagonalizable operators the converse holds.

### Theorem 8.9.5 — Simultaneous Diagonalization

Let  $V$  be a finite-dimensional vector space over  $F$  with  $\dim V \geq 1$ , and let  $S, T \in \mathcal{L}(V)$  be diagonalizable. Then  $S$  and  $T$  are simultaneously diagonalizable if and only if  $ST = TS$ .

For matrices: if  $\mathbf{A}, \mathbf{B} \in M_n(F)$  are diagonalizable over  $F$ , there is an invertible  $\mathbf{P} \in M_n(F)$  with  $\mathbf{P}^{-1}\mathbf{A}\mathbf{P}$  and  $\mathbf{P}^{-1}\mathbf{B}\mathbf{P}$  both diagonal if and only if  $\mathbf{AB} = \mathbf{BA}$ .

*Idea.* Split  $V$  into the eigenspaces of  $T$ . On each piece  $T$  is a scalar, so **any** basis of the piece consists of eigenvectors of  $T$ ; we are free to choose it to suit  $S$ . The piece is  $S$ -invariant because  $S$  commutes with  $T$ , and  $S$  restricted to it is diagonalizable because  $S$  is (Section 8). So each piece has a basis of eigenvectors of  $S$ , and gluing these bases together gives the common basis.

*Proof.* ( $\Rightarrow$ ) Let  $\mathcal{B}$  be a basis with  $[S]_{\mathcal{B}}$  and  $[T]_{\mathcal{B}}$  diagonal. Diagonal matrices commute, so by Corollary 3.6.3,  $[ST]_{\mathcal{B}} = [S]_{\mathcal{B}}[T]_{\mathcal{B}} = [T]_{\mathcal{B}}[S]_{\mathcal{B}} = [TS]_{\mathcal{B}}$ . The map  $\mathbf{R} \mapsto [\mathbf{R}]_{\mathcal{B}}$  is injective (Theorem 3.6.4), so  $ST = TS$ .

( $\Leftarrow$ ) Suppose  $ST = TS$ . Let  $\lambda_1, \dots, \lambda_k$  be the distinct eigenvalues of  $T$ , and put  $U_i = E_{\lambda_i}(T)$ . Since  $T$  is diagonalizable, Theorem 8.4.1 (c) gives

$$V = U_1 \oplus \dots \oplus U_k.$$

Each  $U_i$  is  $S$ -invariant (Theorem 8.9.1), and  $S$  is diagonalizable, so  $S|_{U_i}$  is diagonalizable by Corollary 8.8.3. Hence  $U_i$  has a basis  $\mathcal{B}_i$  of eigenvectors of  $S|_{U_i}$  (Theorem 8.4.1 (b)), which are eigenvectors of  $S$ . They are also eigenvectors of  $T$ , being non-zero vectors of  $U_i = E_{\lambda_i}(T)$ . Let  $\mathcal{B}$  be the list  $\mathcal{B}_1, \dots, \mathcal{B}_k$ . Since the sum is direct,  $\mathcal{B}$  is a basis of  $V$  by Theorem 1.7.5 ((a)  $\Rightarrow$  (d)). Every vector of  $\mathcal{B}$  is an eigenvector of both  $S$  and  $T$ , so  $[S]_{\mathcal{B}}$  and  $[T]_{\mathcal{B}}$  are diagonal (Theorem 8.4.1, (a)  $\Leftrightarrow$  (b)).

For matrices, apply this to  $T_{\mathbf{A}}$  and  $T_{\mathbf{B}}$  on  $F^n$ , which are diagonalizable because  $\mathbf{A}$  and  $\mathbf{B}$  are (Definition 8.4.1), and commute because  $\mathbf{AB} = \mathbf{BA}$ ; let  $\mathbf{P}$  have the common

eigenvectors as columns, as in Section 4.

■

The proof is also an algorithm. Diagonalize  $\mathbf{A}$ ; in each eigenspace of  $\mathbf{A}$  of dimension at least 2, write down the matrix of  $\mathbf{B}$  restricted to it and diagonalize that small matrix; eigenspaces of dimension 1 need no work, since their basis vector is automatically an eigenvector of  $\mathbf{B}$ .

**Example. (Diagonalizing Two Commuting Matrices Together)**

Let

$$\mathbf{A} = \begin{pmatrix} 1 & 0 & 0 \\ -2 & 1 & 2 \\ -2 & 0 & 3 \end{pmatrix}, \quad \mathbf{B} = \begin{pmatrix} 1 & -2 & 2 \\ 0 & 1 & 0 \\ 0 & -2 & 3 \end{pmatrix} \in M_3(\mathbb{R}).$$

Check that  $\mathbf{AB} = \mathbf{BA}$ , and find an invertible  $\mathbf{P}$  such that  $\mathbf{P}^{-1}\mathbf{AP}$  and  $\mathbf{P}^{-1}\mathbf{BP}$  are both diagonal.

*Solution. They commute.* Multiplying out,

$$\mathbf{AB} = \begin{pmatrix} 1 & -2 & 2 \\ -2 & 1 & 2 \\ -2 & -2 & 5 \end{pmatrix} = \mathbf{BA}.$$

*Diagonalize  $\mathbf{A}$ .* Expanding  $\det(x\mathbf{I} - \mathbf{A})$  along the first row, whose only non-zero entry is  $x - 1$ ,

$$p_{\mathbf{A}}(x) = (x - 1) \det \begin{pmatrix} x - 1 & -2 \\ 0 & x - 3 \end{pmatrix} = (x - 1)^2(x - 3).$$

For  $\lambda = 1$ ,  $\mathbf{A} - \mathbf{I} = \begin{pmatrix} 0 & 0 & 0 \\ -2 & 0 & 2 \\ -2 & 0 & 2 \end{pmatrix}$  has rank 1 and reduces to  $x = z$ , so  $E_1(\mathbf{A})$  is the

plane with basis  $\mathbf{u}_1 = (0, 1, 0)$ ,  $\mathbf{u}_2 = (1, 0, 1)$ . For  $\lambda = 3$ ,  $\mathbf{A} - 3\mathbf{I} = \begin{pmatrix} -2 & 0 & 0 \\ -2 & -2 & 2 \\ -2 & 0 & 0 \end{pmatrix}$  gives

$x = 0$  and  $y = z$ , so  $E_3(\mathbf{A}) = \text{span}((0, 1, 1))$ . Since  $g(1) = 2 = a(1)$  and  $g(3) = 1 = a(3)$ ,  $\mathbf{A}$  is diagonalizable. The same computation for  $\mathbf{B}$  gives  $p_{\mathbf{B}} = (x - 1)^2(x - 3)$ , with

$\mathbf{B} - \mathbf{I} = \begin{pmatrix} 0 & -2 & 2 \\ 0 & 0 & 0 \\ 0 & -2 & 2 \end{pmatrix}$  of rank 1 and  $\mathbf{B} - 3\mathbf{I}$  of rank 2, so  $\mathbf{B}$  is diagonalizable as well.

A basis of eigenvectors of  $\mathbf{A}$  need not work for  $\mathbf{B}$ :  $\mathbf{Bu}_1 = (-2, 1, -2)$  is not a multiple of  $\mathbf{u}_1$ . This is where the search inside the eigenspace comes in.

*The one-dimensional eigenspace.*  $E_3(\mathbf{A})$  is  $\mathbf{B}$ -invariant ([Theorem 8.9.1](#)) and one-dimensional, so  $\mathbf{v}_3 = (0, 1, 1)$  is an eigenvector of  $\mathbf{B}$ : indeed  $\mathbf{Bv}_3 = (0, 1, 1) = 1 \cdot \mathbf{v}_3$ .

*The plane  $E_1(\mathbf{A})$ .* It is  $\mathbf{B}$ -invariant, and

$$\mathbf{Bu}_1 = (-2, 1, -2) = \mathbf{u}_1 - 2\mathbf{u}_2, \quad \mathbf{Bu}_2 = (3, 0, 3) = 3\mathbf{u}_2.$$

So the matrix of  $\mathbf{B}|_{E_1(\mathbf{A})}$  in the basis  $(\mathbf{u}_1, \mathbf{u}_2)$  is  $\mathbf{M} = \begin{pmatrix} 1 & 0 \\ -2 & 3 \end{pmatrix}$ , whose columns are the coordinates just found. It is lower triangular with eigenvalues 1 and 3. From  $\mathbf{M} - \mathbf{I} =$

$\begin{pmatrix} 0 & 0 \\ -2 & 2 \end{pmatrix}$  the eigenvector is  $(1, 1)$ , which stands for  $\mathbf{u}_1 + \mathbf{u}_2 = (1, 1, 1)$ ; from  $\mathbf{M} - 3\mathbf{I} = \begin{pmatrix} -2 & 0 \\ -2 & 0 \end{pmatrix}$  it is  $(0, 1)$ , which stands for  $\mathbf{u}_2 = (1, 0, 1)$ .

*The common basis.* Put  $\mathbf{v}_1 = (1, 0, 1)$ ,  $\mathbf{v}_2 = (1, 1, 1)$ ,  $\mathbf{v}_3 = (0, 1, 1)$  as columns:

$$\mathbf{P} = \begin{pmatrix} 1 & 1 & 0 \\ 0 & 1 & 1 \\ 1 & 1 & 1 \end{pmatrix}, \quad \mathbf{P}^{-1}\mathbf{A}\mathbf{P} = \text{diag}(1, 1, 3), \quad \mathbf{P}^{-1}\mathbf{B}\mathbf{P} = \text{diag}(3, 1, 1).$$

*Check.*  $\mathbf{A}\mathbf{v}_1 = \mathbf{v}_1$ ,  $\mathbf{A}\mathbf{v}_2 = \mathbf{v}_2$ ,  $\mathbf{A}\mathbf{v}_3 = (0, 3, 3) = 3\mathbf{v}_3$ ;  $\mathbf{B}\mathbf{v}_1 = (3, 0, 3) = 3\mathbf{v}_1$ ,  $\mathbf{B}\mathbf{v}_2 = \mathbf{v}_2$ ,  $\mathbf{B}\mathbf{v}_3 = \mathbf{v}_3$ . Expanding along the first row,  $\det \mathbf{P} = 1(1 - 1) - 1(0 - 1) + 0 = 1 \neq 0$ , so  $\mathbf{P}$  is invertible, with  $\mathbf{P}^{-1} = \begin{pmatrix} 0 & -1 & 1 \\ 1 & 1 & -1 \\ -1 & 0 & 1 \end{pmatrix}$ .

Notice that neither matrix alone determines  $\mathbf{P}$ :  $\mathbf{A}$  cannot tell  $\mathbf{v}_1$  from  $\mathbf{v}_2$ , and  $\mathbf{B}$  cannot tell  $\mathbf{v}_2$  from  $\mathbf{v}_3$ . Together they pin down all three lines.

**Warning. Commuting does not make an operator diagonalizable.** The theorem assumes that both  $S$  and  $T$  are diagonalizable.  $\mathbf{J} = \begin{pmatrix} 1 & 1 \\ 0 & 1 \end{pmatrix}$  commutes with  $\mathbf{I}_2$ , and  $\mathbf{I}_2$  is diagonal, but  $\mathbf{J}$  is not diagonalizable (Section 4), so no basis makes both diagonal. What survives is [Theorem 8.9.4](#):  $\mathbf{J}$  and  $\mathbf{I}_2$  are already upper triangular together.

The theorem extends from two operators to any number, and even to infinitely many. An infinite family looks hopeless for an induction, but it lives inside the finite-dimensional space  $\mathcal{L}(V)$ , so finitely many of its members already span everything the family spans.

### Theorem 8.9.6 — Simultaneous Diagonalization of a Commuting Family

Let  $V$  be a finite-dimensional vector space over  $F$  with  $\dim V \geq 1$ , and let  $\mathcal{F} \subseteq \mathcal{L}(V)$  be a family of diagonalizable operators. Then  $\mathcal{F}$  is simultaneously diagonalizable if and only if it is commuting.

*Idea.* ① **Finite first.** For finitely many commuting diagonalizable  $T_1, \dots, T_r$ , induct on  $r$ : split  $V$  into eigenspaces of  $T_1$ , and on each piece apply the case  $r - 1$  to the restrictions of  $T_2, \dots, T_r$ . ② **Reduce to finite.** A longest linearly independent list  $T_1, \dots, T_r$  taken from  $\mathcal{F}$  spans every member of  $\mathcal{F}$ , and has length at most  $\dim \mathcal{L}(V)$ . ③ A basis that diagonalizes  $T_1, \dots, T_r$  diagonalizes every linear combination of them.

*Proof.* ( $\Rightarrow$ ) If one basis makes every  $[T]_{\mathcal{B}}$  diagonal, any two of these matrices commute, and the argument of [Theorem 8.9.5](#) ( $\Rightarrow$ ) gives  $ST = TS$  for all  $S, T \in \mathcal{F}$ .

( $\Leftarrow$ ) Suppose  $\mathcal{F}$  is commuting. If  $\mathcal{F}$  is empty, any basis works; assume not.

**Step 1: finitely many operators.** We prove by induction on  $r \geq 1$ : for every non-zero finite-dimensional space  $W$  and all pairwise commuting diagonalizable  $T_1, \dots, T_r \in \mathcal{L}(W)$ , some basis of  $W$  consists of eigenvectors of every  $T_i$ . For  $r = 1$  this is [Theorem 8.4.1](#). Let  $r \geq 2$ , and let  $\lambda_1, \dots, \lambda_k$  be the distinct eigenvalues of  $T_1$ ,

with  $U_j = E_{\lambda_j}(T_1) \neq \{0\}$ . By [Theorem 8.4.1](#) (c),  $W = U_1 \oplus \cdots \oplus U_k$ . Each  $U_j$  is  $T_i$ -invariant for every  $i$  ([Theorem 8.9.1](#)), each  $T_i|_{U_j}$  is diagonalizable ([Corollary 8.8.3](#)), and the restrictions  $T_2|_{U_j}, \dots, T_r|_{U_j}$  commute pairwise, since they agree with  $T_2, \dots, T_r$  on  $U_j$ . By the induction hypothesis,  $U_j$  has a basis  $\mathcal{B}_j$  of common eigenvectors of  $T_2, \dots, T_r$ ; they are eigenvectors of  $T_1$  too, as non-zero vectors of  $U_j$ . The list  $\mathcal{B}_1, \dots, \mathcal{B}_k$  is a basis of  $W$  by [Theorem 1.7.5](#) ((a)  $\Rightarrow$  (d)). This completes the induction.

**Step 2: a finite spanning subfamily.** Let  $N = \dim \mathcal{L}(V) = (\dim V)^2$  ([Theorem 3.6.4](#)). Every linearly independent list in  $\mathcal{L}(V)$  has length at most  $N$  ([Theorem 1.6.5](#)), so among the linearly independent lists of members of  $\mathcal{F}$  there is one of greatest length, say  $(T_1, \dots, T_r)$ , with  $r \geq 1$  because a single non-zero member is independent (if every member of  $\mathcal{F}$  is 0, any basis works). For every  $T \in \mathcal{F}$ , the list  $(T_1, \dots, T_r, T)$  is longer, hence dependent, so  $T \in \text{span}(T_1, \dots, T_r)$  by [Lemma 1.5.3](#).

**Step 3.** By Step 1 there is a basis  $\mathcal{B}$  of  $V$  with every  $[T_i]_{\mathcal{B}}$  diagonal. For  $T \in \mathcal{F}$ , write  $T = c_1 T_1 + \cdots + c_r T_r$ ; then  $[T]_{\mathcal{B}} = c_1 [T_1]_{\mathcal{B}} + \cdots + c_r [T_r]_{\mathcal{B}}$  ([Corollary 3.6.3](#)), a linear combination of diagonal matrices, hence diagonal. So  $\mathcal{B}$  diagonalizes every member of  $\mathcal{F}$ . ■

A typical infinite family is  $\{p(T) : p \in F[x]\}$  for a diagonalizable  $T$ . It is commuting, each  $p(T)$  is diagonalizable (Section 8), and here one basis works for the whole family without any of the theorem's machinery, namely an eigenbasis of  $T$ : an eigenvector of  $T$  for  $\lambda$  is an eigenvector of  $p(T)$  for  $p(\lambda)$  ([Exercise \(B3: Eigenvalues of powers and inverses\)](#)). The theorem says the same for families that are not generated by one operator in so visible a way.

## 8.9.6 Exercises

### 8.9.6.1 A. Check your understanding

#### Exercise A1.

- State the theorem relating commuting operators and eigenspaces.
- True or false: if  $ST = TS$ , then every eigenvector of  $T$  is an eigenvector of  $S$ . Justify your answer.
- True or false: any two commuting matrices in  $M_2(\mathbb{R})$  have a common eigenvector in  $\mathbb{R}^2$ . Justify your answer.
- True or false: if  $\mathbf{A}, \mathbf{B} \in M_n(F)$  are simultaneously diagonalizable, then  $\mathbf{AB} = \mathbf{BA}$ . Justify your answer.
- True or false: if  $\mathbf{AB} = \mathbf{BA}$  and  $\mathbf{A}$  is diagonalizable, then  $\mathbf{B}$  is diagonalizable. Justify your answer.
- In the proof of [Theorem 8.9.5](#), which hypothesis makes each eigenspace of  $T$  invariant under  $S$ , and which makes the restriction of  $S$  diagonalizable?

*Solution.*

- If  $S, T \in \mathcal{L}(V)$  and  $ST = TS$ , then  $E_{\lambda}(T)$  is  $S$ -invariant for every  $\lambda \in F$  ([Theorem 8.9.1](#)).



- (b) False.  $T = \mathbf{I}_2$  commutes with  $S = \begin{pmatrix} 1 & 1 \\ 0 & 1 \end{pmatrix}$ ; the vector  $\mathbf{e}_2$  is an eigenvector of  $T$ , but  $S\mathbf{e}_2 = (1, 1)$  is not a multiple of  $\mathbf{e}_2$ .
- (c) False. The rotation  $\mathbf{R} = \begin{pmatrix} 0 & -1 \\ 1 & 0 \end{pmatrix}$  commutes with itself and has no real eigenvector, since  $p_{\mathbf{R}} = x^2 + 1$  has no real root.
- (d) True. If  $\mathbf{P}^{-1}\mathbf{A}\mathbf{P} = \mathbf{D}$  and  $\mathbf{P}^{-1}\mathbf{B}\mathbf{P} = \mathbf{D}'$  are diagonal, then  $\mathbf{AB} = \mathbf{P}\mathbf{D}\mathbf{P}^{-1}\mathbf{P}\mathbf{D}'\mathbf{P}^{-1} = \mathbf{P}\mathbf{D}\mathbf{D}'\mathbf{P}^{-1} = \mathbf{P}\mathbf{D}'\mathbf{D}\mathbf{P}^{-1} = \mathbf{BA}$ , since diagonal matrices commute.
- (e) False.  $\mathbf{A} = \mathbf{I}_2$  is diagonalizable and commutes with  $\mathbf{B} = \begin{pmatrix} 1 & 1 \\ 0 & 1 \end{pmatrix}$ , which is not diagonalizable.
- (f) The hypothesis  $ST = TS$  gives the invariance ([Theorem 8.9.1](#)). The hypothesis that  $S$  is diagonalizable gives, through [Corollary 8.8.3](#), that  $S|_{E_{\lambda}(T)}$  is diagonalizable.

### 8.9.6.2 B. Practice

#### Exercise B1: Common eigenvectors.

Let

$$\mathbf{A} = \begin{pmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 1 & -1 & 2 \end{pmatrix}, \quad \mathbf{B} = \begin{pmatrix} 2 & 1 & 0 \\ -1 & 4 & 0 \\ 2 & -2 & 5 \end{pmatrix} \in M_3(\mathbb{R}).$$

- (a) Verify that  $\mathbf{AB} = \mathbf{BA}$ .
- (b) Find every common eigenvector of  $\mathbf{A}$  and  $\mathbf{B}$ .
- (c) Hence show that  $\mathbf{A}$  and  $\mathbf{B}$  are not simultaneously diagonalizable, and explain which hypothesis of [Theorem 8.9.5](#) fails.

*Solution.*

- (a) Multiplying out,

$$\mathbf{AB} = \begin{pmatrix} 2 & 1 & 0 \\ -1 & 4 & 0 \\ 7 & -7 & 10 \end{pmatrix} = \mathbf{BA}.$$

For instance, the  $(3, 1)$ -entry of  $\mathbf{AB}$  is row  $(1, -1, 2)$  of  $\mathbf{A}$  times column  $(2, -1, 2)$  of  $\mathbf{B}$ , namely  $2 + 1 + 4 = 7$ , and that of  $\mathbf{BA}$  is row  $(2, -2, 5)$  of  $\mathbf{B}$  times column  $(1, 0, 1)$  of  $\mathbf{A}$ , namely  $2 + 0 + 5 = 7$ .

- (b)  $\mathbf{A}$  is lower triangular, so  $p_{\mathbf{A}} = (x - 1)^2(x - 2)$ .  $\mathbf{A} - \mathbf{I} = \begin{pmatrix} 0 & 0 & 0 \\ 0 & 0 & 0 \\ 1 & -1 & 1 \end{pmatrix}$  has rank 1, so  $E_1(\mathbf{A}) = \{(x, y, z) : x - y + z = 0\}$  with basis  $\mathbf{u}_1 = (1, 1, 0)$ ,  $\mathbf{u}_2 = (-1, 0, 1)$ .  $\mathbf{A} - 2\mathbf{I} = \begin{pmatrix} -1 & 0 & 0 \\ 0 & -1 & 0 \\ 1 & -1 & 0 \end{pmatrix}$  gives  $E_2(\mathbf{A}) = \text{span}(\mathbf{e}_3)$ .

A common eigenvector is an eigenvector of  $\mathbf{A}$ , so it lies in  $E_1(\mathbf{A})$  or  $E_2(\mathbf{A})$ , and it must be an eigenvector of  $\mathbf{B}$  there. On the line  $E_2(\mathbf{A})$ :  $\mathbf{B}\mathbf{e}_3 = (0, 0, 5) = 5\mathbf{e}_3$ , so every non-zero multiple of  $\mathbf{e}_3$  is a common eigenvector. On the plane  $E_1(\mathbf{A})$ ,

which is  $\mathbf{B}$ -invariant by [Theorem 8.9.1](#):

$$\mathbf{B}\mathbf{u}_1 = (3, 3, 0) = 3\mathbf{u}_1, \quad \mathbf{B}\mathbf{u}_2 = (-2, 1, 3) = \mathbf{u}_1 + 3\mathbf{u}_2.$$

So  $[\mathbf{B}|_{E_1(\mathbf{A})}] = \begin{pmatrix} 3 & 1 \\ 0 & 3 \end{pmatrix}$  in the basis  $(\mathbf{u}_1, \mathbf{u}_2)$ . Its only eigenvalue is 3, and its eigenvectors are the non-zero multiples of  $(1, 0)$ , that is, of  $\mathbf{u}_1$ . Hence the common eigenvectors are exactly the non-zero multiples of  $(1, 1, 0)$  and the non-zero multiples of  $(0, 0, 1)$ .

- (c) By (b), any list of common eigenvectors lies in  $\text{span}((1, 1, 0), \mathbf{e}_3)$ , a plane, so no three of them form a basis of  $\mathbb{R}^3$ . Hence no basis diagonalizes both. The failing hypothesis is that  $\mathbf{B}$  is diagonalizable:  $p_{\mathbf{B}} = (x - 3)^2(x - 5)$  (from the block computation,  $p_{\mathbf{B}} = p_{\mathbf{B}|_{E_1(\mathbf{A})}} p_{\bar{\mathbf{B}}}$ , or directly), and  $\mathbf{B} - 3\mathbf{I} = \begin{pmatrix} -1 & 1 & 0 \\ -1 & 1 & 0 \\ 2 & -2 & 2 \end{pmatrix}$  has rank 2, so  $g_{\mathbf{B}}(3) = 1 < 2 = a_{\mathbf{B}}(3)$ .

### Exercise B2: Decide simultaneous diagonalizability.

For each pair of real matrices, determine whether they are simultaneously diagonalizable over  $\mathbb{R}$ . If they are, find a suitable  $\mathbf{P}$ . Justify your answer.

- (a)  $\mathbf{A} = \begin{pmatrix} 1 & 2 \\ 2 & 1 \end{pmatrix}, \mathbf{B} = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}.$   
 (b)  $\mathbf{A} = \begin{pmatrix} 1 & 1 \\ 0 & 2 \end{pmatrix}, \mathbf{B} = \begin{pmatrix} 1 & 0 \\ 0 & 2 \end{pmatrix}.$   
 (c)  $\mathbf{A} = \begin{pmatrix} 2 & 0 \\ 0 & 2 \end{pmatrix}, \mathbf{B} = \begin{pmatrix} 2 & 1 \\ 0 & 2 \end{pmatrix}.$

*Solution.*

- (a) Yes.  $\mathbf{AB} = \begin{pmatrix} 2 & 1 \\ 1 & 2 \end{pmatrix} = \mathbf{BA}$ . Both are diagonalizable:  $\mathbf{A}$  has  $p_{\mathbf{A}} = x^2 - 2x - 3 = (x - 3)(x + 1)$  and  $\mathbf{B}$  has  $p_{\mathbf{B}} = x^2 - 1 = (x - 1)(x + 1)$ , each with distinct eigenvalues ([Corollary 8.4.2](#)). By [Theorem 8.9.5](#) a common  $\mathbf{P}$  exists. The eigenspaces of  $\mathbf{A}$  are lines, so their spanning vectors  $(1, 1)$  and  $(1, -1)$  work:  $\mathbf{P} = \begin{pmatrix} 1 & 1 \\ 1 & -1 \end{pmatrix}$ ,  $\mathbf{P}^{-1}\mathbf{AP} = \text{diag}(3, -1)$ ,  $\mathbf{P}^{-1}\mathbf{BP} = \text{diag}(1, -1)$ , since  $\mathbf{B}(1, 1) = (1, 1)$  and  $\mathbf{B}(1, -1) = (-1, 1)$ .
- (b) No. Both are diagonalizable (distinct eigenvalues 1, 2), but  $\mathbf{AB} = \begin{pmatrix} 1 & 2 \\ 0 & 4 \end{pmatrix} \neq \begin{pmatrix} 1 & 1 \\ 0 & 4 \end{pmatrix} = \mathbf{BA}$ , so by [Theorem 8.9.5](#) ( $\Rightarrow$ ) no common  $\mathbf{P}$  exists.
- (c) No. They commute, since  $\mathbf{A} = 2\mathbf{I}$ . But  $\mathbf{B}$  is not diagonalizable:  $p_{\mathbf{B}} = (x - 2)^2$  and  $\mathbf{B} - 2\mathbf{I}$  has rank 1, so  $g(2) = 1 < 2$ . If  $\mathbf{P}^{-1}\mathbf{BP}$  were diagonal,  $\mathbf{B}$  would be diagonalizable ([Definition 8.4.1](#)).

### Exercise B3: Matrices commuting with a diagonal matrix.

- (a) Let  $\mathbf{D} = \text{diag}(1, 2, 3) \in M_3(\mathbb{R})$ . Prove that a matrix  $\mathbf{B} \in M_3(\mathbb{R})$  commutes with

**D** if and only if **B** is diagonal.

- (b) Let  $\mathbf{D}' = \text{diag}(1, 1, 2) \in M_3(\mathbb{R})$ . Describe all  $\mathbf{B} \in M_3(\mathbb{R})$  with  $\mathbf{B}\mathbf{D}' = \mathbf{D}'\mathbf{B}$ , and find one that is not diagonalizable.

*Solution.*

- (a) ( $\Leftarrow$ ) Diagonal matrices commute. ( $\Rightarrow$ ) Suppose  $\mathbf{B}\mathbf{D} = \mathbf{D}\mathbf{B}$ . The eigenspaces of  $\mathbf{D}$  are  $E_1(\mathbf{D}) = \text{span}(\mathbf{e}_1)$ ,  $E_2(\mathbf{D}) = \text{span}(\mathbf{e}_2)$  and  $E_3(\mathbf{D}) = \text{span}(\mathbf{e}_3)$ . By [Theorem 8.9.1](#) each is  $\mathbf{B}$ -invariant, so  $\mathbf{B}\mathbf{e}_j \in \text{span}(\mathbf{e}_j)$  for each  $j$ . The  $j$ -th column of  $\mathbf{B}$  is  $\mathbf{B}\mathbf{e}_j$ , so it has zeros off the diagonal, and  $\mathbf{B}$  is diagonal. (Directly:  $(\mathbf{D}\mathbf{B})_{ij} = i b_{ij}$  and  $(\mathbf{B}\mathbf{D})_{ij} = b_{ij} j$ , so  $(i - j)b_{ij} = 0$ , and  $b_{ij} = 0$  for  $i \neq j$ .)

- (b) As computed in the text for  $\text{diag}(1, 1, 2)$ ,  $\mathbf{B}\mathbf{D}' = \mathbf{D}'\mathbf{B}$  exactly when  $\mathbf{B} = \begin{pmatrix} b_{11} & b_{12} & 0 \\ b_{21} & b_{22} & 0 \\ 0 & 0 & b_{33} \end{pmatrix}$

with arbitrary entries:  $E_1(\mathbf{D}') = \text{span}(\mathbf{e}_1, \mathbf{e}_2)$  and  $E_2(\mathbf{D}') = \text{span}(\mathbf{e}_3)$  are  $\mathbf{B}$ -invariant, which by [Theorem 8.1.5](#) forces this shape, and conversely  $\mathbf{D}' = \mathbf{I}_2 \oplus (2)$  commutes with every such block matrix ([Theorem 7.1.4](#)). For example  $\mathbf{B} = \begin{pmatrix} 1 & 1 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 0 \end{pmatrix}$  commutes with  $\mathbf{D}'$ , and it is not diagonalizable:  $p_{\mathbf{B}} = x(x - 1)^2$  and  $\mathbf{B} - \mathbf{I}$  has rank 2, so  $g(1) = 1 < 2 = a(1)$ .

### 8.9.6.3 C. Going deeper

**Exercise C1: The commutant of a matrix with distinct eigenvalues.**

Let  $\mathbf{A} \in M_n(F)$ ,  $n \geq 1$ , have  $n$  distinct eigenvalues  $\lambda_1, \dots, \lambda_n \in F$ .

- (a) Prove that if  $\mathbf{A}\mathbf{B} = \mathbf{B}\mathbf{A}$ , then  $\mathbf{B}$  is diagonalizable and every eigenvector of  $\mathbf{A}$  is an eigenvector of  $\mathbf{B}$ .
- (b) Prove that if  $\mathbf{A}\mathbf{B} = \mathbf{B}\mathbf{A}$ , then  $\mathbf{B} = q(\mathbf{A})$  for some  $q \in F[x]$  of degree at most  $n - 1$ .
- (c) Deduce that  $\{\mathbf{B} \in M_n(F) : \mathbf{A}\mathbf{B} = \mathbf{B}\mathbf{A}\}$  is a subspace of  $M_n(F)$  of dimension  $n$ , with basis  $(\mathbf{I}, \mathbf{A}, \dots, \mathbf{A}^{n-1})$ .

*Hint: for (b), use Lagrange interpolation.*

*Solution.*

- (a) By [Corollary 8.4.2](#),  $\mathbf{A}$  is diagonalizable, say  $\mathbf{P}^{-1}\mathbf{A}\mathbf{P} = \mathbf{D}$  with  $\mathbf{D}$  diagonal. A diagonal matrix is upper triangular, so its diagonal entries are the eigenvalues of  $\mathbf{A}$ , each occurring as often as its algebraic multiplicity ([Theorem 8.7.4](#) (b)); here there are  $n$  distinct eigenvalues and  $n$  diagonal places, so each occurs once, and after permuting the columns of  $\mathbf{P}$  and the diagonal entries of  $\mathbf{D}$  together we may assume  $\mathbf{D} = \text{diag}(\lambda_1, \dots, \lambda_n)$ . Let  $\mathbf{v}_i$  be column  $i$  of  $\mathbf{P}$ . Each  $E_{\lambda_i}(\mathbf{A})$  has dimension at least 1, and the dimensions add up to at most  $n$  ([Corollary 8.2.3](#)), so each has dimension exactly 1 and  $E_{\lambda_i}(\mathbf{A}) = \text{span}(\mathbf{v}_i)$ . Every eigenvector of  $\mathbf{A}$  for  $\lambda_i$  is a non-zero multiple of  $\mathbf{v}_i$ . By [Theorem 8.9.1](#),  $\mathbf{B}\mathbf{v}_i \in \text{span}(\mathbf{v}_i)$ , so  $\mathbf{B}\mathbf{v}_i = \mu_i \mathbf{v}_i$  for some  $\mu_i \in F$ , and  $\mathbf{v}_i$ , with every non-zero multiple of it, is an eigenvector of  $\mathbf{B}$ . The columns of  $\mathbf{P}$  form a basis of eigenvectors of  $\mathbf{B}$ , so  $\mathbf{P}^{-1}\mathbf{B}\mathbf{P} = \text{diag}(\mu_1, \dots, \mu_n)$

and  $\mathbf{B}$  is diagonalizable.

- (b) The nodes  $\lambda_1, \dots, \lambda_n$  are distinct, so by [Theorem 5.6.1](#) there is  $q \in F[x]$  with  $\deg q \leq n-1$  and  $q(\lambda_i) = \mu_i$  for every  $i$ . By [Theorem 8.4.3](#),

$$q(\mathbf{A}) = \mathbf{P} \operatorname{diag}(q(\lambda_1), \dots, q(\lambda_n)) \mathbf{P}^{-1} = \mathbf{P} \operatorname{diag}(\mu_1, \dots, \mu_n) \mathbf{P}^{-1} = \mathbf{B}.$$

- (c) If  $\mathbf{AB} = \mathbf{BA}$  and  $\mathbf{AB}' = \mathbf{B}'\mathbf{A}$ , then  $\mathbf{A}(c\mathbf{B} + \mathbf{B}') = c\mathbf{BA} + \mathbf{B}'\mathbf{A} = (c\mathbf{B} + \mathbf{B}')\mathbf{A}$ , and  $\mathbf{A}\mathbf{0} = \mathbf{0}\mathbf{A}$ ; so the set  $\mathcal{C}$  is a subspace. Each  $\mathbf{A}^k$  commutes with  $\mathbf{A}$ , so  $\operatorname{span}(\mathbf{I}, \mathbf{A}, \dots, \mathbf{A}^{n-1}) \subseteq \mathcal{C}$ , and (b) gives the reverse inclusion. For independence, let  $c_0\mathbf{I} + c_1\mathbf{A} + \dots + c_{n-1}\mathbf{A}^{n-1} = \mathbf{0}$  and put  $q = \sum_k c_k x^k$ . By [Theorem 8.4.3](#),  $\mathbf{0} = q(\mathbf{A}) = \mathbf{P} \operatorname{diag}(q(\lambda_1), \dots, q(\lambda_n)) \mathbf{P}^{-1}$ , so  $q(\lambda_i) = 0$  for all  $i$ . A polynomial of degree at most  $n-1$  with  $n$  distinct roots is zero ([Corollary 5.1.4](#)), so every  $c_k = 0$ . Hence  $(\mathbf{I}, \mathbf{A}, \dots, \mathbf{A}^{n-1})$  is a basis of  $\mathcal{C}$ , and  $\dim \mathcal{C} = n$ .

### Exercise C2: Commuting nilpotent matrices.

Recall that  $\mathbf{N} \in M_n(F)$  is **nilpotent** if  $\mathbf{N}^a = \mathbf{0}$  for some  $a \geq 1$  ([Exercise \(C2: Nilpotent operators\)](#)). Let  $\mathbf{N}, \mathbf{M} \in M_n(F)$  with  $\mathbf{NM} = \mathbf{MN}$ ,  $\mathbf{N}^a = \mathbf{0}$  and  $\mathbf{M}^b = \mathbf{0}$  for some  $a, b \geq 1$ .

- Prove that  $\mathbf{NM}$  is nilpotent.
- Prove that  $(\mathbf{N} + \mathbf{M})^{a+b-1} = \mathbf{0}$ , so that  $\mathbf{N} + \mathbf{M}$  is nilpotent.
- Show that both conclusions can fail without  $\mathbf{NM} = \mathbf{MN}$ .
- Now let  $F = \mathbb{C}$ . Give a second proof of (b) using [Theorem 8.9.4](#).

*Hint: for (b), expand the power as a sum of products of  $a+b-1$  factors, each equal to  $\mathbf{N}$  or  $\mathbf{M}$ .*

*Solution.*

- Since  $\mathbf{NM} = \mathbf{MN}$ , induction on  $k$  gives  $(\mathbf{NM})^k = \mathbf{N}^k \mathbf{M}^k$ : if it holds for  $k$ , then  $(\mathbf{NM})^{k+1} = \mathbf{N}^k \mathbf{M}^k \mathbf{NM} = \mathbf{N}^k \mathbf{NM}^k \mathbf{M} = \mathbf{N}^{k+1} \mathbf{M}^{k+1}$ , where  $\mathbf{M}^k \mathbf{N} = \mathbf{NM}^k$  follows from  $\mathbf{MN} = \mathbf{NM}$  by induction as well. With  $k = a$ ,  $(\mathbf{NM})^a = \mathbf{N}^a \mathbf{M}^a = \mathbf{0} \cdot \mathbf{M}^a = \mathbf{0}$ .
- Let  $s = a + b - 1$ . By distributivity,  $(\mathbf{N} + \mathbf{M})^s$  is the sum of all  $2^s$  products  $\mathbf{X}_1 \mathbf{X}_2 \cdots \mathbf{X}_s$  with each  $\mathbf{X}_t \in \{\mathbf{N}, \mathbf{M}\}$ . Since  $\mathbf{N}$  and  $\mathbf{M}$  commute, each such product can be rearranged, one adjacent swap at a time, into  $\mathbf{N}^i \mathbf{M}^{s-i}$ , where  $i$  is the number of factors equal to  $\mathbf{N}$ . If  $i \geq a$ , then  $\mathbf{N}^i = \mathbf{N}^a \mathbf{N}^{i-a} = \mathbf{0}$ . If  $i \leq a-1$ , then  $s-i \geq s-a+1 = b$ , so  $\mathbf{M}^{s-i} = \mathbf{0}$ . Either way the product is  $\mathbf{0}$ , so  $(\mathbf{N} + \mathbf{M})^s = \mathbf{0}$ .
- Let  $\mathbf{N} = \mathbf{E}_{12}$  and  $\mathbf{M} = \mathbf{E}_{21}$  in  $M_2(F)$ . Both square to  $\mathbf{0}$ . But  $\mathbf{NM} = \mathbf{E}_{11}$  satisfies  $\mathbf{E}_{11}^k = \mathbf{E}_{11} \neq \mathbf{0}$  for all  $k \geq 1$ , and  $\mathbf{N} + \mathbf{M} = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}$  satisfies  $(\mathbf{N} + \mathbf{M})^2 = \mathbf{I}$ , so  $(\mathbf{N} + \mathbf{M})^k$  is  $\mathbf{I}$  or  $\mathbf{N} + \mathbf{M}$ , never  $\mathbf{0}$ . Neither is nilpotent, and indeed  $\mathbf{NM} = \mathbf{E}_{11} \neq \mathbf{E}_{22} = \mathbf{MN}$ .
- By [Theorem 8.9.4](#) there is an invertible  $\mathbf{P} \in M_n(\mathbb{C})$  with  $\mathbf{U} = \mathbf{P}^{-1} \mathbf{N} \mathbf{P}$  and  $\mathbf{U}' = \mathbf{P}^{-1} \mathbf{M} \mathbf{P}$  upper triangular. Since  $\mathbf{N}$  is nilpotent,  $\operatorname{spec}(\mathbf{N}) = \{0\}$  ([Exercise \(C2: Nilpotent operators\)](#)), so every diagonal entry of  $\mathbf{U}$  is 0 by [Theorem 8.7.4](#); likewise for  $\mathbf{U}'$ . So  $\mathbf{U} + \mathbf{U}' = \mathbf{P}^{-1}(\mathbf{N} + \mathbf{M})\mathbf{P}$  is upper triangular with zero diagonal. A matrix  $\mathbf{X}$  of this kind is nilpotent:  $\mathbf{X}\mathbf{e}_1 = \mathbf{0}$  and  $\mathbf{X}\mathbf{e}_j \in \operatorname{span}(\mathbf{e}_1, \dots, \mathbf{e}_{j-1})$ , so by

induction  $\mathbf{X}^j \mathbf{e}_j = \mathbf{0}$  and  $\mathbf{X}^n = \mathbf{0}$ . Hence  $(\mathbf{N} + \mathbf{M})^n = \mathbf{P}(\mathbf{U} + \mathbf{U}')^n \mathbf{P}^{-1} = \mathbf{0}$ .

## 8.10 Spectral Mapping, Products in Both Orders, Left Eigenvectors

Once the eigenvalues of  $T$  are known, what can we say about the eigenvalues of  $T^2$ , of  $T^2 - 3T + I$ , or of any polynomial in  $T$ ? And when two matrices are multiplied in the two possible orders, how are the eigenvalues of  $\mathbf{AB}$  and  $\mathbf{BA}$  related? This section answers both questions using the triangular forms of Section 7 and the block matrices of Chapter 7. Along the way it settles two debts: Chapter 6 showed  $p_{\mathbf{AB}} = p_{\mathbf{BA}}$  only when  $\mathbf{A}$  is invertible, and Chapter 7 proved that vanishing traces of all powers force a matrix to be nilpotent by a long computation, with no explanation of why. We end with left eigenvectors, the eigenvectors of the transpose, which pair with ordinary eigenvectors in a useful way and prepare the Markov chains of Section 11.

### 8.10.1 Eigenvalues of a polynomial in $T$

One half of the answer needs no theory. If  $T\mathbf{v} = \lambda\mathbf{v}$  with  $\mathbf{v} \neq \mathbf{0}$ , then  $T^k\mathbf{v} = \lambda^k\mathbf{v}$  for every  $k$ , and so  $q(T)\mathbf{v} = q(\lambda)\mathbf{v}$  for every  $q \in F[x]$  ([Exercise \(B3: Eigenvalues of powers and inverses\)](#)). So  $q$  **maps eigenvalues of  $T$  to eigenvalues of  $q(T)$** , over every field and in every dimension.

The other half, that **every** eigenvalue of  $q(T)$  arises this way, can fail. Part (c) of [Exercise \(B3: Eigenvalues of powers and inverses\)](#) already met the rotation  $\mathbf{R} = \begin{pmatrix} 0 & -1 \\ 1 & 0 \end{pmatrix}$  on  $\mathbb{R}^2$ : it has no real eigenvalue, while  $\mathbf{R}^2 = -\mathbf{I}$  has the eigenvalue  $-1$ . The eigenvalues of  $\mathbf{R}$  that would map to  $-1$ , namely  $\pm i$ , are missing from  $\mathbb{R}$ . So we expect the full statement to need a characteristic polynomial that splits. With that hypothesis we even get the multiplicities, because a triangular form shows all the eigenvalues at once on its diagonal.

*The eigenvalues of  $q(T)$  are the values of  $q$  at the eigenvalues of  $T$ , counted with multiplicity.*

#### Theorem 8.10.1 — Spectral Mapping Theorem

Let  $V$  be a finite-dimensional vector space over  $F$  with  $\dim V = n \geq 1$ , and let  $T \in \mathcal{L}(V)$ . Suppose  $p_T$  splits over  $F$ , say

$$p_T(x) = (x - \lambda_1)(x - \lambda_2) \cdots (x - \lambda_n),$$

where each eigenvalue appears as often as its algebraic multiplicity. Then for every  $q \in F[x]$ ,

$$p_{q(T)}(x) = (x - q(\lambda_1))(x - q(\lambda_2)) \cdots (x - q(\lambda_n)),$$

and consequently

$$\text{spec}(q(T)) = q(\text{spec}(T)) = \{q(\lambda) : \lambda \in \text{spec}(T)\}.$$

In particular this holds for every operator on a non-zero finite-dimensional complex space, and for every  $\mathbf{A} \in M_n(\mathbb{C})$ .

*Idea.* Choose a basis in which  $T$  is upper triangular (Section 7). Its diagonal is  $\lambda_1, \dots, \lambda_n$ . Products and sums of upper triangular matrices are upper triangular, and their diagonals are computed position by position, so  $q([T])$  is upper triangular with diagonal  $q(\lambda_1), \dots, q(\lambda_n)$ . The characteristic polynomial of a triangular matrix is read off its diagonal.

*Proof.* By [Theorem 8.7.2](#) there is a basis  $\mathcal{B}$  of  $V$  such that  $\mathbf{U} = [T]_{\mathcal{B}}$  is upper triangular. By [Theorem 8.7.4](#), its diagonal entries are  $\lambda_1, \dots, \lambda_n$  in some order; renumbering the  $\lambda_i$ , we may assume  $u_{ii} = \lambda_i$ .

*Claim.* If  $\mathbf{X}, \mathbf{Y} \in M_n(F)$  are upper triangular, then so are  $\mathbf{X} + \mathbf{Y}$ ,  $c\mathbf{X}$  for  $c \in F$ , and  $\mathbf{XY}$ , with diagonal entries  $x_{ii} + y_{ii}$ ,  $cx_{ii}$  and  $x_{ii}y_{ii}$ .

*Proof.* The statements for  $\mathbf{X} + \mathbf{Y}$  and  $c\mathbf{X}$  are entrywise. For the product,  $(\mathbf{XY})_{ij} = \sum_{k=1}^n x_{ik}y_{kj}$  ([Definition 0.9.9](#)). If  $i > j$ , every  $k$  has  $k < i$  or  $k \geq i > j$ , so  $x_{ik} = 0$  or  $y_{kj} = 0$ , and the sum is 0. If  $i = j$ , the only term that can be non-zero has  $k \geq i$  and  $k \leq i$ , so  $(\mathbf{XY})_{ii} = x_{ii}y_{ii}$ . ■

By the claim and induction on  $k$ ,  $\mathbf{U}^k$  is upper triangular with diagonal entries  $\lambda_1^k, \dots, \lambda_n^k$  (for  $k = 0$ ,  $\mathbf{U}^0 = \mathbf{I}$ ). Writing  $q = \sum_k a_k x^k$ , the claim again shows that  $q(\mathbf{U}) = \sum_k a_k \mathbf{U}^k$  is upper triangular with diagonal entries  $\sum_k a_k \lambda_i^k = q(\lambda_i)$ . By [Corollary 3.6.3](#),  $[q(T)]_{\mathcal{B}} = q(\mathbf{U})$ , so by [Definition 6.10.2](#) and the examples after [Definition 6.10.1](#),

$$p_{q(T)} = p_{q(\mathbf{U})} = \prod_{i=1}^n (x - q(\lambda_i)).$$

For the spectra: by [Theorem 8.2.1](#),  $\mu \in F$  is an eigenvalue of  $q(T)$  if and only if  $p_{q(T)}(\mu) = \prod_i (\mu - q(\lambda_i)) = 0$ , that is, if and only if  $\mu = q(\lambda_i)$  for some  $i$ , since a product of non-zero field elements is non-zero. The  $\lambda_i$  are exactly the eigenvalues of  $T$ , each listed at least once ([Theorem 8.2.1](#)). Hence  $\text{spec}(q(T)) = \{q(\lambda) : \lambda \in \text{spec}(T)\}$ . Over  $\mathbb{C}$ ,  $p_T$  always splits by [Corollary 5.5.5](#). ■

The formula for  $p_{q(T)}$  carries more than the spectrum. The algebraic multiplicity of  $\mu$  as an eigenvalue of  $q(T)$  is the number of indices  $i$  with  $q(\lambda_i) = \mu$  ([Lemma 5.4.1](#)), so several eigenvalues of  $T$  can merge into one eigenvalue of  $q(T)$ , and their multiplicities add. Combined with [Theorem 8.3.3](#), applied to  $q(T)$ , whose characteristic polynomial splits by the theorem, we get the most used consequence.

### Corollary 8.10.2 — Trace and Determinant of a Polynomial in $T$

Under the hypotheses of [Theorem 8.10.1](#), for every  $q \in F[x]$ ,

$$\text{tr } q(T) = q(\lambda_1) + \dots + q(\lambda_n), \quad \det q(T) = q(\lambda_1) \cdots q(\lambda_n).$$

In particular  $\text{tr } T^k = \lambda_1^k + \dots + \lambda_n^k$  for every  $k \in \mathbb{N}$ .

*Proof.* By Theorem 8.10.1,  $p_{q(T)} = \prod_i (x - q(\lambda_i))$  splits. Group equal values: if  $\mu_1, \dots, \mu_s$  are the distinct values among the  $q(\lambda_i)$ , then  $\mu_j$  occurs exactly  $a_{q(T)}(\mu_j)$  times in the list (Lemma 5.4.1). By Theorem 8.3.3 applied to  $q(T)$ ,  $\text{tr } q(T) = \sum_j a_{q(T)}(\mu_j) \mu_j = \sum_i q(\lambda_i)$  and  $\det q(T) = \prod_j \mu_j^{a_{q(T)}(\mu_j)} = \prod_i q(\lambda_i)$ . Take  $q = x^k$  for the last statement. ■

**Example. (A Polynomial in a Non-Diagonalizable Matrix)**

Let  $\mathbf{A} = \begin{pmatrix} 1 & 1 & 1 \\ 2 & 0 & 1 \\ -3 & 3 & 2 \end{pmatrix} \in M_3(\mathbb{R})$ , for which Exercise (B1: Diagonalize or show impossible) found  $p_{\mathbf{A}} = (x+1)(x-2)^2$  and showed that  $\mathbf{A}$  is not diagonalizable. Without computing any eigenvectors, find  $p_{\mathbf{A}^2}$ ,  $\text{tr } \mathbf{A}^2$  and  $\det \mathbf{A}^2$ . Then find  $p_{q(\mathbf{A})}$  for  $q = x^2 - x - 2$ , and decide whether  $q(\mathbf{A}) = 0$ .

*Solution. The square.*  $p_{\mathbf{A}}$  splits over  $\mathbb{R}$  with the list  $\lambda_1 = -1, \lambda_2 = \lambda_3 = 2$ . By Theorem 8.10.1 with  $q = x^2$ , the list for  $\mathbf{A}^2$  is 1, 4, 4, so

$$p_{\mathbf{A}^2} = (x-1)(x-4)^2, \quad \text{tr } \mathbf{A}^2 = 1 + 4 + 4 = 9, \quad \det \mathbf{A}^2 = 1 \cdot 4 \cdot 4 = 16$$

by Corollary 8.10.2. *Check.* Multiplying out,

$$\mathbf{A}^2 = \begin{pmatrix} 0 & 4 & 4 \\ -1 & 5 & 4 \\ -3 & 3 & 4 \end{pmatrix},$$

whose trace is  $0 + 5 + 4 = 9$ ; and  $\det \mathbf{A} = (-1) \cdot 2 \cdot 2 = -4$  by Theorem 8.3.3, so  $\det \mathbf{A}^2 = (\det \mathbf{A})^2 = 16$  by Theorem 6.5.4.

*The polynomial q.* Here  $q = (x-2)(x+1)$ , so  $q(-1) = 0$  and  $q(2) = 0$ . The list for  $q(\mathbf{A})$  is 0, 0, 0, and  $p_{q(\mathbf{A})} = x^3$ : the only eigenvalue of  $q(\mathbf{A})$  is 0. Yet  $q(\mathbf{A}) \neq 0$ . Indeed

$$q(\mathbf{A}) = \mathbf{A}^2 - \mathbf{A} - 2\mathbf{I} = \begin{pmatrix} -3 & 3 & 3 \\ -3 & 3 & 3 \\ 0 & 0 & 0 \end{pmatrix},$$

and one checks  $q(\mathbf{A})^2 = 0$ . The spectral mapping theorem sees only eigenvalues, and a non-zero matrix can have 0 as its only eigenvalue. That  $q(\mathbf{A}) \neq 0$  here is no accident: if  $q(\mathbf{A}) = 0$ , the minimal polynomial of  $\mathbf{A}$  would divide (Theorem 8.5.1)  $(x-2)(x+1)$ , which has distinct roots, and  $\mathbf{A}$  would be diagonalizable by Theorem 8.8.2.

**Warning. Without splitting, spectral mapping fails.** For the real rotation  $\mathbf{R}$  above,  $\text{spec}(\mathbf{R}) = \emptyset$  in  $\mathbb{R}$ , so  $q(\text{spec}(\mathbf{R})) = \emptyset$  for  $q = x^2$ , while  $\text{spec}(\mathbf{R}^2) = \text{spec}(-\mathbf{I}) = \{-1\}$ . Over  $\mathbb{C}$  the theorem is restored:  $\text{spec}(\mathbf{R}) = \{i, -i\}$ , both square to  $-1$ , and  $p_{\mathbf{R}^2} = (x+1)^2$ . Also, the theorem is about eigenvalues and algebraic multiplicities only:  $\mathbf{A} = \text{diag}(1, -1)$  has two one-dimensional eigenspaces, while  $\mathbf{A}^2 = \mathbf{I}$  has the single eigenspace  $\mathbb{R}^2$ .

*Quick check.* Let  $\mathbf{A} \in M_2(\mathbb{C})$  have eigenvalues  $i$  and  $-i$ . Find  $p_{\mathbf{A}^2}$  and  $\det(\mathbf{A}^2 + \mathbf{I})$ .

*Solution.* The list for  $\mathbf{A}$  is  $i, -i$  (two distinct roots of the degree-2 polynomial  $p_{\mathbf{A}}$ ). By [Theorem 8.10.1](#) with  $q = x^2$ , the list for  $\mathbf{A}^2$  is  $-1, -1$ , so  $p_{\mathbf{A}^2} = (x+1)^2$ . With  $q = x^2+1$ , [Corollary 8.10.2](#) gives  $\det(\mathbf{A}^2 + \mathbf{I}) = q(i)q(-i) = 0 \cdot 0 = 0$ .

## 8.10.2 Traces of powers detect nilpotency

Chapter 7 proved, in [Exercise \(C2: When  \$\mathbf{A}\$  commutes with  \$\[\mathbf{A}, \mathbf{B}\]\$ \)](#), that a matrix  $\mathbf{C}$  over a field of characteristic 0 with  $\operatorname{tr} \mathbf{C}^k = 0$  for every  $k \geq 1$  is nilpotent, by a careful argument with annihilating polynomials that gave no reason for the result. Here is the view through eigenvalues, for complex matrices. The traces of the powers are the **power sums**  $\lambda_1^k + \cdots + \lambda_n^k$  of the eigenvalues ([Corollary 8.10.2](#)), and a nilpotent matrix is one whose eigenvalues are all 0. So the question becomes: if all power sums of  $n$  complex numbers vanish, must the numbers vanish? A Vandermonde determinant says yes.

### Theorem 8.10.3 — Nilpotent Matrices via Eigenvalues and Traces

Let  $n \geq 1$  and  $\mathbf{A} \in M_n(\mathbb{C})$ . The following are equivalent:

- (a)  $\mathbf{A}$  is nilpotent, that is,  $\mathbf{A}^s = \mathbf{0}$  for some  $s \geq 1$ ;
- (b)  $\operatorname{spec}(\mathbf{A}) = \{0\}$ ;
- (c)  $p_{\mathbf{A}} = x^n$ ;
- (d)  $\mathbf{A}^n = \mathbf{0}$ ;
- (e)  $\operatorname{tr} \mathbf{A}^k = 0$  for  $k = 1, 2, \dots, n$ .

The same holds for  $\mathbf{A} \in M_n(F)$  when  $F$  is a subfield of  $\mathbb{C}$ , such as  $\mathbb{Q}$  or  $\mathbb{R}$ , with  $\operatorname{spec}(\mathbf{A})$  in (b) taken over  $\mathbb{C}$ .

*Idea. Plan:* (a)  $\Rightarrow$  (b)  $\Rightarrow$  (c)  $\Rightarrow$  (d)  $\Rightarrow$  (a), and (c)  $\Rightarrow$  (e)  $\Rightarrow$  (b). All but one arrow are short. For (e)  $\Rightarrow$  (b), suppose some eigenvalue is non-zero. Group the non-zero eigenvalues into distinct values  $\mu_1, \dots, \mu_r$  with multiplicities  $m_1, \dots, m_r \geq 1$ . The vanishing traces say  $\sum_j m_j \mu_j^k = 0$  for  $k = 1, \dots, r$ : a square linear system for the vector  $(m_1, \dots, m_r)$ , whose matrix is a Vandermonde matrix times the invertible  $\operatorname{diag}(\mu_j)$ . So the  $m_j$  would be 0, but they are positive integers, and positive integers are non-zero in  $\mathbb{C}$ .

*Proof.*

- (a)  $\Rightarrow$  (b). This is [Exercise \(C2: Nilpotent operators\)](#) (a), applied to  $T_{\mathbf{A}}$  on  $\mathbb{C}^n \neq \{\mathbf{0}\}$ .
- (b)  $\Rightarrow$  (c). By [Corollary 5.5.5](#),  $p_{\mathbf{A}} = (x - z_1) \cdots (x - z_n)$  with  $z_i \in \mathbb{C}$ . Each  $z_i$  is an eigenvalue ([Theorem 8.2.1](#)), hence 0, so  $p_{\mathbf{A}} = x^n$ .
- (c)  $\Rightarrow$  (d). By [Corollary 8.7.3](#),  $\mathbf{A} = \mathbf{P}\mathbf{U}\mathbf{P}^{-1}$  with  $\mathbf{P}$  invertible and  $\mathbf{U}$  upper triangular, and by [Theorem 8.7.4](#) the diagonal entries of  $\mathbf{U}$  are the roots of  $p_{\mathbf{A}} = x^n$ , all 0. Then  $\mathbf{U}\mathbf{e}_1 = \mathbf{0}$  and  $\mathbf{U}\mathbf{e}_j \in \operatorname{span}(\mathbf{e}_1, \dots, \mathbf{e}_{j-1})$  for  $j \geq 2$ , since column  $j$  of  $\mathbf{U}$  has zeros from row  $j$  down. By induction on  $j$ ,  $\mathbf{U}^j \mathbf{e}_j = \mathbf{0}$ , hence  $\mathbf{U}^n \mathbf{e}_j = \mathbf{U}^{n-j} \mathbf{U}^j \mathbf{e}_j = \mathbf{0}$  for every  $j$ , and  $\mathbf{U}^n = \mathbf{0}$ . By [Proposition 3.7.6](#) (c),  $\mathbf{A}^n = \mathbf{P}\mathbf{U}^n\mathbf{P}^{-1} = \mathbf{0}$ .
- (d)  $\Rightarrow$  (a). Take  $s = n$ .
- (e)  $\Rightarrow$  (c). The list of eigenvalues of  $\mathbf{A}$  with multiplicity is  $0, \dots, 0$ , so by [Corol-](#)



lary 8.10.2,  $\text{tr } \mathbf{A}^k = 0^k + \cdots + 0^k = 0$  for every  $k \geq 1$ .

- (f)  $\Rightarrow$  (b). By [Corollary 5.5.5](#),  $p_{\mathbf{A}}$  splits; let  $\lambda_1, \dots, \lambda_n$  be the list of its roots with multiplicity. Since  $n \geq 1$ ,  $\text{spec}(\mathbf{A})$  is non-empty, so it suffices to show that every  $\lambda_i = 0$ . Suppose not, let  $\mu_1, \dots, \mu_r$  ( $1 \leq r \leq n$ ) be the distinct non-zero values among the  $\lambda_i$ , and let  $m_j \geq 1$  be the number of indices  $i$  with  $\lambda_i = \mu_j$ . For  $1 \leq k \leq r$ , the zero eigenvalues contribute  $0^k = 0$ , so by [Corollary 8.10.2](#) and (e),

$$\sum_{j=1}^r m_j \mu_j^k = \text{tr } \mathbf{A}^k = 0.$$

In matrix form,  $\mathbf{W}\mathbf{m} = \mathbf{0}$ , where  $\mathbf{m} = (m_1, \dots, m_r) \in \mathbb{C}^r$  and  $\mathbf{W} \in M_r(\mathbb{C})$  has  $(k, j)$ -entry  $\mu_j^k = \mu_j^{k-1} \mu_j$ . Hence  $\mathbf{W} = V(\mu_1, \dots, \mu_r)^\top \text{diag}(\mu_1, \dots, \mu_r)$ , where  $V(\mu_1, \dots, \mu_r)$  is the Vandermonde matrix with  $(j, k)$ -entry  $\mu_j^{k-1}$  ([Theorem 6.8.1](#)). The  $\mu_j$  are distinct, so  $V$  is invertible ([Corollary 6.8.2](#)), and so is  $V^\top$ , with inverse  $(V^{-1})^\top$  ([Theorem 0.9.5](#)). The  $\mu_j$  are non-zero, so  $\text{diag}(\mu_1, \dots, \mu_r)$  is invertible. Therefore  $\mathbf{W}$  is invertible, and  $\mathbf{m} = \mathbf{W}^{-1}\mathbf{0} = \mathbf{0}$ . But  $m_1$  is a positive integer, and  $m_1 \neq 0$  in  $\mathbb{C}$ . This contradiction shows that every  $\lambda_i = 0$ , so  $\text{spec}(\mathbf{A}) = \{0\}$ .

For a subfield  $F \subseteq \mathbb{C}$ , regard  $\mathbf{A}$  as a complex matrix. Its powers, their traces, its characteristic polynomial and the condition  $\mathbf{A}^s = 0$  are computed by the same arithmetic in  $F$  and in  $\mathbb{C}$ , so (a), (c), (d) and (e) mean the same thing over  $F$  and over  $\mathbb{C}$ , and the equivalences carry over. ■

In the setting of [Exercise \(C2: When  \$\mathbf{A}\$  commutes with  \$\[\mathbf{A}, \mathbf{B}\]\$ \)](#), where  $\mathbf{C} = [\mathbf{A}, \mathbf{B}]$  commutes with  $\mathbf{A}$ , part (a) of that exercise gives  $\text{tr } \mathbf{C}^k = 0$  for all  $k \geq 1$ , and for complex matrices the theorem now shows at once that  $\mathbf{C}^n = 0$ . That exercise remains the route for a general field of characteristic 0.

**Warning. The characteristic matters.** Over  $\mathbb{F}_p$ , the identity  $\mathbf{I}_p \in M_p(\mathbb{F}_p)$  has  $\text{tr } \mathbf{I}_p^k = p \cdot 1 = 0$  for every  $k$ , yet  $\mathbf{I}_p^k = \mathbf{I}_p \neq 0$ . The proof above breaks exactly where it used that the positive integer  $m_1$  is non-zero in the field: here the multiplicity  $m_1 = p$  of the eigenvalue 1 is 0 in  $\mathbb{F}_p$ .

### 8.10.3 AB and BA

Chapter 6, in [Exercise \(B3: AB and BA\)](#), showed that  $p_{\mathbf{AB}} = p_{\mathbf{BA}}$  for square  $\mathbf{A}$  and  $\mathbf{B}$  when  $\mathbf{A}$  is invertible, because then  $\mathbf{BA} = \mathbf{A}^{-1}(\mathbf{AB})\mathbf{A}$  is similar to  $\mathbf{AB}$ . Without invertibility the two products need not be similar, and for rectangular  $\mathbf{A}$  and  $\mathbf{B}$  they do not even have the same size: if  $\mathbf{A}$  is  $m \times n$  and  $\mathbf{B}$  is  $n \times m$ , then  $\mathbf{AB}$  is  $m \times m$  and  $\mathbf{BA}$  is  $n \times n$ . Their characteristic polynomials have degrees  $m$  and  $n$ . Multiplying by powers of  $x$  repairs the degrees, and the identity then holds in general.

**Theorem 8.10.4 — AB and BA Have the Same Non-zero Eigenvalues**

Let  $m, n \geq 1$ ,  $\mathbf{A} \in M_{m \times n}(F)$  and  $\mathbf{B} \in M_{n \times m}(F)$ . Then

$$x^n p_{\mathbf{AB}}(x) = x^m p_{\mathbf{BA}}(x).$$

Consequently:

- (a) for every  $\lambda \neq 0$  in  $F$ ,  $a_{\mathbf{AB}}(\lambda) = a_{\mathbf{BA}}(\lambda)$ ; in particular  $\mathbf{AB}$  and  $\mathbf{BA}$  have the same non-zero eigenvalues;
- (b) if  $m = n$ , then  $p_{\mathbf{AB}} = p_{\mathbf{BA}}$ ; and if  $m \geq n$ , then  $p_{\mathbf{AB}} = x^{m-n} p_{\mathbf{BA}}$ .

*Idea.* We want one big matrix that “contains”  $\mathbf{AB}$  and another that “contains”  $\mathbf{BA}$ , and that are similar to each other. Put them in  $(m+n) \times (m+n)$  block form:

$$\mathbf{M} = \begin{pmatrix} \mathbf{AB} & 0 \\ \mathbf{B} & 0 \end{pmatrix}, \quad \mathbf{N} = \begin{pmatrix} 0 & 0 \\ \mathbf{B} & \mathbf{BA} \end{pmatrix}.$$

Both are block triangular, so their characteristic polynomials are  $p_{\mathbf{AB}} \cdot x^n$  and  $x^m \cdot p_{\mathbf{BA}}$ . The invertible block matrix  $\mathbf{S} = \begin{pmatrix} \mathbf{I}_m & \mathbf{A} \\ 0 & \mathbf{I}_n \end{pmatrix}$  turns one into the other:  $\mathbf{MS}$  and  $\mathbf{SN}$  are both  $\begin{pmatrix} \mathbf{AB} & \mathbf{ABA} \\ \mathbf{B} & \mathbf{BA} \end{pmatrix}$ .

*Proof.* Let  $\mathbf{M}, \mathbf{N}, \mathbf{S} \in M_{m+n}(F)$  be the block matrices displayed in the Idea, partitioned by  $m, n$  in rows and columns. By [Theorem 7.1.1](#),

$$\begin{aligned} \mathbf{MS} &= \begin{pmatrix} \mathbf{AB} \cdot \mathbf{I}_m + 0 & \mathbf{AB} \cdot \mathbf{A} + 0 \\ \mathbf{B} \cdot \mathbf{I}_m + 0 & \mathbf{B} \cdot \mathbf{A} + 0 \end{pmatrix} = \begin{pmatrix} \mathbf{AB} & \mathbf{ABA} \\ \mathbf{B} & \mathbf{BA} \end{pmatrix}, \\ \mathbf{SN} &= \begin{pmatrix} 0 + \mathbf{AB} & 0 + \mathbf{A} \cdot \mathbf{BA} \\ 0 + \mathbf{B} & 0 + \mathbf{BA} \end{pmatrix} = \begin{pmatrix} \mathbf{AB} & \mathbf{ABA} \\ \mathbf{B} & \mathbf{BA} \end{pmatrix}, \end{aligned}$$

using associativity ([Theorem 0.9.4](#)). So  $\mathbf{MS} = \mathbf{SN}$ . The matrix  $\mathbf{S}$  is invertible: by [Theorem 7.1.1](#),  $\begin{pmatrix} \mathbf{I}_m & \mathbf{A} \\ 0 & \mathbf{I}_n \end{pmatrix} \begin{pmatrix} \mathbf{I}_m & -\mathbf{A} \\ 0 & \mathbf{I}_n \end{pmatrix} = \begin{pmatrix} \mathbf{I}_m & -\mathbf{A} + \mathbf{A} \\ 0 & \mathbf{I}_n \end{pmatrix} = \mathbf{I}_{m+n}$ , and [Theorem 2.4.5](#) applies. Hence  $\mathbf{S}^{-1}\mathbf{MS} = \mathbf{N}$ , and  $p_{\mathbf{M}} = p_{\mathbf{N}}$  by [Theorem 6.10.3](#).

Now compute both sides. The matrix  $x\mathbf{I}_{m+n} - \mathbf{M} = \begin{pmatrix} x\mathbf{I}_m - \mathbf{AB} & 0 \\ -\mathbf{B} & x\mathbf{I}_n \end{pmatrix}$  is lower block triangular with square diagonal blocks, so by [Theorem 6.8.4](#), which holds over the commutative ring  $F[x]$  (the remarks before [Example \(A 3 × 3 Characteristic Polynomial\)](#)), and by [Theorem 6.4.6](#) for  $x\mathbf{I}_n$ ,

$$p_{\mathbf{M}}(x) = \det(x\mathbf{I}_m - \mathbf{AB}) \det(x\mathbf{I}_n) = p_{\mathbf{AB}}(x) x^n.$$

Likewise  $x\mathbf{I}_{m+n} - \mathbf{N} = \begin{pmatrix} x\mathbf{I}_m & 0 \\ -\mathbf{B} & x\mathbf{I}_n - \mathbf{BA} \end{pmatrix}$  gives  $p_{\mathbf{N}}(x) = x^m p_{\mathbf{BA}}(x)$ . Since  $p_{\mathbf{M}} = p_{\mathbf{N}}$ , this proves  $x^n p_{\mathbf{AB}} = x^m p_{\mathbf{BA}}$ .

- (a) Let  $\lambda \neq 0$ . Then  $\lambda$  is not a root of  $x^n$ , so  $\text{mult}_{\lambda}(x^n) = 0$ , and similarly for  $x^m$ . By [Theorem 5.4.2](#),

$$a_{\mathbf{AB}}(\lambda) = \text{mult}_{\lambda}(x^n p_{\mathbf{AB}}) = \text{mult}_{\lambda}(x^m p_{\mathbf{BA}}) = a_{\mathbf{BA}}(\lambda).$$

A scalar is an eigenvalue exactly when its algebraic multiplicity is positive ([Theorem 8.2.1](#), [Definition 5.4.1](#)), so the non-zero eigenvalues agree.

- (b) If  $m \geq n$ , then  $x^n p_{\mathbf{AB}} = x^n (x^{m-n} p_{\mathbf{BA}})$ , and canceling the non-zero polynomial  $x^n$  ([Corollary 0.8.4](#)) gives  $p_{\mathbf{AB}} = x^{m-n} p_{\mathbf{BA}}$ . For  $m = n$  this is  $p_{\mathbf{AB}} = p_{\mathbf{BA}}$ . ■

This pays off the promise of [Exercise \(B3:  \$\mathbf{AB}\$  and  \$\mathbf{BA}\$ \)](#):  $p_{\mathbf{AB}} = p_{\mathbf{BA}}$  for **all** square  $\mathbf{A}$  and  $\mathbf{B}$ , invertible or not. Comparing coefficients ([Theorem 6.10.2](#)) recovers  $\text{tr}(\mathbf{AB}) = \text{tr}(\mathbf{BA})$  and  $\det(\mathbf{AB}) = \det(\mathbf{BA})$  once more.

The eigenvectors travel between the two products as well, and by hand. If  $\mathbf{AB}\mathbf{w} = \lambda\mathbf{w}$  with  $\lambda \neq 0$ , then

$$\mathbf{BA}(\mathbf{B}\mathbf{w}) = \mathbf{B}(\mathbf{AB}\mathbf{w}) = \lambda\mathbf{B}\mathbf{w},$$

and  $\mathbf{B}\mathbf{w} \neq \mathbf{0}$ , since  $\mathbf{A}(\mathbf{B}\mathbf{w}) = \lambda\mathbf{w} \neq \mathbf{0}$ . So  $\mathbf{B}\mathbf{w}$  is an eigenvector of  $\mathbf{BA}$  for the same  $\lambda$ : multiplying an eigenvector of  $\mathbf{AB}$  by  $\mathbf{B}$  produces one of  $\mathbf{BA}$ . For  $\lambda = 0$  the recipe can fail, because  $\mathbf{B}\mathbf{w}$  may be  $\mathbf{0}$  — which is the warning below in another guise.

**A rank-one example.** Let  $\mathbf{u}, \mathbf{v} \in F^n$  with  $n \geq 2$ . Take  $\mathbf{A} = \mathbf{u} \in M_{n \times 1}(F)$  and  $\mathbf{B} = \mathbf{v}^\top \in M_{1 \times n}(F)$ . Then  $\mathbf{AB} = \mathbf{uv}^\top$  is  $n \times n$ , and  $\mathbf{BA} = \mathbf{v}^\top \mathbf{u}$  is the  $1 \times 1$  matrix whose entry is the number  $c = \mathbf{v}^\top \mathbf{u}$ , so  $p_{\mathbf{BA}} = x - c$ . Applying the theorem with its  $m$  equal to our  $n$ , and its  $n$  equal to 1, gives  $x p_{\mathbf{uv}^\top} = x^n (x - c)$ , and canceling  $x$ ,

$$p_{\mathbf{uv}^\top}(x) = x^{n-1} (x - \mathbf{v}^\top \mathbf{u}).$$

For instance, the all-ones matrix  $\mathbf{1}\mathbf{1}^\top \in M_n(\mathbb{R})$ , with  $\mathbf{1} = (1, \dots, 1)$ , has  $p = x^{n-1}(x - n)$ : eigenvalue 0 with multiplicity  $n - 1$ , and eigenvalue  $n$ . Evaluating at  $-1$  recovers [Theorem 6.8.5](#):  $\det(\mathbf{I} + \mathbf{uv}^\top) = (-1)^n \det(-\mathbf{I} - \mathbf{uv}^\top) = (-1)^n p_{\mathbf{uv}^\top}(-1) = (-1)^n (-1)^{n-1} (-1 - \mathbf{v}^\top \mathbf{u}) = 1 + \mathbf{v}^\top \mathbf{u}$ .

**Warning.** The zero eigenvalue behaves differently, and  $\mathbf{AB}$  need not be similar to  $\mathbf{BA}$ . Take  $\mathbf{A} = \mathbf{E}_{12}$  and  $\mathbf{B} = \mathbf{E}_{11}$  in  $M_2(F)$ . Then  $\mathbf{AB} = \mathbf{E}_{12}\mathbf{E}_{11} = \mathbf{0}$  and  $\mathbf{BA} = \mathbf{E}_{11}\mathbf{E}_{12} = \mathbf{E}_{12}$ . Both have  $p = x^2$ , as the theorem demands. But  $g_{\mathbf{AB}}(0) = 2$  while  $g_{\mathbf{BA}}(0) = 1$ , and  $\mathbf{AB} = \mathbf{0}$  is similar only to 0, so  $\mathbf{AB} \not\sim \mathbf{BA}$ . For rectangular matrices even the algebraic multiplicities of 0 differ, by  $|m - n|$ .

**Quick check.** Let  $\mathbf{A} \in M_{2 \times 5}(\mathbb{R})$  and  $\mathbf{B} \in M_{5 \times 2}(\mathbb{R})$  with  $p_{\mathbf{AB}} = x^2 - 3x + 2$ . What is  $p_{\mathbf{BA}}$ , and what are  $\text{tr}(\mathbf{BA})$  and  $\det(\mathbf{BA})$ ?

**Solution.** By [Theorem 8.10.4](#) with  $m = 2$ ,  $n = 5$ ,  $x^5 p_{\mathbf{AB}} = x^2 p_{\mathbf{BA}}$ , so  $p_{\mathbf{BA}} = x^3(x^2 - 3x + 2) = x^5 - 3x^4 + 2x^3$ . By [Theorem 6.10.2](#),  $\text{tr}(\mathbf{BA}) = 3$  (equal to  $\text{tr}(\mathbf{AB})$ ), and  $\det(\mathbf{BA}) = (-1)^5 \cdot 0 = 0$ , as it must be, since  $\mathbf{BA}$  has rank at most  $2 < 5$ .

## 8.10.4 Left eigenvectors

An eigenvector multiplies  $\mathbf{A}$  from the right:  $\mathbf{A}\mathbf{v} = \lambda\mathbf{v}$ . Row vectors can multiply from the left, and in applications, especially Markov chains, the row version is often the one with a meaning.

*A left eigenvector is a row that  $\mathbf{A}$  only stretches when it acts from the right of the row.*

### Definition 8.10.1 — Left Eigenvector

Let  $\mathbf{A} \in M_n(F)$  and  $\lambda \in F$ . A **left eigenvector** of  $\mathbf{A}$  for  $\lambda$  is a **non-zero**  $\mathbf{y} \in F^n$  with

$$\mathbf{y}^\top \mathbf{A} = \lambda \mathbf{y}^\top.$$

To distinguish them, the eigenvectors of Definition 8.2.1 are also called **right eigenvectors**.

In words:  $\mathbf{y}^\top$  is a  $1 \times n$  row,  $\mathbf{y}^\top \mathbf{A}$  is again a row, and we ask that it be a multiple of  $\mathbf{y}^\top$ . For example,  $\mathbf{A} = \begin{pmatrix} 1 & 2 \\ 3 & 2 \end{pmatrix}$  and  $\mathbf{y} = (1, 1)$  give  $\mathbf{y}^\top \mathbf{A} = (4 \ 4) = 4\mathbf{y}^\top$ , so  $(1, 1)$  is a left eigenvector for 4, whereas the right eigenvectors for 4 are the multiples of  $(2, 3)$  (Example (Eigenvalues and Eigenspaces of a  $2 \times 2$  Matrix)). For a diagonal matrix,  $\mathbf{e}_i$  is both a left and a right eigenvector for the  $i$ -th diagonal entry. And  $\mathbf{y} = \mathbf{0}$  always satisfies the equation, which is why it is excluded, exactly as for right eigenvectors.

**Non-example by minimal change.** For the same  $\mathbf{A}$ , change  $\mathbf{y} = (1, 1)$  to  $\mathbf{y} = (1, -1)$ . Then  $\mathbf{y}^\top \mathbf{A} = (1 - 3 \ 2 - 2) = (-2 \ 0)$ , which is not a multiple of  $(1 \ -1)$ : the row is non-zero, but the defining equation  $\mathbf{y}^\top \mathbf{A} = \lambda \mathbf{y}^\top$  has no solution  $\lambda$ . Note that  $(1, -1)$  is a right eigenvector of  $\mathbf{A}$ , for  $-1$ ; the two notions do not agree.

Transposing turns the definition into one we know.

### Proposition 8.10.5 — Left Eigenvectors Are Eigenvectors of the Transpose

Let  $\mathbf{A} \in M_n(F)$  and  $\lambda \in F$ .

- (a)  $\mathbf{y}$  is a left eigenvector of  $\mathbf{A}$  for  $\lambda$  if and only if  $\mathbf{y}$  is an eigenvector of  $\mathbf{A}^\top$  for  $\lambda$ .
- (b)  $\mathbf{A}$  has a left eigenvector for  $\lambda$  if and only if  $\lambda \in \text{spec}(\mathbf{A})$ . Moreover  $p_{\mathbf{A}^\top} = p_{\mathbf{A}}$ , and  $a_{\mathbf{A}^\top}(\lambda) = a_{\mathbf{A}}(\lambda)$ ,  $g_{\mathbf{A}^\top}(\lambda) = g_{\mathbf{A}}(\lambda)$ .

*Proof.*

- (a) By Theorem 0.9.5,  $(\mathbf{y}^\top \mathbf{A})^\top = \mathbf{A}^\top \mathbf{y}$  and  $(\lambda \mathbf{y}^\top)^\top = \lambda \mathbf{y}$ . Since transposing twice gives back the original,  $\mathbf{y}^\top \mathbf{A} = \lambda \mathbf{y}^\top$  if and only if  $\mathbf{A}^\top \mathbf{y} = \lambda \mathbf{y}$ .
- (b) By Exercise (B2: Transpose),  $p_{\mathbf{A}^\top} = p_{\mathbf{A}}$ , so  $a_{\mathbf{A}^\top}(\lambda) = a_{\mathbf{A}}(\lambda)$ , and  $\text{spec}(\mathbf{A}^\top) = \text{spec}(\mathbf{A})$  by Theorem 8.2.1. With (a),  $\mathbf{A}$  has a left eigenvector for  $\lambda$  if and only if  $\lambda \in \text{spec}(\mathbf{A}^\top) = \text{spec}(\mathbf{A})$ . Finally,  $\mathbf{A}^\top - \lambda \mathbf{I} = (\mathbf{A} - \lambda \mathbf{I})^\top$ , which has the same rank as  $\mathbf{A} - \lambda \mathbf{I}$  (Theorem 2.5.5), so by Theorem 2.5.7,  $g_{\mathbf{A}^\top}(\lambda) = n - \text{rank}(\mathbf{A} - \lambda \mathbf{I})^\top = n - \text{rank}(\mathbf{A} - \lambda \mathbf{I}) = g_{\mathbf{A}}(\lambda)$ . ■

*Remark.* For an operator  $T$  on  $V$ , left eigenvectors correspond to eigenvectors of the dual map  $T'$  on  $V^*$  (Chapter 4), whose matrix in the dual bases is the transpose (Theorem 4.4.2): an eigenvector of  $T'$  is a functional  $\varphi \neq 0$  with  $\varphi \circ T = \lambda \varphi$ .

Left and right eigenvectors for the **same** eigenvalue are usually unrelated, as  $(1, 1)$  and  $(2, 3)$

show. For **different** eigenvalues, though, they are always orthogonal in the sense of the plain product  $\mathbf{y}^\top \mathbf{v}$ .

### Theorem 8.10.6 — Left and Right Eigenvectors Are Biorthogonal

Let  $\mathbf{A} \in M_n(F)$ , let  $\mathbf{y}$  be a left eigenvector of  $\mathbf{A}$  for  $\mu$ , and let  $\mathbf{v}$  be a right eigenvector of  $\mathbf{A}$  for  $\lambda$ . If  $\lambda \neq \mu$ , then  $\mathbf{y}^\top \mathbf{v} = 0$ .

*Proof.* Compute the  $1 \times 1$  matrix  $\mathbf{y}^\top \mathbf{A} \mathbf{v}$  in two ways, using associativity (Theorem 0.9.4):

$$\mu \mathbf{y}^\top \mathbf{v} = (\mathbf{y}^\top \mathbf{A}) \mathbf{v} = \mathbf{y}^\top \mathbf{A} \mathbf{v} = \mathbf{y}^\top (\mathbf{A} \mathbf{v}) = \lambda \mathbf{y}^\top \mathbf{v}.$$

Hence  $(\lambda - \mu) \mathbf{y}^\top \mathbf{v} = 0$ . Since  $\lambda - \mu \neq 0$ , this forces  $\mathbf{y}^\top \mathbf{v} = 0$  (Theorem 0.7.1). ■

For the example,  $(1, 1)$  is a left eigenvector for 4 and  $(1, -1)$  a right eigenvector for  $-1$ , and indeed  $1 - 1 = 0$ .

For a diagonalizable matrix the left eigenvectors come for free with the diagonalization, and they give a formula for all powers that we use in Section 11.

### Proposition 8.10.7 — Expansion in Left and Right Eigenvectors

Let  $\mathbf{A} = \mathbf{P} \mathbf{D} \mathbf{P}^{-1}$  with  $\mathbf{P} \in M_n(F)$  invertible and  $\mathbf{D} = \text{diag}(\lambda_1, \dots, \lambda_n)$ . Let  $\mathbf{v}_1, \dots, \mathbf{v}_n$  be the columns of  $\mathbf{P}$ , and let  $\mathbf{y}_1^\top, \dots, \mathbf{y}_n^\top$  be the rows of  $\mathbf{P}^{-1}$ . Then:

- (a)  $\mathbf{v}_i$  is a right eigenvector and  $\mathbf{y}_i$  a left eigenvector of  $\mathbf{A}$  for  $\lambda_i$ ;
- (b)  $\mathbf{y}_i^\top \mathbf{v}_j = 1$  if  $i = j$  and 0 if  $i \neq j$ ;
- (c) for every  $q \in F[x]$ ,

$$q(\mathbf{A}) = \sum_{i=1}^n q(\lambda_i) \mathbf{v}_i \mathbf{y}_i^\top, \quad \text{in particular} \quad \mathbf{A}^k = \sum_{i=1}^n \lambda_i^k \mathbf{v}_i \mathbf{y}_i^\top \quad (k \in \mathbb{N}).$$

*Proof.*

- (a) From  $\mathbf{A} \mathbf{P} = \mathbf{P} \mathbf{D}$ , column  $i$  reads  $\mathbf{A} \mathbf{v}_i = \lambda_i \mathbf{v}_i$  (Section 4), and  $\mathbf{v}_i \neq \mathbf{0}$  because  $\mathbf{P}$  is invertible. From  $\mathbf{P}^{-1} \mathbf{A} = \mathbf{D} \mathbf{P}^{-1}$ , row  $i$  reads  $\mathbf{y}_i^\top \mathbf{A} = \lambda_i \mathbf{y}_i^\top$  (Theorem 0.9.2), and  $\mathbf{y}_i \neq \mathbf{0}$  because  $\mathbf{P}^{-1}$  is invertible.
- (b) The  $(i, j)$ -entry of  $\mathbf{P}^{-1} \mathbf{P} = \mathbf{I}$  is row  $i$  of  $\mathbf{P}^{-1}$  times column  $j$  of  $\mathbf{P}$ , that is,  $\mathbf{y}_i^\top \mathbf{v}_j$ .
- (c) By Theorem 8.4.3,  $q(\mathbf{A}) = \mathbf{P} q(\mathbf{D}) \mathbf{P}^{-1}$ . The columns of  $\mathbf{P} q(\mathbf{D})$  are  $q(\lambda_i) \mathbf{v}_i$ , and the rows of  $\mathbf{P}^{-1}$  are  $\mathbf{y}_i^\top$ , so Corollary 7.1.3 gives  $q(\mathbf{A}) = \sum_i q(\lambda_i) \mathbf{v}_i \mathbf{y}_i^\top$ . Take  $q = x^k$ . ■

Part (b) is biorthogonality with a normalization:  $\mathbf{y}_i^\top \mathbf{v}_i = 1$ . Part (c) splits  $\mathbf{A}$  into rank-one pieces  $\mathbf{v}_i \mathbf{y}_i^\top$ , one per eigenvalue, and each piece is simply multiplied by  $\lambda_i^k$  when  $\mathbf{A}$  is raised to the

power  $k$ . In the running example,  $\mathbf{P} = \begin{pmatrix} 2 & 1 \\ 3 & -1 \end{pmatrix}$  and  $\mathbf{D} = \text{diag}(4, -1)$  give  $\mathbf{P}^{-1} = \frac{1}{5} \begin{pmatrix} 1 & 1 \\ 3 & -2 \end{pmatrix}$ , so  $\mathbf{y}_1 = \frac{1}{5}(1, 1)$  and  $\mathbf{y}_2 = \frac{1}{5}(3, -2)$ , and

$$\mathbf{A}^k = \frac{4^k}{5} \begin{pmatrix} 2 & 2 \\ 3 & 3 \end{pmatrix} + \frac{(-1)^k}{5} \begin{pmatrix} 3 & -2 \\ -3 & 2 \end{pmatrix}.$$

For  $k = 1$  this is  $\frac{1}{5} \begin{pmatrix} 8-3 & 8+2 \\ 12+3 & 12-2 \end{pmatrix} = \begin{pmatrix} 1 & 2 \\ 3 & 2 \end{pmatrix}$ , as it should be.

## 8.10.5 Exercises

### 8.10.5.1 A. Check your understanding

#### Exercise A1.

- State the Spectral Mapping Theorem, including its hypothesis on  $p_T$ .
- True or false: for every  $\mathbf{A} \in M_n(\mathbb{R})$ , every real eigenvalue of  $\mathbf{A}^2$  is the square of a real eigenvalue of  $\mathbf{A}$ . Justify your answer.
- True or false: if  $\mathbf{A} \in M_{2 \times 3}(F)$  and  $\mathbf{B} \in M_{3 \times 2}(F)$ , then  $p_{\mathbf{AB}} = p_{\mathbf{BA}}$ . Justify your answer.
- True or false: for all  $\mathbf{A}, \mathbf{B} \in M_n(F)$ ,  $\mathbf{AB}$  is similar to  $\mathbf{BA}$ . Justify your answer.
- Define a left eigenvector of  $\mathbf{A} \in M_n(F)$ , and say how left eigenvectors of  $\mathbf{A}$  are related to  $\mathbf{A}^\top$ .
- True or false: a matrix  $\mathbf{A} \in M_3(\mathbb{C})$  with  $\text{tr } \mathbf{A} = \text{tr } \mathbf{A}^2 = \text{tr } \mathbf{A}^3 = 0$  satisfies  $\mathbf{A}^3 = 0$ . Justify your answer.

*Solution.*

- If  $V$  is finite-dimensional with  $\dim V = n \geq 1$ ,  $T \in \mathcal{L}(V)$ , and  $p_T = \prod_{i=1}^n (x - \lambda_i)$  splits over  $F$ , then for every  $q \in F[x]$ ,  $p_{q(T)} = \prod_i (x - q(\lambda_i))$  and  $\text{spec}(q(T)) = q(\text{spec}(T))$  (Theorem 8.10.1).
- False. The rotation  $\mathbf{R} = \begin{pmatrix} 0 & -1 \\ 1 & 0 \end{pmatrix}$  has no real eigenvalue, while  $\mathbf{R}^2 = -\mathbf{I}$  has the real eigenvalue  $-1$ .
- False. They have different degrees, 2 and 3. The correct identity is  $x^3 p_{\mathbf{AB}} = x^2 p_{\mathbf{BA}}$ , that is,  $p_{\mathbf{BA}} = x p_{\mathbf{AB}}$  (Theorem 8.10.4).
- False.  $\mathbf{A} = \mathbf{E}_{12}$ ,  $\mathbf{B} = \mathbf{E}_{11}$  give  $\mathbf{AB} = 0$  and  $\mathbf{BA} = \mathbf{E}_{12} \neq 0$ , and the only matrix similar to 0 is 0.
- A non-zero  $\mathbf{y} \in F^n$  with  $\mathbf{y}^\top \mathbf{A} = \lambda \mathbf{y}^\top$  for some  $\lambda \in F$ . Equivalently,  $\mathbf{y}$  is an eigenvector of  $\mathbf{A}^\top$  for  $\lambda$  (Proposition 8.10.5).
- True. By Theorem 8.10.3 with  $n = 3$ , (e)  $\Rightarrow$  (d).

### 8.10.5.2 B. Practice

#### Exercise B1: Eigenvalues of a polynomial in $\mathbf{A}$ .

Let  $\mathbf{A} \in M_3(\mathbb{C})$  have  $p_{\mathbf{A}} = (x - 1)^2(x + 2)$ .

- (a) Find  $p_{\mathbf{A}^2 + \mathbf{A} + \mathbf{I}}$ ,  $\text{tr}(\mathbf{A}^2 + \mathbf{A} + \mathbf{I})$  and  $\det(\mathbf{A}^2 + \mathbf{A} + \mathbf{I})$ . Is  $\mathbf{A}^2 + \mathbf{A} + \mathbf{I}$  invertible?  
 (b) Find  $p_{\mathbf{A}^2 + \mathbf{A} - 2\mathbf{I}}$ . Hence show that  $\mathbf{A}^2 + \mathbf{A} - 2\mathbf{I}$  is nilpotent.  
 (c) Find  $\text{tr } \mathbf{A}^5$  and  $\det(\mathbf{A}^{-1})$ .

*Solution.* The list of eigenvalues of  $\mathbf{A}$  with multiplicity is 1, 1,  $-2$ , since  $p_{\mathbf{A}}$  splits.

- (a) With  $q = x^2 + x + 1$ ,  $q(1) = 3$  and  $q(-2) = 4 - 2 + 1 = 3$ . By Theorem 8.10.1,  $p_{q(\mathbf{A})} = (x - 3)^3$ , and by Corollary 8.10.2,  $\text{tr } q(\mathbf{A}) = 9$  and  $\det q(\mathbf{A}) = 27 \neq 0$ . So  $\mathbf{A}^2 + \mathbf{A} + \mathbf{I}$  is invertible (Theorem 6.5.3).  
 (b) With  $r = x^2 + x - 2 = (x - 1)(x + 2)$ ,  $r(1) = r(-2) = 0$ , so  $p_{r(\mathbf{A})} = x^3$  by Theorem 8.10.1. By Theorem 8.10.3 ((c)  $\Rightarrow$  (d)),  $r(\mathbf{A})^3 = 0$ , so  $\mathbf{A}^2 + \mathbf{A} - 2\mathbf{I}$  is nilpotent.  
 (c) By Corollary 8.10.2,  $\text{tr } \mathbf{A}^5 = 1 + 1 + (-2)^5 = -30$ . Also  $\det \mathbf{A} = 1 \cdot 1 \cdot (-2) = -2 \neq 0$  (Theorem 8.3.3), so  $\mathbf{A}$  is invertible and  $\det(\mathbf{A}^{-1}) = (\det \mathbf{A})^{-1} = -\frac{1}{2}$  (Corollary 6.5.6).

### Exercise B2: $\mathbf{AB}$ and $\mathbf{BA}$ for a $2 \times 3$ example.

Let  $\mathbf{A} = \begin{pmatrix} 1 & 0 & 1 \\ 0 & 2 & 0 \end{pmatrix}$  and  $\mathbf{B} = \begin{pmatrix} 1 & 0 \\ 0 & -1 \\ 1 & 1 \end{pmatrix}$ .

- (a) Compute  $\mathbf{AB}$ ,  $\mathbf{BA}$ ,  $p_{\mathbf{AB}}$  and  $p_{\mathbf{BA}}$ , and verify that  $x^3 p_{\mathbf{AB}} = x^2 p_{\mathbf{BA}}$ .  
 (b) Hence list the eigenvalues of  $\mathbf{BA}$  without further computation, and find an eigenvector of  $\mathbf{BA}$  for each non-zero eigenvalue by applying  $\mathbf{B}$  to eigenvectors of  $\mathbf{AB}$ .

*Solution.*

- (a) By row-times-column,

$$\mathbf{AB} = \begin{pmatrix} 2 & 1 \\ 0 & -2 \end{pmatrix}, \quad \mathbf{BA} = \begin{pmatrix} 1 & 0 & 1 \\ 0 & -2 & 0 \\ 1 & 2 & 1 \end{pmatrix}.$$

$\mathbf{AB}$  is upper triangular, so  $p_{\mathbf{AB}} = (x - 2)(x + 2) = x^2 - 4$ . Expanding  $\det(x\mathbf{I} - \mathbf{BA})$  along the second row, whose only non-zero entry is  $x + 2$  in position  $(2, 2)$ ,

$$\begin{aligned} p_{\mathbf{BA}} &= (x + 2) \det \begin{pmatrix} x - 1 & -1 \\ -1 & x - 1 \end{pmatrix} \\ &= (x + 2)((x - 1)^2 - 1) = (x + 2)x(x - 2) = x^3 - 4x. \end{aligned}$$

Then  $x^3 p_{\mathbf{AB}} = x^5 - 4x^3 = x^2(x^3 - 4x) = x^2 p_{\mathbf{BA}}$ .

- (b) The eigenvalues of  $\mathbf{BA}$  are the roots of  $x(x - 2)(x + 2)$ : 0, 2,  $-2$ , matching Theorem 8.10.4 (the non-zero ones are those of  $\mathbf{AB}$ ). If  $\mathbf{AB}\mathbf{w} = \lambda\mathbf{w}$ , then  $\mathbf{BA}(\mathbf{B}\mathbf{w}) = \mathbf{B}(\mathbf{AB}\mathbf{w}) = \lambda\mathbf{B}\mathbf{w}$ , so  $\mathbf{B}\mathbf{w}$  is an eigenvector of  $\mathbf{BA}$  as long as  $\mathbf{B}\mathbf{w} \neq \mathbf{0}$ . For  $\lambda = 2$ :  $\mathbf{w} = (1, 0)$ ,  $\mathbf{B}\mathbf{w} = (1, 0, 1)$ , and  $\mathbf{BA}(1, 0, 1) = (2, 0, 2)$ . For  $\lambda = -2$ :  $\mathbf{AB} + 2\mathbf{I} = \begin{pmatrix} 4 & 1 \\ 0 & 0 \end{pmatrix}$  gives  $\mathbf{w} = (1, -4)$ ,  $\mathbf{B}\mathbf{w} = (1, 4, -3)$ , and  $\mathbf{BA}(1, 4, -3) = (1 - 3, -8, 1 + 8 - 3) = (-2, -8, 6) = -2(1, 4, -3)$ .

**Exercise B3: Left eigenvectors of a  $2 \times 2$  matrix.**

Let  $\mathbf{A} = \begin{pmatrix} 0 & 1 \\ -6 & 5 \end{pmatrix} \in M_2(\mathbb{R})$ , whose right eigenvectors  $(1, 2)$  for 2 and  $(1, 3)$  for 3 were found in [Exercise \(B1: Diagonalize or show impossible\)](#).

- Find a left eigenvector of  $\mathbf{A}$  for each eigenvalue.
- Verify [Theorem 8.10.6](#) for these vectors, and rescale the left eigenvectors so that  $\mathbf{y}_i^\top \mathbf{v}_i = 1$ .
- Hence write  $\mathbf{A}^k$  as a combination of two rank-one matrices, and check the formula for  $k = 0$  and  $k = 1$ .

*Solution.*

- By [Proposition 8.10.5](#), we need eigenvectors of  $\mathbf{A}^\top = \begin{pmatrix} 0 & -6 \\ 1 & 5 \end{pmatrix}$ . For 2:  $\mathbf{A}^\top - 2\mathbf{I} = \begin{pmatrix} -2 & -6 \\ 1 & 3 \end{pmatrix}$  gives  $y_1 = -3y_2$ , so  $(3, -1)$ . For 3:  $\mathbf{A}^\top - 3\mathbf{I} = \begin{pmatrix} -3 & -6 \\ 1 & 2 \end{pmatrix}$  gives  $y_1 = -2y_2$ , so  $(2, -1)$ . Check:  $\begin{pmatrix} 3 & -1 \end{pmatrix} \mathbf{A} = \begin{pmatrix} 6 & -2 \end{pmatrix}$  and  $\begin{pmatrix} 2 & -1 \end{pmatrix} \mathbf{A} = \begin{pmatrix} 6 & -3 \end{pmatrix}$ .
- $\begin{pmatrix} 3 & -1 \end{pmatrix} \cdot \begin{pmatrix} 1 \\ 3 \end{pmatrix} = 0$  and  $\begin{pmatrix} 2 & -1 \end{pmatrix} \cdot \begin{pmatrix} 1 \\ 2 \end{pmatrix} = 0$ , as the theorem predicts for different eigenvalues. Also  $\begin{pmatrix} 3 & -1 \end{pmatrix} \cdot \begin{pmatrix} 1 \\ 2 \end{pmatrix} = 1$  and  $\begin{pmatrix} 2 & -1 \end{pmatrix} \cdot \begin{pmatrix} 1 \\ 3 \end{pmatrix} = -1$ , so take  $\mathbf{y}_1 = \begin{pmatrix} 3 & -1 \end{pmatrix}$  and  $\mathbf{y}_2 = \begin{pmatrix} -2 & 1 \end{pmatrix}$ .
- With  $\mathbf{P} = \begin{pmatrix} 1 & 1 \\ 2 & 3 \end{pmatrix}$ , the rows of  $\mathbf{P}^{-1} = \begin{pmatrix} 3 & -1 \\ -2 & 1 \end{pmatrix}$  are exactly  $\mathbf{y}_1^\top, \mathbf{y}_2^\top$ , and by [Proposition 8.10.7](#),

$$\mathbf{A}^k = 2^k \begin{pmatrix} 1 \\ 2 \end{pmatrix} \begin{pmatrix} 3 & -1 \end{pmatrix} + 3^k \begin{pmatrix} 1 \\ 3 \end{pmatrix} \begin{pmatrix} -2 & 1 \end{pmatrix} = 2^k \begin{pmatrix} 3 & -1 \\ 6 & -2 \end{pmatrix} + 3^k \begin{pmatrix} -2 & 1 \\ -6 & 3 \end{pmatrix}.$$

$$\text{For } k = 0: \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix}. \text{ For } k = 1: \begin{pmatrix} 6 - 6 & -2 + 3 \\ 12 - 18 & -4 + 9 \end{pmatrix} = \begin{pmatrix} 0 & 1 \\ -6 & 5 \end{pmatrix} = \mathbf{A}.$$

**8.10.5.3 C. Going deeper**
**Exercise C1: Traces of powers of  $\mathbf{AB}$  and  $\mathbf{BA}$ .**

Let  $\mathbf{A} \in M_{m \times n}(F)$  and  $\mathbf{B} \in M_{n \times m}(F)$ .

- Prove that  $\mathbf{B}(\mathbf{AB})^{k-1}\mathbf{A} = (\mathbf{BA})^k$  for every  $k \geq 1$ , and deduce that  $\text{tr}((\mathbf{AB})^k) = \text{tr}((\mathbf{BA})^k)$ .
- Now let  $F = \mathbb{C}$ . Give a second proof of the trace identity from [Theorem 8.10.4](#) and [Corollary 8.10.2](#).

*Solution.*

- By induction on  $k$ . For  $k = 1$  both sides are  $\mathbf{BA}$ . If  $\mathbf{B}(\mathbf{AB})^{k-1}\mathbf{A} = (\mathbf{BA})^k$ , then by associativity

$$\begin{aligned} \mathbf{B}(\mathbf{AB})^k \mathbf{A} &= \mathbf{B}(\mathbf{AB})^{k-1}(\mathbf{AB})\mathbf{A} = (\mathbf{B}(\mathbf{AB})^{k-1}\mathbf{A})(\mathbf{BA}) \\ &= (\mathbf{BA})^k(\mathbf{BA}) = (\mathbf{BA})^{k+1}. \end{aligned}$$



Now  $(\mathbf{AB})^k = \mathbf{A}(\mathbf{B}(\mathbf{AB})^{k-1})$ , with  $\mathbf{A}$  of size  $m \times n$  and  $\mathbf{B}(\mathbf{AB})^{k-1}$  of size  $n \times m$ . By [Theorem 0.9.6](#) (3),

$$\operatorname{tr}((\mathbf{AB})^k) = \operatorname{tr}(\mathbf{A} \cdot \mathbf{B}(\mathbf{AB})^{k-1}) = \operatorname{tr}(\mathbf{B}(\mathbf{AB})^{k-1} \cdot \mathbf{A}) = \operatorname{tr}((\mathbf{BA})^k).$$

- (b) Over  $\mathbb{C}$ ,  $p_{\mathbf{AB}}$  and  $p_{\mathbf{BA}}$  split. Suppose  $m \geq n$  (otherwise swap the roles of  $\mathbf{A}$  and  $\mathbf{B}$ ). By [Theorem 8.10.4](#),  $p_{\mathbf{AB}} = x^{m-n}p_{\mathbf{BA}}$ , so if  $\lambda_1, \dots, \lambda_n$  is the list of eigenvalues of  $\mathbf{BA}$  with multiplicity, then the list for  $\mathbf{AB}$  is  $\lambda_1, \dots, \lambda_n$  together with  $m - n$  zeros. For  $k \geq 1$  the zeros contribute  $0^k = 0$ , so by [Corollary 8.10.2](#),  $\operatorname{tr}((\mathbf{AB})^k) = \sum_i \lambda_i^k = \operatorname{tr}((\mathbf{BA})^k)$ .

### Exercise C2: Adding a commuting nilpotent matrix.

Let  $\mathbf{A}, \mathbf{N} \in M_n(\mathbb{C})$ ,  $n \geq 1$ , with  $\mathbf{AN} = \mathbf{NA}$  and  $\mathbf{N}$  nilpotent.

- Prove that  $p_{\mathbf{A}+\mathbf{N}} = p_{\mathbf{A}}$ ; in particular  $\operatorname{spec}(\mathbf{A} + \mathbf{N}) = \operatorname{spec}(\mathbf{A})$  and  $\det(\mathbf{A} + \mathbf{N}) = \det \mathbf{A}$ .
- Deduce that if  $\mathbf{A}$  is invertible, so is  $\mathbf{A} + \mathbf{N}$ .
- Show that (a) fails without  $\mathbf{AN} = \mathbf{NA}$ .

*Hint: triangularize  $\mathbf{A}$  and  $\mathbf{N}$  together.*

*Solution.*

- (a) By [Theorem 8.9.4](#) there is an invertible  $\mathbf{P}$  with  $\mathbf{U} = \mathbf{P}^{-1}\mathbf{A}\mathbf{P}$  and  $\mathbf{U}' = \mathbf{P}^{-1}\mathbf{N}\mathbf{P}$  upper triangular. Since  $\mathbf{N}$  is nilpotent,  $\operatorname{spec}(\mathbf{N}) = \{0\}$  ([Exercise \(C2: Nilpotent operators\)](#)), so every diagonal entry of  $\mathbf{U}'$  is 0 by [Theorem 8.7.4](#). Then  $\mathbf{P}^{-1}(\mathbf{A} + \mathbf{N})\mathbf{P} = \mathbf{U} + \mathbf{U}'$  is upper triangular with the same diagonal as  $\mathbf{U}$ . By [Theorem 6.10.3](#) and the examples after [Definition 6.10.1](#),

$$p_{\mathbf{A}+\mathbf{N}} = p_{\mathbf{U}+\mathbf{U}'} = \prod_i (x - u_{ii}) = p_{\mathbf{U}} = p_{\mathbf{A}}.$$

Equal characteristic polynomials have the same roots ([Theorem 8.2.1](#)) and the same constant term  $(-1)^n \det$  ([Theorem 6.10.2](#)).

- (b) By (a),  $\det(\mathbf{A} + \mathbf{N}) = \det \mathbf{A} \neq 0$ , so  $\mathbf{A} + \mathbf{N}$  is invertible ([Theorem 6.5.3](#)).
- (c) Let  $\mathbf{A} = \begin{pmatrix} 0 & 0 \\ 1 & 0 \end{pmatrix}$  and  $\mathbf{N} = \begin{pmatrix} 0 & 1 \\ 0 & 0 \end{pmatrix}$ . Both are nilpotent, so  $p_{\mathbf{A}} = x^2$ , but  $\mathbf{A} + \mathbf{N} = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}$  has  $p = x^2 - 1$  and eigenvalues  $\pm 1$ . Here  $\mathbf{AN} = \begin{pmatrix} 0 & 0 \\ 0 & 1 \end{pmatrix} \neq \begin{pmatrix} 1 & 0 \\ 0 & 0 \end{pmatrix} = \mathbf{NA}$ .

### Exercise C3: How many traces, and which field?.

- Let  $\omega = e^{2\pi i/3} \in \mathbb{C}$  and  $\mathbf{C} = \operatorname{diag}(1, \omega, \omega^2)$ . Show that  $\operatorname{tr} \mathbf{C} = \operatorname{tr} \mathbf{C}^2 = 0$  but  $\mathbf{C}$  is not nilpotent. Explain why this does not contradict [Theorem 8.10.3](#).
- Let  $n \geq 2$ . Give a matrix  $\mathbf{C} \in M_n(\mathbb{C})$  with  $\operatorname{tr} \mathbf{C}^k = 0$  for  $k = 1, \dots, n-1$  that is not nilpotent.
- Let  $\mathbf{A}, \mathbf{B} \in M_n(\mathbb{C})$  and suppose  $\mathbf{C} = \mathbf{AB} - \mathbf{BA}$  commutes with  $\mathbf{A}$ . Prove that  $\mathbf{C}^n = 0$ .

- (d) Over  $\mathbb{F}_2$ , find a non-nilpotent  $\mathbf{C} \in M_2(\mathbb{F}_2)$  with  $\text{tr } \mathbf{C}^k = 0$  for every  $k \geq 1$ .

*Solution.*

- (a)  $\omega^3 = 1$  and  $\omega \neq 1$ , so  $0 = \omega^3 - 1 = (\omega - 1)(\omega^2 + \omega + 1)$  gives  $1 + \omega + \omega^2 = 0$ . Hence  $\text{tr } \mathbf{C} = 1 + \omega + \omega^2 = 0$ , and  $\text{tr } \mathbf{C}^2 = 1 + \omega^2 + \omega^4 = 1 + \omega^2 + \omega = 0$ . But  $\mathbf{C}^k = \text{diag}(1, \omega^k, \omega^{2k}) \neq 0$  for every  $k$ . No contradiction: the theorem, for  $n = 3$ , requires  $\text{tr } \mathbf{C}^k = 0$  for  $k = 1, 2, 3$ , and  $\text{tr } \mathbf{C}^3 = 1 + 1 + 1 = 3 \neq 0$ .
- (b) Let  $\zeta = e^{2\pi i/n}$  and  $\mathbf{C} = \text{diag}(1, \zeta, \dots, \zeta^{n-1})$ . For  $1 \leq k \leq n-1$ ,  $\zeta^k \neq 1$  and  $(\zeta^k)^n = 1$ , so  $\text{tr } \mathbf{C}^k = \sum_{j=0}^{n-1} (\zeta^k)^j = \frac{(\zeta^k)^n - 1}{\zeta^k - 1} = 0$  by the geometric sum. But  $\mathbf{C}^k$  has the entry 1 in position  $(1, 1)$  for all  $k$ , so  $\mathbf{C}$  is not nilpotent.
- (c) By [Exercise \(C2: When  \$\mathbf{A}\$  commutes with  \$\[\mathbf{A}, \mathbf{B}\]\$ \)](#) (a),  $\text{tr } \mathbf{C}^k = 0$  for every  $k \geq 1$ , in particular for  $k = 1, \dots, n$ . By [Theorem 8.10.3](#) ((e)  $\Rightarrow$  (d)),  $\mathbf{C}^n = 0$ .
- (d)  $\mathbf{C} = \mathbf{I}_2$ :  $\text{tr } \mathbf{C}^k = 1 + 1 = 0$  in  $\mathbb{F}_2$  for every  $k$ , and  $\mathbf{C}^k = \mathbf{I}_2 \neq 0$ .

## 8.11 Powers, Recurrences and Markov Chains

Section 4 promised that diagonalization is a tool for computation, and the Fibonacci numbers gave a first taste. This section collects three situations in which a quantity evolves in discrete steps by a fixed linear rule: a vector recurrence  $\mathbf{x}_{k+1} = \mathbf{A}\mathbf{x}_k$ , a scalar recurrence such as  $a_{k+3} = 2a_{k+2} + a_{k+1} - 2a_k$ , and a system that moves at random between finitely many states. In each case the state after  $k$  steps is a matrix power applied to the starting state, and the eigenvalues decide what happens in the long run. We use only the explicit formula  $\mathbf{A}^k = \mathbf{P}\mathbf{D}^k\mathbf{P}^{-1}$  and limits of sequences of numbers, one entry at a time.

### 8.11.1 Vector recurrences

A **linear recurrence** of first order is a rule  $\mathbf{x}_{k+1} = \mathbf{A}\mathbf{x}_k$  for  $k \in \mathbb{N}$ , with  $\mathbf{A} \in M_n(F)$  fixed and a starting vector  $\mathbf{x}_0 \in F^n$ . By induction on  $k$ , the unique sequence obeying the rule is  $\mathbf{x}_k = \mathbf{A}^k \mathbf{x}_0$ . The question is how to see this power, and diagonalization answers it by changing to coordinates in which  $\mathbf{A}$  acts coordinate by coordinate.

#### Theorem 8.11.1 — Solving a Diagonalizable Recurrence

Let  $\mathbf{A} \in M_n(F)$  be diagonalizable, with  $\mathbf{A} = \mathbf{P}\mathbf{D}\mathbf{P}^{-1}$ ,  $\mathbf{D} = \text{diag}(\lambda_1, \dots, \lambda_n)$ , and let  $\mathbf{v}_1, \dots, \mathbf{v}_n$  be the columns of  $\mathbf{P}$ . Let  $\mathbf{x}_0 \in F^n$ , and let  $(c_1, \dots, c_n) = \mathbf{P}^{-1}\mathbf{x}_0$ , so that  $\mathbf{x}_0 = c_1\mathbf{v}_1 + \dots + c_n\mathbf{v}_n$ . Then the sequence defined by  $\mathbf{x}_{k+1} = \mathbf{A}\mathbf{x}_k$  is

$$\mathbf{x}_k = \mathbf{A}^k \mathbf{x}_0 = c_1 \lambda_1^k \mathbf{v}_1 + c_2 \lambda_2^k \mathbf{v}_2 + \dots + c_n \lambda_n^k \mathbf{v}_n \quad (k \in \mathbb{N}).$$

*Proof.* By induction on  $k$ ,  $\mathbf{x}_k = \mathbf{A}^k \mathbf{x}_0$ : this holds for  $k = 0$  since  $\mathbf{A}^0 = \mathbf{I}$ , and  $\mathbf{x}_{k+1} = \mathbf{A}\mathbf{x}_k = \mathbf{A} \cdot \mathbf{A}^k \mathbf{x}_0 = \mathbf{A}^{k+1} \mathbf{x}_0$ . By [Theorem 8.4.3](#),  $\mathbf{A}^k \mathbf{x}_0 = \mathbf{P}\mathbf{D}^k\mathbf{P}^{-1}\mathbf{x}_0 = \mathbf{P}\mathbf{D}^k\mathbf{c}$  with  $\mathbf{c} = (c_1, \dots, c_n)$ . Now  $\mathbf{D}^k\mathbf{c} = (\lambda_1^k c_1, \dots, \lambda_n^k c_n)$ , and by [Theorem 0.9.3](#),  $\mathbf{P}$  times this

vector is  $\sum_i \lambda_i^k c_i \mathbf{v}_i$ . Finally,  $\mathbf{P}\mathbf{c} = \mathbf{x}_0$  says  $\mathbf{x}_0 = \sum_i c_i \mathbf{v}_i$ , by the same result. ■

In words: expand the starting vector in a basis of eigenvectors; each coordinate is then multiplied by its own eigenvalue at every step, independently of the others. Over  $\mathbb{R}$  or  $\mathbb{C}$ , where absolute values are available, the long-run behavior is visible at once. Terms with  $|\lambda_i| < 1$  die out, terms with  $|\lambda_i| > 1$  grow, and the largest  $|\lambda_i|$  with  $c_i \neq 0$  wins. The worked Fibonacci example (Example (A Closed Formula for the Fibonacci Numbers)) is the case  $n = 2$ .

## 8.11.2 Scalar recurrences and companion matrices

A recurrence such as  $a_{k+3} = 2a_{k+2} + a_{k+1} - 2a_k$  involves one number sequence but several previous terms. The trick of Example (A Closed Formula for the Fibonacci Numbers) turns it into a vector recurrence: remember the last few terms as one vector.

Let  $n \geq 1$  and  $c_0, \dots, c_{n-1} \in F$ , and consider sequences  $(a_k)_{k \in \mathbb{N}}$  in  $F$  with

$$a_{k+n} + c_{n-1}a_{k+n-1} + \dots + c_1a_{k+1} + c_0a_k = 0 \quad (*)$$

for every  $k \in \mathbb{N}$ .

Such a sequence is determined by its **initial values**  $a_0, \dots, a_{n-1}$ , because  $(*)$  computes each later term from the  $n$  before it. The polynomial

$$p(x) = x^n + c_{n-1}x^{n-1} + \dots + c_1x + c_0$$

is called the **characteristic polynomial of the recurrence**; the name is justified below. Put  $\mathbf{x}_k = (a_k, a_{k+1}, \dots, a_{k+n-1}) \in F^n$ . Then  $(*)$  says exactly  $\mathbf{x}_{k+1} = \mathbf{C}\mathbf{x}_k$ , where

$$\mathbf{C} = \begin{pmatrix} 0 & 1 & 0 & \dots & 0 \\ 0 & 0 & 1 & \dots & 0 \\ \vdots & & & \ddots & \vdots \\ 0 & 0 & 0 & \dots & 1 \\ -c_0 & -c_1 & -c_2 & \dots & -c_{n-1} \end{pmatrix} :$$

the first  $n-1$  rows shift the window by one, and the last row is  $(*)$  solved for  $a_{k+n}$ . For  $n \geq 2$ ,  $\mathbf{C}$  is the transpose of the companion matrix  $\mathbf{C}(p)$  of Exercise (C1: Companion matrices), so  $p\mathbf{C} = p\mathbf{C}(p) = p$  by Exercise (B2: Transpose) and Exercise (C1: Companion matrices) (for  $n = 1$ ,  $\mathbf{C} = (-c_0)$  and  $p\mathbf{C} = x + c_0 = p$ ). The eigenvalues of  $\mathbf{C}$  are the roots of  $p$ .

### Theorem 8.11.2 — Scalar Recurrences with Distinct Roots

Let  $n \geq 1$ ,  $c_0, \dots, c_{n-1} \in F$ , and suppose  $p = x^n + c_{n-1}x^{n-1} + \dots + c_0$  has  $n$  **distinct** roots  $\lambda_1, \dots, \lambda_n \in F$ . Then a sequence  $(a_k)_{k \in \mathbb{N}}$  in  $F$  satisfies  $(*)$  if and only if there are  $b_1, \dots, b_n \in F$  with

$$a_k = b_1\lambda_1^k + b_2\lambda_2^k + \dots + b_n\lambda_n^k \quad \text{for every } k \in \mathbb{N}.$$

In that case  $b_1, \dots, b_n$  are uniquely determined by the initial values  $a_0, \dots, a_{n-1}$ .

*Idea.* Why powers of the roots? A geometric sequence  $a_k = \lambda^k$  satisfies  $(*)$  exactly when  $\lambda^k p(\lambda) = 0$ , so the roots of  $p$  give  $n$  solutions. To see that there are no others, use the matrix  $\mathbf{C}$ : its eigenvector for  $\lambda_i$  is the window  $(1, \lambda_i, \dots, \lambda_i^{n-1})$  of the geometric sequence, these eigenvectors form an invertible Vandermonde matrix, and [Theorem 8.11.1](#) does the rest.

*Proof.*  $(\Leftarrow)$  For each  $i$  and  $k$ ,  $\lambda_i^{k+n} + c_{n-1}\lambda_i^{k+n-1} + \dots + c_0\lambda_i^k = \lambda_i^k p(\lambda_i) = 0$ . If  $a_k = \sum_i b_i \lambda_i^k$ , then the left side of  $(*)$  is  $\sum_i b_i \cdot 0 = 0$ , after regrouping the finite sums.

$(\Rightarrow)$  Let  $(a_k)$  satisfy  $(*)$ , and let  $\mathbf{x}_k$  and  $\mathbf{C}$  be as above, so  $\mathbf{x}_{k+1} = \mathbf{C}\mathbf{x}_k$ . For each  $i$ , let  $\mathbf{w}_i = (1, \lambda_i, \lambda_i^2, \dots, \lambda_i^{n-1})$ . Then  $\mathbf{C}\mathbf{w}_i = \lambda_i \mathbf{w}_i$ : for  $r \leq n-1$ , entry  $r$  of  $\mathbf{C}\mathbf{w}_i$  is entry  $r+1$  of  $\mathbf{w}_i$ , which is  $\lambda_i^r = \lambda_i \cdot \lambda_i^{r-1}$ ; and the last entry is  $-(c_0 + c_1\lambda_i + \dots + c_{n-1}\lambda_i^{n-1}) = \lambda_i^n - p(\lambda_i) = \lambda_i^n = \lambda_i \cdot \lambda_i^{n-1}$ . Let  $\mathbf{W}$  be the matrix with columns  $\mathbf{w}_1, \dots, \mathbf{w}_n$ . Its  $(r, i)$ -entry is  $\lambda_i^{r-1}$ , so  $\mathbf{W} = \mathbf{V}(\lambda_1, \dots, \lambda_n)^\top$  in the notation of [Theorem 6.8.1](#), and  $\mathbf{W}$  is invertible, since the  $\lambda_i$  are distinct:  $\det \mathbf{W} = \det \mathbf{V}(\lambda_1, \dots, \lambda_n) \neq 0$  by [Theorem 6.4.5](#) and [Theorem 6.8.1](#), and [Theorem 6.5.2](#) applies. The columns of  $\mathbf{C}\mathbf{W}$  are  $\lambda_i \mathbf{w}_i$ , which are the columns of  $\mathbf{W} \operatorname{diag}(\lambda_1, \dots, \lambda_n)$ , so  $\mathbf{C} = \mathbf{W} \operatorname{diag}(\lambda_1, \dots, \lambda_n) \mathbf{W}^{-1}$ .

By [Theorem 8.11.1](#), with  $(b_1, \dots, b_n) = \mathbf{W}^{-1}\mathbf{x}_0$ ,

$$\mathbf{x}_k = b_1 \lambda_1^k \mathbf{w}_1 + \dots + b_n \lambda_n^k \mathbf{w}_n.$$

The first entry of  $\mathbf{x}_k$  is  $a_k$ , and the first entry of each  $\mathbf{w}_i$  is 1, so  $a_k = \sum_i b_i \lambda_i^k$ .

For uniqueness, suppose  $a_k = \sum_i b_i \lambda_i^k$  for all  $k$ . Taking  $k = 0, \dots, n-1$  gives  $\mathbf{W}\mathbf{b} = (a_0, \dots, a_{n-1}) = \mathbf{x}_0$ , so  $\mathbf{b} = \mathbf{W}^{-1}\mathbf{x}_0$  is determined by the initial values. ■

So solving a recurrence with distinct roots comes down to three steps: find the roots of  $p$ , write the general form  $\sum b_i \lambda_i^k$ , and fit  $b_1, \dots, b_n$  to the initial values by solving a Vandermonde system.

### Example. (A Third-Order Recurrence)

Find a formula for the sequence with  $a_0 = 3$ ,  $a_1 = 2$ ,  $a_2 = 6$  and

$$a_{k+3} = 2a_{k+2} + a_{k+1} - 2a_k \quad (k \in \mathbb{N}).$$

*Solution.* In the form  $(*)$ ,  $a_{k+3} - 2a_{k+2} - a_{k+1} + 2a_k = 0$ , so  $p(x) = x^3 - 2x^2 - x + 2$ . Grouping,  $p = x^2(x-2) - (x-2) = (x-2)(x-1)(x+1)$ , with the distinct roots 1, -1, 2. The window matrix is

$$\mathbf{C} = \begin{pmatrix} 0 & 1 & 0 \\ 0 & 0 & 1 \\ -2 & 1 & 2 \end{pmatrix},$$

with eigenvectors  $(1, 1, 1)$ ,  $(1, -1, 1)$ ,  $(1, 2, 4)$ ; for instance  $\mathbf{C}(1, 2, 4) = (2, 4, -2+2+8) = 2(1, 2, 4)$ .

By [Theorem 8.11.2](#),  $a_k = b_1 + b_2(-1)^k + b_3 2^k$ . The initial values give the Vandermonde system

$$b_1 + b_2 + b_3 = 3, \quad b_1 - b_2 + 2b_3 = 2, \quad b_1 + b_2 + 4b_3 = 6.$$

Subtracting the first equation from the third gives  $3b_3 = 3$ , so  $b_3 = 1$ . Then  $b_1 + b_2 = 2$

and  $b_1 - b_2 = 0$ , so  $b_1 = b_2 = 1$ . Hence

$$a_k = 1 + (-1)^k + 2^k.$$

*Check.* The recurrence gives  $a_3 = 2 \cdot 6 + 2 - 2 \cdot 3 = 8$  and  $a_4 = 2 \cdot 8 + 6 - 2 \cdot 2 = 18$ ; the formula gives  $1 - 1 + 8 = 8$  and  $1 + 1 + 16 = 18$ . For large  $k$ ,  $a_k \approx 2^k$ : the root of largest absolute value dominates.

**Repeated roots.** When  $p$  has a repeated root,  $\mathbf{C}$  is not diagonalizable: for a root  $\lambda$ , every eigenvector  $(w_1, \dots, w_n)$  of  $\mathbf{C}$  satisfies  $w_{r+1} = \lambda w_r$ , so it is a multiple of  $(1, \lambda, \dots, \lambda^{n-1})$ , and  $g_{\mathbf{C}}(\lambda) = 1$  whatever  $a_{\mathbf{C}}(\lambda)$  is. The general answer involves terms  $k\lambda^k, k^2\lambda^k, \dots$ , and it follows from the Jordan canonical form of Chapter 9. The  $2 \times 2$  case can be done by hand.

### Example. (A Repeated Root)

Over  $\mathbb{R}$ , find all sequences with  $a_{k+2} = 4a_{k+1} - 4a_k$  for every  $k \in \mathbb{N}$ .

*Solution.* Here  $p = x^2 - 4x + 4 = (x - 2)^2$ , and the window matrix is  $\mathbf{C} = \begin{pmatrix} 0 & 1 \\ -4 & 4 \end{pmatrix}$ . It is not diagonalizable, since  $\mathbf{C} - 2\mathbf{I} = \begin{pmatrix} -2 & 1 \\ -4 & 2 \end{pmatrix}$  has rank 1, so  $g(2) = 1 < 2 = a(2)$ . But  $\mathbf{N} = \mathbf{C} - 2\mathbf{I}$  satisfies

$$\mathbf{N}^2 = \begin{pmatrix} 4 - 4 & -2 + 2 \\ 8 - 8 & -4 + 4 \end{pmatrix} = \mathbf{0}.$$

*Claim:*  $\mathbf{C}^k = 2^k \mathbf{I} + k2^{k-1} \mathbf{N}$  for all  $k \in \mathbb{N}$ . For  $k = 0$  both sides are  $\mathbf{I}$ . If it holds for  $k$ , then, since  $\mathbf{C} = 2\mathbf{I} + \mathbf{N}$  and  $\mathbf{N}^2 = \mathbf{0}$ ,

$$\begin{aligned} \mathbf{C}^{k+1} &= (2^k \mathbf{I} + k2^{k-1} \mathbf{N})(2\mathbf{I} + \mathbf{N}) \\ &= 2^{k+1} \mathbf{I} + 2^k \mathbf{N} + k2^k \mathbf{N} + k2^{k-1} \mathbf{N}^2 = 2^{k+1} \mathbf{I} + (k+1)2^k \mathbf{N}. \end{aligned}$$

Now  $\mathbf{x}_k = \mathbf{C}^k \mathbf{x}_0$  with  $\mathbf{x}_0 = (a_0, a_1)$ , and  $a_k$  is the first entry. Row 1 of  $\mathbf{C}^k$  is  $(2^k - 2k2^{k-1}, k2^{k-1}) = ((1-k)2^k, k2^{k-1})$ , so

$$a_k = (1-k)2^k a_0 + k2^{k-1} a_1 = \left(a_0 + \left(\frac{a_1}{2} - a_0\right)k\right)2^k.$$

So the solutions are exactly the sequences  $a_k = (b_1 + b_2 k)2^k$  with  $b_1, b_2 \in \mathbb{R}$ : every solution has this form with  $b_1 = a_0$ ,  $b_2 = a_1/2 - a_0$ , and every such sequence is the solution with  $a_0 = b_1$ ,  $a_1 = 2b_1 + 2b_2$ . For example,  $a_0 = 1$ ,  $a_1 = 4$  gives  $a_k = (1+k)2^k$ , and indeed  $a_2 = 4 \cdot 4 - 4 \cdot 1 = 12 = 3 \cdot 4$ . The new term  $k2^k$  is the footprint of the missing eigenvector, just as the entry  $k$  was for  $\begin{pmatrix} 1 & 1 \\ 0 & 1 \end{pmatrix}^k$  in Section 4.

## 8.11.3 Stochastic matrices

Now a system that moves at random. A shared bicycle is parked each night at one of a few stations; a machine is either working or under repair. We do not know the state for sure, but we know the probability of each state, and a fixed table of probabilities for moving from one state to another in one step.

Take states  $1, \dots, n$ . Record the probabilities of the states at step  $k$  as a column  $\mathbf{x}_k \in \mathbb{R}^n$ , and let  $a_{ij}$  be the probability of moving to state  $i$  in one step, given that the system is in state  $j$ . The probability of being in state  $i$  at step  $k+1$  is the sum over the previous state  $j$  of “was in  $j$ ” times “moved from  $j$  to  $i$ ”, that is,  $\sum_j a_{ij}(\mathbf{x}_k)_j$ . So  $\mathbf{x}_{k+1} = \mathbf{A}\mathbf{x}_k$ : a linear recurrence. The matrix  $\mathbf{A}$  has two features: its entries are probabilities, and from each state  $j$  the system must go **somewhere**, so column  $j$  adds up to 1.

*A stochastic matrix is a table of transition probabilities, one column for each starting state.*

### Definition 8.11.1 — Stochastic Matrix, Probability Vector

Let  $n \geq 1$ .

- (a) A vector  $\mathbf{x} \in \mathbb{R}^n$  is a **probability vector** if  $x_i \geq 0$  for every  $i$  and  $x_1 + \dots + x_n = 1$ .
- (b) A matrix  $\mathbf{A} \in M_n(\mathbb{R})$  is **stochastic** if every column of  $\mathbf{A}$  is a probability vector:  $a_{ij} \geq 0$  for **all**  $i, j$ , and  $\sum_{i=1}^n a_{ij} = 1$  for **every**  $j$ .

In words: the entries are non-negative and each **column** sums to 1. With  $\mathbf{1} = (1, \dots, 1) \in \mathbb{R}^n$ , the column condition reads  $\mathbf{1}^\top \mathbf{A} = \mathbf{1}^\top$ , and  $\mathbf{1}^\top \mathbf{x} = 1$  says that the entries of  $\mathbf{x}$  add up to 1.

**Examples.**

- $\begin{pmatrix} 9/10 & 1/5 \\ 1/10 & 4/5 \end{pmatrix}$ : columns  $(9/10, 1/10)$  and  $(1/5, 4/5)$  are probability vectors.
- **Degenerate cases.**  $\mathbf{I}_n$  is stochastic: every state stays put. The  $1 \times 1$  matrix  $(1)$  is the only  $1 \times 1$  stochastic matrix. Every **permutation matrix** is stochastic, since each column is a standard basis vector; it moves the system deterministically.
- $\frac{1}{4} \begin{pmatrix} 2 & 1 & 0 \\ 1 & 2 & 1 \\ 1 & 1 & 3 \end{pmatrix}$ : the columns sum to  $\frac{4}{4}$ , and all entries are non-negative.

**Non-example by minimal change.** Transpose the first example:  $\begin{pmatrix} 9/10 & 1/10 \\ 1/5 & 4/5 \end{pmatrix}$  still has non-negative entries, and its **rows** sum to 1, but its first column sums to  $11/10$ . It fails the column condition. Likewise  $\begin{pmatrix} 3/2 & 0 \\ -1/2 & 1 \end{pmatrix}$  has columns summing to 1 but a negative entry.

**Why this convention.** Many books write probability vectors as **rows** and multiply on the left,  $\mathbf{x}_{k+1}^\top = \mathbf{x}_k^\top \mathbf{M}$ ; their “stochastic” matrices have rows summing to 1, and they are the transposes of ours. We keep column vectors and  $\mathbf{x}_{k+1} = \mathbf{A}\mathbf{x}_k$ , in line with the rest of the book, so our stochastic matrices are **column-stochastic**. Translating between the conventions is a transpose, which changes neither eigenvalues nor their algebraic multiplicities ([Proposition 8.10.5](#)).

The first properties follow from  $\mathbf{1}^\top \mathbf{A} = \mathbf{1}^\top$ .

### Proposition 8.11.3 — Basic Properties of Stochastic Matrices

Let  $\mathbf{A}, \mathbf{B} \in M_n(\mathbb{R})$  be stochastic and  $\mathbf{x} \in \mathbb{R}^n$  a probability vector. Then:

- (a)  $\mathbf{A}\mathbf{x}$  is a probability vector;
- (b)  $\mathbf{A}\mathbf{B}$  is stochastic, and so is  $\mathbf{A}^k$  for every  $k \in \mathbb{N}$ ;

(c)  $\mathbf{1}$  is a left eigenvector of  $\mathbf{A}$  for the eigenvalue 1; in particular  $1 \in \text{spec}(\mathbf{A})$ .

*Proof.*

- (a) Each entry  $(\mathbf{Ax})_i = \sum_j a_{ij}x_j$  is a sum of products of non-negative numbers, so it is non-negative. By associativity,  $\mathbf{1}^\top(\mathbf{Ax}) = (\mathbf{1}^\top\mathbf{A})\mathbf{x} = \mathbf{1}^\top\mathbf{x} = 1$ .
- (b) By [Theorem 0.9.2](#), column  $j$  of  $\mathbf{AB}$  is  $\mathbf{Ab}_j$ , where  $\mathbf{b}_j$  is column  $j$  of  $\mathbf{B}$ , a probability vector. By (a),  $\mathbf{Ab}_j$  is a probability vector, so  $\mathbf{AB}$  is stochastic. By induction on  $k$ ,  $\mathbf{A}^k$  is stochastic:  $\mathbf{A}^0 = \mathbf{I}$  is, and  $\mathbf{A}^{k+1} = \mathbf{A}^k\mathbf{A}$ .
- (c)  $\mathbf{1}^\top\mathbf{A} = \mathbf{1}^\top$  with  $\mathbf{1} \neq \mathbf{0}$  is [Definition 8.10.1](#) with  $\lambda = 1$ , and [Proposition 8.10.5](#) gives  $1 \in \text{spec}(\mathbf{A})$ . ■

*Warning.* The all-ones vector is a left eigenvector, not a right one. For  $\mathbf{A} = \begin{pmatrix} 9/10 & 1/5 \\ 1/10 & 4/5 \end{pmatrix}$ ,  $\mathbf{A}\mathbf{1} = (11/10, 9/10) \neq \mathbf{1}$ . The right eigenvectors for 1 are the multiples of  $(2, 1)$ , and they carry the interesting information: the long-run probabilities. Only when the rows also sum to 1 is  $\mathbf{1}$  a right eigenvector.

## 8.11.4 Markov chains

### Definition 8.11.2 — Markov Chain, Steady State

Let  $\mathbf{A} \in M_n(\mathbb{R})$  be stochastic. The Markov chain with transition matrix  $\mathbf{A}$  and initial probability vector  $\mathbf{x}_0$  is the sequence of probability vectors  $\mathbf{x}_{k+1} = \mathbf{Ax}_k$ , that is,  $\mathbf{x}_k = \mathbf{A}^k\mathbf{x}_0$ . A steady state of  $\mathbf{A}$  is a probability vector  $\mathbf{v}$  with  $\mathbf{Av} = \mathbf{v}$ .

Each  $\mathbf{x}_k$  is a probability vector by [Proposition 8.11.3](#). A steady state is a right eigenvector for the eigenvalue 1, scaled so that its entries add up to 1 and required to have non-negative entries; if the chain starts there, it stays there.

To talk about the long run we need limits, and we use the simplest kind. A sequence of complex matrices  $\mathbf{M}_k \in M_{m \times n}(\mathbb{C})$  converges entrywise to  $\mathbf{M}$  if, for every position  $(r, s)$ , the number sequence  $(\mathbf{M}_k)_{rs}$  converges to  $\mathbf{M}_{rs}$ . We use four facts about sequences of complex numbers from calculus: sums and constant multiples of convergent sequences converge to the sums and multiples of the limits;  $z_k \rightarrow 0$  if and only if  $|z_k| \rightarrow 0$ ;  $r^k \rightarrow 0$  for real  $0 \leq r < 1$ ; and a convergent sequence of real numbers  $\geq 0$  has a real limit  $\geq 0$ . In particular, if  $|\lambda| < 1$ , then  $|\lambda^k| = |\lambda|^k \rightarrow 0$  ([Theorem 0.6.2](#)), so  $\lambda^k \rightarrow 0$ .

#### Example. (A Two-State Chain)

A machine is either working (state 1) or under repair (state 2) each day. A working machine is still working the next day with probability 9/10; a machine under repair is back to work the next day with probability 1/5. So the transition matrix is

$$\mathbf{A} = \begin{pmatrix} 9/10 & 1/5 \\ 1/10 & 4/5 \end{pmatrix}.$$

Find  $\mathbf{A}^k$ , its entrywise limit, and the long-run probability that the machine is working, if it is working on day 0.

*Solution. Eigenvalues.*  $\operatorname{tr} \mathbf{A} = 17/10$  and  $\det \mathbf{A} = 36/50 - 1/50 = 7/10$ , so, by the  $2 \times 2$  formula  $p_{\mathbf{A}} = x^2 - (\operatorname{tr} \mathbf{A})x + \det \mathbf{A}$  (the examples after [Definition 6.10.1](#)),  $p_{\mathbf{A}} = x^2 - \frac{17}{10}x + \frac{7}{10} = (x-1)(x-\frac{7}{10})$ . The eigenvalue 1 was guaranteed by [Proposition 8.11.3](#).

*Eigenvectors.*  $\mathbf{A} - \mathbf{I} = \begin{pmatrix} -1/10 & 1/5 \\ 1/10 & -1/5 \end{pmatrix}$  gives  $x = 2y$ , eigenvector  $(2, 1)$ .  $\mathbf{A} - \frac{7}{10}\mathbf{I} = \begin{pmatrix} 1/5 & 1/5 \\ 1/10 & 1/10 \end{pmatrix}$  gives  $y = -x$ , eigenvector  $(1, -1)$ . The eigenvalues are distinct, so

$$\mathbf{P} = \begin{pmatrix} 2 & 1 \\ 1 & -1 \end{pmatrix}, \quad \mathbf{D} = \operatorname{diag}\left(1, \frac{7}{10}\right), \quad \mathbf{P}^{-1} = \frac{1}{3} \begin{pmatrix} 1 & 1 \\ 1 & -2 \end{pmatrix},$$

using [Theorem 0.9.8](#) with  $\det \mathbf{P} = -3$ .

*Powers.* By [Theorem 8.4.3](#), writing  $t = (7/10)^k$ ,

$$\mathbf{A}^k = \mathbf{P} \begin{pmatrix} 1 & 0 \\ 0 & t \end{pmatrix} \mathbf{P}^{-1} = \frac{1}{3} \begin{pmatrix} 2 & t \\ 1 & -t \end{pmatrix} \begin{pmatrix} 1 & 1 \\ 1 & -2 \end{pmatrix} = \frac{1}{3} \begin{pmatrix} 2+t & 2-2t \\ 1-t & 1+2t \end{pmatrix}.$$

*Check:* for  $k = 1$ ,  $t = 7/10$  and  $\frac{1}{3} \begin{pmatrix} 27/10 & 6/10 \\ 3/10 & 24/10 \end{pmatrix} = \mathbf{A}$ ; each column of  $\mathbf{A}^k$  sums to 1, as [Proposition 8.11.3](#) requires.

*The limit.* Since  $0 \leq 7/10 < 1$ ,  $t \rightarrow 0$ , so entrywise

$$\mathbf{A}^k \rightarrow \frac{1}{3} \begin{pmatrix} 2 & 2 \\ 1 & 1 \end{pmatrix} = \begin{pmatrix} 2/3 \\ 1/3 \end{pmatrix} \begin{pmatrix} 1 & 1 \end{pmatrix}.$$

Starting from  $\mathbf{x}_0 = \mathbf{e}_1$ ,  $\mathbf{x}_k = \mathbf{A}^k \mathbf{e}_1 = \left(\frac{2+t}{3}, \frac{1-t}{3}\right) \rightarrow \left(\frac{2}{3}, \frac{1}{3}\right)$ . In the long run the machine works on two days out of three. The limit does not depend on the start: for any probability vector  $\mathbf{x}_0 = (s, 1-s)$ ,  $\mathbf{A}^k \mathbf{x}_0 \rightarrow \frac{1}{3}(2s + 2(1-s), s + (1-s)) = \left(\frac{2}{3}, \frac{1}{3}\right)$ . The vector  $\mathbf{v} = \left(\frac{2}{3}, \frac{1}{3}\right)$  is the steady state, the eigenvector  $(2, 1)$  scaled to sum 1.

The example shows the general mechanism. The eigenvalue 1 survives in  $\mathbf{D}^k$ , all others fade, and the surviving rank-one piece is (right eigenvector for 1) times (left eigenvector for 1), which for a stochastic matrix is the steady state times  $\mathbf{1}^\top$ .

### Theorem 8.11.4 — Long-Run Behavior of a Diagonalizable Markov Chain

Let  $\mathbf{A} \in M_n(\mathbb{R})$ ,  $n \geq 1$ , be stochastic. Suppose that  $\mathbf{A}$  is diagonalizable over  $\mathbb{C}$ , that  $a_{\mathbf{A}}(1) = 1$ , and that every other eigenvalue  $\lambda \in \mathbb{C}$  of  $\mathbf{A}$  has  $|\lambda| < 1$ . Then:

- (a)  $\mathbf{A}$  has exactly one steady state  $\mathbf{v}$ ;
- (b)  $\mathbf{A}^k \rightarrow \mathbf{v}\mathbf{1}^\top$  entrywise as  $k \rightarrow \infty$ ;
- (c) for every probability vector  $\mathbf{x}_0$ , the Markov chain  $\mathbf{x}_k = \mathbf{A}^k \mathbf{x}_0$  converges entrywise to  $\mathbf{v}$ .



*Idea.* Use the expansion  $\mathbf{A}^k = \sum_i \lambda_i^k \mathbf{v}_i \mathbf{y}_i^\top$  of Section 10. The terms with  $|\lambda_i| < 1$  tend to 0, which leaves the single term for  $\lambda_1 = 1$ . Its left eigenvector  $\mathbf{y}_1$  must be a multiple of  $\mathbf{1}$ , because the left eigenvectors for 1 form a line and  $\mathbf{1}$  is one of them. Rescaling turns  $\mathbf{v}_1 \mathbf{y}_1^\top$  into  $\mathbf{v} \mathbf{1}^\top$ , and the non-negativity of  $\mathbf{v}$  comes for free as a limit of non-negative entries.

*Proof.* Since  $\mathbf{A}$  is diagonalizable over  $\mathbb{C}$ , there are an invertible  $\mathbf{P} \in M_n(\mathbb{C})$  and  $\mathbf{D} = \text{diag}(\lambda_1, \dots, \lambda_n)$  with  $\mathbf{A} = \mathbf{P} \mathbf{D} \mathbf{P}^{-1}$ . A diagonal matrix is upper triangular, so each eigenvalue occurs on the diagonal of  $\mathbf{D}$  as often as its algebraic multiplicity (Theorem 8.7.4 (b)); in particular 1 occurs exactly once, because  $a_{\mathbf{A}}(1) = 1$ . Permuting the columns of  $\mathbf{P}$  and the diagonal entries of  $\mathbf{D}$  together leaves  $\mathbf{A} = \mathbf{P} \mathbf{D} \mathbf{P}^{-1}$  intact, so we may assume  $\lambda_1 = 1$ , and then  $|\lambda_i| < 1$  for  $i \geq 2$  by hypothesis. Let  $\mathbf{v}_i$  be the columns of  $\mathbf{P}$  and  $\mathbf{y}_i^\top$  the rows of  $\mathbf{P}^{-1}$ . By Proposition 8.10.7, for every  $k$ ,

$$\mathbf{A}^k = \mathbf{v}_1 \mathbf{y}_1^\top + \sum_{i=2}^n \lambda_i^k \mathbf{v}_i \mathbf{y}_i^\top.$$

Each entry of the sum is  $\sum_{i \geq 2} \lambda_i^k z_i$  for fixed complex numbers  $z_i$ , and  $\lambda_i^k \rightarrow 0$  for  $i \geq 2$ . Hence  $\mathbf{A}^k \rightarrow \mathbf{L} = \mathbf{v}_1 \mathbf{y}_1^\top$  entrywise.

*The shape of L.* By Proposition 8.10.7,  $\mathbf{y}_1$  is a left eigenvector for 1, and so is  $\mathbf{1}$  (Proposition 8.11.3). Both lie in  $E_1(\mathbf{A}^\top) \subseteq \mathbb{C}^n$  (Proposition 8.10.5 (a)), which has dimension  $g_{\mathbf{A}^\top}(1) = g_{\mathbf{A}}(1) \leq a_{\mathbf{A}}(1) = 1$  by Proposition 8.10.5 (b) and Theorem 8.3.1. Since  $\mathbf{1} \neq \mathbf{0}$ ,  $\mathbf{y}_1 = c\mathbf{1}$  for some  $c \in \mathbb{C}$ . Put  $\mathbf{v} = c\mathbf{v}_1$ . Then  $\mathbf{L} = \mathbf{v} \mathbf{1}^\top$ ,  $\mathbf{A} \mathbf{v} = c \mathbf{A} \mathbf{v}_1 = \mathbf{v}$ , and  $\mathbf{1}^\top \mathbf{v} = \mathbf{y}_1^\top \mathbf{v}_1 = 1$  by Proposition 8.10.7 (b).

*v is a steady state.* Column 1 of  $\mathbf{L} = \mathbf{v} \mathbf{1}^\top$  is  $\mathbf{v}$ . It is the limit of column 1 of  $\mathbf{A}^k$ , and  $\mathbf{A}^k$  is stochastic (Proposition 8.11.3), so every entry of  $\mathbf{v}$  is a limit of real numbers  $\geq 0$ , hence real and  $\geq 0$ . With  $\mathbf{1}^\top \mathbf{v} = 1$ ,  $\mathbf{v}$  is a probability vector, and  $\mathbf{A} \mathbf{v} = \mathbf{v}$ .

*Uniqueness.* If  $\mathbf{w}$  is a steady state, then  $\mathbf{w} \in E_1(\mathbf{A})$ , a subspace of  $\mathbb{C}^n$  of dimension  $g_{\mathbf{A}}(1) \leq 1$  that contains  $\mathbf{v} \neq \mathbf{0}$ . So  $\mathbf{w} = s\mathbf{v}$  for some  $s \in \mathbb{C}$ , and  $1 = \mathbf{1}^\top \mathbf{w} = s \mathbf{1}^\top \mathbf{v} = s$ . Hence  $\mathbf{w} = \mathbf{v}$ . This proves (a) and (b).

(c) Entry  $r$  of  $\mathbf{A}^k \mathbf{x}_0$  is  $\sum_s (\mathbf{A}^k)_{rs} (\mathbf{x}_0)_s$ , which by (b) converges to  $\sum_s v_r (\mathbf{x}_0)_s = v_r (\mathbf{1}^\top \mathbf{x}_0) = v_r$ . ■

In practice: find the eigenvalues, check the hypotheses, and compute  $\mathbf{v}$  by solving  $(\mathbf{A} - \mathbf{I})\mathbf{v} = \mathbf{0}$  and scaling. Then (c) gives the long run **without** computing  $\mathbf{P}^{-1}$  at all.

### Example. (Bicycles at Three Stations)

A shared bicycle is parked each night at station 1, 2 or 3. The transition matrix is

$$\mathbf{A} = \frac{1}{4} \begin{pmatrix} 2 & 1 & 0 \\ 1 & 2 & 1 \\ 1 & 1 & 3 \end{pmatrix};$$

for instance a bicycle at station 3 stays there with probability  $3/4$  and never goes to station 1 in one night. Find  $\mathbf{A}^k$ , its limit, and the long-run distribution of the bicycle.

*Solution. Eigenvectors.* We check three candidates:

$$\mathbf{A} \begin{pmatrix} 1 \\ 2 \\ 3 \end{pmatrix} = \frac{1}{4} \begin{pmatrix} 4 \\ 8 \\ 12 \end{pmatrix}, \quad \mathbf{A} \begin{pmatrix} -1 \\ 0 \\ 1 \end{pmatrix} = \frac{1}{4} \begin{pmatrix} -2 \\ 0 \\ 2 \end{pmatrix}, \quad \mathbf{A} \begin{pmatrix} -1 \\ 1 \\ 0 \end{pmatrix} = \frac{1}{4} \begin{pmatrix} -1 \\ 1 \\ 0 \end{pmatrix}.$$

So  $1, \frac{1}{2}, \frac{1}{4}$  are eigenvalues, with eigenvectors  $\mathbf{v}_1 = (1, 2, 3)$ ,  $\mathbf{v}_2 = (-1, 0, 1)$ ,  $\mathbf{v}_3 = (-1, 1, 0)$ . A  $3 \times 3$  matrix has at most three eigenvalues (Corollary 8.2.3), so these are all; they are distinct, so  $\mathbf{A}$  is diagonalizable (Corollary 8.4.2) and  $a_{\mathbf{A}}(1) = 1$ . As a check,  $\text{tr } \mathbf{A} = \frac{7}{4} = 1 + \frac{1}{2} + \frac{1}{4}$ . The hypotheses of Theorem 8.11.4 hold.

*The expansion.* With  $\mathbf{P} = \begin{pmatrix} 1 & -1 & -1 \\ 2 & 0 & 1 \\ 3 & 1 & 0 \end{pmatrix}$ , whose determinant is  $-6$ ,

$$\mathbf{P}^{-1} = \begin{pmatrix} 1/6 & 1/6 & 1/6 \\ -1/2 & -1/2 & 1/2 \\ -1/3 & 2/3 & -1/3 \end{pmatrix},$$

as multiplying out  $\mathbf{P}^{-1}\mathbf{P} = \mathbf{I}$  confirms. The first row is  $\frac{1}{6}\mathbf{1}^\top$ , a multiple of  $\mathbf{1}^\top$  as the proof of the theorem predicts. By Proposition 8.10.7,

$$\mathbf{A}^k = \frac{1}{6} \begin{pmatrix} 1 & 1 & 1 \\ 2 & 2 & 2 \\ 3 & 3 & 3 \end{pmatrix} + \frac{1}{2^k} \cdot \frac{1}{2} \begin{pmatrix} 1 & 1 & -1 \\ 0 & 0 & 0 \\ -1 & -1 & 1 \end{pmatrix} + \frac{1}{4^k} \cdot \frac{1}{3} \begin{pmatrix} 1 & -2 & 1 \\ -1 & 2 & -1 \\ 0 & 0 & 0 \end{pmatrix}.$$

For example, the  $(1, 1)$ -entry of  $\mathbf{A}^k$  is  $\frac{1}{6} + \frac{1}{2} \cdot 2^{-k} + \frac{1}{3} \cdot 4^{-k}$ , which is 1 for  $k = 0$  and  $\frac{1}{6} + \frac{1}{4} + \frac{1}{12} = \frac{1}{2}$  for  $k = 1$ , matching  $a_{11} = 2/4$ .

*The limit.* As  $k \rightarrow \infty$ ,  $2^{-k} \rightarrow 0$  and  $4^{-k} \rightarrow 0$ , so

$$\mathbf{A}^k \rightarrow \frac{1}{6} \begin{pmatrix} 1 & 1 & 1 \\ 2 & 2 & 2 \\ 3 & 3 & 3 \end{pmatrix} = \mathbf{v}\mathbf{1}^\top, \quad \mathbf{v} = \left(\frac{1}{6}, \frac{1}{3}, \frac{1}{2}\right).$$

Wherever the bicycle starts, in the long run it is at station 1, 2, 3 with probabilities  $\frac{1}{6}, \frac{1}{3}, \frac{1}{2}$ . Directly,  $\mathbf{v} = \frac{1}{6}\mathbf{v}_1$  and  $\mathbf{A}\mathbf{v} = \mathbf{v}$ . The slowest-fading term is the one with  $\frac{1}{2}$ : each entry's gap to its limit roughly halves every night.

**Warning.** A stochastic matrix need not have convergent powers. The swap  $\mathbf{S} = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}$  is stochastic, and  $\mathbf{S}^k$  is  $\mathbf{I}$  for even  $k$  and  $\mathbf{S}$  for odd  $k$ , so  $\mathbf{S}^k$  does not converge. Starting from  $\mathbf{x}_0 = \mathbf{e}_1$ , the chain alternates  $\mathbf{e}_1, \mathbf{e}_2, \mathbf{e}_1, \dots$  forever, although  $\mathbf{S}$  has the steady state  $(\frac{1}{2}, \frac{1}{2})$ . The theorem does not apply:  $\mathbf{S}$  has the eigenvalue  $-1$ , with  $|-1| = 1$ . At the other extreme,  $\mathbf{I}_2$  has convergent powers, but  $a(1) = 2$ , every probability vector is a steady state, and the limit of  $\mathbf{x}_k$  depends on  $\mathbf{x}_0$ .

Chapter 18 proves the Perron–Frobenius theorem, which gives the conclusions of Theorem 8.11.4 for every stochastic matrix with some power having all entries positive, without assuming diagonalizability and without computing eigenvalues. The exercises prove the first ingredient: no

eigenvalue of a stochastic matrix has  $|\lambda| > 1$ .

*Quick check.* Is  $\mathbf{A} = \begin{pmatrix} 1/2 & 1/3 \\ 1/2 & 2/3 \end{pmatrix}$  stochastic? If so, find its steady state.

*Solution.* Yes: the entries are non-negative and the columns sum to 1. From  $(\mathbf{A} - \mathbf{I})\mathbf{v} = \mathbf{0}$ , the first row gives  $-\frac{1}{2}v_1 + \frac{1}{3}v_2 = 0$ , so  $v_2 = \frac{3}{2}v_1$ . Scaling to sum 1:  $v_1 + \frac{3}{2}v_1 = 1$  gives  $\mathbf{v} = (\frac{2}{5}, \frac{3}{5})$ . (The eigenvalues are 1 and  $\text{tr } \mathbf{A} - 1 = \frac{1}{6}$ ; they are distinct, so  $\mathbf{A}$  is diagonalizable with  $a_{\mathbf{A}}(1) = 1$ , and  $\mathbf{v}$  is the only steady state and the long-run distribution, by [Theorem 8.11.4](#).)

## 8.11.5 A population model

Not every matrix of this kind is stochastic. In a population divided into age classes, individuals are born, age, and die, and the total need not be preserved. A **Leslie matrix** records, in its first row, the average number of offspring per individual of each age class, and just below the diagonal, the fraction surviving into the next class. The population vector evolves by  $\mathbf{x}_{k+1} = \mathbf{L}\mathbf{x}_k$ , and [Theorem 8.11.1](#) describes the long run.

### Example. (Three Age Classes)

A species is counted once a year in three age classes: young, juvenile and adult. Half of the young survive to become juveniles, and half of the juveniles survive to become adults; no adult survives another year. Each juvenile has on average  $7/2$  offspring and each adult 3. The Leslie matrix is

$$\mathbf{L} = \begin{pmatrix} 0 & 7/2 & 3 \\ 1/2 & 0 & 0 \\ 0 & 1/2 & 0 \end{pmatrix}.$$

Starting from  $\mathbf{x}_0 = (20, 10, 5)$  (in hundreds of animals), find  $\mathbf{x}_k$  and describe the long-run growth and age distribution.

*Solution. Characteristic polynomial.* Expanding  $\det(x\mathbf{I} - \mathbf{L})$  along the first column,

$$\begin{aligned} p_{\mathbf{L}} &= x \det \begin{pmatrix} x & 0 \\ -1/2 & x \end{pmatrix} + \frac{1}{2} \det \begin{pmatrix} -7/2 & -3 \\ -1/2 & x \end{pmatrix} \\ &= x^3 + \frac{1}{2} \left( -\frac{7}{2}x - \frac{3}{2} \right) = x^3 - \frac{7}{4}x - \frac{3}{4}. \end{aligned}$$

Trying simple values,  $p_{\mathbf{L}}(-1) = -1 + \frac{7}{4} - \frac{3}{4} = 0$ , and dividing,  $p_{\mathbf{L}} = (x+1)(x^2 - x - \frac{3}{4}) = (x+1)(x - \frac{3}{2})(x + \frac{1}{2})$ . The eigenvalues  $\frac{3}{2}, -1, -\frac{1}{2}$  are distinct, so  $\mathbf{L}$  is diagonalizable.

*Eigenvectors.*  $\mathbf{L}(w_1, w_2, w_3) = \lambda(w_1, w_2, w_3)$  forces  $\frac{1}{2}w_1 = \lambda w_2$  and  $\frac{1}{2}w_2 = \lambda w_3$ , so taking  $w_3 = 1$  gives  $w_2 = 2\lambda$ ,  $w_1 = 4\lambda^2$ . Hence, after scaling,

$$\lambda = \frac{3}{2} : (9, 3, 1), \quad \lambda = -1 : (4, -2, 1), \quad \lambda = -\frac{1}{2} : (1, -1, 1).$$

For instance  $\mathbf{L}(9, 3, 1) = (\frac{21}{2} + 3, \frac{9}{2}, \frac{3}{2}) = \frac{3}{2}(9, 3, 1)$ .

*The initial vector.* Solving  $c_1(9, 3, 1) + c_2(4, -2, 1) + c_3(1, -1, 1) = (20, 10, 5)$  gives  $c_1 = 3$ ,

$c_2 = -3, c_3 = 5$ : indeed  $(27 - 12 + 5, 9 + 6 - 5, 3 - 3 + 5) = (20, 10, 5)$ . By [Theorem 8.11.1](#),

$$\mathbf{x}_k = 3\left(\frac{3}{2}\right)^k \begin{pmatrix} 9 \\ 3 \\ 1 \end{pmatrix} - 3(-1)^k \begin{pmatrix} 4 \\ -2 \\ 1 \end{pmatrix} + 5\left(-\frac{1}{2}\right)^k \begin{pmatrix} 1 \\ -1 \\ 1 \end{pmatrix}.$$

*Check.* For  $k = 1$ :  $(\frac{81}{2} + 12 - \frac{5}{2}, \frac{27}{2} - 6 + \frac{5}{2}, \frac{9}{2} + 3 - \frac{5}{2}) = (50, 10, 5)$ , and directly  $L\mathbf{x}_0 = (35 + 15, 10, 5)$ .

*The long run.* Divide by  $(3/2)^k$ :

$$\left(\frac{2}{3}\right)^k \mathbf{x}_k = 3 \begin{pmatrix} 9 \\ 3 \\ 1 \end{pmatrix} - 3\left(-\frac{2}{3}\right)^k \begin{pmatrix} 4 \\ -2 \\ 1 \end{pmatrix} + 5\left(-\frac{1}{3}\right)^k \begin{pmatrix} 1 \\ -1 \\ 1 \end{pmatrix} \rightarrow \begin{pmatrix} 27 \\ 9 \\ 3 \end{pmatrix},$$

since  $|-2/3| < 1$  and  $|-1/3| < 1$ . So the population eventually grows by the factor  $\frac{3}{2}$  each year, and the age classes approach the proportions  $9 : 3 : 1$ , the entries of the eigenvector for the largest eigenvalue. The eigenvalue  $-1$  makes the population oscillate from year to year, but its effect is bounded while the dominant term grows, so its share fades.

## 8.11.6 Exercises

### 8.11.6.1 A. Check your understanding

#### Exercise A1.

- Define a stochastic matrix, and say which convention (rows or columns) this book uses.
- True or false: for every stochastic  $\mathbf{A}$ , the powers  $\mathbf{A}^k$  converge entrywise. Justify your answer.
- True or false:  $\mathbf{1}$  is a right eigenvector of every stochastic matrix. Justify your answer.
- True or false: every stochastic matrix is invertible. Justify your answer.
- Write down the general solution of  $a_{k+2} = 5a_{k+1} - 6a_k$  over  $\mathbb{R}$ .
- In [Theorem 8.11.1](#), what are the numbers  $c_1, \dots, c_n$ ?

*Solution.*

- $\mathbf{A} \in M_n(\mathbb{R})$  is stochastic if all entries are non-negative and every **column** sums to 1 ([Definition 8.11.1](#)). This book uses columns, with  $\mathbf{x}_{k+1} = \mathbf{A}\mathbf{x}_k$ .
- False. The swap  $\mathbf{S} = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}$  has  $\mathbf{S}^k$  alternating between  $\mathbf{I}$  and  $\mathbf{S}$ .
- False.  $\mathbf{A} = \begin{pmatrix} 1 & 1/2 \\ 0 & 1/2 \end{pmatrix}$  is stochastic, and  $\mathbf{A}\mathbf{1} = (3/2, 1/2)$  is not a multiple of  $\mathbf{1}$ . It is a **left** eigenvector ([Proposition 8.11.3](#)).
- False.  $\begin{pmatrix} 1/2 & 1/2 \\ 1/2 & 1/2 \end{pmatrix}$  is stochastic with determinant 0.
- $p = x^2 - 5x + 6 = (x - 2)(x - 3)$ , so by [Theorem 8.11.2](#) the solutions are  $a_k =$

$b_1 2^k + b_2 3^k$  with  $b_1, b_2 \in \mathbb{R}$ .

- (f) The coordinates of  $\mathbf{x}_0$  in the basis of eigenvectors  $\mathbf{v}_1, \dots, \mathbf{v}_n$ :  $(c_1, \dots, c_n) = \mathbf{P}^{-1} \mathbf{x}_0$ .

### 8.11.6.2 B. Practice

#### Exercise B1: Two recurrences.

- (a) Solve  $a_{k+2} = a_{k+1} + 2a_k$  with  $a_0 = 0, a_1 = 3$ .  
 (b) Solve  $a_{k+2} = 6a_{k+1} - 9a_k$  with  $a_0 = 1, a_1 = 6$ , by the method of [Example \(A Repeated Root\)](#).

*Solution.*

- (a)  $p = x^2 - x - 2 = (x - 2)(x + 1)$  has distinct roots, so by [Theorem 8.11.2](#),  $a_k = b_1 2^k + b_2 (-1)^k$ . From  $a_0 = b_1 + b_2 = 0$  and  $a_1 = 2b_1 - b_2 = 3$ ,  $3b_1 = 3$ , so  $b_1 = 1, b_2 = -1$ . Hence  $a_k = 2^k - (-1)^k$ . Check:  $a_2 = 3 + 0 = 3 = 4 - 1$ ,  $a_3 = 3 + 6 = 9 = 8 + 1$ .  
 (b) The window matrix is  $\mathbf{C} = \begin{pmatrix} 0 & 1 \\ -9 & 6 \end{pmatrix}$ , with  $p = (x - 3)^2$ . Put  $\mathbf{N} = \mathbf{C} - 3\mathbf{I} = \begin{pmatrix} -3 & 1 \\ -9 & 3 \end{pmatrix}$ ; then  $\mathbf{N}^2 = \begin{pmatrix} 9 - 9 & -3 + 3 \\ 27 - 27 & -9 + 9 \end{pmatrix} = \mathbf{0}$ . As in the example, by induction  $\mathbf{C}^k = 3^k \mathbf{I} + k 3^{k-1} \mathbf{N}$ : the step is  $(3^k \mathbf{I} + k 3^{k-1} \mathbf{N})(3\mathbf{I} + \mathbf{N}) = 3^{k+1} \mathbf{I} + 3^k \mathbf{N} + k 3^k \mathbf{N}$ . Row 1 of  $\mathbf{C}^k$  is  $(3^k - 3k 3^{k-1}, k 3^{k-1}) = ((1 - k)3^k, k 3^{k-1})$ , so

$$a_k = (1 - k)3^k \cdot 1 + k 3^{k-1} \cdot 6 = (1 - k + 2k)3^k = (1 + k)3^k.$$

Check:  $a_2 = 36 - 9 = 27 = 3 \cdot 9$ ,  $a_3 = 6 \cdot 27 - 9 \cdot 6 = 108 = 4 \cdot 27$ .

#### Exercise B2: A three-state chain.

Let

$$\mathbf{A} = \begin{pmatrix} 1/2 & 1/2 & 1/4 \\ 1/4 & 0 & 1/4 \\ 1/4 & 1/2 & 1/2 \end{pmatrix}.$$

- (a) Verify that  $\mathbf{A}$  is stochastic, and find its steady state.  
 (b) Verify that  $(-1, 0, 1)$  and  $(1, -2, 1)$  are eigenvectors of  $\mathbf{A}$ , and find their eigenvalues.  
 (c) Hence find the entrywise limit of  $\mathbf{A}^k$ , and the limit of the chain started at  $\mathbf{x}_0 = \mathbf{e}_2$ .

*Solution.*

- (a) All entries are non-negative; the columns sum to  $\frac{1}{2} + \frac{1}{4} + \frac{1}{4} = 1$ ,  $\frac{1}{2} + 0 + \frac{1}{2} = 1$  and  $\frac{1}{4} + \frac{1}{4} + \frac{1}{2} = 1$ . For the steady state, solve  $(\mathbf{A} - \mathbf{I})\mathbf{v} = \mathbf{0}$ :

$$\mathbf{A} - \mathbf{I} = \begin{pmatrix} -1/2 & 1/2 & 1/4 \\ 1/4 & -1 & 1/4 \\ 1/4 & 1/2 & -1/2 \end{pmatrix}.$$

Multiplying the rows by 4:  $-2v_1 + 2v_2 + v_3 = 0$ ,  $v_1 - 4v_2 + v_3 = 0$ ,  $v_1 + 2v_2 - 2v_3 = 0$ .

Subtracting the second from the third gives  $6v_2 - 3v_3 = 0$ , so  $v_3 = 2v_2$ ; then the second gives  $v_1 = 4v_2 - 2v_2 = 2v_2$ , and the first holds:  $-4v_2 + 2v_2 + 2v_2 = 0$ . So  $E_1(\mathbf{A}) = \text{span}((2, 1, 2))$ , and scaling to sum 1, the steady state is  $\mathbf{v} = (\frac{2}{5}, \frac{1}{5}, \frac{2}{5})$ .

- (b)  $\mathbf{A}(-1, 0, 1) = (-\frac{1}{2} + \frac{1}{4}, -\frac{1}{4} + \frac{1}{4}, -\frac{1}{4} + \frac{1}{2}) = (-\frac{1}{4}, 0, \frac{1}{4}) = \frac{1}{4}(-1, 0, 1)$ , eigenvalue  $\frac{1}{4}$ .  $\mathbf{A}(1, -2, 1) = (\frac{1}{2} - 1 + \frac{1}{4}, \frac{1}{4} + \frac{1}{4}, \frac{1}{4} - 1 + \frac{1}{2}) = (-\frac{1}{4}, \frac{1}{2}, -\frac{1}{4}) = -\frac{1}{4}(1, -2, 1)$ , eigenvalue  $-\frac{1}{4}$ .
- (c) The eigenvalues  $1, \frac{1}{4}, -\frac{1}{4}$  are distinct, so they are all the eigenvalues of  $\mathbf{A}$ ,  $\mathbf{A}$  is diagonalizable (Corollary 8.4.2),  $a_{\mathbf{A}}(1) = 1$ , and the other two have absolute value  $\frac{1}{4} < 1$ . By Theorem 8.11.4,

$$\mathbf{A}^k \rightarrow \mathbf{v}\mathbf{1}^\top = \frac{1}{5} \begin{pmatrix} 2 & 2 & 2 \\ 1 & 1 & 1 \\ 2 & 2 & 2 \end{pmatrix},$$

and every chain, in particular the one started at  $\mathbf{e}_2$ , converges to  $\mathbf{v} = (\frac{2}{5}, \frac{1}{5}, \frac{2}{5})$ .

### Exercise B3: The limit of a matrix power.

Let  $\mathbf{A} = \begin{pmatrix} 0 & 1 \\ -1/2 & 3/2 \end{pmatrix} \in M_2(\mathbb{R})$ .

- (a) Diagonalize  $\mathbf{A}$ , and find a formula for  $\mathbf{A}^k$ .
- (b) Find the entrywise limit of  $\mathbf{A}^k$ .
- (c) Hence find  $\lim_{k \rightarrow \infty} a_k$  for the sequence with  $a_{k+2} = \frac{3}{2}a_{k+1} - \frac{1}{2}a_k$ , in terms of  $a_0$  and  $a_1$ .

*Solution.*

- (a)  $\text{tr } \mathbf{A} = \frac{3}{2}$  and  $\det \mathbf{A} = \frac{1}{2}$ , so  $p_{\mathbf{A}} = x^2 - \frac{3}{2}x + \frac{1}{2} = (x-1)(x-\frac{1}{2})$ .  $\mathbf{A} - \mathbf{I} = \begin{pmatrix} -1 & 1 \\ -1/2 & 1/2 \end{pmatrix}$  gives the eigenvector  $(1, 1)$ ;  $\mathbf{A} - \frac{1}{2}\mathbf{I} = \begin{pmatrix} -1/2 & 1 \\ -1/2 & 1 \end{pmatrix}$  gives  $(2, 1)$ . So  $\mathbf{P} = \begin{pmatrix} 1 & 2 \\ 1 & 1 \end{pmatrix}$ ,  $\mathbf{D} = \text{diag}(1, \frac{1}{2})$ ,  $\mathbf{P}^{-1} = \begin{pmatrix} -1 & 2 \\ 1 & -1 \end{pmatrix}$  ( $\det \mathbf{P} = -1$ , Theorem 0.9.8). By Theorem 8.4.3, with  $t = 2^{-k}$ ,

$$\mathbf{A}^k = \begin{pmatrix} 1 & 2t \\ 1 & t \end{pmatrix} \begin{pmatrix} -1 & 2 \\ 1 & -1 \end{pmatrix} = \begin{pmatrix} -1+2t & 2-2t \\ -1+t & 2-t \end{pmatrix}.$$

For  $k = 1$ ,  $t = \frac{1}{2}$  gives  $\begin{pmatrix} 0 & 1 \\ -1/2 & 3/2 \end{pmatrix} = \mathbf{A}$ .

- (b) Since  $t = 2^{-k} \rightarrow 0$ ,  $\mathbf{A}^k \rightarrow \begin{pmatrix} -1 & 2 \\ -1 & 2 \end{pmatrix}$  entrywise.
- (c)  $\mathbf{A}$  is the window matrix of this recurrence, so  $(a_k, a_{k+1}) = \mathbf{A}^k(a_0, a_1)$  and  $a_k = (-1+2t)a_0 + (2-2t)a_1$ . Hence  $a_k \rightarrow 2a_1 - a_0$ . (For instance  $a_0 = 0$ ,  $a_1 = 1$  gives  $0, 1, \frac{3}{2}, \frac{7}{4}, \dots \rightarrow 2$ .)

### 8.11.6.3 C. Going deeper

**Exercise C1: Eigenvalues of a stochastic matrix.**

Let  $\mathbf{A} \in M_n(\mathbb{R})$  be stochastic, regarded as a complex matrix, and let  $\lambda \in \mathbb{C}$  be an eigenvalue of  $\mathbf{A}$ .

- (a) Prove that  $|\lambda| \leq 1$ .
- (b) Refine the argument: prove that  $|\lambda - a_{ii}| \leq 1 - a_{ii}$  for some index  $i$ .
- (c) Deduce that if every diagonal entry of  $\mathbf{A}$  is positive, then  $-1$  is not an eigenvalue of  $\mathbf{A}$ .
- (d) Show by an example that (c) fails if a diagonal entry is 0.

*Hint: take a left eigenvector  $\mathbf{y}$  for  $\lambda$  and an index  $i$  with  $|y_i|$  as large as possible.*

*Solution.*

- (a) By [Proposition 8.10.5](#) (over  $\mathbb{C}$ ), there is  $\mathbf{y} \in \mathbb{C}^n$ ,  $\mathbf{y} \neq \mathbf{0}$ , with  $\mathbf{A}^\top \mathbf{y} = \lambda \mathbf{y}$ . Choose  $i$  with  $|y_i| \geq |y_j|$  for all  $j$ ; then  $|y_i| > 0$ , since  $\mathbf{y} \neq \mathbf{0}$ . Entry  $i$  of  $\mathbf{A}^\top \mathbf{y} = \lambda \mathbf{y}$  reads  $\sum_j a_{ji} y_j = \lambda y_i$ , because row  $i$  of  $\mathbf{A}^\top$  is column  $i$  of  $\mathbf{A}$ . By [Theorem 0.6.2](#) (d) and [Theorem 0.6.4](#), applied to the  $n$ -term sum by induction on  $n$ , and since  $a_{ji} \geq 0$  and column  $i$  sums to 1,

$$|\lambda| |y_i| = \left| \sum_j a_{ji} y_j \right| \leq \sum_j a_{ji} |y_j| \leq \sum_j a_{ji} |y_i| = |y_i|.$$

Dividing by  $|y_i| > 0$  gives  $|\lambda| \leq 1$ .

- (b) With the same  $i$ , move the diagonal term across:  $(\lambda - a_{ii})y_i = \sum_{j \neq i} a_{ji} y_j$ . Hence, as in (a),

$$|\lambda - a_{ii}| |y_i| \leq \sum_{j \neq i} a_{ji} |y_j| \leq \left( \sum_{j \neq i} a_{ji} \right) |y_i| = (1 - a_{ii}) |y_i|,$$

and dividing by  $|y_i|$  gives  $|\lambda - a_{ii}| \leq 1 - a_{ii}$ .

- (c) Suppose  $\lambda = -1$ . By (b), some  $i$  has  $|-1 - a_{ii}| \leq 1 - a_{ii}$ . Since  $a_{ii} \geq 0$ ,  $|-1 - a_{ii}| = 1 + a_{ii}$ , so  $1 + a_{ii} \leq 1 - a_{ii}$  and  $a_{ii} \leq 0$ . This contradicts  $a_{ii} > 0$ .
- (d) The swap  $\mathbf{S} = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}$  is stochastic with zero diagonal, and  $\mathbf{S}(1, -1) = (-1, 1)$ , so  $-1$  is an eigenvalue.

**Exercise C2: Two-state chains.**

Let  $0 \leq p, q \leq 1$  and  $\mathbf{A} = \begin{pmatrix} 1-p & q \\ p & 1-q \end{pmatrix}$ .

- (a) Show that  $\mathbf{A}$  is stochastic, with eigenvalues 1 and  $1 - p - q$ .
- (b) Suppose  $0 < p + q < 2$ . Prove that  $\mathbf{A}^k$  converges entrywise, and find the limit and the steady state.
- (c) In particular, if  $\mathbf{A}$  has no zero entry, then  $\mathbf{A}^k$  converges. What happens when  $p + q = 0$  and when  $p + q = 2$ ?

*Solution.*

- (a) The entries lie in  $[0, 1]$ , and the columns sum to  $(1-p)+p=1$  and  $q+(1-q)=1$ . Since  $\text{tr } \mathbf{A} = 2-p-q$  and  $\det \mathbf{A} = (1-p)(1-q)-pq = 1-p-q$ ,  $p_{\mathbf{A}} = x^2 - (2-p-q)x + (1-p-q) = (x-1)(x-(1-p-q))$ , as multiplying out confirms.
- (b) Put  $r = 1-p-q$ . Since  $0 < p+q < 2$ ,  $-1 < r < 1$ , so  $r \neq 1$  and  $|r| < 1$ . The eigenvalues  $1 \neq r$  are distinct, so  $\mathbf{A}$  is diagonalizable ([Corollary 8.4.2](#)). From  $\mathbf{A} - \mathbf{I} = \begin{pmatrix} -p & q \\ p & -q \end{pmatrix}$ ,  $(q, p)$  is an eigenvector for 1, non-zero because  $p+q > 0$ ; from  $\mathbf{A} - r\mathbf{I} = \begin{pmatrix} q & q \\ p & p \end{pmatrix}$ ,  $(1, -1)$  is one for  $r$ . So

$$\mathbf{P} = \begin{pmatrix} q & 1 \\ p & -1 \end{pmatrix}, \quad \mathbf{P}^{-1} = \frac{1}{p+q} \begin{pmatrix} 1 & 1 \\ p & -q \end{pmatrix},$$

with  $\det \mathbf{P} = -(p+q) \neq 0$ . By [Theorem 8.4.3](#),

$$\mathbf{A}^k = \frac{1}{p+q} \begin{pmatrix} q & r^k \\ p & -r^k \end{pmatrix} \begin{pmatrix} 1 & 1 \\ p & -q \end{pmatrix} = \frac{1}{p+q} \begin{pmatrix} q+pr^k & q-qr^k \\ p-pr^k & p+qr^k \end{pmatrix}.$$

Since  $|r| < 1$ ,  $r^k \rightarrow 0$ , and

$$\mathbf{A}^k \rightarrow \frac{1}{p+q} \begin{pmatrix} q & q \\ p & p \end{pmatrix}, \quad \mathbf{v} = \left( \frac{q}{p+q}, \frac{p}{p+q} \right).$$

The limit is  $\mathbf{v}\mathbf{1}^\top$ , and  $\mathbf{v}$  is the steady state, in agreement with [Theorem 8.11.4](#).

- (c) If no entry is zero, then  $1-p > 0$  and  $p > 0$ , so  $0 < p < 1$ , and likewise  $0 < q < 1$ ; hence  $0 < p+q < 2$  and (b) applies. If  $p+q=0$ , then  $p=q=0$  and  $\mathbf{A} = \mathbf{I}$ : the powers converge (to  $\mathbf{I}$ ), but every probability vector is a steady state. If  $p+q=2$ , then  $p=q=1$  and  $\mathbf{A}$  is the swap, whose powers do not converge.



# Chapter 9

## Canonical Forms

Chapter 8 left one question open. Two matrices represent the same operator in different bases exactly when they are similar, and we can now decide similarity in easy cases: diagonalizable matrices are similar precisely when they have the same eigenvalues with the same multiplicities. But the matrices  $\begin{pmatrix} 1 & 1 \\ 0 & 1 \end{pmatrix}$  and  $I_2$  share their characteristic and minimal polynomials, their rank, their trace and their determinant, and they are still not similar. Something is missing.

This chapter supplies it. When the characteristic polynomial splits, every operator has a **Jordan form**: a direct sum of blocks with one eigenvalue on the diagonal and ones just above it, unique up to the order of the blocks. Two matrices are similar exactly when their Jordan forms agree, so the similarity problem is solved. Over a field where the polynomial does not split, the **rational canonical form** does the same job, and the Smith normal form of  $x\mathbf{I} - \mathbf{A}$  computes it. The chapter closes with what these forms are for: functions of matrices, the exponential  $e^{t\mathbf{A}}$  and linear differential equations, and a complete answer to when the powers  $\mathbf{A}^m$  converge.

**What you need.** From Chapter 1, direct sums, basis extension and complexification ([Theorem 1.7.5](#), [Theorem 1.6.5](#), [Definition 1.9.2](#)). From Chapter 5, the arithmetic of  $F[x]$ : division, gcds, Bézout and unique factorization ([Theorem 5.1.1](#), [Corollary 5.2.4](#), [Theorem 5.3.5](#)), and interpolation ([Theorem 5.6.4](#)). From Chapter 6, determinants of block triangular matrices, Cauchy–Binet and the companion matrix of a monic polynomial ([Theorem 6.8.4](#), [Theorem 6.7.2](#), [Exercise \(C1: Companion matrices\)](#)). From Chapter 8, the minimal polynomial, primary decomposition and triangularization ([Definition 8.5.1](#), [Theorem 8.8.1](#), [Theorem 8.7.2](#)).

### 9.0.1 Roadmap

- **Generalized eigenspaces.** Enlarging an eigenspace until it is big enough, and the decomposition it produces.
- **Nilpotent operators.** Chains of vectors, the block structure they give, and how ranks of powers determine it.
- **The Jordan canonical form.** Existence, uniqueness, and the complete answer to the similarity question over a splitting field.
- **The real Jordan form and the Weyr form.** Avoiding complex numbers for a real matrix, and a rearrangement of the Jordan form with its own uses.
- **Cyclic subspaces and companion matrices.** One vector and its images can generate everything.

- **The rational canonical form.** A canonical form over every field, with invariant factors as the complete invariants.
- **Polynomial matrices and the Smith normal form.** Row and column operations on  $x\mathbf{I} - \mathbf{A}$  compute the invariant factors.
- **Semisimple operators and the Jordan–Chevalley decomposition.** Every operator splits uniquely into a diagonalizable part plus a commuting nilpotent part.
- **Functions of matrices, the exponential and linear differential equations.**  $f(\mathbf{A})$  from the Jordan form, and  $x' = \mathbf{A}x$  solved by  $e^{t\mathbf{A}}$ .
- **Convergent and power-bounded matrices.** Exactly when  $\mathbf{A}^m$  converges, and to what.

## 9.0.2 Named moves

- **Enlarge the kernel.** When  $\ker(T - \lambda I)$  is too small, pass to  $\ker(T - \lambda I)^k$  until the chain stops growing.
- **Reduce to the nilpotent case.** On each generalized eigenspace,  $T - \lambda I$  is nilpotent, so the whole structure question is about nilpotent operators.
- **Read the chain off the columns.** The columns of a Jordan block say  $N\mathbf{v}_1 = \mathbf{0}$  and  $N\mathbf{v}_j = \mathbf{v}_{j-1}$ : a chain.
- **Count ranks of powers.** The block sizes are second differences of  $\text{rank } N^j$ , so they can be found without finding a single chain.
- **A cyclic vector gives a companion matrix.** If  $\mathbf{v}, T\mathbf{v}, \dots$  span, the matrix of  $T$  is built from one polynomial.

# 9.1 Generalized Eigenspaces

Chapter 8 left one gap wide open. When an operator has too few eigenvectors, its eigenspaces are too small to fill the space, and diagonalization fails. The remedy of this chapter is to enlarge each eigenspace until the enlarged pieces do fill the space. This section builds the enlargement, shows that the enlarging process stops after finitely many steps, and proves that the pieces split  $V$  whenever the characteristic polynomial splits.

## 9.1.1 Enlarging an eigenspace

Take the standing counterexample of Chapter 8,

$$\mathbf{J} = \begin{pmatrix} 2 & 1 \\ 0 & 2 \end{pmatrix} \in M_2(F),$$

and let  $\mathbf{N} = \mathbf{J} - 2\mathbf{I}$ . The only eigenvalue is 2, and  $E_2(\mathbf{J}) = \ker \mathbf{N} = \text{span}(\mathbf{e}_1)$  is a line: one eigenvector direction where we need two, so  $\mathbf{J}$  is not diagonalizable over any field ([Theorem 8.4.1 \(e\)](#)).

Now look at what  $\mathbf{N}$  does to the missing direction  $\mathbf{e}_2$ . One application does not kill  $\mathbf{e}_2$ , but two do:

$$\mathbf{N}\mathbf{e}_2 = \mathbf{e}_1, \quad \mathbf{N}^2\mathbf{e}_2 = \mathbf{N}\mathbf{e}_1 = \mathbf{0}.$$

So  $\mathbf{e}_2$  is killed by  $\mathbf{N}^2$ , and  $\ker \mathbf{N}^2 = F^2$  is everything. The eigenspace was too small, but the kernel of the **square** is as big as we could wish.

This is the whole idea. Instead of asking for vectors killed by  $T - \lambda \text{id}_V$ , ask for vectors killed by **some power** of  $T - \lambda \text{id}_V$ . The eigenvectors are still there (they are killed at the first step), and there may be more.

*A generalized eigenvector is a vector that some power of  $T - \lambda \text{id}_V$  kills, not necessarily the first power.*

## 9.1.2 Generalized eigenvectors and generalized eigenspaces

### Definition 9.1.1 — Generalized Eigenvector

Let  $V$  be a vector space over  $F$ , let  $T \in \mathcal{L}(V)$  and  $\lambda \in F$ . A **generalized eigenvector** of  $T$  for  $\lambda$  is a vector  $\mathbf{v} \in V$  with  $\mathbf{v} \neq \mathbf{0}$  such that

$$(T - \lambda \text{id}_V)^k \mathbf{v} = \mathbf{0} \quad \text{for some integer } k \geq 1.$$

In words: the vector must be **non-zero**, exactly as for an eigenvector; the scalar  $\lambda$  must lie **in**  $F$ ; and the power  $k$  is not fixed in advance — it may depend on  $\mathbf{v}$ . Taking  $k = 1$  recovers the eigenvector condition, so **every eigenvector for  $\lambda$  is a generalized eigenvector for  $\lambda$** .

Collecting all of them, together with  $\mathbf{0}$ , gives the space we are after.

### Definition 9.1.2 — Generalized Eigenspace

Let  $V$  be a vector space over  $F$ , let  $T \in \mathcal{L}(V)$  and  $\lambda \in F$ . The **generalized eigenspace** of  $T$  for  $\lambda$  is

$$\begin{aligned} G_\lambda(T) &:= \{ \mathbf{v} \in V : (T - \lambda \text{id}_V)^k \mathbf{v} = \mathbf{0} \text{ for some integer } k \geq 1 \} \\ &= \bigcup_{k \geq 1} \ker(T - \lambda \text{id}_V)^k. \end{aligned}$$

For  $\mathbf{A} \in M_n(F)$  we write  $G_\lambda(\mathbf{A}) = G_\lambda(T_\mathbf{A})$ .

So  $G_\lambda(T)$  consists of  $\mathbf{0}$  together with all generalized eigenvectors for  $\lambda$ . Writing  $N = T - \lambda \text{id}_V$  for the rest of this subsection, the definition says  $G_\lambda(T) = \ker N \cup \ker N^2 \cup \ker N^3 \cup \dots$

**Well-definedness: it really is a subspace.** A union of subspaces is almost never a subspace (Proposition 1.2.3), so this needs a check. What saves us is that the union is **increasing**: if  $N^k \mathbf{v} = \mathbf{0}$ , then  $N^{k+1} \mathbf{v} = N(N^k \mathbf{v}) = N\mathbf{0} = \mathbf{0}$ , so

$$\ker N \subseteq \ker N^2 \subseteq \ker N^3 \subseteq \dots$$

Now use the subspace test (Theorem 1.2.1). We have  $\mathbf{0} \in \ker N \subseteq G_\lambda(T)$ , so the set is non-empty. If  $\mathbf{u}, \mathbf{v} \in G_\lambda(T)$ , say  $N^j \mathbf{u} = \mathbf{0}$  and  $N^k \mathbf{v} = \mathbf{0}$ , put  $m = \max(j, k)$ ; then  $\mathbf{u}, \mathbf{v} \in \ker N^m$  by the inclusions above, and  $\ker N^m$  is a subspace, so  $\mathbf{u} + \mathbf{v}$  and  $c\mathbf{u}$  lie in  $\ker N^m \subseteq G_\lambda(T)$  for every  $c \in F$ . Hence  $G_\lambda(T)$  is a subspace of  $V$ .

**Examples.**

- **The standing counterexample.** For  $\mathbf{J} = \begin{pmatrix} 2 & 1 \\ 0 & 2 \end{pmatrix}$ , we computed  $\ker(\mathbf{J} - 2\mathbf{I}) = \text{span}(\mathbf{e}_1)$

and  $\ker(\mathbf{J} - 2\mathbf{I})^2 = F^2$ . So  $G_2(\mathbf{J}) = F^2$ , while  $E_2(\mathbf{J})$  is only a line. The generalized eigenspace is strictly bigger than the eigenspace, and it is all of  $F^2$ .

- **A diagonalizable operator gains nothing.** Let  $\mathbf{A} = \text{diag}(2, 5)$  over  $\mathbb{Q}$ . Then  $(\mathbf{A} - 2\mathbf{I})^k = \text{diag}(0, 3^k)$ , whose kernel is  $\text{span}(\mathbf{e}_1)$  for every  $k \geq 1$ . So  $G_2(\mathbf{A}) = E_2(\mathbf{A}) = \text{span}(\mathbf{e}_1)$ , and likewise  $G_5(\mathbf{A}) = E_5(\mathbf{A}) = \text{span}(\mathbf{e}_2)$ . Enlarging an eigenspace of a diagonalizable operator does nothing, a fact we prove in general below.
- **Differentiation.** Let  $D \in \mathcal{L}(F[x]_{\leq n})$  be differentiation. Then  $D^{n+1} = 0$  ([Example \(Differentiation Is Nilpotent\)](#)), so **every** polynomial is killed by a power of  $D$ , that is, of  $D - 0 \text{ id}$ , and  $G_0(D) = F[x]_{\leq n}$ . By contrast  $E_0(D) = \ker D$  is the line of constants when  $F = \mathbb{R}$ . Here the generalized eigenspace for the single eigenvalue 0 is the whole space, of dimension  $n + 1$ .
- **A scalar not among the eigenvalues.** For the same  $\mathbf{A} = \text{diag}(2, 5)$  and  $\lambda = 7$ , the matrix  $(\mathbf{A} - 7\mathbf{I})^k = \text{diag}((-5)^k, (-2)^k)$  is invertible, so its kernel is  $\{\mathbf{0}\}$  for every  $k$ . Hence  $G_7(\mathbf{A}) = \{\mathbf{0}\}$ . The definition makes sense for every  $\lambda \in F$ , and gives the zero space unless  $\lambda$  is an eigenvalue.
- **A degenerate case.** On a space of dimension 1, every operator is  $\lambda \text{ id}$  for one scalar  $\lambda$ , and  $G_\lambda = V$ ,  $G_\mu = \{\mathbf{0}\}$  for  $\mu \neq \lambda$ . Small as it is, this is the model for a  $1 \times 1$  Jordan block.
- **No eigenvalues at all.** The rotation  $R$  of  $\mathbb{R}^2$  by a right angle has no real eigenvalue ([Example \(A Rotation Has No Invariant Lines\)](#)), and indeed  $G_\lambda(R) = \{\mathbf{0}\}$  for every  $\lambda \in \mathbb{R}$ : if  $(R - \lambda\mathbf{I})^k \mathbf{v} = \mathbf{0}$  with  $\mathbf{v} \neq \mathbf{0}$ , then  $R - \lambda\mathbf{I}$  would not be injective, so  $\lambda$  would be an eigenvalue. Over  $\mathbb{C}$ , where  $p_R = x^2 + 1 = (x - i)(x + i)$  splits,  $G_i(R)$  and  $G_{-i}(R)$  are the two eigenlines.

**Non-example by minimal change.** Keep  $\mathbf{A} = \text{diag}(2, 5)$  over  $\mathbb{Q}$  and take  $\mathbf{v} = (1, 1)$ . This vector is killed by a polynomial in  $\mathbf{A}$  whose roots are eigenvalues:

$$(\mathbf{A} - 2\mathbf{I})(\mathbf{A} - 5\mathbf{I})\mathbf{v} = \mathbf{0},$$

as one checks entry by entry. But it is a generalized eigenvector for **neither** eigenvalue:  $(\mathbf{A} - 2\mathbf{I})^k \mathbf{v} = (0, 3^k) \neq \mathbf{0}$  and  $(\mathbf{A} - 5\mathbf{I})^k \mathbf{v} = ((-3)^k, 0) \neq \mathbf{0}$  for every  $k \geq 1$ . The failing clause is that the definition demands a power of a **single** factor  $T - \lambda \text{ id}_V$ , not a product of different factors. (Of course  $\mathbf{v} = \mathbf{e}_1 + \mathbf{e}_2$  is a *sum* of two generalized eigenvectors; that is the decomposition theorem below, not the definition.)

**Why this definition.** Three clauses are doing work. The vector must be non-zero, or every  $G_\lambda$  would contain a “generalized eigenvector” and the word would carry no information. The exponent must be allowed to grow, because nothing tells us in advance how many steps a vector needs: for  $D$  on  $\mathbb{R}[x]_{\leq n}$  the polynomial  $x^n$  needs  $n + 1$  steps. And the power is a power of one factor, which is what makes  $G_\lambda$  one piece attached to one eigenvalue. The name is the honest one: these are the vectors that behave like eigenvectors after enough applications of  $T - \lambda \text{ id}_V$ .

*Warning.*  $G_\lambda(T)$  is usually bigger than  $E_\lambda(T)$ , and “generalized eigenvector” includes the ordinary ones. The inclusion  $E_\lambda(T) \subseteq G_\lambda(T)$  always holds, and it is strict exactly when  $T$  has too few eigenvectors for  $\lambda$ : for  $\mathbf{J} = \begin{pmatrix} 2 & 1 \\ 0 & 2 \end{pmatrix}$ ,  $\dim E_2 = 1$  and  $\dim G_2 = 2$ . So a generalized eigenvector need not satisfy  $T\mathbf{v} = \lambda\mathbf{v}$ : the vector  $\mathbf{e}_2$  satisfies  $\mathbf{J}\mathbf{e}_2 = \mathbf{e}_1 + 2\mathbf{e}_2$ . Conversely, do not read “generalized” as “not an eigenvector”: every eigenvector is a generalized eigenvector.

### 9.1.3 The kernel chain stabilizes

The definition of  $G_\lambda(T)$  is an infinite union, which is awkward: to decide membership we would have to try every  $k$ . In finite dimension this is an illusion. The chain  $\ker N \subseteq \ker N^2 \subseteq \dots$  is a chain of subspaces of  $V$ , and their dimensions cannot keep growing. Once the chain stops growing, it stops forever.

#### Theorem 9.1.1 — The Kernel Chain Stabilizes

Let  $V$  be a finite-dimensional vector space over  $F$  with  $\dim V = n$ , let  $N \in \mathcal{L}(V)$ , and write  $K_k = \ker N^k$  for  $k \geq 0$ , so that  $K_0 = \{\mathbf{0}\}$ . Then:

- (a)  $K_0 \subseteq K_1 \subseteq K_2 \subseteq \dots$ , and every  $K_k$  is  $N$ -invariant;
- (b) if  $K_m = K_{m+1}$  for some  $m \geq 0$ , then  $K_{m+j} = K_m$  for **every**  $j \geq 0$ ;
- (c) there is a **least** index  $s \geq 0$  with  $K_s = K_{s+1}$ ; it satisfies  $s \leq n$ , the inclusions

$$K_0 \subsetneq K_1 \subsetneq \dots \subsetneq K_s$$

are **strict**, and  $K_k = K_s$  for every  $k \geq s$ ;

- (d) in particular  $\ker N^n = \ker N^{n+j}$  for every  $j \geq 0$ .

*Idea.* The only real point is (b): equality at one step propagates. Read it backwards. To put  $\mathbf{v} \in K_{m+j+1}$  into  $K_{m+j}$ , look at  $\mathbf{w} = N^j \mathbf{v}$ . It lies in  $K_{m+1}$ , which by hypothesis is  $K_m$ , and that is exactly the statement  $N^{m+j} \mathbf{v} = \mathbf{0}$ . Then (c) and (d) are bookkeeping: while the inclusions are strict the dimensions go up by at least 1, so strictness cannot survive past step  $n$ .

*Proof.*

- (a) If  $N^k \mathbf{v} = \mathbf{0}$ , then  $N^{k+1} \mathbf{v} = N(N^k \mathbf{v}) = \mathbf{0}$ , so  $K_k \subseteq K_{k+1}$ . Each  $K_k = \ker N^k$  is the kernel of a polynomial in  $N$ , hence  $N$ -invariant by Theorem 5.7.4 (b).
- (b) Suppose  $K_m = K_{m+1}$ . We prove  $K_{m+j} = K_m$  by induction on  $j \geq 0$ . For  $j = 0$  there is nothing to prove. Assume  $K_{m+j} = K_m$ , and let  $\mathbf{v} \in K_{m+j+1}$ . Then

$$N^{m+1}(N^j \mathbf{v}) = N^{m+j+1} \mathbf{v} = \mathbf{0},$$

so  $N^j \mathbf{v} \in K_{m+1} = K_m$ , which says  $N^{m+j} \mathbf{v} = N^m(N^j \mathbf{v}) = \mathbf{0}$ . Hence  $\mathbf{v} \in K_{m+j} = K_m$ . Together with  $K_m \subseteq K_{m+j+1}$  from (a), this gives  $K_{m+j+1} = K_m$ .

- (c) First, some  $m$  with  $0 \leq m \leq n$  satisfies  $K_m = K_{m+1}$ . Otherwise  $K_m \subsetneq K_{m+1}$  for every  $m = 0, 1, \dots, n$ , so  $\dim K_{m+1} \geq \dim K_m + 1$  for those  $m$ , and hence  $\dim K_{n+1} \geq n + 1$ , contradicting  $\dim K_{n+1} \leq \dim V = n$  (Theorem 1.6.7). By the well-ordering principle (Theorem 0.2.11) there is a least such index; call it  $s$ , and  $s \leq n$ . For  $k < s$  minimality gives  $K_k \subsetneq K_{k+1}$ , so the inclusion  $K_k \subseteq K_{k+1}$  of (a) is strict. For  $k \geq s$ , (b) gives  $K_k = K_s$ .
- (d) Since  $s \leq n$ , (c) gives  $K_n = K_s = K_{n+j}$  for every  $j \geq 0$ .

■

*Quick check.* Can  $\ker N \subsetneq \ker N^2 = \ker N^3$  happen? Can  $\ker N = \ker N^2 \subsetneq \ker N^3$  happen?

*Solution.* The first can: for  $N = \begin{pmatrix} 0 & 1 \\ 0 & 0 \end{pmatrix}$  on  $F^2$  we have  $\ker N = \text{span}(\mathbf{e}_1)$  and  $\ker N^2 = F^2 = \ker N^3$ , since  $N^2 = 0$ . Here  $s = 2$ .

The second cannot, by [Theorem 9.1.1](#) (b) with  $m = 1$ : equality at one step forces equality at every later step. This is why the theorem says the strict inclusions come **first** and the equalities **after**; the chain never resumes growing.

The index  $s$  will come up often enough to name. We call the least  $s \geq 0$  with  $\ker N^s = \ker N^{s+1}$  the **stabilization index** of  $N$ . Part (d) is the useful form: the chain has certainly stopped by step  $n = \dim V$ , whatever  $N$  is. Applying this to  $N = T - \lambda \text{id}_V$  turns the infinite union in [Definition 9.1.2](#) into a single kernel.

### Corollary 9.1.2 — The Generalized Eigenspace Is One Kernel

Let  $V$  be a finite-dimensional vector space over  $F$  with  $\dim V = n$ , let  $T \in \mathcal{L}(V)$  and  $\lambda \in F$ , and put  $N = T - \lambda \text{id}_V$  with stabilization index  $s$ . Then:

- (a)  $G_\lambda(T) = \ker N^s = \ker N^n$ ;
- (b)  $G_\lambda(T)$  is  $T$ -invariant, and  $E_\lambda(T) \subseteq G_\lambda(T)$ ;
- (c)  $G_\lambda(T) \neq \{\mathbf{0}\}$  if and only if  $\lambda$  is an eigenvalue of  $T$ ;
- (d) if  $\mathbf{v} \in G_\lambda(T)$  is non-zero and  $k \geq 1$  is **least** with  $N^k \mathbf{v} = \mathbf{0}$ , then  $N^{k-1} \mathbf{v}$  is an eigenvector of  $T$  for  $\lambda$ .

*Proof.*

- (a) Every  $\ker N^k$  with  $k \geq 1$  is contained in  $\ker N^{\max(k,n)} = \ker N^n$  by [Theorem 9.1.1](#) (a) and (d), so the union  $G_\lambda(T)$  equals  $\ker N^n$ ; and  $\ker N^n = \ker N^s$  by (c) of the same theorem, since  $s \leq n$ .
- (b) By (a),  $G_\lambda(T) = \ker N^n$  is the kernel of a polynomial in  $T$ , so  $T$  maps it into itself ([Theorem 5.7.4](#) (b)). And  $E_\lambda(T) = \ker N \subseteq \ker N^n$ .
- (c) ( $\Leftarrow$ ) If  $\lambda$  is an eigenvalue, then  $E_\lambda(T) \neq \{\mathbf{0}\}$  ([Theorem 8.2.1](#)), and  $E_\lambda(T) \subseteq G_\lambda(T)$  by (b). ( $\Rightarrow$ ) If  $\mathbf{v} \in G_\lambda(T)$  is non-zero, take  $k \geq 1$  least with  $N^k \mathbf{v} = \mathbf{0}$ , which exists by the definition of  $G_\lambda(T)$  and [Theorem 0.2.11](#). Then  $\mathbf{w} = N^{k-1} \mathbf{v} \neq \mathbf{0}$  by minimality of  $k$  (for  $k = 1$  this reads  $\mathbf{w} = \mathbf{v} \neq \mathbf{0}$ ), while  $N\mathbf{w} = N^k \mathbf{v} = \mathbf{0}$ . So  $\mathbf{w} \in \ker N$  is non-zero and  $\lambda$  is an eigenvalue.
- (d) This is the computation just made:  $\mathbf{w} = N^{k-1} \mathbf{v}$  is non-zero and lies in  $\ker N = E_\lambda(T)$ .

■

Part (d) is a move worth remembering: **to get a genuine eigenvector out of a generalized one, apply  $T - \lambda \text{id}_V$  as many times as possible without reaching  $\mathbf{0}$** . It is the reason the enlargement never produces eigenvalues out of thin air, and it will start every Jordan chain in Section 2.

### 9.1.4 The generalized eigenspace decomposition

We can now state the payoff. Chapter 8's primary decomposition ([Theorem 8.8.1](#)) already splits  $V$  into invariant pieces, one for each irreducible factor of  $m_T$ . When  $p_T$  splits, all those factors are linear, the pieces are kernels of powers of  $T - \lambda_i \text{id}_V$  — and those are exactly the generalized eigenspaces. The extra information here is that the piece attached to  $\lambda_i$  has dimension  $a_T(\lambda_i)$ , so the sizes are read off the characteristic polynomial.

*When the characteristic polynomial splits, the generalized eigenspaces fill the space, and the one for  $\lambda$  has dimension the algebraic multiplicity of  $\lambda$ .*

#### Theorem 9.1.3 — Generalized Eigenspace Decomposition

Let  $V$  be a finite-dimensional vector space over  $F$  with  $\dim V = n \geq 1$ , let  $T \in \mathcal{L}(V)$ , and suppose  $p_T$  splits over  $F$ , with distinct eigenvalues  $\lambda_1, \dots, \lambda_k$  and

$$p_T = (x - \lambda_1)^{a_1} \cdots (x - \lambda_k)^{a_k}, \quad m_T = (x - \lambda_1)^{s_1} \cdots (x - \lambda_k)^{s_k},$$

where  $a_i = a_T(\lambda_i)$  and  $1 \leq s_i \leq a_i$ . Write  $G_i = G_{\lambda_i}(T)$  and  $N_i = (T - \lambda_i \text{id}_V)|_{G_i}$ . Then:

(a)  $G_i = \ker(T - \lambda_i \text{id}_V)^{s_i}$ , and

$$V = G_1 \oplus G_2 \oplus \cdots \oplus G_k;$$

(b) each  $G_i$  is  $T$ -invariant and non-zero, with  $E_{\lambda_i}(T) \subseteq G_i$ , and  $E_{\lambda_i}(T) = G_i$  if and only if  $s_i = 1$ ;

(c)  $\dim G_i = a_i$  and  $p_{T|_{G_i}} = (x - \lambda_i)^{a_i}$ ;

(d)  $N_i^{a_i} = 0$ ; that is,  $T|_{G_i} = \lambda_i \text{id}_{G_i} + N_i$  with  $N_i$  nilpotent, of stabilization index  $s_i$ ;

(e) if  $\mathcal{B}$  is a basis of  $V$  obtained by writing bases of  $G_1, \dots, G_k$  one after another, then

$$[T]_{\mathcal{B}} = [T|_{G_1}] \oplus \cdots \oplus [T|_{G_k}]$$

is block diagonal, with the  $i$ -th block of size  $a_i$ .

*Idea.* Almost everything is already proved; the one new step is (a), the identification  $G_i = \ker(T - \lambda_i \text{id}_V)^{s_i}$ . One inclusion is free, since  $\ker(T - \lambda_i)^{s_i}$  sits inside the union defining  $G_i$ . For the other, count. Chapter 8 says the primary piece  $\ker(T - \lambda_i)^{s_i}$  has dimension  $a_i$ , so it is enough to show  $\dim G_i \leq a_i$ . Now  $G_i$  is invariant, and the only eigenvalue of  $T$  on  $G_i$  is  $\lambda_i$ : an eigenvector in  $G_i$  for  $\mu$  is killed by  $(\mu - \lambda_i)^n$ , which forces  $\mu = \lambda_i$ . So the characteristic polynomial of the restriction is a power of  $x - \lambda_i$  dividing  $p_T$ , and its degree  $\dim G_i$  is at most  $a_i$ .

*Proof.* Since  $p_T$  splits and  $m_T \mid p_T$ , [Corollary 8.6.2](#) (b) gives the displayed factorization of  $m_T$  with  $1 \leq s_i \leq a_i$ . Put  $V_i = \ker(T - \lambda_i \text{id}_V)^{s_i}$ . The polynomials  $x - \lambda_i$  are distinct monic irreducibles, so [Theorem 8.8.1](#) applies to  $m_T = \prod_i (x - \lambda_i)^{s_i}$  and gives

$$V = V_1 \oplus \cdots \oplus V_k,$$

with each  $V_i$  invariant and non-zero. By [Corollary 8.8.4](#),  $\dim V_i = a_i$ ,  $p_{T|_{V_i}} = (x - \lambda_i)^{a_i}$ , and  $E_{\lambda_i}(T) \subseteq V_i$  with equality if and only if  $s_i = 1$ .



*Claim:*  $G_i = V_i$ . Fix  $i$  and write  $G = G_i$ ,  $\lambda = \lambda_i$ ,  $a = a_i$ ,  $N = T - \lambda \text{id}_V$ .

$(\supseteq)$   $V_i = \ker N^{s_i} \subseteq \bigcup_{j \geq 1} \ker N^j = G$ .

$(\subseteq)$  It suffices to prove  $\dim G \leq a$ : then  $V_i \subseteq G$  and  $\dim V_i = a \geq \dim G$  force  $\dim V_i = \dim G$ , and [Theorem 1.6.8](#), applied to the subspace  $V_i$  of  $G$ , gives  $V_i = G$ .

First, *the only eigenvalue of  $T|_G$  is  $\lambda$* . Indeed,  $G$  is  $T$ -invariant ([Corollary 9.1.2](#) (b)), so the restriction makes sense ([Definition 8.1.2](#)). Let  $\mu \in F$  be an eigenvalue of  $T|_G$ , with eigenvector  $\mathbf{v} \in G$ ,  $\mathbf{v} \neq \mathbf{0}$ . From  $T\mathbf{v} = \mu\mathbf{v}$  we get  $N\mathbf{v} = (\mu - \lambda)\mathbf{v}$ , hence  $N^n\mathbf{v} = (\mu - \lambda)^n\mathbf{v}$  by induction. But  $G = \ker N^n$  by [Corollary 9.1.2](#) (a), so  $(\mu - \lambda)^n\mathbf{v} = \mathbf{0}$  with  $\mathbf{v} \neq \mathbf{0}$ , which forces the scalar  $(\mu - \lambda)^n$  to be 0 ([Theorem 1.1.9](#)), and so  $\mu = \lambda$ , since a field has no zero divisors ([Theorem 0.7.1](#)).

Now compare characteristic polynomials.  $G \neq \{\mathbf{0}\}$ , since  $\lambda$  is an eigenvalue of  $T$  ([Corollary 9.1.2](#) (c)). If  $G = V$ , then every root of  $p_T$  is an eigenvalue of  $T = T|_G$  ([Theorem 8.2.1](#)), so  $\lambda$  is the only root; as  $p_T$  splits and is monic of degree  $n$  ([Theorem 6.10.2](#)),  $p_T = (x - \lambda)^n$  and  $a = n = \dim G$ . If  $G \neq V$ , then [Theorem 8.1.3](#) (b) applies to the invariant subspace  $G$  and gives

$$p_T = p_{T|_G} p_{\bar{T}},$$

so  $p_{T|_G}$  is a monic divisor of  $p_T = \prod_j (x - \lambda_j)^{a_j}$ . By [Lemma 8.5.2](#),  $p_{T|_G} = \prod_j (x - \lambda_j)^{c_j}$  with  $0 \leq c_j \leq a_j$ . Its roots are the eigenvalues of  $T|_G$  ([Theorem 8.2.1](#), applied to  $T|_G$ ), which we just showed to be  $\lambda$  alone, so  $c_j = 0$  for  $j \neq i$ . Hence  $p_{T|_G} = (x - \lambda)^{c_i}$ , and comparing degrees ([Theorem 6.10.2](#)) gives  $\dim G = c_i \leq a$ . This proves the claim, and with it (a), (b) and (c), since  $G_i = V_i$ .

- (d) By [Theorem 8.8.1](#) (c) and the claim,  $m_{T|_{G_i}} = (x - \lambda_i)^{s_i}$ . For an integer  $j \geq 0$ , the operator  $N_i^j$  is  $(x - \lambda_i)^j$  evaluated at  $T|_{G_i}$ , so  $N_i^j = 0$  if and only if  $(x - \lambda_i)^{s_i}$  divides  $(x - \lambda_i)^j$  ([Theorem 8.5.1](#)), that is, if and only if  $j \geq s_i$ . Since  $s_i \leq a_i$ , this gives  $N_i^{a_i} = 0$ , and it identifies  $s_i$  as the least exponent killing  $N_i$ . Hence  $\ker N_i^{s_i} = G_i$  while  $\ker N_i^{s_i-1} \subsetneq G_i$ , so the stabilization index of  $N_i$  is  $s_i$ .
- (e) Each  $G_i$  is invariant and non-zero, and  $V = \bigoplus_i G_i$ , so [Theorem 8.1.5](#) gives the block diagonal shape, the  $i$ -th block being  $[T|_{G_i}]$  in the chosen basis of  $G_i$ , of size  $\dim G_i = a_i$ .

■

Read (d) again, since it is the plan for the rest of the chapter. **On its generalized eigenspace,  $T$  is a scalar plus a nilpotent operator.** The scalar part is as simple as an operator can be. So the only question left is what a nilpotent operator looks like in a well-chosen basis, and that is the subject of the next section. Once it is answered, (e) assembles the answers for the different eigenvalues into one matrix for  $T$ .

Two consistency checks. Summing dimensions in (a) and using (c) gives  $\sum_i a_i = n$ , which is the degree count for a split  $p_T$ . And if  $T$  is diagonalizable, then every  $s_i = 1$  ([Theorem 8.8.2](#)), so (b) says  $G_i = E_{\lambda_i}(T)$  and (a) becomes the eigenspace decomposition  $V = \bigoplus_i E_{\lambda_i}(T)$  of [Theorem 8.4.1](#) (c). Nothing is lost, and nothing new is gained, in the diagonalizable case.

**Example. (Generalized Eigenspaces of a  $4 \times 4$  Matrix)**



Let

$$\mathbf{A} = \begin{pmatrix} 2 & 1 & -1 & -2 \\ -3 & 3 & -1 & 1 \\ 3 & -1 & 3 & -1 \\ -3 & 1 & -1 & 3 \end{pmatrix} \in M_4(\mathbb{Q}), \quad p_{\mathbf{A}} = (x-2)^3(x-5).$$

Find  $E_2(\mathbf{A})$ ,  $G_2(\mathbf{A})$ ,  $G_5(\mathbf{A})$ , the stabilization index of  $\mathbf{A} - 2\mathbf{I}$ , and the minimal polynomial. Verify [Theorem 9.1.3](#) on this matrix.

*Solution. The eigenspace.* Write  $\mathbf{N} = \mathbf{A} - 2\mathbf{I}$ , so

$$\mathbf{N} = \begin{pmatrix} 0 & 1 & -1 & -2 \\ -3 & 1 & -1 & 1 \\ 3 & -1 & 1 & -1 \\ -3 & 1 & -1 & 1 \end{pmatrix}.$$

Rows 2 and 4 are equal and row 3 is their negative, so the row space is spanned by rows 1 and 2, which are not proportional; hence  $\text{rank } \mathbf{N} = 2$  and nullity  $\mathbf{N} = 2$  ([Theorem 2.5.7](#)). The kernel is cut out by

$$x_2 - x_3 - 2x_4 = 0, \quad -3x_1 + x_2 - x_3 + x_4 = 0.$$

Subtracting the first equation from the second gives  $-3x_1 + 3x_4 = 0$ , so  $x_1 = x_4$ , and then  $x_2 = x_3 + 2x_4$ . With  $x_3 = s$  and  $x_4 = t$  free,

$$E_2(\mathbf{A}) = \{(t, s + 2t, s, t)\} = \text{span}((0, 1, 1, 0), (1, 2, 0, 1)), \quad g_{\mathbf{A}}(2) = 2.$$

Since  $a_{\mathbf{A}}(2) = 3 > 2$ , the matrix is not diagonalizable ([Theorem 8.4.1](#) (e)).

*The generalized eigenspace.* Squaring,

$$\mathbf{N}^2 = 9 \begin{pmatrix} 0 & 0 & 0 & 0 \\ -1 & 0 & 0 & 1 \\ 1 & 0 & 0 & -1 \\ -1 & 0 & 0 & 1 \end{pmatrix},$$

which has rank 1, so  $\ker \mathbf{N}^2$  has dimension 3 and is the hyperplane

$$\ker \mathbf{N}^2 = \{\mathbf{x} : x_1 = x_4\} = \text{span}(\mathbf{e}_2, \mathbf{e}_3, \mathbf{e}_1 + \mathbf{e}_4).$$

The chain of kernels has dimensions 0, 2, 3 so far. Since  $\ker \mathbf{N}^2 \subseteq G_2(\mathbf{A})$  and  $\dim G_2(\mathbf{A}) = a_{\mathbf{A}}(2) = 3 = \dim \ker \mathbf{N}^2$  by part (c) of the theorem, we get  $G_2(\mathbf{A}) = \ker \mathbf{N}^2$ , and the stabilization index is  $s = 2$ . One can see the stabilization by hand as well: multiplying out gives  $\mathbf{N}^3 = 3\mathbf{N}^2$ , so  $\ker \mathbf{N}^3 = \ker \mathbf{N}^2$  directly.

*The second generalized eigenspace.* Here  $a_{\mathbf{A}}(5) = 1$ , so  $\dim G_5(\mathbf{A}) = 1$  and  $G_5(\mathbf{A}) = E_5(\mathbf{A})$  by part (b) — its  $s$  must be 1. Solving  $(\mathbf{A} - 5\mathbf{I})\mathbf{x} = \mathbf{0}$  gives  $G_5(\mathbf{A}) = E_5(\mathbf{A}) = \text{span}((0, 1, -1, 1))$ . As a check,  $\mathbf{A}(0, 1, -1, 1) = (1+1-2, 3+1+1, -1-3-1, 1+1+3) = (0, 5, -5, 5)$ .

*The decomposition.*  $\mathbb{Q}^4 = G_2(\mathbf{A}) \oplus G_5(\mathbf{A})$ , of dimensions  $3+1=4$ , and the sum is direct because  $(0, 1, -1, 1) \notin G_2(\mathbf{A})$ : its first and last coordinates are  $0 \neq 1$ .

*The minimal polynomial.* The stabilization indices are  $s_1 = 2$  at  $\lambda = 2$  and  $s_2 = 1$  at  $\lambda = 5$ , and part (d) of the theorem identifies these indices as the exponents in  $m_{\mathbf{A}}$ , so

$m_{\mathbf{A}} = (x-2)^2(x-5)$ . Consistently,  $(\mathbf{A}-2\mathbf{I})(\mathbf{A}-5\mathbf{I}) \neq 0$ : the two factors are coprime, so the kernel of the product is  $\ker(\mathbf{A}-2\mathbf{I}) \oplus \ker(\mathbf{A}-5\mathbf{I})$  (Theorem 5.7.5 (a)), of dimension  $2+1=3 < 4$ .

*A basis adapted to the splitting.* Take  $\mathcal{B} = (\mathbf{e}_2, \mathbf{e}_3, \mathbf{e}_1 + \mathbf{e}_4, (0, 1, -1, 1))$ , a basis of  $G_2$  followed by one of  $G_5$ . By part (e),  $[T_{\mathbf{A}}]_{\mathcal{B}}$  is block diagonal with a  $3 \times 3$  block and the  $1 \times 1$  block (5).

**Warning.** The dimension of  $G_{\lambda}$  is the algebraic multiplicity, not the number of steps the chain takes. In the example,  $\dim G_2 = 3$  while the chain  $\{\mathbf{0}\} \subsetneq \ker \mathbf{N} \subsetneq \ker \mathbf{N}^2 = G_2$  stabilizes after  $s = 2$  steps. The two numbers measure different things:  $a(\lambda) = \dim G_{\lambda}$  is the total size of the piece, and  $s$  is the exponent of  $x - \lambda$  in the minimal polynomial. All that holds in general is  $1 \leq s \leq a(\lambda)$  and  $g(\lambda) \leq a(\lambda)$  (Theorem 8.3.1).

Finally, a word on what we may not yet assume. The decomposition needs  $p_T$  to split, which is automatic over  $\mathbb{C}$  (Corollary 5.5.5) but fails, for instance, for a rotation of  $\mathbb{R}^2$ , whose generalized eigenspaces over  $\mathbb{R}$  are all  $\{\mathbf{0}\}$  and certainly do not fill  $\mathbb{R}^2$ . Operators over an arbitrary field need the coarser primary decomposition of Chapter 8, whose pieces we return to in Section 6.

## 9.1.5 Exercises

### 9.1.5.1 A. Check your understanding

#### Exercise A1.

- Define a generalized eigenvector and the generalized eigenspace  $G_{\lambda}(T)$ .
- State what Theorem 9.1.1 says about the chain  $\ker N \subseteq \ker N^2 \subseteq \dots$ , including the bound on where it stabilizes.
- True or false: every generalized eigenvector of  $T$  for  $\lambda$  satisfies  $T\mathbf{v} = \lambda\mathbf{v}$ . Justify your answer.
- True or false: if  $\lambda$  is not an eigenvalue of  $T$ , then  $G_{\lambda}(T) = \{\mathbf{0}\}$ . Justify your answer.
- Let  $\dim V = 6$  and let  $p_T = (x-1)^4(x-3)^2$  split over  $F$ . What are  $\dim G_1(T)$  and  $\dim G_3(T)$ ?
- In the situation of (e), what can you say about  $\dim E_1(T)$ ?

*Solution.*

- A generalized eigenvector of  $T$  for  $\lambda \in F$  is a **non-zero**  $\mathbf{v} \in V$  with  $(T - \lambda \operatorname{id}_V)^k \mathbf{v} = \mathbf{0}$  for **some**  $k \geq 1$  (Definition 9.1.1);  $G_{\lambda}(T)$  is the set of all of them together with  $\mathbf{0}$ , that is,  $\bigcup_{k \geq 1} \ker(T - \lambda \operatorname{id}_V)^k$  (Definition 9.1.2).
- The kernels increase; once two consecutive ones are equal, all later ones equal them; the strict inclusions come first, and the chain has stabilized by step  $\dim V$ , so  $G_{\lambda}(T) = \ker(T - \lambda \operatorname{id}_V)^{\dim V}$ .
- False. For  $\mathbf{J} = \begin{pmatrix} 2 & 1 \\ 0 & 2 \end{pmatrix}$ , the vector  $\mathbf{e}_2$  lies in  $G_2(\mathbf{J})$  but  $\mathbf{J}\mathbf{e}_2 = \mathbf{e}_1 + 2\mathbf{e}_2 \neq 2\mathbf{e}_2$ . Only the vectors of  $E_{\lambda}(T) \subseteq G_{\lambda}(T)$  satisfy the eigenvector equation.

- (d) True, by [Corollary 9.1.2](#) (c).
- (e)  $\dim G_1(T) = a_T(1) = 4$  and  $\dim G_3(T) = a_T(3) = 2$  ([Theorem 9.1.3](#) (c)).
- (f) Only  $1 \leq \dim E_1(T) \leq 4$ , since  $E_1(T) \subseteq G_1(T)$  and  $1 \leq g_T(1) \leq a_T(1)$  ([Theorem 8.3.1](#)). All four values occur, as the Jordan forms of Section 3 will show.

### 9.1.5.2 B. Practice

#### Exercise B1: Computing generalized eigenspaces.

For each operator, find all generalized eigenspaces, and compare each with the corresponding eigenspace.

- (a)  $\mathbf{A}_1 = \begin{pmatrix} 3 & 1 \\ 0 & 3 \end{pmatrix} \in M_2(\mathbb{Q})$ .
- (b)  $\mathbf{A}_2 = \begin{pmatrix} 2 & 1 & 1 \\ 0 & 1 & -1 \\ -1 & -1 & 1 \end{pmatrix} \in M_3(\mathbb{R})$ , which has  $m_{\mathbf{A}_2} = (x-1)^2(x-2)$  ([Exercise \(B2: Decomposing  \$\mathbb{R}^3\$ \)](#)).
- (c)  $D \in \mathcal{L}(\mathbb{R}[x]_{\leq 3})$ , differentiation.
- (d) The rotation  $\mathbf{R} = \begin{pmatrix} 0 & -1 \\ 1 & 0 \end{pmatrix}$ , first over  $\mathbb{R}$  and then over  $\mathbb{C}$ .

*Solution.*

- (a)  $p_{\mathbf{A}_1} = (x-3)^2$ , so 3 is the only eigenvalue, and  $G_\lambda = \{\mathbf{0}\}$  for  $\lambda \neq 3$  ([Corollary 9.1.2](#) (c)). Since  $(\mathbf{A}_1 - 3\mathbf{I})^2 = 0$ ,  $G_3(\mathbf{A}_1) = \mathbb{Q}^2$ , while  $E_3(\mathbf{A}_1) = \ker \begin{pmatrix} 0 & 1 \\ 0 & 0 \end{pmatrix} = \text{span}(\mathbf{e}_1)$  is a line.
- (b) From  $m_{\mathbf{A}_2} = (x-1)^2(x-2)$  and [Theorem 9.1.3](#) (a),  $G_1 = \ker(\mathbf{A}_2 - \mathbf{I})^2$  and  $G_2 = \ker(\mathbf{A}_2 - 2\mathbf{I})$ . Chapter 8 computed these:  $G_1 = \text{span}((-1, 1, 0), (0, 0, 1))$ , the plane  $x+y=0$ , and  $G_2 = \text{span}((0, -1, 1))$ . The eigenspace  $E_1(\mathbf{A}_2) = \text{span}((-1, 1, 0))$  is a line strictly inside  $G_1$ , while  $E_2(\mathbf{A}_2) = G_2$  because  $s_2 = 1$ . All other  $G_\lambda$  are  $\{\mathbf{0}\}$ .
- (c)  $D^4 = 0$  ([Example \(Differentiation Is Nilpotent\)](#)), so  $G_0(D) = \mathbb{R}[x]_{\leq 3}$ , of dimension 4, while  $E_0(D) = \ker D = \mathbb{R}$  is the line of constants. Since  $p_D = x^4$  has 0 as its only root, every other generalized eigenspace is  $\{\mathbf{0}\}$ .
- (d) Over  $\mathbb{R}$ ,  $p_{\mathbf{R}} = x^2 + 1$  has no root, so  $\mathbf{R}$  has no eigenvalue and  $G_\lambda(\mathbf{R}) = \{\mathbf{0}\}$  for every  $\lambda \in \mathbb{R}$ . The generalized eigenspaces do **not** fill  $\mathbb{R}^2$ ; the hypothesis that  $p_{\mathbf{R}}$  splits fails. Over  $\mathbb{C}$ ,  $p_{\mathbf{R}} = (x-i)(x+i)$  splits with  $a(i) = a(-i) = 1$ , so  $G_i(\mathbf{R}) = E_i(\mathbf{R}) = \text{span}((1, -i))$  and  $G_{-i}(\mathbf{R}) = E_{-i}(\mathbf{R}) = \text{span}((1, i))$ , and  $\mathbb{C}^2 = G_i \oplus G_{-i}$ . (Check:  $\mathbf{R}(1, -i) = (i, 1) = i(1, -i)$ .)

#### Exercise B2: The image chain stabilizes too.

Let  $V$  be finite-dimensional with  $\dim V = n$ , and let  $N \in \mathcal{L}(V)$ .

- (a) Prove that  $\text{im } N \supseteq \text{im } N^2 \supseteq \text{im } N^3 \supseteq \dots$ , and that each  $\text{im } N^k$  is  $N$ -invariant.
- (b) Prove that  $\text{im } N^m = \text{im } N^{m+1}$  if and only if  $\ker N^m = \ker N^{m+1}$ .
- (c) Deduce that  $\text{im } N^n = \text{im } N^{n+j}$  for every  $j \geq 0$ , and that the image chain stabilizes

at exactly the same index  $s$  as the kernel chain.

*Solution.*

- (a)  $\operatorname{im} N^{k+1} = \{N^{k+1}\mathbf{v} : \mathbf{v} \in V\} = \{N^k(N\mathbf{v}) : \mathbf{v} \in V\} \subseteq \operatorname{im} N^k$ . Each  $\operatorname{im} N^k$  is the image of a polynomial in  $N$ , hence  $N$ -invariant (Theorem 5.7.4 (b)).
- (b) By Theorem 3.3.1,  $\dim \operatorname{im} N^k + \dim \ker N^k = n$  for every  $k$ . The inclusions  $\operatorname{im} N^{m+1} \subseteq \operatorname{im} N^m$  and  $\ker N^m \subseteq \ker N^{m+1}$  are equalities exactly when the dimensions agree (Theorem 1.6.8), and  $\dim \operatorname{im} N^m = \dim \operatorname{im} N^{m+1}$  if and only if  $\dim \ker N^m = \dim \ker N^{m+1}$  by the displayed identity.
- (c) By Theorem 9.1.1 (d),  $\ker N^n = \ker N^{n+j}$ ; iterating (b) at each step from  $n$  to  $n+j$  gives  $\operatorname{im} N^n = \operatorname{im} N^{n+j}$ . And by (b), the least index at which two consecutive images agree is the least index at which two consecutive kernels agree, namely  $s$ .

### Exercise B3: Reading the dimensions off a triangular matrix.

Let

$$\mathbf{A} = \begin{pmatrix} 1 & 1 & 0 & 0 \\ 0 & 1 & 1 & 0 \\ 0 & 0 & 2 & 1 \\ 0 & 0 & 0 & 2 \end{pmatrix} \in M_4(\mathbb{Q}).$$

- (a) Write down  $p_{\mathbf{A}}$  and the algebraic multiplicities.
- (b) Compute  $E_1(\mathbf{A})$ ,  $G_1(\mathbf{A})$ ,  $E_2(\mathbf{A})$  and  $G_2(\mathbf{A})$ , and check  $\dim G_{\lambda} = a_{\mathbf{A}}(\lambda)$ .
- (c) Verify that  $\mathbb{Q}^4 = G_1(\mathbf{A}) \oplus G_2(\mathbf{A})$ , and find  $m_{\mathbf{A}}$ .

*Solution.*

- (a)  $\mathbf{A}$  is upper triangular, so  $p_{\mathbf{A}} = (x-1)^2(x-2)^2$  (Theorem 6.4.6), and  $a_{\mathbf{A}}(1) = a_{\mathbf{A}}(2) = 2$ .

- (b)  $\mathbf{A} - \mathbf{I} = \begin{pmatrix} 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 1 & 1 \\ 0 & 0 & 0 & 1 \end{pmatrix}$  has rank 3 (rows 1, 2, 4 are independent and row 3 is the sum of rows 2 and 4), so  $E_1(\mathbf{A}) = \operatorname{span}(\mathbf{e}_1)$ . Squaring,

$$(\mathbf{A} - \mathbf{I})^2 = \begin{pmatrix} 0 & 0 & 1 & 0 \\ 0 & 0 & 1 & 1 \\ 0 & 0 & 1 & 2 \\ 0 & 0 & 0 & 1 \end{pmatrix},$$

of rank 2, with kernel  $\operatorname{span}(\mathbf{e}_1, \mathbf{e}_2)$ ; so  $G_1(\mathbf{A}) = \operatorname{span}(\mathbf{e}_1, \mathbf{e}_2)$ , of dimension  $2 = a_{\mathbf{A}}(1)$ . (Membership can be seen directly:  $(\mathbf{A} - \mathbf{I})\mathbf{e}_2 = \mathbf{e}_1$  and  $(\mathbf{A} - \mathbf{I})\mathbf{e}_1 = \mathbf{0}$ .)

Similarly  $\mathbf{A} - 2\mathbf{I} = \begin{pmatrix} -1 & 1 & 0 & 0 \\ 0 & -1 & 1 & 0 \\ 0 & 0 & 0 & 1 \\ 0 & 0 & 0 & 0 \end{pmatrix}$  has rank 3, so  $E_2(\mathbf{A})$  is a line; solv-

ing gives  $E_2(\mathbf{A}) = \text{span}((1, 1, 1, 0))$ . And  $(\mathbf{A} - 2\mathbf{I})^2 = \begin{pmatrix} 1 & -2 & 1 & 0 \\ 0 & 1 & -1 & 1 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \end{pmatrix}$  has

rank 2; its kernel is cut out by  $x_1 - 2x_2 + x_3 = 0$  and  $x_2 - x_3 + x_4 = 0$ , so  $G_2(\mathbf{A}) = \text{span}((1, 1, 1, 0), (-2, -1, 0, 1))$ , of dimension  $2 = a_{\mathbf{A}}(2)$ . (Check:  $(\mathbf{A} - 2\mathbf{I})(-2, -1, 0, 1) = (1, 1, 1, 0)$ , the eigenvector.)

- (c) The four vectors  $\mathbf{e}_1, \mathbf{e}_2, (1, 1, 1, 0), (-2, -1, 0, 1)$  form a basis of  $\mathbb{Q}^4$ : the matrix with these columns is upper triangular with diagonal 1, 1, 1, 1, hence invertible. So the sum is direct and equals  $\mathbb{Q}^4$ , as [Theorem 9.1.3](#) (a) predicts. Both stabilization indices are 2 (the eigenspaces are lines, the generalized eigenspaces planes), so  $m_{\mathbf{A}} = (x - 1)^2(x - 2)^2 = p_{\mathbf{A}}$ .

### 9.1.5.3 C. Going deeper

#### Exercise C1: The Fitting decomposition.

Let  $V$  be finite-dimensional with  $\dim V = n$ , and let  $N \in \mathcal{L}(V)$  be **any** operator. Put  $U = \ker N^n$  and  $W = \text{im } N^n$ .

- Prove that  $U \cap W = \{\mathbf{0}\}$ , and deduce  $V = U \oplus W$ .
- Prove that  $U$  and  $W$  are  $N$ -invariant, that  $N|_U$  is nilpotent, and that  $N|_W$  is invertible.
- Identify  $U$  as a generalized eigenspace of  $N$ .
- Compute  $U$  and  $W$  for  $\mathbf{N} = \begin{pmatrix} 0 & 1 \\ 0 & 1 \end{pmatrix} \in M_2(\mathbb{Q})$ .

*Hint: for (a), if  $\mathbf{v} = N^n \mathbf{w}$  and  $N^n \mathbf{v} = \mathbf{0}$ , look at which kernel  $\mathbf{w}$  lies in.*

*Solution.*

- Let  $\mathbf{v} \in U \cap W$ , say  $\mathbf{v} = N^n \mathbf{w}$  with  $N^n \mathbf{v} = \mathbf{0}$ . Then  $N^{2n} \mathbf{w} = N^n \mathbf{v} = \mathbf{0}$ , so  $\mathbf{w} \in \ker N^{2n} = \ker N^n$  by [Theorem 9.1.1](#) (d), and therefore  $\mathbf{v} = N^n \mathbf{w} = \mathbf{0}$ . By [Theorem 3.3.1](#),  $\dim U + \dim W = n$ , so  $\dim(U + W) = \dim U + \dim W - \dim(U \cap W) = n$  ([Theorem 1.7.3](#)), whence  $U + W = V$  ([Theorem 1.6.8](#)) and the sum is direct ([Theorem 1.7.4](#)).
- Both are invariant by [Theorem 5.7.4](#) (b). On  $U$  we have  $(N|_U)^n = 0$  by definition of  $U$ , so  $N|_U$  is nilpotent. On  $W$ , the map  $N|_W: W \rightarrow W$  is surjective:  $N(W) = N(\text{im } N^n) = \text{im } N^{n+1} = \text{im } N^n = W$  by [Exercise \(B2: The image chain stabilizes too\)](#) (c). A surjective operator on a finite-dimensional space is invertible ([Corollary 3.3.2](#) (e)).
- $U = \ker N^n = G_0(N)$ , by [Corollary 9.1.2](#) (a) with  $\lambda = 0$ . So the Fitting decomposition splits  $V$  into the generalized eigenspace for 0 and a part on which  $N$  is invertible.
- Here  $n = 2$  and  $\mathbf{N}^2 = \begin{pmatrix} 0 & 1 \\ 0 & 1 \end{pmatrix} = \mathbf{N}$ , so  $U = \ker \mathbf{N} = \text{span}(\mathbf{e}_1)$  and  $W = \text{im } \mathbf{N} = \text{span}((1, 1))$ . Indeed  $\mathbb{Q}^2 = \text{span}(\mathbf{e}_1) \oplus \text{span}((1, 1))$ ,  $\mathbf{N}$  is zero on  $U$ , and  $\mathbf{N}(1, 1) = (1, 1)$ , so  $\mathbf{N}|_W$  is the identity.

**Exercise C2: Commuting operators preserve generalized eigenspaces.**

Let  $V$  be finite-dimensional, and let  $S, T \in \mathcal{L}(V)$  with  $ST = TS$ .

- (a) Prove that  $S(G_\lambda(T)) \subseteq G_\lambda(T)$  for every  $\lambda \in F$ .
- (b) Deduce that if  $p_T$  splits, then every generalized eigenspace of  $T$  is invariant under every polynomial in  $T$  and under  $S$ , and that  $V$  has a basis in which both  $S$  and  $T$  have block diagonal matrices with blocks of sizes  $a_T(\lambda_1), \dots, a_T(\lambda_k)$ .
- (c) Show by example that part (a) can be vacuous: find commuting  $S, T$  on  $\mathbb{Q}^2$  and a  $\lambda$  with  $E_\lambda(T)$  a line,  $G_\lambda(T) = \mathbb{Q}^2$ , and  $S$  not a scalar multiple of the identity.

*Solution.*

- (a) Since  $ST = TS$ ,  $S$  commutes with every polynomial in  $T$  ([Theorem 5.7.1](#) (d)), in particular with  $N^k$ , where  $N = T - \lambda \text{id}_V$ . Let  $\mathbf{v} \in G_\lambda(T)$ , say  $N^k \mathbf{v} = \mathbf{0}$ . Then

$$N^k(S\mathbf{v}) = S(N^k \mathbf{v}) = S\mathbf{0} = \mathbf{0},$$

so  $S\mathbf{v} \in \ker N^k \subseteq G_\lambda(T)$ . (This is [Theorem 5.7.4](#) (a) with  $R = N^k$ .)

- (b) Invariance under polynomials in  $T$  is [Theorem 5.7.4](#) (b), and invariance under  $S$  is (a). If  $p_T$  splits, then  $V = \bigoplus_i G_{\lambda_i}(T)$  with  $\dim G_{\lambda_i}(T) = a_T(\lambda_i)$  ([Theorem 9.1.3](#)), and a basis assembled from bases of the summands makes both matrices block diagonal, by [Theorem 8.1.5](#) applied to  $T$  and to  $S$ .
- (c) Take  $T = \begin{pmatrix} 2 & 1 \\ 0 & 2 \end{pmatrix}$  and  $S = T$ . They commute,  $E_2(T) = \text{span}(\mathbf{e}_1)$  is a line,  $G_2(T) = \mathbb{Q}^2$ , and  $S$  is not a scalar matrix. Consistently with [Theorem 8.9.1](#),  $S$  maps the line  $\text{span}(\mathbf{e}_1)$  into itself, and by (a) it maps  $G_2(T) = \mathbb{Q}^2$  into itself, which says nothing. The lesson is that (a) is informative only when  $G_\lambda(T) \neq V$ .

## 9.2 Nilpotent Operators

Section 1 reduced our problem to one piece at a time: on the generalized eigenspace  $G_\lambda(T)$ , the operator is  $\lambda \text{id} + N$  with some power of  $N$  equal to zero ([Theorem 9.1.3](#) (d)). The scalar part needs no work, so everything now depends on a single question: what does an operator all of whose powers eventually vanish look like in the right basis? This section answers it completely, and the answer is a list of numbers — the sizes of the staircases the operator builds.

### 9.2.1 Operators killed by a power

The objects have already appeared several times. Differentiation on  $F[x]_{\leq n}$  satisfies  $D^{n+1} = 0$  ([Example \(Differentiation Is Nilpotent\)](#)). The matrix  $\begin{pmatrix} 0 & 1 \\ 0 & 0 \end{pmatrix}$  has square 0. Over a field of characteristic 0, a commutator  $[\mathbf{A}, \mathbf{B}]$  that commutes with  $\mathbf{A}$  has some power equal to 0 ([Exercise \(C2: When  \$\mathbf{A}\$  commutes with  \$\[\mathbf{A}, \mathbf{B}\]\$ \)](#)). Chapter 3 gave these operators their name in passing; here is the definition in full, with the one extra piece of data we shall need.

*A nilpotent operator is one that some power annihilates: repeated application destroys every*

vector.

### Definition 9.2.1 — Nilpotent Operator, Index of Nilpotency

Let  $V$  be a vector space over  $F$  and  $N \in \mathcal{L}(V)$ . We say  $N$  is **nilpotent** if

$$N^s = 0 \quad \text{for some integer } s \geq 1.$$

If  $V \neq \{0\}$  and  $N$  is nilpotent, the **index of nilpotency** of  $N$  is the **least** such  $s$ ; equivalently,  $N^s = 0$  and  $N^{s-1} \neq 0$ . A matrix  $\mathbf{A} \in M_n(F)$  is **nilpotent** if  $\mathbf{A}^s = 0$  for some  $s \geq 1$ , that is, if  $T_{\mathbf{A}}$  is nilpotent.

In words: one **single** exponent must kill **every** vector at once, and the exponent is at least 1, so the zero operator is nilpotent (index 1) but the identity on  $V \neq \{0\}$  is not. The index is well defined: the exponents  $s \geq 1$  with  $N^s = 0$  form a non-empty set of positive integers, which has a least element (Theorem 0.2.11).

#### Examples.

- **The shift.** On  $F^k$ , let  $N\mathbf{e}_1 = 0$  and  $N\mathbf{e}_j = \mathbf{e}_{j-1}$  for  $j \geq 2$ . Then  $N^k$  kills every  $\mathbf{e}_j$ , while  $N^{k-1}\mathbf{e}_k = \mathbf{e}_1 \neq 0$ , so  $N$  is nilpotent of index  $k$ . Its matrix in the standard basis has a 1 in each position  $(j-1, j)$  and zeros elsewhere. This matrix is the model for everything in this section, and we write it

$$\mathbf{J}_k(0) := \begin{pmatrix} 0 & 1 & & \\ & 0 & \ddots & \\ & & \ddots & 1 \\ & & & 0 \end{pmatrix} \in M_k(F),$$

with 1 in every **superdiagonal** entry  $(j, j+1)$  and 0 everywhere else. (Section 3 writes  $\mathbf{J}_k(\lambda) = \lambda\mathbf{I}_k + \mathbf{J}_k(0)$  and calls it a Jordan block.) For  $k=1$ ,  $\mathbf{J}_1(0) = (0)$ .

- **Differentiation.**  $D$  on  $F[x]_{\leq n}$  has  $D^{n+1} = 0$ , and over  $\mathbb{R}$  the index is exactly  $n+1$ , since  $D^n x^n = n! \neq 0$  (Example (Differentiation Is Nilpotent)). Over  $\mathbb{R}$ , in the basis  $(1, x, x^2/2!, \dots, x^n/n!)$ , its matrix is exactly  $\mathbf{J}_{n+1}(0)$ , because  $D(x^j/j!) = x^{j-1}/(j-1)!$ .
- **All entries non-zero.**  $\mathbf{A} = \begin{pmatrix} 2 & 4 \\ -1 & -2 \end{pmatrix}$  has  $\mathbf{A}^2 = \begin{pmatrix} 4-4 & 8-8 \\ -2+2 & -4+4 \end{pmatrix} = 0$ , so  $\mathbf{A}$  is nilpotent of index 2, although no entry of  $\mathbf{A}$  is 0. Nilpotency is not visible from the shape of the entries.
- **The degenerate case.** On a space of dimension 1 the only nilpotent operator is 0: if  $N = c \text{ id}$  and  $N^s = 0$ , then  $c^s = 0$ , so  $c = 0$ . Small as it is, this is why a  $1 \times 1$  block in a nilpotent normal form carries a 0 and nothing else.

**Non-example by minimal change.** Move the 1 in  $\begin{pmatrix} 0 & 1 \\ 0 & 0 \end{pmatrix}$  to the other off-diagonal position

and take  $\mathbf{S} = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}$ , the swap of the two coordinates. It is just as sparse, but  $\mathbf{S}^2 = \mathbf{I}$ , so  $\mathbf{S}^{2k} = \mathbf{I} \neq 0$  and  $\mathbf{S}^{2k+1} = \mathbf{S} \neq 0$  for every  $k$ : **no** power vanishes, and the defining clause fails outright. A gentler non-example is  $\text{diag}(0, 1)$ , which is singular and has 0 as an eigenvalue, yet  $\text{diag}(0, 1)^k = \text{diag}(0, 1) \neq 0$ ; being non-invertible is far from being nilpotent.

**Why this definition.** The clause “for some  $s$ ” cannot be replaced by a fixed exponent without losing the class, since on  $\mathbb{R}[x]_{\leq n}$  differentiation needs  $n+1$  steps. On the other hand the

exponent may always be taken to be  $\dim V$ , as the next proposition shows, so the definition is not as open-ended as it looks. The name says “zero power”: *nil* (nothing) and *potens* (power).

**Warning.** Nilpotent is a statement about one exponent killing everything, not about each vector separately, and not about the entries. Differentiation on  $\mathbb{R}[x]$  (all polynomials) kills every **single** polynomial after enough steps, yet no power of it is 0, because the number of steps needed is unbounded; the space is infinite-dimensional. And the matrix  $\begin{pmatrix} 2 & 4 \\ -1 & -2 \end{pmatrix}$  is nilpotent with no zero entry, while  $\text{diag}(0, 1)$  has a zero entry, a zero eigenvalue, and is not nilpotent.

The first payoff collects the facts we shall use constantly. Note the interplay with Section 1: a nilpotent operator is one whose generalized eigenspace for 0 is everything.

### Proposition 9.2.1 — Basic Properties of Nilpotent Operators

Let  $V$  be a finite-dimensional vector space over  $F$  with  $\dim V = n \geq 1$ , and let  $N \in \mathcal{L}(V)$ . Then:

- (a)  $N$  is nilpotent if and only if  $N^n = 0$ , if and only if  $G_0(N) = V$ ;
- (b) if  $N$  is nilpotent with index  $s$ , then  $s \leq n$ ,  $m_N = x^s$  and  $p_N = x^n$ ;
- (c) if  $N$  is nilpotent, then  $\text{spec}(N) = \{0\}$ , and there is a basis in which the matrix of  $N$  is upper triangular with zero diagonal;
- (d) conversely, if  $p_N = x^n$ , then  $N$  is nilpotent.

*Idea.* For (a) and (b), the kernel chain of Section 1 does the work:  $N^s = 0$  says  $\ker N^s = V$ , and the chain has stopped growing by step  $n$ . For the triangular matrix in (c), build the basis along the chain: take a basis of  $\ker N$ , extend it to one of  $\ker N^2$ , and so on. A vector first appearing at level  $k$  is sent by  $N$  into  $\ker N^{k-1}$ , which is spanned by vectors listed **earlier**, so every column has zeros from the diagonal down.

*Proof.*

- (a) If  $N^s = 0$  for some  $s \geq 1$ , then  $\ker N^s = V$ , so  $V = \ker N^{\max(s,n)} = \ker N^n$  by [Theorem 9.1.1](#) (a) and (d), that is,  $N^n = 0$ . The converse is the definition with  $s = n$ . And  $G_0(N) = \ker N^n$  by [Corollary 9.1.2](#) (a) with  $\lambda = 0$ , so  $G_0(N) = V$  if and only if  $N^n = 0$ .
- (b) Let  $s$  be the index and put  $K_k = \ker N^k$ , so  $\{0\} = K_0 \subseteq K_1 \subseteq \cdots \subseteq K_s = V$  by (a) and [Theorem 9.1.1](#) (a). Choose a basis  $\mathcal{B}_1$  of  $K_1$ ; extend it to a basis  $\mathcal{B}_2$  of  $K_2$ ; and continuing, extend  $\mathcal{B}_{k-1}$  to a basis  $\mathcal{B}_k$  of  $K_k$  for  $k = 2, \dots, s$  ([Theorem 1.6.5](#), legitimate because  $\mathcal{B}_{k-1}$  is an independent list in  $K_k$ ). Then  $\mathcal{B} = \mathcal{B}_s = (\mathbf{v}_1, \dots, \mathbf{v}_n)$  is a basis of  $V$ , and for each  $k$  the first  $\dim K_k$  of its vectors form a basis of  $K_k$ . Let  $1 \leq j \leq n$  and let  $k$  be the level at which  $\mathbf{v}_j$  was added, so  $\mathbf{v}_j \in K_k$  and  $\dim K_{k-1} < j$ . Then  $N\mathbf{v}_j \in K_{k-1} = \text{span}(\mathbf{v}_1, \dots, \mathbf{v}_{\dim K_{k-1}}) \subseteq \text{span}(\mathbf{v}_1, \dots, \mathbf{v}_{j-1})$ . So the  $j$ -th column of  $[N]_{\mathcal{B}}$  has zeros in rows  $j, j+1, \dots, n$  ([Definition 3.6.1](#), [Theorem 1.5.1](#)): the matrix is upper triangular with zero diagonal. Its diagonal entries are, with multiplicity, the eigenvalues of  $N$  ([Theorem 8.7.4](#) (b)), and they are all 0, so  $\text{spec}(N) = \{0\}$ .
- (c) By (a),  $N^n = 0$ , so  $s \leq n$ . The polynomial  $x^s$  annihilates  $N$ , and  $x^{s-1}$  does not, so  $m_N \mid x^s$  and  $m_N \nmid x^{s-1}$  ([Theorem 8.5.1](#)). Every monic divisor of  $x^s$  is  $x^j$  with



$0 \leq j \leq s$  (Lemma 8.5.2), so  $m_N = x^s$ . For  $p_N$ , use the basis  $\mathcal{B}$  of (c): there  $x\mathbf{I}_n - [N]_{\mathcal{B}}$  is upper triangular with every diagonal entry equal to  $x$ , so  $p_N = x^n$  by Theorem 6.4.6 and Definition 6.10.2.

(d) If  $p_N = x^n$ , then  $p_N(N) = N^n = 0$  by Theorem 8.6.1.

■

So nilpotency is detected by the characteristic polynomial:  $N$  is nilpotent exactly when  $p_N = x^n$ , which over  $\mathbb{C}$  is exactly  $\text{spec}(N) = \{0\}$ , as Chapter 8 found by a different route (Theorem 8.10.3). Part (c) also follows abstractly from Theorem 8.7.2, since  $x^n$  splits over every field; the proof above builds the basis explicitly. But a strictly upper triangular matrix still has  $n(n-1)/2$  entries to describe, and almost all of them can be cleared away. That is the content of the structure theorem.

## 9.2.2 What one chain looks like

Look again at  $\mathbf{J}_k(0)$  and read its columns, which is where a matrix always tells us what the operator does to basis vectors. With  $\mathcal{B} = (\mathbf{b}_1, \dots, \mathbf{b}_k)$  the standard basis of  $F^k$ , the matrix  $\mathbf{J}_k(0)$  says

$$N\mathbf{b}_1 = \mathbf{0}, \quad N\mathbf{b}_2 = \mathbf{b}_1, \quad N\mathbf{b}_3 = \mathbf{b}_2, \quad \dots, \quad N\mathbf{b}_k = \mathbf{b}_{k-1}.$$

Reading this from the right, the last basis vector  $\mathbf{b}_k$  generates all the others:  $\mathbf{b}_{k-1} = N\mathbf{b}_k$ ,  $\mathbf{b}_{k-2} = N^2\mathbf{b}_k$ , and so on down to  $\mathbf{b}_1 = N^{k-1}\mathbf{b}_k$ , which is an eigenvector, killed at the next step. So one Jordan block is one vector together with its successive images, a staircase descending to  $\mathbf{0}$ . Such a list deserves a name.

### Definition 9.2.2 — Jordan Chain

Let  $V$  be a vector space over  $F$ , let  $N \in \mathcal{L}(V)$  and let  $\mathbf{u} \in V$  with  $\mathbf{u} \neq \mathbf{0}$ . Suppose  $N^k\mathbf{u} = \mathbf{0}$  and  $N^{k-1}\mathbf{u} \neq \mathbf{0}$  for an integer  $k \geq 1$ . The **Jordan chain generated by  $\mathbf{u}$**  is the ordered list

$$(N^{k-1}\mathbf{u}, N^{k-2}\mathbf{u}, \dots, N\mathbf{u}, \mathbf{u}),$$

of **length  $k$** , and  $\mathbf{u}$  is called the **top** of the chain. For  $T \in \mathcal{L}(V)$  and  $\lambda \in F$ , a Jordan chain of  $T$  for  $\lambda$  means a Jordan chain of  $N = T - \lambda \text{id}_V$ .

The order matters, and we always write it with the **eigenvector first** and the top last, so that  $N$  moves each entry to the one before it and kills the first. The first entry  $N^{k-1}\mathbf{u}$  is indeed an eigenvector for  $0$ , by the argument of Corollary 9.1.2 (d). For example, the chain generated by  $\mathbf{e}_2$  under  $\begin{pmatrix} 0 & 1 \\ 0 & 0 \end{pmatrix}$  is  $(\mathbf{e}_1, \mathbf{e}_2)$ , of length 2.

A chain is automatically independent, which is the first sign that chains are good building blocks.

### Lemma 9.2.2 — A Jordan Chain Is Independent

Let  $N \in \mathcal{L}(V)$  and let  $(N^{k-1}\mathbf{u}, \dots, N\mathbf{u}, \mathbf{u})$  be a Jordan chain, so  $N^k\mathbf{u} = \mathbf{0} \neq N^{k-1}\mathbf{u}$ . Then:

(a) the list is linearly independent, so  $k \leq \dim V$  when  $V$  is finite-dimensional;

- (b)  $C = \text{span}(\mathbf{u}, N\mathbf{u}, \dots, N^{k-1}\mathbf{u})$  is  $N$ -invariant of dimension  $k$ , and the matrix of  $N|_C$  in the chain, taken in the order of [Definition 9.2.2](#), is  $\mathbf{J}_k(0)$ .

*Proof.*

- (a) Suppose  $a_0\mathbf{u} + a_1N\mathbf{u} + \dots + a_{k-1}N^{k-1}\mathbf{u} = \mathbf{0}$  with  $a_j \in F$ . Apply  $N^{k-1}$ . Every term with  $j \geq 1$  becomes  $a_jN^{k-1+j}\mathbf{u} = \mathbf{0}$ , since  $N^k\mathbf{u} = \mathbf{0}$  and  $k-1+j \geq k$ . What remains is  $a_0N^{k-1}\mathbf{u} = \mathbf{0}$ , and  $N^{k-1}\mathbf{u} \neq \mathbf{0}$ , so  $a_0 = 0$  ([Theorem 1.1.9](#)). Now apply  $N^{k-2}$  to the relation with  $a_0 = 0$ : by the same cancellation only  $a_1N^{k-1}\mathbf{u} = \mathbf{0}$  survives, so  $a_1 = 0$ . Repeating this for  $N^{k-3}, \dots, N^0$  gives  $a_2 = \dots = a_{k-1} = 0$ . Hence the list is independent, and an independent list has at most  $\dim V$  entries ([Theorem 1.6.5](#)).
- (b) By (a) the  $k$  spanning vectors are independent, so  $\dim C = k$ . Write  $\mathbf{b}_j = N^{k-j}\mathbf{u}$  for  $j = 1, \dots, k$ , the chain in order. Then  $N\mathbf{b}_1 = N^k\mathbf{u} = \mathbf{0}$  and  $N\mathbf{b}_j = N^{k-j+1}\mathbf{u} = \mathbf{b}_{j-1}$  for  $j \geq 2$ , so all images lie in  $C$  and  $C$  is  $N$ -invariant ([Lemma 8.1.1](#)). The  $j$ -th column of the matrix of  $N|_C$  is the coordinate vector of  $N\mathbf{b}_j$ , which is  $\mathbf{0}$  for  $j = 1$  and  $\mathbf{e}_{j-1}$  for  $j \geq 2$  ([Definition 3.6.1](#)). That is precisely  $\mathbf{J}_k(0)$ . ■

### 9.2.3 The structure theorem

Now the goal. If a nilpotent operator could be split into chains, its matrix would be a direct sum of blocks  $\mathbf{J}_{k_i}(0)$ , and nothing would be left to describe except the list of lengths. That is exactly what happens.

*Every nilpotent operator is a disjoint union of staircases: the space breaks into chains, and the operator shifts each chain one step down.*

#### Theorem 9.2.3 — Structure Theorem for Nilpotent Operators

Let  $V$  be a finite-dimensional vector space over  $F$  with  $\dim V = n \geq 1$ , and let  $N \in \mathcal{L}(V)$  be nilpotent. Then there are an integer  $r \geq 1$ , vectors  $\mathbf{u}_1, \dots, \mathbf{u}_r \in V$  and integers

$$k_1 \geq k_2 \geq \dots \geq k_r \geq 1, \quad k_1 + k_2 + \dots + k_r = n,$$

such that:

- (a)  $N^{k_i}\mathbf{u}_i = \mathbf{0}$  for each  $i$ , and the list  $\mathcal{B}$  obtained by writing the  $r$  Jordan chains one after another,

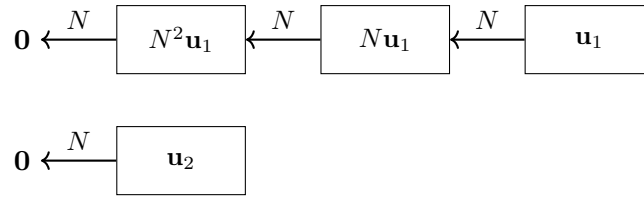
$$\mathcal{B} = \left( \underbrace{N^{k_1-1}\mathbf{u}_1, \dots, N\mathbf{u}_1, \mathbf{u}_1}_{\text{chain 1}}, \dots, \underbrace{N^{k_r-1}\mathbf{u}_r, \dots, N\mathbf{u}_r, \mathbf{u}_r}_{\text{chain } r} \right),$$

is a basis of  $V$ ;

- (b) with  $C_i = \text{span}(\mathbf{u}_i, N\mathbf{u}_i, \dots, N^{k_i-1}\mathbf{u}_i)$ , each  $C_i$  is  $N$ -invariant,  $V = C_1 \oplus \dots \oplus C_r$ , and

$$[N]_{\mathcal{B}} = \mathbf{J}_{k_1}(0) \oplus \mathbf{J}_{k_2}(0) \oplus \dots \oplus \mathbf{J}_{k_r}(0).$$

A basis as in (a) is called a **Jordan basis** for  $N$ . Before the proof, here is the picture the proof builds. Each chain is a row of boxes, the top of the chain on the right, and  $N$  moves one box to the left, off the diagram at the left edge. For lengths  $(3, 1)$ :



the boxes in the left column,  $N^2\mathbf{u}_1$  and  $\mathbf{u}_2$ , form a basis of  $\ker N$

*Idea.* Induction on  $\dim V$ , with the smaller object found inside:  $W = \operatorname{im} N$ . It is  $N$ -invariant,  $N|_W$  is nilpotent, and  $\dim W < \dim V$  because a nilpotent operator on a non-zero space is not injective. So  $W$  already has chains. Each chain of  $W$  is one box short of what we want: its top  $\mathbf{w}_i$  lies in  $\operatorname{im} N$ , so  $\mathbf{w}_i = N\mathbf{u}_i$  for some  $\mathbf{u}_i \in V$ , and we can extend the chain by one box on the right, gaining  $\mathbf{u}_i$ . What is still missing is counted by  $\ker N$ : the ends  $N^{l_i}\mathbf{u}_i$  of the extended chains are independent vectors of  $\ker N$ , and any basis extension of them inside  $\ker N$  supplies the chains of length 1. Then the count comes out exactly right, by Rank–Nullity ([Theorem 3.3.1](#)), and independence follows by applying  $N$  once and using the basis of  $W$ .

*Proof.* We argue by induction on  $n = \dim V \geq 1$ , the statement being: for every nilpotent  $N$  on a space of dimension  $n$ , chains as in (a) exist.

*Base case*  $n = 1$ . Here  $N = 0$ , as noted after [Definition 9.2.1](#). Take  $r = 1$ ,  $k_1 = 1$  and  $\mathbf{u}_1$  any non-zero vector; then  $N^1\mathbf{u}_1 = \mathbf{0}$  and  $\mathcal{B} = (\mathbf{u}_1)$  is a basis.

*Induction step.* Let  $n \geq 2$  and assume the statement for all spaces of dimension less than  $n$ . Let  $N \in \mathcal{L}(V)$  be nilpotent with  $\dim V = n$ .

$N$  is not injective: if it were, then  $N^s$  would be injective for every  $s$  (a composition of injective maps is injective, [Theorem 0.4.3](#) (a)), contradicting  $N^s = 0$  on  $V \neq \{\mathbf{0}\}$ . So nullity  $N \geq 1$ , and with  $W = \operatorname{im} N$  Rank–Nullity gives  $\dim W = n - \text{nullity } N \leq n - 1$  ([Theorem 3.3.1](#)).

**Step 1. Chains inside  $\operatorname{im} N$ .** If  $W = \{\mathbf{0}\}$ , then  $N = 0$ ; take  $r = n$ , all  $k_i = 1$  and  $(\mathbf{u}_1, \dots, \mathbf{u}_n)$  any basis of  $V$ , and (a) holds. So assume  $W \neq \{\mathbf{0}\}$ . Then  $W$  is  $N$ -invariant ([Theorem 5.7.4](#) (b)),  $N|_W$  is nilpotent ( $(N|_W)^s = (N^s)|_W = 0$ ), and  $1 \leq \dim W < n$ . By the induction hypothesis there are  $\mathbf{w}_1, \dots, \mathbf{w}_q \in W$  and  $l_1 \geq \dots \geq l_q \geq 1$  with  $\sum_i l_i = \dim W$ ,  $(N|_W)^{l_i}\mathbf{w}_i = \mathbf{0}$ , and

$$\mathcal{C} = (N^{l_1-1}\mathbf{w}_1, \dots, \mathbf{w}_1, \dots, N^{l_q-1}\mathbf{w}_q, \dots, \mathbf{w}_q)$$

a basis of  $W$ .

**Step 2. Lift each chain by one box.** Each  $\mathbf{w}_i$  lies in  $W = \operatorname{im} N$ , so we may choose  $\mathbf{u}_i \in V$  with  $N\mathbf{u}_i = \mathbf{w}_i$  for  $i = 1, \dots, q$ . Put  $k_i = l_i + 1$ . Then

$$N^j\mathbf{u}_i = N^{j-1}\mathbf{w}_i \quad (j \geq 1), \quad \text{so} \quad N^{k_i}\mathbf{u}_i = N^{l_i}\mathbf{w}_i = \mathbf{0}.$$

**Step 3. Fill up  $\ker N$ .** The  $q$  vectors  $N^{l_i}\mathbf{u}_i = N^{l_i-1}\mathbf{w}_i$  ( $i = 1, \dots, q$ ) are members of the

basis  $\mathcal{C}$ , hence linearly independent, and each lies in  $\ker N$ , because  $N(N^{l_i}\mathbf{u}_i) = N^{k_i}\mathbf{u}_i = \mathbf{0}$ . By [Theorem 1.6.5](#) applied inside  $\ker N$ , there are vectors  $\mathbf{u}_{q+1}, \dots, \mathbf{u}_r \in \ker N$  such that

$$(N^{l_1}\mathbf{u}_1, \dots, N^{l_q}\mathbf{u}_q, \mathbf{u}_{q+1}, \dots, \mathbf{u}_r)$$

is a basis of  $\ker N$ ; in particular  $r = \text{nullity } N$ . Put  $k_i = 1$  for  $q < i \leq r$ , so that  $N^{k_i}\mathbf{u}_i = N\mathbf{u}_i = \mathbf{0}$  for those  $i$  as well.

**Step 4. The combined list is a basis.** Let  $\mathcal{B}$  be the list of all  $N^j\mathbf{u}_i$  with  $1 \leq i \leq r$  and  $0 \leq j \leq k_i - 1$ , the chains written one after another as in (a). Its length is

$$\sum_{i=1}^q (l_i + 1) + (r - q) = \dim W + q + (r - q) = \text{rank } N + \text{nullity } N = n$$

by Step 1 and [Theorem 3.3.1](#). So by [Theorem 1.6.4](#) (a) it is enough to prove independence.

Suppose

$$\sum_{i=1}^r \sum_{j=0}^{k_i-1} a_{ij} N^j \mathbf{u}_i = \mathbf{0} \quad (*)$$

with  $a_{ij} \in F$ . Apply  $N$  to  $(*)$ . For  $i > q$  we have  $k_i = 1$  and  $N\mathbf{u}_i = \mathbf{0}$ , so those terms die. For  $i \leq q$  and  $j = k_i - 1 = l_i$  we get  $N^{l_i+1}\mathbf{u}_i = \mathbf{0}$  by Step 2, so those terms die too. The rest is, using  $N^{j+1}\mathbf{u}_i = N^j\mathbf{w}_i$ ,

$$\sum_{i=1}^q \sum_{j=0}^{l_i-1} a_{ij} N^j \mathbf{w}_i = \mathbf{0}.$$

This is a linear combination of the members of the basis  $\mathcal{C}$  of  $W$ , so

$$a_{ij} = 0 \quad \text{for } 1 \leq i \leq q, 0 \leq j \leq l_i - 1.$$

Substituting these zeros back into  $(*)$  leaves only the terms with  $i \leq q$ ,  $j = l_i$  and those with  $i > q$ ,  $j = 0$ :

$$\sum_{i=1}^q a_{il_i} N^{l_i} \mathbf{u}_i + \sum_{i=q+1}^r a_{i0} \mathbf{u}_i = \mathbf{0}.$$

This is a linear combination of the members of the basis of  $\ker N$  from Step 3, so all the remaining coefficients vanish as well. Hence  $\mathcal{B}$  is independent, and therefore a basis of  $V$ . The lengths satisfy  $\sum_i k_i = n$  by the count above, and reindexing the chains so that the lengths decrease — a permutation of the chains — arranges  $k_1 \geq \dots \geq k_r$ . This proves (a), and completes the induction.

- (b) Each  $C_i$  is  $N$ -invariant of dimension  $k_i$ , with  $[N|_{C_i}] = \mathbf{J}_{k_i}(0)$  in its chain, by [Lemma 9.2.2](#) (b). The chains together form the basis  $\mathcal{B}$  of  $V$ , so  $V = C_1 + \dots + C_r$ . The sum is direct: if  $\mathbf{c}_1 + \dots + \mathbf{c}_r = \mathbf{0}$  with  $\mathbf{c}_i \in C_i$ , then writing each  $\mathbf{c}_i$  in terms of its own chain turns this into a linear combination of the members of the basis  $\mathcal{B}$  equal to  $\mathbf{0}$ , so every coefficient is 0 and every  $\mathbf{c}_i = \mathbf{0}$ ; now apply [Theorem 1.7.5](#) ((b)  $\Rightarrow$  (a)). Finally [Theorem 8.1.5](#) gives that  $[N]_{\mathcal{B}}$  is block diagonal with blocks  $[N|_{C_i}] = \mathbf{J}_{k_i}(0)$ .

■

The theorem says that a nilpotent operator carries no information beyond the list of chain lengths. In matrix terms: **every nilpotent**  $\mathbf{A} \in M_n(F)$  **is similar to**  $\mathbf{J}_{k_1}(0) \oplus \cdots \oplus \mathbf{J}_{k_r}(0)$  **for some**  $k_1 \geq \cdots \geq k_r \geq 1$  **summing to**  $n$  (apply the theorem to  $T_{\mathbf{A}}$  and use [Theorem 3.7.4](#)). The index of nilpotency is  $k_1$ , the largest block (it is also the stabilization index of the kernel chain of [Theorem 9.1.1](#), since  $\ker N^{k_1} = V$ ):  $\mathbf{J}_k(0)^k = 0$  and  $\mathbf{J}_k(0)^{k-1} \neq 0$ , and powers of a block diagonal matrix act blockwise ([Theorem 7.1.4](#) (b)). Two questions remain: are the lengths unique, and how do we find them without finding the chains? One computation answers both.

## 9.2.4 Block sizes from the ranks of the powers

The lengths are visible in the ranks of the powers of  $N$ , which are similarity invariants. This is the practical algorithm and the uniqueness proof at once.

### Theorem 9.2.4 — Block Sizes from Ranks of Powers

Let  $V$  be finite-dimensional with  $\dim V = n \geq 1$ , let  $N \in \mathcal{L}(V)$  be nilpotent, and suppose  $\mathcal{B}$  is a Jordan basis for  $N$  with chain lengths  $k_1 \geq \cdots \geq k_r \geq 1$ , as in [Theorem 9.2.3](#). Write  $r_j = \text{rank } N^j$  for  $j \geq 0$ , so  $r_0 = n$ . Then:

- (a) for every integer  $j \geq 0$ ,

$$\text{rank } N^j = \sum_{i=1}^r \max(k_i - j, 0) \quad \text{and} \quad \dim \ker N^j = \sum_{i=1}^r \min(k_i, j);$$

- (b) for every integer  $j \geq 1$ , the number of chains of length **at least**  $j$  is

$$\#\{i : k_i \geq j\} = r_{j-1} - r_j;$$

- (c) for every integer  $j \geq 1$ , the number of chains of length **exactly**  $j$  is

$$\#\{i : k_i = j\} = r_{j-1} - 2r_j + r_{j+1};$$

- (d) consequently the list  $k_1 \geq \cdots \geq k_r$  is determined by  $N$  alone: any two Jordan bases for  $N$  have the same chain lengths. Moreover  $r = n - \text{rank } N = \text{nullity } N$  and  $k_1$  is the index of nilpotency of  $N$ .

*Idea.* Everything follows from what  $N^j$  does to the basis: it shifts each chain  $j$  boxes to the left, so the surviving images are the boxes at least  $j$  steps from the left edge. They are members of the basis, hence independent, and we can just count them: chain  $i$  contributes  $k_i - j$  of them, or none if it is too short. Then (b) is a telescoping difference, because lengthening the shift by one step loses one box from every chain that still had one.

*Proof.*

- (a) The basis  $\mathcal{B}$  consists of the vectors  $N^m \mathbf{u}_i$  with  $1 \leq i \leq r$  and  $0 \leq m \leq k_i - 1$ . Fix  $j \geq 0$ . Applying  $N^j$  to a member gives  $N^j(N^m \mathbf{u}_i) = N^{m+j} \mathbf{u}_i$ , which is  $\mathbf{0}$  when  $m+j \geq k_i$ , because  $N^{k_i} \mathbf{u}_i = \mathbf{0}$ , and is a member of  $\mathcal{B}$ , hence non-zero, when

$m+j \leq k_i-1$ . Since the image is spanned by the images of a basis (Theorem 3.2.4),

$$\operatorname{im} N^j = \operatorname{span} (N^m \mathbf{u}_i : 1 \leq i \leq r, j \leq m \leq k_i - 1).$$

This spanning list is a sublist of the basis  $\mathcal{B}$ , hence linearly independent, so it is a basis of  $\operatorname{im} N^j$  and  $\operatorname{rank} N^j$  is its length: for each  $i$ , the number of integers  $m$  with  $j \leq m \leq k_i - 1$  is  $k_i - j$  if  $k_i > j$ , and 0 otherwise, that is  $\max(k_i - j, 0)$ . This gives the first formula, and the second follows from Theorem 3.3.1 and  $\sum_i k_i = n$ :

$$\dim \ker N^j = n - \operatorname{rank} N^j = \sum_{i=1}^r (k_i - \max(k_i - j, 0)) = \sum_{i=1}^r \min(k_i, j).$$

- (b) By (a),  $r_{j-1} - r_j = \sum_i (\max(k_i - j + 1, 0) - \max(k_i - j, 0))$ . For a single  $i$ , the bracket is  $(k_i - j + 1) - (k_i - j) = 1$  if  $k_i \geq j$ , and  $0 - 0 = 0$  if  $k_i \leq j - 1$ . Summing over  $i$  counts the indices with  $k_i \geq j$ .
- (c) Subtract the count in (b) for  $j + 1$  from the count for  $j$ : a chain has length exactly  $j$  if and only if its length is at least  $j$  but not at least  $j + 1$ . This gives  $(r_{j-1} - r_j) - (r_j - r_{j+1})$ .
- (d) The numbers  $r_j = \operatorname{rank} N^j$  are defined from  $N$  with no reference to a basis, so by (c) the number of chains of each length is determined by  $N$ ; hence so is the list of lengths in decreasing order. Taking  $j = 1$  in (b) gives  $r = \#\{i : k_i \geq 1\} = r_0 - r_1 = n - \operatorname{rank} N$ , which is nullity  $N$  by Theorem 3.3.1. Finally  $N^{k_1} = 0$  and  $N^{k_1-1} \neq 0$ , since by (a)  $\operatorname{rank} N^{k_1} = 0$  and  $\operatorname{rank} N^{k_1-1} \geq k_1 - (k_1 - 1) = 1$ ; so  $k_1$  is the index.

■

### Corollary 9.2.5 — Similarity of Nilpotent Matrices

Let  $\mathbf{A}, \mathbf{B} \in M_n(F)$ ,  $n \geq 1$ , both nilpotent. Then  $\mathbf{A} \sim \mathbf{B}$  if and only if

$$\operatorname{rank} \mathbf{A}^j = \operatorname{rank} \mathbf{B}^j \quad \text{for every } j = 1, 2, \dots, n-1.$$

Equivalently,  $\mathbf{A} \sim \mathbf{B}$  if and only if their chain lengths, as lists in decreasing order, agree.

*Proof.* ( $\Rightarrow$ ) If  $\mathbf{A} \sim \mathbf{B}$ , say  $\mathbf{B} = \mathbf{P}^{-1} \mathbf{A} \mathbf{P}$ , then  $\mathbf{B}^j = \mathbf{P}^{-1} \mathbf{A}^j \mathbf{P}$  (Proposition 3.7.6 (c)), so  $\operatorname{rank} \mathbf{A}^j = \operatorname{rank} \mathbf{B}^j$  (Proposition 3.7.6 (a)).

( $\Leftarrow$ ) Since  $\operatorname{rank} T_{\mathbf{A}}^j = \operatorname{rank} \mathbf{A}^j$  and  $\operatorname{rank} T_{\mathbf{B}}^j = \operatorname{rank} \mathbf{B}^j$  for every  $j$  (Theorem 3.6.5 (a)), the hypothesis is a statement about the operators. By Theorem 9.2.3 applied to  $T_{\mathbf{A}}$  and to  $T_{\mathbf{B}}$ , together with Theorem 3.7.4,  $\mathbf{A}$  is similar to  $\mathbf{J}_{k_1}(0) \oplus \dots \oplus \mathbf{J}_{k_r}(0)$  and  $\mathbf{B}$  to  $\mathbf{J}_{k'_1}(0) \oplus \dots \oplus \mathbf{J}_{k'_{r'}}(0)$ , with both lists decreasing. Equal ranks of all powers force equal lists, by Theorem 9.2.4 (c) (the ranks for  $j \geq n$  are all 0, so the listed range suffices). Hence  $\mathbf{A}$  and  $\mathbf{B}$  are similar to the same matrix, and  $\mathbf{A} \sim \mathbf{B}$  because similarity is an equivalence relation (Example (Similarity)).

■

**Warning.** The successive differences count the blocks of size at least  $j$ , not the blocks of size exactly  $j$ ; to get the exact counts, subtract twice. Suppose

$n = 5$  and  $\text{rank } N = 3$ ,  $\text{rank } N^2 = 1$ ,  $\text{rank } N^3 = 0$ . Then  $r_0 - r_1 = 2$  chains have length  $\geq 1$ ,  $r_1 - r_2 = 2$  have length  $\geq 2$ , and  $r_2 - r_3 = 1$  has length  $\geq 3$ . Reading these as exact counts would suggest two chains of length 1, two of length 2 and one of length 3, that is, lengths 1, 1, 2, 2, 3, summing to 9 instead of 5. The correct counts are the second differences of [Theorem 9.2.4](#) (c): exactly  $2 - 2 = 0$  chains of length 1, exactly  $2 - 1 = 1$  of length 2, and exactly  $1 - 0 = 1$  of length 3. The partition is  $(3, 2)$ , and  $3 + 2 = 5$  as it must be.

*Quick check.* A nilpotent  $N$  on a space of dimension 6 has  $\text{rank } N = 3$  and  $\text{rank } N^2 = 1$ . What are its chain lengths?

*Solution.* Here  $r_0 = 6$ ,  $r_1 = 3$  and  $r_2 = 1$ , so there are  $r_0 - r_1 = 3$  chains by part (b) of [Theorem 9.2.4](#). Also  $r_3 = 0$ : a chain of length  $\geq 4$  would contribute at least 2 to  $r_2 = \sum_i \max(k_i - 2, 0) = 1$ , so every  $k_i \leq 3$  and  $r_3 = \sum_i \max(k_i - 3, 0) = 0$ . By part (c) the exact counts are  $r_0 - 2r_1 + r_2 = 1$  chain of length 1,  $r_1 - 2r_2 + r_3 = 1$  of length 2, and  $r_2 - 2r_3 + r_4 = 1$  of length 3. So the lengths are  $(3, 2, 1)$ , which indeed sum to 6.

## 9.2.5 Partitions and Young diagrams

The data left over is a list of decreasing positive integers with a given sum. Such lists have a name.

### Definition 9.2.3 — Partition of an Integer, Young Diagram, Dual Partition

Let  $n \geq 1$ . A **partition of  $n$**  is a finite list of integers

$$k_1 \geq k_2 \geq \cdots \geq k_r \geq 1 \quad \text{with} \quad k_1 + k_2 + \cdots + k_r = n.$$

Its **Young diagram** is the left-justified array of  $n$  boxes with  $k_i$  boxes in row  $i$ . The **dual partition**  $k_1^* \geq k_2^* \geq \cdots$  is the list of **column** lengths of the diagram,

$$k_j^* = \#\{i : k_i \geq j\},$$

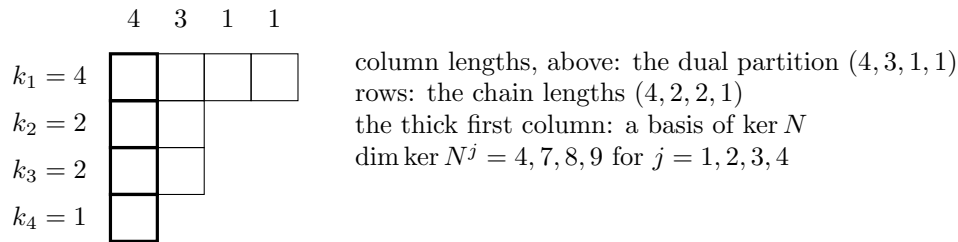
which is again a partition of  $n$ .

(The word *partition* is also used for a partition of a **set** into blocks ([Definition 0.5.4](#)) and for a partition of the row and column indices of a block matrix ([Definition 7.1.1](#)). Those are different notions; here the blocks are boxes in a diagram and the data is a list of numbers.)

For a nilpotent  $N$  on a space of dimension  $n$ , the chain lengths form a partition of  $n$ , and [Theorem 9.2.4](#) (b) identifies the dual partition:

$$k_j^* = \#\{i : k_i \geq j\} = r_{j-1} - r_j = \dim \ker N^j - \dim \ker N^{j-1},$$

using [Theorem 3.3.1](#) for the last equality. **The rows of the diagram are the chains; the columns are the jumps of the kernel chain.** In particular the first column is a basis of  $\ker N$  (its boxes are the vectors  $N^{k_i-1}\mathbf{u}_i$ ), the first two columns span  $\ker N^2$ , and so on. Here is the diagram for the partition  $(4, 2, 2, 1)$  of 9, with the columns marked:



Reading the diagram by rows gives the Jordan blocks; reading it by columns gives the dimensions  $\dim \ker N^j = k_1^* + \cdots + k_j^*$ . Transposing the diagram exchanges a partition with its dual.

*Remark.* For each  $n$ , the nilpotent operators on an  $n$ -dimensional space, up to similarity, correspond exactly to the partitions of  $n$ , and the correspondence does not depend on the field. For  $n = 4$  the partitions are

$$(4), \quad (3, 1), \quad (2, 2), \quad (2, 1, 1), \quad (1, 1, 1, 1),$$

so there are exactly five similarity classes of nilpotent  $4 \times 4$  matrices over any field, with indices of nilpotency 4, 3, 2, 2, 1.

## 9.2.6 Finding a Jordan basis

The proof of [Theorem 9.2.3](#) is an algorithm, but run from the inside out. For small matrices it is easier to work from the longest chain down, using the rank data to know what to look for.

*Algorithm.* **Jordan basis of a nilpotent operator.**

1. Compute  $r_j = \text{rank } N^j$  until  $r_j = 0$ , and read off the chain lengths by [Theorem 9.2.4](#) (c).
2. For each length in decreasing order, choose a top  $\mathbf{u}$  with  $N^{k-1}\mathbf{u} \neq \mathbf{0}$ , keeping the chains collected so far independent, and write down its chain.
3. Chains of length 1 are vectors completing a basis of  $\ker N$  together with the ends  $N^{k_i-1}\mathbf{u}_i$  of the longer chains. Once  $n$  vectors are collected, [Theorem 1.6.4](#) confirms a basis.

**Example.** (A Jordan Basis for a  $4 \times 4$  Nilpotent Matrix)

Let

$$\mathbf{N} = \begin{pmatrix} 1 & 0 & 1 & 1 \\ 1 & -1 & 1 & 1 \\ 1 & -1 & 1 & 1 \\ -1 & 0 & -1 & -1 \end{pmatrix} \in M_4(\mathbb{Q}).$$

Show that  $\mathbf{N}$  is nilpotent, find its chain lengths, and find a Jordan basis together with an invertible  $\mathbf{P}$  with  $\mathbf{P}^{-1}\mathbf{N}\mathbf{P}$  a direct sum of blocks  $\mathbf{J}_k(0)$ .



*Solution. Nilpotency and ranks.* Multiplying out,

$$\mathbf{N}^2 = \begin{pmatrix} 1 & -1 & 1 & 1 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \\ -1 & 1 & -1 & -1 \end{pmatrix}, \quad \mathbf{N}^3 = 0,$$

the last because every row of  $\mathbf{N}^2$  is a multiple of  $(1, -1, 1, 1)$ , while

$$\begin{aligned} (1, -1, 1, 1)\mathbf{N} &= (1 - 1 + 1 - 1, \quad 0 + 1 - 1 + 0, \quad 1 - 1 + 1 - 1, \quad 1 - 1 + 1 - 1) \\ &= 0. \end{aligned}$$

Since  $\mathbf{N}^2 \neq 0$ , the index is 3. Rows 2 and 3 of  $\mathbf{N}$  are equal and row 4 = -row 1, so  $\text{rank } \mathbf{N} = 2$ , and  $\text{rank } \mathbf{N}^2 = 1$ . Thus

$$r_0 = 4, \quad r_1 = 2, \quad r_2 = 1, \quad r_3 = 0.$$

By [Theorem 9.2.4](#) (c), the number of chains of length 1 is  $r_0 - 2r_1 + r_2 = 4 - 4 + 1 = 1$ , of length 2 is  $r_1 - 2r_2 + r_3 = 2 - 2 + 0 = 0$ , and of length 3 is  $r_2 - 2r_3 + r_4 = 1 - 0 + 0 = 1$ . The partition is  $(3, 1)$ : one chain of length 3 and one of length 1, matching the picture drawn after [Theorem 9.2.3](#).

*The long chain.* We need  $\mathbf{u}_1 \notin \ker \mathbf{N}^2$ . Try  $\mathbf{u}_1 = \mathbf{e}_1$ : the first column of  $\mathbf{N}^2$  is  $(1, 0, 0, -1) \neq \mathbf{0}$ , so  $\mathbf{N}^2\mathbf{e}_1 \neq \mathbf{0}$  and  $\mathbf{e}_1$  works. Its chain, read off the first columns of  $\mathbf{N}$  and  $\mathbf{N}^2$ , is

$$\mathbf{N}^2\mathbf{e}_1 = (1, 0, 0, -1), \quad \mathbf{N}\mathbf{e}_1 = (1, 1, 1, -1), \quad \mathbf{e}_1.$$

*The short chain.*  $\ker \mathbf{N}$  has dimension 2 ([Theorem 2.5.7](#)). The equations  $x_1 + x_3 + x_4 = 0$  (row 1) and  $x_1 - x_2 + x_3 + x_4 = 0$  (row 2) give  $x_2 = 0$  and  $x_1 = -x_3 - x_4$ , so

$$\ker \mathbf{N} = \text{span} \{(-1, 0, 1, 0), (-1, 0, 0, 1)\}.$$

The end of the long chain,  $\mathbf{N}^2\mathbf{e}_1 = (1, 0, 0, -1)$ , is the second of these basis vectors with a minus sign, so we complete it to a basis of  $\ker \mathbf{N}$  with  $\mathbf{u}_2 = (-1, 0, 1, 0)$ .

*The basis and the matrix.* Take

$$\begin{aligned} \mathcal{B} &= ((1, 0, 0, -1), (1, 1, 1, -1), \mathbf{e}_1, (-1, 0, 1, 0)), \\ \mathbf{P} &= \begin{pmatrix} 1 & 1 & 1 & -1 \\ 0 & 1 & 0 & 0 \\ 0 & 1 & 0 & 1 \\ -1 & -1 & 0 & 0 \end{pmatrix}, \end{aligned}$$

the matrix whose columns are the vectors of  $\mathcal{B}$ . Its third column has a single non-zero entry, the 1 in row 1, so expanding along that column ([Theorem 6.6.1](#)) gives

$$\det \mathbf{P} = \det \begin{pmatrix} 0 & 1 & 0 \\ 0 & 1 & 1 \\ -1 & -1 & 0 \end{pmatrix} = -1,$$

the last determinant computed by expanding along its own first column. So  $\mathbf{P}$  is invertible and  $\mathcal{B}$  is a basis. Then

$$\mathbf{P}^{-1}\mathbf{N}\mathbf{P} = \mathbf{J}_3(0) \oplus \mathbf{J}_1(0) = \begin{pmatrix} 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \end{pmatrix}.$$

There is no need to invert  $\mathbf{P}$  to check this: the claim  $\mathbf{P}^{-1}\mathbf{N}\mathbf{P} = \mathbf{J}$  is the same as  $\mathbf{N}\mathbf{P} = \mathbf{P}\mathbf{J}$ , which reads column by column as

$$\begin{aligned} \mathbf{N}(1, 0, 0, -1) &= \mathbf{0}, \\ \mathbf{N}(1, 1, 1, -1) &= (1, 0, 0, -1), \\ \mathbf{N}\mathbf{e}_1 &= (1, 1, 1, -1), \\ \mathbf{N}(-1, 0, 1, 0) &= \mathbf{0}, \end{aligned}$$

and these four statements are exactly how the chains were built (the first and last say that the two chain ends lie in  $\ker \mathbf{N}$ ). As a sample verification, the second row of  $\mathbf{N}(1, 1, 1, -1)$  is  $1 - 1 + 1 - 1 = 0$ , matching the second coordinate of  $(1, 0, 0, -1)$ .

## 9.2.7 Exercises

### 9.2.7.1 A. Check your understanding

#### Exercise A1.

- Define a nilpotent operator and its index of nilpotency.
- State the structure theorem for nilpotent operators, and say what a Jordan basis is.
- Write down the formula for the number of chains of length at least  $j$  in terms of the ranks of the powers.
- True or false: a singular matrix is nilpotent. Justify your answer.
- True or false: two nilpotent matrices in  $M_n(F)$  with the same rank are similar. Justify your answer.
- How many similarity classes of nilpotent matrices are there in  $M_3(F)$ , and what are their indices of nilpotency?

*Solution.*

- $N \in \mathcal{L}(V)$  is nilpotent if  $N^s = 0$  for **some** integer  $s \geq 1$ ; the index is the least such  $s$  ([Definition 9.2.1](#)).
- For nilpotent  $N$  on  $V$  with  $\dim V = n \geq 1$ , there are chains generated by  $\mathbf{u}_1, \dots, \mathbf{u}_r$  of lengths  $k_1 \geq \dots \geq k_r \geq 1$  summing to  $n$  whose concatenation is a basis of  $V$  — a Jordan basis — and in it  $[N] = \mathbf{J}_{k_1}(0) \oplus \dots \oplus \mathbf{J}_{k_r}(0)$  ([Theorem 9.2.3](#)).
- $\#\{i : k_i \geq j\} = \text{rank } N^{j-1} - \text{rank } N^j$  ([Theorem 9.2.4](#) (b)).
- False.  $\text{diag}(0, 1)$  is singular, and  $\text{diag}(0, 1)^k = \text{diag}(0, 1) \neq 0$  for every  $k \geq 1$ . What is true is the converse: a nilpotent matrix on a non-zero space is singular, since  $0 \in \text{spec}(N)$  ([Proposition 9.2.1](#) (c)).

- (e) False for  $n \geq 4$ . The partitions  $(3, 1)$  and  $(2, 2)$  of 4 both give rank 2, but  $\text{rank } N^2$  is 1 and 0 respectively, so the matrices are not similar ([Corollary 9.2.5](#)). (For  $n \leq 3$  the rank does determine the partition.)
- (f) Three, one for each partition of 3:  $(3), (2, 1), (1, 1, 1)$ , with indices 3, 2, 1.

### 9.2.7.2 B. Practice

#### Exercise B1: Block sizes from rank data.

In each case  $N$  is nilpotent on a space of dimension  $n$  over  $F$ . Determine the chain lengths, or explain why no such operator exists.

- (a)  $n = 7$ ,  $\text{rank } N = 4$ ,  $\text{rank } N^2 = 2$ ,  $\text{rank } N^3 = 1$ ,  $\text{rank } N^4 = 0$ .
- (b)  $n = 6$ ,  $\text{rank } N = 3$ ,  $\text{rank } N^2 = 1$ ,  $\text{rank } N^3 = 0$ .
- (c)  $n = 6$ ,  $\text{rank } N = 4$ ,  $\text{rank } N^2 = 3$ ,  $\text{rank } N^3 = 0$ .

*Solution.*

- (a) With  $r_0, \dots, r_4 = 7, 4, 2, 1, 0$ , the exact counts from [Theorem 9.2.4](#) (c) are: length 1:  $7 - 8 + 2 = 1$ ; length 2:  $4 - 4 + 1 = 1$ ; length 3:  $2 - 2 + 0 = 0$ ; length 4:  $1 - 0 + 0 = 1$ . The lengths are  $(4, 2, 1)$ , and  $4 + 2 + 1 = 7$ .
- (b) With  $r_0, \dots, r_3 = 6, 3, 1, 0$ : length 1:  $6 - 6 + 1 = 1$ ; length 2:  $3 - 2 + 0 = 1$ ; length 3:  $1 - 0 + 0 = 1$ . The lengths are  $(3, 2, 1)$ , summing to 6.
- (c) No such operator exists. By (b) of the theorem, the number of chains of length  $\geq 2$  would be  $r_1 - r_2 = 1$ , while the number of length  $\geq 3$  would be  $r_2 - r_3 = 3$ , which is impossible: a chain of length  $\geq 3$  has length  $\geq 2$ , so the counts must be non-increasing in  $j$ . (Equivalently, [Theorem 9.2.4](#) (c) would give  $r_1 - 2r_2 + r_3 = 4 - 6 + 0 = -2$  chains of length 2, which is absurd.)

#### Exercise B2: The five classes in dimension four.

Let  $F$  be any field.

- (a) List the five partitions of 4, and for each write down the corresponding nilpotent matrix in  $M_4(F)$  as a direct sum of blocks  $\mathbf{J}_k(0)$ .
- (b) For each, compute  $\text{rank } \mathbf{N}$ ,  $\text{rank } \mathbf{N}^2$ ,  $\text{rank } \mathbf{N}^3$  and the index of nilpotency.
- (c) Which pairs among them share the same rank? Which invariant separates each such pair?
- (d) Deduce that  $\text{rank } \mathbf{N}$  alone is not a complete invariant of nilpotent matrices, but the full list of ranks of powers is.

*Solution.*

- (a)  $(4)$ :  $\mathbf{J}_4(0)$ ;  $(3, 1)$ :  $\mathbf{J}_3(0) \oplus \mathbf{J}_1(0)$ ;  $(2, 2)$ :  $\mathbf{J}_2(0) \oplus \mathbf{J}_2(0)$ ;  $(2, 1, 1)$ :  $\mathbf{J}_2(0) \oplus \mathbf{J}_1(0) \oplus \mathbf{J}_1(0)$ ;  $(1, 1, 1, 1)$ : the zero matrix.
- (b) Using  $\text{rank } \mathbf{N}^j = \sum_i \max(k_i - j, 0)$  ([Theorem 9.2.4](#) (a)), the ranks  $(\text{rank } \mathbf{N}, \text{rank } \mathbf{N}^2, \text{rank } \mathbf{N}^3)$  and the index are:
- $(4)$ : ranks 3, 2, 1; index 4.

- (3, 1): ranks 2, 1, 0; index 3.
  - (2, 2): ranks 2, 0, 0; index 2.
  - (2, 1, 1): ranks 1, 0, 0; index 2.
  - (1, 1, 1, 1): ranks 0, 0, 0; index 1.
- (c) Only (3, 1) and (2, 2) share a rank, namely 2. They are separated by rank  $\mathbf{N}^2$  (1 against 0), equivalently by the index (3 against 2) or the minimal polynomial ( $x^3$  against  $x^2$ , [Proposition 9.2.1](#) (b)).
- (d) The pair in (c) shows that equal rank does not imply similarity, while equality of  $\text{rank } \mathbf{N}^j$  for all  $j$  does ([Corollary 9.2.5](#)).

### Exercise B3: A Jordan basis by hand.

Let

$$\mathbf{M} = \begin{pmatrix} -1 & -1 & 1 & -1 \\ 1 & 2 & -2 & 1 \\ 1 & 2 & -2 & 1 \\ 1 & 1 & -1 & 1 \end{pmatrix} \in M_4(\mathbb{Q}).$$

- (a) Show that  $\mathbf{M}^2 = 0$  and compute  $\text{rank } \mathbf{M}$ .
- (b) Determine the chain lengths of  $\mathbf{M}$ .
- (c) Find a Jordan basis and an invertible  $\mathbf{P}$  with  $\mathbf{P}^{-1}\mathbf{M}\mathbf{P}$  a direct sum of blocks  $\mathbf{J}_k(0)$ . Verify your answer by checking  $\mathbf{M}\mathbf{P} = \mathbf{P}\mathbf{J}$  column by column.

*Hint: for (c), the tops of the chains may be taken among the standard basis vectors.*

*Solution.*

- (a) Write  $\mathbf{m}_1, \dots, \mathbf{m}_4$  for the columns of  $\mathbf{M}$ , so  $\mathbf{m}_1 = (-1, 1, 1, 1)$ ,  $\mathbf{m}_2 = (-1, 2, 2, 1)$ ,  $\mathbf{m}_3 = -\mathbf{m}_2$  and  $\mathbf{m}_4 = \mathbf{m}_1$ . The columns of  $\mathbf{M}^2$  are  $\mathbf{M}\mathbf{m}_1, \dots, \mathbf{M}\mathbf{m}_4$ , so it is enough to check the first two. Row by row,

$$\begin{aligned} \mathbf{M}\mathbf{m}_1 &= (1 - 1 + 1 - 1, -1 + 2 - 2 + 1, -1 + 2 - 2 + 1, -1 + 1 - 1 + 1) \\ &= \mathbf{0}, \end{aligned}$$

$$\begin{aligned} \mathbf{M}\mathbf{m}_2 &= (1 - 2 + 2 - 1, -1 + 4 - 4 + 1, -1 + 4 - 4 + 1, -1 + 2 - 2 + 1) \\ &= \mathbf{0}. \end{aligned}$$

Hence  $\mathbf{M}\mathbf{m}_3 = -\mathbf{M}\mathbf{m}_2 = \mathbf{0}$  and  $\mathbf{M}\mathbf{m}_4 = \mathbf{M}\mathbf{m}_1 = \mathbf{0}$ , so  $\mathbf{M}^2 = 0$ . Also  $\text{im } \mathbf{M} = \text{span}(\mathbf{m}_1, \mathbf{m}_2)$  by the relations among the columns, and  $\mathbf{m}_1, \mathbf{m}_2$  are not proportional, so  $\text{rank } \mathbf{M} = 2$ .

- (b)  $r_0, r_1, r_2 = 4, 2, 0$ . Chains of length 1:  $r_0 - 2r_1 + r_2 = 4 - 4 + 0 = 0$ ; of length 2:  $r_1 - 2r_2 + r_3 = 2$ . So the partition is (2, 2): two chains of length 2.
- (c) We need two tops  $\mathbf{u}, \mathbf{w}$  with  $\mathbf{M}\mathbf{u}, \mathbf{M}\mathbf{w}$  independent; since  $\text{rank } \mathbf{M} = 2$  and  $\text{im } \mathbf{M} = \text{span}(\mathbf{m}_1, \mathbf{m}_2)$ , the vectors  $\mathbf{u} = \mathbf{e}_1$  and  $\mathbf{w} = \mathbf{e}_2$  do it, with  $\mathbf{M}\mathbf{e}_1 = \mathbf{m}_1 = (-1, 1, 1, 1)$  and  $\mathbf{M}\mathbf{e}_2 = \mathbf{m}_2 = (-1, 2, 2, 1)$ . Take

$$\mathcal{B} = ((-1, 1, 1, 1), \mathbf{e}_1, (-1, 2, 2, 1), \mathbf{e}_2), \quad \mathbf{P} = \begin{pmatrix} -1 & 1 & -1 & 0 \\ 1 & 0 & 2 & 1 \\ 1 & 0 & 2 & 0 \\ 1 & 0 & 1 & 0 \end{pmatrix}.$$

The last column of  $\mathbf{P}$  has a single non-zero entry, the 1 in row 2, so expanding along it ([Theorem 6.6.1](#)) gives

$$\det \mathbf{P} = \det \begin{pmatrix} -1 & 1 & -1 \\ 1 & 0 & 2 \\ 1 & 0 & 1 \end{pmatrix} = -1 \cdot (1 \cdot 1 - 2 \cdot 1) = 1 \neq 0,$$

where the second determinant was expanded along its second column. So  $\mathcal{B}$  is a basis, and

$$\mathbf{P}^{-1}\mathbf{M}\mathbf{P} = \mathbf{J}_2(0) \oplus \mathbf{J}_2(0).$$

Column by column,  $\mathbf{M}\mathbf{P} = \mathbf{P}\mathbf{J}$  says:  $\mathbf{M}(-1, 1, 1, 1) = \mathbf{0}$  (first column of  $\mathbf{P}\mathbf{J}$  is  $\mathbf{0}$ );  $\mathbf{M}\mathbf{e}_1 = (-1, 1, 1, 1)$  (second column of  $\mathbf{P}\mathbf{J}$  is the first column of  $\mathbf{P}$ );  $\mathbf{M}(-1, 2, 2, 1) = \mathbf{0}$ ; and  $\mathbf{M}\mathbf{e}_2 = (-1, 2, 2, 1)$ . All four were verified in (a) and (c).

### 9.2.7.3 C. Going deeper

#### Exercise C1: A nilpotent matrix is similar to its transpose.

Let  $\mathbf{A} \in M_n(F)$  be nilpotent.

- (a) Prove that  $\mathbf{A}^\top$  is nilpotent and that  $\text{rank}(\mathbf{A}^\top)^j = \text{rank } \mathbf{A}^j$  for every  $j \geq 1$ .
- (b) Deduce that  $\mathbf{A} \sim \mathbf{A}^\top$ .
- (c) For  $\mathbf{A} = \mathbf{J}_3(0)$ , find an explicit  $\mathbf{P}$  with  $\mathbf{P}^{-1}\mathbf{A}\mathbf{P} = \mathbf{A}^\top$ .

*Solution.*

- (a)  $(\mathbf{A}^\top)^j = (\mathbf{A}^j)^\top$  ([Theorem 0.9.5](#)), so  $(\mathbf{A}^\top)^n = (\mathbf{A}^n)^\top = \mathbf{0}$  by [Proposition 9.2.1](#) (a), and  $\mathbf{A}^\top$  is nilpotent. A matrix and its transpose have the same rank ([Theorem 2.5.5](#)), so  $\text{rank}(\mathbf{A}^\top)^j = \text{rank}(\mathbf{A}^j)^\top = \text{rank } \mathbf{A}^j$ .
- (b) Both matrices are nilpotent with equal ranks of all powers, so [Corollary 9.2.5](#) gives  $\mathbf{A} \sim \mathbf{A}^\top$ .
- (c)  $\mathbf{J}_3(0)$  sends  $\mathbf{e}_1 \mapsto \mathbf{0}$ ,  $\mathbf{e}_2 \mapsto \mathbf{e}_1$ ,  $\mathbf{e}_3 \mapsto \mathbf{e}_2$ , while  $\mathbf{J}_3(0)^\top$  sends  $\mathbf{e}_1 \mapsto \mathbf{e}_2 \mapsto \mathbf{e}_3 \mapsto \mathbf{0}$ , that is, it is the shift in the reversed basis. So take the reversal permutation matrix

$$\mathbf{P} = \begin{pmatrix} 0 & 0 & 1 \\ 0 & 1 & 0 \\ 1 & 0 & 0 \end{pmatrix}, \quad \mathbf{P}^{-1} = \mathbf{P}.$$

Then  $\mathbf{P}^{-1}\mathbf{J}_3(0)\mathbf{P} = \mathbf{J}_3(0)^\top$ , as one checks column by column:  $\mathbf{J}_3(0)\mathbf{P}$  has columns  $\mathbf{J}_3(0)\mathbf{e}_3 = \mathbf{e}_2$ ,  $\mathbf{J}_3(0)\mathbf{e}_2 = \mathbf{e}_1$ ,  $\mathbf{J}_3(0)\mathbf{e}_1 = \mathbf{0}$ , and  $\mathbf{P}\mathbf{J}_3(0)^\top$  has columns  $\mathbf{P}\mathbf{e}_2 = \mathbf{e}_2$ ,  $\mathbf{P}\mathbf{e}_3 = \mathbf{e}_1$ ,  $\mathbf{P}\mathbf{0} = \mathbf{0}$ . (The same reversal works for every  $\mathbf{J}_k(0)$ , and hence, blockwise, for every nilpotent matrix in Jordan form.)

#### Exercise C2: One long chain.

Let  $V$  be finite-dimensional with  $\dim V = n \geq 1$ , and let  $N \in \mathcal{L}(V)$  be nilpotent with  $N^{n-1} \neq 0$ .

- (a) Prove that there is a basis of  $V$  in which the matrix of  $N$  is  $\mathbf{J}_n(0)$ .

- (b) Deduce that  $m_N = p_N = x^n$ , and that nullity  $N = 1$ .
- (c) Show that the conclusion of (a) fails for  $\mathbf{N} = \mathbf{J}_2(0) \oplus \mathbf{J}_1(0)$  on a space of dimension 3, and say which hypothesis fails.

*Hint: for (a), take a vector outside  $\ker N^{n-1}$ , which exists precisely because  $N^{n-1} \neq 0$ .*

*Solution.*

- (a) Since  $N^{n-1} \neq 0$ , there is  $\mathbf{u} \in V$  with  $N^{n-1}\mathbf{u} \neq \mathbf{0}$ , while  $N^n\mathbf{u} = \mathbf{0}$  by [Proposition 9.2.1](#) (a). So  $\mathbf{u}$  generates a Jordan chain of length  $n$  ([Definition 9.2.2](#)), which is a linearly independent list of  $n = \dim V$  vectors ([Lemma 9.2.2](#) (a)), hence a basis of  $V$  ([Theorem 1.6.4](#) (a)). In it the matrix of  $N$  is  $\mathbf{J}_n(0)$ , by [Lemma 9.2.2](#) (b) with  $C = V$ .
- (b) In that basis  $\text{rank } N^j = n - j$  for  $0 \leq j \leq n$  ([Theorem 9.2.4](#) (a) with the single length  $k_1 = n$ ), so nullity  $N = 1$  by [Theorem 3.3.1](#) and the index of nilpotency is  $n$ . Hence  $m_N = x^n$  by [Proposition 9.2.1](#) (b), and  $p_N = x^n$  as well; they coincide, which by [Corollary 8.6.2](#) is the extreme case  $\deg m_N = n$ .
- (c) For  $\mathbf{N} = \mathbf{J}_2(0) \oplus \mathbf{J}_1(0)$  on  $F^3$  we have  $\mathbf{N}^2 = 0$ , so  $\mathbf{N}^{n-1} = \mathbf{N}^2 = 0$  and the hypothesis  $\mathbf{N}^{n-1} \neq 0$  fails. Accordingly  $\mathbf{N}$  is not similar to  $\mathbf{J}_3(0)$ : its rank is 1, while  $\text{rank } \mathbf{J}_3(0) = 2$ , and similar matrices have equal ranks ([Proposition 3.7.6](#) (a)). Its partition is  $(2, 1)$ , not  $(3)$ .

## 9.3 The Jordan Canonical Form

Chapter 3 asked how simple the matrix of an operator can be made by choosing the basis well, and Chapter 8 answered it only for operators with enough eigenvectors. The two previous sections have quietly finished the job: the space splits into generalized eigenspaces, and on each of them the operator is a scalar plus a nilpotent operator, which is a disjoint union of chains. Assembling these pieces gives the best matrix there is, and it is unique. This section states that theorem, proves it, reads the familiar invariants off it, and turns it into a similarity test.

### 9.3.1 The shape we are aiming for

Here is the goal, stated before its parts are named. **If  $p_T$  splits, then  $V$  has a basis in which the matrix of  $T$  is diagonal except for some ones immediately above the diagonal, and that matrix is unique up to the order of its diagonal blocks.** A diagonal matrix is the special case with no ones at all. So the dream of Chapter 3 comes true, with one extra ingredient beyond a diagonal matrix: the superdiagonal ones, one for each step of a Jordan chain.

The building block is the matrix of  $\lambda \text{id} + N$  on a single chain. By [Lemma 9.2.2](#) (b), the nilpotent part contributes  $\mathbf{J}_k(0)$ , so the block is  $\lambda \mathbf{I}_k + \mathbf{J}_k(0)$ .

#### Definition 9.3.1 — Jordan Block

Let  $k \geq 1$  and  $\lambda \in F$ . The **Jordan block** of size  $k$  for  $\lambda$  is the matrix

$$\mathbf{J}_k(\lambda) := \lambda \mathbf{I}_k + \mathbf{J}_k(0) = \begin{pmatrix} \lambda & 1 & & \\ & \lambda & \ddots & \\ & & \ddots & 1 \\ & & & \lambda \end{pmatrix} \in M_k(F),$$

with  $\lambda$  in every diagonal entry, 1 in every **superdiagonal** entry  $(j, j+1)$ , and 0 elsewhere. In particular  $\mathbf{J}_1(\lambda) = (\lambda)$ .

Reading its columns: if  $[T]_{\mathcal{B}} = \mathbf{J}_k(\lambda)$  for a basis  $\mathcal{B} = (\mathbf{b}_1, \dots, \mathbf{b}_k)$ , then  $T\mathbf{b}_1 = \lambda\mathbf{b}_1$  and  $T\mathbf{b}_j = \mathbf{b}_{j-1} + \lambda\mathbf{b}_j$  for  $j \geq 2$ ; equivalently  $(T - \lambda \text{id}_V)\mathbf{b}_j = \mathbf{b}_{j-1}$  and  $(T - \lambda \text{id}_V)\mathbf{b}_1 = \mathbf{0}$ , so  $\mathcal{B}$  is the Jordan chain generated by  $\mathbf{b}_k$  (Definition 9.2.2). The single eigenvalue is  $\lambda$ , with  $a(\lambda) = k$  and  $g(\lambda) = 1$ , since  $\mathbf{J}_k(\lambda) - \lambda\mathbf{I}_k = \mathbf{J}_k(0)$  has rank  $k-1$ .

*A Jordan matrix is a diagonal matrix with some chains glued in: blocks along the diagonal, each block one eigenvalue with ones above it.*

### Definition 9.3.2 — Jordan Matrix, Jordan Basis, Jordan Form

A **Jordan matrix** is a matrix of the form

$$\mathbf{J} = \mathbf{J}_{k_1}(\lambda_1) \oplus \mathbf{J}_{k_2}(\lambda_2) \oplus \cdots \oplus \mathbf{J}_{k_m}(\lambda_m)$$

for some  $m \geq 1$ , sizes  $k_i \geq 1$  and scalars  $\lambda_i \in F$ , **not** assumed distinct. Let  $V$  be a finite-dimensional vector space over  $F$  with  $\dim V \geq 1$  and  $T \in \mathcal{L}(V)$ . A **Jordan form** of  $T$  is a Jordan matrix  $\mathbf{J}$  with  $[T]_{\mathcal{B}} = \mathbf{J}$  for **some** basis  $\mathcal{B}$  of  $V$ ; such a basis is called a **Jordan basis** for  $T$ . For  $\mathbf{A} \in M_n(F)$ , a **Jordan form** of  $\mathbf{A}$  is a Jordan matrix  $\mathbf{J}$  with  $\mathbf{A} \sim \mathbf{J}$ .

In words: the matrix is block diagonal (Definition 7.1.2), and each diagonal block is a single Jordan block. Different blocks may carry the same eigenvalue or different ones, and the sizes may repeat. A Jordan basis is a basis that is a concatenation of Jordan chains, one chain per block. The two descriptions match for  $\mathbf{A} \in M_n(F)$  and  $T = T_{\mathbf{A}}$ , by Theorem 3.7.4.

**The order of the blocks never matters.** Reordering the blocks of  $\mathbf{J}$  permutes the segments of the basis  $\mathcal{B}$ , so the reordered matrix is the matrix of the **same** operator in the permuted basis, hence similar to  $\mathbf{J}$  (Theorem 3.7.4 (a)). This is why the uniqueness statement below can only be “up to the order of the blocks”, and why it costs nothing to list the blocks in any convenient order.

#### Examples.

- **Diagonal matrices.**  $\text{diag}(d_1, \dots, d_n) = \mathbf{J}_1(d_1) \oplus \cdots \oplus \mathbf{J}_1(d_n)$  is a Jordan matrix with all blocks of size 1. So every diagonalizable operator has a Jordan form, namely a diagonal one.
- **The standing counterexample.**  $\begin{pmatrix} 2 & 1 \\ 0 & 2 \end{pmatrix} = \mathbf{J}_2(2)$  is its own Jordan form, and  $(\mathbf{e}_1, \mathbf{e}_2)$  is a Jordan basis:  $\mathbf{e}_1$  is an eigenvector and  $(\mathbf{J} - 2\mathbf{I})\mathbf{e}_2 = \mathbf{e}_1$ .
- **Nilpotent operators.** By Theorem 9.2.3, a nilpotent operator has the Jordan form  $\mathbf{J}_{k_1}(0) \oplus \cdots \oplus \mathbf{J}_{k_r}(0)$ , all eigenvalues 0. Section 2 is exactly the case  $\lambda = 0$  of this section.
- **Mixed blocks.**  $\mathbf{J} = \mathbf{J}_2(3) \oplus \mathbf{J}_1(3) \oplus \mathbf{J}_2(1) \in M_5(F)$  has  $p_{\mathbf{J}} = (x-3)^3(x-1)^2$ ,  $g(3) = 2$

and  $g(1) = 1$ . It is the answer to the worked example at the end of this section.

- **The degenerate case.** On a space of dimension 1, every operator is  $(\lambda) = \mathbf{J}_1(\lambda)$ . One block, size one, no superdiagonal to speak of.

**Non-example by minimal change.** The matrix  $\begin{pmatrix} 2 & 1 \\ 0 & 3 \end{pmatrix}$  is upper triangular with a superdiagonal 1, but it is **not** a Jordan matrix: a 1 above the diagonal may only join two **equal** diagonal entries, because it records the step of a chain for a single eigenvalue. (Indeed this matrix has distinct eigenvalues, so it is diagonalizable, and its Jordan form is  $\text{diag}(2, 3)$ .) Changing the superdiagonal entry instead,  $\begin{pmatrix} 2 & 2 \\ 0 & 2 \end{pmatrix}$  is not a Jordan matrix either, since the entry above the diagonal must be 1; it is similar to  $\mathbf{J}_2(2)$ , by rescaling the second basis vector.

**Why this definition.** Two conventions are being fixed. First, the ones sit **above** the diagonal, which matches the chain order “eigenvector first” of Definition 9.2.2; putting them below the diagonal gives the transposed matrix, which is just as good and appears in some books. Second, the off-diagonal entries are 1 rather than arbitrary non-zero scalars: a non-zero entry  $c$  can be turned into a 1 by scaling one basis vector, so nothing is lost, and the normalization is what makes the form unique. Nothing else is allowed: if any further entry were permitted, the form would no longer be determined by the operator.

*Warning.* **A Jordan form exists only when  $p_T$  splits, so it depends on the field.**

The rotation  $\mathbf{R} = \begin{pmatrix} 0 & -1 \\ 1 & 0 \end{pmatrix} \in M_2(\mathbb{R})$  has no Jordan form over  $\mathbb{R}$ , since  $p_{\mathbf{R}} = x^2 + 1$  has no real root, and a Jordan matrix is triangular with its eigenvalues on the diagonal. Over  $\mathbb{C}$  its Jordan form is  $\text{diag}(i, -i)$ . Over  $\mathbb{Q}$  the matrix  $\begin{pmatrix} 0 & 2 \\ 1 & 0 \end{pmatrix}$  has no Jordan form either, and over  $\mathbb{R}$  it becomes  $\text{diag}(\sqrt{2}, -\sqrt{2})$ . Section 6 gives a canonical form valid over **every** field.

## 9.3.2 The theorem

*Every operator whose characteristic polynomial splits is a direct sum of Jordan blocks, in exactly one way.*

### Theorem 9.3.1 — Jordan Canonical Form Theorem

Let  $V$  be a finite-dimensional vector space over  $F$  with  $\dim V = n \geq 1$ , and let  $T \in \mathcal{L}(V)$ .

- (**Existence**)  $T$  has a Jordan form if and only if  $p_T$  splits over  $F$ .
- (**Uniqueness**) Suppose  $\mathbf{J}$  is a Jordan form of  $T$ . Then for every  $\lambda \in F$  and every integer  $j \geq 1$ , the number of blocks  $\mathbf{J}_k(\lambda)$  in  $\mathbf{J}$  with  $k \geq j$  equals

$$\text{rank}(T - \lambda \text{id}_V)^{j-1} - \text{rank}(T - \lambda \text{id}_V)^j,$$

a number depending only on  $T$ ,  $\lambda$  and  $j$ . Consequently any two Jordan forms of  $T$  have the same blocks, up to the order in which they are listed.

Equivalently, for  $\mathbf{A} \in M_n(F)$  with  $p_{\mathbf{A}}$  split:  $\mathbf{A}$  is similar to a Jordan matrix, and that matrix is unique up to the order of its blocks. We call it **the** Jordan form of  $\mathbf{A}$ .



*Idea.* Existence is an assembly of the two previous sections, in three steps. ① Split  $V = \bigoplus_i G_{\lambda_i}(T)$  ([Theorem 9.1.3](#)). ② On  $G_{\lambda_i}$  write  $T = \lambda_i \text{id} + N_i$  with  $N_i$  nilpotent, and take a Jordan basis for  $N_i$  ([Theorem 9.2.3](#)); adding  $\lambda_i \mathbf{I}$  turns each  $\mathbf{J}_k(0)$  into  $\mathbf{J}_k(\lambda_i)$ . ③ Put the bases together: the matrix is block diagonal. For uniqueness, count ranks. A block  $\mathbf{J}_k(\mu) - \lambda \mathbf{I}$  with  $\mu \neq \lambda$  is invertible, so it contributes its full size to every power and drops out of differences; a block with  $\mu = \lambda$  becomes  $\mathbf{J}_k(0)$ , whose  $j$ -th power has rank  $\max(k - j, 0)$ . So the difference of two consecutive ranks counts the  $\lambda$ -blocks that are still long enough.

*Proof.*

- (a) ( $\Leftarrow$ ) Suppose  $p_T$  splits, with distinct eigenvalues  $\lambda_1, \dots, \lambda_d$ .

**Step 1. Split the space.** By [Theorem 9.1.3](#),  $V = G_1 \oplus \dots \oplus G_d$  with  $G_i = G_{\lambda_i}(T)$  non-zero and  $T$ -invariant,  $\dim G_i = a_i := a_T(\lambda_i)$ , and

$$T|_{G_i} = \lambda_i \text{id}_{G_i} + N_i, \quad N_i = (T - \lambda_i \text{id}_V)|_{G_i} \text{ nilpotent.}$$

**Step 2. One eigenvalue at a time.** Fix  $i$ . By [Theorem 9.2.3](#) applied to  $N_i \in \mathcal{L}(G_i)$ , there is a basis  $\mathcal{B}_i$  of  $G_i$ , a concatenation of Jordan chains of  $N_i$ , with

$$[N_i]_{\mathcal{B}_i} = \mathbf{J}_{k_{i1}}(0) \oplus \dots \oplus \mathbf{J}_{k_{ir_i}}(0), \quad \sum_l k_{il} = a_i.$$

Since  $[\cdot]_{\mathcal{B}_i}$  is linear ([Theorem 3.6.4](#)) and  $[\text{id}_{G_i}]_{\mathcal{B}_i} = \mathbf{I}_{a_i} = \mathbf{I}_{k_{i1}} \oplus \dots \oplus \mathbf{I}_{k_{ir_i}}$ ,

$$\begin{aligned} [T|_{G_i}]_{\mathcal{B}_i} &= \lambda_i \mathbf{I}_{a_i} + [N_i]_{\mathcal{B}_i} \\ &= (\lambda_i \mathbf{I}_{k_{i1}} + \mathbf{J}_{k_{i1}}(0)) \oplus \dots \oplus (\lambda_i \mathbf{I}_{k_{ir_i}} + \mathbf{J}_{k_{ir_i}}(0)) \\ &= \mathbf{J}_{k_{i1}}(\lambda_i) \oplus \dots \oplus \mathbf{J}_{k_{ir_i}}(\lambda_i), \end{aligned}$$

where the middle equality is the blockwise addition of [Theorem 7.1.4](#) (a) and the last is [Definition 9.3.1](#).

**Step 3. Assemble.** Let  $\mathcal{B}$  be the basis of  $V$  obtained by writing  $\mathcal{B}_1, \dots, \mathcal{B}_d$  one after another. By [Theorem 8.1.5](#),

$$[T]_{\mathcal{B}} = [T|_{G_1}]_{\mathcal{B}_1} \oplus \dots \oplus [T|_{G_d}]_{\mathcal{B}_d},$$

which by Step 2 is a direct sum of Jordan blocks, that is, a Jordan matrix. So  $T$  has a Jordan form, and  $\mathcal{B}$  is a Jordan basis.

( $\Rightarrow$ ) Suppose  $[T]_{\mathcal{B}} = \mathbf{J}$  is a Jordan matrix. Each Jordan block is upper triangular, so  $\mathbf{J}$  is upper triangular, and  $T$  is triangularizable ([Definition 8.7.1](#)). By [Theorem 8.7.2](#),  $p_T$  splits over  $F$ .

- (b) Let  $\mathbf{J} = \mathbf{J}_{k_1}(\mu_1) \oplus \dots \oplus \mathbf{J}_{k_m}(\mu_m) = [T]_{\mathcal{B}}$ , with  $\mu_1, \dots, \mu_m \in F$  not necessarily distinct; fix  $\lambda \in F$  and let  $j \geq 0$ . Then  $[T - \lambda \text{id}_V]_{\mathcal{B}} = \mathbf{J} - \lambda \mathbf{I}_n$ , which is block diagonal with blocks  $\mathbf{J}_{k_i}(\mu_i) - \lambda \mathbf{I}_{k_i}$ , so by [Theorem 7.1.4](#) (b),

$$[(T - \lambda \text{id}_V)^j]_{\mathcal{B}} = (\mathbf{J}_{k_1}(\mu_1) - \lambda \mathbf{I}_{k_1})^j \oplus \dots \oplus (\mathbf{J}_{k_m}(\mu_m) - \lambda \mathbf{I}_{k_m})^j,$$

and the rank of a block diagonal matrix is the sum of the ranks of its blocks ([Theorem 7.2.1](#) (b), applied  $m - 1$  times). Also  $\text{rank}(T - \lambda \text{id}_V)^j$  equals the rank of this matrix ([Theorem 3.6.5](#) (a)). Now compute block by block.

*Blocks with  $\mu_i \neq \lambda$ .* Here  $\mathbf{J}_{k_i}(\mu_i) - \lambda \mathbf{I}_{k_i}$  is upper triangular with every diagonal entry equal to  $\mu_i - \lambda \neq 0$ , so its determinant is  $(\mu_i - \lambda)^{k_i} \neq 0$  ([Theorem 6.4.6](#)) and it is

invertible (Theorem 6.5.2). Hence its  $j$ -th power is invertible too, of rank  $k_i$ , for every  $j \geq 0$ .

*Blocks with  $\mu_i = \lambda$ .* Here  $\mathbf{J}_{k_i}(\lambda) - \lambda \mathbf{I}_{k_i} = \mathbf{J}_{k_i}(0)$ , whose standard basis is a single Jordan chain of length  $k_i$ , so  $\text{rank } \mathbf{J}_{k_i}(0)^j = \max(k_i - j, 0)$  by Theorem 9.2.4 (a) with the one-element list of lengths  $(k_i)$ .

Adding up, with  $c = \sum_{\mu_i \neq \lambda} k_i$  independent of  $j$ ,

$$\text{rank}(T - \lambda \text{id}_V)^j = c + \sum_{\mu_i = \lambda} \max(k_i - j, 0).$$

Subtracting the values at  $j$  and  $j - 1$ , the constant  $c$  cancels and each remaining bracket  $\max(k_i - j + 1, 0) - \max(k_i - j, 0)$  equals 1 if  $k_i \geq j$  and 0 otherwise, exactly as in the proof of Theorem 9.2.4 (b). Therefore

$$\text{rank}(T - \lambda \text{id}_V)^{j-1} - \text{rank}(T - \lambda \text{id}_V)^j = \#\{i : \mu_i = \lambda \text{ and } k_i \geq j\}.$$

The left-hand side is defined from  $T$  alone. So for each  $\lambda$  and each  $j$  the number of  $\lambda$ -blocks of size at least  $j$ , and hence by subtraction the number of size exactly  $j$ , is the same for every Jordan form of  $T$ . Two Jordan forms therefore consist of the same blocks with the same multiplicities, and differ only in the order in which the blocks are listed.

For a matrix  $\mathbf{A} \in M_n(F)$  with  $p_{\mathbf{A}}$  split, apply all of this to  $T_{\mathbf{A}}$ , using Theorem 3.7.4 to pass between bases and similarity. ■

Step 1  $\rightarrow$  Step 2  $\rightarrow$  Step 3, and the dream comes true: **for every complex square matrix there is a basis in which the matrix is diagonal apart from some ones on the superdiagonal**, since  $p_{\mathbf{A}}$  always splits over  $\mathbb{C}$  (Corollary 5.5.5). The proof also tells us where the basis comes from: one chain per block, found inside one generalized eigenspace at a time.

*Quick check.* How many Jordan forms, up to the order of the blocks, are possible for an operator with  $p_T = (x - 2)^2(x - 3)$ ? Write them down, and say which invariant tells them apart.

*Solution.* The 3-blocks must have sizes summing to  $a(3) = 1$ , so there is exactly one, of size 1. The 2-blocks have sizes summing to  $a(2) = 2$ , so they form a partition of 2: either (2) or (1, 1). The two possibilities are

$$\mathbf{J}_2(2) \oplus \mathbf{J}_1(3) \quad \text{and} \quad \mathbf{J}_1(2) \oplus \mathbf{J}_1(2) \oplus \mathbf{J}_1(3) = \text{diag}(2, 2, 3).$$

They are told apart by  $g(2)$ , which is the number of 2-blocks: 1 in the first case, 2 in the second. Equivalently by  $m_T = (x - 2)^2(x - 3)$  against  $(x - 2)(x - 3)$ , or by  $\text{rank}(T - 2 \text{id})$ , which is 2 against 1.

### 9.3.3 Reading the invariants off the blocks

The Jordan form makes every invariant of Chapter 8 visible at a glance.

### Corollary 9.3.2 — Invariants from the Jordan Form

Let  $V$  be finite-dimensional with  $\dim V = n \geq 1$ , let  $T \in \mathcal{L}(V)$  with  $p_T$  split, let  $\mathbf{J}$  be a Jordan form of  $T$ , and let  $\lambda \in F$ . Then:

- (a)  $a_T(\lambda)$  is the **sum of the sizes** of the  $\lambda$ -blocks of  $\mathbf{J}$ ; in particular  $\lambda \in \text{spec}(T)$  if and only if  $\mathbf{J}$  has a  $\lambda$ -block;
- (b)  $g_T(\lambda)$  is the **number** of  $\lambda$ -blocks of  $\mathbf{J}$ ;
- (c) the exponent of  $x - \lambda$  in  $m_T$  is the **largest size** of a  $\lambda$ -block of  $\mathbf{J}$  (and 0 when  $\mathbf{J}$  has no  $\lambda$ -block);
- (d)  $\dim \ker(T - \lambda \text{id}_V)^j = \sum \min(k_i, j)$ , summed over the sizes  $k_i$  of the  $\lambda$ -blocks, for every  $j \geq 0$ ;
- (e)  $T$  is diagonalizable if and only if every block of  $\mathbf{J}$  has size 1, that is, if and only if  $\mathbf{J}$  is diagonal.

*Proof.* Write  $\mathbf{J} = \mathbf{J}_{k_1}(\mu_1) \oplus \cdots \oplus \mathbf{J}_{k_m}(\mu_m)$ , with  $\mu_i \in F$  not necessarily distinct, and  $\mathcal{B}$  for a Jordan basis, so  $[T]_{\mathcal{B}} = \mathbf{J}$ .

- (a)  $\mathbf{J}$  is upper triangular with diagonal entries  $\mu_i$  repeated  $k_i$  times, so  $p_T = \prod_{i=1}^m (x - \mu_i)^{k_i}$  (Theorem 6.4.6, Definition 6.10.2). Grouping equal  $\mu_i$  and using that the multiplicity of  $\lambda$  as a root of a product of powers of distinct linear factors is the exponent it carries (Lemma 5.4.1),  $a_T(\lambda) = \sum_{\mu_i=\lambda} k_i$ . The last claim follows since  $\text{spec}(T)$  is the set of roots of  $p_T$  (Theorem 8.2.1).
- (b) Take  $j = 1$  in Theorem 9.3.1 (b): the number of  $\lambda$ -blocks of size  $\geq 1$ , which is all of them, equals  $\text{rank}(T - \lambda \text{id}_V)^0 - \text{rank}(T - \lambda \text{id}_V) = n - \text{rank}(T - \lambda \text{id}_V) = \dim \ker(T - \lambda \text{id}_V) = g_T(\lambda)$ , by Theorem 3.3.1 and Definition 8.3.2.
- (c) First,  $m_{\mathbf{J}_k(\nu)} = (x - \nu)^k$  for every  $\nu \in F$ : the polynomial  $(x - \nu)^k$  annihilates  $\mathbf{J}_k(\nu)$  because  $\mathbf{J}_k(\nu) - \nu \mathbf{I} = \mathbf{J}_k(0)$  and  $\mathbf{J}_k(0)^k = 0$ , while  $\mathbf{J}_k(0)^{k-1} \neq 0$ , so no smaller power of  $x - \nu$  annihilates it; and  $m_{\mathbf{J}_k(\nu)}$  divides  $p_{\mathbf{J}_k(\nu)} = (x - \nu)^k$  (Corollary 8.6.2), hence is a power of  $x - \nu$  (Lemma 8.5.2). Applying Proposition 8.5.6  $m - 1$  times,

$$m_T = m_{\mathbf{J}} = \text{lcm}((x - \mu_1)^{k_1}, \dots, (x - \mu_m)^{k_m}) = \prod_{\nu \in \text{spec}(T)} (x - \nu)^{\max\{k_i : \mu_i = \nu\}},$$

the last equality because the least common multiple of powers of linear factors takes, for each factor, the largest exponent occurring (Definition 5.2.4, Theorem 5.3.5).

- (d) The proof of Theorem 9.3.1 (b) computed

$$\text{rank}(T - \lambda \text{id}_V)^j = c + \sum_{\mu_i=\lambda} \max(k_i - j, 0), \quad c := \sum_{\mu_i \neq \lambda} k_i.$$

Since  $\sum_{i=1}^m k_i = n$ , we have  $n - c = \sum_{\mu_i=\lambda} k_i$ , so by [Theorem 3.3.1](#)

$$\begin{aligned} \dim \ker(T - \lambda \operatorname{id}_V)^j &= n - c - \sum_{\mu_i=\lambda} \max(k_i - j, 0) \\ &= \sum_{\mu_i=\lambda} (k_i - \max(k_i - j, 0)) \\ &= \sum_{\mu_i=\lambda} \min(k_i, j). \end{aligned}$$

- (e) If every block has size 1, then  $\mathbf{J}$  is diagonal and  $T$  is diagonalizable ([Definition 8.4.1](#)). Conversely, if  $T$  is diagonalizable, then  $m_T$  is a product of distinct linear factors ([Theorem 8.8.2](#)), so every exponent in (c) is 1, which says that every block has size 1. ■

Parts (a) and (b) give the practical reading: **the total size of the  $\lambda$ -blocks is  $a(\lambda)$ , and their number is  $g(\lambda)$** . The gap  $a(\lambda) - g(\lambda)$ , which Chapter 8 could only measure, is now explained: it counts the superdiagonal ones belonging to  $\lambda$ .

*Warning.* The multiplicities  $a(\lambda)$  and  $g(\lambda)$  do not always determine the blocks. For  $n = 7$  and one eigenvalue  $\lambda$  with  $a(\lambda) = 7$  and  $g(\lambda) = 3$ , the block sizes form a partition of 7 into 3 parts, and there are four of those:  $(5, 1, 1)$ ,  $(4, 2, 1)$ ,  $(3, 3, 1)$ ,  $(3, 2, 2)$ . Knowing  $a(\lambda)$  and  $g(\lambda)$  means knowing  $\dim \ker(T - \lambda \operatorname{id}_V)^j$  for  $j = 1$  and for  $j$  large; the intermediate values  $j = 2, 3, \dots$  are genuinely new information, and [Theorem 9.3.1](#) (b) needs them all.

## 9.3.4 Deciding similarity

We can now settle the question Chapter 3 raised and Chapter 6 sharpened: which data decide whether two matrices are similar?

### Corollary 9.3.3 — Similarity via the Jordan Form

Let  $\mathbf{A}, \mathbf{B} \in M_n(F)$ ,  $n \geq 1$ , and suppose  $p_{\mathbf{A}}$  splits over  $F$ . The following are equivalent:

- (a)  $\mathbf{A} \sim \mathbf{B}$ ;
- (b)  $p_{\mathbf{B}}$  splits over  $F$ , and  $\mathbf{A}$  and  $\mathbf{B}$  have the same Jordan form, up to the order of the blocks;
- (c)  $p_{\mathbf{A}} = p_{\mathbf{B}}$  and  $\operatorname{rank}(\mathbf{A} - \lambda \mathbf{I})^j = \operatorname{rank}(\mathbf{B} - \lambda \mathbf{I})^j$  for every  $\lambda \in \operatorname{spec}(\mathbf{A})$  and every  $j = 1, \dots, n$ .

*Proof.*

- (a)  $\Rightarrow$  (c). Similar matrices have the same characteristic polynomial ([Theorem 6.10.3](#)). If  $\mathbf{B} = \mathbf{P}^{-1}\mathbf{A}\mathbf{P}$ , then  $(\mathbf{B} - \lambda \mathbf{I})^j = \mathbf{P}^{-1}(\mathbf{A} - \lambda \mathbf{I})^j\mathbf{P}$  for every  $\lambda$  and  $j$  ([Proposition 3.7.6](#) (c) applied to the polynomial  $(x - \lambda)^j$ ), so the ranks agree ([Proposition 3.7.6](#) (a)).

- (b)  $\Rightarrow$  (b). From  $p_{\mathbf{B}} = p_{\mathbf{A}}$  we get that  $p_{\mathbf{B}}$  splits and that  $\text{spec}(\mathbf{B}) = \text{spec}(\mathbf{A})$  ([Theorem 8.2.1](#)). By [Theorem 9.3.1](#) (a) both matrices have Jordan forms, and by (b) of that theorem the number of  $\lambda$ -blocks of each size in each of them is determined by the ranks  $\text{rank}(\cdot - \lambda\mathbf{I})^j$ , with  $j \leq n$  sufficing because  $(\mathbf{A} - \lambda\mathbf{I})^n$  and all higher powers have the same rank ([Theorem 9.1.1](#) (d), [Theorem 3.3.1](#)). These numbers agree by hypothesis for  $\lambda \in \text{spec}(\mathbf{A}) = \text{spec}(\mathbf{B})$ , and for  $\lambda \notin \text{spec}(\mathbf{A})$  neither form has a  $\lambda$ -block ([Corollary 9.3.2](#) (a)). So the two Jordan forms have the same blocks.
- (c)  $\Rightarrow$  (a). Let  $\mathbf{J}$  be a Jordan form of  $\mathbf{A}$  and  $\mathbf{J}'$  one of  $\mathbf{B}$ , with the same blocks in possibly different orders. Then  $\mathbf{J} \sim \mathbf{J}'$ , because reordering blocks is a change of basis, as noted after [Definition 9.3.2](#). Hence  $\mathbf{A} \sim \mathbf{J} \sim \mathbf{J}' \sim \mathbf{B}$ , and similarity is an equivalence relation ([Example \(Similarity\)](#)). ■

This is the promise of Chapter 3 kept, and of Chapter 6: when  $p_{\mathbf{A}}$  splits — always, over  $\mathbb{C}$  — the characteristic polynomial together with the ranks of the powers  $(\mathbf{A} - \lambda\mathbf{I})^j$  is a **complete** set of similarity invariants. Chapter 3's warning pair is decided instantly:  $\mathbf{I}_2$  has Jordan form  $\text{diag}(1, 1)$  and  $\begin{pmatrix} 1 & 1 \\ 0 & 1 \end{pmatrix} = \mathbf{J}_2(1)$  is its own, so they are not similar. And the sharper failure of Chapter 8 becomes transparent.

*Remark.* The matrices  $\mathbf{J}_2(1) \oplus \mathbf{J}_2(1)$  and  $\mathbf{J}_2(1) \oplus \mathbf{J}_1(1) \oplus \mathbf{J}_1(1)$  in  $M_4(F)$  have the same characteristic polynomial  $(x - 1)^4$  and the same minimal polynomial  $(x - 1)^2$ , yet they are not similar: their Jordan forms differ, and  $\text{rank}(\mathbf{A} - \mathbf{I})$  is 2 against 1. So no pair of polynomials can decide similarity; the block sizes carry strictly more information.

When  $p_{\mathbf{A}}$  does not split, the Jordan form does not exist and this test says nothing. Section 6 replaces it by the rational canonical form, which works over every field.

*Warning.* **The Jordan form is not stable under small changes of the matrix.**

For  $\varepsilon \neq 0$ ,  $\mathbf{A}_{\varepsilon} = \begin{pmatrix} 0 & 1 \\ \varepsilon & 0 \end{pmatrix}$  over  $\mathbb{C}$  has  $p = x^2 - \varepsilon$  with two distinct roots  $\pm\sqrt{\varepsilon}$ , so  $\mathbf{A}_{\varepsilon}$  is diagonalizable and its Jordan form is  $\text{diag}(\sqrt{\varepsilon}, -\sqrt{\varepsilon})$ , with **no** superdiagonal one. At  $\varepsilon = 0$  the Jordan form jumps to  $\mathbf{J}_2(0)$ . So arbitrarily small perturbations change the form completely, which is why numerical work avoids it; Chapter 19 returns to this point.

## 9.3.5 Computing a Jordan form

The proof is an algorithm: it says to work one eigenvalue at a time and to find chains inside each generalized eigenspace. The rank data tell us the shape before we look for a single vector.

*Algorithm.* **Jordan form and a Jordan basis of  $\mathbf{A} \in M_n(F)$ .**

1. Factor  $p_{\mathbf{A}}$ . If it does not split over  $F$ , stop: there is no Jordan form.
2. For each eigenvalue  $\lambda$ , compute  $\text{rank}(\mathbf{A} - \lambda\mathbf{I})^j$  for  $j = 0, 1, 2, \dots$  until the rank stops changing, and get the  $\lambda$ -block sizes from [Theorem 9.3.1](#) (b), subtracting twice for the exact counts.
3. For each  $\lambda$ , find a Jordan basis of the nilpotent  $(\mathbf{A} - \lambda\mathbf{I})$  restricted to  $G_{\lambda}(\mathbf{A})$ , as in Section 2: tops of long chains first, then vectors of  $\ker(\mathbf{A} - \lambda\mathbf{I})$  to complete.

4. Let  $\mathbf{P}$  be the matrix whose columns are all the chains, in order; then  $\mathbf{P}^{-1}\mathbf{A}\mathbf{P} = \mathbf{J}$ . Check the answer as  $\mathbf{A}\mathbf{P} = \mathbf{P}\mathbf{J}$ , column by column.

**Example. (The Jordan Form of a  $5 \times 5$  Matrix)**

Let

$$\mathbf{A} = \begin{pmatrix} 3 & 1 & 0 & 0 & -1 \\ -2 & 2 & 2 & -2 & -1 \\ 1 & 0 & 2 & -1 & 1 \\ 1 & 0 & -1 & 2 & 1 \\ -2 & -1 & 2 & -2 & 2 \end{pmatrix} \in M_5(\mathbb{Q}), \quad p_{\mathbf{A}} = (x-3)^3(x-1)^2.$$

Find the Jordan form  $\mathbf{J}$  of  $\mathbf{A}$ , a Jordan basis, and an invertible  $\mathbf{P}$  with  $\mathbf{P}^{-1}\mathbf{A}\mathbf{P} = \mathbf{J}$ . Verify the answer by hand.

*Solution.* The eigenvalue 3. Put  $\mathbf{N} = \mathbf{A} - 3\mathbf{I}$ , so

$$\mathbf{N} = \begin{pmatrix} 0 & 1 & 0 & 0 & -1 \\ -2 & -1 & 2 & -2 & -1 \\ 1 & 0 & -1 & -1 & 1 \\ 1 & 0 & -1 & -1 & 1 \\ -2 & -1 & 2 & -2 & -1 \end{pmatrix}.$$

Rows 3 and 4 are equal, and so are rows 2 and 5, so the row space is spanned by rows 1, 2, 3. Those three are independent: their entries in columns 1, 2, 4 form the matrix  $\begin{pmatrix} 0 & 1 & 0 \\ -2 & -1 & -2 \\ 1 & 0 & -1 \end{pmatrix}$ , of determinant  $-4 \neq 0$ . Hence  $\text{rank } \mathbf{N} = 3$  and nullity  $\mathbf{N} = 2$  (Theorem 2.5.7). Solving  $\mathbf{N}\mathbf{x} = \mathbf{0}$  means solving

$$x_2 - x_5 = 0, \quad -2x_1 - x_2 + 2x_3 - 2x_4 - x_5 = 0, \quad x_1 - x_3 - x_4 + x_5 = 0.$$

The first gives  $x_2 = x_5$ ; substituting into the second gives  $-2x_1 + 2x_3 - 2x_4 - 2x_5 = 0$ , that is,  $x_1 = x_3 - x_4 - x_5$ ; and the third then reads  $(x_3 - x_4 - x_5) - x_3 - x_4 + x_5 = -2x_4 = 0$ , so  $x_4 = 0$ . Hence

$$E_3(\mathbf{A}) = \ker \mathbf{N} = \text{span} \left( (1, 0, 1, 0, 0), (-1, 1, 0, 0, 1) \right), \quad g_{\mathbf{A}}(3) = 2.$$

Squaring,

$$\mathbf{N}^2 = 4 \begin{pmatrix} 0 & 0 & 0 & 0 & 0 \\ 1 & 0 & -1 & 1 & 1 \\ -1 & 0 & 1 & 0 & -1 \\ -1 & 0 & 1 & 0 & -1 \\ 1 & 0 & -1 & 1 & 1 \end{pmatrix},$$

whose non-zero rows span a 2-dimensional space, so  $\text{rank } \mathbf{N}^2 = 2$ . Since  $\dim G_3(\mathbf{A}) = a_{\mathbf{A}}(3) = 3$  and  $\dim \ker \mathbf{N}^2 = 3$  (Theorem 2.5.7), we have  $G_3(\mathbf{A}) = \ker \mathbf{N}^2$ , so  $\text{rank } \mathbf{N}^3 = \text{rank } \mathbf{N}^2 = 2$ . The block sizes for  $\lambda = 3$  follow from Theorem 9.3.1 (b) with  $r_0, r_1, r_2, r_3 = 5, 3, 2, 2$ : blocks of size  $\geq 1$ :  $5 - 3 = 2$ ; of size  $\geq 2$ :  $3 - 2 = 1$ ; of size  $\geq 3$ :  $2 - 2 = 0$ . So the 3-blocks have sizes 2 and 1.

*The chain for 3.* We need a top  $\mathbf{u} \in \ker \mathbf{N}^2 \setminus \ker \mathbf{N}$ . The second column of  $\mathbf{N}^2$  is  $\mathbf{0}$ , so  $\mathbf{e}_2 \in \ker \mathbf{N}^2$ , while the second column of  $\mathbf{N}$  is  $(1, -1, 0, 0, -1) \neq \mathbf{0}$ , so  $\mathbf{e}_2 \notin \ker \mathbf{N}$ . Take

$\mathbf{u} = \mathbf{e}_2$ ; its chain is

$$\mathbf{N}\mathbf{u} = (1, -1, 0, 0, -1), \quad \mathbf{u} = \mathbf{e}_2.$$

For the block of size 1 we need a vector of  $\ker \mathbf{N}$  completing  $\{\mathbf{N}\mathbf{u}\}$  to a basis of  $\ker \mathbf{N}$ . The vectors  $\mathbf{N}\mathbf{u} = (1, -1, 0, 0, -1)$  and  $\mathbf{v} = (1, 0, 1, 0, 0)$  are not proportional (their second entries are  $-1$  and  $0$ , their first entries both  $1$ ), so,  $\ker \mathbf{N}$  being 2-dimensional, they form a basis of it (Theorem 1.6.4 (a)) and  $\mathbf{v}$  does it.

*The eigenvalue 1.* Put  $\mathbf{M} = \mathbf{A} - \mathbf{I}$  and solve  $\mathbf{M}\mathbf{x} = \mathbf{0}$ . Rows 3 and 4 of  $\mathbf{M}$  read  $x_1 + x_3 - x_4 + x_5 = 0$  and  $x_1 - x_3 + x_4 + x_5 = 0$ ; adding and subtracting them gives  $x_1 + x_5 = 0$  and  $x_3 = x_4$ . Row 1 reads  $2x_1 + x_2 - x_5 = 0$ , and row 2, after using  $x_3 = x_4$ , reads  $-2x_1 + x_2 - x_5 = 0$ ; adding and subtracting these two gives  $x_2 = x_5$  and  $x_1 = 0$ , hence  $x_5 = 0$  and  $x_2 = 0$ . So

$$E_1(\mathbf{A}) = \ker \mathbf{M} = \text{span}((0, 0, 1, 1, 0)), \quad g_{\mathbf{A}}(1) = 1.$$

Since  $a_{\mathbf{A}}(1) = 2$ , the only partition of 2 with one part is (2): there is a single 1-block, of size 2. Its chain needs a top  $\mathbf{w} \in \ker \mathbf{M}^2 \setminus \ker \mathbf{M}$ . Computing,

$$\mathbf{M}^2 = 4 \begin{pmatrix} 1 & 1 & 0 & 0 & -1 \\ -1 & 0 & 1 & -1 & 0 \\ 0 & 0 & 1 & -1 & 0 \\ 0 & 0 & 0 & 0 & 0 \\ -1 & -1 & 1 & -1 & 1 \end{pmatrix},$$

whose kernel is cut out by  $x_1 + x_2 - x_5 = 0$ ,  $-x_1 + x_3 - x_4 = 0$  and  $x_3 - x_4 = 0$ , giving  $x_1 = 0$ ,  $x_3 = x_4$ ,  $x_2 = x_5$ ; so  $\ker \mathbf{M}^2 = \text{span}((0, 0, 1, 1, 0), (0, 1, 0, 0, 1))$ , of dimension  $2 = a_{\mathbf{A}}(1)$  as expected. Take  $\mathbf{w} = (0, 1, 0, 0, 1)$ ; then

$$\begin{aligned} \mathbf{M}\mathbf{w} &= (\text{column 2} + \text{column 5 of } \mathbf{M}) \\ &= (1, 1, 0, 0, -1) + (-1, -1, 1, 1, 1) = (0, 0, 1, 1, 0), \end{aligned}$$

which is the eigenvector, as it must be.

*The answer.* Writing the three chains one after another,

$$\mathcal{B} = ((1, -1, 0, 0, -1), \mathbf{e}_2, (1, 0, 1, 0, 0), (0, 0, 1, 1, 0), (0, 1, 0, 0, 1)),$$

$$\mathbf{P} = \begin{pmatrix} 1 & 0 & 1 & 0 & 0 \\ -1 & 1 & 0 & 0 & 1 \\ 0 & 0 & 1 & 1 & 0 \\ 0 & 0 & 0 & 1 & 0 \\ -1 & 0 & 0 & 0 & 1 \end{pmatrix},$$

$$\mathbf{J} = \mathbf{J}_2(3) \oplus \mathbf{J}_1(3) \oplus \mathbf{J}_2(1) = \begin{pmatrix} 3 & 1 & 0 & 0 & 0 \\ 0 & 3 & 0 & 0 & 0 \\ 0 & 0 & 3 & 0 & 0 \\ 0 & 0 & 0 & 1 & 1 \\ 0 & 0 & 0 & 0 & 1 \end{pmatrix}.$$

$\mathbf{P}$  is invertible: its fourth row is  $(0, 0, 0, 1, 0)$ , so expanding  $\det \mathbf{P}$  along that row ([Theorem 6.6.1](#)) leaves

$$\det \mathbf{P} = \det \begin{pmatrix} 1 & 0 & 1 & 0 \\ -1 & 1 & 0 & 1 \\ 0 & 0 & 1 & 0 \\ -1 & 0 & 0 & 1 \end{pmatrix} = \det \begin{pmatrix} 1 & 0 & 0 \\ -1 & 1 & 1 \\ -1 & 0 & 1 \end{pmatrix} = 1,$$

expanding along the third row and then along the first row.

*The check, column by column.* The claim  $\mathbf{P}^{-1}\mathbf{A}\mathbf{P} = \mathbf{J}$  is the same as  $\mathbf{A}\mathbf{P} = \mathbf{P}\mathbf{J}$ . Writing  $\mathbf{p}_1, \dots, \mathbf{p}_5$  for the columns of  $\mathbf{P}$ , the five columns of  $\mathbf{P}\mathbf{J}$  are

$$3\mathbf{p}_1, \quad \mathbf{p}_1 + 3\mathbf{p}_2, \quad 3\mathbf{p}_3, \quad \mathbf{p}_4, \quad \mathbf{p}_4 + \mathbf{p}_5,$$

read straight off the columns of  $\mathbf{J}$ . So the identity  $\mathbf{A}\mathbf{P} = \mathbf{P}\mathbf{J}$  is the list of five statements

$$\mathbf{A}\mathbf{p}_1 = 3\mathbf{p}_1, \quad \mathbf{A}\mathbf{p}_2 = \mathbf{p}_1 + 3\mathbf{p}_2, \quad \mathbf{A}\mathbf{p}_3 = 3\mathbf{p}_3, \quad \mathbf{A}\mathbf{p}_4 = \mathbf{p}_4, \quad \mathbf{A}\mathbf{p}_5 = \mathbf{p}_4 + \mathbf{p}_5,$$

which are exactly how the chains were constructed:  $\mathbf{p}_1, \mathbf{p}_3 \in \ker(\mathbf{A} - 3\mathbf{I})$ ,  $(\mathbf{A} - 3\mathbf{I})\mathbf{p}_2 = \mathbf{p}_1$ ,  $\mathbf{p}_4 \in \ker(\mathbf{A} - \mathbf{I})$  and  $(\mathbf{A} - \mathbf{I})\mathbf{p}_5 = \mathbf{p}_4$ . Two of them verified in full:

$$\mathbf{A}\mathbf{p}_2 = \mathbf{A}\mathbf{e}_2 = (1, 2, 0, 0, -1) = (1, -1, 0, 0, -1) + 3(0, 1, 0, 0, 0) = \mathbf{p}_1 + 3\mathbf{p}_2,$$

reading the second column of  $\mathbf{A}$ ; and

$$\begin{aligned} \mathbf{A}\mathbf{p}_4 &= \mathbf{A}(0, 0, 1, 1, 0) = (0 + 0, 2 - 2, 2 - 1, -1 + 2, 2 - 2) \\ &= (0, 0, 1, 1, 0) = \mathbf{p}_4, \end{aligned}$$

adding the third and fourth columns of  $\mathbf{A}$ . The remaining three are the same kind of one-line computation.

*Cross-checks.* The diagonal of  $\mathbf{J}$  gives  $\text{tr } \mathbf{A} = 3 + 3 + 3 + 1 + 1 = 11$ , and adding the diagonal of  $\mathbf{A}$  gives  $3 + 2 + 2 + 2 + 2 = 11$ . By [Corollary 9.3.2](#),  $a(3) = 2 + 1 = 3$ ,  $g(3) = 2$ ,  $a(1) = 2$ ,  $g(1) = 1$  and  $m_{\mathbf{A}} = (x - 3)^2(x - 1)^2$ , all matching the computations above.

## 9.3.6 Exercises

### 9.3.6.1 A. Check your understanding

#### Exercise A1.

- Define a Jordan block and a Jordan matrix, and state what a Jordan form of  $T$  is.
- State the Jordan Canonical Form Theorem, including the uniqueness clause.
- How do you read  $a(\lambda)$ ,  $g(\lambda)$  and the exponent of  $x - \lambda$  in  $m_T$  off a Jordan form?
- True or false: every matrix in  $M_n(\mathbb{R})$  has a Jordan form over  $\mathbb{R}$ . Justify your answer.
- True or false: if  $\mathbf{A}$  and  $\mathbf{B}$  have the same characteristic and the same minimal polynomial, they are similar. Justify your answer.



- (f) Is  $\begin{pmatrix} 5 & 1 \\ 0 & 4 \end{pmatrix}$  a Jordan matrix? What is its Jordan form?

*Solution.*

- (a)  $\mathbf{J}_k(\lambda) = \lambda \mathbf{I}_k + \mathbf{J}_k(0)$  has  $\lambda$  on the diagonal and 1 on the superdiagonal ([Definition 9.3.1](#)); a Jordan matrix is a direct sum of Jordan blocks; a Jordan form of  $T$  is a Jordan matrix equal to  $[T]_{\mathcal{B}}$  for some basis  $\mathcal{B}$  ([Definition 9.3.2](#)).
- (b)  $T$  has a Jordan form if and only if  $p_T$  splits, and then the number of  $\lambda$ -blocks of size at least  $j$  is  $\text{rank}(T - \lambda \text{id}_V)^{j-1} - \text{rank}(T - \lambda \text{id}_V)^j$ , so the blocks are unique up to their order ([Theorem 9.3.1](#)).
- (c)  $a(\lambda)$  is the total size of the  $\lambda$ -blocks,  $g(\lambda)$  their number, and the exponent in  $m_T$  the largest of their sizes ([Corollary 9.3.2](#)).
- (d) False. A Jordan matrix is triangular, so its existence forces  $p_T$  to split; the rotation by a right angle has  $p = x^2 + 1$ , which does not split over  $\mathbb{R}$ .
- (e) False.  $\mathbf{J}_2(1) \oplus \mathbf{J}_2(1)$  and  $\mathbf{J}_2(1) \oplus \mathbf{J}_1(1) \oplus \mathbf{J}_1(1)$  share  $p = (x-1)^4$  and  $m = (x-1)^2$  but have different Jordan forms, hence are not similar ([Corollary 9.3.3](#)).
- (f) No: a superdiagonal 1 may only stand between two equal diagonal entries. The eigenvalues 5 and 4 are distinct, so the matrix is diagonalizable ([Corollary 8.4.2](#)) and its Jordan form is  $\text{diag}(5, 4)$ .

### 9.3.6.2 B. Practice

#### Exercise B1: Jordan forms from polynomial data.

In each case  $T$  is an operator on a complex vector space with the given characteristic and minimal polynomials. List all possible Jordan forms, up to the order of the blocks, and say what extra data would pin the answer down.

- (a)  $p_T = (x-2)^4$ ,  $m_T = (x-2)^2$ .
- (b)  $p_T = (x-1)^3(x+1)^2$ ,  $m_T = (x-1)^3(x+1)$ .
- (c)  $p_T = (x-5)^5$ ,  $m_T = (x-5)^3$ .

*Solution.* By [Corollary 9.3.2](#), the  $\lambda$ -block sizes form a partition of  $a(\lambda)$  whose largest part is the exponent of  $x - \lambda$  in  $m_T$ .

- (a) Partitions of 4 with largest part 2:  $(2, 2)$  and  $(2, 1, 1)$ . So  $\mathbf{J}_2(2) \oplus \mathbf{J}_2(2)$  or  $\mathbf{J}_2(2) \oplus \mathbf{J}_1(2) \oplus \mathbf{J}_1(2)$ . The value of  $g(2)$  — that is, 2 against 3 — decides.
- (b) For 1: partitions of 3 with largest part 3, only  $(3)$ . For  $-1$ : partitions of 2 with largest part 1, only  $(1, 1)$ . So the Jordan form is  $\mathbf{J}_3(1) \oplus \mathbf{J}_1(-1) \oplus \mathbf{J}_1(-1)$ , and no extra data is needed.
- (c) Partitions of 5 with largest part 3:  $(3, 2)$  and  $(3, 1, 1)$ . So  $\mathbf{J}_3(5) \oplus \mathbf{J}_2(5)$  or  $\mathbf{J}_3(5) \oplus \mathbf{J}_1(5) \oplus \mathbf{J}_1(5)$ , told apart by  $g(5) = 2$  against 3, or by  $\dim \ker(T - 5 \text{id})^2 = 4$  against 5.

#### Exercise B2: Jordan forms and Jordan bases.

For each matrix, find the Jordan form  $\mathbf{J}$  and an invertible  $\mathbf{P}$  with  $\mathbf{P}^{-1}\mathbf{A}\mathbf{P} = \mathbf{J}$ . Verify

your  $\mathbf{P}$  by checking  $\mathbf{AP} = \mathbf{PJ}$  column by column.

$$(a) \mathbf{A}_1 = \begin{pmatrix} -1 & -1 & -2 \\ 1 & 2 & 1 \\ 1 & 0 & 2 \end{pmatrix} \in M_3(\mathbb{Q}), \text{ with } p_{\mathbf{A}_1} = (x-1)^3.$$

$$(b) \mathbf{A}_2 = \begin{pmatrix} 2 & -1 & -1 \\ -3 & 4 & -1 \\ 3 & -2 & 3 \end{pmatrix} \in M_3(\mathbb{Q}), \text{ with } p_{\mathbf{A}_2} = (x-2)^2(x-5).$$

$$(c) \mathbf{A}_3 = \begin{pmatrix} 2 & 1 & 1 & -2 \\ -3 & 1 & -1 & -1 \\ -3 & -1 & 1 & -1 \\ -3 & -1 & -1 & 1 \end{pmatrix} \in M_4(\mathbb{Q}), \text{ with } p_{\mathbf{A}_3} = (x-2)^3(x+1).$$

*Solution.*

$$(a) \mathbf{A}_1 - \mathbf{I} = \begin{pmatrix} -2 & -1 & -2 \\ 1 & 1 & 1 \\ 1 & 0 & 1 \end{pmatrix} \text{ has rank 2, so } g(1) = 1 \text{ and there is one block, of size 3:}$$

$$\mathbf{J} = \mathbf{J}_3(1). \text{ A top must satisfy } (\mathbf{A}_1 - \mathbf{I})^2 \mathbf{u} \neq \mathbf{0}, \text{ and } (\mathbf{A}_1 - \mathbf{I})^2 = \begin{pmatrix} 1 & 1 & 1 \\ 0 & 0 & 0 \\ -1 & -1 & -1 \end{pmatrix}$$

has non-zero first column, so  $\mathbf{u} = \mathbf{e}_1$  works. The chain is

$$(\mathbf{A}_1 - \mathbf{I})^2 \mathbf{e}_1 = (1, 0, -1), \quad (\mathbf{A}_1 - \mathbf{I}) \mathbf{e}_1 = (-2, 1, 1), \quad \mathbf{e}_1,$$

$$\text{so } \mathbf{P} = \begin{pmatrix} 1 & -2 & 1 \\ 0 & 1 & 0 \\ -1 & 1 & 0 \end{pmatrix}, \text{ with } \det \mathbf{P} = 1. \text{ The check } \mathbf{A}_1 \mathbf{P} = \mathbf{P} \mathbf{J}_3(1) \text{ reads } \mathbf{A}_1 \mathbf{p}_1 =$$

$\mathbf{p}_1$ ,  $\mathbf{A}_1 \mathbf{p}_2 = \mathbf{p}_1 + \mathbf{p}_2$ ,  $\mathbf{A}_1 \mathbf{p}_3 = \mathbf{p}_2 + \mathbf{p}_3$ , which are the three chain relations.

$$(b) \mathbf{A}_2 - 2\mathbf{I} = \begin{pmatrix} 0 & -1 & -1 \\ -3 & 2 & -1 \\ 3 & -2 & 1 \end{pmatrix} \text{ has rank 2, so } g(2) = 1 \text{ and the single 2-block has size}$$

$a(2) = 2$ ; the eigenvalue 5 gives one block of size 1. So  $\mathbf{J} = \mathbf{J}_2(2) \oplus \mathbf{J}_1(5)$ . For the chain, the third column of  $\mathbf{A}_2 - 2\mathbf{I}$  is  $(-1, -1, 1) \neq \mathbf{0}$  and  $(\mathbf{A}_2 - 2\mathbf{I})(-1, -1, 1) = \mathbf{0}$ , so  $\mathbf{u} = \mathbf{e}_3$  is a top with  $(\mathbf{A}_2 - 2\mathbf{I})\mathbf{e}_3 = (-1, -1, 1)$ . Solving  $(\mathbf{A}_2 - 5\mathbf{I})\mathbf{x} = \mathbf{0}$  gives  $\text{span}((0, -1, 1))$ . Hence

$$\mathbf{P} = \begin{pmatrix} -1 & 0 & 0 \\ -1 & 0 & -1 \\ 1 & 1 & 1 \end{pmatrix}, \quad \det \mathbf{P} = -1,$$

and  $\mathbf{A}_2 \mathbf{P} = \mathbf{P} \mathbf{J}$  reads  $\mathbf{A}_2 \mathbf{p}_1 = 2\mathbf{p}_1$ ,  $\mathbf{A}_2 \mathbf{p}_2 = \mathbf{p}_1 + 2\mathbf{p}_2$ ,  $\mathbf{A}_2 \mathbf{p}_3 = 5\mathbf{p}_3$ .

$$(c) \mathbf{A}_3 - 2\mathbf{I} = \begin{pmatrix} 0 & 1 & 1 & -2 \\ -3 & -1 & -1 & -1 \\ -3 & -1 & -1 & -1 \\ -3 & -1 & -1 & -1 \end{pmatrix} \text{ has rank 2 (rows 2, 3, 4 are equal), so } g(2) = 2:$$

with  $a(2) = 3$  the 2-blocks have sizes 2 and 1. The eigenvalue  $-1$  has  $a = 1$ , so  $\mathbf{J} = \mathbf{J}_2(2) \oplus \mathbf{J}_1(2) \oplus \mathbf{J}_1(-1)$ . Now  $\ker(\mathbf{A}_3 - 2\mathbf{I}) = \text{span}((0, -1, 1, 0), (-1, 2, 0, 1))$ , and the second column of  $\mathbf{A}_3 - 2\mathbf{I}$  is  $(1, -1, -1, -1)$ , which is non-zero and lies in that kernel, so  $\mathbf{u} = \mathbf{e}_2$  is the top of the 2-chain. Completing  $\{(1, -1, -1, -1)\}$  to a basis of  $\ker(\mathbf{A}_3 - 2\mathbf{I})$  with  $(0, -1, 1, 0)$ , and solving  $(\mathbf{A}_3 + \mathbf{I})\mathbf{x} = \mathbf{0}$  to get

$\text{span}((0, 1, 1, 1))$ , we obtain

$$\mathbf{P} = \begin{pmatrix} 1 & 0 & 0 & 0 \\ -1 & 1 & -1 & 1 \\ -1 & 0 & 1 & 1 \\ -1 & 0 & 0 & 1 \end{pmatrix}, \quad \det \mathbf{P} = 1,$$

and  $\mathbf{A}_3\mathbf{P} = \mathbf{PJ}$  reads  $\mathbf{A}_3\mathbf{p}_1 = 2\mathbf{p}_1$ ,  $\mathbf{A}_3\mathbf{p}_2 = \mathbf{p}_1 + 2\mathbf{p}_2$ ,  $\mathbf{A}_3\mathbf{p}_3 = 2\mathbf{p}_3$ ,  $\mathbf{A}_3\mathbf{p}_4 = -\mathbf{p}_4$ . For instance the second of these is  $\mathbf{A}_3\mathbf{e}_2 = (1, 1, -1, -1) = (1, -1, -1, -1) + 2(0, 1, 0, 0)$ , read off the second column of  $\mathbf{A}_3$ .

### Exercise B3: Similar or not?.

Determine whether each pair of matrices is similar. Justify your answer.

- (a)  $\mathbf{J}_2(4) \oplus \mathbf{J}_1(4)$  and  $\mathbf{J}_1(4) \oplus \mathbf{J}_2(4)$ .
- (b)  $\mathbf{J}_3(0) \oplus \mathbf{J}_1(0)$  and  $\mathbf{J}_2(0) \oplus \mathbf{J}_2(0)$ .
- (c)  $\mathbf{J}_2(1) \oplus \mathbf{J}_2(1)$  and  $\mathbf{J}_2(1) \oplus \mathbf{J}_1(1) \oplus \mathbf{J}_1(1)$ .
- (d)  $\text{diag}(2, 2, 5)$  and the matrix  $\mathbf{A}_2$  of [Exercise \(B2: Jordan forms and Jordan bases\)](#) (b).

*Solution.*

- (a) Similar: the blocks are the same and only their order differs, which is a change of basis, as noted after [Definition 9.3.2](#).
- (b) Not similar. Both are nilpotent of rank 2, but  $\text{rank}(\mathbf{J}_3(0) \oplus \mathbf{J}_1(0))^2 = 1$  while  $(\mathbf{J}_2(0) \oplus \mathbf{J}_2(0))^2 = 0$  ([Theorem 7.1.4](#) (b)). By [Corollary 9.3.3](#) (or [Corollary 9.2.5](#)) they are not similar; their partitions are  $(3, 1)$  and  $(2, 2)$ .
- (c) Not similar, although  $p$  and  $m$  agree, as in the remark after [Corollary 9.3.3](#):  $\text{rank}(\mathbf{A} - \mathbf{I}) = 2$  for the first and 1 for the second, so  $g(1)$  is 2 against 3.
- (d) Not similar.  $\text{diag}(2, 2, 5)$  is diagonal, so  $g(2) = 2$ , while  $\mathbf{A}_2$  has  $g(2) = 1$  by [Exercise \(B2: Jordan forms and Jordan bases\)](#) (b); its Jordan form is  $\mathbf{J}_2(2) \oplus \mathbf{J}_1(5)$ , which is not diagonal. The characteristic polynomials agree, so this pair is another witness that  $p_{\mathbf{A}}$  alone decides nothing.

### 9.3.6.3 C. Going deeper

#### Exercise C1: A matrix is similar to its transpose.

Let  $\mathbf{A} \in M_n(F)$  with  $p_{\mathbf{A}}$  split over  $F$ .

- (a) Let  $\mathbf{R} \in M_k(F)$  be the reversal permutation matrix, with 1 in each position  $(j, k + 1 - j)$  and 0 elsewhere. Prove that  $\mathbf{R}^{-1} = \mathbf{R}$  and  $\mathbf{R}^{-1}\mathbf{J}_k(\lambda)\mathbf{R} = \mathbf{J}_k(\lambda)^{\top}$ .
- (b) Deduce that every Jordan matrix  $\mathbf{J}$  satisfies  $\mathbf{J} \sim \mathbf{J}^{\top}$ .
- (c) Deduce that  $\mathbf{A} \sim \mathbf{A}^{\top}$ .

*Hint: for (c), transpose a similarity  $\mathbf{A} = \mathbf{PJP}^{-1}$ . The conclusion holds over every field; Section 7 removes the hypothesis that  $p_{\mathbf{A}}$  splits.*

*Solution.*

- (a)  $\mathbf{R}\mathbf{e}_j = \mathbf{e}_{k+1-j}$ , so  $\mathbf{R}^2\mathbf{e}_j = \mathbf{e}_j$  for every  $j$  and  $\mathbf{R}^2 = \mathbf{I}_k$ , that is,  $\mathbf{R}^{-1} = \mathbf{R}$ . Since  $\mathbf{J}_k(\lambda) = \lambda\mathbf{I}_k + \mathbf{J}_k(0)$  and  $\mathbf{R}^{-1}(\lambda\mathbf{I}_k)\mathbf{R} = \lambda\mathbf{I}_k$ , it suffices to prove  $\mathbf{R}^{-1}\mathbf{J}_k(0)\mathbf{R} = \mathbf{J}_k(0)^\top$ . Both sides are computed on basis vectors:  $\mathbf{J}_k(0)\mathbf{e}_1 = \mathbf{0}$  and  $\mathbf{J}_k(0)\mathbf{e}_j = \mathbf{e}_{j-1}$  for  $j \geq 2$ , so

$$\mathbf{R}\mathbf{J}_k(0)\mathbf{R}\mathbf{e}_j = \mathbf{R}\mathbf{J}_k(0)\mathbf{e}_{k+1-j} = \begin{cases} \mathbf{R}\mathbf{e}_{k-j} = \mathbf{e}_{j+1}, & j \leq k-1, \\ \mathbf{R}\mathbf{0} = \mathbf{0}, & j = k, \end{cases}$$

which is exactly the action of  $\mathbf{J}_k(0)^\top$ , whose columns satisfy  $\mathbf{J}_k(0)^\top\mathbf{e}_j = \mathbf{e}_{j+1}$  for  $j \leq k-1$  and  $\mathbf{J}_k(0)^\top\mathbf{e}_k = \mathbf{0}$ . Hence  $\mathbf{R}^{-1}\mathbf{J}_k(\lambda)\mathbf{R} = \mathbf{J}_k(\lambda)^\top$ .

- (b) Let  $\mathbf{J} = \mathbf{J}_{k_1}(\lambda_1) \oplus \cdots \oplus \mathbf{J}_{k_m}(\lambda_m)$  and let  $\mathbf{Q} = \mathbf{R}_{k_1} \oplus \cdots \oplus \mathbf{R}_{k_m}$ , the direct sum of the reversal matrices of the block sizes. Then  $\mathbf{Q}$  is invertible with  $\mathbf{Q}^{-1} = \mathbf{R}_{k_1}^{-1} \oplus \cdots \oplus \mathbf{R}_{k_m}^{-1}$  (Theorem 7.1.4 (c)), and multiplying blockwise (Theorem 7.1.4 (a)),

$$\mathbf{Q}^{-1}\mathbf{J}\mathbf{Q} = (\mathbf{R}_{k_1}^{-1}\mathbf{J}_{k_1}(\lambda_1)\mathbf{R}_{k_1}) \oplus \cdots = \mathbf{J}_{k_1}(\lambda_1)^\top \oplus \cdots \oplus \mathbf{J}_{k_m}(\lambda_m)^\top = \mathbf{J}^\top,$$

the last equality because transposing a block diagonal matrix transposes each block (Proposition 7.1.2).

- (c) By Theorem 9.3.1 there are a Jordan matrix  $\mathbf{J}$  and an invertible  $\mathbf{P}$  with  $\mathbf{A} = \mathbf{P}\mathbf{J}\mathbf{P}^{-1}$ . Transposing and using  $(\mathbf{XY})^\top = \mathbf{Y}^\top\mathbf{X}^\top$  and  $(\mathbf{P}^{-1})^\top = (\mathbf{P}^\top)^{-1}$  (Theorem 0.9.5, Theorem 0.9.7),

$$\mathbf{A}^\top = (\mathbf{P}^{-1})^\top\mathbf{J}^\top\mathbf{P}^\top = (\mathbf{P}^\top)^{-1}\mathbf{J}^\top\mathbf{P}^\top,$$

so  $\mathbf{A}^\top \sim \mathbf{J}^\top$ . By (b),  $\mathbf{J}^\top \sim \mathbf{J} \sim \mathbf{A}$ , and similarity is an equivalence relation (Example (Similarity)), so  $\mathbf{A} \sim \mathbf{A}^\top$ .

### Exercise C2: How much do $p$ and $m$ decide?

Work over  $\mathbb{C}$ .

- Find all Jordan forms with  $p_T = (x-1)^3(x-2)^2$  and  $m_T = (x-1)^2(x-2)$ .
- Find all Jordan forms with  $p_T = (x-1)^4(x-2)^2$  and  $m_T = (x-1)^2(x-2)$ .
- In (b), what is the least amount of extra information that determines the Jordan form?
- Prove that  $p_T$  and  $m_T$  determine the Jordan form whenever  $a(\lambda) \leq 3$  for every eigenvalue  $\lambda$ .

*Solution.*

- (a) For  $\lambda = 1$ : partitions of 3 with largest part 2, so  $(2, 1)$  only. For  $\lambda = 2$ : partitions of 2 with largest part 1, so  $(1, 1)$  only. There is exactly one Jordan form,

$$\mathbf{J}_2(1) \oplus \mathbf{J}_1(1) \oplus \mathbf{J}_1(2) \oplus \mathbf{J}_1(2).$$

- (b) For  $\lambda = 1$ : partitions of 4 with largest part 2, namely  $(2, 2)$  and  $(2, 1, 1)$ . For  $\lambda = 2$ :

(1, 1) as before. So there are two Jordan forms,

$$\mathbf{J}_2(1) \oplus \mathbf{J}_2(1) \oplus \mathbf{J}_1(2) \oplus \mathbf{J}_1(2)$$

and  $\mathbf{J}_2(1) \oplus \mathbf{J}_1(1) \oplus \mathbf{J}_1(1) \oplus \mathbf{J}_1(2) \oplus \mathbf{J}_1(2).$

- (c) The single number  $g(1)$ , which is 2 for the first and 3 for the second; equivalently  $\text{rank}(T - \text{id}_V)$  or  $\dim \ker(T - \text{id}_V)$  ([Corollary 9.3.2](#) (b)).
- (d) Fix  $\lambda$  with  $a = a(\lambda) \leq 3$  and let  $s$  be the exponent of  $x - \lambda$  in  $m_T$ , which is the largest block size ([Corollary 9.3.2](#) (c)). The  $\lambda$ -blocks form a partition of  $a$  with largest part  $s$ . For  $a = 1$  the only partition is (1). For  $a = 2$ :  $s = 2$  gives (2), and  $s = 1$  gives (1, 1). For  $a = 3$ :  $s = 3$  gives (3),  $s = 2$  gives (2, 1), and  $s = 1$  gives (1, 1, 1). In every case the pair  $(a, s)$  determines the partition, so the blocks for each eigenvalue, and hence the whole Jordan form, are determined by  $p_T$  and  $m_T$ . The first ambiguity appears at  $a = 4$ ,  $s = 2$ , which is part (a) of [Exercise \(B1: Jordan forms from polynomial data\)](#).

## 9.4 The Real Jordan Form and the Weyr Form

The Jordan form of Section 3 exists only when  $p_T$  splits, and over  $\mathbb{R}$  that often fails: a rotation of the plane has no real eigenvalue at all. But a real matrix is also a complex matrix, and its complex eigenvalues come in conjugate pairs ([Theorem 8.2.5](#)). This section turns a complex Jordan basis into a **real** one by pairing each chain with its conjugate; the price for a conjugate pair is a  $2 \times 2$  rotation-scaling block instead of a scalar, and the result is the best real matrix there is. The second half rearranges the same data in another way, the Weyr form, which is the form to reach for when a second operator commutes with the first.

### 9.4.1 A block for a conjugate pair

The obstruction is visible in the smallest case. The rotation  $\mathbf{R} = \begin{pmatrix} 0 & -1 \\ 1 & 0 \end{pmatrix} \in M_2(\mathbb{R})$  has  $p_{\mathbf{R}} = x^2 + 1$ , so  $\mathbf{R}$  has no real eigenvalue and no real eigenvector ([Example \(A Rotation Has No Invariant Lines\)](#)). Every real matrix similar to  $\mathbf{R}$  therefore has no real eigenvalue either, so it cannot be triangular: whatever basis of  $\mathbb{R}^2$  we choose, the matrix of  $\mathbf{R}$  keeps two off-diagonal entries. So  $\mathbf{R}$  is already as simple as it will ever get over  $\mathbb{R}$ , and the honest thing to do is to promote it to a building block.

More generally, for  $\lambda = a + bi \in \mathbb{C}$  with  $b \neq 0$ , consider

$$\mathbf{\Lambda} = \begin{pmatrix} a & -b \\ b & a \end{pmatrix} \in M_2(\mathbb{R}), \quad p_{\mathbf{\Lambda}} = (x - a)^2 + b^2 = (x - (a + bi))(x - (a - bi)).$$

This matrix carries the conjugate pair  $\lambda, \bar{\lambda}$  as its complex eigenvalues, has trace  $2a = 2 \operatorname{Re} \lambda$  and determinant  $a^2 + b^2 = |\lambda|^2$ , and is a scaled rotation: it stretches by  $|\lambda|$  and turns by the argument of  $\lambda$ . It is the real substitute for the  $1 \times 1$  block  $(\lambda)$ , and it costs one extra dimension.

*A real Jordan block is a Jordan block with the scalar  $\lambda$  replaced by the  $2 \times 2$  rotation-scaling matrix it needs over  $\mathbb{R}$ , and the superdiagonal 1 replaced by  $\mathbf{I}_2$ .*

**Definition 9.4.1 — Rotation-Scaling Block, Real Jordan Block**

Let  $\lambda = a + bi \in \mathbb{C}$  with  $a, b \in \mathbb{R}$  and  $b \neq 0$ . The **rotation-scaling block** of the conjugate pair  $\{\lambda, \bar{\lambda}\}$  is

$$\mathbf{\Lambda}(\lambda) := \begin{pmatrix} a & -b \\ b & a \end{pmatrix} \in M_2(\mathbb{R}).$$

For an integer  $k \geq 1$ , the **real Jordan block** of size  $k$  for the pair  $\{\lambda, \bar{\lambda}\}$  is the matrix  $\mathbf{C}_k(\lambda) \in M_{2k}(\mathbb{R})$ , partitioned into  $k \times k$  blocks of size  $2 \times 2$ , with  $\mathbf{\Lambda}(\lambda)$  in every diagonal block,  $\mathbf{I}_2$  in every **superdiagonal** block, and the zero matrix elsewhere:

$$\mathbf{C}_k(\lambda) := \begin{pmatrix} \mathbf{\Lambda}(\lambda) & \mathbf{I}_2 & & \\ & \mathbf{\Lambda}(\lambda) & \ddots & \\ & & \ddots & \mathbf{I}_2 \\ & & & \mathbf{\Lambda}(\lambda) \end{pmatrix}.$$

A **real Jordan matrix** is a direct sum of ordinary Jordan blocks  $\mathbf{J}_k(\mu)$  with  $\mu \in \mathbb{R}$  and real Jordan blocks  $\mathbf{C}_k(\lambda)$  with  $\lambda \notin \mathbb{R}$  and  $\text{Im } \lambda > 0$ .

In words:  $\mathbf{C}_k(\lambda)$  is built exactly like  $\mathbf{J}_k(\lambda)$  (Definition 9.3.1), but every entry of  $\mathbf{J}_k(\lambda)$  is replaced by a  $2 \times 2$  block — the diagonal  $\lambda$  by  $\mathbf{\Lambda}(\lambda)$ , the superdiagonal 1 by  $\mathbf{I}_2$ , and each 0 by the  $2 \times 2$  zero matrix. It has size  $2k$ , not  $k$ . The last clause matters:  $\mathbf{\Lambda}(\bar{\lambda}) = \mathbf{\Lambda}(\lambda)^\top$ , so the pair  $\{\lambda, \bar{\lambda}\}$  offers two candidate blocks of each size, and a real Jordan matrix is required to use the member with  $b > 0$ , which makes  $\mathbf{\Lambda}(\lambda)$  a rotation through a positive angle. Without that normalization the uniqueness below would fail, since  $\mathbf{C}_k(\lambda)$  and  $\mathbf{C}_k(\bar{\lambda})$  are similar over  $\mathbb{R}$ .

**Examples.**

- **Size one.**  $\mathbf{C}_1(\lambda) = \mathbf{\Lambda}(\lambda)$ . For  $\lambda = i$  this is the rotation  $\mathbf{R}$  above; for  $\lambda = 2 + 3i$  it is  $\begin{pmatrix} 2 & -3 \\ 3 & 2 \end{pmatrix}$ .
- **Size two.** For  $\lambda = 1 + i$ ,

$$\mathbf{C}_2(1 + i) = \begin{pmatrix} 1 & -1 & 1 & 0 \\ 1 & 1 & 0 & 1 \\ 0 & 0 & 1 & -1 \\ 0 & 0 & 1 & 1 \end{pmatrix} \in M_4(\mathbb{R}).$$

Its characteristic polynomial is  $((x-1)^2 + 1)^2 = (x^2 - 2x + 2)^2$ , by Theorem 6.8.4 applied to the block upper triangular shape.

- **A real Jordan matrix mixing the two kinds.**  $\mathbf{J}_2(3) \oplus \mathbf{C}_1(i) \in M_4(\mathbb{R})$  has characteristic polynomial  $(x-3)^2(x^2+1)$ .

**Non-example by minimal change.** Change the sign of the lower-left entry:  $\begin{pmatrix} a & -b \\ -b & a \end{pmatrix}$  is still real, still has trace  $2a$ , and looks just as harmless. But it is symmetric, its characteristic polynomial is  $(x-a)^2 - b^2$ , and its eigenvalues  $a \pm b$  are **real**; for  $a = 0, b = 1$  it is the coordinate swap, which is diagonalizable over  $\mathbb{R}$ . The clause that fails is that the two off-diagonal entries must have **opposite** signs; that is what makes the eigenvalues non-real.

**Why this definition.** Two conventions are being fixed, and both are only conventions. First, the sign of  $b$ : replacing  $\mathbf{\Lambda}(\lambda)$  by  $\mathbf{\Lambda}(\bar{\lambda}) = \mathbf{\Lambda}(\lambda)^\top$  amounts to changing the sign of every second

basis vector, so the two choices give similar matrices. Second, the ones: a real Jordan block could be written with  $\mathbf{I}_2$  below the diagonal instead, giving the transpose. What is **not** a convention is the size: a conjugate pair cannot be carried by a  $1 \times 1$  real block, because a  $1 \times 1$  real matrix has a real eigenvalue.

The following computation is the whole idea of the section, and we will use it twice: once to see what  $\mathbf{C}_k(\lambda)$  is over  $\mathbb{C}$ , and once to build the real basis.

### Lemma 9.4.1 — Realifying a Conjugate Pair of Chains

Let  $\mathbf{A} \in M_n(\mathbb{R})$ , let  $\lambda = a + bi$  with  $a, b \in \mathbb{R}$  and  $b \neq 0$ , and let  $(\mathbf{w}_1, \dots, \mathbf{w}_k)$  be a Jordan chain of  $\mathbf{A}$  for  $\lambda$  in  $\mathbb{C}^n$ , so that

$$(\mathbf{A} - \lambda \mathbf{I})\mathbf{w}_1 = \mathbf{0} \quad \text{and} \quad (\mathbf{A} - \lambda \mathbf{I})\mathbf{w}_j = \mathbf{w}_{j-1} \quad (2 \leq j \leq k).$$

Write  $\mathbf{w}_j = \mathbf{u}_j - i\mathbf{v}_j$  with  $\mathbf{u}_j, \mathbf{v}_j \in \mathbb{R}^n$ , and put  $\mathbf{u}_0 = \mathbf{v}_0 = \mathbf{0}$ . Then:

- (a)  $\mathbf{A}\mathbf{u}_j = a\mathbf{u}_j + b\mathbf{v}_j + \mathbf{u}_{j-1}$  and  $\mathbf{A}\mathbf{v}_j = -b\mathbf{u}_j + a\mathbf{v}_j + \mathbf{v}_{j-1}$  for  $1 \leq j \leq k$ ;
- (b)  $(\overline{\mathbf{w}}_1, \dots, \overline{\mathbf{w}}_k)$  is a Jordan chain of  $\mathbf{A}$  for  $\overline{\lambda}$ ;
- (c) the  $2k$  vectors  $\mathbf{u}_1, \mathbf{v}_1, \dots, \mathbf{u}_k, \mathbf{v}_k$  span, over  $\mathbb{C}$ , the same subspace of  $\mathbb{C}^n$  as the  $2k$  vectors  $\mathbf{w}_1, \dots, \mathbf{w}_k, \overline{\mathbf{w}}_1, \dots, \overline{\mathbf{w}}_k$ , and one of the two lists is linearly independent over  $\mathbb{C}$  if and only if the other is;
- (d) if the  $2k$  vectors  $\mathbf{u}_1, \mathbf{v}_1, \dots, \mathbf{u}_k, \mathbf{v}_k$  are linearly independent over  $\mathbb{R}$ , then  $U = \text{span}(\mathbf{u}_1, \mathbf{v}_1, \dots, \mathbf{u}_k, \mathbf{v}_k) \subseteq \mathbb{R}^n$  is invariant under  $\mathbf{A}$ , and the matrix of  $T_{\mathbf{A}|_U}$  in the ordered basis  $(\mathbf{u}_1, \mathbf{v}_1, \dots, \mathbf{u}_k, \mathbf{v}_k)$  is  $\mathbf{C}_k(\lambda)$ .

*Idea.* For (a), write the chain relation as  $\mathbf{A}\mathbf{w}_j = \lambda\mathbf{w}_j + \mathbf{w}_{j-1}$  and compare real and imaginary parts; since  $\mathbf{A}$  is real,  $\mathbf{A}$  acts on the two parts separately, and the only mixing comes from multiplying by  $a + bi$ . The minus sign in  $\mathbf{w}_j = \mathbf{u}_j - i\mathbf{v}_j$  is chosen so that the answer comes out with the signs of  $\mathbf{A}(\lambda)$  rather than its transpose. Part (c) is bookkeeping:  $(\mathbf{u}_j, \mathbf{v}_j)$  and  $(\mathbf{w}_j, \overline{\mathbf{w}}_j)$  are obtained from each other by an invertible  $2 \times 2$  matrix of scalars. Part (d) just reads off the columns from (a).

*Proof.*

- (a) Every  $\mathbf{z} \in \mathbb{C}^n$  is uniquely  $\mathbf{x} + i\mathbf{y}$  with  $\mathbf{x}, \mathbf{y} \in \mathbb{R}^n$ , by [Definition 0.6.1](#) applied entry by entry; so  $\mathbf{u}_j, \mathbf{v}_j$  are well defined. The chain relation says  $\mathbf{A}\mathbf{w}_j = \lambda\mathbf{w}_j + \mathbf{w}_{j-1}$  for  $1 \leq j \leq k$ . Expanding both sides with  $\mathbf{w}_j = \mathbf{u}_j - i\mathbf{v}_j$  and  $\lambda = a + bi$ ,

$$\begin{aligned} \mathbf{A}\mathbf{u}_j - i\mathbf{A}\mathbf{v}_j &= (a + bi)(\mathbf{u}_j - i\mathbf{v}_j) + \mathbf{u}_{j-1} - i\mathbf{v}_{j-1} \\ &= (a\mathbf{u}_j + b\mathbf{v}_j + \mathbf{u}_{j-1}) + i(b\mathbf{u}_j - a\mathbf{v}_j - \mathbf{v}_{j-1}), \end{aligned}$$

where we used  $i^2 = -1$  and, on the left, that  $\mathbf{A}$  has real entries, so  $\mathbf{A}(\mathbf{u}_j - i\mathbf{v}_j) = \mathbf{A}\mathbf{u}_j - i\mathbf{A}\mathbf{v}_j$  with  $\mathbf{A}\mathbf{u}_j, \mathbf{A}\mathbf{v}_j \in \mathbb{R}^n$ . Comparing real parts gives the first identity, and comparing imaginary parts gives  $-\mathbf{A}\mathbf{v}_j = b\mathbf{u}_j - a\mathbf{v}_j - \mathbf{v}_{j-1}$ , that is, the second.

- (b) Conjugating the identity  $(\mathbf{A} - \lambda \mathbf{I})\mathbf{w}_j = \mathbf{w}_{j-1}$  entry by entry and using that  $\mathbf{A}$  is real,  $\overline{\mathbf{A}\mathbf{z}} = \mathbf{A}\overline{\mathbf{z}}$  and  $\overline{\lambda\mathbf{z}} = \overline{\lambda}\overline{\mathbf{z}}$  for every  $\mathbf{z} \in \mathbb{C}^n$  ([Theorem 0.6.2](#) (a), (b)). Hence  $(\mathbf{A} - \overline{\lambda}\mathbf{I})\overline{\mathbf{w}}_j = \overline{\mathbf{w}}_{j-1}$  for  $j \geq 2$  and  $(\mathbf{A} - \overline{\lambda}\mathbf{I})\overline{\mathbf{w}}_1 = \mathbf{0}$ , which is the chain condition for  $\overline{\lambda}$ . (That  $\overline{\mathbf{w}}_1 \neq \mathbf{0}$  follows from  $\mathbf{w}_1 \neq \mathbf{0}$ .)

(c) Since  $\mathbf{w}_j = \mathbf{u}_j - i\mathbf{v}_j$  and  $\overline{\mathbf{w}_j} = \mathbf{u}_j + i\mathbf{v}_j$ , we have

$$\mathbf{u}_j = \frac{1}{2}(\mathbf{w}_j + \overline{\mathbf{w}_j}), \quad \mathbf{v}_j = \frac{i}{2}(\mathbf{w}_j - \overline{\mathbf{w}_j}).$$

So each vector of either list is a  $\mathbb{C}$ -combination of the other list, and the two spans agree ([Theorem 1.4.2](#)). The passage from  $(\mathbf{w}_j, \overline{\mathbf{w}_j})$  to  $(\mathbf{u}_j, \mathbf{v}_j)$  is given, for each  $j$  separately, by the matrix  $\begin{pmatrix} 1/2 & 1/2 \\ i/2 & -i/2 \end{pmatrix}$  of determinant  $-i/2 \neq 0$ , which is invertible; so the change of lists is an invertible linear substitution of the coefficients, and a non-trivial vanishing combination of one list yields one of the other. Hence one list is independent if and only if the other is.

(d) By (a), every  $\mathbf{A}\mathbf{u}_j$  and every  $\mathbf{A}\mathbf{v}_j$  lies in  $U$ , so  $U$  is  $\mathbf{A}$ -invariant ([Lemma 8.1.1](#)). Order the basis as  $(\mathbf{u}_1, \mathbf{v}_1, \dots, \mathbf{u}_k, \mathbf{v}_k)$  and read off the columns ([Definition 3.6.1](#)). By (a), the column of  $\mathbf{u}_j$  has entry  $a$  at  $\mathbf{u}_j$ ,  $b$  at  $\mathbf{v}_j$ , 1 at  $\mathbf{u}_{j-1}$  and 0 elsewhere; the column of  $\mathbf{v}_j$  has  $-b$  at  $\mathbf{u}_j$ ,  $a$  at  $\mathbf{v}_j$ , 1 at  $\mathbf{v}_{j-1}$  and 0 elsewhere. Grouping the basis into the consecutive pairs  $(\mathbf{u}_j, \mathbf{v}_j)$ , the  $(j, j)$  block is  $\begin{pmatrix} a & -b \\ b & a \end{pmatrix} = \mathbf{\Lambda}(\lambda)$ , the  $(j-1, j)$  block is  $\begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix} = \mathbf{I}_2$ , and all other blocks are zero. That is  $\mathbf{C}_k(\lambda)$  ([Definition 9.4.1](#)).

■

Applying the lemma to the block itself tells us what a real Jordan block looks like over  $\mathbb{C}$ .

### Corollary 9.4.2 — A Real Jordan Block Splits into Two Complex Ones

Let  $\lambda \notin \mathbb{R}$  and  $k \geq 1$ . Then, as matrices over  $\mathbb{C}$ ,

$$\mathbf{C}_k(\lambda) \sim \mathbf{J}_k(\lambda) \oplus \mathbf{J}_k(\overline{\lambda}).$$

In particular  $p_{\mathbf{C}_k(\lambda)} = ((x - \lambda)(x - \overline{\lambda}))^k$ , the polynomial  $((x - a)^2 + b^2)^k \in \mathbb{R}[x]$ , and  $m_{\mathbf{C}_k(\lambda)}$  is the same polynomial.

*Proof.* Put  $\mathbf{A} = \mathbf{C}_k(\lambda) \in M_{2k}(\mathbb{R})$  and let  $(\mathbf{u}_1, \mathbf{v}_1, \dots, \mathbf{u}_k, \mathbf{v}_k)$  be the standard basis of  $\mathbb{R}^{2k}$  in this order. Reading the columns of  $\mathbf{C}_k(\lambda)$  gives exactly the identities of [Lemma 9.4.1](#) (a). Put  $\mathbf{w}_j = \mathbf{u}_j - i\mathbf{v}_j$ , so that  $\mathbf{w}_1 = \mathbf{e}_1 - i\mathbf{e}_2 \neq \mathbf{0}$ . Combining the two real identities,

$$\begin{aligned} \mathbf{A}\mathbf{w}_j &= \mathbf{A}\mathbf{u}_j - i\mathbf{A}\mathbf{v}_j = (a\mathbf{u}_j + b\mathbf{v}_j + \mathbf{u}_{j-1}) - i(-b\mathbf{u}_j + a\mathbf{v}_j + \mathbf{v}_{j-1}) \\ &= (a + bi)\mathbf{u}_j + (b - ai)\mathbf{v}_j + \mathbf{w}_{j-1} \\ &= \lambda\mathbf{w}_j + \mathbf{w}_{j-1}, \end{aligned}$$

since  $\lambda(-i\mathbf{v}_j) = (b - ai)\mathbf{v}_j$ . So  $(\mathbf{w}_1, \dots, \mathbf{w}_k)$  is a Jordan chain of  $\mathbf{A}$  for  $\lambda$ , and by (b) so is  $(\overline{\mathbf{w}_1}, \dots, \overline{\mathbf{w}_k})$  for  $\overline{\lambda}$ . By (c), the  $2k$  vectors  $\mathbf{w}_1, \dots, \mathbf{w}_k, \overline{\mathbf{w}_1}, \dots, \overline{\mathbf{w}_k}$  are independent over  $\mathbb{C}$ , because the standard basis is; being  $2k$  independent vectors in  $\mathbb{C}^{2k}$ , they form a basis ([Theorem 1.6.4](#) (a)). On the span of one chain,  $\mathbf{A} - \lambda\mathbf{I}$  has matrix  $\mathbf{J}_k(0)$  by [Lemma 9.2.2](#) (b), so  $\mathbf{A}$  has matrix  $\lambda\mathbf{I}_k + \mathbf{J}_k(0) = \mathbf{J}_k(\lambda)$  there, the matrix of a sum of operators being the sum of their matrices ([Theorem 3.6.4](#), [Definition 9.3.1](#)); likewise  $\mathbf{J}_k(\overline{\lambda})$  on the span of the conjugate chain. By [Theorem 8.1.5](#) the matrix of  $T_{\mathbf{A}}$  in the combined basis is



$\mathbf{J}_k(\lambda) \oplus \mathbf{J}_k(\bar{\lambda})$ . Hence  $\mathbf{C}_k(\lambda) \sim \mathbf{J}_k(\lambda) \oplus \mathbf{J}_k(\bar{\lambda})$  over  $\mathbb{C}$  (Theorem 3.7.4).

The characteristic polynomial of  $\mathbf{J}_k(\mu)$  is  $(x - \mu)^k$  (Theorem 6.4.6), and similar matrices have the same characteristic polynomial (Theorem 6.10.3), so  $p_{\mathbf{C}_k(\lambda)} = (x - \lambda)^k(x - \bar{\lambda})^k$ . For the minimal polynomial,  $m_{\mathbf{J}_k(\mu)} = (x - \mu)^k$  by Corollary 9.3.2 (c), a single Jordan block being a Jordan form of itself with one block, of size  $k$ ; so  $m_{\mathbf{C}_k(\lambda)} = \text{lcm}((x - \lambda)^k, (x - \bar{\lambda})^k) = (x - \lambda)^k(x - \bar{\lambda})^k$  by Proposition 8.5.6 and Theorem 8.5.5, the factors being powers of distinct irreducibles. ■

*Quick check.* What are the complex eigenvalues of  $\mathbf{C}_2(1 + i)$ , with their algebraic and geometric multiplicities?

*Solution.* By Corollary 9.4.2,  $\mathbf{C}_2(1 + i) \sim \mathbf{J}_2(1 + i) \oplus \mathbf{J}_2(1 - i)$  over  $\mathbb{C}$ . So the eigenvalues are  $1 \pm i$ , each with  $a(\lambda) = 2$  (the total size of its blocks) and  $g(\lambda) = 1$  (the number of its blocks), by Corollary 9.3.2 (a), (b).

## 9.4.2 The real Jordan form

Two facts about real matrices are needed before the theorem. The first is that a real matrix has the same kernel over  $\mathbb{R}$  as over  $\mathbb{C}$ , in the sense that a real basis of the one is a complex basis of the other. Once one knows that  $\tilde{U}$  is the complexification of  $U$  inside  $\mathbb{C}^n = (\mathbb{R}^n)_{\mathbb{C}}$  (Definition 1.9.2, Example (The Complexification of  $\mathbb{R}^n$  Is  $\mathbb{C}^n$ )), this is Theorem 1.9.3. It is worth proving directly, because the identification is itself part of what has to be checked, and because the statement is used both to find the real chains and to compare ranks.

### Lemma 9.4.3 — Real and Complex Null Spaces of a Real Matrix

Let  $\mathbf{M} \in M_n(\mathbb{R})$ , let  $U = \{\mathbf{x} \in \mathbb{R}^n : \mathbf{M}\mathbf{x} = \mathbf{0}\}$  be its null space over  $\mathbb{R}$ , and let  $\tilde{U} = \{\mathbf{z} \in \mathbb{C}^n : \mathbf{M}\mathbf{z} = \mathbf{0}\}$  be its null space over  $\mathbb{C}$ . Then:

- (a) for  $\mathbf{z} = \mathbf{x} + i\mathbf{y}$  with  $\mathbf{x}, \mathbf{y} \in \mathbb{R}^n$ :  $\mathbf{z} \in \tilde{U}$  if and only if  $\mathbf{x} \in U$  and  $\mathbf{y} \in U$ ;
- (b) every basis of  $U$  over  $\mathbb{R}$  is a basis of  $\tilde{U}$  over  $\mathbb{C}$ ; in particular  $\dim_{\mathbb{R}} U = \dim_{\mathbb{C}} \tilde{U}$  and  $\text{rank } \mathbf{M}$  is the same computed over  $\mathbb{R}$  and over  $\mathbb{C}$ .

*Proof.*

- (a) Since  $\mathbf{M}$  is real,  $\mathbf{M}\mathbf{z} = \mathbf{M}\mathbf{x} + i\mathbf{M}\mathbf{y}$  with  $\mathbf{M}\mathbf{x}, \mathbf{M}\mathbf{y} \in \mathbb{R}^n$ . By the uniqueness of real and imaginary parts (Definition 0.6.1),  $\mathbf{M}\mathbf{z} = \mathbf{0}$  if and only if  $\mathbf{M}\mathbf{x} = \mathbf{0}$  and  $\mathbf{M}\mathbf{y} = \mathbf{0}$ .
- (b) Let  $(\mathbf{x}_1, \dots, \mathbf{x}_r)$  be a basis of  $U$  over  $\mathbb{R}$ . *Spanning.* Let  $\mathbf{z} = \mathbf{x} + i\mathbf{y} \in \tilde{U}$ . By (a),  $\mathbf{x}, \mathbf{y} \in U$ , so  $\mathbf{x} = \sum_j c_j \mathbf{x}_j$  and  $\mathbf{y} = \sum_j d_j \mathbf{x}_j$  with  $c_j, d_j \in \mathbb{R}$ ; hence  $\mathbf{z} = \sum_j (c_j + id_j) \mathbf{x}_j$ . *Independence.* Let  $\sum_j \alpha_j \mathbf{x}_j = \mathbf{0}$  with  $\alpha_j = c_j + id_j \in \mathbb{C}$ . Separating real and imaginary parts, which is legitimate because the  $\mathbf{x}_j$  are real, gives  $\sum_j c_j \mathbf{x}_j = \mathbf{0}$  and  $\sum_j d_j \mathbf{x}_j = \mathbf{0}$ ; independence over  $\mathbb{R}$  forces every  $c_j = d_j = 0$ . So  $(\mathbf{x}_1, \dots, \mathbf{x}_r)$  is a basis of  $\tilde{U}$  and  $\dim_{\mathbb{C}} \tilde{U} = r = \dim_{\mathbb{R}} U$ . The rank statement follows from Theorem 2.5.7 over each field.

Now the theorem. Read it as the Jordan form with one amendment: a conjugate pair of complex blocks of size  $k$  is replaced by a single real block of size  $2k$ .

*Every real matrix is similar, over the reals, to a direct sum of Jordan blocks for its real eigenvalues and real Jordan blocks for its conjugate pairs, and that matrix is unique up to the order of its blocks.*

### Theorem 9.4.4 — Real Jordan Form

Let  $\mathbf{A} \in M_n(\mathbb{R})$  with  $n \geq 1$ . List the eigenvalues of  $\mathbf{A}$  in  $\mathbb{C}$  as

$$\begin{aligned} \mu_1, \dots, \mu_p &\in \mathbb{R} \quad (\text{the real ones, distinct}), \\ \lambda_1, \overline{\lambda_1}, \dots, \lambda_q, \overline{\lambda_q} &\quad (\text{the conjugate pairs, with } \operatorname{Im} \lambda_t > 0). \end{aligned}$$

- (a) **(Existence)** There is an invertible  $\mathbf{P} \in M_n(\mathbb{R})$  such that  $\mathbf{P}^{-1}\mathbf{A}\mathbf{P}$  is a real Jordan matrix (Definition 9.4.1), in which the blocks for a real eigenvalue  $\mu_l$  are ordinary Jordan blocks  $\mathbf{J}_k(\mu_l)$  and the blocks for a pair  $\{\lambda_t, \overline{\lambda_t}\}$  are real Jordan blocks  $\mathbf{C}_k(\lambda_t)$ .
- (b) **(Uniqueness)** In any such matrix, the number of blocks  $\mathbf{J}_k(\mu)$  with  $k \geq j$ , for  $\mu \in \mathbb{R}$ , and the number of blocks  $\mathbf{C}_k(\lambda)$  with  $k \geq j$ , for  $\lambda \notin \mathbb{R}$ , both equal

$$\operatorname{rank}(\mathbf{A} - \nu \mathbf{I})^{j-1} - \operatorname{rank}(\mathbf{A} - \nu \mathbf{I})^j \quad (\nu = \mu \text{ resp. } \nu = \lambda),$$

the ranks being computed over  $\mathbb{C}$  in the second case. Consequently the real Jordan matrix is determined by  $\mathbf{A}$  up to the order of its blocks.

*Idea.* Work over  $\mathbb{C}$ , then come home. ① Split  $\mathbb{C}^n = \bigoplus G_\nu(\mathbf{A})$  over all complex eigenvalues  $\nu$  (Theorem 9.1.3). ② For a **real** eigenvalue, the generalized eigenspace is the complexification of a real subspace, by Lemma 9.4.3, and Section 2's structure theorem runs over  $\mathbb{R}$  on that real subspace: nothing complex is needed. ③ For a **pair**, choose the chains in  $G_{\lambda_t}$  freely, and take their conjugates in  $G_{\overline{\lambda_t}}$  — no choice there. ④ Replace each conjugate pair of chains by its real and imaginary parts; Lemma 9.4.1 says the new matrix is  $\mathbf{C}_k(\lambda_t)$ , and a count shows the real vectors collected are exactly  $n$  in number and independent. Uniqueness is the uniqueness of the complex Jordan form, since by Corollary 9.4.2 a real Jordan matrix has a known complex Jordan form.

*Proof.* Since  $p_{\mathbf{A}} \in \mathbb{R}[x] \subseteq \mathbb{C}[x]$  has degree  $n \geq 1$ , it splits over  $\mathbb{C}$  (Corollary 5.5.5), and its non-real roots come in conjugate pairs with equal multiplicities (Theorem 8.2.5 (a)); so the list of eigenvalues has the displayed shape. Write  $a(\nu)$  for the algebraic multiplicity of  $\nu$  in  $p_{\mathbf{A}}$ , so  $a(\lambda_t) = a(\overline{\lambda_t})$  and

$$\sum_{l=1}^p a(\mu_l) + \sum_{t=1}^q 2a(\lambda_t) = n. \quad (*)$$

By Theorem 9.1.3 applied to  $T_{\mathbf{A}}$  over  $\mathbb{C}$ ,

$$\mathbb{C}^n = \bigoplus_{l=1}^p G_{\mu_l}(\mathbf{A}) \oplus \bigoplus_{t=1}^q (G_{\lambda_t}(\mathbf{A}) \oplus G_{\overline{\lambda_t}}(\mathbf{A})),$$

with  $\dim_{\mathbb{C}} G_{\nu}(\mathbf{A}) = a(\nu)$  and  $G_{\nu}(\mathbf{A}) = \ker(\mathbf{A} - \nu \mathbf{I})^n$  (Corollary 9.1.2 (a)).

**Step 1. Real eigenvalues, handled entirely over  $\mathbb{R}$ .** Fix a real eigenvalue  $\mu = \mu_l$ . The matrix  $(\mathbf{A} - \mu \mathbf{I})^n$  is real, so by Lemma 9.4.3 its real null space

$$U_{\mu} := \{ \mathbf{x} \in \mathbb{R}^n : (\mathbf{A} - \mu \mathbf{I})^n \mathbf{x} = \mathbf{0} \}$$

has a real basis which is a  $\mathbb{C}$ -basis of  $G_{\mu}(\mathbf{A})$ ; in particular  $\dim_{\mathbb{R}} U_{\mu} = a(\mu)$ . The subspace  $U_{\mu}$  is  $\mathbf{A}$ -invariant, since  $(\mathbf{A} - \mu \mathbf{I})^n (\mathbf{A} \mathbf{x}) = \mathbf{A} (\mathbf{A} - \mu \mathbf{I})^n \mathbf{x} = \mathbf{0}$  for  $\mathbf{x} \in U_{\mu}$  (Theorem 5.7.4 (b)), and  $N_{\mu} := (\mathbf{A} - \mu \mathbf{I})|_{U_{\mu}}$  satisfies  $N_{\mu}^n = 0$ , so  $N_{\mu}$  is nilpotent (Definition 9.2.1). By Theorem 9.2.3 there is a basis  $\mathcal{B}_{\mu}$  of the real space  $U_{\mu}$ , a concatenation of Jordan chains of  $N_{\mu}$ , with  $[N_{\mu}]_{\mathcal{B}_{\mu}} = \mathbf{J}_{k_1}(0) \oplus \cdots \oplus \mathbf{J}_{k_s}(0)$ . Since  $T_{\mathbf{A}}|_{U_{\mu}} = \mu \text{id}_{U_{\mu}} + N_{\mu}$  and  $[\text{id}_{U_{\mu}}]_{\mathcal{B}_{\mu}} = \mathbf{I}_{k_1} \oplus \cdots \oplus \mathbf{I}_{k_s}$ , adding the matrices (Theorem 3.6.4) blockwise (Theorem 7.1.4 (a), Definition 9.3.1) gives

$$[T_{\mathbf{A}}|_{U_{\mu}}]_{\mathcal{B}_{\mu}} = \mathbf{J}_{k_1}(\mu) \oplus \cdots \oplus \mathbf{J}_{k_s}(\mu),$$

a direct sum of ordinary Jordan blocks. Moreover  $\mathcal{B}_{\mu}$ , viewed in  $\mathbb{C}^n$ , is a Jordan basis of  $G_{\mu}(\mathbf{A})$  for  $T_{\mathbf{A}}$  over  $\mathbb{C}$ : the chain relations are the same equations, and  $\mathcal{B}_{\mu}$  is a  $\mathbb{C}$ -basis of  $G_{\mu}(\mathbf{A})$  by Lemma 9.4.3 (b).

**Step 2. Conjugate pairs, chosen on one side.** Fix  $t$  and write  $\lambda = \lambda_t$ ,  $G = G_{\lambda}(\mathbf{A})$ ,  $\overline{G} = G_{\overline{\lambda}}(\mathbf{A})$ . By Theorem 9.1.3 (d),  $(\mathbf{A} - \lambda \mathbf{I})|_G$  is nilpotent, so Theorem 9.2.3 provides chains whose concatenation is a  $\mathbb{C}$ -basis of  $G$ ; as in Step 1 these are Jordan chains of  $\mathbf{A}$  for  $\lambda$ , of lengths  $k_1, \dots, k_m$  summing to  $a(\lambda)$ .

*Claim: conjugating gives a Jordan basis of  $\overline{G}$ .* Indeed, conjugation maps  $G$  onto  $\overline{G}$ : if  $(\mathbf{A} - \lambda \mathbf{I})^n \mathbf{z} = \mathbf{0}$ , then  $(\mathbf{A} - \overline{\lambda} \mathbf{I})^n \overline{\mathbf{z}} = \mathbf{0}$  by the computation in Lemma 9.4.1 (b), and conjugating twice returns  $\mathbf{z}$ . If  $(\mathbf{z}_1, \dots, \mathbf{z}_d)$  is a  $\mathbb{C}$ -basis of  $G$ , then  $(\overline{\mathbf{z}}_1, \dots, \overline{\mathbf{z}}_d)$  spans  $\overline{G}$  — any  $\mathbf{z} \in \overline{G}$  has  $\overline{\mathbf{z}} \in G$ , say  $\overline{\mathbf{z}} = \sum_j c_j \mathbf{z}_j$ , whence  $\mathbf{z} = \sum_j \overline{c_j} \overline{\mathbf{z}}_j$  — and it has  $d = \dim_{\mathbb{C}} \overline{G}$  entries, so it is a basis (Theorem 1.6.4 (b)). Applying this to the concatenated chains, and using Lemma 9.4.1 (b) chain by chain, the conjugated chains form a Jordan basis of  $\overline{G}$  consisting of chains for  $\overline{\lambda}$  of the same lengths  $k_1, \dots, k_m$ .

**Step 3. Realify each pair of conjugate chains.** Keep  $t$  fixed and let  $(\mathbf{w}_1, \dots, \mathbf{w}_k)$  be one of the chains of Step 2, with  $\mathbf{w}_j = \mathbf{u}_j - i\mathbf{v}_j$ ,  $\mathbf{u}_j, \mathbf{v}_j \in \mathbb{R}^n$ . By Lemma 9.4.1 (c), the  $2k$  real vectors  $\mathbf{u}_1, \mathbf{v}_1, \dots, \mathbf{u}_k, \mathbf{v}_k$  have the same  $\mathbb{C}$ -span as the  $2k$  chain vectors  $\mathbf{w}_1, \dots, \mathbf{w}_k, \overline{\mathbf{w}}_1, \dots, \overline{\mathbf{w}}_k$ . Do this for each of the  $m$  chains attached to  $\lambda_t$ , and for each  $t$ .

**Step 4. The collected real vectors form a basis of  $\mathbb{R}^n$ .** Let  $\mathcal{B}$  be the list obtained by writing, one after another: the real Jordan bases  $\mathcal{B}_{\mu_1}, \dots, \mathcal{B}_{\mu_p}$  of Step 1, and then, for each  $t$  and each chain of Step 2, the block of  $2k$  real vectors of Step 3. Its length is

$$\sum_{l=1}^p a(\mu_l) + \sum_{t=1}^q \sum_{\text{chains}} 2k = \sum_{l=1}^p a(\mu_l) + \sum_{t=1}^q 2a(\lambda_t) = n$$

by (\*), since the chain lengths for  $\lambda_t$  sum to  $a(\lambda_t)$ . Its  $\mathbb{C}$ -span contains a  $\mathbb{C}$ -basis of each  $G_{\mu_l}(\mathbf{A})$  (Step 1) and, for each  $t$ , all the vectors  $\mathbf{w}_j$  and  $\overline{\mathbf{w}}_j$  (Step 3), which together form a  $\mathbb{C}$ -basis of  $G_{\lambda_t}(\mathbf{A}) \oplus G_{\overline{\lambda_t}}(\mathbf{A})$  by Step 2. So the  $\mathbb{C}$ -span of  $\mathcal{B}$  is all of  $\mathbb{C}^n$ , and  $\mathcal{B}$  has exactly  $n$  entries, so  $\mathcal{B}$  is a  $\mathbb{C}$ -basis of  $\mathbb{C}^n$  (Theorem 1.6.4 (b)). In particular  $\mathcal{B}$  is linearly independent over  $\mathbb{C}$ , hence over  $\mathbb{R}$ , and being a list of  $n$  independent vectors in  $\mathbb{R}^n$  it is a basis of  $\mathbb{R}^n$  (Theorem 1.6.4 (a)).

**Step 5. The matrix.** Each segment of  $\mathcal{B}$  spans an  $\mathbf{A}$ -invariant real subspace: the spaces  $U_{\mu_l}$  by Step 1, and the spaces  $U$  of Lemma 9.4.1 (d) by Step 3 and the independence just proved. Their direct sum is  $\mathbb{R}^n$ , since  $\mathcal{B}$  is a basis. By Theorem 8.1.5,  $[T_{\mathbf{A}}]_{\mathcal{B}}$  is block diagonal with the matrices of the restrictions as diagonal blocks, which are  $\mathbf{J}_k(\mu_l)$ 's by Step 1 and  $\mathbf{C}_k(\lambda_t)$ 's by Lemma 9.4.1 (d). Taking  $\mathbf{P}$  to be the real matrix whose columns are the vectors of  $\mathcal{B}$  gives  $\mathbf{P}^{-1}\mathbf{A}\mathbf{P} = [T_{\mathbf{A}}]_{\mathcal{B}}$  (Theorem 3.7.4), which proves (a).

- (b) Let  $\mathbf{B} = \mathbf{P}^{-1}\mathbf{A}\mathbf{P}$  be any real Jordan matrix similar to  $\mathbf{A}$  over  $\mathbb{R}$ , and regard the similarity over  $\mathbb{C}$ . By Corollary 9.4.2 and Theorem 7.1.4, replacing each block  $\mathbf{C}_k(\lambda)$  of  $\mathbf{B}$  by  $\mathbf{J}_k(\lambda) \oplus \mathbf{J}_k(\bar{\lambda})$  produces a Jordan matrix similar to  $\mathbf{B}$ , hence to  $\mathbf{A}$ , over  $\mathbb{C}$ . By Theorem 9.3.1 (b), for every  $\nu \in \mathbb{C}$  and  $j \geq 1$  the number of blocks  $\mathbf{J}_k(\nu)$  with  $k \geq j$  in that Jordan matrix equals  $\text{rank}(\mathbf{A} - \nu\mathbf{I})^{j-1} - \text{rank}(\mathbf{A} - \nu\mathbf{I})^j$ , computed over  $\mathbb{C}$ . For a real  $\mu$ , the blocks  $\mathbf{J}_k(\mu)$  of the Jordan matrix are exactly the blocks  $\mathbf{J}_k(\mu)$  of  $\mathbf{B}$ , since the replaced blocks contribute only non-real eigenvalues; and the ranks over  $\mathbb{C}$  agree with the ranks over  $\mathbb{R}$  by Lemma 9.4.3 (b), applied to the real matrix  $(\mathbf{A} - \mu\mathbf{I})^j$ . For  $\lambda \notin \mathbb{R}$  with  $\text{Im } \lambda > 0$ , the blocks  $\mathbf{J}_k(\lambda)$  of the Jordan matrix come one each from the blocks  $\mathbf{C}_k(\lambda)$  of  $\mathbf{B}$ : a block  $\mathbf{C}_k(\lambda')$  of  $\mathbf{B}$  with  $\lambda' \neq \lambda$  contributes  $\mathbf{J}_k(\lambda')$  and  $\mathbf{J}_k(\bar{\lambda}')$ , and  $\bar{\lambda}' \neq \lambda$  because  $\lambda$  and  $\lambda'$  both have **positive** imaginary part (Definition 9.4.1). In both cases the count is determined by  $\mathbf{A}$ , so any two real Jordan matrices similar to  $\mathbf{A}$  have the same blocks up to order. ■

So over  $\mathbb{R}$  the dream of Chapter 3 comes true in the amended form: **every real square matrix is similar over  $\mathbb{R}$  to a matrix that is block diagonal with blocks  $\mathbf{J}_k(\mu)$  and  $\mathbf{C}_k(\lambda)$** , and the only new feature is the  $2 \times 2$  rotation-scaling block. Counting: a real eigenvalue  $\mu$  contributes  $a(\mu)$  to the size and  $g(\mu)$  blocks; a pair  $\{\lambda, \bar{\lambda}\}$  contributes  $2a(\lambda)$  to the size and  $g(\lambda)$  real blocks. For an operator  $T$  on a real vector space  $V$  the same statement holds, applied to  $[T]_{\mathcal{C}}$  for any basis  $\mathcal{C}$  of  $V$ : the theorem produces a real Jordan matrix similar to it, hence equal to  $[T]_{\mathcal{B}}$  for another basis  $\mathcal{B}$  of  $V$  (Theorem 3.7.4). The complexification  $V_{\mathbb{C}}$  of Definition 1.9.2 is where the complex chains live; for  $V = \mathbb{R}^n$  it is  $\mathbb{C}^n$  (Example (The Complexification of  $\mathbb{R}^n$  Is  $\mathbb{C}^n$ )), which is the case written out above.

*Warning.* **A real Jordan matrix is not triangular, and the real form is not obtained by “ignoring” the imaginary parts.** The real Jordan form of a matrix with a non-real eigenvalue **cannot** be triangular, since a triangular real matrix has its diagonal entries as eigenvalues, all real (Theorem 8.7.4). Nor is  $\mathbf{A}(\lambda)$  a diagonal matrix in disguise: it has no real eigenvector. And the real basis is not the real part of the complex one: each complex chain vector  $\mathbf{w}_j$  contributes **two** real vectors,  $\text{Re } \mathbf{w}_j$  and  $-\text{Im } \mathbf{w}_j$ , and dropping either of them destroys the invariance.

#### Example. (The Real Jordan Form of a $4 \times 4$ Matrix)

Let

$$\mathbf{A} = \begin{pmatrix} 0 & -1 & 1 & 0 \\ 1 & 0 & 0 & 1 \\ -1 & 0 & 2 & -1 \\ 0 & -1 & 1 & 2 \end{pmatrix} \in M_4(\mathbb{R}), \quad p_{\mathbf{A}} = (x^2 - 2x + 2)^2.$$

Find the real Jordan form of  $\mathbf{A}$  (Theorem 9.4.4) and an invertible  $\mathbf{P} \in M_4(\mathbb{R})$  with  $\mathbf{P}^{-1}\mathbf{A}\mathbf{P}$  in that form.

*Solution. The eigenvalues.*  $x^2 - 2x + 2 = (x - 1)^2 + 1$  has roots  $1 \pm i$ , so over  $\mathbb{C}$  the eigenvalues are  $\lambda = 1 + i$  and  $\bar{\lambda} = 1 - i$ , each with  $a(\lambda) = 2$ . There is no real eigenvalue, so the real Jordan form consists of real blocks  $\mathbf{C}_k(1 + i)$  only, with sizes  $k$  summing to  $a(\lambda) = 2$ .

A chain for  $\lambda = 1 + i$ . Put  $\mathbf{M} = \mathbf{A} - (1 + i)\mathbf{I}$ , so that

$$\mathbf{M} = \begin{pmatrix} -1-i & -1 & 1 & 0 \\ 1 & -1-i & 0 & 1 \\ -1 & 0 & 1-i & -1 \\ 0 & -1 & 1 & 1-i \end{pmatrix}.$$

Solve  $\mathbf{M}\mathbf{z} = \mathbf{0}$  using rows 2, 1 and 4 in that order. Row 2 gives  $z_1 = (1 + i)z_2 - z_4$ ; row 1 gives  $z_3 = (1 + i)z_1 + z_2$ . Substituting the first into the second, and writing  $s = z_2$ ,  $u = z_4$ ,

$$z_1 = (1 + i)s - u, \quad z_3 = (1 + i)((1 + i)s - u) + s = (1 + 2i)s - (1 + i)u,$$

using  $(1 + i)^2 = 2i$ . Row 4 reads  $-z_2 + z_3 + (1 - i)z_4 = 0$ , which becomes

$$-s + (1 + 2i)s - (1 + i)u + (1 - i)u = 2is - 2iu = 0,$$

so  $s = u$ . Row 3 is then automatic: with  $s = u = 1$  we get  $\mathbf{z} = (i, 1, i, 1)$ , and row 3 gives  $-i + (1 - i)i - 1 = -i + i + 1 - 1 = 0$ . So

$$\ker \mathbf{M} = \text{span}((i, 1, i, 1)), \quad g(\lambda) = 1.$$

Hence there is exactly one  $\lambda$ -block over  $\mathbb{C}$  (Corollary 9.3.2 (b)), of size  $a(\lambda) = 2$ , and the real Jordan form is  $\mathbf{C}_2(1 + i)$ . For the basis we need the chain: take  $\mathbf{w}_1 = (i, 1, i, 1)$  and solve  $\mathbf{M}\mathbf{w}_2 = \mathbf{w}_1$ . Try  $\mathbf{w}_2 = (0, 0, z_3, z_4)$ , so that only columns 3 and 4 of  $\mathbf{M}$  are used. Row 1 then reads  $z_3 = i$  and row 2 reads  $z_4 = 1$ , so  $\mathbf{w}_2 = (0, 0, i, 1)$  is the only candidate of this shape. Rows 3 and 4 confirm it:

$$(1 - i)i - 1 = (i + 1) - 1 = i \quad \text{and} \quad i + (1 - i) = 1,$$

which are the third and fourth entries of  $\mathbf{w}_1$ . So  $(\mathbf{w}_1, \mathbf{w}_2)$  is a Jordan chain of  $\mathbf{A}$  for  $1 + i$ .

*Normalizing the chain.* If  $(\mathbf{w}_1, \mathbf{w}_2)$  is a chain, so is  $(c\mathbf{w}_1, c\mathbf{w}_2)$  for any  $c \neq 0$ , since the chain relations are linear. Taking  $c = -i$  clears the denominators of the realification:

$$\mathbf{w}'_1 = -i(i, 1, i, 1) = (1, -i, 1, -i), \quad \mathbf{w}'_2 = -i(0, 0, i, 1) = (0, 0, 1, -i).$$

Now read off  $\mathbf{w}'_j = \mathbf{u}_j - i\mathbf{v}_j$  with  $\mathbf{u}_j, \mathbf{v}_j$  real, as Lemma 9.4.1 requires:

$$\mathbf{u}_1 = (1, 0, 1, 0), \quad \mathbf{v}_1 = (0, 1, 0, 1), \quad \mathbf{u}_2 = (0, 0, 1, 0), \quad \mathbf{v}_2 = (0, 0, 0, 1).$$

*The answer.* With  $\mathcal{B} = (\mathbf{u}_1, \mathbf{v}_1, \mathbf{u}_2, \mathbf{v}_2)$ ,

$$\mathbf{P} = \begin{pmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 1 & 0 & 1 & 0 \\ 0 & 1 & 0 & 1 \end{pmatrix}, \quad \mathbf{P}^{-1}\mathbf{A}\mathbf{P} = \mathbf{C}_2(1 + i) = \begin{pmatrix} 1 & -1 & 1 & 0 \\ 1 & 1 & 0 & 1 \\ 0 & 0 & 1 & -1 \\ 0 & 0 & 1 & 1 \end{pmatrix}.$$

$\mathbf{P}$  is lower triangular with ones on the diagonal, so  $\det \mathbf{P} = 1$  ([Theorem 6.4.6](#)) and  $\mathbf{P}$  is invertible.

*The check, column by column.* Writing  $\mathbf{c}_1, \dots, \mathbf{c}_4$  for the columns of  $\mathbf{P}$ , the identity  $\mathbf{A}\mathbf{P} = \mathbf{P}\mathbf{C}_2(1+i)$  says

$$\begin{aligned}\mathbf{A}\mathbf{c}_1 &= \mathbf{c}_1 + \mathbf{c}_2, \\ \mathbf{A}\mathbf{c}_2 &= -\mathbf{c}_1 + \mathbf{c}_2, \\ \mathbf{A}\mathbf{c}_3 &= \mathbf{c}_1 + \mathbf{c}_3 + \mathbf{c}_4, \\ \mathbf{A}\mathbf{c}_4 &= \mathbf{c}_2 - \mathbf{c}_3 + \mathbf{c}_4,\end{aligned}$$

read off the four columns of  $\mathbf{C}_2(1+i)$ ; these are exactly the identities of [Lemma 9.4.1](#) (a) with  $a = b = 1$ . Two of them in full:

$$\begin{aligned}\mathbf{A}\mathbf{c}_1 &= \mathbf{A}(1, 0, 1, 0) = (0 + 1, 1 + 0, -1 + 2, 0 + 1) = (1, 1, 1, 1) \\ &= (1, 0, 1, 0) + (0, 1, 0, 1) = \mathbf{c}_1 + \mathbf{c}_2,\end{aligned}$$

adding columns 1 and 3 of  $\mathbf{A}$ ; and

$$\begin{aligned}\mathbf{A}\mathbf{c}_3 &= \mathbf{A}(0, 0, 1, 0) = (1, 0, 2, 1) = (1, 0, 1, 0) + (0, 0, 1, 0) + (0, 0, 0, 1) \\ &= \mathbf{c}_1 + \mathbf{c}_3 + \mathbf{c}_4,\end{aligned}$$

the third column of  $\mathbf{A}$ . The remaining two are the same kind of one-line computation.

*Cross-checks.*  $\text{tr } \mathbf{A} = 0 + 0 + 2 + 2 = 4$  and  $\text{tr } \mathbf{C}_2(1+i) = 4 \cdot 1 = 4$ . Over  $\mathbb{C}$ , [Corollary 9.4.2](#) gives the complex Jordan form  $\mathbf{J}_2(1+i) \oplus \mathbf{J}_2(1-i)$ , whose characteristic polynomial is  $((x-1)^2 + 1)^2 = p_{\mathbf{A}}$ .

### 9.4.3 The Weyr form

The Jordan form of a nilpotent operator is the Young diagram of Section 2 read **by rows**: one row per chain, one block per row. The same diagram can be read **by columns**, and the columns are the jumps of the kernel chain ([Definition 9.2.3](#) and the discussion after [Theorem 9.2.4](#)). Reordering a Jordan basis accordingly costs nothing and gives a second normal form, the Weyr form. It carries exactly the same information — a partition for each eigenvalue — but it wears it differently, and for one purpose it is strictly better: the matrices that commute with a Weyr matrix are block upper triangular, which is false for the Jordan form.

Here is the diagram for the partition  $(2, 2, 1)$  of 5, whose dual partition is  $(3, 2)$ :

$$\begin{array}{l} w_1 = w_2 = 2 \\ k_1 = 2 \\ k_2 = 2 \\ k_3 = 1 \end{array} \begin{array}{|c|c|} \hline \square & \square \\ \hline \square & \square \\ \hline \square & \\ \hline \end{array} \begin{array}{l} \text{rows: the Jordan block sizes } (2, 2, 1) \\ \text{columns: the Weyr structure } (3, 2) \\ \text{the operator moves each box one step left} \end{array}$$

The basis is the same; only its order changes. Instead of listing chain by chain, list the first column (a basis of the kernel), then the second column, and so on. Since the operator moves each box one step to the left, it maps the  $j$ -th group of basis vectors into the  $(j-1)$ -th, injectively, and the matrix becomes block upper triangular with a very rigid pattern of ones.

### Definition 9.4.2 — Basic Weyr Matrix, Weyr Structure, Weyr Form

Let  $\lambda \in F$  and let  $w_1 \geq w_2 \geq \cdots \geq w_s \geq 1$  be a partition of an integer  $m \geq 1$  (Definition 9.2.3). The **basic Weyr matrix** with eigenvalue  $\lambda$  and **Weyr structure**  $(w_1, \dots, w_s)$  is the matrix  $\mathbf{W} \in M_m(F)$ , partitioned into blocks according to  $m = w_1 + \cdots + w_s$ , whose blocks are

$$\mathbf{W}_{jj} = \lambda \mathbf{I}_{w_j}, \quad \mathbf{W}_{j,j+1} = \begin{pmatrix} \mathbf{I}_{w_{j+1}} \\ 0 \end{pmatrix} \in M_{w_j \times w_{j+1}}(F), \quad \mathbf{W}_{jl} = 0 \text{ otherwise.}$$

A **Weyr matrix** is a direct sum of basic Weyr matrices. A **Weyr form** of  $T \in \mathcal{L}(V)$  is a Weyr matrix  $\mathbf{W}$  with  $[T]_{\mathcal{B}} = \mathbf{W}$  for some basis  $\mathcal{B}$  of  $V$ ; for  $\mathbf{A} \in M_n(F)$ , a Weyr form of  $\mathbf{A}$  is a Weyr matrix similar to  $\mathbf{A}$ .

In words: the diagonal blocks are scalar, of the sizes given by the partition, and the only other non-zero blocks sit **immediately to the right** of the diagonal ones, each being an identity matrix stacked on top of a zero block — so  $\mathbf{W}_{j,j+1}$  has full column rank  $w_{j+1}$ , which is possible exactly because  $w_j \geq w_{j+1}$ . The matrix is upper triangular, with  $\lambda$  repeated  $m$  times on the diagonal.

#### Examples.

- **One column.** If  $s = 1$ , then  $\mathbf{W} = \lambda \mathbf{I}_{w_1}$ : a scalar matrix. This is the diagonalizable case.
- **All parts equal to 1.** If  $w_1 = \cdots = w_s = 1$ , then  $\mathbf{W} = \mathbf{J}_s(\lambda)$ : a single Jordan block. So Jordan blocks are the basic Weyr matrices of the thinnest possible structure.
- **The structure  $(3, 2)$  with  $\lambda = 5$ .** Here  $m = 5$  and

$$\mathbf{W} = \begin{pmatrix} 5 & 0 & 0 & 1 & 0 \\ 0 & 5 & 0 & 0 & 1 \\ 0 & 0 & 5 & 0 & 0 \\ 0 & 0 & 0 & 5 & 0 \\ 0 & 0 & 0 & 0 & 5 \end{pmatrix},$$

with the  $3 \times 2$  block  $\begin{pmatrix} \mathbf{I}_2 \\ 0 \end{pmatrix}$  in the upper right corner. As we will see, this is the Weyr form of  $\mathbf{J}_2(5) \oplus \mathbf{J}_2(5) \oplus \mathbf{J}_1(5)$ , the diagram drawn above.

**Non-example by minimal change.** Put the identity block in the **lower** part of  $\mathbf{W}_{12}$  instead: with structure  $(2, 1)$  and  $\lambda = 0$ , compare

$$\mathbf{W} = \begin{pmatrix} 0 & 0 & 1 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{pmatrix} \quad \text{and} \quad \mathbf{W}' = \begin{pmatrix} 0 & 0 & 0 \\ 0 & 0 & 1 \\ 0 & 0 & 0 \end{pmatrix}.$$

Both are nilpotent of rank 1, and indeed  $\mathbf{W} \sim \mathbf{W}'$  (Corollary 9.2.5): the shape is not an extra invariant, and the two matrices represent the same operator in different bases. But only  $\mathbf{W}$  is a Weyr matrix. The clause that fails for  $\mathbf{W}'$  is that the block  $\mathbf{W}_{j,j+1}$  must be an identity matrix stacked **on top of** a zero block, not below it. The ordering is a normalization, and it is the one the commutant property below is proved with: there the block  $\mathbf{W}_{j,j+1}$  is canceled on the left, which uses that its columns are the **first**  $w_{j+1}$  standard basis vectors, in order. Concretely, every matrix commuting with  $\mathbf{W}$  turns out to be upper triangular, while the commutant of  $\mathbf{W}'$  contains the matrix unit  $\mathbf{E}_{21}$ .



The existence and uniqueness of the form are now a repackaging of Section 3.

### Theorem 9.4.5 — Weyr Form

Let  $V$  be a finite-dimensional vector space over  $F$  with  $\dim V = n \geq 1$ , and let  $T \in \mathcal{L}(V)$ .

- (a) **(Existence)**  $T$  has a Weyr form if and only if  $p_T$  splits over  $F$ . One can be arranged with one basic Weyr block per eigenvalue.
- (b) **(Uniqueness)** In any Weyr form of  $T$  with one basic block per eigenvalue, the block for  $\lambda$  has Weyr structure

$$w_j = \dim \ker(T - \lambda \operatorname{id}_V)^j - \dim \ker(T - \lambda \operatorname{id}_V)^{j-1} \quad (j \geq 1),$$

the non-zero terms of this sequence; equivalently  $(w_1, w_2, \dots)$  is the **dual partition** of the partition formed by the sizes of the  $\lambda$ -blocks of the Jordan form of  $T$ . So the Weyr form is determined by  $T$  up to the order of its basic blocks.

*Idea.* Both parts come from one observation: a Weyr basis is a Jordan basis in a different order. Arrange the  $\lambda$ -chains of a Jordan basis as the rows of the Young diagram, sorted by decreasing length, and read the diagram **column by column**. The operator  $T - \lambda \operatorname{id}$  moves each box one step to the left, so it carries the  $j$ -th column into the  $(j-1)$ -th, matching the  $i$ -th box of one with the  $i$ -th box of the other — which is exactly the block  $\begin{pmatrix} \mathbf{I} \\ 0 \end{pmatrix}$ . For uniqueness, count the boxes: the first  $j$  columns of the diagram are a basis of  $\ker(T - \lambda \operatorname{id})^j$ , so the column lengths are the jumps of the kernel chain.

*Proof.*  $(\Rightarrow)$  A Weyr matrix is upper triangular, so if  $T$  has a Weyr form then  $T$  is triangularizable and  $p_T$  splits (Definition 8.7.1, Theorem 8.7.2).

$(\Leftarrow)$  Suppose  $p_T$  splits, and let  $\lambda$  be an eigenvalue. By Theorem 9.3.1 there is a Jordan basis; fix its part inside  $G_\lambda(T)$ , a concatenation of Jordan chains of  $N = T - \lambda \operatorname{id}_V$  of lengths  $k_1 \geq k_2 \geq \dots \geq k_r \geq 1$  summing to  $a(\lambda)$  (Theorem 9.2.3, after reindexing so that the lengths decrease). Write  $\mathbf{b}_{i,1}, \dots, \mathbf{b}_{i,k_i}$  for the  $i$ -th chain in the order of Definition 9.2.2, so that

$$N\mathbf{b}_{i,1} = \mathbf{0}, \quad N\mathbf{b}_{i,j} = \mathbf{b}_{i,j-1} \quad (2 \leq j \leq k_i).$$

Put  $s = k_1$  and, for  $1 \leq j \leq s$ , let  $w_j = \#\{i : k_i \geq j\}$ , the length of the  $j$ -th column of the Young diagram; then  $w_1 \geq \dots \geq w_s \geq 1$  is the dual partition of  $(k_1, \dots, k_r)$  (Definition 9.2.3), a partition of  $a(\lambda)$ . Because the  $k_i$  decrease,  $\{i : k_i \geq j\} = \{1, 2, \dots, w_j\}$ . Let

$$\mathcal{C}_j = (\mathbf{b}_{1,j}, \mathbf{b}_{2,j}, \dots, \mathbf{b}_{w_j,j})$$

be the  $j$ -th column, and let  $\mathcal{B}_\lambda$  be the list  $\mathcal{C}_1, \mathcal{C}_2, \dots, \mathcal{C}_s$  written one after another: a reordering of the chosen basis of  $G_\lambda(T)$ , hence a basis of it.

Read the columns of  $[T|_{G_\lambda}]_{\mathcal{B}_\lambda}$ . For  $j = 1$ ,  $T\mathbf{b}_{i,1} = \lambda\mathbf{b}_{i,1}$ . For  $j \geq 2$ ,  $T\mathbf{b}_{i,j} = \lambda\mathbf{b}_{i,j} + \mathbf{b}_{i,j-1}$ , and  $\mathbf{b}_{i,j-1}$  is the  $i$ -th entry of  $\mathcal{C}_{j-1}$ , which exists because  $i \leq w_j \leq w_{j-1}$ . In the block partition of  $\mathcal{B}_\lambda$  by the groups  $\mathcal{C}_1, \dots, \mathcal{C}_s$ , this says: the  $(j, j)$  block is  $\lambda\mathbf{I}_{w_j}$ ; the  $(j-1, j)$  block sends the  $i$ -th basis vector of group  $j$  to the  $i$ -th of group  $j-1$  for  $i = 1, \dots, w_j$ , so it is  $\begin{pmatrix} \mathbf{I}_{w_j} \\ 0 \end{pmatrix}$ ; and every other block is 0. That is the basic Weyr matrix with eigenvalue  $\lambda$  and



structure  $(w_1, \dots, w_s)$  (Definition 9.4.2). Writing the lists  $\mathcal{B}_\lambda$  for the distinct eigenvalues one after another and using  $V = \bigoplus_\lambda G_\lambda(T)$  (Theorem 9.1.3) with Theorem 8.1.5, the resulting basis  $\mathcal{B}$  makes  $[T]_{\mathcal{B}}$  a direct sum of basic Weyr matrices, one per eigenvalue.

- (b) Let  $\mathbf{W}$  be a Weyr form of  $T$  with one basic block per eigenvalue, and fix  $\lambda$  with basic block of structure  $(w_1, \dots, w_s)$  and size  $m$ . Let  $\mathcal{B}$  be a basis with  $[T]_{\mathcal{B}} = \mathbf{W}$ , and name the basis vectors inside the  $\lambda$ -block  $\mathbf{c}_{i,j}$ , where  $j$  indexes the groups and  $i = 1, \dots, w_j$  the entries within a group. The blocks of Definition 9.4.2 say precisely that

$$N\mathbf{c}_{i,1} = \mathbf{0}, \quad N\mathbf{c}_{i,j} = \mathbf{c}_{i,j-1} \quad (j \geq 2), \quad N = T - \lambda \text{id}_V,$$

while on the span  $V'$  of the basis vectors belonging to the other basic blocks the operator  $N$  is invertible:  $V'$  is  $T$ -invariant and the matrix of  $N|_{V'}$  is  $\mathbf{W}' - \lambda \mathbf{I}$ , where  $\mathbf{W}'$  is a direct sum of basic Weyr matrices with eigenvalues  $\mu \neq \lambda$ , hence upper triangular with all diagonal entries  $\mu - \lambda \neq 0$ , so of non-zero determinant (Theorem 6.4.6, Theorem 6.5.2). Hence  $N^j$  kills  $\mathbf{c}_{i,l}$  for  $l \leq j$  and sends  $\mathbf{c}_{i,l} \mapsto \mathbf{c}_{i,l-j}$  for  $l > j$ . Write  $V = U \oplus V'$  with  $U = \text{span}(\mathbf{c}_{i,l})$ ; both summands are  $N$ -invariant, so  $\ker N^j = (\ker N^j \cap U) \oplus (\ker N^j \cap V')$ , and the second summand is  $\{\mathbf{0}\}$  because  $N^j|_{V'}$  is invertible. Inside  $U$  the displayed action gives

$$\ker N^j = \text{span}(\mathbf{c}_{i,l} : l \leq j), \quad \dim \ker N^j = w_1 + \dots + w_j$$

for  $0 \leq j \leq s$ , the spanning vectors being members of the basis  $\mathcal{B}$  and hence independent; and  $\ker N^j = \ker N^s$  for  $j \geq s$ . Subtracting consecutive values gives  $w_j = \dim \ker N^j - \dim \ker N^{j-1}$  as claimed, a number depending only on  $T$  and  $\lambda$ . By Corollary 9.3.2 (d) this is also  $\#\{i : k_i \geq j\}$  for the  $\lambda$ -block sizes  $k_i$  of the Jordan form, that is, the dual partition. Since the structures are determined, so is  $\mathbf{W}$  up to the order in which its basic blocks are listed.

■

Existence and uniqueness are thus free; what makes the form worth the trouble is the next property. Here  $\mathbf{K} \in M_n(F)$  is called **block upper triangular** for a partition  $n = n_1 + \dots + n_s$ , taken as both the row and the column partition (Definition 7.1.1), when every block  $\mathbf{K}_{pq}$  with  $p > q$  is zero; for  $s = 2$  this is the shape of Chapter 7.

### Proposition 9.4.6 — The Commutant of a Basic Weyr Matrix Is Block Triangular

Let  $\mathbf{W} \in M_m(F)$  be a basic Weyr matrix with eigenvalue  $\lambda$  and Weyr structure  $(w_1, \dots, w_s)$ , and let  $\mathbf{K} \in M_m(F)$  satisfy  $\mathbf{W}\mathbf{K} = \mathbf{K}\mathbf{W}$ . Then  $\mathbf{K}$  is block upper triangular for the partition  $m = w_1 + \dots + w_s$ : its blocks  $\mathbf{K}_{pq}$  vanish for  $p > q$ .

*Idea.* Write both products blockwise. Because  $\mathbf{W}$  has only one non-zero block in each block row, each block equation is short:  $\mathbf{E}_p \mathbf{K}_{p+1,q} = \mathbf{K}_{p,q-1} \mathbf{E}_{q-1}$ , where  $\mathbf{E}_j = \begin{pmatrix} \mathbf{I}_{w_{j+1}} \\ 0 \end{pmatrix}$ . Every  $\mathbf{E}_j$  has full column rank, so it can be canceled on the left. Now sweep the block columns from left to right: the first block column forces  $\mathbf{K}_{p1} = 0$  for  $p > 1$ , and each new column is forced by the previous one.

*Proof.* Put  $\mathbf{N} = \mathbf{W} - \lambda \mathbf{I}_m$  and  $\mathbf{E}_j = \begin{pmatrix} \mathbf{I}_{w_{j+1}} \\ 0 \end{pmatrix} \in M_{w_j \times w_{j+1}}(F)$  for  $1 \leq j \leq s-1$ , so that  $\mathbf{N}$  has blocks  $\mathbf{N}_{j,j+1} = \mathbf{E}_j$  and all other blocks zero (Definition 9.4.2). Since  $\lambda \mathbf{I}_m$  commutes with everything,  $\mathbf{W}\mathbf{K} = \mathbf{K}\mathbf{W}$  is equivalent to  $\mathbf{N}\mathbf{K} = \mathbf{K}\mathbf{N}$ . Comparing the  $(p, q)$  blocks with Theorem 7.1.1, and reading an undefined symbol as 0,

$$\mathbf{E}_p \mathbf{K}_{p+1,q} = (\mathbf{N}\mathbf{K})_{pq} = (\mathbf{K}\mathbf{N})_{pq} = \mathbf{K}_{p,q-1} \mathbf{E}_{q-1} \quad (*)$$

for all  $1 \leq p, q \leq s$ .

Each  $\mathbf{E}_j$  is injective as a map  $F^{w_{j+1}} \rightarrow F^{w_j}$ , so  $\mathbf{E}_j \mathbf{X} = 0$  forces  $\mathbf{X} = 0$  for every matrix  $\mathbf{X}$  with  $w_{j+1}$  rows.

We prove  $\mathbf{K}_{pq} = 0$  for  $p > q$  by induction on  $q \geq 1$ .

*Case  $q = 1$ .* Let  $p > 1$ . Taking the indices  $(p-1, 1)$  in  $(*)$ , the right-hand side involves  $\mathbf{K}_{p-1,0}$ , which is undefined and therefore read as 0. So  $\mathbf{E}_{p-1} \mathbf{K}_{p,1} = 0$ , and  $p-1 \leq s-1$  makes  $\mathbf{E}_{p-1}$  one of the listed blocks; injectivity gives  $\mathbf{K}_{p,1} = 0$ .

*Induction step.* Let  $q \geq 2$  and assume  $\mathbf{K}_{p'q'} = 0$  whenever  $p' > q'$  and  $q' < q$ . Let  $p > q$ . Taking the indices  $(p-1, q)$  in  $(*)$ ,

$$\mathbf{E}_{p-1} \mathbf{K}_{pq} = \mathbf{K}_{p-1,q-1} \mathbf{E}_{q-1}.$$

Here  $p-1 > q-1$  and  $q-1 < q$ , so  $\mathbf{K}_{p-1,q-1} = 0$  by the induction hypothesis, and the right-hand side is 0. Again  $1 \leq p-1 \leq s-1$ , so injectivity of  $\mathbf{E}_{p-1}$  gives  $\mathbf{K}_{pq} = 0$ . This completes the induction and the proof. ■

*Remark.* The Jordan form has no such property, which is the reason the Weyr form exists at all. Take  $\mathbf{J} = \mathbf{J}_2(0) \oplus \mathbf{J}_1(0)$  and  $\mathbf{W}$  the Weyr matrix of the same operator, of structure  $(2, 1)$ :

$$\mathbf{J} = \begin{pmatrix} 0 & 1 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{pmatrix}, \quad \mathbf{W} = \begin{pmatrix} 0 & 0 & 1 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{pmatrix}.$$

Solving  $\mathbf{J}\mathbf{K} = \mathbf{K}\mathbf{J}$  entry by entry gives the commutant  $\left\{ \begin{pmatrix} a & b & c \\ 0 & a & 0 \\ 0 & h & i \end{pmatrix} \right\}$ , which contains

matrices with a non-zero entry **below** the diagonal, in position  $(3, 2)$ : the commutant of  $\mathbf{J}$  is not contained in the upper triangular matrices, and in particular it is not block upper triangular for the Weyr partition  $3 = 2 + 1$  of this operator. Solving  $\mathbf{W}\mathbf{K} = \mathbf{K}\mathbf{W}$  gives  $\left\{ \begin{pmatrix} a & b & c \\ 0 & e & f \\ 0 & 0 & a \end{pmatrix} \right\}$ , every member of which is upper triangular, in agreement with

Proposition 9.4.6. This is the property that makes the Weyr form the right one when two commuting operators must be brought to normal form at the same time.

*Quick check.* An operator  $T$  on a 7-dimensional complex space has one eigenvalue  $\lambda$ , with Jordan block sizes  $(3, 2, 1, 1)$ . What is the Weyr structure of its Weyr form, and what is  $\dim \ker(T - \lambda \text{id})^2$ ?

*Solution.* The Weyr structure is the dual partition:  $w_1 = \#\{i : k_i \geq 1\} = 4$ ,  $w_2 = \#\{i : k_i \geq 2\} = 2$ ,  $w_3 = \#\{i : k_i \geq 3\} = 1$ , so  $(4, 2, 1)$ , a partition of 7 as it must be. By [Theorem 9.4.5](#) (b),  $\dim \ker(T - \lambda \text{id})^2 = w_1 + w_2 = 6$ . (Equivalently,  $\sum_i \min(k_i, 2) = 2 + 2 + 1 + 1 = 6$ , by [Corollary 9.3.2](#) (d).)

## 9.4.4 Exercises

### 9.4.4.1 A. Check your understanding

#### Exercise A1.

- Write down  $\mathbf{A}(\lambda)$  for  $\lambda = a + bi$  and  $\mathbf{C}_2(\lambda)$ , and say what their sizes are.
- State the Real Jordan Form Theorem.
- True or false: every matrix in  $M_n(\mathbb{R})$  is similar over  $\mathbb{R}$  to an upper triangular real matrix. Justify your answer.
- Let  $\mathbf{A} \in M_6(\mathbb{R})$  have complex eigenvalues  $2, 2, 1 + i, 1 - i, 1 + i, 1 - i$ . What are the possible real Jordan forms?
- Define the Weyr structure of an eigenvalue and say how it is computed from  $T$ .
- What is the Weyr form of  $\mathbf{J}_3(7)$ ? And of  $7\mathbf{I}_3$ ?

*Solution.*

- $\mathbf{A}(\lambda) = \begin{pmatrix} a & -b \\ b & a \end{pmatrix} \in M_2(\mathbb{R})$  and  $\mathbf{C}_2(\lambda) = \begin{pmatrix} \mathbf{A}(\lambda) & \mathbf{I}_2 \\ 0 & \mathbf{A}(\lambda) \end{pmatrix} \in M_4(\mathbb{R})$ ; the real Jordan block  $\mathbf{C}_k(\lambda)$  has size  $2k$  ([Definition 9.4.1](#)).
- Every  $\mathbf{A} \in M_n(\mathbb{R})$  is similar, by a **real** invertible matrix, to a direct sum of blocks  $\mathbf{J}_k(\mu)$  for the real eigenvalues  $\mu$  and  $\mathbf{C}_k(\lambda)$  for the conjugate pairs  $\{\lambda, \bar{\lambda}\}$ , and that matrix is unique up to the order of the blocks ([Theorem 9.4.4](#)).
- False. A triangular real matrix has real eigenvalues ([Theorem 8.7.4](#)), and similar matrices have the same characteristic polynomial ([Theorem 6.10.3](#)); so a real matrix with a non-real eigenvalue, such as the rotation by a right angle, is similar to no real triangular matrix.
- The real eigenvalue 2 has  $a(2) = 2$ , so its blocks are  $\mathbf{J}_2(2)$  or  $\mathbf{J}_1(2) \oplus \mathbf{J}_1(2)$ . The pair  $\{1 \pm i\}$  has  $a(1+i) = 2$ , so its real blocks are  $\mathbf{C}_2(1+i)$  (size 4) or  $\mathbf{C}_1(1+i) \oplus \mathbf{C}_1(1+i)$ . That gives four possible real Jordan forms, of total size  $2 + 4 = 6$  in each case.
- The Weyr structure of  $\lambda$  is the partition  $(w_1, w_2, \dots)$  with  $w_j = \dim \ker(T - \lambda \text{id}_V)^j - \dim \ker(T - \lambda \text{id}_V)^{j-1}$ , that is, the dual partition of the  $\lambda$ -block sizes of the Jordan form ([Theorem 9.4.5](#)).
- $\mathbf{J}_3(7)$  has one 7-block of size 3, so the Weyr structure is  $(1, 1, 1)$  and the Weyr form is  $\mathbf{J}_3(7)$  itself. For  $7\mathbf{I}_3$  the block sizes are  $(1, 1, 1)$ , the dual partition is  $(3)$ , and the Weyr form is  $7\mathbf{I}_3$ .

### 9.4.4.2 B. Practice

**Exercise B1: Real forms of two by two matrices.**

For each real matrix, find the complex eigenvalues, the real Jordan form ([Theorem 9.4.4](#)), and an invertible  $\mathbf{P} \in M_2(\mathbb{R})$  putting it in that form. Verify  $\mathbf{AP} = \mathbf{P}\mathbf{\Lambda}$  column by column.

$$(a) \mathbf{A}_1 = \begin{pmatrix} 1 & 2 \\ -2 & 1 \end{pmatrix}.$$

$$(b) \mathbf{A}_2 = \begin{pmatrix} 3 & -5 \\ 1 & -1 \end{pmatrix}.$$

*Solution.*

- (a)  $p_{\mathbf{A}_1} = x^2 - (\text{tr } \mathbf{A}_1)x + \det \mathbf{A}_1 = x^2 - 2x + 5 = (x - 1)^2 + 4$ , so the eigenvalues are  $1 \pm 2i$  and the real Jordan form is  $\mathbf{\Lambda}(1 + 2i) = \begin{pmatrix} 1 & -2 \\ 2 & 1 \end{pmatrix}$ . For  $\lambda = 1 + 2i$ ,  $\mathbf{A}_1 - \lambda \mathbf{I} = \begin{pmatrix} -2i & 2 \\ -2 & -2i \end{pmatrix}$ , and the first row gives  $z_2 = iz_1$ ; take  $\mathbf{w} = (1, i)$ . Then  $\mathbf{w} = \mathbf{u} - i\mathbf{v}$  with  $\mathbf{u} = (1, 0)$  and  $\mathbf{v} = (0, -1)$ , so

$$\mathbf{P} = \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix}, \quad \mathbf{P}^{-1}\mathbf{A}_1\mathbf{P} = \begin{pmatrix} 1 & -2 \\ 2 & 1 \end{pmatrix}.$$

Check:  $\mathbf{A}_1\mathbf{u} = (1, -2) = \mathbf{u} + 2\mathbf{v}$  and  $\mathbf{A}_1\mathbf{v} = (-2, -1) = -2\mathbf{u} + \mathbf{v}$ , which are the two columns of  $\mathbf{\Lambda}(1 + 2i)$  as [Lemma 9.4.1](#) (a) predicts with  $a = 1, b = 2$ .

- (b)  $p_{\mathbf{A}_2} = x^2 - 2x + (-3 + 5) = x^2 - 2x + 2$ , with roots  $1 \pm i$ ; the real Jordan form is  $\mathbf{\Lambda}(1 + i) = \begin{pmatrix} 1 & -1 \\ 1 & 1 \end{pmatrix}$ . For  $\lambda = 1 + i$ , the second row of  $\mathbf{A}_2 - \lambda \mathbf{I} = \begin{pmatrix} 2 - i & -5 \\ 1 & -2 - i \end{pmatrix}$  gives  $z_1 = (2 + i)z_2$ ; take  $\mathbf{w} = (2 + i, 1)$ . Then  $\mathbf{u} = (2, 1)$  and  $\mathbf{v} = -\text{Im } \mathbf{w} = (-1, 0)$ , so

$$\mathbf{P} = \begin{pmatrix} 2 & -1 \\ 1 & 0 \end{pmatrix}, \quad \det \mathbf{P} = 1, \quad \mathbf{P}^{-1}\mathbf{A}_2\mathbf{P} = \begin{pmatrix} 1 & -1 \\ 1 & 1 \end{pmatrix}.$$

Check:  $\mathbf{A}_2\mathbf{u} = (6 - 5, 2 - 1) = (1, 1) = \mathbf{u} + \mathbf{v}$  and  $\mathbf{A}_2\mathbf{v} = (-3, -1) = -\mathbf{u} + \mathbf{v}$ .

**Exercise B2: Weyr forms of Jordan matrices.**

For each Jordan matrix, write down the block sizes, the Weyr structure of each eigenvalue, and the Weyr form ([Theorem 9.4.5](#)).

- (a)  $\mathbf{J}_3(0) \oplus \mathbf{J}_2(0) \oplus \mathbf{J}_1(0) \in M_6(F)$ .  
 (b)  $\mathbf{J}_2(4) \oplus \mathbf{J}_1(4) \oplus \mathbf{J}_1(9) \in M_4(F)$ .  
 (c)  $\mathbf{J}_2(0) \oplus \mathbf{J}_2(0) \in M_4(F)$ .

*Solution.*

- (a) Block sizes  $(3, 2, 1)$ , a partition of 6. Dual partition:  $w_1 = 3, w_2 = 2, w_3 = 1$ . So

the Weyr form is the basic Weyr matrix with  $\lambda = 0$  and structure  $(3, 2, 1)$ ,

$$\mathbf{W} = \begin{pmatrix} 0 & 0 & 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 1 \\ 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 \end{pmatrix},$$

with the  $3 \times 2$  block  $\begin{pmatrix} \mathbf{I}_2 \\ 0 \end{pmatrix}$  and the  $2 \times 1$  block  $\begin{pmatrix} \mathbf{I}_1 \\ 0 \end{pmatrix}$  above the diagonal. As a check,  $\text{rank } \mathbf{W} = 3 = \text{rank } \mathbf{J}$  and  $\text{rank } \mathbf{W}^2 = 1 = \text{rank } \mathbf{J}^2$ .

- (b) For  $\lambda = 4$ : sizes  $(2, 1)$ , dual  $(2, 1)$ . For  $\lambda = 9$ : sizes  $(1)$ , dual  $(1)$ . The Weyr form is the direct sum of the basic blocks,

$$\begin{pmatrix} 4 & 0 & 1 \\ 0 & 4 & 0 \\ 0 & 0 & 4 \end{pmatrix} \oplus (9).$$

- (c) Sizes  $(2, 2)$ , dual  $w_1 = 2, w_2 = 2$ . The basic Weyr matrix with  $\lambda = 0$  and structure  $(2, 2)$  is

$$\mathbf{W} = \begin{pmatrix} 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \end{pmatrix},$$

which is  $\mathbf{J}_2(0) \oplus \mathbf{J}_2(0)$  with its basis reordered as  $(\mathbf{e}_1, \mathbf{e}_3, \mathbf{e}_2, \mathbf{e}_4)$ .

### Exercise B3: Real forms of three by three matrices from the characteristic polynomial.

For each characteristic polynomial, list all possible real Jordan forms ([Theorem 9.4.4](#)) of a matrix  $\mathbf{A} \in M_3(\mathbb{R})$  with that polynomial.

- (a)  $p_{\mathbf{A}} = (x - 2)(x^2 + 1)$ .
- (b)  $p_{\mathbf{A}} = (x + 1)(x^2 - 2x + 5)$ .
- (c)  $p_{\mathbf{A}} = (x - 1)^3$ .
- (d)  $p_{\mathbf{A}} = (x^2 + 1)(x - 1)$  but with  $\mathbf{A}$  required to be diagonalizable over  $\mathbb{R}$ . Is this possible?

*Solution.*

- (a) The real eigenvalue 2 has  $a(2) = 1$ , giving the block  $\mathbf{J}_1(2) = (2)$ . The pair  $\{i, -i\}$  has  $a(i) = 1$ , giving  $\mathbf{C}_1(i) = \mathbf{A}(i) = \begin{pmatrix} 0 & -1 \\ 1 & 0 \end{pmatrix}$ . So the real Jordan form is  $(2) \oplus \begin{pmatrix} 0 & -1 \\ 1 & 0 \end{pmatrix}$ , the only possibility.
- (b)  $x^2 - 2x + 5$  has roots  $1 \pm 2i$ , so the form is  $(-1) \oplus \mathbf{A}(1 + 2i) = (-1) \oplus \begin{pmatrix} 1 & -2 \\ 2 & 1 \end{pmatrix}$ , again the only possibility.

- (c) All eigenvalues are real, so the real Jordan form is an ordinary Jordan form: the 1-block sizes form a partition of 3, giving  $\mathbf{J}_3(1)$ ,  $\mathbf{J}_2(1) \oplus \mathbf{J}_1(1)$  or  $\mathbf{I}_3$ .
- (d) No. A diagonalizable  $\mathbf{A} \in M_3(\mathbb{R})$  is similar over  $\mathbb{R}$  to a diagonal real matrix, whose characteristic polynomial splits over  $\mathbb{R}$  ([Theorem 6.4.6](#)); but  $x^2 + 1$  has no real root. In fact whenever  $\mathbf{A}$  has a non-real eigenvalue its real Jordan form contains a block  $\mathbf{\Lambda}(\lambda)$ , which is not diagonal.

### 9.4.4.3 C. Going deeper

#### Exercise C1: The real form of a companion matrix.

Let  $p = (x^2 + 1)^2 = x^4 + 2x^2 + 1 \in \mathbb{R}[x]$  and let  $\mathbf{C} = \mathbf{C}(p) \in M_4(\mathbb{R})$  be its companion matrix, with 1 in each position just below the diagonal and last column  $(-1, 0, -2, 0)$  ([Exercise \(C1: Companion matrices\)](#)).

- (a) Write down  $\mathbf{C}$ , and explain why  $p_{\mathbf{C}} = m_{\mathbf{C}} = p$ .
- (b) Determine the complex Jordan form of  $\mathbf{C}$ .
- (c) Deduce the real Jordan form of  $\mathbf{C}$ .
- (d) Verify your answer by comparing traces and characteristic polynomials.

*Hint: for (b), the exponent of  $x - i$  in  $m_{\mathbf{C}}$  is the largest size of an  $i$ -block.*

*Solution.*

- (a) With  $a_0 = 1$ ,  $a_1 = 0$ ,  $a_2 = 2$ ,  $a_3 = 0$ ,

$$\mathbf{C} = \begin{pmatrix} 0 & 0 & 0 & -1 \\ 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & -2 \\ 0 & 0 & 1 & 0 \end{pmatrix}.$$

By [Exercise \(C1: Companion matrices\)](#) (b),  $p_{\mathbf{C}} = p$ , and by [Exercise \(B3: Companion matrices\)](#) (d),  $m_{\mathbf{C}} = p$  as well.

- (b) Over  $\mathbb{C}$ ,  $p = (x - i)^2(x + i)^2$ , so  $a(i) = a(-i) = 2$ . By [Corollary 9.3.2](#) (c), the exponent 2 of  $x - i$  in  $m_{\mathbf{C}}$  is the largest size of an  $i$ -block; the sizes form a partition of 2 with largest part 2, so there is one block  $\mathbf{J}_2(i)$ . Likewise one block  $\mathbf{J}_2(-i)$ . The complex Jordan form is  $\mathbf{J}_2(i) \oplus \mathbf{J}_2(-i)$ .
- (c) By [Theorem 9.4.4](#), the real Jordan form has no ordinary Jordan blocks (there is no real eigenvalue) and one real block per complex  $i$ -block, of the same size parameter. So it is

$$\mathbf{C}_2(i) = \begin{pmatrix} 0 & -1 & 1 & 0 \\ 1 & 0 & 0 & 1 \\ 0 & 0 & 0 & -1 \\ 0 & 0 & 1 & 0 \end{pmatrix}.$$

(Consistently, [Corollary 9.4.2](#) turns  $\mathbf{C}_2(i)$  back into  $\mathbf{J}_2(i) \oplus \mathbf{J}_2(-i)$  over  $\mathbb{C}$ .)

- (d)  $\text{tr } \mathbf{C} = 0 = \text{tr } \mathbf{C}_2(i)$ , and both matrices have characteristic polynomial  $(x^2 + 1)^2$ : for  $\mathbf{C}$  by (a), and for  $\mathbf{C}_2(i)$  by [Corollary 9.4.2](#). Their minimal polynomials agree too, both being  $(x^2 + 1)^2$ .

**Exercise C2: Commuting with a Weyr matrix, and with a Jordan matrix.**

Let  $F$  be a field.

- (a) Let  $\mathbf{W} \in M_3(F)$  be the basic Weyr matrix with  $\lambda = 0$  and structure  $(2, 1)$  (Definition 9.4.2). Find all  $\mathbf{K} \in M_3(F)$  with  $\mathbf{WK} = \mathbf{KW}$ , and check that they are block upper triangular for the partition  $3 = 2 + 1$ .
- (b) Let  $\mathbf{J} = \mathbf{J}_2(0) \oplus \mathbf{J}_1(0)$ . Find all  $\mathbf{K} \in M_3(F)$  with  $\mathbf{JK} = \mathbf{KJ}$ , and exhibit one that is **not** upper triangular. Which partition of 3 does it violate?
- (c) Compute the dimension of each of the two commutants, and explain why they are equal.

*Solution.*

- (a)  $\mathbf{W} = \begin{pmatrix} 0 & 0 & 1 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{pmatrix}$ . Write  $\mathbf{K} = (k_{ij})$ . Then  $\mathbf{WK}$  has first row equal to the third row of  $\mathbf{K}$  and other rows zero, while  $\mathbf{KW}$  has third column equal to the first column of  $\mathbf{K}$  and other columns zero. Equating,

$$(k_{31}, k_{32}, k_{33}) = (0, 0, k_{11}), \quad k_{21} = k_{31} = 0.$$

So the commutant is

$$\left\{ \begin{pmatrix} a & b & c \\ 0 & e & f \\ 0 & 0 & a \end{pmatrix} : a, b, c, e, f \in F \right\},$$

of dimension 5. The lower-left  $1 \times 2$  block is  $(k_{31}, k_{32}) = (0, 0)$ , so every such  $\mathbf{K}$  is block upper triangular for  $3 = 2 + 1$ , as Proposition 9.4.6 asserts.

- (b)  $\mathbf{J} = \begin{pmatrix} 0 & 1 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{pmatrix}$ . Now  $\mathbf{JK}$  has first row equal to the second row of  $\mathbf{K}$  and other rows zero, while  $\mathbf{KJ}$  has second column equal to the first column of  $\mathbf{K}$  and other columns zero. Equating gives  $k_{21} = 0$ ,  $k_{22} = k_{11}$ ,  $k_{23} = 0$  and  $k_{31} = 0$ , so the commutant is

$$\left\{ \begin{pmatrix} a & b & c \\ 0 & a & 0 \\ 0 & h & i \end{pmatrix} \right\},$$

also of dimension 5. Take  $\mathbf{K}_0 = \begin{pmatrix} 0 & 0 & 0 \\ 0 & 0 & 0 \\ 0 & 1 & 0 \end{pmatrix}$ , which commutes with  $\mathbf{J}$  (it is the

case  $a = b = c = i = 0$ ,  $h = 1$ ). Its only non-zero entry sits in position  $(3, 2)$ , below the diagonal, so  $\mathbf{K}_0$  is not upper triangular. The partition it violates is  $3 = 2 + 1$ , the Weyr partition of this operator: the lower-left  $1 \times 2$  block of  $\mathbf{K}_0$  is  $(0, 1) \neq \mathbf{0}$ . So the conclusion of Proposition 9.4.6 genuinely fails for the Jordan form, while (a) confirms it for the Weyr form (Theorem 9.4.5) of the same operator.

- (c) Both have dimension 5. They must agree, because  $\mathbf{J}$  and  $\mathbf{W}$  are similar (both are nilpotent with  $\text{rank} = 1$  and  $\text{rank}^2 = 0$ , so Corollary 9.2.5 applies): if  $\mathbf{W} = \mathbf{P}^{-1}\mathbf{JP}$ , then  $\mathbf{K} \mapsto \mathbf{P}^{-1}\mathbf{KP}$  is a bijection from the commutant of  $\mathbf{J}$  to that of  $\mathbf{W}$ , and it is linear and invertible, hence preserves dimension.

## 9.5 Cyclic Subspaces and Companion Matrices

The Jordan form of Section 3 exists only when the characteristic polynomial splits, because its building block  $\mathbf{J}_k(\lambda)$  needs an eigenvalue  $\lambda$  in  $F$ . To build a normal form over **every** field we need a block that does not name a scalar. Look again at what a Jordan block really is: reading its columns,  $\mathbf{J}_k(\lambda)$  is the matrix of  $T$  on the span of  $\mathbf{u}, T\mathbf{u}, \dots, T^{k-1}\mathbf{u}$  for the top  $\mathbf{u}$  of a chain. That description — *one vector and everything  $T$  can make of it* — mentions no eigenvalue at all. This section studies those subspaces, computes their dimension and their matrix, and decides when a single vector generates the whole space.

### 9.5.1 The subspace generated by one vector

Chapter 8 called a subspace  $U$  **invariant** when  $T\mathbf{u} \in U$  for every  $\mathbf{u} \in U$  (Definition 8.1.1). Given one vector  $\mathbf{v}$ , what is the smallest invariant subspace containing it? It must contain  $\mathbf{v}$ , hence  $T\mathbf{v}$ , hence  $T^2\mathbf{v}$ , and so on, hence every linear combination of these. That list is already closed under  $T$ , so nothing more is forced. So the answer is a span, and it deserves a name.

*The cyclic subspace generated by a vector is everything that can be reached from it by applying the operator and taking combinations.*

#### Definition 9.5.1 — Cyclic Subspace

Let  $V$  be a vector space over  $F$ , let  $T \in \mathcal{L}(V)$  and let  $\mathbf{v} \in V$ . The **cyclic subspace generated by  $\mathbf{v}$**  is

$$Z(\mathbf{v}; T) := \text{span}(\mathbf{v}, T\mathbf{v}, T^2\mathbf{v}, \dots) = \text{span}(\{T^j\mathbf{v} : j \geq 0\}),$$

where  $T^0 = \text{id}_V$ . For  $\mathbf{A} \in M_n(F)$  we write  $Z(\mathbf{v}; \mathbf{A}) = Z(\mathbf{v}; T_{\mathbf{A}})$ .

In words: we apply  $T$  to  $\mathbf{v}$  over and over, collect all the vectors obtained, and take their span — so  $Z(\mathbf{v}; T)$  is a subspace by Theorem 1.3.1, whatever  $\mathbf{v}$  is. The spanning list may be infinite, but we shall see in a moment that in finite dimension its first few entries already suffice. Note  $Z(\mathbf{0}; T) = \text{span}(\emptyset) = \{\mathbf{0}\}$  and  $\mathbf{v} \in Z(\mathbf{v}; T)$ .

#### Proposition 9.5.1 — The Cyclic Subspace Is the Smallest Invariant Subspace Containing the Vector

Let  $V$  be a vector space over  $F$ ,  $T \in \mathcal{L}(V)$  and  $\mathbf{v} \in V$ . Then:

- (a)  $Z(\mathbf{v}; T) = \{p(T)\mathbf{v} : p \in F[x]\}$ ;
- (b)  $Z(\mathbf{v}; T)$  is  $T$ -invariant;
- (c) if  $U$  is a  $T$ -invariant subspace of  $V$  with  $\mathbf{v} \in U$ , then  $Z(\mathbf{v}; T) \subseteq U$ .

*Proof.*



- (a) A linear combination of the vectors  $T^j \mathbf{v}$  is  $\sum_j a_j T^j \mathbf{v} = p(T) \mathbf{v}$  with  $p = \sum_j a_j x^j$ , and conversely (Definition 3.4.2). By Definition 1.3.2, these combinations are exactly the elements of  $Z(\mathbf{v}; T)$ .
- (b) For  $p \in F[x]$ ,  $T(p(T) \mathbf{v}) = (xp)(T) \mathbf{v} \in Z(\mathbf{v}; T)$  by Theorem 5.7.1 (b) and (a). (Or:  $T$  maps each spanning vector  $T^j \mathbf{v}$  to  $T^{j+1} \mathbf{v} \in Z(\mathbf{v}; T)$ , so Lemma 8.1.1 applies.)
- (c) Let  $U$  be invariant with  $\mathbf{v} \in U$ . By induction on  $j$ ,  $T^j \mathbf{v} \in U$ : this holds for  $j = 0$ , and if  $T^j \mathbf{v} \in U$ , then  $T^{j+1} \mathbf{v} = T(T^j \mathbf{v}) \in U$  by invariance. So  $U$  contains the spanning list of  $Z(\mathbf{v}; T)$ , and hence its span (Theorem 1.3.1, Proposition 1.3.4).

■

**Examples.** Throughout,  $V$  is finite-dimensional over  $F$ .

- **An eigenvector generates a line.** If  $T\mathbf{v} = \lambda\mathbf{v}$  with  $\mathbf{v} \neq \mathbf{0}$ , then  $T^j \mathbf{v} = \lambda^j \mathbf{v}$ , so  $Z(\mathbf{v}; T) = \text{span}(\mathbf{v})$ , of dimension 1. The smallest invariant subspace containing an eigenvector is its own line (Proposition 8.1.2).
- **The top of a Jordan chain generates the block.** Let  $T$  have matrix  $\mathbf{J}_k(\lambda)$  in the basis  $\mathcal{B} = (\mathbf{b}_1, \dots, \mathbf{b}_k)$ . Then  $\mathbf{b}_k$  is the top of the chain (Definition 9.2.2), and  $\mathbf{b}_{k-1} = (T - \lambda \text{id}_V) \mathbf{b}_k = T\mathbf{b}_k - \lambda \mathbf{b}_k \in \text{span}(\mathbf{b}_k, T\mathbf{b}_k)$ ; continuing downwards, every  $\mathbf{b}_j$  lies in  $\text{span}(\mathbf{b}_k, T\mathbf{b}_k, \dots, T^{k-j} \mathbf{b}_k)$ . So  $Z(\mathbf{b}_k; T) = V$ : a single Jordan block is one cyclic subspace.
- **Differentiation.** Let  $D \in \mathcal{L}(\mathbb{R}[x]_{\leq 3})$  be differentiation. Then  $Z(x^3; D) = \text{span}(x^3, 3x^2, 6x, 6) = \mathbb{R}[x]_{\leq 3}$ , everything; while  $Z(x^2; D) = \text{span}(x^2, 2x, 2) = \mathbb{R}[x]_{\leq 2}$ , a proper subspace.
- **A companion matrix.** Let  $\mathbf{A} \in M_n(F)$  have 1 in each position just below the diagonal, so that  $\mathbf{A}\mathbf{e}_j = \mathbf{e}_{j+1}$  for  $j \leq n-1$  (Exercise (C1: Companion matrices)). Then  $\mathbf{A}^j \mathbf{e}_1 = \mathbf{e}_{j+1}$  for  $j \leq n-1$ , so  $Z(\mathbf{e}_1; \mathbf{A})$  contains every standard basis vector and equals  $F^n$ .
- **The degenerate case.**  $Z(\mathbf{0}; T) = \{\mathbf{0}\}$ , the smallest invariant subspace of all.

**Non-example by minimal change.** Truncate the spanning list:  $\text{span}(\mathbf{v}, T\mathbf{v})$  need **not** be

invariant. For  $\mathbf{A} = \begin{pmatrix} 0 & 0 & 1 \\ 1 & 0 & 0 \\ 0 & 1 & 0 \end{pmatrix}$  and  $\mathbf{v} = \mathbf{e}_1$  we get  $\text{span}(\mathbf{e}_1, \mathbf{A}\mathbf{e}_1) = \text{span}(\mathbf{e}_1, \mathbf{e}_2)$ , and  $\mathbf{A}\mathbf{e}_2 = \mathbf{e}_3$

is outside it. What fails is closure: cutting the list short loses exactly the vectors that invariance demands. The correct statement is that the list may be cut short at the right place, which is the subject of the next subsection.

*Warning.*  $Z(\mathbf{v}; T)$  depends on  $T$  as much as on  $\mathbf{v}$ , and the spanning list  $(\mathbf{v}, T\mathbf{v}, T^2\mathbf{v}, \dots)$  is almost never independent. For  $\mathbf{v} = \mathbf{e}_1$  and  $T = \text{id}$  the list is  $(\mathbf{e}_1, \mathbf{e}_1, \mathbf{e}_1, \dots)$  and the cyclic subspace is a line; for the same  $\mathbf{v}$  and the companion matrix above it is all of  $F^3$ . In finite dimension the list must become dependent by step  $\dim V$  at the latest (Theorem 1.6.3), and where it first does is the only thing that matters.

## 9.5.2 The annihilator of a vector

The examples suggest measuring  $\mathbf{v}$  by the first dependence in the list  $\mathbf{v}, T\mathbf{v}, T^2\mathbf{v}, \dots$ . A dependence is a polynomial relation:  $\sum_j a_j T^j \mathbf{v} = \mathbf{0}$  says that the polynomial  $p = \sum_j a_j x^j$  kills  $\mathbf{v}$ , not necessarily all of  $V$ . Chapter 5 taught us what to do with such a set of polynomials.

*The  $T$ -annihilator of  $\mathbf{v}$  is the minimal polynomial of  $T$  as seen by the single vector  $\mathbf{v}$ .*

**Definition 9.5.2 — The  $T$ -Annihilator of a Vector**

Let  $V$  be a **finite-dimensional** vector space over  $F$ , let  $T \in \mathcal{L}(V)$  and let  $\mathbf{v} \in V$ . The set

$$I_{\mathbf{v}} := \{ p \in F[x] : p(T)\mathbf{v} = \mathbf{0} \}$$

is a non-zero ideal of  $F[x]$ , and the  $T$ -**annihilator** of  $\mathbf{v}$ , written  $m_{T,\mathbf{v}}$ , is its unique **monic** generator:

$$I_{\mathbf{v}} = \langle m_{T,\mathbf{v}} \rangle.$$

Equivalently,  $m_{T,\mathbf{v}}$  is the monic polynomial of **least degree** with  $m_{T,\mathbf{v}}(T)\mathbf{v} = \mathbf{0}$ . For  $\mathbf{A} \in M_n(F)$  we write  $m_{\mathbf{A},\mathbf{v}}$ .

**Well-definedness.** Three checks. *Ideal:* if  $p, q \in I_{\mathbf{v}}$ , then  $(p+q)(T)\mathbf{v} = p(T)\mathbf{v} + q(T)\mathbf{v} = \mathbf{0}$ , and for any  $h \in F[x]$ ,  $(hp)(T)\mathbf{v} = h(T)(p(T)\mathbf{v}) = \mathbf{0}$ , using [Theorem 5.7.1](#) (a), (b); so  $I_{\mathbf{v}}$  is an ideal ([Definition 5.2.1](#)). *Non-zero:*  $m_T \in I_{\mathbf{v}}$ , since  $m_T(T) = 0$  as an operator ([Definition 8.5.1](#)), and  $m_T \neq 0$ . *Monic generator:* a non-zero ideal of  $F[x]$  has exactly one monic generator, which is its non-zero element of least degree ([Theorem 5.2.2](#)). As with the minimal polynomial, the divisibility statement is the one we use:

$$p(T)\mathbf{v} = \mathbf{0} \iff m_{T,\mathbf{v}} \mid p. \quad (*)$$

In words:  $m_{T,\mathbf{v}}$  records which polynomials in  $T$  kill this one vector. Two degenerate cases:  $m_{T,\mathbf{0}} = 1$ , because every polynomial kills  $\mathbf{0}$ ; and if  $\mathbf{v} \neq \mathbf{0}$ , then  $\deg m_{T,\mathbf{v}} \geq 1$ , since the constant 1 does not kill  $\mathbf{v}$ . If  $\mathbf{v}$  is an eigenvector for  $\lambda$ , then  $m_{T,\mathbf{v}} = x - \lambda$ .

**Why this definition.** The minimal polynomial  $m_T$  is the “annihilator of all of  $V$  at once”:  $m_T$  kills every vector, so  $m_{T,\mathbf{v}} \mid m_T$  for every  $\mathbf{v}$  by (\*). The individual annihilators are finer, and they differ from vector to vector: for  $\mathbf{J} = \begin{pmatrix} 1 & 1 \\ 0 & 1 \end{pmatrix}$  we have  $m_{\mathbf{J},\mathbf{e}_1} = x - 1$  but  $m_{\mathbf{J},\mathbf{e}_2} = (x - 1)^2 = m_{\mathbf{J}}$ . This spread is exactly what will let us split  $V$  into cyclic pieces in Section 6.

Now the payoff: the annihilator computes the dimension of the cyclic subspace, hands us a basis, and hands us the matrix.

**Theorem 9.5.2 — Basis and Matrix of a Cyclic Subspace**

Let  $V$  be a finite-dimensional vector space over  $F$ , let  $T \in \mathcal{L}(V)$  and let  $\mathbf{v} \in V$  with  $\mathbf{v} \neq \mathbf{0}$ . Write  $Z = Z(\mathbf{v}; T)$  and

$$m_{T,\mathbf{v}} = x^k + a_{k-1}x^{k-1} + \cdots + a_1x + a_0, \quad k = \deg m_{T,\mathbf{v}} \geq 1.$$

Then:

- (a)  $(\mathbf{v}, T\mathbf{v}, \dots, T^{k-1}\mathbf{v})$  is a basis of  $Z$ ; in particular  $\dim Z = \deg m_{T,\mathbf{v}}$ ;
- (b)  $T^k\mathbf{v} = -(a_0\mathbf{v} + a_1T\mathbf{v} + \cdots + a_{k-1}T^{k-1}\mathbf{v})$ , and the matrix of  $T|_Z$  in the basis of (a) is the companion matrix of  $m_{T,\mathbf{v}}$  ([Exercise \(C1: Companion matrices\)](#); the definition is repeated in full as [Definition 9.5.3](#) below);
- (c)  $m_{T|_Z} = p_{T|_Z} = m_{T,\mathbf{v}}$ ;
- (d)  $m_{T,\mathbf{v}} \mid m_T$ , and  $\deg m_{T,\mathbf{v}} \leq \deg m_T \leq \dim V$ .

*Idea.* For spanning, divide: any  $p$  is  $q m_{T,\mathbf{v}} + r$  with  $\deg r < k$ , and the  $q m_{T,\mathbf{v}}$  part dies on  $\mathbf{v}$ , so  $p(T)\mathbf{v} = r(T)\mathbf{v}$  uses only the first  $k$  powers. For independence, a shorter dependence would be a non-zero polynomial of degree  $< k$  in  $I_{\mathbf{v}}$ , and  $m_{T,\mathbf{v}}$  has the least degree there. Part (b) is the definition of  $m_{T,\mathbf{v}}$  rearranged, read as the last column of a matrix; (c) says that on  $Z$  the vector  $\mathbf{v}$  already sees everything, so its annihilator is the minimal polynomial of the restriction, and a degree count makes it the characteristic polynomial too.

*Proof.*

- (a) *Spanning.* Let  $\mathbf{z} \in Z$ , say  $\mathbf{z} = p(T)\mathbf{v}$  with  $p \in F[x]$  (Proposition 9.5.1 (a)). Divide by  $m_{T,\mathbf{v}}$ : there are  $q, r \in F[x]$  with  $p = q m_{T,\mathbf{v}} + r$  and  $\deg r < k$  (Theorem 5.1.1). Then

$$\mathbf{z} = p(T)\mathbf{v} = q(T)(m_{T,\mathbf{v}}(T)\mathbf{v}) + r(T)\mathbf{v} = r(T)\mathbf{v},$$

by Theorem 5.7.1 (a), (b) and  $m_{T,\mathbf{v}}(T)\mathbf{v} = \mathbf{0}$ . As  $\deg r < k$ , the vector  $r(T)\mathbf{v}$  is a combination of  $\mathbf{v}, T\mathbf{v}, \dots, T^{k-1}\mathbf{v}$ .

*Independence.* Suppose  $c_0\mathbf{v} + c_1T\mathbf{v} + \dots + c_{k-1}T^{k-1}\mathbf{v} = \mathbf{0}$  with the  $c_j \in F$  not all zero. Then  $q = \sum_{j < k} c_j x^j$  is a non-zero element of  $I_{\mathbf{v}}$  of degree less than  $k$ , contradicting the fact that  $m_{T,\mathbf{v}}$  has least degree among the non-zero elements of  $I_{\mathbf{v}}$  (Theorem 5.2.2). So all  $c_j = 0$ .

- (b) The identity  $m_{T,\mathbf{v}}(T)\mathbf{v} = \mathbf{0}$  reads  $T^k\mathbf{v} + a_{k-1}T^{k-1}\mathbf{v} + \dots + a_0\mathbf{v} = \mathbf{0}$ , which is the display. Write  $\mathbf{z}_j = T^{j-1}\mathbf{v}$  for  $j = 1, \dots, k$ , the basis of (a). For  $j \leq k-1$ ,  $T\mathbf{z}_j = T^j\mathbf{v} = \mathbf{z}_{j+1}$ , so the  $j$ -th column of the matrix is  $\mathbf{e}_{j+1}$ ; and  $T\mathbf{z}_k = T^k\mathbf{v} = -a_0\mathbf{z}_1 - \dots - a_{k-1}\mathbf{z}_k$ , so the last column is  $(-a_0, -a_1, \dots, -a_{k-1})$ . That is exactly the companion matrix of  $m_{T,\mathbf{v}}$  in the sense of Exercise (C1: Companion matrices), whose definition is recalled in full in the next subsection. For  $k = 1$  the matrix is the  $1 \times 1$  matrix  $(-a_0)$ .
- (c) Let  $p \in F[x]$ . If  $p(T)\mathbf{v} = \mathbf{0}$ , then for every  $q \in F[x]$ ,

$$p(T)(q(T)\mathbf{v}) = q(T)(p(T)\mathbf{v}) = \mathbf{0}$$

by Theorem 5.7.1 (c), so  $p(T)$  kills all of  $Z$  by Proposition 9.5.1 (a), that is,  $p(T|_Z) = 0$  (Exercise (C2: Minimal polynomials of the pieces) (a)). Conversely, if  $p(T|_Z) = 0$ , then in particular  $p(T)\mathbf{v} = \mathbf{0}$ , as  $\mathbf{v} \in Z$ . So the annihilating ideals of  $\mathbf{v}$  and of  $T|_Z$  coincide, and their monic generators agree:  $m_{T|_Z} = m_{T,\mathbf{v}}$  (Definition 8.5.1, Theorem 5.2.2). Finally  $m_{T|_Z} \mid p_{T|_Z}$  (Corollary 8.6.2), both are monic, and  $\deg p_{T|_Z} = \dim Z = k = \deg m_{T|_Z}$  by (a) and Theorem 6.10.2; so they are equal (Proposition 5.1.2 (c), (d)).

- (d)  $m_T \in I_{\mathbf{v}}$  as noted after Definition 9.5.2, so  $m_{T,\mathbf{v}} \mid m_T$  by (\*), and the degrees compare by Proposition 5.1.2 (c). And  $\deg m_T \leq \deg p_T = \dim V$  by Corollary 8.6.2 and Theorem 6.10.2.

■

*Quick check.* Let  $T$  be the operator on  $\mathbb{Q}^2$  with matrix  $\text{diag}(2, 5)$ , and let  $\mathbf{v} = (1, 1)$ . Find  $m_{T,\mathbf{v}}$  and  $Z(\mathbf{v}; T)$ . Then do the same for  $\mathbf{v} = \mathbf{e}_1$ .

*Solution.*  $(T - 2 \text{ id})(T - 5 \text{ id})\mathbf{v} = \mathbf{0}$ , as one checks entry by entry, so  $m_{T,\mathbf{v}} \mid (x - 2)(x - 5)$ . Neither factor alone kills  $\mathbf{v}$ :  $(T - 2 \text{ id})\mathbf{v} = (0, 3) \neq \mathbf{0}$  and  $(T - 5 \text{ id})\mathbf{v} = (-3, 0) \neq \mathbf{0}$ . So  $m_{T,\mathbf{v}} = (x - 2)(x - 5)$ , of degree 2, and  $Z(\mathbf{v}; T) = \mathbb{Q}^2$  by Theorem 9.5.2 (a); the basis

it provides is  $(\mathbf{v}, T\mathbf{v}) = ((1, 1), (2, 5))$ . For  $\mathbf{v} = \mathbf{e}_1$ , an eigenvector,  $m_{T, \mathbf{e}_1} = x - 2$  and  $Z(\mathbf{e}_1; T) = \text{span}(\mathbf{e}_1)$  is a line.

### 9.5.3 Companion matrices

The matrix in [Theorem 9.5.2](#) (b) appeared in Chapter 6 as an exercise; it is the block we were looking for, and it deserves a definition of its own.

#### Definition 9.5.3 — Companion Matrix

Let  $n \geq 1$  and let  $p = x^n + a_{n-1}x^{n-1} + \cdots + a_1x + a_0 \in F[x]$  be **monic** of degree  $n$ . The **companion matrix** of  $p$  is the matrix  $\mathbf{C}(p) \in M_n(F)$  with 1 in each position  $(j+1, j)$  just **below** the diagonal, last column  $(-a_0, -a_1, \dots, -a_{n-1})$ , and 0 elsewhere:

$$\mathbf{C}(p) = \begin{pmatrix} 0 & 0 & \cdots & 0 & -a_0 \\ 1 & 0 & \cdots & 0 & -a_1 \\ 0 & 1 & \cdots & 0 & -a_2 \\ \vdots & & \ddots & & \vdots \\ 0 & 0 & \cdots & 1 & -a_{n-1} \end{pmatrix}.$$

For  $n = 1$  and  $p = x + a_0$  this reads  $\mathbf{C}(p) = (-a_0)$ .

This is the convention of [Exercise \(C1: Companion matrices\)](#), where the  $n \geq 2$  case was introduced: the subdiagonal ones shift the standard basis forward,  $\mathbf{C}(p)\mathbf{e}_j = \mathbf{e}_{j+1}$  for  $j \leq n-1$ , and the last column records the coefficients of  $p$  with their signs changed. Some books use the transpose, or put the coefficients in the last row; nothing changes but the bookkeeping, since a matrix and its transpose have the same characteristic polynomial ([Exercise \(B2: Transpose\)](#)).

#### Theorem 9.5.3 — The Companion Matrix Realizes Its Polynomial Twice

Let  $p \in F[x]$  be monic of degree  $n \geq 1$  and  $\mathbf{C} = \mathbf{C}(p)$ . Then

$$p\mathbf{C} = m_{\mathbf{C}} = p,$$

and  $\mathbf{e}_1$  satisfies  $Z(\mathbf{e}_1; \mathbf{C}) = F^n$  with  $m_{\mathbf{C}, \mathbf{e}_1} = p$ . Consequently every monic polynomial over  $F$  is both the characteristic and the minimal polynomial of some square matrix over  $F$ .

*Proof.* For  $n = 1$  the matrix is  $(-a_0)$ , with  $p\mathbf{C} = m_{\mathbf{C}} = x + a_0 = p$ . Let  $n \geq 2$ . Since  $\mathbf{C}\mathbf{e}_j = \mathbf{e}_{j+1}$  for  $j \leq n-1$ , induction gives  $\mathbf{C}^j\mathbf{e}_1 = \mathbf{e}_{j+1}$  for  $0 \leq j \leq n-1$ . Hence  $Z(\mathbf{e}_1; \mathbf{C})$  contains  $\mathbf{e}_1, \dots, \mathbf{e}_n$ , so  $Z(\mathbf{e}_1; \mathbf{C}) = F^n$ , and by [Theorem 9.5.2](#) (a),

$$\deg m_{\mathbf{C}, \mathbf{e}_1} = \dim Z(\mathbf{e}_1; \mathbf{C}) = n.$$

Applying [Theorem 9.5.2](#) (c) with  $Z = F^n$ , so that  $T_{\mathbf{C}}|_Z = T_{\mathbf{C}}$ ,

$$m_{\mathbf{C}} = m_{T_{\mathbf{C}}} = m_{\mathbf{C}, \mathbf{e}_1} = p_{T_{\mathbf{C}}} = p\mathbf{C},$$

where  $m_{\mathbf{C}} = m_{T_{\mathbf{C}}}$  and  $p_{T_{\mathbf{C}}} = p_{\mathbf{C}}$  hold by definition (Definition 8.5.1, Definition 6.10.2), the matrix of  $T_{\mathbf{C}}$  in the standard basis being  $\mathbf{C}$ . Finally  $p_{\mathbf{C}} = p$  by Exercise (C1: Companion matrices) (b). So  $m_{\mathbf{C}} = p_{\mathbf{C}} = p$ . The last sentence follows by taking  $\mathbf{C}(p)$  for the given  $p$ .

■

So the companion matrix is the field-free analogue of a Jordan block:  $\mathbf{J}_k(\lambda)$  and  $\mathbf{C}((x - \lambda)^k)$  both have characteristic and minimal polynomial  $(x - \lambda)^k$ , and they are similar, as we will see in Section 6. The difference is that  $\mathbf{C}(p)$  is available for **every** monic  $p$ , whether or not  $p$  has a root in  $F$ .

## 9.5.4 Cyclic operators

One block is one cyclic subspace. Before assembling many of them, we should know when a single one already fills the space.

### Definition 9.5.4 — Cyclic Vector, Cyclic Operator

Let  $V$  be a finite-dimensional vector space over  $F$  with  $\dim V = n \geq 1$ , and let  $T \in \mathcal{L}(V)$ . A vector  $\mathbf{v} \in V$  is a **cyclic vector** for  $T$  if

$$Z(\mathbf{v}; T) = V.$$

We call  $T$  **cyclic** if it has a cyclic vector, and  $\mathbf{A} \in M_n(F)$  **cyclic** if  $T_{\mathbf{A}}$  is.

By Theorem 9.5.2 (a),  $\mathbf{v}$  is cyclic exactly when  $\deg m_{T, \mathbf{v}} = n$ , and then  $(\mathbf{v}, T\mathbf{v}, \dots, T^{n-1}\mathbf{v})$  is a basis of  $V$ . That is the definition used in Exercise (C1: Cyclic vectors), so the two agree. Reading Theorem 9.5.2 (b) with  $Z = V$ : **in the basis generated by a cyclic vector, the matrix of  $T$  is a companion matrix.**

The key to the existence of cyclic vectors is a lemma about adding vectors whose annihilators have no common factor. It says that annihilators multiply, exactly as one hopes.

### Lemma 9.5.4 — Annihilator of a Sum with Coprime Annihilators

Let  $V$  be finite-dimensional,  $T \in \mathcal{L}(V)$ , and  $\mathbf{u}, \mathbf{w} \in V$  with  $f = m_{T, \mathbf{u}}$  and  $g = m_{T, \mathbf{w}}$  **coprime**. Then

$$m_{T, \mathbf{u} + \mathbf{w}} = fg.$$

More generally, if  $\mathbf{v}_1, \dots, \mathbf{v}_k \in V$  have pairwise coprime  $T$ -annihilators  $f_1, \dots, f_k$ , then  $m_{T, \mathbf{v}_1 + \dots + \mathbf{v}_k} = f_1 f_2 \cdots f_k$ .

*Idea.* Write  $h = m_{T, \mathbf{u} + \mathbf{w}}$ . That  $h$  divides  $fg$  is immediate, since  $fg$  kills both summands. For the other direction, multiply the relation  $h(T)(\mathbf{u} + \mathbf{w}) = \mathbf{0}$  by  $g(T)$ : the  $\mathbf{w}$  part dies, leaving  $(hg)(T)\mathbf{u} = \mathbf{0}$ , so  $f \mid hg$ ; and since  $f$  and  $g$  are coprime we may cancel  $g$  and conclude  $f \mid h$ . Symmetrically  $g \mid h$ , and two coprime divisors multiply.

*Proof.* Polynomials in  $T$  commute (Theorem 5.7.1 (c)). Hence

$$(fg)(T)(\mathbf{u} + \mathbf{w}) = g(T)(f(T)\mathbf{u}) + f(T)(g(T)\mathbf{w}) = \mathbf{0},$$

so  $h \mid fg$ , where  $h = m_{T, \mathbf{u} + \mathbf{w}}$ . Conversely, applying  $g(T)$  to  $h(T)(\mathbf{u} + \mathbf{w}) = \mathbf{0}$  and using  $g(T)\mathbf{w} = \mathbf{0}$ ,

$$\mathbf{0} = g(T)h(T)(\mathbf{u} + \mathbf{w}) = (hg)(T)\mathbf{u} + h(T)(g(T)\mathbf{w}) = (hg)(T)\mathbf{u}.$$

Therefore  $f \mid hg$ , and since  $\gcd(f, g) = 1$ , Theorem 5.2.8 gives  $f \mid h$ . Swapping the roles of  $\mathbf{u}$  and  $\mathbf{w}$  gives  $g \mid h$ . As  $f$  and  $g$  are coprime and both divide  $h$ , their product divides  $h$  (Lemma 5.6.3 (b)). With  $h \mid fg$ , and all three polynomials monic,  $h = fg$  (Proposition 5.1.2 (d)).

For  $k$  vectors, induct on  $k$ . The case  $k = 1$  is trivial. Assume  $m_{T, \mathbf{v}_1 + \dots + \mathbf{v}_{k-1}} = f_1 \cdots f_{k-1}$ . Each  $f_i$  with  $i \leq k-1$  is coprime to  $f_k$ , so  $f_1 \cdots f_{k-1}$  is coprime to  $f_k$  (Lemma 5.6.3 (a)), and the two-vector case applies to  $\mathbf{u} = \mathbf{v}_1 + \dots + \mathbf{v}_{k-1}$  and  $\mathbf{w} = \mathbf{v}_k$ . ■

*Quick check.* Let  $\mathbf{A} = \text{diag}(2, 2, 5)$  on  $\mathbb{Q}^3$ . Find  $m_{\mathbf{A}, \mathbf{e}_1}$ ,  $m_{\mathbf{A}, \mathbf{e}_3}$  and  $m_{\mathbf{A}, \mathbf{e}_1 + \mathbf{e}_3}$ . Is  $\mathbf{e}_1 + \mathbf{e}_3$  a cyclic vector for  $\mathbf{A}$ ?

*Solution.* The vectors  $\mathbf{e}_1$  and  $\mathbf{e}_3$  are eigenvectors, for 2 and for 5, so  $m_{\mathbf{A}, \mathbf{e}_1} = x - 2$  and  $m_{\mathbf{A}, \mathbf{e}_3} = x - 5$ . These are distinct monic linear polynomials, hence coprime, so Lemma 9.5.4 gives

$$m_{\mathbf{A}, \mathbf{e}_1 + \mathbf{e}_3} = (x - 2)(x - 5),$$

of degree 2. By Theorem 9.5.2 (a),  $\dim Z(\mathbf{e}_1 + \mathbf{e}_3; \mathbf{A}) = 2 < 3$ , so  $\mathbf{e}_1 + \mathbf{e}_3$  is **not** cyclic; indeed  $Z(\mathbf{e}_1 + \mathbf{e}_3; \mathbf{A}) = \text{span}(\mathbf{e}_1, \mathbf{e}_3)$ . The lemma makes the annihilator as large as the two pieces allow, and here that is still not all of  $\mathbb{Q}^3$  — the eigenvalue 2 is repeated, so  $m_{\mathbf{A}} = (x - 2)(x - 5)$  has degree 2 and no vector can do better.

The lemma turns the primary decomposition into a supply of vectors with the largest possible annihilator.

### Proposition 9.5.5 — Some Vector Has the Minimal Polynomial as Its Annihilator

Let  $V \neq \{\mathbf{0}\}$  be a finite-dimensional vector space over  $F$  and  $T \in \mathcal{L}(V)$ . Then there is  $\mathbf{v} \in V$  with

$$m_{T, \mathbf{v}} = m_T.$$

*Idea.* The minimal polynomial factors as  $m_T = p_1^{e_1} \cdots p_k^{e_k}$ , and the primary decomposition supplies an invariant piece  $V_i$  on which  $T$  has minimal polynomial exactly  $p_i^{e_i}$ . Inside  $V_i$  some vector must need the full exponent  $e_i$ , or  $p_i^{e_i-1}$  would kill the whole piece. Then add up one such vector from each piece: the annihilators are powers of distinct irreducibles, hence pairwise coprime, so Lemma 9.5.4 multiplies them back into  $m_T$ .

*Proof.* Write  $m_T = p_1^{e_1} \cdots p_k^{e_k}$  with  $p_1, \dots, p_k$  distinct monic irreducibles and  $e_i \geq 1$  (Theorem 5.3.5;  $\deg m_T \geq 1$  because  $V \neq \{\mathbf{0}\}$ , by Theorem 8.5.1). By Theorem 8.8.1,

$V = V_1 \oplus \cdots \oplus V_k$  with  $V_i = \ker p_i(T)^{e_i}$  invariant and  $m_{T|_{V_i}} = p_i^{e_i}$ .

Fix  $i$ . *Claim: some  $\mathbf{v}_i \in V_i$  has  $m_{T, \mathbf{v}_i} = p_i^{e_i}$ .* For  $\mathbf{v} \in V_i$  we have  $p_i^{e_i}(T)\mathbf{v} = \mathbf{0}$  by the definition of  $V_i$ , so  $m_{T, \mathbf{v}} \mid p_i^{e_i}$  and hence  $m_{T, \mathbf{v}} = p_i^f$  with  $0 \leq f \leq e_i$  (Lemma 8.5.2). Suppose no  $\mathbf{v} \in V_i$  has  $f = e_i$ . Then  $m_{T, \mathbf{v}} \mid p_i^{e_i-1}$  for every  $\mathbf{v} \in V_i$ , so  $p_i^{e_i-1}(T)\mathbf{v} = \mathbf{0}$  for every  $\mathbf{v} \in V_i$ , that is,  $p_i^{e_i-1}(T|_{V_i}) = 0$  (Exercise (C2: Minimal polynomials of the pieces) (a)). By Theorem 8.5.1 this forces  $p_i^{e_i} = m_{T|_{V_i}}$  to divide  $p_i^{e_i-1}$ , which is false by degrees (Proposition 5.1.2 (c)). This proves the claim.

Pick such a  $\mathbf{v}_i$  for each  $i$  and put  $\mathbf{v} = \mathbf{v}_1 + \cdots + \mathbf{v}_k$ . The polynomials  $p_1^{e_1}, \dots, p_k^{e_k}$  are pairwise coprime: a monic common divisor of  $p_i^{e_i}$  and  $p_j^{e_j}$  with  $i \neq j$  is both a power of  $p_i$  and a power of  $p_j$  (Lemma 8.5.2), hence is 1 by the uniqueness of factorization (Theorem 5.3.5 (b)). So Lemma 9.5.4 gives

$$m_{T, \mathbf{v}} = p_1^{e_1} \cdots p_k^{e_k} = m_T,$$

as claimed. ■

*An operator is generated by one vector exactly when its minimal polynomial is as large as it can be.*

### Theorem 9.5.6 — Cyclic Operators

Let  $V$  be a finite-dimensional vector space over  $F$  with  $\dim V = n \geq 1$ , and let  $T \in \mathcal{L}(V)$ . The following are equivalent:

- (a)  $T$  is cyclic, that is,  $T$  has a cyclic vector;
- (b)  $\deg m_T = n$ ;
- (c)  $m_T = p_T$ .

In that case  $\mathbf{v}$  is a cyclic vector if and only if  $m_{T, \mathbf{v}} = m_T$ , and in the basis  $(\mathbf{v}, T\mathbf{v}, \dots, T^{n-1}\mathbf{v})$  the matrix of  $T$  is  $\mathbf{C}(m_T)$ . The same holds for  $\mathbf{A} \in M_n(F)$ .

*Proof.*

- (b)  $\Leftrightarrow$  (c). Always  $m_T \mid p_T$  with both monic and  $\deg p_T = n$  (Corollary 8.6.2, Theorem 6.10.2). So  $\deg m_T = n$  if and only if  $m_T = p_T$  (Proposition 5.1.2 (c), (d)).
- (c)  $\Rightarrow$  (b). Let  $\mathbf{v}$  be cyclic. Then  $\mathbf{v} \neq \mathbf{0}$ , since  $Z(\mathbf{0}; T) = \{\mathbf{0}\} \neq V$ , and by Theorem 9.5.2 (a), (d),

$$n = \dim V = \dim Z(\mathbf{v}; T) = \deg m_{T, \mathbf{v}} \leq \deg m_T \leq n.$$

So all these numbers are equal; in particular  $\deg m_T = n$  and  $m_{T, \mathbf{v}} = m_T$ , the two being monic with  $m_{T, \mathbf{v}} \mid m_T$  of equal degree.

- (d)  $\Rightarrow$  (a). By Proposition 9.5.5 there is  $\mathbf{v}$  with  $m_{T, \mathbf{v}} = m_T$ , and then  $\dim Z(\mathbf{v}; T) = \deg m_T = n$  by Theorem 9.5.2 (a). A subspace of the same dimension as  $V$  is  $V$  (Theorem 1.6.8), so  $Z(\mathbf{v}; T) = V$  and  $\mathbf{v}$  is cyclic.

For the final statements, the computation in (a)  $\Rightarrow$  (b) shows a cyclic vector has  $m_{T, \mathbf{v}} =$



$m_T$ ; conversely if  $m_{T,\mathbf{v}} = m_T$  and (b) holds, the argument of (b)  $\Rightarrow$  (a) shows  $\mathbf{v}$  is cyclic. The matrix is then  $\mathbf{C}(m_{T,\mathbf{v}}) = \mathbf{C}(m_T)$  by Theorem 9.5.2 (b). For a matrix  $\mathbf{A}$ , apply all of this to  $T_{\mathbf{A}}$ , using  $m_{\mathbf{A}} = m_{T_{\mathbf{A}}}$  and  $p_{\mathbf{A}} = p_{T_{\mathbf{A}}}$ . ■

The implication (a)  $\Rightarrow$  (b) was Exercise (C1: Cyclic vectors) (a) in Chapter 8, where the converse was promised for this chapter; the promise is now paid off, and the proof shows what the missing vector is made of — one contribution from each primary piece, chosen to need the full exponent. Two consequences are worth naming. First, an operator with  $n$  distinct eigenvalues is cyclic, since then  $m_T = p_T$  has  $n$  distinct roots (Theorem 8.8.2). Second, **a matrix is similar to a companion matrix if and only if  $m_{\mathbf{A}} = p_{\mathbf{A}}$** : one direction is the last sentence of the theorem, and the other follows from Theorem 9.5.3 together with the fact that similar matrices share both polynomials. Section 6 makes this the first case of a general normal form.

*Warning.* **A cyclic vector need not exist, and when it exists it is far from unique.** For  $T = \text{id}$  on  $F^2$  we have  $m_T = x - 1$  of degree  $1 < 2$ , so no vector is cyclic: indeed  $Z(\mathbf{v}; \text{id}) = \text{span}(\mathbf{v})$  is a line for every  $\mathbf{v} \neq \mathbf{0}$ . At the other extreme, if  $\mathbf{v}$  is cyclic, then so is  $c\mathbf{v}$  for every  $c \neq 0$ , and usually many more vectors are: for  $\mathbf{J} = \begin{pmatrix} 1 & 1 \\ 0 & 1 \end{pmatrix}$  over  $\mathbb{Q}$ , a vector  $(s, t)$  is cyclic exactly when  $t \neq 0$ , since  $((s, t), (s + t, t))$  has determinant  $-t^2$ . So “the” cyclic vector is never a well-defined object; only the polynomial  $m_T$  is.

### Example. (A Cyclic Subspace and a Cyclic Vector)

Let

$$\mathbf{A} = \begin{pmatrix} 1 & 1 & 0 \\ 0 & 1 & 0 \\ 1 & 0 & 2 \end{pmatrix} \in M_3(\mathbb{Q}).$$

- Find  $p_{\mathbf{A}}$  and  $m_{\mathbf{A}}$ .
- Find  $m_{\mathbf{A}, \mathbf{e}_1}$ , the subspace  $Z(\mathbf{e}_1; \mathbf{A})$  and the matrix of  $T_{\mathbf{A}}$  restricted to it.
- Find a cyclic vector for  $\mathbf{A}$  and an invertible  $\mathbf{P}$  with  $\mathbf{P}^{-1}\mathbf{A}\mathbf{P}$  a companion matrix.

*Solution.*

- Expanding  $\det(x\mathbf{I} - \mathbf{A})$  along the second row, whose only non-zero entry is  $x - 1$  in position  $(2, 2)$  (Theorem 6.6.1),

$$p_{\mathbf{A}} = (x - 1) \det \begin{pmatrix} x - 1 & 0 \\ -1 & x - 2 \end{pmatrix} = (x - 1)^2(x - 2).$$

For  $m_{\mathbf{A}}$ , the roots 1 and 2 must both occur (Theorem 8.5.3), so the first candidate is  $(x - 1)(x - 2)$ . With

$$\mathbf{A} - \mathbf{I} = \begin{pmatrix} 0 & 1 & 0 \\ 0 & 0 & 0 \\ 1 & 0 & 1 \end{pmatrix}, \quad \mathbf{A} - 2\mathbf{I} = \begin{pmatrix} -1 & 1 & 0 \\ 0 & -1 & 0 \\ 1 & 0 & 0 \end{pmatrix},$$

the product  $(\mathbf{A} - \mathbf{I})(\mathbf{A} - 2\mathbf{I})$  has rows: row 1 of  $\mathbf{A} - \mathbf{I}$  is  $(0, 1, 0)$ , which picks out row 2 of  $\mathbf{A} - 2\mathbf{I}$ , namely  $(0, -1, 0)$ ; row 2 is zero; row 3 is  $(1, 0, 1)$ , which gives row



1 plus row 3 of  $\mathbf{A} - 2\mathbf{I}$ , namely  $(0, 1, 0)$ . So

$$(\mathbf{A} - \mathbf{I})(\mathbf{A} - 2\mathbf{I}) = \begin{pmatrix} 0 & -1 & 0 \\ 0 & 0 & 0 \\ 0 & 1 & 0 \end{pmatrix} \neq 0,$$

and by [Lemma 8.5.2](#) the only remaining candidate is  $m_{\mathbf{A}} = (x - 1)^2(x - 2) = p_{\mathbf{A}}$ . So  $\mathbf{A}$  is cyclic by [Theorem 9.5.6](#).

(b)  $\mathbf{A}\mathbf{e}_1 = (1, 0, 1)$ , the first column, and

$$\mathbf{A}^2\mathbf{e}_1 = \mathbf{A}(1, 0, 1) = (\text{column 1}) + (\text{column 3}) = (1, 0, 1) + (0, 0, 2) = (1, 0, 3).$$

Now  $(1, 0, 3) = -2\mathbf{e}_1 + 3(1, 0, 1)$ , so  $(\mathbf{A}^2 - 3\mathbf{A} + 2\mathbf{I})\mathbf{e}_1 = \mathbf{0}$  and  $m_{\mathbf{A}, \mathbf{e}_1} \mid x^2 - 3x + 2 = (x - 1)(x - 2)$ . Neither factor kills  $\mathbf{e}_1$ :  $(\mathbf{A} - \mathbf{I})\mathbf{e}_1 = (0, 0, 1) \neq \mathbf{0}$  and  $(\mathbf{A} - 2\mathbf{I})\mathbf{e}_1 = (-1, 0, 1) \neq \mathbf{0}$ . Hence

$$m_{\mathbf{A}, \mathbf{e}_1} = (x - 1)(x - 2) = x^2 - 3x + 2,$$

and by [Theorem 9.5.2](#),  $Z(\mathbf{e}_1; \mathbf{A}) = \text{span}(\mathbf{e}_1, (1, 0, 1)) = \text{span}(\mathbf{e}_1, \mathbf{e}_3)$ , of dimension 2, with

$$[T_{\mathbf{A}}|_{Z(\mathbf{e}_1; \mathbf{A})}] = \mathbf{C}(x^2 - 3x + 2) = \begin{pmatrix} 0 & -2 \\ 1 & 3 \end{pmatrix}$$

in the basis  $(\mathbf{e}_1, \mathbf{A}\mathbf{e}_1)$ . In particular  $\mathbf{e}_1$  is **not** cyclic for  $\mathbf{A}$ : its cyclic subspace is a plane, not  $\mathbb{Q}^3$ .

(c) Try  $\mathbf{v} = \mathbf{e}_2$ . Then

$$\mathbf{A}\mathbf{e}_2 = (1, 1, 0), \quad \mathbf{A}^2\mathbf{e}_2 = \mathbf{A}(1, 1, 0) = (1, 0, 1) + (1, 1, 0) = (2, 1, 1).$$

The matrix with these three columns is

$$\mathbf{P} = \begin{pmatrix} 0 & 1 & 2 \\ 1 & 1 & 1 \\ 0 & 0 & 1 \end{pmatrix}, \quad \det \mathbf{P} = -\det \begin{pmatrix} 1 & 2 \\ 0 & 1 \end{pmatrix} = -1 \neq 0,$$

expanding along the first column, whose only non-zero entry is the 1 in position  $(2, 1)$ . So  $(\mathbf{e}_2, \mathbf{A}\mathbf{e}_2, \mathbf{A}^2\mathbf{e}_2)$  is a basis and  $\mathbf{e}_2$  is a cyclic vector. By [Theorem 9.5.6](#),  $m_{\mathbf{A}, \mathbf{e}_2} = m_{\mathbf{A}} = (x - 1)^2(x - 2) = x^3 - 4x^2 + 5x - 2$  and

$$\mathbf{P}^{-1}\mathbf{A}\mathbf{P} = \mathbf{C}(m_{\mathbf{A}}) = \begin{pmatrix} 0 & 0 & 2 \\ 1 & 0 & -5 \\ 0 & 1 & 4 \end{pmatrix}.$$

*The check.* The identity  $\mathbf{A}\mathbf{P} = \mathbf{P}\mathbf{C}(m_{\mathbf{A}})$  reads, column by column,  $\mathbf{A}\mathbf{e}_2 = \mathbf{A}\mathbf{e}_2$ ,  $\mathbf{A}(\mathbf{A}\mathbf{e}_2) = \mathbf{A}^2\mathbf{e}_2$  — both true by construction — and

$$\mathbf{A}(\mathbf{A}^2\mathbf{e}_2) = 2\mathbf{e}_2 - 5\mathbf{A}\mathbf{e}_2 + 4\mathbf{A}^2\mathbf{e}_2,$$

which is the annihilator relation. Verifying it:  $\mathbf{A}(2, 1, 1) = 2(1, 0, 1) + (1, 1, 0) + (0, 0, 2) = (3, 1, 4)$ , while

$$2(0, 1, 0) - 5(1, 1, 0) + 4(2, 1, 1) = (0 - 5 + 8, 2 - 5 + 4, 4) = (3, 1, 4). \checkmark$$

## 9.5.5 Exercises

### 9.5.5.1 A. Check your understanding

#### Exercise A1.

- Define  $Z(\mathbf{v}; T)$  and the  $T$ -annihilator  $m_{T, \mathbf{v}}$ .
- State the formula for  $\dim Z(\mathbf{v}; T)$  and say which basis of  $Z(\mathbf{v}; T)$  the theorem provides.
- Write down  $\mathbf{C}(p)$  for  $p = x^3 - 2x + 5$ , and say what  $p_{\mathbf{C}(p)}$  and  $m_{\mathbf{C}(p)}$  are.
- True or false: every operator on a space of dimension  $n \geq 2$  has a cyclic vector. Justify your answer.
- What is  $m_{T, \mathbf{v}}$  when  $\mathbf{v}$  is an eigenvector of  $T$  for  $\lambda$ ? What is  $Z(\mathbf{v}; T)$  then?
- True or false: if  $m_{T, \mathbf{v}} = m_T$  for some  $\mathbf{v}$ , then  $\mathbf{v}$  is a cyclic vector. Justify your answer.

*Solution.*

- $Z(\mathbf{v}; T) = \text{span}(\mathbf{v}, T\mathbf{v}, T^2\mathbf{v}, \dots) = \{p(T)\mathbf{v} : p \in F[x]\}$  (Definition 9.5.1, Proposition 9.5.1);  $m_{T, \mathbf{v}}$  is the monic generator of the ideal  $\{p : p(T)\mathbf{v} = \mathbf{0}\}$ , equivalently the monic polynomial of least degree killing  $\mathbf{v}$  (Definition 9.5.2).
- $\dim Z(\mathbf{v}; T) = \deg m_{T, \mathbf{v}} = k$ , with basis  $(\mathbf{v}, T\mathbf{v}, \dots, T^{k-1}\mathbf{v})$  (Theorem 9.5.2 (a)).
- Here  $a_0 = 5$ ,  $a_1 = -2$ ,  $a_2 = 0$ , so

$$\mathbf{C}(p) = \begin{pmatrix} 0 & 0 & -5 \\ 1 & 0 & 2 \\ 0 & 1 & 0 \end{pmatrix},$$

and  $p_{\mathbf{C}(p)} = m_{\mathbf{C}(p)} = x^3 - 2x + 5$  (Theorem 9.5.3).

- False.  $\text{id}_V$  has  $m_T = x - 1$  of degree  $1 < n$ , so no cyclic vector (Theorem 9.5.6). More generally any  $T$  with  $\deg m_T < n$  fails.
- $m_{T, \mathbf{v}} = x - \lambda$ , and  $Z(\mathbf{v}; T) = \text{span}(\mathbf{v})$  is the eigenline.
- False. By Theorem 9.5.2 (a),  $\dim Z(\mathbf{v}; T) = \deg m_{T, \mathbf{v}} = \deg m_T$ , which need not be  $n$ . For  $T = \text{id}$  on  $F^2$  and  $\mathbf{v} = \mathbf{e}_1$  we have  $m_{T, \mathbf{v}} = x - 1 = m_T$ , yet  $Z(\mathbf{v}; T) = \text{span}(\mathbf{e}_1)$  is a line. The statement is true when  $\deg m_T = \dim V$  (Theorem 9.5.6).

### 9.5.5.2 B. Practice

#### Exercise B1: Annihilators of vectors.

Let

$$\mathbf{A} = \begin{pmatrix} 1 & 1 & 1 \\ 0 & 1 & 1 \\ 0 & 0 & 2 \end{pmatrix} \in M_3(\mathbb{Q}).$$

- Write down  $p_{\mathbf{A}}$ .
- Find  $m_{\mathbf{A}, \mathbf{e}_1}$ ,  $m_{\mathbf{A}, \mathbf{e}_2}$  and  $m_{\mathbf{A}, \mathbf{e}_3}$ .
- For each of the three vectors, give  $\dim Z(\mathbf{e}_j; \mathbf{A})$  and identify the subspace.

- (d) Write down the matrix of  $T_{\mathbf{A}}$  restricted to  $Z(\mathbf{e}_2; \mathbf{A})$  in the basis  $(\mathbf{e}_2, \mathbf{A}\mathbf{e}_2)$ , and say which companion matrix it is.

*Solution.*

- (a)  $\mathbf{A}$  is upper triangular, so  $p_{\mathbf{A}} = (x-1)^2(x-2)$  (Theorem 6.4.6).

- (b)  $\mathbf{A}\mathbf{e}_1 = \mathbf{e}_1$ , so  $\mathbf{e}_1$  is an eigenvector and  $m_{\mathbf{A}, \mathbf{e}_1} = x-1$ .

$\mathbf{A}\mathbf{e}_2 = (1, 1, 0) = \mathbf{e}_1 + \mathbf{e}_2$ , so  $(\mathbf{A} - \mathbf{I})\mathbf{e}_2 = \mathbf{e}_1 \neq \mathbf{0}$  and  $(\mathbf{A} - \mathbf{I})^2\mathbf{e}_2 = (\mathbf{A} - \mathbf{I})\mathbf{e}_1 = \mathbf{0}$ . Hence  $(x-1)^2$  kills  $\mathbf{e}_2$  but  $x-1$  does not, and  $m_{\mathbf{A}, \mathbf{e}_2} = (x-1)^2$  by Lemma 8.5.2.

$\mathbf{A}\mathbf{e}_3 = (1, 1, 2)$  and  $\mathbf{A}^2\mathbf{e}_3 = \mathbf{A}(1, 1, 2) = (1, 0, 0) + (1, 1, 0) + 2(1, 1, 2) = (4, 3, 4)$ .

The matrix with columns  $\mathbf{e}_3, (1, 1, 2), (4, 3, 4)$  is  $\begin{pmatrix} 0 & 1 & 4 \\ 0 & 1 & 3 \\ 1 & 2 & 4 \end{pmatrix}$ , of determinant  $\det \begin{pmatrix} 1 & 4 \\ 1 & 3 \end{pmatrix} = -1 \neq 0$  by expansion along the first column. So the three vectors are independent,  $\deg m_{\mathbf{A}, \mathbf{e}_3} = 3$ , and since  $m_{\mathbf{A}, \mathbf{e}_3} \mid m_{\mathbf{A}} \mid p_{\mathbf{A}}$  with  $\deg p_{\mathbf{A}} = 3$  we get  $m_{\mathbf{A}, \mathbf{e}_3} = p_{\mathbf{A}} = (x-1)^2(x-2)$ .

- (c)  $\dim Z(\mathbf{e}_1; \mathbf{A}) = 1$  and  $Z(\mathbf{e}_1; \mathbf{A}) = \text{span}(\mathbf{e}_1)$ .  $\dim Z(\mathbf{e}_2; \mathbf{A}) = 2$  and  $Z(\mathbf{e}_2; \mathbf{A}) = \text{span}(\mathbf{e}_2, \mathbf{e}_1 + \mathbf{e}_2) = \text{span}(\mathbf{e}_1, \mathbf{e}_2)$ .  $\dim Z(\mathbf{e}_3; \mathbf{A}) = 3$ , so  $Z(\mathbf{e}_3; \mathbf{A}) = \mathbb{Q}^3$  and  $\mathbf{e}_3$  is a cyclic vector.
- (d)  $(x-1)^2 = x^2 - 2x + 1$ , so by Theorem 9.5.2 (b) the matrix is

$$\mathbf{C}((x-1)^2) = \begin{pmatrix} 0 & -1 \\ 1 & 2 \end{pmatrix}.$$

### Exercise B2: Cyclic or not?

Determine whether each matrix is cyclic. For those that are, find a cyclic vector; for those that are not, say which condition fails.

- (a)  $\text{diag}(1, 2, 3) \in M_3(\mathbb{Q})$ .
- (b)  $\text{diag}(1, 1, 2) \in M_3(\mathbb{Q})$ .
- (c)  $\mathbf{B} = \begin{pmatrix} 2 & 1 & 0 \\ 0 & 2 & 0 \\ 1 & 1 & 1 \end{pmatrix} \in M_3(\mathbb{Q})$ , for which  $p_{\mathbf{B}} = m_{\mathbf{B}} = (x-2)^2(x-1)$ . Check first that none of  $\mathbf{e}_1, \mathbf{e}_2, \mathbf{e}_3$  is cyclic.
- (d)  $\mathbf{J}_2(0) \oplus \mathbf{J}_1(0) \in M_3(\mathbb{Q})$ .

*Hint: for (c), compute the three annihilators  $m_{\mathbf{B}, \mathbf{e}_j}$  and use Lemma 9.5.4.*

*Solution.*

- (a) Cyclic. The three eigenvalues are distinct, so  $m = (x-1)(x-2)(x-3) = p$  (Theorem 8.8.2), of degree 3. A cyclic vector is  $\mathbf{v} = (1, 1, 1)$ : the matrix with columns  $\mathbf{v}, \mathbf{A}\mathbf{v} = (1, 2, 3), \mathbf{A}^2\mathbf{v} = (1, 4, 9)$  is a Vandermonde matrix with distinct nodes 1, 2, 3, hence invertible (Corollary 6.8.2).
- (b) Not cyclic. Here  $m = (x-1)(x-2)$  has degree  $2 < 3$ , so condition (b) of Theorem 9.5.6 fails.
- (c) The standard basis vectors.  $\mathbf{B}\mathbf{e}_1 = (2, 0, 1)$  and  $\mathbf{B}^2\mathbf{e}_1 = \mathbf{B}(2, 0, 1) = 2(2, 0, 1) + (0, 0, 1) = (4, 0, 3)$ . Since  $(4, 0, 3) = -2\mathbf{e}_1 + 3(2, 0, 1)$ , the vector  $\mathbf{e}_1$  is killed by

$x^2 - 3x + 2 = (x - 1)(x - 2)$ , and by neither factor alone  $((\mathbf{B} - \mathbf{I})\mathbf{e}_1 = (1, 0, 1) \neq \mathbf{0}$ ,  $(\mathbf{B} - 2\mathbf{I})\mathbf{e}_1 = (0, 0, 1) \neq \mathbf{0}$ ); so  $m_{\mathbf{B}, \mathbf{e}_1} = (x - 1)(x - 2)$ , of degree 2.

$\mathbf{B}\mathbf{e}_2 = (1, 2, 1)$  and  $\mathbf{B}^2\mathbf{e}_2 = \mathbf{B}(1, 2, 1) = (2, 0, 1) + 2(1, 2, 1) + (0, 0, 1) = (4, 4, 4)$ . As  $(4, 4, 4) = -4\mathbf{e}_2 + 4(1, 2, 1)$ , the vector  $\mathbf{e}_2$  is killed by  $x^2 - 4x + 4 = (x - 2)^2$ , and not by  $x - 2$ ; so  $m_{\mathbf{B}, \mathbf{e}_2} = (x - 2)^2$ .

$\mathbf{B}\mathbf{e}_3 = \mathbf{e}_3$ , so  $m_{\mathbf{B}, \mathbf{e}_3} = x - 1$ .

All three degrees are less than 3, so none of  $\mathbf{e}_1, \mathbf{e}_2, \mathbf{e}_3$  is cyclic (Theorem 9.5.2 (a)).

A *cyclic vector*. The annihilators  $(x - 2)^2$  of  $\mathbf{e}_2$  and  $x - 1$  of  $-\mathbf{e}_3$  — note  $m_{T, -\mathbf{v}} = m_{T, \mathbf{v}}$ , since  $p(T)(-\mathbf{v}) = -p(T)\mathbf{v}$  — are coprime, so Lemma 9.5.4 gives

$$m_{\mathbf{B}, \mathbf{e}_2 - \mathbf{e}_3} = (x - 2)^2(x - 1) = m_{\mathbf{B}}.$$

By Theorem 9.5.6,  $\mathbf{v} = \mathbf{e}_2 - \mathbf{e}_3 = (0, 1, -1)$  is a cyclic vector. As a check,  $\mathbf{B}\mathbf{v} = (1, 2, 1) - (0, 0, 1) = (1, 2, 0)$  and  $\mathbf{B}^2\mathbf{v} = \mathbf{B}(1, 2, 0) = (2, 0, 1) + 2(1, 2, 1) = (4, 4, 3)$ , and the matrix with columns  $(0, 1, -1), (1, 2, 0), (4, 4, 3)$  has determinant

$$-\det \begin{pmatrix} 1 & 4 \\ -1 & 3 \end{pmatrix} + 4 \det \begin{pmatrix} 1 & 2 \\ -1 & 0 \end{pmatrix} = -(3 + 4) + 4 \cdot 2 = 1 \neq 0$$

by expansion along the first row.

(d) Not cyclic:  $m = x^2$  (Proposition 8.5.6) has degree  $2 < 3$ .

### Exercise B3: Cyclic subspaces for differentiation.

Let  $V = \mathbb{R}[x]_{\leq 3}$  and let  $D \in \mathcal{L}(V)$  be differentiation.

- (a) Find  $m_{D, q}$  for  $q = 1$ ,  $q = x$ ,  $q = x^2$  and  $q = x^3$ .
- (b) Find  $Z(x^2; D)$  and the matrix of  $D$  restricted to it in the basis  $(x^2, Dx^2, D^2x^2)$ .
- (c) Is  $x^3$  a cyclic vector for  $D$ ? What is  $m_D$ ?

*Solution.*

- (a)  $D1 = 0$ , so  $m_{D, 1} = x$ .  $Dx = 1 \neq 0$  and  $D^2x = 0$ , so  $m_{D, x} = x^2$ . Likewise  $D^2x^2 = 2 \neq 0$  and  $D^3x^2 = 0$ , so  $m_{D, x^2} = x^3$ ; and  $D^3x^3 = 6 \neq 0$  with  $D^4x^3 = 0$ , so  $m_{D, x^3} = x^4$ . (In each case the annihilator is a monic divisor of  $x^4$ , hence a power of  $x$ , by Lemma 8.5.2.)
- (b)  $Z(x^2; D) = \text{span}(x^2, 2x, 2) = \mathbb{R}[x]_{\leq 2}$ , of dimension  $3 = \deg m_{D, x^2}$ . With  $\mathbf{z}_1 = x^2$ ,  $\mathbf{z}_2 = Dx^2 = 2x$ ,  $\mathbf{z}_3 = D^2x^2 = 2$  we have  $D\mathbf{z}_1 = \mathbf{z}_2$ ,  $D\mathbf{z}_2 = \mathbf{z}_3$ ,  $D\mathbf{z}_3 = 0$ , so the matrix is

$$\mathbf{C}(x^3) = \begin{pmatrix} 0 & 0 & 0 \\ 1 & 0 & 0 \\ 0 & 1 & 0 \end{pmatrix} = \mathbf{J}_3(0)^\top.$$

- (c) Yes:  $\dim Z(x^3; D) = \deg m_{D, x^3} = 4 = \dim V$ . Hence  $m_D = m_{D, x^3} = x^4 = p_D$  by Theorem 9.5.6, which agrees with Example (Differentiation Is Nilpotent).

### 9.5.5.3 C. Going deeper

#### Exercise C1: What commutes with a cyclic operator.

Let  $V$  be finite-dimensional with  $\dim V = n \geq 1$  and let  $T \in \mathcal{L}(V)$  satisfy  $m_T = p_T$ .

- (a) Prove that every  $S \in \mathcal{L}(V)$  with  $ST = TS$  is a polynomial in  $T$ , that is,  $S = q(T)$  for some  $q \in F[x]$ .
- (b) Deduce that the set  $\{S \in \mathcal{L}(V) : ST = TS\}$  is a subspace of  $\mathcal{L}(V)$  of dimension exactly  $n$ .
- (c) Show that (a) fails without the hypothesis  $m_T = p_T$ , by taking  $T = \text{id}_V$  with  $n \geq 2$ .

*Hint for (a): apply  $S$  to a cyclic vector.*

*Solution.*

- (a) By [Theorem 9.5.6](#) there is a cyclic vector  $\mathbf{v}$ , and  $\mathcal{B} = (\mathbf{v}, T\mathbf{v}, \dots, T^{n-1}\mathbf{v})$  is a basis of  $V$ . Since  $\mathcal{B}$  spans  $V$ , there is  $q \in F[x]$  with  $S\mathbf{v} = q(T)\mathbf{v}$  ([Proposition 9.5.1](#) (a)). For each  $j \geq 0$ , using  $ST = TS$  and hence  $ST^j = T^jS$  (induction on  $j$ ), and that polynomials in  $T$  commute with  $T$  ([Theorem 5.7.1](#) (c)),

$$S(T^j\mathbf{v}) = T^j(S\mathbf{v}) = T^jq(T)\mathbf{v} = q(T)(T^j\mathbf{v}).$$

So  $S$  and  $q(T)$  agree on the basis  $\mathcal{B}$ , hence  $S = q(T)$  ([Theorem 3.1.5](#)).

- (b) The set is a subspace: it contains 0, and if  $S_1T = TS_1$  and  $S_2T = TS_2$ , then  $(S_1 + cS_2)T = T(S_1 + cS_2)$ . By (a) it equals  $\{q(T) : q \in F[x]\} = \text{span}(\text{id}_V, T, \dots, T^{n-1})$ , where we used [Theorem 5.1.1](#) to reduce  $q$  modulo  $m_T$ , whose degree is  $n$ . That spanning list is independent: a vanishing combination would give a non-zero polynomial of degree less than  $n = \deg m_T$  annihilating  $T$ , contradicting [Theorem 8.5.1](#). So the dimension is  $n$ .
- (c) For  $T = \text{id}_V$  every  $S$  commutes with  $T$ , so the commutant is all of  $\mathcal{L}(V)$ , of dimension  $n^2 \geq 4$ . But the polynomials in  $\text{id}_V$  are just the scalar operators  $c \text{id}_V$ , a subspace of dimension 1. Here  $m_T = x - 1 \neq (x - 1)^n = p_T$ .

### Exercise C2: How many cyclic vectors?.

Let  $V$  be finite-dimensional with  $\dim V = n \geq 1$  and  $T \in \mathcal{L}(V)$ .

- (a) Prove that  $\mathbf{v}$  is a cyclic vector for  $T$  if and only if  $\mathbf{v}$  lies in no  $T$ -invariant subspace  $U$  with  $U \neq V$ .
- (b) Prove that if  $\mathbf{v}$  is cyclic, then so is  $c\mathbf{v}$  for every  $c \in F$  with  $c \neq 0$ . Deduce that over an infinite field a cyclic operator has infinitely many cyclic vectors.
- (c) Let  $F = \mathbb{F}_2$  and  $T = T_{\mathbf{J}}$  with  $\mathbf{J} = \begin{pmatrix} 1 & 1 \\ 0 & 1 \end{pmatrix}$ . Find **all** cyclic vectors of  $T$ .
- (d) Deduce from (c) that the set of cyclic vectors, together with  $\mathbf{0}$ , need not be a subspace.

*Solution.*

- (a)  $(\Rightarrow)$  Let  $\mathbf{v}$  be cyclic and let  $U$  be  $T$ -invariant with  $\mathbf{v} \in U$ . By [Proposition 9.5.1](#) (c),  $V = Z(\mathbf{v}; T) \subseteq U$ , so  $U = V$ .  $(\Leftarrow)$  If  $\mathbf{v}$  is not cyclic, then  $U = Z(\mathbf{v}; T)$  is itself a  $T$ -invariant subspace containing  $\mathbf{v}$  with  $U \neq V$  ([Proposition 9.5.1](#) (b)).

- (b) For  $c \neq 0$ ,  $p(T)(c\mathbf{v}) = cp(T)\mathbf{v}$ , so  $p(T)(c\mathbf{v}) = \mathbf{0}$  if and only if  $p(T)\mathbf{v} = \mathbf{0}$  (as  $c \neq 0$  and by [Theorem 1.1.9](#)). Hence  $m_{T,c\mathbf{v}} = m_{T,\mathbf{v}}$ , and [Theorem 9.5.6](#) makes  $c\mathbf{v}$  cyclic as well. Over an infinite field the vectors  $c\mathbf{v}$ ,  $c \neq 0$ , are pairwise distinct (if  $c\mathbf{v} = c'\mathbf{v}$  with  $\mathbf{v} \neq \mathbf{0}$ , then  $c = c'$ ), so there are infinitely many.
- (c) Write  $\mathbf{v} = (s, t)$  with  $s, t \in \mathbb{F}_2$ . Then  $\mathbf{J}\mathbf{v} = (s+t, t)$ , and  $(\mathbf{v}, \mathbf{J}\mathbf{v})$  is a basis of  $\mathbb{F}_2^2$  if and only if

$$\det \begin{pmatrix} s & s+t \\ t & t \end{pmatrix} = st - t(s+t) = -t^2 = t^2 \neq 0,$$

that is, if and only if  $t = 1$ . So the cyclic vectors are exactly  $(0, 1)$  and  $(1, 1)$ , two of the four vectors of  $\mathbb{F}_2^2$ .

- (d) The set  $\{\mathbf{0}, (0, 1), (1, 1)\}$  has 3 elements, and a subspace of  $\mathbb{F}_2^2$  has 1, 2 or 4 elements. Concretely,  $(0, 1) + (1, 1) = (1, 0)$ , which is an eigenvector of  $\mathbf{J}$  and hence not cyclic. So the set is not closed under addition.

## 9.6 The Rational Canonical Form

Everything so far has needed the characteristic polynomial to split. Section 5 removed that need from the building block: a companion matrix  $\mathbf{C}(p)$  exists for every monic  $p$ , and it is the matrix of an operator on a cyclic subspace. What is left is to cut an arbitrary space into cyclic pieces. This section does that, over an **arbitrary field**, and the result is a genuine canonical form: a list of polynomials attached to  $T$ , from which the matrix is written down mechanically, and which two operators share exactly when they are similar. Along the way we learn that similarity does not change if the field is enlarged.

### 9.6.1 The plan

The goal, stated before its parts are named:

**Every operator on a non-zero finite-dimensional space decomposes the space into cyclic subspaces  $Z(\mathbf{v}_1; T), \dots, Z(\mathbf{v}_r; T)$  whose annihilators form a divisibility chain  $d_1 \mid d_2 \mid \dots \mid d_r = m_T$ , and that chain is determined by  $T$ .**

In the adapted basis the matrix is then  $\mathbf{C}(d_1) \oplus \dots \oplus \mathbf{C}(d_r)$ , by [Theorem 9.5.2](#) (b) and [Theorem 8.1.5](#). So the whole content is the decomposition, and it is the hardest theorem of the chapter. The route has three steps, and each one is a tool we already have.

- **Step 1 (one prime at a time).** Chapter 8's primary decomposition splits  $V = V_1 \oplus \dots \oplus V_k$ , where  $m_{T|V_i} = p_i^{e_i}$  is a power of a single monic irreducible ([Theorem 8.8.1](#)). Nothing more is needed from the general theory.
- **Step 2 (inside one prime power).** On a piece with  $m_T = p^e$ , the operator  $P = p(T)$  is nilpotent, and the structure theorem for nilpotent operators was proved by induction, with  $\text{im } N$  as the smaller space to which the hypothesis applies ([Theorem 9.2.3](#)). We run the same induction with  $P$  in place of  $N$  and cyclic subspaces  $Z(\mathbf{u}; T)$  in place of the spans of chains. The outcome is a decomposition into cyclic pieces whose annihilators are powers  $p^{e_1} \geq p^{e_2} \geq \dots$  of the one prime.
- **Step 3 (regroup across primes).** A cyclic piece for  $p_1$  and a cyclic piece for  $p_2$  can be **merged**: the sum of their generators has annihilator the product, by [Lemma 9.5.4](#). Merging the largest piece of every prime gives  $d_r = m_T$ , the next largest give  $d_{r-1}$ , and so

on; the divisibility chain is automatic because within one prime the exponents decrease.

Uniqueness is a count, exactly as for the nilpotent case: the dimensions  $\dim \ker p(T)^k$  are visible in the decomposition and see nothing but the exponents.

## 9.6.2 One prime at a time

Here is Step 2. Compare it with [Theorem 9.2.3](#), which is the special case  $p = x$ : there the pieces are spanned by Jordan chains  $\mathbf{u}, N\mathbf{u}, \dots$ , and here by  $\mathbf{u}, T\mathbf{u}, T^2\mathbf{u}, \dots$

### Theorem 9.6.1 — Cyclic Decomposition for a Single Prime Power

Let  $V \neq \{0\}$  be a finite-dimensional vector space over  $F$ , let  $T \in \mathcal{L}(V)$ , and suppose

$$m_T = p^e, \quad p \in F[x] \text{ monic irreducible of degree } d, \quad e \geq 1.$$

Then there are an integer  $m \geq 1$ , vectors  $\mathbf{u}_1, \dots, \mathbf{u}_m \in V$  and exponents

$$e = k_1 \geq k_2 \geq \dots \geq k_m \geq 1$$

such that

$$V = Z(\mathbf{u}_1; T) \oplus Z(\mathbf{u}_2; T) \oplus \dots \oplus Z(\mathbf{u}_m; T), \quad m_{T, \mathbf{u}_i} = p^{k_i} \quad (1 \leq i \leq m).$$

In particular  $\dim V = d(k_1 + \dots + k_m)$ .

*Idea.* Induction on  $\dim V$ , with the smaller object found inside:  $W = \operatorname{im} p(T)$ . It is  $T$ -invariant, the minimal polynomial of  $T|_W$  is  $p^{e-1}$ , and  $\dim W < \dim V$  because  $p(T)$  is not injective. So  $W$  already breaks into cyclic pieces  $Z(\mathbf{w}_i; T)$ . Each is one step short of what we want:  $\mathbf{w}_i = p(T)\mathbf{u}_i$  for some  $\mathbf{u}_i \in V$ , and  $Z(\mathbf{u}_i; T)$  is a longer piece, with annihilator  $p$  times that of  $\mathbf{w}_i$ . What is still missing is counted by  $U = \ker p(T)$ , on which  $T$  has **irreducible** minimal polynomial  $p$ , so that  $U$  is a direct sum of cyclic pieces of dimension  $d$  each and no proper cyclic piece inside one of them exists (Chapter 8, [Exercise \(C2: Irreducible minimal polynomials\)](#)). The ends  $p^{k_i-1}(T)\mathbf{u}_i$  of the lengthened pieces sit inside  $U$ ; complete them to a decomposition of  $U$ , and the extra generators supply the pieces with annihilator  $p$ . Then Rank–Nullity makes the dimensions add up exactly, and directness is proved by applying  $p(T)$  once, as in the nilpotent case.

*Proof.* We argue by induction on  $n = \dim V \geq 1$ , the statement being: for every field  $F$ , every monic irreducible  $p \in F[x]$ , every  $e \geq 1$  and every  $T$  on a space of dimension  $n$  with  $m_T = p^e$ , a decomposition as described exists.

Write  $P = p(T)$  and  $U = \ker P$ , and recall that  $Z(\mathbf{v}; T|_{V'}) = Z(\mathbf{v}; T)$  whenever  $V'$  is a  $T$ -invariant subspace containing  $\mathbf{v}$ , since the two operators agree on  $\mathbf{v}$  and on everything  $T$  produces from it.

*The case  $e = 1$ .* Then  $m_T = p$  is irreducible, and [Exercise \(C2: Irreducible minimal polynomials\)](#) (d) gives  $V = Z(\mathbf{u}_1; T) \oplus \dots \oplus Z(\mathbf{u}_m; T)$  with every  $\mathbf{u}_i \neq 0$  (the subspace called  $W_{\mathbf{v}}$  there is  $Z(\mathbf{v}; T)$ , by [Proposition 9.5.1](#) (a)). For  $\mathbf{u} \neq 0$  we have  $m_{T, \mathbf{u}} \mid m_T = p$  and  $m_{T, \mathbf{u}} \neq 1$ , so  $m_{T, \mathbf{u}} = p$  and every  $k_i = 1 = e$ . This settles  $e = 1$ , and in particular the base case  $n = 1$ , where  $m_T$  is linear.

The case  $e \geq 2$ . Let  $n \geq 2$  and assume the statement for all dimensions less than  $n$ .

**Step 1. Cyclic pieces inside  $\text{im } P$ .** Put  $W = \text{im } P$ . Since  $e \geq 2$ , the polynomial  $p$  does not annihilate  $T$  (else  $m_T = p^e$  would divide  $p$ , [Theorem 8.5.1](#)), so  $P \neq 0$  and  $W \neq \{0\}$ . Also  $P$  is not injective: otherwise  $P^e = m_T(T) = 0$  would be injective on  $V \neq \{0\}$  ([Theorem 0.4.3](#)), which is false. So  $1 \leq \dim W < n$  by [Theorem 3.3.1](#). The subspace  $W$  is  $T$ -invariant ([Theorem 5.7.4](#) (b)), and

$$m_{T|_W} = p^{e-1}.$$

Indeed  $p^{e-1}(T)$  kills  $W$ , because  $p^{e-1}(T)P\mathbf{v} = m_T(T)\mathbf{v} = \mathbf{0}$  for every  $\mathbf{v}$ ; and  $p^{e-2}(T)$  does not, because  $p^{e-2}(T)P \neq 0$  — that operator is  $p^{e-1}(T)$ , which is non-zero since  $m_T = p^e \nmid p^{e-1}$  — so some  $P\mathbf{v} \in W$  survives it. Hence  $m_{T|_W}$  is a monic divisor of  $p^{e-1}$  but not of  $p^{e-2}$ , and [Lemma 8.5.2](#) leaves only  $p^{e-1}$ . By the induction hypothesis applied to  $T|_W$ ,

$$W = Z(\mathbf{w}_1; T) \oplus \cdots \oplus Z(\mathbf{w}_q; T), \quad m_{T, \mathbf{w}_i} = p^{l_i}, \quad e-1 = l_1 \geq \cdots \geq l_q \geq 1.$$

**Step 2. Lengthen each piece by one step.** Each  $\mathbf{w}_i$  lies in  $W = \text{im } P$ , so we may choose  $\mathbf{u}_i \in V$  with  $P\mathbf{u}_i = \mathbf{w}_i$ , for  $i = 1, \dots, q$ . Put  $k_i = l_i + 1$ . Then  $p^{k_i}(T)\mathbf{u}_i = p^{l_i}(T)\mathbf{w}_i = \mathbf{0}$ , so  $m_{T, \mathbf{u}_i} \mid p^{k_i}$ ; and  $p^{l_i}(T)\mathbf{u}_i \neq \mathbf{0}$ , since otherwise  $p^{l_i-1}(T)\mathbf{w}_i = p^{l_i}(T)\mathbf{u}_i = \mathbf{0}$ , contradicting  $m_{T, \mathbf{w}_i} = p^{l_i}$ . By [Lemma 8.5.2](#),

$$m_{T, \mathbf{u}_i} = p^{k_i} = p^{l_i+1} \quad (1 \leq i \leq q).$$

**Step 3. Fill up  $\ker P$ .** Put  $\mathbf{z}_i = p^{l_i}(T)\mathbf{u}_i$  for  $i \leq q$ . Each is non-zero by Step 2, lies in  $U = \ker P$  because  $P\mathbf{z}_i = p^{k_i}(T)\mathbf{u}_i = \mathbf{0}$ , and lies in  $Z(\mathbf{w}_i; T)$ , since  $l_i \geq 1$  gives  $\mathbf{z}_i = p^{l_i-1}(T)p(T)\mathbf{u}_i = p^{l_i-1}(T)\mathbf{w}_i$ . Consequently  $Z(\mathbf{z}_i; T) \subseteq Z(\mathbf{w}_i; T)$  ([Proposition 9.5.1](#) (b), (c)), so the sum  $Z(\mathbf{z}_1; T) + \cdots + Z(\mathbf{z}_q; T)$  is direct: a vanishing sum of vectors  $\mathbf{c}_i \in Z(\mathbf{z}_i; T) \subseteq Z(\mathbf{w}_i; T)$  forces every  $\mathbf{c}_i = \mathbf{0}$  by the directness of Step 1 ([Theorem 1.7.5](#)).

Now  $U \neq \{0\}$ ,  $U$  is  $T$ -invariant ([Theorem 5.7.4](#) (b)), and  $p(T|_U) = 0$  with  $p$  irreducible, so  $m_{T|_U} = p$  as in the case  $e = 1$ . Among all lists  $(\mathbf{z}_1, \dots, \mathbf{z}_q, \mathbf{z}_{q+1}, \dots, \mathbf{z}_s)$  of non-zero vectors of  $U$  extending  $\mathbf{z}_1, \dots, \mathbf{z}_q$  and with  $Z(\mathbf{z}_1; T) + \cdots + Z(\mathbf{z}_s; T)$  direct, choose one with  $s$  as large as possible; this is possible because each summand has dimension  $d$  ([Exercise \(C2: Irreducible minimal polynomials\)](#) (b)), so  $sd \leq \dim U$ . Let  $S$  be that direct sum. If  $S \neq U$ , pick  $\mathbf{y} \in U \setminus S$ . The subspace  $Z(\mathbf{y}; T) \cap S$  is  $T$ -invariant ([Proposition 8.1.4](#)), lies inside  $Z(\mathbf{y}; T)$ , and does not contain  $\mathbf{y}$ ; by [Exercise \(C2: Irreducible minimal polynomials\)](#) (c), applied to  $T|_U$ , it is therefore  $\{0\}$ . Then  $S + Z(\mathbf{y}; T)$  is direct — a relation  $\mathbf{c}_1 + \cdots + \mathbf{c}_s + \mathbf{c} = \mathbf{0}$  with  $\mathbf{c}_j \in Z(\mathbf{z}_j; T)$  and  $\mathbf{c} \in Z(\mathbf{y}; T)$  puts  $\mathbf{c} \in Z(\mathbf{y}; T) \cap S = \{0\}$ , and directness of  $S$  kills the rest ([Theorem 1.7.5](#)) — contradicting the maximality of  $s$ . Hence

$$U = Z(\mathbf{z}_1; T) \oplus \cdots \oplus Z(\mathbf{z}_s; T), \quad m_{T, \mathbf{z}_j} = p \text{ for every } j.$$

Put  $m = s$ , and for  $q < i \leq m$  put  $\mathbf{u}_i = \mathbf{z}_i$  and  $k_i = 1$ , so that  $m_{T, \mathbf{u}_i} = p^{k_i}$  for those  $i$  as well.

**Step 4. The pieces fill  $V$ .** First count. By [Theorem 9.5.2](#) (a),  $\dim Z(\mathbf{u}_i; T) =$



$\deg p^{k_i} = dk_i$ . Hence

$$\begin{aligned} \sum_{i=1}^m \dim Z(\mathbf{u}_i; T) &= \sum_{i=1}^q d(l_i + 1) + (m - q)d \\ &= \left( \sum_{i=1}^q dl_i \right) + md = \dim W + \dim U = n, \end{aligned}$$

where  $\sum_i dl_i = \sum_i \dim Z(\mathbf{w}_i; T) = \dim W$  by Step 1,  $md = \sum_j \dim Z(\mathbf{z}_j; T) = \dim U$  by Step 3, and the last equality is [Theorem 3.3.1](#).

Next, directness. Let  $\mathbf{y}_1 + \cdots + \mathbf{y}_m = \mathbf{0}$  with  $\mathbf{y}_i \in Z(\mathbf{u}_i; T)$ , say  $\mathbf{y}_i = f_i(T)\mathbf{u}_i$  ([Proposition 9.5.1](#) (a)). Apply  $P$ . For  $i > q$  we get  $P\mathbf{y}_i = f_i(T)P\mathbf{u}_i = \mathbf{0}$ , since  $\mathbf{u}_i \in U$ . For  $i \leq q$  we get  $P\mathbf{y}_i = f_i(T)\mathbf{w}_i \in Z(\mathbf{w}_i; T)$ . So

$$\sum_{i=1}^q f_i(T)\mathbf{w}_i = \mathbf{0},$$

and the directness of Step 1 forces  $f_i(T)\mathbf{w}_i = \mathbf{0}$  for every  $i \leq q$ , that is,  $p^{l_i} \mid f_i$  ([Definition 9.5.2](#)). Write  $f_i = p^{l_i}g_i$  for  $i \leq q$ ; then

$$\mathbf{y}_i = g_i(T)p^{l_i}(T)\mathbf{u}_i = g_i(T)\mathbf{z}_i \in Z(\mathbf{z}_i; T) \quad (i \leq q).$$

For  $i > q$ ,  $\mathbf{y}_i \in Z(\mathbf{u}_i; T) = Z(\mathbf{z}_i; T)$  by definition. So the relation  $\mathbf{y}_1 + \cdots + \mathbf{y}_m = \mathbf{0}$  is a relation between vectors of the independent subspaces  $Z(\mathbf{z}_1; T), \dots, Z(\mathbf{z}_m; T)$  of Step 3, and every  $\mathbf{y}_i = \mathbf{0}$ . By [Theorem 1.7.5](#) ((b)  $\Rightarrow$  (a)) the sum  $\sum_i Z(\mathbf{u}_i; T)$  is direct, and by the count its dimension is  $n$ , so it is all of  $V$  ([Theorem 1.6.8](#)).

Finally the exponents. We have  $k_1 = l_1 + 1 = e$ , the  $k_i$  with  $i \leq q$  decrease because the  $l_i$  do, and the remaining ones equal  $1 \leq k_q$ ; so after listing the pieces in this order,  $e = k_1 \geq \cdots \geq k_m \geq 1$ . The dimension formula  $\dim V = d \sum_i k_i$  is the count above. This completes the induction. ■

Two sanity checks. For  $p = x$ , so that  $T$  is nilpotent with  $m_T = x^e$ , the theorem returns [Theorem 9.2.3](#):  $Z(\mathbf{u}; T) = \text{span}(\mathbf{u}, T\mathbf{u}, \dots, T^{k-1}\mathbf{u})$  is the span of a Jordan chain written backwards, and  $d = 1$ , so  $\dim V = \sum_i k_i$ . And for  $e = 1$  it returns [Exercise \(C2: Irreducible minimal polynomials\)](#): all pieces have dimension  $d$ , so  $d \mid \dim V$ .

### 9.6.3 Invariant factors

Now Steps 1 and 3 of the plan. Merging pieces across different primes is where [Lemma 9.5.4](#) earns its keep.

#### Theorem 9.6.2 — Cyclic Decomposition Theorem

Let  $V \neq \{\mathbf{0}\}$  be a finite-dimensional vector space over  $F$  and let  $T \in \mathcal{L}(V)$ . Then there are an integer  $r \geq 1$ , vectors  $\mathbf{v}_1, \dots, \mathbf{v}_r \in V$  and monic **non-constant** polynomials

$d_1, \dots, d_r \in F[x]$  such that

$$V = Z(\mathbf{v}_1; T) \oplus \cdots \oplus Z(\mathbf{v}_r; T), \quad m_{T, \mathbf{v}_j} = d_j,$$

and

- (a)  $d_1 \mid d_2 \mid \cdots \mid d_r$ ;
- (b)  $d_r = m_T$ ;
- (c)  $d_1 d_2 \cdots d_r = p_T$ ; in particular  $\sum_j \deg d_j = \dim V$ ;
- (d) there is a **finer** decomposition  $V = \bigoplus_{i,j} Z(\mathbf{v}_{ij}; T)$  into cyclic subspaces whose annihilators  $m_{T, \mathbf{v}_{ij}}$  are powers of monic irreducibles, and the list of these prime powers, with repetitions, consists exactly of the prime powers occurring in the factorizations of  $d_1, \dots, d_r$ .

*Idea.* Run the plan. Primary decomposition gives one piece per prime; [Theorem 9.6.1](#) breaks each piece into cyclic subspaces with annihilators  $p_i^{e_{i1}}, p_i^{e_{i2}}, \dots$ , the exponents decreasing. Arrange these in an array, one row per prime, with the largest exponent of each prime in the **first** column. Multiplying down a column gives one member of the chain, and the columns are numbered **backwards**: the first column, which carries the largest exponent of every prime, gives  $d_r = m_T$ ; the second gives  $d_{r-1}$ ; the last gives  $d_1$ . The chain  $d_1 \mid \cdots \mid d_r$  comes out for free, because the exponents in a row decrease from left to right. The generator for a column is the sum of the generators in it, whose annihilator is the product by [Lemma 9.5.4](#).

*Proof. Step 1.* By [Theorem 5.3.5](#),  $m_T = p_1^{e_1} \cdots p_k^{e_k}$  with  $p_1, \dots, p_k$  distinct monic irreducibles and  $e_i \geq 1$ ; here  $\deg m_T \geq 1$  since  $V \neq \{\mathbf{0}\}$  ([Theorem 8.5.1](#)). By [Theorem 8.8.1](#),  $V = V_1 \oplus \cdots \oplus V_k$  with  $V_i$  non-zero and  $T$ -invariant and  $m_{T|_{V_i}} = p_i^{e_i}$ .

**Step 2.** Fix  $i$  and apply [Theorem 9.6.1](#) to  $T|_{V_i}$ . Since  $Z(\mathbf{v}; T|_{V_i}) = Z(\mathbf{v}; T)$  for  $\mathbf{v} \in V_i$ , it gives vectors  $\mathbf{v}_{i1}, \dots, \mathbf{v}_{im_i} \in V_i$  and exponents  $e_i = e_{i1} \geq \cdots \geq e_{im_i} \geq 1$  with

$$V_i = Z(\mathbf{v}_{i1}; T) \oplus \cdots \oplus Z(\mathbf{v}_{im_i}; T), \quad m_{T, \mathbf{v}_{ij}} = p_i^{e_{ij}}.$$

Combining with Step 1,  $V$  is the direct sum of **all** the subspaces  $Z(\mathbf{v}_{ij}; T)$ : a basis of  $V$  is obtained by concatenating bases of the  $V_i$ , each of which is a concatenation of bases of its own pieces, so [Theorem 1.7.5](#) ((d)  $\Rightarrow$  (a)) applies.

**Step 3.** Let  $r = \max_i m_i$  and, for  $1 \leq j \leq r$ , put  $j' = r + 1 - j$  and

$$d_j = \prod_{i: m_i \geq j'} p_i^{e_{ij'}}, \quad \mathbf{v}_j = \sum_{i: m_i \geq j'} \mathbf{v}_{ij'}.$$

So  $d_r$  collects the largest exponent of every prime and  $d_1$  the smallest ones still present.

*The divisibility chain.* The exponent of  $p_i$  in  $d_j$  is  $e_{ij'}$  if  $m_i \geq j'$  and 0 otherwise. As  $j$  increases,  $j'$  decreases, so this exponent is non-decreasing in  $j$  (the exponents  $e_{i1} \geq e_{i2} \geq \dots$  decrease in the second index, and a 0 is followed only by values  $\geq 0$ ). Hence  $d_j \mid d_{j+1}$  for every  $j$  ([Lemma 8.5.2](#)), which is (a). Taking  $j = r$ , so  $j' = 1$ , gives  $d_r = \prod_i p_i^{e_{i1}} = \prod_i p_i^{e_i} = m_T$ , which is (b). Each  $d_j$  is non-constant: choosing  $i_0$  with  $m_{i_0} = r$ , we have  $m_{i_0} \geq j'$  for every  $j$ , so the factor  $p_{i_0}^{e_{i_0 j'}}$  occurs in  $d_j$ , with exponent at least 1.

*The annihilators and the pieces.* For fixed  $j$ , the polynomials  $p_i^{e_{ij'}}$  appearing in  $d_j$  are pairwise coprime, being powers of distinct monic irreducibles: a monic common divisor of  $p_i^a$  and  $p_{i'}^b$  with  $i \neq i'$  is a power of each, hence 1 by the uniqueness of factorization (Lemma 8.5.2, Theorem 5.3.5 (b)). So Lemma 9.5.4 gives  $m_{T, \mathbf{v}_j} = d_j$ . Moreover

$$Z(\mathbf{v}_j; T) = \bigoplus_{i: m_i \geq j'} Z(\mathbf{v}_{ij'}; T) :$$

the left side is contained in the right, because the right side is a  $T$ -invariant subspace containing  $\mathbf{v}_j$  (Proposition 8.1.4, Proposition 9.5.1 (c)), and the two have the same dimension,

$$\dim Z(\mathbf{v}_j; T) = \deg d_j = \sum_i \deg p_i^{e_{ij'}} = \sum_i \dim Z(\mathbf{v}_{ij'}; T),$$

by Theorem 9.5.2 (a) and Theorem 1.7.5 ((a)  $\Rightarrow$  (e)); so they are equal (Theorem 1.6.8).

*The decomposition.* Each subspace  $Z(\mathbf{v}_{ij_0}; T)$ , with  $1 \leq j_0 \leq m_i$ , occurs in exactly one of the sums just displayed, namely the one with  $j = r + 1 - j_0$ ; so every pair  $(i, j_0)$  is used exactly once and

$$\sum_{j=1}^r Z(\mathbf{v}_j; T) = \sum_{i,j} Z(\mathbf{v}_{ij}; T) = V,$$

and  $\sum_j \dim Z(\mathbf{v}_j; T) = \sum_{i,j} \dim Z(\mathbf{v}_{ij}; T) = \dim V$ . By Theorem 1.7.5 ((e)  $\Rightarrow$  (a)) the sum over  $j$  is direct, and it equals  $V$ .

- (c) Each  $Z(\mathbf{v}_j; T)$  is  $T$ -invariant and non-zero, so by Theorem 8.1.5 there is a basis  $\mathcal{B}$  of  $V$  with  $[T]_{\mathcal{B}} = [T]_{Z(\mathbf{v}_1; T)} \oplus \cdots \oplus [T]_{Z(\mathbf{v}_r; T)}$ . Then  $x\mathbf{I} - [T]_{\mathcal{B}}$  is block diagonal with the blocks  $x\mathbf{I} - [T]_{Z(\mathbf{v}_j; T)}$ , so Theorem 6.8.4, applied  $r - 1$  times, gives

$$p_T = \det(x\mathbf{I} - [T]_{\mathcal{B}}) = \prod_{j=1}^r p_{T|_{Z(\mathbf{v}_j; T)}}$$

(Definition 6.10.2). By Theorem 9.5.2 (c),  $p_{T|_{Z(\mathbf{v}_j; T)}} = m_{T, \mathbf{v}_j} = d_j$ . Hence  $p_T = d_1 \cdots d_r$ , and comparing degrees gives  $\sum_j \deg d_j = \deg p_T = \dim V$  (Theorem 6.10.2).

- (d) Step 2 already produced the finer decomposition  $V = \bigoplus_{i,j} Z(\mathbf{v}_{ij}; T)$ , with  $m_{T, \mathbf{v}_{ij}} = p_i^{e_{ij}}$  a prime power. By the definition of  $d_j$  in Step 3, the exponent of  $p_i$  in  $d_j$  is  $e_{i, r+1-j}$  when  $m_i \geq r + 1 - j$  and 0 otherwise, and each pair  $(i, j)$  with  $1 \leq j \leq m_i$  is used exactly once. So the prime powers occurring in  $d_1, \dots, d_r$  are exactly the polynomials  $p_i^{e_{ij}}$ , with the same repetitions. ■

The polynomials in the chain, and the prime powers they are built from, are the data we are after — but first we must know that they do not depend on the decomposition we happened to construct. The count that settles this is the count of Section 2 in disguise: there the chain lengths were read off from the ranks of the powers of  $N$ , and here the exponents are read off from the dimensions of the kernels of the powers of  $p(T)$ . Write  $v_p(f)$  for the exponent of a monic irreducible  $p$  in the factorization of a non-zero  $f \in F[x]$ , so that  $v_p(f) \geq 1$  exactly when  $p \mid f$ .

**Proposition 9.6.3 — Counting the Exponents**

Let  $V \neq \{0\}$  be finite-dimensional over  $F$ , let  $T \in \mathcal{L}(V)$ , and suppose

$$V = Z(\mathbf{u}_1; T) \oplus \cdots \oplus Z(\mathbf{u}_N; T), \quad m_{T, \mathbf{u}_i} = g_i,$$

with every  $g_i$  monic and non-constant. Then for every monic irreducible  $p \in F[x]$ :

- (a)  $\dim \ker p(T)^k = \deg p \cdot \sum_{i=1}^N \min(k, v_p(g_i))$  for every integer  $k \geq 0$ ;  
 (b) for every  $k \geq 1$ ,

$$\#\{i : v_p(g_i) \geq k\} = \frac{\dim \ker p(T)^k - \dim \ker p(T)^{k-1}}{\deg p},$$

a number depending only on  $T$ ,  $p$  and  $k$ .

*Idea.* Kernels respect the decomposition, so we may work inside one piece. There a vector is  $h(T)\mathbf{u}$ , and it is killed by  $p(T)^k$  exactly when  $g \mid p^k h$ , which after canceling the common factor  $\gcd(g, p^k)$  says that  $h$  is a multiple of  $g/\gcd(g, p^k)$ . The surviving vectors are therefore a cyclic subspace again, generated by that multiple of  $\mathbf{u}$ , and its annihilator is  $\gcd(g, p^k) = p^{\min(k, v_p(g))}$ . Summing over the pieces and taking differences in  $k$  then counts exponents.

*Proof.*

- (a) Write  $Z_i = Z(\mathbf{u}_i; T)$  and  $S = p(T)^k$ . Each  $Z_i$  is  $T$ -invariant, hence  $S$ -invariant (Theorem 5.7.4 (b)). If  $\mathbf{v} = \mathbf{v}_1 + \cdots + \mathbf{v}_N$  with  $\mathbf{v}_i \in Z_i$ , then  $S\mathbf{v} = S\mathbf{v}_1 + \cdots + S\mathbf{v}_N$  with  $S\mathbf{v}_i \in Z_i$ , so  $S\mathbf{v} = \mathbf{0}$  if and only if every  $S\mathbf{v}_i = \mathbf{0}$ , by the uniqueness of the splitting (Theorem 1.7.5). Therefore

$$\ker S = (\ker S \cap Z_1) \oplus \cdots \oplus (\ker S \cap Z_N),$$

and it suffices to compute each  $\dim(\ker S \cap Z_i)$ . Fix  $i$  and write  $\mathbf{u} = \mathbf{u}_i$ ,  $g = g_i$ ,  $Z = Z_i$ . Put

$$c = \gcd(g, p^k) = p^{\min(k, v_p(g))} \quad \text{and} \quad t = g/c,$$

the formula for  $c$  holding because a monic common divisor of  $g$  and  $p^k$  is a power of  $p$  (Lemma 8.5.2) dividing both, and  $p^s \mid g$  exactly when  $s \leq v_p(g)$  (Theorem 5.3.5). Every element of  $Z$  is  $h(T)\mathbf{u}$  for some  $h \in F[x]$  (Proposition 9.5.1 (a)), and

$$p(T)^k h(T)\mathbf{u} = \mathbf{0} \iff g \mid p^k h \iff t \mid h,$$

where the second equivalence holds because  $g = ct$  and  $p^k = c(p^k/c)$  with  $\gcd(t, p^k/c) = 1$ : from  $ct \mid c(p^k/c)h$  we get  $t \mid (p^k/c)h$ , hence  $t \mid h$  (Theorem 5.2.8), and conversely  $t \mid h$  gives  $g = ct \mid ch \mid p^k h$  since  $c \mid p^k$ . So

$$\ker S \cap Z = \{h(T)\mathbf{u} : t \mid h\} = \{s(T)(t(T)\mathbf{u}) : s \in F[x]\} = Z(t(T)\mathbf{u}; T).$$

Its generator has annihilator  $c$ : for  $s \in F[x]$ ,  $s(T)t(T)\mathbf{u} = \mathbf{0}$  if and only if  $g \mid st$ , that is,  $c \mid s$ , because  $g = ct$ . Taking  $s = 1$  shows that  $t(T)\mathbf{u} \neq \mathbf{0}$  unless  $c = 1$ , so

**Theorem 9.5.2** (a) applies whenever  $c \neq 1$  and gives

$$\dim(\ker S \cap Z_i) = \deg c = \min(k, v_p(g_i)) \deg p,$$

where for  $c = 1$  the cyclic subspace is  $\{0\}$  of dimension 0, in agreement. Adding over  $i$  gives (a).

- (b) Subtract the values of (a) at  $k$  and  $k - 1$ . Each term  $\min(k, v_p(g_i)) - \min(k - 1, v_p(g_i))$  equals 1 if  $v_p(g_i) \geq k$  and 0 otherwise, so the difference is  $\deg p$  times the number of indices  $i$  with  $v_p(g_i) \geq k$ . The left-hand side is defined from  $T$ ,  $p$  and  $k$  alone.

■

### Corollary 9.6.4 — Uniqueness of the Chain and of the Prime Powers

Let  $V \neq \{0\}$  be finite-dimensional over  $F$  and  $T \in \mathcal{L}(V)$ .

- (a) Any two decompositions as in [Theorem 9.6.2](#) give the **same** ordered list  $d_1 \mid \cdots \mid d_r$ .
- (b) Any decomposition  $V = Z(\mathbf{u}_1; T) \oplus \cdots \oplus Z(\mathbf{u}_N; T)$  whose annihilators  $m_{T, \mathbf{u}_i}$  are powers of monic irreducibles gives the same list of annihilators, up to order, namely the prime powers occurring in  $d_1, \dots, d_r$ .
- (c) Explicitly, writing

$$c_p(k) = \frac{\dim \ker p(T)^k - \dim \ker p(T)^{k-1}}{\deg p} \quad (p \text{ monic irreducible, } k \geq 1),$$

the list is recovered by  $r = \max_p c_p(1)$  and

$$v_p(d_j) = \#\{k \geq 1 : c_p(k) \geq r + 1 - j\} \quad (1 \leq j \leq r).$$

In particular the invariant factors of  $T$  are determined by the numbers  $\dim \ker p(T)^k$  alone.

*Idea.* Divisibility puts the zeros first: in the list  $v_p(d_1) \leq \cdots \leq v_p(d_r)$ , the indices where  $p$  actually occurs are a block at the **right** end. So a count of how many  $d_j$  are divisible by  $p^k$  pins down *which* ones they are, and therefore each individual exponent. The counts are supplied by [Proposition 9.6.3](#), which sees only  $T$ .

*Proof.*

- (a) and (c). Let  $V = \bigoplus_{j=1}^r Z(\mathbf{v}_j; T)$  be such a decomposition, with annihilators  $d_1 \mid \cdots \mid d_r$ , all monic and non-constant. Fix a monic irreducible  $p$ . By [Proposition 9.6.3](#) (b), applied to this decomposition,

$$c_p(k) = \#\{j : v_p(d_j) \geq k\} \quad (k \geq 1),$$

so the left-hand side, which is defined from  $T$ ,  $p$  and  $k$  alone, computes the right-hand side. Divisibility makes the exponents non-decreasing,  $v_p(d_1) \leq \cdots \leq v_p(d_r)$ ,

so for each  $k$  the set  $\{j : v_p(d_j) \geq k\}$  is the set of the **last**  $c_p(k)$  indices:

$$v_p(d_j) \geq k \iff j \geq r + 1 - c_p(k) \iff c_p(k) \geq r + 1 - j. \quad (*)$$

First,  $r$  is determined. Each  $d_j$  is non-constant, so  $v_p(d_1) \geq 1$  for some monic irreducible  $p$ ; for that  $p$ ,  $(*)$  with  $k = 1$  and  $j = 1$  gives  $c_p(1) \geq r$ , while  $c_p(1) \leq r$  always. Hence  $r = \max_p c_p(1)$ . Next, for each  $p$  and each  $j$ , counting the  $k \geq 1$  for which  $(*)$  holds gives

$$v_p(d_j) = \#\{k \geq 1 : c_p(k) \geq r + 1 - j\},$$

a finite number, since  $c_p(k) = 0$  for  $k > \deg m_T$ . This proves (c). Finally each  $d_j$  is monic with determined exponents  $v_p(d_j)$  for every  $p$ , so  $d_j$  is determined ([Theorem 5.3.5](#)), which proves (a).

- (b) Let  $V = \bigoplus_i Z(\mathbf{u}_i; T)$  with  $m_{T, \mathbf{u}_i} = q_i^{f_i}$ ,  $q_i$  monic irreducible and  $f_i \geq 1$ . For a monic irreducible  $p$  we have  $v_p(q_i^{f_i}) = f_i$  if  $q_i = p$  and 0 otherwise. Applying [Proposition 9.6.3](#) (b) to this decomposition and to the one in (a) gives, for every  $p$  and every  $k \geq 1$ ,

$$\#\{i : q_i = p, f_i \geq k\} = c_p(k) = \#\{j : v_p(d_j) \geq k\}.$$

Hence, for each  $p$ , the exponents  $f_i$  with  $q_i = p$  and the positive exponents  $v_p(d_j)$  form the same multiset. So the list  $(q_i^{f_i})_i$  is exactly the list of prime powers occurring in  $d_1, \dots, d_r$ , with the same repetitions. By [Theorem 9.6.2](#) (d) such a decomposition exists, so both lists are non-empty and agree. ■

### Definition 9.6.1 — Invariant Factors, Elementary Divisors

Let  $V \neq \{0\}$  be finite-dimensional over  $F$  and  $T \in \mathcal{L}(V)$ . The **invariant factors** of  $T$  are the monic non-constant polynomials

$$d_1 \mid d_2 \mid \cdots \mid d_r$$

of [Theorem 9.6.2](#), which by [Corollary 9.6.4](#) (a) do not depend on the decomposition. The **elementary divisors** of  $T$  are the prime powers occurring in the factorizations of  $d_1, \dots, d_r$ , listed with repetition; by [Corollary 9.6.4](#) (b) they are the annihilators of the pieces in any decomposition of  $V$  into cyclic subspaces with prime-power annihilators. For  $\mathbf{A} \in M_n(F)$  the invariant factors and elementary divisors are those of  $T_{\mathbf{A}}$ .

In words: the invariant factors are the annihilators of the cyclic pieces, ordered so that each divides the next; the largest is  $m_T$ , and their product is  $p_T$ . The elementary divisors are the same information cut along the primes. Each list determines the other: one by factoring, the other by the regrouping recipe of Step 3 in the proof of [Theorem 9.6.2](#), which [Corollary 9.6.4](#) (c) shows to be forced.

**Non-example by minimal change.** Drop the divisibility clause and keep only the product. For an operator on a  $\mathbb{Q}$ -space of dimension 3 with  $p_T = (x - 1)(x - 2)^2$ , the pair

$$d_1 = x - 1, \quad d_2 = (x - 2)^2$$

multiplies to  $p_T$  and consists of monic non-constant polynomials of the right total degree. It is still not a list of invariant factors of anything, because  $d_1 \nmid d_2$ . The clause that fails is (a) of [Theorem 9.6.2](#), and with it (b): the largest member of the list has to be  $m_T$ , and  $m_T(T)$  kills every vector of  $V$ , so  $m_T$  is a multiple of the annihilator of every piece ([Theorem 9.5.2](#) (d)). The genuine lists here are  $((x-1)(x-2)^2)$  and  $(x-2, (x-1)(x-2))$ .

*Quick check.* Let  $T$  be an operator on a  $\mathbb{Q}$ -vector space with  $p_T = (x^2 + 1)^2$ . How many possibilities are there for the list of elementary divisors, and what are the corresponding invariant factors?

*Solution.* The product of the elementary divisors is  $p_T$  (by [Theorem 9.6.2](#) (c) and [Definition 9.6.1](#)), and  $x^2 + 1$  is irreducible over  $\mathbb{Q}$  (no rational root, [Theorem 5.3.1](#) (b)). So the elementary divisors are powers of  $x^2 + 1$  with exponents forming a partition of 2: either  $(x^2 + 1)^2$  alone, or  $x^2 + 1$  twice. The invariant factors are  $d_1 = (x^2 + 1)^2$  in the first case (one piece,  $r = 1$ ) and  $d_1 = d_2 = x^2 + 1$  in the second. The two are told apart by  $m_T$ , which is  $d_r$ :  $(x^2 + 1)^2$  against  $x^2 + 1$ .

## 9.6.4 The rational canonical form

Everything is now assembled. Writing the matrix down is [Theorem 9.5.2](#) (b), one piece at a time.

*Every operator over every field has a block diagonal matrix whose blocks are companion matrices of a divisibility chain, and the chain is unique.*

### Theorem 9.6.5 — Rational Canonical Form

Let  $V \neq \{0\}$  be a finite-dimensional vector space over **any** field  $F$ , and let  $T \in \mathcal{L}(V)$  have invariant factors  $d_1 \mid d_2 \mid \cdots \mid d_r$ . Then there is a basis  $\mathcal{B}$  of  $V$  with

$$[T]_{\mathcal{B}} = \mathbf{C}(d_1) \oplus \mathbf{C}(d_2) \oplus \cdots \oplus \mathbf{C}(d_r),$$

and this matrix is determined by  $T$ . It is called the **rational canonical form** of  $T$ . Equivalently, every  $\mathbf{A} \in M_n(F)$  with  $n \geq 1$  is similar to exactly one matrix of the form  $\mathbf{C}(d_1) \oplus \cdots \oplus \mathbf{C}(d_r)$  with  $d_1 \mid \cdots \mid d_r$  monic and non-constant.

*Proof.* By [Theorem 9.6.2](#),  $V = Z(\mathbf{v}_1; T) \oplus \cdots \oplus Z(\mathbf{v}_r; T)$  with  $m_{T, \mathbf{v}_j} = d_j$ . Each piece is  $T$ -invariant and non-zero, and by [Theorem 9.5.2](#) (a), (b) the list  $\mathcal{B}_j = (\mathbf{v}_j, T\mathbf{v}_j, \dots, T^{\deg d_j - 1}\mathbf{v}_j)$  is a basis of  $Z(\mathbf{v}_j; T)$  in which the matrix of the restriction is  $\mathbf{C}(d_j)$ . Let  $\mathcal{B}$  be  $\mathcal{B}_1, \dots, \mathcal{B}_r$  written one after another; by [Theorem 8.1.5](#),  $\mathcal{B}$  is a basis of  $V$  and  $[T]_{\mathcal{B}} = \mathbf{C}(d_1) \oplus \cdots \oplus \mathbf{C}(d_r)$ . The blocks are determined by  $T$ , since the  $d_j$  are ([Corollary 9.6.4](#) (a)).

For a matrix  $\mathbf{A} \in M_n(F)$ , apply this to  $T_{\mathbf{A}}$  and use [Theorem 3.7.4](#). For the uniqueness in the matrix form, suppose  $\mathbf{A} \sim \mathbf{C}(f_1) \oplus \cdots \oplus \mathbf{C}(f_s)$  with  $f_1 \mid \cdots \mid f_s$  monic and non-constant. By [Theorem 3.7.4](#) there is a basis  $\mathcal{C}$  of  $F^n$  with  $[T_{\mathbf{A}}]_{\mathcal{C}} = \mathbf{C}(f_1) \oplus \cdots \oplus \mathbf{C}(f_s)$ . Let  $U_j$  be the span of the segment of  $\mathcal{C}$  belonging to the  $j$ -th block; then  $F^n = U_1 \oplus \cdots \oplus U_s$  with each  $U_j$  invariant and  $[T_{\mathbf{A}}|_{U_j}] = \mathbf{C}(f_j)$  ([Theorem 8.1.5](#)). By [Theorem 9.5.3](#), [Theorem 8.5.5](#) and [Definition 6.10.2](#),  $m_{T_{\mathbf{A}}|_{U_j}} = p_{T_{\mathbf{A}}|_{U_j}} = f_j$ , so  $T_{\mathbf{A}}|_{U_j}$  is cyclic ([Theorem 9.5.6](#)) and  $U_j = Z(\mathbf{u}_j; T_{\mathbf{A}})$  for some  $\mathbf{u}_j$  with  $m_{T_{\mathbf{A}}, \mathbf{u}_j} = f_j$  ([Theorem 9.5.2](#) (c)). This is a decomposition as in [Theorem 9.6.2](#), so  $(f_1, \dots, f_s) = (d_1, \dots, d_r)$  by [Corollary 9.6.4](#) (a).



Hence the matrix is the rational canonical form of  $\mathbf{A}$ .

■

This pays off the promise made in Section 3: **there is a canonical form over every field.** No hypothesis on  $F$  was used, and none is needed. The form is canonical in the strong sense: the blocks come in a prescribed order, dictated by divisibility, so two operators have the *same* form, not merely similar forms.

### Corollary 9.6.6 — Similarity via the Invariant Factors

Let  $\mathbf{A}, \mathbf{B} \in M_n(F)$  with  $n \geq 1$ . The following are equivalent:

- (a)  $\mathbf{A} \sim \mathbf{B}$ ;
- (b)  $\mathbf{A}$  and  $\mathbf{B}$  have the same invariant factors;
- (c)  $\mathbf{A}$  and  $\mathbf{B}$  have the same elementary divisors, up to order;
- (d)  $\dim \ker p(\mathbf{A})^k = \dim \ker p(\mathbf{B})^k$  for every monic irreducible  $p \in F[x]$  and every  $k \geq 1$ .

*Proof.*

- (a)  $\Rightarrow$  (d). If  $\mathbf{B} = \mathbf{P}^{-1}\mathbf{A}\mathbf{P}$ , then  $p(\mathbf{B})^k = \mathbf{P}^{-1}p(\mathbf{A})^k\mathbf{P}$  (Proposition 3.7.6 (c)), so the two have the same rank (Proposition 3.7.6 (a)) and hence the same nullity (Theorem 2.5.7).
- (b)  $\Rightarrow$  (b). By Corollary 9.6.4 (c), the invariant factors of a matrix are computed from the numbers  $\dim \ker p(\cdot)^k$  alone: those numbers give the counts  $c_p(k)$ , which give the length  $r$  and the exponents  $v_p(d_j)$ . Equal numbers therefore give equal invariant factors.
- (c)  $\Leftrightarrow$  (c). The elementary divisors are the prime powers in the factorizations of the invariant factors (Definition 9.6.1), so (b)  $\Rightarrow$  (c). Conversely,  $c_p(k)$  is the number of elementary divisors of the form  $p^f$  with  $f \geq k$  (Corollary 9.6.4 (b) and (c)), so the list of elementary divisors determines every  $c_p(k)$ , hence the invariant factors by Corollary 9.6.4 (c); so (c)  $\Rightarrow$  (b).
- (d)  $\Rightarrow$  (a). Both matrices are similar to  $\mathbf{C}(d_1) \oplus \cdots \oplus \mathbf{C}(d_r)$  (Theorem 9.6.5), and similarity is an equivalence relation (Example (Similarity)).

■

So the invariant factors are a **complete set of similarity invariants over every field**, which is the promise of Chapter 6 kept in full; Corollary 9.3.3 was the special case in which  $p_{\mathbf{A}}$  splits. Counting similarity classes becomes counting divisibility chains: over any field, the classes of  $n \times n$  matrices with a given characteristic polynomial  $p$  correspond to the ways of writing  $p = d_1 \cdots d_r$  with  $d_1 \mid \cdots \mid d_r$  monic and non-constant.

**Warning.** The invariant factors are not the factors of  $p_{\mathbf{A}}$  in any naive sense, and  $m_{\mathbf{A}}$  alone is only the last of them. For  $\mathbf{A} = \text{diag}(1, 1, 2)$  over  $\mathbb{Q}$  the invariant factors are  $d_1 = x - 1$  and  $d_2 = (x - 1)(x - 2) = m_{\mathbf{A}}$ , and  $d_1 d_2 = (x - 1)^2(x - 2) = p_{\mathbf{A}}$ ; the factorization of  $p_{\mathbf{A}}$  into linear factors plays no role in the chain. Nor does the chain have length  $\deg p_{\mathbf{A}}$ : here  $r = 2$  for a  $3 \times 3$  matrix, and  $r = 1$  exactly when  $\mathbf{A}$  is cyclic



(Theorem 9.5.6).

## 9.6.5 Elementary divisors, and the way back to Jordan

Cutting the pieces as finely as possible gives a second form, usually with more and smaller blocks.

### Theorem 9.6.7 — Elementary Divisor Form

Let  $V \neq \{0\}$  be finite-dimensional over  $F$ , let  $T \in \mathcal{L}(V)$ , and let  $q_1, \dots, q_N$  be the elementary divisors of  $T$ . Then there is a basis  $\mathcal{B}$  of  $V$  with

$$[T]_{\mathcal{B}} = \mathbf{C}(q_1) \oplus \mathbf{C}(q_2) \oplus \cdots \oplus \mathbf{C}(q_N),$$

and the blocks are determined by  $T$  up to their order.

*Proof.* By Theorem 9.6.2 (d) there is a decomposition  $V = \bigoplus_{i,j} Z(\mathbf{v}_{ij}; T)$  into cyclic subspaces whose annihilators are prime powers, and by Corollary 9.6.4 (b) those annihilators are exactly the elementary divisors  $q_1, \dots, q_N$  of  $T$ , up to order. Take the basis of Theorem 9.5.2 (a) in each piece and concatenate; Theorem 9.5.2 (b) and Theorem 8.1.5 give the displayed matrix. The list of blocks depends on  $T$  alone, again by Corollary 9.6.4 (b). ■

When the characteristic polynomial splits, this is the Jordan form in disguise, because a companion block for a power of a linear polynomial is a Jordan block.

### Corollary 9.6.8 — Companion Blocks of Linear Powers Are Jordan Blocks

Let  $\lambda \in F$  and  $k \geq 1$ . Then

$$\mathbf{C}((x - \lambda)^k) \sim \mathbf{J}_k(\lambda).$$

Consequently, if  $p_T$  splits over  $F$ , then the elementary divisors of  $T$  are the polynomials  $(x - \lambda)^k$  for the Jordan blocks  $\mathbf{J}_k(\lambda)$  of  $T$ , and the elementary divisor form of Theorem 9.6.7 and the Jordan form of  $T$  are obtained from one another by replacing each block  $\mathbf{C}((x - \lambda)^k)$  by  $\mathbf{J}_k(\lambda)$  and conversely.

*Proof.* Put  $\mathbf{J} = \mathbf{J}_k(\lambda)$ . Then  $\mathbf{J} - \lambda\mathbf{I} = \mathbf{J}_k(0)$  satisfies  $\mathbf{J}_k(0)^k = 0 \neq \mathbf{J}_k(0)^{k-1}$ , so  $(x - \lambda)^k$  annihilates  $\mathbf{J}$  and no smaller power does; by Lemma 8.5.2 and Theorem 8.5.1,  $m_{\mathbf{J}} = (x - \lambda)^k$ . Also  $p_{\mathbf{J}} = (x - \lambda)^k$  (Theorem 6.4.6), so  $m_{\mathbf{J}} = p_{\mathbf{J}}$  and  $\mathbf{J}$  is cyclic (Theorem 9.5.6). By the last sentence of that theorem,  $\mathbf{J}$  is similar to  $\mathbf{C}(m_{\mathbf{J}}) = \mathbf{C}((x - \lambda)^k)$ .

Now suppose  $p_T$  splits and let  $\mathcal{B}$  be a Jordan basis, with  $[T]_{\mathcal{B}} = \mathbf{J}_{k_1}(\mu_1) \oplus \cdots \oplus \mathbf{J}_{k_m}(\mu_m)$  (Theorem 9.3.1). Each block corresponds to a  $T$ -invariant subspace  $U_l$ , spanned by the segment of  $\mathcal{B}$  belonging to it, with  $V = U_1 \oplus \cdots \oplus U_m$  and  $[T|_{U_l}] = \mathbf{J}_{k_l}(\mu_l)$ . By the

first paragraph,  $m_{T|U_l} = p_{T|U_l} = (x - \mu_l)^{k_l}$ , so  $T|_{U_l}$  is cyclic: there is  $\mathbf{v}_l \in U_l$  with  $U_l = Z(\mathbf{v}_l; T)$  and  $m_{T, \mathbf{v}_l} = (x - \mu_l)^{k_l}$  (Theorem 9.5.6, Theorem 9.5.2 (c)). Since  $x - \mu_l$  is irreducible, being of degree 1, this exhibits  $V$  as a direct sum of cyclic subspaces with prime-power annihilators, so by Corollary 9.6.4 (b) the elementary divisors of  $T$  are exactly the polynomials  $(x - \mu_l)^{k_l}$ . Replacing each Jordan block by the similar companion block, or conversely, therefore turns one form into the other. ■

*Remark.* The two forms carry the same information in different shapes, and each is better for something. The rational canonical form of Theorem 9.6.5 has the fewest blocks and reads off  $m_T = d_r$  at a glance; the elementary divisor form has the most blocks and reads off the primes, and over a splitting field it is the Jordan form. The rational form is also called the **Frobenius normal form**. The word *rational* refers to the field: the form is computed by rational operations in  $F$ , with no root extraction, which is exactly why it exists over every field.

## 9.6.6 Similarity does not depend on the field

Here is a striking consequence, and one that would be hard to guess. Two rational matrices might conceivably be conjugate by a complex matrix and by no rational one. They cannot be.

### Lemma 9.6.9 — Rank Does Not Change with the Field

Let  $K$  be a subfield of a field  $L$  and let  $\mathbf{M} \in M_{m \times n}(K)$ . Then the rank of  $\mathbf{M}$  computed over  $K$  equals its rank computed over  $L$ .

*Proof.* If  $\mathbf{M} = 0$ , both ranks are 0. Otherwise, by Theorem 6.7.1 (a), each rank is the largest  $k$  such that some  $k \times k$  minor of  $\mathbf{M}$  is non-zero. The minors are determinants of submatrices of  $\mathbf{M}$ , computed by the Leibniz formula from the entries of  $\mathbf{M}$  (Definition 6.4.1), so they are the **same elements of  $K$**  whichever field we regard  $\mathbf{M}$  in. And an element of  $K$  is zero in  $K$  if and only if it is zero in  $L$ . So the two largest such  $k$  agree. ■

### Corollary 9.6.10 — Similarity Is Independent of the Field

Let  $K$  be a subfield of  $L$  and let  $\mathbf{A}, \mathbf{B} \in M_n(K)$ ,  $n \geq 1$ . If  $\mathbf{A}$  and  $\mathbf{B}$  are similar over  $L$ , then they are similar over  $K$ :

$$\begin{aligned} \mathbf{A} &= \mathbf{Q}^{-1}\mathbf{B}\mathbf{Q} \text{ for some invertible } \mathbf{Q} \in M_n(L) \\ \implies \mathbf{A} &= \mathbf{P}^{-1}\mathbf{B}\mathbf{P} \text{ for some invertible } \mathbf{P} \in M_n(K). \end{aligned}$$

*Idea.* Similarity over  $K$  is decided by the numbers  $\dim \ker p(\mathbf{A})^k$  for the monic irreducibles  $p$  of  $K[x]$  (Corollary 9.6.6). Each such number is a nullity of a matrix with entries in  $K$ , and nullities do not notice a larger field. Similarity over  $L$  makes the

numbers agree when computed over  $L$ , hence over  $K$ .

*Proof.* Let  $p \in K[x]$  be monic irreducible and  $k \geq 1$ . The matrices  $p(\mathbf{A})^k$  and  $p(\mathbf{B})^k$  have entries in  $K$ . Since  $\mathbf{A}$  and  $\mathbf{B}$  are similar over  $L$ , say  $\mathbf{A} = \mathbf{Q}^{-1}\mathbf{B}\mathbf{Q}$ , we have  $p(\mathbf{A})^k = \mathbf{Q}^{-1}p(\mathbf{B})^k\mathbf{Q}$  (Proposition 3.7.6 (c)), so these two matrices have the same rank over  $L$  (Proposition 3.7.6 (a)). By Lemma 9.6.9 their ranks over  $K$  are the same numbers, hence

$$\dim \ker p(\mathbf{A})^k = n - \text{rank } p(\mathbf{A})^k = n - \text{rank } p(\mathbf{B})^k = \dim \ker p(\mathbf{B})^k$$

over  $K$ , by Theorem 2.5.7. As  $p$  and  $k$  were arbitrary, Corollary 9.6.6 ((d)  $\Rightarrow$  (a)) gives  $\mathbf{A} \sim \mathbf{B}$  over  $K$ . ■

The invariant factors are computed in  $K[x]$  and never need a root; that is the whole reason. Note what the corollary does **not** say: the invariant factors themselves do change with the field, since the irreducible polynomials do. Over  $\mathbb{Q}$  the matrix  $\mathbf{C}((x^2 + 1)^2)$  has the single invariant factor  $(x^2 + 1)^2$  and the single elementary divisor  $(x^2 + 1)^2$ ; over  $\mathbb{C}$  its invariant factor is unchanged, but its elementary divisors are  $(x - i)^2$  and  $(x + i)^2$ , and its Jordan form is  $\mathbf{J}_2(i) \oplus \mathbf{J}_2(-i)$ . What is field-independent is the **answer to the similarity question**, not the bookkeeping.

## 9.6.7 Computing the forms

*Algorithm.* **Invariant factors, elementary divisors and the two forms of  $\mathbf{A} \in M_n(F)$ .**

1. Compute  $p_{\mathbf{A}}$  and factor it into monic irreducibles over  $F$ :  $p_{\mathbf{A}} = p_1^{a_1} \cdots p_k^{a_k}$ .
2. For each  $i$  and each  $k \geq 1$ , compute  $\dim \ker p_i(\mathbf{A})^k$  until it stops growing, and read off the exponents of the elementary divisors for  $p_i$  from Proposition 9.6.3, subtracting twice for exact counts. (Equivalently: find  $m_{\mathbf{A}}$  and, in small cases, use  $\sum_{i,j} e_{ij} \deg p_i = n$  with  $\max_j e_{ij} =$  the exponent of  $p_i$  in  $m_{\mathbf{A}}$ .)
3. Group the exponents of each prime into a row, in decreasing order, and multiply down the columns, numbering them backwards: the first column gives  $d_r = m_{\mathbf{A}}$ , the second  $d_{r-1}$ , the last  $d_1$ .
4. Write  $\mathbf{C}(d_1) \oplus \cdots \oplus \mathbf{C}(d_r)$ , or  $\bigoplus_{i,j} \mathbf{C}(p_i^{e_{ij}})$ .
5. For a basis: find a cyclic generator for each piece, as in Section 5.

### Example. (Invariant Factors from Polynomial Data)

Let  $T$  be an operator on a 6-dimensional  $\mathbb{Q}$ -vector space with

$$p_T = (x - 1)^2(x^2 + 1)^2, \quad m_T = (x - 1)(x^2 + 1)^2.$$

Find the elementary divisors, the invariant factors, the rational canonical form (Theorem 9.6.5) and the elementary divisor form (Theorem 9.6.7).

*Solution.* *The primes.*  $x - 1$  and  $x^2 + 1$  are monic irreducible over  $\mathbb{Q}$  (the second has no rational root, Theorem 5.3.1 (b)), and  $\deg p_T = 2 + 4 = 6$  as it must be.

*The exponents.* The elementary divisors are powers  $(x-1)^a$  and  $(x^2+1)^b$ . For each prime, the largest exponent occurring is its exponent in  $m_T = d_r$  ([Theorem 9.6.2 \(b\)](#), [Definition 9.6.1](#)), and the exponents weighted by the degrees sum to  $\deg p_T$  ([Theorem 9.6.2 \(c\)](#)). For  $x-1$ : the largest exponent is 1 and the total contribution is 2, so there are two elementary divisors  $x-1$ . For  $x^2+1$ : the largest exponent is 2 and the total contribution is  $4 = 2 \cdot 2$ , so there is exactly one elementary divisor  $(x^2+1)^2$ . Hence the elementary divisors are

$$x-1, \quad x-1, \quad (x^2+1)^2.$$

*The invariant factors.* Arrange the exponents by prime in decreasing order:

$$x-1 : 1, 1; \quad x^2+1 : 2.$$

So  $r = \max(2, 1) = 2$ , and multiplying down the columns, the first giving  $d_2$  and the second  $d_1$ ,

$$d_2 = (x-1)(x^2+1)^2 = m_T, \quad d_1 = x-1.$$

Check:  $d_1 \mid d_2$  and  $d_1 d_2 = (x-1)^2(x^2+1)^2 = p_T$ .

*The forms.* With  $d_2 = (x-1)(x^2+1)^2 = (x-1)(x^4+2x^2+1) = x^5-x^4+2x^3-2x^2+x-1$ ,

$$\mathbf{C}(d_1) \oplus \mathbf{C}(d_2) = (1) \oplus \begin{pmatrix} 0 & 0 & 0 & 0 & 1 \\ 1 & 0 & 0 & 0 & -1 \\ 0 & 1 & 0 & 0 & 2 \\ 0 & 0 & 1 & 0 & -2 \\ 0 & 0 & 0 & 1 & 1 \end{pmatrix},$$

of total size  $1 + 5 = 6$ . The elementary divisor form is smaller-blocked:

$$\mathbf{C}(x-1) \oplus \mathbf{C}(x-1) \oplus \mathbf{C}((x^2+1)^2) = (1) \oplus (1) \oplus \begin{pmatrix} 0 & 0 & 0 & -1 \\ 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & -2 \\ 0 & 0 & 1 & 0 \end{pmatrix},$$

of total size  $1 + 1 + 4 = 6$ . Read as a complex matrix, the second form splits further: over  $\mathbb{C}$  we have  $x^2+1 = (x-i)(x+i)$ , so the elementary divisors become  $x-1, x-1, (x-i)^2, (x+i)^2$  — the invariant factors  $d_1, d_2$  are unchanged — and [Corollary 9.6.8](#) turns the elementary divisor form into the Jordan form  $(1) \oplus (1) \oplus \mathbf{J}_2(i) \oplus \mathbf{J}_2(-i)$ . Over  $\mathbb{Q}$  no further splitting is possible, since  $x^2+1$  is irreducible there.

### Example. (The Rational Canonical Form of a Matrix over the Rationals)

Let

$$\mathbf{A} = \begin{pmatrix} 0 & -1 & 1 & 0 \\ 1 & 0 & 0 & 1 \\ 0 & 0 & 0 & -1 \\ 0 & 0 & 1 & 0 \end{pmatrix} \in M_4(\mathbb{Q}), \quad p_{\mathbf{A}} = (x^2+1)^2.$$

Find the invariant factors of  $\mathbf{A}$ , its rational canonical form ([Theorem 9.6.5](#)), and an invertible  $\mathbf{P} \in M_4(\mathbb{Q})$  realizing it.

*Solution. The characteristic polynomial.* In  $2 \times 2$  blocks,  $\mathbf{A} = \begin{pmatrix} \mathbf{R} & \mathbf{I}_2 \\ 0 & \mathbf{R} \end{pmatrix}$  with  $\mathbf{R} =$

$\begin{pmatrix} 0 & -1 \\ 1 & 0 \end{pmatrix}$ , so  $x\mathbf{I} - \mathbf{A}$  is block upper triangular and  $p_{\mathbf{A}} = (\det(x\mathbf{I}_2 - \mathbf{R}))^2 = (x^2 + 1)^2$  (Theorem 6.8.4), as stated.

*The minimal polynomial.* By Theorem 8.5.1 and Lemma 8.5.2,  $m_{\mathbf{A}}$  is  $x^2 + 1$  or  $(x^2 + 1)^2$ . Compute  $\mathbf{A}^2$ : its columns are  $\mathbf{A}$  applied to the columns of  $\mathbf{A}$ . With  $\mathbf{c}_1, \dots, \mathbf{c}_4$  the columns of  $\mathbf{A}$ ,

$$\mathbf{Ac}_1 = \mathbf{A}(0, 1, 0, 0) = (-1, 0, 0, 0), \quad \mathbf{Ac}_2 = \mathbf{A}(-1, 0, 0, 0) = (0, -1, 0, 0),$$

$$\mathbf{Ac}_3 = \mathbf{A}(1, 0, 0, 1) = (0, 1, 0, 0) + (0, 1, -1, 0) = (0, 2, -1, 0),$$

$$\mathbf{Ac}_4 = \mathbf{A}(0, 1, -1, 0) = (-1, 0, 0, 0) + (-1, 0, 0, -1) = (-2, 0, 0, -1).$$

So

$$\mathbf{A}^2 = \begin{pmatrix} -1 & 0 & 0 & -2 \\ 0 & -1 & 2 & 0 \\ 0 & 0 & -1 & 0 \\ 0 & 0 & 0 & -1 \end{pmatrix}, \quad \mathbf{A}^2 + \mathbf{I} = \begin{pmatrix} 0 & 0 & 0 & -2 \\ 0 & 0 & 2 & 0 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \end{pmatrix} \neq 0.$$

Hence  $m_{\mathbf{A}} = (x^2 + 1)^2 = p_{\mathbf{A}}$ , and  $\mathbf{A}$  is cyclic (Theorem 9.5.6). So  $r = 1$ , the single invariant factor is  $d_1 = (x^2 + 1)^2 = x^4 + 2x^2 + 1$ , and the rational canonical form is

$$\mathbf{C}(d_1) = \begin{pmatrix} 0 & 0 & 0 & -1 \\ 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & -2 \\ 0 & 0 & 1 & 0 \end{pmatrix}.$$

The elementary divisor form (Theorem 9.6.7) is the same matrix, since  $x^2 + 1$  is irreducible over  $\mathbb{Q}$  and  $d_1$  is already a prime power.

*A cyclic vector.* By Theorem 9.5.6 we need  $\mathbf{v}$  with  $m_{\mathbf{A}, \mathbf{v}} = m_{\mathbf{A}}$ ; since  $\mathbf{A}^2 + \mathbf{I} \neq 0$ , pick  $\mathbf{v}$  with  $(\mathbf{A}^2 + \mathbf{I})\mathbf{v} \neq \mathbf{0}$ . The third column of  $\mathbf{A}^2 + \mathbf{I}$  is non-zero, so  $\mathbf{v} = \mathbf{e}_3$  works: then  $(\mathbf{A}^2 + \mathbf{I})\mathbf{v} \neq \mathbf{0}$ , so  $m_{\mathbf{A}, \mathbf{v}} \nmid x^2 + 1$ , and as  $m_{\mathbf{A}, \mathbf{v}}$  divides  $(x^2 + 1)^2$  it must equal  $(x^2 + 1)^2$  (Lemma 8.5.2). Its chain is

$$\mathbf{v} = (0, 0, 1, 0), \quad \mathbf{A}\mathbf{v} = (1, 0, 0, 1), \quad \mathbf{A}^2\mathbf{v} = (0, 2, -1, 0),$$

$$\mathbf{A}^3\mathbf{v} = \mathbf{A}(0, 2, -1, 0) = (-2, 0, 0, 0) + (-1, 0, 0, -1) = (-3, 0, 0, -1),$$

reading columns of  $\mathbf{A}$  as before. Hence

$$\mathbf{P} = \begin{pmatrix} 0 & 1 & 0 & -3 \\ 0 & 0 & 2 & 0 \\ 1 & 0 & -1 & 0 \\ 0 & 1 & 0 & -1 \end{pmatrix}, \quad \mathbf{P}^{-1}\mathbf{A}\mathbf{P} = \mathbf{C}((x^2 + 1)^2).$$

*Invertibility of  $\mathbf{P}$ .* Expanding  $\det \mathbf{P}$  along the third row, whose entries are  $(1, 0, -1, 0)$  (Theorem 6.6.1), and then each  $3 \times 3$  determinant along its own first column,

$$\begin{aligned} \det \mathbf{P} &= 1 \cdot \det \begin{pmatrix} 1 & 0 & -3 \\ 0 & 2 & 0 \\ 1 & 0 & -1 \end{pmatrix} + (-1)(-1)^{3+3} \det \begin{pmatrix} 0 & 1 & -3 \\ 0 & 0 & 0 \\ 0 & 1 & -1 \end{pmatrix} \\ &= 1 \cdot (2(-1 + 3)) - 0 = 4 \neq 0, \end{aligned}$$

the second determinant vanishing because it has a zero row, and the first being  $2 \det \begin{pmatrix} 1 & -3 \\ 1 & -1 \end{pmatrix} = 2 \cdot 2 = 4$  by expansion along its second row.

*The check.* The identity  $\mathbf{A}\mathbf{P} = \mathbf{P}\mathbf{C}(d_1)$  reads column by column as  $\mathbf{A}\mathbf{v} = \mathbf{A}\mathbf{v}$ ,  $\mathbf{A}(\mathbf{A}\mathbf{v}) = \mathbf{A}^2\mathbf{v}$ ,  $\mathbf{A}(\mathbf{A}^2\mathbf{v}) = \mathbf{A}^3\mathbf{v}$  — true by construction — and

$$\mathbf{A}(\mathbf{A}^3\mathbf{v}) = -\mathbf{v} - 2\mathbf{A}^2\mathbf{v},$$

which is the annihilator relation  $\mathbf{A}^4 + 2\mathbf{A}^2 + \mathbf{I} = 0$  applied to  $\mathbf{v}$ . Verifying it directly:  $\mathbf{A}(-3, 0, 0, -1) = -3(0, 1, 0, 0) - (0, 1, -1, 0) = (0, -4, 1, 0)$ , while

$$-\mathbf{v} - 2\mathbf{A}^2\mathbf{v} = -(0, 0, 1, 0) - 2(0, 2, -1, 0) = (0, -4, 2 - 1, 0) = (0, -4, 1, 0). \checkmark$$

## 9.6.8 Exercises

### 9.6.8.1 A. Check your understanding

#### Exercise A1.

- State the Cyclic Decomposition Theorem, including the divisibility chain and what  $d_r$  and  $\prod_j d_j$  are.
- Define the invariant factors and the elementary divisors of  $T$ .
- What is the rational canonical form of  $T$ , and over which fields does it exist?
- True or false: two matrices in  $M_n(\mathbb{Q})$  that are similar over  $\mathbb{C}$  are similar over  $\mathbb{Q}$ . Justify your answer.
- An operator on a 5-dimensional  $\mathbb{Q}$ -space has  $p_T = (x - 3)^5$  and  $m_T = (x - 3)^2$ . What are its elementary divisors and invariant factors?
- True or false:  $r = 1$  if and only if  $T$  is cyclic. Justify your answer.

*Solution.*

- For  $V \neq \{0\}$  finite-dimensional and  $T \in \mathcal{L}(V)$ , there are  $\mathbf{v}_1, \dots, \mathbf{v}_r$  with  $V = Z(\mathbf{v}_1; T) \oplus \dots \oplus Z(\mathbf{v}_r; T)$  and  $m_{T, \mathbf{v}_j} = d_j$ , where  $d_1 \mid d_2 \mid \dots \mid d_r$  are monic and non-constant,  $d_r = m_T$  and  $d_1 \cdots d_r = p_T$  ([Theorem 9.6.2](#)).
- The invariant factors are the polynomials  $d_1 \mid \dots \mid d_r$  of that theorem; the elementary divisors are the prime powers occurring in their factorizations, listed with repetition ([Definition 9.6.1](#)). Both are determined by  $T$  ([Corollary 9.6.4](#)).
- The matrix  $\mathbf{C}(d_1) \oplus \dots \oplus \mathbf{C}(d_r)$ , which is  $[T]_{\mathcal{B}}$  for a suitable basis; it exists over **every** field ([Theorem 9.6.5](#)).
- True, by [Corollary 9.6.10](#) with  $K = \mathbb{Q}$  and  $L = \mathbb{C}$ .
- Elementary divisors are powers of  $x - 3$  with largest exponent 2 and exponents summing to 5: so  $(x - 3)^2, (x - 3)^2, x - 3$ . The exponents, in decreasing order, are 2, 2, 1; numbering the columns backwards gives  $d_3 = (x - 3)^2 = m_T$ ,  $d_2 = (x - 3)^2$  and  $d_1 = x - 3$ .
- True.  $r = 1$  means  $V = Z(\mathbf{v}_1; T)$ , that is,  $\mathbf{v}_1$  is a cyclic vector; conversely if  $\mathbf{v}$  is cyclic, then  $d_1 = m_{T, \mathbf{v}} = m_T = p_T$  is the only invariant factor, by [Theorem 9.5.6](#) and the uniqueness in [Corollary 9.6.4](#).

## 9.6.8.2 B. Practice

**Exercise B1: Invariant factors from given data.**

In each case  $T$  is an operator on a  $\mathbb{Q}$ -vector space with the stated characteristic and minimal polynomials. Find the elementary divisors, the invariant factors, and the rational canonical form ([Theorem 9.6.5](#)).

- (a)  $p_T = (x - 2)^4$ ,  $m_T = (x - 2)^2$ .
- (b)  $p_T = (x^2 - 2)^3$ ,  $m_T = (x^2 - 2)^2$ .
- (c)  $p_T = (x - 1)^2(x + 1)^2$ ,  $m_T = (x - 1)(x + 1)$ .

*Solution.*

- (a) Powers of  $x - 2$  with largest exponent 2 and exponents summing to 4: either 2, 2 or 2, 1, 1. So the data do **not** determine the answer; both

$$\text{elementary divisors } (x - 2)^2, (x - 2)^2 \quad \text{and} \quad (x - 2)^2, x - 2, x - 2$$

are possible, with invariant factors  $((x - 2)^2, (x - 2)^2)$  and  $(x - 2, x - 2, (x - 2)^2)$  and rational forms

$$\mathbf{C}((x - 2)^2) \oplus \mathbf{C}((x - 2)^2), \quad (2) \oplus (2) \oplus \mathbf{C}((x - 2)^2),$$

where  $\mathbf{C}((x - 2)^2) = \mathbf{C}(x^2 - 4x + 4) = \begin{pmatrix} 0 & -4 \\ 1 & 4 \end{pmatrix}$ . (The extra datum that decides is  $\dim \ker(T - 2 \text{ id})$ , which is 2 in the first case and 3 in the second, by [Proposition 9.6.3](#).)

- (b)  $x^2 - 2$  is irreducible over  $\mathbb{Q}$ , since  $\sqrt{2} \notin \mathbb{Q}$  ([Theorem 0.2.3](#), [Theorem 5.3.1 \(b\)](#)). The exponents are powers of  $x^2 - 2$  with largest 2 summing to 3: only 2, 1. So the elementary divisors are  $(x^2 - 2)^2$  and  $x^2 - 2$ , the invariant factors are  $d_1 = x^2 - 2$  and  $d_2 = (x^2 - 2)^2 = x^4 - 4x^2 + 4$ , and the rational form is

$$\mathbf{C}(x^2 - 2) \oplus \mathbf{C}((x^2 - 2)^2) = \begin{pmatrix} 0 & 2 \\ 1 & 0 \end{pmatrix} \oplus \begin{pmatrix} 0 & 0 & 0 & -4 \\ 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 4 \\ 0 & 0 & 1 & 0 \end{pmatrix},$$

of size  $2 + 4 = 6 = \deg p_T$ .

- (c) Both primes have largest exponent 1 and total contribution 2, so the elementary divisors are  $x - 1, x - 1, x + 1, x + 1$ . Arranging by prime,  $x - 1 : 1, 1$  and  $x + 1 : 1, 1$ , so  $r = 2$  and  $d_2 = (x - 1)(x + 1) = m_T$ ,  $d_1 = (x - 1)(x + 1)$ . The rational form is

$$\mathbf{C}(x^2 - 1) \oplus \mathbf{C}(x^2 - 1) = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix} \oplus \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}.$$

(This operator is diagonalizable, with  $\text{diag}(1, 1, -1, -1)$  as its Jordan form; the rational form is a different matrix representing the same operator.)

**Exercise B2: Rational forms of matrices.**

For each matrix, find  $p_A$ ,  $m_A$ , the invariant factors, the rational canonical form ([Theo-](#)

rem 9.6.5) and the elementary divisor form (Theorem 9.6.7).

- (a)  $\mathbf{A}_1 = \begin{pmatrix} 0 & 1 & 0 \\ 0 & 0 & 1 \\ 1 & 0 & 0 \end{pmatrix} \in M_3(\mathbb{Q})$ .
- (b)  $\mathbf{A}_2 = \text{diag}(1, 1, 2) \in M_3(\mathbb{Q})$ .
- (c)  $\mathbf{A}_3 = \begin{pmatrix} 0 & -1 & 0 & 0 \\ 1 & 0 & 0 & 0 \\ 0 & 0 & 0 & -1 \\ 0 & 0 & 1 & 0 \end{pmatrix} \in M_4(\mathbb{Q})$ .

*Solution.*

- (a) Reading the columns,  $\mathbf{A}_1 \mathbf{e}_1 = \mathbf{e}_3$ ,  $\mathbf{A}_1 \mathbf{e}_3 = \mathbf{e}_2$  and  $\mathbf{A}_1 \mathbf{e}_2 = \mathbf{e}_1$ , so  $\mathbf{A}_1^3 = \mathbf{I}$  and the polynomial  $x^3 - 1$  annihilates  $\mathbf{A}_1$ . Also  $\mathbf{A}_1^2 \mathbf{e}_1 = \mathbf{A}_1 \mathbf{e}_3 = \mathbf{e}_2$ , so  $(\mathbf{e}_1, \mathbf{A}_1 \mathbf{e}_1, \mathbf{A}_1^2 \mathbf{e}_1) = (\mathbf{e}_1, \mathbf{e}_3, \mathbf{e}_2)$  is a basis of  $\mathbb{Q}^3$  and  $\mathbf{e}_1$  is a cyclic vector. Hence  $\deg m_{\mathbf{A}_1} = 3$  (Theorem 9.5.6), and since  $m_{\mathbf{A}_1}$  is monic of degree 3 dividing  $x^3 - 1$ , we get  $m_{\mathbf{A}_1} = p_{\mathbf{A}_1} = x^3 - 1$ . The single invariant factor is  $d_1 = x^3 - 1$ , and the rational canonical form is

$$\mathbf{C}(x^3 - 1) = \begin{pmatrix} 0 & 0 & 1 \\ 1 & 0 & 0 \\ 0 & 1 & 0 \end{pmatrix}.$$

Over  $\mathbb{Q}$ ,  $x^3 - 1 = (x - 1)(x^2 + x + 1)$  with both factors irreducible (the quadratic has no rational root). So the elementary divisors are  $x - 1$  and  $x^2 + x + 1$ , and the elementary divisor form is

$$(1) \oplus \mathbf{C}(x^2 + x + 1) = (1) \oplus \begin{pmatrix} 0 & -1 \\ 1 & -1 \end{pmatrix}.$$

- (b)  $p_{\mathbf{A}_2} = (x - 1)^2(x - 2)$  and  $m_{\mathbf{A}_2} = (x - 1)(x - 2)$  (Proposition 8.5.6). Elementary divisors:  $x - 1, x - 1, x - 2$ . Invariant factors:  $d_2 = (x - 1)(x - 2) = m_{\mathbf{A}_2}$  and  $d_1 = x - 1$ . Rational form

$$(1) \oplus \mathbf{C}((x - 1)(x - 2)) = (1) \oplus \begin{pmatrix} 0 & -2 \\ 1 & 3 \end{pmatrix},$$

elementary divisor form  $(1) \oplus (1) \oplus (2) = \text{diag}(1, 1, 2) = \mathbf{A}_2$  itself.

- (c)  $\mathbf{A}_3 = \mathbf{R} \oplus \mathbf{R}$  with  $\mathbf{R} = \begin{pmatrix} 0 & -1 \\ 1 & 0 \end{pmatrix}$ , and  $p_{\mathbf{R}} = x^2 + 1$ ,  $m_{\mathbf{R}} = x^2 + 1$ , so  $p_{\mathbf{A}_3} = (x^2 + 1)^2$  (Theorem 6.8.4 applied to  $x\mathbf{I} - \mathbf{A}_3$ , which is block diagonal over  $\mathbb{Q}[x]$ ) and  $m_{\mathbf{A}_3} = \text{lcm}(x^2 + 1, x^2 + 1) = x^2 + 1$  (Proposition 8.5.6). Since  $x^2 + 1$  is irreducible over  $\mathbb{Q}$ , the elementary divisors are  $x^2 + 1$  and  $x^2 + 1$ ;  $r = 2$  and  $d_1 = d_2 = x^2 + 1$ . Both forms are

$$\mathbf{C}(x^2 + 1) \oplus \mathbf{C}(x^2 + 1) = \begin{pmatrix} 0 & -1 \\ 1 & 0 \end{pmatrix} \oplus \begin{pmatrix} 0 & -1 \\ 1 & 0 \end{pmatrix} = \mathbf{A}_3.$$

Compare Example (The Rational Canonical Form of a Matrix over the Rationals): that matrix has the same characteristic polynomial but a different minimal polynomial, so the two are **not** similar.



**Exercise B3: Counting similarity classes over a finite field.**

Work over  $F = \mathbb{F}_2$ . For each polynomial  $p$ , find the number of similarity classes of matrices  $\mathbf{A} \in M_3(\mathbb{F}_2)$  with  $p_{\mathbf{A}} = p$ , and list the invariant factors of each class.

- (a)  $p = (x + 1)^3$ .
- (b)  $p = x(x + 1)^2$ .
- (c)  $p = x^3 + x + 1$ .

*Hint: by Corollary 9.6.6, a class is a way of writing  $p = d_1 \cdots d_r$  with  $d_1 \mid \cdots \mid d_r$ .*

*Solution.*

- (a) The only prime involved is  $x + 1$ , so the elementary divisors are powers  $(x + 1)^a$  with exponents forming a partition of 3. The three partitions (3), (2, 1), (1, 1, 1) give

$$((x + 1)^3), \quad (x + 1, (x + 1)^2), \quad (x + 1, x + 1, x + 1)$$

as lists of invariant factors. So there are 3 classes, with representatives  $\mathbf{C}((x + 1)^3)$ ,  $(1) \oplus \mathbf{C}((x + 1)^2)$  and  $\mathbf{I}_3$  (note  $-1 = 1$  in  $\mathbb{F}_2$ ).

- (b) Now two primes,  $x$  with total exponent 1 and  $x + 1$  with total exponent 2. For  $x$  the only option is the single elementary divisor  $x$ ; for  $x + 1$  the options are  $(x + 1)^2$  or  $x + 1, x + 1$ . So there are 2 classes, with invariant factors

$$(x(x + 1)^2) \quad \text{and} \quad (x + 1, x(x + 1)).$$

In the first case  $r = 1$  and the matrix is cyclic; in the second  $r = 2$ .

- (c)  $x^3 + x + 1$  has no root in  $\mathbb{F}_2$  (its values at 0 and 1 are 1 and 1), so it is irreducible (Theorem 5.3.1 (b)). A prime of degree 3 in a characteristic polynomial of degree 3 can occur only with exponent 1, so the only possible list of elementary divisors is  $(x^3 + x + 1)$  itself: 1 class, represented by  $\mathbf{C}(x^3 + x + 1)$ .

**9.6.8.3 C. Going deeper****Exercise C1: Which matrices are companion matrices.**

Let  $\mathbf{A} \in M_n(F)$  with  $n \geq 1$ .

- (a) Prove that  $\mathbf{A}$  is similar to a companion matrix if and only if  $m_{\mathbf{A}} = p_{\mathbf{A}}$ , and that the companion matrix is then  $\mathbf{C}(p_{\mathbf{A}})$ .
- (b) Deduce that  $\mathbf{A}$  is similar to a companion matrix if and only if its rational canonical form has exactly one block.
- (c) Give an example of a matrix in  $M_2(F)$  that is not similar to any companion matrix.

*Solution.*

- (a) ( $\Leftarrow$ ) If  $m_{\mathbf{A}} = p_{\mathbf{A}}$ , then  $\mathbf{A}$  is cyclic (Theorem 9.5.6) and, in the basis generated by a cyclic vector, its matrix is  $\mathbf{C}(m_{\mathbf{A}}) = \mathbf{C}(p_{\mathbf{A}})$ ; so  $\mathbf{A} \sim \mathbf{C}(p_{\mathbf{A}})$  (Theorem 3.7.4).  
 ( $\Rightarrow$ ) Suppose  $\mathbf{A} \sim \mathbf{C}(q)$  for some monic  $q$ . By Theorem 9.5.3,  $m_{\mathbf{C}(q)} = p_{\mathbf{C}(q)} = q$ . Similar matrices share both polynomials (Theorem 6.10.3, Theorem 8.5.5), so

$m_{\mathbf{A}} = p_{\mathbf{A}} = q$ , and the companion matrix is  $\mathbf{C}(p_{\mathbf{A}})$ .

- (b) By [Theorem 9.6.5](#) the form has one block exactly when  $r = 1$ , that is, when  $d_1 = m_{\mathbf{A}} = p_{\mathbf{A}}$  ([Theorem 9.6.2](#) (b), (c)). Now apply (a).
- (c)  $\mathbf{I}_2$ : here  $m = x - 1 \neq (x - 1)^2 = p$ . Concretely,  $\mathbf{I}_2$  is similar only to itself, and no companion matrix of a monic quadratic is  $\mathbf{I}_2$ , since the  $(2, 1)$  entry of  $\mathbf{C}(q)$  is  $1 \neq 0$ .

**Exercise C2: All similarity classes of two by two matrices over the smallest field.**

Work over  $F = \mathbb{F}_2$ , so that  $M_2(\mathbb{F}_2)$  has 16 elements.

- (a) List the four monic polynomials of degree 2 over  $\mathbb{F}_2$  and factor each into irreducibles.
- (b) For each, list the possible lists of invariant factors of a matrix  $\mathbf{A} \in M_2(\mathbb{F}_2)$  with that characteristic polynomial.
- (c) Conclude the total number of similarity classes in  $M_2(\mathbb{F}_2)$ , and give a representative of each.
- (d) Check your count against the orbit sizes:  $|\mathrm{GL}_2(\mathbb{F}_2)| = 6$ , and the class of  $\mathbf{A}$  has  $6/|\{\mathbf{P} : \mathbf{P}^{-1}\mathbf{A}\mathbf{P} = \mathbf{A}\}|$  elements.

*Solution.*

- (a)  $x^2$ ;  $x^2 + 1 = (x + 1)^2$ ;  $x^2 + x = x(x + 1)$ ; and  $x^2 + x + 1$ , which has no root in  $\mathbb{F}_2$  and is therefore irreducible ([Theorem 5.3.1](#) (b)).
- (b) For  $p = x^2$ : elementary divisors  $x^2$  (invariant factors  $(x^2)$ ) or  $x, x$  (invariant factors  $(x, x)$ ) — two options. For  $p = (x + 1)^2$ : likewise two,  $((x + 1)^2)$  and  $(x + 1, x + 1)$ . For  $p = x(x + 1)$ : each prime occurs once, so the elementary divisors are  $x, x + 1$  and there is a single list of invariant factors,  $(x(x + 1))$ . For the irreducible  $p = x^2 + x + 1$ : one option,  $(x^2 + x + 1)$ .
- (c) Total  $2 + 2 + 1 + 1 = 6$  classes, with representatives

$$\mathbf{C}(x^2) = \begin{pmatrix} 0 & 0 \\ 1 & 0 \end{pmatrix}, \quad 0, \quad \mathbf{C}((x + 1)^2) = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix},$$

$$\mathbf{I}_2, \quad \mathrm{diag}(0, 1), \quad \mathbf{C}(x^2 + x + 1) = \begin{pmatrix} 0 & 1 \\ 1 & 1 \end{pmatrix},$$

using  $-1 = 1$  in  $\mathbb{F}_2$ . (For  $(x + 1)^2 = x^2 + 1$  the coefficients are  $a_0 = 1, a_1 = 0$ , so  $\mathbf{C} = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}$ , the coordinate swap.)

- (d) The two scalar matrices  $0$  and  $\mathbf{I}_2$  commute with everything, so their classes have  $6/6 = 1$  element each. For  $\mathrm{diag}(0, 1)$ , a matrix commutes with it exactly when it is diagonal (compare entries), and the invertible diagonal matrices over  $\mathbb{F}_2$  are only  $\mathbf{I}_2$ , so the class has 6 elements. For  $\mathbf{C}(x^2)$  and  $\mathbf{C}(x^2 + 1)$  the commutant inside  $\mathrm{GL}_2$  has 2 elements — by [Exercise \(C1: What commutes with a cyclic operator\)](#) the commutant of a cyclic matrix is  $\{a\mathbf{I} + b\mathbf{A}\}$ , which here has 4 elements, of which 2 are invertible — so each class has 3 elements. For  $\mathbf{C}(x^2 + x + 1)$ , whose order is 3 in  $\mathrm{GL}_2(\mathbb{F}_2)$ , the commutant is  $\{a\mathbf{I} + b\mathbf{A}\}$  again, with 3 invertible elements, so

the class has 2 elements. Total:

$$1 + 1 + 6 + 3 + 3 + 2 = 16 = |M_2(\mathbb{F}_2)|. \checkmark$$

**Exercise C3: A matrix over the rationals is similar to its transpose.**

- (a) Let  $\mathbf{A}, \mathbf{B} \in M_n(\mathbb{Q})$  have the same Jordan form over  $\mathbb{C}$ . Prove that  $\mathbf{A}$  and  $\mathbf{B}$  are similar over  $\mathbb{Q}$ .
- (b) Deduce that  $\mathbf{A} \sim \mathbf{A}^\top$  over  $\mathbb{Q}$  for every  $\mathbf{A} \in M_n(\mathbb{Q})$ .
- (c) Which step of (b) would fail for a general field  $K$  in place of  $\mathbb{Q}$ ?

*Hint for (b): Exercise (C1: A matrix is similar to its transpose).*

*Solution.*

- (a) A matrix in  $M_n(\mathbb{Q})$  is also a matrix in  $M_n(\mathbb{C})$ , and  $p_{\mathbf{A}}$  splits over  $\mathbb{C}$  (Corollary 5.5.5), so both have Jordan forms there. Having the same Jordan form, they are similar over  $\mathbb{C}$  (Corollary 9.3.3). By Corollary 9.6.10 with  $K = \mathbb{Q}$  and  $L = \mathbb{C}$ , they are similar over  $\mathbb{Q}$ .
- (b) Let  $\mathbf{A} \in M_n(\mathbb{Q})$ . Over  $\mathbb{C}$ ,  $p_{\mathbf{A}}$  splits, so Exercise (C1: A matrix is similar to its transpose) (c) gives  $\mathbf{A} \sim \mathbf{A}^\top$  over  $\mathbb{C}$ . Both matrices lie in  $M_n(\mathbb{Q})$ , so Corollary 9.6.10 gives  $\mathbf{A} \sim \mathbf{A}^\top$  over  $\mathbb{Q}$ .
- (c) The step that fails is the splitting: over a general field  $K$  there is no reason for  $p_{\mathbf{A}}$  to split, and we have not constructed a larger field where it does. (The conclusion is nevertheless true over every field; Section 7 proves it without any splitting, using the Smith normal form of  $x\mathbf{I} - \mathbf{A}$ .) Note that Corollary 9.6.10 itself needs no hypothesis on  $K$  — only the existence of a suitable  $L$  does.

## 9.7 Polynomial Matrices and the Smith Normal Form

The rational canonical form of the previous section is a complete answer to the similarity problem, but the proof did not tell us how to *find* the invariant factors of a given matrix. This section supplies the missing algorithm, and it comes from an unexpected place: instead of working with  $\mathbf{A}$ , we work with the single polynomial matrix  $x\mathbf{I} - \mathbf{A}$ , and we reduce it by row and column operations whose scalars are allowed to be polynomials. The reduced form is diagonal, its non-constant diagonal entries are the invariant factors, and two matrices are similar exactly when their reduced forms agree. On the way we finally pay off a promise from Chapter 3: every square matrix is similar to its transpose, over every field.

### 9.7.1 The space as a module over the polynomials

Everything in this chapter is built by applying **polynomials** in  $T$  to vectors: the  $T$ -annihilator of a vector, the cyclic subspace  $Z(\mathbf{v}; T)$ , the kernels  $\ker p(T)$ . Name that structure once. For  $V$

a vector space over  $F$  and  $T \in \mathcal{L}(V)$ , the product

$$F[x] \times V \rightarrow V, \quad p \cdot \mathbf{v} := p(T) \mathbf{v}$$

satisfies, by [Theorem 5.7.1](#), exactly the vector space axioms with  $F$  replaced by  $F[x]$ : it is additive in each factor,  $(pq) \cdot \mathbf{v} = p \cdot (q \cdot \mathbf{v})$ , and  $1 \cdot \mathbf{v} = \mathbf{v}$ . A set with an addition and a product by the elements of a commutative ring  $R$  obeying these rules is an  $R$ -**module**, so the pair  $(V, T)$  is the same thing as an  $F[x]$ -module that is finite-dimensional over the constants  $F$ . We develop no general theory of modules; the point is the dictionary:

| $F[x]$ -module language                                               | operator language                                                       |
|-----------------------------------------------------------------------|-------------------------------------------------------------------------|
| submodule of $V$                                                      | $T$ -invariant subspace ( <a href="#">Definition 8.1.1</a> )            |
| submodule generated by $\mathbf{v}$ , that is $F[x] \cdot \mathbf{v}$ | cyclic subspace $Z(\mathbf{v}; T)$ ( <a href="#">Definition 9.5.1</a> ) |
| the annihilator of $\mathbf{v}$ in $F[x]$                             | the $T$ -annihilator of $\mathbf{v}$                                    |
| $V$ is generated by one element                                       | $T$ has a cyclic vector                                                 |

A subspace closed under multiplication by  $x$  is closed under multiplication by every polynomial, which is why the first two lines match.

Where does  $x\mathbf{I} - \mathbf{A}$  come from here? Fix  $\mathcal{B} = (\mathbf{b}_1, \dots, \mathbf{b}_n)$  with  $\mathbf{A} = [T]_{\mathcal{B}}$ . As an  $F[x]$ -module,  $V$  is generated by  $\mathbf{b}_1, \dots, \mathbf{b}_n$ , subject to the  $n$  relations  $x \cdot \mathbf{b}_j = \sum_i a_{ij} \mathbf{b}_i$ , whose coefficient vectors are the columns of  $x\mathbf{I} - \mathbf{A}$ . So  $x\mathbf{I} - \mathbf{A}$  is the **matrix of relations**, and changing generators or replacing one relation by a combination of relations should not change  $V$ . That is the intuition behind the operations we now allow; the proofs below are matrix proofs and do not rely on it.

## 9.7.2 Row and column operations with polynomial multipliers

A matrix in  $M_n(F[x])$  will be called a **polynomial matrix**. Over a field we reduced a matrix by three kinds of row operation ([Definition 2.2.1](#)). Over  $F[x]$  we must be careful with one of them: multiplying a row by  $x$  cannot be undone, because  $x$  has no inverse in  $F[x]$ . So the scaling operation keeps its scalar in  $F$ , while the operation that adds a multiple of one row to another may use any polynomial multiplier.

### Definition 9.7.1 — Elementary Operations over $F[x]$

Let  $\mathbf{M} \in M_n(F[x])$ . An **elementary row operation over  $F[x]$**  on  $\mathbf{M}$  is one of:

- (E1) interchanging two rows;
- (E2) multiplying one row by a **non-zero scalar**  $c \in F$ ;
- (E3) adding  $h$  times one row to a **different** row, where  $h \in F[x]$  is **any** polynomial.

**Elementary column operations over  $F[x]$**  are defined the same way with “column” in place of “row”.

In words: (E1) and (E3) are as over a field, except that (E3) now permits a polynomial multiplier; (E2) is restricted, because only the non-zero constants have inverses in  $F[x]$ . Each of the three

is reversible by an operation of the same kind: swap back, scale by  $c^{-1}$ , add  $-h$  times the row back.

Two polynomial matrices reachable from one another by such operations should count as the same. As over a field (Definition 3.8.1), the clean way to say this is with invertible matrices on both sides — but “invertible” must now mean invertible *inside*  $M_n(F[x])$ .

### Proposition 9.7.1 — Invertible Polynomial Matrices

Let  $\mathbf{U} \in M_n(F[x])$ . Then  $\mathbf{U}$  has an inverse in  $M_n(F[x])$  if and only if  $\det \mathbf{U}$  is a **non-zero constant**.

*Proof.* ( $\Rightarrow$ ) Suppose  $\mathbf{U}\mathbf{W} = \mathbf{I}$  with  $\mathbf{W} \in M_n(F[x])$ . Taking determinants and using multiplicativity over the commutative ring  $F[x]$  (Theorem 6.5.4 and the remark on commutative rings after it),  $\det \mathbf{U} \cdot \det \mathbf{W} = 1$ . By Theorem 0.8.3,  $\deg(\det \mathbf{U}) + \deg(\det \mathbf{W}) = \deg 1 = 0$ , and degrees of non-zero polynomials are  $\geq 0$ , so  $\deg(\det \mathbf{U}) = 0$ ; that is,  $\det \mathbf{U} \in F \setminus \{0\}$ .

( $\Leftarrow$ ) Suppose  $\det \mathbf{U} = c \in F \setminus \{0\}$ . The adjugate identity

$$\mathbf{U} \operatorname{adj} \mathbf{U} = \operatorname{adj} \mathbf{U} \mathbf{U} = (\det \mathbf{U}) \mathbf{I}$$

holds over any commutative ring (Theorem 6.6.2 and the remark on commutative rings following it), and  $\operatorname{adj} \mathbf{U} \in M_n(F[x])$ , since its entries are minors of  $\mathbf{U}$ . So  $\mathbf{W} = c^{-1} \operatorname{adj} \mathbf{U} \in M_n(F[x])$  satisfies  $\mathbf{U}\mathbf{W} = \mathbf{W}\mathbf{U} = \mathbf{I}$ . ■

### Definition 9.7.2 — Equivalence over $F[x]$

Let  $\mathbf{M}, \mathbf{M}' \in M_n(F[x])$ . We say  $\mathbf{M}$  and  $\mathbf{M}'$  are **equivalent over  $F[x]$** , written  $\mathbf{M} \approx \mathbf{M}'$ , if there are  $\mathbf{U}, \mathbf{V} \in M_n(F[x])$ , **each invertible in  $M_n(F[x])$** , with

$$\mathbf{M}' = \mathbf{U}\mathbf{M}\mathbf{V}.$$

### Proposition 9.7.2 — Operations and Equivalence

Let  $\mathbf{M}, \mathbf{M}' \in M_n(F[x])$ .

- (a)  $\approx$  is an equivalence relation on  $M_n(F[x])$ .
- (b) If  $\mathbf{M}'$  is obtained from  $\mathbf{M}$  by a finite sequence of elementary row and column operations over  $F[x]$ , then  $\mathbf{M} \approx \mathbf{M}'$ .

*Proof.*

- (a) Take  $\mathbf{U} = \mathbf{V} = \mathbf{I}$  for reflexivity. If  $\mathbf{M}' = \mathbf{U}\mathbf{M}\mathbf{V}$ , then  $\mathbf{M} = \mathbf{U}^{-1}\mathbf{M}'\mathbf{V}^{-1}$ , and  $\mathbf{U}^{-1}, \mathbf{V}^{-1}$  are invertible in  $M_n(F[x])$ ; this gives symmetry. If  $\mathbf{M}' = \mathbf{U}\mathbf{M}\mathbf{V}$  and  $\mathbf{M}'' = \mathbf{U}'\mathbf{M}'\mathbf{V}'$ , then  $\mathbf{M}'' = (\mathbf{U}'\mathbf{U})\mathbf{M}(\mathbf{V}\mathbf{V}')$ , and a product of invertible matrices is invertible, which gives transitivity.

- (b) Each elementary row operation over  $F[x]$  is left multiplication by the matrix obtained by performing that operation on  $\mathbf{I}_n$ , exactly as over a field (Theorem 2.4.1, whose proof uses only the ring operations). Those three matrices are  $\mathbf{P}_{ij}$ ,  $\mathbf{D}_i(c)$  and  $\mathbf{I} + h\mathbf{E}_{ij}$  with  $i \neq j$ , of determinants  $-1$ ,  $c \neq 0$  and  $1$  respectively (Theorem 6.4.6 and Theorem 6.5.1, valid over a commutative ring), hence invertible in  $M_n(F[x])$  by Proposition 9.7.1. Column operations are right multiplication by the same matrices. Now use (a).

■

**Warning.** Multiplying a row by a non-constant polynomial is not an allowed operation. It changes the answer. The matrix  $\mathbf{M} = \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix}$  becomes  $\begin{pmatrix} x & 0 \\ 0 & 1 \end{pmatrix}$  after “multiplying row 1 by  $x$ ”, and the two are not equivalent: their determinants are  $1$  and  $x$ , and by Proposition 9.7.1 the first is invertible over  $F[x]$  while the second is not. The rule to remember is that only the **units** of  $F[x]$  — the non-zero constants — may be used as scaling factors.

### 9.7.3 The Smith normal form

*Every polynomial matrix can be brought to a diagonal form whose diagonal entries divide one another in turn, and that form is unique.*

The existence proof is Euclidean division, run on whichever non-zero entry has the smallest degree: if that entry fails to divide one of its neighbors in the same row or column, the remainder gives a non-zero entry of strictly smaller degree, and degrees cannot decrease forever.

#### Theorem 9.7.3 — Smith Normal Form

Let  $F$  be a field,  $n \geq 1$  and  $\mathbf{M} \in M_n(F[x])$ . Then:

- (a) **(Existence)**  $\mathbf{M}$  is equivalent over  $F[x]$ , by a finite sequence of elementary row and column operations, to a diagonal matrix

$$\mathbf{D} = \text{diag}(d_1, d_2, \dots, d_n)$$

in which every  $d_i$  is **monic or zero** and  $d_i \mid d_{i+1}$  for  $i = 1, \dots, n-1$ .

- (b) **(Uniqueness)** The list  $(d_1, \dots, d_n)$  is determined by  $\mathbf{M}$ : if  $\mathbf{M} \approx \text{diag}(d_1, \dots, d_n) \approx \text{diag}(e_1, \dots, e_n)$  with both lists as in (a), then  $d_i = e_i$  for every  $i$ .

The matrix  $\mathbf{D}$  is called the **Smith normal form** of  $\mathbf{M}$ , and  $d_1, \dots, d_n$  its **invariant divisors**.

Since  $0 \mid f$  holds only for  $f = 0$ , the divisibility chain forces the zero entries, if any, to be the last ones.

**Idea. Step roadmap for existence.** ① Among all matrices reachable from  $\mathbf{M}$  by operations, pick one whose smallest non-zero entry degree is least, and move that entry to the corner. ② That corner entry  $f$  must divide every other entry of its row and column: otherwise divide with remainder and produce an entry of smaller degree, contradicting

the choice in ①. ③ Clear the first row and column with  $f$ ; the same remainder trick, after adding a row to the first one, shows  $f$  divides every entry of what is left. ④ Induct on the size. The divisibility  $d_1 \mid d_2$  survives the induction because operations replace entries by  $F[x]$ -combinations of entries.

For uniqueness, find a quantity that can be read off both  $\mathbf{M}$  and  $\mathbf{D}$ . The greatest common divisor of all  $k \times k$  minors is such a quantity: minors of a product are  $F[x]$ -combinations of minors of the factors (Cauchy–Binet), so the gcd cannot change under equivalence, and for a diagonal matrix with a divisibility chain it is simply  $d_1 d_2 \cdots d_k$ .

*Proof.*

(a) If  $\mathbf{M} = 0$ , it is already of the stated form. So assume  $\mathbf{M} \neq 0$ .

For a non-zero  $\mathbf{N} \in M_n(F[x])$  put

$$\delta(\mathbf{N}) := \min\{\deg n_{ij} : n_{ij} \neq 0\} \in \mathbb{N},$$

the least degree of a non-zero entry. Let  $\mathcal{S}$  be the set of all non-zero matrices obtainable from  $\mathbf{M}$  by finite sequences of elementary row and column operations over  $F[x]$ ; it contains  $\mathbf{M}$ . The set  $\{\delta(\mathbf{N}) : \mathbf{N} \in \mathcal{S}\}$  is a non-empty subset of  $\mathbb{N}$ , so it has a least element; **fix  $\mathbf{N} \in \mathcal{S}$  with  $\delta(\mathbf{N})$  least.**

**Step 1. Put a least-degree entry in the corner, monic.** Choose  $(i, j)$  with  $n_{ij} \neq 0$  and  $\deg n_{ij} = \delta(\mathbf{N})$ . Swapping row  $i$  with row 1 and column  $j$  with column 1 (operations (E1)), and then scaling the new first row by the inverse of the leading coefficient of the corner entry (operation (E2)), we obtain a matrix in  $\mathcal{S}$  whose  $(1, 1)$ -entry  $f$  is **monic** of degree  $\delta(\mathbf{N})$ . These operations permute the entries and rescale some of them, so the new matrix still has least non-zero entry degree  $\delta(\mathbf{N})$ . Rename it  $\mathbf{N}$ .

**Step 2.  $f$  divides every entry of the first row and the first column.** Suppose  $f \nmid n_{1j}$  for some  $j \geq 2$ . By [Theorem 5.1.1](#),  $n_{1j} = qf + r$  with  $r \neq 0$  and  $\deg r < \deg f$ . Adding  $-q$  times column 1 to column  $j$  (operation (E3)) produces a matrix  $\mathbf{N}' \in \mathcal{S}$  whose  $(1, j)$ -entry is  $r \neq 0$  with  $\deg r < \deg f = \delta(\mathbf{N})$ . Then  $\delta(\mathbf{N}') < \delta(\mathbf{N})$ , contradicting the minimality of  $\delta(\mathbf{N})$ . Hence  $f \mid n_{1j}$  for every  $j$ , and the same argument with rows in place of columns gives  $f \mid n_{i1}$  for every  $i$ .

**Step 3. Clear the first row and column.** For  $j \geq 2$  add  $-(n_{1j}/f)$  times column 1 to column  $j$ ; the multipliers are polynomials by Step 2, and after these operations the first row is  $(f, 0, \dots, 0)$  while the first column is unchanged. Then for  $i \geq 2$  add  $-(n_{i1}/f)$  times row 1 to row  $i$ ; since the first row is now  $(f, 0, \dots, 0)$ , this clears the first column and changes nothing else. The result is

$$\begin{pmatrix} f & 0 \\ 0 & \mathbf{M}' \end{pmatrix} \in \mathcal{S}, \quad \mathbf{M}' \in M_{n-1}(F[x]),$$

and its least non-zero entry degree is still  $\delta(\mathbf{N})$ : it is  $\geq \delta(\mathbf{N})$  by minimality, and  $f$  is an entry of degree  $\delta(\mathbf{N})$ .

**Step 4.  $f$  divides every entry of  $\mathbf{M}'$ .** Suppose  $f \nmid (\mathbf{M}')_{kl}$  for some  $k, l$ . Add row  $k+1$  to row 1. The new first row is  $(f, (\mathbf{M}')_{k1}, \dots, (\mathbf{M}')_{k,n-1})$ , so the  $(1, l+1)$ -entry of the resulting matrix  $\mathbf{N}'' \in \mathcal{S}$  is  $(\mathbf{M}')_{kl}$ , which is not divisible by  $f$ , while the  $(1, 1)$ -entry is still  $f$ , monic of degree  $\delta(\mathbf{N})$ . By minimality  $\delta(\mathbf{N}'') = \delta(\mathbf{N})$ , so Step 2 applies to  $\mathbf{N}''$  and says that  $f$  divides every entry of the first row of  $\mathbf{N}''$  — a contradiction.



**Step 5. Induct on  $n$ .** We prove (a) by induction on  $n$ . For  $n = 1$ , scale the single entry to be monic (or leave it 0). For  $n \geq 2$ , Steps 1–4 give a sequence of operations taking  $\mathbf{M}$  to  $f \oplus \mathbf{M}'$  with  $f$  monic and  $f$  dividing every entry of  $\mathbf{M}'$ . By the induction hypothesis there are operations taking  $\mathbf{M}'$  to  $\mathbf{D}' = \text{diag}(d_2, \dots, d_n)$  with every  $d_i$  monic or zero and  $d_2 \mid \dots \mid d_n$ ; performing the corresponding operations on rows and columns 2,  $\dots$ ,  $n$  of  $f \oplus \mathbf{M}'$  takes it to  $f \oplus \mathbf{D}' = \text{diag}(f, d_2, \dots, d_n)$ . It remains to see that  $f \mid d_2$ . By Proposition 9.7.2,  $\mathbf{D}' = \mathbf{U}'\mathbf{M}'\mathbf{V}'$  for some  $\mathbf{U}', \mathbf{V}' \in M_{n-1}(F[x])$ , so every entry of  $\mathbf{D}'$  is a sum of products  $u'_{ik}(\mathbf{M}')_{kl}v'_{lj}$  (Definition 0.9.9) and is therefore divisible by  $f$ , since every  $(\mathbf{M}')_{kl}$  is. In particular  $f \mid d_2$ . Setting  $d_1 = f$  finishes the induction.

- (b) Call a set  $I \subseteq [n]$  with  $|I| = k$  a  $k$ -set. For  $\mathbf{M} \in M_n(F[x])$  and  $1 \leq k \leq n$  define the  $k$ -th **determinantal divisor**

$$\Delta_k(\mathbf{M}) := \text{the monic generator of the ideal } \langle \det \mathbf{M}_{I,J} : I, J \text{ } k\text{-sets} \rangle \subseteq F[x],$$

and  $\Delta_k(\mathbf{M}) := 0$  if all the  $k \times k$  minors of  $\mathbf{M}$  vanish. This is well defined by Proposition 5.2.1 and Theorem 5.2.2: the ideal generated by the finitely many  $k \times k$  minors is  $\langle \Delta_k(\mathbf{M}) \rangle$  for exactly one monic  $\Delta_k(\mathbf{M})$  unless the ideal is  $\{0\}$ . Equivalently,  $\Delta_k(\mathbf{M})$  is the monic greatest common divisor of the  $k \times k$  minors: it divides each of them, and every common divisor of them divides it (Theorem 5.2.3, extended from two polynomials to finitely many by the same ideal argument).

**Claim 1.**  $\mathbf{M} \approx \mathbf{M}'$  implies  $\Delta_k(\mathbf{M}) = \Delta_k(\mathbf{M}')$  for every  $k$ .

*Proof.* Let  $\mathbf{M}' = \mathbf{U}\mathbf{M}\mathbf{V}$  with  $\mathbf{U}, \mathbf{V}$  invertible over  $F[x]$ , and let  $I, J$  be  $k$ -sets. The submatrix of a product on rows  $I$  and columns  $J$  is the product of the submatrix of the left factor on rows  $I$  and all columns with the submatrix of the right factor on all rows and columns  $J$  (Definition 0.9.9, read off entry by entry). Applying this twice and then Theorem 6.7.2 over the commutative ring  $F[x]$  — legitimate by the remark after that theorem — to the  $k \times n$  times  $n \times k$  products involved,

$$\det \mathbf{M}'_{I,J} = \sum_S \det \mathbf{U}_{I,S} \det (\mathbf{M}\mathbf{V})_{S,J} = \sum_S \sum_R \det \mathbf{U}_{I,S} \det \mathbf{M}_{S,R} \det \mathbf{V}_{R,J},$$

with  $S$  and  $R$  running over the  $k$ -sets. So every  $k \times k$  minor of  $\mathbf{M}'$  is an  $F[x]$ -combination of  $k \times k$  minors of  $\mathbf{M}$ , hence lies in  $\langle \det \mathbf{M}_{S,R} \rangle$ . Therefore  $\Delta_k(\mathbf{M})$  divides every  $k \times k$  minor of  $\mathbf{M}'$ , and so  $\Delta_k(\mathbf{M}) \mid \Delta_k(\mathbf{M}')$ . Exchanging the roles of  $\mathbf{M}$  and  $\mathbf{M}'$  using  $\mathbf{M} = \mathbf{U}^{-1}\mathbf{M}'\mathbf{V}^{-1}$  gives  $\Delta_k(\mathbf{M}') \mid \Delta_k(\mathbf{M})$ . Two monic polynomials dividing each other are equal (Proposition 5.1.2), and if one is 0 so is the other. ■

**Claim 2.** If  $\mathbf{D} = \text{diag}(d_1, \dots, d_n)$  with  $d_1 \mid d_2 \mid \dots \mid d_n$ , then  $\Delta_k(\mathbf{D}) = d_1 d_2 \cdots d_k$ .



*Proof.* Let  $I, J$  be  $k$ -sets. If  $I \neq J$ , pick  $i \in I \setminus J$ ; row  $i$  of  $\mathbf{D}$  has its only possibly non-zero entry in column  $i \notin J$ , so the submatrix  $\mathbf{D}_{I,J}$  has a zero row and  $\det \mathbf{D}_{I,J} = 0$ . If  $I = J = \{i_1 < \dots < i_k\}$ , then  $\mathbf{D}_{I,I} = \text{diag}(d_{i_1}, \dots, d_{i_k})$  and  $\det \mathbf{D}_{I,I} = d_{i_1} \cdots d_{i_k}$  (Theorem 6.4.6, valid over the commutative ring  $F[x]$  by the remark after it). Since  $i_j \geq j$ , the chain gives  $d_j \mid d_{i_j}$  for each  $j$ , hence

$$d_1 d_2 \cdots d_k \mid d_{i_1} d_{i_2} \cdots d_{i_k}.$$

So  $d_1 \cdots d_k$  divides every  $k \times k$  minor of  $\mathbf{D}$ , and it is itself the minor for  $I = J = \{1, \dots, k\}$ . Hence the ideal generated by the  $k \times k$  minors is  $\langle d_1 \cdots d_k \rangle$ . If  $d_1 \cdots d_k \neq 0$  it is monic, being a product of monic polynomials, and equals  $\Delta_k(\mathbf{D})$ . If  $d_1 \cdots d_k = 0$ , say  $d_m = 0$  with  $m \leq k$ , then for every  $k$ -set  $I$  we have  $i_k \geq k \geq m$ , so  $d_m \mid d_{i_k}$  forces  $d_{i_k} = 0$  and the minor vanishes; thus  $\Delta_k(\mathbf{D}) = 0 = d_1 \cdots d_k$ . ■

Now let  $\mathbf{M} \approx \text{diag}(d_1, \dots, d_n)$  with the  $d_i$  as in (a). By Claims 1 and 2,

$$\Delta_k(\mathbf{M}) = d_1 d_2 \cdots d_k \quad (k = 1, \dots, n),$$

a quantity depending only on  $\mathbf{M}$ . Reading this backwards recovers the  $d_i$ : first  $d_1 = \Delta_1(\mathbf{M})$ ; and for  $k \geq 2$ , if  $\Delta_{k-1}(\mathbf{M}) \neq 0$  then  $d_k = \Delta_k(\mathbf{M})/\Delta_{k-1}(\mathbf{M})$ , while if  $\Delta_{k-1}(\mathbf{M}) = 0$  then  $d_1 \cdots d_{k-1} = 0$ , so some  $d_m = 0$  with  $m \leq k-1$ , and the chain forces  $d_k = 0$ . Hence any second list  $(e_1, \dots, e_n)$  as in (a) satisfies the same recipe and equals  $(d_1, \dots, d_n)$ . This proves the theorem. ■

The proof is an algorithm, and it is the one to use by hand: hunt for the entry of least degree, divide, and repeat.

*Algorithm. Smith normal form of  $\mathbf{M} \in M_n(F[x])$ .*

1. If  $\mathbf{M} = 0$ , stop. Otherwise find a non-zero entry of least degree and move it to position  $(1, 1)$  by a row and a column swap; scale row 1 so that the entry is monic. Call it  $f$ .
2. For each other entry  $g$  in row 1 or column 1 with  $f \nmid g$ , replace  $g$  by its remainder on division by  $f$  (subtract a multiple of column 1, resp. row 1), move the remainder to the corner, and go back to step 1.
3. When  $f$  divides all of row 1 and column 1, clear them. If some entry  $g$  of the remaining block is not divisible by  $f$ , add its row to row 1 and go back to step 2.
4. Repeat on the remaining  $(n-1) \times (n-1)$  block.
5. Check the answer:  $\Delta_k$  must equal  $d_1 \cdots d_k$ . In particular  $d_1 \cdots d_n$  is  $\det \mathbf{M}$  rescaled by a non-zero constant to be monic, and  $d_1$  is the monic gcd of the entries.

*Quick check.* Two polynomial matrices in  $M_2(F[x])$  have determinants  $x^2$  and  $x^2$ , and in both the greatest common divisor of the four entries is 1. Are they equivalent over  $F[x]$ ?

*Solution.* Yes. Here  $\Delta_1 = 1$  (the gcd of the  $1 \times 1$  minors, that is, of the entries) and  $\Delta_2$  is the determinant rescaled to be monic, so  $\Delta_2 = x^2$  for both. By [Theorem 9.7.3](#) both have Smith normal form  $\text{diag}(1, x^2)$ , and since  $\approx$  is an equivalence relation ([Proposition 9.7.2 \(a\)](#)) they are equivalent to each other. For  $2 \times 2$  matrices the pair  $(\Delta_1, \Delta_2)$  is therefore a complete invariant.

**Example. (A Smith Normal Form Computed by Hand)**

Let

$$\mathbf{A} = \begin{pmatrix} 2 & 1 & -2 \\ 1 & 2 & -2 \\ 1 & 1 & -1 \end{pmatrix} \in M_3(\mathbb{Q}).$$

Find the Smith normal form of  $x\mathbf{I} - \mathbf{A}$  over  $\mathbb{Q}[x]$ , naming every operation, and check the answer with determinantal divisors.

*Solution.* We have

$$x\mathbf{I} - \mathbf{A} = \begin{pmatrix} x-2 & -1 & 2 \\ -1 & x-2 & 2 \\ -1 & -1 & x+1 \end{pmatrix}.$$

*Find the corner.* The entries of least degree are the three entries equal to  $-1$ . Swap columns 1 and 2, then multiply the first row by  $-1$ :

$$\begin{pmatrix} -1 & x-2 & 2 \\ x-2 & -1 & 2 \\ -1 & -1 & x+1 \end{pmatrix} \longrightarrow \begin{pmatrix} 1 & -(x-2) & -2 \\ x-2 & -1 & 2 \\ -1 & -1 & x+1 \end{pmatrix}.$$

The corner is now 1, which divides everything, so no division step is needed.

*Clear the first column.* Add  $-(x-2)$  times row 1 to row 2, and add row 1 to row 3. In row 2 this gives entries  $-1 + (x-2)^2 = (x-1)(x-3)$  and  $2 + 2(x-2) = 2(x-1)$ ; in row 3 it gives  $-1 - (x-2) = -(x-1)$  and  $x+1-2 = x-1$ :

$$\begin{pmatrix} 1 & -(x-2) & -2 \\ 0 & (x-1)(x-3) & 2(x-1) \\ 0 & -(x-1) & x-1 \end{pmatrix}.$$

*Clear the first row.* Add  $(x-2)$  times column 1 to column 2, and 2 times column 1 to column 3. Since the first column below the corner is zero, nothing else changes:

$$\begin{pmatrix} 1 & 0 & 0 \\ 0 & (x-1)(x-3) & 2(x-1) \\ 0 & -(x-1) & x-1 \end{pmatrix} = \text{diag}(1) \oplus \begin{pmatrix} (x-1)(x-3) & 2(x-1) \\ -(x-1) & x-1 \end{pmatrix}.$$

Every entry of the  $2 \times 2$  block is divisible by 1, so we move on to the block.

*The block.* Its entries of least degree are the three of degree 1. Swap rows 2 and 3 and scale the new row 2 by  $-1$ , putting the monic  $x-1$  in the corner of the block:

$$\begin{pmatrix} 1 & 0 & 0 \\ 0 & x-1 & -(x-1) \\ 0 & (x-1)(x-3) & 2(x-1) \end{pmatrix}.$$

Now  $x - 1$  divides every entry of its row and column. Add column 2 to column 3, which clears the block's first row and turns the  $(3, 3)$ -entry into  $2(x - 1) + (x - 1)(x - 3) = (x - 1)(x - 1)$ :

$$\begin{pmatrix} 1 & 0 & 0 \\ 0 & x - 1 & 0 \\ 0 & (x - 1)(x - 3) & (x - 1)^2 \end{pmatrix}.$$

Add  $-(x - 3)$  times row 2 to row 3 to clear the block's first column:

$$\mathbf{D} = \text{diag}(1, x - 1, (x - 1)^2),$$

and  $1 \mid x - 1 \mid (x - 1)^2$ , so this is the Smith normal form.

*The check.* The matrix  $x\mathbf{I} - \mathbf{A}$  has an entry  $-1$ , so  $\Delta_1 = 1$ . Its determinant is  $p_{\mathbf{A}} = (x - 1)^3$  — expanding along the first row,

$$\begin{aligned} (x - 2)[(x - 2)(x + 1) + 2] + [-(x + 1) + 2] + 2[1 + (x - 2)] \\ = (x - 2)(x^2 - x) + (1 - x) + 2(x - 1), \end{aligned}$$

which is  $(x - 2)x(x - 1) + (x - 1) = (x - 1)(x^2 - 2x + 1) = (x - 1)^3$  — so  $\Delta_3 = (x - 1)^3$ . For  $\Delta_2$ , the minor on rows  $\{2, 3\}$  and columns  $\{1, 2\}$  is  $\det \begin{pmatrix} -1 & x - 2 \\ -1 & -1 \end{pmatrix} = 1 + (x - 2) = x - 1$ , so  $\Delta_2 \mid x - 1$ ; and every  $2 \times 2$  minor is divisible by  $x - 1$ , as Claim 2 predicts from  $\mathbf{D}$ . Hence  $\Delta_2 = x - 1$ , and

$$(d_1, d_2, d_3) = \left(\Delta_1, \frac{\Delta_2}{\Delta_1}, \frac{\Delta_3}{\Delta_2}\right) = (1, x - 1, (x - 1)^2),$$

in agreement. As a sanity check on  $\mathbf{A}$  itself:  $\mathbf{A} - \mathbf{I}$  has rank 1, so  $g_{\mathbf{A}}(1) = 2$  and the Jordan form of  $\mathbf{A}$  is  $\mathbf{J}_2(1) \oplus \mathbf{J}_1(1)$  (Corollary 9.3.2).

## 9.7.4 Similarity read off from a polynomial matrix

We can now state the theorem that makes all of this useful. It converts a question about  $\mathbf{A}$  and  $\mathbf{B}$  — is there an invertible  $\mathbf{P} \in M_n(F)$  with  $\mathbf{B} = \mathbf{P}^{-1}\mathbf{A}\mathbf{P}$ ? — into a question about the two polynomial matrices  $x\mathbf{I} - \mathbf{A}$  and  $x\mathbf{I} - \mathbf{B}$ , which the Smith algorithm answers by computation.

One direction is immediate. The other needs a tool: polynomial matrices can be divided by  $x\mathbf{I} - \mathbf{A}$ , with a **constant** remainder.

### Lemma 9.7.4 — Division of a Polynomial Matrix by $x\mathbf{I} - \mathbf{A}$

Let  $\mathbf{A} \in M_n(F)$  and  $\mathbf{M} \in M_n(F[x])$ . Then there are  $\mathbf{Q} \in M_n(F[x])$  and  $\mathbf{R} \in M_n(F)$  with

$$\mathbf{M} = (x\mathbf{I} - \mathbf{A})\mathbf{Q} + \mathbf{R}.$$

*Idea.* Write  $\mathbf{M}$  as a polynomial  $\sum_k \mathbf{M}_k x^k$  with **constant matrix** coefficients. The scalar identity  $x^k - a^k = (x - a)(x^{k-1} + \cdots + a^{k-1})$  has a matrix version  $x^k \mathbf{I} - \mathbf{A}^k = (x\mathbf{I} - \mathbf{A})\mathbf{S}_k$ , in which the second factor is a polynomial in  $x$  and  $\mathbf{A}$  and so commutes with  $x\mathbf{I} - \mathbf{A}$ . Replacing each  $x^k \mathbf{I}$  by  $(x^k \mathbf{I} - \mathbf{A}^k) + \mathbf{A}^k$  puts a factor  $x\mathbf{I} - \mathbf{A}$  in front of everything except a constant matrix.

*Proof.* Collecting the coefficients of each power of  $x$  in the entries of  $\mathbf{M}$ , write  $\mathbf{M} = \sum_{k=0}^m x^k \mathbf{M}_k$  with  $\mathbf{M}_0, \dots, \mathbf{M}_m \in M_n(F)$ ; note that  $x^k$  is a scalar polynomial, so  $x^k \mathbf{M}_k = \mathbf{M}_k x^k$ . For  $k \geq 1$  put

$$\mathbf{S}_k := \sum_{j=0}^{k-1} x^{k-1-j} \mathbf{A}^j \in M_n(F[x]), \quad \mathbf{S}_0 := 0.$$

Expanding the product and telescoping,

$$(x\mathbf{I} - \mathbf{A})\mathbf{S}_k = \sum_{j=0}^{k-1} x^{k-j} \mathbf{A}^j - \sum_{j=0}^{k-1} x^{k-1-j} \mathbf{A}^{j+1} = x^k \mathbf{I} - \mathbf{A}^k$$

for every  $k \geq 0$ . Therefore

$$\begin{aligned} \mathbf{M} &= \sum_{k=0}^m (x^k \mathbf{I}) \mathbf{M}_k = \sum_{k=0}^m (x^k \mathbf{I} - \mathbf{A}^k) \mathbf{M}_k + \sum_{k=0}^m \mathbf{A}^k \mathbf{M}_k \\ &= (x\mathbf{I} - \mathbf{A}) \underbrace{\sum_{k=0}^m \mathbf{S}_k \mathbf{M}_k}_{=\mathbf{Q}} + \underbrace{\sum_{k=0}^m \mathbf{A}^k \mathbf{M}_k}_{=\mathbf{R}}, \end{aligned}$$

with  $\mathbf{Q} \in M_n(F[x])$  and  $\mathbf{R} \in M_n(F)$ , as claimed. ■

### Theorem 9.7.5 — Similarity as Equivalence of $x\mathbf{I} - \mathbf{A}$ and $x\mathbf{I} - \mathbf{B}$

Let  $F$  be a field,  $n \geq 1$  and  $\mathbf{A}, \mathbf{B} \in M_n(F)$ . Then

$$\mathbf{A} \sim \mathbf{B} \text{ over } F \quad \Longleftrightarrow \quad x\mathbf{I} - \mathbf{A} \approx x\mathbf{I} - \mathbf{B} \text{ over } F[x].$$

*Idea.*  $(\Rightarrow)$  is a one-line computation: a constant invertible  $\mathbf{P}$  is also an invertible polynomial matrix.

$(\Leftarrow)$  is the substance, and the whole difficulty is that the matrices  $\mathbf{U}, \mathbf{V}$  witnessing the equivalence have polynomial entries, while we must produce a **constant**  $\mathbf{P}$ . The trick is to divide  $\mathbf{U}$  by  $x\mathbf{I} - \mathbf{B}$  and keep only the constant remainder  $\mathbf{P}$ . Comparing degrees in the resulting identity shows that  $\mathbf{P}$  alone already conjugates:  $\mathbf{P}(x\mathbf{I} - \mathbf{A}) = (x\mathbf{I} - \mathbf{B})\mathbf{P}$ , that is,  $\mathbf{PA} = \mathbf{BP}$ . Running the same argument on the inverted relation produces a second constant  $\mathbf{P}'$  with  $\mathbf{P}'\mathbf{B} = \mathbf{AP}'$ , and multiplying the two expressions for  $\mathbf{V}^{-1}$  and  $\mathbf{V}$  shows  $\mathbf{PP}' = \mathbf{I}$ , so  $\mathbf{P}$  is invertible.

*Proof.*  $(\Rightarrow)$  Suppose  $\mathbf{B} = \mathbf{P}^{-1}\mathbf{AP}$  with  $\mathbf{P} \in M_n(F)$  invertible. Then, inside  $M_n(F[x])$ ,

$$x\mathbf{I} - \mathbf{B} = \mathbf{P}^{-1}(x\mathbf{I})\mathbf{P} - \mathbf{P}^{-1}\mathbf{AP} = \mathbf{P}^{-1}(x\mathbf{I} - \mathbf{A})\mathbf{P},$$

and  $\mathbf{P}^{-1}, \mathbf{P} \in M_n(F) \subseteq M_n(F[x])$  have non-zero constant determinants, hence are invertible over  $F[x]$  (Proposition 9.7.1). So  $x\mathbf{I} - \mathbf{A} \approx x\mathbf{I} - \mathbf{B}$ .

$(\Leftarrow)$  Suppose

$$x\mathbf{I} - \mathbf{B} = \mathbf{U}(x\mathbf{I} - \mathbf{A})\mathbf{V} \tag{*}$$

with  $\mathbf{U}, \mathbf{V} \in M_n(F[x])$  invertible over  $F[x]$ .

**Step 1. Divide  $\mathbf{U}$  by  $x\mathbf{I} - \mathbf{B}$ .** By Lemma 9.7.4 there are  $\mathbf{Q} \in M_n(F[x])$  and  $\mathbf{P} \in M_n(F)$  with  $\mathbf{U} = (x\mathbf{I} - \mathbf{B})\mathbf{Q} + \mathbf{P}$ . Substituting into (\*),

$$x\mathbf{I} - \mathbf{B} = (x\mathbf{I} - \mathbf{B})\mathbf{Q}(x\mathbf{I} - \mathbf{A})\mathbf{V} + \mathbf{P}(x\mathbf{I} - \mathbf{A})\mathbf{V},$$

so  $\mathbf{P}(x\mathbf{I} - \mathbf{A})\mathbf{V} = (x\mathbf{I} - \mathbf{B})[\mathbf{I} - \mathbf{Q}(x\mathbf{I} - \mathbf{A})\mathbf{V}]$ . Multiplying on the right by  $\mathbf{V}^{-1}$  and putting

$$\mathbf{S} := \mathbf{V}^{-1} - \mathbf{Q}(x\mathbf{I} - \mathbf{A}) \in M_n(F[x]),$$

we get

$$\mathbf{P}(x\mathbf{I} - \mathbf{A}) = (x\mathbf{I} - \mathbf{B})\mathbf{S}. \quad (1)$$

**Step 2.  $\mathbf{S}$  is a constant matrix, and  $\mathbf{PA} = \mathbf{BP}$ .** Suppose  $\mathbf{S} \neq 0$  and write  $\mathbf{S} = \sum_{k=0}^d x^k \mathbf{S}_k$  with  $\mathbf{S}_k \in M_n(F)$  and  $\mathbf{S}_d \neq 0$ . The coefficient of  $x^{d+1}$  in  $(x\mathbf{I} - \mathbf{B})\mathbf{S}$  is  $\mathbf{S}_d \neq 0$ , while the left-hand side of (1) is  $x\mathbf{P} - \mathbf{PA}$ , whose coefficients vanish beyond  $x^1$ . Hence  $d+1 \leq 1$ , that is  $d = 0$ . So in every case  $\mathbf{S} \in M_n(F)$ . Comparing the coefficients of  $x^1$  in (1) gives  $\mathbf{P} = \mathbf{S}$ , and comparing the constant terms gives  $-\mathbf{PA} = -\mathbf{BP}$ . Therefore

$$\mathbf{PA} = \mathbf{BP}, \quad \text{equivalently} \quad \mathbf{P}(x\mathbf{I} - \mathbf{A}) = (x\mathbf{I} - \mathbf{B})\mathbf{P}, \quad (2)$$

the second form because  $\mathbf{P}(x\mathbf{I} - \mathbf{A}) = x\mathbf{P} - \mathbf{PA} = x\mathbf{P} - \mathbf{BP} = (x\mathbf{I} - \mathbf{B})\mathbf{P}$ . Moreover  $\mathbf{S} = \mathbf{P}$  turns the definition of  $\mathbf{S}$  into

$$\mathbf{V}^{-1} = \mathbf{P} + \mathbf{Q}(x\mathbf{I} - \mathbf{A}). \quad (3)$$

**Step 3. The same argument, inverted.** From (\*),  $x\mathbf{I} - \mathbf{A} = \mathbf{U}^{-1}(x\mathbf{I} - \mathbf{B})\mathbf{V}^{-1}$ , which is (\*) with  $\mathbf{A}, \mathbf{B}$  interchanged and  $\mathbf{U}, \mathbf{V}$  replaced by the invertible polynomial matrices  $\mathbf{U}^{-1}, \mathbf{V}^{-1}$ . So Steps 1 and 2 apply and give  $\mathbf{Q}' \in M_n(F[x])$  and  $\mathbf{P}' \in M_n(F)$  with

$$\mathbf{P}'(x\mathbf{I} - \mathbf{B}) = (x\mathbf{I} - \mathbf{A})\mathbf{P}' \quad \text{and} \quad \mathbf{V} = \mathbf{P}' + \mathbf{Q}'(x\mathbf{I} - \mathbf{B}). \quad (4)$$

**Step 4.  $\mathbf{P}$  is invertible.** Multiplying (3) by (4),

$$\begin{aligned} \mathbf{I} &= \mathbf{V}^{-1}\mathbf{V} = [\mathbf{P} + \mathbf{Q}(x\mathbf{I} - \mathbf{A})][\mathbf{P}' + \mathbf{Q}'(x\mathbf{I} - \mathbf{B})] \\ &= \mathbf{PP}' + \mathbf{PQ}'(x\mathbf{I} - \mathbf{B}) + \mathbf{Q}(x\mathbf{I} - \mathbf{A})\mathbf{P}' \\ &\quad + \mathbf{Q}(x\mathbf{I} - \mathbf{A})\mathbf{Q}'(x\mathbf{I} - \mathbf{B}). \end{aligned}$$

By the first identity in (4),  $(x\mathbf{I} - \mathbf{A})\mathbf{P}' = \mathbf{P}'(x\mathbf{I} - \mathbf{B})$ , so the third summand is  $\mathbf{QP}'(x\mathbf{I} - \mathbf{B})$  and

$$\mathbf{I} - \mathbf{PP}' = \mathbf{Y}(x\mathbf{I} - \mathbf{B}), \quad \mathbf{Y} := \mathbf{PQ}' + \mathbf{QP}' + \mathbf{Q}(x\mathbf{I} - \mathbf{A})\mathbf{Q}' \in M_n(F[x]).$$

If  $\mathbf{Y} \neq 0$ , write  $\mathbf{Y} = \sum_{k=0}^e x^k \mathbf{Y}_k$  with  $\mathbf{Y}_e \neq 0$ ; the coefficient of  $x^{e+1}$  on the right is  $\mathbf{Y}_e \neq 0$ , while the left-hand side is constant. Hence  $\mathbf{Y} = 0$  and  $\mathbf{PP}' = \mathbf{I}$ . By Theorem 2.4.5,  $\mathbf{P}$  is invertible with  $\mathbf{P}^{-1} = \mathbf{P}'$ .

**Step 5. Conclude.** By (2),  $\mathbf{PA} = \mathbf{BP}$ , so  $\mathbf{B} = \mathbf{PAP}^{-1}$  and  $\mathbf{A} \sim \mathbf{B}$ . This proves the theorem. ■

Two by-products are worth recording. Whenever  $x\mathbf{I} - \mathbf{B} = \mathbf{U}(x\mathbf{I} - \mathbf{A})\mathbf{V}$  with  $\mathbf{U}, \mathbf{V}$  invertible over  $F[x]$ , taking determinants gives  $p_{\mathbf{B}} = (\det \mathbf{U})(\det \mathbf{V})p_{\mathbf{A}}$  with  $\det \mathbf{U}, \det \mathbf{V}$  non-zero constants (Proposition 9.7.1); both  $p_{\mathbf{A}}$  and  $p_{\mathbf{B}}$  are monic of degree  $n$ , so

$$p_{\mathbf{A}} = p_{\mathbf{B}} \quad \text{and} \quad (\det \mathbf{U})(\det \mathbf{V}) = 1,$$

which recovers the invariance of the characteristic polynomial. And, by Theorem 9.7.3, the test is now decidable by a finite computation in  $F[x]$ : reduce  $x\mathbf{I} - \mathbf{A}$  and  $x\mathbf{I} - \mathbf{B}$  to Smith normal form and compare.

### Corollary 9.7.6 — The Smith Form of $x\mathbf{I} - \mathbf{A}$ Carries the Invariant Factors

Let  $\mathbf{A} \in M_n(F)$  with invariant factors  $d_1 \mid d_2 \mid \cdots \mid d_r$  as in Theorem 9.6.5. Then the Smith normal form of  $x\mathbf{I} - \mathbf{A}$  is

$$\text{diag}(\underbrace{1, \dots, 1}_{n-r}, d_1, d_2, \dots, d_r).$$

In particular the invariant factors of  $\mathbf{A}$  are exactly the non-constant diagonal entries of the Smith normal form of  $x\mathbf{I} - \mathbf{A}$ , read from the top down, and two matrices in  $M_n(F)$  are similar if and only if  $x\mathbf{I} - \mathbf{A}$  and  $x\mathbf{I} - \mathbf{B}$  have the same Smith normal form.

*Proof.* By Theorem 9.6.5,  $\mathbf{A} \sim \mathbf{C}(d_1) \oplus \cdots \oplus \mathbf{C}(d_r)$  with  $d_i$  monic,  $\deg d_i = k_i \geq 1$  and  $\sum k_i = n$ . By the ( $\Rightarrow$ ) direction of Theorem 9.7.5 it is enough to find the Smith normal form of

$$x\mathbf{I}_n - (\mathbf{C}(d_1) \oplus \cdots \oplus \mathbf{C}(d_r)) = (x\mathbf{I}_{k_1} - \mathbf{C}(d_1)) \oplus \cdots \oplus (x\mathbf{I}_{k_r} - \mathbf{C}(d_r)),$$

the equality by Theorem 7.1.4 (a).

*One block.* Let  $d$  be monic of degree  $k \geq 1$ . By Definition 9.5.3, the only non-zero entries of  $x\mathbf{I}_k - \mathbf{C}(d)$  are  $x$  on the diagonal (with  $x + a_{k-1}$  in the corner),  $-1$  in each position  $(i+1, i)$ , and the entries of the last column. Delete row 1 and column  $k$ . In the remaining  $(k-1) \times (k-1)$  matrix, the entry in position  $(p, q)$  is the  $(p+1, q)$ -entry of  $x\mathbf{I}_k - \mathbf{C}(d)$ : it is  $-1$  when  $q = p$ , it is  $x$  when  $q = p+1$ , and it is 0 when  $q < p$ . So that matrix is upper triangular with every diagonal entry  $-1$ , and its determinant is  $(-1)^{k-1} \neq 0$  by Theorem 6.4.6, again over  $F[x]$ . Hence  $\Delta_{k-1}(x\mathbf{I}_k - \mathbf{C}(d)) = 1$ . By Claim 2 in the proof of Theorem 9.7.3, the determinantal divisors of any matrix are the partial products  $d_1 \cdots d_j$  of its Smith diagonal, so  $\Delta_j \mid \Delta_{k-1} = 1$  and therefore  $\Delta_j = 1$  for every  $j \leq k-1$ . Also  $\Delta_k = \det(x\mathbf{I}_k - \mathbf{C}(d)) = p_{\mathbf{C}(d)} = d$  by Theorem 9.5.3. By the uniqueness part of Theorem 9.7.3, the Smith normal form of  $x\mathbf{I}_k - \mathbf{C}(d)$  is  $\text{diag}(1, \dots, 1, d)$ . (For  $k = 1$  read this as  $\text{diag}(d)$ .)

*Assembling.* If  $\mathbf{M}_i \approx \mathbf{M}'_i$  with witnesses  $\mathbf{U}_i, \mathbf{V}_i$ , then  $\bigoplus_i \mathbf{M}_i \approx \bigoplus_i \mathbf{M}'_i$  with witnesses  $\bigoplus_i \mathbf{U}_i$  and  $\bigoplus_i \mathbf{V}_i$ , which are invertible with inverses  $\bigoplus_i \mathbf{U}_i^{-1}, \bigoplus_i \mathbf{V}_i^{-1}$  (Theorem 7.1.4 (a), (c)). So  $x\mathbf{I}_n - \mathbf{A}$  is equivalent to the direct sum of the matrices  $\text{diag}(1, \dots, 1, d_i)$ , a diagonal matrix whose diagonal entries are  $n - r$  ones and the polynomials  $d_1, \dots, d_r$ . Simultaneous row and column swaps — each a pair of elementary operations — reorder these entries at will, so  $x\mathbf{I}_n - \mathbf{A}$  is equivalent to

$$\text{diag}(1, \dots, 1, d_1, d_2, \dots, d_r),$$

and this is already in the shape required by Theorem 9.7.3 (a), because  $1 \mid d_1 \mid d_2 \mid \cdots \mid d_r$

and all the entries are monic. By uniqueness it *is* the Smith normal form of  $x\mathbf{I} - \mathbf{A}$ . The final claim combines this with [Theorem 9.7.5](#) and the uniqueness in [Theorem 9.7.3](#). ■

So the determinantal divisors of  $x\mathbf{I} - \mathbf{A}$  compute the invariant factors of  $\mathbf{A}$  without any factorization of polynomials:  $\Delta_k$  is a gcd, obtained from the Euclidean algorithm alone. Moreover  $\Delta_n(x\mathbf{I} - \mathbf{A}) = p_{\mathbf{A}}$  and  $\Delta_n/\Delta_{n-1} = d_r = m_{\mathbf{A}}$ .

## 9.7.5 Every matrix is similar to its transpose

Chapter 3 asked whether  $\mathbf{A}$  and  $\mathbf{A}^\top$  are always similar and postponed the answer; [Exercise \(C1: A matrix is similar to its transpose\)](#) settled it whenever  $p_{\mathbf{A}}$  splits. The Smith form removes that hypothesis, because transposing a matrix does not change its minors.

### Corollary 9.7.7 — A Matrix Is Similar to Its Transpose

Let  $F$  be any field and  $\mathbf{A} \in M_n(F)$ . Then  $\mathbf{A} \sim \mathbf{A}^\top$ .

*Proof.* Let  $\mathbf{M} = x\mathbf{I} - \mathbf{A}$ , so that  $\mathbf{M}^\top = x\mathbf{I} - \mathbf{A}^\top$ . For  $k$ -sets  $I, J$  we have  $(\mathbf{M}^\top)_{I,J} = (\mathbf{M}_{J,I})^\top$  by [Definition 0.9.10](#), so  $\det(\mathbf{M}^\top)_{I,J} = \det \mathbf{M}_{J,I}$  by [Theorem 6.4.5](#), valid over the commutative ring  $F[x]$ . Hence the  $k \times k$  minors of  $\mathbf{M}^\top$  are exactly the  $k \times k$  minors of  $\mathbf{M}$ , listed in a different order, and

$$\Delta_k(x\mathbf{I} - \mathbf{A}^\top) = \Delta_k(x\mathbf{I} - \mathbf{A}) \quad (k = 1, \dots, n).$$

By [Theorem 9.7.3](#), each of  $x\mathbf{I} - \mathbf{A}$  and  $x\mathbf{I} - \mathbf{A}^\top$  is equivalent over  $F[x]$  to the diagonal matrix built from its own determinantal divisors, and these two diagonal matrices coincide. Since  $\approx$  is an equivalence relation ([Proposition 9.7.2 \(a\)](#)),  $x\mathbf{I} - \mathbf{A} \approx x\mathbf{I} - \mathbf{A}^\top$ . By [Theorem 9.7.5](#),  $\mathbf{A} \sim \mathbf{A}^\top$ . ■

Two sentences of consequence. An operator and its dual have the same invariants: by [Theorem 4.4.2](#) the matrix of  $T'$  in the dual basis is  $[T]_{\mathcal{B}}^\top$ , so  $T$  and  $T'$  share their characteristic polynomial, their minimal polynomial and their invariant factors. And a matrix and its transpose always have the same Jordan form when one exists, which says that the *rows* of  $\mathbf{A} - \lambda\mathbf{I}$  are exactly as informative as its columns.

**Warning.** Similar is not the same as “equivalent over  $F[x]$ ” for the matrices themselves. The theorem compares  $x\mathbf{I} - \mathbf{A}$  with  $x\mathbf{I} - \mathbf{B}$ , not  $\mathbf{A}$  with  $\mathbf{B}$ . As polynomial matrices, any two invertible  $\mathbf{A}, \mathbf{B} \in M_n(F)$  are equivalent over  $F[x]$  — take  $\mathbf{U} = \mathbf{A}^{-1}$ ,  $\mathbf{V} = \mathbf{B}$  — whether or not they are similar. The whole content of [Theorem 9.7.5](#) is that inserting the variable  $x$  on the diagonal remembers the conjugation.

## 9.7.6 Exercises

### 9.7.6.1 A. Check your understanding

#### Exercise A1.

- List the three elementary operations over  $F[x]$ , and say which one differs from its counterpart over a field and why.
- State a criterion for a matrix  $\mathbf{U} \in M_n(F[x])$  to be invertible in  $M_n(F[x])$ .
- State the Smith Normal Form theorem, including the uniqueness clause.
- Define the  $k$ -th determinantal divisor of a polynomial matrix, and say why it is unchanged by equivalence.
- True or false: if  $\det(x\mathbf{I} - \mathbf{A}) = \det(x\mathbf{I} - \mathbf{B})$ , then  $x\mathbf{I} - \mathbf{A} \approx x\mathbf{I} - \mathbf{B}$ . Justify your answer.
- How do you read  $m_{\mathbf{A}}$  off the Smith normal form of  $x\mathbf{I} - \mathbf{A}$ ?

*Solution.*

- Swap two rows; multiply a row by a non-zero **scalar** of  $F$ ; add a **polynomial** multiple of one row to a different row (Definition 9.7.1). The scaling operation is restricted, because only the non-zero constants are invertible in  $F[x]$ ; multiplying a row by  $x$  is not reversible.
- $\mathbf{U}$  is invertible over  $F[x]$  if and only if  $\det \mathbf{U}$  is a non-zero constant (Proposition 9.7.1).
- Every  $\mathbf{M} \in M_n(F[x])$  is equivalent over  $F[x]$ , by elementary operations, to a diagonal  $\text{diag}(d_1, \dots, d_n)$  with each  $d_i$  monic or zero and  $d_i \mid d_{i+1}$ ; this diagonal is unique (Theorem 9.7.3).
- $\Delta_k(\mathbf{M})$  is the monic generator of the ideal generated by all  $k \times k$  minors of  $\mathbf{M}$ , equivalently their monic gcd. It is unchanged by equivalence because every  $k \times k$  minor of  $\mathbf{U}\mathbf{M}\mathbf{V}$  is an  $F[x]$ -combination of  $k \times k$  minors of  $\mathbf{M}$ , by the Cauchy–Binet formula (Claim 1 in the proof of Theorem 9.7.3).
- False. Equality of determinants says only  $\Delta_n(x\mathbf{I} - \mathbf{A}) = \Delta_n(x\mathbf{I} - \mathbf{B})$ ; the lower determinantal divisors can differ. For  $\mathbf{A} = \mathbf{J}_2(0) \oplus \mathbf{J}_2(0)$  and  $\mathbf{B} = \mathbf{J}_3(0) \oplus \mathbf{J}_1(0)$  in  $M_4(F)$ , both determinants are  $x^4$ , but the Smith normal forms are  $\text{diag}(1, 1, x^2, x^2)$  and  $\text{diag}(1, 1, x, x^3)$ . (Equivalently, the matrices are not similar, by Corollary 9.3.3.)
- $m_{\mathbf{A}}$  is the **last** diagonal entry of the Smith normal form of  $x\mathbf{I} - \mathbf{A}$ , that is  $\Delta_n/\Delta_{n-1} = p_{\mathbf{A}}/\Delta_{n-1}$  (Corollary 9.7.6, since the last diagonal entry is  $d_r = m_{\mathbf{A}}$  in the notation there).

### 9.7.6.2 B. Practice

#### Exercise B1: Smith normal forms.

Find the Smith normal form over  $\mathbb{Q}[x]$  of each of the following, naming the operations you use.

$$(a) \mathbf{M}_1 = \begin{pmatrix} x & x+1 \\ x-1 & x \end{pmatrix}.$$



$$(b) \mathbf{M}_2 = \begin{pmatrix} x^2 - 1 & 0 \\ 0 & x^2 - x \end{pmatrix}.$$

$$(c) \mathbf{M}_3 = \begin{pmatrix} x & 0 & 1 \\ -1 & x & 0 \\ 0 & -1 & x \end{pmatrix}.$$

*Solution.*

- (a)  $\det \mathbf{M}_1 = x^2 - (x+1)(x-1) = 1$ , so  $\mathbf{M}_1$  is invertible over  $\mathbb{Q}[x]$  ([Proposition 9.7.1](#)) and  $\Delta_1 = \Delta_2 = 1$ . The Smith normal form is  $\text{diag}(1, 1) = \mathbf{I}_2$ . (Concretely: subtract column 1 from column 2 to get  $\begin{pmatrix} x & 1 \\ x-1 & 1 \end{pmatrix}$ , then subtract row 1 from row 2, swap columns, and clear.)
- (b) Here  $\Delta_1 = \gcd(x^2 - 1, x^2 - x) = \gcd((x-1)(x+1), x(x-1)) = x-1$  and  $\Delta_2 = \det \mathbf{M}_2 = x(x-1)^2(x+1)$ . So the Smith normal form is

$$\text{diag}\left(x-1, \frac{x(x-1)^2(x+1)}{x-1}\right) = \text{diag}(x-1, x(x-1)(x+1)),$$

and indeed  $x-1$  divides  $x(x-1)(x+1)$ . By operations: add column 2 to column 1, giving first column  $(x^2 - 1, x^2 - x)$ ; subtract row 1 from row 2 to get  $(x^2 - 1, 1 - x)$  in column 1; the entry  $1 - x$  has least degree, so swap it into the corner, scale by  $-1$ , and continue.

- (c) The  $2 \times 2$  minor on rows  $\{2, 3\}$  and columns  $\{1, 2\}$  is  $\det \begin{pmatrix} -1 & x \\ 0 & -1 \end{pmatrix} = 1$ , so  $\Delta_2 = 1$ , and then  $\Delta_1 = 1$  as well (it divides  $\Delta_2$ ). Expanding  $\det \mathbf{M}_3$  along the first row,  $x(x^2 - 0) + 1 \cdot (1 - 0) = x^3 + 1$ , so  $\Delta_3 = x^3 + 1$  and the Smith normal form is  $\text{diag}(1, 1, x^3 + 1)$ . Note  $\mathbf{M}_3 = x\mathbf{I} - \mathbf{C}$  with  $\mathbf{C} = \begin{pmatrix} 0 & 0 & -1 \\ 1 & 0 & 0 \\ 0 & 1 & 0 \end{pmatrix} = \mathbf{C}(x^3 + 1)$ , and [Corollary 9.7.6](#) predicts exactly this, since a companion matrix has the single invariant factor  $x^3 + 1$ .

### Exercise B2: Invariant factors from determinantal divisors.

For each matrix  $\mathbf{A}$ , compute  $\Delta_1, \dots, \Delta_n$  of  $x\mathbf{I} - \mathbf{A}$ , write down the Smith normal form, and read off the invariant factors,  $m_{\mathbf{A}}$  and the Jordan form.

- (a)  $\mathbf{A} = \mathbf{J}_2(0) \oplus \mathbf{J}_2(0) \in M_4(\mathbb{Q})$ .
- (b)  $\mathbf{A} = \mathbf{J}_3(0) \oplus \mathbf{J}_1(0) \in M_4(\mathbb{Q})$ .
- (c)  $\mathbf{A} = \begin{pmatrix} 3 & 1 \\ 0 & 3 \end{pmatrix} \in M_2(\mathbb{Q})$ .

*Solution.* One remark makes the minors easy. If  $\mathbf{M} = \mathbf{M}_1 \oplus \mathbf{M}_2$  with square blocks, and  $I, J$  are  $k$ -sets, split  $I = I_1 \cup I_2$  and  $J = J_1 \cup J_2$  according to the blocks. If  $|I_1| \neq |J_1|$ , then  $\mathbf{M}_{I,J}$  has a zero row or a zero column and  $\det \mathbf{M}_{I,J} = 0$ ; if  $|I_1| = |J_1|$ , then  $\mathbf{M}_{I,J} = (\mathbf{M}_1)_{I_1, J_1} \oplus (\mathbf{M}_2)_{I_2, J_2}$  and the determinant is the product of the two, by [Theorem 6.8.4](#), whose proof uses only sums and products of entries and so applies over  $F[x]$  (the remark after that theorem). So **the minors of a block diagonal matrix are the products of matching-size minors of the blocks**. Here  $x\mathbf{I} - \mathbf{A}$  is the direct

sum of the blocks  $x\mathbf{I}_k - \mathbf{J}_k(0)$ , each upper triangular with  $x$  on the diagonal and  $-1$  on the superdiagonal.

- (a)  $x\mathbf{I} - \mathbf{A} = \mathbf{N} \oplus \mathbf{N}$  with  $\mathbf{N} = \begin{pmatrix} x & -1 \\ 0 & x \end{pmatrix}$ , whose minors are the entries  $x, -1, 0$  for  $k = 1$  and  $x^2$  for  $k = 2$ . By the remark, the minors of  $\mathbf{N} \oplus \mathbf{N}$  are  $\{x, -1, 0\}$  for  $k = 1$ ; the products of two  $1 \times 1$  minors together with  $x^2$  for  $k = 2$ ;  $x^2$  times a  $1 \times 1$  minor for  $k = 3$ ; and  $x^4$  for  $k = 4$ . Hence

$$\Delta_1 = 1, \quad \Delta_2 = 1 \quad (\text{from } (-1)(-1) = 1), \quad \Delta_3 = x^2, \quad \Delta_4 = x^4,$$

so the Smith normal form is  $\text{diag}(1, 1, x^2, x^2)$ , the invariant factors are  $x^2 \mid x^2$ ,  $m_{\mathbf{A}} = x^2$  is the last of them, and the Jordan form is  $\mathbf{J}_2(0) \oplus \mathbf{J}_2(0)$ .

- (b) Now  $x\mathbf{I} - \mathbf{A} = \mathbf{N}_3 \oplus (x)$  with  $\mathbf{N}_3 = \begin{pmatrix} x & -1 & 0 \\ 0 & x & -1 \\ 0 & 0 & x \end{pmatrix}$ . The minors of  $\mathbf{N}_3$  are  $\{x, -1, 0\}$  for  $k = 1$ ;  $\{x^2, -x, 1, 0\}$  for  $k = 2$ , the value 1 coming from rows  $\{1, 2\}$  and columns  $\{2, 3\}$ ; and  $x^3$  for  $k = 3$ . By the remark, the  $k \times k$  minors of  $x\mathbf{I} - \mathbf{A}$  are those of  $\mathbf{N}_3$  together with  $x$  times its  $(k-1) \times (k-1)$  minors, so

$$\begin{aligned} \Delta_1 &= 1, \\ \Delta_2 &= \gcd(x^2, -x, 1, x \cdot x, -x) = 1, \\ \Delta_3 &= \gcd(x^3, x \cdot x^2, -x \cdot x, x \cdot 1) = x, \\ \Delta_4 &= x^4. \end{aligned}$$

The Smith normal form is  $\text{diag}(1, 1, x, x^3)$ : the invariant factors are  $x \mid x^3$ ,  $m_{\mathbf{A}} = x^3$  is the last of them, and the Jordan form is  $\mathbf{J}_3(0) \oplus \mathbf{J}_1(0)$ . Compared with (a) the determinants agree but  $\Delta_3$  does not, which is exactly why the two matrices are not similar.

- (c)  $x\mathbf{I} - \mathbf{A} = \begin{pmatrix} x-3 & -1 \\ 0 & x-3 \end{pmatrix}$ . Then  $\Delta_1 = 1$  and  $\Delta_2 = (x-3)^2$ , so the Smith normal form is  $\text{diag}(1, (x-3)^2)$ , the only invariant factor is  $(x-3)^2 = m_{\mathbf{A}} = p_{\mathbf{A}}$ , and the Jordan form is  $\mathbf{J}_2(3)$ . Consistently,  $\mathbf{A}$  is a companion matrix up to similarity, since  $m_{\mathbf{A}} = p_{\mathbf{A}}$  (Theorem 9.5.6).

### Exercise B3: Deciding similarity.

Determine whether each pair is similar over the stated field. Justify your answer using Theorem 9.7.5.

- (a)  $\mathbf{A} = \begin{pmatrix} 0 & -1 \\ 1 & 0 \end{pmatrix}$  and  $\mathbf{B} = \begin{pmatrix} 0 & 1 \\ -1 & 0 \end{pmatrix}$  over  $\mathbb{Q}$ .
- (b)  $\mathbf{A} = \begin{pmatrix} 2 & 1 & -2 \\ 1 & 2 & -2 \\ 1 & 1 & -1 \end{pmatrix}$  and  $\mathbf{B} = \mathbf{J}_2(1) \oplus \mathbf{J}_1(1)$  over  $\mathbb{Q}$ .
- (c)  $\mathbf{A} = \begin{pmatrix} 0 & -2 \\ 1 & 0 \end{pmatrix}$  and  $\mathbf{B} = \begin{pmatrix} 0 & -1 \\ 2 & 0 \end{pmatrix}$  over  $\mathbb{Q}$ .

*Solution.*

- (a) Similar. Here  $\mathbf{B} = \mathbf{A}^\top$ , so [Corollary 9.7.7](#) applies. (Explicitly,  $\Delta_1 = 1$  and  $\Delta_2 = x^2 + 1$  for both, so both Smith normal forms are  $\text{diag}(1, x^2 + 1)$ .)
- (b) Similar. By [Example \(A Smith Normal Form Computed by Hand\)](#) the Smith normal form of  $x\mathbf{I} - \mathbf{A}$  is  $\text{diag}(1, x - 1, (x - 1)^2)$ . For  $\mathbf{B}$ ,  $x\mathbf{I} - \mathbf{B} = \begin{pmatrix} x - 1 & -1 \\ 0 & x - 1 \end{pmatrix} \oplus (x - 1)$ . Its entry  $-1$  gives  $\Delta_1 = 1$ . By the block-minor rule in the solution to [Exercise \(B2: Invariant factors from determinantal divisors\)](#), the  $2 \times 2$  minors are  $(x - 1)^2$ , from the  $2 \times 2$  block, together with the products of a  $1 \times 1$  minor of each block, that is  $(x - 1)^2$ ,  $-(x - 1)$  and  $0$ ; so  $\Delta_2 = x - 1$ . Finally  $\Delta_3 = \det(x\mathbf{I} - \mathbf{B}) = (x - 1)^3$ . The Smith normal forms agree, so  $x\mathbf{I} - \mathbf{A} \approx x\mathbf{I} - \mathbf{B}$  and  $\mathbf{A} \sim \mathbf{B}$ .
- (c) Similar. Here

$$x\mathbf{I} - \mathbf{A} = \begin{pmatrix} x & 2 \\ -1 & x \end{pmatrix}, \quad x\mathbf{I} - \mathbf{B} = \begin{pmatrix} x & 1 \\ -2 & x \end{pmatrix},$$

and each has an entry that is a non-zero constant, so  $\Delta_1 = 1$  in both cases; and  $\Delta_2 = \det(x\mathbf{I} - \mathbf{A}) = x^2 + 2 = \det(x\mathbf{I} - \mathbf{B})$ . So both Smith normal forms are  $\text{diag}(1, x^2 + 2)$ , hence  $x\mathbf{I} - \mathbf{A} \approx x\mathbf{I} - \mathbf{B}$  and  $\mathbf{A} \sim \mathbf{B}$  by [Theorem 9.7.5](#). Note that  $x^2 + 2$  is irreducible over  $\mathbb{Q}$ , so no Jordan form exists over  $\mathbb{Q}$  and [Exercise \(C1: A matrix is similar to its transpose\)](#) would not have decided this pair. (Here  $\mathbf{B} \neq \mathbf{A}^\top$ , so [Corollary 9.7.7](#) does not apply directly; the Smith forms settle it anyway.)

### 9.7.6.3 C. Going deeper

#### Exercise C1: The same algorithm over the integers.

The proof of [Theorem 9.7.3](#) used only one property of  $F[x]$ : division with remainder, where the “size” of a non-zero element is its degree. The integers  $\mathbb{Z}$  have the same property, with  $|a|$  in place of  $\deg$ .

- (a) State the version of [Theorem 9.7.3](#) for  $\mathbf{M} \in M_n(\mathbb{Z})$  that the same proof gives, saying what replaces “monic”.
- (b) Compute the Smith normal form over  $\mathbb{Z}$  of  $\mathbf{M} = \begin{pmatrix} 2 & 4 \\ 6 & 4 \end{pmatrix}$ , naming the operations.
- (c) Every finite abelian group  $G$  with  $n$  generators is  $\mathbb{Z}^n$  modulo the column span of some  $\mathbf{M} \in M_n(\mathbb{Z})$ , the matrix of relations, exactly as  $V$  was  $F[x]^n$  modulo the columns of  $x\mathbf{I} - \mathbf{A}$ . Using (b), identify the group with relation matrix  $\mathbf{M}$  as a product of cyclic groups.

*Hint: for (c), a diagonal relation matrix  $\text{diag}(d_1, \dots, d_n)$  presents the product of the cyclic groups  $\mathbb{Z}/d_i\mathbb{Z}$ .*

*Solution.*

- (a) Every  $\mathbf{M} \in M_n(\mathbb{Z})$  can be brought by integer elementary operations — swap two rows or columns, multiply a row or column by  $\pm 1$  (the units of  $\mathbb{Z}$ ), add an **integer** multiple of one row or column to another — to a diagonal matrix  $\text{diag}(d_1, \dots, d_n)$  with every  $d_i \geq 0$  and  $d_i \mid d_{i+1}$ , and this diagonal is unique. “Monic” is replaced by “non-negative”, which is the normalization available because the units of  $\mathbb{Z}$

are  $\pm 1$ . Uniqueness again follows from the determinantal divisors  $\Delta_k$ , now non-negative integer gcds of the  $k \times k$  minors, using the Cauchy–Binet formula over the commutative ring  $\mathbb{Z}$ .

- (b) The least absolute value among the entries is 2, already in the corner. It divides 4 and 6. Subtract 2 times column 1 from column 2, and 3 times row 1 from row 2:

$$\begin{pmatrix} 2 & 4 \\ 6 & 4 \end{pmatrix} \longrightarrow \begin{pmatrix} 2 & 0 \\ 6 & -8 \end{pmatrix} \longrightarrow \begin{pmatrix} 2 & 0 \\ 0 & -8 \end{pmatrix} \longrightarrow \begin{pmatrix} 2 & 0 \\ 0 & 8 \end{pmatrix},$$

the last step multiplying row 2 by  $-1$ . Since  $2 \mid 8$ , the Smith normal form is  $\text{diag}(2, 8)$ . Check:  $\Delta_1 = \gcd(2, 4, 6, 4) = 2$  and  $\Delta_2 = |\det \mathbf{M}| = |8 - 24| = 16 = 2 \cdot 8$ .

- (c) By (b) and the hint, the group is  $\mathbb{Z}/2\mathbb{Z} \times \mathbb{Z}/8\mathbb{Z}$ , of order  $16 = |\det \mathbf{M}|$ . This is the structure theorem for finite abelian groups in miniature: the invariant divisors  $d_1 \mid \cdots \mid d_n$  of the relation matrix are the orders of the cyclic factors.

### Exercise C2: Uniqueness by hand in the smallest case.

Let  $\mathbf{M} \in M_2(F[x])$  be non-zero.

- Prove directly, without using determinantal divisors, that  $\Delta_1(\mathbf{M})$  — the monic gcd of the four entries of  $\mathbf{M}$  — is unchanged by each of the six elementary operations of [Definition 9.7.1](#).
- Prove that  $\det(\mathbf{U}\mathbf{M}\mathbf{V}) = c \det \mathbf{M}$  for a non-zero constant  $c$  whenever  $\mathbf{U}, \mathbf{V}$  are invertible over  $F[x]$ .
- Deduce that if  $\text{diag}(d_1, d_2) \approx \text{diag}(e_1, e_2)$  with all four entries monic or zero,  $d_1 \mid d_2$  and  $e_1 \mid e_2$ , then  $d_1 = e_1$  and  $d_2 = e_2$ .

*Solution.*

- Write  $d = \Delta_1(\mathbf{M})$ , the monic gcd of the entries, that is, the monic generator of the ideal  $\langle m_{11}, m_{12}, m_{21}, m_{22} \rangle$  ([Theorem 5.2.2](#)). A swap permutes the entries, so the ideal is unchanged. Scaling a row by  $c \in F \setminus \{0\}$  replaces two generators  $g$  by  $cg$ , and  $\langle cg \rangle = \langle g \rangle$  since  $c$  is a unit, so the ideal is unchanged. Adding  $h$  times row 1 to row 2 replaces  $m_{2j}$  by  $m_{2j} + hm_{1j}$ , which lies in the old ideal; so the new ideal is contained in the old one, and the operation is reversed by adding  $-h$  times row 1 to row 2, which gives the reverse containment. The same three arguments apply to columns. Hence  $d$  is unchanged.
- By multiplicativity of the determinant over the commutative ring  $F[x]$  ([Theorem 6.5.4](#) and the remark on commutative rings),  $\det(\mathbf{U}\mathbf{M}\mathbf{V}) = \det \mathbf{U} \cdot \det \mathbf{M} \cdot \det \mathbf{V}$ , and  $\det \mathbf{U}, \det \mathbf{V}$  are non-zero constants by [Proposition 9.7.1](#). Take  $c = \det \mathbf{U} \det \mathbf{V}$ .
- By [Proposition 9.7.2](#) and (a),  $\Delta_1$  is an invariant of equivalence as far as elementary operations go; and for a general equivalence  $\mathbf{M}' = \mathbf{U}\mathbf{M}\mathbf{V}$  the argument of (a) applies verbatim with the generators of the ideal replaced using  $m'_{ij} = \sum_{k,l} u_{ik} m_{kl} v_{lj} \in \langle m_{kl} \rangle$ , and symmetrically. For the two diagonal matrices,  $\Delta_1(\text{diag}(d_1, d_2)) = \gcd(d_1, d_2) = d_1$  since  $d_1 \mid d_2$  and  $d_1$  is monic (or  $d_1 = d_2 = 0$ ); likewise  $\Delta_1(\text{diag}(e_1, e_2)) = e_1$ . Hence  $d_1 = e_1$ . By (b),  $d_1 d_2 = c e_1 e_2$  for some non-zero constant  $c$ ; if  $d_1 = e_1 \neq 0$  then  $d_2 = c e_2$ , and two monic polynomials (or two zeros) differing by a constant factor are equal, so  $d_2 = e_2$ . If  $d_1 = e_1 = 0$ , then

$d_2 = e_2 = 0$  by the divisibility chain.

## 9.8 Semisimple Operators and the Jordan–Chevalley Decomposition

The Jordan canonical form says that an operator whose characteristic polynomial splits is “a diagonal matrix plus a nilpotent one” — but only after a basis has been chosen. This section turns that sentence into a statement about the operator itself:  $T = S + N$  with  $S$  diagonalizable,  $N$  nilpotent and, crucially,  $SN = NS$ ; and there is exactly one such pair. Before that we need a name for the operators that deserve to be called diagonalizable even when the field is too small to see it.

### 9.8.1 Diagonalizable once the field is big enough

Two operators fail to be diagonalizable for completely different reasons. The rotation  $\mathbf{R} = \begin{pmatrix} 0 & -1 \\ 1 & 0 \end{pmatrix} \in M_2(\mathbb{R})$  has  $p_{\mathbf{R}} = m_{\mathbf{R}} = x^2 + 1$ , with no real root, so it has no eigenvectors at all (Example (A Real Matrix with Non-Real Eigenvalues)). Yet over  $\mathbb{C}$  it becomes  $\text{diag}(i, -i)$ : the operator was fine, the field was too small. The shear  $\mathbf{J}_2(0)$  has  $m = x^2$  and is not diagonalizable over **any** field, however large: its single eigenvalue 0 has  $a = 2$  and  $g = 1$ , and enlarging the field changes neither.

What separates the two cases is visible in the minimal polynomial. For  $\mathbf{R}$  it is  $x^2 + 1$ , irreducible over  $\mathbb{R}$  and appearing to the first power. For  $\mathbf{J}_2(0)$  it is  $x^2$ , the **square** of the irreducible  $x$ . Chapter 8 proved that  $T$  is diagonalizable exactly when  $m_T$  is a product of **distinct linear** factors (Theorem 8.8.2). Deleting the word “linear” from that criterion gives the notion we want.

*A semisimple operator is one whose minimal polynomial is squarefree: no irreducible factor is repeated.*

#### Definition 9.8.1 — Semisimple Operator

Let  $V$  be a finite-dimensional vector space over  $F$  with  $\dim V \geq 1$ , and let  $T \in \mathcal{L}(V)$ . We say  $T$  is **semisimple** if

$$m_T = p_1 p_2 \cdots p_k$$

for **distinct** monic irreducible  $p_1, \dots, p_k \in F[x]$ ; equivalently, if **no** irreducible polynomial  $p \in F[x]$  satisfies  $p^2 \mid m_T$ . A matrix  $\mathbf{A} \in M_n(F)$  is **semisimple over  $F$**  if  $T_{\mathbf{A}}$  is.

In words: factor  $m_T$  into irreducibles over  $F$  (Theorem 5.3.5). Semisimplicity says every exponent in that factorization is 1. The two descriptions agree: the exponents are all 1 exactly when no  $p^2$  divides  $m_T$ , by uniqueness of factorization. Note the dependence on the field, which is built into the word “irreducible over  $F$ ”.

**Examples.**

- **Diagonalizable operators.** If  $T$  is diagonalizable,  $m_T$  is a product of distinct linear factors (Theorem 8.8.2), and linear polynomials are irreducible (Definition 5.3.1). So  $T$  is semisimple.
- **The rotation.**  $\mathbf{R} \in M_2(\mathbb{R})$  above has  $m_{\mathbf{R}} = x^2 + 1$ , irreducible over  $\mathbb{R}$  since it has no real root (Theorem 5.3.1). One factor, exponent 1: semisimple over  $\mathbb{R}$ , though not diagonalizable over  $\mathbb{R}$ .
- **Any operator with irreducible minimal polynomial,** for the same reason. In particular the companion matrix  $\mathbf{C}(p)$  of an irreducible  $p$  is semisimple, since  $m_{\mathbf{C}(p)} = p$  (Theorem 9.5.3).
- **The degenerate case.** On a space of dimension 1,  $T = \lambda \operatorname{id}_V$  has  $m_T = x - \lambda$ : semisimple. So is  $\operatorname{id}_V$  on any  $V$ , with  $m = x - 1$ .

**Non-example by minimal change.** Over  $\mathbb{R}$ , replace  $m = x^2 + 1$  by  $m = (x^2 + 1)^2$ : the companion matrix  $\mathbf{C}((x^2 + 1)^2) \in M_4(\mathbb{R})$  has  $m = (x^2 + 1)^2$  (Theorem 9.5.3), so the irreducible  $x^2 + 1$  is repeated and the matrix is **not** semisimple. The clause that fails is “distinct”: there is one irreducible factor, but it occurs twice.

**Why this definition.** The word *simple* names the case in which  $V$  has no  $T$ -invariant subspace except  $\{\mathbf{0}\}$  and  $V$ . That happens exactly when  $m_T$  is irreducible **and**  $\dim V = \deg m_T$ , for then every non-zero  $\mathbf{v}$  has  $T$ -annihilator  $m_T$  and so  $Z(\mathbf{v}; T) = V$  (Theorem 9.5.2). *Semisimple* then means “a direct sum of simple pieces”, which is what Theorem 8.8.1 followed by a cyclic decomposition (Theorem 9.6.2) delivers. The definition is the right one because it is exactly diagonalizability after the field has been enlarged enough to split  $m_T$  — provided the field is not pathological. Making that precise needs one small fact: the minimal polynomial does not change when the field grows.

### Lemma 9.8.1 — The Minimal Polynomial Does Not Notice a Larger Field

Let  $K \subseteq L$  be fields and  $\mathbf{A} \in M_n(K)$ ,  $n \geq 1$ . Then the minimal polynomial of  $\mathbf{A}$  computed in  $K[x]$  equals the minimal polynomial of  $\mathbf{A}$  computed in  $L[x]$ .

*Idea.* The degree of  $m_{\mathbf{A}}$  is the position of the first linear dependence in the list  $\mathbf{I}, \mathbf{A}, \mathbf{A}^2, \dots$  (Proposition 8.5.4). Whether a list of matrices with entries in  $K$  is dependent is decided by row reducing one matrix with entries in  $K$ , and row reduction does not care that a bigger field is available.

*Proof.* Write  $m^K$  and  $m^L$  for the two minimal polynomials, and  $d = \deg m^K$ ,  $e = \deg m^L$ .

Since  $m^K \in K[x] \subseteq L[x]$  annihilates  $\mathbf{A}$ , Theorem 8.5.1 gives  $m^L \mid m^K$ , so  $e \leq d$ .

For the reverse inequality, list the entries of  $\mathbf{I}, \mathbf{A}, \dots, \mathbf{A}^e$  as the columns of a matrix  $\mathbf{B} \in M_{n^2 \times (e+1)}(K)$ , one column per power. A linear combination  $\sum_{i=0}^e c_i \mathbf{A}^i$  is the zero matrix exactly when  $\mathbf{B}\mathbf{c} = \mathbf{0}$  for  $\mathbf{c} = (c_0, \dots, c_e)$ . By Proposition 8.5.4 applied over  $L$ , the list  $(\mathbf{I}, \mathbf{A}, \dots, \mathbf{A}^e)$  is dependent over  $L$ , so  $\mathbf{B}\mathbf{c} = \mathbf{0}$  has a non-trivial solution in  $L^{e+1}$ , and hence  $\operatorname{rank}_L \mathbf{B} < e + 1$  (Theorem 2.5.7). Now reduce  $\mathbf{B}$  to reduced row echelon form by row operations performed inside  $K$  (Theorem 2.2.3); the resulting matrix  $\mathbf{R}$  has entries in  $K$ , and the definition of reduced row echelon form (Definition 2.2.4) makes no reference to the field, so  $\mathbf{R}$  is also the reduced row echelon form of  $\mathbf{B}$  over  $L$ , by the uniqueness in Theorem 2.3.4. The rank is the number of pivots of  $\mathbf{R}$  (Corollary 2.5.6),

a number read off  $\mathbf{R}$  alone. Hence

$$\text{rank}_K \mathbf{B} = \text{rank}_L \mathbf{B} < e + 1,$$

so  $\mathbf{B}\mathbf{c} = \mathbf{0}$  has a non-trivial solution in  $K^{e+1}$  as well. That solution gives a non-zero  $f \in K[x]$  of degree at most  $e$  with  $f(\mathbf{A}) = 0$ , whence  $d = \deg m^K \leq e$  by [Theorem 8.5.1](#) and [Proposition 5.1.2](#).

So  $d = e$ , and  $m^L \mid m^K$  with both monic of the same degree forces  $m^L = m^K$ .

■

### Proposition 9.8.2 — Semisimple Means Diagonalizable After Enlarging the Field

Let  $K$  be a field of characteristic 0, let  $\mathbf{A} \in M_n(K)$  with  $n \geq 1$ , and let  $L \supseteq K$  be a field over which  $m_{\mathbf{A}}$  splits. Then

$$\mathbf{A} \text{ is semisimple over } K \iff \mathbf{A} \text{ is diagonalizable over } L.$$

*Proof.* By [Lemma 9.8.1](#) the minimal polynomial is the same polynomial  $m = m_{\mathbf{A}}$  whether computed over  $K$  or over  $L$ . By [Theorem 8.8.2](#),  $\mathbf{A}$  is diagonalizable over  $L$  if and only if  $m$  is a product of distinct linear factors in  $L[x]$ , that is, if and only if  $m$  splits over  $L$  — which it does by hypothesis — with **no repeated root**.

( $\Leftarrow$ ) Suppose  $m$  has no repeated root in  $L$ . If some irreducible  $p \in K[x]$  satisfied  $m = p^2 h$ , then  $p \mid m$  and  $m$  splits over  $L$ , so  $p$  has a root  $\lambda \in L$ ; by [Theorem 5.4.2](#),  $\text{mult}_{\lambda}(m) \geq 2 \text{mult}_{\lambda}(p) \geq 2$ , and  $\lambda$  would be a repeated root of  $m$ . So no such  $p$  exists and  $\mathbf{A}$  is semisimple over  $K$ .

( $\Rightarrow$ ) Suppose  $m = p_1 \cdots p_k$  with  $p_1, \dots, p_k \in K[x]$  distinct monic irreducibles. The formal derivative is computed from the coefficients ([Definition 5.4.3](#)), so each  $p'_i$  is the same polynomial whether read in  $K[x]$  or in  $L[x]$ .

*No  $p_i$  has a repeated root in  $L$ .* Since  $K$  has characteristic 0 and  $\deg p_i \geq 1$ , the derivative  $p'_i$  is non-zero of degree  $\deg p_i - 1$ , so  $p_i \nmid p'_i$ ; as  $p_i$  is irreducible, [Lemma 5.3.2](#) gives  $\gcd(p_i, p'_i) = 1$ , and by [Corollary 5.2.4](#) there are  $a, b \in K[x]$  with  $ap_i + bp'_i = 1$ . If  $\lambda \in L$  were a repeated root of  $p_i$ , then  $p_i(\lambda) = p'_i(\lambda) = 0$  by [Theorem 5.4.5](#) (a), applied in  $L[x]$ ; evaluating the identity at  $\lambda$  would give  $1 = 0$ .

*No two  $p_i$  share a root in  $L$ .* For  $i \neq j$ , a monic divisor of the irreducible  $p_j$  is 1 or  $p_j$ , so  $p_i \nmid p_j$ , and [Lemma 5.3.2](#) gives  $\gcd(p_i, p_j) = 1$ . Again by [Corollary 5.2.4](#),  $ap_i + bp_j = 1$  for some  $a, b \in K[x]$ , and a common root in  $L$  would give  $1 = 0$ .

Therefore, by [Theorem 5.4.2](#), every root of  $m$  in  $L$  has multiplicity 1: it is a root of exactly one  $p_i$ , and a simple one. So  $m$  is a product of distinct linear factors over  $L$ , and  $\mathbf{A}$  is diagonalizable over  $L$ .

■

*Remark.* A field over which every irreducible polynomial has distinct roots in a splitting field is called **perfect**. The proof above shows that every field of characteristic 0 is perfect, and every finite field is perfect as well, although that needs an argument we do



not give here. Over a non-perfect field [Proposition 9.8.2](#) can fail: in characteristic  $p$  there are irreducible polynomials of the form  $q(x^p)$ , whose derivative vanishes identically and which acquire repeated roots in a splitting field. Everything in the rest of this section takes place over a field where  $p_T$  splits, so this subtlety will not arise again.

*Quick check.* Over  $\mathbb{Q}$ , which of  $\mathbf{A} = \mathbf{C}(x^2 + 1) \oplus \mathbf{C}(x^2 + 1) \in M_4(\mathbb{Q})$  and  $\mathbf{B} = \mathbf{C}((x^2 + 1)^2) \in M_4(\mathbb{Q})$  is semisimple? Is either diagonalizable over  $\mathbb{Q}$ ?

*Solution.*  $\mathbf{A}$  is semisimple. Each block has minimal polynomial  $x^2 + 1$  ([Theorem 9.5.3](#)), so  $m_{\mathbf{A}} = \text{lcm}(x^2 + 1, x^2 + 1) = x^2 + 1$  ([Proposition 8.5.6](#)), an irreducible polynomial to the first power.  $\mathbf{B}$  is not:  $m_{\mathbf{B}} = (x^2 + 1)^2$  repeats the irreducible  $x^2 + 1$ . Neither is diagonalizable over  $\mathbb{Q}$ , since  $x^2 + 1$  has no rational root, so neither minimal polynomial is a product of linear factors. Over  $\mathbb{C}$ , however,  $\mathbf{A}$  becomes  $\text{diag}(i, -i, i, -i)$ , while  $\mathbf{B}$  becomes  $\mathbf{J}_2(i) \oplus \mathbf{J}_2(-i)$ , which is not diagonal.

## 9.8.2 Splitting an operator in two

Here is the goal. When  $p_T$  splits,  $T$  is the sum of a diagonalizable operator and a nilpotent operator that commute, in exactly one way, and both summands are polynomials in  $T$ . The Jordan form already shows a splitting: in a Jordan basis,  $\mathbf{J} = \mathbf{D} + \mathbf{M}$  where  $\mathbf{D}$  is the diagonal of  $\mathbf{J}$  and  $\mathbf{M}$  carries the superdiagonal ones, and these commute because  $\mathbf{D}$  is constant on each block. What the theorem adds is that the two pieces do not depend on which Jordan basis was used, and that they can be written down without finding one.

### Theorem 9.8.3 — Jordan–Chevalley Decomposition

Let  $V$  be a finite-dimensional vector space over  $F$  with  $\dim V = n \geq 1$ , and let  $T \in \mathcal{L}(V)$  with  $p_T$  split over  $F$ . Then:

- (a) **(Existence)** There are  $S, N \in \mathcal{L}(V)$  with

$$T = S + N, \quad S \text{ diagonalizable}, \quad N \text{ nilpotent}, \quad SN = NS;$$

moreover  $S = g(T)$  and  $N = h(T)$  for some  $g, h \in F[x]$ .

- (b) **(Uniqueness)** If  $T = S' + N'$  with  $S'$  diagonalizable,  $N'$  nilpotent and  $S'N' = N'S'$ , then  $S' = S$  and  $N' = N$ .

We call  $S$  the **semisimple part** and  $N$  the **nilpotent part** of  $T$ .

*Idea. Existence.* The generalized eigenspace decomposition already puts the answer in front of us: on  $G_\lambda$  the operator is  $\lambda \text{ id} + \text{nilpotent}$ . Declare  $S$  to be  $\lambda \text{ id}$  on  $G_\lambda$  for each  $\lambda$  and let  $N$  be the rest. Then  $S$  is diagonalizable because a basis assembled from bases of the  $G_\lambda$  consists of eigenvectors of  $S$ , and  $N$  is nilpotent because it is nilpotent on each piece. That  $S$  and  $N$  are polynomials in  $T$  is the extra information carried by the primary decomposition: its projections are  $h_i(T)$ .

**Uniqueness.** A second decomposition  $S' + N'$  has  $S'$  commuting with  $T$ , hence with



every polynomial in  $T$ , hence with  $S$  and with  $N$ ; the same for  $N'$ . Then  $S - S' = N' - N$  is simultaneously diagonalizable and nilpotent, and the only such operator is 0.

*Proof.*

(a) Let  $\lambda_1, \dots, \lambda_k$  be the distinct eigenvalues of  $T$  and write

$$m_T = (x - \lambda_1)^{s_1} \cdots (x - \lambda_k)^{s_k}, \quad G_i = G_{\lambda_i}(T),$$

as in [Theorem 9.1.3](#), so that  $V = G_1 \oplus \cdots \oplus G_k$  with  $G_i = \ker(T - \lambda_i \operatorname{id}_V)^{s_i}$ . The factors  $(x - \lambda_i)^{s_i}$  are exactly the primary factors of  $m_T$ , since  $x - \lambda_i$  is irreducible, so  $G_i$  is the  $i$ -th primary component of [Theorem 8.8.1](#). By part (d) of that theorem there are  $h_1, \dots, h_k \in F[x]$  such that

$$\pi_i := h_i(T) \in \mathcal{L}(V)$$

is the projection onto  $G_i$  along  $\bigoplus_{j \neq i} G_j$ . Put

$$g := \sum_{i=1}^k \lambda_i h_i \in F[x],$$

$$S := g(T) = \sum_{i=1}^k \lambda_i \pi_i,$$

$$N := T - S = h(T) \text{ with } h = x - g.$$

*S is diagonalizable.* Fix  $i$  and  $\mathbf{v} \in G_i$ . Then  $\pi_i \mathbf{v} = \mathbf{v}$  and  $\pi_j \mathbf{v} = \mathbf{0}$  for  $j \neq i$  ([Theorem 3.10.1](#)), so  $S\mathbf{v} = \lambda_i \mathbf{v}$ . Choose a basis of each  $G_i$  and let  $\mathcal{B}$  be the concatenation, a basis of  $V$  because  $V = \bigoplus_i G_i$ . Every vector of  $\mathcal{B}$  is an eigenvector of  $S$ , so  $[S]_{\mathcal{B}}$  is diagonal and  $S$  is diagonalizable ([Theorem 8.4.1](#)).

*N is nilpotent.* Fix  $i$  and  $\mathbf{v} \in G_i$ . Then  $N\mathbf{v} = T\mathbf{v} - \lambda_i \mathbf{v} = (T - \lambda_i \operatorname{id}_V)\mathbf{v}$ , which lies in  $G_i$  because  $G_i$  is  $T$ -invariant, so by induction  $N^j \mathbf{v} = (T - \lambda_i \operatorname{id}_V)^j \mathbf{v}$  for every  $j \geq 0$ . With  $s = \max_i s_i$  we get  $N^s \mathbf{v} = (T - \lambda_i \operatorname{id}_V)^s \mathbf{v} = \mathbf{0}$ , since  $(T - \lambda_i \operatorname{id}_V)^{s_i} \mathbf{v} = \mathbf{0}$  and  $s \geq s_i$ . As every  $\mathbf{v} \in V$  is a sum of such vectors,  $N^s = \mathbf{0}$  and  $N$  is nilpotent ([Definition 9.2.1](#)).

*They commute.* Both  $S = g(T)$  and  $N = h(T)$  are polynomials in  $T$ , and any two polynomials in  $T$  commute ([Theorem 5.7.1](#) (c)). This proves (a).

(b) Suppose  $T = S' + N'$  as stated. First,  $S'$  commutes with  $T$ :

$$S'T = S'(S' + N') = S'S' + S'N' = S'S' + N'S' = (S' + N')S' = TS',$$

where the middle equality is the hypothesis  $S'N' = N'S'$ . Hence  $S'$  commutes with every power of  $T$  and so with every polynomial in  $T$ , in particular with  $S = g(T)$  and  $N = h(T)$ . The same computation gives  $N'T = TN'$ , so  $N'$  too commutes with  $S$  and  $N$ .

Now set  $X := S - S'$ . From  $S + N = T = S' + N'$  we get

$$X = S - S' = N' - N.$$

The operators  $S$  and  $S'$  are diagonalizable and commute, so they are simultaneously diagonalizable ([Theorem 8.9.5](#)): there is a basis  $\mathcal{C}$  in which  $[S]_{\mathcal{C}}$  and  $[S']_{\mathcal{C}}$  are both diagonal. Then  $[X]_{\mathcal{C}} = [S]_{\mathcal{C}} - [S']_{\mathcal{C}}$  is diagonal, so  $X$  is diagonalizable. On the other hand  $N$  and

$N'$  are nilpotent and commute, hence so do  $N'$  and  $-N$ , and  $X = N' + (-N)$  is nilpotent by [Exercise \(C2: Commuting nilpotent matrices\)](#) (b), whose binomial argument uses only that the two summands commute and so applies verbatim to operators.

So  $X$  is diagonalizable and nilpotent. Being nilpotent,  $\text{spec}(X) = \{0\}$  ([Proposition 9.2.1](#) (c)); being diagonalizable,  $[X]_{\mathcal{D}}$  is diagonal with the eigenvalues of  $X$  on the diagonal for some basis  $\mathcal{D}$  ([Theorem 8.4.1](#)), so  $[X]_{\mathcal{D}} = 0$  and  $X = 0$ . Hence  $S' = S$  and, subtracting from  $T$ ,  $N' = N$ . This proves the theorem. ■

**Warning. Splitting a matrix into its diagonal part and its off-diagonal part is not the Jordan–Chevalley decomposition.** For  $\mathbf{A} = \begin{pmatrix} 1 & 1 \\ 0 & 2 \end{pmatrix}$ , the “obvious” split is  $\mathbf{D} = \text{diag}(1, 2)$  plus  $\mathbf{E} = \begin{pmatrix} 0 & 1 \\ 0 & 0 \end{pmatrix}$ . Here  $\mathbf{D}$  is diagonal and  $\mathbf{E}$  is nilpotent, but

$$\mathbf{DE} = \begin{pmatrix} 0 & 1 \\ 0 & 0 \end{pmatrix} \neq \begin{pmatrix} 0 & 2 \\ 0 & 0 \end{pmatrix} = \mathbf{ED},$$

so the commuting requirement fails, and the split is not the one the theorem produces. Indeed  $\mathbf{A}$  has distinct eigenvalues, so it is diagonalizable ([Corollary 8.4.2](#)) and its actual decomposition is  $\mathbf{S} = \mathbf{A}$ ,  $\mathbf{N} = 0$ . The off-diagonal part of a matrix is a feature of the basis; the nilpotent part is a feature of the operator.

The formulas in the proof are also the recipe: build the projections  $\pi_i$  onto the generalized eigenspaces — by Bézout, as in [Example \(A Primary Decomposition with an Irreducible Quadratic\)](#) — and set  $S = \sum_i \lambda_i \pi_i$ .

**Example. (The Semisimple and Nilpotent Parts of a  $3 \times 3$  Matrix)**

Let

$$\mathbf{A} = \begin{pmatrix} 2 & 2 & -1 \\ -1 & 3 & -1 \\ -1 & -1 & 2 \end{pmatrix} \in M_3(\mathbb{Q}), \quad p_{\mathbf{A}} = (x-2)^2(x-3).$$

Find the Jordan–Chevalley decomposition  $\mathbf{A} = \mathbf{S} + \mathbf{N}$ , and exhibit  $\mathbf{S}$  and  $\mathbf{N}$  as polynomials in  $\mathbf{A}$ .

*Solution. The minimal polynomial.* Put  $\mathbf{B} = \mathbf{A} - 2\mathbf{I} = \begin{pmatrix} 0 & 2 & -1 \\ -1 & 1 & -1 \\ -1 & -1 & 0 \end{pmatrix}$ . Then

$$\mathbf{B}(\mathbf{A} - 3\mathbf{I}) = \begin{pmatrix} 0 & 2 & -1 \\ -1 & 1 & -1 \\ -1 & -1 & 0 \end{pmatrix} \begin{pmatrix} -1 & 2 & -1 \\ -1 & 0 & -1 \\ -1 & -1 & -1 \end{pmatrix} = \begin{pmatrix} -1 & 1 & -1 \\ 1 & -1 & 1 \\ 2 & -2 & 2 \end{pmatrix} \neq 0,$$

so  $m_{\mathbf{A}} \neq (x-2)(x-3)$ ; since  $m_{\mathbf{A}} \mid p_{\mathbf{A}}$  and  $m_{\mathbf{A}}$  has both 2 and 3 as roots ([Theorem 8.5.3](#)),  $m_{\mathbf{A}} = (x-2)^2(x-3)$ . The primary factors are  $(x-2)^2$  and  $x-3$ .

*Bézout.* Dividing,  $(x-2)^2 = x^2 - 4x + 4 = (x-3)(x-1) + 1$ , so

$$1 \cdot (x-2)^2 - (x-1)(x-3) = 1.$$

Following [Theorem 5.7.5](#) with  $p = (x-2)^2$  and  $q = x-3$ : the projection onto  $\ker q(\mathbf{A}) = G_3(\mathbf{A})$  along  $\ker p(\mathbf{A}) = G_2(\mathbf{A})$  is  $\pi_3 = (\mathbf{A} - 2\mathbf{I})^2 = \mathbf{B}^2$ , and  $\pi_2 = \mathbf{I} - \mathbf{B}^2$ .

*Compute.* Squaring,

$$\mathbf{B}^2 = \begin{pmatrix} -1 & 3 & -2 \\ 0 & 0 & 0 \\ 1 & -3 & 2 \end{pmatrix}, \quad \text{and } \mathbf{B}^3 = \mathbf{B}^2 \text{ (as } \mathbf{B}^2(\mathbf{B} - \mathbf{I}) = 0\text{)}.$$

Hence

$$\begin{aligned} \mathbf{S} &= 2\pi_2 + 3\pi_3 = 2\mathbf{I} + \mathbf{B}^2 = \begin{pmatrix} 1 & 3 & -2 \\ 0 & 2 & 0 \\ 1 & -3 & 4 \end{pmatrix}, \\ \mathbf{N} &= \mathbf{A} - \mathbf{S} = \mathbf{B} - \mathbf{B}^2 = \begin{pmatrix} 1 & -1 & 1 \\ -1 & 1 & -1 \\ -2 & 2 & -2 \end{pmatrix}. \end{aligned}$$

*The checks.* First  $\mathbf{S} + \mathbf{N} = \mathbf{A}$  by construction. Next  $\mathbf{N}^2 = 0$ : reading the rows of  $\mathbf{N}$  as multiples of  $(1, -1, 1)$  shows that  $\mathbf{N} = \mathbf{u}\mathbf{w}^\top$  with  $\mathbf{u} = (1, -1, -2)$  and  $\mathbf{w} = (1, -1, 1)$ , and  $\mathbf{w}^\top \mathbf{u} = 1 + 1 - 2 = 0$ , so  $\mathbf{N}^2 = \mathbf{u}(\mathbf{w}^\top \mathbf{u})\mathbf{w}^\top = 0$ . Next  $\mathbf{S}$  is diagonalizable:  $(\mathbf{S} - 2\mathbf{I})(\mathbf{S} - 3\mathbf{I}) = \mathbf{B}^2(\mathbf{B}^2 - \mathbf{I}) = \mathbf{B}^4 - \mathbf{B}^2 = 0$ , because  $\mathbf{B}^3 = \mathbf{B}^2$  gives  $\mathbf{B}^4 = \mathbf{B}^2$ ; so  $m_{\mathbf{S}}$  divides a product of distinct linear factors and  $\mathbf{S}$  is diagonalizable ([Theorem 8.8.2](#)). Finally  $\mathbf{S}\mathbf{N} = \mathbf{N}\mathbf{S}$ , since both are polynomials in  $\mathbf{A}$ : explicitly  $\mathbf{S} = 2\mathbf{I} + (\mathbf{A} - 2\mathbf{I})^2$  and  $\mathbf{N} = (\mathbf{A} - 2\mathbf{I}) - (\mathbf{A} - 2\mathbf{I})^2$ , that is,

$$\begin{aligned} \mathbf{S} &= g(\mathbf{A}) \text{ with } g = x^2 - 4x + 6, \\ \mathbf{N} &= h(\mathbf{A}) \text{ with } h = -x^2 + 5x - 6 = -(x-2)(x-3). \end{aligned}$$

As a cross-check,  $g + h = x$ , so  $g(\mathbf{A}) + h(\mathbf{A}) = \mathbf{A}$ ; and  $\text{tr } \mathbf{S} = 1 + 2 + 4 = 7 = 2 + 2 + 3$  is the sum of the eigenvalues of  $\mathbf{A}$ , while  $\text{tr } \mathbf{N} = 1 + 1 - 2 = 0$ , as it must be for a nilpotent matrix ([Proposition 9.2.1](#) (c) and [Theorem 8.3.3](#)).

Two consequences are worth naming now. Because  $S$  and  $N$  are polynomials in  $T$ , **every operator commuting with  $T$  automatically commutes with  $S$  and with  $N$** . And the decomposition is exactly what makes functions of matrices computable in the next section: any reasonable function  $f$  will satisfy  $f(T) = f(S + N)$  with the two parts handled separately,  $S$  by its eigenvalues and  $N$  by finitely many terms, since its powers stop.

*Remark.* The splitting exists in greater generality. Over a perfect field, every  $T$  can be written uniquely as  $T = S + N$  with  $S$  **semisimple** in the sense of [Definition 9.8.1](#),  $N$  nilpotent and  $SN = NS$ ; the proof runs through the primary decomposition of  $m_T$  into powers of general irreducibles instead of linear ones. We prove only the split case, which is all we use.

## 9.8.3 Exercises

### 9.8.3.1 A. Check your understanding

#### Exercise A1.

- (a) Define a semisimple operator, and say how the definition specializes to Chapter 8's criterion for diagonalizability.
- (b) State the Jordan–Chevalley Decomposition Theorem, including all three conditions on  $S$  and  $N$  and the uniqueness clause.
- (c) True or false:  $\begin{pmatrix} 0 & -1 \\ 1 & 0 \end{pmatrix} \in M_2(\mathbb{R})$  is semisimple. Justify your answer.
- (d) True or false: if  $\mathbf{A} = \mathbf{D} + \mathbf{E}$  with  $\mathbf{D}$  the diagonal part of  $\mathbf{A}$  and  $\mathbf{E}$  the off-diagonal part, then  $\mathbf{D}$  is the semisimple part of  $\mathbf{A}$ . Justify your answer.
- (e) Which hypothesis of the uniqueness clause does the pair  $(\mathbf{D}, \mathbf{E})$  of part (d) typically violate?
- (f) Why does every operator commuting with  $T$  commute with the semisimple part of  $T$ ?

*Solution.*

- (a)  $T$  is semisimple if  $m_T$  is a product of **distinct monic irreducible** polynomials over  $F$  (Definition 9.8.1). Requiring those irreducibles to be **linear** gives Theorem 8.8.2, the criterion for diagonalizability. So diagonalizable  $\Rightarrow$  semisimple, and the converse holds when  $m_T$  splits.
- (b) If  $p_T$  splits, then  $T = S + N$  with  $S$  diagonalizable,  $N$  nilpotent and  $SN = NS$ ; such  $S$  and  $N$  are unique, and both are polynomials in  $T$  (Theorem 9.8.3).
- (c) True. Its minimal polynomial is  $x^2 + 1$ , which has no real root and so is irreducible over  $\mathbb{R}$  (Theorem 5.3.1); one irreducible factor, exponent 1.
- (d) False. For  $\mathbf{A} = \begin{pmatrix} 1 & 1 \\ 0 & 2 \end{pmatrix}$  the semisimple part is  $\mathbf{A}$  itself, since  $\mathbf{A}$  is diagonalizable, whereas  $\mathbf{D} = \text{diag}(1, 2) \neq \mathbf{A}$ .
- (e) The commuting condition  $\mathbf{DE} = \mathbf{ED}$ ; see the warning after Theorem 9.8.3.
- (f) Because  $S = g(T)$  for some polynomial  $g$  (Theorem 9.8.3 (a)), and an operator commuting with  $T$  commutes with every power of  $T$ , hence with every polynomial in  $T$ .

### 9.8.3.2 B. Practice

#### Exercise B1: Semisimple and nilpotent parts.

For each matrix over  $\mathbb{Q}$ , find the Jordan–Chevalley decomposition  $\mathbf{A} = \mathbf{S} + \mathbf{N}$  and express  $\mathbf{S}$  as a polynomial in  $\mathbf{A}$ .

- (a)  $\mathbf{A}_1 = \begin{pmatrix} 3 & 1 \\ 0 & 3 \end{pmatrix}$ .
- (b)  $\mathbf{A}_2 = \begin{pmatrix} 1 & 2 \\ 0 & 3 \end{pmatrix}$ .
- (c)  $\mathbf{A}_3 = \begin{pmatrix} 2 & 1 & 3 \\ 0 & 2 & 0 \\ 0 & 0 & 5 \end{pmatrix}$ .

*Solution.*

- (a)  $m_{\mathbf{A}_1} = (x - 3)^2$ , so there is one eigenvalue and  $G_3 = \mathbb{Q}^2$ ; the projection onto it is  $\mathbf{I}$ . Hence  $\mathbf{S} = 3\mathbf{I}$  and  $\mathbf{N} = \mathbf{A}_1 - 3\mathbf{I} = \begin{pmatrix} 0 & 1 \\ 0 & 0 \end{pmatrix}$ , with  $\mathbf{N}^2 = 0$ . As a polynomial,  $\mathbf{S} = 3\mathbf{I} = g(\mathbf{A}_1)$  with  $g = 3$  the constant polynomial. (Every scalar multiple of  $\mathbf{I}$  is diagonalizable, and  $\mathbf{S}$  commutes with everything.)
- (b)  $\mathbf{A}_2$  has the distinct eigenvalues 1 and 3, so it is diagonalizable (Corollary 8.4.2). By uniqueness,  $\mathbf{S} = \mathbf{A}_2$  and  $\mathbf{N} = 0$ ; here  $g = x$ . The naive split into  $\text{diag}(1, 3)$  and  $\begin{pmatrix} 0 & 2 \\ 0 & 0 \end{pmatrix}$  is **not** the Jordan–Chevalley decomposition.
- (c)  $p_{\mathbf{A}_3} = (x - 2)^2(x - 5)$ . Put  $\mathbf{M} = \mathbf{A}_3 - 2\mathbf{I} = \begin{pmatrix} 0 & 1 & 3 \\ 0 & 0 & 0 \\ 0 & 0 & 3 \end{pmatrix}$ . Then  $\mathbf{M}(\mathbf{A}_3 - 5\mathbf{I})$  has (1, 2)-entry  $-3 \neq 0$ , so  $m_{\mathbf{A}_3} = (x - 2)^2(x - 5)$ . Bézout:  $(x - 2)^2 = (x - 5)(x + 1) + 9$ , so  $\frac{1}{9}(x - 2)^2 - \frac{1}{9}(x + 1)(x - 5) = 1$ , and by Theorem 5.7.5 the projection onto  $G_5$  along  $G_2$  is  $\pi_5 = \frac{1}{9}\mathbf{M}^2$ . Computing  $\mathbf{M}^2 = \begin{pmatrix} 0 & 0 & 9 \\ 0 & 0 & 0 \\ 0 & 0 & 9 \end{pmatrix}$ ,

$$\mathbf{S} = 2\mathbf{I} + 3\pi_5 = 2\mathbf{I} + \frac{1}{3}\mathbf{M}^2 = \begin{pmatrix} 2 & 0 & 3 \\ 0 & 2 & 0 \\ 0 & 0 & 5 \end{pmatrix}, \quad \mathbf{N} = \mathbf{A}_3 - \mathbf{S} = \begin{pmatrix} 0 & 1 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{pmatrix}.$$

Checks:  $\mathbf{N}^2 = 0$ ;  $\mathbf{S} - 2\mathbf{I}$  has rank 1, so  $g_{\mathbf{S}}(2) = 2 = a_{\mathbf{S}}(2)$  and  $\mathbf{S}$  is diagonalizable (Theorem 8.4.1); and  $\mathbf{SN} = \begin{pmatrix} 0 & 2 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{pmatrix} = \mathbf{NS}$ . As a polynomial,  $\mathbf{S} = 2\mathbf{I} + \frac{1}{3}(\mathbf{A}_3 - 2\mathbf{I})^2 = g(\mathbf{A}_3)$  with  $g = \frac{1}{3}x^2 - \frac{4}{3}x + \frac{10}{3}$ . Note that  $\mathbf{S}$  keeps the entry 3 in position (1, 3): the diagonal part of  $\mathbf{A}_3$  is the wrong answer, and it does not commute with the rest, since  $\text{diag}(2, 2, 5)(\mathbf{A}_3 - \text{diag}(2, 2, 5))$  and the reverse product differ in position (1, 3) by  $-9$ .

### Exercise B2: When is the nilpotent part zero?.

Let  $V$  be finite-dimensional with  $\dim V \geq 1$  and let  $T \in \mathcal{L}(V)$  with  $p_T$  split, with Jordan–Chevalley decomposition  $T = S + N$ . Prove that  $N = 0$  if and only if  $T$  is diagonalizable, and that  $S = 0$  if and only if  $T$  is nilpotent.

*Solution.* If  $N = 0$ , then  $T = S$  is diagonalizable. Conversely, if  $T$  is diagonalizable, then  $T = T + 0$  is a decomposition into a diagonalizable and a nilpotent part that commute ( $T \cdot 0 = 0 = 0 \cdot T$ ), so by the uniqueness in Theorem 9.8.3,  $S = T$  and  $N = 0$ .

If  $S = 0$ , then  $T = N$  is nilpotent. Conversely, if  $T$  is nilpotent, then  $T = 0 + T$  is such a decomposition, since the zero operator is diagonalizable and commutes with  $T$ ; uniqueness gives  $S = 0$  and  $N = T$ .

### Exercise B3: Block diagonal matrices.

Let  $\mathbf{A}_1 \in M_p(F)$  and  $\mathbf{A}_2 \in M_q(F)$  have split characteristic polynomials, with Jordan–Chevalley decompositions  $\mathbf{A}_i = \mathbf{S}_i + \mathbf{N}_i$ . Prove that the Jordan–Chevalley decomposi-

tion of  $\mathbf{A}_1 \oplus \mathbf{A}_2$  is

$$\mathbf{A}_1 \oplus \mathbf{A}_2 = (\mathbf{S}_1 \oplus \mathbf{S}_2) + (\mathbf{N}_1 \oplus \mathbf{N}_2).$$

*Solution.* Write  $\mathbf{S} = \mathbf{S}_1 \oplus \mathbf{S}_2$  and  $\mathbf{N} = \mathbf{N}_1 \oplus \mathbf{N}_2$ . By [Theorem 7.1.4](#) (a),  $\mathbf{S} + \mathbf{N} = (\mathbf{S}_1 + \mathbf{N}_1) \oplus (\mathbf{S}_2 + \mathbf{N}_2) = \mathbf{A}_1 \oplus \mathbf{A}_2$ . Also  $p_{\mathbf{A}_1 \oplus \mathbf{A}_2} = p_{\mathbf{A}_1} p_{\mathbf{A}_2}$  splits ([Theorem 6.8.4](#), applied over  $F[x]$  as the remark after it allows), so [Theorem 9.8.3](#) applies to  $\mathbf{A}_1 \oplus \mathbf{A}_2$  and it suffices, by uniqueness, to check the three conditions.

*S is diagonalizable.* Pick invertible  $\mathbf{P}_i$  with  $\mathbf{P}_i^{-1} \mathbf{S}_i \mathbf{P}_i$  diagonal. Then  $\mathbf{P} = \mathbf{P}_1 \oplus \mathbf{P}_2$  is invertible with  $\mathbf{P}^{-1} = \mathbf{P}_1^{-1} \oplus \mathbf{P}_2^{-1}$ , and  $\mathbf{P}^{-1} \mathbf{S} \mathbf{P} = (\mathbf{P}_1^{-1} \mathbf{S}_1 \mathbf{P}_1) \oplus (\mathbf{P}_2^{-1} \mathbf{S}_2 \mathbf{P}_2)$  is diagonal ([Theorem 7.1.4](#) (a), (c)).

*N is nilpotent.* If  $\mathbf{N}_1^a = 0$  and  $\mathbf{N}_2^b = 0$ , then  $\mathbf{N}^{\max(a,b)} = \mathbf{N}_1^{\max(a,b)} \oplus \mathbf{N}_2^{\max(a,b)} = 0$  by [Theorem 7.1.4](#) (b).

*They commute.*  $\mathbf{S} \mathbf{N} = (\mathbf{S}_1 \mathbf{N}_1) \oplus (\mathbf{S}_2 \mathbf{N}_2) = (\mathbf{N}_1 \mathbf{S}_1) \oplus (\mathbf{N}_2 \mathbf{S}_2) = \mathbf{N} \mathbf{S}$ , again by [Theorem 7.1.4](#) (a).

By the uniqueness in [Theorem 9.8.3](#),  $\mathbf{S}$  and  $\mathbf{N}$  are the semisimple and nilpotent parts of  $\mathbf{A}_1 \oplus \mathbf{A}_2$ .

### 9.8.3.3 C. Going deeper

#### Exercise C1: The multiplicative version.

Call  $U \in \mathcal{L}(V)$  **unipotent** if  $U - \text{id}_V$  is nilpotent. Let  $T \in \mathcal{L}(V)$  be **invertible** with  $p_T$  split, and let  $T = S + N$  be its Jordan–Chevalley decomposition.

- Prove that  $S$  is invertible.
- Prove that  $S^{-1}$  commutes with  $N$ , and that  $S^{-1}N$  is nilpotent.
- Deduce that  $T = SU$  with  $S$  diagonalizable,  $U$  unipotent and  $SU = US$ .
- Prove that this multiplicative decomposition is unique: if  $T = S'U'$  with  $S'$  diagonalizable,  $U'$  unipotent and  $S'U' = U'S'$ , then  $S' = S$  and  $U' = U$ .

*Hint:* for (d), turn the multiplicative statement into an additive one by writing  $T = S' + S'(U' - \text{id}_V)$ .

*Solution.*

- By the proof of [Theorem 9.8.3](#),  $S\mathbf{v} = \lambda_i \mathbf{v}$  for every  $\mathbf{v} \in G_{\lambda_i}(T)$ , so  $\text{spec}(S) = \text{spec}(T)$  — a basis of eigenvectors of  $S$  with these eigenvalues was exhibited there. Since  $T$  is invertible,  $0 \notin \text{spec}(T)$  ([Theorem 8.2.6](#)), so  $0 \notin \text{spec}(S)$  and  $S$  is invertible.
- From  $SN = NS$ , multiply on both sides by  $S^{-1}$ :  $S^{-1}(SN)S^{-1} = S^{-1}(NS)S^{-1}$  gives  $NS^{-1} = S^{-1}N$ . Hence  $(S^{-1}N)^j = S^{-j}N^j$  for every  $j \geq 1$ , by induction using  $NS^{-1} = S^{-1}N$ . With  $N^s = 0$  we get  $(S^{-1}N)^s = 0$ .
- Put  $U = \text{id}_V + S^{-1}N$ . Then  $SU = S + N = T$ , and  $U - \text{id}_V = S^{-1}N$  is nilpotent by (b), so  $U$  is unipotent. They commute:  $SU = S + N = US$ , since  $S(S^{-1}N) = N = (S^{-1}N)S$  by (b).
- Suppose  $T = S'U'$  with  $S'$  diagonalizable,  $U'$  unipotent and  $S'U' = U'S'$ . Put

$N' = S'(U' - \text{id}_V)$ , so that  $T = S' + N'$ . Then  $N'$  is nilpotent:  $S'$  commutes with  $U'$ , hence with  $U' - \text{id}_V$ , so  $(N')^j = (S')^j(U' - \text{id}_V)^j = 0$  once  $j$  is large. And  $S'N' = S'S'(U' - \text{id}_V) = S'(U' - \text{id}_V)S' = N'S'$ . So  $T = S' + N'$  satisfies the hypotheses of [Theorem 9.8.3](#) (b), giving  $S' = S$  and  $N' = N$ . Then  $U' - \text{id}_V = (S')^{-1}N' = S^{-1}N = U - \text{id}_V$ , so  $U' = U$ .

### Exercise C2: The parts inherit every symmetry.

Let  $V$  be finite-dimensional with  $\dim V \geq 1$ , let  $T \in \mathcal{L}(V)$  with  $p_T$  split, and let  $T = S + N$  be its Jordan–Chevalley decomposition.

- Prove that if  $X \in \mathcal{L}(V)$  satisfies  $XT = TX$ , then  $XS = SX$  and  $XN = NX$ .
- Deduce that every  $T$ -invariant subspace is  $S$ -invariant and  $N$ -invariant, and that  $S$  and  $N$  map each  $G_\lambda(T)$  into itself.
- Give an example of a diagonalizable  $S$  and a nilpotent  $N$  with  $SN \neq NS$  for which  $S + N$  is **not** the operator whose semisimple part is  $S$ . (So the commuting hypothesis cannot be dropped from the uniqueness statement.)

*Solution.*

- By [Theorem 9.8.3](#) (a),  $S = g(T)$  and  $N = h(T)$  for some  $g, h \in F[x]$ . From  $XT = TX$  we get  $XT^j = T^jX$  for all  $j \geq 0$  by induction, hence  $Xf(T) = f(T)X$  for every  $f \in F[x]$  by taking linear combinations. Applying this to  $f = g$  and  $f = h$  gives  $XS = SX$  and  $XN = NX$ .
- Let  $U \subseteq V$  be  $T$ -invariant. Then  $U$  is  $T^j$ -invariant for every  $j$ , hence  $f(T)$ -invariant for every  $f \in F[x]$  (a sum of vectors of  $U$  lies in  $U$ ); taking  $f = g, h$  gives that  $U$  is  $S$ - and  $N$ -invariant. In particular  $G_\lambda(T) = \ker(T - \lambda \text{id}_V)^n$  is  $T$ -invariant ([Corollary 9.1.2](#) (b)), so  $S$  and  $N$  map it into itself.
- Take  $\mathbf{S} = \text{diag}(1, 2)$  and  $\mathbf{N} = \begin{pmatrix} 0 & 1 \\ 0 & 0 \end{pmatrix}$  in  $M_2(\mathbb{Q})$ . Then  $\mathbf{S}$  is diagonal,  $\mathbf{N}^2 = 0$ , and  $\mathbf{SN} = \begin{pmatrix} 0 & 1 \\ 0 & 0 \end{pmatrix} \neq \begin{pmatrix} 0 & 2 \\ 0 & 0 \end{pmatrix} = \mathbf{NS}$ . Their sum  $\mathbf{A} = \begin{pmatrix} 1 & 1 \\ 0 & 2 \end{pmatrix}$  has distinct eigenvalues, so it is diagonalizable, and its semisimple part is  $\mathbf{A}$  itself, not  $\mathbf{S}$ , with nilpotent part 0, not  $\mathbf{N}$  ([Exercise \(B2: When is the nilpotent part zero?\)](#)). Without the commuting hypothesis, an operator can be written as “diagonalizable plus nilpotent” in many ways: for any  $c \in \mathbb{Q}$ ,  $\mathbf{A} = \begin{pmatrix} 1 & c \\ 0 & 2 \end{pmatrix} + \begin{pmatrix} 0 & 1-c \\ 0 & 0 \end{pmatrix}$  is such a splitting, the first summand being diagonalizable because its eigenvalues 1, 2 are distinct.

## 9.9 Functions of Matrices, the Exponential and Linear Differential Equations

We have known since Chapter 3 what  $p(\mathbf{A})$  means for a polynomial  $p$ . This section gives a meaning to  $f(\mathbf{A})$  for a much larger class of functions — square roots, logarithms, sines, and above all  $e^{\mathbf{A}}$  — and the Jordan form is what makes it possible. The definition needs a well-

definedness proof, which turns out to be Hermite interpolation from Chapter 5: whatever  $f$  is,  $f(\mathbf{A})$  is really  $p(\mathbf{A})$  for a polynomial  $p$  that agrees with  $f$  to the right order at each eigenvalue. With  $e^{t\mathbf{A}}$  in hand, every linear system of differential equations with constant coefficients is solved in one formula.

Throughout this section  $F = \mathbb{C}$  (or  $\mathbb{R}$ , when we say so), so that every characteristic polynomial splits ([Corollary 5.5.5](#)) and Jordan forms exist.

## 9.9.1 What a polynomial does to a Jordan block

Start with what we already have. A Jordan block is a scalar plus a nilpotent matrix whose powers stop, and a polynomial applied to it can be read off from a Taylor expansion.

### Lemma 9.9.1 — A Polynomial of a Jordan Block

Let  $F$  have characteristic 0, let  $k \geq 1$ ,  $\lambda \in F$  and  $p \in F[x]$ . Then  $p(\mathbf{J}_k(\lambda))$  is the upper triangular matrix whose  $(r, r+j)$ -entry is

$$(p(\mathbf{J}_k(\lambda)))_{r,r+j} = \frac{p^{(j)}(\lambda)}{j!} \quad (0 \leq j \leq k-1, 1 \leq r \leq k-j),$$

with all entries below the diagonal equal to 0. In particular the value  $p(\lambda)$  fills the diagonal,  $p'(\lambda)$  the superdiagonal, and only the first  $k$  derivatives of  $p$  at  $\lambda$  are involved.

*Proof.* Write  $\mathbf{J} = \mathbf{J}_k(\lambda) = \lambda \mathbf{I}_k + \mathbf{M}$  with  $\mathbf{M} = \mathbf{J}_k(0)$  ([Definition 9.3.1](#)). Reading off the columns of  $\mathbf{M}$ , we have  $\mathbf{M}\mathbf{e}_1 = \mathbf{0}$  and  $\mathbf{M}\mathbf{e}_r = \mathbf{e}_{r-1}$  for  $r \geq 2$ , so by induction  $\mathbf{M}^j\mathbf{e}_r = \mathbf{e}_{r-j}$  when  $r > j$  and  $\mathbf{M}^j\mathbf{e}_r = \mathbf{0}$  when  $r \leq j$ . Hence, for  $0 \leq j \leq k-1$ , the matrix  $\mathbf{M}^j$  has 1 in each position  $(r, r+j)$  and 0 elsewhere, and  $\mathbf{M}^k = \mathbf{0}$ . Let  $d = \deg p$ . By [Theorem 5.4.7](#) (b), applied with  $c = \lambda$  and  $n = d$ ,

$$p(x) = \sum_{j=0}^d \frac{p^{(j)}(\lambda)}{j!} (x - \lambda)^j.$$

Substituting  $\mathbf{J}$  for  $x$  — legitimate because evaluation at  $\mathbf{J}$  is a ring homomorphism ([Theorem 5.7.1](#)) — and using  $\mathbf{J} - \lambda \mathbf{I}_k = \mathbf{M}$ ,

$$p(\mathbf{J}) = \sum_{j=0}^d \frac{p^{(j)}(\lambda)}{j!} \mathbf{M}^j = \sum_{j=0}^{\min(d, k-1)} \frac{p^{(j)}(\lambda)}{j!} \mathbf{M}^j,$$

since  $\mathbf{M}^j = \mathbf{0}$  for  $j \geq k$ . Each  $\mathbf{M}^j$  contributes its coefficient to every position  $(r, r+j)$  and to no other position, which is the claim. ■

For  $p = x^m$  this reads  $p^{(j)}(\lambda)/j! = \binom{m}{j} \lambda^{m-j}$ , so

$$(\mathbf{J}_k(\lambda)^m)_{r,r+j} = \binom{m}{j} \lambda^{m-j},$$

which is the formula behind everything that follows, here and in the next section.



## 9.9.2 Functions of a matrix

Reading [Lemma 9.9.1](#) backwards suggests the definition. The formula for  $p(\mathbf{J}_k(\lambda))$  uses nothing about  $p$  except the numbers  $p(\lambda), p'(\lambda), \dots, p^{(k-1)}(\lambda)$ . Any function that has those derivatives can be inserted instead.

*To apply  $f$  to a matrix, apply it to each Jordan block, putting  $f^{(j)}(\lambda)/j!$  on the  $j$ -th superdiagonal.*

### Definition 9.9.1 — Function of a Matrix

Let  $\mathbf{A} \in M_n(\mathbb{C})$  with Jordan form  $\mathbf{J} = \mathbf{J}_{k_1}(\lambda_1) \oplus \dots \oplus \mathbf{J}_{k_m}(\lambda_m)$  and  $\mathbf{A} = \mathbf{P}\mathbf{J}\mathbf{P}^{-1}$ . Let  $f$  be a complex-valued function that is defined and **at least  $k_i - 1$  times differentiable at  $\lambda_i$**  for every  $i$  — for instance, a function given by a convergent power series on a disc around each eigenvalue. Define

$$f(\mathbf{J}_k(\lambda)) := \begin{pmatrix} f(\lambda) & f'(\lambda) & \frac{f''(\lambda)}{2!} & \dots & \frac{f^{(k-1)}(\lambda)}{(k-1)!} \\ & f(\lambda) & f'(\lambda) & \ddots & \vdots \\ & & \ddots & \ddots & \frac{f''(\lambda)}{2!} \\ & & & f(\lambda) & f'(\lambda) \\ & & & & f(\lambda) \end{pmatrix} \in M_k(\mathbb{C}),$$

the upper triangular matrix with  $f^{(j)}(\lambda)/j!$  in every position  $(r, r+j)$ , and

$$f(\mathbf{A}) := \mathbf{P} \left( f(\mathbf{J}_{k_1}(\lambda_1)) \oplus \dots \oplus f(\mathbf{J}_{k_m}(\lambda_m)) \right) \mathbf{P}^{-1}.$$

In words: the diagonal of  $f(\mathbf{J})$  records the values  $f(\lambda_i)$ , and each further superdiagonal records the next derivative, divided by a factorial. For a diagonalizable  $\mathbf{A}$  every block has size 1, no derivatives are needed, and the definition says the familiar thing:  $f(\mathbf{A}) = \mathbf{P} \operatorname{diag}(f(\lambda_1), \dots, f(\lambda_n)) \mathbf{P}^{-1}$ .

**Well-definedness.** Two choices were made: the invertible  $\mathbf{P}$  and the order of the blocks. The Jordan form itself is unique only up to the order of the blocks ([Theorem 9.3.1](#)), and even for a fixed  $\mathbf{J}$  there are many  $\mathbf{P}$  with  $\mathbf{A} = \mathbf{P}\mathbf{J}\mathbf{P}^{-1}$ . The next theorem removes both worries at once, by identifying  $f(\mathbf{A})$  with  $p(\mathbf{A})$  for a suitable **polynomial**  $p$ , an expression in which no choice appears.

### Theorem 9.9.2 — Functions of a Matrix Are Polynomials in It

Let  $\mathbf{A} \in M_n(\mathbb{C})$  with distinct eigenvalues  $\mu_1, \dots, \mu_d$ , and write

$$m_{\mathbf{A}} = (x - \mu_1)^{s_1} \dots (x - \mu_d)^{s_d},$$

so that  $s_l$  is the largest size of a  $\mu_l$ -block of the Jordan form of  $\mathbf{A}$  ([Corollary 9.3.2 \(c\)](#)). Let  $f$  be as in [Definition 9.9.1](#). Then:

- (a) there is a polynomial  $p \in \mathbb{C}[x]$  with  $\deg p < \deg m_{\mathbf{A}}$  and

$$p^{(j)}(\mu_l) = f^{(j)}(\mu_l) \quad \text{for all } l = 1, \dots, d \text{ and } j = 0, \dots, s_l - 1,$$

and  $p$  is the **only** such polynomial of degree less than  $\deg m_{\mathbf{A}}$ ;

- (b) for **every** polynomial  $q \in \mathbb{C}[x]$  satisfying those  $\deg m_{\mathbf{A}}$  equations,  $q(\mathbf{A}) = p(\mathbf{A})$ ;
- (c)  $f(\mathbf{A}) = p(\mathbf{A})$ . In particular  $f(\mathbf{A})$  does not depend on the choice of  $\mathbf{P}$  or on the order of the blocks, and  $f(\mathbf{A})$  is determined by the finitely many numbers  $f^{(j)}(\mu_l)$ .

The polynomial  $p$  of (a) is the **Hermite interpolating polynomial** of  $f$  on the spectrum of  $\mathbf{A}$ .

*Idea.* The list of conditions “match  $f$  and its first  $s_l - 1$  derivatives at  $\mu_l$ ” is exactly a congruence modulo  $(x - \mu_l)^{s_l}$ , by the Taylor expansion; and the moduli  $(x - \mu_l)^{s_l}$  are pairwise coprime. So the Chinese Remainder Theorem produces  $p$  and shows it is unique below the degree of the product  $m_{\mathbf{A}}$  — this is [Example \(Values and Slopes by CRT\)](#) with more nodes and higher orders. Once  $p$  exists, [Lemma 9.9.1](#) says that  $p$  and  $f$  do the same thing to every Jordan block of  $\mathbf{A}$ , because a  $\mu_l$ -block has size at most  $s_l$  and only derivatives up to order  $s_l - 1$  are consulted.

*Proof.*

- (a) For each  $l$  let  $r_l := \sum_{j=0}^{s_l-1} \frac{f^{(j)}(\mu_l)}{j!} (x - \mu_l)^j \in \mathbb{C}[x]$ . For  $g \in \mathbb{C}[x]$ , the Taylor expansion of [Theorem 5.4.7](#) (b) at  $\mu_l$ , together with the uniqueness of the remainder on division by  $(x - \mu_l)^{s_l}$  ([Theorem 5.1.1](#)), gives

$$g \equiv r_l \pmod{(x - \mu_l)^{s_l}} \iff g^{(j)}(\mu_l) = f^{(j)}(\mu_l) \text{ for } j = 0, \dots, s_l - 1,$$

exactly as in [Example \(Values and Slopes by CRT\)](#) (a) for  $s_l = 2$ : the remainder of  $g$  on division by  $(x - \mu_l)^{s_l}$  is  $\sum_{j < s_l} \frac{g^{(j)}(\mu_l)}{j!} (x - \mu_l)^j$ , and two polynomials of degree less than  $s_l$  are equal precisely when their coefficients in the basis  $(1, x - \mu_l, \dots, (x - \mu_l)^{s_l-1})$  agree ([Theorem 5.4.7](#) (a)).

The moduli  $(x - \mu_1)^{s_1}, \dots, (x - \mu_d)^{s_d}$  are pairwise coprime: for  $l \neq l'$ , any common divisor divides both, and a monic common divisor would be a power of  $x - \mu_l$  and a power of  $x - \mu_{l'}$  with  $\mu_l \neq \mu_{l'}$ , hence 1 ([Lemma 8.5.2](#), [Theorem 5.3.5](#)). Their product is  $m_{\mathbf{A}}$ . So [Theorem 5.6.4](#) (c) provides exactly one  $p$  with  $\deg p < \deg m_{\mathbf{A}}$  and  $p \equiv r_l \pmod{(x - \mu_l)^{s_l}}$  for every  $l$ , which by the displayed equivalence is (a).

- (b) Let  $q$  satisfy the same equations. By the equivalence again,  $q \equiv r_l \equiv p \pmod{(x - \mu_l)^{s_l}}$  for every  $l$ , so [Theorem 5.6.4](#) (b) gives  $m_{\mathbf{A}} \mid q - p$ . Writing  $q - p = m_{\mathbf{A}} h$  and evaluating at  $\mathbf{A}$ ,  $q(\mathbf{A}) - p(\mathbf{A}) = m_{\mathbf{A}}(\mathbf{A})h(\mathbf{A}) = 0$  since  $m_{\mathbf{A}}(\mathbf{A}) = 0$  ([Definition 8.5.1](#)). Hence  $q(\mathbf{A}) = p(\mathbf{A})$ .
- (c) Let  $\mathbf{J} = \mathbf{J}_{k_1}(\lambda_1) \oplus \dots \oplus \mathbf{J}_{k_m}(\lambda_m)$  be any Jordan form of  $\mathbf{A}$  and  $\mathbf{A} = \mathbf{PJP}^{-1}$ . Fix  $i$  and let  $l$  be the index with  $\lambda_i = \mu_l$ . By [Corollary 9.3.2](#) (c),  $k_i \leq s_l$ , so [Lemma 9.9.1](#) and the equations in (a) give

$$(p(\mathbf{J}_{k_i}(\lambda_i)))_{r,r+j} = \frac{p^{(j)}(\lambda_i)}{j!} = \frac{f^{(j)}(\lambda_i)}{j!} = (f(\mathbf{J}_{k_i}(\lambda_i)))_{r,r+j} \quad (0 \leq j \leq k_i - 1),$$

the derivatives involved having order at most  $k_i - 1 \leq s_l - 1$ . So  $p(\mathbf{J}_{k_i}(\lambda_i)) = f(\mathbf{J}_{k_i}(\lambda_i))$  for every  $i$ . Since evaluating a polynomial blockwise gives the polynomial of the block diagonal matrix ([Theorem 7.1.4](#) (a), (b)),

$$\begin{aligned} p(\mathbf{A}) &= p(\mathbf{PJP}^{-1}) = \mathbf{P} p(\mathbf{J}) \mathbf{P}^{-1} = \mathbf{P} \left( \bigoplus_i p(\mathbf{J}_{k_i}(\lambda_i)) \right) \mathbf{P}^{-1} \\ &= \mathbf{P} \left( \bigoplus_i f(\mathbf{J}_{k_i}(\lambda_i)) \right) \mathbf{P}^{-1} = f(\mathbf{A}), \end{aligned}$$

the second equality by [Proposition 3.7.6](#) (c). The left-hand side mentions neither  $\mathbf{P}$  nor the order of the blocks, so neither does the right-hand side. This proves the theorem. ■

So the definition is consistent with the old one: if  $f$  happens to be a polynomial, then  $f$  itself satisfies the interpolation conditions, and (b) gives  $f(\mathbf{A})$  in the new sense =  $f(\mathbf{A})$  in the old sense.

**Warning.**  $f(\mathbf{A})$  is not  $f$  applied to the entries of  $\mathbf{A}$ . Even for  $f(x) = x^2$  this fails: with  $\mathbf{A} = \begin{pmatrix} 1 & 1 \\ 0 & 1 \end{pmatrix}$ , squaring each entry gives  $\begin{pmatrix} 1 & 1 \\ 0 & 1 \end{pmatrix}$ , while  $\mathbf{A}^2 = \begin{pmatrix} 1 & 2 \\ 0 & 1 \end{pmatrix}$ . The definition is designed so that  $f(\mathbf{A})$  is a polynomial in  $\mathbf{A}$ , and entrywise application is not.

### Example. (Square Roots of Two Matrices)

Using [Definition 9.9.1](#) with  $f(x) = \sqrt{x}$ , the branch with  $\sqrt{4} = 2$  and  $\sqrt{9} = 3$ , compute a square root of

(a)  $\mathbf{A} = \begin{pmatrix} 5 & 4 \\ 4 & 5 \end{pmatrix};$

(b)  $\mathbf{B} = \begin{pmatrix} 4 & 1 \\ 0 & 4 \end{pmatrix}.$

*Solution.*

(a)  $p_{\mathbf{A}} = (x - 5)^2 - 16 = x^2 - 10x + 9 = (x - 1)(x - 9)$ , so the eigenvalues are 1 and 9, distinct, and  $\mathbf{A}$  is diagonalizable. Solving  $(\mathbf{A} - \mathbf{I})\mathbf{x} = \mathbf{0}$  gives  $\text{span}((1, -1))$  and  $(\mathbf{A} - 9\mathbf{I})\mathbf{x} = \mathbf{0}$  gives  $\text{span}((1, 1))$ . With  $\mathbf{P} = \begin{pmatrix} 1 & 1 \\ -1 & 1 \end{pmatrix}$ ,  $\det \mathbf{P} = 2$  and  $\mathbf{P}^{-1} = \frac{1}{2} \begin{pmatrix} 1 & -1 \\ 1 & 1 \end{pmatrix}$ , we have  $\mathbf{P}^{-1}\mathbf{A}\mathbf{P} = \text{diag}(1, 9)$  and

$$f(\mathbf{A}) = \mathbf{P} \text{diag}(1, 3) \mathbf{P}^{-1} = \frac{1}{2} \begin{pmatrix} 1 & 3 \\ -1 & 3 \end{pmatrix} \begin{pmatrix} 1 & -1 \\ 1 & 1 \end{pmatrix} = \frac{1}{2} \begin{pmatrix} 4 & 2 \\ 2 & 4 \end{pmatrix} = \begin{pmatrix} 2 & 1 \\ 1 & 2 \end{pmatrix}.$$

*Check.*  $\begin{pmatrix} 2 & 1 \\ 1 & 2 \end{pmatrix}^2 = \begin{pmatrix} 5 & 4 \\ 4 & 5 \end{pmatrix} = \mathbf{A}.$

(b)  $\mathbf{B} = \mathbf{J}_2(4)$  is its own Jordan form, with  $\mathbf{P} = \mathbf{I}$ . Here  $f(4) = 2$  and  $f'(x) = \frac{1}{2\sqrt{x}}$ , so  $f'(4) = \frac{1}{4}$ , and [Definition 9.9.1](#) gives

$$f(\mathbf{B}) = \begin{pmatrix} 2 & \frac{1}{4} \\ 0 & 2 \end{pmatrix}.$$

*Check.*  $\begin{pmatrix} 2 & 1/4 \\ 0 & 2 \end{pmatrix}^2 = \begin{pmatrix} 4 & 2 \cdot \frac{1}{4} + \frac{1}{4} \cdot 2 \\ 0 & 4 \end{pmatrix} = \begin{pmatrix} 4 & 1 \\ 0 & 4 \end{pmatrix} = \mathbf{B}.$

*The Hermite polynomial.* For (b),  $m_{\mathbf{B}} = (x - 4)^2$ , and the interpolating polynomial of

degree  $< 2$  with  $p(4) = 2$ ,  $p'(4) = \frac{1}{4}$  is  $p = 2 + \frac{1}{4}(x - 4) = \frac{1}{4}x + 1$ . Indeed  $p(\mathbf{B}) = \frac{1}{4}\mathbf{B} + \mathbf{I} = \begin{pmatrix} 2 & 1/4 \\ 0 & 2 \end{pmatrix}$ , as [Theorem 9.9.2](#) (c) predicts.

*Quick check.* A matrix  $\mathbf{A} \in M_5(\mathbb{C})$  has Jordan form  $\mathbf{J}_3(2) \oplus \mathbf{J}_1(2) \oplus \mathbf{J}_1(7)$ . Which values of  $f$  and its derivatives does  $f(\mathbf{A})$  depend on?

*Solution.* On  $f(2), f'(2), f''(2)$  and  $f(7)$ : four numbers. Indeed  $m_{\mathbf{A}} = (x - 2)^3(x - 7)$  by [Corollary 9.3.2](#) (c), so  $s_1 = 3$  for  $\mu_1 = 2$  and  $s_2 = 1$  for  $\mu_2 = 7$ , and [Theorem 9.9.2](#) lists the derivatives of order 0, 1, 2 at 2 and order 0 at 7. Note  $\deg m_{\mathbf{A}} = 4$ , matching the number of conditions and the degree bound on the interpolating polynomial. Nothing about  $f$  away from  $\{2, 7\}$  matters, and no third derivative is consulted even though  $\dim = 5$ .

### 9.9.3 The matrix exponential

The most important function of a matrix is the exponential, and it has a second description, as the sum of a series. We check that the series converges, and that its sum is the matrix  $\exp(\mathbf{A})$  that [Definition 9.9.1](#) produces. Convergence is meant **entrywise**, as in Chapter 8: a sequence  $\mathbf{M}_1, \mathbf{M}_2, \dots$  in  $M_n(\mathbb{C})$  converges to  $\mathbf{M}$  if for each position  $(r, s)$  the number sequence  $(\mathbf{M}_m)_{rs}$  converges to  $\mathbf{M}_{rs}$ , and a series of matrices converges if its sequence of partial sums does. No notion of length of a matrix is used anywhere in this section.

*Remark. What we assume from analysis.* Everything below uses exactly the following facts about real and complex numbers, and nothing else. Chapter 15 will develop the tools that make such statements routine; here they are quoted.

- (A1) **The scalar exponential.** For every  $z \in \mathbb{C}$  the series  $\sum_{i \geq 0} z^i / i!$  converges; its sum is written  $e^z$ . It satisfies  $e^{z+w} = e^z e^w$ ,  $e^0 = 1$ , and  $e^{i\theta} = \cos \theta + i \sin \theta$  for real  $\theta$ . For real  $s \geq 0$ ,  $e^s \geq 1$ .
- (A2) **Termwise differentiation.** If a power series  $\sum_{m \geq 0} c_m t^m$  with  $c_m \in \mathbb{C}$  converges for **every** real  $t$ , then the function  $t \mapsto \sum_{m \geq 0} c_m t^m$  is differentiable on  $\mathbb{R}$  with derivative  $\sum_{m \geq 1} m c_m t^{m-1}$ .
- (A3) **Limits and products.** Limits of convergent sequences of complex numbers respect sums, scalar multiples and products. The derivative of a product of two differentiable functions  $\mathbb{R} \rightarrow \mathbb{C}$  obeys the product rule, and differentiation is linear.
- (A4) **Zero derivative.** A differentiable function  $\mathbb{R} \rightarrow \mathbb{C}$  whose derivative vanishes everywhere is constant.
- (A5) **Decay.** For every integer  $j \geq 0$  and every real  $c < 0$ ,  $t^j e^{ct} \rightarrow 0$  as  $t \rightarrow \infty$ .

Fact (A5) is used only in the last exercise of this section. Nothing else about  $\exp$ ,  $\cos$  or  $\sin$  is assumed: the derivative  $\frac{d}{dt} e^{zt} = z e^{zt}$  for  $z \in \mathbb{C}$  is the  $1 \times 1$  case of [Theorem 9.9.4](#) (b) below, proved from (A2), and the derivatives of  $\cos$  and  $\sin$  used in the examples follow from it through Euler's formula in (A1).

### Definition 9.9.2 — Matrix Exponential

Let  $\mathbf{A} \in M_n(\mathbb{C})$ . The **matrix exponential** of  $\mathbf{A}$  is

$$e^{\mathbf{A}} := \sum_{m=0}^{\infty} \frac{\mathbf{A}^m}{m!} = \lim_{M \rightarrow \infty} \sum_{m=0}^M \frac{\mathbf{A}^m}{m!},$$

the limit taken entrywise, provided it exists;  $\mathbf{A}^0 = \mathbf{I}$ .

That the limit does exist, and what it is, is the content of the next theorem. The Jordan form reduces the question to a single block, where the series can be summed exactly.

### Theorem 9.9.3 — The Exponential Series Converges to the Jordan Definition

Let  $\mathbf{A} \in M_n(\mathbb{C})$ . Then the series  $\sum_m \mathbf{A}^m/m!$  converges entrywise, and its sum is the matrix  $\exp(\mathbf{A})$  given by Definition 9.9.1 for the function  $f = \exp$ . In particular, for a single Jordan block,

$$e^{\mathbf{J}_k(\lambda)} = e^{\lambda} \begin{pmatrix} 1 & 1 & \frac{1}{2!} & \cdots & \frac{1}{(k-1)!} \\ & 1 & 1 & \ddots & \vdots \\ & & \ddots & \ddots & \frac{1}{2!} \\ & & & 1 & 1 \\ & & & & 1 \end{pmatrix},$$

the upper triangular matrix with  $e^{\lambda}/j!$  in every position  $(r, r+j)$ .

*Idea.* For one block the entries of the partial sums are themselves partial sums of a scalar series, and that series can be recognized: the terms  $\binom{m}{j} \lambda^{m-j}/m!$  simplify to  $\lambda^{m-j}/(j!(m-j)!)$ , which is  $1/j!$  times the  $(m-j)$ -th term of the series for  $e^{\lambda}$ . For general  $\mathbf{A}$ , conjugation by a fixed  $\mathbf{P}$  turns each entry of a partial sum into a fixed finite linear combination of entries of the block partial sums, and limits pass through finite linear combinations.

*Proof. Step 1. One block.* Fix  $k \geq 1$ ,  $\lambda \in \mathbb{C}$  and a position  $(r, r+j)$  with  $0 \leq j \leq k-1$ . By Lemma 9.9.1 applied to  $p = x^m$ ,

$$\left( \frac{\mathbf{J}_k(\lambda)^m}{m!} \right)_{r, r+j} = \frac{1}{m!} \binom{m}{j} \lambda^{m-j} = \begin{cases} \frac{\lambda^{m-j}}{j!(m-j)!}, & m \geq j, \\ 0, & m < j, \end{cases}$$

using  $\binom{m}{j}/m! = 1/(j!(m-j)!)$  for  $m \geq j$  and  $\binom{m}{j} = 0$  for  $m < j$ . Hence the partial sums satisfy

$$\sum_{m=0}^M \left( \frac{\mathbf{J}_k(\lambda)^m}{m!} \right)_{r, r+j} = \frac{1}{j!} \sum_{i=0}^{M-j} \frac{\lambda^i}{i!} \quad (M \geq j),$$

after the substitution  $i = m - j$ . By (A1) the right-hand side converges to  $e^{\lambda}/j!$  as  $M \rightarrow \infty$ . Entries with  $j < 0$ , that is entries below the diagonal, are 0 in every term. So

the series converges entrywise and

$$e^{\mathbf{J}_k(\lambda)} = \left( \text{upper triangular with } \frac{e^\lambda}{j!} \text{ on the } j\text{-th superdiagonal} \right) = \exp(\mathbf{J}_k(\lambda)),$$

the last equality being [Definition 9.9.1](#) for  $f = \exp$ , whose derivatives are  $f^{(j)} = \exp$ , so  $f^{(j)}(\lambda)/j! = e^\lambda/j!$ .

**Step 2. A Jordan matrix.** If  $\mathbf{J} = \mathbf{J}_{k_1}(\lambda_1) \oplus \cdots \oplus \mathbf{J}_{k_m}(\lambda_m)$ , then  $\mathbf{J}^m/m!$  is the direct sum of the corresponding blocks ([Theorem 7.1.4](#) (b)), so each entry of a partial sum of  $\sum_m \mathbf{J}^m/m!$  is either identically 0 or an entry of a partial sum from one block. By Step 1 all of them converge, and  $e^{\mathbf{J}} = \bigoplus_i e^{\mathbf{J}_{k_i}(\lambda_i)}$ .

**Step 3. General A.** By [Theorem 9.3.1](#) there are a Jordan matrix  $\mathbf{J}$  and an invertible  $\mathbf{P}$  with  $\mathbf{A} = \mathbf{P}\mathbf{J}\mathbf{P}^{-1}$ . Then  $\mathbf{A}^m = \mathbf{P}\mathbf{J}^m\mathbf{P}^{-1}$  ([Proposition 3.7.6](#) (c)), so

$$\sum_{m=0}^{\mathbf{M}} \frac{\mathbf{A}^m}{m!} = \mathbf{P} \left( \sum_{m=0}^{\mathbf{M}} \frac{\mathbf{J}^m}{m!} \right) \mathbf{P}^{-1}.$$

Each entry of the left-hand side is a fixed finite linear combination — with coefficients built from the entries of  $\mathbf{P}$  and  $\mathbf{P}^{-1}$ , independent of  $\mathbf{M}$  — of the entries of the middle partial sum. By Step 2 those entries converge, so by (A3) the left-hand side converges entrywise, with

$$e^{\mathbf{A}} = \mathbf{P} e^{\mathbf{J}} \mathbf{P}^{-1} = \mathbf{P} \left( \bigoplus_i e^{\mathbf{J}_{k_i}(\lambda_i)} \right) \mathbf{P}^{-1} = \exp(\mathbf{A}),$$

the last equality by [Definition 9.9.1](#). This proves the theorem. ■

From now on we write  $e^{\mathbf{A}}$  for this matrix and use both descriptions freely. Note one consequence of [Theorem 9.9.2](#):  $e^{\mathbf{A}}$  is a **polynomial** in  $\mathbf{A}$ , namely  $p(\mathbf{A})$  for the Hermite interpolant of  $\exp$  on  $\text{spec}(\mathbf{A})$ . Everything that commutes with  $\mathbf{A}$  therefore commutes with  $e^{\mathbf{A}}$ .

### Theorem 9.9.4 — Properties of the Matrix Exponential

Let  $\mathbf{A}, \mathbf{B} \in M_n(\mathbb{C})$  and let  $\mathbf{P} \in M_n(\mathbb{C})$  be invertible.

- (a)  $e^0 = \mathbf{I}$  and  $e^{\mathbf{P}^{-1}\mathbf{A}\mathbf{P}} = \mathbf{P}^{-1}e^{\mathbf{A}}\mathbf{P}$ .
- (b) The function  $\mathbb{R} \rightarrow M_n(\mathbb{C})$ ,  $t \mapsto e^{t\mathbf{A}}$ , is differentiable entrywise, with

$$\frac{d}{dt} e^{t\mathbf{A}} = \mathbf{A}e^{t\mathbf{A}} = e^{t\mathbf{A}}\mathbf{A}.$$

- (c)  $e^{\mathbf{A}}$  is invertible, with  $(e^{\mathbf{A}})^{-1} = e^{-\mathbf{A}}$ .
- (d) If  $\mathbf{A}\mathbf{B} = \mathbf{B}\mathbf{A}$ , then  $e^{\mathbf{A}+\mathbf{B}} = e^{\mathbf{A}}e^{\mathbf{B}}$ .
- (e)  $\det e^{\mathbf{A}} = e^{\text{tr } \mathbf{A}}$ .

*Idea.* Part (b) is termwise differentiation, entry by entry. Everything after it is the same two-line trick: to prove  $\mathbf{X}(t) = \mathbf{Y}(t)$  for two matrix functions, show that  $\mathbf{Z}(t) = e^{-t\mathbf{C}}\mathbf{X}(t)$  has derivative 0 for a suitable  $\mathbf{C}$ , so  $\mathbf{Z}$  is constant and equals  $\mathbf{Z}(0)$ . That is why (c) and

(d) come after (b), even though the series would also give them. Part (e) is read off the Jordan form:  $e^{\mathbf{J}}$  is triangular with  $e^{\lambda_i}$  on the diagonal.

*Proof.*

- (a)  $0^m = 0$  for  $m \geq 1$  and  $0^0 = \mathbf{I}$ , so the series is  $\mathbf{I}$ . For the second claim, Step 3 of the previous proof applies verbatim with  $\mathbf{J}$  replaced by  $\mathbf{P}^{-1}\mathbf{A}\mathbf{P}$ : the partial sums of  $\sum_m (\mathbf{P}^{-1}\mathbf{A}\mathbf{P})^m/m!$  are  $\mathbf{P}^{-1}(\sum_m \mathbf{A}^m/m!)\mathbf{P}$ , and (A3) lets the limit pass through.
- (b) Fix a position  $(r, s)$ . By Theorem 9.9.3 the series  $\sum_m t^m (\mathbf{A}^m)_{rs}/m!$  converges for every real  $t$ , since  $(t\mathbf{A})^m = t^m \mathbf{A}^m$ . So (A2) applies with  $c_m = (\mathbf{A}^m)_{rs}/m!$  and gives

$$\begin{aligned} \frac{d}{dt} (e^{t\mathbf{A}})_{rs} &= \sum_{m \geq 1} \frac{m t^{m-1}}{m!} (\mathbf{A}^m)_{rs} = \sum_{m \geq 1} \frac{t^{m-1}}{(m-1)!} (\mathbf{A} \cdot \mathbf{A}^{m-1})_{rs} \\ &= \left( \mathbf{A} \sum_{l \geq 0} \frac{t^l \mathbf{A}^l}{l!} \right)_{rs}, \end{aligned}$$

where the last step moves the constant matrix  $\mathbf{A}$  outside the limit: each entry of  $\mathbf{A}(\sum_{l \leq L} t^l \mathbf{A}^l/l!)$  is a fixed linear combination of entries of the partial sum, so (A3) applies. Hence  $\frac{d}{dt} e^{t\mathbf{A}} = \mathbf{A}e^{t\mathbf{A}}$ . The same computation with  $\mathbf{A}^m = \mathbf{A}^{m-1}\mathbf{A}$  gives  $e^{t\mathbf{A}}\mathbf{A}$ .

- (c) Put  $\mathbf{H}(t) = e^{t\mathbf{A}}e^{-t\mathbf{A}}$  for  $t \in \mathbb{R}$ . Each entry of  $\mathbf{H}$  is a finite sum of products of differentiable functions, so by (A3) and (b),

$$\mathbf{H}'(t) = (\mathbf{A}e^{t\mathbf{A}})e^{-t\mathbf{A}} + e^{t\mathbf{A}}(-\mathbf{A}e^{-t\mathbf{A}}) = \mathbf{A}e^{t\mathbf{A}}e^{-t\mathbf{A}} - \mathbf{A}e^{t\mathbf{A}}e^{-t\mathbf{A}} = 0,$$

where the second equality uses  $e^{t\mathbf{A}}\mathbf{A} = \mathbf{A}e^{t\mathbf{A}}$  from (b). By (A4) every entry of  $\mathbf{H}$  is constant, so  $\mathbf{H}(t) = \mathbf{H}(0) = e^0 e^0 = \mathbf{I}$  for all  $t$ . At  $t = 1$ ,  $e^{\mathbf{A}}e^{-\mathbf{A}} = \mathbf{I}$ , and Theorem 2.4.5 finishes it.

- (d) Suppose  $\mathbf{A}\mathbf{B} = \mathbf{B}\mathbf{A}$ . Then  $\mathbf{B}$  commutes with every partial sum of  $\sum_m t^m \mathbf{A}^m/m!$ , so by (A3) it commutes with the limit:  $\mathbf{B}e^{t\mathbf{A}} = e^{t\mathbf{A}}\mathbf{B}$  for every  $t$ . Put

$$\mathbf{G}(t) = e^{-t(\mathbf{A}+\mathbf{B})}e^{t\mathbf{A}}e^{t\mathbf{B}}.$$

Differentiating entrywise with the product rule (A3) and (b),

$$\mathbf{G}'(t) = -(\mathbf{A} + \mathbf{B})e^{-t(\mathbf{A}+\mathbf{B})}e^{t\mathbf{A}}e^{t\mathbf{B}} + e^{-t(\mathbf{A}+\mathbf{B})}\mathbf{A}e^{t\mathbf{A}}e^{t\mathbf{B}} + e^{-t(\mathbf{A}+\mathbf{B})}e^{t\mathbf{A}}\mathbf{B}e^{t\mathbf{B}}.$$

In the third term replace  $e^{t\mathbf{A}}\mathbf{B}$  by  $\mathbf{B}e^{t\mathbf{A}}$ ; in the first, use  $(\mathbf{A} + \mathbf{B})e^{-t(\mathbf{A}+\mathbf{B})} = e^{-t(\mathbf{A}+\mathbf{B})}(\mathbf{A} + \mathbf{B})$  from (b). The three terms become

$$e^{-t(\mathbf{A}+\mathbf{B})}(-(\mathbf{A} + \mathbf{B}) + \mathbf{A} + \mathbf{B})e^{t\mathbf{A}}e^{t\mathbf{B}} = 0.$$

By (A4),  $\mathbf{G}$  is constant, so  $\mathbf{G}(t) = \mathbf{G}(0) = \mathbf{I}$ . Multiplying on the left by  $e^{t(\mathbf{A}+\mathbf{B})}$  and using (c),  $e^{t\mathbf{A}}e^{t\mathbf{B}} = e^{t(\mathbf{A}+\mathbf{B})}$ ; put  $t = 1$ .

- (e) Write  $\mathbf{A} = \mathbf{P}\mathbf{J}\mathbf{P}^{-1}$  with  $\mathbf{J} = \bigoplus_i \mathbf{J}_{k_i}(\lambda_i)$  a Jordan matrix. By (a) and Theorem 9.9.3,

$$\det e^{\mathbf{A}} = \det(\mathbf{P}e^{\mathbf{J}}\mathbf{P}^{-1}) = \det e^{\mathbf{J}}$$

(Corollary 6.5.7), and  $e^{\mathbf{J}} = \bigoplus_i e^{\mathbf{J}_{k_i}(\lambda_i)}$  is upper triangular with  $e^{\lambda_i}$  repeated  $k_i$  times along the diagonal. By Theorem 6.4.6 and (A1),

$$\det e^{\mathbf{J}} = \prod_i (e^{\lambda_i})^{k_i} = e^{\sum_i k_i \lambda_i} = e^{\text{tr } \mathbf{J}} = e^{\text{tr } \mathbf{A}},$$

the last equality because similar matrices have equal traces (Theorem 3.7.5). This proves the theorem. ■

Part (e) says something striking:  $\det e^{\mathbf{A}} \neq 0$  always, which re-proves (c), and if  $\text{tr } \mathbf{A} = 0$  then  $e^{\mathbf{A}} \in \text{SL}_n(\mathbb{C})$ .

The following corollary is the formula actually used in computations. Note that  $t\mathbf{J}_k(\lambda)$  is **not** a Jordan block for  $k \geq 2$  and  $t \neq 1$ , since its superdiagonal entries are  $t$ ; the commuting law does the work instead.

### Corollary 9.9.5 — Exponential of a Jordan Block, with a Parameter

Let  $k \geq 1$ ,  $\lambda \in \mathbb{C}$  and  $t \in \mathbb{R}$ . Then  $e^{t\mathbf{J}_k(\lambda)}$  is the upper triangular matrix with

$$(e^{t\mathbf{J}_k(\lambda)})_{r, r+j} = \frac{t^j e^{\lambda t}}{j!} \quad (0 \leq j \leq k-1)$$

and zeros below the diagonal. Consequently, if  $\mathbf{A} = \mathbf{P}\mathbf{J}\mathbf{P}^{-1}$  with  $\mathbf{J} = \mathbf{J}_{k_1}(\lambda_1) \oplus \cdots \oplus \mathbf{J}_{k_m}(\lambda_m)$ , then

$$e^{t\mathbf{A}} = \mathbf{P} \left( e^{t\mathbf{J}_{k_1}(\lambda_1)} \oplus \cdots \oplus e^{t\mathbf{J}_{k_m}(\lambda_m)} \right) \mathbf{P}^{-1}.$$

In particular, for a diagonal  $\mathbf{D} = \text{diag}(d_1, \dots, d_n)$ ,  $e^{t\mathbf{D}} = \text{diag}(e^{td_1}, \dots, e^{td_n})$ .

*Proof.* Write  $t\mathbf{J}_k(\lambda) = (t\lambda)\mathbf{I}_k + t\mathbf{M}$  with  $\mathbf{M} = \mathbf{J}_k(0)$ . The two summands commute, since  $\mathbf{I}_k$  commutes with everything, so Theorem 9.9.4 (d) gives  $e^{t\mathbf{J}_k(\lambda)} = e^{(t\lambda)\mathbf{I}_k} e^{t\mathbf{M}}$ . For the first factor,  $((t\lambda)\mathbf{I}_k)^m = (t\lambda)^m \mathbf{I}_k$ , so the partial sums are  $(\sum_{m \leq \mathbf{M}} (t\lambda)^m / m!) \mathbf{I}_k$  and  $e^{(t\lambda)\mathbf{I}_k} = e^{\lambda t} \mathbf{I}_k$  by (A1). For the second,  $\mathbf{M}^j = 0$  for  $j \geq k$ , so the series terminates:

$$e^{t\mathbf{M}} = \sum_{j=0}^{k-1} \frac{t^j}{j!} \mathbf{M}^j,$$

and  $\mathbf{M}^j$  has 1 in each position  $(r, r+j)$ , as computed in the proof of Lemma 9.9.1. Multiplying by the scalar  $e^{\lambda t}$  gives the entry formula. The second display follows from  $t\mathbf{A} = \mathbf{P}(t\mathbf{J})\mathbf{P}^{-1}$  with Theorem 9.9.4 (a) and Theorem 7.1.4 (b), exactly as in Step 2 of the proof of Theorem 9.9.3. The diagonal case is the case in which every block has size 1. ■



*Warning.*  $e^{\mathbf{A}+\mathbf{B}} = e^{\mathbf{A}}e^{\mathbf{B}}$  fails when  $\mathbf{A}$  and  $\mathbf{B}$  do not commute. Take

$$\mathbf{A} = \begin{pmatrix} 0 & 1 \\ 0 & 0 \end{pmatrix}, \quad \mathbf{B} = \begin{pmatrix} 0 & 0 \\ 1 & 0 \end{pmatrix}.$$

Both square to 0, so the series stop after two terms:  $e^{\mathbf{A}} = \mathbf{I} + \mathbf{A} = \begin{pmatrix} 1 & 1 \\ 0 & 1 \end{pmatrix}$  and  $e^{\mathbf{B}} = \mathbf{I} + \mathbf{B} = \begin{pmatrix} 1 & 0 \\ 1 & 1 \end{pmatrix}$ . Then

$$e^{\mathbf{A}}e^{\mathbf{B}} = \begin{pmatrix} 2 & 1 \\ 1 & 1 \end{pmatrix} \neq \begin{pmatrix} 1 & 1 \\ 1 & 2 \end{pmatrix} = e^{\mathbf{B}}e^{\mathbf{A}}.$$

Since  $\mathbf{A} + \mathbf{B} = \mathbf{B} + \mathbf{A}$ , at most one of the two products can equal  $e^{\mathbf{A}+\mathbf{B}}$ , so the identity must fail for one of the pairs  $(\mathbf{A}, \mathbf{B})$ ,  $(\mathbf{B}, \mathbf{A})$ . (In fact it fails for both:  $(\mathbf{A} + \mathbf{B})^2 = \mathbf{I}$ , so  $e^{\mathbf{A}+\mathbf{B}}$  has all four entries non-zero and none of them equal to 2.) The moral: check  $\mathbf{AB} = \mathbf{BA}$  before splitting an exponential.

## 9.9.4 Linear systems of differential equations

A system of  $n$  linear differential equations with constant coefficients,

$$x_1' = a_{11}x_1 + \cdots + a_{1n}x_n, \quad \dots, \quad x_n' = a_{n1}x_1 + \cdots + a_{nn}x_n,$$

is the single equation  $\mathbf{x}' = \mathbf{A}\mathbf{x}$  for a vector-valued function  $\mathbf{x}: \mathbb{R} \rightarrow \mathbb{C}^n$ , differentiated entrywise. In one dimension the answer is  $x(t) = e^{ta}x_0$ ; the matrix exponential makes the general answer look exactly the same.

### Theorem 9.9.6 — Solution of a Linear System

Let  $\mathbf{A} \in M_n(\mathbb{C})$  and  $\mathbf{x}_0 \in \mathbb{C}^n$ . Then the function

$$\mathbf{x}(t) = e^{t\mathbf{A}}\mathbf{x}_0 \quad (t \in \mathbb{R})$$

is differentiable and satisfies  $\mathbf{x}'(t) = \mathbf{A}\mathbf{x}(t)$  for all  $t$  and  $\mathbf{x}(0) = \mathbf{x}_0$ ; and it is the **only** differentiable function  $\mathbb{R} \rightarrow \mathbb{C}^n$  with these two properties.

*Idea.* Existence is Theorem 9.9.4 (b) applied to a fixed vector. For uniqueness, take any solution  $\mathbf{y}$  and multiply it by  $e^{-t\mathbf{A}}$ , the one factor that can undo the growth: the product has derivative 0, hence is constant. This is the same move as in the proofs of parts (c) and (d) above.

*Proof. Existence.* Each entry of  $\mathbf{x}(t) = e^{t\mathbf{A}}\mathbf{x}_0$  is a linear combination of entries of  $e^{t\mathbf{A}}$  with constant coefficients, so  $\mathbf{x}$  is differentiable with  $\mathbf{x}'(t) = \left(\frac{d}{dt}e^{t\mathbf{A}}\right)\mathbf{x}_0$  by the linearity of differentiation (A3). By Theorem 9.9.4 (b),  $\mathbf{x}'(t) = \mathbf{A}e^{t\mathbf{A}}\mathbf{x}_0 = \mathbf{A}\mathbf{x}(t)$ . And  $\mathbf{x}(0) = e^0\mathbf{x}_0 = \mathbf{x}_0$  by Theorem 9.9.4 (a).

*Uniqueness.* Let  $\mathbf{y}: \mathbb{R} \rightarrow \mathbb{C}^n$  be differentiable with  $\mathbf{y}' = \mathbf{A}\mathbf{y}$  and  $\mathbf{y}(0) = \mathbf{x}_0$ . Put

$\mathbf{z}(t) = e^{-t\mathbf{A}}\mathbf{y}(t)$ . By the product rule (A3) and Theorem 9.9.4 (b) applied to  $-\mathbf{A}$ ,

$$\mathbf{z}'(t) = (-\mathbf{A}e^{-t\mathbf{A}})\mathbf{y}(t) + e^{-t\mathbf{A}}\mathbf{y}'(t) = -\mathbf{A}e^{-t\mathbf{A}}\mathbf{y}(t) + e^{-t\mathbf{A}}\mathbf{A}\mathbf{y}(t) = \mathbf{0},$$

where the last step uses  $e^{-t\mathbf{A}}\mathbf{A} = \mathbf{A}e^{-t\mathbf{A}}$  from Theorem 9.9.4 (b). By (A4) each entry of  $\mathbf{z}$  is constant, so  $\mathbf{z}(t) = \mathbf{z}(0) = \mathbf{x}_0$  for all  $t$ . Multiplying by  $e^{t\mathbf{A}}$  and using Theorem 9.9.4 (c),

$$\mathbf{y}(t) = e^{t\mathbf{A}}\mathbf{z}(t) = e^{t\mathbf{A}}\mathbf{x}_0 = \mathbf{x}(t).$$

This proves the theorem. ■

The formula is only as useful as our ability to compute  $e^{t\mathbf{A}}$ , and the Jordan form does that. Two shapes of answer occur, and they are worth seeing side by side.

### Example. (Two Two-Dimensional Systems)

Solve  $\mathbf{x}' = \mathbf{A}\mathbf{x}$  with  $\mathbf{x}(0) = (1, 0)$  for

- (a)  $\mathbf{A} = \begin{pmatrix} 1 & -2 \\ 2 & 1 \end{pmatrix}$  (complex eigenvalues);
- (b)  $\mathbf{A} = \begin{pmatrix} 2 & 1 \\ 0 & 2 \end{pmatrix}$  (one eigenvalue, one block of size 2).

*Solution.*

- (a)  $p_{\mathbf{A}} = (x - 1)^2 + 4$ , with roots  $\lambda = 1 \pm 2i$ , distinct, so  $\mathbf{A}$  is diagonalizable over  $\mathbb{C}$  and every block has size 1. Solving  $(\mathbf{A} - (1 + 2i)\mathbf{I})\mathbf{x} = \mathbf{0}$ : the first row reads  $-2ix_1 - 2x_2 = 0$ , so  $x_2 = -ix_1$  and  $(1, -i)$  is an eigenvector;  $(1, i)$  is one for  $1 - 2i$ . With  $\mathbf{P} = \begin{pmatrix} 1 & 1 \\ -i & i \end{pmatrix}$  we get  $\det \mathbf{P} = 2i$  and  $\mathbf{P}^{-1} = \frac{1}{2i} \begin{pmatrix} i & -1 \\ i & 1 \end{pmatrix} = \frac{1}{2} \begin{pmatrix} 1 & i \\ 1 & -i \end{pmatrix}$ , so  $\mathbf{P}^{-1}\mathbf{A}\mathbf{P} = \text{diag}(1 + 2i, 1 - 2i)$ . By Corollary 9.9.5,

$$\begin{aligned} e^{t\mathbf{A}} &= \mathbf{P} \text{diag}(e^{(1+2i)t}, e^{(1-2i)t}) \mathbf{P}^{-1} \\ &= \frac{1}{2} \begin{pmatrix} e^{(1+2i)t} + e^{(1-2i)t} & i(e^{(1+2i)t} - e^{(1-2i)t}) \\ -i(e^{(1+2i)t} - e^{(1-2i)t}) & e^{(1+2i)t} + e^{(1-2i)t} \end{pmatrix}. \end{aligned}$$

By (A1),  $e^{(1\pm 2i)t} = e^t(\cos 2t \pm i \sin 2t)$ , so  $e^{(1+2i)t} + e^{(1-2i)t} = 2e^t \cos 2t$  and  $e^{(1+2i)t} - e^{(1-2i)t} = 2ie^t \sin 2t$ . Substituting, and using  $i \cdot 2i = -2$ ,

$$e^{t\mathbf{A}} = e^t \begin{pmatrix} \cos 2t & -\sin 2t \\ \sin 2t & \cos 2t \end{pmatrix}, \quad \mathbf{x}(t) = e^{t\mathbf{A}} \begin{pmatrix} 1 \\ 0 \end{pmatrix} = e^t \begin{pmatrix} \cos 2t \\ \sin 2t \end{pmatrix}.$$

The solution spirals outwards: rotation at angular speed 2, growth at rate 1. The entries are real, as they must be for a real  $\mathbf{A}$  and a real  $\mathbf{x}_0$ .

*Check.*  $\mathbf{x}'(t) = e^t(\cos 2t - 2\sin 2t, \sin 2t + 2\cos 2t)$ , and  $\mathbf{A}\mathbf{x}(t) = e^t(\cos 2t - 2\sin 2t, 2\cos 2t + \sin 2t)$ . They agree, and  $\mathbf{x}(0) = (1, 0)$ .

- (b)  $\mathbf{A} = \mathbf{J}_2(2)$  is its own Jordan form. Writing  $\mathbf{A} = 2\mathbf{I} + \mathbf{M}$  with  $\mathbf{M} = \begin{pmatrix} 0 & 1 \\ 0 & 0 \end{pmatrix}$ , the

two summands commute and  $\mathbf{M}^2 = 0$ , so [Corollary 9.9.5](#) with  $k = 2$  gives

$$e^{t\mathbf{A}} = e^{2t}(\mathbf{I} + t\mathbf{M}) = e^{2t} \begin{pmatrix} 1 & t \\ 0 & 1 \end{pmatrix}.$$

So  $\mathbf{x}(t) = e^{2t}(1, 0)$ . A different initial vector shows the new phenomenon:  $\mathbf{x}(0) = (0, 1)$  gives  $\mathbf{x}(t) = e^{2t}(t, 1)$ , whose first entry is  $te^{2t}$ . **A repeated eigenvalue with a block of size  $k$  produces solutions containing  $t^j e^{\lambda t}$  for  $j < k$ .** This is the same  $t^j \lambda^m$  pattern that Chapter 8 met for recurrences with repeated roots, now in continuous time.

A scalar equation of higher order is a system in disguise, and the translation is the companion matrix of Section 5.

**Example. (A Second-Order Equation with a Repeated Root)**

Solve  $y'' - 4y' + 4y = 0$  with  $y(0) = 1$  and  $y'(0) = 3$ .

*Solution.* Make it a system. Put  $\mathbf{u} = (y, y')$ . Then  $\mathbf{u}' = (y', y'') = (y', 4y' - 4y)$ , so  $\mathbf{u}' = \mathbf{M}\mathbf{u}$  with

$$\mathbf{M} = \begin{pmatrix} 0 & 1 \\ -4 & 4 \end{pmatrix}, \quad \mathbf{u}(0) = (1, 3).$$

Note that  $p_{\mathbf{M}} = x(x - 4) + 4 = x^2 - 4x + 4$  is the polynomial of the equation, and that  $\mathbf{M} = \mathbf{C}(p_{\mathbf{M}})^{\top}$  for the companion matrix  $\mathbf{C}(p_{\mathbf{M}}) = \begin{pmatrix} 0 & -4 \\ 1 & 4 \end{pmatrix}$  of [Definition 9.5.3](#); by [Corollary 9.7.7](#) the two are similar, so which of them one uses is immaterial.

*Exponentiate.*  $p_{\mathbf{M}} = (x-2)^2$ . Put  $\mathbf{N} = \mathbf{M} - 2\mathbf{I} = \begin{pmatrix} -2 & 1 \\ -4 & 2 \end{pmatrix}$ ; then  $\mathbf{N}^2 = \begin{pmatrix} 4-4 & -2+2 \\ 8-8 & -4+4 \end{pmatrix} = 0$ , so  $2\mathbf{I}$  and  $\mathbf{N}$  commute and [Theorem 9.9.4](#) (d) gives, as in the previous example,

$$e^{t\mathbf{M}} = e^{2t}(\mathbf{I} + t\mathbf{N}) = e^{2t} \begin{pmatrix} 1-2t & t \\ -4t & 1+2t \end{pmatrix}.$$

(No basis change was needed:  $\mathbf{M}$  is a scalar plus a nilpotent matrix on the nose.)

*Read off the answer.*

$$\mathbf{u}(t) = e^{t\mathbf{M}} \begin{pmatrix} 1 \\ 3 \end{pmatrix} = e^{2t} \begin{pmatrix} (1-2t) + 3t \\ -4t + 3(1+2t) \end{pmatrix} = e^{2t} \begin{pmatrix} 1+t \\ 3+2t \end{pmatrix},$$

so  $y(t) = (1+t)e^{2t}$ .

*Check.*  $y' = e^{2t} + 2(1+t)e^{2t} = (3+2t)e^{2t}$ , which is the second entry of  $\mathbf{u}$ , as it should be, and  $y'(0) = 3$ . Then  $y'' = 2e^{2t} + 2(3+2t)e^{2t} = (8+4t)e^{2t}$ , and

$$y'' - 4y' + 4y = [(8+4t) - 4(3+2t) + 4(1+t)]e^{2t} = 0.$$

The repeated root 2 produces the term  $te^{2t}$ , exactly as the size-2 block predicts.

Chapter 15 will return to  $e^{\mathbf{A}}$  with norms available, which give quantitative bounds instead of exact formulas. The next section stays with exact formulas and asks the discrete-time question: what happens to  $\mathbf{A}^m$ ?

## 9.9.5 Exercises

### 9.9.5.1 A. Check your understanding

#### Exercise A1.

- State the formula for  $p(\mathbf{J}_k(\lambda))$  and say where the factorials come from.
- Define  $f(\mathbf{A})$  for  $\mathbf{A} \in M_n(\mathbb{C})$  with a given Jordan form, and list the two choices whose irrelevance has to be checked.
- Which values of  $f$  and its derivatives does  $f(\mathbf{A})$  depend on? How many numbers is that?
- State the five properties of  $e^{\mathbf{A}}$  proved in [Theorem 9.9.4](#), with their hypotheses.
- True or false:  $e^{\mathbf{A}+\mathbf{B}} = e^{\mathbf{A}}e^{\mathbf{B}}$  for all  $\mathbf{A}, \mathbf{B} \in M_n(\mathbb{C})$ . Justify your answer.
- True or false:  $e^{\mathbf{A}}$  can be singular. Justify your answer.

*Solution.*

- $p(\mathbf{J}_k(\lambda))$  is upper triangular with  $p^{(j)}(\lambda)/j!$  in every position  $(r, r+j)$  ([Lemma 9.9.1](#)). The factorials come from the Taylor expansion  $p = \sum_j \frac{p^{(j)}(\lambda)}{j!}(x - \lambda)^j$  ([Theorem 5.4.7](#)), whose  $j$ -th term contributes  $\mathbf{J}_k(0)^j$ , the matrix with ones on the  $j$ -th superdiagonal.
- $f(\mathbf{A}) = \mathbf{P}(\bigoplus_i f(\mathbf{J}_{k_i}(\lambda_i)))\mathbf{P}^{-1}$  with  $f(\mathbf{J}_k(\lambda))$  as in (a) ([Definition 9.9.1](#)). The choices are the matrix  $\mathbf{P}$  and the order in which the blocks are listed; both are irrelevant because  $f(\mathbf{A}) = p(\mathbf{A})$  for the Hermite interpolating polynomial  $p$  ([Theorem 9.9.2](#)).
- On  $f^{(j)}(\mu)$  for each eigenvalue  $\mu$  and each  $j < s_\mu$ , where  $s_\mu$  is the exponent of  $x - \mu$  in  $m_{\mathbf{A}}$ , that is, the largest  $\mu$ -block size. The total number is  $\sum_\mu s_\mu = \deg m_{\mathbf{A}}$  ([Theorem 9.9.2](#)).
- $e^{\mathbf{P}^{-1}\mathbf{A}\mathbf{P}} = \mathbf{P}^{-1}e^{\mathbf{A}}\mathbf{P}$  for invertible  $\mathbf{P}$ ;  $\frac{d}{dt}e^{t\mathbf{A}} = \mathbf{A}e^{t\mathbf{A}} = e^{t\mathbf{A}}\mathbf{A}$ ;  $(e^{\mathbf{A}})^{-1} = e^{-\mathbf{A}}$ ;  $e^{\mathbf{A}+\mathbf{B}} = e^{\mathbf{A}}e^{\mathbf{B}}$  **provided**  $\mathbf{AB} = \mathbf{BA}$ ; and  $\det e^{\mathbf{A}} = e^{\text{tr } \mathbf{A}}$  with no hypothesis.
- False. With  $\mathbf{A} = \begin{pmatrix} 0 & 1 \\ 0 & 0 \end{pmatrix}$  and  $\mathbf{B} = \mathbf{A}^\top$ , the matrices  $e^{\mathbf{A}}e^{\mathbf{B}}$  and  $e^{\mathbf{B}}e^{\mathbf{A}}$  differ, so the identity cannot hold for both orders; see the warning after [Theorem 9.9.4](#).
- False.  $\det e^{\mathbf{A}} = e^{\text{tr } \mathbf{A}} \neq 0$  by [Theorem 9.9.4](#) (e) and (A1), so  $e^{\mathbf{A}}$  is always invertible ([Theorem 6.5.2](#)).

### 9.9.5.2 B. Practice

#### Exercise B1: Computing exponentials.

Compute  $e^{t\mathbf{A}}$  for each of the following.

- $\mathbf{A} = \begin{pmatrix} 1 & 3 \\ 3 & 1 \end{pmatrix}$ .
- $\mathbf{A} = \begin{pmatrix} 5 & 1 \\ 0 & 5 \end{pmatrix}$ .

$$(c) \mathbf{A} = \begin{pmatrix} 3 & 1 & 0 \\ 0 & 3 & 1 \\ 0 & 0 & 3 \end{pmatrix} = \mathbf{J}_3(3).$$

*Solution.*

- (a)  $p_{\mathbf{A}} = (x - 1)^2 - 9$ , with roots 4 and  $-2$ . Eigenvectors:  $(\mathbf{A} - 4\mathbf{I})\mathbf{x} = \mathbf{0}$  gives  $\text{span}((1, 1))$ ;  $(\mathbf{A} + 2\mathbf{I})\mathbf{x} = \mathbf{0}$  gives  $\text{span}((1, -1))$ . With  $\mathbf{P} = \begin{pmatrix} 1 & 1 \\ 1 & -1 \end{pmatrix}$  we have  $\det \mathbf{P} = -2$ ,  $\mathbf{P}^{-1} = \frac{1}{2} \begin{pmatrix} 1 & 1 \\ 1 & -1 \end{pmatrix}$  and  $\mathbf{P}^{-1}\mathbf{A}\mathbf{P} = \text{diag}(4, -2)$ , so by [Corollary 9.9.5](#)

$$e^{t\mathbf{A}} = \mathbf{P} \text{diag}(e^{4t}, e^{-2t}) \mathbf{P}^{-1} = \frac{1}{2} \begin{pmatrix} e^{4t} + e^{-2t} & e^{4t} - e^{-2t} \\ e^{4t} - e^{-2t} & e^{4t} + e^{-2t} \end{pmatrix}.$$

Check at  $t = 0$ : both diagonal entries give 1 and both off-diagonal ones 0, so  $e^0 = \mathbf{I}$ .

- (b)  $\mathbf{A} = \mathbf{J}_2(5)$ , so [Corollary 9.9.5](#) with  $k = 2$  gives

$$e^{t\mathbf{A}} = e^{5t}(\mathbf{I} + t\mathbf{M}) = e^{5t} \begin{pmatrix} 1 & t \\ 0 & 1 \end{pmatrix}.$$

- (c) By [Corollary 9.9.5](#) with  $k = 3$  and  $\lambda = 3$ , or directly from  $\mathbf{A} = 3\mathbf{I} + \mathbf{M}$  with  $\mathbf{M} = \mathbf{J}_3(0)$  and  $\mathbf{M}^3 = 0$ , giving  $e^{t\mathbf{A}} = e^{3t}(\mathbf{I} + t\mathbf{M} + \frac{t^2}{2}\mathbf{M}^2)$ :

$$e^{t\mathbf{A}} = e^{3t} \begin{pmatrix} 1 & t & \frac{t^2}{2} \\ 0 & 1 & t \\ 0 & 0 & 1 \end{pmatrix}.$$

This is [Definition 9.9.1](#) for  $f(x) = e^{tx}$ : the  $j$ -th superdiagonal carries  $f^{(j)}(3)/j! = t^j e^{3t}/j!$ .

### Exercise B2: Solving systems.

Solve each initial value problem, and verify your answer by differentiating.

- (a)  $x' = x + 3y$ ,  $y' = 3x + y$ , with  $x(0) = 2$ ,  $y(0) = 0$ .  
 (b)  $\mathbf{x}' = \begin{pmatrix} 0 & -3 \\ 3 & 0 \end{pmatrix} \mathbf{x}$  with  $\mathbf{x}(0) = (0, 1)$ .  
 (c)  $y''' = 0$  with  $y(0) = 1$ ,  $y'(0) = 0$ ,  $y''(0) = 2$ , by turning it into a system.

*Solution.*

- (a) This is  $\mathbf{x}' = \mathbf{A}\mathbf{x}$  for the matrix of [Exercise \(B1: Computing exponentials\)](#) (a). By [Theorem 9.9.6](#),

$$\mathbf{x}(t) = e^{t\mathbf{A}} \begin{pmatrix} 2 \\ 0 \end{pmatrix} = \begin{pmatrix} e^{4t} + e^{-2t} \\ e^{4t} - e^{-2t} \end{pmatrix},$$

that is,  $x = e^{4t} + e^{-2t}$  and  $y = e^{4t} - e^{-2t}$ . Check.  $x' = 4e^{4t} - 2e^{-2t}$  and  $x + 3y = e^{4t} + e^{-2t} + 3e^{4t} - 3e^{-2t} = 4e^{4t} - 2e^{-2t}$ ; similarly for  $y'$ . And  $x(0) = 2$ ,  $y(0) = 0$ .

- (b) Write  $\mathbf{A} = 3\mathbf{R}$  with  $\mathbf{R} = \begin{pmatrix} 0 & -1 \\ 1 & 0 \end{pmatrix}$ , whose eigenvalues are  $\pm i$  (Example (A Real Matrix with Non-Real Eigenvalues)), so those of  $\mathbf{A}$  are  $\pm 3i$ , distinct. Repeating the computation of Example (Two Two-Dimensional Systems) (a) with  $1 \pm 2i$  replaced by  $\pm 3i$  — eigenvectors  $(1, -i)$  for  $3i$  and  $(1, i)$  for  $-3i$  — gives

$$e^{t\mathbf{A}} = \begin{pmatrix} \cos 3t & -\sin 3t \\ \sin 3t & \cos 3t \end{pmatrix}, \quad \mathbf{x}(t) = \begin{pmatrix} -\sin 3t \\ \cos 3t \end{pmatrix}.$$

*Check.*  $\mathbf{x}'(t) = (-3\cos 3t, -3\sin 3t)$  and  $\mathbf{A}\mathbf{x}(t) = (-3\cos 3t, -3\sin 3t)$ ; and  $\mathbf{x}(0) = (0, 1)$ . The solution runs around the unit circle: no growth, since the eigenvalues are purely imaginary.

- (c) Put  $\mathbf{u} = (y, y', y'')$ . Then  $\mathbf{u}' = (y', y'', 0) = \mathbf{M}\mathbf{u}$  with  $\mathbf{M} = \mathbf{J}_3(0)$ , and  $\mathbf{u}(0) = (1, 0, 2)$ . By Corollary 9.9.5 with  $\lambda = 0$  and  $k = 3$ ,

$$e^{t\mathbf{M}} = \begin{pmatrix} 1 & t & t^2/2 \\ 0 & 1 & t \\ 0 & 0 & 1 \end{pmatrix}, \quad \mathbf{u}(t) = e^{t\mathbf{M}} \begin{pmatrix} 1 \\ 0 \\ 2 \end{pmatrix} = \begin{pmatrix} 1+t^2 \\ 2t \\ 2 \end{pmatrix}.$$

So  $y(t) = 1 + t^2$ . *Check.*  $y' = 2t$ ,  $y'' = 2$ ,  $y''' = 0$ , and the three initial values are 1, 0, 2. The eigenvalue 0 with one block of size 3 gives polynomial solutions  $t^j e^{0t} = t^j$  for  $j < 3$ .

### Exercise B3: A square root and a logarithm.

- (a) Let  $\mathbf{A} = \begin{pmatrix} 2 & 1 \\ 1 & 2 \end{pmatrix}$ . Compute  $f(\mathbf{A})$  for  $f(x) = \sqrt{x}$  (positive branch) and verify that  $f(\mathbf{A})^2 = \mathbf{A}$ .
- (b) Let  $\mathbf{B} = \begin{pmatrix} 1 & 2 \\ 0 & 1 \end{pmatrix}$ . Compute  $f(\mathbf{B})$  for  $f(x) = \log x$  with  $\log 1 = 0$ , and verify that  $e^{f(\mathbf{B})} = \mathbf{B}$ .
- (c) Explain why the same method produces **no** square root of  $\begin{pmatrix} 0 & 1 \\ 0 & 0 \end{pmatrix}$ .

*Solution.*

- (a)  $p_{\mathbf{A}} = (x-2)^2 - 1 = (x-1)(x-3)$ , eigenvalues 1, 3. Eigenvectors  $(1, -1)$  for 1 and  $(1, 1)$  for 3. With  $\mathbf{P} = \begin{pmatrix} 1 & 1 \\ -1 & 1 \end{pmatrix}$  and  $\mathbf{P}^{-1} = \frac{1}{2} \begin{pmatrix} 1 & -1 \\ 1 & 1 \end{pmatrix}$ ,

$$f(\mathbf{A}) = \mathbf{P} \operatorname{diag}(1, \sqrt{3}) \mathbf{P}^{-1} = \frac{1}{2} \begin{pmatrix} 1 + \sqrt{3} & -1 + \sqrt{3} \\ -1 + \sqrt{3} & 1 + \sqrt{3} \end{pmatrix}.$$

Squaring, the diagonal entry is  $\frac{1}{4}[(1+\sqrt{3})^2 + (\sqrt{3}-1)^2] = \frac{1}{4}(4+2\sqrt{3}+4-2\sqrt{3}) = 2$ , and the off-diagonal entry is  $\frac{1}{4} \cdot 2(1+\sqrt{3})(\sqrt{3}-1) = \frac{1}{2}(3-1) = 1$ . So  $f(\mathbf{A})^2 = \mathbf{A}$ .

- (b) Here  $m_{\mathbf{B}} = (x-1)^2$ , since  $(\mathbf{B}-\mathbf{I})^2 = 0 \neq \mathbf{B}-\mathbf{I}$ , so by Theorem 9.9.2 the answer is  $p(\mathbf{B})$  for the polynomial of degree  $< 2$  with  $p(1) = f(1) = 0$  and  $p'(1) = f'(1) = 1$ ,

namely  $p = x - 1$ . Hence

$$f(\mathbf{B}) = \mathbf{B} - \mathbf{I} = \begin{pmatrix} 0 & 2 \\ 0 & 0 \end{pmatrix}.$$

Its square is 0, so  $e^{f(\mathbf{B})} = \mathbf{I} + f(\mathbf{B}) = \begin{pmatrix} 1 & 2 \\ 0 & 1 \end{pmatrix} = \mathbf{B}$ . The Jordan route gives the same thing: with  $\mathbf{Q} = \text{diag}(2, 1)$  we get  $\mathbf{Q}^{-1}\mathbf{B}\mathbf{Q} = \mathbf{J}_2(1)$ , and  $f(\mathbf{J}_2(1)) = \begin{pmatrix} 0 & 1 \\ 0 & 0 \end{pmatrix}$ , so  $f(\mathbf{B}) = \mathbf{Q}f(\mathbf{J}_2(1))\mathbf{Q}^{-1} = \begin{pmatrix} 0 & 2 \\ 0 & 0 \end{pmatrix}$ .

- (c) The matrix is  $\mathbf{J}_2(0)$ , with the single eigenvalue 0 and a block of size 2. [Definition 9.9.1](#) needs  $f$  and  $f'$  at 0, and  $f(x) = \sqrt{x}$  is not differentiable at 0 — the entry  $f'(0)$  does not exist. In fact  $\mathbf{J}_2(0)$  has no square root at all: if  $\mathbf{X}^2 = \mathbf{J}_2(0)$ , then  $\mathbf{X}^4 = \mathbf{J}_2(0)^2 = 0$ , so  $\mathbf{X}$  is nilpotent, so  $\mathbf{X}^2 = 0$  by [Proposition 9.2.1](#) (a) with  $n = 2$ , contradicting  $\mathbf{X}^2 = \mathbf{J}_2(0) \neq 0$ .

### 9.9.5.3 C. Going deeper

**Exercise C1: How badly the exponential law fails.**

Let  $\mathbf{A} = \begin{pmatrix} 0 & 1 \\ 0 & 0 \end{pmatrix}$  and  $\mathbf{B} = \begin{pmatrix} 0 & 0 \\ 1 & 0 \end{pmatrix}$  in  $M_2(\mathbb{R})$ .

- Compute  $e^{\mathbf{A}}$ ,  $e^{\mathbf{B}}$ ,  $e^{\mathbf{A}}e^{\mathbf{B}}$  and  $e^{\mathbf{B}}e^{\mathbf{A}}$ .
- Compute  $e^{\mathbf{A}+\mathbf{B}}$  in closed form. *Hint:*  $(\mathbf{A} + \mathbf{B})^2 = \mathbf{I}$ , so split the series into even and odd terms.
- Verify  $\det e^{\mathbf{A}+\mathbf{B}} = e^{\text{tr}(\mathbf{A}+\mathbf{B})}$  and  $\det(e^{\mathbf{A}}e^{\mathbf{B}}) = e^{\text{tr} \mathbf{A}}e^{\text{tr} \mathbf{B}}$ , and explain why equal determinants do not rescue the identity.
- Show that  $e^{\mathbf{A}}e^{\mathbf{B}} = e^{\mathbf{C}}$  for **some**  $\mathbf{C} \in M_2(\mathbb{R})$ , and find  $\text{tr } \mathbf{C}$ .

*Solution.*

- (a)  $\mathbf{A}^2 = \mathbf{B}^2 = 0$ , so  $e^{\mathbf{A}} = \mathbf{I} + \mathbf{A} = \begin{pmatrix} 1 & 1 \\ 0 & 1 \end{pmatrix}$  and  $e^{\mathbf{B}} = \mathbf{I} + \mathbf{B} = \begin{pmatrix} 1 & 0 \\ 1 & 1 \end{pmatrix}$ . Then

$$e^{\mathbf{A}}e^{\mathbf{B}} = \begin{pmatrix} 2 & 1 \\ 1 & 1 \end{pmatrix}, \quad e^{\mathbf{B}}e^{\mathbf{A}} = \begin{pmatrix} 1 & 1 \\ 1 & 2 \end{pmatrix}.$$

- (b) Let  $\mathbf{K} = \mathbf{A} + \mathbf{B} = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}$ . Then  $\mathbf{K}^2 = \mathbf{I}$ , so  $\mathbf{K}^{2l} = \mathbf{I}$  and  $\mathbf{K}^{2l+1} = \mathbf{K}$ . Splitting the partial sums by parity, which is legitimate because each entry sequence is a sum of two convergent sequences (A3),

$$\begin{aligned} e^{\mathbf{K}} &= \left( \sum_{l \geq 0} \frac{1}{(2l)!} \right) \mathbf{I} + \left( \sum_{l \geq 0} \frac{1}{(2l+1)!} \right) \mathbf{K} \\ &= \frac{e + e^{-1}}{2} \mathbf{I} + \frac{e - e^{-1}}{2} \mathbf{K} = \begin{pmatrix} \frac{e+e^{-1}}{2} & \frac{e-e^{-1}}{2} \\ \frac{e-e^{-1}}{2} & \frac{e+e^{-1}}{2} \end{pmatrix}, \end{aligned}$$

where the two scalar sums were identified by adding and subtracting the series for

$e^1$  and  $e^{-1}$  term by term (A1). Numerically the diagonal entries are about 1.543, not 2, so  $e^{\mathbf{A}+\mathbf{B}} \neq e^{\mathbf{A}}e^{\mathbf{B}}$ .

- (c)  $\text{tr}(\mathbf{A} + \mathbf{B}) = 0$  and  $\det e^{\mathbf{A}+\mathbf{B}} = \left(\frac{e+e^{-1}}{2}\right)^2 - \left(\frac{e-e^{-1}}{2}\right)^2 = \frac{1}{4}[(e+e^{-1})^2 - (e-e^{-1})^2] = \frac{1}{4} \cdot 4 = 1 = e^0$ . Also  $\det(e^{\mathbf{A}}e^{\mathbf{B}}) = 2 \cdot 1 = 1 = e^0 e^0$ . The determinants agree because both matrices are products of exponentials of trace-zero matrices; a determinant is one number and cannot distinguish two matrices, so it gives no information about the identity either way. The entries do.
- (d)  $\mathbf{X} = e^{\mathbf{A}}e^{\mathbf{B}} = \begin{pmatrix} 2 & 1 \\ 1 & 1 \end{pmatrix}$  is symmetric with  $\det \mathbf{X} = 1$  and  $\text{tr} \mathbf{X} = 3$ , so  $p_{\mathbf{X}} = x^2 - 3x + 1$ , whose roots  $\frac{3 \pm \sqrt{5}}{2}$  are real, distinct and positive. So  $\mathbf{X} = \mathbf{P} \text{diag}(\alpha, \beta) \mathbf{P}^{-1}$  with  $\alpha\beta = 1$ ,  $\alpha, \beta > 0$ . Put  $\mathbf{C} = \mathbf{P} \text{diag}(\log \alpha, \log \beta) \mathbf{P}^{-1}$ , a real matrix because  $\mathbf{P}$  can be taken real (the eigenvalues are real) and the logarithms are real. By Definition 9.9.1 applied to  $f = \exp$ ,  $e^{\mathbf{C}} = \mathbf{P} \text{diag}(\alpha, \beta) \mathbf{P}^{-1} = \mathbf{X}$ . Finally  $\text{tr} \mathbf{C} = \log \alpha + \log \beta = \log(\alpha\beta) = \log 1 = 0$ , as Theorem 9.9.4 (e) requires, since  $\det \mathbf{X} = 1$ .

### Exercise C2: When does every solution die out?

Let  $\mathbf{A} \in M_n(\mathbb{C})$ . Prove that the following are equivalent:

- (i) every solution of  $\mathbf{x}' = \mathbf{A}\mathbf{x}$  satisfies  $\mathbf{x}(t) \rightarrow \mathbf{0}$  entrywise as  $t \rightarrow \infty$ ;
- (ii) every eigenvalue  $\lambda$  of  $\mathbf{A}$  has  $\text{Re } \lambda < 0$ .

*Hint: compute the entries of  $e^{t\mathbf{J}_k(\lambda)}$  and use (A5). For the failing direction, an eigenvector gives a solution you can write down.*

*Solution.*

- (ii)  $\Rightarrow$  (i). By Theorem 9.9.6 every solution is  $\mathbf{x}(t) = e^{t\mathbf{A}}\mathbf{x}_0$ , so it suffices to show  $e^{t\mathbf{A}} \rightarrow 0$  entrywise. Write  $\mathbf{A} = \mathbf{P}\mathbf{J}\mathbf{P}^{-1}$  with  $\mathbf{J} = \bigoplus_i \mathbf{J}_{k_i}(\lambda_i)$  (Theorem 9.3.1). Each entry of  $e^{t\mathbf{A}} = \mathbf{P}e^{t\mathbf{J}}\mathbf{P}^{-1}$  is a fixed linear combination of entries of  $e^{t\mathbf{J}}$ , so by (A3) it is enough that  $e^{t\mathbf{J}} \rightarrow 0$ . By Corollary 9.9.5, the non-zero entries of  $e^{t\mathbf{J}_k(\lambda)}$  are

$$\frac{t^j e^{\lambda t}}{j!} \quad (0 \leq j \leq k-1).$$

Writing  $\lambda = c + i\omega$  with  $c = \text{Re } \lambda < 0$ , we get  $|t^j e^{\lambda t}| = t^j e^{ct} |e^{i\omega t}| = t^j e^{ct}$  for  $t \geq 0$ , since  $|e^{i\omega t}| = |\cos \omega t + i \sin \omega t| = 1$  by (A1). By (A5) this tends to 0, and a complex sequence tends to 0 exactly when its moduli do. Hence  $e^{t\mathbf{J}} \rightarrow 0$ .

- (iii)  $\Rightarrow$  (ii). Suppose some eigenvalue  $\lambda$  has  $\text{Re } \lambda = c \geq 0$ , and let  $\mathbf{v} \neq \mathbf{0}$  be an eigenvector. Then  $\mathbf{x}(t) = e^{\lambda t} \mathbf{v}$  is a solution: it is differentiable with  $\mathbf{x}'(t) = \lambda e^{\lambda t} \mathbf{v} = e^{\lambda t} \mathbf{A} \mathbf{v} = \mathbf{A} \mathbf{x}(t)$ , and  $\mathbf{x}(0) = \mathbf{v}$ . Pick a coordinate with  $v_r \neq 0$ ; then

$$|x_r(t)| = |e^{\lambda t}| |v_r| = e^{ct} |v_r| \geq |v_r| > 0$$

for every  $t \geq 0$ , because  $e^{ct} \geq 1$  when  $c \geq 0$ . So  $x_r(t)$  does not tend to 0, and (i) fails. This proves the equivalence.



## 9.10 Convergent and Power-Bounded Matrices

Chapter 8 asked what happens to  $\mathbf{A}^m$  as  $m$  grows, and could answer only for diagonalizable matrices: the powers of  $\mathbf{A}$  behave like the powers of its eigenvalues. For a Markov chain that left a gap, because a transition matrix need not be diagonalizable. The Jordan form closes the gap completely. This section characterizes, in terms of the eigenvalues and the block sizes alone, when  $\mathbf{A}^m$  tends to zero, when it stays bounded, when it converges, and when the geometric series  $\mathbf{I} + \mathbf{A} + \mathbf{A}^2 + \cdots$  converges.

Convergence here means what it meant in Chapter 8: **entrywise**. No notion of the size of a matrix is used. Chapter 15 introduces norms and re-proves several of these results in a way that also gives numerical bounds; the price of doing without them is that we must look at one entry at a time, and the Jordan form is what makes that possible.

### 9.10.1 Entrywise limits and bounds

#### Definition 9.10.1 — Entrywise Convergence and Boundedness

Let  $\mathbf{M}_1, \mathbf{M}_2, \dots$  be matrices in  $M_{p \times q}(\mathbb{C})$  and  $\mathbf{M} \in M_{p \times q}(\mathbb{C})$ .

- (a) The sequence  $(\mathbf{M}_m)$  **converges** to  $\mathbf{M}$ , written  $\mathbf{M}_m \rightarrow \mathbf{M}$ , if for **every** position  $(r, s)$  the sequence of complex numbers  $(\mathbf{M}_m)_{rs}$  converges to  $\mathbf{M}_{rs}$ . We then write  $\mathbf{M} = \lim_m \mathbf{M}_m$ .
- (b) The sequence  $(\mathbf{M}_m)$  is **bounded** if there is a real  $c \geq 0$  with  $|(\mathbf{M}_m)_{rs}| \leq c$  for **all**  $m$  and **all** positions  $(r, s)$ .
- (c) A series  $\sum_{k \geq 0} \mathbf{M}_k$  **converges** to  $\mathbf{M}$  if its sequence of partial sums  $\sum_{k=0}^K \mathbf{M}_k$  converges to  $\mathbf{M}$ .

In words: nothing new is happening. Each of the  $pq$  positions is watched separately, and the words mean for matrices exactly what they mean for the finitely many number sequences at those positions. Since a finite set of bounds can be replaced by their maximum, (b) is the same as asking each entry sequence to be bounded.

*Remark. What we assume from analysis.* Beyond the four facts quoted in Chapter 8 §11, this section uses exactly the following.

- (B1) **Polynomial against geometric decay.** For every integer  $j \geq 0$  and every real  $r$  with  $0 \leq r < 1$ ,  $m^j r^m \rightarrow 0$  as  $m \rightarrow \infty$ .
- (B2) **Growth.** For real  $r > 1$ , the sequence  $(r^m)$  is unbounded.
- (B3) **Limits.** Limits of convergent sequences of complex numbers respect sums, scalar multiples and products; a convergent sequence is bounded; every subsequence of a convergent sequence converges to the same limit; and  $z_m \rightarrow z$  implies  $|z_m| \rightarrow |z|$ .
- (B4) **Terms of a convergent series.** If  $\sum_k z_k$  converges, then  $z_k \rightarrow 0$ .

Two reductions make every proof below short. The first moves the question along a similarity,

so that we may assume  $\mathbf{A}$  is a Jordan matrix. The second computes the powers of a single Jordan block.

### Lemma 9.10.1 — Convergence and Boundedness Survive a Similarity

Let  $\mathbf{A}, \mathbf{B} \in M_n(\mathbb{C})$  with  $\mathbf{B} = \mathbf{P}^{-1}\mathbf{A}\mathbf{P}$  for an invertible  $\mathbf{P} \in M_n(\mathbb{C})$ , and let  $\mathbf{L} \in M_n(\mathbb{C})$ . Then:

- (a)  $\mathbf{A}^m \rightarrow \mathbf{L}$  if and only if  $\mathbf{B}^m \rightarrow \mathbf{P}^{-1}\mathbf{L}\mathbf{P}$ ;
- (b)  $(\mathbf{A}^m)$  is bounded if and only if  $(\mathbf{B}^m)$  is bounded.

*Proof.* By Proposition 3.7.6 (c),  $\mathbf{B}^m = \mathbf{P}^{-1}\mathbf{A}^m\mathbf{P}$  for every  $m \geq 0$ . Writing out the products (Definition 0.9.9), the  $(r, s)$ -entry of  $\mathbf{B}^m$  is

$$(\mathbf{B}^m)_{rs} = \sum_{u,v} (\mathbf{P}^{-1})_{ru} (\mathbf{A}^m)_{uv} \mathbf{P}_{vs},$$

a linear combination of the entries of  $\mathbf{A}^m$  with coefficients that do not depend on  $m$ . So if  $\mathbf{A}^m \rightarrow \mathbf{L}$ , then by (B3) each entry of  $\mathbf{B}^m$  converges to the matching combination of the entries of  $\mathbf{L}$ , that is,  $\mathbf{B}^m \rightarrow \mathbf{P}^{-1}\mathbf{L}\mathbf{P}$ ; and if  $|(\mathbf{A}^m)_{uv}| \leq c$  for all  $m, u, v$ , then  $|(\mathbf{B}^m)_{rs}| \leq c \sum_{u,v} |(\mathbf{P}^{-1})_{ru}| |\mathbf{P}_{vs}|$ , a bound independent of  $m$ , so  $(\mathbf{B}^m)$  is bounded. The converses follow by exchanging the roles of  $\mathbf{A}$  and  $\mathbf{B}$ , using  $\mathbf{A} = \mathbf{P}\mathbf{B}\mathbf{P}^{-1}$ . ■

For the second reduction, recall from Lemma 9.9.1, applied to the polynomial  $x^m$ , that

$$(\mathbf{J}_k(\lambda)^m)_{r, r+j} = \binom{m}{j} \lambda^{m-j} \quad (0 \leq j \leq k-1),$$

with zeros below the diagonal. Everything in this section is read off this one formula. Two remarks about it: the binomial coefficient is bounded by  $m^j$ , since

$$\binom{m}{j} = \frac{m(m-1) \cdots (m-j+1)}{j!} \leq \frac{m^j}{j!} \leq m^j \quad (m \geq j \geq 1),$$

and  $\binom{m}{1} = m$  grows without bound, while  $\binom{m}{0} = 1$  is constant. So the only question is whether the geometric factor  $\lambda^m$  beats the polynomial factor  $m^j$ .

*Quick check.* Compute  $|(\mathbf{J}_3(\lambda)^m)_{1,3}|$  for  $\lambda = \frac{1}{2}$  and for  $\lambda = 1$ , and say in each case whether it tends to 0, stays bounded, or grows without bound.

*Solution.* Here  $j = 2$ , so the entry is  $\binom{m}{2} \lambda^{m-2}$ , of modulus  $\frac{m(m-1)}{2} |\lambda|^{m-2}$ . For  $\lambda = \frac{1}{2}$  this is  $2 \cdot \frac{m(m-1)}{2} 2^{-m} \leq 2m^2 2^{-m}$ , which tends to 0 by (B1) with  $j = 2$  and  $r = \frac{1}{2}$ . For  $\lambda = 1$  it is  $\frac{m(m-1)}{2}$ , which grows without bound. The modulus of  $\lambda$  decides, and when  $|\lambda| = 1$  the block size decides.

## 9.10.2 Powers tending to zero

*The powers of  $\mathbf{A}$  die out exactly when every eigenvalue has modulus less than one.*

**Theorem 9.10.2 — Powers Tending to Zero**

Let  $\mathbf{A} \in M_n(\mathbb{C})$ ,  $n \geq 1$ . Then  $\mathbf{A}^m \rightarrow 0$  if and only if every eigenvalue  $\lambda$  of  $\mathbf{A}$  satisfies  $|\lambda| < 1$ .

*Idea.* ( $\Leftarrow$ ) Pass to the Jordan form and look at one entry: it is  $\binom{m}{j}\lambda^{m-j}$ , a polynomial in  $m$  times a geometric sequence that decays, and geometric decay wins. ( $\Rightarrow$ ) The other direction needs no Jordan form at all: an eigenvector turns  $\mathbf{A}^m$  into  $\lambda^m$ .

*Proof.* ( $\Leftarrow$ ) Suppose  $|\lambda| < 1$  for every  $\lambda \in \text{spec}(\mathbf{A})$ . By Theorem 9.3.1,  $\mathbf{A} = \mathbf{PJP}^{-1}$  with  $\mathbf{J} = \mathbf{J}_{k_1}(\lambda_1) \oplus \cdots \oplus \mathbf{J}_{k_d}(\lambda_d)$  a Jordan matrix, whose diagonal entries  $\lambda_i$  are the eigenvalues of  $\mathbf{A}$  (Corollary 9.3.2 (a)). By Theorem 7.1.4 (b),  $\mathbf{J}^m$  is the direct sum of the blocks  $\mathbf{J}_{k_i}(\lambda_i)^m$ , so each entry of  $\mathbf{J}^m$  is either 0 for every  $m$  or of the form  $\binom{m}{j}\lambda_i^{m-j}$  with  $0 \leq j \leq k_i - 1$ . For the latter, with  $r = |\lambda_i| < 1$ , if  $r = 0$  the entry is 0 for  $m > j$ ; otherwise, for  $m \geq j$ ,

$$\left| \binom{m}{j} \lambda_i^{m-j} \right| \leq m^j r^{m-j} = r^{-j} m^j r^m \rightarrow 0$$

by (B1), the factor  $r^{-j}$  being a constant. So  $\mathbf{J}^m \rightarrow 0$ , and  $\mathbf{A}^m \rightarrow \mathbf{P} \mathbf{0} \mathbf{P}^{-1} = \mathbf{0}$  by Lemma 9.10.1 (a).

( $\Rightarrow$ ) Suppose  $\mathbf{A}^m \rightarrow 0$  and let  $\lambda \in \text{spec}(\mathbf{A})$  with eigenvector  $\mathbf{v} \neq \mathbf{0}$ . Then  $\mathbf{A}^m \mathbf{v} = \lambda^m \mathbf{v}$  for every  $m$  (induction on  $m$ ). Pick a coordinate  $r$  with  $v_r \neq 0$ . The  $r$ -th entry of  $\mathbf{A}^m \mathbf{v}$  is a linear combination of entries of  $\mathbf{A}^m$  with coefficients  $v_1, \dots, v_n$ , so it tends to 0 by (B3); that entry is  $\lambda^m v_r$ , so  $\lambda^m \rightarrow 0$  and hence  $|\lambda|^m = |\lambda^m| \rightarrow 0$  by (B3). If  $|\lambda| \geq 1$ , then  $|\lambda|^m \geq 1$  for every  $m$  and the limit cannot be 0. Hence  $|\lambda| < 1$ . This proves the theorem. ■

Note the asymmetry: the hard direction needs the Jordan form, the easy one needs only an eigenvector. That pattern repeats in the next two theorems.

*Warning.* **The condition is on the eigenvalues, not on the entries.** Every entry of  $\mathbf{A} = \begin{pmatrix} 0.9 & 0.9 \\ 0.9 & 0.9 \end{pmatrix}$  has modulus less than 1, yet  $\mathbf{A}\mathbf{1} = 1.8\mathbf{1}$ , so 1.8 is an eigenvalue and  $\mathbf{A}^m \mathbf{1} = 1.8^m \mathbf{1}$  grows without bound. Conversely, a matrix with huge entries can have all eigenvalues 0: the matrix  $\begin{pmatrix} 0 & 10^6 \\ 0 & 0 \end{pmatrix}$  squares to 0. Nothing can be concluded from the size of the entries alone.

**9.10.3 Powers that stay bounded**

Dropping “tends to zero” to “stays bounded” allows eigenvalues on the unit circle, but only if they carry no chains: a block of size  $\geq 2$  contributes an entry  $\binom{m}{1}\lambda^{m-1}$  of modulus  $m$ .

**Theorem 9.10.3 — Power-Bounded Matrices**

Let  $\mathbf{A} \in M_n(\mathbb{C})$ ,  $n \geq 1$ , with Jordan form  $\mathbf{J}$ . The sequence  $(\mathbf{A}^m)_{m \geq 0}$  is bounded if and only if

- (i) every eigenvalue  $\lambda$  of  $\mathbf{A}$  satisfies  $|\lambda| \leq 1$ , and
- (ii) every eigenvalue  $\lambda$  with  $|\lambda| = 1$  has **all** of its blocks in  $\mathbf{J}$  of size 1, equivalently  $a_{\mathbf{A}}(\lambda) = g_{\mathbf{A}}(\lambda)$ , equivalently  $x - \lambda$  occurs to the first power in  $m_{\mathbf{A}}$ .

*Proof.* Write  $\mathbf{A} = \mathbf{PJP}^{-1}$  with  $\mathbf{J} = \mathbf{J}_{k_1}(\lambda_1) \oplus \cdots \oplus \mathbf{J}_{k_d}(\lambda_d)$  (Theorem 9.3.1). By Lemma 9.10.1 (b),  $(\mathbf{A}^m)$  is bounded if and only if  $(\mathbf{J}^m)$  is, and by Theorem 7.1.4 (b) the latter holds if and only if every block sequence  $(\mathbf{J}_{k_i}(\lambda_i)^m)$  is bounded. The three reformulations in (ii) are Corollary 9.3.2 (a), (b) and (c): all  $\lambda$ -blocks have size 1 exactly when their number equals their total size, and exactly when the largest of their sizes is 1.

( $\Leftarrow$ ) Assume (i) and (ii), and fix a block  $\mathbf{J}_k(\lambda)$  of  $\mathbf{J}$ . If  $|\lambda| < 1$ , then  $\mathbf{J}_k(\lambda)^m \rightarrow 0$  by Theorem 9.10.2 applied to the single block  $\mathbf{J}_k(\lambda)$ , whose only eigenvalue is  $\lambda$ , hence  $(\mathbf{J}_k(\lambda)^m)$  is bounded by (B3). If  $|\lambda| = 1$ , then  $k = 1$  by (ii), so  $\mathbf{J}_k(\lambda)^m = (\lambda^m)$  is a  $1 \times 1$  matrix of modulus  $|\lambda|^m = 1$ . Either way the block sequence is bounded, so  $(\mathbf{J}^m)$  is bounded and so is  $(\mathbf{A}^m)$ .

( $\Rightarrow$ ) Assume  $(\mathbf{A}^m)$  is bounded, hence so is  $(\mathbf{J}^m)$ , hence so is every block sequence.

*Proof of (i).* Let  $\lambda$  be an eigenvalue and  $\mathbf{J}_k(\lambda)$  one of its blocks. The  $(1, 1)$ -entry of  $\mathbf{J}_k(\lambda)^m$  is  $\lambda^m$ , so  $(|\lambda|^m)$  is bounded; by (B2) this forces  $|\lambda| \leq 1$ .

*Proof of (ii).* Suppose some eigenvalue  $\lambda$  with  $|\lambda| = 1$  had a block  $\mathbf{J}_k(\lambda)$  with  $k \geq 2$ . Its  $(1, 2)$ -entry at the  $m$ -th power is  $\binom{m}{1} \lambda^{m-1} = m \lambda^{m-1}$ , of modulus  $m |\lambda|^{m-1} = m$ , which is unbounded — contradicting the boundedness of that block sequence. So every such block has size 1. This proves the theorem. ■

## 9.10.4 Powers that converge

Boundedness is not convergence:  $\begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}^m$  alternates between two matrices forever. Among the eigenvalues of modulus 1, only  $\lambda = 1$  has convergent powers, because  $\lambda^m$  has to sit still.

### Theorem 9.10.4 — Convergent Matrices

Let  $\mathbf{A} \in M_n(\mathbb{C})$ ,  $n \geq 1$ , with Jordan form  $\mathbf{J}$ . Then  $(\mathbf{A}^m)$  converges if and only if every eigenvalue  $\lambda$  of  $\mathbf{A}$  satisfies

- (i)  $|\lambda| < 1$ , or
- (ii)  $\lambda = 1$  and all of its blocks in  $\mathbf{J}$  have size 1.

In that case

$$\lim_{m \rightarrow \infty} \mathbf{A}^m = \mathbf{\Pi},$$

where  $\mathbf{\Pi} \in M_n(\mathbb{C})$  is the matrix, in the standard basis, of the projection onto  $E_1(\mathbf{A})$

along  $\bigoplus_{\lambda \neq 1} G_\lambda(\mathbf{A})$  if  $1 \in \text{spec}(\mathbf{A})$ , and  $\mathbf{\Pi} = 0$  otherwise.

*Idea.* For the shape of the answer, take the Jordan form and let  $m \rightarrow \infty$  in each block: the blocks with  $|\lambda| < 1$  go to 0, the blocks  $\mathbf{J}_1(1) = (1)$  stay at 1. So  $\mathbf{J}^m$  tends to the 0/1 diagonal matrix that marks the 1-blocks, which is the matrix of a projection; conjugating back identifies the two subspaces. For the converse, the  $(1, 1)$ -entry of a block is  $\lambda^m$ , and if that converges to  $z$  then so does  $\lambda^{m+1} = \lambda \cdot \lambda^m$ , giving  $z = \lambda z$  and forcing  $\lambda = 1$  or  $z = 0$ .

*Proof.* Write  $\mathbf{A} = \mathbf{PJP}^{-1}$  with  $\mathbf{J} = \mathbf{J}_{k_1}(\lambda_1) \oplus \cdots \oplus \mathbf{J}_{k_d}(\lambda_d)$  (Theorem 9.3.1), and let  $\mathcal{B} = (\mathbf{b}_1, \dots, \mathbf{b}_n)$  be the corresponding Jordan basis, the columns of  $\mathbf{P}$ . For each  $i$  let  $C_i \subseteq \mathbb{C}^n$  be the span of the  $k_i$  basis vectors belonging to block  $i$ , so that  $\mathbb{C}^n = C_1 \oplus \cdots \oplus C_d$ .

( $\Leftarrow$ ) Suppose every eigenvalue satisfies (i) or (ii). Fix a block. If  $|\lambda_i| < 1$ , then  $\mathbf{J}_{k_i}(\lambda_i)^m \rightarrow 0$  by Theorem 9.10.2 applied to the block  $\mathbf{J}_{k_i}(\lambda_i)$ , whose only eigenvalue is  $\lambda_i$ . If  $\lambda_i = 1$ , then  $k_i = 1$  by (ii) and  $\mathbf{J}_1(1)^m = (1)$  for every  $m$ . By Theorem 7.1.4 (b),  $\mathbf{J}^m$  is the direct sum of these, so

$$\mathbf{J}^m \longrightarrow \mathbf{E},$$

where  $\mathbf{E}$  is the diagonal matrix carrying 1 in the positions of the blocks with  $\lambda_i = 1$  and 0 everywhere else. By Lemma 9.10.1 (a),  $\mathbf{A}^m \rightarrow \mathbf{PEP}^{-1}$ .

It remains to identify  $\mathbf{PEP}^{-1}$ . The matrix  $\mathbf{E}$  is the matrix in the basis  $\mathcal{B}$  of the projection onto

$$U := \bigoplus_{\lambda_i=1} C_i \quad \text{along} \quad W := \bigoplus_{\lambda_i \neq 1} C_i,$$

because  $\mathbf{E}$  fixes each basis vector of the first family and kills each of the second (Theorem 3.10.1). Since the columns of  $\mathbf{P}$  are the vectors of  $\mathcal{B}$ ,  $\mathbf{P}$  is the change-of-coordinates matrix from  $\mathcal{B}$  to the standard basis, so  $\mathbf{PEP}^{-1}$  is the matrix of that projection in the standard basis (Theorem 3.7.3). Now, for any  $\mu \in \text{spec}(\mathbf{A})$ ,

$$\bigoplus_{\lambda_i=\mu} C_i = G_\mu(\mathbf{A}).$$

Indeed each  $C_i$  with  $\lambda_i = \mu$  is killed by  $(\mathbf{A} - \mu\mathbf{I})^{k_i}$ , hence lies in  $G_\mu(\mathbf{A})$  (Corollary 9.1.2 (a)); the sum on the left is direct and has dimension  $\sum_{\lambda_i=\mu} k_i = a_{\mathbf{A}}(\mu)$  by Corollary 9.3.2 (a); and  $\dim G_\mu(\mathbf{A}) = a_{\mathbf{A}}(\mu)$  by Theorem 9.1.3 (c). An inclusion of subspaces with equal dimensions is an equality (Theorem 1.6.8, applied inside  $G_\mu(\mathbf{A})$ ). Hence  $W = \bigoplus_{\lambda \neq 1} G_\lambda(\mathbf{A})$ , and  $U = G_1(\mathbf{A})$  if  $1 \in \text{spec}(\mathbf{A})$ , with  $U = \{0\}$  otherwise. Finally, when  $1 \in \text{spec}(\mathbf{A})$ , condition (ii) says all 1-blocks have size 1, so  $g_{\mathbf{A}}(1) = a_{\mathbf{A}}(1)$  by Corollary 9.3.2 (a), (b); since  $E_1(\mathbf{A}) \subseteq G_1(\mathbf{A})$  with  $\dim E_1(\mathbf{A}) = g_{\mathbf{A}}(1) = a_{\mathbf{A}}(1) = \dim G_1(\mathbf{A})$ , we get  $G_1(\mathbf{A}) = E_1(\mathbf{A})$ . So  $\lim_m \mathbf{A}^m = \mathbf{\Pi}$  as stated.

( $\Rightarrow$ ) Suppose  $(\mathbf{A}^m)$  converges. By Lemma 9.10.1 (a) so does  $(\mathbf{J}^m)$ , hence so does each block sequence. Fix a block  $\mathbf{J}_k(\lambda)$ .

*Case  $|\lambda| > 1$ .* Impossible: a convergent sequence is bounded (B3), so this is excluded by Theorem 9.10.3.

*Case  $|\lambda| = 1$  and  $\lambda \neq 1$ .* The  $(1, 1)$ -entries give a convergent sequence  $\lambda^m \rightarrow z$ . The sequence  $(\lambda^{m+1})_{m \geq 0}$  is a tail of the same sequence, so it also converges to  $z$ ; on the other hand  $\lambda^{m+1} = \lambda \cdot \lambda^m \rightarrow \lambda z$  by (B3). Hence  $z = \lambda z$ , that is,  $(1 - \lambda)z = 0$ , and  $\lambda \neq 1$

forces  $z = 0$ . But  $|\lambda^m| = 1$  for every  $m$ , so  $|z| = 1$  by (B3), a contradiction. So this case does not occur either.

*Case  $\lambda = 1$ .* If  $k \geq 2$ , the  $(1, 2)$ -entry of  $\mathbf{J}_k(1)^m$  is  $\binom{m}{1} 1^{m-1} = m$ , which does not converge. So  $k = 1$ .

The three cases leave exactly (i) and (ii). This proves the theorem. ■

The limit deserves a sentence. It is a projection, so  $\mathbf{\Pi}^2 = \mathbf{\Pi}$  — which had to be true, since  $\mathbf{\Pi}^2 = \lim_m \mathbf{A}^m \lim_m \mathbf{A}^m = \lim_m \mathbf{A}^{2m} = \mathbf{\Pi}$  by (B3) — and it acts as the identity on the eigenvectors for 1 and annihilates everything belonging to the other eigenvalues. In the long run,  $\mathbf{A}$  forgets all of  $\mathbb{C}^n$  except the fixed space  $E_1(\mathbf{A})$ .

### Example. (Deciding and Computing Limits of Powers)

For each matrix, decide whether  $\mathbf{A}^m$  tends to 0, stays bounded, converges, and compute  $\lim \mathbf{A}^m$  when it exists.

- (a)  $\mathbf{A}_1 = \begin{pmatrix} 2 & 1 \\ -1 & 0 \end{pmatrix}$ .
- (b)  $\mathbf{A}_2 = \begin{pmatrix} 0 & -\frac{1}{2} \\ \frac{1}{2} & 0 \end{pmatrix}$ .
- (c)  $\mathbf{A}_3 = \begin{pmatrix} 1 & 0 & 0 \\ 0 & \frac{1}{3} & 1 \\ 0 & 0 & \frac{1}{3} \end{pmatrix}$ .

*Solution.*

- (a)  $p_{\mathbf{A}_1} = x(x - 2) + 1 = (x - 1)^2$ , so the only eigenvalue is 1. Put  $\mathbf{N} = \mathbf{A}_1 - \mathbf{I} = \begin{pmatrix} 1 & 1 \\ -1 & -1 \end{pmatrix}$ ; then  $\mathbf{N} \neq 0$  and  $\mathbf{N}^2 = \begin{pmatrix} 1-1 & 1-1 \\ -1+1 & -1+1 \end{pmatrix} = 0$ , so  $g_{\mathbf{A}_1}(1) = 2 - \text{rank } \mathbf{N} = 1$  and the Jordan form is  $\mathbf{J}_2(1)$ . Condition (ii) of [Theorem 9.10.3](#) fails, so  $(\mathbf{A}_1^m)$  is **unbounded**, and in particular does not converge. Explicitly,  $\mathbf{I}$  and  $\mathbf{N}$  commute and  $\mathbf{N}^2 = 0$ , so the binomial theorem gives

$$\mathbf{A}_1^m = (\mathbf{I} + \mathbf{N})^m = \mathbf{I} + m\mathbf{N} = \begin{pmatrix} 1+m & m \\ -m & 1-m \end{pmatrix},$$

which grows linearly.

- (b)  $p_{\mathbf{A}_2} = x^2 + \frac{1}{4}$ , with roots  $\pm \frac{i}{2}$ , both of modulus  $\frac{1}{2} < 1$ . By [Theorem 9.10.2](#),  $\mathbf{A}_2^m \rightarrow 0$  (so also bounded and convergent). Explicitly  $\mathbf{A}_2^2 = -\frac{1}{4}\mathbf{I}$ , so  $\mathbf{A}_2^{2l} = (-\frac{1}{4})^l \mathbf{I}$  and  $\mathbf{A}_2^{2l+1} = (-\frac{1}{4})^l \mathbf{A}_2$ , and every entry has modulus at most  $4^{-l} \rightarrow 0$ .
- (c)  $\mathbf{A}_3 = (1) \oplus \mathbf{J}_2(\frac{1}{3})$  is already a Jordan matrix. Its eigenvalues are 1, with a single block of size 1, and  $\frac{1}{3}$ , with  $|\frac{1}{3}| < 1$ . So condition (i) or (ii) holds for each eigenvalue and  $\mathbf{A}_3^m$  converges by [Theorem 9.10.4](#). With  $\mathbf{P} = \mathbf{I}$ , the matrix  $\mathbf{E}$  of the proof is  $\text{diag}(1, 0, 0)$ , so

$$\lim_m \mathbf{A}_3^m = \text{diag}(1, 0, 0),$$

the projection onto  $E_1(\mathbf{A}_3) = \text{span}(\mathbf{e}_1)$  along  $G_{1/3}(\mathbf{A}_3) = \text{span}(\mathbf{e}_2, \mathbf{e}_3)$ . Explicitly,

by Lemma 9.9.1,

$$\mathbf{A}_3^m = \begin{pmatrix} 1 & 0 & 0 \\ 0 & 3^{-m} & m 3^{-(m-1)} \\ 0 & 0 & 3^{-m} \end{pmatrix},$$

and  $m 3^{-(m-1)} = 3m 3^{-m} \rightarrow 0$  by (B1).

### 9.10.5 Markov chains, finished

Chapter 8 proved that a Markov chain settles down under three hypotheses:  $\mathbf{A}$  diagonalizable over  $\mathbb{C}$ ,  $a_{\mathbf{A}}(1) = 1$ , and  $|\lambda| < 1$  for every other eigenvalue (Theorem 8.11.4). The first hypothesis can now be dropped, and more: for a stochastic matrix, the block condition of Theorem 9.10.3 holds automatically, because the powers of a stochastic matrix have entries in  $[0, 1]$ .

#### Corollary 9.10.5 — Convergence of a Markov Chain

Let  $\mathbf{A} \in M_n(\mathbb{R})$  be stochastic (Definition 8.11.1),  $n \geq 1$ . Then:

- (a)  $(\mathbf{A}^m)$  is bounded; consequently every eigenvalue  $\lambda \in \mathbb{C}$  of  $\mathbf{A}$  satisfies  $|\lambda| \leq 1$ , and every eigenvalue with  $|\lambda| = 1$  has only  $1 \times 1$  blocks. In particular  $a_{\mathbf{A}}(1) = g_{\mathbf{A}}(1)$ .
- (b)  $(\mathbf{A}^m)$  converges if and only if 1 is the **only** eigenvalue of  $\mathbf{A}$  of modulus 1; and then  $\lim_m \mathbf{A}^m$  is the projection onto  $E_1(\mathbf{A})$  along  $\bigoplus_{\lambda \neq 1} G_{\lambda}(\mathbf{A})$ .

*Proof.*

- (a) By Proposition 8.11.3 (b), every  $\mathbf{A}^m$  is stochastic, so its entries are non-negative and each of its columns sums to 1; hence every entry lies in  $[0, 1]$  and  $(\mathbf{A}^m)$  is bounded with  $c = 1$ . Now apply the  $(\Rightarrow)$  direction of Theorem 9.10.3. Since  $1 \in \text{spec}(\mathbf{A})$  by Proposition 8.11.3 (c), the 1-blocks exist and all have size 1, so  $a_{\mathbf{A}}(1) = g_{\mathbf{A}}(1)$  by Corollary 9.3.2 (a), (b).
- (b) By (a), every eigenvalue has  $|\lambda| \leq 1$  and every eigenvalue of modulus 1 has only  $1 \times 1$  blocks. So the criterion of Theorem 9.10.4 — each eigenvalue has  $|\lambda| < 1$ , or equals 1 with only  $1 \times 1$  blocks — holds precisely when no eigenvalue  $\lambda \neq 1$  has  $|\lambda| = 1$ . The limit is the one given by Theorem 9.10.4. ■

So the only obstruction to a Markov chain settling down is an eigenvalue of modulus 1 other than 1 itself — periodicity, in the language of probability. Chapter 8's warning example, the swap  $\begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}$ , has the eigenvalue  $-1$  and is exactly this obstruction.

#### Example. (A Two-State Chain)

Let  $\mathbf{A} = \begin{pmatrix} 3/4 & 1/2 \\ 1/4 & 1/2 \end{pmatrix}$ . Decide whether  $\mathbf{A}^m$  converges and find the limit.

*Solution.*  $\mathbf{A}$  is stochastic: the entries are non-negative, and each column sums to 1. Its characteristic polynomial is

$$p_{\mathbf{A}} = (x - \frac{3}{4})(x - \frac{1}{2}) - \frac{1}{8} = x^2 - \frac{5}{4}x + \frac{1}{4} = (x - 1)(x - \frac{1}{4}),$$

so  $\text{spec}(\mathbf{A}) = \{1, \frac{1}{4}\}$  and the only eigenvalue of modulus 1 is 1. By [Corollary 9.10.5](#) (b),  $\mathbf{A}^m$  converges.

For the limit,  $E_1(\mathbf{A}) = \ker(\mathbf{A} - \mathbf{I})$ , and  $\mathbf{A} - \mathbf{I} = \begin{pmatrix} -1/4 & 1/2 \\ 1/4 & -1/2 \end{pmatrix}$  gives  $\text{span}((2, 1))$ ; the steady state is  $\mathbf{v} = (\frac{2}{3}, \frac{1}{3})$ , the multiple of  $(2, 1)$  whose entries sum to 1. Also  $E_{1/4}(\mathbf{A}) = \ker(\mathbf{A} - \frac{1}{4}\mathbf{I})$ , and  $\mathbf{A} - \frac{1}{4}\mathbf{I} = \begin{pmatrix} 1/2 & 1/2 \\ 1/4 & 1/4 \end{pmatrix}$  gives  $\text{span}((1, -1))$ . Since the two eigenvalues are distinct,  $G_{1/4}(\mathbf{A}) = E_{1/4}(\mathbf{A})$  and the limit  $\mathbf{\Pi}$  is the projection onto  $\text{span}((2, 1))$  along  $\text{span}((1, -1))$ . Computing it in the basis  $\mathcal{C} = ((2, 1), (1, -1))$ : with  $\mathbf{Q} = \begin{pmatrix} 2 & 1 \\ 1 & -1 \end{pmatrix}$ ,  $\det \mathbf{Q} = -3$  and  $\mathbf{Q}^{-1} = \frac{1}{3} \begin{pmatrix} 1 & 1 \\ 1 & -2 \end{pmatrix}$ , so

$$\mathbf{\Pi} = \mathbf{Q} \text{diag}(1, 0) \mathbf{Q}^{-1} = \begin{pmatrix} 2 & 0 \\ 1 & 0 \end{pmatrix} \cdot \frac{1}{3} \begin{pmatrix} 1 & 1 \\ 1 & -2 \end{pmatrix} = \frac{1}{3} \begin{pmatrix} 2 & 2 \\ 1 & 1 \end{pmatrix} = \begin{pmatrix} 2/3 & 2/3 \\ 1/3 & 1/3 \end{pmatrix}.$$

*Check.*  $\mathbf{\Pi} = \mathbf{v}\mathbf{1}^\top$ , as [Theorem 8.11.4](#) predicts;  $\mathbf{\Pi}^2 = \mathbf{\Pi}$  since  $\mathbf{1}^\top \mathbf{v} = 1$ ; the columns of  $\mathbf{\Pi}$  are the steady state, so every initial probability vector  $\mathbf{x}_0$  gives  $\mathbf{x}_m = \mathbf{A}^m \mathbf{x}_0 \rightarrow \mathbf{\Pi} \mathbf{x}_0 = \mathbf{v}(\mathbf{1}^\top \mathbf{x}_0) = \mathbf{v}$ . And  $\mathbf{A}^2 = \begin{pmatrix} 11/16 & 5/8 \\ 5/16 & 3/8 \end{pmatrix}$  is already close to  $\mathbf{\Pi}$ , the error decaying like  $4^{-m}$ .

## 9.10.6 The geometric series of a matrix

The scalar identity  $\sum_k a^k = (1 - a)^{-1}$  for  $|a| < 1$  has a matrix version, with the same proof and the same hypothesis read off the eigenvalues.

### Theorem 9.10.6 — Neumann Series

Let  $\mathbf{A} \in M_n(\mathbb{C})$ ,  $n \geq 1$ . Then the series  $\sum_{k \geq 0} \mathbf{A}^k$  converges if and only if every eigenvalue  $\lambda$  of  $\mathbf{A}$  satisfies  $|\lambda| < 1$ . In that case  $\mathbf{I} - \mathbf{A}$  is invertible and

$$\sum_{k=0}^{\infty} \mathbf{A}^k = (\mathbf{I} - \mathbf{A})^{-1}.$$

*Idea.* Telescoping gives  $(\mathbf{I} - \mathbf{A})(\mathbf{I} + \mathbf{A} + \cdots + \mathbf{A}^K) = \mathbf{I} - \mathbf{A}^{K+1}$ , an exact identity for every  $K$ . If  $\mathbf{I} - \mathbf{A}$  is invertible, solve it for the partial sum and let  $K \rightarrow \infty$ , where [Theorem 9.10.2](#) kills  $\mathbf{A}^{K+1}$ . Conversely, the terms of a convergent series tend to 0, which is the hypothesis of that theorem again.

*Proof.* ( $\Leftarrow$ ) Suppose  $|\lambda| < 1$  for every  $\lambda \in \text{spec}(\mathbf{A})$ . In particular  $1 \notin \text{spec}(\mathbf{A})$ , so  $0 \notin \text{spec}(\mathbf{I} - \mathbf{A})$  — a vector with  $(\mathbf{I} - \mathbf{A})\mathbf{v} = \mathbf{0}$  is a vector with  $\mathbf{A}\mathbf{v} = \mathbf{v}$  — and  $\mathbf{I} - \mathbf{A}$  is



invertible (Theorem 8.2.6). Put  $\mathbf{S}_K = \sum_{k=0}^K \mathbf{A}^k$ . Multiplying out and canceling,

$$(\mathbf{I} - \mathbf{A})\mathbf{S}_K = \sum_{k=0}^K \mathbf{A}^k - \sum_{k=1}^{K+1} \mathbf{A}^k = \mathbf{I} - \mathbf{A}^{K+1},$$

hence  $\mathbf{S}_K = (\mathbf{I} - \mathbf{A})^{-1}(\mathbf{I} - \mathbf{A}^{K+1})$ . By Theorem 9.10.2,  $\mathbf{A}^{K+1} \rightarrow 0$ . Each entry of  $\mathbf{S}_K$  is a fixed linear combination of the entries of  $\mathbf{I} - \mathbf{A}^{K+1}$ , with coefficients from  $(\mathbf{I} - \mathbf{A})^{-1}$ , so by (B3) it converges to the corresponding combination for  $\mathbf{I}$ , that is,  $\mathbf{S}_K \rightarrow (\mathbf{I} - \mathbf{A})^{-1}\mathbf{I} = (\mathbf{I} - \mathbf{A})^{-1}$ .

( $\Rightarrow$ ) Suppose  $\sum_k \mathbf{A}^k$  converges. Fix a position  $(r, s)$ ; the series of complex numbers  $\sum_k (\mathbf{A}^k)_{rs}$  converges, so  $(\mathbf{A}^k)_{rs} \rightarrow 0$  by (B4). This holds at every position, so  $\mathbf{A}^k \rightarrow 0$ , and Theorem 9.10.2 gives  $|\lambda| < 1$  for every eigenvalue. This proves the theorem. ■

### Example. (Inverting by Summing a Series)

Let  $\mathbf{A} = \begin{pmatrix} 0 & 1/2 \\ 1/2 & 0 \end{pmatrix}$ . Verify that  $\sum_k \mathbf{A}^k$  converges, sum it, and compare with  $(\mathbf{I} - \mathbf{A})^{-1}$ .

*Solution.*  $p_{\mathbf{A}} = x^2 - \frac{1}{4}$ , so  $\text{spec}(\mathbf{A}) = \{\frac{1}{2}, -\frac{1}{2}\}$  and both eigenvalues have modulus  $\frac{1}{2} < 1$ . By Theorem 9.10.6 the series converges.

*Summing it.*  $\mathbf{A}^2 = \frac{1}{4}\mathbf{I}$ , so  $\mathbf{A}^{2l} = 4^{-l}\mathbf{I}$  and  $\mathbf{A}^{2l+1} = 4^{-l}\mathbf{A}$ . Grouping the partial sums by parity, which is legitimate because each entry sequence splits into two convergent pieces (B3),

$$\sum_{k \geq 0} \mathbf{A}^k = \left( \sum_{l \geq 0} 4^{-l} \right) (\mathbf{I} + \mathbf{A}) = \frac{1}{1 - \frac{1}{4}} (\mathbf{I} + \mathbf{A}) = \frac{4}{3} \begin{pmatrix} 1 & 1/2 \\ 1/2 & 1 \end{pmatrix} = \begin{pmatrix} 4/3 & 2/3 \\ 2/3 & 4/3 \end{pmatrix},$$

the scalar sum being the case  $n = 1$  of Theorem 9.10.6 applied to the  $1 \times 1$  matrix  $(\frac{1}{4})$ .

*The inverse.*  $\mathbf{I} - \mathbf{A} = \begin{pmatrix} 1 & -1/2 \\ -1/2 & 1 \end{pmatrix}$  has determinant  $1 - \frac{1}{4} = \frac{3}{4}$ , so by Theorem 0.9.8

$$(\mathbf{I} - \mathbf{A})^{-1} = \frac{4}{3} \begin{pmatrix} 1 & 1/2 \\ 1/2 & 1 \end{pmatrix},$$

in agreement.

*Remark.* The first thing every criterion in this section looks at is the largest modulus of an eigenvalue,

$$\rho(\mathbf{A}) = \max\{|\lambda| : \lambda \in \text{spec}(\mathbf{A})\},$$

called the **spectral radius** of  $\mathbf{A}$ . Two of the criteria need nothing else:  $\mathbf{A}^m \rightarrow 0$  if and only if  $\rho(\mathbf{A}) < 1$ , and  $\sum_k \mathbf{A}^k$  converges if and only if  $\rho(\mathbf{A}) < 1$ . The other two need more when  $\rho(\mathbf{A}) = 1$ : boundedness also asks for the block sizes at modulus 1, and convergence asks in addition that 1 be the only eigenvalue of modulus 1. Chapter 15 measures matrices by norms, proves that the spectral radius never exceeds the norm of  $\mathbf{A}$  induced by any vector norm, and recovers  $\rho(\mathbf{A})$  as a limit built from the norms of the

powers  $\mathbf{A}^m$ ; that is what turns these exact criteria into usable estimates.

## 9.10.7 Exercises

### 9.10.7.1 A. Check your understanding

#### Exercise A1.

- Define entrywise convergence and entrywise boundedness for a sequence of matrices.
- Write down the entries of  $\mathbf{J}_k(\lambda)^m$ , and say which one shows that a block of size  $\geq 2$  with  $|\lambda| = 1$  has unbounded powers.
- State the three criteria:  $\mathbf{A}^m \rightarrow 0$ ,  $(\mathbf{A}^m)$  bounded,  $(\mathbf{A}^m)$  convergent.
- True or false: if every entry of  $\mathbf{A}$  has modulus less than 1, then  $\mathbf{A}^m \rightarrow 0$ . Justify your answer.
- True or false: if  $(\mathbf{A}^m)$  is bounded then it converges. Justify your answer.
- For which  $\mathbf{A}$  does  $\sum_k \mathbf{A}^k$  converge, and what is the sum?

*Solution.*

- $\mathbf{M}_m \rightarrow \mathbf{M}$  if  $(\mathbf{M}_m)_{rs} \rightarrow \mathbf{M}_{rs}$  for every position  $(r, s)$ ;  $(\mathbf{M}_m)$  is bounded if there is one constant  $c$  with  $|(\mathbf{M}_m)_{rs}| \leq c$  for all  $m$  and all  $(r, s)$  (Definition 9.10.1).
- $(\mathbf{J}_k(\lambda)^m)_{r, r+j} = \binom{m}{j} \lambda^{m-j}$  for  $0 \leq j \leq k-1$ , with zeros below the diagonal (Lemma 9.9.1 with  $p = x^m$ ). The entry with  $j = 1$  is  $m\lambda^{m-1}$ , of modulus  $m$  when  $|\lambda| = 1$ .
- $\mathbf{A}^m \rightarrow 0$  iff every eigenvalue has  $|\lambda| < 1$  (Theorem 9.10.2);  $(\mathbf{A}^m)$  is bounded iff every eigenvalue has  $|\lambda| \leq 1$  and those with  $|\lambda| = 1$  have only  $1 \times 1$  blocks (Theorem 9.10.3);  $(\mathbf{A}^m)$  converges iff every eigenvalue has  $|\lambda| < 1$  or equals 1 with only  $1 \times 1$  blocks (Theorem 9.10.4).
- False.  $\mathbf{A} = \begin{pmatrix} 0.9 & 0.9 \\ 0.9 & 0.9 \end{pmatrix}$  has the eigenvalue 1.8, and  $\mathbf{A}^m \mathbf{1} = 1.8^m \mathbf{1}$  is unbounded; see the warning after Theorem 9.10.2.
- False.  $\mathbf{A} = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}$  satisfies  $\mathbf{A}^2 = \mathbf{I}$ , so  $\mathbf{A}^m$  alternates between  $\mathbf{I}$  and  $\mathbf{A}$  and is bounded but not convergent. Its eigenvalues are  $\pm 1$ , and the eigenvalue  $-1$  is what Theorem 9.10.4 excludes.
- Exactly when every eigenvalue has  $|\lambda| < 1$ , and then the sum is  $(\mathbf{I} - \mathbf{A})^{-1}$  (Theorem 9.10.6).

### 9.10.7.2 B. Practice

#### Exercise B1: Decide the behavior.

For each matrix over  $\mathbb{C}$ , decide whether  $\mathbf{A}^m \rightarrow 0$ , whether  $(\mathbf{A}^m)$  is bounded, and whether  $(\mathbf{A}^m)$  converges. Justify your answer.

- (a)  $\mathbf{A}_1 = \begin{pmatrix} 1/2 & 1 \\ 0 & 1/2 \end{pmatrix}$ .
- (b)  $\mathbf{A}_2 = \begin{pmatrix} 0 & -1 \\ 1 & 0 \end{pmatrix}$ .
- (c)  $\mathbf{A}_3 = \mathbf{J}_2(1) \oplus \mathbf{J}_1(\frac{1}{2}) \in M_3(\mathbb{C})$ .
- (d)  $\mathbf{A}_4 = \begin{pmatrix} 0 & 0 & 5 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{pmatrix}$ .

*Solution.*

- (a) The only eigenvalue is  $\frac{1}{2}$ , of modulus  $< 1$ , so  $\mathbf{A}_1^m \rightarrow 0$  by [Theorem 9.10.2](#); hence it is bounded and convergent. (The superdiagonal 1 is irrelevant: by [Lemma 9.9.1](#) the  $(1, 2)$ -entry of  $\mathbf{A}_1^m$  is  $m 2^{-(m-1)} \rightarrow 0$  by (B1).)
- (b)  $p_{\mathbf{A}_2} = x^2 + 1$ , so  $\text{spec}(\mathbf{A}_2) = \{i, -i\}$ , both of modulus 1, and the eigenvalues are distinct, so every block has size 1. By [Theorem 9.10.3](#),  $(\mathbf{A}_2^m)$  is **bounded**; it does not tend to 0 ([Theorem 9.10.2](#)) and does not converge, since the eigenvalue  $i$  has modulus 1 and is not 1 ([Theorem 9.10.4](#)). Concretely  $\mathbf{A}_2^4 = \mathbf{I}$ , so the powers cycle through four matrices.
- (c) The eigenvalue 1 has a block of size 2, so [Theorem 9.10.3](#) (ii) fails:  $(\mathbf{A}_3^m)$  is **unbounded**, hence neither convergent nor tending to 0. The  $(1, 2)$ -entry of  $\mathbf{A}_3^m$  is  $m$ .
- (d)  $\mathbf{A}_4^2 = 0$ , so  $\mathbf{A}_4^m = 0$  for  $m \geq 2$  and  $\mathbf{A}_4^m \rightarrow 0$  trivially: all eigenvalues are 0. The large entry 5 does not matter, illustrating the warning after [Theorem 9.10.2](#).

### Exercise B2: Computing limits.

- (a) Let  $\mathbf{A} = \begin{pmatrix} 1/2 & 1/2 \\ 1/2 & 1/2 \end{pmatrix}$ . Show that  $\mathbf{A}^m$  converges and find the limit, both by [Theorem 9.10.4](#) and by a direct computation of  $\mathbf{A}^m$ .
- (b) Let  $\mathbf{B} = \begin{pmatrix} 1 & 1/3 \\ 0 & 1/3 \end{pmatrix}$ . Find  $\lim_m \mathbf{B}^m$  and identify it as a projection.

*Solution.*

- (a)  $p_{\mathbf{A}} = x^2 - x = x(x-1)$ , so  $\text{spec}(\mathbf{A}) = \{0, 1\}$ . The eigenvalue 0 has  $|0| < 1$  and the eigenvalue 1 has  $a_{\mathbf{A}}(1) = 1$ , so its single block has size 1; by [Theorem 9.10.4](#) the powers converge, and the limit is the projection onto  $E_1(\mathbf{A})$  along  $G_0(\mathbf{A}) = E_0(\mathbf{A})$ . Now  $\mathbf{A} - \mathbf{I} = \begin{pmatrix} -1/2 & 1/2 \\ 1/2 & -1/2 \end{pmatrix}$  gives  $E_1(\mathbf{A}) = \text{span}((1, 1))$  and  $E_0(\mathbf{A}) = \ker \mathbf{A} = \text{span}((1, -1))$ . With  $\mathbf{Q} = \begin{pmatrix} 1 & 1 \\ 1 & -1 \end{pmatrix}$  and  $\mathbf{Q}^{-1} = \frac{1}{2} \begin{pmatrix} 1 & 1 \\ 1 & -1 \end{pmatrix}$ ,

$$\lim_m \mathbf{A}^m = \mathbf{Q} \text{diag}(1, 0) \mathbf{Q}^{-1} = \frac{1}{2} \begin{pmatrix} 1 & 1 \\ 1 & 1 \end{pmatrix} = \mathbf{A}.$$

*Directly.*  $\mathbf{A}^2 = \mathbf{A}$ , so  $\mathbf{A}^m = \mathbf{A}$  for every  $m \geq 1$  and the limit is  $\mathbf{A}$ . ( $\mathbf{A}$  is itself a projection, and the theorem returns it unchanged.)

- (b)  $\mathbf{B}$  is upper triangular, so  $\text{spec}(\mathbf{B}) = \{1, \frac{1}{3}\}$  (Theorem 6.4.6). Both criteria hold:  $|\frac{1}{3}| < 1$ , and  $a_{\mathbf{B}}(1) = 1$  so the 1-block has size 1. Solving  $(\mathbf{B} - \mathbf{I})\mathbf{x} = \mathbf{0}$  with  $\mathbf{B} - \mathbf{I} = \begin{pmatrix} 0 & 1/3 \\ 0 & -2/3 \end{pmatrix}$  gives  $E_1(\mathbf{B}) = \text{span}(\mathbf{e}_1)$ ; solving  $(\mathbf{B} - \frac{1}{3}\mathbf{I})\mathbf{x} = \mathbf{0}$  with  $\mathbf{B} - \frac{1}{3}\mathbf{I} = \begin{pmatrix} 2/3 & 1/3 \\ 0 & 0 \end{pmatrix}$  gives  $E_{1/3}(\mathbf{B}) = \text{span}((1, -2))$ . With  $\mathbf{Q} = \begin{pmatrix} 1 & 1 \\ 0 & -2 \end{pmatrix}$  we have  $\det \mathbf{Q} = -2$  and  $\mathbf{Q}^{-1} = \frac{1}{-2} \begin{pmatrix} -2 & -1 \\ 0 & 1 \end{pmatrix} = \begin{pmatrix} 1 & 1/2 \\ 0 & -1/2 \end{pmatrix}$ . Multiplying in two steps,  $\mathbf{Q} \text{diag}(1, 0) = \begin{pmatrix} 1 & 0 \\ 0 & 0 \end{pmatrix}$  and then

$$\lim_m \mathbf{B}^m = \begin{pmatrix} 1 & 0 \\ 0 & 0 \end{pmatrix} \mathbf{Q}^{-1} = \begin{pmatrix} 1 & 1/2 \\ 0 & 0 \end{pmatrix}.$$

This is the projection onto  $\text{span}(\mathbf{e}_1)$  along  $\text{span}((1, -2))$ : it fixes  $\mathbf{e}_1$  and sends  $(1, -2)$  to  $(1 - 1, 0) = \mathbf{0}$ . *Check.*  $\mathbf{B}^m = \begin{pmatrix} 1 & c_m \\ 0 & 3^{-m} \end{pmatrix}$  with  $c_1 = \frac{1}{3}$  and  $c_{m+1} = \frac{1}{3}(1 + c_m)$ , read off  $\mathbf{B}^{m+1} = \mathbf{B}^m \mathbf{B}$ ; the first values are  $\frac{1}{3}, \frac{4}{9}, \frac{13}{27}$ , climbing towards the fixed point  $c = \frac{1}{3}(1 + c)$ , that is  $c = \frac{1}{2}$ .

### Exercise B3: A Neumann series.

Let  $\mathbf{A} = \begin{pmatrix} 1/3 & 1/3 \\ 0 & 1/3 \end{pmatrix}$ .

- Explain why  $\sum_k \mathbf{A}^k$  converges.
- Compute  $(\mathbf{I} - \mathbf{A})^{-1}$  directly.
- Compute the partial sum  $\mathbf{I} + \mathbf{A} + \mathbf{A}^2$  and compare it with (b).

*Solution.*

- $\mathbf{A}$  is upper triangular with both diagonal entries  $\frac{1}{3}$ , so  $\text{spec}(\mathbf{A}) = \{\frac{1}{3}\}$  (Theorem 6.4.6) and  $|\frac{1}{3}| < 1$ . By Theorem 9.10.6 the series converges, with sum  $(\mathbf{I} - \mathbf{A})^{-1}$ .
- $\mathbf{I} - \mathbf{A} = \begin{pmatrix} 2/3 & -1/3 \\ 0 & 2/3 \end{pmatrix}$ , of determinant  $\frac{4}{9}$ . By Theorem 0.9.8,

$$(\mathbf{I} - \mathbf{A})^{-1} = \frac{9}{4} \begin{pmatrix} 2/3 & 1/3 \\ 0 & 2/3 \end{pmatrix} = \begin{pmatrix} 3/2 & 3/4 \\ 0 & 3/2 \end{pmatrix}.$$

- Write  $\mathbf{A} = \frac{1}{3}(\mathbf{I} + \mathbf{M})$  with  $\mathbf{M} = \begin{pmatrix} 0 & 1 \\ 0 & 0 \end{pmatrix}$ ,  $\mathbf{M}^2 = \mathbf{0}$ ; then  $\mathbf{A}^k = 3^{-k}(\mathbf{I} + k\mathbf{M})$  by the binomial theorem, the two summands commuting. So

$$\mathbf{I} + \mathbf{A} + \mathbf{A}^2 = (1 + \frac{1}{3} + \frac{1}{9})\mathbf{I} + (\frac{1}{3} + \frac{2}{9})\mathbf{M} = \frac{13}{9}\mathbf{I} + \frac{5}{9}\mathbf{M} = \begin{pmatrix} 13/9 & 5/9 \\ 0 & 13/9 \end{pmatrix}.$$

Summing the whole series the same way,  $(\mathbf{I} - \mathbf{A})^{-1} = (\sum_{k \geq 0} 3^{-k})\mathbf{I} + (\sum_{k \geq 0} k 3^{-k})\mathbf{M}$ , so comparing with (b) reads off  $\sum_k 3^{-k} = \frac{3}{2}$  and  $\sum_k k 3^{-k} = \frac{3}{4}$ . The two partial sums  $\frac{13}{9}$  and  $\frac{5}{9}$ , about 1.44 and 0.56, are on their way to  $\frac{3}{2}$  and  $\frac{3}{4}$ ; the convergence is geometric, as the factor  $3^{-k}$  shows.

## 9.10.7.3 C. Going deeper

**Exercise C1: Bounded but not convergent.**

- (a) Give an example of  $\mathbf{A} \in M_2(\mathbb{R})$  with  $(\mathbf{A}^m)$  bounded but not convergent, and one with  $(\mathbf{A}^m)$  convergent but  $\mathbf{A}^m \not\rightarrow 0$ .
- (b) Prove that if  $\mathbf{A}^m \rightarrow \mathbf{L}$ , then  $\mathbf{L}^2 = \mathbf{L}$  and  $\mathbf{A}\mathbf{L} = \mathbf{L}\mathbf{A} = \mathbf{L}$ .
- (c) Deduce that if  $\mathbf{A}^m \rightarrow \mathbf{L}$  with  $\mathbf{L}$  **invertible**, then  $\mathbf{L} = \mathbf{I}$  and  $\mathbf{A} = \mathbf{I}$ .
- (d) Give an invertible  $\mathbf{A}$  with  $\mathbf{A}^m \rightarrow \mathbf{L}$  and  $\mathbf{L} \notin \{0, \mathbf{I}\}$ , so that the invertibility of  $\mathbf{L}$ , not of  $\mathbf{A}$ , is what part (c) needs.

*Solution.*

- (a) For the first,  $\mathbf{A} = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}$ :  $\mathbf{A}^2 = \mathbf{I}$ , so  $\mathbf{A}^m$  alternates between  $\mathbf{I}$  and  $\mathbf{A}$  and is bounded but not convergent; its eigenvalues are  $\pm 1$ . For the second,  $\mathbf{A} = \begin{pmatrix} 1 & 0 \\ 0 & 1/2 \end{pmatrix}$ :  $\mathbf{A}^m = \text{diag}(1, 2^{-m}) \rightarrow \text{diag}(1, 0) \neq 0$ .
- (b) Each entry of  $\mathbf{A}^m \mathbf{A}^m = \mathbf{A}^{2m}$  is a sum of products of entries of  $\mathbf{A}^m$ , so by (B3) it converges to the corresponding expression in the entries of  $\mathbf{L}$ ; hence  $\mathbf{A}^{2m} \rightarrow \mathbf{L}^2$ . But  $(\mathbf{A}^{2m})_m$  is a subsequence of  $(\mathbf{A}^m)_m$  — more precisely, for each position the sequence  $(\mathbf{A}^{2m})_{rs}$  is a subsequence of the convergent sequence  $(\mathbf{A}^m)_{rs}$ , so it has the same limit  $\mathbf{L}_{rs}$ . Hence  $\mathbf{L}^2 = \mathbf{L}$ . Similarly  $\mathbf{A}^{m+1} = \mathbf{A}\mathbf{A}^m = \mathbf{A}^m \mathbf{A}$ , and the left side is a tail of the convergent sequence, so it tends to  $\mathbf{L}$ , while the two right sides tend to  $\mathbf{A}\mathbf{L}$  and  $\mathbf{L}\mathbf{A}$  by (B3). Hence  $\mathbf{A}\mathbf{L} = \mathbf{L}\mathbf{A} = \mathbf{L}$ .
- (c) Suppose  $\mathbf{A}^m \rightarrow \mathbf{L}$  with  $\mathbf{L}$  invertible. By (b),  $\mathbf{L}^2 = \mathbf{L}$ ; multiplying by  $\mathbf{L}^{-1}$  gives  $\mathbf{L} = \mathbf{I}$ . Then  $\mathbf{A}\mathbf{L} = \mathbf{L}$  reads  $\mathbf{A} = \mathbf{I}$ . (Conversely  $\mathbf{A} = \mathbf{I}$  gives  $\mathbf{A}^m = \mathbf{I} \rightarrow \mathbf{I}$ , so this case really occurs.)
- (d)  $\mathbf{A} = \text{diag}(1, \frac{1}{2})$  is invertible, and  $\mathbf{A}^m = \text{diag}(1, 2^{-m}) \rightarrow \text{diag}(1, 0) = \mathbf{L}$ , which is neither 0 nor  $\mathbf{I}$  — and not invertible. So no hypothesis on  $\mathbf{A}$  alone forces  $\mathbf{L} \in \{0, \mathbf{I}\}$ : by [Theorem 9.10.4](#) the limit is the projection onto  $E_1(\mathbf{A})$  along the other generalized eigenspaces, and that projection is 0 or  $\mathbf{I}$  only in the extreme cases  $1 \notin \text{spec}(\mathbf{A})$  and  $E_1(\mathbf{A}) = \mathbb{C}^n$ .

**Exercise C2: Chapter 8's Markov theorem without diagonalizability.**

Let  $\mathbf{A} \in M_n(\mathbb{R})$  be stochastic with  $a_{\mathbf{A}}(1) = 1$  and  $|\lambda| < 1$  for every eigenvalue  $\lambda \neq 1$ .

- (a) Prove that  $\mathbf{1}^\top \mathbf{w} = 0$  for every  $\mathbf{w} \in G_\lambda(\mathbf{A})$  with  $\lambda \neq 1$ . *Hint:*  $\mathbf{1}^\top (\mathbf{A} - \lambda \mathbf{I}) = (1 - \lambda) \mathbf{1}^\top$ .
- (b) Prove that  $(\mathbf{A}^m)$  converges to a **stochastic** matrix  $\mathbf{\Pi}$ , and deduce that  $\mathbf{A}$  has a steady state  $\mathbf{v}$  with  $E_1(\mathbf{A}) = \text{span}(\mathbf{v})$ .
- (c) Deduce that  $\mathbf{\Pi} = \mathbf{v} \mathbf{1}^\top$  and that  $\mathbf{A}^m \mathbf{x}_0 \rightarrow \mathbf{v}$  for every probability vector  $\mathbf{x}_0$ , **without** assuming that  $\mathbf{A}$  is diagonalizable.

*Solution.*

- (a) Since  $\mathbf{A}$  is stochastic,  $\mathbf{1}^\top \mathbf{A} = \mathbf{1}^\top$  ([Proposition 8.11.3](#) (c)), so  $\mathbf{1}^\top (\mathbf{A} - \lambda \mathbf{I}) = \mathbf{1}^\top - \lambda \mathbf{1}^\top = (1 - \lambda) \mathbf{1}^\top$ . Iterating  $n$  times,  $\mathbf{1}^\top (\mathbf{A} - \lambda \mathbf{I})^n = (1 - \lambda)^n \mathbf{1}^\top$ . Let  $\mathbf{w} \in G_\lambda(\mathbf{A})$ ,

so  $(\mathbf{A} - \lambda \mathbf{I})^n \mathbf{w} = \mathbf{0}$  (Corollary 9.1.2 (a)). Then

$$0 = \mathbf{1}^\top (\mathbf{A} - \lambda \mathbf{I})^n \mathbf{w} = (1 - \lambda)^n \mathbf{1}^\top \mathbf{w},$$

and  $\lambda \neq 1$  gives  $(1 - \lambda)^n \neq 0$ , so  $\mathbf{1}^\top \mathbf{w} = 0$ .

- (b) By hypothesis 1 is the only eigenvalue of modulus 1, so  $(\mathbf{A}^m)$  converges by Corollary 9.10.5 (b); call the limit  $\mathbf{\Pi}$ . Every  $\mathbf{A}^m$  is stochastic (Proposition 8.11.3 (b)), so all its entries are  $\geq 0$  and each of its columns sums to 1. Both properties pass to the limit: a limit of real numbers  $\geq 0$  is  $\geq 0$ , and a column sum is a finite sum of entries, so it converges to the sum of the limits (Chapter 8 §11). Hence  $\mathbf{\Pi}$  is stochastic, and in particular  $\mathbf{\Pi} \neq \mathbf{0}$ .

By Theorem 9.10.4,  $\mathbf{\Pi}$  is the projection onto  $E_1(\mathbf{A})$  along  $\bigoplus_{\lambda \neq 1} G_\lambda(\mathbf{A})$ , so  $\text{im } T_{\mathbf{\Pi}} = E_1(\mathbf{A})$ . Since  $a_{\mathbf{A}}(1) = 1$  and  $1 \leq g_{\mathbf{A}}(1) \leq a_{\mathbf{A}}(1)$  (Theorem 8.3.1),  $E_1(\mathbf{A})$  is a line. Each column of  $\mathbf{\Pi}$  lies in  $\text{im } T_{\mathbf{\Pi}} = E_1(\mathbf{A})$  (Definition 0.9.9: column  $j$  is  $\mathbf{\Pi} \mathbf{e}_j$ ) and is a probability vector, hence is fixed by  $\mathbf{A}$  and is a steady state. Any two probability vectors on the same line are equal: if  $\mathbf{w} = c\mathbf{v}$  with both entry sums equal to 1, then  $1 = c$ . So all the columns of  $\mathbf{\Pi}$  are one and the same steady state  $\mathbf{v}$ , and  $E_1(\mathbf{A}) = \text{span}(\mathbf{v})$ .

- (c) All columns of  $\mathbf{\Pi}$  equal  $\mathbf{v}$ , which is exactly the statement  $\mathbf{\Pi} = \mathbf{v} \mathbf{1}^\top$ . (Independently of (b), part (a) confirms it:  $\mathbf{v} \mathbf{1}^\top$  fixes  $\mathbf{v}$  because  $\mathbf{1}^\top \mathbf{v} = 1$ , and kills every  $\mathbf{w} \in G_\lambda(\mathbf{A})$  with  $\lambda \neq 1$  because  $\mathbf{1}^\top \mathbf{w} = 0$ ; by Theorem 9.1.3 (a) those two conditions determine the projection onto  $E_1(\mathbf{A}) = G_1(\mathbf{A})$  along the rest, using  $\dim G_1(\mathbf{A}) = a_{\mathbf{A}}(1) = 1 = \dim E_1(\mathbf{A})$  and Theorem 1.6.8.) For a probability vector  $\mathbf{x}_0$ , each entry of  $\mathbf{A}^m \mathbf{x}_0$  is a fixed linear combination of entries of  $\mathbf{A}^m$ , so by (B3)

$$\mathbf{A}^m \mathbf{x}_0 \longrightarrow \mathbf{\Pi} \mathbf{x}_0 = \mathbf{v} (\mathbf{1}^\top \mathbf{x}_0) = \mathbf{v} (x_{01} + \cdots + x_{0n}) = \mathbf{v}.$$

No diagonalizability was used: the blocks for the eigenvalues  $\lambda \neq 1$  may have any sizes, since Theorem 9.10.2 does not care. This removes the diagonalizability hypothesis from Theorem 8.11.4.

# Chapter 10

## Inner Product Spaces

Nine chapters have described vector spaces without measuring anything in them. We can say that a list spans, that a subspace has dimension four, that two matrices are similar; we cannot say that a vector has length 3, that two vectors are perpendicular, or that one vector is the closest point of a plane to another. Every picture in the book so far has secretly used lengths and angles that the theory did not supply. This chapter supplies them, with a single piece of extra structure: an **inner product**, abstracted from the dot product on  $\mathbb{R}^n$  by keeping exactly the properties that make the law of cosines work.

The gain is out of all proportion to the cost. Length gives the Cauchy–Schwarz and triangle inequalities. Perpendicularity gives orthonormal bases, in which coordinates are read off by a single inner product and Gram–Schmidt produces one from any basis at all. Splitting a space into a subspace and its orthogonal complement gives the nearest point of a subspace to a given vector, and with it least squares, the standard answer to an inconsistent system. The Riesz representation theorem pays off a debt from Chapter 4: an inner product is precisely the extra data that makes  $V$  and  $V^*$  the same space. From Riesz comes the **adjoint**  $T^*$ , which turns the four fundamental subspaces into two orthogonal decompositions, and the isometries, which are the maps that preserve the whole geometry. The chapter ends with the computational and classical faces of the same ideas: the QR factorization, Fourier series and orthogonal polynomials. **Throughout, the field is  $\mathbb{R}$  or  $\mathbb{C}$** , because the construction needs both an ordering, so that a length squared can be positive, and complex conjugation, so that it can be real.

**What you need.** From Chapter 0, the complex conjugate and modulus with their arithmetic (Definition 0.6.2, Theorem 0.6.2) and the trace (Definition 0.9.12, Theorem 0.9.6). From Chapter 1, independence, bases, dimension, basis extension and direct sums (Definition 1.4.1, Theorem 1.6.4, Theorem 1.6.5, Theorem 1.7.4). From Chapter 2, the Invertible Matrix Theorem (Theorem 2.4.4). From Chapter 3, the matrix of a linear map, the coordinate isomorphism and Rank–Nullity (Definition 3.6.1, Corollary 3.5.5, Theorem 3.3.1). From Chapter 4, the dual space, the annihilator and the dual map, together with the warning that an isomorphism  $V \cong V^*$  needs a choice (Definition 4.1.2, Definition 4.2.1, Definition 4.4.1, Theorem 4.4.2). From Chapter 5, the root bound and Lagrange interpolation (Corollary 5.1.4, Theorem 5.6.1). From Chapter 6, multiplicativity of the determinant and the non-negativity of Gram determinants (Theorem 6.5.4, Corollary 6.7.3). Chapter 8 is used only in examples and in one classification (Definition 8.2.1); nothing before Section 7 needs an eigenvalue.

## 10.0.1 Roadmap

- **Inner products and norms.** The axioms, abstracted from the dot product; length, orthogonality, Pythagoras, Cauchy–Schwarz and the triangle inequality.
- **Orthonormal bases and Gram–Schmidt.** Coordinates that cost one inner product each, the process that builds such a basis from any basis, and Gram matrices.
- **Orthogonal complements and projections.** The splitting  $V = U \oplus U^\perp$ , the projection  $P_U$ , and the fact that  $P_U \mathbf{v}$  is the point of  $U$  nearest to  $\mathbf{v}$ .
- **Least squares and minimum-norm solutions.** What to do with an inconsistent system: the normal equations, and the smallest solution of an underdetermined one.
- **The Riesz representation theorem.** Every linear functional is an inner product with one fixed vector, which is the chosen isomorphism  $V \cong V^*$  that Chapter 4 could not supply.
- **The adjoint.** Moving an operator to the other side of an inner product, the matrix  $\mathbf{A}^*$ , and the four fundamental subspaces in orthogonal form.
- **Isometries and unitary maps.** The maps that preserve all lengths, their many equivalent descriptions, and the classification of  $O(2)$ .
- **QR, Householder and Hessenberg.** Gram–Schmidt as a matrix factorization, reflections that do the same job more robustly, and the first step of a serious eigenvalue algorithm.
- **Bessel, Parseval and Fourier series.** What an orthonormal list captures of a vector, and the classical Fourier coefficients as an instance.
- **Orthogonal polynomials.** Gram–Schmidt on  $1, x, x^2, \dots$ , the three-term recurrence, real roots, and Gaussian quadrature.

## 10.0.2 Named moves

- **To show a vector is zero, pair it with itself.** Only  $\mathbf{0}$  has  $\langle \mathbf{z}, \mathbf{z} \rangle = 0$ , so an identity of that form ends the argument.
- **To move  $T$  across an inner product, pay with  $T^*$ .** The adjoint is the price of the transfer, and most proofs about special operators compute one inner product in two ways.
- **Work with the norm squared, then expand.** Square roots do not expand;  $\|\mathbf{u} + \mathbf{v}\|^2 = \langle \mathbf{u} + \mathbf{v}, \mathbf{u} + \mathbf{v} \rangle$  does.
- **Fix an orthonormal basis.** In its coordinates every inner product becomes the dot product, and an abstract statement becomes a matrix statement.
- **Project, then subtract.** The same move orthogonalizes a list (Gram–Schmidt), finds the nearest point of a subspace, and solves a least-squares problem.

# 10.1 Inner Products and Norms

Nine chapters, and we have not measured anything. We can say that a list is independent, that a subspace has dimension four, that an operator is diagonalizable. We cannot yet say that a vector has length 3, or that two vectors meet at a right angle. Geometry has appeared only in pictures, never in the theory. This section puts it into the theory, by abstracting the one operation in  $\mathbb{R}^n$  that already computes lengths and angles: the dot product.



## 10.1.1 The field for this chapter

**Throughout Chapter 10 the field  $F$  is  $\mathbb{R}$  or  $\mathbb{C}$ .** This is not laziness; two features of these fields are used on every page, and no other field has both.

The first is **positivity**. The point of the whole construction is that  $\langle \mathbf{v}, \mathbf{v} \rangle$  should be the square of a length, so it must be a number we can compare with 0. Over  $\mathbb{F}_p$  there is no ordering compatible with the arithmetic at all: in  $\mathbb{F}_5$ , adding 1 to itself five times returns to 0, so no non-zero element can be called “positive” in any useful sense. Over  $\mathbb{Q}$  there is an ordering, but lengths would not exist, since  $\sqrt{2} \notin \mathbb{Q}$ . Over  $\mathbb{R}$  every sum of squares is  $\geq 0$  and every non-negative number has a square root, and that is exactly what we need.

The second is **conjugation**. Over  $\mathbb{C}$  the naive sum of squares fails immediately:  $1^2 + i^2 = 0$ , so “ $z_1^2 + z_2^2$ ” is not a length for  $(1, i) \neq \mathbf{0}$ . The repair is to use  $z\bar{z} = |z|^2$  instead of  $z^2$  ([Theorem 0.6.2 \(c\)](#)), and this needs the complex conjugate. We write  $\bar{a}$  for the conjugate of  $a \in F$ , with the convention that  $\bar{a} = a$  when  $F = \mathbb{R}$ ; every formula below then reads correctly over both fields at once.

## 10.1.2 From the dot product to the axioms

In  $\mathbb{R}^n$  the dot product is

$$\mathbf{x} \cdot \mathbf{y} = x_1 y_1 + \cdots + x_n y_n.$$

It computes lengths, because  $\mathbf{x} \cdot \mathbf{x} = x_1^2 + \cdots + x_n^2$  is the square of the Euclidean length by Pythagoras. It computes angles too. For non-zero  $\mathbf{x}, \mathbf{y} \in \mathbb{R}^2$  at an angle  $\theta$ , the law of cosines applied to the triangle with sides  $\mathbf{x}$ ,  $\mathbf{y}$  and  $\mathbf{x} - \mathbf{y}$  says

$$\|\mathbf{x} - \mathbf{y}\|^2 = \|\mathbf{x}\|^2 + \|\mathbf{y}\|^2 - 2\|\mathbf{x}\|\|\mathbf{y}\|\cos\theta,$$

while expanding the left side entry by entry gives  $\|\mathbf{x}\|^2 + \|\mathbf{y}\|^2 - 2(\mathbf{x} \cdot \mathbf{y})$ . Comparing the two,  $\mathbf{x} \cdot \mathbf{y} = \|\mathbf{x}\|\|\mathbf{y}\|\cos\theta$ . So one bilinear expression carries both length and angle, and perpendicularity is the single equation  $\mathbf{x} \cdot \mathbf{y} = 0$ .

Which properties of the dot product did that argument use? Only three: it is linear in each slot separately, it is symmetric, and  $\mathbf{x} \cdot \mathbf{x} > 0$  for  $\mathbf{x} \neq \mathbf{0}$ . Those three we can demand of an abstract space. Over  $\mathbb{C}$ , as we saw, symmetry has to be weakened to conjugate symmetry, and the price is small: a conjugate bar in the second slot.

*An inner product is a way of multiplying two vectors into a scalar so that a vector paired with itself has positive “length squared”.*

### Definition 10.1.1 — Inner Product

Let  $F$  be  $\mathbb{R}$  or  $\mathbb{C}$ , and let  $V$  be a vector space **over**  $F$ . An **inner product** on  $V$  is a map

$$\langle \cdot, \cdot \rangle: V \times V \rightarrow F$$

such that for all  $\mathbf{u}, \mathbf{v}, \mathbf{w} \in V$  and all  $a, b \in F$ :

- (IP1)  $\langle a\mathbf{u} + b\mathbf{v}, \mathbf{w} \rangle = a\langle \mathbf{u}, \mathbf{w} \rangle + b\langle \mathbf{v}, \mathbf{w} \rangle$  (**linear in the first slot**);
- (IP2)  $\langle \mathbf{v}, \mathbf{u} \rangle = \overline{\langle \mathbf{u}, \mathbf{v} \rangle}$  (**conjugate symmetry**);
- (IP3)  $\langle \mathbf{v}, \mathbf{v} \rangle > 0$  for **every non-zero**  $\mathbf{v} \in V$  (**positive definiteness**).

A vector space equipped with an inner product is an **inner product space**. When  $F = \mathbb{R}$  it is a **real** inner product space, when  $F = \mathbb{C}$  a **complex** one.

In words, clause by clause. (IP1) says that if we freeze the second slot, what is left is a linear functional on  $V$ : the inner product distributes over sums and lets scalars out of the **first** argument untouched. (IP2) says that swapping the two arguments conjugates the answer; over  $\mathbb{R}$  it is plain symmetry,  $\langle \mathbf{v}, \mathbf{u} \rangle = \langle \mathbf{u}, \mathbf{v} \rangle$ . (IP3) says that a non-zero vector pairs with itself to give a **strictly** positive number. For the inequality in (IP3) to mean anything,  $\langle \mathbf{v}, \mathbf{v} \rangle$  must be a **real** number; it is, and the reason is (IP2), as the next result shows. Note also what (IP3) does **not** say: it says nothing about  $\langle \mathbf{u}, \mathbf{v} \rangle$  for  $\mathbf{u} \neq \mathbf{v}$ , which may be negative, or not even real.

The first consequences are small, and we will use all of them constantly.

### Theorem 10.1.1 — First Properties of an Inner Product

Let  $V$  be an inner product space over  $F$ . For all  $\mathbf{u}, \mathbf{v}, \mathbf{w} \in V$  and  $a, b \in F$ :

- (a)  $\langle \mathbf{v}, \mathbf{v} \rangle$  is a **real** number, and  $\langle \mathbf{v}, \mathbf{v} \rangle \geq 0$  with equality if and only if  $\mathbf{v} = \mathbf{0}$ ;
- (b)  $\langle \mathbf{u}, a\mathbf{v} + b\mathbf{w} \rangle = \bar{a}\langle \mathbf{u}, \mathbf{v} \rangle + \bar{b}\langle \mathbf{u}, \mathbf{w} \rangle$  (**conjugate-linear in the second slot**);
- (c)  $\langle \mathbf{0}, \mathbf{v} \rangle = \langle \mathbf{v}, \mathbf{0} \rangle = 0$ ;
- (d) if  $\langle \mathbf{u}, \mathbf{w} \rangle = \langle \mathbf{v}, \mathbf{w} \rangle$  for **every**  $\mathbf{w} \in V$ , then  $\mathbf{u} = \mathbf{v}$ .

*Idea.* Each part is one application of an axiom. For (a), a number equal to its own conjugate is real. For (b), swap slots with (IP2), use (IP1), and swap back. For (d), the only vector orthogonal to everything is the zero vector, and we find that out by pairing the difference **with itself**: that is the move to remember.

*Proof.*

- (a) By (IP2) with  $\mathbf{u} = \mathbf{v}$ ,  $\langle \mathbf{v}, \mathbf{v} \rangle = \overline{\langle \mathbf{v}, \mathbf{v} \rangle}$ , and a complex number equal to its own conjugate has zero imaginary part, so  $\langle \mathbf{v}, \mathbf{v} \rangle \in \mathbb{R}$ . If  $\mathbf{v} \neq \mathbf{0}$ , then  $\langle \mathbf{v}, \mathbf{v} \rangle > 0$  by (IP3). If  $\mathbf{v} = \mathbf{0}$ , then  $\langle \mathbf{0}, \mathbf{0} \rangle = \langle \mathbf{0} \cdot \mathbf{0}, \mathbf{0} \rangle = 0\langle \mathbf{0}, \mathbf{0} \rangle = 0$  by (IP1). Hence  $\langle \mathbf{v}, \mathbf{v} \rangle \geq 0$  always, with equality exactly when  $\mathbf{v} = \mathbf{0}$ .
- (b) Using (IP2), then (IP1), then (IP2) again,

$$\begin{aligned} \langle \mathbf{u}, a\mathbf{v} + b\mathbf{w} \rangle &= \overline{\langle a\mathbf{v} + b\mathbf{w}, \mathbf{u} \rangle} \\ &= \overline{a\langle \mathbf{v}, \mathbf{u} \rangle + b\langle \mathbf{w}, \mathbf{u} \rangle} \\ &= \bar{a}\overline{\langle \mathbf{v}, \mathbf{u} \rangle} + \bar{b}\overline{\langle \mathbf{w}, \mathbf{u} \rangle} \\ &= \bar{a}\langle \mathbf{u}, \mathbf{v} \rangle + \bar{b}\langle \mathbf{u}, \mathbf{w} \rangle, \end{aligned}$$

where the third equality uses Theorem 0.6.2 (a) and (b).

- (c)  $\langle \mathbf{0}, \mathbf{v} \rangle = \langle \mathbf{0} \cdot \mathbf{0}, \mathbf{v} \rangle = 0\langle \mathbf{0}, \mathbf{v} \rangle = 0$  by (IP1), and  $\langle \mathbf{v}, \mathbf{0} \rangle = \overline{\langle \mathbf{0}, \mathbf{v} \rangle} = 0$  by (IP2).
- (d) Put  $\mathbf{z} = \mathbf{u} - \mathbf{v}$ . For every  $\mathbf{w}$ , (IP1) gives  $\langle \mathbf{z}, \mathbf{w} \rangle = \langle \mathbf{u}, \mathbf{w} \rangle - \langle \mathbf{v}, \mathbf{w} \rangle = 0$ . Take  $\mathbf{w} = \mathbf{z}$ : then  $\langle \mathbf{z}, \mathbf{z} \rangle = 0$ , so  $\mathbf{z} = \mathbf{0}$  by (a), that is,  $\mathbf{u} = \mathbf{v}$ .

■

Part (d) is the first appearance of the move that runs through the whole chapter: **to show a**

**vector is zero, pair it with itself.** There is exactly one vector of length zero, so an identity of the form  $\langle \mathbf{z}, \mathbf{z} \rangle = 0$  is a proof that  $\mathbf{z} = \mathbf{0}$ .

Part (b) is where the conjugate becomes visible in practice. An inner product over  $\mathbb{C}$  is **not** bilinear: scalars leave the second slot wearing a bar. A map with properties (IP1) and (b) is called **sesquilinear**, from the Latin for “one and a half times linear”.

### 10.1.3 Examples

**The dot product on  $\mathbb{R}^n$ .** Define  $\langle \mathbf{x}, \mathbf{y} \rangle = x_1 y_1 + \cdots + x_n y_n$ . (IP1) holds because each term is linear in  $x_i$ ; (IP2) is the commutativity of multiplication in  $\mathbb{R}$ , and conjugation is the identity; (IP3) holds because  $\langle \mathbf{x}, \mathbf{x} \rangle = x_1^2 + \cdots + x_n^2$  is a sum of squares of real numbers, which is 0 only when every  $x_i = 0$ . This is the **standard** inner product on  $\mathbb{R}^n$ , and it is the model for everything else.

**The standard inner product on  $\mathbb{C}^n$ .** Define

$$\langle \mathbf{x}, \mathbf{y} \rangle = x_1 \overline{y_1} + \cdots + x_n \overline{y_n}.$$

(IP1) is clear. For (IP2),  $\overline{\langle \mathbf{x}, \mathbf{y} \rangle} = \sum \overline{x_i y_i} = \sum \overline{x_i} \overline{y_i} = \langle \mathbf{y}, \mathbf{x} \rangle$  by Theorem 0.6.2 (a), (b). For (IP3),  $\langle \mathbf{x}, \mathbf{x} \rangle = \sum x_i \overline{x_i} = \sum |x_i|^2$  by Theorem 0.6.2 (c), a sum of non-negative **real** numbers, zero only when every  $x_i = 0$ . The conjugates are doing exactly one job: making the diagonal values real and non-negative.

**Weighted inner products.** Fix real numbers  $w_1, \dots, w_n > 0$  and put  $\langle \mathbf{x}, \mathbf{y} \rangle = \sum_i w_i x_i \overline{y_i}$  on  $F^n$ . The same three checks go through, with  $\langle \mathbf{x}, \mathbf{x} \rangle = \sum w_i |x_i|^2$ ; the weights must be **strictly** positive, or (IP3) fails for a standard basis vector. In the degenerate case  $n = 1$  and  $w_1 = 1$ , this is  $\langle x, y \rangle = x \overline{y}$  on  $F$  itself, the smallest non-zero inner product space there is. It matters because it is where the axioms are just the arithmetic of  $|\cdot|$ .

**Continuous functions.** Let  $V = C[0, 1]$  be the space of continuous functions  $f: [0, 1] \rightarrow F$ , and set

$$\langle f, g \rangle = \int_0^1 f(t) \overline{g(t)} dt.$$

(IP1) is linearity of the integral, and (IP2) follows because conjugation commutes with integration. For (IP3),  $\langle f, f \rangle = \int_0^1 |f(t)|^2 dt \geq 0$ , and we must rule out equality for  $f \neq 0$ . Here is the one fact from analysis we assume: if  $f$  is continuous and  $f(c) \neq 0$  for some  $c$ , then  $|f| \geq |f(c)|/2$  on some interval of positive length  $\delta$  around  $c$ , so  $\int_0^1 |f|^2 \geq \delta |f(c)|^2/4 > 0$ . Continuity is not decoration; the non-example below shows what happens without it.

**Matrices: the Frobenius inner product.** First a piece of notation we will need all chapter.

#### Definition 10.1.2 — Conjugate Transpose

For  $\mathbf{A} \in M_{m \times n}(F)$ , the **conjugate transpose**  $\mathbf{A}^* \in M_{n \times m}(F)$  is the matrix with entries  $(\mathbf{A}^*)_{ij} = \overline{a_{ji}}$ . A square matrix with  $\mathbf{A}^* = \mathbf{A}$  is **Hermitian**.

Over  $\mathbb{R}$ ,  $\mathbf{A}^* = \mathbf{A}^\top$  and Hermitian means symmetric (Definition 0.9.11). Now on  $V = M_{m \times n}(F)$  define

$$\langle \mathbf{A}, \mathbf{B} \rangle = \text{tr}(\mathbf{B}^* \mathbf{A}).$$

Writing out the trace,  $\text{tr}(\mathbf{B}^* \mathbf{A}) = \sum_j \sum_i \overline{b_{ij}} a_{ij}$ , so this is the standard inner product of  $F^{mn}$  after listing the entries in a column. In particular (IP1)–(IP3) hold, with  $\langle \mathbf{A}, \mathbf{A} \rangle = \sum_{i,j} |a_{ij}|^2$ . This is the **Frobenius inner product**; for  $n = 1$  it is the standard inner product on  $F^m$ .

**Polynomials sampled at nodes.** Fix  $n + 1$  **distinct** scalars  $c_0, \dots, c_n \in F$  and define, on  $V = F[x]_{\leq n}$ ,

$$\langle p, q \rangle = \sum_{i=0}^n p(c_i) \overline{q(c_i)}.$$

(IP1) and (IP2) hold because evaluation at  $c_i$  is linear ([Theorem 0.8.5](#)) and the standard inner product on  $F^{n+1}$  has them; indeed  $\langle p, q \rangle$  is the standard inner product of the sample vectors  $(p(c_0), \dots, p(c_n))$  and  $(q(c_0), \dots, q(c_n))$ . Positivity is the interesting clause. If  $\langle p, p \rangle = \sum |p(c_i)|^2 = 0$ , then  $p$  vanishes at all  $n + 1$  distinct nodes. A non-zero polynomial of degree at most  $n$  has at most  $n$  distinct roots ([Corollary 5.1.4](#)), so  $p = 0$ . The **number** of nodes is what makes this work, and the non-example below removes one.

### 10.1.4 Non-examples, by minimal change

**Change a sign: the hyperbolic form on  $\mathbb{R}^2$ .** Let  $\beta(\mathbf{x}, \mathbf{y}) = x_1 y_1 - x_2 y_2$ . Bilinearity survives, since each term is still linear in each slot, and so does symmetry. What fails is (IP3), and the witness is  $\mathbf{e}_2$ :  $\beta(\mathbf{e}_2, \mathbf{e}_2) = -1 < 0$ . The vector  $(1, 1)$  is worse still: it is non-zero, yet  $\beta((1, 1), (1, 1)) = 0$ , so it would be a non-zero vector of length 0. Forms like  $\beta$  are useful (they are the subject of the chapter on bilinear forms), but they are not inner products.

**Drop the conjugate on  $\mathbb{C}^2$ .** Let  $\langle \mathbf{u}, \mathbf{v} \rangle = \mathbf{u}^\top \mathbf{v} = u_1 v_1 + u_2 v_2$  on  $\mathbb{C}^2$ . This map is linear in **both** slots and symmetric, so it satisfies (IP1) and looks tidier than the standard product. Take  $\mathbf{u} = (1, i)$ . Then

$$\langle \mathbf{u}, \mathbf{u} \rangle = 1^2 + i^2 = 1 - 1 = 0,$$

with  $\mathbf{u} \neq \mathbf{0}$ . Clause (IP3) fails. This one example is the whole reason for the conjugate: over  $\mathbb{C}$ , sums of squares vanish on non-zero vectors, sums of squared **moduli** do not.

**Enlarge the space: drop continuity.** Keep the formula  $\langle f, g \rangle = \int_0^1 f g$ , but let  $V$  be all Riemann-integrable functions  $[0, 1] \rightarrow \mathbb{R}$ . Linearity and symmetry are untouched. Let  $f$  be 1 at  $t = \frac{1}{2}$  and 0 elsewhere. It is integrable, non-zero as a function, and  $\int_0^1 f^2 = 0$ . Positivity fails, so  $\langle \cdot, \cdot \rangle$  is only **positive semidefinite** here. There are two standard repairs: restrict to continuous functions, as above, or regard two functions as equal when they differ on a negligible set. We take the first, and every function space in this chapter will be a space of continuous functions.

**Remove a node.** On  $\mathbb{R}[x]_{\leq 2}$ , put  $\langle p, q \rangle = p(0)q(0) + p(1)q(1)$ : the node example with one node too few. Bilinearity and symmetry hold. But  $p = x^2 - x$  is a non-zero polynomial of degree 2 with  $p(0) = p(1) = 0$ , so  $\langle p, p \rangle = 0$ . The clause that fails is (IP3), and it fails because two nodes cannot pin down a quadratic.

**Why this definition.** Each clause earns its place. Without (IP3) there can be non-zero vectors  $\mathbf{v}$  with  $\langle \mathbf{v}, \mathbf{v} \rangle = 0$ , and the proof of [Theorem 10.1.1](#) (d) breaks down, because pairing a vector with itself no longer detects zero. (The form  $x_1 y_1 - x_2 y_2$  below shows that the conclusion of (d) may survive anyway; what is lost is the argument for it, and with it every later proof that runs through  $\langle \mathbf{v}, \mathbf{v} \rangle = 0 \Rightarrow \mathbf{v} = \mathbf{0}$ .) Without (IP2) the diagonal values need not be real, and “ $\langle \mathbf{v}, \mathbf{v} \rangle > 0$ ” is meaningless. Linearity in the first slot rather than the second is a convention, but once chosen it must be kept: with (IP2) in place, linearity in one slot forces conjugate-linearity in the other, so a map cannot be linear in both unless  $F = \mathbb{R}$ .

*Warning.* The conjugate goes in the second slot here, and books differ. With our convention,  $\langle a\mathbf{u}, \mathbf{v} \rangle = a\langle \mathbf{u}, \mathbf{v} \rangle$  and  $\langle \mathbf{u}, a\mathbf{v} \rangle = \bar{a}\langle \mathbf{u}, \mathbf{v} \rangle$ . Many books, especially in physics, put the conjugate in the first slot, so that their  $\langle \mathbf{u}, \mathbf{v} \rangle$  is our  $\langle \mathbf{v}, \mathbf{u} \rangle$ . Every formula involving a lone conjugate then flips. When you read a formula elsewhere, check the convention first; when you write one, say which you use.

*Quick check.* Is  $\langle \mathbf{x}, \mathbf{y} \rangle = x_1y_1 + x_2y_2 + x_1y_2$  an inner product on  $\mathbb{R}^2$ ? If not, name the clause that fails and give a witness.

*Solution.* No. It is bilinear, and  $\langle \mathbf{x}, \mathbf{x} \rangle = x_1^2 + x_2^2 + x_1x_2 = (x_1 + \frac{1}{2}x_2)^2 + \frac{3}{4}x_2^2 > 0$  for  $\mathbf{x} \neq \mathbf{0}$ , so (IP1) and (IP3) both hold. The clause that fails is (IP2): with  $\mathbf{u} = \mathbf{e}_1$  and  $\mathbf{v} = \mathbf{e}_2$  we get  $\langle \mathbf{u}, \mathbf{v} \rangle = 1$  but  $\langle \mathbf{v}, \mathbf{u} \rangle = 0$ . Symmetrizing the cross term, to  $x_1y_1 + x_2y_2 + \frac{1}{2}(x_1y_2 + x_2y_1)$ , repairs it.

## 10.1.5 Norms

By Theorem 10.1.1 (a), the number  $\langle \mathbf{v}, \mathbf{v} \rangle$  is real and  $\geq 0$ , so it has a unique non-negative real square root. That square root is the length we were after.

### Definition 10.1.3 — Induced Norm

Let  $V$  be an inner product space. The **norm** of  $\mathbf{v} \in V$  is

$$\|\mathbf{v}\| := \sqrt{\langle \mathbf{v}, \mathbf{v} \rangle},$$

the **non-negative** real square root. A vector with  $\|\mathbf{v}\| = 1$  is a **unit vector**, and replacing a non-zero  $\mathbf{v}$  by  $\mathbf{v}/\|\mathbf{v}\|$  is called **normalizing** it.

On  $\mathbb{R}^n$  with the dot product this is the Euclidean length  $\sqrt{x_1^2 + \cdots + x_n^2}$ , and on  $\mathbb{C}^n$  it is  $\sqrt{\sum |x_i|^2}$ . On  $C[0, 1]$  it is  $(\int_0^1 |f|^2)^{1/2}$ , and on matrices it is the **Frobenius norm**  $(\sum |a_{ij}|^2)^{1/2}$ . By definition  $\|\mathbf{v}\|^2 = \langle \mathbf{v}, \mathbf{v} \rangle$  always, and this identity, read from right to left, is the most-used step in the chapter: **work with the norm squared, then expand**.

### Theorem 10.1.2 — Properties of the Induced Norm

Let  $V$  be an inner product space over  $F$ ,  $\mathbf{u}, \mathbf{v} \in V$  and  $c \in F$ . Then:

- (a)  $\|\mathbf{v}\| \geq 0$ , with  $\|\mathbf{v}\| = 0$  if and only if  $\mathbf{v} = \mathbf{0}$ ;
- (b)  $\|c\mathbf{v}\| = |c|\|\mathbf{v}\|$ ;
- (c)  $\|\mathbf{u} + \mathbf{v}\|^2 = \|\mathbf{u}\|^2 + 2\operatorname{Re}\langle \mathbf{u}, \mathbf{v} \rangle + \|\mathbf{v}\|^2$ .

*Proof.*

- (a) Immediate from Theorem 10.1.1 (a) and the fact that a non-negative real number has square root 0 only when it is 0.
- (b) By (IP1) and Theorem 10.1.1 (b),  $\langle c\mathbf{v}, c\mathbf{v} \rangle = c\bar{c}\langle \mathbf{v}, \mathbf{v} \rangle = |c|^2\|\mathbf{v}\|^2$  using Theorem 0.6.2 (c). Taking non-negative square roots gives  $\|c\mathbf{v}\| = |c|\|\mathbf{v}\|$ , since  $|c| \geq 0$ .

(c) Expand one slot at a time, by (IP1) and then [Theorem 10.1.1](#) (b):

$$\begin{aligned}\|\mathbf{u} + \mathbf{v}\|^2 &= \langle \mathbf{u} + \mathbf{v}, \mathbf{u} + \mathbf{v} \rangle \\ &= \langle \mathbf{u}, \mathbf{u} \rangle + \langle \mathbf{u}, \mathbf{v} \rangle + \langle \mathbf{v}, \mathbf{u} \rangle + \langle \mathbf{v}, \mathbf{v} \rangle.\end{aligned}$$

By (IP2),  $\langle \mathbf{v}, \mathbf{u} \rangle = \overline{\langle \mathbf{u}, \mathbf{v} \rangle}$ , and  $z + \bar{z} = 2 \operatorname{Re} z$  by [Theorem 0.6.2](#) (c). This gives the stated identity. ■

Part (c) is the workhorse. Over  $\mathbb{R}$  the middle term is just  $2\langle \mathbf{u}, \mathbf{v} \rangle$ , and the identity is the familiar  $(a+b)^2 = a^2 + 2ab + b^2$  for vectors. Replacing  $\mathbf{v}$  by  $-\mathbf{v}$  gives  $\|\mathbf{u} - \mathbf{v}\|^2 = \|\mathbf{u}\|^2 - 2 \operatorname{Re} \langle \mathbf{u}, \mathbf{v} \rangle + \|\mathbf{v}\|^2$ , and adding the two versions gives the **parallelogram law**

$$\|\mathbf{u} + \mathbf{v}\|^2 + \|\mathbf{u} - \mathbf{v}\|^2 = 2\|\mathbf{u}\|^2 + 2\|\mathbf{v}\|^2. \quad (*)$$

We return to  $(*)$  in the last warning of this section.

**Warning.** A squared norm is real and non-negative, never complex. If a computation in a complex space produces  $\|\mathbf{v}\|^2 = 2i$ , or a negative number, something has gone wrong — most often a missing conjugate. For  $\mathbf{v} = (1, i) \in \mathbb{C}^2$ , the correct value is  $\|\mathbf{v}\|^2 = |1|^2 + |i|^2 = 2$ , while the conjugate-free expression  $\mathbf{v}^\top \mathbf{v}$  gives 0. Use the answer as a check on the work.

## 10.1.6 Orthogonality and Pythagoras

With a norm in hand, the special relation  $\langle \mathbf{u}, \mathbf{v} \rangle = 0$  deserves its name.

### Definition 10.1.4 — Orthogonal Vectors

Let  $V$  be an inner product space. Vectors  $\mathbf{u}, \mathbf{v} \in V$  are **orthogonal**, written  $\mathbf{u} \perp \mathbf{v}$ , if  $\langle \mathbf{u}, \mathbf{v} \rangle = 0$ . A list  $(\mathbf{v}_1, \dots, \mathbf{v}_k)$  is **orthogonal** if  $\mathbf{v}_i \perp \mathbf{v}_j$  whenever  $i \neq j$ .

The relation is symmetric:  $\langle \mathbf{u}, \mathbf{v} \rangle = 0$  if and only if  $\langle \mathbf{v}, \mathbf{u} \rangle = \overline{\langle \mathbf{u}, \mathbf{v} \rangle} = 0$ , so “ $\mathbf{u} \perp \mathbf{v}$ ” needs no order. The zero vector is orthogonal to everything ([Theorem 10.1.1](#) (c)), and by part (a) of the same result it is the **only** vector orthogonal to itself. In  $\mathbb{R}^n$  this is perpendicularity in the usual sense, by the law of cosines computation above:  $\cos \theta = 0$  exactly when  $\mathbf{x} \cdot \mathbf{y} = 0$ . In  $\mathbb{C}^2$ , the vectors  $(1, i)$  and  $(1, -i)$  are orthogonal, since  $1 \cdot \bar{1} + i \cdot \overline{(-i)} = 1 + i \cdot i = 0$ .

### Theorem 10.1.3 — Pythagorean Theorem

Let  $(\mathbf{v}_1, \dots, \mathbf{v}_k)$  be an orthogonal list in an inner product space. Then

$$\|\mathbf{v}_1 + \dots + \mathbf{v}_k\|^2 = \|\mathbf{v}_1\|^2 + \dots + \|\mathbf{v}_k\|^2.$$

*Proof.* Expand the left side by (IP1) and [Theorem 10.1.1](#) (b):

$$\left\langle \sum_i \mathbf{v}_i, \sum_j \mathbf{v}_j \right\rangle = \sum_i \sum_j \langle \mathbf{v}_i, \mathbf{v}_j \rangle.$$

Every term with  $i \neq j$  is 0, because the list is orthogonal. What survives is  $\sum_i \langle \mathbf{v}_i, \mathbf{v}_i \rangle = \sum_i \|\mathbf{v}_i\|^2$ , as claimed. ■

So orthogonality is exactly the condition under which squared lengths add. This is why the relation is worth naming, and why the rest of the chapter works so hard to produce orthogonal vectors.

**Warning.** Over  $\mathbb{C}$ , the converse of Pythagoras fails. By Theorem 10.1.2 (c),  $\|\mathbf{u} + \mathbf{v}\|^2 = \|\mathbf{u}\|^2 + \|\mathbf{v}\|^2$  says only that  $\operatorname{Re}\langle \mathbf{u}, \mathbf{v} \rangle = 0$ , not that  $\langle \mathbf{u}, \mathbf{v} \rangle = 0$ . In  $\mathbb{C}$  with  $\langle z, w \rangle = z\bar{w}$ , take  $\mathbf{u} = 1$  and  $\mathbf{v} = i$ : then  $\langle \mathbf{u}, \mathbf{v} \rangle = \bar{i} = -i \neq 0$ , yet  $\|1 + i\|^2 = 2 = \|1\|^2 + \|i\|^2$ . Over  $\mathbb{R}$  the converse does hold, since there  $\operatorname{Re}$  does nothing.

### Theorem 10.1.4 — Orthogonal Non-zero Vectors Are Independent

Let  $(\mathbf{v}_1, \dots, \mathbf{v}_k)$  be an orthogonal list of **non-zero** vectors in an inner product space. Then the list is linearly independent.

*Idea.* Start a dependence relation in the standard way, and then do the one thing an inner product allows: pair the relation with a single  $\mathbf{v}_j$ . Orthogonality kills every term except the  $j$ -th, and what is left is  $a_j \|\mathbf{v}_j\|^2 = 0$  with  $\|\mathbf{v}_j\| \neq 0$ .

*Proof.* Let  $a_1 \mathbf{v}_1 + \dots + a_k \mathbf{v}_k = \mathbf{0}$  with  $a_1, \dots, a_k \in F$ . Fix  $j$  and take the inner product of both sides with  $\mathbf{v}_j$ . By (IP1) and Theorem 10.1.1 (c),

$$0 = \left\langle \sum_i a_i \mathbf{v}_i, \mathbf{v}_j \right\rangle = \sum_i a_i \langle \mathbf{v}_i, \mathbf{v}_j \rangle = a_j \|\mathbf{v}_j\|^2,$$

where the last equality holds because  $\langle \mathbf{v}_i, \mathbf{v}_j \rangle = 0$  for  $i \neq j$ . Since  $\mathbf{v}_j \neq \mathbf{0}$ , Theorem 10.1.1 (a) gives  $\|\mathbf{v}_j\|^2 > 0$ , so  $a_j = 0$ . As  $j$  was arbitrary, all the coefficients vanish, and the list is linearly independent by Definition 1.4.1. ■

The hypothesis “non-zero” cannot be dropped: the list  $(\mathbf{e}_1, \mathbf{0})$  is orthogonal and dependent. With it, the result is a free supply of independent lists, and this is what makes the next section possible.

## 10.1.7 The Cauchy–Schwarz inequality

Angles came from the formula  $\cos \theta = (\mathbf{x} \cdot \mathbf{y}) / (\|\mathbf{x}\| \|\mathbf{y}\|)$ . To define an angle by that formula in an abstract space, we must know that the right-hand side lies in  $[-1, 1]$ . That is exactly the following inequality, the most used inequality in the subject.

### Theorem 10.1.5 — Cauchy–Schwarz Inequality

Let  $V$  be an inner product space and  $\mathbf{u}, \mathbf{v} \in V$ . Then

$$|\langle \mathbf{u}, \mathbf{v} \rangle| \leq \|\mathbf{u}\| \|\mathbf{v}\|,$$

with equality if and only if the list  $(\mathbf{u}, \mathbf{v})$  is linearly dependent.

*Idea.* Picture  $\mathbf{u}$  and  $\mathbf{v}$  in  $\mathbb{R}^2$ . Drop a perpendicular from the tip of  $\mathbf{u}$  onto the line through  $\mathbf{v}$ . The foot of the perpendicular is a multiple  $c\mathbf{v}$ , and the two legs  $c\mathbf{v}$  and  $\mathbf{u} - c\mathbf{v}$  of the resulting right triangle have squared lengths adding to  $\|\mathbf{u}\|^2$ . Since one leg has squared length  $|\langle \mathbf{u}, \mathbf{v} \rangle|^2 / \|\mathbf{v}\|^2$ , the inequality is just “a leg is no longer than the hypotenuse”, and equality means the other leg is  $\mathbf{0}$ . The picture proves nothing in an abstract space, so we check algebraically that  $\mathbf{u} - c\mathbf{v} \perp \mathbf{v}$  for the right  $c$ , and then let Pythagoras do the work.

*Proof.* If  $\mathbf{v} = \mathbf{0}$ , both sides are 0 by Theorem 10.1.1 (c), and  $(\mathbf{u}, \mathbf{0})$  is dependent, so the statement holds. Suppose  $\mathbf{v} \neq \mathbf{0}$ , so  $\|\mathbf{v}\| > 0$ , and put

$$c = \frac{\langle \mathbf{u}, \mathbf{v} \rangle}{\|\mathbf{v}\|^2}, \quad \mathbf{z} = \mathbf{u} - c\mathbf{v}.$$

Then  $\mathbf{z} \perp \mathbf{v}$ , since by (IP1)

$$\langle \mathbf{z}, \mathbf{v} \rangle = \langle \mathbf{u}, \mathbf{v} \rangle - c\langle \mathbf{v}, \mathbf{v} \rangle = \langle \mathbf{u}, \mathbf{v} \rangle - \langle \mathbf{u}, \mathbf{v} \rangle = 0,$$

and hence also  $\mathbf{z} \perp c\mathbf{v}$ . Now  $\mathbf{u} = \mathbf{z} + c\mathbf{v}$  is a sum of two orthogonal vectors, so Theorem 10.1.3 and Theorem 10.1.2 (b) give

$$\|\mathbf{u}\|^2 = \|\mathbf{z}\|^2 + |c|^2 \|\mathbf{v}\|^2 = \|\mathbf{z}\|^2 + \frac{|\langle \mathbf{u}, \mathbf{v} \rangle|^2}{\|\mathbf{v}\|^2}. \quad (\dagger)$$

Since  $\|\mathbf{z}\|^2 \geq 0$ , dropping it gives  $\|\mathbf{u}\|^2 \|\mathbf{v}\|^2 \geq |\langle \mathbf{u}, \mathbf{v} \rangle|^2$ , and taking non-negative square roots proves the inequality.

Equality holds in  $(\dagger)$  exactly when  $\|\mathbf{z}\| = 0$ , that is (Theorem 10.1.2 (a)) exactly when  $\mathbf{u} = c\mathbf{v}$ . If  $\mathbf{u} = c\mathbf{v}$ , the list  $(\mathbf{u}, \mathbf{v})$  is dependent. Conversely, suppose  $(\mathbf{u}, \mathbf{v})$  is dependent, say  $a\mathbf{u} + b\mathbf{v} = \mathbf{0}$  with  $a, b$  not both 0. Here  $a \neq 0$ , since  $a = 0$  would give  $b\mathbf{v} = \mathbf{0}$  with  $b \neq 0$  and hence  $\mathbf{v} = \mathbf{0}$ , which we have excluded. So  $\mathbf{u} = d\mathbf{v}$  with  $d = -b/a$ , and then  $c = d\langle \mathbf{v}, \mathbf{v} \rangle / \|\mathbf{v}\|^2 = d$ , so  $\mathbf{z} = \mathbf{0}$ . This proves the equality case. ■

The proof is worth rereading for its shape: the vector  $\mathbf{z}$  was **found** from a picture and then **checked** from the axioms. That is the standard way to work in an abstract inner product space.

### Corollary 10.1.6 — Triangle Inequality

Let  $V$  be an inner product space and  $\mathbf{u}, \mathbf{v} \in V$ . Then

$$\|\mathbf{u} + \mathbf{v}\| \leq \|\mathbf{u}\| + \|\mathbf{v}\|.$$

*Proof.* For any complex number  $z$  we have  $\operatorname{Re} z \leq |z|$ , as in the proof of Theorem 0.6.4.



Hence, by [Theorem 10.1.2](#) (c) and then [Theorem 10.1.5](#),

$$\begin{aligned}\|\mathbf{u} + \mathbf{v}\|^2 &= \|\mathbf{u}\|^2 + 2\operatorname{Re}\langle \mathbf{u}, \mathbf{v} \rangle + \|\mathbf{v}\|^2 \\ &\leq \|\mathbf{u}\|^2 + 2|\langle \mathbf{u}, \mathbf{v} \rangle| + \|\mathbf{v}\|^2 \\ &\leq \|\mathbf{u}\|^2 + 2\|\mathbf{u}\|\|\mathbf{v}\| + \|\mathbf{v}\|^2 \\ &= (\|\mathbf{u}\| + \|\mathbf{v}\|)^2.\end{aligned}$$

Both  $\|\mathbf{u} + \mathbf{v}\|$  and  $\|\mathbf{u}\| + \|\mathbf{v}\|$  are  $\geq 0$ , so taking square roots preserves the inequality. ■

Together, [Theorem 10.1.2](#) (a), (b) and [Corollary 10.1.6](#) say that  $\|\cdot\|$  is a **norm** in the general sense studied in Chapter 15. Chapter 15 also shows that not every norm arises from an inner product; the last warning of this section explains how to tell.

### Definition 10.1.5 — Angle Between Vectors

Let  $V$  be a **real** inner product space and let  $\mathbf{u}, \mathbf{v} \in V$  be **non-zero**. The **angle** between  $\mathbf{u}$  and  $\mathbf{v}$  is the unique  $\theta \in [0, \pi]$  with

$$\cos \theta = \frac{\langle \mathbf{u}, \mathbf{v} \rangle}{\|\mathbf{u}\| \|\mathbf{v}\|}.$$

The definition needs [Theorem 10.1.5](#) twice over. Cauchy–Schwarz says the quotient has absolute value at most 1, so it lies in  $[-1, 1]$ ; and  $\cos$  is a strictly decreasing bijection from  $[0, \pi]$  onto  $[-1, 1]$ , so exactly one  $\theta$  qualifies. Without the inequality the formula could ask for the angle whose cosine is 7. Note also that  $\theta = \pi/2$  if and only if  $\langle \mathbf{u}, \mathbf{v} \rangle = 0$ , so the definition is consistent with [Definition 10.1.4](#).

Over  $\mathbb{C}$  there is no angle, because  $\langle \mathbf{u}, \mathbf{v} \rangle$  is not real. The quantity  $|\langle \mathbf{u}, \mathbf{v} \rangle|/(\|\mathbf{u}\|\|\mathbf{v}\|) \in [0, 1]$  still makes sense, and it still measures how far the two vectors are from being orthogonal.

#### Example. (Three Angles)

Compute the angle between  $\mathbf{u} = (1, 1, 0)$  and  $\mathbf{v} = (1, 0, 1)$  in  $\mathbb{R}^3$  with the dot product; between  $p = 1$  and  $q = x$  in  $\mathbb{R}[x]_{\leq 1}$  with  $\langle p, q \rangle = \int_0^1 pq$ ; and between  $\mathbf{A} = \begin{pmatrix} 1 & 1 \\ 1 & 1 \end{pmatrix}$  and  $\mathbf{B} = \begin{pmatrix} 1 & -1 \\ 1 & -1 \end{pmatrix}$  in  $M_2(\mathbb{R})$  with the Frobenius inner product.

*Solution.* In  $\mathbb{R}^3$ :  $\langle \mathbf{u}, \mathbf{v} \rangle = 1$ ,  $\|\mathbf{u}\| = \|\mathbf{v}\| = \sqrt{2}$ , so  $\cos \theta = 1/2$  and  $\theta = \pi/3$ .

In  $\mathbb{R}[x]_{\leq 1}$ :  $\langle 1, x \rangle = \int_0^1 t \, dt = \frac{1}{2}$ ,  $\|1\|^2 = \int_0^1 1 = 1$  and  $\|x\|^2 = \int_0^1 t^2 \, dt = \frac{1}{3}$ . Hence

$$\cos \theta = \frac{1/2}{1 \cdot 1/\sqrt{3}} = \frac{\sqrt{3}}{2},$$

so  $\theta = \pi/6$ . The constant 1 and the function  $x$  are far from orthogonal on  $[0, 1]$ , which is believable: both are positive there.

In  $M_2(\mathbb{R})$ :  $\langle \mathbf{A}, \mathbf{B} \rangle = 1 - 1 + 1 - 1 = 0$ , so  $\theta = \pi/2$ . The two matrices are orthogonal, with  $\|\mathbf{A}\| = \|\mathbf{B}\| = 2$ . As a check on Pythagoras,  $\mathbf{A} + \mathbf{B} = \begin{pmatrix} 2 & 0 \\ 2 & 0 \end{pmatrix}$  has  $\|\mathbf{A} + \mathbf{B}\|^2 = 8 = 4 + 4$ .

*Quick check.* In  $\mathbb{C}^2$  with the standard inner product, compute  $\langle (1, i), (1, 1) \rangle$  and  $\langle (1, 1), (1, i) \rangle$ . Check the answers against (IP2), and against Cauchy–Schwarz.

*Solution.*  $\langle (1, i), (1, 1) \rangle = 1 \cdot \bar{1} + i \cdot \bar{1} = 1 + i$ , and  $\langle (1, 1), (1, i) \rangle = 1 \cdot \bar{1} + 1 \cdot \bar{i} = 1 - i$ . They are conjugates, as (IP2) requires. For Cauchy–Schwarz,  $|1 + i| = \sqrt{2}$ , while  $\|(1, i)\| = \sqrt{2}$  and  $\|(1, 1)\| = \sqrt{2}$ , so the inequality reads  $\sqrt{2} \leq 2$ . It is strict, which is right:  $(1, i)$  and  $(1, 1)$  are independent.

*Warning.* **Not every norm comes from an inner product.** A function  $\|\cdot\|$  can satisfy positivity, homogeneity and the inequality of Corollary 10.1.6 without being  $\sqrt{\langle \mathbf{v}, \mathbf{v} \rangle}$  for any inner product. The test is the parallelogram law (\*): every induced norm satisfies it, as we derived from Theorem 10.1.2 (c), so a norm that violates it is induced by nothing. Take  $\|\mathbf{x}\|_\infty = \max(|x_1|, |x_2|)$  on  $\mathbb{R}^2$  with  $\mathbf{u} = \mathbf{e}_1$ ,  $\mathbf{v} = \mathbf{e}_2$ . Then  $\|\mathbf{u} + \mathbf{v}\|_\infty = \|\mathbf{u} - \mathbf{v}\|_\infty = 1$ , so the left side of (\*) is 2, while the right side is 4. Remarkably, the law is not only necessary but sufficient; Exercise (C1: The parallelogram law characterizes inner-product norms) proves the substance of that in  $\mathbb{R}^2$ , and Chapter 15 takes up general norms.

## 10.1.8 Exercises

### 10.1.8.1 A. Check your understanding

#### Exercise A1.

- State the three axioms of an inner product on a vector space over  $F = \mathbb{R}$  or  $\mathbb{C}$ , and say in which slot the map is linear.
- Why must the field be  $\mathbb{R}$  or  $\mathbb{C}$ ? Name the two features used.
- Explain why  $\langle \mathbf{v}, \mathbf{v} \rangle$  is always a real number.
- True or false: an inner product on a complex space is linear in the second slot. Justify your answer.
- State the Cauchy–Schwarz inequality, its equality case, and one place where the equality case is used.
- True or false: if  $\|\mathbf{u} + \mathbf{v}\|^2 = \|\mathbf{u}\|^2 + \|\mathbf{v}\|^2$  in a complex inner product space, then  $\mathbf{u} \perp \mathbf{v}$ . Justify your answer.

*Solution.*

- (IP1)  $\langle a\mathbf{u} + b\mathbf{v}, \mathbf{w} \rangle = a\langle \mathbf{u}, \mathbf{w} \rangle + b\langle \mathbf{v}, \mathbf{w} \rangle$ ; (IP2)  $\langle \mathbf{v}, \mathbf{u} \rangle = \overline{\langle \mathbf{u}, \mathbf{v} \rangle}$ ; (IP3)  $\langle \mathbf{v}, \mathbf{v} \rangle > 0$  for  $\mathbf{v} \neq \mathbf{0}$  (Definition 10.1.1). It is linear in the **first** slot.
- Positivity, so that  $\langle \mathbf{v}, \mathbf{v} \rangle > 0$  has meaning and square roots exist, and conjugation, so that  $\langle \mathbf{v}, \mathbf{v} \rangle$  is real and non-negative over  $\mathbb{C}$ .
- By (IP2) with  $\mathbf{u} = \mathbf{v}$ , the number equals its own conjugate, so its imaginary part is 0 (Theorem 10.1.1 (a)).

- (d) False. It is **conjugate**-linear in the second slot:  $\langle \mathbf{u}, a\mathbf{v} \rangle = \bar{a}\langle \mathbf{u}, \mathbf{v} \rangle$  (Theorem 10.1.1 (b)). For example, in  $\mathbb{C}$  with  $\langle z, w \rangle = z\bar{w}$ ,  $\langle 1, i \rangle = -i$  while  $i\langle 1, 1 \rangle = i$ .
- (e)  $|\langle \mathbf{u}, \mathbf{v} \rangle| \leq \|\mathbf{u}\| \|\mathbf{v}\|$ , with equality if and only if  $(\mathbf{u}, \mathbf{v})$  is dependent (Theorem 10.1.5). The inequality is what makes Definition 10.1.5 well posed: it puts the quotient  $\langle \mathbf{u}, \mathbf{v} \rangle / (\|\mathbf{u}\| \|\mathbf{v}\|)$  inside  $[-1, 1]$ , where arccos is defined. The equality case then says which vectors sit at the ends of that interval:  $\cos \theta = \pm 1$  exactly for dependent pairs, that is,  $\theta = 0$  or  $\theta = \pi$ .
- (f) False. It says only  $\operatorname{Re} \langle \mathbf{u}, \mathbf{v} \rangle = 0$ . In  $\mathbb{C}$  with  $\langle z, w \rangle = z\bar{w}$ , take  $\mathbf{u} = 1$ ,  $\mathbf{v} = i$ : then  $\langle \mathbf{u}, \mathbf{v} \rangle = -i \neq 0$  but  $\|1 + i\|^2 = 2 = 1 + 1$ .

### 10.1.8.2 B. Practice

#### Exercise B1: Determine which are inner products.

Determine which of the following are inner products on the stated space. For those that are, verify the axioms; for those that are not, name the clause that fails and give an explicit witness.

- (a) On  $\mathbb{R}^2$ :  $\langle \mathbf{x}, \mathbf{y} \rangle = x_1 y_1 + 3x_2 y_2$ .
- (b) On  $\mathbb{R}^2$ :  $\langle \mathbf{x}, \mathbf{y} \rangle = x_1 y_1 + x_1 y_2 + x_2 y_1 + 2x_2 y_2$ .
- (c) On  $\mathbb{C}^2$ :  $\langle \mathbf{x}, \mathbf{y} \rangle = x_1 \bar{y}_1 + x_2 y_2$ .
- (d) On  $M_2(\mathbb{R})$ :  $\langle \mathbf{A}, \mathbf{B} \rangle = \operatorname{tr}(\mathbf{AB})$ .
- (e) On  $\mathbb{R}[x]_{\leq 2}$ :  $\langle p, q \rangle = p(0)q(0) + p(1)q(1) + p(2)q(2)$ .

*Solution.*

- (a) An inner product: it is the weighted product with weights  $1, 3 > 0$ . Bilinearity and symmetry are clear, and  $\langle \mathbf{x}, \mathbf{x} \rangle = x_1^2 + 3x_2^2 = 0$  forces  $x_1 = x_2 = 0$ .
- (b) An inner product. It is bilinear term by term, and symmetric because the two cross terms are interchanged by swapping  $\mathbf{x}$  and  $\mathbf{y}$ . For positivity, complete the square:

$$\langle \mathbf{x}, \mathbf{x} \rangle = x_1^2 + 2x_1 x_2 + 2x_2^2 = (x_1 + x_2)^2 + x_2^2,$$

which is 0 only if  $x_2 = 0$  and  $x_1 + x_2 = 0$ , that is  $\mathbf{x} = \mathbf{0}$ .

- (c) Not an inner product. (IP2) fails: with  $\mathbf{x} = (0, 1)$  and  $\mathbf{y} = (0, i)$  we get  $\langle \mathbf{x}, \mathbf{y} \rangle = i$  and  $\langle \mathbf{y}, \mathbf{x} \rangle = i$ , while (IP2) would require  $\langle \mathbf{y}, \mathbf{x} \rangle = \bar{i} = -i$ . Positivity fails too, at  $\mathbf{x} = (0, i)$ :  $\langle \mathbf{x}, \mathbf{x} \rangle = i^2 = -1$ .

- (d) Not an inner product. It is bilinear, and symmetric by Theorem 0.9.6 (3). But (IP3) fails for  $\mathbf{N} = \begin{pmatrix} 0 & 1 \\ 0 & 0 \end{pmatrix} \neq 0$ , since  $\mathbf{N}^2 = 0$  and so  $\langle \mathbf{N}, \mathbf{N} \rangle = \operatorname{tr}(\mathbf{N}^2) = 0$ . Replacing  $\operatorname{tr}(\mathbf{AB})$  by  $\operatorname{tr}(\mathbf{B}^\top \mathbf{A})$  repairs it: that is the Frobenius inner product.

- (e) An inner product: the node example with the three distinct nodes  $0, 1, 2$  on  $\mathbb{R}[x]_{\leq 2}$ . Bilinearity and symmetry come from linearity of evaluation. If  $\langle p, p \rangle = p(0)^2 + p(1)^2 + p(2)^2 = 0$ , then  $p$  has three distinct roots and degree at most 2, so  $p = 0$  by Corollary 5.1.4.

**Exercise B2: Norms and angles.**

- (a) In  $\mathbb{R}^3$  with the dot product, find  $\|\mathbf{u}\|$ ,  $\|\mathbf{v}\|$  and the cosine of the angle between  $\mathbf{u} = (1, 2, 2)$  and  $\mathbf{v} = (2, -1, 2)$ .
- (b) In  $\mathbb{R}[x]_{\leq 2}$  with  $\langle p, q \rangle = \int_{-1}^1 pq$ , compute  $\|1\|$ ,  $\|x\|$  and  $\langle 1, x \rangle$ . What is the angle between 1 and  $x$ ?
- (c) In  $M_2(\mathbb{C})$  with the Frobenius inner product, compute  $\|\mathbf{A}\|$  for  $\mathbf{A} = \begin{pmatrix} 1 & i \\ 0 & 1+i \end{pmatrix}$ .

*Solution.*

- (a)  $\|\mathbf{u}\| = \sqrt{1+4+4} = 3$  and  $\|\mathbf{v}\| = \sqrt{4+1+4} = 3$ . Also  $\langle \mathbf{u}, \mathbf{v} \rangle = 2 - 2 + 4 = 4$ , so  $\cos \theta = 4/9$ . Cauchy–Schwarz reads  $4 \leq 9$ , and it is strict, as it must be for an independent pair.
- (b)  $\|1\|^2 = \int_{-1}^1 1 \, dt = 2$ , so  $\|1\| = \sqrt{2}$ ;  $\|x\|^2 = \int_{-1}^1 t^2 \, dt = \frac{2}{3}$ , so  $\|x\| = \sqrt{2/3}$ ; and  $\langle 1, x \rangle = \int_{-1}^1 t \, dt = 0$ . The angle is  $\pi/2$ : on a symmetric interval, the constants are orthogonal to the odd functions. Compare this with [Example \(Three Angles\)](#), where the interval  $[0, 1]$  gave the angle  $\pi/6$  instead. **The geometry depends on the inner product, not only on the vectors.**
- (c)  $\|\mathbf{A}\|^2 = |1|^2 + |i|^2 + 0 + |1+i|^2 = 1 + 1 + 2 = 4$ , so  $\|\mathbf{A}\| = 2$ .

**Exercise B3: Cauchy–Schwarz, concretely.**

- (a) Verify [Theorem 10.1.5](#) for  $\mathbf{a} = (1, 2, 3)$  and  $\mathbf{b} = (3, 2, 1)$  in  $\mathbb{R}^3$  with the dot product, and say whether the inequality is strict.
- (b) Deduce from [Theorem 10.1.5](#) the classical inequality

$$\left( \sum_{i=1}^n a_i b_i \right)^2 \leq \left( \sum_{i=1}^n a_i^2 \right) \left( \sum_{i=1}^n b_i^2 \right)$$

for all real  $a_i, b_i$ .

- (c) Hence show that  $(a_1 + \cdots + a_n)^2 \leq n(a_1^2 + \cdots + a_n^2)$  for all real  $a_i$ .

*Solution.*

- (a)  $\langle \mathbf{a}, \mathbf{b} \rangle = 3 + 4 + 3 = 10$ , so  $|\langle \mathbf{a}, \mathbf{b} \rangle|^2 = 100$ , while  $\|\mathbf{a}\|^2 \|\mathbf{b}\|^2 = 14 \cdot 14 = 196$ . So  $10 \leq 14$ , strictly. It must be strict, since  $\mathbf{a}$  and  $\mathbf{b}$  are not multiples of each other.
- (b) Apply [Theorem 10.1.5](#) on  $\mathbb{R}^n$  with the dot product to  $\mathbf{a} = (a_1, \dots, a_n)$  and  $\mathbf{b} = (b_1, \dots, b_n)$ . Then  $\langle \mathbf{a}, \mathbf{b} \rangle = \sum a_i b_i$ ,  $\|\mathbf{a}\|^2 = \sum a_i^2$  and  $\|\mathbf{b}\|^2 = \sum b_i^2$ , and squaring the inequality  $|\langle \mathbf{a}, \mathbf{b} \rangle| \leq \|\mathbf{a}\| \|\mathbf{b}\|$  gives the claim.
- (c) Take  $b_i = 1$  for all  $i$  in (b). Then  $\sum a_i b_i = \sum a_i$  and  $\sum b_i^2 = n$ , so  $(\sum a_i)^2 \leq n \sum a_i^2$ . Equality holds exactly when  $\mathbf{a}$  is a multiple of  $(1, \dots, 1)$ , that is, when all the  $a_i$  are equal.

**Exercise B4: A lower bound for a difference of norms.**

Let  $V$  be an inner product space and  $\mathbf{u}, \mathbf{v} \in V$ . Prove that

$$|\|\mathbf{u}\| - \|\mathbf{v}\|| \leq \|\mathbf{u} - \mathbf{v}\|.$$

*Solution.* Write  $\mathbf{u} = (\mathbf{u} - \mathbf{v}) + \mathbf{v}$ . By [Corollary 10.1.6](#),

$$\|\mathbf{u}\| \leq \|\mathbf{u} - \mathbf{v}\| + \|\mathbf{v}\|,$$

so  $\|\mathbf{u}\| - \|\mathbf{v}\| \leq \|\mathbf{u} - \mathbf{v}\|$ . Swapping the roles of  $\mathbf{u}$  and  $\mathbf{v}$  gives  $\|\mathbf{v}\| - \|\mathbf{u}\| \leq \|\mathbf{v} - \mathbf{u}\|$ , and  $\|\mathbf{v} - \mathbf{u}\| = |-1|\|\mathbf{u} - \mathbf{v}\| = \|\mathbf{u} - \mathbf{v}\|$  by [Theorem 10.1.2](#) (b). The quantity  $|\|\mathbf{u}\| - \|\mathbf{v}\||$  is one of the two numbers just bounded, so it is at most  $\|\mathbf{u} - \mathbf{v}\|$ . This proves the inequality.

### 10.1.8.3 C. Going deeper

**Exercise C1: The parallelogram law characterizes inner-product norms.**

Let  $\|\cdot\|$  be a norm on  $\mathbb{R}^2$ , that is, a function  $\mathbb{R}^2 \rightarrow \mathbb{R}$  satisfying the three properties of [Theorem 10.1.2](#) (a), (b) and [Corollary 10.1.6](#). Suppose it satisfies the parallelogram law

$$\|\mathbf{u} + \mathbf{v}\|^2 + \|\mathbf{u} - \mathbf{v}\|^2 = 2\|\mathbf{u}\|^2 + 2\|\mathbf{v}\|^2$$

for all  $\mathbf{u}, \mathbf{v}$ , and define  $\beta(\mathbf{u}, \mathbf{v}) = \frac{1}{4}(\|\mathbf{u} + \mathbf{v}\|^2 - \|\mathbf{u} - \mathbf{v}\|^2)$ .

- Show that  $\beta(\mathbf{u}, \mathbf{u}) = \|\mathbf{u}\|^2$ , that  $\beta(\mathbf{u}, \mathbf{v}) = \beta(\mathbf{v}, \mathbf{u})$ , and that  $\beta(\mathbf{u}, \mathbf{0}) = 0$ .
- Prove that  $\beta(\mathbf{u}, \mathbf{v}) + \beta(\mathbf{u}, \mathbf{w}) = 2\beta(\mathbf{u}, \frac{1}{2}(\mathbf{v} + \mathbf{w}))$  for all  $\mathbf{u}, \mathbf{v}, \mathbf{w}$ .
- Deduce that  $\beta(\mathbf{u}, \mathbf{v}) + \beta(\mathbf{u}, \mathbf{w}) = \beta(\mathbf{u}, \mathbf{v} + \mathbf{w})$ .
- Granting that  $\beta(\mathbf{u}, t\mathbf{v}) = t\beta(\mathbf{u}, \mathbf{v})$  for all  $t \in \mathbb{R}$ , conclude that  $\beta$  is an inner product on  $\mathbb{R}^2$  whose induced norm is  $\|\cdot\|$ .

*Hint: for (b), apply the parallelogram law to the pair  $\mathbf{u} + \frac{1}{2}(\mathbf{v} + \mathbf{w})$ ,  $\frac{1}{2}(\mathbf{v} - \mathbf{w})$ , and then to the pair  $\mathbf{u} - \frac{1}{2}(\mathbf{v} + \mathbf{w})$ ,  $\frac{1}{2}(\mathbf{w} - \mathbf{v})$ .*

*Solution.*

- $\beta(\mathbf{u}, \mathbf{u}) = \frac{1}{4}(\|2\mathbf{u}\|^2 - \|\mathbf{0}\|^2) = \frac{1}{4} \cdot 4\|\mathbf{u}\|^2 = \|\mathbf{u}\|^2$ , using homogeneity. Symmetry holds because  $\|\mathbf{v} - \mathbf{u}\| = \|\mathbf{u} - \mathbf{v}\|$ , again by homogeneity with  $c = -1$ . Finally  $\beta(\mathbf{u}, \mathbf{0}) = \frac{1}{4}(\|\mathbf{u}\|^2 - \|\mathbf{u}\|^2) = 0$ .
- Put  $\mathbf{m} = \frac{1}{2}(\mathbf{v} + \mathbf{w})$  and  $\mathbf{d} = \frac{1}{2}(\mathbf{v} - \mathbf{w})$ , so that  $\mathbf{v} = \mathbf{m} + \mathbf{d}$  and  $\mathbf{w} = \mathbf{m} - \mathbf{d}$ . Apply the parallelogram law to  $\mathbf{u} + \mathbf{m}$  and  $\mathbf{d}$ : since  $(\mathbf{u} + \mathbf{m}) + \mathbf{d} = \mathbf{u} + \mathbf{v}$  and  $(\mathbf{u} + \mathbf{m}) - \mathbf{d} = \mathbf{u} + \mathbf{w}$ ,

$$\|\mathbf{u} + \mathbf{v}\|^2 + \|\mathbf{u} + \mathbf{w}\|^2 = 2\|\mathbf{u} + \mathbf{m}\|^2 + 2\|\mathbf{d}\|^2.$$

Apply it to  $\mathbf{u} - \mathbf{m}$  and  $\mathbf{d}$ : since  $(\mathbf{u} - \mathbf{m}) - \mathbf{d} = \mathbf{u} - \mathbf{v}$  and  $(\mathbf{u} - \mathbf{m}) + \mathbf{d} = \mathbf{u} - \mathbf{w}$ ,

$$\|\mathbf{u} - \mathbf{v}\|^2 + \|\mathbf{u} - \mathbf{w}\|^2 = 2\|\mathbf{u} - \mathbf{m}\|^2 + 2\|\mathbf{d}\|^2.$$

Subtracting the second from the first, the terms  $2\|\mathbf{d}\|^2$  cancel, and dividing by 4 gives  $\beta(\mathbf{u}, \mathbf{v}) + \beta(\mathbf{u}, \mathbf{w}) = 2\beta(\mathbf{u}, \mathbf{m})$ , which is the claim.

- Take  $\mathbf{w} = \mathbf{0}$  in (b). By (a),  $\beta(\mathbf{u}, \mathbf{0}) = 0$ , so  $\beta(\mathbf{u}, \mathbf{v}) = 2\beta(\mathbf{u}, \frac{1}{2}\mathbf{v})$  for all  $\mathbf{v}$ . Applying

this with  $\mathbf{v} + \mathbf{w}$  in place of  $\mathbf{v}$ , we get  $2\beta(\mathbf{u}, \frac{1}{2}(\mathbf{v} + \mathbf{w})) = \beta(\mathbf{u}, \mathbf{v} + \mathbf{w})$ , and the left side is  $\beta(\mathbf{u}, \mathbf{v}) + \beta(\mathbf{u}, \mathbf{w})$  by (b).

- (d) By (c) and the granted homogeneity,  $\beta$  is linear in its second slot, and by the symmetry in (a) it is then linear in the first as well, so (IP1) and (IP2) hold (conjugation is trivial over  $\mathbb{R}$ ). For (IP3),  $\beta(\mathbf{u}, \mathbf{u}) = \|\mathbf{u}\|^2 > 0$  for  $\mathbf{u} \neq \mathbf{0}$ , since a norm vanishes only at  $\mathbf{0}$ . So  $\beta$  is an inner product, and its induced norm is  $\sqrt{\beta(\mathbf{u}, \mathbf{u})} = \|\mathbf{u}\|$ .

*Remark.* The granted step in (d) is where analysis enters: additivity from (c) gives  $\beta(\mathbf{u}, t\mathbf{v}) = t\beta(\mathbf{u}, \mathbf{v})$  for every **rational**  $t$  by induction, and the extension to real  $t$  uses that  $t \mapsto \beta(\mathbf{u}, t\mathbf{v})$  is continuous, which follows from continuity of the norm. Nothing else in the argument needs anything beyond the three norm properties.

### Exercise C2: Recovering the inner product from the norm.

Let  $V$  be an inner product space over  $F$ .

- (a) For  $F = \mathbb{R}$ , prove the polarization identity  $\langle \mathbf{u}, \mathbf{v} \rangle = \frac{1}{4}(\|\mathbf{u} + \mathbf{v}\|^2 - \|\mathbf{u} - \mathbf{v}\|^2)$ .  
 (b) For  $F = \mathbb{C}$ , prove that

$$\langle \mathbf{u}, \mathbf{v} \rangle = \frac{1}{4} \sum_{k=0}^3 i^k \|\mathbf{u} + i^k \mathbf{v}\|^2.$$

- (c) Deduce that two inner products on  $V$  with the same induced norm are equal.

*Solution.*

- (a) By [Theorem 10.1.2](#) (c) and its version with  $-\mathbf{v}$ , and since  $\text{Re}$  is the identity on  $\mathbb{R}$ ,

$$\|\mathbf{u} + \mathbf{v}\|^2 - \|\mathbf{u} - \mathbf{v}\|^2 = 4\langle \mathbf{u}, \mathbf{v} \rangle.$$

Dividing by 4 gives the identity.

- (b) Write  $z = \langle \mathbf{u}, \mathbf{v} \rangle$ . By [Theorem 10.1.2](#) (c) and [Theorem 10.1.1](#) (b),

$$\|\mathbf{u} + i^k \mathbf{v}\|^2 = \|\mathbf{u}\|^2 + \|\mathbf{v}\|^2 + 2\text{Re}(i^k z),$$

where we used  $\|i^k \mathbf{v}\| = |i^k| \|\mathbf{v}\| = \|\mathbf{v}\|$ . Now  $\sum_{k=0}^3 i^k = 1 + i - 1 - i = 0$ , so the terms  $\|\mathbf{u}\|^2 + \|\mathbf{v}\|^2$  contribute nothing. Writing  $z = a + bi$  with  $a, b \in \mathbb{R}$ , the four remaining terms  $2i^k \text{Re}(i^k z)$  are, for  $k = 0, 1, 2, 3$ ,

$$2a, \quad 2ib, \quad 2a, \quad 2ib,$$

since  $\text{Re}(z) = a$ ,  $\text{Re}(-iz) = b$ ,  $\text{Re}(-z) = -a$  and  $\text{Re}(iz) = -b$ . Their sum is  $4a + 4bi = 4z$ , and dividing by 4 gives the identity.

- (c) Both (a) and (b) express  $\langle \mathbf{u}, \mathbf{v} \rangle$  using only the norm. So if two inner products induce the same norm, the right-hand sides agree for all  $\mathbf{u}, \mathbf{v}$ , hence so do the left-hand sides.

**Exercise C3: The equality cases.**

Let  $V$  be an inner product space.

- (a) Prove that  $\|\mathbf{u} + \mathbf{v}\| = \|\mathbf{u}\| + \|\mathbf{v}\|$  if and only if one of  $\mathbf{u}, \mathbf{v}$  is a **non-negative real** multiple of the other.
- (b) Deduce the equality condition in the classical inequality of [Exercise \(B3: Cauchy–Schwarz, concretely\)](#) (b): for real  $a_i, b_i$ , equality holds if and only if one of the vectors  $(a_1, \dots, a_n), (b_1, \dots, b_n)$  is a multiple of the other.
- (c) Give real numbers  $a_i, b_i$ , not all zero, with  $\sum a_i b_i < 0$  and equality in the classical inequality, and explain why this does **not** contradict (a).

*Solution.*

- (a) ( $\Leftarrow$ ) If  $\mathbf{v} = \mathbf{0}$  both sides are  $\|\mathbf{u}\|$ . If  $\mathbf{u} = t\mathbf{v}$  with  $t \geq 0$ , then  $\|\mathbf{u} + \mathbf{v}\| = (t+1)\|\mathbf{v}\| = t\|\mathbf{v}\| + \|\mathbf{v}\| = \|\mathbf{u}\| + \|\mathbf{v}\|$  by [Theorem 10.1.2](#) (b).
- ( $\Rightarrow$ ) Suppose  $\|\mathbf{u} + \mathbf{v}\| = \|\mathbf{u}\| + \|\mathbf{v}\|$ . Squaring and using [Theorem 10.1.2](#) (c),

$$2 \operatorname{Re}\langle \mathbf{u}, \mathbf{v} \rangle = 2\|\mathbf{u}\|\|\mathbf{v}\|.$$

The chain in the proof of [Corollary 10.1.6](#) is therefore an equality throughout, so  $\operatorname{Re}\langle \mathbf{u}, \mathbf{v} \rangle = |\langle \mathbf{u}, \mathbf{v} \rangle| = \|\mathbf{u}\|\|\mathbf{v}\|$ . The second equality and [Theorem 10.1.5](#) make  $\langle \mathbf{u}, \mathbf{v} \rangle$  dependent. Assume  $\mathbf{v} \neq \mathbf{0}$  (otherwise  $\mathbf{v} = 0\mathbf{u}$  and we are done) and write  $\mathbf{u} = c\mathbf{v}$ . Then  $\langle \mathbf{u}, \mathbf{v} \rangle = c\|\mathbf{v}\|^2$ , and the first equality says  $\operatorname{Re}(c)\|\mathbf{v}\|^2 = |c|\|\mathbf{v}\|^2$ , so  $\operatorname{Re} c = |c|$ , which forces  $c$  to be real and  $\geq 0$ .

- (b) Apply [Theorem 10.1.5](#) on  $\mathbb{R}^n$  as in [Exercise \(B3: Cauchy–Schwarz, concretely\)](#) (b). Squaring removes the sign, so equality in the squared inequality is equality in  $|\langle \mathbf{a}, \mathbf{b} \rangle| = \|\mathbf{a}\|\|\mathbf{b}\|$ , which by [Theorem 10.1.5](#) holds exactly when  $(\mathbf{a}, \mathbf{b})$  is dependent, that is, when one of the two vectors is a multiple of the other.
- (c) Take  $n = 1$ ,  $a_1 = 1$ ,  $b_1 = -1$ : then  $\sum a_i b_i = -1$  and both sides of the classical inequality equal 1. There is no contradiction with (a), because (a) is about equality in [Corollary 10.1.6](#), which needs a non-negative multiple, while the classical inequality is the **squared** Cauchy–Schwarz inequality, which is blind to the sign. Here  $\|\mathbf{a} + \mathbf{b}\| = 0 \neq 2 = \|\mathbf{a}\| + \|\mathbf{b}\|$ .

## 10.2 Orthonormal Bases and Gram–Schmidt

A basis lets us write every vector in coordinates, but finding those coordinates costs work: in [Definition 1.5.2](#) the scalars  $a_1, \dots, a_n$  come from solving a linear system. In an inner product space one kind of basis makes that cost disappear. If the basis vectors are mutually orthogonal unit vectors, then the  $i$ -th coordinate of  $\mathbf{v}$  is just  $\langle \mathbf{v}, \mathbf{e}_i \rangle$ , one inner product, computed without reference to the others. This section defines such bases, shows that every finite-dimensional inner product space has one, and gives the algorithm that produces them.

## 10.2.1 Orthonormal lists

The standard basis of  $\mathbb{R}^n$  has two features beyond being a basis: distinct vectors are orthogonal, and each has length 1. The two conditions can be packed into a single formula using the Kronecker delta  $\delta_{ij}$ , which is 1 when  $i = j$  and 0 otherwise.

*An orthonormal list is a list of unit vectors that are pairwise perpendicular.*

### Definition 10.2.1 — Orthonormal List

Let  $V$  be an inner product space. A list  $(\mathbf{e}_1, \dots, \mathbf{e}_k)$  of vectors of  $V$  is **orthonormal** if

$$\langle \mathbf{e}_i, \mathbf{e}_j \rangle = \delta_{ij} \quad \text{for all } i, j \in \{1, \dots, k\},$$

that is, if  $\langle \mathbf{e}_i, \mathbf{e}_j \rangle = 0$  whenever  $i \neq j$  (**orthogonal**) and  $\|\mathbf{e}_i\| = 1$  for every  $i$  (**normalized**).

In words: the single equation  $\langle \mathbf{e}_i, \mathbf{e}_j \rangle = \delta_{ij}$  carries two conditions at once, because the case  $i \neq j$  says the vectors are orthogonal and the case  $i = j$  says  $\|\mathbf{e}_i\|^2 = 1$ . The list must be **ordered**, as every list in this book is, but nothing in the definition depends on the order. Note that no vector of an orthonormal list is  $\mathbf{0}$ , since  $\|\mathbf{0}\| = 0 \neq 1$ ; so by [Theorem 10.1.4](#), **every orthonormal list is linearly independent**. That single sentence is why orthonormality is so useful: independence, which normally takes an argument, comes for free.

### Definition 10.2.2 — Orthonormal Basis

An **orthonormal basis** of an inner product space  $V$  is an orthonormal list that is a basis of  $V$ .

**Examples.**

- **The standard basis.** In  $F^n$  with the standard inner product,  $\langle \mathbf{e}_i, \mathbf{e}_j \rangle = \delta_{ij}$  by inspection. This is the model, and the notation  $\mathbf{e}_i$  is deliberately reused for orthonormal vectors in a general space.
- **A rotated basis of  $\mathbb{R}^2$ .** The list  $(\frac{1}{\sqrt{2}}(1, 1), \frac{1}{\sqrt{2}}(1, -1))$  is orthonormal: the inner product of the two vectors is  $\frac{1}{2}(1 - 1) = 0$ , and each has norm  $\frac{1}{\sqrt{2}}\sqrt{2} = 1$ . Geometrically it is the standard basis turned through  $\pi/4$ .
- **A basis of  $\mathbb{R}^3$ .** The columns of  $\frac{1}{3} \begin{pmatrix} 1 & 2 & 2 \\ 2 & 1 & -2 \\ 2 & -2 & 1 \end{pmatrix}$  form an orthonormal list: each has norm  $\frac{1}{3}\sqrt{1+4+4} = 1$ , and the three pairwise inner products are  $\frac{1}{9}(2+2-4)$ ,  $\frac{1}{9}(2-4+2)$  and  $\frac{1}{9}(4-2-2)$ , all 0. Being independent and of length  $3 = \dim \mathbb{R}^3$ , it is a basis ([Theorem 1.6.4 \(a\)](#)).
- **Matrix units.** In  $M_{m \times n}(F)$  with the Frobenius inner product, the matrix units  $\mathbf{E}_{ij}$  satisfy  $\langle \mathbf{E}_{ij}, \mathbf{E}_{pq} \rangle = 1$  if  $(i, j) = (p, q)$  and 0 otherwise, since only one entry of each is non-zero. So the standard basis of  $M_{m \times n}(F)$  is orthonormal.
- **The Lagrange basis.** On  $F[x]_{\leq n}$  with the node inner product  $\langle p, q \rangle = \sum_{i=0}^n p(c_i) \overline{q(c_i)}$  of the previous section, the Lagrange polynomials  $\ell_0, \dots, \ell_n$  of [Definition 5.6.1](#) vanish at every node but their own, where they take the value 1; that is,  $\ell_i(c_j) = \delta_{ij}$ . Hence  $\langle \ell_i, \ell_j \rangle = \sum_k \ell_i(c_k) \overline{\ell_j(c_k)} = \delta_{ij}$ : the Lagrange basis is an orthonormal basis for this inner



product. An interpolation formula and an orthonormal expansion turn out to be the same statement.

- **The degenerate case.** The empty list is orthonormal, vacuously, and it is an orthonormal basis of  $V = \{\mathbf{0}\}$ . This is not a joke: it is the base case of every induction below.

**Non-example by minimal change.** In  $\mathbb{R}^2$ , the list  $((1, 1), (1, -1))$  is orthogonal, and it is a basis. It is **not** orthonormal, because  $\|(1, 1)\| = \sqrt{2} \neq 1$ . The failure is only in the normalization, and it is repaired by dividing each vector by its norm, which changes neither the spans nor the orthogonality ([Theorem 10.1.2 \(b\)](#)). Scaling is always available; orthogonality is the real content.

## 10.2.2 Coordinates in an orthonormal basis

Here is the payoff promised in the opening.

### Theorem 10.2.1 — Coordinates, Inner Products and Norms in an Orthonormal Basis

Let  $(\mathbf{e}_1, \dots, \mathbf{e}_n)$  be an orthonormal basis of an inner product space  $V$ , and let  $\mathbf{u}, \mathbf{v} \in V$ . Then:

- (a)  $\mathbf{v} = \sum_{i=1}^n \langle \mathbf{v}, \mathbf{e}_i \rangle \mathbf{e}_i$ ;
- (b)  $\langle \mathbf{u}, \mathbf{v} \rangle = \sum_{i=1}^n \langle \mathbf{u}, \mathbf{e}_i \rangle \overline{\langle \mathbf{v}, \mathbf{e}_i \rangle}$ ;
- (c)  $\|\mathbf{v}\|^2 = \sum_{i=1}^n |\langle \mathbf{v}, \mathbf{e}_i \rangle|^2$  (**Parseval's identity** for a finite orthonormal basis).

*Idea.* For (a) we already know that  $\mathbf{v} = \sum a_i \mathbf{e}_i$  for unique scalars  $a_i$ ; the only question is what they are. Pair both sides with  $\mathbf{e}_j$ : orthonormality kills every term except the  $j$ -th, and that one survives with coefficient 1. Parts (b) and (c) then follow by expanding, since the coordinates are now known.

*Proof.*

- (a) Since  $(\mathbf{e}_1, \dots, \mathbf{e}_n)$  is a basis, [Theorem 1.5.1](#) gives scalars  $a_1, \dots, a_n \in F$  with  $\mathbf{v} = \sum_i a_i \mathbf{e}_i$ . Fix  $j$ . Pairing with  $\mathbf{e}_j$  and using (IP1),

$$\langle \mathbf{v}, \mathbf{e}_j \rangle = \sum_{i=1}^n a_i \langle \mathbf{e}_i, \mathbf{e}_j \rangle = \sum_{i=1}^n a_i \delta_{ij} = a_j,$$

by [Definition 10.2.1](#). So  $a_j = \langle \mathbf{v}, \mathbf{e}_j \rangle$  for every  $j$ , which is (a).

- (b) Write  $\mathbf{u} = \sum_i \langle \mathbf{u}, \mathbf{e}_i \rangle \mathbf{e}_i$  and  $\mathbf{v} = \sum_j \langle \mathbf{v}, \mathbf{e}_j \rangle \mathbf{e}_j$  by (a). Expanding the first slot with (IP1) and the second with [Theorem 10.1.1 \(b\)](#),

$$\langle \mathbf{u}, \mathbf{v} \rangle = \sum_i \sum_j \langle \mathbf{u}, \mathbf{e}_i \rangle \overline{\langle \mathbf{v}, \mathbf{e}_j \rangle} \langle \mathbf{e}_i, \mathbf{e}_j \rangle.$$

Only the terms with  $i = j$  survive, and there  $\langle \mathbf{e}_i, \mathbf{e}_i \rangle = 1$ . This is (b).

- (c) Take  $\mathbf{u} = \mathbf{v}$  in (b): the  $i$ -th term becomes  $\langle \mathbf{v}, \mathbf{e}_i \rangle \overline{\langle \mathbf{v}, \mathbf{e}_i \rangle} = |\langle \mathbf{v}, \mathbf{e}_i \rangle|^2$  by [Theorem 0.6.2](#) (c), and the left side is  $\|\mathbf{v}\|^2$ . ■

Read (b) again: in the coordinates supplied by an orthonormal basis, **every** inner product becomes the standard inner product of  $F^n$ . So an orthonormal basis does not merely simplify coordinates; it turns an abstract inner product space into  $F^n$  with the dot product. We record that as a corollary below, and use it repeatedly: **to make an abstract statement concrete, fix an orthonormal basis.**

### Example. (Reading Off Coordinates)

Let  $\mathbf{e}_1 = \frac{1}{3}(1, 2, 2)$ ,  $\mathbf{e}_2 = \frac{1}{3}(2, 1, -2)$ ,  $\mathbf{e}_3 = \frac{1}{3}(2, -2, 1)$ , the orthonormal basis of  $\mathbb{R}^3$  found above. Write  $\mathbf{v} = (3, 0, 3)$  in this basis, and check Parseval's identity.

*Solution.* By [Theorem 10.2.1](#) (a), the coordinates are

$$\begin{aligned}\langle \mathbf{v}, \mathbf{e}_1 \rangle &= \frac{1}{3}(3 + 0 + 6) = 3, \\ \langle \mathbf{v}, \mathbf{e}_2 \rangle &= \frac{1}{3}(6 + 0 - 6) = 0, \\ \langle \mathbf{v}, \mathbf{e}_3 \rangle &= \frac{1}{3}(6 + 0 + 3) = 3,\end{aligned}$$

so  $\mathbf{v} = 3\mathbf{e}_1 + 0\mathbf{e}_2 + 3\mathbf{e}_3$ . No system was solved. Checking directly,  $3\mathbf{e}_1 + 3\mathbf{e}_3 = (1, 2, 2) + (2, -2, 1) = (3, 0, 3)$ . For Parseval,  $\|\mathbf{v}\|^2 = 9 + 0 + 9 = 18$ , and indeed  $3^2 + 0^2 + 3^2 = 18$ .

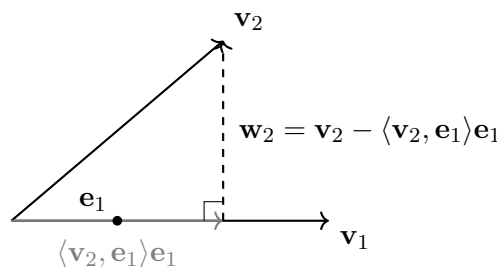
*Quick check.* With the same  $\mathbf{e}_1, \mathbf{e}_2, \mathbf{e}_3$ , find the coordinates of  $\mathbf{v} = (1, 1, 1)$  and verify Parseval's identity.

*Solution.*  $\langle \mathbf{v}, \mathbf{e}_1 \rangle = \frac{1}{3}(1 + 2 + 2) = \frac{5}{3}$ ,  $\langle \mathbf{v}, \mathbf{e}_2 \rangle = \frac{1}{3}(2 + 1 - 2) = \frac{1}{3}$ ,  $\langle \mathbf{v}, \mathbf{e}_3 \rangle = \frac{1}{3}(2 - 2 + 1) = \frac{1}{3}$ . Then  $\frac{25}{9} + \frac{1}{9} + \frac{1}{9} = \frac{27}{9} = 3 = \|\mathbf{v}\|^2$ , as Parseval requires.

## 10.2.3 The Gram–Schmidt process

Everything so far assumed that an orthonormal basis was handed to us. Does one always exist? The answer is yes in finite dimension, and the proof is an algorithm that we will use constantly.

The idea comes from a picture in  $\mathbb{R}^3$ . Given independent  $\mathbf{v}_1, \mathbf{v}_2$ , normalize  $\mathbf{v}_1$  to get  $\mathbf{e}_1$ . The vector  $\mathbf{v}_2$  is generally not perpendicular to  $\mathbf{e}_1$ ; but its “shadow” on the line through  $\mathbf{e}_1$  is the multiple  $\langle \mathbf{v}_2, \mathbf{e}_1 \rangle \mathbf{e}_1$ , and subtracting the shadow leaves a vector  $\mathbf{w}_2$  perpendicular to  $\mathbf{e}_1$ . Normalize it. Then repeat with  $\mathbf{v}_3$ , subtracting its shadows on both  $\mathbf{e}_1$  and  $\mathbf{e}_2$ .



Project, subtract, normalize. The right angle at the foot is the only thing that has to be checked, and one line of algebra checks it.

### Theorem 10.2.2 — Gram–Schmidt Process

Let  $V$  be an inner product space and let  $(\mathbf{v}_1, \dots, \mathbf{v}_m)$  be a **linearly independent** list in  $V$ . Define recursively

$$\mathbf{w}_k = \mathbf{v}_k - \sum_{i=1}^{k-1} \langle \mathbf{v}_k, \mathbf{e}_i \rangle \mathbf{e}_i, \quad \mathbf{e}_k = \frac{\mathbf{w}_k}{\|\mathbf{w}_k\|} \quad (k = 1, \dots, m).$$

Then every  $\mathbf{w}_k$  is non-zero, so the recursion is defined, and:

- (a)  $(\mathbf{e}_1, \dots, \mathbf{e}_m)$  is an orthonormal list;
- (b)  $\text{span}(\mathbf{e}_1, \dots, \mathbf{e}_k) = \text{span}(\mathbf{v}_1, \dots, \mathbf{v}_k)$  for every  $k = 1, \dots, m$ ;
- (c) if  $(\mathbf{v}_1, \dots, \mathbf{v}_r)$  is already orthonormal for some  $r \leq m$ , then  $\mathbf{e}_i = \mathbf{v}_i$  for  $i \leq r$ .

In particular, every finite-dimensional inner product space has an orthonormal basis.

*Idea.* Induction on  $k$ , with the statement “ $(\mathbf{e}_1, \dots, \mathbf{e}_k)$  is orthonormal and spans  $\text{span}(\mathbf{v}_1, \dots, \mathbf{v}_k)$ ” as the hypothesis. Two things need care. First,  $\mathbf{w}_k \neq \mathbf{0}$ : if it were  $\mathbf{0}$ , then  $\mathbf{v}_k$  would lie in  $\text{span}(\mathbf{e}_1, \dots, \mathbf{e}_{k-1}) = \text{span}(\mathbf{v}_1, \dots, \mathbf{v}_{k-1})$ , contradicting independence. That is the only place independence is used, and it is used exactly there. Second,  $\mathbf{w}_k \perp \mathbf{e}_j$  for  $j < k$ : pair and watch the sum collapse.

*Proof.* We prove by induction on  $k$  that  $\mathbf{w}_k \neq \mathbf{0}$ , that  $(\mathbf{e}_1, \dots, \mathbf{e}_k)$  is orthonormal, and that  $\text{span}(\mathbf{e}_1, \dots, \mathbf{e}_k) = \text{span}(\mathbf{v}_1, \dots, \mathbf{v}_k)$ .

*Base case*  $k = 1$ . The empty sum is  $\mathbf{0}$ , so  $\mathbf{w}_1 = \mathbf{v}_1$ , which is non-zero because a list containing  $\mathbf{0}$  is dependent. Then  $\mathbf{e}_1 = \mathbf{v}_1/\|\mathbf{v}_1\|$  has norm 1 by Theorem 10.1.2 (b), and  $\text{span}(\mathbf{e}_1) = \text{span}(\mathbf{v}_1)$  since each vector is a non-zero multiple of the other.

*Induction step.* Let  $2 \leq k \leq m$  and suppose the claim holds for  $k - 1$ . Put  $U = \text{span}(\mathbf{e}_1, \dots, \mathbf{e}_{k-1}) = \text{span}(\mathbf{v}_1, \dots, \mathbf{v}_{k-1})$ .

First,  $\mathbf{w}_k \neq \mathbf{0}$ . If  $\mathbf{w}_k = \mathbf{0}$ , then  $\mathbf{v}_k = \sum_{i < k} \langle \mathbf{v}_k, \mathbf{e}_i \rangle \mathbf{e}_i \in U = \text{span}(\mathbf{v}_1, \dots, \mathbf{v}_{k-1})$ , so  $\mathbf{v}_k$  would be a linear combination of the earlier  $\mathbf{v}_i$ , making  $(\mathbf{v}_1, \dots, \mathbf{v}_m)$  dependent, contrary to hypothesis. Hence  $\|\mathbf{w}_k\| > 0$  and  $\mathbf{e}_k$  is defined, with  $\|\mathbf{e}_k\| = 1$ .

Next,  $\mathbf{e}_k \perp \mathbf{e}_j$  for every  $j < k$ . By (IP1) and the induction hypothesis  $\langle \mathbf{e}_i, \mathbf{e}_j \rangle = \delta_{ij}$ ,

$$\langle \mathbf{w}_k, \mathbf{e}_j \rangle = \langle \mathbf{v}_k, \mathbf{e}_j \rangle - \sum_{i=1}^{k-1} \langle \mathbf{v}_k, \mathbf{e}_i \rangle \delta_{ij} = \langle \mathbf{v}_k, \mathbf{e}_j \rangle - \langle \mathbf{v}_k, \mathbf{e}_j \rangle = 0,$$

and  $\mathbf{e}_k$  is a scalar multiple of  $\mathbf{w}_k$ , so  $\langle \mathbf{e}_k, \mathbf{e}_j \rangle = 0$  as well. Together with the induction hypothesis,  $(\mathbf{e}_1, \dots, \mathbf{e}_k)$  is orthonormal, which is (a).

For (b),  $\mathbf{e}_k$  is a linear combination of  $\mathbf{v}_k$  and  $\mathbf{e}_1, \dots, \mathbf{e}_{k-1}$ , hence lies in  $\text{span}(\mathbf{v}_1, \dots, \mathbf{v}_k)$  by the induction hypothesis; so  $\text{span}(\mathbf{e}_1, \dots, \mathbf{e}_k) \subseteq \text{span}(\mathbf{v}_1, \dots, \mathbf{v}_k)$ . Conversely  $\mathbf{v}_k = \|\mathbf{w}_k\| \mathbf{e}_k + \sum_{i < k} \langle \mathbf{v}_k, \mathbf{e}_i \rangle \mathbf{e}_i \in \text{span}(\mathbf{e}_1, \dots, \mathbf{e}_k)$ , and the earlier  $\mathbf{v}_i$  lie there by the induction hypothesis. The two spans are equal, completing the induction.

For (c), suppose  $(\mathbf{v}_1, \dots, \mathbf{v}_r)$  is orthonormal and that  $\mathbf{e}_i = \mathbf{v}_i$  for all  $i < k$ , where  $k \leq r$ . Then  $\langle \mathbf{v}_k, \mathbf{e}_i \rangle = \langle \mathbf{v}_k, \mathbf{v}_i \rangle = 0$  for  $i < k$ , so  $\mathbf{w}_k = \mathbf{v}_k$  and  $\|\mathbf{w}_k\| = 1$ , giving  $\mathbf{e}_k = \mathbf{v}_k$ . Induction on  $k$  proves (c).

Finally, let  $V$  be finite-dimensional. If  $V = \{\mathbf{0}\}$ , the empty list is an orthonormal basis. Otherwise take any basis  $(\mathbf{v}_1, \dots, \mathbf{v}_n)$  of  $V$  and apply the process: by (a) the output is orthonormal, and by (b) it spans  $\text{span}(\mathbf{v}_1, \dots, \mathbf{v}_n) = V$ , so it is an orthonormal basis. ■

Two remarks on the formulas. The vector subtracted from  $\mathbf{v}_k$  is exactly the expansion of  $\mathbf{v}_k$  in the orthonormal basis  $(\mathbf{e}_1, \dots, \mathbf{e}_{k-1})$  of  $U$ , which by [Theorem 10.2.1](#) (a) is the best available “copy of  $\mathbf{v}_k$  inside  $U$ ”; the next section will call it the orthogonal projection of  $\mathbf{v}_k$  onto  $U$ . And the subtraction is the same move that produced the vector  $\mathbf{z}$  in the proof of [Theorem 10.1.5](#). **Project, then subtract** is one move used twice.

### Corollary 10.2.3 — Extending an Orthonormal List

Let  $V$  be a finite-dimensional inner product space and let  $(\mathbf{e}_1, \dots, \mathbf{e}_r)$  be an orthonormal list in  $V$ . Then there are vectors  $\mathbf{e}_{r+1}, \dots, \mathbf{e}_n \in V$  such that  $(\mathbf{e}_1, \dots, \mathbf{e}_n)$  is an orthonormal basis of  $V$ .

*Proof.* The list  $(\mathbf{e}_1, \dots, \mathbf{e}_r)$  is linearly independent by [Theorem 10.1.4](#), since its vectors are orthogonal and non-zero. By [Theorem 1.6.5](#) there are  $\mathbf{v}_{r+1}, \dots, \mathbf{v}_n \in V$  making  $(\mathbf{e}_1, \dots, \mathbf{e}_r, \mathbf{v}_{r+1}, \dots, \mathbf{v}_n)$  a basis of  $V$ . Apply [Theorem 10.2.2](#) to that basis. By part (c) the first  $r$  output vectors are  $\mathbf{e}_1, \dots, \mathbf{e}_r$  unchanged, and by (a) and (b) the whole output is an orthonormal basis of  $V$ . ■

## 10.2.4 Two worked examples

### Example. (Gram–Schmidt in $\mathbb{R}^3$ )

Apply the Gram–Schmidt process of [Theorem 10.2.2](#) to  $\mathbf{v}_1 = (1, 1, 1)$ ,  $\mathbf{v}_2 = (0, 1, 1)$ ,  $\mathbf{v}_3 = (0, 0, 1)$  in  $\mathbb{R}^3$  with the dot product.

*Solution. Step 1.*  $\|\mathbf{v}_1\|^2 = 3$ , so  $\mathbf{e}_1 = \frac{1}{\sqrt{3}}(1, 1, 1)$ .

*Step 2.*  $\langle \mathbf{v}_2, \mathbf{e}_1 \rangle = \frac{1}{\sqrt{3}}(0 + 1 + 1) = \frac{2}{\sqrt{3}}$ , so

$$\mathbf{w}_2 = (0, 1, 1) - \frac{2}{3}(1, 1, 1) = \frac{1}{3}(-2, 1, 1).$$

Then  $\|\mathbf{w}_2\|^2 = \frac{1}{9}(4 + 1 + 1) = \frac{2}{3}$ , and  $\mathbf{e}_2 = \frac{1}{\sqrt{6}}(-2, 1, 1)$ .

*Step 3.*  $\langle \mathbf{v}_3, \mathbf{e}_1 \rangle = \frac{1}{\sqrt{3}}$  and  $\langle \mathbf{v}_3, \mathbf{e}_2 \rangle = \frac{1}{\sqrt{6}}$ , so

$$\mathbf{w}_3 = (0, 0, 1) - \frac{1}{3}(1, 1, 1) - \frac{1}{6}(-2, 1, 1) = (0, -\frac{1}{2}, \frac{1}{2}).$$

Then  $\|\mathbf{w}_3\|^2 = \frac{1}{2}$  and  $\mathbf{e}_3 = \frac{1}{\sqrt{2}}(0, -1, 1)$ .

*Check.* The three vectors have norm 1, and  $\langle \mathbf{e}_1, \mathbf{e}_2 \rangle = \frac{1}{\sqrt{18}}(-2 + 1 + 1) = 0$ ,  $\langle \mathbf{e}_1, \mathbf{e}_3 \rangle = \frac{1}{\sqrt{6}}(0 - 1 + 1) = 0$ ,  $\langle \mathbf{e}_2, \mathbf{e}_3 \rangle = \frac{1}{\sqrt{12}}(0 - 1 + 1) = 0$ . The span condition is visible too:  $\mathbf{e}_1$  is a multiple of  $\mathbf{v}_1$ , and  $\mathbf{e}_2$  is a combination of  $\mathbf{v}_1$  and  $\mathbf{v}_2$ .

The next example runs the same three steps in a function space, where no picture is available and only the algebra is left.

**Example. (Gram–Schmidt on  $1, x, x^2$ )**

Apply the Gram–Schmidt process of [Theorem 10.2.2](#) to  $(1, x, x^2)$  in  $\mathbb{R}[x]_{\leq 2}$  with

$$\langle p, q \rangle = \int_{-1}^1 p(t)q(t) dt.$$

*Solution. Step 1.*  $\|1\|^2 = \int_{-1}^1 1 dt = 2$ , so  $\mathbf{e}_1 = 1/\sqrt{2}$ .

*Step 2.*  $\langle x, \mathbf{e}_1 \rangle = \frac{1}{\sqrt{2}} \int_{-1}^1 t dt = 0$ , since the integrand is odd. So  $\mathbf{w}_2 = x$ , and  $\|x\|^2 = \int_{-1}^1 t^2 dt = \frac{2}{3}$ , giving  $\mathbf{e}_2 = \sqrt{3/2}x$ .

*Step 3.*  $\langle x^2, \mathbf{e}_1 \rangle = \frac{1}{\sqrt{2}} \cdot \frac{2}{3}$  and  $\langle x^2, \mathbf{e}_2 \rangle = \sqrt{3/2} \int_{-1}^1 t^3 dt = 0$ , again by oddness. Hence

$$\mathbf{w}_3 = x^2 - \frac{1}{\sqrt{2}} \cdot \frac{2}{3} \cdot \frac{1}{\sqrt{2}} = x^2 - \frac{1}{3}.$$

Its squared norm is

$$\int_{-1}^1 \left(t^2 - \frac{1}{3}\right)^2 dt = \frac{2}{5} - \frac{4}{9} + \frac{2}{9} = \frac{8}{45},$$

so  $\mathbf{e}_3 = \frac{\sqrt{10}}{4}(3x^2 - 1)$ .

The monic polynomials produced along the way are  $1$ ,  $x$  and  $x^2 - \frac{1}{3}$ . These are, up to scaling, the first three **Legendre polynomials**; the usual normalization takes  $P_2 = \frac{1}{2}(3x^2 - 1) = \frac{3}{2}(x^2 - \frac{1}{3})$ , fixed by the convention  $P_k(1) = 1$  rather than by the norm. Section 10 of this chapter takes up such families in general.

Notice how much work the oddness of the integrand did: two of the three inner products were 0 before any computation. Symmetry of the weight is worth looking for.

**Warning. Gram–Schmidt depends on the order of the list.** Run the process on  $((0, 1), (1, 1))$  in  $\mathbb{R}^2$ : it returns  $((0, 1), (1, 0))$ . Run it on  $((1, 1), (0, 1))$ , the same two vectors in the other order: it returns  $(\frac{1}{\sqrt{2}}(1, 1), \frac{1}{\sqrt{2}}(-1, 1))$ . Both outputs are orthonormal bases of  $\mathbb{R}^2$ , and they are different. An orthonormal basis is never unique (for  $\dim V \geq 1$  one can always replace  $\mathbf{e}_1$  by  $-\mathbf{e}_1$ ); the process returns **an** orthonormal basis adapted to the given order, not **the** orthonormal basis.

*Quick check.* Run Gram–Schmidt on  $((1, 0), (1, 1))$  in  $\mathbb{R}^2$ , and then on  $((1, 1), (1, 0))$ . Do you get the same list?

*Solution.* First order:  $\mathbf{e}_1 = (1, 0)$ ;  $\langle (1, 1), \mathbf{e}_1 \rangle = 1$ , so  $\mathbf{w}_2 = (1, 1) - (1, 0) = (0, 1)$  and  $\mathbf{e}_2 = (0, 1)$ . Output  $((1, 0), (0, 1))$ .

Second order:  $\mathbf{e}_1 = \frac{1}{\sqrt{2}}(1, 1)$ ;  $\langle (1, 0), \mathbf{e}_1 \rangle = \frac{1}{\sqrt{2}}$ , so  $\mathbf{w}_2 = (1, 0) - \frac{1}{2}(1, 1) = \frac{1}{2}(1, -1)$  and  $\mathbf{e}_2 = \frac{1}{\sqrt{2}}(1, -1)$ . Output  $(\frac{1}{\sqrt{2}}(1, 1), \frac{1}{\sqrt{2}}(1, -1))$ . Not the same list, and not even the same set of lines.

*Warning.* The classical process is numerically fragile. Computed with rounding, the vectors  $\mathbf{e}_1, \mathbf{e}_2, \dots$  drift out of orthogonality, and the drift grows with  $k$  and with how nearly dependent the input list is. Nothing above is wrong — the fault is in the arithmetic, not the mathematics — but in practice one uses a rearranged version or reflections instead. Section 8 of this chapter gives the reflection-based method, and Chapter 23 explains why it is the stable one.

## 10.2.5 All inner product spaces of the same dimension look alike

### Corollary 10.2.4 — Classification of Finite-Dimensional Inner Product Spaces

Let  $V$  be an inner product space over  $F$  with  $\dim V = n$ . Then there is an isomorphism  $T: V \rightarrow F^n$  with

$$\langle T\mathbf{u}, T\mathbf{v} \rangle = \langle \mathbf{u}, \mathbf{v} \rangle \quad \text{for all } \mathbf{u}, \mathbf{v} \in V,$$

where the right-hand inner product is that of  $V$  and the left-hand one is the standard inner product of  $F^n$ . An isomorphism with this property is an **isometric isomorphism**, so every  $n$ -dimensional inner product space over  $F$  is isometrically isomorphic to  $F^n$ .

*Proof.* By Theorem 10.2.2,  $V$  has an orthonormal basis  $\mathcal{E} = (\mathbf{e}_1, \dots, \mathbf{e}_n)$ . Let  $T: V \rightarrow F^n$  be the coordinate map  $T\mathbf{v} = [\mathbf{v}]_{\mathcal{E}}$ , an isomorphism by Corollary 3.5.5. By Theorem 10.2.1 (a), the  $i$ -th entry of  $T\mathbf{v}$  is  $\langle \mathbf{v}, \mathbf{e}_i \rangle$ . Hence the standard inner product of  $T\mathbf{u}$  and  $T\mathbf{v}$  in  $F^n$  is  $\sum_i \langle \mathbf{u}, \mathbf{e}_i \rangle \overline{\langle \mathbf{v}, \mathbf{e}_i \rangle}$ , which equals  $\langle \mathbf{u}, \mathbf{v} \rangle$  by Theorem 10.2.1 (b). ■

So dimension is a complete invariant of finite-dimensional inner product spaces over a fixed field: there is nothing to classify. This should be compared with Theorem 3.5.4, which said the same for vector spaces; adding an inner product adds no new invariant. What becomes interesting is the interaction of the inner product with a given operator, and that is the subject of Chapter 11. Isometries in general are the subject of Section 7 of this chapter.

## 10.2.6 Gram matrices

Given a list of vectors, all of its inner products can be collected into one matrix, and that matrix turns out to answer questions about the list.

### Definition 10.2.3 — Gram Matrix

Let  $(\mathbf{v}_1, \dots, \mathbf{v}_k)$  be a list in an inner product space  $V$  over  $F$ . Its **Gram matrix** is

$\mathbf{G} \in M_k(F)$  with entries

$$(\mathbf{G})_{ij} = \langle \mathbf{v}_j, \mathbf{v}_i \rangle \quad (1 \leq i, j \leq k).$$

The order of the subscripts looks backwards and is chosen on purpose: it makes  $\mathbf{G}$  act on coefficient vectors the right way round, as part (b) below shows. For a real list the distinction disappears, since then  $\langle \mathbf{v}_j, \mathbf{v}_i \rangle = \langle \mathbf{v}_i, \mathbf{v}_j \rangle$ .

For example, the Gram matrix of  $((1, 1, 0), (1, 0, 1), (0, 1, 1))$  in  $\mathbb{R}^3$  is  $\begin{pmatrix} 2 & 1 & 1 \\ 1 & 2 & 1 \\ 1 & 1 & 2 \end{pmatrix}$ , with the squared norms on the diagonal.

Over  $\mathbb{C}$  the order of the subscripts becomes visible. For  $\mathbf{v}_1 = (1, 0)$  and  $\mathbf{v}_2 = (i, 1)$  in  $\mathbb{C}^2$ , the off-diagonal entries are

$$(\mathbf{G})_{12} = \langle \mathbf{v}_2, \mathbf{v}_1 \rangle = i, \quad (\mathbf{G})_{21} = \langle \mathbf{v}_1, \mathbf{v}_2 \rangle = \bar{i} = -i,$$

so  $\mathbf{G} = \begin{pmatrix} 1 & i \\ -i & 2 \end{pmatrix}$ , which is Hermitian but not symmetric. Part (b) below says  $\mathbf{x}^* \mathbf{G} \mathbf{x} = \|x_1 \mathbf{v}_1 + x_2 \mathbf{v}_2\|^2$ ; with  $\mathbf{x} = (1, 1)$  both sides are 3, since  $\mathbf{v}_1 + \mathbf{v}_2 = (1 + i, 1)$ . Writing  $(\mathbf{G})_{ij} = \langle \mathbf{v}_i, \mathbf{v}_j \rangle$  instead would transpose  $\mathbf{G}$  and leave that identity reading  $\mathbf{x}^* \mathbf{G}^\top \mathbf{x}$ .

### Theorem 10.2.5 — Properties of the Gram Matrix

Let  $(\mathbf{v}_1, \dots, \mathbf{v}_k)$  be a list in an inner product space  $V$  over  $F$ , with Gram matrix  $\mathbf{G}$ . Then:

- (a)  $\mathbf{G}$  is **Hermitian**,  $\mathbf{G}^* = \mathbf{G}$ , and its diagonal entries are the real numbers  $\|\mathbf{v}_i\|^2 \geq 0$ ;
- (b)  $\mathbf{x}^* \mathbf{G} \mathbf{x} = \|\sum_j x_j \mathbf{v}_j\|^2$  for every  $\mathbf{x} \in F^k$ ;
- (c)  $\mathbf{G}$  is invertible if and only if  $(\mathbf{v}_1, \dots, \mathbf{v}_k)$  is linearly independent;
- (d)  $\det \mathbf{G}$  is a real number with  $\det \mathbf{G} \geq 0$ , and  $\det \mathbf{G} > 0$  exactly when the list is linearly independent.

*Idea.* Part (b) is the whole theorem: the quadratic expression  $\mathbf{x}^* \mathbf{G} \mathbf{x}$  is a squared length, so it vanishes only for the coefficient vectors of genuine dependence relations. That identifies  $\text{nul}(\mathbf{G})$  with the set of dependence relations and gives (c) at once. For (d), fix an orthonormal basis of the span and write  $\mathbf{G} = \mathbf{A}^* \mathbf{A}$  for a coordinate matrix  $\mathbf{A}$ ; a determinant of that shape is a squared modulus.

*Proof.*

- (a) By (IP2),  $(\mathbf{G}^*)_{ij} = \overline{(\mathbf{G})_{ji}} = \overline{\langle \mathbf{v}_i, \mathbf{v}_j \rangle} = \langle \mathbf{v}_j, \mathbf{v}_i \rangle = (\mathbf{G})_{ij}$ . The diagonal entries are  $\langle \mathbf{v}_i, \mathbf{v}_i \rangle = \|\mathbf{v}_i\|^2$ , real and  $\geq 0$  by Theorem 10.1.1 (a).
- (b) Write  $\mathbf{u} = \sum_j x_j \mathbf{v}_j$ . Expanding by (IP1) and Theorem 10.1.1 (b),

$$\|\mathbf{u}\|^2 = \sum_i \sum_j x_j \bar{x}_i \langle \mathbf{v}_j, \mathbf{v}_i \rangle = \sum_i \bar{x}_i \sum_j (\mathbf{G})_{ij} x_j,$$

and the right-hand side is exactly the entry of the  $1 \times 1$  matrix  $\mathbf{x}^* \mathbf{G} \mathbf{x}$ .

- (c) We show  $\text{nul}(\mathbf{G}) = \{\mathbf{x} \in F^k : \sum_j x_j \mathbf{v}_j = \mathbf{0}\}$ . If  $\sum_j x_j \mathbf{v}_j = \mathbf{0}$ , then for each  $i$ ,

$(\mathbf{G}\mathbf{x})_i = \sum_j \langle \mathbf{v}_j, \mathbf{v}_i \rangle x_j = \langle \sum_j x_j \mathbf{v}_j, \mathbf{v}_i \rangle = 0$  by (IP1), so  $\mathbf{G}\mathbf{x} = \mathbf{0}$ . Conversely, if  $\mathbf{G}\mathbf{x} = \mathbf{0}$ , then  $\mathbf{x}^* \mathbf{G}\mathbf{x} = 0$ , so  $\|\sum_j x_j \mathbf{v}_j\| = 0$  by (b) and hence  $\sum_j x_j \mathbf{v}_j = \mathbf{0}$  by Theorem 10.1.2 (a). Now  $\mathbf{G}$  is invertible if and only if  $\mathbf{G}\mathbf{x} = \mathbf{0}$  forces  $\mathbf{x} = \mathbf{0}$  (Theorem 2.4.4, (a)  $\Leftrightarrow$  (b)), which by the displayed equality of sets says exactly that the only dependence relation among the  $\mathbf{v}_j$  is the trivial one.

- (d) Let  $U = \text{span}(\mathbf{v}_1, \dots, \mathbf{v}_k)$ , let  $r = \dim U$ , and let  $(\mathbf{e}_1, \dots, \mathbf{e}_r)$  be an orthonormal basis of  $U$ , which exists by Theorem 10.2.2. Let  $\mathbf{A} \in M_{r \times k}(F)$  have as its  $j$ -th column the coordinate vector of  $\mathbf{v}_j$ , so that  $\mathbf{v}_j = \sum_p a_{pj} \mathbf{e}_p$ . By Theorem 10.2.1 (b),

$$(\mathbf{G})_{ij} = \langle \mathbf{v}_j, \mathbf{v}_i \rangle = \sum_{p=1}^r a_{pj} \overline{a_{pi}} = (\mathbf{A}^* \mathbf{A})_{ij},$$

so  $\mathbf{G} = \mathbf{A}^* \mathbf{A}$ .

If the list is dependent, then  $\mathbf{G}$  is singular by (c) and  $\det \mathbf{G} = 0$ . If it is independent, then  $r = k$  and  $\mathbf{A}$  is square. By Theorem 6.4.5,  $\det(\mathbf{A}^*) = \det(\overline{\mathbf{A}})$ , and  $\det(\overline{\mathbf{A}}) = \overline{\det \mathbf{A}}$ , because by the Leibniz formula (Theorem 6.4.3) the determinant is a sum of products of entries and conjugation preserves sums and products (Theorem 0.6.2). Hence, by Theorem 6.5.4,

$$\det \mathbf{G} = \det(\mathbf{A}^*) \det(\mathbf{A}) = \overline{\det \mathbf{A}} \det \mathbf{A} = |\det \mathbf{A}|^2.$$

This is real and  $\geq 0$ , and it is  $> 0$  because  $\mathbf{A}$  is the matrix of an isomorphism onto its coordinates, hence invertible with  $\det \mathbf{A} \neq 0$  (Theorem 6.5.2). ■

Over  $\mathbb{R}$ , with  $V = \mathbb{R}^n$  and the dot product, part (d) is the statement Chapter 6 reached from Cauchy–Binet: Corollary 6.7.3 says that  $\det(\mathbf{A}\mathbf{A}^\top) \geq 0$  for a real  $\mathbf{A}$ , with equality exactly when the rows of  $\mathbf{A}$  are dependent. Taking the rows of  $\mathbf{A}$  to be  $\mathbf{v}_1, \dots, \mathbf{v}_k$  makes  $\mathbf{A}\mathbf{A}^\top$  the Gram matrix, so that corollary is the special case of (d) in which the ambient space is  $\mathbb{R}^n$ . The pointer left there is now paid: a Gram matrix is exactly a matrix of pairwise inner products, and its determinant is non-negative for that reason.

### Example. (Deciding Independence with a Gram Matrix)

Decide whether  $((1, 1, 0), (1, 0, 1), (0, 1, -1))$  is independent in  $\mathbb{R}^3$ , using its Gram matrix.

*Solution.* The pairwise dot products give

$$\mathbf{G} = \begin{pmatrix} 2 & 1 & 1 \\ 1 & 2 & -1 \\ 1 & -1 & 2 \end{pmatrix},$$

and expanding along the first row,  $\det \mathbf{G} = 2(4-1) - 1(2+1) + 1(-1-2) = 6-3-3 = 0$ . By Theorem 10.2.5 (d) the list is dependent. Indeed  $\mathbf{G}\mathbf{x} = \mathbf{0}$  for  $\mathbf{x} = (1, -1, -1)$ , and Theorem 10.2.5 (c) says this is a dependence relation:  $(1, 1, 0) - (1, 0, 1) - (0, 1, -1) = \mathbf{0}$ .

Two features of this method are worth noting. It never needs coordinates in the ambient space, so it works verbatim for polynomials and functions, where writing the vectors as columns is not an option. And the answer comes with a witness: a null vector of  $\mathbf{G}$  is a dependence relation,



ready to read off.

## 10.2.7 Exercises

### 10.2.7.1 A. Check your understanding

#### Exercise A1.

- Define an orthonormal list, using the Kronecker delta, and say what an orthonormal basis is.
- Why is every orthonormal list linearly independent? Name the result used.
- Write down the coordinates of  $\mathbf{v}$  in an orthonormal basis  $(\mathbf{e}_1, \dots, \mathbf{e}_n)$ , and state Parseval's identity.
- True or false: the Gram–Schmidt process ([Theorem 10.2.2](#)) applied to a list and to the same list in reverse order produces the same orthonormal list. Justify your answer.
- State the criterion for linear independence in terms of the Gram matrix.
- True or false: every 3-dimensional real inner product space is isometrically isomorphic to  $\mathbb{R}^3$  with the dot product. Justify your answer.

*Solution.*

- A list  $(\mathbf{e}_1, \dots, \mathbf{e}_k)$  with  $\langle \mathbf{e}_i, \mathbf{e}_j \rangle = \delta_{ij}$  for all  $i, j$  ([Definition 10.2.1](#)); an orthonormal basis is such a list that is also a basis ([Definition 10.2.2](#)).
- Its vectors are pairwise orthogonal and non-zero (each has norm 1), so [Theorem 10.1.4](#) applies.
- $\mathbf{v} = \sum_i \langle \mathbf{v}, \mathbf{e}_i \rangle \mathbf{e}_i$ , and  $\|\mathbf{v}\|^2 = \sum_i |\langle \mathbf{v}, \mathbf{e}_i \rangle|^2$  ([Theorem 10.2.1](#)).
- False. On  $((0, 1), (1, 1))$  the process returns  $((0, 1), (1, 0))$ , while on  $((1, 1), (0, 1))$  it returns  $(\frac{1}{\sqrt{2}}(1, 1), \frac{1}{\sqrt{2}}(-1, 1))$ .
- The list is independent if and only if its Gram matrix is invertible, if and only if  $\det \mathbf{G} > 0$  ([Theorem 10.2.5](#) (c), (d)).
- True. By [Corollary 10.2.4](#), the coordinate map with respect to any orthonormal basis is an isometric isomorphism onto  $\mathbb{R}^3$ .

### 10.2.7.2 B. Practice

#### Exercise B1: Gram–Schmidt with numbers.

Apply the Gram–Schmidt process of [Theorem 10.2.2](#) to each list, with the dot product.

- $((1, 0, 1), (1, 1, 0), (0, 1, 1))$  in  $\mathbb{R}^3$ .
- $((1, 1, 0, 0), (1, 0, 1, 0), (0, 0, 1, 1))$  in  $\mathbb{R}^4$ .

*Solution.*

(a)  $\|(1, 0, 1)\|^2 = 2$ , so  $\mathbf{e}_1 = \frac{1}{\sqrt{2}}(1, 0, 1)$ . Next  $\langle(1, 1, 0), \mathbf{e}_1\rangle = \frac{1}{\sqrt{2}}$ , so

$$\mathbf{w}_2 = (1, 1, 0) - \frac{1}{2}(1, 0, 1) = \frac{1}{2}(1, 2, -1),$$

with  $\|\mathbf{w}_2\|^2 = \frac{1}{4} \cdot 6 = \frac{3}{2}$ , giving  $\mathbf{e}_2 = \frac{1}{\sqrt{6}}(1, 2, -1)$ . Finally  $\langle(0, 1, 1), \mathbf{e}_1\rangle = \frac{1}{\sqrt{2}}$  and  $\langle(0, 1, 1), \mathbf{e}_2\rangle = \frac{1}{\sqrt{6}}$ , so

$$\mathbf{w}_3 = (0, 1, 1) - \frac{1}{2}(1, 0, 1) - \frac{1}{6}(1, 2, -1) = \frac{2}{3}(-1, 1, 1),$$

with  $\|\mathbf{w}_3\|^2 = \frac{4}{9} \cdot 3 = \frac{4}{3}$ , giving  $\mathbf{e}_3 = \frac{1}{\sqrt{3}}(-1, 1, 1)$ .

(b)  $\mathbf{e}_1 = \frac{1}{\sqrt{2}}(1, 1, 0, 0)$ . Then  $\langle(1, 0, 1, 0), \mathbf{e}_1\rangle = \frac{1}{\sqrt{2}}$ , so

$$\mathbf{w}_2 = (1, 0, 1, 0) - \frac{1}{2}(1, 1, 0, 0) = \frac{1}{2}(1, -1, 2, 0),$$

with  $\|\mathbf{w}_2\|^2 = \frac{1}{4} \cdot 6 = \frac{3}{2}$  and  $\mathbf{e}_2 = \frac{1}{\sqrt{6}}(1, -1, 2, 0)$ . Finally  $\langle(0, 0, 1, 1), \mathbf{e}_1\rangle = 0$  and  $\langle(0, 0, 1, 1), \mathbf{e}_2\rangle = \frac{2}{\sqrt{6}}$ , so

$$\mathbf{w}_3 = (0, 0, 1, 1) - \frac{1}{3}(1, -1, 2, 0) = \frac{1}{3}(-1, 1, 1, 3),$$

with  $\|\mathbf{w}_3\|^2 = \frac{1}{9} \cdot 12 = \frac{4}{3}$  and  $\mathbf{e}_3 = \frac{1}{2\sqrt{3}}(-1, 1, 1, 3)$ .

### Exercise B2: Gram-Schmidt on polynomials.

In  $\mathbb{R}[x]_{\leq 2}$  with  $\langle p, q \rangle = \int_0^1 p(t)q(t) dt$ , apply the Gram-Schmidt process of [Theorem 10.2.2](#) to  $(1, x, x^2)$ . Compare the monic outputs with those of [Example \(Gram-Schmidt on  \$1, x, x^2\$ \)](#) and explain the difference.

*Solution.*  $\|1\|^2 = 1$ , so  $\mathbf{e}_1 = 1$ . Then  $\langle x, 1 \rangle = \int_0^1 t dt = \frac{1}{2}$ , so  $\mathbf{w}_2 = x - \frac{1}{2}$  and

$$\|\mathbf{w}_2\|^2 = \int_0^1 \left(t - \frac{1}{2}\right)^2 dt = \frac{1}{3} - \frac{1}{2} + \frac{1}{4} = \frac{1}{12},$$

giving  $\mathbf{e}_2 = \sqrt{3}(2x - 1)$ . Next  $\langle x^2, \mathbf{e}_1 \rangle = \frac{1}{3}$  and

$$\langle x^2, \mathbf{e}_2 \rangle = \sqrt{3} \int_0^1 (2t^3 - t^2) dt = \sqrt{3} \left(\frac{1}{2} - \frac{1}{3}\right) = \frac{\sqrt{3}}{6},$$

so  $\mathbf{w}_3 = x^2 - \frac{1}{3} - \frac{\sqrt{3}}{6} \cdot \sqrt{3}(2x - 1) = x^2 - x + \frac{1}{6}$ . Its squared norm is

$$\int_0^1 \left(t^2 - t + \frac{1}{6}\right)^2 dt = \frac{1}{180},$$

so  $\mathbf{e}_3 = \sqrt{5}(6x^2 - 6x + 1)$ .

The monic outputs are  $1, x - \frac{1}{2}, x^2 - x + \frac{1}{6}$ , whereas on  $[-1, 1]$  they were  $1, x, x^2 - \frac{1}{3}$ . Orthogonality is a property of the inner product, not of the polynomials: changing the interval changes which polynomials are orthogonal. These are the shifted Legendre polynomials, and the substitution  $t \mapsto 2t - 1$  carries one family to the other.

**Exercise B3: Coordinates without solving a system.**

Let  $\mathbf{e}_1 = \frac{1}{3}(1, 2, 2)$ ,  $\mathbf{e}_2 = \frac{1}{3}(2, 1, -2)$ ,  $\mathbf{e}_3 = \frac{1}{3}(2, -2, 1)$  in  $\mathbb{R}^3$ .

- (a) Write  $\mathbf{v} = (1, -1, 4)$  in this basis.
- (b) Compute  $\|\mathbf{v}\|^2$  in two ways, and state which theorem says the answers agree.
- (c) Compute  $\langle \mathbf{v}, \mathbf{u} \rangle$  for  $\mathbf{u} = (3, 0, 3)$  from the coordinates of  $\mathbf{v}$  and  $\mathbf{u}$ , using [Example \(Reading Off Coordinates\)](#) for  $\mathbf{u}$ .

*Solution.*

- (a)  $\langle \mathbf{v}, \mathbf{e}_1 \rangle = \frac{1}{3}(1 - 2 + 8) = \frac{7}{3}$ ,  $\langle \mathbf{v}, \mathbf{e}_2 \rangle = \frac{1}{3}(2 - 1 - 8) = -\frac{7}{3}$ ,  $\langle \mathbf{v}, \mathbf{e}_3 \rangle = \frac{1}{3}(2 + 2 + 4) = \frac{8}{3}$ .  
So  $\mathbf{v} = \frac{7}{3}\mathbf{e}_1 - \frac{7}{3}\mathbf{e}_2 + \frac{8}{3}\mathbf{e}_3$ .
- (b) Directly,  $\|\mathbf{v}\|^2 = 1 + 1 + 16 = 18$ . From the coordinates,  $\frac{49}{9} + \frac{49}{9} + \frac{64}{9} = \frac{162}{9} = 18$ . They agree by [Theorem 10.2.1](#) (c).
- (c) The coordinates of  $\mathbf{u}$  are  $(3, 0, 3)$  by [Example \(Reading Off Coordinates\)](#), so by [Theorem 10.2.1](#) (b),

$$\langle \mathbf{v}, \mathbf{u} \rangle = \frac{7}{3} \cdot 3 + \left(-\frac{7}{3}\right) \cdot 0 + \frac{8}{3} \cdot 3 = 7 + 8 = 15.$$

Directly,  $\langle \mathbf{v}, \mathbf{u} \rangle = 3 + 0 + 12 = 15$ .

**Exercise B4: Gram matrices.**

For each list, write down the Gram matrix and use it to decide whether the list is independent.

- (a)  $((1, 1, 0), (1, 0, 1), (0, 1, 1))$  in  $\mathbb{R}^3$  with the dot product.
- (b)  $((1, i), (i, 1))$  in  $\mathbb{C}^2$  with the standard inner product.
- (c)  $(1, x, x^2 - \frac{1}{3})$  in  $\mathbb{R}[x]_{\leq 2}$  with  $\langle p, q \rangle = \int_{-1}^1 pq$ .

*Solution.*

- (a)  $\mathbf{G} = \begin{pmatrix} 2 & 1 & 1 \\ 1 & 2 & 1 \\ 1 & 1 & 2 \end{pmatrix}$ , and expanding along the first row,  $\det \mathbf{G} = 2(4 - 1) - 1(2 - 1) + 1(1 - 2) = 6 - 1 - 1 = 4 > 0$ . By [Theorem 10.2.5](#) (d) the list is independent.
- (b)  $\langle (1, i), (1, i) \rangle = 1 + 1 = 2$  and  $\langle (i, 1), (1, i) \rangle = i\bar{1} + 1\bar{i} = i - i = 0$ , so  $\mathbf{G} = 2\mathbf{I}_2$  by [Theorem 10.2.5](#) (a). Then  $\det \mathbf{G} = 4 > 0$  and the list is independent; in fact it is orthogonal, so [Theorem 10.1.4](#) already gives the answer.
- (c) These are the monic outputs of [Example \(Gram–Schmidt on  \$1, x, x^2\$ \)](#), so they are pairwise orthogonal, and  $\mathbf{G} = \text{diag}(2, \frac{2}{3}, \frac{8}{45})$ . Its determinant is  $2 \cdot \frac{2}{3} \cdot \frac{8}{45} = \frac{32}{135} > 0$ , so the list is independent. A Gram matrix is diagonal exactly when the list is orthogonal.

**10.2.7.3 C. Going deeper**

**Exercise C1: The Gram matrix determines the list.**

Let  $(\mathbf{v}_1, \dots, \mathbf{v}_k)$  be a list in an inner product space  $V$  and  $(\mathbf{w}_1, \dots, \mathbf{w}_k)$  a list in an inner product space  $W$ , both over  $F$ , with the **same** Gram matrix:  $\langle \mathbf{v}_j, \mathbf{v}_i \rangle = \langle \mathbf{w}_j, \mathbf{w}_i \rangle$  for all  $i, j$ .

- (a) Prove that  $\sum_j x_j \mathbf{v}_j = \mathbf{0}$  if and only if  $\sum_j x_j \mathbf{w}_j = \mathbf{0}$ , for every  $\mathbf{x} \in F^k$ .
- (b) Deduce that there is a well-defined isomorphism  $T: \text{span}(\mathbf{v}_1, \dots, \mathbf{v}_k) \rightarrow \text{span}(\mathbf{w}_1, \dots, \mathbf{w}_k)$  with  $T\mathbf{v}_j = \mathbf{w}_j$  for every  $j$ .
- (c) Prove that  $T$  is an isometric isomorphism, that is,  $\langle T\mathbf{u}, T\mathbf{u}' \rangle = \langle \mathbf{u}, \mathbf{u}' \rangle$  for all  $\mathbf{u}, \mathbf{u}'$  in the domain.

*Solution.*

- (a) Let  $\mathbf{G}$  be the common Gram matrix. By [Theorem 10.2.5](#) (b),  $\|\sum_j x_j \mathbf{v}_j\|^2 = \mathbf{x}^* \mathbf{G} \mathbf{x} = \|\sum_j x_j \mathbf{w}_j\|^2$ . By [Theorem 10.1.2](#) (a), one side is 0 if and only if the other is, and a norm is 0 only at the zero vector.
- (b) Every element of  $U = \text{span}(\mathbf{v}_1, \dots, \mathbf{v}_k)$  is  $\sum_j x_j \mathbf{v}_j$  for some  $\mathbf{x}$ . Define  $T(\sum_j x_j \mathbf{v}_j) = \sum_j x_j \mathbf{w}_j$ . This is well defined: if  $\sum_j x_j \mathbf{v}_j = \sum_j y_j \mathbf{v}_j$ , then  $\sum_j (x_j - y_j) \mathbf{v}_j = \mathbf{0}$ , so  $\sum_j (x_j - y_j) \mathbf{w}_j = \mathbf{0}$  by (a), that is  $\sum_j x_j \mathbf{w}_j = \sum_j y_j \mathbf{w}_j$ . It is linear because the formula is linear in  $\mathbf{x}$ , it is surjective onto  $\text{span}(\mathbf{w}_1, \dots, \mathbf{w}_k)$  by construction, and it is injective: if  $T(\sum_j x_j \mathbf{v}_j) = \mathbf{0}$ , then  $\sum_j x_j \mathbf{w}_j = \mathbf{0}$ , so  $\sum_j x_j \mathbf{v}_j = \mathbf{0}$  by (a).
- (c) Let  $\mathbf{u} = \sum_j x_j \mathbf{v}_j$  and  $\mathbf{u}' = \sum_i y_i \mathbf{v}_i$ . Expanding by (IP1) and [Theorem 10.1.1](#) (b),

$$\langle \mathbf{u}, \mathbf{u}' \rangle = \sum_i \sum_j x_j \overline{y_i} \langle \mathbf{v}_j, \mathbf{v}_i \rangle,$$

and the same computation in  $W$  gives  $\langle T\mathbf{u}, T\mathbf{u}' \rangle = \sum_i \sum_j x_j \overline{y_i} \langle \mathbf{w}_j, \mathbf{w}_i \rangle$ . The two sums are equal term by term, since the Gram matrices agree.

**Exercise C2: Gram–Schmidt on a dependent list.**

Let  $(\mathbf{v}_1, \dots, \mathbf{v}_m)$  be a list in an inner product space, **not** assumed independent, and run the recursion of [Theorem 10.2.2](#).

- (a) Let  $k$  be the smallest index with  $\mathbf{v}_k \in \text{span}(\mathbf{v}_1, \dots, \mathbf{v}_{k-1})$ . Prove that the recursion is defined up to  $\mathbf{e}_{k-1}$  and that  $\mathbf{w}_k = \mathbf{0}$ , so the step  $\mathbf{e}_k = \mathbf{w}_k / \|\mathbf{w}_k\|$  fails.
- (b) Repair the process: show that if, whenever  $\mathbf{w}_k = \mathbf{0}$ , we discard  $\mathbf{v}_k$  and continue with  $\mathbf{v}_{k+1}$ , the output is an orthonormal basis of  $\text{span}(\mathbf{v}_1, \dots, \mathbf{v}_m)$ .
- (c) Run the repaired process on  $((1, 1, 0), (2, 2, 0), (1, 0, 1))$  in  $\mathbb{R}^3$ .

*Solution.*

- (a) The list  $(\mathbf{v}_1, \dots, \mathbf{v}_{k-1})$  is independent, since a dependence among its members would, by [Theorem 1.4.1](#), put some  $\mathbf{v}_j$  with  $j < k$  in the span of its predecessors, contradicting the minimality of  $k$ . So [Theorem 10.2.2](#) applies to it and produces  $\mathbf{e}_1, \dots, \mathbf{e}_{k-1}$  with  $\text{span}(\mathbf{e}_1, \dots, \mathbf{e}_{k-1}) = \text{span}(\mathbf{v}_1, \dots, \mathbf{v}_{k-1}) =: U$ . Since  $\mathbf{v}_k \in U$ , [Theorem 10.2.1](#) (a) applied inside  $U$  gives  $\mathbf{v}_k = \sum_{i < k} \langle \mathbf{v}_k, \mathbf{e}_i \rangle \mathbf{e}_i$ , which is precisely the vector subtracted, so  $\mathbf{w}_k = \mathbf{0}$ . Dividing by  $\|\mathbf{w}_k\| = 0$  is impossible.

- (b) Let  $\mathbf{e}_1, \dots, \mathbf{e}_s$  be the vectors that the repaired process outputs. The proof of [Theorem 10.2.2](#) shows, unchanged, that after processing  $\mathbf{v}_1, \dots, \mathbf{v}_k$  the accumulated list is orthonormal and spans  $\text{span}(\mathbf{v}_1, \dots, \mathbf{v}_k)$ : the only step that used independence was the claim  $\mathbf{w}_k \neq \mathbf{0}$ , and in the repaired process a step with  $\mathbf{w}_k = \mathbf{0}$  discards a  $\mathbf{v}_k$  that lies in the span already built, so the span does not change. Taking  $k = m$ , the output is an orthonormal list spanning  $\text{span}(\mathbf{v}_1, \dots, \mathbf{v}_m)$ , hence an orthonormal basis of it.
- (c)  $\mathbf{e}_1 = \frac{1}{\sqrt{2}}(1, 1, 0)$ . For  $\mathbf{v}_2 = (2, 2, 0)$ :  $\langle \mathbf{v}_2, \mathbf{e}_1 \rangle = \frac{4}{\sqrt{2}} = 2\sqrt{2}$ , so  $\mathbf{w}_2 = (2, 2, 0) - 2\sqrt{2} \cdot \frac{1}{\sqrt{2}}(1, 1, 0) = \mathbf{0}$ , and  $\mathbf{v}_2$  is discarded. For  $\mathbf{v}_3 = (1, 0, 1)$ :  $\langle \mathbf{v}_3, \mathbf{e}_1 \rangle = \frac{1}{\sqrt{2}}$ , so  $\mathbf{w}_3 = (1, 0, 1) - \frac{1}{2}(1, 1, 0) = \frac{1}{2}(1, -1, 2)$ , with  $\|\mathbf{w}_3\|^2 = \frac{6}{4} = \frac{3}{2}$  and  $\mathbf{e}_2 = \frac{1}{\sqrt{6}}(1, -1, 2)$ . The output  $(\frac{1}{\sqrt{2}}(1, 1, 0), \frac{1}{\sqrt{6}}(1, -1, 2))$  is an orthonormal basis of the plane spanned by the three given vectors.

## 10.3 Orthogonal Complements and Projections

Chapter 1 proved that every subspace of a finite-dimensional space has a complement, and then warned that it has many: the  $x$ -axis in  $\mathbb{R}^2$  is complemented by every other line through the origin. An inner product breaks the tie. Among all the complements of  $U$  there is now one built from  $U$  alone, the set of vectors perpendicular to everything in  $U$ , and the projection onto  $U$  along it turns out to solve an approximation problem: it finds the vector of  $U$  closest to a given  $\mathbf{v}$ . This section constructs that complement, that projection, and that answer.

Throughout,  $V$  is an inner product space over  $F = \mathbb{R}$  or  $F = \mathbb{C}$ , with inner product  $\langle \cdot, \cdot \rangle$  ([Definition 10.1.1](#)) and induced norm  $\|\cdot\|$  ([Definition 10.1.3](#)). We do **not** assume  $V$  is finite-dimensional; what we assume, each time, is that the subspace we project onto is.

### 10.3.1 Orthogonal complements

In  $\mathbb{R}^3$ , a plane  $U$  through the origin has a distinguished line attached to it: the normal line, the set of vectors perpendicular to the whole plane. Nothing about the plane had to be chosen to produce it. That construction is available in any inner product space, and it is available for any subset, not only for subspaces.

*The orthogonal complement of  $U$  collects every vector that is perpendicular to all of  $U$  at once.*

#### Definition 10.3.1 — Orthogonal complement

Let  $V$  be an inner product space over  $F$  and let  $S \subseteq V$  be **any** subset. The **orthogonal complement** of  $S$  is

$$S^\perp = \{ \mathbf{v} \in V : \langle \mathbf{v}, \mathbf{s} \rangle = 0 \text{ for every } \mathbf{s} \in S \},$$

read “ $S$  perp”.

In words:  $\mathbf{v}$  belongs to  $S^\perp$  when  $\mathbf{v}$  is orthogonal (Definition 10.1.4) to **every** single element of  $S$ , not merely to some of them. The quantifier is what makes the definition useful, and it is also what makes  $S^\perp$  a subspace even when  $S$  is not. Note that the condition could equally be written  $\langle \mathbf{s}, \mathbf{v} \rangle = 0$  for every  $\mathbf{s} \in S$ , since  $\langle \mathbf{s}, \mathbf{v} \rangle = \overline{\langle \mathbf{v}, \mathbf{s} \rangle}$  and a complex number vanishes exactly when its conjugate does. We will use whichever slot is convenient.

### Proposition 10.3.1 — Perp is always a subspace

For every subset  $S$  of an inner product space  $V$ , the set  $S^\perp$  is a subspace of  $V$ . Moreover  $S^\perp = \text{span}(S)^\perp$ .

*Proof.* We apply the Subspace Test (Theorem 1.2.1). First,  $\langle \mathbf{0}, \mathbf{s} \rangle = 0$  for every  $\mathbf{s}$ , because  $\langle \cdot, \mathbf{s} \rangle$  is linear in the first slot and a linear map sends  $\mathbf{0}$  to 0 (Theorem 3.1.1); so  $\mathbf{0} \in S^\perp$ . Next, let  $\mathbf{v}, \mathbf{w} \in S^\perp$  and  $a \in F$ . For every  $\mathbf{s} \in S$ ,

$$\langle \mathbf{v} + \mathbf{w}, \mathbf{s} \rangle = \langle \mathbf{v}, \mathbf{s} \rangle + \langle \mathbf{w}, \mathbf{s} \rangle = 0, \quad \langle a\mathbf{v}, \mathbf{s} \rangle = a\langle \mathbf{v}, \mathbf{s} \rangle = 0,$$

both by linearity in the first slot. Hence  $\mathbf{v} + \mathbf{w} \in S^\perp$  and  $a\mathbf{v} \in S^\perp$ .

For the second claim,  $S \subseteq \text{span}(S)$  gives  $\text{span}(S)^\perp \subseteq S^\perp$ . Conversely let  $\mathbf{v} \in S^\perp$  and let  $\mathbf{u} = a_1\mathbf{s}_1 + \cdots + a_k\mathbf{s}_k$  with  $\mathbf{s}_i \in S$ . Then

$$\langle \mathbf{u}, \mathbf{v} \rangle = a_1\langle \mathbf{s}_1, \mathbf{v} \rangle + \cdots + a_k\langle \mathbf{s}_k, \mathbf{v} \rangle = 0,$$

since each  $\langle \mathbf{s}_i, \mathbf{v} \rangle = \overline{\langle \mathbf{v}, \mathbf{s}_i \rangle} = 0$ . So  $\langle \mathbf{v}, \mathbf{u} \rangle = \overline{\langle \mathbf{u}, \mathbf{v} \rangle} = 0$ , and  $\mathbf{v} \in \text{span}(S)^\perp$ . ■

The second half is what makes computation possible: to test membership in  $U^\perp$  it is enough to test against a spanning list of  $U$ , not against all of  $U$ .

#### Examples.

- **The extremes.**  $\{\mathbf{0}\}^\perp = V$ , since  $\langle \mathbf{v}, \mathbf{0} \rangle = 0$  always. And  $V^\perp = \{\mathbf{0}\}$ : if  $\mathbf{v} \in V^\perp$ , then in particular  $\langle \mathbf{v}, \mathbf{v} \rangle = 0$ , and positive definiteness forces  $\mathbf{v} = \mathbf{0}$ . This is the “pair it with itself” move, and it is the shape in which positivity almost always enters; it is the only place where positivity is used; we will use it again and again.
- **A plane in  $\mathbb{R}^3$ .** Let  $U = \{(x, y, z) : x + y + z = 0\}$  with the dot product. The condition  $x + y + z = 0$  says exactly  $\langle (x, y, z), (1, 1, 1) \rangle = 0$ , so  $U = \{(1, 1, 1)\}^\perp$ . The Orthogonal Decomposition Theorem below will give the converse,  $U^\perp = \text{span}((1, 1, 1))$ .
- **A subset that is not a subspace.** In  $\mathbb{R}^3$ , take  $S = \{(1, 0, 0), (1, 1, 0)\}$ , two vectors and nothing else. Then  $S^\perp = \{(x, y, z) : x = 0, x + y = 0\} = \text{span}((0, 0, 1))$ , a subspace, as Proposition 10.3.1 promises. It equals  $\text{span}(S)^\perp$ , the perp of the  $xy$ -plane.
- **Polynomials.** On  $\mathbb{R}[x]_{\leq 2}$  with  $\langle p, q \rangle = \int_{-1}^1 p(t)q(t) dt$ , let  $U = \text{span}(1)$  be the constants. Then  $p \in U^\perp$  means  $\int_{-1}^1 p(t) dt = 0$ : the polynomials of average value zero. For instance  $x$  and  $x^2 - \frac{1}{3}$  lie in  $U^\perp$ , and they span it, since the Orthogonal Decomposition Theorem below gives  $\dim U^\perp = 3 - 1 = 2$ .

**Non-example by minimal change.** Weaken “for every  $\mathbf{s} \in S$ ” to “for some  $\mathbf{s} \in S$ ”. With  $S = \{\mathbf{e}_1, \mathbf{e}_2\}$  in  $\mathbb{R}^2$ , the weakened set is  $\{\mathbf{v} : \langle \mathbf{v}, \mathbf{e}_1 \rangle = 0 \text{ or } \langle \mathbf{v}, \mathbf{e}_2 \rangle = 0\}$ , the union of the two axes. Closure under scalars still holds, and  $\mathbf{0}$  still belongs; what fails is closure under addition, since  $(1, 0) + (0, 1) = (1, 1)$  lies on neither axis. This is the union of Example (The Union of the

Two Axes) again, in new clothing. The universal quantifier is doing real work.

**Why this definition.** One could try to define “the” complement of  $U$  by choosing a basis of  $U$ , extending it, and spanning the new vectors, which is what [Theorem 1.7.6](#) does. That answer depends on every choice made along the way. The definition above depends only on  $U$  and on the inner product, and the next theorem shows that it really is a complement.

## 10.3.2 The orthogonal decomposition

Here is the theorem that makes  $U^\perp$  more than a definition. Note the hypothesis:  $U$  must be finite-dimensional, but  $V$  need not be. The reason is that the proof needs an orthonormal basis of  $U$ , and Gram–Schmidt supplies one only for a finite list.

### Theorem 10.3.2 — Orthogonal Decomposition Theorem

Let  $V$  be an inner product space over  $F$  and let  $U$  be a **finite-dimensional** subspace of  $V$ . Then:

- (a)  $V = U \oplus U^\perp$ ;
- (b)  $(U^\perp)^\perp = U$ ;
- (c) if in addition  $V$  is finite-dimensional, then  $\dim U^\perp = \dim V - \dim U$ .

*Idea.* For (a) we must produce, from an arbitrary  $\mathbf{v}$ , a piece inside  $U$  and a piece perpendicular to  $U$ . Fix an orthonormal basis of  $U$ . [Theorem 10.2.1](#) says that a vector **of**  $U$  is the sum of its coordinates against that basis; so for a  $\mathbf{v}$  that is not in  $U$ , the same sum is the natural guess for the  $U$ -piece. Then we subtract and check. That is the move of Gram–Schmidt — project, then subtract — used once instead of repeatedly. Uniqueness of the splitting is the “pair with itself” move. For (b), one inclusion is free and the other is (a) applied twice.

*Proof.*

- (a) Since  $U$  is finite-dimensional, it has an orthonormal basis  $(\mathbf{e}_1, \dots, \mathbf{e}_k)$  by [Theorem 10.2.2](#). Let  $\mathbf{v} \in V$  and put

$$\mathbf{u} = \sum_{i=1}^k \langle \mathbf{v}, \mathbf{e}_i \rangle \mathbf{e}_i \in U, \quad \mathbf{w} = \mathbf{v} - \mathbf{u}.$$

For each  $j$ ,

$$\langle \mathbf{w}, \mathbf{e}_j \rangle = \langle \mathbf{v}, \mathbf{e}_j \rangle - \sum_{i=1}^k \langle \mathbf{v}, \mathbf{e}_i \rangle \langle \mathbf{e}_i, \mathbf{e}_j \rangle = \langle \mathbf{v}, \mathbf{e}_j \rangle - \langle \mathbf{v}, \mathbf{e}_j \rangle = 0,$$

where the middle step uses  $\langle \mathbf{e}_i, \mathbf{e}_j \rangle = \delta_{ij}$ . So  $\mathbf{w}$  is orthogonal to each  $\mathbf{e}_j$ , hence to  $\text{span}(\mathbf{e}_1, \dots, \mathbf{e}_k) = U$  by [Proposition 10.3.1](#). Therefore  $\mathbf{v} = \mathbf{u} + \mathbf{w} \in U + U^\perp$ .

The sum is direct: if  $\mathbf{x} \in U \cap U^\perp$ , then  $\langle \mathbf{x}, \mathbf{x} \rangle = 0$  because  $\mathbf{x} \in U^\perp$  and  $\mathbf{x} \in U$ , so  $\mathbf{x} = \mathbf{0}$  by positive definiteness. By [Theorem 1.7.4](#),  $V = U \oplus U^\perp$ .

- (b) ( $\subseteq$ ) Let  $\mathbf{u} \in U$ . For every  $\mathbf{w} \in U^\perp$  we have  $\langle \mathbf{w}, \mathbf{u} \rangle = 0$ , hence  $\langle \mathbf{u}, \mathbf{w} \rangle = \overline{\langle \mathbf{w}, \mathbf{u} \rangle} = 0$ . So  $\mathbf{u} \in (U^\perp)^\perp$ .

( $\supseteq$ ) Let  $\mathbf{v} \in (U^\perp)^\perp$ . By (a), write  $\mathbf{v} = \mathbf{u} + \mathbf{w}$  with  $\mathbf{u} \in U$  and  $\mathbf{w} \in U^\perp$ . By the inclusion just proved,  $\mathbf{u} \in (U^\perp)^\perp$ , so  $\mathbf{w} = \mathbf{v} - \mathbf{u} \in (U^\perp)^\perp$  as well. Thus  $\mathbf{w}$  lies in both  $U^\perp$  and  $(U^\perp)^\perp$ , giving  $\langle \mathbf{w}, \mathbf{w} \rangle = 0$  and  $\mathbf{w} = \mathbf{0}$ . Hence  $\mathbf{v} = \mathbf{u} \in U$ .

(c) By (a),  $U^\perp$  is a complement of  $U$  in  $V$ , so  $\dim U^\perp = \dim V - \dim U$  by [Theorem 1.7.6](#). ■

Part (b) is worth pausing on. For a general subspace of a general vector space there is no operation “ $\perp$ ” to apply twice, and even for the annihilator of Chapter 4 the analogous statement needed the double dual. Here it is a two-line consequence of (a). Part (c) says that  $\perp$  exchanges dimension  $k$  with dimension  $n - k$ , just as the annihilator does ([Theorem 4.2.2](#)); Section 5 explains why the two facts are the same fact.

*Quick check.* Positive definiteness was used twice in the proof of [Theorem 10.3.2](#), in the same way each time. Where? To see that it cannot be dropped, take  $\mathbb{R}^2$  with the indefinite form  $\beta(\mathbf{x}, \mathbf{y}) = x_1y_1 - x_2y_2$ , let  $U = \text{span}((1, 1))$ , and write  $U^\perp$  for the set of  $\mathbf{v}$  with  $\beta(\mathbf{v}, \mathbf{u}) = 0$  for all  $\mathbf{u} \in U$ . Compute  $U \cap U^\perp$ .

*Solution.* It was used to show  $U \cap U^\perp = \{\mathbf{0}\}$  in (a), and again for  $\mathbf{w} = \mathbf{0}$  in (b); both times through “ $\langle \mathbf{x}, \mathbf{x} \rangle = 0$  forces  $\mathbf{x} = \mathbf{0}$ ”.

With  $\beta$ , the vector  $\mathbf{u} = (1, 1)$  satisfies  $\beta(\mathbf{u}, \mathbf{u}) = 1 - 1 = 0$ , so  $\mathbf{u}$  is “orthogonal to itself” and  $\mathbf{u} \in U^\perp$ . Hence  $U \cap U^\perp = U \neq \{\mathbf{0}\}$ , and the sum  $U + U^\perp$  is not direct. Indeed  $U^\perp = \{(x_1, x_2) : x_1 - x_2 = 0\} = U$  here, so  $U + U^\perp = U$  is not all of  $\mathbb{R}^2$ . Both conclusions of the theorem fail.

### 10.3.3 Orthogonal projection

Once  $V = U \oplus U^\perp$ , Chapter 3 hands us an operator: the projection onto  $U$  along  $U^\perp$  ([Theorem 3.10.1](#)). It deserves its own name and symbol, because the complement is no longer a choice.

#### Definition 10.3.2 — Orthogonal projection

Let  $U$  be a finite-dimensional subspace of an inner product space  $V$ . By [Theorem 10.3.2](#) (a), every  $\mathbf{v} \in V$  has a unique splitting  $\mathbf{v} = \mathbf{u} + \mathbf{w}$  with  $\mathbf{u} \in U$  and  $\mathbf{w} \in U^\perp$ . The **orthogonal projection onto  $U$**  is the operator  $P_U \in \mathcal{L}(V)$  defined by

$$P_U \mathbf{v} = \mathbf{u}.$$

The definition makes sense because the splitting exists and is unique, and  $P_U$  is linear by [Theorem 3.10.1](#) (b), which also gives  $P_U^2 = P_U$ ,  $\text{im } P_U = U$  and  $\ker P_U = U^\perp$ . Nothing so far tells us how to compute  $P_U \mathbf{v}$ . The proof of [Theorem 10.3.2](#) does, and that is the next theorem.

#### Theorem 10.3.3 — Projection formula

Let  $U$  be a finite-dimensional subspace of an inner product space  $V$ , with orthonormal



basis  $(\mathbf{e}_1, \dots, \mathbf{e}_k)$ . Then for every  $\mathbf{v} \in V$ :

- (a)  $P_U \mathbf{v} = \sum_{i=1}^k \langle \mathbf{v}, \mathbf{e}_i \rangle \mathbf{e}_i$ ;
- (b)  $\langle P_U \mathbf{u}, \mathbf{v} \rangle = \langle \mathbf{u}, P_U \mathbf{v} \rangle$  for all  $\mathbf{u}, \mathbf{v} \in V$ ;
- (c)  $\|P_U \mathbf{v}\| \leq \|\mathbf{v}\|$ , with equality if and only if  $\mathbf{v} \in U$ .

*Idea.* Part (a) is not a new computation: the vector  $\mathbf{u}$  built in the proof of [Theorem 10.3.2](#) (a) was exactly this sum, and by uniqueness of the splitting it must be  $P_U \mathbf{v}$ . For (b), split **both**  $\mathbf{u}$  and  $\mathbf{v}$ , and watch the cross terms die. For (c), split  $\mathbf{v}$  and use Pythagoras.

*Proof.*

- (a) Put  $\mathbf{u} = \sum_i \langle \mathbf{v}, \mathbf{e}_i \rangle \mathbf{e}_i$  and  $\mathbf{w} = \mathbf{v} - \mathbf{u}$ . The proof of [Theorem 10.3.2](#) (a) showed  $\mathbf{u} \in U$  and  $\mathbf{w} \in U^\perp$ . Since the splitting of  $\mathbf{v}$  in  $U \oplus U^\perp$  is unique ([Definition 1.7.2](#)),  $P_U \mathbf{v} = \mathbf{u}$ .
- (b) Write  $\mathbf{u} = \mathbf{u}_1 + \mathbf{u}_2$  and  $\mathbf{v} = \mathbf{v}_1 + \mathbf{v}_2$  with  $\mathbf{u}_1, \mathbf{v}_1 \in U$  and  $\mathbf{u}_2, \mathbf{v}_2 \in U^\perp$ . Then  $\langle \mathbf{u}_1, \mathbf{v}_2 \rangle = 0$  because  $\mathbf{v}_2 \in U^\perp$  and  $\mathbf{u}_1 \in U$ , and  $\langle \mathbf{u}_2, \mathbf{v}_1 \rangle = 0$  for the same reason with the roles swapped. Hence

$$\langle P_U \mathbf{u}, \mathbf{v} \rangle = \langle \mathbf{u}_1, \mathbf{v}_1 + \mathbf{v}_2 \rangle = \langle \mathbf{u}_1, \mathbf{v}_1 \rangle = \langle \mathbf{u}_1 + \mathbf{u}_2, \mathbf{v}_1 \rangle = \langle \mathbf{u}, P_U \mathbf{v} \rangle.$$

- (c) With  $\mathbf{v} = \mathbf{v}_1 + \mathbf{v}_2$  as above,  $\mathbf{v}_1$  and  $\mathbf{v}_2$  are orthogonal, so  $\|\mathbf{v}\|^2 = \|\mathbf{v}_1\|^2 + \|\mathbf{v}_2\|^2$  by [Theorem 10.1.3](#). Since  $\|\mathbf{v}_2\|^2 \geq 0$ , this gives  $\|P_U \mathbf{v}\|^2 = \|\mathbf{v}_1\|^2 \leq \|\mathbf{v}\|^2$ . Equality holds if and only if  $\|\mathbf{v}_2\| = 0$ , that is  $\mathbf{v}_2 = \mathbf{0}$ , that is  $\mathbf{v} = \mathbf{v}_1 \in U$ .

■

Two comments. First, the right-hand side of (a) visibly depends on the orthonormal basis chosen, while the left-hand side does not; so the sum  $\sum_i \langle \mathbf{v}, \mathbf{e}_i \rangle \mathbf{e}_i$  is the same for every orthonormal basis of  $U$ . That is a genuine piece of information, obtained for free. Second, property (b) says that  $P_U$  can be moved from one slot of the inner product to the other at no cost. Section 6 will call such an operator **self-adjoint** and write the property as  $P_U^* = P_U$ ; we proved the property here so that nothing later depends on the order of the sections.

Orthonormal bases are convenient but not obligatory, and by hand it is usually faster to skip the square roots.

### Corollary 10.3.4 — Projecting with an orthogonal basis

Let  $U$  be a finite-dimensional subspace of an inner product space  $V$ , and let  $(\mathbf{u}_1, \dots, \mathbf{u}_k)$  be an **orthogonal** basis of  $U$ , so that  $\langle \mathbf{u}_i, \mathbf{u}_j \rangle = 0$  for  $i \neq j$  and every  $\mathbf{u}_i \neq \mathbf{0}$ . Then

$$P_U \mathbf{v} = \sum_{i=1}^k \frac{\langle \mathbf{v}, \mathbf{u}_i \rangle}{\langle \mathbf{u}_i, \mathbf{u}_i \rangle} \mathbf{u}_i \quad \text{for every } \mathbf{v} \in V.$$

*Proof.* Each  $\mathbf{u}_i$  is non-zero, so  $\|\mathbf{u}_i\| > 0$  and  $\mathbf{e}_i = \mathbf{u}_i / \|\mathbf{u}_i\|$  is defined. Each  $\mathbf{e}_i$  is a non-zero multiple of  $\mathbf{u}_i$ , so  $(\mathbf{e}_1, \dots, \mathbf{e}_k)$  is again a basis of  $U$ , and it is orthonormal:

$\langle \mathbf{e}_i, \mathbf{e}_j \rangle = \langle \mathbf{u}_i, \mathbf{u}_j \rangle / (\|\mathbf{u}_i\| \|\mathbf{u}_j\|) = \delta_{ij}$ . By Theorem 10.3.3 (a),

$$P_U \mathbf{v} = \sum_{i=1}^k \langle \mathbf{v}, \mathbf{e}_i \rangle \mathbf{e}_i = \sum_{i=1}^k \frac{\langle \mathbf{v}, \mathbf{u}_i \rangle}{\|\mathbf{u}_i\|^2} \mathbf{u}_i,$$

where the two factors  $1/\|\mathbf{u}_i\|$  come out unchanged, the one in the second slot because  $\|\mathbf{u}_i\|$  is real. Finally  $\|\mathbf{u}_i\|^2 = \langle \mathbf{u}_i, \mathbf{u}_i \rangle$  by Definition 10.1.3. ■

Each summand  $\langle \mathbf{v}, \mathbf{u}_i \rangle \mathbf{u}_i / \langle \mathbf{u}_i, \mathbf{u}_i \rangle$  is the projection of  $\mathbf{v}$  onto the line  $\text{span}(\mathbf{u}_i)$ : the case  $k = 1$  of the corollary.

### 10.3.4 Best approximation and distance

Now the payoff. Of all the vectors of  $U$ , which is nearest to a given  $\mathbf{v}$ ? In  $\mathbb{R}^3$ , dropping a perpendicular from a point to a plane is the familiar recipe, and the recipe is correct in every inner product space.

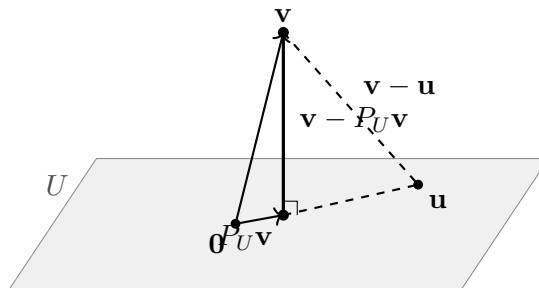
#### Theorem 10.3.5 — Best Approximation Theorem

Let  $U$  be a finite-dimensional subspace of an inner product space  $V$ , and let  $\mathbf{v} \in V$ . Then

$$\|\mathbf{v} - P_U \mathbf{v}\| \leq \|\mathbf{v} - \mathbf{u}\| \quad \text{for every } \mathbf{u} \in U,$$

and equality holds **only** for  $\mathbf{u} = P_U \mathbf{v}$ .

*Idea.* Insert  $P_U \mathbf{v}$  into the difference:  $\mathbf{v} - \mathbf{u} = (\mathbf{v} - P_U \mathbf{v}) + (P_U \mathbf{v} - \mathbf{u})$ . The first bracket is perpendicular to  $U$  and the second lies in  $U$ , so the two are orthogonal, and Pythagoras turns the comparison into “a sum of two squares is at least one of them”. The picture is a right triangle with hypotenuse  $\mathbf{v} - \mathbf{u}$ .



*Proof.* Let  $\mathbf{u} \in U$  and write

$$\mathbf{v} - \mathbf{u} = (\mathbf{v} - P_U \mathbf{v}) + (P_U \mathbf{v} - \mathbf{u}).$$

The first bracket lies in  $U^\perp$ , since  $\mathbf{v} - P_U \mathbf{v}$  is the  $U^\perp$ -part of  $\mathbf{v}$  in Theorem 10.3.2 (a). The second lies in  $U$ , since  $P_U \mathbf{v} \in U$  and  $\mathbf{u} \in U$ . Hence the two brackets are orthogonal,

and [Theorem 10.1.3](#) gives

$$\|\mathbf{v} - \mathbf{u}\|^2 = \|\mathbf{v} - P_U \mathbf{v}\|^2 + \|P_U \mathbf{v} - \mathbf{u}\|^2 \geq \|\mathbf{v} - P_U \mathbf{v}\|^2.$$

Taking square roots proves the inequality, since the norm is non-negative. Equality forces  $\|P_U \mathbf{v} - \mathbf{u}\|^2 = 0$ , hence  $P_U \mathbf{v} - \mathbf{u} = \mathbf{0}$  by positive definiteness, that is  $\mathbf{u} = P_U \mathbf{v}$ . This proves the theorem. ■

So the minimum is attained, and attained at exactly one point. That lets us define a distance.

### Definition 10.3.3 — Distance to a subspace

Let  $U$  be a finite-dimensional subspace of an inner product space  $V$  and let  $\mathbf{v} \in V$ . The **distance from  $\mathbf{v}$  to  $U$**  is

$$d(\mathbf{v}, U) = \min_{\mathbf{u} \in U} \|\mathbf{v} - \mathbf{u}\| = \|\mathbf{v} - P_U \mathbf{v}\|.$$

The word **minimum** is justified, not sloppy: [Theorem 10.3.5](#) says the values  $\|\mathbf{v} - \mathbf{u}\|$  really do have a smallest one. Note also that  $d(\mathbf{v}, U) = 0$  if and only if  $\mathbf{v} \in U$ , and that, when  $V$  is finite-dimensional,  $d(\mathbf{v}, U) = \|P_{U^\perp} \mathbf{v}\|$ , since  $\mathbf{v} - P_U \mathbf{v}$  is the  $U^\perp$ -component of  $\mathbf{v}$ . (The hypothesis is needed for  $P_{U^\perp}$  to be defined at all: [Definition 10.3.2](#) asks the subspace projected onto to be finite-dimensional, and  $U^\perp$  is not, in the infinite-dimensional setting this section is careful to allow.) When  $U^\perp$  is the smaller of the two, computing that component is the faster route.

### Example. (Distance from a point to a plane)

In  $\mathbb{R}^3$  with the dot product, let  $U = \{(x, y, z) : x + y + z = 0\}$  and  $\mathbf{v} = (1, 2, 3)$ . Find  $P_U \mathbf{v}$  and  $d(\mathbf{v}, U)$ , twice: once through  $U^\perp$ , and once through an orthonormal basis of  $U$ .

*Solution.* Through  $U^\perp$ . Put  $\mathbf{n} = (1, 1, 1)$ . Every  $(x, y, z) \in U$  satisfies  $\langle (x, y, z), \mathbf{n} \rangle = x + y + z = 0$ , so  $\mathbf{n} \in U^\perp$ . Since  $U$  is the solution set of one non-trivial equation,  $\dim U = 2$ , so  $\dim U^\perp = 3 - 2 = 1$  by [Theorem 10.3.2](#) (c), and therefore  $U^\perp = \text{span}(\mathbf{n})$ . By [Corollary 10.3.4](#) with the one-element basis  $(\mathbf{n})$ ,

$$P_{U^\perp} \mathbf{v} = \frac{\langle \mathbf{v}, \mathbf{n} \rangle}{\langle \mathbf{n}, \mathbf{n} \rangle} \mathbf{n} = \frac{1 + 2 + 3}{3} (1, 1, 1) = (2, 2, 2),$$

so  $P_U \mathbf{v} = \mathbf{v} - (2, 2, 2) = (-1, 0, 1)$  and  $d(\mathbf{v}, U) = \|(2, 2, 2)\| = 2\sqrt{3}$ .

Through an orthonormal basis of  $U$ . The vectors  $(1, -1, 0)$  and  $(1, 1, -2)$  lie in  $U$  and are orthogonal to each other, so

$$\mathbf{e}_1 = \frac{1}{\sqrt{2}}(1, -1, 0), \quad \mathbf{e}_2 = \frac{1}{\sqrt{6}}(1, 1, -2)$$

is an orthonormal basis of  $U$ . Then  $\langle \mathbf{v}, \mathbf{e}_1 \rangle = (1 - 2)/\sqrt{2} = -1/\sqrt{2}$  and  $\langle \mathbf{v}, \mathbf{e}_2 \rangle = (1 + 2 - 6)/\sqrt{6} = -3/\sqrt{6}$ , so [Theorem 10.3.3](#) (a) gives

$$P_U \mathbf{v} = -\frac{1}{\sqrt{2}} \mathbf{e}_1 - \frac{3}{\sqrt{6}} \mathbf{e}_2 = -\frac{1}{2}(1, -1, 0) - \frac{1}{2}(1, 1, -2) = (-1, 0, 1),$$

the same answer. *Check:*  $(-1) + 0 + 1 = 0$ , so  $P_U \mathbf{v} \in U$ ; and  $\mathbf{v} - P_U \mathbf{v} = (2, 2, 2)$  is a multiple of  $\mathbf{n}$ , so it lies in  $U^\perp$ . The two conditions together identify the splitting, by uniqueness.

The next example is the same theorem in a space of functions, where “closest” is no longer something one can see.

**Example. (The best quadratic approximation to a cubic)**

On  $V = \mathbb{R}[x]_{\leq 3}$  with  $\langle p, q \rangle = \int_{-1}^1 p(t)q(t) dt$ , let  $U = \mathbb{R}[x]_{\leq 2}$ . Find the polynomial of  $U$  closest to  $x^3$ , and the distance.

*Solution.* The list  $(1, x, x^2 - \frac{1}{3})$  is a basis of  $U$ , being three independent polynomials of degrees 0, 1, 2 ([Theorem 1.4.4](#)) in a space of dimension 3. It is **orthogonal**:  $\langle 1, x \rangle = 0$  and  $\langle x, x^2 - \frac{1}{3} \rangle = 0$  because the integrands are odd, and

$$\langle 1, x^2 - \frac{1}{3} \rangle = \int_{-1}^1 \left( t^2 - \frac{1}{3} \right) dt = \frac{2}{3} - \frac{2}{3} = 0.$$

Now apply [Corollary 10.3.4](#) to  $\mathbf{v} = x^3$ . Two of the three coefficients vanish because  $t^3$  is odd:  $\langle x^3, 1 \rangle = 0$  and  $\langle x^3, x^2 - \frac{1}{3} \rangle = 0$ . The remaining one is

$$\frac{\langle x^3, x \rangle}{\langle x, x \rangle} = \frac{\int_{-1}^1 t^4 dt}{\int_{-1}^1 t^2 dt} = \frac{2/5}{2/3} = \frac{3}{5}.$$

Hence  $P_U(x^3) = \frac{3}{5}x$ . The error is  $x^3 - \frac{3}{5}x$ , with

$$\begin{aligned} \|x^3 - \frac{3}{5}x\|^2 &= \int_{-1}^1 \left( t^6 - \frac{6}{5}t^4 + \frac{9}{25}t^2 \right) dt \\ &= \frac{2}{7} - \frac{6}{5} \cdot \frac{2}{5} + \frac{9}{25} \cdot \frac{2}{3} = \frac{2}{7} - \frac{6}{25} = \frac{8}{175}, \end{aligned}$$

so  $d(x^3, U) = \sqrt{8/175} = \frac{2}{35}\sqrt{14} \approx 0.214$ .

*Check.* The error should be orthogonal to  $U$ . Against 1 and  $x^2 - \frac{1}{3}$  this holds because  $t^3 - \frac{3}{5}t$  is odd; against  $x$  it reads  $\int_{-1}^1 (t^4 - \frac{3}{5}t^2) dt = \frac{2}{5} - \frac{3}{5} \cdot \frac{2}{3} = 0$ . Notice that the answer is **not** the Taylor polynomial of  $x^3$  at 0, which is 0: Taylor approximates near one point, while  $P_U$  spreads the error over the whole interval.

## 10.3.5 Three traps

**Warning.**  $U^\perp$  is **not** the annihilator  $U^0$ . Both objects consist of things that “kill”  $U$ , but they live in different places:  $U^\perp \subseteq V$  is made of **vectors** ([Definition 10.3.1](#)), while  $U^0 \subseteq V^*$  is made of **functionals** ([Definition 4.2.1](#)). The annihilator exists over every field and needs no extra structure; the orthogonal complement needs an inner product, so it needs  $F = \mathbb{R}$  or  $F = \mathbb{C}$ . They do have the same dimension,  $\dim V - \dim U$  ([Theorem 4.2.2](#), [Theorem 10.3.2](#) (c)), and that is not a coincidence: Section 5 produces an explicit bijection  $U^\perp \rightarrow U^0$ . Until then, keep them apart.

*Warning.* “Orthogonal projection” is a special case of Chapter 3’s projections, not a synonym. An operator with  $P^2 = P$  is a projection onto  $\text{im } P$  along  $\ker P$  (Definition 3.10.1, Theorem 3.10.1), and Chapter 3, §10 stressed that the direction matters: on  $\mathbb{R}^2$ , both  $\begin{pmatrix} 1 & 0 \\ 0 & 0 \end{pmatrix}$  and  $\begin{pmatrix} 1 & -1 \\ 0 & 0 \end{pmatrix}$  are projections onto the  $x$ -axis. Only the first is *orthogonal*, because only for the first is the kernel the perpendicular line; the second projects along  $\text{span}((1, 1))$  and is called **oblique**. So  $P_U$  carries more information than “ $P^2 = P$  and  $\text{im } P = U$ ”: it also says which complement is used. Theorem 10.3.3 (b) is the extra property, and Exercise C3 below shows that it singles out the orthogonal projections among all projections.

*Warning.*  $U^\perp$  depends on the inner product, not only on  $U$ . In  $\mathbb{R}^2$  let  $U = \text{span}((1, 1))$ . With the dot product,  $U^\perp = \text{span}((1, -1))$ . With the weighted inner product  $\langle \mathbf{x}, \mathbf{y} \rangle = 2x_1y_1 + x_2y_2$ , a vector  $(a, b)$  is orthogonal to  $(1, 1)$  when  $2a + b = 0$ , so  $U^\perp = \text{span}((1, -2))$ . Same subspace, different perpendicular. The symbol  $\perp$  always hides a choice of  $\langle \cdot, \cdot \rangle$ , and when two inner products are in play the notation must say which.

## 10.3.6 Exercises

### 10.3.6.1 A. Check your understanding

#### Exercise A1.

- Define the orthogonal complement  $S^\perp$  of a subset  $S$  of an inner product space.
- True or false:  $S^\perp$  is a subspace only when  $S$  is a subspace. Justify your answer.
- State the Orthogonal Decomposition Theorem, including every hypothesis.
- Let  $U$  be a 3-dimensional subspace of  $\mathbb{R}^7$ . What is  $\dim U^\perp$ , and what is  $\dim(U^\perp)^\perp$ ?
- What quantity does  $P_U \mathbf{v}$  minimize, and over what set?
- Name two differences between  $U^\perp$  and the annihilator  $U^0$ .

*Solution.*

- $S^\perp = \{\mathbf{v} \in V : \langle \mathbf{v}, \mathbf{s} \rangle = 0 \text{ for every } \mathbf{s} \in S\}$  (Definition 10.3.1).
- False.  $S^\perp$  is a subspace for **every** subset  $S$ , by Proposition 10.3.1, and it equals  $\text{span}(S)^\perp$ . For instance  $\{(1, 0, 0), (1, 1, 0)\}^\perp = \text{span}((0, 0, 1))$  in  $\mathbb{R}^3$ .
- If  $V$  is an inner product space over  $\mathbb{R}$  or  $\mathbb{C}$  and  $U$  is a **finite-dimensional** subspace, then  $V = U \oplus U^\perp$  and  $(U^\perp)^\perp = U$ ; if  $V$  is also finite-dimensional, then  $\dim U^\perp = \dim V - \dim U$  (Theorem 10.3.2).  $V$  itself need not be finite-dimensional.
- $\dim U^\perp = 7 - 3 = 4$  and  $\dim(U^\perp)^\perp = \dim U = 3$ , by Theorem 10.3.2 (c) and (b).
- It minimizes  $\|\mathbf{v} - \mathbf{u}\|$  over  $\mathbf{u} \in U$ , and it is the **only** minimizer (Theorem 10.3.5).
- $U^\perp$  consists of vectors of  $V$ ,  $U^0$  of functionals in  $V^*$ ; and  $U^\perp$  needs an inner product (so  $F = \mathbb{R}$  or  $\mathbb{C}$ ), while  $U^0$  is defined over any field. A third difference:  $U^\perp$  is a complement of  $U$  inside  $V$ , whereas  $U^0$  is not a subspace of  $V$  at all.

## 10.3.6.2 B. Practice

**Exercise B1: A complement and a projection in  $\mathbb{R}^4$ .**

In  $\mathbb{R}^4$  with the dot product, let  $U = \text{span}((1, 1, 0, 0), (0, 0, 1, 1))$ .

- (a) Find a basis of  $U^\perp$  and verify the dimension count.
- (b) Compute  $P_U \mathbf{v}$  and  $d(\mathbf{v}, U)$  for  $\mathbf{v} = (1, 2, 3, 4)$ .
- (c) Verify directly that  $\mathbf{v} - P_U \mathbf{v} \in U^\perp$ .

*Solution.*

- (a) By [Proposition 10.3.1](#) it is enough to be orthogonal to the two spanning vectors, so  $(a, b, c, d) \in U^\perp$  means  $a + b = 0$  and  $c + d = 0$ . Hence

$$U^\perp = \text{span}((1, -1, 0, 0), (0, 0, 1, -1)),$$

these two being independent (look at the first and third entries). So  $\dim U^\perp = 2 = 4 - 2$ , as [Theorem 10.3.2](#) (c) requires.

- (b) The spanning list of  $U$  is orthogonal, so [Corollary 10.3.4](#) applies with  $\mathbf{u}_1 = (1, 1, 0, 0)$ ,  $\mathbf{u}_2 = (0, 0, 1, 1)$ , both of squared norm 2:

$$P_U \mathbf{v} = \frac{1+2}{2} \mathbf{u}_1 + \frac{3+4}{2} \mathbf{u}_2 = \left(\frac{3}{2}, \frac{3}{2}, \frac{7}{2}, \frac{7}{2}\right).$$

Then  $\mathbf{v} - P_U \mathbf{v} = (-\frac{1}{2}, \frac{1}{2}, -\frac{1}{2}, \frac{1}{2})$  and  $d(\mathbf{v}, U) = \sqrt{4 \cdot \frac{1}{4}} = 1$ .

- (c) The vector  $(-\frac{1}{2}, \frac{1}{2}, -\frac{1}{2}, \frac{1}{2})$  satisfies  $a + b = 0$  and  $c + d = 0$ , so it lies in  $U^\perp$  by (a).

**Exercise B2: Distance from a point to a plane.**

In  $\mathbb{R}^3$  with the dot product, let  $U = \{(x, y, z) : 2x - y + 2z = 0\}$  and  $\mathbf{v} = (1, 2, 3)$ . Compute  $d(\mathbf{v}, U)$  and  $P_U \mathbf{v}$ , and check that  $P_U \mathbf{v}$  lies in  $U$ . Hence write down the general formula for the distance from  $\mathbf{v}$  to the plane  $\{\mathbf{x} : \langle \mathbf{x}, \mathbf{n} \rangle = 0\}$ ,  $\mathbf{n} \neq \mathbf{0}$ .

*Solution.* The plane is  $\{\mathbf{n}\}^\perp$  for  $\mathbf{n} = (2, -1, 2)$ , and  $\mathbf{n} \in U^\perp$  with  $\dim U^\perp = 3 - 2 = 1$  ([Theorem 10.3.2](#) (c)), so  $U^\perp = \text{span}(\mathbf{n})$ . Here  $\langle \mathbf{v}, \mathbf{n} \rangle = 2 - 2 + 6 = 6$  and  $\langle \mathbf{n}, \mathbf{n} \rangle = 4 + 1 + 4 = 9$ , so

$$P_{U^\perp} \mathbf{v} = \frac{6}{9} \mathbf{n} = \left(\frac{4}{3}, -\frac{2}{3}, \frac{4}{3}\right), \quad P_U \mathbf{v} = \mathbf{v} - P_{U^\perp} \mathbf{v} = \left(-\frac{1}{3}, \frac{8}{3}, \frac{5}{3}\right).$$

*Check:*  $2(-\frac{1}{3}) - \frac{8}{3} + 2 \cdot \frac{5}{3} = \frac{-2-8+10}{3} = 0$ , so  $P_U \mathbf{v} \in U$ . The distance is  $d(\mathbf{v}, U) = \|\frac{6}{9} \mathbf{n}\| = \frac{6}{9} \cdot 3 = 2$ .

In general  $d(\mathbf{v}, U) = \|P_{U^\perp} \mathbf{v}\| = |\langle \mathbf{v}, \mathbf{n} \rangle| \|\mathbf{n}\| / \|\mathbf{n}\|^2 = |\langle \mathbf{v}, \mathbf{n} \rangle| / \|\mathbf{n}\|$ , which for  $\mathbf{n} = (2, -1, 2)$  and  $\mathbf{v} = (1, 2, 3)$  gives  $6/3 = 2$ .

**Exercise B3: The best quadratic approximation to a quartic.**

On  $\mathbb{R}[x]_{\leq 4}$  with  $\langle p, q \rangle = \int_{-1}^1 p(t)q(t) dt$ , let  $U = \mathbb{R}[x]_{\leq 2}$ . Find  $P_U(x^4)$  and  $d(x^4, U)$ .  
*Hint: the orthogonal basis of [Example \(The best quadratic approximation to a cubic\)](#)*

*still works.*

*Solution.* Use the orthogonal basis  $(1, x, x^2 - \frac{1}{3})$  of  $U$  from [Example \(The best quadratic approximation to a cubic\)](#), with  $\langle 1, 1 \rangle = 2$  and

$$\langle x^2 - \frac{1}{3}, x^2 - \frac{1}{3} \rangle = \int_{-1}^1 \left( t^4 - \frac{2}{3}t^2 + \frac{1}{9} \right) dt = \frac{2}{5} - \frac{4}{9} + \frac{2}{9} = \frac{8}{45}.$$

The coefficients against  $x^4$  are  $\langle x^4, 1 \rangle = \frac{2}{5}$ ,  $\langle x^4, x \rangle = 0$  (odd integrand) and

$$\langle x^4, x^2 - \frac{1}{3} \rangle = \int_{-1}^1 \left( t^6 - \frac{1}{3}t^4 \right) dt = \frac{2}{7} - \frac{1}{3} \cdot \frac{2}{5} = \frac{16}{105}.$$

By [Corollary 10.3.4](#),

$$P_U(x^4) = \frac{2/5}{2} \cdot 1 + \frac{16/105}{8/45} \left( x^2 - \frac{1}{3} \right) = \frac{1}{5} + \frac{6}{7} \left( x^2 - \frac{1}{3} \right) = \frac{6}{7}x^2 - \frac{3}{35}.$$

For the distance, [Theorem 10.1.3](#) applied to  $x^4 = P_U(x^4) + (x^4 - P_U(x^4))$  gives  $d^2 = \|x^4\|^2 - \|P_U(x^4)\|^2$ . Here  $\|x^4\|^2 = \int_{-1}^1 t^8 dt = \frac{2}{9}$  and, the basis being orthogonal,

$$\|P_U(x^4)\|^2 = \left( \frac{1}{5} \right)^2 \cdot 2 + \left( \frac{6}{7} \right)^2 \cdot \frac{8}{45} = \frac{2}{25} + \frac{32}{245} = \frac{258}{1225}.$$

Hence  $d^2 = \frac{2}{9} - \frac{258}{1225} = \frac{2450-2322}{11025} = \frac{128}{11025}$  and  $d(x^4, U) = \frac{8\sqrt{2}}{105} \approx 0.108$ .

### 10.3.6.3 C. Going deeper

#### Exercise C1: Perp turns sums into intersections.

Let  $U$  and  $W$  be subspaces of an inner product space  $V$ .

- (a) Prove that  $(U + W)^\perp = U^\perp \cap W^\perp$ . (No finiteness is needed.)
- (b) Assume now that  $V$  is finite-dimensional. Deduce that  $(U \cap W)^\perp = U^\perp + W^\perp$ .

*Solution.*

- (a) ( $\subseteq$ ) Let  $\mathbf{v} \in (U + W)^\perp$ . Since  $U \subseteq U + W$  ([Theorem 1.7.1](#)),  $\mathbf{v}$  is orthogonal to every element of  $U$ , so  $\mathbf{v} \in U^\perp$ ; likewise  $\mathbf{v} \in W^\perp$ .
- ( $\supseteq$ ) Let  $\mathbf{v} \in U^\perp \cap W^\perp$  and let  $\mathbf{u} + \mathbf{w} \in U + W$  with  $\mathbf{u} \in U$ ,  $\mathbf{w} \in W$ . Then  $\langle \mathbf{v}, \mathbf{u} + \mathbf{w} \rangle = \langle \mathbf{v}, \mathbf{u} \rangle + \langle \mathbf{v}, \mathbf{w} \rangle = 0$ , using additivity in the second slot. Hence  $\mathbf{v} \in (U + W)^\perp$ .
- (b) Apply (a) to the subspaces  $U^\perp$  and  $W^\perp$ :

$$(U^\perp + W^\perp)^\perp = (U^\perp)^\perp \cap (W^\perp)^\perp = U \cap W,$$

where the second equality is [Theorem 10.3.2](#) (b), available because  $V$  is finite-dimensional, so all subspaces are ([Theorem 1.6.7](#)). Now take  $\perp$  of both ends and use [Theorem 10.3.2](#) (b) once more, this time on the subspace  $U^\perp + W^\perp$ :

$$U^\perp + W^\perp = \left( (U^\perp + W^\perp)^\perp \right)^\perp = (U \cap W)^\perp.$$

This proves the second identity.

### Exercise C2: A subspace with no orthogonal complement.

Let  $V = \mathbb{R}[x]$ , with the **coefficient inner product**

$$\langle p, q \rangle = \sum_{k \geq 0} a_k b_k \quad \text{for } p = \sum_k a_k x^k, \quad q = \sum_k b_k x^k,$$

a finite sum, since a polynomial has only finitely many non-zero coefficients. Let  $U = \{p \in V : p(1) = 0\}$ .

- Verify that  $\langle \cdot, \cdot \rangle$  is an inner product on  $V$  and that  $U$  is a subspace with  $U \neq V$ .
- Prove that  $U^\perp = \{0\}$ . Conclude that  $U \oplus U^\perp \neq V$ , so [Theorem 10.3.2](#) fails without the hypothesis that  $U$  is finite-dimensional.
- Explain what goes wrong geometrically by exhibiting, for each  $m \geq 1$ , an element  $p_m \in U$  with  $\|p_m - 1\| = 1/\sqrt{m}$ .

*Solution.*

- The form is symmetric and bilinear over  $\mathbb{R}$  because each term  $a_k b_k$  is, and  $\langle p, p \rangle = \sum_k a_k^2 > 0$  whenever some  $a_k \neq 0$ , that is whenever  $p \neq 0$ . So it is an inner product. The map  $\varepsilon: p \mapsto p(1)$  is linear ([Theorem 0.8.5](#)), so  $U = \ker \varepsilon$  is a subspace ([Theorem 3.2.1](#)). It is proper, since the constant polynomial 1 has  $\varepsilon(1) = 1 \neq 0$ .
- Note first that  $x^i - x^j \in U$  for all  $i, j \geq 0$ , since  $1 - 1 = 0$ . Let  $q = \sum_k b_k x^k \in U^\perp$ . Pairing with  $x^i - x^j$  gives  $b_i - b_j = 0$ , so all the coefficients of  $q$  are equal, say to  $c$ . But only finitely many of them are non-zero, and there are infinitely many indices, so  $c = 0$  and  $q = 0$ . Hence  $U^\perp = \{0\}$  and  $U + U^\perp = U \neq V$ .
- Take  $p_m = 1 - \frac{1}{m}(x + x^2 + \cdots + x^m)$ . Then  $p_m(1) = 1 - m \cdot \frac{1}{m} = 0$ , so  $p_m \in U$ , while

$$\|p_m - 1\|^2 = \left\| \frac{1}{m}(x + \cdots + x^m) \right\|^2 = m \cdot \frac{1}{m^2} = \frac{1}{m}.$$

So the polynomial  $1 \notin U$  has elements of  $U$  arbitrarily close to it:  $U$  leaves no room outside itself for a perpendicular direction, and the best approximation to 1 from  $U$  does not exist (the distances  $\|1 - p\|$  get arbitrarily close to 0 without reaching it). The obstruction is not the dimension of  $V$  as such but the fact that  $U$  is not *closed*: filling in the missing limits is exactly what the theory of Hilbert spaces does, and there [Theorem 10.3.2](#) is recovered for closed subspaces. That is analysis, not linear algebra, and we do not pursue it.

### Exercise C3: Which projections are orthogonal.

Let  $V$  be a finite-dimensional inner product space and let  $P \in \mathcal{L}(V)$  satisfy  $P^2 = P$ . Prove that the following are equivalent:

- $P = P_U$  for some subspace  $U$ ;
- $\langle P\mathbf{u}, \mathbf{v} \rangle = \langle \mathbf{u}, P\mathbf{v} \rangle$  for all  $\mathbf{u}, \mathbf{v} \in V$ ;
- $\ker P = (\operatorname{im} P)^\perp$ .



*Hint: for (b)  $\Rightarrow$  (c), pair a vector of  $\ker P$  with  $P\mathbf{v}$ .*

*Solution.*

(a)  $\Rightarrow$  (b) is [Theorem 10.3.3](#) (b).

(b)  $\Rightarrow$  (c). ( $\subseteq$ ) Let  $\mathbf{x} \in \ker P$  and let  $P\mathbf{v} \in \operatorname{im} P$  be arbitrary. Then

$$\langle \mathbf{x}, P\mathbf{v} \rangle = \overline{\langle P\mathbf{v}, \mathbf{x} \rangle} = \overline{\langle \mathbf{v}, P\mathbf{x} \rangle} = \overline{\langle \mathbf{v}, \mathbf{0} \rangle} = 0,$$

using (b) in the second equality. So  $\mathbf{x} \in (\operatorname{im} P)^\perp$ . ( $\supseteq$ ) Let  $\mathbf{x} \in (\operatorname{im} P)^\perp$ . Then

$$\langle P\mathbf{x}, P\mathbf{x} \rangle = \langle \mathbf{x}, P(P\mathbf{x}) \rangle = \langle \mathbf{x}, P\mathbf{x} \rangle = 0,$$

the first equality by (b), the second by  $P^2 = P$ , and the third because  $P\mathbf{x} \in \operatorname{im} P$  while  $\mathbf{x} \in (\operatorname{im} P)^\perp$ . Positive definiteness gives  $P\mathbf{x} = \mathbf{0}$ , so  $\mathbf{x} \in \ker P$ .

(c)  $\Rightarrow$  (a). Put  $U = \operatorname{im} P$ . By [Theorem 3.10.1](#) (a),  $V = \operatorname{im} P \oplus \ker P = U \oplus U^\perp$ , and  $P$  is the identity on  $U$  and zero on  $\ker P = U^\perp$ . By the uniqueness clause of [Theorem 3.10.1](#) (b),  $P$  is the projection onto  $U$  along  $U^\perp$ , which is  $P_U$  by [Definition 10.3.2](#).

## 10.4 Least Squares and Minimum-Norm Solutions

Chapter 2 taught us to decide whether  $\mathbf{Ax} = \mathbf{b}$  has a solution, and to find them all when it does. Real data rarely cooperates. Measure a quantity that depends linearly on two unknowns, take ten measurements, and you have ten equations in two unknowns; each measurement carries a little error, and the ten equations are almost certainly inconsistent. Refusing to answer is not useful. This section replaces the question “which  $\mathbf{x}$  solves  $\mathbf{Ax} = \mathbf{b}$ ?” with “which  $\mathbf{x}$  comes closest?”, and shows that the new question always has an answer, computed by solving a square system.

The other half of the section handles the opposite complaint. When a system is consistent but underdetermined, there are too many solutions, and we want a canonical one. Orthogonality picks it: exactly one solution is perpendicular to the kernel, and it is the shortest.

Throughout,  $F = \mathbb{R}$  or  $F = \mathbb{C}$ , and  $F^n$  and  $F^m$  carry the standard inner product  $\langle \mathbf{x}, \mathbf{y} \rangle = \sum_i x_i \overline{y_i}$ , with  $\|\mathbf{x}\|^2 = \sum_i |x_i|^2$ .

### 10.4.1 The residual, and a piece of notation

Given  $\mathbf{A} \in M_{m \times n}(F)$  and  $\mathbf{b} \in F^m$ , write  $\mathbf{r} = \mathbf{b} - \mathbf{Ax}$  for the **residual** of a candidate  $\mathbf{x}$ . The system is consistent exactly when some  $\mathbf{x}$  makes  $\mathbf{r} = \mathbf{0}$ . When none does, we ask for the  $\mathbf{x}$  making  $\mathbf{r}$  as short as possible, that is, minimizing

$$\|\mathbf{Ax} - \mathbf{b}\|^2 = \sum_{i=1}^m |(\mathbf{Ax})_i - b_i|^2,$$

a **sum of squares**; hence the name **least squares**.

Two observations turn this into geometry. First, as  $\mathbf{x}$  ranges over  $F^n$ , the vector  $\mathbf{Ax}$  ranges over exactly the column space  $\text{col}(\mathbf{A}) \subseteq F^m$  (Theorem 0.9.3), a finite-dimensional subspace. Second, minimizing  $\|\mathbf{Ax} - \mathbf{b}\|$  over all  $\mathbf{x}$  is therefore the same as finding the point of  $\text{col}(\mathbf{A})$  nearest to  $\mathbf{b}$ . We already know that point: it is  $P_{\text{col}(\mathbf{A})}\mathbf{b}$ , by Theorem 10.3.5. All that remains is to turn “the residual is perpendicular to  $\text{col}(\mathbf{A})$ ” into equations, and for that we need one piece of notation.

Recall the conjugate transpose  $\mathbf{A}^* \in M_{n \times m}(F)$  of Definition 10.1.2: the matrix with entries  $(\mathbf{A}^*)_{ij} = \overline{a_{ji}}$ , obtained by transposing and then conjugating every entry, so that  $\mathbf{A}^* = \mathbf{A}^\top$  over  $\mathbb{R}$ . Comparing entries and using Theorem 0.9.5 together with the rules for conjugation (Theorem 0.6.2) gives

$$(\mathbf{A} + \mathbf{B})^* = \mathbf{A}^* + \mathbf{B}^*, \quad (c\mathbf{A})^* = \bar{c}\mathbf{A}^*, \quad (\mathbf{AB})^* = \mathbf{B}^*\mathbf{A}^*, \quad \mathbf{A}^{**} = \mathbf{A},$$

whenever the sizes match. The third is the one worth seeing done:  $((\mathbf{AB})^*)_{ij} = \overline{(\mathbf{AB})_{ji}} = \overline{\sum_k a_{jk}b_{ki}} = \sum_k \overline{b_{ki}}\overline{a_{jk}} = \sum_k (\mathbf{B}^*)_{ik}(\mathbf{A}^*)_{kj} = (\mathbf{B}^*\mathbf{A}^*)_{ij}$ , which is where the order reverses. The one consequence we really use is the following.

### Lemma 10.4.1 — Moving a matrix across the inner product

Let  $\mathbf{A} \in M_{m \times n}(F)$ . Then for all  $\mathbf{x} \in F^n$  and  $\mathbf{y} \in F^m$ ,

$$\langle \mathbf{Ax}, \mathbf{y} \rangle = \langle \mathbf{x}, \mathbf{A}^*\mathbf{y} \rangle, \quad \langle \mathbf{y}, \mathbf{Ax} \rangle = \langle \mathbf{A}^*\mathbf{y}, \mathbf{x} \rangle.$$

*Proof.* Regarding vectors as columns,  $\langle \mathbf{u}, \mathbf{w} \rangle = \mathbf{w}^*\mathbf{u}$ , a  $1 \times 1$  matrix identified with its entry. Hence

$$\langle \mathbf{Ax}, \mathbf{y} \rangle = \mathbf{y}^*(\mathbf{Ax}) = (\mathbf{y}^*\mathbf{A})\mathbf{x} = ((\mathbf{A}^*\mathbf{y})^*)\mathbf{x} = \langle \mathbf{x}, \mathbf{A}^*\mathbf{y} \rangle,$$

where the third equality uses  $(\mathbf{A}^*\mathbf{y})^* = \mathbf{y}^*\mathbf{A}^{**} = \mathbf{y}^*\mathbf{A}$ . The second identity follows by conjugating the first, since  $\langle \mathbf{y}, \mathbf{Ax} \rangle = \overline{\langle \mathbf{Ax}, \mathbf{y} \rangle}$  and  $\overline{\langle \mathbf{x}, \mathbf{A}^*\mathbf{y} \rangle} = \langle \mathbf{A}^*\mathbf{y}, \mathbf{x} \rangle$ . ■

The lemma is the matrix case of a general principle: *to move a map across an inner product, pay with its adjoint*. Section 6 develops that principle for linear maps between abstract inner product spaces; here the two lines above are all we need.

## 10.4.2 The normal equations

### Theorem 10.4.2 — Least Squares Theorem

Let  $\mathbf{A} \in M_{m \times n}(F)$  and  $\mathbf{b} \in F^m$ , and write  $U = \text{col}(\mathbf{A})$ . For  $\mathbf{x} \in F^n$ , the following are equivalent:

- (a)  $\|\mathbf{Ax} - \mathbf{b}\| \leq \|\mathbf{Az} - \mathbf{b}\|$  for every  $\mathbf{z} \in F^n$ ;
- (b)  $\mathbf{Ax} = P_U\mathbf{b}$ ;
- (c)  $\mathbf{A}^*\mathbf{Ax} = \mathbf{A}^*\mathbf{b}$ .

Moreover such an  $\mathbf{x}$  always exists. A vector satisfying these conditions is called a **least-squares solution** of  $\mathbf{Ax} = \mathbf{b}$ , and the equations in (c) are the **normal equations**.

*Idea.* Condition (a) is a minimization over  $\mathbf{x}$ ; condition (b) is the same minimization rewritten in  $F^m$ , where Theorem 10.3.5 has already solved it. So (a)  $\Leftrightarrow$  (b) is a translation, not an argument. The content is (b)  $\Leftrightarrow$  (c): saying  $\mathbf{Ax} = P_U \mathbf{b}$  is the same as saying the residual is perpendicular to  $U$ , and “perpendicular to every column of  $\mathbf{A}$ ” is what  $\mathbf{A}^*(\mathbf{b} - \mathbf{Ax}) = \mathbf{0}$  says. Existence is free, because  $P_U \mathbf{b}$  lies in  $U = \text{col}(\mathbf{A})$  and so is  $\mathbf{Ax}$  for some  $\mathbf{x}$ .

*Proof.* Since  $\{\mathbf{Az} : \mathbf{z} \in F^n\} = \text{col}(\mathbf{A}) = U$  (Theorem 0.9.3), condition (a) says that  $\mathbf{Ax}$  is a vector of  $U$  at least as close to  $\mathbf{b}$  as every other vector of  $U$ . By Theorem 10.3.5 the unique such vector is  $P_U \mathbf{b}$ . Hence (a)  $\Leftrightarrow$  (b).

(b)  $\Leftrightarrow$  (c). Write  $\mathbf{b} = P_U \mathbf{b} + (\mathbf{b} - P_U \mathbf{b})$ , the splitting of Theorem 10.3.2 (a) with  $\mathbf{b} - P_U \mathbf{b} \in U^\perp$ . For  $\mathbf{x} \in F^n$  we have  $\mathbf{Ax} \in U$ , so  $\mathbf{b} - \mathbf{Ax} \in U^\perp$  holds if and only if  $\mathbf{Ax}$  is the  $U$ -part of  $\mathbf{b}$ , that is, if and only if  $\mathbf{Ax} = P_U \mathbf{b}$ . It remains to see that

$$\mathbf{b} - \mathbf{Ax} \in U^\perp \iff \mathbf{A}^* \mathbf{Ax} = \mathbf{A}^* \mathbf{b}.$$

Put  $\mathbf{r} = \mathbf{b} - \mathbf{Ax}$ . By Proposition 10.3.1,  $\mathbf{r} \in U^\perp$  if and only if  $\mathbf{r}$  is orthogonal to every  $\mathbf{Az}$ , and by Lemma 10.4.1,

$$\langle \mathbf{r}, \mathbf{Az} \rangle = \langle \mathbf{A}^* \mathbf{r}, \mathbf{z} \rangle \quad \text{for every } \mathbf{z} \in F^n.$$

If  $\mathbf{A}^* \mathbf{r} = \mathbf{0}$ , all these vanish. Conversely, if all vanish, take  $\mathbf{z} = \mathbf{A}^* \mathbf{r}$  to get  $\langle \mathbf{A}^* \mathbf{r}, \mathbf{A}^* \mathbf{r} \rangle = 0$ , hence  $\mathbf{A}^* \mathbf{r} = \mathbf{0}$  by positive definiteness. So  $\mathbf{r} \in U^\perp$  if and only if  $\mathbf{A}^* \mathbf{r} = \mathbf{0}$ , which rearranges to  $\mathbf{A}^* \mathbf{Ax} = \mathbf{A}^* \mathbf{b}$ .

*Existence.*  $P_U \mathbf{b} \in U = \text{col}(\mathbf{A})$ , so  $P_U \mathbf{b} = \mathbf{Ax}$  for some  $\mathbf{x} \in F^n$ , and that  $\mathbf{x}$  satisfies (b). ■

Three things are worth extracting. The normal equations are a **square**  $n \times n$  system, whatever the shape of  $\mathbf{A}$ . They are **always consistent**, by the existence clause, even when  $\mathbf{Ax} = \mathbf{b}$  is not. And the minimizing *value*  $\mathbf{Ax}$  is unique — it is  $P_U \mathbf{b}$  — even if the minimizing  $\mathbf{x}$  is not.

*Warning.* “Closest” is measured by the inner product, so the answer moves when the inner product does. Least squares is not a coordinate-free notion: it minimizes  $\|\mathbf{Ax} - \mathbf{b}\|$  for a particular norm on  $F^m$ , and rescaling one coordinate — reporting one measurement in centimeters instead of meters, say — is a change of inner product. Concretely, for the data  $(0, 2), (1, 1), (2, 8)$  the least-squares line is  $y = \frac{2}{3} + 3t$ ; if the middle point is trusted twice as much, so that the second coordinate of  $F^3$  is weighted by 2, the best line becomes  $y = 3t$ . Both are correct answers to their own question, and Exercise C1 works the general case out. Before fitting, decide which errors count equally.

*Quick check.* The system  $\mathbf{Ax} = \mathbf{b}$  may have no solution at all, yet the normal equations  $\mathbf{A}^* \mathbf{Ax} = \mathbf{A}^* \mathbf{b}$  always have one. Where does the proof arrange that, and what stops the same argument from solving  $\mathbf{Ax} = \mathbf{b}$  itself?

*Solution.* The existence clause uses that  $P_U \mathbf{b}$  lies in  $U = \text{col}(\mathbf{A})$ , and every vector of  $\text{col}(\mathbf{A})$  is  $\mathbf{A}\mathbf{x}$  for some  $\mathbf{x}$ . So the normal equations ask  $\mathbf{A}\mathbf{x}$  to hit  $P_U \mathbf{b}$ , a target that is inside the column space by construction. The original system asks  $\mathbf{A}\mathbf{x}$  to hit  $\mathbf{b}$ , which need not be in  $\text{col}(\mathbf{A})$  at all. Projecting first is exactly the step that moves the target into reach; what is given up is the requirement  $\mathbf{r} = \mathbf{0}$ , which becomes  $\mathbf{r} \perp \text{col}(\mathbf{A})$ .

### 10.4.3 When is the least-squares solution unique?

The minimizers of  $\|\mathbf{A}\mathbf{x} - \mathbf{b}\|$  are the solutions of the normal equations, so they form the solution set of a linear system: a coset  $\mathbf{x}_0 + \text{nul}(\mathbf{A}^*\mathbf{A})$  (Theorem 2.1.3). To read off uniqueness we therefore need  $\text{nul}(\mathbf{A}^*\mathbf{A})$ , and the following small lemma computes it.

#### Lemma 10.4.3 — The null space of $\mathbf{A}^*\mathbf{A}$

Let  $\mathbf{A} \in M_{m \times n}(F)$ . Then  $\text{nul}(\mathbf{A}^*\mathbf{A}) = \text{nul}(\mathbf{A})$ .

*Proof.* ( $\supseteq$ ) If  $\mathbf{A}\mathbf{x} = \mathbf{0}$ , then  $\mathbf{A}^*\mathbf{A}\mathbf{x} = \mathbf{A}^*\mathbf{0} = \mathbf{0}$ .

( $\subseteq$ ) Suppose  $\mathbf{A}^*\mathbf{A}\mathbf{x} = \mathbf{0}$ . Pair with  $\mathbf{x}$  and move  $\mathbf{A}^*$  across using Lemma 10.4.1:

$$0 = \langle \mathbf{A}^*(\mathbf{A}\mathbf{x}), \mathbf{x} \rangle = \langle \mathbf{A}\mathbf{x}, \mathbf{A}\mathbf{x} \rangle = \|\mathbf{A}\mathbf{x}\|^2.$$

Hence  $\mathbf{A}\mathbf{x} = \mathbf{0}$  by positive definiteness. This proves the lemma. ■

This is the “pair it with itself” move once more, and it is the whole reason the normal equations behave well. It will return in Section 6 in the form  $\ker(T^*T) = \ker T$  for a linear map  $T$  between inner product spaces, with the same two-line proof; the matrix statement above is the case we need now.

#### Corollary 10.4.4 — Uniqueness of the least-squares solution

Let  $\mathbf{A} \in M_{m \times n}(F)$  and  $\mathbf{b} \in F^m$ . The following are equivalent:

- (a)  $\mathbf{A}\mathbf{x} = \mathbf{b}$  has exactly one least-squares solution (for one, equivalently for every,  $\mathbf{b}$ );
- (b) the columns of  $\mathbf{A}$  are linearly independent, that is  $\text{rank } \mathbf{A} = n$ ;
- (c)  $\mathbf{A}^*\mathbf{A}$  is invertible.

In that case the least-squares solution is  $\mathbf{x} = (\mathbf{A}^*\mathbf{A})^{-1}\mathbf{A}^*\mathbf{b}$ .

*Proof.* By Theorem 10.4.2 the least-squares solutions are the solutions of  $\mathbf{A}^*\mathbf{A}\mathbf{x} = \mathbf{A}^*\mathbf{b}$ , a consistent system; by Theorem 2.1.3 its solution set is a coset of  $\text{nul}(\mathbf{A}^*\mathbf{A})$ , which is  $\text{nul}(\mathbf{A})$  by Lemma 10.4.3. So the solution is unique, for one  $\mathbf{b}$  or for all, exactly when  $\text{nul}(\mathbf{A}) = \{\mathbf{0}\}$ . Now  $\text{nul}(\mathbf{A}) = \{\mathbf{0}\}$  says precisely that the only vanishing linear combination of the columns of  $\mathbf{A}$  is the trivial one, that is, that the columns are independent; and by Theorem 2.5.7 this is the same as  $\text{rank } \mathbf{A} = n$ . This proves (a)  $\Leftrightarrow$  (b).

(b)  $\Leftrightarrow$  (c). The matrix  $\mathbf{A}^*\mathbf{A}$  is  $n \times n$ , so it is invertible if and only if  $\text{nul}(\mathbf{A}^*\mathbf{A}) = \{\mathbf{0}\}$  (Theorem 2.4.4). By Lemma 10.4.3 this happens exactly when  $\text{nul}(\mathbf{A}) = \{\mathbf{0}\}$ .

Finally, if  $\mathbf{A}^*\mathbf{A}$  is invertible, multiplying the normal equations on the left by  $(\mathbf{A}^*\mathbf{A})^{-1}$  gives the stated formula. ■

*Warning.*  $(\mathbf{A}^*\mathbf{A})^{-1}$  is not  $\mathbf{A}^{-1}(\mathbf{A}^*)^{-1}$ . The rule  $(\mathbf{BC})^{-1} = \mathbf{C}^{-1}\mathbf{B}^{-1}$  needs  $\mathbf{B}$  and  $\mathbf{C}$  to be invertible, and in least squares  $\mathbf{A}$  is usually not even square. Take

$$\mathbf{A} = \begin{pmatrix} 1 & 0 \\ 1 & 1 \\ 0 & 1 \end{pmatrix}, \quad \mathbf{A}^\top \mathbf{A} = \begin{pmatrix} 2 & 1 \\ 1 & 2 \end{pmatrix}, \quad (\mathbf{A}^\top \mathbf{A})^{-1} = \frac{1}{3} \begin{pmatrix} 2 & -1 \\ -1 & 2 \end{pmatrix}.$$

The left side exists; the symbols  $\mathbf{A}^{-1}$  and  $(\mathbf{A}^\top)^{-1}$  do not exist at all, since  $\mathbf{A}$  is  $3 \times 2$ . The formula  $\mathbf{x} = (\mathbf{A}^*\mathbf{A})^{-1}\mathbf{A}^*\mathbf{b}$  must be read as one object  $(\mathbf{A}^*\mathbf{A})^{-1}$  applied to  $\mathbf{A}^*\mathbf{b}$ , never simplified.

## 10.4.4 Fitting a line and a parabola

**Example.** (The least-squares line through four points)

Find the line  $y = c + dt$  that minimizes the sum of the squared vertical errors at the four data points

$$(1, 3), \quad (2, 1), \quad (3, 5), \quad (4, 5).$$

Compute the residual and the minimum value of the sum of squares.

*Solution.* Asking for  $c + dt_i = y_i$  at the four points is the system  $\mathbf{Ax} = \mathbf{b}$  with

$$\mathbf{A} = \begin{pmatrix} 1 & 1 \\ 1 & 2 \\ 1 & 3 \\ 1 & 4 \end{pmatrix}, \quad \mathbf{x} = \begin{pmatrix} c \\ d \end{pmatrix}, \quad \mathbf{b} = \begin{pmatrix} 3 \\ 1 \\ 5 \\ 5 \end{pmatrix},$$

and  $\|\mathbf{Ax} - \mathbf{b}\|^2$  is exactly the sum of the squared vertical errors. The columns of  $\mathbf{A}$  are independent (they are not multiples of each other), so Corollary 10.4.4 gives a unique answer. Over  $\mathbb{R}$ ,  $\mathbf{A}^* = \mathbf{A}^\top$ , and

$$\mathbf{A}^\top \mathbf{A} = \begin{pmatrix} 4 & 10 \\ 10 & 30 \end{pmatrix}, \quad \mathbf{A}^\top \mathbf{b} = \begin{pmatrix} 3 + 1 + 5 + 5 \\ 3 + 2 + 15 + 20 \end{pmatrix} = \begin{pmatrix} 14 \\ 40 \end{pmatrix}.$$

The normal equations  $4c + 10d = 14$ ,  $10c + 30d = 40$  simplify to  $2c + 5d = 7$  and  $c + 3d = 4$ ; subtracting twice the second from the first gives  $-d = -1$ , so  $d = 1$  and  $c = 1$ . The least-squares line is

$$y = 1 + t.$$

Its values at  $t = 1, 2, 3, 4$  are 2, 3, 4, 5, so the residual is

$$\mathbf{r} = \mathbf{b} - \mathbf{Ax} = (3, 1, 5, 5) - (2, 3, 4, 5) = (1, -2, 1, 0),$$

and the minimum sum of squares is  $\|\mathbf{r}\|^2 = 1 + 4 + 1 + 0 = 6$ .

*Check.* The residual must be orthogonal to both columns of  $\mathbf{A}$ :  $1 - 2 + 1 + 0 = 0$  and  $1 - 4 + 3 + 0 = 0$ . Both hold, which confirms the normal equations.

The same machinery fits any linear combination of prescribed functions; only the columns of  $\mathbf{A}$  change. “Linear” refers to the unknown coefficients, not to the shape of the curve.

### Example. (The least-squares parabola)

Find the quadratic  $y = a + bt + ct^2$  minimizing the sum of squared errors at

$$(-2, 7), \quad (-1, 6), \quad (0, 2), \quad (1, 0), \quad (2, 5).$$

*Solution.* Now  $\mathbf{A}$  has columns  $(1, 1, 1, 1, 1)$ ,  $(t_i)$  and  $(t_i^2)$ :

$$\mathbf{A} = \begin{pmatrix} 1 & -2 & 4 \\ 1 & -1 & 1 \\ 1 & 0 & 0 \\ 1 & 1 & 1 \\ 1 & 2 & 4 \end{pmatrix}, \quad \mathbf{b} = \begin{pmatrix} 7 \\ 6 \\ 2 \\ 0 \\ 5 \end{pmatrix}.$$

Because the nodes are symmetric about 0, every odd power sums to zero:  $\sum t_i = 0$  and  $\sum t_i^3 = 0$ . With  $\sum t_i^2 = 10$  and  $\sum t_i^4 = 34$ ,

$$\mathbf{A}^\top \mathbf{A} = \begin{pmatrix} 5 & 0 & 10 \\ 0 & 10 & 0 \\ 10 & 0 & 34 \end{pmatrix}, \quad \mathbf{A}^\top \mathbf{b} = \begin{pmatrix} 20 \\ -10 \\ 54 \end{pmatrix}.$$

The middle equation decouples:  $10b = -10$ , so  $b = -1$ . The outer two are  $5a + 10c = 20$  and  $10a + 34c = 54$ ; the first gives  $a = 4 - 2c$ , and substituting gives  $40 + 14c = 54$ , so  $c = 1$  and  $a = 2$ . The least-squares parabola is

$$y = 2 - t + t^2.$$

Its values at the nodes are 8, 4, 2, 2, 4, so  $\mathbf{r} = (-1, 2, 0, -2, 1)$  and the minimum sum of squares is  $\|\mathbf{r}\|^2 = 1 + 4 + 0 + 4 + 1 = 10$ .

*Check.*  $\sum r_i = 0$ ,  $\sum t_i r_i = 2 - 2 + 0 - 2 + 2 = 0$  and  $\sum t_i^2 r_i = -4 + 2 + 0 - 2 + 4 = 0$ : the residual is orthogonal to all three columns.

Two practical remarks. Choosing the nodes symmetrically, as above, made  $\mathbf{A}^\top \mathbf{A}$  nearly diagonal and the arithmetic painless; the same trick appears again in Section 9 when the columns are trigonometric functions. And although the normal equations are the right way to *understand* least squares, they are not always the right way to *compute* it: Section 8 solves the same problem through a QR factorization, which avoids forming  $\mathbf{A}^* \mathbf{A}$  at all.

## 10.4.5 The minimum-norm solution

Now the opposite situation:  $\mathbf{Ax} = \mathbf{b}$  is consistent, but  $\mathbf{A}$  has a non-trivial null space, so the solutions form an infinite family  $\mathbf{x}_0 + \text{nul}(\mathbf{A})$  (Theorem 2.1.3). Which one should we report?

Orthogonality answers.

### Theorem 10.4.5 — Minimum-Norm Solution

Let  $\mathbf{A} \in M_{m \times n}(F)$  and let  $\mathbf{b} \in F^m$  be such that  $\mathbf{Ax} = \mathbf{b}$  is consistent. Write  $K = \text{nul}(\mathbf{A})$ . Then:

- (a) exactly one solution  $\mathbf{x}_{\min}$  of  $\mathbf{Ax} = \mathbf{b}$  lies in  $K^\perp$ ;
- (b)  $\|\mathbf{x}_{\min}\| < \|\mathbf{x}\|$  for every other solution  $\mathbf{x}$ .

We call  $\mathbf{x}_{\min}$  the **minimum-norm solution**.

*Idea.* The solution set is a coset  $\mathbf{x}_0 + K$ , and  $F^n = K \oplus K^\perp$ . Throwing away the  $K$ -part of  $\mathbf{x}_0$  keeps it a solution, because that part is in the kernel, and lands it in  $K^\perp$ . Once there, Pythagoras compares it with every other solution: the others differ from it by a kernel vector, which is perpendicular to it and only adds length.

*Proof.*

- (a) *Existence.* Let  $\mathbf{x}_0$  be any solution. Since  $K$  is a subspace of the finite-dimensional space  $F^n$ , [Theorem 10.3.2](#) (a) gives  $F^n = K \oplus K^\perp$ ; write  $\mathbf{x}_0 = \mathbf{k} + \mathbf{x}_{\min}$  with  $\mathbf{k} \in K$  and  $\mathbf{x}_{\min} \in K^\perp$ . Then  $\mathbf{Ax}_{\min} = \mathbf{Ax}_0 - \mathbf{Ak} = \mathbf{b} - \mathbf{0} = \mathbf{b}$ , so  $\mathbf{x}_{\min}$  is a solution lying in  $K^\perp$ .

*Uniqueness.* If  $\mathbf{x}_{\min}$  and  $\mathbf{y}$  are solutions in  $K^\perp$ , then  $\mathbf{A}(\mathbf{x}_{\min} - \mathbf{y}) = \mathbf{0}$ , so  $\mathbf{x}_{\min} - \mathbf{y} \in K \cap K^\perp = \{\mathbf{0}\}$  by [Theorem 10.3.2](#) (a).

- (b) Every solution has the form  $\mathbf{x} = \mathbf{x}_{\min} + \mathbf{k}$  with  $\mathbf{k} \in K$ , by [Theorem 2.1.3](#). Since  $\mathbf{x}_{\min} \in K^\perp$  and  $\mathbf{k} \in K$ , the two are orthogonal, and [Theorem 10.1.3](#) gives

$$\|\mathbf{x}\|^2 = \|\mathbf{x}_{\min}\|^2 + \|\mathbf{k}\|^2.$$

If  $\mathbf{x} \neq \mathbf{x}_{\min}$ , then  $\mathbf{k} \neq \mathbf{0}$ , so  $\|\mathbf{k}\|^2 > 0$  and  $\|\mathbf{x}\| > \|\mathbf{x}_{\min}\|$ . This proves the theorem. ■

The subspace  $K^\perp$  has a matrix description that makes  $\mathbf{x}_{\min}$  computable. For  $\mathbf{x} \in F^n$ ,

$$\begin{aligned} \mathbf{x} \in \text{col}(\mathbf{A}^*)^\perp &\iff \langle \mathbf{x}, \mathbf{A}^* \mathbf{y} \rangle = 0 \text{ for all } \mathbf{y} \in F^m \\ &\iff \langle \mathbf{Ax}, \mathbf{y} \rangle = 0 \text{ for all } \mathbf{y} \in F^m \\ &\iff \mathbf{Ax} = \mathbf{0}, \end{aligned}$$

where the first step is [Definition 10.3.1](#) together with  $\text{col}(\mathbf{A}^*) = \{\mathbf{A}^* \mathbf{y} : \mathbf{y} \in F^m\}$  ([Theorem 0.9.3](#)), the second is [Lemma 10.4.1](#), and the last uses the choice  $\mathbf{y} = \mathbf{Ax}$  with positive definiteness. So  $\text{nul}(\mathbf{A}) = \text{col}(\mathbf{A}^*)^\perp$ , and applying  $\perp$  once more ([Theorem 10.3.2](#) (b)) gives

$$(\text{nul}(\mathbf{A}))^\perp = \text{col}(\mathbf{A}^*).$$

In words: the minimum-norm solution is the unique solution that is a combination of the columns of  $\mathbf{A}^*$ , that is, of the conjugated rows of  $\mathbf{A}$ . Section 6 proves the same statement for linear maps, together with the three companion identities that make up the four fundamental subspaces.

**Example. (A minimum-norm solution)**

Find the solution of smallest norm of

$$\begin{cases} x_1 + x_2 + x_3 = 3, \\ x_1 + 2x_2 + 3x_3 = 7. \end{cases}$$

*Solution.* Here  $\mathbf{A} = \begin{pmatrix} 1 & 1 & 1 \\ 1 & 2 & 3 \end{pmatrix}$ . Subtracting the first equation from the second gives  $x_2 + 2x_3 = 4$ , so  $x_2 = 4 - 2x_3$  and then  $x_1 = 3 - x_2 - x_3 = x_3 - 1$ . With  $t = x_3$  the solutions are

$$\mathbf{x}(t) = (-1, 4, 0) + t(1, -2, 1), \quad t \in \mathbb{R},$$

so  $K = \text{nul}(\mathbf{A}) = \text{span}((1, -2, 1))$ . By [Theorem 10.4.5](#) the answer is the unique  $t$  with  $\mathbf{x}(t) \perp (1, -2, 1)$ :

$$\langle \mathbf{x}(t), (1, -2, 1) \rangle = (-1 + t) - 2(4 - 2t) + t = 6t - 9 = 0, \quad t = \frac{3}{2}.$$

Hence  $\mathbf{x}_{\min} = (\frac{1}{2}, 1, \frac{3}{2})$ , with  $\|\mathbf{x}_{\min}\|^2 = \frac{1}{4} + 1 + \frac{9}{4} = \frac{7}{2}$ .

*Check.* It is a solution, since  $\frac{1}{2} + 1 + \frac{3}{2} = 3$  and  $\frac{1}{2} + 2 + \frac{9}{2} = 7$ , and it is orthogonal to the kernel, since  $\frac{1}{2} - 2 + \frac{3}{2} = 0$ . By the identity  $(\text{nul } \mathbf{A})^\perp = \text{col}(\mathbf{A}^\top)$  above it must also be a combination of the rows of  $\mathbf{A}$ , and it is:  $\mathbf{x}_{\min} = \frac{1}{2}(1, 2, 3)$ , half the second row. Finally, the solution  $\mathbf{x}(0) = (-1, 4, 0)$  has  $\|\mathbf{x}(0)\|^2 = 17 > \frac{7}{2}$ , as [Theorem 10.4.5](#) (b) requires.

The two answers of this section — the least-squares solution and the minimum-norm solution — are usually wanted together: given any  $\mathbf{A}$  and  $\mathbf{b}$ , first minimize  $\|\mathbf{Ax} - \mathbf{b}\|$ , then among the minimizers take the shortest. Chapter 12 packages the combined answer as a single matrix, the **pseudoinverse**  $\mathbf{A}^+$ , with  $\mathbf{x} = \mathbf{A}^+\mathbf{b}$  in all cases at once.

## 10.4.6 Exercises

### 10.4.6.1 A. Check your understanding

#### Exercise A1.

- Define a least-squares solution of  $\mathbf{Ax} = \mathbf{b}$ , and write down the normal equations.
- True or false: if  $\mathbf{Ax} = \mathbf{b}$  is consistent, then its solutions are exactly its least-squares solutions. Justify your answer.
- Explain in one sentence why the normal equations are always consistent.
- Give a condition on the columns of  $\mathbf{A}$  equivalent to uniqueness of the least-squares solution.
- Where does the minimum-norm solution of a consistent system live, and why is it unique there?
- True or false:  $(\mathbf{A}^*\mathbf{A})^{-1} = \mathbf{A}^{-1}(\mathbf{A}^*)^{-1}$ . Justify your answer.

*Solution.*

- A vector  $\mathbf{x} \in F^n$  with  $\|\mathbf{Ax} - \mathbf{b}\| \leq \|\mathbf{Az} - \mathbf{b}\|$  for every  $\mathbf{z} \in F^n$ . The normal



equations are  $\mathbf{A}^* \mathbf{A} \mathbf{x} = \mathbf{A}^* \mathbf{b}$  (Theorem 10.4.2).

- (b) True. If  $\mathbf{A} \mathbf{x}_0 = \mathbf{b}$  then  $\|\mathbf{A} \mathbf{x}_0 - \mathbf{b}\| = 0$ , which is the smallest possible value, so  $\mathbf{x}_0$  is a least-squares solution; conversely a least-squares solution  $\mathbf{x}$  has  $\|\mathbf{A} \mathbf{x} - \mathbf{b}\| \leq 0$ , hence  $\mathbf{A} \mathbf{x} = \mathbf{b}$ . Least squares extends the notion of a solution without changing it when solutions exist.
- (c) Because the target  $P_{\text{col}(\mathbf{A})} \mathbf{b}$  lies in  $\text{col}(\mathbf{A})$  by construction, so  $\mathbf{A} \mathbf{x} = P_{\text{col}(\mathbf{A})} \mathbf{b}$  is solvable (Theorem 10.4.2, existence).
- (d) The columns are linearly independent, equivalently  $\text{rank } \mathbf{A} = n$ , equivalently  $\mathbf{A}^* \mathbf{A}$  is invertible (Corollary 10.4.4).
- (e) In  $(\text{nul } \mathbf{A})^\perp = \text{col}(\mathbf{A}^*)$ . Two solutions there would differ by an element of  $\text{nul}(\mathbf{A}) \cap (\text{nul } \mathbf{A})^\perp = \{\mathbf{0}\}$  (Theorem 10.4.5).
- (f) False, and usually meaningless:  $\mathbf{A}$  need not be square, so  $\mathbf{A}^{-1}$  need not exist, while  $(\mathbf{A}^* \mathbf{A})^{-1}$  can exist perfectly well — see the  $3 \times 2$  example in the warning after Corollary 10.4.4.

### 10.4.6.2 B. Practice

#### Exercise B1: A least-squares line.

Find the least-squares line  $y = c + dt$  for the data  $(0, 3), (1, 2), (2, 3), (3, 6)$ . Compute the residual, verify that it is orthogonal to both columns of  $\mathbf{A}$ , and give the minimum sum of squares.

*Solution.* With  $\mathbf{A}$  having columns  $(1, 1, 1, 1)$  and  $(0, 1, 2, 3)$  and  $\mathbf{b} = (3, 2, 3, 6)$ ,

$$\mathbf{A}^\top \mathbf{A} = \begin{pmatrix} 4 & 6 \\ 6 & 14 \end{pmatrix}, \quad \mathbf{A}^\top \mathbf{b} = \begin{pmatrix} 14 \\ 0 + 2 + 6 + 18 \end{pmatrix} = \begin{pmatrix} 14 \\ 26 \end{pmatrix}.$$

The normal equations are  $2c + 3d = 7$  and  $3c + 7d = 13$ . Multiplying the first by 3 and the second by 2 and subtracting gives  $9d - 14d = 21 - 26$ , so  $d = 1$ , and then  $c = 2$ . The line is  $y = 2 + t$ .

Its values at  $t = 0, 1, 2, 3$  are  $2, 3, 4, 5$ , so  $\mathbf{r} = (1, -1, -1, 1)$ . Orthogonality:  $1 - 1 - 1 + 1 = 0$  against the first column and  $0 - 1 - 2 + 3 = 0$  against the second. The minimum sum of squares is  $\|\mathbf{r}\|^2 = 4$ .

#### Exercise B2: A least-squares parabola.

Find the least-squares quadratic  $y = a + bt + ct^2$  for the data  $(-1, 2), (0, 8), (1, 6), (2, 16)$ , and the minimum sum of squares. *Hint: write down  $\mathbf{A}^\top \mathbf{A}$  from the power sums  $\sum t_i^k$ ,  $k = 0, \dots, 4$ .*

*Solution.* The power sums are  $\sum 1 = 4$ ,  $\sum t_i = 2$ ,  $\sum t_i^2 = 6$ ,  $\sum t_i^3 = 8$ ,  $\sum t_i^4 = 18$ , so

$$\mathbf{A}^\top \mathbf{A} = \begin{pmatrix} 4 & 2 & 6 \\ 2 & 6 & 8 \\ 6 & 8 & 18 \end{pmatrix}, \quad \mathbf{A}^\top \mathbf{b} = \begin{pmatrix} 2 + 8 + 6 + 16 \\ -2 + 0 + 6 + 32 \\ 2 + 0 + 6 + 64 \end{pmatrix} = \begin{pmatrix} 32 \\ 36 \\ 72 \end{pmatrix}.$$

Solving: from the first two equations,  $2a + b + 3c = 16$  and  $a + 3b + 4c = 18$ ; eliminating  $a$  gives  $5b + 5c = 20$ , so  $b + c = 4$ . The third equation is  $3a + 4b + 9c = 36$ . Substituting  $b =$

$4 - c$  into  $2a + b + 3c = 16$  gives  $2a + 2c = 12$ , so  $a = 6 - c$ ; then  $3(6 - c) + 4(4 - c) + 9c = 36$  becomes  $34 + 2c = 36$ , so  $c = 1$ ,  $b = 3$ ,  $a = 5$ . The parabola is

$$y = 5 + 3t + t^2.$$

Its values at  $t = -1, 0, 1, 2$  are  $3, 5, 9, 15$ , so  $\mathbf{r} = (-1, 3, -3, 1)$  and the minimum sum of squares is  $1 + 9 + 9 + 1 = 20$ . Check:  $\sum r_i = 0$ ,  $\sum t_i r_i = 1 + 0 - 3 + 2 = 0$ ,  $\sum t_i^2 r_i = -1 + 0 - 3 + 4 = 0$ .

### Exercise B3: A minimum-norm solution.

Find the minimum-norm solution of

$$\begin{cases} x_1 + x_3 = 2, \\ x_2 + x_3 = 3, \end{cases}$$

and verify that it is orthogonal to  $\text{nul}(\mathbf{A})$ .

*Solution.* With  $x_3 = t$  the solutions are  $\mathbf{x}(t) = (2 - t, 3 - t, t) = (2, 3, 0) + t(-1, -1, 1)$ , so  $\text{nul}(\mathbf{A}) = \text{span}((-1, -1, 1))$ . By Theorem 10.4.5 we need  $\langle \mathbf{x}(t), (-1, -1, 1) \rangle = 0$ :

$$-(2 - t) - (3 - t) + t = 3t - 5 = 0, \quad t = \frac{5}{3}.$$

So  $\mathbf{x}_{\min} = (\frac{1}{3}, \frac{4}{3}, \frac{5}{3})$ , with  $\|\mathbf{x}_{\min}\|^2 = \frac{1+16+25}{9} = \frac{14}{3}$ . It solves the system, since  $\frac{1}{3} + \frac{5}{3} = 2$  and  $\frac{4}{3} + \frac{5}{3} = 3$ , and  $-\frac{1}{3} - \frac{4}{3} + \frac{5}{3} = 0$  confirms the orthogonality. For comparison,  $\|(2, 3, 0)\|^2 = 13 > \frac{14}{3}$ .

### 10.4.6.3 C. Going deeper

#### Exercise C1: Weighted least squares.

Let  $\mathbf{A} \in M_{m \times n}(\mathbb{R})$ ,  $\mathbf{b} \in \mathbb{R}^m$ , and let  $w_1, \dots, w_m > 0$  be weights, with  $\mathbf{W} = \text{diag}(w_1, \dots, w_m)$ .

- Show that  $\langle \mathbf{x}, \mathbf{y} \rangle_{\mathbf{W}} = \mathbf{y}^T \mathbf{W} \mathbf{x}$  is an inner product on  $\mathbb{R}^m$ .
- Prove that the minimizers of  $\sum_{i=1}^m w_i ((\mathbf{A}\mathbf{x})_i - b_i)^2$  are exactly the solutions of  $\mathbf{A}^T \mathbf{W} \mathbf{A} \mathbf{x} = \mathbf{A}^T \mathbf{W} \mathbf{b}$ .
- For the data  $(0, 2), (1, 1), (2, 8)$  with weights  $(1, 2, 1)$ , find the weighted least-squares line, and compare it with the unweighted one.

*Solution.*

- It is bilinear and symmetric because  $\mathbf{y}^T \mathbf{W} \mathbf{x} = \sum_i w_i x_i y_i$ , and  $\langle \mathbf{x}, \mathbf{x} \rangle_{\mathbf{W}} = \sum_i w_i x_i^2 > 0$  whenever some  $x_i \neq 0$ , because every  $w_i > 0$ . (This is where positivity of the weights is used; a zero weight would destroy definiteness.)
- Write  $\|\cdot\|_{\mathbf{W}}$  for the norm induced by  $\langle \cdot, \cdot \rangle_{\mathbf{W}}$ . The quantity to minimize is exactly  $\|\mathbf{A}\mathbf{x} - \mathbf{b}\|_{\mathbf{W}}^2$ . The proof of Theorem 10.4.2 used nothing about the inner product except its axioms, so it applies verbatim with  $\langle \cdot, \cdot \rangle_{\mathbf{W}}$  in place of the standard one: the minimizers are the  $\mathbf{x}$  for which  $\mathbf{A}\mathbf{x} - \mathbf{b}$  is  $\langle \cdot, \cdot \rangle_{\mathbf{W}}$ -orthogonal to  $\text{col}(\mathbf{A})$ , that is,

for which  $\langle \mathbf{Ax} - \mathbf{b}, \mathbf{Az} \rangle_{\mathbf{W}} = 0$  for all  $\mathbf{z} \in \mathbb{R}^n$ . Now

$$\langle \mathbf{Ax} - \mathbf{b}, \mathbf{Az} \rangle_{\mathbf{W}} = (\mathbf{Az})^\top \mathbf{W}(\mathbf{Ax} - \mathbf{b}) = \mathbf{z}^\top (\mathbf{A}^\top \mathbf{W} \mathbf{Ax} - \mathbf{A}^\top \mathbf{W} \mathbf{b}),$$

and this vanishes for all  $\mathbf{z} \in \mathbb{R}^n$  exactly when  $\mathbf{A}^\top \mathbf{W} \mathbf{Ax} - \mathbf{A}^\top \mathbf{W} \mathbf{b} = \mathbf{0}$ , by taking  $\mathbf{z}$  equal to that vector.

- (c) With  $\mathbf{A}$  having columns  $(1, 1, 1)$  and  $(0, 1, 2)$ ,  $\mathbf{b} = (2, 1, 8)$  and  $\mathbf{W} = \text{diag}(1, 2, 1)$ ,

$$\mathbf{A}^\top \mathbf{W} \mathbf{A} = \begin{pmatrix} 4 & 4 \\ 4 & 6 \end{pmatrix}, \quad \mathbf{A}^\top \mathbf{W} \mathbf{b} = \begin{pmatrix} 2 + 2 + 8 \\ 0 + 2 + 16 \end{pmatrix} = \begin{pmatrix} 12 \\ 18 \end{pmatrix},$$

so  $4c + 4d = 12$  and  $4c + 6d = 18$ , giving  $d = 3$  and  $c = 0$ : the weighted line is  $y = 3t$ . Unweighted,  $\mathbf{A}^\top \mathbf{A} = \begin{pmatrix} 3 & 3 \\ 3 & 5 \end{pmatrix}$  and  $\mathbf{A}^\top \mathbf{b} = (11, 17)$ , giving  $3c + 3d = 11$ ,  $3c + 5d = 17$ , so  $d = 3$  and  $c = \frac{2}{3}$ : the unweighted line is  $y = \frac{2}{3} + 3t$ . The middle point is trusted twice as much in the weighted fit, and the line moves down to pass closer to it. The weighted fit has weighted error  $4 + 2 \cdot 4 + 4 = 16$ , smaller than the weighted error  $\frac{160}{9}$  of the unweighted line, while the unweighted fit has plain error  $\frac{32}{3}$ , smaller than the plain error 12 of the weighted line: each line wins the contest it was designed for.

### Exercise C2: The residual, two ways.

Let  $\mathbf{x}$  be a least-squares solution of  $\mathbf{Ax} = \mathbf{b}$  and let  $\mathbf{r} = \mathbf{b} - \mathbf{Ax}$ .

- Prove that  $\|\mathbf{r}\|^2 = \|\mathbf{b}\|^2 - \|\mathbf{Ax}\|^2$ , and also that  $\|\mathbf{r}\|^2 = \langle \mathbf{b}, \mathbf{r} \rangle = \|\mathbf{b}\|^2 - \langle \mathbf{A}^* \mathbf{b}, \mathbf{x} \rangle$ .
- Use the second formula to recompute the minimum sum of squares in [Example \(The least-squares line through four points\)](#) without finding the residual vector.

*Solution.*

- By [Theorem 10.4.2](#),  $\mathbf{Ax} = P_U \mathbf{b}$  with  $U = \text{col}(\mathbf{A})$ , so  $\mathbf{r} = \mathbf{b} - P_U \mathbf{b} \in U^\perp$  while  $\mathbf{Ax} \in U$ . The two are orthogonal, and  $\mathbf{b} = \mathbf{Ax} + \mathbf{r}$ , so [Theorem 10.1.3](#) gives  $\|\mathbf{b}\|^2 = \|\mathbf{Ax}\|^2 + \|\mathbf{r}\|^2$ , which is the first identity. For the second,  $\langle \mathbf{b}, \mathbf{r} \rangle = \langle \mathbf{Ax} + \mathbf{r}, \mathbf{r} \rangle = \langle \mathbf{Ax}, \mathbf{r} \rangle + \|\mathbf{r}\|^2 = \|\mathbf{r}\|^2$ , since  $\langle \mathbf{Ax}, \mathbf{r} \rangle = \langle \mathbf{r}, \mathbf{Ax} \rangle = 0$  by orthogonality. Expanding the other way, and using [Lemma 10.4.1](#),

$$\langle \mathbf{b}, \mathbf{r} \rangle = \langle \mathbf{b}, \mathbf{b} - \mathbf{Ax} \rangle = \|\mathbf{b}\|^2 - \langle \mathbf{b}, \mathbf{Ax} \rangle = \|\mathbf{b}\|^2 - \langle \mathbf{A}^* \mathbf{b}, \mathbf{x} \rangle.$$

- In [Example \(The least-squares line through four points\)](#),  $\|\mathbf{b}\|^2 = 9 + 1 + 25 + 25 = 60$ ,  $\mathbf{A}^\top \mathbf{b} = (14, 40)$  and  $\mathbf{x} = (1, 1)$ , so

$$\|\mathbf{r}\|^2 = 60 - \langle (14, 40), (1, 1) \rangle = 60 - 54 = 6,$$

matching the value found there. Only the data already assembled for the normal equations is needed.

### Exercise C3: A rank-deficient least-squares problem.

Let

$$\mathbf{A} = \begin{pmatrix} 1 & 0 & 1 \\ 0 & 1 & 1 \\ 1 & 1 & 2 \end{pmatrix}, \quad \mathbf{b} = \begin{pmatrix} 1 \\ 2 \\ 0 \end{pmatrix}.$$

- (a) Show that  $\mathbf{Ax} = \mathbf{b}$  is inconsistent and that  $\text{rank } \mathbf{A} = 2$ .
- (b) Describe **all** least-squares solutions.
- (c) Find the one of smallest norm, and the minimum sum of squares.

*Solution.*

- (a) The third column is the sum of the first two, so the columns are dependent and  $\text{rank } \mathbf{A} \leq 2$ ; the first two columns are independent, so  $\text{rank } \mathbf{A} = 2$  and  $\text{col}(\mathbf{A}) = \text{span}((1, 0, 1), (0, 1, 1))$ . A vector  $(b_1, b_2, b_3)$  lies in  $\text{col}(\mathbf{A})$  exactly when  $b_3 = b_1 + b_2$ ; here  $0 \neq 1 + 2$ , so  $\mathbf{b} \notin \text{col}(\mathbf{A})$  and the system is inconsistent.
- (b) By [Theorem 10.4.2](#) the least-squares solutions solve  $\mathbf{A}^\top \mathbf{Ax} = \mathbf{A}^\top \mathbf{b}$ . Here

$$\mathbf{A}^\top \mathbf{A} = \begin{pmatrix} 2 & 1 & 3 \\ 1 & 2 & 3 \\ 3 & 3 & 6 \end{pmatrix}, \quad \mathbf{A}^\top \mathbf{b} = \begin{pmatrix} 1 \\ 2 \\ 3 \end{pmatrix}.$$

Row reducing, the third row is the sum of the first two, and subtracting suitable multiples gives  $x_1 + x_3 = 0$  and  $x_2 + x_3 = 1$ . Hence

$$\mathbf{x}(t) = (0, 1, 0) + t(-1, -1, 1), \quad t \in \mathbb{R},$$

a coset of  $\text{nul}(\mathbf{A}) = \text{span}((-1, -1, 1)) = \text{nul}(\mathbf{A}^\top \mathbf{A})$ , as [Lemma 10.4.3](#) predicts.

- (c) The minimizers form a coset of  $\text{nul}(\mathbf{A})$ , so the argument of [Theorem 10.4.5](#) applies to it verbatim: the shortest is the one orthogonal to  $(-1, -1, 1)$ . From  $-(-t) - (1 - t) + t = 3t - 1 = 0$  we get  $t = \frac{1}{3}$  and

$$\mathbf{x}_{\min} = \left(-\frac{1}{3}, \frac{2}{3}, \frac{1}{3}\right), \quad \|\mathbf{x}_{\min}\|^2 = \frac{1+4+1}{9} = \frac{2}{3}.$$

Every minimizer gives the same  $\mathbf{Ax} = (0, 1, 1) = P_{\text{col}(\mathbf{A})}\mathbf{b}$ , with residual  $\mathbf{r} = (1, 1, -1)$  and minimum sum of squares  $\|\mathbf{r}\|^2 = 3$ . *Check:*  $\mathbf{r}$  is orthogonal to  $(1, 0, 1)$  and to  $(0, 1, 1)$ , since  $1 - 1 = 0$  and  $1 - 1 = 0$ .

## 10.5 The Riesz Representation Theorem

Chapter 4 built the dual space  $V^*$  of all linear functionals on  $V$ , proved that  $\dim V^* = \dim V$  in finite dimension, and then delivered a warning: the resulting isomorphism  $V \cong V^*$  is not canonical. Every basis of  $V$  gives one, different bases give different ones, and no argument singles out a preferred choice ([Corollary 4.1.4](#), and the contrast with the natural map  $\text{ev}_V: V \rightarrow V^{**}$  of [Theorem 4.4.6](#)). The chapter closed by promising that an inner product would supply the missing ingredient. This section keeps the promise.

The mechanism is simple to state. Fix  $\mathbf{w} \in V$  and let  $\varphi_{\mathbf{w}}(\mathbf{v}) = \langle \mathbf{v}, \mathbf{w} \rangle$ . Because the inner product is linear in the **first** slot,  $\varphi_{\mathbf{w}}$  is a linear functional. The Riesz Representation Theorem says that in finite dimension there are no others.

Throughout,  $V$  is a finite-dimensional inner product space over  $F = \mathbb{R}$  or  $F = \mathbb{C}$ , unless the text says otherwise.

## 10.5.1 Every functional is an inner product

In  $F^n$ , [Theorem 4.1.1](#) already told us that every functional has the form  $\mathbf{x} \mapsto a_1x_1 + \cdots + a_nx_n$  for a unique row of coefficients. With the standard inner product  $\langle \mathbf{x}, \mathbf{y} \rangle = \sum_i x_i \overline{y_i}$ , that reads  $\varphi(\mathbf{x}) = \langle \mathbf{x}, \mathbf{w} \rangle$  with  $\mathbf{w} = (\overline{a_1}, \dots, \overline{a_n})$ . So in coordinates the theorem is a restatement of a fact we already have; the content is that it survives the removal of coordinates.

### Theorem 10.5.1 — Riesz Representation Theorem

Let  $V$  be a finite-dimensional inner product space over  $F$  and let  $\varphi \in V^*$ . Then there is a **unique**  $\mathbf{w} \in V$  such that

$$\varphi(\mathbf{v}) = \langle \mathbf{v}, \mathbf{w} \rangle \quad \text{for every } \mathbf{v} \in V.$$

If  $(\mathbf{e}_1, \dots, \mathbf{e}_n)$  is any orthonormal basis of  $V$ , then  $\mathbf{w} = \sum_{i=1}^n \overline{\varphi(\mathbf{e}_i)} \mathbf{e}_i$ .

*Idea.* Fix an orthonormal basis. Then a linear functional is determined by the  $n$  numbers  $\varphi(\mathbf{e}_i)$ , and a vector is determined by its  $n$  coordinates, which [Theorem 10.2.1](#) reads off as inner products. Matching the two lists tells us what  $\mathbf{w}$  must be, and the only care needed is the conjugate:  $\mathbf{w}$  sits in the second slot, so its coordinates come out conjugated. Uniqueness is the standard move — if two vectors pair identically with everything, subtract and pair the difference with itself.

*Proof. Existence.* Let  $(\mathbf{e}_1, \dots, \mathbf{e}_n)$  be an orthonormal basis of  $V$ , which exists by [Theorem 10.2.2](#), and put

$$\mathbf{w} = \sum_{i=1}^n \overline{\varphi(\mathbf{e}_i)} \mathbf{e}_i.$$

For each  $j$ , conjugate-linearity in the second slot and  $\langle \mathbf{e}_j, \mathbf{e}_i \rangle = \delta_{ij}$  give

$$\langle \mathbf{e}_j, \mathbf{w} \rangle = \sum_{i=1}^n \overline{\overline{\varphi(\mathbf{e}_i)}} \langle \mathbf{e}_j, \mathbf{e}_i \rangle = \varphi(\mathbf{e}_j).$$

So the two linear functionals  $\mathbf{v} \mapsto \langle \mathbf{v}, \mathbf{w} \rangle$  and  $\varphi$  agree on a basis, hence are equal by [Theorem 3.1.5](#). (The first is linear because the inner product is linear in the first slot.)

*Uniqueness.* Suppose  $\langle \mathbf{v}, \mathbf{w} \rangle = \langle \mathbf{v}, \mathbf{w}' \rangle$  for every  $\mathbf{v}$ . Then  $\langle \mathbf{v}, \mathbf{w} - \mathbf{w}' \rangle = 0$  for every  $\mathbf{v}$ , by additivity in the second slot. Taking  $\mathbf{v} = \mathbf{w} - \mathbf{w}'$  gives  $\|\mathbf{w} - \mathbf{w}'\|^2 = 0$ , so  $\mathbf{w} = \mathbf{w}'$  by positive definiteness. ■

Call  $\mathbf{w}$  the **Riesz vector** of  $\varphi$ . The formula is worth reading twice: the coordinates of  $\mathbf{w}$  in an orthonormal basis are the conjugates of the values of  $\varphi$  on that basis. Over  $\mathbb{R}$  the conjugation is invisible, which is why real textbooks can omit it and complex ones cannot.

**Example. (Two Riesz vectors)**

- (a) On  $V = \mathbb{R}[x]_{\leq 1}$  with  $\langle p, q \rangle = \int_0^1 p(t)q(t) dt$ , find the Riesz vector of  $\varphi(p) = p(0)$ .
- (b) On  $V = M_n(\mathbb{C})$  with the Frobenius inner product  $\langle \mathbf{A}, \mathbf{B} \rangle = \text{tr}(\mathbf{B}^* \mathbf{A})$ , find the Riesz vector of  $\varphi(\mathbf{A}) = \text{tr } \mathbf{A}$ .

*Solution.*

- (a) We look for  $w = a + bx$  with  $\langle p, w \rangle = p(0)$  for all  $p$ . By linearity it is enough to impose this on the basis  $(1, x)$ :

$$\int_0^1 (a + bt) dt = a + \frac{b}{2} = 1, \quad \int_0^1 t(a + bt) dt = \frac{a}{2} + \frac{b}{3} = 0.$$

The second equation gives  $a = -\frac{2}{3}b$ ; substituting into the first gives  $-\frac{2}{3}b + \frac{1}{2}b = -\frac{1}{6}b = 1$ , so  $b = -6$  and  $a = 4$ . Hence  $w = 4 - 6x$ .

*Check.*  $\int_0^1 (4 - 6t) dt = 4 - 3 = 1$  and  $\int_0^1 t(4 - 6t) dt = 2 - 2 = 0$ , as required; and for a general  $p = c + dx$  the pairing gives  $c \cdot 1 + d \cdot 0 = c = p(0)$ . Note how little  $w$  resembles “evaluation at 0”: the Riesz vector is manufactured by the inner product, not by the point 0.

- (b) We look for  $\mathbf{W}$  with  $\text{tr}(\mathbf{W}^* \mathbf{A}) = \text{tr } \mathbf{A}$  for every  $\mathbf{A}$ . Taking  $\mathbf{W} = \mathbf{I}_n$  works, since  $\mathbf{I}_n^* = \mathbf{I}_n$  and  $\text{tr}(\mathbf{I}_n \mathbf{A}) = \text{tr } \mathbf{A}$ . By uniqueness,  $\mathbf{W} = \mathbf{I}_n$  is the answer. (This example was not solved by the basis formula, but by guessing and then invoking uniqueness — a legitimate and often faster route.)

## 10.5.2 The Riesz map, and the data Chapter 4 was missing

Uniqueness in [Theorem 10.5.1](#) lets us read the theorem as a statement about a map.

### Corollary 10.5.2 — The Riesz map

Let  $V$  be a finite-dimensional inner product space over  $F$ . Define

$$\Phi: V \rightarrow V^*, \quad \Phi(\mathbf{w}) = \langle \cdot, \mathbf{w} \rangle.$$

Then  $\Phi$  is a bijection, and for all  $\mathbf{w}, \mathbf{w}' \in V$  and  $c \in F$ ,

$$\Phi(\mathbf{w} + \mathbf{w}') = \Phi(\mathbf{w}) + \Phi(\mathbf{w}'), \quad \Phi(c\mathbf{w}) = \bar{c}\Phi(\mathbf{w}).$$

In particular  $\Phi$  is an isomorphism of vector spaces when  $F = \mathbb{R}$ , and is **conjugate-linear** when  $F = \mathbb{C}$ .

*Proof.* Each  $\Phi(\mathbf{w})$  lies in  $V^*$  because the inner product is linear in the first slot. Surjectivity is the existence half of [Theorem 10.5.1](#) and injectivity is the uniqueness half (if  $\Phi(\mathbf{w}) = \Phi(\mathbf{w}')$ , then  $\mathbf{w} = \mathbf{w}'$ ). For the two rules, evaluate at an arbitrary  $\mathbf{v}$ : additivity in the second slot gives  $\langle \mathbf{v}, \mathbf{w} + \mathbf{w}' \rangle = \langle \mathbf{v}, \mathbf{w} \rangle + \langle \mathbf{v}, \mathbf{w}' \rangle$ , and conjugate-homogeneity in the second slot gives  $\langle \mathbf{v}, c\mathbf{w} \rangle = \bar{c}\langle \mathbf{v}, \mathbf{w} \rangle$ . Over  $\mathbb{R}$  we have  $\bar{c} = c$ , so  $\Phi$  is linear, and a

linear bijection is an isomorphism (Theorem 0.4.7). ■

This is the payoff Chapter 4 promised. There,  $V \cong V^*$  was proved by choosing a basis  $\mathcal{B}$  and sending  $\mathbf{v}_i \mapsto \varphi_i$ , and the warning was that the isomorphism changes when  $\mathcal{B}$  does: a vector does not come with “its” functional. Here  $\Phi$  is written down with no basis at all. It is not free of choices — it depends on  $\langle \cdot, \cdot \rangle$  — but the choice has been made once, in advance, and it is a choice of *geometry* rather than of coordinates. **The inner product is exactly the extra data that turns  $V \cong V^*$  from a counting fact into a formula.**

The two descriptions fit together neatly. Let  $\mathcal{B} = (\mathbf{v}_1, \dots, \mathbf{v}_n)$  be an **orthonormal** basis of  $V$ , with dual basis  $(\varphi_1, \dots, \varphi_n)$  (Theorem 4.1.3). Then  $\Phi(\mathbf{v}_i)$  is the functional  $\mathbf{v} \mapsto \langle \mathbf{v}, \mathbf{v}_i \rangle$ , which sends  $\mathbf{v}_j$  to  $\delta_{ij}$ ; that is,

$$\Phi(\mathbf{v}_i) = \varphi_i \quad (i = 1, \dots, n),$$

over either field. Over  $\mathbb{R}$ , where  $\Phi$  is linear, this says that  $\Phi$  is Chapter 4’s basis-dependent isomorphism  $\mathbf{v}_i \mapsto \varphi_i$ , for the inner product that declares  $\mathcal{B}$  orthonormal. Choosing a basis and choosing an inner product are two ways of supplying the same missing data.

*Quick check.* Let  $\mathcal{B} = (\mathbf{v}_1, \dots, \mathbf{v}_n)$  be an orthonormal basis of a **complex** inner product space and let  $(\varphi_1, \dots, \varphi_n)$  be its dual basis. What is the Riesz vector of  $i\varphi_1$ ?

*Solution.* By the identity  $\Phi(\mathbf{v}_i) = \varphi_i$  above, which holds over  $\mathbb{C}$  as well,  $\varphi_1$  has Riesz vector  $\mathbf{v}_1$ . For  $i\varphi_1$ , use conjugate-homogeneity in reverse:  $\Phi(-i\mathbf{v}_1) = \overline{(-i)}\Phi(\mathbf{v}_1) = i\varphi_1$ . So the Riesz vector of  $i\varphi_1$  is  $-i\mathbf{v}_1$ , not  $i\mathbf{v}_1$ . Alternatively, apply the formula of Theorem 10.5.1: the coordinates of the Riesz vector are  $\overline{(i\varphi_1)}(\mathbf{v}_j) = i\overline{\delta_{1j}} = -i\delta_{1j}$ .

**Warning.** Over  $\mathbb{C}$ , the Riesz map is not linear, so it is not an isomorphism of complex vector spaces. It is a bijection, and it is linear over  $\mathbb{R}$ , but  $\Phi(i\mathbf{w}) = -i\Phi(\mathbf{w})$ . Complex spaces  $V$  and  $V^*$  are still isomorphic — they have the same dimension — but the isomorphism produced by an inner product is the conjugate-linear  $\Phi$ . Most of what we do with  $\Phi$  (Section 6 builds the adjoint from it) is unaffected, because the conjugation cancels when  $\Phi$  is used twice.

**Warning.** The Riesz vector depends on the inner product, not on  $\varphi$  alone. On  $\mathbb{R}^2$  let  $\varphi(\mathbf{x}) = x_1 + x_2$ . With the dot product,  $\varphi(\mathbf{x}) = \langle \mathbf{x}, (1, 1) \rangle$ , so  $\mathbf{w} = (1, 1)$ . With the weighted inner product  $\langle \mathbf{x}, \mathbf{y} \rangle = 2x_1y_1 + x_2y_2$ , we need  $2w_1 = 1$  and  $w_2 = 1$ , so  $\mathbf{w} = (\frac{1}{2}, 1)$ . Same functional, different representing vector. Whenever two inner products are in play, the notation must say which  $\Phi$  is meant.

### 10.5.3 Annihilators become orthogonal complements

Section 3 warned that the annihilator  $U^0 \subseteq V^*$  and the orthogonal complement  $U^\perp \subseteq V$  are different objects that happen to have the same dimension. The Riesz map explains the coincidence: it carries one onto the other.

#### Theorem 10.5.3 — Perp corresponds to the annihilator

Let  $V$  be a finite-dimensional inner product space over  $F$ , let  $U$  be a subspace and let

$\Phi$  be the Riesz map. Then

$$\Phi(U^\perp) = U^0.$$

Equivalently, a functional  $\varphi \in V^*$  annihilates  $U$  if and only if its Riesz vector lies in  $U^\perp$ .

*Proof.* Let  $\mathbf{w} \in V$ . Then

$$\Phi(\mathbf{w}) \in U^0 \iff \langle \mathbf{u}, \mathbf{w} \rangle = 0 \text{ for every } \mathbf{u} \in U \iff \mathbf{w} \in U^\perp,$$

the first equivalence being the definition of the annihilator (Definition 4.2.1) applied to the functional  $\Phi(\mathbf{w}) = \langle \cdot, \mathbf{w} \rangle$ , and the second the definition of the orthogonal complement (Definition 10.3.1). Since  $\Phi$  is a bijection (Corollary 10.5.2), the set  $\Phi(U^\perp)$  is therefore exactly  $U^0$ . ■

The dimension count now has a reason rather than a coincidence:  $\dim U^0 = \dim V - \dim U$  (Theorem 4.2.2) and  $\dim U^\perp = \dim V - \dim U$  (Theorem 10.3.2 (c)) agree because the two spaces correspond under a bijection. (Over  $\mathbb{C}$  the bijection is only conjugate-linear, but a conjugate-linear bijection still carries subspaces to subspaces of the same dimension, since  $c\Phi(\mathbf{w}) = \Phi(\bar{c}\mathbf{w})$ .)

Here is the correspondence in  $\mathbb{R}^3$ . Let  $U = \text{span}((1, 2, 0))$  with the dot product. Then  $U^\perp = \{(a, b, c) : a + 2b = 0\} = \text{span}((2, -1, 0), (0, 0, 1))$ , and  $U^0$  consists of the functionals  $\mathbf{x} \mapsto \alpha x_1 + \beta x_2 + \gamma x_3$  with  $\alpha + 2\beta = 0$ , spanned by  $2x_1 - x_2$  and  $x_3$ . These are precisely  $\Phi((2, -1, 0))$  and  $\Phi((0, 0, 1))$ .

Section 6 uses Theorem 10.5.3 in the same way for maps: the dual map  $T' : W^* \rightarrow V^*$  of Chapter 4 and the adjoint  $T^* : W \rightarrow V$  are the same map, read through the two Riesz maps.

## 10.5.4 What happens in infinite dimension

The proof used an orthonormal basis of the whole of  $V$ , so finite dimension was essential, and the theorem genuinely fails without it.

### Example. (A functional with no Riesz vector)

Let  $V = \mathbb{R}[x]$  with the coefficient inner product  $\langle p, q \rangle = \sum_{k \geq 0} a_k b_k$  for  $p = \sum_k a_k x^k$  and  $q = \sum_k b_k x^k$  (a finite sum, and an inner product by Exercise (C2: A subspace with no orthogonal complement) (a)). Show that  $\varphi(p) = p(1)$  has no Riesz vector.

*Solution.* Suppose  $w = \sum_k b_k x^k$  satisfied  $\langle p, w \rangle = p(1)$  for every  $p \in V$ . Taking  $p = x^i$  gives

$$b_i = \langle x^i, w \rangle = \varphi(x^i) = 1^i = 1 \quad \text{for every } i \geq 0.$$

So every coefficient of  $w$  equals 1. But a polynomial has only finitely many non-zero coefficients, so no such  $w$  exists.

What fails is not “infinite dimension” as a slogan. Two things are missing at once, and the same example shows both. First,  $V$  has gaps: Exercise (C2: A subspace with no orthogonal complement) exhibits polynomials  $p_m \in \ker \varphi$  with  $\|p_m - 1\| = 1/\sqrt{m}$ , so the vectors  $p_m$  crowd around the polynomial 1 without any limit being forced to lie in  $\ker \varphi$ . Second,  $\varphi$  is wildly



discontinuous with respect to  $\|\cdot\|$ : along the same sequence,  $\varphi(p_m) = 0$  for every  $m$ , while  $\varphi(1) = 1$ .

Repairing both defects — completing the space, and restricting attention to functionals  $\varphi$  with  $|\varphi(\mathbf{v})| \leq C\|\mathbf{v}\|$  — restores the theorem. That statement is the Riesz representation theorem for Hilbert spaces, and it belongs to analysis, not to linear algebra; we record its existence and go no further. What survives here with no hypotheses at all is the easy direction: for every  $\mathbf{w} \in V$ , the map  $\langle \cdot, \mathbf{w} \rangle$  is a linear functional. Only the converse needs finite dimension.

## 10.5.5 Exercises

### 10.5.5.1 A. Check your understanding

#### Exercise A1.

- State the Riesz Representation Theorem, including every hypothesis.
- Write down the formula for the Riesz vector of  $\varphi$  in terms of an orthonormal basis, and say where the conjugate comes from.
- True or false: over  $\mathbb{C}$ , the Riesz map  $\Phi: V \rightarrow V^*$  is a linear isomorphism. Justify your answer.
- In what sense does an inner product answer the question Chapter 4 left open about  $V \cong V^*$ ?
- Under  $\Phi$ , which subspace of  $V^*$  corresponds to  $U^\perp$ ?
- Give an example showing that the theorem fails in infinite dimension.

*Solution.*

- If  $V$  is a **finite-dimensional** inner product space over  $\mathbb{R}$  or  $\mathbb{C}$  and  $\varphi \in V^*$ , then there is a unique  $\mathbf{w} \in V$  with  $\varphi(\mathbf{v}) = \langle \mathbf{v}, \mathbf{w} \rangle$  for all  $\mathbf{v} \in V$  ([Theorem 10.5.1](#)).
- $\mathbf{w} = \sum_i \overline{\varphi(\mathbf{e}_i)} \mathbf{e}_i$  for any orthonormal basis  $(\mathbf{e}_1, \dots, \mathbf{e}_n)$ . The conjugate appears because  $\mathbf{w}$  occupies the second slot, where scalars come out conjugated: to make  $\langle \mathbf{e}_j, \mathbf{w} \rangle$  equal  $\varphi(\mathbf{e}_j)$ , the coefficient of  $\mathbf{e}_j$  in  $\mathbf{w}$  must be  $\overline{\varphi(\mathbf{e}_j)}$ .
- False.  $\Phi$  is a bijection, but  $\Phi(c\mathbf{w}) = \bar{c}\Phi(\mathbf{w})$ , so it is conjugate-linear and not  $\mathbb{C}$ -linear ([Corollary 10.5.2](#)). It is linear over  $\mathbb{R}$ , and over  $\mathbb{R}$  it is an isomorphism.
- Chapter 4 showed  $V \cong V^*$  by counting dimensions, with an isomorphism depending on a chosen basis and no preferred choice available ([Corollary 4.1.4](#)). An inner product supplies a specific map  $\Phi(\mathbf{w}) = \langle \cdot, \mathbf{w} \rangle$ , written without reference to any basis. The inner product is the extra data; picking an orthonormal basis and picking an inner product that makes a given basis orthonormal are the same act.
- $U^0$ , the annihilator of  $U$  ([Theorem 10.5.3](#)).
- On  $\mathbb{R}[x]$  with the coefficient inner product,  $\varphi(p) = p(1)$  has no Riesz vector: such a vector would need every coefficient equal to 1 ([Example \(A functional with no Riesz vector\)](#)).

### 10.5.5.2 B. Practice

**Exercise B1: Riesz vectors for evaluation.**

On  $V = \mathbb{R}[x]_{\leq 2}$  use the node inner product  $\langle p, q \rangle = p(-1)q(-1) + p(0)q(0) + p(1)q(1)$ . Find the Riesz vectors of  $\varphi(p) = p(0)$  and of  $\psi(p) = p(1)$ .

*Solution.* Since  $-1, 0, 1$  are three distinct nodes and  $\dim V = 3$ , a polynomial of  $V$  is determined by its values there (Theorem 2.7.3). Writing  $w$  for a candidate Riesz vector of  $\varphi$ , the requirement  $\langle p, w \rangle = p(0)$  reads

$$p(-1)w(-1) + p(0)w(0) + p(1)w(1) = p(0) \quad \text{for all } p \in V.$$

Taking for  $p$  the three Lagrange polynomials at the nodes, which have value 1 at one node and 0 at the others (Definition 5.6.1), the three equations become  $w(-1) = 0$ ,  $w(0) = 1$ ,  $w(1) = 0$ . The unique quadratic with those values is  $w = 1 - x^2$ .

For  $\psi$  the same argument gives  $w(-1) = 0$ ,  $w(0) = 0$ ,  $w(1) = 1$ , whose interpolant is  $w = \frac{1}{2}x(x+1)$ .

*Check.* For  $p = 3 + 2x - x^2$  we get  $p(-1) = 0$ ,  $p(0) = 3$ ,  $p(1) = 4$ . Pairing with  $1 - x^2$ , whose values are 0, 1, 0, gives  $3 = p(0)$ ; pairing with  $\frac{1}{2}x(x+1)$ , whose values are 0, 0, 1, gives  $4 = p(1)$ .

**Exercise B2: Riesz vectors on matrices.**

On  $M_n(\mathbb{C})$  with the Frobenius inner product  $\langle \mathbf{A}, \mathbf{B} \rangle = \text{tr}(\mathbf{B}^* \mathbf{A})$ , find the Riesz vector of:

- (a)  $\varphi(\mathbf{A}) = a_{12}$  (assume  $n \geq 2$ );
- (b)  $\varphi_{\mathbf{C}}(\mathbf{A}) = \text{tr}(\mathbf{C}\mathbf{A})$ , for a fixed  $\mathbf{C} \in M_n(\mathbb{C})$ .

*Solution.*

- (a) Let  $\mathbf{E}_{12}$  be the matrix unit with a 1 in position  $(1, 2)$  and zeros elsewhere. Then  $\mathbf{E}_{12}^* = \mathbf{E}_{21}$ , and  $(\mathbf{E}_{21}\mathbf{A})_{kk} = \sum_j (\mathbf{E}_{21})_{kj} a_{jk}$ , which is 0 unless  $k = 2$ , where it equals  $a_{12}$ . Hence  $\text{tr}(\mathbf{E}_{12}^* \mathbf{A}) = a_{12} = \varphi(\mathbf{A})$ , and by uniqueness in Theorem 10.5.1 the Riesz vector is  $\mathbf{E}_{12}$ .
- (b) We need  $\mathbf{W}$  with  $\text{tr}(\mathbf{W}^* \mathbf{A}) = \text{tr}(\mathbf{C}\mathbf{A})$  for all  $\mathbf{A}$ . Taking  $\mathbf{W}^* = \mathbf{C}$ , that is  $\mathbf{W} = \mathbf{C}^*$ , works. By uniqueness it is the only answer. Part (a) is the case  $\mathbf{C} = \mathbf{E}_{21}$ , and the example in the text is the case  $\mathbf{C} = \mathbf{I}_n$ .

**Exercise B3: The correspondence in  $\mathbb{R}^4$ .**

In  $\mathbb{R}^4$  with the dot product, let  $U = \text{span}((1, 0, 1, 0), (0, 1, 0, 1))$ . Find a basis of  $U^\perp$  and a basis of  $U^0$ , and verify that  $\Phi$  carries the first to the second.

*Solution.* A vector  $(a, b, c, d)$  lies in  $U^\perp$  exactly when  $a + c = 0$  and  $b + d = 0$  (Proposition 10.3.1, testing against the two spanning vectors). So

$$U^\perp = \text{span}((1, 0, -1, 0), (0, 1, 0, -1)),$$

of dimension  $2 = 4 - 2$ .

A functional  $\varphi(\mathbf{x}) = \alpha x_1 + \beta x_2 + \gamma x_3 + \delta x_4$  lies in  $U^0$  exactly when it kills both spanning vectors, that is, when  $\alpha + \gamma = 0$  and  $\beta + \delta = 0$ . So  $U^0 = \text{span}(\varphi_1, \varphi_2)$  with

$$\varphi_1(\mathbf{x}) = x_1 - x_3, \quad \varphi_2(\mathbf{x}) = x_2 - x_4.$$

Over  $\mathbb{R}$ ,  $\Phi(\mathbf{w})(\mathbf{x}) = \langle \mathbf{x}, \mathbf{w} \rangle = \sum_i x_i w_i$ , so  $\Phi((1, 0, -1, 0)) = \varphi_1$  and  $\Phi((0, 1, 0, -1)) = \varphi_2$ . Since  $\Phi$  is linear over  $\mathbb{R}$  and bijective, it carries the span of the two vectors onto the span of the two functionals, confirming [Theorem 10.5.3](#).

### 10.5.5.3 C. Going deeper

#### Exercise C1: Riesz without an orthonormal basis.

Let  $V$  be a finite-dimensional inner product space and let  $\varphi \in V^*$  be non-zero. Write  $K = \ker \varphi$ .

- (a) Prove that  $\dim K^\perp = 1$ .
- (b) Let  $\mathbf{u}$  span  $K^\perp$  with  $\|\mathbf{u}\| = 1$ . Prove that  $\mathbf{w} = \overline{\varphi(\mathbf{u})} \mathbf{u}$  satisfies  $\varphi(\mathbf{v}) = \langle \mathbf{v}, \mathbf{w} \rangle$  for all  $\mathbf{v} \in V$ .
- (c) Deduce [Theorem 10.5.1](#), including the case  $\varphi = 0$ .

*Hint for (b): split  $\mathbf{v}$  using  $V = K \oplus K^\perp$ .*

*Solution.*

- (a) Since  $\varphi \neq 0$ , its image is a non-zero subspace of the 1-dimensional space  $F$ , so  $\text{im } \varphi = F$  and  $\text{rank } \varphi = 1$ . By [Theorem 3.3.1](#),  $\dim K = \dim V - 1$ , and by [Theorem 10.3.2](#) (c),  $\dim K^\perp = \dim V - \dim K = 1$ .
- (b) By [Theorem 10.3.2](#) (a),  $V = K \oplus K^\perp$ , so each  $\mathbf{v} \in V$  is  $\mathbf{v} = \mathbf{k} + c\mathbf{u}$  with  $\mathbf{k} \in K$  and  $c \in F$ . Then  $\varphi(\mathbf{v}) = \varphi(\mathbf{k}) + c\varphi(\mathbf{u}) = c\varphi(\mathbf{u})$ . On the other side,

$$\langle \mathbf{v}, \mathbf{w} \rangle = \langle \mathbf{k} + c\mathbf{u}, \overline{\varphi(\mathbf{u})} \mathbf{u} \rangle = \varphi(\mathbf{u})(\langle \mathbf{k}, \mathbf{u} \rangle + c\langle \mathbf{u}, \mathbf{u} \rangle) = \varphi(\mathbf{u})c,$$

using conjugate-homogeneity in the second slot (which conjugates  $\overline{\varphi(\mathbf{u})}$  back to  $\varphi(\mathbf{u})$ ), then  $\langle \mathbf{k}, \mathbf{u} \rangle = 0$  because  $\mathbf{u} \in K^\perp$  and  $\mathbf{k} \in K$ , and  $\langle \mathbf{u}, \mathbf{u} \rangle = \|\mathbf{u}\|^2 = 1$ . The two sides agree.

- (c) If  $\varphi = 0$ , then  $\mathbf{w} = \mathbf{0}$  represents it. If  $\varphi \neq 0$ , part (a) supplies a unit vector  $\mathbf{u}$  spanning the line  $K^\perp$  — any non-zero vector of that line divided by its norm — and part (b) supplies  $\mathbf{w}$ . Uniqueness is unchanged: if  $\langle \mathbf{v}, \mathbf{w} \rangle = \langle \mathbf{v}, \mathbf{w}' \rangle$  for all  $\mathbf{v}$ , take  $\mathbf{v} = \mathbf{w} - \mathbf{w}'$  to get  $\|\mathbf{w} - \mathbf{w}'\|^2 = 0$ . Note that this proof never mentions a basis of  $V$ ; it uses only the orthogonal decomposition along the hyperplane  $\ker \varphi$ .

#### Exercise C2: Evaluation at 0 is not represented.

Let  $V = \mathbb{R}[x]$  with  $\langle p, q \rangle = \int_0^1 p(t)q(t) dt$ , and let  $\varphi(p) = p(0)$ . Suppose, for a contradiction, that  $w = \sum_{j=0}^d c_j x^j$  satisfied  $\langle p, w \rangle = p(0)$  for every  $p \in V$ .

- (a) Show that  $\sum_{j=0}^d \frac{c_j}{k+j+1} = 0$  for every integer  $k \geq 1$ , and that the same sum equals 1 for  $k = 0$ .
- (b) Let  $N(s) = \sum_{j=0}^d c_j \prod_{i \neq j} (s+i+1)$ , the product running over  $i \in \{0, \dots, d\} \setminus \{j\}$ . Show that  $N(k) = 0$  for every integer  $k \geq 1$ , and deduce that  $N$  is the zero polynomial.
- (c) Deduce that every  $c_j = 0$ , and hence a contradiction.

*Solution.*

- (a) Take  $p = x^k$ . Then  $p(0) = 0$  for  $k \geq 1$  and  $p(0) = 1$  for  $k = 0$ , while

$$\langle x^k, w \rangle = \int_0^1 t^k \sum_{j=0}^d c_j t^j dt = \sum_{j=0}^d c_j \int_0^1 t^{k+j} dt = \sum_{j=0}^d \frac{c_j}{k+j+1}.$$

- (b) For an integer  $k \geq 1$ , multiply the identity of (a) by  $\prod_{i=0}^d (k+i+1)$ , a non-zero number. Each term  $c_j/(k+j+1)$  becomes  $c_j \prod_{i \neq j} (k+i+1)$ , so the result is exactly  $N(k) = 0$ . Thus the polynomial  $N$ , of degree at most  $d$ , has the infinitely many distinct roots  $1, 2, 3, \dots$ . A non-zero polynomial of degree  $d$  has at most  $d$  distinct roots ([Corollary 5.1.4](#)), so  $N = 0$ .
- (c) Evaluate  $N$  at  $s = -(j_0 + 1)$  for a fixed  $j_0 \in \{0, \dots, d\}$ . Every product  $\prod_{i \neq j} (s+i+1)$  with  $j \neq j_0$  contains the factor  $s+j_0+1 = 0$  and so vanishes; the term  $j = j_0$  contributes  $c_{j_0} \prod_{i \neq j_0} (i-j_0)$ , and that product is non-zero because the factors  $i-j_0$  are non-zero integers. Since  $N = 0$ , we get  $c_{j_0} = 0$ . This holds for every  $j_0$ , so  $w = 0$ . But then the case  $k = 0$  of (a) reads  $0 = 1$ , a contradiction. Hence no Riesz vector exists.

The moral matches [Example \(A functional with no Riesz vector\)](#): point evaluation is a perfectly good linear functional, but on an infinite-dimensional space it is not represented by any vector, whichever of these two inner products we use.

## 10.6 The Adjoint

Chapter 4 attached to every linear map  $T: V \rightarrow W$  a dual map  $T': W^* \rightarrow V^*$  running backwards between the dual spaces ([Definition 4.4.1](#)). An inner product puts a copy of the dual space back inside the space itself, by the Riesz correspondence of [Theorem 10.5.1](#). Composing the two turns the backward map on functionals into a backward map on vectors,  $W \rightarrow V$ . That map is the **adjoint**  $T^*$ , and it is the single most useful construction in this chapter. This section builds it, computes its matrix, uses it to split both  $V$  and  $W$  into orthogonal pieces, and names the two classes of operators it singles out.

Throughout,  $F$  is  $\mathbb{R}$  or  $\mathbb{C}$ , every inner product is linear in the first slot and conjugate-linear in the second, and every inner product space is finite-dimensional over  $F$  unless we say otherwise.

## 10.6.1 Paying to cross the inner product

Here is the situation the adjoint answers. We have a linear map  $T$  and an inner product, and an expression like  $\langle T\mathbf{v}, \mathbf{w} \rangle$  in which  $T$  sits in the first slot. Very often the argument needs  $T$  out of the way: we know something about  $\mathbf{w}$ , or about the second slot, and the  $T$  is in the wrong place. Can we move it across?

For matrices the answer is already on the table. [Lemma 10.4.1](#), proved in Section 4, says that for  $\mathbf{A} \in M_{m \times n}(F)$ ,  $\mathbf{x} \in F^n$  and  $\mathbf{y} \in F^m$ ,

$$\langle \mathbf{A}\mathbf{x}, \mathbf{y} \rangle = \langle \mathbf{x}, \mathbf{A}^*\mathbf{y} \rangle,$$

with  $\mathbf{A}^*$  the conjugate transpose of [Definition 10.1.2](#). The one-line reason is worth repeating, because it is the model for the whole section: writing  $\mathbf{y}^*$  for the row vector  $\bar{\mathbf{y}}^\top$ , the standard inner product is the matrix product  $\langle \mathbf{x}, \mathbf{y} \rangle = \mathbf{y}^*\mathbf{x}$ , so  $\langle \mathbf{A}\mathbf{x}, \mathbf{y} \rangle = (\mathbf{y}^*\mathbf{A})\mathbf{x} = (\mathbf{A}^*\mathbf{y})^*\mathbf{x} = \langle \mathbf{x}, \mathbf{A}^*\mathbf{y} \rangle$ . So a matrix does cross the inner product, and the price is a conjugate transpose. Over  $\mathbb{R}$  the price is a transpose; over  $\mathbb{C}$  a transpose and a conjugation.

*To move a map from the first slot to the second slot, you pay with a new map going the other way.*

That statement is about matrices and the standard inner product. The next theorem says the same map exists for every linear map between finite-dimensional inner product spaces, with no coordinates at all. Riesz does the work: the expression  $\langle T\mathbf{v}, \mathbf{w} \rangle$ , read as a function of  $\mathbf{v}$  with  $\mathbf{w}$  frozen, is a linear functional on  $V$ , and [Theorem 10.5.1](#) says that every such functional is pairing against one fixed vector.

### Theorem 10.6.1 — Existence and Uniqueness of the Adjoint

Let  $V$  and  $W$  be finite-dimensional inner product spaces over  $F$ , and let  $T \in \mathcal{L}(V, W)$ . Then there is exactly one map  $T^*: W \rightarrow V$  such that

$$\langle T\mathbf{v}, \mathbf{w} \rangle_W = \langle \mathbf{v}, T^*\mathbf{w} \rangle_V \quad \text{for all } \mathbf{v} \in V, \mathbf{w} \in W,$$

and this  $T^*$  is linear.

*Idea.* Freeze  $\mathbf{w}$ . Then  $\mathbf{v} \mapsto \langle T\mathbf{v}, \mathbf{w} \rangle$  is a functional on  $V$ , so Riesz hands us one vector that represents it; call it  $T^*\mathbf{w}$ . That is the whole of existence, and it also gives uniqueness, since the Riesz vector is unique. Linearity is not part of the construction and has to be checked, and the way to check an identity between vectors is the signature move of this chapter: show the difference pairs to zero with everything, then pair it with itself.

*Proof. Existence.* Fix  $\mathbf{w} \in W$  and define  $\varphi_{\mathbf{w}}: V \rightarrow F$  by  $\varphi_{\mathbf{w}}(\mathbf{v}) := \langle T\mathbf{v}, \mathbf{w} \rangle$ . It is linear: for  $\mathbf{v}_1, \mathbf{v}_2 \in V$  and  $c \in F$ ,

$$\varphi_{\mathbf{w}}(\mathbf{v}_1 + c\mathbf{v}_2) = \langle T\mathbf{v}_1 + cT\mathbf{v}_2, \mathbf{w} \rangle = \langle T\mathbf{v}_1, \mathbf{w} \rangle + c\langle T\mathbf{v}_2, \mathbf{w} \rangle,$$

using the linearity of  $T$  and linearity in the **first** slot ([Definition 10.1.1](#)). Since  $V$  is finite-dimensional, [Theorem 10.5.1](#) gives a unique  $\mathbf{u} \in V$  with  $\varphi_{\mathbf{w}}(\mathbf{v}) = \langle \mathbf{v}, \mathbf{u} \rangle$  for all  $\mathbf{v} \in V$ . Define  $T^*\mathbf{w} := \mathbf{u}$ . By construction  $\langle T\mathbf{v}, \mathbf{w} \rangle = \langle \mathbf{v}, T^*\mathbf{w} \rangle$  for all  $\mathbf{v}$ , which is the required identity.

**Uniqueness.** Suppose  $S: W \rightarrow V$  also satisfies  $\langle T\mathbf{v}, \mathbf{w} \rangle = \langle \mathbf{v}, S\mathbf{w} \rangle$  for all  $\mathbf{v}, \mathbf{w}$ . Fix  $\mathbf{w}$ .

Then  $\langle \mathbf{v}, T^*\mathbf{w} - S\mathbf{w} \rangle = 0$  for every  $\mathbf{v} \in V$ , by conjugate-additivity in the second slot. Taking  $\mathbf{v} = T^*\mathbf{w} - S\mathbf{w}$  gives  $\|T^*\mathbf{w} - S\mathbf{w}\|^2 = 0$ , so  $T^*\mathbf{w} = S\mathbf{w}$  by positive definiteness (Definition 10.1.1). As  $\mathbf{w}$  was arbitrary,  $S = T^*$ .

**Linearity.** Let  $\mathbf{w}_1, \mathbf{w}_2 \in W$  and  $c \in F$ . For every  $\mathbf{v} \in V$ ,

$$\begin{aligned} \langle \mathbf{v}, T^*(\mathbf{w}_1 + c\mathbf{w}_2) \rangle &= \langle T\mathbf{v}, \mathbf{w}_1 + c\mathbf{w}_2 \rangle = \langle T\mathbf{v}, \mathbf{w}_1 \rangle + \bar{c}\langle T\mathbf{v}, \mathbf{w}_2 \rangle \\ &= \langle \mathbf{v}, T^*\mathbf{w}_1 \rangle + \bar{c}\langle \mathbf{v}, T^*\mathbf{w}_2 \rangle = \langle \mathbf{v}, T^*\mathbf{w}_1 + cT^*\mathbf{w}_2 \rangle, \end{aligned}$$

where the second and the last equalities use conjugate-linearity in the second slot. Hence  $\langle \mathbf{v}, T^*(\mathbf{w}_1 + c\mathbf{w}_2) - T^*\mathbf{w}_1 - cT^*\mathbf{w}_2 \rangle = 0$  for every  $\mathbf{v}$ , and pairing the right-hand vector with itself makes it  $\mathbf{0}$ . So  $T^*$  is linear. This proves the theorem. ■

Notice where the two conjugations went. The scalar  $c$  came out of the second slot as  $\bar{c}$  and went back in as  $c$ ; the two conjugations cancel, which is exactly why  $T^*$  is linear and not conjugate-linear. Notice also that only  $V$  had to be finite-dimensional: the proof never uses a basis of  $W$ . We keep both spaces finite-dimensional because the rest of the section does need it.

## 10.6.2 The adjoint

The theorem produced an object attached to  $T$  and to the two inner products, and it is unique, so it deserves a name and a notation.

*The adjoint of  $T$  is the map you must apply on the other side of the inner product to leave the value unchanged.*

### Definition 10.6.1 — Adjoint

Let  $V$  and  $W$  be finite-dimensional inner product spaces over  $F$  and let  $T \in \mathcal{L}(V, W)$ . The **adjoint** of  $T$  is the unique linear map  $T^* \in \mathcal{L}(W, V)$  satisfying

$$\langle T\mathbf{v}, \mathbf{w} \rangle = \langle \mathbf{v}, T^*\mathbf{w} \rangle \quad \text{for all } \mathbf{v} \in V \text{ and all } \mathbf{w} \in W.$$

In words, clause by clause:

- $T^*$  goes **backwards**: its domain is the codomain of  $T$ . If  $T: V \rightarrow W$  then  $T^*: W \rightarrow V$ , like the dual map and unlike anything else we have attached to  $T$ .
- The defining equation is an identity in **two** variables, and it must hold for **all**  $\mathbf{v}$  and **all**  $\mathbf{w}$ . By linearity in the first slot and conjugate-linearity in the second, it is enough to check it when  $\mathbf{v}$  and  $\mathbf{w}$  range over bases of  $V$  and of  $W$ , and that is usually how  $T^*$  gets computed.
- The left inner product lives in  $W$ , the right one in  $V$ . They are different inner products; only the notation hides it.
- The definition mentions no basis, but it does mention the two inner products. Change an inner product and  $T^*$  changes.

### Example. (Four Adjoints)

Compute the adjoint in each case. In (a) and (b) the spaces carry the standard inner

product; in (c) and (d),  $V$  and  $W$  are arbitrary finite-dimensional inner product spaces.

- (a)  $T_{\mathbf{A}}: \mathbb{C}^3 \rightarrow \mathbb{C}^2$  for  $\mathbf{A} = \begin{pmatrix} 1+i & 0 & 2 \\ i & 1 & -i \end{pmatrix}$ .
- (b) The forward shift  $S: F^3 \rightarrow F^3$ ,  $S(x_1, x_2, x_3) = (0, x_1, x_2)$ .
- (c)  $\text{id}_V$  and the zero map  $V \rightarrow W$ .
- (d) The orthogonal projection  $P_U: V \rightarrow V$  onto a subspace  $U \subseteq V$ .

*Solution.*

- (a) By [Lemma 10.4.1](#),  $\langle \mathbf{A}\mathbf{x}, \mathbf{y} \rangle = \langle \mathbf{x}, \mathbf{A}^*\mathbf{y} \rangle$  for all  $\mathbf{x}, \mathbf{y}$ , which is the defining identity of the adjoint; so  $(T_{\mathbf{A}})^* = T_{\mathbf{A}^*}$  for every matrix, by uniqueness in [Theorem 10.6.1](#). Here

$$\mathbf{A}^* = \begin{pmatrix} 1-i & -i \\ 0 & 1 \\ 2 & i \end{pmatrix},$$

$$\text{so } T_{\mathbf{A}^*}(y_1, y_2) = ((1-i)y_1 - iy_2, y_2, 2y_1 + iy_2).$$

- (b) Directly from the definition, for  $\mathbf{x}, \mathbf{y} \in F^3$ ,

$$\langle S\mathbf{x}, \mathbf{y} \rangle = 0 \cdot \overline{y_1} + x_1 \overline{y_2} + x_2 \overline{y_3}, \quad \langle \mathbf{x}, (y_2, y_3, 0) \rangle = x_1 \overline{y_2} + x_2 \overline{y_3} + x_3 \cdot 0.$$

The two agree for all  $\mathbf{x}$  and  $\mathbf{y}$ . So  $S^*$  is the **backward** shift  $S^*(y_1, y_2, y_3) = (y_2, y_3, 0)$ . The forward shift pushes coordinates up; its adjoint pushes them down and drops the one that falls off the end.

- (c)  $\langle \text{id}_V \mathbf{v}, \mathbf{w} \rangle = \langle \mathbf{v}, \mathbf{w} \rangle = \langle \mathbf{v}, \text{id}_V \mathbf{w} \rangle$ , so  $(\text{id}_V)^* = \text{id}_V$ . For the zero map,  $\langle 0\mathbf{v}, \mathbf{w} \rangle = 0 = \langle \mathbf{v}, 0 \rangle$  for every  $\mathbf{w}$ , so  $0^* = 0$ , the zero map  $W \rightarrow V$ . These are the degenerate cases, and they say the adjoint fixes what should be fixed.
- (d) By [Theorem 10.3.3](#) (b),  $\langle P_U \mathbf{u}, \mathbf{v} \rangle = \langle \mathbf{u}, P_U \mathbf{v} \rangle$  for all  $\mathbf{u}, \mathbf{v} \in V$ . Since  $P_U$  is an operator on  $V$ , this is exactly the defining identity for its own adjoint, so  $P_U^* = P_U$  by uniqueness in [Theorem 10.6.1](#).

Part (d) is the first place where uniqueness earns its keep: we did not construct  $P_U^*$ , we recognized an identity we already had and read the answer off it. That move recurs throughout the section.

### Example. (Differentiation on a Polynomial Space)

On  $V = \mathbb{R}[x]_{\leq 2}$  define  $\langle p, q \rangle := a_0 b_0 + a_1 b_1 + a_2 b_2$  for  $p = a_0 + a_1 x + a_2 x^2$  and  $q = b_0 + b_1 x + b_2 x^2$ . (This is the standard inner product of  $\mathbb{R}^3$  transported by the coordinate map, so it is an inner product, and  $\mathcal{E} = (1, x, x^2)$  is an orthonormal basis.) Find the adjoint of  $D: V \rightarrow V$ ,  $D(p) = p'$ .

*Solution.* For  $p, q$  as above,  $D(p) = a_1 + 2a_2 x$ , so

$$\langle Dp, q \rangle = a_1 b_0 + 2a_2 b_1 + 0 \cdot b_2.$$

We need  $D^*q = c_0 + c_1 x + c_2 x^2$  with  $\langle p, D^*q \rangle = a_0 c_0 + a_1 c_1 + a_2 c_2$  equal to the line above for **all**  $a_0, a_1, a_2$ . Comparing coefficients of  $a_0, a_1, a_2$  gives  $c_0 = 0$ ,  $c_1 = b_0$ ,  $c_2 = 2b_1$ . Hence

$$D^*(b_0 + b_1 x + b_2 x^2) = b_0 x + 2b_1 x^2.$$

So  $D^*$  raises degrees where  $D$  lowers them. It is not an antiderivative: the coefficient 2 and the discarded  $b_2$  both say so.

Here is a non-example by minimal change, and it explains the conjugate. On  $\mathbb{C}^2$ , keep the standard inner product but try the plain transpose: is  $\langle \mathbf{Ax}, \mathbf{y} \rangle = \langle \mathbf{x}, \mathbf{A}^\top \mathbf{y} \rangle$ ? Take  $\mathbf{A} = \begin{pmatrix} i & 0 \\ 0 & 0 \end{pmatrix}$ , which is symmetric, so  $\mathbf{A}^\top = \mathbf{A}$ , and take  $\mathbf{x} = \mathbf{y} = \mathbf{e}_1$ . Then  $\langle \mathbf{A}\mathbf{e}_1, \mathbf{e}_1 \rangle = \langle (i, 0), (1, 0) \rangle = i$ , while  $\langle \mathbf{e}_1, \mathbf{A}^\top \mathbf{e}_1 \rangle = \langle (1, 0), (i, 0) \rangle = 1 \cdot \bar{i} = -i$ . The two sides differ. Everything about the transpose still works — it is linear and it reverses products — but it fails the one clause that matters, the defining identity, and it fails it by a conjugation. Over  $\mathbb{R}$  the conjugation is invisible and the transpose is correct.

**Why this definition.** We could have asked instead for a map  $R$  with  $\langle R\mathbf{w}, \mathbf{v} \rangle = \langle \mathbf{w}, T\mathbf{v} \rangle$ . Conjugating Definition 10.6.1 shows that  $R = T^*$  as well, so nothing new appears; the convention above is the one that reads left to right. The notation  $T^*$  matches the matrix notation  $\mathbf{A}^*$  of Example (Four Adjoints) (a), which is the point of the star. The word *adjoint* is unrelated to the *adjugate* of Definition 6.6.2; the collision of names is historical and the two objects have nothing to do with each other.

**Warning. The adjoint depends on the inner product, not on  $T$  alone.** On  $\mathbb{R}^2$  take  $T(x_1, x_2) = (x_2, 0)$ . With the standard inner product,  $T^*(y_1, y_2) = (0, y_1)$ . With the weighted inner product  $\langle \mathbf{x}, \mathbf{y} \rangle_w := x_1 y_1 + 2x_2 y_2$ , which is also an inner product, we need  $\langle T\mathbf{x}, \mathbf{y} \rangle_w = x_2 y_1$  to equal  $\langle \mathbf{x}, S\mathbf{y} \rangle_w = x_1 (S\mathbf{y})_1 + 2x_2 (S\mathbf{y})_2$  for all  $\mathbf{x}$ , which forces  $(S\mathbf{y})_1 = 0$  and  $(S\mathbf{y})_2 = \frac{1}{2}y_1$ . So the adjoint is now  $\mathbf{y} \mapsto (0, \frac{1}{2}y_1)$ , a different map. Writing  $T^*$  without saying which inner product is meant is as incomplete as writing a matrix of  $T$  without saying which basis.

**Quick check.** Let  $T \in \mathcal{L}(V, W)$  with  $T \neq 0$ . Can  $T^* = 0$ ?

**Solution.** No. If  $T^* = 0$  then  $\langle T\mathbf{v}, \mathbf{w} \rangle = \langle \mathbf{v}, \mathbf{0} \rangle = 0$  for all  $\mathbf{v}, \mathbf{w}$ . Taking  $\mathbf{w} = T\mathbf{v}$  gives  $\|T\mathbf{v}\|^2 = 0$ , so  $T\mathbf{v} = \mathbf{0}$  for every  $\mathbf{v}$ , that is  $T = 0$ . This is the “pair it with itself” move again.

Finally, the finite-dimensional hypothesis is not decoration.

### Example. (A Map With No Adjoint)

On  $V = \mathbb{R}[x]$ , all polynomials, define  $\langle p, q \rangle := \sum_{k \geq 0} a_k b_k$ , a finite sum since the coefficients vanish eventually. Show that  $T: V \rightarrow V$ ,  $T(p) := p(1) \cdot 1$ , has no adjoint.

**Solution.**  $T$  is linear, being evaluation at 1 followed by multiplication by the constant polynomial 1. Suppose  $T^*$  existed. Apply the defining identity with  $p = x^k$  and with  $q = 1$ , the constant polynomial: since  $T(x^k) = 1$ ,

$$\langle x^k, T^*(1) \rangle = \langle T(x^k), 1 \rangle = \langle 1, 1 \rangle = 1 \quad \text{for every } k \geq 0.$$

But  $\langle x^k, T^*(1) \rangle$  is the coefficient of  $x^k$  in  $T^*(1)$ , so every coefficient of the polynomial  $T^*(1)$  equals 1. No polynomial has infinitely many non-zero coefficients, a contradiction. So  $T^*$  does not exist.



What failed is Riesz: the functional  $p \mapsto p(1)$  on  $\mathbb{R}[x]$  is not pairing against any vector, precisely as in the infinite-dimensional discussion of Section 5.

### 10.6.3 Arithmetic of the adjoint

Starring behaves like transposing: it is additive, conjugate-homogeneous, order-reversing on products, and an involution.

#### Theorem 10.6.2 — Properties of the Adjoint

Let  $U, V, W$  be finite-dimensional inner product spaces over  $F$ .

- (a) For  $S, T \in \mathcal{L}(V, W)$ ,  $(S + T)^* = S^* + T^*$ .
- (b) For  $T \in \mathcal{L}(V, W)$  and  $c \in F$ ,  $(cT)^* = \bar{c}T^*$ .
- (c) For  $T \in \mathcal{L}(U, V)$  and  $S \in \mathcal{L}(V, W)$ ,  $(ST)^* = T^*S^*$ .
- (d)  $T^{**} = T$  for every  $T \in \mathcal{L}(V, W)$ , and  $(\text{id}_V)^* = \text{id}_V$ .
- (e) If  $T \in \mathcal{L}(V, W)$  is invertible, then  $T^*$  is invertible and  $(T^*)^{-1} = (T^{-1})^*$ .

*Idea.* Each claim says that two maps  $W \rightarrow V$  agree. By the uniqueness in [Theorem 10.6.1](#), it is enough to check that the proposed right-hand side satisfies the defining identity of the left-hand side. So every proof is one chain of inner products, and nothing is ever solved for. Part (e) needs no chain at all: star the two equations  $T^{-1}T = \text{id}_V$  and  $TT^{-1} = \text{id}_W$  and use (c) and (d).

*Proof.* Throughout,  $\mathbf{v}$  and  $\mathbf{w}$  denote arbitrary vectors of the relevant spaces, and each verification ends by invoking uniqueness in [Theorem 10.6.1](#).

- (a)  $\langle (S + T)\mathbf{v}, \mathbf{w} \rangle = \langle S\mathbf{v}, \mathbf{w} \rangle + \langle T\mathbf{v}, \mathbf{w} \rangle = \langle \mathbf{v}, S^*\mathbf{w} \rangle + \langle \mathbf{v}, T^*\mathbf{w} \rangle = \langle \mathbf{v}, (S^* + T^*)\mathbf{w} \rangle$ , the last step by additivity in the second slot.
- (b)  $\langle (cT)\mathbf{v}, \mathbf{w} \rangle = c\langle T\mathbf{v}, \mathbf{w} \rangle = c\langle \mathbf{v}, T^*\mathbf{w} \rangle = \langle \mathbf{v}, \bar{c}T^*\mathbf{w} \rangle$ , where the last equality puts  $\bar{c}$  into the second slot, which conjugates it back to  $c$ .
- (c) For  $\mathbf{u} \in U$  and  $\mathbf{w} \in W$ , using the defining identity for  $S$  and then for  $T$ ,

$$\langle (ST)\mathbf{u}, \mathbf{w} \rangle = \langle S(T\mathbf{u}), \mathbf{w} \rangle = \langle T\mathbf{u}, S^*\mathbf{w} \rangle = \langle \mathbf{u}, T^*(S^*\mathbf{w}) \rangle = \langle \mathbf{u}, (T^*S^*)\mathbf{w} \rangle.$$

- (d) The map  $T^* \in \mathcal{L}(W, V)$  has its own adjoint  $T^{**} \in \mathcal{L}(V, W)$ , determined by  $\langle T^*\mathbf{w}, \mathbf{v} \rangle = \langle \mathbf{w}, T^{**}\mathbf{v} \rangle$ . Conjugating the defining identity of  $T^*$  and using the conjugate symmetry of the inner product ([Definition 10.1.1](#)),

$$\langle T^*\mathbf{w}, \mathbf{v} \rangle = \overline{\langle \mathbf{v}, T^*\mathbf{w} \rangle} = \overline{\langle T\mathbf{v}, \mathbf{w} \rangle} = \langle \mathbf{w}, T\mathbf{v} \rangle.$$

So  $T$  satisfies the identity that defines  $T^{**}$ , and  $T^{**} = T$ . That  $(\text{id}_V)^* = \text{id}_V$  is [Example \(Four Adjoints\)](#) (c).

- (e) Suppose  $T$  is invertible, with  $T^{-1} \in \mathcal{L}(W, V)$ . Applying (c) and (d) to  $T^{-1}T = \text{id}_V$  and to  $TT^{-1} = \text{id}_W$ ,

$$T^*(T^{-1})^* = (T^{-1}T)^* = (\text{id}_V)^* = \text{id}_V, \quad (T^{-1})^*T^* = (TT^{-1})^* = \text{id}_W.$$

Both composites are identities, so  $T^*$  is invertible with inverse  $(T^{-1})^*$  (Definition 3.5.1). This proves the theorem. ■

Parts (a) and (b) say that  $T \mapsto T^*$  is **conjugate**-linear, not linear: over  $\mathbb{C}$  it is the same half-twist that the Riesz map carries in Corollary 10.5.2, and for the same reason. Part (c) is the order reversal familiar from transposes and from dual maps (Theorem 4.4.1 (b)). Part (d) says that starring is an involution, so no new maps appear by starring twice.

## 10.6.4 The matrix of the adjoint

Now fix bases and compute. The conclusion is the one the opening computation suggests, **provided the bases are orthonormal**; the hypothesis is where all the content sits.

### Theorem 10.6.3 — The Matrix of the Adjoint Is the Conjugate Transpose

Let  $V$  and  $W$  be finite-dimensional inner product spaces over  $F$ , let  $\mathcal{B} = (\mathbf{v}_1, \dots, \mathbf{v}_n)$  be an **orthonormal** basis of  $V$  and  $\mathcal{C} = (\mathbf{w}_1, \dots, \mathbf{w}_m)$  an **orthonormal** basis of  $W$ , and let  $T \in \mathcal{L}(V, W)$ . Then

$$[T^*]_{\mathcal{C}}^{\mathcal{B}} = ([T]_{\mathcal{B}}^{\mathcal{C}})^*.$$

*Idea.* In an orthonormal basis a coordinate is an inner product (Theorem 10.2.1), so every matrix entry is an inner product, and the defining identity of  $T^*$  says precisely how those inner products are related. Write both entries out and compare.

*Proof.* Write  $\mathbf{A} = [T]_{\mathcal{B}}^{\mathcal{C}}$  and  $\mathbf{B} = [T^*]_{\mathcal{C}}^{\mathcal{B}}$ , of sizes  $m \times n$  and  $n \times m$ . By Definition 3.6.1, the  $j$ -th column of  $\mathbf{A}$  holds the coordinates of  $T\mathbf{v}_j$  in  $\mathcal{C}$ , and by Theorem 10.2.1 those coordinates are inner products:

$$a_{ij} = \langle T\mathbf{v}_j, \mathbf{w}_i \rangle, \quad b_{ji} = \langle T^*\mathbf{w}_i, \mathbf{v}_j \rangle.$$

By conjugate symmetry and Definition 10.6.1,

$$b_{ji} = \langle T^*\mathbf{w}_i, \mathbf{v}_j \rangle = \overline{\langle \mathbf{v}_j, T^*\mathbf{w}_i \rangle} = \overline{\langle T\mathbf{v}_j, \mathbf{w}_i \rangle} = \overline{a_{ij}}.$$

So  $\mathbf{B}$  is the matrix whose  $(j, i)$  entry is  $\overline{a_{ij}}$ , which is  $\mathbf{A}^*$ . This proves the theorem. ■

Both hypotheses were used: orthonormality of  $\mathcal{C}$  to read off  $a_{ij}$ , orthonormality of  $\mathcal{B}$  to read off  $b_{ji}$ . Drop either one and the conclusion fails.

**Warning.** The rule  $[T^*]_{\mathcal{B}}^{\mathcal{B}} = ([T]_{\mathcal{B}}^{\mathcal{B}})^*$  is false in a basis that is not orthonormal. The next example exhibits a two-dimensional counterexample with integer entries. The correct general statement involves the Gram matrix and is recorded in the remark after it.

**Example. (The Transpose Rule Breaks)**

Work in  $\mathbb{R}^2$  with the standard inner product and let  $T(x_1, x_2) = (x_2, 0)$ . Let  $\mathcal{B} = (\mathbf{b}_1, \mathbf{b}_2)$  with  $\mathbf{b}_1 = (1, 0)$  and  $\mathbf{b}_2 = (1, 1)$ . Compute  $[T]_{\mathcal{B}}^{\mathcal{B}}$  and  $[T^*]_{\mathcal{B}}^{\mathcal{B}}$ , and compare the second with the transpose of the first.

*Solution.* First,  $\mathcal{B}$  is a basis (two independent vectors in  $\mathbb{R}^2$ ) but it is not orthonormal:  $\langle \mathbf{b}_1, \mathbf{b}_2 \rangle = 1 \neq 0$  and  $\|\mathbf{b}_2\| = \sqrt{2}$ .

By [Example \(Four Adjoints\)](#) (a) with  $\mathbf{A} = \begin{pmatrix} 0 & 1 \\ 0 & 0 \end{pmatrix}$ , the adjoint is  $T^*(y_1, y_2) = (0, y_1)$ .

The matrix of  $T$ .  $T\mathbf{b}_1 = (0, 0) = 0\mathbf{b}_1 + 0\mathbf{b}_2$  and  $T\mathbf{b}_2 = (1, 0) = 1\mathbf{b}_1 + 0\mathbf{b}_2$ , so

$$[T]_{\mathcal{B}}^{\mathcal{B}} = \begin{pmatrix} 0 & 1 \\ 0 & 0 \end{pmatrix}, \quad ([T]_{\mathcal{B}}^{\mathcal{B}})^{\top} = \begin{pmatrix} 0 & 0 \\ 1 & 0 \end{pmatrix}.$$

The matrix of  $T^*$ .  $T^*\mathbf{b}_1 = (0, 1)$  and  $T^*\mathbf{b}_2 = (0, 1)$ . Solving  $(0, 1) = c_1(1, 0) + c_2(1, 1)$  gives  $c_2 = 1$  and  $c_1 = -1$ , so both columns are  $(-1, 1)$ :

$$[T^*]_{\mathcal{B}}^{\mathcal{B}} = \begin{pmatrix} -1 & -1 \\ 1 & 1 \end{pmatrix}.$$

The two displayed matrices are different: the transpose rule fails. A quick sanity check that nothing else went wrong: the traces are 0 and 0, and both matrices have rank 1, as they must, since  $\text{rank } T = \text{rank } T^* = 1$ .

*Remark.* The correct rule. Let  $\mathbf{G}$  be the Gram matrix of  $\mathcal{B}$ , so  $g_{ij} = \langle \mathbf{b}_j, \mathbf{b}_i \rangle$  ([Definition 10.2.3](#)), which is invertible because  $\mathcal{B}$  is independent ([Theorem 10.2.5](#) (c)). Then  $[T^*]_{\mathcal{B}}^{\mathcal{B}} = \mathbf{G}^{-1} ([T]_{\mathcal{B}}^{\mathcal{B}})^* \mathbf{G}$ . In the example,  $\mathbf{G} = \begin{pmatrix} 1 & 1 \\ 1 & 2 \end{pmatrix}$  and  $\mathbf{G}^{-1} = \begin{pmatrix} 2 & -1 \\ -1 & 1 \end{pmatrix}$ , and indeed

$$\begin{pmatrix} 2 & -1 \\ -1 & 1 \end{pmatrix} \begin{pmatrix} 0 & 0 \\ 1 & 0 \end{pmatrix} \begin{pmatrix} 1 & 1 \\ 1 & 2 \end{pmatrix} = \begin{pmatrix} -1 & -1 \\ 1 & 1 \end{pmatrix}.$$

When  $\mathcal{B}$  is orthonormal,  $\mathbf{G} = \mathbf{I}$  and the rule collapses to [Theorem 10.6.3](#). [Exercise \(C3: The adjoint in an arbitrary basis\)](#) asks for the general rule.

## 10.6.5 The adjoint and the dual map

Two backward maps are now attached to  $T$ : the dual map  $T': W^* \rightarrow V^*$  of Chapter 4, which needs no inner product, and the adjoint  $T^*: W \rightarrow V$ , which does. The Riesz map converts one into the other.

Recall from [Corollary 10.5.2](#) the Riesz map, the conjugate-linear bijection

$$\Phi_V: V \rightarrow V^*, \quad \Phi_V(\mathbf{u}) := \langle \cdot, \mathbf{u} \rangle,$$

and likewise  $\Phi_W$  for  $W$ . The following square commutes: the two backward maps run horizontally, the two Riesz maps vertically, and the two routes from  $W$  to  $V^*$  agree.

$$\begin{array}{ccc}
 W & \xrightarrow{T^*} & V \\
 \Phi_W \downarrow & & \downarrow \Phi_V \\
 W^* & \xrightarrow{T'} & V^*
 \end{array}$$

the vertical maps are the Riesz maps, one for each space

### Theorem 10.6.4 — The Adjoint Is the Dual Map Read Through Riesz

Let  $V, W$  be finite-dimensional inner product spaces over  $F$  and  $T \in \mathcal{L}(V, W)$ . Then

$$\Phi_V \circ T^* = T' \circ \Phi_W.$$

*Proof.* Both sides send  $\mathbf{w} \in W$  to a functional on  $V$ , so evaluate them at an arbitrary  $\mathbf{v} \in V$ . By Definition 4.4.1 and the definition of  $\Phi_W$ ,

$$(T'(\Phi_W \mathbf{w}))(\mathbf{v}) = (\Phi_W \mathbf{w})(T\mathbf{v}) = \langle T\mathbf{v}, \mathbf{w} \rangle,$$

while by Definition 10.6.1 and the definition of  $\Phi_V$ ,

$$(\Phi_V(T^* \mathbf{w}))(\mathbf{v}) = \langle \mathbf{v}, T^* \mathbf{w} \rangle = \langle T\mathbf{v}, \mathbf{w} \rangle.$$

The two agree for all  $\mathbf{v}$  and all  $\mathbf{w}$ , so the two maps are equal. ■

Read backwards, the identity says  $T^* = \Phi_V^{-1} T' \Phi_W$ : transport a vector of  $W$  into  $W^*$ , pull it back with the dual map, transport it back into  $V$ . So the adjoint carries no information that  $T'$  does not already carry; what it carries is the inner product used to transport. The matrix statements match too. In dual bases the matrix of  $T'$  is the transpose of the matrix of  $T$  (Theorem 4.4.2); in orthonormal bases the matrix of  $T^*$  is the conjugate transpose (Theorem 10.6.3). The extra conjugation is the price of  $\Phi_V$  being conjugate-linear, and over  $\mathbb{R}$  it disappears, which is why the two constructions are so easily confused there.

## 10.6.6 The four fundamental subspaces

Chapter 2 attached four subspaces to a matrix:  $\text{col}(\mathbf{A})$  and  $\text{nul}(\mathbf{A})$  (Definition 2.5.1, Definition 2.5.3) together with  $\text{row}(\mathbf{A})$  (Definition 2.5.2) and the null space of the transpose. Chapter 4 related two of them, with Theorem 4.6.5: the row space of  $\mathbf{A}$  is the annihilator of  $\text{nul}(\mathbf{A})$ , so every linear equation satisfied by the solution set is a combination of the given ones. That statement lived in the dual space, and Definition 4.2.1 carried a warning with it: over a general field the annihilator has no right to be pictured as a perpendicular. With an inner product, it does. The four subspaces become two orthogonal splittings, one in each of the two spaces.

### Theorem 10.6.5 — The Four Fundamental Subspaces

Let  $V, W$  be finite-dimensional inner product spaces over  $F$  and let  $T \in \mathcal{L}(V, W)$ . Then

- (a)  $\ker T^* = (\operatorname{im} T)^\perp$ ;
- (b)  $\operatorname{im} T^* = (\ker T)^\perp$ ;
- (c)  $V = \ker T \oplus \operatorname{im} T^*$ ;
- (d)  $W = \operatorname{im} T \oplus \ker T^*$ .

*Idea.* Only (a) needs an argument, and it is one chain of equivalences:  $T^*\mathbf{w} = \mathbf{0}$  says a vector is zero, so pair it with everything, and the defining identity turns “everything” into “everything in the image of  $T$ ”. Then (b) is (a) applied to  $T^*$ , using  $T^{**} = T$ , and complemented with  $(U^\perp)^\perp = U$ . Parts (c) and (d) are then just the orthogonal decomposition theorem.

*Proof.*

- (a) Let  $\mathbf{w} \in W$ . Then

$$\begin{aligned} \mathbf{w} \in \ker T^* &\iff T^*\mathbf{w} = \mathbf{0} \iff \langle \mathbf{v}, T^*\mathbf{w} \rangle = 0 \text{ for all } \mathbf{v} \in V \\ &\iff \langle T\mathbf{v}, \mathbf{w} \rangle = 0 \text{ for all } \mathbf{v} \in V \iff \mathbf{w} \in (\operatorname{im} T)^\perp. \end{aligned}$$

The second equivalence is non-degeneracy: forwards it is immediate, and backwards we take  $\mathbf{v} = T^*\mathbf{w}$  and get  $\|T^*\mathbf{w}\|^2 = 0$ . The third is [Definition 10.6.1](#). The fourth is [Definition 10.3.1](#) together with conjugate symmetry, since  $\langle T\mathbf{v}, \mathbf{w} \rangle = 0$  if and only if  $\langle \mathbf{w}, T\mathbf{v} \rangle = 0$ , and the vectors  $T\mathbf{v}$  are exactly the elements of  $\operatorname{im} T$ . Every step is reversible, so both inclusions come at once.

- (b) Apply (a) to  $T^* \in \mathcal{L}(W, V)$  in place of  $T$ . Since  $T^{**} = T$  by [Theorem 10.6.2](#) (d), this reads  $\ker T = (\operatorname{im} T^*)^\perp$ . Taking orthogonal complements and using  $(U^\perp)^\perp = U$  for a subspace  $U$  of a finite-dimensional space ([Theorem 10.3.2](#) (b)),

$$(\ker T)^\perp = ((\operatorname{im} T^*)^\perp)^\perp = \operatorname{im} T^*.$$

- (c)  $\ker T$  is a subspace of  $V$  ([Theorem 3.2.1](#)), so  $V = \ker T \oplus (\ker T)^\perp$  by [Theorem 10.3.2](#) (a), and  $(\ker T)^\perp = \operatorname{im} T^*$  by (b).  
 (d) Likewise  $W = \operatorname{im} T \oplus (\operatorname{im} T)^\perp$  by [Theorem 10.3.2](#) (a), and  $(\operatorname{im} T)^\perp = \ker T^*$  by (a). This proves the theorem. ■

For a matrix  $\mathbf{A} \in M_{m \times n}(F)$  with the standard inner products,  $T_{\mathbf{A}}^* = T_{\mathbf{A}^*}$ , so the theorem reads

$$\operatorname{nul}(\mathbf{A}^*) = \operatorname{col}(\mathbf{A})^\perp, \quad \operatorname{col}(\mathbf{A}^*) = \operatorname{nul}(\mathbf{A})^\perp,$$

with  $F^m = \operatorname{col}(\mathbf{A}) \oplus \operatorname{nul}(\mathbf{A}^*)$  and  $F^n = \operatorname{nul}(\mathbf{A}) \oplus \operatorname{col}(\mathbf{A}^*)$ . Over  $\mathbb{R}$  the columns of  $\mathbf{A}^\top$  are the rows of  $\mathbf{A}$  turned into vectors, so the second identity is the promised statement that **the row space and the null space are orthogonal complements of each other** in  $\mathbb{R}^n$ . This is [Theorem 4.6.5](#) with the annihilator replaced by the orthogonal complement, exactly as [Corollary 10.5.2](#) licenses; and taking dimensions returns Rank–Nullity for matrices ([Theorem 2.5.7](#)) one more time. Section 4 proved the identity  $\operatorname{nul}(\mathbf{A})^\perp = \operatorname{col}(\mathbf{A}^*)$  directly, because the minimum-

norm solution needed it; the theorem above is that computation done once for all four subspaces and for maps rather than matrices.

**Example. (All Four Subspaces of a  $3 \times 3$  Matrix)**

For  $\mathbf{A} = \begin{pmatrix} 1 & 1 & 0 \\ 0 & 1 & 1 \\ 1 & 2 & 1 \end{pmatrix}$  over  $\mathbb{R}$ , find bases of the four subspaces and verify the two orthogonality statements.

*Solution.* Row three is the sum of rows one and two, and rows one and two are independent, so  $\text{rank } \mathbf{A} = 2$  and  $\text{row}(\mathbf{A})$  has the basis  $(1, 1, 0), (0, 1, 1)$  (written as vectors).

*Null space.*  $\mathbf{A}\mathbf{x} = \mathbf{0}$  reduces to  $x_1 + x_2 = 0$  and  $x_2 + x_3 = 0$ , so  $\text{nul}(\mathbf{A}) = \text{span}((1, -1, 1))$ , of dimension  $1 = 3 - 2$ , as [Theorem 2.5.7](#) requires.

*Orthogonality in the domain.*  $(1, 1, 0) \cdot (1, -1, 1) = 0$  and  $(0, 1, 1) \cdot (1, -1, 1) = 0$ . So  $\text{row}(\mathbf{A}) \perp \text{nul}(\mathbf{A})$ , and since  $2 + 1 = 3$  the two are complementary:  $\text{nul}(\mathbf{A})^\perp = \text{row}(\mathbf{A})$ .

*Column space.* Columns one and two are independent and column three is column two minus column one, so  $\text{col}(\mathbf{A}) = \text{span}((1, 0, 1), (1, 1, 2))$ .

*Null space of the transpose.*  $\mathbf{A}^\top \mathbf{y} = \mathbf{0}$  reads  $y_1 + y_3 = 0$ ,  $y_1 + y_2 + 2y_3 = 0$ ,  $y_2 + y_3 = 0$ , giving  $\text{nul}(\mathbf{A}^\top) = \text{span}((-1, -1, 1))$ .

*Orthogonality in the codomain.*  $(1, 0, 1) \cdot (-1, -1, 1) = 0$  and  $(1, 1, 2) \cdot (-1, -1, 1) = 0$ , and  $2 + 1 = 3$ , so  $\text{col}(\mathbf{A})^\perp = \text{nul}(\mathbf{A}^\top)$ .

### Corollary 10.6.6 — Rank of the Adjoint

Let  $V, W$  be finite-dimensional inner product spaces over  $F$  and  $T \in \mathcal{L}(V, W)$ . Then

- (a)  $\text{rank } T^* = \text{rank } T$ ;
- (b)  $\ker(T^*T) = \ker T$ , and consequently  $\text{rank}(T^*T) = \text{rank } T$ .

*Proof.*

- (a) By [Theorem 10.6.5](#) (b) and a dimension count,

$$\text{rank } T^* = \dim \text{im } T^* = \dim(\ker T)^\perp = \dim V - \text{nullity } T = \text{rank } T,$$

the middle equality by [Theorem 10.3.2](#) (c) and the last by the Rank–Nullity Theorem ([Theorem 3.3.1](#)).

- (b) ( $\supseteq$ ) If  $T\mathbf{v} = \mathbf{0}$  then  $T^*T\mathbf{v} = T^*\mathbf{0} = \mathbf{0}$ . ( $\subseteq$ ) Suppose  $T^*T\mathbf{v} = \mathbf{0}$ . Then

$$\|T\mathbf{v}\|^2 = \langle T\mathbf{v}, T\mathbf{v} \rangle = \langle \mathbf{v}, T^*T\mathbf{v} \rangle = \langle \mathbf{v}, \mathbf{0} \rangle = 0,$$

so  $T\mathbf{v} = \mathbf{0}$ . Hence the kernels agree. Both  $T^*T$  and  $T$  are defined on  $V$ , so Rank–Nullity gives  $\text{rank}(T^*T) = \dim V - \text{nullity}(T^*T) = \dim V - \text{nullity } T = \text{rank } T$ .

■

Part (a) is a fourth proof that row rank equals column rank, after the three of Chapters 2, 3 and 4: for a real matrix it says  $\text{rank } \mathbf{A}^\top = \text{rank } \mathbf{A}$ . Part (b) is the engine of the normal equations: it is why  $\mathbf{A}^* \mathbf{A}$  is invertible exactly when  $\mathbf{A}$  has independent columns, which is the hypothesis in [Corollary 10.4.4](#).

*Quick check.* Let  $T \in \mathcal{L}(V, W)$  with  $\dim V = 4$ ,  $\dim W = 7$  and  $\text{rank } T = 3$ . What are  $\dim \ker T^*$  and  $\dim \text{im } T^*$ ?

*Solution.*  $\text{rank } T^* = 3$  by [Corollary 10.6.6](#) (a), so  $\dim \text{im } T^* = 3$ ; and  $T^*: W \rightarrow V$  with  $\dim W = 7$ , so  $\dim \ker T^* = 7 - 3 = 4$ . The same number comes from [Theorem 10.6.5](#) (a):  $\dim(\text{im } T)^\perp = 7 - 3 = 4$ .

## 10.6.7 Self-adjoint and normal operators

For an operator  $T \in \mathcal{L}(V)$ , the adjoint lives in the same space  $\mathcal{L}(V)$ , so we may compare  $T$  with  $T^*$ . Two comparisons matter, and everything in the next chapter rests on them.

*A self-adjoint operator is one that costs nothing to move across the inner product.*

### Definition 10.6.2 — Self-adjoint Operator

An operator  $T \in \mathcal{L}(V)$  on a finite-dimensional inner product space is **self-adjoint** if  $T^* = T$ , that is, if

$$\langle T\mathbf{u}, \mathbf{v} \rangle = \langle \mathbf{u}, T\mathbf{v} \rangle \quad \text{for all } \mathbf{u}, \mathbf{v} \in V.$$

The matrix counterpart is the **Hermitian** matrix of [Definition 10.1.2](#),  $\mathbf{A}^* = \mathbf{A}$ ; over  $\mathbb{R}$  this says  $\mathbf{A}^\top = \mathbf{A}$ , and the matrix is **symmetric**.

In words:  $T^*$  is not a new map but  $T$  over again, so  $T$  must be an **operator**, with the same space and the same inner product on both sides; and the displayed identity is the defining identity of [Definition 10.6.1](#) with  $T^*$  rewritten as  $T$ , still required for **all**  $\mathbf{u}$  and **all**  $\mathbf{v}$ .

By [Theorem 10.6.3](#), an operator on a finite-dimensional inner product space is self-adjoint exactly when its matrix in one (equivalently every) orthonormal basis is Hermitian. Three examples to keep in mind: the orthogonal projection  $P_U$  onto any subspace, self-adjoint by [Example \(Four Adjoints\)](#) (d), which is the geometric case; on  $\mathbb{R}^2$ , the map of the symmetric  $\mathbf{A} = \begin{pmatrix} 2 & 1 \\ 1 & 3 \end{pmatrix}$ , where conjugation does nothing and  $\mathbf{A}^* = \mathbf{A}^\top = \mathbf{A}$ ; and on  $\mathbb{C}^2$ , the map of the Hermitian  $\mathbf{A} = \begin{pmatrix} 2 & 1+i \\ 1-i & 3 \end{pmatrix}$ , where transposing swaps the two off-diagonal entries and conjugating swaps them back, while the diagonal is real and survives both.

Here is a non-example by minimal change: keep the shape and move the conjugate. The matrix  $\mathbf{B} = \begin{pmatrix} 2 & 1+i \\ 1+i & 3 \end{pmatrix}$  is **symmetric**,  $\mathbf{B}^\top = \mathbf{B}$ , but  $\mathbf{B}^* = \begin{pmatrix} 2 & 1-i \\ 1-i & 3 \end{pmatrix} \neq \mathbf{B}$ , so the clause  $T^* = T$  fails. It fails visibly at  $\mathbf{u} = \mathbf{e}_1$  and  $\mathbf{v} = \mathbf{e}_2$ :  $\langle \mathbf{B}\mathbf{e}_1, \mathbf{e}_2 \rangle = 1+i$ , while  $\langle \mathbf{e}_1, \mathbf{B}\mathbf{e}_2 \rangle = \overline{1+i} = 1-i$ . Over  $\mathbb{C}$ , self-adjointness means Hermitian, never merely symmetric.

Two more self-adjoint operators come for free: the combination  $D + D^*$  of [Example \(Differentiation on a Polynomial Space\)](#), and  $T^*T$  for any  $T$ , since  $(T^*T)^* = T^*T^{**} = T^*T$  by [Theorem 10.6.2](#) (c) and (d).

*An operator is normal when it and its adjoint may be applied in either order.*

### Definition 10.6.3 — Normal Operator

An operator  $T \in \mathcal{L}(V)$  on a finite-dimensional inner product space is **normal** if it commutes with its adjoint:

$$T^*T = TT^*.$$

In words: normality is one equation between two operators on  $V$ , with no vectors and no quantifiers in it. It does not ask that  $T$  and  $T^*$  be equal, only that their order may be swapped, so it is strictly weaker than self-adjointness.

Four examples, widening as they go. Every self-adjoint operator is normal, since then both sides are  $T^2$ . So is every invertible  $T$  with  $T^{-1} = T^*$  — Section 7 calls these unitary — since then both sides are  $\text{id}_V$ ; the quarter-turn of  $\mathbb{R}^2$ , with matrix  $\begin{pmatrix} 0 & -1 \\ 1 & 0 \end{pmatrix}$  in the standard basis, is one, and it is not self-adjoint. On  $\mathbb{C}^2$  the operator with matrix  $i\mathbf{I}$  satisfies  $(i\mathbf{I})^* = -i\mathbf{I}$  by Theorem 10.6.2 (b), which commutes with  $i\mathbf{I}$  but is not equal to it: normal and not self-adjoint. Widest of all, every diagonal matrix  $\mathbf{D} = \text{diag}(d_1, \dots, d_n)$  is normal, since  $\mathbf{D}^*\mathbf{D} = \mathbf{D}\mathbf{D}^* = \text{diag}(|d_1|^2, \dots, |d_n|^2)$ .

A non-example by minimal change: take the diagonal matrix  $\text{diag}(1, 0)$  and slide its non-zero entry off the diagonal. The Jordan block  $\mathbf{J}_2(0) = \begin{pmatrix} 0 & 1 \\ 0 & 0 \end{pmatrix}$  on  $\mathbb{C}^2$  has

$$\mathbf{J}_2(0)^*\mathbf{J}_2(0) = \begin{pmatrix} 0 & 0 \\ 0 & 1 \end{pmatrix}, \quad \mathbf{J}_2(0)\mathbf{J}_2(0)^* = \begin{pmatrix} 1 & 0 \\ 0 & 0 \end{pmatrix},$$

and the two differ. The  $(1, 1)$  entries say what went wrong. Writing  $T$  for the map of  $\mathbf{J}_2(0)$ , the  $(1, 1)$  entry of the first matrix is  $\langle T^*T\mathbf{e}_1, \mathbf{e}_1 \rangle = \|T\mathbf{e}_1\|^2 = 0$  and that of the second is  $\langle TT^*\mathbf{e}_1, \mathbf{e}_1 \rangle = \|T^*\mathbf{e}_1\|^2 = 1$ , so  $\|T\mathbf{e}_1\| \neq \|T^*\mathbf{e}_1\|$ : the operator kills  $\mathbf{e}_1$  and its adjoint stretches it to a unit vector.

Over  $\mathbb{C}$ , normality is the weakest hypothesis under which the spectral theorem holds, and that is its reason for existing. Chapter 11 will show that the diagonal example is already the general one.

The facts available now, before any eigenvalue theory, are the ones that follow straight from the definition.

### Proposition 10.6.7 — First Properties of Self-adjoint Operators

Let  $T \in \mathcal{L}(V)$  be self-adjoint on a finite-dimensional inner product space  $V$  over  $F$ .

- (a)  $\langle T\mathbf{v}, \mathbf{v} \rangle \in \mathbb{R}$  for every  $\mathbf{v} \in V$ , even when  $F = \mathbb{C}$ .
- (b) If  $\mathcal{B}$  is an orthonormal basis of  $V$ , the diagonal entries of  $[T]_{\mathcal{B}}^{\mathcal{B}}$  are real.
- (c)  $T$  is normal.

*Proof.*

- (a) By conjugate symmetry (Definition 10.1.1) and self-adjointness,

$$\overline{\langle T\mathbf{v}, \mathbf{v} \rangle} = \langle \mathbf{v}, T\mathbf{v} \rangle = \langle T\mathbf{v}, \mathbf{v} \rangle.$$



A complex number equal to its own conjugate is real.

- (b) Let  $\mathcal{B} = (\mathbf{v}_1, \dots, \mathbf{v}_n)$  be orthonormal. By [Theorem 10.2.1](#) the  $i$ -th coordinate of  $T\mathbf{v}_i$  is  $\langle T\mathbf{v}_i, \mathbf{v}_i \rangle$ , so the  $(i, i)$  entry of  $[T]_{\mathcal{B}}^{\mathcal{B}}$  is  $a_{ii} = \langle T\mathbf{v}_i, \mathbf{v}_i \rangle$ , which is real by (a).
- (c)  $T^*T = TT = TT^*$ .

■

Part (a) is the first hint of the spectral theorem. If  $T\mathbf{v} = \lambda\mathbf{v}$  with  $\mathbf{v} \neq \mathbf{0}$ , then  $\langle T\mathbf{v}, \mathbf{v} \rangle = \lambda\|\mathbf{v}\|^2$ , and since the left side is real and  $\|\mathbf{v}\|^2$  is a positive real,  $\lambda$  must be real: a self-adjoint operator has no non-real eigenvalues. Chapter 11 turns this observation into the spectral theorems: a self-adjoint operator has an orthonormal basis of eigenvectors, with real eigenvalues; on a complex space the operators with an orthonormal basis of eigenvectors are exactly the normal ones, and on a real space exactly the self-adjoint ones. None of that is proved here, and none of it is used before then; but all of it is built from the adjoint and the orthogonal geometry of this chapter.

*Warning.* “Self-adjoint” cannot be read off an arbitrary basis. Being self-adjoint is a property of  $T$  together with the inner product, and the Hermitian test applies only in an **orthonormal** basis. The operator  $T(x_1, x_2) = (x_2, 0)$  of [Example \(The Transpose Rule Breaks\)](#) is not self-adjoint, and its matrix in the standard orthonormal basis,  $\begin{pmatrix} 0 & 1 \\ 0 & 0 \end{pmatrix}$ , is correctly non-symmetric. But the self-adjoint operator  $P(x_1, x_2) = (x_1, 0)$ , the orthogonal projection onto the first axis, has the **non**-symmetric matrix  $\begin{pmatrix} 1 & 1 \\ 0 & 0 \end{pmatrix}$  in the basis  $\mathcal{B}$  of that example, since  $P\mathbf{b}_1 = \mathbf{b}_1$  and  $P\mathbf{b}_2 = (1, 0) = \mathbf{b}_1$ . A non-symmetric matrix is evidence of nothing until the basis is known to be orthonormal.

## 10.6.8 Exercises

### 10.6.8.1 A. Check your understanding

#### Exercise A1.

- (a) State the defining property of the adjoint  $T^*$  of  $T \in \mathcal{L}(V, W)$ , including the domain and codomain of  $T^*$  and all quantifiers.
- (b) Which hypothesis of [Theorem 10.6.1](#) is used to produce  $T^*$ , and which theorem supplies it?
- (c) True or false:  $(cT)^* = cT^*$  for every scalar  $c$ . Justify your answer.
- (d) State the hypothesis needed for  $[T^*]_{\mathcal{C}}^{\mathcal{B}} = ([T]_{\mathcal{B}}^{\mathcal{C}})^*$ , and say what goes wrong without it.
- (e) True or false:  $\ker T^* = (\ker T)^{\perp}$ . Justify your answer.
- (f) Give an example of a normal operator that is not self-adjoint.

*Solution.*

- (a) For  $T: V \rightarrow W$ , the adjoint is the unique  $T^*: W \rightarrow V$  with  $\langle T\mathbf{v}, \mathbf{w} \rangle = \langle \mathbf{v}, T^*\mathbf{w} \rangle$  for **all**  $\mathbf{v} \in V$  and **all**  $\mathbf{w} \in W$  ([Definition 10.6.1](#)).
- (b) Finite-dimensionality of  $V$ , which is what [Theorem 10.5.1](#) needs in order to repre-

sent the functional  $\mathbf{v} \mapsto \langle T\mathbf{v}, \mathbf{w} \rangle$ .

- (c) False:  $(cT)^* = \bar{c}T^*$  ([Theorem 10.6.2](#) (b)). Over  $\mathbb{R}$  the two agree; over  $\mathbb{C}$  take  $T = \text{id}$  and  $c = i$ , so that  $(i \text{id})^* = -i \text{id} \neq i \text{id}$ .
- (d) Both bases must be orthonormal ([Theorem 10.6.3](#)). Without it the rule fails: [Example \(The Transpose Rule Breaks\)](#) has  $([T]_{\mathcal{B}}^{\mathcal{B}})^{\top} = \begin{pmatrix} 0 & 0 \\ 1 & 0 \end{pmatrix}$  but  $[T^*]_{\mathcal{B}}^{\mathcal{B}} = \begin{pmatrix} -1 & -1 \\ 1 & 1 \end{pmatrix}$ .
- (e) False. The correct statements are  $\ker T^* = (\text{im } T)^{\perp}$  and  $\text{im } T^* = (\ker T)^{\perp}$  ([Theorem 10.6.5](#)). The two sides need not even live in the same space when  $V \neq W$ .
- (f)  $T = i \text{id}$  on  $\mathbb{C}^2$ :  $T^* = -i \text{id}$  commutes with  $T$ , but  $T^* \neq T$ .

### 10.6.8.2 B. Practice

#### Exercise B1: Adjoints of matrix maps.

Let  $\mathbf{A} = \begin{pmatrix} 1+i & 0 & 2 \\ i & 1 & -i \end{pmatrix}$  and  $\mathbf{B} = \begin{pmatrix} 0 & 1 \\ i & 0 \end{pmatrix}$ , acting between the complex spaces  $\mathbb{C}^3 \xrightarrow{T_{\mathbf{A}}} \mathbb{C}^2 \xrightarrow{T_{\mathbf{B}}} \mathbb{C}^2$  with the standard inner products.

- (a) Write down  $T_{\mathbf{A}}^*$  and  $T_{\mathbf{B}}^*$  as matrix maps.
- (b) Compute the matrices of  $T_{\mathbf{B}}T_{\mathbf{A}}$  and of  $(T_{\mathbf{B}}T_{\mathbf{A}})^*$ , and verify  $(T_{\mathbf{B}}T_{\mathbf{A}})^* = T_{\mathbf{A}}^*T_{\mathbf{B}}^*$  by direct computation.
- (c) Verify that  $T_{\mathbf{B}}$  is not self-adjoint, and determine whether it is normal.

*Solution.*

- (a)  $T_{\mathbf{A}}^* = T_{\mathbf{A}}^*$  and  $T_{\mathbf{B}}^* = T_{\mathbf{B}}^*$  ([Example \(Four Adjoints\)](#) (a)), with

$$\mathbf{A}^* = \begin{pmatrix} 1-i & -i \\ 0 & 1 \\ 2 & i \end{pmatrix}, \quad \mathbf{B}^* = \begin{pmatrix} 0 & -i \\ 1 & 0 \end{pmatrix}.$$

- (b)  $\mathbf{BA} = \begin{pmatrix} i & 1 & -i \\ i-1 & 0 & 2i \end{pmatrix}$ , so

$$(\mathbf{BA})^* = \begin{pmatrix} -i & -i-1 \\ 1 & 0 \\ i & -2i \end{pmatrix}.$$

On the other side,

$$\mathbf{A}^*\mathbf{B}^* = \begin{pmatrix} 1-i & -i \\ 0 & 1 \\ 2 & i \end{pmatrix} \begin{pmatrix} 0 & -i \\ 1 & 0 \end{pmatrix} = \begin{pmatrix} -i & -i(1-i) \\ 1 & 0 \\ i & -2i \end{pmatrix},$$

and  $-i(1-i) = -i + i^2 = -1 - i$ . The two matrices agree, as [Theorem 10.6.2](#) (c) predicts.

- (c)  $\mathbf{B}^* \neq \mathbf{B}$ , since the  $(1, 2)$  entries are  $-i$  and  $1$ . But

$$\mathbf{B}^*\mathbf{B} = \begin{pmatrix} 0 & -i \\ 1 & 0 \end{pmatrix} \begin{pmatrix} 0 & 1 \\ i & 0 \end{pmatrix} = \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix} = \begin{pmatrix} 0 & 1 \\ i & 0 \end{pmatrix} \begin{pmatrix} 0 & -i \\ 1 & 0 \end{pmatrix} = \mathbf{BB}^*,$$

so  $T_{\mathbf{B}}$  is normal ([Definition 10.6.3](#)) and not self-adjoint.

### Exercise B2: Adjoints on a polynomial space.

Let  $V = \mathbb{R}[x]_{\leq 2}$  and  $W = \mathbb{R}[x]_{\leq 3}$ , each with the coefficient inner product of [Example \(Differentiation on a Polynomial Space\)](#), so that  $\mathcal{E} = (1, x, x^2)$  and  $\mathcal{F} = (1, x, x^2, x^3)$  are orthonormal bases.

- Find the adjoint of  $M: V \rightarrow W$ ,  $M(p) = xp$ .
- Find the adjoint of  $E: V \rightarrow \mathbb{R}$ ,  $E(p) = p(1)$ , where  $\mathbb{R}$  carries  $\langle s, t \rangle = st$ . (Compare [Theorem 10.5.1](#).)
- Verify  $\text{rank } M^* = \text{rank } M$  directly.

*Solution.*

- $M(1) = x$ ,  $M(x) = x^2$ ,  $M(x^2) = x^3$ , so in the two orthonormal bases

$$[M]_{\mathcal{F}}^{\mathcal{E}} = \begin{pmatrix} 0 & 0 & 0 \\ 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{pmatrix}, \quad [M^*]_{\mathcal{E}}^{\mathcal{F}} = ([M]_{\mathcal{F}}^{\mathcal{E}})^{\top} = \begin{pmatrix} 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{pmatrix}$$

by [Theorem 10.6.3](#). Hence  $M^*(c_0 + c_1x + c_2x^2 + c_3x^3) = c_1 + c_2x + c_3x^2$ : delete the constant term and divide by  $x$ .

- $E$  is linear and  $E(a_0 + a_1x + a_2x^2) = a_0 + a_1 + a_2$ . For  $t \in \mathbb{R}$  we need  $\langle p, E^*t \rangle = \langle Ep, t \rangle = (a_0 + a_1 + a_2)t$  for all  $p$ , so comparing coefficients of  $a_0, a_1, a_2$  gives  $E^*(t) = t(1 + x + x^2)$ . In particular  $E^*(1) = 1 + x + x^2$  is the Riesz vector of the evaluation functional at 1.
- $M$  is injective, since  $xp = 0$  forces  $p = 0$ , so  $\text{rank } M = \dim V = 3$  by [Theorem 3.3.1](#). And  $M^*$  is surjective:  $M^*(xq) = q$  for every  $q \in V$ . So  $\text{rank } M^* = 3$  too, agreeing with [Corollary 10.6.6](#) (a).

### Exercise B3: Locating the kernel of the adjoint.

Let  $\mathbf{A} = \begin{pmatrix} 1 & 0 \\ 1 & 1 \\ 0 & 1 \end{pmatrix} \in M_{3 \times 2}(\mathbb{R})$ , and let  $T = T_{\mathbf{A}}: \mathbb{R}^2 \rightarrow \mathbb{R}^3$  with the standard inner products.

- Find a basis of  $\text{im } T$  and a basis of  $\ker T^*$ .
- Verify directly that every vector of  $\ker T^*$  is orthogonal to every vector of  $\text{im } T$ , and that the dimensions add to 3.
- Find  $\ker T$  and  $\text{im } T^*$ , and check [Theorem 10.6.5](#) (b) in this case.

*Solution.*

- The two columns  $(1, 1, 0)$  and  $(0, 1, 1)$  are independent, so they are a basis of  $\text{im } T = \text{col}(\mathbf{A})$  and  $\text{rank } T = 2$ . Since  $T^* = T_{\mathbf{A}^{\top}}$ , the kernel of  $T^*$  is  $\text{nul}(\mathbf{A}^{\top})$ : the equations  $y_1 + y_2 = 0$  and  $y_2 + y_3 = 0$  give  $\ker T^* = \text{span}((1, -1, 1))$ .

- (b)  $(1, 1, 0) \cdot (1, -1, 1) = 1 - 1 + 0 = 0$  and  $(0, 1, 1) \cdot (1, -1, 1) = 0 - 1 + 1 = 0$ . Since these two vectors span  $\text{im } T$ , orthogonality extends to all of  $\text{im } T$  by linearity. The dimensions are  $2 + 1 = 3 = \dim \mathbb{R}^3$ , as [Theorem 10.6.5](#) (d) requires.
- (c)  $A\mathbf{x} = \mathbf{0}$  gives  $x_1 = 0$  and  $x_2 = 0$ , so  $\ker T = \{\mathbf{0}\}$  and  $(\ker T)^\perp = \mathbb{R}^2$ . On the other side,  $\text{rank } T^* = \text{rank } T = 2$  by [Corollary 10.6.6](#) (a), and  $\text{im } T^* \subseteq \mathbb{R}^2$ , so  $\text{im } T^* = \mathbb{R}^2$ . The two agree.

### 10.6.8.3 C. Going deeper

#### Exercise C1: The adjoint of integration.

Let  $V = \mathbb{R}[x]_{\leq 1}$  and  $W = \mathbb{R}[x]_{\leq 2}$  with the coefficient inner products of [Exercise \(B2: Adjoints on a polynomial space\)](#), and let

$$J: V \rightarrow W, \quad J(p)(x) := \int_0^x p(t) dt.$$

- (a) Find  $J^*$ .
- (b) Let  $D: W \rightarrow V$  be differentiation. Show that  $J^* \neq D$ , although  $DJ = \text{id}_V$ .
- (c) Explain in one sentence why (b) does not contradict [Theorem 10.6.2](#) (e).

*Solution.*

- (a)  $J(1) = x$  and  $J(x) = \frac{1}{2}x^2$ , so in the orthonormal bases  $\mathcal{D} = (1, x)$  of  $V$  and  $\mathcal{E} = (1, x, x^2)$  of  $W$ ,

$$[J]_{\mathcal{D}}^{\mathcal{E}} = \begin{pmatrix} 0 & 0 \\ 1 & 0 \\ 0 & \frac{1}{2} \end{pmatrix}, \quad [J^*]_{\mathcal{E}}^{\mathcal{D}} = \begin{pmatrix} 0 & 1 & 0 \\ 0 & 0 & \frac{1}{2} \end{pmatrix}$$

by [Theorem 10.6.3](#). Hence  $J^*(c_0 + c_1x + c_2x^2) = c_1 + \frac{1}{2}c_2x$ .

- (b)  $D(c_0 + c_1x + c_2x^2) = c_1 + 2c_2x$ , so  $J^*$  and  $D$  differ on  $x^2$ :  $J^*(x^2) = \frac{1}{2}x$  while  $D(x^2) = 2x$ . Yet  $D(J(p)) = p$  for every  $p \in V$ , by differentiating the integral, or by multiplying the two matrices:  $[D]_{\mathcal{E}}^{\mathcal{D}} [J]_{\mathcal{D}}^{\mathcal{E}} = \mathbf{I}_2$ .
- (c) Part (e) of [Theorem 10.6.2](#) applies to an **invertible** map, and  $J$  is not invertible: it is injective but not surjective, since  $\dim V = 2 < 3 = \dim W$ . A one-sided inverse carries no information about the adjoint.

#### Exercise C2: Skew-adjoint operators.

Let  $V$  be a finite-dimensional inner product space over  $F$  and let  $T \in \mathcal{L}(V)$  satisfy  $T^* = -T$  (such a  $T$  is called **skew-adjoint**).

- (a) Prove that  $\langle T\mathbf{v}, \mathbf{v} \rangle$  is purely imaginary for every  $\mathbf{v} \in V$  when  $F = \mathbb{C}$ , and that  $\langle T\mathbf{v}, \mathbf{v} \rangle = 0$  for every  $\mathbf{v} \in V$  when  $F = \mathbb{R}$ .
- (b) Prove that every skew-adjoint  $T$  is normal.
- (c) Suppose  $F = \mathbb{C}$ . Prove that  $T$  is skew-adjoint if and only if  $iT$  is self-adjoint, and deduce that every  $S \in \mathcal{L}(V)$  can be written  $S = A + iB$  with  $A, B$  self-adjoint.

*Solution.*

- (a) By conjugate symmetry and  $T^* = -T$ ,

$$\overline{\langle T\mathbf{v}, \mathbf{v} \rangle} = \langle \mathbf{v}, T\mathbf{v} \rangle = \langle T^*\mathbf{v}, \mathbf{v} \rangle = -\langle T\mathbf{v}, \mathbf{v} \rangle.$$

A complex number equal to minus its own conjugate has zero real part, so  $\langle T\mathbf{v}, \mathbf{v} \rangle$  is purely imaginary. When  $F = \mathbb{R}$  the inner product is real, so the same equation reads  $\langle T\mathbf{v}, \mathbf{v} \rangle = -\langle T\mathbf{v}, \mathbf{v} \rangle$ , forcing  $\langle T\mathbf{v}, \mathbf{v} \rangle = 0$ .

- (b)  $T^*T = (-T)T = -T^2$  and  $TT^* = T(-T) = -T^2$ , so the two agree.  
 (c) By [Theorem 10.6.2](#) (b),  $(iT)^* = \bar{i}T^* = -iT^*$ . If  $T^* = -T$  then  $(iT)^* = iT$ , so  $iT$  is self-adjoint; conversely, if  $(iT)^* = iT$  then  $-iT^* = iT$ , so  $T^* = -T$ . For the decomposition put

$$A := \frac{1}{2}(S + S^*), \quad B := \frac{1}{2i}(S - S^*).$$

Then  $A + iB = \frac{1}{2}(S + S^*) + \frac{1}{2}(S - S^*) = S$ . Both are self-adjoint:  $A^* = \frac{1}{2}(S^* + S) = A$  by [Theorem 10.6.2](#) (a), (b) and (d), and  $\frac{1}{2i}(S - S^*)$  is  $-i$  times the skew-adjoint operator  $\frac{1}{2}(S - S^*)$ , hence self-adjoint by the first part of (c). This is the operator version of writing a complex number as  $a + bi$ .

### Exercise C3: The adjoint in an arbitrary basis.

Let  $V$  be a finite-dimensional inner product space, let  $\mathcal{B} = (\mathbf{b}_1, \dots, \mathbf{b}_n)$  be **any** ordered basis of  $V$ , let  $\mathbf{G}$  be its Gram matrix,  $g_{ij} = \langle \mathbf{b}_j, \mathbf{b}_i \rangle$ , and let  $T \in \mathcal{L}(V)$ .

- (a) Prove that  $\langle \mathbf{u}, \mathbf{v} \rangle = [\mathbf{v}]_{\mathcal{B}}^* \mathbf{G} [\mathbf{u}]_{\mathcal{B}}$  for all  $\mathbf{u}, \mathbf{v} \in V$ .  
 (b) Deduce that  $[T^*]_{\mathcal{B}}^{\mathcal{B}} = \mathbf{G}^{-1} ([T]_{\mathcal{B}}^{\mathcal{B}})^* \mathbf{G}$ .  
 (c) Check that (b) returns [Theorem 10.6.3](#) when  $\mathcal{B}$  is orthonormal, and verify it on [Example \(The Transpose Rule Breaks\)](#).

*Hint: for (b), write the defining identity of  $T^*$  in the coordinates of (a) and use that  $\mathbf{x}^* \mathbf{M} \mathbf{y} = \mathbf{x}^* \mathbf{N} \mathbf{y}$  for all  $\mathbf{x}, \mathbf{y}$  forces  $\mathbf{M} = \mathbf{N}$ .*

*Solution.*

- (a) Write  $\mathbf{u} = \sum_j x_j \mathbf{b}_j$  and  $\mathbf{v} = \sum_i y_i \mathbf{b}_i$ , so  $\mathbf{x} = [\mathbf{u}]_{\mathcal{B}}$  and  $\mathbf{y} = [\mathbf{v}]_{\mathcal{B}}$ . Expanding in the first slot and then in the second,

$$\langle \mathbf{u}, \mathbf{v} \rangle = \sum_{i,j} x_j \bar{y}_i \langle \mathbf{b}_j, \mathbf{b}_i \rangle = \sum_{i,j} \bar{y}_i g_{ij} x_j = \mathbf{y}^* \mathbf{G} \mathbf{x}.$$

- (b) Put  $\mathbf{A} = [T]_{\mathcal{B}}^{\mathcal{B}}$  and  $\mathbf{B} = [T^*]_{\mathcal{B}}^{\mathcal{B}}$ . By [Theorem 3.6.1](#),  $[T\mathbf{u}]_{\mathcal{B}} = \mathbf{A}\mathbf{x}$  and  $[T^*\mathbf{v}]_{\mathcal{B}} = \mathbf{B}\mathbf{y}$ . So (a) turns  $\langle T\mathbf{u}, \mathbf{v} \rangle = \langle \mathbf{u}, T^*\mathbf{v} \rangle$  into

$$\mathbf{y}^* \mathbf{G} \mathbf{A} \mathbf{x} = (\mathbf{B}\mathbf{y})^* \mathbf{G} \mathbf{x} = \mathbf{y}^* \mathbf{B}^* \mathbf{G} \mathbf{x} \quad \text{for all } \mathbf{x}, \mathbf{y} \in F^n.$$

Taking  $\mathbf{x} = \mathbf{e}_j$  and  $\mathbf{y} = \mathbf{e}_i$  reads off the  $(i, j)$  entries, so  $\mathbf{G}\mathbf{A} = \mathbf{B}^*\mathbf{G}$ . The Gram matrix is invertible ([Theorem 10.2.5](#) (c)), so  $\mathbf{B}^* = \mathbf{G}\mathbf{A}\mathbf{G}^{-1}$ , and starring both sides gives  $\mathbf{B} = (\mathbf{G}^{-1})^* \mathbf{A}^* \mathbf{G}^* = \mathbf{G}^{-1} \mathbf{A}^* \mathbf{G}$ , using  $\mathbf{G}^* = \mathbf{G}$  ([Theorem 10.2.5](#) (a)) and hence  $(\mathbf{G}^{-1})^* = \mathbf{G}^{-1}$ .

- (c) If  $\mathcal{B}$  is orthonormal then  $g_{ij} = \delta_{ij}$ , so  $\mathbf{G} = \mathbf{I}$  and the formula reads  $\mathbf{B} = \mathbf{A}^*$ , which is [Theorem 10.6.3](#). In [Example \(The Transpose Rule Breaks\)](#) the computation in the remark after it is exactly this formula, with  $\mathbf{G} = \begin{pmatrix} 1 & 1 \\ 1 & 2 \end{pmatrix}$ .

## 10.7 Isometries and Unitary Maps

Chapter 3 asked which linear maps are invertible. Now that we can measure, we can ask a sharper question: which linear maps leave the measurements alone? A map that never changes a length never changes an angle either, and the two facts turn out to be the same fact. This section identifies these maps in five equivalent ways, collects them into the orthogonal and unitary groups, and classifies them completely in dimensions two and three, where they are exactly the rotations and reflections that the words suggest. At the end we drop linearity from the hypothesis and get it back as a conclusion.

As throughout the chapter,  $F$  is  $\mathbb{R}$  or  $\mathbb{C}$ , inner products are linear in the first slot, and spaces are finite-dimensional unless stated otherwise.

### 10.7.1 Maps that do not change length

Take the plane with its usual inner product and ask which linear maps move the picture rigidly. Turning the plane about the origin clearly does not change any length. Flipping it across a line through the origin does not either. Scaling by 2 obviously does. Shearing,  $(x_1, x_2) \mapsto (x_1 + x_2, x_2)$ , also does: it fixes  $\mathbf{e}_1$  but sends  $\mathbf{e}_2$  to  $(1, 1)$ , of length  $\sqrt{2}$ . So “does not change length” is a genuine restriction, and it is worth a name.

*An isometry is a linear map that every ruler agrees with: the image of a vector is exactly as long as the vector.*

#### Definition 10.7.1 — Isometry

Let  $V$  and  $W$  be inner product spaces over  $F$ . A linear map  $T \in \mathcal{L}(V, W)$  is an **isometry** if

$$\|T\mathbf{v}\| = \|\mathbf{v}\| \quad \text{for every } \mathbf{v} \in V.$$

In words: the condition is required for **every** vector, not merely for the vectors of a basis, and it compares a norm in  $W$  with a norm in  $V$ . The word comes from *iso* (equal) and *metron* (measure). Some authors reserve “isometry” for a bijective such map; we do not, and [Definition 10.7.2](#) below names the bijective case separately.

Two consequences are immediate. An isometry is injective, since  $T\mathbf{v} = \mathbf{0}$  forces  $\|\mathbf{v}\| = \|T\mathbf{v}\| = 0$  and hence  $\mathbf{v} = \mathbf{0}$  ([Theorem 3.2.3](#)). And if  $T\mathbf{v} = \lambda\mathbf{v}$  with  $\mathbf{v} \neq \mathbf{0}$ , then  $\|\mathbf{v}\| = \|\lambda\mathbf{v}\| = |\lambda| \|\mathbf{v}\|$  by [Theorem 10.1.2](#) (b), so  $|\lambda| = 1$ . We record the second observation, which the classification below will use twice.

### Proposition 10.7.1 — Eigenvalues of an Isometry Have Modulus 1

Let  $T \in \mathcal{L}(V)$  be an isometry of an inner product space over  $F$ , and let  $\lambda \in F$  be an eigenvalue of  $T$ . Then  $|\lambda| = 1$ . In particular, if  $F = \mathbb{R}$  then  $\lambda = 1$  or  $\lambda = -1$ .

*Proof.* Let  $\mathbf{v} \neq \mathbf{0}$  satisfy  $T\mathbf{v} = \lambda\mathbf{v}$  (Definition 8.2.1). Then  $\|\mathbf{v}\| = \|T\mathbf{v}\| = \|\lambda\mathbf{v}\| = |\lambda| \|\mathbf{v}\|$  by Theorem 10.1.2 (b), and  $\|\mathbf{v}\| \neq 0$ , so dividing gives  $|\lambda| = 1$ . The only real numbers of modulus 1 are  $\pm 1$ . ■

### Example. (Isometries and Near-Isometries)

Decide which of the following linear maps are isometries, with the standard inner products.

- (a)  $T: \mathbb{R}^2 \rightarrow \mathbb{R}^3$ ,  $T(x_1, x_2) = (x_1, x_2, 0)$ .
- (b)  $T_{\mathbf{P}}: F^n \rightarrow F^n$  for a permutation matrix  $\mathbf{P}$  (Definition 2.6.2).
- (c)  $T: \mathbb{C} \rightarrow \mathbb{C}$ ,  $T(z) = \alpha z$ , where  $|\alpha| = 1$ .
- (d)  $T: \mathbb{R}^2 \rightarrow \mathbb{R}^2$ ,  $T(x_1, x_2) = (x_1 + x_2, x_2)$ .
- (e) The orthogonal projection  $P_U$  onto a subspace  $U$  with  $\{\mathbf{0}\} \neq U \neq V$ .

*Solution.*

- (a) Yes:  $\|T\mathbf{x}\|^2 = x_1^2 + x_2^2 + 0 = \|\mathbf{x}\|^2$ . Note that  $T$  is not surjective, so an isometry need not be invertible.
- (b) Yes. A permutation matrix has exactly one 1 in each row and column, so  $\mathbf{P}\mathbf{x}$  is the list  $x_1, \dots, x_n$  in a different order, and  $\|\mathbf{P}\mathbf{x}\|^2 = \sum_i |x_{\sigma(i)}|^2 = \sum_i |x_i|^2$ , the same sum reordered.
- (c) Yes:  $\|\alpha z\| = |\alpha z| = |\alpha||z| = |z|$ .
- (d) No:  $\|T\mathbf{e}_2\| = \|(1, 1)\| = \sqrt{2} \neq 1 = \|\mathbf{e}_2\|$ . One vector is enough to refute.
- (e) No: pick  $\mathbf{v} \in U^\perp$  with  $\mathbf{v} \neq \mathbf{0}$ , which exists because  $U \neq V$  (Theorem 10.3.2 (a)). Then  $P_U\mathbf{v} = \mathbf{0}$  (Theorem 10.3.3 (a)), so  $\|P_U\mathbf{v}\| = 0 \neq \|\mathbf{v}\|$ . A projection destroys length by design.

## 10.7.2 Five ways to say the same thing

The definition mentions only norms, but norms and inner products determine each other. In one direction this is Definition 10.1.3. In the other it is the following identity, which reconstructs the inner product from lengths alone; it is the reason that preserving length automatically preserves angle.

### Lemma 10.7.2 — Polarization Identities

Let  $V$  be an inner product space over  $F$  and let  $\mathbf{u}, \mathbf{v} \in V$ .

- (a) If  $F = \mathbb{R}$ , then  $\langle \mathbf{u}, \mathbf{v} \rangle = \frac{1}{4} (\|\mathbf{u} + \mathbf{v}\|^2 - \|\mathbf{u} - \mathbf{v}\|^2)$ .
- (b) If  $F = \mathbb{C}$ , then  $\langle \mathbf{u}, \mathbf{v} \rangle = \frac{1}{4} \sum_{k=0}^3 i^k \|\mathbf{u} + i^k \mathbf{v}\|^2$ .

*Idea.* Expand every norm squared with  $\|\mathbf{u} + \mathbf{w}\|^2 = \|\mathbf{u}\|^2 + 2\operatorname{Re}\langle \mathbf{u}, \mathbf{w} \rangle + \|\mathbf{w}\|^2$  (Theorem 10.1.2 (c)). In the real case the two expansions differ only in the middle term, which survives doubled. In the complex case the four terms  $\|\mathbf{u}\|^2 + \|\mathbf{w}\|^2$  are the same in all four summands and are killed by  $\sum_k i^k = 0$ ; what is left is a sum in which the  $\langle \mathbf{u}, \mathbf{v} \rangle$  parts reinforce and the  $\overline{\langle \mathbf{u}, \mathbf{v} \rangle}$  parts cancel.

*Proof.*

- (a) By Theorem 10.1.2 (c),  $\|\mathbf{u} \pm \mathbf{v}\|^2 = \|\mathbf{u}\|^2 \pm 2\langle \mathbf{u}, \mathbf{v} \rangle + \|\mathbf{v}\|^2$ , since over  $\mathbb{R}$  the inner product is its own real part. Subtracting,  $\|\mathbf{u} + \mathbf{v}\|^2 - \|\mathbf{u} - \mathbf{v}\|^2 = 4\langle \mathbf{u}, \mathbf{v} \rangle$ .
- (b) Write  $z = \langle \mathbf{u}, \mathbf{v} \rangle$ . Since  $|i^k| = 1$ , Theorem 10.1.2 (b) and (c) give  $\|i^k \mathbf{v}\| = \|\mathbf{v}\|$  and

$$\|\mathbf{u} + i^k \mathbf{v}\|^2 = \|\mathbf{u}\|^2 + \|\mathbf{v}\|^2 + 2\operatorname{Re}\langle \mathbf{u}, i^k \mathbf{v} \rangle = \|\mathbf{u}\|^2 + \|\mathbf{v}\|^2 + 2\operatorname{Re}(i^{-k} z),$$

where  $\langle \mathbf{u}, i^k \mathbf{v} \rangle = \overline{i^k} z = i^{-k} z$  by conjugate-linearity in the second slot. Multiply by  $i^k$  and sum over  $k = 0, 1, 2, 3$ . The first two terms contribute  $(\|\mathbf{u}\|^2 + \|\mathbf{v}\|^2) \sum_k i^k = 0$ , because  $1 + i - 1 - i = 0$ . For the remaining terms use  $2\operatorname{Re}(w) = w + \bar{w}$ :

$$\sum_{k=0}^3 i^k \cdot 2\operatorname{Re}(i^{-k} z) = \sum_{k=0}^3 i^k (i^{-k} z + i^k \bar{z}) = \sum_{k=0}^3 z + \bar{z} \sum_{k=0}^3 i^{2k} = 4z + 0,$$

since  $\sum_k i^{2k} = 1 - 1 + 1 - 1 = 0$ . Dividing by 4 gives the identity. ■

Now the main theorem. It says that length, angle, the adjoint and orthonormal bases all detect the same class of maps, and it is the reason isometries are easy to work with: whichever of the five descriptions a problem hands you, you may switch to whichever one you want.

### Theorem 10.7.3 — Characterizations of Isometries

Let  $V$  and  $W$  be finite-dimensional inner product spaces over  $F$ , and let  $T \in \mathcal{L}(V, W)$ . The following are equivalent.

- (a)  $T$  is an isometry:  $\|T\mathbf{v}\| = \|\mathbf{v}\|$  for every  $\mathbf{v} \in V$ .
- (b)  $T$  preserves inner products:  $\langle T\mathbf{u}, T\mathbf{v} \rangle = \langle \mathbf{u}, \mathbf{v} \rangle$  for all  $\mathbf{u}, \mathbf{v} \in V$ .
- (c)  $T^*T = \operatorname{id}_V$ .
- (d)  $T$  sends **every** orthonormal list in  $V$  to an orthonormal list in  $W$ .
- (e)  $T$  sends **some** orthonormal basis of  $V$  to an orthonormal list in  $W$ .
- (f) For one (equivalently every) pair of orthonormal bases  $\mathcal{B}$  of  $V$  and  $\mathcal{C}$  of  $W$ , the matrix  $\mathbf{A} = [T]_{\mathcal{B}}^{\mathcal{C}} \in M_{m \times n}(F)$ , where  $n = \dim V$  and  $m = \dim W$ , satisfies  $\mathbf{A}^* \mathbf{A} = \mathbf{I}_n$ ; that is, the  $n$  columns of  $\mathbf{A}$  are orthonormal vectors of  $F^m$ .



If in addition  $\dim W = \dim V$ , then (a)–(f) are further equivalent to each of

- (g1)  $T$  is invertible with  $T^{-1} = T^*$ ;
- (g2)  $TT^* = \text{id}_W$ .

*Idea.* Run the cycle  $(a) \Rightarrow (b) \Rightarrow (c) \Rightarrow (a)$ . The first step is polarization: lengths are all we are given, and [Lemma 10.7.2](#) converts them into an inner product. The second is the adjoint doing its job,  $\langle T\mathbf{u}, T\mathbf{v} \rangle = \langle \mathbf{u}, T^*T\mathbf{v} \rangle$ , plus non-degeneracy. The third is the same identity read backwards with  $\mathbf{u} = \mathbf{v}$ . Then  $(b) \Rightarrow (d) \Rightarrow (e) \Rightarrow (b)$  is a second, shorter cycle, and  $(c) \Leftrightarrow (f)$  is [Theorem 10.6.3](#). The last two need a dimension count:  $T^*T = \text{id}$  makes  $T$  injective, and in equal dimensions injective means invertible.

*Proof.*

- (a)  $\Rightarrow$  (b). Suppose  $\|T\mathbf{v}\| = \|\mathbf{v}\|$  for all  $\mathbf{v}$ . If  $F = \mathbb{R}$ , then by [Lemma 10.7.2](#) (a) applied in  $W$  and then in  $V$ , and by the linearity of  $T$ ,

$$\begin{aligned} \langle T\mathbf{u}, T\mathbf{v} \rangle &= \frac{1}{4} (\|T(\mathbf{u} + \mathbf{v})\|^2 - \|T(\mathbf{u} - \mathbf{v})\|^2) \\ &= \frac{1}{4} (\|\mathbf{u} + \mathbf{v}\|^2 - \|\mathbf{u} - \mathbf{v}\|^2) = \langle \mathbf{u}, \mathbf{v} \rangle. \end{aligned}$$

If  $F = \mathbb{C}$ , the same two lines work with [Lemma 10.7.2](#) (b), since  $T(\mathbf{u} + i^k\mathbf{v}) = T\mathbf{u} + i^kT\mathbf{v}$ .

- (b)  $\Rightarrow$  (c). For all  $\mathbf{u}, \mathbf{v} \in V$ , [Definition 10.6.1](#) gives  $\langle \mathbf{u}, T^*T\mathbf{v} \rangle = \langle T\mathbf{u}, T\mathbf{v} \rangle = \langle \mathbf{u}, \mathbf{v} \rangle$ , so  $\langle \mathbf{u}, T^*T\mathbf{v} - \mathbf{v} \rangle = 0$  for every  $\mathbf{u}$ . Taking  $\mathbf{u} = T^*T\mathbf{v} - \mathbf{v}$  makes its norm zero, so  $T^*T\mathbf{v} = \mathbf{v}$  for every  $\mathbf{v}$ .
- (c)  $\Rightarrow$  (a).  $\|T\mathbf{v}\|^2 = \langle T\mathbf{v}, T\mathbf{v} \rangle = \langle \mathbf{v}, T^*T\mathbf{v} \rangle = \langle \mathbf{v}, \mathbf{v} \rangle = \|\mathbf{v}\|^2$ , and norms are non-negative, so  $\|T\mathbf{v}\| = \|\mathbf{v}\|$ .
- (d)  $\Rightarrow$  (d). Let  $(\mathbf{u}_1, \dots, \mathbf{u}_k)$  be orthonormal in  $V$ , so  $\langle \mathbf{u}_i, \mathbf{u}_j \rangle = \delta_{ij}$ . Then  $\langle T\mathbf{u}_i, T\mathbf{u}_j \rangle = \langle \mathbf{u}_i, \mathbf{u}_j \rangle = \delta_{ij}$ , which is exactly what it means for  $(T\mathbf{u}_1, \dots, T\mathbf{u}_k)$  to be orthonormal ([Definition 10.2.1](#)).
- (e)  $\Rightarrow$  (e).  $V$  is finite-dimensional, so it has an orthonormal basis by [Theorem 10.2.2](#), and (d) applies to it.
- (f)  $\Rightarrow$  (b). Let  $\mathcal{B} = (\mathbf{v}_1, \dots, \mathbf{v}_n)$  be an orthonormal basis of  $V$  with  $(T\mathbf{v}_1, \dots, T\mathbf{v}_n)$  orthonormal. Given  $\mathbf{u}, \mathbf{v} \in V$ , expand them in  $\mathcal{B}$  as  $\mathbf{u} = \sum_i a_i \mathbf{v}_i$  and  $\mathbf{v} = \sum_j b_j \mathbf{v}_j$ . Expanding both slots and using orthonormality twice,

$$\langle T\mathbf{u}, T\mathbf{v} \rangle = \sum_{i,j} a_i \bar{b}_j \langle T\mathbf{v}_i, T\mathbf{v}_j \rangle = \sum_i a_i \bar{b}_i = \sum_{i,j} a_i \bar{b}_j \langle \mathbf{v}_i, \mathbf{v}_j \rangle = \langle \mathbf{u}, \mathbf{v} \rangle,$$

since in both double sums only the terms with  $i = j$  survive.

- (g)  $\Leftrightarrow$  (f). Fix orthonormal bases  $\mathcal{B}$  and  $\mathcal{C}$  and put  $\mathbf{A} = [T]_{\mathcal{B}}^{\mathcal{C}}$ . By [Theorem 10.6.3](#),  $[T^*]_{\mathcal{C}}^{\mathcal{B}} = \mathbf{A}^*$ , and by [Theorem 3.6.2](#),  $[T^*T]_{\mathcal{B}}^{\mathcal{B}} = \mathbf{A}^* \mathbf{A}$ . Since a map is determined by its matrix in fixed bases,  $T^*T = \text{id}_V$  if and only if  $\mathbf{A}^* \mathbf{A} = \mathbf{I}_n$ . The  $(i, j)$  entry of  $\mathbf{A}^* \mathbf{A}$  is the standard inner product of column  $j$  with column  $i$ , so this says the columns are orthonormal. As (c) does not mention the bases, the condition holds for one pair of orthonormal bases if and only if it holds for every pair.

Now assume  $\dim W = \dim V$ .

- (c)  $\Rightarrow$  (g1). By (c) and [Theorem 3.2.3](#),  $T$  is injective, so  $\text{rank } T = \dim V = \dim W$

by [Theorem 3.3.1](#), and  $T$  is surjective. Hence  $T$  is invertible, and multiplying  $T^*T = \text{id}_V$  on the right by  $T^{-1}$  gives  $T^* = T^{-1}$ .

(g1)  $\Rightarrow$  (g2).  $TT^* = TT^{-1} = \text{id}_W$ .

(g2)  $\Rightarrow$  (c).  $TT^* = \text{id}_W$  makes  $T$  surjective, hence invertible by the equal dimensions and [Theorem 3.3.1](#), and multiplying on the left by  $T^{-1}$  gives  $T^* = T^{-1}$ , so  $T^*T = \text{id}_V$ . This proves the theorem. ■

Condition (f) is the one to use on a concrete matrix: **check that the columns are orthonormal**. Condition (c) is the one to use in a proof. Condition (e) is the cheapest to verify on an abstract map, since one basis suffices, and (d) is what you get out afterwards.

*Warning.*  $T^*T = \text{id}$  **does not give**  $TT^* = \text{id}$  **when the dimensions differ**. Let

$T: \mathbb{R}^2 \rightarrow \mathbb{R}^3$  be  $T(x_1, x_2) = (x_1, x_2, 0)$ , with matrix  $\mathbf{A} = \begin{pmatrix} 1 & 0 \\ 0 & 1 \\ 0 & 0 \end{pmatrix}$ . Then  $\mathbf{A}^\top \mathbf{A} = \mathbf{I}_2$ , so

$T$  is an isometry; but

$$\mathbf{A}\mathbf{A}^\top = \begin{pmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 0 \end{pmatrix} \neq \mathbf{I}_3.$$

Ask *identity on what?* The composite  $TT^*$  is the orthogonal projection of  $\mathbb{R}^3$  onto  $\text{im } T$ , the  $x_1x_2$ -plane, and it is the identity there and nowhere else. Only when  $T$  is surjective can both products be identities.

*Quick check.* Is  $\begin{pmatrix} 1 & 1 \\ 1 & -1 \end{pmatrix}$  the matrix of an isometry of  $\mathbb{R}^2$ ? If not, is some scalar multiple of it?

*Solution.* No: the first column has norm  $\sqrt{2} \neq 1$ , so the columns are not orthonormal and condition (f) of [Theorem 10.7.3](#) fails. The columns are, however, orthogonal, so dividing by  $\sqrt{2}$  normalizes both and  $\frac{1}{\sqrt{2}} \begin{pmatrix} 1 & 1 \\ 1 & -1 \end{pmatrix}$  is an isometry.

### 10.7.3 Unitary and orthogonal maps

Conditions (g1) and (g2) describe the case that matters most, an isometry of a space with itself. It is the case with a name.

#### Definition 10.7.2 — Unitary and Orthogonal

Let  $V$  be a finite-dimensional inner product space over  $F$ . An operator  $T \in \mathcal{L}(V)$  is **unitary** if

$$T^*T = TT^* = \text{id}_V,$$

equivalently if  $T$  is an invertible isometry. When  $F = \mathbb{R}$ , such a  $T$  is also called **orthogonal**.

A matrix  $\mathbf{A} \in M_n(\mathbb{C})$  with  $\mathbf{A}^* \mathbf{A} = \mathbf{I}_n$  is a **unitary matrix**; a matrix  $\mathbf{A} \in M_n(\mathbb{R})$  with  $\mathbf{A}^\top \mathbf{A} = \mathbf{I}_n$  is an **orthogonal matrix**.

For a **square** matrix the one-sided condition  $\mathbf{A}^* \mathbf{A} = \mathbf{I}$  already forces  $\mathbf{A} \mathbf{A}^* = \mathbf{I}$ , by part (g2) of [Theorem 10.7.3](#), or directly by [Theorem 2.4.5](#). The warning above shows that squareness is not a formality.

### Definition 10.7.3 — The Unitary and Orthogonal Groups

For  $n \geq 1$  set

$$\mathrm{U}(n) := \{\mathbf{A} \in M_n(\mathbb{C}) : \mathbf{A}^* \mathbf{A} = \mathbf{I}_n\}, \quad \mathrm{O}(n) := \{\mathbf{A} \in M_n(\mathbb{R}) : \mathbf{A}^\top \mathbf{A} = \mathbf{I}_n\},$$

and  $\mathrm{SO}(n) := \{\mathbf{A} \in \mathrm{O}(n) : \det \mathbf{A} = 1\}$ , the **special orthogonal group**.

### Proposition 10.7.4 — The Groups and Their Determinants

- (a)  $\mathrm{U}(n)$  and  $\mathrm{O}(n)$  are groups under matrix multiplication, and  $\mathrm{O}(n) = \mathrm{U}(n) \cap M_n(\mathbb{R})$ .  
 (b) If  $\mathbf{A} \in \mathrm{U}(n)$  then  $|\det \mathbf{A}| = 1$ . If  $\mathbf{A} \in \mathrm{O}(n)$  then  $\det \mathbf{A} = 1$  or  $\det \mathbf{A} = -1$ .  
 (c)  $\mathrm{SO}(n)$  is a subgroup of  $\mathrm{O}(n)$ .

*Proof.*

- (a)  $\mathbf{I}^* \mathbf{I} = \mathbf{I}$ , so  $\mathbf{I} \in \mathrm{U}(n)$ . If  $\mathbf{A}, \mathbf{B} \in \mathrm{U}(n)$  then  $(\mathbf{AB})^* (\mathbf{AB}) = \mathbf{B}^* \mathbf{A}^* \mathbf{AB} = \mathbf{B}^* \mathbf{B} = \mathbf{I}$ , so  $\mathbf{AB} \in \mathrm{U}(n)$ . Each  $\mathbf{A} \in \mathrm{U}(n)$  is invertible with  $\mathbf{A}^{-1} = \mathbf{A}^*$ , and  $(\mathbf{A}^*)^* \mathbf{A}^* = \mathbf{AA}^* = \mathbf{I}$ , so  $\mathbf{A}^{-1} \in \mathrm{U}(n)$ . Associativity is inherited from matrix multiplication. The same three lines prove the claim for  $\mathrm{O}(n)$ , with  $\mathbf{A}^* = \mathbf{A}^\top$  for a real matrix, which is also the last assertion.
- (b) The determinant is the sum  $\sum_{\sigma} \operatorname{sgn}(\sigma) a_{\sigma(1)1} \cdots a_{\sigma(n)n}$  of [Definition 6.4.1](#), and conjugation of complex numbers respects sums and products, so  $\det \overline{\mathbf{A}} = \overline{\det \mathbf{A}}$ . With [Theorem 6.4.5](#) this gives  $\det \mathbf{A}^* = \det(\overline{\mathbf{A}^\top}) = \det \overline{\mathbf{A}} = \overline{\det \mathbf{A}}$ . Hence, by [Theorem 6.5.4](#),

$$1 = \det \mathbf{I}_n = \det(\mathbf{A}^* \mathbf{A}) = \overline{\det \mathbf{A}} \det \mathbf{A} = |\det \mathbf{A}|^2.$$

For real  $\mathbf{A}$  the determinant is real, and the real numbers of modulus 1 are  $\pm 1$ .

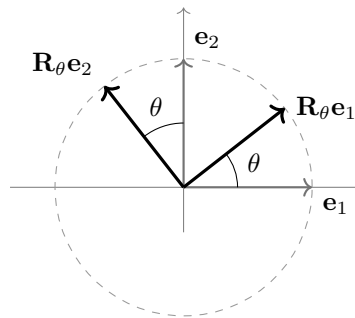
- (c)  $\det \mathbf{I} = 1$ , and  $\det$  is multiplicative ([Theorem 6.5.4](#)), so  $\det(\mathbf{AB}) = 1$  and  $\det(\mathbf{A}^{-1}) = (\det \mathbf{A})^{-1} = 1$  whenever  $\det \mathbf{A} = \det \mathbf{B} = 1$ . ■

*Warning.*  $|\lambda| = 1$  **does not mean**  $\lambda = \pm 1$ , and  $\det \mathbf{A} = 1$  **does not mean** “rotation”. The matrix  $-\mathbf{I}_4 \in \mathrm{SO}(4)$  has determinant  $(-1)^4 = 1$ , but it fixes no non-zero vector, so it is not a rotation about an axis in any sense. Likewise  $-\mathbf{I}_3 \in \mathrm{O}(3)$  has determinant  $-1$  but is not a reflection in a plane: a reflection fixes a plane pointwise, and  $-\mathbf{I}_3$  fixes only  $\mathbf{0}$ . The clean classification below is a feature of dimensions 2 and 3, not a general theorem.

## 10.7.4 Rotations and reflections of the plane

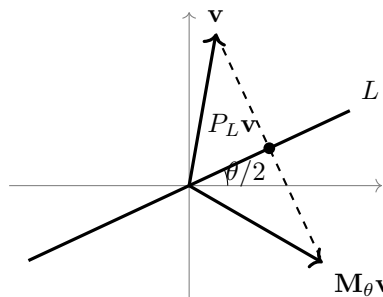
In dimension 2 the group  $O(2)$  can be written down completely, and the two determinant values split it into exactly the two families the words suggest. For  $\theta \in \mathbb{R}$  put

$$\mathbf{R}_\theta := \begin{pmatrix} \cos \theta & -\sin \theta \\ \sin \theta & \cos \theta \end{pmatrix}, \quad \mathbf{M}_\theta := \begin{pmatrix} \cos \theta & \sin \theta \\ \sin \theta & -\cos \theta \end{pmatrix}.$$



the columns of  $\mathbf{R}_\theta$  are the rotated axes

The first matrix rotates the plane by  $\theta$ : its columns are  $\mathbf{e}_1$  and  $\mathbf{e}_2$  turned through  $\theta$ . The second is a reflection, and the picture says in which mirror.



$\mathbf{M}_\theta$  reflects in the line  $L$  at angle  $\theta/2$

### Theorem 10.7.5 — Classification of the Plane Isometries

Let  $\mathbf{A} \in O(2)$ .

- (a) If  $\det \mathbf{A} = 1$ , then  $\mathbf{A} = \mathbf{R}_\theta$  for a unique  $\theta \in [0, 2\pi)$ .
- (b) If  $\det \mathbf{A} = -1$ , then  $\mathbf{A} = \mathbf{M}_\theta$  for a unique  $\theta \in [0, 2\pi)$ . In this case  $\mathbf{A}^2 = \mathbf{I}$ , the eigenvalues of  $\mathbf{A}$  are 1 and  $-1$ , and  $\mathbf{A}$  is the reflection in the line  $L = \text{span}((\cos \frac{\theta}{2}, \sin \frac{\theta}{2}))$ : it fixes  $L$  pointwise and negates  $L^\perp$ .

Consequently  $O(2) = \{\mathbf{R}_\theta\} \cup \{\mathbf{M}_\theta\}$ , and  $SO(2) = \{\mathbf{R}_\theta : \theta \in [0, 2\pi)\}$ .

*Idea.* Condition (f) of Theorem 10.7.3 says the two columns are orthonormal. A unit vector of  $\mathbb{R}^2$  is  $(\cos \theta, \sin \theta)$  for exactly one  $\theta \in [0, 2\pi)$ , and there are exactly two unit vectors orthogonal to it. So there are only two candidates once the first column is chosen,

and the determinant tells them apart. The geometric description of  $\mathbf{M}_\theta$  is then a two-line verification: guess the fixed vector from the picture and check it.

*Proof.* Write  $\mathbf{A} = \begin{pmatrix} a & c \\ b & d \end{pmatrix} \in O(2)$ . By [Theorem 10.7.3](#) (f), the columns are orthonormal:  $a^2 + b^2 = 1$ ,  $c^2 + d^2 = 1$  and  $ac + bd = 0$ . Since  $(a, b)$  is a unit vector of  $\mathbb{R}^2$ , there is a unique  $\theta \in [0, 2\pi)$  with  $a = \cos \theta$  and  $b = \sin \theta$ . The vectors orthogonal to  $(a, b)$  form a line, spanned by  $(-b, a)$ , which is already a unit vector; so the unit vectors on that line are  $\pm(-b, a)$ , and  $(c, d) = \pm(-\sin \theta, \cos \theta)$ . The two choices give

$$\begin{pmatrix} \cos \theta & -\sin \theta \\ \sin \theta & \cos \theta \end{pmatrix} = \mathbf{R}_\theta \quad \text{and} \quad \begin{pmatrix} \cos \theta & \sin \theta \\ \sin \theta & -\cos \theta \end{pmatrix} = \mathbf{M}_\theta,$$

with determinants  $\cos^2 \theta + \sin^2 \theta = 1$  and  $-\cos^2 \theta - \sin^2 \theta = -1$ . This proves (a) and the first sentence of (b), and uniqueness of  $\theta$  in both cases, since  $\theta$  is already determined by the first column.

For the rest of (b), put  $\mathbf{u} = (\cos \frac{\theta}{2}, \sin \frac{\theta}{2})$  and  $\mathbf{w} = (-\sin \frac{\theta}{2}, \cos \frac{\theta}{2})$ , two orthogonal unit vectors. Writing  $\theta = \frac{\theta}{2} + \frac{\theta}{2}$  and using the subtraction formulas for cosine and sine,

$$\mathbf{M}_\theta \mathbf{u} = \begin{pmatrix} \cos \theta \cos \frac{\theta}{2} + \sin \theta \sin \frac{\theta}{2} \\ \sin \theta \cos \frac{\theta}{2} - \cos \theta \sin \frac{\theta}{2} \end{pmatrix} = \begin{pmatrix} \cos \frac{\theta}{2} \\ \sin \frac{\theta}{2} \end{pmatrix} = \mathbf{u},$$

$$\mathbf{M}_\theta \mathbf{w} = \begin{pmatrix} -\cos \theta \sin \frac{\theta}{2} + \sin \theta \cos \frac{\theta}{2} \\ -\sin \theta \sin \frac{\theta}{2} - \cos \theta \cos \frac{\theta}{2} \end{pmatrix} = \begin{pmatrix} \sin \frac{\theta}{2} \\ -\cos \frac{\theta}{2} \end{pmatrix} = -\mathbf{w}.$$

So  $(\mathbf{u}, \mathbf{w})$  is an orthonormal basis of eigenvectors with eigenvalues 1 and  $-1$ , and  $\mathbf{M}_\theta^2$  fixes both of them, hence  $\mathbf{M}_\theta^2 = \mathbf{I}$ . Since  $L = \text{span}(\mathbf{u})$ , we have  $L^\perp = \text{span}(\mathbf{w})$ , because  $\mathbf{w} \perp \mathbf{u}$  and  $\dim L^\perp = 1$  ([Theorem 10.3.2](#) (c)), so  $\mathbf{M}_\theta$  fixes  $L$  pointwise and negates  $L^\perp$ , which is the description of a reflection. This proves the theorem. ■

So  $SO(2)$  is the circle of rotations, and the other half of  $O(2)$  is the circle of reflections. Multiplying two rotations adds the angles,  $\mathbf{R}_\alpha \mathbf{R}_\beta = \mathbf{R}_{\alpha+\beta}$ , so  $SO(2)$  is abelian; [Exercise \(C3: Two reflections make a rotation\)](#) shows what happens when two reflections are multiplied.

### Example. (Naming Two Plane Isometries)

Classify each of the following as a rotation or a reflection, and give the angle or the mirror line.

(a)  $\mathbf{A} = \frac{1}{2} \begin{pmatrix} 1 & -\sqrt{3} \\ \sqrt{3} & 1 \end{pmatrix}.$

(b)  $\mathbf{B} = \frac{1}{5} \begin{pmatrix} 3 & 4 \\ 4 & -3 \end{pmatrix}.$

*Solution.*

- (a) The columns  $\frac{1}{2}(1, \sqrt{3})$  and  $\frac{1}{2}(-\sqrt{3}, 1)$  have norm  $\frac{1}{2}\sqrt{1+3} = 1$  and inner product  $\frac{1}{4}(-\sqrt{3} + \sqrt{3}) = 0$ , so  $\mathbf{A} \in O(2)$ . Its determinant is  $\frac{1}{4}(1+3) = 1$ , so  $\mathbf{A} = \mathbf{R}_\theta$  with

$\cos \theta = \frac{1}{2}$  and  $\sin \theta = \frac{\sqrt{3}}{2}$ : rotation by  $\theta = \pi/3$ .

- (b) The columns  $\frac{1}{5}(3, 4)$  and  $\frac{1}{5}(4, -3)$  are orthonormal, and  $\det \mathbf{B} = \frac{1}{25}(-9 - 16) = -1$ , so  $\mathbf{B} = \mathbf{M}_\theta$  with  $\cos \theta = \frac{3}{5}$  and  $\sin \theta = \frac{4}{5}$ . The mirror is  $L = \text{span}((\cos \frac{\theta}{2}, \sin \frac{\theta}{2}))$ , and rather than halve the angle we find the fixed line directly:  $\mathbf{B}\mathbf{x} = \mathbf{x}$  reads  $3x_1 + 4x_2 = 5x_1$ , that is  $x_1 = 2x_2$ . So  $L = \text{span}((2, 1))$ . Check:  $\mathbf{B}(2, 1) = \frac{1}{5}(6 + 4, 8 - 3) = (2, 1)$ , and  $\mathbf{B}(-1, 2) = \frac{1}{5}(-3 + 8, -4 - 6) = (1, -2) = -(-1, 2)$ , as a reflection must do to  $L^\perp$ .

## 10.7.5 Rotations of space

In dimension 3 the answer is the one physics assumes: every orientation-preserving isometry of  $\mathbb{R}^3$  is a rotation about a line. The proof is where Chapter 8 is needed, for one reason only: to produce an eigenvector.

### Theorem 10.7.6 — Every Element of $\text{SO}(3)$ Is a Rotation

Let  $\mathbf{A} \in \text{SO}(3)$ . Then there is a unit vector  $\mathbf{u} \in \mathbb{R}^3$  with  $\mathbf{A}\mathbf{u} = \mathbf{u}$ , and an orthonormal basis  $\mathcal{B} = (\mathbf{u}, \mathbf{w}_1, \mathbf{w}_2)$  of  $\mathbb{R}^3$  in which

$$[T_{\mathbf{A}}]_{\mathcal{B}} = \begin{pmatrix} 1 & 0 & 0 \\ 0 & \cos \theta & -\sin \theta \\ 0 & \sin \theta & \cos \theta \end{pmatrix}$$

for some  $\theta$ . That is,  $\mathbf{A}$  fixes the **axis**  $\text{span}(\mathbf{u})$  and rotates the plane  $\text{span}(\mathbf{u})^\perp$  by  $\theta$ . Moreover  $\text{tr } \mathbf{A} = 1 + 2 \cos \theta$ .

*Idea.* Three steps. ① Get a real eigenvalue: the characteristic polynomial is real of degree 3, and an odd-degree real polynomial has a real root. ② That eigenvalue is  $\pm 1$ , since  $T_{\mathbf{A}}$  is an isometry. ③ Split off the eigenline: its orthogonal complement is invariant, the restriction there lies in  $\text{O}(2)$ , and Theorem 10.7.5 finishes. If the eigenvalue was  $-1$  the restriction has determinant  $-1$ , so it is a reflection and supplies a  $+1$  eigenvector inside the plane; feed that back into step ③.

*Proof.* Write  $T = T_{\mathbf{A}}$ , an isometry of  $\mathbb{R}^3$  by Theorem 10.7.3.

**Step 1: a real eigenvalue exists.** By Theorem 6.10.2,  $p_{\mathbf{A}} \in \mathbb{R}[x]$  is monic of degree 3, which is odd, so  $p_{\mathbf{A}}$  has a real root  $\lambda$  by Corollary 5.5.9, and  $\lambda$  is an eigenvalue of  $\mathbf{A}$  by Theorem 8.2.1. By Proposition 10.7.1,  $\lambda = 1$  or  $\lambda = -1$ . Fix a unit eigenvector  $\mathbf{u}_0$  for  $\lambda$ , possible after dividing by its norm.

**Step 2: the orthogonal complement of an invariant line is invariant.**

*Claim.* If  $U \subseteq \mathbb{R}^3$  is a subspace with  $T(U) \subseteq U$ , then  $T(U^\perp) \subseteq U^\perp$ .

*Proof.*  $T|_U: U \rightarrow U$  is injective, because  $T$  is, so it is surjective by [Theorem 3.5.7](#) applied on the finite-dimensional space  $U$ . Let  $\mathbf{w} \in U^\perp$  and  $\mathbf{u} \in U$ . Write  $\mathbf{u} = T\mathbf{u}'$  with  $\mathbf{u}' \in U$ . Then  $\langle T\mathbf{w}, \mathbf{u} \rangle = \langle T\mathbf{w}, T\mathbf{u}' \rangle = \langle \mathbf{w}, \mathbf{u}' \rangle = 0$ , the middle step by [Theorem 10.7.3](#) (b) and the last because  $\mathbf{u}' \in U$ . So  $T\mathbf{w} \in U^\perp$ . ■

**Step 3: reduce to the plane.** Put  $U = \text{span}(\mathbf{u}_0)$ , which is  $T$ -invariant. By the claim,  $P := U^\perp$  is  $T$ -invariant, and  $\dim P = 2$  by [Theorem 10.3.2](#) (c). Choose an orthonormal basis  $(\mathbf{w}_1, \mathbf{w}_2)$  of  $P$  ([Theorem 10.2.2](#)); then  $\mathcal{B}_0 = (\mathbf{u}_0, \mathbf{w}_1, \mathbf{w}_2)$  is an orthonormal basis of  $\mathbb{R}^3$ , and in it

$$[T]_{\mathcal{B}_0}^{\mathcal{B}_0} = \begin{pmatrix} \lambda & 0 & 0 \\ 0 & c_{11} & c_{12} \\ 0 & c_{21} & c_{22} \end{pmatrix}, \quad \mathbf{C} := \begin{pmatrix} c_{11} & c_{12} \\ c_{21} & c_{22} \end{pmatrix},$$

the first column because  $T\mathbf{u}_0 = \lambda\mathbf{u}_0$ , and the two zeros in the first row because  $T\mathbf{w}_1, T\mathbf{w}_2 \in P$  have no  $\mathbf{u}_0$ -component. This matrix is orthogonal, being the matrix of an isometry in orthonormal bases ([Theorem 10.7.3](#) (f)), so its second and third columns are orthonormal, which says  $\mathbf{C}^\top \mathbf{C} = \mathbf{I}_2$  and  $\mathbf{C} \in \text{O}(2)$ ; and  $\mathbf{C}$  is the matrix of  $T|_P$  in the basis  $(\mathbf{w}_1, \mathbf{w}_2)$ . Expanding the determinant along the first column, and using that similar matrices have equal determinants ([Corollary 6.5.7](#), with [Theorem 3.7.4](#) (a)),  $1 = \det \mathbf{A} = \lambda \det \mathbf{C}$ , so  $\det \mathbf{C} = \lambda$ .

*Case  $\lambda = 1$ .* Then  $\det \mathbf{C} = 1$ , so  $\mathbf{C} = \mathbf{R}_\theta$  for some  $\theta$  by [Theorem 10.7.5](#) (a), and  $\mathcal{B} = \mathcal{B}_0$  and  $\mathbf{u} = \mathbf{u}_0$  are as required.

*Case  $\lambda = -1$ .* Then  $\det \mathbf{C} = -1$ , so by [Theorem 10.7.5](#) (b) the restriction  $T|_P$  is a reflection of the plane  $P$  and has 1 as an eigenvalue. Let  $\mathbf{u} \in P$  be a unit eigenvector for 1, so  $T\mathbf{u} = \mathbf{u}$ . Now rerun Steps 2 and 3 with  $\mathbf{u}$  in place of  $\mathbf{u}_0$ . Because  $\mathbf{u}$  was chosen with  $T\mathbf{u} = \mathbf{u}$ , the eigenvalue this time is 1, so the rerun lands in the previous case and cannot return here.

Finally,  $\mathbf{A}$  and  $[T]_{\mathcal{B}}^{\mathcal{B}}$  are matrices of the same operator, hence similar ([Theorem 3.7.4](#) (a)), and similar matrices have equal traces ([Theorem 3.7.5](#)), so  $\text{tr } \mathbf{A} = 1 + 2 \cos \theta$ . This proves the theorem. ■

The angle is determined by  $\mathbf{A}$  up to sign, since reversing the order of  $\mathbf{w}_1, \mathbf{w}_2$  replaces  $\theta$  by  $-\theta$ ; the trace formula  $\cos \theta = \frac{1}{2}(\text{tr } \mathbf{A} - 1)$  gives  $|\theta|$  with no computation at all. This settles  $\text{O}(3)$  as well: if  $\det \mathbf{A} = -1$  then  $-\mathbf{A} \in \text{SO}(3)$ , so  $\mathbf{A}$  is a rotation composed with the antipodal map.

#### Example. (Axis and Angle of a Rotation)

Let  $\mathbf{A} = \frac{1}{3} \begin{pmatrix} 2 & -2 & 1 \\ 1 & 2 & 2 \\ -2 & -1 & 2 \end{pmatrix}$ . Verify that  $\mathbf{A} \in \text{SO}(3)$  and find its axis and angle.

*Solution. Orthogonality.* The three columns are  $\frac{1}{3}(2, 1, -2)$ ,  $\frac{1}{3}(-2, 2, -1)$  and  $\frac{1}{3}(1, 2, 2)$ . Their squared norms are  $\frac{1}{9}(4+1+4) = 1$ ,  $\frac{1}{9}(4+4+1) = 1$  and  $\frac{1}{9}(1+4+4) = 1$ , and their pairwise inner products are  $\frac{1}{9}(-4+2+2) = 0$ ,  $\frac{1}{9}(2+2-4) = 0$  and  $\frac{1}{9}(-2+4-2) = 0$ . So the columns are orthonormal and  $\mathbf{A} \in O(3)$  by [Theorem 10.7.3](#) (f).

*Determinant.* Expanding along the first row,  $27 \det \mathbf{A} = 2(4+2) - (-2)(2+4) + 1(-1+4) = 12 + 12 + 3 = 27$ , so  $\det \mathbf{A} = 1$  and  $\mathbf{A} \in SO(3)$ .

*Axis.* Solve  $\mathbf{A}\mathbf{u} = \mathbf{u}$ , that is  $(3\mathbf{I} - 3\mathbf{A})\mathbf{u} = \mathbf{0}$ :

$$\begin{pmatrix} 1 & 2 & -1 \\ -1 & 1 & -2 \\ 2 & 1 & 1 \end{pmatrix} \mathbf{u} = \mathbf{0}.$$

Adding row one to row two gives  $3u_2 - 3u_3 = 0$ , so  $u_2 = u_3$ ; then row one gives  $u_1 = -u_2$ . The axis is  $\text{span}((-1, 1, 1))$ , and indeed  $\mathbf{A}(-1, 1, 1) = \frac{1}{3}(-2-2+1, -1+2+2, 2-1+2) = (-1, 1, 1)$ .

*Angle.*  $\text{tr } \mathbf{A} = \frac{1}{3}(2+2+2) = 2$ , so  $\cos \theta = \frac{1}{2}(2-1) = \frac{1}{2}$  and  $|\theta| = \pi/3$ . The map is a rotation by  $60^\circ$  about the line through  $(-1, 1, 1)$ .

*Quick check.* A matrix  $\mathbf{A} \in SO(3)$  has  $\text{tr } \mathbf{A} = -1$ . What is its angle, and what is  $\mathbf{A}^2$ ?

*Solution.*  $\cos \theta = \frac{1}{2}(-1-1) = -1$ , so  $\theta = \pi$ : a half-turn about the axis. Then  $\mathbf{A}^2$  is the rotation by  $2\pi$  about the same axis, which is  $\mathbf{I}_3$ . (In the basis of [Theorem 10.7.6](#) the matrix is  $\text{diag}(1, -1, -1)$ , whose square is  $\mathbf{I}_3$ .)

## 10.7.6 Rigid motions

We assumed linearity from the start. It is worth knowing that we did not have to: for maps of  $\mathbb{R}^n$  that fix the origin, preserving distance forces linearity.

### Theorem 10.7.7 — Distance-Preserving Maps Are Affine

Let  $f: \mathbb{R}^n \rightarrow \mathbb{R}^n$  be any function, not assumed linear, with

$$\|f(\mathbf{x}) - f(\mathbf{y})\| = \|\mathbf{x} - \mathbf{y}\| \quad \text{for all } \mathbf{x}, \mathbf{y} \in \mathbb{R}^n.$$

- (a) If  $f(\mathbf{0}) = \mathbf{0}$ , then  $f$  is linear and orthogonal.
- (b) In general,  $f(\mathbf{x}) = \mathbf{A}\mathbf{x} + \mathbf{b}$  for a unique  $\mathbf{A} \in O(n)$  and a unique  $\mathbf{b} \in \mathbb{R}^n$ .

*Idea.* Setting  $\mathbf{y} = \mathbf{0}$  turns the hypothesis into “preserves norms”, and expanding the squared distance then turns it into “preserves inner products” without any polarization at all. Once inner products are preserved,  $f$  sends the standard basis to an orthonormal basis, and [Theorem 10.2.1](#) writes  $f(\mathbf{x})$  as an explicit linear expression in the coordinates of  $\mathbf{x}$ . Part (b) is part (a) applied to  $\mathbf{x} \mapsto f(\mathbf{x}) - f(\mathbf{0})$ .

*Proof.*



- (a) Assume  $f(\mathbf{0}) = \mathbf{0}$ . Taking  $\mathbf{y} = \mathbf{0}$  gives  $\|f(\mathbf{x})\| = \|\mathbf{x}\|$  for all  $\mathbf{x}$ . Expanding both sides of the hypothesis with [Theorem 10.1.2](#) (c),

$$\|f(\mathbf{x})\|^2 - 2\langle f(\mathbf{x}), f(\mathbf{y}) \rangle + \|f(\mathbf{y})\|^2 = \|\mathbf{x}\|^2 - 2\langle \mathbf{x}, \mathbf{y} \rangle + \|\mathbf{y}\|^2,$$

and the four outer terms cancel in pairs, leaving

$$\langle f(\mathbf{x}), f(\mathbf{y}) \rangle = \langle \mathbf{x}, \mathbf{y} \rangle \quad \text{for all } \mathbf{x}, \mathbf{y}. \quad (*)$$

Put  $\mathbf{f}_i := f(\mathbf{e}_i)$ . By  $(*)$ ,  $\langle \mathbf{f}_i, \mathbf{f}_j \rangle = \langle \mathbf{e}_i, \mathbf{e}_j \rangle = \delta_{ij}$ , so  $(\mathbf{f}_1, \dots, \mathbf{f}_n)$  is an orthonormal list of  $n$  vectors in  $\mathbb{R}^n$ , hence an orthonormal basis (it is independent by [Theorem 10.1.4](#) and has  $n = \dim \mathbb{R}^n$  elements). Now let  $\mathbf{x} = (x_1, \dots, x_n)$  be arbitrary. By [Theorem 10.2.1](#) applied to the basis  $(\mathbf{f}_1, \dots, \mathbf{f}_n)$ , and then by  $(*)$ ,

$$f(\mathbf{x}) = \sum_{i=1}^n \langle f(\mathbf{x}), \mathbf{f}_i \rangle \mathbf{f}_i = \sum_{i=1}^n \langle f(\mathbf{x}), f(\mathbf{e}_i) \rangle \mathbf{f}_i = \sum_{i=1}^n \langle \mathbf{x}, \mathbf{e}_i \rangle \mathbf{f}_i = \sum_{i=1}^n x_i \mathbf{f}_i.$$

The right-hand side is linear in  $\mathbf{x}$ , so  $f$  is linear; and it preserves norms, so it is an isometry, hence orthogonal by [Definition 10.7.2](#), since it maps  $\mathbb{R}^n$  to  $\mathbb{R}^n$ .

- (b) Put  $\mathbf{b} := f(\mathbf{0})$  and  $g(\mathbf{x}) := f(\mathbf{x}) - \mathbf{b}$ . Then  $g$  preserves distances, since the translation cancels, and  $g(\mathbf{0}) = \mathbf{0}$ ; by (a),  $g$  is orthogonal, say  $g(\mathbf{x}) = \mathbf{A}\mathbf{x}$  with  $\mathbf{A} \in O(n)$ , and  $f(\mathbf{x}) = \mathbf{A}\mathbf{x} + \mathbf{b}$ . For uniqueness, evaluating at  $\mathbf{0}$  recovers  $\mathbf{b}$ , and then  $\mathbf{A}\mathbf{x} = f(\mathbf{x}) - \mathbf{b}$  determines  $\mathbf{A}$ . This proves the theorem. ■

So the rigid motions of  $\mathbb{R}^n$  are exactly the maps “rotate or reflect, then translate”, and the linear ones are exactly the orthogonal maps. That is the sense in which  $O(n)$  is the symmetry group of Euclidean geometry, and it is why the same group reappears whenever a problem has a distance in it.

## 10.7.7 Exercises

### 10.7.7.1 A. Check your understanding

#### Exercise A1.

- Define an isometry  $T \in \mathcal{L}(V, W)$ , and state two conditions equivalent to it.
- What must be checked about a square matrix  $\mathbf{A}$  to decide whether it is unitary? State the check in terms of its columns.
- True or false: every isometry is invertible. Justify your answer.
- True or false: if  $\det \mathbf{A} = 1$  for  $\mathbf{A} \in O(n)$ , then  $\mathbf{A}$  fixes a non-zero vector. Justify your answer.
- Give the two families of matrices making up  $O(2)$ , and say how the determinant distinguishes them.
- An operator  $T$  on a complex inner product space is unitary. Why must every eigenvalue of  $T$  lie on the unit circle?

*Solution.*

- (a)  $\|T\mathbf{v}\| = \|\mathbf{v}\|$  for every  $\mathbf{v} \in V$  (Definition 10.7.1). Equivalent:  $\langle T\mathbf{u}, T\mathbf{v} \rangle = \langle \mathbf{u}, \mathbf{v} \rangle$  for all  $\mathbf{u}, \mathbf{v}$ ; and  $T^*T = \text{id}_V$  (Theorem 10.7.3).
- (b) That its columns are orthonormal in  $F^n$ , that is  $\mathbf{A}^*\mathbf{A} = \mathbf{I}_n$  (Theorem 10.7.3 (f)). For a square matrix this already gives  $\mathbf{A}\mathbf{A}^* = \mathbf{I}_n$ .
- (c) False.  $T: \mathbb{R}^2 \rightarrow \mathbb{R}^3$ ,  $T(x_1, x_2) = (x_1, x_2, 0)$ , is an isometry and is not surjective (Example (Isometries and Near-Isometries) (a)). An isometry is always injective, and it is invertible exactly when  $\dim W = \dim V$ .
- (d) False for even  $n$ :  $-\mathbf{I}_4 \in \text{SO}(4)$  has determinant 1 and fixes only  $\mathbf{0}$ . It is true for odd  $n$ , though not by the argument of Theorem 10.7.6, which uses the  $\text{O}(2)$  classification of a 2-dimensional complement. The general reason is a count: the non-real eigenvalues of a real  $\mathbf{A}$  come in conjugate pairs  $\lambda, \bar{\lambda}$  with  $\lambda\bar{\lambda} = |\lambda|^2 = 1$  for an isometry, so they contribute 1 to the determinant and an even number to the count. With  $n$  odd, an odd number of eigenvalues are real, each  $\pm 1$ , and their product is  $\det \mathbf{A} = 1$ ; an odd number of factors  $\pm 1$  with product 1 cannot all be  $-1$ , so 1 is an eigenvalue and some non-zero vector is fixed.
- (e) The rotations  $\mathbf{R}_\theta$ , with determinant 1, and the reflections  $\mathbf{M}_\theta$ , with determinant  $-1$  (Theorem 10.7.5).
- (f) A unitary operator is an isometry, so Proposition 10.7.1 applies:  $|\lambda| = 1$ .

### 10.7.7.2 B. Practice

#### Exercise B1: Determine which are unitary.

Determine which of the following matrices are orthogonal or unitary. Justify your answer; for those that are not, name the failing condition.

- (a)  $\frac{1}{3} \begin{pmatrix} 2 & -2 & 1 \\ 1 & 2 & 2 \\ -2 & -1 & 2 \end{pmatrix}$  over  $\mathbb{R}$ .
- (b)  $\frac{1}{\sqrt{2}} \begin{pmatrix} 1 & i \\ i & 1 \end{pmatrix}$  over  $\mathbb{C}$ .
- (c)  $\begin{pmatrix} 1 & 1 \\ 0 & 1 \end{pmatrix}$  over  $\mathbb{R}$ .
- (d)  $\frac{1}{5} \begin{pmatrix} 3 & 4 \\ 4 & 3 \end{pmatrix}$  over  $\mathbb{R}$ .

*Solution.*

- (a) Orthogonal, as computed in Example (Axis and Angle of a Rotation): the three columns are orthonormal.
- (b) Unitary. The columns are  $\frac{1}{\sqrt{2}}(1, i)$  and  $\frac{1}{\sqrt{2}}(i, 1)$ , of norm  $\frac{1}{\sqrt{2}}\sqrt{1+1} = 1$ , and their inner product is

$$\frac{1}{2}(1 \cdot \bar{i} + i \cdot \bar{1}) = \frac{1}{2}(-i + i) = 0.$$

So  $\mathbf{A}^*\mathbf{A} = \mathbf{I}_2$ . (It is unitary but not orthogonal, since it is not real.)

- (c) Not orthogonal. The second column  $(1, 1)$  has norm  $\sqrt{2} \neq 1$ , so the columns are not orthonormal. Equivalently, the map stretches  $\mathbf{e}_2$ .

- (d) Not orthogonal. Both columns are unit vectors,  $\frac{1}{5}(3, 4)$  and  $\frac{1}{5}(4, 3)$ , but their inner product is  $\frac{1}{25}(12 + 12) = \frac{24}{25} \neq 0$ . Unit columns are not enough; they must also be orthogonal. (Its determinant is  $\frac{1}{25}(9 - 16) = -\frac{7}{25}$ , neither 1 nor  $-1$ , which confirms it by [Proposition 10.7.4](#).)

### Exercise B2: Name the plane isometry.

Each of the following lies in  $O(2)$ . Decide whether it is a rotation or a reflection, and give the angle of rotation or the mirror line.

- (a)  $\begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}$ .  
 (b)  $\frac{1}{\sqrt{2}} \begin{pmatrix} 1 & 1 \\ -1 & 1 \end{pmatrix}$ .  
 (c)  $\frac{1}{13} \begin{pmatrix} 5 & 12 \\ 12 & -5 \end{pmatrix}$ .

*Solution.*

- (a) Determinant  $-1$ , so a reflection.  $\mathbf{Ax} = \mathbf{x}$  reads  $x_2 = x_1$ , so the mirror is  $L = \text{span}((1, 1))$ , the line  $x_2 = x_1$ . Check:  $\mathbf{A}(1, -1) = (-1, 1) = -(1, -1)$ .  
 (b) Determinant  $\frac{1}{2}(1 + 1) = 1$ , so a rotation  $\mathbf{R}_\theta$ . Comparing with  $\mathbf{R}_\theta$ ,  $\cos \theta = \frac{1}{\sqrt{2}}$  and  $\sin \theta = -\frac{1}{\sqrt{2}}$ , so  $\theta = -\pi/4$ : a clockwise half of a right angle.  
 (c) Determinant  $\frac{1}{169}(-25 - 144) = -1$ , so a reflection.  $\mathbf{Ax} = \mathbf{x}$  reads  $5x_1 + 12x_2 = 13x_1$ , that is  $3x_2 = 2x_1$ , so  $L = \text{span}((3, 2))$ . Check:  $\mathbf{A}(3, 2) = \frac{1}{13}(15 + 24, 36 - 10) = (3, 2)$ .

### Exercise B3: Permutations are orthogonal.

Let  $\sigma \in S_n$  and let  $\mathbf{P}_\sigma \in M_n(\mathbb{R})$  be its permutation matrix ([Definition 2.6.2](#)), whose  $j$ -th column is  $\mathbf{e}_{\sigma(j)}$ .

- (a) Prove that  $\mathbf{P}_\sigma \in O(n)$ .  
 (b) Deduce that  $\det \mathbf{P}_\sigma = \pm 1$ , and identify the sign.  
 (c) Show that  $\mathbf{P}_\sigma^{-1} = \mathbf{P}_\sigma^\top = \mathbf{P}_{\sigma^{-1}}$ .

*Solution.*

- (a) Column  $j$  of  $\mathbf{P}_\sigma$  is  $\mathbf{e}_{\sigma(j)}$ , by [Definition 2.6.2](#). So each column is a unit vector, and for  $i \neq j$  the columns  $\mathbf{e}_{\sigma(i)}$  and  $\mathbf{e}_{\sigma(j)}$  are distinct standard basis vectors, because  $\sigma$  is injective, hence orthogonal. By [Theorem 10.7.3](#) (f),  $\mathbf{P}_\sigma \in O(n)$ .  
 (b) By [Proposition 10.7.4](#) (b),  $\det \mathbf{P}_\sigma = \pm 1$ . The sign is  $\text{sgn}(\sigma)$ : permuting the columns of  $\mathbf{I}_n$  by  $\sigma$  multiplies the determinant by  $\text{sgn}(\sigma)$  ([Theorem 6.2.1](#)), and  $\det \mathbf{I}_n = 1$ .  
 (c)  $\mathbf{P}_\sigma^{-1} = \mathbf{P}_\sigma^\top$  holds because  $\mathbf{P}_\sigma$  is orthogonal, by (a). For the second equality,  $(\mathbf{P}_\sigma)_{ij} = 1$  exactly when  $i = \sigma(j)$ , so  $(\mathbf{P}_\sigma^\top)_{ij} = (\mathbf{P}_\sigma)_{ji} = 1$  exactly when  $j = \sigma(i)$ , that is when  $i = \sigma^{-1}(j)$ ; and that is the defining condition for  $(\mathbf{P}_{\sigma^{-1}})_{ij} = 1$ .

## 10.7.7.3 C. Going deeper

**Exercise C1: Determinant as a homomorphism.**

- (a) Prove that  $\det: O(n) \rightarrow \{1, -1\}$  is a surjective group homomorphism with kernel  $SO(n)$ , and deduce that exactly half of  $O(n)$  has determinant  $-1$ , in the sense that  $O(n)$  is the disjoint union of  $SO(n)$  and one coset  $\mathbf{A}_0 SO(n)$ .
- (b) Prove that  $\det: U(n) \rightarrow \{z \in \mathbb{C} : |z| = 1\}$  is a surjective group homomorphism.
- (c) Describe  $O(1)$  and  $U(1)$  completely.

*Solution.*

- (a)  $\det$  is multiplicative ([Theorem 6.5.4](#)) and takes only the values  $\pm 1$  on  $O(n)$  ([Proposition 10.7.4](#) (b)), so it is a homomorphism into  $\{1, -1\}$ . It is surjective:  $\det \mathbf{I}_n = 1$ , and  $\mathbf{A}_0 := \text{diag}(-1, 1, \dots, 1)$  is orthogonal, since its columns are  $\pm \mathbf{e}_i$ , with  $\det \mathbf{A}_0 = -1$ . Its kernel is  $\{\mathbf{A} : \det \mathbf{A} = 1\} = SO(n)$  by definition. If  $\det \mathbf{A} = -1$  then  $\det(\mathbf{A}_0^{-1} \mathbf{A}) = 1$ , so  $\mathbf{A} \in \mathbf{A}_0 SO(n)$ ; and no matrix has both determinants, so the union is disjoint.
- (b) Multiplicativity again, and  $|\det \mathbf{A}| = 1$  on  $U(n)$  by [Proposition 10.7.4](#) (b). For surjectivity, given  $z$  with  $|z| = 1$ , the matrix  $\text{diag}(z, 1, \dots, 1)$  has orthonormal columns, hence lies in  $U(n)$ , and its determinant is  $z$  ([Theorem 6.4.6](#)).
- (c) A  $1 \times 1$  matrix  $(a)$  satisfies  $\bar{a}a = 1$  exactly when  $|a| = 1$ . So  $U(1)$  is the unit circle of  $\mathbb{C}$  and  $O(1) = \{(1), (-1)\}$ . Consistently,  $SO(1) = \{(1)\}$  is trivial.

**Exercise C2: Linearity is a hypothesis, not a consequence.**

- (a) Let  $c: \mathbb{C} \rightarrow \mathbb{C}$  be  $c(z) = \bar{z}$ , where  $\mathbb{C}$  carries  $\langle z, w \rangle = z\bar{w}$ . Show that  $\|c(z)\| = \|z\|$  for every  $z$ , that  $c$  is not linear over  $\mathbb{C}$ , and that  $c$  does not preserve the inner product. Which step of [Theorem 10.7.3](#) fails?
- (b) Let  $V$  be a **complex** inner product space and let  $T \in \mathcal{L}(V)$  satisfy  $\langle T\mathbf{v}, \mathbf{v} \rangle = 0$  for every  $\mathbf{v} \in V$ . Prove that  $T = 0$ .
- (c) Show that (b) fails over  $\mathbb{R}$  by exhibiting a non-zero  $T \in \mathcal{L}(\mathbb{R}^2)$  with  $\langle T\mathbf{v}, \mathbf{v} \rangle = 0$  for all  $\mathbf{v}$ .

*Hint for (b): apply the hypothesis to  $\mathbf{v} = \mathbf{u} + \mathbf{w}$  and to  $\mathbf{v} = \mathbf{u} + i\mathbf{w}$ .*

*Solution.*

- (a)  $\|c(z)\| = |\bar{z}| = |z| = \|z\|$ . But  $c(iz) = \overline{iz} = -i\bar{z} = -ic(z) \neq ic(z)$  unless  $z = 0$ , so  $c$  is not  $\mathbb{C}$ -linear. And  $\langle c(1), c(i) \rangle = \langle 1, -i \rangle = 1 \cdot \overline{-i} = i$ , while  $\langle 1, i \rangle = -i$ . The proof of (a)  $\Rightarrow$  (b) in [Theorem 10.7.3](#) uses  $T(\mathbf{u} + i^k \mathbf{v}) = T\mathbf{u} + i^k T\mathbf{v}$ , which is exactly what  $c$  does not satisfy.
- (b) Let  $\mathbf{u}, \mathbf{w} \in V$ . Applying the hypothesis to  $\mathbf{u} + \mathbf{w}$  and expanding both slots,

$$\begin{aligned} 0 = \langle T(\mathbf{u} + \mathbf{w}), \mathbf{u} + \mathbf{w} \rangle &= \langle T\mathbf{u}, \mathbf{u} \rangle + \langle T\mathbf{u}, \mathbf{w} \rangle + \langle T\mathbf{w}, \mathbf{u} \rangle + \langle T\mathbf{w}, \mathbf{w} \rangle \\ &= \langle T\mathbf{u}, \mathbf{w} \rangle + \langle T\mathbf{w}, \mathbf{u} \rangle. \end{aligned} \quad (1)$$

Applying it to  $\mathbf{u} + i\mathbf{w}$ , and using  $\langle T\mathbf{u}, i\mathbf{w} \rangle = -i\langle T\mathbf{u}, \mathbf{w} \rangle$ ,  $\langle T(i\mathbf{w}), \mathbf{u} \rangle = i\langle T\mathbf{w}, \mathbf{u} \rangle$

and  $\langle T(i\mathbf{w}), i\mathbf{w} \rangle = i\bar{i}\langle T\mathbf{w}, \mathbf{w} \rangle = 0$ ,

$$0 = -i\langle T\mathbf{u}, \mathbf{w} \rangle + i\langle T\mathbf{w}, \mathbf{u} \rangle.$$

Multiplying by  $i$  gives

$$\langle T\mathbf{u}, \mathbf{w} \rangle - \langle T\mathbf{w}, \mathbf{u} \rangle = 0. \quad (2)$$

Adding (1) and (2),  $2\langle T\mathbf{u}, \mathbf{w} \rangle = 0$ , so  $\langle T\mathbf{u}, \mathbf{w} \rangle = 0$  for all  $\mathbf{u}, \mathbf{w}$ . Taking  $\mathbf{w} = T\mathbf{u}$  gives  $\|T\mathbf{u}\|^2 = 0$ , so  $T\mathbf{u} = \mathbf{0}$  for every  $\mathbf{u}$ , that is  $T = 0$ .

- (c) Let  $T$  be the rotation  $\mathbf{R}_{\pi/2}$ , that is  $T(x_1, x_2) = (-x_2, x_1)$ . Then  $\langle T\mathbf{x}, \mathbf{x} \rangle = -x_2x_1 + x_1x_2 = 0$  for every  $\mathbf{x}$ , and  $T \neq 0$ . The complex proof breaks down because the second equation is unavailable: there is no  $i$  to substitute.

### Exercise C3: Two reflections make a rotation.

Let  $\alpha, \beta \in \mathbb{R}$  and let  $\mathbf{M}_\alpha, \mathbf{M}_\beta \in O(2)$  be the reflections of [Theorem 10.7.5](#), in the lines  $L_\alpha$  and  $L_\beta$  at angles  $\alpha/2$  and  $\beta/2$ .

- Prove that  $\mathbf{M}_\alpha \mathbf{M}_\beta = \mathbf{R}_{\alpha-\beta}$ .
- Deduce that the composite of the reflections in two lines through the origin is the rotation by **twice** the angle from the second line to the first, and that every rotation of the plane is a product of two reflections.
- Deduce that  $O(2)$  is not abelian, by computing  $\mathbf{M}_\beta \mathbf{M}_\alpha$ .

*Solution.*

- (a) Multiplying out,

$$\mathbf{M}_\alpha \mathbf{M}_\beta = \begin{pmatrix} \cos \alpha & \sin \alpha \\ \sin \alpha & -\cos \alpha \end{pmatrix} \begin{pmatrix} \cos \beta & \sin \beta \\ \sin \beta & -\cos \beta \end{pmatrix} = \begin{pmatrix} c & -s \\ s & c \end{pmatrix},$$

where  $c = \cos \alpha \cos \beta + \sin \alpha \sin \beta = \cos(\alpha - \beta)$  and  $s = \sin \alpha \cos \beta - \cos \alpha \sin \beta = \sin(\alpha - \beta)$ , by the subtraction formulas. The top-right entry is  $\cos \alpha \sin \beta - \sin \alpha \cos \beta = -\sin(\alpha - \beta)$ , as displayed. So  $\mathbf{M}_\alpha \mathbf{M}_\beta = \mathbf{R}_{\alpha-\beta}$ .

- The angle from  $L_\beta$  to  $L_\alpha$  is  $\frac{\alpha}{2} - \frac{\beta}{2}$ , and (a) says the composite is the rotation by  $\alpha - \beta$ , twice that. Conversely, given  $\theta$ , choose  $\alpha = \theta$  and  $\beta = 0$ ; then  $\mathbf{R}_\theta = \mathbf{M}_\theta \mathbf{M}_0$ , a product of two reflections.
- By (a) with the roles exchanged,  $\mathbf{M}_\beta \mathbf{M}_\alpha = \mathbf{R}_{\beta-\alpha} = \mathbf{R}_{\alpha-\beta}^{-1}$ . These two agree only when  $\mathbf{R}_{\alpha-\beta}^2 = \mathbf{I}$ , that is when  $\alpha - \beta$  is a multiple of  $\pi$ . Taking  $\alpha = \pi/2$  and  $\beta = 0$  gives  $\mathbf{M}_{\pi/2} \mathbf{M}_0 = \mathbf{R}_{\pi/2} \neq \mathbf{R}_{-\pi/2} = \mathbf{M}_0 \mathbf{M}_{\pi/2}$ , so  $O(2)$  is not abelian.

## 10.8 QR, Householder and Hessenberg

Section 2 turned a basis into an orthonormal basis one vector at a time, and every step was bookkeeping: each new vector recorded in terms of those already produced. Collect that bookkeeping into a matrix and the Gram–Schmidt process becomes a **factorization**  $\mathbf{A} = \mathbf{QR}$  — an orthogonal counterpart of the LU factorization of [Definition 2.6.1](#). This section proves it and

its uniqueness, then builds it a second time out of reflections, a route that survives dependent columns. The same reflections, applied on **both** sides, bring any square matrix to an almost-triangular form by a unitary similarity. Throughout,  $F$  is  $\mathbb{R}$  or  $\mathbb{C}$ , and  $F^m$  carries the standard inner product, linear in the first slot.

### 10.8.1 Gram–Schmidt in matrix form

Let  $\mathbf{A} \in M_{m \times n}(F)$  have columns  $\mathbf{a}_1, \dots, \mathbf{a}_n$ , and suppose they are linearly independent. Apply [Theorem 10.2.2](#) to the list  $(\mathbf{a}_1, \dots, \mathbf{a}_n)$ . It returns an orthonormal list  $(\mathbf{q}_1, \dots, \mathbf{q}_n)$  in  $F^m$  with

$$\text{span}(\mathbf{q}_1, \dots, \mathbf{q}_k) = \text{span}(\mathbf{a}_1, \dots, \mathbf{a}_k)$$

for every  $k$ . In particular  $\mathbf{a}_k$  lies in  $\text{span}(\mathbf{q}_1, \dots, \mathbf{q}_k)$ , and [Theorem 10.2.1](#) (a), applied inside that subspace, expands it:

$$\mathbf{a}_k = \sum_{i=1}^k \langle \mathbf{a}_k, \mathbf{q}_i \rangle \mathbf{q}_i \quad (k = 1, \dots, n). \quad (*)$$

Now read  $(*)$  as a matrix product. Put  $\mathbf{Q} = (\mathbf{q}_1 \mid \dots \mid \mathbf{q}_n) \in M_{m \times n}(F)$  and let  $\mathbf{R} \in M_n(F)$  have entries

$$r_{ik} = \begin{cases} \langle \mathbf{a}_k, \mathbf{q}_i \rangle, & i \leq k, \\ 0, & i > k. \end{cases}$$

Then  $\mathbf{R}$  is upper triangular ([Definition 0.9.5](#)), and  $(*)$  says that the  $k$ -th column of  $\mathbf{A}$  equals  $\mathbf{Q}$  times the  $k$ -th column of  $\mathbf{R}$ . That is exactly  $\mathbf{A} = \mathbf{QR}$ . **Nothing has been computed; the factorization is the Gram–Schmidt output, rewritten.**

Two features of the rewriting matter later. First the diagonal: in the notation of [Theorem 10.2.2](#),  $\mathbf{q}_k = \mathbf{w}_k / \|\mathbf{w}_k\|$  with  $\mathbf{w}_k = \mathbf{a}_k - \sum_{i < k} \langle \mathbf{a}_k, \mathbf{q}_i \rangle \mathbf{q}_i$ , so comparing with  $(*)$  gives

$$r_{kk} = \langle \mathbf{a}_k, \mathbf{q}_k \rangle = \|\mathbf{w}_k\| > 0,$$

**strictly positive** because  $\mathbf{w}_k \neq \mathbf{0}$ . Second,  $\mathbf{R}$  is recoverable from  $\mathbf{Q}$  and  $\mathbf{A}$ : since the columns of  $\mathbf{Q}$  are orthonormal,

$$(\mathbf{Q}^* \mathbf{Q})_{ij} = \sum_{p=1}^m \overline{q_{pi}} q_{pj} = \langle \mathbf{q}_j, \mathbf{q}_i \rangle = \delta_{ij},$$

by [Definition 10.1.2](#), so  $\mathbf{Q}^* \mathbf{Q} = \mathbf{I}_n$  and  $\mathbf{R} = \mathbf{Q}^* \mathbf{A}$ .

#### Theorem 10.8.1 — QR Factorization

Let  $\mathbf{A} \in M_{m \times n}(F)$  have **linearly independent** columns. Then there are matrices  $\mathbf{Q} \in M_{m \times n}(F)$  and  $\mathbf{R} \in M_n(F)$  with

- (a) the columns of  $\mathbf{Q}$  orthonormal, equivalently  $\mathbf{Q}^* \mathbf{Q} = \mathbf{I}_n$ ;
- (b)  $\mathbf{R}$  upper triangular with **positive real** diagonal entries;
- (c)  $\mathbf{A} = \mathbf{QR}$ .

Moreover  $\mathbf{Q}$  and  $\mathbf{R}$  are **unique**: any pair satisfying (a), (b), (c) is this one. The pair  $(\mathbf{Q}, \mathbf{R})$  is the **QR factorization** of  $\mathbf{A}$ .

*Idea.* Existence is the paragraph above: run Gram–Schmidt and read off the coefficients. For uniqueness, suppose  $\mathbf{Q}_1\mathbf{R}_1 = \mathbf{Q}_2\mathbf{R}_2$ . We cannot cancel  $\mathbf{Q}$ , which is not square, but  $\mathbf{Q}^*\mathbf{Q} = \mathbf{I}$  turns  $\mathbf{A}^*\mathbf{A}$  into  $\mathbf{R}^*\mathbf{R}$  and makes the  $\mathbf{Q}$ 's vanish. That leaves  $\mathbf{R}_1^*\mathbf{R}_1 = \mathbf{R}_2^*\mathbf{R}_2$ , so the matrix  $\mathbf{S}$  carrying  $\mathbf{R}_1$  to  $\mathbf{R}_2$  is upper triangular with positive diagonal and satisfies  $\mathbf{S}^*\mathbf{S} = \mathbf{I}$ . Such a matrix has no room to be anything but  $\mathbf{I}$ : column by column, orthogonality to the earlier columns kills every entry above the diagonal.

*Proof. Existence.* Apply [Theorem 10.2.2](#) to the columns of  $\mathbf{A}$ , which is legitimate because they are independent, and define  $\mathbf{Q}$  and  $\mathbf{R}$  as above. Then (a) holds by orthonormality of the output, (b) holds because  $r_{kk} = \|\mathbf{w}_k\| > 0$ , and (c) is (\*).

*Uniqueness.* Suppose  $\mathbf{A} = \mathbf{Q}_1\mathbf{R}_1 = \mathbf{Q}_2\mathbf{R}_2$  with both pairs satisfying (a) and (b). Each  $\mathbf{R}_t$  has non-zero diagonal entries, hence is invertible by [Lemma 2.6.2](#). Using  $\mathbf{Q}_t^*\mathbf{Q}_t = \mathbf{I}_n$  twice,

$$\mathbf{R}_1^*\mathbf{R}_1 = \mathbf{R}_1^*\mathbf{Q}_1^*\mathbf{Q}_1\mathbf{R}_1 = \mathbf{A}^*\mathbf{A} = \mathbf{R}_2^*\mathbf{Q}_2^*\mathbf{Q}_2\mathbf{R}_2 = \mathbf{R}_2^*\mathbf{R}_2.$$

Put  $\mathbf{S} = \mathbf{R}_2\mathbf{R}_1^{-1}$ , so that  $\mathbf{R}_2 = \mathbf{S}\mathbf{R}_1$ . The matrix  $\mathbf{R}_1^{-1}$  is upper triangular: its  $j$ -th column is the solution  $\mathbf{x}$  of  $\mathbf{R}_1\mathbf{x} = \mathbf{e}_j$ , and solving from the bottom row upward gives  $x_i = 0$  for every  $i > j$ , exactly as in the proof of [Lemma 2.6.2](#). Hence  $\mathbf{S}$  is upper triangular, being a product of upper triangular matrices: in  $(\mathbf{S})_{ij} = \sum_k (\mathbf{R}_2)_{ik} (\mathbf{R}_1^{-1})_{kj}$  a term can be non-zero only when  $i \leq k$  and  $k \leq j$ , which is impossible for  $i > j$ , and for  $i = j$  only the term  $k = i$  survives. So its diagonal entries are  $(\mathbf{S})_{kk} = (\mathbf{R}_2)_{kk} (\mathbf{R}_1)_{kk}^{-1} > 0$ , a quotient of positive reals. Substituting  $\mathbf{R}_2 = \mathbf{S}\mathbf{R}_1$  into the display gives  $\mathbf{R}_1^*\mathbf{R}_1 = \mathbf{R}_1^*\mathbf{S}^*\mathbf{S}\mathbf{R}_1$ , and multiplying by  $(\mathbf{R}_1^*)^{-1} = (\mathbf{R}_1^{-1})^*$  on the left and  $\mathbf{R}_1^{-1}$  on the right leaves

$$\mathbf{S}^*\mathbf{S} = \mathbf{I}_n.$$

*Claim.* An upper triangular  $\mathbf{S} \in M_n(F)$  with  $\mathbf{S}^*\mathbf{S} = \mathbf{I}_n$  and positive real diagonal entries is  $\mathbf{I}_n$ .

*Proof.* Write  $\mathbf{s}_1, \dots, \mathbf{s}_n$  for the columns of  $\mathbf{S}$ . As computed before the theorem,  $\mathbf{S}^*\mathbf{S} = \mathbf{I}_n$  says  $\langle \mathbf{s}_j, \mathbf{s}_i \rangle = \delta_{ij}$ . We show  $\mathbf{s}_j = \mathbf{e}_j$  by induction on  $j$ . Let  $1 \leq j \leq n$  and suppose  $\mathbf{s}_i = \mathbf{e}_i$  for every  $i < j$  (no assumption when  $j = 1$ ). For  $i < j$ ,

$$0 = \langle \mathbf{s}_j, \mathbf{s}_i \rangle = \langle \mathbf{s}_j, \mathbf{e}_i \rangle = s_{ij},$$

so every entry of  $\mathbf{s}_j$  above the diagonal vanishes; and  $s_{ij} = 0$  for  $i > j$  because  $\mathbf{S}$  is upper triangular. Hence  $\mathbf{s}_j = s_{jj}\mathbf{e}_j$ , and  $1 = \langle \mathbf{s}_j, \mathbf{s}_j \rangle = |s_{jj}|^2$  forces  $|s_{jj}| = 1$ . Since  $s_{jj} > 0$ , this gives  $s_{jj} = 1$  and  $\mathbf{s}_j = \mathbf{e}_j$ . ■

By the Claim,  $\mathbf{S} = \mathbf{I}_n$ , so  $\mathbf{R}_2 = \mathbf{R}_1$  and therefore  $\mathbf{Q}_2 = \mathbf{A}\mathbf{R}_2^{-1} = \mathbf{A}\mathbf{R}_1^{-1} = \mathbf{Q}_1$ . This proves uniqueness. ■

The positivity in (b) is a normalization and nothing deeper: without it, scaling  $\mathbf{q}_k$  by any  $c$  of modulus 1 and row  $k$  of  $\mathbf{R}$  by  $\bar{c}$  gives another factorization. [Exercise \(C1: All the QR factorizations of a matrix\)](#) describes all of them.

*Warning.* In  $\mathbf{A} = \mathbf{QR}$  the matrix  $\mathbf{Q}$  is usually not square, and  $\mathbf{Q}\mathbf{Q}^* \neq \mathbf{I}$ . Only  $\mathbf{Q}^*\mathbf{Q} = \mathbf{I}_n$  holds. For the  $3 \times 2$  matrix of [Example \(A  \$3 \times 2\$  QR, and a least-squares problem\)](#) below,

$$\mathbf{Q}\mathbf{Q}^* = \frac{1}{18} \begin{pmatrix} 2 & 4 & 4 \\ 4 & 17 & -1 \\ 4 & -1 & 17 \end{pmatrix},$$

whose trace is 2, not 3. This is the “identity on what?” trap of [Theorem 10.7.3](#): the columns of  $\mathbf{Q}$  are an orthonormal basis of a **plane**, and  $\mathbf{Q}\mathbf{Q}^*$  is the orthogonal projection onto it ([Theorem 10.3.3](#)).

*Quick check.* Each factor of a QR factorization is recoverable from the other:  $\mathbf{R} = \mathbf{Q}^*\mathbf{A}$  and  $\mathbf{Q} = \mathbf{A}\mathbf{R}^{-1}$ . One of the two recoveries needs an invertibility hypothesis and the other does not. Which, and why?

*Solution.* The second. Multiplying  $\mathbf{A} = \mathbf{QR}$  on the left by  $\mathbf{Q}^*$  gives  $\mathbf{Q}^*\mathbf{Q}\mathbf{R} = \mathbf{R}$ , which uses only  $\mathbf{Q}^*\mathbf{Q} = \mathbf{I}_n$ : nothing is inverted, and  $\mathbf{Q}$  is not even square. Multiplying on the right by  $\mathbf{R}^{-1}$  presupposes that  $\mathbf{R}$  is invertible — true by [Lemma 2.6.2](#), but only because clause (b) makes the diagonal positive.

## 10.8.2 Least squares through QR

An inconsistent system  $\mathbf{A}\mathbf{x} = \mathbf{b}$  was handled in Section 4 by the normal equations  $\mathbf{A}^*\mathbf{A}\mathbf{x} = \mathbf{A}^*\mathbf{b}$  ([Theorem 10.4.2](#)). When the columns of  $\mathbf{A}$  are independent, the QR factorization simplifies those equations to a triangular system.

Substituting  $\mathbf{A} = \mathbf{QR}$  and using  $\mathbf{Q}^*\mathbf{Q} = \mathbf{I}_n$ ,

$$\mathbf{A}^*\mathbf{A} = \mathbf{R}^*\mathbf{Q}^*\mathbf{Q}\mathbf{R} = \mathbf{R}^*\mathbf{R}, \quad \mathbf{A}^*\mathbf{b} = \mathbf{R}^*\mathbf{Q}^*\mathbf{b},$$

so the normal equations read  $\mathbf{R}^*\mathbf{R}\mathbf{x} = \mathbf{R}^*\mathbf{Q}^*\mathbf{b}$ . Now  $\mathbf{R}$  is invertible by [Lemma 2.6.2](#), hence so is  $\mathbf{R}^*$ , with  $(\mathbf{R}^*)^{-1} = (\mathbf{R}^{-1})^*$  — conjugate-transpose  $\mathbf{R}\mathbf{R}^{-1} = \mathbf{I}_n$ . So  $\mathbf{R}^*$  cancels:

$$\mathbf{R}\mathbf{x} = \mathbf{Q}^*\mathbf{b}. \quad (\dagger)$$

By [Corollary 10.4.4](#) the least-squares solution is unique here, and  $(\dagger)$  is a triangular system, solved by back substitution. The deeper reason to prefer it is not the operation count: forming  $\mathbf{A}^*\mathbf{A}$  squares the sensitivity of the problem to rounding, while  $(\dagger)$  never forms that product at all. Chapter 23 makes “sensitivity” precise and proves the comparison; here we record only the practice.

**Example.** (A  $3 \times 2$  QR, and a least-squares problem)

Let

$$\mathbf{A} = \begin{pmatrix} 1 & 1 \\ 2 & 1 \\ 2 & 3 \end{pmatrix}, \quad \mathbf{b} = \begin{pmatrix} 1 \\ 1 \\ 1 \end{pmatrix}.$$

- Find the QR factorization of  $\mathbf{A}$ .
- Find the  $\mathbf{x} \in \mathbb{R}^2$  minimizing  $\|\mathbf{A}\mathbf{x} - \mathbf{b}\|$ , using (a).



*Solution.*

- (a) Write  $\mathbf{a}_1 = (1, 2, 2)$  and  $\mathbf{a}_2 = (1, 1, 3)$ . They are independent, since neither is a multiple of the other.

*Step 1.*  $\|\mathbf{a}_1\|^2 = 1 + 4 + 4 = 9$ , so  $r_{11} = 3$  and  $\mathbf{q}_1 = \frac{1}{3}(1, 2, 2)$ .

*Step 2.*  $r_{12} = \langle \mathbf{a}_2, \mathbf{q}_1 \rangle = \frac{1}{3}(1 + 2 + 6) = 3$ , so

$$\mathbf{w}_2 = (1, 1, 3) - 3 \cdot \frac{1}{3}(1, 2, 2) = (0, -1, 1),$$

with  $\|\mathbf{w}_2\| = \sqrt{2} = r_{22}$  and  $\mathbf{q}_2 = \frac{1}{\sqrt{2}}(0, -1, 1)$ . Hence

$$\mathbf{Q} = \begin{pmatrix} \frac{1}{3} & 0 \\ \frac{2}{3} & -\frac{1}{\sqrt{2}} \\ \frac{2}{3} & \frac{1}{\sqrt{2}} \end{pmatrix}, \quad \mathbf{R} = \begin{pmatrix} 3 & 3 \\ 0 & \sqrt{2} \end{pmatrix}.$$

*Check.*  $\mathbf{QR}$  has first column  $3\mathbf{q}_1 = (1, 2, 2)$  and second column  $3\mathbf{q}_1 + \sqrt{2}\mathbf{q}_2 = (1, 2, 2) + (0, -1, 1) = (1, 1, 3)$ , which is  $\mathbf{A}$ .

- (b) By (†) we need  $\mathbf{Q}^*\mathbf{b}$ . Here  $\langle \mathbf{b}, \mathbf{q}_1 \rangle = \frac{1}{3}(1+2+2) = \frac{5}{3}$  and  $\langle \mathbf{b}, \mathbf{q}_2 \rangle = \frac{1}{\sqrt{2}}(0-1+1) = 0$ , so the system is

$$\begin{pmatrix} 3 & 3 \\ 0 & \sqrt{2} \end{pmatrix} \mathbf{x} = \begin{pmatrix} \frac{5}{3} \\ 0 \end{pmatrix}.$$

Back substitution gives  $\sqrt{2}x_2 = 0$ , so  $x_2 = 0$ , and then  $3x_1 = \frac{5}{3}$ , so  $x_1 = \frac{5}{9}$ . The minimizer is  $\mathbf{x} = (\frac{5}{9}, 0)$ .

*Check against the normal equations.*  $\mathbf{A}^\top \mathbf{A} = \begin{pmatrix} 9 & 9 \\ 9 & 11 \end{pmatrix}$  and  $\mathbf{A}^\top \mathbf{b} = (5, 5)$ . Subtracting the first equation from the second gives  $2x_2 = 0$ , and then  $9x_1 = 5$ . The same answer, with the same arithmetic done in a worse order.

## 10.8.3 Reflections

Gram–Schmidt builds  $\mathbf{Q}$  out of the columns of  $\mathbf{A}$ . There is an entirely different route to a triangular  $\mathbf{R}$ , modeled on elimination: hit  $\mathbf{A}$  on the left with simple matrices that create zeros, one column at a time, and collect the inverses. Chapter 2 used shears and got  $\mathbf{A} = \mathbf{LU}$ ; if every matrix we hit  $\mathbf{A}$  with is **unitary** instead, the collected product is unitary and we get  $\mathbf{A} = \mathbf{QR}$ . Which unitary matrices are simple enough? A rotation in the plane of two coordinates is one answer, taken up below. The more efficient answer is a reflection, because one reflection can flatten an entire column at once.

*A reflection in a hyperplane sends a vector to its mirror image: keep the part perpendicular to  $\mathbf{w}$ , and reverse the part along  $\mathbf{w}$ .*

### Definition 10.8.1 — Householder Reflection

Let  $\mathbf{w} \in F^m$  be **non-zero**. The **Householder reflection** determined by  $\mathbf{w}$  is

$$\mathbf{H}_{\mathbf{w}} = \mathbf{I}_m - \frac{2\mathbf{w}\mathbf{w}^*}{\|\mathbf{w}\|^2} \in M_m(F).$$

In words:  $\mathbf{w}\mathbf{w}^*$  is an  $m \times m$  matrix (a column times a row), and for any  $\mathbf{v} \in F^m$  the product  $\mathbf{w}^*\mathbf{v}$  is the scalar  $\sum_p \overline{w_p}v_p = \langle \mathbf{v}, \mathbf{w} \rangle$ . So the definition says

$$\mathbf{H}_{\mathbf{w}}\mathbf{v} = \mathbf{v} - 2 \frac{\langle \mathbf{v}, \mathbf{w} \rangle}{\|\mathbf{w}\|^2} \mathbf{w} = \mathbf{v} - 2P_{\text{span}(\mathbf{w})}\mathbf{v}, \quad (\dagger)$$

where the last equality is [Theorem 10.3.3](#) with the orthonormal basis  $(\mathbf{w}/\|\mathbf{w}\|)$  of the line  $\text{span}(\mathbf{w})$ . Writing  $\mathbf{v} = P_{\text{span}(\mathbf{w})}\mathbf{v} + (\mathbf{v} - P_{\text{span}(\mathbf{w})}\mathbf{v})$  as in [Theorem 10.3.2](#), we see that  $\mathbf{H}_{\mathbf{w}}$  keeps the component of  $\mathbf{v}$  in  $\text{span}(\mathbf{w})^\perp$  and reverses the component along  $\mathbf{w}$ . That is a mirror reflection in the hyperplane  $\text{span}(\mathbf{w})^\perp$ .

Note at once that  $\mathbf{H}_{c\mathbf{w}} = \mathbf{H}_{\mathbf{w}}$  for every non-zero scalar  $c$ , since  $(c\mathbf{w})(c\mathbf{w})^* = |c|^2\mathbf{w}\mathbf{w}^*$  and  $\|c\mathbf{w}\|^2 = |c|^2\|\mathbf{w}\|^2$ . **Only the line through  $\mathbf{w}$  matters.**

**Examples.**

- **The smallest case.** In  $F^1$  take  $\mathbf{w} = 1$ : then  $\mathbf{H}_{\mathbf{w}} = -1$ , the map  $x \mapsto -x$ . The “hyperplane” is  $\{0\}$ , and this is the only non-trivial isometry of the line.
- **A coordinate reflection.** In  $\mathbb{R}^2$  take  $\mathbf{w} = \mathbf{e}_2$ , so  $\mathbf{w}\mathbf{w}^\top = \begin{pmatrix} 0 & 0 \\ 0 & 1 \end{pmatrix}$  and  $\mathbf{H}_{\mathbf{w}} = \text{diag}(1, -1)$ : the reflection in the  $x$ -axis.
- **A diagonal reflection.** In  $\mathbb{R}^2$  take  $\mathbf{w} = (1, -1)$ , so  $\|\mathbf{w}\|^2 = 2$  and  $\mathbf{H}_{\mathbf{w}} = \mathbf{I}_2 - \begin{pmatrix} 1 & -1 \\ -1 & 1 \end{pmatrix} = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}$ , which swaps the coordinates. It is the reflection in the line  $y = x$ , and  $\text{span}(\mathbf{w})^\perp$  is that line.
- **A complex one.** In  $\mathbb{C}^m$  take  $\mathbf{w} = \mathbf{e}_1$ : then  $\mathbf{H}_{\mathbf{w}} = \text{diag}(-1, 1, \dots, 1)$ , over either field.

**Non-example by minimal change.** Drop the factor 2: the matrix  $\mathbf{I}_m - \mathbf{w}\mathbf{w}^*/\|\mathbf{w}\|^2$  is  $\mathbf{I}_m - P_{\text{span}(\mathbf{w})}$ , the orthogonal projection onto  $\text{span}(\mathbf{w})^\perp$ . It is Hermitian, as  $\mathbf{H}_{\mathbf{w}}$  is, but **not** unitary and not invertible: it sends  $\mathbf{w}$  to  $\mathbf{0}$ . Halving the coefficient turns “reverse the component along  $\mathbf{w}$ ” into “delete it”, an isometry into a shadow.

### Proposition 10.8.2 — Properties of a Householder Reflection

Let  $\mathbf{w} \in F^m$  be non-zero and  $\mathbf{H} = \mathbf{H}_{\mathbf{w}}$ . Then

- (a)  $\mathbf{H}^* = \mathbf{H}$  ( $\mathbf{H}$  is Hermitian);
- (b)  $\mathbf{H}^2 = \mathbf{I}_m$ , so  $\mathbf{H}^{-1} = \mathbf{H}$ ;
- (c)  $\mathbf{H}$  is unitary;
- (d)  $\mathbf{H}\mathbf{w} = -\mathbf{w}$ , and  $\mathbf{H}\mathbf{v} = \mathbf{v}$  for every  $\mathbf{v} \perp \mathbf{w}$ ;
- (e)  $\det \mathbf{H} = -1$ .

*Proof.*

- (a) By [Definition 10.1.2](#),  $(\mathbf{w}\mathbf{w}^*)^* = \mathbf{w}^{**}\mathbf{w}^* = \mathbf{w}\mathbf{w}^*$ , and  $\|\mathbf{w}\|^2$  is a real number ([Theorem 10.1.1](#) (a)), so  $\mathbf{H}^* = \mathbf{I}_m - 2\mathbf{w}\mathbf{w}^*/\|\mathbf{w}\|^2 = \mathbf{H}$ .
- (b) Since  $\mathbf{w}^*\mathbf{w} = \langle \mathbf{w}, \mathbf{w} \rangle = \|\mathbf{w}\|^2$ ,

$$\mathbf{H}^2 = \mathbf{I}_m - \frac{4\mathbf{w}\mathbf{w}^*}{\|\mathbf{w}\|^2} + \frac{4\mathbf{w}(\mathbf{w}^*\mathbf{w})\mathbf{w}^*}{\|\mathbf{w}\|^4} = \mathbf{I}_m,$$

because the last two terms cancel.

- (c) By (a) and (b),  $\mathbf{H}^*\mathbf{H} = \mathbf{H}^2 = \mathbf{I}_m$ , which is the criterion of [Definition 10.7.2](#).
- (d) Both statements are read off (‡):  $P_{\text{span}(\mathbf{w})}\mathbf{w} = \mathbf{w}$  gives  $\mathbf{H}\mathbf{w} = \mathbf{w} - 2\mathbf{w} = -\mathbf{w}$ , and  $\langle \mathbf{v}, \mathbf{w} \rangle = 0$  gives  $\mathbf{H}\mathbf{v} = \mathbf{v}$ .
- (e) Extend the unit vector  $\mathbf{w}/\|\mathbf{w}\|$  to an orthonormal basis  $(\mathbf{f}_1, \dots, \mathbf{f}_m)$  of  $F^m$  with  $\mathbf{f}_1 = \mathbf{w}/\|\mathbf{w}\|$ , which is possible by [Corollary 10.2.3](#), and let  $\mathbf{P} = (\mathbf{f}_1 \mid \dots \mid \mathbf{f}_m)$ . By (d),  $\mathbf{H}\mathbf{f}_1 = -\mathbf{f}_1$  and  $\mathbf{H}\mathbf{f}_i = \mathbf{f}_i$  for  $i \geq 2$ , that is,  $\mathbf{H}\mathbf{P} = \mathbf{P}\mathbf{D}$  with  $\mathbf{D} = \text{diag}(-1, 1, \dots, 1)$ . The matrix  $\mathbf{P}$  is unitary, hence invertible, so  $\mathbf{H} = \mathbf{P}\mathbf{D}\mathbf{P}^{-1}$ . Taking determinants,  $\det \mathbf{H} = \det \mathbf{D}$  by [Theorem 6.5.4](#) and [Corollary 6.5.6](#), and  $\det \mathbf{D} = -1$  by [Theorem 6.4.6](#).

■

So a Householder reflection is a unitary involution of determinant  $-1$ : cheap to store (one vector) and cheap to apply, by (‡). What makes it useful is the following choice of  $\mathbf{w}$ .

### Theorem 10.8.3 — A Reflection Flattening a Vector

Let  $\mathbf{x} \in F^m$  be non-zero, and let

$$\alpha = \begin{cases} x_1/|x_1|, & x_1 \neq 0, \\ 1, & x_1 = 0, \end{cases} \quad \mathbf{w} = \mathbf{x} + \alpha\|\mathbf{x}\| \mathbf{e}_1.$$

Then  $\mathbf{w} \neq \mathbf{0}$  and

$$\mathbf{H}_{\mathbf{w}}\mathbf{x} = -\alpha\|\mathbf{x}\| \mathbf{e}_1.$$

In particular  $\mathbf{H}_{\mathbf{w}}\mathbf{x}$  is a multiple of  $\mathbf{e}_1$  of length  $\|\mathbf{x}\|$ . Over  $\mathbb{R}$ ,  $\alpha$  is the sign of  $x_1$  (or 1 if  $x_1 = 0$ ), and  $\mathbf{H}_{\mathbf{w}}\mathbf{x} = \mp\|\mathbf{x}\|\mathbf{e}_1$  according to that sign.

*Idea.* Picture the real plane. We want a mirror line taking  $\mathbf{x}$  to a point on the first axis at the same distance from the origin. There are two such points,  $\pm\|\mathbf{x}\|\mathbf{e}_1$ , and for each the mirror is the perpendicular bisector of the segment joining  $\mathbf{x}$  to it — whose normal direction is the **difference**  $\mathbf{x} \mp \|\mathbf{x}\|\mathbf{e}_1$ . So the vector  $\mathbf{w}$  is forced, up to which of the two targets we aim at, and all that remains is to check that the coefficient  $2\langle \mathbf{x}, \mathbf{w} \rangle / \|\mathbf{w}\|^2$  in (‡) comes out exactly 1, so that  $\mathbf{H}_{\mathbf{w}}\mathbf{x} = \mathbf{x} - \mathbf{w}$ . The role of  $\alpha$  is to choose the target on the far side from  $\mathbf{x}$ .

*Proof.* Note first that  $\overline{\alpha}x_1 = |x_1|$  in both cases, and  $|\alpha| = 1$ . Expanding with (IP1) and [Theorem 10.1.1](#) (b),

$$\begin{aligned} \|\mathbf{w}\|^2 &= \|\mathbf{x}\|^2 + \overline{\alpha}\|\mathbf{x}\|\langle \mathbf{x}, \mathbf{e}_1 \rangle + \alpha\|\mathbf{x}\|\langle \mathbf{e}_1, \mathbf{x} \rangle + |\alpha|^2\|\mathbf{x}\|^2 \\ &= 2\|\mathbf{x}\|^2 + 2\|\mathbf{x}\||x_1|, \end{aligned}$$

where the second line uses  $\langle \mathbf{x}, \mathbf{e}_1 \rangle = x_1$ ,  $\langle \mathbf{e}_1, \mathbf{x} \rangle = \overline{x_1}$  and  $\overline{\alpha}x_1 + \alpha\overline{x_1} = 2|x_1|$ . Since  $\mathbf{x} \neq \mathbf{0}$  we have  $\|\mathbf{x}\| > 0$ , so  $\|\mathbf{w}\|^2 > 0$  and  $\mathbf{w} \neq \mathbf{0}$ . Likewise

$$\langle \mathbf{x}, \mathbf{w} \rangle = \|\mathbf{x}\|^2 + \overline{\alpha}\|\mathbf{x}\|x_1 = \|\mathbf{x}\|^2 + \|\mathbf{x}\||x_1|,$$

so that  $2\langle \mathbf{x}, \mathbf{w} \rangle = \|\mathbf{w}\|^2$ . Hence by (‡),

$$\mathbf{H}_{\mathbf{w}}\mathbf{x} = \mathbf{x} - \frac{2\langle \mathbf{x}, \mathbf{w} \rangle}{\|\mathbf{w}\|^2} \mathbf{w} = \mathbf{x} - \mathbf{w} = -\alpha\|\mathbf{x}\| \mathbf{e}_1.$$

Finally  $\|-\alpha\|\mathbf{x}\|\mathbf{e}_1\| = |\alpha|\|\mathbf{x}\| = \|\mathbf{x}\|$  by [Theorem 10.1.2](#) (b), as it must be, since  $\mathbf{H}_{\mathbf{w}}$  is unitary. This proves the theorem. ■

**Why that sign.** The opposite choice  $\mathbf{w}' = \mathbf{x} - \alpha\|\mathbf{x}\|\mathbf{e}_1$  also works whenever  $\mathbf{w}' \neq \mathbf{0}$ , giving  $\mathbf{H}_{\mathbf{w}'}\mathbf{x} = +\alpha\|\mathbf{x}\|\mathbf{e}_1$ ; in exact arithmetic the two are equally good, and in floating point they are not. The first entry of  $\mathbf{w}'$  is  $x_1 - \alpha\|\mathbf{x}\|$ , a difference of two numbers of modulus  $|x_1|$  and  $\|\mathbf{x}\|$  — and when  $\mathbf{x}$  is close to a multiple of  $\mathbf{e}_1$  those two are nearly equal, so the leading digits cancel and the computed first entry keeps only the few digits they did not share. The other entries of  $\mathbf{w}'$  are copied from  $\mathbf{x}$  and stay exact, so the computed  $\mathbf{w}'$  points in a noticeably wrong direction, and so does the reflection built from it. With  $\mathbf{w} = \mathbf{x} + \alpha\|\mathbf{x}\|\mathbf{e}_1$  the first entry has modulus  $|x_1| + \|\mathbf{x}\|$ : like-signed quantities, nothing to cancel. (The same computation shows  $\mathbf{w} \neq \mathbf{0}$  always, whereas  $\mathbf{w}' = \mathbf{0}$  exactly when  $\mathbf{x} = \alpha\|\mathbf{x}\|\mathbf{e}_1$ , the extreme case of that cancellation.) Chapter 23 quantifies the loss; the summary here is that both signs are correct mathematics and only one is correct arithmetic.

*Warning.* **Do not build the matrix  $\mathbf{H}_{\mathbf{w}}$  in order to use it.** Forming it and multiplying costs  $m^2$  operations per column, while  $(\dagger)$  applies the same reflection with one inner product and one scaled subtraction, about  $4m$ . Every “multiply by  $\mathbf{H}$ ” below is really an instance of  $(\dagger)$ ; the matrix is a proof device.

*Quick check.* Let  $\mathbf{x} = (0, 3, 4) \in \mathbb{R}^3$ . Find  $\mathbf{w}$  and  $\mathbf{H}_{\mathbf{w}}\mathbf{x}$  as in [Theorem 10.8.3](#), and say why the sign question does not arise here.

*Solution.*  $\|\mathbf{x}\| = 5$  and  $x_1 = 0$ , so  $\alpha = 1$  and  $\mathbf{w} = (0, 3, 4) + 5(1, 0, 0) = (5, 3, 4)$ . Then  $\mathbf{H}_{\mathbf{w}}\mathbf{x} = -5\mathbf{e}_1 = (-5, 0, 0)$ . With  $x_1 = 0$  there is no cancellation to avoid: the first entry of  $\mathbf{w}$  is  $0 \pm 5$  either way, and both choices are equally accurate. The convention  $\alpha = 1$  is only a convention.

## 10.8.4 QR by reflections

One reflection makes the first column a multiple of  $\mathbf{e}_1$  — exactly the zeros an upper triangular matrix needs in column 1. Repeating on the trailing submatrix triangularizes everything.

### Theorem 10.8.4 — QR by Householder Reflections

Let  $\mathbf{A} \in M_{m \times n}(F)$ . Then there are a **unitary**  $\mathbf{Q} \in M_m(F)$  and an  $\mathbf{R} \in M_{m \times n}(F)$  with  $r_{ij} = 0$  whenever  $i > j$ , such that

$$\mathbf{A} = \mathbf{QR}.$$

Moreover  $\mathbf{Q}$  can be taken to be a product of at most  $\min(n, m-1)$  Householder reflections. **No hypothesis on the columns of  $\mathbf{A}$  is needed.**

*Idea.* Induction on the number of columns. Step one is [Theorem 10.8.3](#) applied to the first column: after it, the first column is  $\rho\mathbf{e}_1$  and what is left to triangularize is the trailing  $(m-1) \times (n-1)$  block. Feed that block to the induction hypothesis, and lift the unitary matrix it returns to  $F^m$  by putting a 1 in the top-left corner — a block matrix of that shape does not touch the first row or the first column, so it cannot spoil the zeros

just created.

*Proof.* We induct on  $n$ , the statement being asserted for every  $m \geq 1$  at once.

If  $m = 1$ , then  $\mathbf{A}$  has no entries below its diagonal, so  $\mathbf{Q} = \mathbf{I}_1$  and  $\mathbf{R} = \mathbf{A}$  work. Assume from now on that  $m \geq 2$ .

Let  $\mathbf{a}$  be the first column of  $\mathbf{A}$ . If  $\mathbf{a} = \mathbf{0}$ , put  $\mathbf{H} = \mathbf{I}_m$  and  $\rho = 0$ ; otherwise let  $\mathbf{H} = \mathbf{H}_{\mathbf{w}}$  be the reflection of [Theorem 10.8.3](#) for  $\mathbf{x} = \mathbf{a}$ , and  $\rho = -\alpha\|\mathbf{a}\|$ . In both cases  $\mathbf{H}$  is unitary and Hermitian with  $\mathbf{H}^{-1} = \mathbf{H}$  — by [Proposition 10.8.2](#) (a)–(c) for a reflection, and because  $\mathbf{I}_m^* = \mathbf{I}_m = \mathbf{I}_m^{-1}$  in the degenerate case — and  $\mathbf{H}\mathbf{a} = \rho\mathbf{e}_1$ . Hence, by [Theorem 7.1.1](#),

$$\mathbf{H}\mathbf{A} = \begin{pmatrix} \rho & \mathbf{b}^* \\ \mathbf{0} & \mathbf{A}' \end{pmatrix}$$

for some  $\mathbf{b} \in F^{n-1}$  and  $\mathbf{A}' \in M_{(m-1) \times (n-1)}(F)$ , where the  $\mathbf{0}$  is a column of  $m-1$  zeros.

*Base case*  $n = 1$ . There is no block  $\mathbf{A}'$ , and  $\mathbf{H}\mathbf{A} = \rho\mathbf{e}_1$  already has  $r_{i1} = 0$  for  $i > 1$ . So  $\mathbf{A} = \mathbf{H}(\mathbf{H}\mathbf{A})$  is of the required form with  $\mathbf{Q} = \mathbf{H}$ .

*Induction step.* Let  $n \geq 2$  and assume the statement for  $n-1$  columns. Applied to  $\mathbf{A}'$ , which has  $m-1 \geq 1$  rows, it gives a unitary  $\mathbf{Q}' \in M_{m-1}(F)$  and an  $\mathbf{R}'$  with zeros below the diagonal and  $\mathbf{A}' = \mathbf{Q}'\mathbf{R}'$ . Put

$$\mathbf{U} = \begin{pmatrix} 1 & \mathbf{0}^* \\ \mathbf{0} & \mathbf{Q}' \end{pmatrix}, \quad \mathbf{R} = \begin{pmatrix} \rho & \mathbf{b}^* \\ \mathbf{0} & \mathbf{R}' \end{pmatrix}.$$

Then  $\mathbf{U}^*\mathbf{U} = \mathbf{I}_m$  by [Theorem 7.1.1](#), so  $\mathbf{U}$  is unitary, and  $\mathbf{U}\mathbf{R} = \mathbf{H}\mathbf{A}$  by the same computation. Also  $\mathbf{R}$  has zeros below its diagonal: in column 1 this is the block  $\mathbf{0}$ ; and for  $j \geq 2$  and  $i > j$ , the  $(i, j)$ -entry of  $\mathbf{R}$  is the  $(i-1, j-1)$ -entry of  $\mathbf{R}'$ , which vanishes because  $i-1 > j-1$ .

Therefore  $\mathbf{A} = \mathbf{H}\mathbf{U}\mathbf{R}$ , and  $\mathbf{Q} = \mathbf{H}\mathbf{U}$  is unitary as a product of unitary matrices ([Proposition 10.7.4](#)). Unwinding the induction,  $\mathbf{Q}$  is a product of one reflection (or  $\mathbf{I}_m$ ) for each of the columns  $1, \dots, \min(n, m-1)$ , since the recursion stops once one row is left. This proves the theorem. ■

### Example. ( $\mathbf{A}$ $3 \times 3$ QR by reflections)

Triangularize

$$\mathbf{A} = \begin{pmatrix} 2 & 0 & 1 \\ 1 & 5 & 2 \\ 2 & 5 & 4 \end{pmatrix}$$

by Householder reflections, and read off  $\mathbf{Q}$  and  $\mathbf{R}$ .

*Solution. Step 1.* The first column is  $\mathbf{a} = (2, 1, 2)$  with  $\|\mathbf{a}\| = 3$  and  $a_1 = 2 > 0$ , so  $\alpha = 1$  and

$$\mathbf{w}_1 = (2, 1, 2) + 3(1, 0, 0) = (5, 1, 2), \quad \|\mathbf{w}_1\|^2 = 30.$$

Apply  $(\ddagger)$  to each column:  $\mathbf{H}_1\mathbf{v} = \mathbf{v} - \frac{1}{15}\langle \mathbf{v}, \mathbf{w}_1 \rangle \mathbf{w}_1$ . All three columns pair with  $\mathbf{w}_1$  to

give 15, so each loses exactly one copy of  $\mathbf{w}_1$ :

$$\mathbf{H}_1 \mathbf{A} = \begin{pmatrix} -3 & -5 & -4 \\ 0 & 4 & 1 \\ 0 & 3 & 2 \end{pmatrix}.$$

(As predicted, the first column is  $-\|\mathbf{a}\|\mathbf{e}_1$ .)

*Step 2.* Work now in the trailing  $2 \times 2$  block, whose first column is  $\mathbf{y} = (4, 3)$  with  $\|\mathbf{y}\| = 5$  and  $y_1 > 0$ . So  $\mathbf{w}_2 = (9, 3)$ , and inside  $\mathbb{R}^3$  we reflect in  $\tilde{\mathbf{w}}_2 = (0, 9, 3)$ , which fixes  $\mathbf{e}_1$  and so leaves column 1 alone; here  $\mathbf{H}_2 \mathbf{v} = \mathbf{v} - \frac{1}{45} \langle \mathbf{v}, \tilde{\mathbf{w}}_2 \rangle \tilde{\mathbf{w}}_2$ . The second and third columns of  $\mathbf{H}_1 \mathbf{A}$  pair with  $\tilde{\mathbf{w}}_2$  to give  $36 + 9 = 45$  and  $9 + 6 = 15$ , so they lose one copy and a third of a copy respectively:

$$\mathbf{R} = \mathbf{H}_2 \mathbf{H}_1 \mathbf{A} = \begin{pmatrix} -3 & -5 & -4 \\ 0 & -5 & -2 \\ 0 & 0 & 1 \end{pmatrix}.$$

*The unitary factor.* By [Proposition 10.8.2](#) (b),  $\mathbf{A} = \mathbf{H}_1 \mathbf{H}_2 \mathbf{R}$ , so  $\mathbf{Q} = \mathbf{H}_1 \mathbf{H}_2$ . Computing the product,

$$\mathbf{Q} = \frac{1}{3} \begin{pmatrix} -2 & 2 & -1 \\ -1 & -2 & -2 \\ -2 & -1 & 2 \end{pmatrix},$$

whose columns have length 1 and are pairwise orthogonal, as one checks from the integer entries: for instance  $(-2)(2) + (-1)(-2) + (-2)(-1) = 0$ .

*Check.*  $\det \mathbf{A} = 15$  by expansion along the first row, and  $\det \mathbf{R} = (-3)(-5)(1) = 15$  by [Theorem 6.4.6](#), consistent with  $\det \mathbf{Q} = (-1)(-1) = 1$  from [Proposition 10.8.2](#) (e).

The diagonal of  $\mathbf{R}$  here is  $(-3, -5, 1)$ , not positive: the reflection route produces a factorization, not the one of [Theorem 10.8.1](#). The two are related by flipping the sign of each row of  $\mathbf{R}$  with a negative diagonal entry, and of the matching column of  $\mathbf{Q}$ ; [Exercise \(B2: The same matrix by reflections\)](#) carries this out on the matrix of [Example \(A  \$3 \times 2\$  QR, and a least-squares problem\)](#).

**Warning.** “The QR factorization” is definite only with the positivity normalization. [Theorem 10.8.1](#) pins the pair down by demanding  $r_{kk} > 0$ ; the reflection algorithm ignores that demand and chooses signs for stability instead. So two correct computations of “the QR factorization of  $\mathbf{A}$ ” can differ by signs. Compare  $|r_{kk}|$ , and the lines spanned by the columns of  $\mathbf{Q}$ .

**Where Gram–Schmidt cannot follow.** [Theorem 10.8.4](#) assumed nothing about the columns, and that is a genuine gain. Let the columns of  $\mathbf{A}$  be dependent, and let  $k$  be the first index with  $\mathbf{a}_k \in \text{span}(\mathbf{a}_1, \dots, \mathbf{a}_{k-1})$ . The Gram–Schmidt recursion then produces  $\mathbf{w}_k = \mathbf{0}$  and stops, because the next step divides by  $\|\mathbf{w}_k\|$ ; [Exercise \(C2: Gram–Schmidt on a dependent list\)](#) works this out. Nor is the failure an artifact of the method: if  $\mathbf{A} = \mathbf{Q}\mathbf{R}$  with  $\mathbf{Q}^* \mathbf{Q} = \mathbf{I}_n$  and  $\mathbf{R}$  upper triangular with positive diagonal, then  $\mathbf{R}$  is invertible by [Lemma 2.6.2](#), so  $\text{rank } \mathbf{A} = \text{rank } \mathbf{Q} = n$  and the columns were independent after all. A QR factorization in the sense of [Theorem 10.8.1](#) exists exactly for matrices of full column rank. The reflection route is untroubled: a dependent column simply puts a zero on the diagonal of  $\mathbf{R}$  where the dependence appears, and the algorithm continues ([Exercise \(C2: Reflections do not mind dependence\)](#)).

## 10.8.5 Givens rotations

A reflection zeroes a whole column at once. Sometimes that is too much: if  $\mathbf{A}$  is nearly triangular already, we would rather kill the few offending entries one at a time and leave the rest alone.

### Definition 10.8.2 — Givens Rotation

Let  $1 \leq i < j \leq m$  and let  $c, s \in \mathbb{R}$  with  $c^2 + s^2 = 1$ . The **Givens rotation**  $\mathbf{G}(i, j; c, s) \in M_m(\mathbb{R})$  is the identity matrix with the four entries in rows and columns  $i, j$  replaced by

$$\begin{pmatrix} g_{ii} & g_{ij} \\ g_{ji} & g_{jj} \end{pmatrix} = \begin{pmatrix} c & s \\ -s & c \end{pmatrix}.$$

The two columns involved are orthonormal, since  $c^2 + s^2 = 1$  and  $cs - sc = 0$ , and every other column of  $\mathbf{G}$  is a standard basis vector, orthogonal to both; so  $\mathbf{G}^\top \mathbf{G} = \mathbf{I}_m$  and  $\mathbf{G}$  is orthogonal, with  $\det \mathbf{G} = c^2 + s^2 = 1$  by expansion along those other columns (Theorem 6.6.1). Moreover  $\mathbf{G}$  acts as the identity on every coordinate other than  $i$  and  $j$ . Given a vector  $\mathbf{v}$  with  $(v_i, v_j) \neq (0, 0)$ , set

$$r = \sqrt{v_i^2 + v_j^2}, \quad c = \frac{v_i}{r}, \quad s = \frac{v_j}{r}.$$

Then the  $i$ -th entry of  $\mathbf{G}\mathbf{v}$  is  $(v_i^2 + v_j^2)/r = r$  and the  $j$ -th is  $(-v_j v_i + v_i v_j)/r = 0$ , while every other entry is unchanged. **One rotation, one new zero.**

Here is one rotation at work. To clear the  $(3, 1)$ -entry of the matrix  $\mathbf{A}$  below, take  $i = 1$ ,  $j = 3$  and for  $\mathbf{v}$  its first column, so that  $(v_1, v_3) = (3, 4)$  and  $r = 5$ ,  $c = \frac{3}{5}$ ,  $s = \frac{4}{5}$ . Then  $\mathbf{G} = \mathbf{G}(1, 3; \frac{3}{5}, \frac{4}{5})$  replaces row 1 by  $\frac{3}{5}(\text{row } 1) + \frac{4}{5}(\text{row } 3)$  and row 3 by  $-\frac{4}{5}(\text{row } 1) + \frac{3}{5}(\text{row } 3)$ , and leaves row 2 alone:

$$\mathbf{A} = \begin{pmatrix} 3 & 5 & 0 \\ 0 & 2 & 1 \\ 4 & 0 & 5 \end{pmatrix}, \quad \mathbf{GA} = \begin{pmatrix} 5 & 3 & 4 \\ 0 & 2 & 1 \\ 0 & -4 & 3 \end{pmatrix}.$$

The new  $(1, 1)$ -entry is  $\frac{1}{5}(3 \cdot 3 + 4 \cdot 4) = 5 = r$  and the new  $(3, 1)$ -entry is  $\frac{1}{5}(-4 \cdot 3 + 3 \cdot 4) = 0$ , as designed; the zero already sitting in position  $(2, 1)$  survives because row 2 was never touched. A reflection would have flattened the whole first column in one step, at the price of rewriting row 2 as well.

Applied in a suitable order, rotations triangularize any real matrix, which is a second proof of Theorem 10.8.4 over  $\mathbb{R}$  at the cost of more steps: a full  $m \times n$  matrix needs about  $mn$  rotations where reflections need  $n$ . For a nearly triangular matrix the count reverses, which is why the Hessenberg matrices below are handled with rotations in practice. Over  $\mathbb{C}$  one takes  $c \in \mathbb{R}$ ,  $s \in \mathbb{C}$  with  $c^2 + |s|^2 = 1$ , and  $-\bar{s}$  in place of  $-s$ ; to zero the  $j$ -th entry of  $\mathbf{v}$  one puts  $r = \sqrt{|v_i|^2 + |v_j|^2}$  and takes  $c = |v_i|/r$  and  $s = (v_i/|v_i|)\bar{v}_j/r$  when  $v_i \neq 0$ , and  $c = 0$ ,  $s = 1$  when  $v_i = 0$ .

## 10.8.6 Hessenberg form

So far the unitary matrices acted on one side. A similarity needs both: to preserve eigenvalues we must pass from  $\mathbf{A}$  to  $\mathbf{Q}^* \mathbf{A} \mathbf{Q}$ , not to  $\mathbf{Q}^* \mathbf{A}$ . What can reflections achieve under that constraint?

Not triangularity, and the reason is worth seeing before the theorem. Suppose we reflect so that the first column becomes  $\rho \mathbf{e}_1$ . Multiplying on the right must not disturb column 1, so the right

factor must fix  $\mathbf{e}_1$  — and so must the left factor, since they are the same matrix. But a reflection fixing  $\mathbf{e}_1$  acts only on the coordinates  $2, \dots, n$ , so in column 1 the best it can do is compress rows  $2, \dots, n$  into row 2. **The entry just below the diagonal survives, by necessity.** That is exactly the pattern we can reach.

### Theorem 10.8.5 — Reduction to Hessenberg Form

Let  $\mathbf{A} \in M_n(F)$ ,  $n \geq 1$ . Then there is a unitary  $\mathbf{Q} \in M_n(F)$  such that  $\mathbf{Z} = \mathbf{Q}^* \mathbf{A} \mathbf{Q}$  is **upper Hessenberg**, that is,

$$z_{ij} = 0 \quad \text{whenever } i > j + 1.$$

The matrix  $\mathbf{Q}$  can be taken to be a product of  $n-2$  factors, each a Householder reflection or an identity matrix (and  $\mathbf{Q} = \mathbf{I}_n$  when  $n \leq 2$ ). For  $n = 4$  the pattern is

$$\mathbf{Z} = \begin{pmatrix} * & * & * & * \\ * & * & * & * \\ 0 & * & * & * \\ 0 & 0 & * & * \end{pmatrix}.$$

*Idea.* Work column by column, as in Theorem 10.8.4, but apply each reflection on both sides. At step  $k$  it is chosen to fix  $\mathbf{e}_1, \dots, \mathbf{e}_k$  and to flatten the part of column  $k$  in rows  $k+1, \dots, n$ . Two observations do the bookkeeping. On the **left** it only recombines rows  $k+1, \dots, n$ , and in the earlier columns those rows are already zero, so a combination of them is still zero. On the **right** it only recombines columns  $k+1, \dots, n$ , so the columns already in shape are untouched.

*Proof.* For  $0 \leq k \leq n-2$  say that  $\mathbf{B} \in M_n(F)$  satisfies  $(*)_k$  if  $b_{ij} = 0$  whenever  $j \leq k$  and  $i \geq j+2$ . Every matrix satisfies  $(*)_0$  vacuously, and  $(*)_{n-2}$  says precisely that  $\mathbf{B}$  is upper Hessenberg, since the conditions with  $j = n-1$  and  $j = n$  are empty.

Put  $\mathbf{A}_0 = \mathbf{A}$ . Let  $1 \leq k \leq n-2$  and suppose  $\mathbf{A}_{k-1}$  satisfies  $(*)_{k-1}$ . Let  $\mathbf{a} \in F^{n-k}$  be the vector of entries in rows  $k+1, \dots, n$  of column  $k$  of  $\mathbf{A}_{k-1}$ . If  $\mathbf{a} = \mathbf{0}$ , put  $\mathbf{H}_k = \mathbf{I}_{n-k}$ ; otherwise let  $\mathbf{H}_k \in M_{n-k}(F)$  be the reflection of Theorem 10.8.3 with  $\mathbf{H}_k \mathbf{a} = \rho \mathbf{e}_1$ , where  $\rho = -\alpha \|\mathbf{a}\|$ . Set

$$\mathbf{Q}_k = \begin{pmatrix} \mathbf{I}_k & \mathbf{0} \\ \mathbf{0} & \mathbf{H}_k \end{pmatrix} \in M_n(F), \quad \mathbf{A}_k = \mathbf{Q}_k \mathbf{A}_{k-1} \mathbf{Q}_k.$$

The block  $\mathbf{H}_k$  is unitary and Hermitian with  $\mathbf{H}_k^{-1} = \mathbf{H}_k$  — by Proposition 10.8.2 (a)–(c) for a reflection, and because  $\mathbf{I}_{n-k}^* = \mathbf{I}_{n-k} = \mathbf{I}_{n-k}^{-1}$  otherwise — so by Theorem 7.1.1  $\mathbf{Q}_k$  is unitary and Hermitian with  $\mathbf{Q}_k^{-1} = \mathbf{Q}_k$ , and  $\mathbf{Q}_k \mathbf{e}_j = \mathbf{e}_j$  for every  $j \leq k$ .

**Claim 1.**  $\mathbf{Q}_k \mathbf{A}_{k-1}$  satisfies  $(*)_{k-1}$ , and its entries in rows  $i \geq k+2$  of column  $k$  are 0.



*Proof.* By [Theorem 7.1.1](#), the rows  $1, \dots, k$  of  $\mathbf{Q}_k \mathbf{A}_{k-1}$  are those of  $\mathbf{A}_{k-1}$ , and each of the rows  $k+1, \dots, n$  is a linear combination of the rows  $k+1, \dots, n$  of  $\mathbf{A}_{k-1}$ . Fix a column  $j \leq k-1$ . By  $(*_k)$ , the entries of that column in rows  $i \geq j+2$  vanish, and  $k+1 \geq j+2$ , so all the entries being combined are 0 and the combination is 0. Hence  $(*_k)$  survives. In column  $k$ , the entries in rows  $k+1, \dots, n$  form the vector  $\mathbf{H}_k \mathbf{a} = \rho \mathbf{e}_1$ , whose entries after the first are 0; these sit in rows  $k+2, \dots, n$ . ■

**Claim 2.**  $\mathbf{A}_k = (\mathbf{Q}_k \mathbf{A}_{k-1}) \mathbf{Q}_k$  has the same first  $k$  columns as  $\mathbf{Q}_k \mathbf{A}_{k-1}$ .

*Proof.* For  $j \leq k$ , the  $j$ -th column of  $\mathbf{BQ}_k$  is  $\mathbf{B}(\mathbf{Q}_k \mathbf{e}_j) = \mathbf{B} \mathbf{e}_j$ , the  $j$ -th column of  $\mathbf{B}$ . ■

Combining the two claims,  $\mathbf{A}_k$  satisfies  $(*_k)$ : columns  $1, \dots, k-1$  keep the zeros guaranteed by Claim 1, and column  $k$  has zeros in rows  $k+2, \dots, n$ , also by Claim 1, and both survive the right multiplication by Claim 2.

Running  $k = 1, \dots, n-2$  produces  $\mathbf{A}_{n-2}$  satisfying  $(*_{n-2})$ , that is, upper Hessenberg. Put  $\mathbf{Q} = \mathbf{Q}_1 \mathbf{Q}_2 \cdots \mathbf{Q}_{n-2}$ , unitary as a product of unitary matrices ([Proposition 10.7.4](#)). Since each  $\mathbf{Q}_k^* = \mathbf{Q}_k$ ,

$$\mathbf{Q}^* \mathbf{A} \mathbf{Q} = \mathbf{Q}_{n-2} \cdots \mathbf{Q}_1 \mathbf{A} \mathbf{Q}_1 \cdots \mathbf{Q}_{n-2} = \mathbf{A}_{n-2},$$

which proves the theorem. (For  $n \leq 2$  there are no steps and every matrix is already Hessenberg.) ■

Because  $\mathbf{Q}^{-1} = \mathbf{Q}^*$ , the matrices  $\mathbf{A}$  and  $\mathbf{Z}$  are similar ([Definition 3.7.2](#)), so they share their characteristic polynomial, their eigenvalues and their trace. The reduction is not merely tidying: it removes  $\binom{n-1}{2}$  entries and keeps **the whole spectrum**.

#### Example. (Hessenberg form of a $3 \times 3$ matrix)

Reduce

$$\mathbf{A} = \begin{pmatrix} 2 & 1 & 0 \\ 0 & 3 & 1 \\ 1 & 0 & 4 \end{pmatrix}$$

to upper Hessenberg form by a unitary similarity.

*Solution.* Only  $k = 1$  occurs, and only the entry  $a_{31}$  has to be killed. The relevant vector is  $\mathbf{a} = (a_{21}, a_{31}) = (0, 1) \in \mathbb{R}^2$ , with  $\|\mathbf{a}\| = 1$  and  $a_1 = 0$ , so  $\alpha = 1$  and

$$\mathbf{w} = (0, 1) + (1, 0) = (1, 1), \quad \|\mathbf{w}\|^2 = 2.$$

Hence  $\mathbf{H}_1 = \mathbf{I}_2 - \mathbf{w}\mathbf{w}^\top = \begin{pmatrix} 0 & -1 \\ -1 & 0 \end{pmatrix}$ , and

$$\mathbf{Q}_1 = \begin{pmatrix} 1 & 0 & 0 \\ 0 & 0 & -1 \\ 0 & -1 & 0 \end{pmatrix}.$$

Left multiplication by  $\mathbf{Q}_1$  exchanges rows 2 and 3 and negates them, and right multiplication does the same to columns 2 and 3:

$$\mathbf{Q}_1\mathbf{A} = \begin{pmatrix} 2 & 1 & 0 \\ -1 & 0 & -4 \\ 0 & -3 & -1 \end{pmatrix}, \quad \mathbf{Q}_1\mathbf{A}\mathbf{Q}_1 = \begin{pmatrix} 2 & 0 & -1 \\ -1 & 4 & 0 \\ 0 & 1 & 3 \end{pmatrix}.$$

Entry  $(3, 1)$  is 0 and entry  $(2, 1)$  is  $-1 = -\|\mathbf{a}\|$ , as [Theorem 10.8.3](#) predicts. Watch column 1 through the second multiplication: it is untouched, because  $\mathbf{Q}_1\mathbf{e}_1 = \mathbf{e}_1$ . Both matrices have trace 9, as similarity requires.

*Warning.* **Hessenberg form is not one small step from triangular form.** A few more reflections will not clear the subdiagonal, and not merely because the argument above breaks. The diagonal of a triangular matrix similar to  $\mathbf{A}$  lists the eigenvalues of  $\mathbf{A}$  ([Theorem 8.7.4](#)), so a finite procedure using only arithmetic and square roots that always produced one would extract the roots of an arbitrary polynomial from its coefficients by those operations alone. No such procedure exists: from degree five on, the roots of a general polynomial cannot be written with arithmetic and radicals at all. That is a theorem of Abel and Ruffini, which this book does not prove and which nothing later depends on; we quote it only to explain why the algorithms of Chapter 23 iterate instead of terminating. Over  $\mathbb{C}$  a triangularizing unitary does exist ([Theorem 8.7.2](#)); what does not exist is a finite algorithm of this kind producing it. Over  $\mathbb{R}$  even the existence fails: a rotation in  $O(2)$  with no real eigenvalue is already Hessenberg and is orthogonally similar to no real triangular matrix.

Hessenberg form is where the practical computation of eigenvalues begins. One applies the reduction once, then iterates a cheap step that preserves the pattern: factor  $\mathbf{Z} = \mathbf{Q}\mathbf{R}$  and replace  $\mathbf{Z}$  by  $\mathbf{R}\mathbf{Q} = \mathbf{Q}^*\mathbf{Z}\mathbf{Q}$ , again a unitary similarity. Chapter 23 develops that iteration and explains why every implementation starts with the reduction proved here.

*Quick check.* In the reduction of a  $5 \times 5$  matrix, how many reflections are used, and how many entries does the second one set to zero?

*Solution.* Three, namely  $k = 1, 2, 3$ . The step  $k = 2$  chooses a reflection acting on the coordinates 3, 4, 5 and sets the entries in rows 4 and 5 of column 2 to zero: two entries. (Step 1 sets three, step 3 sets one, for  $\binom{4}{2} = 6$  in total.)

## 10.8.7 Exercises

### 10.8.7.1 A. Check your understanding

**Exercise A1.**

- (a) State [Theorem 10.8.1](#) in full, including the two normalizations that make  $\mathbf{Q}$  and  $\mathbf{R}$  unique.
- (b) Which hypothesis on  $\mathbf{A}$  does [Theorem 10.8.1](#) need, and which does [Theorem 10.8.4](#) need? Explain the difference in one sentence.
- (c) Write down the Householder reflection determined by  $\mathbf{w}$ , and list three of its properties.
- (d) In [Theorem 10.8.3](#), which of the two possible targets  $\pm\|\mathbf{x}\|\mathbf{e}_1$  is chosen, and why?
- (e) True or false: applying reflections on both sides, as in [Theorem 10.8.5](#), can be pushed to make every complex matrix unitarily similar to an upper triangular one. Justify your answer.
- (f) Why must the reflection used at step  $k$  of [Theorem 10.8.5](#) fix  $\mathbf{e}_1, \dots, \mathbf{e}_k$ ?

*Solution.*

- (a) If the columns of  $\mathbf{A} \in M_{m \times n}(F)$  are independent, then  $\mathbf{A} = \mathbf{QR}$  with  $\mathbf{R}$  upper triangular; the normalizations are that the columns of  $\mathbf{Q}$  are **orthonormal** and that the diagonal of  $\mathbf{R}$  is **positive real**. With both, the pair is unique.
- (b) [Theorem 10.8.1](#) needs independent columns; [Theorem 10.8.4](#) needs nothing. A positive diagonal forces  $\mathbf{R}$  to be invertible and hence  $\text{rank } \mathbf{A} = n$ , while the reflection route is content to put a zero on the diagonal where a dependence occurs.
- (c)  $\mathbf{H}_{\mathbf{w}} = \mathbf{I}_m - 2\mathbf{w}\mathbf{w}^*/\|\mathbf{w}\|^2$  ([Definition 10.8.1](#)); it is Hermitian, unitary, equal to its own inverse, and has determinant  $-1$  ([Proposition 10.8.2](#)).
- (d) The target  $-\alpha\|\mathbf{x}\|\mathbf{e}_1$ , where  $\alpha$  is the phase (over  $\mathbb{R}$ , the sign) of  $x_1$  — that is, the target on the **opposite** side from  $x_1$ . The reason is arithmetic: this makes the first entry of  $\mathbf{w} = \mathbf{x} + \alpha\|\mathbf{x}\|\mathbf{e}_1$  a sum of like-signed quantities, with no cancellation of leading digits.
- (e) False. A reflection preserving the zeros already created must fix  $\mathbf{e}_1, \dots, \mathbf{e}_k$ , and then it cannot touch the entry in row  $k+1$  of column  $k$ . More decisively, such a finite procedure would compute the roots of an arbitrary polynomial by arithmetic and square roots alone, which the theorem of Abel and Ruffini quoted in the warning rules out.
- (f) Because the same matrix multiplies on the right, and right multiplication by  $\mathbf{Q}_k$  changes column  $j$  only if  $\mathbf{Q}_k\mathbf{e}_j \neq \mathbf{e}_j$ . Fixing  $\mathbf{e}_1, \dots, \mathbf{e}_k$  is exactly what leaves columns  $1, \dots, k$ , and the zeros in them, alone.

**10.8.7.2 B. Practice****Exercise B1: QR by Gram–Schmidt.**

Find the QR factorization of

$$\mathbf{A} = \begin{pmatrix} 1 & 1 & 1 \\ 1 & 1 & 0 \\ 1 & 0 & 0 \end{pmatrix}$$

in the sense of [Theorem 10.8.1](#), and verify that  $\mathbf{R} = \mathbf{Q}^\top \mathbf{A}$  for the top-left  $2 \times 2$  entries.

*Solution. Step 1.*  $\mathbf{a}_1 = (1, 1, 1)$  has  $\|\mathbf{a}_1\| = \sqrt{3}$ , so  $r_{11} = \sqrt{3}$  and  $\mathbf{q}_1 = \frac{1}{\sqrt{3}}(1, 1, 1)$ .

*Step 2.*  $r_{12} = \langle \mathbf{a}_2, \mathbf{q}_1 \rangle = \frac{2}{\sqrt{3}}$ , so

$$\mathbf{w}_2 = (1, 1, 0) - \frac{2}{3}(1, 1, 1) = \frac{1}{3}(1, 1, -2),$$

with  $\|\mathbf{w}_2\|^2 = \frac{1}{9} \cdot 6 = \frac{2}{3}$ . Hence  $r_{22} = \sqrt{2/3}$  and  $\mathbf{q}_2 = \frac{1}{\sqrt{6}}(1, 1, -2)$ .

*Step 3.*  $r_{13} = \langle \mathbf{a}_3, \mathbf{q}_1 \rangle = \frac{1}{\sqrt{3}}$  and  $r_{23} = \langle \mathbf{a}_3, \mathbf{q}_2 \rangle = \frac{1}{\sqrt{6}}$ , so

$$\mathbf{w}_3 = (1, 0, 0) - \frac{1}{3}(1, 1, 1) - \frac{1}{6}(1, 1, -2) = \left(\frac{1}{2}, -\frac{1}{2}, 0\right),$$

with  $\|\mathbf{w}_3\| = \frac{1}{\sqrt{2}} = r_{33}$  and  $\mathbf{q}_3 = \frac{1}{\sqrt{2}}(1, -1, 0)$ . Therefore

$$\mathbf{Q} = \begin{pmatrix} \frac{1}{\sqrt{3}} & \frac{1}{\sqrt{6}} & \frac{1}{\sqrt{2}} \\ \frac{1}{\sqrt{3}} & \frac{1}{\sqrt{6}} & -\frac{1}{\sqrt{2}} \\ \frac{1}{\sqrt{3}} & -\frac{2}{\sqrt{6}} & 0 \end{pmatrix}, \quad \mathbf{R} = \begin{pmatrix} \sqrt{3} & \frac{2}{\sqrt{3}} & \frac{1}{\sqrt{3}} \\ 0 & \frac{2}{\sqrt{6}} & \frac{1}{\sqrt{6}} \\ 0 & 0 & \frac{1}{\sqrt{2}} \end{pmatrix},$$

where  $r_{22} = \sqrt{2/3} = 2/\sqrt{6}$ . All three diagonal entries are positive, as required.

For the verification,  $(\mathbf{Q}^\top \mathbf{A})_{11} = \langle \mathbf{a}_1, \mathbf{q}_1 \rangle = 3/\sqrt{3} = \sqrt{3} = r_{11}$  and  $(\mathbf{Q}^\top \mathbf{A})_{12} = \langle \mathbf{a}_2, \mathbf{q}_1 \rangle = 2/\sqrt{3} = r_{12}$ ; also  $(\mathbf{Q}^\top \mathbf{A})_{21} = \langle \mathbf{a}_1, \mathbf{q}_2 \rangle = (1 + 1 - 2)/\sqrt{6} = 0 = r_{21}$  and  $(\mathbf{Q}^\top \mathbf{A})_{22} = \langle \mathbf{a}_2, \mathbf{q}_2 \rangle = 2/\sqrt{6} = r_{22}$ .

### Exercise B2: The same matrix by reflections.

Let  $\mathbf{A} = \begin{pmatrix} 1 & 1 \\ 2 & 1 \\ 2 & 3 \end{pmatrix}$ , the matrix of [Example \(A  \$3 \times 2\$  QR, and a least-squares problem\)](#).

- Find  $\mathbf{w}_1$  and the reflection  $\mathbf{H}_1$  of [Theorem 10.8.3](#) for the first column of  $\mathbf{A}$ , and compute  $\mathbf{H}_1 \mathbf{A}$ .
- Find the second reflection and hence a factorization  $\mathbf{A} = \mathbf{Q} \mathbf{R}$  with  $\mathbf{Q} \in M_3(\mathbb{R})$  orthogonal and  $\mathbf{R} \in M_{3 \times 2}(\mathbb{R})$  upper triangular.
- Compare the  $\mathbf{R}$  of (b) with the  $\mathbf{R}$  of [Example \(A  \$3 \times 2\$  QR, and a least-squares problem\)](#), and say exactly how the two factorizations differ.

*Solution.*

- The first column is  $\mathbf{a} = (1, 2, 2)$  with  $\|\mathbf{a}\| = 3$  and  $a_1 = 1 > 0$ , so  $\alpha = 1$  and  $\mathbf{w}_1 = (1, 2, 2) + 3\mathbf{e}_1 = (4, 2, 2)$ , with  $\|\mathbf{w}_1\|^2 = 24$ . By [\(‡\)](#),  $\mathbf{H}_1 \mathbf{v} = \mathbf{v} - \frac{1}{12} \langle \mathbf{v}, \mathbf{w}_1 \rangle \mathbf{w}_1$ . Since  $\langle \mathbf{a}, \mathbf{w}_1 \rangle = 4 + 4 + 4 = 12$  and  $\langle (1, 1, 3), \mathbf{w}_1 \rangle = 4 + 2 + 6 = 12$ , each column loses one copy of  $\mathbf{w}_1$ :

$$\mathbf{H}_1 \mathbf{A} = \begin{pmatrix} -3 & -3 \\ 0 & -1 \\ 0 & 1 \end{pmatrix}.$$

- The trailing vector is  $\mathbf{y} = (-1, 1)$ , with  $\|\mathbf{y}\| = \sqrt{2}$  and  $y_1 = -1 < 0$ , so  $\alpha = -1$  and  $\mathbf{w}_2 = (-1, 1) - \sqrt{2}(1, 0) = (-1 - \sqrt{2}, 1)$ . By [Theorem 10.8.3](#), the reflection

sends  $\mathbf{y}$  to  $+\sqrt{2}\mathbf{e}_1$ . Embedding it as  $\mathbf{H}_2 = \mathbf{I}_1 \oplus \mathbf{H}_{\mathbf{w}_2}$ , we get

$$\mathbf{R} = \mathbf{H}_2\mathbf{H}_1\mathbf{A} = \begin{pmatrix} -3 & -3 \\ 0 & \sqrt{2} \\ 0 & 0 \end{pmatrix}, \quad \mathbf{Q} = \mathbf{H}_1\mathbf{H}_2,$$

and  $\mathbf{A} = \mathbf{QR}$  because each  $\mathbf{H}_i$  is its own inverse (Proposition 10.8.2 (b)).

- (c) In Example (A  $3 \times 2$  QR, and a least-squares problem),  $\mathbf{R}_0 = \begin{pmatrix} 3 & 3 \\ 0 & \sqrt{2} \end{pmatrix}$ . Deleting the zero row of  $\mathbf{R}$  leaves  $\mathbf{R}_0$  with its **first row negated**, and correspondingly the first column of  $\mathbf{Q}$  is  $-\mathbf{q}_1$  while the second is  $\mathbf{q}_2$ . The reflection route did not aim for a positive diagonal, and by the uniqueness in Theorem 10.8.1 signs are the only discrepancy possible.

### Exercise B3: A least-squares fit through QR.

Fit a line  $y = c_0 + c_1t$  to the data  $(t, y) = (0, 1), (1, 3), (2, 2)$  in the least-squares sense, by finding the QR factorization of the design matrix and solving  $\mathbf{R}\mathbf{x} = \mathbf{Q}^\top \mathbf{b}$ . Compute the residual and check the answer against the normal equations.

*Solution.* The system is  $\mathbf{A}\mathbf{x} = \mathbf{b}$  with

$$\mathbf{A} = \begin{pmatrix} 1 & 0 \\ 1 & 1 \\ 1 & 2 \end{pmatrix}, \quad \mathbf{b} = \begin{pmatrix} 1 \\ 3 \\ 2 \end{pmatrix}, \quad \mathbf{x} = (c_0, c_1).$$

Gram–Schmidt:  $\|\mathbf{a}_1\| = \sqrt{3}$ , so  $r_{11} = \sqrt{3}$  and  $\mathbf{q}_1 = \frac{1}{\sqrt{3}}(1, 1, 1)$ . Then  $r_{12} = \langle \mathbf{a}_2, \mathbf{q}_1 \rangle = 3/\sqrt{3} = \sqrt{3}$  and

$$\mathbf{w}_2 = (0, 1, 2) - (1, 1, 1) = (-1, 0, 1),$$

so  $r_{22} = \sqrt{2}$  and  $\mathbf{q}_2 = \frac{1}{\sqrt{2}}(-1, 0, 1)$ . Now  $\langle \mathbf{b}, \mathbf{q}_1 \rangle = 6/\sqrt{3} = 2\sqrt{3}$  and  $\langle \mathbf{b}, \mathbf{q}_2 \rangle = (-1 + 0 + 2)/\sqrt{2} = 1/\sqrt{2}$ , so (†) reads

$$\begin{pmatrix} \sqrt{3} & \sqrt{3} \\ 0 & \sqrt{2} \end{pmatrix} \mathbf{x} = \begin{pmatrix} 2\sqrt{3} \\ 1/\sqrt{2} \end{pmatrix}.$$

Back substitution gives  $c_1 = \frac{1}{2}$  and then  $\sqrt{3}(c_0 + \frac{1}{2}) = 2\sqrt{3}$ , so  $c_0 = \frac{3}{2}$ . The fitted line is  $y = \frac{3}{2} + \frac{1}{2}t$ , with residual  $\mathbf{A}\mathbf{x} - \mathbf{b} = (\frac{1}{2}, -1, \frac{1}{2})$  and  $\|\mathbf{A}\mathbf{x} - \mathbf{b}\|^2 = \frac{3}{2}$ .

*Check.*  $\mathbf{A}^\top \mathbf{A} = \begin{pmatrix} 3 & 3 \\ 3 & 5 \end{pmatrix}$  and  $\mathbf{A}^\top \mathbf{b} = (6, 7)$ . Subtracting the first normal equation from the second gives  $2c_1 = 1$ , and then  $3c_0 + \frac{3}{2} = 6$ , so  $c_0 = \frac{3}{2}$ . The residual is orthogonal to both columns of  $\mathbf{A}$ :  $\frac{1}{2} - 1 + \frac{1}{2} = 0$  and  $0 \cdot \frac{1}{2} - 1 \cdot 1 + 2 \cdot \frac{1}{2} = 0$ , which is what the normal equations of Theorem 10.4.2 say.

### Exercise B4: Determine which are QR factorizations.

Determine which of the following are QR factorizations of the matrix on the left in the sense of Theorem 10.8.1. Justify your answer; for those that are not, name the clause that fails and repair it if possible.

- (a)  $\begin{pmatrix} 3 & 1 \\ 4 & 7 \end{pmatrix} = \frac{1}{5} \begin{pmatrix} 3 & -4 \\ 4 & 3 \end{pmatrix} \begin{pmatrix} 5 & \frac{31}{5} \\ 0 & \frac{17}{5} \end{pmatrix}.$
- (b)  $\begin{pmatrix} 1 & 0 \\ 1 & 1 \end{pmatrix} = \begin{pmatrix} 1 & -1 \\ 1 & 1 \end{pmatrix} \begin{pmatrix} 1 & \frac{1}{2} \\ 0 & \frac{1}{2} \end{pmatrix}.$
- (c)  $\begin{pmatrix} -1 & 0 \\ 0 & 1 \end{pmatrix} = \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix} \begin{pmatrix} -1 & 0 \\ 0 & 1 \end{pmatrix}.$

*Solution.*

- (a) Yes. The product is

$$\frac{1}{5} \begin{pmatrix} 15 & \frac{93}{5} - \frac{68}{5} \\ 20 & \frac{124}{5} + \frac{51}{5} \end{pmatrix} = \frac{1}{5} \begin{pmatrix} 15 & 5 \\ 20 & 35 \end{pmatrix} = \begin{pmatrix} 3 & 1 \\ 4 & 7 \end{pmatrix},$$

the columns of the left factor are  $\frac{1}{5}(3, 4)$  and  $\frac{1}{5}(-4, 3)$ , both of length 1 and orthogonal, and the diagonal entries 5 and  $\frac{17}{5}$  are positive. All three clauses hold, so by the uniqueness in [Theorem 10.8.1](#) this is **the** QR factorization.

- (b) No. The identity is correct, and  $\mathbf{R}$  is upper triangular with positive diagonal, but the columns of the left factor have length  $\sqrt{2}$ , not 1, so clause (a) of [Theorem 10.8.1](#) fails. Repair it by dividing each column of  $\mathbf{Q}$  by  $\sqrt{2}$  and multiplying the matching row of  $\mathbf{R}$  by  $\sqrt{2}$ :

$$\mathbf{Q} = \frac{1}{\sqrt{2}} \begin{pmatrix} 1 & -1 \\ 1 & 1 \end{pmatrix}, \quad \mathbf{R} = \begin{pmatrix} \sqrt{2} & \frac{1}{\sqrt{2}} \\ 0 & \frac{1}{\sqrt{2}} \end{pmatrix}.$$

- (c) No. The identity is correct and the columns of  $\mathbf{Q} = \mathbf{I}_2$  are orthonormal, but  $r_{11} = -1 < 0$ , so clause (b) fails. Repair it by moving the sign across:  $\mathbf{Q} = \text{diag}(-1, 1)$ , whose columns are still orthonormal, and  $\mathbf{R} = \mathbf{I}_2$ , whose diagonal is positive. This is the move described in [Exercise \(C1: All the QR factorizations of a matrix\)](#) with  $\mathbf{D} = \text{diag}(-1, 1)$ .

### 10.8.7.3 C. Going deeper

#### Exercise C1: All the QR factorizations of a matrix.

Let  $\mathbf{A} \in M_{m \times n}(F)$  have independent columns, with QR factorization  $\mathbf{A} = \mathbf{Q}\mathbf{R}$  as in [Theorem 10.8.1](#).

- (a) Prove that an upper triangular  $\mathbf{S} \in M_n(F)$  with  $\mathbf{S}^* \mathbf{S} = \mathbf{I}_n$  is diagonal, with diagonal entries of modulus 1.
- (b) Deduce that the pairs  $(\mathbf{Q}_1, \mathbf{R}_1)$  with  $\mathbf{Q}_1^* \mathbf{Q}_1 = \mathbf{I}_n$ ,  $\mathbf{R}_1$  upper triangular and invertible, and  $\mathbf{A} = \mathbf{Q}_1 \mathbf{R}_1$ , are exactly the pairs  $(\mathbf{Q} \mathbf{D}^{-1}, \mathbf{D} \mathbf{R})$  for  $\mathbf{D} = \text{diag}(d_1, \dots, d_n)$  with  $|d_k| = 1$ .
- (c) Hence give a second proof of the uniqueness statement of [Theorem 10.8.1](#).

*Solution.*

- (a) Let  $\mathbf{s}_1, \dots, \mathbf{s}_n$  be the columns of  $\mathbf{S}$ ; as in the proof of [Theorem 10.8.1](#),  $\mathbf{S}^* \mathbf{S} = \mathbf{I}_n$  says  $\langle \mathbf{s}_j, \mathbf{s}_i \rangle = \delta_{ij}$ . We show  $\mathbf{s}_j = s_{jj} \mathbf{e}_j$  with  $|s_{jj}| = 1$ , by induction on  $j$ . Suppose

this holds for every  $i < j$ . For such  $i$ ,

$$0 = \langle \mathbf{s}_j, \mathbf{s}_i \rangle = \overline{s_{ii}} \langle \mathbf{s}_j, \mathbf{e}_i \rangle = \overline{s_{ii}} s_{ij},$$

and  $s_{ii} \neq 0$ , so  $s_{ij} = 0$ . Together with upper triangularity this gives  $\mathbf{s}_j = s_{jj} \mathbf{e}_j$ , and  $\|\mathbf{s}_j\| = 1$  gives  $|s_{jj}| = 1$ .

- (b) ( $\Leftarrow$ ) If  $|d_k| = 1$  for all  $k$ , then  $\mathbf{D}^* \mathbf{D} = \mathbf{I}_n$ , so  $(\mathbf{QD}^{-1})^* (\mathbf{QD}^{-1}) = (\mathbf{D}^{-1})^* \mathbf{Q}^* \mathbf{QD}^{-1} = (\mathbf{DD}^*)^{-1} = \mathbf{I}_n$ ; the matrix  $\mathbf{DR}$  is upper triangular and invertible; and  $(\mathbf{QD}^{-1})(\mathbf{DR}) = \mathbf{QR} = \mathbf{A}$ .

( $\Rightarrow$ ) Suppose  $\mathbf{A} = \mathbf{Q}_1 \mathbf{R}_1$  as stated. Exactly as in the proof of [Theorem 10.8.1](#),  $\mathbf{R}_1^* \mathbf{R}_1 = \mathbf{A}^* \mathbf{A} = \mathbf{R}^* \mathbf{R}$ , and  $\mathbf{S} = \mathbf{R}_1 \mathbf{R}^{-1}$  is upper triangular with  $\mathbf{S}^* \mathbf{S} = \mathbf{I}_n$ . By

(a),  $\mathbf{S} = \mathbf{D}$  is diagonal with  $|d_k| = 1$ . Then  $\mathbf{R}_1 = \mathbf{DR}$  and  $\mathbf{Q}_1 = \mathbf{AR}_1^{-1} = \mathbf{QRR}^{-1} \mathbf{D}^{-1} = \mathbf{QD}^{-1}$ .

- (c) If in addition both  $\mathbf{R}$  and  $\mathbf{R}_1 = \mathbf{DR}$  have positive real diagonal entries, then  $d_k = (\mathbf{R}_1)_{kk}/(\mathbf{R})_{kk}$  is a positive real number of modulus 1, so  $d_k = 1$  for every  $k$ . Hence  $\mathbf{D} = \mathbf{I}_n$ ,  $\mathbf{R}_1 = \mathbf{R}$  and  $\mathbf{Q}_1 = \mathbf{Q}$ .

### Exercise C2: Reflections do not mind dependence.

Let

$$\mathbf{A} = \begin{pmatrix} 1 & 1 & 2 \\ 2 & 1 & 3 \\ 2 & 3 & 5 \end{pmatrix},$$

whose third column is the sum of the first two.

- Run the Gram–Schmidt process of [Theorem 10.2.2](#) on the columns of  $\mathbf{A}$  and say precisely which step fails.
- Carry out the Householder reduction of [Theorem 10.8.4](#) on  $\mathbf{A}$  and exhibit  $\mathbf{R}$ .
- Explain, using [Theorem 10.8.1](#), why no factorization  $\mathbf{A} = \mathbf{QR}$  with  $\mathbf{Q}^* \mathbf{Q} = \mathbf{I}_3$  and  $\mathbf{R}$  upper triangular with positive diagonal can exist.

*Solution.*

- Steps 1 and 2 go through:  $\mathbf{q}_1 = \frac{1}{3}(1, 2, 2)$  and, since  $\langle \mathbf{a}_2, \mathbf{q}_1 \rangle = \frac{1}{3}(1 + 2 + 6) = 3$ ,  $\mathbf{w}_2 = (1, 1, 3) - (1, 2, 2) = (0, -1, 1)$  and  $\mathbf{q}_2 = \frac{1}{\sqrt{2}}(0, -1, 1)$ . At step 3,  $\mathbf{a}_3 = \mathbf{a}_1 + \mathbf{a}_2$  lies in  $\text{span}(\mathbf{a}_1, \mathbf{a}_2) = \text{span}(\mathbf{q}_1, \mathbf{q}_2)$ , so by [Theorem 10.2.1](#) (a) the vector subtracted from  $\mathbf{a}_3$  is  $\mathbf{a}_3$  itself and  $\mathbf{w}_3 = \mathbf{0}$ . The normalization  $\mathbf{q}_3 = \mathbf{w}_3/\|\mathbf{w}_3\|$  divides by 0 and fails. This is the situation of [Exercise \(C2: Gram–Schmidt on a dependent list\)](#).
- The first column is  $(1, 2, 2)$ , so as in [Exercise \(B2: The same matrix by reflections\)](#) we take  $\mathbf{w}_1 = (4, 2, 2)$  with  $\|\mathbf{w}_1\|^2 = 24$ , and  $\mathbf{H}_1 \mathbf{v} = \mathbf{v} - \frac{1}{12} \langle \mathbf{v}, \mathbf{w}_1 \rangle \mathbf{w}_1$ . The inner products of the three columns with  $\mathbf{w}_1$  are 12, 12 and 24, so

$$\mathbf{H}_1 \mathbf{A} = \begin{pmatrix} -3 & -3 & -6 \\ 0 & -1 & -1 \\ 0 & 1 & 1 \end{pmatrix}.$$

The trailing vector of column 2 is  $(-1, 1)$ , which by [Exercise \(B2: The same matrix by reflections\)](#) (b) the second reflection sends to  $\sqrt{2} \mathbf{e}_1$ ; it sends the trailing vector

of column 3, which is the same vector  $(-1, 1)$ , to the same place. Hence

$$\mathbf{R} = \begin{pmatrix} -3 & -3 & -6 \\ 0 & \sqrt{2} & \sqrt{2} \\ 0 & 0 & 0 \end{pmatrix},$$

upper triangular, with a zero in position  $(3, 3)$  exactly where the dependence sits. No step of the algorithm divided by zero.

- (c) Such an  $\mathbf{R}$  would be invertible by [Lemma 2.6.2](#), so  $\text{rank } \mathbf{A} = \text{rank}(\mathbf{QR}) = \text{rank } \mathbf{Q} = 3$ . But  $\mathbf{a}_3 = \mathbf{a}_1 + \mathbf{a}_2$ , so the columns are dependent and  $\text{rank } \mathbf{A} \leq 2$ , a contradiction. The positive-diagonal factorization exists precisely for matrices of full column rank.

### Exercise C3: Triangularizing with rotations.

- (a) Let  $(a, b) \in \mathbb{R}^2$  be non-zero and  $r = \sqrt{a^2 + b^2}$ . Verify that the Givens rotation  $\mathbf{G}$  of [Definition 10.8.2](#) with  $c = a/r$  and  $s = b/r$  satisfies  $\mathbf{G}(a, b) = (r, 0)$ , and that  $\mathbf{G} \in \text{SO}(2)$ .
- (b) Use (a) to write  $\mathbf{A} = \begin{pmatrix} 3 & 1 \\ 4 & 2 \end{pmatrix}$  as  $\mathbf{QR}$  with  $\mathbf{Q} \in \text{SO}(2)$  and  $\mathbf{R}$  upper triangular with positive diagonal.
- (c) How many Givens rotations does it take to triangularize a general  $3 \times 3$  real matrix, and which entries does each one kill? Compare with the number of Householder reflections.

*Solution.*

- (a)  $\mathbf{G}(a, b) = (ca + sb, -sa + cb) = ((a^2 + b^2)/r, (-ab + ab)/r) = (r, 0)$ . Also  $c^2 + s^2 = 1$ , so the columns of  $\mathbf{G}$  are orthonormal and  $\det \mathbf{G} = c^2 + s^2 = 1$ ; hence  $\mathbf{G} \in \text{SO}(2)$ .
- (b) The first column is  $(3, 4)$ , so  $r = 5$ ,  $c = \frac{3}{5}$ ,  $s = \frac{4}{5}$  and  $\mathbf{G} = \frac{1}{5} \begin{pmatrix} 3 & 4 \\ -4 & 3 \end{pmatrix}$ . Then

$$\mathbf{GA} = \frac{1}{5} \begin{pmatrix} 9 + 16 & 3 + 8 \\ -12 + 12 & -4 + 6 \end{pmatrix} = \begin{pmatrix} 5 & \frac{11}{5} \\ 0 & \frac{2}{5} \end{pmatrix} =: \mathbf{R},$$

whose diagonal entries 5 and  $\frac{2}{5}$  are positive. Since  $\mathbf{G}^{-1} = \mathbf{G}^\top \in \text{SO}(2)$ , we get

$\mathbf{A} = \mathbf{QR}$  with  $\mathbf{Q} = \mathbf{G}^\top = \frac{1}{5} \begin{pmatrix} 3 & -4 \\ 4 & 3 \end{pmatrix}$ . As a check,  $\det \mathbf{A} = 2 = 5 \cdot \frac{2}{5} = \det \mathbf{R}$ .

- (c) Three: one for  $a_{21}$  (rows 1, 2), one for  $a_{31}$  (rows 1, 3), and one for the new  $(3, 2)$ -entry (rows 2, 3). The last acts on rows 2 and 3, which by then have zeros in column 1, so it does not undo the earlier work. Householder needs two reflections, one per column, since each clears a whole column at once.

## 10.9 Bessel, Parseval and Fourier Series

Section 3 measured a vector against a finite-dimensional subspace and produced the nearest point in it. Read that answer coefficient by coefficient and it becomes an inequality: the numbers  $\langle \mathbf{v}, \mathbf{e}_i \rangle$  collected against an orthonormal list can never carry more squared length than  $\mathbf{v}$  itself



has. That is Bessel's inequality. It costs nothing in dimension, it holds in spaces far too large for Chapter 1's machinery, and its equality case is Parseval's identity. This section proves both, applies them to the sines and cosines, and then stops at the exact place where linear algebra stops and analysis begins.

Throughout,  $V$  is an inner product space over  $F = \mathbb{R}$  or  $F = \mathbb{C}$ , with inner product  $\langle \cdot, \cdot \rangle$  (Definition 10.1.1) and induced norm  $\|\cdot\|$  (Definition 10.1.3). **We do not assume that  $V$  is finite-dimensional.** What we assume each time is that the *list* we compute against is finite.

## 10.9.1 Coefficients against an orthonormal list

In Theorem 10.2.1 the list was an orthonormal **basis**, and the coefficients  $\langle \mathbf{v}, \mathbf{e}_i \rangle$  recovered  $\mathbf{v}$  exactly. Suppose now the list is only a list: too short to be a basis, or sitting inside a space that has no finite basis at all, like the continuous functions on an interval. The numbers  $\langle \mathbf{v}, \mathbf{e}_i \rangle$  are still there. What can be said about them when they no longer determine  $\mathbf{v}$ ?

*Against a shorter list you can only lose length, never gain it.*

### Definition 10.9.1 — Fourier Coefficients With Respect to an Orthonormal List

Let  $(\mathbf{e}_1, \dots, \mathbf{e}_k)$  be an orthonormal list in an inner product space  $V$  and let  $\mathbf{v} \in V$ . The scalars

$$\langle \mathbf{v}, \mathbf{e}_1 \rangle, \dots, \langle \mathbf{v}, \mathbf{e}_k \rangle \in F$$

are the **Fourier coefficients** of  $\mathbf{v}$  with respect to the list.

The name is borrowed from the classical case below, where the  $\mathbf{e}_i$  are sines and cosines; nothing in the definition mentions trigonometry. When the list happens to be a basis, these are exactly the coordinates of  $\mathbf{v}$ , by Theorem 10.2.1 (a). When it is not, they are the coordinates of a *shadow* of  $\mathbf{v}$ , and the next theorem measures that shadow.

### Theorem 10.9.1 — Bessel's Inequality

Let  $V$  be an inner product space over  $F$ , **not assumed finite-dimensional**, and let  $(\mathbf{e}_1, \dots, \mathbf{e}_k)$  be an orthonormal list in  $V$ . Then for **every**  $\mathbf{v} \in V$ ,

$$\sum_{i=1}^k |\langle \mathbf{v}, \mathbf{e}_i \rangle|^2 \leq \|\mathbf{v}\|^2.$$

*Idea.* The left-hand side is not a quantity out of nowhere. Let  $U$  be the span of the list. By the projection formula,  $P_U \mathbf{v}$  is exactly  $\sum_i \langle \mathbf{v}, \mathbf{e}_i \rangle \mathbf{e}_i$ , and its summands are pairwise orthogonal, so Pythagoras turns its squared length into precisely that sum. The inequality then says no more than “a shadow is no longer than what casts it”, which we already proved. The one point needing care is that  $V$  may be infinite-dimensional; it does not matter, because Section 3 only ever asked the **subspace** to be finite-dimensional.

*Proof.* Put  $U = \text{span}(\mathbf{e}_1, \dots, \mathbf{e}_k)$ . The vectors  $\mathbf{e}_i$  are orthogonal and non-zero, so the list is linearly independent by Theorem 10.1.4; being a spanning list of  $U$  as well, it is

an orthonormal basis of  $U$ , and  $\dim U = k$  is finite. Hence [Theorem 10.3.3](#) applies to  $U$  even though  $V$  need not be finite-dimensional, and part (a) of it gives

$$P_U \mathbf{v} = \sum_{i=1}^k \langle \mathbf{v}, \mathbf{e}_i \rangle \mathbf{e}_i.$$

The summands  $\langle \mathbf{v}, \mathbf{e}_i \rangle \mathbf{e}_i$  are pairwise orthogonal, since  $\mathbf{e}_i \perp \mathbf{e}_j$  for  $i \neq j$  and scalars come out of both slots. So [Theorem 10.1.3](#) and [Theorem 10.1.2](#) (b) give

$$\|P_U \mathbf{v}\|^2 = \sum_{i=1}^k |\langle \mathbf{v}, \mathbf{e}_i \rangle|^2 \|\mathbf{e}_i\|^2 = \sum_{i=1}^k |\langle \mathbf{v}, \mathbf{e}_i \rangle|^2,$$

using  $\|\mathbf{e}_i\| = 1$ . By [Theorem 10.3.3](#) (c),  $\|P_U \mathbf{v}\| \leq \|\mathbf{v}\|$ . Both sides are non-negative, so squaring preserves the inequality, and this proves the theorem. ■

The proof gives more than the inequality, and the surplus is worth recording. The vector  $\mathbf{v} - P_U \mathbf{v}$  lies in  $U^\perp$  by [Theorem 10.3.2](#), so it is orthogonal to  $P_U \mathbf{v} \in U$ , and [Theorem 10.1.3](#) turns  $\mathbf{v} = P_U \mathbf{v} + (\mathbf{v} - P_U \mathbf{v})$  into  $\|\mathbf{v}\|^2 = \|P_U \mathbf{v}\|^2 + \|\mathbf{v} - P_U \mathbf{v}\|^2$ . Rearranging, and substituting  $\|P_U \mathbf{v}\|^2 = \sum_{i=1}^k |\langle \mathbf{v}, \mathbf{e}_i \rangle|^2$  from the proof above,

$$\|\mathbf{v} - P_U \mathbf{v}\|^2 = \|\mathbf{v}\|^2 - \sum_{i=1}^k |\langle \mathbf{v}, \mathbf{e}_i \rangle|^2. \quad (*)$$

In words: **the deficit in Bessel's inequality is the squared distance from  $\mathbf{v}$  to the span of the list** ([Definition 10.3.3](#)). An inequality with a known defect is much more useful than an inequality, and we will read  $(*)$  from both sides.

*Remark.* For an infinite orthonormal list  $(\mathbf{e}_1, \mathbf{e}_2, \mathbf{e}_3, \dots)$ , apply the theorem to the first  $k$  terms for each  $k$ . The partial sums  $\sum_{i \leq k} |\langle \mathbf{v}, \mathbf{e}_i \rangle|^2$  increase with  $k$  and are all bounded above by  $\|\mathbf{v}\|^2$ , so they converge, and  $\sum_{i=1}^{\infty} |\langle \mathbf{v}, \mathbf{e}_i \rangle|^2 \leq \|\mathbf{v}\|^2$ . This is the only fact from analysis used in the present paragraph: a bounded increasing sequence of real numbers converges.

Now the equality case, which is where the word *basis* returns.

### Theorem 10.9.2 — Parseval's Identity

Let  $(\mathbf{e}_1, \dots, \mathbf{e}_k)$  be an orthonormal list in an inner product space  $V$  and let  $\mathbf{v} \in V$ . Then

$$\sum_{i=1}^k |\langle \mathbf{v}, \mathbf{e}_i \rangle|^2 = \|\mathbf{v}\|^2 \quad \text{if and only if} \quad \mathbf{v} \in \text{span}(\mathbf{e}_1, \dots, \mathbf{e}_k).$$

In particular, if  $(\mathbf{e}_1, \dots, \mathbf{e}_k)$  is an orthonormal **basis** of  $V$ , the identity holds for every  $\mathbf{v} \in V$ .

*Proof.* Keep the notation of the proof of [Theorem 10.9.1](#), where we showed  $\sum_i |\langle \mathbf{v}, \mathbf{e}_i \rangle|^2 = \|P_U \mathbf{v}\|^2$  with  $U = \text{span}(\mathbf{e}_1, \dots, \mathbf{e}_k)$ . So the stated equality says  $\|P_U \mathbf{v}\| = \|\mathbf{v}\|$ , and by [Theorem 10.3.3](#) (c) this holds if and only if  $\mathbf{v} \in U$ . If the list is a basis of  $V$ , then  $U = V$  and every  $\mathbf{v} \in V$  lies in  $U$ .

For an orthonormal basis of a finite-dimensional space this is [Theorem 10.2.1](#) (c) again; what is new is the equality **case**, which identifies exactly the vectors for which no length is lost. Keep the two statements apart:

- **Bessel** is about an orthonormal **list**, in **any** inner product space, and it is an inequality.
- **Parseval** is the equality, and it holds exactly for the vectors the list already spans.

The whole difference is whether the list reaches  $\mathbf{v}$ .

**Warning. An orthonormal list is not an orthonormal basis, and the deficit in Bessel can be everything.** In  $C[-\pi, \pi]$  below, the list  $(\cos x, \sin x)$  is orthonormal, and for the constant function  $f = 1$  both coefficients are 0, so the left-hand side of Bessel is 0 while  $\|f\|^2 = 2$ . Nothing at all is captured. A short orthonormal list is not a bad basis; it is not a basis.

**Quick check.** In  $\mathbb{R}^4$  with the dot product, let  $\mathbf{e}_1 = \frac{1}{2}(1, 1, 1, 1)$  and  $\mathbf{e}_2 = \frac{1}{2}(1, 1, -1, -1)$ , and let  $\mathbf{v} = (1, 2, 3, 4)$ . Check that  $(\mathbf{e}_1, \mathbf{e}_2)$  is orthonormal, compute both sides of Bessel's inequality, and say what the deficit equals.

**Solution.** Each has norm  $\frac{1}{2}\sqrt{4} = 1$ , and  $\langle \mathbf{e}_1, \mathbf{e}_2 \rangle = \frac{1}{4}(1 + 1 - 1 - 1) = 0$ , so the list is orthonormal. The coefficients are  $\langle \mathbf{v}, \mathbf{e}_1 \rangle = \frac{1}{2}(1 + 2 + 3 + 4) = 5$  and  $\langle \mathbf{v}, \mathbf{e}_2 \rangle = \frac{1}{2}(1 + 2 - 3 - 4) = -2$ , so the left-hand side is  $25 + 4 = 29$ , while  $\|\mathbf{v}\|^2 = 1 + 4 + 9 + 16 = 30$ . The inequality is strict, and by (\*) the deficit 1 is the squared distance from  $\mathbf{v}$  to  $\text{span}(\mathbf{e}_1, \mathbf{e}_2)$ . Indeed  $P_U \mathbf{v} = 5\mathbf{e}_1 - 2\mathbf{e}_2 = (\frac{3}{2}, \frac{3}{2}, \frac{7}{2}, \frac{7}{2})$  and  $\mathbf{v} - P_U \mathbf{v} = \frac{1}{2}(-1, 1, -1, 1)$ , whose squared norm is  $4 \cdot \frac{1}{4} = 1$ .

## 10.9.2 The trigonometric system

The classical home of these two statements is a space of functions, and the first thing to fix is the inner product, constant included.

Let  $V = C[-\pi, \pi]$  be the space of continuous functions  $f: [-\pi, \pi] \rightarrow \mathbb{R}$ , a real vector space, and define

$$\langle f, g \rangle = \frac{1}{\pi} \int_{-\pi}^{\pi} f(t) g(t) dt.$$

This is the continuous-function inner product of [Definition 10.1.1](#) on a different interval and rescaled by the positive constant  $1/\pi$ ; rescaling by a positive constant preserves all three axioms, and positive definiteness is the argument given there. The constant is chosen for one reason only: it makes the sines and cosines unit vectors. With  $1/(2\pi)$  in front they would have norm  $1/\sqrt{2}$ , and with no constant at all, norm  $\sqrt{\pi}$ .

**Warning. Always state the normalizing constant before computing.** Formulas for Fourier coefficients differ between books by exactly such a factor, and a misplaced  $\pi$  or 2 changes every number below. In this section  $\langle f, g \rangle = \frac{1}{\pi} \int_{-\pi}^{\pi} fg$ , and every coefficient, norm and inequality is written for that choice.

Here are the facts we assume. From calculus: the integral of a finite sum is the sum of the integrals;  $\int_{-\pi}^{\pi} dt = 2\pi$ ;  $\int_{-\pi}^{\pi} \sin(mt) dt = 0$  for every integer  $m$  and  $\int_{-\pi}^{\pi} \cos(mt) dt = 0$  for

every **non-zero** integer  $m$  (for  $m \neq 0$  evaluate the antiderivative  $-\cos(mt)/m$ , respectively  $\sin(mt)/m$ , each of which takes the same value at  $t = -\pi$  as at  $t = \pi$ ; for  $m = 0$  the sine integrand is 0); integration by parts, and the substitution  $t \mapsto -t$ , both of which we use later in the section when computing particular Fourier coefficients. From trigonometry: the product-to-sum identities. **No theorem about convergence is used anywhere in this section**; every integral we compute is the integral of a finite sum of sines and cosines.

**Example. (The Trigonometric System Is Orthonormal)**

Fix  $n \geq 1$ . Show that the list of  $2n + 1$  functions

$$\frac{1}{\sqrt{2}}, \cos x, \sin x, \cos 2x, \sin 2x, \dots, \cos nx, \sin nx$$

is orthonormal in  $V = C[-\pi, \pi]$  with  $\langle f, g \rangle = \frac{1}{\pi} \int_{-\pi}^{\pi} fg$ .

*Solution.* There are five kinds of pair. Write  $j, k$  for indices in  $\{1, \dots, n\}$ .

*The constant with itself.*  $\|1/\sqrt{2}\|^2 = \frac{1}{\pi} \int_{-\pi}^{\pi} \frac{1}{2} dt = \frac{1}{\pi} \cdot \pi = 1$ .

*The constant against  $\cos kx$  or  $\sin kx$ .* These are  $\frac{1}{\pi\sqrt{2}} \int_{-\pi}^{\pi} \cos(kt) dt = 0$  and  $\frac{1}{\pi\sqrt{2}} \int_{-\pi}^{\pi} \sin(kt) dt = 0$ , since  $k \neq 0$ .

*Two cosines.* By  $\cos A \cos B = \frac{1}{2}[\cos(A - B) + \cos(A + B)]$ ,

$$\langle \cos jx, \cos kx \rangle = \frac{1}{2\pi} \int_{-\pi}^{\pi} [\cos((j - k)t) + \cos((j + k)t)] dt.$$

If  $j \neq k$ , both  $j - k$  and  $j + k$  are non-zero integers, so both integrals vanish and the value is 0. If  $j = k$ , the first integrand is the constant 1 and the second has non-zero frequency  $2k$ , so the value is  $\frac{1}{2\pi}(2\pi + 0) = 1$ .

*Two sines.* By  $\sin A \sin B = \frac{1}{2}[\cos(A - B) - \cos(A + B)]$ , the same two integrals appear with a minus sign, giving 0 for  $j \neq k$  and  $\frac{1}{2\pi}(2\pi - 0) = 1$  for  $j = k$ .

*A sine against a cosine.* By  $\sin A \cos B = \frac{1}{2}[\sin(A + B) + \sin(A - B)]$ ,

$$\langle \sin jx, \cos kx \rangle = \frac{1}{2\pi} \int_{-\pi}^{\pi} [\sin((j + k)t) + \sin((j - k)t)] dt = 0,$$

because every integral of a sine over  $[-\pi, \pi]$  vanishes, including the case  $j = k$  where the second integrand is  $\sin 0 = 0$ .

So each of the  $2n + 1$  functions is a unit vector and distinct ones are orthogonal, which is [Definition 10.2.1](#).

Write  $W_n = \text{span}(1, \cos x, \sin x, \dots, \cos nx, \sin nx)$ , the space of **trigonometric polynomials of degree at most  $n$** . The list above is an orthonormal list, hence independent ([Theorem 10.1.4](#)), and it spans  $W_n$ , since  $1 = \sqrt{2} \cdot \frac{1}{\sqrt{2}}$ . So it is an orthonormal basis of  $W_n$  and  $\dim W_n = 2n + 1$ .

For  $f \in V$  the classical **Fourier coefficients** of  $f$  are

$$a_k = \frac{1}{\pi} \int_{-\pi}^{\pi} f(t) \cos(kt) dt, \quad b_k = \frac{1}{\pi} \int_{-\pi}^{\pi} f(t) \sin(kt) dt,$$

for  $k \geq 0$  and  $k \geq 1$  respectively. Comparing with the inner product,  $a_k = \langle f, \cos kx \rangle$  and  $b_k = \langle f, \sin kx \rangle$ , while the coefficient against the constant of the orthonormal list is  $\langle f, 1/\sqrt{2} \rangle = a_0/\sqrt{2}$ . These are exactly the Fourier coefficients of [Definition 10.9.1](#) for this list.

### Theorem 10.9.3 — Fourier Partial Sums Are Best Approximations

Let  $f \in C[-\pi, \pi]$ , let  $n \geq 1$ , and let  $a_k, b_k$  be the Fourier coefficients of  $f$ . Put

$$S_n f = \frac{a_0}{2} + \sum_{k=1}^n (a_k \cos kx + b_k \sin kx).$$

Then, with  $\langle f, g \rangle = \frac{1}{\pi} \int_{-\pi}^{\pi} fg$ :

- (a)  $S_n f = P_{W_n} f$ ;
- (b)  $\|f - S_n f\| \leq \|f - g\|$  for every  $g \in W_n$ , with equality **only** for  $g = S_n f$ ;
- (c)  $\frac{a_0^2}{2} + \sum_{k=1}^n (a_k^2 + b_k^2) \leq \frac{1}{\pi} \int_{-\pi}^{\pi} f(t)^2 dt$ .

*Idea.* Nothing new happens here. The subspace  $W_n$  is finite-dimensional and we have an orthonormal basis of it, so Section 3 answers the approximation question and [Theorem 10.9.1](#) answers the coefficient question; the work is only to match the classical notation to ours. The one thing to watch is the factor  $\frac{1}{2}$  in front of  $a_0$ : it is not a convention pulled from the air, it is what  $\langle f, 1/\sqrt{2} \rangle \cdot \frac{1}{\sqrt{2}}$  equals.

*Proof.*

- (a) By [Example \(The Trigonometric System Is Orthonormal\)](#) the list  $(\frac{1}{\sqrt{2}}, \cos x, \sin x, \dots, \cos nx, \sin nx)$  is an orthonormal basis of  $W_n$ . By [Theorem 10.3.3](#) (a),

$$P_{W_n} f = \left\langle f, \frac{1}{\sqrt{2}} \right\rangle \frac{1}{\sqrt{2}} + \sum_{k=1}^n (\langle f, \cos kx \rangle \cos kx + \langle f, \sin kx \rangle \sin kx).$$

The first term is  $\frac{a_0}{\sqrt{2}} \cdot \frac{1}{\sqrt{2}} = \frac{a_0}{2}$ , and the remaining terms are  $a_k \cos kx + b_k \sin kx$ . Hence  $P_{W_n} f = S_n f$ .

- (b) This is [Theorem 10.3.5](#) applied to the subspace  $U = W_n$ , which is legitimate because  $\dim W_n = 2n + 1$  is finite, together with (a).
- (c) This is [Theorem 10.9.1](#) for the same orthonormal list, since  $|\langle f, 1/\sqrt{2} \rangle|^2 = a_0^2/2$  and  $\|f\|^2 = \frac{1}{\pi} \int_{-\pi}^{\pi} f^2$ .

■

So the Fourier partial sum is not a guess at  $f$  that happens to work: it **is** the orthogonal projection of  $f$  onto the trigonometric polynomials of degree at most  $n$ , and by [Theorem 10.3.5](#) it is the unique trigonometric polynomial of that degree closest to  $f$  in this norm. The mysterious  $a_0/2$  of the classical formula is an artifact of normalizing the constant function.

### 10.9.3 A worked computation

**Example. (The Fourier Coefficients of  $f(x) = x$ )**

In  $C[-\pi, \pi]$  with  $\langle f, g \rangle = \frac{1}{\pi} \int_{-\pi}^{\pi} fg$ , let  $f(x) = x$ . Find all  $a_k$  and  $b_k$ , write down  $S_n f$ , and say what [Theorem 10.9.3](#) (c) gives.

*Solution.* We use one further calculus fact: a continuous **odd** function integrates to 0 over  $[-\pi, \pi]$ , and a continuous **even** one integrates to twice its integral over  $[0, \pi]$ . Both come from the substitution  $t \mapsto -t$ .

*The cosine coefficients.* For every  $k \geq 0$  the function  $t \cos(kt)$  is odd, so  $a_k = 0$ . In particular  $a_0 = 0$ .

*The sine coefficients.* The function  $t \sin(kt)$  is even, so for  $k \geq 1$ ,

$$b_k = \frac{1}{\pi} \int_{-\pi}^{\pi} t \sin(kt) dt = \frac{2}{\pi} \int_0^{\pi} t \sin(kt) dt.$$

Integrating by parts, with  $d(-\cos(kt)/k) = \sin(kt) dt$ ,

$$\int_0^{\pi} t \sin(kt) dt = \left[ \frac{-t \cos(kt)}{k} \right]_0^{\pi} + \frac{1}{k} \int_0^{\pi} \cos(kt) dt = \frac{-\pi \cos(k\pi)}{k},$$

since the last integral is  $[\sin(kt)/k^2]_0^{\pi} = 0$ . As  $\cos(k\pi) = (-1)^k$ , this equals  $\pi(-1)^{k+1}/k$ , and therefore

$$b_k = \frac{2(-1)^{k+1}}{k}.$$

*The partial sums.* Hence

$$S_n f = \sum_{k=1}^n \frac{2(-1)^{k+1}}{k} \sin kx = 2 \sin x - \sin 2x + \frac{2}{3} \sin 3x - \dots,$$

and by [Theorem 10.9.3](#) (b) this is the closest trigonometric polynomial of degree at most  $n$  to the function  $x$  in this norm.

*What Bessel's inequality ([Theorem 10.9.1](#)) gives.* First  $\|f\|^2 = \frac{1}{\pi} \int_{-\pi}^{\pi} t^2 dt = \frac{1}{\pi} \cdot \frac{2\pi^3}{3} = \frac{2\pi^2}{3}$ . Since  $a_k = 0$  and  $b_k^2 = 4/k^2$ , part (c) of [Theorem 10.9.3](#) reads

$$\sum_{k=1}^n \frac{4}{k^2} \leq \frac{2\pi^2}{3}, \quad \text{that is} \quad \sum_{k=1}^n \frac{1}{k^2} \leq \frac{\pi^2}{6}$$

for **every**  $n$ . The partial sums on the left increase and are bounded above, so they converge, and  $\sum_{k=1}^{\infty} 1/k^2 \leq \pi^2/6$ .

That bound is sharp: the true value of the sum is  $\pi^2/6$ . But sharpness is not something Bessel can deliver. By [Theorem 10.9.2](#), equality for a fixed finite list holds only when  $f$  lies in the span of that list, and the function  $x$  lies in no  $W_n$  — it is not a trigonometric polynomial, since every element of  $W_n$  takes the same value at  $-\pi$  and at  $\pi$ . Getting the equality requires a statement about the **infinite** system, and that statement is an input from analysis; [Exercise \(C1: Parseval for the full trigonometric system\)](#) carries it out and flags it.

*Quick check.* Let  $g(x) = \cos^2 x$  in  $C[-\pi, \pi]$ . Using  $\cos^2 x = \frac{1}{2} + \frac{1}{2} \cos 2x$ , write down all Fourier coefficients of  $g$ , find  $S_1 g$ , and compute the deficit in Bessel's inequality for  $n = 1$ .

*Solution.* The right-hand side  $\frac{1}{2} + \frac{1}{2} \cos 2x$  already lies in  $W_2$ , and coefficients against an orthonormal basis are unique (Theorem 10.2.1 (a)), so, applying that result inside  $W_2$  and reading the coefficients off:  $a_0 = 1$ ,  $a_2 = \frac{1}{2}$ , and all other  $a_k$  and all  $b_k$  are 0. Hence  $S_1 g = a_0/2 = \frac{1}{2}$ , a constant. For the deficit,  $\|g\|^2 = \frac{1}{\pi} \int_{-\pi}^{\pi} \cos^4 t \, dt = \frac{1}{\pi} \cdot \frac{3\pi}{4} = \frac{3}{4}$ , while the Bessel sum for  $n = 1$  is  $a_0^2/2 = \frac{1}{2}$ . The deficit is  $\frac{1}{4}$ , and by (\*) it should be  $\|g - S_1 g\|^2 = \|\frac{1}{2} \cos 2x\|^2 = \frac{1}{4} \cdot 1 = \frac{1}{4}$ . It is.

## 10.9.4 What best approximation does and does not say

It is worth being exact about the boundary here, because it is easy to read more into Theorem 10.9.3 than it contains.

Identity (\*) applied to  $U = W_n$  says

$$\|f - S_n f\|^2 = \|f\|^2 - \frac{a_0^2}{2} - \sum_{k=1}^n (a_k^2 + b_k^2).$$

The subtracted sum never decreases as  $n$  grows, so the numbers  $\|f - S_n f\|^2$  never increase, and they are bounded below by 0. Hence they converge to some limit  $L \geq 0$ . Everything up to this sentence is linear algebra plus one fact about monotone bounded sequences.

Whether  $L = 0$  is a different kind of question. It asks whether the trigonometric polynomials come arbitrarily close, in this norm, to every continuous function, and no argument in this book decides it. The statement is true, and it is the analytic content of the theory of Fourier series; it is what makes the *infinite* trigonometric system behave like an orthonormal basis even though it is not a basis in the sense of Chapter 1.

And even  $L = 0$  would not be a statement about individual points. Our norm averages the error over the whole interval, so it tolerates a large error on a short interval. The function  $f(x) = x$  shows this at once:

$$S_n f(\pi) = \sum_{k=1}^n \frac{2(-1)^{k+1}}{k} \sin(k\pi) = 0$$

for every  $n$ , while  $f(\pi) = \pi$ . The error at the single point  $x = \pi$  never shrinks at all.

**Warning. Mean-square best approximation is not pointwise approximation.** Theorem 10.9.3 says that  $S_n f$  minimizes  $\int_{-\pi}^{\pi} (f - g)^2$  over  $g \in W_n$ . It does not say that  $S_n f(x)$  is close to  $f(x)$  for a given  $x$ , and for  $f(x) = x$  at  $x = \pi$  it is not. Whether a Fourier series converges at a point is a question in analysis with a delicate answer; it is not decided by anything in this chapter, and a projection theorem cannot decide it.

## 10.9.5 Legendre and Chebyshev

Two families of orthogonal polynomials show the same two theorems at work with a different weight.

**Legendre.** [Example \(Gram–Schmidt on  \$1, x, x^2\$ \)](#) ran Gram–Schmidt on  $(1, x, x^2)$  in  $\mathbb{R}[x]_{\leq 2}$  with  $\langle p, q \rangle = \int_{-1}^1 pq$ , producing the orthogonal list  $1, x, x^2 - \frac{1}{3}$  and, after normalizing, the orthonormal basis

$$\mathbf{e}_1 = \frac{1}{\sqrt{2}}, \quad \mathbf{e}_2 = \sqrt{3/2}x, \quad \mathbf{e}_3 = \frac{\sqrt{10}}{4}(3x^2 - 1).$$

Inside  $\mathbb{R}[x]_{\leq 2}$  this is a **basis**, so [Theorem 10.9.2](#) gives  $\|p\|^2 = \sum_{i=1}^3 |\langle p, \mathbf{e}_i \rangle|^2$  for every quadratic  $p$ . Viewed inside the larger space  $C[-1, 1]$  of continuous functions, the same three vectors are only a **list**, and for an  $f$  outside  $\mathbb{R}[x]_{\leq 2}$  [Theorem 10.9.1](#) is strict, the deficit being the squared distance from  $f$  to  $\mathbb{R}[x]_{\leq 2}$  by  $(*)$ . That deficit is what [Example \(The best quadratic approximation to a cubic\)](#) computed for  $f = x^3$ . One list, two readings, and which reading applies depends on the ambient space, not on the list. In practice one projects with the orthogonal list  $1, x, x^2 - \frac{1}{3}$  through [Corollary 10.3.4](#) and avoids the square roots entirely.

**Chebyshev.** Change the weight. On  $\mathbb{R}[x]$  define

$$\langle p, q \rangle_w = \int_{-1}^1 \frac{p(t)q(t)}{\sqrt{1-t^2}} dt.$$

The integrand is unbounded at  $t = \pm 1$ , so this is an improper integral; it converges for polynomials, because  $|pq|$  is bounded on  $[-1, 1]$  and  $\int_{-1}^1 dt/\sqrt{1-t^2} = \pi$ . The three axioms are checked as in [Definition 10.1.1](#), positivity because  $1/\sqrt{1-t^2} > 0$  on  $(-1, 1)$ .

The **Chebyshev polynomials** are defined by the requirement

$$T_n(\cos \theta) = \cos n\theta \quad \text{for all } \theta,$$

which does determine a polynomial of degree  $n$ , and determines it uniquely, since  $\cos \theta$  runs over the whole of  $[-1, 1]$  and two polynomials agreeing on an infinite set are equal ([Corollary 5.1.4](#)). The first few are  $T_0 = 1, T_1 = x, T_2 = 2x^2 - 1$  (from  $\cos 2\theta = 2\cos^2 \theta - 1$ ), and [Exercise \(C2: Chebyshev orthogonality by substitution\)](#) handles the general case. They are orthogonal for  $\langle \cdot, \cdot \rangle_w$ :

$$\langle T_m, T_n \rangle_w = 0 \quad (m \neq n), \quad \langle T_n, T_n \rangle_w = \begin{cases} \pi, & n = 0, \\ \pi/2, & n \geq 1. \end{cases}$$

For  $m, n \leq 2$  this can be checked by hand, using  $\int_{-1}^1 dt/\sqrt{1-t^2} = \pi$  and  $\int_{-1}^1 t^2 dt/\sqrt{1-t^2} = \pi/2$  (both come from  $t = \cos \theta$ ). The pairs  $(T_0, T_1)$  and  $(T_1, T_2)$  have odd integrands on a symmetric interval, so those integrals are 0 by the substitution  $t \mapsto -t$ . For the remaining pair,

$$\langle T_0, T_2 \rangle_w = \int_{-1}^1 \frac{2t^2 - 1}{\sqrt{1-t^2}} dt = 2 \cdot \frac{\pi}{2} - \pi = 0.$$

Why should such a weight produce orthogonality? Because it is the trigonometric system in disguise. Substituting  $t = \cos \theta$  turns  $\langle p, q \rangle_w$  into  $\int_0^\pi p(\cos \theta)q(\cos \theta) d\theta$ , and on the Chebyshev polynomials that integrand is  $\cos m\theta \cos n\theta$  — the very product whose integral we computed in [Example \(The Trigonometric System Is Orthonormal\)](#). [Exercise \(C2: Chebyshev orthogonality by substitution\)](#) asks for this in full.



Since the  $T_n$  are orthogonal but not normalized, projections onto their span are computed with [Corollary 10.3.4](#) rather than [Theorem 10.3.3](#). Section 10 of this chapter takes up orthogonal polynomials as a subject in their own right, including the recurrence that generates the  $T_n$  and the quadrature rules they produce.

## 10.9.6 Exercises

### 10.9.6.1 A. Check your understanding

#### Exercise A1.

- (a) State Bessel's inequality, saying exactly what is assumed about the space and what is assumed about the list.
- (b) State Parseval's identity, including the condition under which equality holds for a given vector.
- (c) True or false: Bessel's inequality requires the ambient space to be finite-dimensional. Justify your answer.
- (d) Write down the inner product used on  $C[-\pi, \pi]$  in this section, including its normalizing constant, and say what the constant is for.
- (e) State what [Theorem 10.9.3](#) gives about  $S_n f$ , and name one thing it does not give.
- (f) True or false: if  $(\mathbf{e}_1, \dots, \mathbf{e}_k)$  is an orthonormal list and  $\sum_i |\langle \mathbf{v}, \mathbf{e}_i \rangle|^2 = \|\mathbf{v}\|^2$  for one particular  $\mathbf{v} \neq \mathbf{0}$ , then the list is an orthonormal basis of  $V$ . Justify your answer.

*Solution.*

- (a) If  $(\mathbf{e}_1, \dots, \mathbf{e}_k)$  is an orthonormal list in an inner product space  $V$  over  $\mathbb{R}$  or  $\mathbb{C}$ , then  $\sum_{i=1}^k |\langle \mathbf{v}, \mathbf{e}_i \rangle|^2 \leq \|\mathbf{v}\|^2$  for every  $\mathbf{v} \in V$  ([Theorem 10.9.1](#)). The list is assumed **finite and orthonormal**; the space is assumed only to be an inner product space, with no dimension hypothesis.
- (b) Equality  $\sum_i |\langle \mathbf{v}, \mathbf{e}_i \rangle|^2 = \|\mathbf{v}\|^2$  holds if and only if  $\mathbf{v} \in \text{span}(\mathbf{e}_1, \dots, \mathbf{e}_k)$ ; in particular it holds for every  $\mathbf{v}$  when the list is an orthonormal basis of  $V$  ([Theorem 10.9.2](#)).
- (c) False. Only the span of the list has to be finite-dimensional, and it automatically is, being spanned by  $k$  vectors. The proof uses [Theorem 10.3.3](#), whose hypothesis is that the **subspace** projected onto is finite-dimensional. The application to  $C[-\pi, \pi]$  is exactly a case where  $V$  is not finite-dimensional.
- (d)  $\langle f, g \rangle = \frac{1}{\pi} \int_{-\pi}^{\pi} f(t)g(t) dt$ . The constant  $1/\pi$  makes  $\cos kx$  and  $\sin kx$  unit vectors for every  $k \geq 1$ ; without it their norms would be  $\sqrt{\pi}$ .
- (e) It gives that  $S_n f$  is the orthogonal projection of  $f$  onto  $W_n$ , hence the unique element of  $W_n$  minimizing  $\|f - g\|$ , and it gives Bessel's inequality for the coefficients. It does **not** give that  $S_n f(x) \rightarrow f(x)$  at any particular  $x$ , nor even that  $\|f - S_n f\| \rightarrow 0$ .
- (f) False. Equality for one  $\mathbf{v}$  says only that  $\mathbf{v} \in \text{span}(\mathbf{e}_1, \dots, \mathbf{e}_k)$  ([Theorem 10.9.2](#)). In  $\mathbb{R}^3$  take the list  $(\mathbf{e}_1)$  and  $\mathbf{v} = \mathbf{e}_1$ : equality holds, and the list is not a basis.

### 10.9.6.2 B. Practice

**Exercise B1: Fourier coefficients.**

Work in  $C[-\pi, \pi]$  with  $\langle f, g \rangle = \frac{1}{\pi} \int_{-\pi}^{\pi} fg$ . Find all Fourier coefficients  $a_k$  and  $b_k$  of each of the following, and in each case write down  $S_2f$ .

- (a)  $f(x) = |x|$ .
- (b)  $f(x) = x^2$ .
- (c)  $f(x) = 3 - 2\sin x + \sin 3x$ .

*Solution.*

- (a)  $|t| \sin(kt)$  is odd, so  $b_k = 0$  for all  $k$ . For the cosines,  $|t| \cos(kt)$  is even, so  $a_k = \frac{2}{\pi} \int_0^{\pi} t \cos(kt) dt$ . For  $k = 0$  this is  $\frac{2}{\pi} \cdot \frac{\pi^2}{2} = \pi$ . For  $k \geq 1$ , integrating by parts,

$$\int_0^{\pi} t \cos(kt) dt = \left[ \frac{t \sin(kt)}{k} \right]_0^{\pi} - \frac{1}{k} \int_0^{\pi} \sin(kt) dt = \frac{(-1)^k - 1}{k^2},$$

since the bracket vanishes and  $\int_0^{\pi} \sin(kt) dt = (1 - (-1)^k)/k$ . Hence

$$a_k = \frac{2((-1)^k - 1)}{\pi k^2} = \begin{cases} -\frac{4}{\pi k^2}, & k \text{ odd,} \\ 0, & k \text{ even,} \end{cases}$$

and  $S_2f = \frac{\pi}{2} - \frac{4}{\pi} \cos x$ .

- (b)  $t^2 \sin(kt)$  is odd, so  $b_k = 0$ . Also  $a_0 = \frac{2}{\pi} \int_0^{\pi} t^2 dt = \frac{2\pi^2}{3}$ . For  $k \geq 1$ , integrating by parts twice gives

$$\int_0^{\pi} t^2 \cos(kt) dt = \left[ \frac{t^2 \sin(kt)}{k} \right]_0^{\pi} - \frac{2}{k} \int_0^{\pi} t \sin(kt) dt = \frac{2\pi(-1)^k}{k^2},$$

using  $\int_0^{\pi} t \sin(kt) dt = \pi(-1)^{k+1}/k$  from [Example \(The Fourier Coefficients of  \$f\(x\) = x\$ \)](#). Hence  $a_k = \frac{2}{\pi} \cdot \frac{2\pi(-1)^k}{k^2} = \frac{4(-1)^k}{k^2}$ , so  $a_1 = -4$ ,  $a_2 = 1$ , and

$$S_2f = \frac{\pi^2}{3} - 4\cos x + \cos 2x.$$

- (c) The function already lies in  $W_3$ . Coefficients against an orthonormal basis are unique ([Theorem 10.2.1 \(a\)](#)), so, applying that result inside  $W_3$ , we may read them off:  $a_0/2 = 3$ , that is  $a_0 = 6$ , together with  $b_1 = -2$  and  $b_3 = 1$ ; every other coefficient is 0. Hence  $S_2f = 3 - 2\sin x$ , which is **not**  $f$ : the term  $\sin 3x$  lies outside  $W_2$  and is discarded by the projection. (From  $n = 3$  on,  $S_n f = f$ , as [Theorem 10.3.3 \(a\)](#) predicts, since  $f \in W_3$  and a projection fixes its own subspace.)

**Exercise B2: Verifying Bessel for a truncation.**

- (a) In  $\mathbb{R}^4$  with the dot product let  $\mathbf{e}_1 = \frac{1}{2}(1, 1, 1, 1)$ ,  $\mathbf{e}_2 = \frac{1}{2}(1, -1, 1, -1)$  and  $\mathbf{v} = (2, 0, 0, 2)$ . Verify [Theorem 10.9.1](#) for  $\mathbf{v}$  and this list, compute the deficit, and check against (\*) by computing  $\|\mathbf{v} - P_U \mathbf{v}\|$  directly, where  $U = \text{span}(\mathbf{e}_1, \mathbf{e}_2)$ .
- (b) For  $f(x) = |x|$  and  $n = 1$ , write out both sides of [Theorem 10.9.3 \(c\)](#) using [Exercise](#)

(B1: Fourier coefficients) (a), and show that the resulting inequality is equivalent to  $96 \leq \pi^4$ .

*Solution.*

(a) Both vectors have norm  $\frac{1}{2}\sqrt{4} = 1$  and  $\langle \mathbf{e}_1, \mathbf{e}_2 \rangle = \frac{1}{4}(1 - 1 + 1 - 1) = 0$ , so the list is orthonormal. The coefficients are  $\langle \mathbf{v}, \mathbf{e}_1 \rangle = \frac{1}{2}(2 + 0 + 0 + 2) = 2$  and  $\langle \mathbf{v}, \mathbf{e}_2 \rangle = \frac{1}{2}(2 - 0 + 0 - 2) = 0$ , so the left-hand side of Bessel is 4, while  $\|\mathbf{v}\|^2 = 4 + 4 = 8$ . The inequality  $4 \leq 8$  holds, with deficit 4. By [Theorem 10.3.3](#) (a),  $P_U \mathbf{v} = 2\mathbf{e}_1 = (1, 1, 1, 1)$ , so  $\mathbf{v} - P_U \mathbf{v} = (1, -1, -1, 1)$  and  $\|\mathbf{v} - P_U \mathbf{v}\|^2 = 4$ , matching (\*).

(b) With  $a_0 = \pi$  and  $a_1 = -4/\pi$ , and  $b_1 = 0$ , the left-hand side of (c) for  $n = 1$  is

$$\frac{a_0^2}{2} + a_1^2 = \frac{\pi^2}{2} + \frac{16}{\pi^2}.$$

The right-hand side is  $\|f\|^2 = \frac{1}{\pi} \int_{-\pi}^{\pi} t^2 dt = \frac{2\pi^2}{3}$ . So the inequality reads

$$\frac{\pi^2}{2} + \frac{16}{\pi^2} \leq \frac{2\pi^2}{3},$$

that is  $16/\pi^2 \leq 2\pi^2/3 - \pi^2/2 = \pi^2/6$ , that is  $96 \leq \pi^4$ . Since  $\pi > 3.14$  and  $3.14^4 > 97$ , the inequality is true, and it is tight: the deficit is  $\pi^2/6 - 16/\pi^2 \approx 0.024$ , so the single cosine  $\cos x$  already captures almost all of  $|x|$ .

### Exercise B3: The best approximation by $\text{span}(1, \cos x)$ .

In  $C[-\pi, \pi]$  with  $\langle f, g \rangle = \frac{1}{\pi} \int_{-\pi}^{\pi} fg$ , let  $U = \text{span}(1, \cos x)$  and  $f(x) = x^2$ . Find  $P_U f$  and  $d(f, U)^2$ . *Hint: the list  $(1, \cos x)$  is orthogonal but not orthonormal.*

*Solution.* The two functions are orthogonal, since  $\langle 1, \cos x \rangle = \frac{1}{\pi} \int_{-\pi}^{\pi} \cos t dt = 0$ , and both are non-zero, with  $\langle 1, 1 \rangle = 2$  and  $\langle \cos x, \cos x \rangle = 1$  by [Example \(The Trigonometric System Is Orthonormal\)](#). So  $(1, \cos x)$  is an orthogonal basis of  $U$ , and [Corollary 10.3.4](#) gives

$$P_U f = \frac{\langle f, 1 \rangle}{2} \cdot 1 + \frac{\langle f, \cos x \rangle}{1} \cdot \cos x.$$

By [Exercise \(B1: Fourier coefficients\)](#) (b),  $\langle f, 1 \rangle = a_0 = 2\pi^2/3$  and  $\langle f, \cos x \rangle = a_1 = -4$ . Hence

$$P_U f = \frac{\pi^2}{3} - 4 \cos x.$$

For the distance, normalize:  $(1/\sqrt{2}, \cos x)$  is an orthonormal basis of the same  $U$ , with coefficients  $\langle f, 1/\sqrt{2} \rangle = a_0/\sqrt{2}$  and  $\langle f, \cos x \rangle = a_1$ . Now  $\|f\|^2 = \frac{1}{\pi} \int_{-\pi}^{\pi} t^4 dt = \frac{1}{\pi} \cdot \frac{2\pi^5}{5} = \frac{2\pi^4}{5}$  and the Bessel sum is  $a_0^2/2 + a_1^2 = 2\pi^4/9 + 16$ , so (\*) gives

$$d(f, U)^2 = \frac{2\pi^4}{5} - \frac{2\pi^4}{9} - 16 = \frac{8\pi^4}{45} - 16 \approx 1.32.$$

It is positive, as it must be, since  $x^2 \notin U$ .

## 10.9.6.3 C. Going deeper

**Exercise C1: Parseval for the full trigonometric system.**

This exercise uses an **input from analysis that this book does not prove**: the trigonometric system is *complete* in  $C[-\pi, \pi]$ , meaning that for every  $f \in C[-\pi, \pi]$ ,

$$\frac{a_0^2}{2} + \sum_{k=1}^{\infty} (a_k^2 + b_k^2) = \frac{1}{\pi} \int_{-\pi}^{\pi} f(t)^2 dt.$$

- (a) Apply this to  $f(x) = x$  and deduce that  $\sum_{k=1}^{\infty} 1/k^2 = \pi^2/6$ .
- (b) Hence deduce that  $\sum_{k=1}^{\infty} 1/(2k-1)^2 = \pi^2/8$ .
- (c) Say precisely which step used the analysis input, and explain why no result of this section can supply it.

*Solution.*

- (a) By [Example \(The Fourier Coefficients of  \$f\(x\) = x\$ \)](#),  $a_k = 0$  for all  $k$  and  $b_k = 2(-1)^{k+1}/k$ , so  $b_k^2 = 4/k^2$ ; and  $\|f\|^2 = 2\pi^2/3$ . The stated identity becomes

$$\sum_{k=1}^{\infty} \frac{4}{k^2} = \frac{2\pi^2}{3},$$

so  $\sum_{k=1}^{\infty} 1/k^2 = \pi^2/6$ .

- (b) All terms are positive and the series converges, so it may be split into its even-index and odd-index parts. The even part is

$$\sum_{j=1}^{\infty} \frac{1}{(2j)^2} = \frac{1}{4} \sum_{j=1}^{\infty} \frac{1}{j^2} = \frac{\pi^2}{24}.$$

Hence the odd part is  $\pi^2/6 - \pi^2/24 = (4\pi^2 - \pi^2)/24 = \pi^2/8$ , which is the claim.

- (c) The analysis input is used exactly once, at the equality sign in (a). This section proves only the inequality  $\sum_{k \leq n} 4/k^2 \leq 2\pi^2/3$  for each  $n$  ([Theorem 10.9.1](#)), and [Theorem 10.9.2](#) turns that inequality into an equality only when  $f$  lies in the span of the finitely many functions used. The function  $x$  lies in no  $W_n$ , since every element of  $W_n$  takes equal values at  $-\pi$  and  $\pi$  while  $f(-\pi) = -\pi \neq \pi = f(\pi)$ . Passing to the infinite system is a statement about a limit of functions, and nothing in this book proves such a statement.

*Remark.* The classical route to  $\sum 1/(2k-1)^2 = \pi^2/8$  applies the same identity to the square wave, the function equal to  $-1$  on  $[-\pi, 0)$  and  $1$  on  $[0, \pi]$ , whose coefficients are  $b_k = 4/(\pi k)$  for odd  $k$  and  $0$  otherwise. That function is not continuous, so it does not live in the space  $C[-\pi, \pi]$  of this section, and enlarging the space to all integrable functions destroys positive definiteness, as the third non-example of Section 1 shows. Working with  $f(x) = x$  keeps everything inside a genuine inner product space.

**Exercise C2: Chebyshev orthogonality by substitution.**

Let  $\langle p, q \rangle_w = \int_{-1}^1 p(t)q(t)(1-t^2)^{-1/2} dt$  on  $\mathbb{R}[x]$ , the Chebyshev inner product of the text.

- (a) Prove that for every  $n \geq 0$  there is a polynomial  $T_n$  of degree  $n$  with  $T_n(\cos \theta) = \cos n\theta$  for all  $\theta$ , and find  $T_0, T_1, T_2, T_3$ .
- (b) Prove that  $\langle p, q \rangle_w = \int_0^\pi p(\cos \theta) q(\cos \theta) d\theta$  for all polynomials  $p, q$ .
- (c) Deduce that  $\langle T_m, T_n \rangle_w = 0$  for  $m \neq n$ , and compute  $\langle T_n, T_n \rangle_w$ .

*Hint for (a): the identity  $\cos((n+1)\theta) + \cos((n-1)\theta) = 2 \cos \theta \cos n\theta$ .*

*Solution.*

- (a) Induction on  $n$ . Take  $T_0 = 1$  and  $T_1 = x$ , which satisfy  $T_0(\cos \theta) = 1 = \cos 0\theta$  and  $T_1(\cos \theta) = \cos \theta$ , of degrees 0 and 1. Suppose  $T_{n-1}$  and  $T_n$  exist, of degrees  $n-1$  and  $n$ . Put  $T_{n+1} = 2xT_n - T_{n-1}$ . Then for every  $\theta$ ,

$$T_{n+1}(\cos \theta) = 2 \cos \theta \cos n\theta - \cos((n-1)\theta) = \cos((n+1)\theta)$$

by the stated identity. Its degree is  $n+1$ , since the leading term of  $2xT_n$  has degree  $n+1$  and  $T_{n-1}$  has degree  $n-1$ . This completes the induction, and the recurrence gives  $T_2 = 2x^2 - 1$  and  $T_3 = 2x(2x^2 - 1) - x = 4x^3 - 3x$ .

- (b) The integral is improper at both endpoints, so it means the limit as  $\delta \rightarrow 0^+$  of the integrals over  $[-1 + \delta, 1 - \delta]$ . On each such interval the integrand is continuous, and  $t = \cos \theta$  is a continuously differentiable bijection from the corresponding  $\theta$ -interval, so the change-of-variables rule applies there. With  $t = \cos \theta$ ,  $\theta$  running from  $\pi$  down to 0, we have  $dt = -\sin \theta d\theta$  and  $\sqrt{1-t^2} = \sin \theta$ , which is  $\geq 0$  on  $[0, \pi]$ . The two minus signs cancel when the limits are reversed, and letting  $\delta \rightarrow 0^+$ ,

$$\int_{-1}^1 \frac{p(t)q(t)}{\sqrt{1-t^2}} dt = \int_0^\pi p(\cos \theta) q(\cos \theta) d\theta.$$

The right-hand side is a proper integral of a continuous function, so both sides converge, and the limits agree.

- (c) By (a) and (b),  $\langle T_m, T_n \rangle_w = \int_0^\pi \cos m\theta \cos n\theta d\theta$ . Using  $\cos A \cos B = \frac{1}{2}[\cos(A-B) + \cos(A+B)]$  and  $\int_0^\pi \cos(p\theta) d\theta = [\sin(p\theta)/p]_0^\pi = 0$  for every non-zero integer  $p$ :

if  $m \neq n$ , then  $m-n \neq 0$ , and  $m+n \geq 1$  because the two indices differ, so both integrals vanish and the value is 0;

if  $m = n \geq 1$ , the value is  $\frac{1}{2}(\int_0^\pi 1 d\theta + 0) = \pi/2$ ;

if  $m = n = 0$ , the integrand is 1 and the value is  $\pi$ .

So the  $T_n$  are orthogonal for  $\langle \cdot, \cdot \rangle_w$ , with  $\langle T_0, T_0 \rangle_w = \pi$  and  $\langle T_n, T_n \rangle_w = \pi/2$  for  $n \geq 1$ . Consequently  $(T_0/\sqrt{\pi}, T_1/\sqrt{2/\pi}, \dots, T_n/\sqrt{2/\pi})$  is an orthonormal list, and [Theorem 10.9.1](#) applies to it verbatim.

## 10.10 Orthogonal Polynomials

Section 2 ran Gram–Schmidt on  $(1, x, x^2)$  with  $\langle p, q \rangle = \int_{-1}^1 pq$ , and stopped at degree 2 only because the space did ([Example \(Gram–Schmidt on  \$1, x, x^2\$ \)](#)). Nothing in the process cared.

Feed it  $(1, x, x^2, x^3, \dots)$  and it returns an infinite orthogonal sequence, one polynomial of each degree. Three things make such sequences worth a section. Each member is computable from the two before it. Each member of degree  $k$  has  $k$  distinct real roots inside the interval. And those roots are the best possible points at which to sample a function to estimate its integral. The first fact rests on one small identity; the other two rest on the first.

Throughout this section the field is  $\mathbb{R}$ .

## 10.10.1 Two ways to make an inner product on polynomials

Both constructions we need appeared in Section 1.

**From a weight.** Fix an interval with endpoints  $-\infty \leq a < b \leq \infty$  and a **weight**  $w$ : a function continuous and **strictly positive** on the open interval  $(a, b)$ , with  $\int_a^b |t|^k w(t) dt$  finite for every  $k \in \mathbb{N}$ . Define

$$\langle p, q \rangle = \int_a^b p(t) q(t) w(t) dt.$$

Finiteness of the moments makes this a real number for all  $p, q \in \mathbb{R}[x]$ , and bilinearity and symmetry come from linearity of the integral. For positivity, let  $p \neq 0$ . Then  $p^2 w$  is continuous and  $\geq 0$  on  $(a, b)$ , and  $> 0$  off the finitely many roots of  $p$  (Corollary 5.1.4), so  $\langle p, p \rangle > 0$ . That last step is the one calculus fact this section leans on: a continuous non-negative function on an interval whose integral is 0 vanishes identically. As in Definition 10.1.1 it follows from the local fact that a continuous function non-zero at a point stays bounded away from 0 on a small interval around it, which then contributes a positive amount to the integral.

**From nodes.** Fix distinct  $c_0 < \dots < c_n$  in  $\mathbb{R}$  and put  $\langle p, q \rangle = \sum_{i=0}^n p(c_i) q(c_i)$ . Section 1 checked that this is an inner product on  $\mathbb{R}[x]_{\leq n}$ , positivity coming from the root bound. It is not one on  $\mathbb{R}[x]$ : the polynomial  $(x - c_0) \cdots (x - c_n)$  is non-zero and pairs with itself to give 0. So in the node case everything below stops at degree  $n$ .

The two constructions share one property, and it is the engine of the section.

### Proposition 10.10.1 — Multiplication by $x$ is symmetric

Let  $\langle \cdot, \cdot \rangle$  be an inner product on  $\mathbb{R}[x]$  coming from a weight, or the node inner product on  $\mathbb{R}[x]_{\leq n}$ . Then

$$\langle xp, q \rangle = \langle p, xq \rangle$$

for all polynomials  $p, q$  in the space (in the node case, for all  $p, q$  of degree at most  $n-1$ , so that  $xp$  and  $xq$  still lie in it).

*Proof.* In the weight case both sides equal  $\int_a^b t p(t) q(t) w(t) dt$ , and in the node case both sides equal  $\sum_{i=0}^n c_i p(c_i) q(c_i)$ . ■

The proof is one line, but the statement is not a triviality: multiplication by  $x$  moves from one slot to the other for free. That is exactly the identity  $\langle T\mathbf{u}, \mathbf{v} \rangle = \langle \mathbf{u}, T\mathbf{v} \rangle$  defining a **self-adjoint** operator (Definition 10.6.2), and it is the reason the recurrence below has three terms instead

of  $k + 2$ .

*Warning.* **Multiplication by  $x$  is not an operator on  $\mathbb{R}[x]_{\leq n}$ .** It raises degree, so it maps that space out of itself; on  $\mathbb{R}[x]$  it is an operator, but there the space is infinite-dimensional, and Definition 10.6.2 was stated for a finite-dimensional one. “Self-adjoint” is here a name for Proposition 10.10.1, not a literal application of the definition. It is the identity, not the name, that the proofs use.

## 10.10.2 Orthogonal polynomial sequences

Applied to the independent list  $(1, x, x^2, \dots)$ , Gram–Schmidt never stalls. What comes out is one polynomial of each degree, pairwise orthogonal, and that combination deserves a name.

*An orthogonal polynomial sequence is one polynomial of every degree, no two of them pairing to anything but zero.*

### Definition 10.10.1 — Orthogonal Polynomial Sequence

Let  $\langle \cdot, \cdot \rangle$  be an inner product on  $\mathbb{R}[x]$ , or on  $\mathbb{R}[x]_{\leq n}$ . An **orthogonal polynomial sequence** for it is a sequence  $p_0, p_1, p_2, \dots$  of polynomials (a finite list  $p_0, \dots, p_n$  in the second case) such that

(OP1)  $\deg p_k = k$  for **every** index  $k$ ;

(OP2)  $\langle p_j, p_k \rangle = 0$  whenever  $j \neq k$ .

The sequence is **monic** if every  $p_k$  has leading coefficient 1.

In words. (OP1) says one polynomial of each degree: no gaps, no repeats, no accidental drop in degree. (OP2) says the members are pairwise orthogonal. Together they say more than either alone, because by (OP1) the list  $(p_0, \dots, p_{k-1})$  is a basis of  $\mathbb{R}[x]_{\leq k-1}$ , so (OP2) upgrades to

$$\langle p_k, q \rangle = 0 \quad \text{for every } q \in \mathbb{R}[x]_{\leq k-1},$$

and that is the form in which the condition always gets used.

Two things need checking: that such a sequence exists, and that fixing the scale pins it down.

### Proposition 10.10.2 — Existence and uniqueness of the monic sequence

Let  $\langle \cdot, \cdot \rangle$  be an inner product on  $\mathbb{R}[x]$ . For each  $k \in \mathbb{N}$  there is **exactly one** monic  $p_k$  of degree  $k$  with  $\langle p_k, q \rangle = 0$  for every  $q \in \mathbb{R}[x]_{\leq k-1}$ . These  $p_k$  form the unique monic orthogonal polynomial sequence, and  $(p_0, \dots, p_k)$  is an orthogonal basis of  $\mathbb{R}[x]_{\leq k}$  for every  $k$ . The same holds on  $\mathbb{R}[x]_{\leq n}$  for  $k \leq n$ .

*Idea.* Existence is Gram–Schmidt, once we know the  $k$ -th output really has degree  $k$ ; it does, because it lies in  $\mathbb{R}[x]_{\leq k}$  but not in the span of its predecessors, which is  $\mathbb{R}[x]_{\leq k-1}$ . Uniqueness is the standard move: two candidates differ by a polynomial of lower degree that is orthogonal to everything of lower degree, hence to itself.

*Proof. Existence.* The list  $(1, x, \dots, x^k)$  is independent, so [Theorem 10.2.2](#) applies and returns an orthonormal list  $(\mathbf{e}_0, \dots, \mathbf{e}_k)$  with  $\text{span}(\mathbf{e}_0, \dots, \mathbf{e}_j) = \text{span}(1, x, \dots, x^j) = \mathbb{R}[x]_{\leq j}$  for every  $j \leq k$ . Fix  $j$ . Then  $\mathbf{e}_j \in \mathbb{R}[x]_{\leq j}$ , and  $\mathbf{e}_j \notin \mathbb{R}[x]_{\leq j-1} = \text{span}(\mathbf{e}_0, \dots, \mathbf{e}_{j-1})$ , since an orthonormal list is independent ([Theorem 10.1.4](#)). Hence  $\deg \mathbf{e}_j = j$  exactly. Let  $a_j \neq 0$  be its leading coefficient and put  $p_j = \mathbf{e}_j/a_j$ , monic of degree  $j$ . For  $q \in \mathbb{R}[x]_{\leq j-1} = \text{span}(\mathbf{e}_0, \dots, \mathbf{e}_{j-1})$  we get  $\langle p_j, q \rangle = 0$  by orthogonality of the list.

*Uniqueness.* Suppose  $p$  and  $p'$  are both monic of degree  $k$  and orthogonal to every element of  $\mathbb{R}[x]_{\leq k-1}$ . Their leading terms cancel, so  $p - p' \in \mathbb{R}[x]_{\leq k-1}$ . Therefore  $\langle p - p', p - p' \rangle = \langle p, p - p' \rangle - \langle p', p - p' \rangle = 0$ , and positive definiteness gives  $p = p'$ .

Finally,  $(p_0, \dots, p_k)$  is an orthogonal list of non-zero vectors, hence independent ([Theorem 10.1.4](#)), with  $k+1 = \dim \mathbb{R}[x]_{\leq k}$  members, hence a basis ([Theorem 1.6.4](#)). ■

**Examples.** For  $\langle p, q \rangle = \int_{-1}^1 pq$  the first three monic members are  $1, x, x^2 - \frac{1}{3}$  ([Example \(Gram–Schmidt on  \$1, x, x^2\$ \)](#)); these are the monic **Legendre polynomials**. For the node inner product with nodes  $-1, 0, 1$  on  $\mathbb{R}[x]_{\leq 2}$  we have  $\langle 1, 1 \rangle = 3$ ,  $\langle x, 1 \rangle = 0$ ,  $\langle x, x \rangle = 2$ ,  $\langle x^2, 1 \rangle = 2$ ,  $\langle x^2, x \rangle = 0$ , so  $p_0 = 1$ ,  $p_1 = x$ ,  $p_2 = x^2 - \frac{2}{3}$  — and there the sequence ends, since  $\mathbb{R}[x]_{\leq 2}$  holds nothing of degree 3. The degenerate member is always  $p_0 = 1$ , the only monic polynomial of degree 0, with an empty condition on it.

**A non-example by minimal change.** Replace the Legendre  $p_2 = x^2 - \frac{1}{3}$  by  $\tilde{p}_2 = x^2 + x - \frac{1}{3}$ . Still monic of degree 2, so (OP1) survives; but  $\langle \tilde{p}_2, x \rangle = \langle x, x \rangle = \frac{2}{3} \neq 0$ , so (OP2) fails. Adding a multiple of a lower member is exactly the freedom orthogonality removes.

**Why monic?** Rescaling each  $p_k$  by a non-zero  $\lambda_k$  gives another orthogonal polynomial sequence, so a normalization has to be chosen, and traditions differ: the classical Legendre polynomials  $P_k$  are fixed by  $P_k(1) = 1$ , and Gram–Schmidt as run in Section 2 fixes  $\|p_k\| = 1$ . Monic is the convenient choice here, because it is the one for which the recurrence below needs no extra factor.

**Warning.** The sequence belongs to the inner product, not to the polynomials.

“The orthogonal polynomials” is meaningless without a weight and an interval. Exercise B2 of Section 2 ([Exercise \(B2: Gram–Schmidt on polynomials\)](#)) ran the same process with  $\int_0^1$  instead of  $\int_{-1}^1$  and got  $1, x - \frac{1}{2}, x^2 - x + \frac{1}{6}$ : same degrees, different polynomials.

*Quick check.* For  $\langle p, q \rangle = \int_{-1}^1 pq$ , how many polynomials of degree 2 with  $\|p\| = 1$  are orthogonal to  $\mathbb{R}[x]_{\leq 1}$ ? Write them down.

*Solution.* Exactly two. If  $p$  has degree 2 and  $p \perp \mathbb{R}[x]_{\leq 1}$ , let  $\lambda \neq 0$  be its leading coefficient; then  $p/\lambda$  is monic of degree 2 and still orthogonal to  $\mathbb{R}[x]_{\leq 1}$ , so  $p/\lambda = x^2 - \frac{1}{3}$  by [Proposition 10.10.2](#). Thus  $p = \lambda(x^2 - \frac{1}{3})$ , and  $\|p\| = 1$  leaves  $\lambda = \pm 1/\|x^2 - \frac{1}{3}\|$ . Since [Example \(Gram–Schmidt on  \$1, x, x^2\$ \)](#) computed  $\|x^2 - \frac{1}{3}\|^2 = \frac{8}{45}$ , the two answers are  $\pm \frac{\sqrt{10}}{4}(3x^2 - 1)$ . Fixing the scale by the norm leaves a sign; fixing it by “monic” leaves nothing.



### 10.10.3 The three-term recurrence

Computing  $p_n$  by Gram–Schmidt costs one inner product for every earlier member, so reaching degree  $n$  costs about  $n^2/2$  integrals. Almost all of them return 0. The next theorem says which ones do not, and reduces the cost to two integrals per step.

#### Theorem 10.10.3 — Three-Term Recurrence

Let  $(p_k)$  be the monic orthogonal polynomial sequence of [Proposition 10.10.2](#), for an inner product satisfying [Proposition 10.10.1](#). For a weight on an interval this covers every  $k$ ; for the node inner product on  $n$  nodes, where [Proposition 10.10.1](#) holds only up to degree  $n - 1$ , read the statement for  $k + 1 \leq n$ . Put  $p_{-1} = 0$  and, for  $k \geq 0$ ,

$$a_k = \frac{\langle xp_k, p_k \rangle}{\langle p_k, p_k \rangle}, \quad b_k = \frac{\langle p_k, p_k \rangle}{\langle p_{k-1}, p_{k-1} \rangle} \quad (k \geq 1).$$

Then  $p_0 = 1$ ,  $p_1 = x - a_0$ , and

$$p_{k+1} = (x - a_k)p_k - b_k p_{k-1} \quad \text{for every } k \geq 1.$$

Moreover  $b_k > 0$  for every  $k \geq 1$ .

*Idea.* The polynomial  $xp_k$  is monic of degree  $k + 1$ , so it is  $p_{k+1}$  plus something of degree at most  $k$ , and that something expands in the orthogonal basis  $(p_0, \dots, p_k)$  with coefficients  $\langle xp_k, p_j \rangle / \langle p_j, p_j \rangle$  ([Corollary 10.3.4](#)). The point is that almost every coefficient is 0: move the  $x$  across with [Proposition 10.10.1](#), and the coefficient becomes  $\langle p_k, xp_j \rangle$ , which vanishes as soon as  $\deg(xp_j) = j + 1$  is still below  $k$ . Only  $j = k$  and  $j = k - 1$  can survive.

*Proof.*  $p_0 = 1$  is the unique monic polynomial of degree 0, and  $p_1 = x - a_0$  is the unique monic polynomial of degree 1 with  $\langle p_1, p_0 \rangle = 0$ , since  $\langle x - a_0, 1 \rangle = \langle x, 1 \rangle - a_0 \langle 1, 1 \rangle = 0$  exactly for  $a_0 = \langle x \cdot 1, 1 \rangle / \langle 1, 1 \rangle$ .

Now fix  $k \geq 1$ . By [Proposition 10.10.2](#),  $(p_0, \dots, p_{k+1})$  is an orthogonal basis of  $\mathbb{R}[x]_{\leq k+1}$ , and  $xp_k$  lies in that space. By [Corollary 10.3.4](#) applied inside  $\mathbb{R}[x]_{\leq k+1}$ ,

$$xp_k = \sum_{j=0}^{k+1} c_j p_j, \quad c_j = \frac{\langle xp_k, p_j \rangle}{\langle p_j, p_j \rangle}.$$

Both  $xp_k$  and  $p_{k+1}$  are monic of degree  $k + 1$ , and every other term of the sum has degree at most  $k$ ; comparing coefficients of  $x^{k+1}$  gives  $c_{k+1} = 1$ .

Let  $j \leq k - 2$ . By [Proposition 10.10.1](#),  $\langle xp_k, p_j \rangle = \langle p_k, xp_j \rangle$ , and  $\deg(xp_j) = j + 1 \leq k - 1$ , so  $xp_j \in \mathbb{R}[x]_{\leq k-1}$  and the pairing is 0 by [Proposition 10.10.2](#). Hence  $c_j = 0$  for every  $j \leq k - 2$ , and

$$xp_k = p_{k+1} + c_k p_k + c_{k-1} p_{k-1},$$

which rearranges to  $p_{k+1} = (x - c_k)p_k - c_{k-1}p_{k-1}$ .

It remains to identify the two coefficients. By definition  $c_k = \langle xp_k, p_k \rangle / \langle p_k, p_k \rangle = a_k$ . For the other,  $xp_{k-1}$  is monic of degree  $k$ , so  $xp_{k-1} - p_k$  has degree at most  $k - 1$  and

therefore pairs to 0 with  $p_k$ . Hence, using [Proposition 10.10.1](#) again,

$$\langle xp_k, p_{k-1} \rangle = \langle p_k, xp_{k-1} \rangle = \langle p_k, p_k \rangle,$$

so  $c_{k-1} = \langle p_k, p_k \rangle / \langle p_{k-1}, p_{k-1} \rangle = b_k$ . Finally  $b_k$  is a quotient of  $\|p_k\|^2$  by  $\|p_{k-1}\|^2$ , both strictly positive because  $p_k$  and  $p_{k-1}$  are non-zero, so  $b_k > 0$ . ■

Two inner products per step generate the whole sequence. The positivity of  $b_k$  is not decoration either: it is the sign that drives the interlacing argument of [Exercise \(C1: Interlacing roots\)](#).

*Remark.* If the interval is symmetric,  $(a, b) = (-c, c)$ , and the weight is even,  $w(-t) = w(t)$ , then  $a_k = 0$  for every  $k$  and  $p_k(-x) = (-1)^k p_k(x)$ . Induct on  $k$ :  $p_0 = 1$  is even, and  $a_0 = \langle x, 1 \rangle / \langle 1, 1 \rangle = 0$  because  $tw(t)$  is odd, so  $p_1 = x$  is odd. If  $p_{k-1}$  and  $p_k$  have parities  $(-1)^{k-1}$  and  $(-1)^k$ , then  $tp_k(t)^2 w(t)$  is odd, so  $a_k = 0$ , and  $p_{k+1} = xp_k - b_k p_{k-1}$  is a combination of two polynomials of parity  $(-1)^{k+1}$ .

### Example. (The monic Legendre polynomials, continued)

For  $\langle p, q \rangle = \int_{-1}^1 pq$ , [Example \(Gram-Schmidt on  \$1, x, x^2\$ \)](#) gave  $p_0 = 1$ ,  $p_1 = x$ ,  $p_2 = x^2 - \frac{1}{3}$ , with  $\langle p_0, p_0 \rangle = 2$ ,  $\langle p_1, p_1 \rangle = \frac{2}{3}$  and  $\langle p_2, p_2 \rangle = \frac{8}{45}$ . Use [Theorem 10.10.3](#) to obtain  $p_3$ , and check the recurrence reproduces  $p_2$ .

*Solution.* The weight  $w = 1$  is even on  $(-1, 1)$ , so every  $a_k = 0$  by the remark above. Then

$$b_1 = \frac{\langle p_1, p_1 \rangle}{\langle p_0, p_0 \rangle} = \frac{2/3}{2} = \frac{1}{3}, \quad b_2 = \frac{\langle p_2, p_2 \rangle}{\langle p_1, p_1 \rangle} = \frac{8/45}{2/3} = \frac{4}{15}.$$

The recurrence at  $k = 1$  gives  $p_2 = x \cdot x - \frac{1}{3} \cdot 1 = x^2 - \frac{1}{3}$ , which is what Gram-Schmidt produced. At  $k = 2$ ,

$$p_3 = x \left( x^2 - \frac{1}{3} \right) - \frac{4}{15} x = x^3 - \frac{3}{5} x.$$

*Check.*  $\langle p_3, 1 \rangle = \langle p_3, x^2 \rangle = 0$  because the integrands are odd, and

$$\langle p_3, x \rangle = \int_{-1}^1 \left( t^4 - \frac{3}{5} t^2 \right) dt = \frac{2}{5} - \frac{3}{5} \cdot \frac{2}{3} = 0.$$

So  $p_3 \perp \mathbb{R}[x]_{\leq 2}$ , as it should be. Two integrals produced a polynomial that Gram-Schmidt would have charged three for, and the saving grows with the degree.

## 10.10.4 Where the roots are

The roots of  $p_2 = x^2 - \frac{1}{3}$  are  $\pm 1/\sqrt{3} \approx \pm 0.577$ , and those of  $p_3 = x^3 - \frac{3}{5}x$  are 0 and  $\pm\sqrt{3/5} \approx \pm 0.775$ . All five lie inside  $(-1, 1)$ , where the weight lives. No coincidence. The argument first needs one fact about real polynomials, in which  $\text{mult}_r(f)$  is the multiplicity of  $r$  as a root of  $f$  ([Definition 5.4.1](#)).

**Lemma 10.10.4 — Even multiplicities mean no sign change**

Let  $I \subseteq \mathbb{R}$  be an open interval and let  $f \in \mathbb{R}[x]$  be non-zero. If  $\text{mult}_r(f)$  is even for every  $r \in I$ , then  $f \geq 0$  throughout  $I$ , or  $f \leq 0$  throughout  $I$ .

*Proof.* By Theorem 5.4.3 over  $\mathbb{R}$ , write  $f = (x - d_1)^{m_1} \cdots (x - d_l)^{m_l} g$ , where  $d_1, \dots, d_l$  are the distinct real roots of  $f$ ,  $m_i = \text{mult}_{d_i}(f)$ , and  $g$  has no real root. Separate the factors according to whether  $d_i \in I$ :

$$f = h \cdot \left( \prod_{d_i \in I} (x - d_i)^{m_i} \right), \quad h = g \cdot \prod_{d_i \notin I} (x - d_i)^{m_i}.$$

Each  $m_i$  with  $d_i \in I$  is even by hypothesis, so the second factor is the square of a real polynomial and is  $\geq 0$  everywhere. The polynomial  $h$  has no root in  $I$ : its roots are the  $d_i$  outside  $I$ , and  $g$  has none. Being continuous and nowhere zero on the interval  $I$ ,  $h$  has constant sign there by the intermediate value theorem. Hence  $f$  has that same sign throughout  $I$ , weakly, with equality only at the  $d_i$  inside  $I$ . ■

**Theorem 10.10.5 — Roots of an Orthogonal Polynomial**

Let  $w$  be a weight on  $(a, b)$ , let  $\langle p, q \rangle = \int_a^b pqw$ , and let  $(p_k)$  be the monic orthogonal polynomial sequence. For every  $k \geq 1$ , the polynomial  $p_k$  has  $k$  **distinct real** roots, and every one of them lies in the **open** interval  $(a, b)$ .

*Idea.* Suppose  $p_k$  changed sign at only a few points  $c_1, \dots, c_m$  of  $(a, b)$ , with  $m < k$ . Build the polynomial  $q = (x - c_1) \cdots (x - c_m)$ . It is cheap — degree at most  $k - 1$  — so  $p_k$  is orthogonal to it. But  $q$  was designed to change sign at exactly the points where  $p_k$  does, so the product  $p_k q$  changes sign nowhere, and a function of one sign that is not identically zero cannot integrate to zero against a positive weight. The contradiction forces  $m \geq k$ , and  $k$  distinct roots is all a polynomial of degree  $k$  can hold.

*Proof.* Fix  $k \geq 1$ . Let  $c_1 < \cdots < c_m$  be the roots of  $p_k$  in  $(a, b)$  whose multiplicity is **odd** (possibly  $m = 0$ ), and set  $q = (x - c_1) \cdots (x - c_m)$ , with  $q = 1$  when  $m = 0$ .

Suppose, for contradiction, that  $m \leq k - 1$ . Then  $q \in \mathbb{R}[x]_{\leq k-1}$ , so  $\langle p_k, q \rangle = 0$  by Proposition 10.10.2.

Put  $f = p_k q$ , a non-zero polynomial (Theorem 0.8.3). Let  $r \in (a, b)$  be a root of  $f$ . By Theorem 5.4.2,  $\text{mult}_r(f) = \text{mult}_r(p_k) + \text{mult}_r(q)$ . If  $r = c_i$  for some  $i$ , then  $\text{mult}_r(q) = 1$  and  $\text{mult}_r(p_k)$  is odd by the choice of the  $c_i$ , so the sum is even. If  $r$  is none of the  $c_i$ , then  $q(r) \neq 0$ , so  $\text{mult}_r(q) = 0$  and  $\text{mult}_r(p_k)$  is even, again by the choice of the  $c_i$ . Either way  $\text{mult}_r(f)$  is even.

By Lemma 10.10.4,  $f \geq 0$  throughout  $(a, b)$  or  $f \leq 0$  throughout  $(a, b)$ ; replacing  $q$  by  $-q$  if necessary, which changes neither  $\langle p_k, q \rangle = 0$  nor  $\deg q$ , we may assume  $f \geq 0$  there. Then  $fw$  is continuous and  $\geq 0$  on  $(a, b)$ , and it is  $> 0$  off the finitely many roots of  $f$ , because  $w > 0$ . By the calculus fact recalled when the inner product was set up — a continuous non-negative function with integral 0 vanishes identically — an integrand of

that shape cannot integrate to 0. Therefore

$$\langle p_k, q \rangle = \int_a^b f(t) w(t) dt > 0,$$

contradicting  $\langle p_k, q \rangle = 0$ .

Hence  $m \geq k$ . Each  $c_i$  is a root of  $p_k$ , and  $p_k$  is non-zero of degree  $k$ , so  $m \leq k$  by [Corollary 5.1.4](#); thus  $m = k$  and  $p_k$  has  $k$  distinct roots  $c_1, \dots, c_k$ , all in  $(a, b)$ . By [Corollary 5.1.4](#) again,  $(x - c_1) \cdots (x - c_k)$  divides  $p_k$ , and both are monic of degree  $k$ , so  $p_k = (x - c_1) \cdots (x - c_k)$ . In particular each root is simple and  $p_k$  has no roots outside  $(a, b)$ , real or complex. This proves the theorem. ■

So the weight hands us  $k$  points inside its interval, chosen by the inner product rather than by us. The next subsection says what they are good for.

*Quick check.* The monic Laguerre polynomial of degree 2 for the weight  $w(t) = e^{-t}$  on  $(0, \infty)$  is  $p_2 = x^2 - 4x + 2$ . Check that [Theorem 10.10.5](#) is not lying.

*Solution.* The roots are  $2 \pm \sqrt{2}$ , that is  $\approx 0.586$  and  $\approx 3.414$ : real, distinct, and both in  $(0, \infty)$ . An unbounded interval is no obstacle, since the theorem asks only that the weight be positive with finite moments, and  $\int_0^\infty t^k e^{-t} dt = k!$ .

## 10.10.5 Gaussian quadrature

Chapter 5 wrote  $\int_0^1 p$  as a fixed combination of  $p(0)$ ,  $p(\frac{1}{2})$  and  $p(1)$ , valid for every  $p \in \mathbb{R}[x]_{\leq 2}$  ([Example \(Integrals and Slopes from Values\)](#)). The construction works for any prescribed nodes: integrate the Lagrange interpolation formula ([Theorem 5.6.1](#)) and  $k$  distinct nodes give a rule exact on  $\mathbb{R}[x]_{\leq k-1}$ .

But a  $k$ -point rule has  $2k$  numbers at its disposal,  $k$  nodes and  $k$  coefficients, and only  $k$  of them were spent. Spending the nodes too should buy degree  $2k - 1$ . Where to put them is the question, and the answer is the one place we have not chosen: the roots of  $p_k$ .

### Theorem 10.10.6 — Gaussian Quadrature

Let  $w$  be a weight on  $(a, b)$  with  $\langle p, q \rangle = \int_a^b pqw$ , let  $k \geq 1$ , and let  $t_1 < \cdots < t_k$  be the roots of the monic orthogonal polynomial  $p_k$ , which lie in  $(a, b)$  by [Theorem 10.10.5](#). Let  $\ell_1, \dots, \ell_k \in \mathbb{R}[x]_{\leq k-1}$  be the Lagrange polynomials for these nodes ([Definition 5.6.1](#)) and set

$$\alpha_i = \int_a^b \ell_i(t) w(t) dt \quad (i = 1, \dots, k).$$

Then every  $\alpha_i$  is **strictly positive**, and

$$\int_a^b f(t) w(t) dt = \sum_{i=1}^k \alpha_i f(t_i)$$

for **every**  $f \in \mathbb{R}[x]_{\leq 2k-1}$ .

*Idea.* Two moves, one for each side of the identity. On the left, divide  $f$  by  $p_k$ :  $f = qp_k + r$ . The degree bookkeeping —  $\deg f \leq 2k - 1$  against  $\deg p_k = k$  — forces  $\deg q \leq k - 1$ , which is exactly the condition under which  $\int qp_k w = \langle p_k, q \rangle$  vanishes. So the integral of  $f$  is the integral of its remainder. On the right,  $p_k$  vanishes at every node, so the sum also only sees the remainder. And the remainder has degree at most  $k - 1$ , which is precisely the range on which  $k$  nodes reproduce a polynomial exactly. The two sides meet there.

*Proof.* Let  $f \in \mathbb{R}[x]_{\leq 2k-1}$ . Since  $p_k \neq 0$ , [Theorem 5.1.1](#) gives  $q, r \in \mathbb{R}[x]$  with

$$f = qp_k + r, \quad \deg r < k.$$

If  $q = 0$  then  $q \in \mathbb{R}[x]_{\leq k-1}$  trivially. Otherwise  $qp_k = f - r$  has degree at most  $2k - 1$ , while  $\deg(qp_k) = \deg q + k$  by [Theorem 0.8.3](#); hence  $\deg q \leq k - 1$ . Either way  $q \in \mathbb{R}[x]_{\leq k-1}$ , so [Proposition 10.10.2](#) gives  $\langle p_k, q \rangle = 0$ , that is,  $\int_a^b qp_k w = 0$ . Therefore

$$\int_a^b fw = \int_a^b qp_k w + \int_a^b rw = \int_a^b rw. \quad (*)$$

At each node,  $p_k(t_i) = 0$ , so

$$f(t_i) = q(t_i)p_k(t_i) + r(t_i) = r(t_i) \quad (i = 1, \dots, k). \quad (**)$$

Since  $\deg r \leq k - 1$  and  $t_1, \dots, t_k$  are  $k$  distinct points, [Theorem 5.6.1](#) (b) gives  $r = \sum_i r(t_i)\ell_i$ . Integrating against  $w$  and using the definition of  $\alpha_i$ ,

$$\int_a^b rw = \sum_{i=1}^k r(t_i)\alpha_i = \sum_{i=1}^k \alpha_i f(t_i),$$

the last step by (\*\*). Combining with (\*) proves the quadrature identity.

For positivity, fix  $j$  and apply the identity to  $f = \ell_j^2$ , which is legitimate because  $\deg \ell_j^2 = 2k - 2 \leq 2k - 1$ . By [Theorem 5.6.1](#) (a),  $\ell_j(t_i) = \delta_{ij}$ , so the right-hand side collapses to  $\alpha_j$ , while the left-hand side is  $\langle \ell_j, \ell_j \rangle = \|\ell_j\|^2$ . Since  $\ell_j \neq 0$ , positive definiteness gives  $\alpha_j = \|\ell_j\|^2 > 0$ . ■

Each hypothesis did one job. Orthogonality to lower degrees killed the quotient on the left; the nodes being roots of  $p_k$  killed it again on the right. That is also why the nodes cannot be moved: shift one and the second cancellation fails. [Exercise \(C2: The ceiling is  \$2k - 1\$ \)](#) shows  $2k - 1$  is a ceiling, not just what this proof happens to reach.

### Example. (Two-point Gauss–Legendre)

Take  $w = 1$  on  $(-1, 1)$  and  $k = 2$ . Find the nodes and weights, and test the resulting rule on  $f = 1 + 2x + 3x^2 + 4x^3$  and on  $f = x^4$ .

*Solution.* The monic  $p_2$  is  $x^2 - \frac{1}{3}$  (Example (Gram–Schmidt on  $1, x, x^2$ )), so the nodes are  $t_1 = -1/\sqrt{3}$  and  $t_2 = 1/\sqrt{3}$ . The Lagrange polynomials are  $\ell_1 = (x - t_2)/(t_1 - t_2)$  and  $\ell_2 = (x - t_1)/(t_2 - t_1)$ , with  $t_1 - t_2 = -2/\sqrt{3}$ . Since  $\int_{-1}^1 t \, dt = 0$  and  $\int_{-1}^1 1 \, dt = 2$ ,

$$\alpha_1 = \frac{0 - 2t_2}{t_1 - t_2} = \frac{-2/\sqrt{3}}{-2/\sqrt{3}} = 1,$$

and  $\alpha_2 = 1$  by the same computation with the roles swapped. The rule is

$$\int_{-1}^1 f(t) \, dt = f\left(-\frac{1}{\sqrt{3}}\right) + f\left(\frac{1}{\sqrt{3}}\right) \quad \text{for } f \in \mathbb{R}[x]_{\leq 3}.$$

Two evaluations, no coefficients to remember, every cubic exactly.

*The cubic test.* Exactly,  $\int_{-1}^1 (1 + 2t + 3t^2 + 4t^3) \, dt = 2 + 0 + 3 \cdot \frac{2}{3} + 0 = 4$ . By the rule,  $f(-1/\sqrt{3}) = 2 - \frac{10\sqrt{3}}{9}$  and  $f(1/\sqrt{3}) = 2 + \frac{10\sqrt{3}}{9}$ , whose sum is 4. The odd parts cancel and the constant and quadratic parts contribute  $1 + 1$  and  $1 + 1$ .

*The quartic test.* Exactly,  $\int_{-1}^1 t^4 \, dt = \frac{2}{5}$ . By the rule,  $(1/\sqrt{3})^4 + (1/\sqrt{3})^4 = \frac{2}{9}$ . These differ, so degree  $4 = 2k$  is already out of reach. Compare Example (Integrals and Slopes from Values), where the nodes were fixed in advance and **three** of them guaranteed exactness only up to degree 2 — guaranteed by the construction, that is; that particular symmetric rule happens to integrate cubics exactly as well, which is luck of the symmetry and not something the construction promises.

*Warning.* “**Exact**” means **exact on polynomials**. Theorem 10.10.6 says nothing about  $\int_a^b f w$  for a general  $f$ , and nothing about what happens as  $k$  grows; on a non-polynomial the rule returns an approximation whose error is a question for analysis. What linear algebra delivers is exactness on a subspace, and the reason that subspace is the largest available.

## 10.10.6 The classical families

Four weights account for most of the orthogonal polynomials that occur in practice. The table records them; the entries are statements, and Exercise (B2: Two rows of the table) asks you to verify two of the rows by integration.

| Family    | Interval            | Weight $w(t)$      | $p_1$   | $p_2$               |
|-----------|---------------------|--------------------|---------|---------------------|
| Legendre  | $(-1, 1)$           | 1                  | $x$     | $x^2 - \frac{1}{3}$ |
| Chebyshev | $(-1, 1)$           | $(1 - t^2)^{-1/2}$ | $x$     | $x^2 - \frac{1}{2}$ |
| Hermite   | $(-\infty, \infty)$ | $e^{-t^2}$         | $x$     | $x^2 - \frac{1}{2}$ |
| Laguerre  | $(0, \infty)$       | $e^{-t}$           | $x - 1$ | $x^2 - 4x + 2$      |

The last two columns give the monic members of degrees 1 and 2;  $p_0 = 1$  in every row. The Chebyshev weight is unbounded at  $\pm 1$  and the Hermite and Laguerre intervals are unbounded, so in those three rows the integrals defining the inner product are improper; they converge, which is the finiteness condition our definition of a weight demanded. The first three weights are even on symmetric intervals, so their  $a_k$  all vanish and their members alternate in parity; the Laguerre weight is not, and  $a_0 = 1$  there. Section 9 of this chapter meets the Chebyshev

family again from the Fourier side, where the substitution  $t = \cos \theta$  turns its awkward weight into  $d\theta$ .

## 10.10.7 Exercises

### 10.10.7.1 A. Check your understanding

#### Exercise A1.

- Define what it means for  $p_0, p_1, p_2, \dots$  to be a **monic orthogonal polynomial sequence** for an inner product on  $\mathbb{R}[x]$ .
- State the three-term recurrence, including the formulas for  $a_k$  and  $b_k$  and the starting values.
- True or false:  $b_k > 0$  for every  $k \geq 1$ . Justify your answer.
- For  $\langle p, q \rangle = \int_{-1}^1 pq$ , how many roots does  $p_7$  have, of what kind, and where?
- True or false: for **every** inner product on  $\mathbb{R}[x]$  one has  $\langle xp, q \rangle = \langle p, xq \rangle$ . Justify your answer.
- How many nodes does a Gaussian rule need in order to integrate every polynomial of degree at most 11 exactly?

*Solution.*

- A sequence with  $\deg p_k = k$  and leading coefficient 1 for every  $k$ , and  $\langle p_j, p_k \rangle = 0$  whenever  $j \neq k$  ([Definition 10.10.1](#)).
- With  $p_{-1} = 0, p_0 = 1$  and  $p_1 = x - a_0$ , one has  $p_{k+1} = (x - a_k)p_k - b_k p_{k-1}$  for  $k \geq 1$ , where  $a_k = \langle xp_k, p_k \rangle / \langle p_k, p_k \rangle$  and  $b_k = \langle p_k, p_k \rangle / \langle p_{k-1}, p_{k-1} \rangle$  ([Theorem 10.10.3](#)).
- True. Both  $\langle p_k, p_k \rangle = \|p_k\|^2$  and  $\langle p_{k-1}, p_{k-1} \rangle = \|p_{k-1}\|^2$  are strictly positive, since  $p_k$  and  $p_{k-1}$  are non-zero and the inner product is positive definite.
- Seven roots, all real, all distinct, all in the open interval  $(-1, 1)$  ([Theorem 10.10.5](#)). In particular  $p_7$  has no complex roots and no repeated ones.
- False. Take the coefficient inner product  $\langle p, q \rangle = \sum_k u_k v_k$  on  $\mathbb{R}[x]$ , where  $p = \sum_k u_k x^k$  and  $q = \sum_k v_k x^k$  ([Example \(A functional with no Riesz vector\)](#)). Then  $\langle x \cdot 1, x \rangle = \langle x, x \rangle = 1$ , while  $\langle 1, x \cdot x \rangle = \langle 1, x^2 \rangle = 0$ . The identity is a property of the weight and node constructions ([Proposition 10.10.1](#)), not of inner products in general.
- Six. A  $k$ -point rule is exact up to degree  $2k - 1$ , and  $2 \cdot 6 - 1 = 11$ .

### 10.10.7.2 B. Practice

#### Exercise B1: A weighted interval.

On  $\mathbb{R}[x]$  put  $\langle p, q \rangle = \int_0^1 p(t)q(t)t \, dt$ . Compute the monic  $p_0, p_1$  and  $p_2$ , and verify directly that  $p_2 \perp \mathbb{R}[x]_{\leq 1}$ .

*Solution.* The moments are  $\langle x^k, 1 \rangle = \int_0^1 t^{k+1} dt = 1/(k+2)$ , so  $\frac{1}{2}, \frac{1}{3}, \frac{1}{4}, \frac{1}{5}$  for  $k = 0, 1, 2, 3$ .

$p_0 = 1$ . Then  $a_0 = \langle x, 1 \rangle / \langle 1, 1 \rangle = (1/3)/(1/2) = \frac{2}{3}$ , so  $p_1 = x - \frac{2}{3}$ , and

$$\langle p_1, p_1 \rangle = \frac{1}{4} - \frac{4}{3} \cdot \frac{1}{3} + \frac{4}{9} \cdot \frac{1}{2} = \frac{1}{4} - \frac{2}{9} = \frac{1}{36}.$$

Next  $\langle xp_1, p_1 \rangle = \int_0^1 (t^4 - \frac{4}{3}t^3 + \frac{4}{9}t^2) dt = \frac{1}{5} - \frac{1}{3} + \frac{4}{27} = \frac{2}{135}$ , so

$$a_1 = \frac{2/135}{1/36} = \frac{8}{15}, \quad b_1 = \frac{1/36}{1/2} = \frac{1}{18},$$

and [Theorem 10.10.3](#) gives

$$p_2 = \left(x - \frac{8}{15}\right)\left(x - \frac{2}{3}\right) - \frac{1}{18} = x^2 - \frac{6}{5}x + \frac{3}{10}.$$

*Verification.*  $\langle p_2, 1 \rangle = \int_0^1 (t^3 - \frac{6}{5}t^2 + \frac{3}{10}t) dt = \frac{1}{4} - \frac{2}{5} + \frac{3}{20} = 0$ , and  $\langle p_2, x \rangle = \int_0^1 (t^4 - \frac{6}{5}t^3 + \frac{3}{10}t^2) dt = \frac{1}{5} - \frac{3}{10} + \frac{1}{10} = 0$ . Since  $(1, x)$  spans  $\mathbb{R}[x]_{\leq 1}$ , this is the whole check.

### Exercise B2: Two rows of the table.

Verify the degree-1 and degree-2 entries of the table for two of the classical families.

- (a) Chebyshev:  $w(t) = (1 - t^2)^{-1/2}$  on  $(-1, 1)$ . You may use, without proof, that  $\int_{-1}^1 (1 - t^2)^{-1/2} dt = \pi$  and  $\int_{-1}^1 t^2 (1 - t^2)^{-1/2} dt = \pi/2$ .
- (b) Laguerre:  $w(t) = e^{-t}$  on  $(0, \infty)$ . You may use, without proof, that  $\int_0^\infty t^k e^{-t} dt = k!$ .

*Solution.*

- (a) The weight is even, so  $\langle x^j, x^i \rangle = 0$  whenever  $i + j$  is odd. Hence  $\langle x, 1 \rangle = 0$ , and  $x$  is monic of degree 1 and orthogonal to  $\mathbb{R}[x]_{\leq 0}$ , so  $p_1 = x$  by [Proposition 10.10.2](#). Write  $p_2 = x^2 + \beta x + \gamma$ . Then  $\langle p_2, x \rangle = \langle x^2, x \rangle + \beta \langle x, x \rangle + \gamma \langle 1, x \rangle = 0 + \beta \cdot \frac{\pi}{2} + 0$ , forcing  $\beta = 0$ ; and  $\langle p_2, 1 \rangle = \frac{\pi}{2} + 0 + \gamma\pi = 0$ , forcing  $\gamma = -\frac{1}{2}$ . So  $p_2 = x^2 - \frac{1}{2}$ .
- (b) The moments are  $\langle x^k, 1 \rangle = k!$ , so 1, 1, 2, 6 for  $k = 0, 1, 2, 3$ . For  $p_1 = x + \beta$ :  $\langle p_1, 1 \rangle = 1 + \beta = 0$ , so  $p_1 = x - 1$ . For  $p_2 = x^2 + \beta x + \gamma$ , the two conditions  $\langle p_2, 1 \rangle = 0$  and  $\langle p_2, x \rangle = 0$  read

$$2 + \beta + \gamma = 0, \quad 6 + 2\beta + \gamma = 0.$$

Subtracting gives  $\beta = -4$ , then  $\gamma = 2$ . So  $p_2 = x^2 - 4x + 2$ , whose roots  $2 \pm \sqrt{2}$  we already located in the Quick check above.

### Exercise B3: One more Legendre polynomial.

Continue [Example \(The monic Legendre polynomials, continued\)](#): compute  $\langle p_3, p_3 \rangle$  for  $p_3 = x^3 - \frac{3}{5}x$ , then use [Theorem 10.10.3](#) to find  $p_4$ . Verify that  $\langle p_4, 1 \rangle = \langle p_4, x^2 \rangle = 0$ .

*Solution.*

$$\langle p_3, p_3 \rangle = \int_{-1}^1 \left(t^6 - \frac{6}{5}t^4 + \frac{9}{25}t^2\right) dt = \frac{2}{7} - \frac{6}{5} \cdot \frac{2}{5} + \frac{9}{25} \cdot \frac{2}{3} = \frac{2}{7} - \frac{6}{25} = \frac{8}{175}.$$



Since the weight is even,  $a_3 = 0$ , and  $b_3 = \langle p_3, p_3 \rangle / \langle p_2, p_2 \rangle = (8/175)/(8/45) = \frac{9}{35}$ . Hence

$$p_4 = x \left( x^3 - \frac{3}{5}x \right) - \frac{9}{35} \left( x^2 - \frac{1}{3} \right) = x^4 - \frac{6}{7}x^2 + \frac{3}{35}.$$

*Verification.*  $\langle p_4, 1 \rangle = \frac{2}{5} - \frac{6}{7} \cdot \frac{2}{3} + \frac{3}{35} \cdot 2 = \frac{14-20+6}{35} = 0$ , and

$$\langle p_4, x^2 \rangle = \frac{2}{7} - \frac{6}{7} \cdot \frac{2}{5} + \frac{3}{35} \cdot \frac{2}{3} = \frac{10-12+2}{35} = 0.$$

Orthogonality to  $x$  and  $x^3$  is automatic by parity.

#### Exercise B4: A two-point rule on the unit interval.

Work on  $(0, 1)$  with  $w = 1$ . Exercise B2 of Section 2 ([Exercise \(B2: Gram-Schmidt on polynomials\)](#)) found the monic orthogonal polynomials  $1, x - \frac{1}{2}, x^2 - x + \frac{1}{6}$ .

- (a) Write down the two-point Gaussian rule for  $\int_0^1 f$ .
- (b) Check it on  $f = x^3$ , and show that it fails on  $f = x^4$ .

*Solution.*

- (a) The nodes are the roots of  $x^2 - x + \frac{1}{6}$ , namely

$$t_{1,2} = \frac{1}{2} \mp \frac{\sqrt{3}}{6}, \quad t_2 - t_1 = \frac{\sqrt{3}}{3},$$

both in  $(0, 1)$ , as [Theorem 10.10.5](#) promises. With  $\ell_1 = (x - t_2)/(t_1 - t_2)$  and  $\int_0^1 t \, dt = \frac{1}{2}$ ,

$$\alpha_1 = \frac{\frac{1}{2} - t_2}{t_1 - t_2} = \frac{-\sqrt{3}/6}{-\sqrt{3}/3} = \frac{1}{2},$$

and  $\alpha_2 = \frac{1}{2}$  likewise. So  $\int_0^1 f = \frac{1}{2}(f(t_1) + f(t_2))$  for  $f \in \mathbb{R}[x]_{\leq 3}$ .

- (b) Write  $t_{1,2} = \frac{1}{2} \mp u$  with  $u = \sqrt{3}/6$ , so  $u^2 = \frac{1}{12}$ . The odd powers of  $u$  cancel in a sum over  $t_1$  and  $t_2$ , so

$$t_1^3 + t_2^3 = 2 \left( \frac{1}{8} + 3 \cdot \frac{1}{2} \cdot \frac{1}{12} \right) = 2 \cdot \frac{1}{4} = \frac{1}{2},$$

and half of that is  $\frac{1}{4} = \int_0^1 t^3 \, dt$ . For the quartic,

$$t_1^4 + t_2^4 = 2 \left( \frac{1}{16} + 6 \cdot \frac{1}{4} \cdot \frac{1}{12} + \frac{1}{144} \right) = 2 \cdot \frac{28}{144} = \frac{7}{18},$$

so the rule returns  $\frac{7}{36}$ , whereas  $\int_0^1 t^4 \, dt = \frac{1}{5}$ . The rule is exact to degree  $3 = 2k - 1$  and no further.

### 10.10.7.3 C. Going deeper

#### Exercise C1: Interlacing roots.

Let  $w$  be a weight on  $(a, b)$  and let  $(p_k)$  be its monic orthogonal polynomial sequence. The roots of  $p_k$  are said to **interlace** those of  $p_{k+1}$  when exactly one root of  $p_k$  lies strictly between each pair of consecutive roots of  $p_{k+1}$ . This is true for every  $k$ ; prove the first two cases by hand.

- (a) Let  $r$  be the root of  $p_1$  and  $s_1 < s_2$  the roots of  $p_2$ . Prove that  $s_1 < r < s_2$ .
- (b) Let  $u_1 < u_2 < u_3$  be the roots of  $p_3$ . Prove that  $u_1 < s_1 < u_2 < s_2 < u_3$ .

*Hint: evaluate the three-term recurrence at a root.*

*Solution.*

- (a) By [Theorem 10.10.3](#) with  $k = 1$ ,  $p_2 = (x - a_1)p_1 - b_1p_0 = (x - a_1)(x - r) - b_1$ . Evaluating at  $r$  gives  $p_2(r) = -b_1 < 0$ , since  $b_1 > 0$ . By [Theorem 10.10.5](#),  $p_2 = (x - s_1)(x - s_2)$  with  $s_1 < s_2$  real, and this product is negative exactly on  $(s_1, s_2)$ . Hence  $s_1 < r < s_2$ .

- (b) By [Theorem 10.10.3](#) with  $k = 2$ ,  $p_3 = (x - a_2)p_2 - b_2p_1$ . At a root of  $p_2$  the first term dies, so

$$p_3(s_i) = -b_2 p_1(s_i) = -b_2(s_i - r) \quad (i = 1, 2).$$

By (a),  $s_1 - r < 0$  and  $s_2 - r > 0$ , and  $b_2 > 0$ , so  $p_3(s_1) > 0$  and  $p_3(s_2) < 0$ .

By [Theorem 10.10.5](#),  $p_3 = (x - u_1)(x - u_2)(x - u_3)$  with  $u_1 < u_2 < u_3$ . Such a product is negative on  $(-\infty, u_1)$ , positive on  $(u_1, u_2)$ , negative on  $(u_2, u_3)$ , and positive on  $(u_3, \infty)$ . So  $p_3(s_1) > 0$  puts  $s_1$  in  $(u_1, u_2)$  or in  $(u_3, \infty)$ , and  $p_3(s_2) < 0$  puts  $s_2$  in  $(-\infty, u_1)$  or in  $(u_2, u_3)$ . The option  $s_1 > u_3$  is impossible: it would give  $s_2 > s_1 > u_3$ , where  $p_3$  is positive, contradicting  $p_3(s_2) < 0$ . The option  $s_2 < u_1$  is impossible too: it would give  $s_1 < s_2 < u_1$ , where  $p_3$  is negative, contradicting  $p_3(s_1) > 0$ . Hence  $u_1 < s_1 < u_2$  and  $u_2 < s_2 < u_3$ , which is the assertion.

The general case is not proved this way. The coefficients  $a_k$  and  $b_k$  assemble into a symmetric tridiagonal matrix whose characteristic polynomial is  $p_k$ , and the interlacing then follows from the interlacing theorem for the eigenvalues of a symmetric matrix and its leading submatrices, which is Chapter 16's business.

### Exercise C2: The ceiling is $2k - 1$ .

Let  $w$  be a weight on  $(a, b)$  and let  $k \geq 1$ .

- (a) Let  $t_1, \dots, t_k$  be **any**  $k$  distinct real numbers and  $\alpha_1, \dots, \alpha_k$  **any** real numbers. Prove that there is an  $f \in \mathbb{R}[x]_{\leq 2k}$  with  $\int_a^b f w \neq \sum_i \alpha_i f(t_i)$ .
- (b) Hence deduce that the Gaussian rule of [Theorem 10.10.6](#) is exact on  $\mathbb{R}[x]_{\leq 2k-1}$  and on no larger space  $\mathbb{R}[x]_{\leq d}$ .

*Solution.*

- (a) Put  $g = (x - t_1) \cdots (x - t_k)$  and  $f = g^2$ , a polynomial of degree  $2k$ . Every node is a root of  $g$ , so  $f(t_i) = 0$  for every  $i$  and the right-hand side is 0, whatever the  $\alpha_i$  are. The left-hand side is  $\int_a^b g^2 w = \langle g, g \rangle$ , and  $g \neq 0$ , so  $\langle g, g \rangle > 0$  by positive definiteness. The two sides differ.
- (b) Exactness on  $\mathbb{R}[x]_{\leq 2k-1}$  is [Theorem 10.10.6](#). Suppose the rule were exact on  $\mathbb{R}[x]_{\leq d}$  for some  $d \geq 2k$ . Since  $\mathbb{R}[x]_{\leq 2k} \subseteq \mathbb{R}[x]_{\leq d}$ , it would then be exact on every polynomial of degree at most  $2k$ . But its nodes are  $k$  distinct real numbers ([Theorem 10.10.5](#)), so (a) supplies a polynomial of degree  $2k$  on which it is not. Hence no such  $d$  exists, and  $2k - 1$  is the largest degree for which the rule is exact. The concrete instance  $k = 2$ ,  $f = x^4$  is the failure computed in [Example \(Two-point](#)

Gauss–Legendre).

# Chapter 11

## Spectral Theory

Take the unit circle of  $\mathbb{R}^2$  and push it through a linear map. What comes out is an ellipse. The ellipse has two axes, at right angles to each other, and it is natural to ask whether those axes sit along directions that the map merely stretches — whether, in other words, the map's eigenvectors are perpendicular. Usually they are not. Chapter 8 found eigenvectors and asked only that they be independent; a basis of eigenvectors can be as skewed as it likes, and for a matrix like  $\begin{pmatrix} 1 & 1 \\ 0 & 2 \end{pmatrix}$  it is. Perpendicular eigenvectors are a strictly stronger demand, and it is the demand that matters in practice: coordinates in an orthonormal basis are read off by one inner product each, the change of basis is an isometry, and nothing is distorted along the way.

This chapter answers the question completely. There is an orthonormal basis of eigenvectors exactly when the operator commutes with its adjoint over  $\mathbb{C}$ , and exactly when it equals its adjoint over  $\mathbb{R}$ . Those two statements are the **spectral theorems**, and they are the most used theorems in applied linear algebra. Everything else here grows out of them: the triangular form that every complex matrix has in an orthonormal basis (Schur), the shape of a rotation in three dimensions, several commuting operators diagonalized at once, functions of an operator computed on its spectrum, the discrete Fourier transform as one instance of the whole picture, and finally a number that measures how far a matrix is from being normal. **The field is  $\mathbb{R}$  or  $\mathbb{C}$  throughout**, as in Chapter 10, and every statement says which one it needs. The difference between the two is not cosmetic: the Fundamental Theorem of Algebra hands  $\mathbb{C}$  an eigenvalue for free, and over  $\mathbb{R}$  each eigenvalue has to be earned.

**What you need.** From Chapter 5, the Fundamental Theorem of Algebra and the splitting it gives ([Theorem 5.5.1](#), [Corollary 5.5.5](#)), together with Lagrange interpolation ([Theorem 5.6.1](#)). From Chapter 8, eigenvalues and eigenspaces, the characteristic polynomial and multiplicities, the existence of an eigenvalue over  $\mathbb{C}$  ([Definition 8.2.1](#), [Theorem 8.2.1](#), [Theorem 8.2.4](#)), triangularization and what its diagonal records ([Theorem 8.7.2](#), [Corollary 8.7.3](#), [Theorem 8.7.4](#), [Theorem 8.3.3](#)), the conjugate-pair behavior of real matrices ([Theorem 8.2.5](#)), and the two results on commuting operators ([Theorem 8.9.3](#), [Theorem 8.9.5](#)). From Chapter 1, the complexification of a real vector space ([Definition 1.9.2](#), [Theorem 1.9.3](#)). From Chapter 9, the Jordan form and the rotation-scaling block ([Theorem 9.3.1](#), [Definition 9.4.1](#)). From Chapter 7, the Kronecker product and the vec operator ([Definition 7.4.1](#), [Theorem 7.4.1](#), [Definition 7.4.2](#), [Theorem 7.4.4](#)). Above all, from Chapter 10: orthonormal bases and Gram–Schmidt ([Theorem 10.2.2](#), [Corollary 10.2.3](#), [Theorem 10.2.1](#)), orthogonal complements and projections ([Theorem 10.3.2](#), [Definition 10.3.2](#), [Theorem 10.3.3](#)), the adjoint with its arithmetic and its matrix ([Definition 10.6.1](#), [Theorem 10.6.2](#), [Theorem 10.6.3](#), [Theorem 10.6.5](#)), the two classes of operators it singles out

(Definition 10.6.2, Definition 10.6.3, Proposition 10.6.7), and isometries (Definition 10.7.1, Theorem 10.7.3, Definition 10.7.2, Definition 10.7.3).

## 11.0.1 Roadmap

- **Schur's unitary triangularization.** Every complex matrix is upper triangular in some orthonormal basis; the workhorse the rest of the chapter runs on, and the real block version alongside it.
- **Self-adjoint operators.** Real eigenvalues over both fields, orthogonal eigenspaces, invariant complements, and the test  $\langle T\mathbf{v}, \mathbf{v} \rangle = 0$  that forces  $T = 0$ .
- **Normal operators.** The weakest hypothesis that works over  $\mathbb{C}$ : four equivalent descriptions,  $\ker T = \ker T^*$ , and why a normal operator has no Jordan blocks.
- **The complex spectral theorem.** Normal is exactly the same as having an orthonormal eigenbasis; Schur plus one entry-by-entry computation.
- **The real spectral theorem.** Self-adjoint is exactly the same as having an orthonormal eigenbasis over  $\mathbb{R}$ ; the principal axis theorem, and the rotation that shows why "normal" is not enough here.
- **The shape of an orthogonal matrix.** Blocks  $[1]$ ,  $[-1]$  and plane rotations; every element of  $\mathrm{SO}(3)$  is a turn about an axis, and reflections generate.
- **Commuting families.** One orthonormal basis that diagonalizes every member of a commuting family of normal operators at once.
- **Spectral resolutions and functions of a normal operator.** Writing  $T = \sum \lambda_i P_i$  with orthogonal projections, and defining  $f(T)$  by applying  $f$  to the spectrum.
- **Circulants and the discrete Fourier transform.** One unitary matrix diagonalizes every circulant, and multiplication becomes convolution.
- **The Sylvester equation and block diagonalization.** When  $\mathbf{X} \mapsto \mathbf{A}\mathbf{X} - \mathbf{X}\mathbf{B}$  is invertible, and what that buys for block triangular matrices.
- **How far from normal?** Schur's inequality turns normality from a yes-or-no property into a measurement.

## 11.0.2 Named moves

- **Take one eigenvector, then pass to its orthogonal complement.** As soon as the operator is normal or self-adjoint, that complement is invariant, so the induction on dimension runs inside the space instead of in a quotient. This replaces Chapter 8's "extend a basis and pass to the quotient".
- **Compute  $\langle T\mathbf{v}, \mathbf{v} \rangle$  two ways.** Once with the eigenvalue equation, once by moving  $T$  across to  $T^*$ . Comparing the two is almost the whole of the elementary theory.
- **Work with  $\|T\mathbf{v}\|^2 = \langle T^*T\mathbf{v}, \mathbf{v} \rangle$ .** A statement about lengths becomes a statement about the single operator  $T^*T$ , and a difference of two such operators is tested by the zero test.
- **Triangular plus normal forces diagonal.** Compare one diagonal entry of  $\mathbf{A}^*\mathbf{A}$  with the same entry of  $\mathbf{A}\mathbf{A}^*$ , and a row of off-diagonal entries has to vanish.
- **Split into real and imaginary parts.** Write  $T = A + iB$  with  $A$  and  $B$  self-adjoint; then  $T$  is normal exactly when  $A$  and  $B$  commute, and questions about  $T$  become questions about a commuting pair.
- **Diagonalize, then read the answer off the diagonal.** Powers, square roots, exponentials and the whole functional calculus are one line once an orthonormal eigenbasis is in hand.

## 11.1 Schur Triangularization

Chapter 8 proved that every operator on a non-zero complex vector space has an upper triangular matrix in **some** basis (Corollary 8.7.3). That basis was built by extending an eigenvector and passing to a quotient, and it came out as skewed as the operator happened to be. Now we have lengths and angles, so we can be fussier and ask for an **orthonormal** basis. The remarkable thing is that the answer is still yes, at no extra cost in hypotheses, and the proof is the same induction run inside an orthogonal complement rather than in a quotient. This section proves it, reads off the first consequences, and gives the real version, where a conjugate pair of eigenvalues costs one  $2 \times 2$  block.

Throughout,  $V$  is a finite-dimensional inner product space over  $F = \mathbb{R}$  or  $F = \mathbb{C}$ , and every matrix statement uses the standard inner product on  $F^n$ .

### 11.1.1 Why the obvious route fails

Here is the natural first attempt, and it is worth watching it fail, because the repair is the whole idea of the proof.

We want an orthonormal basis  $\mathcal{B} = (\mathbf{v}_1, \dots, \mathbf{v}_n)$  with  $[T]_{\mathcal{B}}^{\mathcal{B}}$  upper triangular, that is, with

$$T\mathbf{v}_j \in \text{span}(\mathbf{v}_1, \dots, \mathbf{v}_j) \quad \text{for every } j.$$

For  $j = 1$  this says  $\mathbf{v}_1$  is an eigenvector of  $T$ , and over  $\mathbb{C}$  one exists (Theorem 8.2.4). Normalize it, set  $U = (\text{span}(\mathbf{v}_1))^{\perp}$ , and try to finish by induction on  $U$ . The attempt stops at once: to speak of  $T$  restricted to  $U$  we need  $U$  to be  $T$ -invariant, and it need not be.

#### Example. (An Eigenline Whose Complement Is Not Invariant)

Let  $\mathbf{A} = \begin{pmatrix} 1 & 1 \\ 0 & 2 \end{pmatrix}$  act on  $\mathbb{C}^2$ . Check that  $\mathbf{e}_1$  is an eigenvector and decide whether  $(\text{span}(\mathbf{e}_1))^{\perp}$  is invariant.

*Solution.*  $\mathbf{A}\mathbf{e}_1 = (1, 0) = \mathbf{e}_1$ , so  $\mathbf{e}_1$  is an eigenvector for the eigenvalue 1. The orthogonal complement of  $\text{span}(\mathbf{e}_1)$  is  $\text{span}(\mathbf{e}_2)$ , and  $\mathbf{A}\mathbf{e}_2 = (1, 2)$ , which is not a multiple of  $\mathbf{e}_2$ . So  $(\text{span}(\mathbf{e}_1))^{\perp}$  is **not** invariant, and the induction has nothing to run on.

So the first basis vector is the wrong end to start from. Look at the target instead. Suppose  $\mathcal{B}$  is orthonormal and  $[T]_{\mathcal{B}}^{\mathcal{B}} = \mathbf{T}$  is upper triangular. By Theorem 10.6.3 the matrix of  $T^*$  in the same basis is  $\mathbf{T}^*$ , which is **lower** triangular; and the last column of a lower triangular matrix has only its bottom entry. Reading that column off,

$$T^*\mathbf{v}_n = \overline{t_{nn}} \mathbf{v}_n.$$

*In the basis we are trying to build, the last vector is an eigenvector of  $T^*$  — not of  $T$ .*

That is the vector to produce first, and it is exactly the one whose orthogonal complement behaves.

### Lemma 11.1.1 — Eigenvectors of the Adjoint Cut Out Invariant Subspaces

Let  $V$  be a finite-dimensional inner product space over  $F$ , let  $T \in \mathcal{L}(V)$ , and let  $W \subseteq V$  be a  $T^*$ -invariant subspace. Then  $W^\perp$  is  $T$ -invariant.

*Proof.* Let  $\mathbf{u} \in W^\perp$  and  $\mathbf{w} \in W$ . Since  $W$  is  $T^*$ -invariant,  $T^*\mathbf{w} \in W$ , so by Definition 10.6.1

$$\langle T\mathbf{u}, \mathbf{w} \rangle = \langle \mathbf{u}, T^*\mathbf{w} \rangle = 0.$$

As  $\mathbf{w} \in W$  was arbitrary,  $T\mathbf{u} \in W^\perp$ . Hence  $T(W^\perp) \subseteq W^\perp$ . ■

The special case we need is  $W = \text{span}(\mathbf{v})$  with  $T^*\mathbf{v} = \mu\mathbf{v}$ : one vector, one line, and an invariant complement of dimension  $n - 1$  to recurse into. And over  $\mathbb{C}$  such a  $\mathbf{v}$  always exists, because  $T^*$  is an operator on a non-zero complex space and the Fundamental Theorem of Algebra gives it an eigenvalue. That is where the field enters, and it is the only place it enters.

## 11.1.2 The theorem

### Theorem 11.1.2 — Schur's Unitary Triangularization

Let  $V$  be a finite-dimensional inner product space over  $\mathbb{C}$  with  $\dim V = n \geq 1$ , and let  $T \in \mathcal{L}(V)$ .

- (a) There is an **orthonormal** basis  $\mathcal{B}$  of  $V$  such that  $[T]_{\mathcal{B}}^{\mathcal{B}}$  is upper triangular.
- (b) More precisely, let  $\lambda_1, \dots, \lambda_n$  be the eigenvalues of  $T$ , each listed as often as its algebraic multiplicity, in **any** order we please. Then  $\mathcal{B}$  can be chosen so that the  $i$ -th diagonal entry of  $[T]_{\mathcal{B}}^{\mathcal{B}}$  is  $\lambda_i$ .

*Idea.* Induction on  $\dim V$ , building the basis from its **last** vector backwards. ① Pick the eigenvalue  $\lambda_n$  we want in the bottom-right corner; the conjugate  $\overline{\lambda_n}$  is then an eigenvalue of  $T^*$ , because starring turns a non-invertible operator into a non-invertible operator. ② Take a unit eigenvector  $\mathbf{e}$  of  $T^*$  for  $\overline{\lambda_n}$  and put  $U = (\text{span}(\mathbf{e}))^\perp$ ; by Lemma 11.1.1 it is  $T$ -invariant, so  $T|_U$  exists and lives on a space of dimension  $n - 1$ . ③ Check that the characteristic polynomial of  $T|_U$  is  $p_T$  with the factor  $x - \lambda_n$  removed, so the inductive hypothesis may be applied with the shortened list. ④ Append  $\mathbf{e}$  to the basis it returns.

*Proof.* We induct on  $n = \dim V$ . Since  $\mathbb{C}$  is algebraically closed,  $p_T$  splits (Corollary 5.5.5), so a list  $\lambda_1, \dots, \lambda_n$  as in (b) exists and  $p_T = (x - \lambda_1) \cdots (x - \lambda_n)$ ; proving (b) proves (a).

For  $n = 1$ , any unit vector  $\mathbf{v}_1$  is a basis, every  $1 \times 1$  matrix is upper triangular, and its single entry is the single eigenvalue  $\lambda_1$ .

Let  $n \geq 2$  and assume the statement for all inner product spaces of dimension  $n - 1$ . Fix a list  $\lambda_1, \dots, \lambda_n$  as in (b).

$\overline{\lambda_n}$  is an eigenvalue of  $T^*$ . Since  $\lambda_n$  is an eigenvalue of  $T$ , the operator  $T - \lambda_n \text{id}_V$  is not invertible (Theorem 8.2.1). By Theorem 10.6.2 (a), (b) and (d), its adjoint is  $T^* - \overline{\lambda_n} \text{id}_V$ , and the adjoint of that is  $T - \lambda_n \text{id}_V$  again. If  $T^* - \overline{\lambda_n} \text{id}_V$  were invertible, then Theorem 10.6.2 (e) would make its adjoint  $T - \lambda_n \text{id}_V$  invertible too. So  $T^* - \overline{\lambda_n} \text{id}_V$  is not invertible, and  $\overline{\lambda_n} \in \text{spec}(T^*)$ .

Choose an eigenvector of  $T^*$  for  $\overline{\lambda_n}$ ; it is non-zero, so dividing by its norm gives a unit vector  $\mathbf{e}$  with  $T^*\mathbf{e} = \overline{\lambda_n}\mathbf{e}$ . Put  $U = (\text{span}(\mathbf{e}))^\perp$ . By Theorem 10.3.2 (c),  $\dim U = n - 1 \geq 1$ , and  $U$  is  $T$ -invariant by Lemma 11.1.1. Write  $S = T|_U \in \mathcal{L}(U)$ , where  $U$  carries the inner product of  $V$ .

**The bottom-right entry is  $\lambda_n$ .** Using Definition 10.6.1 and conjugate-linearity in the second slot,

$$\langle T\mathbf{e}, \mathbf{e} \rangle = \langle \mathbf{e}, T^*\mathbf{e} \rangle = \langle \mathbf{e}, \overline{\lambda_n}\mathbf{e} \rangle = \lambda_n \|\mathbf{e}\|^2 = \lambda_n.$$

**The shortened list belongs to  $S$ .** Take any orthonormal basis  $(\mathbf{f}_1, \dots, \mathbf{f}_{n-1})$  of  $U$  (Theorem 10.2.2) and set  $\mathcal{C} = (\mathbf{f}_1, \dots, \mathbf{f}_{n-1}, \mathbf{e})$ , an orthonormal list of  $n$  vectors, hence an orthonormal basis of  $V$  by Theorem 10.1.4 and Theorem 1.6.4. Since  $U$  is  $T$ -invariant, Theorem 8.1.3 (b) applies with the basis  $\mathcal{C}$  and gives  $p_T = p_S p_{\bar{T}}$ , where  $\bar{T}$  is the induced operator on the one-dimensional quotient  $V/U$ . Its matrix is the single bottom-right entry of  $[T]_{\mathcal{C}}^{\mathcal{C}}$ , which is  $\langle T\mathbf{e}, \mathbf{e} \rangle = \lambda_n$  by Theorem 10.2.1; so  $p_{\bar{T}} = x - \lambda_n$ . Canceling that factor in  $\mathbb{C}[x]$ ,

$$p_S = (x - \lambda_1) \cdots (x - \lambda_{n-1}).$$

So  $\lambda_1, \dots, \lambda_{n-1}$  is a list of the eigenvalues of  $S$  with their algebraic multiplicities.

**Induct.** By the inductive hypothesis there is an orthonormal basis  $(\mathbf{v}_1, \dots, \mathbf{v}_{n-1})$  of  $U$  in which the matrix of  $S$  is upper triangular with diagonal  $\lambda_1, \dots, \lambda_{n-1}$ . Put  $\mathcal{B} = (\mathbf{v}_1, \dots, \mathbf{v}_{n-1}, \mathbf{e})$ , again an orthonormal basis of  $V$  for the reason given above. For  $j \leq n - 1$  we have  $T\mathbf{v}_j = S\mathbf{v}_j \in \text{span}(\mathbf{v}_1, \dots, \mathbf{v}_j)$ , so the  $j$ -th column of  $[T]_{\mathcal{B}}^{\mathcal{B}}$  vanishes below row  $j$  and has  $\lambda_j$  in row  $j$ . The  $n$ -th column has nothing below row  $n$ , and its entry there is  $\langle T\mathbf{e}, \mathbf{e} \rangle = \lambda_n$ . Hence  $[T]_{\mathcal{B}}^{\mathcal{B}}$  is upper triangular with diagonal  $\lambda_1, \dots, \lambda_n$ . This proves the theorem. ■

Two hypotheses did all the work, and it is worth naming which. **Complex** produced the eigenvalue of  $T^*$ , through the Fundamental Theorem of Algebra; over  $\mathbb{R}$  the theorem is false as stated, and the last part of this section says what survives. **Inner product** produced the invariant complement, through Lemma 11.1.1; without one we would be back in Chapter 8's quotient. Notice also what the theorem does **not** claim: nothing was assumed about  $T$ , and nothing beyond triangularity is promised. Diagonal is a different and much stronger request, and Sections 4 and 5 say exactly which operators grant it.

### 11.1.3 The matrix form

Translating through an orthonormal basis turns the theorem into a factorization. Recall that a square complex matrix is **unitary** when  $\mathbf{U}^*\mathbf{U} = \mathbf{I}$  (Definition 10.7.2), and call two matrices **unitarily similar** when  $\mathbf{B} = \mathbf{U}^*\mathbf{A}\mathbf{U}$  for some unitary  $\mathbf{U}$ ; this is similarity by a matrix that is also an isometry, so it preserves the inner product as well as the operator.



### Corollary 11.1.3 — Schur Factorization

Let  $\mathbf{A} \in M_n(\mathbb{C})$  with  $n \geq 1$ , and let  $\lambda_1, \dots, \lambda_n$  be the eigenvalues of  $\mathbf{A}$ , each listed as often as its algebraic multiplicity, in any prescribed order. Then there are a unitary  $\mathbf{U} \in M_n(\mathbb{C})$  and an upper triangular  $\mathbf{T} \in M_n(\mathbb{C})$  with

$$\mathbf{A} = \mathbf{U}\mathbf{T}\mathbf{U}^*, \quad t_{ii} = \lambda_i \quad (1 \leq i \leq n).$$

*Proof.* Apply [Theorem 11.1.2](#) to  $T_{\mathbf{A}}$  on  $\mathbb{C}^n$  with the standard inner product and the given list: there is an orthonormal basis  $\mathcal{B} = (\mathbf{u}_1, \dots, \mathbf{u}_n)$  of  $\mathbb{C}^n$  with  $\mathbf{T} := [T_{\mathbf{A}}]_{\mathcal{B}}^{\mathcal{B}}$  upper triangular and  $t_{ii} = \lambda_i$ . Let  $\mathbf{U}$  be the matrix whose  $j$ -th column is  $\mathbf{u}_j$ . Its columns are orthonormal, so  $\mathbf{U}$  is unitary ([Theorem 10.7.3](#) (f), [Definition 10.7.2](#)), and  $\mathbf{U} = [\text{id}]_{\mathcal{B}}^{\mathcal{E}}$  for the standard basis  $\mathcal{E}$  ([Definition 3.7.1](#)). Since  $[T_{\mathbf{A}}]_{\mathcal{E}}^{\mathcal{E}} = \mathbf{A}$ , [Theorem 3.7.3](#) gives  $\mathbf{T} = \mathbf{U}^{-1}\mathbf{A}\mathbf{U} = \mathbf{U}^*\mathbf{A}\mathbf{U}$ , the last step because  $\mathbf{U}^{-1} = \mathbf{U}^*$  for a unitary matrix. Rearranging gives  $\mathbf{A} = \mathbf{U}\mathbf{T}\mathbf{U}^*$ . ■

The order of the diagonal is free because at each step of the induction we chose which eigenvalue to send to the bottom-right corner, and any of them was available. That freedom is used constantly: it lets us put a distinguished eigenvalue first, or collect equal eigenvalues into a block.

*Remark. How to compute one.* The induction above runs from the bottom-right corner, because that is the corner where an operator on a subspace is available. For a **matrix** there is a second route, running from the top-left, and it is the one to use by hand. Let  $\mathbf{q}_1$  be a unit eigenvector of  $\mathbf{A}$  for an eigenvalue  $\lambda$ , extend it to an orthonormal basis of  $\mathbb{C}^n$  ([Corollary 10.2.3](#)) and let  $\mathbf{Q}$  be the unitary matrix with those columns. Then  $\mathbf{Q}^*\mathbf{A}\mathbf{q}_1 = \lambda\mathbf{Q}^*\mathbf{q}_1 = \lambda\mathbf{e}_1$ , so

$$\mathbf{Q}^*\mathbf{A}\mathbf{Q} = \begin{pmatrix} \lambda & \mathbf{b}^* \\ \mathbf{0} & \mathbf{A}_1 \end{pmatrix}$$

for some  $\mathbf{b} \in \mathbb{C}^{n-1}$  and  $\mathbf{A}_1 \in M_{n-1}(\mathbb{C})$ . Triangularize  $\mathbf{A}_1 = \mathbf{V}\mathbf{T}_1\mathbf{V}^*$  and put  $\mathbf{U} = \mathbf{Q}(1 \oplus \mathbf{V})$ ; the product is unitary and  $\mathbf{U}^*\mathbf{A}\mathbf{U}$  is upper triangular. This is legitimate for matrices precisely because we may replace  $\mathbf{A}$  by the compressed block  $\mathbf{A}_1$ , which is not the restriction of any operator:  $(\text{span}(\mathbf{q}_1))^\perp$  is not invariant, as [Example \(An Eigenline Whose Complement Is Not Invariant\)](#) showed.

#### Example. (A Two-by-Two Schur Factorization)

Find a unitary  $\mathbf{U}$  and an upper triangular  $\mathbf{T}$  with  $\mathbf{U}^*\mathbf{A}\mathbf{U} = \mathbf{T}$ , as in [Corollary 11.1.3](#), for

$$\mathbf{A} = \begin{pmatrix} 2 & -1 \\ 1 & 4 \end{pmatrix}.$$

*Solution.* Here  $\text{tr } \mathbf{A} = 6$  and  $\det \mathbf{A} = 8 + 1 = 9$ , so  $p_{\mathbf{A}} = x^2 - 6x + 9 = (x - 3)^2$  and the only eigenvalue is 3, with algebraic multiplicity 2.

Solving  $(\mathbf{A} - 3\mathbf{I})\mathbf{x} = \mathbf{0}$  with  $\mathbf{A} - 3\mathbf{I} = \begin{pmatrix} -1 & -1 \\ 1 & 1 \end{pmatrix}$  gives the single eigenline  $\text{span}((1, -1))$ .

Take  $\mathbf{q}_1 = \frac{1}{\sqrt{2}}(1, -1)$  and complete with  $\mathbf{q}_2 = \frac{1}{\sqrt{2}}(1, 1)$ , a unit vector orthogonal to  $\mathbf{q}_1$ .

With  $\mathbf{U}$  the matrix of those columns,

$$\mathbf{U} = \frac{1}{\sqrt{2}} \begin{pmatrix} 1 & 1 \\ -1 & 1 \end{pmatrix}, \quad \mathbf{U}^* \mathbf{A} \mathbf{U} = \begin{pmatrix} 3 & -2 \\ 0 & 3 \end{pmatrix}.$$

Both diagonal entries are 3, as they must be. The off-diagonal  $-2$  cannot be removed: a matrix unitarily similar to  $\text{diag}(3, 3) = 3\mathbf{I}$  would equal  $\mathbf{U}(3\mathbf{I})\mathbf{U}^* = 3\mathbf{I}$ , and  $\mathbf{A} \neq 3\mathbf{I}$ .

**Example. (A Three-by-Three Schur Factorization)**

Find a unitary  $\mathbf{U}$  and an upper triangular  $\mathbf{T}$  with  $\mathbf{U}^* \mathbf{A} \mathbf{U} = \mathbf{T}$ , as in [Corollary 11.1.3](#), for

$$\mathbf{A} = \begin{pmatrix} 1 & 0 & 1 \\ 0 & 1 & -2 \\ 4 & -3 & -2 \end{pmatrix}.$$

*Solution. Step 1: peel off one eigenvector.* Expanding  $\det(x\mathbf{I} - \mathbf{A})$  gives  $p_{\mathbf{A}} = (x - 1)(x - 3)(x + 4)$ , so 1 is an eigenvalue. Solving  $(\mathbf{A} - \mathbf{I})\mathbf{x} = \mathbf{0}$  by row reduction gives the eigenline spanned by  $(3, 4, 0)$ , of norm 5. Put

$$\mathbf{q}_1 = \frac{1}{5}(3, 4, 0), \quad \mathbf{q}_2 = \frac{1}{5}(4, -3, 0), \quad \mathbf{q}_3 = (0, 0, 1),$$

an orthonormal basis of  $\mathbb{C}^3$  (each has norm 1 and the three pairwise inner products are 0; the entries are real, so no conjugates intrude). With  $\mathbf{Q} = (\mathbf{q}_1 \quad \mathbf{q}_2 \quad \mathbf{q}_3)$ ,

$$\mathbf{Q}^* \mathbf{A} \mathbf{Q} = \begin{pmatrix} 1 & 0 & -1 \\ 0 & 1 & 2 \\ 0 & 5 & -2 \end{pmatrix}, \quad \mathbf{A}_1 = \begin{pmatrix} 1 & 2 \\ 5 & -2 \end{pmatrix}.$$

*Step 2: triangularize the block.* Here  $\text{tr } \mathbf{A}_1 = -1$  and  $\det \mathbf{A}_1 = -12$ , so  $p_{\mathbf{A}_1} = x^2 + x - 12 = (x - 3)(x + 4)$ . For the eigenvalue 3,  $\mathbf{A}_1 - 3\mathbf{I} = \begin{pmatrix} -2 & 2 \\ 5 & -5 \end{pmatrix}$  gives the eigenvector  $(1, 1)$ . Take  $\mathbf{p}_1 = \frac{1}{\sqrt{2}}(1, 1)$  and  $\mathbf{p}_2 = \frac{1}{\sqrt{2}}(1, -1)$ , and let  $\mathbf{V}$  be the matrix of those columns. Then

$$\mathbf{V}^* \mathbf{A}_1 \mathbf{V} = \begin{pmatrix} 3 & 3 \\ 0 & -4 \end{pmatrix}.$$

*Step 3: assemble.* Put  $\mathbf{U} = \mathbf{Q}(1 \oplus \mathbf{V})$ , whose columns are  $\mathbf{q}_1$ ,  $(\mathbf{q}_2 + \mathbf{q}_3)/\sqrt{2}$  and  $(\mathbf{q}_2 - \mathbf{q}_3)/\sqrt{2}$ :

$$\mathbf{U} = \frac{1}{5\sqrt{2}} \begin{pmatrix} 3\sqrt{2} & 4 & 4 \\ 4\sqrt{2} & -3 & -3 \\ 0 & 5 & -5 \end{pmatrix}, \quad \mathbf{U}^* \mathbf{A} \mathbf{U} = \begin{pmatrix} 1 & -\frac{1}{\sqrt{2}} & \frac{1}{\sqrt{2}} \\ 0 & 3 & 3 \\ 0 & 0 & -4 \end{pmatrix}.$$

The diagonal reads 1, 3,  $-4$ , and indeed  $p_{\mathbf{A}} = (x - 1)(x - 3)(x + 4)$ , with trace  $0 = 1 + 3 - 4$  and determinant  $-12 = 1 \cdot 3 \cdot (-4)$ .

*Quick check.* Why does [Theorem 11.1.2](#) not prove that every complex matrix is unitarily similar to a **diagonal** matrix?

*Solution.* Because the proof controls only the entries below the diagonal. The entries above it are whatever they happen to be, and [Example \(A Two-by-Two Schur Factorization\)](#) shows they need not vanish: the matrix there has a  $-2$  in the corner, and no unitary can remove it. A matrix unitarily similar to a diagonal one satisfies strong conditions — Section 4 identifies them exactly — and most matrices do not.

### 11.1.4 Reading the diagonal

A triangular form with an explicit factorization makes two Chapter 8 facts fall out in a line each.

#### Corollary 11.1.4 — Trace Is the Sum of the Eigenvalues

Let  $\mathbf{A} \in M_n(\mathbb{C})$  have eigenvalues  $\lambda_1, \dots, \lambda_n$ , listed with algebraic multiplicity. Then  $\operatorname{tr} \mathbf{A} = \lambda_1 + \dots + \lambda_n$ .

*Proof.* Write  $\mathbf{A} = \mathbf{U}\mathbf{T}\mathbf{U}^*$  as in [Corollary 11.1.3](#). By [Theorem 0.9.6](#) (3) applied to the pair  $\mathbf{U}$  and  $\mathbf{T}\mathbf{U}^*$ ,

$$\operatorname{tr} \mathbf{A} = \operatorname{tr}(\mathbf{U}(\mathbf{T}\mathbf{U}^*)) = \operatorname{tr}((\mathbf{T}\mathbf{U}^*)\mathbf{U}) = \operatorname{tr} \mathbf{T},$$

since  $\mathbf{U}^*\mathbf{U} = \mathbf{I}$ . The diagonal of  $\mathbf{T}$  is the list  $\lambda_1, \dots, \lambda_n$ . ■

#### Corollary 11.1.5 — Determinant Is the Product of the Eigenvalues

Let  $\mathbf{A} \in M_n(\mathbb{C})$  have eigenvalues  $\lambda_1, \dots, \lambda_n$ , listed with algebraic multiplicity. Then  $\det \mathbf{A} = \lambda_1 \cdots \lambda_n$ .

*Proof.* With  $\mathbf{A} = \mathbf{U}\mathbf{T}\mathbf{U}^*$  as above, [Theorem 6.5.4](#) gives  $\det \mathbf{A} = \det \mathbf{U} \det \mathbf{T} \det \mathbf{U}^*$ , and  $\det \mathbf{U} \det \mathbf{U}^* = \det(\mathbf{U}\mathbf{U}^*) = \det \mathbf{I} = 1$ . So  $\det \mathbf{A} = \det \mathbf{T}$ , which is the product of the diagonal entries by [Theorem 6.4.6](#). ■

Both statements are [Theorem 8.3.3](#) over  $\mathbb{C}$ , recovered from the diagonal of  $\mathbf{T}$  instead of from the coefficients of  $p_{\mathbf{A}}$ . Nothing is new; what is new is how short the argument became once a triangular form was available in hand.

The next consequence is not a restatement of anything. It says that the matrices we know how to handle sit arbitrarily close to the ones we do not.

#### Corollary 11.1.6 — Diagonalizable Matrices Are Everywhere

Let  $\mathbf{A} \in M_n(\mathbb{C})$  and let  $\varepsilon > 0$ . Then there is a diagonalizable  $\mathbf{A}' \in M_n(\mathbb{C})$  with

$$|a_{ij} - a'_{ij}| < \varepsilon \quad \text{for all } i, j.$$

In this sense every complex square matrix is a limit of diagonalizable ones.

*Idea.* Diagonalizability is hard to arrange directly, but  $n$  distinct eigenvalues force it (Corollary 8.4.2), and in a triangular form the eigenvalues are sitting on the diagonal where we can nudge them apart. The only thing to check is that a small nudge inside the triangular form is still a small nudge after conjugating back by  $\mathbf{U}$ , and that is Cauchy–Schwarz on the rows of a unitary matrix.

*Proof.* Write  $\mathbf{A} = \mathbf{U}\mathbf{T}\mathbf{U}^*$  as in Corollary 11.1.3. Choose  $\delta_1, \dots, \delta_n \in \mathbb{C}$  with  $|\delta_k| < \varepsilon$  and with  $t_{11} + \delta_1, \dots, t_{nn} + \delta_n$  pairwise distinct. This is possible one index at a time: having chosen  $\delta_1, \dots, \delta_{k-1}$ , the forbidden values of  $\delta_k$  are the  $k-1$  numbers  $t_{jj} + \delta_j - t_{kk}$ , and the disc  $\{|z| < \varepsilon\}$  has infinitely many points.

Put  $\mathbf{D} = \text{diag}(\delta_1, \dots, \delta_n)$  and  $\mathbf{A}' = \mathbf{U}(\mathbf{T} + \mathbf{D})\mathbf{U}^*$ . The matrix  $\mathbf{T} + \mathbf{D}$  is upper triangular with  $n$  distinct diagonal entries, so by Theorem 8.7.4 (b) it has  $n$  distinct eigenvalues;  $\mathbf{A}'$  is similar to it, hence has the same eigenvalues, and is diagonalizable by Corollary 8.4.2.

For the entries,  $\mathbf{A} - \mathbf{A}' = -\mathbf{U}\mathbf{D}\mathbf{U}^*$ , whose  $(i, j)$  entry is  $-\sum_k u_{ik}\delta_k\overline{u_{jk}}$ . Hence

$$|a_{ij} - a'_{ij}| \leq \max_k |\delta_k| \sum_{k=1}^n |u_{ik}| |u_{jk}| \leq \max_k |\delta_k| < \varepsilon,$$

where the middle inequality is Theorem 10.1.5 applied to the vectors of moduli, together with the fact that the rows of  $\mathbf{U}$  are unit vectors, which is  $\mathbf{U}\mathbf{U}^* = \mathbf{I}$ . This proves the corollary. ■

This is the tool behind a standard strategy: *prove it for diagonalizable matrices, then take a limit*. Chapter 15, where matrices acquire norms and limits can be spoken of properly, runs arguments of exactly this shape; and Exercise (C1: Cayley–Hamilton from the triangular factors) uses the triangular form directly, with no limits at all, to reprove the Cayley–Hamilton theorem over  $\mathbb{C}$ .

## 11.1.5 The real version

Over  $\mathbb{R}$  the theorem fails, and it fails for the reason it always does: a rotation has no real eigenvector, so no real basis makes its matrix triangular (Theorem 8.7.2). The most one can hope for is to let a conjugate pair occupy a  $2 \times 2$  block, and that much is true.

### Theorem 11.1.7 — Real Schur Form

Let  $\mathbf{A} \in M_n(\mathbb{R})$  with  $n \geq 1$ . Then there are an orthogonal  $\mathbf{Q} \in O(n)$  and a matrix  $\mathbf{T} \in M_n(\mathbb{R})$  with

$$\mathbf{A} = \mathbf{Q}\mathbf{T}\mathbf{Q}^\top,$$

where  $\mathbf{T}$  is **block** upper triangular with diagonal blocks of size  $1 \times 1$  and  $2 \times 2$ , and every  $2 \times 2$  diagonal block has a conjugate pair of non-real eigenvalues.

*Idea.* The same induction, with one change. The eigenvector of  $T^*$  may fail to exist over  $\mathbb{R}$ , but an invariant subspace of dimension 1 or 2 always does — that is Exercise

(C2: Real operators have small invariant subspaces), proved in Chapter 8 from a complex eigenvector  $\mathbf{x} + i\mathbf{y}$  by taking  $\text{span}(\mathbf{x}, \mathbf{y})$ . Apply it to  $T^*$ , take the orthogonal complement, and induct. Dimension 1 occurs exactly when  $T^*$  has a real eigenvalue, and that is what keeps the  $2 \times 2$  blocks honest.

*Proof.* Let  $T = T_{\mathbf{A}}$  on  $\mathbb{R}^n$  with the standard inner product, so that  $T^* = T_{\mathbf{A}^\top}$  by [Theorem 10.6.3](#). We induct on  $n$ .

Choose a  $T^*$ -invariant subspace  $W$  as follows. If  $T^*$  has a real eigenvalue, let  $W$  be a corresponding eigenline, so  $d := \dim W = 1$ . If it does not, take a complex eigenvalue  $\mu = a + bi$  of  $\mathbf{A}^\top$  with  $b \neq 0$ , which exists by [Theorem 8.2.4](#) applied to  $\mathbf{A}^\top$  over  $\mathbb{C}$ , and a complex eigenvector  $\mathbf{x} + i\mathbf{y}$  with  $\mathbf{x}, \mathbf{y} \in \mathbb{R}^n$ . Comparing real and imaginary parts of  $\mathbf{A}^\top(\mathbf{x} + i\mathbf{y}) = \mu(\mathbf{x} + i\mathbf{y})$  gives  $\mathbf{A}^\top \mathbf{x} = a\mathbf{x} - b\mathbf{y}$  and  $\mathbf{A}^\top \mathbf{y} = b\mathbf{x} + a\mathbf{y}$ , so  $W := \text{span}(\mathbf{x}, \mathbf{y})$  is  $T^*$ -invariant. It has dimension 2: were  $\mathbf{x}, \mathbf{y}$  dependent,  $W$  would be a  $T^*$ -invariant line and  $T^*$  would have a real eigenvalue, contrary to the case we are in. So  $d = 2$ . ([Exercise \(C2: Real operators have small invariant subspaces\)](#) runs the same argument in Chapter 8.)

Put  $U = W^\perp$ , which is  $T$ -invariant by [Lemma 11.1.1](#) and has dimension  $n - d$  ([Theorem 10.3.2](#) (c)). Pick an orthonormal basis  $(\mathbf{v}_1, \dots, \mathbf{v}_{n-d})$  of  $U$  and one  $(\mathbf{v}_{n-d+1}, \dots, \mathbf{v}_n)$  of  $W$  ([Theorem 10.2.2](#)); together they form an orthonormal basis  $\mathcal{B}$  of  $\mathbb{R}^n$ , since the two blocks are orthogonal to each other. Because  $T(U) \subseteq U$ ,

$$[T]_{\mathcal{B}}^{\mathcal{B}} = \begin{pmatrix} \mathbf{C} & * \\ \mathbf{0} & \mathbf{B} \end{pmatrix}$$

with  $\mathbf{C} \in M_{n-d}(\mathbb{R})$  the matrix of  $T|_U$  and  $\mathbf{B} \in M_d(\mathbb{R})$ . Writing  $\mathbf{Q}_{\mathcal{B}}$  for the matrix whose columns are the vectors of  $\mathcal{B}$ , which is orthogonal because those columns are orthonormal, [Theorem 3.7.3](#) gives  $\mathbf{Q}_{\mathcal{B}}^\top \mathbf{A} \mathbf{Q}_{\mathcal{B}} = [T]_{\mathcal{B}}^{\mathcal{B}}$ .

*The block  $\mathbf{B}$  is of the allowed kind.* If  $d = 1$  it is  $1 \times 1$ . If  $d = 2$ , then by [Theorem 10.6.3](#) the matrix of  $T^*$  in  $\mathcal{B}$  is the transpose of the display, whose bottom-right block is  $\mathbf{B}^\top$ ; as  $W$  is  $T^*$ -invariant,  $\mathbf{B}^\top$  is the matrix of  $T^*|_W$ . A real eigenvalue of  $\mathbf{B}^\top$  would be a real eigenvalue of  $T^*$ , which this case excludes. So  $p_{\mathbf{B}} = p_{\mathbf{B}^\top}$  has no real root, and its two complex roots form a conjugate pair by [Theorem 8.2.5](#).

*Close the induction.* If  $U = \{\mathbf{0}\}$  then  $n = d \leq 2$ , the list  $\mathcal{B}$  is a basis of  $W$ , and  $\mathbf{Q}_{\mathcal{B}}^\top \mathbf{A} \mathbf{Q}_{\mathcal{B}} = \mathbf{B}$  is already a single allowed block; take  $\mathbf{Q} = \mathbf{Q}_{\mathcal{B}}$ . Otherwise  $1 \leq n - d < n$ , and by the inductive hypothesis  $\mathbf{C} = \mathbf{Q}_1 \mathbf{T}_1 \mathbf{Q}_1^\top$  with  $\mathbf{Q}_1 \in O(n-d)$  and  $\mathbf{T}_1$  of the required shape. Put  $\mathbf{Q} = \mathbf{Q}_{\mathcal{B}}(\mathbf{Q}_1 \oplus \mathbf{I}_d)$ , a product of orthogonal matrices and so orthogonal ([Proposition 10.7.4](#) (a)). Then

$$\mathbf{Q}^\top \mathbf{A} \mathbf{Q} = (\mathbf{Q}_1 \oplus \mathbf{I}_d)^\top \begin{pmatrix} \mathbf{C} & * \\ \mathbf{0} & \mathbf{B} \end{pmatrix} (\mathbf{Q}_1 \oplus \mathbf{I}_d) = \begin{pmatrix} \mathbf{T}_1 & * \\ \mathbf{0} & \mathbf{B} \end{pmatrix},$$

which is block upper triangular with diagonal blocks those of  $\mathbf{T}_1$  followed by  $\mathbf{B}$ . This proves the theorem. ■

Two remarks on the shape. The  $1 \times 1$  blocks carry the real eigenvalues of  $\mathbf{A}$  and the  $2 \times 2$  blocks carry the non-real ones in conjugate pairs, so the block sizes are forced by the spectrum, even though the blocks themselves are not. And when all the eigenvalues of  $\mathbf{A}$  are real, every block

is  $1 \times 1$  and the real Schur form is an honest orthogonal triangularization.

*Warning.* **Neither  $\mathbf{T}$  nor  $\mathbf{U}$  is unique; only the diagonal multiset is.** For  $\mathbf{A} = \begin{pmatrix} 1 & 1 \\ 0 & 2 \end{pmatrix}$ , taking  $\mathbf{U} = \mathbf{I}$  gives  $\mathbf{T} = \mathbf{A}$ , with diagonal  $(1, 2)$ . Taking instead  $\mathbf{U} = \frac{1}{\sqrt{2}} \begin{pmatrix} 1 & 1 \\ 1 & -1 \end{pmatrix}$  gives

$$\mathbf{U}^* \mathbf{A} \mathbf{U} = \begin{pmatrix} 2 & -1 \\ 0 & 1 \end{pmatrix},$$

with diagonal  $(2, 1)$  and a different corner entry. So a Schur form is **not** a canonical form: it does not decide similarity, and two matrices can have the same triangular form without being equal. The canonical form for similarity is Jordan's ([Theorem 9.3.1](#)), and Jordan's cannot in general be reached unitarily — [Example \(A Two-by-Two Schur Factorization\)](#) is already a Jordan block in disguise whose off-diagonal entry is  $-2$ , not  $1$ . What the Schur form gives up in uniqueness it buys back in stability: the change of basis is an isometry.

## 11.1.6 Exercises

### 11.1.6.1 A. Check your understanding

#### Exercise A1.

- State [Theorem 11.1.2](#), including the field and every hypothesis.
- In the proof, an eigenvector of  $T^*$  is taken rather than one of  $T$ . Say why, in one sentence, and name the property of  $(\text{span}(\mathbf{v}))^\perp$  that is wanted.
- True or false: if  $\mathbf{A} \in M_n(\mathbb{C})$  is upper triangular in some orthonormal basis, then its eigenvalues are the diagonal entries of that triangular matrix. Justify your answer.
- Which step of the proof of [Theorem 11.1.2](#) fails over  $\mathbb{R}$ ?
- Give a matrix in  $M_2(\mathbb{C})$  that is upper triangular in an orthonormal basis but is **not** diagonal in any.

*Solution.*

- Let  $V$  be a finite-dimensional inner product space over  $\mathbb{C}$  with  $\dim V \geq 1$  and let  $T \in \mathcal{L}(V)$ . Then  $V$  has an orthonormal basis  $\mathcal{B}$  with  $[T]_{\mathcal{B}}^{\mathcal{B}}$  upper triangular, and the diagonal may be made to read the eigenvalues of  $T$ , with algebraic multiplicity, in any prescribed order.
- Because the induction needs  $(\text{span}(\mathbf{v}))^\perp$  to be  $T$ -invariant, and [Lemma 11.1.1](#) gives that exactly when  $\text{span}(\mathbf{v})$  is  $T^*$ -invariant, that is, when  $\mathbf{v}$  is an eigenvector of  $T^*$ .
- True. The matrix of  $T$  in **any** basis has the same characteristic polynomial ([Theorem 6.10.3](#)), and for a triangular matrix that polynomial is the product of  $x - t_{ii}$  ([Theorem 8.7.4](#) (a)). Orthonormality is irrelevant to this particular point.
- The Fundamental Theorem of Algebra, twice: once so that  $p_T$  splits into the list of  $n$  roots the statement speaks of ([Corollary 5.5.5](#)), and once for the existence of an eigenvalue of  $T^*$  at each stage of the induction. It came from [Theorem 8.2.4](#),

whose proof rests on the Fundamental Theorem of Algebra ([Theorem 5.5.1](#)); over  $\mathbb{R}$  the rotation  $\begin{pmatrix} 0 & -1 \\ 1 & 0 \end{pmatrix}$  has no eigenvalue at all. Every other step is valid over  $\mathbb{R}$ .

- (e)  $\mathbf{A} = \begin{pmatrix} 0 & 1 \\ 0 & 0 \end{pmatrix}$ . It is already upper triangular in the standard orthonormal basis. It is not diagonal in any basis, orthonormal or not: its only eigenvalue is 0, so a diagonal form would be the zero matrix, and  $\mathbf{A} \neq 0$ .

### 11.1.6.2 B. Practice

#### Exercise B1: A Schur form with complex entries.

Let  $\mathbf{A} = \begin{pmatrix} 1 & 2 \\ -1 & 3 \end{pmatrix}$ , regarded as a complex matrix.

- Find the eigenvalues of  $\mathbf{A}$ .
- Find a unitary  $\mathbf{U} \in M_2(\mathbb{C})$  with  $\mathbf{U}^* \mathbf{A} \mathbf{U}$  upper triangular, and compute that triangular matrix.
- Explain why no **real** orthogonal  $\mathbf{Q}$  makes  $\mathbf{Q}^\top \mathbf{A} \mathbf{Q}$  upper triangular.

*Solution.*

- $\text{tr } \mathbf{A} = 4$  and  $\det \mathbf{A} = 3 + 2 = 5$ , so  $p_{\mathbf{A}} = x^2 - 4x + 5 = (x - 2)^2 + 1$ , with roots  $2 + i$  and  $2 - i$ .
- For  $\lambda = 2 + i$ ,

$$\mathbf{A} - \lambda \mathbf{I} = \begin{pmatrix} -1 - i & 2 \\ -1 & 1 - i \end{pmatrix}.$$

The second row gives  $-x + (1 - i)y = 0$ , so  $(1 - i, 1)$  is an eigenvector; its norm is  $\sqrt{2 + 1} = \sqrt{3}$ . Put  $\mathbf{q}_1 = \frac{1}{\sqrt{3}}(1 - i, 1)$ . A unit vector orthogonal to it is  $\mathbf{q}_2 = \frac{1}{\sqrt{3}}(1, -1 - i)$ , since

$$\langle \mathbf{q}_2, \mathbf{q}_1 \rangle = \frac{1}{3}(1 \cdot \overline{1 - i} + (-1 - i) \cdot \overline{1}) = \frac{1}{3}((1 + i) - (1 + i)) = 0$$

and  $\|\mathbf{q}_2\|^2 = \frac{1}{3}(1 + 2) = 1$ . With  $\mathbf{U} = (\mathbf{q}_1 \quad \mathbf{q}_2)$ ,

$$\mathbf{U}^* \mathbf{A} \mathbf{U} = \begin{pmatrix} 2 + i & -1 - 2i \\ 0 & 2 - i \end{pmatrix}.$$

- A real upper triangular matrix has its diagonal entries as eigenvalues ([Theorem 8.7.4](#)), and those entries are real. But  $\mathbf{Q}^\top \mathbf{A} \mathbf{Q}$  is similar to  $\mathbf{A}$ , whose eigenvalues  $2 \pm i$  are not real. So no such  $\mathbf{Q}$  exists. (This is [Theorem 11.1.7](#) doing the only thing it can: here  $\mathbf{T}$  is a single  $2 \times 2$  block, and  $\mathbf{Q} = \mathbf{I}$  already works.)

#### Exercise B2: A three-by-three Schur form.

Let  $\mathbf{A} = \begin{pmatrix} 2 & 1 & 1 \\ 1 & 2 & 3 \\ 0 & 0 & 4 \end{pmatrix}$ .

- (a) Verify that  $(1, -1, 0)$  is an eigenvector and find the corresponding eigenvalue.
- (b) Extend it to an orthonormal basis of  $\mathbb{R}^3$  and compute  $\mathbf{Q}^\top \mathbf{A} \mathbf{Q}$  for the resulting  $\mathbf{Q}$ .
- (c) Hence write down the eigenvalues of  $\mathbf{A}$  with their algebraic multiplicities.

*Solution.*

- (a)  $\mathbf{A}(1, -1, 0) = (2 - 1, 1 - 2, 0) = (1, -1, 0)$ , so the eigenvalue is 1.
- (b) Take  $\mathbf{q}_1 = \frac{1}{\sqrt{2}}(1, -1, 0)$ ,  $\mathbf{q}_2 = \frac{1}{\sqrt{2}}(1, 1, 0)$  and  $\mathbf{q}_3 = (0, 0, 1)$ . These are pairwise orthogonal unit vectors, hence an orthonormal basis of  $\mathbb{R}^3$ . With  $\mathbf{Q} = (\mathbf{q}_1 \ \mathbf{q}_2 \ \mathbf{q}_3)$ ,

$$\mathbf{Q}^\top \mathbf{A} \mathbf{Q} = \begin{pmatrix} 1 & 0 & -\sqrt{2} \\ 0 & 3 & 2\sqrt{2} \\ 0 & 0 & 4 \end{pmatrix}.$$

This is already upper triangular, so a single step sufficed and  $\mathbf{U} = \mathbf{Q}$ . (In general the lower-right  $2 \times 2$  block would need a second step, as in [Example \(A Three-by-Three Schur Factorization\)](#).)

- (c) The diagonal is 1, 3, 4, so by [Theorem 8.7.4](#) these are the eigenvalues, each of algebraic multiplicity 1. As a check,  $\text{tr } \mathbf{A} = 8 = 1 + 3 + 4$  and  $\det \mathbf{A} = 4(4 - 1) = 12 = 1 \cdot 3 \cdot 4$ .

### Exercise B3: The field matters.

Let  $\mathbf{A} = \begin{pmatrix} a & -b \\ b & a \end{pmatrix} \in M_2(\mathbb{R})$  with  $b \neq 0$ .

- (a) Find the eigenvalues of  $\mathbf{A}$  over  $\mathbb{C}$  and show that  $\mathbf{A}$  is unitarily similar to a diagonal matrix in  $M_2(\mathbb{C})$ .
- (b) Prove that there is no invertible  $\mathbf{P} \in M_2(\mathbb{R})$  at all, orthogonal or otherwise, with  $\mathbf{P}^{-1} \mathbf{A} \mathbf{P}$  upper triangular.
- (c) Say what [Theorem 11.1.7](#) produces for this  $\mathbf{A}$ .

*Solution.*

- (a)  $\text{tr } \mathbf{A} = 2a$  and  $\det \mathbf{A} = a^2 + b^2$ , so  $p_{\mathbf{A}} = x^2 - 2ax + a^2 + b^2$  with roots  $a \pm bi$ , distinct because  $b \neq 0$ . For  $\lambda = a + bi$ ,  $\mathbf{A}(1, -i) = (a + bi, b - ai) = (a + bi)(1, -i)$ , so  $\mathbf{q}_1 = \frac{1}{\sqrt{2}}(1, -i)$  is a unit eigenvector; likewise  $\mathbf{q}_2 = \frac{1}{\sqrt{2}}(1, i)$  is a unit eigenvector for  $a - bi$ , and

$$\langle \mathbf{q}_1, \mathbf{q}_2 \rangle = \frac{1}{2}(1 \cdot 1 + (-i)\bar{i}) = \frac{1}{2}(1 - 1) = 0.$$

So  $\mathbf{U} = (\mathbf{q}_1 \ \mathbf{q}_2)$  is unitary and  $\mathbf{U}^* \mathbf{A} \mathbf{U} = \text{diag}(a + bi, a - bi)$ .

- (b) Suppose  $\mathbf{P}^{-1} \mathbf{A} \mathbf{P} = \mathbf{T}$  were upper triangular with  $\mathbf{P} \in M_2(\mathbb{R})$  invertible. Then  $\mathbf{T} \in M_2(\mathbb{R})$  is similar to  $\mathbf{A}$ , so  $p_{\mathbf{T}} = p_{\mathbf{A}}$  ([Theorem 6.10.3](#)); but  $p_{\mathbf{T}} = (x - t_{11})(x - t_{22})$  has the real roots  $t_{11}, t_{22}$ , while  $p_{\mathbf{A}}$  has none. Contradiction.
- (c) Since  $\mathbf{A}$  has no real eigenvalue, the real Schur form of  $\mathbf{A}$  is a single  $2 \times 2$  diagonal block;  $\mathbf{Q} = \mathbf{I}_2$  and  $\mathbf{T} = \mathbf{A}$  already work. There is nothing to simplify: over  $\mathbb{R}$  this matrix is as reduced as it gets.



## 11.1.6.3 C. Going deeper

**Exercise C1: Cayley–Hamilton from the triangular factors.**

Let  $\mathbf{T} \in M_n(\mathbb{C})$  be upper triangular with diagonal entries  $\lambda_1, \dots, \lambda_n$ , and set  $\mathbf{N}_k = (\mathbf{T} - \lambda_1 \mathbf{I})(\mathbf{T} - \lambda_2 \mathbf{I}) \cdots (\mathbf{T} - \lambda_k \mathbf{I})$  for  $1 \leq k \leq n$ .

- (a) Prove by induction on  $k$  that the first  $k$  columns of  $\mathbf{N}_k$  are zero.
- (b) Deduce that  $p_{\mathbf{T}}(\mathbf{T}) = 0$ .
- (c) Deduce the Cayley–Hamilton Theorem ([Theorem 8.6.1](#)) for every  $\mathbf{A} \in M_n(\mathbb{C})$ , using [Corollary 11.1.3](#).

*Hint for (a): the  $j$ -th column of  $\mathbf{T} - \lambda_j \mathbf{I}$  has its only possibly non-zero entries in rows  $1, \dots, j-1$ .*

*Solution.*

- (a) For  $k = 1$ : the first column of  $\mathbf{T} - \lambda_1 \mathbf{I}$  is  $(\lambda_1 - \lambda_1, 0, \dots, 0) = \mathbf{0}$ , since  $\mathbf{T}$  is upper triangular. Assume the first  $k-1$  columns of  $\mathbf{N}_{k-1}$  are zero, where  $2 \leq k \leq n$ . Then  $\mathbf{N}_k = \mathbf{N}_{k-1}(\mathbf{T} - \lambda_k \mathbf{I})$ , so the  $j$ -th column of  $\mathbf{N}_k$  is  $\mathbf{N}_{k-1} \mathbf{c}_j$  with  $\mathbf{c}_j$  the  $j$ -th column of  $\mathbf{T} - \lambda_k \mathbf{I}$ . For  $j \leq k-1$  we get  $\mathbf{N}_k \mathbf{e}_j = \mathbf{N}_{k-1}(\mathbf{T} - \lambda_k \mathbf{I}) \mathbf{e}_j$ , and since  $(\mathbf{T} - \lambda_k \mathbf{I}) \mathbf{e}_j \in \text{span}(\mathbf{e}_1, \dots, \mathbf{e}_j)$  by upper triangularity, this is a combination of the first  $j \leq k-1$  columns of  $\mathbf{N}_{k-1}$ , all zero. For  $j = k$  the column  $\mathbf{c}_k$  of  $\mathbf{T} - \lambda_k \mathbf{I}$  has its  $(k, k)$  entry equal to  $\lambda_k - \lambda_k = 0$  and zeros below, so  $\mathbf{c}_k \in \text{span}(\mathbf{e}_1, \dots, \mathbf{e}_{k-1})$ , and  $\mathbf{N}_{k-1} \mathbf{c}_k$  is a combination of the first  $k-1$  columns of  $\mathbf{N}_{k-1}$ , again zero. So all of the first  $k$  columns of  $\mathbf{N}_k$  vanish.
- (b) By [Theorem 8.7.4](#) (a),  $p_{\mathbf{T}} = (x - \lambda_1) \cdots (x - \lambda_n)$ , so  $p_{\mathbf{T}}(\mathbf{T}) = \mathbf{N}_n$ . By (a) with  $k = n$ , all  $n$  columns of  $\mathbf{N}_n$  are zero, that is  $p_{\mathbf{T}}(\mathbf{T}) = 0$ .
- (c) Write  $\mathbf{A} = \mathbf{U} \mathbf{T} \mathbf{U}^*$  by [Corollary 11.1.3](#). Similar matrices have the same characteristic polynomial ([Theorem 6.10.3](#)), so  $p_{\mathbf{A}} = p_{\mathbf{T}}$ , and  $\mathbf{A}^k = \mathbf{U} \mathbf{T}^k \mathbf{U}^*$  for every  $k \geq 0$  because the inner factors  $\mathbf{U}^* \mathbf{U}$  cancel. Hence

$$p_{\mathbf{A}}(\mathbf{A}) = \mathbf{U} p_{\mathbf{T}}(\mathbf{T}) \mathbf{U}^* = \mathbf{U} \mathbf{0} \mathbf{U}^* = \mathbf{0}.$$

Chapter 8 reached the same conclusion from the flag of invariant subspaces ([Exercise \(C1: A second proof of Cayley–Hamilton\)](#)); here the whole argument happens inside one matrix product.

**Exercise C2: The Frobenius norm is unitarily invariant.**

For  $\mathbf{A} \in M_{m \times n}(\mathbb{C})$  write  $\|\mathbf{A}\|_F^2 := \sum_{i,j} |a_{ij}|^2$ ; this is the norm induced by the Frobenius inner product  $\langle \mathbf{A}, \mathbf{B} \rangle = \text{tr}(\mathbf{B}^* \mathbf{A})$  of Chapter 10, Section 1.

- (a) Prove that  $\|\mathbf{A}\|_F^2 = \text{tr}(\mathbf{A}^* \mathbf{A})$ .
- (b) Prove that  $\|\mathbf{U} \mathbf{A}\|_F = \|\mathbf{A}\|_F = \|\mathbf{A} \mathbf{V}\|_F$  whenever  $\mathbf{U} \in M_m(\mathbb{C})$  and  $\mathbf{V} \in M_n(\mathbb{C})$  are unitary.
- (c) Deduce that if  $\mathbf{A} = \mathbf{U} \mathbf{T} \mathbf{U}^*$  is a Schur factorization ([Corollary 11.1.3](#)), then  $\|\mathbf{A}\|_F = \|\mathbf{T}\|_F$ .

*Solution.*

- (a) The  $(j, j)$  entry of  $\mathbf{A}^* \mathbf{A}$  is  $\sum_i \overline{a_{ij}} a_{ij} = \sum_i |a_{ij}|^2$ . Summing over  $j$  gives  $\text{tr}(\mathbf{A}^* \mathbf{A}) = \sum_{i,j} |a_{ij}|^2 = \|\mathbf{A}\|_F^2$ .
- (b) By (a) and  $\mathbf{U}^* \mathbf{U} = \mathbf{I}_m$ ,

$$\|\mathbf{U}\mathbf{A}\|_F^2 = \text{tr}((\mathbf{U}\mathbf{A})^*(\mathbf{U}\mathbf{A})) = \text{tr}(\mathbf{A}^* \mathbf{U}^* \mathbf{U} \mathbf{A}) = \text{tr}(\mathbf{A}^* \mathbf{A}) = \|\mathbf{A}\|_F^2.$$

For the other side, use  $\text{tr}(\mathbf{B}\mathbf{C}) = \text{tr}(\mathbf{C}\mathbf{B})$  (Theorem 0.9.6 (3)) and  $\mathbf{V}\mathbf{V}^* = \mathbf{I}_n$ :

$$\|\mathbf{A}\mathbf{V}\|_F^2 = \text{tr}(\mathbf{V}^* \mathbf{A}^* \mathbf{A} \mathbf{V}) = \text{tr}(\mathbf{A}^* \mathbf{A} \mathbf{V} \mathbf{V}^*) = \text{tr}(\mathbf{A}^* \mathbf{A}) = \|\mathbf{A}\|_F^2.$$

Norms are non-negative, so the squares being equal gives the norms equal.

- (c) Apply (b) twice:  $\|\mathbf{A}\|_F = \|\mathbf{U}\mathbf{T}\mathbf{U}^*\|_F = \|\mathbf{T}\mathbf{U}^*\|_F = \|\mathbf{T}\|_F$ , using that  $\mathbf{U}^*$  is unitary as well. So a Schur factorization redistributes the size of  $\mathbf{A}$  between the diagonal and the strictly upper triangle without changing the total. Section 11 turns that observation into a measurement.

### Exercise C3: Distinct eigenvalues are dense, but not over the reals.

- (a) Sharpen Corollary 11.1.6: prove that for every  $\mathbf{A} \in M_n(\mathbb{C})$  and every  $\varepsilon > 0$  there is  $\mathbf{A}'$  with  $n$  **distinct** eigenvalues and  $|a_{ij} - a'_{ij}| < \varepsilon$  for all  $i, j$ .
- (b) Show that the analogous statement fails over  $\mathbb{R}$ : for  $\mathbf{R} = \begin{pmatrix} 0 & -1 \\ 1 & 0 \end{pmatrix}$  there is an  $\varepsilon > 0$  such that **no** real matrix within  $\varepsilon$  of  $\mathbf{R}$ , entry by entry, has a real eigenvalue at all.

*Hint for (b): a real  $2 \times 2$  matrix with entries  $a, b, c, d$  has real eigenvalues exactly when  $(a - d)^2 + 4bc \geq 0$ .*

*Solution.*

- (a) This is exactly what the proof of Corollary 11.1.6 produced, before the conclusion was weakened. There the  $\delta_k$  were chosen so that the diagonal entries of  $\mathbf{T} + \mathbf{D}$  are pairwise distinct, and by Theorem 8.7.4 (b) those  $n$  distinct numbers are the eigenvalues of  $\mathbf{T} + \mathbf{D}$ , hence of the similar matrix  $\mathbf{A}' = \mathbf{U}(\mathbf{T} + \mathbf{D})\mathbf{U}^*$ . The entrywise estimate is unchanged.
- (b) The characteristic polynomial of  $\begin{pmatrix} a & b \\ c & d \end{pmatrix}$  is  $x^2 - (a + d)x + (ad - bc)$ , whose discriminant is

$$(a + d)^2 - 4(ad - bc) = (a - d)^2 + 4bc,$$

so the eigenvalues are real exactly when this is  $\geq 0$ . Take  $\varepsilon = \frac{1}{4}$  and let  $\mathbf{A}'$  have entries within  $\varepsilon$  of those of  $\mathbf{R}$ , so  $|a|, |d| \leq \frac{1}{4}$ ,  $b \in [-\frac{5}{4}, -\frac{3}{4}]$  and  $c \in [\frac{3}{4}, \frac{5}{4}]$ . Then  $(a - d)^2 \leq \frac{1}{4}$  and  $bc \leq -\frac{9}{16}$ , so

$$(a - d)^2 + 4bc \leq \frac{1}{4} - \frac{9}{4} = -2 < 0.$$

Hence  $\mathbf{A}'$  has no real eigenvalue, let alone two distinct ones. The reason is the one this section keeps meeting: over  $\mathbb{C}$  the eigenvalues move continuously and can

always be separated, while over  $\mathbb{R}$  a conjugate pair cannot be pushed onto the real line by a small change.

## 11.2 Self-Adjoint Operators

Among complex numbers, the real ones are those fixed by conjugation:  $\bar{z} = z$ . Among operators on an inner product space, conjugation is replaced by the adjoint, and the operators fixed by it are the **self-adjoint** ones,  $T^* = T$ . The analogy is not decoration. Self-adjoint operators have real eigenvalues, they come with a real-valued quantity  $\langle T\mathbf{v}, \mathbf{v} \rangle$  attached to every vector, they can be compared and ordered, and over  $\mathbb{R}$  they are exactly the operators the spectral theorem applies to. This section collects everything about them that does not need a spectral theorem, and in doing so assembles every ingredient the real spectral theorem of Section 5 will use.

Throughout,  $V$  is a finite-dimensional inner product space over  $F = \mathbb{R}$  or  $F = \mathbb{C}$ , and  $T \in \mathcal{L}(V)$ .

### 11.2.1 The operators fixed by starring

Chapter 10 already made the definition, so we only recall it:  $T \in \mathcal{L}(V)$  is **self-adjoint** when  $T^* = T$ , equivalently when

$$\langle T\mathbf{u}, \mathbf{v} \rangle = \langle \mathbf{u}, T\mathbf{v} \rangle \quad \text{for all } \mathbf{u}, \mathbf{v} \in V$$

(Definition 10.6.2). In an orthonormal basis this says the matrix is **Hermitian**,  $\mathbf{A}^* = \mathbf{A}$ , which over  $\mathbb{R}$  is **symmetric**,  $\mathbf{A}^\top = \mathbf{A}$  (Theorem 10.6.3). Three facts were proved there and will be used without further comment:  $\langle T\mathbf{v}, \mathbf{v} \rangle$  is real for every  $\mathbf{v}$ , even over  $\mathbb{C}$ ; the diagonal entries of the matrix in an orthonormal basis are real; and every self-adjoint operator is normal (Proposition 10.6.7).

Deciding self-adjointness is always the same job — compute  $T^*$ , or check the displayed identity on a basis — but the traps differ by example.

#### Example. (Four Tests)

Determine which of the following are self-adjoint. Justify your answer; for those that are not, name the failing pair of vectors.

- (a)  $T_{\mathbf{A}}$  on  $\mathbb{C}^2$  with the standard inner product,  $\mathbf{A} = \begin{pmatrix} 2 & 1-i \\ 1+i & 3 \end{pmatrix}$ .
- (b)  $T_{\mathbf{B}}$  on  $\mathbb{C}^2$  with the standard inner product,  $\mathbf{B} = \begin{pmatrix} 1 & i \\ i & 1 \end{pmatrix}$ .
- (c) Differentiation  $D$  on  $V = \mathbb{R}[x]_{\leq 2}$  with  $\langle p, q \rangle = \int_0^1 pq$ .
- (d) The orthogonal projection  $P_U$  onto a subspace  $U$  of any  $V$ .

*Solution.*

- (a) Self-adjoint. Transposing swaps the off-diagonal entries and conjugating swaps them back, while the diagonal entries 2 and 3 are real; so  $\mathbf{A}^* = \mathbf{A}$ . The standard basis of  $\mathbb{C}^2$  is orthonormal, so Theorem 10.6.3 turns this into  $T_{\mathbf{A}}^* = T_{\mathbf{A}}$ .

- (b) Not self-adjoint. It is **symmetric**,  $\mathbf{B}^\top = \mathbf{B}$ , but  $\mathbf{B}^* = \begin{pmatrix} 1 & -i \\ -i & 1 \end{pmatrix} \neq \mathbf{B}$ . The failing pair is  $\mathbf{e}_1, \mathbf{e}_2$ :  $\langle \mathbf{B}\mathbf{e}_1, \mathbf{e}_2 \rangle = i$ , while  $\langle \mathbf{e}_1, \mathbf{B}\mathbf{e}_2 \rangle = \bar{i} = -i$ .
- (c) Not self-adjoint. Take  $p = x$  and  $q = 1$ . Then  $Dp = 1$  and  $Dq = 0$ , so

$$\langle Dp, q \rangle = \int_0^1 1 \, dx = 1, \quad \langle p, Dq \rangle = 0.$$

The obstruction is visible in general: integration by parts gives

$$\langle Dp, q \rangle + \langle p, Dq \rangle = p(1)q(1) - p(0)q(0),$$

so  $D$  is as far from self-adjoint as it could be — the two sides add up rather than agreeing, apart from boundary terms.

- (d) Self-adjoint, for every  $U$ . By [Theorem 10.3.3](#) (b),  $\langle P_U \mathbf{u}, \mathbf{v} \rangle = \langle \mathbf{u}, P_U \mathbf{v} \rangle$  for all  $\mathbf{u}, \mathbf{v}$ , which is the displayed identity.

*Warning.* Over  $\mathbb{C}$ , **symmetric is not the right notion; Hermitian is**. Part (b) above is symmetric and its eigenvalues are not even real:  $\text{tr } \mathbf{B} = 2$  and  $\det \mathbf{B} = 1 - i^2 = 2$ , so  $p_{\mathbf{B}} = x^2 - 2x + 2$  and the eigenvalues are  $1 + i$  and  $1 - i$ . Every theorem in this section would fail for it. The word “symmetric matrix” belongs to the **real** case, where conjugation does nothing and  $\mathbf{A}^* = \mathbf{A}^\top$ ; over  $\mathbb{C}$  always write, and check,  $\mathbf{A}^* = \mathbf{A}$ .

## 11.2.2 Eigenvalues are real

The first theorem is the operator version of “a number equal to its own conjugate is real”, and its proof is the move that runs this whole chapter: compute one inner product two different ways.

### Theorem 11.2.1 — Self-adjoint Operators Have Real Eigenvalues

Let  $V$  be a finite-dimensional inner product space over  $F = \mathbb{R}$  or  $F = \mathbb{C}$  and let  $T \in \mathcal{L}(V)$  be self-adjoint. Then every eigenvalue of  $T$  is real.

*Idea.* Take  $T\mathbf{v} = \lambda\mathbf{v}$  with  $\mathbf{v} \neq \mathbf{0}$  and pair it with  $\mathbf{v}$ . Computing  $\langle T\mathbf{v}, \mathbf{v} \rangle$  straight off the eigenvalue equation produces  $\lambda$ ; computing it by pushing  $T$  into the second slot produces  $\bar{\lambda}$ , because a scalar leaving the second slot picks up a conjugate. The two answers must agree, and  $\|\mathbf{v}\|^2 \neq 0$  lets us cancel.

*Proof.* Let  $\lambda$  be an eigenvalue of  $T$ , with eigenvector  $\mathbf{v} \neq \mathbf{0}$ . On one side,

$$\langle T\mathbf{v}, \mathbf{v} \rangle = \langle \lambda\mathbf{v}, \mathbf{v} \rangle = \lambda\|\mathbf{v}\|^2,$$

by linearity in the first slot. On the other, using [Definition 10.6.2](#) and then conjugate-linearity in the second slot,

$$\langle T\mathbf{v}, \mathbf{v} \rangle = \langle \mathbf{v}, T\mathbf{v} \rangle = \langle \mathbf{v}, \lambda\mathbf{v} \rangle = \bar{\lambda}\|\mathbf{v}\|^2.$$

So  $\lambda\|\mathbf{v}\|^2 = \bar{\lambda}\|\mathbf{v}\|^2$ . Since  $\mathbf{v} \neq \mathbf{0}$  we have  $\|\mathbf{v}\|^2 \neq 0$  by positive definiteness (Definition 10.1.1), and dividing gives  $\lambda = \bar{\lambda}$ . A scalar equal to its own conjugate is real.

■

Chapter 10 already proved this, in the remark after Proposition 10.6.7. It is promoted to a theorem here because Sections 3 to 5 lean on it constantly, and a result cited that often should have a number. Over  $\mathbb{R}$  the conclusion sounds empty, because the eigenvalues of a real operator are real by definition. What is not empty over  $\mathbb{R}$  is the corresponding statement about the **roots of the characteristic polynomial**, and that is the next theorem: it says the roots do not escape into  $\mathbb{C}$ , which is exactly what guarantees that a real self-adjoint operator has an eigenvector at all.

### 11.2.3 Existence over the real numbers

Over  $\mathbb{C}$  an eigenvalue is free (Theorem 8.2.4). Over  $\mathbb{R}$  it is not, and the rotation of  $\mathbb{R}^2$  by a quarter turn has none. Self-adjointness is what rules that out.

#### Theorem 11.2.2 — Real Self-adjoint Operators Have an Eigenvalue

Let  $V \neq \{\mathbf{0}\}$  be a finite-dimensional inner product space over  $\mathbb{R}$  and let  $T \in \mathcal{L}(V)$  be self-adjoint. Then:

- (a) every root of  $p_T$  in  $\mathbb{C}$  is real, so  $p_T$  splits over  $\mathbb{R}$ ;
- (b)  $T$  has an eigenvalue.

*Idea.* There is no complex number in sight, so we put one there. Fix an orthonormal basis; the matrix  $\mathbf{A}$  of  $T$  is real symmetric. The same array of numbers, read as a complex matrix, is Hermitian, and it acts on  $\mathbb{C}^n$ , which is the complexification of  $\mathbb{R}^n$  (Definition 1.9.2, Example (The Complexification of  $\mathbb{R}^n$  Is  $\mathbb{C}^n$ )) — the arena Chapter 1 built, and Chapter 8 put to work, precisely so that a real matrix could borrow complex eigenvalues. Over  $\mathbb{C}$  an eigenvalue exists, and Theorem 11.2.1 says it is real. A real root of  $p_T$  is an eigenvalue of  $T$  itself, and we are back where we started, one eigenvalue richer.

*Proof.* Let  $n = \dim V \geq 1$ . By Theorem 10.2.2,  $V$  has an orthonormal basis  $\mathcal{B}$ ; put  $\mathbf{A} = [T]_{\mathcal{B}}^{\mathcal{B}} \in M_n(\mathbb{R})$ . By Theorem 10.6.3 and Definition 10.6.2,  $\mathbf{A}^* = \mathbf{A}$ , and since  $\mathbf{A}$  is real this says  $\mathbf{A}^{\top} = \mathbf{A}$ .

Now read  $\mathbf{A}$  as an element of  $M_n(\mathbb{C})$ , acting on  $\mathbb{C}^n$  with the standard inner product; this is the extension of the real map  $T_{\mathbf{A}}$  to the complexification  $(\mathbb{R}^n)_{\mathbb{C}} = \mathbb{C}^n$ , which acts on  $\mathbf{x} + i\mathbf{y}$  by  $\mathbf{A}\mathbf{x} + i\mathbf{A}\mathbf{y}$  (Chapter 8, Section 2). Because  $\mathbf{A}$  is real and symmetric,  $\mathbf{A}^* = \overline{\mathbf{A}}^{\top} = \mathbf{A}^{\top} = \mathbf{A}$ , so  $T_{\mathbf{A}}$  is self-adjoint as an operator on  $\mathbb{C}^n$  (Theorem 10.6.3 again, the standard basis of  $\mathbb{C}^n$  being orthonormal).

- (a) By Definition 6.10.2,  $p_T = p_{\mathbf{A}}$ , and the entries of  $x\mathbf{I} - \mathbf{A}$  are real, so this is the same polynomial whether the determinant is expanded over  $\mathbb{R}$  or over  $\mathbb{C}$ ; it lies in  $\mathbb{R}[x] \subseteq \mathbb{C}[x]$ . Let  $\lambda \in \mathbb{C}$  be a root of  $p_{\mathbf{A}}$ . By Theorem 8.2.1,  $\lambda$  is an eigenvalue of

$\mathbf{A}$  over  $\mathbb{C}$ , and  $\mathbf{A}$  is self-adjoint there, so  $\lambda \in \mathbb{R}$  by [Theorem 11.2.1](#). Since every root of  $p_T$  in  $\mathbb{C}$  lies in  $\mathbb{R}$ , and  $p_T$  splits over  $\mathbb{C}$  by [Corollary 5.5.5](#), it splits over  $\mathbb{R}$ .

- (b)  $p_T$  is monic of degree  $n \geq 1$  ([Theorem 6.10.2](#)), so by (a) it has a root  $\lambda \in \mathbb{R}$ . By [Theorem 8.2.1](#) ((e)  $\Rightarrow$  (a)),  $\lambda$  is an eigenvalue of  $T$ . This proves the theorem. ■

*Remark.* There is a second, very different route to (b): the function  $\mathbf{v} \mapsto \langle T\mathbf{v}, \mathbf{v} \rangle$  is real-valued and continuous on the unit sphere of  $V$ , which is closed and bounded, so it attains a maximum, and a vector where it does so turns out to be an eigenvector for the largest eigenvalue. That argument is shorter to state but it is analysis, not algebra, and it needs the extreme value theorem; we keep to the algebraic route so that nothing in this chapter depends on it. The maximization picture returns in Chapter 16, where the Rayleigh quotient and the Courant–Fischer theorem describe every eigenvalue of a self-adjoint operator by optimization.

*Quick check.* The rotation  $\mathbf{R} = \begin{pmatrix} 0 & -1 \\ 1 & 0 \end{pmatrix}$  on  $\mathbb{R}^2$  has no eigenvalue, and over  $\mathbb{C}$  its eigenvalues  $\pm i$  are not real. Which hypothesis of [Theorem 11.2.2](#) does it fail, and how do you see it in one line?

*Solution.* Self-adjointness. In the standard orthonormal basis the matrix of  $\mathbf{R}^*$  is  $\mathbf{R}^\top = \begin{pmatrix} 0 & 1 \\ -1 & 0 \end{pmatrix} = -\mathbf{R}$ , which is not  $\mathbf{R}$ . So  $\mathbf{R}$  is skew-adjoint, not self-adjoint, and neither theorem applies. (It is, however, normal, since  $\mathbf{R}^\top \mathbf{R} = \mathbf{R} \mathbf{R}^\top = \mathbf{I}$  — which is why Section 5 has to insist on self-adjoint rather than normal over  $\mathbb{R}$ .)

## 11.2.4 Eigenvectors for different eigenvalues are orthogonal

The second theorem is the same move with two vectors instead of one, and it is the geometric half of the spectral theorem: the eigenspaces are not merely independent, as in Chapter 8, but perpendicular.

### Theorem 11.2.3 — Distinct Eigenvalues Give Orthogonal Eigenvectors

Let  $T \in \mathcal{L}(V)$  be self-adjoint, and let  $\mathbf{u}, \mathbf{v} \in V$  satisfy  $T\mathbf{u} = \lambda\mathbf{u}$  and  $T\mathbf{v} = \mu\mathbf{v}$  with  $\lambda \neq \mu$ . Then  $\langle \mathbf{u}, \mathbf{v} \rangle = 0$ . Consequently  $E_\lambda(T) \perp E_\mu(T)$  for distinct eigenvalues  $\lambda, \mu$ .

*Proof.* Compute  $\langle T\mathbf{u}, \mathbf{v} \rangle$  two ways. First,  $\langle T\mathbf{u}, \mathbf{v} \rangle = \langle \lambda\mathbf{u}, \mathbf{v} \rangle = \lambda\langle \mathbf{u}, \mathbf{v} \rangle$ . Second, by [Definition 10.6.2](#) and conjugate-linearity in the second slot,

$$\langle T\mathbf{u}, \mathbf{v} \rangle = \langle \mathbf{u}, T\mathbf{v} \rangle = \langle \mathbf{u}, \mu\mathbf{v} \rangle = \bar{\mu}\langle \mathbf{u}, \mathbf{v} \rangle = \mu\langle \mathbf{u}, \mathbf{v} \rangle,$$

the last step because  $\mu$  is real by [Theorem 11.2.1](#). Subtracting,  $(\lambda - \mu)\langle \mathbf{u}, \mathbf{v} \rangle = 0$ , and  $\lambda - \mu \neq 0$ , so  $\langle \mathbf{u}, \mathbf{v} \rangle = 0$ . Since every element of  $E_\lambda(T)$  is such a  $\mathbf{u}$  and every element of

$E_\mu(T)$  such a  $\mathbf{v}$ , the two eigenspaces are orthogonal. ■

Note where the previous theorem was spent: it made  $\bar{\mu} = \mu$ , so that the conjugation which appears on the way out of the second slot does no damage. Without real eigenvalues the computation would end at  $\lambda \langle \mathbf{u}, \mathbf{v} \rangle = \bar{\mu} \langle \mathbf{u}, \mathbf{v} \rangle$ , which says nothing when  $\lambda = \bar{\mu}$ .

**Example. (A Hermitian Two-by-Two)**

For  $\mathbf{A} = \begin{pmatrix} 2 & 1-i \\ 1+i & 3 \end{pmatrix}$  on  $\mathbb{C}^2$ , find the eigenvalues and an eigenvector for each, and verify that the two eigenvectors are orthogonal.

*Solution.* Here  $\text{tr } \mathbf{A} = 5$  and  $\det \mathbf{A} = 6 - (1-i)(1+i) = 6 - 2 = 4$ , so

$$p_{\mathbf{A}} = x^2 - 5x + 4 = (x-1)(x-4).$$

Both eigenvalues are real, as [Theorem 11.2.1](#) promised.

For  $\lambda = 1$ :  $\mathbf{A} - \mathbf{I} = \begin{pmatrix} 1 & 1-i \\ 1+i & 2 \end{pmatrix}$ , and the first row gives  $x + (1-i)y = 0$ , so  $\mathbf{u} = (-1+i, 1)$ . Check the second row:  $(1+i)(-1+i)+2 = (-1+i-i+i^2)+2 = -2+2 = 0$ .

For  $\lambda = 4$ :  $\mathbf{A} - 4\mathbf{I} = \begin{pmatrix} -2 & 1-i \\ 1+i & -1 \end{pmatrix}$ , and the first row gives  $-2x + (1-i)y = 0$ , so  $\mathbf{v} = (1-i, 2)$ .

Finally

$$\langle \mathbf{u}, \mathbf{v} \rangle = (-1+i)\overline{(1-i)} + 1 \cdot \bar{2} = (-1+i)(1+i) + 2 = -2+2 = 0,$$

so  $\mathbf{u} \perp \mathbf{v}$ , as [Theorem 11.2.3](#) requires. Normalizing,  $\|\mathbf{u}\|^2 = 2+1 = 3$  and  $\|\mathbf{v}\|^2 = 2+4 = 6$ , so

$$\left( \frac{1}{\sqrt{3}}(-1+i, 1), \frac{1}{\sqrt{6}}(1-i, 2) \right)$$

is an orthonormal basis of  $\mathbb{C}^2$  consisting of eigenvectors of  $\mathbf{A}$ . Section 4 proves that this always happens, for exactly the operators it happens for.

## 11.2.5 Invariant complements

Section 5 will prove the real spectral theorem by the move named at the front of the chapter: find one eigenvector, pass to its orthogonal complement, induct. For that to work the complement must be invariant, and the restriction must again be self-adjoint so that the induction can be applied to it. Both hold, and neither needs an eigenvector — the statement is about any invariant subspace.

### Theorem 11.2.4 — Self-adjointness Passes to Orthogonal Complements

Let  $T \in \mathcal{L}(V)$  be self-adjoint and let  $U \subseteq V$  be a  $T$ -invariant subspace. Then:

- (a)  $U^\perp$  is  $T$ -invariant;
- (b)  $T|_U \in \mathcal{L}(U)$  and  $T|_{U^\perp} \in \mathcal{L}(U^\perp)$  are self-adjoint, for the inner products inherited from  $V$ .

*Proof.*

- (a) Since  $T$  is self-adjoint,  $T^* = T$ , so  $U$  is  $T^*$ -invariant. By [Lemma 11.1.1](#),  $U^\perp$  is  $T$ -invariant.
- (b)  $U$  is  $T$ -invariant, so  $T|_U$  maps  $U$  to  $U$  and is an operator on  $U$ . For  $\mathbf{u}_1, \mathbf{u}_2 \in U$ ,

$$\langle T|_U \mathbf{u}_1, \mathbf{u}_2 \rangle = \langle T \mathbf{u}_1, \mathbf{u}_2 \rangle = \langle \mathbf{u}_1, T \mathbf{u}_2 \rangle = \langle \mathbf{u}_1, T|_U \mathbf{u}_2 \rangle,$$

where the inner products are those of  $V$ , which is what  $U$  inherits. By [Definition 10.6.2](#),  $T|_U$  is self-adjoint. The same three equalities with  $U^\perp$  in place of  $U$  give the second claim, using (a) to know that  $T|_{U^\perp}$  is an operator on  $U^\perp$ . ■

So the pair  $(U, U^\perp)$  splits a self-adjoint operator into two self-adjoint operators on smaller spaces, and nothing is lost:  $V = U \oplus U^\perp$  by [Theorem 10.3.2](#) (a). Combined with [Theorem 11.2.2](#), which guarantees that a non-zero space always offers an eigenvector to start from, this is a complete induction, and Section 5 will do no more than write it out.

## 11.2.6 Two zero tests, and the gap between the fields

An operator carries the function  $\mathbf{v} \mapsto \langle T\mathbf{v}, \mathbf{v} \rangle$ , sometimes called its quadratic form. How much of  $T$  does that one function remember? The answer depends on the field, and the difference is the sharpest illustration in this book of why  $\mathbb{C}$  is the easier place to work.

### Theorem 11.2.5 — The Zero Test for Self-adjoint Operators

Let  $V$  be a finite-dimensional inner product space over  $F = \mathbb{R}$  or  $F = \mathbb{C}$  and let  $T \in \mathcal{L}(V)$  be **self-adjoint**. If  $\langle T\mathbf{v}, \mathbf{v} \rangle = 0$  for every  $\mathbf{v} \in V$ , then  $T = 0$ .

*Idea.* The hypothesis only ever feeds  $T$  one vector, while the conclusion is about  $\langle T\mathbf{u}, \mathbf{w} \rangle$  for **two**. So manufacture two out of one: apply the hypothesis to  $\mathbf{u} + \mathbf{w}$  and expand. The diagonal terms die by hypothesis and what is left is a statement about the cross terms, which self-adjointness turns into a statement about  $\langle T\mathbf{u}, \mathbf{w} \rangle$  alone.

*Proof.* Let  $\mathbf{u}, \mathbf{w} \in V$ . Expanding  $\langle T(\mathbf{u} + \mathbf{w}), \mathbf{u} + \mathbf{w} \rangle$  in both slots,

$$\begin{aligned} 0 &= \langle T\mathbf{u}, \mathbf{u} \rangle + \langle T\mathbf{u}, \mathbf{w} \rangle + \langle T\mathbf{w}, \mathbf{u} \rangle + \langle T\mathbf{w}, \mathbf{w} \rangle \\ &= \langle T\mathbf{u}, \mathbf{w} \rangle + \langle T\mathbf{w}, \mathbf{u} \rangle, \end{aligned}$$

the two outer terms vanishing by hypothesis. Since  $T$  is self-adjoint,  $\langle T\mathbf{w}, \mathbf{u} \rangle = \langle \mathbf{w}, T\mathbf{u} \rangle =$



$\overline{\langle T\mathbf{u}, \mathbf{w} \rangle}$  by conjugate symmetry. Writing  $z = \langle T\mathbf{u}, \mathbf{w} \rangle$ , the display says  $z + \bar{z} = 0$ , that is

$$\operatorname{Re}\langle T\mathbf{u}, \mathbf{w} \rangle = 0 \quad \text{for all } \mathbf{u}, \mathbf{w} \in V. \quad (*)$$

If  $F = \mathbb{R}$  this already says  $\langle T\mathbf{u}, \mathbf{w} \rangle = 0$  for all  $\mathbf{u}, \mathbf{w}$ . If  $F = \mathbb{C}$ , apply  $(*)$  with  $i\mathbf{w}$  in place of  $\mathbf{w}$ : since  $\langle T\mathbf{u}, i\mathbf{w} \rangle = \bar{i}z = -iz$  and  $\operatorname{Re}(-iz) = \operatorname{Im}(z)$ , we also get  $\operatorname{Im}\langle T\mathbf{u}, \mathbf{w} \rangle = 0$ . Either way  $\langle T\mathbf{u}, \mathbf{w} \rangle = 0$  for all  $\mathbf{u}, \mathbf{w}$ .

Taking  $\mathbf{w} = T\mathbf{u}$  gives  $\|T\mathbf{u}\|^2 = 0$ , so  $T\mathbf{u} = \mathbf{0}$  for every  $\mathbf{u}$ , that is  $T = 0$ . This proves the theorem. ■

Now drop the hypothesis. Over  $\mathbb{C}$  nothing is lost.

### Theorem 11.2.6 — The Zero Test over the Complex Numbers

Let  $V$  be a finite-dimensional **complex** inner product space and let  $T \in \mathcal{L}(V)$  be any operator. If  $\langle T\mathbf{v}, \mathbf{v} \rangle = 0$  for every  $\mathbf{v} \in V$ , then  $T = 0$ .

*Idea.* The previous proof used self-adjointness in exactly one place: to convert  $\langle T\mathbf{w}, \mathbf{u} \rangle$  into  $\overline{\langle T\mathbf{u}, \mathbf{w} \rangle}$ , which gave the real part only. Without it we must extract the imaginary part some other way, and over  $\mathbb{C}$  there is another way — substitute  $\mathbf{u} + i\mathbf{w}$  as well as  $\mathbf{u} + \mathbf{w}$ , and take a combination of the two identities. There is no third substitution available over  $\mathbb{R}$ , which is precisely why the real statement is false.

*Proof.* Let  $\mathbf{u}, \mathbf{w} \in V$ . Applying the hypothesis to  $\mathbf{u} + \mathbf{w}$  and expanding as before,

$$\langle T\mathbf{u}, \mathbf{w} \rangle + \langle T\mathbf{w}, \mathbf{u} \rangle = 0. \quad (1)$$

Applying it to  $\mathbf{u} + i\mathbf{w}$  and using  $\langle T\mathbf{u}, i\mathbf{w} \rangle = -i\langle T\mathbf{u}, \mathbf{w} \rangle$ ,  $\langle T(i\mathbf{w}), \mathbf{u} \rangle = i\langle T\mathbf{w}, \mathbf{u} \rangle$  and  $\langle T(i\mathbf{w}), i\mathbf{w} \rangle = i\bar{i}\langle T\mathbf{w}, \mathbf{w} \rangle = 0$ ,

$$-i\langle T\mathbf{u}, \mathbf{w} \rangle + i\langle T\mathbf{w}, \mathbf{u} \rangle = 0.$$

Multiplying that by  $i$  gives

$$\langle T\mathbf{u}, \mathbf{w} \rangle - \langle T\mathbf{w}, \mathbf{u} \rangle = 0. \quad (2)$$

Adding (1) and (2),  $2\langle T\mathbf{u}, \mathbf{w} \rangle = 0$ , so  $\langle T\mathbf{u}, \mathbf{w} \rangle = 0$  for all  $\mathbf{u}, \mathbf{w}$ . Taking  $\mathbf{w} = T\mathbf{u}$  gives  $\|T\mathbf{u}\|^2 = 0$ , hence  $T = 0$ . ■

This statement appeared once already, as [Exercise \(C2: Linearity is a hypothesis, not a consequence\)](#) (b), where it was an exercise in the arithmetic of complex inner products. It is restated and proved here because Section 3 needs to cite it as a theorem: the four characterizations of a normal operator all come from applying it to  $T^*T - TT^*$ . The proof is the same one; what is new is the company it now keeps.

**Warning.** Over  $\mathbb{R}$ , the quadratic form does not determine the operator. Let  $T$

be the quarter turn of  $\mathbb{R}^2$ ,  $T(x_1, x_2) = (-x_2, x_1)$ . Then

$$\langle T\mathbf{x}, \mathbf{x} \rangle = -x_2x_1 + x_1x_2 = 0 \quad \text{for every } \mathbf{x} \in \mathbb{R}^2,$$

and yet  $T \neq 0$ . So [Theorem 11.2.6](#) genuinely fails over  $\mathbb{R}$ , and [Theorem 11.2.5](#) genuinely needs its hypothesis: this  $T$  is not self-adjoint, since  $T^* = -T$ . The general real statement is that  $\langle T\mathbf{v}, \mathbf{v} \rangle = 0$  for all  $\mathbf{v}$  forces  $T^* = -T$  and no more — [Exercise \(C3: What the real quadratic form does determine\)](#).

*Quick check.* Over  $\mathbb{C}$ , suppose  $\langle T\mathbf{v}, \mathbf{v} \rangle$  is **real** for every  $\mathbf{v} \in V$ . Must  $T$  be self-adjoint?

*Solution.* Yes. Put  $S = T - T^*$ . For every  $\mathbf{v}$ ,  $\langle T^*\mathbf{v}, \mathbf{v} \rangle = \langle \mathbf{v}, T\mathbf{v} \rangle = \overline{\langle T\mathbf{v}, \mathbf{v} \rangle} = \langle T\mathbf{v}, \mathbf{v} \rangle$ , the last step because the number is real. Hence  $\langle S\mathbf{v}, \mathbf{v} \rangle = 0$  for all  $\mathbf{v}$ , and [Theorem 11.2.6](#) gives  $S = 0$ , that is  $T^* = T$ . Combined with [Proposition 10.6.7](#) (a), this says that over  $\mathbb{C}$  “self-adjoint” and “ $\langle T\mathbf{v}, \mathbf{v} \rangle$  always real” are the same condition. Over  $\mathbb{R}$  the criterion is useless, since  $\langle T\mathbf{v}, \mathbf{v} \rangle$  is automatically real for every operator.

## 11.2.7 Exercises

### 11.2.7.1 A. Check your understanding

#### Exercise A1.

- State what it means for  $T \in \mathcal{L}(V)$  to be self-adjoint, in terms of  $T^*$  and in terms of inner products, and say what the matrix condition is and in which bases it may be used.
- True or false: a symmetric matrix in  $M_2(\mathbb{C})$  has real eigenvalues. Justify your answer.
- Which field does [Theorem 11.2.2](#) concern, and which earlier theorem supplies the eigenvalue it borrows?
- True or false: if  $T$  is self-adjoint and  $U$  is any subspace, then  $U^\perp$  is  $T$ -invariant. Justify your answer.
- State the difference between [Theorem 11.2.5](#) and [Theorem 11.2.6](#), and give the standard example showing the second has no real analogue.
- Give a normal operator on  $\mathbb{R}^2$  that is not self-adjoint.

*Solution.*

- $T^* = T$ ; equivalently  $\langle T\mathbf{u}, \mathbf{v} \rangle = \langle \mathbf{u}, T\mathbf{v} \rangle$  for **all**  $\mathbf{u}, \mathbf{v} \in V$  ([Definition 10.6.2](#)). The matrix condition is  $\mathbf{A}^* = \mathbf{A}$ , and by [Theorem 10.6.3](#) it may be used only in an **orthonormal** basis.
- False.  $\mathbf{B} = \begin{pmatrix} 1 & i \\ i & 1 \end{pmatrix}$  is symmetric with eigenvalues  $1 \pm i$  ([Example \(Four Tests\)](#) (b) and the warning after it). Over  $\mathbb{C}$  the relevant condition is  $\mathbf{A}^* = \mathbf{A}$ , not  $\mathbf{A}^\top = \mathbf{A}$ .
- The real field: it says a self-adjoint operator on a non-zero **real** inner product space has an eigenvalue. The eigenvalue is borrowed from [Theorem 8.2.4](#), applied to the matrix read as a complex matrix on the complexification  $\mathbb{C}^n$  of  $\mathbb{R}^n$ , and is

then shown to be real by [Theorem 11.2.1](#).

- (d) False as stated:  $U$  must be  $T$ -invariant ([Theorem 11.2.4](#)). For example,  $T = T_{\mathbf{A}}$  on  $\mathbb{R}^2$  with  $\mathbf{A} = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}$  is self-adjoint, and  $U = \text{span}(\mathbf{e}_1)$  has  $U^\perp = \text{span}(\mathbf{e}_2)$  with  $T\mathbf{e}_2 = \mathbf{e}_1 \notin U^\perp$ . Here  $U$  was not invariant either, since  $T\mathbf{e}_1 = \mathbf{e}_2$ .
- (e) [Theorem 11.2.5](#) holds over  $\mathbb{R}$  and  $\mathbb{C}$  but assumes  $T$  self-adjoint; [Theorem 11.2.6](#) drops that assumption but holds only over  $\mathbb{C}$ . The quarter turn  $T(x_1, x_2) = (-x_2, x_1)$  of  $\mathbb{R}^2$  satisfies  $\langle T\mathbf{x}, \mathbf{x} \rangle = 0$  for every  $\mathbf{x}$  and is not zero.
- (f) The same quarter turn:  $T^* = -T$ , so  $T^*T = TT^* = \text{id}$  and  $T$  is normal, while  $T^* \neq T$ .

### 11.2.7.2 B. Practice

#### Exercise B1: Determine which are self-adjoint.

Determine which of the following operators are self-adjoint. Justify your answer; for those that are not, exhibit a pair  $\mathbf{u}, \mathbf{v}$  with  $\langle T\mathbf{u}, \mathbf{v} \rangle \neq \langle \mathbf{u}, T\mathbf{v} \rangle$ , or compute  $T^*$ .

- (a)  $T_{\mathbf{A}}$  on  $\mathbb{C}^2$ ,  $\mathbf{A} = \begin{pmatrix} 1 & 2+i \\ 2-i & -5 \end{pmatrix}$ , standard inner product.
- (b)  $T_{\mathbf{B}}$  on  $\mathbb{C}^2$ ,  $\mathbf{B} = \begin{pmatrix} 0 & 1 \\ -1 & 0 \end{pmatrix}$ , standard inner product.
- (c)  $T_{\mathbf{C}}$  on  $\mathbb{R}^3$ ,  $\mathbf{C} = \begin{pmatrix} 1 & 2 & 0 \\ 2 & 1 & 3 \\ 0 & 3 & 1 \end{pmatrix}$ , standard inner product.
- (d)  $S^*S$ , for an arbitrary  $S \in \mathcal{L}(V, W)$  between finite-dimensional inner product spaces.
- (e)  $P_U - P_{U^\perp}$ , for a subspace  $U$  of a finite-dimensional  $V$ .
- (f)  $T(x_1, x_2) = (x_2, x_1)$  on  $\mathbb{R}^2$  with the standard inner product, whose matrix in the basis  $\mathcal{C} = ((1, 0), (1, 2))$  is  $\begin{pmatrix} -\frac{1}{2} & \frac{3}{2} \\ \frac{1}{2} & \frac{1}{2} \end{pmatrix}$ , which is not symmetric.

*Solution.*

- (a) Self-adjoint. The diagonal entries 1 and  $-5$  are real and the off-diagonal pair  $2+i$ ,  $2-i$  are conjugates, so  $\mathbf{A}^* = \mathbf{A}$ ; the standard basis is orthonormal, so [Theorem 10.6.3](#) applies.
- (b) Not self-adjoint.  $\mathbf{B}^* = \mathbf{B}^\top = \begin{pmatrix} 0 & -1 \\ 1 & 0 \end{pmatrix} = -\mathbf{B}$ , so  $T_{\mathbf{B}}$  is skew-adjoint. Concretely  $\langle \mathbf{B}\mathbf{e}_1, \mathbf{e}_2 \rangle = -1$  while  $\langle \mathbf{e}_1, \mathbf{B}\mathbf{e}_2 \rangle = 1$ .
- (c) Self-adjoint.  $\mathbf{C}$  is real and  $\mathbf{C}^\top = \mathbf{C}$ , and the standard basis of  $\mathbb{R}^3$  is orthonormal.
- (d) Self-adjoint, for every  $S$ . By [Theorem 10.6.2](#) (c) and (d),  $(S^*S)^* = S^*S^{**} = S^*S$ .
- (e) Self-adjoint. Each of  $P_U$  and  $P_{U^\perp}$  is self-adjoint ([Example \(Four Tests\)](#) (d)), and by [Theorem 10.6.2](#) (a) and (b) the difference of self-adjoint operators is self-adjoint:  $(P_U - P_{U^\perp})^* = P_U^* - P_{U^\perp}^* = P_U - P_{U^\perp}$ .
- (f) Self-adjoint, in spite of the non-symmetric matrix. The basis  $\mathcal{C}$  is **not** orthonormal —  $\langle (1, 0), (1, 2) \rangle = 1 \neq 0$  — so [Theorem 10.6.3](#) does not apply to it, and the shape

of  $[T]_{\mathcal{C}}^{\mathcal{C}}$  says nothing either way. Test the definition instead: for  $\mathbf{x}, \mathbf{y} \in \mathbb{R}^2$ ,

$$\langle T\mathbf{x}, \mathbf{y} \rangle = x_2y_1 + x_1y_2 = \langle \mathbf{x}, T\mathbf{y} \rangle,$$

so  $T^* = T$ . In the standard basis, which *is* orthonormal, the matrix is  $\begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}$ , symmetric as it must be. The moral is the one Chapter 10, Section 6 closes with: self-adjointness is a property of the operator and the inner product, never of a matrix taken on its own.

### Exercise B2: Eigenvalues and orthogonality.

- (a) For  $\mathbf{A} = \begin{pmatrix} 0 & -i \\ i & 0 \end{pmatrix} \in M_2(\mathbb{C})$ , check that  $\mathbf{A}$  is Hermitian, find its eigenvalues and an eigenvector for each, and verify that the eigenvectors are orthogonal.
- (b) For  $\mathbf{S} = \begin{pmatrix} 3 & 1 & 1 \\ 1 & 3 & 1 \\ 1 & 1 & 3 \end{pmatrix} \in M_3(\mathbb{R})$ , find the eigenvalues with their multiplicities and bases of the eigenspaces, and verify that the two eigenspaces are orthogonal.
- (c) Hence write down an orthonormal basis of  $\mathbb{R}^3$  consisting of eigenvectors of  $\mathbf{S}$ .

*Solution.*

- (a)  $\mathbf{A}^* = \overline{\mathbf{A}}^\top = \begin{pmatrix} 0 & -i \\ i & 0 \end{pmatrix} = \mathbf{A}$ , so  $\mathbf{A}$  is Hermitian. Then  $\text{tr } \mathbf{A} = 0$  and  $\det \mathbf{A} = 0 - (-i)(i) = -1$ , so  $p_{\mathbf{A}} = x^2 - 1$  and the eigenvalues are 1 and  $-1$ , both real.  $\mathbf{A}(1, i) = (-i \cdot i, i \cdot 1) = (1, i)$ , so  $\mathbf{u} = (1, i)$  works for  $\lambda = 1$ .  $\mathbf{A}(1, -i) = (-1, i) = -(1, -i)$ , so  $\mathbf{v} = (1, -i)$  works for  $\lambda = -1$ . Finally

$$\langle \mathbf{u}, \mathbf{v} \rangle = 1 \cdot \overline{1} + i \cdot \overline{-i} = 1 + i \cdot i = 0.$$

- (b) Expanding the determinant gives  $p_{\mathbf{S}} = (x - 5)(x - 2)^2$ , so  $\text{spec}(\mathbf{S}) = \{5, 2\}$  with algebraic multiplicities 1 and 2. The eigenspaces are read off from  $\mathbf{S} = 2\mathbf{I} + \mathbf{J}$ , where  $\mathbf{J}$  is the all-ones matrix: since  $\mathbf{J}(1, 1, 1) = (3, 3, 3)$  and  $\mathbf{J}\mathbf{x} = \mathbf{0}$  exactly when  $x_1 + x_2 + x_3 = 0$ ,

$$E_5(\mathbf{S}) = \text{span}((1, 1, 1)), \quad E_2(\mathbf{S}) = \text{span}((1, -1, 0), (1, 0, -1)),$$

of dimensions 1 and 2. Orthogonality:  $(1, 1, 1) \cdot (1, -1, 0) = 0$  and  $(1, 1, 1) \cdot (1, 0, -1) = 0$ , so  $E_5(\mathbf{S}) \perp E_2(\mathbf{S})$ , as [Theorem 11.2.3](#) requires.

- (c) The two vectors spanning  $E_2(\mathbf{S})$  are not orthogonal to each other, so run Gram-Schmidt inside that eigenspace ([Theorem 10.2.2](#)). With  $\mathbf{w}_1 = (1, -1, 0)$ ,

$$(1, 0, -1) - \frac{1}{2}(1, -1, 0) = \frac{1}{2}(1, 1, -2).$$

Normalizing all three vectors gives the orthonormal eigenbasis

$$\frac{1}{\sqrt{3}}(1, 1, 1), \quad \frac{1}{\sqrt{2}}(1, -1, 0), \quad \frac{1}{\sqrt{6}}(1, 1, -2).$$

The eigenvalues are 5, 2, 2 in that order.

**Exercise B3: A self-adjoint square root of zero.**

Let  $T \in \mathcal{L}(V)$  be self-adjoint with  $T^2 = 0$ . Prove that  $T = 0$ . Deduce that a self-adjoint  $T$  with  $T^k = 0$  for some  $k \geq 1$  is zero.

*Solution.* For every  $\mathbf{v} \in V$ , by [Definition 10.6.2](#),

$$\|T\mathbf{v}\|^2 = \langle T\mathbf{v}, T\mathbf{v} \rangle = \langle T^2\mathbf{v}, \mathbf{v} \rangle = \langle \mathbf{0}, \mathbf{v} \rangle = 0,$$

so  $T\mathbf{v} = \mathbf{0}$ . As  $\mathbf{v}$  was arbitrary,  $T = 0$ .

For the deduction, use strong induction on  $k$  ([Theorem 0.2.10](#)). If  $k = 1$  there is nothing to prove. Let  $k \geq 2$  with  $T^k = 0$ , and put  $m = \lceil k/2 \rceil$ , so that  $1 \leq m < k$  and  $2m \geq k$ . Then  $(T^m)^2 = T^{2m} = T^{2m-k}T^k = 0$ , and  $T^m$  is self-adjoint, since  $(T^m)^* = (T^*)^m = T^m$  by [Theorem 10.6.2](#) (c). By the first part,  $T^m = 0$ , and the inductive hypothesis applied to the exponent  $m$  gives  $T = 0$ .

**11.2.7.3 C. Going deeper****Exercise C1: The operator  $S^*S$ .**

Let  $S \in \mathcal{L}(V, W)$  between finite-dimensional inner product spaces over  $F$ , and set  $T = S^*S \in \mathcal{L}(V)$ .

- (a) Prove that  $T$  is self-adjoint and that  $\langle T\mathbf{v}, \mathbf{v} \rangle \geq 0$  for every  $\mathbf{v} \in V$ , with equality if and only if  $S\mathbf{v} = \mathbf{0}$ .
- (b) Deduce that every eigenvalue of  $T$  is a non-negative real number.
- (c) Prove that  $T$  is invertible if and only if  $S$  is injective.

*Hint for (a): the quantity  $\langle T\mathbf{v}, \mathbf{v} \rangle$  is a norm squared in disguise.*

*Solution.*

- (a)  $T^* = (S^*S)^* = S^*S^{**} = S^*S = T$ , by [Theorem 10.6.2](#) (c) and (d). For the second claim, use conjugate symmetry and then [Definition 10.6.1](#):

$$\langle T\mathbf{v}, \mathbf{v} \rangle = \overline{\langle \mathbf{v}, S^*S\mathbf{v} \rangle} = \overline{\langle S\mathbf{v}, S\mathbf{v} \rangle} = \|S\mathbf{v}\|^2,$$

the last step because a norm squared is a non-negative real number and is its own conjugate. So  $\langle T\mathbf{v}, \mathbf{v} \rangle \geq 0$ , with equality exactly when  $\|S\mathbf{v}\| = 0$ , that is  $S\mathbf{v} = \mathbf{0}$  ([Definition 10.1.1](#)).

- (b) Let  $T\mathbf{v} = \lambda\mathbf{v}$  with  $\mathbf{v} \neq \mathbf{0}$ . By [Theorem 11.2.1](#),  $\lambda \in \mathbb{R}$ . By (a),  $0 \leq \langle T\mathbf{v}, \mathbf{v} \rangle = \lambda\|\mathbf{v}\|^2$ , and  $\|\mathbf{v}\|^2 > 0$ , so  $\lambda \geq 0$ .
- (c)  $(\Rightarrow)$  If  $S\mathbf{v} = \mathbf{0}$  then  $T\mathbf{v} = \mathbf{0}$ , so injectivity of  $T$  forces  $\mathbf{v} = \mathbf{0}$ ; and an invertible operator is injective.  $(\Leftarrow)$  If  $S$  is injective and  $T\mathbf{v} = \mathbf{0}$ , then by (a)  $\|S\mathbf{v}\|^2 = \langle T\mathbf{v}, \mathbf{v} \rangle = 0$ , so  $S\mathbf{v} = \mathbf{0}$  and  $\mathbf{v} = \mathbf{0}$ . Thus  $\ker T = \{\mathbf{0}\}$ , so  $T$  is injective, hence bijective by [Corollary 3.3.2](#) (e). This is [Corollary 10.6.6](#) (b) again, seen from the quadratic form.

Operators with the property in (a) are called **positive**, and they are the subject of Chapter 12; part (b) is the first half of the characterization proved there.

### Exercise C2: Sums and products.

Let  $S, T \in \mathcal{L}(V)$  be self-adjoint.

- (a) Prove that  $S + T$  and  $cT$  are self-adjoint for every **real** scalar  $c$ , and show by example that  $cT$  need not be self-adjoint for  $c \in \mathbb{C}$ .
- (b) Prove that  $ST$  is self-adjoint if and only if  $ST = TS$ .
- (c) Give self-adjoint  $S, T \in \mathcal{L}(\mathbb{C}^2)$  with  $ST$  not self-adjoint.

*Solution.*

- (a) By Theorem 10.6.2 (a),  $(S + T)^* = S^* + T^* = S + T$ . By (b) of the same theorem,  $(cT)^* = \bar{c}T^* = \bar{c}T$ , which equals  $cT$  exactly when  $\bar{c} = c$  or  $T = 0$ . For a complex counterexample take  $T = \text{id}$  on  $\mathbb{C}^2$  and  $c = i$ : then  $(i \text{id})^* = -i \text{id} \neq i \text{id}$ .
- (b) By Theorem 10.6.2 (c),  $(ST)^* = T^*S^* = TS$ . So  $(ST)^* = ST$  if and only if  $TS = ST$ .
- (c) Take  $S = T_{\mathbf{A}}$  and  $T = T_{\mathbf{B}}$  on  $\mathbb{C}^2$  with

$$\mathbf{A} = \begin{pmatrix} 1 & 0 \\ 0 & 0 \end{pmatrix}, \quad \mathbf{B} = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}.$$

Both are Hermitian. But  $\mathbf{AB} = \begin{pmatrix} 0 & 1 \\ 0 & 0 \end{pmatrix}$  and  $\mathbf{BA} = \begin{pmatrix} 0 & 0 \\ 1 & 0 \end{pmatrix}$ , which differ, so by (b) the product is not self-adjoint. Indeed  $(\mathbf{AB})^* = \mathbf{BA} \neq \mathbf{AB}$ .

### Exercise C3: What the real quadratic form does determine.

Let  $V$  be a finite-dimensional inner product space over  $\mathbb{R}$  and  $T \in \mathcal{L}(V)$ . Call  $T$  **skew-adjoint** if  $T^* = -T$ .

- (a) Prove that  $\langle T\mathbf{v}, \mathbf{v} \rangle = 0$  for every  $\mathbf{v} \in V$  if and only if  $T$  is skew-adjoint.
- (b) Deduce that every  $T \in \mathcal{L}(V)$  can be written uniquely as  $T = A + B$  with  $A$  self-adjoint and  $B$  skew-adjoint, and that  $\langle T\mathbf{v}, \mathbf{v} \rangle = \langle A\mathbf{v}, \mathbf{v} \rangle$  for every  $\mathbf{v}$ .
- (c) Conclude that over  $\mathbb{R}$  the function  $\mathbf{v} \mapsto \langle T\mathbf{v}, \mathbf{v} \rangle$  determines  $A$  and nothing about  $B$ . Contrast this with the complex case.

*Solution.*

- (a) ( $\Leftarrow$ ) If  $T^* = -T$ , then for every  $\mathbf{v}$ ,  $\langle T\mathbf{v}, \mathbf{v} \rangle = \langle \mathbf{v}, T^*\mathbf{v} \rangle = -\langle \mathbf{v}, T\mathbf{v} \rangle = -\langle T\mathbf{v}, \mathbf{v} \rangle$ , the last step by symmetry of a real inner product. A real number equal to its own negative is 0.
- ( $\Rightarrow$ ) Suppose  $\langle T\mathbf{v}, \mathbf{v} \rangle = 0$  for all  $\mathbf{v}$ . Expanding  $\langle T(\mathbf{u} + \mathbf{w}), \mathbf{u} + \mathbf{w} \rangle = 0$  and canceling the two diagonal terms,

$$\langle T\mathbf{u}, \mathbf{w} \rangle + \langle T\mathbf{w}, \mathbf{u} \rangle = 0.$$

Over  $\mathbb{R}$ ,  $\langle T\mathbf{w}, \mathbf{u} \rangle = \langle \mathbf{u}, T\mathbf{w} \rangle$ , so  $\langle T\mathbf{u}, \mathbf{w} \rangle = \langle \mathbf{u}, -T\mathbf{w} \rangle$  for all  $\mathbf{u}, \mathbf{w}$ . By the unique-

ness in [Theorem 10.6.1](#),  $T^* = -T$ .

- (b) Put  $A = \frac{1}{2}(T + T^*)$  and  $B = \frac{1}{2}(T - T^*)$ . Then  $A + B = T$ , and by [Theorem 10.6.2](#) (a), (b) and (d),  $A^* = A$  and  $B^* = -B$ . For uniqueness, if  $T = A' + B'$  is another such splitting, then  $T^* = A' - B'$ , so  $A' = \frac{1}{2}(T + T^*) = A$  and  $B' = B$ . Finally  $\langle B\mathbf{v}, \mathbf{v} \rangle = 0$  for every  $\mathbf{v}$  by (a), so  $\langle T\mathbf{v}, \mathbf{v} \rangle = \langle A\mathbf{v}, \mathbf{v} \rangle + \langle B\mathbf{v}, \mathbf{v} \rangle = \langle A\mathbf{v}, \mathbf{v} \rangle$ .
- (c) By (b), two operators  $T$  and  $T'$  with the same self-adjoint part have the same quadratic form; and conversely, if  $\langle T\mathbf{v}, \mathbf{v} \rangle = \langle T'\mathbf{v}, \mathbf{v} \rangle$  for all  $\mathbf{v}$ , then  $A - A'$  is self-adjoint with vanishing quadratic form, so  $A = A'$  by [Theorem 11.2.5](#). So the quadratic form sees exactly  $A$ . The skew part is invisible: the quarter turn of  $\mathbb{R}^2$  is skew-adjoint, and adding it to any  $T$  changes nothing. Over  $\mathbb{C}$ , by contrast, [Theorem 11.2.6](#) says the quadratic form determines  $T$  completely — there is no invisible part at all, which is why the complex theory is the easier one.

## 11.3 Normal Operators

Section 2 collected everything a self-adjoint operator gives us: real eigenvalues, orthogonal eigenspaces, an invariant orthogonal complement. That is exactly the equipment an orthonormal basis of eigenvectors needs. But over  $\mathbb{C}$  the hypothesis is too heavy, and this section finds the lighter one that still carries the load. It is a condition already met in Chapter 10, and the rest of the chapter runs on it.

As throughout,  $F$  is  $\mathbb{R}$  or  $\mathbb{C}$ , inner products are linear in the first slot, and every inner product space is finite-dimensional unless we say otherwise.

### 11.3.1 The hypothesis that self-adjointness overshoots

Take the operator on  $\mathbb{C}^2$  whose matrix in the standard basis is  $\mathbf{D} = \text{diag}(1, i)$ . The standard basis is orthonormal and consists of eigenvectors of  $\mathbf{D}$ , so this operator has everything the spectral theorem promises. Yet it is not self-adjoint: one of its eigenvalues is  $i$ , and a self-adjoint operator has only real eigenvalues ([Theorem 11.2.1](#)). So over  $\mathbb{C}$  self-adjointness cannot be the condition we are looking for. It is sufficient and not necessary, and we are throwing away every operator with a non-real eigenvalue — including every unitary operator other than the trivial ones.

What survives? Compare the two computations. For a self-adjoint  $T$  we have  $T^*T = TT = TT^*$ . For  $\mathbf{D} = \text{diag}(1, i)$  we have  $\mathbf{D}^* = \text{diag}(1, -i)$ , which is a different matrix, but

$$\mathbf{D}^*\mathbf{D} = \text{diag}(1, 1) = \mathbf{D}\mathbf{D}^*.$$

Neither operator equals its adjoint; both **commute** with their adjoints. That is the common feature, and Chapter 10 already gave it a name.

*A normal operator is one that never gets in its adjoint's way: the two may be applied in either order.*

Recall [Definition 10.6.3](#): an operator  $T \in \mathcal{L}(V)$  is **normal** when  $T^*T = TT^*$ . That is a single equation between two operators, with no vectors and no quantifiers in it, and it is strictly weaker than  $T^* = T$ . Section 6 of Chapter 10 checked the definition on self-adjoint operators, on unitary

ones, on  $i\mathbf{I}$  and on diagonal matrices, and found the  $2 \times 2$  Jordan block  $\mathbf{J}_2(0)$  failing it. We take those for granted and go one step further: normality is best understood as a statement about **lengths**, and that is the content of the first theorem below.

Before it, a wider catalog, since the class is larger than Chapter 10's four examples suggest.

**Example. (Which of these are normal?)**

Decide which of the following matrices are normal, with the standard inner product on  $\mathbb{C}^n$  or  $\mathbb{R}^n$ . For those that are, say which familiar family they belong to.

(a)  $\mathbf{A} = \begin{pmatrix} 2 & 1+i \\ 1-i & 3 \end{pmatrix}$  over  $\mathbb{C}$ .

(b)  $\mathbf{B} = \begin{pmatrix} 0 & 2 \\ -2 & 0 \end{pmatrix}$  over  $\mathbb{R}$ .

(c)  $\mathbf{U} = \frac{1}{\sqrt{2}} \begin{pmatrix} 1 & i \\ i & 1 \end{pmatrix}$  over  $\mathbb{C}$ .

(d)  $\mathbf{C} = \begin{pmatrix} 1 & 2 & 3 \\ 3 & 1 & 2 \\ 2 & 3 & 1 \end{pmatrix}$  over  $\mathbb{C}$ .

(e)  $\mathbf{M} = \begin{pmatrix} 2 & 1 \\ -1 & 2 \end{pmatrix}$  over  $\mathbb{R}$ .

(f)  $\mathbf{N} = \begin{pmatrix} 1 & 1 \\ 0 & 2 \end{pmatrix}$  over  $\mathbb{R}$ .

*Solution.*

(a) Normal.  $\mathbf{A}^* = \mathbf{A}$ , since transposing swaps the off-diagonal entries and conjugating swaps them back, so  $\mathbf{A}$  is Hermitian and both products equal  $\mathbf{A}^2$ . Every **self-adjoint** operator is normal.

(b) Normal.  $\mathbf{B}^\top = -\mathbf{B}$ , so  $\mathbf{B}^\top \mathbf{B} = -\mathbf{B}^2 = \mathbf{B}\mathbf{B}^\top$ . An operator with  $T^* = -T$  is called **skew-adjoint**, and the same one-line computation makes every such operator normal.

(c) Normal.  $\mathbf{U}^* \mathbf{U} = \frac{1}{2} \begin{pmatrix} 1 & -i \\ -i & 1 \end{pmatrix} \begin{pmatrix} 1 & i \\ i & 1 \end{pmatrix} = \frac{1}{2} \begin{pmatrix} 2 & 0 \\ 0 & 2 \end{pmatrix} = \mathbf{I}$ , so  $\mathbf{U}$  is **unitary** (Definition 10.7.2), and then  $\mathbf{U}^* \mathbf{U} = \mathbf{I} = \mathbf{U}\mathbf{U}^*$ .

(d) Normal. Each row of  $\mathbf{C}$  is the previous one shifted cyclically to the right; such a matrix is called a **circulant**. Writing out the products,

$$\mathbf{C}^\top \mathbf{C} = \mathbf{C}\mathbf{C}^\top = \begin{pmatrix} 14 & 11 & 11 \\ 11 & 14 & 11 \\ 11 & 11 & 14 \end{pmatrix}.$$

Both entries in position  $(i, j)$  are the dot product of two rows of  $\mathbf{C}$ , or of two columns, and cyclic shifting makes those two lists of dot products agree. Section 9 explains circulants properly and diagonalizes all of them at once.

(e) Normal, and in none of the families above:  $\mathbf{M}^\top \neq \pm \mathbf{M}$  and  $\mathbf{M}^\top \mathbf{M} = 5\mathbf{I} \neq \mathbf{I}$ . Still,

$$\mathbf{M}^\top \mathbf{M} = \mathbf{M}\mathbf{M}^\top = \begin{pmatrix} 5 & 0 \\ 0 & 5 \end{pmatrix}.$$



This is  $\sqrt{5}$  times a rotation of the plane, which is the rotation-scaling block  $\mathbf{A}(2+i)$  transposed (Definition 9.4.1). Normality is genuinely wider than “self-adjoint or unitary”.

(f) **Not** normal:

$$\mathbf{N}^\top \mathbf{N} = \begin{pmatrix} 1 & 1 \\ 1 & 5 \end{pmatrix}, \quad \mathbf{N} \mathbf{N}^\top = \begin{pmatrix} 2 & 2 \\ 2 & 4 \end{pmatrix}.$$

The  $(1,1)$  entries already disagree,  $1 \neq 2$ . This matrix is diagonalizable, with eigenvalues 1 and 2; Section 4 shows what it costs to be diagonalizable and not normal.

## 11.3.2 Normality is a statement about lengths

The definition compares two operators. The next theorem turns it into a comparison of two numbers, one vector at a time, and that is the form every later proof uses. The engine is the zero test of Section 2: a self-adjoint operator  $S$  with  $\langle S\mathbf{v}, \mathbf{v} \rangle = 0$  for all  $\mathbf{v}$  is zero.

### Theorem 11.3.1 — Characterizations of Normality

Let  $V$  be a finite-dimensional inner product space over  $F$  and let  $T \in \mathcal{L}(V)$ . The following are equivalent.

- (a)  $T$  is normal.
- (b)  $\|T\mathbf{v}\| = \|T^*\mathbf{v}\|$  for every  $\mathbf{v} \in V$ .
- (c)  $\langle T\mathbf{u}, T\mathbf{v} \rangle = \langle T^*\mathbf{u}, T^*\mathbf{v} \rangle$  for all  $\mathbf{u}, \mathbf{v} \in V$ .

Suppose in addition that  $F = \mathbb{C}$ , and set

$$A = \frac{1}{2}(T + T^*), \quad B = \frac{1}{2i}(T - T^*),$$

so that  $A$  and  $B$  are self-adjoint and  $T = A + iB$ . Then (a)–(c) are further equivalent to

- (d)  $AB = BA$ .

*Idea.* Condition (a) says that the operator  $S = T^*T - TT^*$  is zero. That operator is self-adjoint, so by the zero test of Theorem 11.2.5 it is enough to know  $\langle S\mathbf{v}, \mathbf{v} \rangle = 0$  for every  $\mathbf{v}$  — and  $\langle S\mathbf{v}, \mathbf{v} \rangle$  is exactly  $\|T\mathbf{v}\|^2 - \|T^*\mathbf{v}\|^2$ . So (a) and (b) are the same statement read at two different levels of resolution. Condition (c) is (b) with the two slots unglued, and it costs nothing in one direction and a specialization in the other. Condition (d) is the operator version of writing a complex number as  $a + bi$ ; that decomposition appeared in Exercise (C2: Skew-adjoint operators) (c), and multiplying out  $TT^* - T^*T$  in terms of  $A$  and  $B$  leaves only the commutator.

*Proof.*

(a)  $\Rightarrow$  (c). For all  $\mathbf{u}, \mathbf{v} \in V$ , by Definition 10.6.1 and  $T^{**} = T$  (Theorem 10.6.2 (d)),

$$\langle T\mathbf{u}, T\mathbf{v} \rangle = \langle T^*T\mathbf{u}, \mathbf{v} \rangle, \quad \langle T^*\mathbf{u}, T^*\mathbf{v} \rangle = \langle TT^*\mathbf{u}, \mathbf{v} \rangle.$$

Since  $T^*T = TT^*$ , the two right-hand sides agree.

- (b)  $\Rightarrow$  (b). Take  $\mathbf{u} = \mathbf{v}$ . Then  $\|T\mathbf{v}\|^2 = \|T^*\mathbf{v}\|^2$ , and both norms are non-negative reals, so they are equal.
- (c)  $\Rightarrow$  (a). Put  $S = T^*T - TT^*$ . Then  $S$  is self-adjoint: by [Theorem 10.6.2](#) (a), (c) and (d),

$$S^* = (T^*T)^* - (TT^*)^* = T^*T^{**} - T^{**}T^* = T^*T - TT^* = S.$$

For any  $\mathbf{v} \in V$ , using [Definition 10.6.1](#) twice,

$$\begin{aligned}\langle S\mathbf{v}, \mathbf{v} \rangle &= \langle T^*T\mathbf{v}, \mathbf{v} \rangle - \langle TT^*\mathbf{v}, \mathbf{v} \rangle \\ &= \langle T\mathbf{v}, T\mathbf{v} \rangle - \langle T^*\mathbf{v}, T^*\mathbf{v} \rangle \\ &= \|T\mathbf{v}\|^2 - \|T^*\mathbf{v}\|^2 = 0\end{aligned}$$

by (b). Since  $S$  is self-adjoint, [Theorem 11.2.5](#) gives  $S = 0$ , that is  $T^*T = TT^*$ .

Now let  $F = \mathbb{C}$ . By [Theorem 10.6.2](#) (a), (b) and (d),

$$\begin{aligned}A^* &= \frac{1}{2}(T^* + T) = A, \\ B^* &= \overline{\left(\frac{1}{2i}\right)}(T^* - T) = \frac{1}{2i}(T - T^*) = B,\end{aligned}$$

so both are self-adjoint, and  $A+iB = \frac{1}{2}(T+T^*) + \frac{1}{2}(T-T^*) = T$ ; this is the decomposition of [Exercise \(C2: Skew-adjoint operators\)](#) (c). Expanding  $T = A + iB$  and  $T^* = A - iB$ ,

$$\begin{aligned}TT^* &= (A + iB)(A - iB) = A^2 + B^2 + i(BA - AB), \\ T^*T &= (A - iB)(A + iB) = A^2 + B^2 + i(AB - BA).\end{aligned}$$

Subtracting,  $TT^* - T^*T = 2i(BA - AB)$ . So  $T$  is normal exactly when  $AB = BA$ , which is (a)  $\Leftrightarrow$  (d). This proves the theorem. ■

Condition (b) is the one to remember. It says that  $T$  and  $T^*$  stretch every single vector by the same amount, even though they need not move it in the same direction. Everything in the next three sections is squeezed out of that sentence. Condition (d) records that normality over  $\mathbb{C}$  is the operator analogue of the fact that any two real numbers commute: split  $T$  into a “real part” and an “imaginary part”, and normality is precisely the failure of those two parts to interfere.

*Quick check.* The symmetric matrix  $\begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}$  and the skew-symmetric matrix  $\begin{pmatrix} 0 & 1 \\ -1 & 0 \end{pmatrix}$  are both normal, the first being self-adjoint and the second skew-adjoint. Is their sum normal?

*Solution.* No. The sum is  $\begin{pmatrix} 0 & 2 \\ 0 & 0 \end{pmatrix} = 2\mathbf{J}_2(0)$ , which is twice the standard non-normal example: it kills  $\mathbf{e}_1$  and sends  $\mathbf{e}_2$  to a vector of length 2, while its transpose does the opposite, so condition (b) fails at  $\mathbf{v} = \mathbf{e}_1$ . The normal operators do not form a subspace of  $\mathcal{L}(V)$ , and they are not closed under composition either.

### 11.3.3 What normality buys

Four consequences, all short, all obtained by turning some statement about  $T$  into the same statement about  $T^*$ .

#### Corollary 11.3.2 — Kernel, Image and an Orthogonal Splitting

Let  $T \in \mathcal{L}(V)$  be normal on a finite-dimensional inner product space  $V$ . Then:

- (a)  $\ker T = \ker T^*$ ;
- (b)  $\operatorname{im} T = \operatorname{im} T^*$ ;
- (c)  $V = \ker T \oplus \operatorname{im} T$ , and the two summands are orthogonal to each other.

*Proof.*

- (a) For  $\mathbf{v} \in V$ , [Theorem 11.3.1](#) (b) gives  $\|T\mathbf{v}\| = \|T^*\mathbf{v}\|$ . So  $T\mathbf{v} = \mathbf{0}$  if and only if  $\|T\mathbf{v}\| = 0$ , if and only if  $\|T^*\mathbf{v}\| = 0$ , if and only if  $T^*\mathbf{v} = \mathbf{0}$ .
- (b) By [Theorem 10.6.5](#) (b) applied to  $T^*$  in place of  $T$ , and using  $T^{**} = T$  ([Theorem 10.6.2](#) (d)),

$$\operatorname{im} T = \operatorname{im} T^{**} = (\ker T^*)^\perp = (\ker T)^\perp = \operatorname{im} T^*,$$

where the third equality is (a) and the fourth is [Theorem 10.6.5](#) (b) for  $T$ .

- (c) By (b),  $\operatorname{im} T = (\ker T)^\perp$ , and  $V = \ker T \oplus (\ker T)^\perp$  by [Theorem 10.3.2](#) (a). Orthogonality of the summands is the definition of the orthogonal complement. ■

Part (c) fails badly without normality, and the failure is not subtle: for a general operator  $\ker T$  and  $\operatorname{im} T$  can be **equal**. Take  $T$  with matrix  $\mathbf{J}_2(0) = \begin{pmatrix} 0 & 1 \\ 0 & 0 \end{pmatrix}$  on  $\mathbb{C}^2$ . Then  $T\mathbf{e}_1 = \mathbf{0}$  and  $T\mathbf{e}_2 = \mathbf{e}_1$ , so

$$\ker T = \operatorname{span}(\mathbf{e}_1) = \operatorname{im} T.$$

The sum  $\ker T + \operatorname{im} T$  is one-dimensional, so it is neither direct nor all of  $\mathbb{C}^2$ . Normality is exactly what rules this out.

#### Theorem 11.3.3 — An Eigenvector of a Normal Operator Is an Eigenvector of Its Adjoint

Let  $T \in \mathcal{L}(V)$  be normal, let  $\lambda \in F$  and let  $\mathbf{v} \in V$ . Then

$$T\mathbf{v} = \lambda\mathbf{v} \quad \text{if and only if} \quad T^*\mathbf{v} = \bar{\lambda}\mathbf{v}.$$

In particular  $E_\lambda(T) = E_{\bar{\lambda}}(T^*)$ , and  $\lambda \in \operatorname{spec}(T)$  if and only if  $\bar{\lambda} \in \operatorname{spec}(T^*)$ .

*Idea.* Both sides say that  $\mathbf{v}$  lies in a kernel: the left in  $\ker(T - \lambda \operatorname{id}_V)$ , the right in  $\ker(T^* - \bar{\lambda} \operatorname{id}_V)$ . Those two operators are adjoint to each other, so [Corollary 11.3.2](#) (a) closes the gap — provided  $T - \lambda \operatorname{id}_V$  is itself normal, which is the only thing to check.

*Proof.* Write  $S = T - \lambda \text{id}_V$ . By [Theorem 10.6.2](#) (a), (b) and (d),  $S^* = T^* - \bar{\lambda} \text{id}_V$ . Since  $T^*T = TT^*$  and  $\text{id}_V$  commutes with everything,

$$\begin{aligned} S^*S &= T^*T - \bar{\lambda}T - \lambda T^* + |\lambda|^2 \text{id}_V \\ &= TT^* - \lambda T^* - \bar{\lambda}T + |\lambda|^2 \text{id}_V = SS^*, \end{aligned}$$

so  $S$  is normal. By [Corollary 11.3.2](#) (a),  $\ker S = \ker S^*$ . Now  $T\mathbf{v} = \lambda\mathbf{v}$  says  $\mathbf{v} \in \ker S$ , and  $T^*\mathbf{v} = \bar{\lambda}\mathbf{v}$  says  $\mathbf{v} \in \ker S^*$ , so the two conditions agree. The statement about eigenspaces is the equality  $\ker S = \ker S^*$  rewritten, and the statement about spectra follows because an eigenspace is non-zero exactly when the scalar is an eigenvalue ([Theorem 8.2.1](#)).

■

The theorem is the reason normal operators are so easy to handle:  $T$  and  $T^*$  have the *same* eigenvectors, with conjugate eigenvalues. The proof of the next corollary is the classical self-adjoint argument ([Theorem 11.2.3](#)) with one conjugate bar inserted, and that bar is exactly what [Theorem 11.3.3](#) supplies.

### Corollary 11.3.4 — Eigenvectors for Distinct Eigenvalues Are Orthogonal

Let  $T \in \mathcal{L}(V)$  be normal, let  $\lambda \neq \mu$  be eigenvalues of  $T$ , and let  $T\mathbf{u} = \lambda\mathbf{u}$  and  $T\mathbf{v} = \mu\mathbf{v}$ . Then  $\langle \mathbf{u}, \mathbf{v} \rangle = 0$ . Hence  $E_\lambda(T) \perp E_\mu(T)$  whenever  $\lambda \neq \mu$ .

*Proof.* Compute  $\langle T\mathbf{u}, \mathbf{v} \rangle$  in two ways. First,  $\langle T\mathbf{u}, \mathbf{v} \rangle = \langle \lambda\mathbf{u}, \mathbf{v} \rangle = \lambda\langle \mathbf{u}, \mathbf{v} \rangle$ . Second, by [Definition 10.6.1](#) and then [Theorem 11.3.3](#) applied to  $\mathbf{v}$ ,

$$\langle T\mathbf{u}, \mathbf{v} \rangle = \langle \mathbf{u}, T^*\mathbf{v} \rangle = \langle \mathbf{u}, \bar{\mu}\mathbf{v} \rangle = \bar{\mu}\langle \mathbf{u}, \mathbf{v} \rangle,$$

the last step because a scalar leaves the second slot conjugated. Subtracting,  $(\lambda - \bar{\mu})\langle \mathbf{u}, \mathbf{v} \rangle = 0$ , and  $\lambda - \bar{\mu} \neq 0$ , so  $\langle \mathbf{u}, \mathbf{v} \rangle = 0$ . Every vector of  $E_\lambda(T)$  is orthogonal to every vector of  $E_\mu(T)$  by the same computation.

■

### Theorem 11.3.5 — A Normal Operator Has No Hidden Kernel

Let  $T \in \mathcal{L}(V)$  be normal. Then  $\ker T^k = \ker T$  for every integer  $k \geq 1$ . In particular, a normal nilpotent operator is the zero operator.

*Idea.* The inclusion  $\ker T \subseteq \ker T^2$  is free, and the whole content is the reverse one. If  $T^2\mathbf{v} = \mathbf{0}$ , then the vector  $\mathbf{w} = T\mathbf{v}$  lies in  $\ker T$ , hence also in  $\ker T^*$  by [Corollary 11.3.2](#); and a vector that is killed by  $T^*$  and is itself in the image of  $T$  must be  $\mathbf{0}$ , because pairing it with itself moves  $T$  across the inner product and hits the  $T^*$ .

*Proof.* First take  $k = 2$ . The inclusion  $\ker T \subseteq \ker T^2$  holds for every operator. Conversely, suppose  $T^2\mathbf{v} = \mathbf{0}$  and put  $\mathbf{w} = T\mathbf{v}$ . Then  $T\mathbf{w} = \mathbf{0}$ , so  $T^*\mathbf{w} = \mathbf{0}$  by [Corol-](#)

lary 11.3.2 (a). Therefore, by Definition 10.6.1,

$$\begin{aligned}\|\mathbf{w}\|^2 &= \langle \mathbf{w}, \mathbf{w} \rangle = \langle T\mathbf{v}, \mathbf{w} \rangle \\ &= \langle \mathbf{v}, T^*\mathbf{w} \rangle = \langle \mathbf{v}, \mathbf{0} \rangle = 0,\end{aligned}$$

so  $\mathbf{w} = T\mathbf{v} = \mathbf{0}$  and  $\mathbf{v} \in \ker T$ . Hence  $\ker T^2 = \ker T$ .

Now induct on  $k$ . The cases  $k = 1, 2$  are done. Let  $k \geq 2$  and assume  $\ker T^k = \ker T$ . If  $T^{k+1}\mathbf{v} = \mathbf{0}$ , then  $T^2(T^{k-1}\mathbf{v}) = \mathbf{0}$ , so  $T^k\mathbf{v} = T(T^{k-1}\mathbf{v}) = \mathbf{0}$  by the case  $k = 2$ , and then  $T\mathbf{v} = \mathbf{0}$  by the inductive hypothesis. So  $\ker T^{k+1} \subseteq \ker T$ , and the reverse inclusion is again free.

Finally, if  $T$  is nilpotent (Definition 9.2.1), say  $T^k = 0$ , then  $\ker T = \ker T^k = V$ , so  $T = 0$ . ■

This is the sentence that explains why normal operators never produce Jordan blocks. Over  $\mathbb{C}$ , a Jordan block of size  $m \geq 2$  for the eigenvalue  $\lambda$  contributes a vector killed by  $(T - \lambda \text{id}_V)^m$  but not by  $T - \lambda \text{id}_V$  (Definition 9.2.2). For a normal  $T$  the operator  $T - \lambda \text{id}_V$  is normal, as the proof of Theorem 11.3.3 showed, so Theorem 11.3.5 forbids such a vector. Every block in the Jordan form of a normal operator therefore has size 1, and a normal operator on a complex space is diagonalizable (Theorem 9.3.1, Definition 8.4.1).

*Remark.* That is diagonalizable, not yet **orthogonally** diagonalizable. The eigenspaces of distinct eigenvalues are already mutually orthogonal by Corollary 11.3.4, so all that is missing is an orthonormal basis inside each eigenspace, which Gram–Schmidt supplies (Theorem 10.2.2). Section 4 proves the whole statement in one stroke instead, from Schur’s theorem, because that proof also delivers the converse and works without Chapter 9.

Two converses are worth flagging now, so that you know they are coming and do not try to prove them here. A normal operator whose eigenvalues are all real is self-adjoint, and a normal operator whose eigenvalues all have modulus 1 is unitary. Both are false without normality, and neither can be reached with the tools of this section: they need an orthonormal basis of eigenvectors, which is the business of Section 4.

Before the warning, one structural fact deserves to be stated where later sections can lean on it. It is the reason a normal operator can be taken apart orthogonally at all.

### Proposition 11.3.6 — An Invariant Subspace of a Normal Operator Is Reducing

Let  $T \in \mathcal{L}(V)$  be normal on a finite-dimensional inner product space, and let  $U \subseteq V$  be  $T$ -invariant. Then  $U$  is  $T^*$ -invariant,  $U^\perp$  is  $T$ -invariant, and  $T|_U$  is a normal operator on  $U$ .

*Idea.* Write the matrix in an orthonormal basis adapted to  $U$ . Invariance kills the block below the diagonal, and normality, read on the top-left block alone, kills the block above it. The trace is what converts “these two blocks agree” into “that block is zero”.

*Proof.* Choose an orthonormal basis  $(\mathbf{e}_1, \dots, \mathbf{e}_r)$  of  $U$  and extend it to an orthonormal basis  $\mathcal{B}$  of  $V$  (Corollary 10.2.3). Since  $T\mathbf{e}_j \in U$  for  $j \leq r$ , the first  $r$  columns of  $\mathbf{A} = [T]_{\mathcal{B}}^{\mathcal{B}}$  have zeros below row  $r$ , so

$$\mathbf{A} = \begin{pmatrix} \mathbf{B} & \mathbf{C} \\ \mathbf{0} & \mathbf{D} \end{pmatrix},$$

with  $\mathbf{B}$  of size  $r \times r$ . Because  $\mathcal{B}$  is orthonormal,  $[T^*]_{\mathcal{B}}^{\mathcal{B}} = \mathbf{A}^*$  (Theorem 10.6.3), and  $T$  normal means  $\mathbf{A}^*\mathbf{A} = \mathbf{A}\mathbf{A}^*$ . Comparing the  $(1,1)$  blocks (Theorem 7.1.1),

$$(\mathbf{A}^*\mathbf{A})_{11} = \mathbf{B}^*\mathbf{B}, \quad (\mathbf{A}\mathbf{A}^*)_{11} = \mathbf{B}\mathbf{B}^* + \mathbf{C}\mathbf{C}^*.$$

Taking traces and using  $\text{tr}(\mathbf{B}^*\mathbf{B}) = \text{tr}(\mathbf{B}\mathbf{B}^*)$  (Theorem 0.9.6 (3)) gives  $\text{tr}(\mathbf{C}\mathbf{C}^*) = 0$ . But  $\text{tr}(\mathbf{C}\mathbf{C}^*) = \sum_{i,j} |c_{ij}|^2$ , a sum of non-negative reals, so  $\mathbf{C} = \mathbf{0}$ .

Hence  $\mathbf{A} = \mathbf{B} \oplus \mathbf{D}$ . Reading the last  $n - r$  columns,  $T\mathbf{e}_j \in U^\perp$  for  $j > r$ , so  $U^\perp$  is  $T$ -invariant; reading the first  $r$  columns of  $\mathbf{A}^*$ ,  $U$  is  $T^*$ -invariant. Finally  $[T|_U] = \mathbf{B}$  and the  $(1,1)$  blocks above now read  $\mathbf{B}^*\mathbf{B} = \mathbf{B}\mathbf{B}^*$ , so  $T|_U$  is normal. This shows all three claims. ■

*Warning. Normality cannot be read off a corner.* A principal submatrix of a normal matrix need not be normal. The  $3 \times 3$  cyclic shift

$$\mathbf{S} = \begin{pmatrix} 0 & 0 & 1 \\ 1 & 0 & 0 \\ 0 & 1 & 0 \end{pmatrix}$$

is a permutation matrix, hence orthogonal (Exercise (B3: Permutations are orthogonal)), hence normal; but deleting its last row and column leaves  $\begin{pmatrix} 0 & 0 \\ 1 & 0 \end{pmatrix}$ , the transpose of  $\mathbf{J}_2(0)$ , which is not normal. Equivalently: the operator that  $\mathbf{S}$  compresses to  $\text{span}(\mathbf{e}_1, \mathbf{e}_2)$  is not normal, because that subspace is not  $\mathbf{S}$ -invariant. What **is** true is that an invariant subspace of a normal operator is automatically invariant under the adjoint too, and the restriction there is normal; that is Proposition 11.3.6, proved just above.

## 11.3.4 Exercises

### 11.3.4.1 A. Check your understanding

#### Exercise A1.

- State the definition of a normal operator, and state one equivalent condition phrased in terms of norms.
- True or false: every self-adjoint operator is normal, and every normal operator is self-adjoint. Justify both halves.
- Let  $T$  be normal with  $T\mathbf{v} = 3\mathbf{v}$ . What is  $T^*\mathbf{v}$ ? What if  $T\mathbf{v} = 3i\mathbf{v}$ ?
- Give a normal operator on  $\mathbb{R}^2$  that is neither self-adjoint nor unitary.
- True or false: if  $S$  and  $T$  are normal then  $S + T$  is normal. Justify your answer.

- (f) Explain in one sentence why a normal operator cannot be nilpotent unless it is zero.

*Solution.*

- (a)  $T$  is normal when  $T^*T = TT^*$  (Definition 10.6.3). Equivalently,  $\|T\mathbf{v}\| = \|T^*\mathbf{v}\|$  for every  $\mathbf{v} \in V$  (Theorem 11.3.1 (b)).
- (b) The first half is true: if  $T^* = T$  then both products are  $T^2$ . The second half is false:  $\text{diag}(1, i)$  on  $\mathbb{C}^2$  is normal and has a non-real eigenvalue, so it is not self-adjoint (Theorem 11.2.1).
- (c)  $T^*\mathbf{v} = 3\mathbf{v}$  in the first case and  $T^*\mathbf{v} = -3i\mathbf{v}$  in the second, by Theorem 11.3.3: the adjoint has the conjugate eigenvalue on the same eigenvector.
- (d)  $\begin{pmatrix} 2 & 1 \\ -1 & 2 \end{pmatrix}$  works (Example (Which of these are normal?)) (e): it is not symmetric, and  $\mathbf{M}^\top \mathbf{M} = 5\mathbf{I} \neq \mathbf{I}$ . Any  $\mathbf{A}(\lambda)$  with  $|\lambda| \neq 1$  and  $\text{Im } \lambda \neq 0$  does.
- (e) False.  $\begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix} + \begin{pmatrix} 0 & 1 \\ -1 & 0 \end{pmatrix} = \begin{pmatrix} 0 & 2 \\ 0 & 0 \end{pmatrix}$  is a sum of two normal matrices and is not normal.
- (f) If  $T^k = 0$  then  $\ker T = \ker T^k = V$  by Theorem 11.3.5, so  $T = 0$ .

#### 11.3.4.2 B. Practice

##### Exercise B1: Determine which are normal.

Determine which of the following matrices are normal, with the standard inner product. Justify your answer; for those that are not, exhibit a vector  $\mathbf{v}$  with  $\|\mathbf{A}\mathbf{v}\| \neq \|\mathbf{A}^*\mathbf{v}\|$ .

- (a)  $\begin{pmatrix} 1 & 1 \\ 1 & 1 \end{pmatrix}$  over  $\mathbb{R}$ .
- (b)  $\begin{pmatrix} 1 & 1 \\ 0 & 1 \end{pmatrix}$  over  $\mathbb{R}$ .
- (c)  $\begin{pmatrix} 0 & i \\ i & 0 \end{pmatrix}$  over  $\mathbb{C}$ .
- (d)  $\begin{pmatrix} 1 & 0 & 0 \\ 0 & 0 & -1 \\ 0 & 1 & 0 \end{pmatrix}$  over  $\mathbb{R}$ .
- (e)  $\begin{pmatrix} i & 1 \\ 0 & i \end{pmatrix}$  over  $\mathbb{C}$ .

*Solution.*

- (a) Normal: the matrix is symmetric, so it is self-adjoint over  $\mathbb{R}$ , and self-adjoint implies normal.
- (b) Not normal.  $\mathbf{A}^\top \mathbf{A} = \begin{pmatrix} 1 & 1 \\ 1 & 2 \end{pmatrix}$  and  $\mathbf{A}\mathbf{A}^\top = \begin{pmatrix} 2 & 1 \\ 1 & 1 \end{pmatrix}$ . With  $\mathbf{v} = \mathbf{e}_1$ ,  $\|\mathbf{A}\mathbf{e}_1\| = \|(1, 0)\| = 1$  while  $\|\mathbf{A}^\top \mathbf{e}_1\| = \|(1, 1)\| = \sqrt{2}$ .
- (c) Normal.  $\mathbf{A}^* = \begin{pmatrix} 0 & -i \\ -i & 0 \end{pmatrix} = -\mathbf{A}$ , so  $\mathbf{A}$  is skew-adjoint and  $\mathbf{A}^*\mathbf{A} = -\mathbf{A}^2 = \mathbf{A}\mathbf{A}^*$ .

- (Concretely both products are  $\mathbf{I}$ .)
- (d) Normal: the matrix is  $(1) \oplus \mathbf{A}(i)$ , a direct sum of a  $1 \times 1$  block and a rotation by a quarter turn, and its columns are orthonormal, so it is orthogonal ([Theorem 10.7.3 \(f\)](#)).
- (e) Not normal.  $\mathbf{A}^* \mathbf{A} = \begin{pmatrix} 1 & -i \\ i & 2 \end{pmatrix}$  and  $\mathbf{A} \mathbf{A}^* = \begin{pmatrix} 2 & -i \\ i & 1 \end{pmatrix}$ ; with  $\mathbf{v} = \mathbf{e}_1$  we get  $\|\mathbf{A} \mathbf{e}_1\| = |i| = 1$  and  $\|\mathbf{A}^* \mathbf{e}_1\| = \|(-i, 1)\| = \sqrt{2}$ . Adding  $i\mathbf{I}$  to a non-normal matrix does not repair it:  $\mathbf{A} = i\mathbf{I} + \mathbf{J}_2(0)$  and  $i\mathbf{I}$  is normal, but sums of normal matrices need not be normal.

### Exercise B2: The splitting of a singular normal operator.

Let  $\mathbf{C} = \begin{pmatrix} 1 & -1 & 0 \\ 0 & 1 & -1 \\ -1 & 0 & 1 \end{pmatrix} \in M_3(\mathbb{R})$ , and let  $T = T_{\mathbf{C}}$  be the operator it defines on  $\mathbb{R}^3$ .

- (a) Verify that  $T$  is normal.
- (b) Compute  $\ker T$  and  $\ker T^*$  and check that they agree.
- (c) Compute  $\operatorname{im} T$ , and verify directly that  $\mathbb{R}^3 = \ker T \oplus \operatorname{im} T$  with the two summands orthogonal, as [Corollary 11.3.2 \(c\)](#) predicts.

*Solution.*

- (a) Each row of  $\mathbf{C}$  is the previous one shifted cyclically, so  $\mathbf{C}$  is a circulant. Explicitly,

$$\mathbf{C}^{\top} \mathbf{C} = \mathbf{C} \mathbf{C}^{\top} = \begin{pmatrix} 2 & -1 & -1 \\ -1 & 2 & -1 \\ -1 & -1 & 2 \end{pmatrix},$$

so  $T$  is normal.

- (b)  $\mathbf{C} \mathbf{x} = \mathbf{0}$  reads  $x_1 = x_2 = x_3$ , because the first two rows give  $x_1 = x_2$  and  $x_2 = x_3$ . So  $\ker T = \operatorname{span}((1, 1, 1))$ . The transpose  $\mathbf{C}^{\top}$  has rows  $(1, 0, -1)$ ,  $(-1, 1, 0)$ ,  $(0, -1, 1)$ , and  $\mathbf{C}^{\top} \mathbf{x} = \mathbf{0}$  again reads  $x_1 = x_3$  and  $x_1 = x_2$ . The two kernels agree, as [Corollary 11.3.2 \(a\)](#) requires.
- (c) The rank is  $3 - 1 = 2$  by [Theorem 3.3.1](#), and the first two columns  $(1, 0, -1)$  and  $(-1, 1, 0)$  are independent, so

$$\operatorname{im} T = \operatorname{span}((1, 0, -1), (-1, 1, 0)).$$

Both spanning vectors have coordinate sum 0, hence are orthogonal to  $(1, 1, 1)$ , so  $\operatorname{im} T \perp \ker T$ . Since  $\dim \ker T + \dim \operatorname{im} T = 1 + 2 = 3$  and the two subspaces meet only in  $\mathbf{0}$  (being orthogonal),  $\mathbb{R}^3 = \ker T \oplus \operatorname{im} T$ . Indeed  $\operatorname{im} T = \{\mathbf{x} : x_1 + x_2 + x_3 = 0\} = (\ker T)^{\perp}$ .

### Exercise B3: Triangular and normal, by hand.

Let  $\mathbf{A} = \begin{pmatrix} a & b \\ 0 & d \end{pmatrix} \in M_2(\mathbb{C})$ .



- (a) Compute the  $(1, 1)$  entries of  $\mathbf{A}^*\mathbf{A}$  and of  $\mathbf{A}\mathbf{A}^*$ .
- (b) Hence prove that  $\mathbf{A}$  is normal if and only if  $b = 0$ , that is, if and only if  $\mathbf{A}$  is diagonal.
- (c) Which upper triangular matrices in  $M_2(\mathbb{C})$  are unitary?

*Hint: for (a), you only need one entry of each product.*

*Solution.*

- (a) The  $(1, 1)$  entry of  $\mathbf{A}^*\mathbf{A}$  is the squared norm of the first **column** of  $\mathbf{A}$ , namely  $|a|^2$ . The  $(1, 1)$  entry of  $\mathbf{A}\mathbf{A}^*$  is the squared norm of the first **row**, namely  $|a|^2 + |b|^2$ . In full,

$$\mathbf{A}^*\mathbf{A} = \begin{pmatrix} |a|^2 & \bar{a}b \\ a\bar{b} & |b|^2 + |d|^2 \end{pmatrix},$$

$$\mathbf{A}\mathbf{A}^* = \begin{pmatrix} |a|^2 + |b|^2 & b\bar{d} \\ \bar{b}d & |d|^2 \end{pmatrix}.$$

- (b)  $(\Rightarrow)$  If  $\mathbf{A}$  is normal, comparing the  $(1, 1)$  entries gives  $|a|^2 = |a|^2 + |b|^2$ , so  $|b|^2 = 0$  and  $b = 0$ .  $(\Leftarrow)$  A diagonal matrix is normal, since  $\mathbf{D}^*\mathbf{D} = \mathbf{D}\mathbf{D}^* = \text{diag}(|a|^2, |d|^2)$ .
- (c) A unitary matrix is normal, so by (b) it must be diagonal,  $\text{diag}(a, d)$ , and then  $\mathbf{A}^*\mathbf{A} = \text{diag}(|a|^2, |d|^2) = \mathbf{I}$  forces  $|a| = |d| = 1$ . Conversely every such matrix is unitary. So the upper triangular unitary matrices are exactly  $\text{diag}(a, d)$  with  $|a| = |d| = 1$ .

### 11.3.4.3 C. Going deeper

#### Exercise C1: A normal operator with no proper invariant subspace.

Let  $V$  be a finite-dimensional complex inner product space with  $V \neq \{\mathbf{0}\}$ , and let  $T \in \mathcal{L}(V)$  be normal. Suppose the only  $T$ -invariant subspaces of  $V$  are  $\{\mathbf{0}\}$  and  $V$ .

- (a) Show that  $\dim V = 1$ .
- (b) Show that the conclusion fails over  $\mathbb{R}$ , by exhibiting a normal operator on  $\mathbb{R}^2$  whose only invariant subspaces are  $\{\mathbf{0}\}$  and  $\mathbb{R}^2$ .
- (c) Where exactly does the argument of (a) use that the field is  $\mathbb{C}$ ?

*Hint: an eigenvector spans an invariant subspace.*

*Solution.*

- (a) Since  $V \neq \{\mathbf{0}\}$  and the field is  $\mathbb{C}$ ,  $T$  has an eigenvector  $\mathbf{v} \neq \mathbf{0}$  (Theorem 8.2.4). Then  $\text{span}(\mathbf{v})$  is  $T$ -invariant and non-zero, so by hypothesis  $\text{span}(\mathbf{v}) = V$  and  $\dim V = 1$ . Normality is not needed for this direction.
- (b) The quarter turn  $\mathbf{R}_{\pi/2} = \begin{pmatrix} 0 & -1 \\ 1 & 0 \end{pmatrix}$  is orthogonal, hence normal. A 1-dimensional invariant subspace would be spanned by a real eigenvector, and  $\mathbf{R}_{\pi/2}$  has none: its characteristic polynomial is  $x^2 + 1$ , with no real root. So its only invariant subspaces are  $\{\mathbf{0}\}$  and  $\mathbb{R}^2$ , while  $\dim \mathbb{R}^2 = 2$ .
- (c) Only in the existence of an eigenvector. Over  $\mathbb{C}$  that is the Fundamental Theorem

of Algebra reaching the characteristic polynomial; over  $\mathbb{R}$  the polynomial may have no root, which is exactly what (b) exploits. Note that [Proposition 11.3.6](#) still holds over  $\mathbb{R}$  — it says nothing about invariant subspaces *existing*.

### Exercise C2: Arithmetic of normal operators.

Let  $S, T \in \mathcal{L}(V)$  be normal.

- (a) Prove that  $T^*$  is normal and that  $cT$  is normal for every  $c \in F$ .
- (b) Prove that  $p(T)$  is normal for every polynomial  $p \in F[x]$ .
- (c) Give a concrete  $S$  and  $T$ , both normal, with  $ST$  not normal. (Part (b) shows that this cannot happen when  $S$  is a polynomial in  $T$ .)

*Solution.*

- (a)  $(T^*)^*T^* = TT^* = T^*T = T^*(T^*)^*$ , using  $T^{**} = T$  ([Theorem 10.6.2](#) (d)). For  $cT$ , by [Theorem 10.6.2](#) (b) we have  $(cT)^* = \bar{c}T^*$ , so

$$(cT)^*(cT) = \bar{c}cT^*T = |c|^2 TT^* = (cT)(cT)^*.$$

- (b) Write  $p(x) = \sum_k a_k x^k$ . Then  $p(T)^* = \sum_k \bar{a}_k (T^*)^k$  by [Theorem 10.6.2](#) (a), (b) and (c). Every power of  $T$  commutes with every power of  $T^*$ : from  $TT^* = T^*T$  one moves a  $T^*$  past one  $T$  at a time. Hence  $p(T)$  and  $p(T)^*$ , being sums of products of such powers, commute, and  $p(T)$  is normal.
- (c) Take  $S = \begin{pmatrix} 1 & 0 \\ 0 & 2 \end{pmatrix}$  and  $T = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}$ , both symmetric and hence normal. Then

$$ST = \begin{pmatrix} 0 & 1 \\ 2 & 0 \end{pmatrix},$$

$$(ST)^\top(ST) = \begin{pmatrix} 4 & 0 \\ 0 & 1 \end{pmatrix}, \quad (ST)(ST)^\top = \begin{pmatrix} 1 & 0 \\ 0 & 4 \end{pmatrix},$$

so  $ST$  is not normal.

### Exercise C3: Normality depends on the inner product.

Normality is a property of  $T$  **together with** an inner product, not of  $T$  alone. Work on  $V = \mathbb{R}^2$  with the operator  $T$  whose standard matrix is  $\mathbf{A} = \begin{pmatrix} 0 & -1 \\ 1 & 0 \end{pmatrix}$ .

- (a) Verify that  $T$  is normal for the standard inner product.
- (b) Now use  $\langle \mathbf{x}, \mathbf{y} \rangle_{\mathbf{G}} = \mathbf{y}^\top \mathbf{G} \mathbf{x}$  with  $\mathbf{G} = \text{diag}(1, 2)$ , which is an inner product on  $\mathbb{R}^2$ . Using the rule  $[T^*]_{\mathcal{E}}^{\mathcal{E}} = \mathbf{G}^{-1} \mathbf{A}^\top \mathbf{G}$  in the standard basis  $\mathcal{E}$  ([Exercise \(C3: The adjoint in an arbitrary basis\)](#)), compute the matrix of the adjoint of  $T$  for this inner product.
- (c) Show that  $T$  is **not** normal for  $\langle \cdot, \cdot \rangle_{\mathbf{G}}$ , and reconcile this with (a).

*Solution.*

- (a)  $\mathbf{A}^\top \mathbf{A} = \mathbf{A} \mathbf{A}^\top = \mathbf{I}$ , so  $T$  is even orthogonal, hence normal.  
 (b)  $\mathbf{G}^{-1} = \text{diag}(1, \frac{1}{2})$ , so

$$\begin{aligned} \mathbf{G}^{-1} \mathbf{A}^\top \mathbf{G} &= \begin{pmatrix} 1 & 0 \\ 0 & \frac{1}{2} \end{pmatrix} \begin{pmatrix} 0 & 1 \\ -1 & 0 \end{pmatrix} \begin{pmatrix} 1 & 0 \\ 0 & 2 \end{pmatrix} \\ &= \begin{pmatrix} 0 & 2 \\ -\frac{1}{2} & 0 \end{pmatrix}. \end{aligned}$$

Call this matrix  $\mathbf{A}^*$ .

- (c) Computing both products,

$$\mathbf{A} \mathbf{A}^* = \begin{pmatrix} \frac{1}{2} & 0 \\ 0 & 2 \end{pmatrix}, \quad \mathbf{A}^* \mathbf{A} = \begin{pmatrix} 2 & 0 \\ 0 & \frac{1}{2} \end{pmatrix},$$

which differ, so  $T$  is not normal for  $\langle \cdot, \cdot \rangle_{\mathbf{G}}$ . There is no contradiction: the adjoint is built from the inner product (Definition 10.6.1), so changing the inner product changes  $T^*$ , and with it the equation  $T^*T = TT^*$ . The standard basis is orthonormal for the first inner product and not for the second, which is exactly the hypothesis Theorem 10.6.3 needs.

## 11.4 The Complex Spectral Theorem

Two things are now on the table. Section 1 put every complex operator into upper triangular form in an **orthonormal** basis, and Section 3 identified the operators that commute with their adjoints. Put them together and the chapter's main theorem falls out in a few lines: a triangular matrix that is normal has nothing above the diagonal. This section proves that, reads off the consequences, and says exactly how much of the resulting picture is unique.

The field is  $\mathbb{C}$  throughout this section, and every inner product space is finite-dimensional. The reason for insisting on  $\mathbb{C}$  is Schur's theorem, and behind it the Fundamental Theorem of Algebra (Theorem 5.5.1): over  $\mathbb{R}$  an operator can fail to have a single eigenvector, and Section 5 shows what has to change.

### 11.4.1 Triangular plus normal forces diagonal

Start with the smallest case, where the mechanism is visible at a glance. Let

$$\mathbf{A} = \begin{pmatrix} a & b \\ 0 & d \end{pmatrix} \in M_2(\mathbb{C})$$

be upper triangular. The  $(1,1)$  entry of  $\mathbf{A}^* \mathbf{A}$  is the squared length of the first **column** of  $\mathbf{A}$ , and the  $(1,1)$  entry of  $\mathbf{A} \mathbf{A}^*$  is the squared length of the first **row**:

$$(\mathbf{A}^* \mathbf{A})_{11} = |a|^2, \quad (\mathbf{A} \mathbf{A}^*)_{11} = |a|^2 + |b|^2.$$

If  $\mathbf{A}$  is normal these are equal, so  $|b|^2 = 0$  and  $b = 0$ . One entry of one product did all the work. (This is Exercise (B3: Triangular and normal, by hand), which you may already have done.)

The general case is the same comparison run down the diagonal, one row at a time. Nothing is clever; the only care needed is to use, at step  $j$ , the zeros that steps  $1, \dots, j-1$  have already produced.

### Lemma 11.4.1 — A Normal Triangular Matrix Is Diagonal

Let  $\mathbf{T} = (t_{ij}) \in M_n(\mathbb{C})$  be upper triangular and normal. Then  $\mathbf{T}$  is diagonal.

*Idea.* For any square matrix, the  $j$ -th diagonal entry of  $\mathbf{T}^*\mathbf{T}$  is the squared length of column  $j$ , and the  $j$ -th diagonal entry of  $\mathbf{T}\mathbf{T}^*$  is the squared length of row  $j$ . Normality says those two lengths agree for every  $j$ . For an upper triangular matrix, column  $j$  reaches only down to the diagonal and row  $j$  only rightwards from it, so the equation compares “what sits above  $t_{jj}$ ” with “what sits to the right of  $t_{jj}$ ”. Row 1 has nothing above it, so the right side must be empty too; that clears row 1, which empties the column sums one step further down, and the argument walks down the diagonal.

*Proof.* For each  $j$ , computing the  $(j, j)$  entries of the two products entry by entry,

$$(\mathbf{T}^*\mathbf{T})_{jj} = \sum_{i=1}^n \overline{t_{ij}} t_{ij} = \sum_{i=1}^n |t_{ij}|^2, \quad (\mathbf{T}\mathbf{T}^*)_{jj} = \sum_{k=1}^n |t_{jk}|^2.$$

Since  $\mathbf{T}$  is upper triangular,  $t_{ij} = 0$  whenever  $i > j$ , and  $t_{jk} = 0$  whenever  $k < j$ . So normality of  $\mathbf{T}$  says

$$\sum_{i \leq j} |t_{ij}|^2 = \sum_{k \geq j} |t_{jk}|^2 \quad (j = 1, \dots, n). \quad (11.4.1)$$

We prove by induction on  $j$  that  $t_{jk} = 0$  for every  $k > j$ .

**Base case**  $j = 1$ . The left-hand side of Eq. 11.4.1 has the single term  $|t_{11}|^2$ , so

$$|t_{11}|^2 = |t_{11}|^2 + \sum_{k>1} |t_{1k}|^2.$$

Hence  $\sum_{k>1} |t_{1k}|^2 = 0$ , a sum of non-negative reals, so  $t_{1k} = 0$  for every  $k > 1$ .

**Inductive step.** Let  $1 < j \leq n$  and suppose  $t_{mk} = 0$  whenever  $m < j$  and  $k > m$ . In the left-hand side of Eq. 11.4.1, every term with  $i < j$  is  $|t_{ij}|^2$  with  $i < j$ , hence with  $j > i$ ; by the inductive hypothesis, taking  $m = i$  and  $k = j$ , each such  $t_{ij}$  is 0. So the left-hand side reduces to  $|t_{jj}|^2$  and Eq. 11.4.1 becomes

$$|t_{jj}|^2 = |t_{jj}|^2 + \sum_{k>j} |t_{jk}|^2,$$

whence  $t_{jk} = 0$  for every  $k > j$ , again because a sum of non-negative reals vanishes only when every term does.

By induction, every entry strictly above the diagonal is zero. Every entry strictly below it is zero because  $\mathbf{T}$  is upper triangular. So  $\mathbf{T}$  is diagonal, as claimed. ■

Note what the proof did **not** need: no eigenvector was chosen, no dimension was inducted on, no subspace was split off. The whole argument is an identity between two lists of non-negative numbers. That is why it also survives, unchanged in spirit, into the block version used in Section

5 and the quantitative version in Section 11.

## 11.4.2 The theorem

### Theorem 11.4.2 — Complex Spectral Theorem

Let  $V$  be a finite-dimensional inner product space over  $\mathbb{C}$  and let  $T \in \mathcal{L}(V)$ . The following are equivalent.

- (a)  $T$  is normal.
- (b)  $V$  has an orthonormal basis consisting of eigenvectors of  $T$ .

*Idea.*  $(\Leftarrow)$  is a one-line check: in such a basis the matrix of  $T$  is diagonal, the matrix of  $T^*$  is its conjugate transpose, and two diagonal matrices commute.  $(\Rightarrow)$  is Schur plus the lemma: take an orthonormal basis making the matrix triangular, observe that normality of  $T$  is normality of that matrix because the basis is orthonormal, and apply [Lemma 11.4.1](#). A diagonal matrix in the basis  $\mathcal{B}$  says precisely that each vector of  $\mathcal{B}$  is an eigenvector.

*Proof.*  $(\Leftarrow)$  Suppose  $\mathcal{B} = (\mathbf{e}_1, \dots, \mathbf{e}_n)$  is an orthonormal basis with  $T\mathbf{e}_j = \lambda_j\mathbf{e}_j$  for each  $j$ . Then  $\mathbf{D} := [T]_{\mathcal{B}}^{\mathcal{B}} = \text{diag}(\lambda_1, \dots, \lambda_n)$ , and since  $\mathcal{B}$  is orthonormal,  $[T^*]_{\mathcal{B}}^{\mathcal{B}} = \mathbf{D}^* = \text{diag}(\overline{\lambda_1}, \dots, \overline{\lambda_n})$  by [Theorem 10.6.3](#). Both products are the same diagonal matrix:

$$\mathbf{D}^*\mathbf{D} = \mathbf{D}\mathbf{D}^* = \text{diag}(|\lambda_1|^2, \dots, |\lambda_n|^2).$$

Taking the matrix of an operator in a fixed basis turns composition into matrix multiplication and is injective ([Theorem 3.6.2](#), [Theorem 3.6.4](#)), so  $T^*T = TT^*$ .

$(\Rightarrow)$  Suppose  $T$  is normal. By Schur's theorem ([Theorem 11.1.2](#)) there is an orthonormal basis  $\mathcal{B} = (\mathbf{e}_1, \dots, \mathbf{e}_n)$  of  $V$  in which  $\mathbf{T} := [T]_{\mathcal{B}}^{\mathcal{B}}$  is upper triangular. Because  $\mathcal{B}$  is orthonormal, [Theorem 10.6.3](#) gives  $[T^*]_{\mathcal{B}}^{\mathcal{B}} = \mathbf{T}^*$ , so the equation  $T^*T = TT^*$  reads  $\mathbf{T}^*\mathbf{T} = \mathbf{T}\mathbf{T}^*$ : the matrix  $\mathbf{T}$  is normal. By [Lemma 11.4.1](#),  $\mathbf{T}$  is diagonal, say  $\mathbf{T} = \text{diag}(\lambda_1, \dots, \lambda_n)$ . Reading off the  $j$ -th column ([Definition 3.6.1](#)),  $T\mathbf{e}_j = \lambda_j\mathbf{e}_j$ , and  $\mathbf{e}_j \neq \mathbf{0}$  because  $\|\mathbf{e}_j\| = 1$ . So  $\mathcal{B}$  is an orthonormal basis of eigenvectors of  $T$ . This proves the theorem. ■

### Corollary 11.4.3 — Unitary Diagonalization

Let  $\mathbf{A} \in M_n(\mathbb{C})$ . Then  $\mathbf{A}$  is normal if and only if there exist a unitary  $\mathbf{U} \in U(n)$  and a diagonal  $\mathbf{D} \in M_n(\mathbb{C})$  with

$$\mathbf{A} = \mathbf{U}\mathbf{D}\mathbf{U}^*.$$

A matrix admitting such a factorization is called **unitarily diagonalizable**. The diagonal of  $\mathbf{D}$  is the list of eigenvalues of  $\mathbf{A}$  with multiplicity, in an order that may be prescribed in advance, and the columns of  $\mathbf{U}$  are a corresponding orthonormal basis of eigenvectors.

*Proof.* Apply [Theorem 11.4.2](#) to the operator  $T_{\mathbf{A}}$  on  $\mathbb{C}^n$  with its standard inner product, for which the standard basis is orthonormal.

( $\Rightarrow$ ) Let  $(\mathbf{u}_1, \dots, \mathbf{u}_n)$  be an orthonormal basis of eigenvectors,  $\mathbf{A}\mathbf{u}_j = \lambda_j\mathbf{u}_j$ , and let  $\mathbf{U}$  be the matrix with these columns; it is unitary because its columns are orthonormal ([Theorem 10.7.3](#) (f)). Then  $\mathbf{A}\mathbf{U}$  has  $j$ -th column  $\lambda_j\mathbf{u}_j$ , which is the  $j$ -th column of  $\mathbf{U}\mathbf{D}$  for  $\mathbf{D} = \text{diag}(\lambda_1, \dots, \lambda_n)$ . So  $\mathbf{A}\mathbf{U} = \mathbf{U}\mathbf{D}$  and  $\mathbf{A} = \mathbf{U}\mathbf{D}\mathbf{U}^{-1} = \mathbf{U}\mathbf{D}\mathbf{U}^*$ .

( $\Leftarrow$ ) If  $\mathbf{A} = \mathbf{U}\mathbf{D}\mathbf{U}^*$  then  $\mathbf{A}^* = \mathbf{U}\mathbf{D}^*\mathbf{U}^*$ , and

$$\mathbf{A}^*\mathbf{A} = \mathbf{U}\mathbf{D}^*\mathbf{D}\mathbf{U}^* = \mathbf{U}\mathbf{D}\mathbf{D}^*\mathbf{U}^* = \mathbf{A}\mathbf{A}^*,$$

using  $\mathbf{U}^*\mathbf{U} = \mathbf{I}$  twice and the fact that diagonal matrices commute.

Finally,  $\mathbf{A}$  is similar to  $\mathbf{D}$ , so they have the same characteristic polynomial ([Theorem 6.10.3](#)), and the roots of  $\det(x\mathbf{I} - \mathbf{D}) = \prod_j (x - \lambda_j)$  are the diagonal entries with multiplicity. Reordering the eigenvectors reorders the columns of  $\mathbf{U}$  and the diagonal of  $\mathbf{D}$  in the same way, so any order may be prescribed. ■

#### Example. (Unitarily diagonalizing a Hermitian matrix)

Find a unitary  $\mathbf{U}$  and a diagonal  $\mathbf{D}$  with  $\mathbf{U}^*\mathbf{A}\mathbf{U} = \mathbf{D}$ , for

$$\mathbf{A} = \begin{pmatrix} 2 & 1-i \\ 1+i & 3 \end{pmatrix}.$$

*Solution.*  $\mathbf{A}^* = \mathbf{A}$ , so  $\mathbf{A}$  is Hermitian, hence normal, and [Corollary 11.4.3](#) applies. Its characteristic polynomial is

$$\begin{aligned} x^2 - (\text{tr } \mathbf{A})x + \det \mathbf{A} &= x^2 - 5x + (6 - (1-i)(1+i)) \\ &= x^2 - 5x + 4, \end{aligned}$$

since  $(1-i)(1+i) = 1-i^2 = 2$ . So the eigenvalues are 4 and 1, both real as [Theorem 11.2.1](#) promises.

For  $\lambda = 4$ , the first row of  $\mathbf{A} - 4\mathbf{I} = \begin{pmatrix} -2 & 1-i \\ 1+i & -1 \end{pmatrix}$  gives  $-2z_1 + (1-i)z_2 = 0$ ; taking  $z_2 = 2$  gives  $\mathbf{u} = (1-i, 2)$ , with  $\|\mathbf{u}\|^2 = 2 + 4 = 6$ .

For  $\lambda = 1$ , the first row of  $\mathbf{A} - \mathbf{I} = \begin{pmatrix} 1 & 1-i \\ 1+i & 2 \end{pmatrix}$  gives  $z_1 = -(1-i)z_2$ ; taking  $z_2 = 1$  gives  $\mathbf{w} = (-1+i, 1)$ , with  $\|\mathbf{w}\|^2 = 2 + 1 = 3$ .

The two are orthogonal, as [Corollary 11.3.4](#) predicts:

$$\begin{aligned} \langle \mathbf{u}, \mathbf{w} \rangle &= (1-i)\overline{(-1+i)} + 2 \cdot \bar{1} \\ &= (1-i)(-1-i) + 2 = -2 + 2 = 0. \end{aligned}$$

Normalizing and using them as columns,

$$\mathbf{U} = \begin{pmatrix} \frac{1-i}{\sqrt{6}} & \frac{-1+i}{\sqrt{3}} \\ \frac{2}{\sqrt{6}} & \frac{1}{\sqrt{3}} \end{pmatrix}, \quad \mathbf{D} = \begin{pmatrix} 4 & 0 \\ 0 & 1 \end{pmatrix},$$

and  $\mathbf{U}^*\mathbf{U} = \mathbf{I}$ ,  $\mathbf{U}^*\mathbf{A}\mathbf{U} = \mathbf{D}$ . As a check on the arithmetic,  $\text{tr } \mathbf{D} = 5 = \text{tr } \mathbf{A}$  and  $\det \mathbf{D} = 4 = \det \mathbf{A}$ .

### 11.4.3 Reading the class off the spectrum

Once the eigenvalues sit on a diagonal, every question about  $T$  that is invariant under a unitary change of basis becomes a question about a list of complex numbers. Here are three answers.

#### Corollary 11.4.4 — The Spectrum Decides

Let  $T \in \mathcal{L}(V)$  be normal on a finite-dimensional complex inner product space.

- (a)  $T$  is self-adjoint if and only if every eigenvalue of  $T$  is real.
- (b)  $T$  is unitary if and only if every eigenvalue of  $T$  has modulus 1.
- (c)  $\langle T\mathbf{v}, \mathbf{v} \rangle$  is a non-negative real number for every  $\mathbf{v} \in V$  if and only if every eigenvalue of  $T$  is a non-negative real.

*Idea.* By [Theorem 11.4.2](#) there is an orthonormal basis  $(\mathbf{e}_1, \dots, \mathbf{e}_n)$  with  $T\mathbf{e}_j = \lambda_j\mathbf{e}_j$ , and then  $T^*\mathbf{e}_j = \overline{\lambda_j}\mathbf{e}_j$  by [Theorem 11.3.3](#). Two operators that agree on a basis are equal, so each of the three conditions can be tested one basis vector at a time, where it becomes a statement about the single number  $\lambda_j$ .

*Proof.* Fix an orthonormal basis  $(\mathbf{e}_1, \dots, \mathbf{e}_n)$  of eigenvectors,  $T\mathbf{e}_j = \lambda_j\mathbf{e}_j$  ([Theorem 11.4.2](#)), and recall  $T^*\mathbf{e}_j = \overline{\lambda_j}\mathbf{e}_j$  ([Theorem 11.3.3](#)). Every eigenvalue of  $T$  occurs among  $\lambda_1, \dots, \lambda_n$ , because  $\text{diag}(\lambda_1, \dots, \lambda_n)$  is the matrix of  $T$  and its characteristic polynomial is  $\prod_j (x - \lambda_j)$ .

- (a)  $T = T^*$  holds if and only if  $T\mathbf{e}_j = T^*\mathbf{e}_j$  for every  $j$  ([Theorem 3.1.5](#)), that is  $\lambda_j\mathbf{e}_j = \overline{\lambda_j}\mathbf{e}_j$ , that is  $\lambda_j = \overline{\lambda_j}$  since  $\mathbf{e}_j \neq \mathbf{0}$ . A complex number equals its conjugate exactly when it is real.
- (b)  $T^*T\mathbf{e}_j = T^*(\lambda_j\mathbf{e}_j) = \lambda_j\overline{\lambda_j}\mathbf{e}_j = |\lambda_j|^2\mathbf{e}_j$ , so  $T^*T = \text{id}_V$  if and only if  $|\lambda_j|^2 = 1$  for every  $j$ . That is what it means for  $T$  to be unitary ([Definition 10.7.2](#)): the equation  $T^*T = \text{id}_V$  makes  $T$  injective, hence invertible on a finite-dimensional space ([Theorem 3.5.7](#)), with  $T^{-1} = T^*$  and therefore  $TT^* = \text{id}_V$  as well.
- (c) Write  $\mathbf{v} = \sum_j c_j\mathbf{e}_j$ . Expanding both slots and using  $\langle \mathbf{e}_j, \mathbf{e}_i \rangle = \delta_{ij}$ ,

$$\langle T\mathbf{v}, \mathbf{v} \rangle = \left\langle \sum_j c_j \lambda_j \mathbf{e}_j, \sum_i c_i \mathbf{e}_i \right\rangle = \sum_j \lambda_j |c_j|^2.$$

( $\Leftarrow$ ) If every  $\lambda_j$  is a non-negative real, this is a sum of non-negative reals. ( $\Rightarrow$ ) Taking  $\mathbf{v} = \mathbf{e}_j$  gives  $\langle T\mathbf{e}_j, \mathbf{e}_j \rangle = \lambda_j$ , which is then a non-negative real by hypothesis.

■

Part (c) names a class that Chapter 12 takes as its subject: the **positive** operators. Over  $\mathbb{C}$  the hypothesis of normality is not really needed in (c), because  $\langle T\mathbf{v}, \mathbf{v} \rangle$  real for every  $\mathbf{v}$  already forces  $T = T^*$ : apply [Theorem 11.2.6](#) to the operator  $T - T^*$ , for which  $\langle (T - T^*)\mathbf{v}, \mathbf{v} \rangle = \langle T\mathbf{v}, \mathbf{v} \rangle - \langle T^*\mathbf{v}, \mathbf{v} \rangle = 0$ . Over  $\mathbb{R}$  that shortcut is unavailable, and Section 5 shows the quarter turn defeating it.

The next corollary is a small shock. The adjoint is defined through the inner product and has nothing to do with polynomials. For a normal operator it is one anyway.

### Corollary 11.4.5 — The Adjoint of a Normal Operator Is a Polynomial in It

Let  $T \in \mathcal{L}(V)$  be normal on a finite-dimensional complex inner product space, with distinct eigenvalues  $\lambda_1, \dots, \lambda_k$ . Then there is a polynomial  $p \in \mathbb{C}[x]$  of degree less than  $k$  with  $T^* = p(T)$ .

*Proof.* The scalars  $\lambda_1, \dots, \lambda_k$  are distinct, so by Lagrange interpolation ([Theorem 5.6.1](#)) there is a polynomial  $p \in \mathbb{C}[x]$  of degree less than  $k$  with

$$p(\lambda_i) = \overline{\lambda_i} \quad (i = 1, \dots, k).$$

Let  $(\mathbf{e}_1, \dots, \mathbf{e}_n)$  be an orthonormal basis of eigenvectors ([Theorem 11.4.2](#)), with  $T\mathbf{e}_j = \mu_j\mathbf{e}_j$  where each  $\mu_j$  is one of the  $\lambda_i$ . Then  $p(T)\mathbf{e}_j = p(\mu_j)\mathbf{e}_j = \overline{\mu_j}\mathbf{e}_j = T^*\mathbf{e}_j$ , the first equality because  $\mathbf{e}_j$  is an eigenvector ([Exercise \(B3: Eigenvalues of powers and inverses\)](#) (a)) and the last by [Theorem 11.3.3](#). The operators  $p(T)$  and  $T^*$  agree on a basis, hence are equal ([Theorem 3.1.5](#)). ■

One consequence is worth stating on its own. Any operator  $S$  that commutes with  $T$  also commutes with every polynomial in  $T$ , hence with  $T^*$ . So for normal operators the apparently stronger requirement “commutes with  $T$  and with  $T^*$ ” is no requirement at all, and Section 7 will use this to diagonalize whole commuting families at once. The polynomial depends on  $T$  and not only on its size: for  $T = \text{id}_V$  one may take the constant  $p = 1$ , while for the matrix  $\text{diag}(1, i)$  the conditions  $p(1) = 1$  and  $p(i) = -i$  are met by

$$p(x) = ix + (1 - i),$$

as one checks by substituting  $x = 1$  and  $x = i$ .

## 11.4.4 Exactly what is unique

The factorization  $\mathbf{A} = \mathbf{U}\mathbf{D}\mathbf{U}^*$  is not unique, and it is worth being precise about which parts of it are.

Unique: the **set** of eigenvalues  $\text{spec}(T)$ , and for each  $\lambda \in \text{spec}(T)$  the eigenspace  $E_\lambda(T)$  with its dimension. These are defined from  $T$  alone, without any choice. Consequently the diagonal of  $\mathbf{D}$  is unique **as a list with multiplicities up to reordering**: each  $\lambda$  appears exactly  $\dim E_\lambda(T)$  times.

Not unique: the order of the diagonal, and the matrix  $\mathbf{U}$ . Two freedoms create all the others. First, permuting the eigenvectors permutes the columns of  $\mathbf{U}$  and the diagonal of  $\mathbf{D}$  in



step. Second, within a single eigenspace  $E_\lambda(T)$  of dimension  $m$ , **any** orthonormal basis of that eigenspace will do, and there are many when  $m \geq 2$ : the choices are related by the unitary maps of  $E_\lambda(T)$ . For  $\mathbf{A} = \mathbf{I}_n$ , for instance, every unitary  $\mathbf{U}$  satisfies  $\mathbf{U}^* \mathbf{A} \mathbf{U} = \mathbf{I}_n$ . What no freedom can change is that eigenvectors for distinct eigenvalues stay orthogonal (Corollary 11.3.4), so the decomposition

$$V = E_{\lambda_1}(T) \oplus \cdots \oplus E_{\lambda_k}(T)$$

into mutually orthogonal eigenspaces is itself unique. Section 8 turns that decomposition into the spectral resolution and shows it determines  $T$  completely.

*Quick check.* The matrix  $\mathbf{A} = \text{diag}(1, 1, 2)$  is normal. Write down two different unitary matrices  $\mathbf{U}$  with  $\mathbf{U}^* \mathbf{A} \mathbf{U} = \mathbf{A}$ , and explain why, for  $\text{diag}(1, 2, 3)$  with the order of the diagonal fixed, the freedom shrinks to a single unit-modulus scalar on each column.

*Solution.* Take  $\mathbf{U}_1 = \mathbf{I}_3$  and  $\mathbf{U}_2 = \begin{pmatrix} 0 & 1 & 0 \\ 1 & 0 & 0 \\ 0 & 0 & 1 \end{pmatrix}$ ; more generally  $\mathbf{R} \oplus (1)$  works for **any**

$\mathbf{R} \in \mathbf{U}(2)$ , because  $\mathbf{A} = \mathbf{I}_2 \oplus (2)$  and  $\mathbf{R}^* \mathbf{I}_2 \mathbf{R} = \mathbf{I}_2$ . The freedom comes from the two-dimensional eigenspace  $E_1(\mathbf{A}) = \text{span}(\mathbf{e}_1, \mathbf{e}_2)$ , inside which every orthonormal basis is as good as any other. For  $\text{diag}(1, 2, 3)$  each eigenspace is a line, and a line has exactly two unit vectors over  $\mathbb{R}$  and a circle of them over  $\mathbb{C}$ ; the residual freedom is only the scalar factor of modulus 1 in each column, which changes  $\mathbf{U}$  but not the decomposition into eigenspaces.

*Warning.* “**Diagonalizable**” and “**unitarily diagonalizable**” are different **conditions**. Every unitarily diagonalizable matrix is diagonalizable, because  $\mathbf{U}$  is in particular invertible. The converse fails, and it fails for matrices as small and as ordinary as  $\begin{pmatrix} 1 & 1 \\ 0 & 2 \end{pmatrix}$ . Having  $n$  distinct eigenvalues guarantees diagonalizability (Corollary 8.4.2) and guarantees nothing about orthogonality.

#### Example. (Diagonalizable, and no orthonormal eigenbasis)

For  $\mathbf{A} = \begin{pmatrix} 1 & 1 \\ 0 & 2 \end{pmatrix} \in M_2(\mathbb{R})$ , find an invertible  $\mathbf{P}$  with  $\mathbf{P}^{-1} \mathbf{A} \mathbf{P}$  diagonal, and show that no **orthogonal**  $\mathbf{Q}$  makes  $\mathbf{Q}^\top \mathbf{A} \mathbf{Q}$  diagonal.

*Solution.* The matrix is upper triangular, so its eigenvalues are the diagonal entries 1 and 2 (Theorem 6.4.6). For  $\lambda = 1$ ,  $\mathbf{A} - \mathbf{I} = \begin{pmatrix} 0 & 1 \\ 0 & 1 \end{pmatrix}$  has kernel spanned by  $(1, 0)$ . For  $\lambda = 2$ ,  $\mathbf{A} - 2\mathbf{I} = \begin{pmatrix} -1 & 1 \\ 0 & 0 \end{pmatrix}$  has kernel spanned by  $(1, 1)$ . Two distinct eigenvalues in dimension 2 give a basis of eigenvectors (Corollary 8.4.2), so with

$$\mathbf{P} = \begin{pmatrix} 1 & 1 \\ 0 & 1 \end{pmatrix} \quad \text{we get} \quad \mathbf{P}^{-1} \mathbf{A} \mathbf{P} = \begin{pmatrix} 1 & 0 \\ 0 & 2 \end{pmatrix}.$$

Now suppose  $\mathbf{Q}^\top \mathbf{A} \mathbf{Q} = \mathbf{D}$  were diagonal with  $\mathbf{Q}$  orthogonal. Then  $\mathbf{A} = \mathbf{Q} \mathbf{D} \mathbf{Q}^\top$  would be normal by Corollary 11.4.3. But

$$\mathbf{A}^\top \mathbf{A} = \begin{pmatrix} 1 & 1 \\ 1 & 5 \end{pmatrix} \neq \begin{pmatrix} 2 & 2 \\ 2 & 4 \end{pmatrix} = \mathbf{A} \mathbf{A}^\top,$$

so  $\mathbf{A}$  is not normal and no such  $\mathbf{Q}$  exists. The obstruction is visible without any theory: the eigenvectors are  $(1, 0)$  and  $(1, 1)$ , the eigenspaces are lines and therefore the *only* candidates for eigenvector directions, and  $(1, 0) \cdot (1, 1) = 1 \neq 0$ . There simply is no orthogonal pair of eigenvectors to build  $\mathbf{Q}$  from.

## 11.4.5 Exercises

### 11.4.5.1 A. Check your understanding

#### Exercise A1.

- State the complex spectral theorem, including every hypothesis, and state its matrix form.
- What goes wrong with the proof if the field is  $\mathbb{R}$ ? Name the exact step.
- True or false: a matrix in  $M_n(\mathbb{C})$  with  $n$  distinct eigenvalues is unitarily diagonalizable. Justify your answer.
- Let  $\mathbf{A} \in M_3(\mathbb{C})$  be normal with  $\text{spec}(\mathbf{A}) = \{2, 2, -1\}$  as a list with multiplicity. Is the unitary  $\mathbf{U}$  in  $\mathbf{A} = \mathbf{U}\mathbf{D}\mathbf{U}^*$  determined by  $\mathbf{A}$  and the order of  $\mathbf{D}$ ? What is determined?
- A normal operator  $T$  satisfies  $T^* = T^3$ . What can you say about its eigenvalues?
- Explain why [Lemma 11.4.1](#) cannot be weakened to “a normal matrix with some of its below-diagonal entries zero is diagonal”.

*Solution.*

- For a finite-dimensional inner product space  $V$  over  $\mathbb{C}$  and  $T \in \mathcal{L}(V)$ :  $T$  is normal if and only if  $V$  has an orthonormal basis of eigenvectors of  $T$  ([Theorem 11.4.2](#)). Matrix form:  $\mathbf{A} \in M_n(\mathbb{C})$  is normal if and only if  $\mathbf{A} = \mathbf{U}\mathbf{D}\mathbf{U}^*$  with  $\mathbf{U}$  unitary and  $\mathbf{D}$  diagonal ([Corollary 11.4.3](#)).
- Schur’s theorem. It needs an eigenvalue at every stage of its induction, which over  $\mathbb{C}$  comes from [Theorem 5.5.1](#); over  $\mathbb{R}$  an operator may have none at all, and then there is no triangular form to start from.
- False.  $\begin{pmatrix} 1 & 1 \\ 0 & 2 \end{pmatrix}$  has the distinct eigenvalues 1 and 2 and is not normal, hence is not unitarily diagonalizable ([Example \(Diagonalizable, and no orthonormal eigenbasis\)](#)).
- No. The eigenspace  $E_2(\mathbf{A})$  is two-dimensional, and replacing its orthonormal basis by any other one changes  $\mathbf{U}$  while leaving  $\mathbf{D}$  alone. What is determined:  $\text{spec}(\mathbf{A}) = \{2, -1\}$ , the eigenspaces  $E_2(\mathbf{A})$  and  $E_{-1}(\mathbf{A})$ , their dimensions 2 and 1, and their mutual orthogonality.
- By [Theorem 11.3.3](#),  $T\mathbf{v} = \lambda\mathbf{v}$  gives  $T^*\mathbf{v} = \bar{\lambda}\mathbf{v}$ , while  $T^* = T^3$  gives  $T^*\mathbf{v} = \lambda^3\mathbf{v}$ . Hence  $\bar{\lambda} = \lambda^3$ . Taking moduli,  $|\lambda| = |\lambda|^3$ , so  $|\lambda|$  is 0 or 1. If  $\lambda \neq 0$ , multiplying  $\bar{\lambda} = \lambda^3$  by  $\lambda$  gives  $|\lambda|^2 = \lambda^4$ , that is  $\lambda^4 = 1$ . So every eigenvalue lies in  $\{0, 1, -1, i, -i\}$ .
- The lemma uses triangularity at **every** index, not at a few places: the comparison of row and column lengths at step  $j$  is informative only because the whole of column  $j$  below the diagonal is already known to vanish. Zeros in some of those positions

buy nothing. The circulant  $\begin{pmatrix} 1 & 1 & 0 \\ 0 & 1 & 1 \\ 1 & 0 & 1 \end{pmatrix}$  is normal and has zeros in positions  $(2, 1)$  and  $(3, 2)$ , two of the three places below its diagonal, and it is not diagonal.

### 11.4.5.2 B. Practice

#### Exercise B1: Unitarily diagonalize.

For each matrix, find a unitary  $\mathbf{U}$  and a diagonal  $\mathbf{D}$  with  $\mathbf{U}^* \mathbf{A} \mathbf{U} = \mathbf{D}$ .

(a)  $\mathbf{A} = \begin{pmatrix} 3 & -i \\ i & 3 \end{pmatrix}$ .

(b)  $\mathbf{A} = \frac{1}{\sqrt{2}} \begin{pmatrix} 1 & i \\ i & 1 \end{pmatrix}$ .

*Solution.*

- (a)  $\mathbf{A}^* = \mathbf{A}$ , so  $\mathbf{A}$  is Hermitian and normal. Its characteristic polynomial is  $x^2 - 6x + (9 - (-i)(i)) = x^2 - 6x + 8$ , with roots 4 and 2.

For  $\lambda = 4$ :  $\mathbf{A} - 4\mathbf{I} = \begin{pmatrix} -1 & -i \\ i & -1 \end{pmatrix}$ , and the first row gives  $z_1 = -iz_2$ ; take  $\mathbf{u} = (-i, 1)$ ,

$\|\mathbf{u}\|^2 = 2$ . For  $\lambda = 2$ :  $\mathbf{A} - 2\mathbf{I} = \begin{pmatrix} 1 & -i \\ i & 1 \end{pmatrix}$ , and the first row gives  $z_1 = iz_2$ ; take

$\mathbf{w} = (i, 1)$ ,  $\|\mathbf{w}\|^2 = 2$ . They are orthogonal:  $\langle \mathbf{u}, \mathbf{w} \rangle = (-i)\bar{i} + 1 = (-i)(-i) + 1 = -1 + 1 = 0$ . Hence

$$\mathbf{U} = \frac{1}{\sqrt{2}} \begin{pmatrix} -i & i \\ 1 & 1 \end{pmatrix}, \quad \mathbf{D} = \begin{pmatrix} 4 & 0 \\ 0 & 2 \end{pmatrix}.$$

- (b)  $\mathbf{A}$  is unitary (Example (Which of these are normal?)) (c)), hence normal. Here  $\text{tr } \mathbf{A} = \sqrt{2}$  and  $\det \mathbf{A} = \frac{1}{2}(1 - i^2) = 1$ , so the characteristic polynomial is  $x^2 - \sqrt{2}x + 1$ , with roots

$$x = \frac{\sqrt{2} \pm \sqrt{2-4}}{2} = \frac{1 \pm i}{\sqrt{2}},$$

both of modulus 1, as Corollary 11.4.4 (b) requires. For  $\lambda = (1 + i)/\sqrt{2}$ , the equation  $\mathbf{A}\mathbf{z} = \lambda\mathbf{z}$  reads  $\frac{1}{\sqrt{2}}(z_1 + iz_2) = \frac{1+i}{\sqrt{2}}z_1$ , that is  $iz_2 = iz_1$ , so  $\mathbf{u} = (1, 1)$ .

For  $\lambda = (1 - i)/\sqrt{2}$  the same line gives  $iz_2 = -iz_1$ , so  $\mathbf{w} = (1, -1)$ . These are orthogonal, so

$$\mathbf{U} = \frac{1}{\sqrt{2}} \begin{pmatrix} 1 & 1 \\ 1 & -1 \end{pmatrix}, \quad \mathbf{D} = \frac{1}{\sqrt{2}} \begin{pmatrix} 1+i & 0 \\ 0 & 1-i \end{pmatrix}.$$

Here  $\mathbf{U}$  is real orthogonal even though  $\mathbf{A}$  is not real.

#### Exercise B2: A circulant, diagonalized.

Let  $\omega = e^{2\pi i/3}$ , so  $\omega^3 = 1$  and  $1 + \omega + \omega^2 = 0$ , and let

$$\mathbf{C} = \begin{pmatrix} 1 & 1 & 0 \\ 0 & 1 & 1 \\ 1 & 0 & 1 \end{pmatrix} \in M_3(\mathbb{C}).$$

- (a) Verify that  $\mathbf{C}$  is normal.
- (b) Show that  $\mathbf{f}_j = \frac{1}{\sqrt{3}}(1, \omega^j, \omega^{2j})$  is an eigenvector of  $\mathbf{C}$  for  $j = 0, 1, 2$ , and find the eigenvalue.
- (c) Deduce a unitary  $\mathbf{F}$  with  $\mathbf{F}^*\mathbf{C}\mathbf{F}$  diagonal, and write down the diagonal explicitly.

*Solution.*

- (a) Both products are the same matrix:

$$\mathbf{C}^*\mathbf{C} = \mathbf{C}\mathbf{C}^* = \begin{pmatrix} 2 & 1 & 1 \\ 1 & 2 & 1 \\ 1 & 1 & 2 \end{pmatrix}.$$

(The entries are dot products of pairs of rows, respectively columns, of  $\mathbf{C}$ ; cyclic shifting makes the two lists agree.)

- (b) Write  $\mathbf{f} = (1, \omega^j, \omega^{2j})$  before normalizing. Then

$$\mathbf{C}\mathbf{f} = (1 + \omega^j, \omega^j + \omega^{2j}, 1 + \omega^{2j}).$$

Using  $\omega^{3j} = 1$ , the second entry is  $\omega^j(1 + \omega^j)$  and the third is  $\omega^{2j}(\omega^{-2j} + 1) = \omega^{2j}(\omega^j + 1)$ , since  $\omega^{-2j} = \omega^j$ . So  $\mathbf{C}\mathbf{f} = (1 + \omega^j)\mathbf{f}$ , and the eigenvalue is  $1 + \omega^j$ .

- (c) The three vectors  $\mathbf{f}_0, \mathbf{f}_1, \mathbf{f}_2$  are orthonormal: each has norm  $\frac{1}{\sqrt{3}}\sqrt{3} = 1$ , and for  $j \neq l$ ,

$$\langle \mathbf{f}_j, \mathbf{f}_l \rangle = \frac{1}{3} \sum_{m=0}^2 \omega^{mj} \overline{\omega^{ml}} = \frac{1}{3} \sum_{m=0}^2 (\omega^{j-l})^m = 0,$$

because  $\omega^{j-l} \neq 1$  is a cube root of unity and  $1 + z + z^2 = 0$  for such  $z$ . Taking  $\mathbf{F}$  with columns  $\mathbf{f}_0, \mathbf{f}_1, \mathbf{f}_2$ ,

$$\begin{aligned} \mathbf{F}^*\mathbf{C}\mathbf{F} &= \text{diag}(2, 1 + \omega, 1 + \omega^2) \\ &= \text{diag}\left(2, \frac{1}{2} + \frac{\sqrt{3}}{2}i, \frac{1}{2} - \frac{\sqrt{3}}{2}i\right), \end{aligned}$$

using  $\omega = -\frac{1}{2} + \frac{\sqrt{3}}{2}i$ . Section 9 shows that this one matrix  $\mathbf{F}$  diagonalizes **every**  $3 \times 3$  circulant.

### Exercise B3: Decide unitary diagonalizability.

Determine which of the following matrices are unitarily diagonalizable over  $\mathbb{C}$ , and which are merely diagonalizable. Justify your answer.

- (a)  $\begin{pmatrix} 0 & 1 \\ -1 & 0 \end{pmatrix}.$
- (b)  $\begin{pmatrix} 2 & 3 \\ 0 & 2 \end{pmatrix}.$

- (c)  $\begin{pmatrix} 1 & 2 \\ 0 & 3 \end{pmatrix}$ .
- (d)  $\begin{pmatrix} 0 & 1 & 0 \\ 0 & 0 & 1 \\ 1 & 0 & 0 \end{pmatrix}$ .

*Solution.*

- (a) Unitarily diagonalizable: the matrix is skew-symmetric, hence skew-adjoint over  $\mathbb{C}$  and normal, so [Corollary 11.4.3](#) applies. (Its eigenvalues are  $\pm i$ .)
- (b) Neither. Comparing  $(1,1)$  entries,  $(\mathbf{A}^*\mathbf{A})_{11} = 4$  and  $(\mathbf{A}\mathbf{A}^*)_{11} = 4 + 9 = 13$ , so  $\mathbf{A}$  is not normal. It is not even diagonalizable: the only eigenvalue is 2 and  $\mathbf{A} - 2\mathbf{I} = 3\mathbf{J}_2(0) \neq \mathbf{0}$  has rank 1, so  $\dim E_2(\mathbf{A}) = 1 < 2$  ([Theorem 8.4.1](#)).
- (c) Diagonalizable but not unitarily: the eigenvalues 1 and 3 are distinct ([Corollary 8.4.2](#)), while  $(\mathbf{A}^*\mathbf{A})_{11} = 1 \neq 5 = (\mathbf{A}\mathbf{A}^*)_{11}$ , so  $\mathbf{A}$  is not normal.
- (d) Unitarily diagonalizable: the matrix is a permutation matrix, so its columns are orthonormal and it is unitary, hence normal. (Its eigenvalues are the cube roots of unity.)

### 11.4.5.3 C. Going deeper

#### Exercise C1: Operators of finite order.

Let  $T \in \mathcal{L}(V)$  be normal on a finite-dimensional complex inner product space, and suppose  $T^k = \text{id}_V$  for some integer  $k \geq 1$ .

- (a) Prove that every eigenvalue of  $T$  is a  $k$ -th root of unity.
- (b) Deduce that  $T$  is unitary and unitarily diagonalizable.
- (c) Show that the hypothesis “normal” cannot be dropped from (b) by exhibiting a  $T$  with  $T^2 = \text{id}_V$  that is not unitary.

*Solution.*

- (a) If  $T\mathbf{v} = \lambda\mathbf{v}$  with  $\mathbf{v} \neq \mathbf{0}$ , then  $T^k\mathbf{v} = \lambda^k\mathbf{v}$  ([Exercise \(B3: Eigenvalues of powers and inverses\)](#) (a)) and  $T^k\mathbf{v} = \mathbf{v}$ , so  $(\lambda^k - 1)\mathbf{v} = \mathbf{0}$  and  $\lambda^k = 1$ .
- (b) Every  $k$ -th root of unity has modulus 1, so  $T$  is unitary by [Corollary 11.4.4](#) (b), and it is unitarily diagonalizable by [Theorem 11.4.2](#) because it is normal.
- (c) On  $\mathbb{C}^2$  take  $\mathbf{A} = \begin{pmatrix} 1 & 1 \\ 0 & -1 \end{pmatrix}$ . Then  $\mathbf{A}^2 = \mathbf{I}$ , so  $T_{\mathbf{A}}$  has order 2; but its second column  $(1, -1)$  has norm  $\sqrt{2} \neq 1$ , so  $\mathbf{A}$  is not unitary ([Theorem 10.7.3](#) (f)). It is still diagonalizable, with eigenvalues 1 and  $-1$ ; what fails is only the orthogonality of the eigenvectors.

#### Exercise C2: Same spectrum, same multiplicities.

Let  $\mathbf{A}, \mathbf{B} \in M_n(\mathbb{C})$  be normal, and suppose they have the same eigenvalues with the same multiplicities, that is, the same characteristic polynomial.

- (a) Prove that there is a unitary  $\mathbf{W}$  with  $\mathbf{B} = \mathbf{W}\mathbf{A}\mathbf{W}^*$  (one says  $\mathbf{A}$  and  $\mathbf{B}$  are **unitarily similar**).
- (b) Give two matrices in  $M_2(\mathbb{C})$  with the same characteristic polynomial that are not unitarily similar, and say which hypothesis of (a) they violate.
- (c) Deduce that two normal matrices are unitarily similar if and only if they are similar.

*Solution.*

- (a) By [Corollary 11.4.3](#) write  $\mathbf{A} = \mathbf{U}\mathbf{D}\mathbf{U}^*$  and  $\mathbf{B} = \mathbf{V}\mathbf{D}'\mathbf{V}^*$  with  $\mathbf{U}, \mathbf{V}$  unitary and  $\mathbf{D}, \mathbf{D}'$  diagonal. The diagonals of  $\mathbf{D}$  and  $\mathbf{D}'$  are the eigenvalue lists of  $\mathbf{A}$  and  $\mathbf{B}$  with multiplicity, which agree as multisets; and the order on the diagonal may be prescribed, so we may choose the eigenvector orderings to make  $\mathbf{D}' = \mathbf{D}$ . Then

$$\mathbf{B} = \mathbf{V}\mathbf{D}\mathbf{V}^* = \mathbf{V}\mathbf{U}^*\mathbf{A}\mathbf{U}\mathbf{V}^* = \mathbf{W}\mathbf{A}\mathbf{W}^* \quad \text{with} \quad \mathbf{W} = \mathbf{V}\mathbf{U}^*,$$

and  $\mathbf{W}$  is unitary because  $U(n)$  is closed under products and inverses ([Proposition 10.7.4](#)).

- (b)  $\mathbf{A} = \begin{pmatrix} 0 & 1 \\ 0 & 0 \end{pmatrix}$  and  $\mathbf{B} = \mathbf{0}$  both have characteristic polynomial  $x^2$ ; they are not even similar, since similar matrices have equal rank. A pair that is similar but not unitarily similar:  $\begin{pmatrix} 1 & 1 \\ 0 & 2 \end{pmatrix}$  and  $\text{diag}(1, 2)$ . In both cases the violated hypothesis is normality of one of the two matrices.
- (c)  $(\Rightarrow)$  A unitary similarity is a similarity.  $(\Leftarrow)$  Similar matrices have the same characteristic polynomial ([Theorem 6.10.3](#)), so if both are normal, (a) upgrades the similarity to a unitary one.

### Exercise C3: Normality cannot be dropped.

[Corollary 11.4.4](#) reads the class of a **normal** operator off its spectrum. This exercise shows that each part fails without that hypothesis.

- (a) Give  $\mathbf{A} \in M_2(\mathbb{C})$  with both eigenvalues real that is not Hermitian.
- (b) Give  $\mathbf{A} \in M_2(\mathbb{C})$  with both eigenvalues of modulus 1 that is not unitary.
- (c) Over  $\mathbb{R}$ , give  $\mathbf{A} \in M_2(\mathbb{R})$  with  $\langle \mathbf{A}\mathbf{x}, \mathbf{x} \rangle \geq 0$  for every  $\mathbf{x} \in \mathbb{R}^2$  whose eigenvalues are not non-negative reals. Why does no such example exist over  $\mathbb{C}$ ?

*Solution.*

- (a)  $\mathbf{A} = \begin{pmatrix} 1 & 1 \\ 0 & 2 \end{pmatrix}$  has eigenvalues 1 and 2 and is not Hermitian, since  $\mathbf{A}^* = \begin{pmatrix} 1 & 0 \\ 1 & 2 \end{pmatrix} \neq \mathbf{A}$ . It is not normal ([Example \(Diagonalizable, and no orthonormal eigenbasis\)](#)).
- (b)  $\mathbf{A} = \begin{pmatrix} 1 & 1 \\ 0 & 1 \end{pmatrix}$  has 1 as its only eigenvalue, of modulus 1, and is not unitary: its second column has norm  $\sqrt{2}$ . Again it is not normal, the  $(1, 1)$  entries of  $\mathbf{A}^*\mathbf{A}$  and  $\mathbf{A}\mathbf{A}^*$  being 1 and 2.

(c) Take the quarter turn  $\mathbf{A} = \begin{pmatrix} 0 & -1 \\ 1 & 0 \end{pmatrix}$ . For every  $\mathbf{x} = (x_1, x_2) \in \mathbb{R}^2$ ,

$$\langle \mathbf{A}\mathbf{x}, \mathbf{x} \rangle = -x_2x_1 + x_1x_2 = 0 \geq 0,$$

yet  $\mathbf{A}$  has no real eigenvalue at all; over  $\mathbb{C}$  its eigenvalues are  $\pm i$ , which are not non-negative reals. Over  $\mathbb{C}$  no such example exists: if  $\langle T\mathbf{v}, \mathbf{v} \rangle = 0$  for every  $\mathbf{v}$  then  $T = 0$  by [Theorem 11.2.6](#), and more generally  $\langle T\mathbf{v}, \mathbf{v} \rangle$  real for every  $\mathbf{v}$  forces  $T$  to be self-adjoint, hence normal, by applying [Theorem 11.2.6](#) to  $T - T^*$ . Real scalars are not enough to detect an operator over  $\mathbb{R}$ , and they are over  $\mathbb{C}$ .

## 11.5 The Real Spectral Theorem

Over  $\mathbb{C}$  the answer was clean: an orthonormal basis of eigenvectors exists exactly for the normal operators. Over  $\mathbb{R}$  the same question has a different answer, and a smaller one. The class shrinks from normal to self-adjoint, and the reason is a single example that we have met since Chapter 8 and can finally name as the obstruction: a rotation of the plane is as normal as an operator can be and has no eigenvector at all. This section proves the real theorem, examines that obstruction, and then rescues what can be rescued for real normal operators. It ends with the application that made the theorem famous outside linear algebra: the axes of a conic.

Throughout,  $V$  is a finite-dimensional inner product space over  $\mathbb{R}$  unless stated otherwise, and “symmetric” means  $\mathbf{A}^\top = \mathbf{A}$ .

### 11.5.1 One eigenvector, then its orthogonal complement

The proof is the signature move of the chapter, run as an induction. Section 2 did all the work: it produced an eigenvector, and it showed that the orthogonal complement of an invariant subspace is again invariant with a self-adjoint restriction. Those two facts fit together into an induction on dimension with nothing left over.

#### Theorem 11.5.1 — Real Spectral Theorem

Let  $V$  be a finite-dimensional inner product space over  $\mathbb{R}$  and let  $T \in \mathcal{L}(V)$ . The following are equivalent.

- (a)  $T$  is self-adjoint.
- (b)  $V$  has an orthonormal basis consisting of eigenvectors of  $T$ .

*Idea.*  $(\Leftarrow)$  In such a basis the matrix is diagonal, and a real diagonal matrix is its own transpose.  $(\Rightarrow)$  Induct on  $\dim V$ . Peel off one unit eigenvector  $\mathbf{e}_1$ , which exists by [Theorem 11.2.2](#); the line  $U = \text{span}(\mathbf{e}_1)$  is invariant, so [Theorem 11.2.4](#) makes  $U^\perp$  invariant with a self-adjoint restriction. That restriction lives on a space of dimension  $\dim V - 1$ , so the inductive hypothesis applies to it, and the basis it returns is already orthogonal to  $\mathbf{e}_1$ .

*Proof.* ( $\Leftarrow$ ) Let  $\mathcal{B}$  be an orthonormal basis of eigenvectors, so  $\mathbf{D} := [T]_{\mathcal{B}}^{\mathcal{B}}$  is diagonal with real entries. Since  $\mathcal{B}$  is orthonormal,  $[T^*]_{\mathcal{B}}^{\mathcal{B}} = \mathbf{D}^* = \mathbf{D}^{\top} = \mathbf{D}$  by Theorem 10.6.3, the last equality because a diagonal matrix equals its transpose. Two operators with the same matrix in one basis are equal (Theorem 3.6.4), so  $T^* = T$ .

( $\Rightarrow$ ) Induct on  $n = \dim V$ . If  $n = 0$  the empty list is an orthonormal basis and there is nothing to prove. Let  $n \geq 1$  and suppose the statement holds for every real inner product space of dimension  $n - 1$ .

Since  $V \neq \{\mathbf{0}\}$  and  $T$  is self-adjoint,  $T$  has an eigenvalue  $\lambda \in \mathbb{R}$  by Theorem 11.2.2, say  $T\mathbf{v} = \lambda\mathbf{v}$  with  $\mathbf{v} \neq \mathbf{0}$ . Put  $\mathbf{e}_1 = \mathbf{v}/\|\mathbf{v}\|$ , a unit eigenvector, and  $U = \text{span}(\mathbf{e}_1)$ . Then  $U$  is  $T$ -invariant, because  $T(c\mathbf{e}_1) = c\lambda\mathbf{e}_1 \in U$ .

By Theorem 11.2.4,  $U^{\perp}$  is  $T$ -invariant and the restriction  $T|_{U^{\perp}}$  is a self-adjoint operator on  $U^{\perp}$ , which is an inner product space in its own right under the restricted inner product. Its dimension is  $n - 1$  by Theorem 10.3.2 (c). By the inductive hypothesis,  $U^{\perp}$  has an orthonormal basis  $(\mathbf{e}_2, \dots, \mathbf{e}_n)$  of eigenvectors of  $T|_{U^{\perp}}$ ; each  $\mathbf{e}_j$  satisfies  $T\mathbf{e}_j = \mu_j\mathbf{e}_j$  in  $V$  as well, since the restriction is the same map.

Finally  $(\mathbf{e}_1, \dots, \mathbf{e}_n)$  is orthonormal: the last  $n - 1$  vectors are orthonormal among themselves and each lies in  $U^{\perp}$ , hence is orthogonal to  $\mathbf{e}_1$ , which has norm 1. An orthonormal list is linearly independent (Theorem 10.1.4), and it has  $n = \dim V$  members, so it is a basis (Theorem 1.6.4). Every member is an eigenvector of  $T$ . This proves the theorem. ■

*Remark.* No complexification appears anywhere in that induction, which is worth noticing: the argument is entirely real. The one place where complex numbers were needed is inside Theorem 11.2.2, whose proof complexifies once to borrow an eigenvalue from the Fundamental Theorem of Algebra (Theorem 5.5.1) and then observes that the eigenvalue is real. So  $\mathbb{C}$  is used exactly once, to start the machine, and never again.

### Corollary 11.5.2 — Orthogonal Diagonalization

Let  $\mathbf{A} \in M_n(\mathbb{R})$ . Then  $\mathbf{A}$  is symmetric if and only if there are an orthogonal  $\mathbf{Q} \in O(n)$  and a **real** diagonal  $\mathbf{D}$  with

$$\mathbf{A} = \mathbf{Q}\mathbf{D}\mathbf{Q}^{\top}.$$

The diagonal of  $\mathbf{D}$  is the eigenvalue list of  $\mathbf{A}$  with multiplicity, in any prescribed order, and the columns of  $\mathbf{Q}$  are a corresponding orthonormal basis of eigenvectors.

*Proof.* Apply Theorem 11.5.1 to  $T_{\mathbf{A}}$  on  $\mathbb{R}^n$  with the standard inner product, for which the standard basis is orthonormal, so that  $\mathbf{A}^{\top} = \mathbf{A}$  is exactly self-adjointness (Theorem 10.6.3).

( $\Rightarrow$ ) Let  $(\mathbf{q}_1, \dots, \mathbf{q}_n)$  be an orthonormal basis with  $\mathbf{A}\mathbf{q}_j = \lambda_j\mathbf{q}_j$ , and let  $\mathbf{Q}$  have these columns; it is orthogonal by Theorem 10.7.3 (f). Column  $j$  of  $\mathbf{A}\mathbf{Q}$  is  $\lambda_j\mathbf{q}_j$ , which is column  $j$  of  $\mathbf{Q}\mathbf{D}$  for  $\mathbf{D} = \text{diag}(\lambda_1, \dots, \lambda_n)$ . So  $\mathbf{A}\mathbf{Q} = \mathbf{Q}\mathbf{D}$  and  $\mathbf{A} = \mathbf{Q}\mathbf{D}\mathbf{Q}^{-1} = \mathbf{Q}\mathbf{D}\mathbf{Q}^{\top}$ . The eigenvalues are real because  $\mathbf{A}$  is self-adjoint (Theorem 11.2.1), so  $\mathbf{D}$  is a real matrix.

( $\Leftarrow$ ) If  $\mathbf{A} = \mathbf{Q}\mathbf{D}\mathbf{Q}^{\top}$  with  $\mathbf{D}$  real diagonal, then

$$\mathbf{A}^{\top} = (\mathbf{Q}\mathbf{D}\mathbf{Q}^{\top})^{\top} = \mathbf{Q}\mathbf{D}^{\top}\mathbf{Q}^{\top} = \mathbf{Q}\mathbf{D}\mathbf{Q}^{\top} = \mathbf{A}.$$



The claims about the order and the columns are as in [Corollary 11.4.3](#).

■

## 11.5.2 The negative result: normal is not enough over the reals

Over  $\mathbb{C}$  the class with an orthonormal eigenbasis was the normal operators. Over  $\mathbb{R}$  [Theorem 11.5.1](#) says it is the self-adjoint ones, which is a strictly smaller class: every self-adjoint operator is normal, and the converse now fails. It is worth seeing exactly how it fails, because the failure is not a technicality about multiplicities or about a badly chosen basis. It is the complete absence of an eigenvector.

### Example. (A normal real operator with no eigenvector)

Let  $T \in \mathcal{L}(\mathbb{R}^2)$  be the quarter turn, with matrix

$$\mathbf{R} = \begin{pmatrix} 0 & -1 \\ 1 & 0 \end{pmatrix}$$

in the standard basis. Show that  $T$  is normal, and that  $T$  has no eigenvalue and no eigenvector in  $\mathbb{R}^2$ .

*Solution.* Normality:  $\mathbf{R}^\top \mathbf{R} = \mathbf{R} \mathbf{R}^\top = \mathbf{I}$ , so  $T$  is even orthogonal, hence normal. (One may also read it off the shape:  $\mathbf{R} = \mathbf{A}(i)$  is the rotation-scaling block of [Definition 9.4.1](#).)

No eigenvalue:  $p_{\mathbf{R}}(x) = x^2 + 1$ , which has no root in  $\mathbb{R}$ . So  $\mathbf{R} - \lambda \mathbf{I}$  is invertible for every real  $\lambda$ , and  $\ker(T - \lambda \text{id}) = \{\mathbf{0}\}$  ([Theorem 8.2.1](#)). Geometrically,  $T$  turns every non-zero vector by a right angle, so  $T\mathbf{v}$  is never a real multiple of  $\mathbf{v}$ ; the operator maps no line to itself.

That example rules out every possible improvement of [Theorem 11.5.1](#) to normal operators over  $\mathbb{R}$ , and not by a narrow margin. An orthonormal basis of eigenvectors needs  $n$  eigenvectors; the quarter turn supplies none. So the two spectral theorems are not two versions of one statement with a technical difference: the real one has a genuinely smaller hypothesis, and the only reason it is not smaller still is [Theorem 11.2.2](#), which manufactures the eigenvector that the rotation lacks.

Two further remarks keep the picture honest. First, the quarter turn is not exceptional. Over  $\mathbb{R}$ , *every* normal operator that is not self-adjoint fails to have an orthonormal eigenbasis, because [Theorem 11.5.1](#) is an equivalence: having such a basis and being self-adjoint are the same condition. Second, nothing is wrong with the operator; what is missing is the field. Regard  $\mathbf{R}$  as a complex matrix and it is unitarily diagonalizable, with eigenvalues  $\pm i$  and eigenvectors  $(1, \mp i)$  in  $\mathbb{C}^2$  (the sign of the eigenvalue and of the entry being opposite). The complex spectral theorem never fails; it simply answers a question about  $\mathbb{C}^n$ .

*Quick check.* Is  $\mathbf{R} = \begin{pmatrix} 0 & -1 \\ 1 & 0 \end{pmatrix}$  orthogonally diagonalizable over  $\mathbb{R}$ ? Is it unitarily diagonalizable over  $\mathbb{C}$ ? Are the two answers consistent?

*Solution.* Not over  $\mathbb{R}$ : an orthogonal diagonalization  $\mathbf{R} = \mathbf{Q}\mathbf{D}\mathbf{Q}^\top$  would make  $\mathbf{R}$  symmetric by [Corollary 11.5.2](#), and  $\mathbf{R}^\top = -\mathbf{R} \neq \mathbf{R}$ . Over  $\mathbb{C}$  it is unitarily diagonalizable, because it is normal ([Corollary 11.4.3](#)); explicitly  $\mathbf{U}^*\mathbf{R}\mathbf{U} = \text{diag}(i, -i)$  for  $\mathbf{U} = \frac{1}{\sqrt{2}} \begin{pmatrix} 1 & 1 \\ -i & i \end{pmatrix}$ . There is no conflict: the two statements concern different spaces,  $\mathbb{R}^2$  and  $\mathbb{C}^2$ , and the diagonalizing basis of the second one is not real.

### 11.5.3 What a real normal operator does look like

The rotation is not a defect to be apologized for; it is a building block. Chapter 9 already promoted it to one when it built the real Jordan form, and the same  $2 \times 2$  block returns here as the only thing that can obstruct a real diagonalization of a normal operator.

#### Theorem 11.5.3 — Real Normal Form of a Normal Operator

Let  $V$  be a finite-dimensional inner product space over  $\mathbb{R}$  and let  $T \in \mathcal{L}(V)$  be normal. Then  $V$  has an orthonormal basis in which the matrix of  $T$  is block diagonal,

$$(c_1) \oplus \cdots \oplus (c_p) \oplus \mathbf{\Lambda}(\lambda_1) \oplus \cdots \oplus \mathbf{\Lambda}(\lambda_q),$$

with  $c_1, \dots, c_p \in \mathbb{R}$  and, for each  $j$ ,  $\lambda_j = a_j + b_j i$  with  $b_j > 0$  and

$$\mathbf{\Lambda}(\lambda_j) = \begin{pmatrix} a_j & -b_j \\ b_j & a_j \end{pmatrix}$$

the rotation-scaling block of [Definition 9.4.1](#). In matrix form:  $\mathbf{A} \in M_n(\mathbb{R})$  is normal if and only if  $\mathbf{A} = \mathbf{Q}\mathbf{B}\mathbf{Q}^\top$  with  $\mathbf{Q} \in O(n)$  and  $\mathbf{B}$  of the displayed shape.

*Idea.* The real Schur form of Section 1 gets us to a **block** triangular matrix with  $1 \times 1$  and  $2 \times 2$  diagonal blocks, and the corner blocks above the diagonal are the only obstruction left. The argument that killed them in the complex case, comparing a diagonal entry of  $\mathbf{T}^\top \mathbf{T}$  with the same entry of  $\mathbf{T}\mathbf{T}^\top$ , works verbatim on blocks once “entry” is replaced by “trace of a block”: the identity  $\text{tr}(\mathbf{X}\mathbf{X}^\top) = \sum_{i,j} x_{ij}^2$  turns the comparison into a sum of squares that has to vanish. What remains is to see what a  $2 \times 2$  real normal matrix can be, and there are exactly two answers: symmetric, which splits further, or a rotation-scaling block.

*Proof.* Fix an orthonormal basis of  $V$  and let  $\mathbf{A} \in M_n(\mathbb{R})$  be the matrix of  $T$  in it; then  $\mathbf{A}^\top$  is the matrix of  $T^*$  ([Theorem 10.6.3](#)), and  $T$  normal means  $\mathbf{A}^\top \mathbf{A} = \mathbf{A}\mathbf{A}^\top$ . Every orthogonal change of orthonormal basis replaces  $\mathbf{A}$  by  $\mathbf{Q}^\top \mathbf{A} \mathbf{Q}$ , which is again normal:

$$(\mathbf{Q}^\top \mathbf{A} \mathbf{Q})^\top (\mathbf{Q}^\top \mathbf{A} \mathbf{Q}) = \mathbf{Q}^\top \mathbf{A}^\top \mathbf{A} \mathbf{Q} = \mathbf{Q}^\top \mathbf{A} \mathbf{A}^\top \mathbf{Q},$$

and the same computation in the other order gives  $(\mathbf{Q}^\top \mathbf{A} \mathbf{Q})(\mathbf{Q}^\top \mathbf{A} \mathbf{Q})^\top$ . So we may replace  $\mathbf{A}$  by any orthogonally similar matrix at will.

**Step 1: a block triangular form.** By the real Schur form ([Theorem 11.1.7](#)) there is  $\mathbf{Q}_1 \in O(n)$  with  $\mathbf{T} = \mathbf{Q}_1^\top \mathbf{A} \mathbf{Q}_1$  block upper triangular, its diagonal blocks of size  $1 \times 1$  and  $2 \times 2$ . Write the block partition as  $\mathbf{T} = (\mathbf{T}_{kl})_{1 \leq k, l \leq m}$  with  $\mathbf{T}_{kl} = \mathbf{0}$  for  $k > l$ . By the previous paragraph  $\mathbf{T}$  is normal.

**Step 2: the corners vanish.**

*Claim.* A normal, block upper triangular real matrix is block diagonal.

*Proof.* Induct on the number  $m$  of diagonal blocks;  $m = 1$  is trivial. Comparing the  $(1, 1)$  blocks of the two products ([Theorem 7.1.1](#)) and using  $\mathbf{T}_{k1} = \mathbf{0}$  for  $k > 1$ ,

$$\begin{aligned} (\mathbf{T}^\top \mathbf{T})_{11} &= \mathbf{T}_{11}^\top \mathbf{T}_{11}, \\ (\mathbf{T} \mathbf{T}^\top)_{11} &= \mathbf{T}_{11} \mathbf{T}_{11}^\top + \sum_{l>1} \mathbf{T}_{1l} \mathbf{T}_{1l}^\top. \end{aligned}$$

Take traces. Since  $\text{tr}(\mathbf{X}\mathbf{Y}) = \text{tr}(\mathbf{Y}\mathbf{X})$  ([Theorem 0.9.6](#)), the two terms  $\text{tr}(\mathbf{T}_{11}^\top \mathbf{T}_{11})$  and  $\text{tr}(\mathbf{T}_{11} \mathbf{T}_{11}^\top)$  are equal and cancel, leaving

$$\sum_{l>1} \text{tr}(\mathbf{T}_{1l} \mathbf{T}_{1l}^\top) = 0.$$

For a real matrix  $\mathbf{X}$ ,  $\text{tr}(\mathbf{X}\mathbf{X}^\top) = \sum_{i,j} x_{ij}^2 \geq 0$ , with equality only for  $\mathbf{X} = \mathbf{0}$ . So  $\mathbf{T}_{1l} = \mathbf{0}$  for every  $l > 1$ : the first block row is clear, and  $\mathbf{T} = \mathbf{T}_{11} \oplus \mathbf{T}'$  where  $\mathbf{T}'$  is the trailing block upper triangular matrix with  $m - 1$  diagonal blocks. Then

$$\mathbf{T}^\top \mathbf{T} = \mathbf{T}_{11}^\top \mathbf{T}_{11} \oplus \mathbf{T}'^\top \mathbf{T}', \quad \mathbf{T} \mathbf{T}^\top = \mathbf{T}_{11} \mathbf{T}_{11}^\top \oplus \mathbf{T}' \mathbf{T}'^\top,$$

so normality of  $\mathbf{T}$  gives normality of  $\mathbf{T}'$ , and the inductive hypothesis applies. ■

So after Step 2 the matrix is  $\mathbf{D}_1 \oplus \cdots \oplus \mathbf{D}_m$  with each  $\mathbf{D}_k$  of size 1 or 2. Both products are now computed blockwise, so normality of the whole reads  $\mathbf{D}_k^\top \mathbf{D}_k = \mathbf{D}_k \mathbf{D}_k^\top$  for each  $k$ : every diagonal block is normal.

**Step 3: the shape of a  $2 \times 2$  real normal block.** Let  $\mathbf{D} = \begin{pmatrix} p & q \\ r & s \end{pmatrix}$  be real and normal. Expanding both products,

$$\mathbf{D} \mathbf{D}^\top - \mathbf{D}^\top \mathbf{D} = \begin{pmatrix} q^2 - r^2 & (r - q)(p - s) \\ (r - q)(p - s) & r^2 - q^2 \end{pmatrix},$$

so normality says  $q^2 = r^2$  and  $(r - q)(p - s) = 0$ . The first equation gives  $q = r$  or  $q = -r$ , and when  $q = -r = 0$  both hold, so the following two cases are exhaustive.

*Case 1:*  $q = r$ . Then  $\mathbf{D}$  is symmetric, and [Corollary 11.5.2](#) supplies an orthogonal  $2 \times 2$  matrix turning it into a real diagonal matrix, which is two  $1 \times 1$  blocks.

*Case 2:*  $q = -r$  and  $q \neq 0$ . Then  $r - q = -2q \neq 0$ , so  $p = s$  and

$$\mathbf{D} = \begin{pmatrix} p & q \\ -q & p \end{pmatrix}.$$

If  $q < 0$ , this is  $\mathbf{A}(p - qi)$  with  $-q > 0$ , as required. If  $q > 0$ , conjugate by the orthogonal matrix  $\text{diag}(1, -1)$ :

$$\begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix} \begin{pmatrix} p & q \\ -q & p \end{pmatrix} \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix} = \begin{pmatrix} p & -q \\ q & p \end{pmatrix},$$

which is  $\Lambda(p + qi)$  with  $q > 0$ .

**Step 4: assembling.** Apply Step 3 inside each diagonal block, using the direct sum  $\mathbf{Q}_2$  of the resulting  $1 \times 1$  and  $2 \times 2$  orthogonal matrices, which is orthogonal because its columns are orthonormal. Then  $\mathbf{Q} = \mathbf{Q}_1 \mathbf{Q}_2$  is orthogonal and  $\mathbf{Q}^\top \mathbf{A} \mathbf{Q}$  has the stated shape; reordering the blocks by a permutation matrix, itself orthogonal, puts the  $1 \times 1$  blocks first. The columns of  $\mathbf{Q}$  are the required orthonormal basis.

For the converse, a block diagonal matrix  $\mathbf{B}$  of the displayed shape is normal, since  $\mathbf{B}^\top \mathbf{B}$  and  $\mathbf{B} \mathbf{B}^\top$  are computed blockwise and every block is normal:  $(c)^\top (c) = (c^2) = (c)(c)^\top$ , and  $\Lambda(\lambda)^\top \Lambda(\lambda) = |\lambda|^2 \mathbf{I}_2 = \Lambda(\lambda) \Lambda(\lambda)^\top$ . Then  $\mathbf{A} = \mathbf{Q} \mathbf{B} \mathbf{Q}^\top$  is normal by the first paragraph. This proves the theorem. ■

The two theorems now fit together. A real normal operator is self-adjoint exactly when no  $\Lambda$  block occurs, and then [Theorem 11.5.3](#) is [Theorem 11.5.1](#). The eigenvalues of  $\mathbf{A}$  are the numbers  $c_i$  together with the conjugate pairs  $\lambda_j, \overline{\lambda_j}$ , so the  $\Lambda$  blocks are precisely the non-real part of the spectrum, packaged two real dimensions at a time. Section 6 specializes this to orthogonal operators, where  $|\lambda| = 1$  forces every  $2 \times 2$  block to be a plane rotation.

## 11.5.4 A symmetric matrix as a sum of projections

Writing  $\mathbf{A} = \mathbf{Q} \mathbf{D} \mathbf{Q}^\top$  and expanding the product column by column turns the diagonalization into a sum of very simple matrices.

### Corollary 11.5.4 — Spectral Decomposition of a Symmetric Matrix

Let  $\mathbf{A} \in M_n(\mathbb{R})$  be symmetric, and let  $(\mathbf{q}_1, \dots, \mathbf{q}_n)$  be an orthonormal basis of  $\mathbb{R}^n$  with  $\mathbf{A} \mathbf{q}_i = \lambda_i \mathbf{q}_i$ . Then

$$\mathbf{A} = \sum_{i=1}^n \lambda_i \mathbf{q}_i \mathbf{q}_i^\top,$$

and each  $\mathbf{q}_i \mathbf{q}_i^\top$  is the matrix of the orthogonal projection of  $\mathbb{R}^n$  onto the line  $\text{span}(\mathbf{q}_i)$ .

*Proof.* Let  $\mathbf{Q}$  have columns  $\mathbf{q}_1, \dots, \mathbf{q}_n$  and  $\mathbf{D} = \text{diag}(\lambda_1, \dots, \lambda_n)$ , so  $\mathbf{A} = \mathbf{Q} \mathbf{D} \mathbf{Q}^\top$  by [Corollary 11.5.2](#). The columns of  $\mathbf{Q} \mathbf{D}$  are  $\lambda_i \mathbf{q}_i$ , and the rows of  $\mathbf{Q}^\top$  are  $\mathbf{q}_i^\top$ , so the column-times-row expansion of the product ([Corollary 7.1.3](#)) gives  $\mathbf{A} = \sum_i (\lambda_i \mathbf{q}_i) \mathbf{q}_i^\top$ , which is the stated sum.

For the second claim,  $\mathbf{q}_i$  is a unit vector, so the orthogonal projection onto  $\text{span}(\mathbf{q}_i)$  is  $\mathbf{x} \mapsto \langle \mathbf{x}, \mathbf{q}_i \rangle \mathbf{q}_i$  by [Theorem 10.3.3](#). Since  $\langle \mathbf{x}, \mathbf{q}_i \rangle = \mathbf{q}_i^\top \mathbf{x}$  is a scalar,  $\langle \mathbf{x}, \mathbf{q}_i \rangle \mathbf{q}_i = \mathbf{q}_i (\mathbf{q}_i^\top \mathbf{x}) = (\mathbf{q}_i \mathbf{q}_i^\top) \mathbf{x}$ . ■

**Example.** (Orthogonally diagonalizing a 3 by 3 matrix)

Orthogonally diagonalize

$$\mathbf{A} = \begin{pmatrix} 3 & 1 & 1 \\ 1 & 3 & 1 \\ 1 & 1 & 3 \end{pmatrix},$$

and write it as a sum of rank-one orthogonal projections.

*Solution.* The rows all sum to 5, so  $\mathbf{A}(1, 1, 1) = 5(1, 1, 1)$  and 5 is an eigenvalue. Also  $\mathbf{A} = 2\mathbf{I} + \mathbf{J}$  where  $\mathbf{J}$  is the all-ones matrix, and  $\mathbf{J}\mathbf{x} = \mathbf{0}$  exactly when  $x_1 + x_2 + x_3 = 0$ ; on that plane  $\mathbf{A}\mathbf{x} = 2\mathbf{x}$ . So

$$E_5(\mathbf{A}) = \text{span}((1, 1, 1)), \quad E_2(\mathbf{A}) = \{\mathbf{x} : x_1 + x_2 + x_3 = 0\},$$

of dimensions 1 and 2, which already account for all of  $\mathbb{R}^3$ . The two eigenspaces are orthogonal, as [Theorem 11.2.3](#) requires.

Inside  $E_2(\mathbf{A})$  any orthonormal basis will do. Start from the independent pair  $(1, -1, 0)$  and  $(1, 0, -1)$ , both in the plane, and run one step of Gram–Schmidt ([Theorem 10.2.2](#)): the second vector minus its component along the first is  $(1, 0, -1) - \frac{1}{2}(1, -1, 0) = \frac{1}{2}(1, 1, -2)$ , a multiple of  $(1, 1, -2)$ . Normalizing all three,

$$\mathbf{q}_1 = \frac{1}{\sqrt{3}}(1, 1, 1), \quad \mathbf{q}_2 = \frac{1}{\sqrt{2}}(1, -1, 0), \quad \mathbf{q}_3 = \frac{1}{\sqrt{6}}(1, 1, -2).$$

With  $\mathbf{Q} = (\mathbf{q}_1 \mid \mathbf{q}_2 \mid \mathbf{q}_3)$  we get  $\mathbf{Q}^\top \mathbf{Q} = \mathbf{I}$  and  $\mathbf{Q}^\top \mathbf{A} \mathbf{Q} = \text{diag}(5, 2, 2)$ . By [Corollary 11.5.4](#),

$$\mathbf{A} = 5\mathbf{q}_1\mathbf{q}_1^\top + 2\mathbf{q}_2\mathbf{q}_2^\top + 2\mathbf{q}_3\mathbf{q}_3^\top.$$

As a check,  $\mathbf{q}_2\mathbf{q}_2^\top + \mathbf{q}_3\mathbf{q}_3^\top = \mathbf{I} - \mathbf{q}_1\mathbf{q}_1^\top$  because the three projections sum to the identity, so the right-hand side is  $2\mathbf{I} + 3\mathbf{q}_1\mathbf{q}_1^\top = 2\mathbf{I} + \mathbf{J}$ , which is  $\mathbf{A}$ . Notice that  $\mathbf{q}_2$  and  $\mathbf{q}_3$  were not forced on us, while the two projections  $\mathbf{q}_1\mathbf{q}_1^\top$  and  $\mathbf{q}_2\mathbf{q}_2^\top + \mathbf{q}_3\mathbf{q}_3^\top$  were: they are the projections onto the two eigenspaces. Section 8 makes that observation into a theorem.

## 11.5.5 The principal axis theorem

Here is the application. A **quadratic form** on  $\mathbb{R}^n$  is a function

$$q(\mathbf{x}) = \mathbf{x}^\top \mathbf{A} \mathbf{x} = \sum_{i,j} a_{ij} x_i x_j$$

for some  $\mathbf{A} \in M_n(\mathbb{R})$ . Only the symmetric part of  $\mathbf{A}$  matters, since  $\mathbf{x}^\top \mathbf{A} \mathbf{x} = \mathbf{x}^\top \mathbf{A}^\top \mathbf{x}$  for a  $1 \times 1$  matrix, so we always take  $\mathbf{A}$  symmetric; then  $\mathbf{A}$  is determined by  $q$ , with  $a_{ii}$  the coefficient of  $x_i^2$  and  $a_{ij}$  **half** the coefficient of  $x_i x_j$  for  $i \neq j$ . Chapter 13 studies such forms in their own right.

### Theorem 11.5.5 — Principal Axis Theorem

Let  $\mathbf{A} \in M_n(\mathbb{R})$  be symmetric, let  $q(\mathbf{x}) = \mathbf{x}^\top \mathbf{A} \mathbf{x}$ , and let  $\mathbf{Q} \in O(n)$  with  $\mathbf{Q}^\top \mathbf{A} \mathbf{Q} = \mathbf{D} = \text{diag}(\lambda_1, \dots, \lambda_n)$ , the columns  $\mathbf{q}_1, \dots, \mathbf{q}_n$  of  $\mathbf{Q}$  being an orthonormal basis of eigenvectors. Substituting  $\mathbf{x} = \mathbf{Q}\mathbf{y}$ , that is, taking  $y_i = \langle \mathbf{x}, \mathbf{q}_i \rangle$  to be the coordinates of  $\mathbf{x}$  in that basis,

$$q(\mathbf{x}) = \sum_{i=1}^n \lambda_i y_i^2.$$

In particular, for  $n = 2$  with  $\lambda_1, \lambda_2 > 0$ , the level set  $\{q = 1\}$  is an ellipse whose axes lie along  $\mathbf{q}_1$  and  $\mathbf{q}_2$ , with half-lengths  $1/\sqrt{\lambda_1}$  and  $1/\sqrt{\lambda_2}$ .

*Proof.* Since  $\mathbf{Q}$  is invertible,  $\mathbf{x} = \mathbf{Q}\mathbf{y}$  sets up a bijection between  $\mathbf{x}$  and  $\mathbf{y} = \mathbf{Q}^{-1}\mathbf{x} = \mathbf{Q}^\top \mathbf{x}$ , whose  $i$ -th entry is  $\mathbf{q}_i^\top \mathbf{x} = \langle \mathbf{x}, \mathbf{q}_i \rangle$ . Substituting,

$$q(\mathbf{Q}\mathbf{y}) = (\mathbf{Q}\mathbf{y})^\top \mathbf{A}(\mathbf{Q}\mathbf{y}) = \mathbf{y}^\top (\mathbf{Q}^\top \mathbf{A} \mathbf{Q}) \mathbf{y} = \mathbf{y}^\top \mathbf{D} \mathbf{y} = \sum_i \lambda_i y_i^2,$$

using  $(\mathbf{Q}\mathbf{y})^\top = \mathbf{y}^\top \mathbf{Q}^\top$  (Theorem 0.9.5).

For  $n = 2$  with both  $\lambda_i > 0$ , the condition  $q(\mathbf{x}) = 1$  becomes  $\lambda_1 y_1^2 + \lambda_2 y_2^2 = 1$ , the standard equation of an ellipse in the  $\mathbf{y}$  coordinates, meeting the  $y_1$ -axis where  $y_1 = \pm 1/\sqrt{\lambda_1}$  and the  $y_2$ -axis where  $y_2 = \pm 1/\sqrt{\lambda_2}$ . Those axes are the lines  $\text{span}(\mathbf{q}_1)$  and  $\text{span}(\mathbf{q}_2)$  of the original picture, because  $\mathbf{y} = \mathbf{e}_1$  corresponds to  $\mathbf{x} = \mathbf{Q}\mathbf{e}_1 = \mathbf{q}_1$ . Since  $\mathbf{Q}$  is orthogonal, it preserves lengths (Theorem 10.7.3), so the half-lengths are unchanged by the substitution. ■

### Example. (Finding the axes of a conic)

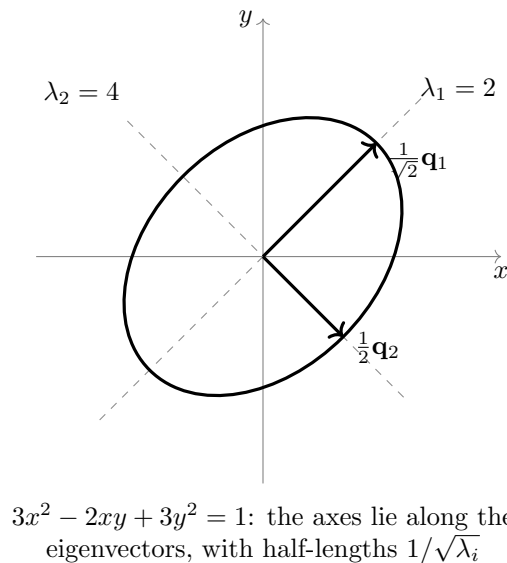
Identify the curve  $3x^2 - 2xy + 3y^2 = 1$  in  $\mathbb{R}^2$ , and find its axes and their half-lengths.

*Solution.* The symmetric matrix of the form is

$$\mathbf{A} = \begin{pmatrix} 3 & -1 \\ -1 & 3 \end{pmatrix},$$

the off-diagonal entries being half of  $-2$ . Its characteristic polynomial is  $x^2 - 6x + 8$ , so the eigenvalues are 2 and 4. For  $\lambda = 2$ ,  $\mathbf{A} - 2\mathbf{I} = \begin{pmatrix} 1 & -1 \\ -1 & 1 \end{pmatrix}$  gives  $x = y$  and the unit eigenvector  $\mathbf{q}_1 = \frac{1}{\sqrt{2}}(1, 1)$ . For  $\lambda = 4$ ,  $\mathbf{A} - 4\mathbf{I} = \begin{pmatrix} -1 & -1 \\ -1 & -1 \end{pmatrix}$  gives  $x = -y$  and  $\mathbf{q}_2 = \frac{1}{\sqrt{2}}(1, -1)$ .

In the coordinates  $y_1 = \langle \mathbf{x}, \mathbf{q}_1 \rangle$ ,  $y_2 = \langle \mathbf{x}, \mathbf{q}_2 \rangle$ , the equation is  $2y_1^2 + 4y_2^2 = 1$ . Both eigenvalues are positive, so the curve is an **ellipse**. Its major axis lies along  $\mathbf{q}_1$ , the line  $y = x$ , with half-length  $1/\sqrt{2}$ ; its minor axis lies along  $\mathbf{q}_2$ , the line  $y = -x$ , with half-length  $1/2$ . The check is that  $\frac{1}{\sqrt{2}}(1, 1)$  scaled to length  $1/\sqrt{2}$  is the point  $(\frac{1}{2}, \frac{1}{2})$ , and  $3 \cdot \frac{1}{4} - 2 \cdot \frac{1}{4} + 3 \cdot \frac{1}{4} = 1$ .



Two features of the picture are worth naming. The axes are **orthogonal** because the eigenvectors are, and that is the whole content of the theorem; a non-symmetric matrix in the same role would produce a pair of distinguished directions that need not be perpendicular. And the bigger eigenvalue gives the **shorter** axis, since the half-length is  $1/\sqrt{\lambda}$ : a large  $\lambda$  makes  $q$  reach the value 1 sooner.

*Remark.* Chapter 13 will diagonalize quadratic forms again, but by **congruence**: replacing  $\mathbf{A}$  by  $\mathbf{P}^\top \mathbf{A} \mathbf{P}$  for an arbitrary invertible  $\mathbf{P}$ , not necessarily orthogonal. That is a weaker relation, and it destroys the eigenvalues while preserving only their signs, which is Sylvester's law of inertia. What makes the present theorem special is that  $\mathbf{Q}^\top = \mathbf{Q}^{-1}$ , so  $\mathbf{Q}^\top \mathbf{A} \mathbf{Q} = \mathbf{Q}^{-1} \mathbf{A} \mathbf{Q}$  and the two relations coincide: an orthogonal change of variables is a similarity *and* a congruence, so it keeps the eigenvalues and the geometry at once.

*Warning.* The theorem needs  $\mathbb{R}$ , and symmetry is not the same as having real eigenvalues. The matrix  $\begin{pmatrix} 1 & 1 \\ 1 & 0 \end{pmatrix}$  is symmetric with rational entries, but its eigenvalues are  $(1 \pm \sqrt{5})/2$ , which are irrational; over  $\mathbb{Q}$  there is no eigenvector at all, and no version of Theorem 11.5.1 survives. Conversely  $\begin{pmatrix} 1 & 1 \\ 0 & 2 \end{pmatrix}$  has the real eigenvalues 1 and 2 and is not symmetric; real eigenvalues alone buy nothing, since they do not make the eigenvectors orthogonal.

## 11.5.6 Exercises

### 11.5.6.1 A. Check your understanding

#### Exercise A1.

- State the real spectral theorem and its matrix form, with every hypothesis.
- Which two results from Section 2 does the induction use, and what does each one supply?
- True or false: a normal operator on a real inner product space has an orthonormal

basis of eigenvectors. Justify your answer.

- (d) What are the possible diagonal blocks in the real normal form of a normal operator, and what does the condition  $b > 0$  rule out?
- (e) Given a symmetric  $\mathbf{A}$  with orthonormal eigenvectors  $\mathbf{q}_1, \mathbf{q}_2$  and eigenvalues 9 and 1, describe the curve  $\mathbf{x}^\top \mathbf{A} \mathbf{x} = 1$ .
- (f) True or false: a real matrix with real eigenvalues is symmetric. Justify your answer.

*Solution.*

- (a) For a finite-dimensional real inner product space  $V$  and  $T \in \mathcal{L}(V)$ :  $T$  is self-adjoint if and only if  $V$  has an orthonormal basis of eigenvectors ([Theorem 11.5.1](#)). Matrix form:  $\mathbf{A} \in M_n(\mathbb{R})$  is symmetric if and only if  $\mathbf{A} = \mathbf{Q} \mathbf{D} \mathbf{Q}^\top$  with  $\mathbf{Q}$  orthogonal and  $\mathbf{D}$  real diagonal ([Corollary 11.5.2](#)).
- (b) [Theorem 11.2.2](#) supplies one eigenvector to peel off, and [Theorem 11.2.4](#) supplies the smaller object to induct on, namely the self-adjoint restriction to the orthogonal complement.
- (c) False. The quarter turn of  $\mathbb{R}^2$  is normal and has no eigenvector at all ([Example \(A normal real operator with no eigenvector\)](#)).
- (d) Blocks (c) with  $c \in \mathbb{R}$ , and  $2 \times 2$  blocks  $\mathbf{A}(\lambda) = \begin{pmatrix} a & -b \\ b & a \end{pmatrix}$  with  $b > 0$  ([Theorem 11.5.3](#)). The condition  $b > 0$  rules out the transposed block  $\mathbf{A}(\bar{\lambda})$ , which is orthogonally similar to it via  $\text{diag}(1, -1)$ ; fixing the sign is what makes the list of blocks well defined.
- (e) An ellipse with axes along  $\mathbf{q}_1$  and  $\mathbf{q}_2$ , of half-lengths  $1/3$  and  $1$  ([Theorem 11.5.5](#)). The longer axis is along  $\mathbf{q}_2$ , the eigenvector for the **smaller** eigenvalue.
- (f) False.  $\begin{pmatrix} 1 & 1 \\ 0 & 2 \end{pmatrix}$  has eigenvalues 1, 2 and is not symmetric. The correct statement is the converse: a symmetric real matrix has real eigenvalues.

### 11.5.6.2 B. Practice

#### Exercise B1: Orthogonally diagonalize.

For each symmetric matrix, find an orthogonal  $\mathbf{Q}$  and a diagonal  $\mathbf{D}$  with  $\mathbf{Q}^\top \mathbf{A} \mathbf{Q} = \mathbf{D}$ .

- (a)  $\mathbf{A} = \begin{pmatrix} 1 & 2 \\ 2 & 1 \end{pmatrix}$ .
- (b)  $\mathbf{A} = \begin{pmatrix} 4 & 0 & 1 \\ 0 & 2 & 0 \\ 1 & 0 & 4 \end{pmatrix}$ .

*Solution.*

- (a)  $p_{\mathbf{A}}(x) = x^2 - 2x - 3 = (x - 3)(x + 1)$ , so the eigenvalues are 3 and  $-1$ . For  $\lambda = 3$ ,  $\mathbf{A} - 3\mathbf{I} = \begin{pmatrix} -2 & 2 \\ 2 & -2 \end{pmatrix}$  gives  $x_1 = x_2$  and  $\mathbf{q}_1 = \frac{1}{\sqrt{2}}(1, 1)$ ; for  $\lambda = -1$ ,  $\mathbf{q}_2 = \frac{1}{\sqrt{2}}(1, -1)$ .



Hence

$$\mathbf{Q} = \frac{1}{\sqrt{2}} \begin{pmatrix} 1 & 1 \\ 1 & -1 \end{pmatrix}, \quad \mathbf{D} = \begin{pmatrix} 3 & 0 \\ 0 & -1 \end{pmatrix}.$$

- (b) The second coordinate is untouched:  $\mathbf{A}\mathbf{e}_2 = 2\mathbf{e}_2$ . On  $\text{span}(\mathbf{e}_1, \mathbf{e}_3)$  the matrix acts as  $\begin{pmatrix} 4 & 1 \\ 1 & 4 \end{pmatrix}$ , with eigenvalues 5 and 3 and eigenvectors  $(1, 1)$  and  $(1, -1)$  there. Lifting back,

$$\mathbf{q}_1 = \frac{1}{\sqrt{2}}(1, 0, 1), \quad \mathbf{q}_2 = \frac{1}{\sqrt{2}}(-1, 0, 1), \quad \mathbf{q}_3 = (0, 1, 0),$$

with eigenvalues 5, 3, 2. These are orthonormal, so  $\mathbf{Q} = (\mathbf{q}_1 \mid \mathbf{q}_2 \mid \mathbf{q}_3)$  is orthogonal and  $\mathbf{Q}^\top \mathbf{A} \mathbf{Q} = \text{diag}(5, 3, 2)$ . As a check,  $5 + 3 + 2 = 10 = \text{tr } \mathbf{A}$ .

### Exercise B2: Identify the conic.

Identify the curve  $5x^2 + 4xy + 5y^2 = 1$ : say what kind of conic it is, find its axes, and find their half-lengths.

*Solution.* The symmetric matrix of the form is  $\mathbf{A} = \begin{pmatrix} 5 & 2 \\ 2 & 5 \end{pmatrix}$ , the off-diagonal entries being half of 4. Its characteristic polynomial is  $x^2 - 10x + 21 = (x - 7)(x - 3)$ , so the eigenvalues are 7 and 3. For  $\lambda = 7$ ,  $\mathbf{A} - 7\mathbf{I} = \begin{pmatrix} -2 & 2 \\ 2 & -2 \end{pmatrix}$  gives  $\mathbf{q}_1 = \frac{1}{\sqrt{2}}(1, 1)$ ; for  $\lambda = 3$ ,  $\mathbf{q}_2 = \frac{1}{\sqrt{2}}(1, -1)$ .

By [Theorem 11.5.5](#) the equation becomes  $7y_1^2 + 3y_2^2 = 1$ . Both eigenvalues are positive, so the curve is an ellipse. Its axes are the lines  $y = x$  and  $y = -x$ , with half-lengths  $1/\sqrt{7}$  along  $\mathbf{q}_1$  and  $1/\sqrt{3}$  along  $\mathbf{q}_2$ ; the major axis is the second one. Check: the point  $\frac{1}{\sqrt{3}}\mathbf{q}_2 = \frac{1}{\sqrt{6}}(1, -1)$  gives  $\frac{5}{6} - \frac{4}{6} + \frac{5}{6} = 1$ .

### Exercise B3: A sum of rank-one projections.

Using your answer to [Exercise \(B1: Orthogonally diagonalize\)](#) (b), write

$$\mathbf{A} = \begin{pmatrix} 4 & 0 & 1 \\ 0 & 2 & 0 \\ 1 & 0 & 4 \end{pmatrix}$$

as  $\sum_i \lambda_i \mathbf{q}_i \mathbf{q}_i^\top$ , with the three matrices  $\mathbf{q}_i \mathbf{q}_i^\top$  written out, and verify the identity by adding them.

*Solution.* With  $\mathbf{q}_1 = \frac{1}{\sqrt{2}}(1, 0, 1)$ ,  $\mathbf{q}_2 = \frac{1}{\sqrt{2}}(-1, 0, 1)$ ,  $\mathbf{q}_3 = (0, 1, 0)$  and  $\lambda = 5, 3, 2$ ,

$$\mathbf{q}_1 \mathbf{q}_1^\top = \frac{1}{2} \begin{pmatrix} 1 & 0 & 1 \\ 0 & 0 & 0 \\ 1 & 0 & 1 \end{pmatrix}, \quad \mathbf{q}_2 \mathbf{q}_2^\top = \frac{1}{2} \begin{pmatrix} 1 & 0 & -1 \\ 0 & 0 & 0 \\ -1 & 0 & 1 \end{pmatrix},$$

and  $\mathbf{q}_3\mathbf{q}_3^\top$  is the matrix with a single 1 in position  $(2, 2)$ . Then

$$5\mathbf{q}_1\mathbf{q}_1^\top + 3\mathbf{q}_2\mathbf{q}_2^\top = \begin{pmatrix} 4 & 0 & 1 \\ 0 & 0 & 0 \\ 1 & 0 & 4 \end{pmatrix},$$

since  $\frac{1}{2}(5+3) = 4$  on the diagonal and  $\frac{1}{2}(5-3) = 1$  off it. Adding  $2\mathbf{q}_3\mathbf{q}_3^\top$  inserts the 2 in position  $(2, 2)$  and gives  $\mathbf{A}$ , as [Corollary 11.5.4](#) requires.

### 11.5.6.3 C. Going deeper

#### Exercise C1: One eigenvalue only.

- Let  $\mathbf{A} \in M_n(\mathbb{R})$  be symmetric with every eigenvalue equal to  $\lambda$ . Prove that  $\mathbf{A} = \lambda\mathbf{I}$ .
- Show that (a) fails without symmetry, by exhibiting a matrix in  $M_2(\mathbb{R})$  whose only eigenvalue is 0 and which is not  $\mathbf{0}$ .
- Deduce that a symmetric  $\mathbf{A}$  with  $\mathbf{A}^2 = \mathbf{A}$  is the matrix of an orthogonal projection.

*Solution.*

- By [Corollary 11.5.2](#),  $\mathbf{A} = \mathbf{Q}\mathbf{D}\mathbf{Q}^\top$  with  $\mathbf{D}$  diagonal carrying the eigenvalues, so  $\mathbf{D} = \lambda\mathbf{I}$ . Hence  $\mathbf{A} = \mathbf{Q}(\lambda\mathbf{I})\mathbf{Q}^\top = \lambda\mathbf{Q}\mathbf{Q}^\top = \lambda\mathbf{I}$ .
- $\mathbf{J}_2(0) = \begin{pmatrix} 0 & 1 \\ 0 & 0 \end{pmatrix}$  has characteristic polynomial  $x^2$ , so 0 is its only eigenvalue, and it is not the zero matrix. It is of course not symmetric.
- If  $\mathbf{A}^2 = \mathbf{A}$  and  $\mathbf{A}\mathbf{x} = \lambda\mathbf{x}$  with  $\mathbf{x} \neq \mathbf{0}$ , then  $\lambda^2\mathbf{x} = \mathbf{A}^2\mathbf{x} = \mathbf{A}\mathbf{x} = \lambda\mathbf{x}$ , so  $\lambda^2 = \lambda$  and  $\lambda \in \{0, 1\}$ . By [Corollary 11.5.2](#),  $\mathbf{A} = \mathbf{Q}\mathbf{D}\mathbf{Q}^\top$  with  $\mathbf{D}$  diagonal of zeros and ones, so  $\mathbf{A} = \sum_{i \in S} \mathbf{q}_i\mathbf{q}_i^\top$  where  $S$  indexes the eigenvalue 1 ([Corollary 11.5.4](#)). That is the sum of the projections onto an orthonormal basis of  $U = E_1(\mathbf{A})$ , which is the orthogonal projection  $P_U$  by [Theorem 10.3.3](#).

#### Exercise C2: Diagonalizing two matrices at once.

Let  $\mathbf{A}, \mathbf{B} \in M_n(\mathbb{R})$  be symmetric with  $\mathbf{AB} = \mathbf{BA}$ .

- Prove that each eigenspace  $E_\lambda(\mathbf{A})$  is  $\mathbf{B}$ -invariant, and that the restriction of  $T_{\mathbf{B}}$  to it is self-adjoint.
- Prove that there is a single  $\mathbf{Q} \in O(n)$  with  $\mathbf{Q}^\top\mathbf{A}\mathbf{Q}$  and  $\mathbf{Q}^\top\mathbf{B}\mathbf{Q}$  both diagonal.
- Show that the conclusion fails without commutativity, using  $\begin{pmatrix} 1 & 0 \\ 0 & 0 \end{pmatrix}$  and  $\begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}$ .

*Hint for (b): apply [Theorem 11.5.1](#) inside each eigenspace of  $\mathbf{A}$ .*

*Solution.*

- Invariance is [Theorem 8.9.1](#): if  $\mathbf{A}\mathbf{x} = \lambda\mathbf{x}$  then  $\mathbf{A}(\mathbf{B}\mathbf{x}) = \mathbf{B}(\mathbf{A}\mathbf{x}) = \lambda\mathbf{B}\mathbf{x}$ . For self-adjointness of the restriction, take  $\mathbf{u}, \mathbf{v} \in E_\lambda(\mathbf{A})$ ; then  $\langle \mathbf{B}\mathbf{u}, \mathbf{v} \rangle = \langle \mathbf{u}, \mathbf{B}\mathbf{v} \rangle$  because  $\mathbf{B}$  is symmetric on all of  $\mathbb{R}^n$ , and both  $\mathbf{B}\mathbf{u}$  and  $\mathbf{B}\mathbf{v}$  lie in  $E_\lambda(\mathbf{A})$  by the first part,

so the identity holds inside the subspace with its restricted inner product.

- (b) Let  $\lambda_1, \dots, \lambda_k$  be the distinct eigenvalues of  $\mathbf{A}$ . By [Theorem 11.5.1](#),  $\mathbb{R}^n$  is the direct sum of the eigenspaces  $E_{\lambda_i}(\mathbf{A})$ , and these are mutually orthogonal ([Theorem 11.2.3](#)). By (a) and [Theorem 11.5.1](#) applied to the self-adjoint operator  $T_{\mathbf{B}}|_{E_{\lambda_i}(\mathbf{A})}$ , each  $E_{\lambda_i}(\mathbf{A})$  has an orthonormal basis of eigenvectors of  $\mathbf{B}$ . Every vector of  $E_{\lambda_i}(\mathbf{A})$  is an eigenvector of  $\mathbf{A}$ , so these basis vectors are common eigenvectors. Stringing the  $k$  bases together gives an orthonormal basis of  $\mathbb{R}^n$ , orthonormal across the blocks because distinct eigenspaces of  $\mathbf{A}$  are orthogonal. Let  $\mathbf{Q}$  have those vectors as columns; then  $\mathbf{Q} \in O(n)$  and both  $\mathbf{Q}^\top \mathbf{A} \mathbf{Q}$  and  $\mathbf{Q}^\top \mathbf{B} \mathbf{Q}$  are diagonal.
- (c) These two do not commute:  $\mathbf{A} \mathbf{B} = \begin{pmatrix} 0 & 1 \\ 0 & 0 \end{pmatrix}$  while  $\mathbf{B} \mathbf{A} = \begin{pmatrix} 0 & 0 \\ 1 & 0 \end{pmatrix}$ . And no common eigenbasis exists: the eigenvectors of  $\mathbf{A}$  are the multiples of  $\mathbf{e}_1$  and of  $\mathbf{e}_2$ , while  $\mathbf{B} \mathbf{e}_1 = \mathbf{e}_2$  is not a multiple of  $\mathbf{e}_1$ . Commutativity is necessary as well as sufficient, since two simultaneously diagonalized matrices commute.

### Exercise C3: A rotation-scaling in normal form.

Let  $\mathbf{A} = \begin{pmatrix} 3 & 4 \\ -4 & 3 \end{pmatrix} \in M_2(\mathbb{R})$ .

- (a) Verify that  $\mathbf{A}$  is normal and not symmetric, and find its complex eigenvalues.
- (b) Find an orthogonal  $\mathbf{Q}$  with  $\mathbf{Q}^\top \mathbf{A} \mathbf{Q} = \mathbf{\Lambda}(\lambda)$  for the eigenvalue  $\lambda$  with positive imaginary part.
- (c) Explain why no orthogonal  $\mathbf{Q}$  makes  $\mathbf{Q}^\top \mathbf{A} \mathbf{Q}$  diagonal, and why this does not contradict [Theorem 11.4.2](#).

*Solution.*

- (a)  $\mathbf{A}^\top \mathbf{A} = \mathbf{A} \mathbf{A}^\top = 25\mathbf{I}$ , so  $\mathbf{A}$  is normal;  $\mathbf{A}^\top \neq \mathbf{A}$  since  $4 \neq -4$ . Its characteristic polynomial is  $x^2 - 6x + 25 = (x - 3)^2 + 16$ , with roots  $3 \pm 4i$ .
- (b) The eigenvalue with positive imaginary part is  $\lambda = 3 + 4i$ , so  $\mathbf{\Lambda}(\lambda) = \begin{pmatrix} 3 & -4 \\ 4 & 3 \end{pmatrix}$ . Take  $\mathbf{Q} = \text{diag}(1, -1)$ , which is orthogonal. Then

$$\mathbf{Q}^\top \mathbf{A} \mathbf{Q} = \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix} \begin{pmatrix} 3 & 4 \\ -4 & 3 \end{pmatrix} \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix} = \begin{pmatrix} 3 & -4 \\ 4 & 3 \end{pmatrix},$$

which is  $\mathbf{\Lambda}(3 + 4i)$ . This is Case 2 of the proof of [Theorem 11.5.3](#) with  $p = 3$  and  $q = 4$ .

- (c) If  $\mathbf{Q}^\top \mathbf{A} \mathbf{Q}$  were diagonal for some  $\mathbf{Q} \in O(2)$ , then  $\mathbf{A} = \mathbf{Q} \mathbf{D} \mathbf{Q}^\top$  would be symmetric by [Corollary 11.5.2](#), and it is not. There is no contradiction with [Theorem 11.4.2](#), which concerns a **complex** inner product space: over  $\mathbb{C}^2$  the matrix  $\mathbf{A}$  is indeed unitarily diagonalizable, to  $\text{diag}(3+4i, 3-4i)$ , but the diagonalizing basis  $\frac{1}{\sqrt{2}}(1, \pm i)$  is not real, so it is unavailable inside  $\mathbb{R}^2$ .

## 11.6 The Shape of an Orthogonal Matrix

An orthogonal matrix is normal, so the real normal form of the previous section applies to it — and the extra information that an orthogonal matrix carries, that every eigenvalue has modulus 1, collapses that form into something very small. What comes out is a complete description of the rigid motions of  $\mathbb{R}^n$  fixing the origin: a few axes reversed, and the rest of the space turned in independent two-dimensional planes. In dimension 3 this is the statement that every rotation has an axis, and we push it as far as an explicit formula for the rotation matrix in terms of its axis and angle.

Throughout,  $\mathbb{R}^n$  carries the standard inner product, and  $\mathbf{Q}$  is a real matrix.

### 11.6.1 From the real normal form to the canonical form

Two pieces of notation, both recalled. For  $\theta \in \mathbb{R}$ , Chapter 10 wrote

$$\mathbf{R}_\theta = \begin{pmatrix} \cos \theta & -\sin \theta \\ \sin \theta & \cos \theta \end{pmatrix}$$

for the matrix that rotates the plane by  $\theta$  ([Theorem 10.7.5](#)). It is exactly the rotation-scaling block of Chapter 9 for a complex number on the unit circle:  $\mathbf{R}_\theta = \mathbf{\Lambda}(e^{i\theta})$  in the notation of [Definition 9.4.1](#), since  $e^{i\theta} = \cos \theta + i \sin \theta$ . And  $\mathbf{A} \oplus \mathbf{B}$  is the block diagonal matrix with diagonal blocks  $\mathbf{A}$  and  $\mathbf{B}$ .

Now the observation that starts everything. If  $\mathbf{Q} \in \mathrm{O}(n)$  then  $\mathbf{Q}^\top \mathbf{Q} = \mathbf{I}_n = \mathbf{Q} \mathbf{Q}^\top$ , so  $\mathbf{Q}$  commutes with its own adjoint: **every orthogonal matrix is normal**. So [Theorem 11.5.3](#) applies. All that remains is to say what the blocks can be, and for that we use the one thing an orthogonal matrix has that a general normal matrix does not.

#### Theorem 11.6.1 — Canonical Form of an Orthogonal Matrix

Let  $\mathbf{Q} \in \mathrm{O}(n)$ . Then there are integers  $p, q, m \geq 0$  with  $p + q + 2m = n$ , and angles  $\theta_1, \dots, \theta_m$  with  $0 < \theta_j < \pi$ , and a matrix  $\mathbf{P} \in \mathrm{O}(n)$ , such that

$$\mathbf{P}^\top \mathbf{Q} \mathbf{P} = \mathbf{I}_p \oplus (-\mathbf{I}_q) \oplus \mathbf{R}_{\theta_1} \oplus \cdots \oplus \mathbf{R}_{\theta_m}.$$

Moreover  $\det \mathbf{Q} = (-1)^q$ , and  $p, q$  and the multiset  $\{\theta_1, \dots, \theta_m\}$  are determined by  $\mathbf{Q}$ .

*Idea.*  $\mathbf{Q}$  is normal, so [Theorem 11.5.3](#) already hands us a block diagonal matrix with  $1 \times 1$  blocks  $[a]$  and  $2 \times 2$  blocks  $\mathbf{\Lambda}(\lambda)$ . The blocks are unconstrained for a general normal matrix; here they are not, because the eigenvalues of an isometry lie on the unit circle. A real eigenvalue on the unit circle is  $\pm 1$ , and a conjugate pair on the unit circle is  $e^{\pm i\theta}$ , which turns the scaling block into a pure rotation. Then sort the  $\pm 1$ s to the front.

*Proof.* The operator  $T_{\mathbf{Q}}$  on  $\mathbb{R}^n$  satisfies  $T_{\mathbf{Q}}^* = T_{\mathbf{Q}^\top}$  by [Theorem 10.6.3](#), since the standard basis is orthonormal, so  $T_{\mathbf{Q}}^* T_{\mathbf{Q}} = T_{\mathbf{Q}^\top \mathbf{Q}} = \mathrm{id}$  and likewise in the other order. Hence  $T_{\mathbf{Q}}$  is normal ([Definition 10.6.3](#)), and by [Theorem 11.5.3](#) there is an orthonormal basis  $\mathcal{B}$  of

$\mathbb{R}^n$  in which

$$[T_{\mathbf{Q}}]_{\mathcal{B}} = [a_1] \oplus \cdots \oplus [a_r] \oplus \mathbf{\Lambda}(\lambda_1) \oplus \cdots \oplus \mathbf{\Lambda}(\lambda_m),$$

where  $a_i \in \mathbb{R}$  and  $\lambda_j = c_j + d_j i$  with  $d_j > 0$ . Let  $\mathbf{P}$  be the matrix whose columns are the vectors of  $\mathcal{B}$ . Its columns are orthonormal, so  $\mathbf{P} \in \mathrm{O}(n)$  by [Theorem 10.7.3](#) (f), and  $\mathbf{P}^\top \mathbf{Q} \mathbf{P} = \mathbf{P}^{-1} \mathbf{Q} \mathbf{P} = [T_{\mathbf{Q}}]_{\mathcal{B}}$  by [Theorem 3.7.3](#).

**The blocks.** A block diagonal matrix is triangular by blocks, so the eigenvalues of  $[T_{\mathbf{Q}}]_{\mathcal{B}}$  over  $\mathbb{C}$  are the  $a_i$  together with the roots of each  $\det(x\mathbf{I}_2 - \mathbf{\Lambda}(\lambda_j)) = (x - c_j)^2 + d_j^2$ , namely  $\lambda_j$  and  $\overline{\lambda_j}$ . Now  $\mathrm{O}(n) = \mathrm{U}(n) \cap M_n(\mathbb{R})$  by [Proposition 10.7.4](#) (a), so  $\mathbf{Q}$  is unitary, and every complex eigenvalue  $\lambda$  of  $\mathbf{Q}$  satisfies  $|\lambda| = 1$  by [Proposition 10.7.1](#) applied to  $T_{\mathbf{Q}}$  on  $\mathbb{C}^n$ . Therefore  $a_i = \pm 1$  for each  $i$ , and  $c_j^2 + d_j^2 = 1$  for each  $j$ . Since  $c_j^2 + d_j^2 = 1$  and  $d_j > 0$ , there is a unique  $\theta_j \in (0, \pi)$  with  $c_j = \cos \theta_j$  and  $d_j = \sin \theta_j$ , and then  $\mathbf{\Lambda}(\lambda_j) = \mathbf{R}_{\theta_j}$ .

**Sorting.** Let  $p$  be the number of  $i$  with  $a_i = 1$  and  $q$  the number with  $a_i = -1$ , so  $p + q + 2m = n$ . Reordering the vectors of  $\mathcal{B}$  so that the  $+1$  blocks come first, then the  $-1$  blocks, then the rotation blocks in pairs, leaves the basis orthonormal and permutes the diagonal blocks; the new  $\mathbf{P}$  is the old one with its columns permuted, still orthogonal. This gives the displayed form.

**The determinant and uniqueness.** By [Theorem 6.5.4](#) and [Corollary 6.5.7](#),  $\det \mathbf{Q} = \det(\mathbf{P}^\top \mathbf{Q} \mathbf{P})$ , and the determinant of a block diagonal matrix is the product of the determinants of its blocks ([Theorem 6.8.4](#)), which are 1,  $-1$  and  $\det \mathbf{R}_{\theta_j} = \cos^2 \theta_j + \sin^2 \theta_j = 1$ . Hence  $\det \mathbf{Q} = (-1)^q$ . Finally, similar matrices have the same eigenvalues with the same multiplicities, and reading the eigenvalues off the displayed form gives 1 with multiplicity  $p$ ,  $-1$  with multiplicity  $q$ , and the pairs  $e^{\pm i\theta_j}$ ; so  $p$ ,  $q$  and the  $\theta_j$  are recovered from  $\mathbf{Q}$  alone. This proves the theorem. ■

So an orthogonal matrix is, up to an orthogonal change of coordinates, nothing but a list of angles together with a count of reversed axes. The number  $q$  of reversed axes is not itself an invariant of the *geometry* — a  $-1$  block and a  $-1$  block together make  $-\mathbf{I}_2 = \mathbf{R}_\pi$ , a rotation by  $\pi$  — which is why the theorem demands  $\theta_j \neq \pi$  and keeps the  $-1$ s separate. What the determinant sees is only the parity of  $q$ .

*Quick check.* What is the canonical form of a  $\mathbf{Q} \in \mathrm{O}(2)$  with  $\det \mathbf{Q} = -1$ ? Compare with [Theorem 10.7.5](#).

*Solution.* Here  $p + q + 2m = 2$  and  $(-1)^q = -1$ , so  $q$  is odd, forcing  $q = 1$ ,  $p = 1$ ,  $m = 0$ . The canonical form is  $\mathbf{I}_1 \oplus (-\mathbf{I}_1) = \mathrm{diag}(1, -1)$ . Chapter 10 found that such a  $\mathbf{Q}$  is the reflection  $\mathbf{M}_\theta$  in a line  $L$ , fixing  $L$  and negating  $L^\perp$ ; in the orthonormal basis of eigenvectors of [Theorem 10.7.5](#) (b) its matrix is exactly  $\mathrm{diag}(1, -1)$ . Note what the canonical form drops: the angle of the mirror. All reflections of the plane are orthogonally similar to one another, and [Exercise \(C1: The plane, re-derived\)](#) makes this comparison precise.

## 11.6.2 Rotations of space

In dimension 3 the arithmetic  $p + q + 2m = 3$  leaves very few possibilities, and they all look the same.

### Corollary 11.6.2 — Orthogonal Matrices in Dimension Three

Let  $\mathbf{Q} \in \mathbf{O}(3)$ . Then there are an orthonormal basis  $(\mathbf{u}, \mathbf{w}_1, \mathbf{w}_2)$  of  $\mathbb{R}^3$  and an angle  $\theta \in [0, \pi]$  such that, in that basis, the matrix of  $T_{\mathbf{Q}}$  is

$$[1] \oplus \mathbf{R}_{\theta} \quad \text{if } \det \mathbf{Q} = 1, \quad [-1] \oplus \mathbf{R}_{\theta} \quad \text{if } \det \mathbf{Q} = -1.$$

In particular every  $\mathbf{Q} \in \mathbf{SO}(3)$  fixes the line  $\text{span}(\mathbf{u})$  pointwise and rotates the plane  $\text{span}(\mathbf{u})^{\perp}$  by  $\theta$ .

*Proof.* Write the canonical form of Theorem 11.6.1, with  $p + q + 2m = 3$  and  $\det \mathbf{Q} = (-1)^q$ .

*Case*  $\det \mathbf{Q} = 1$ . Then  $q$  is even, so  $q \in \{0, 2\}$ . If  $q = 0$  then  $p + 2m = 3$ , giving  $(p, m) = (3, 0)$  or  $(1, 1)$ ; if  $q = 2$  then  $p = 1$  and  $m = 0$ . The three forms are

$$\mathbf{I}_3 = [1] \oplus \mathbf{R}_0, \quad [1] \oplus \mathbf{R}_{\theta_1}, \quad [1] \oplus (-\mathbf{I}_2) = [1] \oplus \mathbf{R}_{\pi},$$

and in every case  $p \geq 1$ , so the form is  $[1] \oplus \mathbf{R}_{\theta}$  with  $\theta \in [0, \pi]$ .

*Case*  $\det \mathbf{Q} = -1$ . Then  $q$  is odd, so  $q \in \{1, 3\}$ . If  $q = 1$  then  $p + 2m = 2$ , giving  $(p, m) = (2, 0)$  or  $(0, 1)$ ; if  $q = 3$  then  $p = m = 0$ . The three forms are  $[-1] \oplus \mathbf{I}_2 = [-1] \oplus \mathbf{R}_0$ , then  $[-1] \oplus \mathbf{R}_{\theta_1}$ , and  $-\mathbf{I}_3 = [-1] \oplus \mathbf{R}_{\pi}$ .

In both cases the first basis vector  $\mathbf{u}$  spans the  $\pm 1$  eigenline and  $(\mathbf{w}_1, \mathbf{w}_2)$  is an orthonormal basis of  $\text{span}(\mathbf{u})^{\perp}$ , which is what the last sentence asserts when  $\det \mathbf{Q} = 1$ . ■

Chapter 10 proved the  $\mathbf{SO}(3)$  statement directly, as Theorem 10.7.6, by producing a real eigenvalue from the odd degree of the characteristic polynomial and then invoking the classification of  $\mathbf{O}(2)$ . We do not repeat that argument: the point here is that the same conclusion falls out of the canonical form with no work at all, and that the same three lines also settle  $\det \mathbf{Q} = -1$ . An orthogonal matrix of determinant  $-1$  in dimension 3 is a **rotatory reflection**: a rotation about an axis followed by the reflection in the plane perpendicular to it. Its two extreme cases are the plane reflection ( $\theta = 0$ ) and the antipodal map  $-\mathbf{I}_3$  ( $\theta = \pi$ ).

*Remark.* The angle  $\theta$  is determined by  $\mathbf{Q}$  only up to sign, because exchanging  $\mathbf{w}_1$  and  $\mathbf{w}_2$  replaces  $\mathbf{R}_{\theta}$  by  $\mathbf{R}_{-\theta}$ . Restricting to  $\theta \in [0, \pi]$ , as above, makes it unique; to speak of a *signed* angle one must also fix which of the two unit vectors on the axis is called  $\mathbf{u}$ .

## 11.6.3 The cross-product matrix

The canonical form tells us that a rotation of  $\mathbb{R}^3$  exists in a suitable basis. For computation we want it in the standard basis, written directly in terms of the axis. The bookkeeping is done by

one matrix.

Recall from Chapter 6 that the cross product of  $\mathbf{a}, \mathbf{b} \in \mathbb{R}^3$  is

$$\mathbf{a} \times \mathbf{b} = (a_2b_3 - a_3b_2, a_3b_1 - a_1b_3, a_1b_2 - a_2b_1).$$

For fixed  $\mathbf{a}$  this is linear in  $\mathbf{b}$ , so it is given by a matrix.

### Definition 11.6.1 — Cross-product Matrix

For  $\mathbf{n} = (n_1, n_2, n_3) \in \mathbb{R}^3$ , the **cross-product matrix** of  $\mathbf{n}$  is

$$\mathbf{K}_{\mathbf{n}} := \begin{pmatrix} 0 & -n_3 & n_2 \\ n_3 & 0 & -n_1 \\ -n_2 & n_1 & 0 \end{pmatrix} \in M_3(\mathbb{R}),$$

so that  $\mathbf{K}_{\mathbf{n}}\mathbf{v} = \mathbf{n} \times \mathbf{v}$  for every  $\mathbf{v} \in \mathbb{R}^3$ .

The last clause is a check of the three entries against the formula above, with  $\mathbf{a} = \mathbf{n}$  and  $\mathbf{b} = \mathbf{v}$ . The map  $\mathbf{n} \mapsto \mathbf{K}_{\mathbf{n}}$  is linear and injective, and its image is exactly the set of skew-symmetric  $3 \times 3$  matrices: a skew matrix has zero diagonal and three free entries below it, and those three entries are  $n_3, -n_2, n_1$ . We record the facts we need, all of them for a **unit** vector.

### Lemma 11.6.3 — Properties of the Cross-product Matrix

Let  $\mathbf{n} \in \mathbb{R}^3$  with  $\|\mathbf{n}\| = 1$ , and write  $\mathbf{K} = \mathbf{K}_{\mathbf{n}}$ . Then

- (a)  $\mathbf{K}^T = -\mathbf{K}$  and  $\mathbf{K}\mathbf{n} = \mathbf{0}$ ;
- (b)  $\mathbf{K}^2 = \mathbf{nn}^T - \mathbf{I}_3$ , so  $\mathbf{K}^2$  is symmetric and  $\mathbf{K}^2\mathbf{v} = -\mathbf{v}$  for every  $\mathbf{v} \perp \mathbf{n}$ ;
- (c)  $\mathbf{K}^3 = -\mathbf{K}$ ;
- (d) for every  $\mathbf{v} \perp \mathbf{n}$ :  $\|\mathbf{K}\mathbf{v}\| = \|\mathbf{v}\|$ ,  $\mathbf{K}\mathbf{v} \perp \mathbf{v}$  and  $\mathbf{K}\mathbf{v} \perp \mathbf{n}$ .

*Idea.* Everything follows from (b), and (b) is one entry of a matrix product together with  $n_1^2 + n_2^2 + n_3^2 = 1$ . Part (c) is (b) multiplied by  $\mathbf{K}$ , where the term  $\mathbf{Knn}^T$  dies by (a). For (d), a norm is computed through its square, and  $\|\mathbf{K}\mathbf{v}\|^2 = \mathbf{v}^T \mathbf{K}^T \mathbf{K} \mathbf{v} = -\mathbf{v}^T \mathbf{K}^2 \mathbf{v}$  turns (d) into (b) as well.

*Proof.*

- (a) Transposing  $\mathbf{K}_{\mathbf{n}}$  changes the sign of each of the three off-diagonal pairs, and the diagonal is zero. For the second claim,  $\mathbf{K}\mathbf{n} = \mathbf{n} \times \mathbf{n} = \mathbf{0}$  directly from the formula, each coordinate being of the form  $n_i n_j - n_j n_i$ .
- (b) The (1,1) entry of  $\mathbf{K}^2$  is  $0 \cdot 0 + (-n_3)n_3 + n_2(-n_2) = -n_2^2 - n_3^2 = n_1^2 - 1$ , using  $\|\mathbf{n}\|^2 = 1$ ; and  $(\mathbf{nn}^T - \mathbf{I}_3)_{11} = n_1^2 - 1$ . The (1,2) entry of  $\mathbf{K}^2$  is  $0(-n_3) + (-n_3)0 + n_2 n_1 = n_1 n_2 = (\mathbf{nn}^T)_{12}$ , and the (1,3) entry is  $0n_2 + (-n_3)(-n_1) + n_2 0 = n_1 n_3$ . The remaining six entries are obtained by cycling the indices  $1 \rightarrow 2 \rightarrow 3 \rightarrow 1$ , under which both  $\mathbf{K}$  and  $\mathbf{nn}^T - \mathbf{I}_3$  are unchanged in form. Hence  $\mathbf{K}^2 = \mathbf{nn}^T - \mathbf{I}_3$ , which is symmetric since  $(\mathbf{nn}^T)^T = \mathbf{nn}^T$ . If  $\mathbf{v} \perp \mathbf{n}$  then  $\mathbf{n}^T \mathbf{v} = 0$ , so  $\mathbf{K}^2 \mathbf{v} = \mathbf{n}(\mathbf{n}^T \mathbf{v}) - \mathbf{v} = -\mathbf{v}$ .
- (c) By (b) and (a),  $\mathbf{K}^3 = \mathbf{K}(\mathbf{nn}^T - \mathbf{I}_3) = (\mathbf{K}\mathbf{n})\mathbf{n}^T - \mathbf{K} = -\mathbf{K}$ .

(d) Let  $\mathbf{v} \perp \mathbf{n}$ . Using (a) and then (b),

$$\|\mathbf{K}\mathbf{v}\|^2 = (\mathbf{K}\mathbf{v})^\top (\mathbf{K}\mathbf{v}) = \mathbf{v}^\top \mathbf{K}^\top \mathbf{K} \mathbf{v} = -\mathbf{v}^\top \mathbf{K}^2 \mathbf{v} = \mathbf{v}^\top \mathbf{v} = \|\mathbf{v}\|^2.$$

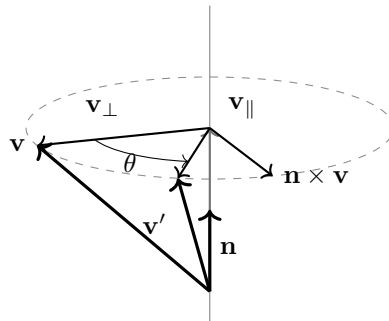
Next,  $\langle \mathbf{K}\mathbf{v}, \mathbf{v} \rangle = \mathbf{v}^\top \mathbf{K} \mathbf{v}$  is a  $1 \times 1$  matrix, hence equal to its own transpose  $\mathbf{v}^\top \mathbf{K}^\top \mathbf{v} = -\mathbf{v}^\top \mathbf{K} \mathbf{v}$ ; a number equal to its own negative is 0, so  $\mathbf{K}\mathbf{v} \perp \mathbf{v}$ . Finally  $\langle \mathbf{K}\mathbf{v}, \mathbf{n} \rangle = \mathbf{n}^\top \mathbf{K} \mathbf{v} = (\mathbf{K}^\top \mathbf{n})^\top \mathbf{v} = -(\mathbf{K}\mathbf{n})^\top \mathbf{v} = 0$  by (a). This proves the lemma. ■

Part (c) is the identity that makes every power of  $\mathbf{K}$  reduce to  $\mathbf{K}$  or  $\mathbf{K}^2$ , and it is the reason Rodrigues' formula has only three terms and no more.

### 11.6.4 Rodrigues' formula

Fix a unit vector  $\mathbf{n}$  and an angle  $\theta$ . We want a formula for “turn space by  $\theta$  about the axis  $\text{span}(\mathbf{n})$ ”. The construction is forced by the picture: split a vector into the part along the axis, which does not move, and the part perpendicular to the axis, which turns inside a single plane.

Write  $\mathbf{v}_\parallel = \langle \mathbf{v}, \mathbf{n} \rangle \mathbf{n} = \mathbf{n} \mathbf{n}^\top \mathbf{v}$  for the component of  $\mathbf{v}$  along the axis and  $\mathbf{v}_\perp = \mathbf{v} - \mathbf{v}_\parallel$  for the component perpendicular to it; this is the splitting of [Theorem 10.3.2](#) for  $U = \text{span}(\mathbf{n})$ , and  $\mathbf{v}_\parallel = P_U \mathbf{v}$  by [Theorem 10.3.3](#) (a) with the orthonormal basis  $(\mathbf{n})$  of  $U$ . Write  $\mathbf{v}'$  for the vector we want, the image of  $\mathbf{v}$  under the turn.



the axis part stays, the perpendicular part turns

By [Lemma 11.6.3](#) (d), the vector  $\mathbf{K}_\mathbf{n} \mathbf{v}_\perp$  has the same length as  $\mathbf{v}_\perp$  and is perpendicular to both  $\mathbf{v}_\perp$  and  $\mathbf{n}$ : it is  $\mathbf{v}_\perp$  turned by a quarter turn inside the plane  $\mathbf{n}^\perp$ . So turning  $\mathbf{v}_\perp$  by  $\theta$  means taking  $\cos \theta \mathbf{v}_\perp + \sin \theta \mathbf{K}_\mathbf{n} \mathbf{v}_\perp$ , and  $\mathbf{v}' = \mathbf{v}_\parallel + \cos \theta \mathbf{v}_\perp + \sin \theta \mathbf{K}_\mathbf{n} \mathbf{v}_\perp$ . Finally  $\mathbf{K}_\mathbf{n} \mathbf{v}_\perp = \mathbf{K}_\mathbf{n} \mathbf{v}$ , since  $\mathbf{K}_\mathbf{n} \mathbf{v}_\parallel$  is a multiple of  $\mathbf{K}_\mathbf{n} \mathbf{n} = \mathbf{0}$ . That is the definition we take, and the theorem says it is the right one.

#### Theorem 11.6.4 — Rodrigues' Rotation Formula

Let  $\mathbf{n} \in \mathbb{R}^3$  be a unit vector, let  $\theta \in \mathbb{R}$ , and write  $\mathbf{K} = \mathbf{K}_\mathbf{n}$ . Define

$$\mathbf{R}_{\mathbf{n},\theta} := \mathbf{I}_3 + \sin \theta \mathbf{K} + (1 - \cos \theta) \mathbf{K}^2.$$



- (a) For every  $\mathbf{v} \in \mathbb{R}^3$ ,

$$\mathbf{R}_{\mathbf{n},\theta}\mathbf{v} = \langle \mathbf{v}, \mathbf{n} \rangle \mathbf{n} + \cos \theta (\mathbf{v} - \langle \mathbf{v}, \mathbf{n} \rangle \mathbf{n}) + \sin \theta (\mathbf{n} \times \mathbf{v}).$$

- (b)  $\mathbf{R}_{\mathbf{n},\theta}\mathbf{n} = \mathbf{n}$ , and for every unit  $\mathbf{w} \perp \mathbf{n}$  the list  $\mathcal{B} = (\mathbf{n}, \mathbf{w}, \mathbf{n} \times \mathbf{w})$  is an orthonormal basis of  $\mathbb{R}^3$  with  $[T_{\mathbf{R}_{\mathbf{n},\theta}}]_{\mathcal{B}} = [1] \oplus \mathbf{R}_{\theta}$ . In particular  $\mathbf{R}_{\mathbf{n},\theta} \in \text{SO}(3)$ .  
 (c) Conversely, every  $\mathbf{Q} \in \text{SO}(3)$  equals  $\mathbf{R}_{\mathbf{n},\theta}$  for some unit vector  $\mathbf{n}$  and some  $\theta \in [0, \pi]$ .

*Idea.* Part (a) is the paragraph above, read backwards: substitute  $\mathbf{nn}^\top = \mathbf{K}^2 + \mathbf{I}_3$  from Lemma 11.6.3 (b) into the geometric expression and collect terms. Part (b) then needs only the images of three basis vectors, and every one of them is a two-line computation with (a) and the lemma; the matrix that comes out is the canonical form of Corollary 11.6.2, which also gives orthogonality and the determinant for free. Part (c) runs Corollary 11.6.2 in reverse: it hands us a basis  $(\mathbf{u}, \mathbf{w}_1, \mathbf{w}_2)$ , and the only question is whether  $\mathbf{w}_2$  is  $\mathbf{u} \times \mathbf{w}_1$  or its negative — and if it is the negative, we change the sign of the angle.

*Proof.*

- (a) By Lemma 11.6.3 (b),  $\mathbf{nn}^\top = \mathbf{K}^2 + \mathbf{I}_3$ . Hence, writing  $\langle \mathbf{v}, \mathbf{n} \rangle \mathbf{n} = \mathbf{nn}^\top \mathbf{v}$  and  $\mathbf{n} \times \mathbf{v} = \mathbf{K}\mathbf{v}$ ,

$$\begin{aligned} \mathbf{nn}^\top \mathbf{v} + \cos \theta (\mathbf{v} - \mathbf{nn}^\top \mathbf{v}) + \sin \theta \mathbf{K}\mathbf{v} \\ &= \cos \theta \mathbf{v} + (1 - \cos \theta) \mathbf{nn}^\top \mathbf{v} + \sin \theta \mathbf{K}\mathbf{v} \\ &= \cos \theta \mathbf{v} + (1 - \cos \theta)(\mathbf{K}^2 + \mathbf{I}_3)\mathbf{v} + \sin \theta \mathbf{K}\mathbf{v} \\ &= (\mathbf{I}_3 + \sin \theta \mathbf{K} + (1 - \cos \theta)\mathbf{K}^2)\mathbf{v}, \end{aligned}$$

since  $\cos \theta + (1 - \cos \theta) = 1$ . The right-hand side is  $\mathbf{R}_{\mathbf{n},\theta}\mathbf{v}$ .

- (b) First  $\mathbf{K}\mathbf{n} = \mathbf{0}$  and  $\mathbf{K}^2\mathbf{n} = \mathbf{K}(\mathbf{K}\mathbf{n}) = \mathbf{0}$  by Lemma 11.6.3 (a), so  $\mathbf{R}_{\mathbf{n},\theta}\mathbf{n} = \mathbf{n}$ . Let  $\mathbf{w}$  be a unit vector with  $\mathbf{w} \perp \mathbf{n}$ , and put  $\mathbf{z} = \mathbf{K}\mathbf{w} = \mathbf{n} \times \mathbf{w}$ . By Lemma 11.6.3 (d),  $\|\mathbf{z}\| = 1$ ,  $\mathbf{z} \perp \mathbf{w}$  and  $\mathbf{z} \perp \mathbf{n}$ , so  $\mathcal{B} = (\mathbf{n}, \mathbf{w}, \mathbf{z})$  is an orthonormal list of three vectors in  $\mathbb{R}^3$ , hence an orthonormal basis (Theorem 10.1.4 and Theorem 1.6.4). Now compute the three images. We have  $\mathbf{R}_{\mathbf{n},\theta}\mathbf{n} = \mathbf{n}$ . Next,  $\mathbf{K}^2\mathbf{w} = -\mathbf{w}$  by Lemma 11.6.3 (b), since  $\mathbf{w} \perp \mathbf{n}$ , so

$$\mathbf{R}_{\mathbf{n},\theta}\mathbf{w} = \mathbf{w} + \sin \theta \mathbf{z} - (1 - \cos \theta)\mathbf{w} = \cos \theta \mathbf{w} + \sin \theta \mathbf{z}.$$

Finally  $\mathbf{z} \perp \mathbf{n}$ , so  $\mathbf{K}^2\mathbf{z} = -\mathbf{z}$ , and  $\mathbf{K}\mathbf{z} = \mathbf{K}^2\mathbf{w} = -\mathbf{w}$ ; hence

$$\mathbf{R}_{\mathbf{n},\theta}\mathbf{z} = \mathbf{z} - \sin \theta \mathbf{w} - (1 - \cos \theta)\mathbf{z} = -\sin \theta \mathbf{w} + \cos \theta \mathbf{z}.$$

Reading off the columns (Definition 3.6.1),  $[T_{\mathbf{R}_{\mathbf{n},\theta}}]_{\mathcal{B}} = [1] \oplus \mathbf{R}_{\theta}$ . This matrix is orthogonal, and it is the matrix of  $T_{\mathbf{R}_{\mathbf{n},\theta}}$  in an orthonormal basis, so  $T_{\mathbf{R}_{\mathbf{n},\theta}}$  is an isometry and  $\mathbf{R}_{\mathbf{n},\theta} \in \text{O}(3)$  by Theorem 10.7.3 (f) and Definition 10.7.2. The two matrices are matrices of the same operator, hence similar (Theorem 3.7.4 (a)), so they have the same determinant (Corollary 6.5.7), namely 1; so  $\mathbf{R}_{\mathbf{n},\theta} \in \text{SO}(3)$ .

- (c) Let  $\mathbf{Q} \in \text{SO}(3)$ . By Corollary 11.6.2 there are an orthonormal basis  $(\mathbf{u}, \mathbf{w}_1, \mathbf{w}_2)$  of  $\mathbb{R}^3$  and  $\theta \in [0, \pi]$  with  $[T_{\mathbf{Q}}]_{(\mathbf{u}, \mathbf{w}_1, \mathbf{w}_2)} = [1] \oplus \mathbf{R}_{\theta}$ . The vector  $\mathbf{u} \times \mathbf{w}_1$  is a unit vector orthogonal to  $\mathbf{u}$  and  $\mathbf{w}_1$  by Lemma 11.6.3 (d), and so is  $\mathbf{w}_2$ ; since  $\text{span}(\mathbf{u}, \mathbf{w}_1)^\perp$  is

a line (Theorem 10.3.2 (c)), its unit vectors are  $\pm \mathbf{u} \times \mathbf{w}_1$ , so  $\mathbf{w}_2 = \varepsilon \mathbf{u} \times \mathbf{w}_1$  with  $\varepsilon = \pm 1$ .

*Case  $\varepsilon = 1$ .* Then (b), applied with  $\mathbf{n} = \mathbf{u}$  and  $\mathbf{w} = \mathbf{w}_1$ , gives  $[T_{\mathbf{R}_{\mathbf{u},\theta}}]_{(\mathbf{u}, \mathbf{w}_1, \mathbf{w}_2)} = [1] \oplus \mathbf{R}_\theta = [T_{\mathbf{Q}}]_{(\mathbf{u}, \mathbf{w}_1, \mathbf{w}_2)}$ . Operators with the same matrix in one basis are equal, so  $\mathbf{Q} = \mathbf{R}_{\mathbf{u},\theta}$ .

*Case  $\varepsilon = -1$ .* Then  $\mathbf{u} \times \mathbf{w}_1 = -\mathbf{w}_2$ , and the matrix of  $T_{\mathbf{Q}}$  in the basis  $(\mathbf{u}, \mathbf{w}_1, -\mathbf{w}_2)$  is obtained from  $[1] \oplus \mathbf{R}_\theta$  by changing the sign of the third basis vector, which changes the sign of the two off-diagonal entries of the  $\mathbf{R}_\theta$  block: it is  $[1] \oplus \mathbf{R}_{-\theta}$ . By the previous case,  $\mathbf{Q} = \mathbf{R}_{\mathbf{u},-\theta}$ . Finally  $\mathbf{K}_{-\mathbf{u}} = -\mathbf{K}_{\mathbf{u}}$ , so

$$\mathbf{R}_{-\mathbf{u},\theta} = \mathbf{I}_3 - \sin \theta \mathbf{K}_{\mathbf{u}} + (1 - \cos \theta) \mathbf{K}_{\mathbf{u}}^2 = \mathbf{R}_{\mathbf{u},-\theta},$$

and  $\mathbf{Q} = \mathbf{R}_{-\mathbf{u},\theta}$  with  $-\mathbf{u}$  a unit vector and  $\theta \in [0, \pi]$ . This proves the theorem. ■

Part (c) is worth restating on its own: **a rotation of  $\mathbb{R}^3$  is nothing but a unit vector and an angle**. That is three numbers of data, one of them redundant, for an object that started life as nine matrix entries subject to six equations.

*Quick check.* Compute  $\mathbf{R}_{\mathbf{n},\pi}$  from Rodrigues' formula and identify the map geometrically.

*Solution.* With  $\sin \pi = 0$  and  $1 - \cos \pi = 2$ , the formula gives  $\mathbf{R}_{\mathbf{n},\pi} = \mathbf{I}_3 + 2\mathbf{K}^2 = \mathbf{I}_3 + 2(\mathbf{nn}^\top - \mathbf{I}_3) = 2\mathbf{nn}^\top - \mathbf{I}_3$ , using Lemma 11.6.3 (b). It fixes  $\mathbf{n}$  and negates every vector perpendicular to  $\mathbf{n}$ : the half-turn about the axis. Equivalently  $\mathbf{R}_{\mathbf{n},\pi} = -(\mathbf{I}_3 - 2\mathbf{nn}^\top) = -\mathbf{H}_{\mathbf{n}}$ , minus the Householder reflection of Definition 10.8.1 in the plane  $\mathbf{n}^\perp$ .

## 11.6.5 Reading off the axis and the angle

Rodrigues' formula runs one way: from  $(\mathbf{n}, \theta)$  to  $\mathbf{Q}$ . The inverse direction is just as cheap, because the formula splits  $\mathbf{Q}$  into a symmetric part and a skew part that carry the two pieces of data separately.

### Theorem 11.6.5 — Axis and Angle from the Matrix

Let  $\mathbf{Q} = \mathbf{R}_{\mathbf{n},\theta} \in \text{SO}(3)$  with  $\mathbf{n}$  a unit vector and  $\theta \in [0, \pi]$ . Then

- (a)  $\text{tr } \mathbf{Q} = 1 + 2 \cos \theta$ , so  $\cos \theta = \frac{1}{2}(\text{tr } \mathbf{Q} - 1)$  and  $\theta$  is determined by  $\mathbf{Q}$ ;
- (b)  $\mathbf{Q} - \mathbf{Q}^\top = 2 \sin \theta \mathbf{K}_{\mathbf{n}}$ ;
- (c) if  $0 < \theta < \pi$ , then  $\sin \theta > 0$  and

$$\mathbf{n} = \frac{1}{2 \sin \theta} (q_{32} - q_{23}, q_{13} - q_{31}, q_{21} - q_{12});$$

if  $\theta = 0$  then  $\mathbf{Q} = \mathbf{I}_3$  and the axis is arbitrary; and if  $\theta = \pi$  then  $\mathbf{Q} + \mathbf{I}_3 = 2\mathbf{nn}^\top$ , whose non-zero columns are the multiples  $2n_i \mathbf{n}$  of  $\mathbf{n}$ .

*Proof.* Write  $\mathbf{K} = \mathbf{K}_{\mathbf{n}}$ . By Lemma 11.6.3 (b),  $\mathbf{K}^2 = \mathbf{n}\mathbf{n}^\top - \mathbf{I}_3$ , so  $\text{tr } \mathbf{K}^2 = \text{tr}(\mathbf{n}\mathbf{n}^\top) - 3 = \|\mathbf{n}\|^2 - 3 = -2$ ; and  $\text{tr } \mathbf{K} = 0$ , the diagonal of  $\mathbf{K}$  being zero.

- (a) Taking traces in Rodrigues' formula, which is legitimate because the trace is linear (Theorem 0.9.6 (1)),

$$\text{tr } \mathbf{Q} = 3 + \sin \theta \cdot 0 + (1 - \cos \theta)(-2) = 1 + 2 \cos \theta.$$

Since  $\cos$  is injective on  $[0, \pi]$ , this determines  $\theta$ .

- (b) By Lemma 11.6.3 (a) and (b),  $\mathbf{K}^\top = -\mathbf{K}$  and  $(\mathbf{K}^2)^\top = \mathbf{K}^2$ . Transposing Rodrigues' formula therefore gives  $\mathbf{Q}^\top = \mathbf{I}_3 - \sin \theta \mathbf{K} + (1 - \cos \theta)\mathbf{K}^2$ , and subtracting,  $\mathbf{Q} - \mathbf{Q}^\top = 2 \sin \theta \mathbf{K}$ .
- (c) For  $\theta \in (0, \pi)$  we have  $\sin \theta > 0$ , so (b) may be divided by  $2 \sin \theta$ , and the three independent entries of  $\mathbf{K}_{\mathbf{n}}$  are  $(\mathbf{K}_{\mathbf{n}})_{32} = n_1$ ,  $(\mathbf{K}_{\mathbf{n}})_{13} = n_2$  and  $(\mathbf{K}_{\mathbf{n}})_{21} = n_3$  by Definition 11.6.1; the displayed formula reads these off  $\mathbf{Q} - \mathbf{Q}^\top$ . If  $\theta = 0$  then Rodrigues' formula gives  $\mathbf{Q} = \mathbf{I}_3$ . If  $\theta = \pi$  then  $\mathbf{Q} = 2\mathbf{n}\mathbf{n}^\top - \mathbf{I}_3$ , as computed in the Quick check above, so  $\mathbf{Q} + \mathbf{I}_3 = 2\mathbf{n}\mathbf{n}^\top$ , whose  $i$ -th column is  $2n_i\mathbf{n}$ ; at least one  $n_i \neq 0$ , so at least one column is a non-zero multiple of  $\mathbf{n}$ . This proves the theorem. ■

The two degenerate angles are exactly the two at which the skew part  $\mathbf{Q} - \mathbf{Q}^\top$  vanishes, and they are exactly the two at which the axis is not determined by a *signed* angle: at  $\theta = 0$  there is no rotation to speak of, and at  $\theta = \pi$  turning by  $+\pi$  and  $-\pi$  about the same axis give the same map. Everywhere else, part (c) is a two-subtraction recipe.

### Example. (Axis and Angle of a Space Rotation)

Let

$$\mathbf{Q} = \frac{1}{9} \begin{pmatrix} 1 & -4 & 8 \\ 8 & 4 & 1 \\ -4 & 7 & 4 \end{pmatrix}.$$

Verify that  $\mathbf{Q} \in \text{SO}(3)$ , find its axis and angle, and confirm the answer with Rodrigues' formula.

*Solution. Orthogonality.* The columns are  $\frac{1}{9}(1, 8, -4)$ ,  $\frac{1}{9}(-4, 4, 7)$  and  $\frac{1}{9}(8, 1, 4)$ . Each has squared norm  $\frac{1}{81}(1 + 64 + 16) = \frac{1}{81}(16 + 16 + 49) = \frac{1}{81}(64 + 1 + 16) = 1$ , and the three pairwise inner products are

$$\frac{1}{81}(-4 + 32 - 28) = \frac{1}{81}(8 + 8 - 16) = \frac{1}{81}(-32 + 4 + 28) = 0.$$

So  $\mathbf{Q} \in \text{O}(3)$  by Theorem 10.7.3 (f).

*Determinant.* Expanding along the first row,  $729 \det \mathbf{Q} = 1(16 - 7) - (-4)(32 + 4) + 8(56 + 16) = 9 + 144 + 576 = 729$ , so  $\det \mathbf{Q} = 1$  and  $\mathbf{Q} \in \text{SO}(3)$ .

*Angle.*  $\text{tr } \mathbf{Q} = \frac{1}{9}(1 + 4 + 4) = 1$ , so  $\cos \theta = \frac{1}{2}(1 - 1) = 0$  and  $\theta = \pi/2$  by Theorem 11.6.5 (a).

*Axis.* Here  $2 \sin \theta = 2$ . The three entries needed by [Theorem 11.6.5](#) (c) are

$$\begin{aligned} q_{32} - q_{23} &= \frac{7-1}{9} = \frac{6}{9}, \\ q_{13} - q_{31} &= q_{21} - q_{12} = \frac{8+4}{9} = \frac{12}{9}, \end{aligned}$$

so  $\mathbf{n} = \frac{1}{2}(\frac{6}{9}, \frac{12}{9}, \frac{12}{9}) = \frac{1}{3}(1, 2, 2)$ , which is a unit vector since  $1 + 4 + 4 = 9$ .

*Check.*  $\mathbf{Q}(1, 2, 2) = \frac{1}{9}(1 - 8 + 16, 8 + 8 + 2, -4 + 14 + 8) = \frac{1}{9}(9, 18, 18) = (1, 2, 2)$ , so the axis is fixed. And Rodrigues' formula with  $\theta = \pi/2$  reads  $\mathbf{R}_{\mathbf{n}, \pi/2} = \mathbf{I}_3 + \mathbf{K} + \mathbf{K}^2$ , where

$$\mathbf{K} = \frac{1}{3} \begin{pmatrix} 0 & -2 & 2 \\ 2 & 0 & -1 \\ -2 & 1 & 0 \end{pmatrix}, \quad \mathbf{K}^2 = \mathbf{nn}^\top - \mathbf{I}_3 = \frac{1}{9} \begin{pmatrix} -8 & 2 & 2 \\ 2 & -5 & 4 \\ 2 & 4 & -5 \end{pmatrix}.$$

Adding  $\mathbf{I}_3 + \mathbf{K} + \mathbf{K}^2$  entry by entry, for instance  $1 + 0 - \frac{8}{9} = \frac{1}{9}$  in position  $(1, 1)$  and  $0 - \frac{2}{3} + \frac{2}{9} = -\frac{4}{9}$  in position  $(1, 2)$ , returns  $\mathbf{Q}$ .

## 11.6.6 Reflections generate

The canonical form is one way to say that an orthogonal matrix is built from simple pieces. Here is another, and the pieces are even simpler: mirrors. Recall from Chapter 10 the Householder reflection  $\mathbf{H}_{\mathbf{w}} = \mathbf{I}_n - 2\mathbf{w}\mathbf{w}^\top/\|\mathbf{w}\|^2$  determined by a non-zero  $\mathbf{w}$  ([Definition 10.8.1](#)); it is orthogonal, equal to its own inverse, and has determinant  $-1$  ([Proposition 10.8.2](#)).

### Lemma 11.6.6 — A Mirror Between Two Vectors of Equal Length

Let  $\mathbf{x}, \mathbf{y} \in \mathbb{R}^n$  with  $\|\mathbf{x}\| = \|\mathbf{y}\|$  and  $\mathbf{x} \neq \mathbf{y}$ . Then  $\mathbf{w} := \mathbf{x} - \mathbf{y}$  is non-zero and  $\mathbf{H}_{\mathbf{w}}\mathbf{x} = \mathbf{y}$ .

*Proof.*  $\mathbf{w} \neq \mathbf{0}$  because  $\mathbf{x} \neq \mathbf{y}$ . Expanding the two inner products and using  $\|\mathbf{y}\|^2 = \|\mathbf{x}\|^2$ ,

$$\begin{aligned} \|\mathbf{w}\|^2 &= \|\mathbf{x}\|^2 - 2\langle \mathbf{x}, \mathbf{y} \rangle + \|\mathbf{y}\|^2 = 2(\|\mathbf{x}\|^2 - \langle \mathbf{x}, \mathbf{y} \rangle), \\ \langle \mathbf{x}, \mathbf{w} \rangle &= \|\mathbf{x}\|^2 - \langle \mathbf{x}, \mathbf{y} \rangle, \end{aligned}$$

so  $2\langle \mathbf{x}, \mathbf{w} \rangle = \|\mathbf{w}\|^2$ . Hence, by the formula  $\mathbf{H}_{\mathbf{w}}\mathbf{v} = \mathbf{v} - 2(\langle \mathbf{v}, \mathbf{w} \rangle/\|\mathbf{w}\|^2)\mathbf{w}$  of [Definition 10.8.1](#),

$$\mathbf{H}_{\mathbf{w}}\mathbf{x} = \mathbf{x} - \mathbf{w} = \mathbf{x} - (\mathbf{x} - \mathbf{y}) = \mathbf{y},$$

as claimed. ■

This is [Theorem 10.8.3](#) with the target chosen for geometry rather than for arithmetic: there,  $\mathbf{y}$  was a prescribed multiple of  $\mathbf{e}_1$  and the sign was picked to avoid cancellation; here  $\mathbf{y}$  is whatever we need, and the mirror is the perpendicular bisector of the segment from  $\mathbf{x}$  to  $\mathbf{y}$ .

### Theorem 11.6.7 — Reflections Generate the Orthogonal Group

Let  $n \geq 1$  and  $\mathbf{Q} \in O(n)$ . Then  $\mathbf{Q}$  is a product of at most  $n$  Householder reflections. (The identity is the empty product, of zero reflections.)

*Idea.* Induct on  $n$ , and make the induction bite by fixing one coordinate at a time. If  $\mathbf{Q}$  already fixes  $\mathbf{e}_1$ , then it maps  $\mathbf{e}_1^\perp$  into itself, and the inductive hypothesis applied inside that  $(n-1)$ -dimensional space produces at most  $n-1$  reflections, each of which extends to a reflection of  $\mathbb{R}^n$  by taking the same  $\mathbf{w}$ . If  $\mathbf{Q}$  does not fix  $\mathbf{e}_1$ , spend one reflection making it do so:  $\mathbf{Q}\mathbf{e}_1$  and  $\mathbf{e}_1$  are two distinct vectors of the same length, and Lemma 11.6.6 is exactly the tool that swaps them.

*Proof.* Induction on  $n$ . For  $n=1$ ,  $O(1) = \{[1], [-1]\}$  by Exercise (C1: Determinant as a homomorphism) (c);  $[1]$  is the empty product and  $[-1] = \mathbf{H}_{\mathbf{e}_1}$ , one reflection.

Let  $n \geq 2$  and assume the statement in dimension  $n-1$ . Let  $\mathbf{Q} \in O(n)$  and set  $\mathbf{x} = \mathbf{Q}\mathbf{e}_1$ , a unit vector since  $T_{\mathbf{Q}}$  is an isometry (Theorem 10.7.3).

**Case 1:**  $\mathbf{x} = \mathbf{e}_1$ . Put  $U = \text{span}(\mathbf{e}_1)^\perp$ , of dimension  $n-1$  by Theorem 10.3.2 (c). For  $\mathbf{v} \in U$  we have  $\langle \mathbf{Q}\mathbf{v}, \mathbf{e}_1 \rangle = \langle \mathbf{Q}\mathbf{v}, \mathbf{Q}\mathbf{e}_1 \rangle = \langle \mathbf{v}, \mathbf{e}_1 \rangle = 0$ , the middle equality by Theorem 10.7.3 (b); so  $U$  is  $T_{\mathbf{Q}}$ -invariant. Identify  $U$  with  $\mathbb{R}^{n-1}$  through the orthonormal basis  $(\mathbf{e}_2, \dots, \mathbf{e}_n)$ . The restriction  $T_{\mathbf{Q}}|_U$  is an isometry of  $U$ , so its matrix  $\mathbf{Q}'$  in that basis lies in  $O(n-1)$  (Theorem 10.7.3 (f)), and by the inductive hypothesis  $\mathbf{Q}' = \mathbf{H}_{\mathbf{w}'_1} \cdots \mathbf{H}_{\mathbf{w}'_k}$  with  $k \leq n-1$  and  $\mathbf{w}'_i \in \mathbb{R}^{n-1}$ . Let  $\mathbf{w}_i \in U \subseteq \mathbb{R}^n$  be the vector with coordinates  $\mathbf{w}'_i$  in the basis  $(\mathbf{e}_2, \dots, \mathbf{e}_n)$ . Then  $\mathbf{H}_{\mathbf{w}_i}$  fixes  $\mathbf{e}_1$ , because  $\mathbf{e}_1 \perp \mathbf{w}_i$  (Proposition 10.8.2 (d)), and on  $U$  it acts as  $\mathbf{H}_{\mathbf{w}'_i}$ , because the defining formula only involves inner products, which agree. Hence  $\mathbf{Q}$  and  $\mathbf{H}_{\mathbf{w}_1} \cdots \mathbf{H}_{\mathbf{w}_k}$  agree on  $\mathbf{e}_1$  and on  $U$ , so they agree on  $\mathbb{R}^n = \text{span}(\mathbf{e}_1) \oplus U$ , and  $\mathbf{Q}$  is a product of  $k \leq n-1 \leq n$  reflections.

**Case 2:**  $\mathbf{x} \neq \mathbf{e}_1$ . Then  $\|\mathbf{x}\| = 1 = \|\mathbf{e}_1\|$ , so  $\mathbf{w} := \mathbf{x} - \mathbf{e}_1$  is non-zero and  $\mathbf{H}_{\mathbf{w}}\mathbf{x} = \mathbf{e}_1$  by Lemma 11.6.6. The matrix  $\mathbf{H}_{\mathbf{w}}\mathbf{Q}$  is orthogonal, being a product of orthogonal matrices (Proposition 10.7.4 (a)), and  $\mathbf{H}_{\mathbf{w}}\mathbf{Q}\mathbf{e}_1 = \mathbf{H}_{\mathbf{w}}\mathbf{x} = \mathbf{e}_1$ . By Case 1,  $\mathbf{H}_{\mathbf{w}}\mathbf{Q}$  is a product of at most  $n-1$  reflections. Multiplying on the left by  $\mathbf{H}_{\mathbf{w}}$  and using  $\mathbf{H}_{\mathbf{w}}^2 = \mathbf{I}_n$  (Proposition 10.8.2 (b)),

$$\mathbf{Q} = \mathbf{H}_{\mathbf{w}}(\mathbf{H}_{\mathbf{w}}\mathbf{Q}),$$

a product of at most  $n$  reflections. This completes the induction. ■

Counting reflections recovers the determinant: each factor contributes  $-1$  (Proposition 10.8.2 (e)), so a product of  $k$  reflections has determinant  $(-1)^k$ , and an element of  $SO(n)$  needs an even number of them. The bound  $n$  is sharp —  $-\mathbf{I}_n$  is a product of  $n$  reflections and no fewer, since a product of  $k$  reflections fixes the intersection of  $k$  hyperplanes, a subspace of dimension at least  $n-k$ , while  $-\mathbf{I}_n$  fixes only  $\mathbf{0}$ .

*Remark.* The general theorem behind this one, due to Cartan and Dieudonné, says the same for the orthogonal group of any non-degenerate symmetric bilinear form over any field of characteristic  $\neq 2$ . Only the Euclidean case is proved here, and only the Euclidean case is used.

## 11.6.7 Composing rotations

Two rotations of the plane about the origin compose into a rotation, and the angles add:  $\mathbf{R}_\alpha \mathbf{R}_\beta = \mathbf{R}_{\alpha+\beta}$ , so  $\text{SO}(2)$  is abelian (Chapter 10). The reason is visible in the canonical form: in dimension 2 there is only one rotation block, so a rotation is a single number, and numbers commute.

In dimension 3 the picture changes completely, because there is an axis to move as well as an angle to add.

### Example. (Two Quarter Turns in Two Orders)

Let  $\mathbf{A} = \mathbf{R}_{\mathbf{e}_1, \pi/2}$  and  $\mathbf{B} = \mathbf{R}_{\mathbf{e}_3, \pi/2}$ , the quarter turns about the first and third coordinate axes. Compute  $\mathbf{AB}$  and  $\mathbf{BA}$ , and find the axis and angle of each.

*Solution.* From [Theorem 11.6.4](#) (b),  $\mathbf{A}$  fixes  $\mathbf{e}_1$  and turns the  $(\mathbf{e}_2, \mathbf{e}_3)$ -plane by  $\pi/2$ , and  $\mathbf{B}$  fixes  $\mathbf{e}_3$  and turns the  $(\mathbf{e}_1, \mathbf{e}_2)$ -plane by  $\pi/2$ :

$$\mathbf{A} = \begin{pmatrix} 1 & 0 & 0 \\ 0 & 0 & -1 \\ 0 & 1 & 0 \end{pmatrix}, \quad \mathbf{B} = \begin{pmatrix} 0 & -1 & 0 \\ 1 & 0 & 0 \\ 0 & 0 & 1 \end{pmatrix}.$$

Multiplying,

$$\mathbf{AB} = \begin{pmatrix} 0 & -1 & 0 \\ 0 & 0 & -1 \\ 1 & 0 & 0 \end{pmatrix}, \quad \mathbf{BA} = \begin{pmatrix} 0 & 0 & 1 \\ 1 & 0 & 0 \\ 0 & 1 & 0 \end{pmatrix}.$$

These are different matrices, so  $\text{SO}(3)$  is not abelian. Both have trace 0, so both are rotations by the angle  $\theta$  with  $\cos \theta = \frac{1}{2}(0 - 1) = -\frac{1}{2}$ , that is  $\theta = 2\pi/3$ . Their axes differ. Solving  $\mathbf{AB}\mathbf{v} = \mathbf{v}$  gives  $-v_2 = v_1$ ,  $-v_3 = v_2$ ,  $v_1 = v_3$ , so the axis of  $\mathbf{AB}$  is  $\text{span}((1, -1, 1))$ ; solving  $\mathbf{BA}\mathbf{v} = \mathbf{v}$  gives  $v_3 = v_1$ ,  $v_1 = v_2$ ,  $v_2 = v_3$ , so the axis of  $\mathbf{BA}$  is  $\text{span}((1, 1, 1))$ . Same angle, different axes: the order of two quarter turns changes where the resulting axis points.

That the product is *again* a rotation is not an accident of this example:  $\text{SO}(3)$  is a group ([Proposition 10.7.4](#)), so any product of rotations is orthogonal with determinant 1, and [Corollary 11.6.2](#) makes it a rotation. What no formula makes obvious is which one; [Exercise \(C2: Products of rotations\)](#) pursues this.

**Warning.** In dimension 4 and above,  $\det \mathbf{Q} = 1$  does not mean “rotation about an axis”. Take

$$\mathbf{Q} = \mathbf{R}_{\pi/2} \oplus \mathbf{R}_{\pi/3} \in \text{O}(4), \quad \det \mathbf{Q} = 1 \cdot 1 = 1.$$

Its canonical form is itself, with  $p = q = 0$  and  $m = 2$ , so 1 is not an eigenvalue and  $\mathbf{Q}$  fixes no non-zero vector at all: there is no axis. Worse, the two angles are different, so  $\mathbf{Q}$  is not a rotation “in a plane” either — it turns two orthogonal planes at two different speeds, and no single angle describes it. Chapter 10 made the same point with  $-\mathbf{I}_4 \in \text{SO}(4)$ ; the canonical form explains it, since for even  $n$  the counts  $p = q = 0$  are allowed. The clean statement “determinant 1 means a rotation about an axis” is a feature of dimension 3, where  $p + q + 2m = 3$  is odd and forces  $p \geq 1$  when  $q$  is even.

## 11.6.8 Exercises

### 11.6.8.1 A. Check your understanding

#### Exercise A1.

- State the canonical form of an element of  $O(n)$ , saying what the blocks are and which angles are allowed.
- Which hypothesis on  $\mathbf{Q}$  turns the rotation-scaling blocks of the real normal form into rotation blocks, and why?
- Write down the cross-product matrix  $\mathbf{K}_{\mathbf{n}}$  and state the identity  $\mathbf{K}_{\mathbf{n}}^3 = ?$  for a unit  $\mathbf{n}$ .
- Given  $\mathbf{Q} \in SO(3)$ , how do you find its angle, and how do you find its axis? Name the two degenerate cases.
- True or false: if  $\mathbf{Q} \in O(n)$  and  $\det \mathbf{Q} = 1$ , then  $\mathbf{Q}$  fixes a non-zero vector. Justify your answer.
- How many Householder reflections can be needed to write an element of  $O(n)$ , and what does the parity of that number say?

*Solution.*

- By [Theorem 11.6.1](#),  $\mathbf{P}^\top \mathbf{Q} \mathbf{P} = \mathbf{I}_p \oplus (-\mathbf{I}_q) \oplus \mathbf{R}_{\theta_1} \oplus \cdots \oplus \mathbf{R}_{\theta_m}$  for some  $\mathbf{P} \in O(n)$ , with  $p + q + 2m = n$  and  $0 < \theta_j < \pi$ . The angles 0 and  $\pi$  are excluded because  $\mathbf{R}_0 = \mathbf{I}_2$  and  $\mathbf{R}_\pi = -\mathbf{I}_2$  are already accounted for by the  $\pm 1$  blocks.
- Orthogonality. An orthogonal matrix is unitary as a complex matrix ([Proposition 10.7.4](#) (a)), so each of its eigenvalues has modulus 1 ([Proposition 10.7.1](#)). A block  $\mathbf{A}(\lambda)$  has eigenvalues  $\lambda, \bar{\lambda}$ , so  $|\lambda| = 1$  and the scaling factor is 1.
- $\mathbf{K}_{\mathbf{n}}$  has rows  $(0, -n_3, n_2)$ ,  $(n_3, 0, -n_1)$ ,  $(-n_2, n_1, 0)$  ([Definition 11.6.1](#)), and  $\mathbf{K}_{\mathbf{n}}^3 = -\mathbf{K}_{\mathbf{n}}$  when  $\|\mathbf{n}\| = 1$  ([Lemma 11.6.3](#) (c)).
- The angle from the trace,  $\cos \theta = \frac{1}{2}(\text{tr } \mathbf{Q} - 1)$  with  $\theta \in [0, \pi]$ ; the axis from the skew part,  $\mathbf{Q} - \mathbf{Q}^\top = 2 \sin \theta \mathbf{K}_{\mathbf{n}}$  ([Theorem 11.6.5](#)). The degenerate cases are  $\theta = 0$ , where  $\mathbf{Q} = \mathbf{I}_3$  and there is no axis to find, and  $\theta = \pi$ , where the skew part vanishes and one reads  $\mathbf{n}$  off  $\mathbf{Q} + \mathbf{I}_3 = 2\mathbf{n}\mathbf{n}^\top$  instead.
- False for even  $n$ . The matrix  $\mathbf{R}_{\pi/2} \oplus \mathbf{R}_{\pi/3} \in SO(4)$  has no eigenvalue 1. It is true for odd  $n$ : then  $p + q + 2m = n$  is odd and  $q$  is even, so  $p = n - q - 2m$  is odd, hence  $p \geq 1$ , and the  $\mathbf{I}_p$  block supplies a fixed vector.
- At most  $n$  ([Theorem 11.6.7](#)). Each reflection has determinant  $-1$ , so the number used has the same parity in every such factorization as the sign of  $\det \mathbf{Q}$  dictates: even for  $\det \mathbf{Q} = 1$ , odd for  $\det \mathbf{Q} = -1$ .

### 11.6.8.2 B. Practice

#### Exercise B1: Axis and angle.

For each of the following, decide whether it lies in  $SO(3)$ ,  $O(3) \setminus SO(3)$ , or neither. For those in  $SO(3)$ , find the axis and the angle; for those in  $O(3)$  with determinant  $-1$ , give the canonical form  $[-1] \oplus \mathbf{R}_\theta$  and say what the map is geometrically.

$$(a) \frac{1}{9} \begin{pmatrix} 4 & -8 & 1 \\ 4 & 1 & -8 \\ 7 & 4 & 4 \end{pmatrix}.$$

$$(b) \frac{1}{9} \begin{pmatrix} -7 & 4 & 4 \\ 4 & -1 & 8 \\ 4 & 8 & -1 \end{pmatrix}.$$

$$(c) \frac{1}{9} \begin{pmatrix} 7 & -4 & -4 \\ -4 & 1 & -8 \\ -4 & -8 & 1 \end{pmatrix}.$$

$$(d) \frac{1}{9} \begin{pmatrix} 4 & -8 & 1 \\ 4 & 1 & -8 \\ 7 & 4 & 5 \end{pmatrix}.$$

*Solution.*

(a) The columns  $\frac{1}{9}(4, 4, 7)$ ,  $\frac{1}{9}(-8, 1, 4)$ ,  $\frac{1}{9}(1, -8, 4)$  have squared norms  $\frac{1}{81}(16 + 16 + 49) = \frac{1}{81}(64 + 1 + 16) = \frac{1}{81}(1 + 64 + 16) = 1$  and pairwise inner products  $\frac{1}{81}(-32 + 4 + 28) = \frac{1}{81}(4 - 32 + 28) = \frac{1}{81}(-8 - 8 + 16) = 0$ , so the matrix is orthogonal. Expanding along the first row,  $729 \det = 4(4 + 32) + 8(16 + 56) + 1(16 - 7) = 144 + 576 + 9 = 729$ , so it lies in  $SO(3)$ . Its trace is  $\frac{1}{9}(4 + 1 + 4) = 1$ , so  $\cos \theta = 0$  and  $\theta = \pi/2$ . For the axis,  $q_{32} - q_{23} = \frac{1}{9}(4 + 8) = \frac{12}{9}$ ,  $q_{13} - q_{31} = \frac{1}{9}(1 - 7) = -\frac{6}{9}$  and  $q_{21} - q_{12} = \frac{1}{9}(4 + 8) = \frac{12}{9}$ ; dividing by  $2 \sin \theta = 2$  gives  $\mathbf{n} = \frac{1}{3}(2, -1, 2)$ . Check:  $\mathbf{Q}(2, -1, 2) = \frac{1}{9}(8 + 8 + 2, 8 - 1 - 16, 14 - 4 + 8) = (2, -1, 2)$ .

(b) The matrix is symmetric. Each column has entries  $\pm 7, \pm 4, \pm 4$  in some order, so each has squared norm  $\frac{1}{81}(49 + 16 + 16) = 1$ , and the pairwise inner products are  $\frac{1}{81}(-28 - 4 + 32) = \frac{1}{81}(-28 + 32 - 4) = \frac{1}{81}(16 - 8 - 8) = 0$ ; so it is orthogonal. Expanding along the first row,  $729 \det = -7(1 - 64) - 4(-4 - 32) + 4(32 + 4) = 441 + 144 + 144 = 729$ , so it lies in  $SO(3)$ . Its trace is  $\frac{1}{9}(-7 - 1 - 1) = -1$ , so  $\cos \theta = -1$  and

$\theta = \pi$ : a half-turn. The skew part is  $\mathbf{0}$ , so use  $\mathbf{Q} + \mathbf{I}_3 = \frac{1}{9} \begin{pmatrix} 2 & 4 & 4 \\ 4 & 8 & 8 \\ 4 & 8 & 8 \end{pmatrix} = 2\mathbf{nn}^\top$  with

$\mathbf{n} = \frac{1}{3}(1, 2, 2)$ . Check:  $\mathbf{Q}(1, 2, 2) = \frac{1}{9}(-7 + 8 + 8, 4 - 2 + 16, 4 + 16 - 2) = (1, 2, 2)$ .

(c) This is the negative of (b), so it is orthogonal with  $\det = (-1)^3 \cdot 1 = -1$ . Its trace is 1, and for the form  $[-1] \oplus \mathbf{R}_\theta$  the trace is  $-1 + 2 \cos \theta$ , so  $\cos \theta = 1$  and  $\theta = 0$ : the canonical form is  $[-1] \oplus \mathbf{I}_2 = \text{diag}(-1, 1, 1)$ , a reflection. Indeed it equals  $\mathbf{I}_3 - 2\mathbf{nn}^\top = \mathbf{H}_\mathbf{n}$  with  $\mathbf{n} = \frac{1}{3}(1, 2, 2)$ , the reflection in the plane  $x_1 + 2x_2 + 2x_3 = 0$ .

(d) Neither. This is (a) with the  $(3, 3)$  entry changed from 4 to 5. The third column  $\frac{1}{9}(1, -8, 5)$  has squared norm  $\frac{1}{81}(1 + 64 + 25) = \frac{90}{81} = \frac{10}{9} \neq 1$ , so the columns are not orthonormal and the matrix is not orthogonal ([Theorem 10.7.3 \(f\)](#)). Its determinant is  $\frac{85}{81}$ , neither 1 nor  $-1$ , which settles it a second way ([Proposition 10.7.4](#)). Note that checking the determinant alone is never enough:  $\det = 1$  does not make a matrix orthogonal.

### Exercise B2: Mirrors.

(a) Write the matrix  $\mathbf{M} = \begin{pmatrix} 0 & 0 & 1 \\ 0 & 1 & 0 \\ 1 & 0 & 0 \end{pmatrix}$  as a Householder reflection  $\mathbf{H}_\mathbf{w}$ , and say which plane is its mirror.



- (b) Following the proof of [Theorem 11.6.7](#), write  $\mathbf{Q} = \begin{pmatrix} 0 & 0 & 1 \\ 1 & 0 & 0 \\ 0 & 1 & 0 \end{pmatrix}$  as a product of Householder reflections, and check the count against  $\det \mathbf{Q}$ .

*Solution.*

- (a)  $\mathbf{M}$  exchanges  $\mathbf{e}_1$  and  $\mathbf{e}_3$  and fixes  $\mathbf{e}_2$ . By [Lemma 11.6.6](#) with  $\mathbf{x} = \mathbf{e}_1$  and  $\mathbf{y} = \mathbf{e}_3$ , take  $\mathbf{w} = \mathbf{e}_1 - \mathbf{e}_3 = (1, 0, -1)$ ; then  $\|\mathbf{w}\|^2 = 2$  and

$$\mathbf{H}_{\mathbf{w}} = \mathbf{I}_3 - \mathbf{w}\mathbf{w}^\top = \mathbf{I}_3 - \begin{pmatrix} 1 & 0 & -1 \\ 0 & 0 & 0 \\ -1 & 0 & 1 \end{pmatrix} = \mathbf{M}.$$

The mirror is  $\text{span}(\mathbf{w})^\perp$ , the plane  $x_1 = x_3$ .

- (b)  $\mathbf{Q}\mathbf{e}_1 = \mathbf{e}_2 \neq \mathbf{e}_1$ , so we are in Case 2 with  $\mathbf{w}_1 = \mathbf{Q}\mathbf{e}_1 - \mathbf{e}_1 = (-1, 1, 0)$  and  $\|\mathbf{w}_1\|^2 = 2$ :

$$\mathbf{H}_{\mathbf{w}_1} = \mathbf{I}_3 - \mathbf{w}_1\mathbf{w}_1^\top = \begin{pmatrix} 0 & 1 & 0 \\ 1 & 0 & 0 \\ 0 & 0 & 1 \end{pmatrix}.$$

Then  $\mathbf{H}_{\mathbf{w}_1}\mathbf{Q} = \begin{pmatrix} 1 & 0 & 0 \\ 0 & 0 & 1 \\ 0 & 1 & 0 \end{pmatrix}$ , which fixes  $\mathbf{e}_1$  and exchanges  $\mathbf{e}_2$  and  $\mathbf{e}_3$ ; the computation of (a), with the indices shifted, shows that it is  $\mathbf{H}_{\mathbf{w}_2}$  for  $\mathbf{w}_2 = \mathbf{e}_2 - \mathbf{e}_3 = (0, 1, -1)$ . Hence  $\mathbf{Q} = \mathbf{H}_{\mathbf{w}_1}\mathbf{H}_{\mathbf{w}_2}$ , a product of two reflections. The count is even, matching  $\det \mathbf{Q} = (-1)^2 = 1$ ; and indeed  $\mathbf{Q}$  is the rotation by  $2\pi/3$  about  $(1, 1, 1)$  found in [Example \(Two Quarter Turns in Two Orders\)](#).

### Exercise B3: Rodrigues in an example.

- (a) Compute  $\mathbf{R}_{\mathbf{e}_3, \theta}$  from Rodrigues' formula and check that it is  $\mathbf{R}_\theta \oplus [1]$ .  
 (b) Compute  $\mathbf{R}_{\mathbf{n}, 2\pi/3}$  for  $\mathbf{n} = \frac{1}{\sqrt{3}}(1, 1, 1)$ , and identify the resulting matrix.

*Solution.*

- (a) With  $\mathbf{n} = \mathbf{e}_3$  we have  $\mathbf{K} = \begin{pmatrix} 0 & -1 & 0 \\ 1 & 0 & 0 \\ 0 & 0 & 0 \end{pmatrix}$  and  $\mathbf{K}^2 = \mathbf{e}_3\mathbf{e}_3^\top - \mathbf{I}_3 = \text{diag}(-1, -1, 0)$ .

Hence

$$\mathbf{R}_{\mathbf{e}_3, \theta} = \mathbf{I}_3 + \sin \theta \mathbf{K} + (1 - \cos \theta) \mathbf{K}^2 = \begin{pmatrix} \cos \theta & -\sin \theta & 0 \\ \sin \theta & \cos \theta & 0 \\ 0 & 0 & 1 \end{pmatrix},$$

the diagonal entries being  $1 - (1 - \cos \theta) = \cos \theta$  twice and 1 once. This is  $\mathbf{R}_\theta \oplus [1]$ , the rotation of the  $(\mathbf{e}_1, \mathbf{e}_2)$ -plane, as it must be.

- (b) Here  $\cos \theta = -\frac{1}{2}$  and  $\sin \theta = \frac{\sqrt{3}}{2}$ . Writing  $\mathbf{1} = (1, 1, 1)$ , so that  $\mathbf{n} = \mathbf{1}/\sqrt{3}$  and

$$\mathbf{nn}^\top = \frac{1}{3}\mathbf{1}\mathbf{1}^\top,$$

$$\sin \theta \mathbf{K} = \frac{\sqrt{3}}{2} \cdot \frac{1}{\sqrt{3}} \mathbf{K}_1 = \frac{1}{2} \begin{pmatrix} 0 & -1 & 1 \\ 1 & 0 & -1 \\ -1 & 1 & 0 \end{pmatrix},$$

$$(1 - \cos \theta) \mathbf{K}^2 = \frac{3}{2} \left( \frac{1}{3} \mathbf{1}\mathbf{1}^\top - \mathbf{I}_3 \right) = \frac{1}{2} \mathbf{1}\mathbf{1}^\top - \frac{3}{2} \mathbf{I}_3.$$

Adding  $\mathbf{I}_3$  to these two gives  $\frac{1}{2}\mathbf{1}\mathbf{1}^\top - \frac{1}{2}\mathbf{I}_3$  plus the skew part. The first piece has 0 on the diagonal and  $\frac{1}{2}$  elsewhere, and adding the skew part turns each off-diagonal pair  $\frac{1}{2}, \frac{1}{2}$  into 0, 1:

$$\mathbf{R}_{\mathbf{n}, 2\pi/3} = \begin{pmatrix} 0 & 0 & 1 \\ 1 & 0 & 0 \\ 0 & 1 & 0 \end{pmatrix}.$$

This is the cyclic shift  $\mathbf{e}_1 \mapsto \mathbf{e}_2 \mapsto \mathbf{e}_3 \mapsto \mathbf{e}_1$ , which is the matrix  $\mathbf{BA}$  of [Example \(Two Quarter Turns in Two Orders\)](#): same axis  $(1, 1, 1)$ , same angle  $2\pi/3$ .

### 11.6.8.3 C. Going deeper

#### Exercise C1: The plane, re-derived.

- Use [Theorem 11.6.1](#) to prove that every  $\mathbf{Q} \in \mathbf{O}(2)$  is either orthogonally similar to  $\mathbf{R}_\theta$  for a unique  $\theta \in [0, \pi]$ , or orthogonally similar to  $\text{diag}(1, -1)$ .
- Chapter 10 proved more:  $\mathbf{Q}$  equals  $\mathbf{R}_\theta$  or  $\mathbf{M}_\theta$  ([Theorem 10.7.5](#)). Explain exactly what information (a) loses, and why the canonical form must lose it.
- Deduce that any two reflections of  $\mathbb{R}^2$  are orthogonally similar, and that  $\mathbf{R}_\alpha$  and  $\mathbf{R}_\beta$  are orthogonally similar if and only if  $\beta = \pm\alpha$ .

*Solution.*

- With  $n = 2$ ,  $p + q + 2m = 2$ . If  $m = 1$  then  $p = q = 0$  and the form is  $\mathbf{R}_{\theta_1}$  with  $\theta_1 \in (0, \pi)$ . If  $m = 0$  then  $(p, q) \in \{(2, 0), (1, 1), (0, 2)\}$ , giving  $\mathbf{I}_2 = \mathbf{R}_0$ ,  $\text{diag}(1, -1)$  and  $-\mathbf{I}_2 = \mathbf{R}_\pi$ . So the possibilities are  $\mathbf{R}_\theta$  with  $\theta \in [0, \pi]$  and  $\text{diag}(1, -1)$ , and they are distinguished by the determinant. Uniqueness of  $\theta$  follows from [Theorem 11.6.1](#), or directly from  $\text{tr } \mathbf{R}_\theta = 2 \cos \theta$  and the injectivity of  $\cos$  on  $[0, \pi]$ .
- Part (a) determines  $\mathbf{Q}$  only up to orthogonal similarity, that is, up to a change of orthonormal coordinates. It therefore cannot see *which* line is the mirror of a reflection, nor the sign of the angle of a rotation. It must lose the mirror, because the mirror moves: if  $\mathbf{P}$  is the rotation by  $\varphi$  then  $\mathbf{P}\mathbf{M}_\theta\mathbf{P}^\top = \mathbf{M}_{\theta+2\varphi}$ , so all the  $\mathbf{M}_\theta$  are conjugate. It must lose the sign of the angle, because conjugating by  $\text{diag}(1, -1)$  sends  $\mathbf{R}_\theta$  to  $\mathbf{R}_{-\theta}$ .
- Every reflection has determinant  $-1$ , so by (a) each is orthogonally similar to  $\text{diag}(1, -1)$ , and orthogonal similarity is an equivalence relation, so any two of them are similar to each other. For rotations: if  $\beta = \pm\alpha$  then  $\mathbf{R}_\beta = \mathbf{R}_\alpha$  or  $\mathbf{R}_\beta = \text{diag}(1, -1)\mathbf{R}_\alpha\text{diag}(1, -1)$ , so they are orthogonally similar. Conversely, similar matrices have equal traces ([Theorem 3.7.5](#)), so  $\cos \alpha = \cos \beta$ , which for real angles means  $\beta \equiv \pm\alpha$  modulo  $2\pi$ .

**Exercise C2: Products of rotations.**

- (a) Prove that the product of two rotations of  $\mathbb{R}^3$  is a rotation of  $\mathbb{R}^3$ , and explain which theorem supplies the axis.
- (b) Let  $\mathbf{A}$  and  $\mathbf{B}$  be the half-turns about two **perpendicular** axes  $\text{span}(\mathbf{a})$  and  $\text{span}(\mathbf{b})$ , with  $\mathbf{a} \perp \mathbf{b}$  unit vectors. Prove that  $\mathbf{AB}$  is the half-turn about the axis  $\text{span}(\mathbf{a} \times \mathbf{b})$ .
- (c) Deduce that every rotation of  $\mathbb{R}^3$  is a product of two half-turns.

*Hint for (c): use Theorem 11.6.7, or the description of a half-turn in the Quick check after Theorem 11.6.4.*

*Solution.*

- (a)  $\text{SO}(3)$  is a group (Proposition 10.7.4 (c)), so if  $\mathbf{A}, \mathbf{B} \in \text{SO}(3)$  then  $\mathbf{AB} \in \text{SO}(3)$ . By Corollary 11.6.2,  $\mathbf{AB}$  fixes a line pointwise and rotates the perpendicular plane; that corollary is what supplies the axis, and it does so through the count  $p \geq 1$  in the canonical form.
- (b) Extend  $(\mathbf{a}, \mathbf{b})$  to an orthonormal basis  $(\mathbf{a}, \mathbf{b}, \mathbf{c})$  of  $\mathbb{R}^3$ , where we may take  $\mathbf{c} = \mathbf{a} \times \mathbf{b}$ : it is a unit vector orthogonal to  $\mathbf{a}$  and  $\mathbf{b}$  by Lemma 11.6.3 (d). By the Quick check after Theorem 11.6.4,  $\mathbf{A} = 2\mathbf{a}\mathbf{a}^\top - \mathbf{I}_3$  fixes  $\mathbf{a}$  and negates  $\mathbf{a}^\perp$ , so in the basis  $(\mathbf{a}, \mathbf{b}, \mathbf{c})$ ,  $\mathbf{A}$  is  $\text{diag}(1, -1, -1)$  and  $\mathbf{B}$  is  $\text{diag}(-1, 1, -1)$ . Their product is  $\text{diag}(-1, -1, 1)$ , which fixes  $\mathbf{c}$  and negates  $\mathbf{c}^\perp$ : the half-turn about  $\text{span}(\mathbf{c}) = \text{span}(\mathbf{a} \times \mathbf{b})$ . (Its trace is  $-1$ , confirming  $\theta = \pi$  through Theorem 11.6.5 (a).)
- (c) Let  $\mathbf{Q} = \mathbf{R}_{\mathbf{n}, \theta} \in \text{SO}(3)$ , with  $\mathbf{n}$  a unit vector and  $\theta \in [0, \pi]$  (Theorem 11.6.4 (c)). Choose a unit  $\mathbf{w} \perp \mathbf{n}$ , and work throughout in the orthonormal basis  $\mathcal{B} = (\mathbf{n}, \mathbf{w}, \mathbf{K}_\mathbf{n}\mathbf{w})$  of Theorem 11.6.4 (b), in which  $\mathbf{Q}$  has matrix  $[1] \oplus \mathbf{R}_\theta$ . Put  $\mathbf{w}' = \mathbf{R}_{\mathbf{n}, \theta/2}\mathbf{w}$ , again a unit vector perpendicular to  $\mathbf{n}$ , since  $\mathbf{R}_{\mathbf{n}, \theta/2}$  fixes  $\mathbf{n}$  and is an isometry; by Theorem 11.6.4 (b) its coordinate vector in  $\mathcal{B}$  is  $\mathbf{d} = (0, \cos \frac{\theta}{2}, \sin \frac{\theta}{2})$ . Let  $\mathbf{A}$  and  $\mathbf{B}$  be the half-turns about  $\text{span}(\mathbf{w}')$  and  $\text{span}(\mathbf{w})$ , so that in  $\mathcal{B}$  their matrices are  $2\mathbf{d}\mathbf{d}^\top - \mathbf{I}_3$  and  $\text{diag}(-1, 1, -1)$  respectively, by the Quick check after Theorem 11.6.4. Using  $2\cos^2 \frac{\theta}{2} - 1 = \cos \theta$ ,  $2\sin^2 \frac{\theta}{2} - 1 = -\cos \theta$  and  $2\cos \frac{\theta}{2} \sin \frac{\theta}{2} = \sin \theta$ ,

$$2\mathbf{d}\mathbf{d}^\top - \mathbf{I}_3 = [-1] \oplus \begin{pmatrix} \cos \theta & \sin \theta \\ \sin \theta & -\cos \theta \end{pmatrix}.$$

Multiplying the two matrices blockwise,  $\mathbf{AB}$  has matrix

$$[1] \oplus \begin{pmatrix} \cos \theta & \sin \theta \\ \sin \theta & -\cos \theta \end{pmatrix} \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix} = [1] \oplus \mathbf{R}_\theta,$$

which is the matrix of  $\mathbf{Q}$  in the same basis. Hence  $\mathbf{Q} = \mathbf{AB}$ .

**Exercise C3: Moving one point of the sphere to another.**

Let  $S = \{\mathbf{x} \in \mathbb{R}^3 : \|\mathbf{x}\| = 1\}$ .

- (a) Prove that for all  $\mathbf{u}, \mathbf{v} \in S$  there is  $\mathbf{Q} \in \text{SO}(3)$  with  $\mathbf{Q}\mathbf{u} = \mathbf{v}$ . (One says that  $\text{SO}(3)$  **acts transitively** on  $S$ .)
- (b) Prove that  $\{\mathbf{Q} \in \text{SO}(3) : \mathbf{Q}\mathbf{u} = \mathbf{u}\} = \{\mathbf{R}_{\mathbf{u}, \theta} : \theta \in \mathbb{R}\}$ , and that this set is an

abelian subgroup of  $\text{SO}(3)$ .

- (c) Is the analogue of (a) true for  $\text{SO}(2)$  acting on the unit circle of  $\mathbb{R}^2$ ? And for  $\text{SO}(4)$  acting on the unit sphere of  $\mathbb{R}^4$ ? Justify your answers.

*Solution.*

- (a) Extend  $\mathbf{u}$  to an orthonormal basis  $(\mathbf{u}, \mathbf{a}_2, \mathbf{a}_3)$  of  $\mathbb{R}^3$  (Corollary 10.2.3), and likewise extend  $\mathbf{v}$  to  $(\mathbf{v}, \mathbf{b}_2, \mathbf{b}_3)$ . Let  $\mathbf{A} = (\mathbf{u} \mid \mathbf{a}_2 \mid \mathbf{a}_3)$  and  $\mathbf{B} = (\mathbf{v} \mid \mathbf{b}_2 \mid \mathbf{b}_3)$ ; both are orthogonal, since their columns are orthonormal (Theorem 10.7.3 (f)). Replacing  $\mathbf{a}_3$  by  $-\mathbf{a}_3$  changes the sign of  $\det \mathbf{A}$  and keeps the basis orthonormal, so we may assume  $\det \mathbf{A} = 1$ , and likewise  $\det \mathbf{B} = 1$ . Put  $\mathbf{Q} = \mathbf{B}\mathbf{A}^\top$ . Then  $\mathbf{Q} \in \text{O}(3)$  and  $\det \mathbf{Q} = \det \mathbf{B} \det \mathbf{A}^\top = 1$  by Theorem 6.5.4 and Theorem 6.4.5, so  $\mathbf{Q} \in \text{SO}(3)$ ; and  $\mathbf{A}^\top \mathbf{u} = \mathbf{e}_1$ , because the entries of  $\mathbf{A}^\top \mathbf{u}$  are the inner products of  $\mathbf{u}$  with the columns of  $\mathbf{A}$ , so  $\mathbf{Q}\mathbf{u} = \mathbf{B}\mathbf{e}_1 = \mathbf{v}$ .
- (b)  $(\supseteq)$  Each  $\mathbf{R}_{\mathbf{u},\theta}$  lies in  $\text{SO}(3)$  and fixes  $\mathbf{u}$ , by Theorem 11.6.4 (b).  $(\subseteq)$  If  $\mathbf{Q} \in \text{SO}(3)$  fixes  $\mathbf{u}$ , write  $\mathbf{Q} = \mathbf{R}_{\mathbf{n},\theta}$  with  $\theta \in [0, \pi]$  (Theorem 11.6.4 (c)). If  $\theta = 0$  then  $\mathbf{Q} = \mathbf{I}_3 = \mathbf{R}_{\mathbf{u},0}$ . Otherwise the fixed space of  $\mathbf{Q}$  is exactly the line  $\text{span}(\mathbf{n})$ , since in the basis of Theorem 11.6.4 (b) the matrix is  $[1] \oplus \mathbf{R}_\theta$  and  $\mathbf{R}_\theta$  fixes no non-zero vector for  $\theta \in (0, \pi]$ ; so  $\mathbf{u} = \pm \mathbf{n}$ , and  $\mathbf{R}_{-\mathbf{n},\theta} = \mathbf{R}_{\mathbf{n},-\theta}$  gives  $\mathbf{Q} = \mathbf{R}_{\mathbf{u},\pm\theta}$ . For the group claim: all these matrices are simultaneously  $[1] \oplus \mathbf{R}_\theta$  in the *same* orthonormal basis  $(\mathbf{u}, \mathbf{w}, \mathbf{K}_\mathbf{u}\mathbf{w})$ , and  $\mathbf{R}_\alpha \mathbf{R}_\beta = \mathbf{R}_{\alpha+\beta} = \mathbf{R}_\beta \mathbf{R}_\alpha$ , so the set is closed under products and inverses and is abelian.
- (c) Yes for  $\text{SO}(2)$ : given unit  $\mathbf{u}, \mathbf{v} \in \mathbb{R}^2$ , write  $\mathbf{u} = (\cos \alpha, \sin \alpha)$  and  $\mathbf{v} = (\cos \beta, \sin \beta)$  and take  $\mathbf{Q} = \mathbf{R}_{\beta-\alpha}$ , which is in  $\text{SO}(2)$  and sends  $\mathbf{u}$  to  $\mathbf{v}$ . Yes for  $\text{SO}(4)$  as well, by the same argument as (a): extend  $\mathbf{u}$  and  $\mathbf{v}$  to orthonormal bases of  $\mathbb{R}^4$ , fix the determinants to 1 by flipping the sign of the last vector if necessary, and take  $\mathbf{Q} = \mathbf{B}\mathbf{A}^\top$ . Nothing in that argument used the dimension 3 — transitivity holds for  $\text{SO}(n)$  with  $n \geq 2$ . What is special to  $n = 3$  is the classification of the maps themselves, not their ability to move one unit vector to another.

## 11.7 Commuting Families

Chapter 8 asked when several operators can be diagonalized in one basis and answered: exactly when they commute, provided each of them is diagonalizable on its own. With an inner product in hand the question sharpens in the same way the spectral theorem sharpened diagonalization. Can one **orthonormal** basis diagonalize all of them at once? For normal operators over  $\mathbb{C}$  the answer is yes, and the proof is the chapter's signature move applied family-wide: cut the space into the eigenspaces of one member and recurse.

Throughout,  $V$  is a finite-dimensional inner product space over  $F = \mathbb{R}$  or  $\mathbb{C}$ , and a family  $\mathcal{F} \subseteq \mathcal{L}(V)$  is called **commuting** if  $ST = TS$  for all  $S, T \in \mathcal{F}$ , as in Chapter 8.

## 11.7.1 What Chapter 8 already gives, and what is missing

Three results from Chapter 8 do the bookkeeping, and it is worth being exact about what each one delivers.

- [Theorem 8.9.1](#): if  $ST = TS$  then every eigenspace  $E_\lambda(T)$  is  $S$ -invariant. No hypothesis at all beyond commuting — not even finite dimension. This is what lets us look for common eigenvectors *inside* an eigenspace.
- [Theorem 8.9.3](#): a commuting family whose characteristic polynomials all split has a common eigenvector. Over  $\mathbb{C}$  the splitting is automatic.
- [Theorem 8.9.5](#) and [Theorem 8.9.6](#): a commuting family of **diagonalizable** operators is simultaneously diagonalizable ([Definition 8.9.1](#)), and conversely.

Two things are missing from that list. First, diagonalizability is a hypothesis there, and it has to be checked separately for each member. Second, and more to the point here, the common basis produced is just a basis: nothing says its vectors are orthogonal, and in general they are not.

Normality repairs both at once, over  $\mathbb{C}$ . It repairs the first because a normal operator is automatically diagonalizable, by [Theorem 11.4.2](#). It repairs the second because that same theorem produces an *orthonormal* eigenbasis, so the pieces we cut the space into are mutually orthogonal.

One more ingredient is needed before the recursion can run: after cutting  $V$  into the eigenspaces of one member, the other members must still be normal on each piece. Section 3 supplies exactly this. An invariant subspace of a normal operator is automatically invariant under the adjoint as well, and the restriction to it is again normal: that is [Proposition 11.3.6](#), proved there by a trace computation on the block form. Over  $\mathbb{R}$  the self-adjoint version is [Theorem 11.2.4](#). (What fails, as the warning of Section 3 records, is the analogous statement for a subspace that is **not** invariant: a principal submatrix of a normal matrix need not be normal.) With that recalled, nothing stands in the way.

## 11.7.2 One orthonormal basis for the whole family

### [Theorem 11.7.1](#) — One Orthonormal Eigenbasis for a Commuting Family

Let  $V \neq \{0\}$  be a finite-dimensional **complex** inner product space and let  $\mathcal{F} \subseteq \mathcal{L}(V)$  be a non-empty commuting family of **normal** operators. Then  $V$  has an orthonormal basis  $\mathcal{B}$  each of whose vectors is an eigenvector of every  $T \in \mathcal{F}$ ; equivalently,  $[T]_{\mathcal{B}}$  is diagonal for every  $T \in \mathcal{F}$ .

Conversely, if some orthonormal basis  $\mathcal{B}$  makes every  $[T]_{\mathcal{B}}$  diagonal, then  $\mathcal{F}$  is commuting and every member of  $\mathcal{F}$  is normal.

*Idea.* Induction on  $\dim V$ , with the two cases that every such induction has. If every member of the family is a scalar multiple of the identity, any orthonormal basis will do and there is nothing to arrange. Otherwise some  $T_0 \in \mathcal{F}$  has at least two distinct eigenvalues, and [Theorem 11.4.2](#) splits  $V$  into the eigenspaces of  $T_0$  — **orthogonally**,

and into pieces strictly smaller than  $V$ . Each piece is invariant under the whole family because the operators commute, and the restriction of a normal operator to an invariant subspace is normal, so each restricted family is again a commuting family of normal operators on a smaller space. Run the induction on each piece and glue.

*Proof.* Induction on  $n = \dim V \geq 1$ .

If  $n = 1$ , any unit vector  $\mathbf{v}$  of  $V$  is a basis, and  $T\mathbf{v} \in V = \text{span}(\mathbf{v})$  makes  $\mathbf{v}$  an eigenvector of every  $T \in \mathcal{L}(V)$ .

Let  $n \geq 2$  and assume the theorem for all complex inner product spaces of dimension less than  $n$ .

**Case 1: every  $T \in \mathcal{F}$  is a scalar multiple of  $\text{id}_V$ .** Then every non-zero vector is an eigenvector of every  $T \in \mathcal{F}$ , and  $V$  has an orthonormal basis by [Theorem 10.2.2](#).

**Case 2: some  $T_0 \in \mathcal{F}$  is not a scalar multiple of  $\text{id}_V$ .** Since  $T_0$  is normal, [Theorem 11.4.2](#) gives an orthonormal basis  $\mathcal{C} = (\mathbf{c}_1, \dots, \mathbf{c}_n)$  of  $V$  with  $T_0\mathbf{c}_j = \mu_j\mathbf{c}_j$ . Let  $\lambda_1, \dots, \lambda_k$  be the distinct values among  $\mu_1, \dots, \mu_n$ , and put

$$V_i = \text{span}\{\mathbf{c}_j : \mu_j = \lambda_i\}, \quad i = 1, \dots, k.$$

The  $V_i$  are spanned by disjoint sub-lists of an orthonormal basis, so they are pairwise orthogonal,  $V = V_1 \oplus \dots \oplus V_k$ , and  $\dim V_1 + \dots + \dim V_k = n$ .

*Claim.*  $V_i = E_{\lambda_i}(T_0)$  for each  $i$ .

*Proof.* The inclusion  $V_i \subseteq E_{\lambda_i}(T_0)$  holds because every spanning vector  $\mathbf{c}_j$  of  $V_i$  satisfies  $T_0\mathbf{c}_j = \lambda_i\mathbf{c}_j$ . Conversely let  $\mathbf{v} \in E_{\lambda_i}(T_0)$  and write  $\mathbf{v} = \mathbf{v}_1 + \dots + \mathbf{v}_k$  with  $\mathbf{v}_j \in V_j$ . Applying  $T_0$ , which acts as  $\lambda_j$  on  $V_j$ , gives  $\sum_j \lambda_j \mathbf{v}_j = T_0\mathbf{v} = \lambda_i\mathbf{v} = \sum_j \lambda_i \mathbf{v}_j$ , so  $\sum_j (\lambda_j - \lambda_i) \mathbf{v}_j = \mathbf{0}$ . The sum being direct, each term vanishes, and  $\lambda_j \neq \lambda_i$  for  $j \neq i$  forces  $\mathbf{v}_j = \mathbf{0}$ . Hence  $\mathbf{v} = \mathbf{v}_i \in V_i$ . ■

Because  $T_0$  is not a scalar multiple of  $\text{id}_V$ , we have  $k \geq 2$ : if  $k = 1$  then  $V = V_1 = E_{\lambda_1}(T_0)$  and  $T_0 = \lambda_1 \text{id}_V$ . Hence  $\dim V_i < n$  for every  $i$ .

Fix  $i$  and let  $S \in \mathcal{F}$ . By [Theorem 8.9.1](#),  $V_i = E_{\lambda_i}(T_0)$  is  $S$ -invariant, and therefore  $S|_{V_i}$  is a normal operator on  $V_i$  by [Proposition 11.3.6](#). Moreover the family

$$\mathcal{F}_i = \{S|_{V_i} : S \in \mathcal{F}\}$$

is commuting, since  $(S|_{V_i})(T|_{V_i}) = (ST)|_{V_i} = (TS)|_{V_i} = (T|_{V_i})(S|_{V_i})$  for  $S, T \in \mathcal{F}$ , all four restrictions being defined because  $V_i$  is invariant.

By the inductive hypothesis applied to  $V_i \neq \{\mathbf{0}\}$  and  $\mathcal{F}_i$ , there is an orthonormal basis  $\mathcal{B}_i$  of  $V_i$  whose vectors are eigenvectors of every  $S|_{V_i}$ ; such a vector  $\mathbf{v} \in V_i$  satisfies  $S\mathbf{v} = (S|_{V_i})\mathbf{v} = \mu\mathbf{v}$ , so it is an eigenvector of  $S$  itself. Let  $\mathcal{B}$  be the concatenation  $\mathcal{B}_1, \dots, \mathcal{B}_k$ . It is orthonormal, because each  $\mathcal{B}_i$  is and because  $V_i \perp V_j$  for  $i \neq j$ , and it has  $\sum_i \dim V_i = n$  vectors, so it is an orthonormal basis of  $V$  ([Theorem 10.1.4](#) and [Theorem 1.6.4](#)). Every vector of  $\mathcal{B}$  is an eigenvector of every  $T \in \mathcal{F}$ . This completes the induction.

For the converse, suppose every  $[T]_{\mathcal{B}}$  is diagonal with  $\mathcal{B}$  orthonormal. Diagonal matrices commute, so  $[ST]_{\mathcal{B}} = [TS]_{\mathcal{B}}$  and hence  $ST = TS$  (Theorem 3.6.4). And  $[T^*]_{\mathcal{B}} = ([T]_{\mathcal{B}})^*$  by Theorem 10.6.3 is diagonal as well, so  $[T^*T]_{\mathcal{B}}$  and  $[TT^*]_{\mathcal{B}}$  are products of diagonal matrices in the two orders, hence equal; so  $T$  is normal. This proves the theorem. ■

Notice which hypothesis did which job. *Commuting* made the eigenspaces of  $T_0$  invariant under the rest of the family; *normal* made those eigenspaces orthogonal and made the restrictions normal again; and *complex* is what makes  $T_0$  have any eigenvalues at all. Drop any one of the three and the argument stops.

In matrix language:

### Corollary 11.7.2 — Commuting Normal Matrices

Let  $\mathbf{A}_1, \dots, \mathbf{A}_r \in M_n(\mathbb{C})$  be normal and pairwise commuting. Then there is a unitary  $\mathbf{U} \in U(n)$  such that  $\mathbf{U}^* \mathbf{A}_i \mathbf{U}$  is diagonal for every  $i$ . The same holds for an arbitrary commuting family of normal matrices in  $M_n(\mathbb{C})$ .

*Proof.* Apply Theorem 11.7.1 to the family  $\{T_{\mathbf{A}_1}, \dots, T_{\mathbf{A}_r}\}$  on  $\mathbb{C}^n$ . Each  $T_{\mathbf{A}_i}$  is normal because  $T_{\mathbf{A}_i}^* = T_{\mathbf{A}_i^*}$  (Theorem 10.6.3, the standard basis being orthonormal) and  $\mathbf{A}_i^* \mathbf{A}_i = \mathbf{A}_i \mathbf{A}_i^*$ ; and the operators commute because the matrices do. Let  $\mathbf{U}$  have the resulting orthonormal basis  $\mathcal{B}$  as its columns; then  $\mathbf{U} \in U(n)$  by Theorem 10.7.3 (f), and  $\mathbf{U}^* \mathbf{A}_i \mathbf{U} = \mathbf{U}^{-1} \mathbf{A}_i \mathbf{U} = [T_{\mathbf{A}_i}]_{\mathcal{B}}$  is diagonal by Theorem 3.7.3. Nothing in the argument used that the family is finite. ■

### Corollary 11.7.3 — Commuting Symmetric Matrices

Let  $V$  be a finite-dimensional **real** inner product space and let  $\mathcal{F} \subseteq \mathcal{L}(V)$  be a non-empty commuting family of self-adjoint operators. Then  $V$  has an orthonormal basis of common eigenvectors. Equivalently, if  $\mathbf{A}_1, \dots, \mathbf{A}_r \in M_n(\mathbb{R})$  are symmetric and pairwise commuting, there is  $\mathbf{Q} \in O(n)$  with every  $\mathbf{Q}^T \mathbf{A}_i \mathbf{Q}$  diagonal.

*Proof.* Repeat the proof of Theorem 11.7.1 word for word, with two substitutions: use Theorem 11.5.1 in place of Theorem 11.4.2 to split  $V$  into the eigenspaces of a member  $T_0$  that is not a scalar multiple of  $\text{id}_V$ , and use Theorem 11.2.4 (b) in place of Proposition 11.3.6 to see that each restriction  $S|_{V_i}$  is again self-adjoint. Every other step is unchanged, and the only property of  $\mathbb{C}$  used in the original was the existence of an eigenvalue, which Theorem 11.5.1 supplies over  $\mathbb{R}$  from self-adjointness. The matrix statement follows as in Corollary 11.7.2, using Corollary 11.5.2 and  $\mathbf{Q}^T = \mathbf{Q}^{-1}$ . ■

**Example. (Two Commuting Symmetric Matrices, Diagonalized Together)**

Let

$$\mathbf{A} = \begin{pmatrix} 0 & 1 & 1 \\ 1 & 0 & 1 \\ 1 & 1 & 0 \end{pmatrix}, \quad \mathbf{B} = \begin{pmatrix} 2 & 0 & 0 \\ 0 & 1 & 1 \\ 0 & 1 & 1 \end{pmatrix} \in M_3(\mathbb{R}).$$

Check that  $\mathbf{AB} = \mathbf{BA}$  and find  $\mathbf{Q} \in O(3)$  with  $\mathbf{Q}^\top \mathbf{A} \mathbf{Q}$  and  $\mathbf{Q}^\top \mathbf{B} \mathbf{Q}$  both diagonal.

*Solution.* They commute. Multiplying out,

$$\mathbf{AB} = \begin{pmatrix} 0 & 2 & 2 \\ 2 & 1 & 1 \\ 2 & 1 & 1 \end{pmatrix} = \mathbf{BA}.$$

Both matrices are symmetric, so [Corollary 11.7.3](#) applies.

*Diagonalize A.* Each row of  $\mathbf{A}$  sums to 2, so  $\mathbf{A}(1, 1, 1) = (2, 2, 2)$  and 2 is an eigenvalue with eigenvector  $(1, 1, 1)$ . Also  $\mathbf{A} = \mathbf{1}\mathbf{1}^\top - \mathbf{I}_3$ , where  $\mathbf{1} = (1, 1, 1)$ ; any  $\mathbf{v}$  with  $\mathbf{1}^\top \mathbf{v} = 0$ , that is with  $v_1 + v_2 + v_3 = 0$ , satisfies  $\mathbf{A}\mathbf{v} = -\mathbf{v}$ . So  $E_2(\mathbf{A}) = \text{span}((1, 1, 1))$  and  $E_{-1}(\mathbf{A})$  is the plane  $v_1 + v_2 + v_3 = 0$ , and these account for all of  $\mathbb{R}^3$ .

*The one-dimensional eigenspace.* By [Theorem 8.9.1](#) the line  $E_2(\mathbf{A})$  is  $\mathbf{B}$ -invariant, so its spanning vector must already be an eigenvector of  $\mathbf{B}$ . Indeed  $\mathbf{B}(1, 1, 1) = (2, 2, 2)$ .

*The plane.* Inside  $E_{-1}(\mathbf{A})$  we must search. The matrix  $\mathbf{B}$  has  $\mathbf{B}(0, 1, -1) = (0, 0, 0)$  and  $\mathbf{B}(-2, 1, 1) = (-4, 2, 2) = 2(-2, 1, 1)$ , and both  $(0, 1, -1)$  and  $(-2, 1, 1)$  have coordinate sum 0, so they lie in  $E_{-1}(\mathbf{A})$ . They are orthogonal, and independent, so they are a basis of the plane.

*The common basis.* Normalizing the three vectors,

$$\mathbf{Q} = \begin{pmatrix} 1/\sqrt{3} & -2/\sqrt{6} & 0 \\ 1/\sqrt{3} & 1/\sqrt{6} & 1/\sqrt{2} \\ 1/\sqrt{3} & 1/\sqrt{6} & -1/\sqrt{2} \end{pmatrix},$$

whose columns are orthonormal, so  $\mathbf{Q} \in O(3)$  by [Theorem 10.7.3](#) (f). Then

$$\mathbf{Q}^\top \mathbf{A} \mathbf{Q} = \text{diag}(2, -1, -1), \quad \mathbf{Q}^\top \mathbf{B} \mathbf{Q} = \text{diag}(2, 2, 0),$$

the diagonal entries being the eigenvalues found above, column by column. Neither matrix alone forces the choice:  $\mathbf{A}$  cannot tell the second column from the third, and  $\mathbf{B}$  cannot tell the first from the second. Together they pin down all three lines.

**Warning. Neither hypothesis can be dropped.**

*Commuting is not enough.* The matrices  $\mathbf{A} = \begin{pmatrix} 1 & 1 \\ 0 & 2 \end{pmatrix}$  and  $\mathbf{B} = \begin{pmatrix} 1 & 3 \\ 0 & 4 \end{pmatrix}$  satisfy  $\mathbf{AB} = \begin{pmatrix} 1 & 7 \\ 0 & 8 \end{pmatrix} = \mathbf{BA}$ , and each is diagonalizable, having two distinct eigenvalues. They even share the eigenbasis  $\mathbf{e}_1, (1, 1)$ :  $\mathbf{A}(1, 1) = 2(1, 1)$  and  $\mathbf{B}(1, 1) = 4(1, 1)$ . But that basis is not orthogonal, and no orthonormal one exists, because an orthonormal eigenbasis would make  $\mathbf{A}$  normal ([Corollary 11.4.3](#)), whereas

$$\mathbf{A}^* \mathbf{A} = \begin{pmatrix} 1 & 1 \\ 1 & 5 \end{pmatrix} \neq \begin{pmatrix} 2 & 2 \\ 2 & 4 \end{pmatrix} = \mathbf{A} \mathbf{A}^*.$$



*Normal is not enough.* The matrices  $\mathbf{X} = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}$  and  $\mathbf{Z} = \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix}$  are real symmetric, hence normal, and each is orthogonally diagonalizable on its own. They do not commute:

$$\mathbf{XZ} = \begin{pmatrix} 0 & -1 \\ 1 & 0 \end{pmatrix} \neq \begin{pmatrix} 0 & 1 \\ -1 & 0 \end{pmatrix} = \mathbf{ZX}.$$

In fact they share no eigenvector at all: the eigenvectors of  $\mathbf{Z}$  are the multiples of  $\mathbf{e}_1$  and of  $\mathbf{e}_2$ , and  $\mathbf{Xe}_1 = \mathbf{e}_2$  is a multiple of neither.

### 11.7.3 Sums and products stay normal

Normality is not preserved by addition or multiplication. It is preserved by both as soon as the two operators commute, and the common eigenbasis makes this a one-line matter.

#### Theorem 11.7.4 — Sums and Products of Commuting Normal Operators

Let  $V$  be a finite-dimensional complex inner product space and let  $S, T \in \mathcal{L}(V)$  be normal with  $ST = TS$ . Then  $S + T$  and  $ST$  are normal.

*Proof.* If  $V = \{\mathbf{0}\}$  there is nothing to prove. Otherwise apply [Theorem 11.7.1](#) to the family  $\{S, T\}$ : there is an orthonormal basis  $\mathcal{B}$  with  $[S]_{\mathcal{B}} = \mathbf{D}$  and  $[T]_{\mathcal{B}} = \mathbf{D}'$  both diagonal. Then  $[S + T]_{\mathcal{B}} = \mathbf{D} + \mathbf{D}'$  and  $[ST]_{\mathcal{B}} = \mathbf{D}\mathbf{D}'$  are diagonal, so every vector of  $\mathcal{B}$  is an eigenvector of both operators, and both are normal by [Theorem 11.4.2](#). ■

*Remark.* Commutativity is essential, not a convenience. Take  $\mathbf{M} = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}$ , which is symmetric, and  $\mathbf{N} = \text{diag}(1, 2i)$ , which is diagonal; both are normal and they do not commute. Then

$$\mathbf{M} + \mathbf{N} = \begin{pmatrix} 1 & 1 \\ 1 & 2i \end{pmatrix}, \quad \mathbf{MN} = \begin{pmatrix} 0 & 2i \\ 1 & 0 \end{pmatrix},$$

and neither is normal: for the sum,  $(\mathbf{M} + \mathbf{N})(\mathbf{M} + \mathbf{N})^*$  has  $(1, 2)$  entry  $1 - 2i$  while  $(\mathbf{M} + \mathbf{N})^*(\mathbf{M} + \mathbf{N})$  has  $1 + 2i$  there; for the product,  $(\mathbf{MN})(\mathbf{MN})^* = \text{diag}(4, 1)$  while  $(\mathbf{MN})^*(\mathbf{MN}) = \text{diag}(1, 4)$ .

*Quick check.* Let  $\mathbf{A} \in M_n(\mathbb{C})$  be normal and let  $p \in \mathbb{C}[x]$ . Explain, without using [Theorem 11.7.1](#), why  $\mathbf{A}$  and  $p(\mathbf{A})$  are simultaneously unitarily diagonalizable.

*Solution.* By [Corollary 11.4.3](#),  $\mathbf{A} = \mathbf{U}\mathbf{D}\mathbf{U}^*$  with  $\mathbf{U}$  unitary and  $\mathbf{D}$  diagonal. Then  $\mathbf{A}^m = \mathbf{U}\mathbf{D}^m\mathbf{U}^*$  for every  $m \geq 0$ , since the inner factors  $\mathbf{U}^*\mathbf{U}$  cancel, so  $p(\mathbf{A}) = \mathbf{U}p(\mathbf{D})\mathbf{U}^*$ , and  $p(\mathbf{D}) = \text{diag}(p(d_1), \dots, p(d_n))$  is diagonal. The *same*  $\mathbf{U}$  diagonalizes both. This is consistent with the theorem, since  $\mathbf{A}$  and  $p(\mathbf{A})$  commute and  $p(\mathbf{A})$  is normal, being unitarily diagonalizable.

## 11.7.4 Without normality: simultaneous triangularization

If we ask only for triangular matrices, normality may be dropped entirely. Chapter 8 proved this with ordinary bases ([Theorem 8.9.4](#)); over  $\mathbb{C}$  an inner product upgrades the basis to an orthonormal one, exactly as [Theorem 11.1.2](#) upgraded [Theorem 8.7.2](#).

### Theorem 11.7.5 — Simultaneous Unitary Triangularization

Let  $V \neq \{0\}$  be a finite-dimensional complex inner product space and let  $\mathcal{F} \subseteq \mathcal{L}(V)$  be a non-empty commuting family. Then there is an orthonormal basis  $\mathcal{B}$  of  $V$  such that  $[T]_{\mathcal{B}}$  is upper triangular for **every**  $T \in \mathcal{F}$ . In particular, if  $\mathbf{A}_1, \dots, \mathbf{A}_r \in M_n(\mathbb{C})$  commute pairwise, there is  $\mathbf{U} \in \mathbf{U}(n)$  with every  $\mathbf{U}^* \mathbf{A}_i \mathbf{U}$  upper triangular.

*Idea.* The same swap that proves Schur's theorem ([Theorem 11.1.2](#)). A common eigenvector  $\mathbf{v}$  of  $\mathcal{F}$  gives an invariant line, but  $\text{span}(\mathbf{v})^\perp$  need not be invariant, so the recursion has nowhere to go. Take instead a common eigenvector  $\mathbf{u}$  of the **adjoints**  $T^*$ : they commute too, and  $\text{span}(\mathbf{u})^\perp$  is invariant under every  $T \in \mathcal{F}$ . Recurse there, and put  $\mathbf{u}$  at the **end** of the basis, where an upper triangular matrix allows an arbitrary column.

*Proof.* Induction on  $n = \dim V \geq 1$ . For  $n = 1$  every  $1 \times 1$  matrix is upper triangular, and a unit vector is an orthonormal basis.

Let  $n \geq 2$  and assume the theorem in dimension  $n - 1$ . The family  $\mathcal{F}^* = \{T^* : T \in \mathcal{F}\}$  is commuting: for  $S, T \in \mathcal{F}$ ,  $S^*T^* = (TS)^* = (ST)^* = T^*S^*$  by [Theorem 10.6.2](#). Since  $V$  is a complex space, [Theorem 8.9.3](#) provides  $\mathbf{u} \neq \mathbf{0}$  with  $T^*\mathbf{u} = \nu_T\mathbf{u}$  for every  $T \in \mathcal{F}$ ; dividing by  $\|\mathbf{u}\|$ , take  $\|\mathbf{u}\| = 1$ .

Put  $W = \text{span}(\mathbf{u})$  and  $U = W^\perp$ , of dimension  $n - 1$  by [Theorem 10.3.2](#) (c). For each  $T \in \mathcal{F}$  the line  $W$  is  $T^*$ -invariant, since  $T^*\mathbf{u} = \nu_T\mathbf{u}$ , so  $U$  is  $T$ -invariant by [Lemma 11.1.1](#). The family  $\{T|_U : T \in \mathcal{F}\}$  is commuting, by the same computation on restrictions as in the proof of [Theorem 11.7.1](#). By the inductive hypothesis there is an orthonormal basis  $(\mathbf{v}_1, \dots, \mathbf{v}_{n-1})$  of  $U$  in which every  $[T|_U]$  is upper triangular.

Let  $\mathcal{B} = (\mathbf{v}_1, \dots, \mathbf{v}_{n-1}, \mathbf{u})$ . It is orthonormal, since  $\mathbf{u}$  is a unit vector orthogonal to  $U$ , and it has  $n$  vectors, so it is an orthonormal basis of  $V$ . Fix  $T \in \mathcal{F}$ . For  $j \leq n - 1$ ,  $T\mathbf{v}_j = T|_U\mathbf{v}_j \in \text{span}(\mathbf{v}_1, \dots, \mathbf{v}_j)$ , because the matrix of  $T|_U$  is upper triangular; so the  $j$ -th column of  $[T]_{\mathcal{B}}$  has zeros below row  $j$ . The last column is unconstrained, and an upper triangular matrix permits any entries there. Hence  $[T]_{\mathcal{B}}$  is upper triangular, which completes the induction.

For the matrix statement, apply this to the operators  $T_{\mathbf{A}_i}$  on  $\mathbb{C}^n$  and let  $\mathbf{U}$  have the vectors of  $\mathcal{B}$  as columns, as in [Corollary 11.7.2](#). ■

The order matters: the common eigenvector of the adjoints goes last, and the triangular shape is built from the bottom right corner upward. Running the same recursion with a common eigenvector of  $\mathcal{F}$  itself and putting it first would work only if  $\text{span}(\mathbf{v})^\perp$  were invariant, which is the extra fact that normality supplies and a bare commuting family does not.

## 11.7.5 Exercises

### 11.7.5.1 A. Check your understanding

#### Exercise A1.

- (a) State [Theorem 11.7.1](#), naming every hypothesis.
- (b) Which result of Chapter 8 makes the eigenspaces of one member invariant under the others, and which hypothesis does it use?
- (c) Why is the restriction of a normal operator to an invariant subspace again normal, and which earlier result says so? Why does the same fail for a subspace that is not invariant?
- (d) True or false: two commuting diagonalizable matrices are simultaneously **unitarily** diagonalizable. Justify your answer.
- (e) True or false: if  $S$  and  $T$  are normal then  $S + T$  is normal. Justify your answer.
- (f) Which hypothesis of [Theorem 11.7.1](#) is dropped in [Theorem 11.7.5](#), and what is lost in the conclusion?

*Solution.*

- (a) Let  $V \neq \{0\}$  be a finite-dimensional **complex** inner product space and  $\mathcal{F} \subseteq \mathcal{L}(V)$  a **non-empty** family of **normal** operators that **commute pairwise**. Then  $V$  has an orthonormal basis of common eigenvectors.
- (b) [Theorem 8.9.1](#): if  $ST = TS$ , then  $E_\lambda(T)$  is  $S$ -invariant. The only hypothesis is that the two operators commute.
- (c) [Proposition 11.3.6](#): in an orthonormal basis adapted to  $U$  the matrix of a normal  $T$  is  $\mathbf{A} = \begin{pmatrix} \mathbf{B} & \mathbf{C} \\ \mathbf{0} & \mathbf{D} \end{pmatrix}$ , and comparing the traces of the  $(1, 1)$  blocks of  $\mathbf{A}^* \mathbf{A}$  and  $\mathbf{A} \mathbf{A}^*$  forces  $\mathbf{C} = \mathbf{0}$ ; the matrix is then block diagonal and its first block is normal. For a subspace that is not invariant the zero block below is absent and the argument collapses: the Section 3 warning compresses the  $3 \times 3$  cyclic shift to  $\text{span}(\mathbf{e}_1, \mathbf{e}_2)$  and gets a non-normal  $2 \times 2$  matrix.
- (d) False.  $\begin{pmatrix} 1 & 1 \\ 0 & 2 \end{pmatrix}$  and  $\begin{pmatrix} 1 & 3 \\ 0 & 4 \end{pmatrix}$  commute and are diagonalizable, but the first is not normal, so it has no orthonormal eigenbasis at all ([Corollary 11.4.3](#)). Simultaneous diagonalization by an invertible matrix is all that Chapter 8 promises ([Theorem 8.9.5](#)).
- (e) False in general, true when they commute ([Theorem 11.7.4](#)). For a counterexample take  $\begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}$  and  $\text{diag}(1, 2i)$ , as in the Remark after that theorem.
- (f) Normality. What is lost is that the common matrices are only upper triangular, not diagonal.

### 11.7.5.2 B. Practice

#### Exercise B1: Diagonalize two at once.

Let

$$\mathbf{A} = \begin{pmatrix} 2 & 1 & 1 \\ 1 & 2 & 1 \\ 1 & 1 & 2 \end{pmatrix}, \quad \mathbf{B} = \begin{pmatrix} 2 & 0 & -2 \\ 0 & 2 & -2 \\ -2 & -2 & 4 \end{pmatrix} \in M_3(\mathbb{R}).$$

- Verify that  $\mathbf{AB} = \mathbf{BA}$  and that both are symmetric.
- Find  $\mathbf{Q} \in O(3)$  with  $\mathbf{Q}^\top \mathbf{A} \mathbf{Q}$  and  $\mathbf{Q}^\top \mathbf{B} \mathbf{Q}$  both diagonal, and write down the two diagonal matrices.
- Explain why  $\mathbf{A}$  on its own does not determine  $\mathbf{Q}$  up to permuting and rescaling columns, but the pair does.

*Solution.*

- Both are equal to their transposes by inspection. Each row of  $\mathbf{B}$  sums to 0 and each column too, so  $\mathbf{1}\mathbf{1}^\top \mathbf{B} = \mathbf{0} = \mathbf{B}\mathbf{1}\mathbf{1}^\top$  where  $\mathbf{1} = (1, 1, 1)$ . Since  $\mathbf{A} = \mathbf{1}\mathbf{1}^\top + \mathbf{I}_3$ , we get  $\mathbf{AB} = \mathbf{B} = \mathbf{BA}$ . (Direct multiplication confirms  $\mathbf{AB} = \mathbf{B}$ .)
- The eigenvectors of  $\mathbf{A} = \mathbf{1}\mathbf{1}^\top + \mathbf{I}_3$  are  $\mathbf{1}$ , with  $\mathbf{A}\mathbf{1} = 3\mathbf{1} + \mathbf{1} = 4\mathbf{1}$ , and every  $\mathbf{v}$  with  $v_1 + v_2 + v_3 = 0$ , with  $\mathbf{A}\mathbf{v} = \mathbf{v}$ . Inside that plane we search with  $\mathbf{B}$ :

$$\mathbf{B}(1, -1, 0) = (2, -2, 0) = 2(1, -1, 0),$$

$$\mathbf{B}(1, 1, -2) = (6, 6, -12) = 6(1, 1, -2),$$

and both vectors have coordinate sum 0. Also  $\mathbf{B}\mathbf{1} = \mathbf{0}$ . The three vectors  $\mathbf{1}$ ,  $(1, -1, 0)$ ,  $(1, 1, -2)$  are pairwise orthogonal, as one checks from  $1 - 1 + 0 = 0$ ,  $1 + 1 - 2 = 0$  and  $1 - 1 + 0 = 0$ . Normalizing,

$$\mathbf{Q} = \begin{pmatrix} 1/\sqrt{3} & 1/\sqrt{2} & 1/\sqrt{6} \\ 1/\sqrt{3} & -1/\sqrt{2} & 1/\sqrt{6} \\ 1/\sqrt{3} & 0 & -2/\sqrt{6} \end{pmatrix},$$

and  $\mathbf{Q}^\top \mathbf{A} \mathbf{Q} = \text{diag}(4, 1, 1)$ ,  $\mathbf{Q}^\top \mathbf{B} \mathbf{Q} = \text{diag}(0, 2, 6)$ .

- The eigenvalue 1 of  $\mathbf{A}$  has a two-dimensional eigenspace, so *every* orthonormal basis of that plane serves  $\mathbf{A}$  equally well, and there are infinitely many, not obtained from one another by permuting and rescaling columns. The matrix  $\mathbf{B}$  has three distinct eigenvalues, so its eigenlines are determined, and the pair therefore determines the three lines of  $\mathbf{Q}$  up to order and sign.

### Exercise B2: Check the two warnings.

- For  $\mathbf{A} = \begin{pmatrix} 1 & 1 \\ 0 & 2 \end{pmatrix}$  and  $\mathbf{B} = \begin{pmatrix} 1 & 3 \\ 0 & 4 \end{pmatrix}$ , find an invertible  $\mathbf{P}$  with  $\mathbf{P}^{-1} \mathbf{A} \mathbf{P}$  and  $\mathbf{P}^{-1} \mathbf{B} \mathbf{P}$  diagonal, and show that no **unitary**  $\mathbf{P}$  does this.
- For  $\mathbf{X} = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}$  and  $\mathbf{Z} = \text{diag}(1, -1)$ , write down an orthogonal  $\mathbf{Q}_1$  diagonalizing  $\mathbf{X}$  and an orthogonal  $\mathbf{Q}_2$  diagonalizing  $\mathbf{Z}$ , and prove that no single invertible matrix diagonalizes both.

*Solution.*

- The common eigenvectors are  $\mathbf{e}_1$  (eigenvalues 1 and 1) and  $(1, 1)$  (eigenvalues

2 and 4):  $\mathbf{A}(1,1) = (2,2)$  and  $\mathbf{B}(1,1) = (4,4)$ . So  $\mathbf{P} = \begin{pmatrix} 1 & 1 \\ 0 & 1 \end{pmatrix}$  works, with  $\mathbf{P}^{-1}\mathbf{A}\mathbf{P} = \text{diag}(1,2)$  and  $\mathbf{P}^{-1}\mathbf{B}\mathbf{P} = \text{diag}(1,4)$ . If a unitary  $\mathbf{P}$  did the same, then  $\mathbf{A} = \mathbf{P}\mathbf{D}\mathbf{P}^*$  with  $\mathbf{D}$  diagonal would be normal by [Corollary 11.4.3](#); but  $\mathbf{A}^*\mathbf{A} = \begin{pmatrix} 1 & 1 \\ 1 & 5 \end{pmatrix}$  and  $\mathbf{A}\mathbf{A}^* = \begin{pmatrix} 2 & 2 \\ 2 & 4 \end{pmatrix}$  differ in every entry.

- (b)  $\mathbf{X}$  has eigenvectors  $(1,1)$  and  $(1,-1)$  for the eigenvalues 1 and  $-1$ , so  $\mathbf{Q}_1 = \frac{1}{\sqrt{2}} \begin{pmatrix} 1 & 1 \\ 1 & -1 \end{pmatrix}$  gives  $\mathbf{Q}_1^\top \mathbf{X} \mathbf{Q}_1 = \text{diag}(1, -1)$ .  $\mathbf{Z}$  is already diagonal, so  $\mathbf{Q}_2 = \mathbf{I}_2$ . No single invertible  $\mathbf{P}$  diagonalizes both: if it did, then  $\mathbf{P}^{-1}\mathbf{X}\mathbf{P}$  and  $\mathbf{P}^{-1}\mathbf{Z}\mathbf{P}$  would be diagonal, hence commuting, and multiplying by  $\mathbf{P}$  and  $\mathbf{P}^{-1}$  would give  $\mathbf{XZ} = \mathbf{ZX}$ . But  $\mathbf{XZ} = \begin{pmatrix} 0 & -1 \\ 1 & 0 \end{pmatrix}$  and  $\mathbf{ZX} = \begin{pmatrix} 0 & 1 \\ -1 & 0 \end{pmatrix}$ .

### Exercise B3: Triangularize two at once.

Let  $\mathbf{A} = \begin{pmatrix} 1 & 1 \\ 0 & 1 \end{pmatrix}$  and  $\mathbf{B} = \begin{pmatrix} 2 & 3 \\ 0 & 2 \end{pmatrix}$  in  $M_2(\mathbb{C})$ .

- Verify that  $\mathbf{AB} = \mathbf{BA}$ .
- Determine whether  $\mathbf{A}$  and  $\mathbf{B}$  are simultaneously diagonalizable, and justify your answer.
- Exhibit a unitary  $\mathbf{U}$  with  $\mathbf{U}^*\mathbf{A}\mathbf{U}$  and  $\mathbf{U}^*\mathbf{B}\mathbf{U}$  both upper triangular, as [Theorem 11.7.5](#) promises.

*Solution.*

- $\mathbf{AB} = \begin{pmatrix} 2 & 5 \\ 0 & 2 \end{pmatrix} = \mathbf{BA}$ .
- No. If one basis made both diagonal, then  $\mathbf{A}$  alone would be diagonalizable; but  $p_{\mathbf{A}} = (x-1)^2$  and  $\mathbf{A} - \mathbf{I}_2 = \begin{pmatrix} 0 & 1 \\ 0 & 0 \end{pmatrix} \neq \mathbf{0}$ , so  $\mathbf{A}$  has the single eigenvalue 1 with a one-dimensional eigenspace, and is not diagonalizable. This is the hypothesis that [Theorem 8.9.5](#) needs and that commuting does not supply.
- $\mathbf{U} = \mathbf{I}_2$ : both matrices are already upper triangular in the standard basis, which is orthonormal. The content of the theorem is that such a basis always exists, not that it is hard to find in this example.

### 11.7.5.3 C. Going deeper

#### Exercise C1: A special case of a theorem of Fuglede.

Let  $V$  be a finite-dimensional complex inner product space and let  $S, T \in \mathcal{L}(V)$  be **both normal**, with  $ST = TS$ .

- Prove that  $ST^* = T^*S$  and  $S^*T = TS^*$ .
- Deduce that  $S + T^*$  and  $ST^*$  are normal.
- Give an example of two commuting operators, **not** both normal, for which the conclusion of (a) fails as stated for the pair — that is, exhibit commuting  $S, T$

with  $ST^* \neq T^*S$ .

Note on the hypotheses: a theorem of Fuglede says that in (a) the operator  $S$  need not be assumed normal — it is enough that  $T$  be normal and that  $S$  commute with  $T$ . That stronger statement is true and is not proved in this book; (a) is only the case in which both are normal, and its proof uses the normality of  $S$  in an essential way.

*Solution.*

- (a) By [Theorem 11.7.1](#) applied to the commuting family  $\{S, T\}$  of normal operators, there is an orthonormal basis  $\mathcal{B}$  with  $[S]_{\mathcal{B}} = \mathbf{D}$  and  $[T]_{\mathcal{B}} = \mathbf{D}'$  diagonal. By [Theorem 10.6.3](#),  $[T^*]_{\mathcal{B}} = (\mathbf{D}')^*$ , which is the diagonal matrix with the conjugated entries. Diagonal matrices commute, so

$$[ST^*]_{\mathcal{B}} = \mathbf{D}(\mathbf{D}')^* = (\mathbf{D}')^*\mathbf{D} = [T^*S]_{\mathcal{B}},$$

and  $ST^* = T^*S$  because a map is determined by its matrix ([Theorem 3.6.4](#)). Exchanging the roles of  $S$  and  $T$  gives  $TS^* = S^*T$ .

- (b)  $T^*$  is normal, since  $(T^*)^*T^* = TT^* = T^*T = T^*(T^*)^*$ . By (a),  $S$  and  $T^*$  commute, and both are normal, so [Theorem 11.7.4](#) applies to the pair  $(S, T^*)$  and gives that  $S + T^*$  and  $ST^*$  are normal.

- (c) Take  $S = \begin{pmatrix} 0 & 1 \\ 0 & 0 \end{pmatrix}$  and  $T = \mathbf{I}_2$ . They commute,  $T$  is normal, and  $S$  is not. Here  $ST^* = S = T^*S$ , so this pair does not refute anything — as the theorem of Fuglede predicts, since  $T$  is normal. Take instead  $T = \begin{pmatrix} 0 & 1 \\ 0 & 0 \end{pmatrix}$  and  $S = \begin{pmatrix} 1 & 1 \\ 0 & 1 \end{pmatrix}$ , which commute, since both are polynomials in  $T$ . Then

$$ST^* = \begin{pmatrix} 1 & 1 \\ 0 & 1 \end{pmatrix} \begin{pmatrix} 0 & 0 \\ 1 & 0 \end{pmatrix} = \begin{pmatrix} 1 & 0 \\ 1 & 0 \end{pmatrix}, \quad T^*S = \begin{pmatrix} 0 & 0 \\ 1 & 1 \end{pmatrix},$$

which differ. Here it is  $T$ , not  $S$ , that fails to be normal, and that is exactly the hypothesis the theorem of Fuglede keeps.

### Exercise C2: A commuting family of orthogonal projections.

Let  $V \neq \{0\}$  be a finite-dimensional inner product space over  $F = \mathbb{R}$  or  $\mathbb{C}$ , and let  $P_1, \dots, P_r \in \mathcal{L}(V)$  be the orthogonal projections onto subspaces  $U_1, \dots, U_r$ , pairwise commuting.

- Prove that there is an orthonormal basis  $\mathcal{B}$  of  $V$  such that every vector of  $\mathcal{B}$  lies in  $U_i$  or in  $U_i^\perp$ , for every  $i$ .
- Deduce that  $P_i P_j$  is the orthogonal projection onto  $U_i \cap U_j$ .
- Prove that  $P_i + P_j$  is an orthogonal projection if and only if  $U_i \perp U_j$ , and identify the subspace in that case.

*Hint for (a): the eigenvalues of an idempotent operator.*

*Solution.*

- (a) Each  $P_i$  is self-adjoint, by [Example \(Four Adjoints\)](#) (d). Over  $\mathbb{R}$ , [Corollary 11.7.3](#)

applies directly to the commuting family  $\{P_1, \dots, P_r\}$ ; over  $\mathbb{C}$  a self-adjoint operator is normal, since  $P^*P = P^2 = PP^*$ , so [Theorem 11.7.1](#) applies. Either way there is an orthonormal basis  $\mathcal{B}$  of common eigenvectors. Let  $\mathbf{v} \in \mathcal{B}$  and fix  $i$ . If  $P_i\mathbf{v} = \mu\mathbf{v}$ , then  $\mu^2\mathbf{v} = P_i^2\mathbf{v} = P_i\mathbf{v} = \mu\mathbf{v}$ , so  $\mu \in \{0, 1\}$ , since  $\mathbf{v} \neq \mathbf{0}$ . If  $\mu = 1$  then  $\mathbf{v} = P_i\mathbf{v} \in \text{im } P_i = U_i$ ; if  $\mu = 0$  then  $\mathbf{v} \in \ker P_i = U_i^\perp$  ([Definition 10.3.2](#)).

(b) Write  $\mathcal{B} = (\mathbf{v}_1, \dots, \mathbf{v}_n)$  and let  $W = U_i \cap U_j$ . By (a) each  $\mathbf{v}_t$  satisfies  $P_i\mathbf{v}_t = a_t\mathbf{v}_t$  and  $P_j\mathbf{v}_t = b_t\mathbf{v}_t$  with  $a_t, b_t \in \{0, 1\}$ , so  $P_iP_j\mathbf{v}_t = a_tb_t\mathbf{v}_t$ , which is  $\mathbf{v}_t$  when  $\mathbf{v}_t \in U_i \cap U_j$  and  $\mathbf{0}$  otherwise. Let  $L = \text{span}\{\mathbf{v}_t : a_tb_t = 1\}$ ; then  $L \subseteq W$ . Conversely, if  $\mathbf{w} \in W$ , expand  $\mathbf{w} = \sum_t c_t\mathbf{v}_t$ ; applying  $P_i$  gives  $\mathbf{w} = \sum_t a_tc_t\mathbf{v}_t$ , so  $c_t = a_tc_t$  and hence  $c_t = 0$  whenever  $a_t = 0$ , and likewise for  $b_t$ . So  $\mathbf{w} \in L$  and  $W = L$ . Now  $P_iP_j$  fixes each  $\mathbf{v}_t \in L$  and kills the rest, and by [Theorem 10.3.3](#) (a) with the orthonormal basis  $\{\mathbf{v}_t : a_tb_t = 1\}$  of  $W$ , so does  $P_W$ ; two operators agreeing on a basis are equal, so  $P_iP_j = P_W$ .

(c) Write  $Q = P_i + P_j$ .

( $\Rightarrow$ ) Suppose  $Q$  is the orthogonal projection onto some subspace  $W$ . Then  $Q^2 = Q$  ([Definition 10.3.2](#)), and expanding,

$$P_i + P_j = Q^2 = P_i^2 + P_iP_j + P_jP_i + P_j^2 = P_i + P_j + 2P_iP_j,$$

using  $P_i^2 = P_i$ ,  $P_j^2 = P_j$  and  $P_jP_i = P_iP_j$ . Hence  $2P_iP_j = 0$ , so  $P_iP_j = 0$ . Now let  $\mathbf{v} \in U_j$ ; then  $P_i\mathbf{v} = P_i(P_j\mathbf{v}) = \mathbf{0}$ , so  $\mathbf{v} \in \ker P_i = U_i^\perp$ . Therefore  $U_j \subseteq U_i^\perp$ , which says  $U_i \perp U_j$ .

( $\Leftarrow$ ) Suppose  $U_i \perp U_j$ , so  $U_j \subseteq U_i^\perp = \ker P_i$  and  $U_i \subseteq \ker P_j$ . For  $\mathbf{v} \in U_i$  we get  $Q\mathbf{v} = \mathbf{v} + \mathbf{0} = \mathbf{v}$ , and symmetrically  $Q\mathbf{v} = \mathbf{v}$  for  $\mathbf{v} \in U_j$ , so  $Q$  is the identity on  $U_i \oplus U_j$ . For  $\mathbf{v} \in (U_i \oplus U_j)^\perp = U_i^\perp \cap U_j^\perp$  we get  $Q\mathbf{v} = \mathbf{0} + \mathbf{0} = \mathbf{0}$ . Since  $V = (U_i \oplus U_j) \oplus (U_i \oplus U_j)^\perp$  by [Theorem 10.3.2](#) (a),  $Q$  agrees with  $P_{U_i \oplus U_j}$  on both summands, hence everywhere. So  $P_i + P_j = P_{U_i \oplus U_j}$ .

## 11.8 Spectral Resolutions and Functions of a Normal Operator

The last two sections produce an orthonormal basis of eigenvectors. A basis, though, is a choice: permute it, or rotate inside the eigenspace of a repeated eigenvalue, and the basis changes while the operator does not. This section rewrites the same information with nothing chosen — a list of eigenvalues, and the orthogonal projections onto their eigenspaces — and then spends it. Once an operator is written as  $\lambda_1 P_1 + \dots + \lambda_k P_k$ , any function whatsoever on those  $k$  numbers produces a new operator, and powers, square roots and exponentials all become one line of arithmetic on the spectrum.

Throughout,  $V$  is a **non-zero** finite-dimensional inner product space over  $F = \mathbb{R}$  or  $F = \mathbb{C}$ , and  $T \in \mathcal{L}(V)$ . We say that  $T$  satisfies

$$(*) \quad V \text{ has an orthonormal basis of eigenvectors of } T.$$

By [Theorem 11.4.2](#) this holds over  $\mathbb{C}$  exactly for the **normal** operators, and by [Theorem 11.5.1](#) it holds over  $\mathbb{R}$  exactly for the **self-adjoint** ones. Stating everything for an operator satisfying

(\*) covers both fields at once, and keeps in view the only property we ever use.

## 11.8.1 Grouping an eigenbasis by eigenvalue

Here is the observation the section runs on. Let  $(\mathbf{e}_1, \dots, \mathbf{e}_n)$  be an orthonormal eigenbasis of  $T$ , with  $T\mathbf{e}_j = \mu_j\mathbf{e}_j$ . The list  $\mu_1, \dots, \mu_n$  repeats: an eigenvalue of geometric multiplicity 3 appears three times. Sort the basis vectors into bundles by the eigenvalue they carry, and each bundle spans one eigenspace. Summing the coordinates of a vector inside one bundle is exactly projecting it onto that eigenspace — orthogonally, because the basis is orthonormal. So  $T$  is a weighted sum of perpendicular projections, one weight per **distinct** eigenvalue, and the weights and the projections no longer remember the basis.

*A normal operator is a sum of perpendicular projections, weighted by its distinct eigenvalues.*

### Definition 11.8.1 — Spectral Resolution

Let  $T \in \mathcal{L}(V)$  satisfy (\*), let  $\lambda_1, \dots, \lambda_k$  be the **distinct** eigenvalues of  $T$ , and write  $E_i = E_{\lambda_i}(T)$  for the corresponding eigenspaces. Let

$$P_i := P_{E_i} \in \mathcal{L}(V)$$

be the **orthogonal** projection onto  $E_i$  (Definition 10.3.2). The identity

$$T = \lambda_1 P_1 + \lambda_2 P_2 + \dots + \lambda_k P_k$$

is the **spectral resolution** of  $T$ , and the family  $(P_1, \dots, P_k)$  is the **resolution of the identity** attached to  $T$ .

Two clauses carry weight. The eigenvalues are listed **once each**, so  $k$  is the number of distinct eigenvalues and not  $\dim V$ ; and the projections are **orthogonal**, which is the whole content of the definition, since every diagonalizable operator has *some* such expression with oblique projections. That the displayed identity is true, and that the  $P_i$  fit together as neatly as one could want, is the next theorem.

### Theorem 11.8.1 — The Spectral Resolution and Its Projections

Let  $T \in \mathcal{L}(V)$  satisfy (\*), with distinct eigenvalues  $\lambda_1, \dots, \lambda_k$ , eigenspaces  $E_i = E_{\lambda_i}(T)$  and  $P_i = P_{E_i}$ . Then:

- (a)  $V = E_1 \oplus \dots \oplus E_k$ , and  $E_i \perp E_j$  whenever  $i \neq j$ ;
- (b)  $P_i^* = P_i$  and  $P_i^2 = P_i$  for every  $i$ , and  $P_i P_j = 0$  whenever  $i \neq j$ ;
- (c)  $P_1 + \dots + P_k = \text{id}_V$ ;
- (d)  $T = \lambda_1 P_1 + \dots + \lambda_k P_k$ ;
- (e)  $T P_i = P_i T = \lambda_i P_i$  for every  $i$ .

*Idea.* Everything is read off one orthonormal eigenbasis. Sorting its vectors by eigenvalue cuts it into  $k$  bundles; the only thing to check is that the  $i$ -th bundle spans the **whole** of  $E_i$ , not just part of it, and that is a coordinate comparison. After that, (a) is the



orthonormality of the basis, (c) and (d) are the coordinate formula  $\mathbf{v} = \sum_j \langle \mathbf{v}, \mathbf{e}_j \rangle \mathbf{e}_j$  with its terms grouped, and (b) and (e) are bookkeeping.

*Proof.* Let  $\mathcal{B} = (\mathbf{e}_1, \dots, \mathbf{e}_n)$  be an orthonormal basis of  $V$  with  $T\mathbf{e}_j = \mu_j \mathbf{e}_j$ , which exists by (\*). Each  $\mathbf{e}_j$  is non-zero, so each  $\mu_j$  is an eigenvalue of  $T$  and therefore equals exactly one  $\lambda_i$ . Put

$$\mathcal{S}_i := \{j : \mu_j = \lambda_i\},$$

so that  $\{1, \dots, n\}$  is the disjoint union of  $\mathcal{S}_1, \dots, \mathcal{S}_k$ .

**Step 1.**  $E_i = \text{span}(\mathbf{e}_j : j \in \mathcal{S}_i)$ . The inclusion ( $\supseteq$ ) is immediate, since  $T\mathbf{e}_j = \lambda_i \mathbf{e}_j$  for  $j \in \mathcal{S}_i$ . For ( $\subseteq$ ), let  $\mathbf{v} \in E_i$  and write  $\mathbf{v} = \sum_j c_j \mathbf{e}_j$  in the basis  $\mathcal{B}$ . Applying  $T$ ,

$$\sum_j c_j \mu_j \mathbf{e}_j = T\mathbf{v} = \lambda_i \mathbf{v} = \sum_j c_j \lambda_i \mathbf{e}_j.$$

Coordinates in a basis are unique, so  $c_j(\mu_j - \lambda_i) = 0$  for every  $j$ , and hence  $c_j = 0$  unless  $j \in \mathcal{S}_i$ . In particular  $\mathcal{S}_i \neq \emptyset$ , because  $E_i \neq \{\mathbf{0}\}$ .

**Step 2 (a).** By Step 1,  $V = \text{span}(\mathcal{B}) = E_1 + \dots + E_k$ . If  $i \neq j$ , then  $\mathcal{S}_i$  and  $\mathcal{S}_j$  are disjoint and  $\mathcal{B}$  is orthonormal, so every spanning vector of  $E_i$  is orthogonal to every spanning vector of  $E_j$ ; hence  $E_i \perp E_j$ . Pairwise orthogonal subspaces are independent: if  $\mathbf{u}_1 + \dots + \mathbf{u}_k = \mathbf{0}$  with  $\mathbf{u}_i \in E_i$ , then pairing with  $\mathbf{u}_l$  kills every term but one and leaves  $\|\mathbf{u}_l\|^2 = 0$ , so  $\mathbf{u}_l = \mathbf{0}$ . By Theorem 1.7.4 the sum is direct.

**Step 3 (b).** Each  $P_i$  is self-adjoint by Theorem 10.3.3 (b), and satisfies  $P_i^2 = P_i$  with  $\text{im } P_i = E_i$  and  $\ker P_i = E_i^\perp$  (Definition 10.3.2 and the discussion following it). For  $i \neq j$ , Step 2 gives  $E_j \subseteq E_i^\perp$ , so  $\text{im } P_j \subseteq \ker P_i$  and  $P_i P_j = 0$ .

**Step 4 (c) and (d).** By Step 1,  $(\mathbf{e}_j : j \in \mathcal{S}_i)$  is an orthonormal basis of  $E_i$ , so Theorem 10.3.3 (a) gives

$$P_i \mathbf{v} = \sum_{j \in \mathcal{S}_i} \langle \mathbf{v}, \mathbf{e}_j \rangle \mathbf{e}_j \quad (\mathbf{v} \in V).$$

Summing over  $i$  collects every index  $j$  exactly once, so by Theorem 10.2.1,

$$\sum_{i=1}^k P_i \mathbf{v} = \sum_{j=1}^n \langle \mathbf{v}, \mathbf{e}_j \rangle \mathbf{e}_j = \mathbf{v},$$

which is (c). Multiplying the  $i$ -th group by  $\lambda_i$  instead,

$$\begin{aligned} \sum_{i=1}^k \lambda_i P_i \mathbf{v} &= \sum_{i=1}^k \sum_{j \in \mathcal{S}_i} \langle \mathbf{v}, \mathbf{e}_j \rangle \lambda_i \mathbf{e}_j \\ &= \sum_{j=1}^n \langle \mathbf{v}, \mathbf{e}_j \rangle \mu_j \mathbf{e}_j = T \left( \sum_{j=1}^n \langle \mathbf{v}, \mathbf{e}_j \rangle \mathbf{e}_j \right) = T\mathbf{v}, \end{aligned}$$

where the second equality uses  $\mu_j = \lambda_i$  for  $j \in \mathcal{S}_i$ , and the third the linearity of  $T$  together with  $T\mathbf{e}_j = \mu_j \mathbf{e}_j$ . This is (d).

**Step 5 (e).** For  $\mathbf{v} \in V$  we have  $P_i \mathbf{v} \in E_i$ , so  $T(P_i \mathbf{v}) = \lambda_i P_i \mathbf{v}$ ; thus  $TP_i = \lambda_i P_i$ . In the other order, (d) and (b) give

$$P_i T = P_i \left( \sum_{j=1}^k \lambda_j P_j \right) = \lambda_i P_i^2 = \lambda_i P_i.$$

This proves the theorem. ■

For matrices the picture is the familiar one, packaged differently. If  $\mathbf{A} \in M_n(\mathbb{C})$  is normal, write  $\mathbf{A} = \mathbf{U} \mathbf{D} \mathbf{U}^*$  with  $\mathbf{U}$  unitary and  $\mathbf{D}$  diagonal ([Corollary 11.4.3](#)). Grouping the columns  $\mathbf{q}_1, \dots, \mathbf{q}_n$  of  $\mathbf{U}$  by their eigenvalue turns this into

$$\mathbf{A} = \sum_{i=1}^k \lambda_i \mathbf{P}_i, \quad \mathbf{P}_i = \sum_{j \in \mathcal{S}_i} \mathbf{q}_j \mathbf{q}_j^*,$$

so each  $\mathbf{P}_i$  is a sum of rank-one projections and  $\text{rank } \mathbf{P}_i = \dim E_i$ .

**Example. (A Symmetric  $3 \times 3$  Matrix)**

Let  $\mathbf{A} = \begin{pmatrix} 1 & 2 & 2 \\ 2 & 1 & 2 \\ 2 & 2 & 1 \end{pmatrix} \in M_3(\mathbb{R})$ . Find the spectral resolution of  $\mathbf{A}$ .

*Solution.* Write  $\mathbf{A} = 2\mathbf{J} - \mathbf{I}$ , where  $\mathbf{J}$  is the all-ones matrix. Since  $\mathbf{J}\mathbf{x} = (x_1 + x_2 + x_3)\mathbf{1}$ , we get  $\mathbf{J}\mathbf{1} = 3\mathbf{1}$  and  $\mathbf{J}\mathbf{x} = \mathbf{0}$  exactly when  $x_1 + x_2 + x_3 = 0$ . So  $\mathbf{J}$  has eigenvalues 3 and 0, and  $\mathbf{A}$  has eigenvalues

$$\lambda_1 = 2 \cdot 3 - 1 = 5, \quad \lambda_2 = 2 \cdot 0 - 1 = -1,$$

with  $E_5 = \text{span}(\mathbf{1})$  of dimension 1 and  $E_{-1} = \mathbf{1}^\perp$  of dimension 2. Since  $\|\mathbf{1}\|^2 = 3$ , the orthogonal projection onto  $E_5$  is  $\mathbf{P}_1 = \frac{1}{3}\mathbf{1}\mathbf{1}^\top = \frac{1}{3}\mathbf{J}$ , and  $\mathbf{P}_2 = \mathbf{I} - \mathbf{P}_1$  by [Theorem 11.8.1](#) (c). Concretely,

$$\mathbf{P}_1 = \frac{1}{3} \begin{pmatrix} 1 & 1 & 1 \\ 1 & 1 & 1 \\ 1 & 1 & 1 \end{pmatrix}, \quad \mathbf{P}_2 = \frac{1}{3} \begin{pmatrix} 2 & -1 & -1 \\ -1 & 2 & -1 \\ -1 & -1 & 2 \end{pmatrix}.$$

*Check.* Both are symmetric, both satisfy  $\mathbf{P}_i^2 = \mathbf{P}_i$ , their product is 0, they sum to  $\mathbf{I}$ , and

$$5\mathbf{P}_1 - \mathbf{P}_2 = \frac{5}{3}\mathbf{J} - \mathbf{I} + \frac{1}{3}\mathbf{J} = 2\mathbf{J} - \mathbf{I} = \mathbf{A}.$$

Note that the resolution has two terms, not three:  $k = 2$  counts **distinct** eigenvalues, while  $\text{rank } \mathbf{P}_2 = 2$  records the multiplicity of  $-1$ .

## 11.8.2 Orthogonal projections versus oblique ones

An expression  $T = \sum \lambda_i \pi_i$  with idempotents summing to the identity is not by itself news. In the proof of [Theorem 9.8.3](#) the semisimple part of  $T$  is written as  $\sum_i \lambda_i \pi_i$ , where  $\pi_i = h_i(T)$

projects onto the generalized eigenspace  $G_{\lambda_i}(T)$  along the sum of the others (Theorem 8.8.1 (d)); when  $T$  is diagonalizable, that is a decomposition of  $T$  itself, and those  $\pi_i$  satisfy  $\pi_i^2 = \pi_i$ ,  $\pi_i\pi_j = 0$  for  $i \neq j$  and  $\sum_i \pi_i = \text{id}_V$ , exactly as in Theorem 11.8.1 (b) and (c). What they do **not** satisfy is  $\pi_i^* = \pi_i$ , and that single missing condition is the whole difference between Chapter 9 and this section.

### Example. (Two Projections That Are Not Orthogonal)

Let  $\mathbf{A} = \begin{pmatrix} 1 & 1 \\ 0 & 2 \end{pmatrix} \in M_2(\mathbb{R})$ , which is diagonalizable with eigenvalues 1 and 2. Compute the projections  $\pi_1, \pi_2$  of Theorem 8.8.1 for  $\mathbf{A}$ , verify that  $\mathbf{A} = \pi_1 + 2\pi_2$ , and decide whether they are orthogonal projections.

*Solution.* The eigenspaces are  $E_1 = \text{span}(\mathbf{e}_1)$  and  $E_2 = \text{span}((1, 1))$ , read off from  $\mathbf{A} - \mathbf{I} = \begin{pmatrix} 0 & 1 \\ 0 & 1 \end{pmatrix}$  and  $\mathbf{A} - 2\mathbf{I} = \begin{pmatrix} -1 & 1 \\ 0 & 0 \end{pmatrix}$ . Since the eigenvalues are distinct, the generalized eigenspaces are the eigenspaces. With  $\mathbf{P} = \begin{pmatrix} 1 & 1 \\ 0 & 1 \end{pmatrix}$  we have  $\mathbf{P}^{-1}\mathbf{A}\mathbf{P} = \text{diag}(1, 2)$ , so

$$\pi_1 = \mathbf{P} \begin{pmatrix} 1 & 0 \\ 0 & 0 \end{pmatrix} \mathbf{P}^{-1} = \begin{pmatrix} 1 & -1 \\ 0 & 0 \end{pmatrix}, \quad \pi_2 = \mathbf{I} - \pi_1 = \begin{pmatrix} 0 & 1 \\ 0 & 1 \end{pmatrix}.$$

Both square to themselves, their products vanish in both orders, they sum to  $\mathbf{I}$ , and

$$1 \cdot \pi_1 + 2 \cdot \pi_2 = \begin{pmatrix} 1 & -1 \\ 0 & 0 \end{pmatrix} + \begin{pmatrix} 0 & 2 \\ 0 & 2 \end{pmatrix} = \mathbf{A}.$$

They are **not** orthogonal projections:  $\pi_1^\top = \begin{pmatrix} 1 & 0 \\ -1 & 0 \end{pmatrix} \neq \pi_1$ . Geometrically,  $\pi_1$  projects onto  $\text{span}(\mathbf{e}_1)$  **along**  $\text{span}((1, 1))$ , and those two lines meet at  $45^\circ$ , not at a right angle. Consistently,  $\mathbf{A}$  is not normal:

$$\mathbf{A}^\top \mathbf{A} = \begin{pmatrix} 1 & 1 \\ 1 & 5 \end{pmatrix} \neq \begin{pmatrix} 2 & 2 \\ 2 & 4 \end{pmatrix} = \mathbf{A}\mathbf{A}^\top.$$

*Warning.* “**Diagonalizable**” gives projections; only “**normal**” makes them **orthogonal**. Every diagonalizable operator splits as  $\sum \lambda_i \pi_i$  with idempotents that annihilate each other and sum to the identity, and Example (Two Projections That Are Not Orthogonal) is such a splitting for a matrix that is not normal. The projections there are not self-adjoint, so they are not the projections of Definition 11.8.1, and the eigenspaces they project onto are not perpendicular. Whenever an argument needs  $\langle P_i \mathbf{u}, \mathbf{v} \rangle = \langle \mathbf{u}, P_i \mathbf{v} \rangle$  — and most of this section does — normality is what pays for it.

## 11.8.3 What makes a resolution the resolution

The projections in Definition 11.8.1 were built from the eigenspaces, so they are visibly determined by  $T$ . It is much more useful to know the converse: that *any* decomposition with the right formal properties has to be that one. This is what lets us recognize a spectral resolution when it turns up by accident, and it is the engine of the uniqueness proof for square roots below.

### Theorem 11.8.2 — Uniqueness of the Spectral Resolution

Let  $V$  be a non-zero finite-dimensional inner product space over  $F$  and let  $T \in \mathcal{L}(V)$ . Suppose

$$T = \mu_1 Q_1 + \mu_2 Q_2 + \cdots + \mu_m Q_m,$$

where  $\mu_1, \dots, \mu_m \in F$  are **distinct** and  $Q_1, \dots, Q_m \in \mathcal{L}(V)$  are **non-zero** operators satisfying

(R1)  $Q_i^* = Q_i$  and  $Q_i^2 = Q_i$  for every  $i$ ;

(R2)  $Q_i Q_j = 0$  whenever  $i \neq j$ ;

(R3)  $Q_1 + \cdots + Q_m = \text{id}_V$ .

Then  $T$  satisfies  $(*)$ ,  $\text{spec}(T) = \{\mu_1, \dots, \mu_m\}$ , and  $Q_i$  is the orthogonal projection onto  $E_{\mu_i}(T)$  for every  $i$ . In particular  $m$  is the number of distinct eigenvalues of  $T$ , and the given decomposition is the spectral resolution of  $T$ .

*Idea.* Three moves. ① A self-adjoint idempotent is the orthogonal projection onto its own image: self-adjointness is used here and nowhere else, which is exactly why [Example \(Two Projections That Are Not Orthogonal\)](#) escapes the theorem. ② Conditions (R2) and (R3) then break  $V$  into the orthogonal direct sum of the images  $U_i = \text{im } Q_i$ . ③ On  $U_i$  the operator  $T$  is multiplication by  $\mu_i$ , so  $U_i \subseteq E_{\mu_i}(T)$ ; and because the  $\mu_i$  are distinct, an eigenvector of  $T$  cannot spread across two pieces, which upgrades the inclusion to an equality.

*Proof.* Write  $U_i = \text{im } Q_i$ , which is non-zero because  $Q_i \neq 0$ .

**Step 1.**  $Q_i = P_{U_i}$ . Fix  $i$  and drop the subscript. If  $\mathbf{u} \in U$ , say  $\mathbf{u} = Q\mathbf{x}$ , then  $Q\mathbf{u} = Q^2\mathbf{x} = Q\mathbf{x} = \mathbf{u}$ , so  $Q$  is the identity on  $U$ . Next,  $\ker Q = U^\perp$ . Indeed, if  $Q\mathbf{w} = \mathbf{0}$ , then for every  $\mathbf{x}$ ,

$$\langle \mathbf{w}, Q\mathbf{x} \rangle = \langle Q\mathbf{w}, \mathbf{x} \rangle = 0$$

by [Definition 10.6.1](#) and (R1), so  $\mathbf{w} \in U^\perp$ ; conversely if  $\mathbf{w} \in U^\perp$ , then  $Q\mathbf{w} \in U$  and

$$\|Q\mathbf{w}\|^2 = \langle Q\mathbf{w}, Q\mathbf{w} \rangle = \langle \mathbf{w}, Q^2\mathbf{w} \rangle = \langle \mathbf{w}, Q\mathbf{w} \rangle = 0,$$

using (R1) twice and then  $Q\mathbf{w} \in U \perp \mathbf{w}$ ; so  $Q\mathbf{w} = \mathbf{0}$ . Finally, for any  $\mathbf{v} \in V$ ,

$$\mathbf{v} = Q\mathbf{v} + (\mathbf{v} - Q\mathbf{v}), \quad Q\mathbf{v} \in U, \quad Q(\mathbf{v} - Q\mathbf{v}) = Q\mathbf{v} - Q^2\mathbf{v} = \mathbf{0},$$

so  $\mathbf{v} - Q\mathbf{v} \in \ker Q = U^\perp$ . This is the splitting of  $\mathbf{v}$  in  $U \oplus U^\perp$  ([Theorem 10.3.2](#)), so  $Q\mathbf{v} = P_U\mathbf{v}$  by [Definition 10.3.2](#).

**Step 2.**  $V = U_1 \oplus \cdots \oplus U_m$ , *orthogonally*. By (R3), every  $\mathbf{v}$  equals  $\sum_i Q_i\mathbf{v} \in \sum_i U_i$ . For  $i \neq j$ ,  $\mathbf{u} = Q_i\mathbf{x}$  and  $\mathbf{w} = Q_j\mathbf{y}$  satisfy

$$\langle \mathbf{u}, \mathbf{w} \rangle = \langle Q_i\mathbf{x}, Q_j\mathbf{y} \rangle = \langle Q_j Q_i\mathbf{x}, \mathbf{y} \rangle = 0$$

by (R1) and (R2), so  $U_i \perp U_j$ . As in Step 2 of the proof of [Theorem 11.8.1](#), pairwise orthogonal subspaces are independent, so the sum is direct.

**Step 3.**  $T$  acts on  $U_i$  as multiplication by  $\mu_i$ . Let  $\mathbf{u} \in U_i$ . By Step 1,  $Q_i\mathbf{u} = \mathbf{u}$ , and for

$j \neq i$ ,  $Q_j \mathbf{u} = Q_j Q_i \mathbf{u} = \mathbf{0}$  by (R2). Hence

$$T\mathbf{u} = \sum_{l=1}^m \mu_l Q_l \mathbf{u} = \mu_i \mathbf{u}.$$

Since  $U_i \neq \{\mathbf{0}\}$ , this makes  $\mu_i$  an eigenvalue of  $T$ , with  $U_i \subseteq E_{\mu_i}(T)$ .

**Step 4.**  $\text{spec}(T) = \{\mu_1, \dots, \mu_m\}$  and  $U_i = E_{\mu_i}(T)$ . Let  $\lambda$  be an eigenvalue of  $T$  with eigenvector  $\mathbf{v} \neq \mathbf{0}$ . By (R3),  $\mathbf{v} = \sum_i Q_i \mathbf{v}$  with  $Q_i \mathbf{v} \in U_i$ , and by Step 3,

$$\sum_{i=1}^m \mu_i Q_i \mathbf{v} = T\mathbf{v} = \lambda \mathbf{v} = \sum_{i=1}^m \lambda Q_i \mathbf{v}.$$

Subtracting,  $\sum_i (\mu_i - \lambda) Q_i \mathbf{v} = \mathbf{0}$  with the  $i$ -th term in  $U_i$ , so by the directness of Step 2 every term vanishes:  $Q_i \mathbf{v} = \mathbf{0}$  whenever  $\mu_i \neq \lambda$ . As  $\mathbf{v} \neq \mathbf{0}$ , some  $Q_i \mathbf{v} \neq \mathbf{0}$ , and for that  $i$  we get  $\lambda = \mu_i$ . The  $\mu_l$  being distinct,  $i$  is the only index with  $\mu_i = \lambda$ , so  $\mathbf{v} = Q_i \mathbf{v} \in U_i$ . Therefore  $\text{spec}(T) \subseteq \{\mu_1, \dots, \mu_m\}$  and  $E_{\mu_i}(T) \subseteq U_i$ ; with Step 3 both are equalities.

**Step 5.** By Steps 1 and 4,  $Q_i = P_{U_i} = P_{E_{\mu_i}(T)}$ . Choosing an orthonormal basis of each  $U_i$  (Theorem 10.2.2) and concatenating gives an orthonormal basis of  $V$  by Step 2, made of eigenvectors of  $T$  by Step 3; so  $T$  satisfies (\*). This proves the theorem. ■

Read the other way round, the theorem is the complex spectral theorem again with no basis in sight: over  $\mathbb{C}$ , an operator is normal **if and only if** it can be written as  $\sum \mu_i Q_i$  with distinct  $\mu_i$  and non-zero self-adjoint idempotents satisfying (R2) and (R3) (Theorem 11.4.2). Conditions (R1)–(R3) are the entire content of the theorem, compressed into three lines of algebra.

*Quick check.* The matrices  $\pi_1, \pi_2$  of Example (Two Projections That Are Not Orthogonal) satisfy  $\pi_i^2 = \pi_i$ ,  $\pi_1 \pi_2 = \pi_2 \pi_1 = 0$  and  $\pi_1 + \pi_2 = \mathbf{I}$ , and the coefficients 1 and 2 are distinct. Which hypothesis of Theorem 11.8.2 fails, and what does its failure cost?

*Solution.* Self-adjointness in (R1):  $\pi_1^\top \neq \pi_1$ . Everything else holds. The cost is Step 1 of the proof:  $\pi_1$  is a projection onto  $\text{span}(\mathbf{e}_1)$ , but along  $\text{span}((1,1))$  rather than along  $\text{span}(\mathbf{e}_1)^\perp = \text{span}(\mathbf{e}_2)$ , so it is not  $P_{\text{span}(\mathbf{e}_1)}$ . Consequently the two images are not perpendicular, and the conclusion fails at its strongest point:  $\begin{pmatrix} 1 & 1 \\ 0 & 2 \end{pmatrix}$  does not satisfy (\*), since it is not normal.

## 11.8.4 Functions of an operator

Powers are now trivial. Multiplying the resolution by itself and using  $P_i P_j = 0$  for  $i \neq j$  leaves only the diagonal terms:

$$T^2 = \left( \sum_i \lambda_i P_i \right) \left( \sum_j \lambda_j P_j \right) = \sum_i \lambda_i^2 P_i,$$

and the same computation repeated gives  $T^m = \sum_i \lambda_i^m P_i$ . So  $p(T) = \sum_i p(\lambda_i) P_i$  for every polynomial  $p$ . Stare at that formula and notice what it does **not** use: nothing about  $p$  except its  $k$  values on the spectrum. Any function at all supplies those.

### Definition 11.8.2 — Function of an Operator

Let  $T \in \mathcal{L}(V)$  satisfy (\*), with distinct eigenvalues  $\lambda_1, \dots, \lambda_k$  and spectral resolution  $T = \sum_i \lambda_i P_i$ . For **any** function  $f: \text{spec}(T) \rightarrow F$ , define

$$f(T) := \sum_{i=1}^k f(\lambda_i) P_i \in \mathcal{L}(V).$$

In words: apply  $f$  to each eigenvalue and reweight the same projections. The function needs no continuity, no differentiability and no formula — it is  $k$  numbers. And the definition is unambiguous: the  $\lambda_i$  and the  $P_i$  are determined by  $T$  alone, by [Theorem 11.8.2](#), so no choice of basis or of eigenvectors enters.

Four instances show the range. Taking  $f(\lambda) = \lambda$  returns  $T$ . Taking  $f \equiv 1$  returns  $\text{id}_V$ , by [Theorem 11.8.1](#) (c). Taking  $f(\lambda) = \bar{\lambda}$  returns  $T^*$ , since  $P_i^* = P_i$ . And if  $0 \notin \text{spec}(T)$ , taking  $f(\lambda) = \lambda^{-1}$  returns  $T^{-1}$ , because

$$\left( \sum_i \lambda_i P_i \right) \left( \sum_j \lambda_j^{-1} P_j \right) = \sum_i P_i = \text{id}_V.$$

The notation is, on its face, dangerous. For a polynomial  $p$  the symbol  $p(T)$  already means something — the operator  $a_0 \text{id}_V + a_1 T + \dots + a_N T^N$  of [Definition 3.4.2](#), which Chapter 5 studied at length. If [Definition 11.8.2](#) disagreed with it, the whole notation would have to be abandoned. Part (a) below says it does not, and that is the reason the rest of the theorem is safe to use.

### Theorem 11.8.3 — Properties of the Functional Calculus

Let  $T \in \mathcal{L}(V)$  satisfy (\*), with distinct eigenvalues  $\lambda_1, \dots, \lambda_k$  and resolution of the identity  $(P_1, \dots, P_k)$ . Let  $f, g: \text{spec}(T) \rightarrow F$  and  $c \in F$ , with  $f + g$ ,  $cf$  and  $fg$  taken pointwise.

- (a) **(Compatibility)** For every  $p \in F[x]$ , the operator  $p(T)$  of [Definition 3.4.2](#) equals  $\sum_i p(\lambda_i) P_i$ ; that is, it agrees with [Definition 11.8.2](#) applied to the restriction of  $p$  to  $\text{spec}(T)$ .
- (b) **(Algebra homomorphism)**  $(f + g)(T) = f(T) + g(T)$ ,  $(cf)(T) = cf(T)$ ,  $(fg)(T) = f(T)g(T) = g(T)f(T)$ , and  $1(T) = \text{id}_V$ .
- (c) **(Adjoint)**  $f(T)^* = \overline{f}(T)$ , where  $\overline{f}(\lambda) = \overline{f(\lambda)}$ . In particular  $f(T)$  is self-adjoint when  $f$  is real-valued, and unitary when  $|f(\lambda_i)| = 1$  for every  $i$ .
- (d) **(Spectral mapping)**  $f(T)$  satisfies (\*), with the same orthonormal eigenbasis as  $T$ , and  $\text{spec}(f(T)) = f(\text{spec}(T))$ .
- (e) **(Commutation)** If  $S \in \mathcal{L}(V)$  commutes with  $T$ , then  $S$  commutes with every  $P_i$  and hence with  $f(T)$ . In particular  $T$  and  $f(T)$  commute.

*Idea.* Everything rests on one multiplication table,  $P_i P_j = \delta_{ij} P_i$ . It says that the operators  $\sum_i c_i P_i$  multiply exactly the way the coefficient lists  $(c_1, \dots, c_k)$  multiply coordinatewise, so  $\mathcal{L}(V)$  contains a copy of the algebra of all  $F$ -valued functions on  $\text{spec}(T)$ , and  $f \mapsto f(T)$  is the obvious map into it. That makes (b) and (c) one-line computations. The one statement needing an argument is (a) — that the copy sitting inside is

the *same* copy Chapter 3 built by substituting  $T$  into polynomials — and the argument is to compute the powers of  $T$ . For (e), a commuting  $S$  preserves each eigenspace, and an operator preserving all the summands of an orthogonal direct sum commutes with the projections onto them.

*Proof.*

(a) We first show that

$$T^m = \sum_{i=1}^k \lambda_i^m P_i \quad \text{for every integer } m \geq 0,$$

with the usual convention  $\lambda^0 = 1$ . For  $m = 0$  both sides equal  $\text{id}_V$ , by [Theorem 11.8.1](#) (c). Assuming the formula for  $m$ , [Theorem 11.8.1](#) (d) and (b) give

$$\begin{aligned} T^{m+1} &= T \cdot T^m = \left( \sum_i \lambda_i P_i \right) \left( \sum_j \lambda_j^m P_j \right) \\ &= \sum_{i,j} \lambda_i \lambda_j^m P_i P_j = \sum_i \lambda_i^{m+1} P_i, \end{aligned}$$

since  $P_i P_j = 0$  for  $i \neq j$  and  $P_i^2 = P_i$ . Now let  $p = a_0 + a_1 x + \cdots + a_N x^N$ . By [Definition 3.4.2](#) and the formula just proved,

$$\begin{aligned} p(T) &= \sum_{m=0}^N a_m T^m = \sum_{m=0}^N a_m \sum_{i=1}^k \lambda_i^m P_i \\ &= \sum_{i=1}^k \left( \sum_{m=0}^N a_m \lambda_i^m \right) P_i = \sum_{i=1}^k p(\lambda_i) P_i, \end{aligned}$$

the middle step being a rearrangement of a finite double sum.

(b) The first two identities are immediate from [Definition 11.8.2](#). For the product, [Theorem 11.8.1](#) (b) gives

$$f(T)g(T) = \sum_{i,j} f(\lambda_i)g(\lambda_j)P_i P_j = \sum_i f(\lambda_i)g(\lambda_i)P_i,$$

which is  $(fg)(T)$ . The right-hand side is symmetric in  $f$  and  $g$ , so  $f(T)g(T) = g(T)f(T)$ . Finally  $1(T) = \sum_i P_i = \text{id}_V$  by [Theorem 11.8.1](#) (c).

(c) By [Theorem 10.6.2](#) (a) and (b), and  $P_i^* = P_i$ ,

$$f(T)^* = \sum_{i=1}^k \overline{f(\lambda_i)} P_i^* = \overline{f}(T).$$

If  $f$  is real-valued then  $\overline{f} = f$  and  $f(T)^* = f(T)$ . If  $|f(\lambda_i)| = 1$  for all  $i$ , then  $\overline{f}f \equiv 1$ , so by (b)  $f(T)^* f(T) = (\overline{f}f)(T) = \text{id}_V$ , and likewise  $f(T)f(T)^* = \text{id}_V$ ; so  $f(T)$  is unitary ([Definition 10.7.2](#)).

(d) Let  $\mathcal{B} = (\mathbf{e}_1, \dots, \mathbf{e}_n)$  be an orthonormal eigenbasis of  $T$ , with  $T\mathbf{e}_j = \mu_j \mathbf{e}_j$ , and fix  $j$ . Let  $i$  be the index with  $\mu_j = \lambda_i$ . Then  $\mathbf{e}_j \in E_i$ , so  $P_i \mathbf{e}_j = \mathbf{e}_j$ , while  $P_l \mathbf{e}_j = \mathbf{0}$

for  $l \neq i$  because  $E_i \subseteq E_l^\perp = \ker P_l$ , by [Theorem 11.8.1](#) (a) and [Definition 10.3.2](#). Hence

$$f(T)\mathbf{e}_j = f(\lambda_i)\mathbf{e}_j = f(\mu_j)\mathbf{e}_j.$$

So  $\mathcal{B}$  is an orthonormal basis of eigenvectors of  $f(T)$ , which is  $(*)$  for  $f(T)$ . The matrix of  $f(T)$  in  $\mathcal{B}$  is  $\text{diag}(f(\mu_1), \dots, f(\mu_n))$ , so its eigenvalues are exactly the numbers  $f(\mu_j)$ , that is the numbers  $f(\lambda_i)$ ; hence  $\text{spec}(f(T)) = f(\text{spec}(T))$ .

- (e) Suppose  $ST = TS$ . Each  $E_i$  is  $S$ -invariant: if  $T\mathbf{v} = \lambda_i\mathbf{v}$ , then  $T(S\mathbf{v}) = S(T\mathbf{v}) = \lambda_i S\mathbf{v}$ . Let  $\mathbf{v} \in V$  and write  $\mathbf{v} = \sum_i P_i\mathbf{v}$  with  $P_i\mathbf{v} \in E_i$  ([Theorem 11.8.1](#) (c)). Applying  $S$  gives  $S\mathbf{v} = \sum_i SP_i\mathbf{v}$  with  $SP_i\mathbf{v} \in E_i$ , and this is the decomposition of  $S\mathbf{v}$  in  $E_1 \oplus \dots \oplus E_k$ , which is unique. Comparing it with  $S\mathbf{v} = \sum_i P_i(S\mathbf{v})$  gives  $P_i S\mathbf{v} = SP_i\mathbf{v}$  for every  $i$ , that is  $SP_i = P_i S$ . Hence  $S$  commutes with  $f(T) = \sum_i f(\lambda_i)P_i$ . Since  $T$  commutes with itself, the last claim follows. This proves the theorem. ■

Part (a) is what makes the notation honest. Writing  $\sqrt{T}$ ,  $e^T$  and  $p(T)$  in the same paragraph is legitimate precisely because the polynomial case is common ground: the new definition extends the old one instead of competing with it. It also reconciles this section with Chapter 9. An operator satisfying  $(*)$  is diagonalizable, so every Jordan block of its matrix has size 1, and [Definition 9.9.1](#) consults no derivatives at all — it too returns  $\sum_i f(\lambda_i)\mathbf{P}_i$ . The two constructions of  $f(\mathbf{A})$  agree for normal  $\mathbf{A}$ , and Chapter 9's extra machinery is precisely the price of dropping normality.

**Warning.** Outside the normal case,  $f$  on the spectrum is not enough data. For  $\mathbf{A} = \begin{pmatrix} 0 & 1 \\ 0 & 0 \end{pmatrix}$  the only eigenvalue is 0, so the recipe “ $f(\mathbf{A}) = \sum_i f(\lambda_i)\mathbf{P}_i$ ” would read  $f(\mathbf{A}) = f(0)\mathbf{I}$  for every  $f$  — and with  $f(x) = x$  it would announce  $\mathbf{A} = 0$ . The recipe is not wrong, it is inapplicable:  $\mathbf{A}$  is not normal, so there is no resolution of the identity to expand in. Chapter 9's definition of  $f(\mathbf{A})$  has to consult  $f'(0)$  as well ([Definition 9.9.1](#)), and that derivative is exactly what the missing orthogonality would have spared us.

**Quick check.** Let  $T \in \mathcal{L}(V)$  satisfy  $(*)$  with  $\text{spec}(T) = \{0, 1\}$ . Identify  $T$ .

**Solution.**  $T$  is the orthogonal projection onto  $E_1(T)$ . Indeed the spectral resolution reads  $T = 0 \cdot P_1 + 1 \cdot P_2 = P_2 = P_{E_1(T)}$ . So the operators satisfying  $(*)$  with spectrum in  $\{0, 1\}$  are exactly the orthogonal projections, together with 0 and  $\text{id}_V$  for the degenerate spectra  $\{0\}$  and  $\{1\}$ .

## 11.8.5 Square roots

The function  $\lambda \mapsto \sqrt{\lambda}$  is defined on  $[0, \infty)$ , so [Definition 11.8.2](#) produces a square root of  $T$  as soon as the spectrum lies there. The interesting part is that the square root is **unique** once we ask it to be of the same kind, and the uniqueness proof is a direct application of [Theorem 11.8.2](#): a square root's own resolution, squared, has to be  $T$ 's.

Note first that over  $\mathbb{C}$  the hypothesis is stronger than it looks. If  $T$  satisfies  $(*)$  and  $\text{spec}(T) \subseteq \mathbb{R}$ , then  $f(\lambda) = \lambda$  is real-valued on the spectrum, so  $T = f(T)$  is self-adjoint by [Theorem 11.8.3](#) (c). A normal operator with real spectrum is self-adjoint.



### Theorem 11.8.4 — The Square Root of a Non-negative Normal Operator

Let  $V$  be a non-zero finite-dimensional inner product space over  $F$  and let  $T \in \mathcal{L}(V)$  satisfy  $(*)$  with  $\text{spec}(T) \subseteq [0, \infty)$ . Then there is **exactly one** operator  $S \in \mathcal{L}(V)$  such that

- (i)  $S$  satisfies  $(*)$ ;
- (ii)  $\text{spec}(S) \subseteq [0, \infty)$ ;
- (iii)  $S^2 = T$ .

It is  $S = \sum_i \sqrt{\lambda_i} P_i$ , where  $T = \sum_i \lambda_i P_i$  is the spectral resolution of  $T$ . This  $S$  is self-adjoint, and it commutes with every operator that commutes with  $T$ .

*Idea.* Existence is the functional calculus applied to  $\sqrt{\cdot}$ : squaring a function squares its values. Uniqueness is where the work is, and the trick is not to compare  $S$  with the candidate directly but to *square  $S$ 's own resolution*. If  $S = \sum_j \mu_j Q_j$ , then  $S^2 = \sum_j \mu_j^2 Q_j$ , and the  $\mu_j^2$  are still distinct because the  $\mu_j$  are distinct **and non-negative**. So  $\sum_j \mu_j^2 Q_j$  meets every condition of Theorem 11.8.2 for  $T$ , which forces it to be  $T$ 's spectral resolution term by term.

*Proof. Existence.* Let  $f(\lambda) = \sqrt{\lambda}$  on  $\text{spec}(T) \subseteq [0, \infty)$ , which takes values in  $[0, \infty) \subseteq F$ , and put  $S = f(T) = \sum_i \sqrt{\lambda_i} P_i$ . Then  $f^2$  is the function  $\lambda \mapsto \lambda$ , so by Theorem 11.8.3 (b) and (a),

$$S^2 = f(T)f(T) = (f^2)(T) = T,$$

giving (iii). By Theorem 11.8.3 (d),  $S$  satisfies  $(*)$  and  $\text{spec}(S) = \{\sqrt{\lambda_1}, \dots, \sqrt{\lambda_k}\} \subseteq [0, \infty)$ , giving (i) and (ii). Since  $f$  is real-valued,  $S$  is self-adjoint by Theorem 11.8.3 (c), and it commutes with everything commuting with  $T$  by (e) of the same theorem.

*Uniqueness.* Let  $S$  satisfy (i), (ii) and (iii), and let

$$S = \mu_1 Q_1 + \dots + \mu_l Q_l$$

be its spectral resolution (Theorem 11.8.1): the  $\mu_j$  are the distinct eigenvalues of  $S$ , and the  $Q_j$  are non-zero self-adjoint idempotents with  $Q_j Q_{j'} = 0$  for  $j \neq j'$  and  $\sum_j Q_j = \text{id}_V$ . By Theorem 11.8.3 (a) applied to  $S$  and  $p = x^2$ ,

$$T = S^2 = \sum_{j=1}^l \mu_j^2 Q_j.$$

The scalars  $\mu_j^2$  are **distinct**: the  $\mu_j$  are distinct by construction, and non-negative by (ii), so  $\mu_j^2 = \mu_{j'}^2$  would force  $\mu_j = \mu_{j'}$ . The family  $(Q_1, \dots, Q_l)$  satisfies (R1), (R2) and (R3) of Theorem 11.8.2, and every  $Q_j$  is non-zero. That theorem therefore identifies the displayed expression as the spectral resolution of  $T$ : after renumbering,  $l = k$ ,  $\mu_j^2 = \lambda_j$  and  $Q_j = P_j$  for every  $j$ . Since  $\mu_j \geq 0$  and  $\mu_j^2 = \lambda_j$ , we get  $\mu_j = \sqrt{\lambda_j}$ , and hence

$$S = \sum_{j=1}^k \sqrt{\lambda_j} P_j,$$

which is the operator constructed above. This proves the theorem. ■

The square root is written  $\sqrt{T}$ , or  $T^{1/2}$ . Chapter 12 takes the hypothesis  $\text{spec}(T) \subseteq [0, \infty)$  as the definition of a **positive** operator, proves that it is equivalent to  $\langle T\mathbf{v}, \mathbf{v} \rangle \geq 0$  for all  $\mathbf{v}$ , and makes this theorem the tool behind polar decomposition and the singular value decomposition; the notation  $\mathbf{A}^{1/2}$  in the notation table refers to exactly the operator built here.

### Example. (A Square Root, and Three Impostors)

Let  $\mathbf{A} = \begin{pmatrix} 10 & 6 \\ 6 & 10 \end{pmatrix} \in M_2(\mathbb{R})$ . Find its spectral resolution, compute  $\sqrt{\mathbf{A}}$  and  $\mathbf{A}^m$ , and find a symmetric square root of  $\mathbf{A}$  other than  $\sqrt{\mathbf{A}}$ .

*Solution. The resolution.*  $\mathbf{A}$  is symmetric, hence self-adjoint on  $\mathbb{R}^2$ . Since  $\mathbf{A}(1, 1) = (16, 16)$  and  $\mathbf{A}(1, -1) = (4, -4)$ , the eigenvalues are 16 and 4, with eigenlines  $\text{span}((1, 1))$  and  $\text{span}((1, -1))$ . Both spanning vectors have squared length 2, so

$$\mathbf{P}_1 = \frac{1}{2} \begin{pmatrix} 1 & 1 \\ 1 & 1 \end{pmatrix}, \quad \mathbf{P}_2 = \frac{1}{2} \begin{pmatrix} 1 & -1 \\ -1 & 1 \end{pmatrix}, \quad \mathbf{A} = 16\mathbf{P}_1 + 4\mathbf{P}_2.$$

*The square root.* Both eigenvalues are positive, so [Theorem 11.8.4](#) applies:

$$\sqrt{\mathbf{A}} = 4\mathbf{P}_1 + 2\mathbf{P}_2 = \begin{pmatrix} 2 & 2 \\ 2 & 2 \end{pmatrix} + \begin{pmatrix} 1 & -1 \\ -1 & 1 \end{pmatrix} = \begin{pmatrix} 3 & 1 \\ 1 & 3 \end{pmatrix},$$

and indeed  $\begin{pmatrix} 3 & 1 \\ 1 & 3 \end{pmatrix}^2 = \begin{pmatrix} 10 & 6 \\ 6 & 10 \end{pmatrix}$ .

*Powers.* By [Theorem 11.8.3](#) (a),  $\mathbf{A}^m = 16^m\mathbf{P}_1 + 4^m\mathbf{P}_2$ , that is

$$\mathbf{A}^m = \frac{1}{2} \begin{pmatrix} 16^m + 4^m & 16^m - 4^m \\ 16^m - 4^m & 16^m + 4^m \end{pmatrix}.$$

For  $m = 3$  this reads  $\begin{pmatrix} 2080 & 2016 \\ 2016 & 2080 \end{pmatrix}$ , which is  $\mathbf{A}^3$ .

*An impostor.* Changing one sign gives  $\mathbf{B} = 4\mathbf{P}_1 - 2\mathbf{P}_2 = \begin{pmatrix} 1 & 3 \\ 3 & 1 \end{pmatrix}$ , and  $\mathbf{B}^2 = \mathbf{A}$  as well.

So  $\mathbf{A}$  has more than one symmetric square root;  $\mathbf{B}$  fails clause (ii) of [Theorem 11.8.4](#), since  $\text{spec}(\mathbf{B}) = \{4, -2\}$ . Flipping the other sign, and both, gives  $-\mathbf{B}$  and  $-\sqrt{\mathbf{A}}$ : four symmetric square roots, and exactly one of them has non-negative spectrum.

*Warning.* “**A square root**” and “**the square root**” are different things. Uniqueness in [Theorem 11.8.4](#) needs all three clauses. Dropping (ii) admits the three sign-flipped roots of [Example \(A Square Root, and Three Impostors\)](#). Dropping (i) admits more still, and the cheapest witness is the zero operator on  $\mathbb{C}^2$ . Take

$$\mathbf{S} = \begin{pmatrix} 0 & 1 \\ 0 & 0 \end{pmatrix}, \quad \mathbf{S}^2 = \mathbf{0}.$$

Here  $\text{spec}(\mathbf{S}) = \{0\} \subseteq [0, \infty)$ , so (ii) holds, and (iii) holds by the display, yet  $\mathbf{S} \neq \mathbf{0} = \sqrt{0}$ . Only (i) fails:  $\mathbf{S}^*\mathbf{S} = \text{diag}(0, 1)$  while  $\mathbf{S}\mathbf{S}^* = \text{diag}(1, 0)$ . The phrase “the square root of  $T$ ” always means the one satisfying all three.

## 11.8.6 The exponential and the unitary group

Chapter 9 defined  $e^{\mathbf{A}}$  for every  $\mathbf{A} \in M_n(\mathbb{C})$  as the sum of the series  $\sum_m \mathbf{A}^m/m!$  (Definition 9.9.2) and proved that the series converges (Theorem 9.9.3). We do not redevelop any of that. What the spectral resolution adds is that for a normal matrix the series can be summed by inspection, and that the answer identifies the unitary matrices as the exponentials of the Hermitian ones — the matrix version of  $|e^{i\theta}| = 1$ .

### Theorem 11.8.5 — Unitary Matrices Are Exponentials of Hermitian Matrices

Let  $n \geq 1$ .

- (a) If  $\mathbf{A} \in M_n(\mathbb{C})$  is normal with spectral resolution  $\mathbf{A} = \sum_i \lambda_i \mathbf{P}_i$ , then  $e^{\mathbf{A}} = \sum_i e^{\lambda_i} \mathbf{P}_i$ .
- (b) If  $\mathbf{H} \in M_n(\mathbb{C})$  is Hermitian, then  $e^{i\mathbf{H}} \in \text{U}(n)$ , and  $\det e^{i\mathbf{H}} = e^{i \text{tr } \mathbf{H}}$ .
- (c) Conversely, every  $\mathbf{U} \in \text{U}(n)$  equals  $e^{i\mathbf{H}}$  for some Hermitian  $\mathbf{H}$ , which may be chosen with  $\text{spec}(\mathbf{H}) \subseteq [0, 2\pi)$ .

*Idea.* For (a), the partial sums of the series collapse:  $\mathbf{A}^m = \sum_i \lambda_i^m \mathbf{P}_i$ , so each partial sum is a fixed combination of the  $k$  scalar partial sums of  $\sum_m \lambda_i^m/m!$ , whose limits are the numbers  $e^{\lambda_i}$ . For (b) and (c), the resolution turns everything into a statement about  $k$  complex numbers:  $\mathbf{H}$  Hermitian means real eigenvalues,  $e^{i\mathbf{H}}$  then has eigenvalues on the unit circle, and a unitary matrix has eigenvalues on the unit circle, each of which is  $e^{i\theta}$  for a unique  $\theta \in [0, 2\pi)$ . Reading the last sentence backwards constructs  $\mathbf{H}$ .

*Proof.*

- (a) By Theorem 11.8.3 (a),  $\mathbf{A}^m = \sum_i \lambda_i^m \mathbf{P}_i$  for every  $m \geq 0$ , so for every  $N \geq 0$ ,

$$\sum_{m=0}^N \frac{\mathbf{A}^m}{m!} = \sum_{i=1}^k \left( \sum_{m=0}^N \frac{\lambda_i^m}{m!} \right) \mathbf{P}_i.$$

Fix a position  $(r, s)$ . Its entry on the right is  $\sum_i c_{N,i} (\mathbf{P}_i)_{rs}$ , where  $c_{N,i} = \sum_{m \leq N} \lambda_i^m/m!$  and the coefficients  $(\mathbf{P}_i)_{rs}$  do not depend on  $N$ . Each  $c_{N,i} \rightarrow e^{\lambda_i}$  as  $N \rightarrow \infty$  by the scalar exponential series, so the entry converges to  $\sum_i e^{\lambda_i} (\mathbf{P}_i)_{rs}$ . Hence the series converges entrywise to  $\sum_i e^{\lambda_i} \mathbf{P}_i$ , and by Definition 9.9.2 this matrix is  $e^{\mathbf{A}}$ .

- (b)  $\mathbf{H}$  is normal, so it has a spectral resolution  $\mathbf{H} = \sum_j \theta_j \mathbf{P}_j$ , and its eigenvalues  $\theta_j$  are real by Theorem 11.2.1. The scalars  $i\theta_j$  are distinct, so  $i\mathbf{H} = \sum_j i\theta_j \mathbf{P}_j$  is the spectral resolution of  $i\mathbf{H}$  by Theorem 11.8.2, and (a) gives

$$e^{i\mathbf{H}} = \sum_j e^{i\theta_j} \mathbf{P}_j.$$

Each  $\theta_j$  is real, so  $\overline{e^{i\theta_j}} e^{i\theta_j} = |e^{i\theta_j}|^2 = 1$ . Hence, using  $\mathbf{P}_j^* = \mathbf{P}_j$  and  $\mathbf{P}_j \mathbf{P}_{j'} = \delta_{jj'} \mathbf{P}_j$  from [Theorem 11.8.1](#) (b),

$$(e^{i\mathbf{H}})^* e^{i\mathbf{H}} = \sum_j \overline{e^{i\theta_j}} e^{i\theta_j} \mathbf{P}_j = \sum_j \mathbf{P}_j = \mathbf{I},$$

so  $e^{i\mathbf{H}} \in \mathbf{U}(n)$  by [Definition 10.7.3](#). For the determinant, [Theorem 9.9.4](#) (e) gives  $\det e^{i\mathbf{H}} = e^{\operatorname{tr}(i\mathbf{H})} = e^{i \operatorname{tr} \mathbf{H}}$ .

- (c) Let  $\mathbf{U} \in \mathbf{U}(n)$ . Then  $\mathbf{U}^* \mathbf{U} = \mathbf{U} \mathbf{U}^* = \mathbf{I}$ , so  $\mathbf{U}$  is normal; let  $\mathbf{U} = \sum_j \nu_j \mathbf{P}_j$  be its spectral resolution. Every eigenvalue has  $|\nu_j| = 1$ : if  $\mathbf{U}\mathbf{x} = \nu_j \mathbf{x}$  with  $\mathbf{x} \neq \mathbf{0}$ , then  $\|\mathbf{x}\| = \|\mathbf{U}\mathbf{x}\| = |\nu_j| \|\mathbf{x}\|$  because  $\mathbf{U}$  is an isometry ([Theorem 10.7.3](#)), and  $\|\mathbf{x}\| \neq 0$ . So there is a unique  $\theta_j \in [0, 2\pi)$  with  $\nu_j = e^{i\theta_j}$ , and the  $\theta_j$  are distinct because the  $\nu_j$  are. Put

$$\mathbf{H} := \sum_j \theta_j \mathbf{P}_j.$$

The  $\theta_j$  are real, so  $\mathbf{H}^* = \sum_j \theta_j \mathbf{P}_j^* = \mathbf{H}$  by [Theorem 10.6.2](#) (a), (b) and [Theorem 11.8.1](#) (b):  $\mathbf{H}$  is Hermitian, with  $\operatorname{spec}(\mathbf{H}) = \{\theta_j\} \subseteq [0, 2\pi)$  by [Theorem 11.8.2](#). By (a) applied to  $i\mathbf{H}$ ,

$$e^{i\mathbf{H}} = \sum_j e^{i\theta_j} \mathbf{P}_j = \sum_j \nu_j \mathbf{P}_j = \mathbf{U}.$$

This proves the theorem. ■

Part (c) says that the exponential map from Hermitian matrices to  $\mathbf{U}(n)$  is **onto**. One consequence belongs to topology rather than algebra, so we only record it: the path  $t \mapsto e^{it\mathbf{H}}$ ,  $t \in [0, 1]$ , stays inside  $\mathbf{U}(n)$  and runs from  $\mathbf{I}$  to  $\mathbf{U}$ , so every unitary matrix can be deformed continuously to the identity. Nothing of the sort holds in  $\mathbf{O}(n)$ , where the determinant is  $\pm 1$  and cannot change sign along a path.

### Example. (Exponentiating a Hermitian Matrix, and Undoing It)

Let  $\mathbf{H} = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}$ . Compute  $e^{i\theta\mathbf{H}}$  for  $\theta \in \mathbb{R}$ . Then find a Hermitian  $\mathbf{K}$  with  $e^{i\mathbf{K}} = \mathbf{Q}$ , where  $\mathbf{Q} = \frac{1}{\sqrt{2}} \begin{pmatrix} 1 & -1 \\ 1 & 1 \end{pmatrix}$ .

*Solution. Forwards.*  $\mathbf{H}$  is symmetric with  $\mathbf{H}(1, 1) = (1, 1)$  and  $\mathbf{H}(1, -1) = -(1, -1)$ , so its eigenvalues are  $\pm 1$  and

$$\mathbf{H} = \mathbf{P}_+ - \mathbf{P}_-, \quad \mathbf{P}_\pm = \frac{1}{2} \begin{pmatrix} 1 & \pm 1 \\ \pm 1 & 1 \end{pmatrix}.$$

Hence  $i\theta\mathbf{H}$  has resolution  $i\theta\mathbf{P}_+ - i\theta\mathbf{P}_-$  (for  $\theta \neq 0$ ), and [Theorem 11.8.5](#) (a) gives

$$\begin{aligned} e^{i\theta\mathbf{H}} &= e^{i\theta\mathbf{P}_+} + e^{-i\theta\mathbf{P}_-} \\ &= \begin{pmatrix} \cos \theta & i \sin \theta \\ i \sin \theta & \cos \theta \end{pmatrix}, \end{aligned}$$

using  $e^{\pm i\theta} = \cos \theta \pm i \sin \theta$ . Each row has squared length  $\cos^2 \theta + \sin^2 \theta = 1$  and the two rows are orthogonal, confirming that the result is unitary.

*Backwards.*  $\mathbf{Q}$  is the rotation by  $\pi/4$ , hence unitary, with  $\mathbf{Q}(1, -i) = e^{i\pi/4}(1, -i)$  and  $\mathbf{Q}(1, i) = e^{-i\pi/4}(1, i)$ . Both eigenvectors have squared length 2, so the spectral projections are

$$\mathbf{R}_+ = \frac{1}{2} \begin{pmatrix} 1 & i \\ -i & 1 \end{pmatrix}, \quad \mathbf{R}_- = \frac{1}{2} \begin{pmatrix} 1 & -i \\ i & 1 \end{pmatrix}.$$

Writing  $e^{-i\pi/4} = e^{i7\pi/4}$  to land in  $[0, 2\pi)$ , the recipe of [Theorem 11.8.5](#) (c) gives

$$\mathbf{K} = \frac{\pi}{4} \mathbf{R}_+ + \frac{7\pi}{4} \mathbf{R}_- = \begin{pmatrix} \pi & -\frac{3\pi i}{4} \\ \frac{3\pi i}{4} & \pi \end{pmatrix},$$

Hermitian with spectrum  $\{\pi/4, 7\pi/4\}$ . Leaving  $[0, 2\pi)$ , the smaller choice  $\frac{\pi}{4}(\mathbf{R}_+ - \mathbf{R}_-) = \frac{\pi}{4} \begin{pmatrix} 0 & i \\ -i & 0 \end{pmatrix}$  works just as well: the Hermitian logarithm is far from unique.

## 11.8.7 Exercises

### 11.8.7.1 A. Check your understanding

#### Exercise A1.

- Define the spectral resolution of  $T$ , and say which operators have one over  $\mathbb{C}$  and which over  $\mathbb{R}$ .
- List the properties of the family  $(P_1, \dots, P_k)$  given by [Theorem 11.8.1](#), and say how many terms the resolution of a self-adjoint  $T \in \mathcal{L}(\mathbb{R}^5)$  with eigenvalues 2, 2, 2, 7, 7 has.
- True or false: if  $T = \mu_1 Q_1 + \mu_2 Q_2$  with  $\mu_1 \neq \mu_2$  and  $Q_1, Q_2$  non-zero operators satisfying  $Q_i^2 = Q_i$ ,  $Q_1 Q_2 = Q_2 Q_1 = 0$  and  $Q_1 + Q_2 = \text{id}_V$ , then  $T$  is normal. Justify your answer.
- Define  $f(T)$  for a function  $f$  on  $\text{spec}(T)$ , and say exactly how much information about  $f$  it uses.
- State [Theorem 11.8.3](#) (a) and explain in one sentence why the rest of the section would be unusable without it.
- Give two different self-adjoint square roots of  $\text{id}_V$  on  $\mathbb{R}^2$ , and say which one [Theorem 11.8.4](#) singles out and why.

*Solution.*

- For  $T$  with an orthonormal basis of eigenvectors, distinct eigenvalues  $\lambda_1, \dots, \lambda_k$  and  $P_i = P_{E_{\lambda_i}(T)}$  the orthogonal projection onto the  $i$ -th eigenspace, the spectral resolution is  $T = \sum_i \lambda_i P_i$  ([Definition 11.8.1](#)). Over  $\mathbb{C}$  the operators with one are exactly the normal operators ([Theorem 11.4.2](#)); over  $\mathbb{R}$ , exactly the self-adjoint ones ([Theorem 11.5.1](#)).
- Each  $P_i$  is self-adjoint and idempotent,  $P_i P_j = 0$  for  $i \neq j$ ,  $\sum_i P_i = \text{id}_V$ , and  $T P_i = P_i T = \lambda_i P_i$ . The resolution has **two** terms,  $T = 2P_1 + 7P_2$ , with  $\text{rank } P_1 = 3$  and  $\text{rank } P_2 = 2$ : the resolution counts distinct eigenvalues, not multiplicities.

- (c) False. Self-adjointness of the  $Q_i$  is missing, and it is the one hypothesis of [Theorem 11.8.2](#) that does the work. The matrices  $\pi_1, \pi_2$  of [Example \(Two Projections That Are Not Orthogonal\)](#) satisfy everything listed, with  $\mu_1 = 1$  and  $\mu_2 = 2$ , and  $\pi_1 + 2\pi_2 = \begin{pmatrix} 1 & 1 \\ 0 & 2 \end{pmatrix}$ , which is not normal.
- (d)  $f(T) = \sum_i f(\lambda_i)P_i$  ([Definition 11.8.2](#)). It uses exactly the  $k$  values  $f(\lambda_1), \dots, f(\lambda_k)$ ; two functions agreeing on  $\text{spec}(T)$  give the same operator, and  $f$  need not be defined, let alone continuous, anywhere else.
- (e) For  $p \in F[x]$ , the operator  $\sum_i p(\lambda_i)P_i$  coincides with the operator  $p(T)$  of [Definition 3.4.2](#). Without it, the symbol  $p(T)$  would carry two unrelated meanings at once, and no computation mixing polynomials with other functions — for instance  $(\sqrt{T})^2 = T$  — could be trusted.
- (f)  $\text{id}$  and the reflection  $T_{\mathbf{D}}$  with  $\mathbf{D} = \text{diag}(1, -1)$ , both self-adjoint with square  $\text{id}$ . [Theorem 11.8.4](#) singles out  $\text{id}$ , because  $\text{spec}(\text{id}) = \{1\} \subseteq [0, \infty)$  while  $\text{spec}(T_{\mathbf{D}}) = \{1, -1\}$  is not. (A third root is  $-\text{id}$ , and a fourth is  $T_{\mathbf{D}'}$  with  $\mathbf{D}' = \text{diag}(-1, 1)$ .)

### 11.8.7.2 B. Practice

**Exercise B1:** Write these as  $\sum \lambda_i \mathbf{P}_i$ .

Find the spectral resolution of each of the following matrices, and check in each case that the projections are self-adjoint, idempotent, mutually annihilating and sum to  $\mathbf{I}$ .

- (a)  $\mathbf{A} = \begin{pmatrix} 2 & 1-i \\ 1+i & 3 \end{pmatrix} \in M_2(\mathbb{C})$ .
- (b)  $\mathbf{S} = \begin{pmatrix} 2 & 0 & 1 \\ 0 & 3 & 0 \\ 1 & 0 & 2 \end{pmatrix} \in M_3(\mathbb{R})$ .
- (c)  $\mathbf{U} = \frac{1}{\sqrt{2}} \begin{pmatrix} 1 & -1 \\ 1 & 1 \end{pmatrix} \in M_2(\mathbb{C})$ .

*Solution.*

- (a)  $\mathbf{A}^* = \mathbf{A}$ ,  $\text{tr } \mathbf{A} = 5$  and  $\det \mathbf{A} = 6 - (1-i)(1+i) = 4$ , so  $p_{\mathbf{A}} = x^2 - 5x + 4$  and  $\text{spec}(\mathbf{A}) = \{1, 4\}$ . From  $\mathbf{A} - 4\mathbf{I} = \begin{pmatrix} -2 & 1-i \\ 1+i & -1 \end{pmatrix}$  the first row gives the eigenvector  $\mathbf{v}_4 = (1-i, 2)$ , with  $\|\mathbf{v}_4\|^2 = 2 + 4 = 6$ ; from  $\mathbf{A} - \mathbf{I}$  likewise  $\mathbf{v}_1 = (-1+i, 1)$ , with  $\|\mathbf{v}_1\|^2 = 3$ . Hence

$$\mathbf{P}_4 = \frac{\mathbf{v}_4 \mathbf{v}_4^*}{6} = \frac{1}{3} \begin{pmatrix} 1 & 1-i \\ 1+i & 2 \end{pmatrix}, \quad \mathbf{P}_1 = \frac{1}{3} \begin{pmatrix} 2 & -1+i \\ -1-i & 1 \end{pmatrix},$$

and  $\mathbf{A} = 1 \cdot \mathbf{P}_1 + 4 \cdot \mathbf{P}_4$ . Both are Hermitian, both square to themselves,  $\mathbf{P}_1 \mathbf{P}_4 = 0$ , and  $\mathbf{P}_1 + \mathbf{P}_4 = \mathbf{I}$ .

- (b)  $\mathbf{S}$  is symmetric, and  $\mathbf{S}(1, 0, 1) = 3(1, 0, 1)$ ,  $\mathbf{S}(0, 1, 0) = 3(0, 1, 0)$ ,  $\mathbf{S}(1, 0, -1) = (1, 0, -1)$ , so  $\text{spec}(\mathbf{S}) = \{3, 1\}$  with  $E_1 = \text{span}((1, 0, -1))$  and  $E_3 = \text{span}((1, 0, 1), (0, 1, 0))$ . Hence

$$\mathbf{P}_1 = \frac{1}{2} \begin{pmatrix} 1 & 0 & -1 \\ 0 & 0 & 0 \\ -1 & 0 & 1 \end{pmatrix}, \quad \mathbf{P}_3 = \mathbf{I} - \mathbf{P}_1 = \frac{1}{2} \begin{pmatrix} 1 & 0 & 1 \\ 0 & 2 & 0 \\ 1 & 0 & 1 \end{pmatrix},$$

and  $\mathbf{S} = \mathbf{P}_1 + 3\mathbf{P}_3$ . The stated properties follow from  $\mathbf{P}_1^\top = \mathbf{P}_1$ ,  $\mathbf{P}_1^2 = \mathbf{P}_1$  and  $\mathbf{P}_3 = \mathbf{I} - \mathbf{P}_1$ .

- (c)  $\mathbf{U}$  is real orthogonal, hence unitary and normal. Since  $\mathbf{U}(1, -i) = \frac{1}{\sqrt{2}}(1+i, 1-i) = e^{i\pi/4}(1, -i)$ , and similarly  $\mathbf{U}(1, i) = e^{-i\pi/4}(1, i)$ , we get  $\text{spec}(\mathbf{U}) = \{e^{\pm i\pi/4}\}$ . Both eigenvectors have squared length 2, so with  $\mathbf{v}_\pm = (1, \mp i)$ ,

$$\mathbf{P}_\pm = \frac{\mathbf{v}_\pm \mathbf{v}_\pm^*}{2} = \frac{1}{2} \begin{pmatrix} 1 & \pm i \\ \mp i & 1 \end{pmatrix}, \quad \mathbf{U} = e^{i\pi/4} \mathbf{P}_+ + e^{-i\pi/4} \mathbf{P}_-.$$

Both are Hermitian and idempotent,  $\mathbf{P}_+ \mathbf{P}_- = 0$  and  $\mathbf{P}_+ + \mathbf{P}_- = \mathbf{I}$ .

### Exercise B2: Computing through the resolution.

Let  $\mathbf{C} = \begin{pmatrix} 13 & 12 \\ 12 & 13 \end{pmatrix} \in M_2(\mathbb{R})$ .

- Find the spectral resolution of  $\mathbf{C}$ .
- Hence compute  $\sqrt{\mathbf{C}}$ , and verify by squaring.
- Hence give a closed formula for  $\mathbf{C}^m$  and write down  $\mathbf{C}^{100}$ .
- Hence compute  $e^{\mathbf{C}}$ .

*Solution.*

- (a)  $\mathbf{C}(1, 1) = 25(1, 1)$  and  $\mathbf{C}(1, -1) = (1, -1)$ , so  $\text{spec}(\mathbf{C}) = \{25, 1\}$  and, both spanning vectors having squared length 2,

$$\mathbf{P}_1 = \frac{1}{2} \begin{pmatrix} 1 & 1 \\ 1 & 1 \end{pmatrix}, \quad \mathbf{P}_2 = \frac{1}{2} \begin{pmatrix} 1 & -1 \\ -1 & 1 \end{pmatrix}, \quad \mathbf{C} = 25\mathbf{P}_1 + \mathbf{P}_2.$$

- (b) Both eigenvalues are positive, so by [Theorem 11.8.4](#),

$$\sqrt{\mathbf{C}} = 5\mathbf{P}_1 + 1 \cdot \mathbf{P}_2 = \begin{pmatrix} 3 & 2 \\ 2 & 3 \end{pmatrix},$$

$$\text{and } \begin{pmatrix} 3 & 2 \\ 2 & 3 \end{pmatrix}^2 = \begin{pmatrix} 13 & 12 \\ 12 & 13 \end{pmatrix} = \mathbf{C}.$$

- (c) By [Theorem 11.8.3](#) (a),  $\mathbf{C}^m = 25^m \mathbf{P}_1 + 1^m \mathbf{P}_2$ , that is

$$\mathbf{C}^m = \frac{1}{2} \begin{pmatrix} 25^m + 1 & 25^m - 1 \\ 25^m - 1 & 25^m + 1 \end{pmatrix},$$

so  $\mathbf{C}^{100} = \frac{1}{2} \begin{pmatrix} 25^{100} + 1 & 25^{100} - 1 \\ 25^{100} - 1 & 25^{100} + 1 \end{pmatrix}$ . (As a check,  $m = 2$  gives  $\frac{1}{2} \begin{pmatrix} 626 & 624 \\ 624 & 626 \end{pmatrix} = \begin{pmatrix} 313 & 312 \\ 312 & 313 \end{pmatrix}$ , which is  $\mathbf{C}^2$ .)

- (d) By [Theorem 11.8.5](#) (a),

$$e^{\mathbf{C}} = e^{25} \mathbf{P}_1 + e \mathbf{P}_2 = \frac{1}{2} \begin{pmatrix} e^{25} + e & e^{25} - e \\ e^{25} - e & e^{25} + e \end{pmatrix}.$$

**Exercise B3: Is it a spectral resolution?.**

Determine which of the following displays the spectral resolution of the operator  $T_{\mathbf{A}}$  on the stated space. Justify your answer; for those that do not, name the condition of [Theorem 11.8.2](#) that fails and say what the correct resolution is.

- (a) On  $\mathbb{R}^2$ :  $\mathbf{A} = \begin{pmatrix} 2 & 0 \\ 0 & -1 \end{pmatrix}$ , with  $\mu_1 = 2$ ,  $\mathbf{Q}_1 = \begin{pmatrix} 1 & 0 \\ 0 & 0 \end{pmatrix}$ ,  $\mu_2 = -1$ ,  $\mathbf{Q}_2 = \begin{pmatrix} 0 & 0 \\ 0 & 1 \end{pmatrix}$ .
- (b) On  $\mathbb{R}^2$ :  $\mathbf{A} = 3\mathbf{I}$ , with  $\mu_1 = \mu_2 = 3$ ,  $\mathbf{Q}_1 = \frac{1}{2} \begin{pmatrix} 1 & 1 \\ 1 & 1 \end{pmatrix}$ ,  $\mathbf{Q}_2 = \frac{1}{2} \begin{pmatrix} 1 & -1 \\ -1 & 1 \end{pmatrix}$ .
- (c) On  $\mathbb{R}^2$ :  $\mathbf{A} = \begin{pmatrix} 3 & 1 \\ 0 & 1 \end{pmatrix}$ , with  $\mu_1 = 3$ ,  $\mathbf{Q}_1 = \frac{1}{2} \begin{pmatrix} 2 & 1 \\ 0 & 0 \end{pmatrix}$ ,  $\mu_2 = 1$ ,  $\mathbf{Q}_2 = \frac{1}{2} \begin{pmatrix} 0 & -1 \\ 0 & 2 \end{pmatrix}$ .

*Solution.*

- (a) Yes. The two matrices are symmetric, idempotent, non-zero, annihilate each other and sum to  $\mathbf{I}$ , and  $2\mathbf{Q}_1 - \mathbf{Q}_2 = \mathbf{A}$ ; the coefficients 2 and  $-1$  are distinct. By [Theorem 11.8.2](#) this is the spectral resolution, and indeed  $E_2 = \text{span}(\mathbf{e}_1)$ ,  $E_{-1} = \text{span}(\mathbf{e}_2)$ .
- (b) No: the scalars  $\mu_1, \mu_2$  are not **distinct**, so the hypothesis of [Theorem 11.8.2](#) fails at once. The identity  $3\mathbf{Q}_1 + 3\mathbf{Q}_2 = 3\mathbf{I} = \mathbf{A}$  is true but uninformative — the eigenvalue 3 has been split artificially across two projections. The spectral resolution of  $3\mathbf{I}$  has a single term,  $3\mathbf{I} = 3 \cdot \text{id}$ , since  $\text{spec}(3\mathbf{I}) = \{3\}$  and  $E_3 = \mathbb{R}^2$ .
- (c) No:  $\mathbf{Q}_1$  is not self-adjoint, since  $\mathbf{Q}_1^\top = \frac{1}{2} \begin{pmatrix} 2 & 0 \\ 1 & 0 \end{pmatrix} \neq \mathbf{Q}_1$ , so (R1) fails. Everything else holds:  $\mathbf{Q}_1^2 = \mathbf{Q}_1$ ,  $\mathbf{Q}_2 = \mathbf{I} - \mathbf{Q}_1$ ,  $\mathbf{Q}_1\mathbf{Q}_2 = \mathbf{Q}_2\mathbf{Q}_1 = 0$  and  $3\mathbf{Q}_1 + \mathbf{Q}_2 = \mathbf{A}$ . There is no correct resolution to offer, because  $\mathbf{A}$  is not normal:  $\mathbf{A}^\top \mathbf{A} = \begin{pmatrix} 9 & 3 \\ 3 & 2 \end{pmatrix}$  while  $\mathbf{A}\mathbf{A}^\top = \begin{pmatrix} 10 & 1 \\ 1 & 1 \end{pmatrix}$ . The projections here are onto  $E_3 = \text{span}(\mathbf{e}_1)$  along  $E_1 = \text{span}((-1, 2))$ , two lines that are not perpendicular.

**11.8.7.3 C. Going deeper**
**Exercise C1: The projections are polynomials in the operator.**

Let  $T \in \mathcal{L}(V)$  satisfy (\*), with distinct eigenvalues  $\lambda_1, \dots, \lambda_k \in F$  and resolution of the identity  $(P_1, \dots, P_k)$ .

- (a) Prove that  $P_i = \ell_i(T)$ , where  $\ell_i \in F[x]$  is the  $i$ -th Lagrange basis polynomial for the nodes  $\lambda_1, \dots, \lambda_k$ .
- (b) Deduce that for every  $f: \text{spec}(T) \rightarrow F$  there is a polynomial  $p$  of degree less than  $k$  with  $f(T) = p(T)$ .
- (c) Deduce that every operator commuting with  $T$  commutes with each  $P_i$ , re-proving [Theorem 11.8.3](#) (e) by a different route.

*Hint for (a):* [Definition 5.6.1](#) produces a polynomial taking prescribed values at prescribed nodes.



*Solution.*

- (a) The  $\lambda_i$  are  $k$  distinct elements of  $F$ , so [Definition 5.6.1](#) gives  $\ell_1, \dots, \ell_k \in F[x]$  of degree  $k - 1$ , and [Theorem 5.6.1](#) (a) gives  $\ell_i(\lambda_j) = \delta_{ij}$ . By [Theorem 11.8.3](#) (a),

$$\ell_i(T) = \sum_{j=1}^k \ell_i(\lambda_j) P_j = \sum_{j=1}^k \delta_{ij} P_j = P_i.$$

- (b) Let  $p$  be the interpolating polynomial of degree less than  $k$  with  $p(\lambda_i) = f(\lambda_i)$  for every  $i$ , which exists by [Theorem 5.6.1](#). By [Theorem 11.8.3](#) (a),

$$p(T) = \sum_i p(\lambda_i) P_i = \sum_i f(\lambda_i) P_i = f(T).$$

Explicitly,  $p = \sum_i f(\lambda_i) \ell_i$ .

- (c) If  $ST = TS$ , then  $S$  commutes with every polynomial in  $T$  by [Theorem 5.7.1](#) (d), in particular with  $\ell_i(T) = P_i$ .

### Exercise C2: When the projections have rank one.

Let  $T \in \mathcal{L}(V)$  satisfy (\*), with  $\dim V = n$  and resolution of the identity  $(P_1, \dots, P_k)$ .

- Prove that  $\text{rank } P_i = \dim E_{\lambda_i}(T)$  and that  $\sum_i \text{rank } P_i = n$ .
- Deduce that every  $P_i$  has rank 1 if and only if  $T$  has  $n$  distinct eigenvalues.
- Suppose every  $P_i$  has rank 1. Prove that every  $S \in \mathcal{L}(V)$  commuting with  $T$  is a polynomial in  $T$ .

*Solution.*

- $\text{im } P_i = E_{\lambda_i}(T)$  by [Definition 10.3.2](#), so  $\text{rank } P_i = \dim E_{\lambda_i}(T)$ . By [Theorem 11.8.1](#) (a),  $V = E_1 \oplus \dots \oplus E_k$ , so the dimensions add to  $n$ .
- If every  $\text{rank } P_i = 1$ , then (a) gives  $k = n$ , so  $T$  has  $n$  distinct eigenvalues. Conversely, if  $T$  has  $n$  distinct eigenvalues, then  $k = n$  and (a) forces each of the  $n$  positive integers  $\text{rank } P_i$  to be 1.
- By (b),  $k = n$  and each  $E_i$  is a line, say  $E_i = \text{span}(\mathbf{e}_i)$  with  $(\mathbf{e}_1, \dots, \mathbf{e}_n)$  an orthonormal eigenbasis. Let  $ST = TS$ . As in the proof of [Theorem 11.8.3](#) (e),  $S$  maps  $E_i$  into itself, so  $S\mathbf{e}_i = c_i\mathbf{e}_i$  for some  $c_i \in F$ . Define  $f: \text{spec}(T) \rightarrow F$  by  $f(\lambda_i) = c_i$ . Then  $f(T)\mathbf{e}_i = c_i\mathbf{e}_i = S\mathbf{e}_i$  for every  $i$ , and two operators agreeing on a basis are equal, so  $S = f(T)$ . By [Exercise \(C1: The projections are polynomials in the operator\)](#) (b),  $f(T)$  is a polynomial in  $T$ .

### Exercise C3: What the functional calculus forgets.

Let  $T \in \mathcal{L}(V)$  satisfy (\*), with distinct eigenvalues  $\lambda_1, \dots, \lambda_k$ , and let  $f, g: \text{spec}(T) \rightarrow F$ .

- Prove that  $f(T) = 0$  if and only if  $f(\lambda_i) = 0$  for every  $i$ . Deduce that  $f(T) = g(T)$  if and only if  $f = g$ .
- Deduce that  $\{p(T) : p \in F[x]\} = \text{span}(P_1, \dots, P_k)$ , a subspace of  $\mathcal{L}(V)$  of dimen-

sion exactly  $k$ .

- (c) Deduce that  $m_T = (x - \lambda_1)(x - \lambda_2) \cdots (x - \lambda_k)$ , so that the minimal polynomial of an operator satisfying  $(*)$  has no repeated root.

*Solution.*

- (a)  $(\Leftarrow)$  is immediate from [Definition 11.8.2](#). For  $(\Rightarrow)$ , fix  $i$  and pick  $\mathbf{v} \in E_{\lambda_i}(T)$  with  $\mathbf{v} \neq \mathbf{0}$ , which is possible because  $\lambda_i$  is an eigenvalue. Then  $P_i \mathbf{v} = \mathbf{v}$  and  $P_j \mathbf{v} = \mathbf{0}$  for  $j \neq i$ , since  $E_i \subseteq \ker P_j$  ([Theorem 11.8.1](#) (a), (b)). Hence

$$\mathbf{0} = f(T)\mathbf{v} = \sum_j f(\lambda_j)P_j \mathbf{v} = f(\lambda_i)\mathbf{v},$$

and  $\mathbf{v} \neq \mathbf{0}$  forces  $f(\lambda_i) = 0$ . For the second claim, apply this to  $f - g$ , using [Theorem 11.8.3](#) (b).

- (b) By [Theorem 11.8.3](#) (a), every  $p(T)$  equals  $\sum_i p(\lambda_i)P_i$ , so the left-hand side is contained in  $\text{span}(P_1, \dots, P_k)$ ; conversely every  $\sum_i c_i P_i$  is  $f(T)$  for the function  $f(\lambda_i) = c_i$ , hence is a polynomial in  $T$  by [Exercise \(C1: The projections are polynomials in the operator\)](#) (b). The  $P_i$  are linearly independent: if  $\sum_i c_i P_i = 0$ , that operator is  $f(T)$  for  $f(\lambda_i) = c_i$ , so every  $c_i = 0$  by (a). Hence the dimension is  $k$ .
- (c) Put  $p = \prod_i (x - \lambda_i)$ , monic of degree  $k$ . Since  $p(\lambda_i) = 0$  for every  $i$ , part (a) gives  $p(T) = 0$ , so  $m_T \mid p$  by [Theorem 8.5.1](#). Conversely  $m_T(T) = 0$ , so  $m_T(\lambda_i) = 0$  for every  $i$  by (a), and hence each  $x - \lambda_i$  divides  $m_T$  ([Theorem 5.1.3](#) (b)). Distinct monic linear polynomials are coprime, since a common divisor would divide their difference, a non-zero constant; so  $p = \prod_i (x - \lambda_i)$  divides  $m_T$  by [Lemma 5.6.3](#) (b). Two monic polynomials dividing each other are equal, so  $m_T = p$ , a product of distinct linear factors.

## 11.9 Circulants and the Discrete Fourier Transform

The spectral theorem of Section 4 gives every normal matrix a unitary of its own, found by solving that matrix's eigenvalue problem. This section does something better for one particular family: it writes down a **single** unitary matrix, before any matrix of the family is named, and that one matrix diagonalizes all of them at once. The family is the circulants, the unitary is built from the  $n$ -th roots of unity, and the resulting dictionary between multiplying matrices and multiplying numbers is the discrete Fourier transform.

Two earlier pages pointed here. In [Example \(Which of these are normal?\)](#) (d) a  $3 \times 3$  matrix whose rows were cyclic shifts of one another turned out to be normal, with the row dot products and the column dot products agreeing entry for entry; [Exercise \(B2: A circulant, diagonalized\)](#) met the same phenomenon again and diagonalized one such matrix by hand. Both times the agreement was described and not explained. It is explained below, and the explanation is the first theorem of the section.

Throughout, the field is  $\mathbb{C}$ , and **rows, columns and vector entries are numbered**  $0, 1, \dots, n-1$ . That is a departure from the rest of the book, and it is worth it: every formula here involves

an index read modulo  $n$ , and modular arithmetic starts at 0.

## 11.9.1 Matrices whose rows walk around

Fix  $n \geq 1$  and let  $\mathbf{S} \in M_n(\mathbb{C})$  be the matrix with  $(\mathbf{S})_{jk} = 1$  when  $k \equiv j + 1 \pmod{n}$  and  $(\mathbf{S})_{jk} = 0$  otherwise: a 1 in each position  $(j, j + 1)$  for  $j \leq n - 2$ , one more in the corner position  $(n - 1, 0)$ , and zeros elsewhere. For  $n = 4$ ,

$$\mathbf{S} = \begin{pmatrix} 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \\ 1 & 0 & 0 & 0 \end{pmatrix}.$$

Applied to a vector,  $\mathbf{S}$  moves every entry one place up and sends the top entry around to the bottom:  $\mathbf{S}\mathbf{x} = (x_1, x_2, \dots, x_{n-1}, x_0)$ . We call  $\mathbf{S}$  the **cyclic shift**. Each row and each column of  $\mathbf{S}$  carries exactly one 1, so  $\mathbf{S}$  is a permutation matrix (Definition 2.6.2), and shifting  $n$  times returns every entry to where it started, so  $\mathbf{S}^n = \mathbf{I}$ .

Now think of  $\mathbf{S}$  as a building block. The powers  $\mathbf{I}, \mathbf{S}, \mathbf{S}^2, \dots, \mathbf{S}^{n-1}$  are the  $n$  distinct “rotations of the identity”, and a general linear combination of them is the object of this section.

*A circulant is a matrix that does not know where the list begins: shift the indices and nothing changes.*

### Definition 11.9.1 — Circulant matrix

Let  $n \geq 1$  and let  $\mathbf{c} = (c_0, c_1, \dots, c_{n-1}) \in \mathbb{C}^n$ . The **circulant** with first row  $\mathbf{c}$  is the matrix  $\mathbf{C} \in M_n(\mathbb{C})$  whose entries are

$$(\mathbf{C})_{jk} = c_{k-j} \quad (0 \leq j, k \leq n-1),$$

where the subscript  $k - j$  is read **modulo**  $n$ . The polynomial

$$p_{\mathbf{c}}(x) = c_0 + c_1x + c_2x^2 + \dots + c_{n-1}x^{n-1} \in \mathbb{C}[x]$$

is called the **symbol** of  $\mathbf{C}$ .

In words: the entry in a given position depends only on how far the column is to the right of the diagonal, counted cyclically. Written out,

$$\mathbf{C} = \begin{pmatrix} c_0 & c_1 & \cdots & c_{n-1} \\ c_{n-1} & c_0 & \cdots & c_{n-2} \\ \vdots & \vdots & \ddots & \vdots \\ c_1 & c_2 & \cdots & c_0 \end{pmatrix},$$

so each row is the row above it moved one step to the right, with the entry that falls off the right-hand end reappearing on the left. The columns behave the same way going down. Note that the data is one vector of length  $n$ , not  $n^2$  numbers: a circulant is a very small object wearing a large costume.

Three examples, simplest first. The vector  $\mathbf{c} = (1, 0, \dots, 0)$  gives  $\mathbf{C} = \mathbf{I}$ , and  $\mathbf{c} = (0, 1, 0, \dots, 0)$  gives  $\mathbf{C} = \mathbf{S}$ ; in both cases the definition is checked by reading off  $(\mathbf{C})_{jk} = c_{k-j}$ , which is 1

exactly when  $k = j$ , respectively  $k = j + 1$ , modulo  $n$ . The vector  $\mathbf{c} = (1, 1, \dots, 1)$  gives the all-ones matrix. The  $3 \times 3$  matrix of [Example \(Which of these are normal?\)](#) (d) is the circulant with first row  $(1, 2, 3)$ , and the matrix of [Exercise \(B2: A circulant, diagonalized\)](#) is the circulant with first row  $(1, 1, 0)$ .

Our running example for the whole section is the  $4 \times 4$  circulant with first row  $(1, 2, 3, 4)$ :

$$\mathbf{C} = \begin{pmatrix} 1 & 2 & 3 & 4 \\ 4 & 1 & 2 & 3 \\ 3 & 4 & 1 & 2 \\ 2 & 3 & 4 & 1 \end{pmatrix}, \quad p_{\mathbf{C}}(x) = 1 + 2x + 3x^2 + 4x^3.$$

A non-example by minimal change. Keep the first row  $(1, 2, 3, 4)$  but let the entries that fall off the end vanish instead of wrapping around, giving a matrix  $\mathbf{B}$  with rows  $(1, 2, 3, 4)$ ,  $(0, 1, 2, 3)$ ,  $(0, 0, 1, 2)$ ,  $(0, 0, 0, 1)$ . Every diagonal of  $\mathbf{B}$  is still constant, and its first row is still  $\mathbf{c}$ , but  $(\mathbf{B})_{10} = 0$  while the definition demands  $c_{0-1} = c_3 = 4$ . The clause that fails is the one that does all the work here: the subscript must be read modulo  $n$ . Matrices with constant diagonals and no wrap-around have their own name and their own theory, and the warning at the end of the section says why they do not fit into this one.

The reason a circulant is worth a name is that it is a polynomial in the shift.

### Theorem 11.9.1 — Circulants Are Exactly the Polynomials in the Shift

Let  $\mathbf{c} \in \mathbb{C}^n$  and let  $\mathbf{C}$  be the circulant with first row  $\mathbf{c}$ , with symbol  $p_{\mathbf{C}}$ . Then:

- (a)  $\mathbf{C} = p_{\mathbf{C}}(\mathbf{S})$ ; conversely,  $q(\mathbf{S})$  is a circulant for every  $q \in \mathbb{C}[x]$ .
- (b)  $\mathbf{C}^*$  is the circulant with first row  $(\overline{c_0}, \overline{c_{n-1}}, \overline{c_{n-2}}, \dots, \overline{c_1})$ .
- (c)  $\mathbf{C}$  is normal, and any two circulants of the same size commute.

*Idea.* For (a), read off the entries of  $\mathbf{S}^m$ : it has a 1 in position  $(j, k)$  exactly when  $k = j + m$  modulo  $n$ . So  $\mathbf{S}^m$  is the circulant that picks out the coefficient  $c_m$ , and summing over  $m$  rebuilds  $\mathbf{C}$  one coefficient at a time. Once (a) is in hand, (b) and (c) are arithmetic with the single relation  $\mathbf{S}^* = \mathbf{S}^{n-1}$ : the adjoint of a polynomial in  $\mathbf{S}$  is again a polynomial in  $\mathbf{S}$ , and two polynomials in one matrix always commute ([Theorem 0.9.9](#)).

*Proof.*

- (a) We claim that  $(\mathbf{S}^m)_{jk} = 1$  if  $k \equiv j + m \pmod{n}$  and  $(\mathbf{S}^m)_{jk} = 0$  otherwise, for every  $m \geq 0$ . For  $m = 0$  this says  $\mathbf{S}^0 = \mathbf{I}$ . Assuming it for  $m$ , expand one product:  $(\mathbf{S}^{m+1})_{jk} = \sum_i (\mathbf{S})_{ji} (\mathbf{S}^m)_{ik}$ , and the only non-zero factor  $(\mathbf{S})_{ji}$  sits at  $i \equiv j + 1$ , so the sum is  $(\mathbf{S}^m)_{j+1,k}$ , which is 1 exactly when  $k \equiv j + 1 + m$ . This proves the claim by induction. Hence, for each  $j$  and  $k$ ,

$$(p_{\mathbf{C}}(\mathbf{S}))_{jk} = \sum_{m=0}^{n-1} c_m (\mathbf{S}^m)_{jk} = c_{k-j},$$

because exactly one value of  $m$  in  $\{0, \dots, n-1\}$  satisfies  $m \equiv k - j \pmod{n}$ . So  $p_{\mathbf{C}}(\mathbf{S}) = \mathbf{C}$ . Conversely, let  $q \in \mathbb{C}[x]$ . Dividing by  $x^n - 1$  ([Theorem 5.1.1](#)) gives  $q = h \cdot (x^n - 1) + r$  with  $\deg r < n$ , and  $\mathbf{S}^n = \mathbf{I}$  gives  $q(\mathbf{S}) = h(\mathbf{S})(\mathbf{S}^n - \mathbf{I}) + r(\mathbf{S}) =$

$r(\mathbf{S})$ , which is the circulant whose first row is the coefficient list of  $r$ .

- (b)  $\mathbf{S}$  is a permutation matrix with real entries, so  $\mathbf{S}^* = \mathbf{S}^\top = \mathbf{S}^{-1}$  (Lemma 2.6.6 (b)), and  $\mathbf{S}^{-1} = \mathbf{S}^{n-1}$  because  $\mathbf{S}^n = \mathbf{I}$ . Therefore

$$\mathbf{C}^* = \sum_{m=0}^{n-1} \overline{c_m} (\mathbf{S}^*)^m = \sum_{m=0}^{n-1} \overline{c_m} \mathbf{S}^{-m}.$$

For  $m = 0$  the term is  $\overline{c_0} \mathbf{I}$ , and for  $1 \leq m \leq n-1$  we have  $\mathbf{S}^{-m} = \mathbf{S}^{n-m}$ . So the coefficient of  $\mathbf{S}^r$  is  $\overline{c_0}$  when  $r = 0$  and  $\overline{c_{n-r}}$  when  $1 \leq r \leq n-1$ , which is the stated first row.

- (c) By (a) and (b), both  $\mathbf{C}$  and  $\mathbf{C}^*$  are polynomials in  $\mathbf{S}$ , and any two polynomials in the same matrix commute (Theorem 0.9.9). Hence  $\mathbf{C}^* \mathbf{C} = \mathbf{C} \mathbf{C}^*$ . The same argument applies to two circulants  $\mathbf{C} = p(\mathbf{S})$  and  $\mathbf{C}' = q(\mathbf{S})$ . ■

That settles the coincidence noticed in Example (Which of these are normal?) (d) and again in Exercise (B2: A circulant, diagonalized) (a). It is worth seeing it once at the level of entries, because the entries are what those pages compared. For any  $\mathbf{A} \in M_n(\mathbb{C})$ , writing  $\mathbf{a}_0, \dots, \mathbf{a}_{n-1}$  for its columns and  $\mathbf{r}_0, \dots, \mathbf{r}_{n-1}$  for its rows, the definition of the matrix product gives

$$(\mathbf{A}^* \mathbf{A})_{jk} = \langle \mathbf{a}_k, \mathbf{a}_j \rangle, \quad (\mathbf{A} \mathbf{A}^*)_{jk} = \langle \mathbf{r}_j, \mathbf{r}_k \rangle.$$

So “normal” says exactly that the table of column inner products and the table of row inner products agree. Now let  $\mathbf{A} = \mathbf{C}$  be a circulant, so that  $(\mathbf{C})_{ik} = c_{k-i}$ . Re-index the first sum by  $t = j - i$  and the second by  $t = i - k$ ; as  $i$  runs through all residues modulo  $n$ , so does  $t$  in each case. This gives

$$\langle \mathbf{a}_k, \mathbf{a}_j \rangle = \sum_{t=0}^{n-1} c_{t+k-j} \overline{c_t} = \langle \mathbf{r}_j, \mathbf{r}_k \rangle,$$

all subscripts modulo  $n$ . Both tables depend only on  $k - j$ , and they are literally the same sum. Cyclic shifting is what makes the re-indexing legal, and that is the whole of the coincidence.

*Quick check.* Which vectors  $\mathbf{c} \in \mathbb{C}^n$  give a circulant that is Hermitian? Answer for  $n = 4$  with real entries.

*Solution.* By Theorem 11.9.1 (b),  $\mathbf{C}^* = \mathbf{C}$  forces  $\overline{c_0} = c_0$  and  $\overline{c_{n-m}} = c_m$  for  $1 \leq m \leq n-1$ . For  $n = 4$  with real entries this reads  $c_1 = c_3$ , with  $c_0$  and  $c_2$  unconstrained: the symmetric real  $4 \times 4$  circulants are exactly those whose first row has the form  $(c_0, c_1, c_2, c_1)$ .

## 11.9.2 The eigenvectors of the shift

Everything now follows from one eigenvalue problem, the one for  $\mathbf{S}$ . Since  $\mathbf{S}^n = \mathbf{I}$ , any eigenvalue  $\lambda$  satisfies  $\lambda^n = 1$ , so the candidates are the  $n$ -th roots of unity of Example (Roots of Unity). There are  $n$  of them and  $\mathbf{S}$  is an  $n \times n$  matrix, which is encouraging. The eigenvectors are equally explicit.

Fix

$$\omega := e^{2\pi i/n},$$

so that  $1, \omega, \omega^2, \dots, \omega^{n-1}$  are the  $n$  distinct  $n$ -th roots of unity, and  $\omega^r = 1$  if and only if  $n$  divides  $r$  ([Example \(Roots of Unity\)](#)).

### Theorem 11.9.2 — Eigenvectors of the Cyclic Shift

For  $j = 0, 1, \dots, n-1$  let

$$\mathbf{f}_j := \frac{1}{\sqrt{n}}(1, \omega^j, \omega^{2j}, \dots, \omega^{(n-1)j}) \in \mathbb{C}^n,$$

that is,  $(\mathbf{f}_j)_m = \omega^{jm}/\sqrt{n}$ . Then  $\mathbf{S}\mathbf{f}_j = \omega^j \mathbf{f}_j$ . Consequently  $\mathbf{S}$  is a unitary matrix with the  $n$  distinct eigenvalues  $1, \omega, \dots, \omega^{n-1}$ , and each eigenspace  $E_{\omega^j}(\mathbf{S})$  is the line  $\text{span}(\mathbf{f}_j)$ .

*Idea.* The shift moves entry  $m+1$  into slot  $m$ . If the entries are the successive powers of a fixed number  $z$ , that is the same as multiplying the whole vector by  $z$  — provided the wrap-around at the end also multiplies by  $z$ , which is exactly the condition  $z^n = 1$ . So the geometric vectors  $(1, z, z^2, \dots)$  with  $z^n = 1$  are forced on us, and there are  $n$  of them.

*Proof.* Fix  $j$  and let  $0 \leq m \leq n-1$ . By the description of  $\mathbf{S}$ , the  $m$ -th entry of  $\mathbf{S}\mathbf{f}_j$  is the  $(m+1)$ -st entry of  $\mathbf{f}_j$ , indices modulo  $n$ . For  $m < n-1$  this is

$$\frac{\omega^{j(m+1)}}{\sqrt{n}} = \omega^j \cdot \frac{\omega^{jm}}{\sqrt{n}} = \omega^j (\mathbf{f}_j)_m,$$

and for  $m = n-1$  it is  $(\mathbf{f}_j)_0 = 1/\sqrt{n}$ , while  $\omega^j (\mathbf{f}_j)_{n-1} = \omega^j \omega^{j(n-1)}/\sqrt{n} = \omega^{jn}/\sqrt{n} = 1/\sqrt{n}$  because  $\omega^n = 1$ . The two agree, so  $\mathbf{S}\mathbf{f}_j = \omega^j \mathbf{f}_j$ . Each  $\mathbf{f}_j$  is non-zero, since its 0-th entry is  $1/\sqrt{n}$ , so  $\omega^j$  is an eigenvalue.

The matrix  $\mathbf{S}$  is a real permutation matrix, so  $\mathbf{S}^\top \mathbf{S} = \mathbf{I}$  by [Lemma 2.6.6](#) (b), and  $\mathbf{S}^* = \mathbf{S}^\top$ ; hence  $\mathbf{S}$  is unitary ([Definition 10.7.2](#)). The  $n$  numbers  $1, \omega, \dots, \omega^{n-1}$  are distinct, so  $\mathbf{S}$  has  $n$  distinct eigenvalues, and the sum of their eigenspaces is direct ([Corollary 8.2.3](#)). Each of those  $n$  eigenspaces has dimension at least 1 and their dimensions add up to at most  $\dim \mathbb{C}^n = n$ , so each has dimension exactly 1. Since  $\mathbf{f}_j \in E_{\omega^j}(\mathbf{S})$  is non-zero, that eigenspace is  $\text{span}(\mathbf{f}_j)$ . ■

The vectors  $\mathbf{f}_0, \dots, \mathbf{f}_{n-1}$  are called the **Fourier vectors**. They are already known to be pairwise orthogonal:  $\mathbf{S}$  is unitary, hence normal, and eigenvectors of a normal matrix for distinct eigenvalues are orthogonal ([Corollary 11.3.4](#)). But the direct computation is short, it produces the normalization at the same time, and it is the computation on which the rest of the section runs. So we do it.

### Theorem 11.9.3 — The Fourier Matrix Is Unitary

The list  $(\mathbf{f}_0, \mathbf{f}_1, \dots, \mathbf{f}_{n-1})$  is an orthonormal basis of  $\mathbb{C}^n$ . Equivalently, the **Fourier**

**matrix**

$$\mathbf{F} := \frac{1}{\sqrt{n}} \begin{pmatrix} 1 & 1 & \cdots & 1 \\ 1 & \omega & \cdots & \omega^{n-1} \\ \vdots & \vdots & & \vdots \\ 1 & \omega^{n-1} & \cdots & \omega^{(n-1)^2} \end{pmatrix}, \quad (\mathbf{F})_{mj} = \frac{\omega^{mj}}{\sqrt{n}},$$

whose  $j$ -th column is  $\mathbf{f}_j$ , is unitary:  $\mathbf{F}^* \mathbf{F} = \mathbf{F} \mathbf{F}^* = \mathbf{I}$ .

*Proof.* Let  $0 \leq j, k \leq n-1$ . Since  $|\omega| = 1$  we have  $\bar{\omega} = \omega^{-1}$ , hence  $\overline{\omega^{km}} = \omega^{-km}$ , and so

$$\langle \mathbf{f}_j, \mathbf{f}_k \rangle = \frac{1}{n} \sum_{m=0}^{n-1} \omega^{jm} \overline{\omega^{km}} = \frac{1}{n} \sum_{m=0}^{n-1} (\omega^{j-k})^m,$$

the inner product being linear in the first slot and conjugate-linear in the second.

*Case  $j = k$ .* Then  $\omega^{j-k} = \omega^0 = 1$ , so every one of the  $n$  terms equals 1 and the sum is  $n$ . Hence  $\langle \mathbf{f}_j, \mathbf{f}_j \rangle = n/n = 1$ , and  $\|\mathbf{f}_j\| = 1$ .

*Case  $j \neq k$ .* Put  $z := \omega^{j-k}$ . Then  $z^n = (\omega^n)^{j-k} = 1$ . Also  $z \neq 1$ : by [Example \(Roots of Unity\)](#),  $\omega^{j-k} = 1$  would force  $n$  to divide  $j-k$ , and  $0 < |j-k| \leq n-1$  makes that impossible. Now use the factorization

$$(z-1)(1+z+z^2+\cdots+z^{n-1}) = z^n - 1 = 0,$$

which is a polynomial identity valid for every complex  $z$ . Since  $z-1 \neq 0$ , we may divide by it, and

$$\sum_{m=0}^{n-1} z^m = 0.$$

Hence  $\langle \mathbf{f}_j, \mathbf{f}_k \rangle = 0$ .

So the list is orthonormal; it has  $n = \dim \mathbb{C}^n$  members and is independent ([Theorem 10.1.4](#)), hence a basis ([Theorem 1.6.4](#)). A matrix whose columns form an orthonormal basis of  $\mathbb{C}^n$  is unitary ([Theorem 10.7.3 \(f\)](#)), which gives  $\mathbf{F}^* \mathbf{F} = \mathbf{F} \mathbf{F}^* = \mathbf{I}$ . ■

The second case is where the whole section lives. A sum of  $n$  unit complex numbers, spread evenly around the circle, cancels — and it cancels for the cheapest possible reason, that a geometric series with ratio  $z \neq 1$  and  $z^n = 1$  telescopes to zero.

### Example. (The Fourier matrix of size four)

Write out  $\mathbf{F}$  for  $n = 4$ , and check one orthogonality relation by hand.

*Solution.* For  $n = 4$ ,  $\omega = e^{2\pi i/4} = i$ , and the powers cycle  $i^0 = 1$ ,  $i^1 = i$ ,  $i^2 = -1$ ,  $i^3 = -i$ . So

$$\mathbf{F} = \frac{1}{2} \begin{pmatrix} 1 & 1 & 1 & 1 \\ 1 & i & -1 & -i \\ 1 & -1 & 1 & -1 \\ 1 & -i & -1 & i \end{pmatrix}.$$

Take  $j = 1, k = 3$ . Then  $\omega^{j-k} = i^{-2} = -1$ , and

$$\langle \mathbf{f}_1, \mathbf{f}_3 \rangle = \frac{1}{4}(1 + (-1) + 1 + (-1)) = 0,$$

matching the geometric-series computation with  $z = -1$ , for which  $1 + z + z^2 + z^3 = 0$ . Note that  $\mathbf{F}$  is symmetric,  $\mathbf{F}^\top = \mathbf{F}$ , but **not** Hermitian:  $\mathbf{F}^* = \overline{\mathbf{F}} \neq \mathbf{F}$ , since  $(\mathbf{F})_{12} = i/2$ .

### 11.9.3 One unitary for every circulant

#### Theorem 11.9.4 — Circulant Diagonalization

Let  $\mathbf{C} \in M_n(\mathbb{C})$  be the circulant with first row  $\mathbf{c}$  and symbol  $p_{\mathbf{c}}$ . Then every Fourier vector is an eigenvector of  $\mathbf{C}$ ,

$$\mathbf{C}\mathbf{f}_j = p_{\mathbf{c}}(\omega^j) \mathbf{f}_j \quad (j = 0, 1, \dots, n-1),$$

and consequently

$$\mathbf{F}^* \mathbf{C} \mathbf{F} = \text{diag}(p_{\mathbf{c}}(1), p_{\mathbf{c}}(\omega), \dots, p_{\mathbf{c}}(\omega^{n-1})).$$

The same unitary  $\mathbf{F}$  works for every circulant of size  $n$ , and  $\det \mathbf{C} = \prod_{j=0}^{n-1} p_{\mathbf{c}}(\omega^j)$ .

*Idea.* There is nothing left to prove. A circulant is a polynomial in  $\mathbf{S}$  by Theorem 11.9.1, and a polynomial in a matrix acts on an eigenvector of that matrix by the same polynomial applied to the eigenvalue. The point is not the proof but the statement:  $\mathbf{F}$  was written down without reference to  $\mathbf{c}$ , so a single change of orthonormal basis handles the entire family at once.

*Proof.* By Theorem 11.9.1 (a),  $\mathbf{C} = p_{\mathbf{c}}(\mathbf{S})$ , and by Theorem 11.9.2,  $\mathbf{S}\mathbf{f}_j = \omega^j \mathbf{f}_j$ . Iterating the latter gives  $\mathbf{S}^m \mathbf{f}_j = \omega^{jm} \mathbf{f}_j$  for every  $m \geq 0$ , so

$$\mathbf{C}\mathbf{f}_j = \sum_{m=0}^{n-1} c_m \mathbf{S}^m \mathbf{f}_j = \left( \sum_{m=0}^{n-1} c_m \omega^{jm} \right) \mathbf{f}_j = p_{\mathbf{c}}(\omega^j) \mathbf{f}_j.$$

The  $j$ -th column of  $\mathbf{F}$  is  $\mathbf{f}_j$ , so the  $j$ -th column of  $\mathbf{C}\mathbf{F}$  is  $p_{\mathbf{c}}(\omega^j) \mathbf{f}_j$ , which is the  $j$ -th column of  $\mathbf{F}\mathbf{D}$  for  $\mathbf{D} = \text{diag}(p_{\mathbf{c}}(1), \dots, p_{\mathbf{c}}(\omega^{n-1}))$ . Hence  $\mathbf{C}\mathbf{F} = \mathbf{F}\mathbf{D}$ , and multiplying on the left by  $\mathbf{F}^* = \mathbf{F}^{-1}$  (Theorem 11.9.3) gives  $\mathbf{F}^* \mathbf{C} \mathbf{F} = \mathbf{D}$ . Taking determinants,  $\det \mathbf{C} = \det \mathbf{D} = \prod_j p_{\mathbf{c}}(\omega^j)$ , since  $\det \mathbf{F}^* \det \mathbf{F} = \det(\mathbf{F}^* \mathbf{F}) = 1$  (Theorem 6.5.4). ■

This re-proves normality from the other side, and closes the account opened in Example (Which of these are normal?) (d):  $\mathbf{C} = \mathbf{F}\mathbf{D}\mathbf{F}^*$  with  $\mathbf{F}$  unitary and  $\mathbf{D}$  diagonal is precisely the factorization that Corollary 11.4.3 reserves for normal matrices. The eigenvalues  $p_{\mathbf{c}}(\omega^j)$  need no characteristic polynomial either: they are  $n$  evaluations of a polynomial of degree less than  $n$ .

**Example.** (The running example, diagonalized)



Find the eigenvalues and the determinant of the circulant  $\mathbf{C}$  with first row  $(1, 2, 3, 4)$ , and write down  $\mathbf{F}^*\mathbf{C}\mathbf{F}$ .

*Solution.* Here  $n = 4$ ,  $\omega = i$  and  $p_{\mathbf{C}}(x) = 1 + 2x + 3x^2 + 4x^3$ . Evaluate at the four fourth roots of unity, using  $i^2 = -1$  and  $i^3 = -i$ :

$$\begin{aligned} p_{\mathbf{C}}(1) &= 1 + 2 + 3 + 4 = 10, \\ p_{\mathbf{C}}(i) &= 1 + 2i - 3 - 4i = -2 - 2i, \\ p_{\mathbf{C}}(-1) &= 1 - 2 + 3 - 4 = -2, \\ p_{\mathbf{C}}(-i) &= 1 - 2i - 3 + 4i = -2 + 2i. \end{aligned}$$

So  $\mathbf{F}^*\mathbf{C}\mathbf{F} = \text{diag}(10, -2 - 2i, -2, -2 + 2i)$  with  $\mathbf{F}$  as in [Example \(The Fourier matrix of size four\)](#), and

$$\det \mathbf{C} = 10 \cdot (-2 - 2i) \cdot (-2) \cdot (-2 + 2i) = -20(4 + 4) = -160,$$

using  $(-2 - 2i)(-2 + 2i) = 4 + 4 = 8$ . As a check on the first eigenvalue: every row of  $\mathbf{C}$  sums to 10, so  $\mathbf{C}\mathbf{1} = 10 \cdot \mathbf{1}$ , and  $\mathbf{f}_0 = \frac{1}{2}\mathbf{1}$ .

*Quick check.* The matrix of [Example \(The running example, diagonalized\)](#) is real, and two of its eigenvalues are not. Why must those two be complex conjugates of each other, and which two are they in general?

*Solution.* If every  $c_m$  is real then  $\overline{p_{\mathbf{C}}(z)} = p_{\mathbf{C}}(\bar{z})$  for every  $z$ . Taking  $z = \omega^j$  and using  $\omega^j = \omega^{-j} = \omega^{n-j}$ ,

$$\overline{p_{\mathbf{C}}(\omega^j)} = p_{\mathbf{C}}(\omega^{n-j}),$$

so the eigenvalue attached to  $\mathbf{f}_j$  and the one attached to  $\mathbf{f}_{n-j}$  are conjugate. Here  $n = 4$  and the pair is  $j = 1, 3$ :  $-2 - 2i = -2 + 2i$ . The eigenvalues for  $j = 0$  and, when  $n$  is even,  $j = n/2$  are their own conjugates, hence real — which is why 10 and  $-2$  came out real. (This is also [Theorem 8.2.5](#), seen here with the eigenvectors named.)

Once every circulant is  $\mathbf{F}\mathbf{D}\mathbf{F}^*$ , the family closes up on itself.

### Corollary 11.9.5 — Circulants Form a Commutative Algebra

Let  $\mathcal{C}_n \subseteq M_n(\mathbb{C})$  be the set of circulants of size  $n$ . Then:

- (a)  $\mathcal{C}_n$  is a subspace of  $M_n(\mathbb{C})$  of dimension  $n$ , with basis  $\mathbf{I}, \mathbf{S}, \dots, \mathbf{S}^{n-1}$ ;
- (b)  $\mathcal{C}_n$  is closed under products, and any two of its members commute;
- (c)  $\mathbf{A} \in \mathcal{C}_n$  if and only if  $\mathbf{F}^*\mathbf{A}\mathbf{F}$  is diagonal;
- (d) a circulant  $\mathbf{C}$  is invertible if and only if  $p_{\mathbf{C}}(\omega^j) \neq 0$  for every  $j$ , and then  $\mathbf{C}^{-1}$  is again a circulant.

*Proof.*

- (a) The map  $\mathbf{c} \mapsto p_{\mathbf{C}}(\mathbf{S})$  from  $\mathbb{C}^n$  to  $M_n(\mathbb{C})$  is linear and, by [Theorem 11.9.1 \(a\)](#), its image is exactly  $\mathcal{C}_n$ . It is injective, because the first row of  $p_{\mathbf{C}}(\mathbf{S})$  is  $\mathbf{c}$ . So  $\mathcal{C}_n$  is a subspace of dimension  $n$ , and the images  $\mathbf{I}, \mathbf{S}, \dots, \mathbf{S}^{n-1}$  of the standard basis form a basis of it.

- (b) If  $\mathbf{C} = p(\mathbf{S})$  and  $\mathbf{C}' = q(\mathbf{S})$ , then  $\mathbf{C}\mathbf{C}' = (pq)(\mathbf{S})$  (Theorem 0.9.9), which is a circulant by Theorem 11.9.1 (a). Commutativity is part (c) of that theorem.
- (c)  $(\Rightarrow)$  is Theorem 11.9.4.  $(\Leftarrow)$  Suppose  $\mathbf{F}^*\mathbf{A}\mathbf{F} = \mathbf{D} = \text{diag}(d_0, \dots, d_{n-1})$ . The  $n$  numbers  $1, \omega, \dots, \omega^{n-1}$  are distinct, so by Lagrange interpolation (Theorem 5.6.1) there is  $q \in \mathbb{C}[x]$  with  $q(\omega^j) = d_j$  for every  $j$ . By Theorem 11.9.4 applied to the circulant  $q(\mathbf{S})$ , which has symbol the remainder of  $q$  on division by  $x^n - 1$ ,

$$\mathbf{F}^*q(\mathbf{S})\mathbf{F} = \text{diag}(q(1), \dots, q(\omega^{n-1})) = \mathbf{D}.$$

Multiplying both sides by  $\mathbf{F}$  on the left and  $\mathbf{F}^*$  on the right gives  $q(\mathbf{S}) = \mathbf{A}$ , so  $\mathbf{A}$  is a circulant.

- (d) From  $\mathbf{C} = \mathbf{F}\mathbf{D}\mathbf{F}^*$ ,  $\det \mathbf{C} = \det \mathbf{D}$ , so  $\mathbf{C}$  is invertible exactly when no diagonal entry  $p_{\mathbf{c}}(\omega^j)$  vanishes (Theorem 6.5.2). In that case  $\mathbf{C}^{-1} = \mathbf{F}\mathbf{D}^{-1}\mathbf{F}^*$ , and  $\mathbf{F}^*\mathbf{C}^{-1}\mathbf{F} = \mathbf{D}^{-1}$  is diagonal, so  $\mathbf{C}^{-1}$  is a circulant by (c). ■

Part (c) is the sharpest statement of the section: being a circulant is not a statement about the pattern of the entries at all, but a statement about which orthonormal basis diagonalizes the matrix. The pattern was only ever a way of recognizing it.

## 11.9.4 Multiplying circulants is convolution

A circulant is its first row. So the product of two circulants must be computable from the two first rows, and it is worth knowing what that computation is.

Given  $\mathbf{c}, \mathbf{d} \in \mathbb{C}^n$ , define their **cyclic convolution**  $\mathbf{c} * \mathbf{d} \in \mathbb{C}^n$  by

$$(\mathbf{c} * \mathbf{d})_r := \sum_{l=0}^{n-1} c_l d_{r-l} \quad (r = 0, \dots, n-1),$$

subscripts modulo  $n$ . In words: to get entry  $r$ , slide the reversed list  $\mathbf{d}$  along  $\mathbf{c}$  so that the indices add up to  $r$ , and add the products. It is the multiplication of polynomials with the exponents read modulo  $n$ , and the next theorem says so.

### Theorem 11.9.6 — Convolution Theorem

Let  $\mathbf{c}, \mathbf{d} \in \mathbb{C}^n$  with circulants  $\mathbf{C}, \mathbf{C}'$  and symbols  $p_{\mathbf{c}}, p_{\mathbf{d}}$ . Then:

- (a)  $\mathbf{C}\mathbf{C}'$  is the circulant with first row  $\mathbf{c} * \mathbf{d}$ ;
- (b) for every  $j$ ,  $p_{\mathbf{c} * \mathbf{d}}(\omega^j) = p_{\mathbf{c}}(\omega^j) p_{\mathbf{d}}(\omega^j)$ .

In words: convolving the first rows corresponds to multiplying the eigenvalue lists entry by entry.

*Idea.* Both parts are the identity  $p_{\mathbf{c}}p_{\mathbf{d}} \equiv p_{\mathbf{c} * \mathbf{d}} \pmod{x^n - 1}$ , looked at twice. Multiplying the two symbols produces a polynomial of degree up to  $2n - 2$ ; reducing the exponents modulo  $n$  folds the coefficients back into  $n$  slots, and the fold is precisely the

convolution sum. Evaluating at a root of  $x^n - 1$  makes the reduction invisible, which is (b); substituting  $\mathbf{S}$ , for which  $\mathbf{S}^n = \mathbf{I}$ , makes it invisible as well, which is (a).

*Proof.* Expand the product of the symbols and collect exponents modulo  $n$ :

$$p_{\mathbf{c}}(x)p_{\mathbf{d}}(x) = \sum_{l=0}^{n-1} \sum_{m=0}^{n-1} c_l d_m x^{l+m}.$$

For each  $r \in \{0, \dots, n-1\}$ , the pairs  $(l, m)$  with  $l + m \equiv r \pmod{n}$  are exactly those with  $m \equiv r - l$ , one value of  $m$  for each  $l$ . Hence, working modulo  $x^n - 1$  and using  $x^{l+m} \equiv x^{(l+m) \bmod n}$ ,

$$p_{\mathbf{c}}(x)p_{\mathbf{d}}(x) \equiv \sum_{r=0}^{n-1} \left( \sum_{l=0}^{n-1} c_l d_{r-l} \right) x^r = p_{\mathbf{c} * \mathbf{d}}(x) \pmod{x^n - 1}.$$

(a) Substitute  $\mathbf{S}$ . Since  $\mathbf{S}^n = \mathbf{I}$ , the multiple of  $x^n - 1$  contributes nothing, so

$$\mathbf{C}\mathbf{C}' = p_{\mathbf{c}}(\mathbf{S})p_{\mathbf{d}}(\mathbf{S}) = p_{\mathbf{c} * \mathbf{d}}(\mathbf{S}),$$

the first equality by [Theorem 11.9.1](#) (a) and [Theorem 0.9.9](#). By [Theorem 11.9.1](#) (a) again,  $p_{\mathbf{c} * \mathbf{d}}(\mathbf{S})$  is the circulant with first row  $\mathbf{c} * \mathbf{d}$ , the coefficient list having length  $n$ .

(b) Substitute  $x = \omega^j$ , a root of  $x^n - 1$ , so that the multiple of  $x^n - 1$  again contributes nothing. This gives  $p_{\mathbf{c}}(\omega^j)p_{\mathbf{d}}(\omega^j) = p_{\mathbf{c} * \mathbf{d}}(\omega^j)$ . ■

Part (b) deserves its own sentence, because it is the reason the discrete Fourier transform is used at all. Define the **discrete Fourier transform** of  $\mathbf{c} \in \mathbb{C}^n$  to be the vector of eigenvalues,

$$\widehat{\mathbf{c}} := (p_{\mathbf{c}}(1), p_{\mathbf{c}}(\omega), \dots, p_{\mathbf{c}}(\omega^{n-1})) = \sqrt{n} \mathbf{F} \mathbf{c},$$

the second expression because the  $j$ -th entry of  $\mathbf{F} \mathbf{c}$  is  $n^{-1/2} \sum_m \omega^{jm} c_m$ . Then (b) reads

$$(\widehat{\mathbf{c} * \mathbf{d}})_j = \widehat{\mathbf{c}}_j \widehat{\mathbf{d}}_j \quad (j = 0, \dots, n-1):$$

the transform turns convolution into entrywise multiplication. Convolution is a tangled sum of  $n^2$  products; after one unitary change of basis it is  $n$  ordinary multiplications. Everything the transform buys comes from this line.

#### Example. (Convolution in size four)

Let  $\mathbf{c} = (1, 2, 3, 4)$  and  $\mathbf{d} = (1, 1, 0, 0)$ . Compute  $\mathbf{c} * \mathbf{d}$ , and verify [Theorem 11.9.6](#) (b) at all four roots of unity.

*Solution.* Since  $d_0 = d_1 = 1$  and  $d_2 = d_3 = 0$ , the sum  $\sum_l c_l d_{r-l}$  keeps only the terms with  $r - l \equiv 0$  or  $1$ , that is  $l = r$  and  $l = r - 1$ . So  $(\mathbf{c} * \mathbf{d})_r = c_r + c_{r-1}$ , subscripts modulo 4:

$$\mathbf{c} * \mathbf{d} = (1 + 4, 2 + 1, 3 + 2, 4 + 3) = (5, 3, 5, 7).$$

The three symbols are  $p_{\mathbf{c}}(x) = 1 + 2x + 3x^2 + 4x^3$ ,  $p_{\mathbf{d}}(x) = 1 + x$  and  $p_{\mathbf{c} * \mathbf{d}}(x) = 5 + 3x + 5x^2 + 7x^3$ . Evaluating each at  $1, i, -1, -i$ , with  $p_{\mathbf{c}}$  already computed in [Example](#)

(The running example, diagonalized):

$$\begin{aligned}
 j = 0 : \quad & 10 \cdot 2 = 20, & 5 + 3 + 5 + 7 = 20; \\
 j = 1 : \quad & (-2 - 2i)(1 + i) = -4i, & 5 + 3i - 5 - 7i = -4i; \\
 j = 2 : \quad & (-2) \cdot 0 = 0, & 5 - 3 + 5 - 7 = 0; \\
 j = 3 : \quad & (-2 + 2i)(1 - i) = 4i, & 5 - 3i - 5 + 7i = 4i.
 \end{aligned}$$

All four agree. For the middle line,  $(-2 - 2i)(1 + i) = -2 - 2i - 2i - 2i^2 = -4i$ .

## 11.9.5 Solving a circulant system, and a word about speed

If  $\mathbf{C}\mathbf{x} = \mathbf{b}$  with  $\mathbf{C}$  an invertible circulant, then in the Fourier basis the system falls apart into  $n$  scalar equations. Writing  $\mathbf{C} = \mathbf{F}\mathbf{D}\mathbf{F}^*$  and setting  $\mathbf{y} = \mathbf{F}^*\mathbf{x}$ ,  $\mathbf{g} = \mathbf{F}^*\mathbf{b}$ , the system becomes  $\mathbf{D}\mathbf{y} = \mathbf{g}$ , that is  $y_j = g_j/\lambda_j$  with  $\lambda_j = p_{\mathbf{C}}(\omega^j)$ . Then  $\mathbf{x} = \mathbf{F}\mathbf{y}$ . Three steps: transform, divide, transform back.

### Example. (A circulant system)

Solve  $\mathbf{C}\mathbf{x} = \mathbf{b}$  for the circulant  $\mathbf{C}$  with first row  $(1, 2, 3, 4)$  and  $\mathbf{b} = (1, 1, -1, -1)$ .

*Solution.* The eigenvalues are  $\lambda = (10, -2 - 2i, -2, -2 + 2i)$  by [Example \(The running example, diagonalized\)](#). First transform  $\mathbf{b}$ : the  $j$ -th entry of  $\mathbf{F}^*\mathbf{b}$  is  $\frac{1}{2} \sum_m \omega^{jm} b_m = \frac{1}{2} \sum_m (-i)^{jm} b_m$ , so

$$\begin{aligned}
 g_0 &= \frac{1}{2}(1 + 1 - 1 - 1) = 0, \\
 g_1 &= \frac{1}{2}(1 + (-i) - (-1) - (i)) = 1 - i, \\
 g_2 &= \frac{1}{2}(1 - 1 - 1 + 1) = 0, \\
 g_3 &= \frac{1}{2}(1 + i + 1 + i) = 1 + i.
 \end{aligned}$$

Now divide. The two zero entries stay zero, and

$$y_1 = \frac{1 - i}{-2 - 2i} = \frac{(1 - i)(-2 + 2i)}{8} = \frac{4i}{8} = \frac{i}{2}, \quad y_3 = \overline{y_1} = -\frac{i}{2},$$

where the middle step multiplies numerator and denominator by  $\overline{-2 - 2i} = -2 + 2i$ , whose product with  $-2 - 2i$  is 8. Finally  $\mathbf{x} = \mathbf{F}\mathbf{y}$ , whose  $m$ -th entry is  $\frac{1}{2}(i^m y_1 + i^{3m} y_3) = \frac{1}{2} \cdot \frac{i}{2}(i^m - (-i)^m)$ :

$$\mathbf{x} = \left(0, -\frac{1}{2}, 0, \frac{1}{2}\right).$$

Check directly: the second row of  $\mathbf{C}$  is  $(4, 1, 2, 3)$ , and  $-\frac{1}{2} + \frac{3}{2} = 1 = b_1$ ; the third row is  $(3, 4, 1, 2)$ , and  $-2 + 1 = -1 = b_2$ . The answer is real, as it must be:  $\mathbf{C}$  and  $\mathbf{b}$  are real, so  $\mathbf{x} = \mathbf{C}^{-1}\mathbf{b}$  is real, and this shows up in the transform as the conjugate symmetry  $y_3 = \overline{y_1}$ .

Counted honestly, the recipe above saves something, but not yet very much. Forming  $\mathbf{F}^*\mathbf{b}$  as a matrix–vector product costs about  $n^2$  multiplications, and so does forming  $\mathbf{F}\mathbf{y}$ , so the whole solve costs on the order of  $n^2$ . That beats general elimination, which is on the order of  $n^3$ , but it is no better than multiplying  $\mathbf{b}$  by  $\mathbf{C}^{-1}$  once that inverse has been found. What changes the

picture is that  $\mathbf{F}$  does not have to be applied as a general matrix. The powers of  $\omega$  repeat, and splitting the sum  $\sum_m \omega^{jm} b_m$  into its even and odd  $m$  reduces one transform of length  $n$  to two of length  $n/2$ ; iterating brings the cost down to about  $n \log n$ . That algorithm is the **fast Fourier transform**. It proves no new theorem: the matrix  $\mathbf{F}$ , the factorization  $\mathbf{C} = \mathbf{FDF}^*$  and everything above are unchanged, and the fast Fourier transform is only a way of applying  $\mathbf{F}$  quickly. Chapter 23 develops it, together with the cost accounting that makes “quickly” precise.

*Warning.* A **Toeplitz** matrix — one with constant diagonals,  $(\mathbf{A})_{jk} = a_{k-j}$  with the subscript **not** reduced modulo  $n$  — is not a circulant, and  $\mathbf{F}$  does not diagonalize it. Take

$$\mathbf{A} = \begin{pmatrix} 0 & 1 & 0 \\ 1 & 0 & 1 \\ 0 & 1 & 0 \end{pmatrix}.$$

Every diagonal of  $\mathbf{A}$  is constant, so  $\mathbf{A}$  is Toeplitz, but the circulant with first row  $(0, 1, 0)$  is  $\mathbf{S}$ , whose second row is  $(0, 0, 1)$ , not  $(1, 0, 1)$ . And  $\mathbf{F}$  fails at the very first column:  $\mathbf{f}_0 = \frac{1}{\sqrt{3}}(1, 1, 1)$  and  $\mathbf{A}\mathbf{f}_0 = \frac{1}{\sqrt{3}}(1, 2, 1)$ , which is not a multiple of  $\mathbf{f}_0$ , so  $\mathbf{f}_0$  is not an eigenvector and  $\mathbf{F}^*\mathbf{A}\mathbf{F}$  is not diagonal. Normality is not the issue:  $\mathbf{A}$  is symmetric, hence normal, and [Theorem 11.5.1](#) does hand it an orthonormal eigenbasis. It is simply a different basis, with the irrational eigenvalues  $0, \pm\sqrt{2}$  in place of evaluations of a polynomial at roots of unity. What the wrap-around buys is not diagonalizability; it is the *same* diagonalizing basis for everybody.

## 11.9.6 Exercises

### 11.9.6.1 A. Check your understanding

#### Exercise A1.

- Define the circulant with first row  $\mathbf{c} = (c_0, \dots, c_{n-1})$ , and write down its symbol.
- State the eigenvalues and the eigenvectors of the cyclic shift  $\mathbf{S}$  on  $\mathbb{C}^n$ .
- Give the eigenvalues of the circulant with first row  $\mathbf{c}$ , and say how many multiplications are needed to compute all of them from  $\mathbf{c}$  by direct evaluation.
- Determine whether the following are true, with a reason. (i) Every Toeplitz matrix is a circulant. (ii) Every circulant is normal. (iii) The Fourier matrix  $\mathbf{F}$  is Hermitian.
- Explain in one sentence why two circulants of the same size commute, without computing any entries.

*Solution.*

- The matrix  $\mathbf{C} \in M_n(\mathbb{C})$  with  $(\mathbf{C})_{jk} = c_{k-j}$ , subscripts modulo  $n$  and indices running from 0 to  $n-1$  ([Definition 11.9.1](#)). Its symbol is  $p_{\mathbf{c}}(x) = c_0 + c_1x + \dots + c_{n-1}x^{n-1}$ .
- With  $\omega = e^{2\pi i/n}$ : the eigenvalues are  $1, \omega, \dots, \omega^{n-1}$ , all distinct, and  $\omega^j$  has the one-dimensional eigenspace spanned by  $\mathbf{f}_j = n^{-1/2}(1, \omega^j, \dots, \omega^{(n-1)j})$  ([Theorem 11.9.2](#)).
- The eigenvalues are  $p_{\mathbf{c}}(\omega^j)$  for  $j = 0, \dots, n-1$  ([Theorem 11.9.4](#)). Each evaluation is a sum of  $n$  products, so all  $n$  of them cost about  $n^2$  multiplications done directly.

- (d) (i) False: the  $3 \times 3$  matrix in the warning above has constant diagonals but is not a circulant. (ii) True, by [Theorem 11.9.1 \(c\)](#); equivalently by [Corollary 11.4.3](#), since  $\mathbf{C} = \mathbf{F}\mathbf{D}\mathbf{F}^*$ . (iii) False.  $\mathbf{F}$  is symmetric, but  $\mathbf{F}^* = \overline{\mathbf{F}}$ , and for  $n = 4$  the entry  $(\mathbf{F})_{12} = i/2$  is not real ([Example \(The Fourier matrix of size four\)](#)).  $\mathbf{F}$  is Hermitian only for  $n = 1, 2$ , where all its entries are real.
- (e) Both are polynomials in the single matrix  $\mathbf{S}$  ([Theorem 11.9.1 \(a\)](#)), and two polynomials in one matrix always commute ([Theorem 0.9.9](#)).

### 11.9.6.2 B. Practice

#### Exercise B1: Eigenvalues of small circulants.

For each vector  $\mathbf{c}$ , write down the symbol, compute all eigenvalues of the circulant with first row  $\mathbf{c}$ , and give its determinant.

- (a)  $\mathbf{c} = (1, 1, 1)$ ,  $n = 3$ .  
 (b)  $\mathbf{c} = (0, 1, 0, 0)$ ,  $n = 4$ .  
 (c)  $\mathbf{c} = (2, -1, 0, -1)$ ,  $n = 4$ .

*Solution.*

- (a)  $p_{\mathbf{c}}(x) = 1 + x + x^2$ . With  $\omega = e^{2\pi i/3}$ , the eigenvalues are  $p_{\mathbf{c}}(1) = 3$  and  $p_{\mathbf{c}}(\omega^j) = 1 + \omega^j + \omega^{2j}$  for  $j = 1, 2$ . For  $j \neq 0$ ,  $z = \omega^j$  satisfies  $z \neq 1$  and  $z^3 = 1$ , so  $1 + z + z^2 = (z^3 - 1)/(z - 1) = 0$  — the geometric-series computation of [Theorem 11.9.3](#) again. The eigenvalues are 3, 0, 0 and  $\det \mathbf{C} = 0$ . (This matrix is the all-ones matrix, of rank 1.)
- (b)  $p_{\mathbf{c}}(x) = x$ , so  $\mathbf{C} = \mathbf{S}$ . The eigenvalues are  $\omega^j = i^j$ , namely 1,  $i$ ,  $-1$ ,  $-i$ , and

$$\det \mathbf{S} = 1 \cdot i \cdot (-1) \cdot (-i) = (i \cdot (-i))(1 \cdot (-1)) = -1.$$

This agrees with the sign of the permutation:  $\mathbf{S}$  permutes the coordinates in one 4-cycle, which is odd.

- (c)  $p_{\mathbf{c}}(x) = 2 - x - x^3$ . Then  $p_{\mathbf{c}}(1) = 0$ ;  $p_{\mathbf{c}}(i) = 2 - i - i^3 = 2 - i + i = 2$ ;  $p_{\mathbf{c}}(-1) = 2 + 1 + 1 = 4$ ;  $p_{\mathbf{c}}(-i) = 2 + i - i = 2$ . The eigenvalues are 0, 2, 4, 2, and  $\det \mathbf{C} = 0$ . The kernel contains  $\mathbf{f}_0$ , that is the all-ones vector, as one sees directly: every row of  $\mathbf{C}$  sums to 0.

#### Exercise B2: Solve a circulant system by transforming.

Let  $\mathbf{C}$  be the  $4 \times 4$  circulant with first row  $(3, 1, 0, 1)$  and let  $\mathbf{b} = (1, 0, 0, 0)$ . Compute the eigenvalues of  $\mathbf{C}$ , and solve  $\mathbf{C}\mathbf{x} = \mathbf{b}$  by transforming, dividing and transforming back. Check your answer against the first two rows of  $\mathbf{C}$ .

*Solution.* The symbol is  $p_{\mathbf{c}}(x) = 3 + x + x^3$ , and with  $\omega = i$ ,

$$p_{\mathbf{c}}(1) = 5, \quad p_{\mathbf{c}}(i) = 3 + i - i = 3, \quad p_{\mathbf{c}}(-1) = 1, \quad p_{\mathbf{c}}(-i) = 3.$$

All four are non-zero, so  $\mathbf{C}$  is invertible ([Corollary 11.9.5 \(d\)](#)), with  $\det \mathbf{C} = 5 \cdot 3 \cdot 1 \cdot 3 = 45$ .

Transform:  $(\mathbf{F}^*\mathbf{b})_j = \frac{1}{2} \sum_m \overline{\omega^{jm}} b_m = \frac{1}{2}$  for every  $j$ , since  $\mathbf{b} = \mathbf{e}_0$ . Divide:  $y_j = \frac{1}{2} \lambda_j^{-1}$ ,

giving  $\mathbf{y} = \frac{1}{2}(\frac{1}{5}, \frac{1}{3}, 1, \frac{1}{3})$ . Transform back: the  $m$ -th entry of  $\mathbf{x} = \mathbf{F}\mathbf{y}$  is

$$x_m = \frac{1}{2} \sum_{j=0}^3 i^{mj} y_j = \frac{1}{4} \left( \frac{1}{5} + \frac{i^m}{3} + (-1)^m + \frac{i^{3m}}{3} \right).$$

For  $m = 1$  and  $m = 3$  the two middle terms cancel, since  $i^m + i^{3m} = i^m + \overline{i^m}$  is 0 when  $i^m = \pm i$ ; the bracket is  $\frac{1}{5} - 1$ , so  $x_1 = x_3 = -\frac{1}{5}$ . For  $m = 0$  the bracket is  $\frac{1}{5} + \frac{1}{3} + 1 + \frac{1}{3} = \frac{28}{15}$ , so  $x_0 = \frac{7}{15}$ . For  $m = 2$  it is  $\frac{1}{5} - \frac{1}{3} + 1 - \frac{1}{3} = \frac{8}{15}$ , so  $x_2 = \frac{2}{15}$ . Hence

$$\mathbf{x} = \left( \frac{7}{15}, -\frac{1}{5}, \frac{2}{15}, -\frac{1}{5} \right).$$

Check: row 0 of  $\mathbf{C}$  is  $(3, 1, 0, 1)$  and  $3 \cdot \frac{7}{15} - \frac{1}{5} - \frac{1}{5} = \frac{21-3-3}{15} = 1$ ; row 1 is  $(1, 3, 1, 0)$  and  $\frac{7}{15} - \frac{3}{5} + \frac{2}{15} = \frac{7-9+2}{15} = 0$ .

### Exercise B3: The convolution theorem in an example.

Let  $\mathbf{c} = (1, 0, 2, 0)$  and  $\mathbf{d} = (0, 1, 0, 3)$  in  $\mathbb{C}^4$ .

- Compute  $\mathbf{c} * \mathbf{d}$ .
- Verify  $p_{\mathbf{c} * \mathbf{d}}(\omega^j) = p_{\mathbf{c}}(\omega^j)p_{\mathbf{d}}(\omega^j)$  for  $j = 0, 1, 2, 3$ .
- Hence write down the first row of the product of the two circulants, without multiplying any matrices.

*Solution.*

- Only  $c_0 = 1$  and  $c_2 = 2$  are non-zero, so  $(\mathbf{c} * \mathbf{d})_r = d_r + 2d_{r-2}$ , subscripts modulo 4. With  $\mathbf{d} = (0, 1, 0, 3)$ :

$$\mathbf{c} * \mathbf{d} = (0 + 0, 1 + 6, 0 + 0, 3 + 2) = (0, 7, 0, 5).$$

- The symbols are  $p_{\mathbf{c}}(x) = 1 + 2x^2$ ,  $p_{\mathbf{d}}(x) = x + 3x^3$  and  $p_{\mathbf{c} * \mathbf{d}}(x) = 7x + 5x^3$ . With  $\omega = i$ :

$$\begin{aligned} j = 0 : & \quad 3 \cdot 4 = 12, & \quad 7 + 5 = 12; \\ j = 1 : & \quad (-1)(-2i) = 2i, & \quad 7i - 5i = 2i; \\ j = 2 : & \quad 3 \cdot (-4) = -12, & \quad -7 - 5 = -12; \\ j = 3 : & \quad (-1)(2i) = -2i, & \quad -7i + 5i = -2i. \end{aligned}$$

Here  $p_{\mathbf{c}}(i) = 1 + 2i^2 = -1$  and  $p_{\mathbf{d}}(i) = i + 3i^3 = i - 3i = -2i$ , and similarly at the other roots. All four rows agree.

- By [Theorem 11.9.6](#) (a), the product is the circulant with first row  $\mathbf{c} * \mathbf{d} = (0, 7, 0, 5)$ .

### 11.9.6.3 C. Going deeper

#### Exercise C1: What commutes with the shift.

Let  $\mathbf{A} \in M_n(\mathbb{C})$  satisfy  $\mathbf{AS} = \mathbf{SA}$ . Prove that  $\mathbf{A}$  is a circulant. *Hint: consider the eigenspaces of  $\mathbf{S}$ .*

*Solution.* By [Theorem 11.9.2](#),  $\mathbf{S}$  has the  $n$  distinct eigenvalues  $\omega^j$ , and  $E_{\omega^j}(\mathbf{S}) = \text{span}(\mathbf{f}_j)$  is one-dimensional. Since  $\mathbf{A}$  commutes with  $\mathbf{S}$ , each eigenspace of  $\mathbf{S}$  is invariant under  $\mathbf{A}$  ([Theorem 8.9.1](#)). So  $\mathbf{A}\mathbf{f}_j \in \text{span}(\mathbf{f}_j)$ , that is  $\mathbf{A}\mathbf{f}_j = d_j\mathbf{f}_j$  for some  $d_j \in \mathbb{C}$ .

Therefore the  $j$ -th column of  $\mathbf{A}\mathbf{F}$  is  $d_j\mathbf{f}_j$ , which is the  $j$ -th column of  $\mathbf{F}\mathbf{D}$  for  $\mathbf{D} = \text{diag}(d_0, \dots, d_{n-1})$ . Hence  $\mathbf{A}\mathbf{F} = \mathbf{F}\mathbf{D}$  and  $\mathbf{F}^*\mathbf{A}\mathbf{F} = \mathbf{D}$  is diagonal, so  $\mathbf{A}$  is a circulant by [Corollary 11.9.5](#) (c). This proves the converse of [Theorem 11.9.1](#) (c) for the single matrix  $\mathbf{S}$ : the commutant of the cyclic shift is exactly the algebra of circulants.

### Exercise C2: Determinant and invertibility.

Let  $\mathbf{C}$  be the circulant with first row  $\mathbf{c} \in \mathbb{C}^n$  and symbol  $p_{\mathbf{c}}$ .

- Prove that  $\det \mathbf{C} = \prod_{j=0}^{n-1} p_{\mathbf{c}}(\omega^j)$ , and use it to compute the determinant of the  $n \times n$  all-ones matrix for every  $n \geq 2$ .
- Prove that  $\mathbf{C}$  is invertible if and only if  $p_{\mathbf{c}}$  and  $x^n - 1$  are coprime in  $\mathbb{C}[x]$ .
- Give an example of a  $4 \times 4$  circulant with integer entries, none of them 0, whose determinant is 0.

*Solution.*

- This is the last claim of [Theorem 11.9.4](#): from  $\mathbf{C} = \mathbf{F}\mathbf{D}\mathbf{F}^*$  and  $\det(\mathbf{F}^*)\det(\mathbf{F}) = \det(\mathbf{F}^*\mathbf{F}) = 1$ , multiplicativity of the determinant ([Theorem 6.5.4](#)) gives  $\det \mathbf{C} = \det \mathbf{D}$ , which is the product of the diagonal entries  $p_{\mathbf{c}}(\omega^j)$ .  
For the all-ones matrix,  $\mathbf{c} = (1, 1, \dots, 1)$  and  $p_{\mathbf{c}}(x) = 1 + x + \dots + x^{n-1}$ . Then  $p_{\mathbf{c}}(1) = n$ , while for  $1 \leq j \leq n-1$  the number  $z = \omega^j$  satisfies  $z^n = 1$  and  $z \neq 1$ , so  $p_{\mathbf{c}}(z) = (z^n - 1)/(z - 1) = 0$ . Hence  $\det \mathbf{C} = n \cdot 0 \cdots 0 = 0$  for every  $n \geq 2$ , as it must be, since all the rows are equal.
- By (a),  $\det \mathbf{C} \neq 0$  if and only if  $p_{\mathbf{c}}(\omega^j) \neq 0$  for every  $j$ , that is, if and only if  $p_{\mathbf{c}}$  and  $x^n - 1$  have no common root, the roots of  $x^n - 1$  being exactly  $1, \omega, \dots, \omega^{n-1}$  ([Example \(Roots of Unity\)](#)). Over  $\mathbb{C}$ , having no common root is the same as having no common non-constant factor: such a factor would have a root by [Theorem 5.5.1](#), and that root would be common to both; conversely a common root  $\alpha$  makes  $x - \alpha$  a common factor. So invertibility of  $\mathbf{C}$  is equivalent to  $p_{\mathbf{c}}$  and  $x^n - 1$  being coprime ([Definition 5.2.3](#)). Combined with [Theorem 6.5.2](#), this is the statement asked for.
- Take  $\mathbf{c} = (1, 2, 1, 2)$ , so

$$\mathbf{C} = \begin{pmatrix} 1 & 2 & 1 & 2 \\ 2 & 1 & 2 & 1 \\ 1 & 2 & 1 & 2 \\ 2 & 1 & 2 & 1 \end{pmatrix}, \quad p_{\mathbf{c}}(x) = 1 + 2x + x^2 + 2x^3.$$

Here  $p_{\mathbf{c}}(i) = 1 + 2i - 1 - 2i = 0$ , so  $\det \mathbf{C} = 0$  by (a). The reason is visible without any of the theory — rows 0 and 2 are equal — and the Fourier picture names a kernel vector:  $\mathbf{C}\mathbf{f}_1 = \mathbf{0}$ , and since  $\mathbf{C}$  is real, the real and imaginary parts of  $\mathbf{f}_1$  are in the kernel too, giving  $\mathbf{C}(1, 0, -1, 0) = \mathbf{0}$ .

### Exercise C3: The real orthogonal form of a real circulant.

Let  $\mathbf{C} \in M_4(\mathbb{R})$  be the circulant with first row  $(1, 2, 3, 4)$ , with eigenvalues  $10, -2 -$



$2i, -2, -2 + 2i$  as in [Example \(The running example, diagonalized\)](#). Put

$$\mathbf{g} := \sqrt{2} \operatorname{Re} \mathbf{f}_1, \quad \mathbf{h} := \sqrt{2} \operatorname{Im} \mathbf{f}_1,$$

the real and imaginary parts taken entrywise.

- Write  $\mathbf{g}$  and  $\mathbf{h}$  out, and show that  $(\mathbf{f}_0, \mathbf{f}_2, \mathbf{g}, \mathbf{h})$  is an orthonormal basis of  $\mathbb{R}^4$ .
- Writing  $a + bi := -2 + 2i$ , prove that  $\mathbf{C}\mathbf{g} = a\mathbf{g} + b\mathbf{h}$  and  $\mathbf{C}\mathbf{h} = -b\mathbf{g} + a\mathbf{h}$ .
- Hence write down the matrix of  $\mathbf{C}$  in the ordered basis of (a), and identify it as an instance of [Theorem 11.5.3](#).

*Hint for (b): apply  $\mathbf{C}$  to  $\mathbf{f}_3 = (\mathbf{g} - i\mathbf{h})/\sqrt{2}$  and compare real and imaginary parts.*

*Solution.*

- With  $\omega = i$ ,  $\mathbf{f}_1 = \frac{1}{2}(1, i, -1, -i)$ , so

$$\mathbf{g} = \frac{1}{\sqrt{2}}(1, 0, -1, 0), \quad \mathbf{h} = \frac{1}{\sqrt{2}}(0, 1, 0, -1).$$

Also  $\mathbf{f}_0 = \frac{1}{2}(1, 1, 1, 1)$  and  $\mathbf{f}_2 = \frac{1}{2}(1, -1, 1, -1)$  are real, so all four vectors lie in  $\mathbb{R}^4$ . Each has norm 1: for instance  $\|\mathbf{g}\|^2 = \frac{1}{2}(1+1) = 1$  and  $\|\mathbf{f}_0\|^2 = \frac{1}{4} \cdot 4 = 1$ . They are pairwise orthogonal for the real dot product. First  $\mathbf{g} \cdot \mathbf{h} = 0$ , because the non-zero entries sit in different slots, and  $\mathbf{f}_0 \cdot \mathbf{f}_2 = \frac{1}{4}(1 - 1 + 1 - 1) = 0$ . The four remaining pairings are

$$\begin{aligned} \mathbf{f}_0 \cdot \mathbf{g} &= \frac{1}{2\sqrt{2}}(1 - 1) = 0, & \mathbf{f}_2 \cdot \mathbf{g} &= \frac{1}{2\sqrt{2}}(1 - 1) = 0, \\ \mathbf{f}_0 \cdot \mathbf{h} &= \frac{1}{2\sqrt{2}}(1 - 1) = 0, & \mathbf{f}_2 \cdot \mathbf{h} &= \frac{1}{2\sqrt{2}}(-1 + 1) = 0. \end{aligned}$$

Four orthonormal vectors are independent ([Theorem 10.1.4](#)) and there are  $\dim \mathbb{R}^4 = 4$  of them, so they form a basis ([Theorem 1.6.4](#)).

- From  $\mathbf{f}_1 = \frac{1}{\sqrt{2}}(\mathbf{g} + i\mathbf{h})$  and  $\mathbf{f}_3 = \overline{\mathbf{f}_1}$  we get  $\mathbf{f}_3 = \frac{1}{\sqrt{2}}(\mathbf{g} - i\mathbf{h})$ . Its eigenvalue is  $p_{\mathbf{C}}(\omega^3) = p_{\mathbf{C}}(-i) = -2 + 2i = a + bi$ , so  $a = -2$  and  $b = 2$ , and  $\mathbf{C}\mathbf{f}_3 = (a + bi)\mathbf{f}_3$  reads

$$\frac{1}{\sqrt{2}}(\mathbf{C}\mathbf{g} - i\mathbf{C}\mathbf{h}) = \frac{1}{\sqrt{2}}(a + bi)(\mathbf{g} - i\mathbf{h}) = \frac{1}{\sqrt{2}}((a\mathbf{g} + b\mathbf{h}) + i(b\mathbf{g} - a\mathbf{h})).$$

Both  $\mathbf{C}\mathbf{g}$  and  $\mathbf{C}\mathbf{h}$  are real vectors, because  $\mathbf{C}$ ,  $\mathbf{g}$  and  $\mathbf{h}$  are real. Comparing real parts gives  $\mathbf{C}\mathbf{g} = a\mathbf{g} + b\mathbf{h}$ , and comparing imaginary parts gives  $-\mathbf{C}\mathbf{h} = b\mathbf{g} - a\mathbf{h}$ , that is  $\mathbf{C}\mathbf{h} = -b\mathbf{g} + a\mathbf{h}$ . With  $a = -2$  and  $b = 2$ ,

$$\mathbf{C}\mathbf{g} = -2\mathbf{g} + 2\mathbf{h}, \quad \mathbf{C}\mathbf{h} = -2\mathbf{g} - 2\mathbf{h}.$$

A direct check:  $\sqrt{2}\mathbf{C}\mathbf{g} = \mathbf{C}(1, 0, -1, 0) = (1-3, 4-2, 3-1, 2-4) = (-2, 2, 2, -2)$ , while  $\sqrt{2}(-2\mathbf{g} + 2\mathbf{h}) = (-2, 2, 2, -2)$ .

- Reading the coordinates of the images off (b), the matrix in the ordered basis  $(\mathbf{f}_0, \mathbf{f}_2, \mathbf{g}, \mathbf{h})$  is

$$\begin{pmatrix} 10 & 0 & 0 & 0 \\ 0 & -2 & 0 & 0 \\ 0 & 0 & -2 & -2 \\ 0 & 0 & 2 & -2 \end{pmatrix} = (10) \oplus (-2) \oplus \mathbf{A}(-2 + 2i),$$

since  $\Lambda(a + bi)$  has rows  $(a, -b)$  and  $(b, a)$ . This is exactly the shape promised by [Theorem 11.5.3](#) for a real normal operator: two  $1 \times 1$  blocks for the real eigenvalues, and one rotation-scaling block for the conjugate pair, with  $b = 2 > 0$ . The change of basis is orthogonal, because the basis is orthonormal.

## 11.10 The Sylvester Equation

Chapter 7 wrote an operator with an invariant subspace as a block triangular matrix, with a corner block  $\mathbf{C}$  recording how the subspace fails to have an invariant complement ([Theorem 7.1.7](#)). Chapter 9 spent its length on what to do when that corner cannot be cleared. The question of *when* it can be cleared has an exact answer, and the answer is a linear equation in the unknown matrix  $\mathbf{X}$ :

$$\mathbf{AX} - \mathbf{XB} = \mathbf{C}.$$

Chapter 7 already turned this equation into an ordinary linear system with the Kronecker product, and stopped one step short of the criterion because that step needs eigenvalues. We have eigenvalues now, and Schur's factorization of Section 1 to organize them. This section proves the criterion, uses it to clear corners, and reads off what it says about the matrices that commute with a given one.

Throughout, the field is  $\mathbb{C}$  unless stated otherwise, and  $\mathbf{A} \in M_m(\mathbb{C})$ ,  $\mathbf{B} \in M_n(\mathbb{C})$ , while the unknown  $\mathbf{X}$  and the data  $\mathbf{C}$  live in  $M_{m \times n}(\mathbb{C})$ .

### 11.10.1 The Sylvester operator

The left-hand side of the equation is linear in  $\mathbf{X}$ : replacing  $\mathbf{X}$  by  $\mathbf{X} + \mathbf{X}'$  or by  $c\mathbf{X}$  does the same to  $\mathbf{AX} - \mathbf{XB}$ , because matrix multiplication is bilinear ([Theorem 0.9.4](#)). So the whole question is about one linear map, and it deserves a name.

*Solving  $\mathbf{AX} - \mathbf{XB} = \mathbf{C}$  means inverting a single linear operator on the space of  $m \times n$  matrices.*

#### Definition 11.10.1 — Sylvester operator

Let  $\mathbf{A} \in M_m(F)$  and  $\mathbf{B} \in M_n(F)$ . The **Sylvester operator** of the pair  $(\mathbf{A}, \mathbf{B})$  is

$$\mathcal{S}_{\mathbf{A}, \mathbf{B}}: M_{m \times n}(F) \rightarrow M_{m \times n}(F), \quad \mathcal{S}_{\mathbf{A}, \mathbf{B}}(\mathbf{X}) = \mathbf{AX} - \mathbf{XB}.$$

The equation  $\mathbf{AX} - \mathbf{XB} = \mathbf{C}$ , for a given  $\mathbf{C} \in M_{m \times n}(F)$ , is the **Sylvester equation**.

In words:  $\mathcal{S}_{\mathbf{A}, \mathbf{B}}$  multiplies by  $\mathbf{A}$  on the left, multiplies by  $\mathbf{B}$  on the right, and subtracts. It is an operator on a space of dimension  $mn$ , not on  $F^m$  or  $F^n$ , and that is the only thing that makes the equation look harder than  $\mathbf{Ax} = \mathbf{b}$ .

Three special cases are worth keeping in view.

1.  $\mathbf{B} = \mathbf{0}$ . Then  $\mathcal{S}_{\mathbf{A}, \mathbf{0}}(\mathbf{X}) = \mathbf{AX}$ , and the equation is  $n$  copies of  $\mathbf{Ax} = \mathbf{b}$ , one for each column of  $\mathbf{X}$  ([Exercise \(B2: One-sided equations\)](#)). It is uniquely solvable for every  $\mathbf{C}$  exactly when  $\mathbf{A}$  is invertible, that is, when  $0 \notin \text{spec}(\mathbf{A})$ . Since  $\text{spec}(\mathbf{0}) = \{0\}$ , this already matches the criterion we are heading for.

2.  $\mathbf{B} = \mathbf{A}$  and  $\mathbf{C} = \mathbf{0}$ . Then  $\ker \mathcal{S}_{\mathbf{A},\mathbf{A}}$  is the **commutant** of  $\mathbf{A}$ , the set of matrices commuting with  $\mathbf{A}$ . This is the question Chapter 7 opened with, and Chapter 9 §04 studied it for a single Jordan or Weyr block.
3.  $m = n = 1$ . Then  $\mathcal{S}_{a,b}(x) = (a - b)x$ , which is invertible exactly when  $a \neq b$ . The general criterion will be this one, run over all pairs of eigenvalues.

**Non-example by minimal change.** The map  $\mathbf{X} \mapsto \mathbf{A}\mathbf{X} - \mathbf{X}\mathbf{B} - \mathbf{X}^\top$  is still linear on  $M_n(F)$ , but it is not a Sylvester operator: the transpose is not of the form “multiply on one side”. Everything below uses the two-sided product shape and nothing else.

Chapter 7 already wrote the matrix of  $\mathcal{S}_{\mathbf{A},\mathbf{B}}$ . Stacking the columns of  $\mathbf{X}$  with the vectorization  $\text{vec}$  of Definition 7.4.2 and applying the  $\text{vec}$  identity Theorem 7.4.4 twice, once with  $\mathbf{B} = \mathbf{I}_n$  and once with  $\mathbf{A} = \mathbf{I}_m$ ,

$$\begin{aligned} \text{vec}(\mathcal{S}_{\mathbf{A},\mathbf{B}}(\mathbf{X})) &= \mathbf{K} \text{vec} \mathbf{X}, \\ \mathbf{K} &:= \mathbf{I}_n \otimes \mathbf{A} - \mathbf{B}^\top \otimes \mathbf{I}_m. \end{aligned} \tag{11.10.1}$$

Here  $\mathbf{K} \in M_{mn}(F)$ , and  $\text{vec}: M_{m \times n}(F) \rightarrow F^{mn}$  is an isomorphism. So Eq. 11.10.1 says that  $\mathbf{K}$  is the matrix of  $\mathcal{S}_{\mathbf{A},\mathbf{B}}$  with respect to the basis of matrix units ordered column by column. In particular  $\mathcal{S}_{\mathbf{A},\mathbf{B}}$  is invertible if and only if  $\mathbf{K}$  is.

That is exactly where Chapter 7 left the problem. What it could not do was compute the eigenvalues of  $\mathbf{K}$ .

## 11.10.2 The spectrum of a Kronecker sum

Chapter 7 proved how the Kronecker product treats rank, trace and determinant (Theorem 7.4.3) and how it treats products, transposes and inverses (Theorem 7.4.1). It proved nothing about eigenvalues, and Exercise (C2: Kronecker sums) got only as far as the case where both factors are already triangular. Schur closes the gap, because over  $\mathbb{C}$  every matrix can be made triangular unitarily, and the mixed product rule carries a pair of unitary conjugations through the Kronecker product in one step.

### Lemma 11.10.1 — The Eigenvalues of a Kronecker Sum

Let  $\mathbf{A} \in M_m(\mathbb{C})$  with eigenvalues  $\lambda_1, \dots, \lambda_m$  and let  $\mathbf{B} \in M_n(\mathbb{C})$  with eigenvalues  $\mu_1, \dots, \mu_n$ , each list repeated according to algebraic multiplicity. Then the  $mn$  eigenvalues of

$$\mathbf{K} = \mathbf{I}_n \otimes \mathbf{A} - \mathbf{B}^\top \otimes \mathbf{I}_m \in M_{mn}(\mathbb{C}),$$

listed with algebraic multiplicity, are the  $mn$  differences  $\lambda_i - \mu_j$  ( $1 \leq i \leq m$ ,  $1 \leq j \leq n$ ).

*Idea.* Triangularize both ingredients at once. Schur (Corollary 11.1.3) writes  $\mathbf{A} = \mathbf{U}\mathbf{T}\mathbf{U}^*$  and  $\mathbf{B}^\top = \mathbf{V}\mathbf{S}\mathbf{V}^*$  with  $\mathbf{T}, \mathbf{S}$  upper triangular. The mixed product rule then conjugates both summands of  $\mathbf{K}$  by the single unitary matrix  $\mathbf{V} \otimes \mathbf{U}$ , because  $\mathbf{I}_n \otimes \mathbf{A}$  and  $\mathbf{B}^\top \otimes \mathbf{I}_m$  each have one trivial factor to absorb the other conjugation. What is left is  $\mathbf{I}_n \otimes \mathbf{T} - \mathbf{S} \otimes \mathbf{I}_m$ , and a Kronecker product with one triangular and one identity factor is triangular, so the eigenvalues are visible on the diagonal.

*Proof.* By [Corollary 11.1.3](#) there are a unitary  $\mathbf{U} \in M_m(\mathbb{C})$  and an upper triangular  $\mathbf{T} \in M_m(\mathbb{C})$  with  $\mathbf{A} = \mathbf{U}\mathbf{T}\mathbf{U}^*$  and  $t_{ii} = \lambda_i$ . By [Proposition 8.10.5](#) (b),  $p_{\mathbf{B}^\top} = p_{\mathbf{B}}$ , so  $\mathbf{B}^\top$  has eigenvalue list  $\mu_1, \dots, \mu_n$ ; applying [Corollary 11.1.3](#) to  $\mathbf{B}^\top$  gives a unitary  $\mathbf{V} \in M_n(\mathbb{C})$  and an upper triangular  $\mathbf{S} \in M_n(\mathbb{C})$  with  $\mathbf{B}^\top = \mathbf{V}\mathbf{S}\mathbf{V}^*$  and  $s_{jj} = \mu_j$ .

Put  $\mathbf{W} = \mathbf{V} \otimes \mathbf{U} \in M_{mn}(\mathbb{C})$ . Conjugating every entry of  $\mathbf{A} \otimes \mathbf{B}$  conjugates every block  $a_{ij}\mathbf{B}$ , so  $\overline{\mathbf{A} \otimes \mathbf{B}} = \overline{\mathbf{A}} \otimes \overline{\mathbf{B}}$  ([Definition 7.4.1](#)), and combining this with [Theorem 7.4.1](#) (d) gives  $(\mathbf{A} \otimes \mathbf{B})^* = \mathbf{A}^* \otimes \mathbf{B}^*$  whenever the product is defined. Hence  $\mathbf{W}^* = \mathbf{V}^* \otimes \mathbf{U}^*$ , and by the mixed product rule [Theorem 7.4.1](#) (c),

$$\mathbf{W}^*\mathbf{W} = (\mathbf{V}^*\mathbf{V}) \otimes (\mathbf{U}^*\mathbf{U}) = \mathbf{I}_n \otimes \mathbf{I}_m = \mathbf{I}_{mn},$$

so  $\mathbf{W}$  is unitary. Using the mixed product rule twice more,

$$\mathbf{W}(\mathbf{I}_n \otimes \mathbf{T})\mathbf{W}^* = (\mathbf{V}\mathbf{I}_n\mathbf{V}^*) \otimes (\mathbf{U}\mathbf{T}\mathbf{U}^*) = \mathbf{I}_n \otimes \mathbf{A},$$

$$\mathbf{W}(\mathbf{S} \otimes \mathbf{I}_m)\mathbf{W}^* = (\mathbf{V}\mathbf{S}\mathbf{V}^*) \otimes (\mathbf{U}\mathbf{I}_m\mathbf{U}^*) = \mathbf{B}^\top \otimes \mathbf{I}_m.$$

Subtracting,  $\mathbf{W}(\mathbf{I}_n \otimes \mathbf{T} - \mathbf{S} \otimes \mathbf{I}_m)\mathbf{W}^* = \mathbf{K}$ , so  $\mathbf{K}$  is similar to

$$\mathbf{R} := \mathbf{I}_n \otimes \mathbf{T} - \mathbf{S} \otimes \mathbf{I}_m.$$

**R is upper triangular.** Label the rows and columns of an  $mn \times mn$  matrix by pairs as in [Definition 7.4.1](#): row  $(j-1)m + i$  with  $1 \leq j \leq n$ ,  $1 \leq i \leq m$ , and likewise column  $(l-1)m + k$ . In those labels the entries of the two summands are

$$(\mathbf{I}_n \otimes \mathbf{T})_{(j-1)m+i, (l-1)m+k} = \delta_{jl} t_{ik},$$

$$(\mathbf{S} \otimes \mathbf{I}_m)_{(j-1)m+i, (l-1)m+k} = s_{jl} \delta_{ik}.$$

The first is non-zero only when  $j = l$  and  $i \leq k$ , since  $\mathbf{T}$  is upper triangular; then  $(j-1)m + i \leq (l-1)m + k$ . The second is non-zero only when  $j \leq l$  and  $i = k$ ; if  $j = l$  the two labels are equal, and if  $j < l$  then

$$(j-1)m + i \leq (j-1)m + m = jm \leq (l-1)m < (l-1)m + k,$$

using  $i \leq m$ ,  $j \leq l-1$  and  $k \geq 1$ . So each summand vanishes below the diagonal, and so does  $\mathbf{R}$ .

**The diagonal.** Taking  $j = l$  and  $i = k$  above, the diagonal entry of  $\mathbf{R}$  in position  $(j-1)m + i$  is  $t_{ii} - s_{jj} = \lambda_i - \mu_j$ . As  $(i, j)$  runs over all  $mn$  pairs, so does the position  $(j-1)m + i$ , by division with remainder.

By [Theorem 8.7.4](#) (a), applied to the operator  $T_{\mathbf{R}}$  on  $\mathbb{C}^{mn}$  and the standard basis,  $p_{\mathbf{R}} = \prod_{i,j} (x - (\lambda_i - \mu_j))$ . Since  $\mathbf{K} \sim \mathbf{R}$ , [Theorem 6.10.3](#) gives  $p_{\mathbf{K}} = p_{\mathbf{R}}$ . This proves the lemma. ■

Two remarks on what the proof used. The hypothesis  $F = \mathbb{C}$  entered once, in Schur, and through Schur it is the Fundamental Theorem of Algebra again. And the unitarity of  $\mathbf{W}$  was a convenience, not a necessity: any similarity triangularizing  $\mathbf{A}$  and  $\mathbf{B}^\top$  would do, but Schur hands us one for free and the Kronecker product of two unitary matrices is unitary, which is worth recording on its own.

## 11.10.3 When the equation is uniquely solvable

The criterion now costs one line.

### Theorem 11.10.2 — Sylvester's Theorem

Let  $\mathbf{A} \in M_m(\mathbb{C})$  and  $\mathbf{B} \in M_n(\mathbb{C})$ . The following are equivalent.

- (a)  $\text{spec}(\mathbf{A}) \cap \text{spec}(\mathbf{B}) = \emptyset$ : the two matrices have **no common eigenvalue**.
- (b) The Sylvester operator  $\mathcal{S}_{\mathbf{A},\mathbf{B}}: \mathbf{X} \mapsto \mathbf{A}\mathbf{X} - \mathbf{X}\mathbf{B}$  is invertible on  $M_{m \times n}(\mathbb{C})$ .
- (c) For every  $\mathbf{C} \in M_{m \times n}(\mathbb{C})$ , the equation  $\mathbf{A}\mathbf{X} - \mathbf{X}\mathbf{B} = \mathbf{C}$  has **exactly one** solution  $\mathbf{X}$ .
- (d) The only  $\mathbf{X} \in M_{m \times n}(\mathbb{C})$  with  $\mathbf{A}\mathbf{X} = \mathbf{X}\mathbf{B}$  is  $\mathbf{X} = \mathbf{0}$ .

*Proof.* By Eq. 11.10.1 the isomorphism  $\text{vec}$  turns  $\mathcal{S}_{\mathbf{A},\mathbf{B}}$  into the matrix  $\mathbf{K} = \mathbf{I}_n \otimes \mathbf{A} - \mathbf{B}^\top \otimes \mathbf{I}_m$ : a matrix  $\mathbf{X}$  solves  $\mathbf{A}\mathbf{X} - \mathbf{X}\mathbf{B} = \mathbf{C}$  if and only if the vector  $\text{vec } \mathbf{X}$  solves  $\mathbf{K}\mathbf{y} = \text{vec } \mathbf{C}$ , and every  $\mathbf{y} \in \mathbb{C}^{mn}$  and every  $\mathbf{z} \in \mathbb{C}^{mn}$  arise this way. Transported through  $\text{vec}$ , statements (b), (c) and (d) say respectively that  $\mathbf{K}$  is invertible, that  $\mathbf{K}\mathbf{y} = \mathbf{z}$  has exactly one solution for every  $\mathbf{z}$ , and that  $\mathbf{K}\mathbf{y} = \mathbf{0}$  forces  $\mathbf{y} = \mathbf{0}$ . These three are equivalent by Theorem 2.4.4, parts (a), (e) and (b): (e) gives existence for every  $\mathbf{z}$ , and (b) upgrades it to uniqueness, since two solutions differ by a member of the null space.

It remains to match (a) with (b). By Theorem 8.2.6,  $\mathbf{K}$  is invertible if and only if  $0 \notin \text{spec}(\mathbf{K})$ . By Lemma 11.10.1,  $\text{spec}(\mathbf{K}) = \{\lambda_i - \mu_j\}$ , where  $\lambda_1, \dots, \lambda_m$  and  $\mu_1, \dots, \mu_n$  list the eigenvalues of  $\mathbf{A}$  and of  $\mathbf{B}$ . So  $0 \in \text{spec}(\mathbf{K})$  if and only if  $\lambda_i = \mu_j$  for some pair  $(i, j)$ , that is, if and only if  $\mathbf{A}$  and  $\mathbf{B}$  have a common eigenvalue. Negating both sides matches (a) with (b), and the proof is complete. ■

This is the statement Chapter 7 promised. At the end of Example (A Sylvester Equation as a Linear System) it was said that over  $\mathbb{C}$  the equation  $\mathbf{A}\mathbf{X} - \mathbf{X}\mathbf{B} = \mathbf{C}$  is uniquely solvable for every  $\mathbf{C}$  exactly when  $\mathbf{A}$  and  $\mathbf{B}$  have no common eigenvalue, and that the criterion needs eigenvalues and would be proved in Chapter 11. Parts (a) and (c) of Theorem 11.10.2 are that sentence, and the eigenvalues it needed were supplied by Lemma 11.10.1. The quantitative version — *how badly* conditioned the equation is when the spectra are close but disjoint — is a separate matter, and belongs with perturbation theory in Chapter 19.

Part (d) is the version used most often in practice, because it is a statement with no data in it: to show that some matrix is zero, exhibit it as a solution of  $\mathbf{A}\mathbf{X} = \mathbf{X}\mathbf{B}$  for a pair with disjoint spectra. The corollary on commutants below is that move and nothing else.

#### Example. (Solving a Sylvester Equation)

Over  $\mathbb{C}$ , solve  $\mathbf{A}\mathbf{X} - \mathbf{X}\mathbf{B} = \mathbf{C}$  for

$$\mathbf{A} = \begin{pmatrix} 2 & 1 \\ 0 & 3 \end{pmatrix}, \quad \mathbf{B} = \begin{pmatrix} 1 & 0 \\ 2 & -1 \end{pmatrix}, \quad \mathbf{C} = \begin{pmatrix} 4 & -2 \\ 0 & 4 \end{pmatrix}.$$

*Solution.* The matrix  $\mathbf{A}$  is upper triangular, so  $\text{spec}(\mathbf{A}) = \{2, 3\}$  (Theorem 8.7.4);  $\mathbf{B}$  is lower triangular, so  $\text{spec}(\mathbf{B}) = \{1, -1\}$ . The spectra are disjoint, and by Theorem 11.10.2 there is exactly one solution.

To find it, use Eq. 11.10.1. Here  $\mathbf{B}^\top = \begin{pmatrix} 1 & 2 \\ 0 & -1 \end{pmatrix}$ , so by Definition 7.4.1

$$\mathbf{I}_2 \otimes \mathbf{A} = \begin{pmatrix} 2 & 1 & 0 & 0 \\ 0 & 3 & 0 & 0 \\ 0 & 0 & 2 & 1 \\ 0 & 0 & 0 & 3 \end{pmatrix}, \quad \mathbf{B}^\top \otimes \mathbf{I}_2 = \begin{pmatrix} 1 & 0 & 2 & 0 \\ 0 & 1 & 0 & 2 \\ 0 & 0 & -1 & 0 \\ 0 & 0 & 0 & -1 \end{pmatrix},$$

and their difference is

$$\mathbf{K} = \begin{pmatrix} 1 & 1 & -2 & 0 \\ 0 & 2 & 0 & -2 \\ 0 & 0 & 3 & 1 \\ 0 & 0 & 0 & 4 \end{pmatrix}.$$

Its diagonal entries are 1, 2, 3, 4, which are the four differences  $2 - 1$ ,  $3 - 1$ ,  $2 - (-1)$ ,  $3 - (-1)$ , as Lemma 11.10.1 predicts; none is zero, confirming invertibility.

With  $\text{vec } \mathbf{X} = (x_{11}, x_{21}, x_{12}, x_{22})$  and  $\text{vec } \mathbf{C} = (4, 0, -2, 4)$ , back substitution in  $\mathbf{K} \text{vec } \mathbf{X} = \text{vec } \mathbf{C}$  gives  $4x_{22} = 4$ , so  $x_{22} = 1$ ; then  $3x_{12} + 1 = -2$ , so  $x_{12} = -1$ ; then  $2x_{21} - 2 = 0$ , so  $x_{21} = 1$ ; then  $x_{11} + 1 + 2 = 4$ , so  $x_{11} = 1$ . Hence

$$\mathbf{X} = \begin{pmatrix} 1 & -1 \\ 1 & 1 \end{pmatrix}.$$

*Check:*  $\mathbf{A}\mathbf{X} = \begin{pmatrix} 3 & -1 \\ 3 & 3 \end{pmatrix}$  and  $\mathbf{X}\mathbf{B} = \begin{pmatrix} -1 & 1 \\ 3 & -1 \end{pmatrix}$ , and their difference is  $\mathbf{C}$ .

*Quick check.* Let  $\mathbf{A} \in M_3(\mathbb{C})$  have eigenvalues 1, 2, 2 and let  $\mathbf{B} \in M_2(\mathbb{C})$  have eigenvalues 0, 3. List the eigenvalues of  $\mathcal{S}_{\mathbf{A}, \mathbf{B}}$  with multiplicity, and decide whether  $\mathbf{A}\mathbf{X} - \mathbf{X}\mathbf{B} = \mathbf{C}$  is uniquely solvable for every  $\mathbf{C} \in M_{3 \times 2}(\mathbb{C})$ .

*Solution.* By Lemma 11.10.1 the six eigenvalues are the differences  $\lambda_i - \mu_j$ :

$$1 - 0 = 1, \quad 2 - 0 = 2, \quad 2 - 0 = 2, \quad 1 - 3 = -2, \quad 2 - 3 = -1, \quad 2 - 3 = -1.$$

None is zero — equivalently  $\text{spec}(\mathbf{A}) = \{1, 2\}$  and  $\text{spec}(\mathbf{B}) = \{0, 3\}$  are disjoint — so by Theorem 11.10.2 the equation has exactly one solution for every  $\mathbf{C}$ .

**Warning. The criterion depends on the sign.** For the equation  $\mathbf{A}\mathbf{X} + \mathbf{X}\mathbf{B} = \mathbf{C}$ , which is the form Chapter 7 used in Example (A Sylvester Equation as a Linear System), the relevant operator is  $\mathcal{S}_{\mathbf{A}, -\mathbf{B}}$ , and  $\text{spec}(-\mathbf{B}) = \{-\mu_j\}$ . So the condition is  $\lambda_i + \mu_j \neq 0$  for all  $i, j$ , **not**  $\text{spec}(\mathbf{A}) \cap \text{spec}(\mathbf{B}) = \emptyset$ . The two conditions are genuinely different: with  $\mathbf{A} = \text{diag}(1, 2)$  and  $\mathbf{B} = \begin{pmatrix} -1 & 5 \\ 0 & 3 \end{pmatrix}$  the spectra  $\{1, 2\}$  and  $\{-1, 3\}$  are disjoint, yet  $1 + (-1) = 0$ , so  $\mathbf{A}\mathbf{X} + \mathbf{X}\mathbf{B} = \mathbf{C}$  is not solvable for every  $\mathbf{C}$ . Read the sign off the equation before quoting the criterion.

### 11.10.4 Clearing the corner

We can now answer the question the section opened with.

#### Corollary 11.10.3 — Block Diagonalization by a Sylvester Solution

Let  $\mathbf{A} \in M_m(F)$ ,  $\mathbf{B} \in M_n(F)$  and  $\mathbf{C} \in M_{m \times n}(F)$ , and put

$$\mathbf{M} = \begin{pmatrix} \mathbf{A} & \mathbf{C} \\ \mathbf{0} & \mathbf{B} \end{pmatrix} \in M_{m+n}(F).$$

- (a) If  $\mathbf{X} \in M_{m \times n}(F)$  satisfies  $\mathbf{A}\mathbf{X} - \mathbf{X}\mathbf{B} = -\mathbf{C}$ , then with  $\mathbf{P} = \begin{pmatrix} \mathbf{I}_m & \mathbf{X} \\ \mathbf{0} & \mathbf{I}_n \end{pmatrix}$  we have  $\mathbf{P}^{-1}\mathbf{M}\mathbf{P} = \mathbf{A} \oplus \mathbf{B}$ . In particular  $\mathbf{M} \sim \mathbf{A} \oplus \mathbf{B}$ .
- (b) Conversely, if  $\mathbf{P}^{-1}\mathbf{M}\mathbf{P} = \mathbf{A} \oplus \mathbf{B}$  for some  $\mathbf{X} \in M_{m \times n}(F)$  and the matrix  $\mathbf{P}$  built from it as in (a), then that  $\mathbf{X}$  satisfies  $\mathbf{A}\mathbf{X} - \mathbf{X}\mathbf{B} = -\mathbf{C}$ .
- (c) If  $F = \mathbb{C}$  and  $\text{spec}(\mathbf{A}) \cap \text{spec}(\mathbf{B}) = \emptyset$ , then a solution  $\mathbf{X}$  exists, and it is unique; so  $\mathbf{M} \sim \mathbf{A} \oplus \mathbf{B}$  for **every** corner  $\mathbf{C}$ .

*Idea.* The matrices  $\begin{pmatrix} \mathbf{I}_m & \mathbf{X} \\ \mathbf{0} & \mathbf{I}_n \end{pmatrix}$  are exactly the changes of basis that add multiples of the first block of basis vectors to the second and leave the first alone. Conjugating  $\mathbf{M}$  by one of them cannot touch the two diagonal blocks or create anything below the diagonal; all it can do is move the corner, and the block multiplication shows it moves the corner by  $\mathbf{A}\mathbf{X} - \mathbf{X}\mathbf{B}$ . So the corner can be driven to  $\mathbf{0}$  precisely when  $-\mathbf{C}$  lies in the image of the Sylvester operator.

*Proof.*

- (a) First,  $\mathbf{P}$  is invertible with

$$\mathbf{P}^{-1} = \begin{pmatrix} \mathbf{I}_m & -\mathbf{X} \\ \mathbf{0} & \mathbf{I}_n \end{pmatrix},$$

since multiplying the two matrices in either order gives  $\mathbf{I}_{m+n}$  by [Theorem 7.1.1](#), the partitions being conformable. The same theorem gives

$$\mathbf{M}\mathbf{P} = \begin{pmatrix} \mathbf{A} & \mathbf{A}\mathbf{X} + \mathbf{C} \\ \mathbf{0} & \mathbf{B} \end{pmatrix},$$

and then

$$\mathbf{P}^{-1}\mathbf{M}\mathbf{P} = \begin{pmatrix} \mathbf{A} & \mathbf{A}\mathbf{X} + \mathbf{C} - \mathbf{X}\mathbf{B} \\ \mathbf{0} & \mathbf{B} \end{pmatrix}.$$

By hypothesis  $\mathbf{A}\mathbf{X} - \mathbf{X}\mathbf{B} = -\mathbf{C}$ , so the corner is  $\mathbf{0}$  and the result is  $\mathbf{A} \oplus \mathbf{B}$  ([Definition 7.1.2](#)).

- (b) The computation in (a) used nothing about  $\mathbf{X}$  until the last sentence. So if that  $\mathbf{P}$  satisfies  $\mathbf{P}^{-1}\mathbf{M}\mathbf{P} = \mathbf{A} \oplus \mathbf{B}$ , comparing the corners of the two sides gives  $\mathbf{A}\mathbf{X} + \mathbf{C} - \mathbf{X}\mathbf{B} = \mathbf{0}$ , which is the equation.
- (c) By [Theorem 11.10.2](#) ((a)  $\Rightarrow$  (c)), the equation  $\mathbf{A}\mathbf{X} - \mathbf{X}\mathbf{B} = -\mathbf{C}$  has exactly one solution. Apply (a).

**Example. (Clearing a Corner)**

Block-diagonalize

$$\mathbf{M} = \begin{pmatrix} 2 & 1 & -4 & 2 \\ 0 & 3 & 0 & -4 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 2 & -1 \end{pmatrix} \in M_4(\mathbb{Q}).$$

*Solution.* Partition  $\mathbf{M}$  into  $2 \times 2$  blocks. The diagonal blocks are the matrices  $\mathbf{A}$  and  $\mathbf{B}$  of [Example \(Solving a Sylvester Equation\)](#), and the corner is

$$\begin{pmatrix} -4 & 2 \\ 0 & -4 \end{pmatrix} = -\mathbf{C}$$

for the  $\mathbf{C}$  of that example. So the equation to solve is  $\mathbf{A}\mathbf{X} - \mathbf{X}\mathbf{B} = -(-\mathbf{C}) = \mathbf{C}$ , which is the one already solved there, with the unique solution  $\mathbf{X} = \begin{pmatrix} 1 & -1 \\ 1 & 1 \end{pmatrix}$ . By [Corollary 11.10.3](#) (a), taking

$$\mathbf{P} = \begin{pmatrix} 1 & 0 & 1 & -1 \\ 0 & 1 & 1 & 1 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{pmatrix}$$

gives  $\mathbf{P}^{-1}\mathbf{M}\mathbf{P} = \mathbf{A} \oplus \mathbf{B}$ , that is,

$$\mathbf{P}^{-1}\mathbf{M}\mathbf{P} = \begin{pmatrix} 2 & 1 & 0 & 0 \\ 0 & 3 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 2 & -1 \end{pmatrix}.$$

The eigenvalues 2, 3, 1,  $-1$  are of course unchanged; what has changed is that the space has been split into two invariant pieces of dimension 2.

*Remark.* Part (b) restricts the shape of  $\mathbf{P}$ , and one may ask whether that is a real restriction: if  $\mathbf{M}$  is similar to  $\mathbf{A} \oplus \mathbf{B}$  by *any* invertible matrix, must the Sylvester equation be solvable? The answer is yes over every field. This is **Roth's removal rule**, and the proof is not a block computation: it goes through the characteristic matrix and the criterion  $\mathbf{A} \sim \mathbf{B}$  if and only if  $x\mathbf{I} - \mathbf{A} \approx x\mathbf{I} - \mathbf{B}$  over  $F[x]$  ([Theorem 9.7.5](#)). We do not need it here and do not prove it; nothing later in the book depends on it.

**Primary decomposition, seen again.** Put [Corollary 11.10.3](#) next to Schur and a familiar theorem drops out with no extra work. Let  $\mathbf{A} \in M_N(\mathbb{C})$  have distinct eigenvalues  $\lambda_1, \dots, \lambda_k$  with algebraic multiplicities  $a_1, \dots, a_k$ . By [Corollary 11.1.3](#) we may prescribe the order of the diagonal in a Schur factorization, so choose the order that lists all copies of  $\lambda_1$  first, then all copies of  $\lambda_2$ , and so on. Partitioning the resulting triangular matrix along  $N = a_1 + \dots + a_k$  makes it block upper triangular with diagonal blocks  $\mathbf{T}_1, \dots, \mathbf{T}_k$ , where  $\mathbf{T}_i$  is triangular with the single diagonal value  $\lambda_i$ , so  $\text{spec}(\mathbf{T}_i) = \{\lambda_i\}$ . Now peel the blocks off one at a time:  $\text{spec}(\mathbf{T}_1)$  is disjoint from the spectrum of the rest, so [Corollary 11.10.3](#) (c) splits off  $\mathbf{T}_1$ , and induction



on  $k$  finishes. The conclusion is that  $\mathbf{A} \sim \mathbf{T}_1 \oplus \cdots \oplus \mathbf{T}_k$  with  $\text{spec}(\mathbf{T}_i) = \{\lambda_i\}$  — the matrix form of the primary decomposition of [Theorem 8.8.1](#), here obtained from a triangular form and a linear equation rather than from the kernel-splitting lemma.

## 11.10.5 Commuting with a matrix of disjoint spectra

Here is the promised use of [Theorem 11.10.2](#) (d). It says that disjoint spectra force a block structure on anything that commutes.

### Corollary 11.10.4 — Disjoint Spectra Split the Commutant

Let  $\mathbf{A}_1 \in M_{n_1}(\mathbb{C}), \dots, \mathbf{A}_r \in M_{n_r}(\mathbb{C})$  have **pairwise disjoint** spectra, and put  $\mathbf{D} = \mathbf{A}_1 \oplus \cdots \oplus \mathbf{A}_r \in M_N(\mathbb{C})$ ,  $N = n_1 + \cdots + n_r$ . Let  $\mathbf{Z} \in M_N(\mathbb{C})$  satisfy  $\mathbf{D}\mathbf{Z} = \mathbf{Z}\mathbf{D}$ , and partition  $\mathbf{Z}$  into blocks  $\mathbf{Z}_{pq} \in M_{n_p \times n_q}(\mathbb{C})$  for the partition  $N = n_1 + \cdots + n_r$ . Then

$$\mathbf{Z}_{pq} = \mathbf{0} \quad \text{for all } p \neq q,$$

so  $\mathbf{Z} = \mathbf{Z}_{11} \oplus \cdots \oplus \mathbf{Z}_{rr}$  is block **diagonal**, and each diagonal block satisfies  $\mathbf{A}_p \mathbf{Z}_{pp} = \mathbf{Z}_{pp} \mathbf{A}_p$ .

*Proof.* By [Theorem 7.1.1](#), the  $(p, q)$  block of  $\mathbf{D}\mathbf{Z}$  is  $\mathbf{A}_p \mathbf{Z}_{pq}$  and the  $(p, q)$  block of  $\mathbf{Z}\mathbf{D}$  is  $\mathbf{Z}_{pq} \mathbf{A}_q$ , because  $\mathbf{D}$  has only one non-zero block in each block row and in each block column. So  $\mathbf{D}\mathbf{Z} = \mathbf{Z}\mathbf{D}$  is equivalent to

$$\mathbf{A}_p \mathbf{Z}_{pq} = \mathbf{Z}_{pq} \mathbf{A}_q \quad (1 \leq p, q \leq r).$$

Let  $p \neq q$ . Then  $\text{spec}(\mathbf{A}_p) \cap \text{spec}(\mathbf{A}_q) = \emptyset$  by hypothesis, so [Theorem 11.10.2](#) ((a)  $\Rightarrow$  (d)) forces  $\mathbf{Z}_{pq} = \mathbf{0}$ . The diagonal equations  $\mathbf{A}_p \mathbf{Z}_{pp} = \mathbf{Z}_{pp} \mathbf{A}_p$  are the remaining cases  $p = q$ , and they carry no information beyond commuting. ■

Read the statement twice, because a similar-sounding claim is false.

**Warning.** It is the disjointness of the spectra that does the work, not the block diagonal shape. Nothing here says that a matrix commuting with a block diagonal matrix is block diagonal, and nothing here says that a commutant consists of triangular matrices. Both fail as soon as two diagonal blocks share an eigenvalue. Chapter 9 §04 computes the commutant of  $\mathbf{J} = \mathbf{J}_2(0) \oplus \mathbf{J}_1(0)$  explicitly and finds it to be

$$\left\{ \begin{pmatrix} a & b & c \\ 0 & a & 0 \\ 0 & h & i \end{pmatrix} : a, b, c, h, i \in F \right\},$$

which contains matrices with a non-zero entry **below** the diagonal, in position  $(3, 2)$ . That does not contradict [Corollary 11.10.4](#): the two blocks  $\mathbf{J}_2(0)$  and  $\mathbf{J}_1(0)$  have the same spectrum  $\{0\}$ , so the corollary has nothing to say about them. What is true for a single repeated eigenvalue is weaker and needs a different normal form: the commutant of a basic **Weyr** matrix is block upper triangular for the Weyr partition ([Proposition 9.4.6](#)),

and the Jordan form has no such property. Note too that even when [Corollary 11.10.4](#) applies, it says nothing about the diagonal blocks  $\mathbf{Z}_{pp}$  themselves, which range over the whole commutant of  $\mathbf{A}_p$ .

The corollary is the reason spectral separation is so useful. Combined with the primary decomposition above, it says that over  $\mathbb{C}$  the commutant of any matrix splits along its distinct eigenvalues: after the change of basis that block-diagonalizes  $\mathbf{A}$  by spectrum, everything commuting with  $\mathbf{A}$  becomes block diagonal too, and the hard part of describing a commutant is always the single-eigenvalue part. The pleasant case of the same phenomenon appeared in [Theorem 11.7.1](#), where the pieces are honest eigenspaces and the splitting is orthogonal.

## 11.10.6 Lyapunov's equation

One specialization is important enough to have its own name.

### Definition 11.10.2 — Lyapunov equation

Let  $\mathbf{A} \in M_n(\mathbb{C})$  and  $\mathbf{Q} \in M_n(\mathbb{C})$ . The **Lyapunov equation** for the pair  $(\mathbf{A}, \mathbf{Q})$  is

$$\mathbf{A}\mathbf{X} + \mathbf{X}\mathbf{A}^* = -\mathbf{Q},$$

with unknown  $\mathbf{X} \in M_n(\mathbb{C})$ . Call  $\mathbf{A}$  **stable** when every  $\lambda \in \text{spec}(\mathbf{A})$  has  $\text{Re } \lambda < 0$ .

This is the Sylvester equation with  $\mathbf{B} = -\mathbf{A}^*$ , since  $\mathbf{A}\mathbf{X} - \mathbf{X}(-\mathbf{A}^*) = \mathbf{A}\mathbf{X} + \mathbf{X}\mathbf{A}^*$ . To apply [Theorem 11.10.2](#) we need  $\text{spec}(-\mathbf{A}^*)$ .

The eigenvalues of  $\mathbf{A}^*$  are the conjugates of those of  $\mathbf{A}$ . Indeed  $\det(\mathbf{M}^*) = \overline{\det \mathbf{M}}$  for every  $\mathbf{M} \in M_n(\mathbb{C})$ , because  $\det(\mathbf{M}^\top) = \det \mathbf{M}$  ([Theorem 6.4.5](#)) and the determinant is a sum of products of entries with coefficients  $\pm 1$  ([Definition 6.4.1](#)), so conjugating every entry conjugates the value. Applying this to  $\mathbf{M} = \lambda \mathbf{I} - \mathbf{A}$ , whose conjugate transpose is  $\bar{\lambda} \mathbf{I} - \mathbf{A}^*$ , gives  $\det(\bar{\lambda} \mathbf{I} - \mathbf{A}^*) = \overline{\det(\lambda \mathbf{I} - \mathbf{A})}$ . So one determinant vanishes exactly when the other does, and  $\lambda \in \text{spec}(\mathbf{A})$  if and only if  $\bar{\lambda} \in \text{spec}(\mathbf{A}^*)$  ([Theorem 8.2.1](#)).

Therefore  $\text{spec}(-\mathbf{A}^*) = \{-\bar{\lambda} : \lambda \in \text{spec}(\mathbf{A})\}$ , and [Theorem 11.10.2](#) says the Lyapunov equation has exactly one solution for every  $\mathbf{Q}$  precisely when

$$\lambda + \bar{\mu} \neq 0 \quad \text{for all } \lambda, \mu \in \text{spec}(\mathbf{A}).$$

If  $\mathbf{A}$  is stable this holds, because then  $\text{Re}(\lambda + \bar{\mu}) = \text{Re } \lambda + \text{Re } \mu < 0$ . So a stable matrix gives a uniquely solvable Lyapunov equation, whatever  $\mathbf{Q}$  is.

What makes the equation important is a statement we are not yet equipped to prove. Call a Hermitian matrix  $\mathbf{X}$  **positive definite** when  $\mathbf{x}^* \mathbf{X} \mathbf{x} > 0$  for every non-zero  $\mathbf{x} \in \mathbb{C}^n$ . Then: *if  $\mathbf{A}$  is stable and  $\mathbf{Q}$  is positive definite, the unique solution  $\mathbf{X}$  of  $\mathbf{A}\mathbf{X} + \mathbf{X}\mathbf{A}^* = -\mathbf{Q}$  is positive definite; and conversely, if some positive definite  $\mathbf{X}$  solves the equation for some positive definite  $\mathbf{Q}$ , then  $\mathbf{A}$  is stable.* Positivity is the subject of Chapter 12, where the tools to handle  $\mathbf{x}^* \mathbf{X} \mathbf{x}$  are built, and the consequence for differential equations — that a stable  $\mathbf{A}$  makes every solution of  $\dot{\mathbf{x}} = \mathbf{A}\mathbf{x}$  decay, with  $\mathbf{x}^* \mathbf{X} \mathbf{x}$  as the quantity that measures the decay — belongs to Chapter 15. We record the equation here because its *solvability*, which is the part Sylvester's theorem settles, is exactly what makes the rest possible.

## 11.10.7 Exercises

### 11.10.7.1 A. Check your understanding

#### Exercise A1.

- Define the Sylvester operator of a pair  $(\mathbf{A}, \mathbf{B})$  with  $\mathbf{A} \in M_m(F)$ ,  $\mathbf{B} \in M_n(F)$ , and state the dimension of the space it acts on.
- Write down the matrix of the Sylvester operator with respect to the vectorization  $\text{vec}$ , and say which Chapter 7 result produces it.
- State the criterion for  $\mathbf{A}\mathbf{X} - \mathbf{X}\mathbf{B} = \mathbf{C}$  to have exactly one solution for every  $\mathbf{C}$ , over  $\mathbb{C}$ .
- True or false: if  $\mathbf{A}\mathbf{X} - \mathbf{X}\mathbf{B} = \mathbf{C}$  has at least one solution for every  $\mathbf{C}$ , then that solution is unique. Justify your answer.
- True or false: if  $\text{spec}(\mathbf{A}) \cap \text{spec}(\mathbf{B}) = \emptyset$ , then  $\mathbf{A}\mathbf{X} + \mathbf{X}\mathbf{B} = \mathbf{C}$  has exactly one solution for every  $\mathbf{C}$ . Justify your answer.

*Solution.*

- $\mathcal{S}_{\mathbf{A}, \mathbf{B}}(\mathbf{X}) = \mathbf{A}\mathbf{X} - \mathbf{X}\mathbf{B}$  for  $\mathbf{X} \in M_{m \times n}(F)$  (Definition 11.10.1). It acts on  $M_{m \times n}(F)$ , of dimension  $mn$ .
- $\mathbf{K} = \mathbf{I}_n \otimes \mathbf{A} - \mathbf{B}^\top \otimes \mathbf{I}_m$ , by Eq. 11.10.1; it comes from the  $\text{vec}$  identity Theorem 7.4.4 applied twice, with  $\text{vec}(\mathbf{A}\mathbf{X}) = (\mathbf{I}_n \otimes \mathbf{A}) \text{vec } \mathbf{X}$  and  $\text{vec}(\mathbf{X}\mathbf{B}) = (\mathbf{B}^\top \otimes \mathbf{I}_m) \text{vec } \mathbf{X}$ .
- Exactly when  $\mathbf{A}$  and  $\mathbf{B}$  have no common eigenvalue, that is  $\text{spec}(\mathbf{A}) \cap \text{spec}(\mathbf{B}) = \emptyset$  (Theorem 11.10.2).
- True. Existence for every  $\mathbf{C}$  says  $\mathcal{S}_{\mathbf{A}, \mathbf{B}}$  is surjective; an operator on the finite-dimensional space  $M_{m \times n}(F)$  is surjective if and only if it is injective (Theorem 2.4.4 through  $\text{vec}$ , or Rank–Nullity), and injectivity is uniqueness.
- False. The criterion for the  $+$  form is  $\lambda + \mu \neq 0$  for all  $\lambda \in \text{spec}(\mathbf{A})$ ,  $\mu \in \text{spec}(\mathbf{B})$ , since  $\mathbf{A}\mathbf{X} + \mathbf{X}\mathbf{B} = \mathcal{S}_{\mathbf{A}, -\mathbf{B}}(\mathbf{X})$ . With  $\mathbf{A} = \text{diag}(1, 2)$  and  $\mathbf{B} = \text{diag}(-1, 3)$  the spectra are disjoint but  $1 + (-1) = 0$ , so the equation is not uniquely solvable.

### 11.10.7.2 B. Practice

#### Exercise B1: A Sylvester equation by vectorization.

Over  $\mathbb{Q}$ , let

$$\mathbf{A} = \begin{pmatrix} 3 & 1 \\ 0 & 1 \end{pmatrix}, \quad \mathbf{B} = \begin{pmatrix} -1 & 0 \\ 1 & 0 \end{pmatrix}, \quad \mathbf{C} = \begin{pmatrix} 2 & 1 \\ 0 & -2 \end{pmatrix}.$$

- Compute  $\text{spec}(\mathbf{A})$  and  $\text{spec}(\mathbf{B})$ , and decide whether  $\mathbf{A}\mathbf{X} - \mathbf{X}\mathbf{B} = \mathbf{C}$  has exactly one solution.
- Write down  $\mathbf{K} = \mathbf{I}_2 \otimes \mathbf{A} - \mathbf{B}^\top \otimes \mathbf{I}_2$ , and check that its diagonal entries are the four differences  $\lambda_i - \mu_j$ .
- Solve the equation.

*Solution.*

- (a)  $\mathbf{A}$  is upper triangular with diagonal 3, 1, so  $\text{spec}(\mathbf{A}) = \{3, 1\}$  (Theorem 8.7.4).

For  $\mathbf{B}$ ,  $p_{\mathbf{B}} = \det \begin{pmatrix} x+1 & 0 \\ -1 & x \end{pmatrix} = x(x+1)$ , so  $\text{spec}(\mathbf{B}) = \{0, -1\}$ . The spectra are disjoint, so by Theorem 11.10.2 there is exactly one solution for this (or any)  $\mathbf{C}$ .

- (b)  $\mathbf{B}^{\top} = \begin{pmatrix} -1 & 1 \\ 0 & 0 \end{pmatrix}$ , so

$$\mathbf{I}_2 \otimes \mathbf{A} = \begin{pmatrix} 3 & 1 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 3 & 1 \\ 0 & 0 & 0 & 1 \end{pmatrix}, \quad \mathbf{B}^{\top} \otimes \mathbf{I}_2 = \begin{pmatrix} -1 & 0 & 1 & 0 \\ 0 & -1 & 0 & 1 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \end{pmatrix},$$

and

$$\mathbf{K} = \begin{pmatrix} 4 & 1 & -1 & 0 \\ 0 & 2 & 0 & -1 \\ 0 & 0 & 3 & 1 \\ 0 & 0 & 0 & 1 \end{pmatrix}.$$

Its diagonal is 4, 2, 3, 1, and the differences  $\lambda_i - \mu_j$  are  $3 - (-1) = 4$ ,  $1 - (-1) = 2$ ,  $3 - 0 = 3$ ,  $1 - 0 = 1$ , as Lemma 11.10.1 asserts.

- (c) With  $\text{vec } \mathbf{X} = (x_{11}, x_{21}, x_{12}, x_{22})$  and  $\text{vec } \mathbf{C} = (2, 0, 1, -2)$ , back substitution gives  $x_{22} = -2$ ; then  $3x_{12} + x_{22} = 1$ , so  $x_{12} = 1$ ; then  $2x_{21} - x_{22} = 0$ , so  $x_{21} = -1$ ; then  $4x_{11} + x_{21} - x_{12} = 2$ , so  $x_{11} = 1$ . Hence

$$\mathbf{X} = \begin{pmatrix} 1 & 1 \\ -1 & -2 \end{pmatrix}.$$

Check:  $\mathbf{A}\mathbf{X} = \begin{pmatrix} 2 & 1 \\ -1 & -2 \end{pmatrix}$  and  $\mathbf{X}\mathbf{B} = \begin{pmatrix} 0 & 0 \\ -1 & 0 \end{pmatrix}$ , and their difference is  $\mathbf{C}$ .

### Exercise B2: Block-diagonalizing a $3 \times 3$ matrix.

Let

$$\mathbf{M} = \begin{pmatrix} 2 & 1 & 2 \\ 0 & 2 & 3 \\ 0 & 0 & 5 \end{pmatrix} \in M_3(\mathbb{Q}).$$

- (a) Partition  $\mathbf{M}$  as  $3 = 2 + 1$ , name the blocks  $\mathbf{A}$ ,  $\mathbf{B}$ ,  $\mathbf{C}$ , and explain why Corollary 11.10.3 applies.  
 (b) Solve the corresponding Sylvester equation.  
 (c) Write down  $\mathbf{P}$  and verify that  $\mathbf{P}^{-1}\mathbf{M}\mathbf{P} = \mathbf{A} \oplus \mathbf{B}$ .

*Solution.*

- (a) The blocks are  $\mathbf{A} = \begin{pmatrix} 2 & 1 \\ 0 & 2 \end{pmatrix}$ ,  $\mathbf{B} = (5)$  and  $\mathbf{C} = \begin{pmatrix} 2 \\ 3 \end{pmatrix}$ . Both diagonal blocks are triangular, so  $\text{spec}(\mathbf{A}) = \{2\}$  and  $\text{spec}(\mathbf{B}) = \{5\}$  (Theorem 8.7.4); these are disjoint, so Corollary 11.10.3 (c) applies and a unique  $\mathbf{X} \in M_{2 \times 1}(\mathbb{Q})$  exists.

- (b) With  $\mathbf{X} = \mathbf{x} = (x_1, x_2)$ , the equation  $\mathbf{A}\mathbf{x} - 5\mathbf{x} = -\mathbf{C}$  reads  $(\mathbf{A} - 5\mathbf{I}_2)\mathbf{x} = -\mathbf{C}$ , that

is

$$\begin{pmatrix} -3 & 1 \\ 0 & -3 \end{pmatrix} \begin{pmatrix} x_1 \\ x_2 \end{pmatrix} = \begin{pmatrix} -2 \\ -3 \end{pmatrix}.$$

The second row gives  $x_2 = 1$ , and the first then gives  $-3x_1 + 1 = -2$ , so  $x_1 = 1$ . Hence  $\mathbf{x} = (1, 1)$ .

(c) Taking

$$\mathbf{P} = \begin{pmatrix} 1 & 0 & 1 \\ 0 & 1 & 1 \\ 0 & 0 & 1 \end{pmatrix}, \quad \mathbf{P}^{-1} = \begin{pmatrix} 1 & 0 & -1 \\ 0 & 1 & -1 \\ 0 & 0 & 1 \end{pmatrix},$$

we get  $\mathbf{MP} = \begin{pmatrix} 2 & 1 & 5 \\ 0 & 2 & 5 \\ 0 & 0 & 5 \end{pmatrix}$  and then

$$\mathbf{P}^{-1}\mathbf{MP} = \begin{pmatrix} 2 & 1 & 0 \\ 0 & 2 & 0 \\ 0 & 0 & 5 \end{pmatrix} = \mathbf{A} \oplus \mathbf{B},$$

as [Corollary 11.10.3](#) (a) predicts.

### Exercise B3: Which equations are uniquely solvable?.

Determine, for each of the following, whether the stated equation has exactly one solution  $\mathbf{X} \in M_2(\mathbb{C})$  for **every** right-hand side  $\mathbf{C} \in M_2(\mathbb{C})$ . Justify your answer.

- (a)  $\mathbf{AX} - \mathbf{XB} = \mathbf{C}$  with  $\mathbf{A} = \begin{pmatrix} 1 & 1 \\ 0 & 1 \end{pmatrix}$ ,  $\mathbf{B} = \begin{pmatrix} 2 & 0 \\ 0 & 3 \end{pmatrix}$ .
- (b)  $\mathbf{AX} - \mathbf{XB} = \mathbf{C}$  with  $\mathbf{A} = \begin{pmatrix} 0 & 1 \\ -1 & 0 \end{pmatrix}$ ,  $\mathbf{B} = \begin{pmatrix} 0 & -1 \\ 1 & 0 \end{pmatrix}$ .
- (c)  $\mathbf{AX} + \mathbf{XB} = \mathbf{C}$  with  $\mathbf{A} = \begin{pmatrix} 1 & 0 \\ 0 & 2 \end{pmatrix}$ ,  $\mathbf{B} = \begin{pmatrix} -1 & 5 \\ 0 & 3 \end{pmatrix}$ .
- (d)  $\mathbf{AX} - \mathbf{XB} = \mathbf{C}$  with  $\mathbf{A} = \begin{pmatrix} 1 & 1 \\ 0 & 1 \end{pmatrix}$ ,  $\mathbf{B} = \begin{pmatrix} 1 & 0 \\ 2 & 1 \end{pmatrix}$ .
- (e)  $\mathbf{AX} - \mathbf{XB} = \mathbf{C}$  with  $\mathbf{A} \in M_2(\mathbb{C})$  nilpotent and  $\mathbf{B} \in M_2(\mathbb{C})$  invertible.

*Solution.*

- (a) Yes.  $\text{spec}(\mathbf{A}) = \{1\}$  and  $\text{spec}(\mathbf{B}) = \{2, 3\}$  are disjoint ([Theorem 8.7.4](#)), so [Theorem 11.10.2](#) applies.
- (b) No. Both characteristic polynomials are  $x^2 + 1$ , so  $\text{spec}(\mathbf{A}) = \text{spec}(\mathbf{B}) = \{i, -i\}$ , which is as far from disjoint as possible. A witness is easy to produce here:  $\mathbf{X} = \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix}$  gives  $\mathbf{AX} = \begin{pmatrix} 0 & -1 \\ -1 & 0 \end{pmatrix} = \mathbf{XB}$ , so  $\mathbf{X}$  is a non-zero member of  $\ker \mathcal{S}_{\mathbf{A}, \mathbf{B}}$  and condition (d) of [Theorem 11.10.2](#) fails.
- (c) No. This is the  $+$  form, so the operator is  $\mathcal{S}_{\mathbf{A}, -\mathbf{B}}$  and the condition is  $\lambda + \mu \neq 0$ . Here  $\text{spec}(\mathbf{A}) = \{1, 2\}$  and  $\text{spec}(\mathbf{B}) = \{-1, 3\}$ , and  $1 + (-1) = 0$ .
- (d) No. Both are triangular with diagonal 1, 1, so  $\text{spec}(\mathbf{A}) = \text{spec}(\mathbf{B}) = \{1\}$ .
- (e) Yes. A nilpotent matrix has  $\text{spec}(\mathbf{A}) = \{0\}$ , since  $\mathbf{A}^k = \mathbf{0}$  forces  $\lambda^k = 0$  for every eigenvalue  $\lambda$ ; and  $0 \notin \text{spec}(\mathbf{B})$  because  $\mathbf{B}$  is invertible ([Theorem 8.2.6](#)). So the

spectra are disjoint.

### 11.10.7.3 C. Going deeper

#### Exercise C1: Eigenvectors of the Sylvester operator.

Let  $\mathbf{A} \in M_m(\mathbb{C})$  and  $\mathbf{B} \in M_n(\mathbb{C})$ , and let  $\mathcal{S} = \mathcal{S}_{\mathbf{A}, \mathbf{B}}$ .

- (a) Suppose  $\mathbf{A}\mathbf{x} = \lambda\mathbf{x}$  with  $\mathbf{x} \in \mathbb{C}^m$  non-zero, and suppose  $\mathbf{y} \in \mathbb{C}^n$  is a left eigenvector of  $\mathbf{B}$  for  $\mu$ , so that  $\mathbf{y}^\top \mathbf{B} = \mu \mathbf{y}^\top$  (Definition 8.10.1). Prove that  $\mathbf{x}\mathbf{y}^\top \neq \mathbf{0}$  and that  $\mathcal{S}(\mathbf{x}\mathbf{y}^\top) = (\lambda - \mu)\mathbf{x}\mathbf{y}^\top$ .
- (b) Deduce that every difference  $\lambda - \mu$  with  $\lambda \in \text{spec}(\mathbf{A})$  and  $\mu \in \text{spec}(\mathbf{B})$  is an eigenvalue of  $\mathcal{S}$ .
- (c) Hence give a second proof that  $\mathcal{S}$  is not invertible when  $\mathbf{A}$  and  $\mathbf{B}$  have a common eigenvalue, one that uses no Kronecker products.

*Solution.*

- (a) Choose  $i$  with  $x_i \neq 0$  and  $j$  with  $y_j \neq 0$ ; these exist because  $\mathbf{x}$  and  $\mathbf{y}$  are non-zero. The  $(i, j)$  entry of  $\mathbf{x}\mathbf{y}^\top$  is  $x_i y_j \neq 0$ , so  $\mathbf{x}\mathbf{y}^\top \neq \mathbf{0}$ . For the eigenvalue equation, associativity of matrix multiplication (Theorem 0.9.4) gives

$$\mathcal{S}(\mathbf{x}\mathbf{y}^\top) = (\mathbf{A}\mathbf{x})\mathbf{y}^\top - \mathbf{x}(\mathbf{y}^\top \mathbf{B}) = \lambda\mathbf{x}\mathbf{y}^\top - \mu\mathbf{x}\mathbf{y}^\top,$$

which is  $(\lambda - \mu)\mathbf{x}\mathbf{y}^\top$ .

- (b) Let  $\lambda \in \text{spec}(\mathbf{A})$  and  $\mu \in \text{spec}(\mathbf{B})$ . A right eigenvector  $\mathbf{x}$  for  $\lambda$  exists by Definition 8.2.1, and a left eigenvector  $\mathbf{y}$  for  $\mu$  exists by Proposition 8.10.5 (b). By (a),  $\mathbf{x}\mathbf{y}^\top$  is a non-zero matrix with  $\mathcal{S}(\mathbf{x}\mathbf{y}^\top) = (\lambda - \mu)\mathbf{x}\mathbf{y}^\top$ , so  $\lambda - \mu$  is an eigenvalue of  $\mathcal{S}$  with eigenvector  $\mathbf{x}\mathbf{y}^\top$ .
- (c) If  $\lambda \in \text{spec}(\mathbf{A}) \cap \text{spec}(\mathbf{B})$ , apply (b) with  $\mu = \lambda$ : then 0 is an eigenvalue of  $\mathcal{S}$ , so  $\ker \mathcal{S} \neq \{\mathbf{0}\}$  and  $\mathcal{S}$  is not invertible. This shows one implication of Theorem 11.10.2 directly. (It does not give the converse: knowing every  $\lambda - \mu$  is an eigenvalue does not by itself show there are no others.)

#### Exercise C2: The kernel when the spectra meet.

Let  $\mathbf{A} \in M_m(\mathbb{C})$  and  $\mathbf{B} \in M_n(\mathbb{C})$  have eigenvalue lists  $\lambda_1, \dots, \lambda_m$  and  $\mu_1, \dots, \mu_n$  with algebraic multiplicity, and let  $d = \#\{(i, j) : \lambda_i = \mu_j\}$ .

- (a) Prove that  $\dim \ker \mathcal{S}_{\mathbf{A}, \mathbf{B}} \geq 1$  if and only if  $d \geq 1$ .
- (b) Prove that  $\dim \ker \mathcal{S}_{\mathbf{A}, \mathbf{B}} \leq d$ .
- (c) Show that the inequality in (b) can be strict, by computing  $\ker \mathcal{S}_{\mathbf{A}, \mathbf{B}}$  for  $\mathbf{A} = \mathbf{B} = \mathbf{J}_2(0)$ .

*Solution.*

- (a)  $d \geq 1$  says exactly that  $\mathbf{A}$  and  $\mathbf{B}$  have a common eigenvalue, and by Theorem 11.10.2 ((a)  $\Leftrightarrow$  (d)) that is exactly the failure of  $\ker \mathcal{S}_{\mathbf{A}, \mathbf{B}} = \{\mathbf{0}\}$ .
- (b) The operator  $\mathcal{S}_{\mathbf{A}, \mathbf{B}}$  has matrix  $\mathbf{K}$  (Eq. 11.10.1), so  $\dim \ker \mathcal{S}_{\mathbf{A}, \mathbf{B}} = g_{\mathbf{K}}(0)$ , the

geometric multiplicity of 0 as an eigenvalue of  $\mathbf{K}$  (when  $0 \notin \text{spec}(\mathbf{K})$  both sides are 0 and there is nothing to prove). By Lemma 11.10.1 the eigenvalue list of  $\mathbf{K}$  consists of the  $mn$  differences  $\lambda_i - \mu_j$ , so the algebraic multiplicity is  $a_{\mathbf{K}}(0) = d$ . By Theorem 8.3.1,  $g_{\mathbf{K}}(0) \leq a_{\mathbf{K}}(0) = d$ .

- (c) Here  $m = n = 2$  and  $\lambda_1 = \lambda_2 = \mu_1 = \mu_2 = 0$ , so  $d = 4$ . The kernel is the commutant of  $\mathbf{J} = \mathbf{J}_2(0)$ : writing  $\mathbf{X} = \begin{pmatrix} p & q \\ r & s \end{pmatrix}$ ,

$$\mathbf{J}\mathbf{X} = \begin{pmatrix} r & s \\ 0 & 0 \end{pmatrix}, \quad \mathbf{X}\mathbf{J} = \begin{pmatrix} 0 & p \\ 0 & r \end{pmatrix},$$

so  $\mathbf{J}\mathbf{X} = \mathbf{X}\mathbf{J}$  forces  $r = 0$  and  $s = p$ , and leaves  $q$  free. Hence  $\ker \mathcal{S}_{\mathbf{J}, \mathbf{J}} = \text{span}(\mathbf{I}_2, \mathbf{J})$ , of dimension  $2 < 4 = d$ .

### Exercise C3: Only the scalars commute with everything.

Let  $F$  be a field,  $n \geq 2$ , and let  $\mathbf{E}_{ij} \in M_n(F)$  be the matrix unit with 1 in position  $(i, j)$  and 0 elsewhere.

- (a) Suppose  $\mathbf{A} \in M_n(F)$  satisfies  $\mathbf{A}\mathbf{E}_{ij} = \mathbf{E}_{ij}\mathbf{A}$  for all  $i \neq j$ . Prove that  $\mathbf{A} = c\mathbf{I}_n$  for some  $c \in F$ .
- (b) Deduce that the only matrices commuting with every  $\mathbf{X} \in M_n(F)$  are the scalar matrices, that is,  $\ker \mathcal{S}_{\mathbf{A}, \mathbf{A}} = M_n(F)$  forces  $\mathbf{A}$  scalar.
- (c) Deduce that the only matrices commuting with every invertible matrix are again the scalar matrices.

*Hint for (c):  $\mathbf{I}_n + \mathbf{E}_{ij}$  is invertible when  $i \neq j$ .*

*Solution.*

- (a) Fix  $i \neq j$ . By Theorem 0.9.2,  $\mathbf{A}\mathbf{E}_{ij}$  has column  $j$  equal to  $\mathbf{A}\mathbf{e}_i$ , that is column  $i$  of  $\mathbf{A}$ , and all other columns zero; while  $\mathbf{E}_{ij}\mathbf{A}$  has row  $i$  equal to row  $j$  of  $\mathbf{A}$ , and all other rows zero. Compare entries. Take  $k \neq i$ . The  $(k, j)$  entry of  $\mathbf{A}\mathbf{E}_{ij}$  is  $a_{ki}$ , and the  $(k, j)$  entry of  $\mathbf{E}_{ij}\mathbf{A}$  is 0, because row  $k \neq i$  of  $\mathbf{E}_{ij}\mathbf{A}$  is zero. Hence  $a_{ki} = 0$  whenever  $k \neq i$ . Since  $n \geq 2$ , every index  $i$  has some  $j \neq i$  available, so this holds for every  $i$ , and  $\mathbf{A}$  is diagonal. Now the  $(i, j)$  entry: on the left it is  $a_{ii}$  (the  $i$ -th entry of column  $i$  of  $\mathbf{A}$ ), and on the right it is  $a_{jj}$  (the  $j$ -th entry of row  $j$  of  $\mathbf{A}$ ). So  $a_{ii} = a_{jj}$  for all  $i \neq j$ . Writing  $c$  for this common value,  $\mathbf{A} = c\mathbf{I}_n$ .
- (b) If  $\mathbf{A}\mathbf{X} = \mathbf{X}\mathbf{A}$  for every  $\mathbf{X} \in M_n(F)$ , then in particular for  $\mathbf{X} = \mathbf{E}_{ij}$  with  $i \neq j$ , so (a) gives  $\mathbf{A} = c\mathbf{I}_n$ . Conversely every scalar matrix commutes with everything. Hence  $\ker \mathcal{S}_{\mathbf{A}, \mathbf{A}} = M_n(F)$  if and only if  $\mathbf{A}$  is scalar.
- (c) For  $i \neq j$  we have  $\mathbf{E}_{ij}^2 = \mathbf{0}$ , so  $(\mathbf{I}_n + \mathbf{E}_{ij})(\mathbf{I}_n - \mathbf{E}_{ij}) = \mathbf{I}_n$  and  $\mathbf{I}_n + \mathbf{E}_{ij}$  is invertible. If  $\mathbf{A}$  commutes with every invertible matrix, it commutes with  $\mathbf{I}_n + \mathbf{E}_{ij}$ , hence with  $\mathbf{E}_{ij}$  (subtract  $\mathbf{A}\mathbf{I}_n = \mathbf{I}_n\mathbf{A}$ ), for all  $i \neq j$ . By (a),  $\mathbf{A} = c\mathbf{I}_n$ .

## 11.11 How Far from Normal

Every theorem so far in this chapter has been a yes-or-no statement. An operator either commutes with its adjoint or it does not; there either is an orthonormal basis of eigenvectors or there is not. That is unsatisfying, because the two matrices  $\begin{pmatrix} 1 & 10^{-6} \\ 0 & 2 \end{pmatrix}$  and  $\begin{pmatrix} 1 & 10^6 \\ 0 & 2 \end{pmatrix}$  are both non-normal, and no one would want to treat them alike. This section replaces the yes-or-no answer with a number. The number comes out of Schur's theorem almost for free, once we know that the Frobenius norm cannot tell one orthonormal basis from another, and it turns out to measure exactly the mass that a Schur factorization is forced to leave above the diagonal.

Throughout,  $\mathbf{A} \in M_n(\mathbb{C})$  and  $\mathbb{C}^n$  carries the standard inner product. Eigenvalues are always counted **in**  $\mathbb{C}$  and **with algebraic multiplicity**, so that an  $n \times n$  matrix always has a list of exactly  $n$  of them (Corollary 5.5.5).

### 11.11.1 The Frobenius norm, officially

Chapter 10 made  $M_{m \times n}(F)$  into an inner product space in its first section, by declaring

$$\langle \mathbf{A}, \mathbf{B} \rangle = \text{tr}(\mathbf{B}^* \mathbf{A}),$$

the **Frobenius inner product**, and observing that it is nothing but the standard inner product of  $F^{mn}$  after the entries are listed in a single column. Its induced norm (Definition 10.1.3) was named there too, but never given a symbol, and we have already borrowed one on the quiet in Exercise (C2: The Frobenius norm is unitarily invariant). It is time to make it official, because the whole of this section is written in it.

For  $\mathbf{A} \in M_{m \times n}(\mathbb{C})$ , the **Frobenius norm** of  $\mathbf{A}$  is

$$\|\mathbf{A}\|_F := \sqrt{\langle \mathbf{A}, \mathbf{A} \rangle} = \left( \sum_{i,j} |a_{ij}|^2 \right)^{1/2}.$$

The subscript  $F$  is there because  $\|\cdot\|$  with no subscript already means the norm of a *vector*, and a matrix will shortly be asked to be both. Two remarks on the formula. First, it treats  $\mathbf{A}$  as a list of  $mn$  numbers and forgets that they are arranged in a rectangle: it is blind to the difference between  $\begin{pmatrix} 1 & 0 \\ 0 & 0 \end{pmatrix}$  and  $\begin{pmatrix} 0 & 1 \\ 0 & 0 \end{pmatrix}$ . Second, it is computable with no eigenvalues, no row reduction and no choices: add up  $mn$  squared moduli.

The trace form of the same quantity is the one that does the work.

*Quick check.* Why is  $\|\mathbf{A}\|_F^2 = \text{tr}(\mathbf{A}^* \mathbf{A})$ ?

*Solution.* The  $(j, j)$  entry of  $\mathbf{A}^* \mathbf{A}$  is  $\sum_i \overline{a_{ij}} a_{ij} = \sum_i |a_{ij}|^2$ , the squared length of the  $j$ -th column. Summing over  $j$  sweeps every entry of  $\mathbf{A}$  exactly once, so  $\text{tr}(\mathbf{A}^* \mathbf{A}) = \sum_{i,j} |a_{ij}|^2$ . Equivalently, it is  $\langle \mathbf{A}, \mathbf{A} \rangle$  read off the definition of the Frobenius inner product.

Now the property that makes the Frobenius norm the right yardstick in a chapter about orthonormal bases: unitary factors are invisible to it.



**Lemma 11.11.1 — Unitary Invariance of the Frobenius Norm**

Let  $\mathbf{A} \in M_{m \times n}(\mathbb{C})$ , let  $\mathbf{U} \in U(m)$  and let  $\mathbf{V} \in U(n)$ . Then

$$\|\mathbf{UA}\|_F = \|\mathbf{A}\|_F = \|\mathbf{AV}\|_F.$$

In particular, if  $\mathbf{A}$  is square and  $\mathbf{U}$  is unitary then  $\|\mathbf{U}^*\mathbf{AU}\|_F = \|\mathbf{A}\|_F$ .

*Proof.* Using  $\|\mathbf{B}\|_F^2 = \text{tr}(\mathbf{B}^*\mathbf{B})$ , then  $(\mathbf{UA})^* = \mathbf{A}^*\mathbf{U}^*$  and  $\mathbf{U}^*\mathbf{U} = \mathbf{I}_m$ ,

$$\|\mathbf{UA}\|_F^2 = \text{tr}(\mathbf{A}^*\mathbf{U}^*\mathbf{UA}) = \text{tr}(\mathbf{A}^*\mathbf{A}) = \|\mathbf{A}\|_F^2.$$

For the right-hand factor, [Theorem 0.9.6](#) (3) lets the  $\mathbf{V}$  travel around the trace, and  $\mathbf{V}\mathbf{V}^* = \mathbf{I}_n$ :

$$\|\mathbf{AV}\|_F^2 = \text{tr}(\mathbf{V}^*\mathbf{A}^*\mathbf{AV}) = \text{tr}(\mathbf{A}^*\mathbf{AV}\mathbf{V}^*) = \|\mathbf{A}\|_F^2.$$

Both norms are non-negative reals, so equality of the squares gives equality of the norms. For the last statement,  $\mathbf{U}^*$  is unitary too, since  $(\mathbf{U}^*)^*\mathbf{U}^* = \mathbf{U}\mathbf{U}^* = \mathbf{I}$ ; apply the first equality to the left factor  $\mathbf{U}^*$  and the second to the right factor  $\mathbf{U}$ . ■

The consequence to keep in mind is this. If  $T \in \mathcal{L}(V)$  is an operator on a finite-dimensional complex inner product space and we compute  $\|[T]_{\mathcal{B}}\|_F$  in an **orthonormal** basis  $\mathcal{B}$ , the answer does not depend on which orthonormal basis we chose, because two such matrices differ by a unitary similarity ([Theorem 3.7.3](#)). So the Frobenius norm is a property of  $T$ , not of our coordinates — exactly like the trace and the determinant, and unlike, say, the largest entry.

**11.11.2 Schur's inequality**

Here is the observation the section is built on. Write a matrix in a Schur form  $\mathbf{A} = \mathbf{UTU}^*$ . The diagonal of  $\mathbf{T}$  holds the eigenvalues, so  $\sum_i |\lambda_i|^2$  is the Frobenius mass sitting *on* the diagonal of  $\mathbf{T}$ . The rest of  $\mathbf{T}$  sits strictly above it. And by [Lemma 11.11.1](#) the total mass of  $\mathbf{T}$  is the total mass of  $\mathbf{A}$ . Rearranged, that is a theorem.

**Theorem 11.11.2 — Schur's Inequality**

Let  $\mathbf{A} \in M_n(\mathbb{C})$  have eigenvalues  $\lambda_1, \dots, \lambda_n$ , listed with algebraic multiplicity. Then

$$\sum_{i=1}^n |\lambda_i|^2 \leq \|\mathbf{A}\|_F^2,$$

with equality if and only if  $\mathbf{A}$  is normal.

*Idea.* Schur-triangularize, then count. ①  $\mathbf{A} = \mathbf{UTU}^*$  with  $\mathbf{T}$  upper triangular and the  $\lambda_i$  on its diagonal ([Corollary 11.1.3](#)). ② The Frobenius norm does not notice the  $\mathbf{U}$ 's, so  $\|\mathbf{A}\|_F = \|\mathbf{T}\|_F$ . ③ Splitting  $\|\mathbf{T}\|_F^2$  into its diagonal part and its strictly upper part gives the inequality on the nose, with the strictly upper part as the exact surplus. ④ For the equality case, that surplus vanishes exactly when  $\mathbf{T}$  is diagonal, and “triangular

and normal” and “diagonal” were shown to be the same thing in [Lemma 11.4.1](#).

*Proof.* By [Corollary 11.1.3](#) there are a unitary  $\mathbf{U}$  and an upper triangular  $\mathbf{T} = (t_{ij})$  with  $\mathbf{A} = \mathbf{U}\mathbf{T}\mathbf{U}^*$  and  $t_{ii} = \lambda_i$  for every  $i$ . Since  $t_{ij} = 0$  whenever  $i > j$ , the entries of  $\mathbf{T}$  fall into the diagonal ones and the strictly upper ones, so by [Lemma 11.11.1](#)

$$\begin{aligned}\|\mathbf{A}\|_F^2 &= \|\mathbf{T}\|_F^2 = \sum_{i=1}^n |t_{ii}|^2 + \sum_{i < j} |t_{ij}|^2 \\ &= \sum_{i=1}^n |\lambda_i|^2 + \sum_{i < j} |t_{ij}|^2.\end{aligned}\tag{11.11.1}$$

The second sum is a sum of non-negative reals, so it is  $\geq 0$ , and the inequality follows. Moreover equality holds if and only if that sum is 0, which happens if and only if  $t_{ij} = 0$  for all  $i < j$ , that is, if and only if **this**  $\mathbf{T}$  is diagonal.

( $\Leftarrow$ ) Suppose  $\mathbf{A}$  is normal. Then so is  $\mathbf{T} = \mathbf{U}^*\mathbf{A}\mathbf{U}$ : using  $\mathbf{U}\mathbf{U}^* = \mathbf{I}$  in the middle of each product,

$$\mathbf{T}^*\mathbf{T} = \mathbf{U}^*\mathbf{A}^*\mathbf{A}\mathbf{U} = \mathbf{U}^*\mathbf{A}\mathbf{A}^*\mathbf{U} = \mathbf{T}\mathbf{T}^*.$$

A normal upper triangular matrix is diagonal ([Lemma 11.4.1](#)), so  $\mathbf{T}$  is diagonal and [Eq. 11.11.1](#) gives equality.

( $\Rightarrow$ ) Suppose equality holds. Then, as noted,  $\mathbf{T}$  is diagonal, so  $\mathbf{A} = \mathbf{U}\mathbf{T}\mathbf{U}^*$  exhibits  $\mathbf{A}$  as unitarily diagonalizable, and [Corollary 11.4.3](#) makes  $\mathbf{A}$  normal.

This proves the inequality and both halves of the equality case. ■

Notice that the proof never had to worry about *which* Schur factorization was used. The two sides of the inequality — the eigenvalue list and  $\|\mathbf{A}\|_F$  — are determined by  $\mathbf{A}$  alone, so the surplus  $\sum_{i < j} |t_{ij}|^2$  in [Eq. 11.11.1](#) comes out the same for every triangularization of  $\mathbf{A}$ , even though the individual entries  $t_{ij}$  certainly do not. That is worth a second look, and we will look at it in a moment.

### Example. (Both sides of Schur’s inequality)

Verify [Theorem 11.11.2](#) for

$$\mathbf{A} = \begin{pmatrix} 0 & 1 \\ 2 & 0 \end{pmatrix},$$

and locate the surplus inside an explicit Schur factorization ([Corollary 11.1.3](#)).

*Solution.* Here  $\text{tr } \mathbf{A} = 0$  and  $\det \mathbf{A} = -2$ , so  $p_{\mathbf{A}} = x^2 - 2$  and the eigenvalues are  $\pm\sqrt{2}$ . Thus

$$\sum_i |\lambda_i|^2 = 2 + 2 = 4, \quad \|\mathbf{A}\|_F^2 = 0 + 1 + 4 + 0 = 5,$$

and indeed  $4 < 5$ . The inequality is strict, so [Theorem 11.11.2](#) already says that  $\mathbf{A}$  is not normal — no product of matrices needed.

For the factorization, solve  $(\mathbf{A} - \sqrt{2}\mathbf{I})\mathbf{x} = \mathbf{0}$ : the first row reads  $-\sqrt{2}x_1 + x_2 = 0$ , giving

the eigenvector  $(1, \sqrt{2})$  of norm  $\sqrt{3}$ . Take

$$\mathbf{q}_1 = \frac{1}{\sqrt{3}}(1, \sqrt{2}), \quad \mathbf{q}_2 = \frac{1}{\sqrt{3}}(-\sqrt{2}, 1),$$

a pair of orthonormal vectors, and let  $\mathbf{U}$  have them as columns. Then

$$\mathbf{U}^* \mathbf{A} \mathbf{U} = \begin{pmatrix} \sqrt{2} & -1 \\ 0 & -\sqrt{2} \end{pmatrix} = \mathbf{T}.$$

Its Frobenius mass is  $2 + 1 + 2 = 5 = \|\mathbf{A}\|_F^2$ , as [Lemma 11.11.1](#) promises, and the split of [Eq. 11.11.1](#) reads  $5 = 4 + 1$ : the whole surplus is the single entry  $-1$  above the diagonal.

*Quick check.* The real matrix  $\mathbf{A} = \begin{pmatrix} 0 & -1 \\ 1 & 0 \end{pmatrix}$  has  $\|\mathbf{A}\|_F^2 = 2$  and no real eigenvalue at all. Does [Theorem 11.11.2](#) fail for it?

*Solution.* No, because the eigenvalues are counted in  $\mathbb{C}$ . They are  $i$  and  $-i$ , so  $\sum_i |\lambda_i|^2 = 1 + 1 = 2 = \|\mathbf{A}\|_F^2$ : equality, which is exactly right, since  $\mathbf{A}^\top \mathbf{A} = \mathbf{A} \mathbf{A}^\top = \mathbf{I}$  makes  $\mathbf{A}$  normal. Restricting the eigenvalue list to  $\mathbb{R}$  would leave the left-hand side empty and the statement meaningless; this is one more place where the Fundamental Theorem of Algebra ([Theorem 5.5.1](#)) is doing silent work.

### 11.11.3 The departure from normality

Schur's inequality has a slack term, and the slack term is the interesting object. It is non-negative for every matrix; it vanishes exactly on the normal ones; and, both of its ingredients being unitary-similarity invariants, it survives a change of orthonormal basis. Those are precisely the properties one wants of a measurement, so we name it.

*The departure from normality is the Frobenius mass that no Schur factorization can push onto the diagonal.*

#### Definition 11.11.1 — Departure from Normality

Let  $\mathbf{A} \in M_n(\mathbb{C})$  have eigenvalues  $\lambda_1, \dots, \lambda_n$ , listed with algebraic multiplicity. The **departure from normality** of  $\mathbf{A}$  is the non-negative real number  $\Delta(\mathbf{A})$  defined by

$$\Delta(\mathbf{A})^2 := \|\mathbf{A}\|_F^2 - \sum_{i=1}^n |\lambda_i|^2.$$

Clause by clause: the quantity on the right is a real number because  $\|\mathbf{A}\|_F^2$  and each  $|\lambda_i|^2$  are real; it is **non-negative** by [Theorem 11.11.2](#), which is what allows us to take its non-negative square root and call the result  $\Delta(\mathbf{A})$ ; and the eigenvalue list, though it comes with no preferred order, contributes a sum that does not care about the order. So  $\Delta(\mathbf{A})$  is well defined, with no choices made anywhere.

Two facts come straight from [Theorem 11.11.2](#) and its proof.

- $\Delta(\mathbf{A}) = 0$  if and only if  $\mathbf{A}$  is normal. This is the equality case, restated.

- $\Delta(\mathbf{A}) = (\sum_{i < j} |t_{ij}|^2)^{1/2}$  **for every** Schur factorization  $\mathbf{A} = \mathbf{U}\mathbf{T}\mathbf{U}^*$ , by [Eq. 11.11.1](#). So  $\Delta(\mathbf{A})$  is the length of the strictly upper triangular part of  $\mathbf{T}$ , and the remark after [Theorem 11.11.2](#) says that this length is the same for all of them, even though the entries are not.

And one more, which is the reason the number deserves to be called a measurement rather than a coincidence:  $\Delta(\mathbf{U}^* \mathbf{A} \mathbf{U}) = \Delta(\mathbf{A})$  for every unitary  $\mathbf{U}$ . Both ingredients are unitary-similarity invariants — the Frobenius norm by [Lemma 11.11.1](#), the eigenvalue list because similar matrices have the same characteristic polynomial ([Theorem 6.10.3](#)). [Exercise \(C1: Unitary similarity preserves the defect, ordinary similarity does not\)](#) asks for the details, and for an example showing that ordinary similarity wrecks the number completely.

### Example. (The defect of a Jordan block)

Compute  $\Delta(\mathbf{J}_k(\lambda))$  for the  $k \times k$  Jordan block with eigenvalue  $\lambda$ .

*Solution.* The matrix  $\mathbf{J}_k(\lambda)$  has  $\lambda$  in each of its  $k$  diagonal positions and 1 in each of the  $k - 1$  superdiagonal positions, and zeros elsewhere, so

$$\|\mathbf{J}_k(\lambda)\|_F^2 = k|\lambda|^2 + (k - 1).$$

It is upper triangular, so its eigenvalue list is  $\lambda$  repeated  $k$  times ([Theorem 8.7.4](#)), giving  $\sum_i |\lambda_i|^2 = k|\lambda|^2$ . Subtracting,

$$\Delta(\mathbf{J}_k(\lambda))^2 = k - 1, \quad \Delta(\mathbf{J}_k(\lambda)) = \sqrt{k - 1}.$$

Three things are worth reading off. The answer does not depend on  $\lambda$ : sliding a Jordan block up and down the complex plane does not make it more or less normal. It is 0 exactly when  $k = 1$ , agreeing with the fact that a  $1 \times 1$  matrix is normal. And it grows with the block, which matches what [Theorem 11.3.5](#) gave us in Section 3: a normal operator has no Jordan block of size larger than 1 ([Theorem 9.3.1](#)). Here the defect is counting exactly the superdiagonal ones that normality forbids.

A minimal-change non-example makes the same point on a smaller scale. The matrix  $\text{diag}(1, 3)$  is normal, so its defect is 0. Change the single entry in position  $(1, 2)$  from 0 to 2:

$$\mathbf{A} = \begin{pmatrix} 1 & 2 \\ 0 & 3 \end{pmatrix}, \quad \|\mathbf{A}\|_F^2 = 14, \quad \sum_i |\lambda_i|^2 = 1 + 9 = 10,$$

so  $\Delta(\mathbf{A}) = 2$ . The eigenvalues did not move — a triangular matrix wears them on its diagonal — and the entire change went into the defect. That is the general picture for triangular matrices: for them,  $\Delta$  is just the length of everything above the diagonal.

The defect also bounds something one might actually want to know. Fix a Schur factorization  $\mathbf{A} = \mathbf{U}\mathbf{T}\mathbf{U}^*$  ([Corollary 11.1.3](#)), write  $\mathbf{D}$  for the diagonal part of  $\mathbf{T}$ , and put  $\mathbf{N} = \mathbf{U}\mathbf{D}\mathbf{U}^*$ , which is normal by [Corollary 11.4.3](#). Then

$$\|\mathbf{A} - \mathbf{N}\|_F = \|\mathbf{T} - \mathbf{D}\|_F = \Delta(\mathbf{A}),$$

the first step by [Lemma 11.11.1](#). So  $\Delta(\mathbf{A})$  is an upper bound for the Frobenius distance from  $\mathbf{A}$  to the set of normal matrices. It is not that distance.

*Warning.*  $\Delta$  is a defect, not a distance. It is tempting to read  $\Delta(\mathbf{A})$  as “how far  $\mathbf{A}$  is from the nearest normal matrix”, and the displayed bound above makes the temptation worse. But the bound can be strict. Take  $\mathbf{A} = \mathbf{J}_2(0) = \begin{pmatrix} 0 & 1 \\ 0 & 0 \end{pmatrix}$ , so that  $\Delta(\mathbf{A}) = 1$  by [Example \(The defect of a Jordan block\)](#). The symmetric — hence normal — matrix  $\mathbf{N} = \frac{1}{2} \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}$  satisfies

$$\|\mathbf{A} - \mathbf{N}\|_F = \left\| \frac{1}{2} \begin{pmatrix} 0 & 1 \\ -1 & 0 \end{pmatrix} \right\|_F = \frac{1}{\sqrt{2}} < 1.$$

So some normal matrix is strictly nearer to  $\mathbf{A}$  than  $\Delta(\mathbf{A})$ . What  $\Delta(\mathbf{A})$  measures is the mass that survives *triangularization*, which is a genuine and useful quantity, but it is an upper bound for the distance and not the distance itself.

## 11.11.4 Eigenvalues that will not sit still

Here is why anyone outside this chapter cares. Eigenvalues are computed, in practice, from data that is only approximately right, so the question “if I move the matrix a little, how far do the eigenvalues move?” is not academic. For a normal matrix the answer is “just as little”. For a matrix with a large defect it can be “enormously further”. The following pair of  $2 \times 2$  matrices shows the whole phenomenon with nothing but the quadratic formula.

**Example.** (A perturbation that moves the eigenvalues a thousand times as far)

Fix a real number  $M > 0$  and a real  $\varepsilon$  with  $0 < \varepsilon < M$ . Compare the effect of adding  $\varepsilon$  to the lower-left entry of each of

$$\mathbf{A} = \begin{pmatrix} 1 & M \\ 0 & 1 \end{pmatrix} \quad \text{and} \quad \mathbf{N} = \begin{pmatrix} 1 & M \\ -M & 1 \end{pmatrix}.$$

Compute the defects of  $\mathbf{A}$  and  $\mathbf{N}$ , the eigenvalues of both perturbed matrices exactly, and the distance each eigenvalue travels.

*Solution.* *The two defects.* The matrix  $\mathbf{A}$  is upper triangular with both diagonal entries 1, so its eigenvalue list is 1, 1 and

$$\Delta(\mathbf{A})^2 = (1 + M^2 + 1) - (1 + 1) = M^2, \quad \Delta(\mathbf{A}) = M.$$

The matrix  $\mathbf{N}$  is normal:  $\mathbf{N}^\top \mathbf{N} = \mathbf{N} \mathbf{N}^\top = (1 + M^2) \mathbf{I}$ , a direct multiplication. So  $\Delta(\mathbf{N}) = 0$ . Its eigenvalues are  $1 \pm Mi$ , since  $p_{\mathbf{N}} = (x - 1)^2 + M^2$ . The two matrices have the same diagonal and the same entry  $M$  in the corner we are leaving alone; where they differ is that one has defect  $M$  and the other defect 0.

*The non-normal one.* Perturbing the corner gives  $\mathbf{A}_\varepsilon = \begin{pmatrix} 1 & M \\ \varepsilon & 1 \end{pmatrix}$ , with

$$p_{\mathbf{A}_\varepsilon} = (x - 1)^2 - M\varepsilon, \quad \lambda_\pm = 1 \pm \sqrt{M\varepsilon}.$$

The perturbation has size  $\|\mathbf{A}_\varepsilon - \mathbf{A}\|_F = \varepsilon$ , and each eigenvalue has moved a distance

$\sqrt{M\varepsilon}$ , which is  $\sqrt{M/\varepsilon}$  times the size of the perturbation. With  $M = 100$  and  $\varepsilon = 10^{-4}$ , the perturbation is one ten-thousandth and the eigenvalues move by  $\sqrt{100 \cdot 10^{-4}} = 1/10$ : a thousandfold amplification.

*The normal one.* Perturbing the same corner gives  $\mathbf{N}_\varepsilon = \begin{pmatrix} 1 & M \\ -M + \varepsilon & 1 \end{pmatrix}$ , with

$$p_{\mathbf{N}_\varepsilon} = (x - 1)^2 + M(M - \varepsilon), \quad \mu_\pm = 1 \pm i\sqrt{M^2 - M\varepsilon},$$

the square root being a real number because  $\varepsilon < M$ . The eigenvalue  $1 + Mi$  has moved to  $1 + i\sqrt{M^2 - M\varepsilon}$ , a distance of

$$M - \sqrt{M^2 - M\varepsilon} = \frac{M\varepsilon}{M + \sqrt{M^2 - M\varepsilon}} \leq \frac{M\varepsilon}{M} = \varepsilon,$$

where the middle equality multiplies top and bottom by  $M + \sqrt{M^2 - M\varepsilon}$  and the last step uses  $\sqrt{M^2 - M\varepsilon} \geq 0$ . So here the eigenvalues move by **at most** the size of the perturbation. The same denominator is at most  $2M$ , so the movement is also at least  $\varepsilon/2$ : with  $M = 100$  and  $\varepsilon = 10^{-4}$  it is a shade over  $\varepsilon/2$ .

The contrast is not an accident of these two matrices, and it is not a theorem we can prove here either. The general statement — that for a diagonalizable  $\mathbf{A}$  every eigenvalue of a perturbed  $\mathbf{A} + \mathbf{E}$  lies within  $\kappa\|\mathbf{E}\|$  of some eigenvalue of  $\mathbf{A}$ , where  $\kappa$  measures how far from unitary the diagonalizing matrix is, and that  $\kappa = 1$  exactly in the normal case — is the Bauer–Fike theorem. It needs matrix norms and a little analysis, and Chapter 19 develops both and proves it. What the example above establishes on its own is modest and exact: of two  $2 \times 2$  matrices with the same diagonal and the same large off-diagonal entry, the one with defect 0 amplifies a corner perturbation by a factor of at most 1, and the one with defect  $M$  amplifies it by  $\sqrt{M/\varepsilon}$ . The defect is not bookkeeping.

*Remark.* The amplification  $\sqrt{M\varepsilon}$  in [Example \(A perturbation that moves the eigenvalues a thousand times as far\)](#) has a square root in it, and that square root is the signature of a repeated eigenvalue. The unperturbed  $\mathbf{A}$  has the single eigenvalue 1 twice over with only one eigenvector, so it is a Jordan block in disguise; splitting a double root is what costs the square root. A matrix with well-separated eigenvalues and a large defect behaves less dramatically.

## 11.11.5 The characterizations collected

Schur's inequality gives a normality test of a new kind. Every earlier test compared two matrix products or ranged over all vectors; this one compares two numbers, and one of them is a trace.

### Theorem 11.11.3 — Normality as a Trace Identity

Let  $\mathbf{A} \in M_n(\mathbb{C})$  have eigenvalues  $\lambda_1, \dots, \lambda_n$ , listed with algebraic multiplicity. Then  $\mathbf{A}$  is normal if and only if

$$\operatorname{tr}(\mathbf{A}^* \mathbf{A}) = \sum_{i=1}^n |\lambda_i|^2.$$

*Proof.* We showed above that  $\text{tr}(\mathbf{A}^*\mathbf{A}) = \|\mathbf{A}\|_F^2$ . So the displayed equation is the equality case of [Theorem 11.11.2](#), which holds exactly when  $\mathbf{A}$  is normal. ■

Small as it is, this is the first criterion in the chapter that never asks for  $\mathbf{A}\mathbf{A}^*$ . If the spectrum is already known — from the characteristic polynomial, or because the matrix arrived in triangular form — normality costs one sum of  $n^2$  squared moduli and one sum of  $n$  more.

We can now put every description of normality in one place. Sections 3 and 4 proved most of them; this section adds the last one; and one equivalence, the converse of [Corollary 11.4.5](#), has not been proved anywhere and is proved here.

### Theorem 11.11.4 — Normality: The Complete List

Let  $\mathbf{A} \in M_n(\mathbb{C})$  have eigenvalues  $\lambda_1, \dots, \lambda_n$ , listed with algebraic multiplicity, and put

$$\mathbf{H} = \frac{1}{2}(\mathbf{A} + \mathbf{A}^*), \quad \mathbf{K} = \frac{1}{2i}(\mathbf{A} - \mathbf{A}^*),$$

both of which are Hermitian, with  $\mathbf{A} = \mathbf{H} + i\mathbf{K}$ . The following are equivalent.

- (a)  $\mathbf{A}^*\mathbf{A} = \mathbf{A}\mathbf{A}^*$ , that is,  $\mathbf{A}$  is normal.
- (b)  $\|\mathbf{A}\mathbf{x}\| = \|\mathbf{A}^*\mathbf{x}\|$  for every  $\mathbf{x} \in \mathbb{C}^n$ .
- (c)  $\langle \mathbf{A}\mathbf{x}, \mathbf{A}\mathbf{y} \rangle = \langle \mathbf{A}^*\mathbf{x}, \mathbf{A}^*\mathbf{y} \rangle$  for all  $\mathbf{x}, \mathbf{y} \in \mathbb{C}^n$ .
- (d)  $\mathbf{H}\mathbf{K} = \mathbf{K}\mathbf{H}$ .
- (e)  $\mathbb{C}^n$  has an orthonormal basis of eigenvectors of  $\mathbf{A}$ .
- (f)  $\mathbf{A} = \mathbf{U}\mathbf{D}\mathbf{U}^*$  for some unitary  $\mathbf{U}$  and some diagonal  $\mathbf{D}$ .
- (g)  $\mathbf{A}^* = p(\mathbf{A})$  for some polynomial  $p \in \mathbb{C}[x]$ .
- (h)  $\sum_{i=1}^n |\lambda_i|^2 = \|\mathbf{A}\|_F^2$ .

*Proof.* Let  $T = T_{\mathbf{A}}$  on  $\mathbb{C}^n$  with its standard inner product, for which the standard basis is orthonormal; then  $T^* = T_{\mathbf{A}^*}$  by [Theorem 10.6.3](#), and  $T$  is normal exactly when  $\mathbf{A}$  is.

(a), (b), (c) and (d) are equivalent by [Theorem 11.3.1](#) applied to  $T$ , whose four conditions are these four read for  $T$  instead of  $\mathbf{A}$ . (a)  $\Leftrightarrow$  (e) is [Theorem 11.4.2](#), again for  $T$ . (a)  $\Leftrightarrow$  (f) is [Corollary 11.4.3](#). (a)  $\Leftrightarrow$  (h) is [Theorem 11.11.3](#), since  $\|\mathbf{A}\|_F^2 = \text{tr}(\mathbf{A}^*\mathbf{A})$ . (a)  $\Rightarrow$  (g) is [Corollary 11.4.5](#), applied to  $T$  and read back as matrices.

It remains to prove (g)  $\Rightarrow$  (a). Suppose  $\mathbf{A}^* = p(\mathbf{A})$  with  $p = c_0 + c_1x + \dots + c_dx^d$ . Every power of  $\mathbf{A}$  commutes with  $\mathbf{A}$ , hence so does any linear combination of powers:

$$p(\mathbf{A})\mathbf{A} = \sum_{k=0}^d c_k \mathbf{A}^{k+1} = \mathbf{A} p(\mathbf{A}).$$

Substituting  $\mathbf{A}^*$  for  $p(\mathbf{A})$  on both sides gives  $\mathbf{A}^*\mathbf{A} = \mathbf{A}\mathbf{A}^*$ , which is (a). This closes the loop and proves the theorem. ■

This list is the summary result of the chapter, and it is worth reading as a whole rather than

as eight separate facts. Conditions (a) and (d) are algebraic: one equation between products. Conditions (b) and (c) are metric:  $\mathbf{A}$  and  $\mathbf{A}^*$  distort lengths and angles identically. Conditions (e) and (f) are geometric: the eigenvectors can be chosen perpendicular. Condition (g) is a surprise, and says that for a normal matrix the adjoint carries no information beyond  $\mathbf{A}$  itself. And condition (h) is quantitative: it is the only one on the list that comes with a *number* attached, so that when it fails one can ask by how much. That number is  $\Delta(\mathbf{A})$ , and it is the reason this section exists.

The operator versions of (a)–(g) hold verbatim on any finite-dimensional complex inner product space, by passing to the matrix in an orthonormal basis. So does (h), once  $\|\cdot\|_F$  is read as the Frobenius norm of that matrix, which [Lemma 11.11.1](#) showed does not depend on which orthonormal basis is used.

## 11.11.6 Exercises

### 11.11.6.1 A. Check your understanding

#### Exercise A1.

- Define the Frobenius norm of a matrix, and give the expression for  $\|\mathbf{A}\|_F^2$  as a trace.
- State [Theorem 11.11.2](#), including the equality case and how the eigenvalues are counted.
- Define  $\Delta(\mathbf{A})$ , and say why the definition needs [Theorem 11.11.2](#) before it can be made.
- True or false:  $\Delta(\mathbf{A})$  equals the Frobenius distance from  $\mathbf{A}$  to the nearest normal matrix. Justify your answer.
- True or false: if  $\mathbf{A}$  and  $\mathbf{B}$  are similar then  $\Delta(\mathbf{A}) = \Delta(\mathbf{B})$ . Justify your answer.
- Name the two facts about the Frobenius norm and about triangular matrices that the proof of [Theorem 11.11.2](#) rests on.

*Solution.*

- $\|\mathbf{A}\|_F = (\sum_{i,j} |a_{ij}|^2)^{1/2}$ , the norm induced by the Frobenius inner product  $\langle \mathbf{A}, \mathbf{B} \rangle = \text{tr}(\mathbf{B}^* \mathbf{A})$  of Chapter 10. As a trace,  $\|\mathbf{A}\|_F^2 = \text{tr}(\mathbf{A}^* \mathbf{A})$ .
- For  $\mathbf{A} \in M_n(\mathbb{C})$  with eigenvalues  $\lambda_1, \dots, \lambda_n$  listed **in  $\mathbb{C}$  and with algebraic multiplicity**,  $\sum_i |\lambda_i|^2 \leq \|\mathbf{A}\|_F^2$ , with equality if and only if  $\mathbf{A}$  is normal.
- $\Delta(\mathbf{A})^2 = \|\mathbf{A}\|_F^2 - \sum_i |\lambda_i|^2$ , and  $\Delta(\mathbf{A})$  is its non-negative square root. Without [Theorem 11.11.2](#) the right-hand side might be negative, and there would be no real square root to take.
- False. It is an upper bound for that distance, as the paragraph before the warning shows, but the warning gives  $\mathbf{A} = \mathbf{J}_2(0)$  with  $\Delta(\mathbf{A}) = 1$  and a normal matrix at distance  $1/\sqrt{2}$ .
- False. The matrices  $\begin{pmatrix} 1 & 1 \\ 0 & 2 \end{pmatrix}$  and  $\text{diag}(1, 2)$  are similar, since the first is diagonalizable with eigenvalues 1 and 2, yet their defects are 1 and 0. What is true is that  $\Delta$  is unchanged by **unitary** similarity; see [Exercise \(C1: Unitary similarity preserves the defect, ordinary similarity does not\)](#).



- (f) That the Frobenius norm is unchanged by unitary factors ([Lemma 11.11.1](#)), so that  $\|\mathbf{A}\|_F = \|\mathbf{T}\|_F$ ; and that a normal upper triangular matrix is diagonal ([Lemma 11.4.1](#)), which is what makes the equality case work in the direction ( $\Leftarrow$ ).

### 11.11.6.2 B. Practice

#### Exercise B1: Computing the defect.

For each matrix below, compute  $\|\mathbf{A}\|_F^2$ , the eigenvalue list, and  $\Delta(\mathbf{A})$ . State in each case whether the matrix is normal, and say how you know without multiplying  $\mathbf{A}$  by  $\mathbf{A}^*$ .

- (a)  $\begin{pmatrix} 3 & 4 \\ 0 & 5 \end{pmatrix}$   
 (b)  $\begin{pmatrix} 1 & 1 \\ -1 & 1 \end{pmatrix}$   
 (c)  $\mathbf{J}_3(2)$   
 (d)  $\begin{pmatrix} 0 & 1 & 0 \\ 0 & 0 & 1 \\ 1 & 0 & 0 \end{pmatrix}$

*Solution.*

- (a)  $\|\mathbf{A}\|_F^2 = 9 + 16 + 25 = 50$ . The matrix is upper triangular, so its eigenvalues are 3 and 5 ([Theorem 8.7.4](#)) and  $\sum_i |\lambda_i|^2 = 34$ . Hence  $\Delta(\mathbf{A})^2 = 16$  and  $\Delta(\mathbf{A}) = 4$ . Since  $\Delta(\mathbf{A}) \neq 0$ , the matrix is not normal ([Theorem 11.11.2](#)).
- (b)  $\|\mathbf{A}\|_F^2 = 1 + 1 + 1 + 1 = 4$ . Here  $\text{tr } \mathbf{A} = 2$  and  $\det \mathbf{A} = 1 + 1 = 2$ , so  $p_{\mathbf{A}} = x^2 - 2x + 2$  and the eigenvalues are  $1 \pm i$ , each of modulus  $\sqrt{2}$ . Then  $\sum_i |\lambda_i|^2 = 2 + 2 = 4$ , so  $\Delta(\mathbf{A}) = 0$  and the matrix **is** normal.
- (c)  $\|\mathbf{J}_3(2)\|_F^2 = 3 \cdot 4 + 2 = 14$ , the eigenvalue is 2 three times, and  $\sum_i |\lambda_i|^2 = 12$ . So  $\Delta^2 = 2$  and  $\Delta = \sqrt{2}$ , which is  $\sqrt{k-1}$  with  $k = 3$ , as [Example \(The defect of a Jordan block\)](#) says. Not normal.
- (d)  $\|\mathbf{A}\|_F^2 = 3$ . The matrix is the cyclic shift, with  $p_{\mathbf{A}} = x^3 - 1$  (expand along the first row, or note  $\mathbf{A}^3 = \mathbf{I}$  and  $\mathbf{A} \neq \mathbf{I}$  with  $\mathbf{A}$  having no repeated eigenvalue). Its eigenvalues are the three cube roots of unity, each of modulus 1, so  $\sum_i |\lambda_i|^2 = 3$  and  $\Delta(\mathbf{A}) = 0$ : normal.

#### Exercise B2: Verifying Schur's inequality.

Let

$$\mathbf{A} = \begin{pmatrix} 1 & 1 & 1 \\ 0 & 2 & 1 \\ 0 & 0 & 3 \end{pmatrix}.$$

- (a) Compute both sides of [Theorem 11.11.2](#) for  $\mathbf{A}$  and verify the inequality.  
 (b) Hence determine whether  $\mathbf{A}$  is normal, without computing  $\mathbf{A}^*\mathbf{A}$  or  $\mathbf{A}\mathbf{A}^*$ .  
 (c) Write down a normal matrix  $\mathbf{N}$  with  $\|\mathbf{A} - \mathbf{N}\|_F = \Delta(\mathbf{A})$ , and verify the equality.

*Solution.*

- (a)  $\|\mathbf{A}\|_F^2 = 1+1+1+4+1+9 = 17$ . The matrix is upper triangular, so its eigenvalues are 1, 2, 3 ([Theorem 8.7.4](#)) and  $\sum_i |\lambda_i|^2 = 1+4+9 = 14$ . Indeed  $14 \leq 17$ .
- (b) The inequality is strict, so by the equality case of [Theorem 11.11.2](#) the matrix is not normal. Equivalently  $\Delta(\mathbf{A}) = \sqrt{17-14} = \sqrt{3} \neq 0$ .
- (c) Since  $\mathbf{A}$  is already upper triangular,  $\mathbf{A} = \mathbf{I}\mathbf{A}\mathbf{I}^*$  is a Schur factorization ([Corollary 11.1.3](#)) with  $\mathbf{U} = \mathbf{I}$  and  $\mathbf{T} = \mathbf{A}$ . The construction before the warning takes  $\mathbf{N}$  to be the diagonal part,  $\mathbf{N} = \text{diag}(1, 2, 3)$ , which is normal. Then  $\mathbf{A} - \mathbf{N}$  has the three entries 1 above the diagonal and zeros elsewhere, so  $\|\mathbf{A} - \mathbf{N}\|_F = \sqrt{3} = \Delta(\mathbf{A})$ .

### Exercise B3: The nearest symmetric matrix.

Let  $\mathbf{A} \in M_n(\mathbb{R})$  and write  $\mathbf{S} = \frac{1}{2}(\mathbf{A} + \mathbf{A}^\top)$  and  $\mathbf{W} = \frac{1}{2}(\mathbf{A} - \mathbf{A}^\top)$ , so that  $\mathbf{A} = \mathbf{S} + \mathbf{W}$  with  $\mathbf{S}^\top = \mathbf{S}$  and  $\mathbf{W}^\top = -\mathbf{W}$ .

- (a) Prove that  $\langle \mathbf{B}, \mathbf{C} \rangle = 0$  in the Frobenius inner product whenever  $\mathbf{B} \in M_n(\mathbb{R})$  is symmetric and  $\mathbf{C} \in M_n(\mathbb{R})$  is skew-symmetric.
- (b) Deduce that  $\|\mathbf{A} - \mathbf{B}\|_F \geq \|\mathbf{W}\|_F$  for every symmetric  $\mathbf{B} \in M_n(\mathbb{R})$ , with equality only for  $\mathbf{B} = \mathbf{S}$ . Hence  $\mathbf{S}$  is the nearest symmetric matrix to  $\mathbf{A}$ .
- (c) Carry this out for  $\mathbf{A} = \begin{pmatrix} 2 & 4 \\ 0 & -2 \end{pmatrix}$ , and compare the resulting distance with  $\Delta(\mathbf{A})$ .

*Solution.*

- (a) For real matrices  $\langle \mathbf{B}, \mathbf{C} \rangle = \text{tr}(\mathbf{C}^\top \mathbf{B})$ . Since  $\mathbf{C}^\top = -\mathbf{C}$ , this is  $-\text{tr}(\mathbf{CB})$ . On the other hand  $\text{tr}(\mathbf{CB}) = \text{tr}((\mathbf{CB})^\top)$  by [Theorem 0.9.6](#) (2), and  $(\mathbf{CB})^\top = \mathbf{B}^\top \mathbf{C}^\top = -\mathbf{BC}$ , so  $\text{tr}(\mathbf{CB}) = -\text{tr}(\mathbf{BC}) = -\text{tr}(\mathbf{CB})$ , the last step by [Theorem 0.9.6](#) (3). A number equal to its own negative is 0, so  $\text{tr}(\mathbf{CB}) = 0$  and  $\langle \mathbf{B}, \mathbf{C} \rangle = 0$ .
- (b) Let  $\mathbf{B}$  be symmetric. Then  $\mathbf{A} - \mathbf{B} = (\mathbf{S} - \mathbf{B}) + \mathbf{W}$ , where  $\mathbf{S} - \mathbf{B}$  is symmetric and  $\mathbf{W}$  is skew-symmetric, so the two are orthogonal by (a) and Pythagoras ([Theorem 10.1.3](#)) gives

$$\|\mathbf{A} - \mathbf{B}\|_F^2 = \|\mathbf{S} - \mathbf{B}\|_F^2 + \|\mathbf{W}\|_F^2 \geq \|\mathbf{W}\|_F^2.$$

Equality forces  $\|\mathbf{S} - \mathbf{B}\|_F = 0$ , that is  $\mathbf{B} = \mathbf{S}$  ([Theorem 10.1.2](#) (a)). So  $\mathbf{S}$  is the unique nearest symmetric matrix, at distance  $\|\mathbf{W}\|_F$ .

- (c) Here  $\mathbf{S} = \begin{pmatrix} 2 & 2 \\ 2 & -2 \end{pmatrix}$  and  $\mathbf{W} = \begin{pmatrix} 0 & 2 \\ -2 & 0 \end{pmatrix}$ , so the distance from  $\mathbf{A}$  to the symmetric matrices is  $\|\mathbf{W}\|_F = \sqrt{8} = 2\sqrt{2}$ . For the defect,  $\|\mathbf{A}\|_F^2 = 4+16+4 = 24$ , and  $\mathbf{A}$  is triangular with eigenvalues 2 and  $-2$ , so  $\sum_i |\lambda_i|^2 = 8$  and  $\Delta(\mathbf{A}) = 4$ . A symmetric matrix is normal, so this again exhibits a normal matrix strictly closer than  $\Delta(\mathbf{A})$ :  $2\sqrt{2} \approx 2.83 < 4$ .

### 11.11.6.3 C. Going deeper

**Exercise C1: Unitary similarity preserves the defect, ordinary similarity does not.**

- (a) Prove that  $\Delta(\mathbf{U}^* \mathbf{A} \mathbf{U}) = \Delta(\mathbf{A})$  for every  $\mathbf{A} \in M_n(\mathbb{C})$  and every unitary  $\mathbf{U}$ .
- (b) Give  $\mathbf{A}, \mathbf{B} \in M_2(\mathbb{C})$  that are similar with  $\Delta(\mathbf{A}) \neq \Delta(\mathbf{B})$ , exhibiting an explicit  $\mathbf{P}$  with  $\mathbf{P}^{-1} \mathbf{A} \mathbf{P} = \mathbf{B}$ .
- (c) Deduce that two similar matrices with different defects cannot be unitarily similar, and use this to show that  $\begin{pmatrix} 1 & 1 \\ 0 & 2 \end{pmatrix}$  and  $\text{diag}(1, 2)$  are not unitarily similar.

*Solution.*

- (a) Put  $\mathbf{B} = \mathbf{U}^* \mathbf{A} \mathbf{U}$ . By [Lemma 11.11.1](#),  $\|\mathbf{B}\|_F = \|\mathbf{A}\|_F$ . Also  $\mathbf{B}$  is similar to  $\mathbf{A}$ , so  $p_{\mathbf{B}} = p_{\mathbf{A}}$  ([Theorem 6.10.3](#)) and the two matrices have the same eigenvalue list with multiplicities. Both terms in [Definition 11.11.1](#) therefore agree, so  $\Delta(\mathbf{B})^2 = \Delta(\mathbf{A})^2$ , and the defects are equal since both are non-negative.
- (b) Take  $\mathbf{A} = \begin{pmatrix} 1 & 1 \\ 0 & 2 \end{pmatrix}$  and  $\mathbf{B} = \text{diag}(1, 2)$ . Then  $\|\mathbf{A}\|_F^2 = 1 + 1 + 4 = 6$  with eigenvalues 1, 2, so  $\Delta(\mathbf{A})^2 = 6 - 5 = 1$  and  $\Delta(\mathbf{A}) = 1$ , while  $\Delta(\mathbf{B}) = 0$  because  $\mathbf{B}$  is diagonal, hence normal. They are similar: eigenvectors of  $\mathbf{A}$  are  $(1, 0)$  for  $\lambda = 1$  and  $(1, 1)$  for  $\lambda = 2$ , so with  $\mathbf{P} = \begin{pmatrix} 1 & 1 \\ 0 & 1 \end{pmatrix}$ , whose inverse is  $\begin{pmatrix} 1 & -1 \\ 0 & 1 \end{pmatrix}$ , we get  $\mathbf{P}^{-1} \mathbf{A} \mathbf{P} = \mathbf{B}$  by direct multiplication.
- (c) By (a), unitarily similar matrices have equal defects. So if  $\Delta(\mathbf{A}) \neq \Delta(\mathbf{B})$  then no unitary  $\mathbf{U}$  can satisfy  $\mathbf{U}^* \mathbf{A} \mathbf{U} = \mathbf{B}$ . The pair in (b) has defects 1 and 0, so it is similar but not unitarily similar. (This is the failure of “diagonalizable” to imply “unitarily diagonalizable”, seen through a number instead of through [Example \(Diagonalizable, and no orthonormal eigenbasis\)](#).)

### Exercise C2: Equality plus a single eigenvalue forces a scalar.

Let  $\mathbf{A} \in M_n(\mathbb{C})$  satisfy  $\|\mathbf{A}\|_F^2 = \sum_{i=1}^n |\lambda_i|^2$ , and suppose  $\mathbf{A}$  has **only one** distinct eigenvalue  $\lambda$ , so that  $p_{\mathbf{A}} = (x - \lambda)^n$ .

- (a) Prove that  $\mathbf{A} = \lambda \mathbf{I}$ .
- (b) Deduce that a normal matrix whose characteristic polynomial is  $x^n$  is the zero matrix.
- (c) Show that the hypothesis of equality cannot be dropped, by computing  $\Delta$  for a matrix with  $p_{\mathbf{A}} = (x - \lambda)^n$  that is not  $\lambda \mathbf{I}$ .

*Solution.*

- (a) By the equality case of [Theorem 11.11.2](#),  $\mathbf{A}$  is normal. So by [Corollary 11.4.3](#) there are a unitary  $\mathbf{U}$  and a diagonal  $\mathbf{D}$  with  $\mathbf{A} = \mathbf{U} \mathbf{D} \mathbf{U}^*$ , the diagonal of  $\mathbf{D}$  being the eigenvalue list of  $\mathbf{A}$ . That list is  $\lambda$  repeated  $n$  times, so  $\mathbf{D} = \lambda \mathbf{I}$  and

$$\mathbf{A} = \mathbf{U}(\lambda \mathbf{I})\mathbf{U}^* = \lambda \mathbf{U} \mathbf{U}^* = \lambda \mathbf{I}.$$

- (b) A normal matrix satisfies the equality hypothesis ([Theorem 11.11.2](#) again), and  $p_{\mathbf{A}} = x^n$  says its only eigenvalue is 0. By (a),  $\mathbf{A} = 0 \cdot \mathbf{I} = 0$ . In words: a normal nilpotent matrix is zero, which is [Theorem 11.3.5](#) seen from the quantitative side.
- (c) Take  $\mathbf{A} = \mathbf{J}_n(\lambda)$  with  $n \geq 2$ . Its characteristic polynomial is  $(x - \lambda)^n$  and it is

not  $\lambda \mathbf{I}$ , and  $\Delta(\mathbf{A}) = \sqrt{n-1} \neq 0$  by [Example \(The defect of a Jordan block\)](#). So equality in [Theorem 11.11.2](#) fails, exactly as it must.

**Exercise C3: The commutator is at least as big as the defect squared.**

For  $\mathbf{A} \in M_n(\mathbb{C})$  write  $[\mathbf{A}, \mathbf{A}^*] = \mathbf{A}\mathbf{A}^* - \mathbf{A}^*\mathbf{A}$ .

- (a) Prove that  $\|[\mathbf{A}, \mathbf{A}^*]\|_F$  is unchanged when  $\mathbf{A}$  is replaced by  $\mathbf{U}^*\mathbf{A}\mathbf{U}$  with  $\mathbf{U}$  unitary.  
 (b) Let  $\mathbf{T} = \begin{pmatrix} \lambda_1 & b \\ 0 & \lambda_2 \end{pmatrix}$ . Compute  $[\mathbf{T}, \mathbf{T}^*]$  and show that

$$\|[\mathbf{T}, \mathbf{T}^*]\|_F^2 = 2|b|^4 + 2|b|^2|\lambda_1 - \lambda_2|^2.$$

- (c) Deduce that every  $\mathbf{A} \in M_2(\mathbb{C})$  satisfies  $\|[\mathbf{A}, \mathbf{A}^*]\|_F \geq \sqrt{2} \Delta(\mathbf{A})^2$ , and determine exactly when equality holds.

*Hint for (c): Schur-triangularize, and use (a) to move the computation to the triangular factor.*

*Solution.*

- (a) Put  $\mathbf{B} = \mathbf{U}^*\mathbf{A}\mathbf{U}$ , so  $\mathbf{B}^* = \mathbf{U}^*\mathbf{A}^*\mathbf{U}$ . Inserting  $\mathbf{U}\mathbf{U}^* = \mathbf{I}$  between the factors,

$$[\mathbf{B}, \mathbf{B}^*] = \mathbf{U}^*(\mathbf{A}\mathbf{A}^* - \mathbf{A}^*\mathbf{A})\mathbf{U} = \mathbf{U}^*[\mathbf{A}, \mathbf{A}^*]\mathbf{U},$$

and [Lemma 11.11.1](#) gives  $\|[\mathbf{B}, \mathbf{B}^*]\|_F = \|[\mathbf{A}, \mathbf{A}^*]\|_F$ .

- (b) Multiplying out,

$$\begin{aligned} \mathbf{T}\mathbf{T}^* &= \begin{pmatrix} |\lambda_1|^2 + |b|^2 & b\bar{\lambda}_2 \\ \lambda_2\bar{b} & |\lambda_2|^2 \end{pmatrix}, \\ \mathbf{T}^*\mathbf{T} &= \begin{pmatrix} |\lambda_1|^2 & \bar{\lambda}_1 b \\ \lambda_1\bar{b} & |b|^2 + |\lambda_2|^2 \end{pmatrix}. \end{aligned}$$

Subtracting,

$$[\mathbf{T}, \mathbf{T}^*] = \begin{pmatrix} |b|^2 & b(\bar{\lambda}_2 - \bar{\lambda}_1) \\ \bar{b}(\lambda_2 - \lambda_1) & -|b|^2 \end{pmatrix}.$$

Its four squared moduli are  $|b|^4$ ,  $|b|^2|\lambda_1 - \lambda_2|^2$ ,  $|b|^2|\lambda_1 - \lambda_2|^2$  and  $|b|^4$ . Adding them gives the stated formula.

- (c) Let  $\mathbf{A} \in M_2(\mathbb{C})$  and write  $\mathbf{A} = \mathbf{U}\mathbf{T}\mathbf{U}^*$  as in [Corollary 11.1.3](#), with  $\mathbf{T}$  upper triangular of the shape in (b) and  $\lambda_1, \lambda_2$  the eigenvalues of  $\mathbf{A}$ . By [Eq. 11.11.1](#),  $\Delta(\mathbf{A})^2 = |b|^2$ . By (a) and (b),

$$\|[\mathbf{A}, \mathbf{A}^*]\|_F^2 = 2|b|^4 + 2|b|^2|\lambda_1 - \lambda_2|^2 \geq 2|b|^4 = 2\Delta(\mathbf{A})^4,$$

and taking non-negative square roots gives  $\|[\mathbf{A}, \mathbf{A}^*]\|_F \geq \sqrt{2} \Delta(\mathbf{A})^2$ .

Equality holds exactly when  $|b|^2|\lambda_1 - \lambda_2|^2 = 0$ , that is, when  $b = 0$  or  $\lambda_1 = \lambda_2$ . The first case says  $\Delta(\mathbf{A}) = 0$ , that is,  $\mathbf{A}$  is normal, and then both sides are 0. So equality holds if and only if  $\mathbf{A}$  is normal or its two eigenvalues coincide.

# Chapter 12

## Positive Matrices and the Singular Value Decomposition

A matrix that is not square has no eigenvalues. There is no equation  $\mathbf{A}\mathbf{v} = \lambda\mathbf{v}$  to write down, because  $\mathbf{A}\mathbf{v}$  and  $\mathbf{v}$  live in different spaces, and everything Chapters 8 to 11 built on that equation is unavailable. Yet the geometric picture that opened Chapter 11 survives untouched: push the unit sphere of  $F^n$  through any  $\mathbf{A} \in M_{m \times n}(F)$ , square or not, and what comes out is an ellipsoid. It has semi-axes, they are perpendicular, and the directions they came from were perpendicular too. Reading that picture as a factorization gives  $\mathbf{A} = \mathbf{U}\mathbf{\Sigma}\mathbf{V}^*$ , the **singular value decomposition**: every matrix whatsoever maps some orthonormal basis to an orthogonal set. It is the most used factorization in applied linear algebra, and this chapter's business is to earn it and then spend it.

The road runs through positivity. A non-square  $\mathbf{A}$  has no eigenvalues, but  $\mathbf{A}^*\mathbf{A}$  does, and  $\mathbf{A}^*\mathbf{A}$  is self-adjoint with a non-negative quadratic form — it is a **positive semidefinite** matrix, the analogue of a non-negative number. Chapter 11 left that class unexamined; here it comes first, with its many equivalent tests, its square root, and the triangular factorization that elimination finds. From the square root comes the absolute value  $|\mathbf{A}|$ , from  $|\mathbf{A}|$  the polar decomposition  $\mathbf{A} = \mathbf{W}|\mathbf{A}|$ , and from the spectral theorem applied to  $\mathbf{A}^*\mathbf{A}$  the singular value decomposition itself. Once it is in hand, a series of questions that look unrelated turn out to have one answer: what are orthonormal bases of the four fundamental subspaces, what is the best rank- $k$  approximation of a matrix, what should  $\mathbf{A}^{-1}$  mean when  $\mathbf{A}$  is not invertible, and when is a projection orthogonal. **The field is  $\mathbb{R}$  or  $\mathbb{C}$  throughout**, as in Chapters 10 and 11, and every statement says which one it needs.

**What you need.** From Chapter 11, the two spectral theorems and their matrix forms ([Theorem 11.4.2](#), [Corollary 11.4.3](#), [Theorem 11.5.1](#), [Corollary 11.5.2](#), [Corollary 11.5.4](#), [Theorem 11.5.5](#)), the elementary theory of self-adjoint operators ([Theorem 11.2.1](#), [Theorem 11.2.3](#), [Theorem 11.2.6](#), [Exercise \(C1: The operator  \$S^\*S\$ \)](#)), commuting families ([Corollary 11.7.2](#), [Corollary 11.7.3](#)), and the functional calculus with its square root ([Definition 11.8.1](#), [Theorem 11.8.1](#), [Theorem 11.8.3](#), [Theorem 11.8.4](#)), together with the Frobenius norm and its unitary invariance ([Lemma 11.11.1](#)). From Chapter 10, the adjoint and what it does to the four subspaces ([Definition 10.6.1](#), [Theorem 10.6.2](#), [Theorem 10.6.3](#), [Theorem 10.6.5](#), [Corollary 10.6.6](#)), orthogonal decompositions and projections ([Theorem 10.3.2](#), [Definition 10.3.2](#), [Theorem 10.3.3](#), [Theorem 10.3.5](#)), least squares ([Theorem 10.4.2](#), [Corollary 10.4.4](#), [Theorem 10.4.5](#), [Lemma 10.4.3](#)), Gram matrices ([Definition 10.2.3](#), [Theorem 10.2.5](#)), isometries ([Definition 10.7.1](#), [Theorem 10.7.3](#), [Definition 10.7.2](#)) and the QR factorization ([Theorem 10.8.1](#)). From Chapter 7, block multiplication and Schur

complements (Theorem 7.1.1, Definition 7.3.1, Theorem 7.3.1, Theorem 7.3.2, Theorem 7.3.3, Theorem 7.3.4). From Chapter 6, multiplicativity of the determinant, triangular determinants and the non-negativity of Gram determinants (Theorem 6.5.4, Theorem 6.4.6, Corollary 6.7.3). From Chapter 5, Lagrange interpolation (Theorem 5.6.1). From Chapter 3, projections along a complement (Definition 3.10.1, Theorem 3.10.1). From Chapter 2, the  $\text{LDL}^\top$  factorization and the invertible matrix theorem it started (Theorem 2.6.8, Theorem 2.4.4).

## 12.0.1 Roadmap

- **Positive semidefinite and positive definite matrices.** The definition, the real-versus-complex asymmetry that trips everyone, and the full list of equivalent tests, ending in Sylvester's criterion and the trap that comes with it.
- **The square root and Cholesky.** The unique positive square root, why it is a polynomial in the matrix, the absolute value  $|\mathbf{A}|$ , and the factorization  $\mathbf{A} = \mathbf{L}\mathbf{L}^*$  whose algorithm is itself the test for definiteness.
- **Gram matrices.** A matrix is a Gram matrix exactly when it is positive semidefinite; the Gram determinant as a squared volume, and distance to a subspace computed two ways.
- **Two forms at once.** A positive definite form and a self-adjoint one can be diagonalized simultaneously — by congruence, not by similarity — and the generalized eigenvalue problem  $\mathbf{B}\mathbf{x} = \lambda\mathbf{A}\mathbf{x}$  follows.
- **The Loewner order.** What  $\mathbf{A} \succeq \mathbf{B}$  means, what monotonicity survives it (traces, determinants, inverses), and the chapter's negative result: squaring does not.
- **Blocks and Schur complements.** Testing a block matrix for positive definiteness one block at a time, and the Fischer and Hadamard determinant inequalities that fall out.
- **Products entry by entry.** The Schur product theorem — the Hadamard product of positive matrices is positive — and Oppenheim's inequality.
- **The singular value decomposition.** Every matrix is  $\mathbf{U}\Sigma\mathbf{V}^*$ ; the singular values are determined, the unitaries are not, and the compact form reads off orthonormal bases of all four fundamental subspaces.
- **Polar decomposition and Takagi.**  $\mathbf{A} = \mathbf{W}|\mathbf{A}|$  is  $z = e^{i\theta}r$  for matrices;  $\mathbf{W}$  is the nearest unitary matrix to  $\mathbf{A}$ ; and a complex symmetric matrix factors as  $\mathbf{U}\Sigma\mathbf{U}^\top$ .
- **Low-rank approximation.** Truncating the singular value sum gives the best approximation of each rank, in the Frobenius norm, with an error one can write down.
- **The pseudoinverse.** Inverting the non-zero singular values defines  $\mathbf{A}^+$ ; it is pinned down by four conditions, and  $\mathbf{A}^+\mathbf{b}$  is the minimum-norm least-squares solution promised in Chapter 10.
- **Projections revisited.** An idempotent is an orthogonal projection exactly when it is self-adjoint; everything else is oblique, and the obliqueness is a measurable angle.

## 12.0.2 Named moves

- **Test with  $\langle \mathbf{A}\mathbf{x}, \mathbf{x} \rangle$ .** Positivity is a statement about one scalar function of one vector. Most proofs in the first half of the chapter consist of evaluating that function at a cleverly chosen  $\mathbf{x}$  — a standard basis vector, a vector supported on a few coordinates, an eigenvector, or  $\mathbf{S}\mathbf{y}$  for a substitution  $\mathbf{S}$ .
- **Factor as  $\mathbf{B}^*\mathbf{B}$ .** Every positive semidefinite matrix is one, and that form turns a statement about the matrix into a statement about lengths:  $\langle \mathbf{B}^*\mathbf{B}\mathbf{x}, \mathbf{x} \rangle = \|\mathbf{B}\mathbf{x}\|^2$ . It is also what makes every positive semidefinite matrix a Gram matrix.
- **Diagonalize, do it to the eigenvalues, come back.** The square root, the pseudoinverse

and the whole functional calculus are this one move. The work is always in checking that the result does not depend on the diagonalization chosen.

- **Apply the spectral theorem to  $\mathbf{A}^*\mathbf{A}$ .** When the matrix in front of you is not square, not normal, or not even an operator, this is the way in. The singular value decomposition is nothing but the spectral theorem for  $\mathbf{A}^*\mathbf{A}$ , read back through  $\mathbf{A}$ .
- **Truncate the sum.** Writing  $\mathbf{A} = \sum_i \sigma_i \mathbf{u}_i \mathbf{v}_i^*$  turns a matrix into a sum of rank-one pieces ordered by importance. Cutting the sum short gives the compact form, the best low-rank approximation and the pseudoinverse, three times from one idea.

## 12.1 Positive Semidefinite and Positive Definite Matrices

Chapter 11 sorted the operators that have an orthonormal eigenbasis. Among them the self-adjoint ones are the matrix analogue of the real numbers, since they are exactly the operators fixed by starring. Inside the real numbers sits a smaller and even more useful set, the non-negative ones, and this section finds its matrix analogue. The answer turns out to have three faces — a sign condition on the spectrum, a sign condition on a scalar function, and a factorization — and the whole of the rest of the chapter is built on the fact that the three agree.

Throughout,  $F = \mathbb{R}$  or  $F = \mathbb{C}$ , and  $F^n$  carries the standard inner product  $\langle \mathbf{x}, \mathbf{y} \rangle = \mathbf{y}^* \mathbf{x}$ .

### 12.1.1 Which self-adjoint matrices deserve to be called non-negative?

A real number  $t$  is non-negative in three equivalent ways:  $t \geq 0$ ;  $tx^2 \geq 0$  for every real  $x$ ; and  $t = s^2$  for some real  $s$ . Each of the three has a matrix reading. “Every eigenvalue is  $\geq 0$ ” copies the first. “ $\langle \mathbf{A}\mathbf{x}, \mathbf{x} \rangle \geq 0$  for every  $\mathbf{x}$ ” copies the second, because  $\langle \mathbf{A}\mathbf{x}, \mathbf{x} \rangle$  is the one scalar a matrix attaches to a single vector. “ $\mathbf{A} = \mathbf{B}^*\mathbf{B}$ ” copies the third, with the star doing what nothing does for numbers but everything does for matrices.

It is not obvious that the three agree, and for a general matrix they do not: they all fail for  $\begin{pmatrix} 0 & 1 \\ 0 & 0 \end{pmatrix}$  in different ways. What makes them agree is self-adjointness, and the middle condition is the one to take as the definition, because it is the one that says something about every vector at once.

*A matrix is positive semidefinite when its quadratic form never goes negative.*

#### Definition 12.1.1 — Positive Semidefinite and Positive Definite

Let  $\mathbf{A} \in M_n(F)$ . Then  $\mathbf{A}$  is **positive semidefinite**, written  $\mathbf{A} \succeq 0$ , if

- (P1)  $\mathbf{A}$  is **Hermitian**, that is  $\mathbf{A}^* = \mathbf{A}$  (over  $\mathbb{R}$ : symmetric,  $\mathbf{A}^\top = \mathbf{A}$ ); and
- (P2)  $\langle \mathbf{A}\mathbf{x}, \mathbf{x} \rangle \geq 0$  for **every**  $\mathbf{x} \in F^n$ .

It is **positive definite**, written  $\mathbf{A} \succ 0$ , if (P1) holds and  $\langle \mathbf{A}\mathbf{x}, \mathbf{x} \rangle > 0$  for every **non-zero**

$\mathbf{x} \in F^n$ .

The same words apply to an operator  $T \in \mathcal{L}(V)$  on a finite-dimensional inner product space:  $T$  is positive semidefinite if  $T^* = T$  and  $\langle T\mathbf{v}, \mathbf{v} \rangle \geq 0$  for every  $\mathbf{v} \in V$ , and positive definite if the inequality is strict for  $\mathbf{v} \neq \mathbf{0}$ .

In words: (P1) puts  $\mathbf{A}$  in the class where the analogy with real numbers is available at all, and (P2) says that the number  $\langle \mathbf{A}\mathbf{x}, \mathbf{x} \rangle = \mathbf{x}^* \mathbf{A} \mathbf{x}$  is never negative. The word **every** in (P2) is the whole content; a single vector proves nothing. In the definite case the vector  $\mathbf{x} = \mathbf{0}$  has to be excluded, since  $\langle \mathbf{A}\mathbf{0}, \mathbf{0} \rangle = 0$  for every  $\mathbf{A}$  whatsoever.

Clause (P1) is also what makes clause (P2) meaningful. For a Hermitian  $\mathbf{A}$ , the number  $\langle \mathbf{A}\mathbf{x}, \mathbf{x} \rangle$  is **real** for every  $\mathbf{x}$ , over  $\mathbb{C}$  as well as over  $\mathbb{R}$  ([Proposition 10.6.7](#)), so comparing it with 0 is legitimate. Without (P1) the number can be a genuine complex number, and “ $\geq 0$ ” would be meaningless.

Positive definite implies positive semidefinite, since a condition that holds strictly for  $\mathbf{x} \neq \mathbf{0}$  holds weakly for  $\mathbf{x} = \mathbf{0}$  too. The reverse fails, and the smallest witness is  $\text{diag}(1, 0)$ .

Before any examples, the trap.

*Warning.* Over  $\mathbb{R}$ , clause (P1) has to be assumed; over  $\mathbb{C}$  it comes free. Let  $\mathbf{A} = \begin{pmatrix} 0 & -1 \\ 1 & 0 \end{pmatrix}$ , the quarter turn of  $\mathbb{R}^2$ . Then  $\langle \mathbf{A}\mathbf{x}, \mathbf{x} \rangle = -x_2x_1 + x_1x_2 = 0$  for every  $\mathbf{x} \in \mathbb{R}^2$ , so (P2) holds in the strongest possible way, and yet  $\mathbf{A}^\top = -\mathbf{A} \neq \mathbf{A}$ . A real matrix can pass the quadratic-form test and be nowhere near symmetric, because over  $\mathbb{R}$  the quadratic form cannot see the skew part of a matrix at all ([Exercise \(C3: What the real quadratic form does determine\)](#)). So a real definition that dropped (P1) would call the quarter turn positive semidefinite, and none of the theorems below would survive.

Over  $\mathbb{C}$  the situation is the opposite, and it is worth recording once.

### Proposition 12.1.1 — Over $\mathbb{C}$ , positivity implies Hermitian

Let  $\mathbf{A} \in M_n(\mathbb{C})$  satisfy  $\langle \mathbf{A}\mathbf{x}, \mathbf{x} \rangle \in \mathbb{R}$  for every  $\mathbf{x} \in \mathbb{C}^n$  — in particular, let  $\langle \mathbf{A}\mathbf{x}, \mathbf{x} \rangle \geq 0$  for every  $\mathbf{x}$ . Then  $\mathbf{A}^* = \mathbf{A}$ .

*Proof.* Put  $\mathbf{S} = \mathbf{A} - \mathbf{A}^*$ . For every  $\mathbf{x}$ , moving the star across ([Lemma 10.4.1](#)) and using conjugate symmetry,

$$\langle \mathbf{A}^*\mathbf{x}, \mathbf{x} \rangle = \langle \mathbf{x}, \mathbf{A}\mathbf{x} \rangle = \overline{\langle \mathbf{A}\mathbf{x}, \mathbf{x} \rangle} = \langle \mathbf{A}\mathbf{x}, \mathbf{x} \rangle,$$

the last equality because the number is real. Hence  $\langle \mathbf{S}\mathbf{x}, \mathbf{x} \rangle = 0$  for every  $\mathbf{x} \in \mathbb{C}^n$ , and [Theorem 11.2.6](#) gives  $\mathbf{S} = \mathbf{0}$ . ■

So over  $\mathbb{C}$  one may state the definition with (P2) alone; over  $\mathbb{R}$  one may not. We keep (P1) in the definition for both fields, so that a single sentence covers them and so that no reader has to remember which field licenses which shortcut.

Now the examples, all checked against (P1) and (P2) in turn.



**Example. (Five Positive Matrices)**

Decide which of the following are positive semidefinite, and which are positive definite.

- (a)  $\mathbf{I}_n$ , and more generally  $\mathbf{D} = \text{diag}(d_1, \dots, d_n)$  with every  $d_i$  real.
- (b)  $\mathbf{x}\mathbf{x}^*$  for a fixed  $\mathbf{x} \in F^n$ .
- (c) The Gram matrix  $\mathbf{G}$  of a list  $(\mathbf{v}_1, \dots, \mathbf{v}_k)$  in an inner product space.
- (d) The zero matrix  $0 \in M_n(F)$ .
- (e)  $\begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix}$ .

*Solution.*

- (a)  $\mathbf{D}^* = \mathbf{D}$  because the  $d_i$  are real, so (P1) holds. For (P2),

$$\langle \mathbf{D}\mathbf{x}, \mathbf{x} \rangle = \mathbf{x}^* \mathbf{D} \mathbf{x} = \sum_{i=1}^n d_i |x_i|^2.$$

So  $\mathbf{D} \succeq 0$  exactly when every  $d_i \geq 0$ , and  $\mathbf{D} \succ 0$  exactly when every  $d_i > 0$ : if some  $d_i < 0$ , the vector  $\mathbf{e}_i$  refutes (P2), and if some  $d_i = 0$ , the vector  $\mathbf{e}_i$  refutes definiteness. In particular  $\mathbf{I}_n \succ 0$ , with  $\langle \mathbf{x}, \mathbf{x} \rangle = \|\mathbf{x}\|^2$ .

- (b)  $(\mathbf{x}\mathbf{x}^*)^* = \mathbf{x}^{**}\mathbf{x}^* = \mathbf{x}\mathbf{x}^*$ , so (P1) holds. For (P2), using  $\langle \mathbf{u}, \mathbf{w} \rangle = \overline{\langle \mathbf{w}, \mathbf{u} \rangle}$  twice,

$$\mathbf{y}^*(\mathbf{x}\mathbf{x}^*)\mathbf{y} = (\mathbf{y}^*\mathbf{x})(\mathbf{x}^*\mathbf{y}) = \langle \mathbf{x}, \mathbf{y} \rangle \overline{\langle \mathbf{x}, \mathbf{y} \rangle} = |\langle \mathbf{x}, \mathbf{y} \rangle|^2 \geq 0.$$

So  $\mathbf{x}\mathbf{x}^* \succeq 0$ . It is never positive definite for  $n \geq 2$ , since any  $\mathbf{y} \neq \mathbf{0}$  orthogonal to  $\mathbf{x}$  makes the value 0; and for  $\mathbf{x} = \mathbf{0}$  it is the zero matrix.

- (c)  $\mathbf{G}^* = \mathbf{G}$  by [Theorem 10.2.5](#) (a), and  $\mathbf{x}^*\mathbf{G}\mathbf{x} = \|\sum_j x_j \mathbf{v}_j\|^2 \geq 0$  by part (b) of the same theorem. So  $\mathbf{G} \succeq 0$  always, and by (c) of that theorem  $\mathbf{G} \succ 0$  exactly when the list is linearly independent.
- (d)  $0^* = 0$  and  $\langle 0\mathbf{x}, \mathbf{x} \rangle = 0 \geq 0$ , so  $0 \succeq 0$ . This is the degenerate case, and it matters: it is the bottom of the order introduced in Section 5, and it shows that (P2) can hold with equality everywhere.
- (e) Symmetric, so (P1) holds; but  $\langle \mathbf{A}\mathbf{e}_2, \mathbf{e}_2 \rangle = -1 < 0$ . Not positive semidefinite. This is (a) with  $d_2 = -1$ .

Example (e) is the non-example by minimal change: keep (P1), change one diagonal entry, and (P2) fails at one standard basis vector. The quarter turn of the warning is the other kind of non-example, keeping (P2) and losing (P1).

Two remarks on the name and the convention. “Definite” is the older word, from the theory of quadratic forms: the form  $q(\mathbf{x}) = \mathbf{x}^* \mathbf{A} \mathbf{x}$  has a *definite* sign. “Semi” weakens the sign to allow zeros. Some authors write “positive” for what we call positive semidefinite and “strictly positive” for positive definite; the words differ but the two classes are the same two classes everywhere.

**Warning. A positive matrix is not a matrix with positive entries.** The all-ones matrix  $\mathbf{J} \in M_2(\mathbb{R})$  has every entry positive and is positive semidefinite, but  $\begin{pmatrix} 1 & 2 \\ 2 & 1 \end{pmatrix}$  has every entry positive and is **not**: at  $\mathbf{x} = (1, -1)$  its form is  $1 - 2 - 2 + 1 = -2$ . Conversely

$\text{diag}(1, 1)$  has a zero entry and is positive definite. Entrywise positivity is a different subject, the one Chapter 8 met through stochastic matrices, and it shares only a word with this one.

## 12.1.2 Positivity survives a change of variable

Before the equivalences, one small result that will be used in almost every proof of this chapter. The quadratic form  $\mathbf{x} \mapsto \mathbf{x}^* \mathbf{A} \mathbf{x}$  is attached to  $\mathbf{A}$ ; substituting  $\mathbf{x} = \mathbf{S} \mathbf{y}$  turns it into the quadratic form of  $\mathbf{S}^* \mathbf{A} \mathbf{S}$ . The operation  $\mathbf{A} \mapsto \mathbf{S}^* \mathbf{A} \mathbf{S}$  is called **congruence**, and Chapter 13 studies it for its own sake. Here we need only that it preserves the sign.

### Proposition 12.1.2 — Congruence preserves positivity

Let  $\mathbf{A} \in M_n(F)$  be Hermitian and  $\mathbf{S} \in M_{n \times m}(F)$ . Then  $\mathbf{S}^* \mathbf{A} \mathbf{S} \in M_m(F)$  is Hermitian, and:

- (a) if  $\mathbf{A} \succeq 0$ , then  $\mathbf{S}^* \mathbf{A} \mathbf{S} \succeq 0$ ;
- (b) if  $\mathbf{A} \succ 0$  and  $\text{nul}(\mathbf{S}) = \{\mathbf{0}\}$ , then  $\mathbf{S}^* \mathbf{A} \mathbf{S} \succ 0$ . In particular, for  $m = n$  and  $\mathbf{S}$  invertible,  $\mathbf{A} \succ 0$  if and only if  $\mathbf{S}^* \mathbf{A} \mathbf{S} \succ 0$ .

*Proof.*  $(\mathbf{S}^* \mathbf{A} \mathbf{S})^* = \mathbf{S}^* \mathbf{A}^* \mathbf{S}^{**} = \mathbf{S}^* \mathbf{A} \mathbf{S}$ , so the matrix is Hermitian. For every  $\mathbf{y} \in F^m$ ,

$$\mathbf{y}^* (\mathbf{S}^* \mathbf{A} \mathbf{S}) \mathbf{y} = (\mathbf{S} \mathbf{y})^* \mathbf{A} (\mathbf{S} \mathbf{y}) = \langle \mathbf{A}(\mathbf{S} \mathbf{y}), \mathbf{S} \mathbf{y} \rangle. \quad (*)$$

- (a) If  $\mathbf{A} \succeq 0$ , the right-hand side of  $(*)$  is  $\geq 0$  for every  $\mathbf{y}$ .
- (b) If  $\mathbf{A} \succ 0$  and  $\text{nul}(\mathbf{S}) = \{\mathbf{0}\}$ , then  $\mathbf{y} \neq \mathbf{0}$  forces  $\mathbf{S} \mathbf{y} \neq \mathbf{0}$ , so the right-hand side of  $(*)$  is  $> 0$ . For the last claim, take  $\mathbf{S}$  invertible: one direction is what we just proved, and the other is the same statement applied to  $\mathbf{S}^{-1}$  and the matrix  $\mathbf{S}^* \mathbf{A} \mathbf{S}$ , since  $(\mathbf{S}^{-1})^* (\mathbf{S}^* \mathbf{A} \mathbf{S}) \mathbf{S}^{-1} = \mathbf{A}$ .

■

## 12.1.3 The equivalent descriptions

The characterization needs one determinant identity, which is worth stating on its own because it also describes every coefficient of a characteristic polynomial. Recall from [Definition 6.7.1](#) that  $\mathbf{A}_{I,I}$  denotes the submatrix of  $\mathbf{A}$  on the rows and columns indexed by  $I$ ; its determinant is a **principal minor** of  $\mathbf{A}$ , and  $\mathbf{A}_{I,I}$  is a **principal submatrix**.

### Lemma 12.1.3 — Shifting by $t\mathbf{I}$ , minor by minor

Let  $\mathbf{A} \in M_n(F)$  and  $t \in F$ . Then

$$\det(\mathbf{A} + t\mathbf{I}_n) = \sum_{k=0}^n E_k(\mathbf{A}) t^{n-k},$$

where  $E_0(\mathbf{A}) = 1$  and, for  $1 \leq k \leq n$ ,  $E_k(\mathbf{A}) = \sum_{|I|=k} \det \mathbf{A}_{I,I}$  is the sum of all  $k \times k$  principal minors of  $\mathbf{A}$ .

*Idea.* Column  $j$  of  $\mathbf{A} + t\mathbf{I}$  is  $\mathbf{a}_j + t\mathbf{e}_j$ , a sum of two columns, and the determinant is linear in each column separately. Expanding all  $n$  columns turns one determinant into  $2^n$  of them, one for each choice of which columns contribute their  $t\mathbf{e}_j$ . A determinant with  $\mathbf{e}_j$  in column  $j$  collapses under cofactor expansion to the determinant with row  $j$  and column  $j$  deleted, and doing that for every chosen column leaves exactly a principal submatrix.

*Proof.* Write  $\mathbf{a}_1, \dots, \mathbf{a}_n$  for the columns of  $\mathbf{A}$ , so that column  $j$  of  $\mathbf{A} + t\mathbf{I}$  is  $\mathbf{a}_j + t\mathbf{e}_j$ . The determinant is linear in each column with the others fixed (Theorem 6.4.3). Applying that in column 1, then in column 2, and so on through column  $n$ ,

$$\det(\mathbf{A} + t\mathbf{I}) = \sum_{S \subseteq \{1, \dots, n\}} t^{|S|} \det \mathbf{M}_S,$$

where  $\mathbf{M}_S$  is the matrix whose  $j$ -th column is  $\mathbf{e}_j$  for  $j \in S$  and  $\mathbf{a}_j$  for  $j \notin S$ .

Fix  $S$  and let  $j \in S$ . Expanding  $\det \mathbf{M}_S$  along column  $j$  (Theorem 6.6.1 (a)) leaves one term, since that column is  $\mathbf{e}_j$ : the entry is 1 in row  $j$  and 0 elsewhere, and its cofactor is  $(-1)^{j+j} M_{jj} = M_{jj}$ . So  $\det \mathbf{M}_S$  equals the determinant of  $\mathbf{M}_S$  with row  $j$  and column  $j$  deleted, which is a matrix of the same shape on the index set  $\{1, \dots, n\} \setminus \{j\}$ . Repeating this for each  $j \in S$  deletes every row and column indexed by  $S$  and leaves

$$\det \mathbf{M}_S = \det \mathbf{A}_{S^c, S^c}, \quad S^c = \{1, \dots, n\} \setminus S.$$

Collecting the subsets by  $k = |S^c|$ , so that  $|S| = n - k$ , gives the stated formula. ■

*Remark.* Running the same expansion over  $F[x]$  with the indeterminate  $-x$  in place of  $t$ , and multiplying by  $(-1)^n$ , turns this into  $p_{\mathbf{A}}(x) = \sum_k (-1)^k E_k(\mathbf{A}) x^{n-k}$ : the coefficients of the characteristic polynomial are, up to sign, the sums of principal minors. The cases  $k = 1$  and  $k = n$  recover  $\text{tr } \mathbf{A}$  and  $\det \mathbf{A}$  from Theorem 6.10.2.

Here is the main theorem of the section.

### Theorem 12.1.4 — Characterizations of Positive Semidefiniteness

Let  $\mathbf{A} \in M_n(F)$  be Hermitian. The following are equivalent.

- (a)  $\mathbf{A} \succeq 0$ .
- (b) Every eigenvalue of  $\mathbf{A}$  is a non-negative real number.
- (c)  $\mathbf{A} = \mathbf{B}^* \mathbf{B}$  for some  $\mathbf{B} \in M_n(F)$ .
- (d)  $\mathbf{A} = \mathbf{C}^2$  for exactly one  $\mathbf{C} \in M_n(F)$  with  $\mathbf{C} \succeq 0$ .
- (e) Every principal minor of  $\mathbf{A}$  is  $\geq 0$ .

*Idea.* The spectral theorem makes (a) and (b) two readings of one diagonal matrix, since an orthonormal change of basis does not change the sign of a quadratic form.

From (b) the square root of Chapter 11 produces  $\mathbf{C}$ , and  $\mathbf{C}$  is its own  $\mathbf{B}$ . The step (c)  $\Rightarrow$  (a) is the identity  $\langle \mathbf{B}^* \mathbf{B} \mathbf{x}, \mathbf{x} \rangle = \|\mathbf{B} \mathbf{x}\|^2$ , already proved in Chapter 11 as an exercise. For (e), a principal submatrix is what the form does to vectors supported on a few coordinates, which gives (a)  $\Rightarrow$  (e) at once; the return trip uses [Lemma 12.1.3](#) to show that  $\det(\mathbf{A} + t\mathbf{I}) > 0$  for every  $t > 0$ , so that no  $-t$  can be an eigenvalue.

*Proof.* (a)  $\Leftrightarrow$  (b). ( $\Rightarrow$ ) Let  $\mathbf{A} \mathbf{v} = \lambda \mathbf{v}$  with  $\mathbf{v} \neq \mathbf{0}$ . By [Theorem 11.2.1](#),  $\lambda \in \mathbb{R}$ , and

$$0 \leq \langle \mathbf{A} \mathbf{v}, \mathbf{v} \rangle = \langle \lambda \mathbf{v}, \mathbf{v} \rangle = \lambda \|\mathbf{v}\|^2,$$

so  $\lambda \geq 0$ , as  $\|\mathbf{v}\|^2 > 0$ . ( $\Leftarrow$ ) By [Corollary 11.4.3](#) (over  $\mathbb{C}$ ) or [Corollary 11.5.2](#) (over  $\mathbb{R}$ ),  $\mathbf{A} = \mathbf{U} \mathbf{D} \mathbf{U}^*$  with  $\mathbf{U}$  unitary and  $\mathbf{D} = \text{diag}(\lambda_1, \dots, \lambda_n)$  the eigenvalues. Put  $\mathbf{y} = \mathbf{U}^* \mathbf{x}$ . Then

$$\langle \mathbf{A} \mathbf{x}, \mathbf{x} \rangle = \mathbf{x}^* \mathbf{U} \mathbf{D} \mathbf{U}^* \mathbf{x} = \mathbf{y}^* \mathbf{D} \mathbf{y} = \sum_i \lambda_i |y_i|^2 \geq 0,$$

every term being a product of two non-negative reals.

(b)  $\Rightarrow$  (d). A Hermitian matrix has an orthonormal basis of eigenvectors, so the operator  $T_{\mathbf{A}}$  on  $F^n$  satisfies the hypothesis of [Theorem 11.8.4](#), whose spectral condition  $\text{spec}(T_{\mathbf{A}}) \subseteq [0, \infty)$  is exactly (b). That theorem gives a unique  $\mathbf{C}$  with an orthonormal eigenbasis, non-negative spectrum and  $\mathbf{C}^2 = \mathbf{A}$ , and it is self-adjoint; by the equivalence just proved, applied to  $\mathbf{C}$ , this says  $\mathbf{C} \succeq 0$ . Conversely any  $\mathbf{C} \succeq 0$  with  $\mathbf{C}^2 = \mathbf{A}$  is Hermitian with non-negative spectrum, hence is that one. So there is exactly one such  $\mathbf{C}$ .

(d)  $\Rightarrow$  (c). Take  $\mathbf{B} = \mathbf{C}$ ; then  $\mathbf{B}^* \mathbf{B} = \mathbf{C}^* \mathbf{C} = \mathbf{C}^2 = \mathbf{A}$ .

(c)  $\Rightarrow$  (a).  $(\mathbf{B}^* \mathbf{B})^* = \mathbf{B}^* \mathbf{B}$ , and  $\langle \mathbf{B}^* \mathbf{B} \mathbf{x}, \mathbf{x} \rangle = \|\mathbf{B} \mathbf{x}\|^2 \geq 0$  for every  $\mathbf{x}$ . Both statements are [Exercise \(C1: The operator  \$S^\* S\$ \)](#) (a), applied to the map  $\mathbf{x} \mapsto \mathbf{B} \mathbf{x}$ .

(a)  $\Rightarrow$  (e). Let  $I = \{i_1 < \dots < i_k\}$ . Given  $\mathbf{y} \in F^k$ , let  $\mathbf{x} \in F^n$  have  $x_{i_p} = y_p$  and all other entries 0. Then  $\mathbf{x}^* \mathbf{A} \mathbf{x} = \mathbf{y}^* \mathbf{A}_{I,I} \mathbf{y}$ , because the terms  $\bar{x}_i a_{ij} x_j$  with  $i \notin I$  or  $j \notin I$  vanish. So  $\mathbf{A}_{I,I}$  is Hermitian and satisfies (P2), that is  $\mathbf{A}_{I,I} \succeq 0$ . By the equivalence (a)  $\Leftrightarrow$  (b) applied to  $\mathbf{A}_{I,I}$ , all its eigenvalues are  $\geq 0$ . Diagonalizing it as  $\mathbf{A}_{I,I} = \mathbf{U} \mathbf{D} \mathbf{U}^*$  as above, with  $\mathbf{U}^* = \mathbf{U}^{-1}$ , and taking determinants ([Theorem 6.5.4](#), [Corollary 6.5.6](#)),

$$\det \mathbf{A}_{I,I} = \det \mathbf{D} = \lambda_1 \cdots \lambda_k \geq 0.$$

(e)  $\Rightarrow$  (b). Let  $t > 0$  be real. Every  $E_k(\mathbf{A}) \geq 0$  by hypothesis and  $E_0(\mathbf{A}) = 1$ , so [Lemma 12.1.3](#) gives

$$\det(\mathbf{A} + t\mathbf{I}) = \sum_{k=0}^n E_k(\mathbf{A}) t^{n-k} \geq t^n > 0.$$

In particular  $\mathbf{A} + t\mathbf{I}$  is invertible ([Theorem 6.5.3](#)), so  $-t$  is not an eigenvalue of  $\mathbf{A}$  ([Theorem 8.2.6](#) applied to  $\mathbf{A} + t\mathbf{I}$ ). As  $t > 0$  was arbitrary and every eigenvalue of  $\mathbf{A}$  is real ([Theorem 11.2.1](#)), no eigenvalue is negative. This closes the cycle and proves the theorem. ■

Item (c) is the one to keep in mind: **every positive semidefinite matrix is a Gram matrix**,

since  $\mathbf{B}^*\mathbf{B}$  is the Gram matrix of the columns of  $\mathbf{B}$  (Definition 10.2.3). Section 3 turns that remark into a theorem. Item (d) names the square root, which Section 2 develops. Item (b) is how one *thinks*; item (e) is, for small matrices, how one *computes*.

## 12.1.4 The definite case, and Sylvester's criterion

The definite versions of (a), (b) and (c) are proved by putting “strict” in front of the same arguments. The new item is a test that can be run by hand on a numerical matrix without finding a single eigenvalue.

### Theorem 12.1.5 — Characterizations of Positive Definiteness, with Sylvester's Criterion

Let  $\mathbf{A} \in M_n(F)$  be Hermitian, and for  $1 \leq k \leq n$  let  $\mathbf{A}_k$  be its leading principal submatrix, the top-left  $k \times k$  corner. The following are equivalent.

- (a)  $\mathbf{A} \succ 0$ .
- (b) Every eigenvalue of  $\mathbf{A}$  is a positive real number.
- (c)  $\mathbf{A} = \mathbf{B}^*\mathbf{B}$  for some invertible  $\mathbf{B} \in M_n(F)$ .
- (d) (**Sylvester's criterion**)  $\det \mathbf{A}_k > 0$  for every  $k = 1, \dots, n$ .

*Idea.* Steps (a)  $\Leftrightarrow$  (b)  $\Leftrightarrow$  (c) repeat the previous proof with strict inequalities. For (a)  $\Rightarrow$  (d), a leading principal submatrix inherits definiteness, and a determinant is a product of eigenvalues. The content is (d)  $\Rightarrow$  (a), and it is an induction on  $n$  that peels off the last row and column. The inductive hypothesis makes the corner  $\mathbf{A}_{n-1}$  definite, hence invertible, and one shear — the congruence by  $\mathbf{S} = \begin{pmatrix} \mathbf{I} & -\mathbf{A}_{n-1}^{-1}\mathbf{b} \\ \mathbf{0}^\top & 1 \end{pmatrix}$ , which is exactly the block elimination of Chapter 7 — clears the border and turns  $\mathbf{A}$  into  $\mathbf{A}_{n-1} \oplus (s)$  with one scalar  $s$  left over. Determinants identify the sign of  $s$ , and Proposition 12.1.2 carries the verdict back.

*Proof.* (a)  $\Leftrightarrow$  (b). As in Theorem 12.1.4, with  $\geq$  replaced by  $>$  throughout: for ( $\Rightarrow$ ),  $0 < \lambda \|\mathbf{v}\|^2$  forces  $\lambda > 0$ ; for ( $\Leftarrow$ ),  $\sum_i \lambda_i |y_i|^2 > 0$  whenever  $\mathbf{y} = \mathbf{U}^*\mathbf{x} \neq \mathbf{0}$ , which happens exactly when  $\mathbf{x} \neq \mathbf{0}$  because  $\mathbf{U}^*$  is invertible.

(b)  $\Rightarrow$  (c). Let  $\mathbf{C} \succeq 0$  with  $\mathbf{C}^2 = \mathbf{A}$  (Theorem 12.1.4 (d)). Since every eigenvalue of  $\mathbf{A}$  is positive,  $0 \notin \text{spec}(\mathbf{A})$  and  $\mathbf{A}$  is invertible (Theorem 8.2.6). If  $\mathbf{C}\mathbf{x} = \mathbf{0}$  then  $\mathbf{A}\mathbf{x} = \mathbf{C}^2\mathbf{x} = \mathbf{0}$ , so  $\mathbf{x} = \mathbf{0}$ ; hence  $\mathbf{C}$  is invertible too (Theorem 2.4.4). Take  $\mathbf{B} = \mathbf{C}$ .

(c)  $\Rightarrow$  (a). For  $\mathbf{x} \neq \mathbf{0}$ ,  $\mathbf{B}\mathbf{x} \neq \mathbf{0}$ , so  $\langle \mathbf{B}^*\mathbf{B}\mathbf{x}, \mathbf{x} \rangle = \|\mathbf{B}\mathbf{x}\|^2 > 0$ .

(a)  $\Rightarrow$  (d). As in the proof of (a)  $\Rightarrow$  (e) of Theorem 12.1.4 with  $I = \{1, \dots, k\}$ , the corner  $\mathbf{A}_k$  satisfies  $\mathbf{y}^*\mathbf{A}_k\mathbf{y} = \mathbf{x}^*\mathbf{A}\mathbf{x} > 0$  for  $\mathbf{y} \neq \mathbf{0}$ , so  $\mathbf{A}_k \succ 0$ . Its eigenvalues are positive, and  $\det \mathbf{A}_k$  is their product, computed as in the proof of Theorem 12.1.4, so  $\det \mathbf{A}_k > 0$ .

(d)  $\Rightarrow$  (a). Induct on  $n$ . For  $n = 1$ ,  $\mathbf{A} = (a_{11})$  with  $a_{11} = \det \mathbf{A}_1 > 0$ , and  $\langle \mathbf{A}\mathbf{x}, \mathbf{x} \rangle = a_{11}|x|^2 > 0$  for  $x \neq 0$ .

Let  $n \geq 2$ , and write

$$\mathbf{A} = \begin{pmatrix} \mathbf{A}' & \mathbf{b} \\ \mathbf{b}^* & d \end{pmatrix}, \quad \mathbf{A}' = \mathbf{A}_{n-1}, \quad \mathbf{b} \in F^{n-1}, \quad d \in \mathbb{R}.$$

The leading principal submatrices of  $\mathbf{A}'$  are  $\mathbf{A}_1, \dots, \mathbf{A}_{n-1}$ , so the inductive hypothesis gives  $\mathbf{A}' \succ 0$ ; in particular  $\mathbf{A}'$  is invertible, since  $\det \mathbf{A}' = \det \mathbf{A}_{n-1} > 0$ . Put

$$\mathbf{S} = \begin{pmatrix} \mathbf{I}_{n-1} & -\mathbf{A}'^{-1}\mathbf{b} \\ \mathbf{0}^\top & 1 \end{pmatrix}, \quad s = d - \mathbf{b}^* \mathbf{A}'^{-1} \mathbf{b}.$$

Block multiplication (Theorem 7.1.1), together with  $(\mathbf{A}'^{-1})^* = \mathbf{A}'^{-1}$  because  $\mathbf{A}'$  is Hermitian, gives

$$\mathbf{S}^* \mathbf{A} \mathbf{S} = \begin{pmatrix} \mathbf{A}' & \mathbf{0} \\ \mathbf{0}^\top & s \end{pmatrix}. \quad (*)$$

Both  $\mathbf{S}$  and  $\mathbf{S}^*$  are triangular with 1s on the diagonal, so  $\det \mathbf{S} = \det \mathbf{S}^* = 1$  by Theorem 6.4.6. Taking determinants in (\*) with Theorem 6.5.4 and Theorem 6.8.4,

$$0 < \det \mathbf{A} = \det(\mathbf{S}^* \mathbf{A} \mathbf{S}) = s \det \mathbf{A}',$$

and  $\det \mathbf{A}' > 0$ , so  $s > 0$ . The right-hand side of (\*) is therefore positive definite: for  $\mathbf{y} = (\mathbf{y}', y_n) \neq \mathbf{0}$ ,

$$\mathbf{y}^* \begin{pmatrix} \mathbf{A}' & \mathbf{0} \\ \mathbf{0}^\top & s \end{pmatrix} \mathbf{y} = \mathbf{y}'^* \mathbf{A}' \mathbf{y}' + s |y_n|^2 > 0,$$

since the two terms are  $\geq 0$  and not both zero. Finally  $\mathbf{S}$  is invertible, so Proposition 12.1.2 (b) turns  $\mathbf{S}^* \mathbf{A} \mathbf{S} \succ 0$  back into  $\mathbf{A} \succ 0$ . This completes the induction and proves the theorem. ■

The number  $s = d - \mathbf{b}^* \mathbf{A}'^{-1} \mathbf{b}$  in that proof is the Schur complement  $\mathbf{A}/\mathbf{A}'$  of Definition 7.3.1, and (\*) is the block LDU factorization of Theorem 7.3.1 written as a congruence. Section 6 replaces the scalar  $s$  by a block and gets the general test.

**Warning.** For semidefiniteness, the leading principal minors are not enough. Sylvester's criterion has no “ $\geq$ ” version. Take

$$\mathbf{A} = \begin{pmatrix} 0 & 0 \\ 0 & -1 \end{pmatrix}.$$

Its leading principal minors are  $\det \mathbf{A}_1 = 0$  and  $\det \mathbf{A}_2 = 0$ , both  $\geq 0$ , and yet  $\langle \mathbf{A} \mathbf{e}_2, \mathbf{e}_2 \rangle = -1$ , so  $\mathbf{A}$  is not positive semidefinite. What Theorem 12.1.4 (e) demands is **all** principal minors, and the one this matrix fails is  $\det \mathbf{A}_{\{2\},\{2\}} = -1 < 0$ . The leading corners simply never look at the second coordinate on its own. This is the single most common error about the subject: for the strict test  $n$  minors suffice, for the weak test one needs all  $2^n - 1$  of them.

*Quick check.* Is  $\mathbf{A} = \begin{pmatrix} 1 & 2 \\ 2 & 4 \end{pmatrix}$  positive semidefinite? Is it positive definite? Which minors did you use?

*Solution.* It is positive semidefinite but not positive definite. The principal minors are 1, 4 and  $\det \mathbf{A} = 4 - 4 = 0$ , all  $\geq 0$ , so [Theorem 12.1.4](#) (e) applies. It is not definite, since  $\det \mathbf{A}_2 = 0$  violates Sylvester's criterion; concretely  $\mathbf{A}(2, -1) = \mathbf{0}$ , so  $\langle \mathbf{A}\mathbf{x}, \mathbf{x} \rangle = 0$  at the non-zero  $\mathbf{x} = (2, -1)$ . One can also see everything at once from  $\mathbf{A} = \mathbf{b}\mathbf{b}^\top$  with  $\mathbf{b} = (1, 2)$ , which is [Example \(Five Positive Matrices\)](#) (b).

## 12.1.5 Two by two, and where these matrices come from

The  $2 \times 2$  case is worth memorizing, because it is the one that appears inside larger arguments.

### Example. (The $2 \times 2$ Test)

Let  $\mathbf{A} = \begin{pmatrix} a & b \\ \bar{b} & d \end{pmatrix}$  be Hermitian, so  $a, d \in \mathbb{R}$  and  $b \in F$ . State and justify the tests for  $\mathbf{A} \succ 0$  and for  $\mathbf{A} \succeq 0$ .

*Solution.*  $\mathbf{A} \succ 0$  if and only if  $a > 0$  and  $ad - |b|^2 > 0$ . This is Sylvester's criterion, [Theorem 12.1.5](#) (d): the two leading principal minors are  $a$  and  $\det \mathbf{A} = ad - |b|^2$ .

$\mathbf{A} \succeq 0$  if and only if  $a \geq 0$ ,  $d \geq 0$  and  $ad - |b|^2 \geq 0$ . This is [Theorem 12.1.4](#) (e): the principal minors are  $a$ ,  $d$  and  $\det \mathbf{A}$ , and all three are needed. Dropping  $d \geq 0$  readmits  $\text{diag}(0, -1)$ .

Note what the second condition says:  $|b|^2 \leq ad$ , so an off-diagonal entry can never be larger in modulus than the geometric mean of the two diagonal entries it sits between.

For  $\mathbf{A} = \begin{pmatrix} 2 & 1+i \\ 1-i & 3 \end{pmatrix}$  over  $\mathbb{C}$ :  $a = 2 > 0$  and  $ad - |b|^2 = 6 - 2 = 4 > 0$ , so  $\mathbf{A} \succ 0$ .

Where do positive semidefinite matrices come from? Almost always from item (c) of [Theorem 12.1.4](#), and the standard source is data.

### Example. (A Covariance-style Matrix)

Three quantities are measured on four occasions, and each column of

$$\mathbf{X} = \begin{pmatrix} 2 & 1 & 0 \\ -1 & 1 & 1 \\ -1 & -1 & 0 \\ 0 & -1 & -1 \end{pmatrix} \in M_{4 \times 3}(\mathbb{R})$$

has been adjusted to sum to 0. Show that  $\mathbf{A} = \mathbf{X}^\top \mathbf{X}$  is positive definite, and check the verdict against Sylvester's criterion.

*Solution.* Multiplying out,  $a_{ij}$  is the dot product of columns  $i$  and  $j$  of  $\mathbf{X}$ :

$$\mathbf{A} = \begin{pmatrix} 6 & 2 & -1 \\ 2 & 4 & 2 \\ -1 & 2 & 2 \end{pmatrix}.$$

The columns of  $\mathbf{X}$  are linearly independent: if  $\mathbf{X}\mathbf{x} = \mathbf{0}$ , rows 1 and 3 read  $2x_1 + x_2 = 0$  and  $-x_1 - x_2 = 0$ , which force  $x_1 = x_2 = 0$ , and then row 2 reads  $x_3 = 0$ . So  $\text{nul}(\mathbf{X}) = \{\mathbf{0}\}$ , and taking  $\mathbf{S} = \mathbf{X}$  in [Proposition 12.1.2](#) (b) with the positive definite matrix  $\mathbf{I}_4$  gives  $\mathbf{A} = \mathbf{X}^\top \mathbf{I}_4 \mathbf{X} \succ 0$ . Equivalently,  $\langle \mathbf{A}\mathbf{x}, \mathbf{x} \rangle = \|\mathbf{X}\mathbf{x}\|^2 > 0$  for  $\mathbf{x} \neq \mathbf{0}$ .

Sylvester's criterion agrees:  $\det \mathbf{A}_1 = 6$ ,  $\det \mathbf{A}_2 = 24 - 4 = 20$ , and expanding along the first row,

$$\det \mathbf{A} = 6(8 - 4) - 2(4 + 2) + (-1)(4 + 4) = 24 - 12 - 8 = 4,$$

all positive. The reason the two routes must agree is that  $\mathbf{A}$  is the Gram matrix of the columns of  $\mathbf{X}$ , and [Theorem 10.2.5](#) (d) already said that its determinant is positive exactly when those columns are independent.

## 12.1.6 The invertible matrix theorem grows

Chapter 2 listed eight conditions equivalent to invertibility ([Theorem 2.4.4](#)); Chapter 6 added the determinant ([Theorem 6.5.3](#)) and Chapter 8 the eigenvalue 0 ([Theorem 8.2.6](#)). Positivity adds a pair, and they are the ones used whenever a square system is attacked through least squares.

### Theorem 12.1.6 — Invertible Matrix Theorem, with Positive Definiteness

Let  $\mathbf{A} \in M_n(F)$  with  $F = \mathbb{R}$  or  $F = \mathbb{C}$ . The following are equivalent, and each is equivalent to each of the conditions of [Theorem 2.4.4](#), [Theorem 6.5.3](#) and [Theorem 8.2.6](#).

- (a)  $\mathbf{A}$  is invertible.
- (b)  $\mathbf{A}^* \mathbf{A} \succ 0$ .
- (c)  $\mathbf{A} \mathbf{A}^* \succ 0$ .

For a general  $\mathbf{A} \in M_{m \times n}(F)$ , one always has  $\mathbf{A}^* \mathbf{A} \succeq 0$ , with  $\mathbf{A}^* \mathbf{A} \succ 0$  if and only if the columns of  $\mathbf{A}$  are linearly independent.

*Proof.* Let  $\mathbf{A} \in M_{m \times n}(F)$ . Then  $\mathbf{A}^* \mathbf{A}$  is positive semidefinite by [Theorem 12.1.4](#) ((c)  $\Rightarrow$  (a)). Whether it is definite is decided by the identity  $\langle \mathbf{A}^* \mathbf{A} \mathbf{x}, \mathbf{x} \rangle = \|\mathbf{A} \mathbf{x}\|^2$  of [Exercise \(C1: The operator  \$S^\* S\$ \)](#) (a): the left side is  $> 0$  for every  $\mathbf{x} \neq \mathbf{0}$  exactly when  $\mathbf{A} \mathbf{x} \neq \mathbf{0}$  for every  $\mathbf{x} \neq \mathbf{0}$ , that is, exactly when  $\text{nul}(\mathbf{A}) = \{\mathbf{0}\}$ , that is, exactly when the columns of  $\mathbf{A}$  are linearly independent. The same conclusion comes from [Lemma 10.4.3](#), which says  $\text{nul}(\mathbf{A}^* \mathbf{A}) = \text{nul}(\mathbf{A})$  and so lets one read the condition off  $\mathbf{A}$  directly.

Now let  $\mathbf{A}$  be square. Then “ $\text{nul}(\mathbf{A}) = \{\mathbf{0}\}$ ” is item (b) of [Theorem 2.4.4](#), which gives (a)  $\Leftrightarrow$  (b). Also  $\mathbf{A}$  is invertible if and only if  $\mathbf{A}^*$  is, since  $(\mathbf{A}^{-1})^*$  is a two-sided inverse of  $\mathbf{A}^*$  and conversely; applying (a)  $\Leftrightarrow$  (b) to  $\mathbf{A}^*$  therefore gives (a)  $\Leftrightarrow$  (c). ■

The two new items look like a detour — why test  $\mathbf{A}^* \mathbf{A}$  when  $\mathbf{A}$  itself is available? Because  $\mathbf{A}^* \mathbf{A}$  is Hermitian and positive, and everything in this chapter applies to it. Chapter 10 already used it once, in the normal equations of least squares; Section 8 will use it again, to build the



singular value decomposition of a matrix that is not even square.

## 12.1.7 Exercises

### 12.1.7.1 A. Check your understanding

#### Exercise A1.

- State the definition of  $\mathbf{A} \succeq 0$  and of  $\mathbf{A} \succ 0$ , including every quantifier and the field.
- Explain why clause (P1) cannot be dropped from the real definition, with a concrete matrix.
- True or false: if  $\mathbf{A}$  is Hermitian and every leading principal minor of  $\mathbf{A}$  is  $\geq 0$ , then  $\mathbf{A} \succeq 0$ . Justify your answer.
- Write down the three conditions of the  $2 \times 2$  test for  $\mathbf{A} \succeq 0$ .
- Name the item of [Theorem 12.1.4](#) that says every positive semidefinite matrix is a Gram matrix.

*Solution.*

- $\mathbf{A} \in M_n(F)$  with  $F = \mathbb{R}$  or  $\mathbb{C}$  is positive semidefinite if  $\mathbf{A}^* = \mathbf{A}$  and  $\langle \mathbf{A}\mathbf{x}, \mathbf{x} \rangle \geq 0$  for **every**  $\mathbf{x} \in F^n$ ; positive definite if  $\mathbf{A}^* = \mathbf{A}$  and  $\langle \mathbf{A}\mathbf{x}, \mathbf{x} \rangle > 0$  for every **non-zero**  $\mathbf{x} \in F^n$  ([Definition 12.1.1](#)).
- Because over  $\mathbb{R}$  the quadratic form does not see the skew part. The quarter turn  $\begin{pmatrix} 0 & -1 \\ 1 & 0 \end{pmatrix}$  has  $\langle \mathbf{A}\mathbf{x}, \mathbf{x} \rangle = 0$  for every  $\mathbf{x} \in \mathbb{R}^2$  and is not symmetric.
- False.  $\text{diag}(0, -1)$  has leading principal minors 0 and 0, and  $\langle \mathbf{A}\mathbf{e}_2, \mathbf{e}_2 \rangle = -1$ . All principal minors are needed ([Theorem 12.1.4](#) (e)).
- With  $\mathbf{A} = \begin{pmatrix} a & b \\ \bar{b} & d \end{pmatrix}$  Hermitian:  $a \geq 0$ ,  $d \geq 0$  and  $ad - |b|^2 \geq 0$ .
- Item (c):  $\mathbf{A} = \mathbf{B}^*\mathbf{B}$ , and  $\mathbf{B}^*\mathbf{B}$  is the Gram matrix of the columns of  $\mathbf{B}$ .

### 12.1.7.2 B. Practice

#### Exercise B1: Determine the definiteness.

For each matrix below, determine whether it is positive definite, positive semidefinite but not positive definite, or neither. Justify your answer.

- $\begin{pmatrix} 2 & 1 \\ 1 & 2 \end{pmatrix}$  over  $\mathbb{R}$ .
- $\begin{pmatrix} 0 & 0 \\ 0 & -1 \end{pmatrix}$  over  $\mathbb{R}$ .
- $\begin{pmatrix} 2 & -1 & 0 \\ -1 & 2 & -1 \\ 0 & -1 & 2 \end{pmatrix}$  over  $\mathbb{R}$ .
- The all-ones matrix  $\mathbf{J} \in M_3(\mathbb{R})$ .

- (e)  $\begin{pmatrix} 0 & 1 \\ -1 & 0 \end{pmatrix}$  over  $\mathbb{R}$ .
- (f)  $\begin{pmatrix} 2 & 1+i \\ 1-i & 1 \end{pmatrix}$  over  $\mathbb{C}$ .

*Solution.*

- (a) Positive definite. Symmetric, with leading principal minors 2 and  $4 - 1 = 3$ , both positive ([Theorem 12.1.5](#) (d)).
- (b) Neither. The leading principal minors are 0 and 0, which tempts one into “semidefinite”, but the principal minor  $\det \mathbf{A}_{\{2\},\{2\}} = -1$  is negative, and  $\langle \mathbf{A}\mathbf{e}_2, \mathbf{e}_2 \rangle = -1 < 0$ .
- (c) Positive definite. Symmetric, with leading principal minors 2,  $4 - 1 = 3$  and  $2 \cdot 3 - (-1)(-2) = 4$ , all positive.
- (d) Positive semidefinite, not positive definite.  $\mathbf{J} = \mathbf{1}\mathbf{1}^\top$ , which is [Example \(Five Positive Matrices\)](#) (b) with  $\mathbf{x} = \mathbf{1}$ ; so  $\langle \mathbf{J}\mathbf{x}, \mathbf{x} \rangle = |\langle \mathbf{x}, \mathbf{1} \rangle|^2 \geq 0$ . It vanishes at the non-zero  $\mathbf{x} = (1, -1, 0)$ , so it is not definite. (Equivalently,  $\det \mathbf{J} = 0$ .)
- (e) Neither: it is not symmetric, so (P1) fails and it is not in the class at all. Its quadratic form is identically 0, which shows how little the form sees over  $\mathbb{R}$ .
- (f) Positive semidefinite, not positive definite. Hermitian, with  $a = 2 \geq 0$ ,  $d = 1 \geq 0$  and  $ad - |b|^2 = 2 - 2 = 0$ . The determinant is 0, so definiteness fails; indeed  $\mathbf{A}(1+i, -2) = \mathbf{0}$ .

### Exercise B2: A parameter.

For which  $t \in \mathbb{R}$  is

$$\mathbf{A}(t) = \begin{pmatrix} 1 & t & 0 \\ t & 2 & t \\ 0 & t & 3 \end{pmatrix}$$

positive definite? Justify your answer.

*Solution.*  $\mathbf{A}(t)$  is symmetric for every real  $t$ , so Sylvester’s criterion applies ([Theorem 12.1.5](#) (d)). The leading principal minors are

$$\det \mathbf{A}_1 = 1, \quad \det \mathbf{A}_2 = 2 - t^2, \quad \det \mathbf{A}_3 = 6 - 4t^2,$$

the last by expansion along the first row:  $1 \cdot (6 - t^2) - t \cdot (3t - 0) = 6 - 4t^2$ . All three are positive exactly when  $t^2 < 2$  and  $t^2 < 3/2$ , that is when  $t^2 < 3/2$ . Hence  $\mathbf{A}(t) \succ 0$  if and only if  $|t| < \sqrt{6}/2$ .

At  $t = \sqrt{6}/2$  the matrix is positive semidefinite but singular, and for  $|t| > \sqrt{6}/2$  the determinant is negative, so  $\mathbf{A}(t)$  has an odd number of negative eigenvalues and is not even semidefinite.

### Exercise B3: The inverse of a positive definite matrix.

Let  $\mathbf{A} \in M_n(F)$  with  $\mathbf{A} \succ 0$ . Prove that  $\mathbf{A}$  is invertible and that  $\mathbf{A}^{-1} \succ 0$ .

*Solution.* By [Theorem 12.1.5](#) (b), every eigenvalue of  $\mathbf{A}$  is positive, so  $0 \notin \text{spec}(\mathbf{A})$  and  $\mathbf{A}$  is invertible ([Theorem 8.2.6](#)). Then  $(\mathbf{A}^{-1})^* = (\mathbf{A}^*)^{-1} = \mathbf{A}^{-1}$ , so  $\mathbf{A}^{-1}$  is Hermitian. Apply [Proposition 12.1.2](#) (b) with  $\mathbf{S} = \mathbf{A}^{-1}$ , which is invertible:

$$\mathbf{S}^* \mathbf{A} \mathbf{S} = \mathbf{A}^{-1} \mathbf{A} \mathbf{A}^{-1} = \mathbf{A}^{-1} \succ 0.$$

(Alternatively: the eigenvalues of  $\mathbf{A}^{-1}$  are the reciprocals  $1/\lambda_i$  of those of  $\mathbf{A}$ , hence positive, and [Theorem 12.1.5](#) (b) applies again.)

### 12.1.7.3 C. Going deeper

#### Exercise C1: Zero trace forces zero.

Let  $\mathbf{A} \in M_n(F)$  with  $\mathbf{A} \succeq 0$ .

- (a) Prove that  $a_{ii} \geq 0$  for every  $i$ , and that  $\text{tr } \mathbf{A} \geq 0$ .
- (b) Prove that if  $\text{tr } \mathbf{A} = 0$  then  $\mathbf{A} = 0$ .
- (c) Deduce that if  $\mathbf{A} \succeq 0$  and  $-\mathbf{A} \succeq 0$ , then  $\mathbf{A} = 0$ .

*Solution.*

- (a)  $a_{ii} = \mathbf{e}_i^* \mathbf{A} \mathbf{e}_i = \langle \mathbf{A} \mathbf{e}_i, \mathbf{e}_i \rangle \geq 0$  by (P2). Summing,  $\text{tr } \mathbf{A} = \sum_i a_{ii} \geq 0$ .
- (b) Write  $\mathbf{A} = \mathbf{U} \mathbf{D} \mathbf{U}^*$  with  $\mathbf{U}$  unitary and  $\mathbf{D} = \text{diag}(\lambda_1, \dots, \lambda_n)$  the eigenvalues ([Corollary 11.4.3](#), [Corollary 11.5.2](#)). Then  $\text{tr } \mathbf{A} = \text{tr}(\mathbf{D} \mathbf{U}^* \mathbf{U}) = \text{tr } \mathbf{D} = \sum_i \lambda_i$  by [Theorem 0.9.6](#), and each  $\lambda_i \geq 0$  by [Theorem 12.1.4](#) (b). A sum of non-negative reals is 0 only if every term is 0, so  $\mathbf{D} = 0$  and hence  $\mathbf{A} = 0$ .  
Alternatively, without eigenvalues: by (a) each  $a_{ii} \geq 0$ , so  $\text{tr } \mathbf{A} = 0$  forces every  $a_{ii} = 0$ . Writing  $\mathbf{A} = \mathbf{B}^* \mathbf{B}$  ([Theorem 12.1.4](#) (c)), the  $i$ -th diagonal entry is  $\|\mathbf{b}_i\|^2$  for the  $i$ -th column  $\mathbf{b}_i$  of  $\mathbf{B}$ ; so every column of  $\mathbf{B}$  is  $\mathbf{0}$ , that is  $\mathbf{B} = 0$  and  $\mathbf{A} = 0$ .
- (c) If both  $\mathbf{A} \succeq 0$  and  $-\mathbf{A} \succeq 0$ , then by (a)  $\text{tr } \mathbf{A} \geq 0$  and  $\text{tr}(-\mathbf{A}) = -\text{tr } \mathbf{A} \geq 0$ , so  $\text{tr } \mathbf{A} = 0$  and (b) gives  $\mathbf{A} = 0$ .

#### Exercise C2: The cone of positive matrices.

Let  $\mathcal{P}_n \subseteq M_n(F)$  be the set of positive semidefinite matrices.

- (a) Prove that if  $\mathbf{A}, \mathbf{B} \in \mathcal{P}_n$  and  $c \geq 0$  is real, then  $\mathbf{A} + \mathbf{B} \in \mathcal{P}_n$  and  $c\mathbf{A} \in \mathcal{P}_n$ . (A set with these two properties is a **convex cone**: it contains  $(1-t)\mathbf{A} + t\mathbf{B}$  for  $0 \leq t \leq 1$  as well.)
- (b) Prove that  $\mathcal{P}_n$  contains no line through 0: if  $\mathbf{A} \in \mathcal{P}_n$  and  $-\mathbf{A} \in \mathcal{P}_n$  then  $\mathbf{A} = 0$ .
- (c) Prove that  $\mathcal{P}_n$  is closed under congruence: if  $\mathbf{A} \in \mathcal{P}_n$  and  $\mathbf{S} \in M_n(F)$ , then  $\mathbf{S}^* \mathbf{A} \mathbf{S} \in \mathcal{P}_n$ . Give an example showing that  $\mathcal{P}_n$  is **not** closed under multiplication.

*Hint for (c): for the example, two positive semidefinite matrices whose product is not even Hermitian will do.*

*Solution.*

- (a) Sums and real multiples of Hermitian matrices are Hermitian. For every  $\mathbf{x}$ ,

$$\langle (\mathbf{A} + \mathbf{B})\mathbf{x}, \mathbf{x} \rangle = \langle \mathbf{A}\mathbf{x}, \mathbf{x} \rangle + \langle \mathbf{B}\mathbf{x}, \mathbf{x} \rangle \geq 0,$$

and  $\langle c\mathbf{A}\mathbf{x}, \mathbf{x} \rangle = c\langle \mathbf{A}\mathbf{x}, \mathbf{x} \rangle \geq 0$  since  $c \geq 0$ . Convexity follows by taking  $c = 1 - t$  and  $c = t$  and adding.

- (b) This is [Exercise \(C1: Zero trace forces zero\)](#) (c).

- (c) Immediate from [Proposition 12.1.2](#) (a). For the failure of multiplicativity, take

$$\mathbf{A} = \begin{pmatrix} 1 & 0 \\ 0 & 0 \end{pmatrix}, \quad \mathbf{B} = \begin{pmatrix} 1 & 1 \\ 1 & 1 \end{pmatrix},$$

both positive semidefinite ( $\mathbf{A} = \mathbf{e}_1\mathbf{e}_1^\top$  and  $\mathbf{B} = \mathbf{1}\mathbf{1}^\top$ ). Then

$$\mathbf{AB} = \begin{pmatrix} 1 & 1 \\ 0 & 0 \end{pmatrix},$$

which is not symmetric, so it is not positive semidefinite — it fails (P1), never mind (P2). Section 2 shows what does survive:  $\text{tr}(\mathbf{AB}) \geq 0$ , and  $\mathbf{AB} \succeq 0$  whenever  $\mathbf{A}$  and  $\mathbf{B}$  commute.

### Exercise C3: The largest entry sits on the diagonal.

Let  $\mathbf{A} \in M_n(F)$  with  $\mathbf{A} \succeq 0$ .

- Prove that  $|a_{ij}|^2 \leq a_{ii}a_{jj}$  for all  $i, j$ .
- Deduce that if  $a_{ii} = 0$  for some  $i$ , then the whole  $i$ -th row and  $i$ -th column of  $\mathbf{A}$  are zero.
- Deduce that  $\max_{i,j} |a_{ij}| = \max_i a_{ii}$ .

*Hint for (a): look at a  $2 \times 2$  principal submatrix.*

*Solution.*

- For  $i = j$  the claim is  $a_{ii}^2 \leq a_{ii}^2$ . For  $i \neq j$ , take  $I = \{i, j\}$ . By the proof of [Theorem 12.1.4](#) ((a)  $\Rightarrow$  (e)),  $\mathbf{A}_{I,I} \succeq 0$ , and [Example \(The  \$2 \times 2\$  Test\)](#) gives  $a_{ii}a_{jj} - |a_{ij}|^2 = \det \mathbf{A}_{I,I} \geq 0$ .
- If  $a_{ii} = 0$ , then (a) gives  $|a_{ij}|^2 \leq 0 \cdot a_{jj} = 0$  for every  $j$ , so  $a_{ij} = 0$ ; and  $a_{ji} = \overline{a_{ij}} = 0$  since  $\mathbf{A}$  is Hermitian.
- Let  $M = \max_i a_{ii}$ , a maximum of non-negative reals by [Exercise \(C1: Zero trace forces zero\)](#) (a). By (a),  $|a_{ij}| \leq \sqrt{a_{ii}a_{jj}} \leq M$  for all  $i, j$ , so  $\max_{i,j} |a_{ij}| \leq M$ . The reverse inequality holds because the diagonal entries are among the entries. Hence the two maxima are equal.

## 12.2 Square Roots and the Cholesky Factorization

A non-negative real number has exactly one non-negative square root, and it is the reason “non-negative” is a useful class of numbers at all: it lets us take lengths, define standard deviations, and undo a squaring. Section 1 produced the matrix analogue as one clause of a theorem. This section makes it an object with a name, adds the one property the clause did not mention — that the square root is a **polynomial** in the matrix — and then turns the same idea into the triangular factorization that elimination discovers, the one Chapter 2 promised and could not yet explain.

Throughout,  $F = \mathbb{R}$  or  $F = \mathbb{C}$ , and all matrices are over  $F$ .

### 12.2.1 The positive square root

**Theorem 12.1.4** (d) already says that a positive semidefinite  $\mathbf{A}$  has exactly one positive semidefinite square root. That is an existence-and-uniqueness statement, so by the usual convention it deserves a notation.

*The square root of a positive matrix is what you get by taking square roots of its eigenvalues and leaving the eigenvectors alone.*

#### Theorem 12.2.1 — The Positive Square Root

Let  $\mathbf{A} \in M_n(F)$  with  $\mathbf{A} \succeq 0$ . Then there is exactly one  $\mathbf{C} \in M_n(F)$  with  $\mathbf{C} \succeq 0$  and  $\mathbf{C}^2 = \mathbf{A}$ . We write  $\mathbf{C} = \mathbf{A}^{1/2}$ , also  $\sqrt{\mathbf{A}}$ . Moreover:

- (a)  $\mathbf{A}^{1/2} = p(\mathbf{A})$  for some polynomial  $p$  with real coefficients (the nodes  $\lambda_i$  and the values  $\sqrt{\lambda_i}$  are real, so the interpolation of [Theorem 5.6.1](#) may be run over  $\mathbb{R}$ , and  $\mathbb{R}[x] \subseteq \mathbb{C}[x]$ );
- (b)  $\mathbf{A}^{1/2}\mathbf{B} = \mathbf{B}\mathbf{A}^{1/2}$  for **every**  $\mathbf{B} \in M_n(F)$  with  $\mathbf{A}\mathbf{B} = \mathbf{B}\mathbf{A}$ ;
- (c)  $\text{nul}(\mathbf{A}^{1/2}) = \text{nul}(\mathbf{A})$  and  $\text{rank } \mathbf{A}^{1/2} = \text{rank } \mathbf{A}$ ; in particular  $\mathbf{A}^{1/2}$  is invertible if and only if  $\mathbf{A}$  is.

The eigenvalues of  $\mathbf{A}^{1/2}$  are the non-negative square roots of those of  $\mathbf{A}$ , with the same eigenvectors.

*Idea.* Existence and uniqueness are Section 1. What is new is (a), and it is the reason (b) is free. On the spectrum, the square root is just a function  $\lambda \mapsto \sqrt{\lambda}$  defined at finitely many points, and a function defined at finitely many points is a polynomial — that is Lagrange interpolation from Chapter 5. The functional calculus of Chapter 11 says that applying a polynomial to  $\mathbf{A}$  and applying the corresponding function to the spectrum are the same operation, so the interpolating polynomial evaluated at  $\mathbf{A}$  is the square root. Once  $\mathbf{A}^{1/2}$  is a polynomial in  $\mathbf{A}$ , anything commuting with  $\mathbf{A}$  commutes with it, with no further work.

*Proof.* Existence and uniqueness are [Theorem 12.1.4](#) (d).

- (a) Let  $\lambda_1, \dots, \lambda_k$  be the **distinct** eigenvalues of  $\mathbf{A}$  and let  $\mathbf{A} = \sum_i \lambda_i \mathbf{P}_i$  be its spectral resolution ([Definition 11.8.1](#), [Theorem 11.8.1](#)), which exists because a Hermitian matrix has an orthonormal basis of eigenvectors. Every  $\lambda_i$  is a non-negative real number by [Theorem 12.1.4](#) (b), so the numbers  $\sqrt{\lambda_i}$  are defined and real. Let  $f(\lambda) = \sqrt{\lambda}$  on  $\text{spec}(\mathbf{A})$ . By [Theorem 11.8.3](#) (b) and (a),

$$f(\mathbf{A})^2 = (f^2)(\mathbf{A}) = \mathbf{A},$$

and by (c) and (d) of the same theorem  $f(\mathbf{A})$  is Hermitian with spectrum  $\{\sqrt{\lambda_1}, \dots, \sqrt{\lambda_k}\} \subseteq [0, \infty)$ , so  $f(\mathbf{A}) \succeq 0$  by [Theorem 12.1.4](#) (b). Uniqueness gives  $f(\mathbf{A}) = \mathbf{A}^{1/2}$ .

The numbers  $\lambda_1, \dots, \lambda_k$  are distinct, so by Lagrange interpolation ([Theorem 5.6.1](#)) there is a polynomial  $p$  of degree at most  $k-1$ , with real coefficients, such that  $p(\lambda_i) = \sqrt{\lambda_i}$  for every  $i$ . By [Theorem 11.8.3](#) (a),  $p(\mathbf{A}) = \sum_i p(\lambda_i) \mathbf{P}_i = f(\mathbf{A}) = \mathbf{A}^{1/2}$ .

- (b) Suppose  $\mathbf{AB} = \mathbf{BA}$ . Then  $\mathbf{B}$  commutes with every power of  $\mathbf{A}$ , hence with every polynomial in  $\mathbf{A}$ , hence with  $p(\mathbf{A}) = \mathbf{A}^{1/2}$ .
- (c) If  $\mathbf{A}^{1/2}\mathbf{x} = \mathbf{0}$  then  $\mathbf{Ax} = \mathbf{A}^{1/2}(\mathbf{A}^{1/2}\mathbf{x}) = \mathbf{0}$ . Conversely, if  $\mathbf{Ax} = \mathbf{0}$ , then since  $\mathbf{A}^{1/2}$  is Hermitian,

$$\|\mathbf{A}^{1/2}\mathbf{x}\|^2 = \langle \mathbf{A}^{1/2}\mathbf{x}, \mathbf{A}^{1/2}\mathbf{x} \rangle = \langle \mathbf{Ax}, \mathbf{x} \rangle = 0,$$

so  $\mathbf{A}^{1/2}\mathbf{x} = \mathbf{0}$ . Hence the two null spaces agree, and the ranks agree by [Theorem 2.5.7](#). Invertibility is the case  $\text{nul} = \{\mathbf{0}\}$  ([Theorem 2.4.4](#)).

The last sentence is the spectral mapping statement of [Theorem 11.8.3](#) (d) for  $f = \sqrt{\cdot}$ . ■

Two routes to the same matrix, then, and both are worth practicing: assemble  $\sum_i \sqrt{\lambda_i} \mathbf{P}_i$  from the projections, or find one interpolating polynomial and substitute.

### Example. (Two Square Roots)

Compute  $\mathbf{A}^{1/2}$  for

$$(a) \quad \mathbf{A} = \begin{pmatrix} 13 & 12 \\ 12 & 13 \end{pmatrix};$$

$$(b) \quad \mathbf{B} = \begin{pmatrix} 2 & 1 & 1 \\ 1 & 2 & 1 \\ 1 & 1 & 2 \end{pmatrix},$$

in each case by the spectral resolution and by a polynomial.

*Solution.*

- (a)  $\mathbf{A}$  is symmetric. Since  $\mathbf{A}(1,1) = (25, 25)$  and  $\mathbf{A}(1,-1) = (1, -1)$ , the eigenvalues are 25 and 1, with eigenlines spanned by  $(1,1)$  and  $(1,-1)$ . Both are positive, so

$\mathbf{A} \succ 0$ . The orthogonal projections onto the two eigenlines are

$$\mathbf{P}_1 = \frac{1}{2} \begin{pmatrix} 1 & 1 \\ 1 & 1 \end{pmatrix}, \quad \mathbf{P}_2 = \frac{1}{2} \begin{pmatrix} 1 & -1 \\ -1 & 1 \end{pmatrix},$$

so  $\mathbf{A} = 25\mathbf{P}_1 + \mathbf{P}_2$  and

$$\mathbf{A}^{1/2} = 5\mathbf{P}_1 + 1 \cdot \mathbf{P}_2 = \begin{pmatrix} 3 & 2 \\ 2 & 3 \end{pmatrix}.$$

By the polynomial route: we need  $p$  with  $p(25) = 5$  and  $p(1) = 1$ . The line through the two points is  $p(x) = \frac{1}{6}x + \frac{5}{6}$ , and

$$\frac{1}{6}\mathbf{A} + \frac{5}{6}\mathbf{I} = \frac{1}{6} \begin{pmatrix} 18 & 12 \\ 12 & 18 \end{pmatrix} = \begin{pmatrix} 3 & 2 \\ 2 & 3 \end{pmatrix}.$$

Check:  $\begin{pmatrix} 3 & 2 \\ 2 & 3 \end{pmatrix}^2 = \begin{pmatrix} 13 & 12 \\ 12 & 13 \end{pmatrix}.$

- (b) Here  $\mathbf{B} = \mathbf{I} + \mathbf{J}$  with  $\mathbf{J}$  the all-ones matrix. Since  $\mathbf{J}\mathbf{1} = 3\mathbf{1}$  and  $\mathbf{J}\mathbf{x} = \mathbf{0}$  for every  $\mathbf{x}$  with entries summing to 0, the eigenvalues of  $\mathbf{J}$  are 3 and 0, so those of  $\mathbf{B}$  are 4 and 1; both are positive, so  $\mathbf{B} \succ 0$ . We need  $p$  with  $p(4) = 2$  and  $p(1) = 1$ , namely  $p(x) = \frac{1}{3}x + \frac{2}{3}$ , giving

$$\mathbf{B}^{1/2} = \frac{1}{3}\mathbf{B} + \frac{2}{3}\mathbf{I} = \frac{1}{3} \begin{pmatrix} 4 & 1 & 1 \\ 1 & 4 & 1 \\ 1 & 1 & 4 \end{pmatrix}.$$

Check by squaring: the diagonal entry of the square is  $\frac{1}{9}(16 + 1 + 1) = 2$  and an off-diagonal entry is  $\frac{1}{9}(4 + 4 + 1) = 1$ , which reproduces  $\mathbf{B}$ . Only two distinct eigenvalues appear, so a polynomial of degree 1 suffices even though  $\mathbf{B}$  is  $3 \times 3$ .

**Warning.**  $\mathbf{A}^{1/2}$  is not the entrywise square root, and it is not the only square root. For  $\mathbf{A} = \begin{pmatrix} 13 & 12 \\ 12 & 13 \end{pmatrix}$  the entrywise roots  $\begin{pmatrix} \sqrt{13} & \sqrt{12} \\ \sqrt{12} & \sqrt{13} \end{pmatrix}$  form a matrix whose square

is not  $\mathbf{A}$ . And  $\mathbf{A}$  has other square roots:  $-\mathbf{A}^{1/2}$  is one, and  $5\mathbf{P}_1 - \mathbf{P}_2 = \begin{pmatrix} 2 & 3 \\ 3 & 2 \end{pmatrix}$  is another symmetric one. The word **positive** in “positive square root” is what makes the object unique; drop it and this  $\mathbf{A}$ , whose two eigenvalues are distinct and positive, has four symmetric square roots, one for each choice of sign on each eigenline. When an eigenvalue repeats there are infinitely many, symmetric ones included: every reflection  $\begin{pmatrix} \cos \theta & \sin \theta \\ \sin \theta & -\cos \theta \end{pmatrix}$  is a symmetric square root of  $\mathbf{I}_2$ .

**Quick check.** Let  $\mathbf{A} \succeq 0$  and let  $\mathbf{U}$  be unitary. What is  $(\mathbf{U}^*\mathbf{A}\mathbf{U})^{1/2}$ ?

**Solution.**  $\mathbf{U}^*\mathbf{A}^{1/2}\mathbf{U}$ . It is positive semidefinite by Proposition 12.1.2 (a), since  $\mathbf{A}^{1/2} \succeq 0$ , and its square is  $\mathbf{U}^*\mathbf{A}^{1/2}\mathbf{U}\mathbf{U}^*\mathbf{A}^{1/2}\mathbf{U} = \mathbf{U}^*\mathbf{A}\mathbf{U}$  because  $\mathbf{U}\mathbf{U}^* = \mathbf{I}$ . Uniqueness in Theorem 12.2.1 does the rest. In words: the square root does not care which orthonormal coordinates we use, which is exactly what one wants of an operation defined on the spectrum.

## 12.2.2 The absolute value of a matrix

A complex number has a modulus  $|z| = \sqrt{\bar{z}z}$ , and the polar form  $z = e^{i\theta}|z|$  splits it into a rotation and a non-negative stretch. Section 9 will split a matrix the same way, and the stretching factor it needs is built now. For **any** matrix, square or not,  $\mathbf{A}^* \mathbf{A}$  is positive semidefinite ([Theorem 12.1.6](#)), so it has a square root.

### Definition 12.2.1 — Absolute Value of a Matrix

Let  $\mathbf{A} \in M_{m \times n}(F)$ . The **absolute value** of  $\mathbf{A}$  is

$$|\mathbf{A}| := (\mathbf{A}^* \mathbf{A})^{1/2} \in M_n(F),$$

the unique positive semidefinite square root of  $\mathbf{A}^* \mathbf{A}$ .

Note the size:  $|\mathbf{A}|$  is  $n \times n$ , not  $m \times n$ , because  $\mathbf{A}^* \mathbf{A}$  acts on the domain of  $\mathbf{A}$ . It is well defined because  $\mathbf{A}^* \mathbf{A} \succeq 0$  and [Theorem 12.2.1](#) gives exactly one root.

### Proposition 12.2.2 — What the absolute value preserves

Let  $\mathbf{A} \in M_{m \times n}(F)$ . Then for every  $\mathbf{x} \in F^n$ ,

$$\| |\mathbf{A}| \mathbf{x} \| = \| \mathbf{A} \mathbf{x} \|.$$

Consequently  $\text{nul} |\mathbf{A}| = \text{nul } \mathbf{A}$  and  $\text{rank} |\mathbf{A}| = \text{rank } \mathbf{A}$ .

*Proof.* Since  $|\mathbf{A}|$  is Hermitian and  $|\mathbf{A}|^2 = \mathbf{A}^* \mathbf{A}$ ,

$$\| |\mathbf{A}| \mathbf{x} \|^2 = \langle |\mathbf{A}|^2 \mathbf{x}, \mathbf{x} \rangle = \langle \mathbf{A}^* \mathbf{A} \mathbf{x}, \mathbf{x} \rangle = \| \mathbf{A} \mathbf{x} \|^2,$$

using [Lemma 10.4.1](#) twice. Both sides are non-negative reals, so the norms agree. Hence  $|\mathbf{A}| \mathbf{x} = \mathbf{0}$  if and only if  $\mathbf{A} \mathbf{x} = \mathbf{0}$ , and the rank statement follows by [Theorem 2.5.7](#). ■

So  $|\mathbf{A}|$  does exactly what  $\mathbf{A}$  does to lengths, and nothing else. That single identity is the whole engine of the polar decomposition: the map  $|\mathbf{A}| \mathbf{x} \mapsto \mathbf{A} \mathbf{x}$  preserves norms, so it extends to a unitary matrix, and  $\mathbf{A} = \mathbf{W} |\mathbf{A}|$  follows.

### Example. (Two Absolute Values)

Compute  $|\mathbf{A}|$  for  $\mathbf{A} = \begin{pmatrix} 0 & -1 \\ 1 & 0 \end{pmatrix}$  and for  $\mathbf{B} = \begin{pmatrix} 3 & 0 \\ 4 & 0 \end{pmatrix}$ , both over  $\mathbb{R}$ .

*Solution.* For the quarter turn,  $\mathbf{A}^\top \mathbf{A} = \mathbf{I}$ , so  $|\mathbf{A}| = \mathbf{I}^{1/2} = \mathbf{I}$ . This is right: a rotation changes no length, so the matrix that does the same thing to lengths and nothing else is the identity.



For  $\mathbf{B}$ ,  $\mathbf{B}^\top \mathbf{B} = \begin{pmatrix} 25 & 0 \\ 0 & 0 \end{pmatrix}$ , whose positive square root is  $|\mathbf{B}| = \begin{pmatrix} 5 & 0 \\ 0 & 0 \end{pmatrix}$ . Check [Proposition 12.2.2](#):  $\|\mathbf{B}\mathbf{x}\| = \|(3x_1, 4x_1)\| = 5|x_1| = \| |\mathbf{B}|\mathbf{x} \|$ . Both have rank 1 and null space  $\text{span}(\mathbf{e}_2)$ .

*Warning.*  $|\mathbf{A}|$  is **not** the entrywise absolute value, and  $|\mathbf{AB}| \neq |\mathbf{A}||\mathbf{B}|$ . For the quarter turn above, the entrywise absolute value is  $\begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}$ , while  $|\mathbf{A}| = \mathbf{I}$ . For the product rule, take

$$\mathbf{A} = \begin{pmatrix} 1 & 0 \\ 0 & 0 \end{pmatrix}, \quad \mathbf{B} = \begin{pmatrix} 0 & 0 \\ 1 & 1 \end{pmatrix}.$$

Then  $\mathbf{AB} = \mathbf{0}$ , so  $|\mathbf{AB}| = \mathbf{0}$ , while  $|\mathbf{A}| = \mathbf{A}$  and  $|\mathbf{B}| = \frac{1}{\sqrt{2}} \begin{pmatrix} 1 & 1 \\ 1 & 1 \end{pmatrix}$ , giving

$$|\mathbf{A}||\mathbf{B}| = \frac{1}{\sqrt{2}} \begin{pmatrix} 1 & 1 \\ 0 & 0 \end{pmatrix} \neq \mathbf{0}.$$

That product is not even Hermitian, so it could not be the absolute value of anything. The modulus of a product of **numbers** is the product of the moduli because numbers commute; matrices do not.

### 12.2.3 Cholesky: elimination on a positive definite matrix

Chapter 2 ran elimination on a symmetric matrix and found that the two triangular factors are secretly the same one:  $\mathbf{A} = \mathbf{L}\mathbf{D}\mathbf{L}^\top$  ([Theorem 2.6.8](#)). It then observed that when every entry of  $\mathbf{D}$  is positive one can split  $\mathbf{D} = \mathbf{D}^{1/2}\mathbf{D}^{1/2}$  and write  $\mathbf{A} = (\mathbf{L}\mathbf{D}^{1/2})(\mathbf{L}\mathbf{D}^{1/2})^\top$ , called this the **Cholesky factorization**, and promised that Chapter 12 would connect it with positive definite matrices. Here is the connection, and it is the best kind: the hypothesis “every entry of  $\mathbf{D}$  is positive” is not an extra assumption at all. It holds exactly when  $\mathbf{A}$  is positive definite, and it can be checked while the elimination runs.

#### Theorem 12.2.3 — Cholesky Factorization

Let  $\mathbf{A} \in M_n(F)$  be Hermitian. Then  $\mathbf{A} \succ 0$  if and only if

$$\mathbf{A} = \mathbf{L}\mathbf{L}^*$$

for some lower triangular  $\mathbf{L} \in M_n(F)$  whose diagonal entries are **real and strictly positive**. In that case  $\mathbf{L}$  is unique.

*Idea.*  $(\Leftarrow)$  is one line: such an  $\mathbf{L}$  is invertible, and  $\mathbf{L}\mathbf{L}^* = (\mathbf{L}^*)^*\mathbf{L}^*$  is [Theorem 12.1.5](#) (c) with  $\mathbf{B} = \mathbf{L}^*$ .

$(\Rightarrow)$  is Chapter 2’s construction with its hypothesis supplied. Sylvester’s criterion makes every leading corner invertible, which is exactly what elimination without row swaps needs; the  $\mathbf{L}\mathbf{D}\mathbf{L}^*$  factorization then appears, and the pivots are the ratios  $\det \mathbf{A}_k / \det \mathbf{A}_{k-1}$ , hence positive. Absorbing  $\mathbf{D}^{1/2}$  into  $\mathbf{L}$  finishes it. For uniqueness, strip the positive di-

agonal off each candidate  $\mathbf{L}$ ; what is left is an LU factorization of  $\mathbf{A}$ , and Chapter 2 already proved those unique.

*Proof.* ( $\Leftarrow$ ) Suppose  $\mathbf{A} = \mathbf{L}\mathbf{L}^*$  with  $\mathbf{L}$  lower triangular and every  $\ell_{kk} > 0$ . Then  $\mathbf{L}$  is invertible by [Lemma 2.6.2](#), hence so is  $\mathbf{L}^*$ , and  $\mathbf{A} = (\mathbf{L}^*)^*\mathbf{L}^*$  is positive definite by [Theorem 12.1.5](#) ((c)  $\Rightarrow$  (a)).

( $\Rightarrow$ ) Suppose  $\mathbf{A} \succ 0$ . By Sylvester's criterion ([Theorem 12.1.5](#) (d)),  $\det \mathbf{A}_k > 0$  for every  $k$ , so every leading principal submatrix  $\mathbf{A}_k$  is invertible ([Theorem 6.5.3](#)). By [Exercise \(C1: Leading principal submatrices\)](#) (c) there is an LU factorization  $\mathbf{A} = \mathbf{M}\mathbf{U}$  with  $\mathbf{M}$  unit lower triangular and  $\mathbf{U}$  upper triangular and invertible. Let  $\mathbf{D} = \text{diag}(u_{11}, \dots, u_{nn})$ , which is invertible by [Lemma 2.6.2](#).

*Claim.*  $\mathbf{A} = \mathbf{M}\mathbf{D}\mathbf{M}^*$ , and the entries  $u_{11}, \dots, u_{nn}$  of  $\mathbf{D}$  are real.

*Proof.* Let  $\mathbf{V} = \mathbf{D}^{-1}\mathbf{U}$ , which is unit upper triangular, so that  $\mathbf{A} = \mathbf{M}\mathbf{D}\mathbf{V}$ . Since  $\mathbf{A}^* = \mathbf{A}$ ,

$$\mathbf{A} = \mathbf{A}^* = (\mathbf{M}\mathbf{D}\mathbf{V})^* = \mathbf{V}^*\mathbf{D}^*\mathbf{M}^* = \mathbf{V}^*(\mathbf{D}^*\mathbf{M}^*).$$

Here  $\mathbf{V}^*$  is unit lower triangular and  $\mathbf{D}^*\mathbf{M}^*$  is upper triangular with diagonal  $\overline{u_{11}}, \dots, \overline{u_{nn}}$ , so this is a second LU factorization of the invertible matrix  $\mathbf{A}$ . By [Theorem 2.6.5](#),  $\mathbf{V}^* = \mathbf{M}$  and  $\mathbf{D}^*\mathbf{M}^* = \mathbf{U} = \mathbf{D}\mathbf{V} = \mathbf{D}\mathbf{M}^*$ . Canceling the invertible  $\mathbf{M}^*$  gives  $\mathbf{D}^* = \mathbf{D}$ , that is, every  $u_{kk}$  is real; and  $\mathbf{A} = \mathbf{M}\mathbf{D}\mathbf{V} = \mathbf{M}\mathbf{D}\mathbf{M}^*$ . ■

Over  $\mathbb{R}$  the claim is [Theorem 2.6.8](#), and the proof just given is that proof with  $*$  in place of  $^\top$ .

Write  $d_k = u_{kk}$ . Leading principal submatrices of a product of a lower and an upper triangular matrix multiply ([Exercise \(C1: Leading principal submatrices\)](#) (a)); applying that twice, first to  $\mathbf{A} = \mathbf{M}(\mathbf{D}\mathbf{M}^*)$  and then to  $\mathbf{D}\mathbf{M}^*$ , gives  $\mathbf{A}_k = \mathbf{M}_k\mathbf{D}_k(\mathbf{M}^*)_k = \mathbf{M}_k\mathbf{D}_k\mathbf{M}_k^*$  for every  $k$ . Taking determinants with [Theorem 6.5.4](#) and [Theorem 6.4.6](#),

$$\det \mathbf{A}_k = 1 \cdot (d_1 \cdots d_k) \cdot 1 = d_1 \cdots d_k.$$

Hence  $d_1 = \det \mathbf{A}_1 > 0$  and, for  $k \geq 2$ ,  $d_k = \det \mathbf{A}_k / \det \mathbf{A}_{k-1} > 0$ . So  $\mathbf{D}^{1/2} := \text{diag}(\sqrt{d_1}, \dots, \sqrt{d_n})$  is a real matrix, and putting

$$\mathbf{L} = \mathbf{M}\mathbf{D}^{1/2}$$

gives a lower triangular matrix — its  $(i, j)$ -entry is  $m_{ij}\sqrt{d_j}$ , which vanishes for  $i < j$  — whose  $k$ -th diagonal entry is  $\sqrt{d_k} > 0$ , and

$$\mathbf{L}\mathbf{L}^* = \mathbf{M}\mathbf{D}^{1/2}(\mathbf{D}^{1/2})^*\mathbf{M}^* = \mathbf{M}\mathbf{D}\mathbf{M}^* = \mathbf{A}.$$

*Uniqueness.* Suppose  $\mathbf{A} = \mathbf{L}_1\mathbf{L}_1^* = \mathbf{L}_2\mathbf{L}_2^*$ , with each  $\mathbf{L}_i$  lower triangular with positive diagonal entries. Split off the diagonal: write  $\mathbf{\Delta}_i = \text{diag}((\mathbf{L}_i)_{11}, \dots, (\mathbf{L}_i)_{nn})$ , a real diagonal matrix with positive entries, and  $\mathbf{M}_i = \mathbf{L}_i\mathbf{\Delta}_i^{-1}$ , which is lower triangular with

every diagonal entry 1. Then  $\mathbf{L}_i = \mathbf{M}_i \mathbf{\Delta}_i$  and

$$\mathbf{A} = \mathbf{M}_i \mathbf{\Delta}_i^2 \mathbf{M}_i^* = \mathbf{M}_i (\mathbf{\Delta}_i^2 \mathbf{M}_i^*).$$

The second factor is upper triangular, its  $(k, l)$ -entry being  $(\mathbf{\Delta}_i^2)_{kk} \overline{(\mathbf{M}_i)_{lk}}$ , which vanishes for  $k > l$ . So for  $i = 1$  and  $i = 2$  this displays an LU factorization of  $\mathbf{A}$ , which is invertible:  $\mathbf{L}_1$  is triangular with non-zero diagonal, hence invertible by [Lemma 2.6.2](#), and  $\mathbf{A} = \mathbf{L}_1 \mathbf{L}_1^*$  is a product of two invertible matrices. By [Theorem 2.6.5](#) the two agree:

$$\mathbf{M}_1 = \mathbf{M}_2 \quad \text{and} \quad \mathbf{\Delta}_1^2 \mathbf{M}_1^* = \mathbf{\Delta}_2^2 \mathbf{M}_2^*.$$

Canceling the invertible  $\mathbf{M}_1^* = \mathbf{M}_2^*$  gives  $\mathbf{\Delta}_1^2 = \mathbf{\Delta}_2^2$ , and since both have positive diagonal entries,  $\mathbf{\Delta}_1 = \mathbf{\Delta}_2$ . Hence  $\mathbf{L}_1 = \mathbf{M}_1 \mathbf{\Delta}_1 = \mathbf{M}_2 \mathbf{\Delta}_2 = \mathbf{L}_2$ . This proves the theorem. ■

Reading the equation  $\mathbf{A} = \mathbf{L} \mathbf{L}^*$  entry by entry turns it into a recipe, and the recipe is the standard way to factor by hand. Comparing the  $(j, j)$  entry of both sides gives  $a_{jj} = \sum_{k \leq j} |\ell_{jk}|^2$ , and comparing the  $(i, j)$  entry for  $i > j$  gives  $a_{ij} = \sum_{k \leq j} \ell_{ik} \overline{\ell_{jk}}$ . Solving each for its one new unknown, column by column:

$$\begin{aligned} \ell_{jj} &= \left( a_{jj} - \sum_{k < j} |\ell_{jk}|^2 \right)^{1/2}, \\ \ell_{ij} &= \left( a_{ij} - \sum_{k < j} \ell_{ik} \overline{\ell_{jk}} \right) / \ell_{jj} \quad (i > j). \end{aligned}$$

### Example. (A Cholesky Factorization, and One That Fails)

Factor

$$\mathbf{A} = \begin{pmatrix} 4 & -2 & 6 \\ -2 & 5 & -1 \\ 6 & -1 & 26 \end{pmatrix}$$

as  $\mathbf{L} \mathbf{L}^\top$ , and run the same algorithm on the matrix  $\mathbf{A}'$  obtained from  $\mathbf{A}$  by changing the  $(3, 3)$  entry to 9.

*Solution.* Column 1.  $\ell_{11} = \sqrt{4} = 2$ ; then  $\ell_{21} = -2/2 = -1$  and  $\ell_{31} = 6/2 = 3$ .

Column 2.  $\ell_{22} = \sqrt{5 - (-1)^2} = \sqrt{4} = 2$ ; then

$$\ell_{32} = \frac{a_{32} - \ell_{31} \ell_{21}}{\ell_{22}} = \frac{-1 - 3(-1)}{2} = 1.$$

Column 3.  $\ell_{33} = \sqrt{26 - 3^2 - 1^2} = \sqrt{16} = 4$ . So

$$\mathbf{L} = \begin{pmatrix} 2 & 0 & 0 \\ -1 & 2 & 0 \\ 3 & 1 & 4 \end{pmatrix},$$

and multiplying back confirms  $\mathbf{L} \mathbf{L}^\top = \mathbf{A}$ . The diagonal entries 2, 2, 4 are positive, so  $\mathbf{A} \succ 0$  by [Theorem 12.2.3](#); their squares 4, 4, 16 are the pivots  $d_k = \det \mathbf{A}_k / \det \mathbf{A}_{k-1}$ , consistent with  $\det \mathbf{A}_1 = 4$ ,  $\det \mathbf{A}_2 = 16$ ,  $\det \mathbf{A} = 256$ .

*The failure.* For  $\mathbf{A}'$ , columns 1 and 2 are unchanged, and the last step asks for

$$\ell_{33} = \sqrt{9 - 3^2 - 1^2} = \sqrt{-1},$$

which is not a positive real number. The algorithm stops, and by [Theorem 12.2.3](#) its stopping is a proof:  $\mathbf{A}'$  is not positive definite. Indeed  $\det \mathbf{A}' = -16 < 0$ , so Sylvester's criterion fails at  $k = 3$ .

The example shows the practical point. **The algorithm is the test.** To decide whether a numerical Hermitian matrix is positive definite, one does not compute eigenvalues and one does not compute  $n$  determinants; one starts the Cholesky recursion and watches whether every quantity under a square root stays positive. Sylvester's criterion guarantees that this verdict is correct, because  $d_1 \cdots d_k = \det \mathbf{A}_k$ .

*Remark.* Two remarks on cost, both of which belong properly to Chapter 23. First, Cholesky does about half the arithmetic of a general LU factorization, because it computes only  $\mathbf{L}$  and never the second triangular factor. Second, and less obvious, it needs no pivoting: the recursion above never divides by something that positive definiteness allows to be zero, so the row swaps that Chapter 2 introduced for stability are simply not required here. Positive definiteness is the one structural hypothesis under which elimination is unconditionally safe.

*Warning.* **Cholesky needs definiteness, not just semidefiniteness, for uniqueness.** The matrix  $\mathbf{A} = \begin{pmatrix} 0 & 0 \\ 0 & 1 \end{pmatrix}$  is positive semidefinite and does factor as  $\mathbf{L}\mathbf{L}^\top$ , for instance with  $\mathbf{L} = \mathbf{A}$ . But the diagonal entry  $\ell_{11} = 0$  is not positive, so [Theorem 12.2.3](#) does not apply — and indeed uniqueness fails, since  $\begin{pmatrix} 0 & 0 \\ 0 & 1 \end{pmatrix}$  and  $\begin{pmatrix} 0 & 0 \\ 0 & -1 \end{pmatrix}$  are two different lower triangular matrices  $\mathbf{L}$  with  $\mathbf{L}\mathbf{L}^\top = \mathbf{A}$ . Also note that the  $\mathbf{L}$  of Cholesky is **not** the  $\mathbf{L}$  of Chapter 2's LU factorization: the latter is unit lower triangular, and the two differ by the factor  $\mathbf{D}^{1/2}$ .

*Quick check.* Let  $\mathbf{A} \succ 0$  with Cholesky factor  $\mathbf{L}$ . What is  $\det \mathbf{A}$  in terms of  $\mathbf{L}$ , and what is the Cholesky factor of  $\mathbf{A}^{-1}$ ?

*Solution.*  $\det \mathbf{A} = \det \mathbf{L} \det \mathbf{L}^* = |\ell_{11} \cdots \ell_{nn}|^2 = (\ell_{11} \cdots \ell_{nn})^2$ , by [Theorem 6.5.4](#) and [Theorem 6.4.6](#), the last step because the  $\ell_{kk}$  are positive reals. For the inverse,  $\mathbf{A}^{-1} = (\mathbf{L}\mathbf{L}^*)^{-1} = (\mathbf{L}^*)^{-1}\mathbf{L}^{-1} = \mathbf{M}\mathbf{M}^*$  with  $\mathbf{M} = (\mathbf{L}^*)^{-1}$ , which is **upper** triangular; so this is not itself a Cholesky factorization. The Cholesky factor of  $\mathbf{A}^{-1}$  is a different matrix, which is why solving  $\mathbf{A}\mathbf{x} = \mathbf{b}$  is done by two triangular substitutions with  $\mathbf{L}$  rather than by inverting anything.

## 12.2.4 Exercises

### 12.2.4.1 A. Check your understanding

#### Exercise A1.

- (a) State what  $\mathbf{A}^{1/2}$  is, including the two conditions that pin it down.

- (b) Define  $|\mathbf{A}|$  for  $\mathbf{A} \in M_{m \times n}(F)$  and say what size it is.
- (c) State the Cholesky factorization theorem.
- (d) True or false: every Hermitian matrix has exactly one Hermitian square root. Justify your answer.
- (e) Explain in one sentence why the Cholesky algorithm is a test for positive definiteness.

*Solution.*

- (a) For  $\mathbf{A} \succeq 0$ ,  $\mathbf{A}^{1/2}$  is the unique matrix  $\mathbf{C}$  with  $\mathbf{C} \succeq 0$  and  $\mathbf{C}^2 = \mathbf{A}$  (Theorem 12.2.1). Both conditions are needed: positivity, and squaring to  $\mathbf{A}$ .
- (b)  $|\mathbf{A}| = (\mathbf{A}^* \mathbf{A})^{1/2}$ , which is  $n \times n$ , the size of the domain of  $\mathbf{A}$  (Definition 12.2.1).
- (c) A Hermitian  $\mathbf{A}$  is positive definite if and only if  $\mathbf{A} = \mathbf{L}\mathbf{L}^*$  with  $\mathbf{L}$  lower triangular with positive real diagonal entries, and then  $\mathbf{L}$  is unique (Theorem 12.2.3).
- (d) False, twice over. A Hermitian matrix with a negative eigenvalue has no Hermitian square root at all, since  $\mathbf{C}^2 \succeq 0$  always. And a positive semidefinite  $\mathbf{A}$  with two distinct eigenvalues has at least two Hermitian square roots,  $\pm \mathbf{A}^{1/2}$ . Uniqueness needs the root itself to be positive semidefinite.
- (e) Because the algorithm computes  $\ell_{jj}^2 = \det \mathbf{A}_j / \det \mathbf{A}_{j-1}$  one step at a time, so it produces a positive real  $\ell_{jj}$  at every step exactly when Sylvester's criterion holds.

#### 12.2.4.2 B. Practice

##### Exercise B1: Square roots.

Compute  $\mathbf{A}^{1/2}$  for each of the following, and in each case give a polynomial  $p$  with  $p(\mathbf{A}) = \mathbf{A}^{1/2}$ .

- (a)  $\mathbf{A} = \begin{pmatrix} 5 & 4 \\ 4 & 5 \end{pmatrix}$ .
- (b)  $\mathbf{A} = \begin{pmatrix} 10 & -6 \\ -6 & 10 \end{pmatrix}$ .

*Solution.*

- (a)  $\mathbf{A}(1,1) = (9,9)$  and  $\mathbf{A}(1,-1) = (1,-1)$ , so the eigenvalues are 9 and 1 with eigenlines  $\text{span}((1,1))$  and  $\text{span}((1,-1))$ ; both positive, so  $\mathbf{A} \succ 0$ . With  $\mathbf{P}_1, \mathbf{P}_2$  the projections onto those lines,

$$\mathbf{A}^{1/2} = 3\mathbf{P}_1 + \mathbf{P}_2 = \begin{pmatrix} 2 & 1 \\ 1 & 2 \end{pmatrix}.$$

For the polynomial, solve  $p(9) = 3$ ,  $p(1) = 1$ :  $p(x) = \frac{1}{4}x + \frac{3}{4}$ , and  $\frac{1}{4}\mathbf{A} + \frac{3}{4}\mathbf{I} = \begin{pmatrix} 2 & 1 \\ 1 & 2 \end{pmatrix}$ . Check:  $\begin{pmatrix} 2 & 1 \\ 1 & 2 \end{pmatrix}^2 = \begin{pmatrix} 5 & 4 \\ 4 & 5 \end{pmatrix}$ .

- (b)  $\mathbf{A}(1,-1) = (16,-16)$  and  $\mathbf{A}(1,1) = (4,4)$ , so the eigenvalues are 16 and 4. Then

$$\mathbf{A}^{1/2} = 4 \cdot \frac{1}{2} \begin{pmatrix} 1 & -1 \\ -1 & 1 \end{pmatrix} + 2 \cdot \frac{1}{2} \begin{pmatrix} 1 & 1 \\ 1 & 1 \end{pmatrix} = \begin{pmatrix} 3 & -1 \\ -1 & 3 \end{pmatrix}.$$

For the polynomial,  $p(16) = 4$  and  $p(4) = 2$  give  $p(x) = \frac{1}{6}x + \frac{4}{3}$ , and  $\frac{1}{6}\mathbf{A} + \frac{4}{3}\mathbf{I} = \begin{pmatrix} 3 & -1 \\ -1 & 3 \end{pmatrix}$ . Check:  $\begin{pmatrix} 3 & -1 \\ -1 & 3 \end{pmatrix}^2 = \begin{pmatrix} 10 & -6 \\ -6 & 10 \end{pmatrix}$ .

**Exercise B2: Cholesky, and a planted failure.**

- (a) Find the Cholesky factor of  $\mathbf{A} = \begin{pmatrix} 1 & 2 & -1 \\ 2 & 5 & 1 \\ -1 & 1 & 14 \end{pmatrix}$ , and hence write down  $\det \mathbf{A}$ .
- (b) Run the same algorithm on  $\mathbf{A}' = \begin{pmatrix} 1 & 2 & -1 \\ 2 & 5 & 1 \\ -1 & 1 & 9 \end{pmatrix}$  and say what the outcome proves.

*Solution.*

- (a)  $\ell_{11} = \sqrt{1} = 1$ ,  $\ell_{21} = 2$ ,  $\ell_{31} = -1$ . Then  $\ell_{22} = \sqrt{5 - 4} = 1$  and

$$\ell_{32} = \frac{1 - (-1)(2)}{1} = 3.$$

Finally  $\ell_{33} = \sqrt{14 - 1 - 9} = \sqrt{4} = 2$ . So

$$\mathbf{L} = \begin{pmatrix} 1 & 0 & 0 \\ 2 & 1 & 0 \\ -1 & 3 & 2 \end{pmatrix},$$

and  $\mathbf{L}\mathbf{L}^\top = \mathbf{A}$ . All diagonal entries are positive, so  $\mathbf{A} \succ 0$ . By the Quick check above,  $\det \mathbf{A} = (1 \cdot 1 \cdot 2)^2 = 4$ .

- (b) Columns 1 and 2 are identical, and the last step asks for  $\ell_{33} = \sqrt{9 - 1 - 9} = \sqrt{-1}$ . The algorithm fails, so  $\mathbf{A}'$  is not positive definite ([Theorem 12.2.3](#)). Confirming by Sylvester:  $\det \mathbf{A}'_1 = 1$ ,  $\det \mathbf{A}'_2 = 1$ , and  $\det \mathbf{A}' = 1 \cdot 44 - 2 \cdot 19 - 7 = -1 < 0$ .

**Exercise B3: An absolute value.**

Let  $\mathbf{A} = \begin{pmatrix} 1 & 1 \\ 1 & 1 \end{pmatrix} \in M_2(\mathbb{R})$ . Compute  $|\mathbf{A}|$ , verify  $\| |\mathbf{A}| \mathbf{x} \| = \| \mathbf{A} \mathbf{x} \|$  directly, and determine whether  $|\mathbf{A}| = \mathbf{A}$ .

*Solution.*  $\mathbf{A}^\top \mathbf{A} = \begin{pmatrix} 2 & 2 \\ 2 & 2 \end{pmatrix}$ , whose eigenvalues are 4 and 0 with eigenlines  $\text{span}((1, 1))$  and  $\text{span}((1, -1))$ . Hence

$$|\mathbf{A}| = 2 \cdot \frac{1}{2} \begin{pmatrix} 1 & 1 \\ 1 & 1 \end{pmatrix} + 0 = \begin{pmatrix} 1 & 1 \\ 1 & 1 \end{pmatrix} = \mathbf{A}.$$

Directly:  $\mathbf{A}\mathbf{x} = (x_1 + x_2)(1, 1)$ , so  $\| \mathbf{A}\mathbf{x} \| = \sqrt{2}|x_1 + x_2|$ , and the same computation applies to  $|\mathbf{A}| = \mathbf{A}$ . The equality  $|\mathbf{A}| = \mathbf{A}$  is no accident:  $\mathbf{A} \succeq 0$  here, and for a positive semidefinite  $\mathbf{A}$  one has  $\mathbf{A}^* \mathbf{A} = \mathbf{A}^2$ , so  $|\mathbf{A}| = (\mathbf{A}^2)^{1/2} = \mathbf{A}$  by uniqueness in [Theorem 12.2.1](#).

## 12.2.4.3 C. Going deeper

**Exercise C1: The trace of a product of positive matrices.**

Let  $\mathbf{A}, \mathbf{B} \in M_n(F)$  with  $\mathbf{A} \succeq 0$  and  $\mathbf{B} \succeq 0$ .

- (a) Prove that  $\operatorname{tr}(\mathbf{AB}) = \operatorname{tr}(\mathbf{A}^{1/2}\mathbf{BA}^{1/2})$ , and deduce that  $\operatorname{tr}(\mathbf{AB}) \geq 0$ .
- (b) Prove that  $\operatorname{tr}(\mathbf{AB}) = 0$  if and only if  $\mathbf{AB} = 0$ .

*Hint for (b): a positive semidefinite matrix of trace 0 is 0.*

*Solution.*

- (a) By [Theorem 0.9.6](#),  $\operatorname{tr}(\mathbf{XY}) = \operatorname{tr}(\mathbf{YX})$ . With  $\mathbf{X} = \mathbf{A}^{1/2}$  and  $\mathbf{Y} = \mathbf{A}^{1/2}\mathbf{B}$ ,

$$\operatorname{tr}(\mathbf{AB}) = \operatorname{tr}(\mathbf{A}^{1/2}\mathbf{A}^{1/2}\mathbf{B}) = \operatorname{tr}(\mathbf{A}^{1/2}\mathbf{BA}^{1/2}).$$

Now  $\mathbf{A}^{1/2}\mathbf{BA}^{1/2} = (\mathbf{A}^{1/2})^*\mathbf{BA}^{1/2} \succeq 0$  by [Proposition 12.1.2](#) (a), since  $\mathbf{A}^{1/2}$  is Hermitian. The trace of a positive semidefinite matrix is  $\geq 0$  by [Exercise \(C1: Zero trace forces zero\)](#) (a). Hence  $\operatorname{tr}(\mathbf{AB}) \geq 0$ .

- (b)  $(\Leftarrow)$  is clear.  $(\Rightarrow)$  Suppose  $\operatorname{tr}(\mathbf{AB}) = 0$ . By (a),  $\mathbf{C} := \mathbf{A}^{1/2}\mathbf{BA}^{1/2} \succeq 0$  and  $\operatorname{tr} \mathbf{C} = 0$ , so  $\mathbf{C} = 0$  by [Exercise \(C1: Zero trace forces zero\)](#) (b). Put  $\mathbf{M} = \mathbf{B}^{1/2}\mathbf{A}^{1/2}$ . Then

$$\mathbf{M}^*\mathbf{M} = \mathbf{A}^{1/2}\mathbf{B}^{1/2}\mathbf{B}^{1/2}\mathbf{A}^{1/2} = \mathbf{A}^{1/2}\mathbf{BA}^{1/2} = \mathbf{C} = 0,$$

so  $\mathbf{M} = 0$  by [Lemma 10.4.3](#) (its null space is all of  $F^n$ , hence so is  $\operatorname{nul} \mathbf{M}$ ). Therefore

$$\mathbf{AB} = \mathbf{A}^{1/2}(\mathbf{A}^{1/2}\mathbf{B}^{1/2})\mathbf{B}^{1/2} = \mathbf{A}^{1/2}\mathbf{M}^*\mathbf{B}^{1/2} = 0.$$

**Exercise C2: Commuting positive matrices.**

Let  $\mathbf{A} \in M_n(F)$  with  $\mathbf{A} \succeq 0$ .

- (a) Prove **without** using [Theorem 12.2.1](#) (a) that if  $\mathbf{BA} = \mathbf{AB}$ , then  $\mathbf{BA}^{1/2} = \mathbf{A}^{1/2}\mathbf{B}$ .  
*Hint: show that  $\mathbf{B}$  maps each eigenspace of  $\mathbf{A}$  into itself.*
- (b) Let  $\mathbf{B} \succeq 0$  with  $\mathbf{AB} = \mathbf{BA}$ . Prove that  $\mathbf{AB} \succeq 0$  and that  $(\mathbf{AB})^{1/2} = \mathbf{A}^{1/2}\mathbf{B}^{1/2}$ .

*Solution.*

- (a) Let  $\lambda \in \operatorname{spec}(\mathbf{A})$  and  $\mathbf{Av} = \lambda\mathbf{v}$ . Then  $\mathbf{A}(\mathbf{Bv}) = \mathbf{B}(\mathbf{Av}) = \lambda\mathbf{Bv}$ , so  $\mathbf{B}$  maps  $E_\lambda(\mathbf{A})$  into itself. Since  $\mathbf{A} \succeq 0$ , its distinct eigenvalues  $\lambda_1, \dots, \lambda_k$  are non-negative reals, and  $\mathbf{A}^{1/2} = \sum_i \sqrt{\lambda_i} \mathbf{P}_i$  with  $\mathbf{P}_i$  the orthogonal projection onto  $E_{\lambda_i}(\mathbf{A})$  (proof of [Theorem 12.2.1](#)). For  $\mathbf{v} \in E_{\lambda_j}(\mathbf{A})$  we have  $\mathbf{P}_i\mathbf{v} = \mathbf{0}$  for  $i \neq j$  and  $\mathbf{P}_j\mathbf{v} = \mathbf{v}$ , so  $\mathbf{A}^{1/2}\mathbf{v} = \sqrt{\lambda_j}\mathbf{v}$ . Now  $F^n = E_{\lambda_1}(\mathbf{A}) \oplus \dots \oplus E_{\lambda_k}(\mathbf{A})$  ([Theorem 11.8.1](#)), so it suffices to check  $\mathbf{BA}^{1/2}\mathbf{v} = \mathbf{A}^{1/2}\mathbf{Bv}$  for  $\mathbf{v} \in E_{\lambda_j}(\mathbf{A})$ . The left side is  $\sqrt{\lambda_j}\mathbf{Bv}$ ; the right side is  $\mathbf{A}^{1/2}(\mathbf{Bv}) = \sqrt{\lambda_j}\mathbf{Bv}$  as well, because  $\mathbf{Bv} \in E_{\lambda_j}(\mathbf{A})$  by the first sentence. Hence  $\mathbf{BA}^{1/2} = \mathbf{A}^{1/2}\mathbf{B}$ .
- (b) By (a),  $\mathbf{A}^{1/2}$  commutes with  $\mathbf{B}$ . Applying (a) once more, this time to  $\mathbf{B} \succeq 0$  with the commuting matrix  $\mathbf{A}^{1/2}$ , gives that  $\mathbf{A}^{1/2}$  commutes with  $\mathbf{B}^{1/2}$ . Put

$\mathbf{C} = \mathbf{A}^{1/2}\mathbf{B}^{1/2}$ . Then

$$\mathbf{C}^* = (\mathbf{B}^{1/2})^*(\mathbf{A}^{1/2})^* = \mathbf{B}^{1/2}\mathbf{A}^{1/2} = \mathbf{C},$$

so  $\mathbf{C}$  is Hermitian, and  $\mathbf{C}^2 = \mathbf{A}^{1/2}\mathbf{B}^{1/2}\mathbf{A}^{1/2}\mathbf{B}^{1/2} = \mathbf{AB}$  after one commutation. It remains to see  $\mathbf{C} \succeq 0$ . The matrices  $\mathbf{A}^{1/2}$  and  $\mathbf{B}^{1/2}$  are commuting Hermitian matrices, so by [Corollary 11.7.2](#) (over  $\mathbb{C}$ ) or [Corollary 11.7.3](#) (over  $\mathbb{R}$ ) there is a unitary  $\mathbf{U}$  with  $\mathbf{U}^*\mathbf{A}^{1/2}\mathbf{U} = \text{diag}(\alpha_i)$  and  $\mathbf{U}^*\mathbf{B}^{1/2}\mathbf{U} = \text{diag}(\beta_i)$ , all  $\alpha_i, \beta_i \geq 0$ . Then  $\mathbf{U}^*\mathbf{C}\mathbf{U} = \text{diag}(\alpha_i\beta_i) \succeq 0$ , so  $\mathbf{C} \succeq 0$  by [Proposition 12.1.2](#) (a). Hence  $\mathbf{AB} = \mathbf{C}^2 \succeq 0$  and, by uniqueness in [Theorem 12.2.1](#),  $(\mathbf{AB})^{1/2} = \mathbf{C} = \mathbf{A}^{1/2}\mathbf{B}^{1/2}$ .

### Exercise C3: Why definiteness is needed.

- (a) Give a positive semidefinite  $\mathbf{A} \in M_2(\mathbb{R})$ , not positive definite, together with two different lower triangular  $\mathbf{L}$  satisfying  $\mathbf{A} = \mathbf{L}\mathbf{L}^\top$ . Which hypothesis of [Theorem 12.2.3](#) fails?
- (b) Prove that if  $\mathbf{A} \succeq 0$  then  $\mathbf{A} + t\mathbf{I} \succ 0$  for every real  $t > 0$ , and hence that every positive semidefinite matrix is a limit of positive definite ones in the entrywise sense.

*Solution.*

- (a) Take  $\mathbf{A} = \text{diag}(0, 1)$ . Then  $\mathbf{L}_1 = \text{diag}(0, 1)$  and  $\mathbf{L}_2 = \text{diag}(0, -1)$  are both lower triangular with  $\mathbf{L}_i\mathbf{L}_i^\top = \mathbf{A}$ . The hypothesis that fails is that the diagonal entries of  $\mathbf{L}$  be strictly positive:  $\ell_{11} = 0$  here, which is exactly the first pivot vanishing. Uniqueness in [Theorem 12.2.3](#) is not available without it.
- (b)  $\mathbf{A} + t\mathbf{I}$  is Hermitian, and for  $\mathbf{x} \neq \mathbf{0}$ ,

$$\langle (\mathbf{A} + t\mathbf{I})\mathbf{x}, \mathbf{x} \rangle = \langle \mathbf{A}\mathbf{x}, \mathbf{x} \rangle + t\|\mathbf{x}\|^2 \geq t\|\mathbf{x}\|^2 > 0,$$

so  $\mathbf{A} + t\mathbf{I} \succ 0$ . Each entry of  $\mathbf{A} + t\mathbf{I}$  differs from the corresponding entry of  $\mathbf{A}$  by  $t$  or by 0, so letting  $t$  decrease to 0 recovers  $\mathbf{A}$  entry by entry. This is the standard device for extending a statement from positive definite to positive semidefinite matrices whenever the statement survives taking limits; [Theorem 12.1.4](#) already used it in the form  $\det(\mathbf{A} + t\mathbf{I}) > 0$ .

## 12.3 Gram Matrices

Chapter 10 attached to every list of vectors the matrix of its pairwise inner products and read three facts off it: it is Hermitian, its quadratic form is a squared length, and it is invertible exactly when the list is independent ([Definition 10.2.3](#), [Theorem 10.2.5](#)). One question was left standing. Given a square matrix, is it the Gram matrix of *something*? The answer is the shortest possible one: exactly when it is positive semidefinite. This section proves that, and then spends it twice — the determinant of a Gram matrix is a squared volume, and a ratio of two Gram determinants is a distance to a subspace.

Throughout,  $F = \mathbb{R}$  or  $F = \mathbb{C}$  and all inner product spaces are over  $F$ . When the list must be named we write  $\mathbf{G}(\mathbf{v}_1, \dots, \mathbf{v}_k)$  for its Gram matrix, whose entries are  $\langle \mathbf{v}_j, \mathbf{v}_i \rangle$  as in [Defini-](#)



tion 10.2.3.

## 12.3.1 Which matrices are Gram matrices?

Half of the answer is already proved. By Theorem 10.2.5 (a) and (b), a Gram matrix  $\mathbf{G}$  is Hermitian and satisfies

$$\mathbf{x}^* \mathbf{G} \mathbf{x} = \left\| \sum_j x_j \mathbf{v}_j \right\|^2 \geq 0$$

for every  $\mathbf{x} \in F^k$ . That is word for word the definition of positive semidefiniteness (Definition 12.1.1). So being positive semidefinite is **necessary**. The content of the next theorem is that nothing else is needed, and this is where the chapter's second signature move earns its keep: a positive semidefinite matrix factors as  $\mathbf{B}^* \mathbf{B}$ , and the columns of  $\mathbf{B}$  are then the list we were looking for.

### Theorem 12.3.1 — Gram Matrices Are Exactly the Positive Semidefinite Matrices

Let  $\mathbf{G} \in M_k(F)$ . The following are equivalent.

- (a)  $\mathbf{G}$  is the Gram matrix of some list of  $k$  vectors in some inner product space over  $F$ ;
- (b)  $\mathbf{G}$  is the Gram matrix of some list of  $k$  vectors in  $F^r$  with the standard inner product, where  $r = \text{rank } \mathbf{G}$ ;
- (c)  $\mathbf{G} \succeq 0$ .

Moreover the list is unique up to an isometry: if  $(\mathbf{v}_1, \dots, \mathbf{v}_k)$  in  $V$  and  $(\mathbf{w}_1, \dots, \mathbf{w}_k)$  in  $W$  have the same Gram matrix, then there is an isometry from  $\text{span}(\mathbf{v}_1, \dots, \mathbf{v}_k)$  onto  $\text{span}(\mathbf{w}_1, \dots, \mathbf{w}_k)$  carrying  $\mathbf{v}_j$  to  $\mathbf{w}_j$  for every  $j$ .

*Idea.*

- (a)  $\Rightarrow$  (c) is the display above. For (c)  $\Rightarrow$  (b), diagonalize: the spectral theorem writes  $\mathbf{G} = \mathbf{U} \mathbf{D} \mathbf{U}^*$  with  $\mathbf{D}$  diagonal and, by Theorem 12.1.4, with non-negative diagonal. Then  $\mathbf{D}^{1/2} \mathbf{U}^*$  is a square root of  $\mathbf{G}$  in the  $\mathbf{B}^* \mathbf{B}$  sense, and its rows beyond the  $r$ -th are zero, so the columns already live in  $F^r$ . For the uniqueness clause, the Gram matrix *is* the recipe for every inner product among the vectors, so the obvious map  $\sum a_j \mathbf{v}_j \mapsto \sum a_j \mathbf{w}_j$  has no choice but to preserve them; the only thing needing care is that it is well defined, and part (b) of Theorem 10.2.5 says precisely that.

*Proof.*

- (b)  $\Rightarrow$  (a) is immediate, since  $F^r$  with the standard inner product is an inner product space.
- (c)  $\Rightarrow$  (c). Let  $\mathbf{G}$  be the Gram matrix of  $(\mathbf{v}_1, \dots, \mathbf{v}_k)$ . By Theorem 10.2.5 (a),  $\mathbf{G}^* = \mathbf{G}$ , and by (b) of the same theorem  $\mathbf{x}^* \mathbf{G} \mathbf{x} = \left\| \sum_j x_j \mathbf{v}_j \right\|^2 \geq 0$  for every  $\mathbf{x} \in F^k$ . Hence  $\mathbf{G} \succeq 0$  by Definition 12.1.1.

(d)  $\Rightarrow$  (b). Since  $\mathbf{G}$  is Hermitian, the spectral theorem for matrices gives a unitary  $\mathbf{U}$  and a diagonal  $\mathbf{D}$  with  $\mathbf{G} = \mathbf{U}\mathbf{D}\mathbf{U}^*$  (Corollary 11.4.3 over  $\mathbb{C}$ ; Corollary 11.5.2 over  $\mathbb{R}$ , with  $\mathbf{U}$  orthogonal and  $\mathbf{D}$  real). The diagonal entries of  $\mathbf{D}$  are the eigenvalues of  $\mathbf{G}$ , so they are real and  $\geq 0$  by Theorem 12.1.4, and we may prescribe their order: let  $d_1, \dots, d_r > 0$  and  $d_{r+1} = \dots = d_k = 0$ . Here  $r$  is the number of non-zero eigenvalues, which is  $\text{rank } \mathbf{G}$ , because  $\mathbf{G}$  and  $\mathbf{D}$  have the same rank.

Put  $\mathbf{B} = \mathbf{D}^{1/2}\mathbf{U}^* \in M_k(F)$ , where  $\mathbf{D}^{1/2} = \text{diag}(\sqrt{d_1}, \dots, \sqrt{d_k})$ . Then

$$\mathbf{B}^*\mathbf{B} = \mathbf{U}\mathbf{D}^{1/2}\mathbf{D}^{1/2}\mathbf{U}^* = \mathbf{U}\mathbf{D}\mathbf{U}^* = \mathbf{G},$$

using  $(\mathbf{D}^{1/2})^* = \mathbf{D}^{1/2}$ . Rows  $r+1, \dots, k$  of  $\mathbf{B}$  are zero, since row  $p$  of  $\mathbf{D}^{1/2}\mathbf{U}^*$  is  $\sqrt{d_p}$  times row  $p$  of  $\mathbf{U}^*$ . Let  $\mathbf{B}' \in M_{r \times k}(F)$  consist of the first  $r$  rows of  $\mathbf{B}$ ; deleting zero rows changes no entry of  $\mathbf{B}^*\mathbf{B}$ , so  $(\mathbf{B}')^*\mathbf{B}' = \mathbf{G}$  as well. Let  $\mathbf{v}_1, \dots, \mathbf{v}_k \in F^r$  be the columns of  $\mathbf{B}'$ . With the standard inner product  $\langle \mathbf{x}, \mathbf{y} \rangle = \mathbf{y}^*\mathbf{x}$  on  $F^r$ ,

$$\langle \mathbf{v}_j, \mathbf{v}_i \rangle = \mathbf{v}_i^*\mathbf{v}_j = ((\mathbf{B}')^*\mathbf{B}')_{ij} = (\mathbf{G})_{ij},$$

so  $\mathbf{G}$  is the Gram matrix of  $(\mathbf{v}_1, \dots, \mathbf{v}_k)$ .

*The uniqueness clause.* Let  $(\mathbf{v}_1, \dots, \mathbf{v}_k)$  in  $V$  and  $(\mathbf{w}_1, \dots, \mathbf{w}_k)$  in  $W$  both have Gram matrix  $\mathbf{G}$ , and put  $U = \text{span}(\mathbf{v}_1, \dots, \mathbf{v}_k)$  and  $U' = \text{span}(\mathbf{w}_1, \dots, \mathbf{w}_k)$ . Define  $T: U \rightarrow U'$  by

$$T\left(\sum_j a_j \mathbf{v}_j\right) = \sum_j a_j \mathbf{w}_j.$$

This is well defined: if  $\sum_j a_j \mathbf{v}_j = \sum_j b_j \mathbf{v}_j$ , let  $\mathbf{x} \in F^k$  be the vector with entries  $x_j = a_j - b_j$ ; then Theorem 10.2.5 (b), applied to each list in turn, gives

$$\left\| \sum_j x_j \mathbf{w}_j \right\|^2 = \mathbf{x}^* \mathbf{G} \mathbf{x} = \left\| \sum_j x_j \mathbf{v}_j \right\|^2 = 0,$$

so  $\sum_j a_j \mathbf{w}_j = \sum_j b_j \mathbf{w}_j$ . The same display with every  $b_j = 0$  shows  $\|T\mathbf{u}\| = \|\mathbf{u}\|$  for every  $\mathbf{u} \in U$ , so  $T$  is an isometry (Definition 10.7.1); it is linear and surjective by its definition on spanning vectors. This proves the theorem. ■

Three consequences are worth naming at once.

First, the corollary promised in Chapter 6 is now proved from the right principle. Corollary 6.7.3 said that  $\det(\mathbf{A}\mathbf{A}^\top) \geq 0$  for real  $\mathbf{A}$ , with equality exactly when the rows of  $\mathbf{A}$  are dependent, and it proved this by writing the determinant as a sum of squares of minors. The structural reason is the theorem above:  $\mathbf{A}\mathbf{A}^\top$  is a Gram matrix, hence positive semidefinite, hence has non-negative eigenvalues, and its determinant is their product. Positivity of the determinant is not an accident of Cauchy–Binet; it is one shadow of a much stronger property, and the other shadows come free — **every** principal minor of a Gram matrix is  $\geq 0$  too, since a principal submatrix of  $\mathbf{G}(\mathbf{v}_1, \dots, \mathbf{v}_k)$  is the Gram matrix of a sublist.

Second, the smallest space that can hold a realization of  $\mathbf{G}$  has dimension  $\text{rank } \mathbf{G}$ , and no smaller one will do: a list of  $k$  vectors in a space of dimension  $s$  has a Gram matrix of rank at most  $s$ , because  $\mathbf{G} = \mathbf{A}^*\mathbf{A}$  for the  $s \times k$  coordinate matrix  $\mathbf{A}$  of the list and  $\text{rank}(\mathbf{A}^*\mathbf{A}) = \text{rank } \mathbf{A} \leq s$  by Corollary 10.6.6 (b).

Third, the theorem is a free source of examples in both directions: any vectors at all produce a positive semidefinite table, and conversely positivity is a real constraint on a table of numbers — a stronger one than most readers expect.

*Warning.* **Plausible-looking symmetric tables need not be Gram matrices.** The condition is not “symmetric with non-negative diagonal”, and it is not even “every  $2 \times 2$  principal minor is  $\geq 0$ ”. Take

$$\mathbf{G} = \begin{pmatrix} 5 & -3 & -3 \\ -3 & 5 & -3 \\ -3 & -3 & 5 \end{pmatrix}.$$

Its three  $2 \times 2$  principal minors are all  $25 - 9 = 16 > 0$ , so any two of the three vectors would be a legitimate pair. But  $\mathbf{G}(1, 1, 1)^\top = (-1, -1, -1)^\top$ , so  $-1$  is an eigenvalue and  $\mathbf{G} \not\geq 0$ : no three vectors in any inner product space have this table. In words, three vectors of squared length 5 cannot be **pairwise** at an angle with cosine  $-3/5$ ; two of them can.

*Quick check.* Is the  $k \times k$  matrix with every entry equal to 1 a Gram matrix? If so, exhibit a list; if not, say which clause fails.

*Solution.* It is. Take  $\mathbf{v}_1 = \cdots = \mathbf{v}_k = \mathbf{v}$  for any unit vector  $\mathbf{v}$ : then  $\langle \mathbf{v}_j, \mathbf{v}_i \rangle = 1$  for all  $i, j$ . Consistently, the matrix has rank 1, and [Theorem 12.3.1](#) (b) predicts a realization in  $F^1$ ; the list  $(1, 1, \dots, 1)$  in  $F$  is one.

## 12.3.2 The Gram determinant is a squared volume

Chapter 6 defined the volume of a parallelepiped spanned by  $n$  vectors of  $\mathbb{R}^n$  as  $|\det|$  of the matrix of their coordinates ([Definition 6.9.1](#)). That definition needs a **square** matrix, so it says nothing about two vectors in  $\mathbb{R}^3$ . The missing notion was already present in Chapter 6’s motivation, though: volume is base times height. Turning that into a definition costs one line, now that we have distances to subspaces.

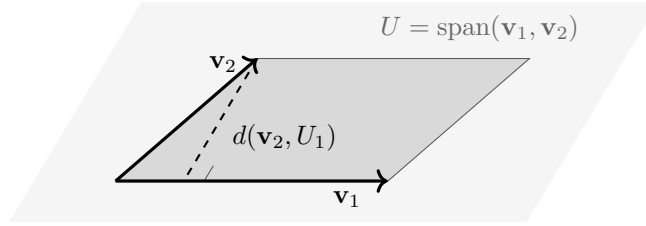
### Definition 12.3.1 — Volume of a list

Let  $(\mathbf{v}_1, \dots, \mathbf{v}_k)$  be a list in an inner product space  $V$  over  $F$ . Its  **$k$ -dimensional volume**  $\text{vol}(\mathbf{v}_1, \dots, \mathbf{v}_k) \geq 0$  is defined recursively by

$$\text{vol}(\mathbf{v}_1) = \|\mathbf{v}_1\|, \quad \text{vol}(\mathbf{v}_1, \dots, \mathbf{v}_k) = \text{vol}(\mathbf{v}_1, \dots, \mathbf{v}_{k-1}) \cdot d(\mathbf{v}_k, U_{k-1}),$$

where  $U_{k-1} = \text{span}(\mathbf{v}_1, \dots, \mathbf{v}_{k-1})$  and  $d(\cdot, \cdot)$  is the distance of [Definition 10.3.3](#).

In words: the volume of the last vector’s parallelepiped is the volume of the one below it, times the height of the last vector above the subspace the others span. For  $k = 2$  this is base times height for a parallelogram, and it is 0 exactly when  $\mathbf{v}_2$  lies on the line through  $\mathbf{v}_1$ .



$$\text{base times height: } \text{vol}(\mathbf{v}_1, \mathbf{v}_2) = \|\mathbf{v}_1\| d(\mathbf{v}_2, U_1)$$

The recursion is easy to state and awkward to compute with, because each step needs a projection. The Gram determinant computes all of it at once.

### Lemma 12.3.2 — Bordering a Gram matrix

Let  $(\mathbf{v}_1, \dots, \mathbf{v}_k)$  be a list in an inner product space  $V$ , let  $U = \text{span}(\mathbf{v}_1, \dots, \mathbf{v}_k)$ , and let  $\mathbf{w} \in V$ . Then

$$\det \mathbf{G}(\mathbf{v}_1, \dots, \mathbf{v}_k, \mathbf{w}) = \det \mathbf{G}(\mathbf{v}_1, \dots, \mathbf{v}_k) \cdot d(\mathbf{w}, U)^2.$$

*Idea.* Replace  $\mathbf{w}$  by  $\mathbf{w} - P_U \mathbf{w}$ . That is a column operation on the list, and a column operation on the list is a congruence  $\mathbf{G} \mapsto \mathbf{E}^* \mathbf{G} \mathbf{E}$  on the Gram matrix, with  $\mathbf{E}$  unitriangular, so the determinant does not move. After the replacement the last vector is orthogonal to all the others, and the Gram matrix has become block diagonal.

*Proof.* First record how the Gram matrix responds to a change of list. If  $\mathbf{u}_j = \sum_t e_{tj} \mathbf{v}_t$  for a matrix  $\mathbf{E} = (e_{tj})$ , then by sesquilinearity

$$\langle \mathbf{u}_j, \mathbf{u}_i \rangle = \sum_t \sum_s e_{tj} \overline{e_{si}} \langle \mathbf{v}_t, \mathbf{v}_s \rangle,$$

which is the  $(i, j)$ -entry of  $\mathbf{E}^* \mathbf{G} \mathbf{E}$ ; that is,  $\mathbf{G}(\mathbf{u}_1, \dots, \mathbf{u}_m) = \mathbf{E}^* \mathbf{G}(\mathbf{v}_1, \dots, \mathbf{v}_k) \mathbf{E}$ .

Now write  $\mathbf{w} = \mathbf{p} + \mathbf{h}$  with  $\mathbf{p} = P_U \mathbf{w} \in U$  and  $\mathbf{h} = \mathbf{w} - \mathbf{p} \in U^\perp$ , which is possible by Theorem 10.3.2 (a). Since  $\mathbf{p} \in U$ , there are scalars  $c_1, \dots, c_k$  with  $\mathbf{p} = \sum_t c_t \mathbf{v}_t$ . Let  $\mathbf{E} \in M_{k+1}(F)$  be the identity matrix with its last column replaced by  $(-c_1, \dots, -c_k, 1)$ . Applying the rule above to the list  $(\mathbf{v}_1, \dots, \mathbf{v}_k, \mathbf{w})$  and this  $\mathbf{E}$  turns it into  $(\mathbf{v}_1, \dots, \mathbf{v}_k, \mathbf{h})$ , so

$$\mathbf{G}(\mathbf{v}_1, \dots, \mathbf{v}_k, \mathbf{h}) = \mathbf{E}^* \mathbf{G}(\mathbf{v}_1, \dots, \mathbf{v}_k, \mathbf{w}) \mathbf{E}.$$

The matrix  $\mathbf{E}$  is triangular with ones on the diagonal, so  $\det \mathbf{E} = 1$  by Theorem 6.4.6, and  $\det(\mathbf{E}^*) = \overline{\det \mathbf{E}} = 1$ . By Theorem 6.5.4 the two Gram determinants are equal.

Finally,  $\mathbf{h} \in U^\perp$  gives  $\langle \mathbf{h}, \mathbf{v}_i \rangle = \langle \mathbf{v}_i, \mathbf{h} \rangle = 0$  for every  $i$ , so

$$\mathbf{G}(\mathbf{v}_1, \dots, \mathbf{v}_k, \mathbf{h}) = \begin{pmatrix} \mathbf{G}(\mathbf{v}_1, \dots, \mathbf{v}_k) & \mathbf{0} \\ \mathbf{0} & \|\mathbf{h}\|^2 \end{pmatrix},$$

whose determinant is  $\det \mathbf{G}(\mathbf{v}_1, \dots, \mathbf{v}_k) \cdot \|\mathbf{h}\|^2$  by cofactor expansion along the last row. Since  $\|\mathbf{h}\| = \|\mathbf{w} - P_U \mathbf{w}\| = d(\mathbf{w}, U)$  by Definition 10.3.3, this proves the lemma. ■

The recursion names the vectors in a particular order, and the last one plays a special role. The value does not depend on that choice: [Theorem 12.3.3](#) below identifies  $\text{vol}(\mathbf{v}_1, \dots, \mathbf{v}_k)^2$  with  $\det \mathbf{G}$ , and permuting the list permutes the rows of  $\mathbf{G}$  and the columns in the same way, which leaves the determinant unchanged ([Theorem 6.5.1](#) (a) applied once to the rows and once to the columns, so the two sign changes cancel). So the volume is an invariant of the list as a set of vectors, and we are free to order it however a computation prefers.

### Theorem 12.3.3 — The Gram Determinant Is the Squared Volume

Let  $(\mathbf{v}_1, \dots, \mathbf{v}_k)$  be a list in an inner product space over  $F$ . Then

$$\det \mathbf{G}(\mathbf{v}_1, \dots, \mathbf{v}_k) = \text{vol}(\mathbf{v}_1, \dots, \mathbf{v}_k)^2.$$

In particular, if  $\mathbf{A} \in M_{m \times k}(F)$  has columns  $\mathbf{v}_1, \dots, \mathbf{v}_k \in F^m$ , then  $\text{vol}(\mathbf{v}_1, \dots, \mathbf{v}_k) = \sqrt{\det(\mathbf{A}^* \mathbf{A})}$ .

*Proof.* Induction on  $k$ . For  $k = 1$ ,  $\mathbf{G}(\mathbf{v}_1) = (\|\mathbf{v}_1\|^2)$  and  $\text{vol}(\mathbf{v}_1) = \|\mathbf{v}_1\|$ .

Let  $k \geq 2$  and assume the statement for  $k - 1$ . With  $U_{k-1} = \text{span}(\mathbf{v}_1, \dots, \mathbf{v}_{k-1})$ , [Lemma 12.3.2](#) and then the induction hypothesis give

$$\begin{aligned} \det \mathbf{G}(\mathbf{v}_1, \dots, \mathbf{v}_k) &= \det \mathbf{G}(\mathbf{v}_1, \dots, \mathbf{v}_{k-1}) \cdot d(\mathbf{v}_k, U_{k-1})^2 \\ &= \text{vol}(\mathbf{v}_1, \dots, \mathbf{v}_{k-1})^2 \cdot d(\mathbf{v}_k, U_{k-1})^2, \end{aligned}$$

and the right-hand side is  $\text{vol}(\mathbf{v}_1, \dots, \mathbf{v}_k)^2$  by [Definition 12.3.1](#).

For the last claim,  $\langle \mathbf{v}_j, \mathbf{v}_i \rangle = \mathbf{v}_i^* \mathbf{v}_j = (\mathbf{A}^* \mathbf{A})_{ij}$  in  $F^m$  with the standard inner product, so  $\mathbf{G}(\mathbf{v}_1, \dots, \mathbf{v}_k) = \mathbf{A}^* \mathbf{A}$ , and the volume is the non-negative square root of its determinant. ■

Two checks that the definition is the right one. If  $k = m$  and  $F = \mathbb{R}$ , then  $\mathbf{A}$  is square and  $\det(\mathbf{A}^\top \mathbf{A}) = (\det \mathbf{A})^2$  by [Theorem 6.5.4](#) and [Theorem 6.4.5](#), so  $\text{vol}(\mathbf{v}_1, \dots, \mathbf{v}_m) = |\det \mathbf{A}|$ : [Definition 12.3.1](#) agrees with [Definition 6.9.1](#) whenever both apply. And for  $k = 2$  in  $\mathbb{R}^3$ , [Exercise \(C1: Area of a triangle in space\)](#) defined the area of a parallelogram in space to be  $\sqrt{\det(\mathbf{A}^\top \mathbf{A})}$  and promised a justification from inner products; the theorem is that justification, since base times height is what an area should be.

The volume vanishes exactly when the list is dependent, which is [Theorem 10.2.5](#) (d) read through the theorem: a flat parallelepiped has no volume.

## 12.3.3 Distance to a subspace, by determinants

Dividing in [Lemma 12.3.2](#) gives a formula for a distance with no projection in sight.

### Theorem 12.3.4 — Distance by Gram Determinants

Let  $U$  be a finite-dimensional subspace of an inner product space  $V$ , let  $(\mathbf{v}_1, \dots, \mathbf{v}_k)$  be

a **basis** of  $U$ , and let  $\mathbf{w} \in V$ . Then

$$d(\mathbf{w}, U)^2 = \frac{\det \mathbf{G}(\mathbf{v}_1, \dots, \mathbf{v}_k, \mathbf{w})}{\det \mathbf{G}(\mathbf{v}_1, \dots, \mathbf{v}_k)}.$$

*Proof.* The list  $(\mathbf{v}_1, \dots, \mathbf{v}_k)$  is independent, so  $\det \mathbf{G}(\mathbf{v}_1, \dots, \mathbf{v}_k) > 0$  by [Theorem 10.2.5](#) (d) and the quotient is defined. Dividing the identity of [Lemma 12.3.2](#) by it gives the formula. ■

No orthonormal basis, no Gram–Schmidt and no projection: two determinants and a division. The price is a  $(k+1) \times (k+1)$  determinant in the numerator, so for large  $k$  the projection route of Chapter 10 is faster. The quotient is also a ratio of squared volumes,  $\text{vol}(\mathbf{v}_1, \dots, \mathbf{v}_k, \mathbf{w})^2 / \text{vol}(\mathbf{v}_1, \dots, \mathbf{v}_k)^2$ : volume over base area, which is the height.

### Example. (The same distance twice)

In  $\mathbb{R}^3$  with the dot product, let  $U = \{(x, y, z) : x + y + z = 0\}$  and  $\mathbf{w} = (1, 2, 3)$ . Compute  $d(\mathbf{w}, U)$  by [Theorem 12.3.4](#), and compare with [Example \(Distance from a point to a plane\)](#).

*Solution.* The vectors  $\mathbf{v}_1 = (1, -1, 0)$  and  $\mathbf{v}_2 = (0, 1, -1)$  lie in  $U$  and are independent, and  $\dim U = 2$ , so they are a basis of  $U$ . Their inner products are  $\langle \mathbf{v}_1, \mathbf{v}_1 \rangle = 2$ ,  $\langle \mathbf{v}_2, \mathbf{v}_2 \rangle = 2$ ,  $\langle \mathbf{v}_1, \mathbf{v}_2 \rangle = -1$ , and with  $\mathbf{w}$ :  $\langle \mathbf{w}, \mathbf{w} \rangle = 14$ ,  $\langle \mathbf{w}, \mathbf{v}_1 \rangle = 1 - 2 = -1$ ,  $\langle \mathbf{w}, \mathbf{v}_2 \rangle = 2 - 3 = -1$ . Hence

$$\mathbf{G}(\mathbf{v}_1, \mathbf{v}_2) = \begin{pmatrix} 2 & -1 \\ -1 & 2 \end{pmatrix}, \quad \mathbf{G}(\mathbf{v}_1, \mathbf{v}_2, \mathbf{w}) = \begin{pmatrix} 2 & -1 & -1 \\ -1 & 2 & -1 \\ -1 & -1 & 14 \end{pmatrix}.$$

The first determinant is  $4 - 1 = 3$ . Expanding the second along the first row,

$$2(28 - 1) + 1(-14 - 1) - 1(1 + 2) = 54 - 15 - 3 = 36.$$

So  $d(\mathbf{w}, U)^2 = 36/3 = 12$  and  $d(\mathbf{w}, U) = 2\sqrt{3}$ . [Example \(Distance from a point to a plane\)](#) obtained  $\|(2, 2, 2)\| = 2\sqrt{3}$  by projecting onto the normal line. The two routes never meet on the page, and they agree.

*Remark.* The numerator 36 is the squared volume of the parallelepiped on  $\mathbf{v}_1, \mathbf{v}_2, \mathbf{w}$ , and the denominator 3 is the squared area of its base. The determinant of the coordinate matrix of  $(\mathbf{v}_1, \mathbf{v}_2, \mathbf{w})$  is 6, and indeed  $6^2 = 36$ : [Theorem 12.3.3](#) in the square case.

## 12.3.4 Exercises

### 12.3.4.1 A. Check your understanding

#### Exercise A1.

- (a) State the criterion for a matrix  $\mathbf{G} \in M_k(F)$  to be the Gram matrix of some list of vectors.

- (b) Determine whether the following statement is correct, and justify your answer: every symmetric matrix with non-negative entries is a Gram matrix.
- (c) A list of 5 vectors in  $\mathbb{R}^3$  has a  $5 \times 5$  Gram matrix. What are the possible values of its rank, and what is its determinant?
- (d) Write down the formula for  $d(\mathbf{w}, U)$  in terms of Gram determinants, saying what list is used.
- (e) Define the  $k$ -dimensional volume of a list, and say what [Theorem 12.3.3](#) asserts about it.

*Solution.*

- (a)  $\mathbf{G}$  is a Gram matrix if and only if  $\mathbf{G} \succeq 0$  ([Theorem 12.3.1](#)). The list may then be taken in  $F^r$  with  $r = \text{rank } \mathbf{G}$ .
- (b) Incorrect.  $\begin{pmatrix} 1 & 2 \\ 2 & 1 \end{pmatrix}$  has non-negative entries and eigenvalues 3 and  $-1$ , so it is not positive semidefinite and by [Theorem 12.3.1](#) it is no Gram matrix. (Concretely: two unit vectors cannot have inner product 2, by Cauchy–Schwarz, [Theorem 10.1.5](#).)
- (c) The rank is  $\text{rank } \mathbf{G} = \dim \text{span}(\mathbf{v}_1, \dots, \mathbf{v}_5) \leq 3$ , so it is 0, 1, 2 or 3; in particular  $\mathbf{G}$  is singular and  $\det \mathbf{G} = 0$ . (A list of 5 vectors in a 3-dimensional space is dependent, so [Theorem 10.2.5](#) (d) gives  $\det \mathbf{G} = 0$  directly.)
- (d) If  $(\mathbf{v}_1, \dots, \mathbf{v}_k)$  is a **basis** of  $U$ , then

$$d(\mathbf{w}, U)^2 = \frac{\det \mathbf{G}(\mathbf{v}_1, \dots, \mathbf{v}_k, \mathbf{w})}{\det \mathbf{G}(\mathbf{v}_1, \dots, \mathbf{v}_k)}$$

by [Theorem 12.3.4](#).

- (e)  $\text{vol}(\mathbf{v}_1) = \|\mathbf{v}_1\|$  and  $\text{vol}(\mathbf{v}_1, \dots, \mathbf{v}_k) = \text{vol}(\mathbf{v}_1, \dots, \mathbf{v}_{k-1}) d(\mathbf{v}_k, \text{span}(\mathbf{v}_1, \dots, \mathbf{v}_{k-1}))$  ([Definition 12.3.1](#)). [Theorem 12.3.3](#) says  $\det \mathbf{G}(\mathbf{v}_1, \dots, \mathbf{v}_k) = \text{vol}(\mathbf{v}_1, \dots, \mathbf{v}_k)^2$ .

### 12.3.4.2 B. Practice

#### Exercise B1: Which are Gram matrices?.

Determine which of the following are Gram matrices of some list of vectors in some real inner product space. For those that are, exhibit a list; for those that are not, give a reason.

- (i)  $\begin{pmatrix} 2 & 1 \\ 1 & 2 \end{pmatrix}$
- (ii)  $\begin{pmatrix} 1 & 1 \\ 0 & 1 \end{pmatrix}$
- (iii)  $\begin{pmatrix} 2 & -1 & -1 \\ -1 & 2 & -1 \\ -1 & -1 & 2 \end{pmatrix}$
- (iv)  $\begin{pmatrix} 1 & 0 & 1 \\ 0 & 1 & 1 \\ 1 & 1 & 1 \end{pmatrix}$

*Solution.*

- (i) Yes. It is symmetric with eigenvalues 1 and 3, both  $> 0$ , so it is positive definite and [Theorem 12.3.1](#) applies. A list:  $\mathbf{v}_1 = (1, 1, 0)$ ,  $\mathbf{v}_2 = (1, 0, 1)$  in  $\mathbb{R}^3$ , whose Gram matrix is exactly this.
- (ii) No. It is not symmetric, so it is not Hermitian and [Definition 12.1.1](#) fails at once; a Gram matrix over  $\mathbb{R}$  is symmetric by [Theorem 10.2.5](#) (a).
- (iii) Yes. Its rank is 2: the rows sum to zero, and the top-left  $2 \times 2$  minor is  $3 \neq 0$ . Its eigenvalues are 0 (eigenvector  $(1, 1, 1)$ ) and 3 (twice, on the plane  $x + y + z = 0$ ), all  $\geq 0$ . A list in  $\mathbb{R}^2$ , as [Theorem 12.3.1](#) (b) promises with  $r = 2$ : the three vectors  $(\sqrt{2}, 0)$ ,  $(-\frac{1}{\sqrt{2}}, \sqrt{\frac{3}{2}})$ ,  $(-\frac{1}{\sqrt{2}}, -\sqrt{\frac{3}{2}})$ , three vectors of length  $\sqrt{2}$  at  $120^\circ$  to each other.
- (iv) No. It is symmetric, but its leading principal minors are 1, 1,  $-1$ , and the determinant of a positive semidefinite matrix is  $\geq 0$  (its eigenvalues are  $\geq 0$ ). So it is not positive semidefinite, and by [Theorem 12.3.1](#) it is not a Gram matrix. (Its eigenvalues are 1 and  $1 \pm \sqrt{2}$ , one of which is negative.)

### Exercise B2: A volume two ways.

In  $\mathbb{R}^3$ , let  $\mathbf{v}_1 = (1, 1, 0)$ ,  $\mathbf{v}_2 = (1, 0, 1)$ ,  $\mathbf{v}_3 = (0, 1, 1)$ . Compute  $\text{vol}(\mathbf{v}_1, \mathbf{v}_2, \mathbf{v}_3)$  with [Theorem 12.3.3](#), and check the answer against [Definition 6.9.1](#).

*Solution.* The pairwise dot products give

$$\mathbf{G} = \begin{pmatrix} 2 & 1 & 1 \\ 1 & 2 & 1 \\ 1 & 1 & 2 \end{pmatrix},$$

and expanding along the first row,  $\det \mathbf{G} = 2(4 - 1) - 1(2 - 1) + 1(1 - 2) = 6 - 1 - 1 = 4$ . By [Theorem 12.3.3](#),  $\text{vol}(\mathbf{v}_1, \mathbf{v}_2, \mathbf{v}_3) = \sqrt{4} = 2$ .

Directly, the matrix  $\mathbf{A}$  with these columns has  $\det \mathbf{A} = 1(0 - 1) - 1(1 - 0) + 0 = -2$ , so [Definition 6.9.1](#) gives  $|-2| = 2$ . The two agree, as they must, since  $\det(\mathbf{A}^\top \mathbf{A}) = (\det \mathbf{A})^2$ .

### Exercise B3: A distance in $\mathbb{R}^4$ .

In  $\mathbb{R}^4$  with the dot product, let  $U = \text{span}((1, 1, 0, 0), (0, 1, 1, 0))$  and  $\mathbf{w} = (1, 0, 0, 1)$ . Find  $d(\mathbf{w}, U)$  using [Theorem 12.3.4](#).

*Solution.* Write  $\mathbf{v}_1 = (1, 1, 0, 0)$  and  $\mathbf{v}_2 = (0, 1, 1, 0)$ ; they are independent, hence a basis of  $U$ . The inner products are  $\langle \mathbf{v}_1, \mathbf{v}_1 \rangle = \langle \mathbf{v}_2, \mathbf{v}_2 \rangle = 2$ ,  $\langle \mathbf{v}_1, \mathbf{v}_2 \rangle = 1$ ,  $\langle \mathbf{w}, \mathbf{w} \rangle = 2$ ,  $\langle \mathbf{w}, \mathbf{v}_1 \rangle = 1$ ,  $\langle \mathbf{w}, \mathbf{v}_2 \rangle = 0$ . So

$$\mathbf{G}(\mathbf{v}_1, \mathbf{v}_2) = \begin{pmatrix} 2 & 1 \\ 1 & 2 \end{pmatrix}, \quad \mathbf{G}(\mathbf{v}_1, \mathbf{v}_2, \mathbf{w}) = \begin{pmatrix} 2 & 1 & 1 \\ 1 & 2 & 0 \\ 1 & 0 & 2 \end{pmatrix},$$

with determinants 3 and  $2(4) - 1(2) + 1(-2) = 4$ . Hence  $d(\mathbf{w}, U)^2 = 4/3$  and  $d(\mathbf{w}, U) = 2/\sqrt{3}$ .



*Check.* The projection is  $P_U \mathbf{w} = \frac{1}{3}(2, 1, -1, 0)$ , so  $\mathbf{w} - P_U \mathbf{w} = \frac{1}{3}(1, -1, 1, 3)$ , of norm  $\frac{1}{3}\sqrt{12} = 2/\sqrt{3}$ .

### 12.3.4.3 C. Going deeper

#### Exercise C1: Rank of a Gram matrix.

Let  $(\mathbf{v}_1, \dots, \mathbf{v}_k)$  be a list in a finite-dimensional inner product space  $V$  over  $F$ , with Gram matrix  $\mathbf{G}$ .

- (a) Prove that  $\text{rank } \mathbf{G} = \dim \text{span}(\mathbf{v}_1, \dots, \mathbf{v}_k)$ .
- (b) Deduce [Theorem 10.2.5](#) (c):  $\mathbf{G}$  is invertible if and only if the list is independent.

*Hint for (a): the coordinate matrix of the list in an orthonormal basis of the span.*

*Solution.*

- (a) Let  $U = \text{span}(\mathbf{v}_1, \dots, \mathbf{v}_k)$ ,  $r = \dim U$ , and let  $(\mathbf{e}_1, \dots, \mathbf{e}_r)$  be an orthonormal basis of  $U$ , which exists by [Theorem 10.2.2](#). Let  $\mathbf{A} \in M_{r \times k}(F)$  have as its  $j$ -th column the coordinate vector of  $\mathbf{v}_j$ . As computed in the proof of [Theorem 10.2.5](#) (d),  $\mathbf{G} = \mathbf{A}^* \mathbf{A}$ . By [Corollary 10.6.6](#) (b),  $\text{rank } \mathbf{G} = \text{rank}(\mathbf{A}^* \mathbf{A}) = \text{rank } \mathbf{A}$ . Finally  $\text{rank } \mathbf{A} = r$ : the columns of  $\mathbf{A}$  span  $F^r$ , because the coordinate map  $U \rightarrow F^r$  is an isomorphism carrying the spanning list  $(\mathbf{v}_1, \dots, \mathbf{v}_k)$  of  $U$  to the columns of  $\mathbf{A}$ .
- (b)  $\mathbf{G}$  is a  $k \times k$  matrix, so it is invertible if and only if  $\text{rank } \mathbf{G} = k$  ([Theorem 2.4.4](#)). By (a) that says  $\dim \text{span}(\mathbf{v}_1, \dots, \mathbf{v}_k) = k$ , which holds exactly when the list is independent.

#### Exercise C2: Why the conjugate is not optional.

Over  $\mathbb{C}$ , one might try to build a table of inner products with the **bilinear** expression  $\mathbf{v}_i^\top \mathbf{v}_j$ , without any conjugate. Let  $\mathbf{v} = (1, i) \in \mathbb{C}^2$ .

- (a) Compute  $\mathbf{v}^\top \mathbf{v}$  and  $\mathbf{v}^* \mathbf{v}$ , and explain why only one of them is a legitimate squared length.
- (b) The list  $(\mathbf{v})$  is independent, yet the  $1 \times 1$  “bilinear Gram matrix”  $(\mathbf{v}^\top \mathbf{v})$  is singular. Which hypothesis of [Theorem 12.3.1](#) is being violated, and which statement of Chapter 6 does this illustrate?

*Solution.*

- (a)  $\mathbf{v}^\top \mathbf{v} = 1^2 + i^2 = 0$ , while  $\mathbf{v}^* \mathbf{v} = |1|^2 + |i|^2 = 2$ . A squared length must be  $\geq 0$  and must vanish only for the zero vector ([Definition 10.1.1](#)), and  $\mathbf{v} \neq \mathbf{0}$ ; so  $\mathbf{v}^\top \mathbf{v}$  cannot be one, and  $\mathbf{v}^* \mathbf{v} = \|\mathbf{v}\|^2$  is the correct value.
- (b) [Theorem 12.3.1](#) is a statement about the Gram matrix of [Definition 10.2.3](#), whose entries are  $\langle \mathbf{v}_j, \mathbf{v}_i \rangle = \mathbf{v}_i^* \mathbf{v}_j$ . The matrix  $(\mathbf{v}^\top \mathbf{v}) = (0)$  is not that matrix, so the theorem simply does not apply; the genuine Gram matrix here is (2), which is positive definite, as it must be for an independent list. This is exactly the failure recorded after [Corollary 6.7.3](#): over  $\mathbb{C}$ ,  $\mathbf{A} = \begin{pmatrix} 1 & i \end{pmatrix}$  has rank 1 while  $\mathbf{A} \mathbf{A}^\top = (0)$ , so the real equality case of that corollary has no complex analogue — unless the

transpose is replaced by the conjugate transpose.

### Exercise C3: Height and volume.

Let  $(\mathbf{v}_1, \dots, \mathbf{v}_k)$  be an independent list in an inner product space over  $F$ . For each  $j$ , write  $U_j$  for the span of  $\mathbf{v}_1, \dots, \mathbf{v}_j$ .

- (a) Prove that  $\text{vol}(\mathbf{v}_1, \dots, \mathbf{v}_k) \leq \|\mathbf{v}_1\| \|\mathbf{v}_2\| \cdots \|\mathbf{v}_k\|$ , with equality if and only if the list is orthogonal.
- (b) Deduce that  $\det \mathbf{G}(\mathbf{v}_1, \dots, \mathbf{v}_k) \leq \|\mathbf{v}_1\|^2 \cdots \|\mathbf{v}_k\|^2$ .

*Hint for (a): compare each height with the corresponding length.*

*Solution.*

- (a) Fix  $j \geq 2$  and write  $\mathbf{v}_j = P_{U_{j-1}} \mathbf{v}_j + \mathbf{h}_j$  with  $\mathbf{h}_j \in U_{j-1}^\perp$ , as in [Theorem 10.3.2](#) (a). The two summands are orthogonal, so [Theorem 10.1.3](#) gives

$$\|\mathbf{v}_j\|^2 = \|P_{U_{j-1}} \mathbf{v}_j\|^2 + \|\mathbf{h}_j\|^2 \geq \|\mathbf{h}_j\|^2 = d(\mathbf{v}_j, U_{j-1})^2,$$

with equality if and only if  $P_{U_{j-1}} \mathbf{v}_j = \mathbf{0}$ . Multiplying the  $k$  inequalities  $d(\mathbf{v}_j, U_{j-1}) \leq \|\mathbf{v}_j\|$  (with  $d(\mathbf{v}_1, U_0)$  read as  $\|\mathbf{v}_1\|$ ) and using [Definition 12.3.1](#) repeatedly gives

$$\text{vol}(\mathbf{v}_1, \dots, \mathbf{v}_k) \leq \|\mathbf{v}_1\| \cdots \|\mathbf{v}_k\|.$$

All factors are  $> 0$ , since the list is independent, so equality holds if and only if it holds in every factor, that is  $P_{U_{j-1}} \mathbf{v}_j = \mathbf{0}$  for every  $j \geq 2$ . That says each  $\mathbf{v}_j$  is orthogonal to  $U_{j-1}$ , hence to every earlier  $\mathbf{v}_i$ ; and conversely an orthogonal list has  $P_{U_{j-1}} \mathbf{v}_j = \mathbf{0}$  by [Theorem 10.3.3](#) (a).

- (b) Square (a) and apply [Theorem 12.3.3](#).

## 12.4 Two Forms at Once

Chapter 11 asked when several operators can be diagonalized in one orthonormal basis, and answered: when they commute and each is normal ([Theorem 11.7.1](#)). That theorem is about **operators**, and its currency is the unitary similarity  $\mathbf{U}^* T \mathbf{U}$ . This section asks a question that looks similar and is not. Given two **forms** — two self-adjoint matrices, each thought of as the recipe  $\mathbf{x} \mapsto \mathbf{x}^* \mathbf{A} \mathbf{x}$  — can one change of variable make both of them diagonal at once? No commuting is assumed, and in general the two matrices do not commute. The price is that the change of variable is no longer unitary, and the hypothesis is that one of the two forms is positive definite.

Throughout,  $F = \mathbb{R}$  or  $F = \mathbb{C}$ , and “self-adjoint” means  $\mathbf{A}^* = \mathbf{A}$  (over  $\mathbb{R}$ : symmetric).

## 12.4.1 Similarity is for operators, congruence is for forms

A matrix can play two roles, and the two roles come with different change-of-variable rules.

As an **operator**,  $\mathbf{A}$  eats a vector and returns a vector. Rewriting  $\mathbf{x} = \mathbf{S}\mathbf{y}$  in the equation  $\mathbf{z} = \mathbf{A}\mathbf{x}$  turns it into  $\mathbf{S}^{-1}\mathbf{z} = (\mathbf{S}^{-1}\mathbf{A}\mathbf{S})\mathbf{y}$ : the new matrix is  $\mathbf{S}^{-1}\mathbf{A}\mathbf{S}$ , and the relation is **similarity**. Eigenvalues, trace, determinant and characteristic polynomial all survive it.

As a **form**,  $\mathbf{A}$  eats a vector and returns the scalar  $\mathbf{x}^*\mathbf{A}\mathbf{x}$ . Rewriting  $\mathbf{x} = \mathbf{S}\mathbf{y}$  turns that scalar into

$$(\mathbf{S}\mathbf{y})^*\mathbf{A}(\mathbf{S}\mathbf{y}) = \mathbf{y}^*(\mathbf{S}^*\mathbf{A}\mathbf{S})\mathbf{y},$$

so the new matrix is  $\mathbf{S}^*\mathbf{A}\mathbf{S}$ . Two matrices  $\mathbf{A}$  and  $\mathbf{S}^*\mathbf{A}\mathbf{S}$  with  $\mathbf{S}$  invertible are called **congruent**; this is the operation of [Proposition 12.1.2](#), where it was already shown to preserve the sign of a form. Congruence is the right relation here for the plain reason that it is the one the question is phrased in: “make both forms diagonal” means “make both  $\mathbf{x}^*\mathbf{A}\mathbf{x}$  and  $\mathbf{x}^*\mathbf{B}\mathbf{x}$  into sums of squares in the same new coordinates”, and that is a statement about  $\mathbf{S}^*\mathbf{A}\mathbf{S}$  and  $\mathbf{S}^*\mathbf{B}\mathbf{S}$ . Chapter 13 studies congruence in its own right, including what it does preserve; for now, note that it certainly does **not** preserve eigenvalues, since  $\mathbf{S}^*\mathbf{I}\mathbf{S} = \mathbf{S}^*\mathbf{S}$  can be any positive definite matrix at all.

Congruence and similarity coincide in exactly one situation: when  $\mathbf{S}$  is unitary,  $\mathbf{S}^* = \mathbf{S}^{-1}$ . That is the situation of Chapter 11, and it is what makes that chapter’s theorems so strong and its hypotheses so restrictive.

*Warning.* This is not Chapter 11’s simultaneous diagonalization. [Theorem 11.7.1](#) needs the operators to **commute** and produces a **unitary**  $\mathbf{U}$  with every  $\mathbf{U}^*\mathbf{T}\mathbf{U}$  diagonal; eigenvalues are preserved. The theorem below needs **no commuting** and one of the two matrices to be **positive definite**, and produces a merely invertible  $\mathbf{S}$ ; the eigenvalues of  $\mathbf{B}$  are destroyed, and even  $\mathbf{A}$  comes out as  $\mathbf{I}$ . The two theorems have no common generalization: neither hypothesis implies the other. When the matrices both commute and one is positive definite, both theorems apply and give different factorizations.

## 12.4.2 Diagonalizing two forms at once

### Theorem 12.4.1 — Diagonalizing Two Forms by Congruence

Let  $\mathbf{A}, \mathbf{B} \in M_n(F)$  be self-adjoint with  $\mathbf{A} \succ 0$ . Then there is an invertible  $\mathbf{S} \in M_n(F)$  with

$$\mathbf{S}^*\mathbf{A}\mathbf{S} = \mathbf{I}_n \quad \text{and} \quad \mathbf{S}^*\mathbf{B}\mathbf{S} = \mathbf{D}$$

for some **real** diagonal  $\mathbf{D}$ . Over  $\mathbb{R}$ , with  $\mathbf{A}, \mathbf{B}$  symmetric,  $\mathbf{S}$  may be taken real.

*Idea.* Two steps, and the first one is the whole idea. ① *Whiten.* The hypothesis  $\mathbf{A} \succ 0$  gives an invertible positive definite square root  $\mathbf{A}^{1/2}$ , and the congruence by  $\mathbf{A}^{-1/2}$  turns  $\mathbf{A}$  into  $\mathbf{I}$ . That uses up the definiteness of  $\mathbf{A}$  and nothing else. ② Now  $\mathbf{I}$  is invariant under every **unitary** congruence, because a unitary congruence is a similarity; so we still have the whole spectral theorem available for the transformed  $\mathbf{B}$ , and it is self-adjoint. Diagonalize it and compose the two changes of variable.

*Proof.* Since  $\mathbf{A} \succ 0$ , [Theorem 12.2.1](#) gives a unique  $\mathbf{A}^{1/2} \succeq 0$  with  $(\mathbf{A}^{1/2})^2 = \mathbf{A}$ . Its eigenvalues are the non-negative square roots of the eigenvalues of  $\mathbf{A}$ , which are  $> 0$  by [Theorem 12.1.5](#); hence  $\mathbf{A}^{1/2} \succ 0$  and in particular  $\mathbf{A}^{1/2}$  is invertible. Write  $\mathbf{A}^{-1/2} = (\mathbf{A}^{1/2})^{-1}$ , which is again self-adjoint, since the inverse of an invertible self-adjoint matrix is self-adjoint.

Put  $\mathbf{C} = \mathbf{A}^{-1/2}\mathbf{B}\mathbf{A}^{-1/2}$ . Then  $\mathbf{C}^* = \mathbf{A}^{-1/2}\mathbf{B}^*\mathbf{A}^{-1/2} = \mathbf{C}$ , so  $\mathbf{C}$  is self-adjoint. By the spectral theorem for matrices there is a unitary  $\mathbf{U}$  and a real diagonal  $\mathbf{D}$  with  $\mathbf{C} = \mathbf{U}\mathbf{D}\mathbf{U}^*$  ([Corollary 11.4.3](#) over  $\mathbb{C}$ ; [Corollary 11.5.2](#) over  $\mathbb{R}$ , with  $\mathbf{U}$  orthogonal and real).

Set  $\mathbf{S} = \mathbf{A}^{-1/2}\mathbf{U}$ , a product of invertible matrices, hence invertible. Then

$$\mathbf{S}^*\mathbf{A}\mathbf{S} = \mathbf{U}^*\mathbf{A}^{-1/2}\mathbf{A}\mathbf{A}^{-1/2}\mathbf{U} = \mathbf{U}^*\mathbf{U} = \mathbf{I}_n,$$

where the middle equality uses  $\mathbf{A}^{-1/2}\mathbf{A}\mathbf{A}^{-1/2} = \mathbf{A}^{-1/2}\mathbf{A}^{1/2}\mathbf{A}^{1/2}\mathbf{A}^{-1/2} = \mathbf{I}_n$ . And

$$\mathbf{S}^*\mathbf{B}\mathbf{S} = \mathbf{U}^*\mathbf{A}^{-1/2}\mathbf{B}\mathbf{A}^{-1/2}\mathbf{U} = \mathbf{U}^*\mathbf{C}\mathbf{U} = \mathbf{D}.$$

Over  $\mathbb{R}$  every matrix named here is real:  $\mathbf{A}^{1/2}$  is real because it is a polynomial in  $\mathbf{A}$  ([Theorem 12.2.1](#)), and  $\mathbf{U}$  is real by [Corollary 11.5.2](#). This proves the theorem. ■

The theorem says more than “both become diagonal”: the definite one becomes the **identity**, which is the strongest normalization available, and it costs nothing extra. Note also which hypothesis did which job. Definiteness of  $\mathbf{A}$  was used exactly once, to invert  $\mathbf{A}^{1/2}$ . Self-adjointness of  $\mathbf{B}$  was used exactly once, to apply the spectral theorem to  $\mathbf{C}$ . Nothing at all was assumed about  $\mathbf{A}\mathbf{B}$  versus  $\mathbf{B}\mathbf{A}$ .

*Quick check.* Why can  $\mathbf{S}$  not be taken unitary in [Theorem 12.4.1](#)?

*Solution.* A unitary  $\mathbf{S}$  has  $\mathbf{S}^* = \mathbf{S}^{-1}$ , so  $\mathbf{S}^*\mathbf{A}\mathbf{S} = \mathbf{I}$  would give  $\mathbf{A} = \mathbf{S}\mathbf{I}\mathbf{S}^* = \mathbf{I}$ . So a unitary  $\mathbf{S}$  can only work when  $\mathbf{A}$  is already the identity — and then the theorem is just the spectral theorem for  $\mathbf{B}$ .

### 12.4.3 The generalized eigenvalue problem

The matrix  $\mathbf{S}$  of the theorem is not an abstract object: its columns solve an eigenvalue problem in which the identity has been replaced by  $\mathbf{A}$ .

#### Definition 12.4.1 — Generalized eigenvalue

Let  $\mathbf{A}, \mathbf{B} \in M_n(F)$ . A scalar  $\lambda \in F$  is a **generalized eigenvalue** of the pair  $(\mathbf{A}, \mathbf{B})$  if

$$\mathbf{B}\mathbf{x} = \lambda\mathbf{A}\mathbf{x} \quad \text{for some } \mathbf{x} \neq \mathbf{0},$$

and such an  $\mathbf{x}$  is a **generalized eigenvector** for  $\lambda$ . The family of matrices  $\mathbf{B} - \lambda\mathbf{A}$  is called the **pencil** of the pair.

Taking  $\mathbf{A} = \mathbf{I}$  recovers Definition 8.2.1, so the word “generalized” is honest. The condition  $\mathbf{B}\mathbf{x} = \lambda\mathbf{A}\mathbf{x}$  with  $\mathbf{x} \neq \mathbf{0}$  says that  $\mathbf{B} - \lambda\mathbf{A}$  is singular, that is  $\det(\mathbf{B} - \lambda\mathbf{A}) = 0$ ; so the generalized eigenvalues are the roots of the polynomial  $\det(\mathbf{B} - \lambda\mathbf{A})$  in  $\lambda$ , the **characteristic polynomial of the pencil**. Its degree is at most  $n$ , and exactly  $n$  when  $\mathbf{A}$  is invertible, since the  $\lambda^n$ -coefficient is  $(-1)^n \det \mathbf{A}$ . When  $\mathbf{A}$  is singular the degree can drop, and the polynomial can even be a non-zero constant, in which case the pair has no generalized eigenvalues at all; exercise C3 below is such a pair. From here on  $\mathbf{A} \succ 0$ , so this does not arise.

Nothing so far forces those roots to be real: for  $\mathbf{A} = \mathbf{I}$  they are just eigenvalues, and a general  $\mathbf{B}$  has complex ones. Self-adjointness of  $\mathbf{B}$  alone is not enough either, as the warning at the end of this section shows. It is definiteness of  $\mathbf{A}$  that makes the problem behave.

### Theorem 12.4.2 — The Definite Generalized Eigenvalue Problem

Let  $\mathbf{A}, \mathbf{B} \in M_n(F)$  be self-adjoint with  $\mathbf{A} \succ 0$ , and let  $\mathbf{S}$  and  $\mathbf{D} = \text{diag}(d_1, \dots, d_n)$  be as in Theorem 12.4.1, with columns  $\mathbf{s}_1, \dots, \mathbf{s}_n$  of  $\mathbf{S}$ . Then:

- (a)  $\mathbf{B}\mathbf{s}_i = d_i\mathbf{A}\mathbf{s}_i$  for every  $i$ , and  $\mathbf{s}_i^* \mathbf{A}\mathbf{s}_j = \delta_{ij}$ : the columns of  $\mathbf{S}$  are a basis of  $F^n$  of generalized eigenvectors, orthonormal for the inner product  $\langle \mathbf{x}, \mathbf{y} \rangle_{\mathbf{A}} = \mathbf{y}^* \mathbf{A} \mathbf{x}$ ;
- (b) the generalized eigenvalues of  $(\mathbf{A}, \mathbf{B})$  are exactly  $d_1, \dots, d_n$ , all **real**, and

$$\det(\mathbf{B} - \lambda\mathbf{A}) = \frac{1}{|\det \mathbf{S}|^2} \prod_{i=1}^n (d_i - \lambda);$$

- (c)  $\mathbf{A}^{-1}\mathbf{B} = \mathbf{S}\mathbf{D}\mathbf{S}^{-1}$ ; in particular  $\mathbf{A}^{-1}\mathbf{B}$  is diagonalizable with real eigenvalues  $d_1, \dots, d_n$ , although it need not be self-adjoint.

*Idea.* Everything is already in the two equations  $\mathbf{S}^* \mathbf{A} \mathbf{S} = \mathbf{I}$  and  $\mathbf{S}^* \mathbf{B} \mathbf{S} = \mathbf{D}$ . Substituting the first into the second turns  $\mathbf{D}$  into  $\mathbf{S}^* \mathbf{A} \mathbf{S} \mathbf{D}$ , and canceling the invertible  $\mathbf{S}^*$  leaves  $\mathbf{B} \mathbf{S} = \mathbf{A} \mathbf{S} \mathbf{D}$ , which read column by column is (a). Then (b) and (c) are two ways of undoing the substitution.

*Proof.*

- (a) From  $\mathbf{S}^* \mathbf{A} \mathbf{S} = \mathbf{I}$  we get  $\mathbf{D} = \mathbf{I} \mathbf{D} = \mathbf{S}^* \mathbf{A} \mathbf{S} \mathbf{D}$ , and also  $\mathbf{D} = \mathbf{S}^* \mathbf{B} \mathbf{S}$ . Therefore  $\mathbf{S}^* (\mathbf{B} \mathbf{S} - \mathbf{A} \mathbf{S} \mathbf{D}) = \mathbf{0}$ , and  $\mathbf{S}^*$  is invertible, so

$$\mathbf{B} \mathbf{S} = \mathbf{A} \mathbf{S} \mathbf{D}.$$

Column  $i$  of the left side is  $\mathbf{B}\mathbf{s}_i$  and of the right side is  $d_i\mathbf{A}\mathbf{s}_i$ , since  $\mathbf{D}$  is diagonal. Each  $\mathbf{s}_i \neq \mathbf{0}$  because  $\mathbf{S}$  is invertible, so each  $d_i$  is a generalized eigenvalue with eigenvector  $\mathbf{s}_i$ . The entry  $(i, j)$  of  $\mathbf{S}^* \mathbf{A} \mathbf{S} = \mathbf{I}$  is  $\mathbf{s}_i^* \mathbf{A} \mathbf{s}_j$ , which is therefore  $\delta_{ij}$ . Finally  $\langle \mathbf{x}, \mathbf{y} \rangle_{\mathbf{A}} = \mathbf{y}^* \mathbf{A} \mathbf{x}$  is an inner product on  $F^n$ , exactly because  $\mathbf{A} \succ 0$ : it is sesquilinear,  $\langle \mathbf{y}, \mathbf{x} \rangle_{\mathbf{A}} = \overline{\langle \mathbf{x}, \mathbf{y} \rangle_{\mathbf{A}}} = \mathbf{y}^* \mathbf{A}^* \mathbf{x} = \langle \mathbf{x}, \mathbf{y} \rangle_{\mathbf{A}}$ , and  $\langle \mathbf{x}, \mathbf{x} \rangle_{\mathbf{A}} = \mathbf{x}^* \mathbf{A} \mathbf{x} > 0$  for  $\mathbf{x} \neq \mathbf{0}$ .

- (b) From  $\mathbf{S}^* \mathbf{A} \mathbf{S} = \mathbf{I}$  and  $\mathbf{S}^* \mathbf{B} \mathbf{S} = \mathbf{D}$  we get  $\mathbf{A} = (\mathbf{S}^*)^{-1} \mathbf{S}^{-1}$  and  $\mathbf{B} = (\mathbf{S}^*)^{-1} \mathbf{D} \mathbf{S}^{-1}$ , hence

$$\mathbf{B} - \lambda\mathbf{A} = (\mathbf{S}^*)^{-1} (\mathbf{D} - \lambda\mathbf{I}) \mathbf{S}^{-1}.$$

Taking determinants with [Theorem 6.5.4](#), and using  $\det(\mathbf{S}^*) = \overline{\det \mathbf{S}}$ ,

$$\det(\mathbf{B} - \lambda \mathbf{A}) = \frac{\det(\mathbf{D} - \lambda \mathbf{I})}{\det \mathbf{S} \det \mathbf{S}} = \frac{1}{|\det \mathbf{S}|^2} \prod_i (d_i - \lambda),$$

the last step by [Theorem 6.4.6](#). Since  $|\det \mathbf{S}|^2 \neq 0$ , the roots of  $\det(\mathbf{B} - \lambda \mathbf{A})$  are exactly  $d_1, \dots, d_n$ ; and  $\lambda$  is a generalized eigenvalue if and only if  $\mathbf{B} - \lambda \mathbf{A}$  is singular, that is if and only if  $\det(\mathbf{B} - \lambda \mathbf{A}) = 0$  ([Theorem 2.4.4](#)). The  $d_i$  are real because  $\mathbf{D} = \mathbf{S}^* \mathbf{B} \mathbf{S}$  is self-adjoint, so its diagonal entries equal their own conjugates.

(c) Inverting  $\mathbf{A} = (\mathbf{S}^*)^{-1} \mathbf{S}^{-1}$  gives  $\mathbf{A}^{-1} = \mathbf{S} \mathbf{S}^*$ , so

$$\mathbf{A}^{-1} \mathbf{B} = \mathbf{S} \mathbf{S}^* (\mathbf{S}^*)^{-1} \mathbf{D} \mathbf{S}^{-1} = \mathbf{S} \mathbf{D} \mathbf{S}^{-1}.$$

This proves the theorem. ■

Part (c) explains the phenomenon. The matrix  $\mathbf{A}^{-1} \mathbf{B}$  is what one would naively form to turn  $\mathbf{B} \mathbf{x} = \lambda \mathbf{A} \mathbf{x}$  into an ordinary eigenvalue problem, and it is usually not self-adjoint, so there is no reason from Chapter 11 for its eigenvalues to be real. But it **is** self-adjoint with respect to the inner product  $\langle \cdot, \cdot \rangle_{\mathbf{A}}$ , and reality and the orthogonal eigenbasis come from that inner product rather than the standard one. The congruence  $\mathbf{S}$  is exactly the change of variable that turns  $\langle \cdot, \cdot \rangle_{\mathbf{A}}$  into the standard inner product.

#### Example. (A pencil that does not commute)

Let

$$\mathbf{A} = \begin{pmatrix} 2 & 1 \\ 1 & 2 \end{pmatrix}, \quad \mathbf{B} = \begin{pmatrix} 0 & 3 \\ 3 & 6 \end{pmatrix}.$$

Check that  $\mathbf{A} \succ 0$ , that  $\mathbf{AB} \neq \mathbf{BA}$ , and that  $\mathbf{B}$  is indefinite. Find the generalized eigenvalues and an  $\mathbf{A}$ -orthonormal basis of generalized eigenvectors, and write down  $\mathbf{S}$ .

*Solution. Setting up.*  $\mathbf{A}$  is symmetric with leading principal minors  $2 > 0$  and  $4 - 1 = 3 > 0$ , so  $\mathbf{A} \succ 0$  by [Theorem 12.1.5](#). Its own eigenvalues are 1 and 3. The products are

$$\mathbf{AB} = \begin{pmatrix} 3 & 12 \\ 6 & 15 \end{pmatrix}, \quad \mathbf{BA} = \begin{pmatrix} 3 & 6 \\ 12 & 15 \end{pmatrix},$$

which differ, so Chapter 11's theorem does not apply. Also  $\det \mathbf{B} = -9 < 0$ , so  $\mathbf{B}$  has one positive and one negative eigenvalue and is neither positive nor negative semidefinite.

*The pencil.*

$$\mathbf{B} - \lambda \mathbf{A} = \begin{pmatrix} -2\lambda & 3 - \lambda \\ 3 - \lambda & 6 - 2\lambda \end{pmatrix},$$

so

$$\begin{aligned} \det(\mathbf{B} - \lambda \mathbf{A}) &= -2\lambda(6 - 2\lambda) - (3 - \lambda)^2 \\ &= 4\lambda^2 - 12\lambda - 9 + 6\lambda - \lambda^2 \\ &= 3\lambda^2 - 6\lambda - 9 = 3(\lambda - 3)(\lambda + 1). \end{aligned}$$

The generalized eigenvalues are  $\lambda = -1$  and  $\lambda = 3$ , both real, as [Theorem 12.4.2](#) (b) promises.

*Eigenvectors.* For  $\lambda = -1$ ,  $\mathbf{B} + \mathbf{A} = \begin{pmatrix} 2 & 4 \\ 4 & 8 \end{pmatrix}$ , whose kernel is spanned by  $\mathbf{x}_1 = (2, -1)$ .

For  $\lambda = 3$ ,  $\mathbf{B} - 3\mathbf{A} = \begin{pmatrix} -6 & 0 \\ 0 & 0 \end{pmatrix}$ , whose kernel is spanned by  $\mathbf{x}_2 = (0, 1)$ . Checking directly:  $\mathbf{B}\mathbf{x}_1 = (-3, 0)$  and  $\mathbf{A}\mathbf{x}_1 = (3, 0)$ , so  $\mathbf{B}\mathbf{x}_1 = -\mathbf{A}\mathbf{x}_1$ ; and  $\mathbf{B}\mathbf{x}_2 = (3, 6) = 3(1, 2) = 3\mathbf{A}\mathbf{x}_2$ .

*Normalizing.*  $\mathbf{x}_1^\top \mathbf{A}\mathbf{x}_1 = (2, -1) \cdot (3, 0) = 6$  and  $\mathbf{x}_2^\top \mathbf{A}\mathbf{x}_2 = (0, 1) \cdot (1, 2) = 2$ , while  $\mathbf{x}_1^\top \mathbf{A}\mathbf{x}_2 = (2, -1) \cdot (1, 2) = 0$ : the two are already  $\mathbf{A}$ -orthogonal, as they must be, having different generalized eigenvalues. Scaling to  $\mathbf{A}$ -length 1,

$$\mathbf{S} = \begin{pmatrix} 2/\sqrt{6} & 0 \\ -1/\sqrt{6} & 1/\sqrt{2} \end{pmatrix}, \quad \mathbf{S}^\top \mathbf{A} \mathbf{S} = \mathbf{I}_2, \quad \mathbf{S}^\top \mathbf{B} \mathbf{S} = \begin{pmatrix} -1 & 0 \\ 0 & 3 \end{pmatrix}.$$

So in the coordinates  $\mathbf{x} = \mathbf{S}\mathbf{y}$  the two forms read  $\mathbf{x}^\top \mathbf{A} \mathbf{x} = y_1^2 + y_2^2$  and  $\mathbf{x}^\top \mathbf{B} \mathbf{x} = -y_1^2 + 3y_2^2$ . The signs  $(-, +)$  match the signs of the eigenvalues of  $\mathbf{B}$  itself, which are  $3 \pm 3\sqrt{2}$ ; Chapter 13 shows that this is no coincidence.

The problem  $\mathbf{B}\mathbf{x} = \lambda\mathbf{A}\mathbf{x}$  with  $\mathbf{A} \succ 0$  is the one an engineer meets first: for a vibrating structure,  $\mathbf{A}$  is the mass matrix and  $\mathbf{B}$  the stiffness matrix, the generalized eigenvalues are the squared natural frequencies, and the  $\mathbf{A}$ -orthonormal eigenvectors are the normal modes, in which the motion decouples into independent one-dimensional oscillations.

## 12.4.4 When neither form is definite

Definiteness of one of the two matrices is not decoration.

**Warning.** Two self-adjoint matrices, neither definite, need not be simultaneously diagonalizable by congruence. Take the real symmetric pair

$$\mathbf{A} = \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix}, \quad \mathbf{B} = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}.$$

Suppose  $\mathbf{S}^\top \mathbf{A} \mathbf{S}$  and  $\mathbf{S}^\top \mathbf{B} \mathbf{S}$  were both diagonal for some invertible real  $\mathbf{S}$ , say  $\mathbf{D}_\mathbf{A}$  and  $\mathbf{D}_\mathbf{B}$ . Since  $\mathbf{A}$  is invertible, so is  $\mathbf{D}_\mathbf{A}$ , and from  $\mathbf{A} = (\mathbf{S}^\top)^{-1} \mathbf{D}_\mathbf{A} \mathbf{S}^{-1}$  and  $\mathbf{B} = (\mathbf{S}^\top)^{-1} \mathbf{D}_\mathbf{B} \mathbf{S}^{-1}$  we get

$$\mathbf{A}^{-1} \mathbf{B} = \mathbf{S} \mathbf{D}_\mathbf{A}^{-1} \mathbf{S}^\top (\mathbf{S}^\top)^{-1} \mathbf{D}_\mathbf{B} \mathbf{S}^{-1} = \mathbf{S} (\mathbf{D}_\mathbf{A}^{-1} \mathbf{D}_\mathbf{B}) \mathbf{S}^{-1},$$

so  $\mathbf{A}^{-1} \mathbf{B}$  would be similar to a real diagonal matrix and would have real eigenvalues. But  $\mathbf{A}^{-1} = \mathbf{A}$  here, and

$$\mathbf{A}^{-1} \mathbf{B} = \begin{pmatrix} 0 & 1 \\ -1 & 0 \end{pmatrix}$$

has characteristic polynomial  $\lambda^2 + 1$ , with roots  $\pm i$ . So no such  $\mathbf{S}$  exists — and the pencil has **no** generalized eigenvalues in  $\mathbb{R}$  at all.

The obstruction in the warning is exactly the failure of part (c) of [Theorem 12.4.2](#). That part is in fact the right way to test a pair: if  $\mathbf{A}$  is invertible, a necessary condition for simultaneous diagonalization by congruence is that  $\mathbf{A}^{-1} \mathbf{B}$  be diagonalizable with real eigenvalues, and the theorem says that  $\mathbf{A} \succ 0$  is enough to guarantee it.

## 12.4.5 Exercises

### 12.4.5.1 A. Check your understanding

#### Exercise A1.

- (a) Define what it means for two matrices to be **congruent**, and say which of similarity and congruence is the natural relation for a form.
- (b) State [Theorem 12.4.1](#), with all hypotheses.
- (c) Name two differences between [Theorem 12.4.1](#) and [Theorem 11.7.1](#).
- (d) Determine whether the following statement is correct, and justify your answer: the generalized eigenvalues of  $(\mathbf{A}, \mathbf{B})$  are eigenvalues of  $\mathbf{B}$ .
- (e) For  $\mathbf{A} \succ 0$  and  $\mathbf{B}$  self-adjoint, what inner product makes  $\mathbf{A}^{-1}\mathbf{B}$  self-adjoint?

*Solution.*

- (a)  $\mathbf{A}$  and  $\mathbf{C}$  are congruent if  $\mathbf{C} = \mathbf{S}^*\mathbf{A}\mathbf{S}$  for some invertible  $\mathbf{S}$ . Congruence is the natural relation for a form, since the substitution  $\mathbf{x} = \mathbf{S}\mathbf{y}$  turns  $\mathbf{x}^*\mathbf{A}\mathbf{x}$  into  $\mathbf{y}^*(\mathbf{S}^*\mathbf{A}\mathbf{S})\mathbf{y}$ ; similarity  $\mathbf{S}^{-1}\mathbf{A}\mathbf{S}$  is the natural relation for an operator.
- (b) If  $\mathbf{A}, \mathbf{B} \in M_n(F)$  are self-adjoint and  $\mathbf{A} \succ 0$ , there is an invertible  $\mathbf{S}$  with  $\mathbf{S}^*\mathbf{A}\mathbf{S} = \mathbf{I}_n$  and  $\mathbf{S}^*\mathbf{B}\mathbf{S}$  real diagonal.
- (c) Chapter 11's theorem assumes the operators commute and are normal, and delivers a **unitary** change of basis preserving eigenvalues; this one assumes nothing about commuting but assumes one matrix is positive definite, and delivers a merely invertible  $\mathbf{S}$  under which eigenvalues are not preserved.
- (d) Incorrect. In [Example \(A pencil that does not commute\)](#) the generalized eigenvalues are  $-1$  and  $3$ , while the eigenvalues of  $\mathbf{B}$  are  $3 \pm 3\sqrt{2}$ . They agree when  $\mathbf{A} = \mathbf{I}$ , and not in general.
- (e)  $\langle \mathbf{x}, \mathbf{y} \rangle_{\mathbf{A}} = \mathbf{y}^*\mathbf{A}\mathbf{x}$  ([Theorem 12.4.2 \(a\)](#)).

### 12.4.5.2 B. Practice

#### Exercise B1: A generalized eigenvalue problem.

Let

$$\mathbf{A} = \begin{pmatrix} 1 & 1 \\ 1 & 2 \end{pmatrix}, \quad \mathbf{B} = \begin{pmatrix} 2 & 2 \\ 2 & 1 \end{pmatrix}.$$

- (a) Verify that  $\mathbf{A} \succ 0$  and find the generalized eigenvalues of  $(\mathbf{A}, \mathbf{B})$ .
- (b) Find generalized eigenvectors and an invertible  $\mathbf{S}$  with  $\mathbf{S}^\top \mathbf{A} \mathbf{S} = \mathbf{I}_2$  and  $\mathbf{S}^\top \mathbf{B} \mathbf{S}$  diagonal.

*Solution.*

- (a) The leading principal minors of  $\mathbf{A}$  are  $1 > 0$  and  $2 - 1 = 1 > 0$ , so  $\mathbf{A} \succ 0$  by



**Theorem 12.1.5.** Then

$$\mathbf{B} - \lambda \mathbf{A} = \begin{pmatrix} 2 - \lambda & 2 - \lambda \\ 2 - \lambda & 1 - 2\lambda \end{pmatrix},$$

so

$$\begin{aligned} \det(\mathbf{B} - \lambda \mathbf{A}) &= (2 - \lambda)(1 - 2\lambda) - (2 - \lambda)^2 \\ &= (2 - \lambda)[(1 - 2\lambda) - (2 - \lambda)] \\ &= (2 - \lambda)(-1 - \lambda) = (\lambda - 2)(\lambda + 1). \end{aligned}$$

The generalized eigenvalues are 2 and  $-1$ .

(b) For  $\lambda = 2$ :  $\mathbf{B} - 2\mathbf{A} = \begin{pmatrix} 0 & 0 \\ 0 & -3 \end{pmatrix}$ , with kernel spanned by  $\mathbf{x}_1 = (1, 0)$ . For  $\lambda = -1$ :

$\mathbf{B} + \mathbf{A} = \begin{pmatrix} 3 & 3 \\ 3 & 3 \end{pmatrix}$ , with kernel spanned by  $\mathbf{x}_2 = (1, -1)$ . Check:  $\mathbf{B}\mathbf{x}_1 = (2, 2) = 2\mathbf{A}\mathbf{x}_1$  since  $\mathbf{A}\mathbf{x}_1 = (1, 1)$ ; and  $\mathbf{B}\mathbf{x}_2 = (0, 1)$  while  $\mathbf{A}\mathbf{x}_2 = (0, -1)$ , so  $\mathbf{B}\mathbf{x}_2 = -\mathbf{A}\mathbf{x}_2$ .

Now  $\mathbf{x}_1^\top \mathbf{A}\mathbf{x}_1 = (1, 0) \cdot (1, 1) = 1$  and  $\mathbf{x}_2^\top \mathbf{A}\mathbf{x}_2 = (1, -1) \cdot (0, -1) = 1$ , and  $\mathbf{x}_1^\top \mathbf{A}\mathbf{x}_2 = (1, 0) \cdot (0, -1) = 0$ . Both are already  $\mathbf{A}$ -unit vectors, so

$$\mathbf{S} = \begin{pmatrix} 1 & 1 \\ 0 & -1 \end{pmatrix}, \quad \mathbf{S}^\top \mathbf{A} \mathbf{S} = \mathbf{I}_2, \quad \mathbf{S}^\top \mathbf{B} \mathbf{S} = \begin{pmatrix} 2 & 0 \\ 0 & -1 \end{pmatrix}.$$

### Exercise B2: Whitening by hand.

Let  $\mathbf{A} = \text{diag}(4, 9)$  and  $\mathbf{B} = \begin{pmatrix} 4 & 6 \\ 6 & 9 \end{pmatrix}$ . Follow the two steps of the proof of [Theorem 12.4.1](#): compute  $\mathbf{A}^{-1/2}$ , then  $\mathbf{C} = \mathbf{A}^{-1/2} \mathbf{B} \mathbf{A}^{-1/2}$ , diagonalize  $\mathbf{C}$ , and write down  $\mathbf{S}$  and  $\mathbf{S}^\top \mathbf{B} \mathbf{S}$ .

*Solution.*  $\mathbf{A}$  is diagonal with positive entries, so  $\mathbf{A}^{1/2} = \text{diag}(2, 3)$  and  $\mathbf{A}^{-1/2} = \text{diag}(\frac{1}{2}, \frac{1}{3})$ . Then

$$\mathbf{C} = \begin{pmatrix} \frac{1}{2} & 0 \\ 0 & \frac{1}{3} \end{pmatrix} \begin{pmatrix} 4 & 6 \\ 6 & 9 \end{pmatrix} \begin{pmatrix} \frac{1}{2} & 0 \\ 0 & \frac{1}{3} \end{pmatrix} = \begin{pmatrix} 1 & 1 \\ 1 & 1 \end{pmatrix}.$$

Its eigenvalues are 2, with unit eigenvector  $\frac{1}{\sqrt{2}}(1, 1)$ , and 0, with unit eigenvector  $\frac{1}{\sqrt{2}}(1, -1)$ .

So  $\mathbf{U} = \frac{1}{\sqrt{2}} \begin{pmatrix} 1 & 1 \\ 1 & -1 \end{pmatrix}$  and  $\mathbf{D} = \text{diag}(2, 0)$ , and

$$\mathbf{S} = \mathbf{A}^{-1/2} \mathbf{U} = \frac{1}{\sqrt{2}} \begin{pmatrix} \frac{1}{2} & \frac{1}{2} \\ \frac{1}{3} & -\frac{1}{3} \end{pmatrix}, \quad \mathbf{S}^\top \mathbf{A} \mathbf{S} = \mathbf{I}_2, \quad \mathbf{S}^\top \mathbf{B} \mathbf{S} = \begin{pmatrix} 2 & 0 \\ 0 & 0 \end{pmatrix}.$$

As a check,  $\det(\mathbf{B} - \lambda \mathbf{A}) = (4 - 4\lambda)(9 - 9\lambda) - 36 = 36\lambda^2 - 72\lambda = 36\lambda(\lambda - 2)$ , whose roots 0 and 2 are the diagonal entries of  $\mathbf{S}^\top \mathbf{B} \mathbf{S}$ .

### Exercise B3: A pair that resists.

Let  $\mathbf{A} = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}$  and  $\mathbf{B} = \begin{pmatrix} 1 & 0 \\ 0 & 0 \end{pmatrix}$ . Determine whether there is an invertible real  $\mathbf{S}$  making  $\mathbf{S}^\top \mathbf{A} \mathbf{S}$  and  $\mathbf{S}^\top \mathbf{B} \mathbf{S}$  both diagonal. Justify your answer.

*Solution.* There is not. Both matrices are symmetric and  $\mathbf{A}$  is invertible, with  $\mathbf{A}^{-1} = \mathbf{A}$ . If such an  $\mathbf{S}$  existed, the computation in the warning above would give  $\mathbf{A}^{-1}\mathbf{B} = \mathbf{S}(\mathbf{D}_\mathbf{A}^{-1}\mathbf{D}_\mathbf{B})\mathbf{S}^{-1}$ , so  $\mathbf{A}^{-1}\mathbf{B}$  would be diagonalizable. But

$$\mathbf{A}^{-1}\mathbf{B} = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix} \begin{pmatrix} 1 & 0 \\ 0 & 0 \end{pmatrix} = \begin{pmatrix} 0 & 0 \\ 1 & 0 \end{pmatrix},$$

which is non-zero with  $(\mathbf{A}^{-1}\mathbf{B})^2 = \mathbf{0}$ , hence has 0 as its only eigenvalue and is not diagonalizable (a diagonalizable matrix with only the eigenvalue 0 is  $\mathbf{0}$ ). Here the generalized eigenvalues are both 0 and real, so reality of the spectrum is not the obstruction; diagonalizability is.

### 12.4.5.3 C. Going deeper

#### Exercise C1: The $\mathbf{A}$ -inner product.

Let  $\mathbf{A} \succ 0$  and  $\mathbf{B}$  be self-adjoint in  $M_n(F)$ , and write  $\langle \mathbf{x}, \mathbf{y} \rangle_\mathbf{A} = \mathbf{y}^* \mathbf{A} \mathbf{x}$ .

- (a) Prove that  $\langle \cdot, \cdot \rangle_\mathbf{A}$  is an inner product on  $F^n$ .
- (b) Prove that  $T = T_{\mathbf{A}^{-1}\mathbf{B}}$  is self-adjoint on  $(F^n, \langle \cdot, \cdot \rangle_\mathbf{A})$ .
- (c) Deduce [Theorem 12.4.2](#) (a) and (b) from the spectral theorem applied to  $T$ , without using [Theorem 12.4.1](#).

*Solution.*

- (a) Sesquilinearity in the first slot is clear from the formula. Conjugate symmetry:  $\overline{\langle \mathbf{y}, \mathbf{x} \rangle_\mathbf{A}} = \overline{\mathbf{x}^* \mathbf{A} \mathbf{y}} = \mathbf{y}^* \mathbf{A}^* \mathbf{x} = \mathbf{y}^* \mathbf{A} \mathbf{x} = \langle \mathbf{x}, \mathbf{y} \rangle_\mathbf{A}$ , using  $\mathbf{A}^* = \mathbf{A}$ . Positive definiteness:  $\langle \mathbf{x}, \mathbf{x} \rangle_\mathbf{A} = \mathbf{x}^* \mathbf{A} \mathbf{x} > 0$  for  $\mathbf{x} \neq \mathbf{0}$ , which is [Definition 12.1.1](#) for  $\mathbf{A} \succ 0$ .
- (b) For all  $\mathbf{x}, \mathbf{y}$ ,

$$\langle T\mathbf{x}, \mathbf{y} \rangle_\mathbf{A} = \mathbf{y}^* \mathbf{A} \mathbf{A}^{-1} \mathbf{B} \mathbf{x} = \mathbf{y}^* \mathbf{B} \mathbf{x},$$

and

$$\langle \mathbf{x}, T\mathbf{y} \rangle_\mathbf{A} = (\mathbf{A}^{-1} \mathbf{B} \mathbf{y})^* \mathbf{A} \mathbf{x} = \mathbf{y}^* \mathbf{B}^* \mathbf{A}^{-1} \mathbf{A} \mathbf{x} = \mathbf{y}^* \mathbf{B} \mathbf{x},$$

using  $(\mathbf{A}^{-1})^* = \mathbf{A}^{-1}$  and  $\mathbf{B}^* = \mathbf{B}$ . The two agree, so  $T$  is self-adjoint for this inner product ([Definition 10.6.1](#)).

- (c) By [Theorem 11.5.1](#) (over  $\mathbb{R}$ ) or [Theorem 11.4.2](#) (over  $\mathbb{C}$ ) applied to the self-adjoint  $T$  on the inner product space  $(F^n, \langle \cdot, \cdot \rangle_\mathbf{A})$ , there is a basis  $\mathbf{s}_1, \dots, \mathbf{s}_n$  of  $F^n$ , orthonormal for  $\langle \cdot, \cdot \rangle_\mathbf{A}$ , with  $T\mathbf{s}_i = d_i \mathbf{s}_i$  and  $d_i \in \mathbb{R}$  ([Theorem 11.2.1](#)). The equation  $\mathbf{A}^{-1} \mathbf{B} \mathbf{s}_i = d_i \mathbf{s}_i$  is  $\mathbf{B} \mathbf{s}_i = d_i \mathbf{A} \mathbf{s}_i$ , and  $\mathbf{s}_i^* \mathbf{A} \mathbf{s}_j = \langle \mathbf{s}_j, \mathbf{s}_i \rangle_\mathbf{A} = \delta_{ij}$ ; this is (a). For (b),  $\lambda$  is a generalized eigenvalue exactly when  $\mathbf{A}^{-1} \mathbf{B} - \lambda \mathbf{I}$  is singular, that is when  $\lambda$  is an eigenvalue of  $\mathbf{A}^{-1} \mathbf{B}$ , and those are  $d_1, \dots, d_n$ .

#### Exercise C2: Signs of the generalized eigenvalues.

Let  $\mathbf{A} \succ 0$  and  $\mathbf{B}$  be self-adjoint in  $M_n(F)$ , with generalized eigenvalues  $d_1, \dots, d_n$ .

- (a) Prove that  $\mathbf{B} \succeq 0$  if and only if every  $d_i \geq 0$ , and  $\mathbf{B} \succ 0$  if and only if every  $d_i > 0$ .
- (b) Deduce that if  $\mathbf{A} \succ 0$  and  $\mathbf{B} \succ 0$ , then  $\det \mathbf{B} / \det \mathbf{A} = \prod_i d_i > 0$ .

*Solution.*

- (a) Let  $\mathbf{S}$  be as in [Theorem 12.4.1](#), so  $\mathbf{S}^*\mathbf{B}\mathbf{S} = \mathbf{D} = \text{diag}(d_1, \dots, d_n)$ . For  $\mathbf{x} \in F^n$  put  $\mathbf{y} = \mathbf{S}^{-1}\mathbf{x}$ ; as  $\mathbf{x}$  runs over  $F^n$  so does  $\mathbf{y}$ , and  $\mathbf{x} \neq \mathbf{0}$  exactly when  $\mathbf{y} \neq \mathbf{0}$ . Then

$$\mathbf{x}^*\mathbf{B}\mathbf{x} = \mathbf{y}^*\mathbf{S}^*\mathbf{B}\mathbf{S}\mathbf{y} = \mathbf{y}^*\mathbf{D}\mathbf{y} = \sum_i d_i |y_i|^2.$$

If every  $d_i \geq 0$ , this is  $\geq 0$  for all  $\mathbf{y}$ , so  $\mathbf{B} \succeq 0$ . Conversely, taking  $\mathbf{y} = \mathbf{e}_i$  gives  $d_i \geq 0$ . The same argument with strict inequalities gives the second claim, using that  $\sum_i d_i |y_i|^2 > 0$  for  $\mathbf{y} \neq \mathbf{0}$  when every  $d_i > 0$ .

- (b) Taking determinants in  $\mathbf{S}^*\mathbf{B}\mathbf{S} = \mathbf{D}$  and  $\mathbf{S}^*\mathbf{A}\mathbf{S} = \mathbf{I}$  gives  $|\det \mathbf{S}|^2 \det \mathbf{B} = \prod_i d_i$  and  $|\det \mathbf{S}|^2 \det \mathbf{A} = 1$ . Dividing,  $\det \mathbf{B} / \det \mathbf{A} = \prod_i d_i$ , which is  $> 0$  by (a).

### Exercise C3: Dropping definiteness of $\mathbf{A}$ .

Let  $\mathbf{A} = \begin{pmatrix} 1 & 0 \\ 0 & 0 \end{pmatrix}$  and  $\mathbf{B} = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}$ , so that  $\mathbf{A} \succeq 0$  but  $\mathbf{A} \neq 0$ .

- (a) Compute  $\det(\mathbf{B} - \lambda\mathbf{A})$  and deduce that the pair has no generalized eigenvalues at all.  
 (b) Prove that no invertible  $\mathbf{S} \in M_2(\mathbb{R})$  makes  $\mathbf{S}^\top \mathbf{A} \mathbf{S}$  and  $\mathbf{S}^\top \mathbf{B} \mathbf{S}$  both diagonal.

*Hint for (b): compare the degree in  $\lambda$  of  $\det(\mathbf{B} - \lambda\mathbf{A})$  on both sides.*

*Solution.*

- (a)  $\mathbf{B} - \lambda\mathbf{A} = \begin{pmatrix} -\lambda & 1 \\ 1 & 0 \end{pmatrix}$ , with determinant  $-1$ . This is never 0, so  $\mathbf{B} - \lambda\mathbf{A}$  is invertible for every  $\lambda$  and there is no  $\mathbf{x} \neq \mathbf{0}$  with  $\mathbf{B}\mathbf{x} = \lambda\mathbf{A}\mathbf{x}$ .  
 (b) Suppose  $\mathbf{S}^\top \mathbf{A} \mathbf{S} = \text{diag}(a_1, a_2)$  and  $\mathbf{S}^\top \mathbf{B} \mathbf{S} = \text{diag}(b_1, b_2)$ . As in the proof of [Theorem 12.4.2](#) (b),

$$\det(\mathbf{B} - \lambda\mathbf{A}) = \frac{1}{(\det \mathbf{S})^2} (b_1 - \lambda a_1)(b_2 - \lambda a_2).$$

By (a) the left side is the constant  $-1$ , so the right side is a non-zero constant. It is non-zero, so  $b_1 b_2 \neq 0$  and hence  $\det(\mathbf{S}^\top \mathbf{B} \mathbf{S}) \neq 0$ ; it is constant in  $\lambda$ , so  $a_1 = a_2 = 0$ , which forces  $\mathbf{S}^\top \mathbf{A} \mathbf{S} = \mathbf{0}$  and therefore  $\mathbf{A} = \mathbf{0}$ . That is false, so no such  $\mathbf{S}$  exists. (Alternatively:  $\text{rank}(\mathbf{S}^\top \mathbf{A} \mathbf{S}) = \text{rank} \mathbf{A} = 1$ , so exactly one  $a_i$  is non-zero, and the right side then has degree 1 in  $\lambda$ , not 0.)

## 12.5 The Loewner Order

Every theorem so far in this chapter has compared a self-adjoint matrix with 0. Comparing it with 0 is only the first case of comparing it with something, and the analogy with real numbers suggests the general one: say that  $\mathbf{A}$  is at least  $\mathbf{B}$  when the difference is non-negative. That is a genuine order, and a surprising amount of ordinary arithmetic survives it. A little does not, and what fails is the point of the section: squaring both sides of an inequality between matrices

is not allowed.

Throughout,  $F = \mathbb{R}$  or  $F = \mathbb{C}$ , all matrices are in  $M_n(F)$ , and  $\lambda_1(\mathbf{A}) \geq \cdots \geq \lambda_n(\mathbf{A})$  are the eigenvalues of a Hermitian  $\mathbf{A}$  in decreasing order.

## 12.5.1 Comparing two self-adjoint matrices

The relation  $\mathbf{A} \succeq 0$  already behaves like “ $\geq 0$ ” for numbers: it is preserved by sums, by non-negative scalars and by congruence (Proposition 12.1.2). For real numbers,  $a \geq b$  means nothing other than  $a - b \geq 0$ . Copying that sentence is the whole definition.

*One self-adjoint matrix dominates another when their difference is positive semidefinite.*

### Definition 12.5.1 — The Loewner order

Let  $\mathbf{A}, \mathbf{B} \in M_n(F)$  be Hermitian. Write

$$\mathbf{A} \succeq \mathbf{B} \quad \text{to mean} \quad \mathbf{A} - \mathbf{B} \succeq 0,$$

and  $\mathbf{A} \succ \mathbf{B}$  to mean  $\mathbf{A} - \mathbf{B} \succ 0$ . This relation on the set of Hermitian matrices is the **Loewner order**.

In words:  $\mathbf{A} \succeq \mathbf{B}$  says that  $\langle \mathbf{A}\mathbf{x}, \mathbf{x} \rangle \geq \langle \mathbf{B}\mathbf{x}, \mathbf{x} \rangle$  for every  $\mathbf{x} \in F^n$ , since  $\langle (\mathbf{A} - \mathbf{B})\mathbf{x}, \mathbf{x} \rangle = \langle \mathbf{A}\mathbf{x}, \mathbf{x} \rangle - \langle \mathbf{B}\mathbf{x}, \mathbf{x} \rangle$ . So the order compares the two quadratic forms pointwise, and it is exactly as strong as that comparison. Note also that the difference of two Hermitian matrices is Hermitian, so clause (P1) of Definition 12.1.1 is automatic here and only (P2) has to be checked. Taking  $\mathbf{B} = \mathbf{0}$  recovers  $\mathbf{A} \succeq 0$ , so the notation is consistent.

Three examples, each of which will be used later.

- **Diagonal matrices.**  $\text{diag}(a_1, \dots, a_n) \succeq \text{diag}(b_1, \dots, b_n)$  if and only if  $a_i \geq b_i$  for every  $i$ , since the difference is diagonal and a diagonal Hermitian matrix is positive semidefinite exactly when its entries, which are its eigenvalues, are  $\geq 0$  (Theorem 12.1.4).
- **Eigenvalue bounds.** For Hermitian  $\mathbf{A}$ ,  $\lambda_1(\mathbf{A})\mathbf{I} \succeq \mathbf{A} \succeq \lambda_n(\mathbf{A})\mathbf{I}$ . Indeed  $\lambda_1(\mathbf{A})\mathbf{I} - \mathbf{A}$  is Hermitian with eigenvalues  $\lambda_1(\mathbf{A}) - \lambda_i(\mathbf{A}) \geq 0$ .
- **Projections.** If  $\mathbf{P} \in M_n(F)$  is the matrix of an orthogonal projection onto a subspace of  $F^n$ , then  $\mathbf{I} \succeq \mathbf{P} \succeq 0$ : it is Hermitian, and its eigenvalues are 0 and 1.

**A non-example by minimal change:** replace  $\text{diag}(1, 1)$  by  $\text{diag}(1, -1)$  in the first item. Then  $\text{diag}(1, -1) \succeq \text{diag}(0, 0)$  fails, and it fails only in the second coordinate — the test is coordinatewise for diagonal matrices, and one bad coordinate is enough.

### Proposition 12.5.1 — The Loewner order is a partial order

On the set of Hermitian matrices in  $M_n(F)$ , the relation  $\succeq$  is reflexive, transitive and antisymmetric:  $\mathbf{A} \succeq \mathbf{A}$ ; if  $\mathbf{A} \succeq \mathbf{B}$  and  $\mathbf{B} \succeq \mathbf{C}$  then  $\mathbf{A} \succeq \mathbf{C}$ ; and if  $\mathbf{A} \succeq \mathbf{B}$  and  $\mathbf{B} \succeq \mathbf{A}$  then  $\mathbf{A} = \mathbf{B}$ .

*Proof.* Reflexivity:  $\mathbf{A} - \mathbf{A} = \mathbf{0}$ , and  $\langle \mathbf{0}\mathbf{x}, \mathbf{x} \rangle = 0 \geq 0$ .

Transitivity:  $\mathbf{A} - \mathbf{C} = (\mathbf{A} - \mathbf{B}) + (\mathbf{B} - \mathbf{C})$ , and for every  $\mathbf{x}$ ,

$$\langle (\mathbf{A} - \mathbf{C})\mathbf{x}, \mathbf{x} \rangle = \langle (\mathbf{A} - \mathbf{B})\mathbf{x}, \mathbf{x} \rangle + \langle (\mathbf{B} - \mathbf{C})\mathbf{x}, \mathbf{x} \rangle \geq 0,$$

both summands being  $\geq 0$  by hypothesis.

Antisymmetry: put  $\mathbf{S} = \mathbf{A} - \mathbf{B}$ , which is Hermitian. From  $\mathbf{A} \succeq \mathbf{B}$  we get  $\langle \mathbf{S}\mathbf{x}, \mathbf{x} \rangle \geq 0$  for every  $\mathbf{x}$ , and from  $\mathbf{B} \succeq \mathbf{A}$  we get  $\langle -\mathbf{S}\mathbf{x}, \mathbf{x} \rangle \geq 0$ , that is  $\langle \mathbf{S}\mathbf{x}, \mathbf{x} \rangle \leq 0$ . Hence  $\langle \mathbf{S}\mathbf{x}, \mathbf{x} \rangle = 0$  for every  $\mathbf{x} \in F^n$ , and  $\mathbf{S}$  is self-adjoint, so  $\mathbf{S} = \mathbf{0}$  by [Theorem 11.2.5](#). Thus  $\mathbf{A} = \mathbf{B}$ . ■

Antisymmetry is the clause that needed an argument, and the argument is the one the chapter keeps reusing: a self-adjoint matrix is determined by its quadratic form. Over  $\mathbb{R}$  this is genuinely a theorem — the quarter turn of [Definition 12.1.1](#)'s warning has vanishing quadratic form and is not  $\mathbf{0}$  — and it is self-adjointness that rules that out.

**Warning.** The Loewner order is partial, and it is not the entrywise order. Two things go wrong if one forgets this. First, most pairs are incomparable:  $\mathbf{A} = \text{diag}(1, 0)$  and  $\mathbf{B} = \text{diag}(0, 1)$  satisfy neither  $\mathbf{A} \succeq \mathbf{B}$  nor  $\mathbf{B} \succeq \mathbf{A}$ , so an argument that splits into “either  $\mathbf{A} \succeq \mathbf{B}$  or  $\mathbf{B} \succeq \mathbf{A}$ ” is not an argument. Second,  $\mathbf{A} \succeq \mathbf{B}$  says nothing entrywise, in either direction:

$$\begin{pmatrix} 2 & 0 \\ 0 & 2 \end{pmatrix} \succeq \begin{pmatrix} 1 & 1 \\ 1 & 1 \end{pmatrix}, \quad \text{yet the off-diagonal entries went down;}$$

the difference is  $\begin{pmatrix} 1 & -1 \\ -1 & 1 \end{pmatrix}$ , with eigenvalues 0 and 2. Conversely  $\begin{pmatrix} 1 & 2 \\ 2 & 1 \end{pmatrix}$  has every entry  $\geq 0$  and is **not**  $\succeq 0$ , its eigenvalues being 3 and  $-1$ .

**Quick check.** Is the set of Hermitian  $\mathbf{A}$  with  $\mathbf{I} \succeq \mathbf{A} \succeq 0$  closed under multiplication?

**Solution.** No. Take  $\mathbf{A} = \begin{pmatrix} 1 & 0 \\ 0 & 0 \end{pmatrix}$  and  $\mathbf{B} = \frac{1}{2} \begin{pmatrix} 1 & 1 \\ 1 & 1 \end{pmatrix}$ , both orthogonal projections, so both lie in the set. Then  $\mathbf{AB} = \frac{1}{2} \begin{pmatrix} 1 & 1 \\ 0 & 0 \end{pmatrix}$ , which is not even Hermitian, so it is not in the set and the order does not apply to it. The order lives on Hermitian matrices, and Hermitian matrices are not closed under multiplication ([Exercise \(C2: Sums and products\)](#)).

## 12.5.2 What survives

One small fact is needed first, and it is worth stating on its own because it is the elementary half of a theme Chapter 16 takes up in full: the largest and smallest eigenvalues of a Hermitian matrix are the largest and smallest values of its quadratic form on the unit sphere.

### Lemma 12.5.2 — The extreme eigenvalues as extreme values

Let  $\mathbf{A} \in M_n(F)$  be Hermitian. Then for every  $\mathbf{x} \in F^n$  with  $\|\mathbf{x}\| = 1$ ,

$$\lambda_n(\mathbf{A}) \leq \langle \mathbf{A}\mathbf{x}, \mathbf{x} \rangle \leq \lambda_1(\mathbf{A}),$$

and both bounds are attained, at unit eigenvectors for  $\lambda_n(\mathbf{A})$  and  $\lambda_1(\mathbf{A})$  respectively.

*Proof.* By [Corollary 11.4.3](#) (over  $\mathbb{C}$ ) or [Corollary 11.5.2](#) (over  $\mathbb{R}$ ),  $F^n$  has an orthonormal basis  $(\mathbf{q}_1, \dots, \mathbf{q}_n)$  with  $\mathbf{A}\mathbf{q}_i = \lambda_i(\mathbf{A})\mathbf{q}_i$ . Write  $\mathbf{x} = \sum_i c_i \mathbf{q}_i$ ; then  $\sum_i |c_i|^2 = \|\mathbf{x}\|^2 = 1$  by [Theorem 10.2.1](#), and

$$\langle \mathbf{A}\mathbf{x}, \mathbf{x} \rangle = \left\langle \sum_i \lambda_i(\mathbf{A}) c_i \mathbf{q}_i, \sum_j c_j \mathbf{q}_j \right\rangle = \sum_i \lambda_i(\mathbf{A}) |c_i|^2,$$

the cross terms vanishing by orthonormality. The right-hand side is a weighted average of the  $\lambda_i(\mathbf{A})$  with non-negative weights summing to 1, hence lies between the smallest and the largest. Taking  $\mathbf{x} = \mathbf{q}_1$  gives  $\lambda_1(\mathbf{A})$  and  $\mathbf{x} = \mathbf{q}_n$  gives  $\lambda_n(\mathbf{A})$ . ■

### Theorem 12.5.3 — Basic Properties of the Loewner Order

Let  $\mathbf{A}, \mathbf{B} \in M_n(F)$  be Hermitian with  $\mathbf{A} \succeq \mathbf{B}$ . Then:

- (a)  $\mathbf{S}^* \mathbf{A} \mathbf{S} \succeq \mathbf{S}^* \mathbf{B} \mathbf{S}$  for every  $\mathbf{S} \in M_{n \times m}(F)$ ;
- (b)  $\lambda_1(\mathbf{A}) \geq \lambda_1(\mathbf{B})$  and  $\lambda_n(\mathbf{A}) \geq \lambda_n(\mathbf{B})$ ;
- (c) if moreover  $\mathbf{B} \succeq 0$ , then  $\text{tr } \mathbf{A} \geq \text{tr } \mathbf{B}$  and  $\det \mathbf{A} \geq \det \mathbf{B}$ ;
- (d) if moreover  $\mathbf{B} \succ 0$ , then  $\mathbf{A} \succ 0$  and  $\mathbf{B}^{-1} \succeq \mathbf{A}^{-1}$ .

*Idea.* Each clause has a one-line mechanism. (a) is the substitution  $\mathbf{x} = \mathbf{S}\mathbf{y}$ . (b) evaluates the quadratic form at the vector that is extremal for the *other* matrix. (c) splits: the trace is a sum of eigenvalues, and for the determinant we whiten by  $\mathbf{B}^{-1/2}$  so that the comparison becomes “ $\succeq \mathbf{I}$ ”, where the determinant is visibly  $\geq 1$ . (d) is the same whitening, plus the observation that inverting reverses the order for *numbers*  $\geq 1$ .

*Proof.*

- (a) This is [Proposition 12.1.2](#) (a) applied to the Hermitian matrix  $\mathbf{A} - \mathbf{B} \succeq 0$ , together with  $\mathbf{S}^* \mathbf{A} \mathbf{S} - \mathbf{S}^* \mathbf{B} \mathbf{S} = \mathbf{S}^* (\mathbf{A} - \mathbf{B}) \mathbf{S}$ .
- (b) Let  $\mathbf{x}$  be a unit eigenvector of  $\mathbf{B}$  for  $\lambda_1(\mathbf{B})$ . By [Lemma 12.5.2](#) applied twice, and  $\mathbf{A} \succeq \mathbf{B}$  in between,

$$\lambda_1(\mathbf{B}) = \langle \mathbf{B}\mathbf{x}, \mathbf{x} \rangle \leq \langle \mathbf{A}\mathbf{x}, \mathbf{x} \rangle \leq \lambda_1(\mathbf{A}).$$

Now let  $\mathbf{y}$  be a unit eigenvector of  $\mathbf{A}$  for  $\lambda_n(\mathbf{A})$ . Then

$$\lambda_n(\mathbf{A}) = \langle \mathbf{A}\mathbf{y}, \mathbf{y} \rangle \geq \langle \mathbf{B}\mathbf{y}, \mathbf{y} \rangle \geq \lambda_n(\mathbf{B}).$$

- (c) The matrix  $\mathbf{A} - \mathbf{B}$  is Hermitian and  $\succeq 0$ , so all its eigenvalues are  $\geq 0$  by [Theorem 12.1.4](#), and  $\text{tr}(\mathbf{A} - \mathbf{B})$ , being their sum ([Theorem 8.3.3](#)), is  $\geq 0$ . Hence  $\text{tr } \mathbf{A} \geq \text{tr } \mathbf{B}$ .

For the determinant, first note  $\mathbf{A} \succeq \mathbf{B} \succeq 0$  gives  $\mathbf{A} \succeq 0$  by [Proposition 12.5.1](#), so  $\det \mathbf{A} \geq 0$  and  $\det \mathbf{B} \geq 0$ , both being products of non-negative eigenvalues. If  $\mathbf{B}$  is singular then  $\det \mathbf{B} = 0 \leq \det \mathbf{A}$  and we are done. Otherwise  $\mathbf{B} \succ 0$  with no zero eigenvalue means  $\mathbf{B} \succ 0$  ([Theorem 12.1.5](#)), so  $\mathbf{B}^{1/2} \succ 0$  is invertible and we may set  $\mathbf{C} = \mathbf{B}^{-1/2} \mathbf{A} \mathbf{B}^{-1/2}$ , where  $\mathbf{B}^{-1/2} = (\mathbf{B}^{1/2})^{-1}$  is Hermitian. By (a) with

$$\mathbf{S} = \mathbf{B}^{-1/2},$$

$$\mathbf{C} = \mathbf{B}^{-1/2} \mathbf{A} \mathbf{B}^{-1/2} \succeq \mathbf{B}^{-1/2} \mathbf{B} \mathbf{B}^{-1/2} = \mathbf{I}_n,$$

so  $\mathbf{C} - \mathbf{I} \succeq 0$  and every eigenvalue of  $\mathbf{C}$  is  $\geq 1$ . Hence  $\det \mathbf{C} \geq 1$ . Since  $\mathbf{A} = \mathbf{B}^{1/2} \mathbf{C} \mathbf{B}^{1/2}$ , Theorem 6.5.4 gives

$$\det \mathbf{A} = \det(\mathbf{B}^{1/2})^2 \det \mathbf{C} = \det \mathbf{B} \cdot \det \mathbf{C} \geq \det \mathbf{B}.$$

- (d) Suppose  $\mathbf{B} \succ 0$ . For  $\mathbf{x} \neq \mathbf{0}$ ,  $\langle \mathbf{A}\mathbf{x}, \mathbf{x} \rangle \geq \langle \mathbf{B}\mathbf{x}, \mathbf{x} \rangle > 0$ , so  $\mathbf{A} \succ 0$  and in particular  $\mathbf{A}$  is invertible (Theorem 12.1.5). Let  $\mathbf{C} = \mathbf{B}^{-1/2} \mathbf{A} \mathbf{B}^{-1/2} \succeq \mathbf{I}$  as in (c). Then  $\mathbf{C} \succ 0$ , so  $\mathbf{C}$  is invertible, and  $\mathbf{C}^{-1}$  is Hermitian with eigenvalues the reciprocals of those of  $\mathbf{C}$ , all lying in  $(0, 1]$ . Therefore  $\mathbf{I} - \mathbf{C}^{-1}$  is Hermitian with non-negative eigenvalues, that is  $\mathbf{I} \succeq \mathbf{C}^{-1}$  (Theorem 12.1.4). Applying (a) with  $\mathbf{S} = \mathbf{B}^{-1/2}$  to this inequality,

$$\mathbf{B}^{-1} = \mathbf{B}^{-1/2} \mathbf{I} \mathbf{B}^{-1/2} \succeq \mathbf{B}^{-1/2} \mathbf{C}^{-1} \mathbf{B}^{-1/2} = \mathbf{B}^{-1/2} \mathbf{B}^{1/2} \mathbf{A}^{-1} \mathbf{B}^{1/2} \mathbf{B}^{-1/2} = \mathbf{A}^{-1},$$

using  $\mathbf{C}^{-1} = (\mathbf{B}^{-1/2} \mathbf{A} \mathbf{B}^{-1/2})^{-1} = \mathbf{B}^{1/2} \mathbf{A}^{-1} \mathbf{B}^{1/2}$ . This proves the theorem. ■

Clause (d) is the memorable one: **the order reverses under inversion**, exactly as  $a \geq b > 0$  gives  $1/b \geq 1/a$  for numbers. Clause (b) is the beginning of a longer story. The full statement is that  $\lambda_i(\mathbf{A}) \geq \lambda_i(\mathbf{B})$  for **every**  $i$ , not only for  $i = 1$  and  $i = n$ ; but the intermediate eigenvalues have no description as a plain maximum over the whole unit sphere, and the min-max description that does the job is the Courant–Fischer theorem of Chapter 16. Nothing in this chapter uses the intermediate cases, so we prove the two extreme ones and leave the rest there.

*Remark.* Clause (c) does not extend to  $\mathbf{A} \succeq \mathbf{B}$  without  $\mathbf{B} \succeq 0$ . Take  $\mathbf{A} = \text{diag}(1, -1)$  and  $\mathbf{B} = \text{diag}(0, -2)$ : then  $\mathbf{A} - \mathbf{B} = \text{diag}(1, 1) \succeq 0$ , so  $\mathbf{A} \succeq \mathbf{B}$ , yet  $\det \mathbf{A} = -1 < 0 = \det \mathbf{B}$ . The hypothesis  $\mathbf{B} \succeq 0$  is what lets “bigger” be read off the eigenvalues one at a time, and so multiplied up into a determinant.

### 12.5.3 What fails: squaring

For positive numbers,  $a \geq b \geq 0$  implies  $a^2 \geq b^2$ . The matrix version is false, and the smallest counterexample has integer entries.

**Example. (Squaring does not preserve the Loewner order)**

Let

$$\mathbf{A} = \begin{pmatrix} 2 & 1 \\ 1 & 1 \end{pmatrix}, \quad \mathbf{B} = \begin{pmatrix} 1 & 1 \\ 1 & 1 \end{pmatrix}.$$

Show that  $\mathbf{A} \succeq \mathbf{B} \succeq 0$  and that  $\mathbf{A}^2 \succeq \mathbf{B}^2$  is false.

*Solution. The hypotheses.* Both matrices are real symmetric.  $\mathbf{B} \succeq 0$ : its leading principal minors are 1 and 0, and directly  $\mathbf{x}^\top \mathbf{B} \mathbf{x} = (x_1 + x_2)^2 \geq 0$ . Its eigenvalues are 0 and 2. Next,

$$\mathbf{A} - \mathbf{B} = \begin{pmatrix} 1 & 0 \\ 0 & 0 \end{pmatrix} \succeq 0,$$

so  $\mathbf{A} \succeq \mathbf{B}$ . (In fact  $\mathbf{A} \succ 0$ :  $\det \mathbf{A} = 1 > 0$  and  $a_{11} = 2 > 0$ , so Sylvester's criterion applies, [Theorem 12.1.5](#).)

*The squares.*

$$\mathbf{A}^2 = \begin{pmatrix} 5 & 3 \\ 3 & 2 \end{pmatrix}, \quad \mathbf{B}^2 = \begin{pmatrix} 2 & 2 \\ 2 & 2 \end{pmatrix}, \quad \mathbf{A}^2 - \mathbf{B}^2 = \begin{pmatrix} 3 & 1 \\ 1 & 0 \end{pmatrix}.$$

Its determinant is  $0 - 1 = -1 < 0$ , so its two eigenvalues have opposite signs and it is not positive semidefinite ([Theorem 12.1.4](#)). A witness is available without computing eigenvalues: with  $\mathbf{x} = (1, -2)$ ,

$$\mathbf{x}^\top (\mathbf{A}^2 - \mathbf{B}^2) \mathbf{x} = 3 - 4 + 0 = -1 < 0.$$

So  $\mathbf{A}^2 \not\succeq \mathbf{B}^2$ .

Where does the argument for numbers break? It uses  $a^2 - b^2 = (a - b)(a + b)$ , a product of two non-negative numbers. For matrices,  $(\mathbf{A} - \mathbf{B})(\mathbf{A} + \mathbf{B}) = \mathbf{A}^2 + \mathbf{AB} - \mathbf{BA} - \mathbf{B}^2$ , which is  $\mathbf{A}^2 - \mathbf{B}^2$  only when  $\mathbf{A}$  and  $\mathbf{B}$  commute — and in the example  $\mathbf{AB} = \begin{pmatrix} 3 & 3 \\ 2 & 2 \end{pmatrix}$  while  $\mathbf{BA} = \begin{pmatrix} 3 & 2 \\ 3 & 2 \end{pmatrix}$ . Non-commutativity is the whole obstruction, and when it is removed the number argument goes through; that is exercise C1.

What *is* true is the same statement for the square **root**, and it is true with no commuting hypothesis at all.

### Proposition 12.5.4 — The square root is order preserving

Let  $\mathbf{A}, \mathbf{B} \in M_n(F)$  with  $\mathbf{A} \succeq \mathbf{B} \succeq 0$ . Then  $\mathbf{A}^{1/2} \succeq \mathbf{B}^{1/2}$ .

*Idea.* Suppose not. Then  $\mathbf{X} - \mathbf{Y}$ , with  $\mathbf{X} = \mathbf{A}^{1/2}$  and  $\mathbf{Y} = \mathbf{B}^{1/2}$ , has a negative eigenvalue  $\mu$ , with unit eigenvector  $\mathbf{u}$ . Two different consequences of  $\mathbf{X}\mathbf{u} = \mathbf{Y}\mathbf{u} + \mu\mathbf{u}$  then collide. Pairing with  $\mathbf{u}$  says  $\langle \mathbf{Y}\mathbf{u}, \mathbf{u} \rangle \geq |\mu|$ , because  $\langle \mathbf{X}\mathbf{u}, \mathbf{u} \rangle \geq 0$ . Taking norms and using  $\|\mathbf{X}\mathbf{u}\|^2 \geq \|\mathbf{Y}\mathbf{u}\|^2$ , which is exactly the hypothesis  $\mathbf{A} \succeq \mathbf{B}$ , says  $|\mu| \geq 2\langle \mathbf{Y}\mathbf{u}, \mathbf{u} \rangle$ . Together:  $|\mu| \geq 2|\mu|$ .

*Proof.* Write  $\mathbf{X} = \mathbf{A}^{1/2}$  and  $\mathbf{Y} = \mathbf{B}^{1/2}$ , both Hermitian and  $\succeq 0$  with  $\mathbf{X}^2 = \mathbf{A}$  and  $\mathbf{Y}^2 = \mathbf{B}$  ([Theorem 12.2.1](#)). Then  $\mathbf{X} - \mathbf{Y}$  is Hermitian. Suppose, for a contradiction, that  $\mathbf{X} - \mathbf{Y} \not\succeq 0$ . By [Theorem 12.1.4](#) it has a negative eigenvalue  $\mu < 0$ ; choose a unit eigenvector  $\mathbf{u}$ , so that

$$\mathbf{X}\mathbf{u} = \mathbf{Y}\mathbf{u} + \mu\mathbf{u}.$$

Pairing this with  $\mathbf{u}$  and using  $\|\mathbf{u}\| = 1$ ,

$$\langle \mathbf{X}\mathbf{u}, \mathbf{u} \rangle = \langle \mathbf{Y}\mathbf{u}, \mathbf{u} \rangle + \mu.$$

All three numbers are real ([Proposition 10.6.7](#)), and  $\langle \mathbf{X}\mathbf{u}, \mathbf{u} \rangle \geq 0$  because  $\mathbf{X} \succeq 0$ . Therefore

$$\langle \mathbf{Y}\mathbf{u}, \mathbf{u} \rangle \geq -\mu = |\mu|. \quad (*)$$

Now take squared norms. Since  $\mathbf{X}$  is Hermitian,  $\|\mathbf{X}\mathbf{u}\|^2 = \langle \mathbf{X}^2\mathbf{u}, \mathbf{u} \rangle = \langle \mathbf{A}\mathbf{u}, \mathbf{u} \rangle$ , and likewise  $\|\mathbf{Y}\mathbf{u}\|^2 = \langle \mathbf{B}\mathbf{u}, \mathbf{u} \rangle$ ; so  $\mathbf{A} \succeq \mathbf{B}$  gives  $\|\mathbf{X}\mathbf{u}\|^2 \geq \|\mathbf{Y}\mathbf{u}\|^2$ . On the other hand,



expanding  $\mathbf{X}\mathbf{u} = \mathbf{Y}\mathbf{u} + \mu\mathbf{u}$ ,

$$\|\mathbf{X}\mathbf{u}\|^2 = \|\mathbf{Y}\mathbf{u}\|^2 + 2\mu\langle\mathbf{Y}\mathbf{u}, \mathbf{u}\rangle + \mu^2,$$

where the cross term is  $2\mu\langle\mathbf{Y}\mathbf{u}, \mathbf{u}\rangle$  because  $\mu$  and  $\langle\mathbf{Y}\mathbf{u}, \mathbf{u}\rangle$  are real. Comparing the two,

$$2\mu\langle\mathbf{Y}\mathbf{u}, \mathbf{u}\rangle + \mu^2 \geq 0.$$

Substituting  $\mu = -|\mu|$  turns this into  $|\mu|^2 \geq 2|\mu|\langle\mathbf{Y}\mathbf{u}, \mathbf{u}\rangle$ , and dividing by  $|\mu| > 0$  gives

$$|\mu| \geq 2\langle\mathbf{Y}\mathbf{u}, \mathbf{u}\rangle. \quad (**)$$

Combining (\*) and (\*\*),  $|\mu| \geq 2|\mu|$ , so  $|\mu| \leq 0$  and  $\mu = 0$ , contradicting  $\mu < 0$ . Hence  $\mathbf{X} - \mathbf{Y} \succeq 0$ , which is the assertion. ■

So among the powers  $t \mapsto t^s$ , the exponent  $s = 1/2$  preserves the Loewner order while  $s = 2$  does not. The dividing line is at  $s = 1$ : the functions preserving the order are called **operator monotone**, and  $t \mapsto t^s$  is operator monotone on  $[0, \infty)$  exactly for  $0 \leq s \leq 1$ . Chapter 20 proves this, along with the companion fact that  $t \mapsto \log t$  is operator monotone while  $t \mapsto e^t$  is not.

## 12.5.4 Exercises

### 12.5.4.1 A. Check your understanding

#### Exercise A1.

- Define  $\mathbf{A} \succeq \mathbf{B}$ , and say on which set of matrices the relation is defined.
- Determine whether the following statement is correct, and justify your answer: for Hermitian  $\mathbf{A}, \mathbf{B}$ , either  $\mathbf{A} \succeq \mathbf{B}$  or  $\mathbf{B} \succeq \mathbf{A}$ .
- Determine whether the following statement is correct, and justify your answer:  $\mathbf{A} \succeq \mathbf{B}$  implies  $a_{ij} \geq b_{ij}$  for all  $i, j$ .
- State what fails when both sides of  $\mathbf{A} \succeq \mathbf{B} \succeq 0$  are squared, and what happens instead when both sides have their square roots taken.
- Which clause of [Theorem 12.5.3](#) reverses an inequality, and what is the analogous fact about real numbers?

*Solution.*

- For Hermitian  $\mathbf{A}, \mathbf{B} \in M_n(F)$ ,  $\mathbf{A} \succeq \mathbf{B}$  means  $\mathbf{A} - \mathbf{B} \succeq 0$ , equivalently  $\langle \mathbf{A}\mathbf{x}, \mathbf{x} \rangle \geq \langle \mathbf{B}\mathbf{x}, \mathbf{x} \rangle$  for every  $\mathbf{x} \in F^n$  ([Definition 12.5.1](#)). It is defined on the Hermitian matrices of a fixed size.
- Incorrect: the order is partial.  $\text{diag}(1, 0)$  and  $\text{diag}(0, 1)$  are incomparable, since  $\text{diag}(1, -1)$  and  $\text{diag}(-1, 1)$  are both indefinite.
- Incorrect.  $\text{diag}(2, 2) \succeq \begin{pmatrix} 1 & 1 \\ 1 & 1 \end{pmatrix}$ , because the difference  $\begin{pmatrix} 1 & -1 \\ -1 & 1 \end{pmatrix}$  has eigenvalues 0 and 2; yet the  $(1, 2)$ -entries are 0 and 1.
- Squaring fails:  $\mathbf{A} \succeq \mathbf{B} \succeq 0$  does not give  $\mathbf{A}^2 \succeq \mathbf{B}^2$  ([Example \(Squaring does not preserve the Loewner order\)](#)). Taking square roots is legitimate:  $\mathbf{A}^{1/2} \succeq \mathbf{B}^{1/2}$

([Proposition 12.5.4](#)).

- (e) Clause (d):  $\mathbf{A} \succeq \mathbf{B} \succ 0$  gives  $\mathbf{B}^{-1} \succeq \mathbf{A}^{-1}$ . For real numbers,  $a \geq b > 0$  gives  $1/b \geq 1/a$ .

#### 12.5.4.2 B. Practice

##### Exercise B1: Decide the relation.

For each pair of real symmetric matrices, determine whether  $\mathbf{A} \succeq \mathbf{B}$ , whether  $\mathbf{B} \succeq \mathbf{A}$ , or neither. Justify your answer.

- (i)  $\mathbf{A} = \begin{pmatrix} 3 & 1 \\ 1 & 3 \end{pmatrix}$ ,  $\mathbf{B} = \mathbf{I}_2$ .
- (ii)  $\mathbf{A} = \begin{pmatrix} 2 & 0 \\ 0 & 1 \end{pmatrix}$ ,  $\mathbf{B} = \begin{pmatrix} 1 & 1 \\ 1 & 1 \end{pmatrix}$ .
- (iii)  $\mathbf{A} = \begin{pmatrix} 1 & 0 \\ 0 & 3 \end{pmatrix}$ ,  $\mathbf{B} = \begin{pmatrix} 2 & 0 \\ 0 & 1 \end{pmatrix}$ .

*Solution.*

- (i)  $\mathbf{A} - \mathbf{B} = \begin{pmatrix} 2 & 1 \\ 1 & 2 \end{pmatrix}$ , with leading principal minors 2 and 3, both  $> 0$ , so  $\mathbf{A} - \mathbf{B} \succ 0$  by [Theorem 12.1.5](#) and  $\mathbf{A} \succ \mathbf{B}$ . Then  $\mathbf{B} \succeq \mathbf{A}$  would force  $\mathbf{A} = \mathbf{B}$  by [Proposition 12.5.1](#), which is false, so  $\mathbf{B} \not\succeq \mathbf{A}$ .
- (ii)  $\mathbf{A} - \mathbf{B} = \begin{pmatrix} 1 & -1 \\ -1 & 0 \end{pmatrix}$ , with determinant  $-1 < 0$ , so it is indefinite and  $\mathbf{A} \not\succeq \mathbf{B}$ . Also  $\mathbf{B} - \mathbf{A}$  has determinant  $-1$ , so  $\mathbf{B} \not\succeq \mathbf{A}$ . Neither.
- (iii)  $\mathbf{A} - \mathbf{B} = \text{diag}(-1, 2)$ , indefinite;  $\mathbf{B} - \mathbf{A} = \text{diag}(1, -2)$ , indefinite. Neither. (For diagonal matrices the comparison is entrywise, and the two entries disagree.)

##### Exercise B2: The order reverses.

Let  $\mathbf{A} = \begin{pmatrix} 3 & 1 \\ 1 & 3 \end{pmatrix}$  and  $\mathbf{B} = \begin{pmatrix} 2 & 1 \\ 1 & 2 \end{pmatrix}$ . Verify that  $\mathbf{A} \succeq \mathbf{B} \succ 0$ , compute  $\mathbf{A}^{-1}$  and  $\mathbf{B}^{-1}$ , and check [Theorem 12.5.3](#) (d) directly. Check the determinant clause as well.

*Solution.*  $\mathbf{A} - \mathbf{B} = \mathbf{I}_2 \succ 0$ , so  $\mathbf{A} \succ \mathbf{B}$ . And  $\mathbf{B} \succ 0$ , its leading principal minors being 2 and 3. Then  $\det \mathbf{A} = 8$ ,  $\det \mathbf{B} = 3$ , so

$$\mathbf{A}^{-1} = \frac{1}{8} \begin{pmatrix} 3 & -1 \\ -1 & 3 \end{pmatrix}, \quad \mathbf{B}^{-1} = \frac{1}{3} \begin{pmatrix} 2 & -1 \\ -1 & 2 \end{pmatrix},$$

and

$$\mathbf{B}^{-1} - \mathbf{A}^{-1} = \frac{1}{24} \begin{pmatrix} 16 - 9 & -8 + 3 \\ -8 + 3 & 16 - 9 \end{pmatrix} = \frac{1}{24} \begin{pmatrix} 7 & -5 \\ -5 & 7 \end{pmatrix}.$$

Its leading principal minors are  $7/24 > 0$  and  $(49 - 25)/576 = 1/24 > 0$ , so  $\mathbf{B}^{-1} \succ \mathbf{A}^{-1}$ , as [Theorem 12.5.3](#) (d) predicts. For the determinant clause,  $\det \mathbf{A} = 8 \geq 3 = \det \mathbf{B}$ ; and for the trace,  $6 \geq 4$ .

**Exercise B3: Another failure of squaring.**

Let  $\mathbf{A} = \begin{pmatrix} 3 & 1 \\ 1 & 1 \end{pmatrix}$  and  $\mathbf{B} = \begin{pmatrix} 1 & 1 \\ 1 & 1 \end{pmatrix}$ . Verify that  $\mathbf{A} \succeq \mathbf{B} \succeq 0$ , and exhibit a vector  $\mathbf{x}$  with  $\mathbf{x}^\top (\mathbf{A}^2 - \mathbf{B}^2) \mathbf{x} < 0$ .

*Solution.*  $\mathbf{A} - \mathbf{B} = \text{diag}(2, 0) \succeq 0$ , so  $\mathbf{A} \succeq \mathbf{B}$ ; and  $\mathbf{x}^\top \mathbf{B} \mathbf{x} = (x_1 + x_2)^2 \geq 0$ , so  $\mathbf{B} \succeq 0$ . Squaring,

$$\mathbf{A}^2 = \begin{pmatrix} 10 & 4 \\ 4 & 2 \end{pmatrix}, \quad \mathbf{B}^2 = \begin{pmatrix} 2 & 2 \\ 2 & 2 \end{pmatrix}, \quad \mathbf{A}^2 - \mathbf{B}^2 = \begin{pmatrix} 8 & 2 \\ 2 & 0 \end{pmatrix},$$

whose determinant is  $-4 < 0$ . With  $\mathbf{x} = (1, -3)$ ,

$$\mathbf{x}^\top (\mathbf{A}^2 - \mathbf{B}^2) \mathbf{x} = 8 - 12 + 0 = -4 < 0.$$

So  $\mathbf{A}^2 \not\succeq \mathbf{B}^2$ . (Any  $\mathbf{x} = (1, t)$  with  $t < -2$  works, since the value is  $8 + 4t$ .)

**12.5.4.3 C. Going deeper****Exercise C1: When squaring does work.**

Let  $\mathbf{A}, \mathbf{B} \in M_n(F)$  with  $\mathbf{A} \succeq \mathbf{B} \succeq 0$  and  $\mathbf{AB} = \mathbf{BA}$ .

- (a) Prove that if  $\mathbf{P}, \mathbf{Q} \succeq 0$  and  $\mathbf{PQ} = \mathbf{QP}$ , then  $\mathbf{PQ} \succeq 0$ .
- (b) Deduce that  $\mathbf{A}^2 \succeq \mathbf{B}^2$ .
- (c) Deduce that  $\mathbf{A}^k \succeq \mathbf{B}^k$  for every integer  $k \geq 1$ .

*Hint for (a):*  $\mathbf{P}^{1/2}$  is a polynomial in  $\mathbf{P}$ .

*Solution.*

- (a) By [Theorem 12.2.1](#),  $\mathbf{P}^{1/2}$  is a polynomial in  $\mathbf{P}$ , so it commutes with every matrix that commutes with  $\mathbf{P}$ ; in particular  $\mathbf{P}^{1/2} \mathbf{Q} = \mathbf{QP}^{1/2}$ . Hence

$$\mathbf{PQ} = \mathbf{P}^{1/2} \mathbf{P}^{1/2} \mathbf{Q} = \mathbf{P}^{1/2} \mathbf{QP}^{1/2} = (\mathbf{P}^{1/2})^* \mathbf{QP}^{1/2},$$

which is  $\succeq 0$  by [Proposition 12.1.2](#) (a), since  $\mathbf{Q} \succeq 0$  and  $\mathbf{P}^{1/2}$  is Hermitian.

- (b) Put  $\mathbf{P} = \mathbf{A} - \mathbf{B}$  and  $\mathbf{Q} = \mathbf{A} + \mathbf{B}$ . Both are  $\succeq 0$ : the first by hypothesis, the second as a sum of two matrices  $\succeq 0$ , using  $\mathbf{A} \succeq \mathbf{B} \succeq 0$  and [Proposition 12.5.1](#). They commute, because  $\mathbf{AB} = \mathbf{BA}$  makes  $\mathbf{PQ} = \mathbf{A}^2 + \mathbf{AB} - \mathbf{BA} - \mathbf{B}^2 = \mathbf{A}^2 - \mathbf{B}^2$  and  $\mathbf{QP} = \mathbf{A}^2 - \mathbf{AB} + \mathbf{BA} - \mathbf{B}^2 = \mathbf{A}^2 - \mathbf{B}^2$  as well. By (a),  $\mathbf{A}^2 - \mathbf{B}^2 = \mathbf{PQ} \succeq 0$ .
- (c) Induction on  $k$ . The case  $k = 1$  is the hypothesis. Assume  $\mathbf{A}^{k-1} \succeq \mathbf{B}^{k-1}$ . Then

$$\mathbf{A}^k - \mathbf{B}^k = \mathbf{A}^{k-1}(\mathbf{A} - \mathbf{B}) + (\mathbf{A}^{k-1} - \mathbf{B}^{k-1})\mathbf{B},$$

and each term is  $\succeq 0$  by (a). Indeed  $\mathbf{A} \succeq 0$  is Hermitian with eigenvalues  $\geq 0$ , so  $\mathbf{A}^{k-1}$  is Hermitian with eigenvalues  $\geq 0$  and hence  $\mathbf{A}^{k-1} \succeq 0$  ([Theorem 12.1.4](#)); it commutes with  $\mathbf{A} - \mathbf{B}$ , since  $\mathbf{AB} = \mathbf{BA}$ . Likewise  $\mathbf{A}^{k-1} - \mathbf{B}^{k-1} \succeq 0$  by the

induction hypothesis and it commutes with  $\mathbf{B} \succeq 0$ . Finally a sum of two matrices  $\succeq 0$  is  $\succeq 0$ , directly from [Definition 12.1.1](#).

### Exercise C2: Kernels and ranks.

Let  $\mathbf{A}, \mathbf{B} \in M_n(F)$  with  $\mathbf{A} \succeq \mathbf{B} \succeq 0$ .

- (a) Prove that  $\text{nul}(\mathbf{A}) \subseteq \text{nul}(\mathbf{B})$ .
- (b) Deduce that  $\text{rank } \mathbf{A} \geq \text{rank } \mathbf{B}$ .
- (c) Give an example with  $\mathbf{A} \succeq \mathbf{B} \succeq 0$  and  $\text{rank } \mathbf{A} = \text{rank } \mathbf{B}$  but  $\mathbf{A} \neq \mathbf{B}$ .

*Hint for (a):* pair  $\mathbf{A}\mathbf{x} = \mathbf{0}$  with  $\mathbf{x}$ , then use  $\mathbf{B}^{1/2}$ .

*Solution.*

- (a) Let  $\mathbf{x} \in \text{nul}(\mathbf{A})$ . Then  $\langle \mathbf{A}\mathbf{x}, \mathbf{x} \rangle = 0$ , and  $0 \leq \langle \mathbf{B}\mathbf{x}, \mathbf{x} \rangle \leq \langle \mathbf{A}\mathbf{x}, \mathbf{x} \rangle = 0$ , so  $\langle \mathbf{B}\mathbf{x}, \mathbf{x} \rangle = 0$ . Writing  $\mathbf{B} = \mathbf{B}^{1/2}\mathbf{B}^{1/2}$  with  $\mathbf{B}^{1/2}$  Hermitian ([Theorem 12.2.1](#)),

$$0 = \langle \mathbf{B}\mathbf{x}, \mathbf{x} \rangle = \langle \mathbf{B}^{1/2}\mathbf{x}, \mathbf{B}^{1/2}\mathbf{x} \rangle = \|\mathbf{B}^{1/2}\mathbf{x}\|^2,$$

so  $\mathbf{B}^{1/2}\mathbf{x} = \mathbf{0}$  and hence  $\mathbf{B}\mathbf{x} = \mathbf{B}^{1/2}(\mathbf{B}^{1/2}\mathbf{x}) = \mathbf{0}$ .

- (b) By (a),  $\text{nullity } \mathbf{A} \leq \text{nullity } \mathbf{B}$ , and by Rank–Nullity ([Theorem 3.3.1](#)) applied to both matrices on  $F^n$ ,  $\text{rank } \mathbf{A} = n - \text{nullity } \mathbf{A} \geq n - \text{nullity } \mathbf{B} = \text{rank } \mathbf{B}$ .
- (c)  $\mathbf{A} = \text{diag}(2, 2)$  and  $\mathbf{B} = \text{diag}(1, 1)$ : both have rank 2,  $\mathbf{A} - \mathbf{B} = \mathbf{I}_2 \succeq 0$ , and  $\mathbf{A} \neq \mathbf{B}$ .

### Exercise C3: Largest eigenvalue of a sum.

Let  $\mathbf{A}, \mathbf{B} \in M_n(F)$  be Hermitian.

- (a) Prove that  $\lambda_1(\mathbf{A})\mathbf{I} \succeq \mathbf{A}$ , and deduce  $\lambda_1(\mathbf{A} + \mathbf{B}) \leq \lambda_1(\mathbf{A}) + \lambda_1(\mathbf{B})$ .
- (b) Deduce that if  $\mathbf{A} \succeq \mathbf{B}$ , then  $0 \leq \lambda_1(\mathbf{A}) - \lambda_1(\mathbf{B}) \leq \lambda_1(\mathbf{A} - \mathbf{B})$ .

*Solution.*

- (a) The matrix  $\lambda_1(\mathbf{A})\mathbf{I} - \mathbf{A}$  is Hermitian with eigenvalues  $\lambda_1(\mathbf{A}) - \lambda_i(\mathbf{A}) \geq 0$ , so it is  $\succeq 0$  by [Theorem 12.1.4](#); that is  $\lambda_1(\mathbf{A})\mathbf{I} \succeq \mathbf{A}$ . Adding the same inequality for  $\mathbf{B}$ , and using that a sum of two matrices  $\succeq 0$  is again  $\succeq 0$  ([Exercise \(C2: The cone of positive matrices\)](#) (a)),

$$(\lambda_1(\mathbf{A}) + \lambda_1(\mathbf{B}))\mathbf{I} \succeq \mathbf{A} + \mathbf{B}.$$

By [Theorem 12.5.3](#) (b) applied to this pair,  $\lambda_1(\mathbf{A}) + \lambda_1(\mathbf{B}) = \lambda_1((\lambda_1(\mathbf{A}) + \lambda_1(\mathbf{B}))\mathbf{I}) \geq \lambda_1(\mathbf{A} + \mathbf{B})$ .

- (b) Write  $\mathbf{A} = \mathbf{B} + (\mathbf{A} - \mathbf{B})$ . Applying (a) to the pair  $\mathbf{B}$  and  $\mathbf{A} - \mathbf{B}$ ,

$$\lambda_1(\mathbf{A}) \leq \lambda_1(\mathbf{B}) + \lambda_1(\mathbf{A} - \mathbf{B}),$$

so  $\lambda_1(\mathbf{A}) - \lambda_1(\mathbf{B}) \leq \lambda_1(\mathbf{A} - \mathbf{B})$ . For the lower bound,  $\mathbf{A} \succeq \mathbf{B}$  gives  $\lambda_1(\mathbf{A}) \geq \lambda_1(\mathbf{B})$

by [Theorem 12.5.3](#) (b). Note that (a) itself used no order hypothesis at all; only the lower bound needs  $\mathbf{A} \succeq \mathbf{B}$ .

## 12.6 Positive Matrices in Blocks

Chapter 7 cut a matrix into blocks, ran one step of elimination, and gave a name to what was left in the far corner: the Schur complement. Section 1 of this chapter met that same quantity as a single number, in the induction that proved Sylvester's criterion. This section puts the two together. For a Hermitian matrix split into blocks, positive definiteness can be decided one block at a time, and the deciding quantity is the Schur complement; two determinant inequalities, Fischer's and Hadamard's, then come almost for free.

Throughout,  $F = \mathbb{R}$  or  $F = \mathbb{C}$ , and  $F^n$  carries the standard inner product.

### 12.6.1 What the blocks of a positive matrix must satisfy

A Hermitian matrix cut into blocks of sizes  $k$  and  $l$  has a shape forced on it. Suppose

$$\mathbf{M} = \begin{pmatrix} \mathbf{A} & \mathbf{B} \\ \mathbf{D} & \mathbf{C} \end{pmatrix} \in M_{k+l}(F),$$

with  $\mathbf{A} \in M_k(F)$ ,  $\mathbf{B} \in M_{k \times l}(F)$ ,  $\mathbf{D} \in M_{l \times k}(F)$  and  $\mathbf{C} \in M_l(F)$ . Starring a block matrix transposes the arrangement and stars each block, so  $\mathbf{M}^* = \mathbf{M}$  says exactly three things:  $\mathbf{A}^* = \mathbf{A}$ ,  $\mathbf{C}^* = \mathbf{C}$ , and  $\mathbf{D} = \mathbf{B}^*$ . A Hermitian  $\mathbf{M}$  is therefore always of the form

$$\mathbf{M} = \begin{pmatrix} \mathbf{A} & \mathbf{B} \\ \mathbf{B}^* & \mathbf{C} \end{pmatrix}$$

with  $\mathbf{A}$  and  $\mathbf{C}$  Hermitian and  $\mathbf{B}$  unrestricted, and we write it this way from now on.

Two necessary conditions for positivity are immediate. The vectors of  $F^{k+l}$  whose last  $l$  coordinates vanish form a copy of  $F^k$ , and on them the quadratic form of  $\mathbf{M}$  is the quadratic form of  $\mathbf{A}$ :

$$\begin{pmatrix} \mathbf{x} \\ \mathbf{0} \end{pmatrix}^* \mathbf{M} \begin{pmatrix} \mathbf{x} \\ \mathbf{0} \end{pmatrix} = \mathbf{x}^* \mathbf{A} \mathbf{x}, \quad \begin{pmatrix} \mathbf{0} \\ \mathbf{y} \end{pmatrix}^* \mathbf{M} \begin{pmatrix} \mathbf{0} \\ \mathbf{y} \end{pmatrix} = \mathbf{y}^* \mathbf{C} \mathbf{y}.$$

So  $\mathbf{M} \succeq 0$  forces  $\mathbf{A} \succeq 0$  and  $\mathbf{C} \succeq 0$ . This is the statement that a principal submatrix of a positive semidefinite matrix is positive semidefinite, used already in the proof of [Theorem 12.1.4](#), here for the two index blocks  $\{1, \dots, k\}$  and  $\{k+1, \dots, k+l\}$ .

The two conditions are not sufficient, and the smallest counterexample has  $k = l = 1$ :

$$\mathbf{M} = \begin{pmatrix} 0 & 1 \\ 1 & 1 \end{pmatrix}, \quad \mathbf{A} = (0) \succeq 0, \quad \mathbf{C} = (1) \succ 0.$$

Here  $\det \mathbf{M} = -1 < 0$ , so  $\mathbf{M} \not\geq 0$  by [Theorem 12.1.4](#) (e). Something must couple  $\mathbf{B}$  to  $\mathbf{A}$  and  $\mathbf{C}$ , and the  $2 \times 2$  case already says what. For  $\mathbf{M} = \begin{pmatrix} a & b \\ \bar{b} & c \end{pmatrix}$  Hermitian, [Example \(The  \$2 \times 2\$  Test\)](#) gives  $\mathbf{M} \succ 0$  if and only if  $a > 0$  and  $ac - |b|^2 > 0$ . Divide the second condition by the positive number  $a$ :

$$c - \bar{b}a^{-1}b > 0.$$

Written with  $\mathbf{B} = (b)$ , so that  $\mathbf{B}^* = (\bar{b})$ , the left-hand side is the Schur complement  $\mathbf{M}/\mathbf{A}$  of [Definition 7.3.1](#). The general statement now writes itself: replace the scalars by blocks.

## 12.6.2 Testing one block at a time

### Theorem 12.6.1 — Positive Definiteness Block by Block

Let  $k, l \geq 1$  and let

$$\mathbf{M} = \begin{pmatrix} \mathbf{A} & \mathbf{B} \\ \mathbf{B}^* & \mathbf{C} \end{pmatrix} \in M_{k+l}(F)$$

be Hermitian, with  $\mathbf{A} \in M_k(F)$ ,  $\mathbf{B} \in M_{k \times l}(F)$  and  $\mathbf{C} \in M_l(F)$ .

- (a) If  $\mathbf{M} \geq 0$ , then  $\mathbf{A} \geq 0$  and  $\mathbf{C} \geq 0$ . If  $\mathbf{M} \succ 0$ , then  $\mathbf{A} \succ 0$  and  $\mathbf{C} \succ 0$ .
- (b) Suppose  $\mathbf{A} \succ 0$ . Then  $\mathbf{A}$  is invertible, the Schur complement

$$\mathbf{M}/\mathbf{A} = \mathbf{C} - \mathbf{B}^* \mathbf{A}^{-1} \mathbf{B} \in M_l(F)$$

is Hermitian, and

$$\mathbf{M} \succ 0 \iff \mathbf{M}/\mathbf{A} \succ 0, \quad \mathbf{M} \geq 0 \iff \mathbf{M}/\mathbf{A} \geq 0.$$

*Idea.* Everything is one congruence. Chapter 7's block elimination clears  $\mathbf{B}$  and  $\mathbf{B}^*$  at once and leaves  $\mathbf{A}$  and  $\mathbf{M}/\mathbf{A}$  on the diagonal; for a Hermitian  $\mathbf{M}$  the two outer factors of that elimination are adjoint to each other, because the row operation and the column operation are the same operation seen twice. So the elimination is a congruence  $\mathbf{S}^* \mathbf{M} \mathbf{S}$ , and [Proposition 12.1.2](#) says that a congruence by an invertible matrix neither creates nor destroys positivity. All that remains is to read the verdict off a block **diagonal** matrix, where the quadratic form splits into two independent pieces. This is the step (\*) of Section 1's proof of Sylvester's criterion, with the last coordinate replaced by a block of  $l$  coordinates.

*Proof.*

- (a) Let  $\mathbf{x} \in F^k$  and put  $\mathbf{v} = (\mathbf{x}, \mathbf{0}) \in F^{k+l}$ . By [Theorem 7.1.1](#),  $\mathbf{M}\mathbf{v} = (\mathbf{A}\mathbf{x}, \mathbf{B}^*\mathbf{x})$ , and hence  $\mathbf{v}^* \mathbf{M} \mathbf{v} = \mathbf{x}^* \mathbf{A} \mathbf{x}$ . If  $\mathbf{M} \geq 0$ , the left side is  $\geq 0$  for every  $\mathbf{x}$ , so  $\mathbf{A} \geq 0$ ; if  $\mathbf{M} \succ 0$ , then  $\mathbf{x} \neq \mathbf{0}$  forces  $\mathbf{v} \neq \mathbf{0}$  and the left side is  $> 0$ , so  $\mathbf{A} \succ 0$ . The same argument with  $\mathbf{v} = (\mathbf{0}, \mathbf{y})$  gives the statements for  $\mathbf{C}$ .
- (b) Since  $\mathbf{A} \succ 0$ , every eigenvalue of  $\mathbf{A}$  is positive ([Theorem 12.1.5](#) (b)), so  $0 \notin \text{spec}(\mathbf{A})$  and  $\mathbf{A}$  is invertible ([Theorem 8.2.6](#)). Also  $(\mathbf{A}^{-1})^* = (\mathbf{A}^*)^{-1} = \mathbf{A}^{-1}$ , so

$$(\mathbf{M}/\mathbf{A})^* = \mathbf{C}^* - \mathbf{B}^*(\mathbf{A}^{-1})^*\mathbf{B} = \mathbf{C} - \mathbf{B}^*\mathbf{A}^{-1}\mathbf{B} = \mathbf{M}/\mathbf{A},$$

and  $\mathbf{M}/\mathbf{A}$  is Hermitian. Put

$$\mathbf{S} = \begin{pmatrix} \mathbf{I}_k & -\mathbf{A}^{-1}\mathbf{B} \\ 0 & \mathbf{I}_l \end{pmatrix}, \quad \mathbf{S}^* = \begin{pmatrix} \mathbf{I}_k & 0 \\ -\mathbf{B}^*\mathbf{A}^{-1} & \mathbf{I}_l \end{pmatrix},$$

the second because  $\mathbf{A}^{-1}$  is Hermitian. By [Theorem 7.1.5](#),  $\mathbf{S}$  is invertible. Two applications of [Theorem 7.1.1](#) give

$$\begin{aligned} \mathbf{MS} &= \begin{pmatrix} \mathbf{A} & \mathbf{B} - \mathbf{A}\mathbf{A}^{-1}\mathbf{B} \\ \mathbf{B}^* & \mathbf{C} - \mathbf{B}^*\mathbf{A}^{-1}\mathbf{B} \end{pmatrix} = \begin{pmatrix} \mathbf{A} & 0 \\ \mathbf{B}^* & \mathbf{M}/\mathbf{A} \end{pmatrix}, \\ \mathbf{S}^*(\mathbf{MS}) &= \begin{pmatrix} \mathbf{A} & 0 \\ \mathbf{B}^* - \mathbf{B}^*\mathbf{A}^{-1}\mathbf{A} & \mathbf{M}/\mathbf{A} \end{pmatrix} = \begin{pmatrix} \mathbf{A} & 0 \\ 0 & \mathbf{M}/\mathbf{A} \end{pmatrix}. \end{aligned} \tag{*}$$

Write  $\mathbf{N} = \mathbf{A} \oplus (\mathbf{M}/\mathbf{A})$  for that block diagonal matrix. For  $\mathbf{y} = (\mathbf{y}_1, \mathbf{y}_2) \in F^k \times F^l$ ,

$$\mathbf{y}^*\mathbf{N}\mathbf{y} = \mathbf{y}_1^*\mathbf{A}\mathbf{y}_1 + \mathbf{y}_2^*(\mathbf{M}/\mathbf{A})\mathbf{y}_2. \tag{\dagger}$$

*Claim.* Given  $\mathbf{A} \succ 0$ :  $\mathbf{N} \succ 0$  if and only if  $\mathbf{M}/\mathbf{A} \succ 0$ , and  $\mathbf{N} \succeq 0$  if and only if  $\mathbf{M}/\mathbf{A} \succeq 0$ .

*Proof.* Taking  $\mathbf{y}_1 = \mathbf{0}$  in  $(\dagger)$  shows that positivity of  $\mathbf{N}$ , weak or strict, passes to  $\mathbf{M}/\mathbf{A}$ , since  $\mathbf{y}_2 \neq \mathbf{0}$  forces  $\mathbf{y} \neq \mathbf{0}$ . Conversely, suppose  $\mathbf{M}/\mathbf{A} \succeq 0$ ; then both terms of  $(\dagger)$  are  $\geq 0$ , so  $\mathbf{N} \succeq 0$ . If moreover  $\mathbf{M}/\mathbf{A} \succ 0$ , then for  $\mathbf{y} \neq \mathbf{0}$  at least one of  $\mathbf{y}_1, \mathbf{y}_2$  is non-zero, so at least one term of  $(\dagger)$  is  $> 0$  while the other is  $\geq 0$ ; hence  $\mathbf{N} \succ 0$ . ■

Finally  $\mathbf{S}$  is invertible, so [Proposition 12.1.2](#) (b) gives  $\mathbf{M} \succ 0$  if and only if  $\mathbf{S}^*\mathbf{M}\mathbf{S} = \mathbf{N} \succ 0$ . For the weak version, [Proposition 12.1.2](#) (a) applied to  $\mathbf{S}$  turns  $\mathbf{M} \succeq 0$  into  $\mathbf{N} \succeq 0$ , and applied to  $\mathbf{S}^{-1}$ , together with  $(\mathbf{S}^{-1})^*\mathbf{N}\mathbf{S}^{-1} = \mathbf{M}$ , turns  $\mathbf{N} \succeq 0$  back into  $\mathbf{M} \succeq 0$ . Combining with the claim proves (b). ■

The identity  $(*)$  is [Theorem 7.3.1](#) (a) in disguise. That theorem factors  $\mathbf{M} = \mathbf{L}\mathbf{N}\mathbf{U}$  with  $\mathbf{L}$  lower and  $\mathbf{U}$  upper block triangular; here  $\mathbf{L}$  and  $\mathbf{U}$  are  $(\mathbf{S}^{-1})^*$  and  $\mathbf{S}^{-1}$ , so the two outer factors are adjoint to each other and the factorization is a **congruence**, not merely an equivalence. Taking determinants in  $(*)$  recovers [Theorem 7.3.2](#),

$$\det \mathbf{M} = \det \mathbf{A} \cdot \det(\mathbf{M}/\mathbf{A}),$$

which is the form in which we use it below.

Chapter 7 closed its treatment of Schur complements by observing that when  $\mathbf{M}$  is real sym-

metric the complement  $\mathbf{M}/\mathbf{A}$  is symmetric again, and promising that “Chapter 12 uses Schur complements to test whether  $\mathbf{M}$  is positive definite one block at a time”. [Theorem 12.6.1](#) is that test, and part (b) says exactly what the promise says: once the corner block  $\mathbf{A}$  is known to be positive definite, the whole of  $\mathbf{M}$  is positive definite precisely when the smaller matrix  $\mathbf{M}/\mathbf{A}$  is. A question about  $k + l$  coordinates has become a question about  $l$  of them.

It also finishes what Section 1 started. There, in the induction proving Sylvester’s criterion, the matrix was split with  $k = n - 1$  and  $l = 1$ ; the shear  $\mathbf{S}$  was the matrix above with the column  $\mathbf{b}$  in place of  $\mathbf{B}$ ; and the congruence came out as  $\mathbf{S}^* \mathbf{A} \mathbf{S} = \mathbf{A}' \oplus (s)$  with the single number  $s = d - \mathbf{b}^* \mathbf{A}'^{-1} \mathbf{b}$  in the corner. That section promised that Section 6 would replace the scalar by a block. The scalar has become  $\mathbf{M}/\mathbf{A}$ , and the argument is word for word the same one.

**Example. ( $\mathbf{A}$   $4 \times 4$  Tested in  $2 \times 2$  Blocks)**

Decide whether each of

$$\mathbf{M} = \left( \begin{array}{cc|cc} 2 & 1 & 1 & 0 \\ 1 & 1 & 1 & 1 \\ \hline 1 & 1 & 3 & 2 \\ 0 & 1 & 2 & 4 \end{array} \right), \quad \mathbf{M}' = \left( \begin{array}{cc|cc} 2 & 1 & 1 & 0 \\ 1 & 1 & 1 & 1 \\ \hline 1 & 1 & 2 & 0 \\ 0 & 1 & 0 & 2 \end{array} \right)$$

is positive definite, using the displayed partition. For the one that is not, exhibit a vector on which the quadratic form is negative.

*Solution.* Both matrices are real symmetric with the same corner and border,

$$\mathbf{A} = \begin{pmatrix} 2 & 1 \\ 1 & 1 \end{pmatrix}, \quad \mathbf{B} = \begin{pmatrix} 1 & 0 \\ 1 & 1 \end{pmatrix}.$$

*Step 1: the corner.* The leading principal minors of  $\mathbf{A}$  are 2 and  $2 - 1 = 1$ , both positive, so  $\mathbf{A} \succ 0$  by Sylvester’s criterion ([Theorem 12.1.5](#) (d)) and [Theorem 12.6.1](#) (b) applies. Since  $\det \mathbf{A} = 1$ , [Theorem 0.9.8](#) gives  $\mathbf{A}^{-1} = \begin{pmatrix} 1 & -1 \\ -1 & 2 \end{pmatrix}$ .

*Step 2: the coupling term.* It is the same for both matrices:

$$\mathbf{B}^\top \mathbf{A}^{-1} = \begin{pmatrix} 0 & 1 \\ -1 & 2 \end{pmatrix}, \quad \mathbf{B}^\top \mathbf{A}^{-1} \mathbf{B} = \begin{pmatrix} 1 & 1 \\ 1 & 2 \end{pmatrix}.$$

*Step 3: the complements.* For  $\mathbf{M}$ , with  $\mathbf{C} = \begin{pmatrix} 3 & 2 \\ 2 & 4 \end{pmatrix}$ ,

$$\mathbf{M}/\mathbf{A} = \begin{pmatrix} 3 & 2 \\ 2 & 4 \end{pmatrix} - \begin{pmatrix} 1 & 1 \\ 1 & 2 \end{pmatrix} = \begin{pmatrix} 2 & 1 \\ 1 & 2 \end{pmatrix},$$

whose leading principal minors are 2 and 3. So  $\mathbf{M}/\mathbf{A} \succ 0$ , hence  $\mathbf{M} \succ 0$ . As a by-product,  $\det \mathbf{M} = \det \mathbf{A} \cdot \det(\mathbf{M}/\mathbf{A}) = 1 \cdot 3 = 3$ .

For  $\mathbf{M}'$ , with  $\mathbf{C}' = \begin{pmatrix} 2 & 0 \\ 0 & 2 \end{pmatrix}$ ,



$$\mathbf{M}'/\mathbf{A} = \begin{pmatrix} 2 & 0 \\ 0 & 2 \end{pmatrix} - \begin{pmatrix} 1 & 1 \\ 1 & 2 \end{pmatrix} = \begin{pmatrix} 1 & -1 \\ -1 & 0 \end{pmatrix},$$

whose determinant is  $-1$ . So  $\mathbf{M}'/\mathbf{A}$  is not even positive semidefinite, and  $\mathbf{M}'$  is not positive definite.

*Step 4: the witness.* The congruence  $(*)$  turns a bad vector for  $\mathbf{M}'/\mathbf{A}$  into a bad vector for  $\mathbf{M}'$ . Here  $\mathbf{y}_2 = (1, 1)$  gives  $\mathbf{y}_2^\top (\mathbf{M}'/\mathbf{A}) \mathbf{y}_2 = 1 - 1 - 1 + 0 = -1$ , so  $\mathbf{y} = (0, 0, 1, 1)$  satisfies  $\mathbf{y}^\top (\mathbf{S}^\top \mathbf{M}' \mathbf{S}) \mathbf{y} = -1$ , and  $\mathbf{x} = \mathbf{S} \mathbf{y}$  satisfies  $\mathbf{x}^\top \mathbf{M}' \mathbf{x} = -1$ . Since  $-\mathbf{A}^{-1} \mathbf{B} = \begin{pmatrix} 0 & 1 \\ -1 & 2 \end{pmatrix}$ , we get  $\mathbf{x} = (1, -3, 1, 1)$ . Checking directly,  $\mathbf{M}' \mathbf{x} = (0, 0, 0, -1)$  and  $\mathbf{x}^\top (\mathbf{M}' \mathbf{x}) = 0 + 0 + 0 - 1 = -1$ .

One  $2 \times 2$  inverse and one  $2 \times 2$  product replaced four leading principal minors of a  $4 \times 4$  matrix. And when the answer is “no”, the congruence hands back an explicit vector, which a determinant test never does.

*Quick check.* Let  $\mathbf{A} \succ 0$ . What is  $\mathbf{M}/\mathbf{A}$  when  $\mathbf{B} = 0$ , and what does [Theorem 12.6.1](#) (b) say in that case?

*Solution.* When  $\mathbf{B} = 0$  the Schur complement is  $\mathbf{M}/\mathbf{A} = \mathbf{C} - 0 = \mathbf{C}$ , so the criterion reads:  $\mathbf{M} \succ 0$  if and only if  $\mathbf{C} \succ 0$ . That is the right answer, since  $\mathbf{M} = \mathbf{A} \oplus \mathbf{C}$  is then block diagonal and  $(\dagger)$  says its quadratic form is the sum of the two separate quadratic forms. The general case is this case after the change of variable  $\mathbf{S}$ .

*Warning.* **The corner must be definite, not merely semidefinite.** If  $\mathbf{A}$  is singular, then  $\mathbf{A}^{-1}$  does not exist,  $\mathbf{M}/\mathbf{A}$  is undefined, and [Theorem 12.6.1](#) (b) says nothing whatsoever. This is a real obstruction, not a gap in the proof. The matrix  $\begin{pmatrix} 0 & 1 \\ 1 & 1 \end{pmatrix}$  above has  $\mathbf{A} \succeq 0$  and  $\mathbf{C} \succ 0$  and is **not** positive semidefinite, while  $\begin{pmatrix} 0 & 0 \\ 0 & 1 \end{pmatrix}$ , with the same  $\mathbf{A}$  and the same  $\mathbf{C}$ , **is**. The two differ only in  $\mathbf{B}$ , so no criterion that ignores  $\mathbf{B}$  can separate them.

*Remark.* There is a criterion for the semidefinite case with a singular corner, and it has three clauses instead of one: for Hermitian  $\mathbf{M}$ , one has  $\mathbf{M} \succeq 0$  exactly when  $\mathbf{A} \succeq 0$ , every column of  $\mathbf{B}$  lies in the column space of  $\mathbf{A}$ , and  $\mathbf{C} - \mathbf{B}^* \mathbf{A}^+ \mathbf{B} \succeq 0$ . The middle clause is what the warning’s first matrix violates. The matrix  $\mathbf{A}^+$  in the third clause is the Moore–Penrose pseudoinverse; it agrees with  $\mathbf{A}^{-1}$  when  $\mathbf{A}$  is invertible, and it is built from the singular values of  $\mathbf{A}$  in Section 11 of this chapter. The statement and its proof belong there, with the tools they need. This section keeps the definite case, which is the one the rest of the chapter consumes.

### 12.6.3 Fischer’s inequality

The congruence  $(*)$  computes  $\det \mathbf{M}$  as  $\det \mathbf{A} \cdot \det(\mathbf{M}/\mathbf{A})$ . If the Schur complement could be compared with  $\mathbf{C}$ , that would bound  $\det \mathbf{M}$  by  $\det \mathbf{A} \det \mathbf{C}$  — and it can be, because  $\mathbf{C}$  is the Schur complement plus something positive:

$$\mathbf{C} = (\mathbf{M}/\mathbf{A}) + \mathbf{B}^* \mathbf{A}^{-1} \mathbf{B}.$$

So what is needed is that adding a positive semidefinite matrix never decreases a determinant.

That is the next lemma, and it is where the square root of Section 2 earns its keep.

**Lemma 12.6.2 — Adding a positive matrix does not decrease the determinant**

Let  $\mathbf{P}, \mathbf{Q} \in M_n(F)$  with  $\mathbf{P} \succ 0$  and  $\mathbf{Q} \succeq 0$ . Then

$$\det(\mathbf{P} + \mathbf{Q}) \geq \det \mathbf{P} > 0,$$

with equality if and only if  $\mathbf{Q} = 0$ .

*Idea.* Whiten: divide the problem by  $\mathbf{P}$  on both sides, using the inverse of  $\mathbf{P}^{1/2}$ . The sum  $\mathbf{P} + \mathbf{Q}$  becomes  $\mathbf{I} + \mathbf{K}$  with  $\mathbf{K} \succeq 0$ , and the determinant of  $\mathbf{I} + \mathbf{K}$  is  $\prod_i (1 + \mu_i)$  over the eigenvalues  $\mu_i \geq 0$  of  $\mathbf{K}$ . A product of numbers  $\geq 1$  is  $\geq 1$ , and it equals 1 only when every factor does.

*Proof.* By Theorem 12.2.1 the matrix  $\mathbf{P}^{1/2} \succeq 0$  exists, and it is invertible because  $\mathbf{P}$  is (Theorem 12.2.1 (c)). Write  $\mathbf{R} = (\mathbf{P}^{1/2})^{-1}$ , which is Hermitian, being the inverse of a Hermitian matrix. Since  $\mathbf{P}^{1/2}\mathbf{R} = \mathbf{I}$  and  $(\mathbf{P}^{1/2})^2 = \mathbf{P}$ ,

$$\mathbf{P}^{1/2}(\mathbf{I} + \mathbf{RQR})\mathbf{P}^{1/2} = \mathbf{P} + \mathbf{Q}.$$

Put  $\mathbf{K} = \mathbf{R}^*\mathbf{QR} = \mathbf{RQR}$ , so that  $\mathbf{K} \succeq 0$  by Proposition 12.1.2 (a). Taking determinants with Theorem 6.5.4, and using  $(\det \mathbf{P}^{1/2})^2 = \det \mathbf{P}$ ,

$$\det(\mathbf{P} + \mathbf{Q}) = \det \mathbf{P} \cdot \det(\mathbf{I} + \mathbf{K}).$$

Now  $\mathbf{K}$  is Hermitian, so  $\mathbf{K} = \mathbf{U} \operatorname{diag}(\mu_1, \dots, \mu_n) \mathbf{U}^*$  with  $\mathbf{U}$  unitary, by Corollary 11.4.3 over  $\mathbb{C}$  and Corollary 11.5.2 over  $\mathbb{R}$ ; and every  $\mu_i \geq 0$  by Theorem 12.1.4 (b). Then  $\mathbf{I} + \mathbf{K} = \mathbf{U} \operatorname{diag}(1 + \mu_1, \dots, 1 + \mu_n) \mathbf{U}^*$ , so

$$\det(\mathbf{I} + \mathbf{K}) = \prod_{i=1}^n (1 + \mu_i) \geq 1,$$

each factor being  $\geq 1$ . Also  $\det \mathbf{P} > 0$ , since the eigenvalues of  $\mathbf{P}$  are positive (Theorem 12.1.5 (b)) and the determinant is their product. Hence  $\det(\mathbf{P} + \mathbf{Q}) \geq \det \mathbf{P} > 0$ .

For the equality case, the product equals 1 exactly when every  $\mu_i = 0$ , that is when  $\mathbf{K} = \mathbf{U} \mathbf{0} \mathbf{U}^* = \mathbf{0}$ , and then  $\mathbf{Q} = \mathbf{P}^{1/2} \mathbf{K} \mathbf{P}^{1/2} = \mathbf{0}$ . Conversely  $\mathbf{Q} = \mathbf{0}$  gives equality. This proves the lemma. ■

*Remark.* Section 5 states this in its natural generality: the determinant is monotone for the Loewner order, so  $\mathbf{A} \succeq \mathbf{B} \succeq 0$  implies  $\det \mathbf{A} \geq \det \mathbf{B}$ . The special case proved here, in which the smaller of the two matrices is definite, is all this section and the next one need, and its proof is short enough to keep local.

**Corollary 12.6.3 — Fischer's Inequality**

Let  $\mathbf{M} = \begin{pmatrix} \mathbf{A} & \mathbf{B} \\ \mathbf{B}^* & \mathbf{C} \end{pmatrix} \in M_{k+l}(F)$  with  $\mathbf{M} \succeq 0$ , where  $\mathbf{A} \in M_k(F)$  and  $\mathbf{C} \in M_l(F)$ . Then

$$\det \mathbf{M} \leq \det \mathbf{A} \cdot \det \mathbf{C}.$$

If  $\mathbf{M} \succ 0$ , equality holds if and only if  $\mathbf{B} = 0$ .

*Idea.* Do the definite case first and reach the general case by a limit. For  $\mathbf{M} \succ 0$  the Schur complement exists, the determinant splits as  $\det \mathbf{A} \det(\mathbf{M}/\mathbf{A})$ , and  $\det \mathbf{C} \geq \det(\mathbf{M}/\mathbf{A})$  by Lemma 12.6.2 applied to  $\mathbf{C} = (\mathbf{M}/\mathbf{A}) + \mathbf{B}^* \mathbf{A}^{-1} \mathbf{B}$ . For  $\mathbf{M} \succeq 0$ , apply the definite case to  $\mathbf{M} + t\mathbf{I}$  and let  $t$  shrink to 0: both sides are polynomials in  $t$ .

*Proof. Case 1:*  $\mathbf{M} \succ 0$ . By Theorem 12.6.1 (a),  $\mathbf{A} \succ 0$ , and by (b),  $\mathbf{M}/\mathbf{A} \succ 0$ . By Theorem 7.3.2,

$$\det \mathbf{M} = \det \mathbf{A} \cdot \det(\mathbf{M}/\mathbf{A}).$$

The matrix  $\mathbf{A}^{-1}$  is positive definite (Exercise (B3: The inverse of a positive definite matrix)), so  $\mathbf{Q} := \mathbf{B}^* \mathbf{A}^{-1} \mathbf{B} \succeq 0$  by Proposition 12.1.2 (a) with  $\mathbf{S} = \mathbf{B}$ . Since  $\mathbf{C} = (\mathbf{M}/\mathbf{A}) + \mathbf{Q}$ , Lemma 12.6.2 with  $\mathbf{P} = \mathbf{M}/\mathbf{A}$  gives  $\det \mathbf{C} \geq \det(\mathbf{M}/\mathbf{A})$ , with equality if and only if  $\mathbf{Q} = 0$ . Multiplying by  $\det \mathbf{A} > 0$ ,

$$\det \mathbf{M} = \det \mathbf{A} \det(\mathbf{M}/\mathbf{A}) \leq \det \mathbf{A} \det \mathbf{C},$$

with equality if and only if  $\mathbf{B}^* \mathbf{A}^{-1} \mathbf{B} = 0$ .

It remains to see that  $\mathbf{B}^* \mathbf{A}^{-1} \mathbf{B} = 0$  holds only for  $\mathbf{B} = 0$ . Let  $\mathbf{R} = (\mathbf{A}^{1/2})^{-1}$ , a Hermitian matrix with  $\mathbf{R}^2 = \mathbf{A}^{-1}$ , as in the proof of Lemma 12.6.2. Then  $\mathbf{B}^* \mathbf{A}^{-1} \mathbf{B} = (\mathbf{R}\mathbf{B})^* (\mathbf{R}\mathbf{B})$ , whose  $(j, j)$  entry is the squared norm of the  $j$ -th column of  $\mathbf{R}\mathbf{B}$ . If the whole matrix is 0, every column of  $\mathbf{R}\mathbf{B}$  is  $\mathbf{0}$ , so  $\mathbf{R}\mathbf{B} = 0$  and  $\mathbf{B} = \mathbf{A}^{1/2}(\mathbf{R}\mathbf{B}) = 0$ . The converse is clear.

*Case 2:*  $\mathbf{M} \succeq 0$ . For real  $t > 0$  the matrix  $\mathbf{M} + t\mathbf{I}_{k+l}$  is Hermitian with  $\mathbf{v}^*(\mathbf{M} + t\mathbf{I})\mathbf{v} = \mathbf{v}^*\mathbf{M}\mathbf{v} + t\|\mathbf{v}\|^2 > 0$  for  $\mathbf{v} \neq \mathbf{0}$ , so  $\mathbf{M} + t\mathbf{I} \succ 0$ . Its blocks are  $\mathbf{A} + t\mathbf{I}_k$ ,  $\mathbf{B}$  and  $\mathbf{C} + t\mathbf{I}_l$ , so Case 1 gives

$$\det(\mathbf{M} + t\mathbf{I}) \leq \det(\mathbf{A} + t\mathbf{I}_k) \cdot \det(\mathbf{C} + t\mathbf{I}_l)$$

for every  $t > 0$ . A determinant is a polynomial in the entries (Theorem 6.4.3), and the entries here are polynomials in  $t$ , so both sides are polynomial, hence continuous, functions of  $t$ . Letting  $t$  decrease to 0 preserves the inequality, and at  $t = 0$  it reads  $\det \mathbf{M} \leq \det \mathbf{A} \det \mathbf{C}$ . This proves the corollary. ■

The inequality says that for a positive matrix the off-diagonal blocks are a liability: they can only shrink the determinant, and they cost nothing only when they are absent. Applying it repeatedly, to finer and finer partitions, gives the classical case in which every block is  $1 \times 1$ .

That is the next result.

## 12.6.4 Hadamard's inequality

**Theorem 12.6.4 — Hadamard's Inequality** (a) Let  $\mathbf{A} \in M_n(F)$  with  $\mathbf{A} \succeq \mathbf{0}$ . Then

$$\det \mathbf{A} \leq a_{11}a_{22} \cdots a_{nn},$$

with equality if and only if  $\mathbf{A}$  is diagonal or some  $a_{ii} = 0$ .

(b) Let  $\mathbf{B} \in M_n(F)$  have columns  $\mathbf{b}_1, \dots, \mathbf{b}_n$ . Then

$$|\det \mathbf{B}| \leq \|\mathbf{b}_1\| \|\mathbf{b}_2\| \cdots \|\mathbf{b}_n\|,$$

with equality if and only if the columns are pairwise orthogonal or some column is  $\mathbf{0}$ .

*Idea.* Part (a) is Fischer's inequality with  $k = 1$ , applied again and again: peel off the first row and column, bound  $\det \mathbf{A}$  by  $a_{11} \det \mathbf{A}'$ , and induct on the size. Part (b) is part (a) for the Gram matrix  $\mathbf{A} = \mathbf{B}^* \mathbf{B}$ , whose diagonal entries are the squared lengths  $\|\mathbf{b}_j\|^2$  and whose determinant is  $|\det \mathbf{B}|^2$ . The equality conditions transfer along the same dictionary:  $\mathbf{A}$  diagonal says the columns of  $\mathbf{B}$  are orthogonal, and  $a_{jj} = 0$  says  $\mathbf{b}_j = \mathbf{0}$ .

*Proof.*

(a) Induct on  $n$ . For  $n = 1$  both sides are  $a_{11}$ .

Let  $n \geq 2$  and split  $\mathbf{A}$  with  $k = 1$ :

$$\mathbf{A} = \begin{pmatrix} a_{11} & \mathbf{b}^* \\ \mathbf{b} & \mathbf{A}' \end{pmatrix}, \quad \mathbf{b} \in F^{n-1}, \quad \mathbf{A}' \in M_{n-1}(F),$$

so that  $\mathbf{A}'$  is the submatrix on rows and columns  $2, \dots, n$ . By [Theorem 12.6.1 \(a\)](#),  $\mathbf{A}' \succeq \mathbf{0}$ , and by [Corollary 12.6.3](#),

$$\det \mathbf{A} \leq a_{11} \det \mathbf{A}' \leq a_{11} (a_{22} \cdots a_{nn}),$$

the second step being the inductive hypothesis applied to  $\mathbf{A}'$ . This is the inequality. For equality, suppose first that some  $a_{ii} = 0$ . Then [Exercise \(C3: The largest entry sits on the diagonal\)](#) (b) makes row  $i$  and column  $i$  of  $\mathbf{A}$  zero, so  $\det \mathbf{A} = 0$ ; and the right-hand side is 0 too, so equality holds. If  $\mathbf{A}$  is diagonal, both sides are the product of the diagonal entries. Conversely, suppose equality holds and every  $a_{ii} > 0$ . Then  $\det \mathbf{A} = a_{11} \cdots a_{nn} > 0$ , so  $\mathbf{A}$  is invertible, and a positive semidefinite matrix with no zero eigenvalue is positive definite ([Theorem 12.1.5 \(b\)](#)). Both inequalities in the display are therefore equalities. The first is the equality case of [Corollary 12.6.3](#) for the positive definite  $\mathbf{A}$ , so  $\mathbf{b} = \mathbf{0}$ : the first row and column vanish off the diagonal. The second is the equality case of the inductive hypothesis for  $\mathbf{A}'$ , whose diagonal entries are positive, so  $\mathbf{A}'$  is diagonal. Hence  $\mathbf{A}$  is diagonal, and the induction is complete.

(b) Put  $\mathbf{A} = \mathbf{B}^* \mathbf{B}$ , which is positive semidefinite by [Theorem 12.1.4 \(c\)](#). Its  $(j, j)$  entry is  $\mathbf{b}_j^* \mathbf{b}_j = \|\mathbf{b}_j\|^2$ . By [Theorem 6.5.4](#), and because  $\det(\mathbf{B}^*) = \overline{\det \mathbf{B}}$  — the star transposes, which does not change the determinant ([Theorem 6.4.5](#)), and

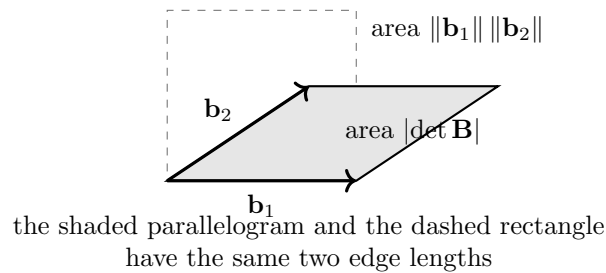
conjugates every entry, which conjugates the determinant —

$$\det \mathbf{A} = \overline{\det \mathbf{B}} \det \mathbf{B} = |\det \mathbf{B}|^2.$$

Part (a) now reads  $|\det \mathbf{B}|^2 \leq \|\mathbf{b}_1\|^2 \cdots \|\mathbf{b}_n\|^2$ , and taking non-negative square roots gives the inequality.

For equality, note that  $a_{jj} = 0$  says  $\mathbf{b}_j = \mathbf{0}$ , and that  $\mathbf{A}$  is diagonal exactly when  $\langle \mathbf{b}_j, \mathbf{b}_i \rangle = a_{ij} = 0$  for all  $i \neq j$ , that is, exactly when the columns are pairwise orthogonal. So the equality condition of (a) translates into the stated one. This proves the theorem. ■

*Remark.* Part (b) has a geometric reading, and it is the reason the inequality is remembered. Over  $\mathbb{R}$ ,  $|\det \mathbf{B}|$  is the volume of the parallelepiped with edges  $\mathbf{b}_1, \dots, \mathbf{b}_n$  (Definition 6.9.1), and the right-hand side is the volume of the box with the same edge lengths. So among all parallelepipeds with prescribed edge lengths, the rectangular one has the largest volume, and up to degenerate cases it is the only one. Tilting an edge costs volume.



### Example. (Two Determinant Bounds)

Bound  $|\det \mathbf{B}|$  using Theorem 12.6.4 (b) for

$$\mathbf{B}_1 = \begin{pmatrix} 1 & 1 \\ 1 & -1 \end{pmatrix}, \quad \mathbf{B}_2 = \begin{pmatrix} 2 & 1 & 0 \\ 1 & 2 & 1 \\ 0 & 1 & 2 \end{pmatrix},$$

and compare each bound with the exact value.

*Solution.* For  $\mathbf{B}_1$ , both columns have norm  $\sqrt{2}$ , so the bound is 2. The determinant is  $-1 - 1 = -2$ , so  $|\det \mathbf{B}_1| = 2$  and the bound is attained. The equality case predicts this:  $\langle (1, -1), (1, 1) \rangle = 1 - 1 = 0$ , so the columns are orthogonal.

For  $\mathbf{B}_2$ , the columns have norms  $\sqrt{5}$ ,  $\sqrt{6}$  and  $\sqrt{5}$ , so the bound is  $5\sqrt{6} \approx 12.25$ . Expanding along the first row,  $\det \mathbf{B}_2 = 2(4 - 1) - 1(2 - 0) = 4$ . The bound holds and is far from sharp, as it must be, since  $\langle (1, 2, 1), (2, 1, 0) \rangle = 4 \neq 0$  means the columns are not orthogonal.

*Warning.* **Part (a) needs positivity; without it the inequality can fail in either direction.** The symmetric matrix

$$\mathbf{A} = \begin{pmatrix} 1 & 1 & 1 \\ 1 & -1 & 0 \\ 1 & 0 & -1 \end{pmatrix}$$

has  $\det \mathbf{A} = 3$ , while  $a_{11}a_{22}a_{33} = 1 \cdot (-1) \cdot (-1) = 1$ . So  $\det \mathbf{A} > \prod_i a_{ii}$ , and a diagonal entry of an indefinite matrix carries no information about the size of the determinant. Part (b), by contrast, has no hypothesis at all — but it bounds by **column** norms. For rows, apply the same result to  $\mathbf{B}^\top$ , whose determinant is unchanged ([Theorem 6.4.5](#)).

## 12.6.5 Exercises

### 12.6.5.1 A. Check your understanding

#### Exercise A1.

- Write down the general form of a Hermitian matrix in  $2 \times 2$  block form, and say what each block must satisfy.
- State [Theorem 12.6.1](#) (b) in full, including its hypothesis.
- True or false: if  $\mathbf{M} = \begin{pmatrix} \mathbf{A} & \mathbf{B} \\ \mathbf{B}^* & \mathbf{C} \end{pmatrix}$  is Hermitian with  $\mathbf{A} \succeq 0$  and  $\mathbf{C} \succeq 0$ , then  $\mathbf{M} \succeq 0$ . Justify your answer.
- State both forms of Hadamard's inequality, and say what hypothesis each one needs.
- In the proof of Fischer's inequality, which matrix plays the role of  $\mathbf{Q}$  in [Lemma 12.6.2](#), and which plays the role of  $\mathbf{P}$ ?

*Solution.*

- $\mathbf{M} = \begin{pmatrix} \mathbf{A} & \mathbf{B} \\ \mathbf{B}^* & \mathbf{C} \end{pmatrix}$  with  $\mathbf{A} \in M_k(F)$  and  $\mathbf{C} \in M_l(F)$  Hermitian and  $\mathbf{B} \in M_{k \times l}(F)$  arbitrary. The condition  $\mathbf{M}^* = \mathbf{M}$  forces the lower-left block to be  $\mathbf{B}^*$  and both diagonal blocks to be Hermitian.
- If  $\mathbf{A} \succ 0$ , then  $\mathbf{A}$  is invertible,  $\mathbf{M}/\mathbf{A} = \mathbf{C} - \mathbf{B}^*\mathbf{A}^{-1}\mathbf{B}$  is Hermitian,  $\mathbf{M} \succ 0$  if and only if  $\mathbf{M}/\mathbf{A} \succ 0$ , and  $\mathbf{M} \succeq 0$  if and only if  $\mathbf{M}/\mathbf{A} \succeq 0$ .
- False. Take  $\mathbf{A} = (0)$ ,  $\mathbf{B} = (1)$  and  $\mathbf{C} = (1)$ , so  $\mathbf{M} = \begin{pmatrix} 0 & 1 \\ 1 & 1 \end{pmatrix}$ . Both diagonal blocks are positive semidefinite, but  $\det \mathbf{M} = -1 < 0$ , so  $\mathbf{M} \not\succeq 0$  by [Theorem 12.1.4](#) (e).
- By [Theorem 12.6.4](#): for  $\mathbf{A} \succeq 0$ ,  $\det \mathbf{A} \leq a_{11} \cdots a_{nn}$ , and the hypothesis is positive semidefiniteness. For any square  $\mathbf{B}$ :  $|\det \mathbf{B}| \leq \|\mathbf{b}_1\| \cdots \|\mathbf{b}_n\|$  over the columns, with no hypothesis.
- In [Corollary 12.6.3](#),  $\mathbf{P} = \mathbf{M}/\mathbf{A}$  and  $\mathbf{Q} = \mathbf{B}^*\mathbf{A}^{-1}\mathbf{B}$ , so that  $\mathbf{P} + \mathbf{Q} = \mathbf{C}$ .

### 12.6.5.2 B. Practice

**Exercise B1: Definiteness by blocks.**

Each matrix below is real symmetric and is displayed with a 2, 2 partition. Determine whether it is positive definite. Justify your answer, and for any that is not, give a vector  $\mathbf{x}$  with  $\mathbf{x}^\top \mathbf{M} \mathbf{x} < 0$ .

(a)

$$\mathbf{M}_1 = \left( \begin{array}{cc|cc} 1 & 0 & 1 & 1 \\ 0 & 1 & 1 & -1 \\ \hline 1 & 1 & 3 & 0 \\ 1 & -1 & 0 & 3 \end{array} \right)$$

(b)

$$\mathbf{M}_2 = \left( \begin{array}{cc|cc} 1 & 0 & 1 & 1 \\ 0 & 1 & 1 & -1 \\ \hline 1 & 1 & 2 & 0 \\ 1 & -1 & 0 & 1 \end{array} \right)$$

*Solution.* In both cases  $\mathbf{A} = \mathbf{I}_2 \succ 0$  and  $\mathbf{B} = \begin{pmatrix} 1 & 1 \\ 1 & -1 \end{pmatrix}$ , so [Theorem 12.6.1](#) (b) applies with  $\mathbf{A}^{-1} = \mathbf{I}_2$  and

$$\mathbf{B}^\top \mathbf{A}^{-1} \mathbf{B} = \mathbf{B}^\top \mathbf{B} = \begin{pmatrix} 2 & 0 \\ 0 & 2 \end{pmatrix}.$$

- (a)  $\mathbf{M}_1/\mathbf{A} = \begin{pmatrix} 3 & 0 \\ 0 & 3 \end{pmatrix} - \begin{pmatrix} 2 & 0 \\ 0 & 2 \end{pmatrix} = \mathbf{I}_2 \succ 0$ . Hence  $\mathbf{M}_1 \succ 0$ , and  $\det \mathbf{M}_1 = \det \mathbf{A} \det(\mathbf{M}_1/\mathbf{A}) = 1$ .
- (b)  $\mathbf{M}_2/\mathbf{A} = \begin{pmatrix} 2 & 0 \\ 0 & 1 \end{pmatrix} - \begin{pmatrix} 2 & 0 \\ 0 & 2 \end{pmatrix} = \begin{pmatrix} 0 & 0 \\ 0 & -1 \end{pmatrix}$ . This is the matrix of Section 1's warning: its leading principal minors are 0 and 0, and yet it is not positive semidefinite. So  $\mathbf{M}_2$  is not positive definite, and not even positive semidefinite.

For the witness,  $\mathbf{y}_2 = (0, 1)$  gives  $\mathbf{y}_2^\top (\mathbf{M}_2/\mathbf{A}) \mathbf{y}_2 = -1$ . With  $\mathbf{S} = \begin{pmatrix} \mathbf{I}_2 & -\mathbf{B} \\ 0 & \mathbf{I}_2 \end{pmatrix}$  and  $\mathbf{y} = (0, 0, 0, 1)$ , the vector  $\mathbf{x} = \mathbf{S} \mathbf{y} = (-1, 1, 0, 1)$  satisfies  $\mathbf{x}^\top \mathbf{M}_2 \mathbf{x} = -1$ .

**Exercise B2: A bordered matrix.**

Let

$$\mathbf{M} = \begin{pmatrix} 2 & 1 & 1 \\ 1 & 2 & 0 \\ 1 & 0 & 2 \end{pmatrix}.$$

- (a) Using the partition with  $k = 1$ , compute  $\mathbf{M}/\mathbf{A}$  and decide whether  $\mathbf{M} \succ 0$ .
- (b) Hence compute  $\det \mathbf{M}$ , and check Fischer's inequality for this partition.

*Solution.*

- (a) Here  $\mathbf{A} = (2) \succ 0$ ,  $\mathbf{B} = \begin{pmatrix} 1 & 1 \end{pmatrix}$  and  $\mathbf{C} = \begin{pmatrix} 2 & 0 \\ 0 & 2 \end{pmatrix}$ . Then

$$\mathbf{B}^* \mathbf{A}^{-1} \mathbf{B} = \frac{1}{2} \begin{pmatrix} 1 \\ 1 \end{pmatrix} (1 \quad 1) = \frac{1}{2} \begin{pmatrix} 1 & 1 \\ 1 & 1 \end{pmatrix}, \quad \mathbf{M}/\mathbf{A} = \begin{pmatrix} 3/2 & -1/2 \\ -1/2 & 3/2 \end{pmatrix}.$$

The leading principal minors of  $\mathbf{M}/\mathbf{A}$  are  $3/2$  and  $9/4 - 1/4 = 2$ , both positive, so  $\mathbf{M}/\mathbf{A} \succ 0$  and  $\mathbf{M} \succ 0$  by [Theorem 12.6.1](#) (b).

- (b) By [Theorem 7.3.2](#),  $\det \mathbf{M} = 2 \cdot 2 = 4$ . Fischer's inequality ([Corollary 12.6.3](#)) for this partition reads  $\det \mathbf{M} \leq \det \mathbf{A} \det \mathbf{C} = 2 \cdot 4 = 8$ , and indeed  $4 \leq 8$ . The inequality is strict, as it must be, since  $\mathbf{B} \neq 0$ .

### Exercise B3: Hadamard bounds.

- (a) Bound  $|\det \mathbf{B}|$  for  $\mathbf{B} = \begin{pmatrix} 1 & 2 \\ 3 & 4 \end{pmatrix}$ , and compare with the exact value.  
 (b) Let

$$\mathbf{H} = \begin{pmatrix} 1 & 1 & 1 & 1 \\ 1 & 1 & -1 & -1 \\ 1 & -1 & 1 & -1 \\ 1 & -1 & -1 & 1 \end{pmatrix}.$$

Compute  $\mathbf{H}^\top \mathbf{H}$ , and deduce  $|\det \mathbf{H}|$  without expanding the determinant.

*Solution.*

- (a) The columns are  $(1, 3)$  and  $(2, 4)$ , with norms  $\sqrt{10}$  and  $\sqrt{20}$ , so the bound is  $\sqrt{200} = 10\sqrt{2} \approx 14.14$ . The determinant is  $4 - 6 = -2$ , so  $|\det \mathbf{B}| = 2$ . The bound is true and very weak, because the columns are far from orthogonal.  
 (b) Every column of  $\mathbf{H}$  has norm 2, and any two distinct columns are orthogonal: each of the six pairs agrees in two positions and disagrees in the other two, so the dot product is  $1 + 1 - 1 - 1 = 0$ . Hence  $\mathbf{H}^\top \mathbf{H} = 4\mathbf{I}_4$ . Taking determinants with [Theorem 6.5.4](#) and [Theorem 6.4.5](#),  $(\det \mathbf{H})^2 = 4^4 = 256$ , so  $|\det \mathbf{H}| = 16$ . This is exactly the Hadamard bound  $\|\mathbf{h}_1\| \cdots \|\mathbf{h}_4\| = 2^4 = 16$ , as the equality case of [Theorem 12.6.4](#) (b) predicts for orthogonal columns.

### 12.6.5.3 C. Going deeper

#### Exercise C1: A zero corner kills its border.

- (a) Prove that if  $\mathbf{P} \succeq 0$  and  $\mathbf{x}^* \mathbf{P} \mathbf{x} = 0$ , then  $\mathbf{P} \mathbf{x} = \mathbf{0}$ .  
 (b) Deduce that if  $\mathbf{M} = \begin{pmatrix} \mathbf{A} & \mathbf{B} \\ \mathbf{B}^* & \mathbf{C} \end{pmatrix} \succeq 0$  and  $\mathbf{A} = \mathbf{0}$ , then  $\mathbf{B} = \mathbf{0}$ .  
 (c) Deduce that if  $\mathbf{M} \succeq 0$  and  $\mathbf{A}$  is singular, then  $\mathbf{M}$  is singular. What does Fischer's inequality say in that case?

*Hint for (a): write  $\mathbf{P} = \mathbf{N}^* \mathbf{N}$ .*

*Solution.*

- (a) By [Theorem 12.1.4](#) (c),  $\mathbf{P} = \mathbf{N}^* \mathbf{N}$  for some  $\mathbf{N}$ . Then  $0 = \mathbf{x}^* \mathbf{P} \mathbf{x} = \|\mathbf{N} \mathbf{x}\|^2$ , so  $\mathbf{N} \mathbf{x} = \mathbf{0}$  and  $\mathbf{P} \mathbf{x} = \mathbf{N}^* (\mathbf{N} \mathbf{x}) = \mathbf{0}$ .  
 (b) Let  $\mathbf{x} \in F^k$  and put  $\mathbf{v} = (\mathbf{x}, \mathbf{0})$ . As computed in the proof of [Theorem 12.6.1](#) (a),



$\mathbf{v}^* \mathbf{M} \mathbf{v} = \mathbf{x}^* \mathbf{A} \mathbf{x} = 0$ . By (a),  $\mathbf{M} \mathbf{v} = \mathbf{0}$ . But  $\mathbf{M} \mathbf{v} = (\mathbf{A} \mathbf{x}, \mathbf{B}^* \mathbf{x}) = (\mathbf{0}, \mathbf{B}^* \mathbf{x})$ , so  $\mathbf{B}^* \mathbf{x} = \mathbf{0}$  for every  $\mathbf{x} \in F^k$ , that is  $\mathbf{B}^* = \mathbf{0}$  and  $\mathbf{B} = \mathbf{0}$ .

- (c) Choose  $\mathbf{x} \neq \mathbf{0}$  with  $\mathbf{A} \mathbf{x} = \mathbf{0}$ , which is possible since  $\mathbf{A}$  is singular (Theorem 2.4.4), and put  $\mathbf{v} = (\mathbf{x}, \mathbf{0}) \neq \mathbf{0}$ . Then  $\mathbf{v}^* \mathbf{M} \mathbf{v} = \mathbf{x}^* \mathbf{A} \mathbf{x} = 0$ , so  $\mathbf{M} \mathbf{v} = \mathbf{0}$  by (a), and  $\mathbf{M}$  is singular. Hence  $\det \mathbf{M} = 0$ , while  $\det \mathbf{A} \det \mathbf{C} = 0$  as well, so Corollary 12.6.3 holds with both sides equal to 0. It is true, and it says nothing; the informative case is  $\mathbf{A} \succ \mathbf{0}$ .

### Exercise C2: A Cauchy–Schwarz inequality for blocks.

Let  $\mathbf{M} = \begin{pmatrix} \mathbf{A} & \mathbf{B} \\ \mathbf{B}^* & \mathbf{C} \end{pmatrix} \succeq \mathbf{0}$  with  $\mathbf{A} \in M_k(F)$  and  $\mathbf{C} \in M_l(F)$ .

- (a) Prove that for all  $\mathbf{x} \in F^k$  and  $\mathbf{y} \in F^l$ ,

$$|\mathbf{x}^* \mathbf{B} \mathbf{y}|^2 \leq (\mathbf{x}^* \mathbf{A} \mathbf{x}) (\mathbf{y}^* \mathbf{C} \mathbf{y}).$$

- (b) Deduce the Cauchy–Schwarz inequality  $|\langle \mathbf{y}, \mathbf{x} \rangle| \leq \|\mathbf{x}\| \|\mathbf{y}\|$  in  $F^n$ .

*Hint for (a): apply the definition of  $\mathbf{M} \succeq \mathbf{0}$  to the vector  $(\mathbf{x}, t\mathbf{y})$ , and choose the scalar  $t$  well.*

*Solution.*

- (a) Write  $p = \mathbf{x}^* \mathbf{A} \mathbf{x} \geq 0$ ,  $q = \mathbf{y}^* \mathbf{C} \mathbf{y} \geq 0$  and  $z = \mathbf{x}^* \mathbf{B} \mathbf{y}$ . For a scalar  $t \in F$ , block multiplication (Theorem 7.1.1) gives, for  $\mathbf{v} = (\mathbf{x}, t\mathbf{y})$ ,

$$\mathbf{v}^* \mathbf{M} \mathbf{v} = p + \bar{t} \bar{z} + tz + |t|^2 q = p + 2 \operatorname{Re}(tz) + |t|^2 q \geq 0,$$

where the middle step uses  $\bar{t} \bar{z} = \overline{tz}$  and  $w + \bar{w} = 2 \operatorname{Re}(w)$ . Now take  $t = r\omega$  with  $r \geq 0$  real and  $\omega$  a scalar of modulus 1 chosen so that  $\omega z = -|z|$  (over  $\mathbb{R}$ ,  $\omega = \pm 1$ ). Then

$$p - 2r|z| + r^2 q \geq 0 \quad \text{for every } r \geq 0. \quad (*)$$

If  $q = 0$ , then  $(*)$  says  $p \geq 2r|z|$  for all  $r \geq 0$ , which forces  $|z| = 0$ , and the claim reads  $0 \leq 0$ . If  $q > 0$ , take  $r = |z|/q \geq 0$  in  $(*)$ :

$$p - \frac{2|z|^2}{q} + \frac{|z|^2}{q} = p - \frac{|z|^2}{q} \geq 0,$$

so  $|z|^2 \leq pq$ , as claimed.

- (b) Take  $k = l = n$  and  $\mathbf{A} = \mathbf{B} = \mathbf{C} = \mathbf{I}_n$ , so that  $\mathbf{M} = \begin{pmatrix} \mathbf{I}_n & \mathbf{I}_n \\ \mathbf{I}_n & \mathbf{I}_n \end{pmatrix}$ . This  $\mathbf{M}$  is positive semidefinite, since  $\mathbf{M} = \mathbf{W}^* \mathbf{W}$  for  $\mathbf{W} = \begin{pmatrix} \mathbf{I}_n & \mathbf{I}_n \\ \mathbf{0} & \mathbf{0} \end{pmatrix}$  (Theorem 12.1.4 (c)). Part (a) then reads  $|\mathbf{x}^* \mathbf{y}|^2 \leq \|\mathbf{x}\|^2 \|\mathbf{y}\|^2$ , and  $\mathbf{x}^* \mathbf{y} = \langle \mathbf{y}, \mathbf{x} \rangle$  by the definition of the standard inner product. This is Theorem 10.1.5 for  $F^n$ .

### Exercise C3: How large can a determinant be?.

- (a) Let  $\mathbf{B} \in M_n(F)$  have every entry of modulus at most 1. Prove that  $|\det \mathbf{B}| \leq n^{n/2}$ .  
 (b) Show that the bound is attained for  $n = 1$ ,  $n = 2$  and  $n = 4$ .  
 (c) Let  $\mathbf{B} \in M_3(\mathbb{R})$  have every entry equal to 1 or  $-1$ . Prove that  $\det \mathbf{B}$  is a multiple

of 4, and deduce that  $|\det \mathbf{B}| \leq 4$ . Hence the bound of (a) is **not** attained for  $n = 3$ .

*Hint for (c): normalize the first row and the first column, then subtract the first column from the other two.*

*Solution.*

- (a) Each column  $\mathbf{b}_j$  has  $n$  entries of modulus  $\leq 1$ , so  $\|\mathbf{b}_j\|^2 = \sum_i |b_{ij}|^2 \leq n$  and  $\|\mathbf{b}_j\| \leq \sqrt{n}$ . By [Theorem 12.6.4](#) (b),  $|\det \mathbf{B}| \leq (\sqrt{n})^n = n^{n/2}$ .
- (b) For  $n = 1$ ,  $\mathbf{B} = (1)$  gives  $1 = 1^{1/2}$ . For  $n = 2$ ,  $\mathbf{B} = \begin{pmatrix} 1 & 1 \\ 1 & -1 \end{pmatrix}$  has  $|\det \mathbf{B}| = 2 = 2^{2/2}$ . For  $n = 4$ , the matrix  $\mathbf{H}$  of [Exercise \(B3: Hadamard bounds\)](#) (b) has  $|\det \mathbf{H}| = 16 = 4^{4/2}$ . In each case all entries have modulus 1 and the columns are orthogonal, which is exactly what is needed for equality both in  $\|\mathbf{b}_j\| \leq \sqrt{n}$  and in [Theorem 12.6.4](#) (b).
- (c) Multiplying a row or a column by  $-1$  multiplies  $\det \mathbf{B}$  by  $-1$  ([Theorem 6.5.1](#) (b) and (d)) and keeps all entries  $\pm 1$ , so we may assume that the first row **and** the first column both consist of 1s: first scale the three columns to make the first row  $(1, 1, 1)$ , then scale rows 2 and 3 to make the first column  $(1, 1, 1)$ , which leaves the first row alone. Subtracting column 1 from columns 2 and 3 leaves the determinant unchanged ([Theorem 6.5.1](#) (c) and (d)) and gives a matrix whose first row is  $(1, 0, 0)$  and whose remaining entries are  $c_{ij} = b_{ij} - 1$  for  $i, j \in \{2, 3\}$ ; each lies in  $\{0, -2\}$ , since  $b_{ij} \in \{1, -1\}$ . Expanding along the first row ([Theorem 6.6.1](#)),

$$\det \mathbf{B} = c_{22}c_{33} - c_{23}c_{32}$$

with every  $c_{ij} \in \{0, -2\}$ . Each product lies in  $\{0, 4\}$ , so  $\det \mathbf{B} \in \{-4, 0, 4\}$  and in particular  $|\det \mathbf{B}| \leq 4$ . Since  $3^{3/2} = 3\sqrt{3} \approx 5.196 > 4$ , the bound of (a) is not attained. The value 4 does occur, for instance for  $\begin{pmatrix} 1 & 1 & 1 \\ 1 & 1 & 1 \\ 1 & 1 & -1 \end{pmatrix}$ .

## 12.7 Products Entry by Entry

Matrix multiplication is not the only way to multiply two matrices of the same size. There is a much simpler product — multiply the entries in matching positions — which is useless for composing linear maps and extremely useful everywhere else. This section introduces it and proves the one theorem that makes it interesting: it preserves positivity. A determinant inequality follows, and it contains Hadamard's inequality of Section 6 as a special case.

Throughout,  $F = \mathbb{R}$  or  $F = \mathbb{C}$ .

### 12.7.1 The entrywise product

Two matrices of the same shape can be multiplied the way two lists of numbers are: position by position. The operation has no interpretation in terms of linear maps, which is one reason it looks like a curiosity at first sight, and it is written with a small circle to keep it away from

ordinary multiplication.

*The Hadamard product multiplies matrices the way a spreadsheet would.*

### Definition 12.7.1 — Hadamard Product

Let  $\mathbf{A}, \mathbf{B} \in M_{m \times n}(F)$ . Their **Hadamard product**, or **entrywise product**, is the matrix  $\mathbf{A} \circ \mathbf{B} \in M_{m \times n}(F)$  with

$$(\mathbf{A} \circ \mathbf{B})_{ij} = a_{ij}b_{ij} \quad \text{for all } i, j.$$

Both factors must have **the same shape**, and the result has that shape too; there is no rule like “columns of the first match rows of the second”. The operation is commutative and associative, and it is bilinear —  $(\mathbf{A} + \mathbf{A}') \circ \mathbf{B} = \mathbf{A} \circ \mathbf{B} + \mathbf{A}' \circ \mathbf{B}$  and  $(c\mathbf{A}) \circ \mathbf{B} = c(\mathbf{A} \circ \mathbf{B})$  — because each of these is a statement about numbers, checked one entry at a time. Its identity element is the all-ones matrix  $\mathbf{J}$ , not  $\mathbf{I}$ .

#### Examples.

- $\begin{pmatrix} 1 & 2 \\ 3 & 4 \end{pmatrix} \circ \begin{pmatrix} 5 & 0 \\ -1 & 2 \end{pmatrix} = \begin{pmatrix} 5 & 0 \\ -3 & 8 \end{pmatrix}$ .
- $\mathbf{A} \circ \mathbf{I} = \text{diag}(a_{11}, \dots, a_{nn})$ : multiplying entrywise by the identity **erases** everything off the diagonal, where ordinary multiplication by  $\mathbf{I}$  changes nothing.
- $\mathbf{A} \circ \mathbf{J} = \mathbf{A}$ , so  $\mathbf{J}$  is the identity for  $\circ$ .
- For two diagonal matrices,  $\mathbf{D} \circ \mathbf{D}' = \mathbf{DD}'$ : both products are  $\text{diag}(d_1d'_1, \dots, d_nd'_n)$ . For matrices in general the two have nothing to do with each other, though they can coincide by accident:  $\begin{pmatrix} 1 & 0 \\ 0 & 0 \end{pmatrix}$  and  $\begin{pmatrix} 1 & 0 \\ 0 & 0 \end{pmatrix}$  have equal Hadamard and matrix products without either being diagonal.
- For column vectors  $\mathbf{u}, \mathbf{w} \in F^n$ , regarded as  $n \times 1$  matrices,  $\mathbf{u} \circ \mathbf{w} = (u_1w_1, \dots, u_nw_n)$ .

**Non-example by minimal change.** Take  $\mathbf{A} = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}$  and  $\mathbf{B} = \mathbf{A}$ . Then  $\mathbf{AB} = \mathbf{I}$ , but  $\mathbf{A} \circ \mathbf{B} = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix} \circ \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix} = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}$ . The two products differ, and they differ in the most basic way:  $\mathbf{A}$  is its own inverse and its own Hadamard square.

*Warning.*  $\mathbf{A} \circ \mathbf{B}$  is not  $\mathbf{AB}$ , and almost nothing carries over. Two invertible matrices can have a singular entrywise product:

$$\begin{pmatrix} 1 & 2 \\ 2 & 1 \end{pmatrix} \circ \begin{pmatrix} 2 & 1 \\ 1 & 2 \end{pmatrix} = \begin{pmatrix} 2 & 2 \\ 2 & 2 \end{pmatrix},$$

where the two factors have determinants  $-3$  and  $3$  and the product has determinant  $0$ . There is no formula for  $\det(\mathbf{A} \circ \mathbf{B})$  in terms of  $\det \mathbf{A}$  and  $\det \mathbf{B}$ , no relation between the eigenvalues, and no rule  $(\mathbf{A} \circ \mathbf{B})^{-1} = \mathbf{A}^{-1} \circ \mathbf{B}^{-1}$ . The one structural property that does survive is the subject of this section, and it is a genuine surprise.

The name is the same one that Section 6 attached to a determinant inequality. That is a coincidence of a busy mathematician, not of the mathematics: Hadamard’s inequality and the Hadamard product are unrelated statements, and this section will nonetheless prove one from the other.

## 12.7.2 The Schur product theorem

Positivity is a statement about the quadratic form  $\mathbf{x}^* \mathbf{A} \mathbf{x}$ , and the quadratic form of  $\mathbf{A} \circ \mathbf{B}$  has no visible relation to those of  $\mathbf{A}$  and  $\mathbf{B}$ . So the following theorem cannot be proved by staring at the form. What works is to break both matrices into the simplest positive pieces there are — rank-one matrices  $\mathbf{u}\mathbf{u}^*$  — and to notice that the entrywise product of two such pieces is again one.

### Lemma 12.7.1 — Entrywise product of two rank-one positive matrices

Let  $\mathbf{u}, \mathbf{w} \in F^n$ . Then

$$(\mathbf{u}\mathbf{u}^*) \circ (\mathbf{w}\mathbf{w}^*) = (\mathbf{u} \circ \mathbf{w})(\mathbf{u} \circ \mathbf{w})^*.$$

*Proof.* Compare the  $(i, j)$  entries. The  $(i, j)$  entry of  $\mathbf{u}\mathbf{u}^*$  is  $u_i \overline{u_j}$  and that of  $\mathbf{w}\mathbf{w}^*$  is  $w_i \overline{w_j}$ , so the left-hand side has  $(i, j)$  entry

$$u_i \overline{u_j} w_i \overline{w_j} = (u_i w_i) \overline{(u_j w_j)},$$

the rearrangement using commutativity of  $F$  and  $\overline{u_j} \overline{w_j} = \overline{u_j w_j}$ . The vector  $\mathbf{u} \circ \mathbf{w}$  has  $i$ -th entry  $u_i w_i$ , so the right-hand side has  $(i, j)$  entry  $(u_i w_i) \overline{(u_j w_j)}$  as well. The two matrices agree entry by entry. ■

That one line is the whole theorem; everything else is bookkeeping over the spectral decompositions.

### Theorem 12.7.2 — Schur Product Theorem

Let  $\mathbf{A}, \mathbf{B} \in M_n(F)$ .

- (a) If  $\mathbf{A} \succeq 0$  and  $\mathbf{B} \succeq 0$ , then  $\mathbf{A} \circ \mathbf{B} \succeq 0$ .
- (b) If  $\mathbf{A} \succ 0$  and  $\mathbf{B} \succ 0$ , then  $\mathbf{A} \circ \mathbf{B} \succ 0$ .

*Idea.* For (a): the spectral theorem writes  $\mathbf{A} = \sum_i \lambda_i \mathbf{u}_i \mathbf{u}_i^*$  and  $\mathbf{B} = \sum_j \mu_j \mathbf{w}_j \mathbf{w}_j^*$  with all  $\lambda_i, \mu_j \geq 0$ . Expand  $\mathbf{A} \circ \mathbf{B}$  by bilinearity into  $n^2$  terms; by Lemma 12.7.1 each term is a non-negative multiple of a matrix  $\mathbf{z}\mathbf{z}^*$ , and those are positive semidefinite. A sum of positive semidefinite matrices is positive semidefinite, so we are done.

For (b), do not redo the argument with strict inequalities: the vectors  $\mathbf{u}_i \circ \mathbf{w}_j$  need not span, and the bookkeeping becomes unpleasant. Instead peel a multiple of the identity off  $\mathbf{B}$ . If  $\varepsilon > 0$  is the smallest eigenvalue of  $\mathbf{B}$ , then  $\mathbf{B} = (\mathbf{B} - \varepsilon \mathbf{I}) + \varepsilon \mathbf{I}$  with  $\mathbf{B} - \varepsilon \mathbf{I} \succeq 0$ , so

$$\mathbf{A} \circ \mathbf{B} = \mathbf{A} \circ (\mathbf{B} - \varepsilon \mathbf{I}) + \varepsilon (\mathbf{A} \circ \mathbf{I}),$$

and  $\mathbf{A} \circ \mathbf{I}$  is the diagonal matrix of the  $a_{ii}$ , which are strictly positive. Part (a) handles

the first term, and a positive definite matrix plus a positive semidefinite one is positive definite.

*Proof.*

- (a) Both  $\mathbf{A}$  and  $\mathbf{B}$  are Hermitian, so each has an orthonormal basis of eigenvectors. Let  $(\mathbf{u}_1, \dots, \mathbf{u}_n)$  be one for  $\mathbf{A}$ , with  $\mathbf{A}\mathbf{u}_i = \lambda_i \mathbf{u}_i$ , and let  $\mathbf{U}$  be the unitary matrix with these columns, so that  $\mathbf{A} = \mathbf{U}\mathbf{D}\mathbf{U}^*$  with  $\mathbf{D} = \text{diag}(\lambda_1, \dots, \lambda_n)$  by [Corollary 11.4.3](#) over  $\mathbb{C}$  and [Corollary 11.5.2](#) over  $\mathbb{R}$ . The columns of  $\mathbf{U}\mathbf{D}$  are  $\lambda_i \mathbf{u}_i$  and the rows of  $\mathbf{U}^*$  are  $\mathbf{u}_i^*$ , so [Corollary 7.1.3](#) gives

$$\mathbf{A} = \sum_{i=1}^n \lambda_i \mathbf{u}_i \mathbf{u}_i^*,$$

which over  $\mathbb{R}$  is [Corollary 11.5.4](#). In the same way  $\mathbf{B} = \sum_j \mu_j \mathbf{w}_j \mathbf{w}_j^*$ . Every  $\lambda_i \geq 0$  and every  $\mu_j \geq 0$ , by [Theorem 12.1.4](#) (b).

The Hadamard product is bilinear, so expanding both sums,

$$\mathbf{A} \circ \mathbf{B} = \sum_{i=1}^n \sum_{j=1}^n \lambda_i \mu_j [(\mathbf{u}_i \mathbf{u}_i^*) \circ (\mathbf{w}_j \mathbf{w}_j^*)] = \sum_{i,j} \lambda_i \mu_j \mathbf{z}_{ij} \mathbf{z}_{ij}^*,$$

where  $\mathbf{z}_{ij} = \mathbf{u}_i \circ \mathbf{w}_j$  and the second equality is [Lemma 12.7.1](#). Each  $\mathbf{z}_{ij} \mathbf{z}_{ij}^*$  is positive semidefinite by [Example \(Five Positive Matrices\)](#) (b), and each coefficient  $\lambda_i \mu_j$  is a non-negative real number. A non-negative combination of positive semidefinite matrices is positive semidefinite ([Exercise \(C2: The cone of positive matrices\)](#) (a)), so  $\mathbf{A} \circ \mathbf{B} \succeq 0$ .

- (b) Let  $\varepsilon > 0$  be the smallest eigenvalue of  $\mathbf{B}$ , which is positive by [Theorem 12.1.5](#) (b). The matrix  $\mathbf{B} - \varepsilon \mathbf{I}$  is Hermitian with eigenvalues  $\mu_j - \varepsilon \geq 0$ , so  $\mathbf{B} - \varepsilon \mathbf{I} \succeq 0$  by [Theorem 12.1.4](#) (b). Since  $\circ$  is bilinear and  $\mathbf{A} \circ \mathbf{I} = \text{diag}(a_{11}, \dots, a_{nn})$ ,

$$\mathbf{A} \circ \mathbf{B} = \mathbf{A} \circ (\mathbf{B} - \varepsilon \mathbf{I}) + \varepsilon \text{diag}(a_{11}, \dots, a_{nn}).$$

The first term is positive semidefinite by (a). In the second,  $a_{ii} = \langle \mathbf{A}\mathbf{e}_i, \mathbf{e}_i \rangle > 0$  because  $\mathbf{A} \succ 0$  and  $\mathbf{e}_i \neq \mathbf{0}$ , so  $\varepsilon \text{diag}(a_{11}, \dots, a_{nn}) \succ 0$  by [Example \(Five Positive Matrices\)](#) (a). For  $\mathbf{x} \neq \mathbf{0}$ , the quadratic form of the sum is  $\geq 0$  plus  $> 0$ , hence  $> 0$ . So  $\mathbf{A} \circ \mathbf{B} \succ 0$ , and the theorem is proved. ■

The theorem is a genuine addition to the list of operations that preserve positivity. Sums and non-negative multiples were easy ([Exercise \(C2: The cone of positive matrices\)](#)), congruence was easy ([Proposition 12.1.2](#)), and the ordinary product is not on the list at all, since  $\mathbf{AB}$  need not even be Hermitian. The entrywise product is on the list, and nothing in the definition of positivity suggests it should be.

#### Example. ( $3 \times 3$ Entrywise Product)

Let

$$\mathbf{A} = \begin{pmatrix} 2 & 1 & 0 \\ 1 & 2 & 1 \\ 0 & 1 & 2 \end{pmatrix}, \quad \mathbf{B} = \begin{pmatrix} 1 & -1 & 1 \\ -1 & 2 & -1 \\ 1 & -1 & 3 \end{pmatrix}.$$

Check that  $\mathbf{A} \succ 0$  and  $\mathbf{B} \succ 0$ , compute  $\mathbf{A} \circ \mathbf{B}$ , and verify directly that it is positive definite.

*Solution.* The leading principal minors of  $\mathbf{A}$  are 2,  $4 - 1 = 3$  and, expanding along the first row,  $2(4 - 1) - 1(2 - 0) = 4$ . All are positive, so  $\mathbf{A} \succ 0$  by Sylvester's criterion (Theorem 12.1.5 (d)).

Those of  $\mathbf{B}$  are 1,  $2 - 1 = 1$  and  $1(6 - 1) + 1(-3 + 1) + 1(1 - 2) = 5 - 2 - 1 = 2$ . All positive, so  $\mathbf{B} \succ 0$ .

Entry by entry,

$$\mathbf{A} \circ \mathbf{B} = \begin{pmatrix} 2 & -1 & 0 \\ -1 & 4 & -1 \\ 0 & -1 & 6 \end{pmatrix},$$

whose leading principal minors are 2,  $8 - 1 = 7$  and  $2(24 - 1) + 1(-6 - 0) = 46 - 6 = 40$ . All positive, so  $\mathbf{A} \circ \mathbf{B} \succ 0$ , as Theorem 12.7.2 (b) promised. Note that the eigenvalues of  $\mathbf{A} \circ \mathbf{B}$  have nothing to do with those of  $\mathbf{A}$  and  $\mathbf{B}$ , and that  $\det(\mathbf{A} \circ \mathbf{B}) = 40$  is much larger than  $\det \mathbf{A} \det \mathbf{B} = 4 \cdot 2 = 8$ . The next theorem explains part of that: it forces  $\det(\mathbf{A} \circ \mathbf{B}) \geq \det \mathbf{A} \prod_i b_{ii} = 24$ . Getting down to the product  $\det \mathbf{A} \det \mathbf{B}$  needs Hadamard's inequality as well, which is Exercise (C2: The product of the determinants).

*Quick check.* Let  $\mathbf{A} = \begin{pmatrix} 1 & 1 \\ 1 & 1 \end{pmatrix}$  and  $\mathbf{B} = \begin{pmatrix} 2 & -1 \\ -1 & 2 \end{pmatrix}$ . Compute  $\mathbf{A} \circ \mathbf{B}$  and decide whether it is positive definite. Does this contradict Theorem 12.7.2 (b)?

*Solution.*  $\mathbf{A} \circ \mathbf{B} = \begin{pmatrix} 2 & -1 \\ -1 & 2 \end{pmatrix}$ , with leading principal minors 2 and 3, so it is positive definite. There is no contradiction: Theorem 12.7.2 (b) is an implication, not an equivalence. Here  $\mathbf{A}$  is only positive semidefinite ( $\mathbf{A} = \mathbf{1}\mathbf{1}^\top$ , so  $\det \mathbf{A} = 0$ ), and the theorem's hypothesis fails — but the conclusion happens to hold anyway. Part (a) is what applies, and it correctly predicts  $\mathbf{A} \circ \mathbf{B} \succeq 0$ .

### 12.7.3 Oppenheim's inequality

Hadamard's inequality of Section 6 compared  $\det \mathbf{A}$  with the product of the diagonal entries of  $\mathbf{A}$ . Oppenheim's inequality does the same for a Hadamard product, and it contains Hadamard's as the special case  $\mathbf{B} = \mathbf{I}$ .

#### Theorem 12.7.3 — Oppenheim's Inequality

Let  $\mathbf{A}, \mathbf{B} \in M_n(F)$  with  $\mathbf{A} \succeq 0$  and  $\mathbf{B} \succeq 0$ . Then

$$\det(\mathbf{A} \circ \mathbf{B}) \geq \det \mathbf{A} \cdot b_{11}b_{22} \cdots b_{nn}.$$

*Idea.* Induct on  $n$ , taking both matrices positive definite first. Split off the  $(1, 1)$  entry of each: the Schur complements  $\mathbf{A}/\alpha$  and  $\mathbf{B}/\beta$  are positive definite of size  $n - 1$ , and a short computation, whose only real step is [Lemma 12.7.1](#) again, shows that the Schur complement of the corner in  $\mathbf{A} \circ \mathbf{B}$  is

$$(\mathbf{A} \circ \mathbf{B})/(\alpha\beta) = (\mathbf{A}/\alpha) \circ \mathbf{B}' + \frac{1}{\alpha} (\mathbf{a}\mathbf{a}^*) \circ (\mathbf{B}/\beta),$$

a positive definite matrix plus a positive semidefinite one. So its determinant is at least that of the first term, by [Lemma 12.6.2](#), and the inductive hypothesis handles that. The semidefinite case follows by replacing  $\mathbf{A}$  and  $\mathbf{B}$  with  $\mathbf{A} + t\mathbf{I}$  and  $\mathbf{B} + t\mathbf{I}$  and letting  $t$  shrink to 0.

*Proof. Case 1:  $\mathbf{A} \succ 0$  and  $\mathbf{B} \succ 0$ .* Induct on  $n$ . For  $n = 1$  both sides are  $a_{11}b_{11}$ .

Let  $n \geq 2$ , and split both matrices with a  $1 \times (n - 1)$  partition:

$$\mathbf{A} = \begin{pmatrix} \alpha & \mathbf{a}^* \\ \mathbf{a} & \mathbf{A}' \end{pmatrix}, \quad \mathbf{B} = \begin{pmatrix} \beta & \mathbf{b}^* \\ \mathbf{b} & \mathbf{B}' \end{pmatrix},$$

with  $\alpha = a_{11} > 0$ ,  $\beta = b_{11} > 0$  and  $\mathbf{a}, \mathbf{b} \in F^{n-1}$ . The Schur complements are

$$\mathbf{A}/\alpha = \mathbf{A}' - \frac{1}{\alpha} \mathbf{a}\mathbf{a}^*, \quad \mathbf{B}/\beta = \mathbf{B}' - \frac{1}{\beta} \mathbf{b}\mathbf{b}^*,$$

and both are positive definite by [Theorem 12.6.1](#) (b). Since conjugation acts entrywise,  $\mathbf{a}^* \circ \mathbf{b}^* = (\mathbf{a} \circ \mathbf{b})^*$ , so

$$\mathbf{A} \circ \mathbf{B} = \begin{pmatrix} \alpha\beta & (\mathbf{a} \circ \mathbf{b})^* \\ \mathbf{a} \circ \mathbf{b} & \mathbf{A}' \circ \mathbf{B}' \end{pmatrix},$$

and  $\alpha\beta > 0$ , so this matrix too has a Schur complement at its corner:

$$(\mathbf{A} \circ \mathbf{B})/(\alpha\beta) = \mathbf{A}' \circ \mathbf{B}' - \frac{1}{\alpha\beta} (\mathbf{a} \circ \mathbf{b})(\mathbf{a} \circ \mathbf{b})^*.$$

By [Lemma 12.7.1](#) the subtracted term is  $\frac{1}{\alpha\beta} (\mathbf{a}\mathbf{a}^*) \circ (\mathbf{b}\mathbf{b}^*)$ . Substituting  $\mathbf{A}' = \mathbf{A}/\alpha + \frac{1}{\alpha} \mathbf{a}\mathbf{a}^*$  and expanding by bilinearity,

$$\begin{aligned} \mathbf{A}' \circ \mathbf{B}' &= (\mathbf{A}/\alpha) \circ \mathbf{B}' + \frac{1}{\alpha} (\mathbf{a}\mathbf{a}^*) \circ \mathbf{B}' \\ &= (\mathbf{A}/\alpha) \circ \mathbf{B}' + \frac{1}{\alpha} (\mathbf{a}\mathbf{a}^*) \circ \left( \mathbf{B}/\beta + \frac{1}{\beta} \mathbf{b}\mathbf{b}^* \right), \end{aligned}$$

and the last term of this is exactly  $\frac{1}{\alpha\beta} (\mathbf{a}\mathbf{a}^*) \circ (\mathbf{b}\mathbf{b}^*)$ . It therefore cancels, leaving

$$(\mathbf{A} \circ \mathbf{B})/(\alpha\beta) = \underbrace{(\mathbf{A}/\alpha) \circ \mathbf{B}'}_{\succ 0} + \underbrace{\frac{1}{\alpha}(\mathbf{a}\mathbf{a}^*) \circ (\mathbf{B}/\beta)}_{\succeq 0}. \quad (*)$$

The first term is positive definite by [Theorem 12.7.2](#) (b), since  $\mathbf{A}/\alpha \succ 0$  and  $\mathbf{B}' \succ 0$  (the latter by [Theorem 12.6.1](#) (a) applied to  $\mathbf{B}$ ). The second is positive semidefinite by [Theorem 12.7.2](#) (a), since  $\mathbf{a}\mathbf{a}^* \succeq 0$  by [Example \(Five Positive Matrices\)](#) (b) and  $\mathbf{B}/\beta \succ 0$ , and  $1/\alpha > 0$ .

Now compute. By [Theorem 7.3.2](#) applied to  $\mathbf{A} \circ \mathbf{B}$ , then  $(*)$  with [Lemma 12.6.2](#), then the inductive hypothesis applied to the pair  $\mathbf{A}/\alpha$  and  $\mathbf{B}'$ :

$$\begin{aligned} \det(\mathbf{A} \circ \mathbf{B}) &= \alpha\beta \cdot \det[(\mathbf{A} \circ \mathbf{B})/(\alpha\beta)] \\ &\geq \alpha\beta \cdot \det[(\mathbf{A}/\alpha) \circ \mathbf{B}'] \\ &\geq \alpha\beta \cdot \det(\mathbf{A}/\alpha) \cdot b_{22} \cdots b_{nn}. \end{aligned}$$

By [Theorem 7.3.2](#) again, now for  $\mathbf{A}$ , we have  $\det(\mathbf{A}/\alpha) = \det \mathbf{A}/\alpha$ . Substituting and canceling  $\alpha$ ,

$$\det(\mathbf{A} \circ \mathbf{B}) \geq \beta \det \mathbf{A} \cdot b_{22} \cdots b_{nn} = \det \mathbf{A} \cdot b_{11} b_{22} \cdots b_{nn},$$

which completes the induction.

*Case 2:  $\mathbf{A} \succeq 0$  and  $\mathbf{B} \succeq 0$ .* For real  $t > 0$ , the matrices  $\mathbf{A} + t\mathbf{I}$  and  $\mathbf{B} + t\mathbf{I}$  are positive definite, as in the proof of [Corollary 12.6.3](#). Case 1 gives

$$\det[(\mathbf{A} + t\mathbf{I}) \circ (\mathbf{B} + t\mathbf{I})] \geq \det(\mathbf{A} + t\mathbf{I}) \cdot \prod_{i=1}^n (b_{ii} + t).$$

Every entry of  $(\mathbf{A} + t\mathbf{I}) \circ (\mathbf{B} + t\mathbf{I})$  is a polynomial in  $t$ , so both sides are polynomial, hence continuous, functions of  $t$  ([Theorem 6.4.3](#)). Letting  $t$  decrease to 0 preserves the inequality and gives the statement. This proves the theorem. ■

*Remark.* Taking  $\mathbf{B} = \mathbf{I}$  recovers Hadamard's inequality. Indeed  $\mathbf{A} \circ \mathbf{I} = \text{diag}(a_{11}, \dots, a_{nn})$  and  $\prod_i b_{ii} = 1$ , so the conclusion reads  $a_{11} \cdots a_{nn} \geq \det \mathbf{A}$ , which is [Theorem 12.6.4](#) (a). Swapping the roles of  $\mathbf{A}$  and  $\mathbf{B}$  gives the companion bound  $\det(\mathbf{A} \circ \mathbf{B}) \geq \det \mathbf{B} \cdot \prod_i a_{ii}$ , and combining either one with Hadamard's inequality gives  $\det(\mathbf{A} \circ \mathbf{B}) \geq \det \mathbf{A} \det \mathbf{B}$ ; that last consequence is [Exercise \(C2: The product of the determinants\)](#).

## 12.7.4 Where this gets used

The natural home of the Hadamard product is data. A covariance matrix is a Gram matrix ([Theorem 10.2.5](#)), hence positive semidefinite, and rescaling each variable to unit variance turns it into a **correlation matrix**: positive semidefinite with every diagonal entry equal to 1. The Schur product theorem says that the entrywise product of two correlation matrices is another



one — the diagonal entries stay  $1 \cdot 1 = 1$  — so one can damp the correlations of one model by those of another and still have a legitimate covariance structure. Nothing in the definition of “positive semidefinite” makes that obvious, and it is the reason the theorem is quoted far outside linear algebra.

**Warning. Entrywise products preserve positivity; entrywise inverses do not.** Let  $\mathbf{A} = \begin{pmatrix} 2 & 1 \\ 1 & 2 \end{pmatrix}$ , which is positive definite, with leading principal minors 2 and 3. Replacing each entry by its reciprocal gives  $\begin{pmatrix} 1/2 & 1 \\ 1 & 1/2 \end{pmatrix}$ , whose determinant is  $\frac{1}{4} - 1 = -\frac{3}{4} < 0$ : not positive semidefinite. The entrywise reciprocal is not the Hadamard inverse of anything useful, and the matrix inverse  $\mathbf{A}^{-1}$ , which **is** positive definite ([Exercise \(B3: The inverse of a positive definite matrix\)](#)), has nothing to do with it.

## 12.7.5 Exercises

### 12.7.5.1 A. Check your understanding

#### Exercise A1.

- Define  $\mathbf{A} \circ \mathbf{B}$ , and say what shapes the two factors must have.
- What is  $\mathbf{A} \circ \mathbf{I}$ ? What is  $\mathbf{A} \circ \mathbf{J}$ ?
- State the Schur product theorem.
- True or false: if  $\mathbf{A} \succ 0$ , then the matrix with entries  $1/a_{ij}$  (assuming no entry is 0) is positive definite. Justify your answer.
- What does Oppenheim’s inequality say when  $\mathbf{B} = \mathbf{I}$ ?

*Solution.*

- For  $\mathbf{A}, \mathbf{B} \in M_{m \times n}(F)$  of the **same** shape,  $(\mathbf{A} \circ \mathbf{B})_{ij} = a_{ij}b_{ij}$ , and  $\mathbf{A} \circ \mathbf{B}$  has that same shape ([Definition 12.7.1](#)).
- $\mathbf{A} \circ \mathbf{I} = \text{diag}(a_{11}, \dots, a_{nn})$ , and  $\mathbf{A} \circ \mathbf{J} = \mathbf{A}$ , since  $\mathbf{J}$  has every entry 1.
- If  $\mathbf{A} \succeq 0$  and  $\mathbf{B} \succeq 0$ , then  $\mathbf{A} \circ \mathbf{B} \succeq 0$ ; if both are positive definite, so is  $\mathbf{A} \circ \mathbf{B}$  ([Theorem 12.7.2](#)).
- False. For  $\mathbf{A} = \begin{pmatrix} 2 & 1 \\ 1 & 2 \end{pmatrix} \succ 0$ , the entrywise reciprocal is  $\begin{pmatrix} 1/2 & 1 \\ 1 & 1/2 \end{pmatrix}$ , with determinant  $-3/4 < 0$ .
- By [Theorem 12.7.3](#) it says  $\det(\mathbf{A} \circ \mathbf{I}) \geq \det \mathbf{A} \cdot 1$ , that is  $a_{11} \cdots a_{nn} \geq \det \mathbf{A}$ : Hadamard’s inequality, [Theorem 12.6.4](#) (a).

### 12.7.5.2 B. Practice

#### Exercise B1: Compute and decide.

For each pair below, compute  $\mathbf{A} \circ \mathbf{B}$  and determine whether it is positive definite, positive semidefinite but not definite, or neither. Say in each case whether [Theorem 12.7.2](#) applies.

- $\mathbf{A} = \begin{pmatrix} 1 & 1 \\ 1 & 1 \end{pmatrix}$ ,  $\mathbf{B} = \begin{pmatrix} 1 & 1 \\ 1 & 1 \end{pmatrix}$ .

- (b)  $\mathbf{A} = \begin{pmatrix} 1 & 1 \\ 1 & 1 \end{pmatrix}$ ,  $\mathbf{B} = \begin{pmatrix} 1 & 2 \\ 2 & 1 \end{pmatrix}$ .  
 (c)  $\mathbf{A} = \begin{pmatrix} 2 & 1 \\ 1 & 2 \end{pmatrix}$ ,  $\mathbf{B} = \begin{pmatrix} 3 & -1 \\ -1 & 1 \end{pmatrix}$ .

*Solution.*

- (a)  $\mathbf{A} \circ \mathbf{B} = \begin{pmatrix} 1 & 1 \\ 1 & 1 \end{pmatrix}$ . Both factors are  $\mathbf{1}\mathbf{1}^\top \succeq 0$  (Example (Five Positive Matrices) (b)), so Theorem 12.7.2 (a) applies and predicts  $\succeq 0$ . The product has determinant 0, so it is positive semidefinite but not definite.
- (b)  $\mathbf{A} \circ \mathbf{B} = \begin{pmatrix} 1 & 2 \\ 2 & 1 \end{pmatrix}$ , whose determinant is  $1 - 4 = -3 < 0$ : neither. The theorem does not apply, since  $\mathbf{B}$  is not positive semidefinite (its determinant is  $-3$ ). This is what the hypothesis is for.
- (c)  $\mathbf{A} \succ 0$  (minors 2, 3) and  $\mathbf{B} \succ 0$  (minors 3, 2), so Theorem 12.7.2 (b) applies. Indeed  $\mathbf{A} \circ \mathbf{B} = \begin{pmatrix} 6 & -1 \\ -1 & 2 \end{pmatrix}$ , with minors 6 and 11, is positive definite. Theorem 12.7.3 also checks out:  $11 \geq \det \mathbf{A} \cdot b_{11}b_{22} = 3 \cdot 3 = 9$ .

### Exercise B2: Entrywise squares.

Let  $\mathbf{A} = \begin{pmatrix} 1 & 1 & 1 \\ 1 & 2 & 2 \\ 1 & 2 & 3 \end{pmatrix}$ .

- (a) Show that  $\mathbf{A} \succ 0$ .  
 (b) Compute  $\mathbf{A} \circ \mathbf{A}$ , the matrix of squared entries, and verify that it is positive definite.  
 (c) Explain why (b) needed no computation.

*Solution.*

- (a) The leading principal minors are 1,  $2 - 1 = 1$  and, expanding along the first row,  $1(6 - 4) - 1(3 - 2) + 1(2 - 2) = 1$ . All positive, so  $\mathbf{A} \succ 0$  by Theorem 12.1.5 (d).
- (b)  $\mathbf{A} \circ \mathbf{A} = \begin{pmatrix} 1 & 1 & 1 \\ 1 & 4 & 4 \\ 1 & 4 & 9 \end{pmatrix}$ , with leading principal minors 1,  $4 - 1 = 3$  and  $1(36 - 16) - 1(9 - 4) + 1(4 - 4) = 20 - 5 = 15$ . All positive, so it is positive definite.
- (c) Theorem 12.7.2 (b) with  $\mathbf{B} = \mathbf{A}$  gives  $\mathbf{A} \circ \mathbf{A} \succ 0$  immediately.

### 12.7.5.3 C. Going deeper

#### Exercise C1: Polynomials applied entrywise.

Let  $\mathbf{A} \in M_n(F)$  with  $\mathbf{A} \succeq 0$ .

- (a) Prove that the entrywise power  $\mathbf{A}^{\circ m}$ , with  $(i, j)$  entry  $a_{ij}^m$ , is positive semidefinite for every integer  $m \geq 1$ .  
 (b) Let  $p(x) = c_0 + c_1x + \cdots + c_dx^d$  have **non-negative real** coefficients. Prove that the matrix  $p^\circ(\mathbf{A})$  with  $(i, j)$  entry  $p(a_{ij})$  is positive semidefinite.  
 (c) Give an example of  $\mathbf{A} \succeq 0$  and a polynomial  $p$  with a negative coefficient for which  $p^\circ(\mathbf{A})$  is not positive semidefinite.

*Solution.*

- (a) Induct on  $m$ . For  $m = 1$  the claim is the hypothesis. If  $\mathbf{A}^{\circ m} \succeq 0$ , then  $\mathbf{A}^{\circ(m+1)} =$

- $\mathbf{A}^{\circ m} \circ \mathbf{A} \succeq 0$  by [Theorem 12.7.2](#) (a).
- (b) By definition of the entries,  $p^\circ(\mathbf{A}) = c_0 \mathbf{J} + c_1 \mathbf{A} + c_2 \mathbf{A}^{\circ 2} + \cdots + c_d \mathbf{A}^{\circ d}$ , since each side has  $(i, j)$  entry  $\sum_k c_k a_{ij}^k$  (with  $a_{ij}^0 = 1$ , which is the  $(i, j)$  entry of  $\mathbf{J}$ ). Now  $\mathbf{J} = \mathbf{1}\mathbf{1}^\top \succeq 0$  by [Example \(Five Positive Matrices\)](#) (b), each  $\mathbf{A}^{\circ k} \succeq 0$  by (a), and each  $c_k \geq 0$ . A non-negative combination of positive semidefinite matrices is positive semidefinite ([Exercise \(C2: The cone of positive matrices\)](#) (a)).
- (c) Take  $\mathbf{A} = \mathbf{J} \in M_2(\mathbb{R})$ , which is positive semidefinite, and  $p(x) = x - 2$ . Then  $p^\circ(\mathbf{A})$  has every entry  $-1$ , that is  $p^\circ(\mathbf{A}) = -\mathbf{J}$ , and  $\langle -\mathbf{J}\mathbf{e}_1, \mathbf{e}_1 \rangle = -1 < 0$ . So the hypothesis on the coefficients cannot be dropped.

### Exercise C2: The product of the determinants.

Let  $\mathbf{A}, \mathbf{B} \in M_n(F)$  with  $\mathbf{A} \succeq 0$  and  $\mathbf{B} \succeq 0$ . Prove that

$$\det(\mathbf{A} \circ \mathbf{B}) \geq \det \mathbf{A} \cdot \det \mathbf{B}.$$

Is the inequality ever an equality with both matrices positive definite?

*Solution.* By [Theorem 12.7.3](#),  $\det(\mathbf{A} \circ \mathbf{B}) \geq \det \mathbf{A} \cdot \prod_i b_{ii}$ . By [Theorem 12.6.4](#) (a) applied to  $\mathbf{B} \succeq 0$ ,  $\prod_i b_{ii} \geq \det \mathbf{B}$ . Both  $\det \mathbf{A} \geq 0$  and  $\prod_i b_{ii} \geq 0$  (the diagonal entries of a positive semidefinite matrix are  $\geq 0$  by [Exercise \(C1: Zero trace forces zero\)](#) (a)), so multiplying the second inequality by  $\det \mathbf{A} \geq 0$  and chaining,

$$\det(\mathbf{A} \circ \mathbf{B}) \geq \det \mathbf{A} \prod_i b_{ii} \geq \det \mathbf{A} \det \mathbf{B}.$$

Yes, equality can occur: take  $\mathbf{A} = \mathbf{B} = \mathbf{I}_n$ , so that  $\mathbf{A} \circ \mathbf{B} = \mathbf{I}_n$  and all three quantities are 1. More generally, take  $\mathbf{A} \succ 0$  diagonal and  $\mathbf{B} = \mathbf{I}_n$ .

### Exercise C3: The rank of an entrywise product.

Let  $\mathbf{A}, \mathbf{B} \in M_n(F)$  with  $\mathbf{A} \succeq 0$  and  $\mathbf{B} \succeq 0$ , of ranks  $r$  and  $s$ . Prove that

$$\text{rank}(\mathbf{A} \circ \mathbf{B}) \leq rs.$$

*Hint: how many terms in the proof of [Theorem 12.7.2](#) (a) are actually non-zero?*

*Solution.* Write  $\mathbf{A} = \sum_i \lambda_i \mathbf{u}_i \mathbf{u}_i^*$  and  $\mathbf{B} = \sum_j \mu_j \mathbf{w}_j \mathbf{w}_j^*$  as in the proof of [Theorem 12.7.2](#) (a). The number of non-zero eigenvalues of  $\mathbf{A}$  is  $\text{rank } \mathbf{A} = r$ : diagonalizing,  $\mathbf{A} = \mathbf{U} \mathbf{D} \mathbf{U}^*$  with  $\mathbf{U}$  invertible, so  $\text{rank } \mathbf{A} = \text{rank } \mathbf{D}$  by [Theorem 2.5.9](#), and  $\text{rank } \mathbf{D}$  counts the non-zero  $\lambda_i$ . Likewise exactly  $s$  of the  $\mu_j$  are non-zero. Dropping the vanishing terms,

$$\mathbf{A} \circ \mathbf{B} = \sum_{\lambda_i \neq 0} \sum_{\mu_j \neq 0} \lambda_i \mu_j \mathbf{z}_{ij} \mathbf{z}_{ij}^*, \quad \mathbf{z}_{ij} = \mathbf{u}_i \circ \mathbf{w}_j,$$

a sum of  $rs$  matrices, each of rank at most 1, since every column of  $\mathbf{z}_{ij} \mathbf{z}_{ij}^*$  lies in  $\text{span}(\mathbf{z}_{ij})$ .

The rank of a sum of two matrices is at most the sum of their ranks ([Exercise \(C1: Rank of a sum\)](#)); applying that repeatedly to the  $rs$  summands gives  $\text{rank}(\mathbf{A} \circ \mathbf{B}) \leq rs$ .

## 12.8 The Singular Value Decomposition

Chapter 11 answered one question completely: an operator has an orthonormal basis of eigenvectors exactly when it is normal. That answer is useless for a  $3 \times 5$  matrix, which has no eigenvalues at all — the equation  $\mathbf{A}\mathbf{x} = \lambda\mathbf{x}$  is not even type-correct, because  $\mathbf{A}\mathbf{x}$  and  $\mathbf{x}$  live in different spaces. This section shows that every matrix, of every shape, still admits an orthonormal-to-orthogonal description, and that the numbers in it are the eigenvalues of a positive semidefinite matrix we already understand. It is the most-used factorization in applied linear algebra, and the whole of it comes from applying the spectral theorem to  $\mathbf{A}^*\mathbf{A}$ .

Throughout,  $F$  is  $\mathbb{R}$  or  $\mathbb{C}$ ,  $\mathbf{A}^*$  is the conjugate transpose (the transpose when  $F = \mathbb{R}$ ), and *unitary* means unitary over  $\mathbb{C}$  and orthogonal over  $\mathbb{R}$ , as in [Definition 10.7.2](#). Both  $F^m$  and  $F^n$  carry their standard inner products.

### 12.8.1 Where to look when there are no eigenvalues

Let  $\mathbf{A} \in M_{m \times n}(F)$ . There is exactly one square matrix that can be built from  $\mathbf{A}$  at no cost, and Chapter 10 has already used it twice: the  $n \times n$  matrix  $\mathbf{A}^*\mathbf{A}$  of the normal equations. It is self-adjoint, since  $(\mathbf{A}^*\mathbf{A})^* = \mathbf{A}^*\mathbf{A}^{**} = \mathbf{A}^*\mathbf{A}$ , and it is positive semidefinite for the reason that makes every argument in this chapter work:

$$\langle \mathbf{A}^*\mathbf{A}\mathbf{x}, \mathbf{x} \rangle = \langle \mathbf{A}\mathbf{x}, \mathbf{A}\mathbf{x} \rangle = \|\mathbf{A}\mathbf{x}\|^2 \geq 0. \quad (12.8.1)$$

So  $\mathbf{A}^*\mathbf{A} \succeq 0$ , and [Theorem 12.1.4](#) says its eigenvalues are non-negative real numbers, while the spectral theorem supplies an orthonormal basis of  $F^n$  consisting of its eigenvectors. Those eigenvalues are attached to  $\mathbf{A}$ , they exist for every shape of  $\mathbf{A}$ , and [Eq. 12.8.1](#) says what they measure: how much  $\mathbf{A}$  stretches.

*A rectangular matrix has no eigenvalues of its own, but  $\mathbf{A}^*\mathbf{A}$  does, and their square roots are the stretching factors of  $\mathbf{A}$ .*

#### Definition 12.8.1 — Singular Values

Let  $\mathbf{A} \in M_{m \times n}(F)$  and put  $p = \min(m, n)$ . Let

$$\lambda_1 \geq \lambda_2 \geq \cdots \geq \lambda_p \geq 0$$

be the eigenvalues of  $\mathbf{A}^*\mathbf{A} \in M_n(F)$ , listed **with multiplicity** in decreasing order. The **singular values** of  $\mathbf{A}$  are

$$\sigma_i(\mathbf{A}) := \sqrt{\lambda_i} \quad (i = 1, \dots, p),$$

so that  $\sigma_1(\mathbf{A}) \geq \sigma_2(\mathbf{A}) \geq \cdots \geq \sigma_p(\mathbf{A}) \geq 0$ . We write  $\sigma_i$  when  $\mathbf{A}$  is clear.

Clause by clause. The list has exactly  $p = \min(m, n)$  entries, not  $n$ : when  $n > m$  the matrix  $\mathbf{A}^*\mathbf{A}$  has rank  $\text{rank } \mathbf{A} \leq m < n$  by [Corollary 10.6.6](#) (b), so  $\lambda_{m+1} = \cdots = \lambda_n = 0$  and the discarded eigenvalues are all zero. The square root is the ordinary non-negative real square root, available because every  $\lambda_i \geq 0$ . The ordering is **decreasing**, matching the book's convention  $\lambda_1(\mathbf{A}) \geq \cdots \geq \lambda_n(\mathbf{A})$  for the eigenvalues of a Hermitian matrix, and it is part of the definition: “the second singular value” means a definite number. Finally, the singular values are attached to  $\mathbf{A}$  itself, with no choices made, because the eigenvalue list of  $\mathbf{A}^*\mathbf{A}$  is.

Four examples, simplest first.

1. **The identity.**  $\mathbf{A} = \mathbf{I}_n$  gives  $\mathbf{A}^*\mathbf{A} = \mathbf{I}_n$ , so every  $\sigma_i = 1$ . More generally a unitary  $\mathbf{U} \in M_n(F)$  has  $\mathbf{U}^*\mathbf{U} = \mathbf{I}$ , so all its singular values are 1. A unitary matrix stretches nothing in any direction, and the list of singular values says exactly that.
2. **A rectangular diagonal matrix.** If  $\mathbf{A} \in M_{m \times n}(F)$  has  $a_{ii} = d_i$  for  $i \leq p$  and all other entries 0, then  $\mathbf{A}^*\mathbf{A} = \text{diag}(|d_1|^2, \dots)$  and the singular values are  $|d_1|, \dots, |d_p|$  **sorted downwards**. The absolute value matters: the singular values of  $\text{diag}(1, -2)$  are 2, 1.
3. **A positive semidefinite matrix.** If  $\mathbf{A} \succeq 0$  then  $\mathbf{A}^*\mathbf{A} = \mathbf{A}^2$ , whose eigenvalues are  $\lambda_i(\mathbf{A})^2$  with  $\lambda_i(\mathbf{A}) \geq 0$ , so  $\sigma_i(\mathbf{A}) = \lambda_i(\mathbf{A})$ . For the matrices this chapter started with, singular values and eigenvalues coincide.
4. **A single row.** For  $\mathbf{A} = \mathbf{a}^* \in M_{1 \times n}(F)$  with  $\mathbf{a} \neq \mathbf{0}$ ,  $p = 1$  and  $\mathbf{A}^*\mathbf{A} = \mathbf{a}\mathbf{a}^*$ , whose non-zero eigenvalue is  $\|\mathbf{a}\|^2$ . So  $\sigma_1 = \|\mathbf{a}\|$ . The degenerate case  $\mathbf{A} = \mathbf{0}$  has every  $\sigma_i = 0$ , and it matters: it is the only matrix with  $\sigma_1 = 0$ , since  $\sigma_1 = 0$  forces  $\mathbf{A}^*\mathbf{A} = \mathbf{0}$  and then  $\|\mathbf{A}\mathbf{x}\|^2 = 0$  for every  $\mathbf{x}$  by [Eq. 12.8.1](#).

Now the non-example, by minimal change. Replace “square roots of the eigenvalues of  $\mathbf{A}^*\mathbf{A}$ ” by “absolute values of the eigenvalues of  $\mathbf{A}$ ”. For  $\mathbf{A} \succeq 0$ , and more generally for any normal  $\mathbf{A}$ , the two agree ([Exercise \(C3: Singular values equal to eigenvalue moduli\)](#)). But the second recipe is undefined for a non-square matrix, and for

$$\mathbf{N} = \begin{pmatrix} 0 & 1 \\ 0 & 0 \end{pmatrix}$$

it returns 0, 0, while  $\mathbf{N}^*\mathbf{N} = \text{diag}(0, 1)$  gives the singular values 1, 0. The clause that fails is the one that does all the work:  $\mathbf{A}^*\mathbf{A}$  is positive semidefinite for **every**  $\mathbf{A}$ , whereas  $\mathbf{A}$  itself may have no useful spectrum.

**Warning. Singular values are not eigenvalues.** The matrix  $\mathbf{N}$  above has both eigenvalues equal to 0 and largest singular value 1: it annihilates  $\mathbf{e}_1$  but sends  $\mathbf{e}_2$  to a unit vector, so calling it “small” because its spectrum is  $\{0\}$  is wrong. Singular values are always real and non-negative; eigenvalues need not be either. The two lists agree, after taking absolute values, exactly for normal matrices — this is [Exercise \(C3: Singular values equal to eigenvalue moduli\)](#), and Chapter 11’s Schur inequality is what makes the converse true.

**Quick check.** What are the singular values of an orthogonal projection  $\mathbf{P} \in M_n(F)$  onto an  $r$ -dimensional subspace?

**Solution.**  $\mathbf{P}^*\mathbf{P} = \mathbf{P}^2 = \mathbf{P}$ , because an orthogonal projection is self-adjoint and idempotent. So the eigenvalues of  $\mathbf{P}^*\mathbf{P}$  are those of  $\mathbf{P}$ , namely 1 with multiplicity  $r$  and 0 with

multiplicity  $n - r$ . Sorted downwards, the singular values are  $\sigma_1 = \cdots = \sigma_r = 1$  and  $\sigma_{r+1} = \cdots = \sigma_n = 0$ . This is also Example 3 above, since  $\mathbf{P} \succeq 0$ .

## 12.8.2 The theorem

Here is the payoff. The eigenvectors of  $\mathbf{A}^* \mathbf{A}$  are an orthonormal basis of the domain; the claim is that  $\mathbf{A}$  carries them to an **orthogonal** list in the codomain, whose lengths are the singular values.

### Theorem 12.8.1 — Singular Value Decomposition

Let  $\mathbf{A} \in M_{m \times n}(F)$ , let  $p = \min(m, n)$  and let  $\sigma_1 \geq \cdots \geq \sigma_p \geq 0$  be its singular values. Then there are unitary matrices  $\mathbf{U} \in M_m(F)$  and  $\mathbf{V} \in M_n(F)$  such that

$$\mathbf{A} = \mathbf{U} \mathbf{\Sigma} \mathbf{V}^*,$$

where  $\mathbf{\Sigma} \in M_{m \times n}(\mathbb{R})$  has  $(\mathbf{\Sigma})_{ii} = \sigma_i$  for  $i = 1, \dots, p$  and every other entry 0. Moreover the number of non-zero  $\sigma_i$  equals  $\text{rank } \mathbf{A}$ .

*Idea.* Picture the  $2 \times 2$  real case. A matrix carries the unit circle to an ellipse, and an ellipse has two perpendicular axes; the theorem says that the two unit vectors sent to the ends of those axes are themselves perpendicular. That is exactly what Chapter 11's principal axis theorem said about the level curves of a quadratic form, and the link is that  $\|\mathbf{A}\mathbf{x}\|^2 = \mathbf{x}^*(\mathbf{A}^* \mathbf{A})\mathbf{x}$  is a quadratic form, with matrix  $\mathbf{A}^* \mathbf{A}$ . So the plan is: ① diagonalize  $\mathbf{A}^* \mathbf{A}$  by the spectral theorem, getting an orthonormal basis  $\mathbf{v}_1, \dots, \mathbf{v}_n$  with  $\mathbf{A}^* \mathbf{A} \mathbf{v}_i = \sigma_i^2 \mathbf{v}_i$ ; ② compute  $\langle \mathbf{A} \mathbf{v}_i, \mathbf{A} \mathbf{v}_j \rangle$  and watch the eigenvalue equation force it to be  $\sigma_i^2 \delta_{ij}$ , so the images are already orthogonal; ③ normalize the non-zero ones to unit vectors  $\mathbf{u}_i$ , and, if there are fewer than  $m$  of them, pad the list out to an orthonormal basis of  $F^m$ ; ④ read  $\mathbf{A} \mathbf{V} = \mathbf{U} \mathbf{\Sigma}$  off column by column.

*Proof. Step 1: diagonalize  $\mathbf{A}^* \mathbf{A}$ .* The matrix  $\mathbf{A}^* \mathbf{A} \in M_n(F)$  is self-adjoint, and Eq. 12.8.1 makes it positive semidefinite in the sense of Definition 12.1.1. By the spectral theorem in matrix form — Corollary 11.5.2 when  $F = \mathbb{R}$ , Corollary 11.4.3 when  $F = \mathbb{C}$ , each of which allows the eigenvalues to be listed in a prescribed order — there is a unitary  $\mathbf{V} \in M_n(F)$  with columns  $\mathbf{v}_1, \dots, \mathbf{v}_n$  such that

$$\mathbf{A}^* \mathbf{A} \mathbf{v}_i = \lambda_i \mathbf{v}_i, \quad \lambda_1 \geq \cdots \geq \lambda_n \geq 0,$$

the eigenvalues being non-negative by Theorem 12.1.4. Set  $\sigma_i = \sqrt{\lambda_i}$  for all  $i \leq n$ ; for  $i \leq p$  these are the singular values of Definition 12.8.1.

**Step 2: the images are orthogonal.** For all  $i, j$ , using  $\langle \mathbf{x}, \mathbf{y} \rangle = \mathbf{y}^* \mathbf{x}$  and the eigenvalue equation,

$$\langle \mathbf{A} \mathbf{v}_i, \mathbf{A} \mathbf{v}_j \rangle = \langle \mathbf{A}^* \mathbf{A} \mathbf{v}_i, \mathbf{v}_j \rangle = \lambda_i \langle \mathbf{v}_i, \mathbf{v}_j \rangle = \lambda_i \delta_{ij},$$

the first equality by Definition 10.6.1 and the last because  $\mathbf{V}$  is unitary, so its columns are orthonormal (Theorem 10.7.3 (f)). In particular  $\|\mathbf{A} \mathbf{v}_i\|^2 = \lambda_i = \sigma_i^2$ , so

$$\mathbf{A} \mathbf{v}_i = \mathbf{0} \iff \sigma_i = 0. \tag{12.8.2}$$

**Step 3: normalize, and count.** Let  $r$  be the number of indices with  $\sigma_i > 0$ ; since the  $\sigma_i$  decrease, these are  $i = 1, \dots, r$ . For  $i \leq r$  put

$$\mathbf{u}_i := \frac{1}{\sigma_i} \mathbf{A} \mathbf{v}_i,$$

which is legitimate because  $\sigma_i \neq 0$ . By Step 2,  $\langle \mathbf{u}_i, \mathbf{u}_j \rangle = \lambda_i \delta_{ij} / (\sigma_i \sigma_j) = \delta_{ij}$  for  $i, j \leq r$ , so  $(\mathbf{u}_1, \dots, \mathbf{u}_r)$  is an orthonormal list in  $F^m$ .

We claim  $r = \text{rank } \mathbf{A}$ . Any  $\mathbf{x} \in F^n$  is  $\mathbf{x} = \sum_i c_i \mathbf{v}_i$  since  $(\mathbf{v}_1, \dots, \mathbf{v}_n)$  is a basis, and then  $\mathbf{A} \mathbf{x} = \sum_{i \leq r} c_i \sigma_i \mathbf{u}_i$  by Eq. 12.8.2. Hence  $\text{im } \mathbf{A} = \text{span}(\mathbf{u}_1, \dots, \mathbf{u}_r)$ , and the  $\mathbf{u}_i$  are orthonormal, hence independent (Theorem 10.1.4), so  $\text{rank } \mathbf{A} = r$ . In particular  $r \leq \min(m, n) = p$ , so every index  $i$  with  $\sigma_i > 0$  satisfies  $i \leq p$ , and the non-zero singular values are exactly  $\sigma_1, \dots, \sigma_r$ .

**Step 4: pad the list.** If  $r = m$  the list  $(\mathbf{u}_1, \dots, \mathbf{u}_r)$  is already an orthonormal basis of  $F^m$ , and we set  $\mathbf{U} = (\mathbf{u}_1 \mid \dots \mid \mathbf{u}_m)$ . If  $r < m$ , apply Corollary 10.2.3 to the orthonormal list  $(\mathbf{u}_1, \dots, \mathbf{u}_r)$  in  $F^m$ : there are  $\mathbf{u}_{r+1}, \dots, \mathbf{u}_m$  making  $(\mathbf{u}_1, \dots, \mathbf{u}_m)$  an orthonormal basis of  $F^m$ . Either way let  $\mathbf{U} \in M_m(F)$  have these columns; it is unitary because its columns are orthonormal (Theorem 10.7.3 (f)). Note that the padding vectors are **not** determined by  $\mathbf{A}$ , and nothing below uses any property of them beyond orthonormality.

**Step 5: read off the columns.** Let  $\Sigma \in M_{m \times n}(\mathbb{R})$  be as in the statement. Column  $i$  of  $\mathbf{U}\Sigma$  is  $\mathbf{U}$  times column  $i$  of  $\Sigma$ , which is  $\sigma_i \mathbf{e}_i$  for  $i \leq p$  and  $\mathbf{0}$  for  $i > p$ ; so column  $i$  of  $\mathbf{U}\Sigma$  is  $\sigma_i \mathbf{u}_i$  for  $i \leq p$  and  $\mathbf{0}$  for  $i > p$ . Column  $i$  of  $\mathbf{A}\mathbf{V}$  is  $\mathbf{A} \mathbf{v}_i$ . Compare:

- for  $i \leq r$ ,  $\mathbf{A} \mathbf{v}_i = \sigma_i \mathbf{u}_i$  by the definition of  $\mathbf{u}_i$ ;
- for  $r < i \leq p$ ,  $\sigma_i = 0$ , so both columns are  $\mathbf{0}$  by Eq. 12.8.2;
- for  $i > p$  (possible only when  $n > m$ ), we have  $i > p \geq r$ , so  $\sigma_i = 0$  and  $\mathbf{A} \mathbf{v}_i = \mathbf{0}$ , and both columns are  $\mathbf{0}$ .

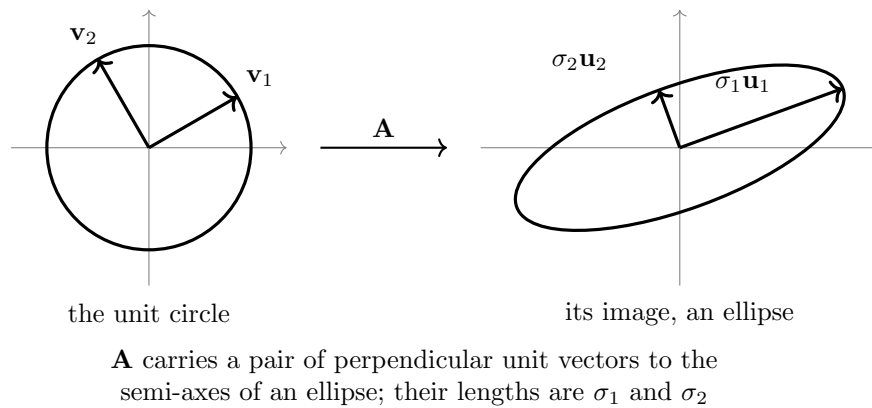
Hence  $\mathbf{A}\mathbf{V} = \mathbf{U}\Sigma$ . Since  $\mathbf{V}$  is unitary,  $\mathbf{V}^{-1} = \mathbf{V}^*$ , so  $\mathbf{A} = \mathbf{U}\Sigma\mathbf{V}^*$ . This proves the theorem. ■

A factorization  $\mathbf{A} = \mathbf{U}\Sigma\mathbf{V}^*$  as in the theorem is called a **singular value decomposition** of  $\mathbf{A}$ , the columns  $\mathbf{u}_i$  of  $\mathbf{U}$  are **left singular vectors** and the columns  $\mathbf{v}_i$  of  $\mathbf{V}$  are **right singular vectors**. The two equations to keep in mind are the ones the proof produced,

$$\mathbf{A} \mathbf{v}_i = \sigma_i \mathbf{u}_i, \quad \mathbf{A}^* \mathbf{u}_i = \sigma_i \mathbf{v}_i \quad (i \leq r), \quad (12.8.3)$$

the second obtained by applying  $\mathbf{A}^*$  to the first and using  $\mathbf{A}^* \mathbf{A} \mathbf{v}_i = \sigma_i^2 \mathbf{v}_i$ . In words:  $\mathbf{A}$  matches up two orthonormal bases, one in each space, stretching the  $i$ -th direction by  $\sigma_i$ , and  $\mathbf{A}^*$  runs the same correspondence backwards with the same factors.

Here is the picture the Idea described, in the real  $2 \times 2$  case.



Compare the principal axes figure of Chapter 11 §05. There the ellipse was a level curve  $\mathbf{x}^\top \mathbf{A} \mathbf{x} = 1$  of a quadratic form and the axes lay along the eigenvectors of a symmetric  $\mathbf{A}$ ; here the ellipse is the *image* of the unit circle under an arbitrary  $\mathbf{A}$ . The two pictures are the same picture, because  $\|\mathbf{A}\mathbf{x}\|^2 = \mathbf{x}^*(\mathbf{A}^*\mathbf{A})\mathbf{x}$  is a quadratic form with matrix  $\mathbf{A}^*\mathbf{A}$ , and the right singular vectors of  $\mathbf{A}$  are its principal axes. A large  $\sigma_i$  now means a **long** semi-axis, whereas in [Theorem 11.5.5](#) a large  $\lambda_i$  meant a short one; the reason is that the two pictures are related by taking reciprocals of square roots.

### 12.8.3 What is unique and what is not

The singular values were defined before the theorem, so they cannot depend on any choice. What needs saying is the converse: any factorization of the stated shape recovers them.

#### Theorem 12.8.2 — The Singular Values Are Determined

Let  $\mathbf{A} \in M_{m \times n}(F)$  and suppose  $\mathbf{A} = \mathbf{U}\mathbf{\Sigma}\mathbf{V}^*$  with  $\mathbf{U}, \mathbf{V}$  unitary and  $\mathbf{\Sigma} \in M_{m \times n}(\mathbb{R})$  of the shape in [Theorem 12.8.1](#), with diagonal entries  $s_1 \geq \dots \geq s_p \geq 0$ . Then  $s_i = \sigma_i(\mathbf{A})$  for every  $i$ . Moreover:

- (a) the columns of  $\mathbf{V}$  are forced to be an orthonormal basis of  $F^n$  consisting of eigenvectors of  $\mathbf{A}^*\mathbf{A}$ , with  $\mathbf{A}^*\mathbf{A}\mathbf{v}_i = \sigma_i^2\mathbf{v}_i$  for every  $i \leq n$ , where  $\sigma_i := 0$  for  $p < i \leq n$ ; and any such basis occurs;
- (b) once  $\mathbf{V}$  is chosen,  $\mathbf{u}_i = \mathbf{A}\mathbf{v}_i/\sigma_i$  is forced for every  $i \leq r = \text{rank } \mathbf{A}$ , while  $\mathbf{u}_{r+1}, \dots, \mathbf{u}_m$  may be any orthonormal basis of  $(\text{im } \mathbf{A})^\perp$ .

*Proof.* From  $\mathbf{A} = \mathbf{U}\mathbf{\Sigma}\mathbf{V}^*$  and  $\mathbf{U}^*\mathbf{U} = \mathbf{I}_m$ ,

$$\mathbf{A}^*\mathbf{A} = \mathbf{V}\mathbf{\Sigma}^*\mathbf{U}^*\mathbf{U}\mathbf{\Sigma}\mathbf{V}^* = \mathbf{V}(\mathbf{\Sigma}^*\mathbf{\Sigma})\mathbf{V}^*,$$

and  $\mathbf{\Sigma}^*\mathbf{\Sigma} \in M_n(\mathbb{R})$  is the diagonal matrix with entries  $s_1^2, \dots, s_p^2$  followed by zeros. So  $\mathbf{A}^*\mathbf{A}$  is unitarily diagonalized with this diagonal, and its eigenvalue list in decreasing order is therefore  $s_1^2 \geq \dots \geq s_p^2 \geq 0 \geq \dots$  — already sorted, since the  $s_i$  are. Comparing with [Definition 12.8.1](#) gives  $s_i^2 = \sigma_i(\mathbf{A})^2$ , hence  $s_i = \sigma_i(\mathbf{A})$  because both are non-negative.

- (a) The displayed equation says the  $i$ -th column of  $\mathbf{V}$  satisfies  $\mathbf{A}^*\mathbf{A}\mathbf{v}_i = \sigma_i^2\mathbf{v}_i$ , and the columns of a unitary matrix are orthonormal. Conversely, given any orthonormal



basis of eigenvectors of  $\mathbf{A}^*\mathbf{A}$  ordered by decreasing eigenvalue, Steps 2–5 of the proof of [Theorem 12.8.1](#) build a factorization from it.

- (b) For  $i \leq r$  we have  $\sigma_i \neq 0$ , and comparing column  $i$  of  $\mathbf{A}\mathbf{V} = \mathbf{U}\mathbf{\Sigma}$  gives  $\mathbf{A}\mathbf{v}_i = \sigma_i\mathbf{u}_i$ , so  $\mathbf{u}_i = \mathbf{A}\mathbf{v}_i/\sigma_i$ . For  $i > r$  the corresponding column of  $\mathbf{\Sigma}$  is zero, so  $\mathbf{u}_i$  is unconstrained by the equation  $\mathbf{A}\mathbf{V} = \mathbf{U}\mathbf{\Sigma}$ ; the only constraint left is that  $(\mathbf{u}_1, \dots, \mathbf{u}_m)$  be orthonormal, which says exactly that  $(\mathbf{u}_{r+1}, \dots, \mathbf{u}_m)$  is an orthonormal basis of  $\text{span}(\mathbf{u}_1, \dots, \mathbf{u}_r)^\perp = (\text{im } \mathbf{A})^\perp$ .

■

So  $\mathbf{\Sigma}$  is unique and  $\mathbf{U}, \mathbf{V}$  are not. The freedom is of two kinds, and both are visible in small cases. If  $\mathbf{A}^*\mathbf{A}$  has a repeated eigenvalue, any orthonormal basis of that eigenspace may be used:  $\mathbf{A} = \mathbf{I}_2$  has  $\mathbf{\Sigma} = \mathbf{I}_2$  and  $\mathbf{I}_2 = \mathbf{U}\mathbf{I}_2\mathbf{U}^*$  for **every** unitary  $\mathbf{U}$ , with  $\mathbf{V} = \mathbf{U}$ . If the eigenvalues are distinct, each  $\mathbf{v}_i$  is still free up to a scalar of modulus 1, and replacing  $\mathbf{v}_i$  by  $c\mathbf{v}_i$  replaces  $\mathbf{u}_i$  by  $c\mathbf{u}_i$ , leaving  $\sigma_i\mathbf{u}_i\mathbf{v}_i^*$  unchanged because  $c\bar{c} = 1$ .

*Remark.* That last observation is why the factorization is worth having even though it is not unique: the *products*  $\sigma_i\mathbf{u}_i\mathbf{v}_i^*$  are what later sections use, and those are unchanged by the phase freedom. Section 11 will define the pseudoinverse through a singular value decomposition and will have to check exactly this.

## 12.8.4 The compact form, and the four subspaces

Steps 3 and 5 of the proof produced more than the statement recorded: they located  $\text{im } \mathbf{A}$  and  $\ker \mathbf{A}$  inside the two bases. Writing the factorization so that only the non-zero singular values appear makes this visible.

### Theorem 12.8.3 — Compact Singular Value Decomposition

Let  $\mathbf{A} \in M_{m \times n}(F)$  have rank  $r$  and let  $\mathbf{A} = \mathbf{U}\mathbf{\Sigma}\mathbf{V}^*$  be as in [Theorem 12.8.1](#), with columns  $\mathbf{u}_1, \dots, \mathbf{u}_m$  and  $\mathbf{v}_1, \dots, \mathbf{v}_n$ . Then

$$\mathbf{A} = \sum_{i=1}^r \sigma_i \mathbf{u}_i \mathbf{v}_i^*,$$

a sum of  $r$  matrices of rank 1. Equivalently  $\mathbf{A} = \mathbf{U}_r \mathbf{\Sigma}_r \mathbf{V}_r^*$ , where  $\mathbf{U}_r \in M_{m \times r}(F)$  and  $\mathbf{V}_r \in M_{n \times r}(F)$  hold the first  $r$  columns of  $\mathbf{U}$  and  $\mathbf{V}$ , and  $\mathbf{\Sigma}_r = \text{diag}(\sigma_1, \dots, \sigma_r)$  is square and **invertible**.

*Proof.* Write  $\mathbf{A} = (\mathbf{U}\mathbf{\Sigma})\mathbf{V}^*$ . The columns of  $\mathbf{U}\mathbf{\Sigma}$  are  $\sigma_1\mathbf{u}_1, \dots, \sigma_p\mathbf{u}_p$  followed by zero columns, as computed in Step 5 of [Theorem 12.8.1](#), and the rows of  $\mathbf{V}^*$  are  $\mathbf{v}_1^*, \dots, \mathbf{v}_n^*$ . By the outer product expansion of a matrix product ([Corollary 7.1.3](#)),

$$\mathbf{A} = \sum_{i=1}^p (\sigma_i \mathbf{u}_i) \mathbf{v}_i^* = \sum_{i=1}^r \sigma_i \mathbf{u}_i \mathbf{v}_i^*,$$

the terms with  $i > r$  vanishing because  $\sigma_i = 0$  there. Each  $\mathbf{u}_i\mathbf{v}_i^*$  is non-zero with all columns multiples of  $\mathbf{u}_i$ , hence of rank 1. The same computation with the truncated matrices gives  $\mathbf{U}_r \mathbf{\Sigma}_r \mathbf{V}_r^* = \sum_{i \leq r} \sigma_i \mathbf{u}_i \mathbf{v}_i^*$ , and  $\mathbf{\Sigma}_r$  is invertible because its diagonal entries

$\sigma_1, \dots, \sigma_r$  are non-zero.

■

### Corollary 12.8.4 — The Four Fundamental Subspaces from the Compact Form

With the notation of [Theorem 12.8.3](#):

- (a)  $(\mathbf{u}_1, \dots, \mathbf{u}_r)$  is an orthonormal basis of  $\text{im } \mathbf{A} = \text{col}(\mathbf{A})$ ;
- (b)  $(\mathbf{u}_{r+1}, \dots, \mathbf{u}_m)$  is an orthonormal basis of  $\ker \mathbf{A}^*$ ;
- (c)  $(\mathbf{v}_1, \dots, \mathbf{v}_r)$  is an orthonormal basis of  $\text{im } \mathbf{A}^*$ ;
- (d)  $(\mathbf{v}_{r+1}, \dots, \mathbf{v}_n)$  is an orthonormal basis of  $\ker \mathbf{A}$ .

*Idea.* Two of the four were proved on the way to [Theorem 12.8.1](#). The other two come for free by applying the same statements to  $\mathbf{A}^*$ , whose singular value decomposition is  $\mathbf{A}^* = \mathbf{V}\Sigma^*\mathbf{U}^*$  — the roles of  $\mathbf{u}$  and  $\mathbf{v}$  simply swap.

*Proof. Part (a).* Step 3 of [Theorem 12.8.1](#) showed  $\text{im } \mathbf{A} = \text{span}(\mathbf{u}_1, \dots, \mathbf{u}_r)$ , and these  $r$  vectors are orthonormal.

*Part (d).* By [Eq. 12.8.2](#),  $\mathbf{A}\mathbf{v}_i = \mathbf{0}$  exactly for  $i > r$ , and for  $\mathbf{x} = \sum_i c_i \mathbf{v}_i$  we computed  $\mathbf{A}\mathbf{x} = \sum_{i \leq r} c_i \sigma_i \mathbf{u}_i$ . Since the  $\mathbf{u}_i$  are independent and the  $\sigma_i$  non-zero for  $i \leq r$ ,  $\mathbf{A}\mathbf{x} = \mathbf{0}$  if and only if  $c_1 = \dots = c_r = 0$ , that is,  $\mathbf{x} \in \text{span}(\mathbf{v}_{r+1}, \dots, \mathbf{v}_n)$ .

*Parts (b) and (c).* Taking conjugate transposes,  $\mathbf{A}^* = \mathbf{V}\Sigma^*\mathbf{U}^*$ , and  $\Sigma^* \in M_{n \times m}(\mathbb{R})$  has the same diagonal  $\sigma_1, \dots, \sigma_p$ . So this is a singular value decomposition of  $\mathbf{A}^*$ , with the columns of  $\mathbf{V}$  as left singular vectors and those of  $\mathbf{U}$  as right singular vectors, and  $\text{rank } \mathbf{A}^* = \text{rank } \mathbf{A} = r$  by [Corollary 10.6.6](#) (a). Now the two parts just proved, applied to  $\mathbf{A}^*$  instead of  $\mathbf{A}$ , give exactly (c) and (b).

■

This is the cleanest possible form of [Theorem 10.6.5](#). That theorem said the four subspaces come in two orthogonal pairs,  $F^n = \ker \mathbf{A} \oplus \text{im } \mathbf{A}^*$  and  $F^m = \text{im } \mathbf{A} \oplus \ker \mathbf{A}^*$ . The corollary hands over **orthonormal bases of all four at once**, out of two matrices, and adds the missing piece of information: how  $\mathbf{A}$  maps one pair to the other. Restricted to  $\text{im } \mathbf{A}^*$ , the map  $\mathbf{A}$  is a bijection onto  $\text{im } \mathbf{A}$ , and in the bases  $(\mathbf{v}_1, \dots, \mathbf{v}_r)$  and  $(\mathbf{u}_1, \dots, \mathbf{u}_r)$  its matrix is the diagonal  $\Sigma_r$ . Every matrix, after an orthonormal change of basis on each side, is a positive diagonal matrix padded with zeros.

*Quick check.* Let  $\mathbf{A} \in M_{m \times n}(F)$  have rank  $r$ . Using [Corollary 12.8.4](#), what is  $\dim \ker \mathbf{A}^*$ , and why does the answer not depend on which singular value decomposition was used?

*Solution.* The list  $(\mathbf{u}_{r+1}, \dots, \mathbf{u}_m)$  has  $m - r$  members, so  $\dim \ker \mathbf{A}^* = m - r$ . It cannot depend on the factorization because  $r = \text{rank } \mathbf{A}$  is determined by  $\mathbf{A}$  ([Theorem 12.8.2](#)), and in any case the answer agrees with  $\dim(\text{im } \mathbf{A})^\perp = m - \text{rank } \mathbf{A}$  from [Theorem 10.6.5](#) and [Theorem 10.3.2](#).

## 12.8.5 Two computations and one comparison

**The Frobenius norm.** From  $\|\mathbf{A}\|_F^2 = \text{tr}(\mathbf{A}^* \mathbf{A})$  and  $\mathbf{A}^* \mathbf{A} = \mathbf{V}(\mathbf{\Sigma}^* \mathbf{\Sigma}) \mathbf{V}^*$ , or directly from Lemma 11.11.1,

$$\|\mathbf{A}\|_F^2 = \|\mathbf{U} \mathbf{\Sigma} \mathbf{V}^*\|_F^2 = \|\mathbf{\Sigma}\|_F^2 = \sum_{i=1}^p \sigma_i^2.$$

So the Frobenius norm is the Euclidean length of the singular value list. Section 10 turns this one identity into the best low-rank approximation theorem.

**The largest stretch.** Let  $\mathbf{x} = \sum_i c_i \mathbf{v}_i$  be a unit vector, so  $\sum_i |c_i|^2 = 1$  by Theorem 10.9.2. Then  $\mathbf{A}\mathbf{x} = \sum_{i \leq r} c_i \sigma_i \mathbf{u}_i$ , and since the  $\mathbf{u}_i$  are orthonormal, Pythagoras (Theorem 10.1.3) gives

$$\|\mathbf{A}\mathbf{x}\|^2 = \sum_{i \leq r} \sigma_i^2 |c_i|^2 \leq \sigma_1^2 \sum_i |c_i|^2 = \sigma_1^2,$$

with equality at  $\mathbf{x} = \mathbf{v}_1$ . So  $\sigma_1 = \max\{\|\mathbf{A}\mathbf{x}\| : \|\mathbf{x}\| = 1\}$ : the largest singular value is the largest factor by which  $\mathbf{A}$  can lengthen a vector. Chapter 15 gives that quantity a name, the operator norm  $\|\mathbf{A}\|_2$ , and this computation is why  $\|\mathbf{A}\|_2 = \sigma_1$ .

**Against the eigenvalues.** For a normal  $\mathbf{A} \in M_n(\mathbb{C})$ , write  $\mathbf{A} = \mathbf{U} \mathbf{D} \mathbf{U}^*$  with  $\mathbf{D} = \text{diag}(\lambda_1, \dots, \lambda_n)$  by Corollary 11.4.3. Then

$$\mathbf{A}^* \mathbf{A} = \mathbf{U} \mathbf{D}^* \mathbf{D} \mathbf{U}^* = \mathbf{U} \text{diag}(|\lambda_1|^2, \dots, |\lambda_n|^2) \mathbf{U}^*,$$

so the singular values of  $\mathbf{A}$  are the moduli  $|\lambda_i|$ , sorted downwards. For a general matrix there is no relation at all in one direction —  $\mathbf{N} = \begin{pmatrix} 0 & 1 \\ 0 & 0 \end{pmatrix}$  has spectrum  $\{0\}$  and  $\sigma_1 = 1$  — and only the crude bound  $|\lambda| \leq \sigma_1$  in the other, since  $\mathbf{A}\mathbf{x} = \lambda\mathbf{x}$  with  $\|\mathbf{x}\| = 1$  gives  $|\lambda| = \|\mathbf{A}\mathbf{x}\| \leq \sigma_1$ .

### Example. (A singular value decomposition of a 2 by 3 matrix)

Find a singular value decomposition of

$$\mathbf{A} = \begin{pmatrix} 1 & 0 & 1 \\ 0 & 1 & -1 \end{pmatrix} \in M_{2 \times 3}(\mathbb{R}),$$

and read off orthonormal bases of its four fundamental subspaces.

*Solution.* Here  $m = 2$ ,  $n = 3$ ,  $p = 2$ . The proof asks for the eigenvectors of the  $3 \times 3$  matrix  $\mathbf{A}^\top \mathbf{A}$ , but there is a shortcut worth taking whenever  $m < n$ : run the proof on  $\mathbf{A}^\top$  instead. It produces  $\mathbf{A}^\top = \mathbf{V} \mathbf{\Sigma}^\top \mathbf{U}^\top$ , and transposing that equation gives  $\mathbf{A} = \mathbf{U} \mathbf{\Sigma} \mathbf{V}^\top$ , which is what we want. The eigenvalue computation then lives in the smaller matrix

$$\mathbf{A} \mathbf{A}^\top = \begin{pmatrix} 2 & -1 \\ -1 & 2 \end{pmatrix}.$$

*Step 1.* The characteristic polynomial of  $\mathbf{A} \mathbf{A}^\top$  is  $x^2 - 4x + 3 = (x - 1)(x - 3)$ , so the eigenvalues are 3 and 1 and the singular values are  $\sigma_1 = \sqrt{3}$ ,  $\sigma_2 = 1$ . For  $\lambda = 3$ ,  $\mathbf{A} \mathbf{A}^\top - 3\mathbf{I} = \begin{pmatrix} -1 & -1 \\ -1 & -1 \end{pmatrix}$  gives  $x_1 = -x_2$ , and for  $\lambda = 1$  we get  $x_1 = x_2$ . Normalizing,

$$\mathbf{u}_1 = \frac{1}{\sqrt{2}}(1, -1), \quad \mathbf{u}_2 = \frac{1}{\sqrt{2}}(1, 1).$$

Both singular values are non-zero, so  $r = \text{rank } \mathbf{A} = 2$ .

*Steps 2–4.* The roles of  $\mathbf{u}$  and  $\mathbf{v}$  are swapped along with the roles of  $\mathbf{A}$  and  $\mathbf{A}^\top$ , so here  $\mathbf{v}_i = \mathbf{A}^\top \mathbf{u}_i / \sigma_i$ :

$$\mathbf{v}_1 = \frac{\mathbf{A}^\top(1,-1)}{\sqrt{2}\sqrt{3}} = \frac{1}{\sqrt{6}}(1, -1, 2), \quad \mathbf{v}_2 = \frac{\mathbf{A}^\top(1,1)}{\sqrt{2}} = \frac{1}{\sqrt{2}}(1, 1, 0).$$

This time the padding step is needed, since  $r = 2 < 3$ : a unit vector orthogonal to both is

$$\mathbf{v}_3 = \frac{1}{\sqrt{3}}(-1, 1, 1).$$

*Step 5.* Assembling,

$$\mathbf{U} = \frac{1}{\sqrt{2}} \begin{pmatrix} 1 & 1 \\ -1 & 1 \end{pmatrix}, \quad \mathbf{\Sigma} = \begin{pmatrix} \sqrt{3} & 0 & 0 \\ 0 & 1 & 0 \end{pmatrix},$$

$$\mathbf{V} = \begin{pmatrix} \frac{1}{\sqrt{6}} & \frac{1}{\sqrt{2}} & -\frac{1}{\sqrt{3}} \\ -\frac{1}{\sqrt{6}} & \frac{1}{\sqrt{2}} & \frac{1}{\sqrt{3}} \\ \frac{2}{\sqrt{6}} & 0 & \frac{1}{\sqrt{3}} \end{pmatrix}, \quad \mathbf{A} = \mathbf{U}\mathbf{\Sigma}\mathbf{V}^\top.$$

As a check,  $\|\mathbf{A}\|_F^2 = 1 + 0 + 1 + 0 + 1 + 1 = 4$  and  $\sigma_1^2 + \sigma_2^2 = 3 + 1 = 4$ .

The four subspaces (Corollary 12.8.4), with  $r = 2$ :  $\text{col}(\mathbf{A})$  has the orthonormal basis  $\mathbf{u}_1, \mathbf{u}_2$ , which is all of  $\mathbb{R}^2$ , so  $\text{nul}(\mathbf{A}^\top) = \{\mathbf{0}\}$ ;  $\text{row}(\mathbf{A}) = \text{col}(\mathbf{A}^\top)$  has the orthonormal basis  $\mathbf{v}_1, \mathbf{v}_2$ ; and  $\text{nul}(\mathbf{A}) = \text{span}(\mathbf{v}_3)$  is the line through  $(-1, 1, 1)$ . The last is easy to verify directly:  $\mathbf{A}(-1, 1, 1) = (-1 + 1, 1 - 1) = \mathbf{0}$ .

## 12.8.6 Exercises

### 12.8.6.1 A. Check your understanding

#### Exercise A1.

- Define the singular values of a matrix  $\mathbf{A} \in M_{m \times n}(F)$ . How many are there?
- State the singular value decomposition, with the size and the defining property of each factor.
- True or false: the singular values of  $\mathbf{A}$  are the absolute values of its eigenvalues. Justify your answer.
- How many non-zero singular values does  $\mathbf{A}$  have, and why?
- Given a compact singular value decomposition  $\mathbf{A} = \sum_{i \leq r} \sigma_i \mathbf{u}_i \mathbf{v}_i^*$ , write down orthonormal bases of all four fundamental subspaces of  $\mathbf{A}$ .
- True or false: in  $\mathbf{A} = \mathbf{U}\mathbf{\Sigma}\mathbf{V}^*$  the three factors are determined by  $\mathbf{A}$ . Justify your answer.

*Solution.*

- They are  $\sigma_i = \sqrt{\lambda_i}$  for  $i = 1, \dots, \min(m, n)$ , where  $\lambda_1 \geq \dots \geq \lambda_n \geq 0$  are the eigenvalues of  $\mathbf{A}^* \mathbf{A}$  with multiplicity (Definition 12.8.1). There are  $\min(m, n)$  of them.

- (b) [Theorem 12.8.1](#):  $\mathbf{A} = \mathbf{U}\mathbf{\Sigma}\mathbf{V}^*$  with  $\mathbf{U} \in M_m(F)$  unitary,  $\mathbf{V} \in M_n(F)$  unitary and  $\mathbf{\Sigma} \in M_{m \times n}(\mathbb{R})$  having  $\sigma_1 \geq \cdots \geq \sigma_p \geq 0$  in positions  $(i, i)$  and zeros elsewhere.
- (c) False.  $\mathbf{N} = \begin{pmatrix} 0 & 1 \\ 0 & 0 \end{pmatrix}$  has both eigenvalues 0 and singular values 1, 0. The statement is true for normal matrices ([Exercise \(C3: Singular values equal to eigenvalue moduli\)](#)), and for a non-square matrix it does not even make sense.
- (d) Exactly rank  $\mathbf{A}$ , by [Theorem 12.8.1](#). The reason is that  $\mathbf{A}\mathbf{v}_i = \mathbf{0}$  precisely when  $\sigma_i = 0$ , so the  $\mathbf{v}_i$  with  $\sigma_i = 0$  span  $\ker \mathbf{A}$ .
- (e)  $(\mathbf{u}_1, \dots, \mathbf{u}_r)$  for  $\operatorname{im} \mathbf{A}$ ;  $(\mathbf{u}_{r+1}, \dots, \mathbf{u}_m)$  for  $\ker \mathbf{A}^*$ ;  $(\mathbf{v}_1, \dots, \mathbf{v}_r)$  for  $\operatorname{im} \mathbf{A}^*$ ;  $(\mathbf{v}_{r+1}, \dots, \mathbf{v}_n)$  for  $\ker \mathbf{A}$  ([Corollary 12.8.4](#)).
- (f) False for  $\mathbf{U}$  and  $\mathbf{V}$ , true for  $\mathbf{\Sigma}$  ([Theorem 12.8.2](#)). For instance  $\mathbf{I}_2 = \mathbf{U}\mathbf{I}_2\mathbf{U}^*$  for every unitary  $\mathbf{U}$ . The freedom is: any orthonormal eigenbasis of  $\mathbf{A}^*\mathbf{A}$  may be used for  $\mathbf{V}$ , and the last  $m - r$  columns of  $\mathbf{U}$  may be any orthonormal basis of  $(\operatorname{im} \mathbf{A})^\perp$ .

### 12.8.6.2 B. Practice

#### Exercise B1: A 2 by 2 decomposition.

Find a singular value decomposition  $\mathbf{A} = \mathbf{U}\mathbf{\Sigma}\mathbf{V}^\top$  of

$$\mathbf{A} = \begin{pmatrix} 3 & 0 \\ 4 & 5 \end{pmatrix} \in M_2(\mathbb{R}),$$

and compare the singular values with the eigenvalues of  $\mathbf{A}$ .

*Solution.*  $\mathbf{A}^\top \mathbf{A} = \begin{pmatrix} 25 & 20 \\ 20 & 25 \end{pmatrix}$ , with characteristic polynomial  $x^2 - 50x + (625 - 400) = x^2 - 50x + 225$ , whose roots are 45 and 5. So  $\sigma_1 = 3\sqrt{5}$  and  $\sigma_2 = \sqrt{5}$ .

For  $\lambda = 45$  the equation  $-20x_1 + 20x_2 = 0$  gives  $\mathbf{v}_1 = \frac{1}{\sqrt{2}}(1, 1)$ ; for  $\lambda = 5$ ,  $\mathbf{v}_2 = \frac{1}{\sqrt{2}}(1, -1)$ . Then

$$\mathbf{u}_1 = \frac{\mathbf{A}\mathbf{v}_1}{3\sqrt{5}} = \frac{(3, 9)}{3\sqrt{10}} = \frac{1}{\sqrt{10}}(1, 3), \quad \mathbf{u}_2 = \frac{\mathbf{A}\mathbf{v}_2}{\sqrt{5}} = \frac{(3, -1)}{\sqrt{10}}.$$

Hence

$$\mathbf{U} = \frac{1}{\sqrt{10}} \begin{pmatrix} 1 & 3 \\ 3 & -1 \end{pmatrix}, \quad \mathbf{\Sigma} = \begin{pmatrix} 3\sqrt{5} & 0 \\ 0 & \sqrt{5} \end{pmatrix}, \quad \mathbf{V} = \frac{1}{\sqrt{2}} \begin{pmatrix} 1 & 1 \\ 1 & -1 \end{pmatrix},$$

and  $\mathbf{U}\mathbf{\Sigma}\mathbf{V}^\top = \mathbf{A}$ . Check:  $\sigma_1^2 + \sigma_2^2 = 45 + 5 = 50 = 9 + 16 + 25 = \|\mathbf{A}\|_F^2$ .

The eigenvalues of  $\mathbf{A}$  are 3 and 5, the diagonal entries of a triangular matrix. They are not the singular values, not even after reordering:  $\mathbf{A}$  is not normal, since  $\mathbf{A}^\top \mathbf{A} \neq \mathbf{A}\mathbf{A}^\top = \begin{pmatrix} 9 & 12 \\ 12 & 41 \end{pmatrix}$ . Note, however, that  $\sigma_1\sigma_2 = 15 = |\det \mathbf{A}|$ , as [Exercise \(C2: The adjoint and the determinant\)](#) explains.

#### Exercise B2: When the left singular vectors must be padded.

Find a singular value decomposition of

$$\mathbf{A} = \begin{pmatrix} 1 & 0 \\ 0 & 1 \\ 1 & 0 \end{pmatrix} \in M_{3 \times 2}(\mathbb{R}),$$

saying explicitly where the padding step of [Theorem 12.8.1](#) is used.

*Solution.*  $\mathbf{A}^\top \mathbf{A} = \text{diag}(2, 1)$ , already diagonal with decreasing diagonal, so  $\mathbf{V} = \mathbf{I}_2$ ,  $\mathbf{v}_1 = \mathbf{e}_1$ ,  $\mathbf{v}_2 = \mathbf{e}_2$ , and  $\sigma_1 = \sqrt{2}$ ,  $\sigma_2 = 1$ . Both are non-zero, so  $r = 2$ . Then

$$\mathbf{u}_1 = \frac{1}{\sqrt{2}} \mathbf{A} \mathbf{e}_1 = \frac{1}{\sqrt{2}}(1, 0, 1), \quad \mathbf{u}_2 = \mathbf{A} \mathbf{e}_2 = (0, 1, 0).$$

Here  $r = 2 < 3 = m$ , so Step 4 of [Theorem 12.8.1](#) applies: extend  $(\mathbf{u}_1, \mathbf{u}_2)$  to an orthonormal basis of  $\mathbb{R}^3$ . A vector orthogonal to both is  $(1, 0, -1)$ , so take  $\mathbf{u}_3 = \frac{1}{\sqrt{2}}(1, 0, -1)$ . Then

$$\mathbf{U} = \begin{pmatrix} \frac{1}{\sqrt{2}} & 0 & \frac{1}{\sqrt{2}} \\ 0 & 1 & 0 \\ \frac{1}{\sqrt{2}} & 0 & -\frac{1}{\sqrt{2}} \end{pmatrix}, \quad \mathbf{\Sigma} = \begin{pmatrix} \sqrt{2} & 0 \\ 0 & 1 \\ 0 & 0 \end{pmatrix}, \quad \mathbf{V} = \mathbf{I}_2,$$

and  $\mathbf{U} \mathbf{\Sigma} \mathbf{V}^\top = \mathbf{U} \mathbf{\Sigma} = \mathbf{A}$ . The vector  $\mathbf{u}_3$  is not determined by  $\mathbf{A}$  beyond its line:  $-\mathbf{u}_3$  would do just as well. By [Corollary 12.8.4](#),  $\text{nul}(\mathbf{A}^\top) = \text{span}(\mathbf{u}_3)$ , which is confirmed by  $\mathbf{A}^\top(1, 0, -1) = (1 - 1, 0) = \mathbf{0}$ .

### Exercise B3: Compact form and the four subspaces.

For

$$\mathbf{A} = \begin{pmatrix} 2 & 2 \\ 1 & 1 \\ 2 & 2 \end{pmatrix} \in M_{3 \times 2}(\mathbb{R}),$$

find the compact singular value decomposition  $\mathbf{A} = \sum_{i \leq r} \sigma_i \mathbf{u}_i \mathbf{v}_i^\top$ , and hence orthonormal bases of all four fundamental subspaces.

*Solution.*  $\mathbf{A}^\top \mathbf{A} = \begin{pmatrix} 9 & 9 \\ 9 & 9 \end{pmatrix}$ , with eigenvalues 18 and 0, so  $\sigma_1 = 3\sqrt{2}$ ,  $\sigma_2 = 0$  and  $r = 1$ .

An eigenvector for 18 is  $\mathbf{v}_1 = \frac{1}{\sqrt{2}}(1, 1)$ , and for 0 it is  $\mathbf{v}_2 = \frac{1}{\sqrt{2}}(1, -1)$ . Then

$$\mathbf{u}_1 = \frac{\mathbf{A} \mathbf{v}_1}{3\sqrt{2}} = \frac{(4, 2, 4)}{\sqrt{2} \cdot 3\sqrt{2}} = \frac{1}{3}(2, 1, 2),$$

a unit vector since  $4 + 1 + 4 = 9$ . The compact form is  $\mathbf{A} = 3\sqrt{2} \mathbf{u}_1 \mathbf{v}_1^\top$ , and multiplying out gives back  $\mathbf{A}$ .

Four subspaces:  $\text{col}(\mathbf{A}) = \text{span}(\frac{1}{3}(2, 1, 2))$  and  $\text{row}(\mathbf{A}) = \text{span}(\frac{1}{\sqrt{2}}(1, 1))$ , both one-dimensional;  $\text{nul}(\mathbf{A}) = \text{span}(\frac{1}{\sqrt{2}}(1, -1))$ ; and  $\text{nul}(\mathbf{A}^\top)$  is the plane  $\{\mathbf{y} : 2y_1 + y_2 + 2y_3 = 0\}$ , for which an orthonormal basis is  $\frac{1}{\sqrt{2}}(1, 0, -1)$  and  $\frac{1}{3\sqrt{2}}(1, -4, 1)$ . (Any orthonormal basis of that plane is a legitimate  $(\mathbf{u}_2, \mathbf{u}_3)$ , which is the freedom described in [Theorem 12.8.2](#) (b).)

**Exercise B4: Determine whether.**

Determine which of the following statements are correct. For those that are correct, prove them; for those that are not, give a counterexample.

- (a) If every eigenvalue of  $\mathbf{A} \in M_n(\mathbb{C})$  is 0, then every singular value of  $\mathbf{A}$  is 0.
- (b)  $\sigma_i(c\mathbf{A}) = |c| \sigma_i(\mathbf{A})$  for every scalar  $c$ .
- (c)  $\sigma_i(\mathbf{A}^2) = \sigma_i(\mathbf{A})^2$  for every square  $\mathbf{A}$ .
- (d)  $\mathbf{A}$  is invertible if and only if  $n = m$  and every singular value of  $\mathbf{A}$  is non-zero.

*Solution.*

- (a) Incorrect.  $\mathbf{N} = \begin{pmatrix} 0 & 1 \\ 0 & 0 \end{pmatrix}$  has spectrum  $\{0\}$ , while  $\mathbf{N}^* \mathbf{N} = \text{diag}(0, 1)$  gives  $\sigma_1 = 1$ .
- (b) Correct.  $(c\mathbf{A})^*(c\mathbf{A}) = \bar{c}c \mathbf{A}^* \mathbf{A} = |c|^2 \mathbf{A}^* \mathbf{A}$ , whose eigenvalues are  $|c|^2 \lambda_i$ ; taking square roots and noting that multiplying by  $|c|^2 \geq 0$  preserves the decreasing order gives the claim.
- (c) Incorrect, with the same  $\mathbf{N}$ :  $\mathbf{N}^2 = \mathbf{0}$  has both singular values 0, while  $\sigma_1(\mathbf{N})^2 = 1$ . (It is correct for  $\mathbf{A} \succeq 0$ , and more generally for normal  $\mathbf{A}$ , because then the singular values are the moduli of the eigenvalues and  $\mathbf{A}^2$  has eigenvalues  $\lambda_i^2$ .)
- (d) Correct. If  $\mathbf{A} \in M_n(F)$  then, by [Theorem 12.8.1](#), the number of non-zero singular values is  $\text{rank } \mathbf{A}$ , and  $\mathbf{A}$  is invertible exactly when  $\text{rank } \mathbf{A} = n$  ([Theorem 2.4.4](#)). A non-square matrix is never invertible.

**12.8.6.3 C. Going deeper****Exercise C1: The Frobenius norm and the largest singular value.**

Let  $\mathbf{A} \in M_{m \times n}(F)$  with singular values  $\sigma_1 \geq \cdots \geq \sigma_p$ .

- (a) The text obtained  $\|\mathbf{A}\|_F^2 = \sum_{i=1}^p \sigma_i^2$  from unitary invariance. Prove it a second way, directly from  $\|\mathbf{A}\|_F^2 = \text{tr}(\mathbf{A}^* \mathbf{A})$  and the eigenvalues of  $\mathbf{A}^* \mathbf{A}$ , without invoking [Lemma 11.11.1](#).
- (b) Deduce that  $\sigma_1 \leq \|\mathbf{A}\|_F$ , and determine exactly when equality holds.

*Solution.*

- (a) Take a singular value decomposition  $\mathbf{A} = \mathbf{U} \mathbf{\Sigma} \mathbf{V}^*$  ([Theorem 12.8.1](#)). By [Lemma 11.11.1](#), applied once on each side,  $\|\mathbf{A}\|_F = \|\mathbf{U} \mathbf{\Sigma} \mathbf{V}^*\|_F = \|\mathbf{\Sigma}\|_F$ . The only non-zero entries of  $\mathbf{\Sigma}$  are  $\sigma_1, \dots, \sigma_p$  on the diagonal, so  $\|\mathbf{\Sigma}\|_F^2 = \sum_i \sigma_i^2$ .
- (b) Every term of  $\sum_i \sigma_i^2$  is non-negative, so  $\sigma_1^2 \leq \sum_i \sigma_i^2 = \|\mathbf{A}\|_F^2$ , and taking non-negative square roots gives  $\sigma_1 \leq \|\mathbf{A}\|_F$ . Equality holds if and only if  $\sigma_2 = \cdots = \sigma_p = 0$ , that is, if and only if  $\text{rank } \mathbf{A} \leq 1$  ([Theorem 12.8.1](#)).

**Exercise C2: The adjoint and the determinant.**

Let  $\mathbf{A} \in M_{m \times n}(F)$ .

- (a) Prove that  $\mathbf{A}$  and  $\mathbf{A}^*$  have the same non-zero singular values, with the same multiplicities.
- (b) Suppose  $m = n$ . Prove that  $|\det \mathbf{A}| = \sigma_1 \sigma_2 \cdots \sigma_n$ .
- (c) Hence give a formula for  $|\det \mathbf{A}|$  in terms of  $\mathbf{A}^* \mathbf{A}$  only, and check it on  $\mathbf{A} = \begin{pmatrix} 3 & 0 \\ 4 & 5 \end{pmatrix}$ .

*Solution.*

- (a) If  $\mathbf{A} = \mathbf{U} \mathbf{\Sigma} \mathbf{V}^*$  is a singular value decomposition, then  $\mathbf{A}^* = \mathbf{V} \mathbf{\Sigma}^* \mathbf{U}^*$  with  $\mathbf{V}, \mathbf{U}$  unitary and  $\mathbf{\Sigma}^* \in M_{n \times m}(\mathbb{R})$  carrying the same diagonal entries  $\sigma_1, \dots, \sigma_p$ . By [Theorem 12.8.2](#) this is a singular value decomposition of  $\mathbf{A}^*$ , so the singular value list of  $\mathbf{A}^*$  is  $\sigma_1, \dots, \sigma_p$  as well. (The two lists have the same length  $p = \min(m, n)$ , so they are in fact equal, not merely equal in their non-zero parts.)
- (b) Take determinants in  $\mathbf{A} = \mathbf{U} \mathbf{\Sigma} \mathbf{V}^*$ . By [Theorem 6.5.4](#),  $\det \mathbf{A} = \det \mathbf{U} \cdot \det \mathbf{\Sigma} \cdot \det \mathbf{V}^*$ . A unitary  $\mathbf{U}$  has  $|\det \mathbf{U}| = 1$ , since  $\det(\mathbf{U}^*) \det(\mathbf{U}) = \det(\mathbf{I}) = 1$  and  $\det(\mathbf{U}^*) = \overline{\det \mathbf{U}}$ , so  $|\det \mathbf{U}|^2 = 1$ . The same holds for  $\mathbf{V}^*$ . And  $\det \mathbf{\Sigma} = \sigma_1 \cdots \sigma_n$  by [Theorem 6.4.6](#). Taking absolute values gives  $|\det \mathbf{A}| = \sigma_1 \cdots \sigma_n$ .
- (c) Squaring,  $|\det \mathbf{A}|^2 = \prod_i \sigma_i^2 = \det(\mathbf{A}^* \mathbf{A})$ , so  $|\det \mathbf{A}| = \sqrt{\det(\mathbf{A}^* \mathbf{A})}$ . For the given  $\mathbf{A}$ ,  $\det \mathbf{A} = 15$  and  $\det(\mathbf{A}^\top \mathbf{A}) = 625 - 400 = 225 = 15^2$ .

### Exercise C3: Singular values equal to eigenvalue moduli.

Let  $\mathbf{A} \in M_n(\mathbb{C})$  have eigenvalues  $\lambda_1, \dots, \lambda_n$  listed with algebraic multiplicity and ordered so that  $|\lambda_1| \geq \cdots \geq |\lambda_n|$ .

- (a) Prove that if  $\mathbf{A}$  is normal, then  $\sigma_i(\mathbf{A}) = |\lambda_i|$  for every  $i$ .
- (b) Prove the converse: if  $\sigma_i(\mathbf{A}) = |\lambda_i|$  for every  $i$ , then  $\mathbf{A}$  is normal.

*Hint for (b): compare two expressions for  $\|\mathbf{A}\|_F^2$ .*

*Solution.*

- (a) Suppose  $\mathbf{A}$  is normal. By [Corollary 11.4.3](#),  $\mathbf{A} = \mathbf{U} \mathbf{D} \mathbf{U}^*$  with  $\mathbf{U}$  unitary and  $\mathbf{D} = \text{diag}(\lambda_1, \dots, \lambda_n)$ . Then

$$\mathbf{A}^* \mathbf{A} = \mathbf{U} \mathbf{D}^* \mathbf{U}^* \mathbf{U} \mathbf{D} \mathbf{U}^* = \mathbf{U} \text{diag}(|\lambda_i|^2) \mathbf{U}^*,$$

so the eigenvalues of  $\mathbf{A}^* \mathbf{A}$  are  $|\lambda_1|^2, \dots, |\lambda_n|^2$  with multiplicity. These are already in decreasing order by hypothesis, so  $\sigma_i(\mathbf{A}) = |\lambda_i|$  for every  $i$  by [Definition 12.8.1](#).

- (b) Suppose  $\sigma_i(\mathbf{A}) = |\lambda_i|$  for every  $i$ . By [Exercise \(C1: The Frobenius norm and the largest singular value\)](#) (a),

$$\sum_{i=1}^n |\lambda_i|^2 = \sum_{i=1}^n \sigma_i^2 = \|\mathbf{A}\|_F^2.$$

Schur's inequality ([Theorem 11.11.2](#)) says  $\sum_i |\lambda_i|^2 \leq \|\mathbf{A}\|_F^2$  with equality **if and only if**  $\mathbf{A}$  is normal. We have equality, so  $\mathbf{A}$  is normal.



## 12.9 Polar Decomposition and Takagi's Factorization

Every non-zero complex number is a rotation times a stretch,  $z = e^{i\theta}r$  with  $r = |z| > 0$ . Section 8 has put the matrix version within reach: if  $\mathbf{A} = \mathbf{U}\mathbf{\Sigma}\mathbf{V}^*$ , then inserting  $\mathbf{V}^*\mathbf{V} = \mathbf{I}$  in the middle splits the factorization into a unitary part and a positive semidefinite part. This section carries that out, proves that the unitary part is the closest unitary matrix to  $\mathbf{A}$ , and then treats a factorization that looks like the singular value decomposition but is not: Takagi's, for complex **symmetric** matrices.

The field is  $\mathbb{C}$  unless stated otherwise, and  $|\mathbf{A}| = (\mathbf{A}^*\mathbf{A})^{1/2}$  is the matrix absolute value of [Definition 12.2.1](#).

### 12.9.1 Rotation times stretch

#### Theorem 12.9.1 — Polar Decomposition

Let  $\mathbf{A} \in M_n(\mathbb{C})$ . Then there is a unitary  $\mathbf{W} \in M_n(\mathbb{C})$  with

$$\mathbf{A} = \mathbf{W}|\mathbf{A}|.$$

The positive semidefinite factor  $|\mathbf{A}|$  is determined by  $\mathbf{A}$ . The unitary factor  $\mathbf{W}$  is determined by  $\mathbf{A}$  if and only if  $\mathbf{A}$  is invertible, and then  $\mathbf{W} = \mathbf{A}|\mathbf{A}|^{-1}$ . Moreover the same  $\mathbf{W}$  satisfies the right-handed identity

$$\mathbf{A} = |\mathbf{A}^*| \mathbf{W}, \quad |\mathbf{A}^*| = \mathbf{W}|\mathbf{A}|\mathbf{W}^*.$$

*Idea.* The singular value decomposition already separates the two kinds of behavior:  $\mathbf{U}$  and  $\mathbf{V}^*$  rotate, and  $\mathbf{\Sigma}$  stretches along the axes. Writing  $\mathbf{U}\mathbf{\Sigma}\mathbf{V}^* = (\mathbf{U}\mathbf{V}^*)(\mathbf{V}\mathbf{\Sigma}\mathbf{V}^*)$  puts all the rotating in the first bracket and all the stretching in the second, and the second bracket is a positive semidefinite matrix whose square is  $\mathbf{A}^*\mathbf{A}$ . Uniqueness of the positive semidefinite square root does the rest. For the uniqueness of  $\mathbf{W}$ , notice that  $\mathbf{W}$  is pinned down only where  $|\mathbf{A}|$  is invertible; when  $\mathbf{A} = \mathbf{0}$ , every unitary works.

*Proof. Existence.* By [Theorem 12.8.1](#) write  $\mathbf{A} = \mathbf{U}\mathbf{\Sigma}\mathbf{V}^*$  with  $\mathbf{U}, \mathbf{V} \in M_n(\mathbb{C})$  unitary and  $\mathbf{\Sigma} = \text{diag}(\sigma_1, \dots, \sigma_n)$  with  $\sigma_i \geq 0$ . Put

$$\mathbf{W} := \mathbf{U}\mathbf{V}^*, \quad \mathbf{P} := \mathbf{V}\mathbf{\Sigma}\mathbf{V}^*.$$

Then  $\mathbf{W}$  is unitary, being a product of unitary matrices, and

$$\mathbf{W}\mathbf{P} = \mathbf{U}\mathbf{V}^*\mathbf{V}\mathbf{\Sigma}\mathbf{V}^* = \mathbf{U}\mathbf{\Sigma}\mathbf{V}^* = \mathbf{A}.$$

The matrix  $\mathbf{P}$  is self-adjoint, since  $\mathbf{P}^* = \mathbf{V}\mathbf{\Sigma}^*\mathbf{V}^* = \mathbf{P}$ , and positive semidefinite: its eigenvalues are the  $\sigma_i \geq 0$  because  $\mathbf{V}^*\mathbf{P}\mathbf{V} = \mathbf{\Sigma}$ , so [Theorem 12.1.4](#) applies. Finally

$$\mathbf{P}^2 = \mathbf{V}\mathbf{\Sigma}\mathbf{V}^*\mathbf{V}\mathbf{\Sigma}\mathbf{V}^* = \mathbf{V}\mathbf{\Sigma}^2\mathbf{V}^* = \mathbf{A}^*\mathbf{A},$$

the last equality because  $\mathbf{A}^* \mathbf{A} = \mathbf{V} \Sigma \mathbf{U}^* \mathbf{U} \Sigma \mathbf{V}^* = \mathbf{V} \Sigma^2 \mathbf{V}^*$ . A positive semidefinite matrix has exactly one positive semidefinite square root ([Theorem 12.2.1](#)), so  $\mathbf{P} = (\mathbf{A}^* \mathbf{A})^{1/2} = |\mathbf{A}|$ , and  $\mathbf{A} = \mathbf{W} |\mathbf{A}|$ .

**The factor  $|\mathbf{A}|$  is determined**, since  $\mathbf{A}^* \mathbf{A}$  is and the square root is unique.

**Uniqueness of  $\mathbf{W}$ .** ( $\Rightarrow$ ) Suppose  $\mathbf{A}$  is invertible. Then  $|\mathbf{A}|^2 = \mathbf{A}^* \mathbf{A}$  is invertible, so  $|\mathbf{A}|$  is, and any  $\mathbf{W}$  with  $\mathbf{A} = \mathbf{W} |\mathbf{A}|$  satisfies  $\mathbf{W} = \mathbf{A} |\mathbf{A}|^{-1}$ : there is only one. ( $\Leftarrow$ ) Suppose  $\mathbf{A}$  is not invertible, so there is a non-zero  $\mathbf{z} \in \ker \mathbf{A}$ . Let  $\mathbf{H} = \mathbf{H}_{\mathbf{z}}$  be the Householder reflection of [Definition 10.8.1](#), which is unitary with  $\mathbf{H}\mathbf{z} = -\mathbf{z}$  and  $\mathbf{H}\mathbf{v} = \mathbf{v}$  for every  $\mathbf{v} \perp \mathbf{z}$  ([Proposition 10.8.2](#) (c), (d)). By [Proposition 12.2.2](#),  $\text{nul} |\mathbf{A}| = \text{nul} \mathbf{A}$ , and  $|\mathbf{A}|$  is self-adjoint, so

$$\text{col} |\mathbf{A}| = \text{col} (|\mathbf{A}|^*) = (\text{nul} |\mathbf{A}|)^{\perp} = (\text{nul} \mathbf{A})^{\perp} \subseteq \mathbf{z}^{\perp}$$

by [Theorem 10.6.5](#) (b). Hence  $\mathbf{H}$  fixes every vector of  $\text{col} |\mathbf{A}|$ , that is,  $\mathbf{H} |\mathbf{A}| = |\mathbf{A}|$ , and therefore  $(\mathbf{W}\mathbf{H}) |\mathbf{A}| = \mathbf{A}$  as well, with  $\mathbf{W}\mathbf{H}$  unitary and  $\mathbf{W}\mathbf{H} \neq \mathbf{W}$  because  $\mathbf{H} \neq \mathbf{I}$  and  $\mathbf{W}$  is invertible. So  $\mathbf{W}$  is not determined.

**The right-handed form.** Set  $\mathbf{Q} := \mathbf{W} |\mathbf{A}| \mathbf{W}^*$ . It is self-adjoint and positive semidefinite, being a unitary congruence of a positive semidefinite matrix:  $\mathbf{x}^* \mathbf{Q} \mathbf{x} = (\mathbf{W}^* \mathbf{x})^* |\mathbf{A}| (\mathbf{W}^* \mathbf{x}) \geq 0$ . And

$$\mathbf{Q}^2 = \mathbf{W} |\mathbf{A}|^2 \mathbf{W}^* = \mathbf{W} \mathbf{A}^* \mathbf{A} \mathbf{W}^* = \mathbf{A} \mathbf{A}^*,$$

using  $\mathbf{A} = \mathbf{W} |\mathbf{A}|$ , so  $\mathbf{A}^* = |\mathbf{A}| \mathbf{W}^*$  and  $\mathbf{A} \mathbf{A}^* = \mathbf{W} |\mathbf{A}|^2 \mathbf{W}^*$ . By uniqueness of the square root again,  $\mathbf{Q} = (\mathbf{A} \mathbf{A}^*)^{1/2} = |\mathbf{A}^*|$ . Therefore  $|\mathbf{A}^*| \mathbf{W} = \mathbf{W} |\mathbf{A}| \mathbf{W}^* \mathbf{W} = \mathbf{W} |\mathbf{A}| = \mathbf{A}$ . This proves the theorem. ■

*Remark.* There is a second route to existence, the one [Proposition 12.2.2](#) was pointing at: the identity  $\| |\mathbf{A}| \mathbf{x} \| = \| \mathbf{A} \mathbf{x} \|$  says that  $|\mathbf{A}| \mathbf{x} \mapsto \mathbf{A} \mathbf{x}$  is a well-defined isometry from  $\text{col} |\mathbf{A}|$  onto  $\text{col} \mathbf{A}$ , and extending it by any isometry between the orthogonal complements produces  $\mathbf{W}$  directly. That proof shows more clearly *where* the freedom in  $\mathbf{W}$  lives, but it needs the extension step written out; the route above borrows all of that from [Theorem 12.8.1](#).

For  $n = 1$  the theorem is exactly  $z = e^{i\theta} |z|$ : a  $1 \times 1$  unitary matrix is a complex number of modulus 1, and  $|\mathbf{A}|$  is the modulus. The non-uniqueness for singular  $\mathbf{A}$  is the matrix version of the fact that  $\theta$  is undefined when  $z = 0$ .

### Example. (A polar decomposition)

Find the left and right polar decompositions of  $\mathbf{A} = \begin{pmatrix} 3 & 0 \\ 4 & 5 \end{pmatrix}$ .

*Solution.* [Exercise \(B1: A 2 by 2 decomposition\)](#) gives the singular value decomposition

$$\mathbf{U} = \frac{1}{\sqrt{10}} \begin{pmatrix} 1 & 3 \\ 3 & -1 \end{pmatrix}, \quad \Sigma = \sqrt{5} \begin{pmatrix} 3 & 0 \\ 0 & 1 \end{pmatrix}, \quad \mathbf{V} = \frac{1}{\sqrt{2}} \begin{pmatrix} 1 & 1 \\ 1 & -1 \end{pmatrix}.$$

Hence

$$\mathbf{W} = \mathbf{U}\mathbf{V}^\top = \frac{1}{\sqrt{20}} \begin{pmatrix} 1 & 3 \\ 3 & -1 \end{pmatrix} \begin{pmatrix} 1 & 1 \\ 1 & -1 \end{pmatrix} = \frac{1}{\sqrt{5}} \begin{pmatrix} 2 & -1 \\ 1 & 2 \end{pmatrix},$$

$$|\mathbf{A}| = \mathbf{V}\mathbf{\Sigma}\mathbf{V}^\top = \sqrt{5} \begin{pmatrix} 2 & 1 \\ 1 & 2 \end{pmatrix}.$$

Check:  $|\mathbf{A}|^2 = 5 \begin{pmatrix} 5 & 4 \\ 4 & 5 \end{pmatrix} = \begin{pmatrix} 25 & 20 \\ 20 & 25 \end{pmatrix} = \mathbf{A}^\top \mathbf{A}$ , the columns of  $\mathbf{W}$  are orthonormal, and  $\mathbf{W}|\mathbf{A}| = \mathbf{A}$ . Here  $\mathbf{W}$  is a rotation, since  $\det \mathbf{W} = (4+1)/5 = 1$ .

For the right-handed form,  $\mathbf{A}\mathbf{A}^\top = \begin{pmatrix} 9 & 12 \\ 12 & 41 \end{pmatrix}$  and

$$|\mathbf{A}^*| = \mathbf{W}|\mathbf{A}|\mathbf{W}^\top = \frac{\sqrt{5}}{5} \begin{pmatrix} 6 & 3 \\ 3 & 14 \end{pmatrix},$$

whose square is indeed  $\frac{1}{5} \begin{pmatrix} 45 & 60 \\ 60 & 205 \end{pmatrix} = \mathbf{A}\mathbf{A}^\top$ . The two absolute values differ, and that is not an accident.

*Remark.*  $|\mathbf{A}| = |\mathbf{A}^*|$  exactly when  $\mathbf{A}$  is normal. Indeed  $\mathbf{A}^*\mathbf{A} = \mathbf{A}\mathbf{A}^*$  if and only if the two positive semidefinite square roots agree, by uniqueness. So for a normal matrix — and only then — the stretching looks the same before and after the rotation, and  $\mathbf{W}$  commutes with  $|\mathbf{A}|$  by the identity  $|\mathbf{A}^*| = \mathbf{W}|\mathbf{A}|\mathbf{W}^*$ .

*Quick check.* What is the polar decomposition of a unitary matrix? Of a positive semidefinite matrix?

*Solution.* If  $\mathbf{A}$  is unitary then  $\mathbf{A}^*\mathbf{A} = \mathbf{I}$ , so  $|\mathbf{A}| = \mathbf{I}$  and  $\mathbf{W} = \mathbf{A}$ : all rotation, no stretch. If  $\mathbf{A} \succeq 0$  then  $\mathbf{A}^*\mathbf{A} = \mathbf{A}^2$  and  $\mathbf{A}$  is itself a positive semidefinite square root of  $\mathbf{A}^2$ , so  $|\mathbf{A}| = \mathbf{A}$  by uniqueness (Theorem 12.2.1) and  $\mathbf{W} = \mathbf{I}$  whenever  $\mathbf{A}$  is invertible: all stretch, no rotation.

## 12.9.2 The nearest unitary matrix

The polar factor is not merely a unitary matrix attached to  $\mathbf{A}$ . It is the one closest to  $\mathbf{A}$ , measured in the Frobenius norm of Chapter 11 §11. This is the reason the polar decomposition is used in practice: given a matrix that ought to be unitary but has drifted,  $\mathbf{W}$  is how to put it back.

### Corollary 12.9.2 — The Polar Factor Is the Nearest Unitary Matrix

Let  $\mathbf{A} \in M_n(\mathbb{C})$  with polar decomposition  $\mathbf{A} = \mathbf{W}|\mathbf{A}|$ . Then

$$\|\mathbf{A} - \mathbf{W}\|_F \leq \|\mathbf{A} - \mathbf{Q}\|_F \quad \text{for every unitary } \mathbf{Q} \in M_n(\mathbb{C}),$$

and the minimum value is  $(\|\mathbf{A}\|_F^2 + n - 2 \sum_i \sigma_i)^{1/2}$ . If  $\mathbf{A}$  is invertible,  $\mathbf{W}$  is the **only** unitary matrix attaining the minimum.

*Idea.* Two moves. First, the Frobenius norm does not see unitary factors (Lemma 11.11.1), so sandwiching by  $\mathbf{U}^*$  and  $\mathbf{V}$  turns the problem into: among unitary  $\mathbf{Z}$ , which is closest to the diagonal matrix  $\mathbf{\Sigma}$ ? Second, expanding  $\|\mathbf{\Sigma} - \mathbf{Z}\|_F^2$  leaves only the diagonal entries of  $\mathbf{Z}$  to be chosen, and each of them has modulus at most 1 because it sits in a unit column. So the best possible  $\mathbf{Z}$  is  $\mathbf{I}$ , and translating back gives  $\mathbf{Q} = \mathbf{U}\mathbf{V}^* = \mathbf{W}$ .

*Proof.* Fix a singular value decomposition  $\mathbf{A} = \mathbf{U}\mathbf{\Sigma}\mathbf{V}^*$  as in Theorem 12.8.1, so that  $\mathbf{W} = \mathbf{U}\mathbf{V}^*$  as in the proof of Theorem 12.9.1. Let  $\mathbf{Q}$  be unitary and put  $\mathbf{Z} := \mathbf{U}^*\mathbf{Q}\mathbf{V}$ , which is unitary; as  $\mathbf{Q}$  runs over all unitary matrices so does  $\mathbf{Z}$ , since  $\mathbf{Q} = \mathbf{U}\mathbf{Z}\mathbf{V}^*$ . By Lemma 11.11.1, applied with the unitary factors  $\mathbf{U}^*$  on the left and  $\mathbf{V}$  on the right,

$$\|\mathbf{A} - \mathbf{Q}\|_F = \|\mathbf{U}^*(\mathbf{A} - \mathbf{Q})\mathbf{V}\|_F = \|\mathbf{\Sigma} - \mathbf{Z}\|_F.$$

Expanding the square in the Frobenius inner product  $\langle \mathbf{X}, \mathbf{Y} \rangle = \text{tr}(\mathbf{Y}^*\mathbf{X})$  of Chapter 10 §01,

$$\begin{aligned} \|\mathbf{\Sigma} - \mathbf{Z}\|_F^2 &= \|\mathbf{\Sigma}\|_F^2 + \|\mathbf{Z}\|_F^2 - 2 \text{Re} \text{tr}(\mathbf{Z}^*\mathbf{\Sigma}) \\ &= \sum_{i=1}^n \sigma_i^2 + n - 2 \sum_{i=1}^n \sigma_i \text{Re } z_{ii}, \end{aligned} \quad (12.9.1)$$

where  $\|\mathbf{Z}\|_F^2 = \text{tr}(\mathbf{Z}^*\mathbf{Z}) = \text{tr} \mathbf{I}_n = n$ , and  $\text{tr}(\mathbf{Z}^*\mathbf{\Sigma}) = \sum_i (\mathbf{Z}^*)_{ii} \sigma_i = \sum_i \overline{z_{ii}} \sigma_i$ , whose real part is  $\sum_i \sigma_i \text{Re } z_{ii}$ .

Column  $i$  of  $\mathbf{Z}$  is a unit vector, so  $|z_{ii}| \leq 1$  and hence  $\text{Re } z_{ii} \leq 1$ . Since every  $\sigma_i \geq 0$ ,

$$\sum_{i=1}^n \sigma_i \text{Re } z_{ii} \leq \sum_{i=1}^n \sigma_i,$$

so Eq. 12.9.1 gives  $\|\mathbf{A} - \mathbf{Q}\|_F^2 \geq \sum_i \sigma_i^2 + n - 2 \sum_i \sigma_i$  for every unitary  $\mathbf{Q}$ . Taking  $\mathbf{Q} = \mathbf{W} = \mathbf{U}\mathbf{V}^*$  makes  $\mathbf{Z} = \mathbf{U}^*\mathbf{U}\mathbf{V}^*\mathbf{V} = \mathbf{I}_n$ , for which  $\text{Re } z_{ii} = 1$  for every  $i$  and the bound is attained. Since  $\|\mathbf{A}\|_F^2 = \sum_i \sigma_i^2$  by Exercise (C1: The Frobenius norm and the largest singular value), the minimum value is as stated.

Now suppose  $\mathbf{A}$  is invertible, so  $\sigma_i > 0$  for every  $i$ , and suppose  $\mathbf{Q}$  attains the minimum. Then  $\sum_i \sigma_i(1 - \text{Re } z_{ii}) = 0$  is a sum of non-negative terms, so  $\text{Re } z_{ii} = 1$  for every  $i$ ; with  $|z_{ii}| \leq 1$  this forces  $z_{ii} = 1$ . Column  $i$  of  $\mathbf{Z}$  then has a coordinate equal to 1 and total squared length 1, so all its other coordinates vanish and the column is  $\mathbf{e}_i$ . Hence  $\mathbf{Z} = \mathbf{I}_n$  and  $\mathbf{Q} = \mathbf{U}\mathbf{Z}\mathbf{V}^* = \mathbf{W}$ . This proves the corollary. ■

For  $\mathbf{A} = \begin{pmatrix} 3 & 0 \\ 4 & 5 \end{pmatrix}$  of Example (A polar decomposition), with  $\sigma_1 = 3\sqrt{5}$ ,  $\sigma_2 = \sqrt{5}$  and  $\|\mathbf{A}\|_F^2 = 50$ , the nearest unitary matrix is  $\mathbf{W} = \frac{1}{\sqrt{5}} \begin{pmatrix} 2 & -1 \\ 1 & 2 \end{pmatrix}$  and the distance to the unitary group is  $\sqrt{52 - 8\sqrt{5}}$ , about 5.84.

*Warning.* The corollary is a statement about the **Frobenius** norm. A different way of measuring distance can have a different minimizer, and “nearest” always needs its

yardstick named. It is true, though not proved here, that  $\mathbf{W}$  also minimizes the distance in every unitarily invariant norm; Chapter 20 takes up that family of norms.

### 12.9.3 Takagi's factorization

A complex matrix can be symmetric,  $\mathbf{A}^\top = \mathbf{A}$ , without being Hermitian,  $\mathbf{A}^* = \mathbf{A}$ . Over  $\mathbb{R}$  the two coincide and Chapter 11 handled them; over  $\mathbb{C}$  they are genuinely different conditions, and symmetry is much the weaker one. It does not imply normality, it does not imply diagonalizability, and it says nothing about the eigenvalues. Here is the standing example:

$$\mathbf{N} = \begin{pmatrix} 1 & i \\ i & -1 \end{pmatrix}, \quad \mathbf{N}^\top = \mathbf{N}, \quad \mathbf{N}^2 = \mathbf{0}.$$

A non-zero matrix with  $\mathbf{N}^2 = \mathbf{0}$  has minimal polynomial  $x^2$  and is therefore not diagonalizable, so no version of the spectral theorem can apply to  $\mathbf{N}$ . Nevertheless something survives, and it is a statement about singular values rather than eigenvalues.

#### Theorem 12.9.3 — Takagi's Factorization

Let  $\mathbf{A} \in M_n(\mathbb{C})$  be **symmetric**,  $\mathbf{A}^\top = \mathbf{A}$ , with singular values  $\sigma_1 \geq \cdots \geq \sigma_n \geq 0$ . Then there is a unitary  $\mathbf{U} \in M_n(\mathbb{C})$  with

$$\mathbf{A} = \mathbf{U}\mathbf{\Sigma}\mathbf{U}^\top, \quad \mathbf{\Sigma} = \text{diag}(\sigma_1, \dots, \sigma_n).$$

Read the statement against the two theorems it resembles.

- **Against the singular value decomposition.** Theorem 12.8.1 gives  $\mathbf{A} = \mathbf{U}\mathbf{\Sigma}\mathbf{V}^*$  with two unrelated unitary matrices. Since  $\mathbf{U}^\top = (\overline{\mathbf{U}})^*$ , Takagi's identity says  $\mathbf{A} = \mathbf{U}\mathbf{\Sigma}(\overline{\mathbf{U}})^*$ : it is a singular value decomposition in which the right factor is forced to be  $\mathbf{V} = \overline{\mathbf{U}}$ . So the content is not a new diagonal matrix but a relation between  $\mathbf{U}$  and  $\mathbf{V}$ .
- **Against the spectral theorem.** Corollary 11.4.3 writes a normal  $\mathbf{A}$  as  $\mathbf{U}\mathbf{D}\mathbf{U}^*$ , a *similarity*, with the eigenvalues of  $\mathbf{A}$  on the diagonal. Takagi writes a symmetric  $\mathbf{A}$  as  $\mathbf{U}\mathbf{\Sigma}\mathbf{U}^\top$ , which is a *congruence*  $\mathbf{X} \mapsto \mathbf{U}\mathbf{X}\mathbf{U}^\top$ , with the singular values on the diagonal. The hypotheses are different (symmetric, not normal), the diagonal holds different numbers, and the transpose is not the conjugate transpose.

The difference is visible even for a real matrix. Take  $\mathbf{A} = \begin{pmatrix} 3 & 4 \\ 4 & -3 \end{pmatrix}$ , which is real symmetric with eigenvalues 5 and  $-5$ . Corollary 11.5.2 gives  $\mathbf{A} = \mathbf{Q} \text{diag}(5, -5) \mathbf{Q}^\top$  with  $\mathbf{Q}$  real orthogonal. Takagi insists on the **non-negative** diagonal  $\text{diag}(5, 5)$ , and no real  $\mathbf{U}$  can achieve that, since  $\mathbf{U} \text{diag}(5, 5) \mathbf{U}^\top = 5\mathbf{U}\mathbf{U}^\top = 5\mathbf{I}$  for real orthogonal  $\mathbf{U}$ . A genuinely complex  $\mathbf{U}$  is needed, and Example (Takagi for a real symmetric matrix) finds one.

*Idea.* Four steps, and the first two are the polar decomposition doing the work. ① Symmetry makes  $\mathbf{A}$  intertwine  $\mathbf{A}^*\mathbf{A}$  with its conjugate:  $\mathbf{A}(\mathbf{A}^*\mathbf{A}) = \overline{(\mathbf{A}^*\mathbf{A})}\mathbf{A}$ . Because the positive square root is a polynomial in the matrix, the same relation passes to  $|\mathbf{A}|$ . ② That relation, combined with  $\mathbf{A}^\top = \mathbf{A}$ , forces the polar factor  $\mathbf{W}$  to be a **symmetric** unitary matrix. ③ A symmetric unitary matrix factors as  $\mathbf{V}\mathbf{V}^\top$ : split it into real and imaginary parts, which turn out to be commuting real symmetric matrices, and use Chapter 11's simultaneous diagonalization. ④ Conjugating  $\mathbf{A}$  by that  $\mathbf{V}$  in the right way produces a matrix that is both symmetric and Hermitian, hence **real** symmetric,

and the real spectral theorem finishes the job. We prove all of this for invertible  $\mathbf{A}$ , where the polar factor is unique and  $|\mathbf{A}|^{-1}$  exists.

*Proof.* We prove the theorem for **invertible**  $\mathbf{A}$ . Throughout,  $\mathbf{A}^\top = \mathbf{A}$ , so  $\mathbf{A}^* = \overline{\mathbf{A}}^\top = \overline{\mathbf{A}}$ . Write  $\mathbf{M} := \mathbf{A}^* \mathbf{A} = \overline{\mathbf{A}} \mathbf{A}$  and  $\mathbf{P} := |\mathbf{A}| = \mathbf{M}^{1/2}$ . Since  $\mathbf{A}$  is invertible, so are  $\mathbf{M}$  and  $\mathbf{P}$ , and  $\mathbf{M} \succ 0$ .

**Step 1:  $\mathbf{A}\mathbf{M} = \overline{\mathbf{M}}\mathbf{A}$  and  $\mathbf{A}\mathbf{P} = \overline{\mathbf{P}}\mathbf{A}$ .** For the first,  $\overline{\mathbf{M}} = \overline{\overline{\mathbf{A}}\mathbf{A}} = \mathbf{A}\overline{\mathbf{A}}$ , so

$$\mathbf{A}\mathbf{M} = \mathbf{A}\overline{\mathbf{A}}\mathbf{A} = \overline{\mathbf{M}}\mathbf{A}.$$

For the second we need a polynomial.

*Claim.* There is a polynomial  $p$  with **real** coefficients such that  $p(\mathbf{M}) = \mathbf{P}$  and  $p(\overline{\mathbf{M}}) = \overline{\mathbf{P}}$ .

*Proof.* By [Theorem 12.2.1](#) (a) there is a polynomial  $p$  with real coefficients such that  $p(\mathbf{M}) = \mathbf{M}^{1/2} = \mathbf{P}$ . Conjugating every entry of the identity  $\mathbf{P} = p(\mathbf{M})$ , and using that the coefficients of  $p$  are real so that conjugation passes through them, gives  $\overline{\mathbf{P}} = p(\overline{\mathbf{M}})$ . ■

Iterating Step 1's first identity gives  $\mathbf{A}\mathbf{M}^j = \overline{\mathbf{M}}^j \mathbf{A}$  for every  $j \geq 0$ , hence  $\mathbf{A}p(\mathbf{M}) = p(\overline{\mathbf{M}}) \mathbf{A}$  for every polynomial  $p$ , and with the claim's  $p$  this reads

$$\mathbf{A}\mathbf{P} = \overline{\mathbf{P}}\mathbf{A}. \tag{12.9.2}$$

**Step 2: the polar factor is symmetric.** By [Theorem 12.9.1](#),  $\mathbf{A} = \mathbf{W}\mathbf{P}$  with  $\mathbf{W} = \mathbf{A}\mathbf{P}^{-1}$  unitary. Substituting  $\mathbf{A} = \mathbf{W}\mathbf{P}$  into [Eq. 12.9.2](#) gives  $\mathbf{W}\mathbf{P}^2 = \overline{\mathbf{P}}\mathbf{W}\mathbf{P}$ , and canceling the invertible  $\mathbf{P}$  on the right,

$$\mathbf{W}\mathbf{P} = \overline{\mathbf{P}}\mathbf{W}.$$

On the other hand  $\mathbf{P}$  is Hermitian, so  $\mathbf{P}^\top = \overline{\mathbf{P}}$ , and transposing  $\mathbf{A} = \mathbf{W}\mathbf{P}$  while using  $\mathbf{A}^\top = \mathbf{A}$  gives

$$\mathbf{W}\mathbf{P} = \mathbf{A} = \mathbf{A}^\top = \mathbf{P}^\top \mathbf{W}^\top = \overline{\mathbf{P}} \mathbf{W}^\top.$$

Comparing the two displays,  $\overline{\mathbf{P}}\mathbf{W} = \overline{\mathbf{P}}\mathbf{W}^\top$ , and  $\overline{\mathbf{P}}$  is invertible, so  $\mathbf{W}^\top = \mathbf{W}$ .

**Step 3: a symmetric unitary matrix is  $\mathbf{V}\mathbf{V}^\top$ .**

*Claim.* Let  $\mathbf{W} \in M_n(\mathbb{C})$  be unitary with  $\mathbf{W}^\top = \mathbf{W}$ . Then  $\mathbf{W} = \mathbf{V}\mathbf{V}^\top$  for some unitary  $\mathbf{V}$ .

*Proof.* From  $\mathbf{W}^{-1} = \mathbf{W}^* = \overline{\mathbf{W}}^\top = \overline{\mathbf{W}}$  we get  $\overline{\mathbf{W}}\mathbf{W} = \mathbf{I}$ . Write  $\mathbf{W} = \mathbf{X} + i\mathbf{Y}$  with  $\mathbf{X} = \frac{1}{2}(\mathbf{W} + \overline{\mathbf{W}})$  and  $\mathbf{Y} = \frac{1}{2i}(\mathbf{W} - \overline{\mathbf{W}})$ , both **real** matrices, and both symmetric because  $\mathbf{W}$  and  $\overline{\mathbf{W}}$  are. Then

$$\mathbf{I} = \overline{\mathbf{W}}\mathbf{W} = (\mathbf{X} - i\mathbf{Y})(\mathbf{X} + i\mathbf{Y}) = \mathbf{X}^2 + \mathbf{Y}^2 + i(\mathbf{X}\mathbf{Y} - \mathbf{Y}\mathbf{X}),$$

and since  $\mathbf{X}^2 + \mathbf{Y}^2$  and  $\mathbf{X}\mathbf{Y} - \mathbf{Y}\mathbf{X}$  are real matrices, comparing real and imaginary parts gives

$$\mathbf{X}^2 + \mathbf{Y}^2 = \mathbf{I}, \quad \mathbf{X}\mathbf{Y} = \mathbf{Y}\mathbf{X}.$$

So  $\mathbf{X}$  and  $\mathbf{Y}$  are commuting real symmetric matrices, and [Corollary 11.7.3](#) supplies  $\mathbf{Q} \in O(n)$  with  $\mathbf{Q}^\top \mathbf{X} \mathbf{Q} = \text{diag}(x_1, \dots, x_n)$  and  $\mathbf{Q}^\top \mathbf{Y} \mathbf{Q} = \text{diag}(y_1, \dots, y_n)$ . Hence  $\mathbf{Q}^\top \mathbf{W} \mathbf{Q} = \text{diag}(x_1 + iy_1, \dots, x_n + iy_n)$ , and  $x_j^2 + y_j^2 = 1$  for every  $j$  by the first relation. Each  $x_j + iy_j$  therefore has modulus 1. The polynomial  $x^2 - (x_j + iy_j)$  has a root  $c_j \in \mathbb{C}$  by [Theorem 5.5.1](#), and  $|c_j|^2 = |c_j^2| = 1$ , so  $|c_j| = 1$ . Put  $\mathbf{C} = \text{diag}(c_1, \dots, c_n)$  and  $\mathbf{V} := \mathbf{Q}\mathbf{C}$ , which is unitary as a product of unitary matrices. Since  $\mathbf{Q}$  is real orthogonal,  $\mathbf{Q}^\top = \mathbf{Q}^{-1}$ , and  $\mathbf{C}^\top = \mathbf{C}$ , so

$$\mathbf{V}\mathbf{V}^\top = \mathbf{Q}\mathbf{C}\mathbf{C}^\top \mathbf{Q}^\top = \mathbf{Q}\mathbf{C}^2 \mathbf{Q}^\top = \mathbf{Q} \text{diag}(x_j + iy_j) \mathbf{Q}^\top = \mathbf{W}.$$

This proves the claim. ■

**Step 4: conjugate to a real matrix.** Take  $\mathbf{V}$  from Step 3, so  $\mathbf{W} = \mathbf{V}\mathbf{V}^\top$ , and set

$$\mathbf{B} := \mathbf{V}^* \mathbf{A} \overline{\mathbf{V}}.$$

Three observations. First,  $\mathbf{B}^\top = \mathbf{B}$ : using  $(\overline{\mathbf{V}})^\top = \mathbf{V}^*$ ,  $(\mathbf{V}^*)^\top = \overline{\mathbf{V}}$  and  $\mathbf{A}^\top = \mathbf{A}$ ,

$$\mathbf{B}^\top = (\overline{\mathbf{V}})^\top \mathbf{A}^\top (\mathbf{V}^*)^\top = \mathbf{V}^* \mathbf{A} \overline{\mathbf{V}} = \mathbf{B}.$$

Second,  $\mathbf{B} \succ 0$ : since  $\mathbf{A} = \mathbf{W}\mathbf{P} = \mathbf{V}\mathbf{V}^\top \mathbf{P}$  and  $\mathbf{V}^* \mathbf{V} = \mathbf{I}$ ,

$$\mathbf{B} = \mathbf{V}^* \mathbf{V} \mathbf{V}^\top \mathbf{P} \overline{\mathbf{V}} = \mathbf{V}^\top \mathbf{P} \overline{\mathbf{V}} = (\overline{\mathbf{V}})^* \mathbf{P} (\overline{\mathbf{V}}),$$

a unitary congruence of the positive definite  $\mathbf{P}$ , so  $\mathbf{x}^* \mathbf{B} \mathbf{x} = (\overline{\mathbf{V}} \mathbf{x})^* \mathbf{P} (\overline{\mathbf{V}} \mathbf{x}) > 0$  for  $\mathbf{x} \neq \mathbf{0}$ ; in particular  $\mathbf{B}^* = \mathbf{B}$ . Third,  $\mathbf{B}$  is **real**, because

$$\overline{\mathbf{B}} = \overline{\mathbf{B}^\top} = \mathbf{B}^* = \mathbf{B},$$

the first equality using  $\mathbf{B}^\top = \mathbf{B}$ , the second the definition of the conjugate transpose, and the third Hermitian symmetry.

So  $\mathbf{B}$  is a real symmetric positive definite matrix. By [Corollary 11.5.2](#) there is  $\mathbf{Q}_0 \in O(n)$  with  $\mathbf{B} = \mathbf{Q}_0 \mathbf{\Sigma} \mathbf{Q}_0^\top$ , where  $\mathbf{\Sigma}$  is the diagonal matrix of eigenvalues of  $\mathbf{B}$ , which we list in decreasing order. Finally, using  $\mathbf{V}\mathbf{V}^* = \mathbf{I}$  and its conjugate  $\overline{\mathbf{V}}\mathbf{V}^\top = \mathbf{I}$ ,

$$\mathbf{V}\mathbf{B}\mathbf{V}^\top = \mathbf{V}\mathbf{V}^* \mathbf{A} \overline{\mathbf{V}}\mathbf{V}^\top = \mathbf{A},$$

so with  $\mathbf{U} := \mathbf{V}\mathbf{Q}_0$ , which is unitary,

$$\mathbf{A} = \mathbf{V}\mathbf{Q}_0 \mathbf{\Sigma} \mathbf{Q}_0^\top \mathbf{V}^\top = \mathbf{U} \mathbf{\Sigma} \mathbf{U}^\top.$$

It remains to identify the diagonal. Conjugate-transposing,  $\mathbf{A}^* = \overline{\mathbf{U}}\mathbf{\Sigma}\mathbf{U}^*$ , so

$$\mathbf{A}^*\mathbf{A} = \overline{\mathbf{U}}\mathbf{\Sigma}\mathbf{U}^*\mathbf{U}\mathbf{\Sigma}\mathbf{U}^\top = \overline{\mathbf{U}}\mathbf{\Sigma}^2(\overline{\mathbf{U}})^*.$$

This exhibits  $\mathbf{\Sigma}^2$  as the eigenvalue list of  $\mathbf{A}^*\mathbf{A}$  in decreasing order, so  $\mathbf{\Sigma} = \text{diag}(\sigma_1, \dots, \sigma_n)$  by [Definition 12.8.1](#). This proves the theorem for invertible  $\mathbf{A}$ . ■

*Remark. What is left out.* The theorem as stated is true for every complex symmetric  $\mathbf{A}$ , singular ones included, and the example below exhibits a singular case. Our proof used invertibility twice: to know that  $\mathbf{P}$  is invertible, so that  $\mathbf{W} = \mathbf{A}\mathbf{P}^{-1}$  is determined and the cancellations of Step 2 are legal. One way to remove the hypothesis is to apply the proved case to  $\mathbf{A} + t\mathbf{I}$ , which is symmetric and invertible for all but finitely many  $t$ , and let  $t \rightarrow 0$  along a sequence for which the unitary factors converge — a step that needs the compactness of the unitary group, which is analysis this book has not set up. We state the general theorem and prove the invertible case.

**Example. (Takagi for a real symmetric matrix)**

Find a Takagi factorization of  $\mathbf{A} = \begin{pmatrix} 3 & 4 \\ 4 & -3 \end{pmatrix}$ .

*Solution.* First the singular values:  $\mathbf{A}^*\mathbf{A} = \mathbf{A}^\top\mathbf{A} = \mathbf{A}^2 = 25\mathbf{I}$ , so  $\sigma_1 = \sigma_2 = 5$  and  $\mathbf{\Sigma} = 5\mathbf{I}$ . We therefore need a unitary  $\mathbf{U}$  with

$$\mathbf{A} = 5\mathbf{U}\mathbf{U}^\top, \quad \text{that is} \quad \mathbf{U}\mathbf{U}^\top = \frac{1}{5}\mathbf{A}.$$

The matrix  $\frac{1}{5}\mathbf{A}$  is real symmetric and orthogonal, with eigenvalues 1 and  $-1$ . Its eigenvectors:  $(\mathbf{A} - 5\mathbf{I})\mathbf{x} = \mathbf{0}$  reads  $-2x_1 + 4x_2 = 0$ , giving  $\mathbf{q}_1 = \frac{1}{\sqrt{5}}(2, 1)$ , and orthogonally  $\mathbf{q}_2 = \frac{1}{\sqrt{5}}(1, -2)$ . So with  $\mathbf{Q} = \frac{1}{\sqrt{5}}\begin{pmatrix} 2 & 1 \\ 1 & -2 \end{pmatrix} \in \text{O}(2)$  we have  $\frac{1}{5}\mathbf{A} = \mathbf{Q} \text{diag}(1, -1)\mathbf{Q}^\top$ . The diagonal entries 1 and  $-1$  have square roots 1 and  $i$ , so, following Step 3 of [Theorem 12.9.3](#), set

$$\mathbf{U} = \mathbf{Q} \text{diag}(1, i) = \frac{1}{\sqrt{5}}\begin{pmatrix} 2 & i \\ 1 & -2i \end{pmatrix}.$$

Then  $\mathbf{U}$  is unitary — its columns  $\frac{1}{\sqrt{5}}(2, 1)$  and  $\frac{i}{\sqrt{5}}(1, -2)$  are orthonormal, since multiplying a real orthonormal pair by unit scalars keeps it orthonormal — and

$$\mathbf{U}\mathbf{\Sigma}\mathbf{U}^\top = 5\mathbf{U}\mathbf{U}^\top = \mathbf{Q} \text{diag}(5, -5)\mathbf{Q}^\top = \mathbf{A}.$$

Note where the complex entries were needed:  $\text{diag}(1, i)$  converted the eigenvalue  $-5$  of  $\mathbf{A}$  into the singular value  $+5$ .

For a singular example, return to  $\mathbf{N} = \begin{pmatrix} 1 & i \\ i & -1 \end{pmatrix}$ . Here  $\mathbf{N}^*\mathbf{N} = \begin{pmatrix} 2 & 2i \\ -2i & 2 \end{pmatrix}$ , with eigenvalues 4 and 0, so  $\sigma_1 = 2$  and  $\sigma_2 = 0$ . With  $\mathbf{U} = \frac{1}{\sqrt{2}}\begin{pmatrix} 1 & i \\ i & 1 \end{pmatrix}$ , which is unitary, one checks  $\mathbf{U} \text{diag}(2, 0)\mathbf{U}^\top = 2\mathbf{u}_1\mathbf{u}_1^\top = \mathbf{N}$ , where  $\mathbf{u}_1 = \frac{1}{\sqrt{2}}(1, i)$  is the first column of  $\mathbf{U}$ . The theorem holds here even though our proof did not reach it, and even though  $\mathbf{N}$  has no eigenbasis at all.



*Warning.* **Symmetric is not Hermitian, and  $\mathbf{U}^\top$  is not  $\mathbf{U}^*$ .** Takagi's hypothesis is  $\mathbf{A}^\top = \mathbf{A}$  with no conjugation, which over  $\mathbb{C}$  is a different and weaker condition than  $\mathbf{A}^* = \mathbf{A}$ ; the matrix  $\mathbf{N}$  above is symmetric, is not Hermitian, is not normal and is not diagonalizable. The conclusion likewise carries  $\mathbf{U}^\top$ , not  $\mathbf{U}^*$ : writing  $\mathbf{U}\Sigma\mathbf{U}^*$  by mistake would assert that  $\mathbf{A}$  is positive semidefinite, which  $\mathbf{N}$  certainly is not.

## 12.9.4 Exercises

### 12.9.4.1 A. Check your understanding

#### Exercise A1.

- State the polar decomposition, and say which factor is unique and when.
- What is  $|\mathbf{A}|$ , and what identity relates  $\|\mathbf{A}\mathbf{x}\|$  to  $\|\mathbf{A}\mathbf{x}\|$ ?
- What does the polar decomposition say when  $n = 1$ ?
- True or false:  $|\mathbf{A}| = |\mathbf{A}^*|$  for every square  $\mathbf{A}$ . Justify your answer.
- State Takagi's factorization, and name two ways in which it differs from the singular value decomposition and from the complex spectral theorem.
- True or false: a complex symmetric matrix is diagonalizable. Justify your answer.

*Solution.*

- Every  $\mathbf{A} \in M_n(\mathbb{C})$  is  $\mathbf{A} = \mathbf{W}|\mathbf{A}|$  with  $\mathbf{W}$  unitary (Theorem 12.9.1). The positive semidefinite factor  $|\mathbf{A}|$  is always unique;  $\mathbf{W}$  is unique exactly when  $\mathbf{A}$  is invertible, and then  $\mathbf{W} = \mathbf{A}|\mathbf{A}|^{-1}$ .
- $|\mathbf{A}| = (\mathbf{A}^*\mathbf{A})^{1/2}$ , the unique positive semidefinite square root of  $\mathbf{A}^*\mathbf{A}$  (Definition 12.2.1), and  $\|\mathbf{A}\mathbf{x}\| = \|\mathbf{A}\mathbf{x}\|$  for every  $\mathbf{x}$ .
- It says  $z = e^{i\theta}|z|$ : a complex number is a number of modulus 1 times its modulus, with  $\theta$  undetermined exactly when  $z = 0$ .
- False in general, true exactly for normal  $\mathbf{A}$ . For  $\mathbf{A} = \begin{pmatrix} 3 & 0 \\ 4 & 5 \end{pmatrix}$ ,  $\mathbf{A}^\top \mathbf{A} = \begin{pmatrix} 25 & 20 \\ 20 & 25 \end{pmatrix} \neq \begin{pmatrix} 9 & 12 \\ 12 & 41 \end{pmatrix} = \mathbf{A}\mathbf{A}^\top$ , so the two square roots differ.
- Theorem 12.9.3: a symmetric  $\mathbf{A} \in M_n(\mathbb{C})$  is  $\mathbf{U}\Sigma\mathbf{U}^\top$  with  $\mathbf{U}$  unitary and  $\Sigma$  the diagonal matrix of singular values. Against the singular value decomposition: the right factor is not free but forced to be  $\bar{\mathbf{U}}$ . Against the spectral theorem: the hypothesis is symmetry rather than normality, the operation is the congruence  $\mathbf{X} \mapsto \mathbf{U}\mathbf{X}\mathbf{U}^\top$  rather than a similarity, and the diagonal carries singular values rather than eigenvalues.
- False.  $\mathbf{N} = \begin{pmatrix} 1 & i \\ i & -1 \end{pmatrix}$  is symmetric and non-zero with  $\mathbf{N}^2 = \mathbf{0}$ , so its minimal polynomial is  $x^2$ , which has a repeated root; it is not diagonalizable.

### 12.9.4.2 B. Practice

**Exercise B1: A polar decomposition by hand.**

Find both polar decompositions  $\mathbf{A} = \mathbf{W}|\mathbf{A}| = |\mathbf{A}^*|\mathbf{W}$  of

$$\mathbf{A} = \begin{pmatrix} 0 & -2 \\ 3 & 0 \end{pmatrix},$$

and say what  $\mathbf{W}$  is geometrically.

*Solution.*  $\mathbf{A}^\top \mathbf{A} = \begin{pmatrix} 9 & 0 \\ 0 & 4 \end{pmatrix}$ , which is diagonal with non-negative entries, so  $|\mathbf{A}| = \text{diag}(3, 2)$ : this is positive semidefinite and squares to  $\mathbf{A}^\top \mathbf{A}$ , which by [Theorem 12.2.1](#) identifies it. Since  $\mathbf{A}$  is invertible,

$$\mathbf{W} = \mathbf{A}|\mathbf{A}|^{-1} = \begin{pmatrix} 0 & -2 \\ 3 & 0 \end{pmatrix} \begin{pmatrix} \frac{1}{3} & 0 \\ 0 & \frac{1}{2} \end{pmatrix} = \begin{pmatrix} 0 & -1 \\ 1 & 0 \end{pmatrix},$$

the quarter turn of  $\mathbb{R}^2$ . Check:  $\mathbf{W}^\top \mathbf{W} = \mathbf{I}$  and  $\mathbf{W}|\mathbf{A}| = \mathbf{A}$ .

For the right-handed form,  $\mathbf{A}\mathbf{A}^\top = \text{diag}(4, 9)$ , so  $|\mathbf{A}^*| = \text{diag}(2, 3)$ , and indeed

$$|\mathbf{A}^*|\mathbf{W} = \begin{pmatrix} 2 & 0 \\ 0 & 3 \end{pmatrix} \begin{pmatrix} 0 & -1 \\ 1 & 0 \end{pmatrix} = \begin{pmatrix} 0 & -2 \\ 3 & 0 \end{pmatrix} = \mathbf{A}.$$

The two absolute values are different, which is consistent with  $\mathbf{A}$  not being normal. In words:  $\mathbf{A}$  stretches by 3 horizontally and 2 vertically, then rotates by a quarter turn; equivalently it rotates first and then stretches by 2 horizontally and 3 vertically.

**Exercise B2: The nearest unitary matrix.**

Let  $\mathbf{A} = \text{diag}(2, -3) \in M_2(\mathbb{R})$ . Find the unitary matrix nearest to  $\mathbf{A}$  in the Frobenius norm, compute the distance, and verify the value against the formula of [Corollary 12.9.2](#).

*Solution.*  $\mathbf{A}^* \mathbf{A} = \text{diag}(4, 9)$ , so  $|\mathbf{A}| = \text{diag}(2, 3)$  and  $\mathbf{W} = \mathbf{A}|\mathbf{A}|^{-1} = \text{diag}(1, -1)$ , which is unitary. By [Corollary 12.9.2](#),  $\mathbf{W}$  is the nearest unitary matrix, and it is the only one since  $\mathbf{A}$  is invertible. Directly,

$$\mathbf{A} - \mathbf{W} = \text{diag}(1, -2), \quad \|\mathbf{A} - \mathbf{W}\|_F^2 = 1 + 4 = 5.$$

The formula gives  $\|\mathbf{A}\|_F^2 + n - 2(\sigma_1 + \sigma_2) = 13 + 2 - 2(3 + 2) = 5$ , the same. So the distance is  $\sqrt{5}$ .

**Exercise B3: A Takagi factorization.**

Find a Takagi factorization of  $\mathbf{A} = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}$ , and explain why no **real**  $\mathbf{U}$  can work.

*Solution.*  $\mathbf{A}$  is symmetric and  $\mathbf{A}^* \mathbf{A} = \mathbf{A}^2 = \mathbf{I}$ , so  $\sigma_1 = \sigma_2 = 1$  and  $\Sigma = \mathbf{I}$ . We need a unitary  $\mathbf{U}$  with  $\mathbf{U}\mathbf{U}^\top = \mathbf{A}$ .

Diagonalize  $\mathbf{A}$  over  $\mathbb{R}$ : its eigenvalues are 1 and  $-1$ , with unit eigenvectors  $\frac{1}{\sqrt{2}}(1, 1)$

and  $\frac{1}{\sqrt{2}}(1, -1)$ , so  $\mathbf{A} = \mathbf{Q} \operatorname{diag}(1, -1) \mathbf{Q}^\top$  with  $\mathbf{Q} = \frac{1}{\sqrt{2}} \begin{pmatrix} 1 & 1 \\ 1 & -1 \end{pmatrix}$ . Following Step 3 of [Theorem 12.9.3](#), take square roots 1 and  $i$  of the diagonal entries:

$$\mathbf{U} = \mathbf{Q} \operatorname{diag}(1, i) = \frac{1}{\sqrt{2}} \begin{pmatrix} 1 & i \\ 1 & -i \end{pmatrix}, \quad \mathbf{U} \mathbf{U}^\top = \mathbf{Q} \operatorname{diag}(1, -1) \mathbf{Q}^\top = \mathbf{A}.$$

$\mathbf{U}$  is unitary, and  $\mathbf{U} \Sigma \mathbf{U}^\top = \mathbf{U} \mathbf{U}^\top = \mathbf{A}$ .

No real  $\mathbf{U}$  works: a real  $\mathbf{U}$  with  $\mathbf{U}^* \mathbf{U} = \mathbf{I}$  is orthogonal, and then  $\mathbf{U} \Sigma \mathbf{U}^\top = \mathbf{U} \mathbf{U}^\top = \mathbf{I} \neq \mathbf{A}$ .

### 12.9.4.3 C. Going deeper

**Exercise C1: The positive factor is forced.**

Let  $\mathbf{A} \in M_n(\mathbb{C})$ .

- (a) Suppose  $\mathbf{A} = \mathbf{Q} \mathbf{P}$  with  $\mathbf{Q}$  unitary and  $\mathbf{P} \succeq 0$ . Prove that  $\mathbf{P} = |\mathbf{A}|$ . (Invertibility is **not** assumed.)
- (b) Deduce that  $\mathbf{A}$  is unitary if and only if  $|\mathbf{A}| = \mathbf{I}$ .

*Solution.*

- (a) From  $\mathbf{A} = \mathbf{Q} \mathbf{P}$  we get  $\mathbf{A}^* \mathbf{A} = \mathbf{P}^* \mathbf{Q}^* \mathbf{Q} \mathbf{P} = \mathbf{P}^* \mathbf{P} = \mathbf{P}^2$ , using  $\mathbf{Q}^* \mathbf{Q} = \mathbf{I}$  and  $\mathbf{P}^* = \mathbf{P}$ . So  $\mathbf{P}$  is a positive semidefinite square root of  $\mathbf{A}^* \mathbf{A}$ . Such a square root is unique ([Theorem 12.2.1](#)), and  $|\mathbf{A}|$  is one by definition ([Definition 12.2.1](#)). Hence  $\mathbf{P} = |\mathbf{A}|$ .
- (b) ( $\Rightarrow$ ) If  $\mathbf{A}$  is unitary then  $\mathbf{A}^* \mathbf{A} = \mathbf{I}$ , and  $\mathbf{I} \succeq 0$  with  $\mathbf{I}^2 = \mathbf{I}$ , so  $|\mathbf{A}| = \mathbf{I}$  by uniqueness.  
 ( $\Leftarrow$ ) If  $|\mathbf{A}| = \mathbf{I}$  then  $\mathbf{A}^* \mathbf{A} = |\mathbf{A}|^2 = \mathbf{I}$ , which for a square matrix says  $\mathbf{A}$  is unitary ([Definition 10.7.2](#)).

**Exercise C2: Normality through the polar factors.**

Let  $\mathbf{A} \in M_n(\mathbb{C})$  be invertible with polar decomposition  $\mathbf{A} = \mathbf{W} |\mathbf{A}|$ . Prove that the following are equivalent.

- (a)  $\mathbf{A}$  is normal.
- (b)  $|\mathbf{A}| = |\mathbf{A}^*|$ .
- (c)  $\mathbf{W} |\mathbf{A}| = |\mathbf{A}| \mathbf{W}$ .

*Solution.*

- (a)  $\Rightarrow$  (b). If  $\mathbf{A}^* \mathbf{A} = \mathbf{A} \mathbf{A}^*$ , then  $|\mathbf{A}|$  and  $|\mathbf{A}^*|$  are positive semidefinite square roots of the same matrix, hence equal by [Theorem 12.2.1](#).
- (b)  $\Rightarrow$  (c). By [Theorem 12.9.1](#),  $|\mathbf{A}^*| = \mathbf{W} |\mathbf{A}| \mathbf{W}^*$ . If this equals  $|\mathbf{A}|$ , multiply on the right by  $\mathbf{W}$  to get  $|\mathbf{A}| \mathbf{W} = \mathbf{W} |\mathbf{A}|$ .
- (c)  $\Rightarrow$  (a). Write  $\mathbf{P} = |\mathbf{A}|$ , so  $\mathbf{A} = \mathbf{W} \mathbf{P} = \mathbf{P} \mathbf{W}$ . Then

$$\mathbf{A} \mathbf{A}^* = \mathbf{W} \mathbf{P} \mathbf{P}^* \mathbf{W}^* = \mathbf{W} \mathbf{P}^2 \mathbf{W}^* = \mathbf{P}^2 \mathbf{W} \mathbf{W}^* = \mathbf{P}^2,$$

using  $\mathbf{P}^* = \mathbf{P}$ , then  $\mathbf{WP}^2 = \mathbf{P}^2\mathbf{W}$  (which follows from  $\mathbf{WP} = \mathbf{PW}$ ), then  $\mathbf{WW}^* = \mathbf{I}$ . And  $\mathbf{A}^*\mathbf{A} = \mathbf{P}^*\mathbf{W}^*\mathbf{WP} = \mathbf{P}^2$ . So  $\mathbf{AA}^* = \mathbf{A}^*\mathbf{A}$ .

### Exercise C3: Complex symmetric matrices are transposed squares.

Let  $\mathbf{A} \in M_n(\mathbb{C})$  be symmetric and invertible.

- (a) Using [Theorem 12.9.3](#), prove that  $\mathbf{A} = \mathbf{B}^\top \mathbf{B}$  for some invertible  $\mathbf{B} \in M_n(\mathbb{C})$ .
- (b) Show that over  $\mathbb{R}$  the analogous statement is false unless  $\mathbf{A}$  is positive definite, and give a  $1 \times 1$  witness.
- (c) Explain in one sentence why (a) and (b) do not contradict each other.

*Solution.*

- (a) By [Theorem 12.9.3](#),  $\mathbf{A} = \mathbf{U}\mathbf{\Sigma}\mathbf{U}^\top$  with  $\mathbf{U}$  unitary and  $\mathbf{\Sigma} = \text{diag}(\sigma_1, \dots, \sigma_n)$ , where every  $\sigma_i > 0$  because  $\mathbf{A}$  is invertible ([Theorem 12.8.1](#)). Let  $\mathbf{\Sigma}^{1/2} = \text{diag}(\sqrt{\sigma_1}, \dots, \sqrt{\sigma_n})$  and put  $\mathbf{B} := \mathbf{\Sigma}^{1/2}\mathbf{U}^\top$ , which is invertible as a product of invertible matrices. Then

$$\mathbf{B}^\top \mathbf{B} = \mathbf{U} \mathbf{\Sigma}^{1/2} \mathbf{\Sigma}^{1/2} \mathbf{U}^\top = \mathbf{U} \mathbf{\Sigma} \mathbf{U}^\top = \mathbf{A},$$

using  $(\mathbf{\Sigma}^{1/2})^\top = \mathbf{\Sigma}^{1/2}$  and  $(\mathbf{U}^\top)^\top = \mathbf{U}$ .

- (b) Over  $\mathbb{R}$ , if  $\mathbf{A} = \mathbf{B}^\top \mathbf{B}$  with  $\mathbf{B} \in M_n(\mathbb{R})$  then  $\mathbf{x}^\top \mathbf{A} \mathbf{x} = \|\mathbf{B} \mathbf{x}\|^2 \geq 0$  for every  $\mathbf{x}$ , so  $\mathbf{A} \succeq 0$ , and  $\mathbf{A} \succ 0$  when  $\mathbf{B}$  is invertible. So no real  $\mathbf{A}$  with a negative eigenvalue is of this form: for  $n = 1$ ,  $\mathbf{A} = (-1)$  is symmetric and invertible but  $b^2 = -1$  has no real solution.
- (c) The factor  $\mathbf{B}$  produced in (a) is complex, and over  $\mathbb{C}$  the quantity  $\mathbf{x}^\top \mathbf{A} \mathbf{x}$  is not a sum of squared lengths — the transpose carries no conjugation, so  $\mathbf{B}^\top \mathbf{B}$  says nothing about positivity. Indeed  $b = i$  solves  $b^2 = -1$ .

## 12.10 Low-Rank Approximation

The compact singular value decomposition writes a matrix as a sum of rank-one pieces arranged in decreasing order of size. A sum can be cut short. This section shows that cutting it short after  $k$  terms produces the **best** approximation to  $\mathbf{A}$  by a matrix of rank  $k$ , measured in the Frobenius norm, and that the error it leaves is a quantity we can write down before doing any work. One theorem answers three questions at once: how to compress a matrix, which  $k$ -dimensional subspace best fits a cloud of data, and how far a given matrix is from being singular.

Throughout,  $F = \mathbb{R}$  or  $F = \mathbb{C}$ ,  $\mathbf{A} \in M_{m \times n}(F)$ ,  $p = \min(m, n)$ ,  $r = \text{rank } \mathbf{A}$ , and

$$\sigma_1 \geq \sigma_2 \geq \dots \geq \sigma_p \geq 0$$

are the singular values of  $\mathbf{A}$  ([Definition 12.8.1](#)), so that  $\sigma_i > 0$  exactly for  $i \leq r$ . We fix a singular value decomposition  $\mathbf{A} = \mathbf{U}\mathbf{\Sigma}\mathbf{V}^*$  ([Theorem 12.8.1](#)) and write  $\mathbf{u}_1, \dots, \mathbf{u}_m$  for the columns of  $\mathbf{U}$  and  $\mathbf{v}_1, \dots, \mathbf{v}_n$  for the columns of  $\mathbf{V}$ .

## 12.10.1 Cutting the sum short

Multiplying out  $\mathbf{U}\Sigma\mathbf{V}^*$  column by column gives the rank-one expansion

$$\mathbf{A} = \sum_{i=1}^p \sigma_i \mathbf{u}_i \mathbf{v}_i^* = \sum_{i=1}^r \sigma_i \mathbf{u}_i \mathbf{v}_i^*,$$

the second equality because the terms with  $i > r$  have  $\sigma_i = 0$ ; this is the compact form [Theorem 12.8.3](#). Each summand is a matrix of rank one, and the coefficients decrease. The natural thing to do with such a sum is to keep the large terms and discard the small ones.

*Keep the first  $k$  rank-one pieces and throw the rest away.*

### Definition 12.10.1 — Truncated singular value decomposition

For  $0 \leq k \leq p$ , the **rank- $k$  truncation** of  $\mathbf{A}$  relative to the chosen singular value decomposition  $\mathbf{A} = \mathbf{U}\Sigma\mathbf{V}^*$  ([Theorem 12.8.1](#)) is

$$\mathbf{A}_k := \sum_{i=1}^k \sigma_i \mathbf{u}_i \mathbf{v}_i^*,$$

with  $\mathbf{A}_0 = \mathbf{0}$ ; for  $k \geq p$  we set  $\mathbf{A}_k = \mathbf{A}$ .

Two clauses need comment. The subscript counts **terms kept**, not rows and columns:  $\mathbf{A}_k$  is the same size as  $\mathbf{A}$ , not a submatrix of it. (Section 1 used  $\mathbf{A}_k$  for the top-left  $k \times k$  corner; that meaning is gone for the rest of this section.) And the definition says “relative to the chosen decomposition”, because [Theorem 12.8.2](#) leaves the unitary factors some freedom, and different choices can give different truncations. Everything proved below holds for **every** choice; the warning at the end of the section shows that the dependence is real.

The rank of  $\mathbf{A}_k$  is exactly  $\min(k, r)$ : the vectors  $\mathbf{u}_1, \dots, \mathbf{u}_k$  are orthonormal, hence independent, so the columns of  $\mathbf{A}_k$  span  $\text{span}(\mathbf{u}_1, \dots, \mathbf{u}_{\min(k, r)})$  once the vanishing terms are dropped.

### Example. (Truncating a diagonal matrix)

Let  $\mathbf{A} = \text{diag}(5, 3, 1) \in M_3(\mathbb{R})$ . Write down  $\mathbf{A}_1$  and  $\mathbf{A}_2$ .

*Solution.* A diagonal matrix with positive decreasing entries is already in the form  $\mathbf{U}\Sigma\mathbf{V}^*$  with  $\mathbf{U} = \mathbf{V} = \mathbf{I}_3$ , so  $\sigma_i = 5, 3, 1$  and  $\mathbf{u}_i = \mathbf{v}_i = \mathbf{e}_i$ . Hence

$$\mathbf{A}_1 = 5\mathbf{e}_1\mathbf{e}_1^* = \text{diag}(5, 0, 0), \quad \mathbf{A}_2 = \text{diag}(5, 3, 0).$$

Truncation of a diagonal matrix is just deleting its smallest diagonal entries. Everything in this section is this statement, conjugated by two unitary matrices.

The first thing to compute is how much was thrown away.

### Lemma 12.10.1 — The Frobenius norm counts singular values

Let  $\mathbf{A} \in M_{m \times n}(F)$  have singular values  $\sigma_1 \geq \dots \geq \sigma_p$ . Then (the first identity is the

one Section 8 already obtained, recorded here with a label because the proof of the main theorem uses it repeatedly)

$$\|\mathbf{A}\|_F^2 = \sum_{i=1}^p \sigma_i^2, \quad \|\mathbf{A} - \mathbf{A}_k\|_F^2 = \sum_{i>k} \sigma_i^2$$

for every  $0 \leq k \leq p$ .

*Proof.* The Frobenius norm does not see unitary factors (Lemma 11.11.1), so from  $\mathbf{A} = \mathbf{U}\mathbf{\Sigma}\mathbf{V}^*$ ,

$$\|\mathbf{A}\|_F = \|\mathbf{\Sigma}\mathbf{V}^*\|_F = \|\mathbf{\Sigma}\|_F.$$

The only non-zero entries of  $\mathbf{\Sigma}$  are  $\sigma_1, \dots, \sigma_p$  on the diagonal, so  $\|\mathbf{\Sigma}\|_F^2 = \sum_i \sigma_i^2$ .

For the second identity, subtract the first  $k$  terms of the rank-one expansion:

$$\mathbf{A} - \mathbf{A}_k = \sum_{i>k} \sigma_i \mathbf{u}_i \mathbf{v}_i^* = \mathbf{U}\mathbf{\Sigma}^{(k)}\mathbf{V}^*,$$

where  $\mathbf{\Sigma}^{(k)}$  is  $\mathbf{\Sigma}$  with its first  $k$  diagonal entries replaced by 0. Applying the first identity to this matrix — the display is a singular value decomposition of it in the sense of Theorem 12.8.1, up to reordering the diagonal — gives  $\|\mathbf{A} - \mathbf{A}_k\|_F^2 = \sum_{i>k} \sigma_i^2$ . ■

So the truncation error is decided in advance by the singular values, and it is small exactly when the discarded  $\sigma_i$  are small. The content of the section is that no other matrix of rank  $k$  does better.

*Quick check.* A matrix has singular values 5, 4, 3. What is the smallest  $k$  for which the relative error  $\|\mathbf{A} - \mathbf{A}_k\|_F / \|\mathbf{A}\|_F$  is at most  $\frac{1}{2}$ ?

*Solution.* By Lemma 12.10.1,  $\|\mathbf{A}\|_F^2 = 25 + 16 + 9 = 50$ . For  $k = 1$  the error squared is  $16 + 9 = 25$ , so the relative error is  $\sqrt{25/50} = 1/\sqrt{2} \approx 0.71$ , too big. For  $k = 2$  the error squared is 9, and  $\sqrt{9/50} \approx 0.42 \leq \frac{1}{2}$ . So  $k = 2$ . Note that the test is on the *squares*: halving the error means keeping three quarters of  $\sum \sigma_i^2$ .

## 12.10.2 How much a subspace can hold

The proof of the theorem needs one inequality, and it is the only place where any work happens. The question it answers is: if we are allowed to keep only an  $s$ -dimensional subspace of  $F^m$ , how much of  $\mathbf{A}$  can we capture? The answer is  $\sigma_1^2 + \dots + \sigma_s^2$ , and the best subspace is the span of the leading left singular vectors.

**Lemma 12.10.2 — An orthonormal list captures at most the leading singular values**

Let  $\mathbf{A} \in M_{m \times n}(F)$  with singular values  $\sigma_1 \geq \dots \geq \sigma_p$ , extended by  $\sigma_i = 0$  for  $p < i \leq m$ .

Let  $\mathbf{q}_1, \dots, \mathbf{q}_s$  be an orthonormal list in  $F^m$ , so that  $s \leq m$ . Then

$$\sum_{j=1}^s \|\mathbf{A}^* \mathbf{q}_j\|^2 \leq \sum_{i=1}^s \sigma_i^2.$$

*Idea.* Write each  $\mathbf{q}_j$  in the orthonormal basis  $(\mathbf{u}_1, \dots, \mathbf{u}_m)$ . The quantity  $\|\mathbf{A}^* \mathbf{q}_j\|^2$  then becomes  $\sum_i \sigma_i^2 |c_{ji}|^2$ , so the total is  $\sum_i \sigma_i^2 t_i$  with weights  $t_i = \sum_j |c_{ji}|^2$ . Two facts pin the weights down: each  $t_i \leq 1$ , because the  $\mathbf{q}_j$  are orthonormal and Bessel's inequality applies to  $\mathbf{u}_i$ ; and  $\sum_i t_i = s$ , because each  $\mathbf{q}_j$  is a unit vector. A weighted sum of a decreasing sequence with weights in  $[0, 1]$  that sum to  $s$  is largest when the first  $s$  weights are 1 — which is the last display of the proof.

*Proof.* If  $s = 0$  both sides are empty sums, so assume  $s \geq 1$ . Fix  $j$  and put  $c_{ji} = \langle \mathbf{q}_j, \mathbf{u}_i \rangle$  for  $1 \leq i \leq m$ , so that the  $i$ -th entry of  $\mathbf{U}^* \mathbf{q}_j$  is  $c_{ji}$ . Since  $\mathbf{A}^* = \mathbf{V} \mathbf{\Sigma}^* \mathbf{U}^*$  and  $\mathbf{V}$  is unitary, hence an isometry (Definition 10.7.2, Theorem 10.7.3),

$$\|\mathbf{A}^* \mathbf{q}_j\|^2 = \|\mathbf{\Sigma}^* \mathbf{U}^* \mathbf{q}_j\|^2 = \sum_{i=1}^m \sigma_i^2 |c_{ji}|^2,$$

because  $\mathbf{\Sigma}^*$  multiplies the  $i$ -th entry by  $\sigma_i$  and discards the entries beyond the  $p$ -th, which is the same as using  $\sigma_i = 0$  there. Summing over  $j$  and exchanging the two finite sums,

$$\sum_{j=1}^s \|\mathbf{A}^* \mathbf{q}_j\|^2 = \sum_{i=1}^m \sigma_i^2 t_i, \quad t_i := \sum_{j=1}^s |c_{ji}|^2.$$

Now bound the weights. Since  $|c_{ji}| = |\langle \mathbf{u}_i, \mathbf{q}_j \rangle|$  and  $(\mathbf{q}_1, \dots, \mathbf{q}_s)$  is an orthonormal list, Bessel's inequality (Theorem 10.9.1) applied to the vector  $\mathbf{u}_i$  gives

$$t_i = \sum_{j=1}^s |\langle \mathbf{u}_i, \mathbf{q}_j \rangle|^2 \leq \|\mathbf{u}_i\|^2 = 1.$$

Since  $(\mathbf{u}_1, \dots, \mathbf{u}_m)$  is an orthonormal **basis** of  $F^m$ , Parseval's identity (Theorem 10.9.2) gives  $\sum_{i=1}^m |c_{ji}|^2 = \|\mathbf{q}_j\|^2 = 1$  for each  $j$ , and therefore

$$\sum_{i=1}^m t_i = \sum_{j=1}^s \sum_{i=1}^m |c_{ji}|^2 = s.$$

Finally, using  $\sigma_i^2 \geq \sigma_s^2$  together with  $t_i - 1 \leq 0$  for  $i \leq s$ , and  $\sigma_i^2 \leq \sigma_s^2$  together with  $t_i \geq 0$  for  $i > s$ ,

$$\begin{aligned} \sum_{i=1}^m \sigma_i^2 t_i - \sum_{i=1}^s \sigma_i^2 &= \sum_{i \leq s} \sigma_i^2 (t_i - 1) + \sum_{i > s} \sigma_i^2 t_i \\ &\leq \sigma_s^2 \left( \sum_{i \leq s} (t_i - 1) + \sum_{i > s} t_i \right) \\ &= \sigma_s^2 \left( \sum_{i=1}^m t_i - s \right) = 0. \end{aligned}$$

This is the stated inequality. ■

Equality is attainable: take  $\mathbf{q}_j = \mathbf{u}_j$  for  $j \leq s$ . Then  $\mathbf{A}^* \mathbf{u}_j = \sigma_j \mathbf{v}_j$  by the rank-one expansion, so  $\|\mathbf{A}^* \mathbf{u}_j\|^2 = \sigma_j^2$  and the two sides agree. The lemma is the extreme case of a family of statements about sums of the largest eigenvalues of a Hermitian matrix; Chapter 16 develops the family through the Courant–Fischer min–max theorem, of which the inequality above is the crudest and most useful consequence.

### 12.10.3 The Eckart–Young theorem

Now the theorem. Read it as a statement about a minimization problem: over the set of matrices of rank at most  $k$  — which is not a subspace, so no earlier optimization theorem applies to it directly — the minimum of  $\|\mathbf{A} - \mathbf{B}\|_F$  exists, is attained at a matrix we can write down, and has a value we already computed.

#### Theorem 12.10.3 — Eckart–Young Theorem, Frobenius case

Let  $\mathbf{A} \in M_{m \times n}(F)$  with singular values  $\sigma_1 \geq \cdots \geq \sigma_p$ , and let  $0 \leq k \leq p$ . Then for every  $\mathbf{B} \in M_{m \times n}(F)$  with  $\text{rank } \mathbf{B} \leq k$ ,

$$\|\mathbf{A} - \mathbf{B}\|_F \geq \left( \sum_{i>k} \sigma_i^2 \right)^{1/2},$$

and equality holds for  $\mathbf{B} = \mathbf{A}_k$ . In words: the truncation  $\mathbf{A}_k$  minimizes  $\|\mathbf{A} - \mathbf{B}\|_F$  over all matrices of rank at most  $k$ , and the minimum value is  $(\sum_{i>k} \sigma_i^2)^{1/2}$ .

*Idea.* The difficulty is that  $\mathbf{B}$  is an arbitrary matrix of rank  $\leq k$ , and there is no formula for such a thing. The rescue is to stop looking at  $\mathbf{B}$  and look only at its column space  $S$ , a subspace of dimension  $s \leq k$ . ① Every column of  $\mathbf{B}$  lies in  $S$ , so column by column the Best Approximation Theorem says  $\mathbf{B}$  cannot beat  $P_S \mathbf{A}$ , the matrix obtained by projecting every column of  $\mathbf{A}$  onto  $S$ . So it suffices to minimize over subspaces. ② For a subspace, Pythagoras splits the error into “all of  $\mathbf{A}$ ” minus “the part  $S$  captures”, and the captured part is exactly the quantity Lemma 12.10.2 bounds. ③ The bound is  $\sigma_1^2 + \cdots + \sigma_k^2$ , and subtracting it from  $\|\mathbf{A}\|_F^2 = \sum_i \sigma_i^2$  leaves  $\sum_{i>k} \sigma_i^2$ , which Lemma 12.10.1 says is exactly what  $\mathbf{A}_k$  achieves.

*Proof. Step 1: reduce to a projection.* Let  $\mathbf{B}$  have rank  $\leq k$  and put  $S = \text{col}(\mathbf{B})$ , so  $s := \dim S = \text{rank } \mathbf{B} \leq k$ . Write  $\mathbf{a}_1, \dots, \mathbf{a}_n$  and  $\mathbf{b}_1, \dots, \mathbf{b}_n$  for the columns of  $\mathbf{A}$  and of  $\mathbf{B}$ . By the definition of the Frobenius norm, summing squares of entries column by column,

$$\|\mathbf{A} - \mathbf{B}\|_F^2 = \sum_{j=1}^n \|\mathbf{a}_j - \mathbf{b}_j\|^2.$$

Each  $\mathbf{b}_j$  lies in  $S$ , so Theorem 10.3.5 gives  $\|\mathbf{a}_j - \mathbf{b}_j\| \geq \|\mathbf{a}_j - P_S \mathbf{a}_j\|$  for every  $j$ , where  $P_S$  is the orthogonal projection onto  $S$  (Definition 10.3.2), identified throughout with its matrix in the standard basis of  $F^m$ . Hence

$$\|\mathbf{A} - \mathbf{B}\|_F^2 \geq \sum_{j=1}^n \|\mathbf{a}_j - P_S \mathbf{a}_j\|^2 = \|\mathbf{A} - P_S \mathbf{A}\|_F^2,$$

the last equality because the  $j$ -th column of  $P_S \mathbf{A}$  is  $P_S \mathbf{a}_j$ .



**Step 2: what a subspace captures.** Choose an orthonormal basis  $(\mathbf{q}_1, \dots, \mathbf{q}_s)$  of  $S$ . For each  $j$ , the vectors  $\mathbf{a}_j - P_S \mathbf{a}_j$  and  $P_S \mathbf{a}_j$  are orthogonal, so [Theorem 10.1.3](#) gives  $\|\mathbf{a}_j - P_S \mathbf{a}_j\|^2 = \|\mathbf{a}_j\|^2 - \|P_S \mathbf{a}_j\|^2$ , and summing over  $j$ ,

$$\|\mathbf{A} - P_S \mathbf{A}\|_F^2 = \|\mathbf{A}\|_F^2 - \|P_S \mathbf{A}\|_F^2.$$

By [Theorem 10.3.3](#) (a),  $P_S \mathbf{a}_j = \sum_{l \leq s} \langle \mathbf{a}_j, \mathbf{q}_l \rangle \mathbf{q}_l$ , whose summands are pairwise orthogonal, so [Theorem 10.1.3](#) gives  $\|P_S \mathbf{a}_j\|^2 = \sum_{l=1}^s |\langle \mathbf{a}_j, \mathbf{q}_l \rangle|^2$ . Exchanging the sums,

$$\|P_S \mathbf{A}\|_F^2 = \sum_{l=1}^s \sum_{j=1}^n |\langle \mathbf{a}_j, \mathbf{q}_l \rangle|^2 = \sum_{l=1}^s \|\mathbf{A}^* \mathbf{q}_l\|^2,$$

where the last equality holds because  $\langle \mathbf{a}_j, \mathbf{q}_l \rangle = \mathbf{q}_l^* \mathbf{a}_j$  is the  $j$ -th entry of the row vector  $\mathbf{q}_l^* \mathbf{A}$ , whose entries are the conjugates of those of  $\mathbf{A}^* \mathbf{q}_l$ .

**Step 3: apply the bound.** By [Lemma 12.10.2](#) and  $s \leq k$ , and since all  $\sigma_i^2 \geq 0$ ,

$$\|P_S \mathbf{A}\|_F^2 \leq \sum_{i=1}^s \sigma_i^2 \leq \sum_{i=1}^k \sigma_i^2.$$

Combining the three steps with  $\|\mathbf{A}\|_F^2 = \sum_i \sigma_i^2$  ([Lemma 12.10.1](#)),

$$\|\mathbf{A} - \mathbf{B}\|_F^2 \geq \|\mathbf{A}\|_F^2 - \sum_{i=1}^k \sigma_i^2 = \sum_{i>k} \sigma_i^2.$$

**Equality.** The matrix  $\mathbf{A}_k$  has rank  $\min(k, r) \leq k$ , and  $\|\mathbf{A} - \mathbf{A}_k\|_F^2 = \sum_{i>k} \sigma_i^2$  by [Lemma 12.10.1](#). So the lower bound is attained, and it is the minimum. This proves the theorem. ■

Three remarks are worth extracting. First, the theorem is a genuine minimum, not an infimum: the minimizing matrix is exhibited. Second, the minimum value depends on  $\mathbf{A}$  only through its singular values, so it is unchanged if  $\mathbf{A}$  is replaced by  $\mathbf{U}' \mathbf{A} \mathbf{V}'$  with  $\mathbf{U}', \mathbf{V}'$  unitary. Third, the case  $k = r - 1$  reads: the Frobenius distance from  $\mathbf{A}$  to the matrices of rank  $< r$  is  $\sigma_r$ , the smallest non-zero singular value. Exercise C1 turns that into a statement about how far an invertible matrix is from being singular.

*Remark.* The same matrix  $\mathbf{A}_k$  is also optimal in the **spectral norm**  $\|\cdot\|_2$ , where the minimum value is  $\sigma_{k+1}$  rather than  $(\sum_{i>k} \sigma_i^2)^{1/2}$ . We cannot state that here, because the spectral norm is defined only in Chapter 15. In fact both statements are instances of one theorem:  $\mathbf{A}_k$  is a best rank- $k$  approximation in **every** unitarily invariant norm. Chapter 20 proves that, and the proof is genuinely different from the one above — it needs inequalities between the singular values of  $\mathbf{A}$  and of  $\mathbf{A} - \mathbf{B}$ , not a projection argument.

## 12.10.4 A worked example

**Example. (The best rank-one approximation of a  $3 \times 2$  matrix)**

Let

$$\mathbf{A} = \begin{pmatrix} 1 & 0 \\ 0 & 1 \\ 1 & 1 \end{pmatrix} \in M_{3 \times 2}(\mathbb{R}).$$

Find the best rank-one approximation of  $\mathbf{A}$  in the Frobenius norm, and compute the error in two ways.

*Solution. Singular values and vectors.* Since  $\mathbf{A}$  has more rows than columns, work with the  $2 \times 2$  matrix

$$\mathbf{A}^\top \mathbf{A} = \begin{pmatrix} 2 & 1 \\ 1 & 2 \end{pmatrix},$$

whose eigenvalues are 3 and 1 with orthonormal eigenvectors

$$\mathbf{v}_1 = \frac{1}{\sqrt{2}}(1, 1), \quad \mathbf{v}_2 = \frac{1}{\sqrt{2}}(1, -1).$$

Hence  $\sigma_1 = \sqrt{3}$  and  $\sigma_2 = 1$ , and the left singular vectors are  $\mathbf{u}_i = \mathbf{A}\mathbf{v}_i/\sigma_i$ :

$$\mathbf{u}_1 = \frac{1}{\sqrt{6}}(1, 1, 2), \quad \mathbf{u}_2 = \frac{1}{\sqrt{2}}(1, -1, 0).$$

(Both are unit vectors, as they must be:  $1 + 1 + 4 = 6$  and  $1 + 1 = 2$ .)

*The approximation.* By [Theorem 12.10.3](#) the answer is  $\mathbf{A}_1 = \sigma_1 \mathbf{u}_1 \mathbf{v}_1^\top$ . The scalar in front is  $\sqrt{3}/(\sqrt{6}\sqrt{2}) = \sqrt{3}/\sqrt{12} = \frac{1}{2}$ , so

$$\mathbf{A}_1 = \frac{1}{2} \begin{pmatrix} 1 & 1 \\ 1 & 1 \\ 2 & 2 \end{pmatrix}.$$

It has rank one, its two columns being equal and non-zero.

*The error, twice.* Directly,

$$\mathbf{A} - \mathbf{A}_1 = \frac{1}{2} \begin{pmatrix} 1 & -1 \\ -1 & 1 \\ 0 & 0 \end{pmatrix}, \quad \|\mathbf{A} - \mathbf{A}_1\|_F^2 = 4 \cdot \frac{1}{4} = 1.$$

From the theorem, the error squared is  $\sigma_2^2 = 1$ . The two agree. As a further check,  $\|\mathbf{A}\|_F^2 = 1 + 1 + 1 + 1 = 4$  and  $\sigma_1^2 + \sigma_2^2 = 3 + 1 = 4$ , as [Lemma 12.10.1](#) requires; the relative error of the rank-one approximation is  $\sqrt{1/4} = \frac{1}{2}$ .

## 12.10.5 Best-fitting subspaces

Step 1 of the proof did more than its job. It showed that the optimal  $\mathbf{B}$  may be taken of the form  $P_S \mathbf{A}$ , and Steps 2 and 3 identified the best  $S$ . That identification is a theorem in its own right, and it is the statement people mean when they speak of fitting a low-dimensional structure to data.

### Corollary 12.10.4 — The best-fitting subspace

Let  $\mathbf{a}_1, \dots, \mathbf{a}_n \in F^m$  be the columns of  $\mathbf{A}$ , and let  $1 \leq k \leq p$ . Among all subspaces  $S \subseteq F^m$  with  $\dim S \leq k$ , the total squared distance

$$D(S) := \sum_{j=1}^n d(\mathbf{a}_j, S)^2$$

is smallest for  $S = \text{span}(\mathbf{u}_1, \dots, \mathbf{u}_k)$ , and its smallest value is  $\sum_{i>k} \sigma_i^2$ .

*Proof.* By Definition 10.3.3,  $d(\mathbf{a}_j, S) = \|\mathbf{a}_j - P_S \mathbf{a}_j\|$ , so  $D(S) = \|\mathbf{A} - P_S \mathbf{A}\|_F^2$  as in Step 1 of Theorem 12.10.3. Since  $\text{rank}(P_S \mathbf{A}) \leq \dim S \leq k$ , the theorem gives  $D(S) \geq \sum_{i>k} \sigma_i^2$ . For  $S_0 = \text{span}(\mathbf{u}_1, \dots, \mathbf{u}_k)$  the columns  $\mathbf{u}_1, \dots, \mathbf{u}_k$  are orthonormal, so Theorem 10.3.3 (a) gives

$$P_{S_0} \mathbf{A} = \sum_{i=1}^k \mathbf{u}_i \mathbf{u}_i^* \mathbf{A} = \sum_{i=1}^k \sigma_i \mathbf{u}_i \mathbf{v}_i^* = \mathbf{A}_k,$$

using  $\mathbf{u}_i^* \mathbf{A} = \sigma_i \mathbf{v}_i^*$ , which is the rank-one expansion read one row at a time. Hence  $D(S_0) = \|\mathbf{A} - \mathbf{A}_k\|_F^2 = \sum_{i>k} \sigma_i^2$ . ■

**What this is used for, honestly.** Two applications are within reach of what we have proved, and a third is not.

The first is **compression**. Storing  $\mathbf{A}$  costs  $mn$  numbers; storing  $\mathbf{A}_k$  as the list  $\sigma_1, \mathbf{u}_1, \mathbf{v}_1, \dots, \sigma_k, \mathbf{u}_k, \mathbf{v}_k$  costs  $k(m+n+1)$ . When  $k$  is much smaller than  $m$  and  $n$  that is a large saving, and Theorem 12.10.3 says no other rank- $k$  object stored the same way is closer to  $\mathbf{A}$  in the Frobenius norm. The relative error is  $(\sum_{i>k} \sigma_i^2 / \sum_i \sigma_i^2)^{1/2}$ , computable before choosing  $k$ . Whether that error is acceptable is not a mathematical question.

The second is **principal components**. Take  $n$  data points in  $\mathbb{R}^m$  and place them as the columns of  $\mathbf{A}$ . Then Corollary 12.10.4 says: the  $k$ -dimensional subspace through the origin that minimizes the total squared distance to the data is spanned by  $\mathbf{u}_1, \dots, \mathbf{u}_k$ , the leading left singular vectors. These directions are the **principal components** of the data, and the numbers  $\sigma_i^2$  measure how much of  $\sum_j \|\mathbf{a}_j\|^2$  each one accounts for. That is a theorem of linear algebra, and it is the whole of what this book has proved.

What is *not* proved here is the statistics. The usual procedure subtracts the mean of the data first, so that  $\frac{1}{n} \mathbf{A} \mathbf{A}^\top$  becomes a sample covariance matrix, and then interprets  $\sigma_i^2 / \sum_l \sigma_l^2$  as a fraction of variance explained. Those steps need a probabilistic model, a reason to prefer squared error, and an argument that the sample answer says something about the population. Chapter 23 supplies them, along with how a singular value decomposition is actually computed for a large matrix — which is never by forming  $\mathbf{A}^* \mathbf{A}$  and finding its eigenvalues, the route taken in every hand computation in this chapter.

**Warning.** The truncation is optimal for this norm, and low rank does not mean small. Two mistakes to avoid.

*Other measures of error give other answers.* Let  $\mathbf{A} = \text{diag}(2, 1)$ , so  $\sigma_1 = 2$ ,  $\sigma_2 = 1$ , and  $\mathbf{A}_1 = \text{diag}(2, 0)$ , whose error matrix  $\text{diag}(0, 1)$  has a largest entry of modulus 1. The

rank-one matrix

$$\mathbf{B} = \frac{1}{2} \begin{pmatrix} 3 & \sqrt{3} \\ \sqrt{3} & 1 \end{pmatrix}, \quad \mathbf{A} - \mathbf{B} = \frac{1}{2} \begin{pmatrix} 1 & -\sqrt{3} \\ -\sqrt{3} & 1 \end{pmatrix},$$

has  $\det \mathbf{B} = \frac{3}{4} - \frac{3}{4} = 0$ , so  $\text{rank } \mathbf{B} = 1$ , and every entry of  $\mathbf{A} - \mathbf{B}$  has modulus at most  $\sqrt{3}/2 \approx 0.87 < 1$ . So  $\mathbf{B}$  beats  $\mathbf{A}_1$  if the error is measured entrywise. It loses in the Frobenius norm, as it must:  $\|\mathbf{A} - \mathbf{B}\|_F^2 = 2 > 1 = \|\mathbf{A} - \mathbf{A}_1\|_F^2$ .

*Small rank and small size are unrelated.* The  $n \times n$  all-ones matrix  $\mathbf{J}$  has rank 1 and  $\|\mathbf{J}\|_F = n$ , while  $\varepsilon \mathbf{I}_n$  has rank  $n$  and  $\|\varepsilon \mathbf{I}_n\|_F = \varepsilon \sqrt{n}$ , as small as we like. “Approximating by a matrix of low rank” is a statement about structure, never about magnitude.

## 12.10.6 Exercises

### 12.10.6.1 A. Check your understanding

#### Exercise A1.

- State the Eckart–Young Theorem, including the value of the minimum.
- Express  $\|\mathbf{A}\|_F$  in terms of the singular values of  $\mathbf{A}$ .
- Let  $\text{rank } \mathbf{A} = r$ . What is  $\mathbf{A}_k$  for  $k \geq r$ , and what is the minimum of  $\|\mathbf{A} - \mathbf{B}\|_F$  over  $\text{rank } \mathbf{B} \leq k$  in that case?
- True or false: the truncation  $\mathbf{A}_k$  is the *unique* matrix of rank at most  $k$  nearest to  $\mathbf{A}$  in the Frobenius norm. Justify your answer.
- A matrix of rank 1 is close to  $\mathbf{0}$  in the Frobenius norm. True or false? Justify your answer.

*Solution.*

- For  $\mathbf{A} \in M_{m \times n}(F)$  with singular values  $\sigma_1 \geq \dots \geq \sigma_p$  and  $0 \leq k \leq p$ : every  $\mathbf{B}$  with  $\text{rank } \mathbf{B} \leq k$  satisfies  $\|\mathbf{A} - \mathbf{B}\|_F \geq (\sum_{i>k} \sigma_i^2)^{1/2}$ , with equality for  $\mathbf{B} = \mathbf{A}_k = \sum_{i \leq k} \sigma_i \mathbf{u}_i \mathbf{v}_i^*$ .
- $\|\mathbf{A}\|_F = (\sum_{i=1}^p \sigma_i^2)^{1/2}$  (Lemma 12.10.1).
- $\mathbf{A}_k = \mathbf{A}$ , since the terms with  $i > r$  are zero; the minimum is 0, attained at  $\mathbf{B} = \mathbf{A}$  itself, which has rank  $r \leq k$ .
- False in general. Take  $\mathbf{A} = \mathbf{I}_2$ , so  $\sigma_1 = \sigma_2 = 1$ . Both  $\mathbf{B}_1 = \text{diag}(1, 0)$  and  $\mathbf{B}_2 = \frac{1}{2} \begin{pmatrix} 1 & 1 \\ 1 & 1 \end{pmatrix}$  have rank 1, and  $\|\mathbf{I}_2 - \mathbf{B}_1\|_F^2 = 1 = \|\mathbf{I}_2 - \mathbf{B}_2\|_F^2$ , which is the minimum  $\sigma_2^2$ . (Both are truncations of  $\mathbf{I}_2$ , for two different singular value decompositions.)
- False. Rank is unrelated to size: the all-ones matrix  $\mathbf{J} \in M_n(\mathbb{R})$  has rank 1 and  $\|\mathbf{J}\|_F = n$ .

### 12.10.6.2 B. Practice

#### Exercise B1: Errors from the singular values alone.

A matrix  $\mathbf{A}$  has singular values 6, 3, 2, 1.

- (a) Compute  $\|\mathbf{A}\|_F$  and the minimum of  $\|\mathbf{A} - \mathbf{B}\|_F$  over matrices  $\mathbf{B}$  of rank at most  $k$ , for  $k = 1, 2, 3$ .
- (b) Find the smallest  $k$  for which the relative error  $\|\mathbf{A} - \mathbf{A}_k\|_F / \|\mathbf{A}\|_F$  is at most 0.3.

*Solution.*

- (a)  $\|\mathbf{A}\|_F^2 = 36 + 9 + 4 + 1 = 50$ , so  $\|\mathbf{A}\|_F = \sqrt{50} = 5\sqrt{2}$ . By [Theorem 12.10.3](#) the minima are
- $$\sqrt{9+4+1} = \sqrt{14}, \quad \sqrt{4+1} = \sqrt{5}, \quad \sqrt{1} = 1$$
- for  $k = 1, 2, 3$ .
- (b) The condition is  $\sum_{i>k} \sigma_i^2 \leq 0.09 \cdot 50 = 4.5$ . For  $k = 2$  the left side is  $5 > 4.5$ ; for  $k = 3$  it is  $1 \leq 4.5$ . So  $k = 3$ .

### Exercise B2: A best rank-one approximation.

Let  $\mathbf{A} = \begin{pmatrix} 3 & 1 \\ 1 & 3 \end{pmatrix} \in M_2(\mathbb{R})$ . Find the singular values of  $\mathbf{A}$ , the best rank-one approximation  $\mathbf{A}_1$  in the Frobenius norm, and the error, computed both directly and from the theorem.

*Solution.*  $\mathbf{A}$  is symmetric with eigenvalues 4 and 2 (trace 6, determinant 8) and orthonormal eigenvectors  $\mathbf{q}_1 = \frac{1}{\sqrt{2}}(1, 1)$ ,  $\mathbf{q}_2 = \frac{1}{\sqrt{2}}(1, -1)$ . Both eigenvalues are positive, so  $\mathbf{A} \succ 0$ , and  $\mathbf{A}^\top \mathbf{A} = \mathbf{A}^2$  has eigenvalues 16 and 4; hence  $\sigma_1 = 4$ ,  $\sigma_2 = 2$ , with  $\mathbf{u}_i = \mathbf{v}_i = \mathbf{q}_i$ .

Therefore

$$\mathbf{A}_1 = 4\mathbf{q}_1\mathbf{q}_1^\top = 4 \cdot \frac{1}{2} \begin{pmatrix} 1 & 1 \\ 1 & 1 \end{pmatrix} = \begin{pmatrix} 2 & 2 \\ 2 & 2 \end{pmatrix},$$

which has rank one. Directly,

$$\mathbf{A} - \mathbf{A}_1 = \begin{pmatrix} 1 & -1 \\ -1 & 1 \end{pmatrix}, \quad \|\mathbf{A} - \mathbf{A}_1\|_F^2 = 4.$$

From [Theorem 12.10.3](#) the error squared is  $\sigma_2^2 = 4$ . The two agree, and  $\|\mathbf{A}\|_F^2 = 9 + 1 + 1 + 9 = 20 = 16 + 4$  checks [Lemma 12.10.1](#).

### Exercise B3: Pythagoras for the truncation.

Let  $\mathbf{A} \in M_{m \times n}(F)$  and  $0 \leq k \leq p$ . Prove that

$$\|\mathbf{A}\|_F^2 = \|\mathbf{A}_k\|_F^2 + \|\mathbf{A} - \mathbf{A}_k\|_F^2,$$

and deduce that  $k \mapsto \|\mathbf{A} - \mathbf{A}_k\|_F$  is non-increasing.

*Solution.* By [Lemma 12.10.1](#) applied to  $\mathbf{A}$  and to  $\mathbf{A} - \mathbf{A}_k$ , and by the same lemma applied to  $\mathbf{A}_k$ , whose singular values are  $\sigma_1, \dots, \sigma_k$  followed by zeros (it equals  $\mathbf{U}\Sigma'\mathbf{V}^*$  with  $\Sigma'$  the matrix keeping only the first  $k$  diagonal entries of  $\Sigma$ ),

$$\|\mathbf{A}_k\|_F^2 + \|\mathbf{A} - \mathbf{A}_k\|_F^2 = \sum_{i \leq k} \sigma_i^2 + \sum_{i > k} \sigma_i^2 = \|\mathbf{A}\|_F^2.$$

Since  $\|\mathbf{A} - \mathbf{A}_k\|_F^2 = \sum_{i>k} \sigma_i^2$  and the  $\sigma_i^2$  are non-negative, increasing  $k$  by one removes a non-negative term, so the sequence is non-increasing. This proves both claims.

### 12.10.6.3 C. Going deeper

#### Exercise C1: How far is a matrix from being singular?

Let  $\mathbf{A} \in M_n(F)$  be invertible, with singular values  $\sigma_1 \geq \dots \geq \sigma_n > 0$ .

- Prove that  $\|\mathbf{A} - \mathbf{B}\|_F \geq \sigma_n$  for every singular  $\mathbf{B} \in M_n(F)$ .
- Exhibit a singular  $\mathbf{B}$  with  $\|\mathbf{A} - \mathbf{B}\|_F = \sigma_n$ .
- Deduce that the distance from  $\mathbf{A}$  to the set of singular matrices, measured in the Frobenius norm, is exactly  $\sigma_n$ , and illustrate this with  $\mathbf{A} = \begin{pmatrix} 3 & 1 \\ 1 & 3 \end{pmatrix}$ .

*Solution.*

- A singular  $\mathbf{B} \in M_n(F)$  has  $\text{rank } \mathbf{B} \leq n - 1$  (Theorem 2.4.4). By Theorem 12.10.3 with  $k = n - 1$ ,

$$\|\mathbf{A} - \mathbf{B}\|_F \geq \left( \sum_{i>n-1} \sigma_i^2 \right)^{1/2} = \sigma_n.$$

- Take  $\mathbf{B} = \mathbf{A}_{n-1} = \sum_{i \leq n-1} \sigma_i \mathbf{u}_i \mathbf{v}_i^*$ . Its rank is  $n - 1 < n$ , so it is singular, and  $\|\mathbf{A} - \mathbf{B}\|_F = \sigma_n$  by Lemma 12.10.1.
- By (a) the distance is at least  $\sigma_n$  and by (b) it is at most  $\sigma_n$ , so it equals  $\sigma_n$ ; in particular the minimum is attained. For the given  $\mathbf{A}$ , Exercise B2 found  $\sigma_2 = 2$  and  $\mathbf{A}_1 = \begin{pmatrix} 2 & 2 \\ 2 & 2 \end{pmatrix}$ , which is singular and at Frobenius distance 2. So no singular matrix is closer to  $\mathbf{A}$  than 2. The smallest singular value is therefore a quantitative version of invertibility:  $\mathbf{A}$  is invertible exactly when  $\sigma_n > 0$ , and it is *far* from singular exactly when  $\sigma_n$  is large.

#### Exercise C2: The best line through a cloud of points.

The four points  $(1, 1)$ ,  $(2, 2)$ ,  $(1, -1)$  and  $(0, 0)$  of  $\mathbb{R}^2$  are to be fitted by a line through the origin, minimizing the total squared distance from the points to the line.

- Write the points as the columns of a matrix  $\mathbf{A}$  and compute  $\mathbf{A}\mathbf{A}^\top$  and the singular values of  $\mathbf{A}$ .
- Use Corollary 12.10.4 to find the best line and the minimum total squared distance.
- Verify the answer by computing the four distances directly.

*Solution.*

- With the points as columns,

$$\mathbf{A} = \begin{pmatrix} 1 & 2 & 1 & 0 \\ 1 & 2 & -1 & 0 \end{pmatrix}, \quad \mathbf{A}\mathbf{A}^\top = \begin{pmatrix} 6 & 4 \\ 4 & 6 \end{pmatrix}.$$

The eigenvalues of  $\mathbf{A}\mathbf{A}^\top$  are 10 and 2 (trace 12, determinant 20), with orthonormal eigenvectors  $\mathbf{u}_1 = \frac{1}{\sqrt{2}}(1, 1)$  and  $\mathbf{u}_2 = \frac{1}{\sqrt{2}}(1, -1)$ . The eigenvalues of  $\mathbf{A}\mathbf{A}^\top$  are the

squares of the singular values, since  $\mathbf{A}\mathbf{A}^\top = \mathbf{U}\mathbf{\Sigma}\mathbf{\Sigma}^*\mathbf{U}^*$ ; hence  $\sigma_1 = \sqrt{10}$  and  $\sigma_2 = \sqrt{2}$ .

- (b) By [Corollary 12.10.4](#) with  $k = 1$ , the best line is  $S_0 = \text{span}(\mathbf{u}_1)$ , the line  $y = x$ , and the minimum total squared distance is  $\sigma_2^2 = 2$ .
- (c) The points  $(1, 1)$ ,  $(2, 2)$  and  $(0, 0)$  lie on  $y = x$ , contributing 0. For  $(1, -1)$ , the projection onto  $\text{span}((1, 1))$  is  $\frac{1-1}{2}(1, 1) = (0, 0)$ , so the distance squared is  $1 + 1 = 2$ . The total is 2, as predicted. (The point  $(0, 0)$  contributes nothing to any subspace through the origin; it is there as a reminder that the corollary counts all columns, including zero ones.)

### Exercise C3: Truncation respects unitaries but not sums.

Let  $\mathbf{A} \in M_{m \times n}(F)$ , and let  $\mathbf{Q} \in M_m(F)$  and  $\mathbf{R} \in M_n(F)$  be unitary (orthogonal, when  $F = \mathbb{R}$ ).

- (a) Prove that  $\mathbf{Q}\mathbf{A}_k\mathbf{R}^*$  is a best rank- $k$  approximation of  $\mathbf{Q}\mathbf{A}\mathbf{R}^*$  in the Frobenius norm.
- (b) Deduce that the function  $\mathbf{A} \mapsto \min\{\|\mathbf{A} - \mathbf{B}\|_F : \text{rank } \mathbf{B} \leq k\}$  is unchanged by  $\mathbf{A} \mapsto \mathbf{Q}\mathbf{A}\mathbf{R}^*$ .
- (c) Give  $\mathbf{A}, \mathbf{B} \in M_2(\mathbb{R})$  with  $(\mathbf{A} + \mathbf{B})_1 \neq \mathbf{A}_1 + \mathbf{B}_1$  for every choice of the truncations involved. *Hint: two rank-one matrices whose sum is a multiple of the identity.*

*Solution.*

- (a) If  $\mathbf{A} = \mathbf{U}\mathbf{\Sigma}\mathbf{V}^*$  is a singular value decomposition ([Theorem 12.8.1](#)), then

$$\mathbf{Q}\mathbf{A}\mathbf{R}^* = (\mathbf{Q}\mathbf{U})\mathbf{\Sigma}(\mathbf{R}\mathbf{V})^*$$

is one as well, since a product of unitary matrices is unitary ([Proposition 10.7.4](#) (a)) and  $\mathbf{\Sigma}$  is unchanged. Its left and right singular vectors are  $\mathbf{Q}\mathbf{u}_i$  and  $\mathbf{R}\mathbf{v}_i$ , so its rank- $k$  truncation for this decomposition is

$$\sum_{i \leq k} \sigma_i (\mathbf{Q}\mathbf{u}_i)(\mathbf{R}\mathbf{v}_i)^* = \mathbf{Q} \left( \sum_{i \leq k} \sigma_i \mathbf{u}_i \mathbf{v}_i^* \right) \mathbf{R}^* = \mathbf{Q}\mathbf{A}_k\mathbf{R}^*.$$

By [Theorem 12.10.3](#) applied to  $\mathbf{Q}\mathbf{A}\mathbf{R}^*$ , this is a best rank- $k$  approximation.

- (b) The two matrices have the same singular values, so by [Theorem 12.10.3](#) both minima equal  $(\sum_{i > k} \sigma_i^2)^{1/2}$ .
- (c) Take  $\mathbf{A} = \text{diag}(1, 0)$  and  $\mathbf{B} = \text{diag}(0, 1)$ . Each has rank one, so  $\mathbf{A}_1 = \mathbf{A}$  and  $\mathbf{B}_1 = \mathbf{B}$  for every singular value decomposition, and  $\mathbf{A}_1 + \mathbf{B}_1 = \mathbf{I}_2$ . But  $\mathbf{A} + \mathbf{B} = \mathbf{I}_2$ , whose rank-one truncations all have rank one, hence are never  $\mathbf{I}_2$ . So  $(\mathbf{A} + \mathbf{B})_1 \neq \mathbf{A}_1 + \mathbf{B}_1$ . Truncation is not a linear operation, which is why “keep the biggest pieces” cannot be analyzed one summand at a time.

## 12.11 The Moore–Penrose Pseudoinverse

Chapter 10 solved  $\mathbf{Ax} = \mathbf{b}$  twice over: when there is no solution it produced a least-squares solution, and when there are too many it produced the shortest one. It then left an IOU. The two answers “are usually wanted together: given any  $\mathbf{A}$  and  $\mathbf{b}$ , first minimize  $\|\mathbf{Ax} - \mathbf{b}\|$ , then among the minimizers take the shortest. Chapter 12 packages the combined answer as a single matrix, the **pseudoinverse**  $\mathbf{A}^+$ , with  $\mathbf{x} = \mathbf{A}^+\mathbf{b}$  in all cases at once.” This section builds that matrix and pays the debt.

Throughout,  $F = \mathbb{R}$  or  $F = \mathbb{C}$ ,  $\mathbf{A} \in M_{m \times n}(F)$ ,  $p = \min(m, n)$  and  $r = \text{rank } \mathbf{A}$ , with singular values  $\sigma_1 \geq \cdots \geq \sigma_p \geq 0$  (Definition 12.8.1), so  $\sigma_i > 0$  exactly for  $i \leq r$ .

### 12.11.1 One matrix instead of three formulas

Here is the state of play at the end of Chapter 10. To answer “what is the best  $\mathbf{x}$ ?” we had three different recipes, and choosing between them required knowing something about  $\mathbf{A}$  first.

- If  $\mathbf{A}$  is square and invertible,  $\mathbf{x} = \mathbf{A}^{-1}\mathbf{b}$ .
- If the columns of  $\mathbf{A}$  are independent,  $\mathbf{x} = (\mathbf{A}^*\mathbf{A})^{-1}\mathbf{A}^*\mathbf{b}$  (Corollary 10.4.4).
- If  $\mathbf{Ax} = \mathbf{b}$  is consistent but  $\mathbf{A}$  has a kernel,  $\mathbf{x} = \mathbf{x}_{\min}$ , the unique solution orthogonal to  $\text{nul}(\mathbf{A})$  (Theorem 10.4.5).

Three recipes, three hypotheses, and a gap: when  $\mathbf{A}$  has both a kernel *and* a column space missing  $\mathbf{b}$ , none of them applies. That is not an exotic case. It is what happens whenever a large table of measurements has both redundant columns and noise.

The goal, stated before we know how to reach it, is a single matrix  $\mathbf{A}^+$ , depending on  $\mathbf{A}$  alone, for which  $\mathbf{x} = \mathbf{A}^+\mathbf{b}$  is the right answer in every case. Reading the goal backwards tells us what  $\mathbf{A}^+$  must do, and the singular value decomposition tells us how to build it.

### 12.11.2 Inverting what can be inverted

Why is there no inverse in general? Two obstructions, and they are the two halves of Theorem 10.6.5. The map  $\mathbf{x} \mapsto \mathbf{Ax}$  may fail to be injective, because  $\text{nul}(\mathbf{A}) \neq \{\mathbf{0}\}$ ; and it may fail to be surjective, because  $\text{col}(\mathbf{A}) \neq F^m$ . Neither obstruction is fatal if we are willing to sacrifice part of the domain and part of the codomain.

Restrict the map to  $\text{col}(\mathbf{A}^*) = \text{nul}(\mathbf{A})^\perp$ , which throws away the kernel. What is left is a bijection onto  $\text{col}(\mathbf{A})$ , and a bijection has an inverse. Extend that inverse to all of  $F^m$  by declaring it  $\mathbf{0}$  on  $\text{col}(\mathbf{A})^\perp$ , where nothing in the image can tell us anything. That is the whole construction, and it is the same trick that turns  $\sin$  into  $\arcsin$ : shrink the domain until the function is injective, shrink the codomain until it is surjective, invert, and say clearly what was given up.

*Invert what can be inverted, and send the rest to zero.*

The singular value decomposition performs all three operations at once, because it presents  $\mathbf{A}$  as a diagonal matrix in orthonormal coordinates, and a diagonal matrix is inverted one entry at a time.



**Definition 12.11.1 — Moore–Penrose pseudoinverse**

Let  $\mathbf{A} \in M_{m \times n}(F)$  and let  $\mathbf{A} = \mathbf{U}\mathbf{\Sigma}\mathbf{V}^*$  be a singular value decomposition (Theorem 12.8.1), so that  $\mathbf{U} \in M_m(F)$  and  $\mathbf{V} \in M_n(F)$  are unitary and  $\mathbf{\Sigma} \in M_{m \times n}(F)$  has the singular values  $\sigma_1, \dots, \sigma_p$  on its diagonal and zeros elsewhere. Let  $\mathbf{\Sigma}^+ \in M_{n \times m}(F)$  be the matrix with

$$(\mathbf{\Sigma}^+)_{ii} = \begin{cases} \sigma_i^{-1}, & \sigma_i > 0, \\ 0, & \sigma_i = 0, \end{cases}$$

for  $1 \leq i \leq p$ , and all other entries 0. The **Moore–Penrose pseudoinverse** of  $\mathbf{A}$  is

$$\mathbf{A}^+ := \mathbf{V}\mathbf{\Sigma}^+\mathbf{U}^* \in M_{n \times m}(F).$$

Clause by clause. The matrix  $\mathbf{\Sigma}^+$  is  $\mathbf{\Sigma}$  **transposed and then inverted where inversion is legal**: the shape flips from  $m \times n$  to  $n \times m$ , each non-zero diagonal entry is replaced by its reciprocal, and each zero diagonal entry is left alone — **not** replaced by anything, since  $0^{-1}$  does not exist. The outer factors are exchanged and starred, exactly as they would be in  $(\mathbf{U}\mathbf{\Sigma}\mathbf{V}^*)^{-1} = \mathbf{V}\mathbf{\Sigma}^{-1}\mathbf{U}^*$  if  $\mathbf{\Sigma}$  were invertible. And the result is  $n \times m$ , the shape an inverse of  $\mathbf{A}$  would have to be.

**Well-definedness is a real question here, not a formality.** The singular *values* are determined by  $\mathbf{A}$ , but the unitary factors are not (Theorem 12.8.2), so the recipe above is applied to a choice, and nothing so far says that two choices give the same answer. They do.

**Proposition 12.11.1 — The pseudoinverse does not depend on the decomposition**

Let  $\mathbf{A} \in M_{m \times n}(F)$  and let  $\mathbf{A} = \mathbf{U}\mathbf{\Sigma}\mathbf{V}^*$  be any singular value decomposition (Theorem 12.8.1). Then  $\mathbf{X} = \mathbf{V}\mathbf{\Sigma}^+\mathbf{U}^*$  is the unique matrix in  $M_{n \times m}(F)$  such that

- (i) for every  $\mathbf{y} \in \text{col}(\mathbf{A})$ ,  $\mathbf{X}\mathbf{y}$  is the unique vector of  $\text{col}(\mathbf{A}^*)$  that  $\mathbf{A}$  sends to  $\mathbf{y}$ ;
- (ii)  $\mathbf{X}\mathbf{y} = \mathbf{0}$  for every  $\mathbf{y} \in \text{col}(\mathbf{A})^\perp$ .

In particular  $\mathbf{A}^+$  depends on  $\mathbf{A}$  alone.

*Idea.* Conditions (i) and (ii) mention only  $\mathbf{A}$ , so if one matrix satisfies them, that matrix is determined by  $\mathbf{A}$ . The work is to check that they can be satisfied at all — which is the “restrict, invert, extend” story of the previous page written out — and that the matrix built from an arbitrary decomposition is the one that satisfies them. For the second part, the singular vectors are orthonormal bases of the subspaces involved, so both conditions can be checked on basis vectors.

*Proof.* **The conditions determine at most one matrix.** First,  $\mathbf{A}$  maps  $\text{col}(\mathbf{A}^*)$  bijectively onto  $\text{col}(\mathbf{A})$ . Injectivity: if  $\mathbf{x} \in \text{col}(\mathbf{A}^*)$  and  $\mathbf{A}\mathbf{x} = \mathbf{0}$ , then  $\mathbf{x} \in \text{nul}(\mathbf{A}) \cap \text{nul}(\mathbf{A})^\perp = \{\mathbf{0}\}$ , since  $\text{col}(\mathbf{A}^*) = \text{nul}(\mathbf{A})^\perp$  by Theorem 10.6.5 (b). Surjectivity: the image is contained in  $\text{col}(\mathbf{A})$ , and by injectivity it has dimension  $\dim \text{col}(\mathbf{A}^*) = \text{rank } \mathbf{A}^* = \text{rank } \mathbf{A} = \dim \text{col}(\mathbf{A})$ , the middle equality by Corollary 10.6.6 (a) applied to  $\mathbf{x} \mapsto \mathbf{A}\mathbf{x}$ . So (i) prescribes  $\mathbf{X}$  on  $\text{col}(\mathbf{A})$  and (ii) prescribes it on  $\text{col}(\mathbf{A})^\perp$ ; since  $F^m = \text{col}(\mathbf{A}) \oplus \text{col}(\mathbf{A})^\perp$  by Theorem 10.3.2 (a), a linear map is determined by the two prescriptions.

**The matrix  $\mathbf{X} = \mathbf{V}\mathbf{\Sigma}^+\mathbf{U}^*$  satisfies them.** Write  $\mathbf{u}_1, \dots, \mathbf{u}_m$  and  $\mathbf{v}_1, \dots, \mathbf{v}_n$  for the

columns of  $\mathbf{U}$  and  $\mathbf{V}$ . Since rank is unchanged by invertible factors (Theorem 2.5.9),  $r = \text{rank } \mathbf{A} = \text{rank } \Sigma$  is the number of non-zero  $\sigma_i$ . By Corollary 12.8.4,  $(\mathbf{u}_1, \dots, \mathbf{u}_r)$  is an orthonormal basis of  $\text{col}(\mathbf{A})$ , the remaining  $\mathbf{u}_j$  are an orthonormal basis of  $\text{col}(\mathbf{A})^\perp$ , and  $(\mathbf{v}_1, \dots, \mathbf{v}_r)$  is an orthonormal basis of  $\text{col}(\mathbf{A}^*)$ . Multiplying out the definition,

$$\mathbf{X} = \sum_{i=1}^r \sigma_i^{-1} \mathbf{v}_i \mathbf{u}_i^*,$$

so  $\mathbf{X}\mathbf{u}_i = \sigma_i^{-1} \mathbf{v}_i$  for  $i \leq r$  and  $\mathbf{X}\mathbf{u}_j = \mathbf{0}$  for  $j > r$ . The second statement gives (ii) by linearity.

For (i), let  $\mathbf{y} = \sum_{i \leq r} c_i \mathbf{u}_i \in \text{col}(\mathbf{A})$ . Then  $\mathbf{X}\mathbf{y} = \sum_{i \leq r} c_i \sigma_i^{-1} \mathbf{v}_i \in \text{col}(\mathbf{A}^*)$ , and since  $\mathbf{A}\mathbf{v}_i = \sigma_i \mathbf{u}_i$  (the rank-one expansion of Theorem 12.8.3 read one column at a time),

$$\mathbf{A}(\mathbf{X}\mathbf{y}) = \sum_{i \leq r} c_i \sigma_i^{-1} \sigma_i \mathbf{u}_i = \mathbf{y}.$$

By the injectivity proved above,  $\mathbf{X}\mathbf{y}$  is the *unique* vector of  $\text{col}(\mathbf{A}^*)$  with this property. This proves (i), and with it the proposition. ■

So  $\mathbf{A}^+$  is an honest function of  $\mathbf{A}$ . Five examples, simplest first.

**Example. (Five pseudoinverses)**

- (a) **A invertible.** Then  $m = n = r$ ,  $\Sigma$  is invertible and  $\Sigma^+ = \Sigma^{-1}$ , so  $\mathbf{A}^+ = \mathbf{V}\Sigma^{-1}\mathbf{U}^* = (\mathbf{U}\Sigma\mathbf{V}^*)^{-1} = \mathbf{A}^{-1}$ . The pseudoinverse extends the inverse, which is the first thing it had to do.
- (b)  **$\mathbf{A} = \mathbf{0} \in M_{m \times n}(F)$ .** Every singular value is 0, so  $\Sigma^+ = \mathbf{0}$  and  $\mathbf{A}^+ = \mathbf{0} \in M_{n \times m}(F)$ . The degenerate case is worth keeping in mind: the pseudoinverse of the worst possible matrix is not undefined, it is zero, which is what “invert nothing, kill everything” means.
- (c) **Diagonal.**  $\text{diag}(2, 0, 5)^+ = \text{diag}(\frac{1}{2}, 0, \frac{1}{5})$ . The zero is left where it is.
- (d) **A single non-zero column  $\mathbf{a} \in M_{n \times 1}(F)$ .** Here  $\sigma_1 = \|\mathbf{a}\|$ ,  $\mathbf{u}_1 = \mathbf{a}/\|\mathbf{a}\|$  and  $\mathbf{v}_1 = 1$ , so  $\mathbf{a}^+ = \mathbf{a}^*/\|\mathbf{a}\|^2$ , a row vector. Then  $\mathbf{a}^+\mathbf{a} = 1$  but  $\mathbf{a}\mathbf{a}^+ = \mathbf{a}\mathbf{a}^*/\|\mathbf{a}\|^2$  is the orthogonal projection onto  $\text{span}(\mathbf{a})$ .
- (e)  **$\mathbf{A} = \begin{pmatrix} 1 & 1 \\ 1 & 1 \end{pmatrix}$ .** This is  $\sigma_1 \mathbf{u}_1 \mathbf{v}_1^*$  with  $\sigma_1 = 2$  and  $\mathbf{u}_1 = \mathbf{v}_1 = \frac{1}{\sqrt{2}}(1, 1)$ , since  $2\mathbf{u}_1 \mathbf{v}_1^* = 2 \cdot \frac{1}{2} \begin{pmatrix} 1 & 1 \\ 1 & 1 \end{pmatrix} = \mathbf{A}$ . Hence

$$\mathbf{A}^+ = \frac{1}{2} \mathbf{v}_1 \mathbf{u}_1^* = \frac{1}{4} \begin{pmatrix} 1 & 1 \\ 1 & 1 \end{pmatrix}.$$

**A non-example by minimal change.** Item (a) of the list is often written  $\mathbf{A}^+ = (\mathbf{A}^* \mathbf{A})^{-1} \mathbf{A}^*$ , and Theorem 12.11.3 below confirms that formula whenever the columns of  $\mathbf{A}$  are independent. Change one column of a  $3 \times 2$  matrix into a multiple of the other, and the formula stops existing:  $\mathbf{A}^* \mathbf{A}$  is then singular (Lemma 10.4.3), so  $(\mathbf{A}^* \mathbf{A})^{-1}$  is not a matrix at all. The pseudoinverse survives the change; the formula does not. That is the difference between a definition and a recipe.

**Why this definition.** It is forced. Any reasonable “inverse” must undo  $\mathbf{A}$  where  $\mathbf{A}$  can be

undone; there is exactly one subspace on which  $\mathbf{A}$  is a bijection and which meets  $\text{nul}(\mathbf{A})$  only in  $\mathbf{0}$  while being determined by  $\mathbf{A}$  alone, namely  $\text{nul}(\mathbf{A})^\perp$ ; and setting the answer to  $\mathbf{0}$  on  $\text{col}(\mathbf{A})^\perp$  is the only choice that keeps the map linear without inventing information. The names are those of the two people who published the idea, thirty-five years apart: Moore in 1920 through a projection formula, Penrose in 1955 through the four conditions of the next theorem.

**Warning.**  $\mathbf{A}^+\mathbf{A}$  is not the identity, and  $\mathbf{A}^+$  is  $n \times m$ . Two habits from the invertible case have to go. First the shape: if  $\mathbf{A}$  is  $3 \times 5$  then  $\mathbf{A}^+$  is  $5 \times 3$ , so  $\mathbf{A}\mathbf{A}^+$  is  $3 \times 3$  and  $\mathbf{A}^+\mathbf{A}$  is  $5 \times 5$  — they are not even the same size, let alone both  $\mathbf{I}$ . Second the identity: for  $\mathbf{A} = \begin{pmatrix} 1 & 1 \\ 1 & 1 \end{pmatrix}$  of [Example \(Five pseudoinverses\)](#) (e),

$$\mathbf{A}^+\mathbf{A} = \frac{1}{4} \begin{pmatrix} 1 & 1 \\ 1 & 1 \end{pmatrix} \begin{pmatrix} 1 & 1 \\ 1 & 1 \end{pmatrix} = \frac{1}{2} \begin{pmatrix} 1 & 1 \\ 1 & 1 \end{pmatrix} \neq \mathbf{I}_2.$$

The right question is not “is it the identity?” but “the identity on *what*?”, and [Theorem 12.11.3](#) answers it:  $\mathbf{A}^+\mathbf{A}$  is the identity on  $\text{col}(\mathbf{A}^*)$  and zero on  $\text{nul}(\mathbf{A})$ .

**Quick check.** Let  $\mathbf{A} = \begin{pmatrix} 1 & 1 \\ 0 & 0 \end{pmatrix}$ . Find  $\mathbf{A}^+$ , then compute  $\mathbf{A}\mathbf{A}^+$  and  $\mathbf{A}^+\mathbf{A}$  and identify each as a projection.

**Solution.** Here  $\mathbf{A} = \mathbf{e}_1(1, 1)$  has rank one, so  $\sigma_1 = \|\mathbf{e}_1\| \|(1, 1)\| = \sqrt{2}$  with  $\mathbf{u}_1 = \mathbf{e}_1$  and  $\mathbf{v}_1 = \frac{1}{\sqrt{2}}(1, 1)$ , and

$$\mathbf{A}^+ = \sigma_1^{-1} \mathbf{v}_1 \mathbf{u}_1^* = \frac{1}{2} \begin{pmatrix} 1 & 0 \\ 1 & 0 \end{pmatrix}.$$

Therefore

$$\mathbf{A}\mathbf{A}^+ = \begin{pmatrix} 1 & 0 \\ 0 & 0 \end{pmatrix}, \quad \mathbf{A}^+\mathbf{A} = \frac{1}{2} \begin{pmatrix} 1 & 1 \\ 1 & 1 \end{pmatrix}.$$

Neither is  $\mathbf{I}_2$ . The first is the orthogonal projection onto  $\text{span}(\mathbf{e}_1) = \text{col}(\mathbf{A})$ ; the second is the orthogonal projection onto  $\text{span}((1, 1)) = \text{col}(\mathbf{A}^\top)$ , the row space. Each is the identity exactly on the subspace named, and zero on its orthogonal complement — which is what “as close to the identity as the shapes allow” means.

### 12.11.3 The four Penrose conditions

The definition builds  $\mathbf{A}^+$  out of a decomposition. The following theorem characterizes it by four equations that mention nothing but  $\mathbf{A}$ ,  $\mathbf{X}$  and the star. It is the second, and better, answer to the well-definedness question, and it is how one *verifies* a candidate pseudoinverse in practice: guess, then check four identities.

#### Theorem 12.11.2 — The Penrose Conditions

Let  $\mathbf{A} \in M_{m \times n}(F)$ . Then  $\mathbf{X} = \mathbf{A}^+$  is the **unique** matrix in  $M_{n \times m}(F)$  satisfying all four of

(P1)  $\mathbf{A}\mathbf{X}\mathbf{A} = \mathbf{A}$ ;

(P2)  $\mathbf{X}\mathbf{A}\mathbf{X} = \mathbf{X}$ ;

$$(P3) \quad (\mathbf{AX})^* = \mathbf{AX};$$

$$(P4) \quad (\mathbf{XA})^* = \mathbf{XA}.$$

*Idea.* Existence is a computation with  $\Sigma$ : each condition collapses to an identity between  $\Sigma$ ,  $\Sigma^+$  and the diagonal matrix of  $r$  ones. Uniqueness is a puzzle. Given two solutions  $\mathbf{X}_1$  and  $\mathbf{X}_2$ , the goal is a chain of equalities starting at  $\mathbf{X}_1$  and ending at  $\mathbf{X}_2$ , so the plan is to walk to a symmetric midpoint: get  $\mathbf{X}_1 = \mathbf{X}_1 \mathbf{A} \mathbf{X}_2$  by pushing  $\mathbf{X}_2$  in from the right, then run the mirror image of the same argument with  $\mathbf{X}_2$  as the subject. Two preliminary identities make the walk possible, and both come from starring one of the conditions — the move that manufactures an  $\mathbf{A}^*$  when the hypotheses offer only  $\mathbf{A}$ .

*Proof. Existence.* Let  $\mathbf{A} = \mathbf{U} \Sigma \mathbf{V}^*$  and  $\mathbf{X} = \mathbf{V} \Sigma^+ \mathbf{U}^*$ . Using  $\mathbf{V}^* \mathbf{V} = \mathbf{I}_n$  and  $\mathbf{U}^* \mathbf{U} = \mathbf{I}_m$ ,

$$\mathbf{AX} = \mathbf{U} \Sigma \Sigma^+ \mathbf{U}^*, \quad \mathbf{XA} = \mathbf{V} \Sigma^+ \Sigma \mathbf{V}^*.$$

Now  $\Sigma \Sigma^+ \in M_m(F)$  is diagonal with  $\sigma_i \sigma_i^{-1} = 1$  in the first  $r$  diagonal places and 0 elsewhere, and  $\Sigma^+ \Sigma \in M_n(F)$  is diagonal with the same pattern. Both are real diagonal, hence Hermitian, so  $\mathbf{AX}$  and  $\mathbf{XA}$  are Hermitian, which is (P3) and (P4). Moreover  $\Sigma \Sigma^+ \Sigma = \Sigma$  and  $\Sigma^+ \Sigma \Sigma^+ = \Sigma^+$ , because multiplying a diagonal matrix by a diagonal matrix of ones and zeros keeps exactly the entries in the positions marked 1, and those are the positions where  $\Sigma$  and  $\Sigma^+$  are non-zero anyway. Hence

$$\mathbf{AXA} = \mathbf{U} \Sigma \Sigma^+ \Sigma \mathbf{V}^* = \mathbf{A}, \quad \mathbf{XAX} = \mathbf{V} \Sigma^+ \Sigma \Sigma^+ \mathbf{U}^* = \mathbf{X},$$

which is (P1) and (P2).

**Uniqueness.** Suppose  $\mathbf{X}$  satisfies (P1)–(P4). Two consequences:

$$\mathbf{XX}^* \mathbf{A}^* = \mathbf{X}, \quad \mathbf{AA}^* \mathbf{X}^* = \mathbf{A}. \quad (*)$$

For the first,  $\mathbf{X} = \mathbf{XAX}$  by (P2), and  $\mathbf{AX} = (\mathbf{AX})^* = \mathbf{X}^* \mathbf{A}^*$  by (P3), so  $\mathbf{X} = \mathbf{X}(\mathbf{AX}) = \mathbf{XX}^* \mathbf{A}^*$ . For the second,  $\mathbf{A} = \mathbf{AXA}$  by (P1), and  $\mathbf{XA} = (\mathbf{XA})^* = \mathbf{A}^* \mathbf{X}^*$  by (P4), so  $\mathbf{A} = \mathbf{A}(\mathbf{XA}) = \mathbf{AA}^* \mathbf{X}^*$ . Starring the second identity of  $(*)$  gives the form we also need,

$$\mathbf{XAA}^* = \mathbf{A}^*. \quad (**)$$

Now let  $\mathbf{X}_1$  and  $\mathbf{X}_2$  both satisfy (P1)–(P4). Starting from  $\mathbf{X}_1$  and working towards a term containing  $\mathbf{X}_2$ ,

$$\begin{aligned} \mathbf{X}_1 &= \mathbf{X}_1 \mathbf{X}_1^* \mathbf{A}^* = \mathbf{X}_1 \mathbf{X}_1^* (\mathbf{AX}_2 \mathbf{A})^* \\ &= \mathbf{X}_1 \mathbf{X}_1^* \mathbf{A}^* (\mathbf{AX}_2)^* = \mathbf{X}_1 \mathbf{X}_1^* \mathbf{A}^* \mathbf{AX}_2 \\ &= \mathbf{X}_1 \mathbf{AX}_2. \end{aligned}$$

The first equality is  $(*)$  for  $\mathbf{X}_1$ ; the second replaces  $\mathbf{A}$  by  $\mathbf{AX}_2 \mathbf{A}$ , which is (P1) for  $\mathbf{X}_2$ ; the third reverses the product under the star; the fourth is (P3) for  $\mathbf{X}_2$ ; and the last is  $(*)$  for  $\mathbf{X}_1$  again, applied to the first three factors.

Now the mirror argument, with  $\mathbf{X}_2$  as the subject:

$$\begin{aligned} \mathbf{X}_1 \mathbf{AX}_2 &= \mathbf{X}_1 \mathbf{AA}^* \mathbf{X}_2^* \mathbf{X}_2 = \mathbf{A}^* \mathbf{X}_2^* \mathbf{X}_2 \\ &= (\mathbf{X}_2 \mathbf{A})^* \mathbf{X}_2 = \mathbf{X}_2 \mathbf{AX}_2 = \mathbf{X}_2. \end{aligned}$$

Here the first equality replaces  $\mathbf{A}$  by  $\mathbf{A}\mathbf{A}^*\mathbf{X}_2^*$ , which is  $(*)$  for  $\mathbf{X}_2$ ; the second is  $(**)$  for  $\mathbf{X}_1$ ; the fourth is (P4) for  $\mathbf{X}_2$ ; and the last is (P2) for  $\mathbf{X}_2$ .

Hence  $\mathbf{X}_1 = \mathbf{X}_2$ . Since  $\mathbf{A}^+$  satisfies the four conditions, every matrix satisfying them equals  $\mathbf{A}^+$ . This proves the theorem. ■

Both halves of the argument are the same two moves: star a condition to trade  $\mathbf{A}$  for  $\mathbf{A}^*$ , then use  $(*)$  to collapse a triple product. The conditions are also exactly what one should check when an explicit candidate is on the table, and several of the proofs below do nothing else.

*Remark.* The four conditions are not independent decoration. Dropping (P2) leaves the **generalized inverses**, matrices with  $\mathbf{A}\mathbf{X}\mathbf{A} = \mathbf{A}$ , of which there are many when  $\mathbf{A}$  is singular; dropping (P3) and (P4) leaves more still. Each condition removes freedom, and only all four together remove all of it. Exercise C3 identifies the matrices for which  $\mathbf{A}^*$  itself already satisfies them.

## 12.11.4 What the pseudoinverse does

### Theorem 12.11.3 — Properties of the pseudoinverse

Let  $\mathbf{A} \in M_{m \times n}(F)$ . Then:

- (a)  $\mathbf{A}\mathbf{A}^+ = P_{\text{col}(\mathbf{A})}$  and  $\mathbf{A}^+\mathbf{A} = P_{\text{col}(\mathbf{A}^*)}$ , the orthogonal projections onto the column space of  $\mathbf{A}$  in  $F^m$  and onto the column space of  $\mathbf{A}^*$  in  $F^n$ ;
- (b)  $(\mathbf{A}^+)^+ = \mathbf{A}$ ;
- (c)  $(\mathbf{A}^+)^* = (\mathbf{A}^*)^+$ ;
- (d)  $\text{rank } \mathbf{A}^+ = \text{rank } \mathbf{A}$ ;
- (e)  $\mathbf{A}^+ = \mathbf{A}^{-1}$  if  $\mathbf{A}$  is invertible;  $\mathbf{A}^+ = (\mathbf{A}^*\mathbf{A})^{-1}\mathbf{A}^*$  if  $\text{rank } \mathbf{A} = n$ ; and  $\mathbf{A}^+ = \mathbf{A}^*(\mathbf{A}\mathbf{A}^*)^{-1}$  if  $\text{rank } \mathbf{A} = m$ .

*Idea.* Clause (a) is the computation already done in the existence half of Theorem 12.11.2, read through Theorem 10.3.3. Clauses (b) and (c) are one line each, because the Penrose conditions are almost symmetric in  $\mathbf{A}$  and  $\mathbf{X}$ , and starring all four of them produces the four conditions for  $\mathbf{A}^*$ . Clause (e) is three checks of the same kind: write down the candidate and confirm (P1)–(P4).

*Proof.*

- (a) With  $\mathbf{A} = \mathbf{U}\Sigma\mathbf{V}^*$  and the notation of Proposition 12.11.1, the existence computation gave  $\mathbf{A}\mathbf{A}^+ = \mathbf{U}\Sigma\Sigma^+\mathbf{U}^*$ , where  $\Sigma\Sigma^+$  is diagonal with  $r$  ones followed by zeros. Multiplying out,

$$\mathbf{A}\mathbf{A}^+ = \sum_{i=1}^r \mathbf{u}_i \mathbf{u}_i^*.$$

For  $\mathbf{y} \in F^m$  this sends  $\mathbf{y}$  to  $\sum_{i \leq r} \langle \mathbf{y}, \mathbf{u}_i \rangle \mathbf{u}_i$ , which by Theorem 10.3.3 (a) is the orthogonal projection of  $\mathbf{y}$  onto  $\text{span}(\mathbf{u}_1, \dots, \mathbf{u}_r) = \text{col}(\mathbf{A})$  (Corollary 12.8.4). The same computation with  $\Sigma^+\Sigma$  gives  $\mathbf{A}^+\mathbf{A} = \sum_{i \leq r} \mathbf{v}_i \mathbf{v}_i^* = P_{\text{col}(\mathbf{A}^*)}$ .

- (b) The four Penrose conditions for the pair  $(\mathbf{A}, \mathbf{X})$  become, when the roles of  $\mathbf{A}$  and  $\mathbf{X}$  are exchanged, the list (P2), (P1), (P4), (P3) — the same four conditions. So  $\mathbf{A}$  satisfies the Penrose conditions for  $\mathbf{A}^+$ , and [Theorem 12.11.2](#) gives  $(\mathbf{A}^+)^+ = \mathbf{A}$ .
- (c) Put  $\mathbf{X} = (\mathbf{A}^+)^*$  and check the four conditions for  $\mathbf{A}^*$ . Using  $(\mathbf{BC})^* = \mathbf{C}^*\mathbf{B}^*$  throughout,

$$\mathbf{A}^*\mathbf{X}\mathbf{A}^* = (\mathbf{A}\mathbf{A}^+\mathbf{A})^* = \mathbf{A}^*, \quad \mathbf{X}\mathbf{A}^*\mathbf{X} = (\mathbf{A}^+\mathbf{A}\mathbf{A}^+)^* = \mathbf{X},$$

which are (P1) and (P2). Next,  $\mathbf{A}^*\mathbf{X} = (\mathbf{A}^+\mathbf{A})^* = \mathbf{A}^+\mathbf{A}$  by (P4) for  $\mathbf{A}$ , and  $\mathbf{A}^+\mathbf{A}$  is Hermitian, so (P3) holds; and  $\mathbf{X}\mathbf{A}^* = (\mathbf{A}\mathbf{A}^+)^* = \mathbf{A}\mathbf{A}^+$  by (P3) for  $\mathbf{A}$ , which is Hermitian, so (P4) holds. By uniqueness,  $\mathbf{X} = (\mathbf{A}^*)^+$ .

- (d)  $\mathbf{A}^+ = \mathbf{V}\mathbf{\Sigma}^+\mathbf{U}^*$  with  $\mathbf{U}, \mathbf{V}$  invertible, so  $\text{rank } \mathbf{A}^+ = \text{rank } \mathbf{\Sigma}^+ = r$  by [Theorem 2.5.9](#), the last equality because  $\mathbf{\Sigma}^+$  has exactly  $r$  non-zero entries, all in distinct rows and columns.
- (e) The invertible case is [Example \(Five pseudoinverses\)](#) (a). Suppose  $\text{rank } \mathbf{A} = n$ . Then  $\mathbf{A}^*\mathbf{A}$  is invertible ([Corollary 10.4.4](#)), so  $\mathbf{X} = (\mathbf{A}^*\mathbf{A})^{-1}\mathbf{A}^*$  is defined, and  $((\mathbf{A}^*\mathbf{A})^{-1})^* = (\mathbf{A}^*\mathbf{A})^{-1}$  since  $\mathbf{A}^*\mathbf{A}$  is Hermitian. Then

$$\mathbf{X}\mathbf{A} = (\mathbf{A}^*\mathbf{A})^{-1}\mathbf{A}^*\mathbf{A} = \mathbf{I}_n,$$

which is Hermitian, giving (P4); and (P1), (P2) follow at once, since  $\mathbf{A}\mathbf{X}\mathbf{A} = \mathbf{A}\mathbf{I}_n = \mathbf{A}$  and  $\mathbf{X}\mathbf{A}\mathbf{X} = \mathbf{I}_n\mathbf{X} = \mathbf{X}$ . For (P3),  $\mathbf{A}\mathbf{X} = \mathbf{A}(\mathbf{A}^*\mathbf{A})^{-1}\mathbf{A}^*$ , whose conjugate transpose is itself. So  $\mathbf{X} = \mathbf{A}^+$ .

For the last formula, suppose  $\text{rank } \mathbf{A} = m$ . Then  $\text{rank } \mathbf{A}^* = m$  is the number of columns of  $\mathbf{A}^*$ , so the case just proved applies to  $\mathbf{A}^*$  and gives  $(\mathbf{A}^*)^+ = (\mathbf{A}\mathbf{A}^*)^{-1}\mathbf{A}$ . By (c),

$$\mathbf{A}^+ = ((\mathbf{A}^*)^+)^* = ((\mathbf{A}\mathbf{A}^*)^{-1}\mathbf{A})^* = \mathbf{A}^*(\mathbf{A}\mathbf{A}^*)^{-1},$$

using that  $\mathbf{A}\mathbf{A}^*$  is Hermitian, hence so is its inverse. This proves the theorem. ■

Clause (a) is the one to remember, and it answers the question the warning raised. The products  $\mathbf{A}\mathbf{A}^+$  and  $\mathbf{A}^+\mathbf{A}$  are not identities; they are the identity **restricted to the subspaces where  $\mathbf{A}$  is invertible**, and zero on the complements. Clause (e) shows that the three recipes of Chapter 10 were all the pseudoinverse in disguise, each written in the notation that its hypothesis made available.

*Warning.*  $(\mathbf{AB})^+ \neq \mathbf{B}^+\mathbf{A}^+$  in general. The rule  $(\mathbf{AB})^{-1} = \mathbf{B}^{-1}\mathbf{A}^{-1}$  does not survive. Take

$$\mathbf{A} = \begin{pmatrix} 1 & 1 \\ 0 & 0 \end{pmatrix}, \quad \mathbf{B} = \begin{pmatrix} 1 & 0 \\ 0 & 0 \end{pmatrix}.$$

The Quick check above found  $\mathbf{A}^+ = \frac{1}{2}\begin{pmatrix} 1 & 0 \\ 1 & 0 \end{pmatrix}$ , and one may confirm it against [Theorem 12.11.2](#): the two products computed there are Hermitian, and  $\mathbf{A}\mathbf{A}^+\mathbf{A} = \mathbf{A}$ ,  $\mathbf{A}^+\mathbf{A}\mathbf{A}^+ = \mathbf{A}^+$ . Meanwhile  $\mathbf{B}$  is Hermitian and idempotent, so  $\mathbf{X} = \mathbf{B}$  satisfies (P1)–(P4) and  $\mathbf{B}^+ = \mathbf{B}$ . Now  $\mathbf{AB} = \begin{pmatrix} 1 & 0 \\ 0 & 0 \end{pmatrix} = \mathbf{B}$ , so

$$(\mathbf{AB})^+ = \begin{pmatrix} 1 & 0 \\ 0 & 0 \end{pmatrix}, \quad \mathbf{B}^+\mathbf{A}^+ = \frac{1}{2}\begin{pmatrix} 1 & 0 \\ 0 & 0 \end{pmatrix}.$$

The two differ by a factor of 2. The reversal rule does hold under extra hypotheses —

for instance when  $\mathbf{A}$  has independent columns and  $\mathbf{B}$  has independent rows, which is Exercise C2 — but never assume it.

## 12.11.5 Least squares and minimum norm, in one matrix

Everything is now in place to pay Chapter 10's debt.

### Theorem 12.11.4 — The pseudoinverse solves both problems at once

Let  $\mathbf{A} \in M_{m \times n}(F)$  and  $\mathbf{b} \in F^m$ , and put  $\mathbf{x}^+ = \mathbf{A}^+\mathbf{b}$ . Then:

- (a)  $\mathbf{x}^+$  is a least-squares solution of  $\mathbf{Ax} = \mathbf{b}$ , that is,  $\|\mathbf{Ax}^+ - \mathbf{b}\| \leq \|\mathbf{Az} - \mathbf{b}\|$  for every  $\mathbf{z} \in F^n$ ;
- (b)  $\|\mathbf{x}^+\| < \|\mathbf{x}\|$  for every other least-squares solution  $\mathbf{x}$ ;
- (c) if  $\mathbf{Ax} = \mathbf{b}$  is consistent,  $\mathbf{x}^+$  is its minimum-norm solution in the sense of Theorem 10.4.5; if  $\mathbf{A}$  is invertible,  $\mathbf{x}^+ = \mathbf{A}^{-1}\mathbf{b}$ .

*Idea.* Both halves are already proved; the pseudoinverse only has to be recognized inside them. For (a),  $\mathbf{Ax}^+ = \mathbf{AA}^+\mathbf{b}$  is the projection of  $\mathbf{b}$  onto  $\text{col}(\mathbf{A})$ , which is precisely condition (b) of the Least Squares Theorem. For (b), the least-squares solutions are the solutions of one consistent system, so the Minimum-Norm Solution Theorem applies to that system and singles out the solution lying in  $\text{nul}(\mathbf{A})^\perp$ ; and  $\mathbf{A}^+\mathbf{b}$  lies there, because every column of  $\mathbf{A}^+$  does.

*Proof.*

- (a) By Theorem 12.11.3 (a),  $\mathbf{Ax}^+ = \mathbf{AA}^+\mathbf{b} = P_U\mathbf{b}$  with  $U = \text{col}(\mathbf{A})$ . This is condition (b) of Theorem 10.4.2, so  $\mathbf{x}^+$  is a least-squares solution.
- (b) By Theorem 10.4.2, the least-squares solutions of  $\mathbf{Ax} = \mathbf{b}$  are exactly the solutions of the system  $\mathbf{Ax} = P_U\mathbf{b}$ , which is consistent because  $P_U\mathbf{b} \in U = \text{col}(\mathbf{A})$ . Apply Theorem 10.4.5 to that system: exactly one of its solutions lies in  $\text{nul}(\mathbf{A})^\perp$ , and it is strictly shorter than all the others.  
It remains to see that  $\mathbf{x}^+$  is that one. By (P2),  $\mathbf{A}^+ = \mathbf{A}^+\mathbf{AA}^+ = P_{\text{col}(\mathbf{A}^*)}\mathbf{A}^+$  using Theorem 12.11.3 (a), so every column of  $\mathbf{A}^+$ , and hence every vector  $\mathbf{A}^+\mathbf{b}$ , lies in  $\text{col}(\mathbf{A}^*)$ . By Theorem 10.6.5 (b),  $\text{col}(\mathbf{A}^*) = \text{nul}(\mathbf{A})^\perp$ . So  $\mathbf{x}^+$  is the minimum-norm solution of  $\mathbf{Ax} = P_U\mathbf{b}$ , which by the previous paragraph is what (b) asserts.
- (c) If  $\mathbf{Ax} = \mathbf{b}$  is consistent then  $\mathbf{b} \in \text{col}(\mathbf{A})$ , so  $P_U\mathbf{b} = \mathbf{b}$  and the least-squares solutions are the actual solutions; (b) then says  $\mathbf{x}^+$  is the shortest of them, which is Theorem 10.4.5's  $\mathbf{x}_{\min}$ . If  $\mathbf{A}$  is invertible,  $\mathbf{A}^+ = \mathbf{A}^{-1}$  by Theorem 12.11.3 (e). This proves the theorem.

■

**The promise, discharged.** Section 4 of Chapter 10 said that the two answers are usually wanted together — first minimize  $\|\mathbf{Ax} - \mathbf{b}\|$ , then among the minimizers take the shortest —

and that Chapter 12 would package the combined answer as a single matrix  $\mathbf{A}^+$ , with  $\mathbf{x} = \mathbf{A}^+\mathbf{b}$  in all cases at once. That is [Theorem 12.11.4](#), word for word: clause (a) does the minimizing, clause (b) takes the shortest, and clause (c) plus [Theorem 12.11.3](#) (e) checks that the three special recipes of Chapter 10 are recovered as special cases. There is no hypothesis on  $\mathbf{A}$  anywhere in the statement.

**The notation, settled.** Chapter 10 wrote  $\mathbf{x}_{\min}$  for the minimum-norm solution of a *consistent* system and left the symbol without a formula. From now on it has one, and its meaning extends to every system:

$$\mathbf{x}_{\min} = \mathbf{A}^+\mathbf{b},$$

the minimum-norm least-squares solution of  $\mathbf{A}\mathbf{x} = \mathbf{b}$ . For a consistent system this agrees with Chapter 10's  $\mathbf{x}_{\min}$ , by [Theorem 12.11.4](#) (c); for an inconsistent one it is the shortest of the minimizers of  $\|\mathbf{A}\mathbf{x} - \mathbf{b}\|$ , by clauses (a) and (b). One symbol, one formula, no case distinction.

**Example. (Chapter 10's minimum-norm solution, recomputed)**

In Chapter 10 the system

$$x_1 + x_2 + x_3 = 3, \quad x_1 + 2x_2 + 3x_3 = 7$$

was solved for its shortest solution by intersecting the solution line with  $\text{nul}(\mathbf{A})^\perp$ , as [Theorem 10.4.5](#) prescribes. Recompute  $\mathbf{x}_{\min}$  as  $\mathbf{A}^+\mathbf{b}$ .

*Solution.* Here

$$\mathbf{A} = \begin{pmatrix} 1 & 1 & 1 \\ 1 & 2 & 3 \end{pmatrix}, \quad \mathbf{b} = (3, 7).$$

The two rows are independent, so  $\text{rank } \mathbf{A} = 2 = m$  and [Theorem 12.11.3](#) (e) gives  $\mathbf{A}^+ = \mathbf{A}^\top(\mathbf{A}\mathbf{A}^\top)^{-1}$ . Now

$$\mathbf{A}\mathbf{A}^\top = \begin{pmatrix} 3 & 6 \\ 6 & 14 \end{pmatrix}, \quad (\mathbf{A}\mathbf{A}^\top)^{-1} = \frac{1}{6} \begin{pmatrix} 14 & -6 \\ -6 & 3 \end{pmatrix},$$

since the determinant is  $42 - 36 = 6$ . Therefore

$$\mathbf{A}^+ = \frac{1}{6} \begin{pmatrix} 1 & 1 \\ 1 & 2 \\ 1 & 3 \end{pmatrix} \begin{pmatrix} 14 & -6 \\ -6 & 3 \end{pmatrix} = \frac{1}{6} \begin{pmatrix} 8 & -3 \\ 2 & 0 \\ -4 & 3 \end{pmatrix},$$

and

$$\mathbf{x}_{\min} = \mathbf{A}^+\mathbf{b} = \frac{1}{6} \begin{pmatrix} 24 - 21 \\ 6 + 0 \\ -12 + 21 \end{pmatrix} = \left(\frac{1}{2}, 1, \frac{3}{2}\right).$$

This is exactly the answer Chapter 10 obtained by hand, and  $\|\mathbf{x}_{\min}\|^2 = \frac{1}{4} + 1 + \frac{9}{4} = \frac{7}{2}$  as before. The system is consistent —  $\frac{1}{2} + 1 + \frac{3}{2} = 3$  and  $\frac{1}{2} + 2 + \frac{9}{2} = 7$  — so no projection was needed; what the pseudoinverse did was replace a geometric argument by one matrix product.

The next example is the case no formula of Chapter 10 reaches: the system is inconsistent **and** underdetermined at the same time.



**Example. (Inconsistent and underdetermined at once)**

Find  $\mathbf{x}_{\min}$  for

$$\mathbf{A} = \begin{pmatrix} 1 & 1 \\ 1 & 1 \end{pmatrix}, \quad \mathbf{b} = (1, 3),$$

and verify directly that it minimizes the residual and is the shortest minimizer.

*Solution.* The system reads  $x_1 + x_2 = 1$  and  $x_1 + x_2 = 3$ , so it is inconsistent; and  $\text{nul}(\mathbf{A}) = \text{span}((1, -1)) \neq \{\mathbf{0}\}$ , so the columns are dependent. Neither  $\mathbf{A}^{-1}$  nor  $(\mathbf{A}^* \mathbf{A})^{-1} \mathbf{A}^*$  exists. From [Example \(Five pseudoinverses\)](#) (e),  $\mathbf{A}^+ = \frac{1}{4} \begin{pmatrix} 1 & 1 \\ 1 & 1 \end{pmatrix}$ , so

$$\mathbf{x}_{\min} = \mathbf{A}^+ \mathbf{b} = \frac{1}{4}(1 + 3, 1 + 3) = (1, 1).$$

*Check.*  $\text{col}(\mathbf{A}) = \text{span}((1, 1))$ , and the projection of  $\mathbf{b} = (1, 3)$  onto it is  $\frac{1+3}{2}(1, 1) = (2, 2)$ . Indeed  $\mathbf{A}(1, 1) = (2, 2)$ , so the residual is  $\mathbf{b} - (2, 2) = (-1, 1)$ , of squared length 2 — the smallest possible. The least-squares solutions are all  $\mathbf{x}$  with  $x_1 + x_2 = 2$ , that is  $(1, 1) + t(1, -1)$ , and

$$\|(1, 1) + t(1, -1)\|^2 = 2 + 2t^2,$$

which is smallest exactly at  $t = 0$ . So  $(1, 1)$  is the shortest least-squares solution, as [Theorem 12.11.4](#) (b) predicts.

## 12.11.6 The block criterion, finished

Section 6 tested a block matrix for positive definiteness one block at a time, and had to assume the corner block invertible: the Schur complement  $\mathbf{M}/\mathbf{A} = \mathbf{C} - \mathbf{B}^* \mathbf{A}^{-1} \mathbf{B}$  is meaningless otherwise. The pseudoinverse removes that hypothesis, and the promise Section 6 made is paid here.

### Proposition 12.11.5 — The Semidefinite Block Criterion

Let  $\mathbf{A} \in M_k(F)$  and  $\mathbf{C} \in M_l(F)$  be self-adjoint and put

$$\mathbf{M} = \begin{pmatrix} \mathbf{A} & \mathbf{B} \\ \mathbf{B}^* & \mathbf{C} \end{pmatrix} \in M_{k+l}(F).$$

Then  $\mathbf{M} \succeq 0$  if and only if  $\mathbf{A} \succeq 0$ , every column of  $\mathbf{B}$  lies in  $\text{col}(\mathbf{A})$ , and  $\mathbf{C} - \mathbf{B}^* \mathbf{A}^+ \mathbf{B} \succeq 0$ .

*Idea.* Section 6 cleared the corner with the shear  $\begin{pmatrix} \mathbf{I} & -\mathbf{A}^{-1} \mathbf{B} \\ \mathbf{0} & \mathbf{I} \end{pmatrix}$ . Write  $\mathbf{A}^+$  in place of  $\mathbf{A}^{-1}$  and the same shear still clears it — but only because  $\mathbf{A} \mathbf{A}^+$  is the projection onto  $\text{col}(\mathbf{A})$ , so  $\mathbf{A} \mathbf{A}^+ \mathbf{B} = \mathbf{B}$  exactly when the columns of  $\mathbf{B}$  are already in  $\text{col}(\mathbf{A})$ . That condition is not a technicality: it is what the two-sided test forces.

*Proof.* Suppose first that  $\mathbf{M} \succeq 0$ . Taking  $\mathbf{v} = (\mathbf{x}, \mathbf{0})$  gives  $\langle \mathbf{A} \mathbf{x}, \mathbf{x} \rangle \geq 0$ , so  $\mathbf{A} \succeq 0$ . For the column condition, let  $\mathbf{x} \in \text{nul}(\mathbf{A})$  and  $\mathbf{y} \in F^l$ , and put  $\mathbf{v}_t = (t\mathbf{x}, \mathbf{y})$  for  $t \in \mathbb{R}$ . Then

$$\langle \mathbf{M} \mathbf{v}_t, \mathbf{v}_t \rangle = t^2 \langle \mathbf{A} \mathbf{x}, \mathbf{x} \rangle + 2t \Re \langle \mathbf{B} \mathbf{y}, \mathbf{x} \rangle + \langle \mathbf{C} \mathbf{y}, \mathbf{y} \rangle = 2t \Re \langle \mathbf{B} \mathbf{y}, \mathbf{x} \rangle + \langle \mathbf{C} \mathbf{y}, \mathbf{y} \rangle,$$

using  $\mathbf{Ax} = \mathbf{0}$ . A real affine function of  $t$  that is non-negative for every  $t$  has zero slope, so  $\Re\langle \mathbf{By}, \mathbf{x} \rangle = 0$  for all  $\mathbf{y}$ ; replacing  $\mathbf{y}$  by  $i\mathbf{y}$  over  $\mathbb{C}$  kills the imaginary part too. Hence  $\langle \mathbf{By}, \mathbf{x} \rangle = 0$  for every  $\mathbf{y}$ , that is  $\mathbf{B}^*\mathbf{x} = \mathbf{0}$ . So  $\text{nul}(\mathbf{A}) \subseteq \text{nul}(\mathbf{B}^*)$ , and taking orthogonal complements with [Theorem 10.6.5](#) gives  $\text{col}(\mathbf{B}) \subseteq \text{col}(\mathbf{A})$ , since  $\mathbf{A}$  is self-adjoint.

Now assume  $\mathbf{A} \succeq 0$  and  $\text{col}(\mathbf{B}) \subseteq \text{col}(\mathbf{A})$ ; we show  $\mathbf{M} \succeq 0$  is equivalent to  $\mathbf{C} - \mathbf{B}^*\mathbf{A}^+\mathbf{B} \succeq 0$ , which finishes both directions. By [Theorem 12.11.3](#) (a),  $\mathbf{AA}^+ = P_{\text{col}(\mathbf{A})}$ , so  $\mathbf{AA}^+\mathbf{B} = \mathbf{B}$ ; taking adjoints, and using that  $\mathbf{A}$  and  $\mathbf{A}^+$  are self-adjoint ([Theorem 12.11.3](#) (c)), also  $\mathbf{B}^*\mathbf{A}^+\mathbf{A} = \mathbf{B}^*$ . Put

$$\mathbf{S} = \begin{pmatrix} \mathbf{I}_k & -\mathbf{A}^+\mathbf{B} \\ \mathbf{0} & \mathbf{I}_l \end{pmatrix},$$

which is invertible. Multiplying out and canceling with the two identities just recorded,

$$\mathbf{S}^*\mathbf{MS} = \begin{pmatrix} \mathbf{A} & \mathbf{0} \\ \mathbf{0} & \mathbf{C} - \mathbf{B}^*\mathbf{A}^+\mathbf{B} \end{pmatrix}.$$

Since  $\mathbf{S}$  is invertible, [Proposition 12.1.2](#) (a) applied to  $\mathbf{S}$  and then to  $\mathbf{S}^{-1}$  shows that  $\mathbf{M} \succeq 0$  if and only if  $\mathbf{S}^*\mathbf{MS} \succeq 0$ , and a block diagonal matrix is positive semidefinite exactly when both blocks are. This shows the three conditions together are equivalent to  $\mathbf{M} \succeq 0$ . ■

The column condition cannot be dropped. With

$$\mathbf{A} = \begin{pmatrix} 2 & 0 \\ 0 & 0 \end{pmatrix}, \quad \mathbf{C} = (1),$$

the choice  $\mathbf{B} = (1, 0)^\top$  has its column in  $\text{col}(\mathbf{A})$  and gives  $\mathbf{C} - \mathbf{B}^*\mathbf{A}^+\mathbf{B} = (\frac{1}{2}) \succeq 0$ , so  $\mathbf{M} \succeq 0$ . The choice  $\mathbf{B} = (0, 1)^\top$  does not, and there  $\mathbf{M}$  is not positive semidefinite even though  $\mathbf{A} \succeq 0$ ,  $\mathbf{C} \succeq 0$  and  $\mathbf{B}^*\mathbf{A}^+\mathbf{B} = \mathbf{0}$  makes the third condition hold trivially.

## 12.11.7 Exercises

### 12.11.7.1 A. Check your understanding

#### Exercise A1.

- Define  $\mathbf{A}^+$  in terms of a singular value decomposition of  $\mathbf{A}$  ([Theorem 12.8.1](#)), and say what size it is.
- State the four Penrose conditions, and say what [Theorem 12.11.2](#) asserts about them.
- What are  $\mathbf{AA}^+$  and  $\mathbf{A}^+\mathbf{A}$ ? Under what condition on  $\mathbf{A}$  is the second of them  $\mathbf{I}_n$ ?
- Write down  $\mathbf{0}^+$  for the zero matrix  $\mathbf{0} \in M_{m \times n}(F)$ , and  $\mathbf{A}^+$  for an invertible  $\mathbf{A}$ .
- True or false:  $(\mathbf{AB})^+ = \mathbf{B}^+\mathbf{A}^+$  whenever the product  $\mathbf{AB}$  is defined. Justify your answer.

*Solution.*

- If  $\mathbf{A} = \mathbf{U}\Sigma\mathbf{V}^*$ , then  $\mathbf{A}^+ = \mathbf{V}\Sigma^+\mathbf{U}^*$ , where  $\Sigma^+ \in M_{n \times m}(F)$  has  $\sigma_i^{-1}$  in the  $(i, i)$  place for each  $\sigma_i > 0$  and zeros elsewhere. It is  $n \times m$ , the transpose of the shape

of  $\mathbf{A}$ .

- (b)  $\mathbf{A}\mathbf{X}\mathbf{A} = \mathbf{A}$ ,  $\mathbf{X}\mathbf{A}\mathbf{X} = \mathbf{X}$ ,  $(\mathbf{A}\mathbf{X})^* = \mathbf{A}\mathbf{X}$ ,  $(\mathbf{X}\mathbf{A})^* = \mathbf{X}\mathbf{A}$ . The theorem says  $\mathbf{A}^+$  satisfies all four and is the only matrix that does; so the four conditions may be used as a definition, and checking them is enough to identify a candidate.
- (c)  $\mathbf{A}\mathbf{A}^+ = P_{\text{col}(\mathbf{A})}$  and  $\mathbf{A}^+\mathbf{A} = P_{\text{col}(\mathbf{A}^*)}$  (Theorem 12.11.3 (a)). The second is  $\mathbf{I}_n$  exactly when  $\text{col}(\mathbf{A}^*) = F^n$ , that is when  $\text{rank } \mathbf{A} = n$ , that is when the columns of  $\mathbf{A}$  are independent.
- (d)  $\mathbf{0}^+ = \mathbf{0} \in M_{n \times m}(F)$ ;  $\mathbf{A}^+ = \mathbf{A}^{-1}$  when  $\mathbf{A}$  is invertible.
- (e) False. With  $\mathbf{A} = \begin{pmatrix} 1 & 0 \\ 0 & 0 \end{pmatrix}$  and  $\mathbf{B} = \begin{pmatrix} 1 & 0 \\ 0 & 0 \end{pmatrix}$  the two sides are  $\begin{pmatrix} 1 & 0 \\ 0 & 0 \end{pmatrix}$  and  $\frac{1}{2}\begin{pmatrix} 1 & 0 \\ 0 & 0 \end{pmatrix}$ , as computed in the warning above.

### 12.11.7.2 B. Practice

#### Exercise B1: Four pseudoinverses.

Compute  $\mathbf{A}^+$  for each of the following, stating which method you used.

- (a)  $\mathbf{A} = \text{diag}(3, 0, 0, 4) \in M_4(\mathbb{R})$ ;
- (b)  $\mathbf{A} = (1, 2, 2)^\top \in M_{3 \times 1}(\mathbb{R})$ ;
- (c)  $\mathbf{A} = \begin{pmatrix} 1 & 2 \\ 2 & 4 \end{pmatrix} \in M_2(\mathbb{R})$ ;
- (d)  $\mathbf{A} = \begin{pmatrix} 1 & 0 & 1 \\ 0 & 1 & 1 \end{pmatrix} \in M_{2 \times 3}(\mathbb{R})$ .

*Solution.*

- (a) A diagonal matrix is its own singular value decomposition (Theorem 12.8.1) up to ordering, so invert the non-zero entries in place:  $\mathbf{A}^+ = \text{diag}(\frac{1}{3}, 0, 0, \frac{1}{4})$ .
- (b) A single non-zero column, so Example (Five pseudoinverses) (d) applies with  $\|\mathbf{A}\|^2 = 1 + 4 + 4 = 9$ :

$$\mathbf{A}^+ = \frac{1}{9}(1, 2, 2) \in M_{1 \times 3}(\mathbb{R}).$$

- (c) Rank one:  $\mathbf{A} = \mathbf{u}\mathbf{v}^\top$  with  $\mathbf{u} = (1, 2)$  and  $\mathbf{v} = (1, 2)$ . Then  $\sigma_1 = \|\mathbf{u}\| \|\mathbf{v}\| = 5$  with  $\mathbf{u}_1 = \mathbf{u}/\sqrt{5}$  and  $\mathbf{v}_1 = \mathbf{v}/\sqrt{5}$ , so

$$\mathbf{A}^+ = \frac{1}{5}\mathbf{v}_1\mathbf{u}_1^\top = \frac{1}{25}\begin{pmatrix} 1 & 2 \\ 2 & 4 \end{pmatrix}.$$

(Check:  $\mathbf{A}\mathbf{A}^+ = \frac{1}{25}\begin{pmatrix} 5 & 10 \\ 10 & 20 \end{pmatrix}$  is Hermitian and idempotent, and  $\mathbf{A}\mathbf{A}^+\mathbf{A} = \mathbf{A}$ .)

- (d) The rows are independent, so  $\text{rank } \mathbf{A} = 2 = m$  and Theorem 12.11.3 (e) gives  $\mathbf{A}^+ = \mathbf{A}^\top(\mathbf{A}\mathbf{A}^\top)^{-1}$ . Here  $\mathbf{A}\mathbf{A}^\top = \begin{pmatrix} 2 & 1 \\ 1 & 2 \end{pmatrix}$  with inverse  $\frac{1}{3}\begin{pmatrix} 2 & -1 \\ -1 & 2 \end{pmatrix}$ , so

$$\mathbf{A}^+ = \frac{1}{3}\begin{pmatrix} 1 & 0 \\ 0 & 1 \\ 1 & 1 \end{pmatrix}\begin{pmatrix} 2 & -1 \\ -1 & 2 \end{pmatrix} = \frac{1}{3}\begin{pmatrix} 2 & -1 \\ -1 & 2 \\ 1 & 1 \end{pmatrix}.$$

#### Exercise B2: A system that is both inconsistent and underdetermined.

Let

$$\mathbf{A} = \begin{pmatrix} 1 & 2 \\ 2 & 4 \\ 1 & 2 \end{pmatrix}, \quad \mathbf{b} = (1, 4, 1).$$

Compute  $\mathbf{A}^+$  and  $\mathbf{x}_{\min} = \mathbf{A}^+\mathbf{b}$ . Verify that  $\mathbf{A}\mathbf{x}_{\min}$  is the orthogonal projection of  $\mathbf{b}$  onto  $\text{col}(\mathbf{A})$ , and compare  $\|\mathbf{x}_{\min}\|$  with the norm of one other least-squares solution.

*Solution.* Both columns are multiples of  $\mathbf{u} = (1, 2, 1)$ , so  $\mathbf{A} = \mathbf{u}\mathbf{v}^\top$  with  $\mathbf{v} = (1, 2)$ , and  $\text{rank } \mathbf{A} = 1$ . Then  $\|\mathbf{u}\|^2 = 6$ ,  $\|\mathbf{v}\|^2 = 5$  and  $\sigma_1 = \sqrt{30}$ , with  $\mathbf{u}_1 = \mathbf{u}/\sqrt{6}$  and  $\mathbf{v}_1 = \mathbf{v}/\sqrt{5}$ . Hence

$$\mathbf{A}^+ = \sigma_1^{-1} \mathbf{v}_1 \mathbf{u}_1^\top = \frac{1}{30} \mathbf{v} \mathbf{u}^\top = \frac{1}{30} \begin{pmatrix} 1 & 2 & 1 \\ 2 & 4 & 2 \end{pmatrix}.$$

Therefore

$$\mathbf{x}_{\min} = \mathbf{A}^+\mathbf{b} = \frac{1}{30}(1 + 8 + 1, 2 + 16 + 2) = \left(\frac{1}{3}, \frac{2}{3}\right).$$

*Projection.*  $\text{col}(\mathbf{A}) = \text{span}(\mathbf{u})$  and  $\langle \mathbf{b}, \mathbf{u} \rangle = 1 + 8 + 1 = 10$ , so  $P_{\text{col}(\mathbf{A})}\mathbf{b} = \frac{10}{6}\mathbf{u} = \frac{5}{3}(1, 2, 1)$ . And  $\mathbf{A}\mathbf{x}_{\min} = \mathbf{u}(\mathbf{v}^\top \mathbf{x}_{\min}) = \mathbf{u}\left(\frac{1}{3} + \frac{4}{3}\right) = \frac{5}{3}\mathbf{u}$ . They agree.

*Shortest.* The least-squares solutions are the  $\mathbf{x}$  with  $\mathbf{v}^\top \mathbf{x} = x_1 + 2x_2 = \frac{5}{3}$ . Another one is  $(\frac{5}{3}, 0)$ , of squared norm  $\frac{25}{9}$ , against  $\|\mathbf{x}_{\min}\|^2 = \frac{1}{9} + \frac{4}{9} = \frac{5}{9}$ . The pseudoinverse solution is shorter, as [Theorem 12.11.4](#) (b) requires.

### Exercise B3: Checking a candidate.

Let  $\mathbf{A} = \begin{pmatrix} 1 & 0 \\ 1 & 1 \\ 0 & 1 \end{pmatrix}$  and  $\mathbf{X} = \frac{1}{3} \begin{pmatrix} 2 & 1 & -1 \\ -1 & 1 & 2 \end{pmatrix}$ . Verify the four Penrose conditions for  $\mathbf{X}$ , and hence find the minimum-norm least-squares solution of  $\mathbf{A}\mathbf{x} = (1, 2, 3)$ .

*Solution.* Compute the two products first:

$$\mathbf{X}\mathbf{A} = \frac{1}{3} \begin{pmatrix} 3 & 0 \\ 0 & 3 \end{pmatrix} = \mathbf{I}_2, \quad \mathbf{A}\mathbf{X} = \frac{1}{3} \begin{pmatrix} 2 & 1 & -1 \\ 1 & 2 & 1 \\ -1 & 1 & 2 \end{pmatrix}.$$

(P4):  $\mathbf{X}\mathbf{A} = \mathbf{I}_2$  is Hermitian. (P3):  $\mathbf{A}\mathbf{X}$  is visibly symmetric. (P1):  $\mathbf{A}\mathbf{X}\mathbf{A} = \mathbf{A}(\mathbf{X}\mathbf{A}) = \mathbf{A}\mathbf{I}_2 = \mathbf{A}$ . (P2):  $\mathbf{X}\mathbf{A}\mathbf{X} = \mathbf{I}_2\mathbf{X} = \mathbf{X}$ . All four hold, so  $\mathbf{X} = \mathbf{A}^+$  by [Theorem 12.11.2](#).

Then

$$\mathbf{x}_{\min} = \mathbf{X}(1, 2, 3) = \frac{1}{3}(2 + 2 - 3, -1 + 2 + 6) = \left(\frac{1}{3}, \frac{7}{3}\right).$$

(The columns of  $\mathbf{A}$  are independent, so this is also  $(\mathbf{A}^\top \mathbf{A})^{-1} \mathbf{A}^\top \mathbf{b}$ , and it is the unique least-squares solution; the normal equations  $\begin{pmatrix} 2 & 1 \\ 1 & 2 \end{pmatrix} \mathbf{x} = (3, 5)$  confirm it.)

### 12.11.7.3 C. Going deeper

#### Exercise C1: Pseudoinverses of $\mathbf{A}^* \mathbf{A}$ .

Let  $\mathbf{A} \in M_{m \times n}(F)$ .

- (a) Prove that  $(\mathbf{A}^*\mathbf{A})^+ = \mathbf{A}^+(\mathbf{A}^+)^*$ .  
 (b) Deduce that  $\mathbf{A}^+ = (\mathbf{A}^*\mathbf{A})^+\mathbf{A}^*$ , the formula  $(\mathbf{A}^*\mathbf{A})^{-1}\mathbf{A}^*$  with the inverse replaced by the pseudoinverse.

*Hint: use a singular value decomposition and reduce each identity to an identity between diagonal matrices.*

*Solution.* Write  $\mathbf{A} = \mathbf{U}\mathbf{\Sigma}\mathbf{V}^*$ .

- (a) Then  $\mathbf{A}^*\mathbf{A} = \mathbf{V}\mathbf{\Sigma}^*\mathbf{\Sigma}\mathbf{V}^*$ , and  $\mathbf{\Sigma}^*\mathbf{\Sigma} \in M_n(F)$  is diagonal with entries  $\sigma_1^2, \dots, \sigma_p^2$  followed by zeros. That display is a singular value decomposition of  $\mathbf{A}^*\mathbf{A}$  in the sense of [Theorem 12.8.1](#) (both outer factors are  $\mathbf{V}$ , which is unitary, and the middle factor is diagonal with non-negative entries), so

$$(\mathbf{A}^*\mathbf{A})^+ = \mathbf{V}(\mathbf{\Sigma}^*\mathbf{\Sigma})^+\mathbf{V}^*,$$

where  $(\mathbf{\Sigma}^*\mathbf{\Sigma})^+$  is diagonal with entries  $\sigma_i^{-2}$  for  $i \leq r$  and 0 elsewhere. On the other side,  $(\mathbf{A}^+)^* = (\mathbf{V}\mathbf{\Sigma}^+\mathbf{U}^*)^* = \mathbf{U}(\mathbf{\Sigma}^+)^*\mathbf{V}^*$ , so

$$\mathbf{A}^+(\mathbf{A}^+)^* = \mathbf{V}\mathbf{\Sigma}^+\mathbf{U}^*\mathbf{U}(\mathbf{\Sigma}^+)^*\mathbf{V}^* = \mathbf{V}\mathbf{\Sigma}^+(\mathbf{\Sigma}^+)^*\mathbf{V}^*.$$

The middle factor  $\mathbf{\Sigma}^+(\mathbf{\Sigma}^+)^*$  is diagonal with entries  $\sigma_i^{-1}\sigma_i^{-1} = \sigma_i^{-2}$  for  $i \leq r$  and 0 elsewhere, which is  $(\mathbf{\Sigma}^*\mathbf{\Sigma})^+$ . Hence the two sides agree.

- (b) Using (a) and  $\mathbf{A}^* = \mathbf{V}\mathbf{\Sigma}^*\mathbf{U}^*$ ,

$$(\mathbf{A}^*\mathbf{A})^+\mathbf{A}^* = \mathbf{V}\mathbf{\Sigma}^+(\mathbf{\Sigma}^+)^*\mathbf{\Sigma}^*\mathbf{U}^*.$$

Now  $(\mathbf{\Sigma}^+)^*\mathbf{\Sigma}^* = (\mathbf{\Sigma}\mathbf{\Sigma}^+)^* = \mathbf{\Sigma}\mathbf{\Sigma}^+$ , which is the  $m \times m$  diagonal matrix with  $r$  ones and then zeros; and  $\mathbf{\Sigma}^+$  multiplied by it on the right is  $\mathbf{\Sigma}^+$ , since  $\mathbf{\Sigma}^+$  has non-zero entries only in the first  $r$  columns. So the right-hand side is  $\mathbf{V}\mathbf{\Sigma}^+\mathbf{U}^* = \mathbf{A}^+$ .

### Exercise C2: When the reversal rule does hold.

Let  $\mathbf{A} \in M_{m \times n}(F)$  with  $\text{rank } \mathbf{A} = n$  and  $\mathbf{B} \in M_{n \times q}(F)$  with  $\text{rank } \mathbf{B} = n$ . Prove that

$$(\mathbf{AB})^+ = \mathbf{B}^+\mathbf{A}^+.$$

*Hint: write both pseudoinverses with the formulas of [Theorem 12.11.3](#) (e) and verify the Penrose conditions for the product.*

*Solution.* By [Theorem 12.11.3](#) (e),  $\mathbf{A}^+ = (\mathbf{A}^*\mathbf{A})^{-1}\mathbf{A}^*$  and  $\mathbf{B}^+ = \mathbf{B}^*(\mathbf{BB}^*)^{-1}$ , the two inverses existing because  $\text{rank } \mathbf{A} = n$  and  $\text{rank } \mathbf{B}^* = n$ . Put

$$\mathbf{X} = \mathbf{B}^+\mathbf{A}^+ = \mathbf{B}^*(\mathbf{BB}^*)^{-1}(\mathbf{A}^*\mathbf{A})^{-1}\mathbf{A}^*.$$

Throughout we use  $\mathbf{A}^+\mathbf{A} = \mathbf{I}_n$  and  $\mathbf{BB}^+ = \mathbf{I}_n$ , which the two formulas give at once.

$$(P1): (\mathbf{AB})\mathbf{X}(\mathbf{AB}) = \mathbf{A}(\mathbf{BB}^+)(\mathbf{A}^+\mathbf{A})\mathbf{B} = \mathbf{AI}_n\mathbf{I}_n\mathbf{B} = \mathbf{AB}.$$

$$(P2): \mathbf{X}(\mathbf{AB})\mathbf{X} = \mathbf{B}^+(\mathbf{A}^+\mathbf{A})(\mathbf{BB}^+)\mathbf{A}^+ = \mathbf{B}^+\mathbf{A}^+ = \mathbf{X}.$$

$$(P3): (\mathbf{AB})\mathbf{X} = \mathbf{A}(\mathbf{BB}^+)\mathbf{A}^+ = \mathbf{AA}^+ = \mathbf{A}(\mathbf{A}^*\mathbf{A})^{-1}\mathbf{A}^*, \text{ which equals its own conjugate transpose, since } \mathbf{A}^*\mathbf{A} \text{ is Hermitian and so is its inverse.}$$

(P4):  $\mathbf{X}(\mathbf{AB}) = \mathbf{B}^+(\mathbf{A}^+\mathbf{A})\mathbf{B} = \mathbf{B}^+\mathbf{B} = \mathbf{B}^*(\mathbf{BB}^*)^{-1}\mathbf{B}$ , Hermitian for the same reason.

All four hold, so  $\mathbf{X} = (\mathbf{AB})^+$  by [Theorem 12.11.2](#).

### Exercise C3: When the pseudoinverse is the adjoint.

Let  $\mathbf{A} \in M_{m \times n}(F)$ .

- (a) Prove that  $\mathbf{A}^+ = \mathbf{A}^*$  if and only if  $\mathbf{AA}^*\mathbf{A} = \mathbf{A}$ .
- (b) Deduce that  $\mathbf{A}^+ = \mathbf{A}^*$  if and only if every non-zero singular value of  $\mathbf{A}$  equals 1.
- (c) Give an example of such an  $\mathbf{A}$  in  $M_2(\mathbb{R})$  which is neither unitary nor an orthogonal projection.

*Solution.*

- (a) ( $\Rightarrow$ ) If  $\mathbf{A}^+ = \mathbf{A}^*$ , then (P1) reads  $\mathbf{AA}^*\mathbf{A} = \mathbf{A}$ .  
 ( $\Leftarrow$ ) Suppose  $\mathbf{AA}^*\mathbf{A} = \mathbf{A}$  and put  $\mathbf{X} = \mathbf{A}^*$ . Then (P1) is the hypothesis; (P2) is  $\mathbf{A}^*\mathbf{AA}^* = (\mathbf{AA}^*\mathbf{A})^* = \mathbf{A}^*$ ; (P3) is  $(\mathbf{AA}^*)^* = \mathbf{AA}^*$ , true for every  $\mathbf{A}$ ; and (P4) is  $(\mathbf{A}^*\mathbf{A})^* = \mathbf{A}^*\mathbf{A}$ , likewise. By [Theorem 12.11.2](#),  $\mathbf{A}^+ = \mathbf{A}^*$ .
- (b) With  $\mathbf{A} = \mathbf{U}\mathbf{\Sigma}\mathbf{V}^*$  we have  $\mathbf{A}^* = \mathbf{V}\mathbf{\Sigma}^*\mathbf{U}^*$  and  $\mathbf{A}^+ = \mathbf{V}\mathbf{\Sigma}^+\mathbf{U}^*$ . Since  $\mathbf{U}$  and  $\mathbf{V}$  are invertible,  $\mathbf{A}^+ = \mathbf{A}^*$  if and only if  $\mathbf{\Sigma}^+ = \mathbf{\Sigma}^*$ , that is if and only if  $\sigma_i^{-1} = \sigma_i$  for every  $i \leq r$ , that is if and only if every non-zero singular value is 1 (the equation  $\sigma^2 = 1$  with  $\sigma > 0$  forces  $\sigma = 1$ ).
- (c) Take

$$\mathbf{A} = \frac{1}{\sqrt{2}} \begin{pmatrix} 1 & 1 \\ 0 & 0 \end{pmatrix}.$$

Then  $\mathbf{AA}^\top = \begin{pmatrix} 1 & 0 \\ 0 & 0 \end{pmatrix}$ , so  $\mathbf{AA}^\top\mathbf{A} = \mathbf{A}$  and (a) applies. Its singular values are 1 and 0, matching (b). It is not unitary, since it is singular, and not an orthogonal projection, since  $\mathbf{A}^\top \neq \mathbf{A}$ . Such a matrix maps  $\text{col}(\mathbf{A}^*)$  isometrically onto  $\text{col}(\mathbf{A})$  and sends  $\text{nul}(\mathbf{A})$  to  $\mathbf{0}$ : a unitary matrix on the part of the space where it is invertible, and nothing at all elsewhere.

## 12.12 Projections Revisited

Chapter 3 built projections out of nothing but a direct sum: given  $V = U \oplus W$ , the operator that keeps the  $U$ -piece and discards the  $W$ -piece satisfies  $P^2 = P$ , and every operator satisfying  $P^2 = P$  arises this way. Chapter 10 built a second kind by taking  $W = U^\perp$ , the orthogonal projection  $P_U$ . The second is a special case of the first, and Chapter 3 left the comparison open, saying only that its projections “are often called **oblique** projections, to distinguish them from the orthogonal projections met later, in the chapters on inner product spaces”. This section makes the distinction exact, measures how far an oblique projection is from being orthogonal, and closes Part III.

Throughout,  $V$  is a finite-dimensional inner product space over  $F = \mathbb{R}$  or  $\mathbb{C}$ , and  $n = \dim V$ .

## 12.12.1 Which projections are orthogonal?

Two recollections, because the section is about the gap between them. From Chapter 3: an operator  $P \in \mathcal{L}(V)$  with  $P^2 = P$  is a **projection**, also called an **idempotent** (Definition 3.10.1); it satisfies  $V = \operatorname{im} P \oplus \ker P$ , it is the identity on  $\operatorname{im} P$  and zero on  $\ker P$ , and each splitting  $V = U \oplus W$  has exactly one projection with image  $U$  and kernel  $W$  (Theorem 3.10.1). No inner product appears there, and the field was arbitrary.

From Chapter 10: the splitting  $V = U \oplus U^\perp$  of Theorem 10.3.2 is one of the many available, and  $P_U$  is the projection belonging to it (Definition 10.3.2). Chapter 3 warned that a subspace is the image of infinitely many projections, one per complement. Which one does the inner product pick out, and how is it recognized without being told the kernel?

*A projection is orthogonal exactly when it throws away perpendicularly — and that is the same as costing nothing to move across the inner product.*

### Theorem 12.12.1 — Orthogonal Projections Among the Idempotents

Let  $V$  be a finite-dimensional inner product space over  $F$  and let  $P \in \mathcal{L}(V)$  satisfy  $P^2 = P$ . The following are equivalent.

- (a)  $P$  is the orthogonal projection onto its image:  $P = P_U$  with  $U = \operatorname{im} P$ .
- (b)  $\ker P = (\operatorname{im} P)^\perp$ .
- (c)  $P^* = P$ .
- (d)  $P$  is normal.
- (e)  $\|P\mathbf{v}\| \leq \|\mathbf{v}\|$  for every  $\mathbf{v} \in V$ .

*Idea.* Conditions (a) and (b) say one thing twice, once about the operator and once about its two subspaces. Going from (b) to (c) is a computation Chapter 10 has already done for  $P_U$ . Coming back from (c) is the four-subspaces theorem:  $\ker P^*$  is always  $(\operatorname{im} P)^\perp$ , so once  $P^* = P$  the kernel is forced; normality is the same remark with  $\ker P^*$  replaced by  $\ker P$ . The one direction with work in it is (e)  $\Rightarrow$  (b), where a bound on lengths has to become a statement about angles. Feed  $P$  a vector  $\mathbf{u} + t\mathbf{w}$  with  $\mathbf{u} \in \operatorname{im} P$  and  $\mathbf{w} \in \ker P$ : its image is  $\mathbf{u}$  whatever  $t$  is, so choosing  $t$  to make  $\|\mathbf{u} + t\mathbf{w}\|$  as small as possible squeezes  $\langle \mathbf{u}, \mathbf{w} \rangle$  to zero.

*Proof.* (a)  $\Leftrightarrow$  (b). Write  $U = \operatorname{im} P$ . ( $\Rightarrow$ ) If  $P = P_U$ , then  $\ker P = \ker P_U = U^\perp$ , since  $P_U$  is the projection onto  $U$  along  $U^\perp$  (Definition 10.3.2 and Theorem 3.10.1 (b)). ( $\Leftarrow$ ) If  $\ker P = U^\perp$ , then  $P$  is a projection with image  $U$  and kernel  $U^\perp$ , and so is  $P_U$ ; by the uniqueness in Theorem 3.10.1 (b), applied to  $V = U \oplus U^\perp$ , the two are equal.

(b)  $\Rightarrow$  (c). By the equivalence just proved,  $P = P_U$  with  $U = \operatorname{im} P$ . By Theorem 10.3.3 (b),  $\langle P\mathbf{u}, \mathbf{v} \rangle = \langle \mathbf{u}, P\mathbf{v} \rangle$  for all  $\mathbf{u}, \mathbf{v} \in V$ . That is the defining identity of the adjoint, so  $P^* = P$  by the uniqueness in Definition 10.6.1.

(c)  $\Rightarrow$  (d). If  $P^* = P$  then  $P^*P = P^2 = PP^*$ , which is Definition 10.6.3.

(d)  $\Rightarrow$  (b). Since  $P$  is normal,  $\ker P = \ker P^*$  by Corollary 11.3.2 (a). By Theorem 10.6.5 (a),  $\ker P^* = (\operatorname{im} P)^\perp$ . Combining the two gives (b).

(b)  $\Rightarrow$  (e). Let  $\mathbf{v} \in V$  and split it as  $\mathbf{v} = P\mathbf{v} + (\mathbf{v} - P\mathbf{v})$ . The first piece lies in  $\text{im } P$ , and the second lies in  $\ker P$ , because  $P(\mathbf{v} - P\mathbf{v}) = P\mathbf{v} - P^2\mathbf{v} = \mathbf{0}$ . By (b) the two pieces are orthogonal, so [Theorem 10.1.3](#) gives

$$\|\mathbf{v}\|^2 = \|P\mathbf{v}\|^2 + \|\mathbf{v} - P\mathbf{v}\|^2 \geq \|P\mathbf{v}\|^2,$$

and taking non-negative square roots gives  $\|P\mathbf{v}\| \leq \|\mathbf{v}\|$ .

(e)  $\Rightarrow$  (b). We first show  $\langle \mathbf{u}, \mathbf{w} \rangle = 0$  for every  $\mathbf{u} \in \text{im } P$  and every  $\mathbf{w} \in \ker P$ . There is nothing to prove when  $\mathbf{w} = \mathbf{0}$ , so assume  $\mathbf{w} \neq \mathbf{0}$ , write  $z = \langle \mathbf{u}, \mathbf{w} \rangle$ , and put

$$t = -\frac{z}{\|\mathbf{w}\|^2}, \quad \mathbf{v} = \mathbf{u} + t\mathbf{w},$$

which is legitimate because  $\|\mathbf{w}\| \neq 0$ . Then  $P\mathbf{v} = P\mathbf{u} + tP\mathbf{w} = \mathbf{u}$ , since  $P$  is the identity on  $\text{im } P$  and zero on  $\ker P$  ([Theorem 3.10.1](#) (a)). Expanding the norm squared and using conjugate symmetry,

$$\begin{aligned} \|\mathbf{v}\|^2 &= \|\mathbf{u}\|^2 + \bar{t}\langle \mathbf{u}, \mathbf{w} \rangle + t\langle \mathbf{w}, \mathbf{u} \rangle + |t|^2\|\mathbf{w}\|^2 \\ &= \|\mathbf{u}\|^2 - \frac{|z|^2}{\|\mathbf{w}\|^2} - \frac{|z|^2}{\|\mathbf{w}\|^2} + \frac{|z|^2}{\|\mathbf{w}\|^2} = \|\mathbf{u}\|^2 - \frac{|z|^2}{\|\mathbf{w}\|^2}. \end{aligned}$$

Now (e) applied to this  $\mathbf{v}$  says  $\|\mathbf{u}\|^2 = \|P\mathbf{v}\|^2 \leq \|\mathbf{v}\|^2$ , which forces  $|z|^2 \leq 0$  and hence  $z = 0$ .

So  $\ker P \subseteq (\text{im } P)^\perp$ . The two subspaces have the same dimension:  $\dim \ker P = n - \text{rank } P$  by [Theorem 3.3.1](#), and  $\dim(\text{im } P)^\perp = n - \dim \text{im } P = n - \text{rank } P$  by [Theorem 10.3.2](#) (c). A subspace contained in another of the same finite dimension equals it ([Theorem 1.6.8](#)), so  $\ker P = (\text{im } P)^\perp$ . This closes the cycle and proves the theorem. ■

The theorem is the promised comparison. Every projection of Chapter 3 that is **not** self-adjoint is oblique, and now “oblique” has a precise meaning: its kernel is a complement of its image that is not the perpendicular one. Chapter 11 has already met a pair of such projections, in [Example \(Two Projections That Are Not Orthogonal\)](#), and the Quick check after [Theorem 11.8.2](#) traced exactly what their obliqueness cost.

Clause (c) also settles Chapter 3’s warning that “projection onto  $U$ ” is incomplete information.

### Corollary 12.12.2 — One Orthogonal Projection per Subspace

Let  $U$  be a subspace of a finite-dimensional inner product space  $V$ . Among the projections of  $V$  with image  $U$  — one for each complement of  $U$ , by [Theorem 3.10.1](#) — exactly one is self-adjoint, namely  $P_U$ .

*Proof.*  $P_U$  has image  $U$  and is self-adjoint by [Theorem 12.12.1](#), (a)  $\Rightarrow$  (c). Conversely, let  $P$  be a self-adjoint projection with  $\text{im } P = U$ . By the same theorem, (c)  $\Rightarrow$  (b),  $\ker P = U^\perp$ , which is also  $\ker P_U$ . Two projections with the same image and the same kernel are equal, by the uniqueness in [Theorem 3.10.1](#) (b). ■



Chapter 3's own example is now decided by inspection. On  $\mathbb{R}^2$  with the dot product,  $P(x, y) = (x, 0)$  and  $P'(x, y) = (x - y, 0)$  are both projections onto the  $x$ -axis, with standard matrices

$$\mathbf{P} = \begin{pmatrix} 1 & 0 \\ 0 & 0 \end{pmatrix}, \quad \mathbf{P}' = \begin{pmatrix} 1 & -1 \\ 0 & 0 \end{pmatrix}.$$

The standard basis is orthonormal, so [Theorem 10.6.3](#) says self-adjointness is symmetry of these matrices. The first is symmetric, so  $P = P_U$  for the  $x$ -axis  $U$ . The second is not, so  $P'$  is oblique — as the picture in Chapter 3 showed, it slides along the line  $y = x$  rather than straight down. Clause (e) fails visibly at  $\mathbf{v} = (1, -1)$ : there  $P'\mathbf{v} = (2, 0)$ , and  $\|P'\mathbf{v}\| = 2$  while  $\|\mathbf{v}\| = \sqrt{2}$ .

*Quick check.* Does  $\mathbf{A} = \begin{pmatrix} 1 & 0 \\ 1 & 0 \end{pmatrix}$  act on  $\mathbb{R}^2$  as a projection? If so, is it an orthogonal projection?

*Solution.*  $\mathbf{A}^2 = \begin{pmatrix} 1 & 0 \\ 1 & 0 \end{pmatrix} = \mathbf{A}$ , so yes, it is a projection. Its image is the span of its columns,  $\text{span}((1, 1))$ , and its kernel is  $\text{span}(\mathbf{e}_2)$ . Since  $\langle (1, 1), \mathbf{e}_2 \rangle = 1 \neq 0$ , the kernel is not  $(\text{im } \mathbf{A})^\perp$ , so the projection is oblique by [Theorem 12.12.1](#) (b). The quickest route is clause (c):  $\mathbf{A}^\top \neq \mathbf{A}$ .

*Warning.* **Clause (e) is a statement about every vector, and one vector proves nothing.** For the oblique  $\mathbf{P}'$  above,  $\|\mathbf{P}'\mathbf{e}_1\| = 1 = \|\mathbf{e}_1\|$  and  $\|\mathbf{P}'(1, 1)\| = 0 \leq \sqrt{2}$ ; both are consistent with the bound, and both are worthless as evidence. The bound fails only in the direction  $(1, -1)$ . If you want to certify a projection as orthogonal by hand, check  $\mathbf{P}^* = \mathbf{P}$  in an orthonormal basis, not the length of one image.

*Warning.* **An orthogonal projection is almost never an orthogonal matrix.** If  $\mathbf{P}$  is unitary and idempotent, it is invertible, so multiplying  $\mathbf{P}^2 = \mathbf{P}$  by  $\mathbf{P}^{-1}$  gives  $\mathbf{P} = \mathbf{I}$ . Apart from the identity, every orthogonal projection is singular. “Orthogonal” in  $P_U$  names the direction of the discarding, not membership in  $O(n)$ .

## 12.12.2 The block that measures obliqueness

Chapter 3 already knows the shape of a projection up to a change of basis: with a basis of  $\text{im } P$  followed by a basis of  $\ker P$ , the matrix is  $\text{diag}(1, \dots, 1, 0, \dots, 0)$  with  $\text{rank } P$  ones ([Corollary 3.10.2](#)). That basis is built to fit  $P$  and is usually not orthonormal, so the normal form erases exactly the information we now want. Insisting on an orthonormal basis costs one extra block, and that block is the answer.

### Theorem 12.12.3 — Canonical Form of an Idempotent in an Orthonormal Basis

Let  $V$  be an  $n$ -dimensional inner product space over  $F$  and let  $P \in \mathcal{L}(V)$  be a projection of rank  $r$ . Let  $(\mathbf{f}_1, \dots, \mathbf{f}_r)$  be any orthonormal basis of  $\text{im } P$  and  $(\mathbf{f}_{r+1}, \dots, \mathbf{f}_n)$  any orthonormal basis of  $(\text{im } P)^\perp$ . Then  $\mathcal{B} = (\mathbf{f}_1, \dots, \mathbf{f}_n)$  is an orthonormal basis of  $V$ , and

$$[P]_{\mathcal{B}} = \begin{pmatrix} \mathbf{I}_r & \mathbf{X} \\ 0 & 0 \end{pmatrix}$$

for some  $\mathbf{X} \in M_{r \times (n-r)}(F)$ . Moreover  $P$  is an orthogonal projection if and only if  $\mathbf{X} = \mathbf{0}$ .

*Idea.* Only two facts are used, and both are already proved. First,  $P$  fixes its image, so the first  $r$  columns are forced to be  $\mathbf{e}_1, \dots, \mathbf{e}_r$ . Second, every output of  $P$  lies in  $\text{im } P$ , so the last  $n - r$  columns have nothing below row  $r$ . What sits **above** row  $r$  in those columns is unconstrained, and that is  $\mathbf{X}$ : it records where  $P$  sends the directions perpendicular to its image, which an orthogonal projection would send to  $\mathbf{0}$ .

*Proof.* Write  $U = \text{im } P$ , so  $\dim U = r$  and  $\dim U^\perp = n - r$  by Theorem 10.3.2 (c). Each  $\mathbf{f}_i$  with  $i \leq r$  is orthogonal to each  $\mathbf{f}_j$  with  $j > r$ , since the first lies in  $U$  and the second in  $U^\perp$ ; so  $\mathcal{B}$  is an orthonormal list of  $n$  vectors, hence an orthonormal basis of  $V$  (Theorem 10.1.4).

For  $i \leq r$  we have  $\mathbf{f}_i \in \text{im } P$ , so  $P\mathbf{f}_i = \mathbf{f}_i$  by Theorem 3.10.1 (a), and column  $i$  of  $[P]_{\mathcal{B}}$  is  $\mathbf{e}_i$ . For  $j > r$  we have  $P\mathbf{f}_j \in \text{im } P = U = \text{span}(\mathbf{f}_1, \dots, \mathbf{f}_r)$ , so  $P\mathbf{f}_j = \sum_{i \leq r} x_{ij} \mathbf{f}_i$  for scalars  $x_{ij} \in F$ , and column  $j$  has zeros in rows  $r + 1, \dots, n$ . Collecting the  $x_{ij}$  into  $\mathbf{X}$  gives the displayed shape.

For the last claim,  $\mathcal{B}$  is orthonormal, so  $[P^*]_{\mathcal{B}} = ([P]_{\mathcal{B}})^*$  by Theorem 10.6.3, and

$$([P]_{\mathcal{B}})^* = \begin{pmatrix} \mathbf{I}_r & \mathbf{0} \\ \mathbf{X}^* & \mathbf{0} \end{pmatrix}.$$

The two matrices agree if and only if  $\mathbf{X} = \mathbf{0}$ . Since an operator is determined by its matrix in a basis,  $P^* = P$  if and only if  $\mathbf{X} = \mathbf{0}$ , and  $P^* = P$  says  $P$  is an orthogonal projection by Theorem 12.12.1. ■

The block  $\mathbf{X}$  is not unique, because the two orthonormal bases were not. Replacing them changes  $\mathbf{X}$  into  $\mathbf{A}^* \mathbf{X} \mathbf{B}$  for some unitary  $\mathbf{A} \in M_r(F)$  and  $\mathbf{B} \in M_{n-r}(F)$ , and that leaves the singular values of  $\mathbf{X}$  alone: since

$$(\mathbf{A}^* \mathbf{X} \mathbf{B})^* (\mathbf{A}^* \mathbf{X} \mathbf{B}) = \mathbf{B}^* (\mathbf{X}^* \mathbf{X}) \mathbf{B},$$

the two matrices  $(\mathbf{A}^* \mathbf{X} \mathbf{B})^* (\mathbf{A}^* \mathbf{X} \mathbf{B})$  and  $\mathbf{X}^* \mathbf{X}$  are unitarily similar, so they have the same eigenvalues, and Definition 12.8.1 reads the singular values off those. So the list  $\sigma_1(\mathbf{X}) \geq \sigma_2(\mathbf{X}) \geq \dots$  belongs to  $P$  itself, and  $P$  is an orthogonal projection exactly when that list is all zeros.

### 12.12.2.1 The two-dimensional case, in full

Take  $n = 2$  and  $r = 1$ , so that  $\mathbf{X}$  is a single scalar  $x \in F$ , and

$$[P]_{\mathcal{B}} = \begin{pmatrix} 1 & x \\ 0 & 0 \end{pmatrix}, \quad \mathcal{B} = (\mathbf{f}_1, \mathbf{f}_2),$$

with  $\mathbf{f}_1$  spanning the line  $\text{im } P$  and  $\mathbf{f}_2$  spanning the perpendicular line  $(\text{im } P)^\perp$ . Everything about  $P$  can be read off this one number.

**The kernel.**  $P(a\mathbf{f}_1 + b\mathbf{f}_2) = (a + xb)\mathbf{f}_1$ , which is  $\mathbf{0}$  exactly when  $a = -xb$ . So

$$\ker P = \text{span}(-x\mathbf{f}_1 + \mathbf{f}_2),$$

a line that meets  $(\text{im } P)^\perp = \text{span}(\mathbf{f}_2)$  only at  $\mathbf{0}$  unless  $x = 0$ .

**The angle.** Suppose  $F = \mathbb{R}$ . The two lines  $\text{im } P$  and  $\ker P$  are distinct, since their intersection is  $\{\mathbf{0}\}$ ; let  $\theta \in (0, \pi/2]$  be the angle between them, meaning the angle of Definition 10.1.5 between  $\mathbf{f}_1$  and whichever of  $\pm(-x\mathbf{f}_1 + \mathbf{f}_2)$  makes it acute. With  $\mathbf{k} = -x\mathbf{f}_1 + \mathbf{f}_2$  we have  $\|\mathbf{k}\| = \sqrt{1+x^2}$  and  $\langle \mathbf{f}_1, \mathbf{k} \rangle = -x$ , so

$$\cos \theta = \frac{|x|}{\sqrt{1+x^2}}, \quad \sin \theta = \frac{1}{\sqrt{1+x^2}}, \quad |x| = \cot \theta.$$

The dictionary is complete:  $x = 0$  is  $\theta = \pi/2$ , the orthogonal case; and as the kernel tips toward the image,  $\theta \rightarrow 0$  and  $|x| \rightarrow \infty$ . The single entry  $x$  is the obliqueness, in the units of a cotangent.

**The stretching.** For a unit vector  $\mathbf{v} = a\mathbf{f}_1 + b\mathbf{f}_2$ , we have  $\|P\mathbf{v}\| = |a + xb| = |\langle (a, b), (1, \bar{x}) \rangle|$  in  $F^2$  with the standard inner product, so Theorem 10.1.5 gives

$$\|P\mathbf{v}\| \leq \sqrt{1+|x|^2},$$

with equality exactly when  $(a, b)$  is a multiple of  $(1, \bar{x})$ , for instance  $(a, b) = (1, \bar{x})/\sqrt{1+|x|^2}$ . So the largest value of  $\|P\mathbf{v}\|$  over unit vectors  $\mathbf{v}$  is  $\sqrt{1+|x|^2}$ , which over  $\mathbb{R}$  is  $1/\sin \theta$ . Chapter 15 will give this number a name, the operator norm  $\|P\|$ ; here it is enough to say that an orthogonal projection never lengthens a vector, while an oblique one lengthens some vector by  $1/\sin \theta$ , a factor that is unbounded as the kernel closes in on the image.

### Example. (Two Oblique Projections, Measured)

For each of

$$\mathbf{P}' = \begin{pmatrix} 1 & -1 \\ 0 & 0 \end{pmatrix}, \quad \mathbf{Q} = \begin{pmatrix} 2 & -1 \\ 2 & -1 \end{pmatrix}$$

on  $\mathbb{R}^2$ , verify that it is a projection, put it in the form of Theorem 12.12.3, and find the angle between its image and its kernel and the largest length of an image of a unit vector.

*Solution.* The matrix  $\mathbf{P}'$ .  $(\mathbf{P}')^2 = \mathbf{P}'$  by direct multiplication, and  $\text{im } \mathbf{P}' = \text{span}(\mathbf{e}_1)$ ,  $\ker \mathbf{P}' = \text{span}((1, 1))$ . Here  $\mathbf{f}_1 = \mathbf{e}_1$  already spans the image and  $\mathbf{f}_2 = \mathbf{e}_2$  spans its perpendicular, and the standard basis is orthonormal, so the given matrix is the canonical form, with  $x = -1$ . Hence  $\cot \theta = 1$ , that is  $\theta = \pi/4$ , and the largest value of  $\|\mathbf{P}'\mathbf{v}\|$  on the unit circle is  $\sqrt{1+1} = \sqrt{2}$ , attained at  $\mathbf{v} = (1, -1)/\sqrt{2}$ . Both readings match Chapter 3's figure, where  $\mathbf{P}'$  projects along the line  $y = x$ .

The matrix  $\mathbf{Q}$ .  $\mathbf{Q}^2 = \begin{pmatrix} 4-2 & -2+1 \\ 4-2 & -2+1 \end{pmatrix} = \mathbf{Q}$ , so  $\mathbf{Q}$  is a projection, of rank 1. Its image is  $\text{span}((1, 1))$  and its kernel is  $\text{span}((1, 2))$ , since  $\mathbf{Q}(1, 2) = (0, 0)$ . Take

$$\mathbf{f}_1 = \frac{1}{\sqrt{2}}(1, 1), \quad \mathbf{f}_2 = \frac{1}{\sqrt{2}}(1, -1),$$

an orthonormal basis of  $\text{im } \mathbf{Q}$  and of  $(\text{im } \mathbf{Q})^\perp$  respectively. Then  $\mathbf{Q}\mathbf{f}_1 = \mathbf{f}_1$  and  $\mathbf{Q}\mathbf{f}_2 = \frac{1}{\sqrt{2}}(3, 3) = 3\mathbf{f}_1$ , so  $x = 3$  and

$$[\mathbf{Q}]_{\mathcal{B}} = \begin{pmatrix} 1 & 3 \\ 0 & 0 \end{pmatrix}.$$

Hence  $\cot \theta = 3$ . Checking against the two lines directly: the angle between  $(1, 1)$  and  $(1, 2)$  has  $\cos \theta = 3/\sqrt{10}$  and  $\sin \theta = 1/\sqrt{10}$ , so  $\cot \theta = 3$ , and the largest value of  $\|\mathbf{Q}\mathbf{v}\|$  on the unit circle is  $\sqrt{1+9} = \sqrt{10} = 1/\sin \theta$ . The two routes agree.

In higher rank the same block  $\mathbf{X}$  does the same job, and [Exercise \(C3: How far an oblique projection can stretch\)](#) turns its largest singular value into the largest value of  $\|P\mathbf{v}\|$  over unit  $\mathbf{v}$ , by the argument that just ran in the  $2 \times 2$  case. The reason the singular values appear is the one this chapter has used throughout:  $\mathbf{X}$  is a rectangular matrix, so its eigenvalues do not exist, and [Theorem 12.8.1](#) is what measures it instead.

### 12.12.3 Looking back at Part III

Part III added one piece of structure to the vector spaces of Parts I and II, and spent three chapters finding out what it buys.

**Chapter 10 added the inner product**, and with it length, orthogonality and angle. Orthonormal bases made coordinates free, orthogonal complements split every space canonically, and projection onto a subspace solved the nearest-point problem, hence least squares. Its deepest construction was the **adjoint**, the price of moving an operator from one slot of the inner product to the other; the adjoint is what turned the kernel and the image into two orthogonal pairs of subspaces.

**Chapter 11 asked which operators the inner product can diagonalize**, and the adjoint answered: over  $\mathbb{C}$  the normal ones, over  $\mathbb{R}$  the self-adjoint ones. Written without bases, the spectral theorem says  $T = \sum_i \lambda_i P_i$  with the  $P_i$  orthogonal projections onto the eigenspaces — which is why the present section had to exist, and why [Theorem 11.8.1](#) needed clause (b) at all. Once a matrix is diagonal in an orthonormal basis, anything can be done to its eigenvalues and undone afterwards, and that is the functional calculus.

**Chapter 12 asked which self-adjoint operators behave like non-negative numbers**, collected the equivalent answers in [Theorem 12.1.4](#), and used them twice: to build the square root and the absolute value  $|\mathbf{A}| = (\mathbf{A}^* \mathbf{A})^{1/2}$  of [Definition 12.2.1](#), and then to get past the restriction that had hung over the whole part. The spectral theorem needs a square operator, and most matrices are not square — but  $\mathbf{A}^* \mathbf{A}$  always is square, self-adjoint and positive semidefinite, whatever  $\mathbf{A}$  is. Applying the spectral theorem to it and reading the result back through  $\mathbf{A}$  gives [Theorem 12.8.1](#), and with it orthonormal bases of all four fundamental subspaces ([Corollary 12.8.4](#)), the polar decomposition of [Theorem 12.9.1](#), the best low-rank approximation and the pseudoinverse. A rectangular matrix has no eigenvalues; it has singular values, and they do the work.

So the part is one chain, each link an application of the last: **inner product**  $\rightarrow$  **adjoint**  $\rightarrow$  **spectral theorem**  $\rightarrow$  **singular value decomposition**. Its named moves travel. To show a vector is zero, pair it with itself. To move an operator across the inner product, pay with the adjoint. Work with the norm squared. Fix an orthonormal basis. Project, then subtract. Test positivity with the single scalar  $\langle \mathbf{A}\mathbf{x}, \mathbf{x} \rangle$ , and factor a positive matrix as  $\mathbf{B}^* \mathbf{B}$ . Diagonalize, work on the eigenvalues, come back. When the matrix is not square, apply the spectral theorem to  $\mathbf{A}^* \mathbf{A}$ . And when the answer is a sum ordered by size, truncate it.

What comes next takes the quadratic form  $\mathbf{x} \mapsto \langle \mathbf{A}\mathbf{x}, \mathbf{x} \rangle$ , which was only a test in this chapter, and makes it the object of study: Chapter 13 classifies forms up to congruence, by inertia, over  $\mathbb{R}$  and over  $\mathbb{C}$ . Chapter 15 gives the operator norm promised twice above, and Chapter 16 characterizes the eigenvalues of a Hermitian matrix as maxima and minima of  $\langle \mathbf{A}\mathbf{x}, \mathbf{x} \rangle$  — at which point the test becomes the theory.

## 12.12.4 Exercises

### 12.12.4.1 A. Check your understanding

#### Exercise A1.

- Define what it means for  $P \in \mathcal{L}(V)$  to be a projection, and what it means for it to be the orthogonal projection onto a subspace  $U$ . Which of the two definitions needs an inner product?
- State three conditions on a projection  $P$ , each equivalent to “ $P$  is an orthogonal projection”.
- Determine whether the following is correct, and justify your answer: if  $\|P\mathbf{v}\| \leq \|\mathbf{v}\|$  for some non-zero  $\mathbf{v}$ , then the projection  $P$  is orthogonal.
- Determine whether the following is correct, and justify your answer: every projection of rank  $r$  on an  $n$ -dimensional inner product space has matrix  $\mathbf{I}_r \oplus 0$  in some orthonormal basis.

*Solution.*

- $P$  is a projection when  $P^2 = P$  (Definition 3.10.1);  $P$  is the orthogonal projection onto  $U$  when it is the projection onto  $U$  along  $U^\perp$  (Definition 10.3.2). Only the second needs an inner product, since  $U^\perp$  is undefined without one.
- By Theorem 12.12.1:  $\ker P = (\operatorname{im} P)^\perp$ ;  $P^* = P$ ;  $P$  is normal;  $\|P\mathbf{v}\| \leq \|\mathbf{v}\|$  for every  $\mathbf{v}$ . Any three of these will do.
- Incorrect. Take  $\mathbf{P}' = \begin{pmatrix} 1 & -1 \\ 0 & 0 \end{pmatrix}$  on  $\mathbb{R}^2$  and  $\mathbf{v} = (1, 1)$ . Then  $\mathbf{P}'\mathbf{v} = \mathbf{0}$ , so  $\|\mathbf{P}'\mathbf{v}\| = 0 \leq \sqrt{2}$ , yet  $\mathbf{P}'$  is oblique because  $(\mathbf{P}')^\top \neq \mathbf{P}'$ . Clause (e) of Theorem 12.12.1 quantifies over **every**  $\mathbf{v}$ .
- Incorrect in general, and correct exactly for the orthogonal projections. By Theorem 12.12.3 the matrix in a suitable orthonormal basis is  $\begin{pmatrix} \mathbf{I}_r & \mathbf{X} \\ 0 & 0 \end{pmatrix}$ , and it is  $\mathbf{I}_r \oplus 0$  only when  $\mathbf{X} = \mathbf{0}$ . If some orthonormal basis gave  $\mathbf{I}_r \oplus 0$ , that matrix would be Hermitian, so  $P$  would be self-adjoint by Theorem 10.6.3. The matrix  $\mathbf{P}'$  of part (c) is a counterexample. (In a basis that is merely a basis,  $\mathbf{I}_r \oplus 0$  is always attainable: that is Corollary 3.10.2.)

### 12.12.4.2 B. Practice

#### Exercise B1: Which are orthogonal projections?.

Determine which of the following matrices are projections, and among those, which are orthogonal projections. Justify your answer. For the oblique ones over  $\mathbb{R}$ , find the angle between image and kernel.

- $\mathbf{A} = \frac{1}{5} \begin{pmatrix} 1 & 2 \\ 2 & 4 \end{pmatrix}$  on  $\mathbb{R}^2$ .
- $\mathbf{B} = \begin{pmatrix} 1 & 0 \\ -2 & 0 \end{pmatrix}$  on  $\mathbb{R}^2$ .
- $\mathbf{C} = \begin{pmatrix} 1 & 1 \\ 1 & 1 \end{pmatrix}$  on  $\mathbb{R}^2$ .

$$(d) \mathbf{D} = \begin{pmatrix} 1 & i \\ 0 & 0 \end{pmatrix} \text{ on } \mathbb{C}^2.$$

*Solution.*

- (a)  $\mathbf{A}^2 = \frac{1}{25} \begin{pmatrix} 5 & 10 \\ 10 & 20 \end{pmatrix} = \mathbf{A}$ , so  $\mathbf{A}$  is a projection, and  $\mathbf{A}^\top = \mathbf{A}$ . By [Theorem 12.12.1](#) it is the orthogonal projection onto  $\text{im } \mathbf{A} = \text{span}((1, 2))$ .
- (b)  $\mathbf{B}^2 = \mathbf{B}$ , so  $\mathbf{B}$  is a projection, but  $\mathbf{B}^\top \neq \mathbf{B}$ , so it is oblique. Its image is  $\text{span}((1, -2))$  and its kernel is  $\text{span}(\mathbf{e}_2)$ . The angle between these lines has  $\cos \theta = |\langle (1, -2), (0, 1) \rangle| / (\sqrt{5} \cdot 1) = 2/\sqrt{5}$ , so  $\sin \theta = 1/\sqrt{5}$  and  $\cot \theta = 2$ ; that is  $\theta = \arctan(1/2)$ , about  $26.6^\circ$ .
- (c)  $\mathbf{C}^2 = \begin{pmatrix} 2 & 2 \\ 2 & 2 \end{pmatrix} = 2\mathbf{C} \neq \mathbf{C}$ , so  $\mathbf{C}$  is not a projection at all, although it is symmetric. Symmetry alone is not idempotence;  $\frac{1}{2}\mathbf{C}$  is the orthogonal projection onto  $\text{span}((1, 1))$ .
- (d)  $\mathbf{D}^2 = \mathbf{D}$ , so  $\mathbf{D}$  is a projection, and  $\mathbf{D}^* = \begin{pmatrix} 1 & 0 \\ -i & 0 \end{pmatrix} \neq \mathbf{D}$ , so it is oblique. It is already in the canonical form of [Theorem 12.12.3](#), with  $x = i$ . There is no angle over  $\mathbb{C}$  ([Definition 10.1.5](#) is stated for real spaces), but  $|x| = 1$ , and the largest value of  $\|\mathbf{D}\mathbf{v}\|$  over unit  $\mathbf{v}$  is  $\sqrt{1+1} = \sqrt{2}$ , the same number the real matrix  $\mathbf{P}'$  produced.

### Exercise B2: Putting a projection in canonical form.

Let  $\mathbf{A} = \begin{pmatrix} 0 & 1 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{pmatrix}$  act on  $\mathbb{R}^3$  with the dot product.

- (a) Verify that  $\mathbf{A}$  is a projection and find  $\text{im } \mathbf{A}$  and  $\ker \mathbf{A}$ .
- (b) Find an orthonormal basis  $\mathcal{B}$  of  $\mathbb{R}^3$  as in [Theorem 12.12.3](#) and write down  $[\mathbf{A}]_{\mathcal{B}}$ .
- (c) Hence determine whether  $\mathbf{A}$  is an orthogonal projection, and find  $\text{rank } \mathbf{A}$  without row reducing.

*Solution.*

- (a)  $\mathbf{A}^2 = \mathbf{A}$ , by multiplying out: the second column of  $\mathbf{A}^2$  is  $\mathbf{A}(1, 1, 0) = (1, 1, 0)$ , and the first and third columns are  $\mathbf{0}$  and  $\mathbf{e}_3$ , unchanged. The columns of  $\mathbf{A}$  span  $\text{im } \mathbf{A} = \text{span}((1, 1, 0), \mathbf{e}_3)$ , which is 2-dimensional. And  $\mathbf{A}\mathbf{x} = (x_2, x_2, x_3)$  is  $\mathbf{0}$  exactly when  $x_2 = x_3 = 0$ , so  $\ker \mathbf{A} = \text{span}(\mathbf{e}_1)$ .
- (b)  $\text{im } \mathbf{A}$  has the orthonormal basis  $\mathbf{f}_1 = \frac{1}{\sqrt{2}}(1, 1, 0)$ ,  $\mathbf{f}_2 = \mathbf{e}_3$ , and  $(\text{im } \mathbf{A})^\perp = \text{span}((1, -1, 0))$  has the orthonormal basis  $\mathbf{f}_3 = \frac{1}{\sqrt{2}}(1, -1, 0)$ . Now  $\mathbf{A}\mathbf{f}_1 = \mathbf{f}_1$  and  $\mathbf{A}\mathbf{f}_2 = \mathbf{f}_2$ , while

$$\mathbf{A}\mathbf{f}_3 = \frac{1}{\sqrt{2}}(-1, -1, 0) = -\mathbf{f}_1.$$

Hence

$$[\mathbf{A}]_{\mathcal{B}} = \begin{pmatrix} 1 & 0 & -1 \\ 0 & 1 & 0 \\ 0 & 0 & 0 \end{pmatrix}, \quad \mathbf{X} = \begin{pmatrix} -1 \\ 0 \end{pmatrix}.$$

- (c)  $\mathbf{X} \neq \mathbf{0}$ , so  $\mathbf{A}$  is oblique by [Theorem 12.12.3](#); indeed  $\ker \mathbf{A} = \text{span}(\mathbf{e}_1)$  is not  $(\text{im } \mathbf{A})^\perp = \text{span}((1, -1, 0))$ . The rank is the number of ones in the canonical form, namely 2; equivalently  $\text{rank } \mathbf{A} = \text{tr } \mathbf{A} = 0 + 1 + 1 = 2$  by [Theorem 3.10.4](#).

**Exercise B3: The complementary projection.**

Let  $P \in \mathcal{L}(V)$  be a projection on a finite-dimensional inner product space, with  $U = \text{im } P$ .

- (a) Prove that  $P$  is an orthogonal projection if and only if  $\text{id}_V - P$  is.
- (b) Prove that in that case  $\text{id}_V - P = P_{U^\perp}$ .

*Hint: Exercise (B2: The complementary projection) identifies the image and the kernel of  $\text{id}_V - P$ .*

*Solution.*

- (a) By Exercise (B2: The complementary projection) (a),  $\text{id}_V - P$  is a projection. By Theorem 10.6.2 (a), (b) and (d),  $(\text{id}_V - P)^* = \text{id}_V - P^*$ . Hence  $(\text{id}_V - P)^* = \text{id}_V - P$  if and only if  $P^* = P$ , and Theorem 12.12.1 converts each side into the corresponding statement about orthogonal projections.
- (b) Suppose  $P$  is an orthogonal projection. By Exercise (B2: The complementary projection) (b),  $\text{im}(\text{id}_V - P) = \ker P$ , which is  $U^\perp$  by Theorem 12.12.1 (b). By (a),  $\text{id}_V - P$  is an orthogonal projection, so by clause (a) of that theorem it is the orthogonal projection onto its own image, that is  $P_{U^\perp}$ .

**12.12.4.3 C. Going deeper****Exercise C1: Products of orthogonal projections.**

Let  $P$  and  $Q$  be orthogonal projections on a finite-dimensional inner product space  $V$ , with images  $U$  and  $W$ .

- (a) Prove that  $PQ$  is an orthogonal projection if and only if  $PQ = QP$ .
- (b) Suppose  $PQ = QP$ . Prove that  $PQ = P_{U \cap W}$ .
- (c) Give an explicit example of two orthogonal projections on  $\mathbb{R}^2$  whose product is not a projection.

*Solution.*

- (a)  $(\Leftarrow)$  Suppose  $PQ = QP$ . Then  $(PQ)^2 = P(QP)Q = P(PQ)Q = P^2Q^2 = PQ$ , so  $PQ$  is a projection, and  $(PQ)^* = Q^*P^* = QP = PQ$  by Theorem 10.6.2 (c) and self-adjointness of  $P$  and  $Q$ . By Theorem 12.12.1,  $PQ$  is an orthogonal projection.  $(\Rightarrow)$  Suppose  $PQ$  is an orthogonal projection. Then  $PQ = (PQ)^* = Q^*P^* = QP$ .
- (b) Assume  $PQ = QP$ , so  $PQ$  is an orthogonal projection by (a), and by clause (a) of Theorem 12.12.1 it equals  $P_{\text{im}(PQ)}$ . It remains to show  $\text{im}(PQ) = U \cap W$ .  $(\subseteq)$  Every value  $PQ\mathbf{v}$  lies in  $\text{im } P = U$ ; and  $PQ\mathbf{v} = QP\mathbf{v}$  lies in  $\text{im } Q = W$ .  $(\supseteq)$  If  $\mathbf{v} \in U \cap W$ , then  $Q\mathbf{v} = \mathbf{v}$  and  $P\mathbf{v} = \mathbf{v}$ , since a projection fixes its image (Theorem 3.10.1 (a)), so  $PQ\mathbf{v} = \mathbf{v}$  and  $\mathbf{v} \in \text{im}(PQ)$ .
- (c) Project onto the  $x$ -axis and onto the line  $y = x$ , with standard matrices

$$\mathbf{P} = \begin{pmatrix} 1 & 0 \\ 0 & 0 \end{pmatrix}, \quad \mathbf{Q} = \frac{1}{2} \begin{pmatrix} 1 & 1 \\ 1 & 1 \end{pmatrix}.$$

Both are symmetric and idempotent, hence orthogonal projections. But  $\mathbf{PQ} = \frac{1}{2} \begin{pmatrix} 1 & 1 \\ 0 & 0 \end{pmatrix}$  and  $(\mathbf{PQ})^2 = \frac{1}{4} \begin{pmatrix} 1 & 1 \\ 0 & 0 \end{pmatrix} \neq \mathbf{PQ}$ , so the product is not even a projection. Consistently with (a),  $\mathbf{QP} = \frac{1}{2} \begin{pmatrix} 1 & 0 \\ 1 & 0 \end{pmatrix} \neq \mathbf{PQ}$ .

### Exercise C2: The Frobenius norm detects obliqueness.

Let  $\mathbf{P} \in M_n(F)$  be idempotent, of rank  $r$ .

- (a) Prove that  $\|\mathbf{P}\|_F^2 \geq r$ .
- (b) Prove that equality holds if and only if  $\mathbf{P}$  is an orthogonal projection.

*Hint: the canonical form of Theorem 12.12.3, together with the invariance of the Frobenius norm under a unitary change of orthonormal basis.*

*Solution.* Apply Theorem 12.12.3 to the operator  $\mathbf{x} \mapsto \mathbf{P}\mathbf{x}$  on  $F^n$ , and let  $\mathbf{U}$  be the matrix whose columns are the vectors of the orthonormal basis it produces. Then  $\mathbf{U}$  is unitary (Theorem 10.7.3 (f)), and the change-of-basis formula gives

$$\mathbf{U}^* \mathbf{P} \mathbf{U} = \mathbf{U}^{-1} \mathbf{P} \mathbf{U} = \begin{pmatrix} \mathbf{I}_r & \mathbf{X} \\ 0 & 0 \end{pmatrix}.$$

By Lemma 11.11.1,  $\|\mathbf{P}\|_F = \|\mathbf{U}^* \mathbf{P} \mathbf{U}\|_F$ .

- (a) Summing the squared moduli of the entries of the block matrix,

$$\|\mathbf{P}\|_F^2 = \|\mathbf{I}_r\|_F^2 + \|\mathbf{X}\|_F^2 = r + \|\mathbf{X}\|_F^2 \geq r,$$

since  $\|\mathbf{X}\|_F^2 \geq 0$ .

- (b) Equality holds if and only if  $\|\mathbf{X}\|_F = 0$ , that is  $\mathbf{X} = 0$ , which by Theorem 12.12.3 is exactly the condition that  $\mathbf{P}$  be an orthogonal projection. (Note that  $r = \text{rank } \mathbf{P} = \text{tr } \mathbf{P}$  by Theorem 3.10.4, so the criterion may also be written  $\text{tr}(\mathbf{P}^* \mathbf{P}) = \text{tr } \mathbf{P}$ .)

### Exercise C3: How far an oblique projection can stretch.

Let  $P$  be a projection of rank  $r$  on an  $n$ -dimensional inner product space  $V$  over  $F$ , with  $0 < r < n$ , and let  $\mathbf{X}$  be the block of Theorem 12.12.3, with largest singular value  $\sigma_1 = \sigma_1(\mathbf{X})$ .

- (a) Prove that  $\|\mathbf{X}\mathbf{z}\| \leq \sigma_1 \|\mathbf{z}\|$  for every  $\mathbf{z} \in F^{n-r}$ , with equality for some  $\mathbf{z} \neq \mathbf{0}$ .
- (b) Prove that the largest value of  $\|P\mathbf{v}\|$  over unit vectors  $\mathbf{v} \in V$  is  $\sqrt{1 + \sigma_1^2}$ .
- (c) Deduce the two-dimensional formula  $\sqrt{1 + |x|^2}$  of the text, and state what the answer becomes when  $P$  is an orthogonal projection.

*Solution.*

- (a) Write  $\mathbf{X} = \mathbf{U}\Sigma\mathbf{V}^*$  as in Theorem 12.8.1, with  $\mathbf{U}, \mathbf{V}$  unitary, and let  $p$  be the length of the list of singular values. For  $\mathbf{z} \in F^{n-r}$  put  $\mathbf{w} = \mathbf{V}^* \mathbf{z}$ , so that  $\|\mathbf{w}\| = \|\mathbf{z}\|$  and  $\|\mathbf{X}\mathbf{z}\| = \|\Sigma\mathbf{w}\|$ , a unitary matrix preserving norms (Theorem 10.7.3). The vector



$\Sigma \mathbf{w}$  has entries  $\sigma_i w_i$  for  $i \leq p$  and 0 beyond, so

$$\|\Sigma \mathbf{w}\|^2 = \sum_{i \leq p} \sigma_i^2 |w_i|^2 \leq \sigma_1^2 \sum_{i \leq p} |w_i|^2 \leq \sigma_1^2 \|\mathbf{w}\|^2 = \sigma_1^2 \|\mathbf{z}\|^2,$$

since  $\sigma_1$  is the largest of the  $\sigma_i$ . Taking  $\mathbf{z} = \mathbf{V}\mathbf{e}_1$ , a unit vector, gives  $\mathbf{w} = \mathbf{e}_1$  and  $\|\mathbf{X}\mathbf{z}\| = \sigma_1$ , so the bound is attained.

- (b) Use the orthonormal basis  $\mathcal{B}$  of [Theorem 12.12.3](#) and write a vector of  $V$  by its coordinate vector  $(\mathbf{y}, \mathbf{z})$  with  $\mathbf{y} \in F^r$  and  $\mathbf{z} \in F^{n-r}$ ; since  $\mathcal{B}$  is orthonormal, [Theorem 10.2.1](#) (c) gives  $\|\mathbf{v}\|^2 = \|\mathbf{y}\|^2 + \|\mathbf{z}\|^2$ , and  $P\mathbf{v}$  has coordinates  $(\mathbf{y} + \mathbf{X}\mathbf{z}, \mathbf{0})$ . By [Corollary 10.1.6](#), then (a), then [Theorem 10.1.5](#) in  $\mathbb{R}^2$  applied to  $(\|\mathbf{y}\|, \|\mathbf{z}\|)$  and  $(1, \sigma_1)$ ,

$$\begin{aligned} \|P\mathbf{v}\| &= \|\mathbf{y} + \mathbf{X}\mathbf{z}\| \leq \|\mathbf{y}\| + \sigma_1 \|\mathbf{z}\| \\ &\leq \sqrt{1 + \sigma_1^2} \sqrt{\|\mathbf{y}\|^2 + \|\mathbf{z}\|^2} = \sqrt{1 + \sigma_1^2} \|\mathbf{v}\|. \end{aligned}$$

For equality, let  $\mathbf{z}_0$  be the unit vector of (a), so  $\mathbf{X}\mathbf{z}_0 = \sigma_1 \mathbf{u}_0$  with  $\|\mathbf{u}_0\| = 1$  (take  $\mathbf{u}_0 = \mathbf{X}\mathbf{z}_0/\sigma_1$  if  $\sigma_1 > 0$ ; if  $\sigma_1 = 0$  take any unit  $\mathbf{u}_0$ ). Put

$$\mathbf{y} = \frac{\mathbf{u}_0}{\sqrt{1 + \sigma_1^2}}, \quad \mathbf{z} = \frac{\sigma_1 \mathbf{z}_0}{\sqrt{1 + \sigma_1^2}}.$$

Then  $\|\mathbf{y}\|^2 + \|\mathbf{z}\|^2 = 1$  and  $\mathbf{y} + \mathbf{X}\mathbf{z} = (1 + \sigma_1^2)\mathbf{u}_0/\sqrt{1 + \sigma_1^2}$ , whose norm is  $\sqrt{1 + \sigma_1^2}$ .

- (c) When  $n = 2$  and  $r = 1$ ,  $\mathbf{X}$  is the  $1 \times 1$  matrix  $(x)$ , whose only singular value is  $|x|$ ; the formula becomes  $\sqrt{1 + |x|^2}$ . When  $P$  is an orthogonal projection,  $\mathbf{X} = 0$ , so  $\sigma_1 = 0$  and the largest value is 1 — an orthogonal projection never lengthens a vector, which is clause (e) of [Theorem 12.12.1](#) with the sharpest constant.

# Chapter 13

## Bilinear and Quadratic Forms

An inner product does two jobs at once. It **pairs** two vectors and returns a scalar, linearly in each of them; and it **measures**, because  $\langle \mathbf{v}, \mathbf{v} \rangle$  is a positive number whenever  $\mathbf{v} \neq \mathbf{0}$ . Chapters 10 to 12 leaned on the second job constantly: lengths, angles, orthonormal bases, least squares, positive definiteness and the singular value decomposition are all consequences of positivity. This chapter keeps the pairing and throws the measuring away. What is left is a **bilinear form**, a function  $\beta: V \times V \rightarrow F$  linear in each slot, with no sign condition, no symmetry required, and no field restriction. The question of the chapter is what survives.

A surprising amount does. Forms still have matrices, and the matrices still change by a rule when the basis changes — but the rule is  $\mathbf{A} \mapsto \mathbf{P}^\top \mathbf{A} \mathbf{P}$ , not  $\mathbf{A} \mapsto \mathbf{P}^{-1} \mathbf{A} \mathbf{P}$ . The organizing relation is therefore **congruence**, not similarity, and the two are genuinely different: congruent matrices need not share a single eigenvalue. What they do share is a rank, a symmetry type, and — over  $\mathbb{R}$  — a triple of signs, which is Sylvester’s law of inertia and the central classification theorem of the chapter. Beyond  $\mathbb{R}$  the classification depends on the arithmetic of the field, and beyond symmetry there is a second world, the alternating forms, where every non-degenerate example looks like every other of the same size. Along the way we meet vectors on which a form vanishes without being zero, the groups that preserve a form, the geometry that physics runs on, and one algebra built to make a quadratic form into a square.

**The field is a general  $F$  here.** Since Chapter 9 the book has worked over  $\mathbb{R}$  or  $\mathbb{C}$ , because inner products need them; this chapter opens the field back up, and that is the point rather than an afterthought. Chapters 10 to 12 turn out to have been the positive definite special case of one line of this theory. Two consequences of the general setting are worth flagging in advance. First, “positive” has no meaning over a field with no order, so the definiteness arguments of Chapter 12 simply cannot be run; what replaces them is either a square-class computation or nothing at all. Second, when  $1 + 1 = 0$  — characteristic 2, [Definition 0.7.2](#) — the identity  $\beta(\mathbf{u}, \mathbf{v}) = \frac{1}{2}(q(\mathbf{u} + \mathbf{v}) - q(\mathbf{u}) - q(\mathbf{v}))$  is unavailable, and symmetric and alternating forms stop being complementary. Every statement below that needs a characteristic hypothesis carries one, and every statement without one holds over every field.

**What you need.** From Chapter 6, multilinearity and alternation, which this chapter reuses rather than redefines ([Definition 6.2.1](#), [Definition 6.2.2](#), [Definition 6.2.3](#), [Theorem 6.2.1](#), [Theorem 6.2.2](#)), together with the determinant and its multiplicativity ([Definition 6.4.1](#), [Theorem 6.5.4](#), [Theorem 6.4.5](#)). From Chapter 3, the matrix of a map, change of coordinates and similarity ([Definition 3.6.1](#), [Theorem 3.6.1](#), [Definition 3.7.1](#), [Theorem 3.7.1](#), [Theorem 3.7.3](#), [Definition 3.7.2](#)), together with rank, equivalence of matrices and the rank normal form ([Definition 3.3.2](#), [Definition 3.8.1](#), [Theorem 3.8.3](#)). From Chapter 4, the dual space and its bases ([Def-](#)

inition 4.1.1, Definition 4.1.2, Theorem 4.1.3, Corollary 4.1.4, Theorem 4.5.1, Theorem 4.3.2). From Chapter 0, fields and characteristic (Definition 0.7.1, Definition 0.7.2, Theorem 0.7.5), the transpose and symmetry (Definition 0.9.10, Definition 0.9.11, Theorem 0.9.5), and groups (Definition 0.10.1, Definition 0.10.2). From Chapter 2, the  $\mathbf{LDL}^\top$  factorization (Theorem 2.6.8). From Chapters 10 to 12, everything used for the real and complex specializations: the orthogonal and unitary groups (Definition 10.7.2, Definition 10.7.3), the real spectral theorem and principal axes (Theorem 11.5.1, Corollary 11.5.2, Theorem 11.5.5), the orthogonal group in low dimension (Proposition 10.7.4, Theorem 11.6.1), and positivity (Definition 12.1.1, Theorem 12.1.4, Theorem 12.1.5, Proposition 12.1.2, Theorem 12.4.1). Those last results are used **only** where the field is  $\mathbb{R}$  or  $\mathbb{C}$ ; they are never hypotheses of the general theory.

## 13.0.1 Roadmap

- **Bilinear forms and their matrices.** What a form is, the Gram matrix that records it, the bijection between forms and matrices, and the form's companion map  $V \rightarrow V^*$ , whose kernel is the radical and whose invertibility is non-degeneracy.
- **Congruence.** Changing the basis replaces  $\mathbf{A}$  by  $\mathbf{P}^\top \mathbf{A} \mathbf{P}$ . What that relation preserves — rank, symmetry type, non-degeneracy, the square class of the determinant — and what it destroys, starting with the eigenvalues.
- **Three kinds of form.** Symmetric, alternating, skew-symmetric and Hermitian; why the middle two coincide exactly when the characteristic is not 2; polarization, and the quadratic form.
- **Completing the square.** Over any field of characteristic  $\neq 2$ , every symmetric form has an orthogonal basis. The proof is an induction, and the algorithm is symmetric row-and-column elimination.
- **Inertia.** Over  $\mathbb{R}$  the number of positive, negative and zero diagonal entries does not depend on the diagonalization. This classifies real symmetric matrices up to congruence and reconciles congruence with the spectral theorem.
- **Definiteness revisited.** Chapter 12's tests, restated as statements about signatures, and the second-derivative test they decide.
- **Where the form vanishes.** Isotropic vectors, totally isotropic subspaces, hyperbolic planes, and the Witt decomposition of a non-degenerate space.
- **Witt's theorems.** An isometry between subspaces extends to the whole space; one summand of an orthogonal sum may be canceled from both sides.
- **Symplectic forms.** A non-degenerate alternating form lives only in even dimension and has exactly one shape, over every field, with no characteristic hypothesis anywhere.
- **The classical groups.**  $O(n)$ ,  $O(p, q)$ ,  $U(n)$  and the symplectic group, all read off one equation  $\mathbf{P}^\top \mathbf{A} \mathbf{P} = \mathbf{A}$ .
- **Minkowski space.** The signature  $(1, 3)$  form, the light cone, hyperbolic rotations, and the triangle inequality running backwards.
- **A first look at Clifford algebras.** Making a quadratic form into a square, and how  $\mathbb{C}$  and the quaternions come out of it.

## 13.0.2 Named moves

- **Polarize.** A symmetric form and its quadratic function  $q(\mathbf{v}) = \beta(\mathbf{v}, \mathbf{v})$  carry the same information, because  $\beta$  can be recovered from  $q$  by expanding  $q(\mathbf{u} + \mathbf{v})$  and dividing by 2. That division is the whole reason characteristic 2 is excluded whenever it is excluded.
- **Complete the square.** Every diagonalization theorem for symmetric forms in this chap-

ter is this one manipulation, applied by induction on the dimension: isolate a variable that actually appears, absorb its cross terms into a perfect square, and recurse on what is left.

- **Split off a hyperbolic plane.** For indefinite forms this replaces “peel off an eigenvector”. A single non-zero isotropic vector forces a two-dimensional piece of standard shape to split off orthogonally, and the induction runs on the rest.
- **Count signs, not values.** The numbers on a diagonalized form are not invariants; their signs are. Whenever a quantity refuses to be an invariant, ask whether some coarser feature of it is.
- **Congruence, not similarity.** At every theorem, ask which relation is in play. If the matrix is eaten twice by the same vector, the relation is congruence and eigenvalues are not available; if it is eaten once and returns a vector, the relation is similarity and they are.

## 13.1 Bilinear Forms and Their Matrices

Chapter 6 needed a function of several vectors that was linear in each of them separately, and called it a multilinear form (Definition 6.2.1). Chapter 10 needed a function of two vectors that was linear in the first, conjugate-linear in the second, and positive on the diagonal, and called it an inner product. This section takes the common skeleton of the two — two slots, linear in each, nothing else assumed — and studies it on its own. The payoff is immediate: such a function is recorded by a square matrix, exactly as a linear map is, and the whole rest of the chapter is about how that matrix behaves.

Throughout,  $V$  is a vector space over a field  $F$ . From the second subsection on,  $V$  is finite-dimensional with  $\dim V = n \geq 1$ , and  $\mathcal{B} = (\mathbf{v}_1, \dots, \mathbf{v}_n)$  is an ordered basis of  $V$ . **No hypothesis on the characteristic of  $F$  is used anywhere in this section**, and none is used in the next one either.

### 13.1.1 Pairing without measuring

The dot product on  $F^n$  is a machine with two inputs and one scalar output. Over  $\mathbb{R}$  it does two different jobs. It **pairs**: it takes  $\mathbf{x}$  and  $\mathbf{y}$  and returns  $\sum_i x_i y_i$ , linearly in each of them. And it **measures**:  $\mathbf{x} \cdot \mathbf{x}$  is a sum of squares, so it is positive unless  $\mathbf{x} = \mathbf{0}$ , and from that one fact Chapter 10 extracted lengths, angles, orthonormal bases and everything built on them.

Measuring is the fragile half. It needs an order on  $F$ , so it is meaningless over  $\mathbb{C}$  without a conjugation and meaningless over  $\mathbb{F}_5$  altogether. Pairing needs nothing:  $\sum_i x_i y_i$  makes sense over every field, and so do plenty of other two-slot expressions, such as  $x_1 y_2 - x_2 y_1$  or  $x_1 y_1 - x_2 y_2$ . Chapter 6 already met the first of these as the signed area, and met the general notion as the case  $k = 2$  of a multilinear form. We give that case its own name, because from here on it is the object of study rather than a step toward the determinant.

*A bilinear form pairs two vectors of the same space into a scalar, linearly in each slot separately.*

#### Definition 13.1.1 — Bilinear Form

Let  $V$  be a vector space over a field  $F$ . A **bilinear form** on  $V$  is a function  $\beta: V \times V \rightarrow F$  that is linear in each argument when the other is held fixed: **for all  $\mathbf{u}, \mathbf{u}', \mathbf{v}, \mathbf{v}' \in V$  and**

all  $c \in F$ ,

(B1)  $\beta(\mathbf{u} + \mathbf{u}', \mathbf{v}) = \beta(\mathbf{u}, \mathbf{v}) + \beta(\mathbf{u}', \mathbf{v})$  and  $\beta(c\mathbf{u}, \mathbf{v}) = c\beta(\mathbf{u}, \mathbf{v})$ ;

(B2)  $\beta(\mathbf{u}, \mathbf{v} + \mathbf{v}') = \beta(\mathbf{u}, \mathbf{v}) + \beta(\mathbf{u}, \mathbf{v}')$  and  $\beta(\mathbf{u}, c\mathbf{v}) = c\beta(\mathbf{u}, \mathbf{v})$ .

Equivalently,  $\beta$  is a 2-linear form on  $V$  in the sense of [Definition 6.2.1](#).

In words: (B1) says that for each fixed  $\mathbf{v}$ , the function  $\mathbf{u} \mapsto \beta(\mathbf{u}, \mathbf{v})$  is a linear functional on  $V$  ([Definition 4.1.1](#)), and (B2) says the same for  $\mathbf{v} \mapsto \beta(\mathbf{u}, \mathbf{v})$  with  $\mathbf{u}$  fixed. The word **fixed** is the whole content. Nothing is assumed about how the two slots compare:  $\beta(\mathbf{u}, \mathbf{v})$  and  $\beta(\mathbf{v}, \mathbf{u})$  may be unrelated, and  $\beta(\mathbf{v}, \mathbf{v})$  may be any scalar at all, including 0 for a non-zero  $\mathbf{v}$ .

Two consequences come free from linearity in one slot. Taking  $c = 0$  in (B1) gives  $\beta(\mathbf{0}, \mathbf{v}) = 0$  for every  $\mathbf{v}$ , and likewise  $\beta(\mathbf{u}, \mathbf{0}) = 0$ . And by induction on the number of terms,

$$\beta\left(\sum_i a_i \mathbf{u}_i, \sum_j b_j \mathbf{v}_j\right) = \sum_{i,j} a_i b_j \beta(\mathbf{u}_i, \mathbf{v}_j),$$

which is the identity everything in this section rests on: a bilinear form is determined by its values on pairs drawn from a spanning set.

### Example. (Six two-slot functions)

Decide which of the following are bilinear forms on the given space over the given field.

- (a) The dot product on  $F^n$ :  $\beta(\mathbf{x}, \mathbf{y}) = x_1 y_1 + \cdots + x_n y_n$ .
- (b) On  $F^n$ , for a fixed  $\mathbf{A} \in M_n(F)$ :  $\beta_{\mathbf{A}}(\mathbf{x}, \mathbf{y}) = \mathbf{x}^\top \mathbf{A} \mathbf{y}$ .
- (c) On  $F^2$ :  $\beta(\mathbf{x}, \mathbf{y}) = x_1 y_2 - x_2 y_1$ .
- (d) On  $\mathbb{R}[x]_{\leq 2}$ :  $\beta(f, g) = \int_0^1 f(t)g(t) dt$ .
- (e) On any  $V$ : the zero function  $\beta(\mathbf{u}, \mathbf{v}) = 0$ .
- (f) On  $\mathbb{R}^n$ :  $\beta(\mathbf{x}, \mathbf{y}) = \|\mathbf{x}\| \|\mathbf{y}\|$ , the product of the Euclidean lengths.

*Solution.*

- (a) Fix  $\mathbf{y}$ . Then  $\mathbf{x} \mapsto \sum_i x_i y_i$  is a linear functional on  $F^n$ , since it is a fixed linear combination of the coordinate functionals ([Theorem 4.1.1](#)). Fixing  $\mathbf{x}$  gives the same statement in  $\mathbf{y}$ . So (B1) and (B2) hold and the dot product is bilinear.
- (b) With  $\mathbf{A} = \mathbf{I}_n$  this is (a). In general, write out the entries:  $\mathbf{x}^\top \mathbf{A} \mathbf{y} = \sum_{i,j} a_{ij} x_i y_j$ . For fixed  $\mathbf{y}$  this is  $\sum_i (\sum_j a_{ij} y_j) x_i$ , a fixed linear combination of the coordinates of  $\mathbf{x}$ , hence a linear functional; symmetrically in  $\mathbf{y}$ . So  $\beta_{\mathbf{A}}$  is bilinear for **every**  $\mathbf{A}$ , symmetric or not. Part (c) below is the case  $\mathbf{A} = \begin{pmatrix} 0 & 1 \\ -1 & 0 \end{pmatrix}$ .
- (c) This is the signed area of Chapter 6, and it is bilinear by [Example \(Alternating Forms\)](#) (a). It is worth noticing what it does on the diagonal:  $\beta(\mathbf{x}, \mathbf{x}) = x_1 x_2 - x_2 x_1 = 0$  for **every**  $\mathbf{x}$ . A bilinear form may vanish identically on the diagonal without being the zero form.
- (d) For fixed  $g$ , the map  $f \mapsto \int_0^1 f g$  is linear, because the integral of a sum is the sum of the integrals and constants come out. Symmetrically in  $g$ . So  $\beta$  is bilinear. Here  $\beta(f, f) = \int_0^1 f^2 \geq 0$ , so over  $\mathbb{R}$  this form does measure: it is the inner product of Chapter 10.
- (e) Both clauses read  $0 = 0 + 0$  and  $0 = c \cdot 0$ . The zero form is bilinear on every space

over every field. It is the degenerate case, and it matters: it shows that a bilinear form can carry no information whatsoever, and it is the form whose matrix below is the zero matrix.

- (f) Not bilinear. Fix  $\mathbf{y} = \mathbf{e}_1$ . Then  $\beta(\mathbf{e}_1, \mathbf{e}_1) + \beta(-\mathbf{e}_1, \mathbf{e}_1) = 1 + 1 = 2$ , while  $\beta(\mathbf{e}_1 + (-\mathbf{e}_1), \mathbf{e}_1) = \beta(\mathbf{0}, \mathbf{e}_1) = 0$ . Additivity in the first slot fails, so (B1) fails. The homogeneity half of (B1) also fails, since  $\beta(-\mathbf{e}_1, \mathbf{e}_1) = 1 \neq -\beta(\mathbf{e}_1, \mathbf{e}_1)$ .

Item (f) is the non-example by minimal change worth remembering: the Cauchy–Schwarz inequality says that  $|\mathbf{x} \cdot \mathbf{y}| \leq \|\mathbf{x}\| \|\mathbf{y}\|$ , so  $\|\mathbf{x}\| \|\mathbf{y}\|$  sits right next to the dot product numerically, and yet it is not a form at all, because the absolute value hidden in the norm destroys additivity. A second minimal change makes the same point more cheaply:  $\beta(\mathbf{x}, \mathbf{y}) = x_1 y_1 + 1$  fails (B1) because  $\beta(\mathbf{0}, \mathbf{0}) = 1 \neq 0$ .

**Warning. Bilinear is not linear.** A bilinear form is not a linear function of the pair  $(\mathbf{u}, \mathbf{v})$ , and the two properties are not even compatible in general. For the dot product on  $\mathbb{R}^2$ , doubling both arguments multiplies the value by 4:  $\beta(2\mathbf{e}_1, 2\mathbf{e}_1) = 4$ , not  $2\beta(\mathbf{e}_1, \mathbf{e}_1) = 2$ . Sums may only be split **one slot at a time**, with the other slot held frozen. Chapter 6 gave this warning for multilinear forms in general; it is repeated here because it is the mistake readers make most often when a form is written with the familiar-looking symbol  $\mathbf{x}^\top \mathbf{A} \mathbf{y}$ .

**Remark.** The same definition with two different spaces,  $\beta: V \times W \rightarrow F$  linear in each slot, is called a **bilinear pairing** of  $V$  and  $W$ . The basic example is the evaluation pairing  $V^* \times V \rightarrow F$ ,  $(\varphi, \mathbf{v}) \mapsto \varphi(\mathbf{v})$ , which is linear in  $\varphi$  by the definition of the operations on  $V^*$  (Definition 4.1.2) and linear in  $\mathbf{v}$  because  $\varphi$  is. We insist on  $V = W$  in Definition 13.1.1 because the questions of this chapter — is  $\beta(\mathbf{u}, \mathbf{v})$  equal to  $\beta(\mathbf{v}, \mathbf{u})$ , what is  $\beta(\mathbf{v}, \mathbf{v})$ , which vectors are orthogonal to which — only make sense when both slots hold vectors of the same space.

## 13.1.2 The matrix of a form

The displayed expansion above says that a bilinear form on a finite-dimensional space is determined by finitely many numbers: the values  $\beta(\mathbf{v}_i, \mathbf{v}_j)$  on pairs of basis vectors. There are  $n^2$  of them, they are indexed by a row and a column, and that is a matrix. Chapter 3 recorded a linear map by a matrix in exactly the same spirit (Definition 3.6.1); here the recipe is even simpler, because there is no basis of an output space to choose.

*The matrix of a form is its table of values on pairs of basis vectors.*

### Definition 13.1.2 — Matrix of a Bilinear Form

Let  $\beta$  be a bilinear form on a finite-dimensional  $V$  with ordered basis  $\mathcal{B} = (\mathbf{v}_1, \dots, \mathbf{v}_n)$ . The **matrix of  $\beta$  with respect to  $\mathcal{B}$** , also called its **Gram matrix**, is the matrix  $[\beta]_{\mathcal{B}} \in M_n(F)$  with entries

$$([\beta]_{\mathcal{B}})_{ij} := \beta(\mathbf{v}_i, \mathbf{v}_j).$$

In words: the entry in row  $i$  and column  $j$  is the value of  $\beta$  with the  $i$ -th basis vector in the **left** slot and the  $j$ -th in the **right** slot. The order matters, because  $\beta(\mathbf{v}_i, \mathbf{v}_j)$  and  $\beta(\mathbf{v}_j, \mathbf{v}_i)$  need not

agree; the convention chosen here is the one that makes the recovery formula below carry no transpose on the **matrix**.

Chapter 10 also called a matrix of pairings a Gram matrix, and the two conventions differ, so it is worth saying how. [Definition 10.2.3](#) set  $(\mathbf{G})_{ij} = \langle \mathbf{v}_j, \mathbf{v}_i \rangle$ , with the indices crossed, because an inner product is conjugate-linear in its second slot and the crossing is what makes  $\mathbf{G}$  Hermitian and  $\mathbf{x}^* \mathbf{G} \mathbf{x}$  come out as a squared norm. A bilinear form has no conjugation to accommodate, so here the natural order is the uncrossed one, which is what makes the recovery formula below free of transposes. For a real symmetric form the two agree; in general each is the transpose of the other.

Nothing needs checking for well-definedness: once  $\mathcal{B}$  is fixed, each entry is a single value of a function. What does need saying is that no information is lost.

### Theorem 13.1.1 — Forms and Matrices Correspond

Let  $V$  be an  $n$ -dimensional vector space over  $F$  with ordered basis  $\mathcal{B}$ , and let  $\beta$  be a bilinear form on  $V$  with  $\mathbf{A} = [\beta]_{\mathcal{B}}$ .

- (a) For all  $\mathbf{u}, \mathbf{v} \in V$ ,

$$\beta(\mathbf{u}, \mathbf{v}) = [\mathbf{u}]_{\mathcal{B}}^{\top} \mathbf{A} [\mathbf{v}]_{\mathcal{B}}.$$

- (b)  $\mathbf{A}$  is the **only** matrix in  $M_n(F)$  with this property.  
 (c) The assignment  $\beta \mapsto [\beta]_{\mathcal{B}}$  is a bijection from the set of bilinear forms on  $V$  onto  $M_n(F)$ , and it respects sums and scalar multiples:  $[\beta + \gamma]_{\mathcal{B}} = [\beta]_{\mathcal{B}} + [\gamma]_{\mathcal{B}}$  and  $[c\beta]_{\mathcal{B}} = c[\beta]_{\mathcal{B}}$ .

*Idea.* Part (a) is the expansion  $\beta(\sum a_i \mathbf{v}_i, \sum b_j \mathbf{v}_j) = \sum_{i,j} a_i b_j \beta(\mathbf{v}_i, \mathbf{v}_j)$  with the double sum recognized as a matrix product. For (b), feed the property a pair of basis vectors and watch it return a single entry. For (c), (b) gives injectivity, and surjectivity is [Example \(Six two-slot functions\)](#) (b) transported to  $V$  through coordinates.

*Proof.*

- (a) Write  $[\mathbf{u}]_{\mathcal{B}} = (a_1, \dots, a_n)$  and  $[\mathbf{v}]_{\mathcal{B}} = (b_1, \dots, b_n)$ , so that  $\mathbf{u} = \sum_i a_i \mathbf{v}_i$  and  $\mathbf{v} = \sum_j b_j \mathbf{v}_j$ . Expanding one slot at a time, by (B1) and then (B2),

$$\beta(\mathbf{u}, \mathbf{v}) = \sum_{i=1}^n \sum_{j=1}^n a_i b_j \beta(\mathbf{v}_i, \mathbf{v}_j) = \sum_{i,j} a_i a_{ij} b_j,$$

writing  $a_{ij} = \beta(\mathbf{v}_i, \mathbf{v}_j)$  for the entries of  $\mathbf{A}$ . The last sum is the  $1 \times 1$  matrix product  $[\mathbf{u}]_{\mathcal{B}}^{\top} \mathbf{A} [\mathbf{v}]_{\mathcal{B}}$ , by the definition of matrix multiplication ([Definition 0.9.9](#)).

- (b) Suppose  $\mathbf{M} \in M_n(F)$  also satisfies  $\beta(\mathbf{u}, \mathbf{v}) = [\mathbf{u}]_{\mathcal{B}}^{\top} \mathbf{M} [\mathbf{v}]_{\mathcal{B}}$  for all  $\mathbf{u}, \mathbf{v}$ . Take  $\mathbf{u} = \mathbf{v}_i$  and  $\mathbf{v} = \mathbf{v}_j$ . Then  $[\mathbf{v}_i]_{\mathcal{B}} = \mathbf{e}_i$  and  $[\mathbf{v}_j]_{\mathcal{B}} = \mathbf{e}_j$ , so the right-hand side is  $\mathbf{e}_i^{\top} \mathbf{M} \mathbf{e}_j = m_{ij}$ , while the left-hand side is  $\beta(\mathbf{v}_i, \mathbf{v}_j) = a_{ij}$ . Hence  $m_{ij} = a_{ij}$  for all  $i, j$ , that is,  $\mathbf{M} = \mathbf{A}$ .

- (c) *Injective.* If  $[\beta]_{\mathcal{B}} = [\gamma]_{\mathcal{B}}$ , then by (a) the two forms agree at every pair  $(\mathbf{u}, \mathbf{v})$ , so  $\beta = \gamma$ .

*Surjective.* Given  $\mathbf{M} \in M_n(F)$ , define  $\gamma(\mathbf{u}, \mathbf{v}) := [\mathbf{u}]_{\mathcal{B}}^{\top} \mathbf{M} [\mathbf{v}]_{\mathcal{B}}$ . The coordinate map  $\mathbf{v} \mapsto [\mathbf{v}]_{\mathcal{B}}$  is linear ([Corollary 3.5.5](#)), and  $(\mathbf{x}, \mathbf{y}) \mapsto \mathbf{x}^{\top} \mathbf{M} \mathbf{y}$  is bilinear on  $F^n$  by [Example](#)



(Six two-slot functions) (b); composing a bilinear form with a linear map in each slot leaves it bilinear, so  $\gamma$  is a bilinear form on  $V$ . By (b) its matrix is  $\mathbf{M}$ .

*Linear.* The forms  $\beta + \gamma$  and  $c\beta$  are defined pointwise, so their values at  $(\mathbf{v}_i, \mathbf{v}_j)$  are  $\beta(\mathbf{v}_i, \mathbf{v}_j) + \gamma(\mathbf{v}_i, \mathbf{v}_j)$  and  $c\beta(\mathbf{v}_i, \mathbf{v}_j)$ . These are the entries of  $[\beta]_{\mathcal{B}} + [\gamma]_{\mathcal{B}}$  and of  $c[\beta]_{\mathcal{B}}$ . This proves the theorem. ■

So bilinear forms on an  $n$ -dimensional space are the same thing as  $n \times n$  matrices, once a basis is fixed — and in particular they form a vector space of dimension  $n^2$ . The qualification “once a basis is fixed” is not decoration; the next section is entirely about what happens when the basis is changed.

### Example. (Four Gram matrices)

Compute  $[\beta]_{\mathcal{B}}$  in each case.

- (a) The dot product on  $F^n$ , with  $\mathcal{B} = \mathcal{E}$  the standard basis.
- (b)  $\beta_{\mathbf{A}}(\mathbf{x}, \mathbf{y}) = \mathbf{x}^{\top} \mathbf{A} \mathbf{y}$  on  $F^n$ , with  $\mathcal{B} = \mathcal{E}$ .
- (c) The signed area  $\beta(\mathbf{x}, \mathbf{y}) = x_1 y_2 - x_2 y_1$  on  $F^2$ , with  $\mathcal{B} = \mathcal{E}$ .
- (d)  $\beta(f, g) = \int_0^1 fg$  on  $\mathbb{R}[x]_{\leq 2}$ , with  $\mathcal{B} = (1, x, x^2)$ .

*Solution.*

- (a)  $\beta(\mathbf{e}_i, \mathbf{e}_j) = \delta_{ij}$ , so  $[\beta]_{\mathcal{E}} = \mathbf{I}_n$ . The dot product is the form whose matrix is the identity.
- (b)  $\beta_{\mathbf{A}}(\mathbf{e}_i, \mathbf{e}_j) = \mathbf{e}_i^{\top} \mathbf{A} \mathbf{e}_j = a_{ij}$ , so  $[\beta_{\mathbf{A}}]_{\mathcal{E}} = \mathbf{A}$ . Together with [Theorem 13.1.1](#) (c) this says that on  $F^n$  with the standard basis, “form” and “matrix” are two words for the same data, and  $\mathbf{x}^{\top} \mathbf{A} \mathbf{y}$  is the general form.
- (c)  $\beta(\mathbf{e}_1, \mathbf{e}_1) = 0$ ,  $\beta(\mathbf{e}_1, \mathbf{e}_2) = 1$ ,  $\beta(\mathbf{e}_2, \mathbf{e}_1) = -1$ ,  $\beta(\mathbf{e}_2, \mathbf{e}_2) = 0$ , so

$$[\beta]_{\mathcal{E}} = \begin{pmatrix} 0 & 1 \\ -1 & 0 \end{pmatrix}.$$

The matrix is not symmetric, and its diagonal is zero — the two features that will separate this form from the dot product throughout the chapter.

- (d) With  $\mathcal{B} = (1, x, x^2)$  the  $(i, j)$  entry is  $\int_0^1 t^{i-1} t^{j-1} dt = 1/(i+j-1)$ , so

$$[\beta]_{\mathcal{B}} = \begin{pmatrix} 1 & \frac{1}{2} & \frac{1}{3} \\ \frac{1}{2} & \frac{1}{3} & \frac{1}{4} \\ \frac{1}{3} & \frac{1}{4} & \frac{1}{5} \end{pmatrix}.$$

This matrix is symmetric, as it must be, since  $\beta(f, g) = \beta(g, f)$ . Its determinant is  $1/2160$ , so it is invertible; what that means for  $\beta$  is the subject of the last part of this section.

*Quick check.* A bilinear form  $\beta$  on  $F^2$  satisfies  $\beta(\mathbf{e}_1, \mathbf{e}_1) = 2$ ,  $\beta(\mathbf{e}_1, \mathbf{e}_2) = -1$ ,  $\beta(\mathbf{e}_2, \mathbf{e}_1) = 3$  and  $\beta(\mathbf{e}_2, \mathbf{e}_2) = 0$ . What is  $\beta((1, 1), (1, -1))$ ?



*Solution.* The matrix is  $\mathbf{A} = \begin{pmatrix} 2 & -1 \\ 3 & 0 \end{pmatrix}$ , and by [Theorem 13.1.1](#) (a),

$$\beta((1, 1), (1, -1)) = \begin{pmatrix} 1 & 1 \end{pmatrix} \begin{pmatrix} 2 & -1 \\ 3 & 0 \end{pmatrix} \begin{pmatrix} 1 \\ -1 \end{pmatrix} = 6.$$

Expanding by hand gives the same. The pair is  $(\mathbf{e}_1 + \mathbf{e}_2, \mathbf{e}_1 - \mathbf{e}_2)$ , so the four terms are

$$\beta(\mathbf{e}_1, \mathbf{e}_1) - \beta(\mathbf{e}_1, \mathbf{e}_2) + \beta(\mathbf{e}_2, \mathbf{e}_1) - \beta(\mathbf{e}_2, \mathbf{e}_2),$$

which is  $2 - (-1) + 3 - 0 = 6$ . Note that the two off-diagonal values enter with **opposite** roles:  $\beta(\mathbf{e}_1, \mathbf{e}_2)$  comes from the minus sign in the right slot,  $\beta(\mathbf{e}_2, \mathbf{e}_1)$  from the plus sign in the left one. Since  $\beta(\mathbf{e}_1, \mathbf{e}_2) \neq \beta(\mathbf{e}_2, \mathbf{e}_1)$  here, swapping them would change the answer.

Now the warning that the rest of the chapter exists to manage.

**Warning.** The matrix of a form and the matrix of a map are different objects with the same name. Both are square arrays attached to  $V$  and a basis, both are written  $\mathbf{A}$ , and both change when the basis changes — by **different** rules. If  $\mathbf{P}$  is the change-of-coordinates matrix between the old and the new basis, then

$$\text{a map: } \mathbf{A} \mapsto \mathbf{P}^{-1}\mathbf{A}\mathbf{P}, \quad \text{a form: } \mathbf{A} \mapsto \mathbf{P}^T\mathbf{A}\mathbf{P}.$$

The first is [Theorem 3.7.3](#); the second is proved in the next section. They agree only when  $\mathbf{P}^T = \mathbf{P}^{-1}$ . Concretely, take  $\mathbf{A} = \text{diag}(1, 2)$  and  $\mathbf{P} = \begin{pmatrix} 1 & 1 \\ 0 & 1 \end{pmatrix}$ . Then

$$\mathbf{P}^{-1}\mathbf{A}\mathbf{P} = \begin{pmatrix} 1 & -1 \\ 0 & 2 \end{pmatrix}, \quad \mathbf{P}^T\mathbf{A}\mathbf{P} = \begin{pmatrix} 1 & 1 \\ 1 & 3 \end{pmatrix}.$$

The first still has eigenvalues 1 and 2; the second has eigenvalues  $2 \pm \sqrt{2}$ . Same  $\mathbf{A}$ , same  $\mathbf{P}$ , two answers with nothing in common. Before writing  $\mathbf{A}$  down, ask what it eats: a map eats one vector and returns a vector, so  $\mathbf{A}\mathbf{x}$  is meaningful; a form eats two vectors and returns a scalar, so  $\mathbf{x}^T\mathbf{A}\mathbf{y}$  is meaningful and  $\mathbf{A}\mathbf{x}$  is a meaningless intermediate. Almost every error in this chapter is this confusion.

### 13.1.3 The form as a map into the dual space

Fixing the right slot of  $\beta$  leaves a function of the left slot, and clause (B1) says that function is a linear functional. So each  $\mathbf{v} \in V$  produces an element of  $V^*$  ([Definition 4.1.2](#)), and clause (B2) says the production is itself linear. That is a linear map  $V \rightarrow V^*$ , and it is the cleanest way to say what it means for a form to be non-degenerate.

Write  $R_\beta: V \rightarrow V^*$  for the map sending  $\mathbf{v}$  to the functional  $\beta(\cdot, \mathbf{v})$ , that is,

$$(R_\beta(\mathbf{v}))(\mathbf{u}) = \beta(\mathbf{u}, \mathbf{v}) \quad (\mathbf{u}, \mathbf{v} \in V).$$

The letter is local to this section and the subscript names the form;  $R$  is for the **right** slot, the one that is fixed. That  $R_\beta(\mathbf{v})$  lies in  $V^*$  is (B1), and that  $R_\beta$  is linear is (B2). Its kernel has a name.

### Definition 13.1.3 — Radical of a Form

Let  $\beta$  be a bilinear form on  $V$ . The **radical** of  $\beta$  is

$$\text{rad}(\beta) := \{ \mathbf{v} \in V : \beta(\mathbf{u}, \mathbf{v}) = 0 \text{ for every } \mathbf{u} \in V \} = \ker R_\beta,$$

a subspace of  $V$ . Its elements are the vectors that **every** vector is orthogonal to.

### Definition 13.1.4 — Non-degenerate Form

A bilinear form  $\beta$  on  $V$  is **non-degenerate** if  $\text{rad}(\beta) = \{\mathbf{0}\}$ , that is, if for every **non-zero**  $\mathbf{v} \in V$  there **exists**  $\mathbf{u} \in V$  with  $\beta(\mathbf{u}, \mathbf{v}) \neq 0$ . Otherwise  $\beta$  is **degenerate**.

In words: non-degeneracy says that no non-zero vector is invisible to the form. It does not say that  $\beta(\mathbf{v}, \mathbf{v}) \neq 0$ , and it does not say that  $\beta$  is positive in any sense; it only says that a vector which the form cannot distinguish from  $\mathbf{0}$  must be  $\mathbf{0}$ . On a finite-dimensional space it is a rank condition.

### Proposition 13.1.2 — Non-degeneracy is invertibility of the Gram matrix

Let  $\beta$  be a bilinear form on an  $n$ -dimensional  $V$  with ordered basis  $\mathcal{B}$ , and put  $\mathbf{A} = [\beta]_{\mathcal{B}}$ . Then:

- (a)  $\mathbf{A}$  is the matrix of  $R_\beta$  from the basis  $\mathcal{B}$  to the dual basis  $\mathcal{B}^*$ ; that is,  $\mathbf{A} = [R_\beta]_{\mathcal{B}}^{\mathcal{B}^*}$ .
- (b)  $\dim \text{rad}(\beta) = n - \text{rank } \mathbf{A}$ .
- (c) The following are equivalent:  $\beta$  is non-degenerate;  $\mathbf{A}$  is invertible;  $R_\beta$  is an isomorphism  $V \rightarrow V^*$ .

*Idea.* Only (a) has content, and it is one line once the right formula is recalled: the coordinates of a functional in the dual basis are its values on  $\mathcal{B}$  (Theorem 4.1.3 (c)), and the values of  $R_\beta(\mathbf{v}_j)$  on  $\mathcal{B}$  are the entries of the  $j$ -th column of  $\mathbf{A}$ . Parts (b) and (c) are then Rank–Nullity and the invertible matrix theorem, applied to a map we now know the matrix of.

*Proof.*

- (a) By Definition 3.6.1, the  $j$ -th column of  $[R_\beta]_{\mathcal{B}}^{\mathcal{B}^*}$  is  $[R_\beta(\mathbf{v}_j)]_{\mathcal{B}^*}$ . By Theorem 4.1.3 (c), a functional  $\varphi \in V^*$  has  $\varphi = \sum_i \varphi(\mathbf{v}_i) \varphi_i$  in the dual basis  $\mathcal{B}^* = (\varphi_1, \dots, \varphi_n)$ , so the  $i$ -th coordinate of  $R_\beta(\mathbf{v}_j)$  is  $(R_\beta(\mathbf{v}_j))(\varphi_i) = \beta(\mathbf{v}_i, \mathbf{v}_j) = a_{ij}$ . Hence the  $(i, j)$  entry of  $[R_\beta]_{\mathcal{B}}^{\mathcal{B}^*}$  is  $a_{ij}$ , as claimed.
- (b) By (a) and Theorem 3.6.5,  $\text{rank } R_\beta = \text{rank } \mathbf{A}$ . Since  $\text{rad}(\beta) = \ker R_\beta$ , Rank–Nullity (Theorem 3.3.1) gives  $\dim \text{rad}(\beta) = n - \text{rank } \mathbf{A}$ .
- (c) By (b),  $\text{rad}(\beta) = \{\mathbf{0}\}$  if and only if  $\text{rank } \mathbf{A} = n$ , which by Theorem 2.4.4 is exactly invertibility of  $\mathbf{A}$ . Since  $\dim V^* = n = \dim V$  (Corollary 4.1.4), the map  $R_\beta$  is an isomorphism if and only if it is injective (Corollary 3.3.2 (e)), that is, if and only if its kernel is  $\{\mathbf{0}\}$  (Theorem 3.2.3). This proves the proposition.



*Remark.* Freezing the **left** slot instead gives a second map  $\mathbf{u} \mapsto \beta(\mathbf{u}, \cdot)$ , whose kernel is the **left radical**  $\{\mathbf{u} : \beta(\mathbf{u}, \mathbf{v}) = 0 \text{ for all } \mathbf{v}\}$ . The same computation shows its matrix is  $\mathbf{A}^\top$ , so by [Theorem 2.5.5](#) and [Theorem 3.3.1](#) the two radicals have the same dimension — although, as [Exercise \(B3: Two radicals\)](#) shows, they need not be the same subspace. In particular “non-degenerate” means the same thing read from either side, and for the symmetric forms that occupy most of this chapter the two radicals coincide.

### Example. (Three radicals)

Find  $\text{rad}(\beta)$  and decide degeneracy.

- (a)  $\beta(\mathbf{x}, \mathbf{y}) = x_1 y_1$  on  $F^2$ .
- (b)  $\beta(\mathbf{x}, \mathbf{y}) = x_1 y_1 - x_2 y_2$  on  $\mathbb{R}^2$ .
- (c) The zero form on any  $V \neq \{\mathbf{0}\}$ .

*Solution.*

- (a)  $[\beta]_{\mathcal{E}} = \begin{pmatrix} 1 & 0 \\ 0 & 0 \end{pmatrix}$ , of rank 1, so the radical is one-dimensional by [Proposition 13.1.2](#).
- (b). Explicitly,  $\beta(\mathbf{u}, \mathbf{v}) = u_1 v_1$  vanishes for all  $\mathbf{u}$  exactly when  $v_1 = 0$ , so  $\text{rad}(\beta) = \text{span}(\mathbf{e}_2)$ . Degenerate.
- (b)  $[\beta]_{\mathcal{E}} = \text{diag}(1, -1)$ , invertible, so  $\beta$  is non-degenerate. Yet  $\beta((1, 1), (1, 1)) = 1 - 1 = 0$ : a non-zero vector on which the form vanishes when paired **with itself**. Non-degeneracy forbids a vector orthogonal to everything; it says nothing about a vector orthogonal to itself. This form is the one this chapter returns to twice, in the sections on isotropic vectors and on Minkowski space.
- (c) The matrix is 0, of rank 0, and  $\text{rad}(\beta) = V$ . This is the extreme degenerate case: every vector is invisible.

*Warning.* **Non-degenerate does not mean definite, and the radical is not the set where  $\beta(\mathbf{v}, \mathbf{v}) = 0$ .** Example (b) above is non-degenerate, and the set  $\{\mathbf{v} : \beta(\mathbf{v}, \mathbf{v}) = 0\}$  is the pair of lines  $x_2 = \pm x_1$  — not a subspace, and not the radical, which is  $\{\mathbf{0}\}$ . Example (c) of [Example \(Six two-slot functions\)](#) has  $\beta(\mathbf{v}, \mathbf{v}) = 0$  for **every**  $\mathbf{v}$  and is still non-degenerate, since its matrix  $\begin{pmatrix} 0 & 1 \\ -1 & 0 \end{pmatrix}$  is invertible. An inner-product-trained reader expects “ $\beta(\mathbf{v}, \mathbf{v}) = 0 \Rightarrow \mathbf{v} = \mathbf{0}$ ”, and that expectation is a theorem about positive definite forms over  $\mathbb{R}$ , not about forms.

*Quick check.* Let  $\beta$  be a bilinear form on a finite-dimensional  $V$ , and let  $U = \text{rad}(\beta)$ . Is  $\beta$  non-degenerate on the subspace  $U$ ?

*Solution.* No, unless  $U = \{\mathbf{0}\}$  — and then only for a silly reason. The restriction of  $\beta$  to  $U \times U$  is the **zero** form, since  $\beta(\mathbf{u}, \mathbf{v}) = 0$  already for  $\mathbf{u} \in V$  and  $\mathbf{v} \in U$ . The zero form on a non-zero space is degenerate, so the restriction is non-degenerate exactly when  $U = \{\mathbf{0}\}$ , that is, exactly when  $\beta$  was non-degenerate to begin with. The point of the question is that “non-degenerate” is a property of a form **on a specified space**, and restricting changes the space.

## 13.1.4 Exercises

### 13.1.4.1 A. Check your understanding

#### Exercise A1.

- Define what it means for  $\beta: V \times V \rightarrow F$  to be a bilinear form, stating both clauses in full.
- Define the matrix  $[\beta]_{\mathcal{B}}$  of a bilinear form with respect to an ordered basis  $\mathcal{B}$ , and state how  $\beta(\mathbf{u}, \mathbf{v})$  is recovered from it.
- Determine whether the following is correct, with a reason: every bilinear form on  $F^n$  is  $\beta(\mathbf{x}, \mathbf{y}) = \mathbf{x}^\top \mathbf{A} \mathbf{y}$  for exactly one  $\mathbf{A} \in M_n(F)$ .
- Define the radical of  $\beta$  and state what it means for  $\beta$  to be non-degenerate.
- Determine whether the following is correct, with a reason: if  $\beta(\mathbf{v}, \mathbf{v}) = 0$  for some  $\mathbf{v} \neq \mathbf{0}$ , then  $\beta$  is degenerate.

*Solution.*

- [Definition 13.1.1](#): linearity in the left slot for each fixed right argument, (B1), and linearity in the right slot for each fixed left argument, (B2).
- [Definition 13.1.2](#):  $([\beta]_{\mathcal{B}})_{ij} = \beta(\mathbf{v}_i, \mathbf{v}_j)$ . Recovery is [Theorem 13.1.1](#) (a):  $\beta(\mathbf{u}, \mathbf{v}) = [\mathbf{u}]_{\mathcal{B}}^\top [\beta]_{\mathcal{B}} [\mathbf{v}]_{\mathcal{B}}$ .
- Correct. Existence and uniqueness are [Theorem 13.1.1](#) (c) and (b) with  $\mathcal{B} = \mathcal{E}$ , for which  $[\mathbf{x}]_{\mathcal{E}} = \mathbf{x}$ .
- [Definition 13.1.3](#) and [Definition 13.1.4](#):  $\text{rad}(\beta) = \{\mathbf{v} : \beta(\mathbf{u}, \mathbf{v}) = 0 \text{ for all } \mathbf{u}\}$ , and  $\beta$  is non-degenerate when this is  $\{\mathbf{0}\}$ .
- Incorrect. The two conditions are unrelated. The signed area on  $F^2$  has  $\beta(\mathbf{v}, \mathbf{v}) = 0$  for every  $\mathbf{v}$  and is non-degenerate, since its matrix  $\begin{pmatrix} 0 & 1 \\ -1 & 0 \end{pmatrix}$  is invertible ([Proposition 13.1.2](#)).

### 13.1.4.2 B. Practice

#### Exercise B1: Which are forms?.

Determine which of the following are bilinear forms on the given space. Justify your answer; for those that fail, name the clause that fails and give a witness.

- On  $\mathbb{R}^2$ :  $\beta(\mathbf{x}, \mathbf{y}) = x_1 y_1 + 2x_1 y_2 - 3x_2 y_1$ .
- On  $\mathbb{R}^2$ :  $\beta(\mathbf{x}, \mathbf{y}) = x_1 y_1 + 1$ .
- On  $\mathbb{R}[x]_{\leq 2}$ :  $\beta(f, g) = f(1)g(2) - f(0)g(0)$ .
- On  $\mathbb{R}^2$ :  $\beta(\mathbf{x}, \mathbf{y}) = x_1^2 y_1$ .
- On  $M_2(\mathbb{R})$ :  $\beta(\mathbf{X}, \mathbf{Y}) = \text{tr}(\mathbf{X}^\top \mathbf{Y})$ .

*Solution.*

- Bilinear: it is  $\mathbf{x}^\top \mathbf{A} \mathbf{y}$  with  $\mathbf{A} = \begin{pmatrix} 1 & 2 \\ -3 & 0 \end{pmatrix}$ , so [Example \(Six two-slot functions\)](#) (b) applies.

- (b) Not bilinear. Clause (B1) fails:  $\beta(\mathbf{0}, \mathbf{0}) = 1$ , whereas homogeneity with  $c = 0$  forces  $\beta(\mathbf{0}, \mathbf{y}) = 0$  for any bilinear form.
- (c) Bilinear. Evaluation at a point is linear ([Theorem 0.8.5](#)), so for fixed  $g$  the map  $f \mapsto g(2)f(1) - g(0)f(0)$  is a linear combination of two linear functionals, hence linear; symmetrically in  $g$ .
- (d) Not bilinear. Clause (B1) fails in the additive half: with  $\mathbf{y} = \mathbf{e}_1$ ,  $\beta(\mathbf{e}_1 + \mathbf{e}_1, \mathbf{e}_1) = 4$  while  $\beta(\mathbf{e}_1, \mathbf{e}_1) + \beta(\mathbf{e}_1, \mathbf{e}_1) = 2$ . The homogeneity half fails too, since  $\beta(c\mathbf{x}, \mathbf{y}) = c^2 x_1^2 y_1$ .
- (e) Bilinear. By [Theorem 0.9.5](#) and linearity of the trace ([Theorem 0.9.6](#)), for fixed  $\mathbf{Y}$  the map  $\mathbf{X} \mapsto \text{tr}(\mathbf{X}^\top \mathbf{Y})$  is linear, and for fixed  $\mathbf{X}$  the map  $\mathbf{Y} \mapsto \text{tr}(\mathbf{X}^\top \mathbf{Y})$  is linear. This is the Frobenius inner product of Chapter 10, seen as a form.

### Exercise B2: A Gram matrix from sample values.

On  $V = \mathbb{R}[x]_{\leq 2}$ , define  $\beta(f, g) = f(0)g(0) + f(1)g(1) + f(2)g(2)$ .

- (a) Verify that  $\beta$  is a bilinear form and compute  $[\beta]_{\mathcal{B}}$  for  $\mathcal{B} = (1, x, x^2)$ .
- (b) Compute  $\beta(1+x, x^2-1)$  directly from the definition, and again from the matrix. Check that the answers agree.
- (c) Hence decide whether  $\beta$  is non-degenerate.

*Solution.*

- (a) Each of the three terms is a product of two evaluations, hence bilinear as in [Exercise \(B1: Which are forms?\)](#) (c), and a sum of bilinear forms is bilinear by [Theorem 13.1.1](#) (c). Numbering rows and columns 1, 2, 3 for  $1, x, x^2$ , the  $(i, j)$  entry is  $\sum_{t \in \{0, 1, 2\}} t^{i-1} t^{j-1}$ , with  $0^0 = 1$ . So

$$[\beta]_{\mathcal{B}} = \begin{pmatrix} 3 & 3 & 5 \\ 3 & 5 & 9 \\ 5 & 9 & 17 \end{pmatrix}.$$

- (b) Directly:  $f = 1+x$  takes the values 1, 2, 3 at 0, 1, 2, and  $g = x^2-1$  takes the values  $-1, 0, 3$ . So  $\beta(f, g) = 1(-1) + 2 \cdot 0 + 3 \cdot 3 = 8$ . From the matrix,  $[f]_{\mathcal{B}} = (1, 1, 0)$  and  $[g]_{\mathcal{B}} = (-1, 0, 1)$ , and

$$[\beta]_{\mathcal{B}}[g]_{\mathcal{B}} = (2, 6, 12), \quad (1, 1, 0) \cdot (2, 6, 12) = 8.$$

- (c) Let  $\mathbf{M}$  be the matrix with rows  $(1, t, t^2)$  for  $t = 0, 1, 2$ . Then  $[\beta]_{\mathcal{B}} = \mathbf{M}^\top \mathbf{M}$ , and  $\det \mathbf{M} = 2$  by the Vandermonde formula ([Theorem 6.8.1](#)), so  $\det[\beta]_{\mathcal{B}} = 4 \neq 0$  by [Theorem 6.5.4](#) and [Theorem 6.4.5](#). By [Proposition 13.1.2](#),  $\beta$  is non-degenerate.

### Exercise B3: Two radicals.

On  $F^3$ , let  $\beta(\mathbf{x}, \mathbf{y}) = \mathbf{x}^\top \mathbf{A} \mathbf{y}$  with

$$\mathbf{A} = \begin{pmatrix} 1 & 2 & 3 \\ 2 & 4 & 6 \\ 1 & 2 & 3 \end{pmatrix}.$$

Find  $\text{rad}(\beta)$  and the left radical, give a basis of each, and decide whether they are equal.

*Solution.* Every row of  $\mathbf{A}$  is a multiple of  $(1, 2, 3)$ , so  $\text{rank } \mathbf{A} = 1$  and both radicals have dimension  $3 - 1 = 2$ , by Proposition 13.1.2 (b) and the remark following it.

The radical is  $\text{nul}(\mathbf{A}) = \{\mathbf{y} : y_1 + 2y_2 + 3y_3 = 0\}$ , with basis  $(-2, 1, 0)$  and  $(-3, 0, 1)$ .

The left radical is  $\text{nul}(\mathbf{A}^\top)$ . Every row of  $\mathbf{A}^\top$  is a multiple of  $(1, 2, 1)$ , so it is  $\{\mathbf{x} : x_1 + 2x_2 + x_3 = 0\}$ , with basis  $(-2, 1, 0)$  and  $(-1, 0, 1)$ .

They are **not** equal:  $(-3, 0, 1)$  lies in the radical but  $-3 + 0 + 1 = -2 \neq 0$ , so it is not in the left radical. Equal dimensions, different subspaces. In particular  $\beta$  is degenerate.

### 13.1.4.3 C. Going deeper

#### Exercise C1: Over $\mathbb{F}_2$ .

Let  $\beta(\mathbf{x}, \mathbf{y}) = x_1y_1 + x_2y_2$  on  $V = \mathbb{F}_2^2$ .

- (a) Write down  $[\beta]_{\mathcal{E}}$  and prove that  $\beta$  is non-degenerate.
- (b) Find every  $\mathbf{v} \in V$  with  $\beta(\mathbf{v}, \mathbf{v}) = 0$ .
- (c) Hence explain why the argument “ $\beta(\mathbf{v}, \mathbf{v}) = 0$  only for  $\mathbf{v} = \mathbf{0}$ ” cannot be used to prove non-degeneracy over a general field, even for the dot product.

*Solution.*

- (a)  $[\beta]_{\mathcal{E}} = \mathbf{I}_2$ , which is invertible, so  $\beta$  is non-degenerate by Proposition 13.1.2.
- (b)  $\beta(\mathbf{v}, \mathbf{v}) = v_1^2 + v_2^2$ . In  $\mathbb{F}_2$  every element satisfies  $t^2 = t$ , so this is  $v_1 + v_2$ , which vanishes for  $\mathbf{v} = (0, 0)$  and  $\mathbf{v} = (1, 1)$ . So the non-zero vector  $(1, 1)$  satisfies  $\beta(\mathbf{v}, \mathbf{v}) = 0$ .
- (c) Over  $\mathbb{R}$ ,  $\beta(\mathbf{v}, \mathbf{v}) = v_1^2 + v_2^2$  is a sum of squares and vanishes only at  $\mathbf{0}$ , so the implication holds there; the proof uses that a sum of non-negative reals is zero only when each term is. Over  $\mathbb{F}_2$  there is no order, nothing is non-negative, and the conclusion is false for  $(1, 1)$  even though the form is the dot product and is non-degenerate. Non-degeneracy is a statement about the rank of a matrix and survives the change of field; the diagonal statement is a statement about positivity and does not.

#### Exercise C2: Orthogonal complements for a form.

Let  $\beta$  be a bilinear form on a vector space  $V$  with  $\dim V = n$ , and for a subspace  $U \subseteq V$  put

$$U^{\perp_\beta} := \{\mathbf{v} \in V : \beta(\mathbf{u}, \mathbf{v}) = 0 \text{ for every } \mathbf{u} \in U\}.$$

- (a) Prove that  $U^{\perp_\beta}$  is a subspace and that  $\dim U^{\perp_\beta} \geq n - \dim U$ .
- (b) Prove that equality holds when  $\beta$  is non-degenerate.
- (c) Give an example with  $\dim U = 1$  in which the inequality in (a) is strict.

*Hint:* consider the map  $V \rightarrow U^*$  that sends  $\mathbf{v}$  to the restriction of  $\beta(\cdot, \mathbf{v})$  to  $U$ .

*Solution.*

- (a) Let  $\rho: V^* \rightarrow U^*$  be the restriction map and  $S = \rho R_\beta: V \rightarrow U^*$ , so  $S(\mathbf{v}) = \beta(\cdot, \mathbf{v})|_U$ . Both factors are linear, so  $S$  is linear, and  $U^{\perp_\beta} = \ker S$  by the definition of  $S$ ; hence  $U^{\perp_\beta}$  is a subspace. By Rank–Nullity (Theorem 3.3.1),

$$\dim U^{\perp_\beta} = n - \text{rank } S \geq n - \dim U^* = n - \dim U,$$

using  $\text{im } S \subseteq U^*$  and  $\dim U^* = \dim U$  (Corollary 4.1.4).

- (b) If  $\beta$  is non-degenerate then  $R_\beta$  is an isomorphism (Proposition 13.1.2 (c)), and  $\rho$  is surjective (Theorem 4.5.1). So  $S = \rho R_\beta$  is surjective,  $\text{rank } S = \dim U$ , and the inequality in (a) is an equality.
- (c) Take  $\beta(\mathbf{x}, \mathbf{y}) = x_1 y_1$  on  $F^2$  and  $U = \text{span}(\mathbf{e}_2)$ . Then  $\beta(\mathbf{e}_2, \mathbf{v}) = 0$  for every  $\mathbf{v}$ , so  $U^{\perp_\beta} = F^2$  has dimension 2, while  $n - \dim U = 1$ . The form is degenerate, as (b) requires it to be.

### Exercise C3: The evaluation pairing.

Let  $V$  be finite-dimensional and let  $W = V^* \times V$ . Define  $\beta$  on  $W$  by

$$\beta((\varphi, \mathbf{v}), (\psi, \mathbf{w})) := \psi(\mathbf{v}) + \varphi(\mathbf{w}).$$

- (a) Prove that  $\beta$  is a bilinear form on  $W$ .
- (b) Prove that  $\beta$  is non-degenerate. *Hint: the two summands of an element of the radical can be killed separately.*
- (c) Deduce that if  $\mathbf{v} \in V$  satisfies  $\varphi(\mathbf{v}) = 0$  for every  $\varphi \in V^*$ , then  $\mathbf{v} = \mathbf{0}$ , and name the result of Chapter 4 this recovers.

*Solution.*

- (a) Fix  $(\psi, \mathbf{w})$ . The map  $(\varphi, \mathbf{v}) \mapsto \psi(\mathbf{v}) + \varphi(\mathbf{w})$  is the sum of  $(\varphi, \mathbf{v}) \mapsto \psi(\mathbf{v})$ , which is linear because  $\psi$  is and because projection onto the second summand is linear, and  $(\varphi, \mathbf{v}) \mapsto \varphi(\mathbf{w}) = \text{ev}_{\mathbf{w}}(\varphi)$ , which is linear because  $\text{ev}_{\mathbf{w}} \in V^{**}$  (Theorem 4.3.1). So clause (B1) holds. Clause (B2) follows by the same argument with the roles of the two arguments exchanged, since the defining expression is symmetric in them.
- (b) Suppose  $(\psi, \mathbf{w}) \in \text{rad}(\beta)$ , so  $\psi(\mathbf{v}) + \varphi(\mathbf{w}) = 0$  for all  $\varphi \in V^*$  and all  $\mathbf{v} \in V$ . Taking  $\varphi = 0$  gives  $\psi(\mathbf{v}) = 0$  for every  $\mathbf{v}$ , so  $\psi = 0$ . Taking  $\mathbf{v} = \mathbf{0}$  then gives  $\varphi(\mathbf{w}) = 0$  for every  $\varphi \in V^*$ , so  $\text{ev}_{\mathbf{w}} = 0$  in  $V^{**}$ , and  $\mathbf{w} = \mathbf{0}$  because  $\text{ev}_V$  is injective (Theorem 4.3.1). Hence the radical is  $\{\mathbf{0}\}$ .
- (c) The second half of (b) is exactly that statement:  $\varphi(\mathbf{v}) = 0$  for all  $\varphi \in V^*$  forces  $\mathbf{v} = \mathbf{0}$ . It recovers the injectivity of the evaluation map  $\text{ev}_V: V \rightarrow V^{**}$  of Theorem 4.3.1, which is the first half of Theorem 4.3.2. So the non-degeneracy of this form is the statement that functionals separate points (Theorem 4.1.5).

## 13.2 Congruence

The matrix of a form in [Definition 13.1.2](#) depends on a basis, and a basis is a choice. As with linear maps in Chapter 3, the useful question is therefore not “what is the matrix?” but “which matrices arise from the same form?”. The answer here is  $\mathbf{P}^\top \mathbf{A} \mathbf{P}$  rather than  $\mathbf{P}^{-1} \mathbf{A} \mathbf{P}$ , and one symbol’s difference changes the whole theory: the new relation keeps the rank and the symmetry type of a matrix and throws its eigenvalues away. This section proves the change-of-basis rule, names the relation, sorts the invariants from the casualties, and settles the debt Chapters 11 and 12 ran up when they kept postponing all of this.

Throughout,  $V$  is a vector space over an arbitrary field  $F$  with  $\dim V = n \geq 1$ , and  $\beta$  is a bilinear form on  $V$ . **No hypothesis on the characteristic of  $F$  is needed in this section.**

### 13.2.1 The change-of-basis rule for a form

Two bases  $\mathcal{B}$  and  $\mathcal{B}'$  of  $V$  give two matrices for one form. The link between them is the change-of-coordinates matrix of [Definition 3.7.1](#), and the computation is three lines, because [Theorem 13.1.1](#) (a) already expresses the form through coordinates.

#### Theorem 13.2.1 — Change of Basis for a Form

Let  $\beta$  be a bilinear form on a finite-dimensional  $V$ , let  $\mathcal{B}$  and  $\mathcal{B}'$  be ordered bases of  $V$ , and let  $\mathbf{P} = [\text{id}]_{\mathcal{B}'}^{\mathcal{B}}$  be the change-of-coordinates matrix from  $\mathcal{B}'$  to  $\mathcal{B}$ . Then

$$[\beta]_{\mathcal{B}'} = \mathbf{P}^\top [\beta]_{\mathcal{B}} \mathbf{P}.$$

*Idea.* There is nothing to discover: write the form in  $\mathcal{B}$ -coordinates, substitute  $[\mathbf{v}]_{\mathcal{B}} = \mathbf{P}[\mathbf{v}]_{\mathcal{B}'}$  in both slots, and read off the matrix that appears between the two coordinate vectors. The uniqueness clause of [Theorem 13.1.1](#) then says that matrix is the one we want, with no further checking.

*Proof.* Write  $\mathbf{A} = [\beta]_{\mathcal{B}}$ . By [Theorem 3.7.1](#) (a),  $[\mathbf{v}]_{\mathcal{B}} = \mathbf{P}[\mathbf{v}]_{\mathcal{B}'}$  for every  $\mathbf{v} \in V$ . So for all  $\mathbf{u}, \mathbf{v} \in V$ , using [Theorem 13.1.1](#) (a) and then [Theorem 0.9.5](#) (d),

$$\begin{aligned} \beta(\mathbf{u}, \mathbf{v}) &= [\mathbf{u}]_{\mathcal{B}}^\top \mathbf{A} [\mathbf{v}]_{\mathcal{B}} \\ &= (\mathbf{P}[\mathbf{u}]_{\mathcal{B}'})^\top \mathbf{A} (\mathbf{P}[\mathbf{v}]_{\mathcal{B}'}) \\ &= [\mathbf{u}]_{\mathcal{B}'}^\top (\mathbf{P}^\top \mathbf{A} \mathbf{P}) [\mathbf{v}]_{\mathcal{B}'}. \end{aligned}$$

Thus  $\mathbf{P}^\top \mathbf{A} \mathbf{P}$  is a matrix with the property that characterizes  $[\beta]_{\mathcal{B}'}$ , and by the uniqueness in [Theorem 13.1.1](#) (b) it is  $[\beta]_{\mathcal{B}'}$ . This proves the theorem. ■

Notice that  $\mathbf{P}$  is the same matrix as in [Theorem 3.7.3](#): the change-of-coordinates matrix from the **new** basis to the **old** one, whose columns are the new basis vectors written in old coordinates. Only the way it acts has changed, from  $\mathbf{P}^{-1} \mathbf{A} \mathbf{P}$  to  $\mathbf{P}^\top \mathbf{A} \mathbf{P}$ .



**Example. (Killing the cross term)**

On  $\mathbb{R}^2$ , let  $\beta(\mathbf{x}, \mathbf{y}) = x_1y_1 + 2x_1y_2 + 2x_2y_1 + x_2y_2$ , so that

$$\mathbf{A} = [\beta]_{\mathcal{E}} = \begin{pmatrix} 1 & 2 \\ 2 & 1 \end{pmatrix}.$$

Compute  $[\beta]_{\mathcal{B}'}$  for  $\mathcal{B}' = ((1, 0), (-2, 1))$ , both by [Theorem 13.2.1](#) and directly from [Definition 13.1.2](#).

*Solution.* The columns of  $\mathbf{P} = [\text{id}]_{\mathcal{B}'}$  are the vectors of  $\mathcal{B}'$  in standard coordinates, so

$$\mathbf{P} = \begin{pmatrix} 1 & -2 \\ 0 & 1 \end{pmatrix}, \quad \mathbf{A}\mathbf{P} = \begin{pmatrix} 1 & 0 \\ 2 & -3 \end{pmatrix},$$

and therefore

$$\mathbf{P}^{\top} \mathbf{A} \mathbf{P} = \begin{pmatrix} 1 & 0 \\ -2 & 1 \end{pmatrix} \begin{pmatrix} 1 & 0 \\ 2 & -3 \end{pmatrix} = \begin{pmatrix} 1 & 0 \\ 0 & -3 \end{pmatrix}.$$

Directly: with  $\mathbf{v}'_1 = (1, 0)$  and  $\mathbf{v}'_2 = (-2, 1)$ , the four values are  $\beta(\mathbf{v}'_1, \mathbf{v}'_1) = 1$ ,  $\beta(\mathbf{v}'_1, \mathbf{v}'_2) = \beta(\mathbf{v}'_2, \mathbf{v}'_1) = 0$  and  $\beta(\mathbf{v}'_2, \mathbf{v}'_2) = -3$ . The two computations agree.

The new basis was not guessed. In coordinates the form reads  $q(\mathbf{x}) = x_1^2 + 4x_1x_2 + x_2^2 = (x_1 + 2x_2)^2 - 3x_2^2$ , a perfect square plus a leftover, and the substitution  $y_1 = x_1 + 2x_2$ ,  $y_2 = x_2$  — that is,  $\mathbf{x} = \mathbf{P}\mathbf{y}$  — is what completing the square asks for. A later section turns this into a theorem and an algorithm.

## 13.2.2 Congruent matrices

Every invertible  $\mathbf{P}$  is a change-of-coordinates matrix between some pair of bases ([Proposition 3.7.2](#)), so [Theorem 13.2.1](#) says exactly this: the matrices of one form, over all bases, are the matrices  $\mathbf{P}^{\top} \mathbf{A} \mathbf{P}$  with  $\mathbf{P}$  invertible. That set deserves a name, and the name is the organizing idea of the chapter.

*Two matrices are congruent when they record the same form in two bases.*

### Definition 13.2.1 — Congruent Matrices

Let  $\mathbf{A}, \mathbf{B} \in M_n(F)$ . We say  $\mathbf{A}$  is **congruent** to  $\mathbf{B}$ , written  $\mathbf{A} \simeq \mathbf{B}$ , if there **exists** an **invertible**  $\mathbf{P} \in M_n(F)$  such that

$$\mathbf{B} = \mathbf{P}^{\top} \mathbf{A} \mathbf{P}.$$

In words: congruence allows one matrix to be sandwiched by an invertible matrix and its transpose. The word **invertible** is essential; without it,  $\mathbf{A} \mapsto \mathbf{P}^{\top} \mathbf{A} \mathbf{P}$  with  $\mathbf{P} = 0$  would relate everything to 0. The **transpose**, not the inverse, is what distinguishes the relation from similarity ([Definition 3.7.2](#)), and the symbol  $\simeq$  is chosen to sit near  $\sim$  without being it.

Three immediate readings. First, by [Theorem 13.2.1](#),  $\mathbf{A} \simeq \mathbf{B}$  if and only if  $\mathbf{A}$  and  $\mathbf{B}$  are the matrices of a single bilinear form in two bases. Second, in the language of quadratic expressions on  $F^n$ ,  $\mathbf{A} \simeq \mathbf{B}$  says that the substitution  $\mathbf{x} = \mathbf{P}\mathbf{y}$  turns  $\mathbf{x}^{\top} \mathbf{A} \mathbf{x}$  into  $\mathbf{y}^{\top} \mathbf{B} \mathbf{y}$ : congruence is

invertible change of variables in a quadratic expression. Third, over  $\mathbb{R}$  it is the relation  $\mathbf{A} \mapsto \mathbf{S}^* \mathbf{A} \mathbf{S}$  of Chapter 12, because there the star is the transpose.

**Example. (Congruences and a non-congruence)**

- (a)  $\mathbf{I}_2 \simeq \begin{pmatrix} 1 & 1 \\ 1 & 2 \end{pmatrix}$  over any field.
- (b)  $\mathbf{I}_2 \simeq \text{diag}(4, 1)$  over  $\mathbb{Q}$ .
- (c)  $\mathbf{A} \simeq \mathbf{A}$  for every  $\mathbf{A}$ .
- (d)  $\mathbf{I}_2 \not\simeq \begin{pmatrix} 1 & 1 \\ 0 & 1 \end{pmatrix}$  over any field.

*Solution.*

- (a) Take  $\mathbf{P} = \begin{pmatrix} 1 & 1 \\ 0 & 1 \end{pmatrix}$ , which is invertible with determinant 1. Then  $\mathbf{P}^\top \mathbf{I}_2 \mathbf{P} = \mathbf{P}^\top \mathbf{P} = \begin{pmatrix} 1 & 1 \\ 1 & 2 \end{pmatrix}$ .
- (b) Take  $\mathbf{P} = \text{diag}(2, 1)$ , invertible over  $\mathbb{Q}$ . Then  $\mathbf{P}^\top \mathbf{I}_2 \mathbf{P} = \text{diag}(4, 1)$ . Scaling a basis vector by  $c$  scales the corresponding diagonal entry by  $c^2$ , which is the single most useful congruence to have in mind.
- (c) Take  $\mathbf{P} = \mathbf{I}_n$ .
- (d) Suppose  $\begin{pmatrix} 1 & 1 \\ 0 & 1 \end{pmatrix} = \mathbf{P}^\top \mathbf{I}_2 \mathbf{P} = \mathbf{P}^\top \mathbf{P}$  for some  $\mathbf{P}$ . Then, by [Theorem 0.9.5](#),  $(\mathbf{P}^\top \mathbf{P})^\top = \mathbf{P}^\top \mathbf{P}$ , so the matrix would be symmetric; it is not. This is the non-example by minimal change: one entry of  $\begin{pmatrix} 1 & 1 \\ 1 & 2 \end{pmatrix}$  in (a) is altered, and congruence to  $\mathbf{I}_2$  is lost — not for a subtle reason, but because the altered matrix is not the matrix of a symmetric form in any basis. The general statement is [Proposition 13.2.4](#) below.

Before using the word “congruent” in both directions, we owe the usual check.

**Proposition 13.2.2 — Congruence is an equivalence relation**

Congruence is an equivalence relation on  $M_n(F)$  ([Definition 0.5.2](#)).

*Proof. Reflexive:*  $\mathbf{A} = \mathbf{I}_n^\top \mathbf{A} \mathbf{I}_n$ .

*Symmetric:* suppose  $\mathbf{B} = \mathbf{P}^\top \mathbf{A} \mathbf{P}$  with  $\mathbf{P}$  invertible, and put  $\mathbf{Q} = \mathbf{P}^{-1}$ , which is invertible. By [Theorem 0.9.5](#) (d),  $\mathbf{Q}^\top \mathbf{P}^\top = (\mathbf{P} \mathbf{Q})^\top = \mathbf{I}_n^\top = \mathbf{I}_n$ , so

$$\mathbf{Q}^\top \mathbf{B} \mathbf{Q} = (\mathbf{Q}^\top \mathbf{P}^\top) \mathbf{A} (\mathbf{P} \mathbf{Q}) = \mathbf{I}_n \mathbf{A} \mathbf{I}_n = \mathbf{A}.$$

*Transitive:* if  $\mathbf{B} = \mathbf{P}^\top \mathbf{A} \mathbf{P}$  and  $\mathbf{C} = \mathbf{Q}^\top \mathbf{B} \mathbf{Q}$  with  $\mathbf{P}, \mathbf{Q}$  invertible, then  $\mathbf{P} \mathbf{Q}$  is invertible and  $\mathbf{C} = \mathbf{Q}^\top \mathbf{P}^\top \mathbf{A} \mathbf{P} \mathbf{Q} = (\mathbf{P} \mathbf{Q})^\top \mathbf{A} (\mathbf{P} \mathbf{Q})$ . This proves the proposition. ■

So the matrices in  $M_n(F)$  fall into congruence classes, and the classification problem of this chapter is to describe them. The rest of the section collects the tools for telling classes apart.

### 13.2.3 What congruence preserves

Three invariants come out of the definition with almost no work. The first is the rank, which we already know is the co-dimension of the radical ([Proposition 13.1.2](#)).

#### Theorem 13.2.3 — Congruence preserves rank

Let  $\mathbf{A}, \mathbf{B} \in M_n(F)$  with  $\mathbf{A} \simeq \mathbf{B}$ . Then  $\text{rank } \mathbf{A} = \text{rank } \mathbf{B}$ . Consequently, whether the associated form is non-degenerate is a property of the congruence class, and it is legitimate to speak of the **rank of a bilinear form**, meaning the rank of its matrix in any basis.

*Proof.* Write  $\mathbf{B} = \mathbf{P}^\top \mathbf{A} \mathbf{P}$  with  $\mathbf{P}$  invertible. Then  $\mathbf{P}^\top$  is invertible, with inverse  $(\mathbf{P}^{-1})^\top$  ([Theorem 0.9.5](#)). Multiplying by invertible matrices on either side does not change the rank ([Theorem 2.5.9](#)), so  $\text{rank } \mathbf{B} = \text{rank}(\mathbf{A} \mathbf{P}) = \text{rank } \mathbf{A}$ . Non-degeneracy is the condition  $\text{rank } \mathbf{A} = n$  by [Proposition 13.1.2](#) (c), so it too depends only on the class. The last sentence follows because, by [Theorem 13.2.1](#), any two matrices of one form are congruent. ■

The second invariant is the symmetry type. Here “alternating” is used in the matrix sense that [Definition 6.2.2](#) suggests: a **zero diagonal** on top of skew-symmetry. The next section separates the three conditions carefully; for now we only need that congruence respects each of them.

#### Proposition 13.2.4 — Congruence preserves the symmetry type

Let  $\mathbf{A}, \mathbf{B} \in M_n(F)$  with  $\mathbf{B} = \mathbf{P}^\top \mathbf{A} \mathbf{P}$  for an invertible  $\mathbf{P}$ .

- (a) If  $\mathbf{A}^\top = \mathbf{A}$ , then  $\mathbf{B}^\top = \mathbf{B}$ .
- (b) If  $\mathbf{A}^\top = -\mathbf{A}$ , then  $\mathbf{B}^\top = -\mathbf{B}$ .
- (c) If  $\mathbf{A}^\top = -\mathbf{A}$  and every diagonal entry of  $\mathbf{A}$  is 0, the same holds for  $\mathbf{B}$ .

Each implication is an equivalence, since congruence is symmetric ([Proposition 13.2.2](#)).

*Proof.* By [Theorem 0.9.5](#),  $\mathbf{B}^\top = (\mathbf{P}^\top \mathbf{A} \mathbf{P})^\top = \mathbf{P}^\top \mathbf{A}^\top \mathbf{P}$ .

- (a) If  $\mathbf{A}^\top = \mathbf{A}$ , this is  $\mathbf{P}^\top \mathbf{A} \mathbf{P} = \mathbf{B}$ .
- (b) If  $\mathbf{A}^\top = -\mathbf{A}$ , it is  $\mathbf{P}^\top (-\mathbf{A}) \mathbf{P} = -\mathbf{B}$ .
- (c) Assume the hypothesis of (c). Then  $\mathbf{B}$  is skew-symmetric by (b), so only the diagonal is left to check. For any  $\mathbf{x} \in F^n$ , grouping the terms of the double sum in pairs,

$$\mathbf{x}^\top \mathbf{A} \mathbf{x} = \sum_{i < j} (a_{ij} + a_{ji}) x_i x_j + \sum_i a_{ii} x_i^2 = 0,$$

since  $a_{ij} + a_{ji} = 0$  by skew-symmetry and  $a_{ii} = 0$  by hypothesis. Writing  $\mathbf{p}_1, \dots, \mathbf{p}_n$  for the columns of  $\mathbf{P}$ , the  $i$ -th diagonal entry of  $\mathbf{B} = \mathbf{P}^\top \mathbf{A} \mathbf{P}$  is  $b_{ii} = \mathbf{e}_i^\top \mathbf{P}^\top \mathbf{A} \mathbf{P} \mathbf{e}_i = \mathbf{p}_i^\top \mathbf{A} \mathbf{p}_i$ , which is 0 by the display. This proves the proposition.

The third invariant is the determinant, but only in a weakened sense: congruence multiplies it by a square. ■

### Theorem 13.2.5 — Congruence changes the determinant by a square

Let  $\mathbf{A}, \mathbf{B} \in M_n(F)$  with  $\mathbf{B} = \mathbf{P}^\top \mathbf{A} \mathbf{P}$  for an invertible  $\mathbf{P}$ . Then

$$\det \mathbf{B} = (\det \mathbf{P})^2 \det \mathbf{A},$$

with  $\det \mathbf{P} \neq 0$ . In particular  $\det \mathbf{A} = 0$  if and only if  $\det \mathbf{B} = 0$ , and when they are non-zero,  $\det \mathbf{B} = c^2 \det \mathbf{A}$  for some  $c \in F \setminus \{0\}$ .

*Proof.* By Theorem 6.5.4 and Theorem 6.4.5,

$$\det \mathbf{B} = \det(\mathbf{P}^\top) \det \mathbf{A} \det \mathbf{P} = (\det \mathbf{P})^2 \det \mathbf{A}.$$

Since  $\mathbf{P}$  is invertible,  $\det \mathbf{P} \neq 0$  (Theorem 6.5.2), so  $(\det \mathbf{P})^2 \neq 0$  and the two determinants vanish together. Taking  $c = \det \mathbf{P}$  gives the last claim. ■

The determinant itself is therefore **not** a congruence invariant — Example (Congruences and a non-congruence) (b) moved it from 1 to 4. What is invariant is the determinant read modulo squares. Write  $F^\times = F \setminus \{0\}$ , and for  $a, b \in F^\times$  declare  $a$  and  $b$  equivalent when  $a = c^2 b$  for some  $c \in F^\times$ . This is an equivalence relation on  $F^\times$  (Definition 0.5.2): it is reflexive with  $c = 1$ , symmetric with  $c^{-1}$ , and transitive by multiplying the two scalars. Its classes are called **square classes**, and the set of them is commonly written  $F^\times / (F^\times)^2$ .

### Definition 13.2.2 — Discriminant

Let  $\beta$  be a **non-degenerate** bilinear form on an  $n$ -dimensional  $V$ . The **discriminant** of  $\beta$  is the square class of  $\det[\beta]_{\mathcal{B}}$  in  $F^\times$ , for any ordered basis  $\mathcal{B}$  of  $V$ .

This is well defined by Theorem 13.2.5 and Theorem 13.2.1: another basis changes the determinant by the square  $(\det \mathbf{P})^2$ , hence not its class. The discriminant is coarse but genuinely useful, and how much it sees depends entirely on the field. Over  $\mathbb{C}$  every non-zero number is a square, so there is only one class and the discriminant says nothing. Over  $\mathbb{R}$  there are two,  $\{t > 0\}$  and  $\{t < 0\}$ , so the discriminant records one bit: the sign of the determinant. Over  $\mathbb{Q}$  there are infinitely many, and the discriminant is a strong invariant.

*Quick check.* Over  $\mathbb{R}$ , are  $\text{diag}(1, 1)$  and  $\text{diag}(-1, -1)$  distinguished by their discriminants?

*Solution.* No. Both determinants equal 1, so the discriminants agree. And yet the two matrices are not congruent over  $\mathbb{R}$ . Suppose  $\text{diag}(-1, -1) = \mathbf{P}^\top \mathbf{I}_2 \mathbf{P} = \mathbf{P}^\top \mathbf{P}$  for an

invertible  $\mathbf{P}$ . Evaluating both sides on  $\mathbf{x} = \mathbf{e}_1$  gives

$$-1 = \mathbf{e}_1^\top \text{diag}(-1, -1)\mathbf{e}_1 = \mathbf{e}_1^\top \mathbf{P}^\top \mathbf{P} \mathbf{e}_1 = \|\mathbf{P}\mathbf{e}_1\|^2 \geq 0,$$

a contradiction. So the discriminant is not a complete invariant, even over  $\mathbb{R}$  and even for  $2 \times 2$  matrices. Proving that the full “sign pattern” *is* an invariant, and that it is the complete one over  $\mathbb{R}$ , is the law of inertia, proved later in this chapter.

### 13.2.4 What congruence destroys

Now the negative result of the section, and the reason the previous three chapters could not have been this one.

**Example. (Congruent, with no eigenvalue in common)**

- (a)  $\mathbf{I}_2 \simeq \begin{pmatrix} 1 & 1 \\ 1 & 2 \end{pmatrix}$  by [Example \(Congruences and a non-congruence\)](#) (a). Compare their eigenvalues.
- (b)  $\begin{pmatrix} 1 & 2 \\ 2 & 1 \end{pmatrix} \simeq \text{diag}(1, -3)$  by [Example \(Killing the cross term\)](#). Compare their eigenvalues.

*Solution.*

- (a)  $\mathbf{I}_2$  has the single eigenvalue 1, twice. The characteristic polynomial of  $\begin{pmatrix} 1 & 1 \\ 1 & 2 \end{pmatrix}$  is  $x^2 - 3x + 1$ , whose roots are  $(3 \pm \sqrt{5})/2$ , that is, approximately 2.618 and 0.382. The two matrices share no eigenvalue. They do share a rank, 2; a symmetry type, symmetric; and a discriminant, since  $\det = 1$  for both.
- (b) The characteristic polynomial of  $\begin{pmatrix} 1 & 2 \\ 2 & 1 \end{pmatrix}$  is  $x^2 - 2x - 3 = (x - 3)(x + 1)$ , so its eigenvalues are 3 and  $-1$ . Those of  $\text{diag}(1, -3)$  are 1 and  $-3$ . Again no eigenvalue in common — but this time one positive and one negative on each side, and the determinants are  $-3$  both times. The signs survived; the numbers did not. That is the pattern the law of inertia will explain.

Chapter 12 introduced this operation twice and both times deferred it to here. Section 1 of that chapter wrote that “the operation  $\mathbf{A} \mapsto \mathbf{S}^* \mathbf{A} \mathbf{S}$  is called **congruence**, and Chapter 13 studies it for its own sake”, keeping only that it preserves the sign of a positive form ([Proposition 12.1.2](#)). Section 4 repeated the promise in fuller form: “Chapter 13 studies congruence in its own right, including what it does preserve; for now, note that it certainly does **not** preserve eigenvalues, since  $\mathbf{S}^* \mathbf{I} \mathbf{S} = \mathbf{S}^* \mathbf{S}$  can be any positive definite matrix at all.”

Both debts are now paid, and it is worth saying the answer in one place. **What congruence preserves:** the rank ([Theorem 13.2.3](#)), and with it non-degeneracy and the dimension of the radical; the symmetry type ([Proposition 13.2.4](#)); and the square class of the determinant when that is non-zero ([Theorem 13.2.5](#)), the discriminant. Over  $\mathbb{R}$  it preserves one thing more, the triple of signs, and that is the deepest of the four; it is proved in the section on inertia. **What congruence destroys:** the eigenvalues, and therefore the characteristic polynomial, the trace and the determinant itself. Not every similarity invariant is lost — rank and invertibility, parts (a) and (b) of [Proposition 3.7.6](#), survive both relations — but every invariant that reads the eigenvalues is. Chapter 12’s parenthetical remark is exactly right and can be made sharper: taking  $\mathbf{A} = \mathbf{I}_n$  in [Definition 13.2.1](#), the congruence class of  $\mathbf{I}_n$  over  $\mathbb{R}$  is the set of all matrices  $\mathbf{P}^\top \mathbf{P}$  with  $\mathbf{P}$  invertible, which by [Theorem 12.1.5](#) (c) is the set of **all** positive definite matrices.

One class, every spectrum with all eigenvalues positive.

*Warning.* **Congruent and similar are different relations, and neither implies the other.** By [Example \(Congruent, with no eigenvalue in common\)](#) (a),  $\mathbf{I}_2 \simeq \begin{pmatrix} 1 & 1 \\ 1 & 2 \end{pmatrix}$ , while the only matrix similar to  $\mathbf{I}_2$  is  $\mathbf{I}_2$  itself, since  $\mathbf{P}^{-1}\mathbf{I}_2\mathbf{P} = \mathbf{I}_2$  for every invertible  $\mathbf{P}$ . In the other direction,  $\text{diag}(1, 2)$  is similar to  $\begin{pmatrix} 1 & -1 \\ 0 & 2 \end{pmatrix}$  — conjugate by  $\begin{pmatrix} 1 & 1 \\ 0 & 1 \end{pmatrix}$  — but not congruent to it, because  $\text{diag}(1, 2)$  is symmetric and the other matrix is not ([Proposition 13.2.4](#)). So do not carry any habit across. In particular, “diagonalizable” means two different things in this chapter, and a matrix that is diagonalizable by congruence need not be diagonalizable by similarity, nor conversely.

### 13.2.5 Where the two relations meet

There is exactly one situation in which the confusion is harmless: when the change-of-basis matrix satisfies  $\mathbf{P}^\top = \mathbf{P}^{-1}$ , that is, when  $\mathbf{P}$  is orthogonal ([Definition 10.7.3](#)).

#### Proposition 13.2.6 — An orthogonal change of basis is both

Let  $\mathbf{A} \in M_n(\mathbb{R})$  and  $\mathbf{Q} \in O(n)$ . Then  $\mathbf{Q}^\top \mathbf{A} \mathbf{Q} = \mathbf{Q}^{-1} \mathbf{A} \mathbf{Q}$ , so this matrix is at once congruent and similar to  $\mathbf{A}$ . Consequently, if  $\mathbf{A}$  is symmetric, then  $\mathbf{A}$  is congruent to the diagonal matrix  $\mathbf{D}$  of its eigenvalues, listed with multiplicity in any prescribed order.

*Proof.* The first statement is the definition of an orthogonal matrix,  $\mathbf{Q}^\top \mathbf{Q} = \mathbf{I}_n$ , which gives  $\mathbf{Q}^\top = \mathbf{Q}^{-1}$ ; the matrix  $\mathbf{Q}^\top \mathbf{A} \mathbf{Q}$  is then congruent to  $\mathbf{A}$  by [Definition 13.2.1](#), since  $\mathbf{Q}$  is invertible, and similar to  $\mathbf{A}$  by [Definition 3.7.2](#).

For the second statement, let  $\mathbf{A}$  be symmetric. By [Corollary 11.5.2](#) there are  $\mathbf{Q} \in O(n)$  and a real diagonal  $\mathbf{D}$ , carrying the eigenvalues of  $\mathbf{A}$  in any prescribed order, with  $\mathbf{A} = \mathbf{Q} \mathbf{D} \mathbf{Q}^\top$ . Multiplying on the left by  $\mathbf{Q}^\top$  and on the right by  $\mathbf{Q}$  gives  $\mathbf{D} = \mathbf{Q}^\top \mathbf{A} \mathbf{Q}$ , which is a congruence. This proves the proposition. ■

This is why the real spectral theorem is so strong, and it is what Chapter 11 meant when it said, of the principal axis theorem, that “an orthogonal change of variables is a similarity *and* a congruence, so it keeps the eigenvalues and the geometry at once”. The general theory of this chapter has no orthogonal matrices to call on — a general field has no inner product, hence no orthogonal group in that sense — so the two relations part company and only congruence remains. What [Proposition 13.2.6](#) gives over  $\mathbb{R}$  is a bridge: it says that every real symmetric matrix is congruent to a diagonal matrix, and it identifies the diagonal entries as eigenvalues. The section on inertia will show that the entries are free to move but their signs are not, which is how the two pictures are finally reconciled.

*Quick check.* Over  $\mathbb{R}$ , which matrices are congruent to  $\mathbf{I}_n$ ?

*Solution.* Exactly the positive definite symmetric matrices. If  $\mathbf{B} = \mathbf{P}^\top \mathbf{I}_n \mathbf{P} = \mathbf{P}^\top \mathbf{P}$  with  $\mathbf{P}$  invertible, then  $\mathbf{B}$  is symmetric and  $\mathbf{x}^\top \mathbf{B} \mathbf{x} = \|\mathbf{P} \mathbf{x}\|^2 > 0$  for  $\mathbf{x} \neq \mathbf{0}$ , since  $\mathbf{P} \mathbf{x} \neq \mathbf{0}$ . Conversely every positive definite  $\mathbf{B}$  is  $\mathbf{P}^\top \mathbf{P}$  for an invertible  $\mathbf{P}$  by [Theorem 12.1.5](#). Compare the similarity class of  $\mathbf{I}_n$ , which contains  $\mathbf{I}_n$  alone: one relation collapses this

class to a point, the other spreads it over every positive definite matrix there is.

## 13.2.6 Exercises

### 13.2.6.1 A. Check your understanding

#### Exercise A1.

- State the change-of-basis rule for the matrix of a bilinear form, saying which matrix  $\mathbf{P}$  appears.
- Define what it means for  $\mathbf{A}, \mathbf{B} \in M_n(F)$  to be congruent.
- For each of rank, symmetry, determinant, eigenvalues and the square class of a non-zero determinant, say whether it is a congruence invariant.
- Determine whether the following is correct, with a reason: congruent matrices are similar.
- State the one circumstance in which a change of basis is simultaneously a similarity and a congruence.

*Solution.*

- [Theorem 13.2.1](#):  $[\beta]_{\mathcal{B}'} = \mathbf{P}^\top [\beta]_{\mathcal{B}} \mathbf{P}$ , where  $\mathbf{P} = [\text{id}]_{\mathcal{B}'}^{\mathcal{B}}$  is the change-of-coordinates matrix from the new basis to the old one.
- [Definition 13.2.1](#):  $\mathbf{B} = \mathbf{P}^\top \mathbf{A} \mathbf{P}$  for some invertible  $\mathbf{P} \in M_n(F)$ .
- Invariant: rank ([Theorem 13.2.3](#)); symmetry ([Proposition 13.2.4](#)); the square class of a non-zero determinant ([Theorem 13.2.5](#)). Not invariant: the determinant itself, and the eigenvalues ([Example \(Congruent, with no eigenvalue in common\)](#)).
- Incorrect.  $\mathbf{I}_2 \simeq \begin{pmatrix} 1 & 1 \\ 1 & 2 \end{pmatrix}$ , but the only matrix similar to  $\mathbf{I}_2$  is  $\mathbf{I}_2$ .
- When the change-of-coordinates matrix  $\mathbf{P}$  satisfies  $\mathbf{P}^\top = \mathbf{P}^{-1}$ , that is, when  $\mathbf{P}$  is orthogonal ([Proposition 13.2.6](#)).

### 13.2.6.2 B. Practice

#### Exercise B1: A congruence in $\mathbb{Q}$ .

Let  $\mathbf{A} = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}$  and  $\mathbf{P} = \begin{pmatrix} 1 & 1 \\ 1 & -1 \end{pmatrix}$ , both over  $\mathbb{Q}$ .

- Compute  $\mathbf{P}^\top \mathbf{A} \mathbf{P}$ , and hence write down a matrix congruent to  $\mathbf{A}$  that is diagonal.
- Find the eigenvalues of both matrices and compare them.
- Compute the discriminant of the form  $\mathbf{x}^\top \mathbf{A} \mathbf{y}$  and check that it agrees for the two matrices.

*Solution.*

- $\mathbf{P}$  is invertible, with  $\det \mathbf{P} = -2 \neq 0$ . Then

$$\mathbf{A} \mathbf{P} = \begin{pmatrix} 1 & -1 \\ 1 & 1 \end{pmatrix}, \quad \mathbf{P}^\top \mathbf{A} \mathbf{P} = \begin{pmatrix} 2 & 0 \\ 0 & -2 \end{pmatrix}.$$

So  $\mathbf{A} \simeq \text{diag}(2, -2)$  over  $\mathbb{Q}$ .

- (b) The characteristic polynomial of  $\mathbf{A}$  is  $x^2 - 1$ , so its eigenvalues are 1 and  $-1$ ; those of  $\text{diag}(2, -2)$  are 2 and  $-2$ . No eigenvalue is shared, and each matrix has one positive and one negative eigenvalue.
- (c)  $\det \mathbf{A} = -1$  and  $\det \text{diag}(2, -2) = -4 = (-2)^2 \cdot (-1)$ , which is the relation  $\det \mathbf{B} = (\det \mathbf{P})^2 \det \mathbf{A}$  of [Theorem 13.2.5](#). The two determinants differ by the square 4, so they lie in the same square class of  $\mathbb{Q}^\times$  and the discriminants agree, as [Definition 13.2.2](#) requires.

### Exercise B2: A non-congruence over $\mathbb{Q}$ .

- (a) Prove that  $\text{diag}(1, 1)$  and  $\text{diag}(1, 2)$  are **not** congruent over  $\mathbb{Q}$ .
- (b) Prove that they **are** congruent over  $\mathbb{R}$ , by exhibiting  $\mathbf{P}$ .
- (c) Explain in one sentence why (a) and (b) are consistent.

*Solution.*

- (a) Suppose  $\text{diag}(1, 2) = \mathbf{P}^\top \text{diag}(1, 1) \mathbf{P}$  with  $\mathbf{P} \in M_2(\mathbb{Q})$  invertible. By [Theorem 13.2.5](#),  $2 = (\det \mathbf{P})^2 \cdot 1$ , so  $\det \mathbf{P} \in \mathbb{Q}$  would satisfy  $(\det \mathbf{P})^2 = 2$ . No rational number squares to 2 ([Theorem 0.2.3](#)), a contradiction. Hence the two are not congruent over  $\mathbb{Q}$ .
- (b) Take  $\mathbf{P} = \text{diag}(1, 1/\sqrt{2})$ , which is invertible over  $\mathbb{R}$ . Then  $\mathbf{P}^\top \text{diag}(1, 2) \mathbf{P} = \text{diag}(1, 1)$ , and congruence is symmetric ([Proposition 13.2.2](#)).
- (c) Congruence is a relation **over a specified field**: the matrix  $\mathbf{P}$  is required to have entries in that field, and  $1/\sqrt{2}$  is available in  $\mathbb{R}$  but not in  $\mathbb{Q}$ . Equivalently, 2 is a square in  $\mathbb{R}^\times$  and not in  $\mathbb{Q}^\times$ , so the discriminant separates the two matrices over  $\mathbb{Q}$  and not over  $\mathbb{R}$ .

### Exercise B3: Completing the square, again.

On  $\mathbb{R}^2$ , let  $\beta(\mathbf{x}, \mathbf{y}) = x_1 y_1 - x_1 y_2 - x_2 y_1 + 2x_2 y_2$ .

- (a) Write down  $\mathbf{A} = [\beta]_{\mathcal{E}}$  and expand  $q(\mathbf{x}) = \beta(\mathbf{x}, \mathbf{x})$  as a polynomial in  $x_1, x_2$ .
- (b) Complete the square to find a basis  $\mathcal{B}'$  with  $[\beta]_{\mathcal{B}'} = \mathbf{I}_2$ . Verify by computing  $\mathbf{P}^\top \mathbf{A} \mathbf{P}$ .
- (c) Hence find the eigenvalues of  $\mathbf{A}$  and compare them with those of  $\mathbf{I}_2$ .

*Solution.*

- (a)  $\mathbf{A} = \begin{pmatrix} 1 & -1 \\ -1 & 2 \end{pmatrix}$ , and  $q(\mathbf{x}) = x_1^2 - 2x_1 x_2 + 2x_2^2$ .
- (b)  $q(\mathbf{x}) = (x_1 - x_2)^2 + x_2^2$ . So the substitution  $y_1 = x_1 - x_2$ ,  $y_2 = x_2$ , that is  $x_1 = y_1 + y_2$  and  $x_2 = y_2$ , makes  $q = y_1^2 + y_2^2$ . This is  $\mathbf{x} = \mathbf{P} \mathbf{y}$  with

$$\mathbf{P} = \begin{pmatrix} 1 & 1 \\ 0 & 1 \end{pmatrix}, \quad \mathbf{A} \mathbf{P} = \begin{pmatrix} 1 & 0 \\ -1 & 1 \end{pmatrix}, \quad \mathbf{P}^\top \mathbf{A} \mathbf{P} = \mathbf{I}_2.$$

By [Theorem 13.2.1](#) the basis is  $\mathcal{B}' = ((1, 0), (1, 1))$ , the columns of  $\mathbf{P}$ .



- (c) The characteristic polynomial of  $\mathbf{A}$  is  $x^2 - 3x + 1$ , with roots  $(3 \pm \sqrt{5})/2$ ;  $\mathbf{I}_2$  has the eigenvalue 1 twice. The two matrices are congruent with no eigenvalue in common. What they share is that all eigenvalues are positive on both sides, which by the check above is no accident: the congruence class of  $\mathbf{I}_2$  over  $\mathbb{R}$  consists of exactly the positive definite matrices.

### 13.2.6.3 C. Going deeper

#### Exercise C1: Congruence versus equivalence.

- (a) Prove that if  $\mathbf{A} \simeq \mathbf{B}$  then  $\mathbf{A}$  and  $\mathbf{B}$  are equivalent in the sense of [Definition 3.8.1](#).  
 (b) Give two matrices in  $M_2(\mathbb{Q})$  that are equivalent but not congruent, and justify your example.  
 (c) Hence explain what “the congruence classes refine the equivalence classes” means, and say what extra information a congruence class carries that a rank does not.

*Solution.*

- (a) If  $\mathbf{B} = \mathbf{P}^\top \mathbf{A} \mathbf{P}$  with  $\mathbf{P}$  invertible, then  $\mathbf{P}^\top$  is invertible ([Theorem 0.9.5](#)), so  $\mathbf{B} = \mathbf{Q} \mathbf{A} \mathbf{P}$  with  $\mathbf{Q} = \mathbf{P}^\top$  and  $\mathbf{P}$  both invertible, which is [Definition 3.8.1](#).  
 (b) Take  $\text{diag}(1, 1)$  and  $\text{diag}(1, 2)$ . Both have rank 2, so they are equivalent by [Theorem 3.8.3](#) (c). They are not congruent over  $\mathbb{Q}$  by [Exercise \(B2: A non-congruence over  \$\mathbb{Q}\$ \)](#) (a).  
 (c) By (a), each congruence class lies inside a single equivalence class, and by (b) an equivalence class may split into more than one congruence class; that is what “refine” means. An equivalence class is named by the rank alone ([Theorem 3.8.3](#) (c)). A congruence class remembers the rank, and in addition the symmetry type and the discriminant — and over  $\mathbb{R}$ , the signs. Equivalence allows two independent changes of basis, one in each slot; congruence allows only one, used twice, and the extra invariants are the price of that constraint.

#### Exercise C2: Orthogonal congruence.

Let  $\mathbf{A} \in M_n(\mathbb{R})$  be symmetric with eigenvalues  $\lambda_1, \dots, \lambda_n$ , listed with multiplicity.

- (a) Prove that  $\mathbf{A} \simeq \text{diag}(\lambda_1, \dots, \lambda_n)$ .  
 (b) Deduce that if every  $\lambda_i$  is non-zero, then  $\mathbf{A}$  is congruent to a diagonal matrix all of whose entries are  $\pm 1$ .  
 (c) Give two real symmetric  $2 \times 2$  matrices that are congruent and have no eigenvalue in common.

*Hint for (b): recall from [Example \(Congruences and a non-congruence\)](#) (b) what scaling a basis vector does to a diagonal entry.*

*Solution.*

- (a) This is [Proposition 13.2.6](#): by [Corollary 11.5.2](#) there is  $\mathbf{Q} \in O(n)$  with  $\mathbf{Q}^\top \mathbf{A} \mathbf{Q} = \mathbf{D} = \text{diag}(\lambda_1, \dots, \lambda_n)$ , and  $\mathbf{Q}$  is invertible, so  $\mathbf{A} \simeq \mathbf{D}$ .

- (b) Let  $\mathbf{S} = \text{diag}(c_1, \dots, c_n)$  with  $c_i = 1/\sqrt{|\lambda_i|}$ , which is defined and non-zero because  $\lambda_i \neq 0$ . Then  $\mathbf{S}$  is invertible and  $\mathbf{S}^\top \mathbf{D} \mathbf{S} = \text{diag}(c_1^2 \lambda_1, \dots, c_n^2 \lambda_n)$ , whose  $i$ -th entry is  $\lambda_i/|\lambda_i| = \pm 1$  according to the sign of  $\lambda_i$ . By (a) and transitivity ([Proposition 13.2.2](#)),  $\mathbf{A}$  is congruent to that matrix.
- (c) By (b),  $\text{diag}(1, -3)$  and  $\text{diag}(2, -1)$  are both congruent to  $\text{diag}(1, -1)$ , hence to each other by [Proposition 13.2.2](#); explicitly,  $\mathbf{P} = \text{diag}(\sqrt{2}, 1/\sqrt{3})$  gives  $\mathbf{P}^\top \text{diag}(1, -3) \mathbf{P} = \text{diag}(2, -1)$ . Their eigenvalues are  $\{1, -3\}$  and  $\{2, -1\}$ , which are disjoint. Note that the **signs** match, one positive and one negative on each side; part (b) is the reason, and the law of inertia will say it is forced.

### Exercise C3: How much the discriminant sees.

- (a) Show that  $\mathbb{C}^\times$  has exactly one square class, and that  $\mathbb{R}^\times$  has exactly two.
- (b) List the squares in  $\mathbb{F}_5^\times$  and show that  $\mathbb{F}_5^\times$  has exactly two square classes.
- (c) Hence determine, for each of  $\mathbb{C}$ ,  $\mathbb{R}$  and  $\mathbb{F}_5$ , the largest number of pairwise non-congruent invertible matrices in  $M_n$  that the discriminant alone can distinguish.

*Solution.*

- (a) Over  $\mathbb{C}$ , every  $a \neq 0$  has a square root ([Example \(Square Roots of 5 – 12i\)](#)), so  $a = c^2 \cdot 1$  and every element is equivalent to 1: one class. Over  $\mathbb{R}$ , squares of non-zero reals are exactly the positive reals, so  $a$  and  $b$  are equivalent precisely when  $a/b > 0$ , which happens exactly when they have the same sign. The classes are the positive reals and the negative reals: two classes, and they are distinct since 1 and  $-1$  are not related.
- (b) The squares are  $1^2 = 1$ ,  $2^2 = 4$ ,  $3^2 = 4$ ,  $4^2 = 1$ , so the set of non-zero squares is  $\{1, 4\}$ . Multiplying  $\{1, 4\}$  by 2 gives  $\{2, 3\}$ , so the classes are  $\{1, 4\}$  and  $\{2, 3\}$ : two classes, each of size 2, covering all four elements of  $\mathbb{F}_5^\times$ .
- (c) The discriminant takes one value per square class, so it can separate at most as many classes as there are square classes: one over  $\mathbb{C}$ , two over  $\mathbb{R}$  and two over  $\mathbb{F}_5$ . Over  $\mathbb{C}$  it is therefore useless, and over  $\mathbb{R}$  and  $\mathbb{F}_5$  it sorts invertible matrices into at most two piles. By [Exercise \(C2: Orthogonal congruence\)](#) (b), over  $\mathbb{R}$  every invertible symmetric matrix is congruent to one of the  $n+1$  matrices  $\mathbf{I}_p \oplus -\mathbf{I}_{n-p}$ , whose determinant is  $(-1)^{n-p}$ ; so the discriminant sees only the parity of  $n-p$ , and for  $n \geq 2$  it cannot separate, say,  $\mathbf{I}_2$  from  $-\mathbf{I}_2$ . That the  $n+1$  matrices really are pairwise non-congruent is the law of inertia, not anything the discriminant can deliver.

## 13.3 Three Kinds of Form

A bilinear form has two slots, and so far nothing has related them. Almost every form that occurs in practice does relate them: the dot product does not care which vector goes first, while the determinant of a  $2 \times 2$  matrix changes sign when its columns are swapped. This section names the three symmetry conditions the rest of the chapter runs on, shows that two of them are the same condition exactly when  $1 + 1 \neq 0$  in the field, and extracts from a symmetric form the single-variable function  $q(\mathbf{v}) = \beta(\mathbf{v}, \mathbf{v})$  that carries all of its information.

Throughout,  $V$  is a finite-dimensional vector space over a field  $F$ , and  $\beta$  is a bilinear form on  $V$  (Definition 13.1.1). The characteristic hypothesis is stated wherever it is needed and nowhere else.

### 13.3.1 Symmetric forms

The dot product on  $F^n$  satisfies  $\mathbf{x} \cdot \mathbf{y} = \mathbf{y} \cdot \mathbf{x}$ , and so does every integral pairing  $\int_0^1 fg$ , and so does  $\mathbf{x}^\top \mathbf{A} \mathbf{y}$  whenever  $\mathbf{A}$  happens to satisfy  $\mathbf{A}^\top = \mathbf{A}$ . The condition recurs, so it gets a name.

*A symmetric form cannot tell its two arguments apart.*

#### Definition 13.3.1 — Symmetric Bilinear Form

A bilinear form  $\beta: V \times V \rightarrow F$  is **symmetric** if

$$\beta(\mathbf{u}, \mathbf{v}) = \beta(\mathbf{v}, \mathbf{u})$$

for all  $\mathbf{u}, \mathbf{v} \in V$ .

In words: the value is unchanged by swapping the two inputs. The condition is imposed for **every** pair, not just for pairs of basis vectors — although, as the next result shows, checking it on basis vectors is enough.

Three examples and one non-example.

- **The dot product** on  $F^n$ ,  $\beta(\mathbf{x}, \mathbf{y}) = x_1y_1 + \cdots + x_ny_n$ . Each product  $x_iy_i$  is unchanged by the swap because multiplication in  $F$  is commutative.
- **A symmetric matrix**,  $\beta(\mathbf{x}, \mathbf{y}) = \mathbf{x}^\top \mathbf{A} \mathbf{y}$  with  $\mathbf{A}^\top = \mathbf{A}$  (Definition 0.9.11). Here  $\mathbf{x}^\top \mathbf{A} \mathbf{y}$  is a  $1 \times 1$  matrix, so it equals its own transpose  $\mathbf{y}^\top \mathbf{A}^\top \mathbf{x} = \mathbf{y}^\top \mathbf{A} \mathbf{x}$  by Theorem 0.9.5.
- **Evaluation pairings**. On  $F[x]_{\leq n}$ , fix scalars  $t_1, \dots, t_k \in F$  and set  $\beta(p, q) = \sum_i p(t_i)q(t_i)$ . Symmetry is again commutativity of  $F$ .
- **A non-example by minimal change**. On  $F^2$  put  $\beta(\mathbf{x}, \mathbf{y}) = x_1y_2$ . It is bilinear, by the one-slot check of Definition 6.2.1. But  $\beta(\mathbf{e}_1, \mathbf{e}_2) = 1$  and  $\beta(\mathbf{e}_2, \mathbf{e}_1) = 0$ , so the symmetry clause fails at that one pair. Exactly one clause has been broken: bilinearity survives untouched.

The matrix of a form (Definition 13.1.2) turns the condition into a matrix condition, and turns “for all  $\mathbf{u}, \mathbf{v}$ ” into a finite check.

#### Proposition 13.3.1 — Symmetry Is Visible in the Matrix

Let  $\mathcal{B}$  be a basis of  $V$  and  $\mathbf{A} = [\beta]_{\mathcal{B}}$ . Then  $\beta$  is symmetric if and only if  $\mathbf{A}^\top = \mathbf{A}$ .

*Proof.* ( $\Rightarrow$ ) Suppose  $\beta$  is symmetric. Writing  $\mathcal{B} = (\mathbf{v}_1, \dots, \mathbf{v}_n)$ , the entries of  $\mathbf{A}$  are  $a_{ij} = \beta(\mathbf{v}_i, \mathbf{v}_j)$ , so  $a_{ij} = \beta(\mathbf{v}_j, \mathbf{v}_i) = a_{ji}$ , which is Definition 0.9.11.

( $\Leftarrow$ ) Suppose  $\mathbf{A}^\top = \mathbf{A}$ , and let  $\mathbf{u}, \mathbf{v} \in V$ . By Theorem 13.1.1,

$$\beta(\mathbf{u}, \mathbf{v}) = [\mathbf{u}]_{\mathcal{B}}^\top \mathbf{A} [\mathbf{v}]_{\mathcal{B}}.$$

The right-hand side is a  $1 \times 1$  matrix, hence equal to its transpose, which by [Theorem 0.9.5](#) is  $[\mathbf{v}]_{\mathcal{B}}^{\top} \mathbf{A}^{\top} [\mathbf{u}]_{\mathcal{B}} = [\mathbf{v}]_{\mathcal{B}}^{\top} \mathbf{A} [\mathbf{u}]_{\mathcal{B}} = \beta(\mathbf{v}, \mathbf{u})$ . This proves the equivalence. ■

Because congruence sends  $\mathbf{A}$  to  $\mathbf{P}^{\top} \mathbf{A} \mathbf{P}$  and  $(\mathbf{P}^{\top} \mathbf{A} \mathbf{P})^{\top} = \mathbf{P}^{\top} \mathbf{A}^{\top} \mathbf{P}$  by [Theorem 0.9.5](#), a congruent copy of a symmetric matrix is symmetric. So the condition  $\mathbf{A}^{\top} = \mathbf{A}$  does not depend on which basis was used to build  $\mathbf{A}$  — as it must not, since by [Proposition 13.3.1](#) it says something about  $\beta$  alone.

### 13.3.2 Alternating and skew-symmetric forms

Chapter 6 needed two ways of saying “the value collapses when two arguments coincide”, and found that they are not the same condition. Both were defined there for  $k$ -linear forms, and a bilinear form is the case  $k = 2$ , so nothing needs redefining. Read at  $k = 2$ :

- $\beta$  is **alternating** ([Definition 6.2.2](#)) if  $\beta(\mathbf{v}, \mathbf{v}) = 0$  for **every**  $\mathbf{v} \in V$ ;
- $\beta$  is **skew-symmetric** ([Definition 6.2.3](#)) if  $\beta(\mathbf{u}, \mathbf{v}) = -\beta(\mathbf{v}, \mathbf{u})$  for **all**  $\mathbf{u}, \mathbf{v} \in V$ .

The standard example of both is the signed area  $\beta(\mathbf{x}, \mathbf{y}) = x_1 y_2 - x_2 y_1$  on  $F^2$ , which is the determinant of the matrix with columns  $\mathbf{x}$  and  $\mathbf{y}$ . A second family, the one Section 9 will be about, is the **standard alternating form** on  $F^{2m}$ ,

$$\beta(\mathbf{x}, \mathbf{y}) = \sum_{i=1}^m (x_i y_{m+i} - x_{m+i} y_i),$$

which pairs the first block of coordinates against the second.

Each condition reads off the matrix, and the two readings differ by the diagonal.

#### Proposition 13.3.2 — Alternation and Skewness in the Matrix

Let  $\mathcal{B} = (\mathbf{v}_1, \dots, \mathbf{v}_n)$  be a basis of  $V$  and  $\mathbf{A} = [\beta]_{\mathcal{B}}$ . Then:

- $\beta$  is skew-symmetric if and only if  $\mathbf{A}^{\top} = -\mathbf{A}$ ;
- $\beta$  is alternating if and only if  $\mathbf{A}^{\top} = -\mathbf{A}$  and  $a_{ii} = 0$  for every  $i$ .

*Proof.*

- repeats the proof of [Proposition 13.3.1](#) with a minus sign throughout.
- ( $\Rightarrow$ ) If  $\beta$  is alternating then  $a_{ii} = \beta(\mathbf{v}_i, \mathbf{v}_i) = 0$ , and  $\beta$  is skew-symmetric by [Theorem 6.2.1](#) (a), so  $\mathbf{A}^{\top} = -\mathbf{A}$  by (a).

( $\Leftarrow$ ) Suppose  $\mathbf{A}^{\top} = -\mathbf{A}$  with zero diagonal, and let  $\mathbf{v} \in V$  have coordinates  $[\mathbf{v}]_{\mathcal{B}} = (c_1, \dots, c_n)$ . By [Theorem 13.1.1](#) and bilinearity,

$$\beta(\mathbf{v}, \mathbf{v}) = \sum_{i,j} c_i c_j a_{ij} = \sum_i c_i^2 a_{ii} + \sum_{i < j} c_i c_j (a_{ij} + a_{ji}),$$

where the second sum collects the terms  $(i, j)$  and  $(j, i)$  together. Every  $a_{ii}$  is 0, and every  $a_{ij} + a_{ji}$  is 0 because  $\mathbf{A}^{\top} = -\mathbf{A}$ . So  $\beta(\mathbf{v}, \mathbf{v}) = 0$ . This proves the equivalence.

Part (b) is the whole difficulty in one line: a skew-symmetric matrix has  $a_{ii} = -a_{ii}$ , and *usually* that forces  $a_{ii} = 0$ . Usually, but not always.

### 13.3.3 Where characteristic 2 earns its keep

Chapter 0 warned that in a field of characteristic 2 the argument “ $x = -x$ , so  $2x = 0$ , so  $x = 0$ ” divides by  $2 = 1 + 1$ , which is 0 there, and it promised that this is why later results on symmetric bilinear forms would carry the hypothesis “characteristic  $\neq 2$ ” (Definition 0.7.2). This is the theorem the promise was about.

#### Theorem 13.3.3 — Alternating Versus Skew-Symmetric

Let  $F$  be a field and let  $V$  be a vector space over  $F$ .

- (a) Every alternating bilinear form on  $V$  is skew-symmetric. No hypothesis on  $F$  or on  $V$  is needed.
- (b) If  $\text{char } F \neq 2$ , every skew-symmetric bilinear form on  $V$  is alternating. Again  $V$  is arbitrary.
- (c) If  $\text{char } F = 2$  and  $V \neq \{0\}$ , there is a skew-symmetric bilinear form on  $V$  that is **not** alternating.

So the converse of (a) holds **if and only if**  $\text{char } F \neq 2$ .

*Idea.* For (a), feed  $\mathbf{u} + \mathbf{v}$  into both slots and expand: the two diagonal terms vanish by hypothesis, and what is left is exactly the skew identity. For (b), feed  $\mathbf{v}$  into both slots of the skew identity, which gives  $\beta(\mathbf{v}, \mathbf{v}) = -\beta(\mathbf{v}, \mathbf{v})$ ; canceling needs  $1 + 1$  to be invertible, and that is precisely what characteristic  $\neq 2$  supplies. For (c), notice that in characteristic 2 the words “skew-symmetric” and “symmetric” mean the same thing, so any symmetric form that is non-zero on some vector **paired with itself** is a witness — and the dot product is one.

*Proof.*

- (a) Let  $\beta$  be alternating and let  $\mathbf{u}, \mathbf{v} \in V$ . Expanding by bilinearity,

$$0 = \beta(\mathbf{u} + \mathbf{v}, \mathbf{u} + \mathbf{v}) = \beta(\mathbf{u}, \mathbf{u}) + \beta(\mathbf{u}, \mathbf{v}) + \beta(\mathbf{v}, \mathbf{u}) + \beta(\mathbf{v}, \mathbf{v}),$$

and the first and last terms are 0 because  $\beta$  is alternating. Hence  $\beta(\mathbf{u}, \mathbf{v}) = -\beta(\mathbf{v}, \mathbf{u})$ . (This is Theorem 6.2.1 (a) at  $k = 2$ .)

- (b) Let  $\beta$  be skew-symmetric and  $\mathbf{v} \in V$ . Taking  $\mathbf{u} = \mathbf{v}$  in the defining identity gives  $\beta(\mathbf{v}, \mathbf{v}) = -\beta(\mathbf{v}, \mathbf{v})$ , that is,  $(1 + 1)\beta(\mathbf{v}, \mathbf{v}) = 0$ . Since  $\text{char } F \neq 2$ , we have  $1 + 1 \neq 0$  in  $F$ : otherwise the smallest positive  $n$  with  $n \cdot 1 = 0$  would be 2, which is Definition 0.7.2. A non-zero element of a field is invertible, so multiplying by  $(1 + 1)^{-1}$  gives  $\beta(\mathbf{v}, \mathbf{v}) = 0$ . (This is Theorem 6.2.2 at  $k = 2$ .)
- (c) Suppose  $\text{char } F = 2$ , so that  $-1 = 1$  in  $F$ . Since  $V \neq \{0\}$  is finite-dimensional, it has a basis with at least one vector, and we may take  $\varphi \in V^*$  to be the coordinate

functional of its first vector  $\mathbf{v}_1$ , so that  $\varphi(\mathbf{v}_1) = 1$ . Define

$$\beta(\mathbf{u}, \mathbf{v}) = \varphi(\mathbf{u}) \varphi(\mathbf{v}).$$

This is bilinear, being a product of two linear functionals, and it is symmetric. Because  $-1 = 1$ , symmetric and skew-symmetric say the same thing, so  $\beta$  is skew-symmetric. But  $\beta(\mathbf{v}_1, \mathbf{v}_1) = 1 \neq 0$ , so  $\beta$  is not alternating.

Finally, (b) and (c) together say that the converse of (a) holds exactly when  $\text{char } F \neq 2$ . This proves the theorem. ■

The concrete witness is worth writing down.

**Example. (The dot product over the field with two elements)**

Let  $F = \mathbb{F}_2$  and  $V = \mathbb{F}_2^2$ , and let  $\beta$  be the dot product,  $\beta(\mathbf{x}, \mathbf{y}) = x_1y_1 + x_2y_2$ . Decide whether  $\beta$  is symmetric, skew-symmetric and alternating, and compute the function  $\mathbf{v} \mapsto \beta(\mathbf{v}, \mathbf{v})$  on all four vectors of  $V$ .

*Solution.* The matrix of  $\beta$  in the standard basis is  $\mathbf{I}_2$ . It is symmetric, so  $\beta$  is symmetric by Proposition 13.3.1. Over  $\mathbb{F}_2$  we have  $-\mathbf{I}_2 = \mathbf{I}_2$ , so the matrix also satisfies  $\mathbf{I}_2^\top = -\mathbf{I}_2$ , and  $\beta$  is skew-symmetric by Proposition 13.3.2 (a). It is **not** alternating: the diagonal entries of  $\mathbf{I}_2$  are  $1 \neq 0$ , and indeed  $\beta(\mathbf{e}_1, \mathbf{e}_1) = 1$ .

On the four vectors,  $\beta(\mathbf{v}, \mathbf{v}) = v_1^2 + v_2^2$ . In  $\mathbb{F}_2$  both elements satisfy  $c^2 = c$ , so this equals  $v_1 + v_2$ :

$$\beta(\mathbf{0}, \mathbf{0}) = 0, \quad \beta(\mathbf{e}_1, \mathbf{e}_1) = 1, \quad \beta(\mathbf{e}_2, \mathbf{e}_2) = 1, \quad \beta(\mathbf{1}, \mathbf{1}) = 0.$$

Notice what that list is: the function  $\mathbf{v} \mapsto \beta(\mathbf{v}, \mathbf{v})$  is the **linear** functional  $v_1 + v_2$ . In characteristic 2 a “quadratic” function coming from a bilinear form can be linear, which is the first sign that the notions below will not behave.

*Warning.* In characteristic 2, “skew-symmetric” carries no information beyond “symmetric”. Since  $-1 = 1$ , the two conditions are literally the same condition, and the useful notion is the strictly stronger one, alternating. This is why Chapter 6 built the determinant on alternation rather than skewness, and why every statement below that pairs symmetric against alternating needs  $\text{char } F \neq 2$ .

### 13.3.4 Splitting a form into two halves

In characteristic  $\neq 2$  the two conditions are complementary in the strongest possible sense: every bilinear form is a symmetric one plus an alternating one, in exactly one way. The recipe is the one that splits a function on  $\mathbb{R}$  into its even and odd parts.

**Theorem 13.3.4 — Symmetric Plus Alternating**

Let  $\text{char } F \neq 2$  and let  $\beta$  be a bilinear form on  $V$ . Then there are a **unique** symmetric

form  $\beta_s$  and a **unique** alternating form  $\beta_a$  on  $V$  with  $\beta = \beta_s + \beta_a$ , namely

$$\begin{aligned}\beta_s(\mathbf{u}, \mathbf{v}) &= \frac{1}{2}(\beta(\mathbf{u}, \mathbf{v}) + \beta(\mathbf{v}, \mathbf{u})), \\ \beta_a(\mathbf{u}, \mathbf{v}) &= \frac{1}{2}(\beta(\mathbf{u}, \mathbf{v}) - \beta(\mathbf{v}, \mathbf{u})).\end{aligned}$$

In matrices:  $\mathbf{A} = \frac{1}{2}(\mathbf{A} + \mathbf{A}^\top) + \frac{1}{2}(\mathbf{A} - \mathbf{A}^\top)$ .

*Proof.* Since  $\text{char } F \neq 2$ , the element  $1 + 1$  is non-zero, hence invertible, and  $\frac{1}{2}$  denotes  $(1 + 1)^{-1}$ . Both displayed formulas are bilinear, being combinations of the bilinear maps  $(\mathbf{u}, \mathbf{v}) \mapsto \beta(\mathbf{u}, \mathbf{v})$  and  $(\mathbf{u}, \mathbf{v}) \mapsto \beta(\mathbf{v}, \mathbf{u})$ , and their sum is  $\beta$ . Swapping  $\mathbf{u}$  and  $\mathbf{v}$  leaves  $\beta_s$  unchanged, so  $\beta_s$  is symmetric; and  $\beta_a(\mathbf{v}, \mathbf{v}) = \frac{1}{2}(\beta(\mathbf{v}, \mathbf{v}) - \beta(\mathbf{v}, \mathbf{v})) = 0$ , so  $\beta_a$  is alternating.

For uniqueness, suppose  $\beta_s + \beta_a = \beta'_s + \beta'_a$  with  $\beta_s, \beta'_s$  symmetric and  $\beta_a, \beta'_a$  alternating, and put  $\delta = \beta_s - \beta'_s = \beta'_a - \beta_a$ . Then  $\delta$  is symmetric, being a difference of symmetric forms, and alternating, being a difference of alternating forms. By [Theorem 13.3.3 \(a\)](#) it is also skew-symmetric, so for all  $\mathbf{u}, \mathbf{v}$ ,

$$\delta(\mathbf{u}, \mathbf{v}) = \delta(\mathbf{v}, \mathbf{u}) = -\delta(\mathbf{u}, \mathbf{v}),$$

giving  $(1 + 1)\delta(\mathbf{u}, \mathbf{v}) = 0$  and hence  $\delta(\mathbf{u}, \mathbf{v}) = 0$ . Therefore  $\beta_s = \beta'_s$  and  $\beta_a = \beta'_a$ , as claimed. ■

Both halves of the argument divided by  $1 + 1$ , and both failures are real. In characteristic 2 the formulas are meaningless, and more than that, the statement itself is false: alternating forms are then a *subset* of the symmetric ones, not a complement to them.

**Warning.** In characteristic 2, symmetric and alternating forms are not complementary. Over  $\mathbb{F}_2$  take  $\beta(\mathbf{x}, \mathbf{y}) = x_1y_2$  on  $\mathbb{F}_2^2$ . An alternating form is skew-symmetric ([Theorem 13.3.3 \(a\)](#)), and skew-symmetric means symmetric here, so **every** sum of a symmetric form and an alternating form is symmetric. But  $\beta(\mathbf{e}_1, \mathbf{e}_2) = 1$  while  $\beta(\mathbf{e}_2, \mathbf{e}_1) = 0$ , so  $\beta$  is not symmetric and therefore cannot be written as such a sum at all. The decomposition does not merely lose uniqueness; existence fails.

### 13.3.5 The quadratic form

A bilinear form is a function of two vectors, which makes it awkward to plot, to differentiate, or to write on a page. Feeding the same vector into both slots turns it into a function of one vector. Chapter 11 met this function on  $\mathbb{R}^n$  under the name **quadratic form** and used it to identify conics; here is the general definition, and the surprising fact that nothing is lost.

*A quadratic form is what a bilinear form sees when both of its arguments are the same vector.*

#### Definition 13.3.2 — Quadratic Form

Let  $\beta$  be a bilinear form on  $V$ . The **quadratic form associated with  $\beta$**  is the function

$$q: V \rightarrow F, \quad q(\mathbf{v}) = \beta(\mathbf{v}, \mathbf{v}).$$

A function  $q: V \rightarrow F$  is called a **quadratic form on  $V$**  if  $q(\mathbf{v}) = \beta(\mathbf{v}, \mathbf{v})$  for some **symmetric** bilinear form  $\beta$  on  $V$ .

Two clauses deserve separate readings. The first says how to build  $q$  from a given  $\beta$ , for **any**  $\beta$  over **any** field. The second says which functions  $V \rightarrow F$  are entitled to the name, and it insists on a **symmetric**  $\beta$ ; the polarization theorem below will show that over a field of characteristic  $\neq 2$  this symmetric  $\beta$  is then unique, so that “the” bilinear form of a quadratic form is well defined.

Insisting on symmetry costs nothing in characteristic  $\neq 2$ . If  $q(\mathbf{v}) = \beta(\mathbf{v}, \mathbf{v})$  for some bilinear  $\beta$ , then by [Theorem 13.3.4](#) we may split  $\beta = \beta_s + \beta_a$ , and  $\beta_a(\mathbf{v}, \mathbf{v}) = 0$ , so  $q(\mathbf{v}) = \beta_s(\mathbf{v}, \mathbf{v})$  with  $\beta_s$  symmetric. The alternating part is invisible to  $q$ .

In coordinates, with  $\mathbf{A} = [\beta]_{\mathcal{B}}$  symmetric and  $\mathbf{x} = [\mathbf{v}]_{\mathcal{B}}$ ,

$$q(\mathbf{v}) = \mathbf{x}^\top \mathbf{A} \mathbf{x} = \sum_i a_{ii} x_i^2 + 2 \sum_{i < j} a_{ij} x_i x_j, \quad (13.3.1)$$

which is a homogeneous polynomial of degree 2 in the coordinates: hence the name. Reading [Eq. 13.3.1](#) backwards recovers  $\mathbf{A}$  from the polynomial:  $a_{ii}$  is the coefficient of  $x_i^2$ , and  $a_{ij}$  for  $i \neq j$  is **half** the coefficient of  $x_i x_j$ . That is word for word the recipe Chapter 11 gave for the symmetric matrix of a real quadratic form, and the halving is the same division by  $1 + 1$  as everywhere else in this section.

**Example. (From a polynomial to its matrix and back)**

Over  $\mathbb{Q}$ , let  $q(x_1, x_2, x_3) = 2x_1^2 - x_2^2 + 6x_1x_2 - 4x_2x_3$ . Find the symmetric matrix  $\mathbf{A}$  with  $q(\mathbf{x}) = \mathbf{x}^\top \mathbf{A} \mathbf{x}$ , and evaluate  $q(1, 1, 1)$  both from the polynomial and from  $\mathbf{A}$ .

*Solution.* The squared terms give  $a_{11} = 2$ ,  $a_{22} = -1$  and  $a_{33} = 0$ . The cross terms give  $a_{12} = a_{21} = 3$  (half of 6),  $a_{23} = a_{32} = -2$  (half of  $-4$ ), and  $a_{13} = a_{31} = 0$  since  $x_1x_3$  does not appear. So

$$\mathbf{A} = \begin{pmatrix} 2 & 3 & 0 \\ 3 & -1 & -2 \\ 0 & -2 & 0 \end{pmatrix}.$$

From the polynomial,  $q(1, 1, 1) = 2 - 1 + 6 - 4 = 3$ . From the matrix,  $\mathbf{A}(1, 1, 1) = (5, 0, -2)$  and  $(1, 1, 1) \cdot (5, 0, -2) = 3$ . The two agree.

Now the theorem that makes  $q$  as good as  $\beta$ .

**Theorem 13.3.5 — Polarization**

Let  $\text{char } F \neq 2$ . The map  $\beta \mapsto q$ ,  $q(\mathbf{v}) = \beta(\mathbf{v}, \mathbf{v})$ , is a bijection from the set of symmetric bilinear forms on  $V$  onto the set of quadratic forms on  $V$ . Its inverse is given by either of

$$\begin{aligned} \beta(\mathbf{u}, \mathbf{v}) &= \frac{1}{2}(q(\mathbf{u} + \mathbf{v}) - q(\mathbf{u}) - q(\mathbf{v})) \\ &= \frac{1}{4}(q(\mathbf{u} + \mathbf{v}) - q(\mathbf{u} - \mathbf{v})). \end{aligned}$$

In particular a symmetric form is determined by its quadratic form: if  $\beta(\mathbf{v}, \mathbf{v}) = \beta'(\mathbf{v}, \mathbf{v})$  for all  $\mathbf{v}$ , then  $\beta = \beta'$ .



*Idea.* Only one computation is needed, and it is the expansion of  $q(\mathbf{u} \pm \mathbf{v})$  by bilinearity. Symmetry merges the two cross terms into one, so the cross term appears with a factor  $1 + 1$ , and dividing by it is the whole trick. Surjectivity is the definition of “quadratic form”; injectivity is the displayed formula.

*Proof.* Let  $\beta$  be symmetric with quadratic form  $q$ , and let  $\mathbf{u}, \mathbf{v} \in V$ . By bilinearity and then symmetry,

$$q(\mathbf{u} \pm \mathbf{v}) = \beta(\mathbf{u}, \mathbf{u}) \pm \beta(\mathbf{u}, \mathbf{v}) \pm \beta(\mathbf{v}, \mathbf{u}) + \beta(\mathbf{v}, \mathbf{v}),$$

so  $q(\mathbf{u} \pm \mathbf{v}) = q(\mathbf{u}) + q(\mathbf{v}) \pm (1 + 1)\beta(\mathbf{u}, \mathbf{v})$ . Since  $\text{char } F \neq 2$ , the element  $1 + 1$  is invertible, and taking the upper sign gives the first formula. Subtracting the two versions gives  $q(\mathbf{u} + \mathbf{v}) - q(\mathbf{u} - \mathbf{v}) = (1 + 1)^2\beta(\mathbf{u}, \mathbf{v})$ , which is the second.

The map  $\beta \mapsto q$  is onto the set of quadratic forms by [Definition 13.3.2](#), which defines that set as the image. It is injective because the first displayed formula recovers  $\beta(\mathbf{u}, \mathbf{v})$  from  $q$  alone; so if two symmetric forms have the same  $q$ , the formula returns the same value for each, and they are equal. Hence the map is a bijection with the stated inverse. This proves the theorem. ■

*Quick check.* Which bilinear forms are both symmetric and alternating, over a field of characteristic  $\neq 2$ ?

*Solution.* Only the zero form. If  $\beta$  is alternating then  $q = 0$ , and if  $\beta$  is also symmetric then [Theorem 13.3.5](#) recovers  $\beta(\mathbf{u}, \mathbf{v}) = \frac{1}{2}(0 - 0 - 0) = 0$ . (Over  $\mathbb{F}_2$  the answer is different: every alternating form is symmetric there, as [Theorem 13.3.3](#) shows, and there are plenty of non-zero ones.)

The second formula of [Theorem 13.3.5](#) should look familiar. Over  $\mathbb{R}$ , an inner product is a symmetric bilinear form whose quadratic form is  $q(\mathbf{v}) = \|\mathbf{v}\|^2$ , and the formula reads

$$\langle \mathbf{u}, \mathbf{v} \rangle = \frac{1}{4}(\|\mathbf{u} + \mathbf{v}\|^2 - \|\mathbf{u} - \mathbf{v}\|^2),$$

which is exactly [Lemma 10.7.2](#) (a). Chapter 10 proved it to show that an isometry automatically preserves angles; the content here is the same, stripped of positivity and of  $\mathbb{R}$ . What Chapter 10 needed positivity for was the *norm*, not the identity.

### 13.3.6 Hermitian forms

Over  $\mathbb{C}$  the pattern changes, because  $\langle \mathbf{v}, \mathbf{v} \rangle$  is required to be a positive real number and no bilinear form can deliver that: bilinearity gives  $\beta(i\mathbf{v}, i\mathbf{v}) = i^2\beta(\mathbf{v}, \mathbf{v}) = -\beta(\mathbf{v}, \mathbf{v})$ , so the values at  $\mathbf{v}$  and  $i\mathbf{v}$  have opposite signs. Chapter 10 solved this by making the second slot conjugate-linear. That fix has nothing to do with positivity, so it can be made here too.

#### Definition 13.3.3 — Hermitian Form

Let  $V$  be a vector space over  $\mathbb{C}$ . A function  $\beta: V \times V \rightarrow \mathbb{C}$  is a **Hermitian form** if it

is **linear in the first argument**, and

$$\beta(\mathbf{v}, \mathbf{u}) = \overline{\beta(\mathbf{u}, \mathbf{v})} \quad \text{for all } \mathbf{u}, \mathbf{v} \in V.$$

Two remarks on the clauses. First, the two conditions together force conjugate-linearity in the second argument:  $\beta(\mathbf{u}, c\mathbf{v}) = \overline{\beta(c\mathbf{v}, \mathbf{u})} = \overline{c\beta(\mathbf{v}, \mathbf{u})} = \bar{c}\beta(\mathbf{u}, \mathbf{v})$ . So a Hermitian form is **not** bilinear, and [Definition 13.1.1](#) does not cover it. Second, taking  $\mathbf{u} = \mathbf{v}$  gives  $\beta(\mathbf{v}, \mathbf{v}) = \overline{\beta(\mathbf{v}, \mathbf{v})}$ , so the associated function  $q(\mathbf{v}) = \beta(\mathbf{v}, \mathbf{v})$  takes **real** values even though  $\beta$  does not.

The examples are the ones from Chapter 10 with positivity dropped:  $\beta(\mathbf{x}, \mathbf{y}) = \mathbf{y}^* \mathbf{A} \mathbf{x}$  on  $\mathbb{C}^n$  for any  $\mathbf{A}$  with  $\mathbf{A}^* = \mathbf{A}$ , including indefinite choices such as  $\mathbf{A} = \text{diag}(1, -1)$ . The matrix criterion is the same proof as [Proposition 13.3.1](#) with stars in place of transposes.

### Proposition 13.3.6 — Hermitian Forms and Hermitian Matrices

Let  $\beta$  be linear in the first argument and conjugate-linear in the second, let  $\mathcal{B} = (\mathbf{v}_1, \dots, \mathbf{v}_n)$  be a basis of  $V$ , and let  $\mathbf{A}$  be the matrix with entries  $a_{ij} = \beta(\mathbf{v}_j, \mathbf{v}_i)$ . Then  $\beta$  is Hermitian if and only if  $\mathbf{A}^* = \mathbf{A}$ , and in that case  $\beta(\mathbf{u}, \mathbf{v}) = [\mathbf{v}]_{\mathcal{B}}^* \mathbf{A} [\mathbf{u}]_{\mathcal{B}}$ .

*Proof.* The index order  $a_{ij} = \beta(\mathbf{v}_j, \mathbf{v}_i)$  is the one used for Gram matrices in Chapter 10 ( $\mathbf{G}$  in [Definition 10.2.3](#)), chosen so that the displayed formula comes out with no transposing of the answer. Expanding  $\mathbf{u} = \sum_j c_j \mathbf{v}_j$  and  $\mathbf{v} = \sum_i d_i \mathbf{v}_i$  by linearity in the first slot and conjugate-linearity in the second,

$$\beta(\mathbf{u}, \mathbf{v}) = \sum_{i,j} c_j \bar{d}_i \beta(\mathbf{v}_j, \mathbf{v}_i) = \sum_{i,j} \bar{d}_i a_{ij} c_j,$$

which is the matrix product  $[\mathbf{v}]_{\mathcal{B}}^* \mathbf{A} [\mathbf{u}]_{\mathcal{B}}$ . If  $\beta$  is Hermitian then in particular  $a_{ji} = \beta(\mathbf{v}_i, \mathbf{v}_j) = \overline{\beta(\mathbf{v}_j, \mathbf{v}_i)} = \overline{a_{ij}}$ , which is  $\mathbf{A}^* = \mathbf{A}$ . Conversely, if  $a_{ji} = \overline{a_{ij}}$  then conjugating the display gives

$$\overline{\beta(\mathbf{u}, \mathbf{v})} = \sum_{i,j} d_i \overline{a_{ij}} \bar{c}_j = \sum_{i,j} \bar{c}_j a_{ji} d_i = \beta(\mathbf{v}, \mathbf{u}),$$

so  $\beta$  is Hermitian. ■

Polarization works for Hermitian forms too, and it is the four-term identity rather than the two-term one: by [Lemma 10.7.2](#) (b) with  $q(\mathbf{v}) = \beta(\mathbf{v}, \mathbf{v})$ ,

$$\beta(\mathbf{u}, \mathbf{v}) = \frac{1}{4} \sum_{k=0}^3 i^k q(\mathbf{u} + i^k \mathbf{v}).$$

The computation in Chapter 10 expanded each term using linearity in the first slot, conjugate-linearity in the second and  $\beta(\mathbf{v}, \mathbf{u}) = \overline{\beta(\mathbf{u}, \mathbf{v})}$ , and used positivity nowhere, so it proves this identity too. Note the contrast with the real case: over  $\mathbb{C}$  there is no characteristic to worry about, since  $\text{char } \mathbb{C} = 0$ , and the reason a Hermitian form is recoverable from  $q$  is the extra room provided by  $i$ .

**Warning.** A quadratic form over a field of characteristic 2 is a different object, and this book does not define it. Two things go wrong there, and [Theorem 13.3.5](#) repairs neither. The form is not determined by  $q$ . Over  $\mathbb{F}_2$ , the symmetric form  $\beta(\mathbf{x}, \mathbf{y}) =$

$x_1y_2 + x_2y_1$  has  $q(\mathbf{x}) = 2x_1x_2 = 0$ , the same  $q$  as the zero form, so  $\beta \mapsto q$  is not injective on symmetric forms. And [Definition 13.3.2](#) names too few functions. The function  $q(\mathbf{x}) = x_1x_2$  on  $\mathbb{F}_2^2$  is  $\beta(\mathbf{x}, \mathbf{x})$  for the bilinear  $\beta(\mathbf{x}, \mathbf{y}) = x_1y_2$ , yet it is  $\beta(\mathbf{x}, \mathbf{x})$  for **no symmetric**  $\beta$ : for a symmetric  $\mathbf{A}$ , [Eq. 13.3.1](#) gives  $a_{11}x_1^2 + a_{22}x_2^2 = a_{11}x_1 + a_{22}x_2$  over  $\mathbb{F}_2$ , whose values at  $(1, 1)$ ,  $(1, 0)$  and  $(0, 1)$  are  $a_{11} + a_{22}$ ,  $a_{11}$  and  $a_{22}$ , and no choice makes those 1, 0, 0. The theory of quadratic forms in characteristic 2 therefore defines  $q$  first, by the axioms  $q(c\mathbf{v}) = c^2q(\mathbf{v})$  and “ $q(\mathbf{u} + \mathbf{v}) - q(\mathbf{u}) - q(\mathbf{v})$  is bilinear”, and treats  $\beta$  as derived data. Everything that follows assumes  $\text{char } F \neq 2$  precisely to stay out of that theory.

Three kinds of form, then, and three matrix shapes: symmetric, alternating, Hermitian. The next two sections take the first of them and classify it up to congruence, first over any field of characteristic  $\neq 2$ , then over  $\mathbb{R}$ .

## 13.3.7 Exercises

### 13.3.7.1 A. Check your understanding

#### Exercise A1.

- Define what it means for a bilinear form  $\beta$  on  $V$  to be **symmetric**, **alternating** and **skew-symmetric**.
- State the relationship between alternating and skew-symmetric forms, including the exact role of the characteristic.
- Write down the matrix condition, in a fixed basis, for each of the three notions in (a).
- Determine whether the following statement is correct, and justify your answer: over  $\mathbb{Q}$ , a bilinear form is determined by the function  $\mathbf{v} \mapsto \beta(\mathbf{v}, \mathbf{v})$ .
- Why is a Hermitian form not an example of [Definition 13.1.1](#)?

*Solution.*

- Symmetric:  $\beta(\mathbf{u}, \mathbf{v}) = \beta(\mathbf{v}, \mathbf{u})$  for all  $\mathbf{u}, \mathbf{v}$ . Alternating:  $\beta(\mathbf{v}, \mathbf{v}) = 0$  for all  $\mathbf{v}$ . Skew-symmetric:  $\beta(\mathbf{u}, \mathbf{v}) = -\beta(\mathbf{v}, \mathbf{u})$  for all  $\mathbf{u}, \mathbf{v}$ .
- Alternating always implies skew-symmetric ([Theorem 13.3.3](#) (a)). The converse holds if and only if  $\text{char } F \neq 2$ ; in characteristic 2 the dot product on  $\mathbb{F}_2^2$  is skew-symmetric and not alternating.
- With  $\mathbf{A} = [\beta]_{\mathcal{B}}$ : symmetric means  $\mathbf{A}^\top = \mathbf{A}$ ; skew-symmetric means  $\mathbf{A}^\top = -\mathbf{A}$ ; alternating means  $\mathbf{A}^\top = -\mathbf{A}$  together with  $a_{ii} = 0$  for all  $i$  ([Proposition 13.3.1](#), [Proposition 13.3.2](#)).
- Incorrect. Since  $\text{char } \mathbb{Q} = 0 \neq 2$ , only the **symmetric** part is determined: any alternating form has  $\beta(\mathbf{v}, \mathbf{v}) = 0$  identically, so  $\beta$  and  $\beta + \beta_a$  have the same quadratic form for every alternating  $\beta_a$ . What is true is [Theorem 13.3.5](#), which determines  $\beta$  among *symmetric* forms.
- It is not linear in the second argument:  $\beta(\mathbf{u}, c\mathbf{v}) = \bar{c}\beta(\mathbf{u}, \mathbf{v})$ , and  $\bar{c} \neq c$  for  $c \notin \mathbb{R}$ .

## 13.3.7.2 B. Practice

**Exercise B1: Determine which of the following.**

Over  $\mathbb{Q}$ , with  $V = \mathbb{Q}^2$  and  $\mathbf{x} = (x_1, x_2)$ ,  $\mathbf{y} = (y_1, y_2)$ , determine for each of the following bilinear forms whether it is symmetric, whether it is alternating, and whether it is skew-symmetric. Justify each answer.

- (a)  $\beta(\mathbf{x}, \mathbf{y}) = x_1y_1 - x_2y_2$ .
- (b)  $\beta(\mathbf{x}, \mathbf{y}) = 3x_1y_2 - 3x_2y_1$ .
- (c)  $\beta(\mathbf{x}, \mathbf{y}) = x_1y_2 + 2x_2y_1$ .
- (d)  $\beta(\mathbf{x}, \mathbf{y}) = 0$ .

*Solution.* Write down the matrix  $\mathbf{A}$  in the standard basis in each case and apply [Proposition 13.3.1](#) and [Proposition 13.3.2](#).

- (a)  $\mathbf{A} = \text{diag}(1, -1)$ , which satisfies  $\mathbf{A}^\top = \mathbf{A}$ : symmetric. Not skew-symmetric, since  $\mathbf{A}^\top = -\mathbf{A}$  would need  $1 = -1$  in  $\mathbb{Q}$ . Not alternating, since  $\beta(\mathbf{e}_1, \mathbf{e}_1) = 1$ .
- (b)  $\mathbf{A} = \begin{pmatrix} 0 & 3 \\ -3 & 0 \end{pmatrix}$ , with  $\mathbf{A}^\top = -\mathbf{A}$  and zero diagonal: skew-symmetric and alternating. Not symmetric, since  $3 \neq -3$ .
- (c)  $\mathbf{A} = \begin{pmatrix} 0 & 1 \\ 2 & 0 \end{pmatrix}$ . Symmetry would need  $1 = 2$  and skewness would need  $1 = -2$ , both false in  $\mathbb{Q}$ ; so it is neither, and it is not alternating either, since an alternating form is skew-symmetric. (Its diagonal entries do vanish, so  $\beta(\mathbf{e}_i, \mathbf{e}_i) = 0$  for the two basis vectors — but  $\beta(\mathbf{1}, \mathbf{1}) = 1 + 2 = 3 \neq 0$ . Checking the basis vectors alone is not enough.)
- (d) The zero form is symmetric, skew-symmetric and alternating at once; its matrix is  $\mathbf{0}$ , which satisfies every condition. By the Quick check above it is the only form that is both symmetric and alternating over  $\mathbb{Q}$ .

**Exercise B2: Polarizing a quadratic form.**

Over  $\mathbb{R}$ , let  $q(x_1, x_2, x_3) = x_1^2 + 4x_1x_2 - 2x_2x_3 + 5x_3^2$ .

- (a) Find the symmetric matrix  $\mathbf{A}$  with  $q(\mathbf{x}) = \mathbf{x}^\top \mathbf{A} \mathbf{x}$ .
- (b) Compute  $\beta(\mathbf{e}_1 + \mathbf{e}_2, \mathbf{e}_3)$  directly from  $\mathbf{A}$ , and again from the polarization formula  $\frac{1}{2}(q(\mathbf{u} + \mathbf{v}) - q(\mathbf{u}) - q(\mathbf{v}))$ .

*Solution.*

- (a) The squared terms give  $a_{11} = 1$ ,  $a_{22} = 0$ ,  $a_{33} = 5$ ; the cross terms give  $a_{12} = a_{21} = 2$ ,  $a_{23} = a_{32} = -1$ ,  $a_{13} = a_{31} = 0$ . So

$$\mathbf{A} = \begin{pmatrix} 1 & 2 & 0 \\ 2 & 0 & -1 \\ 0 & -1 & 5 \end{pmatrix}.$$

- (b) Put  $\mathbf{u} = (1, 1, 0)$  and  $\mathbf{v} = (0, 0, 1)$ . Directly,  $\mathbf{A}\mathbf{v} = (0, -1, 5)$  and  $\mathbf{u}^\top(\mathbf{A}\mathbf{v}) = 0 - 1 = -1$ .

By polarization,  $q(\mathbf{u}) = 1 + 4 = 5$  and  $q(\mathbf{v}) = 5$ , while  $\mathbf{u} + \mathbf{v} = (1, 1, 1)$  gives  $q(\mathbf{u} + \mathbf{v}) =$

$1 + 4 - 2 + 5 = 8$ . Hence  $\frac{1}{2}(8 - 5 - 5) = -1$ . The two agree, as [Theorem 13.3.5](#) says they must.

### Exercise B3: Splitting a form.

Let  $\mathbf{A} = \begin{pmatrix} 1 & 5 \\ -1 & 4 \end{pmatrix}$  over  $\mathbb{Q}$ , and let  $\beta(\mathbf{x}, \mathbf{y}) = \mathbf{x}^\top \mathbf{A} \mathbf{y}$ . Write  $\beta = \beta_s + \beta_a$  as in [Theorem 13.3.4](#), giving the matrix of each part, and verify that  $\beta$  and  $\beta_s$  have the same quadratic form.

*Solution.* We have  $\mathbf{A}^\top = \begin{pmatrix} 1 & -1 \\ 5 & 4 \end{pmatrix}$ , so

$$\frac{1}{2}(\mathbf{A} + \mathbf{A}^\top) = \begin{pmatrix} 1 & 2 \\ 2 & 4 \end{pmatrix}, \quad \frac{1}{2}(\mathbf{A} - \mathbf{A}^\top) = \begin{pmatrix} 0 & 3 \\ -3 & 0 \end{pmatrix}.$$

The first is symmetric and the second is alternating, and they sum to  $\mathbf{A}$ . For the quadratic forms,  $\mathbf{x}^\top \mathbf{A} \mathbf{x} = x_1^2 + 5x_1x_2 - x_2x_1 + 4x_2^2 = x_1^2 + 4x_1x_2 + 4x_2^2$ , and the symmetric part gives  $x_1^2 + 2x_1x_2 + 2x_2x_1 + 4x_2^2$ , the same polynomial. The alternating part contributes  $3x_1x_2 - 3x_2x_1 = 0$ .

### 13.3.7.3 C. Going deeper

#### Exercise C1: Is this still true over $\mathbb{F}_2$ ?

Let  $V$  be a vector space over  $\mathbb{F}_2$  with  $\dim V = n \geq 1$ .

- Prove that every alternating bilinear form on  $V$  is symmetric.
- Prove that the alternating forms on  $V$  form a subspace of the space of all bilinear forms, and compute its dimension.
- Deduce that for  $n = 2$  there is exactly one non-zero alternating form on  $V$ , and write down its matrix.

*Hint:* for (b), use [Proposition 13.3.2](#).

*Solution.*

- Let  $\beta$  be alternating. By [Theorem 13.3.3](#) (a),  $\beta(\mathbf{u}, \mathbf{v}) = -\beta(\mathbf{v}, \mathbf{u})$ . Over  $\mathbb{F}_2$  we have  $-1 = 1$ , so  $-\beta(\mathbf{v}, \mathbf{u}) = \beta(\mathbf{v}, \mathbf{u})$  and  $\beta$  is symmetric.
- Fix a basis  $\mathcal{B}$ . The map  $\beta \mapsto [\beta]_{\mathcal{B}}$  is a linear bijection from bilinear forms to  $M_n(\mathbb{F}_2)$  ([Theorem 13.1.1](#)). By [Proposition 13.3.2](#) (b),  $\beta$  is alternating exactly when its matrix satisfies  $\mathbf{A}^\top = -\mathbf{A}$  with zero diagonal; over  $\mathbb{F}_2$  the first condition is  $\mathbf{A}^\top = \mathbf{A}$ , so the image is the set of symmetric matrices with zero diagonal. Such a matrix is determined freely by its entries  $a_{ij}$  with  $i < j$ , of which there are  $\binom{n}{2}$ . These conditions are linear, so the set of alternating forms is a subspace, of dimension  $n(n-1)/2$ .
- For  $n = 2$  the dimension is 1, so the space of alternating forms is  $\{\mathbf{0}, \beta\}$  for a single non-zero  $\beta$ , whose matrix must be  $\begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}$  — the only symmetric  $\mathbb{F}_2$ -matrix with zero diagonal other than  $\mathbf{0}$ . Concretely  $\beta(\mathbf{x}, \mathbf{y}) = x_1y_2 + x_2y_1$ , and indeed  $\beta(\mathbf{x}, \mathbf{x}) = 2x_1x_2 = 0$ .

**Exercise C2: How much of a form does  $q$  see?**

Let  $\text{char } F \neq 2$  and let  $\beta$  be any bilinear form on  $V$ , with  $q(\mathbf{v}) = \beta(\mathbf{v}, \mathbf{v})$ .

- (a) Prove that two bilinear forms have the same quadratic form if and only if their difference is alternating.
- (b) Deduce that the set of bilinear forms with a given quadratic form  $q$  contains exactly one symmetric form.
- (c) For  $\dim V = n$ , how many bilinear forms share a given quadratic form when  $F = \mathbb{F}_3$ ?

*Solution.*

- (a)  $(\Rightarrow)$  Suppose  $\beta_1(\mathbf{v}, \mathbf{v}) = \beta_2(\mathbf{v}, \mathbf{v})$  for all  $\mathbf{v}$ . Then  $\delta = \beta_1 - \beta_2$  is bilinear with  $\delta(\mathbf{v}, \mathbf{v}) = 0$  for all  $\mathbf{v}$ , which is [Definition 6.2.2](#) at  $k = 2$ .  
 $(\Leftarrow)$  If  $\delta = \beta_1 - \beta_2$  is alternating then  $\beta_1(\mathbf{v}, \mathbf{v}) - \beta_2(\mathbf{v}, \mathbf{v}) = \delta(\mathbf{v}, \mathbf{v}) = 0$  for every  $\mathbf{v}$ .
- (b) Existence: by [Theorem 13.3.4](#) write  $\beta = \beta_s + \beta_a$ ; then  $\beta_s$  is symmetric with the same quadratic form, since  $\beta_a(\mathbf{v}, \mathbf{v}) = 0$ . Uniqueness: two symmetric forms with the same  $q$  are equal by [Theorem 13.3.5](#).
- (c) By (a), the forms sharing  $q$  are exactly  $\beta_s + \beta_a$  with  $\beta_a$  alternating, so there are as many as there are alternating forms. Over a field with 3 elements, an alternating form corresponds by [Proposition 13.3.2](#) (b) to a matrix with  $\mathbf{A}^\top = -\mathbf{A}$  and zero diagonal, which is determined freely by the  $\binom{n}{2}$  entries above the diagonal. So there are  $3^{n(n-1)/2}$  of them.

**Exercise C3: A Hermitian form is a real symmetric form in disguise.**

Let  $\beta$  be a Hermitian form on  $\mathbb{C}^n$  with matrix  $\mathbf{A} = \mathbf{A}^*$  as in [Proposition 13.3.6](#), and regard  $\mathbb{C}^n$  as a real vector space of dimension  $2n$  by forgetting multiplication by  $i$ .

- (a) Prove that  $\tau(\mathbf{u}, \mathbf{v}) = \text{Re } \beta(\mathbf{u}, \mathbf{v})$  is a symmetric  $\mathbb{R}$ -bilinear form on that real space, with  $\tau(\mathbf{v}, \mathbf{v}) = \beta(\mathbf{v}, \mathbf{v})$ .
- (b) Prove that  $\omega(\mathbf{u}, \mathbf{v}) = \text{Im } \beta(\mathbf{u}, \mathbf{v})$  is an alternating  $\mathbb{R}$ -bilinear form.
- (c) For  $n = 1$  and  $\mathbf{A} = (1)$ , write both  $\tau$  and  $\omega$  as explicit forms on  $\mathbb{R}^2$ .

*Solution.*

- (a) Taking real parts of an  $\mathbb{R}$ -linear identity is  $\mathbb{R}$ -linear, and  $\beta$  is additive in each slot and commutes with real scalars in each slot (conjugation fixes  $\mathbb{R}$ ), so  $\tau$  is  $\mathbb{R}$ -bilinear. Symmetry:  $\beta(\mathbf{v}, \mathbf{u}) = \overline{\beta(\mathbf{u}, \mathbf{v})}$ , and conjugate numbers have equal real parts, so  $\tau(\mathbf{v}, \mathbf{u}) = \tau(\mathbf{u}, \mathbf{v})$ . Finally  $\beta(\mathbf{v}, \mathbf{v})$  is real, as noted after [Definition 13.3.3](#), so  $\tau(\mathbf{v}, \mathbf{v}) = \beta(\mathbf{v}, \mathbf{v})$ .
- (b)  $\mathbb{R}$ -bilinearity is the same argument with imaginary parts. Conjugate numbers have opposite imaginary parts, so  $\omega(\mathbf{v}, \mathbf{u}) = -\omega(\mathbf{u}, \mathbf{v})$ :  $\omega$  is skew-symmetric, and since  $\text{char } \mathbb{R} = 0 \neq 2$ , [Theorem 13.3.3](#) (b) makes it alternating. (Directly:  $\beta(\mathbf{v}, \mathbf{v})$  is real, so its imaginary part is 0.)
- (c) Here  $\beta(u, v) = u\bar{v}$  for  $u, v \in \mathbb{C}$ . Writing  $u = x_1 + ix_2$  and  $v = y_1 + iy_2$ , we get  $u\bar{v} = (x_1y_1 + x_2y_2) + i(x_2y_1 - x_1y_2)$ . So  $\tau$  is the dot product on  $\mathbb{R}^2$  and

$\omega(\mathbf{x}, \mathbf{y}) = x_2 y_1 - x_1 y_2$ , the negative of the signed area.

## 13.4 Completing the Square

Every symmetric matrix over  $\mathbb{R}$  can be diagonalized by an orthogonal matrix, which is Chapter 11's spectral theorem. That theorem needs  $\mathbb{R}$ , an inner product and the existence of eigenvalues. This section proves a weaker conclusion under far weaker hypotheses: over **any** field of characteristic  $\neq 2$ , with no inner product, no order and no eigenvalues available, every symmetric form still becomes diagonal after a change of basis — provided we ask only for congruence, not for similarity. The proof is one algebraic manipulation, completing the square, run by induction; and the induction has a case that a careless proof loses.

Throughout,  $F$  is a field with  $\text{char } F \neq 2$  (Definition 0.7.2),  $V$  is a finite-dimensional vector space over  $F$ ,  $\beta$  is a **symmetric** bilinear form on  $V$  (Definition 13.3.1), and  $q(\mathbf{v}) = \beta(\mathbf{v}, \mathbf{v})$  is its quadratic form (Definition 13.3.2).

### 13.4.1 Orthogonality without an inner product

Chapter 10 called two vectors orthogonal when their inner product vanished. The word survives the loss of positivity unchanged, because only the vanishing was ever used.

#### Definition 13.4.1 — Orthogonal Basis for a Form

Let  $\beta$  be a bilinear form on  $V$ . Vectors  $\mathbf{u}, \mathbf{v} \in V$  are **orthogonal with respect to  $\beta$**  if  $\beta(\mathbf{u}, \mathbf{v}) = 0$ . An ordered basis  $\mathcal{B} = (\mathbf{v}_1, \dots, \mathbf{v}_n)$  of  $V$  is an **orthogonal basis for  $\beta$**  if

$$\beta(\mathbf{v}_i, \mathbf{v}_j) = 0 \quad \text{whenever } i \neq j.$$

The definition is engineered to say exactly one thing about matrices: by Definition 13.1.2,  $\mathcal{B}$  is an orthogonal basis for  $\beta$  precisely when  $[\beta]_{\mathcal{B}}$  is a **diagonal** matrix, with  $q(\mathbf{v}_1), \dots, q(\mathbf{v}_n)$  down the diagonal. Two warnings come free. The vectors of such a basis are not required to satisfy  $q(\mathbf{v}_i) = 1$ , and over a general field they usually cannot be rescaled to. And a vector may be orthogonal to itself:  $q(\mathbf{v}_i) = 0$  is allowed, and for a degenerate form it is unavoidable.

What an orthogonal basis buys is that  $q$  becomes a sum of squares with no cross terms. If  $\mathbf{v} = c_1 \mathbf{v}_1 + \dots + c_n \mathbf{v}_n$  and  $d_i = q(\mathbf{v}_i)$ , then expanding by bilinearity and deleting the  $i \neq j$  terms,

$$q(\mathbf{v}) = \sum_{i=1}^n d_i c_i^2. \tag{13.4.1}$$

That is the “sum of squares” a reader has met when completing the square in two variables, and Eq. 13.4.1 is the reason the whole section is called what it is.

## 13.4.2 The theorem, and the case that breaks it

Here is the naive argument, and it is worth seeing it fail. To make  $q(\mathbf{v}) = ax_1^2 + 2bx_1x_2 + cx_2^2$  into a sum of squares, write

$$q = a\left(x_1 + \frac{b}{a}x_2\right)^2 + \left(c - \frac{b^2}{a}\right)x_2^2,$$

which is exactly a change of variable. The step divides by  $a = q(\mathbf{e}_1)$ . If  $a = 0$  there is nothing to complete, and one may hope to start with  $x_2$  instead — but  $q(\mathbf{x}) = 2x_1x_2$  has  $q(\mathbf{e}_1) = q(\mathbf{e}_2) = 0$ , and neither variable offers a foothold. The repair is to change the vector, not the variable:  $q(\mathbf{e}_1 + \mathbf{e}_2) = 2 \neq 0$ . That repair is a lemma, and it is where  $\text{char } F \neq 2$  is spent.

### Lemma 13.4.1 — A Non-Zero Form Is Non-Zero Somewhere on the Diagonal

Let  $\text{char } F \neq 2$  and let  $\beta$  be a symmetric bilinear form on  $V$  with  $\beta \neq 0$ . Then there exists  $\mathbf{v} \in V$  with  $q(\mathbf{v}) \neq 0$ .

*Proof.* Since  $\beta \neq 0$ , there are  $\mathbf{u}, \mathbf{w} \in V$  with  $a = \beta(\mathbf{u}, \mathbf{w}) \neq 0$ . If  $q(\mathbf{u}) \neq 0$  or  $q(\mathbf{w}) \neq 0$  we are done. So suppose  $q(\mathbf{u}) = q(\mathbf{w}) = 0$ . Expanding by bilinearity and using symmetry,

$$q(\mathbf{u} + \mathbf{w}) = q(\mathbf{u}) + 2\beta(\mathbf{u}, \mathbf{w}) + q(\mathbf{w}) = (1 + 1)a.$$

Since  $\text{char } F \neq 2$ , the element  $1 + 1$  is non-zero, and a product of two non-zero elements of a field is non-zero, so  $q(\mathbf{u} + \mathbf{w}) \neq 0$ . This proves the lemma. ■

Contrapositively: if  $q$  vanishes identically then  $\beta = 0$ . That is the case the induction must handle separately, and the lemma is what lets it be handled in one line.

### Theorem 13.4.2 — Diagonalization of a Symmetric Form

Let  $F$  be a field with  $\text{char } F \neq 2$ , let  $V$  be a finite-dimensional vector space over  $F$ , and let  $\beta$  be a symmetric bilinear form on  $V$ . Then  $V$  has an orthogonal basis for  $\beta$ .

Equivalently: every symmetric  $\mathbf{A} \in M_n(F)$  is congruent to a diagonal matrix, that is, there is an invertible  $\mathbf{P} \in M_n(F)$  with  $\mathbf{P}^\top \mathbf{A} \mathbf{P}$  diagonal.

*Idea.* Induction on  $\dim V$ . The step wants to peel off one basis vector and recurse on a complement, exactly as “peel off an eigenvector” does in Chapter 8 — but the complement here is the set of vectors orthogonal to the chosen one, not an invariant subspace, and it is available only if the chosen  $\mathbf{v}_1$  has  $q(\mathbf{v}_1) \neq 0$ . So the proof splits into two steps: ① if  $q$  vanishes everywhere, the form is zero and **every** basis is orthogonal; ② otherwise Lemma 13.4.1 supplies a  $\mathbf{v}_1$  with  $q(\mathbf{v}_1) \neq 0$ , the linear functional  $\mathbf{w} \mapsto \beta(\mathbf{v}_1, \mathbf{w})$  is non-zero, its kernel  $W$  is a hyperplane meeting  $\text{span}(\mathbf{v}_1)$  trivially, and the induction hypothesis applies to  $\beta$  restricted to  $W$ . Step ① is not a formality: without it the argument has no base for the recursion when the form degenerates, and the word “equivalently” in the statement would be unearned.



*Proof.* We induct on  $n = \dim V$ . If  $n = 0$  the empty basis is orthogonal, vacuously; if  $n = 1$  any basis  $(\mathbf{v}_1)$  is orthogonal, since there is no pair  $i \neq j$ . Let  $n \geq 2$  and suppose the result holds for all spaces of dimension  $n - 1$  over  $F$ .

**Step 1: the form vanishes on the diagonal.** Suppose  $q(\mathbf{v}) = 0$  for every  $\mathbf{v} \in V$ . By [Lemma 13.4.1](#), read contrapositively,  $\beta = 0$ . Then  $\beta(\mathbf{v}_i, \mathbf{v}_j) = 0$  for every pair of vectors whatsoever, so **any** basis of  $V$  is an orthogonal basis for  $\beta$ , and its matrix is  $\mathbf{0}$ , which is diagonal.

**Step 2: the form does not vanish on the diagonal.** Otherwise there is  $\mathbf{v}_1 \in V$  with  $c = q(\mathbf{v}_1) \neq 0$ ; in particular  $\mathbf{v}_1 \neq \mathbf{0}$ . Define

$$\varphi: V \rightarrow F, \quad \varphi(\mathbf{w}) = \beta(\mathbf{v}_1, \mathbf{w}).$$

Then  $\varphi$  is linear, because  $\beta$  is linear in its second argument, and  $\varphi(\mathbf{v}_1) = c \neq 0$ , so  $\varphi \neq 0$ . A non-zero functional is surjective onto  $F$  ([Proposition 4.1.2](#)), so  $\text{rank } \varphi = 1$  and, by the Rank–Nullity Theorem ([Theorem 3.3.1](#)),  $W = \ker \varphi$  has  $\dim W = n - 1$ .

Moreover  $\text{span}(\mathbf{v}_1) \cap W = \{\mathbf{0}\}$ : if  $a\mathbf{v}_1 \in W$  then  $0 = \varphi(a\mathbf{v}_1) = ac$ , and  $c \neq 0$  forces  $a = 0$ . So the sum  $\text{span}(\mathbf{v}_1) + W$  is direct, and  $\dim(\text{span}(\mathbf{v}_1) + W) = 1 + (n - 1) = n$ , both by [Theorem 1.7.4](#). A subspace of  $V$  of dimension  $\dim V$  is all of  $V$  ([Theorem 1.6.8](#)), so

$$V = \text{span}(\mathbf{v}_1) \oplus W.$$

The restriction of  $\beta$  to  $W \times W$  is a bilinear form on  $W$ , and it is symmetric because  $\beta$  is. As  $\dim W = n - 1$ , the induction hypothesis provides an orthogonal basis  $(\mathbf{v}_2, \dots, \mathbf{v}_n)$  of  $W$  for that restriction.

Concatenating bases of the summands of a direct sum gives a basis of it ([Theorem 1.7.5](#), (a)  $\Leftrightarrow$  (d)), so  $\mathcal{B} = (\mathbf{v}_1, \mathbf{v}_2, \dots, \mathbf{v}_n)$  is a basis of  $V$ . It is orthogonal for  $\beta$ : for  $j \geq 2$  we have  $\mathbf{v}_j \in W = \ker \varphi$ , so  $\beta(\mathbf{v}_1, \mathbf{v}_j) = \varphi(\mathbf{v}_j) = 0$ , and  $\beta(\mathbf{v}_j, \mathbf{v}_1) = 0$  by symmetry; and for  $2 \leq i \neq j$  the value  $\beta(\mathbf{v}_i, \mathbf{v}_j)$  is 0 by the choice of the basis of  $W$ . This completes the induction.

For the matrix statement, let  $\mathbf{A} \in M_n(F)$  be symmetric and let  $\beta(\mathbf{x}, \mathbf{y}) = \mathbf{x}^\top \mathbf{A} \mathbf{y}$  on  $V = F^n$ , which is symmetric by [Proposition 13.3.1](#) and has  $[\beta]_{\mathcal{E}} = \mathbf{A}$  in the standard basis  $\mathcal{E}$ . Take an orthogonal basis  $\mathcal{B}$  as above and let  $\mathbf{P} = [\text{id}]_{\mathcal{B}}^{\mathcal{E}}$ , the matrix whose columns are the vectors of  $\mathcal{B}$ , which is invertible. By [Theorem 13.2.1](#),  $[\beta]_{\mathcal{B}} = \mathbf{P}^\top \mathbf{A} \mathbf{P}$ , and this matrix is diagonal. This proves the theorem. ■

Two hypotheses did visible work. Symmetry was used twice: to make  $\beta(\mathbf{v}_j, \mathbf{v}_1)$  vanish along with  $\beta(\mathbf{v}_1, \mathbf{v}_j)$ , and to keep the restriction to  $W$  symmetric so the induction could run. The characteristic hypothesis was used exactly once, inside [Lemma 13.4.1](#) — and that single use is essential: Exercise C2 below exhibits a symmetric matrix over  $\mathbb{F}_2$  that is congruent to no diagonal matrix at all.

**Warning.** **Diagonal does not mean “the diagonal entries are eigenvalues”.** The matrix  $\mathbf{P}^\top \mathbf{A} \mathbf{P}$  of [Theorem 13.4.2](#) is congruent to  $\mathbf{A}$ , not similar to it, and its entries are the values  $q(\mathbf{v}_i)$  at whatever basis the algorithm happened to produce. Over  $\mathbb{R}$  they can be scaled at will: replacing  $\mathbf{v}_i$  by  $t\mathbf{v}_i$  replaces  $d_i$  by  $t^2 d_i$ . There is no canonical list of numbers here. The next section finds what there is.

*Quick check.* In [Theorem 13.4.2](#), how many of the diagonal entries  $d_i$  are zero?

*Solution.* Exactly  $n - \text{rank } \mathbf{A}$  of them. A diagonal matrix has rank equal to its number of non-zero entries, and congruence preserves rank ([Theorem 13.2.3](#)), so the count is the same for every orthogonal basis even though the individual  $d_i$  are not. In the language of Section 1, those  $\mathbf{v}_i$  with  $d_i = 0$  span the radical  $\text{rad}(\beta)$  of [Definition 13.1.3](#), since in an orthogonal basis such a  $\mathbf{v}_i$  pairs to zero with every basis vector; and  $\beta$  is non-degenerate ([Definition 13.1.4](#)) exactly when no  $d_i$  is zero.

### 13.4.3 The algorithm: symmetric row and column operations

The proof is constructive, but running it by hand means chasing kernels of functionals. There is a bookkeeping device that does the same work on the matrix directly, and it is the practical content of this section.

The key observation is that a congruence by an elementary matrix is a **pair** of operations. Let  $\mathbf{E}$  be an elementary matrix ([Definition 2.4.1](#)). Right multiplication  $\mathbf{A}\mathbf{E}$  performs a column operation, and left multiplication  $\mathbf{E}^\top \mathbf{A}$  performs the corresponding row operation. So  $\mathbf{A} \mapsto \mathbf{E}^\top \mathbf{A} \mathbf{E}$  performs both, and the result is congruent to  $\mathbf{A}$  and again symmetric. Doing the operations in pairs is exactly what keeps the matrix symmetric, and that is the whole idea.

*Algorithm.* To diagonalize a symmetric  $\mathbf{A} \in M_n(F)$  by congruence, write  $\mathbf{A}$  and  $\mathbf{I}_n$  side by side and repeat, for  $i = 1, 2, \dots, n$  in turn:

- **If**  $a_{ii} = 0$  **but some**  $a_{jj} \neq 0$  **with**  $j > i$ , swap rows  $i, j$  and then columns  $i, j$ .
- **If**  $a_{ii} = 0$  **and every**  $a_{jj} = 0$  **for**  $j \geq i$ , look for an entry  $a_{ij} \neq 0$  with  $j > i$ . If there is none, then row  $i$  and column  $i$  are already zero in every position, so move on to  $i + 1$ . If there is one, add row  $j$  to row  $i$ , then column  $j$  to column  $i$ ; the new  $a_{ii}$  is  $2a_{ij} \neq 0$ .
- **Now**  $a_{ii} \neq 0$ . For each  $j > i$  with  $a_{ij} \neq 0$ , subtract  $(a_{ij}/a_{ii})$  times row  $i$  from row  $j$ , then  $(a_{ij}/a_{ii})$  times column  $i$  from column  $j$ .

Over a field where division is inconvenient —  $\mathbb{Q}$  or  $\mathbb{Z}$ -valued data by hand — the clearing step may instead be done as row  $j \mapsto a_{ii} \text{ row } j - a_{ij} \text{ row } i$ , with the matching column operation. That is the same move composed with scaling row and column  $j$  by  $a_{ii} \neq 0$ , which is itself a congruence, so it is legitimate; it keeps the entries integral and changes the diagonal result only by square factors. The worked example below takes this route.

Perform only the **column** half of each operation on the matrix standing to the right. When  $\mathbf{A}$  has become diagonal, that matrix is  $\mathbf{P}$ , and  $\mathbf{P}^\top \mathbf{A} \mathbf{P}$  is the diagonal result.

The second bullet is [Lemma 13.4.1](#) made mechanical, and the factor  $2 = 1 + 1$  in it is where the algorithm would stop in characteristic 2. The right-hand matrix works because applying the column operations  $\mathbf{E}_1, \mathbf{E}_2, \dots, \mathbf{E}_k$  in order turns  $\mathbf{I}$  into  $\mathbf{E}_1 \mathbf{E}_2 \cdots \mathbf{E}_k$ , which is the product whose transpose and self sandwich  $\mathbf{A}$ .

**Example. (Diagonalizing a symmetric matrix with zero diagonal)**

Over  $\mathbb{Q}$ , diagonalize

$$\mathbf{A} = \begin{pmatrix} 0 & 1 & 1 \\ 1 & 0 & 1 \\ 1 & 1 & 0 \end{pmatrix}$$

by congruence, producing an invertible  $\mathbf{P}$  with  $\mathbf{P}^\top \mathbf{A} \mathbf{P}$  diagonal.

*Solution.* The form is  $q(\mathbf{x}) = 2x_1x_2 + 2x_1x_3 + 2x_2x_3$ : every diagonal entry is 0, so there is no square to complete yet, and the second bullet applies.

*Step 1 (create a pivot).* Add row 2 to row 1, then column 2 to column 1. The matrix becomes

$$\begin{pmatrix} 1 & 1 & 2 \\ 1 & 0 & 1 \\ 1 & 1 & 0 \end{pmatrix} \longrightarrow \begin{pmatrix} 2 & 1 & 2 \\ 1 & 0 & 1 \\ 2 & 1 & 0 \end{pmatrix},$$

the left matrix after the row operation and the right one after the matching column operation. The new corner entry is  $q(\mathbf{e}_1 + \mathbf{e}_2) = 2$ .

*Step 2 (clear the first row and column, in two paired moves).* Rather than divide by 2, use the operation “row 2  $\mapsto 2 \cdot$  row 2 – row 1”, which is a scaling followed by an addition, hence a congruence by a product of two elementary matrices, and legitimate because  $2 \neq 0$  in  $\mathbb{Q}$ . Apply it and its column partner. This keeps every entry an integer:

$$\begin{pmatrix} 2 & 1 & 2 \\ 0 & -1 & 0 \\ 2 & 1 & 0 \end{pmatrix} \longrightarrow \begin{pmatrix} 2 & 0 & 2 \\ 0 & -2 & 0 \\ 2 & 0 & 0 \end{pmatrix}.$$

Then “row 3  $\mapsto$  row 3 – row 1” and its column partner:

$$\begin{pmatrix} 2 & 0 & 2 \\ 0 & -2 & 0 \\ 0 & 0 & -2 \end{pmatrix} \longrightarrow \begin{pmatrix} 2 & 0 & 0 \\ 0 & -2 & 0 \\ 0 & 0 & -2 \end{pmatrix}.$$

*The transforming matrix.* Applying the three **column** operations to  $\mathbf{I}_3$  in the same order gives

$$\mathbf{I}_3 \rightarrow \begin{pmatrix} 1 & 0 & 0 \\ 1 & 1 & 0 \\ 0 & 0 & 1 \end{pmatrix} \rightarrow \begin{pmatrix} 1 & -1 & 0 \\ 1 & 1 & 0 \\ 0 & 0 & 1 \end{pmatrix} \rightarrow \begin{pmatrix} 1 & -1 & -1 \\ 1 & 1 & -1 \\ 0 & 0 & 1 \end{pmatrix} = \mathbf{P}.$$

Here  $\det \mathbf{P} = 2 \neq 0$ , so  $\mathbf{P}$  is invertible, and

$$\mathbf{P}^\top \mathbf{A} \mathbf{P} = \text{diag}(2, -2, -2).$$

The check is one column at a time: with  $\mathbf{p}_1 = (1, 1, 0)$ ,  $\mathbf{p}_2 = (-1, 1, 0)$  and  $\mathbf{p}_3 = (-1, -1, 1)$ , we get  $q(\mathbf{p}_1) = 2$ ,  $q(\mathbf{p}_2) = -2$  and  $q(\mathbf{p}_3) = 2 - 2 - 2 = -2$ , while  $\beta(\mathbf{p}_1, \mathbf{p}_2) = \mathbf{p}_1^\top \mathbf{A} \mathbf{p}_2 = (1, 1, 0) \cdot (1, -1, 0) = 0$  and similarly for the other two pairs.

Two observations about that example. The determinants are  $\det \mathbf{A} = 2$  and  $\det(\mathbf{P}^\top \mathbf{A} \mathbf{P}) = 8$ , and they differ by  $(\det \mathbf{P})^2 = 4$ : a congruence multiplies the determinant by a square, which is the discriminant statement of Section 2. And the eigenvalues of  $\mathbf{A}$  are  $2, -1, -1$ , whose signs are  $+, -, -$  — the same pattern of signs as  $2, -2, -2$ , with none of the same numbers. That

coincidence is Section 5.

### 13.4.4 Over the complex numbers, and over a general field

The diagonal entries produced by the algorithm are not canonical, but there is a precise statement of how much freedom remains.

#### Proposition 13.4.3 — Diagonal Entries Are Defined Only up to Squares

Let  $\mathcal{B} = (\mathbf{v}_1, \dots, \mathbf{v}_n)$  be an orthogonal basis for  $\beta$  with  $d_i = q(\mathbf{v}_i)$ , and let  $t_1, \dots, t_n \in F$  be non-zero. Then  $(t_1\mathbf{v}_1, \dots, t_n\mathbf{v}_n)$  is again an orthogonal basis for  $\beta$ , with diagonal entries  $t_i^2 d_i$ .

*Proof.* Scaling basis vectors by non-zero scalars again gives a basis. For  $i \neq j$ , bilinearity gives  $\beta(t_i\mathbf{v}_i, t_j\mathbf{v}_j) = t_i t_j \beta(\mathbf{v}_i, \mathbf{v}_j) = 0$ , so the new basis is orthogonal; and  $q(t_i\mathbf{v}_i) = t_i^2 q(\mathbf{v}_i) = t_i^2 d_i$ . ■

So each non-zero  $d_i$  can be changed to any other element of its coset in the group  $F^\times / (F^\times)^2$  of **square classes**, and no further, by this move. How much that costs depends entirely on the arithmetic of  $F$ :

- Over  $\mathbb{C}$ , every element is a square, so there is exactly one non-zero square class and every non-zero  $d_i$  can be turned into 1.
- Over  $\mathbb{R}$ , the squares are the positive numbers, so there are two classes and every non-zero  $d_i$  can be turned into 1 or  $-1$ , according to its sign — the subject of the next section.
- Over  $\mathbb{Q}$  there are infinitely many classes ( $2, 3, 5, \dots$  are pairwise inequivalent), so the 1-dimensional forms  $q(x) = x^2$  and  $q(x) = 2x^2$  are already non-congruent.
- Over  $\mathbb{F}_p$  with  $p$  odd there are exactly two classes, as for  $\mathbb{R}$ , but “which class” is not a matter of sign.

The complex case collapses completely.

#### Corollary 13.4.4 — Classification of Complex Symmetric Matrices by Rank

Let  $\mathbf{A} \in M_n(\mathbb{C})$  be symmetric of rank  $r$ . Then  $\mathbf{A}$  is congruent to

$$\mathbf{I}_r \oplus \mathbf{0}_{n-r} = \text{diag}(\underbrace{1, \dots, 1}_r, 0, \dots, 0).$$

Consequently two symmetric matrices in  $M_n(\mathbb{C})$  are congruent if and only if they have the same rank, and there are exactly  $n + 1$  congruence classes.

*Proof.* By [Theorem 13.4.2](#) (with  $\text{char } \mathbb{C} = 0$ ) there is an invertible  $\mathbf{P}_1$  with  $\mathbf{P}_1^\top \mathbf{A} \mathbf{P}_1 = \mathbf{D} = \text{diag}(d_1, \dots, d_n)$  diagonal. Congruence preserves rank ([Theorem 13.2.3](#)), and the rank of a diagonal matrix is its number of non-zero entries, so exactly  $r$  of the  $d_i$  are non-zero. Permuting the basis vectors is a congruence by a permutation matrix  $\mathbf{P}_2$ , so we may assume  $d_1, \dots, d_r \neq 0$  and  $d_{r+1} = \dots = d_n = 0$ .

Every non-zero complex number has a square root, so choose  $s_i \in \mathbb{C}$  with  $s_i^2 = d_i$  for  $i \leq r$ , and set  $t_i = 1/s_i$  for  $i \leq r$  and  $t_i = 1$  otherwise. By [Proposition 13.4.3](#) the congruence by  $\mathbf{P}_3 = \text{diag}(t_1, \dots, t_n)$  replaces  $d_i$  by  $t_i^2 d_i$ , which is 1 for  $i \leq r$  and 0 otherwise. So  $\mathbf{P}^\top \mathbf{A} \mathbf{P} = \mathbf{I}_r \oplus \mathbf{0}_{n-r}$  with  $\mathbf{P} = \mathbf{P}_1 \mathbf{P}_2 \mathbf{P}_3$ .

For the classification: congruent matrices have equal rank by [Theorem 13.2.3](#), and matrices of equal rank are both congruent to the same normal form, hence to each other by [Proposition 13.2.2](#). The possible ranks are  $0, 1, \dots, n$ , giving  $n + 1$  classes. This proves the corollary. ■

Contrast this with similarity, where complex symmetric matrices are not classified by anything so simple: the matrix  $\begin{pmatrix} 1 & i \\ i & -1 \end{pmatrix}$  is symmetric, non-zero, and satisfies  $\mathbf{A}^2 = \mathbf{0}$ , so it is not even diagonalizable as an operator. Congruence is a coarser relation than similarity on symmetric matrices, and over  $\mathbb{C}$  it is coarse enough to be finite.

## 13.4.5 Exercises

### 13.4.5.1 A. Check your understanding

#### Exercise A1.

- Define what it means for an ordered basis to be an **orthogonal basis for a bilinear form**  $\beta$ , and say what that means about  $[\beta]_{\mathcal{B}}$ .
- State [Theorem 13.4.2](#) with all of its hypotheses.
- Which step of the proof uses  $\text{char } F \neq 2$ , and what exactly is divided by?
- Determine whether the following statement is correct, and justify your answer: if  $\mathbf{P}^\top \mathbf{A} \mathbf{P} = \text{diag}(d_1, \dots, d_n)$ , then the  $d_i$  are the eigenvalues of  $\mathbf{A}$ .
- Describe the pair of operations that a congruence by an elementary matrix performs.

*Solution.*

- $\mathcal{B} = (\mathbf{v}_1, \dots, \mathbf{v}_n)$  is orthogonal for  $\beta$  if  $\beta(\mathbf{v}_i, \mathbf{v}_j) = 0$  whenever  $i \neq j$ . Equivalently  $[\beta]_{\mathcal{B}}$  is diagonal, with entries  $q(\mathbf{v}_i)$ .
- If  $\text{char } F \neq 2$ ,  $V$  is finite-dimensional over  $F$  and  $\beta$  is a symmetric bilinear form on  $V$ , then  $V$  has an orthogonal basis for  $\beta$ ; equivalently every symmetric  $\mathbf{A} \in M_n(F)$  is congruent to a diagonal matrix.
- [Lemma 13.4.1](#), inside Step 2 of the proof. It produces  $q(\mathbf{u} + \mathbf{w}) = (1 + 1)\beta(\mathbf{u}, \mathbf{w})$  and needs  $1 + 1 \neq 0$  to conclude that this is non-zero. Equivalently, the algorithm's repair move multiplies by 2, so it needs  $2 \neq 0$ .

- (d) Incorrect. The relation is congruence, not similarity. In [Example \(Diagonalizing a symmetric matrix with zero diagonal\)](#) the diagonal entries are 2, −2, −2 while the eigenvalues are 2, −1, −1. Only the signs matched there, and Section 5 explains why.
- (e) A column operation and the matching row operation:  $\mathbf{A} \mapsto \mathbf{E}^\top \mathbf{A} \mathbf{E}$ . Performing both keeps the matrix symmetric and keeps it in the same congruence class.

### 13.4.5.2 B. Practice

#### Exercise B1: Symmetric elimination.

Over  $\mathbb{Q}$ , let

$$\mathbf{A} = \begin{pmatrix} 1 & 2 & 0 \\ 2 & 1 & 3 \\ 0 & 3 & 1 \end{pmatrix}.$$

Diagonalize  $\mathbf{A}$  by paired row and column operations, exhibiting  $\mathbf{P}$  and  $\mathbf{P}^\top \mathbf{A} \mathbf{P}$ .

*Solution.* Here  $a_{11} = 1 \neq 0$ , so the pivot is already available.

*Clear the first row and column.* Subtract  $2 \times$  row 1 from row 2, then  $2 \times$  column 1 from column 2. The entry  $a_{13}$  is already 0, so nothing else is needed in this round:

$$\begin{pmatrix} 1 & 2 & 0 \\ 0 & -3 & 3 \\ 0 & 3 & 1 \end{pmatrix} \longrightarrow \begin{pmatrix} 1 & 0 & 0 \\ 0 & -3 & 3 \\ 0 & 3 & 1 \end{pmatrix}.$$

*Clear the second row and column.* The pivot is  $-3$ , and  $a_{23} = 3$ , so add row 2 to row 3 and then column 2 to column 3:

$$\begin{pmatrix} 1 & 0 & 0 \\ 0 & -3 & 3 \\ 0 & 0 & 4 \end{pmatrix} \longrightarrow \begin{pmatrix} 1 & 0 & 0 \\ 0 & -3 & 0 \\ 0 & 0 & 4 \end{pmatrix}.$$

*The transforming matrix.* Applying the two column operations to  $\mathbf{I}_3$  gives

$$\mathbf{P} = \begin{pmatrix} 1 & -2 & -2 \\ 0 & 1 & 1 \\ 0 & 0 & 1 \end{pmatrix}, \quad \mathbf{P}^\top \mathbf{A} \mathbf{P} = \text{diag}(1, -3, 4).$$

As a check,  $\det \mathbf{P} = 1$  and  $\det \mathbf{A} = -12 = \det \text{diag}(1, -3, 4)$ .

#### Exercise B2: No pivot on the diagonal.

Over  $\mathbb{Q}$ , let  $\mathbf{A} = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}$  and  $q(\mathbf{x}) = 2x_1x_2$ .

- (a) Find a vector  $\mathbf{v}$  with  $q(\mathbf{v}) \neq 0$ , as in [Lemma 13.4.1](#).
- (b) Diagonalize  $\mathbf{A}$  by paired operations and exhibit  $\mathbf{P}$ .
- (c) Write  $q$  as a difference of two squares in the new coordinates.

*Solution.*

- (a) Both  $\mathbf{e}_1$  and  $\mathbf{e}_2$  give  $q = 0$ , so take  $\mathbf{v} = \mathbf{e}_1 + \mathbf{e}_2$ :  $q(\mathbf{v}) = 2 \neq 0$ .  
 (b) Add row 2 to row 1, then column 2 to column 1, reaching  $\begin{pmatrix} 2 & 1 \\ 1 & 0 \end{pmatrix}$ . Then apply “row  $2 \mapsto 2 \cdot \text{row } 2 - \text{row } 1$ ” and its column partner, reaching  $\text{diag}(2, -2)$ . Applying the same two column operations to  $\mathbf{I}_2$  gives

$$\mathbf{P} = \begin{pmatrix} 1 & -1 \\ 1 & 1 \end{pmatrix}, \quad \mathbf{P}^\top \mathbf{A} \mathbf{P} = \text{diag}(2, -2).$$

Check directly:  $\mathbf{P}^\top \mathbf{A} = \begin{pmatrix} 1 & 1 \\ 1 & -1 \end{pmatrix}$  and multiplying by  $\mathbf{P}$  gives  $\begin{pmatrix} 2 & 0 \\ 0 & -2 \end{pmatrix}$ .

- (c) With  $\mathbf{x} = \mathbf{P}\mathbf{y}$ , that is  $x_1 = y_1 - y_2$  and  $x_2 = y_1 + y_2$ , we get  $q = 2(y_1 - y_2)(y_1 + y_2) = 2y_1^2 - 2y_2^2$ . This is the familiar identity  $ab = \left(\frac{a+b}{2}\right)^2 - \left(\frac{a-b}{2}\right)^2$  read as a change of variable, and it needs 2 to be invertible.

### Exercise B3: Reading off the rank.

Let  $\mathbf{A} \in M_4(\mathbb{C})$  be symmetric with  $\text{rank } \mathbf{A} = 3$ . Write down the normal form of  $\mathbf{A}$  under congruence, and say how many congruence classes of symmetric matrices there are in  $M_4(\mathbb{C})$ .

*Solution.* By [Corollary 13.4.4](#),  $\mathbf{A}$  is congruent to  $\mathbf{I}_3 \oplus \mathbf{0}_1 = \text{diag}(1, 1, 1, 0)$ . Since rank is a complete invariant there, the classes correspond to the possible ranks 0, 1, 2, 3, 4, so there are 5 of them.

### 13.4.5.3 C. Going deeper

#### Exercise C1: Square classes over a finite field.

Work over  $\mathbb{F}_5$ , whose non-zero squares are 1 and 4.

- (a) Prove that the  $1 \times 1$  matrices (1) and (2) are not congruent over  $\mathbb{F}_5$ .  
 (b) Prove that  $\mathbf{I}_2$  and  $2\mathbf{I}_2$  are congruent over  $\mathbb{F}_5$ , by exhibiting a  $\mathbf{P}$ .  
 (c) Explain why (a) and (b) are consistent with [Proposition 13.4.3](#).

*Hint:* for (b), look for two vectors  $\mathbf{p}_1, \mathbf{p}_2$  with  $\mathbf{p}_i \cdot \mathbf{p}_i = 2$  and  $\mathbf{p}_1 \cdot \mathbf{p}_2 = 0$ .

*Solution.*

- (a) A congruence of  $1 \times 1$  matrices is  $(a) \mapsto (p^2 a)$  with  $p \neq 0$ . So (1) and (2) are congruent exactly when  $2 = p^2$  for some  $p \in \mathbb{F}_5^\times$ . The squares of 1, 2, 3, 4 are 1, 4, 4, 1, so 2 is not a square and the two are not congruent.  
 (b) Take  $\mathbf{p}_1 = (1, 1)$  and  $\mathbf{p}_2 = (1, 4)$ . Then  $\mathbf{p}_1 \cdot \mathbf{p}_1 = 2$ ,  $\mathbf{p}_2 \cdot \mathbf{p}_2 = 1 + 16 = 17 = 2$  and  $\mathbf{p}_1 \cdot \mathbf{p}_2 = 1 + 4 = 5 = 0$  in  $\mathbb{F}_5$ . With

$$\mathbf{P} = \begin{pmatrix} 1 & 1 \\ 1 & 4 \end{pmatrix}, \quad \det \mathbf{P} = 4 - 1 = 3 \neq 0,$$

we get  $\mathbf{P}^\top \mathbf{I}_2 \mathbf{P} = \mathbf{P}^\top \mathbf{P} = \text{diag}(2, 2) = 2\mathbf{I}_2$ .

- (c) [Proposition 13.4.3](#) only says what **rescaling a single basis vector** can do, and that move cannot change a square class. Part (b) does not contradict it, because the congruence there is not diagonal: it mixes the two coordinates, and both diagonal entries change square class at once. What is invariant is the class of their product, the **determinant**:  $\det \mathbf{I}_2 = 1$  and  $\det(2\mathbf{I}_2) = 4$  differ by  $4 = 2^2$ , a square.

**Exercise C2: Is this still true over  $\mathbb{F}_2$ ?**

Let  $\mathbf{A} = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}$  over  $\mathbb{F}_2$ , and let  $\beta(\mathbf{x}, \mathbf{y}) = \mathbf{x}^\top \mathbf{A} \mathbf{y}$  on  $\mathbb{F}_2^2$ .

- Prove that  $\beta$  is alternating.
- Prove that if  $\mathbf{P} \in M_2(\mathbb{F}_2)$  is invertible, then every diagonal entry of  $\mathbf{P}^\top \mathbf{A} \mathbf{P}$  is 0.
- Deduce that  $\mathbf{A}$  is congruent to **no** diagonal matrix except the zero matrix, and hence that [Theorem 13.4.2](#) fails over  $\mathbb{F}_2$ .

*Solution.*

- By [Proposition 13.3.2](#) (b): over  $\mathbb{F}_2$  we have  $-\mathbf{A} = \mathbf{A}$ , so  $\mathbf{A}^\top = \mathbf{A} = -\mathbf{A}$ , and the diagonal entries of  $\mathbf{A}$  are 0. (Directly:  $\beta(\mathbf{x}, \mathbf{x}) = 2x_1x_2 = 0$ .)
- The  $i$ -th diagonal entry of  $\mathbf{P}^\top \mathbf{A} \mathbf{P}$  is  $\mathbf{p}_i^\top \mathbf{A} \mathbf{p}_i = q(\mathbf{p}_i)$ , where  $\mathbf{p}_i$  is the  $i$ -th column of  $\mathbf{P}$ . By (a),  $q$  vanishes identically, so every diagonal entry is 0.
- A diagonal matrix congruent to  $\mathbf{A}$  has all diagonal entries 0 by (b), hence is the zero matrix. But congruence preserves rank ([Theorem 13.2.3](#)) and  $\text{rank } \mathbf{A} = 2 \neq 0 = \text{rank } \mathbf{0}$ . So no diagonal matrix is congruent to  $\mathbf{A}$ . Since  $\mathbf{A}$  is symmetric over  $\mathbb{F}_2$ , the conclusion of [Theorem 13.4.2](#) fails there, and the hypothesis  $\text{char } F \neq 2$  cannot be dropped.

**Exercise C3: Diagonalizing the restriction.**

Let  $\text{char } F \neq 2$ , let  $\beta$  be a symmetric bilinear form on  $V$ , and let  $U \subseteq V$  be a subspace.

- Prove that  $U$  has an orthogonal basis for the restriction of  $\beta$  to  $U$ .
- Give an example, with  $\dim V = 2$ , in which  $\beta$  is non-degenerate but its restriction to some 1-dimensional  $U$  is the zero form.
- Deduce that an orthogonal basis of  $U$  cannot always be extended to an orthogonal basis of  $V$  by adding vectors of  $U$  alone, and say in one sentence what [Theorem 13.4.2](#) does guarantee instead.

*Solution.*

- The restriction  $\beta|_{U \times U}$  is bilinear, since it is the same function on a smaller domain, and symmetric for the same reason.  $U$  is finite-dimensional, being a subspace of a finite-dimensional space ([Theorem 1.6.7](#)). So [Theorem 13.4.2](#) applies to  $(U, \beta|_{U \times U})$  directly.
- Over  $\mathbb{Q}$  take  $\beta(\mathbf{x}, \mathbf{y}) = x_1y_2 + x_2y_1$  on  $\mathbb{Q}^2$ , with matrix  $\begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}$ , which is invertible, so  $\beta$  is non-degenerate. Put  $U = \text{span}(\mathbf{e}_1)$ . Then  $\beta(a\mathbf{e}_1, b\mathbf{e}_1) = 0$  for all  $a, b$ , so the restriction is the zero form even though  $\beta$  is not.



- (c) In (b) the single vector  $\mathbf{e}_1$  is an orthogonal basis of  $U$ , but no vector of  $U$  can be added to it, and any orthogonal basis of  $V$  for this  $\beta$  must consist of vectors  $\mathbf{v}$  with  $q(\mathbf{v}) \neq 0$  (its matrix is diagonal and invertible), so  $\mathbf{e}_1$  belongs to none of them. What the theorem guarantees is only the existence of *some* orthogonal basis of  $V$ , built from scratch; it promises nothing about extending a given orthogonal list, and Section 8 is where a correct extension statement finally appears.

## 13.5 Sylvester's Law of Inertia

The previous section produced a diagonal matrix and then took it away again: the diagonal entries are not invariants, because rescaling the  $i$ -th basis vector by  $t$  multiplies the  $i$ -th entry by  $t^2$ . Over  $\mathbb{R}$  the squares are exactly the positive numbers, so rescaling can change the size of an entry but never its sign. This section proves that the signs are all that survives, and that they survive completely: the number of positive, negative and zero entries is the same for every diagonalization. That is the classification of real symmetric matrices up to congruence, and it is also the missing link between congruence and the spectral theorem.

**Throughout this section the field is  $\mathbb{R}$ .** Every form is a symmetric bilinear form on a finite-dimensional real vector space  $V$ , and every matrix is a real symmetric matrix. The words “positive” and “negative” have no meaning over a general field, and almost nothing in this section survives the move to one; [Corollary 13.4.4](#) already showed that over  $\mathbb{C}$  the answer is different, since rank alone classifies there.

### 13.5.1 Normalizing the diagonal

Start from [Theorem 13.4.2](#), which applies because  $\text{char } \mathbb{R} = 0$ . It gives an orthogonal basis  $(\mathbf{v}_1, \dots, \mathbf{v}_n)$  for  $\beta$ , with  $q(\mathbf{v}_i) = d_i$ . Over  $\mathbb{R}$  we may go one step further and scale.

By [Proposition 13.4.3](#), replacing  $\mathbf{v}_i$  by  $t_i \mathbf{v}_i$  replaces  $d_i$  by  $t_i^2 d_i$ . Choosing  $t_i = 1/\sqrt{|d_i|}$  when  $d_i \neq 0$ , and  $t_i = 1$  when  $d_i = 0$ , turns every diagonal entry into 1,  $-1$  or 0. A permutation of the basis then sorts them, so after a further congruence by a permutation matrix the matrix of  $\beta$  is

$$\mathbf{I}_{n_+} \oplus (-\mathbf{I}_{n_-}) \oplus \mathbf{0}_{n_0}, \quad (13.5.1)$$

where  $n_+$  counts the positive  $d_i$ ,  $n_-$  the negative ones and  $n_0$  the zeros, and  $n_+ + n_- + n_0 = n$ . Nothing yet says that these three counts do not depend on the orthogonal basis we started from. That is the theorem.

*Quick check.* One of the three counts is already known to be an invariant, by a result of Section 2. Which one, and why?

*Solution.* The count  $n_0$ . A diagonal matrix has rank equal to its number of non-zero entries, so  $n_+ + n_- = \text{rank}[\beta]_{\mathcal{B}}$ , and congruence preserves rank ([Theorem 13.2.3](#)). Hence  $n_+ + n_-$  is the same for every orthogonal basis, and so is  $n_0 = n - (n_+ + n_-)$ . What is **not** yet known is how that total splits into positives and negatives — which is exactly the work left for the theorem below.

## 13.5.2 The law of inertia

### Theorem 13.5.1 — Sylvester's Law of Inertia

Let  $\beta$  be a symmetric bilinear form on a real vector space  $V$  with  $\dim V = n$ , and let  $\mathcal{B}$  and  $\mathcal{B}'$  be two orthogonal bases for  $\beta$ . Let  $(n_+, n_-, n_0)$  be the numbers of positive, negative and zero diagonal entries of  $[\beta]_{\mathcal{B}}$ , and  $(n'_+, n'_-, n'_0)$  the corresponding numbers for  $[\beta]_{\mathcal{B}'}$ . Then

$$n_+ = n'_+, \quad n_- = n'_-, \quad n_0 = n'_0.$$

*Idea.* Rank has already delivered  $n_+ + n_- = n'_+ + n'_-$ , so it is enough to prove  $n_+ = n'_+$ , and for that we may suppose  $n_+ > n'_+$  and look for a disaster. The two bases carve out two subspaces with incompatible behavior: on  $U = \text{span}(\mathbf{v}_1, \dots, \mathbf{v}_{n_+})$  the form is **strictly positive** away from  $\mathbf{0}$ , because Eq. 13.4.1 has only positive coefficients there; on  $U' = \text{span}(\mathbf{v}'_{n'_++1}, \dots, \mathbf{v}'_n)$  it is  $\leq 0$  everywhere, because there the coefficients are all  $\leq 0$ . A non-zero vector in both would have  $q(\mathbf{v}) > 0$  and  $q(\mathbf{v}) \leq 0$  at once, so the two subspaces meet only in  $\mathbf{0}$ . But their dimensions add up to more than  $n$ , and two such subspaces of an  $n$ -dimensional space cannot meet only in  $\mathbf{0}$ . Contradiction plus a dimension count.

*Proof.* Write  $\mathcal{B} = (\mathbf{v}_1, \dots, \mathbf{v}_n)$  and  $\mathcal{B}' = (\mathbf{v}'_1, \dots, \mathbf{v}'_n)$ , and order each so that the positive entries come first, then the negative ones, then the zeros; reordering a basis does not change the three counts. Put  $d_i = q(\mathbf{v}_i)$  and  $d'_j = q(\mathbf{v}'_j)$ , so that

$$d_1, \dots, d_{n_+} > 0, \quad d'_{n'_++1}, \dots, d'_n \leq 0.$$

As in the Quick check,  $n_+ + n_- = \text{rank}[\beta]_{\mathcal{B}} = \text{rank}[\beta]_{\mathcal{B}'} = n'_+ + n'_-$  by Theorem 13.2.3, and  $n_0 = n - (n_+ + n_-)$ . So it suffices to prove  $n_+ = n'_+$ ; the other two equalities then follow.

Suppose not. Swapping the names of the two bases if necessary, we may assume  $n_+ > n'_+$ . Put

$$U = \text{span}(\mathbf{v}_1, \dots, \mathbf{v}_{n_+}), \quad U' = \text{span}(\mathbf{v}'_{n'_++1}, \dots, \mathbf{v}'_n),$$

so that  $\dim U = n_+$  and  $\dim U' = n - n'_+$ .

*Claim:*  $U \cap U' = \{\mathbf{0}\}$ . Let  $\mathbf{v} \in U \cap U'$  with  $\mathbf{v} \neq \mathbf{0}$ . Writing  $\mathbf{v} = \sum_{i \leq n_+} c_i \mathbf{v}_i$  and using Eq. 13.4.1 for the basis  $\mathcal{B}$ , whose later coordinates of  $\mathbf{v}$  are zero,

$$q(\mathbf{v}) = \sum_{i=1}^{n_+} d_i c_i^2 > 0,$$

because every  $d_i$  with  $i \leq n_+$  is positive, every  $c_i^2 \geq 0$ , and at least one  $c_i \neq 0$  since  $\mathbf{v} \neq \mathbf{0}$ . Writing instead  $\mathbf{v} = \sum_{j > n'_+} b_j \mathbf{v}'_j$  and using Eq. 13.4.1 for  $\mathcal{B}'$ ,

$$q(\mathbf{v}) = \sum_{j=n'_++1}^n d'_j b_j^2 \leq 0,$$

because every  $d'_j$  with  $j > n'_+$  is  $\leq 0$  and every  $b_j^2 \geq 0$ . A real number cannot be both  $> 0$  and  $\leq 0$ , so no such  $\mathbf{v}$  exists, which proves the claim.

On the other hand,  $U + U'$  is a subspace of  $V$ , so  $\dim(U + U') \leq n$ , and the dimension formula ([Theorem 1.7.3](#)) gives

$$\begin{aligned}\dim(U \cap U') &= \dim U + \dim U' - \dim(U + U') \\ &\geq n_+ + (n - n'_+) - n \\ &= n_+ - n'_+ > 0.\end{aligned}$$

So  $U \cap U'$  contains a non-zero vector, contradicting the claim. Therefore  $n_+ = n'_+$ , and with it  $n_- = n'_-$  and  $n_0 = n'_0$ . This proves the theorem. ■

The name comes from mechanics, where the form is an inertia tensor and the theorem says that a change of coordinates cannot turn a direction of positive inertia into one of negative inertia. The invariant has a name of its own.

### Definition 13.5.1 — Inertia and Signature

Let  $\beta$  be a symmetric bilinear form on a real vector space  $V$  of dimension  $n$ , diagonalized in some orthogonal basis. The triple

$$(n_+, n_-, n_0)$$

of numbers of positive, negative and zero diagonal entries is the **inertia** of  $\beta$ , and it is well defined by [Theorem 13.5.1](#). The pair  $(n_+, n_-)$  is the **signature** of  $\beta$ , and  $n_+ + n_- = \text{rank } \beta$  while  $n_0 = n - \text{rank } \beta$ . The same words are used for a real symmetric matrix  $\mathbf{A}$ , through the form  $\beta(\mathbf{x}, \mathbf{y}) = \mathbf{x}^\top \mathbf{A} \mathbf{y}$ , and we write  $n_+(\mathbf{A})$ ,  $n_-(\mathbf{A})$ ,  $n_0(\mathbf{A})$ .

Two conventions are in circulation and both are called the signature: the pair  $(n_+, n_-)$ , used here, and the single integer  $n_+ - n_-$ . They carry the same information once the rank is known, and the difference is only ever notational; where a source writes “signature 2” it means  $n_+ - n_- = 2$ . The triple  $(n_+, n_-, n_0)$  is the safest thing to quote, since it also records the degeneracy.

## 13.5.3 The classification over the reals

### Corollary 13.5.2 — Classification of Real Symmetric Matrices by Signature

Let  $\mathbf{A}, \mathbf{B} \in M_n(\mathbb{R})$  be symmetric. Then:

- (a)  $\mathbf{A}$  is congruent to  $\mathbf{I}_{n_+} \oplus (-\mathbf{I}_{n_-}) \oplus \mathbf{0}_{n_0}$ , where  $(n_+, n_-, n_0)$  is the inertia of  $\mathbf{A}$ , and to no other matrix of this shape;
- (b)  $\mathbf{A} \simeq \mathbf{B}$  if and only if  $\mathbf{A}$  and  $\mathbf{B}$  have the same inertia;
- (c) the number of congruence classes of symmetric matrices in  $M_n(\mathbb{R})$  is  $\frac{1}{2}(n+1)(n+2)$ .

*Proof.*

- (a) The congruence to  $\mathbf{I}_{n_+} \oplus (-\mathbf{I}_{n_-}) \oplus \mathbf{0}_{n_0}$  is [Eq. 13.5.1](#), obtained from [Theorem 13.4.2](#)

followed by the scaling and sorting described above. That matrix is itself diagonal with inertia  $(n_+, n_-, n_0)$ , so by [Theorem 13.5.1](#) no matrix of the same shape with a different triple can be congruent to  $\mathbf{A}$ .

- (b)  $(\Rightarrow)$  A congruence  $\mathbf{B} = \mathbf{P}^\top \mathbf{A} \mathbf{P}$  with  $\mathbf{P}$  invertible is a change of basis for the form ([Theorem 13.2.1](#)), so an orthogonal basis diagonalizing  $\mathbf{B}$  yields one diagonalizing  $\mathbf{A}$  with the same diagonal, and [Theorem 13.5.1](#) makes the inertia equal.  
 $(\Leftarrow)$  If  $\mathbf{A}$  and  $\mathbf{B}$  have the same inertia, each is congruent to the same matrix  $\mathbf{I}_{n_+} \oplus (-\mathbf{I}_{n_-}) \oplus \mathbf{0}_{n_0}$  by (a), hence to each other, since congruence is an equivalence relation ([Proposition 13.2.2](#)).
- (c) By (b) the classes correspond to the triples  $(n_+, n_-, n_0)$  of non-negative integers summing to  $n$ , equivalently to the pairs  $(n_+, n_-)$  with  $n_+ + n_- \leq n$ . For each value  $k = n_+ + n_- \in \{0, 1, \dots, n\}$  there are  $k + 1$  choices of  $n_+$ , so the total is  $1 + 2 + \dots + (n + 1) = \frac{1}{2}(n + 1)(n + 2)$ . This proves the corollary. ■

So congruence over  $\mathbb{R}$  has a complete and finite list of invariants, in sharp contrast to similarity, where even  $2 \times 2$  real symmetric matrices fall into infinitely many classes (one for each unordered pair of eigenvalues). Congruence forgets a great deal; the point of the law of inertia is that what it forgets is precisely everything except the signs.

### 13.5.4 Reconciling congruence with the spectral theorem

Chapter 11 diagonalized a real symmetric matrix a different way, by an **orthogonal**  $\mathbf{Q}$ , and got the eigenvalues on the diagonal. Chapter 11 also pointed out why that theorem is special: because  $\mathbf{Q}^\top = \mathbf{Q}^{-1}$ , the one equation  $\mathbf{Q}^\top \mathbf{A} \mathbf{Q} = \mathbf{D}$  can be read twice, and “an orthogonal change of variables is a similarity *and* a congruence, so it keeps the eigenvalues and the geometry at once”. Reading the same equation twice is all that is needed to identify the inertia.

#### Theorem 13.5.3 — Inertia Is the Sign Pattern of the Spectrum

Let  $\mathbf{A} \in M_n(\mathbb{R})$  be symmetric. Then  $n_+(\mathbf{A})$  is the number of positive eigenvalues of  $\mathbf{A}$ , counted with multiplicity,  $n_-(\mathbf{A})$  the number of negative ones, and  $n_0(\mathbf{A})$  the multiplicity of 0.

*Proof.* By [Corollary 11.5.2](#) there is an orthogonal  $\mathbf{Q} \in O(n)$  and a real diagonal  $\mathbf{D} = \text{diag}(\lambda_1, \dots, \lambda_n)$  with  $\mathbf{A} = \mathbf{Q} \mathbf{D} \mathbf{Q}^\top$ , where  $\lambda_1, \dots, \lambda_n$  is the eigenvalue list of  $\mathbf{A}$  with multiplicity. Equivalently  $\mathbf{D} = \mathbf{Q}^\top \mathbf{A} \mathbf{Q}$ .

Read as a **similarity**:  $\mathbf{Q}^\top = \mathbf{Q}^{-1}$ , so  $\mathbf{D} = \mathbf{Q}^{-1} \mathbf{A} \mathbf{Q}$  and the diagonal entries of  $\mathbf{D}$  are the eigenvalues of  $\mathbf{A}$ .

Read as a **congruence**:  $\mathbf{Q}$  is invertible, so  $\mathbf{D} = \mathbf{Q}^\top \mathbf{A} \mathbf{Q}$  exhibits  $\mathbf{D}$  as congruent to  $\mathbf{A}$  ([Definition 13.2.1](#)). Being diagonal,  $\mathbf{D}$  is the matrix of the form of  $\mathbf{A}$  in an orthogonal basis, namely the columns of  $\mathbf{Q}$ . By [Theorem 13.5.1](#) its numbers of positive, negative and zero entries are the inertia of  $\mathbf{A}$ .

The same list of numbers is being counted both times, so the counts agree. This proves

the theorem. ■

This is what Chapter 12 promised when, having diagonalized a pencil and found  $\mathbf{S}^\top \mathbf{B} \mathbf{S} = \text{diag}(-1, 3)$  while the eigenvalues of  $\mathbf{B}$  were  $3 \pm 3\sqrt{2}$ , it observed that the signs  $(-, +)$  “match the signs of the eigenvalues of  $\mathbf{B}$  itself” and said that Chapter 13 would show this is no coincidence. It is not: the congruence  $\mathbf{S}^\top \mathbf{B} \mathbf{S}$  has the inertia of  $\mathbf{B}$  by [Theorem 13.5.1](#), and that inertia is the sign pattern of the spectrum by [Theorem 13.5.3](#). Here  $3 + 3\sqrt{2} > 0$  and  $3 - 3\sqrt{2} < 0$  — since  $3\sqrt{2} > 4$  — so the inertia of  $\mathbf{B}$  is  $(1, 1, 0)$ , and any congruent diagonal matrix must show exactly one positive and one negative entry, as  $\text{diag}(-1, 3)$  does.

The same reading settles the definiteness tests of Chapter 12 in one line:  $\mathbf{A} \succ 0$  means  $\mathbf{x}^\top \mathbf{A} \mathbf{x} > 0$  for all  $\mathbf{x} \neq \mathbf{0}$ , which by [Theorem 12.1.5](#) is the statement that every eigenvalue is positive, which by [Theorem 13.5.3](#) is the statement that the inertia is  $(n, 0, 0)$ . Section 6 takes that dictionary further.

**Warning.** Congruent matrices have the same inertia and unrelated eigenvalues. Take  $\mathbf{A} = \text{diag}(1, -1)$  and  $\mathbf{P} = \text{diag}(2, 3)$ . Then

$$\mathbf{P}^\top \mathbf{A} \mathbf{P} = \text{diag}(4, -9),$$

so  $\mathbf{A} \simeq \text{diag}(4, -9)$ , yet the eigenvalues have moved from  $1, -1$  to  $4, -9$ . Nothing about their sizes, their ratios, their sum or their product is preserved —  $\text{tr}$  goes from  $0$  to  $-5$ ,  $\det$  from  $-1$  to  $-36$  — and only the pattern  $(+, -)$  survives. For a second witness with no scaling in sight,  $\mathbf{I}_2$  is congruent to  $\mathbf{S}^\top \mathbf{S} = \begin{pmatrix} 1 & 1 \\ 1 & 2 \end{pmatrix}$  with  $\mathbf{S} = \begin{pmatrix} 1 & 1 \\ 0 & 1 \end{pmatrix}$ ; the eigenvalues change from  $1, 1$  to  $(3 \pm \sqrt{5})/2$ , and the inertia stays  $(2, 0, 0)$ .

### 13.5.5 Computing a signature

[Theorem 13.5.3](#) gives a method — find the eigenvalues and count signs — but it is the expensive one, since it needs the roots of a degree- $n$  polynomial. The cheap method is the algorithm of the previous section, which needs only arithmetic: diagonalize by paired row and column operations and read off the signs. The eigenvalues are never computed and never needed.

**Example. (A signature by symmetric elimination)**

Find the inertia of

$$\mathbf{A} = \begin{pmatrix} 1 & 1 & 2 \\ 1 & 2 & 3 \\ 2 & 3 & 1 \end{pmatrix},$$

and say whether the form  $q(\mathbf{x}) = \mathbf{x}^\top \mathbf{A} \mathbf{x}$  is positive definite.

*Solution.* The corner entry is 1, so no repair move is needed.

*Clear the first row and column.* Subtract row 1 from row 2 and  $2 \times$  row 1 from row 3,

then do the same two operations on columns:

$$\begin{pmatrix} 1 & 1 & 2 \\ 0 & 1 & 1 \\ 0 & 1 & -3 \end{pmatrix} \longrightarrow \begin{pmatrix} 1 & 0 & 0 \\ 0 & 1 & 1 \\ 0 & 1 & -3 \end{pmatrix}.$$

*Clear the second row and column.* The pivot is 1 and  $a_{23} = 1$ , so subtract row 2 from row 3 and then column 2 from column 3:

$$\begin{pmatrix} 1 & 0 & 0 \\ 0 & 1 & 1 \\ 0 & 0 & -4 \end{pmatrix} \longrightarrow \text{diag}(1, 1, -4).$$

So the inertia is  $(2, 1, 0)$  and the signature is  $(2, 1)$ . The form is **not** positive definite: it is negative at the third basis vector of the new basis, namely the third column of

$$\mathbf{P} = \begin{pmatrix} 1 & -1 & -1 \\ 0 & 1 & -1 \\ 0 & 0 & 1 \end{pmatrix},$$

which is  $\mathbf{p}_3 = (-1, -1, 1)$ . Checking directly,  $\mathbf{A}\mathbf{p}_3 = (0, 0, -4)$  and  $\mathbf{p}_3 \cdot (0, 0, -4) = -4 < 0$ .

The eigenvalue route would have to factor  $\det(x\mathbf{I} - \mathbf{A}) = x^3 - 4x^2 - 9x + 4$ , whose roots are irrational; the elimination never met them. As a partial check on the arithmetic,  $\det \mathbf{A} = -4$  and  $\det \text{diag}(1, 1, -4) = -4$ , consistent with  $\det \mathbf{P} = 1$ .

Over  $\mathbb{R}$ , then, a symmetric matrix carries exactly two pieces of congruence data, and both are cheap: its rank and how that rank splits into positive and negative directions. The next section reads the same two numbers as a statement about definiteness, and uses them to decide maxima and minima.

## 13.5.6 Exercises

### 13.5.6.1 A. Check your understanding

#### Exercise A1.

- Define the **inertia** and the **signature** of a real symmetric matrix.
- State [Theorem 13.5.1](#), with all hypotheses.
- Which part of the inertia is already an invariant before the law of inertia is proved, and by which earlier result?
- Determine whether the following statement is correct, and justify your answer: congruent real symmetric matrices have the same trace.
- Explain in one sentence why an orthogonal change of basis lets the spectral theorem compute an inertia.

*Solution.*

- (a) Diagonalize the associated form in an orthogonal basis. The inertia is the triple  $(n_+, n_-, n_0)$  of numbers of positive, negative and zero diagonal entries, and the signature is the pair  $(n_+, n_-)$ .
- (b) If  $\beta$  is a symmetric bilinear form on a finite-dimensional real vector space and  $\mathcal{B}, \mathcal{B}'$  are two orthogonal bases for  $\beta$ , then the two matrices  $[\beta]_{\mathcal{B}}$  and  $[\beta]_{\mathcal{B}'}$  have the same numbers of positive, negative and zero diagonal entries.
- (c) The count  $n_0$ , equivalently the sum  $n_+ + n_-$ , because that sum is the rank and congruence preserves rank ([Theorem 13.2.3](#)).
- (d) Incorrect.  $\text{diag}(1, -1)$  and  $\text{diag}(4, -9)$  are congruent, by the congruence with  $\mathbf{P} = \text{diag}(2, 3)$ , and their traces are 0 and  $-5$ . Trace is a similarity invariant ([Theorem 3.7.5](#)), not a congruence invariant.
- (e) For an orthogonal  $\mathbf{Q}$  we have  $\mathbf{Q}^\top = \mathbf{Q}^{-1}$ , so the single equation  $\mathbf{Q}^\top \mathbf{A} \mathbf{Q} = \mathbf{D}$  is simultaneously a similarity, which puts the eigenvalues on the diagonal, and a congruence, which makes that diagonal compute the inertia.

### 13.5.6.2 B. Practice

#### Exercise B1: Signatures by elimination.

Find the inertia of each of the following real symmetric matrices by paired row and column operations. Do not compute any eigenvalues.

- (a)  $\mathbf{A} = \begin{pmatrix} 1 & 2 & 0 \\ 2 & 1 & 3 \\ 0 & 3 & 1 \end{pmatrix}$ .
- (b)  $\mathbf{B} = \begin{pmatrix} 0 & 1 & 1 \\ 1 & 0 & 1 \\ 1 & 1 & 0 \end{pmatrix}$ .
- (c)  $\mathbf{C} = \begin{pmatrix} 1 & 2 \\ 2 & 4 \end{pmatrix}$ .

*Solution.*

- (a) By [Exercise \(B1: Symmetric elimination\)](#),  $\mathbf{A}$  is congruent to  $\text{diag}(1, -3, 4)$ . Inertia  $(2, 1, 0)$ .
- (b) By [Example \(Diagonalizing a symmetric matrix with zero diagonal\)](#),  $\mathbf{B}$  is congruent to  $\text{diag}(2, -2, -2)$ . Inertia  $(1, 2, 0)$ .
- (c) Subtract  $2 \times$  row 1 from row 2, then  $2 \times$  column 1 from column 2:

$$\begin{pmatrix} 1 & 2 \\ 0 & 0 \end{pmatrix} \longrightarrow \begin{pmatrix} 1 & 0 \\ 0 & 0 \end{pmatrix}.$$

Inertia  $(1, 0, 1)$ : the form  $q(\mathbf{x}) = (x_1 + 2x_2)^2$  is degenerate, vanishing on the line  $x_1 = -2x_2$ .

#### Exercise B2: Which of these are congruent?.

Sort the following real symmetric matrices into congruence classes, justifying each an-

swer.

$$\mathbf{A}_1 = \text{diag}(1, 1, -1), \quad \mathbf{A}_2 = \text{diag}(5, 2, -7), \quad \mathbf{A}_3 = \text{diag}(1, -1, -1),$$

$$\mathbf{A}_4 = \begin{pmatrix} 0 & 1 & 0 \\ 1 & 0 & 0 \\ 0 & 0 & 1 \end{pmatrix}, \quad \mathbf{A}_5 = \begin{pmatrix} 2 & 0 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & -3 \end{pmatrix}.$$

*Solution.* Compute each inertia and apply [Corollary 13.5.2](#) (b).

$\mathbf{A}_1$ : (2, 1, 0).  $\mathbf{A}_2$ : (2, 1, 0).  $\mathbf{A}_3$ : (1, 2, 0).  $\mathbf{A}_5$ : (1, 1, 1).

For  $\mathbf{A}_4$ , the top-left  $2 \times 2$  block is the form  $2x_1x_2$ , which [Exercise \(B2: No pivot on the diagonal\)](#) diagonalizes to  $\text{diag}(2, -2)$ ; so  $\mathbf{A}_4$  is congruent to  $\text{diag}(2, -2, 1)$ , with inertia (2, 1, 0).

Therefore the classes are  $\{\mathbf{A}_1, \mathbf{A}_2, \mathbf{A}_4\}$ ,  $\{\mathbf{A}_3\}$  and  $\{\mathbf{A}_5\}$ . Note that  $\mathbf{A}_1$  and  $\mathbf{A}_3$  have the same rank and are still not congruent, so rank alone does not classify over  $\mathbb{R}$  — unlike over  $\mathbb{C}$  ([Corollary 13.4.4](#)).

### Exercise B3: Inertia from a characteristic polynomial.

Let  $\mathbf{A} \in M_3(\mathbb{R})$  be symmetric with  $\det(x\mathbf{I} - \mathbf{A}) = x^3 - 2x^2 - 5x + 6$ .

- (a) Find the inertia of  $\mathbf{A}$ .
- (b) Is  $\mathbf{A}$  congruent to  $\text{diag}(1, 1, -1)$ ? Is it similar to it?

*Solution.*

- (a) The polynomial factors as  $(x - 1)(x + 2)(x - 3)$ , since 1 is a root and dividing gives  $x^2 - x - 6$ . The eigenvalues are 1, 3, -2, so by [Theorem 13.5.3](#) the inertia is (2, 1, 0).
- (b) Congruent: yes, since  $\text{diag}(1, 1, -1)$  also has inertia (2, 1, 0) and [Corollary 13.5.2](#) (b) applies. Similar: no, because similar matrices have equal characteristic polynomials, and  $\text{diag}(1, 1, -1)$  has  $(x - 1)^2(x + 1) \neq (x - 1)(x - 3)(x + 2)$ .

### 13.5.6.3 C. Going deeper

#### Exercise C1: An intrinsic description of $n_+$ .

Let  $\beta$  be a symmetric bilinear form on a real vector space  $V$  of dimension  $n$ , with quadratic form  $q$ . Call a subspace  $U \subseteq V$  **positive** if  $q(\mathbf{v}) > 0$  for every  $\mathbf{v} \in U$  with  $\mathbf{v} \neq \mathbf{0}$ .

- (a) Prove that  $n_+$  is the largest dimension of a positive subspace of  $V$ .
- (b) Deduce [Theorem 13.5.1](#) again from (a), in two lines.
- (c) State and prove the corresponding description of  $n_-$ .

*Hint: for (a), use an orthogonal basis to produce one positive subspace of dimension  $n_+$ , and the dimension formula to rule out a larger one.*



*Solution.*

- (a) Fix an orthogonal basis  $(\mathbf{v}_1, \dots, \mathbf{v}_n)$  ordered so that  $d_1, \dots, d_{n_+} > 0$  and  $d_j \leq 0$  for  $j > n_+$ .

A positive subspace of dimension  $n_+$ . Put  $U = \text{span}(\mathbf{v}_1, \dots, \mathbf{v}_{n_+})$ . For  $\mathbf{v} = \sum_{i \leq n_+} c_i \mathbf{v}_i$  non-zero, Eq. 13.4.1 gives  $q(\mathbf{v}) = \sum_{i \leq n_+} d_i c_i^2 > 0$ , since all  $d_i > 0$  and some  $c_i \neq 0$ . So  $U$  is positive and  $\dim U = n_+$ .

No larger one. Let  $X$  be positive with  $\dim X = k$ , and put  $W = \text{span}(\mathbf{v}_{n_++1}, \dots, \mathbf{v}_n)$ , of dimension  $n - n_+$ . Every  $\mathbf{w} \in W$  has  $q(\mathbf{w}) = \sum_{j > n_+} d_j c_j^2 \leq 0$ . If  $k > n_+$  then by Theorem 1.7.3,

$$\dim(X \cap W) \geq k + (n - n_+) - n = k - n_+ > 0,$$

so there is a non-zero  $\mathbf{v} \in X \cap W$  with  $q(\mathbf{v}) > 0$  and  $q(\mathbf{v}) \leq 0$ , which is impossible. Hence  $k \leq n_+$ , and the maximum is  $n_+$ .

- (b) The right-hand side of (a) — the largest dimension of a positive subspace — is defined from  $\beta$  alone and mentions no basis. So the number  $n_+$  computed from any orthogonal basis equals that basis-free quantity, hence is the same for all of them. Then  $n_+ + n_- = \text{rank } \beta$  is basis-free too (Theorem 13.2.3), so  $n_-$  and  $n_0$  are determined as well.
- (c)  $n_-$  is the largest dimension of a **negative** subspace, one on which  $q(\mathbf{v}) < 0$  for every  $\mathbf{v} \neq \mathbf{0}$ . Apply (a) to the form  $-\beta$ , whose diagonal entries in any orthogonal basis are the negatives of those of  $\beta$ , so that  $n_+(-\beta) = n_-(\beta)$ , and whose positive subspaces are exactly the negative subspaces of  $\beta$ .

### Exercise C2: Congruence cannot be read off the eigenvalues, and inertia can.

- (a) Give two real symmetric  $2 \times 2$  matrices that are congruent but have no eigenvalue in common.
- (b) Prove that if  $\mathbf{A}$  and  $\mathbf{B}$  are real symmetric and **similar**, then they are congruent.
- (c) Give two real symmetric  $2 \times 2$  matrices that are congruent but not similar, so that the converse of (b) fails.

*Solution.*

- (a)  $\text{diag}(1, -1)$  and  $\text{diag}(4, -9)$ , congruent by  $\mathbf{P} = \text{diag}(2, 3)$ . Their spectra are  $\{1, -1\}$  and  $\{4, -9\}$ , which are disjoint.
- (b) Similar matrices have the same characteristic polynomial (Theorem 6.10.3), hence the same eigenvalues with multiplicity. By Theorem 13.5.3 they then have the same inertia, and by Corollary 13.5.2 (b) they are congruent. (Alternatively and more directly: by Corollary 11.5.2 each is orthogonally congruent to the diagonal matrix of its eigenvalues, and those diagonal matrices agree up to a permutation.)
- (c) The pair in (a) again:  $\text{diag}(1, -1)$  and  $\text{diag}(4, -9)$  are congruent, but their characteristic polynomials are  $x^2 - 1$  and  $x^2 + 5x - 36$ , so they are not similar (Theorem 6.10.3). Congruence is strictly coarser than similarity on real symmetric matrices.

**Exercise C3: Inertia of a block matrix.**

Let  $\mathbf{A} \in M_m(\mathbb{R})$  and  $\mathbf{B} \in M_k(\mathbb{R})$  be symmetric. Let also  $\mathbf{D} \in M_n(\mathbb{R})$  be symmetric and invertible, let  $\mathbf{b} \in \mathbb{R}^n$ , let  $c \in \mathbb{R}$ , and put

$$\mathbf{M} = \begin{pmatrix} \mathbf{D} & \mathbf{b} \\ \mathbf{b}^\top & c \end{pmatrix} \in M_{n+1}(\mathbb{R}).$$

- (a) Prove that the inertia of  $\mathbf{A} \oplus \mathbf{B}$  is the entrywise sum of the inertias of  $\mathbf{A}$  and  $\mathbf{B}$ .
- (b) Prove that  $\mathbf{M}$  is congruent to  $\mathbf{D} \oplus (c - \mathbf{b}^\top \mathbf{D}^{-1} \mathbf{b})$ .
- (c) Hence find the inertia of  $\mathbf{M}$  when  $\mathbf{D} = \text{diag}(1, -1)$ ,  $\mathbf{b} = (1, 1)$  and  $c = 0$ .

*Hint: for (b), the congruence is the block elimination of Chapter 7, and the leftover scalar is a Schur complement.*

*Solution.*

- (a) Let  $\mathbf{P}^\top \mathbf{A} \mathbf{P}$  and  $\mathbf{Q}^\top \mathbf{B} \mathbf{Q}$  be the canonical forms of [Corollary 13.5.2](#) (a). Then  $\mathbf{P} \oplus \mathbf{Q}$  is invertible and

$$(\mathbf{P} \oplus \mathbf{Q})^\top (\mathbf{A} \oplus \mathbf{B}) (\mathbf{P} \oplus \mathbf{Q}) = (\mathbf{P}^\top \mathbf{A} \mathbf{P}) \oplus (\mathbf{Q}^\top \mathbf{B} \mathbf{Q}),$$

a diagonal matrix whose positive, negative and zero entries are those of the two blocks together. By [Theorem 13.5.1](#) those counts are the inertia of  $\mathbf{A} \oplus \mathbf{B}$ .

- (b) Put

$$\mathbf{S} = \begin{pmatrix} \mathbf{I}_n & -\mathbf{D}^{-1} \mathbf{b} \\ \mathbf{0}^\top & 1 \end{pmatrix},$$

which is invertible, being unit upper triangular. Multiplying, the first block row of  $\mathbf{M} \mathbf{S}$  is  $(\mathbf{D} \mid \mathbf{D}(-\mathbf{D}^{-1} \mathbf{b}) + \mathbf{b}) = (\mathbf{D} \mid \mathbf{0})$ , and then

$$\mathbf{S}^\top \mathbf{M} \mathbf{S} = \begin{pmatrix} \mathbf{D} & \mathbf{0} \\ \mathbf{0}^\top & c - \mathbf{b}^\top \mathbf{D}^{-1} \mathbf{b} \end{pmatrix},$$

using  $\mathbf{D}^\top = \mathbf{D}$ . This is  $\mathbf{D} \oplus (c - \mathbf{b}^\top \mathbf{D}^{-1} \mathbf{b})$ , the Schur complement  $\mathbf{M}/\mathbf{D}$  sitting in the corner.

- (c) Here  $\mathbf{D}^{-1} = \text{diag}(1, -1)$ , so  $\mathbf{b}^\top \mathbf{D}^{-1} \mathbf{b} = 1 - 1 = 0$  and the leftover scalar is  $c - 0 = 0$ . By (b) and (a), the inertia of  $\mathbf{M}$  is the inertia of  $\text{diag}(1, -1)$  plus that of the  $1 \times 1$  zero matrix, namely  $(1, 1, 0) + (0, 0, 1) = (1, 1, 1)$ . In particular  $\mathbf{M}$  is singular.

## 13.6 Definiteness and Second Derivatives

Chapter 12 studied positive definite matrices at length, and the previous section showed that the signature is what congruence remembers. Putting the two together costs almost nothing and buys two things: a vocabulary that covers the forms Chapter 12 had no name for, the indefinite and the semidefinite ones, and the classical application that made quadratic forms worth studying in the first place. Nothing from Chapter 12 is reproved here. The field is  $\mathbb{R}$  throughout, since none of the five words below means anything without an order.

Throughout,  $V$  is a finite-dimensional real vector space,  $\beta$  is a **symmetric** bilinear form on  $V$  (Definition 13.3.1),  $q(\mathbf{v}) = \beta(\mathbf{v}, \mathbf{v})$  is its quadratic form (Definition 13.3.2), and  $\mathbf{A} = [\beta]_{\mathcal{B}}$  is its matrix in a basis  $\mathcal{B}$  (Definition 13.1.2), a real symmetric matrix.

### 13.6.1 Five words for the sign of a form

Chapter 12 named two of the possible behaviors of  $q$ : never negative, and always positive away from  $\mathbf{0}$ . There are three more, and the missing ones are not exotic. The form  $q(x_1, x_2) = x_1^2 - x_2^2$  is positive at  $\mathbf{e}_1$ , negative at  $\mathbf{e}_2$ , and nothing in Chapter 12's vocabulary describes it.

*A form is definite when  $q$  has one strict sign, semidefinite when it has one sign but is allowed to vanish, and indefinite when it has both signs.*

#### Definition 13.6.1 — Definiteness of a Real Symmetric Form

Let  $\beta$  be a symmetric bilinear form on a real vector space  $V$ , with quadratic form  $q$ . Then  $\beta$  is:

- (a) **positive definite** if  $q(\mathbf{v}) > 0$  for every **non-zero**  $\mathbf{v} \in V$ ;
- (b) **positive semidefinite** if  $q(\mathbf{v}) \geq 0$  for every  $\mathbf{v} \in V$ ;
- (c) **negative definite** if  $q(\mathbf{v}) < 0$  for every non-zero  $\mathbf{v} \in V$ ;
- (d) **negative semidefinite** if  $q(\mathbf{v}) \leq 0$  for every  $\mathbf{v} \in V$ ;
- (e) **indefinite** if there exist  $\mathbf{u}, \mathbf{w} \in V$  with  $q(\mathbf{u}) > 0$  and  $q(\mathbf{w}) < 0$ .

The same five words apply to a real symmetric matrix  $\mathbf{A}$ , through the form  $\beta(\mathbf{x}, \mathbf{y}) = \mathbf{x}^\top \mathbf{A} \mathbf{y}$  on  $\mathbb{R}^n$ .

The small words carry the weight. In (a) and (c) the inequality is **strict** and the vector must be **non-zero**; in (b) and (d) it is weak and  $\mathbf{v} = \mathbf{0}$  is allowed, where  $q(\mathbf{0}) = 0$  makes the condition automatic. Definite implies semidefinite of the same sign. Clause (e) is not the negation of (b):  $q(x_1, x_2) = -x_1^2$  is neither positive semidefinite nor indefinite. For matrices, (a) and (b) are Definition 12.1.1 restricted to  $F = \mathbb{R}$ , so  $\mathbf{A} \succ 0$  and  $\mathbf{A} \succeq 0$  keep their meanings, and the negative versions are the positive ones for  $-\mathbf{A}$ .

Four examples on  $\mathbb{R}^2$ , all diagonal so that  $q$  can be read off Eq. 13.4.1:  $x_1^2 + x_2^2$  is positive definite;  $x_1^2$  is positive semidefinite but not definite, since it vanishes at  $\mathbf{e}_2 \neq \mathbf{0}$ ;  $-x_1^2 - x_2^2$  is negative definite;  $x_1^2 - x_2^2$  is indefinite. The zero form is the degenerate case, satisfying (b) and (d) and nothing else — which is why the five classes overlap rather than partition. Changing one sign in the third gives the non-example  $-x_1^2 + x_2^2$ : it is no longer negative semidefinite, and the clause that fails is (d) at  $\mathbf{v} = \mathbf{e}_2$ , where  $q = 1 > 0$ .

Now the reading through the signature, which makes all five conditions into statements about three integers.

#### Theorem 13.6.1 — Definiteness Is the Signature

Let  $\beta$  be a symmetric bilinear form on a real vector space  $V$  of dimension  $n$ , with inertia  $(n_+, n_-, n_0)$  (Definition 13.5.1). Then:

- (a)  $\beta$  is positive definite  $\iff (n_+, n_-, n_0) = (n, 0, 0)$ ;
- (b)  $\beta$  is positive semidefinite  $\iff n_- = 0$ ;
- (c)  $\beta$  is negative definite  $\iff (n_+, n_-, n_0) = (0, n, 0)$ ;
- (d)  $\beta$  is negative semidefinite  $\iff n_+ = 0$ ;
- (e)  $\beta$  is indefinite  $\iff n_+ > 0$  and  $n_- > 0$ .

*Idea.* Fix an orthogonal basis. Then  $q$  is  $\sum_i d_i c_i^2$  with no cross terms, and each of the five conditions becomes a visible condition on the signs of the  $d_i$ : plugging in a basis vector reads off one  $d_i$ , and a general  $\mathbf{v}$  mixes the  $d_i$  with non-negative weights. Sylvester's law is what lets us fix one basis and still be talking about  $\beta$ .

*Proof.* By [Theorem 13.4.2](#) there is an orthogonal basis  $\mathcal{B} = (\mathbf{v}_1, \dots, \mathbf{v}_n)$  for  $\beta$ ; put  $d_i = q(\mathbf{v}_i)$ . By [Theorem 13.5.1](#) the numbers of positive, negative and zero  $d_i$  are  $n_+$ ,  $n_-$  and  $n_0$ , whichever orthogonal basis is used. For  $\mathbf{v} = \sum_i c_i \mathbf{v}_i$ , [Eq. 13.4.1](#) gives

$$q(\mathbf{v}) = \sum_{i=1}^n d_i c_i^2.$$

- (b)  $(\implies)$  If some  $d_j < 0$  then  $q(\mathbf{v}_j) = d_j < 0$ , so  $\beta$  is not positive semidefinite.  $(\impliedby)$  If no  $d_i$  is negative then every term  $d_i c_i^2$  is  $\geq 0$ , so  $q(\mathbf{v}) \geq 0$ .
- (c)  $(\implies)$  A positive definite  $\beta$  is positive semidefinite, so  $n_- = 0$  by (b); and if some  $d_j = 0$  then  $q(\mathbf{v}_j) = 0$  with  $\mathbf{v}_j \neq \mathbf{0}$ , a contradiction. Hence  $n_0 = 0$  and  $n_+ = n$ .  $(\impliedby)$  If every  $d_i > 0$  and  $\mathbf{v} \neq \mathbf{0}$ , some  $c_j \neq 0$ , so  $q(\mathbf{v}) \geq d_j c_j^2 > 0$ , the other terms being  $\geq 0$ .
- (d) and (d) follow by applying (a) and (b) to  $-\beta$ , whose diagonal entries are  $-d_i$  and whose inertia is therefore  $(n_-, n_+, n_0)$ .
- (e)  $(\implies)$  Suppose  $q(\mathbf{u}) > 0$  and  $q(\mathbf{w}) < 0$ . Then  $\beta$  is neither positive semidefinite nor negative semidefinite, so  $n_- > 0$  by (b) and  $n_+ > 0$  by (d).  $(\impliedby)$  If  $n_+ > 0$  and  $n_- > 0$ , some  $d_i > 0$  and some  $d_j < 0$ , and the basis vectors  $\mathbf{v}_i, \mathbf{v}_j$  are the required  $\mathbf{u}, \mathbf{w}$ . This proves the theorem. ■

So the five words sort the  $\frac{1}{2}(n+1)(n+2)$  congruence classes of [Corollary 13.5.2](#) into named families, and every one of them is decided by counting signs. Two of these five verdicts were already settled in Chapter 12 by other means, and it is worth putting the tests side by side.

### Proposition 13.6.2 — Chapter 12's Tests, in the Language of Forms

Let  $\beta$  be a symmetric bilinear form on an  $n$ -dimensional real vector space,  $\mathbf{A} = [\beta]_{\mathcal{B}}$  its matrix in any basis, and  $\mathbf{A}_k$  the top-left  $k \times k$  corner of  $\mathbf{A}$ . Then:

- (a)  $\beta$  is positive definite  $\iff$  every eigenvalue of  $\mathbf{A}$  is positive  $\iff \det \mathbf{A}_k > 0$  for  $k = 1, \dots, n$ ;
- (b)  $\beta$  is negative definite  $\iff$  every eigenvalue of  $\mathbf{A}$  is negative  $\iff (-1)^k \det \mathbf{A}_k > 0$  for  $k = 1, \dots, n$ .

*Proof.*

- (a) By [Definition 13.1.2](#) and [Theorem 13.1.1](#),  $q(\mathbf{v}) = \mathbf{x}^\top \mathbf{A} \mathbf{x}$  where  $\mathbf{x} = [\mathbf{v}]_{\mathcal{B}}$ , and  $\mathbf{v} \neq \mathbf{0}$  exactly when  $\mathbf{x} \neq \mathbf{0}$ . So  $\beta$  is positive definite exactly when  $\mathbf{A} \succ 0$ , and the two stated tests are [Theorem 12.1.5](#) (b) and (d), the second being Sylvester's criterion.
- (b) Apply (a) to  $-\beta$ , whose matrix is  $-\mathbf{A}$ . The eigenvalues of  $-\mathbf{A}$  are the negatives of those of  $\mathbf{A}$ , and  $(-\mathbf{A})_k = -\mathbf{A}_k$  has  $\det(-\mathbf{A}_k) = (-1)^k \det \mathbf{A}_k$ , since scaling each of the  $k$  rows by  $-1$  multiplies the determinant by  $-1$  once per row ([Theorem 6.5.1](#) (b)). This proves the proposition. ■

The eigenvalue test is the one to think with, the minor test the one to compute with; cheapest of all is the symmetric elimination of Section 4, which returns the whole signature rather than a verdict. Nothing is said about the semidefinite cases on purpose: Sylvester's criterion has no weak version, as Chapter 12 showed with  $\text{diag}(0, -1)$ , whose leading minors are both 0.

*Quick check.* A real symmetric  $\mathbf{A} \in M_3(\mathbb{R})$  has  $\det \mathbf{A}_1 = -2$ ,  $\det \mathbf{A}_2 = 5$ ,  $\det \mathbf{A}_3 = -1$ . What is its definiteness?

*Solution.* Negative definite. The signs alternate starting with a negative one, so  $(-1)^k \det \mathbf{A}_k > 0$  for  $k = 1, 2, 3$ : indeed  $(-1)(-2) = 2 > 0$ ,  $(+1)(5) = 5 > 0$ ,  $(-1)(-1) = 1 > 0$ . By [Proposition 13.6.2](#) (b) the form is negative definite, and by [Theorem 13.6.1](#) (c) its inertia is  $(0, 3, 0)$ . Note that  $\det \mathbf{A}_3 = \det \mathbf{A} < 0$  on its own would have been no evidence at all.

## 13.6.2 The second-derivative test

Here is the use. A differentiable function  $f: \mathbb{R}^n \rightarrow \mathbb{R}$  is constant to first order at a critical point, so whether the point is a minimum, a maximum or neither is decided — usually — by the second-order behavior. That behavior is a quadratic form, and [Theorem 13.6.1](#) decides it. The linear algebra is the whole content; what has to be imported is one theorem of analysis, and we state it rather than hide it.

### Definition 13.6.2 — Hessian

Let  $f$  be a real-valued function on an open set  $U \subseteq \mathbb{R}^n$ , twice continuously differentiable on  $U$ , and let  $\mathbf{a} \in U$ . The **Hessian** of  $f$  at  $\mathbf{a}$  is the matrix

$$\mathbf{H}_f(\mathbf{a}) \in M_n(\mathbb{R}), \quad (\mathbf{H}_f(\mathbf{a}))_{ij} = \frac{\partial^2 f}{\partial x_i \partial x_j}(\mathbf{a}).$$

A point  $\mathbf{a} \in U$  with  $\nabla f(\mathbf{a}) = \mathbf{0}$  is a **critical point** of  $f$ .

Two facts about  $\mathbf{H}_f(\mathbf{a})$  are needed, and neither is linear algebra. They are exactly the analysis this book imports, so we set them out as assumptions rather than let them pass unnoticed. Let  $f$  be twice continuously differentiable on an open  $U \subseteq \mathbb{R}^n$  and let  $\mathbf{a} \in U$ .

- (A1) (*Symmetry of the second derivatives.*) The mixed partial derivatives satisfy  $\partial^2 f / \partial x_i \partial x_j = \partial^2 f / \partial x_j \partial x_i$  on  $U$ , so  $\mathbf{H}_f(\mathbf{a})$  is a **symmetric** matrix.

(A2) (*Taylor's theorem with a second-order remainder.*) For every  $\mathbf{h}$  small enough that  $\mathbf{a} + \mathbf{h} \in U$ ,

$$f(\mathbf{a} + \mathbf{h}) = f(\mathbf{a}) + \nabla f(\mathbf{a}) \cdot \mathbf{h} + \frac{1}{2} \mathbf{h}^\top \mathbf{H}_f(\mathbf{a}) \mathbf{h} + r(\mathbf{h}),$$

where the remainder satisfies  $r(\mathbf{h})/\|\mathbf{h}\|^2 \rightarrow 0$  as  $\mathbf{h} \rightarrow \mathbf{0}$ .

These are two standard theorems of multivariable calculus. This book proves neither, and nothing later depends on either; they are quoted only so that the theorem below has hypotheses to stand on, exactly as the theorem of Abel and Ruffini was quoted in Chapter 10. Everything after them is linear algebra.

By (A1) the Hessian is a real symmetric matrix, so it has an inertia, and by [Theorem 13.6.1](#) that inertia decides the definiteness of the quadratic form  $\mathbf{h} \mapsto \mathbf{h}^\top \mathbf{H}_f(\mathbf{a}) \mathbf{h}$ . One small lemma converts definiteness into a quantitative bound, and it is the only place where a real number is needed rather than a sign.

### Lemma 13.6.3 — A Positive Definite Form Is Bounded Below by a Multiple of the Norm Squared

Let  $\mathbf{A} \in M_n(\mathbb{R})$  be symmetric and positive definite, and let  $\lambda$  be its smallest eigenvalue. Then  $\lambda > 0$  and

$$\mathbf{x}^\top \mathbf{A} \mathbf{x} \geq \lambda \|\mathbf{x}\|^2 \quad \text{for every } \mathbf{x} \in \mathbb{R}^n.$$

*Proof.* By [Corollary 11.5.2](#) there are an orthogonal  $\mathbf{Q} \in O(n)$  and a diagonal  $\mathbf{D} = \text{diag}(\lambda_1, \dots, \lambda_n)$  listing the eigenvalues of  $\mathbf{A}$  with  $\mathbf{A} = \mathbf{Q} \mathbf{D} \mathbf{Q}^\top$ . By [Theorem 12.1.5](#) (b) every  $\lambda_i > 0$ , so  $\lambda = \min_i \lambda_i > 0$ . Put  $\mathbf{y} = \mathbf{Q}^\top \mathbf{x}$ . Then

$$\mathbf{x}^\top \mathbf{A} \mathbf{x} = \mathbf{y}^\top \mathbf{D} \mathbf{y} = \sum_{i=1}^n \lambda_i y_i^2 \geq \lambda \sum_{i=1}^n y_i^2 = \lambda \|\mathbf{y}\|^2,$$

using  $\lambda_i \geq \lambda$  and  $y_i^2 \geq 0$  termwise. Finally  $\mathbf{Q} \mathbf{Q}^\top = \mathbf{I}$ , so  $\|\mathbf{y}\|^2 = \mathbf{x}^\top \mathbf{Q} \mathbf{Q}^\top \mathbf{x} = \|\mathbf{x}\|^2$ . This proves the lemma. ■

### Theorem 13.6.4 — The Second-Derivative Test

Let  $f$  be a real-valued function, twice continuously differentiable on an open set  $U \subseteq \mathbb{R}^n$ , let  $\mathbf{a} \in U$  be a critical point of  $f$ , and let  $(n_+, n_-, n_0)$  be the inertia of  $\mathbf{H}_f(\mathbf{a})$ . Then:

- (a) if  $\mathbf{H}_f(\mathbf{a})$  is positive definite, that is  $(n_+, n_-, n_0) = (n, 0, 0)$ , then  $\mathbf{a}$  is a strict local minimum of  $f$ ;
- (b) if  $\mathbf{H}_f(\mathbf{a})$  is negative definite, that is  $(n_+, n_-, n_0) = (0, n, 0)$ , then  $\mathbf{a}$  is a strict local maximum of  $f$ ;
- (c) if  $\mathbf{H}_f(\mathbf{a})$  is indefinite, that is  $n_+ > 0$  and  $n_- > 0$ , then  $\mathbf{a}$  is neither a local minimum nor a local maximum: every neighborhood of  $\mathbf{a}$  contains points where  $f > f(\mathbf{a})$  and points where  $f < f(\mathbf{a})$ . Such an  $\mathbf{a}$  is a **saddle point**.

If  $n_0 > 0$  and  $\mathbf{a}$  is not covered by (a)–(c), the test gives no information.

*Idea.* Taylor says  $f(\mathbf{a} + \mathbf{h}) - f(\mathbf{a})$  is  $\frac{1}{2}q(\mathbf{h})$  plus a remainder that is smaller than  $\|\mathbf{h}\|^2$  near  $\mathbf{0}$ , where  $q(\mathbf{h}) = \mathbf{h}^\top \mathbf{H}_f(\mathbf{a})\mathbf{h}$ . For (a) the quadratic term is at least  $\frac{1}{2}\lambda\|\mathbf{h}\|^2$  by Lemma 13.6.3, which beats a remainder eventually smaller than  $\frac{1}{4}\lambda\|\mathbf{h}\|^2$ ; so the difference is positive. For (c) there is no bound in every direction, and none is needed: walk along a single line  $\mathbf{h} = t\mathbf{u}$  where  $q(\mathbf{u}) > 0$ , and the whole thing is  $t^2$  times something tending to  $\frac{1}{2}q(\mathbf{u}) > 0$ . One direction up, one direction down, and no extremum.

*Proof.* Write  $\mathbf{H} = \mathbf{H}_f(\mathbf{a})$  and  $q(\mathbf{h}) = \mathbf{h}^\top \mathbf{H}\mathbf{h}$ . Since  $\mathbf{a}$  is critical,  $\nabla f(\mathbf{a}) = \mathbf{0}$ , so (A2) reads

$$f(\mathbf{a} + \mathbf{h}) - f(\mathbf{a}) = \frac{1}{2}q(\mathbf{h}) + r(\mathbf{h}), \quad \frac{r(\mathbf{h})}{\|\mathbf{h}\|^2} \rightarrow 0. \quad (*)$$

- (a) Let  $\lambda > 0$  be the smallest eigenvalue of  $\mathbf{H}$ , so that  $q(\mathbf{h}) \geq \lambda\|\mathbf{h}\|^2$  by Lemma 13.6.3. Since  $r(\mathbf{h})/\|\mathbf{h}\|^2 \rightarrow 0$ , there is  $\delta > 0$  with  $\mathbf{a} + \mathbf{h} \in U$  and  $|r(\mathbf{h})| \leq \frac{1}{4}\lambda\|\mathbf{h}\|^2$  whenever  $0 < \|\mathbf{h}\| < \delta$ . For such  $\mathbf{h}$ , by (\*),

$$f(\mathbf{a} + \mathbf{h}) - f(\mathbf{a}) \geq \frac{1}{2}\lambda\|\mathbf{h}\|^2 - \frac{1}{4}\lambda\|\mathbf{h}\|^2 = \frac{1}{4}\lambda\|\mathbf{h}\|^2 > 0.$$

So  $f(\mathbf{a} + \mathbf{h}) > f(\mathbf{a})$  for every  $\mathbf{h} \neq \mathbf{0}$  with  $\|\mathbf{h}\| < \delta$ , which is what “strict local minimum” says.

- (b) The Hessian of  $-f$  at  $\mathbf{a}$  is  $-\mathbf{H}$ , whose inertia is  $(n, 0, 0)$  because that of  $\mathbf{H}$  is  $(0, n, 0)$ ; so  $-\mathbf{H}$  is positive definite by Theorem 13.6.1 (a). Part (a) applied to  $-f$  makes  $\mathbf{a}$  a strict local minimum of  $-f$ , that is a strict local maximum of  $f$ .
- (c) By Theorem 13.6.1 (e) there are  $\mathbf{u}, \mathbf{w} \in \mathbb{R}^n$  with  $q(\mathbf{u}) > 0$  and  $q(\mathbf{w}) < 0$ ; both are non-zero, since  $q(\mathbf{0}) = 0$ . Fix  $t \neq 0$  small and put  $\mathbf{h} = t\mathbf{u}$ . Then  $q(t\mathbf{u}) = t^2q(\mathbf{u})$ , so (\*) gives

$$\frac{f(\mathbf{a} + t\mathbf{u}) - f(\mathbf{a})}{t^2} = \frac{1}{2}q(\mathbf{u}) + \frac{r(t\mathbf{u})}{t^2}.$$

Now  $r(t\mathbf{u})/t^2 = \|\mathbf{u}\|^2 \cdot r(t\mathbf{u})/\|t\mathbf{u}\|^2 \rightarrow 0$  as  $t \rightarrow 0$ , because  $\|t\mathbf{u}\| \rightarrow 0$  and  $\|\mathbf{u}\|^2$  is a constant. So for all small enough  $t \neq 0$  the right-hand side is at least  $\frac{1}{4}q(\mathbf{u}) > 0$ , and hence  $f(\mathbf{a} + t\mathbf{u}) > f(\mathbf{a})$ . The same computation with  $\mathbf{w}$  in place of  $\mathbf{u}$  gives  $f(\mathbf{a} + t\mathbf{w}) < f(\mathbf{a})$  for all small enough  $t \neq 0$ . Every neighborhood of  $\mathbf{a}$  contains points of both kinds, so  $\mathbf{a}$  is neither a local minimum nor a local maximum. This proves the theorem. ■

Cases (a)–(c) exhaust the non-degenerate ones: if  $n_0 = 0$  then  $n_+ + n_- = n$ , so either  $n_+ = n$ , or  $n_- = n$ , or both are positive. What is left undecided is exactly  $n_0 > 0$ , and there the test is silent because it has to be.

### Example. (Two critical points of a cubic surface)

Find the critical points of  $f(x, y) = x^3 - 3xy + y^3$  on  $\mathbb{R}^2$  and classify each one.

*Solution.* The partial derivatives are  $f_x = 3x^2 - 3y$  and  $f_y = -3x + 3y^2$ . Setting both to zero gives  $y = x^2$  and  $x = y^2$ , hence  $x = x^4$ , so  $x(x^3 - 1) = 0$  and  $x \in \{0, 1\}$ . The real critical points are  $(0, 0)$  and  $(1, 1)$ .

Differentiating again,

$$\mathbf{H}_f(x, y) = \begin{pmatrix} 6x & -3 \\ -3 & 6y \end{pmatrix}.$$

At  $(0, 0)$  the Hessian is  $\begin{pmatrix} 0 & -3 \\ -3 & 0 \end{pmatrix}$ , with  $\det = -9 < 0$ . A  $2 \times 2$  symmetric matrix with negative determinant has eigenvalue product  $-9$ , hence one eigenvalue of each sign, so its inertia is  $(1, 1, 0)$  by [Theorem 13.5.3](#) and the form is indefinite. By [Theorem 13.6.4](#) (c),  $(0, 0)$  is a saddle point. Concretely  $f(t, t) = 2t^3 - 3t^2 < 0$  and  $f(t, -t) = 3t^2 > 0$  for small  $t > 0$ .

At  $(1, 1)$  the Hessian is  $\begin{pmatrix} 6 & -3 \\ -3 & 6 \end{pmatrix}$ , whose leading principal minors are  $6 > 0$  and  $36 - 9 = 27 > 0$ . By [Proposition 13.6.2](#) (a) it is positive definite, so  $(1, 1)$  is a strict local minimum, with value  $f(1, 1) = -1$ .

### Example. (One Hessian, three answers)

The three functions

$$g_1 = x^2 + y^4, \quad g_2 = x^2 - y^4, \quad g_3 = x^2$$

all have  $\nabla g_i(0, 0) = \mathbf{0}$  and the same Hessian at the origin. What is that Hessian, and what happens at the origin in each case?

*Solution.* Every second derivative of  $y^4$  vanishes at  $y = 0$ , so all three have

$$\mathbf{H}_{g_i}(0, 0) = \begin{pmatrix} 2 & 0 \\ 0 & 0 \end{pmatrix},$$

with inertia  $(1, 0, 1)$ : positive semidefinite, not definite, not indefinite. The test is silent, and it must be, because the three answers differ. For  $g_1$  the origin is a strict local minimum, since  $x^2 + y^4 > 0$  away from  $\mathbf{0}$ . For  $g_2$  it is a saddle:  $g_2(t, 0) = t^2 > 0$  and  $g_2(0, t) = -t^4 < 0$  for  $t \neq 0$ . For  $g_3$  it is a minimum but not a strict one, since  $g_3$  vanishes along the whole  $y$ -axis. The quadratic term decided nothing here; the quartic term decided everything.

**Warning.** The test reads the signature, not the determinant. In two variables the familiar rule is “ $\det \mathbf{H} > 0$  and  $f_{xx} > 0$  means a minimum”, and the second half is not decoration:  $\text{diag}(-1, -1)$  has  $\det = 1 > 0$ , and  $-x^2 - y^2$  has a strict local **maximum** at the origin. All  $\det \mathbf{H} > 0$  says in two variables is that  $n_0 = 0$  and  $n_-$  is even — definite, of a sign the determinant cannot see. In three variables it says even less:  $\text{diag}(1, -1, -1)$  has  $\det = 1 > 0$  and is indefinite.

## 13.6.3 Exercises

### 13.6.3.1 A. Check your understanding

#### Exercise A1.

- (a) Define what it means for a symmetric bilinear form on a real vector space to be



**indefinite.**

- (b) State the signature condition equivalent to negative semidefiniteness, and the one equivalent to positive definiteness.
- (c) Determine whether the following statement is correct, and justify your answer: a symmetric form that is not positive semidefinite is indefinite.
- (d) Name the two results of analysis that [Theorem 13.6.4](#) assumes, and say which part of its proof uses each.
- (e) A critical point has  $n_0 = 2$  and  $n_- = 0$ . What does [Theorem 13.6.4](#) conclude?

*Solution.*

- (a)  $\beta$  is indefinite if there exist  $\mathbf{u}$  and  $\mathbf{w}$  in  $V$  with  $q(\mathbf{u}) > 0$  and  $q(\mathbf{w}) < 0$ , where  $q(\mathbf{v}) = \beta(\mathbf{v}, \mathbf{v})$ .
- (b) Negative semidefinite  $\iff n_+ = 0$ ; positive definite  $\iff (n_+, n_-, n_0) = (n, 0, 0)$ , equivalently  $n_+ = n$ . Both are [Theorem 13.6.1](#).
- (c) Incorrect. The form  $q(x_1, x_2) = -x_1^2$  has inertia  $(0, 1, 1)$ ; it is not positive semidefinite, since  $q(\mathbf{e}_1) = -1$ , and not indefinite, since  $q$  is never positive. Failing (b) of [Definition 13.6.1](#) means only  $n_- > 0$ ; being indefinite needs  $n_+ > 0$  as well.
- (d) Symmetry of the mixed second partial derivatives, which makes the Hessian symmetric so that it has an inertia at all; and Taylor's theorem with a second-order remainder, which is the identity (\*) that every part of the proof manipulates.
- (e) Nothing. Here  $n_+ = n - 2$  and  $n_- = 0$ , so the Hessian is positive semidefinite but not definite and not indefinite, and none of (a), (b), (c) applies. [Example \(One Hessian, three answers\)](#) shows that all of “strict minimum”, “non-strict minimum” and “saddle” really do occur in this situation.

### 13.6.3.2 B. Practice

#### Exercise B1: Classify by the signature.

Determine the definiteness of each of the following real symmetric matrices, and give the inertia of each. Justify your answers.

$$\mathbf{A}_1 = \begin{pmatrix} 2 & 1 \\ 1 & 2 \end{pmatrix}, \quad \mathbf{A}_2 = \begin{pmatrix} -1 & 2 \\ 2 & -5 \end{pmatrix}, \quad \mathbf{A}_3 = \begin{pmatrix} 1 & 2 \\ 2 & 1 \end{pmatrix}.$$

*Solution.* For  $\mathbf{A}_1$ :  $\det(\mathbf{A}_1)_1 = 2 > 0$  and  $\det \mathbf{A}_1 = 4 - 1 = 3 > 0$ , so  $\mathbf{A}_1$  is positive definite by [Proposition 13.6.2](#) (a), with inertia  $(2, 0, 0)$ .

For  $\mathbf{A}_2$ :  $(-1)^1 \det(\mathbf{A}_2)_1 = 1 > 0$  and  $(-1)^2 \det \mathbf{A}_2 = 5 - 4 = 1 > 0$ , so  $\mathbf{A}_2$  is negative definite by [Proposition 13.6.2](#) (b), with inertia  $(0, 2, 0)$ .

For  $\mathbf{A}_3$ :  $\det \mathbf{A}_3 = 1 - 4 = -3 < 0$ , so the two eigenvalues have product  $-3$  and hence opposite signs. The inertia is  $(1, 1, 0)$  by [Theorem 13.5.3](#), and  $\mathbf{A}_3$  is indefinite by [Theorem 13.6.1](#) (e). Explicitly  $q(1, 1) = 6 > 0$  and  $q(1, -1) = -2 < 0$ .

#### Exercise B2: Critical points of a quartic.

Find all critical points of  $f(x, y) = x^4 + y^4 - 4xy$  and classify each with [Theorem 13.6.4](#).

*Solution.*  $f_x = 4x^3 - 4y$  and  $f_y = 4y^3 - 4x$ , so  $y = x^3$  and  $x = y^3 = x^9$ . Then  $x(x^8 - 1) = 0$ , so  $x \in \{0, 1, -1\}$  and the critical points are  $(0, 0)$ ,  $(1, 1)$ ,  $(-1, -1)$ . The Hessian is

$$\mathbf{H}_f(x, y) = \begin{pmatrix} 12x^2 & -4 \\ -4 & 12y^2 \end{pmatrix}.$$

At  $(0, 0)$  it is  $\begin{pmatrix} 0 & -4 \\ -4 & 0 \end{pmatrix}$ , with determinant  $-16 < 0$ , hence inertia  $(1, 1, 0)$  and a saddle point. At  $(\pm 1, \pm 1)$  it is  $\begin{pmatrix} 12 & -4 \\ -4 & 12 \end{pmatrix}$ , with leading minors  $12 > 0$  and  $144 - 16 = 128 > 0$ , hence positive definite: both are strict local minima, each of value  $f = 1 + 1 - 4 = -2$ .

### 13.6.3.3 C. Going deeper

#### Exercise C1: The signature measures positive subspaces.

Let  $\beta$  be a symmetric bilinear form on a real vector space  $V$  of dimension  $n$ , with inertia  $(n_+, n_-, n_0)$ . Call a subspace  $U \subseteq V$  **positive** if  $q(\mathbf{v}) > 0$  for every non-zero  $\mathbf{v} \in U$ .

- Prove that  $V$  has a positive subspace of dimension  $n_+$ .
- Prove that every positive subspace has dimension at most  $n_+$ . *Hint: consider a subspace on which  $q \leq 0$  of the largest available dimension.*
- Hence deduce that  $n_+$  equals the largest dimension of a positive subspace, and that this gives a second proof that  $n_+$  does not depend on the orthogonal basis.

*Solution.* Fix an orthogonal basis  $\mathcal{B} = (\mathbf{v}_1, \dots, \mathbf{v}_n)$  for  $\beta$ , ordered so that  $d_i = q(\mathbf{v}_i) > 0$  for  $i \leq n_+$ , then negative, then zero.

- Put  $U = \text{span}(\mathbf{v}_1, \dots, \mathbf{v}_{n_+})$ . For a non-zero  $\mathbf{v} = \sum_{i \leq n_+} c_i \mathbf{v}_i$ , [Eq. 13.4.1](#) gives  $q(\mathbf{v}) = \sum_{i \leq n_+} d_i c_i^2 > 0$ , since every  $d_i > 0$  and some  $c_i \neq 0$ . So  $U$  is positive of dimension  $n_+$ .
- Put  $U' = \text{span}(\mathbf{v}_{n_++1}, \dots, \mathbf{v}_n)$ , of dimension  $n - n_+$ . For  $\mathbf{v} \in U'$ , [Eq. 13.4.1](#) gives  $q(\mathbf{v}) = \sum_{i > n_+} d_i c_i^2 \leq 0$ , every  $d_i$  there being  $\leq 0$ . Let  $W$  be any positive subspace. Then  $W \cap U' = \{\mathbf{0}\}$ , since a non-zero vector in the intersection would have  $q > 0$  and  $q \leq 0$ . By the dimension formula ([Theorem 1.7.3](#)),

$$\dim W + (n - n_+) = \dim(W + U') \leq n,$$

so  $\dim W \leq n_+$ .

- By (a) and (b) the largest dimension of a positive subspace is exactly  $n_+$ . That description mentions only  $\beta$  and  $V$ , no basis, so  $n_+$  is an invariant of  $\beta$ . Together with the invariance of  $\text{rank } \beta = n_+ + n_-$  under congruence ([Theorem 13.2.3](#)) this recovers [Theorem 13.5.1](#).

#### Exercise C2: A saddle the determinant hides.

Let  $f(x, y, z) = x^2 - y^2 - z^2 + xyz$ .

- Verify that the origin is a critical point and compute  $\mathbf{H}_f(\mathbf{0})$  and its determinant.
- Classify the origin, and explain why the sign of  $\det \mathbf{H}_f(\mathbf{0})$  alone would have been

useless.

- (c) Prove that no twice continuously differentiable  $g: \mathbb{R}^3 \rightarrow \mathbb{R}$  has a strict local maximum at a critical point  $\mathbf{0}$  with  $\det \mathbf{H}_g(\mathbf{0}) > 0$ , and give a  $g: \mathbb{R}^2 \rightarrow \mathbb{R}$  that does. Which feature of the dimension is responsible?

*Solution.*

- (a)  $f_x = 2x + yz$ ,  $f_y = -2y + xz$ ,  $f_z = -2z + xy$ , all zero at  $\mathbf{0}$ . The second derivatives at  $\mathbf{0}$  are  $f_{xx} = 2$ ,  $f_{yy} = f_{zz} = -2$ , and  $f_{xy} = z$ ,  $f_{xz} = y$ ,  $f_{yz} = x$ , all zero at  $\mathbf{0}$ . So  $\mathbf{H}_f(\mathbf{0}) = \text{diag}(2, -2, -2)$ , with  $\det = 8 > 0$ .
- (b) The inertia is  $(1, 2, 0)$ , so the Hessian is indefinite and  $\mathbf{0}$  is a saddle point by [Theorem 13.6.4](#) (c). The determinant is the product of the eigenvalues ([Corollary 11.1.5](#)) and therefore records only the parity of  $n_-$ ; here  $\det > 0$  says  $n_-$  is even, which is equally true of the negative definite  $\text{diag}(-2, -2, -2)$  and of the positive definite  $\mathbf{I}_3$ . It cannot distinguish them.
- (c) Suppose  $\mathbf{0}$  is a local maximum of  $g$  and write  $q(\mathbf{h}) = \mathbf{h}^\top \mathbf{H}_g(\mathbf{0}) \mathbf{h}$ . If  $q(\mathbf{u}) > 0$  for some  $\mathbf{u}$ , the computation in the proof of [Theorem 13.6.4](#) (c) gives  $g(t\mathbf{u}) > g(\mathbf{0})$  for all small  $t \neq 0$ , contradicting the maximum. So  $q(\mathbf{u}) \leq 0$  for every  $\mathbf{u}$ , that is  $\mathbf{H}_g(\mathbf{0})$  is negative semidefinite and  $n_+ = 0$  by [Theorem 13.6.1](#) (d). If moreover  $\det \mathbf{H}_g(\mathbf{0}) > 0$  then 0 is not an eigenvalue, so  $n_0 = 0$  and the inertia is  $(0, 3, 0)$ . But then  $\det \mathbf{H}_g(\mathbf{0}) = \lambda_1 \lambda_2 \lambda_3$  is a product of three negative numbers, hence negative — a contradiction. On  $\mathbb{R}^2$  there is no obstruction:  $g(x, y) = -x^2 - y^2$  has  $\mathbf{H}_g(\mathbf{0}) = -2\mathbf{I}_2$ , determinant  $4 > 0$ , and a strict local maximum at  $\mathbf{0}$ . The responsible feature is the parity of the dimension: the determinant is the product of  $n$  eigenvalues, so a negative definite Hessian has  $\det$  of sign  $(-1)^n$ .

## 13.7 Where the Form Vanishes

Sections 4 and 5 diagonalized a symmetric form and counted the signs on the diagonal. Over  $\mathbb{R}$  that is the whole story, but it hides the feature that makes indefinite forms geometrically different from inner products: there are non-zero vectors on which the form vanishes. This section takes those vectors seriously. They are not defects to be quarantined; they organize the space, because each one forces a two-dimensional piece of completely standard shape to split off. The result is a decomposition of any non-degenerate space into a pile of identical planes plus a remainder on which the form vanishes nowhere.

Throughout,  $F$  is a field with  $\text{char } F \neq 2$  ([Definition 0.7.2](#)),  $V$  is a finite-dimensional vector space over  $F$  with  $\dim V = n$ , and  $\beta$  is a **symmetric** bilinear form on  $V$  ([Definition 13.3.1](#)) with quadratic form  $q(\mathbf{v}) = \beta(\mathbf{v}, \mathbf{v})$  ([Definition 13.3.2](#)). Two vectors are orthogonal when  $\beta$  pairs them to 0 ([Definition 13.4.1](#)). For a subspace  $U \subseteq V$  we write, as in [Exercise \(C2: Orthogonal complements for a form\)](#),

$$U^{\perp\beta} = \{ \mathbf{v} \in V : \beta(\mathbf{u}, \mathbf{v}) = 0 \text{ for every } \mathbf{u} \in U \},$$

and we write  $V = U \perp W$  to say that  $V = U \oplus W$  with every vector of  $U$  orthogonal to every vector of  $W$  — an **orthogonal direct sum**. The subscript  $\beta$  is dropped when only one form is in play.

### 13.7.1 An intuition to give up

Thirteen chapters have trained one reflex above all others:  $\langle \mathbf{v}, \mathbf{v} \rangle = 0$  forces  $\mathbf{v} = \mathbf{0}$ . It is how we proved that an isometry is injective, that an orthogonal list is independent, that a Gram matrix determines its vectors. It is also the first thing that goes.

Take  $V = \mathbb{R}^2$  with

$$q(x_1, x_2) = x_1^2 - x_2^2, \quad \mathbf{A} = \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix}.$$

The vector  $\mathbf{v} = (1, 1)$  has  $q(\mathbf{v}) = 1 - 1 = 0$ . It is not  $\mathbf{0}$ , and the form is not defective:  $\det \mathbf{A} = -1 \neq 0$ , so  $\beta$  is non-degenerate (Proposition 13.1.2). Indeed  $\beta(\mathbf{v}, (1, -1)) = 1 - (1)(-1) = 2 \neq 0$ , so  $\mathbf{v}$  is perfectly visible to  $\beta$  — it is only invisible to *itself*. The line  $\text{span}(\mathbf{v})$  is a line of “length zero” inside a space where length is otherwise perfectly well behaved.

It gets worse, and the worse case is the one worth internalizing. Take  $V = \mathbb{R}^4$  with

$$q(\mathbf{x}) = x_1^2 + x_2^2 - x_3^2 - x_4^2,$$

of signature  $(2, 2)$  and therefore non-degenerate. Let  $W = \text{span}(\mathbf{e}_1 + \mathbf{e}_3, \mathbf{e}_2 + \mathbf{e}_4)$ , a plane. For  $\mathbf{w} = a(\mathbf{e}_1 + \mathbf{e}_3) + b(\mathbf{e}_2 + \mathbf{e}_4)$ ,

$$q(\mathbf{w}) = a^2 + b^2 - a^2 - b^2 = 0,$$

and in fact  $\beta$  vanishes on every pair from  $W$ , since  $\beta(\mathbf{e}_1 + \mathbf{e}_3, \mathbf{e}_2 + \mathbf{e}_4) = 0$  too. So  $\beta$  restricted to  $W$  is the **zero form** on a two-dimensional subspace of a four-dimensional space on which  $\beta$  is non-degenerate.

Say that again, because it is the sentence an inner-product-trained reader will refuse. A non-degenerate form can vanish identically on a subspace of half the dimension of the whole space. Non-degeneracy is a statement about  $V$ ; it says that no vector is orthogonal to *everything*. It says nothing whatsoever about what  $\beta$  does inside a subspace. The vectors of  $W$  above are orthogonal to each other and to themselves; they are not orthogonal to  $\mathbf{e}_1 - \mathbf{e}_3$ , and that is all non-degeneracy ever promised.

Once that is accepted, the rest of the section is straightforward. The vanishing vectors get a name, and then a structure theorem.

### 13.7.2 Isotropic vectors and totally isotropic subspaces

*An isotropic vector is a non-zero vector that the form cannot tell from zero when it looks at it twice.*

#### Definition 13.7.1 — Isotropic and Anisotropic

Let  $\beta$  be a symmetric bilinear form on  $V$  with quadratic form  $q$ .

- (a) A vector  $\mathbf{v} \in V$  is **isotropic** if  $\mathbf{v} \neq \mathbf{0}$  and  $q(\mathbf{v}) = 0$ , and **anisotropic** if  $q(\mathbf{v}) \neq 0$ .
- (b) The form  $\beta$  is **anisotropic** if it has no isotropic vectors: that is, if  $q(\mathbf{v}) = 0$  implies  $\mathbf{v} = \mathbf{0}$ . Otherwise  $\beta$  is **isotropic**.

Both clauses hinge on small words. In (a) the vector is required to be **non-zero**, so  $\mathbf{0}$  is neither isotropic nor anisotropic; the zero vector always has  $q(\mathbf{0}) = 0$  and carries no information. In (b), “anisotropic” is a property of the **form on a particular space**. Anisotropy passes down to subspaces, since a subspace has fewer vectors to test; isotropy does not, and a form isotropic on  $V$  is very often anisotropic on a subspace. The word comes from *iso* (equal) and *tropos* (direction): a vector along which the form does not distinguish one direction from another.

Examples, over several fields.

- **A positive definite form has no isotropic vectors.** If  $q(\mathbf{v}) > 0$  for  $\mathbf{v} \neq \mathbf{0}$  then  $q(\mathbf{v}) = 0$  forces  $\mathbf{v} = \mathbf{0}$ . So every inner product is anisotropic, which is exactly the reflex described above. The same holds for negative definite forms.
- **Over  $\mathbb{R}$ , signature  $(1, 1)$ .** For  $q = x_1^2 - x_2^2$  the isotropic vectors are the non-zero multiples of  $(1, 1)$  and of  $(1, -1)$ , since  $q = (x_1 - x_2)(x_1 + x_2)$ . Two lines, and nothing else.
- **Over  $\mathbb{C}$ .** For  $q = x_1^2 + x_2^2$  on  $\mathbb{C}^2$  the factorization  $q = (x_1 + ix_2)(x_1 - ix_2)$  is available, so  $(1, i)$  and  $(1, -i)$  are isotropic. Over  $\mathbb{R}$  the same  $q$  is positive definite. **Isotropy depends on the field**, not only on the polynomial.
- **Over  $\mathbb{F}_3$  and  $\mathbb{F}_5$ .** For  $q = x_1^2 + x_2^2$  the squares in  $\mathbb{F}_3$  are 0 and 1, so  $q(\mathbf{x}) = 0$  needs  $x_1 = x_2 = 0$  and the form is anisotropic. In  $\mathbb{F}_5$ , however,  $1^2 + 2^2 = 5 = 0$ , so  $(1, 2)$  is isotropic. The same formula, two finite fields, two answers.
- **The degenerate case.** Every non-zero vector of the radical is isotropic, since  $\mathbf{v} \in \text{rad}(\beta)$  gives  $\beta(\mathbf{u}, \mathbf{v}) = 0$  for every  $\mathbf{u}$ , in particular for  $\mathbf{u} = \mathbf{v}$ . So a degenerate form is automatically isotropic. The converse fails, and the  $\mathbb{R}^2$  example above is the witness.

When a whole subspace consists of isotropic vectors, more is true than the definition says, and it deserves its own name.

### Definition 13.7.2 — Totally Isotropic Subspace

A subspace  $W \subseteq V$  is **totally isotropic** for  $\beta$  if  $\beta(\mathbf{u}, \mathbf{w}) = 0$  for **all**  $\mathbf{u}, \mathbf{w} \in W$ ; equivalently, if  $W \subseteq W^{\perp_\beta}$ . The zero subspace is totally isotropic, trivially.

The two formulations agree because  $W \subseteq W^{\perp_\beta}$  says exactly that every  $\mathbf{w} \in W$  is orthogonal to every  $\mathbf{u} \in W$ . A third formulation is available in characteristic  $\neq 2$  and is the one to use in practice:  $W$  is totally isotropic if and only if  $q(\mathbf{w}) = 0$  for every  $\mathbf{w} \in W$ . One direction is immediate; the other is [Theorem 13.3.5](#) applied to the restricted form, which recovers  $\beta$  on  $W$  from  $q$  on  $W$  and so makes it zero. In the  $\mathbb{R}^4$  example above we checked exactly this:  $q$  vanished on  $W$ , and therefore so did  $\beta$ .

Totally isotropic subspaces cannot be too large.

### Proposition 13.7.1 — Half the Dimension, at Most

Let  $\beta$  be a non-degenerate symmetric form on  $V$  with  $\dim V = n$ , and let  $W \subseteq V$  be totally isotropic. Then  $\dim W \leq n/2$ .

*Proof.* Since  $\beta$  is non-degenerate,  $\dim W^{\perp_\beta} = n - \dim W$  by [Exercise \(C2: Orthogonal complements for a form\)](#) (b). Since  $W$  is totally isotropic,  $W \subseteq W^{\perp_\beta}$ , so  $\dim W \leq n - \dim W$ , which is the claim.



**Warning.** An isotropic vector is not a zero vector, and a form that vanishes on a subspace need not be degenerate. These are two different errors and both are common. The first confuses  $q(\mathbf{v}) = 0$  with  $\mathbf{v} = \mathbf{0}$ ; over  $\mathbb{R}^2$  with  $q = x_1^2 - x_2^2$  the vector  $(1, 1)$  separates them. The second confuses “ $\beta$  vanishes on  $W \times W$ ” with “ $\beta$  vanishes on  $\{\mathbf{v}\} \times V$ ”. Degeneracy is about the **second**: the radical consists of vectors killed by pairing with *everything*. The plane  $W$  of the  $\mathbb{R}^4$  example carries the zero form and contains no radical vector at all.

**Quick check.** Is the plane  $\text{span}(\mathbf{e}_1, \mathbf{e}_2) \subseteq \mathbb{R}^4$  totally isotropic for  $q(\mathbf{x}) = x_1^2 + x_2^2 - x_3^2 - x_4^2$ ? What about  $\text{span}(\mathbf{e}_1 + \mathbf{e}_3, \mathbf{e}_1 - \mathbf{e}_3)$ ?

**Solution.** No to both. On  $\text{span}(\mathbf{e}_1, \mathbf{e}_2)$  the form restricts to  $x_1^2 + x_2^2$ , which is positive definite, so that plane is as far from totally isotropic as possible: it is anisotropic. The second plane is  $\text{span}(\mathbf{e}_1, \mathbf{e}_3)$ , on which  $q$  restricts to  $x_1^2 - x_3^2$ ; its two spanning vectors  $\mathbf{e}_1 \pm \mathbf{e}_3$  are indeed isotropic, but  $\beta(\mathbf{e}_1 + \mathbf{e}_3, \mathbf{e}_1 - \mathbf{e}_3) = 1 - (-1) = 2 \neq 0$ , so the plane is not totally isotropic. Containing isotropic vectors is not the same as being totally isotropic.

### 13.7.3 Splitting off a non-degenerate piece

Everything below rests on one mechanism: if a subspace carries a non-degenerate restriction, it splits off orthogonally and the rest of the space is again non-degenerate. That is the analogue of “restrict to an invariant complement”, and it is what makes induction possible.

#### Proposition 13.7.2 — A Non-degenerate Subspace Splits Off

Let  $\beta$  be a non-degenerate symmetric form on  $V$  with  $\dim V = n$ , and let  $U \subseteq V$  be a subspace such that  $\beta|_{U \times U}$  is non-degenerate. Then

$$V = U \perp U^{\perp\beta},$$

and  $\beta$  restricted to  $U^{\perp\beta}$  is non-degenerate.

**Idea.** Two things to check and one dimension count. The intersection  $U \cap U^{\perp\beta}$  is precisely the radical of the restricted form, which the hypothesis kills; the dimension of  $U^{\perp\beta}$  is  $n - \dim U$  because the global form is non-degenerate; and a vector of  $U^{\perp\beta}$  orthogonal to  $U^{\perp\beta}$  is orthogonal to all of  $V$ , hence zero.

**Proof.** Write  $U^\perp = U^{\perp\beta}$ . A vector  $\mathbf{v} \in U \cap U^\perp$  lies in  $U$  and satisfies  $\beta(\mathbf{u}, \mathbf{v}) = 0$  for every  $\mathbf{u} \in U$ , so  $\mathbf{v} \in \text{rad}(\beta|_{U \times U})$ , which is  $\{\mathbf{0}\}$  by hypothesis (Definition 13.1.4). Hence  $U \cap U^\perp = \{\mathbf{0}\}$  and the sum  $U + U^\perp$  is direct (Theorem 1.7.4).

Since  $\beta$  is non-degenerate on  $V$ , Exercise (C2: Orthogonal complements for a form) (b) gives  $\dim U^\perp = n - \dim U$ . So  $\dim(U \oplus U^\perp) = n$ , and a subspace of  $V$  of dimension  $n$  is  $V$  (Theorem 1.6.8). The sum is orthogonal by the definition of  $U^\perp$  together with the symmetry of  $\beta$ , so  $V = U \perp U^\perp$ .

Finally let  $\mathbf{w} \in U^\perp$  satisfy  $\beta(\mathbf{x}, \mathbf{w}) = 0$  for every  $\mathbf{x} \in U^\perp$ . For  $\mathbf{u} \in U$  we also have  $\beta(\mathbf{u}, \mathbf{w}) = 0$ , since  $\mathbf{w} \in U^\perp$ . Every  $\mathbf{v} \in V$  is  $\mathbf{u} + \mathbf{x}$  with  $\mathbf{u} \in U$  and  $\mathbf{x} \in U^\perp$ , so  $\beta(\mathbf{v}, \mathbf{w}) = 0$  for every  $\mathbf{v} \in V$ , that is  $\mathbf{w} \in \text{rad}(\beta) = \{\mathbf{0}\}$ . This proves the proposition.  $\blacksquare$

## 13.7.4 Hyperbolic planes

Which two-dimensional forms are isotropic? Over  $\mathbb{R}$  with signature  $(1, 1)$  we found two isotropic lines, and the example  $q = x_1^2 - x_2^2$  can be rewritten by the substitution  $u = x_1 + x_2$ ,  $w = (x_1 - x_2)/2$  as  $q = 2uw$  — the cross term alone, with no squares at all. That shape is the model, and it is the same over every field.

### Definition 13.7.3 — Hyperbolic Pair, Hyperbolic Plane

Let  $\beta$  be a symmetric bilinear form on  $V$ . A **hyperbolic pair** is an ordered pair  $(\mathbf{e}, \mathbf{f})$  of vectors of  $V$  with

$$q(\mathbf{e}) = 0, \quad q(\mathbf{f}) = 0, \quad \beta(\mathbf{e}, \mathbf{f}) = 1.$$

The subspace  $H = \text{span}(\mathbf{e}, \mathbf{f})$  is then called a **hyperbolic plane**, and  $(\mathbf{e}, \mathbf{f})$  is a **hyperbolic basis** of it.

Four things follow at once. First,  $\mathbf{e}$  and  $\mathbf{f}$  are linearly independent: if  $\mathbf{f} = c\mathbf{e}$  then  $\beta(\mathbf{e}, \mathbf{f}) = cq(\mathbf{e}) = 0 \neq 1$ . So  $H$  really is a plane,  $\dim H = 2$ . Second, in the basis  $(\mathbf{e}, \mathbf{f})$  the Gram matrix of  $\beta|_H$  is

$$[\beta|_H]_{(\mathbf{e}, \mathbf{f})} = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix},$$

whose determinant is  $-1 \neq 0$ , so  $\beta|_H$  is **non-degenerate** (Proposition 13.1.2) and Proposition 13.7.2 applies to it. Third,  $q(a\mathbf{e} + b\mathbf{f}) = 2ab$ , so on a hyperbolic plane  $q$  takes **every** value in  $F$ : given  $c \in F$ , take  $a = c/2$  and  $b = 1$ . Fourth, over  $\mathbb{R}$  the Gram matrix has eigenvalues 1 and  $-1$ , so a hyperbolic plane has signature  $(1, 1)$  and is the unique real example up to congruence.

Now the workhorse of this section and the next. It says that one isotropic vector is enough to manufacture a whole hyperbolic plane around it.

### Lemma 13.7.3 — Every Isotropic Vector Lies in a Hyperbolic Plane

Let  $\text{char } F \neq 2$ , let  $\beta$  be a **non-degenerate** symmetric bilinear form on  $V$ , and let  $\mathbf{e} \in V$  be isotropic. Then there is  $\mathbf{f} \in V$  such that  $(\mathbf{e}, \mathbf{f})$  is a hyperbolic pair. In particular  $\mathbf{e}$  lies in a hyperbolic plane  $H = \text{span}(\mathbf{e}, \mathbf{f})$ , and  $\beta|_H$  is non-degenerate.

*Idea.* Non-degeneracy hands us some  $\mathbf{u}$  with  $\beta(\mathbf{e}, \mathbf{u}) \neq 0$ , and rescaling makes  $\beta(\mathbf{e}, \mathbf{u}) = 1$ . Only one condition is still missing,  $q(\mathbf{u}) = 0$ , and  $\mathbf{u}$  can be corrected without disturbing what has been achieved: adding a multiple of  $\mathbf{e}$  changes  $q(\mathbf{u})$  but not  $\beta(\mathbf{e}, \mathbf{u})$ , because  $q(\mathbf{e}) = 0$ . Solving for the multiple is one line, and it divides by 2 — which is



where the characteristic hypothesis is spent, for the third time in this chapter.

*Proof.* Since  $\mathbf{e} \neq \mathbf{0}$  and  $\beta$  is non-degenerate, there is  $\mathbf{u}_0 \in V$  with  $c = \beta(\mathbf{e}, \mathbf{u}_0) \neq 0$  (Definition 13.1.4). Put  $\mathbf{u} = c^{-1}\mathbf{u}_0$ , so that

$$\beta(\mathbf{e}, \mathbf{u}) = 1.$$

Now set

$$\mathbf{f} = \mathbf{u} - \frac{1}{2}q(\mathbf{u})\mathbf{e},$$

which is legitimate because  $\text{char } F \neq 2$  makes  $2 = 1+1$  invertible. Then, using bilinearity, symmetry and  $q(\mathbf{e}) = 0$ ,

$$\beta(\mathbf{e}, \mathbf{f}) = \beta(\mathbf{e}, \mathbf{u}) - \frac{1}{2}q(\mathbf{u})q(\mathbf{e}) = 1 - 0 = 1,$$

and

$$\begin{aligned} q(\mathbf{f}) &= q(\mathbf{u}) - 2 \cdot \frac{1}{2}q(\mathbf{u})\beta(\mathbf{u}, \mathbf{e}) + \frac{1}{4}q(\mathbf{u})^2q(\mathbf{e}) \\ &= q(\mathbf{u}) - q(\mathbf{u}) + 0 = 0. \end{aligned}$$

Together with  $q(\mathbf{e}) = 0$ , this shows that  $(\mathbf{e}, \mathbf{f})$  is a hyperbolic pair. As noted after Definition 13.7.3,  $\mathbf{e}$  and  $\mathbf{f}$  are independent, so  $H = \text{span}(\mathbf{e}, \mathbf{f})$  is a plane, and the Gram matrix  $\begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}$  is invertible, so  $\beta|_H$  is non-degenerate. This proves the lemma. ■

*Remark.* Non-degeneracy of  $\beta$  is not decoration here. On  $\mathbb{R}^2$  with  $q(\mathbf{x}) = x_1^2$ , the vector  $\mathbf{e}_2$  is isotropic and lies in no hyperbolic plane, because  $\beta(\mathbf{e}_2, \mathbf{v}) = 0$  for every  $\mathbf{v}$  and a hyperbolic pair needs  $\beta(\mathbf{e}, \mathbf{f}) = 1$ . What fails is the very first line of the proof.

## 13.7.5 The Witt decomposition

Peeling off a hyperbolic plane and repeating is now a two-line induction, and it is the structural theorem of the section: it does for indefinite forms what “peel off an eigenvector” did for operators in Chapter 8.

### Theorem 13.7.4 — Witt Decomposition

Let  $\text{char } F \neq 2$  and let  $\beta$  be a non-degenerate symmetric bilinear form on a finite-dimensional  $V$ . Then there are an integer  $m \geq 0$ , hyperbolic planes  $H_1, \dots, H_m \subseteq V$  and a subspace  $V_0 \subseteq V$  such that

$$V = H_1 \perp H_2 \perp \dots \perp H_m \perp V_0$$

and  $\beta$  restricted to  $V_0$  is **anisotropic**. In particular  $\dim V = 2m + \dim V_0$ .

*Idea.* Induction on  $\dim V$ . Either  $\beta$  is anisotropic, in which case take  $m = 0$  and  $V_0 = V$  and stop; or there is an isotropic vector, in which case Lemma 13.7.3 wraps a hyperbolic plane  $H_1$  around it, Proposition 13.7.2 peels  $H_1$  off with a non-degenerate remainder  $H_1^\perp$  of dimension  $\dim V - 2$ , and the induction hypothesis finishes the job there.



*Proof.* Induct on  $\dim V$ . If  $\dim V = 0$ , take  $m = 0$  and  $V_0 = \{\mathbf{0}\}$ , on which the form is anisotropic vacuously. Let  $\dim V \geq 1$  and suppose the statement holds for every non-degenerate symmetric form on a space of smaller dimension.

If  $\beta$  is anisotropic, take  $m = 0$  and  $V_0 = V$ .

Otherwise  $\beta$  has an isotropic vector  $\mathbf{e}$ . By [Lemma 13.7.3](#) there is  $\mathbf{f}$  with  $(\mathbf{e}, \mathbf{f})$  a hyperbolic pair, and  $H_1 = \text{span}(\mathbf{e}, \mathbf{f})$  is a hyperbolic plane with  $\beta|_{H_1}$  non-degenerate. By [Proposition 13.7.2](#),

$$V = H_1 \perp H_1^{\perp\beta},$$

and  $\beta$  restricted to  $V' = H_1^{\perp\beta}$  is non-degenerate, with  $\dim V' = \dim V - 2 < \dim V$ . The induction hypothesis applied to  $(V', \beta|_{V'})$  gives hyperbolic planes  $H_2, \dots, H_m \subseteq V'$  and an anisotropic  $V_0 \subseteq V'$  with  $V' = H_2 \perp \dots \perp H_m \perp V_0$ . Each  $H_i$  with  $i \geq 2$  and  $V_0$  lie inside  $V' = H_1^{\perp\beta}$ , so all the summands are pairwise orthogonal, and

$$V = H_1 \perp H_2 \perp \dots \perp H_m \perp V_0.$$

Counting dimensions,  $\dim V = 2m + \dim V_0$ . This completes the induction. ■

Two questions are now unavoidable: is  $m$  the same for every such decomposition, and is  $V_0$  determined? The first has an answer here, and it comes from a quantity that mentions no decomposition at all.

### Definition 13.7.4 — Witt Index

Let  $\beta$  be a non-degenerate symmetric bilinear form on a finite-dimensional  $V$ . The **Witt index** of  $\beta$  is

$$\text{ind}(\beta) := \max\{\dim W : W \subseteq V \text{ totally isotropic}\},$$

the largest dimension of a totally isotropic subspace. By [Proposition 13.7.1](#) it satisfies  $\text{ind}(\beta) \leq n/2$ , and it is 0 exactly when  $\beta$  is anisotropic.

The maximum exists because the dimensions in question form a non-empty finite set of integers:  $W = \{\mathbf{0}\}$  is always totally isotropic, and all dimensions are at most  $n$ . Nothing in the definition refers to a basis or a decomposition, so the Witt index is an invariant of  $(V, \beta)$  by construction. It is also exactly the number of planes.

### Proposition 13.7.5 — The Number of Hyperbolic Planes Is the Witt Index

In any decomposition  $V = H_1 \perp \dots \perp H_m \perp V_0$  as in [Theorem 13.7.4](#),  $m = \text{ind}(\beta)$ . In particular  $m$  is the same for every such decomposition, and so is  $\dim V_0 = n - 2m$ .

*Idea.* One inequality is a construction: the first vectors  $\mathbf{e}_1, \dots, \mathbf{e}_m$  of the  $m$  hyperbolic bases span a totally isotropic subspace of dimension  $m$ , since each is isotropic and different planes are orthogonal. The other inequality is an induction that trades one plane for one dimension: given a totally isotropic  $W$ , the vectors of  $W$  orthogonal to  $\mathbf{e}_1$  project into  $H_1^{\perp}$  as a totally isotropic subspace of dimension at least  $\dim W - 1$ . Repeating  $m$

times lands inside the anisotropic  $V_0$ , where the only totally isotropic subspace is  $\{0\}$ .

*Proof.* Let  $(\mathbf{e}_i, \mathbf{f}_i)$  be a hyperbolic basis of  $H_i$  for each  $i$ .

**Step 1:**  $\text{ind}(\beta) \geq m$ . Put  $W = \text{span}(\mathbf{e}_1, \dots, \mathbf{e}_m)$ . The  $\mathbf{e}_i$  lie in different summands of a direct sum and are non-zero, so they are independent and  $\dim W = m$ . For  $i \neq j$  we have  $\beta(\mathbf{e}_i, \mathbf{e}_j) = 0$  because  $H_i \perp H_j$ , and  $\beta(\mathbf{e}_i, \mathbf{e}_i) = q(\mathbf{e}_i) = 0$ . By bilinearity  $\beta$  vanishes on  $W \times W$ , so  $W$  is totally isotropic of dimension  $m$ .

**Step 2: a plane can be traded for a dimension.**

*Claim.* Let  $\beta$  be non-degenerate on  $V$ , let  $H = \text{span}(\mathbf{e}, \mathbf{f}) \subseteq V$  be a hyperbolic plane, let  $V' = H^\perp_\beta$ , and let  $W \subseteq V$  be totally isotropic. Then  $V'$  contains a totally isotropic subspace of dimension at least  $\dim W - 1$ .

*Proof.* By [Proposition 13.7.2](#),  $V = H \perp V'$ ; let  $P: V \rightarrow V'$  be the projection with kernel  $H$ , so that  $\mathbf{v} - P(\mathbf{v}) \in H$  for every  $\mathbf{v}$ . Put

$$W_1 = \{ \mathbf{w} \in W : \beta(\mathbf{e}, \mathbf{w}) = 0 \},$$

the kernel of a linear functional on  $W$ , so  $\dim W_1 \geq \dim W - 1$ .

For  $\mathbf{w} \in V$  write  $\mathbf{w} - P(\mathbf{w}) = a\mathbf{e} + b\mathbf{f}$ . Since  $\mathbf{e} \in H$  is orthogonal to  $V'$ ,

$$\beta(\mathbf{e}, \mathbf{w}) = \beta(\mathbf{e}, a\mathbf{e} + b\mathbf{f}) = a q(\mathbf{e}) + b = b.$$

So  $\mathbf{w} \in W_1$  forces  $\mathbf{w} - P(\mathbf{w}) \in \text{span}(\mathbf{e})$ . Consequently, for  $\mathbf{w}, \mathbf{w}' \in W_1$  the vectors  $\mathbf{w} - P(\mathbf{w})$  and  $\mathbf{w}' - P(\mathbf{w}')$  both lie in  $\text{span}(\mathbf{e})$ , where  $\beta$  vanishes because  $q(\mathbf{e}) = 0$ ; hence

$$\beta(P(\mathbf{w}), P(\mathbf{w}')) = \beta(\mathbf{w}, \mathbf{w}') - \beta(\mathbf{w} - P(\mathbf{w}), \mathbf{w}' - P(\mathbf{w}')) = 0 - 0 = 0,$$

the first equality using the orthogonality of  $H$  and  $V'$ . So  $P(W_1)$  is a totally isotropic subspace of  $V'$ .

It remains to bound its dimension. By Rank–Nullity ([Theorem 3.3.1](#)) applied to  $P|_{W_1}$ , whose kernel is  $W_1 \cap H$ ,

$$\dim P(W_1) = \dim W_1 - \dim(W_1 \cap H).$$

If  $W_1 \cap H = \{0\}$ , then  $\dim P(W_1) = \dim W_1 \geq \dim W - 1$  and we are done. Otherwise pick  $\mathbf{x} \neq 0$  in  $W_1 \cap H$ . Then  $P(\mathbf{x}) = 0$ , so  $\mathbf{x} = \mathbf{x} - P(\mathbf{x}) \in \text{span}(\mathbf{e})$  by the paragraph above, whence  $\mathbf{e} \in W$ . But then  $\beta(\mathbf{e}, \mathbf{w}) = 0$  for every  $\mathbf{w} \in W$ , because  $W$  is totally isotropic and  $\mathbf{e} \in W$ ; so  $W_1 = W$ . Moreover every  $\mathbf{x} \in W_1 \cap H$  lies in  $\text{span}(\mathbf{e})$  by the same argument, so  $W_1 \cap H = \text{span}(\mathbf{e})$  has dimension 1, and  $\dim P(W_1) = \dim W - 1$ . Either way the claim holds. ■

**Step 3:**  $\text{ind}(\beta) \leq m$ . Let  $W \subseteq V$  be totally isotropic. Apply Step 2 to  $H = H_1$ : since  $H_1^\perp_\beta = H_2 \perp \dots \perp H_m \perp V_0$  carries a non-degenerate form, we obtain a totally isotropic subspace of it of dimension at least  $\dim W - 1$ . Repeating with  $H_2, \dots, H_m$  in turn produces a totally isotropic subspace of  $V_0$  of dimension at least  $\dim W - m$ . But  $\beta|_{V_0}$  is anisotropic, so a non-zero vector of  $V_0$  has  $q \neq 0$  and the only totally isotropic

subspace of  $V_0$  is  $\{0\}$ . Hence  $\dim W - m \leq 0$ .

Steps 1 and 3 give  $\text{ind}(\beta) = m$ , and  $\dim V_0 = n - 2m$  follows by counting. This proves the proposition. ■

So the number of planes is pinned down. What is not yet pinned down is  $V_0$  itself: different decompositions produce different subspaces, and the claim one wants is that they all carry *the same form* in a suitable sense. Making that precise needs the word “isometry” for forms, and proving it needs Witt’s theorems; both are the business of the next section.

## 13.7.6 The two familiar fields

Over  $\mathbb{R}$  and  $\mathbb{C}$  everything can be computed, because the classification is already known.

### Proposition 13.7.6 — The Witt Index over the Reals and the Complexes (a)

Let  $\beta$  be a non-degenerate symmetric form on a real  $V$  with signature  $(n_+, n_-)$  (Definition 13.5.1), so  $n_+ + n_- = n$ . Then  $\text{ind}(\beta) = \min(n_+, n_-)$ , and the anisotropic part has dimension  $|n_+ - n_-|$  and is definite.

- (b) Let  $\beta$  be a non-degenerate symmetric form on a complex  $V$  with  $\dim V = n$ . Then  $\text{ind}(\beta) = \lfloor n/2 \rfloor$ . In particular every such form with  $n \geq 2$  has an isotropic vector, and the anisotropic part has dimension 0 or 1.

*Proof.*

- (a) Put  $k = \min(n_+, n_-)$ . By Theorem 13.4.2 and the rescaling that produced Eq. 13.5.1, together with  $n_0 = 0$ , there is an orthogonal basis  $(\mathbf{v}_1, \dots, \mathbf{v}_n)$  with  $q(\mathbf{v}_i) = 1$  for  $i \leq n_+$ ,  $q(\mathbf{v}_i) = -1$  for  $n_+ < i \leq n$ , and distinct basis vectors orthogonal. For  $1 \leq i \leq k$  set

$$\mathbf{e}_i = \mathbf{v}_i + \mathbf{v}_{n_++i}, \quad \mathbf{f}_i = \frac{1}{2}(\mathbf{v}_i - \mathbf{v}_{n_++i}).$$

Then  $q(\mathbf{e}_i) = 1 + (-1) = 0$  and  $q(\mathbf{f}_i) = \frac{1}{4}(1 + (-1)) = 0$ , while

$$\beta(\mathbf{e}_i, \mathbf{f}_i) = \frac{1}{2}(q(\mathbf{v}_i) - q(\mathbf{v}_{n_++i})) = \frac{1}{2}(1 + 1) = 1,$$

so  $(\mathbf{e}_i, \mathbf{f}_i)$  is a hyperbolic pair and  $H_i = \text{span}(\mathbf{e}_i, \mathbf{f}_i)$  a hyperbolic plane. These planes are pairwise orthogonal, since for  $i \neq j$  their spanning vectors involve disjoint sets of the  $\mathbf{v}$ ’s. Let  $V_0$  be the span of the remaining  $|n_+ - n_-|$  basis vectors, all of the same sign; then  $V = H_1 \perp \dots \perp H_k \perp V_0$ , and  $\beta|_{V_0}$  is definite, hence anisotropic. By Proposition 13.7.5,  $\text{ind}(\beta) = k$ .

- (b) By Corollary 13.4.4 and non-degeneracy, there is a basis  $(\mathbf{v}_1, \dots, \mathbf{v}_n)$  with  $q(\mathbf{v}_j) = 1$  for every  $j$  and distinct basis vectors orthogonal. Write  $n = 2k + r$  with  $r \in \{0, 1\}$ , and let  $i$  denote the imaginary unit. For  $1 \leq j \leq k$  set

$$\mathbf{e}_j = \mathbf{v}_{2j-1} + i \mathbf{v}_{2j}, \quad \mathbf{f}_j = \frac{1}{2}(\mathbf{v}_{2j-1} - i \mathbf{v}_{2j}).$$

Then  $q(\mathbf{e}_j) = 1 + i^2 = 0$  and likewise  $q(\mathbf{f}_j) = \frac{1}{4}(1 + i^2) = 0$ , while  $\beta(\mathbf{e}_j, \mathbf{f}_j) = \frac{1}{2}(1 + 1) = 1$ . As in (a) these span  $k$  pairwise orthogonal hyperbolic planes, and what is left is the span of  $\mathbf{v}_n$  when  $r = 1$ , on which  $q(c\mathbf{v}_n) = c^2 \neq 0$  for  $c \neq 0$ , and

$\{\mathbf{0}\}$  when  $r = 0$ . So  $\text{ind}(\beta) = k = \lfloor n/2 \rfloor$  by [Proposition 13.7.5](#). This proves the proposition. ■

Two readings are worth recording. Over  $\mathbb{R}$  the Witt index is the smaller of the two counts of signs: a definite form has index 0, and a form of signature  $(m, m)$  has the largest possible index  $m = n/2$  — the  $\mathbb{R}^4$  example that opened the section, with  $n_+ = n_- = 2$ , sits exactly there. Over  $\mathbb{C}$  the index is as large as it can be for every non-degenerate form, because  $-1$  is a square and so a sum of two squares always factors; the only anisotropic complex spaces are  $\{\mathbf{0}\}$  and the lines.

*Quick check.* A non-degenerate real form has signature  $(3, 1)$ . What is its Witt index, and what is the anisotropic part?

*Solution.* The index is  $\min(3, 1) = 1$ , so exactly one hyperbolic plane splits off and  $\dim V_0 = 4 - 2 = 2$ . By [Proposition 13.7.6](#) (a),  $V_0$  carries a definite form of dimension  $|3 - 1| = 2$ , positive definite here since the positive signs are the ones in surplus. Concretely  $\mathbb{R}^{3,1}$  with  $q = x_1^2 + x_2^2 + x_3^2 - x_4^2$  splits as one hyperbolic plane, spanned by  $\mathbf{e}_3 \pm \mathbf{e}_4$  after rescaling, orthogonal to the positive definite plane  $\text{span}(\mathbf{e}_1, \mathbf{e}_2)$ . The largest totally isotropic subspace is a line, and the lines of this kind sweep out the light cone that Section 11 studies.

## 13.7.7 Exercises

### 13.7.7.1 A. Check your understanding

#### Exercise A1.

- Define **isotropic vector**, **totally isotropic subspace** and **hyperbolic pair**.
- Determine whether the following statement is correct, and justify your answer: a non-degenerate form has no isotropic vectors.
- What is the Witt index of a positive definite form on a space of dimension  $n$ ? Of a form of signature  $(4, 4)$ ?
- State the Gram matrix of a hyperbolic plane in a hyperbolic basis, and say why it shows the restricted form is non-degenerate.
- Which hypothesis of [Lemma 13.7.3](#) fails for the form  $q(\mathbf{x}) = x_1^2$  on  $\mathbb{R}^2$  at the isotropic vector  $\mathbf{e}_2$ ?

*Solution.*

- A vector  $\mathbf{v}$  is isotropic if  $\mathbf{v} \neq \mathbf{0}$  and  $q(\mathbf{v}) = 0$ . A subspace  $W$  is totally isotropic if  $\beta(\mathbf{u}, \mathbf{w}) = 0$  for all  $\mathbf{u}, \mathbf{w} \in W$ . A hyperbolic pair is a pair  $(\mathbf{e}, \mathbf{f})$  with  $q(\mathbf{e}) = q(\mathbf{f}) = 0$  and  $\beta(\mathbf{e}, \mathbf{f}) = 1$ .
- Incorrect. Non-degeneracy says  $\text{rad}(\beta) = \{\mathbf{0}\}$ , a statement about pairing with **every** vector. On  $\mathbb{R}^2$  with  $q = x_1^2 - x_2^2$  the form is non-degenerate, since its Gram matrix has determinant  $-1$ , and  $(1, 1)$  is isotropic.
- 0 for a positive definite form, since it is anisotropic, so the only totally isotropic sub-

space is  $\{\mathbf{0}\}$ . For signature  $(4, 4)$  the index is  $\min(4, 4) = 4$  by [Proposition 13.7.6](#) (a), the largest value [Proposition 13.7.1](#) permits in dimension 8.

- (d) It is  $\begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}$ , of determinant  $-1 \neq 0$ , and a form is non-degenerate exactly when its Gram matrix is invertible ([Proposition 13.1.2](#)).
- (e) Non-degeneracy of  $\beta$ . Here  $\mathbf{e}_2 \in \text{rad}(\beta)$ , so there is no  $\mathbf{u}$  with  $\beta(\mathbf{e}_2, \mathbf{u}) \neq 0$  and the first line of the proof has nothing to choose.

### 13.7.7.2 B. Practice

#### Exercise B1: Find the isotropic vectors.

For each of the following real quadratic forms, find all isotropic vectors, give the signature, and state the Witt index.

- (a)  $q(x_1, x_2) = x_1^2 + 2x_1x_2$  on  $\mathbb{R}^2$ .
- (b)  $q(x_1, x_2, x_3) = x_1x_2$  on  $\mathbb{R}^3$ . (*Is this form non-degenerate?*)
- (c)  $q(x_1, x_2, x_3) = x_1^2 + x_2^2 + x_3^2$  on  $\mathbb{R}^3$ .

*Solution.*

- (a)  $q = x_1(x_1 + 2x_2)$ , so  $q = 0$  exactly on the two lines  $x_1 = 0$  and  $x_1 = -2x_2$ . The isotropic vectors are the non-zero multiples of  $(0, 1)$  and of  $(-2, 1)$ . The Gram matrix is  $\begin{pmatrix} 1 & 1 \\ 1 & 0 \end{pmatrix}$ , of determinant  $-1 < 0$ , so the eigenvalues have opposite signs and the signature is  $(1, 1)$ . The Witt index is  $\min(1, 1) = 1$ ; indeed  $(0, 1)$  is isotropic and by [Lemma 13.7.3](#) the whole space is one hyperbolic plane.
- (b) The Gram matrix is  $\begin{pmatrix} 0 & 1/2 & 0 \\ 1/2 & 0 & 0 \\ 0 & 0 & 0 \end{pmatrix}$ , of rank 2, so the form is **degenerate** with  $\text{rad}(\beta) = \text{span}(\mathbf{e}_3)$  and the notion of Witt index does not apply as defined. The isotropic vectors are those with  $x_1x_2 = 0$ , namely the union of the two planes  $x_1 = 0$  and  $x_2 = 0$ , minus  $\mathbf{0}$ . On the non-degenerate part  $\text{span}(\mathbf{e}_1, \mathbf{e}_2)$  the signature is  $(1, 1)$  and the index is 1.
- (c) There are none:  $q$  is positive definite, so  $q(\mathbf{x}) = 0$  forces  $\mathbf{x} = \mathbf{0}$ . The signature is  $(3, 0)$  and the Witt index is 0.

#### Exercise B2: Build a hyperbolic basis.

- (a) For  $q(x_1, x_2) = x_1^2 + x_2^2$  on  $\mathbb{C}^2$ , find a hyperbolic pair.
- (b) For  $q(x_1, x_2, x_3) = x_1^2 + x_2^2 - x_3^2$  on  $\mathbb{R}^3$ , find a hyperbolic pair  $(\mathbf{e}, \mathbf{f})$  with  $\mathbf{e} = \mathbf{e}_1 + \mathbf{e}_3$ , and identify the anisotropic part  $\text{span}(\mathbf{e}, \mathbf{f})^{\perp_\beta}$ .

*Solution.*

- (a) Take  $\mathbf{e} = (1, i)$ , so  $q(\mathbf{e}) = 1 + i^2 = 0$ . The vector  $\mathbf{u} = (1, -i)$  has  $\beta(\mathbf{e}, \mathbf{u}) = 1 \cdot 1 + i \cdot (-i) = 2$ , so rescale:  $\mathbf{f} = \frac{1}{2}(1, -i)$  gives  $\beta(\mathbf{e}, \mathbf{f}) = 1$  and  $q(\mathbf{f}) = \frac{1}{4}(1 + i^2) = 0$ . So  $((1, i), \frac{1}{2}(1, -i))$  is a hyperbolic pair, and  $\mathbb{C}^2$  is a single hyperbolic plane.
- (b)  $q(\mathbf{e}) = 1 - 1 = 0$ , so  $\mathbf{e}$  is isotropic. Follow the proof of [Lemma 13.7.3](#):  $\mathbf{u}_0 = \mathbf{e}_1$

gives  $\beta(\mathbf{e}, \mathbf{e}_1) = 1$ , so take  $\mathbf{u} = \mathbf{e}_1$ , and  $q(\mathbf{u}) = 1$ , whence

$$\mathbf{f} = \mathbf{e}_1 - \frac{1}{2}(\mathbf{e}_1 + \mathbf{e}_3) = \frac{1}{2}(\mathbf{e}_1 - \mathbf{e}_3).$$

Check:  $q(\mathbf{f}) = \frac{1}{4}(1 - 1) = 0$  and  $\beta(\mathbf{e}, \mathbf{f}) = \frac{1}{2}(1 + 1) = 1$ . For the complement,  $\mathbf{v} = (v_1, v_2, v_3)$  is orthogonal to both  $\mathbf{e}$  and  $\mathbf{f}$  exactly when  $v_1 - v_3 = 0$  and  $v_1 + v_3 = 0$ , that is  $v_1 = v_3 = 0$ . So  $\text{span}(\mathbf{e}, \mathbf{f})^{\perp\beta} = \text{span}(\mathbf{e}_2)$ , on which  $q = x_2^2$  is positive definite. This is the decomposition of [Theorem 13.7.4](#) with  $m = 1$ , matching  $\min(2, 1) = 1$ .

### Exercise B3: A totally isotropic subspace of the largest size.

Let  $q(\mathbf{x}) = x_1x_4 + x_2x_3$  on  $\mathbb{R}^4$ .

- Write down the Gram matrix and verify that  $\beta$  is non-degenerate.
- Exhibit a totally isotropic subspace of dimension 2, and explain why none of dimension 3 exists.
- Determine the signature of  $q$  and confirm the Witt index by [Proposition 13.7.6](#) (a).

*Solution.*

- Halving the cross-term coefficients ([Eq. 13.3.1](#)),

$$\mathbf{A} = \frac{1}{2} \begin{pmatrix} 0 & 0 & 0 & 1 \\ 0 & 0 & 1 & 0 \\ 0 & 1 & 0 & 0 \\ 1 & 0 & 0 & 0 \end{pmatrix},$$

whose determinant is  $\frac{1}{16} \det$  of the reversal permutation matrix, namely  $\frac{1}{16} \cdot 1 \neq 0$ . So  $\beta$  is non-degenerate by [Proposition 13.1.2](#).

- Take  $W = \text{span}(\mathbf{e}_1, \mathbf{e}_2)$ . Every  $\mathbf{w} = (a, b, 0, 0)$  has  $q(\mathbf{w}) = 0$ , and  $\beta(\mathbf{e}_1, \mathbf{e}_2) = 0$  from the matrix, so  $W$  is totally isotropic of dimension 2. No totally isotropic subspace has dimension 3, because [Proposition 13.7.1](#) caps it at  $4/2 = 2$ .
- The two planes  $\text{span}(\mathbf{e}_1, \mathbf{e}_4)$  and  $\text{span}(\mathbf{e}_2, \mathbf{e}_3)$  are orthogonal, and  $(\mathbf{e}_1, 2\mathbf{e}_4)$  is a hyperbolic pair, since  $q(\mathbf{e}_1) = q(2\mathbf{e}_4) = 0$  and  $\beta(\mathbf{e}_1, 2\mathbf{e}_4) = 2 \cdot \frac{1}{2} = 1$ ; the same works for  $(\mathbf{e}_2, 2\mathbf{e}_3)$ . So each plane is hyperbolic, of signature  $(1, 1)$ . Hence  $q$  has signature  $(2, 2)$  and, by [Proposition 13.7.6](#) (a), Witt index  $\min(2, 2) = 2$ , attained by the  $W$  of part (b).

### 13.7.7.3 C. Going deeper

#### Exercise C1: Isotropy makes the form surjective.

Let  $\beta$  be a non-degenerate symmetric form on  $V$  over a field  $F$  with  $\text{char } F \neq 2$ .

- Prove that if  $\beta$  has an isotropic vector, then  $q: V \rightarrow F$  is surjective.
- Deduce that if some  $c \in F$  is not of the form  $q(\mathbf{v})$ , then  $\beta$  is anisotropic.
- Use (b) to show that the form  $q(x_1, x_2) = x_1^2 + x_2^2$  on  $\mathbb{F}_3^2$  is anisotropic, and give

its Witt index.

*Solution.*

- (a) Let  $\mathbf{e}$  be isotropic. By [Lemma 13.7.3](#) there is  $\mathbf{f}$  with  $(\mathbf{e}, \mathbf{f})$  a hyperbolic pair. For  $c \in F$ , put  $\mathbf{v} = \frac{c}{2}\mathbf{e} + \mathbf{f}$ ; this is legitimate since  $\text{char } F \neq 2$ . Then

$$q(\mathbf{v}) = \frac{c^2}{4}q(\mathbf{e}) + 2 \cdot \frac{c}{2}\beta(\mathbf{e}, \mathbf{f}) + q(\mathbf{f}) = 0 + c + 0 = c.$$

So every  $c \in F$  is a value of  $q$ .

- (b) Contrapositive of (a): if  $q$  is not surjective then  $\beta$  has no isotropic vector.
- (c) The squares in  $\mathbb{F}_3$  are  $0^2 = 0$ ,  $1^2 = 1$ ,  $2^2 = 1$ , so the values of  $x_1^2 + x_2^2$  are 0, 1, 2 — which is all of  $\mathbb{F}_3$ , so (b) gives nothing directly. Argue instead from the definition:  $q(\mathbf{x}) = 0$  needs  $x_1^2 = -x_2^2 = 2x_2^2$ . If  $x_2 \neq 0$  then  $x_2^2 = 1$ , so  $x_1^2 = 2$ , which is not a square in  $\mathbb{F}_3$ ; hence  $x_2 = 0$  and then  $x_1 = 0$ . The form is anisotropic, its Witt index is 0, and the Gram matrix  $\mathbf{I}_2$  shows it is non-degenerate. Over  $\mathbb{F}_5$  the same form is isotropic, since  $1^2 + 2^2 = 0$ .

### Exercise C2: A maximal totally isotropic subspace and its complement.

Let  $\beta$  be non-degenerate symmetric on  $V$  with  $\dim V = n$ , and let  $W \subseteq V$  be totally isotropic with  $\dim W = \text{ind}(\beta) = m$ .

- (a) Prove that  $\dim W^{\perp\beta} = n - m$  and  $W \subseteq W^{\perp\beta}$ .
- (b) Prove that the form induced by  $\beta$  on the quotient  $W^{\perp\beta}/W$  is well defined and anisotropic. *Hint: a coset with an isotropic representative would enlarge  $W$ .*
- (c) Deduce that  $n - 2m = \dim(W^{\perp\beta}/W)$  agrees with the dimension of the anisotropic part in [Theorem 13.7.4](#).

*Solution.*

- (a) The dimension is [Exercise \(C2: Orthogonal complements for a form\)](#) (b), using non-degeneracy, and the inclusion is [Definition 13.7.2](#).
- (b) For  $\mathbf{u}, \mathbf{v} \in W^{\perp\beta}$  and  $\mathbf{w}, \mathbf{w}' \in W$ ,

$$\beta(\mathbf{u} + \mathbf{w}, \mathbf{v} + \mathbf{w}') = \beta(\mathbf{u}, \mathbf{v}),$$

because each of the three remaining terms pairs a vector of  $W$  with a vector of  $W^{\perp\beta} \supseteq W$ , hence vanishes. So  $\tilde{\beta}(\mathbf{u} + W, \mathbf{v} + W) := \beta(\mathbf{u}, \mathbf{v})$  is well defined, and it is bilinear and symmetric. Suppose  $\tilde{q}(\mathbf{u} + W) = 0$  with  $\mathbf{u} \notin W$ , that is  $q(\mathbf{u}) = 0$  with  $\mathbf{u} \in W^{\perp\beta} \setminus W$ . Then  $W + \text{span}(\mathbf{u})$  is totally isotropic: it is spanned by isotropic vectors,  $\beta(\mathbf{u}, \mathbf{w}) = 0$  for  $\mathbf{w} \in W$  because  $\mathbf{u} \in W^{\perp\beta}$ , and  $\beta$  vanishes on  $W \times W$ . Its dimension is  $m + 1 > \text{ind}(\beta)$ , a contradiction. So  $\tilde{\beta}$  is anisotropic.

- (c) By (a),  $\dim(W^{\perp\beta}/W) = (n - m) - m = n - 2m$ . By [Proposition 13.7.5](#) the anisotropic part  $V_0$  of a Witt decomposition also has dimension  $n - 2m$ , since the index is  $m$ . (That the two anisotropic forms agree, and not merely their dimensions, is proved in the next section.)

**Exercise C3: Where characteristic 2 breaks.**

Work over  $\mathbb{F}_2$  and let  $\beta$  be the dot product on  $\mathbb{F}_2^2$ , with Gram matrix  $\mathbf{I}_2$ .

- (a) Verify that  $\beta$  is symmetric and non-degenerate, and find all  $\mathbf{v}$  with  $q(\mathbf{v}) = 0$ .
- (b) Show that  $\beta$  has a vector with  $q(\mathbf{v}) = 0$  and  $\mathbf{v} \neq \mathbf{0}$  that lies in no hyperbolic plane, and identify which line of the proof of [Lemma 13.7.3](#) fails.
- (c) Explain in one sentence why the third description of “totally isotropic” given after [Definition 13.7.2](#) is unavailable here.

*Solution.*

- (a) The Gram matrix  $\mathbf{I}_2$  is symmetric and invertible, so  $\beta$  is symmetric and non-degenerate ([Proposition 13.1.2](#)). Here  $q(\mathbf{x}) = x_1^2 + x_2^2 = (x_1 + x_2)^2$  over  $\mathbb{F}_2$ , which vanishes exactly when  $x_1 = x_2$ . So  $q(\mathbf{v}) = 0$  for  $\mathbf{v} = \mathbf{0}$  and for  $\mathbf{v} = (1, 1)$ .
- (b) Take  $\mathbf{v} = (1, 1)$ . A hyperbolic plane containing it would need  $\mathbf{f}$  with  $q(\mathbf{f}) = 0$  and  $\beta(\mathbf{v}, \mathbf{f}) = 1$ . The only non-zero vector with  $q = 0$  is  $\mathbf{v}$  itself, and  $\beta(\mathbf{v}, \mathbf{v}) = q(\mathbf{v}) = 0 \neq 1$ . The step that fails is the correction  $\mathbf{f} = \mathbf{u} - \frac{1}{2}q(\mathbf{u})\mathbf{e}$ : the element  $2 = 1 + 1$  is 0 in  $\mathbb{F}_2$  and is not invertible, so  $\frac{1}{2}$  does not exist.
- (c) Because it is [Theorem 13.3.5](#), which needs  $\text{char } F \neq 2$  to divide by 2; over  $\mathbb{F}_2$  the line  $\text{span}((1, 1))$  has  $q \equiv 0$  on it and is indeed totally isotropic here, but the implication “ $q$  vanishes on  $W$  implies  $\beta$  vanishes on  $W \times W$ ” is not available in general.

## 13.8 Witt's Extension and Cancellation Theorems

The previous section left one question open. A non-degenerate space splits as hyperbolic planes plus an anisotropic remainder, and the number of planes is pinned down by the Witt index — but the remainder is a subspace chosen along the way, and different choices give different subspaces. What should be true is that they all carry the same form. Proving it needs a theorem with a much wider reach: any isometry between two subspaces of a non-degenerate space is the restriction of an isometry of the whole space. That theorem is the longest proof of the chapter, and the cancellation statement that finishes Section 7 falls out of it in four lines.

Throughout,  $F$  is a field with  $\text{char } F \neq 2$  ([Definition 0.7.2](#)), all spaces are finite-dimensional over  $F$ , and all forms are **symmetric** bilinear forms ([Definition 13.3.1](#)). Orthogonal complements  $U^{\perp_\beta}$ , orthogonal direct sums  $U \perp W$ , and isotropic and hyperbolic vocabulary are as in Section 7.

### 13.8.1 Isometries of forms

Chapter 10 called a linear map an isometry when it preserved the inner product ([Definition 10.7.1](#)). The same word works here, with the form in place of the inner product and with no positivity anywhere.



*An isometry is a change of variables that the form does not notice.*

### Definition 13.8.1 — Isometry of Bilinear Spaces

Let  $\beta$  be a bilinear form on  $V$  and  $\beta'$  one on  $V'$ . An **isometry** from  $(V, \beta)$  to  $(V', \beta')$  is a linear **bijection**  $\sigma: V \rightarrow V'$  with

$$\beta'(\sigma(\mathbf{u}), \sigma(\mathbf{v})) = \beta(\mathbf{u}, \mathbf{v}) \quad \text{for all } \mathbf{u}, \mathbf{v} \in V.$$

The two spaces are **isometric**, written  $(V, \beta) \cong (V', \beta')$  or just  $V \cong V'$ , if such a  $\sigma$  exists. An isometry of  $(V, \beta)$  with itself is an **isometry of  $V$** .

When  $U \subseteq V$  and  $U' \subseteq V'$  are subspaces, “an isometry  $U \rightarrow U'$ ” always means an isometry for the **restricted** forms  $\beta|_{U \times U}$  and  $\beta'|_{U' \times U'}$ .

Two clauses need pointing at. The map must be a **bijection**, so that the inverse is again an isometry; and the identity must hold for **all** pairs, though by bilinearity it is enough to check it on a pair of bases. The symbol  $\cong$  is the one used for isomorphism throughout the book, and no confusion arises: an isometry is an isomorphism that also respects the form, and within this chapter  $\cong$  between spaces carrying forms always means isometric.

Three readings make the notion concrete.

- **Isometry is congruence in disguise.** If  $\mathcal{B}$  is a basis of  $V$  and  $\mathcal{B}'$  one of  $V'$ , then  $\sigma$  is an isometry exactly when the Gram matrices satisfy  $[\beta]_{\mathcal{B}} = \mathbf{P}^{\top} [\beta']_{\mathcal{B}'} \mathbf{P}$  with  $\mathbf{P} = [\sigma]_{\mathcal{B}}^{\mathcal{B}'}$ . So  $(V, \beta) \cong (V', \beta')$  if and only if their Gram matrices are congruent (Definition 13.2.1). Every classification theorem of this chapter is therefore a classification up to isometry.
- **Over  $\mathbb{R}$ .** Two real spaces with non-degenerate symmetric forms are isometric exactly when they have the same signature, by Corollary 13.5.2. So, writing  $\mathbb{R}^{p,q}$  for  $\mathbb{R}^{p+q}$  carrying the form of signature  $(p, q)$ , we have  $\mathbb{R}^{2,1} \not\cong \mathbb{R}^{1,2}$  even though both are three-dimensional and both non-degenerate.
- **The degenerate case.** An isometry carries  $\text{rad}(\beta)$  onto  $\text{rad}(\beta')$ , because  $\beta'(\sigma\mathbf{u}, \sigma\mathbf{v}) = \beta(\mathbf{u}, \mathbf{v})$  and  $\sigma$  is onto. In particular isometric spaces have radicals of equal dimension.

The isometries of a fixed  $(V, \beta)$  form a group under composition; Section 10 names the classical examples. For now only two kinds are needed, and the second is built from the first.

## 13.8.2 Reflections

Chapter 10 §08 built Householder reflections out of an inner product, to clear a column. The same formula works for any symmetric form, provided the vector we reflect in is anisotropic, and it is the only explicit isometry this section needs.

### Definition 13.8.2 — Reflection in an Anisotropic Vector

Let  $\beta$  be a symmetric bilinear form on  $V$  and let  $\mathbf{w} \in V$  be anisotropic, that is  $q(\mathbf{w}) \neq 0$ .

The **reflection in  $\mathbf{w}$**  is the map

$$\tau_{\mathbf{w}}: V \rightarrow V, \quad \tau_{\mathbf{w}}(\mathbf{v}) = \mathbf{v} - \frac{2\beta(\mathbf{v}, \mathbf{w})}{q(\mathbf{w})} \mathbf{w}.$$

The formula is [Definition 10.8.1](#) with  $\langle \mathbf{v}, \mathbf{w} \rangle$  replaced by  $\beta(\mathbf{v}, \mathbf{w})$  and  $\|\mathbf{w}\|^2$  by  $q(\mathbf{w})$ ; the division is legitimate precisely because  $\mathbf{w}$  is anisotropic.

### Proposition 13.8.1 — Reflections Are Involutive Isometries

Let  $\beta$  be a symmetric bilinear form on  $V$  and  $\mathbf{w} \in V$  anisotropic. Then  $\tau_{\mathbf{w}}$  is linear, and:

- (a)  $\tau_{\mathbf{w}}(\mathbf{w}) = -\mathbf{w}$ , and  $\tau_{\mathbf{w}}(\mathbf{v}) = \mathbf{v}$  for every  $\mathbf{v}$  with  $\beta(\mathbf{v}, \mathbf{w}) = 0$ ;
- (b)  $\tau_{\mathbf{w}} \circ \tau_{\mathbf{w}} = \text{id}_V$ ; in particular  $\tau_{\mathbf{w}}$  is bijective;
- (c)  $\beta(\tau_{\mathbf{w}}\mathbf{u}, \tau_{\mathbf{w}}\mathbf{v}) = \beta(\mathbf{u}, \mathbf{v})$  for all  $\mathbf{u}, \mathbf{v} \in V$ .

Hence  $\tau_{\mathbf{w}}$  is an isometry of  $V$ .

*Proof.* Linearity holds because  $\mathbf{v} \mapsto \beta(\mathbf{v}, \mathbf{w})$  is linear and  $q(\mathbf{w})$  is a fixed non-zero scalar. Write  $c(\mathbf{v}) = 2\beta(\mathbf{v}, \mathbf{w})/q(\mathbf{w})$ .

- (a)  $c(\mathbf{w}) = 2q(\mathbf{w})/q(\mathbf{w}) = 2$ , so  $\tau_{\mathbf{w}}(\mathbf{w}) = \mathbf{w} - 2\mathbf{w} = -\mathbf{w}$ . If  $\beta(\mathbf{v}, \mathbf{w}) = 0$  then  $c(\mathbf{v}) = 0$  and  $\tau_{\mathbf{w}}(\mathbf{v}) = \mathbf{v}$ .
- (b) By (a) and linearity,  $\beta(\tau_{\mathbf{w}}\mathbf{v}, \mathbf{w}) = \beta(\mathbf{v}, \mathbf{w}) - c(\mathbf{v})q(\mathbf{w}) = -\beta(\mathbf{v}, \mathbf{w})$ , so  $c(\tau_{\mathbf{w}}\mathbf{v}) = -c(\mathbf{v})$  and

$$\tau_{\mathbf{w}}(\tau_{\mathbf{w}}\mathbf{v}) = (\mathbf{v} - c(\mathbf{v})\mathbf{w}) + c(\mathbf{v})\mathbf{w} = \mathbf{v}.$$

- (c) Expanding by bilinearity and using symmetry,

$$\begin{aligned} \beta(\tau_{\mathbf{w}}\mathbf{u}, \tau_{\mathbf{w}}\mathbf{v}) &= \beta(\mathbf{u}, \mathbf{v}) - c(\mathbf{v})\beta(\mathbf{u}, \mathbf{w}) - c(\mathbf{u})\beta(\mathbf{w}, \mathbf{v}) \\ &\quad + c(\mathbf{u})c(\mathbf{v})q(\mathbf{w}). \end{aligned}$$

Now  $c(\mathbf{u})c(\mathbf{v})q(\mathbf{w}) = 4\beta(\mathbf{u}, \mathbf{w})\beta(\mathbf{v}, \mathbf{w})/q(\mathbf{w})$ , while the two middle terms together contribute  $-4\beta(\mathbf{u}, \mathbf{w})\beta(\mathbf{v}, \mathbf{w})/q(\mathbf{w})$ . They cancel, leaving  $\beta(\mathbf{u}, \mathbf{v})$ . Together with (b) this makes  $\tau_{\mathbf{w}}$  a bijective linear map preserving  $\beta$ , that is an isometry. ■

The use of reflections is that they move one anisotropic vector to any other of the same value — the one explicit construction the long proof below will need.

### Lemma 13.8.2 — Moving One Anisotropic Vector to Another

Let  $\text{char } F \neq 2$ , let  $\beta$  be a symmetric bilinear form on  $V$ , and let  $\mathbf{u}, \mathbf{u}' \in V$  satisfy

$$q(\mathbf{u}) = q(\mathbf{u}') \neq 0.$$

Then there is an isometry  $\rho$  of  $V$  with  $\rho(\mathbf{u}) = \mathbf{u}'$ , and  $\rho$  may be taken to be a reflection or a product of two reflections.

*Idea.* The reflection in  $\mathbf{u} - \mathbf{u}'$  is the natural candidate, and it works whenever  $\mathbf{u} - \mathbf{u}'$  is anisotropic. It need not be — but  $\mathbf{u} + \mathbf{u}'$  then is, because the two values  $q(\mathbf{u} - \mathbf{u}')$  and  $q(\mathbf{u} + \mathbf{u}')$  add up to  $4q(\mathbf{u}) \neq 0$  and so cannot both vanish. Reflecting in  $\mathbf{u} + \mathbf{u}'$  overshoots to  $-\mathbf{u}'$ , and one more reflection, in  $\mathbf{u}'$  itself, brings it back.

*Proof.* Expanding both by bilinearity and adding,

$$q(\mathbf{u} - \mathbf{u}') + q(\mathbf{u} + \mathbf{u}') = 2q(\mathbf{u}) + 2q(\mathbf{u}') = 4q(\mathbf{u}) \neq 0,$$

where the last step uses  $q(\mathbf{u}) = q(\mathbf{u}') \neq 0$ . So at least one of  $q(\mathbf{u} - \mathbf{u}')$  and  $q(\mathbf{u} + \mathbf{u}')$  is non-zero.

*Case 1:*  $q(\mathbf{u} - \mathbf{u}') \neq 0$ . Put  $\mathbf{w} = \mathbf{u} - \mathbf{u}'$ . Then

$$q(\mathbf{w}) = q(\mathbf{u}) - 2\beta(\mathbf{u}, \mathbf{u}') + q(\mathbf{u}') = 2(q(\mathbf{u}) - \beta(\mathbf{u}, \mathbf{u}')),$$

while  $\beta(\mathbf{u}, \mathbf{w}) = q(\mathbf{u}) - \beta(\mathbf{u}, \mathbf{u}')$ . Hence  $2\beta(\mathbf{u}, \mathbf{w})/q(\mathbf{w}) = 1$ , and

$$\tau_{\mathbf{w}}(\mathbf{u}) = \mathbf{u} - \mathbf{w} = \mathbf{u}'.$$

Take  $\rho = \tau_{\mathbf{w}}$ .

*Case 2:*  $q(\mathbf{u} - \mathbf{u}') = 0$ . Then  $q(\mathbf{u} + \mathbf{u}') = 4q(\mathbf{u}) \neq 0$ ; put  $\mathbf{w} = \mathbf{u} + \mathbf{u}'$ . The same computation with  $-\mathbf{u}'$  in place of  $\mathbf{u}'$  gives  $q(\mathbf{w}) = 2(q(\mathbf{u}) + \beta(\mathbf{u}, \mathbf{u}'))$  and  $\beta(\mathbf{u}, \mathbf{w}) = q(\mathbf{u}) + \beta(\mathbf{u}, \mathbf{u}')$ , so again  $2\beta(\mathbf{u}, \mathbf{w})/q(\mathbf{w}) = 1$  and

$$\tau_{\mathbf{w}}(\mathbf{u}) = \mathbf{u} - \mathbf{w} = -\mathbf{u}'.$$

Since  $q(\mathbf{u}') \neq 0$ , the reflection  $\tau_{\mathbf{u}'}$  is defined, and  $\tau_{\mathbf{u}'}(-\mathbf{u}') = \mathbf{u}'$  by [Proposition 13.8.1 \(a\)](#). Take  $\rho = \tau_{\mathbf{u}'} \circ \tau_{\mathbf{w}}$ , a composition of two isometries and hence an isometry. This proves the lemma. ■

*Quick check.* In  $\mathbb{R}^3$  with  $q(\mathbf{x}) = x_1^2 + x_2^2 - x_3^2$ , find a reflection carrying  $\mathbf{e}_1$  to  $\mathbf{e}_2$ .

*Solution.* Both have  $q = 1 \neq 0$ , so [Lemma 13.8.2](#) applies. Case 1 works:  $\mathbf{w} = \mathbf{e}_1 - \mathbf{e}_2$  has  $q(\mathbf{w}) = 1 + 1 = 2 \neq 0$ , so

$$\tau_{\mathbf{w}}(\mathbf{v}) = \mathbf{v} - \beta(\mathbf{v}, \mathbf{w}) \mathbf{w},$$

since  $2/q(\mathbf{w}) = 1$ . Then  $\beta(\mathbf{e}_1, \mathbf{w}) = 1$  and  $\tau_{\mathbf{w}}(\mathbf{e}_1) = \mathbf{e}_1 - (\mathbf{e}_1 - \mathbf{e}_2) = \mathbf{e}_2$ , as required. The reflection fixes  $\mathbf{e}_1 + \mathbf{e}_2$  and  $\mathbf{e}_3$ , which are the vectors orthogonal to  $\mathbf{w}$ .

### 13.8.3 Witt's extension theorem

Here is the chapter's long theorem. Read the statement carefully: the two subspaces live in the **same** space  $V$ , and the form on  $V$  is required to be non-degenerate, but nothing at all is assumed about the restrictions to  $U$  and  $U'$  — they may be degenerate, they may be zero.

**Theorem 13.8.3 — Witt's Extension Theorem**

Let  $F$  be a field with  $\text{char } F \neq 2$ , let  $V$  be a finite-dimensional vector space over  $F$ , and let  $\beta$  be a **non-degenerate** symmetric bilinear form on  $V$ . Let  $U, U' \subseteq V$  be subspaces and let

$$\sigma: U \rightarrow U'$$

be an isometry for the restricted forms. Then there is an isometry  $\varphi$  of  $V$  with  $\varphi|_U = \sigma$ .

*Idea.* Induction, but on what? Two different moves are available, and each improves a different number.

① If  $\beta|_U$  is **non-degenerate**, then  $U$  contains an anisotropic vector  $\mathbf{u}_1$ . A reflection moves  $\sigma(\mathbf{u}_1)$  back to  $\mathbf{u}_1$ , so we may assume  $\sigma$  fixes  $\mathbf{u}_1$ ; then everything of  $U$  orthogonal to  $\mathbf{u}_1$  stays inside the hyperplane  $\mathbf{u}_1^\perp$ , which is again non-degenerate and **one dimension smaller**. Recurse there and glue.

② If  $\beta|_U$  is **degenerate**, pick a non-zero  $\mathbf{z}$  in its radical. Non-degeneracy of  $\beta$  on  $V$  produces a partner  $\mathbf{w}$  making  $(\mathbf{z}, \mathbf{w})$  a hyperbolic pair orthogonal to the rest of  $U$ , and  $\sigma$  extends to  $U \oplus F\mathbf{w}$  by sending  $\mathbf{w}$  to the matching partner on the other side. The subspace has grown, but its **radical has shrunk by one**.

So the induction runs on the pair  $(\dim V, \dim \text{rad}(\beta|_U))$  read lexicographically: ① lowers the first entry, ② keeps it and lowers the second. Step 0 disposes of  $U = \{\mathbf{0}\}$ .

*Proof.* We prove the statement for every triple  $(V, U, \sigma)$  satisfying the hypotheses, by induction on the pair

$$(\dim V, \dim \text{rad}(\beta|_{U \times U}))$$

ordered lexicographically: one pair precedes another if its first entry is smaller, or the first entries agree and the second is smaller. This order on pairs of non-negative integers has no infinite descending chain, so the induction is legitimate. Write  $R = \text{rad}(\beta|_{U \times U}) = U \cap U^\perp_\beta$ .

**Step 0:**  $U = \{\mathbf{0}\}$ . Then  $U' = \{\mathbf{0}\}$  and  $\varphi = \text{id}_V$  works.

**Step 1:**  $U \neq \{\mathbf{0}\}$  and  $R = \{\mathbf{0}\}$ . Since  $\beta|_U$  is non-degenerate and  $U \neq \{\mathbf{0}\}$ , the form  $\beta|_U$  is not the zero form. By [Lemma 13.4.1](#), applied to  $\beta|_U$  on  $U$  and using  $\text{char } F \neq 2$ , there is  $\mathbf{u}_1 \in U$  with  $q(\mathbf{u}_1) \neq 0$ .

*Claim.* We may assume  $\sigma(\mathbf{u}_1) = \mathbf{u}_1$ .

*Proof.* Put  $\mathbf{u}'_1 = \sigma(\mathbf{u}_1)$ . Then  $q(\mathbf{u}'_1) = \beta(\sigma\mathbf{u}_1, \sigma\mathbf{u}_1) = q(\mathbf{u}_1) \neq 0$ , so [Lemma 13.8.2](#) gives an isometry  $\rho$  of  $V$  with  $\rho(\mathbf{u}_1) = \mathbf{u}'_1$ . The composite  $\tau = \rho^{-1} \circ \sigma$  is an isometry from  $U$  onto  $\rho^{-1}(U')$  with  $\tau(\mathbf{u}_1) = \mathbf{u}_1$ , and if  $\varphi_0$  is an isometry of  $V$  extending  $\tau$ , then  $\rho \circ \varphi_0$  is an isometry of  $V$  extending  $\sigma$ . So it suffices to treat  $\tau$ . ■

Assume then that  $\sigma(\mathbf{u}_1) = \mathbf{u}_1$ . The map  $\mathbf{u} \mapsto \beta(\mathbf{u}_1, \mathbf{u})$  is a linear functional on  $U$  taking the non-zero value  $q(\mathbf{u}_1)$  at  $\mathbf{u}_1$ , so its kernel

$$U_1 = \{ \mathbf{u} \in U : \beta(\mathbf{u}_1, \mathbf{u}) = 0 \}$$

has  $\dim U_1 = \dim U - 1$  by Rank-Nullity ([Theorem 3.3.1](#)), and  $U = \text{span}(\mathbf{u}_1) \oplus U_1$ , since  $a\mathbf{u}_1 \in U_1$  forces  $a q(\mathbf{u}_1) = 0$  and hence  $a = 0$ .

Because  $q(\mathbf{u}_1) \neq 0$ , the form restricted to  $\text{span}(\mathbf{u}_1)$  is non-degenerate, so [Proposition 13.7.2](#) gives

$$V = \text{span}(\mathbf{u}_1) \perp W, \quad W = \text{span}(\mathbf{u}_1)^{\perp_\beta},$$

with  $\beta|_W$  non-degenerate and  $\dim W = \dim V - 1$ . Now  $U_1 \subseteq W$  by definition, and  $\sigma(U_1) \subseteq W$  as well: for  $\mathbf{u} \in U_1$ ,

$$\beta(\mathbf{u}_1, \sigma\mathbf{u}) = \beta(\sigma\mathbf{u}_1, \sigma\mathbf{u}) = \beta(\mathbf{u}_1, \mathbf{u}) = 0.$$

So  $\sigma|_{U_1}: U_1 \rightarrow \sigma(U_1)$  is an isometry between two subspaces of  $W$ , and  $(W, \beta|_W)$  satisfies the hypotheses of the theorem with  $\dim W < \dim V$ . By the induction hypothesis there is an isometry  $\theta$  of  $W$  with  $\theta|_{U_1} = \sigma|_{U_1}$ .

Define  $\varphi: V \rightarrow V$  by

$$\varphi(a\mathbf{u}_1 + \mathbf{x}) = a\mathbf{u}_1 + \theta(\mathbf{x}) \quad (a \in F, \mathbf{x} \in W),$$

which is well defined and linear because  $V = \text{span}(\mathbf{u}_1) \oplus W$ , and bijective because  $\theta$  is. It is an isometry: using the orthogonality of  $\text{span}(\mathbf{u}_1)$  and  $W$  on both sides,

$$\begin{aligned} \beta(\varphi(a\mathbf{u}_1 + \mathbf{x}), \varphi(b\mathbf{u}_1 + \mathbf{y})) &= ab q(\mathbf{u}_1) + \beta(\theta\mathbf{x}, \theta\mathbf{y}) \\ &= ab q(\mathbf{u}_1) + \beta(\mathbf{x}, \mathbf{y}) \\ &= \beta(a\mathbf{u}_1 + \mathbf{x}, b\mathbf{u}_1 + \mathbf{y}). \end{aligned}$$

Finally  $\varphi$  extends  $\sigma$ : a vector  $\mathbf{u} \in U$  is  $a\mathbf{u}_1 + \mathbf{x}$  with  $\mathbf{x} \in U_1 \subseteq W$ , and

$$\varphi(\mathbf{u}) = a\mathbf{u}_1 + \theta(\mathbf{x}) = a\sigma(\mathbf{u}_1) + \sigma(\mathbf{x}) = \sigma(\mathbf{u}).$$

**Step 2:**  $R \neq \{0\}$ . Choose  $\mathbf{z} \in R$  with  $\mathbf{z} \neq 0$  and a complement  $U_0$  of  $\text{span}(\mathbf{z})$  in  $U$ , so that  $U = \text{span}(\mathbf{z}) \oplus U_0$  ([Theorem 1.7.6](#)). Since  $\sigma$  is an isometry of  $U$  onto  $U'$ , it carries  $R$  onto  $\text{rad}(\beta|_{U'})$ ; write  $\mathbf{z}' = \sigma(\mathbf{z})$  and  $U'_0 = \sigma(U_0)$ , so that  $U' = \text{span}(\mathbf{z}') \oplus U'_0$  and  $\mathbf{z}'$  lies in the radical of  $\beta|_{U'}$ . Note  $q(\mathbf{z}) = 0$ , because  $\mathbf{z}$  pairs to zero with everything in  $U$ , in particular with itself; likewise  $q(\mathbf{z}') = 0$ .

*Claim.* There is  $\mathbf{w} \in V$  with  $q(\mathbf{w}) = 0$ ,  $\beta(\mathbf{z}, \mathbf{w}) = 1$  and  $\beta(\mathbf{u}_0, \mathbf{w}) = 0$  for every  $\mathbf{u}_0 \in U_0$ .

*Proof.* Consider the linear map  $S: V \rightarrow U^*$  sending  $\mathbf{v}$  to the restriction of  $\beta(\cdot, \mathbf{v})$  to  $U$ . Its kernel is  $U^{\perp_\beta}$ , which has dimension  $\dim V - \dim U$  by [Exercise \(C2: Orthogonal complements for a form\)](#) (b), since  $\beta$  is non-degenerate. By Rank–Nullity ([Theorem 3.3.1](#)),  $\text{rank } S = \dim U = \dim U^*$ , so  $S$  is onto. Let  $\psi_{\mathbf{z}} \in U^*$  be the functional with  $\psi_{\mathbf{z}}(\mathbf{z}) = 1$  and  $\psi_{\mathbf{z}}(U_0) = 0$ , which exists because  $U = \text{span}(\mathbf{z}) \oplus U_0$ . Pick  $\mathbf{w}_0 \in V$  with  $S(\mathbf{w}_0) = \psi_{\mathbf{z}}$ , that is  $\beta(\mathbf{z}, \mathbf{w}_0) = 1$  and  $\beta(\mathbf{u}_0, \mathbf{w}_0) = 0$  for  $\mathbf{u}_0 \in U_0$ .

Put  $\mathbf{w} = \mathbf{w}_0 - \frac{1}{2}q(\mathbf{w}_0)\mathbf{z}$ , which uses  $\text{char } F \neq 2$ . Since  $q(\mathbf{z}) = 0$  and  $\beta(\mathbf{u}, \mathbf{z}) = 0$  for every  $\mathbf{u} \in U$ ,

$$\beta(\mathbf{z}, \mathbf{w}) = 1 - \frac{1}{2}q(\mathbf{w}_0)q(\mathbf{z}) = 1, \quad \beta(\mathbf{u}_0, \mathbf{w}) = 0,$$

and

$$q(\mathbf{w}) = q(\mathbf{w}_0) - q(\mathbf{w}_0)\beta(\mathbf{w}_0, \mathbf{z}) + \frac{1}{4}q(\mathbf{w}_0)^2q(\mathbf{z}) = 0,$$

using  $\beta(\mathbf{w}_0, \mathbf{z}) = \beta(\mathbf{z}, \mathbf{w}_0) = 1$ .

■

Apply the claim on both sides: it gives  $\mathbf{w}$  for  $(\mathbf{z}, U_0)$  and, by the same argument with  $\mathbf{z}'$  and  $U'_0$ , a vector  $\mathbf{w}'$  with  $q(\mathbf{w}') = 0$ ,  $\beta(\mathbf{z}', \mathbf{w}') = 1$  and  $\beta(U'_0, \mathbf{w}') = 0$ .

The vector  $\mathbf{w}$  is not in  $U$ , since  $\beta(\mathbf{z}, \mathbf{w}) = 1$  while  $\beta(\mathbf{z}, \mathbf{u}) = 0$  for every  $\mathbf{u} \in U$ ; likewise  $\mathbf{w}' \notin U'$ . So

$$\tilde{U} = U \oplus F\mathbf{w}, \quad \tilde{U}' = U' \oplus F\mathbf{w}'$$

are subspaces of  $V$  of dimension  $\dim U + 1$ . Define  $\tilde{\sigma}(\mathbf{u} + c\mathbf{w}) = \sigma(\mathbf{u}) + c\mathbf{w}'$ , a linear bijection  $\tilde{U} \rightarrow \tilde{U}'$ . It is an isometry: for  $\mathbf{u} = a\mathbf{z} + \mathbf{u}_0 \in U$  we have

$$\beta(\mathbf{u}, \mathbf{w}) = a\beta(\mathbf{z}, \mathbf{w}) + \beta(\mathbf{u}_0, \mathbf{w}) = a,$$

and  $\sigma(\mathbf{u}) = a\mathbf{z}' + \sigma(\mathbf{u}_0)$  with  $\sigma(\mathbf{u}_0) \in U'_0$ , so  $\beta(\sigma\mathbf{u}, \mathbf{w}') = a$  as well. Hence, expanding by bilinearity and using  $q(\mathbf{w}) = q(\mathbf{w}') = 0$ ,

$$\begin{aligned} \beta(\tilde{\sigma}(\mathbf{u} + c\mathbf{w}), \tilde{\sigma}(\mathbf{y} + d\mathbf{w}')) &= \beta(\sigma\mathbf{u}, \sigma\mathbf{y}) + d\beta(\sigma\mathbf{u}, \mathbf{w}') + c\beta(\sigma\mathbf{y}, \mathbf{w}') \\ &= \beta(\mathbf{u}, \mathbf{y}) + d\beta(\mathbf{u}, \mathbf{w}) + c\beta(\mathbf{y}, \mathbf{w}) \\ &= \beta(\mathbf{u} + c\mathbf{w}, \mathbf{y} + d\mathbf{w}). \end{aligned}$$

*Claim.*  $\text{rad}(\beta|_{\tilde{U}}) = R \cap U_0$ , and its dimension is  $\dim R - 1$ .

*Proof.* Let  $\mathbf{x} = a\mathbf{z} + \mathbf{u}_0 + c\mathbf{w}$  lie in  $\text{rad}(\beta|_{\tilde{U}})$ , with  $a, c \in F$  and  $\mathbf{u}_0 \in U_0$ . Pairing  $\mathbf{x}$  with  $\mathbf{w}$  gives  $a\beta(\mathbf{z}, \mathbf{w}) + \beta(\mathbf{u}_0, \mathbf{w}) + cq(\mathbf{w}) = a$ , so  $a = 0$ . Pairing  $\mathbf{x}$  with  $\mathbf{z}$  gives  $\beta(\mathbf{u}_0, \mathbf{z}) + c\beta(\mathbf{w}, \mathbf{z}) = c$ , since  $\mathbf{z} \in R$  kills  $\mathbf{u}_0$ ; so  $c = 0$ . Hence  $\mathbf{x} = \mathbf{u}_0 \in U_0$ , and  $\mathbf{x}$  pairs to zero with all of  $U \subseteq \tilde{U}$ , so  $\mathbf{x} \in R \cap U_0$ . Conversely a vector of  $R \cap U_0$  pairs to zero with  $U$  and with  $\mathbf{w}$ , hence with  $\tilde{U}$ .

For the dimension: every  $\mathbf{r} \in R \subseteq U$  is  $a\mathbf{z} + \mathbf{u}_0$ , and  $\mathbf{u}_0 = \mathbf{r} - a\mathbf{z} \in R$  because  $\mathbf{z} \in R$ ; so  $R = \text{span}(\mathbf{z}) \oplus (R \cap U_0)$  and  $\dim(R \cap U_0) = \dim R - 1$ .

■

The pair attached to the triple  $(V, \tilde{U}, \tilde{\sigma})$  is therefore  $(\dim V, \dim R - 1)$ , which precedes  $(\dim V, \dim R)$  in the lexicographic order. By the induction hypothesis there is an isometry  $\varphi$  of  $V$  with  $\varphi|_{\tilde{U}} = \tilde{\sigma}$ . Since  $U \subseteq \tilde{U}$  and  $\tilde{\sigma}|_U = \sigma$ , this  $\varphi$  extends  $\sigma$ .

Steps 0, 1 and 2 cover every case, so the induction is complete and the theorem is proved. ■

Three places earned their hypotheses. Non-degeneracy of  $\beta$  on  $V$  was used twice, once through [Proposition 13.7.2](#) in Step 1 and once through the surjectivity of  $S$  in Step 2. Characteristic  $\neq 2$  was used twice as well, in [Lemma 13.4.1](#) and in the correction  $\mathbf{w} = \mathbf{w}_0 - \frac{1}{2}q(\mathbf{w}_0)\mathbf{z}$ , which is the same correction that built a hyperbolic pair in [Lemma 13.7.3](#). And symmetry is everywhere: without it, “the” radical and “the” orthogonal complement would each split into a left and a right version.

### 13.8.4 Cancellation

The corollary is what the theorem is usually used for, and it is where a careless statement goes wrong. Witt cancellation removes a summand from an **orthogonal** decomposition of two spaces that are **already known to be isometric**, and what it produces is an isometry of the complements.

#### Corollary 13.8.4 — Witt's Cancellation Theorem

Let  $\text{char } F \neq 2$ .

- (a) Let  $\beta$  be a non-degenerate symmetric form on  $V$ , and let  $U, U' \subseteq V$  be isometric subspaces. Then  $U^{\perp_\beta} \cong U'^{\perp_\beta}$ .
- (b) Let  $(V, \beta)$  and  $(V', \beta')$  be finite-dimensional spaces with non-degenerate symmetric forms, and suppose

$$V = U \perp W, \quad V' = U' \perp W'.$$

If  $V \cong V'$  and  $U \cong U'$ , then  $W \cong W'$ .

*Idea.* For (a), extend the isometry  $U \rightarrow U'$  to the whole of  $V$  and watch where the complement goes: an isometry of  $V$  carrying  $U$  onto  $U'$  has no choice but to carry  $U^\perp$  onto  $U'^\perp$ . For (b), transport everything into the single space  $V$  along an isometry  $V \rightarrow V'$ , and apply (a).

*Proof.*

- (a) Let  $\sigma: U \rightarrow U'$  be an isometry. By [Theorem 13.8.3](#) there is an isometry  $\varphi$  of  $V$  with  $\varphi|_U = \sigma$ ; in particular  $\varphi(U) = U'$ . If  $\mathbf{x} \in U^{\perp_\beta}$  and  $\mathbf{u}' \in U'$ , write  $\mathbf{u}' = \varphi(\mathbf{u})$  with  $\mathbf{u} \in U$ ; then

$$\beta(\mathbf{u}', \varphi\mathbf{x}) = \beta(\varphi\mathbf{u}, \varphi\mathbf{x}) = \beta(\mathbf{u}, \mathbf{x}) = 0,$$

so  $\varphi(U^{\perp_\beta}) \subseteq U'^{\perp_\beta}$ . Both subspaces have dimension  $\dim V - \dim U$  by [Exercise \(C2: Orthogonal complements for a form\)](#) (b), using  $\dim U = \dim U'$ , and  $\varphi$  is injective, so the inclusion is an equality. Restricting  $\varphi$  gives an isometry  $U^{\perp_\beta} \rightarrow U'^{\perp_\beta}$ .

- (b) First,  $W = U^{\perp_\beta}$ : the orthogonality in  $V = U \perp W$  gives  $W \subseteq U^{\perp_\beta}$ , and both have

dimension  $\dim V - \dim U$ , so they are equal ([Theorem 1.6.8](#)). Likewise  $W' = U'^{\perp_{\beta'}}$ .

Let  $\psi: V \rightarrow V'$  be an isometry. Then  $\psi^{-1}(U')$  is a subspace of  $V$  isometric to  $U'$ , hence isometric to  $U$ . Moreover  $\psi^{-1}(U'^{\perp_{\beta'}}) = (\psi^{-1}(U'))^{\perp_{\beta}}$ , because  $\psi$  preserves the forms and is onto. Applying (a) inside  $V$  to the isometric subspaces  $U$  and  $\psi^{-1}(U')$ ,

$$W = U^{\perp_{\beta}} \cong (\psi^{-1}(U'))^{\perp_{\beta}} = \psi^{-1}(W') \cong W',$$

the last isometry being the restriction of  $\psi$ . This proves the corollary. ■

**Warning. Cancellation needs the whole spaces to be isometric, and the sums to be orthogonal.** Neither hypothesis can be dropped.

Drop the first: inside  $\mathbb{R}^2$  with  $q = x_1^2 + x_2^2$  and inside  $\mathbb{R}^2$  with  $q' = x_1^2 - x_2^2$ , the lines  $\text{span}(\mathbf{e}_1)$  are isometric, each carrying the form  $\langle 1 \rangle$ . Their complements are  $\text{span}(\mathbf{e}_2)$  with  $q = x_2^2$  and  $\text{span}(\mathbf{e}_2)$  with  $q' = -x_2^2$ , which are **not** isometric: one is positive definite, the other negative definite. Isometric subspaces of **different** ambient spaces say nothing about complements.

Drop the second: in  $\mathbb{R}^2$  with  $q = x_1^2 - x_2^2$ , both

$$V = \text{span}(\mathbf{e}_1) \oplus \text{span}(\mathbf{e}_1 + \mathbf{e}_2) \quad \text{and} \quad V = \text{span}(\mathbf{e}_1) \oplus \text{span}(\mathbf{e}_2)$$

are direct sums with the same first summand. The second summands carry  $q(\mathbf{e}_1 + \mathbf{e}_2) = 0$  and  $q(\mathbf{e}_2) = -1$ , so they are not isometric. The first decomposition is not orthogonal, since  $\beta(\mathbf{e}_1, \mathbf{e}_1 + \mathbf{e}_2) = 1 \neq 0$ , and that is exactly what breaks.

### 13.8.5 The Witt decomposition is unique

We can now close the question Section 7 left open.

#### Corollary 13.8.5 — The Anisotropic Part Is Determined

Let  $\text{char } F \neq 2$ , let  $\beta$  be a non-degenerate symmetric form on a finite-dimensional  $V$ , and let

$$V = H_1 \perp \cdots \perp H_m \perp V_0 = H'_1 \perp \cdots \perp H'_{m'} \perp V'_0$$

be two decompositions as in [Theorem 13.7.4](#). Then  $m = m' = \text{ind}(\beta)$  and  $V_0 \cong V'_0$ .

*Proof.* That  $m = m' = \text{ind}(\beta)$  is [Proposition 13.7.5](#). Put  $U = H_1 \perp \cdots \perp H_m$  and  $U' = H'_1 \perp \cdots \perp H'_{m'}$ . Choosing a hyperbolic basis of each plane gives bases of  $U$  and of  $U'$  in which both Gram matrices are the same block diagonal matrix, namely  $m$  copies of  $\begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}$  down the diagonal. The linear bijection  $U \rightarrow U'$  matching these bases in order is therefore an isometry.

Both  $V_0$  and  $V'_0$  are complements:  $V_0 \subseteq U^{\perp_{\beta}}$  by orthogonality, and the dimensions agree, so  $V_0 = U^{\perp_{\beta}}$  ([Theorem 1.6.8](#)), and likewise  $V'_0 = U'^{\perp_{\beta}}$ . By [Corollary 13.8.4](#) (a),  $V_0 \cong V'_0$ . This proves the corollary.



So a non-degenerate symmetric form is completely described by two pieces of data: an integer, the Witt index, and an anisotropic form, determined up to isometry. Classifying all forms over a field  $F$  therefore reduces to classifying the **anisotropic** ones, which is where the arithmetic of  $F$  enters and where the subject stops being linear algebra. Over  $\mathbb{R}$  the anisotropic forms are the definite ones, so the data  $(\text{ind}, V_0)$  is just the signature again. Over  $\mathbb{C}$  the anisotropic forms have dimension 0 or 1, so the index and the parity of  $n$  are everything. Over  $\mathbb{Q}$  the classification is a theorem of number theory and is not in this book. ■

## 13.8.6 Exercises

### 13.8.6.1 A. Check your understanding

#### Exercise A1.

- (a) State Witt's extension theorem, with all its hypotheses.
- (b) Define the reflection  $\tau_{\mathbf{w}}$  and say which vectors it fixes.
- (c) Determine whether the following statement is correct, and justify your answer: if  $U \subseteq V$  and  $U' \subseteq V'$  are isometric subspaces of non-degenerate spaces, then  $U^\perp$  and  $U'^\perp$  are isometric.
- (d) Where is  $\text{char } F \neq 2$  used in the proof of [Theorem 13.8.3](#)?
- (e) Two real non-degenerate forms of dimension 6 both have Witt index 2. Must they be isometric?

*Solution.*

- (a) Let  $\text{char } F \neq 2$ , let  $\beta$  be a non-degenerate symmetric bilinear form on a finite-dimensional  $V$  over  $F$ , let  $U, U' \subseteq V$  be subspaces and  $\sigma: U \rightarrow U'$  an isometry for the restricted forms. Then  $\sigma$  extends to an isometry of  $V$ .
- (b)  $\tau_{\mathbf{w}}(\mathbf{v}) = \mathbf{v} - (2\beta(\mathbf{v}, \mathbf{w})/q(\mathbf{w}))\mathbf{w}$ , defined for anisotropic  $\mathbf{w}$ . It fixes every  $\mathbf{v}$  with  $\beta(\mathbf{v}, \mathbf{w}) = 0$  and sends  $\mathbf{w}$  to  $-\mathbf{w}$ .
- (c) Incorrect. The warning after [Corollary 13.8.4](#) gives the witness: the line  $\text{span}(\mathbf{e}_1)$  inside  $(\mathbb{R}^2, x_1^2 + x_2^2)$  and inside  $(\mathbb{R}^2, x_1^2 - x_2^2)$  are isometric, with complements of opposite definiteness. Cancellation requires the ambient spaces to be isometric too.
- (d) Twice. In Step 1, through [Lemma 13.4.1](#), which produces an anisotropic vector in a non-zero subspace where the form is not zero. In Step 2, through the correction  $\mathbf{w} = \mathbf{w}_0 - \frac{1}{2}q(\mathbf{w}_0)\mathbf{z}$ , which divides by 2.
- (e) No. By [Corollary 13.8.5](#) each is two hyperbolic planes orthogonal to an anisotropic part of dimension  $6 - 4 = 2$ , and over  $\mathbb{R}$  an anisotropic form is definite, so that part has signature  $(2, 0)$  or  $(0, 2)$ . The two possibilities give signatures  $(4, 2)$  and  $(2, 4)$  for the whole space, which are different, so the forms need not be isometric. What [Corollary 13.8.5](#) says is that the anisotropic part is determined by the form, not that the index determines the form.

### 13.8.6.2 B. Practice

**Exercise B1: Extend an isometry by hand.**

Let  $V = \mathbb{R}^3$  with  $q(\mathbf{x}) = x_1^2 + x_2^2 - x_3^2$ , let  $U = \text{span}(\mathbf{e}_1)$ ,  $U' = \text{span}(\mathbf{e}_2)$ , and let  $\sigma(\mathbf{e}_1) = \mathbf{e}_2$ .

- (a) Verify that  $\sigma$  is an isometry of  $U$  onto  $U'$ .
- (b) Extend  $\sigma$  to an isometry of  $V$ , and write down its matrix in the standard basis.
- (c) Determine  $U^{\perp_\beta}$  and  $U'^{\perp_\beta}$  and exhibit the isometry between them produced by your extension.

*Solution.*

- (a)  $\beta(a\mathbf{e}_1, b\mathbf{e}_1) = ab$  and  $\beta(\sigma(a\mathbf{e}_1), \sigma(b\mathbf{e}_1)) = \beta(a\mathbf{e}_2, b\mathbf{e}_2) = ab$ , and  $\sigma$  is a linear bijection  $U \rightarrow U'$ .
- (b) Use the Quick check:  $\mathbf{w} = \mathbf{e}_1 - \mathbf{e}_2$  has  $q(\mathbf{w}) = 2$ , and  $\tau_{\mathbf{w}}(\mathbf{v}) = \mathbf{v} - \beta(\mathbf{v}, \mathbf{w})\mathbf{w}$  sends  $\mathbf{e}_1$  to  $\mathbf{e}_2$ . In coordinates  $\beta(\mathbf{v}, \mathbf{w}) = v_1 - v_2$ , so

$$\tau_{\mathbf{w}}(\mathbf{v}) = (v_2, v_1, v_3), \quad \mathbf{P} = \begin{pmatrix} 0 & 1 & 0 \\ 1 & 0 & 0 \\ 0 & 0 & 1 \end{pmatrix}.$$

This swaps the first two coordinates and indeed satisfies  $\mathbf{P}^\top \text{diag}(1, 1, -1)\mathbf{P} = \text{diag}(1, 1, -1)$ .

- (c)  $U^{\perp_\beta} = \{\mathbf{v} : v_1 = 0\} = \text{span}(\mathbf{e}_2, \mathbf{e}_3)$  and  $U'^{\perp_\beta} = \text{span}(\mathbf{e}_1, \mathbf{e}_3)$ . The extension maps  $\mathbf{e}_2 \mapsto \mathbf{e}_1$  and  $\mathbf{e}_3 \mapsto \mathbf{e}_3$ , an isometry between them; both carry the form  $\text{diag}(1, -1)$ , as [Corollary 13.8.4](#) (a) predicts.

**Exercise B2: Cancel a hyperbolic plane.**

Let  $H$  denote a hyperbolic plane over  $\mathbb{R}$ , and let  $A$  and  $B$  be non-degenerate real spaces with forms, with  $\dim A = \dim B = 3$ . Suppose  $H \perp A \cong H \perp B$ .

- (a) Prove that  $A \cong B$ .
- (b) If  $A$  has signature  $(2, 1)$ , what is the signature of  $B$ ?
- (c) Does  $A \perp B \cong A \perp B$  with  $A \cong A$  allow us to conclude  $B \cong B$ ? Say which part of [Corollary 13.8.4](#) is being used and why the statement is not vacuous.

*Solution.*

- (a) Put  $V = H \perp A$  and  $V' = H \perp B$ , both non-degenerate. The two copies of  $H$  are isometric, and  $V \cong V'$  by hypothesis, so [Corollary 13.8.4](#) (b) gives  $A \cong B$ .
- (b) By (a),  $B \cong A$ , so  $B$  has signature  $(2, 1)$  as well; isometric real spaces have equal signatures by [Corollary 13.5.2](#).
- (c) It uses (b) of [Corollary 13.8.4](#) with  $U = U' = A$  and  $W = W' = B$ ; the conclusion  $B \cong B$  is true but empty. The content of (b) appears only when the two decompositions are genuinely different, as in (a) above, where the hypothesis  $V \cong V'$  is an assumption about the **whole** spaces and not about  $A$  and  $B$ .

**Exercise B3: Reflections in an indefinite space.**

In  $\mathbb{R}^2$  with  $q(\mathbf{x}) = x_1^2 - x_2^2$ :

- Compute the matrix of  $\tau_{\mathbf{e}_1}$  and of  $\tau_{\mathbf{e}_2}$  in the standard basis, and check each satisfies  $\mathbf{P}^\top \mathbf{A} \mathbf{P} = \mathbf{A}$  with  $\mathbf{A} = \text{diag}(1, -1)$ .
- Explain why  $\tau_{\mathbf{v}}$  is undefined for  $\mathbf{v} = (1, 1)$ , and say what fails.
- Find an isometry carrying  $(1, 0)$  to  $(-1, 0)$  and one carrying  $(2, \sqrt{3})$  to  $(-2, \sqrt{3})$ . (Check the values of  $q$  first.)

*Solution.*

- $q(\mathbf{e}_1) = 1$  and  $\beta(\mathbf{v}, \mathbf{e}_1) = v_1$ , so  $\tau_{\mathbf{e}_1}(\mathbf{v}) = \mathbf{v} - 2v_1\mathbf{e}_1 = (-v_1, v_2)$ , with matrix  $\text{diag}(-1, 1)$ . Similarly  $q(\mathbf{e}_2) = -1$  and  $\beta(\mathbf{v}, \mathbf{e}_2) = -v_2$ , so the coefficient is  $2(-v_2)/(-1) = 2v_2$  and  $\tau_{\mathbf{e}_2}(\mathbf{v}) = \mathbf{v} - 2v_2\mathbf{e}_2 = (v_1, -v_2)$ , with matrix  $\text{diag}(1, -1)$ . Both matrices  $\mathbf{P}$  are diagonal with entries  $\pm 1$ , so  $\mathbf{P}^\top \mathbf{A} \mathbf{P} = \mathbf{P} \mathbf{A} \mathbf{P} = \mathbf{A}$ , as each sign is squared.
- $q(1, 1) = 1 - 1 = 0$ , so  $(1, 1)$  is isotropic and the denominator  $q(\mathbf{w})$  in [Definition 13.8.2](#) is zero. The formula divides by 0; there is no reflection in an isotropic vector.
- Both  $(1, 0)$  and  $(-1, 0)$  have  $q = 1 \neq 0$ . Here  $\mathbf{u} - \mathbf{u}' = (2, 0)$  with  $q = 4 \neq 0$ , so Case 1 of [Lemma 13.8.2](#) applies:  $\tau_{(2,0)} = \tau_{\mathbf{e}_1}$  works, and indeed  $\tau_{\mathbf{e}_1}(1, 0) = (-1, 0)$ . For the second pair,  $q(2, \sqrt{3}) = 4 - 3 = 1 = q(-2, \sqrt{3})$ . Now  $\mathbf{u} - \mathbf{u}' = (4, 0)$ , with  $q = 16 \neq 0$ , so  $\tau_{\mathbf{e}_1}$  again works: it negates the first coordinate and fixes the second.

**13.8.6.3 C. Going deeper****Exercise C1: Inertia from cancellation.**

Work over  $\mathbb{R}$  and let  $V$  be an  $n$ -dimensional space with a non-degenerate symmetric form.

- Prove that if  $\mathbb{R}^{p,q}$  denotes  $\mathbb{R}^{p+q}$  with the form  $x_1^2 + \cdots + x_p^2 - x_{p+1}^2 - \cdots - x_{p+q}^2$ , then  $\mathbb{R}^{p,q} \cong \mathbb{R}^{p',q'}$  forces  $p = p'$  and  $q = q'$ , using only [Corollary 13.8.4](#) and [Proposition 13.7.6](#).
- Explain in one sentence how this compares with the proof of [Theorem 13.5.1](#) in Section 5.

*Solution.*

- Suppose  $\mathbb{R}^{p,q} \cong \mathbb{R}^{p',q'}$ . Equal dimension gives  $p + q = p' + q' = n$ . By [Proposition 13.7.6](#) (a) the Witt index is an isometry invariant, so  $\min(p, q) = \min(p', q') = k$ . By [Corollary 13.8.5](#) each space is  $k$  hyperbolic planes orthogonal to an anisotropic part, and those anisotropic parts are isometric. The anisotropic part of  $\mathbb{R}^{p,q}$  is positive definite of dimension  $p - q$  when  $p \geq q$  and negative definite of dimension  $q - p$  when  $q > p$ . A positive definite form is never isometric to a negative definite one of the same dimension, since  $q > 0$  on one and  $q < 0$  on the other away from  $\mathbf{0}$  — unless both dimensions are 0. So the two anisotropic parts have the same dimension and the same sign, giving  $p - q = p' - q'$ . With  $p + q = p' + q'$  this forces

$p = p'$  and  $q = q'$ .

- (b) Section 5 proved this directly, by producing a positive subspace and a non-positive subspace whose dimensions are too large to intersect trivially; the route here instead invokes the whole Witt machinery, so it is longer but shows that inertia is the real instance of a statement that holds over every field of characteristic  $\neq 2$ .

### Exercise C2: The isometry group acts transitively.

Let  $\beta$  be a non-degenerate symmetric form on  $V$ , with  $\text{char } F \neq 2$ , and let  $G$  be the group of isometries of  $V$ .

- (a) Prove that if  $q(\mathbf{u}) = q(\mathbf{u}') \neq 0$  then some  $\varphi \in G$  has  $\varphi(\mathbf{u}) = \mathbf{u}'$ .  
 (b) Prove that if  $\mathbf{u}$  and  $\mathbf{u}'$  are both isotropic and non-zero, then some  $\varphi \in G$  has  $\varphi(\mathbf{u}) = \mathbf{u}'$ . *Hint: an isotropic line is a one-dimensional subspace on which the form is zero.*  
 (c) Deduce that  $G$  acts transitively on each “level set”  $\{\mathbf{v} \neq \mathbf{0} : q(\mathbf{v}) = c\}$  for  $c \neq 0$ , and on the set of non-zero isotropic vectors.

*Solution.*

- (a) This is [Lemma 13.8.2](#), which already produces such an isometry as a product of at most two reflections. Alternatively,  $\text{span}(\mathbf{u}) \rightarrow \text{span}(\mathbf{u}')$ ,  $\mathbf{u} \mapsto \mathbf{u}'$ , is an isometry of one-dimensional subspaces because  $q(\mathbf{u}) = q(\mathbf{u}')$ , so [Theorem 13.8.3](#) extends it.  
 (b) The map  $\sigma: \text{span}(\mathbf{u}) \rightarrow \text{span}(\mathbf{u}')$  with  $\sigma(a\mathbf{u}) = a\mathbf{u}'$  is a linear bijection, and

$$\beta(\sigma(a\mathbf{u}), \sigma(b\mathbf{u})) = abq(\mathbf{u}') = 0 = abq(\mathbf{u}) = \beta(a\mathbf{u}, b\mathbf{u}),$$

so it is an isometry of the restricted forms — both of which are zero. By [Theorem 13.8.3](#) it extends to  $\varphi \in G$ . Note the restricted forms here are **degenerate**, which is exactly the case Step 2 of the proof was built for.

- (c) Immediate from (a) and (b): any two members of such a set are carried to one another by some  $\varphi \in G$ . In particular, over  $\mathbb{R}^{1,3}$  the isometry group moves any non-zero null vector to any other, which is the statement that the light cone has no distinguished direction.

### Exercise C3: What extension does not say.

- (a) Give an example of a non-degenerate  $(V, \beta)$  and subspaces  $U, U'$  of the **same dimension** that are not isometric, and say why [Theorem 13.8.3](#) does not apply.  
 (b) Let  $\beta$  be non-degenerate on  $V$  and let  $U \subseteq V$  be totally isotropic with  $\dim U = \text{ind}(\beta)$ . Prove that any two such  $U$  are carried onto one another by an isometry of  $V$ .  
 (c) Deduce that all maximal totally isotropic subspaces of a non-degenerate space have the same dimension, and compare with [Proposition 13.7.1](#).

*Solution.*

- (a) Take  $V = \mathbb{R}^2$  with  $q = x_1^2 - x_2^2$ ,  $U = \text{span}(\mathbf{e}_1)$  and  $U' = \text{span}(\mathbf{e}_2)$ . Both are lines,

but  $q$  is positive on  $U \setminus \{0\}$  and negative on  $U' \setminus \{0\}$ , so no linear bijection can preserve the form. [Theorem 13.8.3](#) has nothing to extend: its hypothesis is that an isometry  $U \rightarrow U'$  is **given**, and here none exists.

- (b) Let  $U$  and  $U'$  be totally isotropic of the same dimension  $k$ . Any linear bijection  $\sigma: U \rightarrow U'$  is an isometry, since both restricted forms are identically zero, so  $\beta(\sigma\mathbf{u}, \sigma\mathbf{v}) = 0 = \beta(\mathbf{u}, \mathbf{v})$  for all  $\mathbf{u}, \mathbf{v} \in U$ . By [Theorem 13.8.3](#),  $\sigma$  extends to an isometry  $\varphi$  of  $V$ , and  $\varphi(U) = U'$ .
- (c) Let  $U$  be maximal among totally isotropic subspaces, and let  $U_1$  be one of the largest dimension  $m = \text{ind}(\beta)$ . If  $\dim U < m$ , choose any  $k$ -dimensional subspace  $U_2 \subseteq U_1$  with  $k = \dim U$ ; it is totally isotropic, so by (b) there is an isometry  $\varphi$  of  $V$  with  $\varphi(U) = U_2$ . Then  $\varphi^{-1}(U_1)$  is totally isotropic of dimension  $m > k$  and contains  $\varphi^{-1}(U_2) = U$ , contradicting maximality. So  $\dim U = m$  for every maximal  $U$ . [Proposition 13.7.1](#) says only  $m \leq n/2$ ; the present statement is sharper, since it says that no totally isotropic subspace can be maximal below the top dimension.

## 13.9 Alternating Forms and Darboux Bases

Almost every classification theorem so far in this chapter has carried the same asterisk — the exceptions being the crude invariants of Section 2, rank and the square class of the determinant, which need nothing of the field. Diagonalize a symmetric form: characteristic  $\neq 2$ . Split a form into a symmetric and an alternating half: characteristic  $\neq 2$ . Recover a form from its quadratic form: characteristic  $\neq 2$ . This section has no asterisk. Alternating forms admit a complete classification over **every** field, with no hypothesis on the characteristic at all, and the classification is as clean as a classification gets: one congruence class for each even rank.

Throughout,  $F$  is an arbitrary field,  $V$  is a finite-dimensional vector space over  $F$ , and  $\beta$  is an **alternating** bilinear form on  $V$  ([Definition 6.2.2](#)), meaning  $\beta(\mathbf{v}, \mathbf{v}) = 0$  for **every**  $\mathbf{v} \in V$ . Two facts from Section 3 are used constantly. First,  $\beta$  is then skew-symmetric,  $\beta(\mathbf{u}, \mathbf{v}) = -\beta(\mathbf{v}, \mathbf{u})$ , by [Theorem 13.3.3](#) (a) — and that direction needs no hypothesis on  $F$ . Second, by [Proposition 13.3.2](#) (b) the matrix of  $\beta$  in any basis is skew-symmetric with **zero diagonal**, and we call such a matrix **alternating**. Symmetry of the situation is worth noting once: since  $\beta(\mathbf{u}, \mathbf{v}) = 0$  if and only if  $\beta(\mathbf{v}, \mathbf{u}) = 0$ , “orthogonal” needs no left or right, and the radical  $\text{rad}(\beta)$  of [Definition 13.1.3](#) is what one would want it to be from either side.

### 13.9.1 The two-dimensional model

A symmetric form has a one-dimensional model to build from: on a line  $\text{span}(\mathbf{v})$  it is  $q(c\mathbf{v}) = c^2q(\mathbf{v})$ , a scalar multiple of the squaring function, and Section 4 assembled a general symmetric form out of such lines. An alternating form offers nothing on a line at all. For any  $\mathbf{v}$  and any scalars  $c, d$ ,

$$\beta(c\mathbf{v}, d\mathbf{v}) = cd\beta(\mathbf{v}, \mathbf{v}) = 0,$$

so  $\beta$  restricted to any line is zero. The one-dimensional pieces are all identical and all trivial. The smallest place where an alternating form has something to say is a plane, and there is essentially only one thing it can say there.

**Example. (Alternating Forms on a Plane)**

Describe all alternating bilinear forms on  $F^2$ , and decide which are non-degenerate.

*Solution.* An alternating  $2 \times 2$  matrix has zero diagonal and  $a_{21} = -a_{12}$ , so it is

$$\mathbf{A} = \begin{pmatrix} 0 & a \\ -a & 0 \end{pmatrix}$$

for a single scalar  $a \in F$ , and every such matrix is alternating. By [Theorem 13.1.1](#) the corresponding form is  $\beta(\mathbf{x}, \mathbf{y}) = a(x_1y_2 - x_2y_1)$ . Its determinant is  $a^2$ , so by [Proposition 13.1.2](#) the form is non-degenerate exactly when  $a \neq 0$ .

When  $a \neq 0$ , replacing the basis vector  $\mathbf{e}_2$  by  $a^{-1}\mathbf{e}_2$  replaces  $a$  by 1. So up to a change of basis there are exactly **two** alternating forms on  $F^2$ : the zero form and the one with matrix  $\begin{pmatrix} 0 & 1 \\ -1 & 0 \end{pmatrix}$ , whatever the field.

That last matrix is the atom of the theory, and the pair of vectors producing it deserves a name. It is the alternating counterpart of the hyperbolic plane of Section 7: a two-dimensional space carrying a non-degenerate form on which the associated quadratic function vanishes identically. Here the vanishing is automatic, because *every* alternating form vanishes on the diagonal, so the whole content is in the pairing.

*A hyperbolic pair is two vectors that pair to 1 and to nothing else.*

**Definition 13.9.1 — Hyperbolic Pair**

Let  $\beta$  be an alternating form on  $V$ . A **hyperbolic pair** for  $\beta$  is an ordered pair  $(\mathbf{u}, \mathbf{w})$  of vectors of  $V$  with

$$\beta(\mathbf{u}, \mathbf{w}) = 1.$$

Its **span**  $H = \text{span}(\mathbf{u}, \mathbf{w})$  is called a **hyperbolic plane** for  $\beta$ .

The definition looks as though it asks for one condition, but it quietly supplies three more. Alternation already gives  $\beta(\mathbf{u}, \mathbf{u}) = \beta(\mathbf{w}, \mathbf{w}) = 0$ , and skewness gives  $\beta(\mathbf{w}, \mathbf{u}) = -1$ . So the matrix of  $\beta$  restricted to  $H$  in the ordered basis  $(\mathbf{u}, \mathbf{w})$  is exactly  $\begin{pmatrix} 0 & 1 \\ -1 & 0 \end{pmatrix}$ . Also,  $\mathbf{u}$  and  $\mathbf{w}$  are automatically linearly independent: if  $\mathbf{w} = c\mathbf{u}$  then  $\beta(\mathbf{u}, \mathbf{w}) = c\beta(\mathbf{u}, \mathbf{u}) = 0 \neq 1$ , and  $\mathbf{u} \neq \mathbf{0}$  for the same reason. So  $H$  really is a plane,  $\dim H = 2$ , and  $\beta$  restricted to  $H$  is non-degenerate.

*Warning.* **An alternating form is never diagonalizable, except when it is zero.** Section 4 found an orthogonal basis for every symmetric form in characteristic  $\neq 2$ ; no such basis exists here. If  $\mathcal{B}$  is any basis and  $[\beta]_{\mathcal{B}}$  is diagonal, then its off-diagonal entries vanish by assumption and its diagonal entries vanish because  $\beta$  is alternating, so  $[\beta]_{\mathcal{B}} = \mathbf{0}$  and  $\beta = 0$ . The standard form below is a block form, and it has to be: the  $2 \times 2$  block is irreducible.

**13.9.2 Splitting off a hyperbolic pair**

The move that drives the whole section is the one Section 7 used for isotropic vectors: find a small non-degenerate piece, and cut the space along it. Here the small piece is a hyperbolic

plane, and it can always be found as soon as  $\beta$  is not identically zero.

### Lemma 13.9.1 — Peeling Off a Hyperbolic Plane

Let  $\beta$  be an alternating form on a finite-dimensional  $V$ , and suppose  $\beta \neq 0$ .

- (a) There is a hyperbolic pair  $(\mathbf{u}, \mathbf{w})$  for  $\beta$ .
- (b) For any such pair, with  $H = \text{span}(\mathbf{u}, \mathbf{w})$  and

$$H^{\perp\beta} = \{ \mathbf{v} \in V : \beta(\mathbf{u}, \mathbf{v}) = \beta(\mathbf{w}, \mathbf{v}) = 0 \},$$

we have  $V = H \oplus H^{\perp\beta}$  and  $\dim H^{\perp\beta} = \dim V - 2$ .

- (c)  $\text{rad}(\beta) = \text{rad}(\beta|_{H^{\perp\beta}})$ . In particular, if  $\beta$  is non-degenerate then so is its restriction to  $H^{\perp\beta}$ .

*Idea.* For (a), the hypothesis  $\beta \neq 0$  hands us two vectors pairing to something non-zero; divide one of them by that scalar. For (b), the two conditions cutting out  $H^{\perp\beta}$  are two linear conditions, so the count should be  $\dim V - 2$ , and Rank–Nullity will say so once we check that the two conditions are independent — which is exactly what  $\beta(\mathbf{u}, \mathbf{w}) = 1$  provides. Part (c) is the bookkeeping that lets the induction in the theorem keep its hypothesis.

*Proof.*

- (a) Since  $\beta \neq 0$  there are  $\mathbf{u}_0, \mathbf{w}_0 \in V$  with  $c := \beta(\mathbf{u}_0, \mathbf{w}_0) \neq 0$ . Put  $\mathbf{u} = \mathbf{u}_0$  and  $\mathbf{w} = c^{-1}\mathbf{w}_0$ , which is legitimate because  $c \neq 0$  and  $F$  is a field. Then  $\beta(\mathbf{u}, \mathbf{w}) = c^{-1}\beta(\mathbf{u}_0, \mathbf{w}_0) = 1$  by linearity in the second argument.
- (b) Define  $\varphi: V \rightarrow F^2$  by  $\varphi(\mathbf{v}) = (\beta(\mathbf{u}, \mathbf{v}), \beta(\mathbf{w}, \mathbf{v}))$ . It is linear, because  $\beta$  is linear in its second argument, and  $\ker \varphi = H^{\perp\beta}$  by the definition of  $H^{\perp\beta}$ . It is surjective:

$$\varphi(\mathbf{w}) = (\beta(\mathbf{u}, \mathbf{w}), \beta(\mathbf{w}, \mathbf{w})) = (1, 0), \quad \varphi(\mathbf{u}) = (0, -1),$$

using  $\beta(\mathbf{w}, \mathbf{w}) = \beta(\mathbf{u}, \mathbf{u}) = 0$  and  $\beta(\mathbf{w}, \mathbf{u}) = -1$ , and these two values span  $F^2$ . Hence  $\text{rank } \varphi = 2$  and  $\dim H^{\perp\beta} = \dim V - 2$  by Rank–Nullity (Theorem 3.3.1).

Next,  $H \cap H^{\perp\beta} = \{0\}$ . Let  $\mathbf{v} = a\mathbf{u} + b\mathbf{w}$  lie in  $H^{\perp\beta}$ . Then  $0 = \beta(\mathbf{u}, \mathbf{v}) = a\beta(\mathbf{u}, \mathbf{u}) + b\beta(\mathbf{u}, \mathbf{w}) = b$ , and  $0 = \beta(\mathbf{w}, \mathbf{v}) = a\beta(\mathbf{w}, \mathbf{u}) = -a$ , so  $a = b = 0$ . Hence the sum  $H + H^{\perp\beta}$  is direct (Theorem 1.7.4), and by Theorem 1.7.3 its dimension is  $2 + (\dim V - 2) = \dim V$ , so it is all of  $V$  (Theorem 1.6.8).

- (c) ( $\supseteq$ ) Let  $\mathbf{v} \in H^{\perp\beta}$  satisfy  $\beta(\mathbf{z}, \mathbf{v}) = 0$  for every  $\mathbf{z} \in H^{\perp\beta}$ . Any  $\mathbf{y} \in V$  decomposes as  $\mathbf{y} = \mathbf{h} + \mathbf{z}$  with  $\mathbf{h} \in H$  and  $\mathbf{z} \in H^{\perp\beta}$  by (b), and  $\beta(\mathbf{h}, \mathbf{v}) = -\beta(\mathbf{v}, \mathbf{h}) = 0$  because  $\mathbf{v} \in H^{\perp\beta}$  and  $\mathbf{h}$  is a combination of  $\mathbf{u}$  and  $\mathbf{w}$ . Hence  $\beta(\mathbf{y}, \mathbf{v}) = 0$  for every  $\mathbf{y} \in V$ , so  $\mathbf{v} \in \text{rad}(\beta)$ .

( $\subseteq$ ) Let  $\mathbf{v} \in \text{rad}(\beta)$ . Then in particular  $\beta(\mathbf{u}, \mathbf{v}) = \beta(\mathbf{w}, \mathbf{v}) = 0$ , so  $\mathbf{v} \in H^{\perp\beta}$ , and  $\beta(\mathbf{z}, \mathbf{v}) = 0$  for every  $\mathbf{z}$  at all, so  $\mathbf{v}$  lies in the radical of the restriction. This proves the lemma. ■

Notice what the proof did **not** use: it never divided by 2, never distinguished  $\beta$  from  $-\beta$ , and never needed a vector with  $\beta(\mathbf{v}, \mathbf{v}) \neq 0$  — which is fortunate, since no such vector exists. Com-



pare the corresponding step for symmetric forms, [Lemma 13.4.1](#) in Section 4: there the whole difficulty was producing a vector on which the quadratic form did not vanish, and producing it cost the hypothesis  $\text{char } F \neq 2$ . Here the quadratic form vanishes everywhere and we do not care.

### 13.9.3 The standard form

With the lemma in hand the classification is an induction that peels two dimensions at a time.

#### Theorem 13.9.2 — Standard Form of an Alternating Form

Let  $F$  be **any** field, let  $V$  be a vector space over  $F$  with  $\dim V = n$ , and let  $\beta$  be an alternating bilinear form on  $V$ . Then there are an integer  $m \geq 0$  with  $2m \leq n$  and a basis

$$\mathcal{B} = (\mathbf{u}_1, \dots, \mathbf{u}_m, \mathbf{w}_1, \dots, \mathbf{w}_m, \mathbf{z}_1, \dots, \mathbf{z}_{n-2m})$$

of  $V$  such that, for all  $i, j$ ,

$$\beta(\mathbf{u}_i, \mathbf{w}_j) = \delta_{ij}, \quad \beta(\mathbf{u}_i, \mathbf{u}_j) = \beta(\mathbf{w}_i, \mathbf{w}_j) = 0,$$

and  $\beta(\mathbf{z}_k, \mathbf{v}) = 0$  for every  $k$  and every  $\mathbf{v} \in V$ . Equivalently, in this basis

$$[\beta]_{\mathcal{B}} = \mathbf{\Omega}_{2m} \oplus \mathbf{0}_{n-2m}, \quad \mathbf{\Omega}_{2m} := \begin{pmatrix} \mathbf{0} & \mathbf{I}_m \\ -\mathbf{I}_m & \mathbf{0} \end{pmatrix}. \quad (13.9.1)$$

In particular:

- (a) every alternating  $\mathbf{A} \in M_n(F)$  is congruent to  $\mathbf{\Omega}_{2m} \oplus \mathbf{0}_{n-2m}$  for exactly one  $m$ ;
- (b) if  $\beta$  is non-degenerate then  $n = 2m$  is **even** and  $[\beta]_{\mathcal{B}} = \mathbf{\Omega}_n$ .

*Idea.* Induction on  $n$ , two dimensions at a time. ① If  $\beta = 0$ , take  $m = 0$  and any basis; this is the base case and also the case that stops the recursion. ② Otherwise [Lemma 13.9.1](#) produces a hyperbolic pair  $(\mathbf{u}_1, \mathbf{w}_1)$  and splits  $V = H \oplus H^{\perp\beta}$  with  $\dim H^{\perp\beta} = n - 2$ ; apply the induction hypothesis to  $\beta$  restricted to  $H^{\perp\beta}$ , and put the pieces together. ③ Finally reorder the basis so that all the  $\mathbf{u}$ 's come before all the  $\mathbf{w}$ 's, which turns a string of  $2 \times 2$  blocks into the one block matrix displayed. The radical, by part (c) of the lemma, is exactly what the last stage of the recursion leaves behind.

*Proof. Existence, by induction on  $n = \dim V$ .* For  $n = 0$  and  $n = 1$  every alternating form is zero: in dimension 1,  $\beta(c\mathbf{v}, d\mathbf{v}) = cd\beta(\mathbf{v}, \mathbf{v}) = 0$ . Take  $m = 0$ .

Let  $n \geq 2$  and assume the statement for all spaces of dimension less than  $n$ .

*Case 1:*  $\beta = 0$ . Take  $m = 0$  and any basis of  $V$ ; every required equation reads  $0 = 0$ .

*Case 2:*  $\beta \neq 0$ . By [Lemma 13.9.1](#) (a) there is a hyperbolic pair  $(\mathbf{u}_1, \mathbf{w}_1)$ , and by (b) we have  $V = H \oplus W$  with  $H = \text{span}(\mathbf{u}_1, \mathbf{w}_1)$  and  $W = H^{\perp\beta}$  of dimension  $n - 2$ . The restriction  $\beta|_W$  is again alternating, being a restriction of an alternating form, so the induction hypothesis applies to it: there is a basis

$$(\mathbf{u}_2, \dots, \mathbf{u}_m, \mathbf{w}_2, \dots, \mathbf{w}_m, \mathbf{z}_1, \dots, \mathbf{z}_{n-2m})$$



of  $W$  in which  $\beta|_W(\mathbf{u}_i, \mathbf{w}_j) = \delta_{ij}$ , all other pairings among the  $\mathbf{u}$ 's and  $\mathbf{w}$ 's vanish, and each  $\mathbf{z}_k$  lies in  $\text{rad}(\beta|_W)$ .

Putting  $\mathbf{u}_1$  and  $\mathbf{w}_1$  in front gives a basis of  $V = H \oplus W$ . The required equations hold: within  $H$  they are the definition of a hyperbolic pair together with alternation; within  $W$  they are the induction hypothesis; and across the two, every pairing of  $\mathbf{u}_1$  or  $\mathbf{w}_1$  with a vector of  $W$  vanishes because  $W = H^{\perp\beta}$  and  $\beta$  is skew-symmetric. Finally each  $\mathbf{z}_k$  lies in  $\text{rad}(\beta|_W) = \text{rad}(\beta)$  by [Lemma 13.9.1](#) (c), which is the last assertion.

**The matrix.** Order the basis as  $\mathcal{B} = (\mathbf{u}_1, \dots, \mathbf{u}_m, \mathbf{w}_1, \dots, \mathbf{w}_m, \mathbf{z}_1, \dots, \mathbf{z}_{n-2m})$ . By [Definition 13.1.2](#) the  $(i, j)$  entry of  $[\beta]_{\mathcal{B}}$  is the pairing of the  $i$ -th and  $j$ -th basis vectors, and the equations above say precisely that the top-left  $m \times m$  block and the  $\mathbf{w}$ - $\mathbf{w}$  block are  $\mathbf{0}$ , the  $\mathbf{u}$ - $\mathbf{w}$  block is  $\mathbf{I}_m$ , the  $\mathbf{w}$ - $\mathbf{u}$  block is  $-\mathbf{I}_m$ , and every row and column indexed by a  $\mathbf{z}_k$  is zero. That is [Eq. 13.9.1](#).

(a). Given alternating  $\mathbf{A} \in M_n(F)$ , let  $\beta(\mathbf{x}, \mathbf{y}) = \mathbf{x}^\top \mathbf{A} \mathbf{y}$  on  $F^n$ ; it is alternating by [Proposition 13.3.2](#) (b). The basis just constructed gives  $\mathbf{P}^\top \mathbf{A} \mathbf{P} = \mathbf{\Omega}_{2m} \oplus \mathbf{0}_{n-2m}$  with  $\mathbf{P}$  the matrix whose columns are the basis vectors, by [Theorem 13.2.1](#), so  $\mathbf{A} \simeq \mathbf{\Omega}_{2m} \oplus \mathbf{0}_{n-2m}$ . Uniqueness of  $m$ : congruence preserves rank ([Theorem 13.2.3](#)), and  $\text{rank}(\mathbf{\Omega}_{2m} \oplus \mathbf{0}_{n-2m}) = 2m$  because  $\mathbf{\Omega}_{2m}$  is invertible, its square being  $-\mathbf{I}_{2m}$ . So  $2m = \text{rank } \mathbf{A}$  is determined by  $\mathbf{A}$ .

(b). If  $\beta$  is non-degenerate then  $\text{rad}(\beta) = \{\mathbf{0}\}$  by [Definition 13.1.4](#), so there are no  $\mathbf{z}_k$ , that is  $n = 2m$ . This proves the theorem. ■

### Definition 13.9.2 — Symplectic Basis

A basis of  $V$  as in [Theorem 13.9.2](#) is a **symplectic basis**, or a **Darboux basis**, for the alternating form  $\beta$ . A non-degenerate alternating form on a finite-dimensional space is called a **symplectic form**, and the pair  $(V, \beta)$  a **symplectic vector space**.

Now read the hypotheses of the theorem again, and notice what is missing. There is no “ $\text{char } F \neq 2$ ”, and there is no place one could be inserted, because no step of the proof divided by 2. Compare the statements this chapter has accumulated:

- [Theorem 13.3.3](#): skew implies alternating only when  $\text{char } F \neq 2$ .
- [Theorem 13.3.4](#): the splitting into halves exists only when  $\text{char } F \neq 2$ .
- [Theorem 13.3.5](#): a symmetric form is recovered from its quadratic form only when  $\text{char } F \neq 2$ .
- [Theorem 13.4.2](#): a symmetric form has an orthogonal basis only when  $\text{char } F \neq 2$ .
- [Theorem 13.9.2](#): no hypothesis.

Over  $\mathbb{F}_2$ , where every one of the first four statements is false, the last one is true and gives a complete classification. The reason is visible in the proof: the symmetric theory needs a vector with  $q(\mathbf{v}) \neq 0$ , and that vector is manufactured by a polarization identity that divides by 2. The alternating theory needs a *pair* with  $\beta(\mathbf{u}, \mathbf{w}) \neq 0$ , and a pair is handed over by the assumption  $\beta \neq 0$  itself, free of charge. Symmetric forms are hard in characteristic 2; alternating forms are not hard anywhere.

*Quick check.* Over  $\mathbb{F}_2$  we have  $-1 = 1$ , so  $\Omega_{2m} = \begin{pmatrix} \mathbf{0} & \mathbf{I}_m \\ \mathbf{I}_m & \mathbf{0} \end{pmatrix}$ , which is a **symmetric** matrix. Does that contradict the warning above, that an alternating form is never diagonalizable?

*Solution.* No. Over  $\mathbb{F}_2$  every alternating matrix is symmetric, since skew-symmetric and symmetric are the same condition there ([Theorem 13.3.3](#) and the warning after it). What distinguishes alternating from merely symmetric over  $\mathbb{F}_2$  is the zero diagonal, and  $\Omega_{2m}$  does have a zero diagonal. The warning says a **diagonal** alternating matrix must be  $\mathbf{0}$ , and  $\Omega_{2m}$  is symmetric but not diagonal. Concretely, over  $\mathbb{F}_2$  the matrix  $\begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}$  is the Gram matrix of the alternating form  $\beta(\mathbf{x}, \mathbf{y}) = x_1y_2 + x_2y_1$ , and no change of basis makes it diagonal.

### 13.9.4 Running the algorithm

The proof is an algorithm, and it is short enough to carry out by hand. Each round of the induction does three things: find a pair with  $\beta(\mathbf{u}, \mathbf{w}) \neq 0$ , rescale to make the value 1, and then correct every remaining basis vector so that it becomes orthogonal to both. The correction is a formula worth recording. If  $\beta(\mathbf{u}, \mathbf{w}) = 1$ , put

$$\mathbf{v}' := \mathbf{v} - \beta(\mathbf{v}, \mathbf{w}) \mathbf{u} + \beta(\mathbf{v}, \mathbf{u}) \mathbf{w}. \quad (13.9.2)$$

Expanding and using  $\beta(\mathbf{u}, \mathbf{u}) = \beta(\mathbf{w}, \mathbf{w}) = 0$ ,  $\beta(\mathbf{u}, \mathbf{w}) = 1$  and  $\beta(\mathbf{w}, \mathbf{u}) = -1$ ,

$$\begin{aligned} \beta(\mathbf{v}', \mathbf{u}) &= \beta(\mathbf{v}, \mathbf{u}) - \beta(\mathbf{v}, \mathbf{w}) \cdot 0 + \beta(\mathbf{v}, \mathbf{u}) \cdot (-1) = 0, \\ \beta(\mathbf{v}', \mathbf{w}) &= \beta(\mathbf{v}, \mathbf{w}) - \beta(\mathbf{v}, \mathbf{w}) \cdot 1 + \beta(\mathbf{v}, \mathbf{u}) \cdot 0 = 0, \end{aligned}$$

so  $\mathbf{v}' \in H^{\perp\beta}$ . It is the alternating analogue of subtracting a projection in Gram–Schmidt, with one difference: nothing is divided by a length, so it works over any field.

#### Example. (A Symplectic Basis for a $4 \times 4$ Form)

Let  $\beta(\mathbf{x}, \mathbf{y}) = \mathbf{x}^\top \mathbf{A} \mathbf{y}$  on  $\mathbb{Q}^4$ , where

$$\mathbf{A} = \begin{pmatrix} 0 & 1 & 2 & 3 \\ -1 & 0 & 1 & 1 \\ -2 & -1 & 0 & 0 \\ -3 & -1 & 0 & 0 \end{pmatrix}.$$

Find a symplectic basis, and the matrix  $\mathbf{P}$  with  $\mathbf{P}^\top \mathbf{A} \mathbf{P} = \Omega_4$ .

*Solution.* The matrix is skew-symmetric with zero diagonal, so  $\beta$  is alternating ([Proposition 13.3.2](#) (b)).

**Round 1.** The entry  $a_{12} = \beta(\mathbf{e}_1, \mathbf{e}_2) = 1$  is already 1, so  $(\mathbf{u}_1, \mathbf{w}_1) = (\mathbf{e}_1, \mathbf{e}_2)$  is a hyperbolic pair and no rescaling is needed. Correct  $\mathbf{e}_3$  and  $\mathbf{e}_4$  by [Eq. 13.9.2](#), reading the needed values off the matrix:  $\beta(\mathbf{e}_3, \mathbf{e}_2) = a_{32} = -1$ ,  $\beta(\mathbf{e}_3, \mathbf{e}_1) = a_{31} = -2$ ,  $\beta(\mathbf{e}_4, \mathbf{e}_2) =$

$a_{42} = -1$ ,  $\beta(\mathbf{e}_4, \mathbf{e}_1) = a_{41} = -3$ . So

$$\begin{aligned}\mathbf{v}_3 &= \mathbf{e}_3 + \mathbf{e}_1 - 2\mathbf{e}_2 = (1, -2, 1, 0), \\ \mathbf{v}_4 &= \mathbf{e}_4 + \mathbf{e}_1 - 3\mathbf{e}_2 = (1, -3, 0, 1).\end{aligned}$$

**Round 2.** Both  $\mathbf{v}_3$  and  $\mathbf{v}_4$  pair to zero with  $\mathbf{e}_1$  and  $\mathbf{e}_2$ , as the formula guarantees, so the recursion continues inside  $\text{span}(\mathbf{v}_3, \mathbf{v}_4)$ . Compute  $\mathbf{A}\mathbf{v}_4$  row by row with  $\mathbf{v}_4 = (1, -3, 0, 1)$ : row 1 gives  $0 - 3 + 0 + 3 = 0$ , row 2 gives  $-1 + 0 + 0 + 1 = 0$ , row 3 gives  $-2 + 3 + 0 + 0 = 1$ , row 4 gives  $-3 + 3 + 0 + 0 = 0$ . So  $\mathbf{A}\mathbf{v}_4 = (0, 0, 1, 0)^\top$ , and

$$\beta(\mathbf{v}_3, \mathbf{v}_4) = \mathbf{v}_3^\top (\mathbf{A}\mathbf{v}_4) = 1 \cdot 0 + (-2) \cdot 0 + 1 \cdot 1 + 0 \cdot 0 = 1.$$

The pair  $(\mathbf{u}_2, \mathbf{w}_2) = (\mathbf{v}_3, \mathbf{v}_4)$  is therefore hyperbolic, and the recursion stops: the two remaining dimensions are used up.

**The answer.** The symplectic basis is  $\mathcal{B} = (\mathbf{e}_1, \mathbf{v}_3, \mathbf{e}_2, \mathbf{v}_4)$ , listing the  $\mathbf{u}$ 's first and then the  $\mathbf{w}$ 's. The matrix whose columns are these four vectors is

$$\mathbf{P} = \begin{pmatrix} 1 & 1 & 0 & 1 \\ 0 & -2 & 1 & -3 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 1 \end{pmatrix}, \quad \mathbf{P}^\top \mathbf{A} \mathbf{P} = \Omega_4.$$

A check of two entries confirms the pattern: the  $(1, 3)$  entry of  $\mathbf{P}^\top \mathbf{A} \mathbf{P}$  is  $\beta(\mathbf{e}_1, \mathbf{e}_2) = 1$  and the  $(2, 4)$  entry is  $\beta(\mathbf{v}_3, \mathbf{v}_4) = 1$ , while the  $(1, 2)$  entry is  $\beta(\mathbf{e}_1, \mathbf{v}_3) = a_{11} + a_{12}(-2) + a_{13} = 0 - 2 + 2 = 0$ .

### 13.9.5 What the standard form classifies

The theorem is a classification, and it is worth reading off exactly what it classifies and how coarsely.

#### Corollary 13.9.3 — Alternating Matrices Are Classified by Rank

Let  $F$  be any field and let  $\mathbf{A}, \mathbf{B} \in M_n(F)$  be alternating.

- (a) rank  $\mathbf{A}$  is **even**.
- (b)  $\mathbf{A} \simeq \mathbf{B}$  if and only if rank  $\mathbf{A} = \text{rank } \mathbf{B}$ .
- (c) There are exactly  $\lfloor n/2 \rfloor + 1$  congruence classes of alternating matrices in  $M_n(F)$ , one for each even rank  $0, 2, 4, \dots$  up to  $n$ .
- (d) If  $n$  is odd, every alternating  $\mathbf{A} \in M_n(F)$  is singular.

*Proof.*

- (a) By Theorem 13.9.2 (a),  $\mathbf{A} \simeq \Omega_{2m} \oplus \mathbf{0}_{n-2m}$  for some  $m$ , and congruence preserves rank (Theorem 13.2.3), so rank  $\mathbf{A} = 2m$ .
- (b)  $(\Rightarrow)$  is Theorem 13.2.3.  $(\Leftarrow)$  If rank  $\mathbf{A} = \text{rank } \mathbf{B} = 2m$  then by (a) and Theorem 13.9.2 both are congruent to  $\Omega_{2m} \oplus \mathbf{0}_{n-2m}$ , and congruence is an equivalence relation (Proposition 13.2.2), so  $\mathbf{A} \simeq \mathbf{B}$ .

- (c) By (a) and (b) the classes are in bijection with the even integers  $2m$  satisfying  $0 \leq 2m \leq n$ , and there are  $\lfloor n/2 \rfloor + 1$  of those.
- (d) By (a),  $\text{rank } \mathbf{A}$  is even and at most  $n$ ; since  $n$  is odd,  $\text{rank } \mathbf{A} \leq n - 1 < n$ , so  $\mathbf{A}$  is not invertible. ■

Part (d) deserves a second look, because there is a one-line proof of it that works only half the time. If  $\mathbf{A}^\top = -\mathbf{A}$  with  $n$  odd, then  $\det \mathbf{A} = \det(\mathbf{A}^\top) = \det(-\mathbf{A}) = (-1)^n \det \mathbf{A} = -\det \mathbf{A}$ , so  $(1 + 1) \det \mathbf{A} = 0$  and  $\det \mathbf{A} = 0$  — provided  $1 + 1 \neq 0$ . In characteristic 2 that argument collapses, and it collapses for a real reason: over  $\mathbb{F}_2$  the matrix  $\text{diag}(1, 0, 0)$  is skew-symmetric (because skew means symmetric there) and has odd rank 1. Part (d) survives because it is a statement about *alternating* matrices, and the standard form proves it without ever touching the determinant.

**Warning.** “Skew-symmetric” is not enough. Every statement in [Corollary 13.9.3](#) is about alternating matrices: skew-symmetric **with zero diagonal**. Over  $\mathbb{F}_2$ ,  $\mathbf{I}_2$  is skew-symmetric and has rank 2, which is harmless; but  $\text{diag}(1, 0) \in M_2(\mathbb{F}_2)$  is skew-symmetric of rank 1, and  $\mathbf{I}_3 \in M_3(\mathbb{F}_2)$  is skew-symmetric and invertible. Both would contradict the corollary if “skew-symmetric” could be substituted for “alternating”. In characteristic  $\neq 2$  the two words agree ([Theorem 13.3.3](#) (b)) and the distinction can be forgotten; nowhere else.

### 13.9.6 The determinant, and the Pfaffian

Part (a) of [Theorem 13.9.2](#) says that in even size an invertible alternating matrix is congruent to  $\Omega_n$ . Congruence multiplies the determinant by a square ([Theorem 13.2.5](#)), so the determinant of an alternating matrix is heavily constrained.

#### Corollary 13.9.4 — The Determinant of an Alternating Matrix Is a Square

Let  $F$  be any field and let  $\mathbf{A} \in M_n(F)$  be alternating. Then  $\det \mathbf{A}$  is a square in  $F$ : there is  $c \in F$  with  $\det \mathbf{A} = c^2$ .

*Proof.* If  $\mathbf{A}$  is singular then  $\det \mathbf{A} = 0 = 0^2$ . So assume  $\mathbf{A}$  is invertible. Then  $\text{rank } \mathbf{A} = n$  is even by [Corollary 13.9.3](#) (a), say  $n = 2m$ , and [Theorem 13.9.2](#) (a) supplies an invertible  $\mathbf{P}$  with  $\mathbf{P}^\top \mathbf{A} \mathbf{P} = \Omega_{2m}$ . Taking determinants and using [Theorem 13.2.5](#),

$$\det \Omega_{2m} = (\det \mathbf{P})^2 \det \mathbf{A}.$$

Now  $\det \Omega_{2m} = 1$ . Indeed, performing the  $m$  row swaps  $R_i \leftrightarrow R_{m+i}$  for  $i = 1, \dots, m$  turns  $\Omega_{2m}$  into  $(-\mathbf{I}_m) \oplus \mathbf{I}_m$ , so by [Theorem 6.5.1](#) (a) and [Theorem 6.8.4](#),

$$(-1)^m \det \Omega_{2m} = \det((-\mathbf{I}_m) \oplus \mathbf{I}_m) = (-1)^m,$$

and  $(-1)^m$  is invertible, so  $\det \Omega_{2m} = 1$ . Hence  $\det \mathbf{A} = (\det \mathbf{P})^{-2} = ((\det \mathbf{P})^{-1})^2$ , a

square, since  $\det \mathbf{P} \neq 0$ . This proves the corollary. ■

The corollary says a square root of  $\det \mathbf{A}$  exists; it does not produce one. In fact there is a canonical choice. For every even  $n$  there is a polynomial  $\text{Pf}$  in the entries  $a_{ij}$  with  $i < j$ , with integer coefficients, called the **Pfaffian**, satisfying  $\det \mathbf{A} = \text{Pf}(\mathbf{A})^2$  for every alternating  $\mathbf{A}$ . Constructing it in general is a piece of multilinear algebra that we do not carry out, so we record only the name and the two smallest cases, both of which can be checked by expanding the determinant.

**Example. (The Pfaffian in Sizes 2 and 4)**

For  $n = 2$ ,

$$\mathbf{A} = \begin{pmatrix} 0 & a \\ -a & 0 \end{pmatrix}, \quad \det \mathbf{A} = a^2, \quad \text{Pf}(\mathbf{A}) = a.$$

For  $n = 4$ , write the six independent entries as  $a = a_{12}$ ,  $b = a_{13}$ ,  $c = a_{14}$ ,  $d = a_{23}$ ,  $e = a_{24}$ ,  $f = a_{34}$ . Expanding the  $4 \times 4$  determinant gives

$$\det \mathbf{A} = (af - be + cd)^2, \quad \text{Pf}(\mathbf{A}) = af - be + cd.$$

The three terms pair up  $\{1, 2\}\{3, 4\}$ ,  $\{1, 3\}\{2, 4\}$  and  $\{1, 4\}\{2, 3\}$ : the Pfaffian sums over the ways of splitting  $\{1, 2, 3, 4\}$  into two pairs, with a sign. For the matrix  $\mathbf{A}$  of [Example \(A Symplectic Basis for a  \$4 \times 4\$  Form\)](#) this reads  $\text{Pf}(\mathbf{A}) = 1 \cdot 0 - 2 \cdot 1 + 3 \cdot 1 = 1$ , and indeed  $\det \mathbf{A} = 1$ .

*Remark.* The Pfaffian is not merely a square root: it is the one that behaves well under congruence, satisfying  $\text{Pf}(\mathbf{P}^\top \mathbf{A} \mathbf{P}) = \det(\mathbf{P}) \text{Pf}(\mathbf{A})$ . That identity is the reason the Pfaffian matters. Section 10 records one of its consequences for the symplectic group, without proof and without leaning on it. We do not prove the identity either.

## 13.9.7 Symmetric against alternating

Put the two halves of the chapter side by side. Both are theories of a bilinear form under congruence; the answers could hardly be more different.

|                              | Symmetric forms                                                                                          | Alternating forms                      |
|------------------------------|----------------------------------------------------------------------------------------------------------|----------------------------------------|
| Field hypothesis             | $\text{char } F \neq 2$ for every theorem                                                                | none                                   |
| Canonical form               | diagonal, entries not canonical                                                                          | $\Omega_{2m} \oplus \mathbf{0}_{n-2m}$ |
| Complete invariants          | rank <b>and</b> more: over $\mathbb{R}$ the signature, over $\mathbb{Q}$ the discriminant and local data | rank alone                             |
| Classes in $M_n(\mathbb{R})$ | $\frac{1}{2}(n+1)(n+2)$                                                                                  | $\lfloor n/2 \rfloor + 1$              |
| Classes in $M_n(\mathbb{C})$ | $n+1$                                                                                                    | $\lfloor n/2 \rfloor + 1$              |
| Classes in $M_n(\mathbb{Q})$ | infinitely many already for $n=1$                                                                        | $\lfloor n/2 \rfloor + 1$              |
| Possible ranks               | every $r$ with $0 \leq r \leq n$                                                                         | even $r$ only                          |
| Effect of the field          | decisive                                                                                                 | none                                   |

Two entries carry the point. Over  $\mathbb{Q}$ , the  $1 \times 1$  symmetric matrices  $[1], [2], [3], [5], \dots$  are pairwise non-congruent, because congruence in size 1 multiplies by a non-zero square ([Theorem 13.2.5](#)) and distinct squarefree positive integers lie in distinct classes of  $\mathbb{Q}^\times/(\mathbb{Q}^\times)^2$ . So the symmetric classification over  $\mathbb{Q}$  is an infinite problem in number theory — Section 4 said as much. The alternating classification over  $\mathbb{Q}$  has four classes in size 6, and the same four over  $\mathbb{F}_2$ , over  $\mathbb{R}$ , and over any field whatsoever.

The reason, one last time, is the two-dimensional model. A symmetric form on a line is  $c \mapsto ac^2$ , and rescaling changes  $a$  by a square, so the diagonal entry carries the field's arithmetic with it. An alternating form on a plane is  $\begin{pmatrix} 0 & a \\ -a & 0 \end{pmatrix}$ , and rescaling *one* of the two basis vectors changes  $a$  to  $ca$  for any  $c \neq 0$  — an arbitrary non-zero scalar, not a square. There is nothing left for the field to remember.

## 13.9.8 Exercises

### 13.9.8.1 A. Check your understanding

#### Exercise A1.

- Define a hyperbolic pair for an alternating form, and say why its two vectors are automatically linearly independent.
- State [Theorem 13.9.2](#), and say which hypothesis on  $F$  it needs.
- Determine whether the following is correct: every skew-symmetric matrix over every field has even rank. Justify your answer.
- Why can an alternating form never have a diagonal Gram matrix, unless the form is zero?
- How many congruence classes of alternating matrices are there in  $M_5(\mathbb{R})$ ? In  $M_5(\mathbb{F}_7)$ ?

*Solution.*

- A pair  $(\mathbf{u}, \mathbf{w})$  with  $\beta(\mathbf{u}, \mathbf{w}) = 1$  ([Definition 13.9.1](#)). If  $\mathbf{w} = c\mathbf{u}$ , then  $\beta(\mathbf{u}, \mathbf{w}) = c\beta(\mathbf{u}, \mathbf{u}) = 0 \neq 1$ , so  $\mathbf{w} \notin \text{span}(\mathbf{u})$ ; and  $\mathbf{u} \neq \mathbf{0}$  since  $\beta(\mathbf{0}, \mathbf{w}) = 0$ . Two vectors, neither in the span of the other, are independent.
- Every alternating form on an  $n$ -dimensional  $V$  has a basis in which its matrix is  $\Omega_{2m} \oplus \mathbf{0}_{n-2m}$ , with  $2m = \text{rank } \beta$ ; equivalently every alternating matrix is congruent to exactly one such matrix. **No** hypothesis on  $F$  is needed — in particular none on  $\text{char } F$ .
- Incorrect. Over  $\mathbb{F}_2$ ,  $\text{diag}(1, 0)$  is skew-symmetric, because  $-1 = 1$  makes skew-symmetric and symmetric the same condition, and it has rank 1. The correct statement is about **alternating** matrices, [Corollary 13.9.3](#) (a). In characteristic  $\neq 2$  the two notions coincide ([Theorem 13.3.3](#) (b)) and the statement is true.
- The off-diagonal entries of a diagonal matrix are 0 by definition, and the diagonal entries of the Gram matrix of an alternating form are  $\beta(\mathbf{v}_i, \mathbf{v}_i) = 0$ . So the whole matrix is  $\mathbf{0}$ , and then  $\beta = 0$  by [Theorem 13.1.1](#) (b).
- $[5/2] + 1 = 3$  in both cases, by [Corollary 13.9.3](#) (c): the ranks 0, 2, 4. The field makes no difference.

## 13.9.8.2 B. Practice

**Exercise B1: A symplectic basis.**

Let  $\beta(\mathbf{x}, \mathbf{y}) = \mathbf{x}^\top \mathbf{B} \mathbf{y}$  on  $\mathbb{Q}^4$ , with

$$\mathbf{B} = \begin{pmatrix} 0 & 0 & 1 & 1 \\ 0 & 0 & 2 & 1 \\ -1 & -2 & 0 & 0 \\ -1 & -1 & 0 & 0 \end{pmatrix}.$$

Find a symplectic basis for  $\beta$  and an invertible  $\mathbf{P} \in M_4(\mathbb{Q})$  with  $\mathbf{P}^\top \mathbf{B} \mathbf{P} = \Omega_4$ .

*Solution.*  $\mathbf{B}$  is skew-symmetric with zero diagonal, hence alternating (Proposition 13.3.2 (b)).

**Round 1.** Here  $\beta(\mathbf{e}_1, \mathbf{e}_2) = 0$ , so the first two standard basis vectors are **not** a hyperbolic pair; this is the step the worked example did not need. Look for any non-zero entry:  $\beta(\mathbf{e}_1, \mathbf{e}_3) = 1$ . Take  $\mathbf{u}_1 = \mathbf{e}_1$ ,  $\mathbf{w}_1 = \mathbf{e}_3$ .

Correct the other two by Eq. 13.9.2, using  $\beta(\mathbf{e}_2, \mathbf{e}_3) = 2$ ,  $\beta(\mathbf{e}_2, \mathbf{e}_1) = 0$ ,  $\beta(\mathbf{e}_4, \mathbf{e}_3) = 0$ ,  $\beta(\mathbf{e}_4, \mathbf{e}_1) = -1$ :

$$\mathbf{v}_2 = \mathbf{e}_2 - 2\mathbf{e}_1 + 0 \cdot \mathbf{e}_3 = (-2, 1, 0, 0),$$

$$\mathbf{v}_4 = \mathbf{e}_4 - 0 \cdot \mathbf{e}_1 - \mathbf{e}_3 = (0, 0, -1, 1).$$

**Round 2.** Compute  $\mathbf{B} \mathbf{v}_4$  with  $\mathbf{v}_4 = (0, 0, -1, 1)$ : row 1 gives  $-1 + 1 = 0$ , row 2 gives  $-2 + 1 = -1$ , rows 3 and 4 give 0. So  $\mathbf{B} \mathbf{v}_4 = (0, -1, 0, 0)^\top$  and

$$\beta(\mathbf{v}_2, \mathbf{v}_4) = \mathbf{v}_2^\top (\mathbf{B} \mathbf{v}_4) = (-2) \cdot 0 + 1 \cdot (-1) = -1.$$

This is non-zero but not 1, so rescale: replace  $\mathbf{v}_4$  by  $\mathbf{w}_2 := -\mathbf{v}_4 = (0, 0, 1, -1)$ , giving  $\beta(\mathbf{v}_2, \mathbf{w}_2) = 1$ . Set  $\mathbf{u}_2 = \mathbf{v}_2$ .

**The answer.** The symplectic basis is  $(\mathbf{u}_1, \mathbf{u}_2, \mathbf{w}_1, \mathbf{w}_2) = (\mathbf{e}_1, (-2, 1, 0, 0), \mathbf{e}_3, (0, 0, 1, -1))$ , so

$$\mathbf{P} = \begin{pmatrix} 1 & -2 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 1 \\ 0 & 0 & 0 & -1 \end{pmatrix}, \quad \mathbf{P}^\top \mathbf{B} \mathbf{P} = \Omega_4.$$

$\mathbf{P}$  is invertible, being block upper triangular with invertible diagonal blocks, and  $\det \mathbf{P} = -1 \neq 0$ .

**Exercise B2: Pfaffians and determinants.**

For each matrix below, compute the Pfaffian using the formula of Example (The Pfaffian in Sizes 2 and 4), and verify that its square is the determinant.

$$\mathbf{A}_1 = \begin{pmatrix} 0 & 2 & 0 & 1 \\ -2 & 0 & 1 & 0 \\ 0 & -1 & 0 & 3 \\ -1 & 0 & -3 & 0 \end{pmatrix}, \quad \mathbf{A}_2 = \begin{pmatrix} 0 & 1 & 1 & 1 \\ -1 & 0 & 1 & 1 \\ -1 & -1 & 0 & 1 \\ -1 & -1 & -1 & 0 \end{pmatrix}.$$

Hence write down, for each, an invertible matrix congruence class in the sense of [Corollary 13.9.3](#).

*Solution.* For  $\mathbf{A}_1$ :  $a = 2, b = 0, c = 1, d = 1, e = 0, f = 3$ , so  $\text{Pf}(\mathbf{A}_1) = af - be + cd = 6 - 0 + 1 = 7$  and  $\det \mathbf{A}_1 = 49$ . Expanding the determinant along the first row confirms 49.

For  $\mathbf{A}_2$ :  $a = b = c = d = e = f = 1$ , so  $\text{Pf}(\mathbf{A}_2) = 1 - 1 + 1 = 1$  and  $\det \mathbf{A}_2 = 1$ .

Both determinants are non-zero, so both matrices have rank 4, and by [Corollary 13.9.3](#) (b) both lie in the single congruence class of invertible alternating matrices in  $M_4(\mathbb{Q})$ : each is congruent to  $\Omega_4$ , and therefore  $\mathbf{A}_1 \simeq \mathbf{A}_2$ . Note that the two have different determinants, 49 and 1; congruence changes the determinant by a square ([Theorem 13.2.5](#)), and  $49 = 7^2$ .

### Exercise B3: Determine which of the following.

Determine which of the following matrices are alternating, and for those that are, give the rank and the standard form  $\Omega_{2m} \oplus \mathbf{0}_{n-2m}$  it is congruent to. Justify your answer.

- (a)  $\begin{pmatrix} 0 & 5 \\ -5 & 0 \end{pmatrix}$  over  $\mathbb{Q}$ .
- (b)  $\begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}$  over  $\mathbb{Q}$ .
- (c)  $\begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}$  over  $\mathbb{F}_2$ .
- (d) The  $5 \times 5$  matrix with  $a_{ij} = j - i$  over  $\mathbb{Q}$ .

*Solution.*

- (a) Alternating: zero diagonal and  $a_{21} = -a_{12}$ . Its determinant is  $25 \neq 0$ , so the rank is 2 and it is congruent to  $\Omega_2$ . (Explicitly, scaling the second basis vector by  $1/5$  does it.)
- (b) Not alternating. It is symmetric with  $a_{12} = 1 \neq -1 = -a_{21}$ , so it is not even skew-symmetric over  $\mathbb{Q}$ .
- (c) Alternating. Over  $\mathbb{F}_2$  we have  $1 = -1$ , so the matrix is skew-symmetric, and its diagonal is zero. Its determinant is  $-1 = 1 \neq 0$ , so the rank is 2 and it is congruent to  $\Omega_2 = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}$  — it **is**  $\Omega_2$ , since  $-1 = 1$ . Comparing (b) and (c): the same array of symbols is alternating over one field and not over another.
- (d) Alternating:  $a_{ii} = 0$  and  $a_{ji} = i - j = -a_{ij}$ . Its rank is even by [Corollary 13.9.3](#) (a), and it is not zero, so the rank is 2 or 4. Every row is a combination of  $(1, 2, 3, 4, 5)$  and  $(1, 1, 1, 1, 1)$ : indeed row  $i$  is  $(j - i)_j = (1, 2, 3, 4, 5) - i(1, 1, 1, 1, 1)$ . So the row space has dimension at most 2, and rows 1 and 2 are independent, giving rank exactly 2. Hence the matrix is congruent to  $\Omega_2 \oplus \mathbf{0}_3$ . In particular it is singular, as [Corollary 13.9.3](#) (d) requires for odd size.

### 13.9.8.3 C. Going deeper

#### Exercise C1: Lagrangian subspaces.

Let  $\beta$  be a symplectic form on  $V$  with  $\dim V = 2m$  ([Definition 13.9.2](#)), and call a subspace  $U \subseteq V$  **totally isotropic** if  $\beta(\mathbf{u}, \mathbf{u}') = 0$  for all  $\mathbf{u}, \mathbf{u}' \in U$ .



- (a) Prove that  $\text{span}(\mathbf{u}_1, \dots, \mathbf{u}_m)$  is totally isotropic, for any symplectic basis.
- (b) Prove that every totally isotropic subspace satisfies  $\dim U \leq m$ .
- (c) Deduce that a symplectic form on a space of dimension  $2m$  has a totally isotropic subspace of dimension exactly  $m$ , and no larger one. (These are the **Lagrangian** subspaces.)

*Hint for (b): compare  $U$  with  $U^{\perp\beta}$  of [Exercise \(C2: Orthogonal complements for a form\)](#).*

*Solution.*

- (a) By [Theorem 13.9.2](#),  $\beta(\mathbf{u}_i, \mathbf{u}_j) = 0$  for all  $i, j$ . A general pair of elements of the span is  $\sum_i c_i \mathbf{u}_i$  and  $\sum_j d_j \mathbf{u}_j$ , and bilinearity gives  $\sum_{i,j} c_i d_j \beta(\mathbf{u}_i, \mathbf{u}_j) = 0$ .
- (b) Let  $U$  be totally isotropic. By definition  $\beta(\mathbf{u}, \mathbf{v}) = 0$  for all  $\mathbf{u} \in U$  and all  $\mathbf{v} \in U$ , so  $U \subseteq U^{\perp\beta}$ . Since  $\beta$  is non-degenerate,  $\dim U^{\perp\beta} = 2m - \dim U$  by [Exercise \(C2: Orthogonal complements for a form\)](#) (b). Therefore

$$\dim U \leq 2m - \dim U,$$

that is  $2 \dim U \leq 2m$ . Both sides are integers, so  $\dim U \leq m$ . (No division by 2 inside the field occurs here: the inequality is between integers, not field elements, so the argument is valid in every characteristic.)

- (c) The subspace of (a) has dimension  $m$ , since the  $\mathbf{u}_i$  are part of a basis, and it is totally isotropic; by (b) no totally isotropic subspace has dimension greater than  $m$ .

### Exercise C2: The inverse of an alternating matrix.

Let  $F$  be any field.

- (a) Prove that every alternating matrix of **odd** size over  $F$  is singular, without using determinants.
- (b) Let  $\mathbf{A} \in M_n(F)$  be alternating and invertible. Prove that  $\mathbf{A}^{-1}$  is alternating.

*Hint for (b): the adjugate. The  $(i, i)$  entry of  $\text{adj } \mathbf{A}$  is a minor of  $\mathbf{A}$  of size  $n - 1$ .*

*Solution.*

- (a) This is [Corollary 13.9.3](#) (d): by [Theorem 13.9.2](#) the rank is even, and an even number that is at most an odd  $n$  is at most  $n - 1$ , so the matrix is not invertible. The proof of the standard form uses no determinants.
- (b) Since  $\mathbf{A}$  is invertible, [Theorem 6.6.2](#) gives  $\mathbf{A}^{-1} = (\det \mathbf{A})^{-1} \text{adj } \mathbf{A}$ . First, skewness:  $(\mathbf{A}^{-1})^\top = (\mathbf{A}^\top)^{-1} = (-\mathbf{A})^{-1} = -\mathbf{A}^{-1}$ , using [Theorem 0.9.5](#) and the fact that inversion turns  $-\mathbf{A} = (-1)\mathbf{A}$  into  $(-1)^{-1}\mathbf{A}^{-1} = -\mathbf{A}^{-1}$ .

Now the diagonal. By [Definition 6.6.2](#) the  $(i, i)$  entry of  $\text{adj } \mathbf{A}$  is the cofactor  $C_{ii} = (-1)^{2i} M_{ii} = M_{ii}$ , the determinant of the matrix obtained from  $\mathbf{A}$  by deleting row  $i$  and column  $i$ . Deleting a row and the matching column of an alternating matrix leaves an alternating matrix, since the surviving entries keep their positions relative to the diagonal. Its size is  $n - 1$ , and  $n$  is even by [Corollary 13.9.3](#) (a) applied to the invertible

$\mathbf{A}$ ; so  $n - 1$  is odd and  $M_{ii} = 0$  by (a). Hence every diagonal entry of  $\mathbf{A}^{-1}$  is 0, and with skewness this makes  $\mathbf{A}^{-1}$  alternating.

### Exercise C3: Counting over a finite field.

Let  $q$  be a power of an **odd** prime.

- How many alternating bilinear forms are there on  $\mathbb{F}_q^{2m}$ ? How many symmetric ones?
- How many of the alternating forms on  $\mathbb{F}_q^2$  are non-degenerate, and how many congruence classes do they fall into?
- How many of the symmetric forms on  $\mathbb{F}_q^1$  are non-degenerate, and how many congruence classes do they fall into? *Hint: the squaring map  $\mathbb{F}_q^\times \rightarrow \mathbb{F}_q^\times$  has kernel  $\{1, -1\}$ .*
- Explain in one or two sentences why the answers to (b) and (c) differ, in terms of what a change of basis can do to the single free parameter in each case.

*Solution.*

- By [Theorem 13.1.1](#) (c), forms correspond bijectively to matrices, so we count matrices. An alternating matrix of size  $n = 2m$  is determined freely by its entries above the diagonal, of which there are  $\binom{n}{2} = m(2m - 1)$ ; so there are  $q^{m(2m-1)}$  alternating forms. A symmetric matrix is determined freely by its entries on and above the diagonal,  $\binom{n}{2} + n = m(2m + 1)$  of them, giving  $q^{m(2m+1)}$ .
- The alternating forms on  $\mathbb{F}_q^2$  have matrices  $\begin{pmatrix} 0 & a \\ -a & 0 \end{pmatrix}$ , so there are  $q$  of them, and the non-degenerate ones are those with  $a \neq 0$ , of which there are  $q - 1$ . By [Corollary 13.9.3](#) (b) all of these have rank 2 and so lie in a **single** congruence class.
- A symmetric form on  $\mathbb{F}_q^1$  has matrix  $[d]$  and is non-degenerate exactly when  $d \neq 0$ , so there are  $q - 1$  of them. Congruence in size 1 replaces  $d$  by  $p^2d$  with  $p \neq 0$  ([Definition 13.2.1](#)), so the classes are the cosets of the subgroup of squares in  $\mathbb{F}_q^\times$ . The squaring homomorphism  $\mathbb{F}_q^\times \rightarrow \mathbb{F}_q^\times$  has kernel  $\{x : x^2 = 1\} = \{1, -1\}$ , which has **two** elements because  $q$  is odd; so its image has index 2, and there are exactly **two** congruence classes. For  $q = 5$ , for instance, the squares are  $\{1, 4\}$  and the non-squares  $\{2, 3\}$ , and  $[1] \not\sim [2]$ .
- In the alternating case a change of basis may replace  $a$  by  $ca$  for **any**  $c \neq 0$ , because the two slots of  $\beta(\mathbf{u}, \mathbf{w})$  hold different basis vectors and only one of them need be rescaled. In the symmetric case the parameter sits in  $\beta(\mathbf{v}, \mathbf{v})$ , where both slots hold the same vector, so rescaling changes  $d$  to  $c^2d$  — by a **square** only. The square class then survives as an invariant, and over  $\mathbb{F}_q$  it has two values.

## 13.10 Groups Defined by Forms

Chapter 10 defined the orthogonal and unitary groups by a sentence: they consist of the operators that preserve the inner product. Read that sentence again and notice that positivity never appears in it. “Preserves  $\langle \cdot, \cdot \rangle$ ” makes sense for any bilinear or Hermitian form at all, and it

always produces a group. This section takes the definition apart and puts it back together over an arbitrary form, so that the four families of **classical groups** — orthogonal, indefinite orthogonal, unitary, symplectic — become four instances of one construction, chosen by the four kinds of form this chapter has classified.

Everything is recalled, not reproved.  $O(n)$ ,  $SO(n)$  and  $U(n)$  were defined in Chapter 10 (Definition 10.7.2, Definition 10.7.3), shown there to be groups with the expected determinants (Proposition 10.7.4), and classified element by element in the real case in Chapter 11 (Theorem 11.6.1). None of that is repeated. What is new here is the pattern behind it.

### 13.10.1 The isometry group of a form

An operator preserves a form when applying it to both arguments changes nothing. This is the same idea as an isometry of an inner product space (Definition 10.7.1), with the norm removed from the statement because a general form has no norm.

*The isometry group of a form is the set of invertible changes of coordinates that the form cannot detect.*

#### Definition 13.10.1 — Isometry Group of a Form

Let  $V$  be a finite-dimensional vector space over  $F$  and  $\beta$  a **non-degenerate** bilinear form on  $V$  (Definition 13.1.4). An operator  $T \in \mathcal{L}(V)$  is an **isometry of  $\beta$**  if

$$\beta(T\mathbf{u}, T\mathbf{v}) = \beta(\mathbf{u}, \mathbf{v}) \quad \text{for all } \mathbf{u}, \mathbf{v} \in V,$$

and the set of all of them is written  $\text{Isom}(\beta)$ .

For a **non-singular**  $\mathbf{A} \in M_n(F)$ , the corresponding matrix group is

$$\text{Isom}(\mathbf{A}) := \{ \mathbf{P} \in M_n(F) : \mathbf{P}^\top \mathbf{A} \mathbf{P} = \mathbf{A} \}.$$

When  $F = \mathbb{C}$  and  $\mathbf{A}^* = \mathbf{A}$  is non-singular Hermitian,  $\text{Isom}(\mathbf{A})$  means instead  $\{ \mathbf{P} : \mathbf{P}^* \mathbf{A} \mathbf{P} = \mathbf{A} \}$ .

Read the clauses. The condition is required **for all** pairs, not just for a basis — though checking it on a basis is enough, since both sides are bilinear in  $\mathbf{u}$  and in  $\mathbf{v}$ . The matrix condition  $\mathbf{P}^\top \mathbf{A} \mathbf{P} = \mathbf{A}$  is the matrix condition of Theorem 13.2.1 with the new Gram matrix required to be the *old* one: an isometry is a change of basis that leaves the Gram matrix alone. And the two versions match, because if  $\mathbf{A} = [\beta]_{\mathcal{B}}$  then  $T \in \text{Isom}(\beta)$  if and only if  $[T]_{\mathcal{B}} \in \text{Isom}(\mathbf{A})$ , by Theorem 13.1.1 (a) applied twice.

Two clauses need defending. First, **non-degeneracy**. Without it the definition is worthless: for  $\mathbf{A} = \mathbf{0}$  the condition  $\mathbf{P}^\top \mathbf{0} \mathbf{P} = \mathbf{0}$  holds for every  $\mathbf{P}$ , including  $\mathbf{P} = \mathbf{0}$ , and the “group” is all of  $M_n(F)$ , which has no inverses. Second, the definition does **not** require  $\mathbf{P}$  to be invertible. It does not have to: invertibility comes free, as the next result records, and this is exactly what non-degeneracy buys.

#### Theorem 13.10.1 — Isometry Groups Are Groups

Let  $\mathbf{A} \in M_n(F)$  be non-singular. Then every  $\mathbf{P} \in \text{Isom}(\mathbf{A})$  is invertible, and  $\text{Isom}(\mathbf{A})$

is a subgroup of  $GL_n(F)$ . The same holds for  $\text{Isom}(\beta)$  inside the group of invertible operators on  $V$ , and for the Hermitian version.

*Proof.* Let  $\mathbf{P} \in \text{Isom}(\mathbf{A})$ . Taking determinants in  $\mathbf{P}^\top \mathbf{A} \mathbf{P} = \mathbf{A}$  and using [Theorem 6.5.4](#) and [Theorem 6.4.5](#),

$$(\det \mathbf{P})^2 \det \mathbf{A} = \det \mathbf{A}.$$

Since  $\mathbf{A}$  is non-singular,  $\det \mathbf{A} \neq 0$  ([Theorem 6.5.2](#)), and canceling it in the field  $F$  gives  $(\det \mathbf{P})^2 = 1$ . Hence  $\det \mathbf{P} \neq 0$  and  $\mathbf{P}$  is invertible, again by [Theorem 6.5.2](#).

Now the three subgroup checks.  $\mathbf{I}_n^\top \mathbf{A} \mathbf{I}_n = \mathbf{A}$ , so  $\mathbf{I}_n \in \text{Isom}(\mathbf{A})$ . If  $\mathbf{P}, \mathbf{Q} \in \text{Isom}(\mathbf{A})$  then

$$(\mathbf{PQ})^\top \mathbf{A} (\mathbf{PQ}) = \mathbf{Q}^\top (\mathbf{P}^\top \mathbf{A} \mathbf{P}) \mathbf{Q} = \mathbf{Q}^\top \mathbf{A} \mathbf{Q} = \mathbf{A},$$

using [Theorem 0.9.5](#) for the first equality, so  $\mathbf{PQ} \in \text{Isom}(\mathbf{A})$ . And if  $\mathbf{P} \in \text{Isom}(\mathbf{A})$ , multiply  $\mathbf{P}^\top \mathbf{A} \mathbf{P} = \mathbf{A}$  on the left by  $(\mathbf{P}^\top)^{-1} = (\mathbf{P}^{-1})^\top$  and on the right by  $\mathbf{P}^{-1}$ , which is legitimate because  $\mathbf{P}$  is invertible; this gives  $\mathbf{A} = (\mathbf{P}^{-1})^\top \mathbf{A} \mathbf{P}^{-1}$ , so  $\mathbf{P}^{-1} \in \text{Isom}(\mathbf{A})$ . Associativity is inherited from  $M_n(F)$ .

For the Hermitian version replace  $^\top$  by  $*$  throughout; the determinant step becomes  $\overline{\det \mathbf{P}} \det \mathbf{P} \det \mathbf{A} = \det \mathbf{A}$ , that is  $|\det \mathbf{P}|^2 = 1$ , which again forces  $\det \mathbf{P} \neq 0$ . The operator version is the matrix version read through a basis. This proves the theorem. ■

The proof is the one Chapter 10 gave for  $O(n)$  and  $U(n)$  in [Proposition 10.7.4](#) (a), with  $\mathbf{I}_n$  replaced by  $\mathbf{A}$ ; nothing in it used what  $\mathbf{A}$  was, only that it was non-singular. That is the whole point of the section: the group structure never depended on positivity, on symmetry, or on the field.

## 13.10.2 Congruent forms give conjugate groups

Before listing the families, one observation makes the list a classification rather than a list of unrelated examples. Congruent matrices have the “same” isometry group, in the strongest sense available.

### Proposition 13.10.2 — Congruence Conjugates the Isometry Group

Let  $\mathbf{A}, \mathbf{B} \in M_n(F)$  be non-singular with  $\mathbf{B} = \mathbf{Q}^\top \mathbf{A} \mathbf{Q}$  for an invertible  $\mathbf{Q}$  ([Definition 13.2.1](#)). Then

$$\text{Isom}(\mathbf{B}) = \mathbf{Q}^{-1} \text{Isom}(\mathbf{A}) \mathbf{Q},$$

so the two groups are conjugate subgroups of  $GL_n(F)$ ; in particular they are isomorphic as groups, by  $\mathbf{P} \mapsto \mathbf{QPQ}^{-1}$ .

*Proof.* Let  $\mathbf{P} \in M_n(F)$ . Substituting  $\mathbf{B} = \mathbf{Q}^\top \mathbf{A} \mathbf{Q}$  and writing  $\mathbf{R} = \mathbf{QPQ}^{-1}$ ,

$$\begin{aligned} \mathbf{P}^\top \mathbf{B} \mathbf{P} = \mathbf{B} &\iff \mathbf{P}^\top \mathbf{Q}^\top \mathbf{A} \mathbf{Q} \mathbf{P} = \mathbf{Q}^\top \mathbf{A} \mathbf{Q} \\ &\iff (\mathbf{QPQ}^{-1})^\top \mathbf{A} (\mathbf{QPQ}^{-1}) = \mathbf{A}, \end{aligned}$$

where the second step multiplies on the left by  $(\mathbf{Q}^\top)^{-1}$  and on the right by  $\mathbf{Q}^{-1}$  and uses

**Theorem 0.9.5**; both operations are reversible because  $\mathbf{Q}$  is invertible. So  $\mathbf{P} \in \text{Isom}(\mathbf{B})$  if and only if  $\mathbf{R} = \mathbf{QPQ}^{-1} \in \text{Isom}(\mathbf{A})$ , which is the claimed equality of sets. Conjugation by  $\mathbf{Q}$  is an isomorphism of groups, being a bijection that respects products.

■

So the isometry group depends only on the congruence class of the form, and this chapter has already listed the congruence classes. Over  $\mathbb{R}$ , [Corollary 13.5.2](#) says a non-degenerate symmetric form is congruent to exactly one  $\mathbf{I}_{n+} \oplus (-\mathbf{I}_{n-})$ ; over  $\mathbb{C}$ , [Corollary 13.4.4](#) says a non-degenerate symmetric form is congruent to  $\mathbf{I}_n$  and nothing else; and [Theorem 13.9.2](#) says a symplectic form is congruent to  $\mathbf{\Omega}_{2m}$  and nothing else. Each classification theorem of this chapter therefore doubles as a list of groups.

### 13.10.3 The four families

#### Definition 13.10.2 — The Indefinite Orthogonal and Symplectic Groups

For  $p, q \geq 0$  with  $p + q = n$ , set  $\mathbf{I}_{p,q} := \mathbf{I}_p \oplus (-\mathbf{I}_q) \in M_n(\mathbb{R})$  and

$$\text{O}(p, q) := \text{Isom}(\mathbf{I}_{p,q}) = \{ \mathbf{P} \in M_n(\mathbb{R}) : \mathbf{P}^\top \mathbf{I}_{p,q} \mathbf{P} = \mathbf{I}_{p,q} \},$$

the **indefinite orthogonal group** of signature  $(p, q)$ . For  $m \geq 1$  and any field  $F$ , set

$$\text{Sp}(2m, F) := \text{Isom}(\mathbf{\Omega}_{2m}) = \{ \mathbf{P} \in M_{2m}(F) : \mathbf{P}^\top \mathbf{\Omega}_{2m} \mathbf{P} = \mathbf{\Omega}_{2m} \},$$

the **symplectic group**, where  $\mathbf{\Omega}_{2m}$  is the standard alternating matrix of [Eq. 13.9.1](#).

With Chapter 10's groups recalled, the four families line up as four choices of form.

| Family             | Form preserved                       | Kind of form                  | Section   |
|--------------------|--------------------------------------|-------------------------------|-----------|
| $\text{O}(n)$      | $\mathbf{I}_n$ over $\mathbb{R}$     | symmetric, positive definite  | Ch 10 §07 |
| $\text{O}(p, q)$   | $\mathbf{I}_{p,q}$ over $\mathbb{R}$ | symmetric, signature $(p, q)$ | §05       |
| $\text{U}(n)$      | $\mathbf{I}_n$ over $\mathbb{C}$     | Hermitian, positive definite  | Ch 10 §07 |
| $\text{Sp}(2m, F)$ | $\mathbf{\Omega}_{2m}$ over $F$      | alternating, non-degenerate   | §09       |

Two remarks on the list.  $\text{O}(n) = \text{O}(n, 0)$ , so the ordinary orthogonal group is the extreme case of the indefinite one; and  $\text{O}(p, q)$  and  $\text{O}(q, p)$  are conjugate, because negating a form does not change which maps preserve it, and  $-\mathbf{I}_{p,q} = (-\mathbf{I}_p) \oplus \mathbf{I}_q$  becomes  $\mathbf{I}_{q,p}$  after a permutation of the coordinates, which is a congruence. There is no  $\text{Sp}$  for odd size: a non-degenerate alternating form exists only in even dimension ([Theorem 13.9.2](#) (b)).

**Example. (Members of Each Family)**

Verify the following memberships, with  $\Omega_{2m}$  as above.

- (a)  $\begin{pmatrix} \cosh t & \sinh t \\ \sinh t & \cosh t \end{pmatrix} \in O(1, 1)$  for every  $t \in \mathbb{R}$ .
- (b)  $\begin{pmatrix} \mathbf{M} & \mathbf{0} \\ \mathbf{0} & (\mathbf{M}^\top)^{-1} \end{pmatrix} \in \text{Sp}(2m, F)$  for every invertible  $\mathbf{M} \in M_m(F)$ .
- (c)  $\begin{pmatrix} \mathbf{I}_m & \mathbf{S} \\ \mathbf{0} & \mathbf{I}_m \end{pmatrix} \in \text{Sp}(2m, F)$  if and only if  $\mathbf{S}^\top = \mathbf{S}$ .

*Solution.*

- (a) With  $\mathbf{A} = \mathbf{I}_{1,1} = \text{diag}(1, -1)$ ,  $c = \cosh t$  and  $s = \sinh t$ ,

$$\begin{pmatrix} c & s \\ s & c \end{pmatrix} \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix} \begin{pmatrix} c & s \\ s & c \end{pmatrix} = \begin{pmatrix} c^2 - s^2 & 0 \\ 0 & s^2 - c^2 \end{pmatrix} = \mathbf{I}_{1,1},$$

using  $\cosh^2 t - \sinh^2 t = 1$  and the symmetry of the outer matrix.

- (b) Write  $\mathbf{P} = \mathbf{M} \oplus \mathbf{N}$  with  $\mathbf{N} = (\mathbf{M}^\top)^{-1}$ . Multiplying the blocks,

$$\mathbf{P}^\top \Omega_{2m} \mathbf{P} = \begin{pmatrix} \mathbf{M}^\top & \mathbf{0} \\ \mathbf{0} & \mathbf{N}^\top \end{pmatrix} \begin{pmatrix} \mathbf{0} & \mathbf{N} \\ -\mathbf{M} & \mathbf{0} \end{pmatrix} = \begin{pmatrix} \mathbf{0} & \mathbf{M}^\top \mathbf{N} \\ -\mathbf{N}^\top \mathbf{M} & \mathbf{0} \end{pmatrix},$$

which equals  $\Omega_{2m}$  exactly when  $\mathbf{M}^\top \mathbf{N} = \mathbf{I}_m$ , that is  $\mathbf{N} = (\mathbf{M}^\top)^{-1}$ .

- (c) With  $\mathbf{P} = \begin{pmatrix} \mathbf{I}_m & \mathbf{S} \\ \mathbf{0} & \mathbf{I}_m \end{pmatrix}$ ,

$$\mathbf{P}^\top \Omega_{2m} \mathbf{P} = \begin{pmatrix} \mathbf{I}_m & \mathbf{0} \\ \mathbf{S}^\top & \mathbf{I}_m \end{pmatrix} \begin{pmatrix} \mathbf{0} & \mathbf{I}_m \\ -\mathbf{I}_m & -\mathbf{S} \end{pmatrix} = \begin{pmatrix} \mathbf{0} & \mathbf{I}_m \\ -\mathbf{I}_m & \mathbf{S}^\top - \mathbf{S} \end{pmatrix},$$

which equals  $\Omega_{2m}$  exactly when  $\mathbf{S}^\top = \mathbf{S}$ . Such a  $\mathbf{P}$  is a **symplectic shear**; part (c) is the first sign that symmetric matrices appear inside the symplectic group, and the dimension count below will make that precise.

### 13.10.4 Determinants

For the orthogonal family the determinant argument is already inside the proof of [Theorem 13.10.1](#), and it costs nothing to state it over any field.

#### Theorem 13.10.3 — Determinants in the Classical Groups

- (a) Let  $F$  be any field and  $\mathbf{A} \in M_n(F)$  non-singular. Then  $\det \mathbf{P} = \pm 1$  for every  $\mathbf{P} \in \text{Isom}(\mathbf{A})$ . In particular  $\det \mathbf{P} = \pm 1$  for  $\mathbf{P} \in O(p, q)$ , and both signs occur when  $n \geq 1$ .
- (b) If  $\mathbf{A} \in M_n(\mathbb{C})$  is Hermitian and non-singular, then  $|\det \mathbf{P}| = 1$  for every  $\mathbf{P} \in \text{Isom}(\mathbf{A})$ ; in particular for  $\mathbf{P} \in U(n)$ .
- (c)  $\text{Sp}(2, F) = \text{SL}_2(F)$ . In particular every  $\mathbf{P} \in \text{Sp}(2, F)$  has  $\det \mathbf{P} = 1$  exactly, and  $-1$  does not occur.

*Proof.*

- (a) The proof of [Theorem 13.10.1](#) established  $(\det \mathbf{P})^2 = 1$ . The polynomial  $x^2 - 1 = (x - 1)(x + 1)$  has at most two roots in a field ([Corollary 5.1.4](#)), and 1 and  $-1$  are roots, so  $\det \mathbf{P} \in \{1, -1\}$ . For  $O(p, q)$ , the identity has determinant 1; and  $\mathbf{D} := \text{diag}(-1, 1, \dots, 1)$  satisfies  $\mathbf{D}^\top \mathbf{I}_{p,q} \mathbf{D} = \mathbf{I}_{p,q}$ , since  $\mathbf{D}$  is diagonal and squaring its entries gives  $\mathbf{I}_n$ , while  $\det \mathbf{D} = -1$ .
- (b) Also established in that proof:  $|\det \mathbf{P}|^2 = 1$ , and a non-negative real with square 1 is 1. This is [Proposition 10.7.4](#) (b) for  $\mathbf{A} = \mathbf{I}_n$ .
- (c) Write  $\mathbf{J} = \Omega_2 = \begin{pmatrix} 0 & 1 \\ -1 & 0 \end{pmatrix}$  and let  $\mathbf{P} = \begin{pmatrix} a & b \\ c & d \end{pmatrix}$  be arbitrary. Multiplying out,

$$\mathbf{P}^\top \mathbf{J} \mathbf{P} = \begin{pmatrix} a & c \\ b & d \end{pmatrix} \begin{pmatrix} c & d \\ -a & -b \end{pmatrix} = \begin{pmatrix} 0 & ad - bc \\ bc - ad & 0 \end{pmatrix} = (\det \mathbf{P}) \mathbf{J}.$$

Since  $\mathbf{J} \neq \mathbf{0}$ , the equation  $\mathbf{P}^\top \mathbf{J} \mathbf{P} = \mathbf{J}$  holds if and only if  $\det \mathbf{P} = 1$ . This proves the theorem. ■

**Warning.** Every symplectic matrix has determinant  $+1$ , not  $\pm 1$ . The general argument of part (a) applies to  $\Omega_{2m}$  as well and gives only  $(\det \mathbf{P})^2 = 1$ , so it leaves  $-1$  open. It is nevertheless true that  $\text{Sp}(2m, F) \subseteq \text{SL}_{2m}(F)$  for every  $m$ , so  $-1$  never occurs. Part (c) proves this for  $m = 1$ ; **the general case is not proved in this book**. The clean route runs through the Pfaffian of Section 9 and its transformation rule  $\text{Pf}(\mathbf{P}^\top \mathbf{A} \mathbf{P}) = \det(\mathbf{P}) \text{Pf}(\mathbf{A})$ : applied to  $\mathbf{A} = \Omega_{2m}$ , whose Pfaffian is 1, it turns  $\mathbf{P}^\top \Omega_{2m} \mathbf{P} = \Omega_{2m}$  directly into  $\det \mathbf{P} = 1$ . We stated that rule without proof, so we state its consequence without proof too. Do not use the statement as a step in an argument elsewhere in this book; treat it as information about the wider subject, like the fact that the general polynomial of degree five has no solution by radicals.

There is a consequence worth noticing.  $O(n)$  has the proper subgroup  $SO(n)$  cut out by  $\det = 1$ , and Chapter 11 made heavy use of it: rotations against reflections. The symplectic group has no such subgroup, because — by the fact just recorded, which this book does not prove — the determinant condition is already satisfied by everything in it. A symplectic form has no notion of reflection to detect.

### 13.10.5 Counting parameters

How big is each group? “Big” has an honest answer in this book only as a count, so here is the count, and here is what it does and does not mean.

The equation  $\mathbf{P}^\top \mathbf{A} \mathbf{P} = \mathbf{A}$  is a system of scalar equations in the  $n^2$  unknown entries of  $\mathbf{P}$ . If the system consisted of  $k$  independent equations we would expect the solutions to form a family described by  $n^2 - k$  free parameters, and that number is what everybody calls the **dimension** of the group. The key observation is that the  $n^2$  equations are far from independent, because the matrix  $\mathbf{P}^\top \mathbf{A} \mathbf{P} - \mathbf{A}$  is not an arbitrary matrix: it has the same symmetry type as  $\mathbf{A}$ , by [Proposition 13.2.4](#). So the equations repeat, and only the independent ones should be counted.

- If  $\mathbf{A}$  is **symmetric**, then  $\mathbf{P}^\top \mathbf{A} \mathbf{P} - \mathbf{A}$  is symmetric, and a symmetric  $n \times n$  matrix has  $\frac{1}{2}n(n+1)$  independent entries. Expected dimension:  $n^2 - \frac{1}{2}n(n+1) = \frac{1}{2}n(n-1)$ .
- If  $\mathbf{A}$  is **alternating**, then  $\mathbf{P}^\top \mathbf{A} \mathbf{P} - \mathbf{A}$  is alternating, and an alternating  $n \times n$  matrix has  $\frac{1}{2}n(n-1)$  independent entries. Expected dimension:  $n^2 - \frac{1}{2}n(n-1) = \frac{1}{2}n(n+1)$ .

- If  $\mathbf{A}$  is **Hermitian** over  $\mathbb{C}$ , count real parameters:  $\mathbf{P}$  has  $2n^2$  of them, and a Hermitian matrix has  $n$  real diagonal entries and  $\frac{1}{2}n(n-1)$  complex entries above the diagonal, in total  $n^2$  real conditions. Expected real dimension:  $2n^2 - n^2 = n^2$ .

| Group                               | Entries in $\mathbf{P}$ | Independent conditions | Dimension           |
|-------------------------------------|-------------------------|------------------------|---------------------|
| $O(n)$ , $O(p, q)$ ,<br>$p + q = n$ | $n^2$                   | $\frac{1}{2}n(n+1)$    | $\frac{1}{2}n(n-1)$ |
| $U(n)$ (real parameters)            | $2n^2$                  | $n^2$                  | $n^2$               |
| $Sp(2m, \mathbb{R})$                | $4m^2$                  | $m(2m-1)$              | $m(2m+1)$           |

Small cases are a sanity check.  $O(2)$  comes out at dimension 1, and Chapter 10 found that its elements are described by a single angle.  $O(3)$  comes out at 3, matching the axis-and-angle description of [Corollary 11.6.2](#): two parameters for the axis direction, one for the angle.  $U(1)$  comes out at 1, and  $U(1)$  is the unit circle in  $\mathbb{C}$ . And  $Sp(2, \mathbb{R}) = SL_2(\mathbb{R})$  comes out at 3, which is right: four entries subject to the single equation  $ad - bc = 1$ .

Compare the orthogonal and symplectic rows of the table at  $n = 2m$ . The orthogonal group is smaller than the symplectic group of the same size, by exactly  $\frac{1}{2}n(n+1) - \frac{1}{2}n(n-1) = n$ . The gap is the gap between symmetric and alternating matrices, which is the  $n$  diagonal entries — the same diagonal that Section 3 found separating the two notions in the first place.

*Warning.* **This count is a heuristic, not a theorem of this book.** “Dimension” here means the number of free parameters one expects a solution set to have, and the step from “ $k$  independent-looking equations” to “ $n^2 - k$  parameters” is not justified by anything proved above; for a non-linear system it can fail. Making it precise requires the notion of a smooth manifold and the implicit function theorem, which belong to analysis and differential geometry, not here. What is proved here is only the count of conditions. Every number in the table is nevertheless correct.

*Quick check.* The family  $O(p, q)$  is indexed by two integers, yet the table gives it a dimension depending only on  $n = p + q$ , not on how  $n$  splits. Does that mean  $O(3, 0)$  and  $O(2, 1)$  are the same group?

*Solution.* No. The count of conditions only sees that  $\mathbf{I}_{p,q}$  is symmetric, not which symmetric matrix it is, so it cannot distinguish the two. They are genuinely different groups:  $O(3) = O(3, 0)$  consists of matrices all of whose entries have absolute value at most 1, since its columns are unit vectors, whereas  $O(2, 1)$  contains matrices with arbitrarily large entries, as the next warning shows. Conjugate groups come from **congruent** forms ([Proposition 13.10.2](#)), and  $\mathbf{I}_{3,0}$  and  $\mathbf{I}_{2,1}$  are not congruent, because their signatures differ ([Corollary 13.5.2](#)).

### 13.10.6 Boundedness and its absence

*Warning.*  $O(p, q)$  is **not compact when  $p, q \geq 1$ , unlike  $O(n)$** . Compactness is a topological notion that this book does not develop, so the statement is **recorded here, not proved**. What can be seen without topology is the underlying reason, which is that the entries are unbounded. Take the hyperbolic plane  $\mathbf{A} = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}$ , a non-degenerate



symmetric real form of signature  $(1, 1)$ , hence congruent to  $\mathbf{I}_{1,1}$ . For every real  $t \neq 0$ ,

$$\begin{pmatrix} t & 0 \\ 0 & t^{-1} \end{pmatrix}^\top \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix} \begin{pmatrix} t & 0 \\ 0 & t^{-1} \end{pmatrix} = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix},$$

so  $\text{diag}(t, t^{-1})$  lies in  $\text{Isom}(\mathbf{A})$ , and its entries are unbounded as  $t \rightarrow \infty$ . By [Proposition 13.10.2](#) the same unbounded family sits inside  $O(1, 1)$ , and padding with an identity block puts it inside every  $O(p, q)$  with  $p, q \geq 1$ . By contrast every entry of a matrix in  $O(n)$  has absolute value at most 1, because each column is a unit vector for the Euclidean norm.

The same phenomenon shows up in  $\text{Sp}(2m, \mathbb{R})$ , by [Example \(Members of Each Family\)](#) (b) with  $\mathbf{M} = [t]$ : the symplectic group is unbounded too. Only the two positive definite families,  $O(n)$  and  $U(n)$ , are bounded, and positivity is exactly why — the form itself controls the size of the columns. Definiteness, which this chapter spent nine sections learning to do without, turns out to be what made Chapter 10's groups small.

Section 11 takes the indefinite case that physics uses, signature  $(1, 3)$ , and works out what its isometry group does to the geometry of spacetime.

## 13.10.7 Exercises

### 13.10.7.1 A. Check your understanding

#### Exercise A1.

- Define  $\text{Isom}(\mathbf{A})$  for a non-singular  $\mathbf{A} \in M_n(F)$ , and explain why its elements are automatically invertible.
- Which form does  $O(p, q)$  preserve, and which does  $\text{Sp}(2m, F)$  preserve?
- Determine whether the following is correct: if  $\mathbf{A} \simeq \mathbf{B}$ , then  $\text{Isom}(\mathbf{A}) = \text{Isom}(\mathbf{B})$ . Justify your answer.
- State what is known here about  $\det \mathbf{P}$  for  $\mathbf{P} \in \text{Sp}(2m, F)$ , distinguishing what is proved from what is only recorded.
- In the parameter count, what exactly does the word “dimension” mean, and what is not being claimed?

*Solution.*

- $\text{Isom}(\mathbf{A}) = \{\mathbf{P} : \mathbf{P}^\top \mathbf{A} \mathbf{P} = \mathbf{A}\}$  ([Definition 13.10.1](#)). Taking determinants gives  $(\det \mathbf{P})^2 \det \mathbf{A} = \det \mathbf{A}$ , and  $\det \mathbf{A} \neq 0$ , so  $(\det \mathbf{P})^2 = 1$  and  $\det \mathbf{P} \neq 0$  ([Theorem 13.10.1](#)).
- $O(p, q)$  preserves the real symmetric form with matrix  $\mathbf{I}_{p,q} = \mathbf{I}_p \oplus (-\mathbf{I}_q)$ , of signature  $(p, q)$ .  $\text{Sp}(2m, F)$  preserves the standard alternating form with matrix  $\boldsymbol{\Omega}_{2m}$  ([Definition 13.10.2](#)).
- Incorrect as stated: the two groups are **conjugate**,  $\text{Isom}(\mathbf{B}) = \mathbf{Q}^{-1} \text{Isom}(\mathbf{A}) \mathbf{Q}$ , not equal ([Proposition 13.10.2](#)). For a witness take  $\mathbf{A} = \mathbf{I}_2$  and  $\mathbf{B} = \text{diag}(1, 4) = \mathbf{Q}^\top \mathbf{A} \mathbf{Q}$  with  $\mathbf{Q} = \text{diag}(1, 2)$ . Then  $\begin{pmatrix} 0 & 1 \\ -1 & 0 \end{pmatrix} \in O(2) = \text{Isom}(\mathbf{A})$ , but the corresponding element  $\mathbf{Q}^{-1} \begin{pmatrix} 0 & 1 \\ -1 & 0 \end{pmatrix} \mathbf{Q} = \begin{pmatrix} 0 & 2 \\ -1/2 & 0 \end{pmatrix}$  of  $\text{Isom}(\mathbf{B})$  is not in  $O(2)$ . What is always

true is that the two groups are isomorphic.

- (d) Proved here:  $(\det \mathbf{P})^2 = 1$ , hence  $\det \mathbf{P} = \pm 1$ , for every  $m$  ([Theorem 13.10.3 \(a\)](#)); and  $\det \mathbf{P} = 1$  exactly, for  $m = 1$ , since  $\mathrm{Sp}(2, F) = \mathrm{SL}_2(F)$  (part (c)). Recorded but not proved:  $\det \mathbf{P} = 1$  for every  $m$ , which follows from the transformation rule of the Pfaffian.
- (e) The number of free parameters expected in the solution set of  $\mathbf{P}^\top \mathbf{A} \mathbf{P} = \mathbf{A}$ , computed as  $n^2$  minus the number of independent scalar conditions. It is not claimed that the solution set really is a family described by that many free parameters; that statement needs the implicit function theorem and is not proved here.

### 13.10.7.2 B. Practice

#### Exercise B1: Determine which of the following.

For each matrix, determine whether it lies in  $\mathrm{O}(2)$ , in  $\mathrm{O}(1, 1)$ , in  $\mathrm{Sp}(2, \mathbb{R})$  — where the relevant forms are  $\mathbf{I}_2$ ,  $\mathbf{I}_{1,1} = \mathrm{diag}(1, -1)$  and  $\mathbf{\Omega}_2$  — listing all that apply. Justify each answer.

- (a)  $\begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix}$ .
- (b)  $\begin{pmatrix} 0 & -1 \\ 1 & 0 \end{pmatrix}$ .
- (c)  $\begin{pmatrix} 3 & 0 \\ 0 & 1/3 \end{pmatrix}$ .
- (d)  $\begin{pmatrix} 5/3 & 4/3 \\ 4/3 & 5/3 \end{pmatrix}$ .

*Solution.* Throughout, membership in  $\mathrm{Sp}(2, \mathbb{R})$  is decided by  $\det \mathbf{P} = 1$  alone ([Theorem 13.10.3 \(c\)](#)), and membership in  $\mathrm{O}(2)$  by orthonormality of the columns.

- (a)  $\mathrm{O}(2)$ : yes, the columns are  $\mathbf{e}_1$  and  $-\mathbf{e}_2$ , orthonormal.  $\mathrm{O}(1, 1)$ : yes,  $\mathbf{P}$  is diagonal with entries  $\pm 1$ , so  $\mathbf{P}^\top \mathbf{I}_{1,1} \mathbf{P} = \mathbf{I}_{1,1}$ .  $\mathrm{Sp}(2, \mathbb{R})$ : no,  $\det \mathbf{P} = -1$ .
- (b)  $\mathrm{O}(2)$ : yes, a rotation by  $\pi/2$ .  $\mathrm{O}(1, 1)$ : no —  $\mathbf{P}^\top \mathbf{I}_{1,1} \mathbf{P} = \begin{pmatrix} -1 & 0 \\ 0 & 1 \end{pmatrix} = -\mathbf{I}_{1,1} \neq \mathbf{I}_{1,1}$ .  $\mathrm{Sp}(2, \mathbb{R})$ : yes,  $\det \mathbf{P} = 1$ .
- (c)  $\mathrm{O}(2)$ : no, the first column has norm 3.  $\mathrm{O}(1, 1)$ : no — for diagonal  $\mathbf{P} = \mathrm{diag}(s, t)$  the condition is  $\mathrm{diag}(s^2, -t^2) = \mathrm{diag}(1, -1)$ , that is  $s, t = \pm 1$ , and  $3 \neq \pm 1$ .  $\mathrm{Sp}(2, \mathbb{R})$ : yes,  $\det \mathbf{P} = 3 \cdot \frac{1}{3} = 1$ .
- (d)  $\mathrm{O}(2)$ : no, the first column has norm  $\sqrt{25/9 + 16/9} = \sqrt{41}/3 \neq 1$ .  $\mathrm{O}(1, 1)$ : yes — this is  $\begin{pmatrix} \cosh t & \sinh t \\ \sinh t & \cosh t \end{pmatrix}$  with  $\cosh t = 5/3$ ,  $\sinh t = 4/3$  and  $(5/3)^2 - (4/3)^2 = 1$ , so [Example \(Members of Each Family\)](#) (a) applies.  $\mathrm{Sp}(2, \mathbb{R})$ : yes,  $\det \mathbf{P} = 25/9 - 16/9 = 1$ .

#### Exercise B2: A symplectic matrix of size four.

Let  $\mathbf{M} = \begin{pmatrix} 1 & 2 \\ 0 & 1 \end{pmatrix}$  and  $\mathbf{S} = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}$ , and put

$$\mathbf{P}_1 = \begin{pmatrix} \mathbf{M} & \mathbf{0} \\ \mathbf{0} & (\mathbf{M}^\top)^{-1} \end{pmatrix}, \quad \mathbf{P}_2 = \begin{pmatrix} \mathbf{I}_2 & \mathbf{S} \\ \mathbf{0} & \mathbf{I}_2 \end{pmatrix}.$$

- (a) Write out  $\mathbf{P}_1$  explicitly and verify  $\mathbf{P}_1 \in \mathrm{Sp}(4, \mathbb{Q})$  directly from the definition.
- (b) Explain why  $\mathbf{P}_1 \mathbf{P}_2 \in \mathrm{Sp}(4, \mathbb{Q})$  without computing the product.

- (c) Compute  $\det \mathbf{P}_1$  and  $\det \mathbf{P}_2$ , and check them against [Theorem 13.10.3](#).

*Solution.*

- (a)  $\mathbf{M}^\top = \begin{pmatrix} 1 & 0 \\ 2 & 1 \end{pmatrix}$ , so  $(\mathbf{M}^\top)^{-1} = \begin{pmatrix} 1 & 0 \\ -2 & 1 \end{pmatrix}$  and

$$\mathbf{P}_1 = \begin{pmatrix} 1 & 2 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & -2 & 1 \end{pmatrix}.$$

By the block computation of [Example \(Members of Each Family\)](#) (b),  $\mathbf{P}_1^\top \Omega_4 \mathbf{P}_1 = \begin{pmatrix} \mathbf{0} & \mathbf{M}^\top \mathbf{N} \\ -\mathbf{N}^\top \mathbf{M} & \mathbf{0} \end{pmatrix}$  with  $\mathbf{N} = (\mathbf{M}^\top)^{-1}$ , and  $\mathbf{M}^\top \mathbf{N} = \mathbf{M}^\top (\mathbf{M}^\top)^{-1} = \mathbf{I}_2$ . So  $\mathbf{P}_1^\top \Omega_4 \mathbf{P}_1 = \Omega_4$ .

- (b)  $\mathbf{S}$  is symmetric, so  $\mathbf{P}_2 \in \mathrm{Sp}(4, \mathbb{Q})$  by [Example \(Members of Each Family\)](#) (c). A product of two elements of a group is again in the group, and  $\mathrm{Sp}(4, \mathbb{Q})$  is a group by [Theorem 13.10.1](#).
- (c)  $\mathbf{P}_1 = \mathbf{M} \oplus (\mathbf{M}^\top)^{-1}$  is block diagonal, so by [Theorem 6.8.4](#) and [Corollary 6.5.6](#),

$$\det \mathbf{P}_1 = \det \mathbf{M} \det((\mathbf{M}^\top)^{-1}) = \det \mathbf{M} (\det \mathbf{M})^{-1} = 1,$$

using  $\det \mathbf{M}^\top = \det \mathbf{M}$  ([Theorem 6.4.5](#)). And  $\mathbf{P}_2$  is block upper triangular with both diagonal blocks equal to  $\mathbf{I}_2$ , so  $\det \mathbf{P}_2 = 1$  by [Theorem 6.8.4](#). Both equal  $+1$ , consistent with the recorded fact  $\mathrm{Sp}(2m, F) \subseteq \mathrm{SL}_{2m}(F)$ ; part (a) of the theorem only guarantees  $\pm 1$ .

### Exercise B3: Counting parameters.

- (a) Compute the expected dimensions of  $\mathrm{O}(4)$ ,  $\mathrm{O}(2, 2)$ ,  $\mathrm{U}(3)$  and  $\mathrm{Sp}(4, \mathbb{R})$ .
- (b) For which  $n$  and  $m$  do  $\mathrm{O}(n)$  and  $\mathrm{Sp}(2m, \mathbb{R})$  have the same expected dimension? Solve the resulting equation.
- (c) The group  $\mathrm{SO}(n)$  is cut out of  $\mathrm{O}(n)$  by the single equation  $\det \mathbf{P} = 1$ . Why does this **not** reduce the expected dimension by one?

*Solution.*

- (a)  $\mathrm{O}(4)$  and  $\mathrm{O}(2, 2)$  both have  $n = 4$ , so  $\frac{1}{2} \cdot 4 \cdot 3 = 6$ .  $\mathrm{U}(3)$ :  $3^2 = 9$  real parameters.  $\mathrm{Sp}(4, \mathbb{R})$ :  $m = 2$ , so  $m(2m + 1) = 10$ .
- (b) The equation is  $\frac{1}{2}n(n - 1) = m(2m + 1)$ , that is  $n(n - 1) = (2m + 1)(2m)$ . Reading the right-hand side as  $n(n - 1)$  with  $n = 2m + 1$  shows that  $n = 2m + 1$  is a solution for **every**  $m$ , and it is the only one:  $n \mapsto n(n - 1)$  is strictly increasing for  $n \geq 1$ , so it takes each value at most once. Hence

$$\dim \mathrm{O}(2m + 1) = \dim \mathrm{Sp}(2m, \mathbb{R}) = m(2m + 1),$$

and no other pairing of the two families matches. The smallest case is  $\mathrm{O}(3)$  and  $\mathrm{Sp}(2, \mathbb{R}) = \mathrm{SL}_2(\mathbb{R})$ , both of dimension 3.

- (c) Because  $\det \mathbf{P} = \pm 1$  already, by [Theorem 13.10.3](#) (a). The condition  $\det \mathbf{P} = 1$  does not shrink a continuum of possibilities to a smaller one; it selects one of two

discrete alternatives. So  $\mathrm{SO}(n)$  has the same expected dimension  $\frac{1}{2}n(n-1)$  as  $\mathrm{O}(n)$ , and the two groups differ only in how many separate pieces they have — which is a topological statement of the kind this section does not prove.

### 13.10.7.3 C. Going deeper

#### Exercise C1: Isometries of a degenerate form.

Let  $\mathbf{A} = \begin{pmatrix} 1 & 0 \\ 0 & 0 \end{pmatrix} \in M_2(\mathbb{R})$  and let  $G = \{ \mathbf{P} \in M_2(\mathbb{R}) : \mathbf{P}^\top \mathbf{A} \mathbf{P} = \mathbf{A} \}$ , the set produced by dropping the non-degeneracy hypothesis from [Definition 13.10.1](#).

- Determine  $G$  explicitly.
- Exhibit an element of  $G$  that is not invertible, and explain which step of the proof of [Theorem 13.10.1](#) it defeats.
- Prove that the invertible elements of  $G$  do form a group.

*Solution.*

- Write  $\mathbf{P} = \begin{pmatrix} a & b \\ c & d \end{pmatrix}$ . Then

$$\mathbf{P}^\top \mathbf{A} \mathbf{P} = \begin{pmatrix} a & c \\ b & d \end{pmatrix} \begin{pmatrix} a & b \\ 0 & 0 \end{pmatrix} = \begin{pmatrix} a^2 & ab \\ ab & b^2 \end{pmatrix},$$

so the condition is  $a^2 = 1$ ,  $ab = 0$ ,  $b^2 = 0$ . The last two say  $b = 0$ , and the first says  $a = \pm 1$ . Hence

$$G = \left\{ \begin{pmatrix} \pm 1 & 0 \\ c & d \end{pmatrix} : c, d \in \mathbb{R} \right\}.$$

- Take  $c = d = 0$  and  $a = 1$ : the matrix  $\begin{pmatrix} 1 & 0 \\ 0 & 0 \end{pmatrix}$  lies in  $G$  and has determinant 0. The step it defeats is the cancellation of  $\det \mathbf{A}$  in  $(\det \mathbf{P})^2 \det \mathbf{A} = \det \mathbf{A}$ . Here  $\det \mathbf{A} = 0$ , the equation reads  $0 = 0$ , and it says nothing about  $\det \mathbf{P}$ .
- Let  $G^\times$  be the set of invertible elements of  $G$ ; from (a) these are the matrices above with  $d \neq 0$ . It contains  $\mathbf{I}_2$ . It is closed under products, since  $G$  is (the computation in [Theorem 13.10.1](#) needs only  $\mathbf{P}^\top \mathbf{A} \mathbf{P} = \mathbf{A}$  and  $\mathbf{Q}^\top \mathbf{A} \mathbf{Q} = \mathbf{A}$ , not invertibility) and a product of invertible matrices is invertible. And it is closed under inverses: if  $\mathbf{P} \in G^\times$  then multiplying  $\mathbf{P}^\top \mathbf{A} \mathbf{P} = \mathbf{A}$  by  $(\mathbf{P}^{-1})^\top$  and  $\mathbf{P}^{-1}$  gives  $\mathbf{P}^{-1} \in G$ , and  $\mathbf{P}^{-1}$  is invertible. So  $G^\times$  is a group — but it is *not* all of  $G$ , which is the reason [Definition 13.10.1](#) assumes non-degeneracy rather than repairing the damage afterwards.

#### Exercise C2: The symplectic group of a two-dimensional form, revisited.

Let  $F$  be any field and let  $\beta$  be a non-degenerate alternating form on a two-dimensional vector space  $V$  over  $F$ .

- Prove that  $\mathrm{Isom}(\beta) = \{ T \in \mathcal{L}(V) : \det T = 1 \}$ .
- Deduce that for  $F = \mathbb{F}_2$  the group  $\mathrm{Sp}(2, \mathbb{F}_2)$  has exactly 6 elements, and that  $\mathrm{Sp}(2, \mathbb{F}_2) = \mathrm{GL}_2(\mathbb{F}_2)$ .
- Is  $\mathrm{Sp}(2, \mathbb{F}_3) = \mathrm{GL}_2(\mathbb{F}_3)$ ? Justify your answer by a count.

*Hint for (b) and (c): count the invertible matrices over  $\mathbb{F}_q$  by choosing the columns in turn.*

*Solution.*

- (a) Choose a symplectic basis  $\mathcal{B}$  for  $\beta$ , which exists by [Theorem 13.9.2](#) (b); then  $[\beta]_{\mathcal{B}} = \Omega_2$ . By the correspondence after [Definition 13.10.1](#),  $T \in \text{Isom}(\beta)$  if and only if  $\mathbf{P} := [T]_{\mathcal{B}}$  lies in  $\text{Isom}(\Omega_2) = \text{Sp}(2, F)$ , which by [Theorem 13.10.3](#) (c) says  $\det \mathbf{P} = 1$ . Finally  $\det T = \det \mathbf{P}$  by [Definition 6.5.1](#), since the determinant of an operator is that of its matrix in any basis.
- (b) By (a),  $\text{Sp}(2, \mathbb{F}_2) = \text{SL}_2(\mathbb{F}_2)$ . Count  $\text{GL}_2(\mathbb{F}_2)$  first: the first column may be any of the  $4 - 1 = 3$  non-zero vectors, and the second any of the  $4 - 2 = 2$  vectors outside the span of the first, giving 6. Over  $\mathbb{F}_2$  the only non-zero scalar is 1, so every invertible matrix has  $\det = 1$ ; hence  $\text{SL}_2(\mathbb{F}_2) = \text{GL}_2(\mathbb{F}_2)$  has 6 elements.
- (c) No. The same count over  $\mathbb{F}_3$  gives  $|\text{GL}_2(\mathbb{F}_3)| = (9-1)(9-3) = 48$ . The determinant is a surjective homomorphism  $\text{GL}_2(\mathbb{F}_3) \rightarrow \mathbb{F}_3^\times$ , whose target has 2 elements, and its fibers all have the same size; so  $|\text{SL}_2(\mathbb{F}_3)| = 24 \neq 48$ . Concretely  $\text{diag}(1, -1)$  is invertible with determinant  $-1 \neq 1$ , so it lies in  $\text{GL}_2(\mathbb{F}_3)$  and not in  $\text{Sp}(2, \mathbb{F}_3)$ .

### Exercise C3: Bounded and unbounded.

- (a) Prove that every entry of a matrix  $\mathbf{P} \in \text{O}(n)$  satisfies  $|p_{ij}| \leq 1$ .
- (b) Prove that for every  $n \geq 2$  and every real  $M$ , the group  $\text{O}(n-1, 1)$  contains a matrix with an entry larger than  $M$ .
- (c) Explain in two sentences why (a) and (b) together make it plausible that  $\text{O}(n)$  is compact and  $\text{O}(n-1, 1)$  is not, and say precisely what has and has not been proved.

*Solution.*

- (a) The columns of  $\mathbf{P} \in \text{O}(n)$  are orthonormal in  $\mathbb{R}^n$  ([Theorem 10.7.3](#) (f)), so for each  $j$  we have  $\sum_i p_{ij}^2 = 1$ . Each term of a sum of non-negative reals is at most the sum, so  $p_{ij}^2 \leq 1$  and  $|p_{ij}| \leq 1$ .
- (b) Put  $\mathbf{B}_t := \begin{pmatrix} \cosh t & \sinh t \\ \sinh t & \cosh t \end{pmatrix}$ , which lies in  $\text{O}(1, 1)$  for every real  $t$  by [Example \(Members of Each Family\)](#) (a), and consider

$$\mathbf{P}_t := \mathbf{I}_{n-2} \oplus \mathbf{B}_t \in M_n(\mathbb{R}).$$

(For  $n = 2$  read  $\mathbf{P}_t = \mathbf{B}_t$ .) As a block diagonal matrix  $\mathbf{I}_{n-1,1} = \mathbf{I}_{n-2} \oplus \mathbf{I}_{1,1}$ , and each block of  $\mathbf{P}_t$  preserves its own block of the form:  $\mathbf{I}_{n-2}^\top \mathbf{I}_{n-2} \mathbf{I}_{n-2} = \mathbf{I}_{n-2}$ , and  $\mathbf{B}_t^\top \mathbf{I}_{1,1} \mathbf{B}_t = \mathbf{I}_{1,1}$  by [Example \(Members of Each Family\)](#) (a). Multiplying block by block gives  $\mathbf{P}_t^\top \mathbf{I}_{n-1,1} \mathbf{P}_t = \mathbf{I}_{n-1,1}$ , so  $\mathbf{P}_t \in \text{O}(n-1, 1)$ . Its  $(n, n)$  entry is  $\cosh t = \frac{1}{2}(e^t + e^{-t}) \geq \frac{1}{2}e^t$ , and  $e^t$  grows without bound, so some  $t$  makes this entry exceed the given  $M$ .

- (c) A compact subset of  $M_n(\mathbb{R}) \cong \mathbb{R}^{n^2}$  must be bounded, so (b) rules out compactness for  $\text{O}(n-1, 1)$ , while (a) supplies half of what  $\text{O}(n)$  would need. What has been **proved** here is only the statement about entries:  $\text{O}(n)$  is bounded and  $\text{O}(n-1, 1)$  is not. That “bounded and closed implies compact” in  $\mathbb{R}^{n^2}$ , and that  $\text{O}(n)$  is closed, are theorems of analysis that this book does not prove, so the compactness

claim itself remains recorded rather than established.

## 13.11 Minkowski Space

Sylvester’s law of inertia says that a real symmetric form is nothing but a pair of counts, and that any two forms with the same counts are the same form in disguise. One pair of counts has a claim on our attention that none of the others has: the form of signature  $(1, 3)$  is the one the physical world appears to run on. This section does the linear algebra of that form. Everything in it is this chapter’s machinery specialized, and the payoff is a picture — a cone dividing space into three regions — together with a classification theorem that is the hyperbolic twin of a circular one we already proved.

### 13.11.1 The form of special relativity

Fix  $n_+ = 1$  and  $n_- = 3$  in the canonical form of [Corollary 13.5.2](#), and there is only one form to write down.

#### Definition 13.11.1 — Minkowski Space

**Minkowski space**  $\mathbb{R}^{1,3}$  is the real vector space  $\mathbb{R}^4$  equipped with the symmetric bilinear form

$$\eta(\mathbf{x}, \mathbf{y}) := x_1y_1 - x_2y_2 - x_3y_3 - x_4y_4,$$

whose quadratic form is  $q(\mathbf{x}) = x_1^2 - x_2^2 - x_3^2 - x_4^2$ . The first coordinate is the **time** coordinate and the remaining three are the **space** coordinates. More generally, for  $p, m \geq 0$ ,  $\mathbb{R}^{p,m}$  is  $\mathbb{R}^{p+m}$  with the form whose matrix in the standard basis is  $\mathbf{I}_p \oplus (-\mathbf{I}_m)$ .

In words:  $\eta$  is the dot product of  $\mathbb{R}^4$  with three of its four signs reversed. Its matrix in the standard basis  $\mathcal{E}$  is

$$[\eta]_{\mathcal{E}} = \text{diag}(1, -1, -1, -1),$$

which is symmetric, so  $\eta$  is a symmetric form ([Definition 13.3.1](#)), and it is diagonal, so  $\mathcal{E}$  is an orthogonal basis for  $\eta$  ([Definition 13.4.1](#)). Reading off the signs, the signature of  $\eta$  is  $(1, 3)$  and its inertia is  $(1, 3, 0)$  ([Definition 13.5.1](#)). The form is **non-degenerate**:  $\eta(\mathbf{e}_1, \mathbf{x}) = x_1$  and  $\eta(\mathbf{e}_j, \mathbf{x}) = -x_j$  for  $j = 2, 3, 4$ , so a vector paired to zero with everything has all four coordinates zero ([Definition 13.1.4](#)).

*Warning.* **Two sign conventions are in use, and both are called “the” Minkowski form.** We take  $(+, -, -, -)$ , signature  $(1, 3)$ ; many sources take  $(-, +, +, +)$ , signature  $(3, 1)$ , which is our  $-\eta$ . The two are not congruent — congruence preserves the signature ([Theorem 13.5.1](#)) — but every statement below transfers by reversing each inequality on  $q$ , because  $q$  changes sign. The one thing both conventions agree on is the set  $\{q = 0\}$ , which is why the picture of this section is convention-free. State your convention before computing, and never mix the two.

### 13.11.2 Timelike, spacelike and null

Nothing in this section is a new notion. Section 7 studied the vectors on which a form vanishes, and called them isotropic; over  $\mathbb{R}$  a non-degenerate indefinite form has plenty of them, and the ones belonging to  $\eta$  have names of their own.

*The form sorts non-zero vectors into three classes by the sign of  $q$ , and the middle class — where  $q$  vanishes — is exactly the isotropy of Section 7.*

#### Definition 13.11.2 — Timelike, Spacelike, Null

Let  $\mathbf{x} \in \mathbb{R}^{1,3}$  with  $\mathbf{x} \neq \mathbf{0}$ . Then  $\mathbf{x}$  is

- (a) **timelike** if  $q(\mathbf{x}) > 0$ ;
- (b) **null**, or **lightlike**, if  $q(\mathbf{x}) = 0$ ;
- (c) **spacelike** if  $q(\mathbf{x}) < 0$ .

The set  $\{\mathbf{x} \in \mathbb{R}^{1,3} : q(\mathbf{x}) = 0\}$  is the **light cone**. A timelike  $\mathbf{x}$  is **future-pointing** if  $x_1 > 0$  and **past-pointing** if  $x_1 < 0$ .

The word **null** is a second name for something already defined: a null vector is precisely a non-zero **isotropic** vector for  $\eta$  in the sense of Definition 13.7.1, and the light cone is the isotropy locus of  $\eta$  together with  $\mathbf{0}$ . So “the light cone” is not new geometry; it is the locus Section 7 already forced us to take seriously, drawn for one particular form. Timelike and spacelike are then just names for the two sides that the isotropy locus separates.

The last clause of the definition is legitimate because a timelike vector cannot have  $x_1 = 0$ : if it did, then  $q(\mathbf{x}) = -(x_2^2 + x_3^2 + x_4^2) \leq 0$ . So every timelike vector is future-pointing or past-pointing, and exactly one of the two.

#### Example. (Reading off the three types)

Classify  $\mathbf{e}_1$ ,  $\mathbf{e}_2$ ,  $\mathbf{e}_1 + \mathbf{e}_2$  and  $(2, 1, 1, 1)$  in  $\mathbb{R}^{1,3}$ , and decide which of them are isotropic.

*Solution.*  $q(\mathbf{e}_1) = 1 > 0$ , so  $\mathbf{e}_1$  is timelike and future-pointing.  $q(\mathbf{e}_2) = -1 < 0$ , so  $\mathbf{e}_2$  is spacelike.  $q(\mathbf{e}_1 + \mathbf{e}_2) = 1 - 1 = 0$ , so  $\mathbf{e}_1 + \mathbf{e}_2$  is null, hence isotropic. Finally  $q(2, 1, 1, 1) = 4 - 3 = 1 > 0$ , timelike and future-pointing. Only  $\mathbf{e}_1 + \mathbf{e}_2$  is isotropic; the other three are anisotropic.

Section 7’s structure theorem applies verbatim. A non-zero isotropic vector sits in a hyperbolic plane, and here we can point to one: take  $\mathbf{u} = \mathbf{e}_1 + \mathbf{e}_2$  and  $\mathbf{w} = \frac{1}{2}(\mathbf{e}_1 - \mathbf{e}_2)$ . Then  $q(\mathbf{u}) = q(\mathbf{w}) = 0$  and

$$\eta(\mathbf{u}, \mathbf{w}) = \frac{1}{2}(1 - (-1)) = 1,$$

so the Gram matrix of  $(\mathbf{u}, \mathbf{w})$  is  $\begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}$  and  $\text{span}(\mathbf{u}, \mathbf{w})$  is a hyperbolic plane (Definition 13.7.3). Splitting it off, Theorem 13.7.4 writes

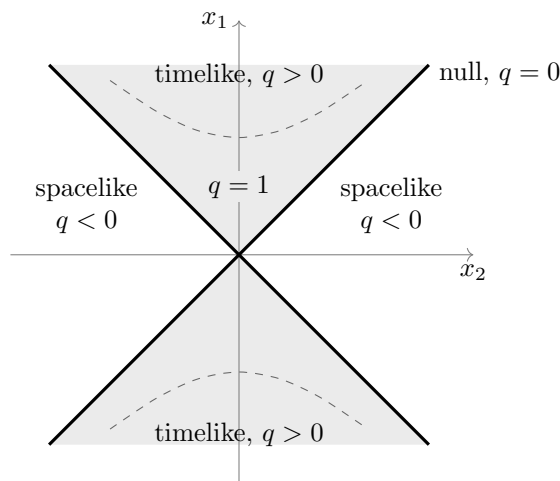
$$\mathbb{R}^{1,3} = H \perp W,$$

with  $H$  hyperbolic of signature  $(1, 1)$  and  $W$  anisotropic of signature  $(0, 2)$ , that is, negative definite. Only one plane splits off, because the Witt index of a real form of signature  $(n_+, n_-)$  is  $\min(n_+, n_-)$  (Proposition 13.7.6), which here is  $\min(1, 3) = 1$ .

*Quick check.* Is there a **plane** in  $\mathbb{R}^{1,3}$  on which  $\eta$  vanishes identically?

*Solution.* No. Suppose  $\mathbf{x} = (a, \mathbf{u})$  and  $\mathbf{y} = (b, \mathbf{w})$  span such a plane, where  $a, b \in \mathbb{R}$  and  $\mathbf{u}, \mathbf{w} \in \mathbb{R}^3$ . Then  $q(\mathbf{x}) = q(\mathbf{y}) = 0$  gives  $a^2 = \|\mathbf{u}\|^2$  and  $b^2 = \|\mathbf{w}\|^2$ , while  $\eta(\mathbf{x}, \mathbf{y}) = 0$  gives  $ab = \mathbf{u} \cdot \mathbf{w}$ . Hence  $|\mathbf{u} \cdot \mathbf{w}| = |a||b| = \|\mathbf{u}\|\|\mathbf{w}\|$ , which is the equality case of [Theorem 10.1.5](#), so  $(\mathbf{u}, \mathbf{w})$  is dependent. If  $\mathbf{u} = \mathbf{0}$  then  $a = 0$  and  $\mathbf{x} = \mathbf{0}$ ; otherwise  $\mathbf{w} = \lambda \mathbf{u}$ , so  $b^2 = \lambda^2 a^2$  and  $ab = \lambda a^2$ , which force  $b = \lambda a$  and  $\mathbf{y} = \lambda \mathbf{x}$ . Either way  $(\mathbf{x}, \mathbf{y})$  is dependent, so no plane exists. Structurally this is the statement that the Witt index of  $\mathbb{R}^{1,3}$  is  $\min(1, 3) = 1$  ([Proposition 13.7.6](#)), since by [Definition 13.7.4](#) no totally isotropic subspace has dimension larger than the index.

The picture is easiest to draw with one space coordinate instead of three, in  $\mathbb{R}^{1,1}$ , where the light cone degenerates to the two lines  $x_2 = \pm x_1$ .



The light cone  $q = 0$  in the Minkowski plane separates the timelike vectors from the spacelike ones; the “unit sphere”  $q = 1$  is the dashed hyperbola, and it is unbounded

Two features of the picture deserve to be said out loud. The shaded region is **not** connected: the future-pointing and past-pointing timelike vectors form two separate pieces, and no continuous path of timelike vectors joins them. That last claim is worth one line, since it is the only place in this section where a topological word is used. Suppose  $\gamma$  were a continuous path of timelike vectors from a future-pointing vector to a past-pointing one. Its first coordinate  $x_1$  is a continuous real function that is positive at one end and negative at the other, so by the intermediate value theorem — an analysis input, quoted and not proved here, as in Section 6 — it vanishes somewhere along the path. But  $x_1 = 0$  forces  $q(\mathbf{x}) = -x_2^2 - x_3^2 - x_4^2 \leq 0$ , so that point of the path is not timelike, a contradiction. And the level set  $q = 1$ , which plays the role the unit circle plays for the dot product, is unbounded. Both facts come straight from the mixed signs, and both are invisible to anyone whose intuition was trained on positive definite forms.

**Warning. Orthogonality for  $\eta$  is not the perpendicularity you can see.** A null vector is  $\eta$ -orthogonal to *itself*, and the consequence is that Chapter 10’s decomposition  $V = U \oplus U^\perp$  fails. Take  $U = \text{span}((1, 1, 0, 0))$ . Since  $\eta((1, 1, 0, 0), \mathbf{x}) = x_1 - x_2$ , the  $\eta$ -orthogonal complement of  $U$  is the hyperplane  $\{x_1 = x_2\}$ , which **contains**  $U$ . So  $U + U^\perp = U^\perp$  is three-dimensional, not four, and  $U \cap U^\perp = U \neq \{\mathbf{0}\}$ . Nothing has gone wrong; positivity was doing that work in Chapter 10, and it is gone.



### 13.11.3 Hyperbolic rotations

Now the isometries. The group attached to  $\eta$  is the isometry group of its matrix (Definition 13.10.1), which is  $O(1,3)$  in the notation of Definition 13.10.2: the matrices  $\mathbf{P}$  with  $\mathbf{P}^\top[\eta]_{\mathcal{E}}\mathbf{P} = [\eta]_{\mathcal{E}}$ . Its elements are the **Lorentz transformations**. We classify them in dimension 2, where  $\eta$  becomes

$$\eta_2(\mathbf{x}, \mathbf{y}) = x_1y_1 - x_2y_2, \quad [\eta_2]_{\mathcal{E}} = \text{diag}(1, -1).$$

The classification of  $O(2)$  in Theorem 10.7.5 rested on one geometric fact: a unit vector of the Euclidean plane is  $(\cos \theta, \sin \theta)$  for exactly one  $\theta \in [0, 2\pi)$ . Replace the circle by the hyperbola and the same proof runs. We need the corresponding parametrization, and we take it from calculus:

$$\begin{aligned} \cosh t &= \frac{1}{2}(e^t + e^{-t}), & \sinh t &= \frac{1}{2}(e^t - e^{-t}), \\ \cosh^2 t - \sinh^2 t &= 1, & \cosh t &> 0, \\ \cosh(s+t) &= \cosh s \cosh t + \sinh s \sinh t, \\ \sinh(s+t) &= \sinh s \cosh t + \cosh s \sinh t, \end{aligned}$$

and  $\sinh: \mathbb{R} \rightarrow \mathbb{R}$  is a strictly increasing bijection. Those facts are the only analysis used in this section.

#### Definition 13.11.3 — Boost

For  $t \in \mathbb{R}$ , the **boost of rapidity**  $t$  is

$$\mathbf{B}_t := \begin{pmatrix} \cosh t & \sinh t \\ \sinh t & \cosh t \end{pmatrix} \in M_2(\mathbb{R}).$$

A boost is the hyperbolic analogue of the rotation  $\mathbf{R}_\theta$  of Chapter 10: same shape, hyperbolic functions in place of circular ones, and no minus sign in the corner.

#### Theorem 13.11.1 — Classification of the Plane Lorentz Transformations

Let  $\mathbf{P} \in O(1,1)$ , that is,  $\mathbf{P} \in M_2(\mathbb{R})$  with  $\mathbf{P}^\top \text{diag}(1, -1)\mathbf{P} = \text{diag}(1, -1)$ . Then there are a unique  $t \in \mathbb{R}$  and unique signs  $\varepsilon, \mu \in \{1, -1\}$  with

$$\mathbf{P} = \begin{pmatrix} \varepsilon \cosh t & \mu \sinh t \\ \sinh t & \varepsilon \mu \cosh t \end{pmatrix},$$

and  $\det \mathbf{P} = \mu$ . Writing  $\mathbf{D} = \text{diag}(1, -1)$ , this says

$$O(1,1) = \{ \mathbf{B}_t, \mathbf{B}_t\mathbf{D}, -\mathbf{B}_t, -\mathbf{B}_t\mathbf{D} : t \in \mathbb{R} \},$$

a disjoint union of four families. Moreover  $\mathbf{B}_s\mathbf{B}_t = \mathbf{B}_{s+t}$  for all  $s, t$ , so  $\{\mathbf{B}_t : t \in \mathbb{R}\}$  is a subgroup of  $O(1,1)$  isomorphic to  $(\mathbb{R}, +)$ .

*Idea.* The congruence  $\mathbf{P}^\top \mathbf{D} \mathbf{P} = \mathbf{D}$  says exactly that the two columns of  $\mathbf{P}$  are an orthogonal basis for  $\eta_2$  with  $q$ -values 1 and  $-1$ . So: parametrize the vectors with  $q = 1$ , which is the hyperbola; then the second column is forced, because the  $\eta_2$ -orthogonal complement of a non-isotropic vector is a line, and on that line only two vectors have  $q = -1$ . This is [Theorem 10.7.5](#)'s proof with the circle replaced by the hyperbola.

*Proof.* Write  $\mathbf{P} = \begin{pmatrix} a & c \\ b & d \end{pmatrix}$  with columns  $\mathbf{p}_1 = (a, b)$  and  $\mathbf{p}_2 = (c, d)$ . The  $(i, j)$  entry of  $\mathbf{P}^\top \mathbf{D} \mathbf{P}$  is  $\eta_2(\mathbf{p}_i, \mathbf{p}_j)$ , so the hypothesis says

$$q(\mathbf{p}_1) = 1, \quad q(\mathbf{p}_2) = -1, \quad \eta_2(\mathbf{p}_1, \mathbf{p}_2) = 0.$$

**The first column.** Since  $\sinh$  is a bijection of  $\mathbb{R}$ , there is a unique  $t \in \mathbb{R}$  with  $b = \sinh t$ . Then  $a^2 = 1 + b^2 = 1 + \sinh^2 t = \cosh^2 t$ , so  $a = \varepsilon \cosh t$  for a unique sign  $\varepsilon$ , unique because  $\cosh t > 0$ .

**The second column.** The vector  $\mathbf{z} = (\sinh t, \varepsilon \cosh t)$  satisfies  $\eta_2(\mathbf{p}_1, \mathbf{z}) = \varepsilon \cosh t \sinh t - \sinh t \varepsilon \cosh t = 0$ , and  $\mathbf{z} \neq \mathbf{0}$  because  $\varepsilon \cosh t \neq 0$ . The set  $\{\mathbf{y} : \eta_2(\mathbf{p}_1, \mathbf{y}) = 0\}$  is the null space of the non-zero row vector  $(a, -b)$ , hence a line, so it equals  $\text{span}(\mathbf{z})$ . Therefore  $\mathbf{p}_2 = \mu' \mathbf{z}$  for some  $\mu' \in \mathbb{R}$ , and  $q(\mathbf{z}) = \sinh^2 t - \cosh^2 t = -1$  gives  $-1 = q(\mathbf{p}_2) = (\mu')^2 q(\mathbf{z}) = -(\mu')^2$ . Hence  $\mu' = \mu$  is a sign, and it is determined by  $\mathbf{P}$ , since  $d = \mu \varepsilon \cosh t$  and  $\cosh t > 0$ . This is the displayed matrix, and expanding,

$$\det \mathbf{P} = \mu(\varepsilon^2 \cosh^2 t - \sinh^2 t) = \mu.$$

**The four families.** For  $\varepsilon = \mu = 1$  the matrix is  $\mathbf{B}_t$ ; for  $\varepsilon = 1, \mu = -1$  it is  $\mathbf{B}_t \mathbf{D}$ ; for  $\varepsilon = -1, \mu = 1$  it is  $-\mathbf{B}_{-t}$ ; and for  $\varepsilon = \mu = -1$  it is  $-\mathbf{B}_{-t} \mathbf{D}$ , each by multiplying out and using  $\sinh(-t) = -\sinh t$ ,  $\cosh(-t) = \cosh t$ . As  $t$  runs over  $\mathbb{R}$  so does  $-t$ , which gives the displayed set equality. The families are disjoint because  $\varepsilon$  is the sign of the  $(1, 1)$  entry and  $\mu = \det \mathbf{P}$ , and both are determined by  $\mathbf{P}$ .

**The subgroup.** By the addition formulas for  $\cosh$  and  $\sinh$ ,

$$\mathbf{B}_s \mathbf{B}_t = \begin{pmatrix} \cosh(s+t) & \sinh(s+t) \\ \sinh(s+t) & \cosh(s+t) \end{pmatrix} = \mathbf{B}_{s+t}.$$

So  $t \mapsto \mathbf{B}_t$  is a group homomorphism  $(\mathbb{R}, +) \rightarrow \text{O}(1, 1)$ ; it is injective because  $\sinh$  is, and its image is a subgroup by [Definition 0.10.2](#). This proves the theorem. ■

The four families are sorted by two signs, and each sign has a reading. The determinant  $\mu$  says whether the space coordinate is reversed, exactly as in  $\text{O}(2)$ . The sign  $\varepsilon$  of the  $(1, 1)$  entry says whether the future cone is sent to the future cone: for  $\mathbf{B}_t$  the image of  $\mathbf{e}_1$  is  $(\cosh t, \sinh t)$ , with positive first coordinate, while  $-\mathbf{B}_t$  sends it to  $(-\cosh t, -\sinh t)$ . So the identity component consists of the boosts alone, and the other three families are obtained from it by reversing space, time, or both.

### 13.11.4 Circular and hyperbolic, side by side

The comparison is the point of this section, so here it is in one table. In each column, the isometry group of the named form on  $\mathbb{R}^2$ .

|                     | O(2): form $\mathbf{I}_2$                                                                                   | O(1, 1): form $\text{diag}(1, -1)$                                                    |
|---------------------|-------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------|
| level set $q = 1$   | circle $x_1^2 + x_2^2 = 1$                                                                                  | hyperbola $x_1^2 - x_2^2 = 1$ , two branches                                          |
| canonical element   | $\mathbf{R}_\theta = \begin{pmatrix} \cos \theta & -\sin \theta \\ \sin \theta & \cos \theta \end{pmatrix}$ | $\mathbf{B}_t = \begin{pmatrix} \cosh t & \sinh t \\ \sinh t & \cosh t \end{pmatrix}$ |
| parameter           | $\theta \in [0, 2\pi)$ , periodic                                                                           | $t \in \mathbb{R}$ , unbounded                                                        |
| addition law        | $\mathbf{R}_\alpha \mathbf{R}_\beta = \mathbf{R}_{\alpha+\beta}$                                            | $\mathbf{B}_s \mathbf{B}_t = \mathbf{B}_{s+t}$                                        |
| determinant +1 part | $\text{SO}(2) = \{\mathbf{R}_\theta\}$                                                                      | $\{\mathbf{B}_t\} \cup \{-\mathbf{B}_t\}$                                             |
| determinant -1 part | reflections $\mathbf{M}_\theta$                                                                             | $\{\mathbf{B}_t \mathbf{D}\} \cup \{-\mathbf{B}_t \mathbf{D}\}$                       |
| number of pieces    | 2                                                                                                           | 4                                                                                     |
| entries             | all of modulus $\leq 1$                                                                                     | unbounded                                                                             |

The two canonical forms differ in one place only:  $\mathbf{R}_\theta$  carries a minus sign that  $\mathbf{B}_t$  does not, and that single sign is the difference between  $\cos^2 + \sin^2 = 1$  and  $\cosh^2 - \sinh^2 = 1$ . Everything else in the table follows from it. In particular the last row is the unboundedness that Section 10 recorded when it observed, without proof, that  $O(p, q)$  is not compact for  $p, q \geq 1$  while  $O(n)$  is.

*Remark.* In the physical reading,  $\tanh t$  is a velocity measured in units of the speed of light, and the additivity  $\mathbf{B}_s \mathbf{B}_t = \mathbf{B}_{s+t}$  of rapidities becomes the relativistic law for adding velocities. [Exercise \(C1\)](#) derives the formula from the theorem. The book verifies the algebra and stops there.

### 13.11.5 The triangle inequality, reversed

For a timelike  $\mathbf{x}$  the number  $q(\mathbf{x})$  is positive, so it has a positive square root, and the analogy with a norm is irresistible. Write

$$\tau(\mathbf{x}) := \sqrt{q(\mathbf{x})} \quad (\mathbf{x} \text{ timelike}).$$

The analogy holds, but with every inequality upside down. First the tool.

#### Lemma 13.11.2 — Reversed Cauchy–Schwarz Inequality

Let  $\mathbf{x}, \mathbf{y} \in \mathbb{R}^{1,3}$  be timelike and both future-pointing. Then

$$\eta(\mathbf{x}, \mathbf{y}) \geq \tau(\mathbf{x})\tau(\mathbf{y}) > 0,$$

with equality if and only if  $\mathbf{y} = \lambda \mathbf{x}$  for some  $\lambda > 0$ .

*Idea.* Split each vector as a time coordinate plus a space part,  $\mathbf{x} = (a, \mathbf{u})$ . Timelike and future-pointing says  $a > \|\mathbf{u}\| \geq 0$ . Then  $\eta(\mathbf{x}, \mathbf{y}) = ab - \mathbf{u} \cdot \mathbf{w}$ , and [Theorem 10.1.5](#) bounds  $\mathbf{u} \cdot \mathbf{w}$  by  $\|\mathbf{u}\|\|\mathbf{w}\|$  — in the helpful direction, since it appears with a minus sign. What is left is to compare  $ab - \|\mathbf{u}\|\|\mathbf{w}\|$  with  $\tau(\mathbf{x})\tau(\mathbf{y})$ , and squaring both sides turns

the difference into a perfect square.

*Proof.* Write  $\mathbf{x} = (a, \mathbf{u})$  and  $\mathbf{y} = (b, \mathbf{w})$  with  $a, b \in \mathbb{R}$  and  $\mathbf{u}, \mathbf{w} \in \mathbb{R}^3$ , so that  $q(\mathbf{x}) = a^2 - \|\mathbf{u}\|^2$  and  $\eta(\mathbf{x}, \mathbf{y}) = ab - \mathbf{u} \cdot \mathbf{w}$ . Timelike gives  $a^2 > \|\mathbf{u}\|^2$  and  $b^2 > \|\mathbf{w}\|^2$ , and future-pointing gives  $a, b > 0$ ; hence  $a > \|\mathbf{u}\| \geq 0$  and  $b > \|\mathbf{w}\| \geq 0$ , so  $ab > \|\mathbf{u}\|\|\mathbf{w}\| \geq 0$ . By [Theorem 10.1.5](#) in  $\mathbb{R}^3$ ,

$$\eta(\mathbf{x}, \mathbf{y}) = ab - \mathbf{u} \cdot \mathbf{w} \geq ab - \|\mathbf{u}\|\|\mathbf{w}\| > 0.$$

Both sides of the inequality to be proved are now known positive, so it suffices to compare squares. Expanding,

$$\begin{aligned} (ab - \|\mathbf{u}\|\|\mathbf{w}\|)^2 - q(\mathbf{x})q(\mathbf{y}) &= a^2\|\mathbf{w}\|^2 + b^2\|\mathbf{u}\|^2 - 2ab\|\mathbf{u}\|\|\mathbf{w}\| \\ &= (a\|\mathbf{w}\| - b\|\mathbf{u}\|)^2 \geq 0. \end{aligned}$$

Therefore  $ab - \|\mathbf{u}\|\|\mathbf{w}\| \geq \sqrt{q(\mathbf{x})q(\mathbf{y})} = \tau(\mathbf{x})\tau(\mathbf{y})$ , and combining with the previous display gives  $\eta(\mathbf{x}, \mathbf{y}) \geq \tau(\mathbf{x})\tau(\mathbf{y}) > 0$ .

For the equality case, equality forces both steps to be equalities:  $\mathbf{u} \cdot \mathbf{w} = \|\mathbf{u}\|\|\mathbf{w}\|$  and  $a\|\mathbf{w}\| = b\|\mathbf{u}\|$ . Suppose first  $\mathbf{u} = \mathbf{0}$ . Then  $a\|\mathbf{w}\| = 0$  and  $a > 0$ , so  $\mathbf{w} = \mathbf{0}$ , and  $\mathbf{y} = (b, \mathbf{0}) = (b/a)\mathbf{x}$  with  $b/a > 0$ . Suppose instead  $\mathbf{u} \neq \mathbf{0}$ . The equality case of [Theorem 10.1.5](#) makes  $(\mathbf{u}, \mathbf{w})$  dependent, so  $\mathbf{w} = \lambda\mathbf{u}$  with  $\lambda \in \mathbb{R}$ , and then  $\mathbf{u} \cdot \mathbf{w} = \lambda\|\mathbf{u}\|^2 = \|\mathbf{u}\|\|\mathbf{w}\| = |\lambda|\|\mathbf{u}\|^2$  forces  $\lambda \geq 0$ . Now  $a\|\mathbf{w}\| = b\|\mathbf{u}\|$  reads  $a\lambda\|\mathbf{u}\| = b\|\mathbf{u}\|$ , so  $b = \lambda a$  and  $\mathbf{y} = \lambda\mathbf{x}$ ; and  $\lambda > 0$  because  $b > 0$ . Conversely, if  $\mathbf{y} = \lambda\mathbf{x}$  with  $\lambda > 0$  then  $\eta(\mathbf{x}, \mathbf{y}) = \lambda q(\mathbf{x})$  and  $\tau(\mathbf{x})\tau(\mathbf{y}) = \sqrt{\lambda^2 q(\mathbf{x})^2} = \lambda q(\mathbf{x})$ , so equality holds. This proves the lemma. ■

### Theorem 13.11.3 — Reversed Triangle Inequality

Let  $\mathbf{x}, \mathbf{y} \in \mathbb{R}^{1,3}$  be timelike and both future-pointing. Then  $\mathbf{x} + \mathbf{y}$  is timelike and future-pointing, and

$$\tau(\mathbf{x} + \mathbf{y}) \geq \tau(\mathbf{x}) + \tau(\mathbf{y}),$$

with equality if and only if  $\mathbf{y}$  is a positive multiple of  $\mathbf{x}$ .

*Proof.* By [Theorem 13.3.5](#),  $q(\mathbf{x} + \mathbf{y}) = q(\mathbf{x}) + 2\eta(\mathbf{x}, \mathbf{y}) + q(\mathbf{y})$ . By [Lemma 13.11.2](#),  $\eta(\mathbf{x}, \mathbf{y}) \geq \tau(\mathbf{x})\tau(\mathbf{y})$ , so

$$q(\mathbf{x} + \mathbf{y}) \geq q(\mathbf{x}) + 2\tau(\mathbf{x})\tau(\mathbf{y}) + q(\mathbf{y}) = (\tau(\mathbf{x}) + \tau(\mathbf{y}))^2,$$

where the last step uses  $q = \tau^2$  on timelike vectors. The right-hand side is positive, so  $\mathbf{x} + \mathbf{y}$  is timelike; its first coordinate is a sum of two positive numbers, so it is future-pointing. Both sides of the last display are non-negative, so taking square roots preserves the inequality and gives  $\tau(\mathbf{x} + \mathbf{y}) \geq \tau(\mathbf{x}) + \tau(\mathbf{y})$ . Equality holds exactly when it holds in [Lemma 13.11.2](#), that is, exactly when  $\mathbf{y} = \lambda\mathbf{x}$  with  $\lambda > 0$ . This proves the theorem. ■

*Remark.* Compare [Corollary 10.1.6](#), which gives  $\|\mathbf{u} + \mathbf{v}\| \leq \|\mathbf{u}\| + \|\mathbf{v}\|$  in an inner product space, and [Theorem 0.6.4](#), its ancestor in  $\mathbb{C}$ . Same shape, opposite direction: the reversal is produced by the single sign change in the form, and by nothing else.

Read as a statement about clocks, the reversed inequality says that a journey made in two straight legs accumulates *less* elapsed time than the single straight leg between the same end-points, which is the algebra behind the twin paradox; whether that algebra describes the world is a question for physics, and this book answers only the algebra.

*Quick check.* Where exactly would the proof of [Theorem 13.11.3](#) break if  $\mathbf{x}$  were future-pointing and  $\mathbf{y}$  past-pointing?

*Solution.* At [Lemma 13.11.2](#), whose proof uses  $a, b > 0$  to get  $ab > \|\mathbf{u}\|\|\mathbf{w}\| \geq 0$ . With  $b < 0$  we have  $ab < 0$ , and  $\eta(\mathbf{x}, \mathbf{y}) = ab - \mathbf{u} \cdot \mathbf{w}$  can be negative, so the chain collapses at its first step. The conclusion genuinely fails: take  $\mathbf{x} = (1, 0, 0, 0)$  and  $\mathbf{y} = (-1, 0, 0, 0)$ , both timelike with  $\tau = 1$ , whose sum is  $\mathbf{0}$  and is not timelike at all.

## 13.11.6 Exercises

### 13.11.6.1 A. Check your understanding

#### Exercise A1.

- State the sign convention used in this section, give the matrix of  $\eta$  in the standard basis, and say what its signature and inertia are.
- Define timelike, null and spacelike, and say which earlier notion the word “null” renames.
- True or false: the sum of two spacelike vectors is spacelike. Justify your answer.
- Write down  $\det \mathbf{B}_t$  and  $\mathbf{B}_t^{-1}$ , and say which row of the comparison table each answer sits in.
- True or false:  $O(1, 1)$  is abelian. Justify your answer.

*Solution.*

- The convention is  $(+, -, -, -)$ , so  $[\eta]_{\mathcal{E}} = \text{diag}(1, -1, -1, -1)$ , with signature  $(1, 3)$  and inertia  $(1, 3, 0)$  ([Definition 13.5.1](#)). The opposite convention  $(-, +, +, +)$  uses  $-\eta$  and has signature  $(3, 1)$ .
- For  $\mathbf{x} \neq \mathbf{0}$ : timelike means  $q(\mathbf{x}) > 0$ , null means  $q(\mathbf{x}) = 0$ , spacelike means  $q(\mathbf{x}) < 0$  ([Definition 13.11.2](#)). “Null” renames “non-zero isotropic vector” ([Definition 13.7.1](#)).
- False. Take  $\mathbf{x} = (1, 2, 0, 0)$  and  $\mathbf{y} = (1, -2, 0, 0)$ , with  $q(\mathbf{x}) = q(\mathbf{y}) = 1 - 4 = -3 < 0$ . Their sum is  $(2, 0, 0, 0)$ , with  $q = 4 > 0$ , which is timelike.
- $\det \mathbf{B}_t = 1$  by [Theorem 13.11.1](#) with  $\mu = 1$ , and  $\mathbf{B}_t^{-1} = \mathbf{B}_{-t}$ , since  $\mathbf{B}_t \mathbf{B}_{-t} = \mathbf{B}_0 = \mathbf{I}_2$ . The first sits in the “determinant +1 part” row, the second in the “addition law” row, which is what makes  $t \mapsto \mathbf{B}_t$  a group homomorphism.
- False.  $\mathbf{B}_1$  and  $\mathbf{D} = \text{diag}(1, -1)$  both lie in  $O(1, 1)$ , and  $\mathbf{B}_1 \mathbf{D}$  has  $(1, 2)$  entry  $-\sinh 1$  while  $\mathbf{D} \mathbf{B}_1$  has  $(1, 2)$  entry  $\sinh 1$ ; these differ since  $\sinh 1 \neq 0$ .

## 13.11.6.2 B. Practice

**Exercise B1.**

Determine which of the following lie in  $O(1, 1)$ . For each one that does, give its rapidity  $t$  and the signs  $\varepsilon, \mu$  of [Theorem 13.11.1](#). Justify your answer.

$$\mathbf{A}_1 = \begin{pmatrix} 5/3 & 4/3 \\ 4/3 & 5/3 \end{pmatrix}, \quad \mathbf{A}_2 = \begin{pmatrix} 2 & 1 \\ 1 & 2 \end{pmatrix}, \quad \mathbf{A}_3 = \begin{pmatrix} 5/3 & -4/3 \\ 4/3 & -5/3 \end{pmatrix}.$$

*Solution.* Write  $\mathbf{D} = \text{diag}(1, -1)$  and test  $\mathbf{P}^\top \mathbf{D} \mathbf{P} = \mathbf{D}$  column by column, using  $q(\mathbf{p}_1) = 1$ ,  $q(\mathbf{p}_2) = -1$ ,  $\eta_2(\mathbf{p}_1, \mathbf{p}_2) = 0$ .

$\mathbf{A}_1$ :  $(5/3)^2 - (4/3)^2 = (25-16)/9 = 1$ ;  $(4/3)^2 - (5/3)^2 = -1$ ;  $(5/3)(4/3) - (4/3)(5/3) = 0$ . So  $\mathbf{A}_1 \in O(1, 1)$ . Here  $\sinh t = 4/3$  and  $\cosh t = 5/3$ , so  $\varepsilon = \mu = 1$  and  $\mathbf{A}_1 = \mathbf{B}_t$  with  $t = \log 3$ , since  $\cosh(\log 3) = \frac{1}{2}(3 + \frac{1}{3}) = 5/3$  and  $\sinh(\log 3) = \frac{1}{2}(3 - \frac{1}{3}) = 4/3$ .

$\mathbf{A}_2$ :  $2^2 - 1^2 = 3 \neq 1$ , so the first column already fails and  $\mathbf{A}_2 \notin O(1, 1)$ . (Indeed  $\mathbf{A}_2^\top \mathbf{D} \mathbf{A}_2 = \text{diag}(3, -3) = 3\mathbf{D}$ , so  $\mathbf{A}_2$  scales the form rather than preserving it.)

$\mathbf{A}_3$ : first column as for  $\mathbf{A}_1$ ; second column  $(-4/3)^2 - (-5/3)^2 = -1$ ; and  $(5/3)(-4/3) - (4/3)(-5/3) = 0$ . So  $\mathbf{A}_3 \in O(1, 1)$ , with  $t = \log 3$ ,  $\varepsilon = 1$  and  $\mu = \det \mathbf{A}_3 = -25/9 + 16/9 = -1$ ; that is,  $\mathbf{A}_3 = \mathbf{B}_{\log 3} \mathbf{D}$ .

**Exercise B2.**

- Classify  $\mathbf{x} = (3, 1, 2, 2)$ ,  $\mathbf{y} = (1, 0, 1, 0)$  and  $\mathbf{z} = (0, 1, 1, 1)$  in  $\mathbb{R}^{1,3}$  as timelike, null or spacelike.
- The vector  $\mathbf{y}$  is null. Find a  $\mathbf{w} \in \mathbb{R}^{1,3}$  such that  $(\mathbf{y}, \mathbf{w})$  is a basis of a hyperbolic plane, that is,  $q(\mathbf{w}) = 0$  and  $\eta(\mathbf{y}, \mathbf{w}) = 1$ .

*Solution.*

- $q(\mathbf{x}) = 9 - 1 - 4 - 4 = 0$ , so  $\mathbf{x}$  is null.  $q(\mathbf{y}) = 1 - 0 - 1 - 0 = 0$ , so  $\mathbf{y}$  is null.  $q(\mathbf{z}) = 0 - 1 - 1 - 1 = -3 < 0$ , so  $\mathbf{z}$  is spacelike.
- Try  $\mathbf{w} = (c, 0, -c, 0)$ , which has  $q(\mathbf{w}) = c^2 - c^2 = 0$  for every  $c$ . Then  $\eta(\mathbf{y}, \mathbf{w}) = 1 \cdot c - 1 \cdot (-c) = 2c$ , so  $c = \frac{1}{2}$  works:  $\mathbf{w} = (\frac{1}{2}, 0, -\frac{1}{2}, 0)$ . The Gram matrix of  $(\mathbf{y}, \mathbf{w})$  is  $\begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}$ , so  $\text{span}(\mathbf{y}, \mathbf{w})$  is a hyperbolic plane ([Definition 13.7.3](#)).

**Exercise B3.**

Let  $\mathbf{x} = (5, 3, 0, 0)$  and  $\mathbf{y} = (13, 12, 0, 0)$  in  $\mathbb{R}^{1,3}$ . Compute  $\tau(\mathbf{x})$ ,  $\tau(\mathbf{y})$ ,  $\eta(\mathbf{x}, \mathbf{y})$  and  $\tau(\mathbf{x} + \mathbf{y})$ , and check [Lemma 13.11.2](#) and [Theorem 13.11.3](#) on this pair. Is either inequality an equality?

*Solution.*  $q(\mathbf{x}) = 25 - 9 = 16$ , so  $\mathbf{x}$  is timelike and future-pointing with  $\tau(\mathbf{x}) = 4$ .  $q(\mathbf{y}) = 169 - 144 = 25$ , so  $\tau(\mathbf{y}) = 5$ . Next  $\eta(\mathbf{x}, \mathbf{y}) = 5 \cdot 13 - 3 \cdot 12 = 65 - 36 = 29$ , and  $\tau(\mathbf{x})\tau(\mathbf{y}) = 20$ , so  $29 \geq 20$  as [Lemma 13.11.2](#) requires. Finally  $\mathbf{x} + \mathbf{y} = (18, 15, 0, 0)$  with  $q = 324 - 225 = 99$ , so  $\tau(\mathbf{x} + \mathbf{y}) = \sqrt{99}$ . Since  $9^2 = 81 < 99$ , we get  $\tau(\mathbf{x} + \mathbf{y}) > 9 = \tau(\mathbf{x}) + \tau(\mathbf{y})$ , as [Theorem 13.11.3](#) requires. Neither is an equality, and indeed  $\mathbf{y}$  is not a

multiple of  $\mathbf{x}$ , since  $13/5 \neq 12/3$ .

### 13.11.6.3 C. Going deeper

#### Exercise C1.

For  $t \in \mathbb{R}$  put  $v(t) = \tanh t = \sinh t / \cosh t$ .

- (a) Prove that  $v: \mathbb{R} \rightarrow (-1, 1)$  is a bijection, and that  $\mathbf{B}_t \mathbf{e}_1$  is a positive multiple of  $(1, v(t), 0, 0)$  read in  $\mathbb{R}^{1,1}$ .
- (b) Hence deduce that  $v(s+t) = \frac{v(s) + v(t)}{1 + v(s)v(t)}$ , and say which part of [Theorem 13.11.1](#) does the work.

*Solution.*

- (a) Since  $\cosh t > 0$  and  $\cosh^2 t - \sinh^2 t = 1$ , we have  $|\sinh t| < \cosh t$ , so  $|v(t)| < 1$  and  $v$  maps into  $(-1, 1)$ . For injectivity and surjectivity, solve  $v(t) = c$  with  $|c| < 1$ : the equations  $\sinh t = c \cosh t$  and  $\cosh^2 t - \sinh^2 t = 1$  give  $\cosh^2 t(1 - c^2) = 1$ , so  $\cosh t = (1 - c^2)^{-1/2}$ , which is a legitimate value since it is  $\geq 1$ , and then  $\sinh t = c(1 - c^2)^{-1/2}$  determines  $t$  uniquely because  $\sinh$  is a bijection. So each  $c$  has exactly one preimage.

In  $\mathbb{R}^{1,1}$ ,  $\mathbf{B}_t \mathbf{e}_1 = (\cosh t, \sinh t) = \cosh t(1, v(t))$ , and  $\cosh t > 0$ .

- (b) By [Theorem 13.11.1](#),  $\mathbf{B}_s \mathbf{B}_t = \mathbf{B}_{s+t}$ . Comparing entries,

$$v(s+t) = \frac{\sinh(s+t)}{\cosh(s+t)} = \frac{\sinh s \cosh t + \cosh s \sinh t}{\cosh s \cosh t + \sinh s \sinh t},$$

and dividing numerator and denominator by  $\cosh s \cosh t > 0$  gives  $(v(s)+v(t))/(1+v(s)v(t))$ . The work is done by the addition law  $\mathbf{B}_s \mathbf{B}_t = \mathbf{B}_{s+t}$ ; the parametrization merely renames  $t$  as  $v(t)$ , and because  $v$  is a bijection onto  $(-1, 1)$  no information is lost in the renaming.

#### Exercise C2.

- (a) Prove that  $\{\mathbf{P} \mathbf{e}_1 : \mathbf{P} \in O(1, 1)\}$  is the set of all  $\mathbf{x} \in \mathbb{R}^{1,1}$  with  $q(\mathbf{x}) = 1$ , and that this set is unbounded.
- (b) Deduce that no bound  $|p_{ij}| \leq c$  holds for all  $\mathbf{P} \in O(1, 1)$ , and contrast this with  $O(2)$ , where  $|p_{ij}| \leq 1$  always. Which hypothesis of Chapter 10's  $O(2)$  theory is responsible?

*Solution.*

- (a) ( $\subseteq$ ) If  $\mathbf{P} \in O(1, 1)$  then  $q(\mathbf{P} \mathbf{e}_1) = (\mathbf{P} \mathbf{e}_1)^\top \mathbf{D}(\mathbf{P} \mathbf{e}_1) = \mathbf{e}_1^\top \mathbf{P}^\top \mathbf{D} \mathbf{P} \mathbf{e}_1 = \mathbf{e}_1^\top \mathbf{D} \mathbf{e}_1 = 1$ .  
 ( $\supseteq$ ) If  $q(\mathbf{x}) = 1$ , the proof of [Theorem 13.11.1](#) shows  $\mathbf{x} = (\varepsilon \cosh t, \sinh t)$  for some  $t$  and sign  $\varepsilon$ ; then  $\mathbf{x}$  is the first column of  $\mathbf{B}_t$  when  $\varepsilon = 1$ , and of  $-\mathbf{B}_{-t}$  when  $\varepsilon = -1$ , and both lie in  $O(1, 1)$ . The set is unbounded because  $\cosh t \rightarrow \infty$  as  $t \rightarrow \infty$ .
- (b) The first column of  $\mathbf{P}$  ranges over an unbounded set by (a), so its entries are unbounded, and no such  $c$  exists; this is the unboundedness Section 10 recorded, obtained here from the orbit rather than from a one-parameter family. In  $O(2)$

the columns are unit vectors for the Euclidean norm, so every entry has modulus at most 1 and the group sits inside a bounded region. The responsible hypothesis is **positive definiteness** of the form: it is what makes  $q(\mathbf{x}) = 1$  a bounded set, namely the unit sphere. Once one sign is reversed, the level set is a hyperbola and nothing is bounded.

### Exercise C3.

The Quick check above showed that  $\mathbb{R}^{1,3}$  has no plane on which the form vanishes identically. Show that  $\mathbb{R}^{2,2}$  does have one, by exhibiting two vectors that span it, and reconcile the two answers with [Theorem 13.7.4](#).

*Hint: pair each positive coordinate with a negative one.*

*Solution.* In  $\mathbb{R}^{2,2}$  the form is  $\eta'(\mathbf{x}, \mathbf{y}) = x_1y_1 + x_2y_2 - x_3y_3 - x_4y_4$ . Put  $\mathbf{u} = (1, 0, 1, 0)$  and  $\mathbf{w} = (0, 1, 0, 1)$ . Then

$$q(\mathbf{u}) = 1 - 1 = 0, \quad q(\mathbf{w}) = 1 - 1 = 0, \quad \eta'(\mathbf{u}, \mathbf{w}) = 0,$$

the last because every term pairs a zero with something. The list  $(\mathbf{u}, \mathbf{w})$  is independent, since a relation  $a\mathbf{u} + b\mathbf{w} = \mathbf{0}$  reads  $(a, b, a, b) = \mathbf{0}$ . So  $\text{span}(\mathbf{u}, \mathbf{w})$  is a totally isotropic plane, and by bilinearity  $\eta'$  vanishes on all of it.

The reconciliation: by [Definition 13.7.4](#) the Witt index bounds the dimension of a totally isotropic subspace, it counts the hyperbolic planes of [Theorem 13.7.4](#), and over  $\mathbb{R}$  it equals  $\min(n_+, n_-)$  ([Proposition 13.7.6](#)). For  $\mathbb{R}^{1,3}$  this is  $\min(1, 3) = 1$ , so totally isotropic subspaces are lines at most; for  $\mathbb{R}^{2,2}$  it is  $\min(2, 2) = 2$ , so a totally isotropic plane is exactly what one should expect, and the two vectors above realize it. What differs is not the dimension 4 but how evenly the four signs are split.

## 13.12 A First Look at Clifford Algebras

The chapter ends with a construction rather than a classification. Everything so far has taken a quadratic form apart; this section asks instead for an algebra in which the form becomes a *square*, so that  $q(\mathbf{v})$  is literally  $\mathbf{v}$  times  $\mathbf{v}$  for a multiplication we invent. The answer is a Clifford algebra, and two of the most familiar algebras in mathematics are its first two examples. This is an introduction only: we define the object, prove what the definition alone gives, work three cases out completely, and stop.

### 13.12.1 Making a form into a square

Take  $V = \mathbb{R}^2$  with the negative definite form  $q(\mathbf{x}) = -(x_1^2 + x_2^2)$ , and suppose we had an associative multiplication on some larger space containing  $V$ , with an identity 1, in which every vector satisfies  $\mathbf{v}^2 = q(\mathbf{v}) \cdot 1$ . Write  $\epsilon_1, \epsilon_2$  for the two standard basis vectors sitting inside that algebra. Expanding  $\mathbf{v} = x_1\epsilon_1 + x_2\epsilon_2$  and using only bilinearity of the product,

$$\mathbf{v}^2 = x_1^2\epsilon_1^2 + x_1x_2(\epsilon_1\epsilon_2 + \epsilon_2\epsilon_1) + x_2^2\epsilon_2^2.$$



We want this to equal  $-(x_1^2 + x_2^2) \cdot 1$  for **every**  $x_1, x_2$ . Taking  $x_2 = 0$  forces  $\epsilon_1^2 = -1$ , and taking  $x_1 = 0$  forces  $\epsilon_2^2 = -1$ . What is left is  $x_1 x_2 (\epsilon_1 \epsilon_2 + \epsilon_2 \epsilon_1) = 0$  for all  $x_1, x_2$ , which forces

$$\epsilon_1 \epsilon_2 = -\epsilon_2 \epsilon_1.$$

So the two basis vectors must square to  $-1$  and must **anticommute**. That last relation hands us a third square root of  $-1$  we did not ask for, since  $(\epsilon_1 \epsilon_2)^2 = \epsilon_1 \epsilon_2 \epsilon_1 \epsilon_2 = -\epsilon_1^2 \epsilon_2^2 = -1$ . Three anticommuting square roots of  $-1$ : those are Hamilton's quaternions. Run the same argument on a one-dimensional  $V$  with  $q(x) = -x^2$  and the single relation  $\epsilon_1^2 = -1$  produces  $\mathbb{C}$ .

*A Clifford algebra is the smallest place where the vectors of  $V$  can be multiplied and each one squares to its own  $q$ -value.*

## 13.12.2 Algebras

The objects above are algebras, a word the book has used informally and now needs precisely.

### Definition 13.12.1 — Algebra over a Field

An **algebra over  $F$** , or an  **$F$ -algebra**, is a vector space  $A$  over  $F$  together with a map  $A \times A \rightarrow A$ , written  $(x, y) \mapsto xy$ , which is

- (A1) **bilinear**: linear in each argument with the other fixed;
- (A2) **associative**:  $(xy)z = x(yz)$  for all  $x, y, z \in A$ ;
- (A3) **unital**: there is an element  $1_A \in A$  with  $1_A x = x 1_A = x$  for every  $x \in A$ .

A **homomorphism of  $F$ -algebras** is an  $F$ -linear map  $\varphi: A \rightarrow B$  with  $\varphi(xy) = \varphi(x)\varphi(y)$  for all  $x, y$  and  $\varphi(1_A) = 1_B$ . An **isomorphism** is a bijective homomorphism.

Every algebra here is associative and unital, and we stop saying so. The examples are old friends:  $F$  itself;  $M_n(F)$ ;  $F[x]$ ;  $\mathcal{L}(V)$  under composition; and  $\mathbb{C}$  over  $\mathbb{R}$ , of dimension 2. A non-example by minimal change:  $\mathbb{R}^3$  with the cross product is bilinear but fails (A2) and (A3), since  $(\mathbf{e}_1 \times \mathbf{e}_1) \times \mathbf{e}_2 = \mathbf{0}$  while  $\mathbf{e}_1 \times (\mathbf{e}_1 \times \mathbf{e}_2) = -\mathbf{e}_2$ , and no vector acts as an identity.

One consequence of (A1) is used constantly below: a bilinear map is determined by its values on a basis, so two products agree, and a linear map is multiplicative, as soon as this is checked on basis elements.

## 13.12.3 The definition

### Definition 13.12.2 — Clifford Algebra

Let  $F$  be a field with  $\text{char } F \neq 2$ , let  $V$  be a finite-dimensional  $F$ -vector space, let  $\beta$  be a symmetric bilinear form on  $V$  and let  $q(\mathbf{v}) = \beta(\mathbf{v}, \mathbf{v})$  be its quadratic form (Definition 13.3.2). A **Clifford algebra** of  $(V, q)$  is a pair  $(C, \iota)$ , where  $C$  is an  $F$ -algebra and  $\iota: V \rightarrow C$  is a linear map, such that

- (C1)  $\iota(\mathbf{v})^2 = q(\mathbf{v})1_C$  for **every**  $\mathbf{v} \in V$ ;

(C2) (**universal property**) for every  $F$ -algebra  $A$  and every linear map  $f: V \rightarrow A$  with  $f(\mathbf{v})^2 = q(\mathbf{v})1_A$  for every  $\mathbf{v} \in V$ , there is a **unique** algebra homomorphism  $\bar{f}: C \rightarrow A$  with  $\bar{f} \circ \iota = f$ .

We write  $\text{Cl}(q)$ , or  $\text{Cl}(V, q)$ , for a Clifford algebra of  $(V, q)$ , and  $\text{Cl}_{p,m}(\mathbb{R})$  when  $V$  is the space  $\mathbb{R}^{p,m}$  of Definition 13.11.1, that is,  $\mathbb{R}^{p+m}$  with the real form of signature  $(p, m)$ .

Clause by clause. (C1) is the demand we started from: inside  $C$  every vector has a square, and that square is a *number* times the identity, not a vector. (C2) says  $C$  is the most economical algebra with that property. Any other algebra where the vectors square correctly receives a map from  $C$ , so  $C$  has no relations beyond those (C1) forces; and the map is unique, so  $C$  has no room to spare either. What is defined is the pair  $(C, \iota)$ , not  $C$  alone.

*Remark.* The subscripts in  $\text{Cl}_{p,m}(\mathbb{R})$  are usually written  $p, q$ , but  $q$  already names the quadratic form here, so the second index is  $m$ .

**Existence is deferred.** The definition says what a Clifford algebra would be; it does not produce one. The standard construction is not available yet: one takes the **tensor algebra** of  $V$  — the infinite-dimensional algebra whose elements are formal sums of strings of vectors, multiplied by concatenation — and divides out by the relations  $\mathbf{v} \otimes \mathbf{v} - q(\mathbf{v})1$ . Chapter 14 builds tensor algebras and carries this out, for every  $V$  and  $q$  at once. So everything below is of one of two kinds: a consequence of the definition, valid whenever a Clifford algebra exists, or an explicit algebra written down by hand, which proves existence in that case. We say which each time.

The first consequence is that “the” Clifford algebra is legitimate language.

### Proposition 13.12.1 — Uniqueness of the Clifford Algebra

Let  $(C, \iota)$  and  $(C', \iota')$  both be Clifford algebras of  $(V, q)$ . Then there is a unique algebra isomorphism  $\varphi: C \rightarrow C'$  with  $\varphi \circ \iota = \iota'$ .

*Idea.* Each one’s universal property produces a map to the other. The two composites are self-maps that do nothing to  $\iota$ , and the uniqueness clause says the only such self-map is the identity.

*Proof.* Since  $(C', \iota')$  satisfies (C1), the universal property (C2) of  $C$ , applied with  $A = C'$  and  $f = \iota'$ , gives an algebra homomorphism  $\varphi: C \rightarrow C'$  with  $\varphi \iota = \iota'$ . Symmetrically, (C2) for  $C'$  gives  $\psi: C' \rightarrow C$  with  $\psi \iota' = \iota$ . Then  $\psi \varphi: C \rightarrow C$  is an algebra homomorphism with  $\psi \varphi \iota = \psi \iota' = \iota$ . The identity  $\text{id}_C$  is another one. By the uniqueness clause of (C2) for  $C$ , applied with  $A = C$  and  $f = \iota$ , the two coincide:  $\psi \varphi = \text{id}_C$ . Swapping the roles of  $C$  and  $C'$  gives  $\varphi \psi = \text{id}_{C'}$ . Hence  $\varphi$  is an isomorphism, and it is unique because (C2) already delivers it uniquely. This proves the proposition. ■

Next, the identity that does all the computing.

### Lemma 13.12.2 — The Anticommutation Relation

Let  $(C, \iota)$  be a Clifford algebra of  $(V, q)$ . Then for all  $\mathbf{u}, \mathbf{v} \in V$ ,

$$\iota(\mathbf{u})\iota(\mathbf{v}) + \iota(\mathbf{v})\iota(\mathbf{u}) = 2\beta(\mathbf{u}, \mathbf{v})1_C.$$

In particular  $\iota(\mathbf{u})$  and  $\iota(\mathbf{v})$  anticommute whenever  $\beta(\mathbf{u}, \mathbf{v}) = 0$ .

*Proof.* By (C1) applied to  $\mathbf{u} + \mathbf{v}$ , and then bilinearity of the product together with linearity of  $\iota$ ,

$$q(\mathbf{u} + \mathbf{v})1_C = \iota(\mathbf{u} + \mathbf{v})^2 = \iota(\mathbf{u})^2 + \iota(\mathbf{u})\iota(\mathbf{v}) + \iota(\mathbf{v})\iota(\mathbf{u}) + \iota(\mathbf{v})^2.$$

By [Theorem 13.3.5](#),  $q(\mathbf{u} + \mathbf{v}) = q(\mathbf{u}) + 2\beta(\mathbf{u}, \mathbf{v}) + q(\mathbf{v})$ , and by (C1) again  $\iota(\mathbf{u})^2 = q(\mathbf{u})1_C$  and  $\iota(\mathbf{v})^2 = q(\mathbf{v})1_C$ . Canceling those two terms from both sides leaves the stated identity. The last sentence is the case  $\beta(\mathbf{u}, \mathbf{v}) = 0$ . ■

So an **orthogonal** basis for  $\beta$  becomes a list of pairwise anticommuting elements of  $C$ , each with a scalar square. That is the whole computational content of a Clifford algebra, and [Theorem 13.4.2](#) guarantees such a basis exists whenever  $\text{char } F \neq 2$  — which is where the characteristic hypothesis in [Definition 13.12.2](#) earns its place, since [Theorem 13.3.5](#) needs it too.

### 13.12.4 Three algebras, worked out

#### Example. (The zero space)

Let  $V = \{\mathbf{0}\}$ , so  $q = 0$  and  $n = 0$ . Show that  $(F, \iota)$  with  $\iota = 0$  is a Clifford algebra of  $(V, q)$ , so that  $\text{Cl}(0) = F$ .

*Solution.* (C1): the only vector is  $\mathbf{0}$ , and  $\iota(\mathbf{0})^2 = 0 = q(\mathbf{0})1_F$ .

(C2): let  $A$  be an  $F$ -algebra and  $f: V \rightarrow A$  linear, necessarily  $f = 0$ , so the condition on  $f$  reads  $0 = 0$ . Define  $\bar{f}: F \rightarrow A$  by  $\bar{f}(a) = a1_A$ : it is linear, sends 1 to  $1_A$ , and is multiplicative because  $\bar{f}(ab) = (ab)1_A = (a1_A)(b1_A)$  by bilinearity of the product and  $1_A1_A = 1_A$ ; and  $\bar{f}\iota = 0 = f$ . It is unique, since any algebra homomorphism  $\varphi: F \rightarrow A$  is linear with  $\varphi(1) = 1_A$ , hence  $\varphi(a) = a1_A$ .

So  $\text{Cl}(0) = F$ , of dimension  $1 = 2^0$ : the construction adds nothing when there is nothing to add.

#### Example. (The complex numbers)

Let  $V = \mathbb{R}$  with  $q(x) = -x^2$ , the form of signature  $(0, 1)$ . Show that  $\mathbb{C}$ , with  $\iota: \mathbb{R} \rightarrow \mathbb{C}$  given by  $\iota(x) = xi$ , is a Clifford algebra of  $(V, q)$ , so that  $\text{Cl}_{0,1}(\mathbb{R}) = \mathbb{C}$ .

*Solution.* Here  $\mathbb{C}$  is an algebra over  $\mathbb{R}$  of dimension 2, with basis  $(1, i)$ .

(C1):  $\iota(x)^2 = (xi)^2 = x^2i^2 = -x^2 = q(x) \cdot 1$ .

(C2): let  $A$  be a real algebra and  $f: \mathbb{R} \rightarrow A$  linear with  $f(x)^2 = -x^21_A$ . Put  $u = f(1)$ ,

so  $f(x) = xu$  by linearity and  $u^2 = -1_A$ . Define  $\bar{f}: \mathbb{C} \rightarrow A$  by  $\bar{f}(a + bi) = a1_A + bu$ , which is  $\mathbb{R}$ -linear and sends 1 to  $1_A$ . To see it is multiplicative, note that  $(z, w) \mapsto \bar{f}(zw)$  and  $(z, w) \mapsto \bar{f}(z)\bar{f}(w)$  are both  $\mathbb{R}$ -bilinear, so it suffices to compare them on the basis  $(1, i)$ : the pairs involving 1 are immediate, and  $\bar{f}(i \cdot i) = \bar{f}(-1) = -1_A = u^2 = \bar{f}(i)\bar{f}(i)$ . Finally  $\bar{f}\iota(x) = \bar{f}(xi) = xu = f(x)$ . For uniqueness, an algebra homomorphism  $\varphi$  with  $\varphi\iota = f$  has  $\varphi(1) = 1_A$  and  $\varphi(i) = \varphi(\iota(1)) = f(1) = u$ , and  $(1, i)$  spans  $\mathbb{C}$ , so  $\varphi = \bar{f}$ .

So  $\text{Cl}_{0,1}(\mathbb{R}) = \mathbb{C}$ , of dimension  $2 = 2^1$ . Read this backwards:  $\mathbb{C}$  is not primarily “ $\mathbb{R}$  with a square root of  $-1$  attached”, it is the smallest algebra in which the number  $x$  squares to  $-x^2$ .

For the two-dimensional case we need the quaternions, and we need them to be associative. The cheapest way to get associativity is to find them inside a matrix algebra, where it is free.

### Definition 13.12.3 — The Quaternions

Inside  $M_2(\mathbb{C})$ , regarded as an algebra over  $\mathbb{R}$ , put

$$\mathbf{i} = \begin{pmatrix} i & 0 \\ 0 & -i \end{pmatrix}, \quad \mathbf{j} = \begin{pmatrix} 0 & 1 \\ -1 & 0 \end{pmatrix}, \quad \mathbf{k} = \begin{pmatrix} 0 & i \\ i & 0 \end{pmatrix}.$$

The **quaternions**  $\mathbb{H}$  are the real span of  $\mathbf{I}_2, \mathbf{i}, \mathbf{j}, \mathbf{k}$  in  $M_2(\mathbb{C})$ .

The three matrices carry Hamilton’s names and are written bold lowercase, as basis vectors of  $\mathbb{H}$ ; the plain  $i$  in their entries is the complex unit, which is a different object from the matrix  $\mathbf{i}$ . Multiplying them out gives

$$\mathbf{i}^2 = \mathbf{j}^2 = \mathbf{k}^2 = -\mathbf{I}_2, \quad \mathbf{ij} = \mathbf{k} = -\mathbf{ji},$$

and then  $\mathbf{jk} = \mathbf{i} = -\mathbf{kj}$  and  $\mathbf{ki} = \mathbf{j} = -\mathbf{ik}$ . A real combination  $a\mathbf{I}_2 + b\mathbf{i} + c\mathbf{j} + d\mathbf{k}$  is the matrix with rows  $(a + bi, c + di)$  and  $(-c + di, a - bi)$ , which vanishes only when  $a = b = c = d = 0$ ; so  $\dim_{\mathbb{R}} \mathbb{H} = 4$ . The relations show the span is closed under multiplication, so  $\mathbb{H}$  is a real subalgebra of  $M_2(\mathbb{C})$  — and it is associative for free, because matrix multiplication is.

### Example. (The quaternions)

Let  $V = \mathbb{R}^2$  with  $q(\mathbf{x}) = -(x_1^2 + x_2^2)$ , the form of signature  $(0, 2)$ . Show that  $\mathbb{H}$ , with  $\iota(\mathbf{x}) = x_1\mathbf{i} + x_2\mathbf{j}$ , is a Clifford algebra of  $(V, q)$ , so that  $\text{Cl}_{0,2}(\mathbb{R}) = \mathbb{H}$ .

*Solution.* (C1): using  $\mathbf{ij} + \mathbf{ji} = \mathbf{0}$  and  $\mathbf{i}^2 = \mathbf{j}^2 = -\mathbf{I}_2$ ,

$$\iota(\mathbf{x})^2 = x_1^2\mathbf{i}^2 + x_1x_2(\mathbf{ij} + \mathbf{ji}) + x_2^2\mathbf{j}^2 = -(x_1^2 + x_2^2)\mathbf{I}_2,$$

which is  $q(\mathbf{x})1_{\mathbb{H}}$ .

(C2): let  $A$  be a real algebra and  $f: \mathbb{R}^2 \rightarrow A$  linear with  $f(\mathbf{x})^2 = q(\mathbf{x})1_A$ . Put  $u = f(\mathbf{e}_1)$  and  $w = f(\mathbf{e}_2)$ , so  $u^2 = w^2 = -1_A$ , and by Lemma 13.12.2 — or directly, by expanding  $f(\mathbf{e}_1 + \mathbf{e}_2)^2 = -2 \cdot 1_A$  — we get  $uw + wu = 0$ , that is,  $wu = -uw$ . Define the linear map

$$\bar{f}(a\mathbf{I}_2 + b\mathbf{i} + c\mathbf{j} + d\mathbf{k}) = a1_A + bu + cw + d uw,$$

which is well defined because  $(\mathbf{I}_2, \mathbf{i}, \mathbf{j}, \mathbf{k})$  is a basis of  $\mathbb{H}$ , and sends  $1_{\mathbb{H}} = \mathbf{I}_2$  to  $1_A$ . Both  $(z, z') \mapsto \bar{f}(zz')$  and  $(z, z') \mapsto \bar{f}(z)\bar{f}(z')$  are  $\mathbb{R}$ -bilinear, so multiplicativity need only be

checked on the basis. The pairs involving  $\mathbf{I}_2$  are immediate, and the remaining nine are

$$\begin{aligned} \mathbf{ii} &= -\mathbf{I}_2 \mapsto -1_A = u^2, & \mathbf{jj} &= -\mathbf{I}_2 \mapsto -1_A = w^2, \\ \mathbf{ij} &= \mathbf{k} \mapsto uw, & \mathbf{ji} &= -\mathbf{k} \mapsto -uw = wu, \\ \mathbf{ik} &= -\mathbf{j} \mapsto -w = u(uw), & \mathbf{ki} &= \mathbf{j} \mapsto w = (uw)u, \\ \mathbf{jk} &= \mathbf{i} \mapsto u = w(uw), & \mathbf{kj} &= -\mathbf{i} \mapsto -u = (uw)w, \\ \mathbf{kk} &= -\mathbf{I}_2 \mapsto -1_A = (uw)^2. \end{aligned}$$

Each right-hand claim is one line of algebra in  $A$ , using associativity,  $wu = -uw$  and  $u^2 = w^2 = -1_A$ . For instance  $(uw)u = u(wu) = -u(uw) = -u^2w = w$ , and  $(uw)^2 = u(wu)w = -u^2w^2 = -1_A$ .

Finally  $\bar{f}\iota(\mathbf{x}) = x_1u + x_2w = f(\mathbf{x})$ . For uniqueness, an algebra homomorphism  $\varphi$  with  $\varphi\iota = f$  is forced on  $\mathbf{I}_2$ , on  $\mathbf{i} = \iota(\mathbf{e}_1)$  and on  $\mathbf{j} = \iota(\mathbf{e}_2)$ , hence on  $\mathbf{k} = \mathbf{ij}$ , and these four span  $\mathbb{H}$ .

So  $\text{Cl}_{0,2}(\mathbb{R}) = \mathbb{H}$ , of dimension  $4 = 2^2$ .

**Warning.** A Clifford algebra is not commutative, and a product of two vectors is not a vector. In  $\text{Cl}_{0,2}(\mathbb{R}) = \mathbb{H}$  the image  $\iota(\mathbb{R}^2) = \text{span}(\mathbf{i}, \mathbf{j})$  is a plane, and  $\mathbf{ij} = \mathbf{k}$  lies outside it; worse,  $\mathbf{ij} = -\mathbf{ji}$ , so the order matters. Two habits have to go: that multiplying two elements of  $V$  keeps you in  $V$ , and that  $xy = yx$ . Beyond the scalars  $F \cdot 1_C$ , nothing may be assumed to commute.

### 13.12.5 How big is it

The three examples all came out with  $\dim = 2^n$ . Half of that is a theorem we can prove.

#### Proposition 13.12.3 — The Clifford Algebra Is Spanned by Sorted Products

Let  $(C, \iota)$  be a Clifford algebra of  $(V, q)$ , where  $\dim V = n$  and  $\text{char } F \neq 2$ , and let  $(\mathbf{e}_1, \dots, \mathbf{e}_n)$  be an orthogonal basis of  $V$  for  $\beta$ . Then  $C$  is spanned over  $F$  by the  $2^n$  products

$$\iota(\mathbf{e}_{i_1})\iota(\mathbf{e}_{i_2}) \cdots \iota(\mathbf{e}_{i_k}), \quad 1 \leq i_1 < i_2 < \cdots < i_k \leq n,$$

one for each subset  $\{i_1, \dots, i_k\}$  of  $\{1, \dots, n\}$ , the empty product being  $1_C$ . In particular  $\dim_F C \leq 2^n$ .

*Idea.* Two steps. ① Show that  $C$  is generated as an algebra by  $1_C$  and  $\iota(V)$  — this is not automatic, and it is where the *uniqueness* half of the universal property does its work. ② Sort: any product of basis vectors can be put in increasing order by swapping neighbors at the cost of a sign (Lemma 13.12.2), and a repeated neighbor collapses to a scalar ( $C1$ ).

*Proof.* An orthogonal basis exists by Theorem 13.4.2, since  $\text{char } F \neq 2$ . Write  $d_i = q(\mathbf{e}_i)$ .

**Step 1:**  $C$  is generated by  $1_C$  and  $\iota(V)$ . Let  $C' \subseteq C$  be the set of all  $F$ -linear combinations of  $1_C$  and of finite products  $\iota(\mathbf{v}_1) \cdots \iota(\mathbf{v}_m)$  with  $\mathbf{v}_1, \dots, \mathbf{v}_m \in V$ . It is a

subspace closed under multiplication and contains  $1_C$ , hence is itself an  $F$ -algebra, and  $\iota$  followed by the change of codomain gives a linear map  $\iota': V \rightarrow C'$  satisfying (C1). By the universal property of  $C'$ , applied with  $A = C'$  and  $f = \iota'$ , there is an algebra homomorphism  $g: C \rightarrow C'$  with  $g\iota = \iota'$ . Let  $j: C' \rightarrow C$  be the inclusion, an algebra homomorphism. Then  $kg: C \rightarrow C$  is an algebra homomorphism with  $kg\iota = \iota$ , and so is  $\text{id}_C$ ; by the uniqueness clause of (C2), applied with  $A = C$  and  $f = \iota$ , we get  $kg = \text{id}_C$ . Hence  $j$  is surjective, so  $C' = C$ .

**Step 2: sorting.** By Step 1 and linearity of  $\iota$ ,  $C$  is spanned by  $1_C$  together with the products  $\iota(\mathbf{e}_{j_1}) \cdots \iota(\mathbf{e}_{j_m})$  over all finite index strings  $(j_1, \dots, j_m)$ . Fix such a product and induct on the pair  $(m, \text{inv}(j_1, \dots, j_m))$ , ordered lexicographically, where  $\text{inv}$  counts the pairs  $r < s$  with  $j_r > j_s$ . If the string is strictly increasing, the product is on the list and we are done. Otherwise some neighboring pair has  $j_r \geq j_{r+1}$ . If  $j_r = j_{r+1}$ , then (C1) replaces  $\iota(\mathbf{e}_{j_r})^2$  by the scalar  $d_{j_r}$ , producing a scalar multiple of a product of length  $m - 2$ . If  $j_r > j_{r+1}$ , then Lemma 13.12.2 with  $\beta(\mathbf{e}_{j_r}, \mathbf{e}_{j_{r+1}}) = 0$  lets us swap the two factors at the cost of a sign, producing a product of the same length with one fewer inversion. Either way the induction applies, and the product is a scalar multiple of a listed one.

There are  $2^n$  subsets of  $\{1, \dots, n\}$ , so the spanning list has  $2^n$  members and  $\dim_F C \leq 2^n$ . This proves the proposition. ■

*Quick check.* Where exactly does Proposition 13.12.3 use the word **unique** in (C2), and what would go wrong without it?

*Solution.* In Step 1, at the sentence “by the uniqueness clause of (C2) ... we get  $kg = \text{id}_C$ ”. Existence alone gives an algebra homomorphism  $kg: C \rightarrow C$  fixing  $\iota$ , but nothing then makes it the identity, and the argument cannot conclude that  $j$  is surjective. The conclusion really would be false. Take  $V = \mathbb{R}$  with  $q(x) = -x^2$ , let  $A = \mathbb{C} \times \mathbb{C}$  with componentwise multiplication, and set  $\iota(x) = (xi, xi)$ . Then  $\iota(x)^2 = q(x)1_A$ , so (C1) holds, and so does the existence half of (C2): the first projection, followed by the homomorphism  $\mathbb{C} \rightarrow B$  built in Example (The complex numbers), does the job for every target  $B$ . But  $A$  is **not** spanned by  $1_A$  and products of vectors, since every such product lies in the diagonal  $\{(z, z)\}$ , of dimension 2. What fails is uniqueness — the second projection is a second map — and that is the clause the proof needs.

The other half of the count is a different matter.

### Theorem 13.12.4 — Dimension of a Clifford Algebra

Let  $\text{char } F \neq 2$ , let  $\dim V = n$ , and let  $(C, \iota)$  be a Clifford algebra of  $(V, q)$ . Then the  $2^n$  sorted products of Proposition 13.12.3 form a **basis** of  $C$ , so  $\dim_F C = 2^n$ . In particular  $\iota$  is injective.

**What is proved here, and what is not.** The inequality  $\dim_F C \leq 2^n$  is Proposition 13.12.3, proved in full. The reverse inequality — that the  $2^n$  sorted products are linearly independent — is **not proved in this section**, and it cannot be, because the definition alone does not even produce an algebra: independence and existence have to be established together, by constructing

one algebra of dimension exactly  $2^n$  that satisfies (C1) and (C2). Chapter 14 does that, from the tensor algebra. What this section does establish is the theorem in the three cases above:  $n = 0$  with  $\dim = 1$ ,  $n = 1$  with  $\dim = 2$ , and the case  $q = -x_1^2 - x_2^2$  of  $n = 2$  with  $\dim = 4$ , each by writing the algebra down and checking (C1) and (C2) by hand. For  $n \geq 3$ , read the statement above as quoted from Chapter 14 and not as proved here.

Granting the theorem, the subsets of  $\{1, \dots, n\}$  indexing the basis are the first sign that  $\text{Cl}(q)$  is assembled from the exterior powers  $\Lambda^k V$ , one for each  $k$ . Chapter 14 constructs both and makes that precise.

### 13.12.6 What Clifford algebras are for

One computation shows where the subject goes. Recall from Section 10 that the isometries of  $q$  form a group (Definition 13.10.1). Inside  $\text{Cl}(q)$ , some of them are conjugations.

#### Proposition 13.12.5 — Reflections Are Conjugations

Let  $(C, \iota)$  be a Clifford algebra of  $(V, q)$  and let  $\mathbf{u} \in V$  with  $q(\mathbf{u}) \neq 0$ . Then  $\iota(\mathbf{u})$  is invertible in  $C$ , with inverse  $q(\mathbf{u})^{-1}\iota(\mathbf{u})$ , and for every  $\mathbf{v} \in V$ ,

$$-\iota(\mathbf{u})\iota(\mathbf{v})\iota(\mathbf{u})^{-1} = \iota(s_{\mathbf{u}}(\mathbf{v})), \quad s_{\mathbf{u}}(\mathbf{v}) := \mathbf{v} - \frac{2\beta(\mathbf{u}, \mathbf{v})}{q(\mathbf{u})} \mathbf{u}.$$

The map  $s_{\mathbf{u}}$  fixes  $\{\mathbf{v} : \beta(\mathbf{u}, \mathbf{v}) = 0\}$  pointwise, sends  $\mathbf{u}$  to  $-\mathbf{u}$ , and satisfies  $q(s_{\mathbf{u}}(\mathbf{v})) = q(\mathbf{v})$ .

*Proof.* By (C1),  $\iota(\mathbf{u})^2 = q(\mathbf{u})1_C$  with  $q(\mathbf{u}) \neq 0$ , so  $\iota(\mathbf{u})(q(\mathbf{u})^{-1}\iota(\mathbf{u})) = 1_C$  and likewise on the other side. By Lemma 13.12.2,  $\iota(\mathbf{u})\iota(\mathbf{v}) = 2\beta(\mathbf{u}, \mathbf{v})1_C - \iota(\mathbf{v})\iota(\mathbf{u})$ , so multiplying on the right by  $\iota(\mathbf{u})$ ,

$$\iota(\mathbf{u})\iota(\mathbf{v})\iota(\mathbf{u}) = 2\beta(\mathbf{u}, \mathbf{v})\iota(\mathbf{u}) - q(\mathbf{u})\iota(\mathbf{v}).$$

Dividing by  $q(\mathbf{u})$  and negating gives  $-\iota(\mathbf{u})\iota(\mathbf{v})\iota(\mathbf{u})^{-1} = \iota(\mathbf{v}) - \frac{2\beta(\mathbf{u}, \mathbf{v})}{q(\mathbf{u})}\iota(\mathbf{u})$ , which is  $\iota(s_{\mathbf{u}}(\mathbf{v}))$  by linearity of  $\iota$ . If  $\beta(\mathbf{u}, \mathbf{v}) = 0$  then  $s_{\mathbf{u}}(\mathbf{v}) = \mathbf{v}$ ; and  $s_{\mathbf{u}}(\mathbf{u}) = \mathbf{u} - 2\mathbf{u} = -\mathbf{u}$ . Finally, writing  $c = 2\beta(\mathbf{u}, \mathbf{v})/q(\mathbf{u})$ ,

$$q(\mathbf{v} - c\mathbf{u}) = q(\mathbf{v}) - 2c\beta(\mathbf{u}, \mathbf{v}) + c^2q(\mathbf{u}) = q(\mathbf{v}),$$

since  $c^2q(\mathbf{u}) = c \cdot 2\beta(\mathbf{u}, \mathbf{v})$ . This proves the proposition. ■

So the reflection in a hyperplane, the basic isometry of a quadratic form, is carried out inside  $\text{Cl}(q)$  by conjugating with a single vector. Composing such conjugations reaches more isometries, and the invertible elements of  $\text{Cl}(q)$  acting on  $V$  this way form a group mapping onto the isometry group of  $q$  — two-to-one, since  $\iota(\mathbf{u})$  and  $-\iota(\mathbf{u})$  conjugate alike. That double covering is where **spinors** live: objects the covering group acts on and the isometry group does not, because going once around a rotation returns them with a sign. This book will not build them. What it will do is Chapter 14: construct tensor algebras, construct  $\text{Cl}(q)$  from one, and prove Theorem 13.12.4. Nothing later depends on this section, and the chapter ends here.



## 13.12.7 Exercises

### 13.12.7.1 A. Check your understanding

#### Exercise A1.

- (a) State the definition of an  $F$ -algebra, and give one example of dimension 1 and one of dimension 4.
- (b) State the universal property (C2), with every quantifier.
- (c) What is  $\text{Cl}(0)$  when  $V = \{\mathbf{0}\}$ , and what is its dimension?
- (d) True or false:  $\text{Cl}_{0,2}(\mathbb{R})$  is commutative. Justify your answer.
- (e) What does this book prove about  $\dim \text{Cl}(q)$  when  $\dim V = 3$ , and what does it only quote?

*Solution.*

- (a) A vector space over  $F$  with a bilinear, associative, unital multiplication ([Definition 13.12.1](#)). Dimension 1:  $F$  itself. Dimension 4:  $M_2(F)$ , or  $\mathbb{H}$  over  $\mathbb{R}$ .
- (b) For every  $F$ -algebra  $A$  and every linear  $f: V \rightarrow A$  such that  $f(\mathbf{v})^2 = q(\mathbf{v})1_A$  for every  $\mathbf{v} \in V$ , there exists a unique algebra homomorphism  $\bar{f}: C \rightarrow A$  such that  $\bar{f} \circ \iota = f$ .
- (c)  $\text{Cl}(0) = F$ , of dimension  $1 = 2^0$  ([Example \(The zero space\)](#)).
- (d) False. It is  $\mathbb{H}$  by [Example \(The quaternions\)](#), and  $\mathbf{ij} = \mathbf{k} \neq -\mathbf{k} = \mathbf{ji}$ .
- (e) It proves  $\dim \text{Cl}(q) \leq 2^3 = 8$ , by [Proposition 13.12.3](#). The equality  $\dim \text{Cl}(q) = 8$ , and indeed the existence of  $\text{Cl}(q)$  at all, is quoted from Chapter 14.

### 13.12.7.2 B. Practice

#### Exercise B1: The split case.

Let  $V = \mathbb{R}$  with  $q(x) = x^2$ , the form of signature  $(1, 0)$ . Let  $A = \mathbb{R} \times \mathbb{R}$  with componentwise addition and multiplication, and  $\iota: \mathbb{R} \rightarrow A$ ,  $\iota(x) = (x, -x)$ . Prove that  $(A, \iota)$  is a Clifford algebra of  $(V, q)$ , and state its dimension.

*Solution.*  $A$  is a real algebra of dimension 2 with identity  $1_A = (1, 1)$ , and  $(1_A, \iota(1)) = ((1, 1), (1, -1))$  is a basis, since  $a(1, 1) + b(1, -1) = (a + b, a - b)$  vanishes only for  $a = b = 0$ .

(C1):  $\iota(x)^2 = (x^2, x^2) = q(x)1_A$ .

(C2): let  $B$  be a real algebra and  $f: \mathbb{R} \rightarrow B$  linear with  $f(x)^2 = x^2 1_B$ . Put  $u = f(1)$ , so  $f(x) = xu$  and  $u^2 = 1_B$ . Define  $\bar{f}(a, b) = \frac{a+b}{2}1_B + \frac{a-b}{2}u$ , which is linear and sends  $1_A$  to  $1_B$  and  $\iota(1)$  to  $u$ . By bilinearity, multiplicativity need only be checked on that basis: the pairs involving  $1_A$  are immediate, and  $\iota(1)^2 = 1_A \mapsto 1_B = u^2$ . Also  $\bar{f}\iota(x) = xu = f(x)$ , and any such homomorphism is forced on  $1_A$  and  $\iota(1)$ , which span  $A$ .

So  $\text{Cl}_{1,0}(\mathbb{R}) = \mathbb{R} \times \mathbb{R}$ , of dimension  $2 = 2^1$ . Comparing with [Example \(The complex](#)



numbers): the same one-dimensional  $V$  with the opposite sign of  $q$  gives  $\mathbb{C}$  instead, so the Clifford algebra sees the sign of the form, not just its rank.

### Exercise B2.

Work in  $\text{Cl}_{0,2}(\mathbb{R}) = \mathbb{H}$ , with  $\iota(\mathbf{x}) = x_1\mathbf{i} + x_2\mathbf{j}$ . Let  $\mathbf{u} = (1, 2)$  and  $\mathbf{v} = (3, -1)$ .

- (a) Compute  $\iota(\mathbf{u})\iota(\mathbf{v})$  and  $\iota(\mathbf{v})\iota(\mathbf{u})$ , and verify [Lemma 13.12.2](#) on this pair.
- (b) Compute  $q(\mathbf{u})$ , verify  $\iota(\mathbf{u})^2 = q(\mathbf{u})1_{\mathbb{H}}$ , and write down  $\iota(\mathbf{u})^{-1}$ .

*Solution.*

- (a) Using  $\mathbf{i}^2 = \mathbf{j}^2 = -1$  and  $\mathbf{ij} = \mathbf{k} = -\mathbf{ji}$ , and writing 1 for  $\mathbf{I}_2$ ,

$$\begin{aligned}\iota(\mathbf{u})\iota(\mathbf{v}) &= (\mathbf{i} + 2\mathbf{j})(3\mathbf{i} - \mathbf{j}) = -3 - \mathbf{k} - 6\mathbf{k} + 2 = -1 - 7\mathbf{k}, \\ \iota(\mathbf{v})\iota(\mathbf{u}) &= (3\mathbf{i} - \mathbf{j})(\mathbf{i} + 2\mathbf{j}) = -3 + 6\mathbf{k} + \mathbf{k} + 2 = -1 + 7\mathbf{k}.\end{aligned}$$

Their sum is  $-2$ . On the other side,  $\beta(\mathbf{u}, \mathbf{v}) = -(1 \cdot 3 + 2 \cdot (-1)) = -1$ , so  $2\beta(\mathbf{u}, \mathbf{v})1_{\mathbb{H}} = -2$ . The two agree.

- (b)  $q(\mathbf{u}) = -(1+4) = -5$ . And  $\iota(\mathbf{u})^2 = (\mathbf{i} + 2\mathbf{j})^2 = -1 + 2\mathbf{k} - 2\mathbf{k} - 4 = -5$ , as required. Hence  $\iota(\mathbf{u})^{-1} = q(\mathbf{u})^{-1}\iota(\mathbf{u}) = -\frac{1}{5}(\mathbf{i} + 2\mathbf{j})$  by [Proposition 13.12.5](#).

### Exercise B3.

In  $\mathbb{R}^{0,2}$  let  $\mathbf{u} = \mathbf{e}_1$  and  $\mathbf{v} = \mathbf{e}_1 + \mathbf{e}_2$ . Compute the reflection  $s_{\mathbf{u}}(\mathbf{v})$  directly from the formula in [Proposition 13.12.5](#), then compute  $-\iota(\mathbf{u})\iota(\mathbf{v})\iota(\mathbf{u})^{-1}$  inside  $\mathbb{H}$ , and check that the two agree.

*Solution.* Here  $\beta(\mathbf{x}, \mathbf{y}) = -(x_1y_1 + x_2y_2)$ , so  $\beta(\mathbf{u}, \mathbf{v}) = -1$  and  $q(\mathbf{u}) = -1$ . The formula gives

$$s_{\mathbf{u}}(\mathbf{v}) = \mathbf{v} - \frac{2(-1)}{-1}\mathbf{u} = \mathbf{v} - 2\mathbf{u} = -\mathbf{e}_1 + \mathbf{e}_2.$$

In  $\mathbb{H}$ :  $\iota(\mathbf{u}) = \mathbf{i}$ ,  $\iota(\mathbf{v}) = \mathbf{i} + \mathbf{j}$ , and  $\iota(\mathbf{u})^{-1} = q(\mathbf{u})^{-1}\mathbf{i} = -\mathbf{i}$ . Hence

$$-\mathbf{i}(\mathbf{i} + \mathbf{j})(-\mathbf{i}) = \mathbf{i}(\mathbf{i} + \mathbf{j})\mathbf{i} = (-1 + \mathbf{k})\mathbf{i} = -\mathbf{i} + \mathbf{ki} = -\mathbf{i} + \mathbf{j},$$

using  $\mathbf{ki} = \mathbf{j}$ . That is  $\iota(-\mathbf{e}_1 + \mathbf{e}_2)$ , which agrees with  $\iota(s_{\mathbf{u}}(\mathbf{v}))$ . Geometrically  $s_{\mathbf{u}}$  negates the  $\mathbf{e}_1$  coordinate and fixes the  $\mathbf{e}_2$  one, which is what the answer shows.

### 13.12.7.3 C. Going deeper

#### Exercise C1: A Clifford algebra of matrices.

In  $M_2(\mathbb{R})$  put  $\mathbf{E}_1 = \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix}$  and  $\mathbf{E}_2 = \begin{pmatrix} 0 & 1 \\ -1 & 0 \end{pmatrix}$ , and let  $V = \mathbb{R}^2$  carry the form  $q(\mathbf{x}) = x_1^2 - x_2^2$  of signature  $(1, 1)$ , with  $f(\mathbf{x}) = x_1\mathbf{E}_1 + x_2\mathbf{E}_2$ .

- (a) Prove that  $f(\mathbf{x})^2 = q(\mathbf{x})\mathbf{I}_2$  for every  $\mathbf{x}$ , and that  $(\mathbf{I}_2, \mathbf{E}_1, \mathbf{E}_2, \mathbf{E}_1\mathbf{E}_2)$  is a basis of  $M_2(\mathbb{R})$ .
- (b) Assume a Clifford algebra  $(C, \iota)$  of  $(V, q)$  exists. Deduce that  $C \cong M_2(\mathbb{R})$ , and

hence that  $\dim C = 4$ .

*Hint for (b): the homomorphism the universal property hands you has a surjective image; then count.*

*Solution.*

- (a) A direct multiplication gives  $\mathbf{E}_1^2 = \mathbf{I}_2$ ,  $\mathbf{E}_2^2 = -\mathbf{I}_2$  and

$$\mathbf{E}_1\mathbf{E}_2 = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix} = -\mathbf{E}_2\mathbf{E}_1,$$

so  $\mathbf{E}_1\mathbf{E}_2 + \mathbf{E}_2\mathbf{E}_1 = \mathbf{0}$  and

$$f(\mathbf{x})^2 = x_1^2\mathbf{E}_1^2 + x_1x_2(\mathbf{E}_1\mathbf{E}_2 + \mathbf{E}_2\mathbf{E}_1) + x_2^2\mathbf{E}_2^2 = (x_1^2 - x_2^2)\mathbf{I}_2.$$

For the basis:  $a\mathbf{I}_2 + b\mathbf{E}_1 + c\mathbf{E}_2 + d\mathbf{E}_1\mathbf{E}_2$  is the matrix with rows  $(a+b, c+d)$  and  $(-c+d, a-b)$ , and setting all four entries to zero forces  $a = b = c = d = 0$ . Four independent vectors in the 4-dimensional space  $M_2(\mathbb{R})$  form a basis.

- (b) By (a) and the universal property (C2) with  $A = M_2(\mathbb{R})$ , there is an algebra homomorphism  $\bar{f}: C \rightarrow M_2(\mathbb{R})$  with  $\bar{f}\iota = f$ . Its image is a subalgebra containing  $\mathbf{I}_2$ ,  $\mathbf{E}_1 = f(\mathbf{e}_1)$  and  $\mathbf{E}_2 = f(\mathbf{e}_2)$ , hence also  $\mathbf{E}_1\mathbf{E}_2$ ; by (a) those four span  $M_2(\mathbb{R})$ , so  $\bar{f}$  is surjective and  $\dim C \geq 4$ . On the other hand  $\dim C \leq 2^2 = 4$  by [Proposition 13.12.3](#). Therefore  $\dim C = 4$  and  $\bar{f}$  is a surjective linear map between spaces of equal finite dimension, hence bijective, hence an algebra isomorphism. So  $\text{Cl}_{1,1}(\mathbb{R}) \cong M_2(\mathbb{R})$ .

### Exercise C2: Why characteristic 2 is excluded.

Explain why [Definition 13.12.2](#) assumes  $\text{char } F \neq 2$ , by naming the first statement of this section that fails without it and saying what replaces the missing input.

*Solution.* The first casualty is [Lemma 13.12.2](#), whose proof cites [Theorem 13.3.5](#) for  $q(\mathbf{u} + \mathbf{v}) = q(\mathbf{u}) + 2\beta(\mathbf{u}, \mathbf{v}) + q(\mathbf{v})$  — and that identity is how  $\beta$  is recovered from  $q$ , by a division by 2 that is unavailable in characteristic 2. Without it, the relation  $\iota(\mathbf{u})\iota(\mathbf{v}) + \iota(\mathbf{v})\iota(\mathbf{u}) = 2\beta(\mathbf{u}, \mathbf{v})1_C$  collapses to  $\iota(\mathbf{u})\iota(\mathbf{v}) + \iota(\mathbf{v})\iota(\mathbf{u}) = 0$  and carries no information about the form at all. [Proposition 13.12.3](#) then loses its sorting step in the form given, and [Theorem 13.4.2](#), which supplied the orthogonal basis, has the same hypothesis.

What is missing is a definition. In characteristic 2 a quadratic form is not the same data as a symmetric bilinear form — Section 3 warned exactly this — so a quadratic form must be defined in its own right, as a function  $q: V \rightarrow F$  with  $q(a\mathbf{v}) = a^2q(\mathbf{v})$  whose associated  $\beta(\mathbf{u}, \mathbf{v}) = q(\mathbf{u} + \mathbf{v}) - q(\mathbf{u}) - q(\mathbf{v})$  is bilinear. Clifford algebras are defined in that setting too, by the same (C1) and (C2), but this book does not define quadratic forms there and so does not go on.

# Chapter 14

## Tensors and Exterior Algebra

Three times already this book has built an object out of one construction and then walked past the construction without naming it. Chapter 6 characterized the determinant as the **unique** alternating  $n$ -linear form sending the standard basis to 1 ([Theorem 6.4.2](#)), and got a number out of a function of  $n$  vectors. Chapter 7 defined the Kronecker product  $\mathbf{A} \otimes \mathbf{B}$  by writing  $\mathbf{A}$  with its entries replaced by blocks, borrowed the symbol  $\otimes$ , and admitted that the symbol was borrowed from something it had not yet met. Chapter 13 defined a Clifford algebra by saying what maps out of it must look like, proved that at most one such algebra exists, and then could not produce one: the construction needed an algebra of “formal sums of strings of vectors”, and no such algebra had been built. The missing object is the same in all three places. It is the **tensor product**, and this chapter builds it.

The problem it solves is stated in one line. A bilinear map  $\beta: V \times W \rightarrow Z$  is not a linear map, so nothing from Chapters 3 to 5 applies to it: it has no matrix, no kernel, no rank, and  $\beta(2\mathbf{v}, 2\mathbf{w})$  is  $4\beta(\mathbf{v}, \mathbf{w})$  rather than  $2\beta(\mathbf{v}, \mathbf{w})$ . Multilinearity is everywhere in the book — the determinant, the dot product, matrix multiplication, every bilinear form of Chapter 13 — and each time we have had to develop it by hand, because the general theory of linear maps was unavailable. The tensor product ends that. It manufactures a single space  $V \otimes W$  and a single bilinear map  $(\mathbf{v}, \mathbf{w}) \mapsto \mathbf{v} \otimes \mathbf{w}$  into it, through which **every** bilinear map out of  $V \times W$  factors, uniquely, as a linear map. Bilinear maps on  $V \times W$  become linear maps on  $V \otimes W$ . From there the whole chapter is bookkeeping: repeat the construction  $k$  times to get the tensor algebra, then divide it by the relation  $\mathbf{v} \otimes \mathbf{w} = \mathbf{w} \otimes \mathbf{v}$  to get symmetric powers, or by the relation  $\mathbf{v} \otimes \mathbf{v} = 0$  to get exterior powers, where the determinant reappears as an induced map and the Clifford algebra finally gets built.

**The method of this chapter is new, and §02 is where it is learned.** Until now every object in the book was defined by a formula or by a construction: the determinant by the Leibniz sum, the quotient by cosets, the Jordan form by an algorithm. From here on we define objects by what maps out of them look like — a **universal property** — and the order of business inverts. First we say what the object must do; then we prove that any two objects doing it are isomorphic by a *unique* isomorphism, so that “the” object is legitimate; and only then do we build one, to know that the definition is not empty. Chapter 13 §12 already ran this argument once, for Clifford algebras ([Proposition 13.12.1](#)), without saying that it was a pattern. It is a pattern. Sections 02, 06, 07, 08 and 10 each execute it in exactly the same three moves, on five different objects. [Theorem 14.2.1](#) in §02 is the template, written out at full length; the four later repetitions are compressed, and they are compressed on purpose, so that the reader who has absorbed the template recognizes it rather than rereads it. It is worth learning here because it

is the only way to define these objects that does not depend on a choice of basis, and because a proof from a universal property is usually three lines long where a proof from a construction is a page.

The field is a general  $F$  throughout. Where a statement genuinely needs  $\text{char } F = 0$  or  $\text{char } F \neq 2$  — and some do, in §§07, 08 and 10 — the hypothesis is stated and its job is named, as Chapter 13 did. Everything else holds over every field.

**What you need.** From Chapter 3, quotient spaces and their universal property, which is the engine of every construction here (Definition 3.9.3, Lemma 3.9.3, Theorem 3.9.4, Theorem 3.9.10, Theorem 3.9.5, Theorem 3.9.7), together with the space of linear maps and its dimension (Definition 3.4.1, Theorem 3.4.1, Theorem 3.6.4), the matrix of a map (Definition 3.6.1, Theorem 3.6.1), products of spaces (Definition 3.9.1, Theorem 3.9.1), rank (Definition 3.3.2, Theorem 3.3.1) and the trace (Definition 3.10.2, Theorem 3.7.5). From Chapter 1, bases in any dimension and the freedom to prescribe a linear map on one (Definition 1.5.1, Theorem 1.6.4, Theorem 1.8.2, Theorem 3.1.6, Theorem 1.2.1). From Chapter 4, duality, used constantly: Definition 4.1.2, Definition 4.1.1, Theorem 4.1.3, Theorem 4.1.5, Definition 4.4.1, Theorem 4.4.2, Theorem 4.4.6. From Chapter 6, multilinearity and alternation, which §01 extends rather than redefines (Definition 6.2.1, Definition 6.2.2, Definition 6.2.3, Theorem 6.2.1, Theorem 6.2.2), and the determinant with its Leibniz formula and multiplicativity (Definition 6.4.1, Theorem 6.4.3, Theorem 6.4.2, Theorem 6.5.4, Theorem 6.4.5, Theorem 6.6.1, Theorem 6.7.1). From Chapter 7, the Kronecker product, whose two open promises §03 pays (Definition 7.4.1, Theorem 7.4.1, Theorem 7.4.3, Definition 7.4.2, Theorem 7.4.4). From Chapter 12, the singular value decomposition, which §11 turns out to be reading in another language (Theorem 12.8.1). From Chapter 13, algebras, forms and the Clifford algebra whose existence §10 supplies (Definition 13.12.1, Definition 13.1.1, Definition 13.3.2, Definition 13.3.1, Definition 13.12.2, Proposition 13.12.1, Proposition 13.12.3, Theorem 13.12.4). From Chapter 5, the polynomial ring, which §07 identifies with a symmetric algebra (Definition 0.8.2).

## 14.0.1 Roadmap

- **Multilinear maps.** Chapter 6's multilinear forms, extended from  $V^k \rightarrow F$  to  $V_1 \times \cdots \times V_k \rightarrow W$ ; the space they form, its dimension  $\dim W \cdot \prod_i \dim V_i$ , and the problem that they are not linear maps.
- **The tensor product.** The universal property, uniqueness up to unique isomorphism, the construction as a free space modulo the bilinearity relations, the basis  $(\mathbf{v}_i \otimes \mathbf{w}_j)$ , and three warnings about what a tensor is not.
- **Tensor products of maps.**  $S \otimes T$  built from the universal property; its matrix is exactly the Kronecker product of Chapter 7, which settles two debts; and the identification  $V^* \otimes W \cong \mathcal{L}(V, W)$  that everything later uses.
- **Contraction.** Evaluation  $V^* \otimes V \rightarrow F$ , extended to mixed tensors, and the discovery that under the previous identification it is the trace — the basis-free definition of trace the book has wanted since Chapter 3.
- **Index notation.** Components, upper and lower indices, the summation convention, the transformation law derived rather than asserted, and the older definition of a tensor proved equivalent to ours.
- **The tensor algebra.**  $T(V) = \bigoplus_k V^{\otimes k}$ : graded, associative, non-commutative, infinite-dimensional, and free. Its universal property is what the next three sections quotient.
- **Symmetric powers.**  $\text{Sym}^k V$  by killing  $\mathbf{v} \otimes \mathbf{w} - \mathbf{w} \otimes \mathbf{v}$ ; a basis indexed by non-decreasing tuples; the symmetric algebra is the polynomial ring of Chapter 5; and what goes wrong in characteristic  $p$ .

- **Exterior powers.**  $\Lambda^k V$  by killing  $\mathbf{v} \otimes \mathbf{v}$  — the characteristic-free choice, exactly Chapter 13’s distinction between alternating and skew. A basis indexed by increasing tuples,  $\dim \Lambda^k V = \binom{n}{k}$ , and the test that  $\mathbf{v}_1 \wedge \cdots \wedge \mathbf{v}_k \neq 0$  exactly when the list is independent.
- **The determinant, again.**  $\Lambda^n V$  is a line, and  $\Lambda^n T$  is multiplication by  $\det T$ . Multiplicativity becomes a one-line consequence, Laplace expansion falls out of  $\Lambda^k$ , and the coordinates of a wedge are the minors.
- **Grassmann and Clifford.** The exterior algebra  $\Lambda V$ , of dimension  $2^n$ ; elements of  $\Lambda^2 V$  that are not decomposable; and  $\text{Cl}(q) = T(V)/\langle \mathbf{v} \otimes \mathbf{v} - q(\mathbf{v})1 \rangle$ , which constructs the Clifford algebra of Chapter 13 and proves its dimension is  $2^n$ .
- **Tensors in quantum mechanics.** Composite systems as tensor products, entanglement as failure of simplicity, and the Schmidt decomposition, which is the singular value decomposition of Chapter 12 wearing different notation.

## 14.0.2 Named moves

- **Define by universal property, then build a model.** State what maps out of the object must look like. Prove that any two objects with that property are isomorphic by a unique isomorphism, using nothing but the property itself. Then construct one object, so that the definition has content. The uniqueness proof is always the same four lines: each object’s property produces a map to the other, each composite is a self-map fixing the structure map, and the uniqueness clause says the identity is the only such self-map.
- **A quotient kills exactly the relations you want.** To impose relations on a space, take the free object, take the subspace spanned by the relations written as “left side minus right side”, and divide. A map respecting the relations kills that subspace, so it factors through the quotient. This chapter does it three times: tensor, symmetric, exterior.
- **To define a map out of a quotient, define it upstairs first.** Never define a map on cosets, or on simple tensors, by a formula and hope it is well defined. Define it on the free object, where a map may be prescribed arbitrarily on a basis, check that it kills the relations, and let the universal property hand you the induced map.
- **Count a basis by counting index tuples.** Every dimension formula in this chapter comes from counting tuples: all  $k$ -tuples for  $V^{\otimes k}$ , giving  $n^k$ ; non-decreasing ones for  $\text{Sym}^k V$ , giving  $\binom{n+k-1}{k}$ ; strictly increasing ones for  $\Lambda^k V$ , giving  $\binom{n}{k}$ . Spanning is a rewriting argument, independence is a supply of linear functionals, and the count is combinatorics.

## 14.1 Multilinear Maps

Chapter 6 needed functions of several vectors that were linear in each vector separately, and called them multilinear forms (Definition 6.2.1). Every one of them landed in the field  $F$ : the determinant, the signed area, a product of functionals. This chapter needs the same idea with the output allowed to be any vector space, because three of the most familiar operations in the book — matrix multiplication, composition of linear maps, the cross product — are linear in each argument and return something that is not a scalar. So we begin by extending Chapter 6’s definition, count how many such maps there are, and then state the problem that the rest of the chapter exists to solve: a multilinear map is **not** a linear map, so none of the machinery of Chapters 3 to 5 is available to it.

Throughout,  $F$  is a field,  $k \geq 1$  is an integer, and all vector spaces are over  $F$ . From the third

subsection on, every space named is finite-dimensional.

## 14.1.1 From forms to maps

Here are three operations the book has used constantly.

- **Matrix multiplication.** For fixed sizes,  $(\mathbf{A}, \mathbf{B}) \mapsto \mathbf{AB}$  satisfies  $(\mathbf{A} + \mathbf{A}')\mathbf{B} = \mathbf{AB} + \mathbf{A}'\mathbf{B}$  and  $\mathbf{A}(\mathbf{B} + \mathbf{B}') = \mathbf{AB} + \mathbf{AB}'$ , and scalars pull out of either factor.
- **Composition of linear maps.**  $(S, T) \mapsto ST$  satisfies the same four rules.
- **The cross product** on  $\mathbb{R}^3$ , which is linear in each factor and returns a vector.

Each is linear in one argument while the other is held fixed. Each therefore looks exactly like [Definition 6.2.1](#) with  $k = 2$  — except that the value is a matrix, a linear map or a vector, not a scalar. Chapter 6 had no reason to allow that, because it was building the determinant. We do, so we widen the definition. Nothing about Chapter 6's statements changes; a multilinear **form** will simply be the special case where the target is  $F$ , and we will keep using [Definition 6.2.1](#) by that name whenever the target really is  $F$ .

*A multilinear map is a function of several vectors that is linear in each one while the others are held fixed, with values in a vector space rather than in the field.*

### Definition 14.1.1 — Multilinear Map

Let  $V_1, \dots, V_k$  and  $W$  be vector spaces over the **same** field  $F$ , with  $k \geq 1$ . A function

$$\beta: V_1 \times V_2 \times \cdots \times V_k \rightarrow W$$

is  **$k$ -linear**, or a **multilinear map**, if **for every** position  $i \in \{1, \dots, k\}$  and **every** choice of fixed vectors  $\mathbf{v}_j \in V_j$  for the positions  $j \neq i$ , the function

$$V_i \rightarrow W, \quad \mathbf{x} \mapsto \beta(\mathbf{v}_1, \dots, \mathbf{v}_{i-1}, \mathbf{x}, \mathbf{v}_{i+1}, \dots, \mathbf{v}_k),$$

is linear. A 2-linear map is called **bilinear**.

When  $V_1 = \cdots = V_k = V$  we say  $\beta$  is  $k$ -linear **on**  $V$ ; when in addition  $W = F$ , the definition is word for word [Definition 6.2.1](#), and we call  $\beta$  a multilinear **form**.

In words, clause by clause. **Same field:** all  $k + 1$  spaces are over one  $F$ , since otherwise “linear” would mean different things in different slots. **For every position:** the condition is imposed  $k$  separate times, once per slot. **Every choice of fixed vectors:** the other  $k - 1$  arguments are frozen at arbitrary values, not at convenient ones. **Is linear:** written out, for all  $\mathbf{x}, \mathbf{y} \in V_i$  and all  $c \in F$ ,

$$\begin{aligned} \beta(\dots, \mathbf{x} + \mathbf{y}, \dots) &= \beta(\dots, \mathbf{x}, \dots) + \beta(\dots, \mathbf{y}, \dots), \\ \beta(\dots, c\mathbf{x}, \dots) &= c\beta(\dots, \mathbf{x}, \dots), \end{aligned}$$

where the dots stand for the same frozen vectors on both sides. The spaces  $V_1, \dots, V_k$  are allowed to be different from each other and from  $W$ ; nothing forces  $k$  to be related to any dimension.

Two consequences are free. Putting  $c = 0$  in slot  $i$  gives  $\beta(\dots, \mathbf{0}, \dots) = \mathbf{0}$ : a multilinear map kills any tuple with a zero entry. And by induction on the number of terms, expanding one slot

at a time,

$$\beta\left(\sum_{i_1} a_{i_1} \mathbf{x}_{i_1}, \dots, \sum_{i_k} a_{i_k} \mathbf{y}_{i_k}\right) = \sum_{i_1, \dots, i_k} a_{i_1} \cdots a_{i_k} \beta(\mathbf{x}_{i_1}, \dots, \mathbf{y}_{i_k}),$$

a sum over all tuples of indices, one index per slot. This identity is the whole of the section: it says that a multilinear map is determined by its values on tuples drawn from spanning sets, and that the number of such tuples is a product, not a sum.

**Example. (Seven functions of several vectors)**

Decide which of the following are multilinear, and with which  $k$ .

- (a) Any linear map  $T: V \rightarrow W$ , viewed as a function of one vector.
- (b) Scalar multiplication  $F \times V \rightarrow V$ ,  $(c, \mathbf{v}) \mapsto c\mathbf{v}$ , with  $F$  regarded as a vector space over itself.
- (c) Matrix multiplication  $M_{m \times n}(F) \times M_{n \times p}(F) \rightarrow M_{m \times p}(F)$ ,  $(\mathbf{A}, \mathbf{B}) \mapsto \mathbf{AB}$ .
- (d) Composition  $\mathcal{L}(W, Z) \times \mathcal{L}(V, W) \rightarrow \mathcal{L}(V, Z)$ ,  $(S, T) \mapsto ST$ .
- (e) The determinant  $(F^n)^n \rightarrow F$  sending  $(\mathbf{a}_1, \dots, \mathbf{a}_n)$  to  $\det \mathbf{A}$ , where  $\mathbf{A}$  has columns  $\mathbf{a}_1, \dots, \mathbf{a}_n$ .
- (f) The evaluation pairing  $V^* \times V \rightarrow F$ ,  $(\varphi, \mathbf{v}) \mapsto \varphi(\mathbf{v})$ .
- (g) The cross product  $\mathbb{R}^3 \times \mathbb{R}^3 \rightarrow \mathbb{R}^3$ .

*Solution.*

- (a) With  $k = 1$  there is one position and no other argument to freeze, so the condition reads:  $\mathbf{x} \mapsto T\mathbf{x}$  is linear. That is the hypothesis. So a 1-linear map is **exactly** a linear map (Definition 3.1.1), and Definition 14.1.1 generalizes it.
- (b) Fix  $\mathbf{v}$ . Then  $c \mapsto c\mathbf{v}$  is linear by (VS6) and (VS7) of Definition 1.1.1, which give  $(a + b)\mathbf{v} = a\mathbf{v} + b\mathbf{v}$  and  $a(c\mathbf{v}) = (ac)\mathbf{v}$ . Fix  $c$ . Then  $\mathbf{v} \mapsto c\mathbf{v}$  is linear by (VS5) and (VS7), the second of which gives  $c(a\mathbf{v}) = (ca)\mathbf{v} = (ac)\mathbf{v} = a(c\mathbf{v})$ , using that  $F$  is commutative. So scalar multiplication is bilinear. It is the smallest interesting example, and it shows that the two input spaces need not be equal.
- (c) Fix  $\mathbf{B}$ . Then  $\mathbf{A} \mapsto \mathbf{AB}$  is linear, since  $(\mathbf{A} + \mathbf{A}')\mathbf{B} = \mathbf{AB} + \mathbf{A}'\mathbf{B}$  and  $(c\mathbf{A})\mathbf{B} = c(\mathbf{AB})$  by Theorem 0.9.4. Fixing  $\mathbf{A}$  gives the other slot. So matrix multiplication is bilinear, with three different spaces in play when  $m, n, p$  differ.
- (d) Same check, with Theorem 3.4.2 (a) supplying linearity of  $ST$  and Theorem 3.4.2 (d) supplying the two slots. Bilinear.
- (e) The determinant is  $n$ -linear in the columns, and alternating, by Theorem 6.4.3. It is the standard example of a multilinear form with  $k$  equal to the dimension. Here  $k = n$  is forced by the shape of a square matrix, not by the definition.
- (f) Fix  $\mathbf{v}$ . Then  $\varphi \mapsto \varphi(\mathbf{v})$  is linear because the operations on  $V^*$  are pointwise (Definition 4.1.2). Fix  $\varphi$ . Then  $\mathbf{v} \mapsto \varphi(\mathbf{v})$  is linear because  $\varphi$  is a linear functional. Bilinear. This one-line example is the source of the contraction operation of a later section.
- (g) The cross product of Chapter 6 has components  $a_2b_3 - a_3b_2$  and its two cousins, each bilinear as in Example (Alternating Forms) (b). A vector of bilinear functions is bilinear, since linearity in a slot may be checked coordinate by coordinate. So it is bilinear, with values in  $\mathbb{R}^3$  — an example that genuinely needs Definition 14.1.1 rather than Definition 6.2.1.



**Non-example, by minimal change.** Keep  $k = 2$ , keep  $V_1 = V_2 = W = F$ , and change multiplication to addition:  $\sigma(a, b) = a + b$ . Freeze  $b = 1$ . The resulting function  $a \mapsto a + 1$  sends 0 to 1, so it is not linear (Theorem 3.1.1), and the clause “is linear” fails in the first slot. What still works is that  $\sigma$  is linear as a function on the product space  $F \times F$  of Definition 3.9.1. That is precisely the distinction the next subsection is about.

**Why this definition.** One could try to define a “map of several variables” by asking for linearity in all slots at once, that is, by asking for a linear map on  $V_1 \times \cdots \times V_k$ . The result would be a different and much poorer notion: it would exclude multiplication, composition, the determinant and every inner product, and would include only sums of linear maps applied to the separate slots. Freezing all but one slot is what keeps products in the theory. The name says the same thing: *multi*, once per slot; *linear*, in that slot alone.

## 14.1.2 Multilinear is not linear

The product  $V \times W$  is itself a vector space, with  $(\mathbf{v}, \mathbf{w}) + (\mathbf{v}', \mathbf{w}') = (\mathbf{v} + \mathbf{v}', \mathbf{w} + \mathbf{w}')$  and  $c(\mathbf{v}, \mathbf{w}) = (c\mathbf{v}, c\mathbf{w})$  (Definition 3.9.1). So a function  $\beta: V \times W \rightarrow Z$  has two different conditions it might satisfy, and they are stated on the same domain. They are not the same condition, and confusing them is the standard first error with tensors.

Side by side, for  $\beta: V \times W \rightarrow Z$ :

- **Linear on the product** means  $\beta((\mathbf{v}, \mathbf{w}) + (\mathbf{v}', \mathbf{w}')) = \beta(\mathbf{v}, \mathbf{w}) + \beta(\mathbf{v}', \mathbf{w}')$  and  $\beta(c\mathbf{v}, c\mathbf{w}) = c\beta(\mathbf{v}, \mathbf{w})$ . Both slots move together.
- **Bilinear** means  $\beta(\mathbf{v} + \mathbf{v}', \mathbf{w}) = \beta(\mathbf{v}, \mathbf{w}) + \beta(\mathbf{v}', \mathbf{w})$ , and the same in the second slot with the first frozen. One slot moves at a time.

*Warning.* **Do not add in both slots at once.** For a bilinear  $\beta$ , the correct expansion has **four** terms:

$$\beta(\mathbf{v} + \mathbf{v}', \mathbf{w} + \mathbf{w}') = \beta(\mathbf{v}, \mathbf{w}) + \beta(\mathbf{v}, \mathbf{w}') + \beta(\mathbf{v}', \mathbf{w}) + \beta(\mathbf{v}', \mathbf{w}').$$

Concretely, with  $\beta(a, b) = ab$  on  $F \times F$ :  $\beta(1 + 1, 1 + 1) = 4$ , not  $\beta(1, 1) + \beta(1, 1) = 2$ . Scaling is worse:  $\beta(c\mathbf{v}, c\mathbf{w}) = c^2\beta(\mathbf{v}, \mathbf{w})$ , so a bilinear map scales by the **square**. Pull out scalars one slot at a time, always.

*Quick check.* Is  $\beta(\mathbf{x}, \mathbf{y}) = x_1y_1 + x_2$  a bilinear map  $F^2 \times F^2 \rightarrow F$ ?

*Solution.* No. Freeze  $\mathbf{y}$ . Then  $\mathbf{x} \mapsto y_1x_1 + x_2$  is a linear functional, so the first slot is fine. Freeze  $\mathbf{x} = \mathbf{e}_2$ . Then  $\mathbf{y} \mapsto 0 \cdot y_1 + 1 = 1$  is the constant function 1, which sends  $\mathbf{0}$  to 1 and so is not linear. The clause that fails is “is linear” at position  $i = 2$ . The extra summand  $x_2$  does not involve  $\mathbf{y}$  at all, and a bilinear map must vanish whenever an argument is  $\mathbf{0}$ .

## 14.1.3 Values on basis tuples

A linear map may be prescribed arbitrarily on a basis and is determined by those values (Theorem 3.1.6). The corresponding statement for multilinear maps replaces “basis vectors” by “tuples of basis vectors, one from each slot”, and the proof is the expansion identity above read in both directions.



**Theorem 14.1.1 — Multilinear Maps from Values on Basis Tuples**

Let  $V_1, \dots, V_k$  be finite-dimensional, with ordered bases  $\mathcal{B}_i = (\mathbf{v}_1^{(i)}, \dots, \mathbf{v}_{n_i}^{(i)})$  for  $i = 1, \dots, k$ , and let  $W$  be any vector space over  $F$ . For every family of vectors

$$\mathbf{w}_{i_1 i_2 \dots i_k} \in W, \quad 1 \leq i_1 \leq n_1, \dots, 1 \leq i_k \leq n_k,$$

there is **exactly one**  $k$ -linear map  $\beta: V_1 \times \dots \times V_k \rightarrow W$  with  $\beta(\mathbf{v}_{i_1}^{(1)}, \dots, \mathbf{v}_{i_k}^{(k)}) = \mathbf{w}_{i_1 \dots i_k}$  for every tuple of indices.

*Idea.* Uniqueness is the expansion identity: write each argument in its basis, expand one slot at a time, and every value of  $\beta$  is a combination of the prescribed ones. For existence we need a formula, and the coefficients in that expansion are exactly the coordinates, which [Theorem 4.1.3](#) supplies as linear functionals. So multiply the right functionals together and use them as coefficients.

*Proof. Uniqueness.* Let  $\beta$  and  $\beta'$  be  $k$ -linear with the same values on all basis tuples, and let  $\mathbf{x}_i \in V_i$ . Write  $\mathbf{x}_i = \sum_j a_j^{(i)} \mathbf{v}_j^{(i)}$ . Expanding each slot in turn by [Definition 14.1.1](#),

$$\beta(\mathbf{x}_1, \dots, \mathbf{x}_k) = \sum_{i_1, \dots, i_k} a_{i_1}^{(1)} \cdots a_{i_k}^{(k)} \mathbf{w}_{i_1 \dots i_k},$$

and the same computation applies to  $\beta'$ . Hence  $\beta = \beta'$ .

*Existence.* For each  $i$ , let  $(\varphi_1^{(i)}, \dots, \varphi_{n_i}^{(i)})$  be the basis of  $V_i^*$  dual to  $\mathcal{B}_i$ , so that  $\varphi_a^{(i)}(\mathbf{v}_b^{(i)}) = \delta_{ab}$  and  $\varphi_j^{(i)}(\mathbf{x})$  is the  $j$ -th coordinate of  $\mathbf{x}$  ([Theorem 4.1.3](#)). Define

$$\beta(\mathbf{x}_1, \dots, \mathbf{x}_k) := \sum_{i_1, \dots, i_k} \varphi_{i_1}^{(1)}(\mathbf{x}_1) \cdots \varphi_{i_k}^{(k)}(\mathbf{x}_k) \mathbf{w}_{i_1 \dots i_k},$$

a finite sum of vectors of  $W$ , so a well-defined element of  $W$ .

It is  $k$ -linear: freeze every slot but the  $m$ -th. Each summand is then the vector  $\mathbf{w}_{i_1 \dots i_k}$  multiplied by a fixed scalar times  $\varphi_{i_m}^{(m)}(\mathbf{x}_m)$ , and  $\varphi_{i_m}^{(m)}$  is linear; a finite sum of linear functions of  $\mathbf{x}_m$ , each scaled by a fixed vector, is linear in  $\mathbf{x}_m$ . This holds for every  $m$ , which is the condition of [Definition 14.1.1](#).

Its values are the prescribed ones: at  $\mathbf{x}_i = \mathbf{v}_{j_i}^{(i)}$  the coefficient  $\varphi_{i_1}^{(1)}(\mathbf{v}_{j_1}^{(1)}) \cdots \varphi_{i_k}^{(k)}(\mathbf{v}_{j_k}^{(k)})$  is 1 when  $i_1 = j_1, \dots, i_k = j_k$  and 0 otherwise, so only the term  $\mathbf{w}_{j_1 \dots j_k}$  survives. This proves the theorem. ■

So a multilinear map is “nothing but” a family of vectors of  $W$  indexed by tuples of basis indices, exactly as a linear map is nothing but a family indexed by single basis indices. The count of tuples is where the product in the next theorem comes from.

## 14.1.4 The space of multilinear maps

Two multilinear maps with the same slots and the same target can be added and scaled pointwise, and the results are again multilinear. That makes them a vector space, and we name it before counting it.

### Definition 14.1.2 — The Space of Multilinear Maps

Let  $V_1, \dots, V_k$  and  $W$  be vector spaces over  $F$ . Write

$$\mathcal{M}(V_1, \dots, V_k; W)$$

for the set of all  $k$ -linear maps  $V_1 \times \dots \times V_k \rightarrow W$ , with the **pointwise** operations

$$(\beta + \gamma)(\mathbf{x}_1, \dots, \mathbf{x}_k) := \beta(\mathbf{x}_1, \dots, \mathbf{x}_k) + \gamma(\mathbf{x}_1, \dots, \mathbf{x}_k),$$

and  $(c\beta)(\mathbf{x}_1, \dots, \mathbf{x}_k) := c\beta(\mathbf{x}_1, \dots, \mathbf{x}_k)$ . The semicolon separates the inputs from the output.

**Well-definedness.** The set of *all* functions  $V_1 \times \dots \times V_k \rightarrow W$  is a vector space under the pointwise operations, by the argument of [Example \(The Standard Vector Spaces\)](#) (d) with  $F$  replaced by  $W$ : two functions are equal when they agree at every point, so each axiom is checked one point at a time and reduces to the corresponding axiom in  $W$ . Inside it,  $\mathcal{M}(V_1, \dots, V_k; W)$  passes [Theorem 1.2.1](#). The zero function is  $k$ -linear, since freezing slots leaves the zero map. If  $\beta, \gamma$  are  $k$ -linear and  $c \in F$ , then freezing all slots but the  $m$ -th turns  $\beta + \gamma$  and  $c\beta$  into a sum and a scalar multiple of linear maps  $V_m \rightarrow W$ , which are linear by [Theorem 3.4.1](#). So  $\mathcal{M}(V_1, \dots, V_k; W)$  is a subspace, hence a vector space.

Two special cases are old friends. For  $k = 1$ ,  $\mathcal{M}(V; W) = \mathcal{L}(V, W)$  by [Example \(Seven functions of several vectors\)](#) (a). For  $W = F$ , the elements are the multilinear forms of [Definition 6.2.1](#); Chapter 6 wrote the space of alternating  $n$ -linear forms on  $F^n$  and proved it one-dimensional ([Corollary 6.4.4](#)), which is a subspace of  $\mathcal{M}(F^n, \dots, F^n; F)$ .

### Theorem 14.1.2 — Dimension of the Space of Multilinear Maps

Let  $V_1, \dots, V_k$  and  $W$  be finite-dimensional, with  $\dim V_i = n_i$  and  $\dim W = m$ . Then  $\mathcal{M}(V_1, \dots, V_k; W)$  is finite-dimensional, of dimension

$$\dim \mathcal{M}(V_1, \dots, V_k; W) = (\dim W) \prod_{i=1}^k \dim V_i = m n_1 n_2 \cdots n_k. \quad (14.1.1)$$

Explicitly, fix bases  $\mathcal{B}_i$  of  $V_i$  with dual bases  $(\varphi_j^{(i)})$ , and a basis  $(\mathbf{w}_1, \dots, \mathbf{w}_m)$  of  $W$ . For a tuple  $I = (i_1, \dots, i_k)$  and an index  $r$ , let

$$E_I^r(\mathbf{x}_1, \dots, \mathbf{x}_k) := \varphi_{i_1}^{(1)}(\mathbf{x}_1) \cdots \varphi_{i_k}^{(k)}(\mathbf{x}_k) \mathbf{w}_r.$$

The  $m n_1 \cdots n_k$  maps  $E_I^r$  form a basis of  $\mathcal{M}(V_1, \dots, V_k; W)$ .

*Idea.* [Theorem 14.1.1](#) says a multilinear map is the same thing as a list of vectors of  $W$ , one per index tuple. So count the lists: there are  $n_1 \cdots n_k$  tuples and each entry needs  $m$  coordinates. The maps  $E_I^r$  are the maps whose list has a single  $\mathbf{w}_r$  in position  $I$  and zeros elsewhere, which is why they should be the basis.

*Proof.* Each  $E_I^r$  is  $k$ -linear by the existence half of [Theorem 14.1.1](#), being the map prescribed to take the value  $\mathbf{w}_r$  at the basis tuple  $(\mathbf{v}_{i_1}^{(1)}, \dots, \mathbf{v}_{i_k}^{(k)})$  and  $\mathbf{0}$  at every other basis tuple.

*Spanning.* Let  $\beta \in \mathcal{M}(V_1, \dots, V_k; W)$ . For each tuple  $I$  write its value on the corresponding basis tuple in the basis of  $W$ :

$$\beta(\mathbf{v}_{i_1}^{(1)}, \dots, \mathbf{v}_{i_k}^{(k)}) = \sum_{r=1}^m c_{I,r} \mathbf{w}_r,$$

which is possible, and with unique coefficients, since  $(\mathbf{w}_1, \dots, \mathbf{w}_m)$  is a basis. The map  $\sum_{I,r} c_{I,r} E_I^r$  is  $k$ -linear and takes exactly these values on the basis tuples. By the uniqueness half of [Theorem 14.1.1](#) it equals  $\beta$ .

*Independence.* Suppose  $\sum_{I,r} c_{I,r} E_I^r = 0$ , the zero map. Evaluate at a basis tuple indexed by  $J$ . By the displayed formula for  $E_I^r$ , the coefficient  $\varphi_{i_1}^{(1)}(\mathbf{v}_{j_1}^{(1)}) \cdots \varphi_{i_k}^{(k)}(\mathbf{v}_{j_k}^{(k)})$  is 1 for  $I = J$  and 0 otherwise, so the value is  $\sum_r c_{J,r} \mathbf{w}_r = \mathbf{0}$ . Since  $(\mathbf{w}_r)$  is independent,  $c_{J,r} = 0$  for every  $r$ . As  $J$  was arbitrary, every coefficient vanishes.

Hence the  $E_I^r$  form a basis. There are  $n_1 n_2 \cdots n_k$  index tuples  $I$  and  $m$  choices of  $r$ , so the count is  $m n_1 \cdots n_k$ , which is [Eq. 14.1.1](#). This proves the theorem. ■

**Aftermath.** Two sanity checks. With  $k = 1$ , [Eq. 14.1.1](#) reads  $\dim \mathcal{L}(V, W) = (\dim W)(\dim V)$ , which is [Theorem 3.6.4](#). With  $W = F$  and  $k = 2$ , it reads  $\dim \mathcal{M}(V_1, V_2; F) = n_1 n_2$ , which is the count of  $n_1 \times n_2$  matrices — and indeed Chapter 13 recorded a bilinear form on a single space by its Gram matrix ([Definition 13.1.2](#)). The  $n_i$  enter as a **product**, once per slot, and that is the number to remember.

### Example. (Counting bilinear maps)

- (a) Find  $\dim \mathcal{M}(F^2, F^3; F^2)$ .
- (b) Find  $\dim \mathcal{M}(F^3, F^3, F^3; F)$ , and the dimension of the subspace of alternating forms inside it when  $F = \mathbb{R}$ .
- (c) Write the bilinear map  $\beta: F^2 \times F^2 \rightarrow F^2$ ,  $\beta(\mathbf{x}, \mathbf{y}) = (x_1 y_1, x_1 y_2 + x_2 y_1)$ , in the basis of [Theorem 14.1.2](#) for the standard bases.

*Solution.*

- (a)  $\dim W \cdot \dim V_1 \cdot \dim V_2 = 2 \cdot 2 \cdot 3 = 12$ .
- (b)  $1 \cdot 3 \cdot 3 \cdot 3 = 27$ . The alternating ones form a subspace of dimension 1 by [Corollary 6.4.4](#), spanned by the determinant. So alternation is a very strong condition: it cuts 27 down to 1.
- (c) Here  $m = n_1 = n_2 = 2$ , so there are  $2 \cdot 2 \cdot 2 = 8$  basis maps  $E_{(i,j)}^r$ , and  $E_{(i,j)}^r(\mathbf{x}, \mathbf{y}) =$

$x_i y_j \mathbf{e}_r$ . Evaluate  $\beta$  on the four basis pairs:

$$\begin{aligned}\beta(\mathbf{e}_1, \mathbf{e}_1) &= (1, 0) = \mathbf{e}_1, & \beta(\mathbf{e}_1, \mathbf{e}_2) &= (0, 1) = \mathbf{e}_2, \\ \beta(\mathbf{e}_2, \mathbf{e}_1) &= (0, 1) = \mathbf{e}_2, & \beta(\mathbf{e}_2, \mathbf{e}_2) &= (0, 0) = \mathbf{0}.\end{aligned}$$

By the spanning argument,  $\beta = E_{(1,1)}^1 + E_{(1,2)}^2 + E_{(2,1)}^2$ , with the other five coefficients zero. Reading the formula back,  $\beta(\mathbf{x}, \mathbf{y}) = x_1 y_1 \mathbf{e}_1 + x_1 y_2 \mathbf{e}_2 + x_2 y_1 \mathbf{e}_2$ , which is the given map.

*Quick check.* What does Eq. 14.1.1 give for  $\mathcal{M}(V_1, V_2; W)$  when  $V_2 = \{\mathbf{0}\}$ ? Does the answer make sense?

*Solution.* It gives  $(\dim W) \cdot (\dim V_1) \cdot 0 = 0$ , so the space is  $\{0\}$ . That is right: a bilinear map must satisfy  $\beta(\mathbf{v}, \mathbf{0}) = \mathbf{0}$ , and  $\mathbf{0}$  is the only vector of  $V_2$ , so the only bilinear map is the zero map. The degenerate case is worth checking because it is where a formula with a product behaves differently from one with a sum.

## 14.1.5 The problem this chapter solves

We now have a good supply of multilinear maps and we know how many there are. What we do not have is any way to *work* with them. Consider what Chapters 3 to 5 give a linear map  $T: V \rightarrow W$ : a matrix, a kernel, an image, a rank, Rank–Nullity, a dual map, a determinant when  $V = W$ , eigenvalues, a Jordan form. A bilinear map  $\beta: V \times W \rightarrow Z$  has none of these. It has no matrix, because Definition 3.6.1 applies to linear maps. It has no kernel that is a subspace: the set where  $\beta$  vanishes contains  $(\mathbf{v}, \mathbf{0})$  and  $(\mathbf{0}, \mathbf{w})$  for all  $\mathbf{v}, \mathbf{w}$  but usually not their sum. It is not an element of  $\mathcal{L}(V \times W, Z)$ , as the previous subsection insisted. Every fact we know about it has had to be proved by hand, one slot at a time.

The wish is therefore easy to state. **We want to replace multilinearity by linearity.** Concretely, for two spaces  $V$  and  $W$  we want a single vector space — call it  $X$  for now — and a single bilinear map  $\tau: V \times W \rightarrow X$ , so universal that every bilinear map out of  $V \times W$  is  $\tau$  followed by a **linear** map. If that works, then bilinear maps  $V \times W \rightarrow Z$  correspond to linear maps  $X \rightarrow Z$ , and the entire toolkit of Chapters 3 to 5 applies to the latter.

Before building anything, we can predict the size of  $X$ , and the prediction is a good reason to believe the object exists. Suppose such an  $X$  and  $\tau$  existed, and suppose the correspondence “ $\beta$  on  $V \times W \leftrightarrow$  linear map on  $X$ ” is a bijection for every target  $Z$ . Take  $Z = F$ . Then Eq. 14.1.1 gives

$$\dim \mathcal{M}(V, W; F) = 1 \cdot \dim V \cdot \dim W,$$

while Theorem 3.6.4 gives  $\dim \mathcal{L}(X, F) = \dim X$ . A bijection that respects addition and scalars forces these to agree, so

$$\dim X = (\dim V)(\dim W).$$

The product of the dimensions, not the sum. This already rules out the naive guess  $X = V \times W$ , whose dimension is  $\dim V + \dim W$  by Theorem 3.9.1, and it tells us what to aim for. The next section constructs  $X$ , calls it  $V \otimes W$ , and proves that its dimension is exactly this product.

One more remark before we go. The prediction used only the case  $Z = F$ , but the same count works for every  $Z$ : Eq. 14.1.1 gives  $\dim Z \cdot \dim V \cdot \dim W$  on one side and  $\dim Z \cdot \dim X$  on the other, and the two agree precisely when  $\dim X = (\dim V)(\dim W)$ . When the next section

produces a basis of  $V \otimes W$  indexed by pairs  $(i, j)$  of basis indices, that is the same count seen a second time, and the agreement is not a coincidence — it is the universal property, counted.

## 14.1.6 Exercises

### 14.1.6.1 A. Check your understanding

#### Exercise A1.

- Define what it means for  $\beta: V_1 \times \cdots \times V_k \rightarrow W$  to be  $k$ -linear, with every quantifier.
- State the difference between “ $\beta$  is bilinear” and “ $\beta$  is linear on the product space  $V \times W$ ”, and give a function that is one but not the other.
- Write out  $\beta(\mathbf{u} + \mathbf{u}', \mathbf{v} + \mathbf{v}')$  for a bilinear  $\beta$ . How many terms are there?
- State  $\dim \mathcal{M}(V_1, \dots, V_k; W)$  in terms of the dimensions of the spaces involved.
- True or false: every bilinear map  $\beta: V \times W \rightarrow Z$  satisfies  $\beta(\mathbf{v}, \mathbf{0}) = \mathbf{0}$ . Justify your answer.
- What is a 1-linear map? What is a  $k$ -linear map with  $W = F$  called?

*Solution.*

- It is  $k$ -linear if for **every** position  $i$  and **every** choice of fixed  $\mathbf{v}_j \in V_j$  with  $j \neq i$ , the function  $\mathbf{x} \mapsto \beta(\mathbf{v}_1, \dots, \mathbf{x}, \dots, \mathbf{v}_k)$  with  $\mathbf{x}$  in position  $i$  is a linear map  $V_i \rightarrow W$  (Definition 14.1.1). All spaces are over the same field.
- Bilinear means linear in each slot with the other **frozen**; linear on the product means linear when both slots move **together**. The multiplication  $\beta(a, b) = ab$  on  $F \times F$  is bilinear but not linear on the product, since  $\beta(2 \cdot (1, 1)) = 4 \neq 2\beta(1, 1)$ . The sum  $\sigma(a, b) = a + b$  is linear on the product but not bilinear, since  $a \mapsto a + 1$  is not linear.
- $\beta(\mathbf{u}, \mathbf{v}) + \beta(\mathbf{u}, \mathbf{v}') + \beta(\mathbf{u}', \mathbf{v}) + \beta(\mathbf{u}', \mathbf{v}')$ : four terms.
- $(\dim W) \prod_{i=1}^k \dim V_i$  (Eq. 14.1.1).
- True. Freeze  $\mathbf{v}$ ; the map  $\mathbf{y} \mapsto \beta(\mathbf{v}, \mathbf{y})$  is linear, and a linear map sends  $\mathbf{0}$  to  $\mathbf{0}$  by Theorem 3.1.1.
- A 1-linear map is exactly a linear map. A  $k$ -linear map into  $F$  is a multilinear form (Definition 6.2.1).

### 14.1.6.2 B. Practice

#### Exercise B1: Determine which are multilinear.

Determine which of the following are multilinear maps. For those that are, say what  $k$  is; for those that are not, name the clause that fails and give a concrete witness. Justify your answer.

- $\beta: M_n(F) \times M_n(F) \rightarrow M_n(F)$ ,  $\beta(\mathbf{A}, \mathbf{B}) = \mathbf{AB} - \mathbf{BA}$ .
- $\beta: \mathbb{R}^2 \times \mathbb{R}^2 \rightarrow \mathbb{R}$ ,  $\beta(\mathbf{x}, \mathbf{y}) = x_1 y_2 - x_2 y_1 + 1$ .
- $\beta: F^n \times F^n \rightarrow M_n(F)$ ,  $\beta(\mathbf{x}, \mathbf{y}) = \mathbf{xy}^\top$ .

- (d)  $\beta: M_n(F) \times M_n(F) \rightarrow F$ ,  $\beta(\mathbf{A}, \mathbf{B}) = \det(\mathbf{AB})$ , for  $n \geq 2$ .  
 (e)  $\beta: V^* \times V^* \times V \times V \rightarrow F$ ,  $\beta(\varphi, \psi, \mathbf{u}, \mathbf{v}) = \varphi(\mathbf{u})\psi(\mathbf{v})$ .

*Solution.*

- (a) Bilinear,  $k = 2$ . This is the commutator  $[\mathbf{A}, \mathbf{B}]$  of Definition 7.5.1. Freeze  $\mathbf{B}$ : then  $\mathbf{A} \mapsto \mathbf{AB} - \mathbf{BA}$  is a difference of two linear maps of  $\mathbf{A}$ , hence linear (Theorem 3.4.1). Freezing  $\mathbf{A}$  is the same computation.  
 (b) Not multilinear. Freeze  $\mathbf{y} = \mathbf{0}$ ; then  $\mathbf{x} \mapsto 1$  is constant, so it sends  $\mathbf{0}$  to 1 and the clause “is linear” fails at position 1. Removing the  $+1$  leaves the signed area, which is bilinear.  
 (c) Bilinear,  $k = 2$ . The  $(i, j)$  entry of  $\mathbf{xy}^\top$  is  $x_i y_j$ . Freeze  $\mathbf{y}$ : each entry is a linear functional of  $\mathbf{x}$  and the entries are the coordinates of the output in the basis of matrix units, so  $\mathbf{x} \mapsto \mathbf{xy}^\top$  is linear. Symmetrically in  $\mathbf{y}$ . This map will reappear as the model of the tensor product of two coordinate spaces.  
 (d) Not multilinear for  $n \geq 2$ . By Theorem 6.5.4,  $\beta(\mathbf{A}, \mathbf{B}) = \det \mathbf{A} \det \mathbf{B}$ . Freeze  $\mathbf{B} = \mathbf{I}_n$ ; then  $\mathbf{A} \mapsto \det \mathbf{A}$  is not linear, since  $\det(c\mathbf{A}) = c^n \det \mathbf{A}$  and  $n \geq 2$ . Concretely with  $n = 2$ ,  $\det(2\mathbf{I}_2) = 4 \neq 2 \det \mathbf{I}_2 = 2$ . The failing clause is again “is linear”.  
 (e) 4-linear. Freeze three of the four arguments; in each case what remains is one of  $\varphi \mapsto \varphi(\mathbf{u}) \cdot \psi(\mathbf{v})$ ,  $\psi \mapsto \psi(\mathbf{v}) \cdot \varphi(\mathbf{u})$ ,  $\mathbf{u} \mapsto \varphi(\mathbf{u}) \cdot \psi(\mathbf{v})$ ,  $\mathbf{v} \mapsto \psi(\mathbf{v}) \cdot \varphi(\mathbf{u})$ , each a fixed scalar times a linear functional, hence linear. Note that the four slots do **not** all lie in the same space, which Definition 14.1.1 allows and Definition 6.2.1 does not.

### Exercise B2: Dimension counts.

Compute the dimension of each space. The field is  $F$  throughout.

- (a)  $\mathcal{M}(F^4, F^5; F^3)$ .  
 (b)  $\mathcal{M}(F^2, F^2, F^2; F^2)$ .  
 (c)  $\mathcal{M}(M_2(F), F^3; F)$ .  
 (d)  $\mathcal{M}(F^n; W)$  for  $\dim W = m$ , and the space  $\mathcal{L}(F^n, W)$ . Compare.

*Solution.*

- (a)  $3 \cdot 4 \cdot 5 = 60$  by Eq. 14.1.1.  
 (b)  $2 \cdot 2 \cdot 2 \cdot 2 = 16$ .  
 (c)  $\dim M_2(F) = 4$ , so  $1 \cdot 4 \cdot 3 = 12$ .  
 (d) Both are  $mn$ . With  $k = 1$  the two spaces are equal, by Example (Seven functions of several vectors) (a), and Eq. 14.1.1 reduces to Theorem 3.6.4. The formula is consistent with what Chapter 3 already proved.

### Exercise B3: Expanding in the basis.

Let  $\beta: F^2 \times F^3 \rightarrow F^2$  be given by

$$\beta(\mathbf{x}, \mathbf{y}) = (x_1 y_1 - 2x_2 y_3, x_1 y_2 + x_2 y_1).$$

- (a) Verify that  $\beta$  is bilinear.
- (b) Compute the six values  $\beta(\mathbf{e}_i, \mathbf{f}_j)$ , where  $(\mathbf{e}_1, \mathbf{e}_2)$  and  $(\mathbf{f}_1, \mathbf{f}_2, \mathbf{f}_3)$  are the standard bases.
- (c) Write  $\beta$  in the basis  $\{E_{(i,j)}^r\}$  of [Theorem 14.1.2](#), and say how many of the 12 coefficients are non-zero.

*Solution.*

- (a) Freeze  $\mathbf{y}$ . Each output coordinate is a fixed linear combination of  $x_1$  and  $x_2$ , hence a linear functional of  $\mathbf{x}$  ([Theorem 4.1.1](#)), and a map into  $F^2$  whose coordinates are linear is linear. Freeze  $\mathbf{x}$ : each output coordinate is a fixed linear combination of  $y_1, y_2, y_3$ , and the same argument applies. So  $\beta$  is bilinear.
- (b) Substituting the standard basis vectors:

$$\begin{aligned}\beta(\mathbf{e}_1, \mathbf{f}_1) &= (1, 0), & \beta(\mathbf{e}_1, \mathbf{f}_2) &= (0, 1), & \beta(\mathbf{e}_1, \mathbf{f}_3) &= (0, 0), \\ \beta(\mathbf{e}_2, \mathbf{f}_1) &= (0, 1), & \beta(\mathbf{e}_2, \mathbf{f}_2) &= (0, 0), & \beta(\mathbf{e}_2, \mathbf{f}_3) &= (-2, 0).\end{aligned}$$

- (c) Reading the coordinates off,  $c_{(1,1),1} = 1$ ,  $c_{(1,2),2} = 1$ ,  $c_{(2,1),2} = 1$  and  $c_{(2,3),1} = -2$ , all others zero. Hence

$$\beta = E_{(1,1)}^1 + E_{(1,2)}^2 + E_{(2,1)}^2 - 2E_{(2,3)}^1,$$

so 4 of the  $2 \cdot 2 \cdot 3 = 12$  coefficients are non-zero.

### 14.1.6.3 C. Going deeper

**Exercise C1: Bilinear forms on coordinate spaces are matrices.**

Let  $m, n \geq 1$ .

- (a) Prove that for every bilinear  $\beta: F^m \times F^n \rightarrow F$  there is a **unique**  $\mathbf{A} \in M_{m \times n}(F)$  with  $\beta(\mathbf{x}, \mathbf{y}) = \mathbf{x}^\top \mathbf{A} \mathbf{y}$  for all  $\mathbf{x}, \mathbf{y}$ .
- (b) Hence deduce that  $\dim \mathcal{M}(F^m, F^n; F) = mn$ , and check that this agrees with [Eq. 14.1.1](#).

*Hint for (a): the entries of  $\mathbf{A}$  can only be the values of  $\beta$  on pairs of standard basis vectors.*

*Solution.*

- (a) *Existence.* Put  $a_{ij} := \beta(\mathbf{e}_i, \mathbf{f}_j)$ , where  $(\mathbf{e}_i)$  and  $(\mathbf{f}_j)$  are the standard bases of  $F^m$  and  $F^n$ , and let  $\mathbf{A} = (a_{ij})$ . For  $\mathbf{x} = \sum_i x_i \mathbf{e}_i$  and  $\mathbf{y} = \sum_j y_j \mathbf{f}_j$ , expanding one slot at a time by [Definition 14.1.1](#) gives

$$\beta(\mathbf{x}, \mathbf{y}) = \sum_{i,j} x_i y_j \beta(\mathbf{e}_i, \mathbf{f}_j) = \sum_{i,j} x_i a_{ij} y_j = \mathbf{x}^\top \mathbf{A} \mathbf{y}.$$

*Uniqueness.* If  $\mathbf{x}^\top \mathbf{A} \mathbf{y} = \mathbf{x}^\top \mathbf{A}' \mathbf{y}$  for all  $\mathbf{x}, \mathbf{y}$ , take  $\mathbf{x} = \mathbf{e}_i$  and  $\mathbf{y} = \mathbf{f}_j$ : the left side is  $a_{ij}$  and the right side is  $a'_{ij}$ . Hence  $\mathbf{A} = \mathbf{A}'$ .

- (b) By (a) the map  $\mathcal{M}(F^m, F^n; F) \rightarrow M_{m \times n}(F)$ ,  $\beta \mapsto \mathbf{A}$ , is a bijection, and it is linear because the entries  $\beta(\mathbf{e}_i, \mathbf{f}_j)$  depend linearly on  $\beta$  by the definition of the pointwise

operations. So it is an isomorphism, and  $\dim \mathcal{M}(F^m, F^n; F) = \dim M_{m \times n}(F) = mn$  by [Theorem 3.5.4](#). [Eq. 14.1.1](#) gives  $1 \cdot m \cdot n = mn$ , the same value.

### Exercise C2: Linear and bilinear at once.

Let  $V, W, Z$  be vector spaces over  $F$  and let  $\beta: V \times W \rightarrow Z$  be a function.

- Prove that if  $\beta$  is **both** bilinear and linear as a map on the product space  $V \times W$ , then  $\beta$  is the zero map.
- Give an example, with  $V = W = Z = F$ , of a non-zero bilinear map and a non-zero linear map on the product, and say which property each one fails.
- Does (a) still hold when  $V = \{\mathbf{0}\}$ ? Explain what the statement says in that case.

*Solution.*

- Let  $\mathbf{v} \in V$  and  $\mathbf{w} \in W$ . In the product space,  $(\mathbf{v}, \mathbf{w}) = (\mathbf{v}, \mathbf{0}) + (\mathbf{0}, \mathbf{w})$ . Since  $\beta$  is linear on the product,

$$\beta(\mathbf{v}, \mathbf{w}) = \beta(\mathbf{v}, \mathbf{0}) + \beta(\mathbf{0}, \mathbf{w}).$$

Since  $\beta$  is bilinear, freezing the first slot at  $\mathbf{v}$  makes  $\mathbf{y} \mapsto \beta(\mathbf{v}, \mathbf{y})$  linear, so  $\beta(\mathbf{v}, \mathbf{0}) = \mathbf{0}$  by [Theorem 3.1.1](#); freezing the second slot at  $\mathbf{w}$  gives  $\beta(\mathbf{0}, \mathbf{w}) = \mathbf{0}$  likewise. Therefore  $\beta(\mathbf{v}, \mathbf{w}) = \mathbf{0}$  for all  $\mathbf{v}, \mathbf{w}$ , that is,  $\beta = \mathbf{0}$ .

- The product  $\mu(a, b) = ab$  is bilinear and fails linearity on the product, since  $\mu(2 \cdot (1, 1)) = 4 \neq 2\mu(1, 1)$ . The sum  $\sigma(a, b) = a + b$  is linear on the product and fails bilinearity, since  $\sigma(0, 1) = 1 \neq 0$  while a bilinear map must vanish when a slot is  $\mathbf{0}$ .
- Yes, but it says nothing: if  $V = \{\mathbf{0}\}$  then  $\mathbf{v} = \mathbf{0}$  always, and the only function  $\{\mathbf{0}\} \times W \rightarrow Z$  that is bilinear is already the zero map, since  $\beta(\mathbf{0}, \mathbf{w}) = \mathbf{0}$ . The content of (a) appears only when both  $V$  and  $W$  are non-zero and  $Z \neq \{\mathbf{0}\}$ .

### Exercise C3: Counting over a finite field.

Work over  $\mathbb{F}_2$ , the field with two elements.

- How many bilinear maps  $\mathbb{F}_2^2 \times \mathbb{F}_2^2 \rightarrow \mathbb{F}_2$  are there? How many trilinear maps  $\mathbb{F}_2^2 \times \mathbb{F}_2^2 \times \mathbb{F}_2^2 \rightarrow \mathbb{F}_2$ ?
- How many of the bilinear maps in (a) are symmetric, that is, satisfy  $\beta(\mathbf{x}, \mathbf{y}) = \beta(\mathbf{y}, \mathbf{x})$  for all  $\mathbf{x}, \mathbf{y}$ ? How many are alternating in the sense of [Definition 6.2.2](#)?
- Is every alternating bilinear form on  $\mathbb{F}_2^2$  symmetric? Is every symmetric one alternating?

*Solution.*

- By [Eq. 14.1.1](#),  $\dim \mathcal{M}(\mathbb{F}_2^2, \mathbb{F}_2^2; \mathbb{F}_2) = 1 \cdot 2 \cdot 2 = 4$ . A vector space of dimension  $d$  over  $\mathbb{F}_2$  has exactly  $2^d$  elements, since each of the  $d$  coordinates is 0 or 1. So there are  $2^4 = 16$  bilinear maps. For three slots the dimension is  $2^3 = 8$ , giving  $2^8 = 256$  trilinear maps.
- By [Exercise \(C1: Bilinear forms on coordinate spaces are matrices\)](#) a bilinear form on  $\mathbb{F}_2^2$  is  $\mathbf{x}^\top \mathbf{A} \mathbf{y}$  for a unique  $\mathbf{A} \in M_2(\mathbb{F}_2)$ , and  $\beta$  is symmetric exactly when



$\mathbf{A} = \mathbf{A}^\top$  (compare the values at  $(\mathbf{e}_i, \mathbf{e}_j)$  and  $(\mathbf{e}_j, \mathbf{e}_i)$ ). A symmetric  $2 \times 2$  matrix is determined by  $a_{11}, a_{12}, a_{22}$ , so there are  $2^3 = 8$  symmetric ones. Alternating means  $\beta(\mathbf{x}, \mathbf{x}) = 0$  for all  $\mathbf{x}$ , which forces  $a_{11} = \beta(\mathbf{e}_1, \mathbf{e}_1) = 0$ ,  $a_{22} = 0$ , and then  $\beta(\mathbf{e}_1 + \mathbf{e}_2, \mathbf{e}_1 + \mathbf{e}_2) = a_{12} + a_{21} = 0$ . Over  $\mathbb{F}_2$  that says  $a_{21} = a_{12}$ , so  $\mathbf{A} = \begin{pmatrix} 0 & a \\ a & 0 \end{pmatrix}$  and there are 2 alternating forms: the zero form and the signed area.

- (c) Both alternating forms found in (b) have symmetric matrices, so yes, every alternating form here is symmetric — which is [Theorem 13.3.3](#) in characteristic 2, where  $-1 = 1$ . The converse fails: the dot product  $\beta(\mathbf{x}, \mathbf{y}) = x_1y_1 + x_2y_2$  is symmetric, and  $\beta(\mathbf{e}_1, \mathbf{e}_1) = 1 \neq 0$ , so it is not alternating. Of the 8 symmetric forms only 2 are alternating.

## 14.2 The Tensor Product

The previous section ended with a wish: a space through which every bilinear map out of  $V \times W$  becomes a **linear** map. This section grants it. The object is called the tensor product, and building it is the main technical work of the chapter — but the way we go about it matters more than the object itself, because the same three moves recur four more times, for the tensor algebra, the symmetric powers, the exterior powers and the Clifford algebra.

The three moves are: **state what the object must do**, in the form of a property of the maps out of it; **prove that any two objects doing it are isomorphic by a unique isomorphism**, using nothing but that property; and only then **construct one**, to know that the definition is not empty. The middle move is what licenses the word “the”. [Theorem 14.2.1](#) below is written out at full length and is the template for the rest of the chapter, so it is worth reading twice.

Throughout,  $F$  is a field and all vector spaces are over  $F$ . No finite-dimensionality is assumed until we reach bases, where it is stated.

### 14.2.1 What the object has to do

Here is the goal, stated before its terms are defined. We want a space  $X$  and a bilinear map  $\tau: V \times W \rightarrow X$  such that *every bilinear map  $\beta$  out of  $V \times W$  can be pushed through  $\tau$ , and what comes out the other side is linear*. Two demands hide in that sentence. There must be **at least one** way to push  $\beta$  through, or the property is useless; and **at most one**, or we cannot recover  $\beta$  from its linear shadow. So we ask for existence and uniqueness together.

*A tensor product is the most efficient bilinear map out of  $V \times W$ : every other one is it, followed by a single linear map.*

#### Definition 14.2.1 — Tensor Product

Let  $V$  and  $W$  be vector spaces over  $F$ . A **tensor product of  $V$  and  $W$**  is a pair  $(X, \tau)$ , where  $X$  is a vector space over  $F$  and  $\tau: V \times W \rightarrow X$  is a bilinear map, satisfying:

- (T1) (**universal property**) **For every** vector space  $Z$  over  $F$  and **every** bilinear map  $\beta: V \times W \rightarrow Z$ , there is a **unique** linear map  $\bar{\beta}: X \rightarrow Z$  with  $\bar{\beta} \circ \tau = \beta$ .

In words, clause by clause. **For every**  $Z$ : the property is a demand about *all* targets at once, not about one favorite target. **Every bilinear**  $\beta$ : the arbitrary bilinear map is the input; nothing is assumed about it beyond Definition 14.1.1 with  $k = 2$ . **There is a unique linear**  $\bar{\beta}$ : exactly one, so both existence and uniqueness are being asserted. **With**  $\bar{\beta} \circ \tau = \beta$ : this is an equality of functions on  $V \times W$ , namely  $\bar{\beta}(\tau(\mathbf{v}, \mathbf{w})) = \beta(\mathbf{v}, \mathbf{w})$  for all  $\mathbf{v}, \mathbf{w}$ . We say  $\beta$  **factors through**  $\tau$ , and picture it as a triangle: the two routes from  $V \times W$  to  $Z$ , one direct and one through  $X$ , give the same answer.

Notice what the definition does **not** say. It says nothing about the elements of  $X$ , nothing about a formula, and nothing about a basis: it is a statement purely about maps. That is deliberate, since a definition naming elements would depend on a construction, and there are several.

*Warning.* The definition does not say that every element of  $X$  has the form  $\tau(\mathbf{v}, \mathbf{w})$ . It says that the values  $\tau(\mathbf{v}, \mathbf{w})$  are enough to *determine* a linear map on  $X$  — which, as we prove below, means they span  $X$ , not that they exhaust it. Sums of them are generally not of that form. This is the single most common misreading of the tensor product, and we return to it with a proof at the end of the section.

## 14.2.2 Any two tensor products are the same

Before building anything, we settle the word “the”. If two pairs both satisfy (T1), each one’s property produces a map to the other, and the composites have no choice but to be identities. This is the argument Chapter 13 used for the Clifford algebra (Proposition 13.12.1), and it is the argument we will use in four more sections.

### Theorem 14.2.1 — Uniqueness of the Tensor Product

Let  $V$  and  $W$  be vector spaces over  $F$ , and let  $(X, \tau)$  and  $(X', \tau')$  both be tensor products of  $V$  and  $W$ . Then there is a **unique** isomorphism  $\varphi: X \rightarrow X'$  with  $\varphi \circ \tau = \tau'$ .

*Idea.* Three steps. ① Each pair’s universal property, applied with the *other* pair’s bilinear map as input, produces a linear map; so we get  $\varphi: X \rightarrow X'$  and  $\psi: X' \rightarrow X$  at once, for free. ② Compose them. The composite  $\psi\varphi$  is a linear self-map of  $X$  that does nothing to  $\tau$  — and so is the identity. ③ The uniqueness half of (T1), applied with  $Z = X$  and  $\beta = \tau$ , says there is only **one** linear self-map doing nothing to  $\tau$ . So the two coincide. The uniqueness half is the engine; without it the argument produces two maps and no reason for them to be inverse.

*Proof.* **Step 1.** The map  $\tau': V \times W \rightarrow X'$  is bilinear. Apply (T1) for  $(X, \tau)$  with  $Z = X'$  and  $\beta = \tau'$ : there is a unique linear  $\varphi: X \rightarrow X'$  with  $\varphi \circ \tau = \tau'$ . Symmetrically, applying (T1) for  $(X', \tau')$  with  $Z = X$  and  $\beta = \tau$  gives a unique linear  $\psi: X' \rightarrow X$  with  $\psi \circ \tau' = \tau$ .

**Step 2.** The composite  $\psi\varphi: X \rightarrow X$  is linear (Theorem 3.4.2 (a)), and

$$(\psi\varphi) \circ \tau = \psi \circ (\varphi \circ \tau) = \psi \circ \tau' = \tau.$$

The identity  $\text{id}_X$  is also linear and satisfies  $\text{id}_X \circ \tau = \tau$ .

**Step 3.** Apply (T1) for  $(X, \tau)$  with  $Z = X$  and  $\beta = \tau$ , which is bilinear by hypothesis. It says there is **exactly one** linear map  $X \rightarrow X$  whose composite with  $\tau$  is  $\tau$ . Both

$\psi\varphi$  and  $\text{id}_X$  are such maps. Hence  $\psi\varphi = \text{id}_X$ . Swapping the roles of  $(X, \tau)$  and  $(X', \tau')$  gives  $\varphi\psi = \text{id}_{X'}$ .

So  $\varphi$  has a two-sided inverse and is a bijection, hence an isomorphism (Theorem 3.5.1). It is the unique isomorphism with  $\varphi \circ \tau = \tau'$ , because (T1) already produced it as the unique **linear** map with that property, and an isomorphism is in particular linear. This proves the theorem. ■

**Aftermath, and the template.** Read the proof again with the names removed. It says: *an object defined by a universal property is determined up to a unique isomorphism compatible with its structure map, because each object's property hands you a comparison map, and the uniqueness clause forces the composites to be identities.* Nothing in it used bilinearity, or vector spaces, or two factors. Sections 06, 07, 08 and 10 define the tensor algebra, the symmetric power, the exterior power and the Clifford algebra by universal properties of exactly this shape, and each time the uniqueness proof is these three steps with different letters. When you meet them, do not reread the argument; recognize it.

Because of Theorem 14.2.1 we may now speak of **the** tensor product and fix notation.

### Definition 14.2.2 — Notation for Tensors

When a tensor product of  $V$  and  $W$  exists, we write it  $V \otimes W$ , write  $\mathbf{v} \otimes \mathbf{w}$  for  $\tau(\mathbf{v}, \mathbf{w})$ , and call an element of the form  $\mathbf{v} \otimes \mathbf{w}$  a **simple tensor** (also called a *pure* or *decomposable* tensor). The defining property then reads: every bilinear  $\beta: V \times W \rightarrow Z$  has a unique linear  $\bar{\beta}: V \otimes W \rightarrow Z$  with  $\bar{\beta}(\mathbf{v} \otimes \mathbf{w}) = \beta(\mathbf{v}, \mathbf{w})$ .

Bilinearity of  $\tau$ , rewritten in this notation, gives the rules we compute with:

$$\begin{aligned}(\mathbf{v} + \mathbf{v}') \otimes \mathbf{w} &= \mathbf{v} \otimes \mathbf{w} + \mathbf{v}' \otimes \mathbf{w}, \\ \mathbf{v} \otimes (\mathbf{w} + \mathbf{w}') &= \mathbf{v} \otimes \mathbf{w} + \mathbf{v} \otimes \mathbf{w}', \\ (c\mathbf{v}) \otimes \mathbf{w} &= c(\mathbf{v} \otimes \mathbf{w}) = \mathbf{v} \otimes (c\mathbf{w}).\end{aligned}$$

The third line is the one to remember: **a scalar moves freely across the  $\otimes$  sign.** That is the whole reason  $\mathbf{v} \otimes \mathbf{w}$  does not determine  $\mathbf{v}$  and  $\mathbf{w}$ .

## 14.2.3 Building one: a free space modulo relations

Uniqueness is worthless if nothing satisfies (T1), so we construct a tensor product. The strategy is the one named in the chapter introduction: *take a space in which the symbols  $\mathbf{v} \otimes \mathbf{w}$  are completely unrelated, then divide out exactly the relations bilinearity demands.* Both halves need a tool.

The first tool is a space with a basis indexed by an arbitrary set. It is the space of finitely supported functions, which Chapter 1 met for  $S = \mathbb{N}$  (Exercise (B1: Finitely Supported Sequences)).

**Definition 14.2.3 — Free Vector Space on a Set**

Let  $S$  be a set and  $F$  a field. The **free vector space on  $S$  over  $F$**  is

$$F^{(S)} := \{ f: S \rightarrow F : f(s) = 0 \text{ for all but finitely many } s \},$$

a subset of the space  $F^S$  of all functions  $S \rightarrow F$  (Example (The Standard Vector Spaces) (d)), with the pointwise operations. For  $s \in S$ , let  $\delta_s \in F^{(S)}$  be the function with  $\delta_s(s) = 1$  and  $\delta_s(t) = 0$  for  $t \neq s$ .

**Lemma 14.2.2 — Universal Property of the Free Vector Space**

Let  $S$  be a set and  $F$  a field.

- (a)  $F^{(S)}$  is a subspace of  $F^S$ , and  $\{\delta_s : s \in S\}$  is a basis of it.
- (b) For **every** vector space  $Z$  over  $F$  and **every** function  $g: S \rightarrow Z$  — no structure on  $Z$  is assumed, since  $S$  is only a set — there is **exactly one** linear map  $\bar{g}: F^{(S)} \rightarrow Z$  with  $\bar{g}(\delta_s) = g(s)$  for all  $s \in S$ .

*Proof.*

- (a) We use Theorem 1.2.1. The zero function vanishes everywhere, so it lies in  $F^{(S)}$ . If  $f, h \in F^{(S)}$  vanish outside the finite sets  $A$  and  $B$ , then  $f + h$  vanishes outside  $A \cup B$ , which is finite, and  $cf$  vanishes outside  $A$ . So  $F^{(S)}$  is a subspace. Each  $\delta_s$  lies in  $F^{(S)}$ . *Spanning:* if  $f \in F^{(S)}$  is non-zero exactly on the finite set  $\{s_1, \dots, s_r\}$ , then  $f$  and  $\sum_i f(s_i)\delta_{s_i}$  take the same value at every point of  $S$ , hence are equal. *Independence:* if  $\sum_i a_i\delta_{s_i} = 0$  for distinct  $s_1, \dots, s_r$ , evaluate at  $s_j$  to get  $a_j = 0$ . By Definition 1.5.1,  $\{\delta_s\}$  is a basis, possibly infinite in the sense of Theorem 1.8.2.
- (b) Immediate from (a) and Theorem 3.1.6, which prescribes a linear map arbitrarily on a basis, in one and only one way.

■

In words: **a function on a set becomes a linear map on the free space, for free.** That is the point of the construction, and the whole of it:  $F^{(S)}$  imposes no relations among the  $\delta_s$ , so nothing can obstruct a prescription.

The second tool is the quotient. Chapter 3 recorded exactly the statement we need: a linear map that kills a subspace  $U$  passes, in exactly one way, to a linear map on  $V/U$  (Theorem 3.9.10 (a)). We use it in the form “to define a map out of a quotient, define it upstairs and check it kills the relations”.

**Theorem 14.2.3 — Existence of the Tensor Product**

For every pair of vector spaces  $V, W$  over  $F$ , a tensor product  $(X, \tau)$  of  $V$  and  $W$  exists.

*Idea.* Write down the symbols  $\mathbf{v} \otimes \mathbf{w}$  as a basis of a huge space  $P$ , one basis vector for each **pair** of vectors — so at this stage  $\mathbf{v} \otimes \mathbf{w}$  has no properties at all. The four bilinearity rules are then four families of elements of  $P$  that we want to be zero, so divide by the subspace they span. A bilinear  $\beta$  becomes a linear map on  $P$  by [Lemma 14.2.2](#), kills each relation because it is bilinear, and therefore passes to the quotient by [Theorem 3.9.10](#). Uniqueness of the induced map is spanning: the cosets of the basis vectors span the quotient.

*Proof.* Let  $S = V \times W$ , regarded as a **set**, and let  $P = F^{(S)}$  be the free vector space on it ([Definition 14.2.3](#)), with basis  $\{\delta_{(\mathbf{v}, \mathbf{w})}\}$ . Let  $R \subseteq P$  be the span of all elements of the four forms

$$\begin{aligned} \delta_{(\mathbf{v}+\mathbf{v}', \mathbf{w})} - \delta_{(\mathbf{v}, \mathbf{w})} - \delta_{(\mathbf{v}', \mathbf{w})}, & \quad \delta_{(\mathbf{v}, \mathbf{w}+\mathbf{w}')} - \delta_{(\mathbf{v}, \mathbf{w})} - \delta_{(\mathbf{v}, \mathbf{w}')}, \\ \delta_{(c\mathbf{v}, \mathbf{w})} - c\delta_{(\mathbf{v}, \mathbf{w})}, & \quad \delta_{(\mathbf{v}, c\mathbf{w})} - c\delta_{(\mathbf{v}, \mathbf{w})}, \end{aligned}$$

taken over all  $\mathbf{v}, \mathbf{v}' \in V$ , all  $\mathbf{w}, \mathbf{w}' \in W$  and all  $c \in F$ . A span is a subspace, so  $R$  is one. Put

$$X := P/R, \quad \tau(\mathbf{v}, \mathbf{w}) := \pi(\delta_{(\mathbf{v}, \mathbf{w})}),$$

where  $\pi: P \rightarrow P/R$  is the quotient map, which is linear with kernel  $R$  ([Theorem 3.9.4](#)).

$\tau$  is bilinear. The element  $\delta_{(\mathbf{v}+\mathbf{v}', \mathbf{w})} - \delta_{(\mathbf{v}, \mathbf{w})} - \delta_{(\mathbf{v}', \mathbf{w})}$  lies in  $R = \ker \pi$ , so applying  $\pi$  and using its linearity gives  $\tau(\mathbf{v} + \mathbf{v}', \mathbf{w}) - \tau(\mathbf{v}, \mathbf{w}) - \tau(\mathbf{v}', \mathbf{w}) = \mathbf{0}$ . The other three families give, in the same way, additivity in the second slot and the two scalar rules. These are the clauses of [Definition 14.1.1](#) for  $k = 2$ .

(T1), *existence.* Let  $Z$  be a vector space and  $\beta: V \times W \rightarrow Z$  bilinear. Regarding  $\beta$  as a **function** on the set  $S$ , [Lemma 14.2.2](#) (b) gives a unique linear  $b: P \rightarrow Z$  with  $b(\delta_{(\mathbf{v}, \mathbf{w})}) = \beta(\mathbf{v}, \mathbf{w})$ . It kills every generator of  $R$ : for instance

$$b(\delta_{(\mathbf{v}+\mathbf{v}', \mathbf{w})} - \delta_{(\mathbf{v}, \mathbf{w})} - \delta_{(\mathbf{v}', \mathbf{w})}) = \beta(\mathbf{v} + \mathbf{v}', \mathbf{w}) - \beta(\mathbf{v}, \mathbf{w}) - \beta(\mathbf{v}', \mathbf{w}) = \mathbf{0}$$

because  $\beta$  is linear in its first slot, and the other three families vanish by the other three clauses of bilinearity. Since  $b$  is linear and vanishes on a spanning set of  $R$ , it vanishes on  $R$  ([Theorem 3.1.4](#)); that is,  $R \subseteq \ker b$ . By [Theorem 3.9.10](#) (a) there is exactly one linear  $\bar{\beta}: X \rightarrow Z$  with  $\bar{\beta} \circ \pi = b$ , and then

$$\bar{\beta}(\tau(\mathbf{v}, \mathbf{w})) = \bar{\beta}(\pi(\delta_{(\mathbf{v}, \mathbf{w})})) = b(\delta_{(\mathbf{v}, \mathbf{w})}) = \beta(\mathbf{v}, \mathbf{w}),$$

so  $\bar{\beta} \circ \tau = \beta$ .

(T1), *uniqueness.* Suppose  $L: X \rightarrow Z$  is linear with  $L \circ \tau = \beta$ . Since  $\pi$  is surjective and the  $\delta_{(\mathbf{v}, \mathbf{w})}$  span  $P$ , the elements  $\tau(\mathbf{v}, \mathbf{w}) = \pi(\delta_{(\mathbf{v}, \mathbf{w})})$  span  $X$ . On each of them,  $L$  and  $\bar{\beta}$  agree, both giving  $\beta(\mathbf{v}, \mathbf{w})$ . Two linear maps that agree on a spanning set agree everywhere, by [Theorem 3.1.4](#). Hence  $L = \bar{\beta}$ .

So  $(X, \tau)$  satisfies (T1), which proves the theorem. ■

**Aftermath.** The construction is ugly and the object is not. That is normal for a universal property: the model is a scaffolding, and after this proof we never open it again. Every later fact about  $V \otimes W$  in this book is derived from (T1), never from cosets of formal sums — which is also why the identification  $F^m \otimes F^n \cong M_{m \times n}(F)$  below is legitimate as an alternative model,

and much more pleasant to compute in.

One property is worth extracting at once, because we will use it constantly and because it is the second appearance of the self-map trick.

### Proposition 14.2.4 — Simple Tensors Span

Let  $(V \otimes W, \tau)$  be a tensor product. Then  $V \otimes W = \text{span}\{\mathbf{v} \otimes \mathbf{w} : \mathbf{v} \in V, \mathbf{w} \in W\}$ .

*Proof.* Let  $Y = \text{span}\{\mathbf{v} \otimes \mathbf{w}\}$ , a subspace of  $V \otimes W$ , and let  $j: Y \rightarrow V \otimes W$  be the inclusion, which is linear. Restricting the codomain of  $\tau$  gives a bilinear map  $\tau_Y: V \times W \rightarrow Y$ ,  $(\mathbf{v}, \mathbf{w}) \mapsto \mathbf{v} \otimes \mathbf{w}$ ; it is bilinear because  $\tau$  is and because  $Y$  is a subspace, so the identities hold already in  $Y$ . By (T1) with  $Z = Y$  there is a linear  $L: V \otimes W \rightarrow Y$  with  $L \circ \tau = \tau_Y$ . Then  $jL$  is a linear self-map of  $V \otimes W$  with  $jL\tau = j\tau_Y = \tau$ , and so is  $\text{id}$ . By the uniqueness clause of (T1) with  $Z = V \otimes W$  and  $\beta = \tau$ , we get  $jL = \text{id}$ . Hence  $j$  is surjective, so  $Y = V \otimes W$ . ■

*Remark.* That proof used no basis, no construction and no finite dimension — only (T1). It is the same three lines as Step 1 of Proposition 13.12.3’s proof in Chapter 13, where the generated subalgebra was shown to be everything. Once you see “a subobject that receives the structure map must be the whole thing”, you can reuse it verbatim.

## 14.2.4 A basis and a dimension

Now we can count. The prediction at the end of the previous section was  $\dim(V \otimes W) = (\dim V)(\dim W)$ , and here it is, with a basis attached.

### Theorem 14.2.5 — Basis of a Tensor Product

Let  $V$  and  $W$  be finite-dimensional, with bases  $(\mathbf{v}_1, \dots, \mathbf{v}_n)$  of  $V$  and  $(\mathbf{w}_1, \dots, \mathbf{w}_m)$  of  $W$ . Then the  $nm$  simple tensors

$$\mathbf{v}_i \otimes \mathbf{w}_j, \quad 1 \leq i \leq n, \quad 1 \leq j \leq m,$$

form a basis of  $V \otimes W$ . In particular  $\dim(V \otimes W) = (\dim V)(\dim W)$ .

*Idea.* Spanning is bilinearity: expand both slots of a simple tensor and use Proposition 14.2.4. Independence needs linear functionals on  $V \otimes W$  that pick out one coefficient — and (T1) manufactures functionals out of bilinear forms, while Theorem 4.1.3 supplies all the bilinear forms we could want, namely  $(\mathbf{x}, \mathbf{y}) \mapsto \varphi_p(\mathbf{x})\psi_q(\mathbf{y})$ .

*Proof. Spanning.* By Proposition 14.2.4 it suffices to write each simple tensor in terms of the listed ones. Let  $\mathbf{v} = \sum_i a_i \mathbf{v}_i$  and  $\mathbf{w} = \sum_j b_j \mathbf{w}_j$ . Expanding one slot at a time by

the rules following [Definition 14.2.2](#),

$$\mathbf{v} \otimes \mathbf{w} = \left( \sum_i a_i \mathbf{v}_i \right) \otimes \left( \sum_j b_j \mathbf{w}_j \right) = \sum_{i,j} a_i b_j (\mathbf{v}_i \otimes \mathbf{w}_j).$$

*Independence.* Let  $(\varphi_1, \dots, \varphi_n)$  and  $(\psi_1, \dots, \psi_m)$  be the bases of  $V^*$  and  $W^*$  dual to the given ones, so  $\varphi_p(\mathbf{v}_i) = \delta_{pi}$  and  $\psi_q(\mathbf{w}_j) = \delta_{qj}$  ([Theorem 4.1.3](#)). Fix  $p$  and  $q$ . The function

$$\beta_{pq}: V \times W \rightarrow F, \quad \beta_{pq}(\mathbf{x}, \mathbf{y}) = \varphi_p(\mathbf{x}) \psi_q(\mathbf{y}),$$

is bilinear: freezing  $\mathbf{y}$  leaves  $\psi_q(\mathbf{y})\varphi_p$ , a scalar multiple of a linear functional, and symmetrically. By (T1) there is a linear  $f_{pq}: V \otimes W \rightarrow F$  with  $f_{pq}(\mathbf{x} \otimes \mathbf{y}) = \varphi_p(\mathbf{x})\psi_q(\mathbf{y})$ ; in particular  $f_{pq}(\mathbf{v}_i \otimes \mathbf{w}_j) = \delta_{pi}\delta_{qj}$ .

Now suppose  $\sum_{i,j} c_{ij} (\mathbf{v}_i \otimes \mathbf{w}_j) = \mathbf{0}$ . Apply  $f_{pq}$ . By linearity the left side becomes  $\sum_{i,j} c_{ij} \delta_{pi} \delta_{qj} = c_{pq}$ , and the right side is 0. Hence  $c_{pq} = 0$ . Since  $p$  and  $q$  were arbitrary, all coefficients vanish.

So the  $nm$  listed tensors form a basis, and  $\dim(V \otimes W) = nm$ . This proves the theorem. ■

**Aftermath: the count agrees.** In [Theorem 14.1.2](#) we found  $\dim \mathcal{M}(V, W; Z) = (\dim Z)(\dim V)(\dim W)$ , and we argued that a universal object  $X$  would have to satisfy  $\dim \mathcal{L}(X, Z) = \dim \mathcal{M}(V, W; Z)$ , forcing  $\dim X = (\dim V)(\dim W)$ . [Theorem 14.2.5](#) delivers exactly that number, and the agreement is not a coincidence — it is the universal property, counted. Here is the statement behind it.

### Corollary 14.2.6 — Bilinear Maps Are Linear Maps on the Tensor Product

Let  $V, W, Z$  be vector spaces over  $F$ . The map

$$\Xi: \mathcal{L}(V \otimes W, Z) \rightarrow \mathcal{M}(V, W; Z), \quad L \mapsto L \circ \tau,$$

is an isomorphism of vector spaces. When  $V, W$  and  $Z$  are finite-dimensional, both sides have dimension  $(\dim Z)(\dim V)(\dim W)$ .

*Proof.*  $\Xi$  lands where claimed:  $L \circ \tau$  is bilinear, since freezing one slot of  $\tau$  gives a linear map and composing with the linear  $L$  keeps it linear ([Theorem 3.4.2](#) (a)). It is linear in  $L$  by [Theorem 3.4.2](#) (d), which gives  $(L_1 + L_2) \circ \tau = L_1 \tau + L_2 \tau$  and  $(cL) \circ \tau = c(L \tau)$ . It is surjective by the existence half of (T1) and injective by the uniqueness half: if  $L \tau = 0$ , then both  $L$  and the zero map are linear maps composing with  $\tau$  to give the zero bilinear map, so  $L = 0$ , and [Theorem 3.2.3](#) applies. The dimension count is [Theorem 14.1.2](#) on one side and [Theorem 3.6.4](#) together with [Theorem 14.2.5](#) on the other. ■

*Quick check.* What is  $V \otimes W$  when  $W = F$ , viewed as a one-dimensional space over itself? What is it when  $W = \{\mathbf{0}\}$ ?



*Solution.* For  $W = F$ :  $\dim(V \otimes F) = (\dim V) \cdot 1 = \dim V$  by [Theorem 14.2.5](#), and indeed  $\mathbf{v} \mapsto \mathbf{v} \otimes 1$  is an isomorphism  $V \rightarrow V \otimes F$ , since it carries a basis  $(\mathbf{v}_i)$  to the basis  $(\mathbf{v}_i \otimes 1)$ . For  $W = \{\mathbf{0}\}$ : the dimension is 0, so  $V \otimes \{\mathbf{0}\} = \{\mathbf{0}\}$ . Both are the degenerate cases one should be able to predict from the product formula, and both come out right.

## 14.2.5 Coordinates: the tensor product of coordinate spaces

An abstract object is easier to trust once it has a concrete model. For coordinate spaces there is a very good one:  $F^m \otimes F^n$  is the space of  $m \times n$  matrices, with the simple tensor  $\mathbf{x} \otimes \mathbf{y}$  corresponding to the outer product  $\mathbf{xy}^\top$ .

### Theorem 14.2.7 — $F^m \otimes F^n \cong M_{m \times n}(F)$

Let  $m, n \geq 1$ . There is a unique linear map

$$\Theta: F^m \otimes F^n \rightarrow M_{m \times n}(F), \quad \Theta(\mathbf{x} \otimes \mathbf{y}) = \mathbf{xy}^\top,$$

and it is an isomorphism, carrying  $\mathbf{e}_i \otimes \mathbf{f}_j$  to the matrix unit  $\mathbf{E}_{ij}$ . Under  $\Theta$ , the simple tensors correspond exactly to the matrices of rank at most 1.

*Proof.* The map  $(\mathbf{x}, \mathbf{y}) \mapsto \mathbf{xy}^\top$  is bilinear, since its  $(i, j)$  entry is  $x_i y_j$  ([Exercise \(B1: Determine which are multilinear\)](#) (c)). By (T1) there is a unique linear  $\Theta$  with  $\Theta(\mathbf{x} \otimes \mathbf{y}) = \mathbf{xy}^\top$ . On standard basis vectors,  $\mathbf{e}_i \mathbf{f}_j^\top$  has a single 1 in position  $(i, j)$ , so  $\Theta(\mathbf{e}_i \otimes \mathbf{f}_j) = \mathbf{E}_{ij}$ .

By [Theorem 14.2.5](#) the  $mn$  tensors  $\mathbf{e}_i \otimes \mathbf{f}_j$  form a basis of  $F^m \otimes F^n$ , and the  $mn$  matrix units form a basis of  $M_{m \times n}(F)$  ([Example \(Standard Bases\)](#) (c)). By [Theorem 3.1.5](#) there is a linear  $\Psi: M_{m \times n}(F) \rightarrow F^m \otimes F^n$  with  $\Psi(\mathbf{E}_{ij}) = \mathbf{e}_i \otimes \mathbf{f}_j$ . Then  $\Psi\Theta$  and  $\Theta\Psi$  are linear maps fixing a basis, hence are the identities by the uniqueness half of [Theorem 3.1.5](#). So  $\Theta$  is an isomorphism.

For the last claim, a non-zero  $\mathbf{xy}^\top$  is a product of an  $m \times 1$  and a  $1 \times n$  matrix, so it has rank 1 by [Proposition 3.8.2](#); and  $\mathbf{xy}^\top = \mathbf{0}$  when  $\mathbf{x} = \mathbf{0}$  or  $\mathbf{y} = \mathbf{0}$ , of rank 0. Conversely, a matrix  $\mathbf{A}$  of rank 1 factors as  $\mathbf{A} = \mathbf{xy}^\top$  with  $\mathbf{x} \in F^m$  and  $\mathbf{y}^\top \in M_{1 \times n}(F)$ , again by [Proposition 3.8.2](#), and the zero matrix is  $\mathbf{0} \otimes \mathbf{0}$ . This proves the theorem. ■

### Example. (Computing in $F^2 \otimes F^3$ )

Work in  $F^2 \otimes F^3$  with standard bases  $(\mathbf{e}_1, \mathbf{e}_2)$  and  $(\mathbf{f}_1, \mathbf{f}_2, \mathbf{f}_3)$ .

- Expand  $(\mathbf{e}_1 + 2\mathbf{e}_2) \otimes (3\mathbf{f}_1 - \mathbf{f}_2)$  in the basis of [Theorem 14.2.5](#), and write the corresponding matrix.
- Decide whether  $t = \mathbf{e}_1 \otimes \mathbf{f}_1 + 2\mathbf{e}_1 \otimes \mathbf{f}_2 + 3\mathbf{e}_2 \otimes \mathbf{f}_1 + 6\mathbf{e}_2 \otimes \mathbf{f}_2$  is a simple tensor, and if so write it as one.
- Decide whether  $u = \mathbf{e}_1 \otimes \mathbf{f}_1 + \mathbf{e}_2 \otimes \mathbf{f}_2$  is a simple tensor.



*Solution.*

- (a) Expanding one slot at a time,

$$(\mathbf{e}_1 + 2\mathbf{e}_2) \otimes (3\mathbf{f}_1 - \mathbf{f}_2) = 3\mathbf{e}_1 \otimes \mathbf{f}_1 - \mathbf{e}_1 \otimes \mathbf{f}_2 + 6\mathbf{e}_2 \otimes \mathbf{f}_1 - 2\mathbf{e}_2 \otimes \mathbf{f}_2.$$

The coefficient of  $\mathbf{e}_i \otimes \mathbf{f}_j$  is the  $(i, j)$  entry of the matrix, so  $\Theta$  sends it to

$$\begin{pmatrix} 3 & -1 & 0 \\ 6 & -2 & 0 \end{pmatrix} = \begin{pmatrix} 1 \\ 2 \end{pmatrix} \begin{pmatrix} 3 & -1 & 0 \end{pmatrix},$$

which is the outer product predicted by [Theorem 14.2.7](#).

- (b) The coefficients give  $\Theta(t) = \begin{pmatrix} 1 & 2 & 0 \\ 3 & 6 & 0 \end{pmatrix}$ . Its second row is 3 times its first, so every  $2 \times 2$  minor vanishes and the rank is 1 ([Theorem 6.7.1](#)). By [Theorem 14.2.7](#),  $t$  is simple. Factoring,  $\Theta(t) = (1, 3)^\top (1, 2, 0)$ , so

$$t = (\mathbf{e}_1 + 3\mathbf{e}_2) \otimes (\mathbf{f}_1 + 2\mathbf{f}_2),$$

which expands back to  $t$ .

- (c) Here  $\Theta(u) = \begin{pmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \end{pmatrix}$ , whose first two columns are independent, so its rank is 2. By [Theorem 14.2.7](#),  $u$  is **not** simple. No amount of rewriting will turn it into a single  $\mathbf{v} \otimes \mathbf{w}$ .

## 14.2.6 Three things a tensor is not

Part (c) above is not a curiosity; it is the central negative fact about tensor products, and the first of three misreadings worth naming before they happen.

### 14.2.6.1 Not every tensor is simple

#### Example. (A tensor that is not simple)

In  $F^2 \otimes F^2$ , the element  $u = \mathbf{e}_1 \otimes \mathbf{e}_1 + \mathbf{e}_2 \otimes \mathbf{e}_2$  is not a simple tensor, for any field  $F$ .

*Idea.* Two routes, and both are worth having. The quick one:  $\Theta(u) = \mathbf{I}_2$ , of rank 2, while simple tensors have rank at most 1. The portable one: suppose  $u = \mathbf{v} \otimes \mathbf{w}$ , hit it with the four coordinate functionals  $f_{pq}$  built in the proof of [Theorem 14.2.5](#), and read off four scalar equations that cannot all hold.

*Proof.* Suppose  $u = \mathbf{v} \otimes \mathbf{w}$  for some  $\mathbf{v}, \mathbf{w} \in F^2$ . Let  $(\varphi_1, \varphi_2)$  be the basis of  $(F^2)^*$  dual to  $(\mathbf{e}_1, \mathbf{e}_2)$ , and for  $p, q \in \{1, 2\}$  let  $f_{pq}: F^2 \otimes F^2 \rightarrow F$  be the linear map with  $f_{pq}(\mathbf{x} \otimes \mathbf{y}) = \varphi_p(\mathbf{x})\varphi_q(\mathbf{y})$ , supplied by (T1) as in the proof of [Theorem 14.2.5](#). Applying  $f_{pq}$  to both sides of  $u = \mathbf{v} \otimes \mathbf{w}$  gives

$$v_p w_q = f_{pq}(u) = \delta_{p1}\delta_{q1} + \delta_{p2}\delta_{q2},$$

that is,  $v_1 w_1 = 1$ ,  $v_2 w_2 = 1$ ,  $v_1 w_2 = 0$  and  $v_2 w_1 = 0$ . From  $v_1 w_1 = 1$  we get  $v_1 \neq 0$ , so  $v_1 w_2 = 0$  forces  $w_2 = 0$  — and then  $v_2 w_2 = 0 \neq 1$ . This contradiction shows that no

such  $\mathbf{v}, \mathbf{w}$  exist, so  $u$  is not simple. ■

*Warning.* A general element of  $V \otimes W$  is a *sum* of simple tensors, not a simple tensor. Proposition 14.2.4 says the simple tensors span; Example (A tensor that is not simple) says they do not exhaust. A statement proved only for simple tensors is therefore proved for nothing until you extend it by linearity — which is legitimate precisely when the statement is about a linear map, and illegitimate when it is about, say, ranks or products. Whenever you want to define something on  $V \otimes W$ , do not write “define it on  $\mathbf{v} \otimes \mathbf{w}$  and extend”: use (T1), which is that move done correctly.

The smallest number of simple tensors needed to write a given tensor is a genuine invariant, and for coordinate spaces Theorem 14.2.7 identifies it: it is the rank of the corresponding matrix. So “not simple” means “rank at least 2”. A later section meets the same failure for elements of  $\Lambda^2 V$  that are not wedges of two vectors, and the closing section meets it again under the name *entanglement*.

#### 14.2.6.2 A simple tensor vanishes only for an obvious reason

##### Proposition 14.2.8 — When a Simple Tensor Is Zero

Let  $V$  and  $W$  be finite-dimensional,  $\mathbf{v} \in V$  and  $\mathbf{w} \in W$ . Then  $\mathbf{v} \otimes \mathbf{w} = \mathbf{0}$  if and only if  $\mathbf{v} = \mathbf{0}$  or  $\mathbf{w} = \mathbf{0}$ .

*Proof.* ( $\Leftarrow$ ) If  $\mathbf{v} = \mathbf{0}$ , then  $\mathbf{v} \otimes \mathbf{w} = (\mathbf{0} \cdot \mathbf{0}) \otimes \mathbf{w} = \mathbf{0}(\mathbf{0} \otimes \mathbf{w}) = \mathbf{0}$ , using that a scalar passes across  $\otimes$ . The case  $\mathbf{w} = \mathbf{0}$  is the same with the slots swapped.

( $\Rightarrow$ ) Suppose  $\mathbf{v} \neq \mathbf{0}$  and  $\mathbf{w} \neq \mathbf{0}$ . By Theorem 4.1.5 (a) there are  $\varphi \in V^*$  and  $\psi \in W^*$  with  $\varphi(\mathbf{v}) = 1$  and  $\psi(\mathbf{w}) = 1$ . The map  $(\mathbf{x}, \mathbf{y}) \mapsto \varphi(\mathbf{x})\psi(\mathbf{y})$  is bilinear, so (T1) gives a linear  $f: V \otimes W \rightarrow F$  with  $f(\mathbf{x} \otimes \mathbf{y}) = \varphi(\mathbf{x})\psi(\mathbf{y})$ . Then  $f(\mathbf{v} \otimes \mathbf{w}) = 1 \neq 0$ , while  $f(\mathbf{0}) = 0$ . Hence  $\mathbf{v} \otimes \mathbf{w} \neq \mathbf{0}$ . This proves the proposition. ■

*Warning.* A product of non-zero things can be zero in many algebraic settings, but not here. It is tempting to expect  $V \otimes W$  to behave like a ring with zero divisors, so that some  $\mathbf{v} \otimes \mathbf{w}$  vanishes accidentally. It does not: Proposition 14.2.8 says the only way to get  $\mathbf{0}$  is to feed in  $\mathbf{0}$ . The zero divisors appear one level later — in the algebras of the following sections, where products of non-zero *sums* of simple tensors can vanish.

#### 14.2.6.3 A simple tensor does not remember its factors

##### Proposition 14.2.9 — When Two Simple Tensors Agree

Let  $V$  and  $W$  be finite-dimensional, and let  $\mathbf{v}, \mathbf{v}' \in V$  and  $\mathbf{w}, \mathbf{w}' \in W$  with  $\mathbf{v} \otimes \mathbf{w} \neq \mathbf{0}$ . Then

$$\mathbf{v} \otimes \mathbf{w} = \mathbf{v}' \otimes \mathbf{w}'$$

if and only if there is a scalar  $c \neq 0$  with  $\mathbf{v}' = c\mathbf{v}$  and  $\mathbf{w}' = c^{-1}\mathbf{w}$ .

*Idea.* The “if” half is the scalar rule. For “only if”, the danger is that  $\mathbf{v}'$  points somewhere new. Suppose it does; then  $(\mathbf{v}, \mathbf{v}')$  is independent, so there is a functional killing  $\mathbf{v}$  and not  $\mathbf{v}'$ , and the corresponding  $f$  on  $V \otimes W$  separates the two sides.

*Proof.* ( $\Leftarrow$ ) If  $\mathbf{v}' = c\mathbf{v}$  and  $\mathbf{w}' = c^{-1}\mathbf{w}$ , then  $\mathbf{v}' \otimes \mathbf{w}' = (c\mathbf{v}) \otimes (c^{-1}\mathbf{w}) = cc^{-1}(\mathbf{v} \otimes \mathbf{w}) = \mathbf{v} \otimes \mathbf{w}$ , moving each scalar across  $\otimes$ .

( $\Rightarrow$ ) Suppose  $\mathbf{v} \otimes \mathbf{w} = \mathbf{v}' \otimes \mathbf{w}' \neq \mathbf{0}$ . By Proposition 14.2.8 all four vectors are non-zero. Suppose, for a contradiction, that  $\mathbf{v}' \notin \text{span}(\mathbf{v})$ . Then  $(\mathbf{v}, \mathbf{v}')$  is linearly independent, so it extends to a basis of  $V$  (Theorem 1.6.5), and the dual basis (Theorem 4.1.3) provides  $\varphi \in V^*$  with  $\varphi(\mathbf{v}) = 0$  and  $\varphi(\mathbf{v}') = 1$ . Since  $\mathbf{w}' \neq \mathbf{0}$ , Theorem 4.1.5 (a) provides  $\psi \in W^*$  with  $\psi(\mathbf{w}') = 1$ . By (T1) there is a linear  $f: V \otimes W \rightarrow F$  with  $f(\mathbf{x} \otimes \mathbf{y}) = \varphi(\mathbf{x})\psi(\mathbf{y})$ . Evaluating both sides of  $\mathbf{v} \otimes \mathbf{w} = \mathbf{v}' \otimes \mathbf{w}'$ ,

$$0 = \varphi(\mathbf{v})\psi(\mathbf{w}) = f(\mathbf{v} \otimes \mathbf{w}) = f(\mathbf{v}' \otimes \mathbf{w}') = \varphi(\mathbf{v}')\psi(\mathbf{w}') = 1,$$

which is absurd. Hence  $\mathbf{v}' = c\mathbf{v}$  for some  $c \in F$ , and  $c \neq 0$  because  $\mathbf{v}' \neq \mathbf{0}$ .

Then  $\mathbf{v} \otimes \mathbf{w} = (c\mathbf{v}) \otimes \mathbf{w}' = \mathbf{v} \otimes (c\mathbf{w}')$ , so  $\mathbf{v} \otimes (\mathbf{w} - c\mathbf{w}') = \mathbf{0}$  by bilinearity. Since  $\mathbf{v} \neq \mathbf{0}$ , Proposition 14.2.8 forces  $\mathbf{w} - c\mathbf{w}' = \mathbf{0}$ , that is,  $\mathbf{w}' = c^{-1}\mathbf{w}$ . This proves the proposition. ■

*Warning.*  $\mathbf{v} \otimes \mathbf{w} = \mathbf{v}' \otimes \mathbf{w}'$  does not mean  $\mathbf{v} = \mathbf{v}'$  and  $\mathbf{w} = \mathbf{w}'$ . Already  $2\mathbf{v} \otimes \mathbf{w} = \mathbf{v} \otimes 2\mathbf{w}$ , with different factors on the two sides. So “the first factor of a simple tensor” is not a well-defined function: writing  $t = \mathbf{v} \otimes \mathbf{w}$  and then speaking of “the  $\mathbf{v}$  of  $t$ ” is an error. What Proposition 14.2.9 salvages is that the *lines*  $\text{span}(\mathbf{v})$  and  $\text{span}(\mathbf{w})$  are determined, which is exactly the information carried by a rank-one matrix: its column space and its row space.

*Quick check.* In  $\mathbb{R}^2 \otimes \mathbb{R}^2$ , is  $\mathbf{e}_1 \otimes \mathbf{e}_2 + \mathbf{e}_2 \otimes \mathbf{e}_1$  a simple tensor?

*Solution.* No. Its matrix under  $\Theta$  is  $\begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}$ , of determinant  $-1 \neq 0$ , hence of rank 2 (Theorem 6.5.2). By Theorem 14.2.7 a simple tensor would have rank at most 1. Note how little the symmetry of the expression helps: being a sum of two simple tensors is the generic situation, not a special one.

## 14.2.7 Exercises

### 14.2.7.1 A. Check your understanding

#### Exercise A1.

- State the universal property (T1) of the tensor product, with every quantifier.
- What does Theorem 14.2.1 assert, and which clause of (T1) makes the composites identities?
- Give  $\dim(V \otimes W)$  in terms of  $\dim V$  and  $\dim W$ , and a basis.
- True or false: every element of  $V \otimes W$  is of the form  $\mathbf{v} \otimes \mathbf{w}$ . Justify your answer.
- True or false: if  $\mathbf{v} \otimes \mathbf{w} = \mathbf{0}$  then  $\mathbf{v} = \mathbf{0}$ . Justify your answer.

- (f) Explain in one sentence why the tensor product is **not** defined by a formula for its elements.

*Solution.*

- (a) A pair  $(X, \tau)$  with  $\tau: V \times W \rightarrow X$  bilinear such that for **every** vector space  $Z$  and **every** bilinear  $\beta: V \times W \rightarrow Z$  there is a **unique** linear  $\bar{\beta}: X \rightarrow Z$  with  $\bar{\beta} \circ \tau = \beta$  (Definition 14.2.1).
- (b) That any two tensor products of  $V$  and  $W$  are isomorphic by a unique isomorphism commuting with the two structure maps. The **uniqueness** clause: it says there is only one linear self-map  $X \rightarrow X$  composing with  $\tau$  to give  $\tau$ , and since both  $\psi\varphi$  and  $\text{id}_X$  are such maps, they are equal.
- (c)  $(\dim V)(\dim W)$ , with basis  $(\mathbf{v}_i \otimes \mathbf{w}_j)$  for bases  $(\mathbf{v}_i)$  of  $V$  and  $(\mathbf{w}_j)$  of  $W$  (Theorem 14.2.5).
- (d) False. In  $F^2 \otimes F^2$ ,  $\mathbf{e}_1 \otimes \mathbf{e}_1 + \mathbf{e}_2 \otimes \mathbf{e}_2$  is not simple (Example (A tensor that is not simple)).
- (e) False as stated: it says  $\mathbf{v} = \mathbf{0}$  or  $\mathbf{w} = \mathbf{0}$  (Proposition 14.2.8). For instance  $\mathbf{v} \otimes \mathbf{0} = \mathbf{0}$  for every  $\mathbf{v}$ .
- (f) Because a formula would require choosing a construction, and the object is characterized by what maps out of it do, not by what its elements look like; different constructions give different-looking elements and the same space up to a unique isomorphism.

### 14.2.7.2 B. Practice

#### Exercise B1: Simple or not.

For each element of  $F^3 \otimes F^3$  below, determine whether it is a simple tensor. If it is, write it as  $\mathbf{v} \otimes \mathbf{w}$ ; if it is not, justify your answer. Use the standard basis  $(\mathbf{e}_1, \mathbf{e}_2, \mathbf{e}_3)$  in both factors.

- (a)  $t_1 = \mathbf{e}_1 \otimes \mathbf{e}_2 - 2\mathbf{e}_1 \otimes \mathbf{e}_3 + 3\mathbf{e}_3 \otimes \mathbf{e}_2 - 6\mathbf{e}_3 \otimes \mathbf{e}_3$ .
- (b)  $t_2 = \mathbf{e}_1 \otimes \mathbf{e}_1 + \mathbf{e}_2 \otimes \mathbf{e}_2 + \mathbf{e}_3 \otimes \mathbf{e}_3$ .
- (c)  $t_3 = \mathbf{e}_1 \otimes \mathbf{e}_2 + \mathbf{e}_2 \otimes \mathbf{e}_1$ .
- (d)  $t_4 = \mathbf{0}$ .

*Solution.* Throughout we use Theorem 14.2.7:  $t$  is simple exactly when the matrix  $\Theta(t)$ , whose  $(i, j)$  entry is the coefficient of  $\mathbf{e}_i \otimes \mathbf{e}_j$ , has rank at most 1.

- (a)  $\Theta(t_1) = \begin{pmatrix} 0 & 1 & -2 \\ 0 & 0 & 0 \\ 0 & 3 & -6 \end{pmatrix}$ . Row 3 is 3 times row 1 and row 2 is zero, so the rank is 1. Factoring,  $\Theta(t_1) = (1, 0, 3)^\top (0, 1, -2)$ , so

$$t_1 = (\mathbf{e}_1 + 3\mathbf{e}_3) \otimes (\mathbf{e}_2 - 2\mathbf{e}_3),$$

which expands to the four given terms.

- (b)  $\Theta(t_2) = \mathbf{I}_3$ , of rank 3. Not simple. This is Example (A tensor that is not simple) one size larger.
- (c)  $\Theta(t_3)$  has 1 in positions  $(1, 2)$  and  $(2, 1)$  and zeros elsewhere. Its top-left  $2 \times 2$

minor is  $0 \cdot 0 - 1 \cdot 1 = -1 \neq 0$ , so its rank is 2 ([Theorem 6.7.1](#)). Not simple.

- (d)  $\Theta(t_4) = \mathbf{0}$ , of rank  $0 \leq 1$ . So  $t_4$  is simple, and indeed  $\mathbf{0} = \mathbf{0} \otimes \mathbf{0}$ . The degenerate case is simple for a boring reason, and it is worth keeping in the statement of [Theorem 14.2.7](#).

### Exercise B2: Inducing maps from bilinear maps.

In each case, a bilinear map is given. Use (T1) to produce the induced linear map on the tensor product, and compute the requested value.

- (a)  $\beta: F^2 \times F^2 \rightarrow F$ ,  $\beta(\mathbf{x}, \mathbf{y}) = x_1y_1 + x_2y_2$ . Compute  $\bar{\beta}(\mathbf{e}_1 \otimes \mathbf{e}_1 + \mathbf{e}_2 \otimes \mathbf{e}_2)$ .  
 (b)  $\beta: F^n \times F^n \rightarrow M_n(F)$ ,  $\beta(\mathbf{x}, \mathbf{y}) = \mathbf{x}\mathbf{y}^\top - \mathbf{y}\mathbf{x}^\top$ . Compute  $\bar{\beta}(\mathbf{v} \otimes \mathbf{v})$  for any  $\mathbf{v}$ .  
 (c)  $\beta: V^* \times V \rightarrow F$ ,  $\beta(\varphi, \mathbf{v}) = \varphi(\mathbf{v})$ , for  $V$  finite-dimensional. Compute  $\bar{\beta}(\varphi_1 \otimes \mathbf{v}_1 + \varphi_2 \otimes \mathbf{v}_2)$  for a basis  $(\mathbf{v}_i)$  with dual basis  $(\varphi_i)$ , assuming  $\dim V \geq 2$ .

*Solution.*

- (a)  $\beta$  is the dot product, bilinear by [Example \(Six two-slot functions\)](#) (a). By (T1) there is a unique linear  $\bar{\beta}$  with  $\bar{\beta}(\mathbf{x} \otimes \mathbf{y}) = x_1y_1 + x_2y_2$ . By linearity,

$$\bar{\beta}(\mathbf{e}_1 \otimes \mathbf{e}_1 + \mathbf{e}_2 \otimes \mathbf{e}_2) = \bar{\beta}(\mathbf{e}_1 \otimes \mathbf{e}_1) + \bar{\beta}(\mathbf{e}_2 \otimes \mathbf{e}_2) = 1 + 1 = 2.$$

- (b) Each entry of  $\mathbf{x}\mathbf{y}^\top - \mathbf{y}\mathbf{x}^\top$  is  $x_iy_j - y_ix_j$ , linear in each slot, so  $\beta$  is bilinear and (T1) applies. Then  $\bar{\beta}(\mathbf{v} \otimes \mathbf{v}) = \mathbf{v}\mathbf{v}^\top - \mathbf{v}\mathbf{v}^\top = \mathbf{0}$  for every  $\mathbf{v}$ , so the induced map kills every “diagonal” simple tensor without being zero:  $\bar{\beta}(\mathbf{e}_1 \otimes \mathbf{e}_2) = \mathbf{E}_{12} - \mathbf{E}_{21} \neq \mathbf{0}$  for  $n \geq 2$ .  
 (c) The evaluation pairing is bilinear by [Example \(Seven functions of several vectors\)](#) (f), so  $\bar{\beta}: V^* \otimes V \rightarrow F$  exists with  $\bar{\beta}(\varphi \otimes \mathbf{v}) = \varphi(\mathbf{v})$ . Hence

$$\bar{\beta}(\varphi_1 \otimes \mathbf{v}_1 + \varphi_2 \otimes \mathbf{v}_2) = \varphi_1(\mathbf{v}_1) + \varphi_2(\mathbf{v}_2) = 1 + 1 = 2.$$

This map is the contraction, and the answer 2 is a first glimpse of a later section: it is the trace of a rank-two projection.

### Exercise B3: Dimensions and bases.

Let  $V$ ,  $W$  and  $U$  be finite-dimensional over  $F$ , with  $\dim V = 3$ ,  $\dim W = 4$  and  $\dim U = 2$ .

- (a) Find  $\dim(V \otimes W)$  and  $\dim((V \otimes W) \otimes U)$ .  
 (b) Find  $\dim \mathcal{L}(V \otimes W, U)$  and  $\dim \mathcal{M}(V, W; U)$ , and explain why they agree.  
 (c) Let  $V = F^3$  with basis  $(\mathbf{e}_1, \mathbf{e}_2, \mathbf{e}_3)$  and  $W = F^4$  with basis  $(\mathbf{f}_1, \dots, \mathbf{f}_4)$ . Write down a basis of  $V \otimes W$  and say which basis vector corresponds to the matrix unit  $\mathbf{E}_{23}$  under  $\Theta$ .

*Solution.*

- (a)  $\dim(V \otimes W) = 3 \cdot 4 = 12$  by [Theorem 14.2.5](#), and then  $\dim((V \otimes W) \otimes U) = 12 \cdot 2 = 24$ , applying the theorem again to the pair  $(V \otimes W, U)$ .

- (b)  $\dim \mathcal{L}(V \otimes W, U) = (\dim U) \dim(V \otimes W) = 2 \cdot 12 = 24$  by [Theorem 3.6.4](#), and  $\dim \mathcal{M}(V, W; U) = 2 \cdot 3 \cdot 4 = 24$  by [Eq. 14.1.1](#). They agree because [Corollary 14.2.6](#) exhibits an isomorphism between the two spaces; the equality of numbers is that isomorphism, counted.
- (c) The 12 tensors  $\mathbf{e}_i \otimes \mathbf{f}_j$  with  $1 \leq i \leq 3$  and  $1 \leq j \leq 4$ . Since  $\Theta(\mathbf{e}_i \otimes \mathbf{f}_j) = \mathbf{E}_{ij}$ , the basis vector corresponding to  $\mathbf{E}_{23}$  is  $\mathbf{e}_2 \otimes \mathbf{f}_3$ .

### 14.2.7.3 C. Going deeper

**Exercise C1: The tensor product is commutative, up to a unique isomorphism.**

Let  $V$  and  $W$  be vector spaces over  $F$ .

- (a) Prove that there is a unique linear map  $\sigma: V \otimes W \rightarrow W \otimes V$  with  $\sigma(\mathbf{v} \otimes \mathbf{w}) = \mathbf{w} \otimes \mathbf{v}$  for all  $\mathbf{v}, \mathbf{w}$ , and that it is an isomorphism.
- (b) Deduce that  $\sigma^2 = \text{id}$  when  $V = W$ , where  $\sigma^2$  means  $\sigma$  applied twice to  $V \otimes V$ .
- (c) Explain why the argument in (a) never mentions a basis, and why that matters.

*Hint for (a): the map  $(\mathbf{v}, \mathbf{w}) \mapsto \mathbf{w} \otimes \mathbf{v}$  is bilinear; then build the inverse the same way and use the uniqueness clause of (T1).*

*Solution.*

- (a) The map  $V \times W \rightarrow W \otimes V$ ,  $(\mathbf{v}, \mathbf{w}) \mapsto \mathbf{w} \otimes \mathbf{v}$ , is bilinear, because  $\otimes$  is bilinear and swapping the slots exchanges the two clauses of [Definition 14.1.1](#). By (T1) for  $V \otimes W$  there is a unique linear  $\sigma$  with  $\sigma(\mathbf{v} \otimes \mathbf{w}) = \mathbf{w} \otimes \mathbf{v}$ . Symmetrically, (T1) for  $W \otimes V$  gives a unique linear  $\sigma': W \otimes V \rightarrow V \otimes W$  with  $\sigma'(\mathbf{w} \otimes \mathbf{v}) = \mathbf{v} \otimes \mathbf{w}$ . Then  $\sigma'\sigma$  is a linear self-map of  $V \otimes W$  with  $(\sigma'\sigma)(\mathbf{v} \otimes \mathbf{w}) = \mathbf{v} \otimes \mathbf{w}$ , and so is the identity; by the uniqueness clause of (T1) applied with  $Z = V \otimes W$  and  $\beta = \tau$ , the two agree, so  $\sigma'\sigma = \text{id}$ . Symmetrically  $\sigma\sigma' = \text{id}$ . Hence  $\sigma$  is an isomorphism ([Theorem 3.5.1](#)).
- (b) With  $V = W$ , part (a) gives  $\sigma' = \sigma$ , since both are the unique linear map sending  $\mathbf{v} \otimes \mathbf{w}$  to  $\mathbf{w} \otimes \mathbf{v}$ . Hence  $\sigma^2 = \sigma'\sigma = \text{id}$ .
- (c) Every step used only (T1) and the bilinearity of  $\otimes$ . So the argument works for infinite-dimensional spaces, where [Theorem 14.2.5](#) is unavailable, and it produces a **canonical** isomorphism, whereas a basis-dependent construction would give one isomorphism per basis and no reason to prefer any of them.

**Exercise C2: Recovering the factors from a simple tensor.**

Let  $V$  and  $W$  be finite-dimensional and let  $t = \mathbf{v} \otimes \mathbf{w} \neq \mathbf{0}$ .

- (a) Prove that the subspace  $\text{span}(\mathbf{v}) \subseteq V$  depends only on  $t$ , not on the chosen factorization.
- (b) Prove the corresponding statement for  $\text{span}(\mathbf{w}) \subseteq W$ .
- (c) For  $V = F^m$  and  $W = F^n$ , identify these two subspaces in terms of the matrix

$\Theta(t)$  of [Theorem 14.2.7](#), and say why (a) and (b) are then statements you already knew.

*Solution.*

- (a) Suppose  $t = \mathbf{v} \otimes \mathbf{w} = \mathbf{v}' \otimes \mathbf{w}'$  with  $t \neq \mathbf{0}$ . By [Proposition 14.2.9](#) there is  $c \neq 0$  with  $\mathbf{v}' = c\mathbf{v}$ . A non-zero scalar multiple spans the same subspace, so  $\text{span}(\mathbf{v}') = \text{span}(\mathbf{v})$ . Hence the line is determined by  $t$  alone.
- (b) The same proposition gives  $\mathbf{w}' = c^{-1}\mathbf{w}$  with  $c^{-1} \neq 0$ , so  $\text{span}(\mathbf{w}') = \text{span}(\mathbf{w})$ .
- (c) Under  $\Theta$ ,  $t$  corresponds to  $\mathbf{A} = \mathbf{v}\mathbf{w}^\top$ , a matrix of rank 1 ([Theorem 14.2.7](#)). Every column of  $\mathbf{A}$  is a scalar multiple of  $\mathbf{v}$ , and  $\mathbf{A}$  is non-zero, so  $\text{col}(\mathbf{A}) = \text{span}(\mathbf{v})$ ; every row is a multiple of  $\mathbf{w}^\top$ , so  $\text{row}(\mathbf{A}) = \text{span}(\mathbf{w}^\top)$ . Thus (a) and (b) say that a matrix has a well-defined column space and row space, which Chapter 3 established long ago. The tensor language adds nothing here except the reason the statement looks surprising: it is surprising only if one expects  $\mathbf{v}$  itself, rather than its line, to be recoverable.

## 14.3 Tensor Products of Maps

Section 02 built the space  $V \otimes W$ . A construction on spaces is worth little until it acts on maps as well: the philosophy of the whole book has been objects first, then the maps between them. So given  $S: V \rightarrow V'$  and  $T: W \rightarrow W'$ , we want a single map  $V \otimes W \rightarrow V' \otimes W'$  built from the two. This section builds it, computes its matrix, and discovers that the matrix is the Kronecker product of Chapter 7 — which pays both of the debts that Chapter 7 §04 left open. It ends with the identification  $V^* \otimes W \cong \mathcal{L}(V, W)$ , which every later section of the chapter uses.

### 14.3.1 A map on each factor gives a map on the product

The definition writes itself, and that is the danger. We want a map that sends  $\mathbf{v} \otimes \mathbf{w}$  to  $S\mathbf{v} \otimes T\mathbf{w}$ , so the temptation is to write

“Define  $(S \otimes T)(\mathbf{v} \otimes \mathbf{w}) := S\mathbf{v} \otimes T\mathbf{w}$ , and extend linearly.”

Nothing in that sentence is a definition. Simple tensors span  $V \otimes W$  ([Proposition 14.2.4](#)), but they do not exhaust it, and worse, the expression of an element as a combination of them is wildly non-unique:  $\mathbf{v} \otimes \mathbf{w} = (2\mathbf{v}) \otimes (\frac{1}{2}\mathbf{w})$ , and  $\mathbf{v} \otimes \mathbf{w} + \mathbf{v} \otimes \mathbf{w}' = \mathbf{v} \otimes (\mathbf{w} + \mathbf{w}')$ . A formula prescribed on simple tensors may therefore assign two different values to one element, and then it defines nothing at all. Here is a formula that really does fail. Try to define  $g: V \otimes W \rightarrow V$  by  $g(\mathbf{v} \otimes \mathbf{w}) := \mathbf{v}$ . Taking  $\mathbf{w} = \mathbf{0}$  gives  $\mathbf{v} \otimes \mathbf{0} = \mathbf{0}$  ([Proposition 14.2.8](#)), so  $g$  would have to send  $\mathbf{0}$  to every  $\mathbf{v}$  at once.

*To build a map out of  $V \otimes W$ , never write a formula on simple tensors: write down a bilinear map and let the universal property produce the linear one.*

That is the discipline of this chapter, and applied here it costs three lines.

### Theorem 14.3.1 — Tensor Product of Two Linear Maps

Let  $V, V', W, W'$  be vector spaces over  $F$ , and let  $S \in \mathcal{L}(V, V')$  and  $T \in \mathcal{L}(W, W')$ . Then there is **exactly one** linear map

$$f: V \otimes W \rightarrow V' \otimes W'$$

with  $f(\mathbf{v} \otimes \mathbf{w}) = S\mathbf{v} \otimes T\mathbf{w}$  for **all**  $\mathbf{v} \in V$  and  $\mathbf{w} \in W$ .

*Idea.* The assignment  $(\mathbf{v}, \mathbf{w}) \mapsto S\mathbf{v} \otimes T\mathbf{w}$  is bilinear, because  $S$  and  $T$  are linear and  $\otimes$  is bilinear in its two slots. A bilinear map out of  $V \times W$  is exactly what the universal property consumes, and what it returns is a unique linear map out of  $V \otimes W$ . There is nothing else to check.

*Proof.* Define  $\beta: V \times W \rightarrow V' \otimes W'$  by  $\beta(\mathbf{v}, \mathbf{w}) := S\mathbf{v} \otimes T\mathbf{w}$ . Fix  $\mathbf{w} \in W$ . The map  $\mathbf{v} \mapsto S\mathbf{v} \otimes T\mathbf{w}$  is the composite of  $S$ , which is linear, with the map  $\mathbf{u} \mapsto \mathbf{u} \otimes T\mathbf{w}$ , which is linear because the map  $\otimes: V' \times W' \rightarrow V' \otimes W'$  of Definition 14.2.1 is bilinear, hence linear in its first slot with the second slot held fixed. A composite of linear maps is linear (Theorem 3.4.2), so  $\beta$  is linear in its first argument. Fixing  $\mathbf{v}$  instead and swapping the roles of the two slots gives linearity in the second. Hence  $\beta$  is bilinear.

By the universal property in Definition 14.2.1 there is a unique linear  $f: V \otimes W \rightarrow V' \otimes W'$  with  $f(\mathbf{v} \otimes \mathbf{w}) = \beta(\mathbf{v}, \mathbf{w}) = S\mathbf{v} \otimes T\mathbf{w}$  for all  $\mathbf{v}, \mathbf{w}$ . This proves both existence and uniqueness. ■

Uniqueness is what licenses a name.

### Definition 14.3.1 — Tensor product of linear maps

Let  $S \in \mathcal{L}(V, V')$  and  $T \in \mathcal{L}(W, W')$ . The **tensor product** of  $S$  and  $T$  is the unique linear map

$$S \otimes T: V \otimes W \rightarrow V' \otimes W'$$

of Theorem 14.3.1, characterized by

$$(S \otimes T)(\mathbf{v} \otimes \mathbf{w}) = S\mathbf{v} \otimes T\mathbf{w} \quad (\mathbf{v} \in V, \mathbf{w} \in W).$$

The symbol  $\otimes$  now has three jobs — a space, a vector, a map — and they agree:  $S \otimes T$  applied to  $\mathbf{v} \otimes \mathbf{w}$  is  $S\mathbf{v} \otimes T\mathbf{w}$ , which is the only thing the notation could mean.

**Examples.**

1. **The identity.**  $\text{id}_V \otimes \text{id}_W$  sends  $\mathbf{v} \otimes \mathbf{w}$  to  $\mathbf{v} \otimes \mathbf{w}$ , and so does  $\text{id}_{V \otimes W}$ . Both are linear and both are the factorization of the same bilinear map, so  $\text{id}_V \otimes \text{id}_W = \text{id}_{V \otimes W}$  by the uniqueness clause. Degenerate, and used in almost every proof below.
2. **Acting on one slot.**  $S \otimes \text{id}_W$  sends  $\mathbf{v} \otimes \mathbf{w}$  to  $S\mathbf{v} \otimes \mathbf{w}$ : it does  $S$  in the first slot and leaves the second alone. If  $V = W = F[x]$  and  $S = D$  is differentiation,  $D \otimes \text{id}$  is “differentiate the first factor”.
3. **Scalars.** For  $c, d \in F$ ,  $(c \text{id}_V) \otimes (d \text{id}_W)$  sends  $\mathbf{v} \otimes \mathbf{w}$  to  $c\mathbf{v} \otimes d\mathbf{w} = cd(\mathbf{v} \otimes \mathbf{w})$ , so it is



$cd \operatorname{id}_{V \otimes W}$ . In particular a scalar may be moved from one slot to the other, which is the map-level shadow of  $c\mathbf{v} \otimes \mathbf{w} = \mathbf{v} \otimes c\mathbf{w}$ .

4. **Zero.** If  $S = 0$ , then  $(S \otimes T)(\mathbf{v} \otimes \mathbf{w}) = \mathbf{0} \otimes T\mathbf{w} = \mathbf{0}$ , so  $S \otimes T = 0$ . The converse also holds; see [Exercise \(C1: When a tensor product of maps vanishes\)](#).

**Non-example by minimal change.** Keep  $S$  and  $T$  linear, but try to send  $\mathbf{v} \otimes \mathbf{w}$  to  $S\mathbf{v} \otimes T\mathbf{w} + \mathbf{v} \otimes \mathbf{w}$ . This one *is* legitimate: the assignment  $(\mathbf{v}, \mathbf{w}) \mapsto S\mathbf{v} \otimes T\mathbf{w} + \mathbf{v} \otimes \mathbf{w}$  is a sum of two bilinear maps, hence bilinear, and the universal property applies. What is not legitimate is  $(\mathbf{v}, \mathbf{w}) \mapsto S\mathbf{v} \otimes T\mathbf{w} + \mathbf{u}_0$  for a fixed non-zero  $\mathbf{u}_0 \in V' \otimes W'$ : the added constant destroys linearity in each slot, so no linear map on  $V \otimes W$  does this, and indeed the prescription contradicts itself at  $\mathbf{w} = \mathbf{0}$ . The exact clause that fails is bilinearity, and it fails for the same reason as in the map  $g$  above.

We will lean repeatedly on the following consequence of the uniqueness clause. It replaces the phrase “and extend linearly”, which we are not allowed to use.

### Lemma 14.3.2 — Simple Tensors Determine a Map

Let  $Z$  be a vector space over  $F$  and let  $f, g: V \otimes W \rightarrow Z$  be linear. If  $f(\mathbf{v} \otimes \mathbf{w}) = g(\mathbf{v} \otimes \mathbf{w})$  for all  $\mathbf{v} \in V$  and  $\mathbf{w} \in W$ , then  $f = g$ .

*Proof.* Put  $\beta(\mathbf{v}, \mathbf{w}) := f(\mathbf{v} \otimes \mathbf{w})$ . Since  $\otimes$  is bilinear and  $f$  is linear,  $\beta$  is bilinear. By hypothesis  $\beta(\mathbf{v}, \mathbf{w}) = g(\mathbf{v} \otimes \mathbf{w})$  as well, so  $f$  and  $g$  are both linear maps factoring  $\beta$  through  $\otimes$ . The universal property of [Definition 14.2.1](#) says there is only one such map. Hence  $f = g$ . ■

No finite-dimensionality is needed, and no basis is chosen. With it, the algebra of  $\otimes$  on maps is a formality.

### Proposition 14.3.3 — Algebra of Tensor Products of Maps

Let  $S, S_1, S_2 \in \mathcal{L}(V, V')$ ,  $T, T_1, T_2 \in \mathcal{L}(W, W')$  and  $c \in F$ . Let also  $R \in \mathcal{L}(V', V'')$  and  $U \in \mathcal{L}(W', W'')$ . Then:

- (a)  $(S_1 + S_2) \otimes T = S_1 \otimes T + S_2 \otimes T$  and  $S \otimes (T_1 + T_2) = S \otimes T_1 + S \otimes T_2$ ;
- (b)  $(cS) \otimes T = c(S \otimes T) = S \otimes (cT)$ ;
- (c)  $(R \otimes U)(S \otimes T) = RS \otimes UT$ ;
- (d)  $\operatorname{id}_V \otimes \operatorname{id}_W = \operatorname{id}_{V \otimes W}$ .

*Proof.* In each part both sides are linear maps out of  $V \otimes W$ , so by [Lemma 14.3.2](#) it suffices to compare their values at a simple tensor  $\mathbf{v} \otimes \mathbf{w}$ .

- (a) The left side gives  $(S_1 + S_2)\mathbf{v} \otimes T\mathbf{w} = (S_1\mathbf{v} + S_2\mathbf{v}) \otimes T\mathbf{w}$ , which equals  $S_1\mathbf{v} \otimes T\mathbf{w} + S_2\mathbf{v} \otimes T\mathbf{w}$  because  $\otimes$  is additive in its first slot. That is the value of the right side. The second identity is the same argument in the other slot.
- (b)  $(cS)\mathbf{v} \otimes T\mathbf{w} = (cS\mathbf{v}) \otimes T\mathbf{w} = c(S\mathbf{v} \otimes T\mathbf{w}) = S\mathbf{v} \otimes (cT\mathbf{w})$ , where both outer equalities use bilinearity of  $\otimes$ .

- (c)  $(R \otimes U)((S \otimes T)(\mathbf{v} \otimes \mathbf{w})) = (R \otimes U)(S\mathbf{v} \otimes T\mathbf{w}) = RS\mathbf{v} \otimes UT\mathbf{w}$ , which is the value of  $RS \otimes UT$ .
- (d) Done in Example 1.

■

Part (c) is the statement that  $\otimes$  respects composition: it is *functoriality*, and we will see in a moment that it is the mixed product rule of [Theorem 7.4.1](#) in disguise.

Two more structural facts are needed before §04 can speak of tensors with several slots. The first extends the universal property from two factors to any number, and the second says that the brackets do not matter.

For  $k \geq 2$  define the iterated tensor product by

$$V_1 \otimes \cdots \otimes V_k := (V_1 \otimes \cdots \otimes V_{k-1}) \otimes V_k,$$

starting from  $V_1$  itself when  $k = 1$ , and write  $\mathbf{v}_1 \otimes \cdots \otimes \mathbf{v}_k$  for  $(\mathbf{v}_1 \otimes \cdots \otimes \mathbf{v}_{k-1}) \otimes \mathbf{v}_k$ . Multilinear maps of  $k$  arguments were defined in [Definition 14.1.1](#).

### Lemma 14.3.4 — Universal Property for $k$ Factors

Let  $V_1, \dots, V_k$  and  $Z$  be vector spaces over  $F$ , and let  $\beta: V_1 \times \cdots \times V_k \rightarrow Z$  be multilinear. Then there is **exactly one** linear map

$$\bar{\beta}: V_1 \otimes \cdots \otimes V_k \rightarrow Z$$

with  $\bar{\beta}(\mathbf{v}_1 \otimes \cdots \otimes \mathbf{v}_k) = \beta(\mathbf{v}_1, \dots, \mathbf{v}_k)$  for all  $\mathbf{v}_1, \dots, \mathbf{v}_k$ . Moreover the tensors  $\mathbf{v}_1 \otimes \cdots \otimes \mathbf{v}_k$  span  $V_1 \otimes \cdots \otimes V_k$ .

*Idea.* Induction on  $k$ , freezing the last argument. With  $\mathbf{v}_k$  held fixed,  $\beta$  is multilinear in the first  $k - 1$  arguments, so the inductive hypothesis converts it into a linear map on  $X = V_1 \otimes \cdots \otimes V_{k-1}$ . Letting  $\mathbf{v}_k$  move again gives a bilinear map on  $X \times V_k$ , and (T1) finishes. The spanning claim is the same induction.

*Proof.* For  $k = 1$  take  $\bar{\beta} = \beta$ , which is linear, and the claim is trivial; for  $k = 2$  the statement is [Definition 14.2.1](#) together with [Proposition 14.2.4](#). Let  $k \geq 3$  and assume the lemma for  $k - 1$ . Put  $X := V_1 \otimes \cdots \otimes V_{k-1}$ .

Fix  $\mathbf{v}_k \in V_k$ . The map  $(\mathbf{v}_1, \dots, \mathbf{v}_{k-1}) \mapsto \beta(\mathbf{v}_1, \dots, \mathbf{v}_k)$  is multilinear, so the inductive hypothesis gives a unique linear  $g_{\mathbf{v}_k}: X \rightarrow Z$  with  $g_{\mathbf{v}_k}(\mathbf{v}_1 \otimes \cdots \otimes \mathbf{v}_{k-1}) = \beta(\mathbf{v}_1, \dots, \mathbf{v}_k)$ .

The map  $X \times V_k \rightarrow Z$ ,  $(x, \mathbf{v}_k) \mapsto g_{\mathbf{v}_k}(x)$ , is bilinear. It is linear in  $x$  because each  $g_{\mathbf{v}_k}$  is. For linearity in  $\mathbf{v}_k$ , fix  $\mathbf{v}_k, \mathbf{v}'_k \in V_k$  and  $c \in F$ : the linear maps  $g_{\mathbf{v}_k + c\mathbf{v}'_k}$  and  $g_{\mathbf{v}_k} + c g_{\mathbf{v}'_k}$  agree at every  $\mathbf{v}_1 \otimes \cdots \otimes \mathbf{v}_{k-1}$ , because  $\beta$  is linear in its last argument, and those tensors span  $X$  by the inductive hypothesis; so the two maps are equal.

By [Definition 14.2.1](#) there is a unique linear  $\bar{\beta}$  on  $X \otimes V_k = V_1 \otimes \cdots \otimes V_k$  with  $\bar{\beta}(x \otimes \mathbf{v}_k) = g_{\mathbf{v}_k}(x)$ ; taking  $x = \mathbf{v}_1 \otimes \cdots \otimes \mathbf{v}_{k-1}$  gives  $\bar{\beta}(\mathbf{v}_1 \otimes \cdots \otimes \mathbf{v}_k) = \beta(\mathbf{v}_1, \dots, \mathbf{v}_k)$ .

Finally,  $X \otimes V_k$  is spanned by the simple tensors  $x \otimes \mathbf{v}_k$  ([Proposition 14.2.4](#)), each  $x$  is a combination of  $\mathbf{v}_1 \otimes \cdots \otimes \mathbf{v}_{k-1}$  by the inductive hypothesis, and  $\otimes$  is linear in its first slot; so the  $k$ -fold tensors span. A linear map is determined by its values on a spanning

set, which gives uniqueness of  $\bar{\beta}$ . ■

### Proposition 14.3.5 — Associativity of the Tensor Product

Let  $U, V, W$  be vector spaces over  $F$ . There is a unique isomorphism

$$\alpha: (U \otimes V) \otimes W \rightarrow U \otimes (V \otimes W)$$

with  $\alpha((\mathbf{u} \otimes \mathbf{v}) \otimes \mathbf{w}) = \mathbf{u} \otimes (\mathbf{v} \otimes \mathbf{w})$  for all  $\mathbf{u}, \mathbf{v}, \mathbf{w}$ .

*Proof.* The map  $(\mathbf{u}, \mathbf{v}, \mathbf{w}) \mapsto \mathbf{u} \otimes (\mathbf{v} \otimes \mathbf{w})$  is trilinear, since  $\otimes$  is bilinear in each of its two uses. By Lemma 14.3.4 with  $k = 3$  there is a unique linear  $\alpha$  on  $(U \otimes V) \otimes W$  with the stated values. Symmetrically, freezing the *first* argument instead and running the two-factor property twice produces a linear  $\alpha': U \otimes (V \otimes W) \rightarrow (U \otimes V) \otimes W$  with  $\alpha'(\mathbf{u} \otimes (\mathbf{v} \otimes \mathbf{w})) = (\mathbf{u} \otimes \mathbf{v}) \otimes \mathbf{w}$ . The composite  $\alpha'\alpha$  fixes every  $(\mathbf{u} \otimes \mathbf{v}) \otimes \mathbf{w}$ , and those span by Lemma 14.3.4, so  $\alpha'\alpha = \text{id}$ ; symmetrically  $\alpha\alpha' = \text{id}$ . Hence  $\alpha$  is an isomorphism. ■

From now on we drop the brackets and write  $U \otimes V \otimes W$ , with  $\mathbf{u} \otimes \mathbf{v} \otimes \mathbf{w}$  for the element that both ways of bracketing name, and likewise for any number of factors.

## 14.3.2 The matrix of a tensor product of maps

Now we compute. A matrix needs **ordered** bases, and  $V \otimes W$  has a basis indexed by *pairs*  $(i, k)$ , so the whole content of the computation is which order we put the pairs in.

*Order the basis  $\mathbf{v}_i \otimes \mathbf{w}_k$  by dictionary order on the pair  $(i, k)$ : the index of the first factor changes slowly, the index of the second factor runs fastest.*

### Definition 14.3.2 — The product basis, ordered

Let  $\mathcal{B} = (\mathbf{v}_1, \dots, \mathbf{v}_n)$  be an ordered basis of  $V$  and  $\mathcal{C} = (\mathbf{w}_1, \dots, \mathbf{w}_q)$  an ordered basis of  $W$ . Write  $\mathcal{B} \otimes \mathcal{C}$  for the basis  $(\mathbf{v}_i \otimes \mathbf{w}_k)$  of  $V \otimes W$  given by Theorem 14.2.5, **ordered** so that  $\mathbf{v}_i \otimes \mathbf{w}_k$  occupies position

$$(i-1)q + k \quad (1 \leq i \leq n, 1 \leq k \leq q),$$

that is, in the order

$$\mathbf{v}_1 \otimes \mathbf{w}_1, \dots, \mathbf{v}_1 \otimes \mathbf{w}_q, \mathbf{v}_2 \otimes \mathbf{w}_1, \dots, \mathbf{v}_n \otimes \mathbf{w}_q.$$

Division with remainder writes every position  $1, \dots, nq$  as  $(i-1)q + k$  for exactly one pair, so this really is an ordering of the whole basis. It is the dictionary order on pairs, and it is the same rule that Chapter 7 used to label the rows and columns of a Kronecker product in the formula (\*) after Definition 7.4.1. That is not a coincidence, and the next theorem says why.

### Theorem 14.3.6 — The Kronecker Product Is the Matrix of a Tensor Product of Maps

Let  $V, V', W, W'$  be finite-dimensional vector spaces over  $F$  with ordered bases

$$\begin{aligned} \mathcal{B} &= (\mathbf{v}_1, \dots, \mathbf{v}_n) \text{ of } V, & \mathcal{B}' &= (\mathbf{v}'_1, \dots, \mathbf{v}'_m) \text{ of } V', \\ \mathcal{C} &= (\mathbf{w}_1, \dots, \mathbf{w}_q) \text{ of } W, & \mathcal{C}' &= (\mathbf{w}'_1, \dots, \mathbf{w}'_p) \text{ of } W'. \end{aligned}$$

Let  $S \in \mathcal{L}(V, V')$  and  $T \in \mathcal{L}(W, W')$ . Then, with the orderings of Definition 14.3.2 on both sides,

$$[S \otimes T]_{\mathcal{B} \otimes \mathcal{C}}^{\mathcal{B}' \otimes \mathcal{C}'} = [S]_{\mathcal{B}}^{\mathcal{B}'} \otimes [T]_{\mathcal{C}}^{\mathcal{C}'},$$

the Kronecker product of Definition 7.4.1.

*Idea.* A matrix is read off column by column, and column  $(j, l)$  of the left side holds the coordinates of  $(S \otimes T)(\mathbf{v}_j \otimes \mathbf{w}_l) = S\mathbf{v}_j \otimes T\mathbf{w}_l$ . Expanding  $S\mathbf{v}_j$  and  $T\mathbf{w}_l$  in their bases and multiplying out by bilinearity produces the coefficient  $a_{ij}b_{kl}$  on  $\mathbf{v}'_i \otimes \mathbf{w}'_k$  — a product of one entry of each matrix. The entry formula (\*) of Chapter 7 says the Kronecker product puts exactly that number in exactly that place, provided the pairs are read in dictionary order. So the theorem is the entry formula and bilinearity, nothing more.

*Proof.* Write  $\mathbf{A} = (a_{ij}) = [S]_{\mathcal{B}}^{\mathcal{B}'} \in M_{m \times n}(F)$  and  $\mathbf{B} = (b_{kl}) = [T]_{\mathcal{C}}^{\mathcal{C}'} \in M_{p \times q}(F)$ , so that by Definition 3.6.1

$$S\mathbf{v}_j = \sum_{i=1}^m a_{ij}\mathbf{v}'_i, \quad T\mathbf{w}_l = \sum_{k=1}^p b_{kl}\mathbf{w}'_k.$$

Fix  $j$  and  $l$ . Using Definition 14.3.1 and then bilinearity of  $\otimes$  in each slot,

$$\begin{aligned} (S \otimes T)(\mathbf{v}_j \otimes \mathbf{w}_l) &= S\mathbf{v}_j \otimes T\mathbf{w}_l \\ &= \left( \sum_i a_{ij}\mathbf{v}'_i \right) \otimes \left( \sum_k b_{kl}\mathbf{w}'_k \right) \\ &= \sum_{i=1}^m \sum_{k=1}^p a_{ij}b_{kl} (\mathbf{v}'_i \otimes \mathbf{w}'_k). \end{aligned}$$

By Definition 14.3.2,  $\mathbf{v}_j \otimes \mathbf{w}_l$  is the basis vector in position  $(j-1)q + l$  of  $\mathcal{B} \otimes \mathcal{C}$ , and  $\mathbf{v}'_i \otimes \mathbf{w}'_k$  is the basis vector in position  $(i-1)p + k$  of  $\mathcal{B}' \otimes \mathcal{C}'$ . So the displayed expansion says that the entry of  $[S \otimes T]_{\mathcal{B} \otimes \mathcal{C}}^{\mathcal{B}' \otimes \mathcal{C}'}$  in row  $(i-1)p + k$  and column  $(j-1)q + l$  is  $a_{ij}b_{kl}$ .

By the entry formula (\*) accompanying Definition 7.4.1, that is exactly the entry of  $\mathbf{A} \otimes \mathbf{B}$  in the same position. Every row index of an  $mp \times nq$  matrix is  $(i-1)p + k$  for exactly one pair  $(i, k)$ , and every column index is  $(j-1)q + l$  for exactly one pair  $(j, l)$ , so the two matrices agree in every position. This proves the theorem. ■

**The ordering is the whole content, so it is worth being exact about it.** Suppose we had ordered the product basis the other way, putting  $\mathbf{v}_i \otimes \mathbf{w}_k$  in position  $(k-1)n + i$ , so that the *second* index changes slowly. Running the proof again, the entry in row  $(k-1)m + i$  and column  $(l-1)n + j$  is still  $a_{ij}b_{kl}$ , which by (\*) is the entry of  $\mathbf{B} \otimes \mathbf{A}$  in that position. So the

other ordering gives

$$[T]_{\mathcal{C}}^{\mathcal{C}'} \otimes [S]_{\mathcal{B}}^{\mathcal{B}'},$$

the Kronecker product with the factors **swapped** — not the transpose, and not the same matrix. For square factors the two answers are related exactly as [Proposition 7.4.2](#) says, by conjugating with the permutation matrix that relabels dictionary order into its reverse — which is precisely the change-of-coordinates matrix between the two orderings of the product basis. In the rectangular setting of [Theorem 14.3.6](#) the relation is not a conjugation at all: the row and column relabelings are different permutations, so  $\mathbf{B} \otimes \mathbf{A} = \mathbf{P}_{\sigma}^{\top}(\mathbf{A} \otimes \mathbf{B})\mathbf{P}_{\tau}$  with two matrices, as Chapter 7 notes just after that proposition’s proof.

**Warning. Do not reverse the two factors of the basis ordering.** The theorem needs the index of the **first** tensor factor to be the slow one, matching Chapter 7’s dictionary order in (\*). With the fast and slow indices exchanged the matrix of  $S \otimes T$  is  $[T] \otimes [S]$ . For  $\mathbf{A} = \begin{pmatrix} 1 & 2 \\ 3 & 4 \end{pmatrix}$  and  $\mathbf{B} = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}$  these two differ in the  $(1, 2)$  entry, which is  $a_{11}b_{12} = 1$  in the first and  $b_{11}a_{12} = 0$  in the second (the non-example after [Definition 7.4.1](#)).

**Chapter 7’s two promises, paid.** At [Definition 7.4.1](#), Chapter 7 wrote that the symbol  $\otimes$  “anticipates the tensor product of Chapter 14, of which this is the matrix version”. [Theorem 14.3.6](#) is that statement, made precise and proved: the Kronecker product of two matrices is the matrix of the tensor product of the two maps they represent, taken in the product bases. At the end of the same section, Chapter 7 wrote that the Kronecker product is “the coordinate shadow of a basis-free construction: the tensor product of linear maps, in Chapter 14, whose matrix in suitable bases is exactly  $\mathbf{A} \otimes \mathbf{B}$ ”, and promised that “the dictionary order in (\*) will reappear there as an ordering of a basis  $\mathbf{v}_i \otimes \mathbf{w}_k$ ”. The suitable bases are  $\mathcal{B} \otimes \mathcal{C}$  and  $\mathcal{B}' \otimes \mathcal{C}'$ ; the dictionary order has reappeared, as [Definition 14.3.2](#); and the shadow is cast by  $S \otimes T$ . The second promise gets a sharper payment at the end of this section, where the vec identity turns out to be this same theorem applied to a space of matrices.

### Example. (Computing with a tensor product of maps)

Let  $S: F^2 \rightarrow F^2$  and  $T: F^3 \rightarrow F^2$  have matrices, in the standard bases,

$$\mathbf{A} = \begin{pmatrix} 2 & 1 \\ 0 & 3 \end{pmatrix}, \quad \mathbf{B} = \begin{pmatrix} 1 & 0 & -1 \\ 2 & 1 & 0 \end{pmatrix}.$$

Compute  $(S \otimes T)(\mathbf{e}_1 \otimes \mathbf{f}_2 + \mathbf{e}_2 \otimes \mathbf{f}_3)$  directly from [Definition 14.3.1](#), and again from the matrix of [Theorem 14.3.6](#). Here  $(\mathbf{e}_1, \mathbf{e}_2)$  is the standard basis of  $F^2$  and  $(\mathbf{f}_1, \mathbf{f}_2, \mathbf{f}_3)$  that of  $F^3$ .

**Solution. Directly.** Columns of  $\mathbf{A}$  and  $\mathbf{B}$  give  $S\mathbf{e}_1 = 2\mathbf{e}_1$ ,  $S\mathbf{e}_2 = \mathbf{e}_1 + 3\mathbf{e}_2$ ,  $T\mathbf{f}_2 = \mathbf{e}_2$  and  $T\mathbf{f}_3 = -\mathbf{e}_1$ . Hence

$$\begin{aligned} (S \otimes T)(\mathbf{e}_1 \otimes \mathbf{f}_2 + \mathbf{e}_2 \otimes \mathbf{f}_3) &= 2\mathbf{e}_1 \otimes \mathbf{e}_2 + (\mathbf{e}_1 + 3\mathbf{e}_2) \otimes (-\mathbf{e}_1) \\ &= -\mathbf{e}_1 \otimes \mathbf{e}_1 + 2\mathbf{e}_1 \otimes \mathbf{e}_2 - 3\mathbf{e}_2 \otimes \mathbf{e}_1, \end{aligned}$$

using bilinearity of  $\otimes$  in both slots.

**By matrix.** The domain basis, ordered as in [Definition 14.3.2](#), is  $(\mathbf{e}_1 \otimes \mathbf{f}_1, \mathbf{e}_1 \otimes \mathbf{f}_2, \mathbf{e}_1 \otimes \mathbf{f}_3, \mathbf{e}_2 \otimes \mathbf{f}_1, \mathbf{e}_2 \otimes \mathbf{f}_2, \mathbf{e}_2 \otimes \mathbf{f}_3)$ , so the input has coordinate column  $(0, 1, 0, 0, 0, 1)$ . By [The-](#)

orem 14.3.6 the matrix of  $S \otimes T$  is

$$\mathbf{A} \otimes \mathbf{B} = \begin{pmatrix} 2 & 0 & -2 & 1 & 0 & -1 \\ 4 & 2 & 0 & 2 & 1 & 0 \\ 0 & 0 & 0 & 3 & 0 & -3 \\ 0 & 0 & 0 & 6 & 3 & 0 \end{pmatrix},$$

and multiplying it by that column gives  $(-1, 2, -3, 0)$ . In the codomain basis  $(\mathbf{e}_1 \otimes \mathbf{e}_1, \mathbf{e}_1 \otimes \mathbf{e}_2, \mathbf{e}_2 \otimes \mathbf{e}_1, \mathbf{e}_2 \otimes \mathbf{e}_2)$  this is the same element as before.

### 14.3.3 What Chapter 7 was computing

Every property of the Kronecker product proved by entry-chasing in Chapter 7 is now a property of maps, and several of them become one-liners. We record the three that the rest of the chapter uses.

The mixed product rule comes first, because it is the one Chapter 7 called “the engine of everything that follows”.

#### Corollary 14.3.7 — The Mixed Product Rule Is Functoriality

Let  $\mathbf{A} \in M_{m \times n}(F)$ ,  $\mathbf{C} \in M_{n \times t}(F)$ ,  $\mathbf{B} \in M_{p \times q}(F)$  and  $\mathbf{D} \in M_{q \times u}(F)$ . Then

$$(\mathbf{A} \otimes \mathbf{B})(\mathbf{C} \otimes \mathbf{D}) = (\mathbf{AC}) \otimes (\mathbf{BD}).$$

*Proof.* Let  $S, R, T, U$  be the linear maps with matrices  $\mathbf{A}, \mathbf{C}, \mathbf{B}, \mathbf{D}$  in the standard bases of the relevant spaces  $F^k$ . By Theorem 14.3.6, with all product bases ordered as in Definition 14.3.2,  $\mathbf{A} \otimes \mathbf{B}$  and  $\mathbf{C} \otimes \mathbf{D}$  are the matrices of  $S \otimes T$  and  $R \otimes U$ . By Theorem 3.6.2 the product of the matrices is the matrix of the composite  $(S \otimes T)(R \otimes U)$ , which is  $SR \otimes TU$  by Proposition 14.3.3 (c). Applying Theorem 14.3.6 once more, and Theorem 3.6.2 again to each factor, the matrix of  $SR \otimes TU$  is  $(\mathbf{AC}) \otimes (\mathbf{BD})$ . ■

This is Theorem 7.4.1 (c), which Chapter 7 proved by partitioning both matrices into conformable blocks and multiplying blockwise. The proof above says what the identity *means*: doing two things one after the other in each slot separately is the same as doing them one after the other in both slots at once. In the same way, the bilinearity of Theorem 7.4.1 (a) is parts (a) and (b) of Proposition 14.3.3, its associativity (b) is Proposition 14.3.5, and its inverse rule (e) is parts (c) and (d) together with invertibility of a composite.

Next, the numerical invariants.

#### Corollary 14.3.8 — Rank, Determinant and Trace of a Tensor Product of Maps

Let  $V, V', W, W'$  be finite-dimensional over  $F$ ,  $S \in \mathcal{L}(V, V')$ ,  $T \in \mathcal{L}(W, W')$ . Then

$$\text{rank}(S \otimes T) = \text{rank } S \cdot \text{rank } T.$$

If moreover  $V = V'$  and  $W = W'$ , with  $n = \dim V \geq 1$  and  $m = \dim W \geq 1$ , then

$$\det(S \otimes T) = (\det S)^m (\det T)^n, \quad \operatorname{tr}(S \otimes T) = \operatorname{tr} S \cdot \operatorname{tr} T.$$

*Idea.* For the rank, choose bases adapted to the two kernels: then  $S \otimes T$  kills most product basis vectors and sends the remaining  $\operatorname{rank} S \cdot \operatorname{rank} T$  of them to independent vectors. For the determinant, split  $S \otimes T = (S \otimes \operatorname{id})(\operatorname{id} \otimes T)$ . The factor  $\operatorname{id}_V \otimes T$  preserves each of the  $n$  subspaces  $\mathbf{v}_i \otimes W$  and acts on each as  $T$ ; the factor  $S \otimes \operatorname{id}_W$  preserves each of the  $m$  subspaces  $V \otimes \mathbf{w}_k$  and acts as  $S$ . So each factor is block diagonal with repeated blocks, and the exponents count the *other* space's dimension, which is where the exponents come from. The trace we take from the matrix.

*Proof. Rank.* Let  $r = \operatorname{rank} S$  and  $s = \operatorname{rank} T$ . Choose a basis  $(\mathbf{v}_1, \dots, \mathbf{v}_n)$  of  $V$  whose last  $n - r$  vectors form a basis of  $\ker S$ ; this is possible by [Theorem 1.8.2](#) applied to a basis of  $\ker S$ , after reordering. Then  $S\mathbf{v}_1, \dots, S\mathbf{v}_r$  span  $\operatorname{im} S$  by [Theorem 3.2.4](#), and being  $r = \operatorname{rank} S$  spanning vectors of an  $r$ -dimensional space they form a basis of  $\operatorname{im} S$  ([Theorem 1.6.4](#)). Choose a basis  $(\mathbf{w}_1, \dots, \mathbf{w}_m)$  of  $W$  the same way, so that  $T\mathbf{w}_1, \dots, T\mathbf{w}_s$  is a basis of  $\operatorname{im} T$  and  $T\mathbf{w}_l = \mathbf{0}$  for  $l > s$ .

By [Theorem 3.2.4](#),  $\operatorname{im}(S \otimes T)$  is spanned by the vectors  $S\mathbf{v}_j \otimes T\mathbf{w}_l$ , and those with  $j > r$  or  $l > s$  are  $\mathbf{0}$ . Extend  $(S\mathbf{v}_1, \dots, S\mathbf{v}_r)$  to a basis of  $V'$  and  $(T\mathbf{w}_1, \dots, T\mathbf{w}_s)$  to a basis of  $W'$  ([Theorem 1.8.2](#)). By [Theorem 14.2.5](#) the products of these basis vectors form a basis of  $V' \otimes W'$ ; in particular the  $rs$  vectors  $S\mathbf{v}_j \otimes T\mathbf{w}_l$  with  $j \leq r$ ,  $l \leq s$  are linearly independent. They therefore form a basis of  $\operatorname{im}(S \otimes T)$ , and  $\operatorname{rank}(S \otimes T) = rs$ .

**Determinant.** By [Proposition 14.3.3](#) (c) and (d),

$$S \otimes T = (S \otimes \operatorname{id}_W)(\operatorname{id}_V \otimes T),$$

so  $\det(S \otimes T) = \det(S \otimes \operatorname{id}_W) \det(\operatorname{id}_V \otimes T)$  by [Theorem 6.5.8](#). Fix bases  $\mathcal{B}$  of  $V$  and  $\mathcal{C}$  of  $W$  and use the ordering of [Definition 14.3.2](#). For each  $i$ , the subspace  $\mathbf{v}_i \otimes W := \operatorname{span}(\mathbf{v}_i \otimes \mathbf{w}_1, \dots, \mathbf{v}_i \otimes \mathbf{w}_m)$  is carried into itself by  $\operatorname{id}_V \otimes T$ , which acts on it as  $T$  does on  $W$ ; and  $V \otimes W$  is the direct sum of these  $n$  subspaces, because their bases together form the basis  $\mathcal{B} \otimes \mathcal{C}$ . Consecutive blocks of the ordering are exactly these subspaces, so

$$[\operatorname{id}_V \otimes T]_{\mathcal{B} \otimes \mathcal{C}}^{\mathcal{B} \otimes \mathcal{C}} = \underbrace{[T]_{\mathcal{C}}^{\mathcal{C}} \oplus \dots \oplus [T]_{\mathcal{C}}^{\mathcal{C}}}_{n \text{ blocks}},$$

whose determinant is  $(\det T)^n$  by [Theorem 7.1.4](#) (d) and induction on the number of blocks. The same argument for  $S \otimes \operatorname{id}_W$ , using instead the  $m$  subspaces  $V \otimes \mathbf{w}_k$ , gives  $\det(S \otimes \operatorname{id}_W) = (\det S)^m$ . (This time the relevant basis vectors are not consecutive in the ordering of [Definition 14.3.2](#), so reorder the basis to make them so; matrices of one operator in two bases are similar, by [Theorem 3.7.3](#), hence have the same determinant, by [Corollary 6.5.7](#).) Multiplying the two gives the formula.

**Trace.** By [Theorem 14.3.6](#) the matrix of  $S \otimes T$  in  $\mathcal{B} \otimes \mathcal{C}$  is  $[S]_{\mathcal{B}}^{\mathcal{B}} \otimes [T]_{\mathcal{C}}^{\mathcal{C}}$ , so by [Definition 3.10.2](#) and [Theorem 7.4.3](#) (b),  $\operatorname{tr}(S \otimes T) = \operatorname{tr} S \cdot \operatorname{tr} T$ .

■

These are parts (a), (c) and (b) of [Theorem 7.4.3](#), transported to operators. Two remarks on what has changed. First, the exponents in the determinant formula are no longer something to memorize:  $\det S$  appears to the power  $\dim W$  because  $S \otimes \operatorname{id}_W$  runs  $S$  once on each of the  $\dim W$



copies of  $V$  inside  $V \otimes W$ . Second, the proof of the trace formula above is still a computation in a basis. There is a genuinely basis-free reason, and §04 gives it. The determinant, too, has a basis-free explanation, which has to wait for the top exterior power in §09.

**Warning. Most operators on  $V \otimes W$  are not of the form  $S \otimes T$ .** The operators that are of this form are heavily constrained: by Corollary 14.3.8 their rank is a product  $rs$  with  $r \leq \dim V$  and  $s \leq \dim W$ . Take  $V = W = F^2$ , so  $\dim(V \otimes W) = 4$ , and let  $P$  be any operator on  $V \otimes W$  of rank 3, for instance the projection onto the span of the first three basis vectors of Definition 14.3.2. Then  $3 = rs$  with  $r, s \in \{0, 1, 2\}$  is impossible, so  $P \neq S \otimes T$  for every  $S$  and  $T$ . The maps  $S \otimes T$  do **span**  $\mathcal{L}(V \otimes W)$ , which is a different and weaker statement; this is the same distinction as between simple tensors and general tensors in §02.

*Quick check.* Let  $S \in \mathcal{L}(V)$  and  $T \in \mathcal{L}(W)$  with  $\dim V = 3$  and  $\dim W = 2$ , and suppose  $\det S = 2$  and  $\det T = -1$ . What are  $\det(S \otimes T)$  and  $\det(T \otimes S)$ , where the second is an operator on  $W \otimes V$ ?

*Solution.* By Corollary 14.3.8,  $\det(S \otimes T) = (\det S)^{\dim W} (\det T)^{\dim V} = 2^2(-1)^3 = -4$ . For  $T \otimes S$  the roles swap, giving  $(\det T)^{\dim V} (\det S)^{\dim W} = (-1)^3 2^2 = -4$  as well. The two agree, as they must: a determinant is a product of scalars, and the exponent always belongs to the *other* factor's dimension. The tempting wrong answer,  $(\det S)^3 (\det T)^2 = 8$ , puts each map's own dimension in its exponent.

## 14.3.4 Maps as tensors

We come to the identification that the rest of the chapter runs on. A functional  $\varphi \in V^*$  and a vector  $\mathbf{w} \in W$  together determine a linear map  $V \rightarrow W$ : measure with  $\varphi$ , then scale  $\mathbf{w}$  by the answer. That recipe is bilinear in  $(\varphi, \mathbf{w})$ , which is now a reflex: it must come from a linear map on  $V^* \otimes W$ .

### Theorem 14.3.9 — Tensors Are Maps

Let  $V$  and  $W$  be **finite-dimensional** vector spaces over  $F$ . There is a unique linear map

$$\Theta: V^* \otimes W \rightarrow \mathcal{L}(V, W), \quad \Theta(\varphi \otimes \mathbf{w})(\mathbf{v}) = \varphi(\mathbf{v}) \mathbf{w},$$

and it is an isomorphism. Moreover, for  $R \in \mathcal{L}(V_0, V)$  and  $U \in \mathcal{L}(W, W_0)$  with  $V_0, W_0$  finite-dimensional,

$$\Theta((R' \otimes U)(\xi)) = U \Theta(\xi) R \quad (\xi \in V^* \otimes W),$$

where  $R': V^* \rightarrow V_0^*$  is the dual map of Definition 4.4.1 and the  $\Theta$  on the left is the one for the pair  $(V_0, W_0)$ .

*Idea.* Existence is the universal property again. For bijectivity, count: both sides have dimension  $\dim V \cdot \dim W$ , so it is enough to see that  $\Theta$  carries a basis onto a basis. It carries  $\varphi_j \otimes \mathbf{w}_i$  to the map whose matrix is the matrix unit  $\mathbf{E}_{ij}$ , and the matrix units are a basis. The last formula is the statement that  $\Theta$  turns the tensor product of maps into composition on the two sides; it is checked on simple tensors.



*Proof. Existence and uniqueness.* For fixed  $\varphi$  and  $\mathbf{w}$ , the assignment  $\mathbf{v} \mapsto \varphi(\mathbf{v})\mathbf{w}$  is linear, so it is an element  $\varphi \cdot \mathbf{w}$  of  $\mathcal{L}(V, W)$ . The map  $(\varphi, \mathbf{w}) \mapsto \varphi \cdot \mathbf{w}$  is bilinear: for every  $\mathbf{v}$ ,

$$\begin{aligned}(\varphi + \psi)(\mathbf{v})\mathbf{w} &= \varphi(\mathbf{v})\mathbf{w} + \psi(\mathbf{v})\mathbf{w}, \\ \varphi(\mathbf{v})(\mathbf{w} + \mathbf{w}') &= \varphi(\mathbf{v})\mathbf{w} + \varphi(\mathbf{v})\mathbf{w}', \\ (c\varphi)(\mathbf{v})\mathbf{w} &= c\varphi(\mathbf{v})\mathbf{w} = \varphi(\mathbf{v})(c\mathbf{w}),\end{aligned}$$

and two maps agreeing at every  $\mathbf{v}$  are equal. By [Definition 14.2.1](#) there is a unique linear  $\Theta$  with  $\Theta(\varphi \otimes \mathbf{w}) = \varphi \cdot \mathbf{w}$ .

**Bijectivity.** Let  $\mathcal{B} = (\mathbf{v}_1, \dots, \mathbf{v}_n)$  be a basis of  $V$  with dual basis  $\mathcal{B}^* = (\varphi_1, \dots, \varphi_n)$  ([Definition 4.1.3](#)), and  $\mathcal{C} = (\mathbf{w}_1, \dots, \mathbf{w}_m)$  a basis of  $W$ . By [Theorem 14.2.5](#) the products  $\varphi_j \otimes \mathbf{w}_i$  form a basis of  $V^* \otimes W$ , of size  $nm$ . Now  $\Theta(\varphi_j \otimes \mathbf{w}_i)$  sends  $\mathbf{v}_k$  to  $\varphi_j(\mathbf{v}_k)\mathbf{w}_i = \delta_{jk}\mathbf{w}_i$ , so it sends  $\mathbf{v}_j$  to  $\mathbf{w}_i$  and every other basis vector to  $\mathbf{0}$ . By [Definition 3.6.1](#) its matrix is the matrix unit  $\mathbf{E}_{ij}$ . The matrix units form a basis of  $M_{m \times n}(F)$  ([Example \(Standard Bases\)](#)) and  $T \mapsto [T]_{\mathcal{B}}^{\mathcal{C}}$  is an isomorphism ([Theorem 3.6.4](#)), so the  $nm$  maps  $\Theta(\varphi_j \otimes \mathbf{w}_i)$  form a basis of  $\mathcal{L}(V, W)$ .

Hence  $\text{im } \Theta$  contains a basis of  $\mathcal{L}(V, W)$ , so  $\Theta$  is surjective ([Theorem 3.2.4](#)). Since  $\dim(V^* \otimes W) = nm = \dim \mathcal{L}(V, W)$ , [Theorem 3.3.1](#) gives nullity  $\Theta = 0$ , so  $\Theta$  is injective ([Theorem 3.2.3](#)). Therefore  $\Theta$  is an isomorphism.

**Naturality.** Both  $\xi \mapsto \Theta((R' \otimes U)\xi)$  and  $\xi \mapsto U\Theta(\xi)R$  are linear in  $\xi$ , so by [Lemma 14.3.2](#) it suffices to check them at  $\xi = \varphi \otimes \mathbf{w}$ . On the left,  $(R' \otimes U)(\varphi \otimes \mathbf{w}) = R'\varphi \otimes U\mathbf{w}$ , and for  $\mathbf{v} \in V_0$ ,

$$\Theta(R'\varphi \otimes U\mathbf{w})(\mathbf{v}) = (R'\varphi)(\mathbf{v})U\mathbf{w} = \varphi(R\mathbf{v})U\mathbf{w},$$

using [Definition 4.4.1](#). On the right,  $(U\Theta(\varphi \otimes \mathbf{w})R)(\mathbf{v}) = U(\varphi(R\mathbf{v})\mathbf{w}) = \varphi(R\mathbf{v})U\mathbf{w}$  by linearity of  $U$ . The two agree. ■

**Warning. Finite dimension is a hypothesis, not decoration.** The map  $\Theta$  is defined and injective for any  $V$  and  $W$ , but it is surjective only onto the maps of finite rank:  $\Theta(\sum_{t=1}^N \varphi_t \otimes \mathbf{w}_t)$  has image inside  $\text{span}(\mathbf{w}_1, \dots, \mathbf{w}_N)$ , so its rank is at most  $N$ . On  $V = W = F[x]$ , the identity map has infinite rank and is therefore not  $\Theta(\xi)$  for any  $\xi \in V^* \otimes V$ . Everything below uses the finite-dimensional statement.

The letter  $\Theta$  is deliberate: taking  $V = F^n$  and  $W = F^m$  and using the dual basis to identify  $(F^n)^*$  with  $F^n$ , this  $\Theta$  is the isomorphism  $F^m \otimes F^n \cong M_{m \times n}(F)$  of [Theorem 14.2.7](#), with the two factors written in the order that makes the source a space of *maps* rather than of matrices. Section 02's model was the special case; this is the general statement.

Two uses of the theorem, immediately. First, it names the matrix units: under  $\Theta$ , the map with matrix  $\mathbf{E}_{ij}$  is the tensor  $\varphi_j \otimes \mathbf{w}_i$ , so a matrix is a sum  $\sum_{i,j} a_{ij} \varphi_j \otimes \mathbf{w}_i$ . Note the index order — the functional carries the column index, the vector the row index — and compare it with Chapter 7's vectorization, which stacks columns.

Second, the naturality formula is the vec identity.

**Example.** (The vec identity, read as a tensor product of maps)

Let  $\mathbf{A} \in M_{k \times m}(F)$  and  $\mathbf{B} \in M_{n \times l}(F)$ , and let

$$\Psi: M_{m \times n}(F) \rightarrow M_{k \times l}(F), \quad \mathbf{X} \mapsto \mathbf{A}\mathbf{X}\mathbf{B},$$

which is linear. Identify  $M_{m \times n}(F)$  with  $\mathcal{L}(F^n, F^m)$  and  $M_{k \times l}(F)$  with  $\mathcal{L}(F^l, F^k)$ . Explain why [Theorem 7.4.4](#),  $\text{vec}(\mathbf{A}\mathbf{X}\mathbf{B}) = (\mathbf{B}^\top \otimes \mathbf{A}) \text{vec } \mathbf{X}$ , is [Theorem 14.3.6](#) applied to  $\Psi$ .

*Solution.* Let  $R: F^l \rightarrow F^n$  and  $U: F^m \rightarrow F^k$  be the maps with matrices  $\mathbf{B}$  and  $\mathbf{A}$  in the standard bases. Under the identification,  $\Psi$  becomes the map  $\mathcal{L}(F^n, F^m) \rightarrow \mathcal{L}(F^l, F^k)$  sending  $T$  to  $UTR$ , so by the naturality clause of [Theorem 14.3.9](#) it corresponds, through  $\Theta$ , to

$$R' \otimes U: (F^n)^* \otimes F^m \rightarrow (F^l)^* \otimes F^k.$$

Order the basis of  $(F^n)^* \otimes F^m$  as in [Definition 14.3.2](#), using the dual standard basis  $(\varphi_1, \dots, \varphi_n)$  first and the standard basis  $(\mathbf{e}_1, \dots, \mathbf{e}_m)$  second. Then  $\varphi_j \otimes \mathbf{e}_i$  sits in position  $(j-1)m + i$ , and  $\Theta(\varphi_j \otimes \mathbf{e}_i) = \mathbf{E}_{ij}$ . By [Definition 7.4.2](#),  $\text{vec } \mathbf{E}_{ij} = \mathbf{e}_{(j-1)m+i}$ . So  $\text{vec}$  is the coordinate map of this ordered basis: the dictionary order with the *column* index slow is exactly the order in which vectorization stacks the columns.

Now apply [Theorem 14.3.6](#). The matrix of  $R'$  in the dual standard bases is  $\mathbf{B}^\top$ , because the matrix of a dual map is the transpose ([Theorem 4.4.2](#)), and the matrix of  $U$  is  $\mathbf{A}$ . Hence the matrix of  $R' \otimes U$  in the product bases above is  $\mathbf{B}^\top \otimes \mathbf{A}$ . Reading that equality of matrices through the coordinate map  $\text{vec}$  gives  $\text{vec}(\mathbf{A}\mathbf{X}\mathbf{B}) = (\mathbf{B}^\top \otimes \mathbf{A}) \text{vec } \mathbf{X}$ , which is [Theorem 7.4.4](#).

For a numerical check, take  $\mathbf{A} = \begin{pmatrix} 1 & 2 \\ 0 & 1 \end{pmatrix}$  and  $\mathbf{B} = \begin{pmatrix} 3 & 0 \\ 1 & -1 \end{pmatrix}$ . Writing the four images  $\mathbf{A}\mathbf{E}_{ij}\mathbf{B}$  in the order  $\mathbf{E}_{11}, \mathbf{E}_{21}, \mathbf{E}_{12}, \mathbf{E}_{22}$  and stacking the four columns  $\text{vec}(\mathbf{A}\mathbf{E}_{ij}\mathbf{B})$  side by side gives

$$\begin{pmatrix} 3 & 6 & 1 & 2 \\ 0 & 3 & 0 & 1 \\ 0 & 0 & -1 & -2 \\ 0 & 0 & 0 & -1 \end{pmatrix},$$

which is  $\mathbf{B}^\top \otimes \mathbf{A}$  written out.

So the transpose in the  $\text{vec}$  identity, which Chapter 7 had to warn readers not to get wrong, is not a bookkeeping accident: it is there because  $\Psi$  multiplies  $\mathbf{X}$  on the right by  $\mathbf{B}$ , and right multiplication is *precomposition*, which reverses arrows. The dual map is what reverses them, and the matrix of a dual map is a transpose. That is the sharper form of Chapter 7's second promise, and it is why the wrong guesses  $\mathbf{A} \otimes \mathbf{B}^\top$  and  $\mathbf{B} \otimes \mathbf{A}$  are wrong.

## 14.3.5 Exercises

### 14.3.5.1 A. Check your understanding

#### Exercise A1.

- (a) State the defining property of  $S \otimes T$ , and say why writing “ $(S \otimes T)(\mathbf{v} \otimes \mathbf{w}) = S\mathbf{v} \otimes T\mathbf{w}$ , extended linearly” is not by itself a definition.

- (b) Let  $\dim V = 3$ ,  $\dim W = 4$ ,  $\dim V' = 2$ ,  $\dim W' = 5$ . What are the sizes of  $[S]_{\mathcal{B}}^{\mathcal{B}'}$ ,  $[T]_{\mathcal{C}}^{\mathcal{C}'}$  and  $[S \otimes T]_{\mathcal{B} \otimes \mathcal{C}}^{\mathcal{B}' \otimes \mathcal{C}'}$ ?
- (c) In the ordering of [Definition 14.3.2](#) with  $\dim V = 3$  and  $\dim W = 4$ , which basis vector of  $V \otimes W$  sits in position 7?
- (d) True or false: every operator on  $V \otimes W$  is of the form  $S \otimes T$ . Justify your answer.
- (e) State [Theorem 14.3.9](#), including its hypothesis, and say what  $\Theta(\varphi \otimes \mathbf{w})$  does to a vector  $\mathbf{v}$ .

*Solution.*

- (a)  $S \otimes T$  is the **unique** linear map  $V \otimes W \rightarrow V' \otimes W'$  with  $(S \otimes T)(\mathbf{v} \otimes \mathbf{w}) = S\mathbf{v} \otimes T\mathbf{w}$  for all  $\mathbf{v}, \mathbf{w}$  ([Definition 14.3.1](#)). The quoted phrase is not a definition because a general element of  $V \otimes W$  is a sum of simple tensors in many different ways, so a prescription on simple tensors may be inconsistent;  $\mathbf{v} \otimes \mathbf{0} = \mathbf{0}$  already forces consistency conditions. What makes it work is that  $(\mathbf{v}, \mathbf{w}) \mapsto S\mathbf{v} \otimes T\mathbf{w}$  is bilinear, so [Definition 14.2.1](#) supplies the map.
- (b)  $2 \times 3$ ,  $5 \times 4$ , and  $10 \times 12$  ([Theorem 14.3.6](#): an  $m \times n$  and a  $p \times q$  matrix give an  $mp \times nq$  one).
- (c) With  $q = 4$ , position  $7 = (i - 1) \cdot 4 + k$  forces  $i = 2$ ,  $k = 3$ . So it is  $\mathbf{v}_2 \otimes \mathbf{w}_3$ .
- (d) False. By [Corollary 14.3.8](#),  $\text{rank}(S \otimes T) = \text{rank } S \cdot \text{rank } T$ . With  $\dim V = \dim W = 2$ , an operator of rank 3 on the 4-dimensional space  $V \otimes W$  cannot have its rank factored as a product of two integers each at most 2.
- (e) For  $V, W$  **finite-dimensional**, there is a unique linear  $\Theta: V^* \otimes W \rightarrow \mathcal{L}(V, W)$  with  $\Theta(\varphi \otimes \mathbf{w})(\mathbf{v}) = \varphi(\mathbf{v})\mathbf{w}$ , and it is an isomorphism. So  $\Theta(\varphi \otimes \mathbf{w})$  measures  $\mathbf{v}$  with  $\varphi$  and scales  $\mathbf{w}$  by the result.

### 14.3.5.2 B. Practice

#### Exercise B1: A tensor product of maps, two ways.

Let  $S: \mathbb{Q}^2 \rightarrow \mathbb{Q}^2$  and  $T: \mathbb{Q}^2 \rightarrow \mathbb{Q}^3$  have matrices, in the standard bases,

$$\mathbf{A} = \begin{pmatrix} 1 & -1 \\ 2 & 0 \end{pmatrix}, \quad \mathbf{B} = \begin{pmatrix} 1 & 0 \\ 0 & 2 \\ -1 & 1 \end{pmatrix}.$$

- (a) Write down  $[S \otimes T]_{\mathcal{B} \otimes \mathcal{C}}^{\mathcal{B}' \otimes \mathcal{C}'}$  for the standard bases, ordered as in [Definition 14.3.2](#).
- (b) Compute  $(S \otimes T)(\mathbf{e}_1 \otimes \mathbf{e}_2 - \mathbf{e}_2 \otimes \mathbf{e}_1)$  directly from [Definition 14.3.1](#).
- (c) Check (b) against (a) by multiplying the matrix by the coordinate column of the input.

*Solution.*

- (a) By [Theorem 14.3.6](#) the matrix is  $\mathbf{A} \otimes \mathbf{B}$ , of size  $6 \times 4$ :

$$\mathbf{A} \otimes \mathbf{B} = \begin{pmatrix} 1 & 0 & -1 & 0 \\ 0 & 2 & 0 & -2 \\ -1 & 1 & 1 & -1 \\ 2 & 0 & 0 & 0 \\ 0 & 4 & 0 & 0 \\ -2 & 2 & 0 & 0 \end{pmatrix}.$$

- (b) The columns of  $\mathbf{A}$  give  $S\mathbf{e}_1 = \mathbf{e}_1 + 2\mathbf{e}_2$  and  $S\mathbf{e}_2 = -\mathbf{e}_1$ ; the columns of  $\mathbf{B}$  give  $T\mathbf{e}_1 = \mathbf{f}_1 - \mathbf{f}_3$  and  $T\mathbf{e}_2 = 2\mathbf{f}_2 + \mathbf{f}_3$ , where  $(\mathbf{f}_1, \mathbf{f}_2, \mathbf{f}_3)$  is the standard basis of  $\mathbb{Q}^3$ . Hence

$$\begin{aligned} (S \otimes T)(\mathbf{e}_1 \otimes \mathbf{e}_2 - \mathbf{e}_2 \otimes \mathbf{e}_1) &= (\mathbf{e}_1 + 2\mathbf{e}_2) \otimes (2\mathbf{f}_2 + \mathbf{f}_3) - (-\mathbf{e}_1) \otimes (\mathbf{f}_1 - \mathbf{f}_3) \\ &= \mathbf{e}_1 \otimes \mathbf{f}_1 + 2\mathbf{e}_1 \otimes \mathbf{f}_2 + 4\mathbf{e}_2 \otimes \mathbf{f}_2 + 2\mathbf{e}_2 \otimes \mathbf{f}_3, \end{aligned}$$

where the  $\mathbf{e}_1 \otimes \mathbf{f}_3$  terms cancel:  $+1$  from the first product and  $-1$  from the second.

- (c) The input has coordinates  $(0, 1, -1, 0)$  in the basis  $(\mathbf{e}_1 \otimes \mathbf{e}_1, \mathbf{e}_1 \otimes \mathbf{e}_2, \mathbf{e}_2 \otimes \mathbf{e}_1, \mathbf{e}_2 \otimes \mathbf{e}_2)$ . Multiplying gives  $(1, 2, 0, 0, 4, 2)$ , which in the basis  $(\mathbf{e}_1 \otimes \mathbf{f}_1, \mathbf{e}_1 \otimes \mathbf{f}_2, \mathbf{e}_1 \otimes \mathbf{f}_3, \mathbf{e}_2 \otimes \mathbf{f}_1, \mathbf{e}_2 \otimes \mathbf{f}_2, \mathbf{e}_2 \otimes \mathbf{f}_3)$  is exactly the answer to (b).

### Exercise B2: Invariants without multiplying matrices.

Let  $S \in \mathcal{L}(V)$  and  $T \in \mathcal{L}(W)$  over  $\mathbb{R}$ , with  $\dim V = 4$ ,  $\dim W = 3$ ,  $\text{rank } S = 2$ ,  $\text{rank } T = 3$ ,  $\text{tr } S = 5$ ,  $\text{tr } T = -2$ ,  $\det T = 6$ .

- Find  $\text{rank}(S \otimes T)$  and  $\text{nullity}(S \otimes T)$ .
- Find  $\text{tr}(S \otimes T)$ .
- Find  $\det(S \otimes T)$ . Justify the value from the ranks alone.
- Find  $\text{tr}(S \otimes \text{id}_W)$  and  $\text{tr}(\text{id}_V \otimes T)$ , and check that their product is **not**  $\text{tr}(S \otimes T)$  in general by comparing with (b).

*Solution.*

- By [Corollary 14.3.8](#),  $\text{rank}(S \otimes T) = 2 \cdot 3 = 6$ . Since  $\dim(V \otimes W) = 12$  by [Theorem 14.2.5](#), [Theorem 3.3.1](#) gives  $\text{nullity}(S \otimes T) = 12 - 6 = 6$ .
- $\text{tr}(S \otimes T) = 5 \cdot (-2) = -10$ .
- $\text{rank } S = 2 < 4 = \dim V$ , so  $S$  is not invertible and  $\det S = 0$  by [Theorem 2.4.4](#) and [Definition 6.5.1](#). Hence  $\det(S \otimes T) = 0^3 \cdot 6^4 = 0$ . Directly:  $\text{rank}(S \otimes T) = 6 < 12$ , so  $S \otimes T$  is not invertible.
- $\text{tr}(S \otimes \text{id}_W) = \text{tr } S \cdot \text{tr } \text{id}_W = 5 \cdot 3 = 15$  and  $\text{tr}(\text{id}_V \otimes T) = 4 \cdot (-2) = -8$ , using  $\text{tr id} = (\dim) \cdot 1$  from [Theorem 3.10.3](#). Their product is  $-120 \neq -10$ . There is no contradiction:  $S \otimes T$  is the *composite* of the two, and the trace of a composite is not the product of the traces.

### Exercise B3: Which statements hold?.

Let  $V, W$  be finite-dimensional over  $F$ . Determine which of the following hold for **all**  $S, S_1, S_2 \in \mathcal{L}(V)$  and  $T \in \mathcal{L}(W)$ . Justify your answer.

- (a)  $(S_1 S_2) \otimes T = (S_1 \otimes T)(S_2 \otimes T)$ .
- (b)  $(S_1 S_2) \otimes T^2 = (S_1 \otimes T)(S_2 \otimes T)$ .
- (c)  $S \otimes T = 0$  implies  $S = 0$  or  $T = 0$ .
- (d)  $S \otimes \text{id}_W$  and  $\text{id}_V \otimes T$  commute.
- (e) If  $S$  and  $T$  are invertible, so is  $S \otimes T$ , with inverse  $S^{-1} \otimes T^{-1}$ .

*Solution.*

- (a) Fails. By [Proposition 14.3.3](#) (c) the right side is  $S_1 S_2 \otimes T^2$ , and with  $S_1 = S_2 = \text{id}_V$  and  $T = 2 \text{id}_W$  over  $\mathbb{Q}$  the two sides are  $2 \text{id}$  and  $4 \text{id}$  on  $V \otimes W$ , by [Example 3](#) after [Definition 14.3.1](#).
- (b) Holds: it is exactly [Proposition 14.3.3](#) (c).
- (c) Holds. If  $S \neq 0$  and  $T \neq 0$ , then  $\text{rank } S \geq 1$  and  $\text{rank } T \geq 1$ , so  $\text{rank}(S \otimes T) \geq 1$  by [Corollary 14.3.8](#), and  $S \otimes T \neq 0$ .
- (d) Holds. By [Proposition 14.3.3](#) (c) and (d), both composites equal  $S \otimes T$ .
- (e) Holds. By [Proposition 14.3.3](#) (c) and (d),  $(S \otimes T)(S^{-1} \otimes T^{-1}) = \text{id}_V \otimes \text{id}_W = \text{id}_{V \otimes W}$ , and likewise in the other order.

### 14.3.5.3 C. Going deeper

#### Exercise C1: When a tensor product of maps vanishes.

Let  $V, V', W, W'$  be finite-dimensional over  $F$ ,  $S \in \mathcal{L}(V, V')$  and  $T \in \mathcal{L}(W, W')$ .

- (a) Prove that  $S \otimes T = 0$  if and only if  $S = 0$  or  $T = 0$ .
- (b) Prove that  $\ker(S \otimes \text{id}_W) = \ker S \otimes W$ , meaning the span of all  $\mathbf{v} \otimes \mathbf{w}$  with  $\mathbf{v} \in \ker S$ .
- (c) Deduce  $\text{nullity}(S \otimes \text{id}_W) = \text{nullity } S \cdot \dim W$ , and check it against [Corollary 14.3.8](#).

*Solution.*

- (a)  $(\Leftarrow)$  If  $S = 0$ , then  $(S \otimes T)(\mathbf{v} \otimes \mathbf{w}) = \mathbf{0} \otimes T\mathbf{w} = \mathbf{0}$  for all  $\mathbf{v}, \mathbf{w}$ , so  $S \otimes T = 0$  by [Lemma 14.3.2](#). Likewise if  $T = 0$ .  $(\Rightarrow)$  Suppose  $S \neq 0$  and  $T \neq 0$ . Then  $\text{rank } S \geq 1$  and  $\text{rank } T \geq 1$ , so  $\text{rank}(S \otimes T) = \text{rank } S \cdot \text{rank } T \geq 1$  by [Corollary 14.3.8](#), and  $S \otimes T \neq 0$ .
- (b)  $(\supseteq)$  For  $\mathbf{v} \in \ker S$  and  $\mathbf{w} \in W$ ,  $(S \otimes \text{id}_W)(\mathbf{v} \otimes \mathbf{w}) = \mathbf{0} \otimes \mathbf{w} = \mathbf{0}$ ; a kernel is a subspace ([Theorem 3.2.1](#)), so it contains the span of these elements.  $(\subseteq)$  Choose a basis  $(\mathbf{v}_1, \dots, \mathbf{v}_n)$  of  $V$  whose last  $n - r$  vectors span  $\ker S$ , where  $r = \text{rank } S$ , and such that  $S\mathbf{v}_1, \dots, S\mathbf{v}_r$  are independent (as in the proof of [Corollary 14.3.8](#)). Let  $(\mathbf{w}_1, \dots, \mathbf{w}_q)$  be a basis of  $W$  and let  $z = \sum_{j,l} c_{jl} \mathbf{v}_j \otimes \mathbf{w}_l$  lie in the kernel. Then

$$\mathbf{0} = (S \otimes \text{id}_W)(z) = \sum_{j \leq r} \sum_l c_{jl} S\mathbf{v}_j \otimes \mathbf{w}_l,$$

and the vectors  $S\mathbf{v}_j \otimes \mathbf{w}_l$  with  $j \leq r$  are linearly independent by [Theorem 14.2.5](#), after extending  $(S\mathbf{v}_1, \dots, S\mathbf{v}_r)$  to a basis of  $V'$ . Hence  $c_{jl} = 0$  for all  $j \leq r$ , so  $z$  is a combination of the  $\mathbf{v}_j \otimes \mathbf{w}_l$  with  $j > r$ , all of which lie in  $\ker S \otimes W$ .

- (c) By (b) and [Theorem 14.2.5](#) applied to  $\ker S$  and  $W$ ,  $\text{nullity}(S \otimes \text{id}_W) = \dim \ker S \cdot \dim W = \text{nullity } S \cdot \dim W$ . Consistency: [Corollary 14.3.8](#) gives  $\text{rank}(S \otimes \text{id}_W) =$

$\text{rank } S \cdot \dim W$ , and adding the two gives  $(\text{rank } S + \text{nullity } S) \dim W = \dim V \cdot \dim W = \dim(V \otimes W)$ , as [Theorem 3.3.1](#) requires.

### Exercise C2: Rank-one maps and simple tensors.

Let  $V, W$  be finite-dimensional over  $F$  and let  $\Theta$  be the isomorphism of [Theorem 14.3.9](#).

- Prove that  $\Theta(\xi)$  has rank at most 1 if and only if  $\xi$  is a simple tensor  $\varphi \otimes \mathbf{w}$ .
- Deduce that “not every tensor is simple” (§02) says exactly that not every linear map has rank at most 1, and identify the tensor corresponding to  $\text{id}_V$  in a basis.
- Let  $\dim V = \dim W = 2$ . Exhibit  $\xi \in V^* \otimes W$  that is not simple, by exhibiting a map of rank 2.

*Solution.*

- ( $\Leftarrow$ )  $\Theta(\varphi \otimes \mathbf{w})$  sends every  $\mathbf{v}$  into  $\text{span}(\mathbf{w})$ , so its image has dimension at most 1.  
( $\Rightarrow$ ) Suppose  $R := \Theta(\xi)$  has rank at most 1. If  $R = 0$ , then  $\xi = 0 = 0 \otimes \mathbf{0}$  since  $\Theta$  is injective. Otherwise  $\text{im } R = \text{span}(\mathbf{w})$  for some  $\mathbf{w} \neq \mathbf{0}$ , so for each  $\mathbf{v}$  there is a unique scalar  $\varphi(\mathbf{v})$  with  $R\mathbf{v} = \varphi(\mathbf{v})\mathbf{w}$ ; uniqueness and linearity of  $R$  make  $\varphi$  linear, so  $\varphi \in V^*$ . Then  $R = \Theta(\varphi \otimes \mathbf{w})$ , and  $\xi = \varphi \otimes \mathbf{w}$  because  $\Theta$  is injective.
- By (a) and bijectivity of  $\Theta$ , the non-simple tensors in  $V^* \otimes W$  correspond exactly to the maps of rank at least 2. Taking  $W = V$  with basis  $\mathcal{B} = (\mathbf{v}_1, \dots, \mathbf{v}_n)$  and dual basis  $(\varphi_1, \dots, \varphi_n)$ , the matrix of  $\text{id}_V$  is  $\mathbf{I}_n = \sum_i \mathbf{E}_{ii}$ , so  $\text{id}_V = \Theta(\sum_{i=1}^n \varphi_i \otimes \mathbf{v}_i)$ . For  $n \geq 2$  this has rank  $n \geq 2$ , hence is not simple. For  $n = 2$  it is the witness  $\varphi_1 \otimes \mathbf{v}_1 + \varphi_2 \otimes \mathbf{v}_2$ , which is [Example \(A tensor that is not simple\)](#) read through  $\Theta$ .
- Take  $\xi = \varphi_1 \otimes \mathbf{w}_1 + \varphi_2 \otimes \mathbf{w}_2$  for bases  $(\mathbf{v}_1, \mathbf{v}_2)$  of  $V$  and  $(\mathbf{w}_1, \mathbf{w}_2)$  of  $W$ . Then  $\Theta(\xi)$  sends  $\mathbf{v}_1 \mapsto \mathbf{w}_1$  and  $\mathbf{v}_2 \mapsto \mathbf{w}_2$ , so it has rank 2, and by (a)  $\xi$  is not simple.

### Exercise C3: Repeating the swap.

Let  $V$  and  $W$  be finite-dimensional over  $F$  with ordered bases  $\mathcal{B}$  and  $\mathcal{C}$ .

- Prove that there is a unique isomorphism  $\tau: V \otimes W \rightarrow W \otimes V$  with  $\tau(\mathbf{v} \otimes \mathbf{w}) = \mathbf{w} \otimes \mathbf{v}$ .
- Prove that  $\tau(S \otimes T) = (T \otimes S)\tau$  for all  $S \in \mathcal{L}(V)$ ,  $T \in \mathcal{L}(W)$ .
- Deduce [Proposition 7.4.2](#): for  $\mathbf{A} \in M_n(F)$  and  $\mathbf{B} \in M_m(F)$  there is a permutation matrix  $\mathbf{P}$  with  $\mathbf{B} \otimes \mathbf{A} = \mathbf{P}^{-1}(\mathbf{A} \otimes \mathbf{B})\mathbf{P}$ , and identify  $\mathbf{P}$ .

*Hint for (c): think of  $\tau$  as renaming basis vectors rather than as moving them.*

*Solution.*

- The map  $(\mathbf{v}, \mathbf{w}) \mapsto \mathbf{w} \otimes \mathbf{v}$  is bilinear, so [Definition 14.2.1](#) gives a unique linear  $\tau$  with  $\tau(\mathbf{v} \otimes \mathbf{w}) = \mathbf{w} \otimes \mathbf{v}$ . The same construction with the roles swapped gives  $\sigma: W \otimes V \rightarrow V \otimes W$  with  $\sigma(\mathbf{w} \otimes \mathbf{v}) = \mathbf{v} \otimes \mathbf{w}$ . Then  $\sigma\tau$  fixes every simple tensor, so  $\sigma\tau = \text{id}$  by [Lemma 14.3.2](#), and likewise  $\tau\sigma = \text{id}$ . Hence  $\tau$  is an isomorphism.
- Both sides are linear maps out of  $V \otimes W$ , so by [Lemma 14.3.2](#) compare at  $\mathbf{v} \otimes \mathbf{w}$ : the left gives  $\tau(S\mathbf{v} \otimes T\mathbf{w}) = T\mathbf{w} \otimes S\mathbf{v}$ , and the right gives  $(T \otimes S)(\mathbf{w} \otimes \mathbf{v}) = T\mathbf{w} \otimes S\mathbf{v}$ .

- (c) Let  $S, T$  be the operators on  $F^n, F^m$  with matrices  $\mathbf{A}, \mathbf{B}$ . Both  $\mathcal{B} \otimes \mathcal{C}$  and  $\tau^{-1}(\mathcal{C} \otimes \mathcal{B})$  are orderings of the same basis of  $F^n \otimes F^m$ , differing by the permutation  $\sigma$  that sends position  $(k-1)n+i$  to position  $(i-1)m+k$  — the direction matters, since  $\sigma$  is not an involution unless  $n=m$  — and let  $\mathbf{P}$  be its matrix, which is exactly the  $\mathbf{P}_\sigma$  built in the proof of [Proposition 7.4.2](#).

## 14.4 Contraction and the Trace

Chapter 3 defined the trace of an operator by choosing a basis, writing down the matrix, and adding the diagonal entries; it then proved that the answer does not depend on the choice ([Definition 3.10.2](#)). That is a correct definition and an unsatisfying one. It says how to compute the number, not what the number is, and the independence of the basis arrives afterwards, as a theorem about similar matrices. Section 03 offers a way out. An operator on  $V$  is an element of  $V^* \otimes V$  ([Theorem 14.3.9](#)), and on  $V^* \otimes V$  there is one obvious scalar-valued map that uses no basis at all: apply the functional to the vector. This section builds that map, proves that it is the trace, and extends it to tensors with many slots.

### 14.4.1 Evaluation, made linear

The dual space was invented so that its elements could eat vectors. The eating is a function of two arguments,

$$V^* \times V \rightarrow F, \quad (\varphi, \mathbf{v}) \mapsto \varphi(\mathbf{v}),$$

and it is bilinear: linear in  $\mathbf{v}$  because  $\varphi$  is a linear map, and linear in  $\varphi$  because that is how  $V^*$  is added and scaled ([Definition 4.1.2](#)). A bilinear map is exactly what the tensor product converts into a linear one, and by now this should be a reflex.

*Contraction is the evaluation pairing, turned into a single linear map by the tensor product.*

#### Definition 14.4.1 — Contraction

Let  $V$  be a vector space over  $F$ . The **contraction** of  $V$  is the unique linear map

$$C: V^* \otimes V \rightarrow F, \quad C(\varphi \otimes \mathbf{v}) = \varphi(\mathbf{v}) \quad (\varphi \in V^*, \mathbf{v} \in V).$$

**Well-definedness.** The pairing  $(\varphi, \mathbf{v}) \mapsto \varphi(\mathbf{v})$  is bilinear, so [Definition 14.2.1](#) supplies exactly one linear map on  $V^* \otimes V$  taking those values. Nothing else is chosen: no basis, no finite dimension, no extra structure on  $V$ .

#### Examples.

1. **In coordinates.** Let  $\dim V = n$ , let  $\mathcal{B} = (\mathbf{v}_1, \dots, \mathbf{v}_n)$  be a basis and  $\mathcal{B}^* = (\varphi^1, \dots, \varphi^n)$  its dual basis ([Definition 4.1.3](#)), so  $\varphi^j(\mathbf{v}_i) = \delta_i^j$ . A general  $\xi \in V^* \otimes V$  is  $\xi = \sum_{i,j} a_{ij} \varphi^j \otimes \mathbf{v}_i$  by [Theorem 14.2.5](#), and

$$C(\xi) = \sum_{i,j} a_{ij} \varphi^j(\mathbf{v}_i) = \sum_{i=1}^n a_{ii}.$$

The contraction sums the coefficients whose two indices agree. That is a diagonal sum, which is the first hint of what is coming.



2. **The smallest case.** If  $\dim V = 1$ , pick  $\mathbf{v} \neq \mathbf{0}$  and let  $\varphi$  be dual to it. Every element of  $V^* \otimes V$  is  $a\varphi \otimes \mathbf{v}$ , and  $C(a\varphi \otimes \mathbf{v}) = a$ . So  $C$  is the isomorphism  $V^* \otimes V \cong F$ . Degenerate, and worth seeing: the contraction is the only thing that can happen here.
3. **A functional times a vector, in  $F^n$ .** With  $V = F^n$ ,  $\varphi = \mathbf{y}^\top$  and  $\mathbf{v} = \mathbf{x}$ ,  $C(\varphi \otimes \mathbf{v}) = \mathbf{y}^\top \mathbf{x}$ , the dot product. Chapter 10's inner product on  $\mathbb{R}^n$  is this contraction composed with the identification of  $\mathbb{R}^n$  with its dual — which is extra structure, and is why the dot product is not a contraction of  $V \otimes V$ .

**Non-example by minimal change.** Replace  $\varphi(\mathbf{v})$  by  $\varphi(\mathbf{v})^2$ . The assignment  $(\varphi, \mathbf{v}) \mapsto \varphi(\mathbf{v})^2$  is not bilinear — doubling  $\mathbf{v}$  multiplies it by 4 — so no linear map on  $V^* \otimes V$  agrees with it, and [Definition 14.2.1](#) does not apply. The exact clause that fails is linearity in each slot.

**Why one upper slot and one lower slot.** Contraction needs a functional *and* a vector: the functional supplies the mouth. There is no analogous map  $V \otimes V \rightarrow F$ .

*Warning.* **There is no canonical contraction of two vectors.** A linear map  $V \otimes V \rightarrow F$  is the same thing as a bilinear form on  $V$  ([Definition 14.2.1](#) with  $Z = F$ ), and Chapter 13 spent a chapter on the fact that a vector space carries *many* bilinear forms and prefers none of them. Each basis of  $V$  supplies one,  $(\mathbf{v}, \mathbf{w}) \mapsto \sum_i \varphi^i(\mathbf{v})\varphi^i(\mathbf{w})$ , and different bases supply different ones: on  $\mathbb{R}^2$  the standard basis gives the dot product, while  $((1, 0), (1, 1))$  gives a form under which  $(1, 0)$  and  $(0, 1)$  are not orthogonal. Only the pairing between  $V^*$  and  $V$  is free of charge.

## 14.4.2 Contraction is the trace

We can now say what the trace is.

### Theorem 14.4.1 — The Trace Is a Contraction

Let  $V$  be a finite-dimensional vector space over  $F$ , and let

$$\Theta: V^* \otimes V \rightarrow \mathcal{L}(V)$$

be the isomorphism of [Theorem 14.3.9](#). Then

$$\mathrm{tr}(\Theta(\xi)) = C(\xi) \quad \text{for every } \xi \in V^* \otimes V,$$

that is,  $\mathrm{tr} = C \circ \Theta^{-1}$  as maps  $\mathcal{L}(V) \rightarrow F$ .

*Idea.* Both sides are linear in  $\xi$ , so by [Lemma 14.3.2](#) it is enough to check simple tensors. And  $\Theta(\varphi \otimes \mathbf{v})$  is the rank-at-most-one map  $\mathbf{u} \mapsto \varphi(\mathbf{u})\mathbf{v}$ , whose trace can be read off from a single well-chosen basis: put  $\mathbf{v}$  first.

*Proof.* Both  $\xi \mapsto \mathrm{tr}(\Theta(\xi))$  and  $\xi \mapsto C(\xi)$  are linear, being composites of linear maps ([Theorem 3.10.3](#) for the first). By [Lemma 14.3.2](#) it suffices to prove  $\mathrm{tr}(\Theta(\varphi \otimes \mathbf{v})) = \varphi(\mathbf{v})$  for all  $\varphi \in V^*$ ,  $\mathbf{v} \in V$ .

Write  $R := \Theta(\varphi \otimes \mathbf{v})$ , so  $R\mathbf{u} = \varphi(\mathbf{u})\mathbf{v}$  for every  $\mathbf{u}$ . If  $\mathbf{v} = \mathbf{0}$ , then  $R = 0$  and  $\varphi(\mathbf{v}) = 0$ , so both sides vanish. Otherwise  $(\mathbf{v})$  is a linearly independent list, so by [Theorem 1.8.2](#)



it extends to a basis  $\mathcal{B} = (\mathbf{u}_1, \dots, \mathbf{u}_n)$  of  $V$  with  $\mathbf{u}_1 = \mathbf{v}$ . For each  $j$ ,

$$R\mathbf{u}_j = \varphi(\mathbf{u}_j) \mathbf{v} = \varphi(\mathbf{u}_j) \mathbf{u}_1,$$

so column  $j$  of  $[R]_{\mathcal{B}}^{\mathcal{B}}$  has  $\varphi(\mathbf{u}_j)$  in row 1 and zeros below it (Definition 3.6.1). Hence the only non-zero diagonal entry is the one in position  $(1, 1)$ , equal to  $\varphi(\mathbf{u}_1) = \varphi(\mathbf{v})$ . By Definition 3.10.2,  $\text{tr } R = \varphi(\mathbf{v})$ , as required. ■

**What has changed.** Chapter 3's definition named a basis and then had to prove that the answer survived a change of basis, using Theorem 3.7.5. Here nothing was named. The map  $C$  was manufactured by the universal property out of the evaluation pairing, which exists before any basis does, and  $\Theta$  was manufactured the same way. So basis-independence is not a theorem about  $C \circ \Theta^{-1}$ ; there is no basis in the construction to be independent of. Chapter 3's theorem is still true, and it is what makes the two descriptions agree — but it is now an observation about a *computation* of the trace, not part of its definition. The single basis used in the proof above was a convenience for reading off one number, not a choice the answer could depend on.

Two consequences come free. First, the trace of a rank-one map.

### Corollary 14.4.2 — Trace of a Rank-One Map

Let  $V$  be finite-dimensional,  $\varphi \in V^*$  and  $\mathbf{v} \in V$ , and let  $R \in \mathcal{L}(V)$  be  $R\mathbf{u} = \varphi(\mathbf{u})\mathbf{v}$ . Then  $\text{tr } R = \varphi(\mathbf{v})$ . In particular, with a basis  $\mathcal{B}$  and its dual basis,  $\text{id}_V = \Theta(\sum_i \varphi^i \otimes \mathbf{v}_i)$  and  $\text{tr id}_V = \sum_i \varphi^i(\mathbf{v}_i) = n \cdot 1$ .

*Proof.* The first claim was proved inside Theorem 14.4.1. For the second,  $\Theta(\sum_i \varphi^i \otimes \mathbf{v}_i)$  sends  $\mathbf{v}_j$  to  $\sum_i \varphi^i(\mathbf{v}_j)\mathbf{v}_i = \mathbf{v}_j$ , so it is  $\text{id}_V$  (Theorem 3.1.5); applying  $C$  gives  $\sum_i \varphi^i(\mathbf{v}_i) = \sum_i 1 = n \cdot 1$ . ■

Second, the promise made in §03: the trace of a tensor product of maps multiplies, and now we can see why. The key is that a functional on  $V \otimes W$  is built from one on each factor.

### Proposition 14.4.3 — The Dual of a Tensor Product

Let  $V$  and  $W$  be finite-dimensional over  $F$ . There is a unique linear map

$$\kappa: V^* \otimes W^* \rightarrow (V \otimes W)^*, \quad \kappa(\varphi \otimes \psi)(\mathbf{v} \otimes \mathbf{w}) = \varphi(\mathbf{v})\psi(\mathbf{w}),$$

and it is an isomorphism.

*Proof.* For fixed  $\varphi, \psi$ , the map  $(\mathbf{v}, \mathbf{w}) \mapsto \varphi(\mathbf{v})\psi(\mathbf{w})$  is bilinear, so Definition 14.2.1 produces a unique functional  $\varphi \cdot \psi \in (V \otimes W)^*$  with the stated values. The assignment  $(\varphi, \psi) \mapsto \varphi \cdot \psi$  is itself bilinear, since two functionals agreeing on all simple tensors are equal (Lemma 14.3.2), so Definition 14.2.1 gives the unique linear  $\kappa$ .

Let  $\mathcal{B} = (\mathbf{v}_1, \dots, \mathbf{v}_n)$  and  $\mathcal{C} = (\mathbf{w}_1, \dots, \mathbf{w}_m)$  be bases with dual bases  $(\varphi^i)$  and  $(\psi^j)$ . Then  $\kappa(\varphi^i \otimes \psi^j)(\mathbf{v}_k \otimes \mathbf{w}_l) = \delta_k^i \delta_l^j$ , so  $\kappa$  carries the basis  $(\varphi^i \otimes \psi^j)$  of  $V^* \otimes W^*$  (Theorem 14.2.5)

to the dual basis of  $\mathcal{B} \otimes \mathcal{C}$  (Theorem 4.1.3 (a), which characterizes it by these values). A linear map carrying a basis to a basis is an isomorphism, by Theorem 3.2.4 and Theorem 3.3.1. ■

### Corollary 14.4.4 — Why the Trace Multiplies

Let  $V, W$  be finite-dimensional over  $F$ ,  $S \in \mathcal{L}(V)$  and  $T \in \mathcal{L}(W)$ . Then  $\text{tr}(S \otimes T) = \text{tr } S \cdot \text{tr } T$ .

*Proof.* Both  $(S, T) \mapsto \text{tr}(S \otimes T)$  and  $(S, T) \mapsto \text{tr } S \cdot \text{tr } T$  are bilinear on  $\mathcal{L}(V) \times \mathcal{L}(W)$ , by Proposition 14.3.3 (a), (b) and Theorem 3.10.3. The maps of rank at most one span  $\mathcal{L}(V)$ , since  $\Theta$  is onto and simple tensors span  $V^* \otimes V$  (Proposition 14.2.4); likewise for  $\mathcal{L}(W)$ . So it suffices to check the identity when  $S = \Theta(\varphi \otimes \mathbf{v})$  and  $T = \Theta(\psi \otimes \mathbf{w})$ .

For such  $S$  and  $T$  and any  $\mathbf{a} \otimes \mathbf{b} \in V \otimes W$ ,

$$(S \otimes T)(\mathbf{a} \otimes \mathbf{b}) = \varphi(\mathbf{a})\mathbf{v} \otimes \psi(\mathbf{b})\mathbf{w} = \kappa(\varphi \otimes \psi)(\mathbf{a} \otimes \mathbf{b})(\mathbf{v} \otimes \mathbf{w}),$$

so  $S \otimes T = \Theta(\kappa(\varphi \otimes \psi) \otimes (\mathbf{v} \otimes \mathbf{w}))$  by Lemma 14.3.2. By Theorem 14.4.1 and Proposition 14.4.3,

$$\text{tr}(S \otimes T) = \kappa(\varphi \otimes \psi)(\mathbf{v} \otimes \mathbf{w}) = \varphi(\mathbf{v})\psi(\mathbf{w}),$$

which is  $\text{tr } S \cdot \text{tr } T$  by Corollary 14.4.2. ■

So the trace multiplies because *evaluation* multiplies: a functional on a tensor product is a product of functionals, and contraction is evaluation. Chapter 7 proved this by summing  $a_{ii}b_{kk}$  over all pairs (Theorem 7.4.3 (b)), which is the same fact with the sum written out.

## 14.4.3 Tensors with many slots

Contraction is not confined to one upper and one lower slot; it can be applied to a chosen pair inside a tensor with any number of them. First we name those tensors. Write

$$V^{\otimes p} := \underbrace{V \otimes \cdots \otimes V}_p, \quad V^{\otimes 0} := F,$$

which is unambiguous by Proposition 14.3.5.

### Definition 14.4.2 — Tensor of type $(p, q)$

Let  $V$  be a vector space over  $F$  and  $p, q \geq 0$ . A **tensor of type  $(p, q)$**  on  $V$  is an element of

$$T_q^p(V) := V^{\otimes p} \otimes (V^*)^{\otimes q}.$$

The  $p$  copies of  $V$  are the **contravariant** slots and the  $q$  copies of  $V^*$  the **covariant**

slots. The number  $p + q$  is the **rank** or **order** of the tensor.

If  $\dim V = n$ , then  $\dim T_q^p(V) = n^{p+q}$  by [Theorem 14.2.5](#) and [Corollary 4.1.4](#). The small cases are all things we have met:

- $T_0^0(V) = F$ : scalars.
- $T_0^1(V) = V$  and  $T_1^0(V) = V^*$ : vectors and functionals.
- $T_1^1(V) = V \otimes V^* \cong \mathcal{L}(V)$ : operators, by [Theorem 14.3.9](#) together with the swap of [Exercise \(C3: Proving the swap\)](#), which sends  $\mathbf{v} \otimes \varphi$  to  $\varphi \otimes \mathbf{v}$ .
- $T_2^0(V) \cong (V \otimes V)^*$  by [Proposition 14.4.3](#), and a functional on  $V \otimes V$  is a bilinear form on  $V$  by [Definition 14.2.1](#). So a type  $(0, 2)$  tensor is exactly a bilinear form — the whole subject matter of Chapter 13.

*Warning.* The word “**rank**” is overloaded here. The rank of a tensor in the sense of [Definition 14.4.2](#) is the number of slots,  $p + q$ ; it has nothing to do with  $\text{rank } T$  for a linear map, nor with the smallest number of simple tensors needed to write an element. A type  $(1, 1)$  tensor always has rank 2 in this sense, whatever the rank of the corresponding operator. Where confusion is possible we say **order**.

Now the general contraction. Contracting slot  $a$  against slot  $b$  means feeding the  $b$ -th functional the  $a$ -th vector and keeping everything else.

### Definition 14.4.3 — Contraction on a pair of slots

Let  $V$  be a vector space over  $F$ , let  $p, q \geq 1$ , and fix  $1 \leq a \leq p$  and  $1 \leq b \leq q$ . The **contraction on slots  $a$  and  $b$**  is the unique linear map

$$C_b^a: T_q^p(V) \rightarrow T_{q-1}^{p-1}(V)$$

with, for all  $\mathbf{v}_1, \dots, \mathbf{v}_p \in V$  and  $\varphi^1, \dots, \varphi^q \in V^*$ ,

$$\begin{aligned} C_b^a(\mathbf{v}_1 \otimes \dots \otimes \mathbf{v}_p \otimes \varphi^1 \otimes \dots \otimes \varphi^q) \\ = \varphi^b(\mathbf{v}_a) \mathbf{v}_1 \otimes \dots \widehat{\mathbf{v}_a} \dots \otimes \mathbf{v}_p \otimes \varphi^1 \otimes \dots \widehat{\varphi^b} \dots \otimes \varphi^q, \end{aligned}$$

where a hat means that the factor is omitted.

**Well-definedness.** The right-hand side is multilinear in the  $p + q$  arguments: each argument appears exactly once, either inside the scalar  $\varphi^b(\mathbf{v}_a)$  or as one factor of a tensor product, and both are linear in it. By [Lemma 14.3.4](#) there is exactly one linear map on  $T_q^p(V)$  with these values. For  $p = q = 1$  it lands in  $T_0^0(V) = F$  and sends  $\mathbf{v} \otimes \varphi$  to  $\varphi(\mathbf{v})$ : it is [Definition 14.4.1](#) read on  $V \otimes V^*$  instead of  $V^* \otimes V$ , that is,  $C$  composed with the swap of [Exercise \(C3: Proving the swap\)](#). We use that identification silently from here on, and [Theorem 14.4.1](#) applies to either reading.

*Warning.* Which slots you contract matters. Let  $\dim V = 2$  with basis  $(\mathbf{v}_1, \mathbf{v}_2)$  and dual basis  $(\varphi^1, \varphi^2)$ , and take the type  $(2, 2)$  tensor

$$t = \mathbf{v}_1 \otimes \mathbf{v}_2 \otimes \varphi^1 \otimes \varphi^2.$$

Then  $C_1^1(t) = \varphi^1(\mathbf{v}_1) \mathbf{v}_2 \otimes \varphi^2 = \mathbf{v}_2 \otimes \varphi^2$ , while  $C_2^1(t) = \varphi^2(\mathbf{v}_1) \mathbf{v}_2 \otimes \varphi^1 = 0$ . The two contractions of the same tensor differ, one of them being zero. There is no “the”

contraction of a tensor of order  $\geq 3$ : the slots must be named.

*Quick check.* Let  $\dim V = 3$  with basis  $(\mathbf{v}_1, \mathbf{v}_2, \mathbf{v}_3)$  and dual basis  $(\varphi^1, \varphi^2, \varphi^3)$ , and let

$$t = \mathbf{v}_1 \otimes \varphi^1 + 2\mathbf{v}_2 \otimes \varphi^2 - \mathbf{v}_1 \otimes \varphi^3 \in T_1^1(V).$$

Compute  $C_1^1(t)$ , and say which operator  $t$  corresponds to.

*Solution.*  $C_1^1(t) = \varphi^1(\mathbf{v}_1) + 2\varphi^2(\mathbf{v}_2) - \varphi^3(\mathbf{v}_1) = 1 + 2 - 0 = 3$ . The corresponding operator sends  $\mathbf{v}_1 \mapsto \mathbf{v}_1$ ,  $\mathbf{v}_2 \mapsto 2\mathbf{v}_2$ ,  $\mathbf{v}_3 \mapsto -\mathbf{v}_1$ , so its matrix in  $(\mathbf{v}_1, \mathbf{v}_2, \mathbf{v}_3)$  is  $\begin{pmatrix} 1 & 0 & -1 \\ 0 & 2 & 0 \\ 0 & 0 & 0 \end{pmatrix}$ , of trace 3, in agreement with [Theorem 14.4.1](#).

Index notation, in the next section, makes contraction visibly what it looks like in physics: set an upper index equal to a lower one and sum.

## 14.4.4 Exercises

### 14.4.4.1 A. Check your understanding

#### Exercise A1.

- Define the contraction  $C: V^* \otimes V \rightarrow F$ , and say what makes it well defined.
- State the relation between the trace and the contraction, with its hypothesis.
- Explain why there is no canonical linear map  $V \otimes V \rightarrow F$ .
- For  $\dim V = 4$ , what is  $\dim T_3^2(V)$ ?
- True or false: for  $t \in T_2^2(V)$ , the four contractions  $C_b^a(t)$  all agree. Justify your answer.

*Solution.*

- $C$  is the unique linear map with  $C(\varphi \otimes \mathbf{v}) = \varphi(\mathbf{v})$  ([Definition 14.4.1](#)). It is well defined because  $(\varphi, \mathbf{v}) \mapsto \varphi(\mathbf{v})$  is bilinear, so [Definition 14.2.1](#) produces exactly one such linear map.
- If  $V$  is **finite-dimensional**, then  $\text{tr}(\Theta(\xi)) = C(\xi)$  for all  $\xi \in V^* \otimes V$ , where  $\Theta$  is the isomorphism of [Theorem 14.3.9](#) ([Theorem 14.4.1](#)).
- Such a map is the same thing as a bilinear form on  $V$ , and a vector space has many bilinear forms with none distinguished; each basis produces a different one. Contraction works only because  $V^*$  comes with the evaluation pairing built in.
- $4^{2+3} = 4^5 = 1024$ .
- False. [Definition 14.4.3](#)'s warning gives  $t = \mathbf{v}_1 \otimes \mathbf{v}_2 \otimes \varphi^1 \otimes \varphi^2$  with  $C_1^1(t) = \mathbf{v}_2 \otimes \varphi^2 \neq 0$  and  $C_2^1(t) = 0$ .

### 14.4.4.2 B. Practice

**Exercise B1: Contracting a mixed tensor.**

Let  $\dim V = 3$  with basis  $\mathcal{B} = (\mathbf{v}_1, \mathbf{v}_2, \mathbf{v}_3)$  and dual basis  $(\varphi^1, \varphi^2, \varphi^3)$ . Let

$$t = 2\mathbf{v}_1 \otimes \varphi^1 - \mathbf{v}_1 \otimes \varphi^2 + 3\mathbf{v}_2 \otimes \varphi^2 + \mathbf{v}_3 \otimes \varphi^1 - 4\mathbf{v}_3 \otimes \varphi^3.$$

- Write the matrix of the operator  $S$  corresponding to  $t$ , in the basis  $\mathcal{B}$ .
- Compute  $C_1^1(t)$  from [Definition 14.4.3](#), and check it against  $\text{tr } S$ .
- Compute  $\text{tr}(S \otimes S)$  without writing down a  $9 \times 9$  matrix.

*Solution.*

- The tensor  $\mathbf{v}_i \otimes \varphi^j$  corresponds to the operator  $\mathbf{u} \mapsto \varphi^j(\mathbf{u})\mathbf{v}_i$ , whose matrix is the matrix unit  $\mathbf{E}_{ij}$  ([Theorem 14.3.9](#)). Adding up,

$$[S]_{\mathcal{B}}^{\mathcal{B}} = \begin{pmatrix} 2 & -1 & 0 \\ 0 & 3 & 0 \\ 1 & 0 & -4 \end{pmatrix}.$$

- $C_1^1(t) = 2\varphi^1(\mathbf{v}_1) - \varphi^2(\mathbf{v}_1) + 3\varphi^2(\mathbf{v}_2) + \varphi^1(\mathbf{v}_3) - 4\varphi^3(\mathbf{v}_3) = 2 - 0 + 3 + 0 - 4 = 1$ , and the diagonal of the matrix in (a) sums to  $2 + 3 - 4 = 1$ .
- By [Corollary 14.4.4](#),  $\text{tr}(S \otimes S) = (\text{tr } S)^2 = 1$ .

**Exercise B2: Two contractions of one tensor.**

Let  $\dim V = 2$  with basis  $(\mathbf{v}_1, \mathbf{v}_2)$  and dual basis  $(\varphi^1, \varphi^2)$ , and let

$$t = \mathbf{v}_1 \otimes \varphi^1 \otimes \varphi^2 + \mathbf{v}_2 \otimes \varphi^2 \otimes \varphi^2 \in T_2^1(V).$$

Compute  $C_1^1(t)$  and  $C_2^1(t)$ , and state which space each lies in.

*Solution.* Both lie in  $T_1^0(V) = V^*$ . Contracting the single upper slot against the **first** lower slot,

$$C_1^1(t) = \varphi^1(\mathbf{v}_1)\varphi^2 + \varphi^2(\mathbf{v}_2)\varphi^2 = \varphi^2 + \varphi^2 = 2\varphi^2.$$

Against the **second** lower slot, the remaining factor is the first functional:

$$C_2^1(t) = \varphi^2(\mathbf{v}_1)\varphi^1 + \varphi^2(\mathbf{v}_2)\varphi^2 = 0 + \varphi^2 = \varphi^2.$$

So  $C_1^1(t) = 2\varphi^2 \neq \varphi^2 = C_2^1(t)$ .

**Exercise B3: Which statements hold?.**

Let  $V, W$  be finite-dimensional over  $F$ . Determine which of the following hold. Justify your answer.

- $C(\xi) = 0$  implies  $\xi = 0$ .
- $\text{tr}(\Theta(\varphi \otimes \mathbf{v})) = \varphi(\mathbf{v})$  for all  $\varphi \in V^*$ ,  $\mathbf{v} \in V$ .
- Every element of  $T_2^0(V)$  is a bilinear form on  $V$ , and conversely.
- $\text{tr}(S \otimes T) = \text{tr}(T \otimes S)$  for  $S \in \mathcal{L}(V)$ ,  $T \in \mathcal{L}(W)$ .

(e)  $C: V^* \otimes V \rightarrow F$  is surjective whenever  $V \neq \{0\}$ .

*Solution.*

- (a) Fails for  $\dim V \geq 2$ . With a basis and its dual,  $\xi = \varphi^1 \otimes \mathbf{v}_2 \neq 0$  (Proposition 14.2.8) but  $C(\xi) = \varphi^1(\mathbf{v}_2) = 0$ . Contraction is a functional, not an injection; its kernel has dimension  $n^2 - 1$ .
- (b) Holds: Corollary 14.4.2.
- (c) Holds. By Proposition 14.4.3,  $T_2^0(V) = V^* \otimes V^* \cong (V \otimes V)^*$ , and by Definition 14.2.1 the linear maps  $V \otimes V \rightarrow F$  correspond exactly to the bilinear maps  $V \times V \rightarrow F$ , which are the bilinear forms of Definition 13.1.1.
- (d) Holds: both equal  $\text{tr } S \cdot \text{tr } T$  by Corollary 14.4.4, the two products of scalars being equal.
- (e) Holds. Take  $\mathbf{v} \neq 0$ ; by Theorem 4.1.5 there is  $\varphi$  with  $\varphi(\mathbf{v}) = 1$ , so  $C(c\varphi \otimes \mathbf{v}) = c$  for every  $c \in F$ .

#### 14.4.4.3 C. Going deeper

##### Exercise C1: Composition, in tensor language.

Let  $V$  be finite-dimensional over  $F$  and  $\Theta: V^* \otimes V \rightarrow \mathcal{L}(V)$  the isomorphism of Theorem 14.3.9.

- (a) Prove that  $\Theta(\varphi \otimes \mathbf{v}) \Theta(\psi \otimes \mathbf{w}) = \varphi(\mathbf{w}) \Theta(\psi \otimes \mathbf{v})$  for all  $\varphi, \psi \in V^*$  and  $\mathbf{v}, \mathbf{w} \in V$ . Recall that  $ST$  means first  $T$ , then  $S$ .
- (b) Hence compute  $\text{tr}(\Theta(\varphi \otimes \mathbf{v}) \Theta(\psi \otimes \mathbf{w}))$ , and observe that the answer is unchanged when  $(\varphi, \mathbf{v})$  and  $(\psi, \mathbf{w})$  are exchanged.
- (c) Deduce that  $\text{tr}(ST) = \text{tr}(TS)$  for all  $S, T \in \mathcal{L}(V)$ .

*Solution.*

- (a) Both sides are operators on  $V$ , so evaluate at an arbitrary  $\mathbf{u} \in V$ . Applying  $\Theta(\psi \otimes \mathbf{w})$  first gives  $\psi(\mathbf{u})\mathbf{w}$ , and then

$$\Theta(\varphi \otimes \mathbf{v})(\psi(\mathbf{u})\mathbf{w}) = \psi(\mathbf{u})\varphi(\mathbf{w})\mathbf{v},$$

using linearity of  $\Theta(\varphi \otimes \mathbf{v})$ . The right side gives  $\varphi(\mathbf{w})\psi(\mathbf{u})\mathbf{v}$ , the same scalar times the same vector. Hence the two operators agree at every  $\mathbf{u}$ .

- (b) By (a) and Corollary 14.4.2,

$$\text{tr}(\Theta(\varphi \otimes \mathbf{v}) \Theta(\psi \otimes \mathbf{w})) = \varphi(\mathbf{w})\psi(\mathbf{v}).$$

Exchanging  $(\varphi, \mathbf{v})$  with  $(\psi, \mathbf{w})$  turns this into  $\psi(\mathbf{v})\varphi(\mathbf{w})$ , which is the same product of scalars.

- (c) The maps  $(S, T) \mapsto \text{tr}(ST)$  and  $(S, T) \mapsto \text{tr}(TS)$  are bilinear on  $\mathcal{L}(V) \times \mathcal{L}(V)$ , and the rank-one maps  $\Theta(\varphi \otimes \mathbf{v})$  span  $\mathcal{L}(V)$  (as  $\Theta$  is onto and simple tensors span, by Proposition 14.2.4). By (b) the two maps agree on such pairs, hence everywhere.

**Exercise C2: The contraction of the identity, over any field.**

Let  $V$  be finite-dimensional over  $F$ ,  $n = \dim V \geq 1$ , and let  $\iota \in V^* \otimes V$  be the tensor with  $\Theta(\iota) = \text{id}_V$ .

- (a) Prove that  $\iota = \sum_{i=1}^n \varphi^i \otimes \mathbf{v}_i$  for **every** basis  $\mathcal{B}$  of  $V$  with dual basis  $\mathcal{B}^*$ . (This tensor is basis-free, though each expression for it is not.)
- (b) Prove that  $C(\iota) = n \cdot 1$  in  $F$ .
- (c) Over  $\mathbb{F}_2$  with  $n = 2$ , find a tensor  $\xi \neq \iota$  in  $V^* \otimes V$  with  $C(\xi) = C(\iota)$ , and relate this to Chapter 3's observation that the trace does not determine the rank of a projection over  $\mathbb{F}_2$ .

*Solution.*

- (a) By [Corollary 14.4.2](#),  $\Theta(\sum_i \varphi^i \otimes \mathbf{v}_i) = \text{id}_V$  for any basis and its dual basis. Since  $\Theta$  is injective ([Theorem 14.3.9](#)), all these tensors are equal, and each equals  $\iota$ .
- (b)  $C(\iota) = \sum_i \varphi^i(\mathbf{v}_i) = \sum_i 1 = n \cdot 1$  by [Definition 4.1.3](#).
- (c) Take  $\xi = 0$ . Then  $C(\xi) = 0 = 2 \cdot 1 = C(\iota)$  in  $\mathbb{F}_2$ , while  $\xi \neq \iota$  because  $\Theta(\iota) = \text{id} \neq 0$ . Under  $\Theta$  these are the identity and the zero operator, both projections, of ranks 2 and 0; this is exactly [Theorem 3.10.4](#)'s warning that over a field of characteristic 2 the integer rank  $P$  cannot be recovered from  $\text{tr } P$ . The contraction is faithful to the tensor, but the field can be too small for the resulting scalar to distinguish them.

## 14.5 Index Notation and the Transformation Law

Outside mathematics, a tensor is usually not a vector in  $T_q^p(V)$  at all: it is an array of numbers with some indices written upstairs and some downstairs, together with a rule saying how the array changes when the coordinates change. That older definition is still the working one in physics and in differential geometry, and a reader who meets it after this chapter deserves a dictionary. This section supplies one. We write the components of a tensor, derive — not assert — how they transform under a change of basis, and then prove that the classical definition and ours describe the same objects in finite dimensions. The section ends with the warning the classical definition exists to make: an array of numbers is not automatically a tensor.

### 14.5.1 Components, upstairs and downstairs

Fix a finite-dimensional  $V$  over  $F$  with  $n = \dim V \geq 1$ , an ordered basis  $\mathcal{B} = (\mathbf{v}_1, \dots, \mathbf{v}_n)$ , and its dual basis  $\mathcal{B}^* = (\varphi^1, \dots, \varphi^n)$ , characterized by  $\varphi^i(\mathbf{v}_j) = \delta_j^i$  ([Definition 4.1.3](#)). By [Theorem 14.2.5](#), the  $n^{p+q}$  tensors

$$\mathbf{v}_{i_1} \otimes \dots \otimes \mathbf{v}_{i_p} \otimes \varphi^{j_1} \otimes \dots \otimes \varphi^{j_q}$$

form a basis of  $T_q^p(V)$ , so every tensor has unique coordinates in it.

**Definition 14.5.1 — Components of a tensor**

Let  $t \in T_q^p(V)$ . The **components of  $t$  in the basis  $\mathcal{B}$**  are the unique scalars  $T_{j_1 \dots j_q}^{i_1 \dots i_p} \in F$ , for  $1 \leq i_1, \dots, i_p, j_1, \dots, j_q \leq n$ , with

$$t = \sum T_{j_1 \dots j_q}^{i_1 \dots i_p} \mathbf{v}_{i_1} \otimes \dots \otimes \mathbf{v}_{i_p} \otimes \varphi^{j_1} \otimes \dots \otimes \varphi^{j_q},$$

the sum running over all  $p + q$  indices from 1 to  $n$ . An index written **upstairs** belongs to a  $V$ -slot and is called **contravariant**; an index written **downstairs** belongs to a  $V^*$ -slot and is called **covariant**.

In words: the position of an index records which kind of slot it labels, so that the shape of a symbol already tells you the type of the tensor. Nothing here is new mathematics; it is the coordinate vector of [Theorem 14.2.5](#), with the indices arranged to be readable.

Two pieces of shorthand come with it. The first is the placement rule: a vector has its coordinates upstairs,  $\mathbf{v} = \sum_i v^i \mathbf{v}_i$ , while a functional has them downstairs,  $\xi = \sum_j \xi_j \varphi^j$  — which is [Theorem 4.1.3](#) (c) written in the new notation. The second is the **summation convention**: an index appearing once upstairs and once downstairs in a product is summed over, and the  $\sum$  is not written. With it, the displayed formula above reads

$$t = T_{j_1 \dots j_q}^{i_1 \dots i_p} \mathbf{v}_{i_1} \otimes \dots \otimes \varphi^{j_q},$$

and the evaluation of a functional on a vector is  $\xi(\mathbf{v}) = \xi_i v^i$ . The rest of this book writes its sums out; this section is the one place where the convention is used, and it is used because the transformation law is unreadable without it.

**Examples.**

1. **Type (1, 1): an operator.** If  $t \in T_1^1(V)$  corresponds to  $S \in \mathcal{L}(V)$  under [Theorem 14.3.9](#), then  $T_j^i$  is the  $(i, j)$  entry of  $[S]_{\mathcal{B}}^{\mathcal{B}}$ : the tensor  $\mathbf{v}_i \otimes \varphi^j$  is the operator  $\mathbf{u} \mapsto \varphi^j(\mathbf{u}) \mathbf{v}_i$ , whose matrix is  $\mathbf{E}_{ij}$ . Upper index = row, lower index = column.
2. **Type (0, 2): a bilinear form.** If  $t \in T_2^0(V)$  corresponds to the bilinear form  $\beta$  ([Proposition 14.4.3](#) and [Exercise \(B3: Which statements hold?\)](#)), then  $T_{ij} = \beta(\mathbf{v}_i, \mathbf{v}_j)$ , which is the Gram matrix  $[\beta]_{\mathcal{B}}$  of [Definition 13.1.2](#). Both indices downstairs.
3. **Contraction.** In components, [Definition 14.4.3](#) is “set an upper index equal to a lower one and sum”. For  $t \in T_1^1(V)$ ,

$$C_1^1(t) = T_i^i \quad \left( = \sum_{i=1}^n T_i^i \right),$$

which is the trace, by [Theorem 14.4.1](#). This is the whole reason the convention pairs an upper with a lower index: a repeated index is a contraction, and contraction is the only basis-free way to remove slots.

*Remark.* The names are historically backwards, and it is worth saying so once. A *contravariant* index sits upstairs, on a  $V$ -slot, and its components transform by the **inverse** of the matrix that transforms the basis vectors; a *covariant* index sits downstairs and transforms by the same matrix as the basis vectors. The names describe the components, not the slots. [Theorem 14.5.2](#) below is what they refer to.



## 14.5.2 How the components change

Let  $\mathcal{B} = (\mathbf{v}_1, \dots, \mathbf{v}_n)$  and  $\mathcal{B}' = (\mathbf{v}'_1, \dots, \mathbf{v}'_n)$  be two ordered bases, and let

$$\mathbf{P} = [\text{id}]_{\mathcal{B}'}^{\mathcal{B}} = (p_j^i)$$

be the change-of-coordinates matrix from  $\mathcal{B}'$  to  $\mathcal{B}$  (Definition 3.7.1), whose  $j$ -th column is  $[\mathbf{v}'_j]_{\mathcal{B}}$ . That is,

$$\mathbf{v}'_j = \sum_{i=1}^n p_j^i \mathbf{v}_i,$$

with the row index upstairs and the column index downstairs. Write  $\mathbf{Q} = \mathbf{P}^{-1} = (q_j^i)$ , which exists because a change-of-coordinates matrix is invertible (Proposition 3.7.2).

The dual bases move by the inverse matrix. This is the one computation the whole section rests on, so we do it first.

### Lemma 14.5.1 — Dual Bases Move by the Inverse Matrix

With the notation above,  $(\varphi')^k = \sum_{l=1}^n q_l^k \varphi^l$  for each  $k$ , where  $(\varphi')^1, \dots, (\varphi')^n$  is the dual basis of  $\mathcal{B}'$  supplied by Theorem 4.1.3.

*Proof.* Both sides are linear maps on  $V$ , so by the uniqueness clause of Theorem 3.1.5 it is enough to check that they agree at each vector of the basis  $\mathcal{B}'$ . By Definition 4.1.3 the left side gives  $(\varphi')^k(\mathbf{v}'_j) = \delta_j^k$ . The right side gives

$$\sum_l q_l^k \varphi^l \left( \sum_i p_j^i \mathbf{v}_i \right) = \sum_l \sum_i q_l^k p_j^i \delta_i^l = \sum_i q_i^k p_j^i,$$

which is the  $(k, j)$  entry of  $\mathbf{QP} = \mathbf{I}_n$ , that is  $\delta_j^k$ . The two agree. ■

So the dual basis vectors carry the entries of  $\mathbf{P}^{-1}$ , while the basis vectors carry the entries of  $\mathbf{P}$ . Substituting both into the expansion of a tensor gives the transformation law.

### Theorem 14.5.2 — Transformation Law for Components

Let  $t \in T_q^p(V)$  have components  $T_{j_1 \dots j_q}^{i_1 \dots i_p}$  in  $\mathcal{B}$  and  $(T')_{l_1 \dots l_q}^{k_1 \dots k_p}$  in  $\mathcal{B}'$ , and let  $\mathbf{P} = (p_j^i)$  and  $\mathbf{Q} = \mathbf{P}^{-1} = (q_j^i)$  be as above. Then, with the summation convention,

$$(T')_{l_1 \dots l_q}^{k_1 \dots k_p} = q_{i_1}^{k_1} \dots q_{i_p}^{k_p} \times p_{l_1}^{j_1} \dots p_{l_q}^{j_q} T_{j_1 \dots j_q}^{i_1 \dots i_p}.$$

Each **upper** index is contracted with a factor of  $\mathbf{P}^{-1}$ , each **lower** index with a factor of  $\mathbf{P}$ .

*Idea.* There is only one move: write  $t$  in the new basis, replace every new basis vector and every new dual basis vector by its expression in the old ones, multiply out using multilinearity of  $\otimes$ , and compare with the expansion of  $t$  in the old basis. Coordinates in a basis are unique, so the comparison is an equality of coefficients. Doing it once for type  $(1, 0)$  shows the whole pattern; the general case is the same computation with more indices.

*Proof.* Start with the expansion of  $t$  in the new basis:

$$t = (T')_{l_1 \dots l_q}^{k_1 \dots k_p} \mathbf{v}'_{k_1} \otimes \dots \otimes \mathbf{v}'_{k_p} \otimes (\varphi')^{l_1} \otimes \dots \otimes (\varphi')^{l_q}.$$

Substitute  $\mathbf{v}'_k = p_k^i \mathbf{v}_i$  and, by [Lemma 14.5.1](#),  $(\varphi')^l = q_j^l \varphi^j$ . Since  $\otimes$  is linear in each of its slots, the scalars may be pulled out of all  $p + q$  factors at once, giving

$$\begin{aligned} t = (T')_{l_1 \dots l_q}^{k_1 \dots k_p} p_{k_1}^{i_1} \dots p_{k_p}^{i_p} q_{j_1}^{l_1} \dots q_{j_q}^{l_q} \\ \times \mathbf{v}_{i_1} \otimes \dots \otimes \mathbf{v}_{i_p} \otimes \varphi^{j_1} \otimes \dots \otimes \varphi^{j_q}. \end{aligned}$$

The tensors on the right form a basis of  $T_q^p(V)$  ([Theorem 14.2.5](#)), so coordinates in them are unique ([Theorem 1.5.1](#)). Comparing with [Definition 14.5.1](#),

$$T_{j_1 \dots j_q}^{i_1 \dots i_p} = p_{k_1}^{i_1} \dots p_{k_p}^{i_p} q_{j_1}^{l_1} \dots q_{j_q}^{l_q} (T')_{l_1 \dots l_q}^{k_1 \dots k_p}. \quad (*)$$

It remains to invert  $(*)$ . Contract both sides of  $(*)$  with  $q_{i_1}^{m_1} \dots q_{i_p}^{m_p} p_{r_1}^{j_1} \dots p_{r_q}^{j_q}$  and sum over  $i_1, \dots, i_p, j_1, \dots, j_q$ . On the right, each pair  $q_i^m p_k^j$  sums to  $(\mathbf{QP})_k^m = \delta_k^m$ , and each pair  $q_j^l p_r^j$  sums to  $\delta_r^l$ ; every delta then collapses one summation. What survives on the right is  $(T')_{r_1 \dots r_q}^{m_1 \dots m_p}$ , and what stands on the left is the asserted expression. Renaming  $m$  to  $k$  and  $r$  to  $l$  gives the theorem. ■

The two smallest cases are results the book already has, and they are worth reading off as a check.

- **Type  $(1, 1)$ .** The law reads  $(T')_l^k = q_i^k p_l^j T_{ij}^i$ , that is  $\mathbf{T}' = \mathbf{P}^{-1} \mathbf{T} \mathbf{P}$ . This is [Theorem 3.7.3](#): matrices of an operator in two bases are **similar**.
- **Type  $(0, 2)$ .** The law reads  $(T')_{kl} = p_k^i p_l^j T_{ij}$ , that is  $\mathbf{T}' = \mathbf{P}^\top \mathbf{T} \mathbf{P}$ . This is [Theorem 13.2.1](#): Gram matrices of a bilinear form in two bases are **congruent**.

So similarity and congruence are not two unrelated relations that happen to look alike. They are the transformation laws of type  $(1, 1)$  and type  $(0, 2)$ , and the difference between  $\mathbf{P}^{-1}$  and  $\mathbf{P}^\top$  is exactly the difference between an upper and a lower index.

### 14.5.3 The classical definition

The classical definition turns the theorem into a definition: a tensor *is* a rule producing arrays that obey the law.

#### Definition 14.5.2 — Classical tensor

Let  $V$  be a vector space over  $F$  with  $\dim V = n \geq 1$ , and let  $p, q \geq 0$ . A **classical tensor of type  $(p, q)$**  on  $V$  is a rule  $\mathcal{B} \mapsto A_{\mathcal{B}}$  assigning to **every** ordered basis  $\mathcal{B}$  of  $V$  an array

$$A_{\mathcal{B}} = (A_{j_1 \dots j_q}^{i_1 \dots i_p})$$

of  $n^{p+q}$  scalars, such that for **any two** ordered bases  $\mathcal{B}, \mathcal{B}'$ , with  $\mathbf{P} = [\text{id}]_{\mathcal{B}'}^{\mathcal{B}}$  and  $\mathbf{Q} = \mathbf{P}^{-1}$ ,

$$(A_{\mathcal{B}'})_{l_1 \dots l_q}^{k_1 \dots k_p} = q_{i_1}^{k_1} \cdots q_{i_p}^{k_p} \times p_{l_1}^{j_1} \cdots p_{l_q}^{j_q} (A_{\mathcal{B}})_{j_1 \dots j_q}^{i_1 \dots i_p}.$$

### Theorem 14.5.3 — The Two Definitions Agree

Let  $V$  be finite-dimensional over  $F$  with  $\dim V = n \geq 1$ , and let  $p, q \geq 0$ . The map sending  $t \in T_q^p(V)$  to the rule

$$\mathcal{B} \longmapsto (\text{components of } t \text{ in } \mathcal{B})$$

is a bijection from  $T_q^p(V)$  onto the set of classical tensors of type  $(p, q)$  on  $V$ . It is linear for the entrywise operations on arrays.

*Idea.* Well defined is [Theorem 14.5.2](#). Injective because a tensor is determined by its coordinates in one basis. Surjective is the only step with content, and it is short: fix one basis  $\mathcal{B}_0$ , build the tensor whose components there are the given array, and then observe that its components in any other basis obey the same law as the rule's array does, starting from the same array — so the two agree.

*Proof.* **Well defined.** If  $t \in T_q^p(V)$ , its components in the various bases satisfy the required identity, by [Theorem 14.5.2](#). So the rule attached to  $t$  is a classical tensor.

**Injective.** Fix a basis  $\mathcal{B}_0$ . If two tensors give the same rule, they have the same components in  $\mathcal{B}_0$ , hence are equal, since coordinates in the basis of [Theorem 14.2.5](#) are unique ([Theorem 1.5.1](#)).

**Surjective.** Let  $\mathcal{B} \mapsto A_{\mathcal{B}}$  be a classical tensor. Fix an ordered basis  $\mathcal{B}_0 = (\mathbf{v}_1, \dots, \mathbf{v}_n)$  with dual basis  $(\varphi^1, \dots, \varphi^n)$ , and define

$$t := (A_{\mathcal{B}_0})_{j_1 \dots j_q}^{i_1 \dots i_p} \mathbf{v}_{i_1} \otimes \cdots \otimes \varphi^{j_q} \in T_q^p(V),$$

so that the components of  $t$  in  $\mathcal{B}_0$  are exactly  $A_{\mathcal{B}_0}$ . Let  $\mathcal{B}'$  be any other ordered basis and put  $\mathbf{P} = [\text{id}]_{\mathcal{B}'}^{\mathcal{B}_0}$ . By [Theorem 14.5.2](#) the components of  $t$  in  $\mathcal{B}'$  are obtained from  $A_{\mathcal{B}_0}$  by the law with this  $\mathbf{P}$ ; by [Definition 14.5.2](#),  $A_{\mathcal{B}'}$  is obtained from  $A_{\mathcal{B}_0}$  by the law with the same  $\mathbf{P}$ . Both are the value of the same expression, so they are equal. Hence the rule attached to  $t$  is the given one.

**Linearity.** The components of  $t + ct'$  in each basis are  $A + cA'$ , because coordinate maps are linear ([Corollary 3.5.5](#)). ■

**Why the older definition survives.** It is tempting to file [Definition 14.5.2](#) as a historical curiosity, and that would be a mistake. Three honest reasons keep it in use.

- **Components are what gets measured and computed.** A stress tensor, a metric, an inertia tensor: the physicist has numbers in a chosen frame, and the question that matters is what those numbers become in another frame. The transformation law is not a consequence to be derived afterwards; it is the datum.
- **There is often no global basis to be abstract about.** On a curved space, the objects of interest are tensor *fields*, and a manifold generally admits no single coordinate system covering it. What it does admit is many local coordinate patches with prescribed rules for passing between them. In that setting “an array in each chart, transforming correctly on overlaps” is a definition that works, while “an element of  $T_q^p(V)$ ” needs a  $V$  that varies from point to point, which is a later and heavier construction.
- **It is a usable test.** Given a recipe that produces an array from a basis, checking the transformation law is a finite computation, and it is the standard way to certify that a quantity is geometric rather than an artifact of the coordinates. The next warning is exactly such a test, failed.

### 14.5.4 An array is not a tensor

*Warning.* **A matrix is not automatically a tensor.** Consider the rule that assigns to **every** ordered basis of  $V$  the same array, written as a matrix,

$$\mathbf{A}_{\mathcal{B}} = \begin{pmatrix} 1 & 0 \\ 0 & 0 \end{pmatrix}, \quad \dim V = 2,$$

read as a type  $(1, 1)$  array. Take  $\mathcal{B} = (\mathbf{v}_1, \mathbf{v}_2)$  and  $\mathcal{B}' = (\mathbf{v}_1, \mathbf{v}_1 + \mathbf{v}_2)$ , so that

$$\mathbf{P} = \begin{pmatrix} 1 & 1 \\ 0 & 1 \end{pmatrix}, \quad \mathbf{P}^{-1} = \begin{pmatrix} 1 & -1 \\ 0 & 1 \end{pmatrix}.$$

The law demands  $\mathbf{A}_{\mathcal{B}'} = \mathbf{P}^{-1} \mathbf{A}_{\mathcal{B}} \mathbf{P}$ , which is  $\begin{pmatrix} 1 & 1 \\ 0 & 0 \end{pmatrix}$ , not  $\mathbf{A}_{\mathcal{B}}$ . So this rule is **not** a classical tensor of type  $(1, 1)$ : it is a matrix, and nothing more. The operator it describes in  $\mathcal{B}$  — projection onto  $\text{span}(\mathbf{v}_1)$  along  $\text{span}(\mathbf{v}_2)$  — is a perfectly good tensor, but then its array in  $\mathcal{B}'$  is the second matrix, not the first. “Write down the same numbers in every basis” is a rule, and almost never a tensor.

Two refinements of the same point are worth having.

**The same array can be two different tensors.** Take the array  $\mathbf{I}_2$ . As a type  $(1, 1)$  rule it *is* a tensor:  $\mathbf{P}^{-1} \mathbf{I}_2 \mathbf{P} = \mathbf{I}_2$  for every  $\mathbf{P}$ , and the tensor is the identity operator, which is the  $\iota$  of [Exercise \(C2: The contraction of the identity, over any field\)](#). As a type  $(0, 2)$  rule, the same constant array is **not** a tensor: the law demands  $\mathbf{P}^\top \mathbf{I}_2 \mathbf{P}$ , and for the  $\mathbf{P}$  above that is  $\begin{pmatrix} 1 & 1 \\ 1 & 2 \end{pmatrix} \neq \mathbf{I}_2$ . Geometrically: “the identity operator” is basis-free, while “the bilinear form whose Gram matrix is the identity”, that is, the choice of an inner product declaring the basis orthonormal, is not — it depends on the basis, and orthonormality, as Chapter 10 defines it, is relative to a chosen inner product.

*Warning.* **Which type an array has is not visible in the array.** The array  $\begin{pmatrix} 1 & 0 \\ 0 & 0 \end{pmatrix}$  transforms as  $\mathbf{P}^{-1} \mathbf{A} \mathbf{P}$  if it is the matrix of an operator and as  $\mathbf{P}^\top \mathbf{A} \mathbf{P}$  if it is the Gram

matrix of a form, and for the  $\mathbf{P}$  above those give  $\begin{pmatrix} 1 & 1 \\ 0 & 0 \end{pmatrix}$  and  $\begin{pmatrix} 1 & 1 \\ 1 & 1 \end{pmatrix}$ . Same numbers, same basis change, different answers. This is why index position is not decoration: writing  $A_j^i$  rather than  $A_{ij}$  is a statement about what the object is.

*Quick check.* Let  $\dim V = 2$ . Is the rule assigning to every ordered basis the array  $A_{ij} = \delta_{ij}$  of type  $(0,2)$  ever a classical tensor — say, over  $\mathbb{F}_2$ , or if we only allow bases obtained from one another by permutation? Decide, with a reason.

*Solution.* Over  $\mathbb{F}_2$  it still fails: with  $\mathbf{P} = \begin{pmatrix} 1 & 1 \\ 0 & 1 \end{pmatrix}$ , the law demands  $\mathbf{P}^\top \mathbf{I}_2 \mathbf{P} = \begin{pmatrix} 1 & 1 \\ 1 & 0 \end{pmatrix}$  (since  $2 = 0$  in  $\mathbb{F}_2$ ), which is not  $\mathbf{I}_2$ . If the bases are restricted to permutations of one fixed basis, then  $\mathbf{P}$  is a permutation matrix, so  $\mathbf{P}^\top = \mathbf{P}^{-1}$  and  $\mathbf{P}^\top \mathbf{I}_2 \mathbf{P} = \mathbf{I}_2$ : the rule survives. But [Definition 14.5.2](#) quantifies over **every** ordered basis, and a rule that works only for a restricted family is exactly the kind of coordinate-dependent object the definition is designed to exclude. The lesson is that the law is a demand about all changes of basis, not about some.

## 14.5.5 Exercises

### 14.5.5.1 A. Check your understanding

#### Exercise A1.

- Define the components of a tensor of type  $(p, q)$  in an ordered basis, and say which indices go upstairs.
- State the summation convention, and write  $\xi(\mathbf{v})$  with it.
- For a type  $(1,1)$  tensor, write out the transformation law and name the relation it expresses between the two matrices.
- Give the transformation law for type  $(0,2)$  and name the relation.
- True or false: any  $n \times n$  array of scalars, attached to every basis, is a tensor of type  $(1,1)$ . Justify your answer.

*Solution.*

- They are the unique coordinates of  $t$  in the basis of [Theorem 14.2.5](#) built from  $\mathcal{B}$  and  $\mathcal{B}^*$  ([Definition 14.5.1](#)). An index belonging to a  $V$ -slot goes upstairs; one belonging to a  $V^*$ -slot goes downstairs.
- An index occurring once upstairs and once downstairs in a product is summed from 1 to  $n$ , with the summation sign suppressed. So  $\xi(\mathbf{v}) = \xi_i v^i$ .
- $(T')_l^k = q_i^k p_l^j T_j^i$ , that is  $\mathbf{T}' = \mathbf{P}^{-1} \mathbf{T} \mathbf{P}$ : the two matrices are **similar** ([Theorem 3.7.3](#)).
- $(T')_{kl} = p_k^i p_l^j T_{ij}$ , that is  $\mathbf{T}' = \mathbf{P}^\top \mathbf{T} \mathbf{P}$ : the two matrices are **congruent** ([Theorem 13.2.1](#), [Definition 13.2.1](#)).

- (e) False. The rule must satisfy the transformation law; the constant rule  $\begin{pmatrix} 1 & 0 \\ 0 & 0 \end{pmatrix}$  does not, as the warning above shows.

### 14.5.5.2 B. Practice

#### Exercise B1: Transforming components.

Let  $V$  be a 2-dimensional space over  $\mathbb{Q}$  with basis  $\mathcal{B} = (\mathbf{v}_1, \mathbf{v}_2)$ , and let  $\mathcal{B}' = (\mathbf{v}_1 + \mathbf{v}_2, \mathbf{v}_1 + 2\mathbf{v}_2)$ .

- Write down  $\mathbf{P} = [\text{id}]_{\mathcal{B}'}^{\mathcal{B}}$  and  $\mathbf{Q} = \mathbf{P}^{-1}$ .
- Let  $t \in T_1^1(V)$  have components  $T_1^1 = 1, T_2^1 = 2, T_1^2 = 3, T_2^2 = 4$  in  $\mathcal{B}$ . Find its components in  $\mathcal{B}'$ .
- Let  $s \in T_2^0(V)$  have the same array  $S_{ij}$  of components in  $\mathcal{B}$ . Find its components in  $\mathcal{B}'$ , and compare with (b).
- Compute  $C_1^1(t)$  in both bases and check that the two answers agree.

*Solution.*

- (a) The columns of  $\mathbf{P}$  are the  $\mathcal{B}$ -coordinates of the vectors of  $\mathcal{B}'$ , so

$$\mathbf{P} = \begin{pmatrix} 1 & 1 \\ 1 & 2 \end{pmatrix}, \quad \mathbf{Q} = \mathbf{P}^{-1} = \begin{pmatrix} 2 & -1 \\ -1 & 1 \end{pmatrix},$$

using [Theorem 0.9.8](#) with  $\det \mathbf{P} = 1$ .

- (b) By [Theorem 14.5.2](#),  $\mathbf{T}' = \mathbf{P}^{-1}\mathbf{T}\mathbf{P}$ . With  $\mathbf{T} = \begin{pmatrix} 1 & 2 \\ 3 & 4 \end{pmatrix}$ ,

$$\mathbf{P}^{-1}\mathbf{T} = \begin{pmatrix} -1 & 0 \\ 2 & 2 \end{pmatrix}, \quad \mathbf{T}' = \begin{pmatrix} -1 & -1 \\ 4 & 6 \end{pmatrix}.$$

- (c) Now the law gives  $\mathbf{S}' = \mathbf{P}^\top \mathbf{S} \mathbf{P}$ . With  $\mathbf{P}^\top = \begin{pmatrix} 1 & 1 \\ 1 & 2 \end{pmatrix}$  (this  $\mathbf{P}$  is symmetric),

$$\mathbf{P}^\top \mathbf{S} = \begin{pmatrix} 4 & 6 \\ 7 & 10 \end{pmatrix}, \quad \mathbf{S}' = \begin{pmatrix} 10 & 16 \\ 17 & 27 \end{pmatrix}.$$

The same array in the same two bases produces completely different answers, because the two objects are of different types.

- (d) In  $\mathcal{B}$ :  $T_1^1 + T_2^2 = 1 + 4 = 5$ . In  $\mathcal{B}'$ :  $-1 + 6 = 5$ . They agree, as they must, since [Definition 14.4.3](#) is defined without reference to a basis. (By contrast, the “trace” of the type  $(0, 2)$  arrays is  $1 + 4 = 5$  against  $10 + 27 = 37$ : there is no contraction of two lower indices, and summing  $S_{ii}$  is not a basis-free operation.)

#### Exercise B2: Tensor or not?

Let  $\dim V = 2$ . For each rule below, determine whether it is a classical tensor of the stated type. Justify your answer, using  $\mathcal{B} = (\mathbf{v}_1, \mathbf{v}_2)$  and  $\mathcal{B}' = (\mathbf{v}_2, \mathbf{v}_1)$  or  $\mathcal{B}' = (\mathbf{v}_1, \mathbf{v}_1 + \mathbf{v}_2)$  where a counterexample is needed.

- (a) Type  $(1, 1)$ :  $A_{\mathcal{B}} = \mathbf{I}_2$  for every  $\mathcal{B}$ .
- (b) Type  $(1, 1)$ :  $A_{\mathcal{B}} = \begin{pmatrix} 0 & 1 \\ 0 & 0 \end{pmatrix}$  for every  $\mathcal{B}$ .
- (c) Type  $(0, 2)$ :  $A_{\mathcal{B}} = \mathbf{0}$  for every  $\mathcal{B}$ .
- (d) Type  $(1, 1)$ :  $A_{\mathcal{B}} = (\text{tr } \mathbf{M}_{\mathcal{B}})\mathbf{I}_2$ , where  $\mathbf{M}_{\mathcal{B}}$  is the matrix in  $\mathcal{B}$  of one fixed operator  $M \in \mathcal{L}(V)$ .

*Solution.*

- (a) Yes. The law demands  $\mathbf{P}^{-1}\mathbf{I}_2\mathbf{P} = \mathbf{I}_2$ , which holds for every invertible  $\mathbf{P}$ . The tensor is  $\text{id}_V$ .
- (b) No. With  $\mathcal{B}' = (\mathbf{v}_2, \mathbf{v}_1)$  we get  $\mathbf{P} = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix} = \mathbf{P}^{-1}$ , and  $\mathbf{P}^{-1}A\mathbf{P} = \begin{pmatrix} 0 & 0 \\ 1 & 0 \end{pmatrix} \neq A$ . (This is the standard non-diagonalizable matrix of Chapter 8; here it is failing a different test.)
- (c) Yes. The law reads  $\mathbf{0} = \mathbf{P}^T\mathbf{0}\mathbf{P}$ , which holds. The tensor is the zero tensor, and the zero array is the one array that is a tensor of every type.
- (d) Yes. By [Theorem 3.7.5](#) the scalar  $c = \text{tr } \mathbf{M}_{\mathcal{B}}$  is the same for every  $\mathcal{B}$ , so the rule is the constant rule  $c\mathbf{I}_2$ , and  $\mathbf{P}^{-1}(c\mathbf{I}_2)\mathbf{P} = c\mathbf{I}_2$ . The tensor is  $c \text{id}_V$ . The point of contrast with (b) is that  $c$  was itself built basis-freely.

### 14.5.5.3 C. Going deeper

#### Exercise C1: Contraction in indices.

Let  $\dim V = n$  and let  $t \in T_1^1(V)$  have components  $T_j^i$  in  $\mathcal{B}$  and  $(T')_l^k$  in  $\mathcal{B}'$ .

- (a) Prove directly from [Theorem 14.5.2](#) that  $(T')_k^k = T_i^i$ , without using [Theorem 14.4.1](#).
- (b) Let  $t \in T_1^2(V)$  with components  $T_k^{ij}$ . Show that  $U^j := T_i^{ij}$  are the components of a type  $(1, 0)$  tensor, and identify it in terms of [Definition 14.4.3](#).
- (c) Let  $s \in T_0^2(V)$  have components  $S^{ij}$ . Show that the scalar  $\sum_i S^{ii}$  is **not** basis-independent, and say which step of (b) fails for it.

*Solution.*

- (a) By the law,  $(T')_l^k = q_i^k p_l^j T_j^i$ . Setting  $l = k$  and summing,

$$(T')_k^k = \sum_k \sum_{i,j} q_i^k p_k^j T_j^i = \sum_{i,j} \left( \sum_k p_k^j q_i^k \right) T_j^i = \sum_{i,j} \delta_i^j T_j^i = \sum_i T_i^i,$$

where the inner sum is the  $(j, i)$  entry of  $\mathbf{PQ} = \mathbf{I}_n$ .

- (b) By [Theorem 14.5.2](#) for type  $(2, 1)$ ,  $(T')_l^{km} = q_i^k q_j^m p_l^r T_r^{ij}$ . Setting  $l = k$  and summing over  $k$ ,

$$(U')^m = \sum_k (T')_k^{km} = q_j^m \left( \sum_{k,i,r} q_i^k p_k^r T_r^{ij} \right) = q_j^m \delta_i^r T_r^{ij} = q_j^m U^j,$$

which is the transformation law for a single upper index. So  $(U^j)$  is a classical tensor of type  $(1, 0)$ , and by [Theorem 14.5.3](#) it is a vector; it is  $C_1^1(t)$ , the contraction

of the first upper slot against the lower one (Definition 14.4.3).

- (c) Take  $n = 2$  and  $s = \mathbf{v}_1 \otimes \mathbf{v}_1$ , so  $S^{11} = 1$  and the other components vanish; then  $\sum_i S^{ii} = 1$ . Let  $\mathcal{B}' = (\mathbf{v}_1 + \mathbf{v}_2, \mathbf{v}_2)$ , so that  $\mathbf{P} = \begin{pmatrix} 1 & 0 \\ 1 & 1 \end{pmatrix}$  and  $\mathbf{v}_1 = \mathbf{v}'_1 - \mathbf{v}'_2$ . Expanding,

$$s = (\mathbf{v}'_1 - \mathbf{v}'_2) \otimes (\mathbf{v}'_1 - \mathbf{v}'_2),$$

so  $(S')^{11} = (S')^{22} = 1$  and  $(S')^{12} = (S')^{21} = -1$ . Hence  $\sum_k (S')^{kk} = 2 \neq 1$ . The step that fails is the collapse in (b): setting the two indices equal pairs a factor  $q_i^k$  with another factor  $q_j^k$ , and  $\sum_k q_i^k q_j^k$  is an entry of  $\mathbf{Q}^\top \mathbf{Q}$ , not of  $\mathbf{PQ} = \mathbf{I}_n$ , so nothing cancels. Only an upper index repeated against a **lower** one is a contraction.

### Exercise C2: Raising an index needs a form.

Let  $V$  be finite-dimensional over  $F$  and let  $\beta$  be a non-degenerate bilinear form on  $V$ , with components  $g_{ij} = \beta(\mathbf{v}_i, \mathbf{v}_j)$  in  $\mathcal{B}$ .

- Prove that the matrix  $\mathbf{G} = (g_{ij})$  is invertible, and let  $(g^{ij})$  denote the entries of  $\mathbf{G}^{-1}$ .
- Let  $t \in T_1^0(V)$  have components  $T_j$ . Show that  $U^i := g^{ij} T_j$  satisfies the type  $(1, 0)$  transformation law, so that “raising the index with  $\beta$ ” produces a genuine tensor.
- Explain in one or two sentences why this does **not** contradict the warning that there is no canonical map  $V^* \rightarrow V$ .

*Solution.*

- By Proposition 13.1.2,  $\beta$  is non-degenerate exactly when  $\mathbf{G} = [\beta]_{\mathcal{B}}$  is invertible.
- By Theorem 13.2.1,  $\mathbf{G}' = \mathbf{P}^\top \mathbf{G} \mathbf{P}$ , so  $(\mathbf{G}')^{-1} = \mathbf{P}^{-1} \mathbf{G}^{-1} (\mathbf{P}^\top)^{-1} = \mathbf{Q} \mathbf{G}^{-1} \mathbf{Q}^\top$ , that is  $(g')^{kl} = q_i^k q_j^l g^{ij}$ : the inverse array carries two upper indices. Also  $(T')_l = p_l^r T_r$  by Theorem 14.5.2. Hence

$$\begin{aligned} (U')^k &= (g')^{kl} (T')_l = q_i^k g^{ij} \left( \sum_l q_j^l p_l^r \right) T_r \\ &= q_i^k g^{ij} \delta_j^r T_r = q_i^k g^{ij} T_j = q_i^k U^i, \end{aligned}$$

using  $\sum_l p_l^r q_j^l = (\mathbf{PQ})_j^r = \delta_j^r$ . That is the type  $(1, 0)$  law.

- The map is canonical *given*  $\beta$ , not given  $V$  alone: a different non-degenerate form raises indices differently, and there is no distinguished choice. This is the same point as the warning after Definition 14.4.1: the isomorphism  $V \cong V^*$  is extra structure, which Chapter 10 supplies with an inner product and Chapter 13 with a form.

## 14.6 The Tensor Algebra

Sections 03 and 04 already iterated the tensor product:  $V^{\otimes k}$  is  $V \otimes \cdots \otimes V$  with  $k$  factors, its brackets do not matter (Proposition 14.3.5), and  $k$ -linear maps out of  $V^k$  become linear maps



on it (Lemma 14.3.4). Two things are still missing, and Chapter 13 needs both. The first is a **basis** of  $V^{\otimes k}$ , which the next three sections will cut down in two different ways. The second is a single algebra holding all the powers at once: Chapter 13 §12 described it as the algebra of “formal sums of strings of vectors, multiplied by concatenation”, said no such algebra had been built, and deferred the Clifford algebra until it was. This section builds it, calls it  $T(V)$ , and proves the universal property that Sections 07, 08 and 10 will each quotient.

Throughout,  $F$  is a field and  $V$  is a finite-dimensional vector space over  $F$ , with  $n = \dim V$ . The letter  $k$  always denotes a non-negative integer.

## 14.6.1 The $k$ -th tensor power

Section 04 wrote  $V^{\otimes k}$  for the  $k$ -fold tensor product and  $V^{\otimes 0} = F$ , and Section 03 proved what maps out of it look like. Before using that space three more times it is worth restating the property as a **characterization**, in the form the rest of the chapter will copy: *say what maps out of the object look like, and demand that the object be the most economical one with that property*. The definition below is Definition 14.2.1 with  $k$  slots in place of two, and nothing else changes.

*The  $k$ -th tensor power is the space in which  $k$ -linear maps become linear.*

### Definition 14.6.1 — $k$ -th Tensor Power

Let  $V$  be a vector space over  $F$  and let  $k \geq 1$ . A  **$k$ -th tensor power of  $V$**  is a pair  $(P, \mu)$ , where  $P$  is a vector space over  $F$  and  $\mu: V^k \rightarrow P$  is a  $k$ -linear map, such that

(T1) (**universal property**) for **every** vector space  $W$  over  $F$  and **every**  $k$ -linear map  $f: V^k \rightarrow W$ , there is a **unique** linear map  $\bar{f}: P \rightarrow W$  with  $\bar{f} \circ \mu = f$ .

We write  $V^{\otimes k}$  for  $P$  and

$$\mathbf{v}_1 \otimes \mathbf{v}_2 \otimes \cdots \otimes \mathbf{v}_k := \mu(\mathbf{v}_1, \dots, \mathbf{v}_k)$$

for the value of  $\mu$ , and we call such an element a **simple tensor**. By convention  $V^{\otimes 0} = F$ , with  $\mu$  the empty product  $1 \in F$ .

Clause by clause. The pair is what is defined, not the space alone: without  $\mu$  there is nothing for  $f$  to factor through, and a bare vector space of the right dimension satisfies nothing. **Every** vector space  $W$  is allowed as a target, including  $W = F$  and including  $P$  itself; the property is a demand about all of them simultaneously. **Unique** is the clause that does the work later: existence of  $\bar{f}$  says  $P$  is big enough to record  $f$ , and uniqueness says  $P$  is not bigger than that.

At  $k = 1$  a 1-linear map is a linear map, and the pair  $(V, \text{id}_V)$  satisfies (T1) with  $\bar{f} = f$ . So  $V^{\otimes 1} = V$ , and the notation agrees with itself. At  $k = 2$  the definition is Definition 14.2.1 with  $W$  replaced by  $V$ , so  $V^{\otimes 2} = V \otimes V$ . The convention  $V^{\otimes 0} = F$  is a convention in the same sense as  $\text{span}(\emptyset) = \{\mathbf{0}\}$  and  $0! = 1$ : it is the choice that keeps the formulas below true at the boundary, as the dimension count  $\dim V^{\otimes k} = n^k$  and the multiplication rule will both confirm.

Uniqueness first, exactly as in Section 02.

### Theorem 14.6.1 — Uniqueness of the Tensor Power

Let  $(P, \mu)$  and  $(P', \mu')$  both be  $k$ -th tensor powers of  $V$ . Then there is a **unique** linear isomorphism  $\varphi: P \rightarrow P'$  with  $\varphi \circ \mu = \mu'$ .

*Proof.* This is the proof of Theorem 14.2.1 with  $k$  slots in place of two, and we run it once more so that the pattern is visible. Since  $\mu'$  is  $k$ -linear, (T1) for  $P$  gives a linear  $\varphi: P \rightarrow P'$  with  $\varphi\mu = \mu'$ . Since  $\mu$  is  $k$ -linear, (T1) for  $P'$  gives a linear  $\psi: P' \rightarrow P$  with  $\psi\mu' = \mu$ . Then  $\psi\varphi: P \rightarrow P$  is linear with  $\psi\varphi\mu = \psi\mu' = \mu$ , and  $\text{id}_P$  is another such map. By the uniqueness clause of (T1) for  $P$ , applied to the  $k$ -linear map  $\mu$  itself, the two coincide:  $\psi\varphi = \text{id}_P$ . Swapping the roles of  $P$  and  $P'$  gives  $\varphi\psi = \text{id}_{P'}$ . Hence  $\varphi$  is an isomorphism, and (T1) delivered it uniquely. This proves the theorem. ■

Four lines, and they used nothing but the property itself. That is the point of defining objects this way, and it is why the remaining uniqueness proofs in this chapter will be compressed to a citation of this one.

Now existence. Here the chapter can be lazy: a  $k$ -th tensor power does not need a new construction, because iterating the binary one already produces it.

### Theorem 14.6.2 — Existence of the Tensor Power

Let  $k \geq 1$  and let  $V^{\otimes k} = V \otimes \cdots \otimes V$  be the iterated tensor product of Section 03, with

$$\mu_k(\mathbf{v}_1, \dots, \mathbf{v}_k) = \mathbf{v}_1 \otimes \cdots \otimes \mathbf{v}_k.$$

Then  $(V^{\otimes k}, \mu_k)$  is a  $k$ -th tensor power of  $V$ , and it is spanned by its simple tensors.

*Proof.* The map  $\mu_k$  is  $k$ -linear, being  $\otimes$  applied  $k - 1$  times and  $\otimes$  bilinear in each use. Property (T1) and the spanning claim are exactly Lemma 14.3.4, read with  $V_1 = \cdots = V_k = V$ . This proves the theorem. ■

The proof is one sentence because Section 03 did the induction: freeze the last argument, apply the  $(k - 1)$ -factor statement, let the last argument move again, and finish with the two-factor property. Two consequences are used in every argument below, so they are worth saying in words.  $V^{\otimes k}$  **is spanned by its simple tensors**, so two linear maps out of  $V^{\otimes k}$  are equal as soon as they agree on simple tensors; and **a  $k$ -linear recipe on  $V^k$  defines a linear map on  $V^{\otimes k}$** , which is how every map out of a tensor power gets built, here and in Sections 07 and 08.

From now on  $V^{\otimes k}$  denotes this space, and by Theorem 14.6.1 any other  $k$ -th tensor power is identified with it by a unique isomorphism respecting the simple tensors. So it is legitimate to speak of *the*  $k$ -th tensor power, and the model may be forgotten: every proof below argues from (T1).

## 14.6.2 A basis, and the dimension

The count is the one Section 01 predicted. A  $k$ -linear map on an  $n$ -dimensional space may be prescribed freely on the  $n^k$  tuples of basis vectors and is determined by those values ([Theorem 14.1.1](#)); so  $V^{\otimes k}$  had better have dimension  $n^k$ , and it does.

### Theorem 14.6.3 — Basis of a Tensor Power

Let  $\mathcal{B} = (\mathbf{v}_1, \dots, \mathbf{v}_n)$  be a basis of  $V$  and let  $k \geq 1$ . Then the list of simple tensors

$$\mathbf{v}_{i_1} \otimes \mathbf{v}_{i_2} \otimes \cdots \otimes \mathbf{v}_{i_k}, \quad (i_1, \dots, i_k) \in \{1, \dots, n\}^k,$$

is a basis of  $V^{\otimes k}$ . In particular  $\dim V^{\otimes k} = n^k$ .

*Idea.* Spanning is multilinear expansion: write each argument in the basis and expand one slot at a time. Independence needs, for each tuple  $J$ , a linear functional on  $V^{\otimes k}$  that is 1 on the tensor indexed by  $J$  and 0 on the others. Build it upstairs, as a product of dual basis functionals, and let the universal property carry it down.

*Proof. Spanning.* By [Theorem 14.6.2](#),  $V^{\otimes k}$  is spanned by its simple tensors, so it suffices to expand one of them in the basis  $\mathcal{B}$ . Write  $\mathbf{u}_t = \sum_{i=1}^n c_{ti} \mathbf{v}_i$  for  $t = 1, \dots, k$ . Expanding  $\mathbf{u}_1 \otimes \cdots \otimes \mathbf{u}_k$  in slot 1, then in slot 2, and so on, using  $k$ -linearity of  $\mu_k$  once per slot, gives

$$\mathbf{u}_1 \otimes \cdots \otimes \mathbf{u}_k = \sum_{i_1, \dots, i_k} c_{1i_1} \cdots c_{ki_k} \mathbf{v}_{i_1} \otimes \cdots \otimes \mathbf{v}_{i_k},$$

a linear combination of the listed tensors.

*Independence.* Let  $(\varphi_1, \dots, \varphi_n)$  be the dual basis of  $\mathcal{B}$  ([Theorem 4.1.3](#)), so  $\varphi_i(\mathbf{v}_j) = \delta_{ij}$ . Fix a tuple  $J = (j_1, \dots, j_k)$  and define

$$f_J(\mathbf{u}_1, \dots, \mathbf{u}_k) = \varphi_{j_1}(\mathbf{u}_1) \varphi_{j_2}(\mathbf{u}_2) \cdots \varphi_{j_k}(\mathbf{u}_k).$$

This is  $k$ -linear: in position  $t$  it is the functional  $\varphi_{j_t}$  times a fixed scalar. By (T1) it induces a linear  $\bar{f}_J: V^{\otimes k} \rightarrow F$  with

$$\bar{f}_J(\mathbf{v}_{i_1} \otimes \cdots \otimes \mathbf{v}_{i_k}) = \delta_{j_1 i_1} \cdots \delta_{j_k i_k},$$

which is 1 if  $I = J$  and 0 otherwise. Now suppose  $\sum_I a_I \mathbf{v}_{i_1} \otimes \cdots \otimes \mathbf{v}_{i_k} = \mathbf{0}$ , the sum over all  $n^k$  tuples  $I$ . Applying  $\bar{f}_J$  to both sides leaves  $a_J = 0$ . Since  $J$  was arbitrary, every coefficient vanishes.

Hence the list is a basis. It has one member per tuple in  $\{1, \dots, n\}^k$ , and there are  $n^k$  of those, so  $\dim V^{\otimes k} = n^k$ . This proves the theorem. ■

The count also confirms the convention  $V^{\otimes 0} = F$ : the formula  $n^k$  at  $k = 0$  reads  $n^0 = 1 = \dim F$ , and the only 0-tuple is the empty one.

*Quick check.* Let  $\dim V = 3$ . What is  $\dim V^{\otimes 4}$ , and how many of its basis tensors have all four indices distinct?

*Solution.*  $\dim V^{\otimes 4} = 3^4 = 81$ . None: an index tuple of length 4 drawn from  $\{1, 2, 3\}$  must repeat a value, by the pigeonhole principle. This is the first sign that the tensors with a repeated index are not a negligible part of  $V^{\otimes k}$  — Section 08 will divide by the span of the simple tensors  $\mathbf{u}_1 \otimes \cdots \otimes \mathbf{u}_k$  that repeat a **vector**, a subspace containing every one of these.

**Example. (A tensor cube, in coordinates)**

Let  $V = F^2$  with standard basis  $(\mathbf{e}_1, \mathbf{e}_2)$ . Write out a basis of  $V^{\otimes 3}$ , and expand  $(\mathbf{e}_1 + \mathbf{e}_2) \otimes (\mathbf{e}_1 - \mathbf{e}_2) \otimes \mathbf{e}_1$  in it.

*Solution.* By [Theorem 14.6.3](#) a basis is indexed by the  $2^3 = 8$  triples of indices from  $\{1, 2\}$ :

$$\begin{array}{llll} \mathbf{e}_1 \otimes \mathbf{e}_1 \otimes \mathbf{e}_1, & \mathbf{e}_1 \otimes \mathbf{e}_1 \otimes \mathbf{e}_2, & \mathbf{e}_1 \otimes \mathbf{e}_2 \otimes \mathbf{e}_1, & \mathbf{e}_1 \otimes \mathbf{e}_2 \otimes \mathbf{e}_2, \\ \mathbf{e}_2 \otimes \mathbf{e}_1 \otimes \mathbf{e}_1, & \mathbf{e}_2 \otimes \mathbf{e}_1 \otimes \mathbf{e}_2, & \mathbf{e}_2 \otimes \mathbf{e}_2 \otimes \mathbf{e}_1, & \mathbf{e}_2 \otimes \mathbf{e}_2 \otimes \mathbf{e}_2. \end{array}$$

Expanding slot by slot, first in slot 1 and then in slot 2, and noting that slot 3 is already a basis vector,

$$\begin{aligned} (\mathbf{e}_1 + \mathbf{e}_2) \otimes (\mathbf{e}_1 - \mathbf{e}_2) \otimes \mathbf{e}_1 &= \mathbf{e}_1 \otimes \mathbf{e}_1 \otimes \mathbf{e}_1 - \mathbf{e}_1 \otimes \mathbf{e}_2 \otimes \mathbf{e}_1 \\ &\quad + \mathbf{e}_2 \otimes \mathbf{e}_1 \otimes \mathbf{e}_1 - \mathbf{e}_2 \otimes \mathbf{e}_2 \otimes \mathbf{e}_1. \end{aligned}$$

Four of the eight coordinates are non-zero and the other four are 0. Note also that  $\mathbf{e}_1 \otimes \mathbf{e}_2 \otimes \mathbf{e}_1$  and  $\mathbf{e}_2 \otimes \mathbf{e}_1 \otimes \mathbf{e}_1$  are **different** basis vectors: the order of the slots is part of the data.

### 14.6.3 Stacking the powers: the tensor algebra

We now have one space for each  $k$ . Chapter 13 wanted a single algebra containing all of them, in which  $\mathbf{v}_1 \otimes \mathbf{v}_2$  and  $\mathbf{v}_3 \otimes \mathbf{v}_4 \otimes \mathbf{v}_5$  could be multiplied to give  $\mathbf{v}_1 \otimes \mathbf{v}_2 \otimes \mathbf{v}_3 \otimes \mathbf{v}_4 \otimes \mathbf{v}_5$ . The multiplication is concatenation of strings; the only work is to define it without choosing a basis.

First the multiplication on a single pair of powers.

**Proposition 14.6.4 — Concatenation of Tensors**

Let  $k, l \geq 1$ . There is a unique bilinear map

$$V^{\otimes k} \times V^{\otimes l} \rightarrow V^{\otimes(k+l)}, \quad (x, y) \mapsto x \cdot y,$$

such that for all  $\mathbf{u}_1, \dots, \mathbf{u}_k, \mathbf{w}_1, \dots, \mathbf{w}_l \in V$ ,

$$(\mathbf{u}_1 \otimes \cdots \otimes \mathbf{u}_k) \cdot (\mathbf{w}_1 \otimes \cdots \otimes \mathbf{w}_l) = \mathbf{u}_1 \otimes \cdots \otimes \mathbf{u}_k \otimes \mathbf{w}_1 \otimes \cdots \otimes \mathbf{w}_l.$$

*Idea.* The same peeling move as in [Theorem 14.6.2](#), applied to the two blocks of slots in turn: first make the last  $l$  slots into a linear map, then make the first  $k$  slots into a linear map whose values are those linear maps.

*Proof.* Fix  $\mathbf{u}_1, \dots, \mathbf{u}_k \in V$ . The map  $V^l \rightarrow V^{\otimes(k+l)}$  sending  $(\mathbf{w}_1, \dots, \mathbf{w}_l)$  to  $\mathbf{u}_1 \otimes \dots \otimes \mathbf{u}_k \otimes \mathbf{w}_1 \otimes \dots \otimes \mathbf{w}_l$  is  $l$ -linear, since  $\mu_{k+l}$  is  $(k+l)$ -linear and the first  $k$  arguments are fixed. By (T1) for  $V^{\otimes l}$  it induces a unique linear map  $R(\mathbf{u}_1, \dots, \mathbf{u}_k): V^{\otimes l} \rightarrow V^{\otimes(k+l)}$ .

Next,  $(\mathbf{u}_1, \dots, \mathbf{u}_k) \mapsto R(\mathbf{u}_1, \dots, \mathbf{u}_k)$  is  $k$ -linear as a map into  $\mathcal{L}(V^{\otimes l}, V^{\otimes(k+l)})$ . Indeed, two linear maps  $V^{\otimes l} \rightarrow V^{\otimes(k+l)}$  are equal as soon as they agree on simple tensors, and on a simple tensor both sides of each linearity identity reduce to linearity of  $\mu_{k+l}$  in one of its first  $k$  slots. By (T1) for  $V^{\otimes k}$  we get a linear map

$$M: V^{\otimes k} \rightarrow \mathcal{L}(V^{\otimes l}, V^{\otimes(k+l)}),$$

and we set  $x \cdot y = M(x)(y)$ . This is linear in  $x$  because  $M$  is linear, and linear in  $y$  because each  $M(x)$  is linear, so it is bilinear, and on simple tensors it is the displayed concatenation.

Uniqueness: a bilinear map is determined by its values on pairs of spanning sets, and each factor is spanned by its simple tensors. This proves the proposition. ■

With  $k = 0$  or  $l = 0$  we let  $\cdot$  be scalar multiplication, which is the only bilinear option on  $F \times V^{\otimes l}$  agreeing with “prepend nothing”. Now stack.

### Definition 14.6.2 — Tensor Algebra

Let  $V$  be a vector space over  $F$ . The **tensor algebra** of  $V$  is the vector space

$$T(V) = \bigoplus_{k \geq 0} V^{\otimes k},$$

whose elements are the sequences  $x = (x_0, x_1, x_2, \dots)$  with  $x_k \in V^{\otimes k}$  and  $x_k = \mathbf{0}$  for **all but finitely many**  $k$ , with componentwise addition and scalar multiplication, and whose multiplication is

$$(xy)_m = \sum_{k+l=m} x_k \cdot y_l,$$

with  $\cdot$  as in Proposition 14.6.4. We identify  $V^{\otimes k}$  with the set of sequences supported in position  $k$ , so that  $F = V^{\otimes 0}$  and  $V = V^{\otimes 1}$  are subspaces of  $T(V)$ . An element of  $V^{\otimes k} \subseteq T(V)$  is called **homogeneous of degree  $k$** .

The “all but finitely many” clause is the same one that makes  $F[x]$  a space of polynomials rather than power series (Definition 0.8.1): an element of  $T(V)$  is a **finite** sum  $x_0 + x_1 + \dots + x_N$ , not a series. The formula for  $(xy)_m$  is then a finite sum, so the product is defined.

### Theorem 14.6.5 — The Tensor Algebra Is an Algebra

$T(V)$  with the above operations is an associative unital  $F$ -algebra (Definition 13.12.1), with identity  $1 \in F = V^{\otimes 0}$ . Moreover  $V^{\otimes k} \cdot V^{\otimes l} \subseteq V^{\otimes(k+l)}$  for all  $k, l$ .

*Proof.* Bilinearity of the multiplication is immediate from bilinearity of each  $\cdot$  in Proposition 14.6.4, since the components of  $xy$  are finite sums of bilinear expressions in the

components of  $x$  and  $y$ .

For associativity it suffices, by bilinearity in each of the three arguments and because the homogeneous components span, to check  $(xy)z = x(yz)$  for homogeneous  $x, y, z$ , and then, because each power is spanned by its simple tensors, for simple tensors. Both sides are then the single simple tensor obtained by writing the three strings of vectors one after another, so they agree.

For the identity,  $1 \in F = V^{\otimes 0}$  acts on each  $V^{\otimes k}$  as scalar multiplication by 1, hence as the identity. The grading statement is the target of [Proposition 14.6.4](#). This proves the theorem. ■

From here on we drop the dot and write  $xy$ , and for simple tensors we write  $\mathbf{u}_1 \otimes \cdots \otimes \mathbf{u}_k$  for both the element and the product  $\mathbf{u}_1 \mathbf{u}_2 \cdots \mathbf{u}_k$  of  $k$  elements of  $V$  — they are the same thing, since concatenating  $k$  strings of length 1 gives the string of length  $k$ .

**Warning.**  $T(V)$  is **not commutative**, and **not finite-dimensional**. If  $\dim V \geq 2$ , pick basis vectors  $\mathbf{v}_1 \neq \mathbf{v}_2$ ; then  $\mathbf{v}_1 \otimes \mathbf{v}_2$  and  $\mathbf{v}_2 \otimes \mathbf{v}_1$  are **distinct basis vectors** of  $V^{\otimes 2}$  by [Theorem 14.6.3](#), so  $\mathbf{v}_1 \mathbf{v}_2 \neq \mathbf{v}_2 \mathbf{v}_1$ . And  $\dim T(V) = \sum_k n^k$  is infinite as soon as  $n \geq 1$ , even though  $V$  is finite-dimensional. Every individual element still has only finitely many non-zero components.

#### Example. (The tensor algebra of a line)

Let  $\dim V = 1$ , say  $V = F\mathbf{v}$ . Identify  $T(V)$  with a familiar algebra.

**Solution.** By [Theorem 14.6.3](#),  $V^{\otimes k}$  has the single basis vector  $\mathbf{v}^{\otimes k} = \mathbf{v} \otimes \cdots \otimes \mathbf{v}$ , so  $\dim V^{\otimes k} = 1^k = 1$ . An element of  $T(V)$  is therefore a finite sum  $a_0 + a_1 \mathbf{v} + a_2 \mathbf{v}^{\otimes 2} + \cdots$  with  $a_k \in F$ , and concatenation gives  $\mathbf{v}^{\otimes k} \mathbf{v}^{\otimes l} = \mathbf{v}^{\otimes (k+l)}$ . Matching  $\mathbf{v}^{\otimes k}$  with  $x^k$  turns these into exactly the addition and multiplication rules of [Definition 0.8.2](#), so  $T(V) \cong F[x]$  as  $F$ -algebras.

In particular  $T(V)$  is commutative when  $\dim V = 1$ : there is only one letter, so no two letters can fail to commute. The warning above needed  $\dim V \geq 2$ , and it needed it for a reason.

### 14.6.4 The universal property of the tensor algebra

The tensor powers were characterized by what  $k$ -linear maps do. The algebra assembled from them has its own characterization, and it is the one Sections 07, 08 and 10 will use: **to define an algebra homomorphism out of  $T(V)$ , you only have to say what it does to the vectors.**

#### Theorem 14.6.6 — Universal Property of the Tensor Algebra

Let  $V$  be a vector space over  $F$ , let  $A$  be an associative unital  $F$ -algebra, and let

$f: V \rightarrow A$  be a **linear** map. Then there is a **unique** algebra homomorphism

$$\bar{f}: T(V) \rightarrow A \quad \text{with} \quad \bar{f}|_V = f.$$

Explicitly,  $\bar{f}(\mathbf{u}_1 \otimes \cdots \otimes \mathbf{u}_k) = f(\mathbf{u}_1) \cdots f(\mathbf{u}_k)$  and  $\bar{f}(a) = a1_A$  for  $a \in F$ .

*Idea.* Degree by degree. In degree  $k$ , the recipe  $(\mathbf{u}_1, \dots, \mathbf{u}_k) \mapsto f(\mathbf{u}_1) \cdots f(\mathbf{u}_k)$  is  $k$ -linear because the product of  $A$  is bilinear and  $f$  is linear, so (T1) converts it into a linear map on  $V^{\otimes k}$ . Assemble the pieces and check multiplicativity on simple tensors, which span.

*Proof. Existence.* For  $k \geq 1$  the map  $V^k \rightarrow A$  given by  $(\mathbf{u}_1, \dots, \mathbf{u}_k) \mapsto f(\mathbf{u}_1)f(\mathbf{u}_2) \cdots f(\mathbf{u}_k)$  is  $k$ -linear: with all arguments but the  $t$ -th fixed, it is  $f$  followed by multiplication on the left and on the right by fixed elements of  $A$ , and all three of those maps are linear, the last two by bilinearity of the product (A1). By (T1) there is a unique linear  $\bar{f}_k: V^{\otimes k} \rightarrow A$  with  $\bar{f}_k(\mathbf{u}_1 \otimes \cdots \otimes \mathbf{u}_k) = f(\mathbf{u}_1) \cdots f(\mathbf{u}_k)$ . Put  $\bar{f}_0: F \rightarrow A$ ,  $a \mapsto a1_A$ , and define  $\bar{f}(x) = \sum_k \bar{f}_k(x_k)$ , a finite sum. This is linear, and  $\bar{f}(1) = 1_A$ .

Multiplicativity: the maps  $(x, y) \mapsto \bar{f}(xy)$  and  $(x, y) \mapsto \bar{f}(x)\bar{f}(y)$  are both bilinear, so it suffices to compare them on a spanning set of each argument, and the simple tensors of all degrees span  $T(V)$ . For two simple tensors,

$$\begin{aligned} \bar{f}((\mathbf{u}_1 \otimes \cdots \otimes \mathbf{u}_k)(\mathbf{w}_1 \otimes \cdots \otimes \mathbf{w}_l)) &= f(\mathbf{u}_1) \cdots f(\mathbf{u}_k)f(\mathbf{w}_1) \cdots f(\mathbf{w}_l) \\ &= \bar{f}(\mathbf{u}_1 \otimes \cdots \otimes \mathbf{u}_k) \bar{f}(\mathbf{w}_1 \otimes \cdots \otimes \mathbf{w}_l), \end{aligned}$$

the first equality by [Proposition 14.6.4](#) and the definition of  $\bar{f}_{k+l}$ , the second by the definition of  $\bar{f}_k$  and  $\bar{f}_l$  together with associativity in  $A$ . The degree-0 cases are the identity  $\bar{f}(ax) = a\bar{f}(x)$ , which holds by linearity. Finally  $\bar{f}|_V = \bar{f}_1 = f$ .

*Uniqueness.* Let  $\varphi$  be an algebra homomorphism with  $\varphi|_V = f$ . Then  $\varphi(1) = 1_A$ , so  $\varphi(a) = a1_A$  on  $V^{\otimes 0}$ , and for a simple tensor, multiplicativity gives

$$\varphi(\mathbf{u}_1 \otimes \cdots \otimes \mathbf{u}_k) = \varphi(\mathbf{u}_1) \cdots \varphi(\mathbf{u}_k) = f(\mathbf{u}_1) \cdots f(\mathbf{u}_k).$$

So  $\varphi$  agrees with  $\bar{f}$  on the simple tensors, which span  $T(V)$ , and both are linear. Hence  $\varphi = \bar{f}$ . This proves the theorem. ■

In words:  $T(V)$  is the **free associative unital algebra on  $V$** . “Free” means that a linear map out of  $V$  extends to an algebra map in exactly one way, with no condition on  $f$  whatsoever — no multiplicativity is asked of  $f$ , because  $V$  carries no multiplication to preserve. Compare the corresponding statement one level down, [Theorem 3.1.6](#): a *function* on a basis extends to a linear map in exactly one way. The tensor algebra plays the same role one level up.

*Remark.* The hypothesis on  $f$  is empty, and that is precisely what makes  $T(V)$  too big to be useful on its own. Every interesting algebra built from  $V$  is a quotient of  $T(V)$  by the relations one wants to impose, and [Theorem 14.6.6](#) is what shows that the quotient has the expected universal property. Section 07 imposes  $\mathbf{uw} = \mathbf{wu}$ , Section 08 imposes  $\mathbf{vv} = 0$ , and Section 10 imposes  $\mathbf{vv} = q(\mathbf{v})1$ , which is the construction Chapter 13 §12 said was unavailable.

*Warning.* **Do not confuse the degree with the dimension.**  $V^{\otimes k}$  is the space of tensors of degree  $k$ ; its dimension is  $n^k$ , not  $k$ . A general element of  $V^{\otimes k}$  is a sum of simple tensors, and — exactly as in Section 02 — usually not a single one. A general element of  $T(V)$  is not even homogeneous:  $1 + \mathbf{v}_1 + \mathbf{v}_1 \otimes \mathbf{v}_2$  is a perfectly good element of degrees 0, 1, 2 at once.

## 14.6.5 Exercises

### 14.6.5.1 A. Check your understanding

#### Exercise A1.

- State the universal property (T1) of the  $k$ -th tensor power, with every quantifier.
- Let  $\dim V = 4$ . Compute  $\dim V^{\otimes 0}$ ,  $\dim V^{\otimes 1}$  and  $\dim V^{\otimes 3}$ .
- Give a spanning set of  $V^{\otimes k}$ , and explain why it lets you check an identity between linear maps out of  $V^{\otimes k}$  on simple tensors only.
- Determine whether  $T(V)$  is commutative. Justify your answer, distinguishing the possible values of  $\dim V$ .
- State what data is needed to define an algebra homomorphism out of  $T(V)$ , and what must be checked about it.

*Solution.*

- For every vector space  $W$  over  $F$  and every  $k$ -linear map  $f: V^k \rightarrow W$ , there exists exactly one linear map  $\bar{f}: V^{\otimes k} \rightarrow W$  such that  $\bar{f} \circ \mu = f$ , where  $\mu(\mathbf{v}_1, \dots, \mathbf{v}_k) = \mathbf{v}_1 \otimes \dots \otimes \mathbf{v}_k$ .
- $\dim V^{\otimes 0} = \dim F = 1$ ;  $\dim V^{\otimes 1} = 4$ ;  $\dim V^{\otimes 3} = 4^3 = 64$ , by [Theorem 14.6.3](#).
- The simple tensors  $\mathbf{u}_1 \otimes \dots \otimes \mathbf{u}_k$  span  $V^{\otimes k}$  ([Theorem 14.6.2](#)); so does the smaller list of [Theorem 14.6.3](#). Two linear maps that agree on a spanning set agree on every linear combination of it, hence everywhere.
- It is commutative if  $\dim V \leq 1$  and non-commutative if  $\dim V \geq 2$ . For  $\dim V = 0$ ,  $T(V) = F$ ; for  $\dim V = 1$ ,  $T(V) \cong F[x]$  by [Example \(The tensor algebra of a line\)](#). For  $\dim V \geq 2$ ,  $\mathbf{v}_1 \otimes \mathbf{v}_2$  and  $\mathbf{v}_2 \otimes \mathbf{v}_1$  are distinct members of a basis by [Theorem 14.6.3](#), so they are unequal.
- A **linear** map  $f: V \rightarrow A$  into an associative unital algebra  $A$ , and nothing else: no condition on  $f$  needs checking, by [Theorem 14.6.6](#). What must be checked is that  $A$  really is an associative unital algebra and that  $f$  really is linear.

### 14.6.5.2 B. Practice

#### Exercise B1: Expanding a simple tensor.

Let  $V = F^2$ . Expand each of the following in the basis of [Theorem 14.6.3](#).

- $(2\mathbf{e}_1 - \mathbf{e}_2) \otimes (\mathbf{e}_1 + 3\mathbf{e}_2)$  in  $V^{\otimes 2}$ .
- $\mathbf{e}_1 \otimes (\mathbf{e}_1 - \mathbf{e}_2) \otimes (\mathbf{e}_1 + \mathbf{e}_2)$  in  $V^{\otimes 3}$ .



*Solution.*

(a) Expanding in slot 1, then slot 2,

$$(2\mathbf{e}_1 - \mathbf{e}_2) \otimes (\mathbf{e}_1 + 3\mathbf{e}_2) = 2\mathbf{e}_1 \otimes \mathbf{e}_1 + 6\mathbf{e}_1 \otimes \mathbf{e}_2 \\ - \mathbf{e}_2 \otimes \mathbf{e}_1 - 3\mathbf{e}_2 \otimes \mathbf{e}_2.$$

(b) Slot 1 is already a basis vector, so expand slots 2 and 3:

$$\mathbf{e}_1 \otimes (\mathbf{e}_1 - \mathbf{e}_2) \otimes (\mathbf{e}_1 + \mathbf{e}_2) = \mathbf{e}_1 \otimes \mathbf{e}_1 \otimes \mathbf{e}_1 + \mathbf{e}_1 \otimes \mathbf{e}_1 \otimes \mathbf{e}_2 \\ - \mathbf{e}_1 \otimes \mathbf{e}_2 \otimes \mathbf{e}_1 - \mathbf{e}_1 \otimes \mathbf{e}_2 \otimes \mathbf{e}_2.$$

The four basis tensors beginning with  $\mathbf{e}_2$  have coefficient 0, as they must: every term of the expansion carries  $\mathbf{e}_1$  in slot 1.

### Exercise B2: A product in the tensor algebra.

Let  $V = F^2$  and let  $x = 1 + \mathbf{e}_1$  and  $y = \mathbf{e}_2 - \mathbf{e}_1 \otimes \mathbf{e}_1$  in  $T(V)$ . Compute  $xy$  and  $yx$ , sort the answers by degree, and say whether  $x$  and  $y$  commute.

*Solution.* Multiply out, using that 1 is the identity and that concatenation is [Proposition 14.6.4](#):

$$xy = \mathbf{e}_2 - \mathbf{e}_1 \otimes \mathbf{e}_1 + \mathbf{e}_1 \otimes \mathbf{e}_2 - \mathbf{e}_1 \otimes \mathbf{e}_1 \otimes \mathbf{e}_1, \\ yx = \mathbf{e}_2 - \mathbf{e}_1 \otimes \mathbf{e}_1 + \mathbf{e}_2 \otimes \mathbf{e}_1 - \mathbf{e}_1 \otimes \mathbf{e}_1 \otimes \mathbf{e}_1.$$

Both have degree-1 part  $\mathbf{e}_2$  and degree-3 part  $-\mathbf{e}_1 \otimes \mathbf{e}_1 \otimes \mathbf{e}_1$ . Their degree-2 parts differ:  $-\mathbf{e}_1 \otimes \mathbf{e}_1 + \mathbf{e}_1 \otimes \mathbf{e}_2$  against  $-\mathbf{e}_1 \otimes \mathbf{e}_1 + \mathbf{e}_2 \otimes \mathbf{e}_1$ . Since  $\mathbf{e}_1 \otimes \mathbf{e}_2 \neq \mathbf{e}_2 \otimes \mathbf{e}_1$  by [Theorem 14.6.3](#),  $xy \neq yx$ : they do not commute.

### Exercise B3: A homomorphism built from the universal property.

Let  $V = F^2$  and let  $A = M_2(F)$ . Define  $f: V \rightarrow A$  by

$$f(x_1\mathbf{e}_1 + x_2\mathbf{e}_2) = \begin{pmatrix} 0 & x_1 \\ x_2 & 0 \end{pmatrix}.$$

Prove that there is a unique algebra homomorphism  $\bar{f}: T(V) \rightarrow M_2(F)$  with  $\bar{f}|_V = f$ , and compute  $\bar{f}(\mathbf{e}_1 \otimes \mathbf{e}_2)$ ,  $\bar{f}(\mathbf{e}_2 \otimes \mathbf{e}_1)$  and  $\bar{f}(\mathbf{e}_1 \otimes \mathbf{e}_1)$ .

*Solution.*  $M_2(F)$  is an associative unital  $F$ -algebra ([Definition 13.12.1](#)) and  $f$  is linear, since each entry of  $f(\mathbf{x})$  is a linear function of  $\mathbf{x}$ . By [Theorem 14.6.6](#) there is exactly one algebra homomorphism  $\bar{f}$  with  $\bar{f}|_V = f$ .

On simple tensors  $\bar{f}$  is the product of the images, so with  $\mathbf{A} = f(\mathbf{e}_1)$  and  $\mathbf{B} = f(\mathbf{e}_2)$ ,

$$\mathbf{A} = \begin{pmatrix} 0 & 1 \\ 0 & 0 \end{pmatrix}, \quad \mathbf{B} = \begin{pmatrix} 0 & 0 \\ 1 & 0 \end{pmatrix},$$

$$\bar{f}(\mathbf{e}_1 \otimes \mathbf{e}_2) = \mathbf{AB} = \begin{pmatrix} 1 & 0 \\ 0 & 0 \end{pmatrix}, \quad \bar{f}(\mathbf{e}_2 \otimes \mathbf{e}_1) = \mathbf{BA} = \begin{pmatrix} 0 & 0 \\ 0 & 1 \end{pmatrix},$$

and  $\bar{f}(\mathbf{e}_1 \otimes \mathbf{e}_1) = \mathbf{A}^2 = \mathbf{0}$ . The first two differ, which is another way to see that

$\mathbf{e}_1 \otimes \mathbf{e}_2 \neq \mathbf{e}_2 \otimes \mathbf{e}_1$ : a homomorphism cannot separate equal elements. The third shows  $\bar{f}$  is far from injective — as it must be, since  $T(V)$  is infinite-dimensional and  $M_2(F)$  is not.

### 14.6.5.3 C. Going deeper

#### Exercise C1: The grading is a direct sum of subspaces.

- (a) Prove that if  $x \in V^{\otimes k}$  and  $y \in V^{\otimes l}$  with  $k \neq l$  are both non-zero, then  $x + y$  lies in no single  $V^{\otimes m}$ .
- (b) Deduce that an algebra homomorphism  $\varphi: T(V) \rightarrow A$  is determined by its restrictions  $\varphi|_{V^{\otimes k}}$ , and that these are in turn determined by  $\varphi|_V$ .

*Solution.*

- (a) Under the identification of [Definition 14.6.2](#),  $x$  is the sequence with  $x$  in position  $k$  and  $\mathbf{0}$  elsewhere, and similarly for  $y$ . Their sum has non-zero entries in both positions  $k$  and  $l$ . An element of  $V^{\otimes m}$  has at most one non-zero entry, in position  $m$ . Since  $k \neq l$ , the sum has two, so it is in no  $V^{\otimes m}$ .
- (b) Every element of  $T(V)$  is the finite sum of its homogeneous components, so a linear map is determined by its restrictions to the  $V^{\otimes k}$ . By multiplicativity,  $\varphi(\mathbf{u}_1 \otimes \cdots \otimes \mathbf{u}_k) = \varphi(\mathbf{u}_1) \cdots \varphi(\mathbf{u}_k)$ , which depends only on  $\varphi|_V$ ; and  $V^{\otimes k}$  is spanned by its simple tensors, so  $\varphi|_{V^{\otimes k}}$  is determined by  $\varphi|_V$ . (This is the uniqueness half of [Theorem 14.6.6](#), isolated.)

#### Exercise C2: Extending an operator to the algebra.

Let  $T: V \rightarrow V$  be linear.

- (a) Prove that there is a unique algebra homomorphism  $T(T): T(V) \rightarrow T(V)$  with  $T(T)|_V = T$ , and that it maps  $V^{\otimes k}$  into  $V^{\otimes k}$ .
- (b) Prove that  $T(ST) = T(S)T(T)$  for linear  $S, T: V \rightarrow V$ , and that  $T(\text{id}_V) = \text{id}$ .
- (c) Deduce that if  $T$  is invertible then so is  $T(T)$ .

*Solution.*

- (a)  $T(V)$  is an associative unital algebra and the composite  $V \xrightarrow{T} V \hookrightarrow T(V)$  is linear, so [Theorem 14.6.6](#) gives a unique algebra homomorphism  $T(T)$  restricting to  $T$  on  $V$ . On a simple tensor it is  $\mathbf{u}_1 \otimes \cdots \otimes \mathbf{u}_k \mapsto T\mathbf{u}_1 \otimes \cdots \otimes T\mathbf{u}_k$ , which lies in  $V^{\otimes k}$ ; those tensors span  $V^{\otimes k}$ , so  $T(T)(V^{\otimes k}) \subseteq V^{\otimes k}$ .
- (b)  $T(S)T(T)$  is a composite of algebra homomorphisms, hence an algebra homomorphism, and on  $V$  it is  $ST$ . By the uniqueness clause of [Theorem 14.6.6](#) applied to the linear map  $ST: V \rightarrow T(V)$ , it equals  $T(ST)$ . Likewise  $\text{id}_{T(V)}$  is an algebra homomorphism restricting to  $\text{id}_V$ , so it equals  $T(\text{id}_V)$ .
- (c) By (b),  $T(T)T(T^{-1}) = T(TT^{-1}) = T(\text{id}_V) = \text{id}$ , and the same with the factors swapped. Hence  $T(T)$  is invertible with inverse  $T(T^{-1})$ .

**Exercise C3: Why the empty degree is  $F$ .**

Suppose we had instead set  $V^{\otimes 0} = \{0\}$  and kept every other definition.

- (a) Explain why [Theorem 14.6.3](#) would then fail at  $k = 0$ .
- (b) Explain why  $T(V)$  would have no identity element, and hence would not be an algebra in the sense of [Definition 13.12.1](#).
- (c) Give one further formula in this section that the convention  $V^{\otimes 0} = F$  keeps true at  $k = 0$ .

*Solution.*

- (a) The theorem counts one basis vector per  $k$ -tuple of indices. There is exactly one 0-tuple, the empty one, so the count gives  $n^0 = 1$ , the dimension of  $F$ , not of  $\{0\}$ .
- (b) The identity of  $T(V)$  is  $1 \in V^{\otimes 0}$ . With  $V^{\otimes 0} = \{0\}$  there is no candidate: a homogeneous element of degree  $m \geq 1$  multiplies  $V^{\otimes k}$  into  $V^{\otimes(k+m)}$ , so it cannot fix a non-zero element of  $V^{\otimes k}$ , and neither can a sum of such elements, by [Exercise \(C1: The grading is a direct sum of subspaces\)](#) (a). Hypothesis (A3) of [Definition 13.12.1](#) would fail.
- (c) The universal property [Theorem 14.6.6](#): its formula  $\bar{f}(a) = a1_A$  for  $a \in F$  is the degree-0 case, and it is what makes  $\bar{f}$  unital. Equally acceptable:  $\dim T(V) = \sum_{k \geq 0} n^k$ , or the multiplication rule  $V^{\otimes 0} \cdot V^{\otimes l} \subseteq V^{\otimes l}$ , which reads “multiplying by a scalar does not change the degree”.

## 14.7 Symmetric Powers

The tensor algebra is free, and being free it is useless on its own: it records the order of the slots even when nothing in the problem cares about the order. A symmetric bilinear form does not care ([Definition 13.3.1](#)); nor does a quadratic form; nor does the monomial  $x_1x_2$ , which is the same polynomial as  $x_2x_1$ . This section forces  $T(V)$  to forget the order, by dividing the tensor power  $V^{\otimes k}$  by the span of the differences  $\mathbf{u} \otimes \mathbf{w} - \mathbf{w} \otimes \mathbf{u}$ . The result, the symmetric power  $\text{Sym}^k V$ , turns out to have a basis indexed by non-decreasing index tuples, and the algebra assembled from all the  $\text{Sym}^k V$  turns out to be the polynomial algebra of Chapter 0, one variable per basis vector.

The construction repeats Section 02’s three moves for the second time: universal property, uniqueness, then a model built as a quotient. If the shape of the argument below feels familiar, that is the intention.

Throughout,  $F$  is a field,  $V$  is a finite-dimensional vector space over  $F$  with  $n = \dim V \geq 1$ , and  $k \geq 1$  unless stated otherwise. **The field is arbitrary**, and the single place where the characteristic matters is isolated in the last part of the section.

### 14.7.1 Symmetric multilinear maps

Before asking for a space, we name the maps it is supposed to linearize. Chapter 13 called a *bilinear form* symmetric when swapping its two arguments changed nothing ([Definition 13.3.1](#)).

With  $k$  arguments there are more ways to rearrange them, and we demand invariance under all of them.

*A symmetric map cannot tell in which order its arguments were handed to it.*

### Definition 14.7.1 — Symmetric Multilinear Map

Let  $W$  be a vector space over  $F$ . A  $k$ -linear map  $f: V^k \rightarrow W$  (Definition 14.1.1) is **symmetric** if

$$f(\mathbf{u}_{\sigma(1)}, \mathbf{u}_{\sigma(2)}, \dots, \mathbf{u}_{\sigma(k)}) = f(\mathbf{u}_1, \mathbf{u}_2, \dots, \mathbf{u}_k)$$

for every permutation  $\sigma \in S_k$  and for all  $\mathbf{u}_1, \dots, \mathbf{u}_k \in V$ .

At  $k = 2$  the only non-identity permutation is the swap, so this is exactly Definition 13.3.1 with a general target  $W$  in place of  $F$ . At  $k = 1$  the condition is empty and every linear map is symmetric.

Three examples and one non-example.

- **A symmetric bilinear form**,  $\beta: V \times V \rightarrow F$  with  $\beta(\mathbf{u}, \mathbf{w}) = \beta(\mathbf{w}, \mathbf{u})$ . This is the case  $k = 2$ ,  $W = F$ .
- **Products of one fixed functional**. For  $\varphi \in V^*$ , the map  $(\mathbf{u}_1, \dots, \mathbf{u}_k) \mapsto \varphi(\mathbf{u}_1)\varphi(\mathbf{u}_2) \cdots \varphi(\mathbf{u}_k)$  is  $k$ -linear, by the third example after Definition 6.2.1, and symmetric because multiplication in  $F$  is commutative: rearranging the factors does not change the product.
- **Multiplication in a commutative algebra**. If  $A$  is a commutative  $F$ -algebra and  $f: V \rightarrow A$  is linear, then  $(\mathbf{u}_1, \dots, \mathbf{u}_k) \mapsto f(\mathbf{u}_1) \cdots f(\mathbf{u}_k)$  is symmetric, by commutativity of  $A$ .
- **A non-example by minimal change**. On  $V = F^2$ , the map  $(\mathbf{u}, \mathbf{w}) \mapsto u_1 w_2$  is bilinear but not symmetric: it sends  $(\mathbf{e}_1, \mathbf{e}_2)$  to 1 and  $(\mathbf{e}_2, \mathbf{e}_1)$  to 0. Only the symmetry clause fails; bilinearity survives untouched.

Note which permutations are allowed: **all** of  $S_k$ , not only the swaps of adjacent slots. Nothing is lost by asking for all of them, and asking for all of them spares us a lemma about generating  $S_k$ .

## 14.7.2 The symmetric power

Now the object. The demand is Definition 14.6.1's, restricted to the symmetric maps.

*The  $k$ -th symmetric power is the space in which symmetric  $k$ -linear maps become linear.*

### Definition 14.7.2 — $k$ -th Symmetric Power

A  **$k$ -th symmetric power of  $V$**  is a pair  $(P, \pi)$ , where  $P$  is a vector space over  $F$  and  $\pi: V^k \rightarrow P$  is a **symmetric**  $k$ -linear map, such that

- (S1) (**universal property**) for **every** vector space  $W$  over  $F$  and **every symmetric**  $k$ -linear map  $f: V^k \rightarrow W$ , there is a **unique** linear map  $\tilde{f}: P \rightarrow W$  with  $\tilde{f} \circ \pi = f$ .

We write  $\text{Sym}^k V$  for  $P$  and denote the value  $\pi(\mathbf{u}_1, \dots, \mathbf{u}_k)$  by juxtaposition,

$$\mathbf{u}_1 \mathbf{u}_2 \cdots \mathbf{u}_k := \pi(\mathbf{u}_1, \dots, \mathbf{u}_k),$$

calling it a **symmetric product**. By convention  $\text{Sym}^0 V = F$ .

The one difference from (T1) is the word **symmetric**, in two places: the structure map  $\pi$  is required to be symmetric, and only symmetric maps  $f$  are required to factor. Both changes are needed. If  $\pi$  were not symmetric, the notation  $\mathbf{u}\mathbf{w} = \mathbf{w}\mathbf{u}$  would be false; if non-symmetric maps had to factor, no space could do the job, since  $\bar{f} \circ \pi$  is symmetric whenever  $\pi$  is.

Uniqueness is the template of [Theorem 14.2.1](#) for the third time, and we no longer write it out.

### Theorem 14.7.1 — Uniqueness of the Symmetric Power

Let  $(P, \pi)$  and  $(P', \pi')$  both be  $k$ -th symmetric powers of  $V$ . Then there is a unique linear isomorphism  $\varphi: P \rightarrow P'$  with  $\varphi \circ \pi = \pi'$ .

*Proof.* Word for word the proof of [Theorem 14.6.1](#), with “symmetric  $k$ -linear” in place of “ $k$ -linear” throughout: (S1) for  $P$  applied to the symmetric map  $\pi'$  gives  $\varphi$ , (S1) for  $P'$  applied to  $\pi$  gives  $\psi$ , the composite  $\psi\varphi$  is a linear self-map of  $P$  with  $\psi\varphi\pi = \pi$ , and the uniqueness clause of (S1) applied to  $\pi$  forces  $\psi\varphi = \text{id}_P$ . Symmetrically  $\varphi\psi = \text{id}_{P'}$ . This proves the theorem. ■

And now the model: the tensor power, divided by the relations we want to impose.

### Theorem 14.7.2 — Existence of the Symmetric Power

Let  $C_k \subseteq V^{\otimes k}$  be the span of all the elements

$$\mathbf{u}_1 \otimes \cdots \otimes \mathbf{u}_k - \mathbf{u}_{\sigma(1)} \otimes \cdots \otimes \mathbf{u}_{\sigma(k)}, \quad \mathbf{u}_i \in V, \sigma \in S_k,$$

and let  $q: V^{\otimes k} \rightarrow V^{\otimes k}/C_k$  be the quotient map ([Definition 3.9.3](#)). Then  $V^{\otimes k}/C_k$ , together with

$$\pi(\mathbf{u}_1, \dots, \mathbf{u}_k) = q(\mathbf{u}_1 \otimes \cdots \otimes \mathbf{u}_k),$$

is a  $k$ -th symmetric power of  $V$ . At  $k = 2$  the subspace  $C_2$  is the span of the elements  $\mathbf{u} \otimes \mathbf{w} - \mathbf{w} \otimes \mathbf{u}$ .

*Idea.* Two things to check, and both are one line each once the right tool is named. That  $\pi$  is symmetric is the definition of  $C_k$ : permuting the arguments changes the tensor by an element of  $C_k$ , which  $q$  kills. That symmetric maps factor is [Theorem 3.9.10](#): turn  $f$  into a linear map on  $V^{\otimes k}$  by (T1), observe that symmetry of  $f$  makes it vanish on the generators of  $C_k$ , and let the quotient’s universal property do the rest. This is the move the chapter uses three times: **a quotient kills exactly the relations you want**.

*Proof.*  $C_k$  is a subspace, being a span. The map  $\pi$  is  $k$ -linear, being the  $k$ -linear  $\mu_k$  of

[Theorem 14.6.2](#) followed by the linear map  $q$ . It is symmetric: for  $\sigma \in S_k$ , the difference

$$\mathbf{u}_1 \otimes \cdots \otimes \mathbf{u}_k - \mathbf{u}_{\sigma(1)} \otimes \cdots \otimes \mathbf{u}_{\sigma(k)}$$

lies in  $C_k$  by definition, so the two tensors have the same image under  $q$ , that is,  $\pi(\mathbf{u}_{\sigma(1)}, \dots, \mathbf{u}_{\sigma(k)}) = \pi(\mathbf{u}_1, \dots, \mathbf{u}_k)$ .

Now let  $f: V^k \rightarrow W$  be symmetric  $k$ -linear. By (T1) there is a unique linear  $\tilde{f}: V^{\otimes k} \rightarrow W$  with  $\tilde{f}(\mathbf{u}_1 \otimes \cdots \otimes \mathbf{u}_k) = f(\mathbf{u}_1, \dots, \mathbf{u}_k)$ . On a generator of  $C_k$ ,

$$\tilde{f}(\mathbf{u}_1 \otimes \cdots \otimes \mathbf{u}_k - \mathbf{u}_{\sigma(1)} \otimes \cdots \otimes \mathbf{u}_{\sigma(k)}) = f(\mathbf{u}_1, \dots, \mathbf{u}_k) - f(\mathbf{u}_{\sigma(1)}, \dots, \mathbf{u}_{\sigma(k)}),$$

which is 0 because  $f$  is symmetric. A linear map vanishing on a spanning set of  $C_k$  vanishes on  $C_k$ , so  $C_k \subseteq \ker \tilde{f}$ . By [Theorem 3.9.10](#) (a) there is exactly one linear  $\bar{f}: V^{\otimes k}/C_k \rightarrow W$  with  $\bar{f} \circ q = \tilde{f}$ , and then

$$\bar{f} \circ \pi = \bar{f} \circ q \circ \mu_k = \tilde{f} \circ \mu_k = f.$$

For uniqueness, suppose  $h$  is linear with  $h \circ \pi = f$ . Since  $q$  is surjective and  $V^{\otimes k}$  is spanned by its simple tensors ([Theorem 14.6.2](#)), the elements  $\pi(\mathbf{u}_1, \dots, \mathbf{u}_k)$  span  $V^{\otimes k}/C_k$ . So  $h$  and  $\bar{f}$  agree on a spanning set, hence everywhere. This proves the theorem. ■

*Warning.*  $\text{Sym}^k V$  is a quotient of  $V^{\otimes k}$ , not a subspace of it. Its elements are cosets, and the identity  $\mathbf{u}\mathbf{w} = \mathbf{w}\mathbf{u}$  in  $\text{Sym}^2 V$  says that  $\mathbf{u} \otimes \mathbf{w}$  and  $\mathbf{w} \otimes \mathbf{u}$  lie in the **same** coset. It does **not** say  $\mathbf{u} \otimes \mathbf{w} = \mathbf{w} \otimes \mathbf{u}$  in  $V^{\otimes 2}$ , which is false for independent  $\mathbf{u}, \mathbf{w}$  by [Theorem 14.6.3](#). Confusing the two is the commonest error here; the safe habit is never to write an element of  $\text{Sym}^k V$  as a tensor.

### 14.7.3 A basis, and the dimension

Sorting an index tuple is now legal, so the basis of [Theorem 14.6.3](#) collapses: only the sorted tuples survive. Counting them is the substance of the section.

#### Theorem 14.7.3 — Basis of a Symmetric Power

Let  $\mathcal{B} = (\mathbf{v}_1, \dots, \mathbf{v}_n)$  be a basis of  $V$  and  $k \geq 1$ . Then the symmetric products

$$\mathbf{v}_{i_1} \mathbf{v}_{i_2} \cdots \mathbf{v}_{i_k}, \quad 1 \leq i_1 \leq i_2 \leq \cdots \leq i_k \leq n,$$

one for each **non-decreasing** tuple, form a basis of  $\text{Sym}^k V$ . Consequently

$$\dim \text{Sym}^k V = \binom{n+k-1}{k},$$

over **every** field  $F$ .

*Idea.* Spanning is sorting: the image of any index tuple equals the image of its sorted version, so the sorted ones already span. Independence needs one linear functional per non-decreasing tuple, and it must be built **without dividing by anything**, since we want the theorem over every field. The functional that works is a sum over all ways of dealing out the  $n$  dual basis functionals to the  $k$  slots with prescribed multiplicities: every coefficient in it is 1. Counting the tuples is then the stars-and-bars bijection, which turns a non-decreasing tuple into a strictly increasing one by adding  $0, 1, 2, \dots$  to its entries.

*Proof. Spanning.* By Theorem 14.6.3 the tensors  $\mathbf{v}_{j_1} \otimes \dots \otimes \mathbf{v}_{j_k}$ , over all tuples  $J$ , form a basis of  $V^{\otimes k}$ , and  $q$  is surjective, so their images  $\mathbf{v}_{j_1} \cdots \mathbf{v}_{j_k}$  span  $\text{Sym}^k V$ . Given any  $J$ , let  $\sigma \in S_k$  sort it, so that  $(j_{\sigma(1)}, \dots, j_{\sigma(k)})$  is non-decreasing. Since  $\pi$  is symmetric,  $\mathbf{v}_{j_1} \cdots \mathbf{v}_{j_k} = \mathbf{v}_{j_{\sigma(1)}} \cdots \mathbf{v}_{j_{\sigma(k)}}$ . So the non-decreasing tuples already give a spanning set.

*Independence.* Let  $(\varphi_1, \dots, \varphi_n)$  be the dual basis of  $\mathcal{B}$  (Theorem 4.1.3). For a non-decreasing tuple  $I = (i_1, \dots, i_k)$ , let

$$m_j = |\{t : i_t = j\}|, \quad j = 1, \dots, n,$$

be its **multiplicity vector**, so  $m_1 + \dots + m_n = k$ , and let  $\Phi(I)$  be the set of functions  $\tau: \{1, \dots, k\} \rightarrow \{1, \dots, n\}$  taking the value  $j$  exactly  $m_j$  times, for each  $j$ . Define

$$g_I(\mathbf{u}_1, \dots, \mathbf{u}_k) = \sum_{\tau \in \Phi(I)} \varphi_{\tau(1)}(\mathbf{u}_1) \varphi_{\tau(2)}(\mathbf{u}_2) \cdots \varphi_{\tau(k)}(\mathbf{u}_k).$$

Each summand is  $k$ -linear, being a product of one functional per slot, so  $g_I$  is  $k$ -linear.

It is symmetric. Let  $\rho \in S_k$ . Substituting and then reindexing the product by  $s = \rho(t)$ ,

$$g_I(\mathbf{u}_{\rho(1)}, \dots, \mathbf{u}_{\rho(k)}) = \sum_{\tau \in \Phi(I)} \prod_{s=1}^k \varphi_{(\tau \rho^{-1})(s)}(\mathbf{u}_s).$$

The map  $\tau \mapsto \tau \rho^{-1}$  is a bijection of  $\Phi(I)$  onto itself, since composing with a permutation of the domain does not change how often a value is taken. So the sum is unchanged, and  $g_I$  is symmetric.

By (S1),  $g_I$  induces a linear  $\bar{g}_I: \text{Sym}^k V \rightarrow F$ . Evaluate it on a basis candidate: for any tuple  $J$ ,

$$\bar{g}_I(\mathbf{v}_{j_1} \cdots \mathbf{v}_{j_k}) = \sum_{\tau \in \Phi(I)} \prod_{t=1}^k \delta_{\tau(t) j_t}.$$

A summand is 1 exactly when  $\tau(t) = j_t$  for every  $t$ , and 0 otherwise; so the whole sum is 1 if the function  $t \mapsto j_t$  belongs to  $\Phi(I)$ , and 0 if it does not. For non-decreasing  $I$  and  $J$ , that function lies in  $\Phi(I)$  exactly when  $J$  has the same multiplicity vector as  $I$ , which for non-decreasing tuples means  $J = I$ . Hence  $\bar{g}_I(\mathbf{v}_{j_1} \cdots \mathbf{v}_{j_k}) = \delta_{IJ}$ .

Now suppose  $\sum_I a_I \mathbf{v}_{i_1} \cdots \mathbf{v}_{i_k} = \mathbf{0}$ , the sum over non-decreasing tuples. Applying  $\bar{g}_J$  leaves  $a_J = 0$ . Since  $J$  was arbitrary, every coefficient is 0, and the list is independent. So it is a basis.

*The count.* Send a non-decreasing tuple  $(i_1, \dots, i_k)$  to

$$(i_1, i_2 + 1, i_3 + 2, \dots, i_k + k - 1).$$

Consecutive entries satisfy  $i_t + t - 1 < i_{t+1} + t$  because  $i_t \leq i_{t+1}$ , so the image is **strictly** increasing, and its entries lie between 1 and  $n + k - 1$ . Conversely, subtracting  $0, 1, \dots, k-1$  from a strictly increasing tuple in  $\{1, \dots, n+k-1\}$  returns a non-decreasing tuple in  $\{1, \dots, n\}$ : the differences  $j_{t+1} - j_t \geq 1$  become  $i_{t+1} - i_t \geq 0$ , and  $1 \leq j_1, j_k \leq n + k - 1$  give  $1 \leq i_1$  and  $i_k \leq n$ . The two assignments are mutually inverse, so the non-decreasing  $k$ -tuples from  $\{1, \dots, n\}$  are in bijection with the  $k$ -element subsets of  $\{1, \dots, n + k - 1\}$ , and there are  $\binom{n+k-1}{k}$  of those. This proves the theorem. ■

Nothing in the proof divided by an integer, which is why the conclusion holds over  $\mathbb{F}_2$  as it does over  $\mathbb{C}$ . Keep that in view: the last part of this section shows that a *different* and very common description of  $\text{Sym}^k V$  does not survive over  $\mathbb{F}_2$ , and the difference between the two is exactly one division.

*Quick check.* Let  $\dim V = 3$ . Compute  $\dim \text{Sym}^2 V$  and  $\dim V^{\otimes 2}$ , and say which basis tensors of  $V^{\otimes 2}$  become equal in  $\text{Sym}^2 V$ .

*Solution.*  $\dim \text{Sym}^2 V = \binom{4}{2} = 6$  and  $\dim V^{\otimes 2} = 3^2 = 9$ . The six surviving basis elements are  $\mathbf{v}_1^2, \mathbf{v}_1\mathbf{v}_2, \mathbf{v}_1\mathbf{v}_3, \mathbf{v}_2^2, \mathbf{v}_2\mathbf{v}_3, \mathbf{v}_3^2$ . The three pairs  $\mathbf{v}_i \otimes \mathbf{v}_j$  and  $\mathbf{v}_j \otimes \mathbf{v}_i$  with  $i < j$  become equal, which removes 3 from the count:  $9 - 3 = 6$ .

#### Example. (A symmetric square, in coordinates)

Let  $V = F^2$ . Write down a basis of  $\text{Sym}^2 V$  and expand  $(\mathbf{e}_1 + 2\mathbf{e}_2)(3\mathbf{e}_1 - \mathbf{e}_2)$  in it.

*Solution.* By Theorem 14.7.3 a basis is  $(\mathbf{e}_1^2, \mathbf{e}_1\mathbf{e}_2, \mathbf{e}_2^2)$ , of size  $\binom{3}{2} = 3$ . Expanding one slot at a time and then using  $\mathbf{e}_2\mathbf{e}_1 = \mathbf{e}_1\mathbf{e}_2$ ,

$$\begin{aligned} (\mathbf{e}_1 + 2\mathbf{e}_2)(3\mathbf{e}_1 - \mathbf{e}_2) &= 3\mathbf{e}_1\mathbf{e}_1 - \mathbf{e}_1\mathbf{e}_2 + 6\mathbf{e}_2\mathbf{e}_1 - 2\mathbf{e}_2\mathbf{e}_2 \\ &= 3\mathbf{e}_1^2 + 5\mathbf{e}_1\mathbf{e}_2 - 2\mathbf{e}_2^2. \end{aligned}$$

Compare with expanding  $(x + 2y)(3x - y) = 3x^2 + 5xy - 2y^2$  in ordinary algebra. That is not a coincidence, and the next part of this section says exactly what it is.

*Warning.* **Not every element of  $\text{Sym}^k V$  is a product of  $k$  vectors.** Take  $F = \mathbb{R}$ ,  $V = \mathbb{R}^2$  and  $\omega = \mathbf{e}_1^2 + \mathbf{e}_2^2 \in \text{Sym}^2 V$ . If  $\omega = \mathbf{u}\mathbf{w}$  with  $\mathbf{u} = a\mathbf{e}_1 + b\mathbf{e}_2$  and  $\mathbf{w} = c\mathbf{e}_1 + d\mathbf{e}_2$ , then comparing coordinates in the basis  $(\mathbf{e}_1^2, \mathbf{e}_1\mathbf{e}_2, \mathbf{e}_2^2)$  gives  $ac = 1$ ,  $ad + bc = 0$  and  $bd = 1$ . The first and last force  $a, b, c, d \neq 0$ , and multiplying  $ad = -bc$  by  $bc$  gives  $-(bc)^2 = (ac)(bd) = 1$ , impossible in  $\mathbb{R}$ . This is the same phenomenon as the non-simple tensors of Section 02, and it is why  $\text{Sym}^k V$  is much larger than the set of products.

### 14.7.4 The symmetric algebra is a polynomial algebra

Stacking the symmetric powers gives an algebra, exactly as stacking the tensor powers did, and the multiplication descends from concatenation.



### Proposition 14.7.4 — The Symmetric Product of Two Symmetric Tensors

Let  $k, l \geq 1$ . There is a unique bilinear map  $\text{Sym}^k V \times \text{Sym}^l V \rightarrow \text{Sym}^{k+l} V$  with

$$(\mathbf{u}_1 \cdots \mathbf{u}_k)(\mathbf{w}_1 \cdots \mathbf{w}_l) = \mathbf{u}_1 \cdots \mathbf{u}_k \mathbf{w}_1 \cdots \mathbf{w}_l.$$

*Proof.* Concatenation carries the relations into the relations. Let  $x \in C_k$  and  $y \in V^{\otimes l}$ . For a generator  $x = \mathbf{u}_1 \otimes \cdots \otimes \mathbf{u}_k - \mathbf{u}_{\sigma(1)} \otimes \cdots \otimes \mathbf{u}_{\sigma(k)}$  and a simple  $y = \mathbf{w}_1 \otimes \cdots \otimes \mathbf{w}_l$ , the product  $x \cdot y$  of Proposition 14.6.4 is again a generator of  $C_{k+l}$ , for the permutation acting as  $\sigma$  on the first  $k$  slots and fixing the last  $l$ . Since generators span  $C_k$ , since  $V^{\otimes l}$  is spanned by its simple tensors, and since  $\cdot$  is bilinear,  $C_k \cdot V^{\otimes l} \subseteq C_{k+l}$ . Swapping the roles of the two blocks gives  $V^{\otimes k} \cdot C_l \subseteq C_{k+l}$ .

*Descending, one argument at a time.* Let  $q_m: V^{\otimes m} \rightarrow \text{Sym}^m V$  denote the quotient maps. Fix  $y \in V^{\otimes l}$ . The linear map  $x \mapsto q_{k+l}(x \cdot y)$  kills  $C_k$  by the first inclusion, so Theorem 3.9.10 gives a unique linear  $\beta_y: \text{Sym}^k V \rightarrow \text{Sym}^{k+l} V$  with  $\beta_y(q_k x) = q_{k+l}(x \cdot y)$ . The assignment  $y \mapsto \beta_y$  is linear, since  $\beta_{ay+y'}$  and  $a\beta_y + \beta_{y'}$  agree on the spanning set  $q_k(V^{\otimes k}) = \text{Sym}^k V$ ; and it kills  $C_l$ , since  $y \in C_l$  gives  $x \cdot y \in C_{k+l}$  and hence  $\beta_y = 0$ . By Theorem 3.9.10 again it descends to a linear map  $\text{Sym}^l V \rightarrow \mathcal{L}(\text{Sym}^k V, \text{Sym}^{k+l} V)$ , and evaluating gives the required bilinear map. It is unique because the products of vectors span each factor. When  $k = 0$  or  $l = 0$  the product is scalar multiplication, as in Section 06. This proves the proposition. ■

### Definition 14.7.3 — Symmetric Algebra

The **symmetric algebra** of  $V$  is the vector space

$$\text{Sym}(V) = \bigoplus_{k \geq 0} \text{Sym}^k V,$$

of sequences with finitely many non-zero components, with the multiplication assembled from Proposition 14.7.4 by  $(xy)_m = \sum_{k+l=m} x_k y_l$ .

### Proposition 14.7.5 — $\text{Sym}(V)$ Is a Commutative Algebra

$\text{Sym}(V)$  is an associative, **commutative**, unital  $F$ -algebra with identity  $1 \in \text{Sym}^0 V = F$ , and  $\text{Sym}^k V \cdot \text{Sym}^l V \subseteq \text{Sym}^{k+l} V$ .

*Proof.* Bilinearity, associativity and the identity are proved exactly as in Theorem 14.6.5: all three are identities between multilinear expressions, so it suffices to check them on products of vectors, which span each  $\text{Sym}^k V$ , and there all three read “concatenate the strings”.

Commutativity is the new point. Both  $(\mathbf{u}_1 \cdots \mathbf{u}_k)(\mathbf{w}_1 \cdots \mathbf{w}_l)$  and  $(\mathbf{w}_1 \cdots \mathbf{w}_l)(\mathbf{u}_1 \cdots \mathbf{u}_k)$  equal the symmetric product of the same  $k + l$  vectors in two different orders, and  $\pi$  is symmetric, so they are equal. Two bilinear maps agreeing on spanning sets of both

arguments agree everywhere, so  $xy = yx$  for all  $x, y$ . This proves the proposition. ■

To name what  $\text{Sym}(V)$  is, we need polynomials in more than one variable. Chapter 0 built  $F[x]$ ; the several-variable version is the same construction with monomials indexed by exponent vectors.

### Definition 14.7.4 — The Polynomial Algebra $F[x_1, \dots, x_n]$

Let  $n \geq 1$ . The **polynomial algebra in  $n$  variables** over  $F$  is the vector space  $F[x_1, \dots, x_n]$  with basis the **monomials**

$$x^{\mathbf{a}} = x_1^{a_1} x_2^{a_2} \cdots x_n^{a_n}, \quad \mathbf{a} = (a_1, \dots, a_n) \in \mathbb{N}^n,$$

with the unique bilinear multiplication satisfying  $x^{\mathbf{a}} x^{\mathbf{b}} = x^{\mathbf{a}+\mathbf{b}}$ . The **degree** of  $x^{\mathbf{a}}$  is  $|\mathbf{a}| = a_1 + \cdots + a_n$ , and  $F[x_1, \dots, x_n]_k$  denotes the span of the monomials of degree  $k$ .

The bilinear extension exists and is unique because a bilinear map may be prescribed freely on a pair of bases, by the argument used twice in the proof of [Proposition 14.6.4](#). The multiplication is associative, commutative and unital, with identity the monomial all of whose exponents are 0, because exponent addition in  $\mathbb{N}^n$  is associative, commutative and has an identity. For  $n = 1$  the monomials are  $1, x, x^2, \dots$  and the rule is  $x^a x^b = x^{a+b}$ , so  $F[x_1]$  is the ring  $F[x]$  of [Definition 0.8.2](#), whose basis of monomials is [Example \(A Basis of  \$F\[x\]\$ \)](#). Everything Chapter 5 proved — division, greatest common divisors, unique factorization — lives in that one-variable case.

### Theorem 14.7.6 — The Symmetric Algebra Is a Polynomial Algebra

Let  $\mathcal{B} = (\mathbf{v}_1, \dots, \mathbf{v}_n)$  be a basis of  $V$ . Then there is an isomorphism of  $F$ -algebras

$$\Psi: \text{Sym}(V) \xrightarrow{\sim} F[x_1, \dots, x_n], \quad \Psi(\mathbf{v}_i) = x_i,$$

and it carries  $\text{Sym}^k V$  onto  $F[x_1, \dots, x_n]_k$ . In particular

$$\dim F[x_1, \dots, x_n]_k = \binom{n+k-1}{k}.$$

*Idea.* Both sides have a basis indexed by the same combinatorial object: a non-decreasing  $k$ -tuple on the left, a degree- $k$  monomial on the right, and each is determined by its multiplicity vector  $\mathbf{m} \in \mathbb{N}^n$  with  $|\mathbf{m}| = k$ . So match the bases, which gives a linear isomorphism for free, and then check multiplicativity on basis elements, where both products are “add the exponent vectors”.

*Proof.* Write  $|\mathbf{m}| = m_1 + \cdots + m_n$  for  $\mathbf{m} \in \mathbb{N}^n$ . Given  $\mathbf{m}$  with  $|\mathbf{m}| = k$ , let  $I(\mathbf{m})$  be the non-decreasing tuple taking the value  $j$  exactly  $m_j$  times, and write  $\mathbf{v}^{\mathbf{m}}$  for the corresponding symmetric product  $\mathbf{v}_{i_1} \cdots \mathbf{v}_{i_k}$ . The assignment  $\mathbf{m} \mapsto I(\mathbf{m})$  is a bijection onto the non-decreasing  $k$ -tuples, since a non-decreasing tuple is recovered from its multiplicity vector and conversely. So by [Theorem 14.7.3](#) the family  $(\mathbf{v}^{\mathbf{m}})_{|\mathbf{m}|=k}$  is a basis

of  $\text{Sym}^k V$ , and hence  $(\mathbf{v}^{\mathbf{m}})_{\mathbf{m} \in \mathbb{N}^n}$  is a basis of  $\text{Sym}(V)$ .

Define  $\Psi$  to be the unique linear map with  $\Psi(\mathbf{v}^{\mathbf{m}}) = x^{\mathbf{m}}$ , which exists by [Theorem 3.1.6](#). It sends a basis of  $\text{Sym}(V)$  bijectively onto the basis of monomials, so it is a linear isomorphism, and it sends the degree- $k$  basis onto the degree- $k$  monomials, so  $\Psi(\text{Sym}^k V) = F[x_1, \dots, x_n]_k$ .

Multiplicativity: the maps  $(y, z) \mapsto \Psi(yz)$  and  $(y, z) \mapsto \Psi(y)\Psi(z)$  are bilinear, so it is enough to compare them on the basis. By [Proposition 14.7.4](#),  $\mathbf{v}^{\mathbf{m}}\mathbf{v}^{\mathbf{m}'}$  is the symmetric product of the  $|\mathbf{m}| + |\mathbf{m}'|$  vectors of the two strings, which after sorting is  $\mathbf{v}^{\mathbf{m}+\mathbf{m}'}$ ; and  $x^{\mathbf{m}}x^{\mathbf{m}'} = x^{\mathbf{m}+\mathbf{m}'}$  by [Definition 14.7.4](#). Also  $\Psi$  sends the empty product 1 to the monomial with all exponents 0, which is 1. Hence  $\Psi$  is an algebra isomorphism. Finally  $\mathbf{v}_i$  is the basis element  $\mathbf{v}^{\mathbf{m}}$  whose  $\mathbf{m}$  has a single 1 in position  $i$ , so  $\Psi(\mathbf{v}_i) = x_i$ .

The dimension formula follows from [Theorem 14.7.3](#), since  $\Psi$  restricts to an isomorphism in each degree. This proves the theorem. ■

So the symmetric algebra is nothing exotic: *it is the polynomial algebra, with one variable per basis vector of  $V$* . The isomorphism depends on the basis, exactly as the coordinate isomorphism  $V \cong F^n$  does;  $\text{Sym}(V)$  is the basis-free version of “polynomials in  $n$  variables”, and  $\text{Sym}^k V$  is the basis-free version of “homogeneous polynomials of degree  $k$ ”. Read backwards, the theorem also computes something Chapter 5 never did: the number of monomials of degree  $k$  in  $n$  variables is  $\binom{n+k-1}{k}$ .

*Remark.* The case  $k = 2$  recovers a count from Chapter 13. A symmetric bilinear form on  $V$  is a symmetric 2-linear map into  $F$ , so by (S1) the symmetric forms on  $V$  are in bijection with the linear functionals on  $\text{Sym}^2 V$ , and there are  $\binom{n+1}{2} = n(n+1)/2$  dimensions of them — the number of entries on and above the diagonal of a symmetric matrix, which is what [Proposition 13.3.1](#) would give by counting.

## 14.7.5 Symmetric powers versus symmetric tensors

Here is the trap, and it is worth the space. Many treatments define the  $k$ -th symmetric power as a **subspace** of  $V^{\otimes k}$ : the tensors left fixed by every permutation of the slots. That is a different construction from the quotient above, and over a field of characteristic 0 the two agree. Over  $\mathbb{F}_2$  they do not, and the reason is a single division by  $k!$ .

We first need the permutation operators.

### Lemma 14.7.7 — Permuting the Slots of a Tensor

For each  $\sigma \in S_k$  there is a unique linear operator  $P_\sigma$  on  $V^{\otimes k}$  with

$$P_\sigma(\mathbf{u}_1 \otimes \cdots \otimes \mathbf{u}_k) = \mathbf{u}_{\sigma(1)} \otimes \cdots \otimes \mathbf{u}_{\sigma(k)},$$

and  $P_\sigma P_\tau = P_{\tau\sigma}$  for all  $\sigma, \tau \in S_k$ , with  $P_{\text{id}} = \text{id}$ .

*Proof.* The map  $(\mathbf{u}_1, \dots, \mathbf{u}_k) \mapsto \mathbf{u}_{\sigma(1)} \otimes \cdots \otimes \mathbf{u}_{\sigma(k)}$  is  $k$ -linear, since in slot  $t$  it is  $\mu_k$  in slot  $\sigma^{-1}(t)$ . So (T1) gives a unique linear  $P_\sigma$  with the stated values, and uniqueness

holds because those tensors span  $V^{\otimes k}$ . For the composition rule, put  $\mathbf{w}_t = \mathbf{u}_{\tau(t)}$ ; then

$$P_\sigma P_\tau(\mathbf{u}_1 \otimes \cdots \otimes \mathbf{u}_k) = P_\sigma(\mathbf{w}_1 \otimes \cdots \otimes \mathbf{w}_k) = \mathbf{w}_{\sigma(1)} \otimes \cdots \otimes \mathbf{w}_{\sigma(k)},$$

and  $\mathbf{w}_{\sigma(t)} = \mathbf{u}_{\tau(\sigma(t))}$ , which is the  $t$ -th argument of  $P_{\tau\sigma}$ . Both sides are linear and agree on simple tensors, hence are equal. This proves the lemma. ■

### Definition 14.7.5 — Symmetric Tensor

A tensor  $x \in V^{\otimes k}$  is **symmetric** if  $P_\sigma x = x$  for **every**  $\sigma \in S_k$ . The symmetric tensors form a subspace of  $V^{\otimes k}$ , which we write  $(V^{\otimes k})^{S_k}$ ; it is the intersection of the kernels of the linear maps  $P_\sigma - \text{id}$ .

### Theorem 14.7.8 — Symmetric Tensors and the Symmetric Power

Let  $n = \dim V \geq 1$  and  $k \geq 1$ , and let  $\iota: (V^{\otimes k})^{S_k} \rightarrow \text{Sym}^k V$  be the restriction of the quotient map  $q$ .

- (a) Over **every** field,  $\dim(V^{\otimes k})^{S_k} = \binom{n+k-1}{k} = \dim \text{Sym}^k V$ .
- (b) If  $\text{char } F = 0$  or  $\text{char } F > k$ , then  $\iota$  is an isomorphism.
- (c) If  $\text{char } F = 2$ ,  $k = 2$  and  $n \geq 2$ , then  $\iota$  is **neither** injective nor surjective.

*Idea.* For (a), a symmetric tensor must have equal coordinates along each orbit of basis tensors, so the **orbit sums** are a basis; orbits correspond to multisets, which [Theorem 14.7.3](#) already counted. For (b), average:  $s = \sum_\sigma P_\sigma$  maps everything into the symmetric tensors and kills  $C_k$ , so  $\frac{1}{k!}s$  is an inverse for  $\iota$  — provided  $k!$  is invertible, which is exactly the hypothesis. For (c), the tensor  $\mathbf{v}_1 \otimes \mathbf{v}_2 + \mathbf{v}_2 \otimes \mathbf{v}_1$  is symmetric and non-zero, but its image is  $2\mathbf{v}_1\mathbf{v}_2 = 0$ .

*Proof.*

- (a) Let  $\mathcal{B} = (\mathbf{v}_1, \dots, \mathbf{v}_n)$  be a basis of  $V$ . For a tuple  $J$  write  $\mathbf{v}_J = \mathbf{v}_{j_1} \otimes \cdots \otimes \mathbf{v}_{j_k}$ , so that  $(\mathbf{v}_J)_J$  is a basis of  $V^{\otimes k}$  ([Theorem 14.6.3](#)), and note  $P_\sigma \mathbf{v}_J = \mathbf{v}_{J\sigma}$ , where  $J\sigma$  is the tuple  $t \mapsto j_{\sigma(t)}$ . Two tuples lie in the same **orbit** when one is a rearrangement of the other, that is, when they have the same multiplicity vector. For each non-decreasing  $I$  let  $s_I = \sum_{J \in O_I} \mathbf{v}_J$ , the sum over the orbit of  $I$ . Each  $s_I$  is symmetric, because  $J \mapsto J\sigma$  permutes  $O_I$ . The  $s_I$  are independent, because distinct orbits involve disjoint sets of basis vectors. They span: if  $x = \sum_J a_J \mathbf{v}_J$  satisfies  $P_\sigma x = x$  for every  $\sigma$ , then comparing coordinates in  $\sum_J a_J \mathbf{v}_{J\sigma} = \sum_J a_J \mathbf{v}_J$  gives  $a_J = a_{J\sigma}$  for all  $J$  and  $\sigma$ , so  $J \mapsto a_J$  is constant on orbits and  $x = \sum_I a_I s_I$ . Hence  $(s_I)$  is a basis, with one member per orbit, that is, per multiplicity vector of total  $k$ . By the bijection in the proof of [Theorem 14.7.3](#) there are  $\binom{n+k-1}{k}$  of those. No division occurred, so this holds over every field.

- (b) Let  $s = \sum_{\sigma \in S_k} P_\sigma$ , a linear operator on  $V^{\otimes k}$ . By [Lemma 14.7.7](#), for any  $\tau$ ,

$$P_\tau s = \sum_{\sigma} P_\tau P_\sigma = \sum_{\sigma} P_{\sigma\tau} = s, \quad s P_\tau = \sum_{\sigma} P_\sigma P_\tau = \sum_{\sigma} P_{\tau\sigma} = s,$$

since  $\sigma \mapsto \sigma\tau$  and  $\sigma \mapsto \tau\sigma$  are bijections of  $S_k$ . The first identity says  $\text{im } s \subseteq (V^{\otimes k})^{S_k}$ . The second gives  $s(\text{id} - P_\tau) = 0$ , and the generators of  $C_k$  are exactly the elements  $(\text{id} - P_\sigma)(\mathbf{u}_1 \otimes \cdots \otimes \mathbf{u}_k)$ , so  $C_k \subseteq \ker s$ .

By hypothesis  $k! \cdot 1 \neq 0$  in  $F$ . Indeed  $k! \cdot 1 = (1 \cdot 1)(2 \cdot 1) \cdots (k \cdot 1)$  in  $F$ , a product of  $k$  factors; if  $\text{char } F = 0$  none of them is 0, and if  $\text{char } F = p > k$  then  $i \cdot 1 \neq 0$  for  $1 \leq i \leq k < p$  by [Definition 0.7.2](#). A field has no zero divisors, so the product is non-zero, hence invertible. Since  $C_k \subseteq \ker s$ , [Theorem 3.9.10](#) gives a linear map  $\theta: \text{Sym}^k V \rightarrow (V^{\otimes k})^{S_k}$  with  $\theta(q(x)) = \frac{1}{k!} s(x)$ .

Now compute both composites. For  $x \in V^{\otimes k}$ , each  $P_\sigma x$  differs from  $x$  by an element of  $C_k$ , so  $q(s(x)) = \sum_{\sigma} q(P_\sigma x) = k! q(x)$ , whence

$$\iota\theta(q(x)) = \frac{1}{k!} q(s(x)) = q(x),$$

and  $q$  is surjective, so  $\iota\theta = \text{id}$ . Conversely, if  $x$  is symmetric then  $s(x) = \sum_{\sigma} P_\sigma x = k! x$ , so  $\theta\iota(x) = \theta(q(x)) = \frac{1}{k!} k! x = x$ . Hence  $\iota$  is an isomorphism with inverse  $\theta$ .

- (c) Let  $\text{char } F = 2$ ,  $k = 2$ ,  $n \geq 2$ , and put  $t = \mathbf{v}_1 \otimes \mathbf{v}_2 + \mathbf{v}_2 \otimes \mathbf{v}_1$ . It is symmetric, since the only non-identity element of  $S_2$  swaps the two summands, and it is non-zero, being a sum of two distinct basis vectors of  $V^{\otimes 2}$  ([Theorem 14.6.3](#)). But

$$\iota(t) = \mathbf{v}_1 \mathbf{v}_2 + \mathbf{v}_2 \mathbf{v}_1 = 2 \mathbf{v}_1 \mathbf{v}_2 = 0$$

because  $1 + 1 = 0$  in  $F$ . So  $\ker \iota \neq \{0\}$  and  $\iota$  is not injective. By (a) the two spaces have the same finite dimension, so a surjective  $\iota$  would be injective by [Theorem 3.3.1](#); hence  $\iota$  is not surjective either. This proves the theorem. ■

Part (b) is where the division lives, and part (c) is what it costs. The operator  $s$  still exists over  $\mathbb{F}_2$ ; what fails is the *rescaling*  $\frac{1}{k!} s$ , and with it the splitting of the quotient.

**Warning.** In characteristic  $p \leq k$ , “symmetric power” and “symmetric tensors” are different objects. They have the same dimension by [Theorem 14.7.8](#) (a), so they are abstractly isomorphic, but the *natural* map between them is not an isomorphism, and no natural one is available. So: do not define  $\text{Sym}^k V$  as  $(V^{\otimes k})^{S_k}$ , do not write  $\frac{1}{k!} \sum_{\sigma} P_\sigma$  without saying that  $k!$  is invertible, and do not transport a statement proved for one to the other. This is Chapter 13’s discipline again, where alternating and skew-symmetric forms coincided only away from characteristic 2 ([Theorem 13.3.3](#)): the same clause, the same division, and the same smallest witness,  $\mathbb{F}_2^2$ .

Everything else in this section is characteristic-free. In particular [Theorem 14.7.3](#), [Theorem 14.7.6](#) and the dimension  $\binom{n+k-1}{k}$  hold over every field, because the quotient construction never divides. The next section takes the same care with the same reward: the exterior power is built from a relation chosen so that no division is ever needed.

## 14.7.6 Exercises

### 14.7.6.1 A. Check your understanding

#### Exercise A1.

- State the universal property (S1), with every quantifier, and say which maps are required to factor.
- Which subspace of  $V^{\otimes k}$  is divided out to build  $\text{Sym}^k V$ ? Write down its generators for  $k = 3$ .
- Let  $\dim V = 4$ . Compute  $\dim \text{Sym}^3 V$  and  $\dim V^{\otimes 3}$ .
- Determine whether  $\text{Sym}^2 V$  is a subspace of  $V^{\otimes 2}$ . Justify your answer.
- What hypothesis on  $F$  makes the symmetric tensors of degree  $k$  naturally isomorphic to  $\text{Sym}^k V$ , and which step of the proof needs it?

*Solution.*

- For every vector space  $W$  over  $F$  and every **symmetric**  $k$ -linear map  $f: V^k \rightarrow W$ , there exists exactly one linear  $\bar{f}: \text{Sym}^k V \rightarrow W$  with  $\bar{f} \circ \pi = f$ . Only the symmetric maps factor; a non-symmetric one cannot, since  $\bar{f} \circ \pi$  is symmetric for every linear  $\bar{f}$ .
- The subspace  $C_k$  of [Theorem 14.7.2](#), spanned by the differences  $\mathbf{u}_1 \otimes \cdots \otimes \mathbf{u}_k - \mathbf{u}_{\sigma(1)} \otimes \cdots \otimes \mathbf{u}_{\sigma(k)}$ . For  $k = 3$  the generators are the six differences obtained from the six  $\sigma \in S_3$ , for all  $\mathbf{u}_1, \mathbf{u}_2, \mathbf{u}_3 \in V$ .
- $\dim \text{Sym}^3 V = \binom{4+3-1}{3} = \binom{6}{3} = 20$  and  $\dim V^{\otimes 3} = 4^3 = 64$ .
- No. It is a quotient of  $V^{\otimes 2}$ , and its elements are cosets of  $C_2$ , not tensors. (Over a field of characteristic 0 there is a subspace of  $V^{\otimes 2}$  naturally isomorphic to it, namely the symmetric tensors, but the isomorphism is not an equality, and it does not exist in characteristic 2.)
- $\text{char } F = 0$  or  $\text{char } F > k$ , so that  $k!$  is invertible in  $F$  ([Theorem 14.7.8](#) (b)). The step that needs it is the construction of the inverse map  $\theta$ , which divides the symmetrization  $s = \sum_{\sigma} P_{\sigma}$  by  $k!$ .

### 14.7.6.2 B. Practice

#### Exercise B1: Expanding symmetric products.

Let  $V = F^2$ .

- Expand  $(2\mathbf{e}_1 - \mathbf{e}_2)(\mathbf{e}_1 + 3\mathbf{e}_2)$  in the basis  $(\mathbf{e}_1^2, \mathbf{e}_1\mathbf{e}_2, \mathbf{e}_2^2)$  of  $\text{Sym}^2 V$ .
- Expand  $(\mathbf{e}_1 + \mathbf{e}_2)(\mathbf{e}_1 - \mathbf{e}_2)\mathbf{e}_1$  in the basis of  $\text{Sym}^3 V$  given by [Theorem 14.7.3](#).

*Solution.*

- Expanding one slot at a time and then using  $\mathbf{e}_2\mathbf{e}_1 = \mathbf{e}_1\mathbf{e}_2$ ,

$$(2\mathbf{e}_1 - \mathbf{e}_2)(\mathbf{e}_1 + 3\mathbf{e}_2) = 2\mathbf{e}_1^2 + 6\mathbf{e}_1\mathbf{e}_2 - \mathbf{e}_1\mathbf{e}_2 - 3\mathbf{e}_2^2 = 2\mathbf{e}_1^2 + 5\mathbf{e}_1\mathbf{e}_2 - 3\mathbf{e}_2^2.$$

- (b) The basis is  $(\mathbf{e}_1^3, \mathbf{e}_1^2\mathbf{e}_2, \mathbf{e}_1\mathbf{e}_2^2, \mathbf{e}_2^3)$ , of size  $\binom{4}{3} = 4$ . Expanding the first two slots,

$$(\mathbf{e}_1 + \mathbf{e}_2)(\mathbf{e}_1 - \mathbf{e}_2)\mathbf{e}_1 = \mathbf{e}_1^2\mathbf{e}_1 - \mathbf{e}_1\mathbf{e}_2\mathbf{e}_1 + \mathbf{e}_2\mathbf{e}_1\mathbf{e}_1 - \mathbf{e}_2^2\mathbf{e}_1,$$

and the two middle terms are equal after sorting, so they cancel:

$$(\mathbf{e}_1 + \mathbf{e}_2)(\mathbf{e}_1 - \mathbf{e}_2)\mathbf{e}_1 = \mathbf{e}_1^3 - \mathbf{e}_1\mathbf{e}_2^2.$$

In the polynomial language of [Theorem 14.7.6](#) this is  $(x + y)(x - y)x = x^3 - xy^2$ .

### Exercise B2: Counting.

- Compute  $\dim \operatorname{Sym}^k V$  for  $\dim V = 3$  and  $k = 0, 1, 2, 3, 4$ .
- For  $\dim V = n$ , compute  $\dim \operatorname{Sym}^2 V$  in closed form and check it against the number of entries on or above the diagonal of an  $n \times n$  matrix.
- Determine all pairs  $(n, k)$  with  $k \geq 1$  for which  $\dim \operatorname{Sym}^k V = \dim V^{\otimes k}$ . Justify your answer.

*Solution.*

- $\binom{k+2}{k}$  for  $n = 3$ , giving 1, 3, 6, 10, 15 for  $k = 0, 1, 2, 3, 4$ .
- $\binom{n+1}{2} = n(n+1)/2$ . An  $n \times n$  matrix has  $n$  diagonal entries and  $(n^2 - n)/2$  entries strictly above the diagonal, and  $n + (n^2 - n)/2 = n(n+1)/2$ . The agreement is the remark after [Theorem 14.7.6](#): a symmetric bilinear form is the same data as a functional on  $\operatorname{Sym}^2 V$ , and also the same data as a symmetric matrix.
- Exactly those with  $n = 1$ , or with  $k = 1$ . If  $k = 1$  both sides are  $n$ . If  $n = 1$  both sides are 1. Otherwise  $n \geq 2$  and  $k \geq 2$ , and then  $\mathbf{v}_1 \otimes \mathbf{v}_2 \neq \mathbf{v}_2 \otimes \mathbf{v}_1$  are two basis tensors with the same image, so  $C_k \neq \{0\}$  and  $\dim \operatorname{Sym}^k V = n^k - \dim C_k < n^k$  by [Theorem 3.9.5](#).

### Exercise B3: A form factored through the symmetric square.

Let  $\beta$  be a symmetric bilinear form on  $V$ . Prove that there is a unique linear functional  $\bar{\beta} \in (\operatorname{Sym}^2 V)^*$  with  $\bar{\beta}(\mathbf{u}\mathbf{w}) = \beta(\mathbf{u}, \mathbf{w})$  for all  $\mathbf{u}, \mathbf{w} \in V$ , and compute  $\bar{\beta}$  on the basis of [Theorem 14.7.3](#) when  $V = F^2$  and  $\beta(\mathbf{x}, \mathbf{y}) = x_1y_1 - x_2y_2$ .

*Solution.*  $\beta$  is a symmetric 2-linear map  $V^2 \rightarrow F$  ([Definition 14.7.1](#) with  $k = 2$ ,  $W = F$ ), so (S1) gives exactly one linear  $\bar{\beta}: \operatorname{Sym}^2 V \rightarrow F$  with  $\bar{\beta} \circ \pi = \beta$ , which is the displayed identity.

For the given  $\beta$ :  $\bar{\beta}(\mathbf{e}_1^2) = \beta(\mathbf{e}_1, \mathbf{e}_1) = 1$ ,  $\bar{\beta}(\mathbf{e}_1\mathbf{e}_2) = \beta(\mathbf{e}_1, \mathbf{e}_2) = 0$  and  $\bar{\beta}(\mathbf{e}_2^2) = \beta(\mathbf{e}_2, \mathbf{e}_2) = -1$ . So in the coordinates given by the basis  $(\mathbf{e}_1^2, \mathbf{e}_1\mathbf{e}_2, \mathbf{e}_2^2)$  the functional reads  $(1, 0, -1)$ , and under [Theorem 14.7.6](#) it is the rule “read off the coefficients of  $x^2$  and  $y^2$  and subtract”.

#### 14.7.6.3 C. Going deeper

**Exercise C1: The characteristic-two witness in full.**

Let  $F = \mathbb{F}_2$  and  $V = \mathbb{F}_2^2$ , and take  $k = 2$ .

- (a) Write down a basis of  $(V^{\otimes 2})^{S_2}$  and verify that its dimension is 3.
- (b) Compute  $\dim C_2$  and  $\dim \operatorname{Sym}^2 V$ , and identify  $\ker \iota$  and  $\operatorname{im} \iota$  for the map  $\iota$  of [Theorem 14.7.8](#).
- (c) Are  $(V^{\otimes 2})^{S_2}$  and  $\operatorname{Sym}^2 V$  isomorphic as vector spaces? Explain why this does not repair the situation.

*Solution.*

- (a) Here  $S_2 = \{\operatorname{id}, \sigma\}$  with  $\sigma$  the swap, and  $P_\sigma$  exchanges  $\mathbf{e}_1 \otimes \mathbf{e}_2$  with  $\mathbf{e}_2 \otimes \mathbf{e}_1$  and fixes  $\mathbf{e}_1 \otimes \mathbf{e}_1$  and  $\mathbf{e}_2 \otimes \mathbf{e}_2$ . So  $x = a\mathbf{e}_1 \otimes \mathbf{e}_1 + b\mathbf{e}_1 \otimes \mathbf{e}_2 + c\mathbf{e}_2 \otimes \mathbf{e}_1 + d\mathbf{e}_2 \otimes \mathbf{e}_2$  is symmetric if and only if  $b = c$ . A basis is

$$\mathbf{e}_1 \otimes \mathbf{e}_1, \quad \mathbf{e}_2 \otimes \mathbf{e}_2, \quad \mathbf{e}_1 \otimes \mathbf{e}_2 + \mathbf{e}_2 \otimes \mathbf{e}_1,$$

the orbit sums of [Theorem 14.7.8](#) (a), so the dimension is  $3 = \binom{3}{2}$ .

- (b)  $C_2$  is spanned by the elements  $\mathbf{u} \otimes \mathbf{w} - \mathbf{w} \otimes \mathbf{u}$ . Writing  $\mathbf{u} = a\mathbf{e}_1 + b\mathbf{e}_2$  and  $\mathbf{w} = c\mathbf{e}_1 + d\mathbf{e}_2$  and expanding both slots, the  $\mathbf{e}_1 \otimes \mathbf{e}_1$  and  $\mathbf{e}_2 \otimes \mathbf{e}_2$  terms cancel and

$$\mathbf{u} \otimes \mathbf{w} - \mathbf{w} \otimes \mathbf{u} = (ad - bc)(\mathbf{e}_1 \otimes \mathbf{e}_2 - \mathbf{e}_2 \otimes \mathbf{e}_1).$$

So  $C_2$  is the line spanned by  $\mathbf{e}_1 \otimes \mathbf{e}_2 + \mathbf{e}_2 \otimes \mathbf{e}_1$ , the sign being irrelevant over  $\mathbb{F}_2$ , and  $\dim C_2 = 1$  and  $\dim \operatorname{Sym}^2 V = 4 - 1 = 3$  by [Theorem 3.9.5](#). Now  $\ker \iota = (V^{\otimes 2})^{S_2} \cap C_2 = C_2$ , of dimension 1, and  $\operatorname{im} \iota$  is therefore 2-dimensional, spanned by  $\mathbf{e}_1^2$  and  $\mathbf{e}_2^2$ ; it misses  $\mathbf{e}_1\mathbf{e}_2$ .

- (c) Yes, they are isomorphic: both have dimension 3 over  $\mathbb{F}_2$ , and [Theorem 3.5.4](#) applies. It does not repair anything, because the isomorphism is not  $\iota$  and does not respect the structure maps: an isomorphism chosen by hand would not satisfy  $\varphi(\mathbf{u} \otimes \mathbf{w}) = \mathbf{u}\mathbf{w}$ , so it could not be used to transport statements about products of vectors. A construction is only as good as its universal property, and  $(V^{\otimes 2})^{S_2}$  has none here.

**Exercise C2:  $\operatorname{Sym}(V)$  is the free commutative algebra.**

- (a) Prove that the products  $\mathbf{u}_1 \cdots \mathbf{u}_k$  with  $\mathbf{u}_i \in V$  span  $\operatorname{Sym}^k V$ , and deduce that an algebra homomorphism out of  $\operatorname{Sym}(V)$  is determined by its restriction to  $V$ .
- (b) Prove that for every **commutative** associative unital  $F$ -algebra  $A$  and every linear map  $f: V \rightarrow A$  there is a unique algebra homomorphism  $\bar{f}: \operatorname{Sym}(V) \rightarrow A$  with  $\bar{f}|_V = f$ .
- (c) Explain in one sentence why (b) fails if the word “commutative” is deleted, and name the algebra that has the corresponding property for all  $A$ .

*Solution.*

- (a)  $V^{\otimes k}$  is spanned by its simple tensors ([Theorem 14.6.2](#)) and  $q$  is surjective and linear, so their images, the products  $\mathbf{u}_1 \cdots \mathbf{u}_k$ , span  $\operatorname{Sym}^k V$ . If  $\varphi, \psi$  are algebra homomorphisms agreeing on  $V$ , then  $\varphi(\mathbf{u}_1 \cdots \mathbf{u}_k) = \varphi(\mathbf{u}_1) \cdots \varphi(\mathbf{u}_k) = \psi(\mathbf{u}_1 \cdots \mathbf{u}_k)$ , and



both send 1 to  $1_A$ ; since these elements span  $\text{Sym}(V)$  and both maps are linear,  $\varphi = \psi$ .

- (b) For  $k \geq 1$  the map  $(\mathbf{u}_1, \dots, \mathbf{u}_k) \mapsto f(\mathbf{u}_1) \cdots f(\mathbf{u}_k)$  is  $k$ -linear, as in the proof of [Theorem 14.6.6](#), and it is **symmetric** because  $A$  is commutative (the third example after [Definition 14.7.1](#)). By (S1) it induces a linear  $\bar{f}_k: \text{Sym}^k V \rightarrow A$ ; set  $\bar{f}_0(a) = a1_A$  and  $\bar{f} = \sum_k \bar{f}_k$  on the direct sum. Multiplicativity is checked on products of vectors, which span, exactly as in [Theorem 14.6.6](#), and  $\bar{f}|_V = f$ . Uniqueness is (a).
- (c) It fails because  $\bar{f}$  would have to satisfy  $f(\mathbf{u})f(\mathbf{w}) = \bar{f}(\mathbf{uw}) = \bar{f}(\mathbf{wu}) = f(\mathbf{w})f(\mathbf{u})$ , so the images of  $V$  would be forced to commute, which need not happen in a non-commutative  $A$ . The algebra with the property for **all**  $A$  is the tensor algebra  $T(V)$ , by [Theorem 14.6.6](#).

### Exercise C3: One variable.

Let  $\dim V = 1$ .

- (a) Prove that  $C_k = \{\mathbf{0}\}$  for every  $k$ , and deduce that the quotient map  $T(V) \rightarrow \text{Sym}(V)$  assembled from the  $q$ 's is an isomorphism of algebras.
- (b) Hence deduce that  $T(V) \cong \text{Sym}(V) \cong F[x]$ , and reconcile this with the fact that  $T(V)$  is non-commutative for  $\dim V \geq 2$ .

*Solution.*

- (a) Write  $V = F\mathbf{v}$ . By [Theorem 14.6.3](#),  $V^{\otimes k}$  is spanned by  $\mathbf{v}^{\otimes k}$ . A generator of  $C_k$  is  $\mathbf{u}_1 \otimes \cdots \otimes \mathbf{u}_k - \mathbf{u}_{\sigma(1)} \otimes \cdots \otimes \mathbf{u}_{\sigma(k)}$ ; writing  $\mathbf{u}_t = c_t \mathbf{v}$  and expanding, both terms equal  $c_1 c_2 \cdots c_k \mathbf{v}^{\otimes k}$ , since the scalars multiply in the same way whatever the order. So every generator is  $\mathbf{0}$  and  $C_k = \{\mathbf{0}\}$ . Hence each  $q: V^{\otimes k} \rightarrow \text{Sym}^k V$  is an isomorphism ([Theorem 3.9.5](#), or directly: its kernel is  $C_k$ ), and the assembled map is a bijective algebra homomorphism, so an isomorphism.
- (b) By [Example \(The tensor algebra of a line\)](#),  $T(V) \cong F[x]$ , so (a) gives  $\text{Sym}(V) \cong F[x]$  as well; the same conclusion comes from [Theorem 14.7.6](#) with  $n = 1$ . There is no tension with non-commutativity:  $T(V)$  is commutative precisely when  $\dim V \leq 1$ , because non-commutativity needs two independent vectors to put in the two slots (the warning after [Theorem 14.6.5](#)). With one variable there is nothing to reorder, so imposing the symmetry relations imposes nothing.

## 14.8 Exterior Powers

The symmetric power made the slots of a tensor interchangeable. This section makes them *hostile*: we divide  $V^{\otimes k}$  by a subspace chosen so that a tensor dies as soon as two of its slots hold the same vector. The result, the exterior power  $\Lambda^k V$ , has a basis indexed by **strictly increasing** index tuples, so  $\dim \Lambda^k V = \binom{n}{k}$  and  $\Lambda^k V = \{\mathbf{0}\}$  once  $k$  exceeds  $n$ . That collapse is the whole reason the determinant exists, and Section 09 will read the determinant off the top exterior power.

One choice in the construction is delicate, and it is the choice Chapter 13 prepared us for. There

are two natural ways to say “the value flips when two slots are exchanged”, and they are not the same condition: [Theorem 13.3.3](#) showed that *alternating* implies *skew-symmetric* over every field, while the converse needs  $\text{char } F \neq 2$ . We must therefore decide which relation to divide by, and the answer is the one that works over every field.

This is Section 02’s method for the third time: universal property, uniqueness, then a model built as a quotient of the tensor power. Throughout,  $F$  is a field,  $V$  is a finite-dimensional vector space over  $F$  with  $n = \dim V$ , and  $k \geq 1$  unless stated otherwise. **No hypothesis on the characteristic is made anywhere in this section**, and the construction is arranged so that none is needed.

## 14.8.1 Alternating multilinear maps

Chapter 6 defined alternating  $k$ -linear **forms**, valued in  $F$  ([Definition 6.2.2](#)), because that is all the determinant needed. The target plays no role in the definition, so we extend it, as Section 01 extended multilinearity.

*An alternating map dies the moment two of its arguments coincide.*

### Definition 14.8.1 — Alternating Multilinear Map

Let  $W$  be a vector space over  $F$ . A  $k$ -linear map  $f: V^k \rightarrow W$  ([Definition 14.1.1](#)) is **alternating** if

$$f(\mathbf{u}_1, \dots, \mathbf{u}_k) = \mathbf{0} \quad \text{whenever } \mathbf{u}_i = \mathbf{u}_j \text{ for some pair of positions } i \neq j.$$

The condition is on **some** pair of positions, adjacent or not, and it says nothing about inputs whose entries are all distinct. For  $W = F$  this is [Definition 6.2.2](#) verbatim. The examples of Chapter 6 are all available: the determinant  $\det(\mathbf{a}_1, \dots, \mathbf{a}_n)$  as a function of the columns; the signed area  $(\mathbf{u}, \mathbf{w}) \mapsto u_1 w_2 - u_2 w_1$  on  $F^2$ ; any linear functional, vacuously, at  $k = 1$ . A non-example by minimal change: the dot product on  $F^2$  is bilinear but  $\mathbf{e}_1 \cdot \mathbf{e}_1 = 1 \neq 0$ , so the alternating clause fails at that one input while bilinearity survives.

Everything Chapter 6 deduced from the definition used only the vector-space structure of the target, so it carries over.

### Lemma 14.8.1 — Properties of Alternating Maps

Let  $f: V^k \rightarrow W$  be alternating  $k$ -linear, let  $\mathbf{u}_1, \dots, \mathbf{u}_k \in V$ , and let  $\sigma \in S_k$ .

- (a) Exchanging two arguments multiplies the value by  $-1$ ; in particular  $f$  is skew-symmetric.
- (b)  $f(\mathbf{u}_{\sigma(1)}, \dots, \mathbf{u}_{\sigma(k)}) = \text{sgn}(\sigma) f(\mathbf{u}_1, \dots, \mathbf{u}_k)$ .
- (c) If  $(\mathbf{u}_1, \dots, \mathbf{u}_k)$  is linearly dependent, then  $f(\mathbf{u}_1, \dots, \mathbf{u}_k) = \mathbf{0}$ .

*Proof.*

- (a) Fix positions  $i < j$ , freeze the other arguments, and let  $g(\mathbf{x}, \mathbf{y}) \in W$  be the value of  $f$  with  $\mathbf{x}$  in position  $i$  and  $\mathbf{y}$  in position  $j$ . Then  $g$  is bilinear and  $g(\mathbf{x}, \mathbf{x}) = \mathbf{0}$ ,

so expanding

$$\mathbf{0} = g(\mathbf{u}_i + \mathbf{u}_j, \mathbf{u}_i + \mathbf{u}_j) = g(\mathbf{u}_i, \mathbf{u}_j) + g(\mathbf{u}_j, \mathbf{u}_i)$$

gives the swap rule. This is the computation of [Theorem 6.2.1](#) (a), which used nothing about  $F$  as a target beyond its being a vector space.

- (b) For  $k = 1$  the only permutation is the identity and there is nothing to prove. For  $k \geq 2$ , use [Theorem 0.10.4](#) to write  $\sigma = \tau_1 \cdots \tau_r$  as a product of transpositions. Applying (a) once per transposition multiplies the value by  $(-1)^r$ , and  $(-1)^r = \text{sgn}(\sigma)$  by [Theorem 6.3.1](#) and [Corollary 6.3.2](#). This is [Lemma 6.4.1](#), with the same proof.
- (c) By the Linear Dependence Lemma ([Theorem 1.4.1](#)) some  $\mathbf{u}_j$  lies in  $\text{span}(\mathbf{u}_1, \dots, \mathbf{u}_{j-1})$ . Expanding  $f$  in slot  $j$  writes the value as a combination of values in which some  $\mathbf{u}_m$  with  $m < j$  sits both in slot  $m$  and in slot  $j$ , and each such value is  $\mathbf{0}$ . This is [Theorem 6.2.1](#) (d). If  $j = 1$  then  $\mathbf{u}_1 = \mathbf{0}$  and linearity in slot 1 gives  $\mathbf{0}$  at once. ■

## 14.8.2 Which relation to divide by

Now the delicate point, stated before the definition so that the definition looks inevitable. We want a quotient of  $V^{\otimes k}$  in which the image of  $\mathbf{u}_1 \otimes \cdots \otimes \mathbf{u}_k$  vanishes exactly when it should. There are two candidate subspaces to divide by:

$$A_k = \text{span}\{\mathbf{u}_1 \otimes \cdots \otimes \mathbf{u}_k : \mathbf{u}_i = \mathbf{u}_j \text{ for some } i \neq j\},$$

$$N_k = \text{span}\{x + P_\tau x : x \text{ a simple tensor, } \tau \text{ a transposition}\},$$

where  $P_\tau$  permutes the slots as in [Lemma 14.7.7](#). Dividing by  $A_k$  imposes “repeat a vector and you get zero”; dividing by  $N_k$  imposes “swap two slots and the sign flips”. The two are not the same subspace.

### Proposition 14.8.2 — The Repetition Relations Are the Stronger Ones

For every field  $F$  and every  $k \geq 2$  we have  $N_k \subseteq A_k$ . If  $\text{char } F \neq 2$ , then  $N_k = A_k$ . If  $\text{char } F = 2$ ,  $k = 2$  and  $n \geq 1$ , the inclusion is **strict**.

*Idea.* For the inclusion, feed the sum  $\mathbf{u}_i + \mathbf{u}_j$  into **both** slots  $i$  and  $j$ : the resulting tensor is in  $A_k$ , and expanding it produces the two “square” terms, also in  $A_k$ , plus the two cross terms, whose sum is therefore in  $A_k$  too. That is [Theorem 13.3.3](#) (a) transcribed into tensors. For the reverse, a tensor with a repeat is its own swap, so  $x + P_\tau x = 2x$ , and recovering  $x$  is a division by 2 — the same division, in the same place, as in [Theorem 13.3.3](#) (b).

*Proof.* ( $\subseteq$ ) Let  $x = \mathbf{u}_1 \otimes \cdots \otimes \mathbf{u}_k$  be simple and  $\tau$  the transposition of positions  $i < j$ . Let  $y$  be the tensor obtained from  $x$  by putting  $\mathbf{u}_i + \mathbf{u}_j$  in **both** positions  $i$  and  $j$ , which lies in  $A_k$  because those two slots now agree. Expanding  $y$  by multilinearity in those two slots gives four terms:

$$y = [\mathbf{u}_i, \mathbf{u}_i] + [\mathbf{u}_i, \mathbf{u}_j] + [\mathbf{u}_j, \mathbf{u}_i] + [\mathbf{u}_j, \mathbf{u}_j],$$

where  $[\mathbf{p}, \mathbf{q}]$  denotes  $x$  with  $\mathbf{p}$  in position  $i$  and  $\mathbf{q}$  in position  $j$ . The first and last terms lie in  $A_k$ , again by a repeat, and  $[\mathbf{u}_i, \mathbf{u}_j] = x$  while  $[\mathbf{u}_j, \mathbf{u}_i] = P_\tau x$ . Hence  $x + P_\tau x = y - [\mathbf{u}_i, \mathbf{u}_i] - [\mathbf{u}_j, \mathbf{u}_j] \in A_k$ . Generators of  $N_k$  lie in  $A_k$ , so  $N_k \subseteq A_k$ .

( $\supseteq$ ), assuming  $\text{char } F \neq 2$ . Let  $x = \mathbf{u}_1 \otimes \cdots \otimes \mathbf{u}_k$  with  $\mathbf{u}_i = \mathbf{u}_j$ ,  $i \neq j$ , and let  $\tau$  swap  $i$  and  $j$ . Then  $P_\tau x = x$ , so  $x + P_\tau x = (1 + 1)x$  lies in  $N_k$ . Since  $\text{char } F \neq 2$ , the scalar  $1 + 1$  is invertible (Definition 0.7.2), so  $x \in N_k$ . Generators of  $A_k$  lie in  $N_k$ , so  $A_k \subseteq N_k$ .

*Strictness in characteristic 2,  $k = 2$ .* Suppose  $\text{char } F = 2$  and  $n \geq 1$ , and fix a basis  $(\mathbf{v}_1, \dots, \mathbf{v}_n)$ . Expand a generator of  $N_2$  in the basis of Theorem 14.6.3: writing  $\mathbf{u} = \sum_i a_i \mathbf{v}_i$  and  $\mathbf{w} = \sum_j b_j \mathbf{v}_j$ ,

$$\mathbf{u} \otimes \mathbf{w} + \mathbf{w} \otimes \mathbf{u} = \sum_{i,j} (a_i b_j + a_j b_i) \mathbf{v}_i \otimes \mathbf{v}_j,$$

and the coefficient of a **diagonal** basis tensor  $\mathbf{v}_i \otimes \mathbf{v}_i$  is  $2a_i b_i = 0$ . Every element of  $N_2$  therefore has coefficient 0 on  $\mathbf{v}_1 \otimes \mathbf{v}_1$ . But  $\mathbf{v}_1 \otimes \mathbf{v}_1$  is itself a generator of  $A_2$ , and it is a basis vector of  $V^{\otimes 2}$ , hence non-zero. So  $\mathbf{v}_1 \otimes \mathbf{v}_1 \in A_2 \setminus N_2$  and the inclusion is strict. This proves the proposition. ■

So the two relations differ, and we must choose. **We divide by  $A_k$ .** Three reasons, and they are the same three that made Chapter 6 build the determinant on alternation rather than skewness:

- The quotient by  $A_k$  is the object whose maps are the **alternating** ones, and alternating is the condition the determinant satisfies over every field.
- $A_k$  is the larger subspace, so the quotient by it is the smaller space; and by Proposition 14.8.2 the two quotients agree whenever  $\text{char } F \neq 2$ . So nothing is lost outside characteristic 2, and something is gained inside it.
- The dimension  $\binom{n}{k}$  comes out right over every field. Dividing by  $N_2$  instead would give, for  $V = \mathbb{F}_2^2$ , a space of dimension 3 where  $\binom{2}{2} = 1$  is wanted; Exercise C1 carries out that count.

### 14.8.3 The exterior power

*The  $k$ -th exterior power is the space in which alternating  $k$ -linear maps become linear.*

#### Definition 14.8.2 — $k$ -th Exterior Power

A  **$k$ -th exterior power of  $V$**  is a pair  $(P, \wedge)$ , where  $P$  is a vector space over  $F$  and  $\wedge: V^k \rightarrow P$  is an **alternating**  $k$ -linear map, such that

(E1) (**universal property**) for **every** vector space  $W$  over  $F$  and **every alternating**  $k$ -linear map  $f: V^k \rightarrow W$ , there is a **unique** linear map  $\tilde{f}: P \rightarrow W$  with  $\tilde{f} \circ \wedge = f$ .

We write  $\Lambda^k V$  for  $P$  and

$$\mathbf{v}_1 \wedge \mathbf{v}_2 \wedge \cdots \wedge \mathbf{v}_k := \wedge(\mathbf{v}_1, \dots, \mathbf{v}_k),$$

calling it a **wedge** of the list. By convention  $\Lambda^0 V = F$ .

Clause by clause, and the small words matter. The structure map  $\wedge$  must itself be **alternating**, which is what makes  $\mathbf{v} \wedge \mathbf{v} = \mathbf{0}$  a fact rather than a wish. Only **alternating** maps are required to factor, and that is forced:  $\tilde{f} \circ \wedge$  is alternating for every linear  $\tilde{f}$ , so no space could linearize a non-alternating map. **Unique** does the usual two jobs: existence of  $\tilde{f}$  says  $\Lambda^k V$  is large enough, uniqueness says it is no larger.

At  $k = 1$  the condition is vacuous and  $(V, \text{id}_V)$  works, so  $\Lambda^1 V = V$ . At  $k = 0$  the convention  $\Lambda^0 V = F$  is the one that makes  $\binom{n}{k}$  correct at  $k = 0$ , where it reads  $\binom{n}{0} = 1$ .

### Theorem 14.8.3 — Uniqueness of the Exterior Power

Let  $(P, \wedge)$  and  $(P', \wedge')$  both be  $k$ -th exterior powers of  $V$ . Then there is a unique linear isomorphism  $\varphi: P \rightarrow P'$  with  $\varphi \circ \wedge = \wedge'$ .

*Proof.* The proof of Theorem 14.6.1, with “alternating  $k$ -linear” in place of “ $k$ -linear”: (E1) for  $P$  applied to  $\wedge'$  gives  $\varphi$ , (E1) for  $P'$  applied to  $\wedge$  gives  $\psi$ , and  $\psi\varphi$  is a linear self-map of  $P$  with  $\psi\varphi\wedge = \wedge$ , so the uniqueness clause of (E1) applied to  $\wedge$  forces  $\psi\varphi = \text{id}_P$ ; symmetrically  $\varphi\psi = \text{id}_{P'}$ . This proves the theorem. ■

### Theorem 14.8.4 — Existence of the Exterior Power

Let  $A_k \subseteq V^{\otimes k}$  be as above and let  $q: V^{\otimes k} \rightarrow V^{\otimes k}/A_k$  be the quotient map (Definition 3.9.3). Then  $V^{\otimes k}/A_k$ , together with

$$\mathbf{v}_1 \wedge \cdots \wedge \mathbf{v}_k = q(\mathbf{v}_1 \otimes \cdots \otimes \mathbf{v}_k),$$

is a  $k$ -th exterior power of  $V$ .

*Proof.*  $A_k$  is a subspace, being a span, and  $\wedge$  is  $k$ -linear, being  $\mu_k$  followed by the linear map  $q$ . It is alternating: if  $\mathbf{u}_i = \mathbf{u}_j$  with  $i \neq j$ , then  $\mathbf{u}_1 \otimes \cdots \otimes \mathbf{u}_k$  is by definition a generator of  $A_k$ , so its image under  $q$  is  $\mathbf{0}$ .

Let  $f: V^k \rightarrow W$  be alternating  $k$ -linear. By (T1) there is a unique linear  $\tilde{f}: V^{\otimes k} \rightarrow W$  with  $\tilde{f}(\mathbf{u}_1 \otimes \cdots \otimes \mathbf{u}_k) = f(\mathbf{u}_1, \dots, \mathbf{u}_k)$ . On a generator of  $A_k$ , that is, on a simple tensor with  $\mathbf{u}_i = \mathbf{u}_j$  for some  $i \neq j$ , the value is  $\tilde{f}(\mathbf{u}_1 \otimes \cdots \otimes \mathbf{u}_k) = \mathbf{0}$  because  $f$  is alternating. A linear map vanishing on a spanning set of  $A_k$  vanishes on  $A_k$ , so  $A_k \subseteq \ker \tilde{f}$ , and Theorem 3.9.10 (a) gives exactly one linear  $\bar{f}$  on the quotient with  $\bar{f} \circ q = \tilde{f}$ , whence  $\bar{f} \circ \wedge = f$ .

Uniqueness:  $q$  is surjective and  $V^{\otimes k}$  is spanned by its simple tensors (Theorem 14.6.2), so the wedges  $\mathbf{u}_1 \wedge \cdots \wedge \mathbf{u}_k$  span  $V^{\otimes k}/A_k$ , and two linear maps agreeing on them agree everywhere. This proves the theorem. ■

From Lemma 14.8.1 applied to the alternating map  $\wedge$  itself we get the rules used in every

computation below, for all  $\mathbf{u}_i \in V$  and  $\sigma \in S_k$ :

$$\begin{aligned}\mathbf{u}_{\sigma(1)} \wedge \cdots \wedge \mathbf{u}_{\sigma(k)} &= \operatorname{sgn}(\sigma) \mathbf{u}_1 \wedge \cdots \wedge \mathbf{u}_k, \\ \mathbf{u}_1 \wedge \cdots \wedge \mathbf{u}_k &= \mathbf{0} \quad \text{if the list is dependent.}\end{aligned}$$

Four instances, smallest first, and one look-alike that is not one.

- **One slot.**  $\Lambda^1 V = V$ , and the wedge of a single vector is the vector itself. The alternating condition asks about pairs of positions, and with one position there are none.
- **The plane.** For  $V = F^2$ , expanding both slots and canceling the two repeats gives  $\mathbf{u} \wedge \mathbf{w} = (u_1 w_2 - u_2 w_1) \mathbf{e}_1 \wedge \mathbf{e}_2$ . The coefficient is the signed area of Chapter 6 §01; the wedge is that number, moved out of  $F$  and into a line inside  $\Lambda^2 V$ .
- **Polynomials.** On  $V = F[x]_{\leq 2}$  with basis  $(1, x, x^2)$ , the same expansion gives  $(1+x) \wedge (x+x^2) = 1 \wedge x + 1 \wedge x^2 + x \wedge x^2$ , the term  $x \wedge x$  having died.
- **A degenerate case.** If  $V = \{\mathbf{0}\}$  then every wedge of  $k \geq 1$  vectors is  $\mathbf{0}$ , so  $\Lambda^k V = \{\mathbf{0}\}$ . It matters because it is the base of the count below:  $\binom{0}{k} = 0$  for  $k \geq 1$ .
- **Not an example: the symmetric product.** In  $\operatorname{Sym}^2 V$  the product  $\mathbf{u}\mathbf{w}$  is bilinear and satisfies  $\mathbf{u}\mathbf{w} = \mathbf{w}\mathbf{u}$ , but  $\mathbf{e}_1 \mathbf{e}_1 \neq 0$  by Theorem 14.7.3, so the symmetric product map is **not** alternating. Exactly one clause differs between the two constructions, and it is this one.

*Warning.* The relation is  $\mathbf{v} \wedge \mathbf{v} = \mathbf{0}$ , not  $\mathbf{u} \wedge \mathbf{w} = -\mathbf{w} \wedge \mathbf{u}$ . The second follows from the first (Lemma 14.8.1 (a)), but not conversely: over  $\mathbb{F}_2$  the two say different things, since  $-1 = 1$  there makes “ $\mathbf{u} \wedge \mathbf{w} = -\mathbf{w} \wedge \mathbf{u}$ ” mean “ $\mathbf{u} \wedge \mathbf{w} = \mathbf{w} \wedge \mathbf{u}$ ”, which is no condition at all on a repeated slot. This is exactly Theorem 13.3.3, transcribed from forms to tensors by Proposition 14.8.2, and it is why the construction above divides by  $A_k$ . A reader who remembers that theorem should recognize the choice as the only characteristic-free one.

*Quick check.* In  $\Lambda^2 V$ , simplify  $(\mathbf{u} + \mathbf{w}) \wedge (\mathbf{u} - \mathbf{w})$ .

*Solution.* Expand both slots:  $\mathbf{u} \wedge \mathbf{u} - \mathbf{u} \wedge \mathbf{w} + \mathbf{w} \wedge \mathbf{u} - \mathbf{w} \wedge \mathbf{w}$ . The first and last terms are  $\mathbf{0}$  because  $\wedge$  is alternating, and  $\mathbf{w} \wedge \mathbf{u} = -\mathbf{u} \wedge \mathbf{w}$  by Lemma 14.8.1 (a). So the value is  $-2\mathbf{u} \wedge \mathbf{w}$ . Over  $\mathbb{F}_2$  that is  $\mathbf{0}$ , which is correct there and not a sign of anything going wrong: over  $\mathbb{F}_2$ ,  $\mathbf{u} + \mathbf{w}$  and  $\mathbf{u} - \mathbf{w}$  are the same vector, so the wedge of the pair must vanish.

## 14.8.4 A basis, and the dimension

Sorting is again the move, but now it costs a sign, and tuples with a repeat are simply gone. What survives is the strictly increasing tuples.

### Theorem 14.8.5 — Basis of an Exterior Power

Let  $\mathcal{B} = (\mathbf{v}_1, \dots, \mathbf{v}_n)$  be a basis of  $V$  and  $k \geq 1$ . Then the wedges

$$\mathbf{v}_{i_1} \wedge \mathbf{v}_{i_2} \wedge \cdots \wedge \mathbf{v}_{i_k}, \quad 1 \leq i_1 < i_2 < \cdots < i_k \leq n,$$

one for each **strictly increasing** tuple, form a basis of  $\Lambda^k V$ . Consequently

$$\dim \Lambda^k V = \binom{n}{k}, \quad \text{and} \quad \Lambda^k V = \{\mathbf{0}\} \quad \text{for } k > n,$$

over **every** field  $F$ .

*Idea.* Spanning: start from the basis of [Theorem 14.6.3](#), throw away every tuple with a repeated index — those wedges are  $\mathbf{0}$  — and sort the rest, paying a sign. Independence: one linear functional per increasing tuple  $I$ , and the functional must be **alternating**, so the natural candidate is a determinant: read the  $k$  arguments through the  $k$  dual functionals  $\varphi_{i_1}, \dots, \varphi_{i_k}$ , assemble the  $k \times k$  matrix of values, and take its determinant. Determinants are alternating in their columns by construction, which is exactly what is needed, and the value at the basis wedge indexed by  $J$  is  $\det \mathbf{I}_k = 1$  if  $I = J$  and 0 otherwise, because some row is then zero.

*Proof. Spanning.* By [Theorem 14.6.3](#) the tensors  $\mathbf{v}_{j_1} \otimes \cdots \otimes \mathbf{v}_{j_k}$ , over all tuples  $J$ , form a basis of  $V^{\otimes k}$ , and  $q$  is surjective, so the wedges  $\mathbf{v}_{j_1} \wedge \cdots \wedge \mathbf{v}_{j_k}$  span  $\Lambda^k V$ . If  $J$  has two equal entries, the corresponding wedge is  $\mathbf{0}$  because  $\wedge$  is alternating. If the entries of  $J$  are distinct, let  $\sigma \in S_k$  sort them into increasing order; then by [Lemma 14.8.1](#) (b) the wedge equals  $\pm$  the wedge of the sorted tuple. So the increasing tuples already span, and if  $k > n$  there are no such tuples at all, whence  $\Lambda^k V = \{\mathbf{0}\}$ .

*Independence.* Let  $(\varphi_1, \dots, \varphi_n)$  be the dual basis of  $\mathcal{B}$  ([Theorem 4.1.3](#)). For an increasing tuple  $I = (i_1 < \cdots < i_k)$  define

$$g_I(\mathbf{u}_1, \dots, \mathbf{u}_k) = \det \left( \varphi_{i_a}(\mathbf{u}_b) \right)_{a,b=1}^k \in F,$$

the determinant of the  $k \times k$  matrix whose  $(a, b)$  entry is  $\varphi_{i_a}(\mathbf{u}_b)$ . Column  $b$  of that matrix depends linearly on  $\mathbf{u}_b$ , because each  $\varphi_{i_a}$  is linear, and  $\det$  is linear in each column and vanishes when two columns coincide ([Theorem 6.4.3](#)). Hence  $g_I$  is  $k$ -linear, and alternating: if  $\mathbf{u}_a = \mathbf{u}_b$  with  $a \neq b$ , two columns of the matrix are equal and the determinant is 0.

By (E1),  $g_I$  induces a linear  $\bar{g}_I: \Lambda^k V \rightarrow F$ . Evaluate it at the wedge of an increasing tuple  $J$ : the matrix has  $(a, b)$  entry  $\varphi_{i_a}(\mathbf{v}_{j_b}) = \delta_{i_a j_b}$ . If  $I = J$  this matrix is  $\mathbf{I}_k$  and the determinant is 1. If  $I \neq J$ , then since  $I$  and  $J$  each list  $k$  distinct indices and are not the same increasing tuple, some  $i_a$  is not among  $j_1, \dots, j_k$ ; row  $a$  is then zero, so the rows are dependent, and by [Theorem 6.4.5](#) together with [Theorem 6.2.1](#) (d) the determinant is 0. Hence  $\bar{g}_I(\mathbf{v}_{j_1} \wedge \cdots \wedge \mathbf{v}_{j_k}) = \delta_{IJ}$ .

Now let  $\sum_I a_I \mathbf{v}_{i_1} \wedge \cdots \wedge \mathbf{v}_{i_k} = \mathbf{0}$ , the sum over increasing tuples. Applying  $\bar{g}_J$  leaves  $a_J = 0$ , and  $J$  was arbitrary. So the list is independent, hence a basis.

*The count.* An increasing  $k$ -tuple in  $\{1, \dots, n\}$  is the same thing as a  $k$ -element subset of  $\{1, \dots, n\}$ , listed in its only increasing order. There are  $\binom{n}{k}$  such subsets, and none at all when  $k > n$ . No step divided by an integer, so the conclusion holds over every field. This proves the theorem. ■

### Corollary 14.8.6 — The Top Exterior Power Is a Line

If  $\dim V = n \geq 1$ , then  $\dim \Lambda^n V = 1$ , with basis the single wedge  $\mathbf{v}_1 \wedge \cdots \wedge \mathbf{v}_n$  of any basis of  $V$ . For  $k > n$ ,  $\Lambda^k V = \{\mathbf{0}\}$ .



*Proof.* There is exactly one increasing  $n$ -tuple in  $\{1, \dots, n\}$ , namely  $(1, 2, \dots, n)$ , so [Theorem 14.8.5](#) gives a basis with one member, and  $\binom{n}{n} = 1$ . The second statement is the last clause of that theorem. ■

That one-dimensionality is the source of the determinant, and Section 09 is about it. For now, notice the shape of the numbers:  $\dim \Lambda^k V$  rises from 1 at  $k = 0$  to a maximum in the middle and back to 1 at  $k = n$ , and then stops. Compare  $\dim \operatorname{Sym}^k V = \binom{n+k-1}{k}$ , which grows forever. The symmetric power forgets the order; the exterior power forgets the order **and** forbids repetition, and forbidding repetition is what makes the tower finite.

*Remark.* The two counts fit together at  $k = 2$ :  $\binom{n+1}{2} + \binom{n}{2} = n^2 = \dim V^{\otimes 2}$  over every field. In characteristic  $\neq 2$  this reflects an actual decomposition of  $V^{\otimes 2}$  into symmetric and alternating tensors, in the manner of [Theorem 13.3.4](#). Over  $\mathbb{F}_2$  the identity between the numbers survives while the decomposition does not, exactly as in [Theorem 14.7.8](#).

*Warning.*  $\Lambda^k V$  is a quotient of  $V^{\otimes k}$ , not a subspace. Its elements are cosets of  $A_k$ , and  $\mathbf{u} \wedge \mathbf{w} = -\mathbf{w} \wedge \mathbf{u}$  is an identity between cosets, not between tensors. It is also a mistake to replace it by a subspace of  $V^{\otimes k}$  — say “the tensors that change sign under every swap of two slots”. Over  $\mathbb{F}_2$  that subspace is the space of symmetric tensors of [Definition 14.7.5](#), because  $-1 = 1$  there; for  $V = \mathbb{F}_2^2$  and  $k = 2$  it has dimension  $\binom{3}{2} = 3$  by [Theorem 14.7.8](#) (a), while  $\Lambda^2 V$  has dimension  $\binom{2}{2} = 1$ .

### Example. (A wedge in $\Lambda^2(F^3)$ )

Let  $V = F^3$ ,  $\mathbf{u} = \mathbf{e}_1 + 2\mathbf{e}_2 + 3\mathbf{e}_3$  and  $\mathbf{w} = 4\mathbf{e}_1 + 5\mathbf{e}_2 + 6\mathbf{e}_3$ . Write out a basis of  $\Lambda^2 V$  and expand  $\mathbf{u} \wedge \mathbf{w}$  in it.

*Solution.* By [Theorem 14.8.5](#) a basis is  $(\mathbf{e}_1 \wedge \mathbf{e}_2, \mathbf{e}_1 \wedge \mathbf{e}_3, \mathbf{e}_2 \wedge \mathbf{e}_3)$ , of size  $\binom{3}{2} = 3$ . Expanding both slots gives nine terms; the three with a repeated index vanish, and each remaining pair combines by  $\mathbf{e}_j \wedge \mathbf{e}_i = -\mathbf{e}_i \wedge \mathbf{e}_j$ . The coefficient of  $\mathbf{e}_i \wedge \mathbf{e}_j$  is therefore  $u_i w_j - u_j w_i$ :

$$\begin{aligned} \mathbf{u} \wedge \mathbf{w} &= (1 \cdot 5 - 2 \cdot 4) \mathbf{e}_1 \wedge \mathbf{e}_2 + (1 \cdot 6 - 3 \cdot 4) \mathbf{e}_1 \wedge \mathbf{e}_3 \\ &\quad + (2 \cdot 6 - 3 \cdot 5) \mathbf{e}_2 \wedge \mathbf{e}_3 \\ &= -3 \mathbf{e}_1 \wedge \mathbf{e}_2 - 6 \mathbf{e}_1 \wedge \mathbf{e}_3 - 3 \mathbf{e}_2 \wedge \mathbf{e}_3. \end{aligned}$$

The three coefficients are the  $2 \times 2$  minors of the  $3 \times 2$  matrix with columns  $\mathbf{u}$  and  $\mathbf{w}$ . That is not an accident, and Section 09 proves it in general.

## 14.8.5 Independence detects wedges

The last result of the section is small to state and is the tool everything after it uses: **a wedge is non-zero exactly when its factors are independent**. One direction is [Lemma 14.8.1](#) (c); the other needs [Theorem 14.8.5](#), which is why it comes last.



**Theorem 14.8.7 — Wedges Detect Independence**

Let  $\mathbf{u}_1, \dots, \mathbf{u}_k \in V$  with  $1 \leq k \leq n$ . Then

$$\mathbf{u}_1 \wedge \mathbf{u}_2 \wedge \cdots \wedge \mathbf{u}_k \neq \mathbf{0}$$

if and only if  $(\mathbf{u}_1, \dots, \mathbf{u}_k)$  is independent.

For  $k > n$  the wedge is always  $\mathbf{0}$ .

*Idea.*  $(\Rightarrow)$  is the contrapositive of “alternating maps kill dependent lists”.  $(\Leftarrow)$  is the reason [Theorem 14.8.5](#) was worth proving: an independent list extends to a basis of  $V$ , and that theorem then says its wedge is one of the basis vectors of  $\Lambda^k V$ , hence non-zero.

*Proof.*  $(\Rightarrow)$  We prove the contrapositive. If  $(\mathbf{u}_1, \dots, \mathbf{u}_k)$  is dependent, then  $\mathbf{u}_1 \wedge \cdots \wedge \mathbf{u}_k = \mathbf{0}$  by [Lemma 14.8.1](#) (c), applied to the alternating map  $\wedge$ .

$(\Leftarrow)$  Suppose the list is independent. By the Basis Extension Theorem ([Theorem 1.6.5](#)) it extends to a basis  $\mathcal{B} = (\mathbf{u}_1, \dots, \mathbf{u}_k, \mathbf{u}_{k+1}, \dots, \mathbf{u}_n)$  of  $V$ , which is possible because  $k \leq n$ . Apply [Theorem 14.8.5](#) to  $\mathcal{B}$ : the wedges indexed by increasing  $k$ -tuples form a basis of  $\Lambda^k V$ , and  $\mathbf{u}_1 \wedge \cdots \wedge \mathbf{u}_k$  is the one indexed by  $(1, 2, \dots, k)$ . A member of a basis is non-zero, so  $\mathbf{u}_1 \wedge \cdots \wedge \mathbf{u}_k \neq \mathbf{0}$ .

For  $k > n$ ,  $\Lambda^k V = \{\mathbf{0}\}$  by [Theorem 14.8.5](#), so every wedge vanishes; consistently, any  $k > n$  vectors in  $V$  are dependent. This proves the theorem. ■

*Remark.* The theorem turns a question about a list into a question about a single vector, and that is its value. Section 09 uses it to prove that  $\Lambda^n T$  is multiplication by  $\det T$  and that  $T$  is invertible exactly when that scalar is non-zero; Section 10 uses it to decide which elements of  $\Lambda^2 V$  are wedges at all. Note the hypothesis it does **not** need: no assumption on  $F$ , because neither half divided by anything.

**14.8.6 Exercises****14.8.6.1 A. Check your understanding****Exercise A1.**

- (a) State the universal property (E1), with every quantifier, and say which maps are required to factor.
- (b) Which subspace of  $V^{\otimes k}$  is divided out, and why is the other candidate  $N_k$  rejected?
- (c) Let  $\dim V = 5$ . Compute  $\dim \Lambda^k V$  for  $k = 0, 1, 2, 3, 4, 5, 6$ .
- (d) Determine whether  $\mathbf{u} \wedge \mathbf{w} \wedge \mathbf{u}$  can be non-zero. Justify your answer.
- (e) State the test for  $\mathbf{u}_1 \wedge \cdots \wedge \mathbf{u}_k \neq \mathbf{0}$ , and say which hypothesis on the field it needs.

*Solution.*

- (a) For every vector space  $W$  over  $F$  and every **alternating**  $k$ -linear map  $f: V^k \rightarrow W$ , there exists exactly one linear  $\bar{f}: \Lambda^k V \rightarrow W$  with  $\bar{f} \circ \wedge = f$ . Only alternating maps factor; a non-alternating one cannot, since  $\bar{f} \circ \wedge$  is alternating for every linear  $\bar{f}$ .
- (b) The subspace  $A_k$  spanned by the simple tensors with a repeated entry ([Theorem 14.8.4](#)). The candidate  $N_k$ , spanned by the elements  $x + P_\tau x$ , is rejected because  $N_k \subsetneq A_k$  in characteristic 2 ([Proposition 14.8.2](#)), so the quotient by  $N_k$  would have the wrong dimension there and its maps would be the skew-symmetric ones rather than the alternating ones.
- (c)  $\binom{5}{k}$ , giving 1, 5, 10, 10, 5, 1, 0 for  $k = 0, \dots, 6$ .
- (d) No: the list  $(\mathbf{u}, \mathbf{w}, \mathbf{u})$  repeats  $\mathbf{u}$  in positions 1 and 3, so the wedge is  $\mathbf{0}$  because  $\wedge$  is alternating. (Equivalently, the list is dependent, so [Theorem 14.8.7](#) applies.)
- (e)  $\mathbf{u}_1 \wedge \cdots \wedge \mathbf{u}_k \neq \mathbf{0}$  if and only if the list is linearly independent ([Theorem 14.8.7](#)). It needs no hypothesis on  $F$ .

### 14.8.6.2 B. Practice

#### Exercise B1: Computing wedges.

Let  $V = F^3$  with  $F = \mathbb{Q}$ .

- (a) Expand  $(2\mathbf{e}_1 - \mathbf{e}_2 + \mathbf{e}_3) \wedge (\mathbf{e}_1 + \mathbf{e}_2 - \mathbf{e}_3)$  in the basis  $(\mathbf{e}_1 \wedge \mathbf{e}_2, \mathbf{e}_1 \wedge \mathbf{e}_3, \mathbf{e}_2 \wedge \mathbf{e}_3)$ .
- (b) Expand  $\mathbf{e}_1 \wedge (\mathbf{e}_1 + \mathbf{e}_2) \wedge (\mathbf{e}_1 + \mathbf{e}_2 + \mathbf{e}_3)$  in the basis of  $\Lambda^3 V$ .

*Solution.*

- (a) With  $\mathbf{u} = (2, -1, 1)$  and  $\mathbf{w} = (1, 1, -1)$ , the coefficient of  $\mathbf{e}_i \wedge \mathbf{e}_j$  is  $u_i w_j - u_j w_i$ , as in [Example \(A wedge in  \$\Lambda^2\(F^3\)\$ \)](#):

$$\begin{aligned} 2 \cdot 1 - (-1) \cdot 1 &= 3, & 2 \cdot (-1) - 1 \cdot 1 &= -3, \\ (-1) \cdot (-1) - 1 \cdot 1 &= 0, \end{aligned}$$

so the wedge is  $3\mathbf{e}_1 \wedge \mathbf{e}_2 - 3\mathbf{e}_1 \wedge \mathbf{e}_3$ .

- (b) By [Corollary 14.8.6](#) the basis is the single wedge  $\mathbf{e}_1 \wedge \mathbf{e}_2 \wedge \mathbf{e}_3$ . Expand from the left: in the second slot,  $\mathbf{e}_1 \wedge \mathbf{e}_1 \wedge \cdots = \mathbf{0}$ , so only  $\mathbf{e}_2$  survives; then in the third slot, both  $\mathbf{e}_1$  and  $\mathbf{e}_2$  already occur, so only  $\mathbf{e}_3$  survives. Hence the wedge is  $\mathbf{e}_1 \wedge \mathbf{e}_2 \wedge \mathbf{e}_3$ . (Equivalently, the three vectors are the columns of a unit upper triangular matrix, of determinant 1; Exercise C3 explains why the answer had to be that determinant.)

#### Exercise B2: Independence by wedging.

Determine whether each list is linearly independent, by computing a wedge. Justify your answer.

- (a)  $((1, 2), (3, 6))$  in  $\mathbb{Q}^2$ .
- (b)  $((1, 0, 1), (0, 1, 1), (1, 1, 2))$  in  $\mathbb{Q}^3$ .
- (c)  $((1, 0, 0), (0, 1, 0), (0, 0, 1), (1, 1, 1))$  in  $\mathbb{Q}^3$ .

*Solution.*

- (a)  $(1, 2) \wedge (3, 6) = (1 \cdot 6 - 2 \cdot 3) \mathbf{e}_1 \wedge \mathbf{e}_2 = \mathbf{0}$ . By [Theorem 14.8.7](#) the list is dependent, as it visibly is: the second vector is three times the first.
- (b) The minor computation  $(1 \cdot 1 - 0 \cdot 0, 1 \cdot 1 - 1 \cdot 0, 0 \cdot 1 - 1 \cdot 1) = (1, 1, -1)$  gives, for the first two vectors,  $\mathbf{e}_1 \wedge \mathbf{e}_2 + \mathbf{e}_1 \wedge \mathbf{e}_3 - \mathbf{e}_2 \wedge \mathbf{e}_3$ . Wedging with the third vector  $(1, 1, 2)$  and keeping only the terms with all three indices distinct,

$$\begin{aligned} & (\mathbf{e}_1 \wedge \mathbf{e}_2) \wedge 2\mathbf{e}_3 + (\mathbf{e}_1 \wedge \mathbf{e}_3) \wedge \mathbf{e}_2 - (\mathbf{e}_2 \wedge \mathbf{e}_3) \wedge \mathbf{e}_1 \\ &= (2 - 1 - 1) \mathbf{e}_1 \wedge \mathbf{e}_2 \wedge \mathbf{e}_3 = \mathbf{0}, \end{aligned}$$

using  $\mathbf{e}_1 \wedge \mathbf{e}_3 \wedge \mathbf{e}_2 = -\mathbf{e}_1 \wedge \mathbf{e}_2 \wedge \mathbf{e}_3$  and  $\mathbf{e}_2 \wedge \mathbf{e}_3 \wedge \mathbf{e}_1 = \mathbf{e}_1 \wedge \mathbf{e}_2 \wedge \mathbf{e}_3$  from [Lemma 14.8.1](#) (b). So the list is dependent; indeed the third vector is the sum of the first two.

- (c) Here  $k = 4 > 3 = n$ , so  $\Lambda^4(\mathbb{Q}^3) = \{\mathbf{0}\}$  by [Theorem 14.8.5](#) and the wedge is  $\mathbf{0}$  without any computation. The list is dependent, as any 4 vectors in a 3-dimensional space must be.

### Exercise B3: A functional built from the universal property.

Let  $V = F^3$  and let  $f: V^2 \rightarrow F$  be  $f(\mathbf{u}, \mathbf{w}) = u_1 w_3 - u_3 w_1$ .

- (a) Prove that  $f$  is alternating and bilinear, and deduce that there is a unique linear  $\bar{f}: \Lambda^2 V \rightarrow F$  with  $\bar{f}(\mathbf{u} \wedge \mathbf{w}) = f(\mathbf{u}, \mathbf{w})$ .
- (b) Compute  $\bar{f}$  on the basis  $(\mathbf{e}_1 \wedge \mathbf{e}_2, \mathbf{e}_1 \wedge \mathbf{e}_3, \mathbf{e}_2 \wedge \mathbf{e}_3)$ , and evaluate  $\bar{f}$  on the answer to [Example \(A wedge in  \$\Lambda^2\(F^3\)\$ \)](#).

*Solution.*

- (a) Each slot is a linear functional with the other fixed, so  $f$  is bilinear, and  $f(\mathbf{u}, \mathbf{u}) = u_1 u_3 - u_3 u_1 = 0$ , so  $f$  is alternating ([Definition 14.8.1](#) with  $k = 2$ ). By (E1) there is exactly one linear  $\bar{f}: \Lambda^2 V \rightarrow F$  with  $\bar{f} \circ \wedge = f$ .
- (b)  $\bar{f}(\mathbf{e}_1 \wedge \mathbf{e}_2) = f(\mathbf{e}_1, \mathbf{e}_2) = 0$ ;  $\bar{f}(\mathbf{e}_1 \wedge \mathbf{e}_3) = 1$ ;  $\bar{f}(\mathbf{e}_2 \wedge \mathbf{e}_3) = 0$ . So  $\bar{f}$  reads off the  $\mathbf{e}_1 \wedge \mathbf{e}_3$  coordinate. On  $\mathbf{u} \wedge \mathbf{w} = -3\mathbf{e}_1 \wedge \mathbf{e}_2 - 6\mathbf{e}_1 \wedge \mathbf{e}_3 - 3\mathbf{e}_2 \wedge \mathbf{e}_3$  it gives  $-6$ , which agrees with  $f(\mathbf{u}, \mathbf{w}) = 1 \cdot 6 - 3 \cdot 4 = -6$ .

### 14.8.6.3 C. Going deeper

#### Exercise C1: The characteristic-two witness.

Let  $F = \mathbb{F}_2$ ,  $V = \mathbb{F}_2^2$  and  $k = 2$ .

- (a) List the elements of  $A_2$  and of  $N_2$ , and compute both dimensions.
- (b) Compute  $\dim \Lambda^2 V$  from (a) and check it against [Theorem 14.8.5](#).
- (c) Compute  $\dim(V^{\otimes 2}/N_2)$  and explain, in one sentence, what would go wrong if the exterior power had been defined as that quotient.

*Solution.*

- (a) By [Theorem 14.6.3](#),  $V^{\otimes 2}$  has basis  $\mathbf{e}_1 \otimes \mathbf{e}_1, \mathbf{e}_1 \otimes \mathbf{e}_2, \mathbf{e}_2 \otimes \mathbf{e}_1, \mathbf{e}_2 \otimes \mathbf{e}_2$ . Now  $A_2$  is

spanned by the tensors  $\mathbf{u} \otimes \mathbf{u}$ , and the three choices  $\mathbf{u} = \mathbf{e}_1, \mathbf{e}_2, \mathbf{e}_1 + \mathbf{e}_2$  give

$$\mathbf{e}_1 \otimes \mathbf{e}_1, \quad \mathbf{e}_2 \otimes \mathbf{e}_2, \quad \mathbf{e}_1 \otimes \mathbf{e}_1 + \mathbf{e}_1 \otimes \mathbf{e}_2 + \mathbf{e}_2 \otimes \mathbf{e}_1 + \mathbf{e}_2 \otimes \mathbf{e}_2,$$

whose span also contains  $t = \mathbf{e}_1 \otimes \mathbf{e}_2 + \mathbf{e}_2 \otimes \mathbf{e}_1$ . Those three vectors  $\mathbf{e}_1 \otimes \mathbf{e}_1, \mathbf{e}_2 \otimes \mathbf{e}_2, t$  are independent, being distinct disjoint sums of basis vectors, and they span all three listed generators, so  $\dim A_2 = 3$ ; there are no further generators, since  $\mathbf{0}$  is the only remaining vector of  $\mathbb{F}_2^2$ . For  $N_2$ , the coefficient formula in the proof of [Proposition 14.8.2](#) gives

$$\mathbf{u} \otimes \mathbf{w} + \mathbf{w} \otimes \mathbf{u} = (a_1 b_2 + a_2 b_1) t,$$

since the diagonal coefficients vanish and the two off-diagonal ones are equal. Hence  $N_2 = \text{span}(t)$  and  $\dim N_2 = 1$ .

- (b)  $\dim \Lambda^2 V = 4 - 3 = 1$  by [Theorem 3.9.5](#), matching  $\binom{2}{2} = 1$ .
- (c)  $\dim(V^{\otimes 2}/N_2) = 4 - 1 = 3$ . Defining  $\Lambda^2 V$  as that quotient would make the image of  $\mathbf{e}_1 \otimes \mathbf{e}_1$  non-zero, so  $\mathbf{v} \wedge \mathbf{v} \neq \mathbf{0}$  for some  $\mathbf{v}$ , the structure map would be skew-symmetric but not alternating, and the dimension would be 3 instead of  $\binom{2}{2} = 1$ .

### Exercise C2: Alternating forms, recounted.

- (a) Prove that the alternating  $k$ -linear forms on  $V$  are in bijection with the linear functionals on  $\Lambda^k V$ , and that the bijection is an isomorphism of vector spaces.
- (b) Deduce that the space of alternating  $k$ -linear forms on  $V$  has dimension  $\binom{n}{k}$ .
- (c) Hence recover [Corollary 6.4.4](#): the alternating  $n$ -linear forms on  $F^n$  form a line.

*Solution.*

- (a) Write  $\mathcal{A}$  for the space of alternating  $k$ -linear forms on  $V$ , a vector space by [Theorem 6.2.3](#). The map  $\Theta: (\Lambda^k V)^* \rightarrow \mathcal{A}$ ,  $\Theta(\lambda) = \lambda \circ \wedge$ , is well defined, since a composite of the alternating  $\wedge$  with a linear map is alternating and  $k$ -linear, and it is linear in  $\lambda$  because composition with a fixed map is. It is injective: if  $\lambda \circ \wedge = 0$  then  $\lambda$  vanishes on the wedges, which span  $\Lambda^k V$  ([Theorem 14.8.4](#)), so  $\lambda = 0$ . It is surjective: given  $f \in \mathcal{A}$ , (E1) supplies  $\bar{f}$  with  $\bar{f} \circ \wedge = f$ . So  $\Theta$  is an isomorphism.
- (b) By (a) and [Corollary 4.1.4](#),  $\dim \mathcal{A} = \dim(\Lambda^k V)^* = \dim \Lambda^k V = \binom{n}{k}$ , using [Theorem 14.8.5](#).
- (c) Take  $V = F^n$  and  $k = n$ : the dimension is  $\binom{n}{n} = 1$ . Since  $\det$  is a non-zero alternating  $n$ -linear form ([Theorem 6.4.3](#), with  $\det \mathbf{I}_n = 1$ ), it spans that line, which is [Corollary 6.4.4](#).

### Exercise C3: A wedge sees only the subspace.

Let  $(\mathbf{u}_1, \dots, \mathbf{u}_k)$  and  $(\mathbf{w}_1, \dots, \mathbf{w}_k)$  be two lists in  $V$  with  $\mathbf{w}_b = \sum_{a=1}^k p_{ab} \mathbf{u}_a$  for a matrix  $\mathbf{P} = (p_{ab}) \in M_k(F)$ .

- (a) Prove that  $\mathbf{w}_1 \wedge \dots \wedge \mathbf{w}_k = (\det \mathbf{P}) \mathbf{u}_1 \wedge \dots \wedge \mathbf{u}_k$ .
- (b) Deduce that if  $(\mathbf{u}_1, \dots, \mathbf{u}_k)$  and  $(\mathbf{w}_1, \dots, \mathbf{w}_k)$  are two bases of the **same**  $k$ -dimensional subspace  $U \subseteq V$ , then their wedges differ by a non-zero scalar factor.

- (c) Deduce that the line  $\text{span}(\mathbf{u}_1 \wedge \cdots \wedge \mathbf{u}_k) \subseteq \Lambda^k V$  depends only on  $U$ , not on the basis chosen for it.

*Hint: for (a), expand every slot and use [Lemma 14.8.1](#) (b) together with [Definition 6.4.1](#).*

*Solution.*

- (a) Expanding slot by slot,

$$\mathbf{w}_1 \wedge \cdots \wedge \mathbf{w}_k = \sum_{a_1, \dots, a_k} p_{a_1 1} \cdots p_{a_k k} \mathbf{u}_{a_1} \wedge \cdots \wedge \mathbf{u}_{a_k},$$

the sum over all  $k$ -tuples  $(a_1, \dots, a_k)$  from  $\{1, \dots, k\}$ . A term with two equal indices has a repeated argument, so its wedge is  $\mathbf{0}$ . A term with all indices distinct has  $(a_1, \dots, a_k) = (\sigma(1), \dots, \sigma(k))$  for a unique  $\sigma \in S_k$ , and then  $\mathbf{u}_{\sigma(1)} \wedge \cdots \wedge \mathbf{u}_{\sigma(k)} = \text{sgn}(\sigma) \mathbf{u}_1 \wedge \cdots \wedge \mathbf{u}_k$  by [Lemma 14.8.1](#) (b). Hence

$$\mathbf{w}_1 \wedge \cdots \wedge \mathbf{w}_k = \left( \sum_{\sigma \in S_k} \text{sgn}(\sigma) p_{\sigma(1)1} \cdots p_{\sigma(k)k} \right) \mathbf{u}_1 \wedge \cdots \wedge \mathbf{u}_k,$$

and the bracket is  $\det \mathbf{P}$  by [Definition 6.4.1](#).

- (b) Both lists are bases of  $U$ , so each  $\mathbf{w}_b$  is a combination of the  $\mathbf{u}_a$ , giving  $\mathbf{P}$  as above, and  $\mathbf{P}$  is the change-of-coordinates matrix between two bases of  $U$ , hence invertible ([Proposition 3.7.2](#)). By [Theorem 6.5.2](#),  $\det \mathbf{P} \neq 0$ , and (a) gives the claim. Both wedges are non-zero, by [Theorem 14.8.7](#).
- (c) By (b) the two wedges are non-zero scalar multiples of each other, so they span the same one-dimensional subspace of  $\Lambda^k V$ . Any basis of  $U$  therefore determines the same line, which consequently depends on  $U$  alone. (This line is the object that Section 09's coordinates — the  $k \times k$  minors — describe, and it is how a  $k$ -dimensional subspace is turned into a single point.)

## 14.9 The Determinant as a Top Exterior Power

Chapter 6 built the determinant twice. First it proved that an alternating  $n$ -linear form on  $F^n$  is determined by its value at the standard basis ([Theorem 6.4.2](#)), which says *at most one* normalized such form exists; then it wrote the Leibniz sum down and checked the three rules ([Theorem 6.4.3](#)), which says *at least one* does. The number  $\det \mathbf{A}$  of [Definition 6.4.1](#) is that form's value. This section builds the determinant a third time, and the honest way to describe what it adds is to say up front what it does not: no new value, no faster computation, and no theorem Chapter 6 was missing. What it adds is an **object**. The exterior powers of the previous section turn  $\det T$  from a number attached to a matrix into the operator that  $T$  induces on a one-dimensional space, and from that description multiplicativity stops being a computation and becomes the statement that composing two induced maps induces the composite.

Throughout,  $F$  is a field,  $V$  and  $W$  are finite-dimensional vector spaces over  $F$ , and  $\Lambda^k V$ , the wedge  $\mathbf{v}_1 \wedge \cdots \wedge \mathbf{v}_k$ , and the universal property for alternating maps are as in [Definition 14.8.2](#). We use freely that the wedges  $\mathbf{v}_{i_1} \wedge \cdots \wedge \mathbf{v}_{i_k}$  over strictly increasing index tuples form a basis of  $\Lambda^k V$ , so that  $\dim \Lambda^k V = \binom{n}{k}$  when  $\dim V = n$  ([Theorem 14.8.5](#)), and that a wedge of vectors is non-zero exactly when the list is linearly independent ([Theorem 14.8.7](#)).

## 14.9.1 Wedging with a fixed vector

Two facts from the previous section will be used on every page, and one small tool is still missing.

The first fact is the **reordering rule**: the wedge map  $(\mathbf{v}_1, \dots, \mathbf{v}_k) \mapsto \mathbf{v}_1 \wedge \dots \wedge \mathbf{v}_k$  is alternating (Definition 14.8.2), and an alternating multilinear map satisfies

$$f(\mathbf{v}_{\sigma(1)}, \dots, \mathbf{v}_{\sigma(k)}) = \text{sgn}(\sigma) f(\mathbf{v}_1, \dots, \mathbf{v}_k)$$

for every  $\sigma \in S_k$ , by Lemma 14.8.1 (b). Reordering a wedge costs the sign of the permutation, and nothing else. The second is that  $\Lambda^n V$  is one-dimensional when  $\dim V = n \geq 1$  (Corollary 14.8.6); that is the whole engine of this section.

The missing tool is the ability to wedge a fixed vector onto a whole exterior power, not just onto a list of vectors. Section 10 builds the product on all of  $\Lambda V$ ; here one vector at a time is enough.

### Lemma 14.9.1 — Wedging with a Fixed Vector

Let  $\mathbf{v} \in V$  and  $k \geq 1$ . There is a **unique** linear map

$$\varepsilon_{\mathbf{v}}: \Lambda^k V \rightarrow \Lambda^{k+1} V, \quad \varepsilon_{\mathbf{v}}(\mathbf{w}_1 \wedge \dots \wedge \mathbf{w}_k) = \mathbf{v} \wedge \mathbf{w}_1 \wedge \dots \wedge \mathbf{w}_k.$$

*Proof.* The map  $V^k \rightarrow \Lambda^{k+1} V$  sending  $(\mathbf{w}_1, \dots, \mathbf{w}_k)$  to  $\mathbf{v} \wedge \mathbf{w}_1 \wedge \dots \wedge \mathbf{w}_k$  is  $k$ -linear, because the wedge of  $k+1$  vectors is linear in each slot and the first slot is frozen at  $\mathbf{v}$ ; and it is alternating, because if  $\mathbf{w}_p = \mathbf{w}_r$  with  $p \neq r$  then the wedge of  $k+1$  vectors has two equal arguments and is  $\mathbf{0}$ . By the universal property in Definition 14.8.2 it factors through  $\Lambda^k V$ , giving a linear  $\varepsilon_{\mathbf{v}}$  with the stated values. Uniqueness holds because the wedges  $\mathbf{w}_1 \wedge \dots \wedge \mathbf{w}_k$  span  $\Lambda^k V$  (Theorem 14.8.5), and a linear map is determined on a spanning set. ■

## 14.9.2 Induced maps on exterior powers

A linear map should act on everything built from its domain. A map  $T: V \rightarrow W$  sends a list  $(\mathbf{v}_1, \dots, \mathbf{v}_k)$  to the list  $(T\mathbf{v}_1, \dots, T\mathbf{v}_k)$ , so it ought to send the wedge  $\mathbf{v}_1 \wedge \dots \wedge \mathbf{v}_k$  to  $T\mathbf{v}_1 \wedge \dots \wedge T\mathbf{v}_k$ . The only question is whether that rule defines a map at all: an element of  $\Lambda^k V$  can be written as a wedge in many ways, and an assignment made on wedges must be checked. The universal property answers it in two lines, and that is the point of having a universal property.

*The  $k$ -th exterior power of a map wedges its values together.*

### Definition 14.9.1 — Exterior Power of a Linear Map

Let  $T: V \rightarrow W$  be linear and let  $k \geq 1$ . The  **$k$ -th exterior power** of  $T$  is the unique linear map

$$\Lambda^k T: \Lambda^k V \rightarrow \Lambda^k W$$

satisfying  $\Lambda^k T(\mathbf{v}_1 \wedge \dots \wedge \mathbf{v}_k) = T\mathbf{v}_1 \wedge \dots \wedge T\mathbf{v}_k$  for all  $\mathbf{v}_1, \dots, \mathbf{v}_k \in V$ .

**Well-definedness.** The map  $V^k \rightarrow \Lambda^k W$  sending  $(\mathbf{v}_1, \dots, \mathbf{v}_k)$  to  $T\mathbf{v}_1 \wedge \dots \wedge T\mathbf{v}_k$  is  $k$ -linear, since  $T$  is linear and the wedge in  $W$  is linear in each slot; and it is alternating, since  $\mathbf{v}_p = \mathbf{v}_r$  forces  $T\mathbf{v}_p = T\mathbf{v}_r$  and a wedge with two equal arguments is  $\mathbf{0}$ . So Definition 14.8.2 produces exactly one linear map on  $\Lambda^k V$  with the stated values, and the wedges span  $\Lambda^k V$  (Theorem 14.8.5), so no other linear map has them.

### Examples.

- $k = 1$ .  $\Lambda^1 V = V$  and  $\Lambda^1 T = T$ : the definition says nothing new in degree one.
- **The identity.**  $\Lambda^k(\text{id}_V) = \text{id}$ , since both sides send  $\mathbf{v}_1 \wedge \dots \wedge \mathbf{v}_k$  to itself.
- **A scalar.** For  $T = c \text{id}_V$ ,  $\Lambda^k T(\mathbf{v}_1 \wedge \dots \wedge \mathbf{v}_k) = c^k \mathbf{v}_1 \wedge \dots \wedge \mathbf{v}_k$ , so  $\Lambda^k T = c^k \text{id}$ . One factor of  $c$  comes out of each of the  $k$  slots.
- **A degenerate case.** If  $k > \dim V$  then  $\Lambda^k V = \{\mathbf{0}\}$  (Theorem 14.8.5) and  $\Lambda^k T$  is the zero map between zero spaces. Worth recording so that statements below need no size caveat.

**Non-example by minimal change.** Replace “alternating” by nothing: the rule  $\mathbf{v}_1 \otimes \dots \otimes \mathbf{v}_k \mapsto T\mathbf{v}_1 \wedge \dots \wedge T\mathbf{v}_k$  on  $V^{\otimes k}$  is perfectly well defined, but it is **not** the map above, because  $\Lambda^k V$  is a quotient of  $V^{\otimes k}$  and not a subspace of it. What makes Definition 14.9.1 legal is precisely that the rule kills the relations that were divided out.

The reason for having the notation is the next sentence.

### Theorem 14.9.2 — Exterior Powers Respect Composition

Let  $T: U \rightarrow V$  and  $S: V \rightarrow W$  be linear, and let  $k \geq 1$ . Then

$$\Lambda^k(ST) = (\Lambda^k S)(\Lambda^k T), \quad \Lambda^k(\text{id}_V) = \text{id}_{\Lambda^k V}.$$

*Proof.* Both  $\Lambda^k(ST)$  and  $(\Lambda^k S)(\Lambda^k T)$  are linear maps  $\Lambda^k U \rightarrow \Lambda^k W$ , so it suffices to compare them on the wedges, which span  $\Lambda^k U$  (Theorem 14.8.5). By Definition 14.9.1 applied three times,

$$(\Lambda^k S)(\Lambda^k T)(\mathbf{u}_1 \wedge \dots \wedge \mathbf{u}_k) = ST\mathbf{u}_1 \wedge \dots \wedge ST\mathbf{u}_k = \Lambda^k(ST)(\mathbf{u}_1 \wedge \dots \wedge \mathbf{u}_k).$$

The second identity was checked above. This proves the theorem. ■

### Example. (The second exterior power of a $3 \times 3$ matrix)

Let  $T: \mathbb{Q}^3 \rightarrow \mathbb{Q}^3$  have matrix

$$\mathbf{A} = \begin{pmatrix} 2 & 1 & 3 \\ 0 & 4 & 1 \\ 5 & 2 & 1 \end{pmatrix}$$

in the standard basis. Find the matrix of  $\Lambda^2 T$  in the basis  $(\mathbf{e}_1 \wedge \mathbf{e}_2, \mathbf{e}_1 \wedge \mathbf{e}_3, \mathbf{e}_2 \wedge \mathbf{e}_3)$  of  $\Lambda^2 \mathbb{Q}^3$ , and compute its determinant.

*Solution.* By Definition 14.9.1,  $\Lambda^2 T(\mathbf{e}_i \wedge \mathbf{e}_j) = \mathbf{a}_i \wedge \mathbf{a}_j$ , where  $\mathbf{a}_1, \mathbf{a}_2, \mathbf{a}_3$  are the columns

of  $\mathbf{A}$ . Expand one column:

$$\begin{aligned}\mathbf{a}_1 \wedge \mathbf{a}_2 &= (2\mathbf{e}_1 + 5\mathbf{e}_3) \wedge (\mathbf{e}_1 + 4\mathbf{e}_2 + 2\mathbf{e}_3) \\ &= 8\mathbf{e}_1 \wedge \mathbf{e}_2 + 4\mathbf{e}_1 \wedge \mathbf{e}_3 + 5\mathbf{e}_3 \wedge \mathbf{e}_1 + 20\mathbf{e}_3 \wedge \mathbf{e}_2 \\ &= 8\mathbf{e}_1 \wedge \mathbf{e}_2 - \mathbf{e}_1 \wedge \mathbf{e}_3 - 20\mathbf{e}_2 \wedge \mathbf{e}_3,\end{aligned}$$

where the terms  $2\mathbf{e}_1 \wedge \mathbf{e}_1$  and  $10\mathbf{e}_3 \wedge \mathbf{e}_3$  were dropped in the second line because a wedge with a repeated vector is  $\mathbf{0}$ , and the third line used  $\mathbf{e}_3 \wedge \mathbf{e}_1 = -\mathbf{e}_1 \wedge \mathbf{e}_3$  and  $\mathbf{e}_3 \wedge \mathbf{e}_2 = -\mathbf{e}_2 \wedge \mathbf{e}_3$  (Lemma 14.8.1 (b)). The coefficients 8,  $-1$ ,  $-20$  are the three  $2 \times 2$  minors of the first two columns of  $\mathbf{A}$ , on rows  $\{1, 2\}, \{1, 3\}, \{2, 3\}$  — a pattern the last part of this section explains. Doing the same for the other two column pairs gives

$$[\Lambda^2 T] = \begin{pmatrix} 8 & 2 & -11 \\ -1 & -13 & -5 \\ -20 & -5 & 2 \end{pmatrix},$$

whose determinant is 2601. Since  $\det \mathbf{A} = -51$ , this is  $(\det \mathbf{A})^2$ .

*Quick check.* In the example, why is  $\Lambda^2 T$  **not** multiplication by a scalar, while  $\Lambda^3 T$  is?

*Solution.*  $\dim \Lambda^2 \mathbb{Q}^3 = \binom{3}{2} = 3$  by Theorem 14.8.5, and the displayed matrix is not a scalar multiple of  $\mathbf{I}_3$ , so  $\Lambda^2 T$  is an operator on a three-dimensional space with no reason to be scalar. But  $\dim \Lambda^3 \mathbb{Q}^3 = \binom{3}{3} = 1$ , and every linear operator on a one-dimensional space is multiplication by a scalar. Being a line is what makes the top power special.

### 14.9.3 The top exterior power

That last observation is the whole theorem. When  $k$  equals  $\dim V$ , the space  $\Lambda^k V$  is a line, an operator on a line is a scalar, and the scalar can be read off by feeding in a basis.

#### Theorem 14.9.3 — The Determinant Is the Top Exterior Power

Let  $V$  be a vector space over  $F$  with  $\dim V = n \geq 1$ , and let  $T \in \mathcal{L}(V)$ . Then  $\dim \Lambda^n V = 1$ , and  $\Lambda^n T$  is multiplication by  $\det T$ :

$$\Lambda^n T(\omega) = (\det T) \omega \quad \text{for every } \omega \in \Lambda^n V.$$

Equivalently, for all  $\mathbf{v}_1, \dots, \mathbf{v}_n \in V$ ,

$$T\mathbf{v}_1 \wedge \cdots \wedge T\mathbf{v}_n = (\det T) \mathbf{v}_1 \wedge \cdots \wedge \mathbf{v}_n. \quad (14.9.1)$$

*Idea.* Three moves. ①  $\Lambda^n V$  is one-dimensional, so  $\Lambda^n T$  is multiplication by one unknown scalar  $c$ . ② To find  $c$ , evaluate on a basis wedge  $\mathbf{v}_1 \wedge \cdots \wedge \mathbf{v}_n$  and expand each  $T\mathbf{v}_j$  in the basis. ③ The expansion produces one term for every function from slots to basis indices; the non-injective ones die because a wedge with a repeat is zero, and what survives is a sum over permutations with signs — the Leibniz formula, arriving on its own.



*Proof.* Fix a basis  $\mathcal{B} = (\mathbf{v}_1, \dots, \mathbf{v}_n)$  of  $V$ . By [Corollary 14.8.6](#),  $\dim \Lambda^n V = 1$  with basis the single wedge  $\omega_0 = \mathbf{v}_1 \wedge \dots \wedge \mathbf{v}_n$ . Since  $\Lambda^n T$  is a linear operator on a one-dimensional space, there is a unique  $c \in F$  with  $\Lambda^n T(\omega_0) = c\omega_0$ , and then  $\Lambda^n T(a\omega_0) = ac\omega_0$  for every  $a \in F$ , so  $\Lambda^n T = c \text{ id}$  on all of  $\Lambda^n V$ .

It remains to show  $c = \det T$ . Let  $\mathbf{A} = [T]_{\mathcal{B}}^{\mathcal{B}}$  have entries  $a_{ij}$ , so that  $T\mathbf{v}_j = \sum_{i=1}^n a_{ij}\mathbf{v}_i$  ([Definition 3.6.1](#)). The wedge of  $n$  vectors is linear in each slot, so expanding slot by slot gives one term for each function  $\varphi: \{1, \dots, n\} \rightarrow \{1, \dots, n\}$ :

$$T\mathbf{v}_1 \wedge \dots \wedge T\mathbf{v}_n = \sum_{\varphi} a_{\varphi(1)1} \dots a_{\varphi(n)n} \mathbf{v}_{\varphi(1)} \wedge \dots \wedge \mathbf{v}_{\varphi(n)}.$$

If  $\varphi$  is not injective, two slots hold the same vector and the wedge is  $\mathbf{0}$ . A non-injective  $\varphi$  is possible only because the domain and codomain are finite of the same size, so the surviving  $\varphi$  are exactly the bijections, that is, the permutations  $\sigma \in S_n$ . For those, [Lemma 14.8.1](#) (b) gives  $\mathbf{v}_{\sigma(1)} \wedge \dots \wedge \mathbf{v}_{\sigma(n)} = \text{sgn}(\sigma)\omega_0$ . Therefore

$$\Lambda^n T(\omega_0) = \left( \sum_{\sigma \in S_n} \text{sgn}(\sigma) a_{\sigma(1)1} \dots a_{\sigma(n)n} \right) \omega_0,$$

and the bracket is  $\det \mathbf{A}$  by [Definition 6.4.1](#), which is  $\det T$  by [Definition 6.5.1](#). So  $c = \det T$ .

For the last display, note that both sides of [Eq. 14.9.1](#) are  $n$ -linear in  $(\mathbf{v}_1, \dots, \mathbf{v}_n)$  and agree when the list is the basis  $\mathcal{B}$ ; but in fact no extra argument is needed, since  $\mathbf{v}_1 \wedge \dots \wedge \mathbf{v}_n \in \Lambda^n V$  for **any** list, and the first display applies to it. This proves the theorem. ■

**Warning. Only the top power is a line.** For  $1 \leq k < n$  the operator  $\Lambda^k T$  is not multiplication by a scalar in general: [Example \(The second exterior power of a  \$3 \times 3\$  matrix\)](#) exhibits a  $\Lambda^2 T$  with three different diagonal entries. The determinant appears because  $\binom{n}{n} = 1$ , not because exterior powers are scalars.

**Warning. A map between different spaces still has no determinant.** If  $T: V \rightarrow W$  is linear with  $\dim V = \dim W = n$ , then  $\Lambda^n T$  is a linear map between two **different** one-dimensional spaces, and there is no scalar attached to it until bases of  $\Lambda^n V$  and  $\Lambda^n W$  are chosen. Changing the basis of  $\Lambda^n W$  alone rescales the number arbitrarily. This is the same obstruction Chapter 6 recorded after [Definition 6.5.1](#), now visible as the failure of a line to have a canonical generator.

**Quick check.** Let  $T$  be the operator on  $F^3$  with  $T\mathbf{e}_1 = \mathbf{e}_2$ ,  $T\mathbf{e}_2 = \mathbf{e}_3$  and  $T\mathbf{e}_3 = \mathbf{e}_1$ . Read  $\det T$  off [Eq. 14.9.1](#), without writing down a matrix.

**Solution.**  $T\mathbf{e}_1 \wedge T\mathbf{e}_2 \wedge T\mathbf{e}_3 = \mathbf{e}_2 \wedge \mathbf{e}_3 \wedge \mathbf{e}_1$ . The permutation carrying  $(1, 2, 3)$  to  $(2, 3, 1)$  is a 3-cycle, of sign  $+1$  ([Proposition 6.3.4](#)), so by [Lemma 14.8.1](#) (b) this equals  $\mathbf{e}_1 \wedge \mathbf{e}_2 \wedge \mathbf{e}_3$ . Comparing with [Eq. 14.9.1](#) on the non-zero element  $\mathbf{e}_1 \wedge \mathbf{e}_2 \wedge \mathbf{e}_3$  gives  $\det T = 1$ .

## 14.9.4 What the third construction adds

It is worth saying plainly, because the temptation to oversell a slick construction is real.

**It adds no new value.** The scalar produced in the proof is the Leibniz sum, arrived at by the same expansion that proved [Theorem 6.4.2](#) in Chapter 6: expand multilinearly, discard the terms with a repeated basis vector, sort the survivors and collect signs. Anyone who has read Chapter 6 §04 has read that computation. The exterior power did not compute the determinant; it stored the same computation in a different place.

**It adds no faster method.** Nothing here helps evaluate a determinant. Elimination remains the way to do that ([Example \(A  \$4 \times 4\$  Determinant by Elimination\)](#)).

**What it does add is that  $\det$  is a map, not just a number.** Chapter 6 defined  $\det \mathbf{A}$  by a formula in the entries and then had to *prove* that operators inherit it: the formula depends on a basis, so [Corollary 6.5.7](#) was needed before [Definition 6.5.1](#) could be written down. Here  $\Lambda^n T$  is defined with no basis in sight, and the scalar exists because  $\Lambda^n V$  is a line. Basis-independence is not a theorem about the determinant; it is the absence of any choice to be independent of. One honest qualification: the route is not self-contained. [Theorem 14.8.5](#), which is what makes  $\Lambda^n V$  a line in the first place, proves independence with a  $k \times k$  determinant, and the identification of the scalar with  $\det T$  goes through [Definition 6.5.1](#). Chapter 6 is not being replaced here; it is being re-read.

And from that, multiplicativity is free.

### Corollary 14.9.4 — Multiplicativity and Invertibility, from Functoriality

Let  $\dim V = n \geq 1$  and  $S, T \in \mathcal{L}(V)$ .

- (a)  $\det(ST) = \det S \det T$ , and  $\det \text{id}_V = 1$ .
- (b)  $T$  is invertible if and only if  $\det T \neq 0$ .

*Proof.*

- (a) By [Theorem 14.9.2](#),  $\Lambda^n(ST) = (\Lambda^n S)(\Lambda^n T)$ . By [Theorem 14.9.3](#) all three maps are multiplications by scalars on the line  $\Lambda^n V$ , namely by  $\det(ST)$ ,  $\det S$  and  $\det T$ ; and multiplication by  $\det T$  followed by multiplication by  $\det S$  is multiplication by  $\det S \det T$ . Two scalar multiplications on a non-zero space are equal only if the scalars are, so  $\det(ST) = \det S \det T$ . Likewise  $\Lambda^n(\text{id}_V) = \text{id}$ , so  $\det \text{id}_V = 1$ .
- (b)  $(\Rightarrow)$  If  $T$  is invertible, apply (a) to  $TT^{-1} = \text{id}_V$ : then  $\det T \cdot \det(T^{-1}) = 1$ , so  $\det T \neq 0$ .  $(\Leftarrow)$  Suppose  $T$  is not invertible. Then  $T$  is not injective, so  $\ker T \neq \{0\}$ ; choose  $\mathbf{v}_1 \in \ker T$  with  $\mathbf{v}_1 \neq 0$  and extend to a basis  $(\mathbf{v}_1, \dots, \mathbf{v}_n)$  of  $V$  ([Theorem 1.6.5](#)). Since  $T\mathbf{v}_1 = 0$  and the wedge is linear in its first slot,  $T\mathbf{v}_1 \wedge \dots \wedge T\mathbf{v}_n = 0$ . By [Eq. 14.9.1](#) this is  $(\det T) \mathbf{v}_1 \wedge \dots \wedge \mathbf{v}_n$ , and the wedge of a basis is non-zero ([Theorem 14.8.7](#)), so  $\det T = 0$ .

■

Part (b) is the promise [Theorem 14.8.7](#) made: the test “is this list independent?” and the test “is this determinant non-zero?” are the same test, once the determinant is read as an induced

map.

Compare the work this replaces. Chapter 6 proved [Theorem 6.5.4](#) by fixing  $\mathbf{A}$ , checking that  $\mathbf{B} \mapsto \det(\mathbf{AB})$  is alternating and  $n$ -linear in the columns of  $\mathbf{B}$ , and applying uniqueness. That proof is short and correct; what the present one shows is that the content is not really about determinants at all. Multiplicativity is the statement that  $\Lambda^n$  turns composition into composition — and  $\Lambda^n$  does that for the same reason that  $\Lambda^k$  does it for every  $k$ , where no scalar is available and no multiplicativity statement can even be phrased.

*Remark.* What is **not** improved: [Theorem 6.4.5](#). The identity  $\det \mathbf{A}^\top = \det \mathbf{A}$  is a statement about rows versus columns, and  $\Lambda^n$  sees only the map  $T$ , which has no rows. Chapter 6's proof, reindexing the Leibniz sum by  $\sigma^{-1}$ , remains the natural one.

## 14.9.5 The coordinates of a wedge

The example above produced the minors of a matrix as the coefficients of a wedge, apparently by accident. It was not an accident, and the general statement is the bridge between this chapter and Chapter 6 §07.

Fix  $n \geq 1$ , write  $[n] = \{1, \dots, n\}$  as Chapter 6 §07 does, and recall the notation of [Definition 6.7.1](#): for  $\mathbf{A} \in M_{n \times k}(F)$  and a set  $I = \{i_1 < \dots < i_k\} \subseteq [n]$  of size  $k$ , the submatrix  $\mathbf{A}_{I,[k]} \in M_k(F)$  keeps the rows in  $I$  and all  $k$  columns, and its determinant is a  $k \times k$  minor of  $\mathbf{A}$ . Write

$$\mathbf{e}_I := \mathbf{e}_{i_1} \wedge \mathbf{e}_{i_2} \wedge \dots \wedge \mathbf{e}_{i_k} \in \Lambda^k F^n,$$

so that the  $\mathbf{e}_I$ , over all  $k$ -element  $I \subseteq [n]$ , form the basis of [Theorem 14.8.5](#).

### Theorem 14.9.5 — The Coordinates of a Wedge Are the Minors

Let  $1 \leq k \leq n$ , let  $\mathbf{v}_1, \dots, \mathbf{v}_k \in F^n$ , and let  $\mathbf{A} \in M_{n \times k}(F)$  be the matrix with these vectors as its columns. Then

$$\mathbf{v}_1 \wedge \mathbf{v}_2 \wedge \dots \wedge \mathbf{v}_k = \sum_I (\det \mathbf{A}_{I,[k]}) \mathbf{e}_I,$$

the sum running over all subsets  $I \subseteq [n]$  with  $|I| = k$ . That is, the coordinate of  $\mathbf{v}_1 \wedge \dots \wedge \mathbf{v}_k$  at  $\mathbf{e}_I$  is the  $k \times k$  minor of  $\mathbf{A}$  on the rows  $I$ .

*Idea.* The same expansion as in [Theorem 14.9.3](#), but now the index functions run into a bigger set. An injective  $\varphi: [k] \rightarrow [n]$  is a choice of image set  $I$  followed by an ordering of it, so the terms group into one bunch per  $I$ , and each bunch is a signed sum over  $S_k$ : a determinant of size  $k$ .

*Proof.* Write  $\mathbf{A} = (a_{ij})$ , so  $\mathbf{v}_j = \sum_{i=1}^n a_{ij} \mathbf{e}_i$ . Expanding the wedge in each of its  $k$  slots gives one term per function  $\varphi: [k] \rightarrow [n]$ :

$$\mathbf{v}_1 \wedge \dots \wedge \mathbf{v}_k = \sum_{\varphi} a_{\varphi(1)1} \dots a_{\varphi(k)k} \mathbf{e}_{\varphi(1)} \wedge \dots \wedge \mathbf{e}_{\varphi(k)}.$$

A non-injective  $\varphi$  repeats a basis vector, so its wedge is  $\mathbf{0}$  and its term drops out. An injective  $\varphi$  is determined by its image  $I = \{i_1 < \dots < i_k\}$  together with the permutation  $\sigma \in S_k$  defined by  $\varphi(j) = i_{\sigma(j)}$ ; conversely every pair  $(I, \sigma)$  arises from exactly one

injective  $\varphi$ . For such a  $\varphi$ , [Lemma 14.8.1](#) (b) gives

$$\mathbf{e}_{\varphi(1)} \wedge \cdots \wedge \mathbf{e}_{\varphi(k)} = \mathbf{e}_{i_{\sigma(1)}} \wedge \cdots \wedge \mathbf{e}_{i_{\sigma(k)}} = \operatorname{sgn}(\sigma) \mathbf{e}_I.$$

Grouping the terms by  $I$  therefore gives

$$\mathbf{v}_1 \wedge \cdots \wedge \mathbf{v}_k = \sum_I \left( \sum_{\sigma \in S_k} \operatorname{sgn}(\sigma) a_{i_{\sigma(1)}1} \cdots a_{i_{\sigma(k)}k} \right) \mathbf{e}_I.$$

By [Definition 6.7.1](#) the  $(p, q)$ -entry of  $\mathbf{A}_{I,[k]}$  is  $a_{i_p q}$ , so the inner sum is exactly the Leibniz sum of [Definition 6.4.1](#) for the  $k \times k$  matrix  $\mathbf{A}_{I,[k]}$ , that is,  $\det \mathbf{A}_{I,[k]}$ . This proves the theorem. ■

### Example. (A wedge of two vectors in four-dimensional space)

In  $\mathbb{Q}^4$ , let  $\mathbf{v}_1 = (1, 0, 2, 1)$  and  $\mathbf{v}_2 = (0, 1, 1, 3)$ . Write  $\mathbf{v}_1 \wedge \mathbf{v}_2$  in the basis of  $\Lambda^2 \mathbb{Q}^4$ .

*Solution.* Let  $\mathbf{A} \in M_{4 \times 2}(\mathbb{Q})$  have these two vectors as its columns, so that  $\mathbf{A}_{\{1,2\},[2]} = \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix}$  and the other five  $2 \times 2$  submatrices read off the same way. Their determinants are

$$\begin{aligned} \det \mathbf{A}_{\{1,2\}} &= 1, & \det \mathbf{A}_{\{1,3\}} &= 1, & \det \mathbf{A}_{\{1,4\}} &= 3, \\ \det \mathbf{A}_{\{2,3\}} &= -2, & \det \mathbf{A}_{\{2,4\}} &= -1, & \det \mathbf{A}_{\{3,4\}} &= 5, \end{aligned}$$

abbreviating  $\mathbf{A}_{I,[2]}$  to  $\mathbf{A}_I$ . By [Theorem 14.9.5](#),

$$\begin{aligned} \mathbf{v}_1 \wedge \mathbf{v}_2 &= \mathbf{e}_{12} + \mathbf{e}_{13} + 3\mathbf{e}_{14} \\ &\quad - 2\mathbf{e}_{23} - \mathbf{e}_{24} + 5\mathbf{e}_{34}. \end{aligned}$$

As a check, some coordinate is non-zero, so  $\mathbf{v}_1 \wedge \mathbf{v}_2 \neq \mathbf{0}$ , so  $(\mathbf{v}_1, \mathbf{v}_2)$  is linearly independent by [Theorem 14.8.7](#) — which is visible directly, since neither vector is a multiple of the other.

That check generalizes at once, and what comes out is a theorem of Chapter 6.

### Corollary 14.9.6 — Independence via Minors

Let  $\mathbf{v}_1, \dots, \mathbf{v}_k \in F^n$  and let  $\mathbf{A} \in M_{n \times k}(F)$  have these vectors as its columns. Then  $(\mathbf{v}_1, \dots, \mathbf{v}_k)$  is linearly independent if and only if some  $k \times k$  minor of  $\mathbf{A}$  is non-zero.

*Proof.* By [Theorem 14.8.7](#) the list is independent exactly when  $\mathbf{v}_1 \wedge \cdots \wedge \mathbf{v}_k \neq \mathbf{0}$ . By [Theorem 14.9.5](#) the coordinates of that wedge in the basis  $(\mathbf{e}_I)$  of [Theorem 14.8.5](#) are the  $k \times k$  minors of  $\mathbf{A}$ , and a vector is  $\mathbf{0}$  exactly when all of its coordinates in a basis vanish. (For  $k > n$  both sides fail:  $\Lambda^k F^n = \{\mathbf{0}\}$  and  $\mathbf{A}$  has no  $k \times k$  submatrix.) ■

This is [Theorem 6.7.1](#) in the case of full column rank, and the general case follows from it:  $\operatorname{rank} \mathbf{A}$  is the largest number of independent columns, hence the largest  $k$  for which some  $k$

columns carry a non-zero  $k \times k$  minor. Chapter 6 proved it with two careful steps involving pivot columns and row rank. The wedge proves it by naming the coordinates.

## 14.9.6 Laplace expansion

Cofactor expansion also comes out of the coordinate theorem, once the wedge of  $n - 1$  vectors is available. The route is: pull the chosen column out of the wedge, recognize what is left as an element of  $\Lambda^{n-1}F^n$  whose coordinates are the  $(n - 1) \times (n - 1)$  minors, and wedge  $\mathbf{e}_i$  back on.

### Corollary 14.9.7 — Cofactor Expansion, from a Wedge

Let  $n \geq 2$ , let  $\mathbf{A} \in M_n(F)$  with minors  $M_{ij}$  as in Definition 6.6.1, and fix  $j \in [n]$ . Then

$$\det \mathbf{A} = \sum_{i=1}^n (-1)^{i+j} a_{ij} M_{ij}.$$

*Idea.* Write  $\det \mathbf{A}$  as the wedge of the columns. Linearity in the  $j$ -th slot splits it into  $n$  pieces, one per  $i$ , with  $\mathbf{e}_i$  sitting in that slot. Move  $\mathbf{e}_i$  to the front at a cost of  $(-1)^{j-1}$ ; the remaining  $n - 1$  columns form a wedge whose coordinates are the minors of the matrix with column  $j$  deleted, and wedging  $\mathbf{e}_i$  on kills every coordinate except the one whose index set misses  $i$ .

*Proof.* Let  $\mathbf{a}_1, \dots, \mathbf{a}_n$  be the columns of  $\mathbf{A}$  and  $\omega_0 = \mathbf{e}_1 \wedge \dots \wedge \mathbf{e}_n$ . By Theorem 14.9.3 applied to the operator  $\mathbf{x} \mapsto \mathbf{A}\mathbf{x}$ , whose matrix in the standard basis is  $\mathbf{A}$ ,

$$\mathbf{a}_1 \wedge \dots \wedge \mathbf{a}_n = (\det \mathbf{A}) \omega_0.$$

Substituting  $\mathbf{a}_j = \sum_{i=1}^n a_{ij} \mathbf{e}_i$  and using linearity in slot  $j$ ,

$$(\det \mathbf{A}) \omega_0 = \sum_{i=1}^n a_{ij} \mathbf{a}_1 \wedge \dots \wedge \underset{\text{slot } j}{\mathbf{e}_i} \wedge \dots \wedge \mathbf{a}_n.$$

Fix  $i$ . Moving  $\mathbf{e}_i$  from slot  $j$  to the front means  $j - 1$  exchanges of neighbors, so by Lemma 14.8.1 (b) the  $i$ -th wedge above equals  $(-1)^{j-1} \varepsilon_{\mathbf{e}_i}(\eta)$ , where  $\eta = \mathbf{a}_1 \wedge \dots \wedge \widehat{\mathbf{a}_j} \wedge \dots \wedge \mathbf{a}_n \in \Lambda^{n-1}F^n$  omits the  $j$ -th column, and  $\varepsilon_{\mathbf{e}_i}$  is the map of Lemma 14.9.1.

Let  $\mathbf{B} \in M_{n \times (n-1)}(F)$  be  $\mathbf{A}$  with column  $j$  deleted. By Theorem 14.9.5,

$$\eta = \sum_{l=1}^n (\det \mathbf{B}_{[n] \setminus \{l\}, [n-1]}) \mathbf{e}_{[n] \setminus \{l\}},$$

since the  $(n - 1)$ -element subsets of  $[n]$  are exactly the complements of single elements. Deleting row  $l$  from  $\mathbf{B}$  deletes row  $l$  and column  $j$  from  $\mathbf{A}$ , so  $\det \mathbf{B}_{[n] \setminus \{l\}, [n-1]} = M_{lj}$  by Definition 6.6.1. Now apply  $\varepsilon_{\mathbf{e}_i}$ . For  $l \neq i$  the index set  $[n] \setminus \{l\}$  contains  $i$ , so  $\mathbf{e}_i \wedge \mathbf{e}_{[n] \setminus \{l\}}$  repeats  $\mathbf{e}_i$  and vanishes; for  $l = i$ , moving  $\mathbf{e}_i$  into its place costs  $i - 1$  exchanges, so  $\mathbf{e}_i \wedge \mathbf{e}_{[n] \setminus \{i\}} = (-1)^{i-1} \omega_0$  by Lemma 14.8.1 (b). Hence  $\varepsilon_{\mathbf{e}_i}(\eta) = (-1)^{i-1} M_{ij} \omega_0$ , and

$$(\det \mathbf{A}) \omega_0 = \sum_{i=1}^n a_{ij} (-1)^{j-1} (-1)^{i-1} M_{ij} \omega_0.$$

Since  $(-1)^{i+j-2} = (-1)^{i+j}$  and  $\omega_0 \neq \mathbf{0}$  (Theorem 14.8.7), comparing coefficients gives the stated formula. ■

This is Theorem 6.6.1 (a). The row version follows from it by Theorem 6.4.5, exactly as in Chapter 6. The gain is again structural rather than practical: the sign  $(-1)^{i+j}$ , which in Chapter 6 arrived from a careful count of row and column moves, is here the price of two reversals,  $j - 1$  in one direction and  $i - 1$  in the other.

## 14.9.7 Exercises

### 14.9.7.1 A. Check your understanding

#### Exercise A1.

- (a) State the defining property of  $\Lambda^k T$ , and say why a rule given on wedges needs an argument before it defines a map.
- (b) State Theorem 14.9.3.
- (c) Name one thing this section's construction of the determinant adds to Chapter 6's, and one thing it does not.
- (d) True or false: for every  $T \in \mathcal{L}(V)$  and every  $k$ , the map  $\Lambda^k T$  is multiplication by a scalar. Justify your answer.
- (e) What are the coordinates of  $\mathbf{v}_1 \wedge \cdots \wedge \mathbf{v}_k$  in  $\Lambda^k F^n$ , in the basis  $(\mathbf{e}_I)$ ?

*Solution.*

- (a)  $\Lambda^k T$  is the unique linear map  $\Lambda^k V \rightarrow \Lambda^k W$  with  $\Lambda^k T(\mathbf{v}_1 \wedge \cdots \wedge \mathbf{v}_k) = T\mathbf{v}_1 \wedge \cdots \wedge T\mathbf{v}_k$  (Definition 14.9.1). An element of  $\Lambda^k V$  has many expressions as a sum of wedges, so a rule stated on wedges might assign two different values to one element; the universal property of Definition 14.8.2 removes the danger by producing the map from an alternating multilinear map upstairs.
- (b) If  $\dim V = n \geq 1$  and  $T \in \mathcal{L}(V)$ , then  $\dim \Lambda^n V = 1$  and  $\Lambda^n T(\omega) = (\det T)\omega$  for every  $\omega \in \Lambda^n V$ .
- (c) It adds that the determinant is the induced map on a line, so that no basis and no similarity-invariance argument is needed to define  $\det T$ , and multiplicativity becomes Theorem 14.9.2. It does not add a new value, a new method of computation, or any improvement to  $\det \mathbf{A}^\top = \det \mathbf{A}$ .
- (d) False. It is true when  $k = n = \dim V$ , and when  $k > n$  for the trivial reason that  $\Lambda^k V = \{\mathbf{0}\}$ . For  $1 \leq k < n$  it usually fails: see Example (The second exterior power of a  $3 \times 3$  matrix), where  $\Lambda^2 T$  has matrix with rows  $(8, 2, -11)$ ,  $(-1, -13, -5)$ ,  $(-20, -5, 2)$ , not a scalar matrix.
- (e) The  $k \times k$  minors of the matrix whose columns are  $\mathbf{v}_1, \dots, \mathbf{v}_k$ : the coordinate at  $\mathbf{e}_I$  is  $\det \mathbf{A}_{I,[k]}$  (Theorem 14.9.5).

### 14.9.7.2 B. Practice

**Exercise B1: A second exterior power.**

Let  $T: \mathbb{Q}^3 \rightarrow \mathbb{Q}^3$  have standard matrix

$$\mathbf{A} = \begin{pmatrix} 1 & 2 & 0 \\ 0 & 1 & 3 \\ 2 & 0 & 1 \end{pmatrix}.$$

Find the matrix of  $\Lambda^2 T$  in the basis  $(\mathbf{e}_1 \wedge \mathbf{e}_2, \mathbf{e}_1 \wedge \mathbf{e}_3, \mathbf{e}_2 \wedge \mathbf{e}_3)$ . Hence verify that  $\det(\Lambda^2 T) = (\det T)^2$ .

*Solution.* By [Definition 14.9.1](#) and [Theorem 14.9.5](#), the entry of  $[\Lambda^2 T]$  in row  $I$  and column  $J$  is  $\det \mathbf{A}_{I,J}$ , with rows and columns indexed by  $\{1, 2\}, \{1, 3\}, \{2, 3\}$  in that order. Computing the nine  $2 \times 2$  minors,

$$[\Lambda^2 T] = \begin{pmatrix} 1 & 3 & 6 \\ -4 & 1 & 2 \\ -2 & -6 & 1 \end{pmatrix}.$$

Its determinant is 169. Expanding  $\det \mathbf{A}$  along the first column ([Theorem 6.6.1](#)) gives  $1 \cdot (1 - 0) + 2 \cdot (6 - 0) = 13$ , and  $13^2 = 169$ .

**Exercise B2: Independence by minors.**

In  $\mathbb{Q}^4$ , let  $\mathbf{v}_1 = (1, 2, 0, 1)$ ,  $\mathbf{v}_2 = (0, 1, 1, 1)$  and  $\mathbf{v}_3 = (1, 0, -2, -1)$ . Compute the coordinates of  $\mathbf{v}_1 \wedge \mathbf{v}_2 \wedge \mathbf{v}_3$  in  $\Lambda^3 \mathbb{Q}^4$ , and hence determine whether the list is linearly independent. Justify your answer.

*Solution.* Let  $\mathbf{A} \in M_{4 \times 3}(\mathbb{Q})$  have these columns. By [Theorem 14.9.5](#) the four coordinates are the  $3 \times 3$  minors on the row sets  $\{1, 2, 3\}, \{1, 2, 4\}, \{1, 3, 4\}, \{2, 3, 4\}$ . Computing them,

$$\det \mathbf{A}_{\{1,2,3\}} = \det \begin{pmatrix} 1 & 0 & 1 \\ 2 & 1 & 0 \\ 0 & 1 & -2 \end{pmatrix} = 0,$$

and the same for the other three:  $\det \mathbf{A}_{\{1,2,4\}} = \det \mathbf{A}_{\{1,3,4\}} = \det \mathbf{A}_{\{2,3,4\}} = 0$ . So  $\mathbf{v}_1 \wedge \mathbf{v}_2 \wedge \mathbf{v}_3 = \mathbf{0}$ , and by [Theorem 14.8.7](#) the list is **dependent**. Indeed  $\mathbf{v}_3 = \mathbf{v}_1 - 2\mathbf{v}_2$ .

**Exercise B3: Determinant of an inverse.**

Let  $\dim V = n \geq 1$  and let  $T \in \mathcal{L}(V)$  be invertible. Using [Theorem 14.9.2](#) and [Theorem 14.9.3](#), and no formula for the determinant, prove that  $\Lambda^n T$  is invertible and that  $\det(T^{-1}) = (\det T)^{-1}$ .

*Solution.* By [Theorem 14.9.2](#),  $(\Lambda^n T)(\Lambda^n T^{-1}) = \Lambda^n(TT^{-1}) = \Lambda^n(\text{id}_V) = \text{id}$ , and symmetrically in the other order. So  $\Lambda^n T$  is invertible with inverse  $\Lambda^n T^{-1}$ . By [Theorem 14.9.3](#) these two maps are multiplication by  $\det T$  and by  $\det(T^{-1})$  on the line  $\Lambda^n V$ , and their composite is multiplication by  $\det T \cdot \det(T^{-1})$ , which equals  $\text{id}$ . Since  $\Lambda^n V \neq \{\mathbf{0}\}$ , this forces  $\det T \cdot \det(T^{-1}) = 1$ . In particular  $\det T \neq 0$ , and  $\det(T^{-1}) = (\det T)^{-1}$ .

## 14.9.7.3 C. Going deeper

**Exercise C1: The trace of an exterior power.**

Let  $\mathbf{A} \in M_n(F)$  and  $1 \leq k \leq n$ , and let  $T: F^n \rightarrow F^n$  be  $\mathbf{x} \mapsto \mathbf{A}\mathbf{x}$ .

- (a) Prove that the matrix of  $\Lambda^k T$  in the basis  $(\mathbf{e}_I)$  has  $(I, J)$ -entry  $\det \mathbf{A}_{I,J}$ .
- (b) Deduce that  $\text{tr}(\Lambda^k T)$  is the sum of the  $\binom{n}{k}$  **principal**  $k \times k$  minors of  $\mathbf{A}$ , those with  $I = J$ .
- (c) Compute  $\text{tr}(\Lambda^2 T)$  for the matrix  $\mathbf{A}$  of [Example \(The second exterior power of a  \$3 \times 3\$  matrix\)](#), and check it against the matrix found there.

*Solution.*

- (a) Column  $J = \{j_1 < \dots < j_k\}$  of the matrix is the coordinate vector of  $\Lambda^k T(\mathbf{e}_J) = \mathbf{a}_{j_1} \wedge \dots \wedge \mathbf{a}_{j_k}$ , where  $\mathbf{a}_j = T\mathbf{e}_j$  is the  $j$ -th column of  $\mathbf{A}$  ([Definition 14.9.1](#)). By [Theorem 14.9.5](#) applied to the  $n \times k$  matrix with columns  $\mathbf{a}_{j_1}, \dots, \mathbf{a}_{j_k}$ , the coordinate at  $\mathbf{e}_I$  is the determinant of its submatrix on rows  $I$ , which is  $\det \mathbf{A}_{I,J}$  ([Definition 6.7.1](#)).
- (b) The trace is the sum of the diagonal entries, that is, of the entries with  $I = J$ , and by (a) those are the principal minors  $\det \mathbf{A}_{I,I}$ .
- (c) The three principal  $2 \times 2$  minors of  $\mathbf{A}$  are  $\det(\begin{smallmatrix} 2 & 1 \\ 0 & 4 \end{smallmatrix}) = 8$ ,  $\det(\begin{smallmatrix} 2 & 3 \\ 5 & 1 \end{smallmatrix}) = -13$  and  $\det(\begin{smallmatrix} 4 & 1 \\ 2 & 1 \end{smallmatrix}) = 2$ . Their sum is  $-3$ , which is the sum of the diagonal entries  $8, -13, 2$  of the matrix computed in [Example \(The second exterior power of a  \$3 \times 3\$  matrix\)](#).

**Exercise C2: Cauchy–Binet from a wedge.**

Let  $m \leq n$ , let  $\mathbf{A} \in M_{m \times n}(F)$  and  $\mathbf{B} \in M_{n \times m}(F)$ , and let  $T_{\mathbf{A}}: F^n \rightarrow F^m$  and  $T_{\mathbf{B}}: F^m \rightarrow F^n$  be the associated maps.

- (a) Show that  $\Lambda^m T_{\mathbf{B}}(\mathbf{e}_1 \wedge \dots \wedge \mathbf{e}_m) = \sum_J (\det \mathbf{B}_{J,[m]}) \mathbf{e}_J$ , the sum over  $m$ -element subsets  $J \subseteq [n]$ .
- (b) Show that  $\Lambda^m T_{\mathbf{A}}(\mathbf{e}_J) = (\det \mathbf{A}_{[m],J}) \mathbf{e}_1 \wedge \dots \wedge \mathbf{e}_m$  for each such  $J$ .
- (c) Hence deduce the Cauchy–Binet formula  $\det(\mathbf{AB}) = \sum_J \det \mathbf{A}_{[m],J} \det \mathbf{B}_{J,[m]}$  ([Theorem 6.7.2](#)).

*Solution.*

- (a)  $\Lambda^m T_{\mathbf{B}}(\mathbf{e}_1 \wedge \dots \wedge \mathbf{e}_m) = \mathbf{B}\mathbf{e}_1 \wedge \dots \wedge \mathbf{B}\mathbf{e}_m$  by [Definition 14.9.1](#), and the columns of  $\mathbf{B}$  are  $\mathbf{B}\mathbf{e}_1, \dots, \mathbf{B}\mathbf{e}_m$ . Apply [Theorem 14.9.5](#) in  $F^n$  with  $k = m$ .
- (b) Writing  $J = \{j_1 < \dots < j_m\}$ , we get  $\Lambda^m T_{\mathbf{A}}(\mathbf{e}_J) = \mathbf{A}\mathbf{e}_{j_1} \wedge \dots \wedge \mathbf{A}\mathbf{e}_{j_m}$ , a wedge of  $m$  vectors of  $F^m$ . By [Theorem 14.9.5](#) with  $k = m = n$  there is only one index set,  $[m]$  itself, and the coefficient is the determinant of the matrix with those columns, which is  $\det \mathbf{A}_{[m],J}$ .
- (c) By [Theorem 14.9.2](#),  $\Lambda^m(T_{\mathbf{A}}T_{\mathbf{B}}) = (\Lambda^m T_{\mathbf{A}})(\Lambda^m T_{\mathbf{B}})$ . The map  $T_{\mathbf{A}}T_{\mathbf{B}}$  is the operator on  $F^m$  with matrix  $\mathbf{AB}$ , so by [Theorem 14.9.3](#) the left side sends  $\mathbf{e}_1 \wedge \dots \wedge \mathbf{e}_m$  to  $\det(\mathbf{AB}) \mathbf{e}_1 \wedge \dots \wedge \mathbf{e}_m$ . By (a) and then (b), the right side sends it to  $\sum_J \det \mathbf{B}_{J,[m]} \det \mathbf{A}_{[m],J} \mathbf{e}_1 \wedge \dots \wedge \mathbf{e}_m$ . Comparing coefficients on the basis vector  $\mathbf{e}_1 \wedge \dots \wedge \mathbf{e}_m \neq \mathbf{0}$  gives the formula.



**Exercise C3: Exterior powers of an injection.**

Let  $T: V \rightarrow W$  be an **injective** linear map between finite-dimensional spaces and let  $1 \leq k \leq \dim V$ . Prove that  $\Lambda^k T$  is injective. Is the analogous statement true with “injective” replaced by “surjective”?

*Hint: choose a basis of  $V$  and use Theorem 14.8.5 in both spaces.*

*Solution.* Let  $(\mathbf{v}_1, \dots, \mathbf{v}_m)$  be a basis of  $V$ . Since  $T$  is injective,  $(T\mathbf{v}_1, \dots, T\mathbf{v}_m)$  is linearly independent in  $W$ , so it extends to a basis  $(T\mathbf{v}_1, \dots, T\mathbf{v}_m, \mathbf{w}_{m+1}, \dots)$  of  $W$  (Theorem 1.6.5). By Theorem 14.8.5, the wedges  $\mathbf{v}_{i_1} \wedge \dots \wedge \mathbf{v}_{i_k}$  over increasing tuples form a basis of  $\Lambda^k V$ , and  $\Lambda^k T$  sends each of them to  $T\mathbf{v}_{i_1} \wedge \dots \wedge T\mathbf{v}_{i_k}$ , which is one of the basis vectors of  $\Lambda^k W$  provided by the extended basis. So  $\Lambda^k T$  carries a basis to a linearly independent list, hence is injective.

Yes for surjective as well, and more cheaply: if  $T$  is surjective then every  $\mathbf{w}_1 \wedge \dots \wedge \mathbf{w}_k$  equals  $T\mathbf{v}_1 \wedge \dots \wedge T\mathbf{v}_k$  for preimages  $\mathbf{v}_i$ , so the image of  $\Lambda^k T$  contains a spanning set of  $\Lambda^k W$  (Theorem 14.8.5) and is therefore all of it.

## 14.10 The Exterior Algebra, and Clifford at Last

The exterior powers have so far been used one degree at a time, and the wedge of two of them has had nowhere to live:  $\Lambda^2 V$  and  $\Lambda^3 V$  are different spaces, and until now nothing said what  $(\mathbf{u} \wedge \mathbf{w}) \wedge \mathbf{x}$  means as an operation. Stacking them fixes that, exactly as stacking the tensor powers produced the tensor algebra (Definition 14.6.2) and stacking the symmetric powers produced the symmetric algebra (Definition 14.7.3). The result is a  $2^n$ -dimensional algebra, and it is the last object this chapter needs. With it we can finally pay the debt Chapter 13 recorded in writing: a Clifford algebra of  $(V, q)$  exists, and its dimension is  $2^n$ .

Throughout,  $F$  is a field and  $V$  is a vector space over  $F$  with  $\dim V = n$ . From the subsection on decomposability on,  $\text{char } F \neq 2$ , because Definition 13.12.2 assumes it. We use the exterior powers of Definition 14.8.2 with their universal property for alternating maps, their bases of increasing wedges (Theorem 14.8.5), the test Theorem 14.8.7, and the reordering rule Lemma 14.8.1 (b) of the previous section.

### 14.10.1 Stacking the exterior powers

The pieces are already in place; only the multiplication is missing. The pattern is the one Proposition 14.7.4 used for symmetric powers: a product of a  $k$ -fold wedge and an  $l$ -fold wedge ought to be the  $(k + l)$ -fold wedge of all the vectors, and the only thing to check is that this depends on the two factors and not on the way each was written.

#### Proposition 14.10.1 — The Wedge of Two Exterior Powers

Let  $k, l \geq 1$ . There is a **unique** bilinear map

$$\Lambda^k V \times \Lambda^l V \rightarrow \Lambda^{k+l} V, \quad (\omega, \eta) \mapsto \omega \wedge \eta,$$

such that for all  $\mathbf{u}_1, \dots, \mathbf{u}_k, \mathbf{w}_1, \dots, \mathbf{w}_l \in V$ ,

$$(\mathbf{u}_1 \wedge \dots \wedge \mathbf{u}_k) \wedge (\mathbf{w}_1 \wedge \dots \wedge \mathbf{w}_l) = \mathbf{u}_1 \wedge \dots \wedge \mathbf{u}_k \wedge \mathbf{w}_1 \wedge \dots \wedge \mathbf{w}_l.$$

*Idea.* Freeze the second factor's vectors and use the universal property once, in the first  $k$  slots; that turns the left-hand factor into an element of  $\Lambda^k V$ . Then let the second factor's vectors vary: what we have built depends on them  $l$ -linearly, and vanishes as soon as two of them coincide, with values in a space of linear maps, so the universal property applies a second time.

*Proof.* Fix  $\mathbf{w}_1, \dots, \mathbf{w}_l \in V$ . The map  $V^k \rightarrow \Lambda^{k+l} V$  given by

$$(\mathbf{u}_1, \dots, \mathbf{u}_k) \mapsto \mathbf{u}_1 \wedge \dots \wedge \mathbf{u}_k \wedge \mathbf{w}_1 \wedge \dots \wedge \mathbf{w}_l$$

is  $k$ -linear, since the wedge of  $k+l$  vectors is linear in each slot and the last  $l$  are frozen, and alternating, since a repeat among the first  $k$  arguments is a repeat among all  $k+l$ . By [Definition 14.8.2](#) it induces a unique linear map

$$R(\mathbf{w}_1, \dots, \mathbf{w}_l): \Lambda^k V \rightarrow \Lambda^{k+l} V$$

with  $R(\mathbf{w}_1, \dots, \mathbf{w}_l)(\mathbf{u}_1 \wedge \dots \wedge \mathbf{u}_k)$  equal to the displayed wedge.

Now let the  $\mathbf{w}$ 's vary. The assignment  $(\mathbf{w}_1, \dots, \mathbf{w}_l) \mapsto R(\mathbf{w}_1, \dots, \mathbf{w}_l)$ , with values in the vector space  $\mathcal{L}(\Lambda^k V, \Lambda^{k+l} V)$ , is  $l$ -linear and alternating: two linear maps on  $\Lambda^k V$  are equal as soon as they agree on the wedges  $\mathbf{u}_1 \wedge \dots \wedge \mathbf{u}_k$ , which span  $\Lambda^k V$  ([Theorem 14.8.5](#)), and on those the required identities are the linearity and the alternation of the wedge of  $k+l$  vectors in its last  $l$  slots. So [Definition 14.8.2](#) applies again and gives a unique linear map

$$\Lambda^l V \rightarrow \mathcal{L}(\Lambda^k V, \Lambda^{k+l} V), \quad \mathbf{w}_1 \wedge \dots \wedge \mathbf{w}_l \mapsto R(\mathbf{w}_1, \dots, \mathbf{w}_l).$$

Reading a linear map into a space of linear maps as a bilinear map gives the required product, with the stated values. It is unique because the wedges span both factors and a bilinear map is determined on a pair of spanning sets. This proves the proposition. ■

With the product in hand, the algebra assembles itself.

*The exterior algebra is all the exterior powers at once, multiplied by wedging.*

### Definition 14.10.1 — Exterior Algebra

Let  $V$  be a vector space over  $F$ , with  $\Lambda^0 V = F$  and  $\Lambda^1 V = V$  as in [Definition 14.8.2](#). The **exterior algebra**, or **Grassmann algebra**, of  $V$  is the vector space

$$\Lambda V := \bigoplus_{k \geq 0} \Lambda^k V,$$

whose elements are the sequences  $x = (x_0, x_1, \dots)$  with  $x_k \in \Lambda^k V$  and only finitely many  $x_k$  non-zero, with componentwise operations, and whose multiplication is

$$(x \wedge y)_m = \sum_{k+l=m} x_k \wedge y_l,$$

where  $\wedge$  on the right is [Proposition 14.10.1](#) for  $k, l \geq 1$ , and is scalar multiplication when  $k = 0$  or  $l = 0$ . We identify each  $\Lambda^k V$  with the sequences supported in position  $k$ , so that  $F$  and  $V$  are subspaces of  $\Lambda V$ , and call an element of  $\Lambda^k V \subseteq \Lambda V$  **homogeneous of degree  $k$** .

In words: an element of  $\Lambda V$  is a finite sum  $x_0 + x_1 + \dots + x_N$  of pieces of different degrees, the scalars sit inside as the degree-0 piece, and **the product adds degrees**. The finiteness clause is automatic here, since  $\Lambda^k V = \{0\}$  once  $k > n$  ([Theorem 14.8.5](#)).

### Theorem 14.10.2 — Structure of the Exterior Algebra

Let  $\dim V = n$  with basis  $(\mathbf{e}_1, \dots, \mathbf{e}_n)$ . For a subset  $S = \{i_1 < \dots < i_k\} \subseteq [n]$  write  $\mathbf{e}_S = \mathbf{e}_{i_1} \wedge \dots \wedge \mathbf{e}_{i_k}$ , with  $\mathbf{e}_\emptyset = 1$ . Then

- (a)  $\Lambda V$  is an associative unital  $F$ -algebra with identity  $1 \in \Lambda^0 V = F$ , and  $\Lambda^k V \wedge \Lambda^l V \subseteq \Lambda^{k+l} V$ .
- (b) The  $2^n$  elements  $\mathbf{e}_S$ , one for each subset  $S \subseteq [n]$ , form a basis of  $\Lambda V$ . In particular  $\dim_F \Lambda V = 2^n$ .
- (c)  $\Lambda V$  is generated as an algebra by 1 and  $V$ .

*Proof.*

- (a) Bilinearity of the product holds componentwise, by [Proposition 14.10.1](#). For associativity, both  $(x \wedge y) \wedge z$  and  $x \wedge (y \wedge z)$  are trilinear in  $(x, y, z)$ , so it is enough to compare them on homogeneous elements, and then on the wedges of vectors, which span each  $\Lambda^k V$  for  $k \geq 1$  ([Theorem 14.8.5](#)) while  $\Lambda^0 V$  is spanned by 1. On those both sides are the single wedge of all the listed vectors in order, by the defining identity of [Proposition 14.10.1](#) applied twice. The scalar 1 acts on each  $\Lambda^k V$  as multiplication by 1, hence as the identity. The grading statement is built into the definition of the product.
- (b) By [Theorem 14.8.5](#) the  $\mathbf{e}_S$  with  $|S| = k$  form a basis of  $\Lambda^k V$ , and  $\Lambda^k V = \{0\}$  for  $k > n$ . A basis of a direct sum is the union of bases of the summands, so the  $\mathbf{e}_S$  over all  $S \subseteq [n]$  form a basis of  $\Lambda V$ . Their number is  $\sum_{k=0}^n \binom{n}{k} = 2^n$ , by the binomial theorem applied to  $(1 + 1)^n$ .
- (c) Each  $\mathbf{e}_S$  is a product of elements of  $V$  by the defining identity of the product, and the  $\mathbf{e}_S$  span by (b).

■

The one structural fact that is not inherited from the tensor algebra is how the order of a product matters.

**Theorem 14.10.3 — Graded Commutativity**

Let  $\omega \in \Lambda^k V$  and  $\eta \in \Lambda^l V$ . Then

$$\omega \wedge \eta = (-1)^{kl} \eta \wedge \omega.$$

In particular  $\omega \wedge \omega = \mathbf{0}$  whenever  $k$  is **odd**, over every field.

*Idea.* Count the swaps. To turn  $\mathbf{w}_1 \wedge \cdots \wedge \mathbf{w}_l \wedge \mathbf{u}_1 \wedge \cdots \wedge \mathbf{u}_k$  into  $\mathbf{u}_1 \wedge \cdots \wedge \mathbf{u}_k \wedge \mathbf{w}_1 \wedge \cdots \wedge \mathbf{w}_l$ , each of the  $k$  vectors  $\mathbf{u}_i$  must cross each of the  $l$  vectors  $\mathbf{w}_j$ , and every crossing costs a sign.

*Proof.* Both sides are bilinear in  $(\omega, \eta)$ , so by Theorem 14.8.5 it suffices to take  $\omega = \mathbf{u}_1 \wedge \cdots \wedge \mathbf{u}_k$  and  $\eta = \mathbf{w}_1 \wedge \cdots \wedge \mathbf{w}_l$ , and the cases  $k = 0$  or  $l = 0$  are scalar multiplication and are clear. The permutation of  $\{1, \dots, k+l\}$  carrying the list  $(\mathbf{w}_1, \dots, \mathbf{w}_l, \mathbf{u}_1, \dots, \mathbf{u}_k)$  to  $(\mathbf{u}_1, \dots, \mathbf{u}_k, \mathbf{w}_1, \dots, \mathbf{w}_l)$  has exactly  $kl$  inversions, one for each pair consisting of a  $\mathbf{u}$  and a  $\mathbf{w}$ , so its sign is  $(-1)^{kl}$  (Definition 6.3.2). By Lemma 14.8.1 (b) the two wedges differ by that factor, which is the claim.

For the last sentence let  $k$  be odd and  $\omega \in \Lambda^k V$ . Write  $\omega = \sum_S c_S \mathbf{e}_S$  in the basis of Theorem 14.10.2, the sum over the  $S$  with  $|S| = k$ . Then

$$\omega \wedge \omega = \sum_S \sum_T c_S c_T \mathbf{e}_S \wedge \mathbf{e}_T.$$

A term with  $S = T$  has a repeated vector, so  $\mathbf{e}_S \wedge \mathbf{e}_S = \mathbf{0}$ . A pair  $S \neq T$  occurs twice, and the two occurrences are  $c_S c_T \mathbf{e}_S \wedge \mathbf{e}_T$  and  $c_T c_S \mathbf{e}_T \wedge \mathbf{e}_S = -c_S c_T \mathbf{e}_S \wedge \mathbf{e}_T$ , since  $(-1)^{kl} = (-1)^{k^2} = -1$  for odd  $k$ . So the two cancel, and nothing divides by 2 anywhere. ■

**Warning. Graded commutative is not commutative.** In  $\Lambda F^2$  we have  $\mathbf{e}_1 \wedge \mathbf{e}_2 = -\mathbf{e}_2 \wedge \mathbf{e}_1$ , and these are different elements unless  $\text{char } F = 2$ . What is true in all degrees is only the signed rule. And the sign is  $+1$  as soon as one of the degrees is even, so an element of  $\Lambda^2 V$  commutes with everything, while two elements of  $\Lambda^1 V = V$  anticommute.

**Warning.  $\omega \wedge \omega$  need not be zero.** The identity  $\mathbf{v} \wedge \mathbf{v} = \mathbf{0}$  is a statement about degree 1. For  $\omega = \mathbf{e}_1 \wedge \mathbf{e}_2 + \mathbf{e}_3 \wedge \mathbf{e}_4$  in  $\Lambda F^4$ , a direct expansion gives  $\omega \wedge \omega = 2 \mathbf{e}_1 \wedge \mathbf{e}_2 \wedge \mathbf{e}_3 \wedge \mathbf{e}_4$ , which is non-zero whenever  $\text{char } F \neq 2$ . Squaring to zero is the privilege of odd degree.

**Example. (The exterior algebra of a three-dimensional space)**

Describe  $\Lambda F^3$  completely, and compute  $(\mathbf{e}_1 + \mathbf{e}_2 \wedge \mathbf{e}_3) \wedge (\mathbf{e}_2 + \mathbf{e}_1 \wedge \mathbf{e}_3)$ .

*Solution.* By Theorem 14.10.2 it has dimension  $2^3 = 8$ , with basis

$$1; \quad \mathbf{e}_1, \mathbf{e}_2, \mathbf{e}_3; \quad \mathbf{e}_{12}, \mathbf{e}_{13}, \mathbf{e}_{23}; \quad \mathbf{e}_{123},$$

writing  $\mathbf{e}_{ij}$  for  $\mathbf{e}_i \wedge \mathbf{e}_j$  and  $\mathbf{e}_{123}$  for  $\mathbf{e}_1 \wedge \mathbf{e}_2 \wedge \mathbf{e}_3$ . The degrees have dimensions 1, 3, 3, 1. Every product of two elements of degree  $\geq 2$  lands in degree  $\geq 4$ , hence is  $\mathbf{0}$ .

For the computation, expand by bilinearity into four products:

$$\begin{aligned} \mathbf{e}_1 \wedge \mathbf{e}_2 &= \mathbf{e}_{12}, \\ \mathbf{e}_1 \wedge (\mathbf{e}_1 \wedge \mathbf{e}_3) &= \mathbf{0} \quad (\text{repeated } \mathbf{e}_1), \\ (\mathbf{e}_2 \wedge \mathbf{e}_3) \wedge \mathbf{e}_2 &= \mathbf{0} \quad (\text{repeated } \mathbf{e}_2), \\ (\mathbf{e}_2 \wedge \mathbf{e}_3) \wedge (\mathbf{e}_1 \wedge \mathbf{e}_3) &= \mathbf{0} \quad (\text{degree } 4). \end{aligned}$$

So the product is  $\mathbf{e}_{12}$ , while the reversed product is  $-\mathbf{e}_{12}$  plus the same three zeros.

*Quick check.* In  $\Lambda F^4$ , what is  $(\mathbf{e}_1 \wedge \mathbf{e}_2) \wedge (\mathbf{e}_2 \wedge \mathbf{e}_3)$ , and what is  $(\mathbf{e}_1 \wedge \mathbf{e}_2) \wedge (\mathbf{e}_3 \wedge \mathbf{e}_4)$ ?

*Solution.* The first is  $\mathbf{e}_1 \wedge \mathbf{e}_2 \wedge \mathbf{e}_2 \wedge \mathbf{e}_3 = \mathbf{0}$ , since  $\mathbf{e}_2$  is repeated. The second is  $\mathbf{e}_1 \wedge \mathbf{e}_2 \wedge \mathbf{e}_3 \wedge \mathbf{e}_4$ , a basis vector of  $\Lambda^4 F^4$ , so it is non-zero. Wedging two 2-vectors gives zero exactly when the four vectors involved are dependent.

## 14.10.2 Decomposable elements

Section 2 warned that not every element of  $V \otimes W$  is a simple tensor  $\mathbf{v} \otimes \mathbf{w}$ , and proved it with  $\mathbf{e}_1 \otimes \mathbf{e}_1 + \mathbf{e}_2 \otimes \mathbf{e}_2$  (Example (A tensor that is not simple)). The same thing happens one floor up, and for the same reason: the simple objects form a low-dimensional cone inside a large space, and a sum of two of them usually escapes it.

### Definition 14.10.2 — Decomposable Element

Let  $k \geq 1$ . An element  $\omega \in \Lambda^k V$  is **decomposable**, or a  **$k$ -blade**, if there exist  $\mathbf{v}_1, \dots, \mathbf{v}_k \in V$  with

$$\omega = \mathbf{v}_1 \wedge \mathbf{v}_2 \wedge \cdots \wedge \mathbf{v}_k.$$

**Examples.** Every element of  $\Lambda^1 V = V$  is decomposable, with  $k = 1$ ; so is  $\mathbf{0}$ , being  $\mathbf{v} \wedge \mathbf{v} \wedge \mathbf{v}_3 \wedge \cdots \wedge \mathbf{v}_k$ ; and so is every element of the line  $\Lambda^n V$ , since  $c\mathbf{e}_1 \wedge \cdots \wedge \mathbf{e}_n = (c\mathbf{e}_1) \wedge \mathbf{e}_2 \wedge \cdots \wedge \mathbf{e}_n$ .

**Why the word.** A decomposable  $\omega \neq \mathbf{0}$  carries a  $k$ -dimensional subspace: its factors are independent by Theorem 14.8.7, and they span one. Non-decomposable elements are exactly those that come from no single subspace, which is why the failure below is a phenomenon and not a notational accident.

### Theorem 14.10.4 — Not Every 2-Vector Is Decomposable

Let  $\dim V = n \geq 4$ , let  $(\mathbf{e}_1, \dots, \mathbf{e}_n)$  be a basis of  $V$ , and set

$$\omega = \mathbf{e}_1 \wedge \mathbf{e}_2 + \mathbf{e}_3 \wedge \mathbf{e}_4 \in \Lambda^2 V.$$

Then  $\omega$  is **not** decomposable.

*Idea.* A decomposable 2-vector  $\mathbf{u} \wedge \mathbf{w}$  squares to zero, because the four-fold wedge  $\mathbf{u} \wedge \mathbf{w} \wedge \mathbf{u} \wedge \mathbf{w}$  repeats a vector. So compute  $\omega \wedge \omega$  and find it non-zero. That test needs

$\text{char } F \neq 2$ ; the proof below takes a second route that does not, by asking which vectors annihilate  $\omega$ .

*Proof.* Suppose  $\omega = \mathbf{u} \wedge \mathbf{w}$  for some  $\mathbf{u}, \mathbf{w} \in V$ . Since  $\mathbf{e}_1 \wedge \mathbf{e}_2$  and  $\mathbf{e}_3 \wedge \mathbf{e}_4$  are distinct basis vectors of  $\Lambda^2 V$  (Theorem 14.8.5),  $\omega \neq \mathbf{0}$ , so  $(\mathbf{u}, \mathbf{w})$  is linearly independent by Theorem 14.8.7; in particular  $\mathbf{u} \neq \mathbf{0}$ .

Consider  $\mathbf{u} \wedge \omega = \mathbf{u} \wedge \mathbf{u} \wedge \mathbf{w} = \mathbf{0}$ , the wedge having a repeated argument. We show this is impossible. Let  $\mathbf{v} = \sum_{i=1}^n x_i \mathbf{e}_i$  be any vector and compute  $\mathbf{v} \wedge \omega$ , using bilinearity and dropping every term with a repeated basis vector:

$$\begin{aligned} \mathbf{v} \wedge \omega &= x_3 \mathbf{e}_3 \wedge \mathbf{e}_1 \wedge \mathbf{e}_2 + x_4 \mathbf{e}_4 \wedge \mathbf{e}_1 \wedge \mathbf{e}_2 \\ &\quad + x_1 \mathbf{e}_1 \wedge \mathbf{e}_3 \wedge \mathbf{e}_4 + x_2 \mathbf{e}_2 \wedge \mathbf{e}_3 \wedge \mathbf{e}_4 \\ &\quad + \sum_{i \geq 5} x_i (\mathbf{e}_i \wedge \mathbf{e}_1 \wedge \mathbf{e}_2 + \mathbf{e}_i \wedge \mathbf{e}_3 \wedge \mathbf{e}_4). \end{aligned}$$

Sorting each wedge by Lemma 14.8.1 (b) turns this into a combination of the distinct basis vectors

$$\mathbf{e}_{123}, \mathbf{e}_{124}, \mathbf{e}_{134}, \mathbf{e}_{234}, \mathbf{e}_{12i}, \mathbf{e}_{34i} \quad (i \geq 5)$$

of  $\Lambda^3 V$ , with coefficients  $x_3, x_4, x_1, x_2$  and  $x_i, x_i$  respectively, up to signs. Since these basis vectors are distinct and the  $\mathbf{e}_S$  are independent (Theorem 14.8.5),  $\mathbf{v} \wedge \omega = \mathbf{0}$  forces  $x_1 = x_2 = \cdots = x_n = 0$ , that is,  $\mathbf{v} = \mathbf{0}$ . This contradicts  $\mathbf{u} \neq \mathbf{0}$  with  $\mathbf{u} \wedge \omega = \mathbf{0}$ . Hence  $\omega$  is not decomposable. ■

*Remark.* When  $\text{char } F \neq 2$  there is a one-line version:  $\omega \wedge \omega = 2 \mathbf{e}_1 \wedge \mathbf{e}_2 \wedge \mathbf{e}_3 \wedge \mathbf{e}_4 \neq \mathbf{0}$ , whereas  $(\mathbf{u} \wedge \mathbf{w}) \wedge (\mathbf{u} \wedge \mathbf{w}) = \mathbf{0}$  always. The proof above was written to avoid that division, because the test fails over  $\mathbb{F}_2$  — see the warning next.

*Warning.* **The square test is blind in characteristic 2.** Over  $\mathbb{F}_2$  every  $\omega \in \Lambda^2 V$  satisfies  $\omega \wedge \omega = \mathbf{0}$ : in the expansion  $\sum_{S,T} c_{S,T} \mathbf{e}_S \wedge \mathbf{e}_T$  the diagonal terms vanish and each off-diagonal pair contributes  $2c_{S,T} \mathbf{e}_S \wedge \mathbf{e}_T = \mathbf{0}$ . So over  $\mathbb{F}_2$  the vanishing of  $\omega \wedge \omega$  says nothing at all, while  $\mathbf{e}_1 \wedge \mathbf{e}_2 + \mathbf{e}_3 \wedge \mathbf{e}_4$  is still not decomposable, by the proof above. A criterion and a phenomenon are different things.

*Quick check.* Why does Theorem 14.10.4 need  $n \geq 4$ ? Show that every element of  $\Lambda^2 V$  is decomposable when  $\dim V = 3$ .

*Solution.* With  $n \leq 3$  there is no room to write  $\mathbf{e}_1 \wedge \mathbf{e}_2 + \mathbf{e}_3 \wedge \mathbf{e}_4$  at all. For  $n = 3$ , take  $\omega = a \mathbf{e}_{12} + b \mathbf{e}_{13} + c \mathbf{e}_{23}$ . If  $a = b = c = 0$  then  $\omega = \mathbf{0}$ , which is decomposable. If  $a \neq 0$ , then

$$\left( \mathbf{e}_1 - \frac{c}{a} \mathbf{e}_3 \right) \wedge (a \mathbf{e}_2 + b \mathbf{e}_3) = a \mathbf{e}_{12} + b \mathbf{e}_{13} + c \mathbf{e}_{23} = \omega,$$

using  $\mathbf{e}_3 \wedge \mathbf{e}_2 = -\mathbf{e}_{23}$ . If  $a = 0$ , relabel the basis so that the surviving non-zero coefficient sits in the first slot and repeat. So the phenomenon starts exactly at  $n = 4$ .

Together with Example (A tensor that is not simple) this is the chapter's negative result, stated in its two forms: a general tensor is not a simple tensor, and a general  $k$ -vector is not a blade. The optimistic reading — that every element of a space built from  $V$  is built from finitely many vectors of  $V$  in the obvious way — is false in both places, and Section 11 will find the first form

of it wearing a physical name.

### 14.10.3 Algebras modulo relations

One tool is still missing. The Clifford algebra is to be built by imposing relations on  $T(V)$ , and imposing relations on an **algebra** means dividing by more than a subspace: the relations must stay relations after multiplication on either side. Chapter 5 met the commutative version of the idea for  $F[x]$  (Definition 5.2.1).

#### Definition 14.10.3 — Two-Sided Ideal, and the Ideal Generated by a Set

Let  $A$  be an  $F$ -algebra (Definition 13.12.1). A **two-sided ideal** of  $A$  is a subspace  $\mathcal{I} \subseteq A$  such that  $yx \in \mathcal{I}$  and  $xy \in \mathcal{I}$  for **every**  $x \in \mathcal{I}$  and **every**  $y \in A$ . For a subset  $S \subseteq A$ , the **ideal generated by  $S$**  is

$$\langle S \rangle := \text{span}\{ ysz : y, z \in A, s \in S \}.$$

**Examples.** In  $F[x]$  the ideals are the sets  $\langle p \rangle$  of Definition 5.2.1; in any  $A$ , both  $\{0\}$  and  $A$  are ideals. **Non-example by minimal change.** The diagonal matrices in  $M_2(F)$  form a subspace closed under multiplication, but not an ideal: it contains  $\mathbf{E}_{11}$ , while  $\mathbf{E}_{21}\mathbf{E}_{11} = \mathbf{E}_{21}$  is not diagonal. Being closed among one's own elements is not the condition; absorbing products with the **whole** algebra is.

The span above is a subspace by construction, and it is a two-sided ideal because  $a(ysz) = (ay)sz$  and  $(ysz)a = ys(za)$  are again of the listed form. It contains  $S$ , taking  $y = z = 1_A$ , and every two-sided ideal containing  $S$  contains each  $ysz$ ; so  $\langle S \rangle$  is the smallest two-sided ideal containing  $S$ , which is what “generated” means.

#### Lemma 14.10.5 — Quotient of an Algebra by an Ideal

Let  $A$  be an  $F$ -algebra and  $\mathcal{I}$  a two-sided ideal of  $A$ .

- (a) The quotient space  $A/\mathcal{I}$  carries exactly one multiplication making the quotient map  $\pi: A \rightarrow A/\mathcal{I}$  an algebra homomorphism, namely  $(x + \mathcal{I})(y + \mathcal{I}) = xy + \mathcal{I}$ ; with it  $A/\mathcal{I}$  is an  $F$ -algebra with identity  $1_A + \mathcal{I}$ , and  $\ker \pi = \mathcal{I}$ .
- (b) If  $\varphi: A \rightarrow B$  is an algebra homomorphism with  $\mathcal{I} \subseteq \ker \varphi$ , then there is exactly one algebra homomorphism  $\bar{\varphi}: A/\mathcal{I} \rightarrow B$  with  $\bar{\varphi} \circ \pi = \varphi$ .
- (c) The kernel of any algebra homomorphism is a two-sided ideal.

*Proof.*

- (a) The rule is well defined: if  $x' = x + u$  and  $y' = y + w$  with  $u, w \in \mathcal{I}$ , then

$$x'y' = xy + uy + xw + uw \in xy + \mathcal{I},$$

since  $\mathcal{I}$  absorbs multiplication on both sides and is closed under addition. Bilinearity, associativity and the identity are then inherited from  $A$ , because each is an identity among cosets that follows by applying  $\pi$  to the corresponding identity in

- A. Any multiplication making  $\pi$  multiplicative must satisfy the displayed rule, and  $\pi$  is surjective, so the multiplication is unique. Finally  $\ker \pi = \mathcal{I}$  by [Lemma 3.9.3](#).
- (b) Since  $\varphi$  is linear with  $\mathcal{I} \subseteq \ker \varphi$ , [Theorem 3.9.10](#) (a) gives exactly one **linear**  $\bar{\varphi}$  with  $\bar{\varphi}\pi = \varphi$ . It is multiplicative: every element of  $A/\mathcal{I}$  is  $\pi(x)$  for some  $x$ , and

$$\bar{\varphi}(\pi(x)\pi(y)) = \bar{\varphi}(\pi(xy)) = \varphi(xy) = \varphi(x)\varphi(y),$$

which is  $\bar{\varphi}(\pi(x))\bar{\varphi}(\pi(y))$ . Also  $\bar{\varphi}(\pi(1_A)) = \varphi(1_A) = 1_B$ . Uniqueness is already contained in the uniqueness of the linear map.

- (c) If  $\varphi(x) = 0$  and  $y \in A$ , then  $\varphi(xy) = \varphi(x)\varphi(y) = 0$  and  $\varphi(yx) = 0$ ; and  $\ker \varphi$  is a subspace because  $\varphi$  is linear.

■

## 14.10.4 The Clifford algebra, constructed

Chapter 13 §12 defined a Clifford algebra of  $(V, q)$  as a pair  $(C, \iota)$  with  $\iota: V \rightarrow C$  linear satisfying

- (C1)  $\iota(\mathbf{v})^2 = q(\mathbf{v})1_C$  for every  $\mathbf{v} \in V$ , and
- (C2) the universal property: every linear  $f: V \rightarrow A$  into an  $F$ -algebra with  $f(\mathbf{v})^2 = q(\mathbf{v})1_A$  factors as  $f = \bar{f} \circ \iota$  for a unique algebra homomorphism  $\bar{f}$ ,

and then said in as many words that it could not produce one: “The definition says what a Clifford algebra would be; it does not produce one.” What it needed was an algebra of formal strings of vectors, which is now [Definition 14.6.2](#), and a way to impose the relation  $\mathbf{v} \otimes \mathbf{v} = q(\mathbf{v})1$ , which is now [Lemma 14.10.5](#).

### Theorem 14.10.6 — Existence of the Clifford Algebra

Let  $F$  be a field with  $\text{char } F \neq 2$ , let  $V$  be a finite-dimensional  $F$ -vector space, let  $\beta$  be a symmetric bilinear form on  $V$  and  $q(\mathbf{v}) = \beta(\mathbf{v}, \mathbf{v})$ . Let

$$\mathcal{I}_q := \langle \mathbf{v} \otimes \mathbf{v} - q(\mathbf{v})1 : \mathbf{v} \in V \rangle \subseteq T(V)$$

be the two-sided ideal generated by these elements, put  $\text{Cl}(q) := T(V)/\mathcal{I}_q$ , and let  $\iota: V \rightarrow \text{Cl}(q)$  be the restriction to  $V \subseteq T(V)$  of the quotient map  $\pi$ . Then  $(\text{Cl}(q), \iota)$  is a Clifford algebra of  $(V, q)$  in the sense of [Definition 13.12.2](#).

In particular a Clifford algebra exists for every  $(V, q)$ , and by [Proposition 13.12.1](#) any two are isomorphic by a unique isomorphism compatible with the structure maps.

*Idea.* The tensor algebra is the free associative algebra on  $V$ : a linear map out of  $V$  extends uniquely to it ([Theorem 14.6.6](#)), with no relations in the way. Dividing by the smallest ideal containing  $\mathbf{v} \otimes \mathbf{v} - q(\mathbf{v})1$  imposes exactly the relation (C1) and nothing else, and “nothing else” is precisely what the universal property (C2) asserts.

*Proof.* By [Lemma 14.10.5](#) (a),  $\text{Cl}(q)$  is an  $F$ -algebra and  $\pi$  is a surjective algebra homomorphism with  $\ker \pi = \mathcal{I}_q$ . The map  $\iota = \pi|_V$  is linear, being a restriction of a linear map.



(C1). Let  $\mathbf{v} \in V$ . In  $T(V)$  the product of  $\mathbf{v}$  with itself is  $\mathbf{v} \otimes \mathbf{v}$  (Definition 14.6.2), and  $\mathbf{v} \otimes \mathbf{v} - q(\mathbf{v})1$  is a generator of  $\mathcal{I}_q = \ker \pi$ . Since  $\pi$  is multiplicative and linear,

$$\iota(\mathbf{v})^2 = \pi(\mathbf{v})\pi(\mathbf{v}) = \pi(\mathbf{v} \otimes \mathbf{v}) = \pi(q(\mathbf{v})1) = q(\mathbf{v})1_{\text{Cl}(q)}.$$

(C2). Let  $A$  be an  $F$ -algebra and let  $f: V \rightarrow A$  be linear with  $f(\mathbf{v})^2 = q(\mathbf{v})1_A$  for every  $\mathbf{v} \in V$ . By Theorem 14.6.6 there is a unique algebra homomorphism  $\tilde{f}: T(V) \rightarrow A$  with  $\tilde{f}|_V = f$ . For every  $\mathbf{v} \in V$ ,

$$\tilde{f}(\mathbf{v} \otimes \mathbf{v} - q(\mathbf{v})1) = f(\mathbf{v})f(\mathbf{v}) - q(\mathbf{v})1_A = 0,$$

so  $\ker \tilde{f}$  contains every generator of  $\mathcal{I}_q$ . By Lemma 14.10.5 (c),  $\ker \tilde{f}$  is a two-sided ideal, and  $\mathcal{I}_q$  is the **smallest** one containing those generators (Definition 14.10.3), so  $\mathcal{I}_q \subseteq \ker \tilde{f}$ . By Lemma 14.10.5 (b) there is exactly one algebra homomorphism  $\bar{f}: \text{Cl}(q) \rightarrow A$  with  $\bar{f} \circ \pi = \tilde{f}$ , and restricting to  $V$  gives  $\bar{f} \circ \iota = \tilde{f}|_V = f$ .

For the uniqueness clause of (C2), let  $g: \text{Cl}(q) \rightarrow A$  be any algebra homomorphism with  $g \circ \iota = f$ . Then  $g \circ \pi$  is an algebra homomorphism  $T(V) \rightarrow A$  whose restriction to  $V$  is  $g\iota = f$ , so  $g\pi = \tilde{f}$  by the uniqueness clause of Theorem 14.6.6. Hence  $g\pi = \bar{f}\pi$ , and  $\pi$  is surjective, so  $g = \bar{f}$ . This proves the theorem. ■

*Remark.* Neither  $\text{char } F \neq 2$  nor finite dimension was used in this proof; the construction and both checks work for any  $V$  and any quadratic form. The hypotheses are carried because Definition 13.12.2 states them, and because the dimension count below needs an orthogonal basis.

## 14.10.5 The dimension, at last

Chapter 13 proved half of the count. Its Proposition 13.12.3 showed that for an orthogonal basis  $(\mathbf{e}_1, \dots, \mathbf{e}_n)$  of  $\beta$  the  $2^n$  **sorted products**

$$\iota(\mathbf{e}_{i_1})\iota(\mathbf{e}_{i_2}) \cdots \iota(\mathbf{e}_{i_k}), \quad i_1 < i_2 < \cdots < i_k,$$

span  $C$ , so that  $\dim_F C \leq 2^n$ . The other half — their independence — is what this section owes. The idea is to make  $\Lambda V$ , which has exactly the right dimension, into a place where the vectors of  $V$  act with square  $q(\mathbf{v})$ ; the universal property then sends  $C$  into the operators on  $\Lambda V$ , and evaluating at 1 turns the sorted products into distinct basis vectors.

Two families of operators on  $\Lambda V$  do the work: wedging a basis vector on, and deleting it again.

### Lemma 14.10.7 — Creation and Deletion Operators

Let  $(\mathbf{e}_1, \dots, \mathbf{e}_n)$  be a basis of  $V$ , and for  $S \subseteq [n]$  let  $\mathbf{e}_S$  be as in Theorem 14.10.2. For  $i \in [n]$  write  $\sigma(i, S) = |\{j \in S : j < i\}|$ . Let  $E_i \in \mathcal{L}(\Lambda V)$  be left multiplication by  $\mathbf{e}_i$ , that is,  $E_i(\omega) = \mathbf{e}_i \wedge \omega$ , and let  $D_i \in \mathcal{L}(\Lambda V)$  be the unique linear map with

$$D_i(\mathbf{e}_S) = \begin{cases} (-1)^{\sigma(i, S)} \mathbf{e}_{S \setminus \{i\}}, & i \in S, \\ 0, & i \notin S, \end{cases}$$

which exists and is unique because the  $\mathbf{e}_S$  form a basis (Theorem 14.10.2). Then for all  $i, j \in [n]$ ,

$$\begin{aligned} E_i E_j + E_j E_i &= 0, \\ D_i D_j + D_j D_i &= 0, \\ D_i E_j + E_j D_i &= \delta_{ij} \operatorname{id}_{\Lambda V}. \end{aligned}$$

*Idea.*  $E_i$  inserts the index  $i$  into the set  $S$  and  $D_i$  removes it, each at the cost of the sign needed to move  $\mathbf{e}_i$  past the smaller members of  $S$ . All three identities are then a comparison of two signs, and in the third the two cases “ $i \in S$ ” and “ $i \notin S$ ” are exactly the two ways the identity can be satisfied: one term is zero and the other is  $\mathbf{e}_S$ .

*Proof.* First record what  $E_i$  does on the basis. If  $i \in S$  then  $\mathbf{e}_i \wedge \mathbf{e}_S$  repeats  $\mathbf{e}_i$  and is  $\mathbf{0}$ . If  $i \notin S$ , then sorting  $\mathbf{e}_i \wedge \mathbf{e}_S$  moves  $\mathbf{e}_i$  past the  $\sigma(i, S)$  members of  $S$  smaller than  $i$ , so by Lemma 14.8.1 (b)

$$E_i(\mathbf{e}_S) = (-1)^{\sigma(i, S)} \mathbf{e}_{S \cup \{i\}}.$$

**First identity.** For every  $\omega \in \Lambda V$ ,  $E_i E_j(\omega) = \mathbf{e}_i \wedge \mathbf{e}_j \wedge \omega$  and  $E_j E_i(\omega) = \mathbf{e}_j \wedge \mathbf{e}_i \wedge \omega$ , and  $\mathbf{e}_j \wedge \mathbf{e}_i = -\mathbf{e}_i \wedge \mathbf{e}_j$  by Theorem 14.10.3. Adding gives 0. (For  $i = j$  each term is separately 0.)

**Third identity, case  $i = j$ .** If  $i \notin S$ , then  $D_i(\mathbf{e}_S) = \mathbf{0}$  and  $E_i(\mathbf{e}_S) = (-1)^{\sigma(i, S)} \mathbf{e}_{S \cup \{i\}}$ ; since  $\sigma(i, S \cup \{i\}) = \sigma(i, S)$ , applying  $D_i$  returns  $(-1)^{2\sigma(i, S)} \mathbf{e}_S = \mathbf{e}_S$ . If  $i \in S$ , then  $E_i(\mathbf{e}_S) = \mathbf{0}$  and  $D_i(\mathbf{e}_S) = (-1)^{\sigma(i, S)} \mathbf{e}_{S \setminus \{i\}}$ ; since  $\sigma(i, S \setminus \{i\}) = \sigma(i, S)$ , applying  $E_i$  returns  $\mathbf{e}_S$ . Either way the sum is  $\mathbf{e}_S$ .

**Third identity, case  $i \neq j$ .** If  $j \in S$  then  $E_j(\mathbf{e}_S) = \mathbf{0}$ , and  $D_i(\mathbf{e}_S)$  is either  $\mathbf{0}$  or a multiple of  $\mathbf{e}_{S \setminus \{i\}}$ , which still contains  $j$ , so  $E_j$  kills it; both terms vanish. If  $j \notin S$  and  $i \notin S$  then  $D_i(\mathbf{e}_S) = \mathbf{0}$  and  $D_i(\mathbf{e}_{S \cup \{j\}}) = \mathbf{0}$  as well, since  $i \neq j$ ; again both terms vanish. So let  $j \notin S$  and  $i \in S$ . Then

$$\begin{aligned} D_i E_j(\mathbf{e}_S) &= (-1)^{\sigma(j, S) + \sigma(i, S \cup \{j\})} \mathbf{e}_{(S \cup \{j\}) \setminus \{i\}}, \\ E_j D_i(\mathbf{e}_S) &= (-1)^{\sigma(i, S) + \sigma(j, S \setminus \{i\})} \mathbf{e}_{(S \setminus \{i\}) \cup \{j\}}, \end{aligned}$$

and the two index sets agree. Now  $\sigma(i, S \cup \{j\}) = \sigma(i, S) + [j < i]$  and  $\sigma(j, S \setminus \{i\}) = \sigma(j, S) - [i < j]$ , writing  $[P]$  for 1 if  $P$  holds and 0 otherwise. So the two exponents differ by  $[j < i] + [i < j] = 1$ , which is odd because  $i \neq j$ . The two terms therefore cancel.

**Second identity.** Both  $D_i D_j(\mathbf{e}_S)$  and  $D_j D_i(\mathbf{e}_S)$  vanish unless  $i, j \in S$  and  $i \neq j$ , and in that case they are  $(-1)^{\sigma(j, S) + \sigma(i, S \setminus \{j\})}$  and  $(-1)^{\sigma(i, S) + \sigma(j, S \setminus \{i\})}$  times the same  $\mathbf{e}_{S \setminus \{i, j\}}$ . The same two identities for  $\sigma$  show the exponents differ by an odd number, so the terms cancel. For  $i = j$ ,  $D_i^2(\mathbf{e}_S) = \mathbf{0}$  because  $i \notin S \setminus \{i\}$ . This proves the lemma. ■

### Theorem 14.10.8 — Dimension of a Clifford Algebra

Let  $F$  be a field with  $\operatorname{char} F \neq 2$ , let  $\dim V = n$ , let  $\beta$  be a symmetric bilinear form on  $V$  with quadratic form  $q$ , and let  $(C, \iota)$  be a Clifford algebra of  $(V, q)$ . Let  $(\mathbf{e}_1, \dots, \mathbf{e}_n)$  be an orthogonal basis of  $V$  for  $\beta$ . Then the  $2^n$  sorted products of Proposition 13.12.3

form a **basis** of  $C$ . Consequently

$$\dim_F C = 2^n,$$

the map  $\iota$  is injective, and  $C$  is isomorphic to  $\Lambda V$  as a **vector space**. This is [Theorem 13.12.4](#).

*Idea.* Build a representation. ① Turn each basis vector  $\mathbf{e}_i$  into the operator  $c_i = E_i + d_i D_i$  on  $\Lambda V$ , where  $d_i = q(\mathbf{e}_i)$ ; [Lemma 14.10.7](#) makes these anticommute and gives  $c_i^2 = d_i \text{id}$ , which is exactly (C1). ② The universal property then hands us an algebra homomorphism  $C \rightarrow \mathcal{L}(\Lambda V)$ . ③ Evaluate at the element  $1 \in \Lambda^0 V$ : the sorted product for the index set  $S$  is sent to  $\mathbf{e}_S$ , because at each step the new index is smaller than all the ones already there, so the deletion half contributes nothing. Distinct basis vectors come out, so the sorted products were independent.

*Proof.* An orthogonal basis exists by [Theorem 13.4.2](#), since  $\text{char } F \neq 2$ . Write  $d_i = q(\mathbf{e}_i)$  and  $c_i = E_i + d_i D_i \in \mathcal{L}(\Lambda V)$ , with  $E_i, D_i$  as in [Lemma 14.10.7](#). By [Proposition 13.12.3](#) the sorted products span  $C$  and  $\dim_F C \leq 2^n$ , so only independence is left.

**Step 1: the  $c_i$  satisfy the Clifford relations.** Expanding the product and collecting the four brackets,

$$\begin{aligned} c_i c_j + c_j c_i &= (E_i E_j + E_j E_i) + d_j (E_i D_j + D_j E_i) \\ &\quad + d_i (D_i E_j + E_j D_i) + d_i d_j (D_i D_j + D_j D_i). \end{aligned}$$

By [Lemma 14.10.7](#) the first and last brackets are 0 and each middle bracket is  $\delta_{ij} \text{id}$ . So  $c_i c_j = -c_j c_i$  for  $i \neq j$ . Setting  $i = j$  in the display would give  $2c_i^2$ , so expand  $c_i^2$  directly instead; with  $E_i^2 = D_i^2 = 0$ , which were checked separately in the lemma,

$$c_i^2 = E_i^2 + d_i (E_i D_i + D_i E_i) + d_i^2 D_i^2 = d_i \text{id},$$

with no division by 2 anywhere. Now  $\mathcal{L}(\Lambda V)$  is an  $F$ -algebra under composition ([Definition 13.12.1](#)). Let  $f: V \rightarrow \mathcal{L}(\Lambda V)$  be the linear map with  $f(\mathbf{e}_i) = c_i$ , which exists and is unique by [Theorem 3.1.6](#). For  $\mathbf{v} = \sum_i x_i \mathbf{e}_i$ ,

$$f(\mathbf{v})^2 = \sum_i x_i^2 c_i^2 + \sum_{i < j} x_i x_j (c_i c_j + c_j c_i) = \left( \sum_i d_i x_i^2 \right) \text{id},$$

the cross terms vanishing because  $i \neq j$  there. Since the basis is orthogonal,  $\beta(\mathbf{e}_i, \mathbf{e}_j) = 0$  for  $i \neq j$ , so  $q(\mathbf{v}) = \beta(\mathbf{v}, \mathbf{v}) = \sum_i d_i x_i^2$ . Hence  $f(\mathbf{v})^2 = q(\mathbf{v}) \text{id}$ , which is (C1) for  $f$ .

**Step 2: the induced homomorphism.** By the universal property (C2) of [Definition 13.12.2](#) there is an algebra homomorphism  $\Psi: C \rightarrow \mathcal{L}(\Lambda V)$  with  $\Psi\iota = f$ . Define the linear map

$$\Phi: C \rightarrow \Lambda V, \quad \Phi(x) = \Psi(x)(1),$$

where  $1 \in \Lambda^0 V = F \subseteq \Lambda V$ . It is linear because  $\Psi$  is linear and evaluation at a fixed element is linear.

**Step 3:  $\Phi$  sends sorted products to basis vectors.** Let  $S = \{i_1 < \cdots < i_k\} \subseteq [n]$ . Since  $\Psi$  is multiplicative and  $\Psi\iota = f$ ,

$$\Psi(\iota(\mathbf{e}_{i_1}) \cdots \iota(\mathbf{e}_{i_k})) = c_{i_1} c_{i_2} \cdots c_{i_k}.$$

We claim  $c_{i_r} c_{i_{r+1}} \cdots c_{i_k}(1) = \mathbf{e}_{T_r}$  with  $T_r = \{i_r, \dots, i_k\}$ , by downward induction on  $r$ . For  $r = k + 1$  the product is empty and the value is  $1 = \mathbf{e}_\emptyset$ . Assume the claim for  $r + 1$ . Because  $i_r < i_{r+1}$  and the indices increase,  $i_r \notin T_{r+1}$ , so  $D_{i_r}(\mathbf{e}_{T_{r+1}}) = \mathbf{0}$ ; and  $i_r$  is smaller than every element of  $T_{r+1}$ , so  $\sigma(i_r, T_{r+1}) = 0$  and  $E_{i_r}(\mathbf{e}_{T_{r+1}}) = \mathbf{e}_{T_r}$ . Hence  $c_{i_r}(\mathbf{e}_{T_{r+1}}) = \mathbf{e}_{T_r}$ , completing the induction. Taking  $r = 1$ ,

$$\Phi(\iota(\mathbf{e}_{i_1}) \cdots \iota(\mathbf{e}_{i_k})) = \mathbf{e}_S,$$

and for  $S = \emptyset$ ,  $\Phi(1_C) = \text{id}(1) = 1 = \mathbf{e}_\emptyset$ .

**Step 4: conclusion.** Suppose a linear combination of the sorted products is 0. Applying  $\Phi$  gives the same combination of the  $\mathbf{e}_S$ , which are linearly independent by [Theorem 14.10.2](#) (b); so all coefficients vanish. The sorted products are therefore linearly independent, and being also a spanning list they form a basis of  $C$ , of size  $2^n$ . Thus  $\dim_F C = 2^n$ .

Since  $1_C$  and  $\iota(\mathbf{e}_1), \dots, \iota(\mathbf{e}_n)$  are among the basis elements just found, the list  $(\iota(\mathbf{e}_1), \dots, \iota(\mathbf{e}_n))$  is independent, so  $\iota$  carries a basis of  $V$  to an independent list and is injective. Finally  $\Phi$  carries the basis of  $C$  bijectively onto the basis  $(\mathbf{e}_S)$  of  $\Lambda V$ , so  $\Phi$  is a vector space isomorphism. This proves the theorem. ■

**The debt, and how it was paid.** Chapter 13 §12 stated [Theorem 13.12.4](#) and then wrote, under the heading “What is proved here, and what is not”, that the inequality  $\dim_F C \leq 2^n$  was proved in full, while “the reverse inequality — that the  $2^n$  sorted products are linearly independent — is **not proved in this section**”, and that the two halves would have to be established together by exhibiting one algebra of dimension exactly  $2^n$  satisfying (C1) and (C2). Both halves are now supplied. The missing one was independence, and [Theorem 14.10.8](#) supplies it; the prior question, whether any Clifford algebra exists at all, is [Theorem 14.10.6](#). Chapter 13 read the statement for  $n \geq 3$  “as quoted from Chapter 14 and not as proved here”; it may now be read as proved.

The prediction was right in substance and slightly off in shape. The algebra of dimension  $2^n$  that settled the matter is the exterior algebra, but it is not the Clifford algebra: it is only a space that  $\text{Cl}(q)$  acts on, and independence came from the action rather than from an identification. Chapter 13 also guessed that “ $\text{Cl}(q)$  is assembled from the exterior powers  $\Lambda^k V$ , one for each  $k$ ”, and  $\Phi$  is that assembly, matching the sorted product on  $S$  with the wedge  $\mathbf{e}_S$ .

**Warning.**  $\text{Cl}(q)$  and  $\Lambda V$  are the same size, not the same algebra. The map  $\Phi$  above is an isomorphism of **vector spaces** only. It is not multiplicative, and no other map is either, unless  $q = 0$ : in  $\Lambda V$  the square of a vector is  $\mathbf{0}$ , while in  $\text{Cl}(q)$  it is  $q(\mathbf{v})1$ . Writing an element of  $\text{Cl}(q)$  as a sum of wedges is a choice of coordinates, not a description of the multiplication.

That warning deserves a proof rather than a slogan.

### Proposition 14.10.9 — Comparing the Two Algebras

Let  $\text{char } F \neq 2$  and  $\dim V = n \geq 1$ , and let  $q$  be the quadratic form of a symmetric bilinear form on  $V$ .

- (a) If  $q = 0$ , then  $\text{Cl}(0)$  and  $\Lambda V$  are isomorphic as  $F$ -algebras.
- (b) If  $q \neq 0$ , they are **not** isomorphic as  $F$ -algebras, although they are isomorphic as vector spaces.

*Idea.* For (b), find a property of  $\Lambda V$  that  $\text{Cl}(q)$  fails. In  $\Lambda V$  everything of positive degree is nilpotent, so an element whose square is a **non-zero scalar** can have no positive-degree part at all; but in  $\text{Cl}(q)$  a vector  $\mathbf{v}$  with  $q(\mathbf{v}) \neq 0$  squares to a non-zero scalar and is not a scalar itself.

*Proof.*

- (a) Let  $q = 0$  and let  $j: V \rightarrow \Lambda V$  be the inclusion  $V = \Lambda^1 V \subseteq \Lambda V$ . Then  $j(\mathbf{v})^2 = \mathbf{v} \wedge \mathbf{v} = \mathbf{0} = q(\mathbf{v})1$ , so (C1) holds for  $j$ . By the universal property (C2) of  $\text{Cl}(0)$  there is an algebra homomorphism  $\psi: \text{Cl}(0) \rightarrow \Lambda V$  with  $\psi\iota = j$ . Its image is a subalgebra containing 1 and  $V$ , hence all of  $\Lambda V$  by [Theorem 14.10.2](#) (c); so  $\psi$  is surjective. Both spaces have dimension  $2^n$ , by [Theorem 14.10.8](#) and [Theorem 14.10.2](#) (b), so  $\psi$  is bijective and is an algebra isomorphism.
- (b) *Claim.* If  $x \in \Lambda V$  and  $x \wedge x = a \cdot 1$  with  $a \in F$ ,  $a \neq 0$ , then  $x \in F \cdot 1$ .

*Proof of claim.* Throughout,  $y^t$  means the  $t$ -fold wedge product of  $y$  with itself. Write  $x = x_0 + m$  with  $x_0 \in \Lambda^0 V = F$  and  $m$  the sum of the components of positive degree. Then

$$x \wedge x = x_0^2 + 2x_0m + m \wedge m,$$

and the only degree-0 term is  $x_0^2$ , so  $x_0^2 = a \neq 0$  and hence  $x_0 \neq 0$ . Canceling that term leaves

$$m \wedge (2x_0 \cdot 1 + m) = \mathbf{0}.$$

Every product of  $n+1$  elements of positive degree lies in  $\bigoplus_{k>n} \Lambda^k V = \{\mathbf{0}\}$ , so  $m^{n+1} = \mathbf{0}$ . Put  $c = 2x_0$ , which is non-zero because  $\text{char } F \neq 2$  and  $x_0 \neq 0$ , and  $u = -c^{-1}m$ , so that  $u^{n+1} = \mathbf{0}$  and  $c \cdot 1 + m = c(1 - u)$ . Then

$$(1 - u) \wedge (1 + u + u^2 + \cdots + u^n) = 1 - u^{n+1} = 1,$$

and likewise in the other order, so  $c \cdot 1 + m$  is invertible. Multiplying  $m \wedge (c \cdot 1 + m) = \mathbf{0}$  by its inverse gives  $m = \mathbf{0}$ , so  $x = x_0 \in F \cdot 1$ , proving the claim.

Now suppose  $q \neq 0$  and let  $\varphi: \text{Cl}(q) \rightarrow \Lambda V$  be an algebra isomorphism. Choose  $\mathbf{v} \in V$  with  $q(\mathbf{v}) \neq 0$  and put  $x = \varphi(\iota(\mathbf{v}))$ . Then

$$x \wedge x = \varphi(\iota(\mathbf{v})^2) = \varphi(q(\mathbf{v})1) = q(\mathbf{v}) \cdot 1,$$

a non-zero scalar, so  $x \in F \cdot 1$  by the claim, say  $x = c \cdot 1 = \varphi(c \cdot 1_{\text{Cl}(q)})$ . Since  $\varphi$  is injective,  $\iota(\mathbf{v}) = c \cdot 1_{\text{Cl}(q)}$ . But by [Theorem 14.10.8](#) the elements  $1_{\text{Cl}(q)}, \iota(\mathbf{e}_1), \dots, \iota(\mathbf{e}_n)$  are part of a basis of  $\text{Cl}(q)$ , and  $\iota(\mathbf{v})$  is a combination of  $\iota(\mathbf{e}_1), \dots, \iota(\mathbf{e}_n)$  alone; comparing coordinates forces  $c = 0$  and  $\iota(\mathbf{v}) = 0$ , contradicting  $\iota(\mathbf{v})^2 = q(\mathbf{v})1 \neq 0$ . So no such  $\varphi$  exists. The vector space isomorphism is [Theorem 14.10.8](#). ■

The dimension count also makes Clifford algebras easy to recognize, which is how one identifies them in practice.

**Corollary 14.10.10 — Recognizing a Clifford Algebra**

Let  $\text{char } F \neq 2$ ,  $\dim V = n$ , and let  $q$  be as above. Let  $A$  be an  $F$ -algebra with  $\dim_F A = 2^n$ , and let  $\iota_A: V \rightarrow A$  be linear with  $\iota_A(\mathbf{v})^2 = q(\mathbf{v})1_A$  for every  $\mathbf{v} \in V$ . If  $A$  is generated as an algebra by  $1_A$  and  $\iota_A(V)$ , then  $(A, \iota_A)$  is a Clifford algebra of  $(V, q)$ .

*Proof.* Let  $(C, \iota)$  be the Clifford algebra of [Theorem 14.10.6](#). By its universal property there is an algebra homomorphism  $\psi: C \rightarrow A$  with  $\psi\iota = \iota_A$ . Its image is a subalgebra of  $A$  containing  $1_A$  and  $\iota_A(V)$ , hence is all of  $A$  by the generation hypothesis. So  $\psi$  is surjective, and  $\dim_F C = 2^n = \dim_F A$  by [Theorem 14.10.8](#), so  $\psi$  is an isomorphism.

It remains to transport (C2). Let  $f: V \rightarrow B$  be linear with  $f(\mathbf{v})^2 = q(\mathbf{v})1_B$ . Then  $\bar{f}\psi^{-1}: A \rightarrow B$  is an algebra homomorphism with  $\bar{f}\psi^{-1}\iota_A = \bar{f}\iota = f$ , where  $\bar{f}: C \rightarrow B$  comes from (C2) for  $C$ ; and if  $g: A \rightarrow B$  is another one, then  $g\psi: C \rightarrow B$  satisfies  $g\psi\iota = g\iota_A = f$ , so  $g\psi = \bar{f}$  by uniqueness for  $C$ , whence  $g = \bar{f}\psi^{-1}$ . ■

**Example. (The Clifford algebra of the Euclidean plane)**

Let  $V = \mathbb{R}^2$  with  $q(\mathbf{x}) = x_1^2 + x_2^2$ , the form of signature  $(2, 0)$ . Show that  $\text{Cl}_{2,0}(\mathbb{R}) \cong M_2(\mathbb{R})$ .

*Solution.* Put

$$\mathbf{A} = \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix}, \quad \mathbf{B} = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}$$

in  $M_2(\mathbb{R})$ , and let  $\iota(\mathbf{x}) = x_1\mathbf{A} + x_2\mathbf{B}$ , a linear map  $\mathbb{R}^2 \rightarrow M_2(\mathbb{R})$ . Multiplying out,  $\mathbf{A}^2 = \mathbf{B}^2 = \mathbf{I}_2$  and  $\mathbf{AB} = -\mathbf{BA}$ , the products being  $\begin{pmatrix} 0 & 1 \\ -1 & 0 \end{pmatrix}$  and its negative. Hence

$$\iota(\mathbf{x})^2 = x_1^2\mathbf{A}^2 + x_1x_2(\mathbf{AB} + \mathbf{BA}) + x_2^2\mathbf{B}^2 = q(\mathbf{x})\mathbf{I}_2,$$

which is (C1). Next,  $\dim_{\mathbb{R}} M_2(\mathbb{R}) = 4 = 2^2$ , and the four matrices  $\mathbf{I}_2, \mathbf{A}, \mathbf{B}, \mathbf{AB}$  are linearly independent: the first two span the diagonal matrices and the last two span the off-diagonal ones, and the diagonal and off-diagonal matrices meet only in 0. So they are a basis, and  $M_2(\mathbb{R})$  is generated by  $\mathbf{I}_2$  and  $\iota(\mathbb{R}^2)$ . By [Corollary 14.10.10](#),  $(M_2(\mathbb{R}), \iota)$  is a Clifford algebra of  $(\mathbb{R}^2, q)$ , so  $\text{Cl}_{2,0}(\mathbb{R}) \cong M_2(\mathbb{R})$ .

Compare Chapter 13's  $\text{Cl}_{0,2}(\mathbb{R}) = \mathbb{H}$  ([Example \(The quaternions\)](#)). All three of  $\mathbb{H}$ ,  $M_2(\mathbb{R})$  and  $\Lambda\mathbb{R}^2$  have dimension 4, and no two are isomorphic:  $\mathbb{H}$  has no zero divisors,  $M_2(\mathbb{R})$  has plenty, and  $\Lambda\mathbb{R}^2$  differs from both by [Proposition 14.10.9](#) (b). Changing a sign in  $q$  changes the algebra completely and never changes its size.

*Quick check.* Where in the proof of [Theorem 14.10.8](#) is the hypothesis  $\text{char } F \neq 2$  used?

*Solution.* Twice, both times through the orthogonal basis. It is needed to produce one at all: [Theorem 13.4.2](#) assumes  $\text{char } F \neq 2$ . And it is needed for Chapter 13's [Proposition 13.12.3](#), which supplies the spanning half, since that proof uses [Lemma 13.12.2](#), which comes from polarization. The operator identities of [Lemma 14.10.7](#) and the induction of Step 3 use no such hypothesis.

With that, Chapter 13's last open statement is closed and the construction side of this chapter is complete. The final section spends the machinery on one application, where the failure of decomposability proved above reappears under another name.

## 14.10.6 Exercises

### 14.10.6.1 A. Check your understanding

#### Exercise A1.

- (a) Define the exterior algebra  $\Lambda V$  and state its dimension when  $\dim V = n$ .
- (b) State the graded commutativity rule, and say what it gives when both degrees are odd.
- (c) Define what it means for  $\omega \in \Lambda^k V$  to be decomposable, and give one decomposable and one non-decomposable element of  $\Lambda^2 F^4$ .
- (d) Which half of [Theorem 13.12.4](#) was left unproved in Chapter 13, and which result of this section supplies it?
- (e) True or false:  $\text{Cl}(q)$  and  $\Lambda V$  are isomorphic algebras. Justify your answer.

*Solution.*

- (a)  $\Lambda V = \bigoplus_{k \geq 0} \Lambda^k V$  with the product of [Proposition 14.10.1](#) extended by bilinearity ([Definition 14.10.1](#)). Its dimension is  $\sum_k \binom{n}{k} = 2^n$  ([Theorem 14.10.2](#)).
- (b)  $\omega \wedge \eta = (-1)^{kl} \eta \wedge \omega$  for  $\omega \in \Lambda^k V$ ,  $\eta \in \Lambda^l V$  ([Theorem 14.10.3](#)). Both odd gives  $\omega \wedge \eta = -\eta \wedge \omega$ : odd-degree elements anticommute.
- (c)  $\omega$  is decomposable if  $\omega = \mathbf{v}_1 \wedge \cdots \wedge \mathbf{v}_k$  for some  $\mathbf{v}_i \in V$  ([Definition 14.10.2](#)). Decomposable:  $\mathbf{e}_1 \wedge \mathbf{e}_2$ . Non-decomposable:  $\mathbf{e}_1 \wedge \mathbf{e}_2 + \mathbf{e}_3 \wedge \mathbf{e}_4$  ([Theorem 14.10.4](#)).
- (d) The independence of the  $2^n$  sorted products, that is, the inequality  $\dim_F C \geq 2^n$ . It is supplied by [Theorem 14.10.8](#), which needs [Theorem 14.10.6](#) first so that the object exists.
- (e) False in general. They are isomorphic as vector spaces, and as algebras exactly when  $q = 0$  ([Proposition 14.10.9](#)).

### 14.10.6.2 B. Practice

#### Exercise B1: Multiplying in the exterior algebra.

In  $\Lambda F^4$  with  $F = \mathbb{Q}$ , put  $x = \mathbf{e}_1 + \mathbf{e}_2 \wedge \mathbf{e}_3$  and  $y = \mathbf{e}_2 + \mathbf{e}_1 \wedge \mathbf{e}_4$ . Compute the following, expressed in the basis of [Theorem 14.10.2](#).

- (a)  $x \wedge y$ .
- (b)  $y \wedge x$ .
- (c)  $x \wedge x$ .

*Solution.* Write  $\mathbf{e}_S$  for the sorted wedge on  $S$ .



(a) Expanding into four products:

$$\begin{aligned} \mathbf{e}_1 \wedge \mathbf{e}_2 &= \mathbf{e}_{12}, & \mathbf{e}_1 \wedge (\mathbf{e}_1 \wedge \mathbf{e}_4) &= \mathbf{0}, \\ (\mathbf{e}_2 \wedge \mathbf{e}_3) \wedge \mathbf{e}_2 &= \mathbf{0}, & (\mathbf{e}_2 \wedge \mathbf{e}_3) \wedge (\mathbf{e}_1 \wedge \mathbf{e}_4) &= \mathbf{e}_{1234}. \end{aligned}$$

The last one sorts as  $\mathbf{e}_2 \wedge \mathbf{e}_3 \wedge \mathbf{e}_1 \wedge \mathbf{e}_4 = \mathbf{e}_{1234}$ , since moving  $\mathbf{e}_1$  two places to the left costs  $(-1)^2$ . So  $x \wedge y = \mathbf{e}_{12} + \mathbf{e}_{1234}$ .

(b) The surviving products are  $\mathbf{e}_2 \wedge \mathbf{e}_1 = -\mathbf{e}_{12}$  and  $(\mathbf{e}_1 \wedge \mathbf{e}_4) \wedge (\mathbf{e}_2 \wedge \mathbf{e}_3) = \mathbf{e}_{1234}$ , the latter because  $\mathbf{e}_1 \wedge \mathbf{e}_4 \wedge \mathbf{e}_2 \wedge \mathbf{e}_3$  sorts with two transpositions. So  $y \wedge x = -\mathbf{e}_{12} + \mathbf{e}_{1234}$ . This agrees with [Theorem 14.10.3](#) applied degree by degree: the  $(1, 1)$  part changes sign, the  $(1, 2)$  and  $(2, 2)$  parts do not.

(c)  $x \wedge x = \mathbf{e}_1 \wedge \mathbf{e}_1 + \mathbf{e}_1 \wedge \mathbf{e}_2 \wedge \mathbf{e}_3 + \mathbf{e}_2 \wedge \mathbf{e}_3 \wedge \mathbf{e}_1 + \mathbf{0} = 2\mathbf{e}_{123}$ , since  $\mathbf{e}_2 \wedge \mathbf{e}_3 \wedge \mathbf{e}_1 = \mathbf{e}_{123}$ . An inhomogeneous element need not square to zero.

### Exercise B2: Decomposable or not.

In  $\Lambda^2 \mathbb{Q}^4$ , writing  $\mathbf{e}_{ij} = \mathbf{e}_i \wedge \mathbf{e}_j$ , determine whether each of the following is decomposable. Justify your answer.

$$\omega_1 = \mathbf{e}_{12} + \mathbf{e}_{13} + 3\mathbf{e}_{14} - 2\mathbf{e}_{23} - \mathbf{e}_{24} + 5\mathbf{e}_{34}, \quad \omega_2 = \mathbf{e}_{12} + \mathbf{e}_{34} + \mathbf{e}_{14}.$$

*Solution.* For  $\omega = \sum_{i < j} c_{ij} \mathbf{e}_{ij}$  in  $\Lambda^2 \mathbb{Q}^4$ , expanding gives

$$\omega \wedge \omega = 2(c_{12}c_{34} - c_{13}c_{24} + c_{14}c_{23}) \mathbf{e}_{1234},$$

since the only pairs of index sets with empty intersection are  $\{12\}, \{34\}$  and  $\{13\}, \{24\}$  and  $\{14\}, \{23\}$ , each occurring in two orders, and  $\mathbf{e}_{13} \wedge \mathbf{e}_{24} = -\mathbf{e}_{1234}$  while the other two products are  $+\mathbf{e}_{1234}$ .

For  $\omega_1$ :  $1 \cdot 5 - 1 \cdot (-1) + 3 \cdot (-2) = 5 + 1 - 6 = 0$ , so  $\omega_1 \wedge \omega_1 = \mathbf{0}$  and the test is inconclusive. In fact  $\omega_1$  is decomposable: it is  $\mathbf{v}_1 \wedge \mathbf{v}_2$  for  $\mathbf{v}_1 = (1, 0, 2, 1)$  and  $\mathbf{v}_2 = (0, 1, 1, 3)$ , by [Example \(A wedge of two vectors in four-dimensional space\)](#).

For  $\omega_2$ :  $1 \cdot 1 - 0 \cdot 0 + 1 \cdot 0 = 1$ , so  $\omega_2 \wedge \omega_2 = 2\mathbf{e}_{1234} \neq \mathbf{0}$ . A decomposable  $\mathbf{u} \wedge \mathbf{w}$  satisfies  $(\mathbf{u} \wedge \mathbf{w}) \wedge (\mathbf{u} \wedge \mathbf{w}) = \mathbf{0}$ , so  $\omega_2$  is **not** decomposable.

### Exercise B3: A Clifford algebra of a line.

Let  $V = \mathbb{R}$  with  $q(x) = x^2$ , the form of signature  $(1, 0)$ . Let  $A = \mathbb{R} \times \mathbb{R}$  with componentwise operations, and  $\iota_A(x) = (x, -x)$ . Prove that  $(A, \iota_A)$  is a Clifford algebra of  $(V, q)$ , so that  $\text{Cl}_{1,0}(\mathbb{R}) \cong \mathbb{R} \times \mathbb{R}$ . Contrast this with  $\text{Cl}_{0,1}(\mathbb{R})$ .

*Solution.*  $A$  is an associative unital  $\mathbb{R}$ -algebra with identity  $(1, 1)$  and  $\dim_{\mathbb{R}} A = 2 = 2^1$ . The map  $\iota_A$  is linear, and

$$\iota_A(x)^2 = (x, -x)(x, -x) = (x^2, x^2) = q(x)(1, 1),$$

which is (C1). The subalgebra generated by  $(1, 1)$  and  $\iota_A(\mathbb{R}) = \text{span}\{(1, -1)\}$  contains the independent pair  $(1, 1), (1, -1)$ , hence is all of  $A$ . By [Corollary 14.10.10](#),  $(A, \iota_A)$  is a Clifford algebra of  $(V, q)$ .



The contrast: Chapter 13 showed  $\text{Cl}_{0,1}(\mathbb{R}) = \mathbb{C}$  (Example (The complex numbers)), where  $q(x) = -x^2$ . Both algebras have dimension 2, but  $\mathbb{C}$  is a field and  $\mathbb{R} \times \mathbb{R}$  has zero divisors,  $(1,0)(0,1) = (0,0)$ . A single sign in  $q$  decides which.

### 14.10.6.3 C. Going deeper

#### Exercise C1: The Plücker relation.

Let  $\dim V = 4$  with basis  $(\mathbf{e}_1, \dots, \mathbf{e}_4)$ , let  $\text{char } F \neq 2$ , and let  $\omega = \sum_{i < j} c_{ij} \mathbf{e}_{ij} \in \Lambda^2 V$ .

- (a) Prove that  $\omega \wedge \omega = 2(c_{12}c_{34} - c_{13}c_{24} + c_{14}c_{23}) \mathbf{e}_{1234}$ .
- (b) Deduce that if  $c_{12}c_{34} - c_{13}c_{24} + c_{14}c_{23} \neq 0$  then  $\omega$  is not decomposable.
- (c) Prove the converse: if the expression vanishes, then  $\omega$  is decomposable. *Hint: if  $\omega \neq \mathbf{0}$ , some  $c_{ij} \neq 0$ ; after relabeling the basis assume  $c_{12} \neq 0$  and look for  $\omega$  as a wedge of two vectors whose first two coordinates you may normalize.*

*Solution.*

- (a) By bilinearity,  $\omega \wedge \omega = \sum_{S,T} c_{SCT} \mathbf{e}_S \wedge \mathbf{e}_T$  over 2-element sets  $S, T$ . A term vanishes unless  $S \cap T = \emptyset$ , so the surviving pairs are  $(\{1,2\}, \{3,4\})$ ,  $(\{1,3\}, \{2,4\})$ ,  $(\{1,4\}, \{2,3\})$  and their reverses. By Theorem 14.10.3 with  $k = l = 2$ , reversing a pair does not change the product, so each contributes twice. Sorting,  $\mathbf{e}_{12} \wedge \mathbf{e}_{34} = \mathbf{e}_{1234}$ ;  $\mathbf{e}_{13} \wedge \mathbf{e}_{24} = \mathbf{e}_1 \wedge \mathbf{e}_3 \wedge \mathbf{e}_2 \wedge \mathbf{e}_4 = -\mathbf{e}_{1234}$ , one transposition; and  $\mathbf{e}_{14} \wedge \mathbf{e}_{23} = \mathbf{e}_1 \wedge \mathbf{e}_4 \wedge \mathbf{e}_2 \wedge \mathbf{e}_3 = \mathbf{e}_{1234}$ , two transpositions. Collecting gives the stated formula.
- (b) If  $\omega = \mathbf{u} \wedge \mathbf{w}$  then  $\omega \wedge \omega = \mathbf{u} \wedge \mathbf{w} \wedge \mathbf{u} \wedge \mathbf{w} = \mathbf{0}$ , a wedge with a repeated vector. Since  $\text{char } F \neq 2$ , (a) then forces the bracket to vanish. Contrapositively, a non-zero bracket rules out decomposability.
- (c) If  $\omega = \mathbf{0}$  it is decomposable. Otherwise relabel the basis so that  $c_{12} \neq 0$ ; this is legitimate because a permutation of the basis permutes the six coefficients up to sign and changes the bracket by at most a sign, so both the hypothesis and the conclusion are preserved. Put

$$\mathbf{u} = c_{12}\mathbf{e}_1 - c_{23}\mathbf{e}_3 - c_{24}\mathbf{e}_4, \quad \mathbf{w} = \mathbf{e}_2 + \frac{c_{13}}{c_{12}}\mathbf{e}_3 + \frac{c_{14}}{c_{12}}\mathbf{e}_4.$$

Expanding and sorting each wedge,

$$\begin{aligned} \mathbf{u} \wedge \mathbf{w} &= c_{12}\mathbf{e}_{12} + c_{13}\mathbf{e}_{13} + c_{14}\mathbf{e}_{14} \\ &\quad + c_{23}\mathbf{e}_{23} + c_{24}\mathbf{e}_{24} + \frac{1}{c_{12}}(c_{13}c_{24} - c_{14}c_{23})\mathbf{e}_{34}. \end{aligned}$$

Each of the first five coefficients is read off a single product: for instance the  $\mathbf{e}_{23}$  term comes from  $(-c_{23}\mathbf{e}_3) \wedge \mathbf{e}_2 = c_{23}\mathbf{e}_{23}$ , and the  $\mathbf{e}_{13}$  term from  $c_{12}\mathbf{e}_1 \wedge \frac{c_{13}}{c_{12}}\mathbf{e}_3$ . The last comes from

$$(-c_{23}\mathbf{e}_3 - c_{24}\mathbf{e}_4) \wedge \left( \frac{c_{13}}{c_{12}}\mathbf{e}_3 + \frac{c_{14}}{c_{12}}\mathbf{e}_4 \right) = \frac{c_{13}c_{24} - c_{14}c_{23}}{c_{12}} \mathbf{e}_{34}.$$

By hypothesis  $c_{13}c_{24} - c_{14}c_{23} = c_{12}c_{34}$ , so that coefficient is  $c_{34}$  and  $\mathbf{u} \wedge \mathbf{w} = \omega$ . Hence  $\omega$  is decomposable.

**Exercise C2: Characteristic two.**

Let  $F = \mathbb{F}_2$  and  $\dim V = 4$  with basis  $(\mathbf{e}_1, \dots, \mathbf{e}_4)$ .

- (a) Prove that  $\omega \wedge \omega = \mathbf{0}$  for **every**  $\omega \in \Lambda^2 V$ .
- (b) Explain why (a) does not make every element of  $\Lambda^2 V$  decomposable, and name the element and the argument that settle it.
- (c) Is the statement “ $\omega \wedge \omega = \mathbf{0}$  for every  $\omega$  of odd degree” also special to characteristic 2? Justify your answer.

*Solution.*

- (a) By the computation of [Exercise \(C1: The Plücker relation\)](#) (a),  $\omega \wedge \omega$  is 2 times an element of  $\Lambda^4 V$ , and  $2 = 0$  in  $\mathbb{F}_2$ . (Equivalently: in  $\sum_{S,T} c_{S,T} \mathbf{e}_S \wedge \mathbf{e}_T$  the diagonal terms vanish and the off-diagonal ones pair up, each pair summing to  $2c_{S,T} \mathbf{e}_S \wedge \mathbf{e}_T = \mathbf{0}$ .)
- (b) Because  $\omega \wedge \omega = \mathbf{0}$  is only a **necessary** condition for decomposability, and over  $\mathbb{F}_2$  everything satisfies it, so it carries no information at all. The element  $\omega = \mathbf{e}_1 \wedge \mathbf{e}_2 + \mathbf{e}_3 \wedge \mathbf{e}_4$  is still not decomposable, by [Theorem 14.10.4](#), whose proof computes  $\mathbf{v} \wedge \omega$  for a general  $\mathbf{v}$  and never divides by 2.
- (c) No, that one is characteristic-free: [Theorem 14.10.3](#) proves  $\omega \wedge \omega = \mathbf{0}$  for odd degree over every field, by pairing off the terms with  $S \neq T$  rather than by dividing. Over  $\mathbb{F}_2$  the statement is true but vacuous as a test, for the same reason as in (a).

**Exercise C3: The even part of a Clifford algebra.**

Let  $\text{char } F \neq 2$ ,  $\dim V = n \geq 1$ , and let  $(C, \iota)$  be a Clifford algebra of  $(V, q)$ , and let  $(\mathbf{e}_1, \dots, \mathbf{e}_n)$  be an orthogonal basis of  $V$  for  $\beta$ . Let  $C^+ \subseteq C$  be the span of the sorted products with  $k$  **even**, and  $C^-$  the span of those with  $k$  odd. Prove that  $C = C^+ \oplus C^-$ , that  $C^+$  is a subalgebra of  $C$ , and that  $\dim_F C^+ = \dim_F C^- = 2^{n-1}$ . Is  $C^-$  a subalgebra?

*Hint: what does the sorting procedure of [Proposition 13.12.3](#) do to the parity of the length of a product?*

*Solution.* The sorted products form a basis of  $C$  by [Theorem 14.10.8](#), and they are partitioned by the parity of  $|S|$ , so  $C = C^+ \oplus C^-$  and the two dimensions are  $\sum_{k \text{ even}} \binom{n}{k}$  and  $\sum_{k \text{ odd}} \binom{n}{k}$ . These are equal and sum to  $2^n$ , since  $\sum_k (-1)^k \binom{n}{k} = (1 - 1)^n = 0$  for  $n \geq 1$ ; so each is  $2^{n-1}$ .

For the subalgebra claim, note that the two moves in the proof of [Proposition 13.12.3](#) — swapping two neighboring distinct factors, and replacing  $\iota(\mathbf{e}_i)^2$  by the scalar  $d_i$  — change the length of a product by 0 and by 2 respectively. Both preserve parity. So the product of two sorted products of lengths  $k$  and  $l$  is a combination of sorted products of length congruent to  $k + l$  modulo 2. Hence  $C^+ C^+ \subseteq C^+$ , and since  $1_C \in C^+$  (the empty product, of length 0),  $C^+$  is a subalgebra.

$C^-$  is not:  $1_C \notin C^-$ , and  $C^- C^- \subseteq C^+$  by the same parity count, so  $C^-$  fails to be closed as soon as it holds an element with non-zero square, such as  $\iota(\mathbf{e}_i)$  with  $q(\mathbf{e}_i) \neq 0$ .

## 14.11 Tensors in Quantum Mechanics

The chapter ends outside mathematics, and the arrangement is the one Chapter 13 used for Minkowski space and for the second derivative test: physics supplies a modeling assumption, the book quotes it and proves nothing about it, and everything after it is linear algebra. The assumption is that a composite quantum system is the tensor product of its parts. Two things follow. One is a phenomenon, **entanglement**, which turns out to be nothing but §02's observation that not every element of  $V \otimes W$  is a simple tensor. The other is a theorem, the **Schmidt decomposition**, which turns out to be Chapter 12's singular value decomposition wearing different notation. The second is the real payoff, and it is why this section closes Part IV: it joins the two halves of the book's later chapters at a single point.

### 14.11.1 What is quoted and what is proved

Quantum mechanics, as a physical theory, is not derived here and cannot be. What we take from it is a short list of modeling conventions, stated once so that the reader can see exactly where the mathematics starts.

- (Q1) The states of a quantum system are the **unit vectors** of a complex inner product space  $V$ , where two unit vectors differing by a scalar of modulus 1 describe the same state.
- (Q2) If a system  $A$  has state space  $V$  and a system  $B$  has state space  $W$ , then the composite system “ $A$  together with  $B$ ” has state space  $V \otimes W$ .
- (Q3) If  $A$  is in the state  $\mathbf{v}$  and  $B$  is in the state  $\mathbf{w}$ , then the composite is in the state  $\mathbf{v} \otimes \mathbf{w}$ .

None of (Q1)–(Q3) is a theorem, and this book does not argue for any of them. They are quoted exactly as Taylor's theorem was quoted in Chapter 13 §06 — as hypotheses that give the mathematics below something to stand on. Everything proved in this section is a statement about finite-dimensional complex inner product spaces, true whether or not (Q1)–(Q3) describe the world.

Two remarks make the list less arbitrary. First,  $\otimes$  rather than  $\oplus$  is visible in a dimension count: a spin has a two-dimensional state space, a **qubit**, and two spins together have a four-dimensional one, not a three-dimensional one. Of the two standard constructions, only  $\otimes$  produces the observed count, since  $\dim(V \otimes W) = \dim V \cdot \dim W$  by [Theorem 14.2.5](#). Second, (Q3) very nearly forces (Q2): the assignment  $(\mathbf{v}, \mathbf{w}) \mapsto (\text{joint state})$  is required to be bilinear, and by [Definition 14.2.1](#) every bilinear map out of  $V \times W$  factors uniquely through  $V \otimes W$ . The universal property is the precise statement that the joint state depends on the parts bilinearly and in no other way.

Genuine quantum mechanics uses infinite-dimensional spaces with a completeness condition. Everything below is finite-dimensional, which is where this book lives, and where qubits live too.

Before states can be unit vectors,  $V \otimes W$  needs an inner product. There is only one reasonable candidate, and it exists.

#### Proposition 14.11.1 — The Inner Product on a Tensor Product

Let  $V$  and  $W$  be finite-dimensional inner product spaces over  $F = \mathbb{C}$  (or over  $\mathbb{R}$ ). Then

there is **exactly one** inner product on  $V \otimes W$  satisfying

$$\langle \mathbf{v} \otimes \mathbf{w}, \mathbf{v}' \otimes \mathbf{w}' \rangle = \langle \mathbf{v}, \mathbf{v}' \rangle \langle \mathbf{w}, \mathbf{w}' \rangle$$

for all  $\mathbf{v}, \mathbf{v}' \in V$  and  $\mathbf{w}, \mathbf{w}' \in W$ . Moreover, if  $\mathcal{E}$  and  $\mathcal{F}$  are orthonormal bases of  $V$  and  $W$ , then  $(\mathbf{e}_i \otimes \mathbf{f}_j)$  is an orthonormal basis of  $V \otimes W$ .

*Proof. Uniqueness.* The simple tensors span  $V \otimes W$  (Proposition 14.2.4), and an inner product is additive in the first slot and conjugate-additive in the second, so its values on all pairs of vectors are determined by its values on pairs of simple tensors.

*Existence.* Fix orthonormal bases  $\mathcal{E} = (\mathbf{e}_1, \dots, \mathbf{e}_m)$  of  $V$  and  $\mathcal{F} = (\mathbf{f}_1, \dots, \mathbf{f}_n)$  of  $W$ , which exist by Theorem 10.2.2. By Theorem 14.2.5 the  $mn$  vectors  $\mathbf{e}_i \otimes \mathbf{f}_j$  form a basis of  $V \otimes W$ , so the rule

$$\left\langle \sum_{i,j} c_{ij} \mathbf{e}_i \otimes \mathbf{f}_j, \sum_{i,j} d_{ij} \mathbf{e}_i \otimes \mathbf{f}_j \right\rangle := \sum_{i,j} c_{ij} \overline{d_{ij}}$$

is well defined; it is the standard inner product of coordinate vectors, hence satisfies (IP1)–(IP3) of Definition 10.1.1. To check the product formula, write  $\mathbf{v} = \sum_i a_i \mathbf{e}_i$  and  $\mathbf{w} = \sum_j b_j \mathbf{f}_j$ . Bilinearity of  $\otimes$  gives  $\mathbf{v} \otimes \mathbf{w} = \sum_{i,j} a_i b_j \mathbf{e}_i \otimes \mathbf{f}_j$ , and likewise for  $\mathbf{v}' \otimes \mathbf{w}'$ , so

$$\langle \mathbf{v} \otimes \mathbf{w}, \mathbf{v}' \otimes \mathbf{w}' \rangle = \sum_{i,j} a_i b_j \overline{a'_i b'_j} = \left( \sum_i a_i \overline{a'_i} \right) \left( \sum_j b_j \overline{b'_j} \right),$$

which is  $\langle \mathbf{v}, \mathbf{v}' \rangle \langle \mathbf{w}, \mathbf{w}' \rangle$  by Theorem 10.2.1. Finally, the product formula gives  $\langle \mathbf{e}_i \otimes \mathbf{f}_j, \mathbf{e}_k \otimes \mathbf{f}_l \rangle = \delta_{ik} \delta_{jl}$  for **any** orthonormal  $\mathcal{E}$  and  $\mathcal{F}$ , so the displayed construction did not depend on the pair chosen. This proves the proposition. ■

## 14.11.2 Entanglement is failure of simplicity

The whole phenomenon fits in the language the chapter already has. A state of the composite is a unit vector of  $V \otimes W$ , and a state of the form  $\mathbf{v} \otimes \mathbf{w}$  is one in which each part has a state of its own. Section 02 proved that not every vector of a tensor product has that form; a composite state that does not describes a system whose parts have no separate states at all.

*A state is entangled when it is a vector of  $V \otimes W$  that is not a simple tensor.*

### Definition 14.11.1 — Product State, Entangled State

Let  $V$  and  $W$  be finite-dimensional complex inner product spaces, and let  $\mathbf{z} \in V \otimes W$  be a unit vector. Then  $\mathbf{z}$  is a **product state**, also called **separable**, if

$$\mathbf{z} = \mathbf{v} \otimes \mathbf{w} \quad \text{for some } \mathbf{v} \in V, \mathbf{w} \in W,$$

and  $\mathbf{z}$  is **entangled** otherwise. The same words are used for any non-zero  $\mathbf{z} \in V \otimes W$ , the unit-length condition playing no part in the distinction.

Clause by clause. **Product state** is a synonym for “simple tensor”, the term of §02; only the context is new. The definition is relative to the **splitting**  $V \otimes W$ , not to the vector alone: “entangled” is a statement about a vector *and* a choice of two factors. And  $\mathbf{v}$  and  $\mathbf{w}$  need not be unit vectors, though they can be taken so:  $\|\mathbf{v} \otimes \mathbf{w}\| = \|\mathbf{v}\|\|\mathbf{w}\|$  by Proposition 14.11.1, so rescaling one factor up and the other down costs nothing.

**Example. (The Bell state)**

In  $\mathbb{C}^2 \otimes \mathbb{C}^2$ , write  $\mathcal{E} = (\mathbf{e}_1, \mathbf{e}_2)$  for the standard orthonormal basis of the first factor and  $\mathcal{F} = (\mathbf{f}_1, \mathbf{f}_2)$  for the same basis regarded as sitting in the second factor, the two names being there only to keep the factors apart. Let

$$\mathbf{z}_0 := \frac{1}{\sqrt{2}}(\mathbf{e}_1 \otimes \mathbf{f}_1 + \mathbf{e}_2 \otimes \mathbf{f}_2).$$

Show that  $\mathbf{z}_0$  is a unit vector, and that it is entangled.

*Solution.* By Proposition 14.11.1 the four vectors  $\mathbf{e}_i \otimes \mathbf{f}_j$  are orthonormal, so  $\|\mathbf{z}_0\|^2 = \frac{1}{2}(1 + 1) = 1$ .

For entanglement there is nothing to prove that §02 has not already proved. Once the two bases are recognized as the same one,  $\mathbf{e}_1 \otimes \mathbf{f}_1 + \mathbf{e}_2 \otimes \mathbf{f}_2$  is exactly the tensor of Example (A tensor that is not simple), and  $\mathbf{z}_0$  is a non-zero multiple of it: if  $\mathbf{z}_0 = \mathbf{v} \otimes \mathbf{w}$  then  $\mathbf{e}_1 \otimes \mathbf{f}_1 + \mathbf{e}_2 \otimes \mathbf{f}_2 = (\sqrt{2}\mathbf{v}) \otimes \mathbf{w}$ , which Example (A tensor that is not simple) rules out. So  $\mathbf{z}_0$  is entangled.

The physical reading is one paragraph long, and the book stops there. Under (Q1)–(Q3), a composite in the state  $\mathbf{z}_0$  is a pair of qubits of which two things are true at once: the pair has a completely specified state, and neither member has a state at all, since  $\mathbf{z}_0$  is not of the form  $\mathbf{v} \otimes \mathbf{w}$ . Nothing about the individual qubits is being suppressed; the postulates simply do not assign them states. That is what is meant by calling the two qubits entangled, and it is why the word attaches to a piece of linear algebra rather than to a measurement. Whether the postulates are true of the world is a question this book does not address.

*Warning.* **A sum of simple tensors can still be simple.** “Entangled” does not mean “written as a sum of more than one term”. In  $\mathbb{C}^2 \otimes \mathbb{C}^2$ ,

$$\frac{1}{2}(\mathbf{e}_1 \otimes \mathbf{f}_1 + \mathbf{e}_1 \otimes \mathbf{f}_2 + \mathbf{e}_2 \otimes \mathbf{f}_1 + \mathbf{e}_2 \otimes \mathbf{f}_2) = \frac{\mathbf{e}_1 + \mathbf{e}_2}{\sqrt{2}} \otimes \frac{\mathbf{f}_1 + \mathbf{f}_2}{\sqrt{2}},$$

a four-term expression for a product state. Entanglement is a property of the **vector**, not of an expression for it, and deciding it needs an invariant. Supplying one is the job of the rest of this section.

*Quick check.* Is  $\mathbf{z} = \frac{1}{\sqrt{3}}(\mathbf{e}_1 \otimes \mathbf{f}_1 + \mathbf{e}_1 \otimes \mathbf{f}_2 + \mathbf{e}_2 \otimes \mathbf{f}_1)$  in  $\mathbb{C}^2 \otimes \mathbb{C}^2$  entangled?

*Solution.* Yes. Suppose  $\mathbf{z} = \mathbf{v} \otimes \mathbf{w}$  with  $\mathbf{v} = a_1\mathbf{e}_1 + a_2\mathbf{e}_2$  and  $\mathbf{w} = b_1\mathbf{f}_1 + b_2\mathbf{f}_2$ . Expanding,  $\mathbf{v} \otimes \mathbf{w} = \sum_{i,j} a_i b_j \mathbf{e}_i \otimes \mathbf{f}_j$ , and the  $\mathbf{e}_i \otimes \mathbf{f}_j$  are a basis (Theorem 14.2.5), so comparing coefficients gives

$$a_1 b_1 = a_1 b_2 = a_2 b_1 = \frac{1}{\sqrt{3}}, \quad a_2 b_2 = 0.$$

Then  $(a_1 b_1)(a_2 b_2) = 0$  while  $(a_1 b_2)(a_2 b_1) = \frac{1}{3}$ ; but both products equal  $a_1 a_2 b_1 b_2$ . So no such  $\mathbf{v}, \mathbf{w}$  exist.

### 14.11.3 The Schmidt decomposition

The Quick check used a  $2 \times 2$  determinant on an array of coefficients, and that is a hint about where to look. Section 03 already said what the array is: by [Theorem 14.3.9](#) a tensor **is** a linear map, and a linear map between finite-dimensional spaces is a matrix. Chapter 12 knows exactly one thing about an arbitrary matrix, and it knows it completely.

Write out the identification for the case at hand. Fix bases  $\mathcal{E} = (\mathbf{e}_1, \dots, \mathbf{e}_m)$  of  $V$  and  $\mathcal{F} = (\mathbf{f}_1, \dots, \mathbf{f}_n)$  of  $W$ , and let  $(\varphi_1, \dots, \varphi_n)$  be the basis of  $W^*$  dual to  $\mathcal{F}$  ([Theorem 4.1.3](#)). The map

$$V \times W \rightarrow \mathcal{L}(W^*, V), \quad (\mathbf{v}, \mathbf{w}) \mapsto (\varphi \mapsto \varphi(\mathbf{w}) \mathbf{v}),$$

is bilinear in  $(\mathbf{v}, \mathbf{w})$ , so by [Definition 14.2.1](#) it induces a linear  $\Theta: V \otimes W \rightarrow \mathcal{L}(W^*, V)$ . It sends  $\mathbf{e}_i \otimes \mathbf{f}_j$  to the map  $\varphi_k \mapsto \delta_{jk} \mathbf{e}_i$ , whose matrix from  $(\varphi_1, \dots, \varphi_n)$  to  $\mathcal{E}$  is the matrix unit  $\mathbf{E}_{ij}$ . The matrix units are a basis of  $M_{m \times n}(\mathbb{C})$  and taking the matrix of a map in fixed bases is an isomorphism ([Theorem 3.6.4](#)), so  $\Theta$  carries a basis to a basis and is itself an isomorphism. It is [Theorem 14.3.9](#) with the two factors in the other order and  $W$  identified with  $W^{**}$  ([Theorem 4.4.6](#)), and for coordinate spaces it is §02's matrix model [Theorem 14.2.7](#); the letter is shared for that reason. What matters below is the consequence: the matrix of  $\Theta(\mathbf{z})$  is the array of coefficients of  $\mathbf{z}$  in the basis  $(\mathbf{e}_i \otimes \mathbf{f}_j)$ .

#### Theorem 14.11.2 — Schmidt Decomposition

Let  $V$  and  $W$  be finite-dimensional complex inner product spaces of dimensions  $m$  and  $n$ , and let  $\mathbf{z} \in V \otimes W$  be non-zero. Then there are an integer  $r$  with  $1 \leq r \leq \min(m, n)$ , real numbers

$$\sigma_1 \geq \sigma_2 \geq \dots \geq \sigma_r > 0,$$

an orthonormal list  $(\mathbf{x}_1, \dots, \mathbf{x}_r)$  in  $V$  and an orthonormal list  $(\mathbf{y}_1, \dots, \mathbf{y}_r)$  in  $W$  such that

$$\mathbf{z} = \sum_{k=1}^r \sigma_k \mathbf{x}_k \otimes \mathbf{y}_k. \quad (14.11.1)$$

Moreover  $r$  and the numbers  $\sigma_1, \dots, \sigma_r$  are determined by  $\mathbf{z}$ : they do not depend on the lists chosen, nor on any basis. Finally  $\|\mathbf{z}\|^2 = \sum_k \sigma_k^2$ , so  $\mathbf{z}$  is a unit vector exactly when  $\sum_k \sigma_k^2 = 1$ .

*Idea.* The statement is the singular value decomposition of [Theorem 12.8.1](#) with the words changed, so the plan is to change the words back. ① Fix orthonormal bases on both sides; [Theorem 14.2.5](#) turns  $\mathbf{z}$  into a matrix  $\mathbf{C}$  of coefficients, which by the paragraph above is the matrix of the linear map  $\Theta(\mathbf{z})$ . ② Feed  $\mathbf{C}$  to the compact singular value decomposition, [Theorem 12.8.3](#), which over  $\mathbb{C}$  still returns **real positive** numbers  $\sigma_k$  and two orthonormal lists of complex column vectors. ③ Read the column vectors back as vectors of  $V$  and  $W$ ; orthonormal columns become orthonormal vectors because coordinates in an orthonormal basis compute inner products. For uniqueness, run ③ backwards: any expression of the shape [Eq. 14.11.1](#) turns into a factorization  $\mathbf{C} = \mathbf{X}\mathbf{D}\mathbf{Y}^\top$  with  $\mathbf{X}, \mathbf{Y}$  having orthonormal columns, and then  $\mathbf{C}\mathbf{C}^* = \mathbf{X}\mathbf{D}^2\mathbf{X}^*$  hands over the  $\sigma_k^2$  as the non-zero eigenvalues of a matrix built from  $\mathbf{z}$  alone.

*Proof. Step 1: turn  $\mathbf{z}$  into a matrix.* Fix orthonormal bases  $\mathcal{E} = (\mathbf{e}_1, \dots, \mathbf{e}_m)$  of  $V$  and  $\mathcal{F} = (\mathbf{f}_1, \dots, \mathbf{f}_n)$  of  $W$  (Theorem 10.2.2). By Theorem 14.2.5 there are unique scalars  $c_{ij} \in \mathbb{C}$  with  $\mathbf{z} = \sum_{i,j} c_{ij} \mathbf{e}_i \otimes \mathbf{f}_j$ . Let  $\mathbf{C} \in M_{m \times n}(\mathbb{C})$  be the matrix with entries  $c_{ij}$ ; it is the matrix of the linear map  $\Theta(\mathbf{z})$  described above, and it is non-zero because  $\mathbf{z}$  is.

**Step 2: decompose the matrix.** Put  $r = \text{rank } \mathbf{C} \geq 1$ . By Theorem 12.8.3 there are orthonormal lists  $(\mathbf{u}_1, \dots, \mathbf{u}_r)$  in  $\mathbb{C}^m$  and  $(\mathbf{v}_1, \dots, \mathbf{v}_r)$  in  $\mathbb{C}^n$ , and the non-zero singular values  $\sigma_1 \geq \dots \geq \sigma_r > 0$  of  $\mathbf{C}$  (Definition 12.8.1), with

$$\mathbf{C} = \sum_{k=1}^r \sigma_k \mathbf{u}_k \mathbf{v}_k^*, \quad \text{that is} \quad c_{ij} = \sum_{k=1}^r \sigma_k (\mathbf{u}_k)_i \overline{(\mathbf{v}_k)_j}.$$

The  $\sigma_k$  are real and positive even though  $\mathbf{C}$  is complex, since they are square roots of eigenvalues of the positive semidefinite matrix  $\mathbf{C}^* \mathbf{C}$ . Also  $r = \text{rank } \mathbf{C} \leq \min(m, n)$ .

**Step 3: read the columns back.** Set

$$\begin{aligned} \mathbf{x}_k &:= \sum_{i=1}^m (\mathbf{u}_k)_i \mathbf{e}_i \in V, \\ \mathbf{y}_k &:= \sum_{j=1}^n \overline{(\mathbf{v}_k)_j} \mathbf{f}_j \in W. \end{aligned}$$

Substituting the formula for  $c_{ij}$  into  $\mathbf{z} = \sum_{i,j} c_{ij} \mathbf{e}_i \otimes \mathbf{f}_j$  and interchanging the three finite sums, bilinearity of  $\otimes$  gives

$$\mathbf{z} = \sum_{k=1}^r \sigma_k \left( \sum_i (\mathbf{u}_k)_i \mathbf{e}_i \right) \otimes \left( \sum_j \overline{(\mathbf{v}_k)_j} \mathbf{f}_j \right) = \sum_{k=1}^r \sigma_k \mathbf{x}_k \otimes \mathbf{y}_k.$$

Since  $\mathcal{E}$  is orthonormal, Theorem 10.2.1 gives

$$\langle \mathbf{x}_k, \mathbf{x}_l \rangle = \sum_i (\mathbf{u}_k)_i \overline{(\mathbf{u}_l)_i} = \langle \mathbf{u}_k, \mathbf{u}_l \rangle = \delta_{kl},$$

and likewise, since  $\mathcal{F}$  is orthonormal,

$$\langle \mathbf{y}_k, \mathbf{y}_l \rangle = \sum_j \overline{(\mathbf{v}_k)_j} (\mathbf{v}_l)_j = \langle \mathbf{v}_l, \mathbf{v}_k \rangle = \delta_{kl}.$$

So both lists are orthonormal, and Eq. 14.11.1 holds.

**Step 4: uniqueness.** Suppose  $\mathbf{z} = \sum_{k=1}^s \tau_k \mathbf{x}'_k \otimes \mathbf{y}'_k$  is **any** expression of the stated shape, with  $\tau_1 \geq \dots \geq \tau_s > 0$  and both lists orthonormal. Keep the bases  $\mathcal{E}, \mathcal{F}$  and write  $\mathbf{x}'_k = \sum_i \xi_{ik} \mathbf{e}_i$  and  $\mathbf{y}'_k = \sum_j \eta_{jk} \mathbf{f}_j$ . Let  $\mathbf{X} \in M_{m \times s}(\mathbb{C})$  have entries  $\xi_{ik}$  and  $\mathbf{Y} \in M_{n \times s}(\mathbb{C})$  have entries  $\eta_{jk}$ . Orthonormality of the two lists says, by Theorem 10.2.1 again, that the columns of  $\mathbf{X}$  and of  $\mathbf{Y}$  are orthonormal, that is

$$\mathbf{X}^* \mathbf{X} = \mathbf{I}_s, \quad \mathbf{Y}^* \mathbf{Y} = \mathbf{I}_s.$$



Expanding the expression for  $\mathbf{z}$  in the basis  $(\mathbf{e}_i \otimes \mathbf{f}_j)$  and comparing with the unique coefficients  $c_{ij}$  of Step 1,

$$c_{ij} = \sum_{k=1}^s \tau_k \xi_{ik} \eta_{jk}, \quad \text{i.e.} \quad \mathbf{C} = \mathbf{X} \mathbf{D} \mathbf{Y}^\top,$$

where  $\mathbf{D} = \text{diag}(\tau_1, \dots, \tau_s)$ . Taking conjugate transposes and using  $\mathbf{Y}^\top \overline{\mathbf{Y}} = \overline{\mathbf{Y}^* \mathbf{Y}} = \mathbf{I}_s$ ,

$$\mathbf{C} \mathbf{C}^* = \mathbf{X} \mathbf{D} (\mathbf{Y}^\top \overline{\mathbf{Y}}) \mathbf{D} \mathbf{X}^* = \mathbf{X} \mathbf{D}^2 \mathbf{X}^*.$$

Now read off the eigenvalues of  $\mathbf{C} \mathbf{C}^* \in M_m(\mathbb{C})$ . For each  $k$ , the  $k$ -th column  $\mathbf{X} \mathbf{e}_k$  satisfies

$$\mathbf{C} \mathbf{C}^* (\mathbf{X} \mathbf{e}_k) = \mathbf{X} \mathbf{D}^2 \mathbf{X}^* \mathbf{X} \mathbf{e}_k = \mathbf{X} \mathbf{D}^2 \mathbf{e}_k = \tau_k^2 (\mathbf{X} \mathbf{e}_k),$$

and these  $s$  columns are orthonormal. If instead  $\mathbf{z}' \in \mathbb{C}^m$  is orthogonal to every column of  $\mathbf{X}$  then  $\mathbf{X}^* \mathbf{z}' = \mathbf{0}$ , so  $\mathbf{C} \mathbf{C}^* \mathbf{z}' = \mathbf{0}$ . Since  $\mathbb{C}^m = \text{col}(\mathbf{X}) \oplus \text{col}(\mathbf{X})^\perp$  by [Theorem 10.3.2](#), putting an orthonormal basis of  $\text{col}(\mathbf{X})^\perp$  beside the columns of  $\mathbf{X}$  produces an orthonormal basis of  $\mathbb{C}^m$  consisting of eigenvectors of  $\mathbf{C} \mathbf{C}^*$ . Hence the eigenvalue list of  $\mathbf{C} \mathbf{C}^*$ , with multiplicity, is  $\tau_1^2, \dots, \tau_s^2$  together with  $m - s$  zeros.

Every  $\tau_k^2$  is positive, so  $s$  is the number of non-zero eigenvalues of  $\mathbf{C} \mathbf{C}^*$ , and  $(\tau_1^2, \dots, \tau_s^2)$  is the list of those eigenvalues in decreasing order. Both are determined by  $\mathbf{C}$ , which is determined by  $\mathbf{z}$  once  $\mathcal{E}$  and  $\mathcal{F}$  are fixed. So any two expressions of the shape [Eq. 14.11.1](#) have the same  $s$  and the same  $\tau_k$ ; in particular they agree with the  $r$  and  $\sigma_k$  of Steps 2 and 3, and since the statement “ $\mathbf{z}$  admits [Eq. 14.11.1](#) with these  $r, \sigma_k$ ” mentions no basis, the numbers cannot depend on  $\mathcal{E}$  and  $\mathcal{F}$  either.

**Step 5: the norm.** By [Proposition 14.11.1](#),  $\langle \mathbf{x}_k \otimes \mathbf{y}_k, \mathbf{x}_l \otimes \mathbf{y}_l \rangle = \langle \mathbf{x}_k, \mathbf{x}_l \rangle \langle \mathbf{y}_k, \mathbf{y}_l \rangle = \delta_{kl}$ , so the  $r$  vectors  $\mathbf{x}_k \otimes \mathbf{y}_k$  are orthonormal in  $V \otimes W$ . By [Theorem 10.1.3](#),  $\|\mathbf{z}\|^2 = \sum_k \sigma_k^2$ . This proves the theorem. ■

### Definition 14.11.2 — Schmidt Coefficients, Schmidt Rank

Let  $\mathbf{z} \in V \otimes W$  be non-zero. The numbers  $\sigma_1 \geq \dots \geq \sigma_r > 0$  of [Theorem 14.11.2](#) are the **Schmidt coefficients** of  $\mathbf{z}$ , and the integer  $r$  is the **Schmidt rank** of  $\mathbf{z}$ . By convention the Schmidt rank of  $\mathbf{0}$  is 0.

Both are well defined precisely because of Step 4, and the proof identifies them: the Schmidt coefficients are the non-zero singular values of the coefficient matrix of  $\mathbf{z}$  in any pair of orthonormal bases, and the Schmidt rank is the rank of that matrix. That is the invariant the warning above asked for, and it is also §02’s: the smallest number of simple tensors summing to  $\mathbf{z}$  is the rank of  $\Theta(\mathbf{z})$ , and [Theorem 14.11.2](#) now exhibits a shortest such sum, with its terms orthogonal on both sides at once.

### Corollary 14.11.3 — Entanglement Is Schmidt Rank at Least Two

A non-zero  $\mathbf{z} \in V \otimes W$  is a product state if and only if its Schmidt rank is 1. Hence a state is entangled exactly when its Schmidt rank is at least 2.



*Proof.* ( $\Leftarrow$ ) If  $r = 1$  then  $\mathbf{z} = \sigma_1 \mathbf{x}_1 \otimes \mathbf{y}_1 = (\sigma_1 \mathbf{x}_1) \otimes \mathbf{y}_1$ , a simple tensor.

( $\Rightarrow$ ) Suppose  $\mathbf{z} = \mathbf{v} \otimes \mathbf{w}$ . Neither factor is  $\mathbf{0}$ , since  $\mathbf{z} \neq \mathbf{0}$ . Put  $\mathbf{x}_1 = \mathbf{v}/\|\mathbf{v}\|$  and  $\mathbf{y}_1 = \mathbf{w}/\|\mathbf{w}\|$ , which is legitimate because  $\|\mathbf{v}\|, \|\mathbf{w}\| > 0$ . Then  $\mathbf{z} = \sigma_1 \mathbf{x}_1 \otimes \mathbf{y}_1$  with  $\sigma_1 = \|\mathbf{v}\|\|\mathbf{w}\| > 0$ , and the one-term lists  $(\mathbf{x}_1), (\mathbf{y}_1)$  are orthonormal. This is an expression of the shape Eq. 14.11.1 with one term, so the uniqueness clause of Theorem 14.11.2 gives  $r = 1$ . ■

**Warning.** The Schmidt coefficients are determined; the vectors are not. For the Bell state  $\mathbf{z}_0$  of Example (The Bell state) the coefficient matrix is  $\frac{1}{\sqrt{2}}\mathbf{I}_2$ , whose singular values are  $\frac{1}{\sqrt{2}}, \frac{1}{\sqrt{2}}$ . Repeated coefficients leave the lists free, exactly as a repeated eigenvalue of  $\mathbf{A}^*\mathbf{A}$  leaves  $\mathbf{V}$  free in Theorem 12.8.2: for every orthonormal basis  $(\mathbf{x}_1, \mathbf{x}_2)$  of  $\mathbb{C}^2$  one has  $\mathbf{z}_0 = \frac{1}{\sqrt{2}}(\mathbf{x}_1 \otimes \mathbf{y}_1 + \mathbf{x}_2 \otimes \mathbf{y}_2)$  for a suitable orthonormal  $(\mathbf{y}_1, \mathbf{y}_2)$ . Never speak of *the* Schmidt vectors; speak of *the* Schmidt coefficients.

### Example. (A Schmidt decomposition in two by three)

In  $\mathbb{C}^2 \otimes \mathbb{C}^3$ , find the Schmidt decomposition, the Schmidt rank and the Schmidt coefficients of

$$\mathbf{z} = \frac{1}{2}(\mathbf{e}_1 \otimes \mathbf{f}_1 + \mathbf{e}_1 \otimes \mathbf{f}_3 + \mathbf{e}_2 \otimes \mathbf{f}_2 - \mathbf{e}_2 \otimes \mathbf{f}_3),$$

where  $\mathcal{E}$  and  $\mathcal{F}$  are the standard orthonormal bases.

*Solution.* The coefficient matrix is

$$\mathbf{C} = \frac{1}{2} \begin{pmatrix} 1 & 0 & 1 \\ 0 & 1 & -1 \end{pmatrix} \in M_{2 \times 3}(\mathbb{R}).$$

This is half the matrix of Example (A singular value decomposition of a 2 by 3 matrix), so by Exercise (B4: Determine whether) (b) its singular values are half of those found there:  $\sigma_1 = \frac{\sqrt{3}}{2}$  and  $\sigma_2 = \frac{1}{2}$ . Both are non-zero, so  $\text{rank } \mathbf{C} = 2$  and the Schmidt rank is 2; by Corollary 14.11.3,  $\mathbf{z}$  is entangled. As a check,  $\sigma_1^2 + \sigma_2^2 = \frac{3}{4} + \frac{1}{4} = 1$ , and indeed  $\|\mathbf{z}\|^2 = 4 \cdot \frac{1}{4} = 1$ .

For the vectors, Theorem 12.8.3 asks for an orthonormal eigenbasis of the  $3 \times 3$  matrix  $\mathbf{C}^*\mathbf{C}$ ; as in Example (A singular value decomposition of a 2 by 3 matrix) it is cheaper to work with the smaller  $\mathbf{C}\mathbf{C}^\top = \frac{1}{4} \begin{pmatrix} 2 & -1 \\ -1 & 2 \end{pmatrix}$ , whose orthonormal eigenvectors are the left singular vectors  $\mathbf{u}_k$ . They are  $\frac{1}{\sqrt{2}}(1, -1)$  for the eigenvalue  $\frac{3}{4}$  and  $\frac{1}{\sqrt{2}}(1, 1)$  for  $\frac{1}{4}$ . So  $\mathbf{u}_1 = \frac{1}{\sqrt{2}}(1, -1)$  and  $\mathbf{u}_2 = \frac{1}{\sqrt{2}}(1, 1)$ , and the relation  $\mathbf{C}^*\mathbf{u}_i = \sigma_i \mathbf{v}_i$  of Eq. 12.8.3 gives

$$\mathbf{v}_1 = \frac{\mathbf{C}^\top \mathbf{u}_1}{\sigma_1} = \frac{1}{\sqrt{6}}(1, -1, 2), \quad \mathbf{v}_2 = \frac{\mathbf{C}^\top \mathbf{u}_2}{\sigma_2} = \frac{1}{\sqrt{2}}(1, 1, 0).$$

All entries are real, so the conjugation in Step 3 of Theorem 14.11.2 does nothing here, and reading the columns back gives

$$\begin{aligned} \mathbf{x}_1 &= \frac{1}{\sqrt{2}}(\mathbf{e}_1 - \mathbf{e}_2), & \mathbf{y}_1 &= \frac{1}{\sqrt{6}}(\mathbf{f}_1 - \mathbf{f}_2 + 2\mathbf{f}_3), \\ \mathbf{x}_2 &= \frac{1}{\sqrt{2}}(\mathbf{e}_1 + \mathbf{e}_2), & \mathbf{y}_2 &= \frac{1}{\sqrt{2}}(\mathbf{f}_1 + \mathbf{f}_2), \end{aligned}$$

so that  $\mathbf{z} = \frac{\sqrt{3}}{2} \mathbf{x}_1 \otimes \mathbf{y}_1 + \frac{1}{2} \mathbf{x}_2 \otimes \mathbf{y}_2$ . Multiplying out recovers  $\mathbf{z}$ : the two terms are

$\frac{1}{4}(\mathbf{e}_1 - \mathbf{e}_2) \otimes (\mathbf{f}_1 - \mathbf{f}_2 + 2\mathbf{f}_3)$  and  $\frac{1}{4}(\mathbf{e}_1 + \mathbf{e}_2) \otimes (\mathbf{f}_1 + \mathbf{f}_2)$ , whose coefficients at  $\mathbf{e}_1 \otimes \mathbf{f}_2$  and at  $\mathbf{e}_2 \otimes \mathbf{f}_1$  cancel, leaving  $\frac{1}{2}$  at  $\mathbf{e}_1 \otimes \mathbf{f}_1$ ,  $\mathbf{e}_1 \otimes \mathbf{f}_3$  and  $\mathbf{e}_2 \otimes \mathbf{f}_2$  and  $-\frac{1}{2}$  at  $\mathbf{e}_2 \otimes \mathbf{f}_3$ .

*Remark.* Truncating Eq. 14.11.1 after  $k$  terms is truncating a singular value decomposition, so Theorem 12.10.3 applies verbatim: the closest state of Schmidt rank at most  $k$  is the truncated sum, at distance  $(\sum_{i>k} \sigma_i^2)^{1/2}$ . Exercise (C2: Best approximation by a state of small Schmidt rank) works this out. The remark also marks the limit of the theorem: everything here concerns **two** factors, because a matrix has two indices. For  $V \otimes W \otimes U$  a general vector admits no expression as a sum of simple tensors built from three orthonormal families; the book records that without proof.

## 14.11.4 Looking back at Part IV

Part IV asked one question in two ways. Chapter 13 took a single bilinear form and asked what it is up to a change of basis; Chapter 14 asked what all multilinear maps have in common, and answered by building the space that turns every one of them into a linear map.

The two chapters answer each other. Chapter 13 worked one form at a time and classified: congruence rather than similarity, complete the square, split off a hyperbolic plane, count signs rather than values. Every invariant it produced — the inertia of Theorem 13.5.1, the Witt index of Definition 13.7.4, the discriminant of Definition 13.2.2 — was a number attached to a form that survives a change of basis, and every classification theorem said that a short list of such numbers is all there is. Chapter 14 worked with all forms at once and constructed: define by universal property and then build a model, impose relations by a quotient, define a map out of a quotient upstairs first, and count a basis by counting index tuples. Where Chapter 13 diagonalized a symmetric form by completing the square, Chapter 14 produced the space  $\text{Sym}^2 V$  on which that form is a single linear functional; where Chapter 13 proved that alternating is the characteristic-free strengthening of skew, Chapter 14 built  $\Lambda^k V$  by killing  $\mathbf{v} \otimes \mathbf{v}$  for exactly that reason; and where Chapter 13 could only say that a Clifford algebra is unique if it exists, Chapter 14 built one out of  $T(V)$  and settled its dimension.

This last section adds a third connection, backwards rather than sideways. The Schmidt decomposition is not a new theorem; it is Theorem 12.8.3 with  $V \otimes W$  written in place of  $M_{m \times n}$ . Part III spent a chapter on what a matrix does to two orthonormal bases, one in each of two spaces; Part IV spent a chapter on the space in which a pair of vectors, one from each of two spaces, becomes a single object. Those are the same statement read in two directions, and Theorem 14.3.9 is the sentence that says so. Asked years from now what the tensor product is for, a reader may answer that it is where a bilinear map becomes linear; the shortest true answer is that  $V \otimes W$  is where the two sides of a matrix live.

## 14.11.5 Exercises

### 14.11.5.1 A. Check your understanding

#### Exercise A1.

- (a) State the three modeling assumptions (Q1)–(Q3) of this section, and say which of them this book proves.

- (b) Define **product state** and **entangled state**, and say what data the words depend on besides the vector itself.
- (c) State the Schmidt decomposition, with every hypothesis, and say what is unique in it and what is not.
- (d) Define the Schmidt rank, and give the criterion for entanglement in terms of it.
- (e) True or false: a state written as a sum of three simple tensors has Schmidt rank 3. Justify your answer.
- (f) Which theorem of Chapter 12 is the Schmidt decomposition, and which theorem of §03 licenses the translation?

*Solution.*

- (a) (Q1) states are unit vectors of a complex inner product space, up to a scalar of modulus 1; (Q2) the state space of a composite system is the tensor product of the state spaces of the parts; (Q3) if the parts are in states  $\mathbf{v}$  and  $\mathbf{w}$ , the composite is in the state  $\mathbf{v} \otimes \mathbf{w}$ . The book proves none of them: they are quoted from physics.
- (b) A unit vector  $\mathbf{z} \in V \otimes W$  is a product state if  $\mathbf{z} = \mathbf{v} \otimes \mathbf{w}$  for some  $\mathbf{v} \in V$  and  $\mathbf{w} \in W$ , and entangled otherwise (Definition 14.11.1). The words depend on the **splitting**: the pair of factors  $V$  and  $W$ , not the vector alone.
- (c) Theorem 14.11.2: for  $V, W$  finite-dimensional complex inner product spaces and  $\mathbf{z} \in V \otimes W$  non-zero, there are  $r \geq 1$ , reals  $\sigma_1 \geq \cdots \geq \sigma_r > 0$  and orthonormal lists  $(\mathbf{x}_k)$  in  $V$  and  $(\mathbf{y}_k)$  in  $W$  with  $\mathbf{z} = \sum_k \sigma_k \mathbf{x}_k \otimes \mathbf{y}_k$ . The number  $r$  and the  $\sigma_k$  are unique; the lists are not.
- (d) The Schmidt rank is that  $r$  (Definition 14.11.2). By Corollary 14.11.3,  $\mathbf{z}$  is entangled exactly when  $r \geq 2$ .
- (e) False. The Schmidt rank is at most 3, but need not equal 3: the warning after Definition 14.11.1 displays a four-term sum whose Schmidt rank is 1.
- (f) It is Theorem 12.8.3, the compact form of the Singular Value Decomposition Theorem 12.8.1. The translation is licensed by Theorem 14.3.9, which identifies a tensor with a linear map and hence, in bases, with a matrix.

### 14.11.5.2 B. Practice

#### Exercise B1: Determine which are entangled.

Work in  $\mathbb{C}^2 \otimes \mathbb{C}^2$  with the standard orthonormal bases. Determine which of the following unit vectors are entangled. Justify your answer, and for each product state exhibit the two factors.

- (a)  $\frac{1}{2}(\mathbf{e}_1 \otimes \mathbf{f}_1 + \mathbf{e}_1 \otimes \mathbf{f}_2 + \mathbf{e}_2 \otimes \mathbf{f}_1 + \mathbf{e}_2 \otimes \mathbf{f}_2)$
- (b)  $\frac{1}{\sqrt{2}}(\mathbf{e}_1 \otimes \mathbf{f}_2 - \mathbf{e}_2 \otimes \mathbf{f}_1)$
- (c)  $\frac{1}{\sqrt{3}}(\mathbf{e}_1 \otimes \mathbf{f}_1 + \mathbf{e}_1 \otimes \mathbf{f}_2 + \mathbf{e}_2 \otimes \mathbf{f}_2)$
- (d)  $\frac{1}{\sqrt{10}}(\mathbf{e}_1 \otimes \mathbf{f}_1 + 2\mathbf{e}_1 \otimes \mathbf{f}_2 - \mathbf{e}_2 \otimes \mathbf{f}_1 - 2\mathbf{e}_2 \otimes \mathbf{f}_2)$

*Solution.* By Corollary 14.11.3 and the identification of the Schmidt rank with the rank of the coefficient matrix, it suffices to compute rank  $\mathbf{C}$  in each case, and for a  $2 \times 2$  matrix that is decided by the determinant.

- (a)  $\mathbf{C} = \frac{1}{2} \begin{pmatrix} 1 & 1 \\ 1 & 1 \end{pmatrix}$ , with  $\det \mathbf{C} = 0$  and  $\mathbf{C} \neq \mathbf{0}$ , so  $\text{rank } \mathbf{C} = 1$ : a product state. Indeed it equals  $\frac{1}{\sqrt{2}}(\mathbf{e}_1 + \mathbf{e}_2) \otimes \frac{1}{\sqrt{2}}(\mathbf{f}_1 + \mathbf{f}_2)$ .
- (b)  $\mathbf{C} = \frac{1}{\sqrt{2}} \begin{pmatrix} 0 & 1 \\ -1 & 0 \end{pmatrix}$ , with  $\det \mathbf{C} = \frac{1}{2} \neq 0$ , so  $\text{rank } \mathbf{C} = 2$ : entangled.
- (c)  $\mathbf{C} = \frac{1}{\sqrt{3}} \begin{pmatrix} 1 & 1 \\ 0 & 1 \end{pmatrix}$ , with  $\det \mathbf{C} = \frac{1}{3} \neq 0$ , so  $\text{rank } \mathbf{C} = 2$ : entangled.
- (d)  $\mathbf{C} = \frac{1}{\sqrt{10}} \begin{pmatrix} 1 & 2 \\ -1 & -2 \end{pmatrix}$ , with  $\det \mathbf{C} = 0$  and  $\mathbf{C} \neq \mathbf{0}$ , so  $\text{rank } \mathbf{C} = 1$ : a product state, equal to  $\frac{1}{\sqrt{2}}(\mathbf{e}_1 - \mathbf{e}_2) \otimes \frac{1}{\sqrt{5}}(\mathbf{f}_1 + 2\mathbf{f}_2)$ .

### Exercise B2: A Schmidt decomposition.

In  $\mathbb{C}^2 \otimes \mathbb{C}^2$ , let

$$\mathbf{z} = \frac{1}{\sqrt{10}}(\mathbf{e}_1 \otimes \mathbf{f}_1 + 2\mathbf{e}_1 \otimes \mathbf{f}_2 + 2\mathbf{e}_2 \otimes \mathbf{f}_1 + \mathbf{e}_2 \otimes \mathbf{f}_2).$$

Find the Schmidt coefficients, the Schmidt rank and a Schmidt decomposition of  $\mathbf{z}$ , and verify that  $\sum_k \sigma_k^2 = 1$ .

*Solution.* The coefficient matrix is  $\mathbf{C} = \frac{1}{\sqrt{10}} \begin{pmatrix} 1 & 2 \\ 2 & 1 \end{pmatrix}$ , which is real symmetric. Then

$$\mathbf{C}\mathbf{C}^\top = \frac{1}{10} \begin{pmatrix} 5 & 4 \\ 4 & 5 \end{pmatrix},$$

with characteristic polynomial  $x^2 - x + \frac{9}{100}$  and eigenvalues  $\frac{9}{10}$  and  $\frac{1}{10}$ . So the Schmidt coefficients are  $\sigma_1 = \frac{3}{\sqrt{10}}$  and  $\sigma_2 = \frac{1}{\sqrt{10}}$ , the Schmidt rank is 2, and  $\sigma_1^2 + \sigma_2^2 = \frac{9}{10} + \frac{1}{10} = 1$ .

Eigenvectors of  $\mathbf{C}\mathbf{C}^\top$  are  $\mathbf{u}_1 = \frac{1}{\sqrt{2}}(1, 1)$  for  $\frac{9}{10}$  and  $\mathbf{u}_2 = \frac{1}{\sqrt{2}}(1, -1)$  for  $\frac{1}{10}$ . Then

$$\mathbf{v}_1 = \frac{\mathbf{C}^\top \mathbf{u}_1}{\sigma_1} = \frac{1}{\sqrt{2}}(1, 1), \quad \mathbf{v}_2 = \frac{\mathbf{C}^\top \mathbf{u}_2}{\sigma_2} = \frac{1}{\sqrt{2}}(-1, 1).$$

All entries are real, so reading the columns back as in Step 3 of [Theorem 14.11.2](#),

$$\begin{aligned} \mathbf{x}_1 &= \frac{\mathbf{e}_1 + \mathbf{e}_2}{\sqrt{2}}, & \mathbf{y}_1 &= \frac{\mathbf{f}_1 + \mathbf{f}_2}{\sqrt{2}}, \\ \mathbf{x}_2 &= \frac{\mathbf{e}_1 - \mathbf{e}_2}{\sqrt{2}}, & \mathbf{y}_2 &= \frac{-\mathbf{f}_1 + \mathbf{f}_2}{\sqrt{2}}, \end{aligned}$$

and  $\mathbf{z} = \frac{3}{\sqrt{10}} \mathbf{x}_1 \otimes \mathbf{y}_1 + \frac{1}{\sqrt{10}} \mathbf{x}_2 \otimes \mathbf{y}_2$ . Expanding the two terms gives  $\frac{1}{2\sqrt{10}}(3-1)\mathbf{e}_1 \otimes \mathbf{f}_1$  and similarly for the other three coefficients, recovering  $\mathbf{z}$ .

### Exercise B3: How large can the Schmidt rank be?.

Let  $V$  and  $W$  be complex inner product spaces with  $\dim V = m$  and  $\dim W = n$ .

- (a) Prove that every non-zero  $\mathbf{z} \in V \otimes W$  has Schmidt rank at most  $\min(m, n)$ .
- (b) Give, for each  $r$  with  $1 \leq r \leq \min(m, n)$ , a unit vector of Schmidt rank exactly  $r$ .

*Solution.*

- (a) Fix orthonormal bases and let  $\mathbf{C} \in M_{m \times n}(\mathbb{C})$  be the coefficient matrix of  $\mathbf{z}$ . By [Theorem 14.11.2](#) the Schmidt rank is  $\text{rank } \mathbf{C}$ , and the rank of an  $m \times n$  matrix is at

most  $\min(m, n)$  because  $\text{rank } \mathbf{C} = \dim \text{col}(\mathbf{C}) \leq m$  and  $\text{rank } \mathbf{C} = \dim \text{row}(\mathbf{C}) \leq n$ .

- (b) Take  $\mathbf{z}_r = \frac{1}{\sqrt{r}} \sum_{k=1}^r \mathbf{e}_k \otimes \mathbf{f}_k$ . The lists  $(\mathbf{e}_1, \dots, \mathbf{e}_r)$  and  $(\mathbf{f}_1, \dots, \mathbf{f}_r)$  are orthonormal and every coefficient  $1/\sqrt{r}$  is positive, so this is already an expression of the shape [Eq. 14.11.1](#); by the uniqueness clause the Schmidt rank is  $r$ . It is a unit vector since  $r \cdot (1/\sqrt{r})^2 = 1$ .

### 14.11.5.3 C. Going deeper

#### Exercise C1: Acting on one part at a time.

Let  $S \in \mathcal{L}(V)$  and  $T \in \mathcal{L}(W)$  be unitary, and let  $\mathbf{z} \in V \otimes W$  be non-zero.

- (a) Prove that  $(S \otimes T)(\mathbf{z})$  has the same Schmidt coefficients, and hence the same Schmidt rank, as  $\mathbf{z}$ .  
 (b) Hence deduce that no operator of the form  $S \otimes T$  with  $S, T$  unitary carries the Bell state  $\mathbf{z}_0$  of [Example \(The Bell state\)](#) to a product state.

*Hint for (a): apply  $S \otimes T$  to a Schmidt decomposition.*

*Solution.*

- (a) Write  $\mathbf{z} = \sum_{k=1}^r \sigma_k \mathbf{x}_k \otimes \mathbf{y}_k$  as in [Theorem 14.11.2](#). By [Definition 14.3.1](#),  $(S \otimes T)(\mathbf{x}_k \otimes \mathbf{y}_k) = S\mathbf{x}_k \otimes T\mathbf{y}_k$ , so by linearity

$$(S \otimes T)(\mathbf{z}) = \sum_{k=1}^r \sigma_k (S\mathbf{x}_k) \otimes (T\mathbf{y}_k).$$

A unitary operator preserves inner products ([Theorem 10.7.3](#)), so  $\langle S\mathbf{x}_k, S\mathbf{x}_l \rangle = \langle \mathbf{x}_k, \mathbf{x}_l \rangle = \delta_{kl}$  and likewise for  $T$ . Both lists are therefore orthonormal, the  $\sigma_k$  are still positive and decreasing, and the display is an expression of the shape [Eq. 14.11.1](#) for  $(S \otimes T)(\mathbf{z})$ . By the uniqueness clause of [Theorem 14.11.2](#), its Schmidt coefficients are  $\sigma_1, \dots, \sigma_r$  and its Schmidt rank is  $r$ .

- (b) The Schmidt rank of  $\mathbf{z}_0$  is 2, as computed in the warning after [Corollary 14.11.3](#). By (a),  $(S \otimes T)(\mathbf{z}_0)$  also has Schmidt rank 2, and by [Corollary 14.11.3](#) a product state has Schmidt rank 1. So  $(S \otimes T)(\mathbf{z}_0)$  is entangled.

#### Exercise C2: Best approximation by a state of small Schmidt rank.

Let  $\mathbf{z} \in V \otimes W$  be non-zero with Schmidt decomposition  $\mathbf{z} = \sum_{k=1}^r \sigma_k \mathbf{x}_k \otimes \mathbf{y}_k$ , and let  $0 \leq k \leq r$ . Put  $\mathbf{z}_{(k)} = \sum_{i=1}^k \sigma_i \mathbf{x}_i \otimes \mathbf{y}_i$ .

- (a) Fix orthonormal bases of  $V$  and  $W$  and let  $\mathbf{C}$  be the coefficient matrix of  $\mathbf{z}$ . Prove that  $\|\mathbf{z}\| = \|\mathbf{C}\|_F$ , and that the Schmidt rank of a vector equals the rank of its coefficient matrix.  
 (b) Hence deduce that for every  $\mathbf{z}' \in V \otimes W$  of Schmidt rank at most  $k$ ,

$$\|\mathbf{z} - \mathbf{z}'\| \geq \left( \sum_{i>k} \sigma_i^2 \right)^{1/2},$$

with equality for  $\mathbf{z}' = \mathbf{z}_{(k)}$ .

*Solution.*

- (a) By [Proposition 14.11.1](#) the basis  $(\mathbf{e}_i \otimes \mathbf{f}_j)$  of  $V \otimes W$  is orthonormal, so  $\|\mathbf{z}\|^2 = \sum_{i,j} |c_{ij}|^2$  by [Theorem 10.9.2](#), and that sum is  $\|\mathbf{C}\|_F^2$  by the definition of the Frobenius norm. For the second claim, Step 2 of [Theorem 14.11.2](#) produced the Schmidt coefficients as the non-zero singular values of  $\mathbf{C}$ , and [Theorem 12.8.1](#) says there are  $\text{rank } \mathbf{C}$  of them.
- (b) The map  $\mathbf{z} \mapsto \mathbf{C}$  is the coordinate isomorphism attached to the basis  $(\mathbf{e}_i \otimes \mathbf{f}_j)$  ([Corollary 3.5.5](#)), with the  $mn$  coordinates arranged in a rectangle, so it is linear and bijective; and by (a) it carries  $\|\cdot\|$  to  $\|\cdot\|_F$  and Schmidt rank to matrix rank. Let  $\mathbf{B}$  be the coefficient matrix of  $\mathbf{z}'$ , so  $\text{rank } \mathbf{B} \leq k$ . By [Theorem 12.10.3](#),

$$\|\mathbf{z} - \mathbf{z}'\| = \|\mathbf{C} - \mathbf{B}\|_F \geq \left( \sum_{i>k} \sigma_i^2 \right)^{1/2},$$

since the non-zero  $\sigma_i$  are the singular values of  $\mathbf{C}$ . Running Step 3 of [Theorem 14.11.2](#) backwards turns  $\mathbf{z}_{(k)}$  into the rank- $k$  truncation  $\mathbf{C}_k$  ([Definition 12.10.1](#)) of a compact singular value decomposition of  $\mathbf{C}$  ([Theorem 12.8.3](#)), and [Theorem 12.10.3](#) attains equality there.

### Exercise C3: Does the field matter?.

Let  $V$  and  $W$  be **real** inner product spaces of dimensions  $m$  and  $n$ , with orthonormal bases  $\mathcal{E}$  and  $\mathcal{F}$ , and let  $\mathbf{z} \in V \otimes W$  have coefficient matrix  $\mathbf{C} \in M_{m \times n}(\mathbb{R})$ . Let  $\mathbf{z}_{\mathbb{C}} \in \mathbb{C}^m \otimes \mathbb{C}^n$  be the vector whose coefficient matrix in the standard orthonormal bases is the same  $\mathbf{C}$ .

- (a) Prove that [Theorem 14.11.2](#) holds verbatim over  $\mathbb{R}$ , with orthonormal lists of real vectors.
- (b) Prove that  $\mathbf{z}$  and  $\mathbf{z}_{\mathbb{C}}$  have the same Schmidt coefficients and the same Schmidt rank.
- (c) Is the same true of the Schmidt **vectors**? Justify your answer.

*Solution.*

- (a) Every step of the proof runs over  $F = \mathbb{R}$ . [Theorem 10.2.2](#), [Theorem 14.2.5](#), [Theorem 12.8.3](#), [Theorem 10.2.1](#) and [Theorem 10.3.2](#) are all stated for  $F = \mathbb{R}$  or  $\mathbb{C}$ , and over  $\mathbb{R}$  the conjugations in Step 3 and Step 4 are the identity, so  $\mathbf{y}_k = \sum_j (\mathbf{v}_k)_j \mathbf{f}_j$  and  $\mathbf{C}\mathbf{C}^\top = \mathbf{X}\mathbf{D}^2\mathbf{X}^\top$ .
- (b) By Step 4 of [Theorem 14.11.2](#), read over each field in turn, the Schmidt coefficients of  $\mathbf{z}$  and of  $\mathbf{z}_{\mathbb{C}}$  are both times the positive square roots of the non-zero eigenvalues of  $\mathbf{C}\mathbf{C}^\top$ , and both Schmidt ranks equal  $\text{rank } \mathbf{C}$ . This is one and the same real symmetric matrix with one and the same characteristic polynomial, so the eigenvalues agree; and  $\text{rank } \mathbf{C}$  is unchanged because row reduction over  $\mathbb{R}$  is row reduction over  $\mathbb{C}$ .
- (c) No. The warning after [Corollary 14.11.3](#) says why: a repeated Schmidt coefficient leaves the vectors free, and over  $\mathbb{C}$  there is more freedom to use. For  $\mathbf{z}_0$  the

orthonormal pair  $\mathbf{x}_1 = \frac{1}{\sqrt{2}}(\mathbf{e}_1 + i\mathbf{e}_2)$ ,  $\mathbf{x}_2 = \frac{1}{\sqrt{2}}(\mathbf{e}_1 - i\mathbf{e}_2)$  occurs in a complex Schmidt decomposition — with  $\mathbf{y}_1 = \frac{1}{\sqrt{2}}(\mathbf{f}_1 - i\mathbf{f}_2)$  and  $\mathbf{y}_2 = \frac{1}{\sqrt{2}}(\mathbf{f}_1 + i\mathbf{f}_2)$  — and has no real counterpart. What is invariant is the coefficient list, not the vectors.

# Chapter 15

## Norms and Matrix Analysis

Nine chapters have been writing promissory notes. Chapter 8 asked what happens to  $\mathbf{A}^m$  and Chapter 9 §10 answered exactly, entry by entry, with no number attached to the answer: [Theorem 9.10.2](#) says *whether* the powers die out, never *how fast*. Chapter 9 §09 built  $e^{\mathbf{A}}$  out of a series ([Definition 9.9.2](#)) and promised that the estimates would come once there was something to estimate with. Chapter 10 solved the least-squares problem ([Theorem 10.4.2](#)) and then, at [Theorem 10.8.1](#), recommended one algorithm over another for a reason it could not yet state. Chapter 11 §01 proved that every matrix has a diagonalizable matrix arbitrarily close to it ([Corollary 11.1.6](#)) and left the word “close” doing work no definition supported. Chapter 12 §08 met a number, the largest stretching factor of a matrix, and said that this chapter would give it a name. Every one of those debts is the same debt: the book has had no way to say **how big** a vector or a matrix is, except through one particular inner product.

A **norm** is that way. It keeps exactly what measurement needs — a length is zero only for the zero vector, it scales with the scalar, and a detour is never shorter than the direct route — and it throws away the inner product that Chapter 10 used to produce it. There are many norms, and the first surprise is that in finite dimension it does not matter which one you pick: any two of them are trapped between constant multiples of each other, so they agree about every limit, every bounded set and every continuous function. The second surprise is that the constants cannot be made independent of the dimension, and that the whole theorem collapses in infinite dimension. From there the chapter measures matrices, and then uses the measurements: the spectral radius as the infimum of all the norms, a geometric series that inverts  $\mathbf{I} - \mathbf{A}$ , quantitative bounds for the exponential, the continuity of eigenvalues, and finally the condition number, which is the exact constant in the sentence “a small change in the data produces a small change in the answer”.

**The field in this chapter is  $\mathbb{R}$  or  $\mathbb{C}$ .** A norm returns a real number and its homogeneity clause uses  $|c|$ , so the scalars must have an absolute value; no other field in this book has one. This is the same restriction Chapters 10 to 12 worked under, and for the same reason.

### 15.0.1 The analysis this chapter imports

This chapter leans on analysis harder than any before it, and the book’s practice is to name what it borrows rather than let it pass unnoticed — as Chapter 13 §06 did with Taylor’s theorem, and Chapter 10 with the theorem of Abel and Ruffini. Here is the complete list. Each item is standard, each is quoted without proof, and **each is named again at the point where it is used**.



- (A1) (*Completeness.*) Every non-empty set of real numbers that is bounded above has a least upper bound; equivalently, every Cauchy sequence of real or complex numbers converges. One standard consequence is used as well: every real number is the limit of a sequence of rational numbers.
- (A2) (*Monotone convergence.*) An increasing sequence of real numbers that is bounded above converges, and its limit is its least upper bound.
- (A3) (*Compactness of the closed unit ball.*) In  $\mathbb{R}^n$  and in  $\mathbb{C}^n$  with the Euclidean norm, every sequence lying in a closed bounded set has a subsequence converging to a point **of that set**. In particular the closed unit ball and the unit sphere are compact.
- (A4) (*Extreme value theorem.*) A continuous real-valued function on a non-empty compact set attains a maximum value and a minimum value; the bounds are not merely approached, they are reached.
- (A5) (*Intermediate value theorem.*) A continuous real-valued function on an interval takes every value between any two of its values.
- (A6) (*Mean value theorem.*) A real-valued function differentiable on an interval satisfies  $f(b) - f(a) = f'(c)(b - a)$  for some  $c$  strictly between  $a$  and  $b$ . Two consequences are used: a function with negative derivative throughout an interval is strictly decreasing there, and a function whose derivative is positive to the left of a point and negative to its right attains its maximum at that point.

Alongside these, the **elementary algebra of limits** is used throughout without further comment: limits respect sums, products and quotients with non-vanishing denominator; sums, products and composites of continuous functions are continuous; and a non-strict inequality survives a limit, so that  $a_k \leq M$  for every  $k$  forces  $\lim a_k \leq M$ . Chapter 9 §09 quoted the same package as its fact (A3). What the list above records is everything beyond that.

Several of these speak of **continuity**, which the book has not needed until now, so fix the meaning here. A function  $f$  defined on a subset of a normed space is **continuous at  $\mathbf{x}$**  when  $\|\mathbf{x}_k - \mathbf{x}\| \rightarrow 0$  forces  $f(\mathbf{x}_k) \rightarrow f(\mathbf{x})$  for every sequence  $(\mathbf{x}_k)$  in its domain, and **continuous** when it is continuous at every point of its domain. This is the sequential form, matching the way (A3) is stated. Every continuity claim in this chapter is established the same cheap way: by exhibiting a constant  $M$  with  $|f(\mathbf{x}) - f(\mathbf{y})| \leq M\|\mathbf{x} - \mathbf{y}\|$ , which gives the implication at once. §02 proves the first such bound.

Nothing else analytic enters silently. Where a section needs a fact outside this list — the differentiability of the entries of a matrix-valued function, or the continuous dependence of the roots of a polynomial on its coefficients — it is recorded as an assumption in the same manner, with a counterexample proved where one is available, and never used as a step in a proof that pretends to be self-contained.

## 15.0.2 What you need

The chapter is built on results the book already has. Gathered by source, so that a reader can go back to one place at a time:

- **Chapter 1 and Chapter 3** — bases, dimension and coordinates: [Definition 1.5.1](#), [Definition 1.6.1](#), [Definition 1.5.2](#), [Theorem 1.5.2](#), [Corollary 3.5.5](#).
- **Chapter 10 §01** — the inner product and the norm it induces, with their three properties and the two inequalities that follow: [Definition 10.1.1](#), [Definition 10.1.3](#), [Theorem 10.1.2](#), [Theorem 10.1.5](#), [Corollary 10.1.6](#), together with the two exercises that recover an inner

product from its norm, [Exercise \(C1: The parallelogram law characterizes inner-product norms\)](#) and [Exercise \(C2: Recovering the inner product from the norm\)](#).

- **Chapter 10 §04 and §08** — least squares and the **QR** factorization whose stability claim was deferred: [Theorem 10.4.2](#), [Corollary 10.4.4](#), [Theorem 10.8.1](#).
- **Chapter 9 §10** — the whole convergence apparatus that this chapter re-proves with numbers attached: [Definition 9.10.1](#), [Lemma 9.10.1](#), [Theorem 9.10.2](#), [Theorem 9.10.3](#), [Theorem 9.10.4](#), [Theorem 9.10.6](#), [Corollary 9.10.5](#).
- **Chapter 9 §09 and §03** — the matrix exponential and the Jordan form: [Definition 9.9.2](#), [Theorem 9.9.3](#), [Theorem 9.9.4](#), [Theorem 9.3.1](#).
- **Chapter 11** — Schur triangularization with its density corollary, and the Frobenius norm with the defect it measures: [Theorem 11.1.2](#), [Corollary 11.1.6](#), [Lemma 11.11.1](#), [Theorem 11.11.2](#), [Definition 11.11.1](#).
- **Chapter 12** — the singular value decomposition and the two constructions built on it: [Theorem 12.8.1](#), [Definition 12.8.1](#), [Definition 12.11.1](#).

### 15.0.3 Roadmap

- **Norms.** The three axioms, the 1-, 2- and  $\infty$ -norms with their unit balls drawn, and the test that decides which norms come from an inner product: the parallelogram law. This pays off a promise made in Chapter 10 §01.
- **All norms agree.** On a finite-dimensional space any two norms are trapped between constant multiples of each other. Compactness enters the book here, and the payoff is that Chapter 9 §10's entrywise convergence is simply *the* convergence, whatever norm one prefers.
- **Measuring a matrix.** Submultiplicativity, and the operator norm  $\|\mathbf{A}\| = \max\{\|\mathbf{Ax}\| : \|\mathbf{x}\| = 1\}$  induced by a vector norm. Its formulas for  $p = 1, 2, \infty$ , including  $\|\mathbf{A}\|_2 = \sigma_1(\mathbf{A})$ , which is the name Chapter 12 §08 asked for.
- **The spectral radius.**  $\rho(\mathbf{A}) \leq \|\mathbf{A}\|$  for every matrix norm, a norm that comes within  $\varepsilon$  of  $\rho(\mathbf{A})$ , and Gelfand's formula  $\rho(\mathbf{A}) = \lim_{k \rightarrow \infty} \|\mathbf{A}^k\|^{1/k}$ .
- **Inverting a perturbation of the identity.** The Neumann series, the openness of the invertible matrices, and the first perturbation estimate.
- **The exponential, with bounds; and derivatives.**  $\|e^{\mathbf{A}}\| \leq e^{\|\mathbf{A}\|}$ , decay when the spectrum sits in the left half plane, and the derivatives of the determinant and of the inverse.
- **Eigenvalues move continuously.** Continuously as a multiset, and no better: a Jordan block's eigenvalues move like  $\varepsilon^{1/k}$ .
- **How wrong can the answer be?** The condition number  $\kappa(\mathbf{A}) = \|\mathbf{A}\| \|\mathbf{A}^{-1}\|$ , what it predicts about solving  $\mathbf{Ax} = \mathbf{b}$ , and why the normal equations square it.

### 15.0.4 Named moves

- **Compactness turns “for all unit vectors” into a maximum.** A supremum over the unit sphere is hard to control; a maximum is attained at a vector you can name and feed back into the argument. Both the equivalence of norms and the very existence of the operator norm are this one move.
- **Ask for submultiplicativity.** A matrix is a vector, so a matrix norm is a vector norm; what makes it useful is the extra clause  $\|\mathbf{AB}\| \leq \|\mathbf{A}\| \|\mathbf{B}\|$ , which is the only reason a bound on  $\|\mathbf{A}\|$  says anything about  $\mathbf{A}^k$ . Test every candidate against it, and expect natural-looking candidates to fail.

- **Bound the tail of a series.** Neumann, the exponential and Gelfand all reduce to estimating  $\sum_{k \geq K} \|\mathbf{A}\|^k$  or  $\sum_{k \geq K} \|\mathbf{A}\|^k / k!$  by something that visibly tends to zero.
- **Perturb and compare.** Replace  $\mathbf{A}$  by  $\mathbf{A} + \mathbf{E}$ , subtract the two conclusions, and read the result as a bound on the relative change in the answer by a constant times the relative change in the data. The constant is the condition number.
- **Reduce to the 2-norm.** Equivalence of norms means any single norm may be used as the hub: prove the estimate for the Euclidean norm, where the inner product and the singular value decomposition are available, and transfer.

## 15.1 Norms

Chapter 10 measured vectors, and measured them in exactly one way:  $\|\mathbf{v}\| = \sqrt{\langle \mathbf{v}, \mathbf{v} \rangle}$ . Every length in Chapters 10 to 12 came out of that formula. This section keeps the measuring and throws the inner product away. The reason is not economy. The quantities this part of the book owes — how fast a sequence of matrices settles down, how far a solution moves when the data moves — are often measured most naturally by a size that no inner product produces, and the arguments that use them turn out to need only three properties of a length.

**The field here is  $\mathbb{R}$  or  $\mathbb{C}$ ,** as it is throughout the chapter. A length is a real number, and the scaling rule below multiplies it by  $|c|$ ; so the scalars must carry an absolute value, and these are the only fields in this book that do.

### 15.1.1 What measurement actually needs

Look back at how  $\|\cdot\|$  was used after Chapter 10 defined it. Cauchy–Schwarz needed the inner product. So did Pythagoras, orthogonality and the adjoint. But a long list of arguments did not: that a convergent sequence is bounded, that  $\|\mathbf{A}\mathbf{x}\|$  is small when  $\mathbf{x}$  is small, that a limit of unit vectors is a unit vector, that  $\|\mathbf{u} - \mathbf{w}\| \leq \|\mathbf{u} - \mathbf{v}\| + \|\mathbf{v} - \mathbf{w}\|$ . Every one of those used only that  $\|\mathbf{v}\|$  vanishes just for  $\mathbf{v} = \mathbf{0}$ , that  $\|c\mathbf{v}\| = |c|\|\mathbf{v}\|$ , and the triangle inequality of [Corollary 10.1.6](#). Three properties, used over and over. Following the book’s habit, we give the three of them a name and see what else satisfies them.

*A norm is a length: zero only at the zero vector, scaling with the scalar, and never rewarding a detour.*

#### Definition 15.1.1 — Norm

Let  $F$  be  $\mathbb{R}$  or  $\mathbb{C}$  and let  $V$  be a vector space **over**  $F$ . A **norm** on  $V$  is a function

$$\|\cdot\|: V \rightarrow \mathbb{R}$$

such that for **all**  $\mathbf{u}, \mathbf{v} \in V$  and **all**  $c \in F$ :

- (N1)  $\|\mathbf{v}\| \geq 0$ , and  $\|\mathbf{v}\| = 0$  **only** for  $\mathbf{v} = \mathbf{0}$  (**positive definiteness**);
- (N2)  $\|c\mathbf{v}\| = |c| \|\mathbf{v}\|$  (**absolute homogeneity**);
- (N3)  $\|\mathbf{u} + \mathbf{v}\| \leq \|\mathbf{u}\| + \|\mathbf{v}\|$  (**the triangle inequality**).

A vector space carrying a norm is a **normed space**. A vector with  $\|\mathbf{v}\| = 1$  is a **unit**

**vector**, and replacing a non-zero  $\mathbf{v}$  by  $\mathbf{v}/\|\mathbf{v}\|$  is **normalizing** it.

In words, clause by clause. (N1) says the output is a genuine non-negative size, and that only the zero vector has size zero; the word **only** is the whole content, since the reverse implication is free (take  $c = 0$  in (N2) to get  $\|\mathbf{0}\| = 0 \cdot \|\mathbf{0}\| = 0$ ). (N2) says that stretching a vector by  $c$  stretches its length by  $|c|$  — the **absolute value** of  $c$ , so that  $\|-\mathbf{v}\| = \|\mathbf{v}\|$  and, over  $\mathbb{C}$ ,  $\|i\mathbf{v}\| = \|\mathbf{v}\|$ . (N3) says that going from  $\mathbf{0}$  to  $\mathbf{u} + \mathbf{v}$  directly is at most as long as going by way of  $\mathbf{u}$ . Note what the definition does not mention: no second vector is paired with the first, no angle appears, and there is no inner product anywhere.

Two remarks on the shape of the definition. First, the codomain is  $\mathbb{R}$  even when  $F = \mathbb{C}$ : a length is never complex. Second, the three clauses are not independent of the arithmetic of  $\mathbb{R}$  — (N3) is an inequality, and inequalities are why the field had to be restricted.

**Example. (The 1-, 2- and  $\infty$ -norms on  $F^n$ )**

For  $\mathbf{x} = (x_1, \dots, x_n) \in F^n$  put

$$\begin{aligned}\|\mathbf{x}\|_1 &= |x_1| + \dots + |x_n|, \\ \|\mathbf{x}\|_2 &= (|x_1|^2 + \dots + |x_n|^2)^{1/2}, \\ \|\mathbf{x}\|_\infty &= \max\{|x_1|, \dots, |x_n|\}.\end{aligned}$$

Check each of them against [Definition 15.1.1](#).

*Solution.* (N1) holds for all three: each is a sum, a square root of a sum, or a maximum of the non-negative numbers  $|x_i|$ , so each is  $\geq 0$ ; and each vanishes only when every  $|x_i| = 0$ , that is, only for  $\mathbf{x} = \mathbf{0}$ . (N2) holds because  $|cx_i| = |c||x_i|$  for every  $i$ , and the common factor  $|c|$  comes out of a sum, out of a square root of a sum of squares, and out of a maximum of non-negative numbers alike.

For (N3), the 1-norm inherits it one coordinate at a time:  $|x_i + y_i| \leq |x_i| + |y_i|$  for each  $i$  by the triangle inequality in  $F$ , and summing over  $i$  gives  $\|\mathbf{x} + \mathbf{y}\|_1 \leq \|\mathbf{x}\|_1 + \|\mathbf{y}\|_1$ . The  $\infty$ -norm inherits it too, but from one coordinate rather than all: for each  $i$ ,

$$|x_i + y_i| \leq |x_i| + |y_i| \leq \|\mathbf{x}\|_\infty + \|\mathbf{y}\|_\infty,$$

and a bound valid for every  $i$  bounds the maximum. The 2-norm is the norm induced by the standard inner product on  $F^n$  ([Definition 10.1.3](#)), so its triangle inequality is [Corollary 10.1.6](#) and needs no fresh argument.

So one space carries at least three different norms. For  $\mathbf{x} = (3, -4, 12) \in \mathbb{R}^3$  they give  $\|\mathbf{x}\|_1 = 19$ ,  $\|\mathbf{x}\|_2 = 13$  and  $\|\mathbf{x}\|_\infty = 12$  — three honest answers to “how big is  $\mathbf{x}$ ”, and no reason yet to prefer one.

*Remark.* There is a  $p$ -norm  $\|\mathbf{x}\|_p = (\sum_i |x_i|^p)^{1/p}$  for every real  $p \geq 1$ , and  $\|\mathbf{x}\|_\infty$  is its limit as  $p \rightarrow \infty$ . Its triangle inequality is Minkowski’s inequality, which this book does not prove and never uses; only  $p = 1, 2, \infty$  appear below, and each of those was verified above from scratch.

**Example. (Inner products, weights, and a function space)**

Three further families.

- (a) **Every inner product gives a norm.** If  $V$  is an inner product space, then  $\|\mathbf{v}\| = \sqrt{\langle \mathbf{v}, \mathbf{v} \rangle}$  satisfies (N1) and (N2) by [Theorem 10.1.2](#) (a), (b) and (N3) by [Corollary 10.1.6](#). In particular the Frobenius norm  $\|\mathbf{A}\|_F$  on  $M_{m \times n}(F)$  and the norm  $(\int_0^1 |f|^2)^{1/2}$  on  $C[0, 1]$  are norms.
- (b) **Weighted norms.** Fix real weights  $w_1, \dots, w_n > 0$  and put  $\|\mathbf{x}\|_w = \max_i w_i |x_i|$ . All three clauses go through exactly as for  $\|\cdot\|_\infty$ , the weights riding along untouched; positivity of every  $w_i$  is what (N1) needs.
- (c) **The sup norm.** On  $C[0, 1]$ , the real continuous functions on  $[0, 1]$ , put  $\|f\|_\infty = \max_{t \in [0, 1]} |f(t)|$ .

For (c) the maximum has to exist before the formula defines anything, and that is the first place this chapter borrows:  $[0, 1]$  is compact and  $|f|$  is continuous, so by **the extreme value theorem, fact (A4) of the chapter introduction**, the maximum is attained. Granting that, (N1) says a continuous function with  $\max |f| = 0$  is the zero function, which is true pointwise; (N2) and (N3) follow from the corresponding facts about  $|\cdot|$  at each point, exactly as for  $\|\cdot\|_\infty$  on  $F^n$ . Note that  $C[0, 1]$  is infinite-dimensional — it contains all the polynomials — and it is kept in view deliberately, because §02's main theorem will fail on it.

A degenerate case is worth one line. On  $V = F$  itself,  $\|c\| = |c|$  is a norm, and (N2) forces every norm on  $F$  to be  $\|c\| = |c|\|1\|$ , a positive multiple of that one. On the zero space  $V = \{\mathbf{0}\}$  the only function to  $\mathbb{R}$  is  $\|\mathbf{0}\| = 0$ , and it is a norm. Both matter: they say that the theory is not empty at the bottom, and the first of them is the model for the general statement that any two norms differ by bounded factors.

**Example. (Three near misses)**

Change one feature of an example above and watch a single clause break.

- (a) **Drop the square root.** On  $F^n$  put  $N(\mathbf{x}) = |x_1|^2 + \dots + |x_n|^2$ . Then (N1) still holds. But  $N(c\mathbf{x}) = |c|^2 N(\mathbf{x})$ , so (N2) fails as soon as  $|c| \neq 0, 1$ : for  $\mathbf{x} = \mathbf{e}_1$  and  $c = 2$  it gives  $N(2\mathbf{e}_1) = 4 \neq 2 = |2|N(\mathbf{e}_1)$ . The square root in  $\|\cdot\|_2$  is there for exactly this reason.
- (b) **Forget a coordinate.** On  $F^2$  put  $N(\mathbf{x}) = |x_1|$ . Then (N2) and (N3) both hold, inherited from  $|\cdot|$  on  $F$ . What fails is the word **only** in (N1):  $N(\mathbf{e}_2) = 0$  while  $\mathbf{e}_2 \neq \mathbf{0}$ . A function satisfying (N1) with “only” deleted, together with (N2) and (N3), is called a **seminorm**; it measures a quotient of  $V$  rather than  $V$ .
- (c) **Bend the exponent the other way.** On  $\mathbb{R}^2$  put  $N(\mathbf{x}) = (|x_1|^{1/2} + |x_2|^{1/2})^2$ . Then (N1) holds, and (N2) holds because  $|c|^{1/2}$  factors out of the inner sum and is squared back to  $|c|$ . Only (N3) fails:  $N(\mathbf{e}_1) = N(\mathbf{e}_2) = 1$  but  $N(\mathbf{e}_1 + \mathbf{e}_2) = (1 + 1)^2 = 4 > 2$ .

Item (c) is the one to remember, because it shows the three clauses are genuinely independent: two of them can hold exactly while the third fails, and it is the failing one that carries all the geometry.

*Warning.* A norm is not determined by what it does to the standard basis. Both  $\|\cdot\|_1$  and  $\|\cdot\|_\infty$  give every  $\mathbf{e}_i$  length 1, and so does  $\|\cdot\|_2$ , yet the three disagree everywhere else: on  $\mathbf{1} = (1, \dots, 1)$  they return  $n$ , 1 and  $\sqrt{n}$ . Knowing the lengths of a basis pins down nothing, and this is worth holding on to, because §02's theorem is sometimes misremembered as saying that norms which agree on a basis agree everywhere. It says no such thing.

One small consequence of (N3) will be used twice below and again in §02, so it gets a name here. It is [Exercise \(B4: A lower bound for a difference of norms\)](#) with the inner product removed from the proof.

### Lemma 15.1.1 — Reverse Triangle Inequality

Let  $\|\cdot\|$  be a norm on  $V$ . Then for all  $\mathbf{u}, \mathbf{v} \in V$ ,

$$| \|\mathbf{u}\| - \|\mathbf{v}\| | \leq \|\mathbf{u} - \mathbf{v}\|.$$

*Proof.* By (N3) applied to  $\mathbf{u} = (\mathbf{u} - \mathbf{v}) + \mathbf{v}$ , we have  $\|\mathbf{u}\| \leq \|\mathbf{u} - \mathbf{v}\| + \|\mathbf{v}\|$ , so  $\|\mathbf{u}\| - \|\mathbf{v}\| \leq \|\mathbf{u} - \mathbf{v}\|$ . Swapping the roles of  $\mathbf{u}$  and  $\mathbf{v}$  gives  $\|\mathbf{v}\| - \|\mathbf{u}\| \leq \|\mathbf{v} - \mathbf{u}\|$ , and  $\|\mathbf{v} - \mathbf{u}\| = |-1|\|\mathbf{u} - \mathbf{v}\| = \|\mathbf{u} - \mathbf{v}\|$  by (N2). The number  $|\|\mathbf{u}\| - \|\mathbf{v}\||$  is one of the two quantities just bounded, so it is at most  $\|\mathbf{u} - \mathbf{v}\|$ . This proves the inequality. ■

Read with  $\mathbf{v}$  replaced by  $\mathbf{v}'$ , it says that the function  $\|\cdot\|$  changes by at most  $\|\mathbf{u} - \mathbf{v}\|$  when its argument moves from  $\mathbf{u}$  to  $\mathbf{v}$ . That is a Lipschitz estimate, and it is the reason a norm is continuous with respect to itself — a fact §02 turns into the main theorem.

## 15.1.2 Unit balls

The quickest way to see the difference between two norms is to draw the set of vectors they call short.

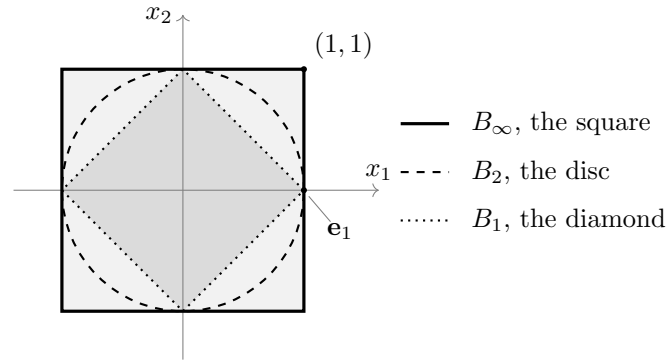
### Definition 15.1.2 — Closed Unit Ball

Let  $\|\cdot\|$  be a norm on  $V$ . Its **closed unit ball** is

$$B_{\|\cdot\|} = \{\mathbf{v} \in V : \|\mathbf{v}\| \leq 1\}.$$

In  $F^n$  we write  $B_1$ ,  $B_2$  and  $B_\infty$  for the balls of [Example \(The 1-, 2- and  \$\infty\$ -norms on  \$F^n\$ \)](#).

In  $\mathbb{R}^2$  the three are a diamond, a disc and a square, and they are nested.



The closed unit balls of the three norms on  $\mathbb{R}^2$ . A smaller ball belongs to a larger norm:  $B_1 \subseteq B_2 \subseteq B_\infty$  says exactly that  $\|\mathbf{x}\|_\infty \leq \|\mathbf{x}\|_2 \leq \|\mathbf{x}\|_1$ , and all three agree at  $\mathbf{e}_1$

Two features of the picture are general. Each ball is **symmetric**,  $\mathbf{v} \in B$  if and only if  $-\mathbf{v} \in B$ , which is (N2) with  $c = -1$ ; and each is **convex**, since if  $\|\mathbf{u}\|, \|\mathbf{v}\| \leq 1$  and  $0 \leq t \leq 1$  then

$$\|t\mathbf{u} + (1-t)\mathbf{v}\| \leq t\|\mathbf{u}\| + (1-t)\|\mathbf{v}\| \leq 1$$

by (N3) and (N2). The diamond has corners and the square has corners; convexity is what the triangle inequality buys, and it is all it buys. The disc is rounder than the other two for a reason that is the subject of the last part of this section.

The nesting in the picture is an inequality, and it holds in every dimension, with a matching chain running the other way.

### Proposition 15.1.2 — Comparing the Three Norms

For every  $\mathbf{x} \in F^n$ ,

$$\|\mathbf{x}\|_\infty \leq \|\mathbf{x}\|_2 \leq \|\mathbf{x}\|_1$$

and

$$\|\mathbf{x}\|_1 \leq \sqrt{n} \|\mathbf{x}\|_2, \quad \|\mathbf{x}\|_2 \leq \sqrt{n} \|\mathbf{x}\|_\infty.$$

Each of the four inequalities is attained: the first two by  $\mathbf{x} = \mathbf{e}_1$ , the last two by  $\mathbf{x} = \mathbf{1}$ .

*Idea.* Four inequalities, two moves. The ones **without** a  $\sqrt{n}$  compare a sum against its largest term, or a sum of squares against the square of a sum, and need nothing but non-negativity. The ones **with** a  $\sqrt{n}$  are a trade of  $n$  terms against one, and the sharp constant comes from Cauchy–Schwarz against the all-ones vector  $\mathbf{1}$  — which is exactly why  $\mathbf{1}$  is the vector that attains them.

*Proof.* Write  $a_i = |x_i| \geq 0$  and let  $a_k = \max_i a_i$ . Then  $a_k^2 \leq \sum_i a_i^2$ , and taking non-negative square roots gives  $\|\mathbf{x}\|_\infty \leq \|\mathbf{x}\|_2$ . Next,

$$\|\mathbf{x}\|_1^2 = \left( \sum_i a_i \right)^2 = \sum_i a_i^2 + \sum_{i \neq j} a_i a_j \geq \|\mathbf{x}\|_2^2,$$

since every cross term  $a_i a_j$  is  $\geq 0$ ; taking square roots gives  $\|\mathbf{x}\|_2 \leq \|\mathbf{x}\|_1$ .

For the third inequality, apply [Theorem 10.1.5](#) in  $\mathbb{R}^n$  with the dot product to the vectors



$(a_1, \dots, a_n)$  and  $\mathbf{1}$ :

$$\|\mathbf{x}\|_1 = \sum_i a_i \cdot 1 \leq \left( \sum_i a_i^2 \right)^{1/2} \left( \sum_i 1 \right)^{1/2} = \sqrt{n} \|\mathbf{x}\|_2.$$

For the fourth,  $\sum_i a_i^2 \leq n \max_i a_i^2$ , and taking square roots gives  $\|\mathbf{x}\|_2 \leq \sqrt{n} \|\mathbf{x}\|_\infty$ .

For the equality cases,  $\mathbf{e}_1$  has  $\|\mathbf{e}_1\|_1 = \|\mathbf{e}_1\|_2 = \|\mathbf{e}_1\|_\infty = 1$ , and  $\mathbf{1}$  has  $\|\mathbf{1}\|_1 = n$ ,  $\|\mathbf{1}\|_2 = \sqrt{n}$  and  $\|\mathbf{1}\|_\infty = 1$ , so  $\|\mathbf{1}\|_1 = \sqrt{n} \|\mathbf{1}\|_2$  and  $\|\mathbf{1}\|_2 = \sqrt{n} \|\mathbf{1}\|_\infty$ . This proves the proposition. ■

So the three norms never disagree by a factor worse than  $\sqrt{n}$ , and the factor  $\sqrt{n}$  really occurs. Hold on to both halves of that sentence: §02 proves the first half for *every* pair of norms, and the second half is why its constants cannot be made to forget the dimension.

*Quick check.* Sketch the set  $\{\mathbf{x} \in \mathbb{R}^2 : \|\mathbf{x}\|_w \leq 1\}$  for the weighted norm  $\|\mathbf{x}\|_w = \max(|x_1|, 2|x_2|)$  of [Example \(Inner products, weights, and a function space\)](#) (b), and find the smallest  $C$  with  $\|\mathbf{x}\|_\infty \leq C\|\mathbf{x}\|_w$  for all  $\mathbf{x}$ .

*Solution.* The ball is the rectangle  $|x_1| \leq 1, |x_2| \leq \frac{1}{2}$ : the square  $B_\infty$  squashed vertically by a factor 2. Since  $|x_1| \leq \|\mathbf{x}\|_w$  and  $|x_2| \leq \frac{1}{2}\|\mathbf{x}\|_w$ , we get  $\|\mathbf{x}\|_\infty \leq \|\mathbf{x}\|_w$ , so  $C = 1$  works, and it is smallest because  $\mathbf{x} = \mathbf{e}_1$  gives  $\|\mathbf{x}\|_\infty = \|\mathbf{x}\|_w = 1$ . The other direction costs a factor:  $\|\mathbf{x}\|_w \leq 2\|\mathbf{x}\|_\infty$ , attained at  $\mathbf{e}_2$ . Squashing a ball inward enlarges the norm, and the largest squash factor is the constant.

### 15.1.3 Which norms come from an inner product

Chapter 10 §01 closed with a promise. It observed that [Theorem 10.1.2](#) (a), (b) and [Corollary 10.1.6](#) make every induced norm a norm in the general sense, warned that the converse fails, and named the test: the **parallelogram law**

$$\|\mathbf{u} + \mathbf{v}\|^2 + \|\mathbf{u} - \mathbf{v}\|^2 = 2\|\mathbf{u}\|^2 + 2\|\mathbf{v}\|^2. \quad (*)$$

It then proved one direction outright — expand both sides with [Theorem 10.1.2](#) (c) and add — and left the converse to [Exercise \(C1: The parallelogram law characterizes inner-product norms\)](#), which carries it out in  $\mathbb{R}^2$  while **granting** one step, the homogeneity  $\beta(\mathbf{u}, t\mathbf{v}) = t\beta(\mathbf{u}, \mathbf{v})$  for real  $t$ . That exercise's own remark says where the gap is: additivity gives homogeneity for rational  $t$  by induction, and passing to real  $t$  needs a limit.

Here is what this section adds. The dimension 2 in [Exercise \(C1: The parallelogram law characterizes inner-product norms\)](#) was never used, so the computation runs in any real space; the granted step is supplied by a limit, using the completeness of  $\mathbb{R}$ ; and the complex case, which Chapter 10 did not touch at all, is built from the real one. The result is the theorem the promise described, with no hypothesis on the dimension.

#### Theorem 15.1.3 — The Parallelogram Law Characterizes Inner-Product Norms

Let  $F$  be  $\mathbb{R}$  or  $\mathbb{C}$ , let  $V$  be a vector space over  $F$ , and let  $\|\cdot\|$  be a norm on  $V$ . Then there



is an inner product on  $V$  with  $\langle \mathbf{v}, \mathbf{v} \rangle = \|\mathbf{v}\|^2$  for every  $\mathbf{v}$  **if and only if**  $\|\cdot\|$  satisfies the parallelogram law (\*) for all  $\mathbf{u}, \mathbf{v} \in V$ . When it exists, that inner product is unique.

*Idea.* The forward direction is Chapter 10's computation. For the converse there is only one candidate for the inner product, because [Exercise \(C2: Recovering the inner product from the norm\)](#) already shows that an inner product is determined by its norm through polarization; so write down the polarization formula and check the axioms. ① Over  $\mathbb{R}$  the symmetry and the additivity are the computation of [Exercise \(C1: The parallelogram law characterizes inner-product norms\)](#), which never mentions the dimension. ② Homogeneity comes free for rational scalars and is pushed to real scalars by a limit, which is the one analytic step. ③ Over  $\mathbb{C}$ , apply ① and ② to  $V$  regarded as a real space, then glue the real form  $\beta$  and its twist  $\beta(\mathbf{u}, i\mathbf{v})$  into a complex inner product; the single identity  $\beta(i\mathbf{u}, i\mathbf{v}) = \beta(\mathbf{u}, \mathbf{v})$  does all the work.

*Proof.* ( $\Rightarrow$ ) Suppose  $\langle \mathbf{v}, \mathbf{v} \rangle = \|\mathbf{v}\|^2$  for an inner product on  $V$ . By [Theorem 10.1.2](#) (c) and the same identity with  $-\mathbf{v}$  in place of  $\mathbf{v}$ ,

$$\begin{aligned}\|\mathbf{u} + \mathbf{v}\|^2 &= \|\mathbf{u}\|^2 + 2\operatorname{Re}\langle \mathbf{u}, \mathbf{v} \rangle + \|\mathbf{v}\|^2, \\ \|\mathbf{u} - \mathbf{v}\|^2 &= \|\mathbf{u}\|^2 - 2\operatorname{Re}\langle \mathbf{u}, \mathbf{v} \rangle + \|\mathbf{v}\|^2.\end{aligned}$$

Adding them gives (\*).

( $\Leftarrow$ ) Suppose (\*) holds. Regard  $V$  as a vector space over  $\mathbb{R}$ , by restricting which scalars are allowed. This changes nothing about  $\|\cdot\|$ : the vectors, the addition and the values of the norm are the same, and (N2) for real  $c$  is a special case of (N2) for  $c \in F$ . Define

$$\beta(\mathbf{u}, \mathbf{v}) = \frac{1}{4}(\|\mathbf{u} + \mathbf{v}\|^2 - \|\mathbf{u} - \mathbf{v}\|^2).$$

**Step 1.**  $\beta$  is symmetric and additive in each slot. [Exercise \(C1: The parallelogram law characterizes inner-product norms\)](#) runs this computation in  $\mathbb{R}^2$ ; rather than transport an exercise past the hypotheses it was stated under, here it is in full, for an arbitrary  $V$ .

Setting  $\mathbf{v} = \mathbf{u}$  gives  $\beta(\mathbf{u}, \mathbf{u}) = \frac{1}{4}\|2\mathbf{u}\|^2 = \|\mathbf{u}\|^2$  by (N2). Swapping the arguments replaces  $\|\mathbf{u} - \mathbf{v}\|$  by  $\|\mathbf{v} - \mathbf{u}\|$ , equal by (N2) with  $c = -1$ , so  $\beta(\mathbf{u}, \mathbf{v}) = \beta(\mathbf{v}, \mathbf{u})$ ; and  $\beta(\mathbf{u}, \mathbf{0}) = 0$ .

For additivity, apply (\*) twice. To the pair  $\mathbf{u} + \frac{1}{2}(\mathbf{v} + \mathbf{w})$ ,  $\frac{1}{2}(\mathbf{v} - \mathbf{w})$ , whose sum is  $\mathbf{u} + \mathbf{v}$  and whose difference is  $\mathbf{u} + \mathbf{w}$ :

$$\|\mathbf{u} + \mathbf{v}\|^2 + \|\mathbf{u} + \mathbf{w}\|^2 = 2\|\mathbf{u} + \frac{1}{2}(\mathbf{v} + \mathbf{w})\|^2 + 2\|\frac{1}{2}(\mathbf{v} - \mathbf{w})\|^2.$$

Then to the pair  $\mathbf{u} - \frac{1}{2}(\mathbf{v} + \mathbf{w})$ ,  $\frac{1}{2}(\mathbf{w} - \mathbf{v})$ , whose sum is  $\mathbf{u} - \mathbf{v}$  and whose difference is  $\mathbf{u} - \mathbf{w}$ :

$$\|\mathbf{u} - \mathbf{v}\|^2 + \|\mathbf{u} - \mathbf{w}\|^2 = 2\|\mathbf{u} - \frac{1}{2}(\mathbf{v} + \mathbf{w})\|^2 + 2\|\frac{1}{2}(\mathbf{w} - \mathbf{v})\|^2.$$

Subtract the second from the first. On the left, the two differences of squares are  $4\beta(\mathbf{u}, \mathbf{v})$  and  $4\beta(\mathbf{u}, \mathbf{w})$ , by the definition of  $\beta$ . On the right, the last terms cancel, since  $\|\frac{1}{2}(\mathbf{v} - \mathbf{w})\| = \|\frac{1}{2}(\mathbf{w} - \mathbf{v})\|$  by (N2) with  $c = -1$ , and the remaining difference of squares is  $4\beta(\mathbf{u}, \frac{1}{2}(\mathbf{v} + \mathbf{w}))$ . So  $4\beta(\mathbf{u}, \mathbf{v}) + 4\beta(\mathbf{u}, \mathbf{w}) = 8\beta(\mathbf{u}, \frac{1}{2}(\mathbf{v} + \mathbf{w}))$ , that is,

$$\beta(\mathbf{u}, \mathbf{v}) + \beta(\mathbf{u}, \mathbf{w}) = 2\beta(\mathbf{u}, \frac{1}{2}(\mathbf{v} + \mathbf{w})).$$

Taking  $\mathbf{w} = \mathbf{0}$  and using  $\beta(\mathbf{u}, \mathbf{0}) = 0$  turns the right-hand side into  $2\beta(\mathbf{u}, \frac{1}{2}\mathbf{v})$ , so  $\beta(\mathbf{u}, \mathbf{v}) = 2\beta(\mathbf{u}, \frac{1}{2}\mathbf{v})$  for every  $\mathbf{v}$ ; substituting that back with  $\mathbf{v} + \mathbf{w}$  in place of  $\mathbf{v}$  gives

$$\beta(\mathbf{u}, \mathbf{v}) + \beta(\mathbf{u}, \mathbf{w}) = \beta(\mathbf{u}, \mathbf{v} + \mathbf{w})$$

for all  $\mathbf{u}, \mathbf{v}, \mathbf{w} \in V$ . Nothing in this argument mentions a dimension. Additivity in the first slot follows from the symmetry.

**Step 2.**  $\beta(\mathbf{u}, t\mathbf{v}) = t\beta(\mathbf{u}, \mathbf{v})$  for every real  $t$ . This is the step [Exercise \(C1: The parallelogram law characterizes inner-product norms\)](#) granted. First,  $\beta(\mathbf{u}, -\mathbf{v}) = -\beta(\mathbf{u}, \mathbf{v})$ , directly from the definition of  $\beta$ , since replacing  $\mathbf{v}$  by  $-\mathbf{v}$  exchanges the two terms. Next, induction on  $m \geq 0$  using Step 1 gives  $\beta(\mathbf{u}, m\mathbf{v}) = m\beta(\mathbf{u}, \mathbf{v})$ , the case  $m = 0$  being  $\beta(\mathbf{u}, \mathbf{0}) = 0$ ; combined with the previous sentence this covers every integer  $m$ . For a rational  $q = m/n$  with  $n \geq 1$ ,

$$n\beta(\mathbf{u}, q\mathbf{v}) = \beta(\mathbf{u}, nq\mathbf{v}) = \beta(\mathbf{u}, m\mathbf{v}) = m\beta(\mathbf{u}, \mathbf{v}),$$

so  $\beta(\mathbf{u}, q\mathbf{v}) = q\beta(\mathbf{u}, \mathbf{v})$ .

Now fix a real  $t$  and put  $M = \|\mathbf{u}\| + (|t| + 1)\|\mathbf{v}\|$ . For any rational  $q$  with  $|q - t| \leq 1$ , both  $\|\mathbf{u} \pm t\mathbf{v}\|$  and  $\|\mathbf{u} \pm q\mathbf{v}\|$  are at most  $M$  by (N2) and (N3), while [Lemma 15.1.1](#) and (N2) give

$$|\|\mathbf{u} \pm t\mathbf{v}\| - \|\mathbf{u} \pm q\mathbf{v}\|| \leq |t - q| \|\mathbf{v}\|.$$

Since  $|a^2 - b^2| = |a - b|(a + b)$  for  $a, b \geq 0$ , each of the two squared terms in  $\beta$  moves by at most  $2M\|\mathbf{v}\||t - q|$ , and therefore

$$|\beta(\mathbf{u}, t\mathbf{v}) - \beta(\mathbf{u}, q\mathbf{v})| \leq M\|\mathbf{v}\| |t - q|.$$

By the completeness of  $\mathbb{R}$ , fact (A1) of the chapter introduction, there is a sequence of rationals  $q_k \rightarrow t$ , and we may assume  $|q_k - t| \leq 1$ . Then

$$\begin{aligned} & |\beta(\mathbf{u}, t\mathbf{v}) - t\beta(\mathbf{u}, \mathbf{v})| \\ & \leq |\beta(\mathbf{u}, t\mathbf{v}) - \beta(\mathbf{u}, q_k\mathbf{v})| + |q_k - t| |\beta(\mathbf{u}, \mathbf{v})| \\ & \leq (M\|\mathbf{v}\| + |\beta(\mathbf{u}, \mathbf{v})|) |q_k - t|, \end{aligned}$$

where the middle term used  $\beta(\mathbf{u}, q_k\mathbf{v}) = q_k\beta(\mathbf{u}, \mathbf{v})$ . The left-hand side does not depend on  $k$ , and the right-hand side tends to 0, so the left-hand side is 0.

If  $F = \mathbb{R}$ , we are done:  $\beta$  is symmetric, bilinear by Steps 1 and 2, and  $\beta(\mathbf{v}, \mathbf{v}) = \|\mathbf{v}\|^2 > 0$  for  $\mathbf{v} \neq \mathbf{0}$  by (N1), so (IP1), (IP2) and (IP3) of [Definition 10.1.1](#) all hold and the induced norm is  $\|\cdot\|$ .

**Step 3. The complex case.** Suppose  $F = \mathbb{C}$ . Steps 1 and 2 apply to  $V$  as a real space, so  $\beta$  is a symmetric  $\mathbb{R}$ -bilinear form with  $\beta(\mathbf{v}, \mathbf{v}) = \|\mathbf{v}\|^2$ . Define

$$\langle \mathbf{u}, \mathbf{v} \rangle = \beta(\mathbf{u}, \mathbf{v}) + i\beta(\mathbf{u}, i\mathbf{v}).$$

Everything follows from one identity:  $\beta(i\mathbf{u}, i\mathbf{v}) = \beta(\mathbf{u}, \mathbf{v})$ , which holds because  $\|i\mathbf{u} \pm i\mathbf{v}\| = \|i(\mathbf{u} \pm \mathbf{v})\| = |i|\|\mathbf{u} \pm \mathbf{v}\| = \|\mathbf{u} \pm \mathbf{v}\|$  by (N2). Applying it twice,  $\beta(i\mathbf{u}, \mathbf{v}) = \beta(i \cdot i\mathbf{u}, i\mathbf{v}) = -\beta(\mathbf{u}, i\mathbf{v})$ .

(IP3) and the norm:  $\beta(\mathbf{u}, i\mathbf{u}) = \frac{1}{4}(\|\mathbf{u} + i\mathbf{u}\|^2 - \|\mathbf{u} - i\mathbf{u}\|^2)$ , and (N2) turns this into  $\frac{1}{4}(|1 + i|^2 - |1 - i|^2)\|\mathbf{u}\|^2 = 0$ . Hence  $\langle \mathbf{u}, \mathbf{u} \rangle = \beta(\mathbf{u}, \mathbf{u}) = \|\mathbf{u}\|^2$ , which is  $> 0$  for  $\mathbf{u} \neq \mathbf{0}$ , and the induced norm is  $\|\cdot\|$ .

(IP1): additivity in the first slot and homogeneity for real scalars come from Steps 1 and 2 applied to both  $\beta(\cdot, \mathbf{v})$  and  $\beta(\cdot, i\mathbf{v})$ . For the scalar  $i$ ,

$$\begin{aligned}\langle i\mathbf{u}, \mathbf{v} \rangle &= \beta(i\mathbf{u}, \mathbf{v}) + i\beta(i\mathbf{u}, i\mathbf{v}) \\ &= -\beta(\mathbf{u}, i\mathbf{v}) + i\beta(\mathbf{u}, \mathbf{v}) = i\langle \mathbf{u}, \mathbf{v} \rangle.\end{aligned}$$

Every complex scalar is  $a + bi$  with  $a, b$  real, so (IP1) follows.

(IP2): using  $\beta(\mathbf{v}, i\mathbf{u}) = \beta(i\mathbf{v}, i \cdot i\mathbf{u}) = -\beta(\mathbf{u}, i\mathbf{v})$  and the symmetry of  $\beta$ ,

$$\langle \mathbf{v}, \mathbf{u} \rangle = \beta(\mathbf{u}, \mathbf{v}) - i\beta(\mathbf{u}, i\mathbf{v}) = \overline{\langle \mathbf{u}, \mathbf{v} \rangle}.$$

So  $\langle \cdot, \cdot \rangle$  is an inner product inducing  $\|\cdot\|$ .

**Uniqueness.** Suppose  $\langle \cdot, \cdot \rangle'$  is any inner product on  $V$  with  $\langle \mathbf{v}, \mathbf{v} \rangle' = \|\mathbf{v}\|^2$  for every  $\mathbf{v}$ . The two displays of the  $(\Rightarrow)$  direction apply to it verbatim, and subtracting them instead of adding them gives

$$\|\mathbf{u} + \mathbf{v}\|^2 - \|\mathbf{u} - \mathbf{v}\|^2 = 4 \operatorname{Re} \langle \mathbf{u}, \mathbf{v} \rangle',$$

that is,  $\operatorname{Re} \langle \mathbf{u}, \mathbf{v} \rangle' = \beta(\mathbf{u}, \mathbf{v})$ : the real part of  $\langle \cdot, \cdot \rangle'$  is forced to be the  $\beta$  built above. Over  $\mathbb{R}$  an inner product is real-valued, so  $\langle \mathbf{u}, \mathbf{v} \rangle' = \beta(\mathbf{u}, \mathbf{v})$  and there is nothing left to choose. Over  $\mathbb{C}$ , apply the same identity to the pair  $\mathbf{u}, i\mathbf{v}$ . By (IP1) and (IP2) together the form is conjugate-linear in its second slot, so  $\langle \mathbf{u}, i\mathbf{v} \rangle' = \bar{i} \langle \mathbf{u}, \mathbf{v} \rangle' = -i \langle \mathbf{u}, \mathbf{v} \rangle'$ , whose real part is  $\operatorname{Im} \langle \mathbf{u}, \mathbf{v} \rangle'$ . Hence  $\operatorname{Im} \langle \mathbf{u}, \mathbf{v} \rangle' = \beta(\mathbf{u}, i\mathbf{v})$ , and the two parts together give  $\langle \mathbf{u}, \mathbf{v} \rangle' = \beta(\mathbf{u}, \mathbf{v}) + i\beta(\mathbf{u}, i\mathbf{v})$ , which is exactly the inner product constructed above. So it is the only one. ([Exercise \(C2: Recovering the inner product from the norm\)](#) set this computation as an exercise in Chapter 10; it is carried out here because the theorem rests on it.)

This proves the theorem. ■

The theorem says that the parallelogram law is not one property of induced norms among many — it is the whole difference. A norm either obeys it, and then it is secretly an inner product with an inner product's entire apparatus of angles, orthogonality and adjoints; or it violates it once, and then no inner product will ever produce it.

**Warning. Neither  $\|\cdot\|_1$  nor  $\|\cdot\|_\infty$  comes from an inner product.** Test  $(*)$  on  $\mathbf{u} = \mathbf{e}_1$  and  $\mathbf{v} = \mathbf{e}_2$  in  $F^2$ . Both  $\mathbf{u} + \mathbf{v} = (1, 1)$  and  $\mathbf{u} - \mathbf{v} = (1, -1)$  have  $\|\cdot\|_1 = 2$ , so the left side of  $(*)$  is  $4 + 4 = 8$ , while the right side is  $2 + 2 = 4$ . For  $\|\cdot\|_\infty$  both have norm 1, so the left side is 2 and the right side is again 4. One failing pair is enough, and there is no repair: by [Theorem 15.1.3](#) no inner product whatsoever induces these norms, in any dimension  $\geq 2$ . The same two vectors  $f(t) = t$  and  $g(t) = 1 - t$  in  $C[0, 1]$  refute it for the sup norm, since  $\|f\|_\infty = \|g\|_\infty = \|f + g\|_\infty = \|f - g\|_\infty = 1$ .

**Quick check.** The function  $N(\mathbf{x}) = (x_1^2 + 4x_2^2)^{1/2}$  is a norm on  $\mathbb{R}^2$ , and it is lopsided:  $N(\mathbf{e}_1) = 1$  while  $N(\mathbf{e}_2) = 2$ . Does it come from an inner product?

**Solution.** Yes. It is the norm induced by  $\langle \mathbf{x}, \mathbf{y} \rangle = x_1y_1 + 4x_2y_2$ , which satisfies (IP1) and (IP2) by inspection and (IP3) because  $x_1^2 + 4x_2^2 > 0$  for  $\mathbf{x} \neq \mathbf{0}$ . One may also check

(\*) directly: both sides expand to  $2x_1^2 + 8x_2^2 + 2y_1^2 + 8y_2^2$ . The lesson is that a norm can be visibly lopsided — its unit ball here is an ellipse, not a circle — and still be an inner-product norm. What matters is not roundness but the parallelogram law.

So the norms of this book split in two. On one side sit  $\|\cdot\|_2$ , the Frobenius norm and every weighted Euclidean norm, where Chapters 10 to 12 apply in full. On the other sit  $\|\cdot\|_1$ ,  $\|\cdot\|_\infty$  and the sup norm, where there is no orthogonality and no adjoint, and the only tools are the three axioms. The next section shows that for questions about convergence the split does not matter at all.

## 15.1.4 Exercises

### 15.1.4.1 A. Check your understanding

#### Exercise A1.

- State the three clauses of [Definition 15.1.1](#), and say which one fails for  $N(\mathbf{x}) = |x_1|$  on  $F^2$ .
- Deduce  $\|\mathbf{0}\| = 0$  from (N2) alone, and explain why (N1) is therefore stated with the word **only**.
- True or false: a function satisfying (N1) and (N3) but not (N2) cannot exist. Justify your answer.
- Name the test that decides whether a norm is induced by an inner product, and state what [Exercise \(C1: The parallelogram law characterizes inner-product norms\)](#) had already established before this section.
- Order  $\|\mathbf{x}\|_1$ ,  $\|\mathbf{x}\|_2$ ,  $\|\mathbf{x}\|_\infty$  for a general  $\mathbf{x} \in F^n$ , and give a vector for which all three are equal.

*Solution.*

- (N1) positive definiteness, (N2) absolute homogeneity, (N3) the triangle inequality ([Definition 15.1.1](#)). For  $N(\mathbf{x}) = |x_1|$  only (N1) fails, and only its “only” half:  $N(\mathbf{e}_2) = 0$  with  $\mathbf{e}_2 \neq \mathbf{0}$  ([Example \(Three near misses\)](#) (b)).
- Take  $c = 0$  and any  $\mathbf{v}$ : (N2) gives  $\|\mathbf{0}\| = \|\mathbf{0}\mathbf{v}\| = |0|\|\mathbf{v}\| = 0$ . So “ $\mathbf{v} = \mathbf{0} \Rightarrow \|\mathbf{v}\| = 0$ ” is automatic, and the only content left in (N1) is the converse, which is why it is the clause that is stated.
- False. Take  $N(\mathbf{x}) = \min(\|\mathbf{x}\|_2, 1)$  on  $\mathbb{R}^2$ . It is  $\geq 0$  and vanishes only at  $\mathbf{0}$ , so (N1) holds. For (N3), note first that  $\min(a+b, 1) \leq \min(a, 1) + \min(b, 1)$  for all  $a, b \geq 0$ : if either  $a \geq 1$  or  $b \geq 1$  the right side is at least 1, which bounds the left side; and otherwise the right side is  $a+b$ , which is at least  $\min(a+b, 1)$ . Applying this to  $a = \|\mathbf{u}\|_2$ ,  $b = \|\mathbf{v}\|_2$  and using that  $s \mapsto \min(s, 1)$  is increasing gives (N3). But  $N(2\mathbf{e}_1) = 1 \neq 2 = 2N(\mathbf{e}_1)$ , so (N2) fails.
- The parallelogram law (\*): a norm is induced by an inner product exactly when  $\|\mathbf{u} + \mathbf{v}\|^2 + \|\mathbf{u} - \mathbf{v}\|^2 = 2\|\mathbf{u}\|^2 + 2\|\mathbf{v}\|^2$  always ([Theorem 15.1.3](#)). Before this section, [Exercise \(C1: The parallelogram law characterizes inner-product norms\)](#) had proved the converse direction on  $\mathbb{R}^2$ , and only modulo the granted homogeneity  $\beta(\mathbf{u}, t\mathbf{v}) = t\beta(\mathbf{u}, \mathbf{v})$  for real  $t$ ; the general dimension, the real homogeneity step and the complex case are what [Theorem 15.1.3](#) adds.

- (e)  $\|\mathbf{x}\|_\infty \leq \|\mathbf{x}\|_2 \leq \|\mathbf{x}\|_1$  ([Proposition 15.1.2](#)). All three agree at  $\mathbf{x} = \mathbf{e}_1$ , and more generally at any  $\mathbf{x}$  with at most one non-zero coordinate.

### 15.1.4.2 B. Practice

#### Exercise B1: Three measurements.

For  $\mathbf{x} = (1, -1, 2, -2) \in \mathbb{R}^4$ , compute  $\|\mathbf{x}\|_1$ ,  $\|\mathbf{x}\|_2$  and  $\|\mathbf{x}\|_\infty$ , and verify all four inequalities of [Proposition 15.1.2](#). Which of them is closest to being an equality?

*Solution.*  $\|\mathbf{x}\|_1 = 1 + 1 + 2 + 2 = 6$ ;  $\|\mathbf{x}\|_2 = (1 + 1 + 4 + 4)^{1/2} = \sqrt{10}$ ;  $\|\mathbf{x}\|_\infty = 2$ . Here  $n = 4$ , so  $\sqrt{n} = 2$ . The four inequalities read

$$2 \leq \sqrt{10} \leq 6, \quad 6 \leq 2\sqrt{10}, \quad \sqrt{10} \leq 4,$$

and  $\sqrt{10} \approx 3.162$ , so all four hold. The tightest is  $\|\mathbf{x}\|_1 \leq \sqrt{n}\|\mathbf{x}\|_2$ , that is  $6 \leq 6.32$ ; this is as expected, since equality there needs all coordinates of equal modulus, and  $\mathbf{x}$  has only two distinct moduli.

#### Exercise B2: Determine which are norms.

Determine which of the following define norms on  $\mathbb{R}^2$ . For those that do, verify all three clauses of [Definition 15.1.1](#); for those that do not, name the exact clause that fails and give a witness. Justify your answer.

- (a)  $N(\mathbf{x}) = |x_1| + 2|x_2|$ .
- (b)  $N(\mathbf{x}) = |x_1 - x_2|$ .
- (c)  $N(\mathbf{x}) = \max(|x_1|, |x_2|) - \frac{1}{4} \min(|x_1|, |x_2|)$ .
- (d)  $N(\mathbf{x}) = |x_1| + |x_2| - \frac{1}{2} \min(|x_1|, |x_2|)$ .

*Solution.*

- (a) A norm. It is the weighted norm of [Example \(Inner products, weights, and a function space\)](#) (b) in its 1-norm form: (N1) because both terms are  $\geq 0$  and both vanish only for  $\mathbf{x} = \mathbf{0}$ ; (N2) because  $|cx_i| = |c||x_i|$ ; (N3) coordinatewise, exactly as for  $\|\cdot\|_1$ .
- (b) Not a norm. (N2) and (N3) hold, inherited from  $|\cdot|$  on  $\mathbb{R}$ . **(N1) fails:**  $N((1, 1)) = 0$  with  $(1, 1) \neq \mathbf{0}$ .
- (c) Not a norm. (N1) holds, since  $N(\mathbf{x}) \geq \frac{3}{4} \max(|x_1|, |x_2|)$ , which is  $> 0$  for  $\mathbf{x} \neq \mathbf{0}$ ; and (N2) holds because max and min of the scaled moduli both pick up the factor  $|c|$ . **(N3) fails:** with  $\mathbf{u} = (1, 1)$  and  $\mathbf{v} = (1, -1)$  we get  $N(\mathbf{u}) = N(\mathbf{v}) = 1 - \frac{1}{4} = \frac{3}{4}$ , while  $\mathbf{u} + \mathbf{v} = (2, 0)$  has  $N(\mathbf{u} + \mathbf{v}) = 2 - 0 = 2 > \frac{3}{2}$ . Subtracting from the largest coordinate is exactly what a cancellation in the sum can punish.
- (d) A norm, despite the minus sign. Since  $|x_1| + |x_2| = \max + \min$ , the formula is  $N(\mathbf{x}) = \max + \frac{1}{2} \min$ , which is

$$N(\mathbf{x}) = \frac{1}{2}\|\mathbf{x}\|_1 + \frac{1}{2}\|\mathbf{x}\|_\infty.$$

A sum of two norms, each scaled by a positive number, is a norm: (N1), (N2)

and (N3) are each inherited term by term, and (N1) needs only one of the two summands to detect  $\mathbf{x} \neq \mathbf{0}$ . Compare (c): there the subtracted term was large enough to break (N3), here it is not, and the way to tell is to rewrite the formula rather than to guess.

### Exercise B3: Which of these come from an inner product?.

For each norm, determine whether it is induced by an inner product. Justify your answer, using [Theorem 15.1.3](#) in each case.

- (a)  $\|\mathbf{x}\| = (x_1^2 + x_1x_2 + x_2^2)^{1/2}$  on  $\mathbb{R}^2$ .
- (b)  $\|\mathbf{A}\|_F$  on  $M_2(\mathbb{R})$ .
- (c)  $N(\mathbf{A}) = \max_{i,j} |a_{ij}|$  on  $M_2(\mathbb{R})$ .

*Solution.*

- (a) Yes. Put  $\langle \mathbf{x}, \mathbf{y} \rangle = x_1y_1 + \frac{1}{2}(x_1y_2 + x_2y_1) + x_2y_2$ . It is bilinear and symmetric by inspection, and  $\langle \mathbf{x}, \mathbf{x} \rangle = x_1^2 + x_1x_2 + x_2^2 = (x_1 + \frac{1}{2}x_2)^2 + \frac{3}{4}x_2^2$ , which is  $> 0$  unless  $x_2 = 0$  and  $x_1 = 0$ ; so (IP3) holds and the induced norm is the given one. (Completing the square is also how one sees that the given function is a norm at all.)
- (b) Yes. The Frobenius norm is induced by the Frobenius inner product  $\langle \mathbf{A}, \mathbf{B} \rangle = \text{tr}(\mathbf{B}^\top \mathbf{A})$  ([Example \(Inner products, weights, and a function space\)](#) (a)), so (\*) holds by the forward direction of [Theorem 15.1.3](#).
- (c) No. Take  $\mathbf{A} = \mathbf{E}_{11}$  and  $\mathbf{B} = \mathbf{E}_{12}$ , the matrix units. Then  $\mathbf{A} \pm \mathbf{B}$  has entries 1 and  $\pm 1$  in the first row and zeros elsewhere, so  $N(\mathbf{A} + \mathbf{B}) = N(\mathbf{A} - \mathbf{B}) = 1$  and the left side of (\*) is 2, while the right side is  $2 \cdot 1 + 2 \cdot 1 = 4$ . By [Theorem 15.1.3](#) no inner product induces  $N$ .

### 15.1.4.3 C. Going deeper

#### Exercise C1: Sharpness of the comparison constants.

- (a) Prove that  $\|\mathbf{x}\|_1 = \sqrt{n} \|\mathbf{x}\|_2$  holds if and only if all the numbers  $|x_1|, \dots, |x_n|$  are equal. *Hint: the equality case of [Theorem 10.1.5](#).*
- (b) Hence deduce that no constant  $C$  independent of  $n$  satisfies  $\|\mathbf{x}\|_1 \leq C \|\mathbf{x}\|_2$  for all  $n$  and all  $\mathbf{x} \in \mathbb{R}^n$ .

*Solution.*

- (a) With  $\mathbf{a} = (|x_1|, \dots, |x_n|)$ , the proof of [Proposition 15.1.2](#) reads  $\|\mathbf{x}\|_1 = \langle \mathbf{a}, \mathbf{1} \rangle \leq \|\mathbf{a}\|_2 \|\mathbf{1}\|_2 = \sqrt{n} \|\mathbf{x}\|_2$ , an application of [Theorem 10.1.5](#) to  $\mathbf{a}$  and  $\mathbf{1}$ . Equality in [Theorem 10.1.5](#) holds exactly when  $(\mathbf{a}, \mathbf{1})$  is dependent. Since  $\mathbf{1} \neq \mathbf{0}$ , that means  $\mathbf{a} = t\mathbf{1}$  for some real  $t$ , which is  $t \geq 0$  because the entries of  $\mathbf{a}$  are; and  $\mathbf{a} = t\mathbf{1}$  says precisely that all the  $|x_i|$  are equal.
- (b) By (a),  $\mathbf{x} = \mathbf{1} \in \mathbb{R}^n$  gives  $\|\mathbf{1}\|_1 = n$  and  $\|\mathbf{1}\|_2 = \sqrt{n}$ , so any valid  $C$  satisfies  $C \geq n/\sqrt{n} = \sqrt{n}$ . As  $n$  is arbitrary, no single  $C$  works for all  $n$ . The comparison constants between two norms are genuinely functions of the dimension.

**Exercise C2: A real norm that is not a complex norm.**

Regard  $\mathbb{C}$  as a one-dimensional complex vector space, and also as  $\mathbb{R}^2$  by  $z = a + bi \leftrightarrow (a, b)$ . Define  $N(z) = 2|a| + |b|$ .

- (a) Prove that  $N$  is a norm on  $\mathbb{R}^2$ .
- (b) Prove that  $N$  is **not** a norm on  $\mathbb{C}$  as a complex vector space, and name the exact clause that fails.
- (c) Prove that if  $\|\cdot\|$  is any norm on a complex vector space  $V$ , then it is also a norm on  $V$  regarded as a real vector space. Explain why (b) does not contradict this.

*Solution.*

- (a) It is the weighted 1-norm  $|x_1| \cdot 2 + |x_2| \cdot 1$  of [Exercise \(B2: Determine which are norms\)](#) (a) with the weights swapped, so all three clauses hold as there.
- (b) **(N2) fails** for the complex scalar  $c = i$ . Indeed  $N(1) = 2$  and  $N(i) = 1$ , while (N2) would demand  $N(i \cdot 1) = |i|N(1) = 2$ .
- (c) Restricting the scalars from  $\mathbb{C}$  to  $\mathbb{R}$  leaves the vectors, the addition and the zero vector unchanged, so (N1) and (N3) are the same statements. (N2) for real  $c$  is a special case of (N2) for complex  $c$ , the absolute value of a real number being the same in  $\mathbb{R}$  and in  $\mathbb{C}$ . There is no contradiction with (b) because the implication runs one way only: (N2) over  $\mathbb{C}$  is a **strictly stronger** demand than (N2) over  $\mathbb{R}$ , the extra content being  $\|i\mathbf{v}\| = \|\mathbf{v}\|$ , and  $N$  fails exactly that.

**Exercise C3: The ball determines the norm.**

Let  $\|\cdot\|$  and  $N$  be two norms on a vector space  $V$  with the **same** closed unit ball, in the sense of [Definition 15.1.2](#):

$$\{\mathbf{v} : \|\mathbf{v}\| \leq 1\} = \{\mathbf{v} : N(\mathbf{v}) \leq 1\}.$$

- (a) Prove that for every  $\mathbf{v} \in V$  and every real  $t > 0$ , we have  $\|\mathbf{v}\| \leq t$  if and only if  $\mathbf{v}/t$  lies in the closed unit ball of  $\|\cdot\|$ .
- (b) Hence prove that  $\|\mathbf{v}\| = N(\mathbf{v})$  for every  $\mathbf{v} \in V$ .
- (c) Give two different norms on  $\mathbb{R}^2$  whose closed unit balls have the same area, and say why (b) does not apply to them.

*Solution.*

- (a) Since  $t > 0$ , (N2) gives  $\|\mathbf{v}/t\| = \|\mathbf{v}\|/t$ . So  $\mathbf{v}/t$  lies in the ball exactly when  $\|\mathbf{v}\|/t \leq 1$ , that is, exactly when  $\|\mathbf{v}\| \leq t$ .
- (b) For  $\mathbf{v} = \mathbf{0}$  both sides are 0 by (N1). Let  $\mathbf{v} \neq \mathbf{0}$  and put  $t = \|\mathbf{v}\| > 0$ . By (a),  $\mathbf{v}/t$  lies in the ball of  $\|\cdot\|$ , hence in the ball of  $N$ , hence  $N(\mathbf{v}) \leq t = \|\mathbf{v}\|$  by (a) applied to  $N$ . Swapping the roles of  $\|\cdot\|$  and  $N$  gives  $\|\mathbf{v}\| \leq N(\mathbf{v})$ . Therefore  $\|\mathbf{v}\| = N(\mathbf{v})$ .
- (c) Take  $\|\cdot\|_1$ , whose ball is the diamond with vertices  $\pm \mathbf{e}_1$  and  $\pm \mathbf{e}_2$ , of area 2; and  $N(\mathbf{x}) = 2|x_1| + \frac{1}{2}|x_2|$ , whose ball is the diamond with vertices  $\pm \frac{1}{2}\mathbf{e}_1$  and  $\pm 2\mathbf{e}_2$ , also of area 2. They differ at  $\mathbf{e}_1$ , where they give 1 and 2. Part (b) does not apply

because equal **area** is far weaker than equal **set**: the recipe in (a) reads the norm off the ball as a set of vectors, and the area of that set retains almost none of it.

## 15.2 All Norms Agree

§01 left a worry. There are many norms on  $F^n$ , they give different answers, and every statement of the form “this quantity is small” now has to say small in which norm. This section removes the worry in finite dimension, in the strongest way available: any two norms on a finite-dimensional space are trapped between fixed multiples of each other, so they agree about every limit, every bounded set and every continuous function. It also says what the theorem costs — constants that grow with the dimension — and where it stops being true.

### 15.2.1 Trapped between multiples

Look again at [Proposition 15.1.2](#). Its four inequalities say that on  $F^n$ ,

$$\frac{1}{\sqrt{n}} \|\mathbf{x}\|_1 \leq \|\mathbf{x}\|_2 \leq \|\mathbf{x}\|_1,$$

so  $\|\cdot\|_2$  never differs from  $\|\cdot\|_1$  by a factor worse than  $\sqrt{n}$ , in either direction. Whatever  $\|\cdot\|_1$  calls small,  $\|\cdot\|_2$  calls small too, at the cost of a fixed factor. That is a relation worth naming.

*Two norms are equivalent when each is sandwiched between two fixed multiples of the other, so that neither can be small while the other is large.*

#### Definition 15.2.1 — Equivalent Norms

Let  $V$  be a vector space over  $F = \mathbb{R}$  or  $\mathbb{C}$ , and let  $\|\cdot\|_a$  and  $\|\cdot\|_b$  be norms on  $V$ . They are **equivalent** if there exist real constants  $c, C > 0$ , **not depending on the vector**, such that

$$c \|\mathbf{v}\|_b \leq \|\mathbf{v}\|_a \leq C \|\mathbf{v}\|_b \quad \text{for every } \mathbf{v} \in V.$$

In words. The two inequalities are read separately: the right one says  $\|\cdot\|_a$  is *no larger than* a fixed multiple of  $\|\cdot\|_b$ , and the left one says it is *no smaller than* a fixed positive multiple. The phrase **not depending on the vector** is the whole content, since for each individual  $\mathbf{v} \neq \mathbf{0}$  the ratio  $\|\mathbf{v}\|_a / \|\mathbf{v}\|_b$  is some positive number and one could always take that number as the constant. What is being asked is that one pair of constants works for all  $\mathbf{v}$  at once. Note also that the definition says nothing about the two norms being close;  $c$  may be  $10^{-6}$ .

#### Example. (The three $p$ -norms, with best constants)

On  $F^n$ , [Proposition 15.1.2](#) gives

$$\begin{aligned} \|\mathbf{x}\|_\infty &\leq \|\mathbf{x}\|_2 \leq \sqrt{n} \|\mathbf{x}\|_\infty, \\ \|\mathbf{x}\|_\infty &\leq \|\mathbf{x}\|_1 \leq n \|\mathbf{x}\|_\infty, \\ \frac{1}{\sqrt{n}} \|\mathbf{x}\|_1 &\leq \|\mathbf{x}\|_2 \leq \|\mathbf{x}\|_1, \end{aligned}$$

so the three are pairwise equivalent. The middle line's upper constant comes from



$\|\mathbf{x}\|_1 \leq \sqrt{n}\|\mathbf{x}\|_2 \leq \sqrt{n} \cdot \sqrt{n}\|\mathbf{x}\|_\infty$ . Every constant shown is attained, so none of them can be improved. On the first two lines the lower constant is attained at  $\mathbf{e}_1$  and the upper one at  $\mathbf{1}$ . **The third line runs the other way:** at  $\mathbf{1}$  the lower inequality reads  $\frac{1}{\sqrt{n}} \cdot n = \sqrt{n} \leq \sqrt{n}$ , and at  $\mathbf{e}_1$  the upper one reads  $1 \leq 1$ .

Before the theorem, one small fact makes the proof strategy possible: it is enough to compare every norm with a single one of our choosing.

### Proposition 15.2.1 — Equivalence of Norms Is an Equivalence Relation

On a fixed vector space  $V$  over  $F$ , the relation of Definition 15.2.1 is reflexive, symmetric and transitive.

*Proof.* Reflexive: take  $c = C = 1$ . Symmetric: if  $c\|\mathbf{v}\|_b \leq \|\mathbf{v}\|_a \leq C\|\mathbf{v}\|_b$  for all  $\mathbf{v}$ , then dividing through by the positive constants gives  $C^{-1}\|\mathbf{v}\|_a \leq \|\mathbf{v}\|_b \leq c^{-1}\|\mathbf{v}\|_a$ . Transitive: if in addition  $c'\|\mathbf{v}\|_d \leq \|\mathbf{v}\|_b \leq C'\|\mathbf{v}\|_d$ , then multiplying the first chain's outer inequalities by the positive constants  $c$  and  $C$  gives

$$cc'\|\mathbf{v}\|_d \leq c\|\mathbf{v}\|_b \leq \|\mathbf{v}\|_a \leq C\|\mathbf{v}\|_b \leq CC'\|\mathbf{v}\|_d,$$

and  $cc', CC' > 0$ . This proves all three properties. ■

So to prove that *any two* norms on  $V$  are equivalent, it suffices to prove that *every* norm is equivalent to one fixed reference norm. We take the reference to be the Euclidean norm of a chosen basis, because that is where compactness lives.

## 15.2.2 Compactness enters the book

### Theorem 15.2.2 — Equivalence of Norms in Finite Dimension

Let  $V$  be a vector space over  $F = \mathbb{R}$  or  $\mathbb{C}$  with  $\dim V = n < \infty$ . Then **any two** norms on  $V$  are equivalent. Explicitly, if  $\mathcal{B} = (\mathbf{v}_1, \dots, \mathbf{v}_n)$  is a basis of  $V$ , if  $\|\cdot\|_2$  denotes the Euclidean norm of the coordinate vector,

$$\left\| \sum_i x_i \mathbf{v}_i \right\|_2 := (|x_1|^2 + \dots + |x_n|^2)^{1/2},$$

and if  $N$  is any norm on  $V$ , then there are constants  $0 < m \leq M$  with

$$m \|\mathbf{v}\|_2 \leq N(\mathbf{v}) \leq M \|\mathbf{v}\|_2 \quad \text{for every } \mathbf{v} \in V.$$

*Idea.* The two inequalities are not symmetric, and it is worth seeing why before starting. The **upper** bound is cheap: write  $\mathbf{v}$  in the basis, apply the triangle inequality once, and the constant  $M$  is built out of the numbers  $N(\mathbf{v}_i)$ . The **lower** bound cannot be obtained the same way; there is no inequality running from  $N(\sum x_i \mathbf{v}_i)$  down to the individual coordinates, and the temptation to assume that the two norms are comparable on the

basis and then “extend by linearity” is exactly the error the theorem exists to forbid — [Example \(Agreeing on a basis is not enough\)](#) below shows two norms agreeing on a basis and differing by a factor of  $n$  elsewhere.

The route is therefore ① upper bound by the triangle inequality; ② use that upper bound to show  $N$  is continuous as a function of the coordinates; ③ restrict  $N$  to the Euclidean unit sphere, which is compact, so  $N$  attains a minimum  $m$  there; ④ the minimum is positive because  $N$  vanishes only at  $\mathbf{0}$ , which is not on the sphere; ⑤ scale back. Compactness is what turns the infimum over infinitely many unit vectors into a value attained at a particular unit vector, and a value attained at a vector can be tested for positivity one vector at a time.

*Proof.* By [Proposition 15.2.1](#) it suffices to prove the displayed two-sided bound, since then any two norms  $N_1, N_2$  are each equivalent to  $\|\cdot\|_2$  and hence to each other. If  $n = 0$  then  $V = \{\mathbf{0}\}$ , both norms are the zero function and  $m = M = 1$  works; assume  $n \geq 1$ . The coordinate map  $\Phi: F^n \rightarrow V$ ,  $\mathbf{x} \mapsto \sum_i x_i \mathbf{v}_i$ , is an isomorphism ([Corollary 3.5.5](#)), and by construction  $\|\Phi(\mathbf{x})\|_2 = \|\mathbf{x}\|_2$  is the Euclidean norm of [Example \(The 1-, 2- and  \$\infty\$ -norms on  \$F^n\$ \)](#). It is enough to prove the two bounds for  $\mathbf{x} \in F^n$  with  $N$  replaced by  $f(\mathbf{x}) := N(\Phi(\mathbf{x}))$ .

**Step 1. The upper bound.** Let  $M = (\sum_{i=1}^n N(\mathbf{v}_i)^2)^{1/2}$ . For  $\mathbf{x} \in F^n$ , the triangle inequality (N3) applied  $n - 1$  times and then (N2) give

$$f(\mathbf{x}) = N\left(\sum_i x_i \mathbf{v}_i\right) \leq \sum_i |x_i| N(\mathbf{v}_i),$$

and [Theorem 10.1.5](#) in  $\mathbb{R}^n$ , applied to the vectors  $(|x_1|, \dots, |x_n|)$  and  $(N(\mathbf{v}_1), \dots, N(\mathbf{v}_n))$ , bounds the right-hand side by  $M\|\mathbf{x}\|_2$ . Hence  $f(\mathbf{x}) \leq M\|\mathbf{x}\|_2$  for every  $\mathbf{x}$ . Also  $M > 0$ , since  $N(\mathbf{v}_1) > 0$  by (N1) and  $\mathbf{v}_1 \neq \mathbf{0}$ .

**Step 2.  $f$  is continuous on  $F^n$  for the Euclidean norm.** By [Lemma 15.1.1](#) applied to  $N$ , and then Step 1 applied to  $\mathbf{x} - \mathbf{y}$ ,

$$|f(\mathbf{x}) - f(\mathbf{y})| \leq N(\Phi(\mathbf{x} - \mathbf{y})) \leq M \|\mathbf{x} - \mathbf{y}\|_2.$$

So  $f$  changes by at most  $M$  times the distance moved: it is continuous, and the continuity was **deduced** from Step 1 rather than assumed.

**Step 3. The minimum on the unit sphere.** Let

$$S = \{\mathbf{x} \in F^n : \|\mathbf{x}\|_2 = 1\},$$

which is non-empty because  $n \geq 1$  (it contains  $\mathbf{e}_1$ ). It is bounded, and it is closed: if  $\|\mathbf{x}_k - \mathbf{x}\|_2 \rightarrow 0$  with every  $\|\mathbf{x}_k\|_2 = 1$ , then [Lemma 15.1.1](#) gives  $|\|\mathbf{x}\|_2 - 1| \leq \|\mathbf{x} - \mathbf{x}_k\|_2 \rightarrow 0$ , so  $\|\mathbf{x}\|_2 = 1$  and  $\mathbf{x} \in S$ . By **the compactness of closed bounded sets in  $F^n$ , fact (A3) of the chapter introduction**,  $S$  is compact; by **the extreme value theorem, fact (A4)**, the continuous function  $f$  of Step 2 attains a minimum value on  $S$ . Say  $m = f(\mathbf{x}_0)$  with  $\mathbf{x}_0 \in S$ .

**Step 4. The minimum is positive.** Since  $\mathbf{x}_0 \in S$  we have  $\|\mathbf{x}_0\|_2 = 1 \neq 0$ , so  $\mathbf{x}_0 \neq \mathbf{0}$  and hence  $\Phi(\mathbf{x}_0) \neq \mathbf{0}$ ,  $\Phi$  being injective. By (N1) for  $N$ ,  $m = N(\Phi(\mathbf{x}_0)) > 0$ . This is the step that needs a minimum rather than an infimum: an infimum of positive numbers can be 0, while a minimum is the value at an actual vector, and there (N1) can be applied.

**Step 5. Scaling back.** Let  $\mathbf{x} \neq \mathbf{0}$ . Then  $\mathbf{x}/\|\mathbf{x}\|_2 \in S$ , so  $f(\mathbf{x}/\|\mathbf{x}\|_2) \geq m$ , and (N2) for  $N$  gives  $f(\mathbf{x}) \geq m\|\mathbf{x}\|_2$ . For  $\mathbf{x} = \mathbf{0}$  both sides are 0. Combining with Step 1,

$$m\|\mathbf{x}\|_2 \leq f(\mathbf{x}) \leq M\|\mathbf{x}\|_2 \quad (\mathbf{x} \in F^n),$$

and transporting along  $\Phi$  gives the statement for  $V$ . Since  $m \leq f(\mathbf{x}_0) \leq M\|\mathbf{x}_0\|_2 = M$ , the constants satisfy  $0 < m \leq M$ . This proves the theorem. ■

The proof used the finite dimension twice, and it is worth saying where: once in Step 1, where the triangle inequality was applied finitely many times to a finite basis expansion, and once in Step 3, where  $F^n$  is the space in which closed bounded sets are compact. Remove either and the argument collapses, as the last part of this section shows.

### Example. (Agreeing on a basis is not enough)

On  $F^n$ , the norms  $\|\cdot\|_1$  and  $\|\cdot\|_\infty$  both give every standard basis vector the value 1. Yet  $\|\mathbf{1}\|_1 = n$  and  $\|\mathbf{1}\|_\infty = 1$ . So the data “ $N_1(\mathbf{v}_i) = N_2(\mathbf{v}_i)$  for every basis vector” implies no inequality between  $N_1$  and  $N_2$  beyond what [Theorem 15.2.2](#) gives for any pair at all.

*Quick check.* In Step 1 the constant was  $M = (\sum_i N(\mathbf{v}_i)^2)^{1/2}$ . Where would the proof break if we tried the symmetric guess  $m = \min_i N(\mathbf{v}_i)$  for the lower bound?

*Solution.* It breaks immediately, and [Example \(Agreeing on a basis is not enough\)](#) is the witness. Take  $V = F^2$  with the basis  $(\mathbf{e}_1, \mathbf{e}_2)$ ,  $N = \|\cdot\|_\infty$  and  $\|\cdot\|_2$  the Euclidean norm. Then  $\min_i N(\mathbf{e}_i) = 1$ , and the guessed inequality  $\|\mathbf{x}\|_\infty \geq \|\mathbf{x}\|_2$  fails at  $\mathbf{x} = \mathbf{1}$ , where the two sides are 1 and  $\sqrt{2}$ . The reason is that the triangle inequality only ever bounds  $N(\sum x_i \mathbf{v}_i)$  from **above**; there is no lower bound in the axioms, because a sum of long vectors can be short. Getting a lower bound needs a genuinely different mechanism, and compactness is that mechanism.

## 15.2.3 What the theorem buys

The point of the theorem is that whole sentences may now drop the words “in which norm”.

### Corollary 15.2.3 — Convergence and Boundedness Do Not Depend on the Norm

Let  $V$  be finite-dimensional over  $\mathbb{R}$  or  $\mathbb{C}$ , let  $\|\cdot\|_a$  and  $\|\cdot\|_b$  be norms on  $V$ , and let  $(\mathbf{u}_k)$  be a sequence in  $V$  with  $\mathbf{u} \in V$ . Then:

- (a)  $\|\mathbf{u}_k - \mathbf{u}\|_a \rightarrow 0$  if and only if  $\|\mathbf{u}_k - \mathbf{u}\|_b \rightarrow 0$ ;
- (b)  $(\mathbf{u}_k)$  is bounded for  $\|\cdot\|_a$  if and only if it is bounded for  $\|\cdot\|_b$ ;
- (c)  $(\mathbf{u}_k)$  is Cauchy for  $\|\cdot\|_a$  if and only if it is Cauchy for  $\|\cdot\|_b$ .

*Proof.* By [Theorem 15.2.2](#) there are  $c, C > 0$  with  $c\|\mathbf{v}\|_b \leq \|\mathbf{v}\|_a \leq C\|\mathbf{v}\|_b$  for every  $\mathbf{v} \in V$ . For (a), apply this to  $\mathbf{v} = \mathbf{u}_k - \mathbf{u}$ : if  $\|\mathbf{u}_k - \mathbf{u}\|_b \rightarrow 0$  then  $0 \leq \|\mathbf{u}_k - \mathbf{u}\|_a \leq$

$C\|\mathbf{u}_k - \mathbf{u}\|_b \rightarrow 0$ , and conversely  $\|\mathbf{u}_k - \mathbf{u}\|_b \leq c^{-1}\|\mathbf{u}_k - \mathbf{u}\|_a \rightarrow 0$ . For (b), a bound  $\|\mathbf{u}_k\|_b \leq K$  for all  $k$  gives  $\|\mathbf{u}_k\|_a \leq CK$ , and conversely with  $c^{-1}$ . For (c), apply the same two estimates to  $\mathbf{v} = \mathbf{u}_k - \mathbf{u}_l$ . This proves all three. ■

Now the connection this chapter was built for. Chapter 9 §10 defined convergence of matrices **entrywise** (Definition 9.10.1) and said, in its opening paragraph, that no notion of the size of a matrix was used and that Chapter 15 would supply one. Here is the reconciliation.

### Corollary 15.2.4 — Entrywise Convergence Is Convergence in Every Norm

Let  $\mathbf{M}_1, \mathbf{M}_2, \dots$  and  $\mathbf{M}$  lie in  $M_{p \times q}(\mathbb{C})$ , and let  $\|\cdot\|$  be **any** norm on  $M_{p \times q}(\mathbb{C})$ . Then  $\mathbf{M}_k \rightarrow \mathbf{M}$  entrywise in the sense of Definition 9.10.1 if and only if  $\|\mathbf{M}_k - \mathbf{M}\| \rightarrow 0$ . Likewise  $(\mathbf{M}_k)$  is bounded in the sense of Definition 9.10.1 if and only if  $(\|\mathbf{M}_k\|)$  is a bounded sequence of real numbers.

*Idea.* Entrywise convergence is convergence in one particular norm, the entrywise maximum. Theorem 15.2.2 says every other norm is trapped between two multiples of it, and a quantity squeezed between two multiples of something tending to 0 tends to 0 as well. Both directions are that one sentence.

*Proof.* The space  $M_{p \times q}(\mathbb{C})$  is a complex vector space of dimension  $pq$ , and

$$N_\infty(\mathbf{M}) = \max_{r,s} |m_{rs}|$$

is a norm on it: it is the  $\infty$ -norm of Example (The 1-, 2- and  $\infty$ -norms on  $F^n$ ) read on the list of entries, through the coordinate isomorphism of Corollary 3.5.5 that sends a matrix to its  $pq$  entries. By Definition 9.10.1 (a),  $\mathbf{M}_k \rightarrow \mathbf{M}$  entrywise means  $(\mathbf{M}_k)_{rs} \rightarrow \mathbf{M}_{rs}$  for each of the finitely many positions  $(r, s)$ , and since a maximum of finitely many non-negative sequences tends to 0 exactly when each of them does, this says precisely  $N_\infty(\mathbf{M}_k - \mathbf{M}) \rightarrow 0$ . By Theorem 15.2.2 the norms  $N_\infty$  and  $\|\cdot\|$  are equivalent, so Corollary 15.2.3 (a) converts this into  $\|\mathbf{M}_k - \mathbf{M}\| \rightarrow 0$ . The statement about boundedness is Definition 9.10.1 (b) together with Corollary 15.2.3 (b), read the same way. ■

### Example. (Watching one sequence converge)

In  $M_2(\mathbb{R})$  let

$$\mathbf{M}_k = \begin{pmatrix} \frac{1}{k} & 2 \\ 0 & 1 - \frac{1}{k} \end{pmatrix}, \quad k \geq 1.$$

Entry by entry,  $\frac{1}{k} \rightarrow 0$ ,  $2 \rightarrow 2$ ,  $0 \rightarrow 0$  and  $1 - \frac{1}{k} \rightarrow 1$ , so  $\mathbf{M}_k \rightarrow \mathbf{M} = \begin{pmatrix} 0 & 2 \\ 0 & 1 \end{pmatrix}$  in the sense of Definition 9.10.1. The corollary says this is convergence in *every* norm, and here one can see it directly:  $\mathbf{M}_k - \mathbf{M} = \text{diag}(\frac{1}{k}, -\frac{1}{k})$ , so

$$\|\mathbf{M}_k - \mathbf{M}\|_F = \sqrt{\frac{1}{k^2} + \frac{1}{k^2}} = \frac{\sqrt{2}}{k} \rightarrow 0,$$

and  $\|\mathbf{M}_k - \mathbf{M}\|_\infty$ , the largest absolute row sum, is  $\frac{1}{k}$ . Different norms, different constants, the same verdict — which is the content of the corollary.

So Chapter 9 §10 lost nothing by working entrywise. Its [Theorem 9.10.2](#), [Theorem 9.10.3](#), [Theorem 9.10.4](#) and [Theorem 9.10.6](#) are statements about *the* convergence of matrices, and every norm this chapter introduces — the Frobenius norm, the operator norms of §03, anything at all — measures the same thing. What the norms add is not a new notion of limit but a **rate**: a single number  $\|\mathbf{A}^k\|$  that can be bounded, compared and summed, where an entrywise statement gives  $pq$  separate sequences and no way to combine them.

One more corollary, which §03 will use to know that the operator norm exists at all.

### Corollary 15.2.5 — Closed and Bounded Is Compact, in Every Norm

Let  $V$  be finite-dimensional over  $\mathbb{R}$  or  $\mathbb{C}$ . Call a subset  $K \subseteq V$  **bounded** for a norm if  $K \subseteq \{\mathbf{v} : \|\mathbf{v}\| \leq R\}$  for some real  $R$ , and **closed** for that norm if whenever  $\|\mathbf{u}_k - \mathbf{u}\| \rightarrow 0$  with every  $\mathbf{u}_k \in K$ , also  $\mathbf{u} \in K$ . Then a subset that is closed and bounded for one norm is closed and bounded for every norm, and is **compact**: every sequence in it has a subsequence converging to a point of it. In particular, when  $V \neq \{\mathbf{0}\}$ , the unit sphere  $\{\mathbf{v} : \|\mathbf{v}\| = 1\}$  of any norm is non-empty and compact, and a continuous real-valued function on it attains a maximum and a minimum.

*Idea.* Both words in “closed and bounded” are defined by a norm, so the first thing to check is that changing the norm changes neither. That is [Theorem 15.2.2](#) again. Once the words are norm-free, the statement is [Theorem 15.2.2](#)’s own compactness step read in a coordinate-free way: carry  $K$  to  $F^n$  by coordinates, apply (A3), and carry the limit back.

*Proof.* Fix a basis and transport everything to  $F^n$  by the coordinate isomorphism ([Corollary 3.5.5](#)), as in the proof of [Theorem 15.2.2](#). Let  $\|\cdot\|_a$  and  $\|\cdot\|_b$  be norms, with  $c\|\cdot\|_b \leq \|\cdot\|_a \leq C\|\cdot\|_b$  from [Theorem 15.2.2](#). A set bounded for one is bounded for the other by the same two estimates as in [Corollary 15.2.3](#) (b). A set closed for one is closed for the other, because by [Corollary 15.2.3](#) (a) the two norms have the same convergent sequences with the same limits, and the notion of closed defined in the statement asks exactly that limits of convergent sequences stay in the set.

For compactness, let  $K \subseteq F^n$  be closed and bounded for some norm; by the previous paragraph it is closed and bounded for  $\|\cdot\|_2$ . By **the compactness of closed bounded sets in  $F^n$ , fact (A3) of the chapter introduction**, every sequence in  $K$  has a subsequence converging in  $\|\cdot\|_2$  to a point of  $K$ , and by [Corollary 15.2.3](#) (a) that subsequence converges in every norm to the same point. The unit sphere of a norm  $N$  is bounded, and it is closed because  $|N(\mathbf{v}) - N(\mathbf{u}_k)| \leq N(\mathbf{v} - \mathbf{u}_k)$  by [Lemma 15.1.1](#), so a limit of vectors with  $N = 1$  has  $N = 1$ ; it is non-empty when  $V \neq \{\mathbf{0}\}$ , since  $N(\mathbf{v})^{-1}\mathbf{v}$  lies on it for any  $\mathbf{v} \neq \mathbf{0}$ . The last sentence is then **the extreme value theorem, fact (A4)**, applied to that non-empty compact set. This proves the corollary. ■

## 15.2.4 Two warnings, and where the theorem stops

The theorem is not permission to forget which norm is in use. Here is the first reason.

*Warning. The constants depend on the dimension, and grow.* Example (The three  $p$ -norms, with best constants) gives  $\|\mathbf{x}\|_1 \leq n\|\mathbf{x}\|_\infty$  on  $F^n$ , and Exercise (C1: Sharpness of the comparison constants) shows the constant  $\sqrt{n}$  in  $\|\mathbf{x}\|_1 \leq \sqrt{n}\|\mathbf{x}\|_2$  is attained, hence best possible. So a statement such as “the error is at most  $10^{-6}$  in the  $\infty$ -norm” translates into the 1-norm as “at most  $n \cdot 10^{-6}$ ”, which is a different statement when  $n$  is a million. Theorem 15.2.2 says the two norms cannot disagree about **whether** something tends to zero; it says nothing about **how fast**, and every quantitative estimate in the rest of this chapter names its norm for that reason.

The second reason is that the theorem is false without the finite dimension, and not by a narrow margin.

### Example. (Two inequivalent norms on a space of polynomials)

Let  $V = \mathbb{R}[x]$ , the real polynomials — equivalently, the sequences of real coefficients with only finitely many non-zero terms. For  $p = a_0 + a_1x + \cdots + a_dx^d$  put

$$\|p\|_1 = |a_0| + \cdots + |a_d|, \quad \|p\|_\infty = \max_i |a_i|.$$

Both are norms, by the same coordinatewise checks as in Example (The 1-, 2- and  $\infty$ -norms on  $F^n$ ); the only difference is that the number of coordinates is unbounded. One inequality survives:  $\|p\|_\infty \leq \|p\|_1$  always. The other fails completely. Put

$$p_k = 1 + x + x^2 + \cdots + x^{k-1}.$$

Then  $\|p_k\|_1 = k$  while  $\|p_k\|_\infty = 1$ , so a constant  $C$  with  $\|p\|_1 \leq C\|p\|_\infty$  for all  $p$  would have to satisfy  $C \geq k$  for every  $k$ . No such  $C$  exists, and the two norms are not equivalent.

Look at what the sequence  $(\frac{1}{k}p_k)$  does: it tends to 0 in  $\|\cdot\|_\infty$ , since  $\|\frac{1}{k}p_k\|_\infty = 1/k$ , while  $\|\frac{1}{k}p_k\|_1 = 1$  for every  $k$ . So the two norms disagree about which sequences converge — not about the rate, about the fact. Everything in Corollary 15.2.3 and Corollary 15.2.5 fails here at once, and it fails because Step 3 of the proof of Theorem 15.2.2 has nothing to stand on. The general reason is recorded, and not proved, in the warning below.

*Warning. Infinite dimension is outside the scope of this book.* The closed unit ball of a normed space is compact **only** when the space is finite-dimensional. That is a genuine theorem, due to Riesz, and it is **recorded here, not proved**: it lies outside the four imported facts of the chapter introduction, this book never uses it as a step in an argument, and it is stated only so that Example (Two inequivalent norms on a space of polynomials) is not mistaken for an accident of a badly chosen pair. Every result of this chapter carries the hypothesis  $\dim V < \infty$ , and on  $C[0, 1]$  or  $\mathbb{R}[x]$  none of them is available. The study of norms on infinite-dimensional spaces is functional analysis, and there the choice of norm is the main content rather than an irrelevance.

*Quick check.* On  $C[0, 1]$ , compare the sup norm  $\|f\|_\infty$  of Example (Inner products, weights, and a function space) (c) with  $\|f\|_1 = \int_0^1 |f|$ , which is a norm because a contin-

uous  $f \geq 0$  with  $\int_0^1 f = 0$  is the zero function — the one fact from analysis that Chapter 10 §01 recorded when it put an inner product on  $C[0, 1]$ , quoted again here. Which of the two inequalities of [Definition 15.2.1](#) holds, and which fails?

*Solution.* The inequality  $\|f\|_1 \leq \|f\|_\infty$  holds, since  $\int_0^1 |f| \leq \int_0^1 \|f\|_\infty = \|f\|_\infty$ . The other fails. Take  $f_k(t) = \max(0, 1 - kt)$  for  $k \geq 1$ , the spike that drops from 1 to 0 on  $[0, 1/k]$ . Then  $\|f_k\|_\infty = f_k(0) = 1$ , while

$$\|f_k\|_1 = \int_0^{1/k} (1 - kt) dt = \frac{1}{2k},$$

the area of a triangle of base  $1/k$  and height 1, so no integration theory is needed to evaluate it. A constant  $C$  with  $\|f\|_\infty \leq C\|f\|_1$  would need  $C \geq 2k$  for every  $k$ . A tall thin spike has small integral and large maximum, and in infinite dimension nothing stops it from being arbitrarily thin.

With the choice of norm made harmless in finite dimension, the next section can measure matrices without apologizing for which measurement it uses — and then, when the estimates start to matter, name the norm on purpose.

## 15.2.5 Exercises

### 15.2.5.1 A. Check your understanding

#### Exercise A1.

- State [Definition 15.2.1](#), and explain why the phrase “not depending on the vector” cannot be dropped.
- State [Theorem 15.2.2](#), including its hypothesis on  $V$ .
- Which two of the four imported analysis facts does the proof of [Theorem 15.2.2](#) use, and at which step is each used?
- True or false: if two norms on  $\mathbb{R}^n$  are equivalent, then some sequence converges in one and not the other. Justify your answer.
- True or false: equivalent norms give the same value to every vector. Justify your answer.

*Solution.*

- Two norms  $\|\cdot\|_a, \|\cdot\|_b$  on  $V$  are equivalent if there are constants  $c, C > 0$ , independent of the vector, with  $c\|\mathbf{v}\|_b \leq \|\mathbf{v}\|_a \leq C\|\mathbf{v}\|_b$  for every  $\mathbf{v} \in V$  ([Definition 15.2.1](#)). Without the phrase the condition is vacuous: for each  $\mathbf{v} \neq \mathbf{0}$  the number  $\|\mathbf{v}\|_a/\|\mathbf{v}\|_b$  is positive and could be taken as both  $c$  and  $C$  at that one vector, so every pair of norms would qualify.
- On a finite-dimensional vector space over  $\mathbb{R}$  or  $\mathbb{C}$ , any two norms are equivalent ([Theorem 15.2.2](#)). Finite dimension is essential — see [Example \(Two inequivalent norms on a space of polynomials\)](#).
- Fact (A3), the compactness of closed bounded sets in  $F^n$ , at Step 3, to know the Euclidean unit sphere is compact; and fact (A4), the extreme value theorem, also



at Step 3, to know that the continuous function  $f$  attains a minimum there rather than merely approaching an infimum.

- (d) False, and this is the content of [Corollary 15.2.3](#) (a): equivalent norms have exactly the same convergent sequences with exactly the same limits.
- (e) False.  $\|\cdot\|_1$  and  $\|\cdot\|_\infty$  on  $\mathbb{R}^2$  are equivalent by [Example \(The three  \$p\$ -norms, with best constants\)](#), yet  $\|\mathbf{1}\|_1 = 2 \neq 1 = \|\mathbf{1}\|_\infty$ . Equivalence constrains ratios, not values.

### 15.2.5.2 B. Practice

#### Exercise B1: Best constants for a weighted norm.

On  $\mathbb{R}^2$  let  $N(\mathbf{x}) = (x_1^2 + 4x_2^2)^{1/2}$ . Find the **best** constants  $c, C > 0$  with

$$c \|\mathbf{x}\|_\infty \leq N(\mathbf{x}) \leq C \|\mathbf{x}\|_\infty \quad (\mathbf{x} \in \mathbb{R}^2),$$

and give a vector attaining each. Hence deduce that  $N$  and  $\|\cdot\|_\infty$  are equivalent.

*Solution.* Write  $a = |x_1|$ ,  $b = |x_2|$  and  $M = \max(a, b) = \|\mathbf{x}\|_\infty$ . Then  $N(\mathbf{x})^2 = a^2 + 4b^2$ .

Upper:  $a^2 + 4b^2 \leq M^2 + 4M^2 = 5M^2$ , so  $N(\mathbf{x}) \leq \sqrt{5} \|\mathbf{x}\|_\infty$ . Equality needs  $a = b = M$ , and  $\mathbf{x} = (1, 1)$  gives  $N(\mathbf{x}) = \sqrt{5}$  and  $\|\mathbf{x}\|_\infty = 1$ . So  $C = \sqrt{5}$  and it cannot be lowered.

Lower: if  $M = a$  then  $a^2 + 4b^2 \geq a^2 = M^2$ , and if  $M = b$  then  $a^2 + 4b^2 \geq 4b^2 \geq M^2$ . Either way  $N(\mathbf{x}) \geq \|\mathbf{x}\|_\infty$ , so  $c = 1$  works, and  $\mathbf{x} = \mathbf{e}_1$  gives  $N(\mathbf{x}) = \|\mathbf{x}\|_\infty = 1$ , so it cannot be raised.

Since 1 and  $\sqrt{5}$  are positive constants independent of  $\mathbf{x}$ , the two norms are equivalent by [Definition 15.2.1](#).

#### Exercise B2: Reading a convergence off the entries.

In  $M_2(\mathbb{R})$  let

$$\mathbf{M}_k = \begin{pmatrix} 1 + \frac{1}{k} & \frac{(-1)^k}{k^2} \\ \frac{1}{2^k} & 3 - \frac{5}{k} \end{pmatrix}.$$

Determine  $\lim_k \mathbf{M}_k$  in the sense of [Definition 9.10.1](#), and compute  $\|\mathbf{M}_k - \mathbf{M}\|_F$  explicitly. Hence confirm [Corollary 15.2.4](#) for this sequence and the Frobenius norm.

*Solution.* Each entry converges:  $1 + 1/k \rightarrow 1$ ,  $(-1)^k/k^2 \rightarrow 0$ ,  $2^{-k} \rightarrow 0$ ,  $3 - 5/k \rightarrow 3$ . So  $\mathbf{M}_k \rightarrow \mathbf{M} = \begin{pmatrix} 1 & 0 \\ 0 & 3 \end{pmatrix}$  entrywise. Then

$$\|\mathbf{M}_k - \mathbf{M}\|_F = \left( \frac{1}{k^2} + \frac{1}{k^4} + \frac{1}{4^k} + \frac{25}{k^2} \right)^{1/2},$$

which is at most  $(28/k^2)^{1/2} = \sqrt{28}/k$  for  $k \geq 1$ , since  $1/k^4 \leq 1/k^2$  and  $4^{-k} \leq 1/k^2$  for  $k \geq 1$ . Hence  $\|\mathbf{M}_k - \mathbf{M}\|_F \rightarrow 0$ , as [Corollary 15.2.4](#) predicts. Note what the norm added: a single sequence  $\sqrt{28}/k$  bounding the whole matrix, where the entrywise statement gave four unrelated sequences.



**Exercise B3: An equivalence with the constants written down.**

Let  $V = \mathbb{R}[x]_{\leq 2}$  with basis  $\mathcal{B} = (1, x, x^2)$ . For  $p \in V$  let  $\|p\|_{\mathcal{B}}$  be the Euclidean norm of the coefficient vector, and let  $N(p) = |p(0)| + |p(1)| + |p(-1)|$ .

- (a) Prove that  $N$  is a norm on  $V$ .
- (b) Find an explicit  $M$  with  $N(p) \leq M\|p\|_{\mathcal{B}}$ , following Step 1 of the proof of [Theorem 15.2.2](#).
- (c) Explain, without computing it, why a constant  $m > 0$  with  $N(p) \geq m\|p\|_{\mathcal{B}}$  exists.

*Solution.*

- (a) (N2) and (N3) are inherited pointwise from  $|\cdot|$  on  $\mathbb{R}$ , exactly as for  $\|\cdot\|_1$  in [Example \(The 1-, 2- and  \$\infty\$ -norms on  \$F^n\$ \)](#). For (N1):  $N(p) = 0$  forces  $p(0) = p(1) = p(-1) = 0$ , so  $p$  has three distinct roots while  $\deg p \leq 2$ , hence  $p = 0$ . This is the only clause with any content, and it is where  $\dim V = 3$  and the three sample points match up.
- (b) Step 1 takes  $M = (\sum_i N(\mathbf{v}_i)^2)^{1/2}$  over the basis. Here  $N(1) = 1 + 1 + 1 = 3$ ,  $N(x) = 0 + 1 + 1 = 2$  and  $N(x^2) = 0 + 1 + 1 = 2$ , so

$$M = (9 + 4 + 4)^{1/2} = \sqrt{17}.$$

- (c)  $V$  is finite-dimensional, so [Theorem 15.2.2](#) applies to the pair  $N, \|\cdot\|_{\mathcal{B}}$  and delivers both constants at once. Only the upper one is computable by hand from the basis; the lower one comes from compactness and the proof gives no formula for it.

**15.2.5.3 C. Going deeper****Exercise C1: Every linear map out of a finite-dimensional space is continuous.**

Let  $V$  be finite-dimensional over  $F$  with norm  $\|\cdot\|_V$ , let  $W$  be a normed space over  $F$  with norm  $\|\cdot\|_W$ , and let  $T: V \rightarrow W$  be linear. Fix a basis  $\mathcal{B} = (\mathbf{v}_1, \dots, \mathbf{v}_n)$  of  $V$  and write  $\|\cdot\|_1$  for the 1-norm of the coordinate vector in the basis  $\mathcal{B}$ .

- (a) Prove that  $\|T\mathbf{v}\|_W \leq K\|\mathbf{v}\|_1$  for every  $\mathbf{v} \in V$ , where  $K = \max_i \|T\mathbf{v}_i\|_W$ .
- (b) Hence prove that there is a constant  $L$  with  $\|T\mathbf{u} - T\mathbf{v}\|_W \leq L\|\mathbf{u} - \mathbf{v}\|_V$  for all  $\mathbf{u}, \mathbf{v} \in V$ .
- (c) Explain why (b) may fail when  $V$  is infinite-dimensional, using [Example \(Two inequivalent norms on a space of polynomials\)](#).

*Solution.*

- (a) Write  $\mathbf{v} = \sum_i x_i \mathbf{v}_i$ . By linearity of  $T$ , then (N3) and (N2) in  $W$ ,

$$\|T\mathbf{v}\|_W = \left\| \sum_i x_i T\mathbf{v}_i \right\|_W \leq \sum_i |x_i| \|T\mathbf{v}_i\|_W \leq K\|\mathbf{v}\|_1.$$

- (b) The function  $\|\cdot\|_1$  is a norm on  $V$  (it is  $\|\cdot\|_1$  of [Example \(The 1-, 2- and  \$\infty\$ -norms on  \$F^n\$ \)](#) transported by the coordinate isomorphism [Corollary 3.5.5](#)), and  $V$  is finite-

dimensional, so by [Theorem 15.2.2](#) there is  $C > 0$  with  $\|\mathbf{v}\|_1 \leq C\|\mathbf{v}\|_V$  for all  $\mathbf{v}$ . Put  $L = KC$ . By linearity,  $T\mathbf{u} - T\mathbf{v} = T(\mathbf{u} - \mathbf{v})$ , so (a) gives

$$\|T\mathbf{u} - T\mathbf{v}\|_W \leq K\|\mathbf{u} - \mathbf{v}\|_1 \leq L\|\mathbf{u} - \mathbf{v}\|_V.$$

- (c) Part (a) needs a **finite** basis expansion, and part (b) needs [Theorem 15.2.2](#), which is false in infinite dimension. Concretely, take  $V = \mathbb{R}[x]$  with the coefficient norm  $\|\cdot\|_\infty$  of [Example \(Two inequivalent norms on a space of polynomials\)](#),  $W = \mathbb{R}$  with  $|\cdot|$ , and the linear functional  $T(p) = p(1)$ , which adds up the coefficients. With  $p_k = 1 + x + \cdots + x^{k-1}$  as there, the polynomials  $q_k = \frac{1}{k}p_k$  satisfy  $\|q_k\|_\infty = 1/k \rightarrow 0$  while  $T(q_k) = 1$  for every  $k$ . So  $T$  sends a sequence tending to 0 to a sequence that does not, and no constant  $L$  can exist.

### Exercise C2: No dimension-free constants.

For each  $n \geq 1$  let  $c_n, C_n > 0$  be the best constants with

$$c_n\|\mathbf{x}\|_\infty \leq \|\mathbf{x}\|_1 \leq C_n\|\mathbf{x}\|_\infty \quad (\mathbf{x} \in F^n).$$

- Determine  $c_n$  and  $C_n$ , with a vector attaining each.
- Deduce that there is no pair of constants  $c, C > 0$  working simultaneously for every  $n$ .
- A statement about  $\mathbb{R}^n$  says “the error  $\mathbf{e}$  satisfies  $\|\mathbf{e}\|_\infty \leq 10^{-6}$ ”. What is the strongest bound on  $\|\mathbf{e}\|_1$  that follows, and for which  $n$  does it stop being informative if the entries of  $\mathbf{e}$  are themselves of size about  $10^{-6}$ ?

*Solution.*

- $c_n = 1$  and  $C_n = n$ . Indeed  $\|\mathbf{x}\|_\infty = |x_k|$  for some  $k$ , and  $|x_k| \leq \sum_i |x_i| = \|\mathbf{x}\|_1$ , with equality at  $\mathbf{x} = \mathbf{e}_1$ ; and  $\|\mathbf{x}\|_1 = \sum_i |x_i| \leq n \max_i |x_i| = n\|\mathbf{x}\|_\infty$ , with equality at  $\mathbf{x} = \mathbf{1}$ . Both constants are therefore attained and cannot be improved.
- A dimension-free  $C$  would need  $C \geq C_n = n$  for every  $n$ , which is impossible. (The constant  $c = 1$  does happen to work for all  $n$ ; it is the upper constant that blows up, and one blown-up constant is enough.)
- The strongest bound is  $\|\mathbf{e}\|_1 \leq n \cdot 10^{-6}$ , by (a), and it is attained when all entries have modulus  $10^{-6}$  — which is exactly the situation described. So for  $n = 10^6$  the bound reads  $\|\mathbf{e}\|_1 \leq 1$ , which says nothing useful about a quantity that is itself of order 1. The two norms agree that the error tends to 0 as the computation improves; they disagree, by a factor of  $n$ , about whether it is currently small.

### Exercise C3: A finite-dimensional subspace is closed.

Let  $W$  be a normed space over  $F$  with norm  $\|\cdot\|$ , and let  $U \subseteq W$  be a **finite-dimensional** subspace. Let  $(\mathbf{u}_k)$  be a sequence in  $U$  with  $\|\mathbf{u}_k - \mathbf{w}\| \rightarrow 0$  for some  $\mathbf{w} \in W$ .

- Prove that  $(\mathbf{u}_k)$  is bounded in  $U$  for the restricted norm.
- Using [Corollary 15.2.5](#), produce a subsequence  $(\mathbf{u}_{k_j})$  converging to some  $\mathbf{u} \in U$ .
- Deduce that  $\mathbf{w} = \mathbf{u}$ , so that  $U$  contains the limit of every convergent sequence of its own elements. *Hint: the triangle inequality, through  $\mathbf{u}_{k_j}$ .*

*Solution.*

- (a) By Lemma 15.1.1,  $|\|\mathbf{u}_k\| - \|\mathbf{w}\|| \leq \|\mathbf{u}_k - \mathbf{w}\|$ , and the right side tends to 0. So there is  $K$  with  $\|\mathbf{u}_k - \mathbf{w}\| \leq 1$  for  $k \geq K$ , whence  $\|\mathbf{u}_k\| \leq \|\mathbf{w}\| + 1$  for those  $k$ ; the finitely many earlier terms are bounded by their maximum. Hence  $(\mathbf{u}_k)$  is bounded.
- (b) The restriction of  $\|\cdot\|$  to  $U$  is a norm on  $U$ , and  $U$  is finite-dimensional. The set  $K = \{\mathbf{u} \in U : \|\mathbf{u}\| \leq R\}$ , with  $R$  the bound from (a), is closed and bounded in  $U$ , hence compact by Corollary 15.2.5. The sequence  $(\mathbf{u}_k)$  lies in  $K$ , so some subsequence  $(\mathbf{u}_{k_j})$  converges to a point  $\mathbf{u} \in K \subseteq U$ .
- (c) For every  $j$ , by (N3),

$$\|\mathbf{w} - \mathbf{u}\| \leq \|\mathbf{w} - \mathbf{u}_{k_j}\| + \|\mathbf{u}_{k_j} - \mathbf{u}\|.$$

Both terms on the right tend to 0 as  $j \rightarrow \infty$ : the first because  $(\mathbf{u}_k)$  converges to  $\mathbf{w}$  and a subsequence of a convergent sequence has the same limit, the second by (b). The left side does not depend on  $j$ , so  $\|\mathbf{w} - \mathbf{u}\| = 0$ , and (N1) gives  $\mathbf{w} = \mathbf{u} \in U$ . Hence  $U$  contains every such limit, which is what it means for  $U$  to be closed in  $W$ . Note that  $W$  itself was never assumed finite-dimensional; only  $U$  was, and that is exactly where Corollary 15.2.5 was applied.

## 15.3 Measuring a Matrix

Chapter 11 §11 gave the Frobenius norm its own symbol and explained the subscript: a bare  $\|\cdot\|$  already meant the norm of a *vector*, and a matrix, it warned, would shortly be asked to carry both kinds of measurement. This is that moment. A matrix is a list of  $mn$  numbers, and it is also a map; measuring the list is easy, measuring the map is what we usually want, and the two answers are not the same number. This section builds the second measurement, computes it in the three cases anyone computes, and says exactly where the first one fails.

**Throughout,**  $F$  is  $\mathbb{R}$  or  $\mathbb{C}$ , all spaces are finite-dimensional, and  $\|\cdot\|$  with no subscript is a norm in the sense of Definition 15.1.1.

### 15.3.1 What a matrix norm has to do

Chapter 9 §10 decided whether  $\mathbf{A}^m \rightarrow 0$  by looking at one entry of one Jordan block at a time. A single number attached to  $\mathbf{A}$  would be a far better instrument — but only if it interacts with matrix multiplication, because the questions we want to answer are questions about products:  $\mathbf{A}^m$ ,  $\sum_k \mathbf{A}^k$ ,  $e^{\mathbf{A}}$ . A norm that satisfies

$$\|\mathbf{AB}\| \leq \|\mathbf{A}\| \|\mathbf{B}\|$$

immediately gives  $\|\mathbf{A}^m\| \leq \|\mathbf{A}\|^m$ , and with that one line a geometric series is in reach. A norm that does not satisfy it gives nothing at all about powers. So the property is not a technical nicety; it is the entire reason for the definition.

*A matrix norm is a norm on matrices that never lets a product be bigger than the product of the sizes.*

**Definition 15.3.1 — Matrix Norm**

Let  $n \geq 1$ . A **matrix norm** on  $M_n(F)$  is a function  $\|\cdot\|: M_n(F) \rightarrow \mathbb{R}$  which is a norm on  $M_n(F)$  as a vector space ([Definition 15.1.1](#)) and which is in addition **submultiplicative**: for all  $\mathbf{A}, \mathbf{B} \in M_n(F)$ ,

$$\|\mathbf{AB}\| \leq \|\mathbf{A}\| \|\mathbf{B}\|.$$

In words: the first requirement is the ordinary one — clauses (N1), (N2), (N3) of [Definition 15.1.1](#) — and it sees  $M_n(F)$  only as the vector space  $F^{n^2}$  in disguise. The second requirement is the one that notices the multiplication. Note that submultiplicativity is an inequality, not an equality: it is easy for a norm to make  $\|\mathbf{AB}\|$  far smaller than  $\|\mathbf{A}\| \|\mathbf{B}\|$ , and impossible for a matrix norm to make it larger.

The condition needs a square matrix only so that  $\mathbf{AB}$  is defined for every pair; nothing else in the definition cares about the shape.

**Example. (The Frobenius norm is submultiplicative)**

Show that  $\|\mathbf{AB}\|_F \leq \|\mathbf{A}\|_F \|\mathbf{B}\|_F$  for all  $\mathbf{A}, \mathbf{B} \in M_n(F)$ , so that  $\|\cdot\|_F$  is a matrix norm.

*Solution.* The entry  $(\mathbf{AB})_{ij} = \sum_k a_{ik} b_{kj}$  is the inner product in  $F^n$  of the  $i$ -th row of  $\mathbf{A}$  with the  $j$ -th column of  $\mathbf{B}$  (up to a conjugation, which does not change moduli). By [Theorem 10.1.5](#),

$$|(\mathbf{AB})_{ij}|^2 \leq \left( \sum_k |a_{ik}|^2 \right) \left( \sum_k |b_{kj}|^2 \right).$$

Summing over all  $i$  and  $j$ , and noting that the two bracketed factors depend only on  $i$  and only on  $j$  respectively,

$$\|\mathbf{AB}\|_F^2 \leq \left( \sum_{i,k} |a_{ik}|^2 \right) \left( \sum_{k,j} |b_{kj}|^2 \right) = \|\mathbf{A}\|_F^2 \|\mathbf{B}\|_F^2.$$

Both sides are non-negative, so taking square roots gives the claim. That  $\|\cdot\|_F$  is a norm was settled in Chapter 10: it is the norm induced by the Frobenius inner product ([Definition 10.1.3](#)), so [Theorem 10.1.2](#) and [Corollary 10.1.6](#) apply.

Now the minimal change that breaks it. Keep the idea “read the matrix as a list of numbers” but measure the list with the maximum instead of the Euclidean length.

**Example. (The entrywise maximum is not submultiplicative)**

For  $\mathbf{A} \in M_n(F)$  put  $N(\mathbf{A}) = \max_{i,j} |a_{ij}|$ . Show that  $N$  is a norm on  $M_n(F)$  but not a matrix norm for  $n \geq 2$ .

*Solution.*  $N$  is the  $\infty$ -norm of the list of  $n^2$  entries, so it is a norm (Definition 15.1.1): it vanishes only on the zero matrix,  $N(c\mathbf{A}) = |c|N(\mathbf{A})$ , and (N3) holds entry by entry.

Submultiplicativity fails already for  $n = 2$ . Take  $\mathbf{A} = \mathbf{B} = \mathbf{J} = \begin{pmatrix} 1 & 1 \\ 1 & 1 \end{pmatrix}$ . Then  $N(\mathbf{J}) = 1$ , while

$$\mathbf{J}^2 = \begin{pmatrix} 2 & 2 \\ 2 & 2 \end{pmatrix}, \quad N(\mathbf{J}^2) = 2 > 1 = N(\mathbf{J})N(\mathbf{J}).$$

The mechanism is plain: an entry of a product is a sum of  $n$  terms, each as large as the largest entry, and a maximum taken entry by entry has no way of charging for the length of that sum. For  $n \times n$  the same matrix gives  $N(\mathbf{J}^2) = n$ .

*Warning.* **Submultiplicativity is a property of the norm, not of matrices.** The failure above is not caused by  $\mathbf{J}$  being special; it is caused by  $N$  being the wrong yardstick. Exercise (C2: Repairing the entrywise maximum) shows that the single rescaling  $\mathbf{A} \mapsto nN(\mathbf{A})$  repairs it, which is a good way to see that nothing deep is being violated — only a normalization.

## 15.3.2 The norm a vector norm induces

The definition above is a filter, not a construction: it says which norms are usable, not where to find one. The construction comes from remembering what  $\mathbf{A}$  is for. It sends vectors to vectors, and if both are already measured, we can ask how much  $\mathbf{A}$  magnifies.

This is exactly the question Chapter 12 §08 asked of the singular value decomposition, and the question Chapter 12 §12 asked of an oblique projection, in each case computing  $\max\{\|\mathbf{A}\mathbf{x}\| : \|\mathbf{x}\| = 1\}$  and then promising that the number would get a name here.

Before naming it, we must know that the maximum exists. “Take the largest value of a function over infinitely many vectors” is not something one may write without a reason, and the reason is compactness, already packaged for us in Corollary 15.2.5.

### Lemma 15.3.1 — The Largest Stretch Is Attained

Let  $n, m \geq 1$ , let  $\|\cdot\|$  be a norm on  $F^n$  and also a norm on  $F^m$ , and let  $\mathbf{A} \in M_{m \times n}(F)$ . Then the function  $\mathbf{x} \mapsto \|\mathbf{A}\mathbf{x}\|$  attains a maximum on the unit sphere  $S = \{\mathbf{x} \in F^n : \|\mathbf{x}\| = 1\}$ .

*Idea.* Two steps, and the second is already packaged for us. ① The function is Lipschitz, hence continuous: Lemma 15.1.1 converts a difference of norms into the norm of a difference, and a crude expansion in the standard basis bounds  $\|\mathbf{A}\mathbf{z}\|$  by a constant times  $\|\mathbf{z}\|$ . ② The unit sphere is closed and bounded, hence compact, and a continuous real function on it attains its bounds — which is exactly what Corollary 15.2.5 says.

*Proof.* **Step 1: a Lipschitz bound.** Write  $\mathbf{z} = \sum_{j=1}^n z_j \mathbf{e}_j$ . By subadditivity and homogeneity,

$$\|\mathbf{Az}\| \leq \sum_{j=1}^n |z_j| \|\mathbf{Ae}_j\| \leq \left( \max_j \|\mathbf{Ae}_j\| \right) \|\mathbf{z}\|_1.$$

By [Theorem 15.2.2](#), applied on  $F^n$  to the norms  $\|\cdot\|_1$  and  $\|\cdot\|$ , there is a constant  $c > 0$  with  $\|\mathbf{z}\|_1 \leq c\|\mathbf{z}\|$  for every  $\mathbf{z}$ . Hence  $\|\mathbf{Az}\| \leq M\|\mathbf{z}\|$  with  $M = c \max_j \|\mathbf{Ae}_j\|$ , a constant depending on  $\mathbf{A}$  but not on  $\mathbf{z}$ .

**Step 2: continuity.** Let  $\mathbf{x}, \mathbf{y} \in F^n$ . By [Lemma 15.1.1](#), applied to the vectors  $\mathbf{Ax}$  and  $\mathbf{Ay}$  of  $F^m$ , and then by Step 1,

$$\left| \|\mathbf{Ax}\| - \|\mathbf{Ay}\| \right| \leq \|\mathbf{A}(\mathbf{x} - \mathbf{y})\| \leq M\|\mathbf{x} - \mathbf{y}\|.$$

So  $\mathbf{x} \mapsto \|\mathbf{Ax}\|$  is Lipschitz with constant  $M$ , and in particular continuous.

**Step 3: compactness.** The set  $S$  is non-empty, since  $n \geq 1$  and  $\mathbf{e}_1/\|\mathbf{e}_1\| \in S$ , the division being legal because  $\mathbf{e}_1 \neq \mathbf{0}$ . By [Corollary 15.2.5](#) the unit sphere of any norm on a finite-dimensional space is compact, and a continuous real-valued function on it attains a maximum. That corollary is where this chapter's imported facts (A3), the compactness of closed bounded sets, and (A4), the extreme value theorem, are cashed in; here we only apply it, to the continuous function of Step 2. This proves the lemma. ■

*The operator norm of a matrix is the worst stretching it can do to a unit vector.*

### Definition 15.3.2 — Operator Norm

Let  $n, m \geq 1$  and let a norm  $\|\cdot\|$  be given on  $F^n$  and on  $F^m$ ; we write both with the same symbol, the argument saying which is meant. The **operator norm** of  $\mathbf{A} \in M_{m \times n}(F)$  **induced by** those norms is

$$\|\mathbf{A}\| := \max \{ \|\mathbf{Ax}\| : \mathbf{x} \in F^n, \|\mathbf{x}\| = 1 \},$$

which exists by [Lemma 15.3.1](#). When the vector norm is  $\|\cdot\|_p$  on both sides we write  $\|\mathbf{A}\|_p$ , and  $\|\mathbf{A}\|_2$  is also called the **spectral norm**.

Clause by clause: the maximum is over **unit** vectors only, so a matrix that stretches one long vector enormously is not thereby given a large norm — what counts is the *factor*. The norm on the source and the norm on the target may be different, and then  $\|\mathbf{A}\|$  depends on both; the notation hides that, which is why the vector norm must always be named. And the word **induced** is doing real work: not every matrix norm arises this way, as the end of this section shows.

Three equivalent descriptions are worth having at once. For  $\mathbf{A} \neq \mathbf{0}$ ,

$$\|\mathbf{A}\| = \max_{\|\mathbf{x}\|=1} \|\mathbf{Ax}\| = \max_{\mathbf{x} \neq \mathbf{0}} \frac{\|\mathbf{Ax}\|}{\|\mathbf{x}\|} = \max_{\|\mathbf{x}\| \leq 1} \|\mathbf{Ax}\|.$$

The middle expression equals the first because  $\|\mathbf{Ax}\|/\|\mathbf{x}\| = \|\mathbf{A}(\mathbf{x}/\|\mathbf{x}\|)\|$  by homogeneity, and  $\mathbf{x}/\|\mathbf{x}\|$  runs over  $S$  as  $\mathbf{x}$  runs over the non-zero vectors. The third equals the first because scaling any  $\mathbf{x}$  with  $0 < \|\mathbf{x}\| \leq 1$  up to the sphere only increases  $\|\mathbf{Ax}\|$ , and  $\mathbf{x} = \mathbf{0}$  contributes 0.

The smallest cases are worth seeing before any real example, because they pin the scale. For  $\mathbf{A} = \mathbf{0}$  every  $\mathbf{Ax}$  is  $\mathbf{0}$ , so  $\|\mathbf{0}\| = 0$ , as (N1) demands. For  $m = n = 1$ , a matrix is a scalar ( $a$ ), the only unit vectors are the scalars of modulus 1, and  $\|(a)\| = |a|$ : the operator norm of a number is its absolute value, so the definition extends the one-dimensional case rather than replacing it. And for  $\mathbf{A} = \mathbf{I}$ , every unit vector satisfies  $\|\mathbf{Ix}\| = \|\mathbf{x}\| = 1$ , so  $\|\mathbf{I}\| = 1$  **for every choice of vector norm** — a fact used so often below that it is worth fixing now.

*Quick check.* Why is the maximum in Definition 15.3.2 taken over  $\|\mathbf{x}\| = 1$  rather than over all  $\mathbf{x}$ ? What would the supremum over all  $\mathbf{x} \in F^n$  be?

*Solution.* For a non-zero  $\mathbf{A}$  it would be infinite: pick  $\mathbf{x}_0$  with  $\mathbf{Ax}_0 \neq \mathbf{0}$ ; then  $\|\mathbf{A}(t\mathbf{x}_0)\| = t\|\mathbf{Ax}_0\| \rightarrow \infty$  as  $t \rightarrow \infty$ . Restricting to unit vectors measures the *ratio* of output size to input size, which is what “how much does  $\mathbf{A}$  magnify?” means, and by homogeneity the ratio is already known on all of  $F^n$  once it is known on the sphere.

### Theorem 15.3.2 — Properties of the Operator Norm

Let  $\|\cdot\|$  be a norm on each of  $F^n$ ,  $F^m$ ,  $F^\ell$ , and let  $\mathbf{A} \in M_{m \times n}(F)$ . Then:

- (a)  $\|\mathbf{Ax}\| \leq \|\mathbf{A}\| \|\mathbf{x}\|$  for every  $\mathbf{x} \in F^n$ , and  $\|\mathbf{A}\|$  is the smallest constant with this property;
- (b)  $\|\cdot\|$  is a norm on the vector space  $M_{m \times n}(F)$ ;
- (c)  $\|\mathbf{I}_n\| = 1$ ;
- (d)  $\|\mathbf{AB}\| \leq \|\mathbf{A}\| \|\mathbf{B}\|$  for every  $\mathbf{B} \in M_{n \times \ell}(F)$ . In particular the operator norm induced on  $M_n(F)$  by a single norm on  $F^n$  is a matrix norm in the sense of Definition 15.3.1.

*Idea.* Everything is the definition plus homogeneity. The one move worth naming is in (a): a general  $\mathbf{x} \neq \mathbf{0}$  is  $\|\mathbf{x}\|$  times a unit vector, so the inequality on the sphere upgrades to an inequality everywhere; and once (a) is available, (d) is two applications of it.

*Proof.*

- (a) For  $\mathbf{x} = \mathbf{0}$  both sides are 0. For  $\mathbf{x} \neq \mathbf{0}$ , the vector  $\mathbf{u} = \mathbf{x}/\|\mathbf{x}\|$  satisfies  $\|\mathbf{u}\| = 1$ , so  $\|\mathbf{Au}\| \leq \|\mathbf{A}\|$  by Definition 15.3.2; multiplying by  $\|\mathbf{x}\|$  and using homogeneity gives  $\|\mathbf{Ax}\| \leq \|\mathbf{A}\| \|\mathbf{x}\|$ . If  $K$  is any constant with  $\|\mathbf{Ax}\| \leq K \|\mathbf{x}\|$  for all  $\mathbf{x}$ , then taking a unit  $\mathbf{x}_0$  at which the maximum is attained (Lemma 15.3.1) gives  $\|\mathbf{A}\| = \|\mathbf{Ax}_0\| \leq K$ .
- (b) Each value is a maximum of non-negative numbers, so  $\|\mathbf{A}\| \geq 0$ . If  $\|\mathbf{A}\| = 0$  then  $\mathbf{Au} = \mathbf{0}$  for every unit  $\mathbf{u}$ , hence  $\mathbf{Ax} = \mathbf{0}$  for every  $\mathbf{x}$  by (a), hence  $\mathbf{A} = \mathbf{0}$ . For a scalar  $c$ ,  $\|c\mathbf{A}\| = \max_{\|\mathbf{x}\|=1} |c| \|\mathbf{Ax}\| = |c| \|\mathbf{A}\|$ . For subadditivity, let  $\mathbf{x}$  be a unit vector; then  $\|(\mathbf{A} + \mathbf{B})\mathbf{x}\| \leq \|\mathbf{Ax}\| + \|\mathbf{Bx}\| \leq \|\mathbf{A}\| + \|\mathbf{B}\|$ , and taking the maximum over  $\mathbf{x}$  gives  $\|\mathbf{A} + \mathbf{B}\| \leq \|\mathbf{A}\| + \|\mathbf{B}\|$ .
- (c)  $\|\mathbf{I}_n \mathbf{x}\| = \|\mathbf{x}\| = 1$  for every unit  $\mathbf{x}$ , so the maximum is 1.
- (d) Let  $\mathbf{x} \in F^\ell$  be a unit vector. By (a) applied twice, first to  $\mathbf{A}$  and then to  $\mathbf{B}$ ,

$$\|\mathbf{A}\mathbf{B}\mathbf{x}\| \leq \|\mathbf{A}\| \|\mathbf{B}\mathbf{x}\| \leq \|\mathbf{A}\| \|\mathbf{B}\| \|\mathbf{x}\| = \|\mathbf{A}\| \|\mathbf{B}\|.$$

Taking the maximum over unit  $\mathbf{x}$  gives  $\|\mathbf{A}\mathbf{B}\| \leq \|\mathbf{A}\| \|\mathbf{B}\|$ . This proves the theorem. ■

Part (c) is small and will do a large amount of work: it is the single property that separates the operator norms from the rest of the matrix norms.

### 15.3.3 The three formulas

A definition by maximization is not yet a computation. For the three standard vector norms the maximization can be carried out completely, and in two of the three cases the answer is read straight off the entries.

#### Theorem 15.3.3 — Computing the 1-, 2- and Infinity-Norms

Let  $\mathbf{A} = (a_{ij}) \in M_{m \times n}(F)$  with  $m, n \geq 1$ . Then

- (a)  $\|\mathbf{A}\|_1 = \max_{1 \leq j \leq n} \sum_{i=1}^m |a_{ij}|$ , the largest absolute column sum;
- (b)  $\|\mathbf{A}\|_\infty = \max_{1 \leq i \leq m} \sum_{j=1}^n |a_{ij}|$ , the largest absolute row sum;
- (c)  $\|\mathbf{A}\|_2 = \sigma_1(\mathbf{A})$ , the largest singular value of  $\mathbf{A}$  (Definition 12.8.1).

*Idea.* Each part is an upper bound and a witness. ① For  $\|\cdot\|_1$  the upper bound is subadditivity of the modulus with the column sums pulled out, and the witness is a standard basis vector  $\mathbf{e}_k$  selecting the heaviest column. ② For  $\|\cdot\|_\infty$  the upper bound is the same estimate read along a row, and the witness is a vector of unimodular entries chosen to rotate every term of the heaviest row into the positive reals, so that no cancellation occurs. ③ For  $\|\cdot\|_2$  the witness is the first right singular vector, and the upper bound is the singular value decomposition: unitary factors are invisible to the Euclidean norm, so only the diagonal matrix  $\mathbf{\Sigma}$  is left, and a diagonal matrix stretches most along its largest entry, because it scales each coordinate separately.

*Proof.* Write  $C = \max_j \sum_i |a_{ij}|$  and  $R = \max_i \sum_j |a_{ij}|$ .

(a). Let  $\|\mathbf{x}\|_1 = 1$ . Then

$$\|\mathbf{A}\mathbf{x}\|_1 = \sum_i \left| \sum_j a_{ij} x_j \right| \leq \sum_j |x_j| \sum_i |a_{ij}| \leq C \sum_j |x_j| = C,$$

where the first inequality is subadditivity of the modulus together with an exchange of two finite sums. So  $\|\mathbf{A}\|_1 \leq C$ . For the reverse, choose  $k$  with  $\sum_i |a_{ik}| = C$ . The vector  $\mathbf{e}_k$  has  $\|\mathbf{e}_k\|_1 = 1$  and  $\mathbf{A}\mathbf{e}_k$  is the  $k$ -th column of  $\mathbf{A}$ , so  $\|\mathbf{A}\mathbf{e}_k\|_1 = C$ . Hence  $\|\mathbf{A}\|_1 \geq C$ .

(b). Let  $\|\mathbf{x}\|_\infty = 1$ . For each  $i$ ,



$$\left| \sum_j a_{ij} x_j \right| \leq \sum_j |a_{ij}| |x_j| \leq \sum_j |a_{ij}| \leq R,$$

so  $\|\mathbf{Ax}\|_\infty \leq R$  and  $\|\mathbf{A}\|_\infty \leq R$ . For the reverse, choose  $k$  with  $\sum_j |a_{kj}| = R$  and define  $\mathbf{x}$  by  $x_j = \overline{a_{kj}}/|a_{kj}|$  when  $a_{kj} \neq 0$  and  $x_j = 1$  otherwise. Every  $x_j$  has modulus 1, so  $\|\mathbf{x}\|_\infty = 1$ ; and  $a_{kj}x_j = |a_{kj}|$  for every  $j$ , the choice being legal exactly because we checked  $a_{kj} \neq 0$  before dividing. Hence the  $k$ -th entry of  $\mathbf{Ax}$  is  $\sum_j |a_{kj}| = R$ , giving  $\|\mathbf{Ax}\|_\infty \geq R$ .

(c). Let  $\mathbf{A} = \mathbf{U}\mathbf{\Sigma}\mathbf{V}^*$  be a singular value decomposition (Theorem 12.8.1), with  $\mathbf{U}, \mathbf{V}$  unitary and  $\mathbf{\Sigma}$  carrying  $\sigma_1 \geq \cdots \geq \sigma_p \geq 0$  on its diagonal,  $p = \min(m, n)$ . Let  $\|\mathbf{x}\|_2 = 1$  and put  $\mathbf{y} = \mathbf{V}^*\mathbf{x}$ . A unitary matrix is an isometry for  $\|\cdot\|_2$  (Theorem 10.7.3, (a) and (f)), so  $\|\mathbf{y}\|_2 = 1$  and  $\|\mathbf{Ax}\|_2 = \|\mathbf{U}\mathbf{\Sigma}\mathbf{y}\|_2 = \|\mathbf{\Sigma}\mathbf{y}\|_2$ . The vector  $\mathbf{\Sigma}\mathbf{y}$  has entries  $\sigma_i y_i$  for  $i \leq p$  and zeros below, so

$$\|\mathbf{Ax}\|_2^2 = \sum_{i=1}^p \sigma_i^2 |y_i|^2 \leq \sigma_1^2 \sum_{i=1}^n |y_i|^2 = \sigma_1^2.$$

Hence  $\|\mathbf{A}\|_2 \leq \sigma_1$ . For the reverse, take  $\mathbf{x} = \mathbf{v}_1$ , the first column of  $\mathbf{V}$ ; then  $\mathbf{y} = \mathbf{V}^*\mathbf{v}_1 = \mathbf{e}_1$ , so  $\|\mathbf{Av}_1\|_2 = \|\mathbf{\Sigma}\mathbf{e}_1\|_2 = \sigma_1$ , and  $\|\mathbf{v}_1\|_2 = 1$  because the columns of a unitary matrix are orthonormal. This proves the theorem. ■

Part (c) is the promise this section was written to keep. Chapter 12 §08 computed  $\sigma_1 = \max\{\|\mathbf{Ax}\| : \|\mathbf{x}\| = 1\}$  and said that Chapter 15 would give that quantity a name, the operator norm  $\|\mathbf{A}\|_2$ , “and this computation is why  $\|\mathbf{A}\|_2 = \sigma_1$ ”. The name is Definition 15.3.2; the reason has now been separated from the name, so that the equality is a theorem about two independently defined numbers rather than a definition dressed up as one. Chapter 12 §12 measured an oblique projection by the same maximum and deferred the word twice; Example (The oblique projection, finally measured) below supplies it. And Chapter 11 §11, when it wrote  $\|\mathbf{A}\|_F$  with an obligatory subscript, explained that the bare symbol was reserved because a matrix would soon be asked to be measured both as a list and as a map: parts (a)–(c) are that second measurement, and the rest of this section is the difference between the two.

#### Example. (One matrix, four numbers)

For  $\mathbf{A} = \begin{pmatrix} 3 & 0 \\ 4 & 5 \end{pmatrix} \in M_2(\mathbb{R})$ , compute  $\|\mathbf{A}\|_1$ ,  $\|\mathbf{A}\|_\infty$ ,  $\|\mathbf{A}\|_F$  and  $\|\mathbf{A}\|_2$ .

*Solution.* The column sums of absolute values are  $3 + 4 = 7$  and  $0 + 5 = 5$ , so  $\|\mathbf{A}\|_1 = 7$  by Theorem 15.3.3 (a). The row sums are 3 and 9, so  $\|\mathbf{A}\|_\infty = 9$  by (b). The Frobenius norm is  $\sqrt{9 + 16 + 25} = 5\sqrt{2}$ .

For the spectral norm, form

$$\mathbf{A}^\top \mathbf{A} = \begin{pmatrix} 25 & 20 \\ 20 & 25 \end{pmatrix}.$$

This matrix is  $25\mathbf{I} + 20\begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}$ , whose eigenvalues are  $25 \pm 20$ , namely 45 and 5. So the singular values are  $\sigma_1 = 3\sqrt{5}$  and  $\sigma_2 = \sqrt{5}$ , and  $\|\mathbf{A}\|_2 = 3\sqrt{5} \approx 6.708$  by (c).

Two checks come for free. The product  $\sigma_1\sigma_2 = 15$  equals  $|\det \mathbf{A}| = 15$ , and  $\sigma_1^2 + \sigma_2^2 = 50 = \|\mathbf{A}\|_F^2$ . The four numbers 7, 9,  $5\sqrt{2} \approx 7.071$  and  $3\sqrt{5} \approx 6.708$  are all different, which is the point: “the size of  $\mathbf{A}$ ” is not a well-posed phrase until the norm is named.

### Example. (The oblique projection, finally measured)

Chapter 12 §12 studied a projection  $P$  of  $F^2$  onto a line, with matrix  $\begin{pmatrix} 1 & x \\ 0 & 0 \end{pmatrix}$  in a suitable orthonormal basis, and found that the largest value of  $\|P\mathbf{v}\|$  over unit  $\mathbf{v}$  is  $\sqrt{1 + |x|^2}$ , which over  $\mathbb{R}$  is  $1/\sin \theta$  for the angle  $\theta$  between image and kernel. Confirm this with [Theorem 15.3.3](#) (c) for  $x = -1$ , the matrix  $\mathbf{P}' = \begin{pmatrix} 1 & -1 \\ 0 & 0 \end{pmatrix}$  of [Example \(Two Oblique Projections, Measured\)](#).

*Solution.*  $(\mathbf{P}')^\top \mathbf{P}' = \begin{pmatrix} 1 & -1 \\ -1 & 1 \end{pmatrix}$ , with eigenvalues 2 and 0; so  $\sigma_1 = \sqrt{2}$  and  $\|\mathbf{P}'\|_2 = \sqrt{2}$ . This agrees with  $\sqrt{1 + |-1|^2} = \sqrt{2}$ , and with  $1/\sin \theta$  for  $\theta = \pi/4$ , the angle between the two lines. An orthogonal projection, by contrast, has  $\|P_U\|_2 = 1$  whenever  $U \neq \{\mathbf{0}\}$ , since it fixes every unit vector of  $U$  and shortens the rest.

The two matrix norms now in play compare in a way worth recording, because the Frobenius norm is far cheaper to compute and the spectral norm is usually the one wanted.

### Proposition 15.3.4 — Frobenius Against Spectral

Let  $\mathbf{A} \in M_{m \times n}(F)$  be non-zero, of rank  $r$ . Then

$$\|\mathbf{A}\|_2 \leq \|\mathbf{A}\|_F \leq \sqrt{r} \|\mathbf{A}\|_2,$$

and the left inequality is an equality if and only if  $r = 1$ .

*Idea.* Write both norms in terms of the singular values:  $\|\mathbf{A}\|_2 = \sigma_1$  and  $\|\mathbf{A}\|_F^2 = \sigma_1^2 + \cdots + \sigma_r^2$ . The claim is then the statement that one term of a sum of  $r$  non-negative numbers is at most the sum, which is at most  $r$  times the largest — and that the first is an equality exactly when there is only one term.

*Proof.* By [Lemma 11.11.1](#) and  $\mathbf{A} = \mathbf{U}\Sigma\mathbf{V}^*$  ([Theorem 12.8.1](#)),  $\|\mathbf{A}\|_F = \|\Sigma\|_F$ , so  $\|\mathbf{A}\|_F^2 = \sum_{i=1}^p \sigma_i^2$ . Exactly  $r$  of the  $\sigma_i$  are non-zero ([Theorem 12.8.1](#)), and each satisfies  $\sigma_i \leq \sigma_1$ . Hence

$$\sigma_1^2 \leq \sum_{i=1}^p \sigma_i^2 \leq r\sigma_1^2,$$

which is the displayed chain after taking square roots and using  $\|\mathbf{A}\|_2 = \sigma_1$  ([Theorem 15.3.3](#) (c)). The left inequality is an equality exactly when  $\sigma_i = 0$  for all  $i \geq 2$ , that

is, when  $r = 1$ .

■

So the two norms never differ by a factor worse than  $\sqrt{r} \leq \sqrt{\min(m, n)}$ , and they agree exactly on the rank-one matrices. What they do *not* agree on is more interesting.

### 15.3.4 What submultiplicativity does not buy

The Frobenius norm is a matrix norm: [Example \(The Frobenius norm is submultiplicative\)](#) proved it submultiplicative. It is tempting to conclude that it is therefore the operator norm induced by some vector norm — perhaps a cleverly chosen one. It is not, and the obstruction is a single line.

*Warning.* **Submultiplicative does not mean induced.** Every operator norm satisfies  $\|\mathbf{I}_n\| = 1$  by [Theorem 15.3.2](#) (c), whereas

$$\|\mathbf{I}_n\|_F = \sqrt{\underbrace{1 + 1 + \cdots + 1}_n} = \sqrt{n}.$$

So for  $n \geq 2$  the Frobenius norm is **not** induced by any norm on  $F^n$  whatsoever — not by a bad choice, but by no choice at all. It is a matrix norm and not an operator norm, and the two phrases must be kept apart.

The identity is the cheapest witness, and it is decisive because  $\|\mathbf{I}\| = 1$  is forced: the identity stretches nothing, so a norm that measures stretching must give it 1. A norm that assigns the identity the value  $\sqrt{n}$  is measuring something else — the amount of material in the matrix, which for  $\mathbf{I}_n$  grows with  $n$ .

*Warning.* **Two different objects are called  $\|\mathbf{A}\|_\infty$  in the wild.** In this book  $\|\mathbf{A}\|_\infty$  is the operator norm induced by  $\|\cdot\|_\infty$  on vectors, that is, the largest row sum. The entrywise maximum  $N(\mathbf{A}) = \max_{i,j} |a_{ij}|$  of [Example \(The entrywise maximum is not submultiplicative\)](#) is a different function, and it is not even submultiplicative. Whenever a source writes  $\|\mathbf{A}\|_\infty$ , check which one is meant by evaluating it at  $\mathbf{I}_2$ : the row-sum norm gives 1, the entrywise maximum gives 1 as well — so evaluate instead at  $\mathbf{J}$ , where the row-sum norm gives 2 and the entrywise maximum gives 1.

*Quick check.* The Frobenius norm satisfies  $\|\mathbf{A}\mathbf{x}\|_2 \leq \|\mathbf{A}\|_F \|\mathbf{x}\|_2$  for every  $\mathbf{x}$ , by [Proposition 15.3.4](#) and [Theorem 15.3.2](#) (a). Does that make it an operator norm?

*Solution.* No. Being *compatible* with a vector norm — satisfying  $\|\mathbf{A}\mathbf{x}\| \leq \|\mathbf{A}\| \|\mathbf{x}\|$  — is weaker than being induced by it. The induced norm is the **smallest** constant that works ([Theorem 15.3.2](#) (a)), and for  $\mathbf{A} = \mathbf{I}_n$  with  $n \geq 2$  the smallest constant is 1 while  $\|\mathbf{I}_n\|_F = \sqrt{n}$ . Compatibility says the number is big enough; being induced says it is exactly right.

Where this leaves us: we have a supply of matrix norms ([Definition 15.3.1](#)), among them a distinguished family, the induced ones ([Definition 15.3.2](#)), and inside that family three that are computable ([Theorem 15.3.3](#)). The next section asks which quantity all of these norms are

simultaneously bounding from above, and finds that the answer is the spectral radius.

## 15.3.5 Exercises

### 15.3.5.1 A. Check your understanding

#### Exercise A1.

- Define a **matrix norm** on  $M_n(F)$ , and define the **operator norm** induced by a vector norm.
- State the three formulas of [Theorem 15.3.3](#).
- Which analysis facts make [Definition 15.3.2](#) legitimate, and where is each used?
- Determine whether the following statement is correct, and justify your answer: every matrix norm on  $M_n(\mathbb{R})$  is the operator norm induced by some norm on  $\mathbb{R}^n$ .
- Determine whether the following statement is correct, and justify your answer: for every  $\mathbf{A} \in M_n(F)$  and every operator norm,  $\|\mathbf{A}^2\| = \|\mathbf{A}\|^2$ .

*Solution.*

- A matrix norm is a norm on the vector space  $M_n(F)$  that also satisfies  $\|\mathbf{AB}\| \leq \|\mathbf{A}\|\|\mathbf{B}\|$  for all  $\mathbf{A}, \mathbf{B}$  ([Definition 15.3.1](#)). Given norms on  $F^n$  and  $F^m$ , the operator norm of  $\mathbf{A} \in M_{m \times n}(F)$  is  $\|\mathbf{A}\| = \max\{\|\mathbf{Ax}\| : \|\mathbf{x}\| = 1\}$  ([Definition 15.3.2](#)).
- $\|\mathbf{A}\|_1$  is the largest absolute column sum,  $\|\mathbf{A}\|_\infty$  the largest absolute row sum, and  $\|\mathbf{A}\|_2 = \sigma_1(\mathbf{A})$ .
- Two: the compactness of closed bounded sets, fact (A3) of the chapter introduction, used for the unit sphere  $S$ ; and the extreme value theorem, fact (A4), used for  $\mathbf{x} \mapsto \|\mathbf{Ax}\|$ . Both enter through [Corollary 15.2.5](#), quoted in Step 3 of [Lemma 15.3.1](#). The continuity needed for the second is proved, not imported, and rests on [Theorem 15.2.2](#) and [Lemma 15.1.1](#).
- Incorrect. The Frobenius norm is a matrix norm, and  $\|\mathbf{I}_n\|_F = \sqrt{n} \neq 1$  for  $n \geq 2$ , whereas every operator norm has  $\|\mathbf{I}_n\| = 1$  by [Theorem 15.3.2](#) (c).
- Incorrect; only  $\leq$  holds, by [Theorem 15.3.2](#) (d). Take  $\mathbf{A} = \begin{pmatrix} 0 & 1 \\ 0 & 0 \end{pmatrix}$ , for which  $\|\mathbf{A}\|_2 = 1$  but  $\mathbf{A}^2 = 0$ , so  $\|\mathbf{A}^2\|_2 = 0$ .

### 15.3.5.2 B. Practice

#### Exercise B1: Four numbers for one matrix.

Let  $\mathbf{A} = \begin{pmatrix} 2 & -1 \\ 2 & 2 \end{pmatrix} \in M_2(\mathbb{R})$ . Compute  $\|\mathbf{A}\|_1$ ,  $\|\mathbf{A}\|_\infty$ ,  $\|\mathbf{A}\|_F$  and  $\|\mathbf{A}\|_2$ , and verify that  $\|\mathbf{A}\|_2 \leq \|\mathbf{A}\|_F$ .

*Solution.* Absolute column sums:  $2 + 2 = 4$  and  $1 + 2 = 3$ , so  $\|\mathbf{A}\|_1 = 4$ . Absolute row sums:  $2 + 1 = 3$  and  $2 + 2 = 4$ , so  $\|\mathbf{A}\|_\infty = 4$ . Also  $\|\mathbf{A}\|_F = \sqrt{4 + 1 + 4 + 4} = \sqrt{13}$ .

For the spectral norm,

$$\mathbf{A}^\top \mathbf{A} = \begin{pmatrix} 8 & 2 \\ 2 & 5 \end{pmatrix},$$

with trace 13 and determinant  $40 - 4 = 36$ , so its eigenvalues solve  $\lambda^2 - 13\lambda + 36 = 0$ , giving  $\lambda = 9$  and  $\lambda = 4$ . Hence  $\sigma_1 = 3$ ,  $\sigma_2 = 2$  and  $\|\mathbf{A}\|_2 = 3$  by [Theorem 15.3.3](#) (c). Finally  $3 \leq \sqrt{13}$ , as [Proposition 15.3.4](#) requires; the inequality is strict because  $\text{rank } \mathbf{A} = 2$ .

### Exercise B2: A rank-one matrix.

Let  $\mathbf{u} \in F^m$  and  $\mathbf{v} \in F^n$  be non-zero and put  $\mathbf{A} = \mathbf{u}\mathbf{v}^*$ .

- (a) Prove that  $\|\mathbf{A}\|_2 = \|\mathbf{u}\|_2 \|\mathbf{v}\|_2$ .
- (b) Verify the formula for  $\mathbf{u} = (1, 2)$  and  $\mathbf{v} = (2, -1)$  in  $\mathbb{R}^2$  by computing  $\sigma_1(\mathbf{A})$  directly.

*Solution.*

- (a) For  $\mathbf{x} \in F^n$  we have  $\mathbf{A}\mathbf{x} = (\mathbf{v}^*\mathbf{x})\mathbf{u}$ , so  $\|\mathbf{A}\mathbf{x}\|_2 = |\mathbf{v}^*\mathbf{x}| \|\mathbf{u}\|_2$ . By [Theorem 10.1.5](#),  $|\mathbf{v}^*\mathbf{x}| = |\langle \mathbf{x}, \mathbf{v} \rangle| \leq \|\mathbf{x}\|_2 \|\mathbf{v}\|_2$ , with equality when  $\mathbf{x} = \mathbf{v}/\|\mathbf{v}\|_2$ . Hence for unit  $\mathbf{x}$  the quantity  $\|\mathbf{A}\mathbf{x}\|_2$  is at most  $\|\mathbf{u}\|_2 \|\mathbf{v}\|_2$  and attains that value, so  $\|\mathbf{A}\|_2 = \|\mathbf{u}\|_2 \|\mathbf{v}\|_2$ .
- (b) Here  $\mathbf{A} = \begin{pmatrix} 2 & -1 \\ 4 & -2 \end{pmatrix}$  and  $\mathbf{A}^\top \mathbf{A} = \begin{pmatrix} 20 & -10 \\ -10 & 5 \end{pmatrix}$ , of trace 25 and determinant  $100 - 100 = 0$ . So the eigenvalues are 25 and 0, giving  $\sigma_1 = 5$ . And  $\|\mathbf{u}\|_2 \|\mathbf{v}\|_2 = \sqrt{5} \cdot \sqrt{5} = 5$ , in agreement. Note also  $\|\mathbf{A}\|_F = 5 = \|\mathbf{A}\|_2$ , as [Proposition 15.3.4](#) predicts for a rank-one matrix.

### Exercise B3: The star trick.

Let  $\mathbf{A} \in M_{m \times n}(F)$ . Prove that  $\|\mathbf{A}^* \mathbf{A}\|_2 = \|\mathbf{A}\|_2^2$ . *Hint: the eigenvalues of  $\mathbf{A}^* \mathbf{A}$ .*

*Solution.* By [Definition 12.8.1](#) the eigenvalues of  $\mathbf{A}^* \mathbf{A}$  are  $\sigma_1^2 \geq \dots \geq \sigma_n^2 \geq 0$ , where the  $\sigma_i$  are the singular values of  $\mathbf{A}$  padded with zeros if  $n > m$ . The matrix  $\mathbf{A}^* \mathbf{A}$  is positive semidefinite, hence Hermitian, so by [Corollary 11.4.3](#) it is  $\mathbf{W}\mathbf{D}\mathbf{W}^*$  with  $\mathbf{W}$  unitary and  $\mathbf{D} = \text{diag}(\sigma_1^2, \dots, \sigma_n^2)$ . Then  $(\mathbf{A}^* \mathbf{A})^*(\mathbf{A}^* \mathbf{A}) = \mathbf{W}\mathbf{D}^2\mathbf{W}^*$ , whose eigenvalues are  $\sigma_i^4$ , so the singular values of  $\mathbf{A}^* \mathbf{A}$  are the  $\sigma_i^2$  and its largest is  $\sigma_1^2$ . By [Theorem 15.3.3](#) (c), applied to  $\mathbf{A}^* \mathbf{A}$  and then to  $\mathbf{A}$ ,

$$\|\mathbf{A}^* \mathbf{A}\|_2 = \sigma_1^2 = \|\mathbf{A}\|_2^2.$$

This proves the claim. Notice that it is an equality, whereas [Theorem 15.3.2](#) (d) alone would give only  $\|\mathbf{A}^* \mathbf{A}\|_2 \leq \|\mathbf{A}^*\|_2 \|\mathbf{A}\|_2$ .

### 15.3.5.3 C. Going deeper

#### Exercise C1: The spectral norm between the other two.

Let  $\mathbf{A} \in M_{m \times n}(F)$ .

- (a) Prove that  $\|\mathbf{A}\|_2 \leq \sqrt{\|\mathbf{A}\|_1 \|\mathbf{A}\|_\infty}$ .  
 (b) Check the inequality on the matrix of [Example \(One matrix, four numbers\)](#), and say how far it is from equality.

*Hint for (a): apply [Theorem 10.1.5](#) to the splitting  $|a_{ij}| = |a_{ij}|^{1/2} \cdot |a_{ij}|^{1/2} |x_j|$  inside one row.*

*Solution.*

- (a) Write  $C = \|\mathbf{A}\|_1$  and  $R = \|\mathbf{A}\|_\infty$  for the largest column and row sums ([Theorem 15.3.3](#)). Let  $\|\mathbf{x}\|_2 = 1$ . For each  $i$ , [Theorem 10.1.5](#) applied to the vectors with entries  $|a_{ij}|^{1/2}$  and  $|a_{ij}|^{1/2} |x_j|$  gives

$$\left| \sum_j a_{ij} x_j \right|^2 \leq \left( \sum_j |a_{ij}| \right) \left( \sum_j |a_{ij}| |x_j|^2 \right) \leq R \sum_j |a_{ij}| |x_j|^2.$$

Summing over  $i$  and exchanging the two finite sums,

$$\|\mathbf{Ax}\|_2^2 \leq R \sum_j |x_j|^2 \sum_i |a_{ij}| \leq RC \sum_j |x_j|^2 = RC.$$

Taking the maximum over unit  $\mathbf{x}$  gives  $\|\mathbf{A}\|_2^2 \leq RC$ , as claimed.

- (b) There  $\|\mathbf{A}\|_1 = 7$  and  $\|\mathbf{A}\|_\infty = 9$ , so the bound is  $\sqrt{63} \approx 7.937$ , while  $\|\mathbf{A}\|_2 = 3\sqrt{5} \approx 6.708$ . The bound overestimates by about 18 per cent. It is sharp in general: for  $\mathbf{A} = \mathbf{I}_n$  all three numbers are 1.

### Exercise C2: Repairing the entrywise maximum.

For  $\mathbf{A} \in M_n(F)$  let  $N(\mathbf{A}) = \max_{i,j} |a_{ij}|$ , as in [Example \(The entrywise maximum is not submultiplicative\)](#).

- (a) Prove that  $N(\mathbf{AB}) \leq nN(\mathbf{A})N(\mathbf{B})$ , and deduce that  $\mathbf{A} \mapsto nN(\mathbf{A})$  is a matrix norm.  
 (b) Deduce that  $nN(\cdot)$  is not an operator norm for any  $n \geq 2$ .  
 (c) Explain why (a) and (b) together show that submultiplicativity and  $\|\mathbf{I}\| = 1$  are independent requirements.

*Solution.*

- (a) For all  $i, j$ ,

$$|(\mathbf{AB})_{ij}| = \left| \sum_k a_{ik} b_{kj} \right| \leq \sum_k |a_{ik}| |b_{kj}| \leq n N(\mathbf{A}) N(\mathbf{B}),$$

so  $N(\mathbf{AB}) \leq nN(\mathbf{A})N(\mathbf{B})$ . Put  $\|\mathbf{A}\| = nN(\mathbf{A})$ . It is a norm, being a positive multiple of the norm  $N$ , and

$$\|\mathbf{AB}\| = nN(\mathbf{AB}) \leq n^2 N(\mathbf{A}) N(\mathbf{B}) = \|\mathbf{A}\| \|\mathbf{B}\|.$$

- (b)  $\|\mathbf{I}_n\| = nN(\mathbf{I}_n) = n \neq 1$  for  $n \geq 2$ , while every operator norm gives the identity the value 1 ([Theorem 15.3.2](#) (c)).  
 (c)  $N$  itself is a norm with  $N(\mathbf{I}_n) = 1$  that is not submultiplicative;  $nN$  is submulti-

plicative with  $nN(\mathbf{I}_n) = n$ . So neither property implies the other, and an operator norm is precisely a matrix norm that also passes the identity test — a necessary condition which, as  $\|\cdot\|_F$  already showed, is still not sufficient on its own for  $n$  large.

### Exercise C3: When is a bound an equality?.

Let  $\mathbf{A} \in M_{m \times n}(F)$  be non-zero.

- (a) Prove that  $\|\mathbf{A}\|_2 = \|\mathbf{A}\|_F$  if and only if  $\text{rank } \mathbf{A} = 1$ .
- (b) Give an example with  $m = n = 2$  where  $\|\mathbf{A}\|_F = \sqrt{2} \|\mathbf{A}\|_2$ , the worst case allowed by [Proposition 15.3.4](#).
- (c) Is there a non-zero  $\mathbf{A}$  with  $\|\mathbf{A}\|_F < \|\mathbf{A}\|_2$ ? Justify your answer.

*Solution.*

- (a) This is the equality case of [Proposition 15.3.4](#), proved there:  $\|\mathbf{A}\|_F^2 = \sum_i \sigma_i^2$  and  $\|\mathbf{A}\|_2^2 = \sigma_1^2$ , so the two agree exactly when  $\sigma_i = 0$  for every  $i \geq 2$ , which by [Theorem 12.8.1](#) says  $\text{rank } \mathbf{A} = 1$ .
- (b) Take  $\mathbf{A} = \mathbf{I}_2$ . Its singular values are 1, 1, so  $\|\mathbf{A}\|_2 = 1$  and  $\|\mathbf{A}\|_F = \sqrt{2}$ . Any matrix with two equal non-zero singular values does the same, for instance any  $2 \times 2$  unitary matrix.
- (c) No.  $\|\mathbf{A}\|_2 = \sigma_1 \leq (\sum_i \sigma_i^2)^{1/2} = \|\mathbf{A}\|_F$  always, since every term of the sum is non-negative. The Frobenius norm counts all the singular values and the spectral norm only the largest, so the Frobenius norm can never be the smaller of the two.

## 15.4 The Spectral Radius

Every criterion in Chapter 9 §10 began by looking at the largest modulus of an eigenvalue, and that section gave the number a name in passing — the spectral radius — without a numbered definition and without a single norm available to compare it with. It also left a note: “Chapter 15 measures matrices by norms, proves that the spectral radius never exceeds the norm of  $\mathbf{A}$  induced by any vector norm, and recovers  $\rho(\mathbf{A})$  as a limit built from the norms of the powers  $\mathbf{A}^m$ .” This section pays that note in full, and is careful about what is new when it does.

**Throughout, matrices are complex.** A real matrix is read inside  $M_n(\mathbb{C})$ , exactly as Chapter 9 §10 read it when it counted eigenvalues; over  $\mathbb{R}$  a matrix may have no eigenvalue at all, and then there is nothing to take the largest modulus of.

### 15.4.1 The number Chapter 9 named

*The spectral radius is the radius of the smallest disk about the origin that holds the whole spectrum.*

**Definition 15.4.1 — Spectral Radius**

Let  $\mathbf{A} \in M_n(\mathbb{C})$  with  $n \geq 1$ . The **spectral radius** of  $\mathbf{A}$  is

$$\rho(\mathbf{A}) := \max\{|\lambda| : \lambda \in \text{spec}(\mathbf{A})\}.$$

For an operator  $T$  on a non-zero finite-dimensional complex vector space,  $\rho(T)$  is defined by the same formula with  $\text{spec}(T)$  in place of  $\text{spec}(\mathbf{A})$ .

The maximum is over a **finite non-empty** set, so it exists with no analysis at all:  $\text{spec}(\mathbf{A})$  is the root set of the characteristic polynomial  $p_{\mathbf{A}}$ , which is non-empty by the Fundamental Theorem of Algebra ([Theorem 5.5.1](#)) and has at most  $n$  elements. It is the largest modulus of an eigenvalue, **not** the largest eigenvalue: over  $\mathbb{C}$  there is no order to take a largest in, and even for a real spectrum  $\rho(\text{diag}(-5, 1)) = 5$ , not 1. This is the number Chapter 9 §10 introduced in prose, now with a label so that later sections can cite it.

Some values, simplest first.

- **Diagonal or triangular.** The eigenvalues of a triangular matrix are its diagonal entries ([Theorem 8.7.4](#) (b)), so  $\rho$  is the largest modulus on the diagonal. For  $\text{diag}(d_1, \dots, d_n)$  it is  $\max_i |d_i|$ .
- **Normal.** If  $\mathbf{A}$  is normal then  $\mathbf{A} = \mathbf{U}\mathbf{D}\mathbf{U}^*$  with  $\mathbf{D} = \text{diag}(\lambda_1, \dots, \lambda_n)$  ([Corollary 11.4.3](#)), so  $\mathbf{A}^*\mathbf{A} = \mathbf{U} \text{diag}(|\lambda_i|^2) \mathbf{U}^*$  and the singular values of  $\mathbf{A}$  are the moduli  $|\lambda_i|$ . Hence  $\rho(\mathbf{A}) = \sigma_1(\mathbf{A}) = \|\mathbf{A}\|_2$  by [Theorem 15.3.3](#) (c). For a normal matrix the spectral radius is the spectral norm, which is where the second name comes from.
- **A rotation.**  $\begin{pmatrix} 0 & -1 \\ 1 & 0 \end{pmatrix}$  has eigenvalues  $\pm i$ , so  $\rho = 1$ , although no real vector is left in place.
- **Nilpotent.**  $\mathbf{J}_k(0)$  is triangular with zero diagonal, so by the first bullet  $\rho(\mathbf{J}_k(0)) = 0$ . The matrix itself is not 0 when  $k \geq 2$ .

The last entry is the degenerate case, and it is the important one: it says at once that  $\rho$  is not a norm.

*Warning.*  $\rho$  is not a norm, and it is neither subadditive nor submultiplicative.

Take

$$\mathbf{A} = \begin{pmatrix} 0 & 1 \\ 0 & 0 \end{pmatrix}, \quad \mathbf{B} = \begin{pmatrix} 0 & 0 \\ 1 & 0 \end{pmatrix}.$$

Both are nilpotent, so  $\rho(\mathbf{A}) = \rho(\mathbf{B}) = 0$  although neither is the zero matrix. Their sum  $\begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}$  has eigenvalues  $\pm 1$  and their product  $\mathbf{AB} = \begin{pmatrix} 1 & 0 \\ 0 & 0 \end{pmatrix}$  has eigenvalues 1 and 0. So  $\rho(\mathbf{A} + \mathbf{B}) = 1 > 0 = \rho(\mathbf{A}) + \rho(\mathbf{B})$  and  $\rho(\mathbf{AB}) = 1 > 0 = \rho(\mathbf{A})\rho(\mathbf{B})$ . So positive definiteness and subadditivity both fail, and so does submultiplicativity, all on the same small pair. Only absolute homogeneity survives, since  $\rho(c\mathbf{A}) = |c|\rho(\mathbf{A})$  for every scalar  $c$ .

So  $\rho$  is a genuinely different kind of measurement from the ones of the previous section. The relation between the two is the subject of the rest of this section, and it is one-sided.



## 15.4.2 Every norm dominates the spectral radius

### Theorem 15.4.1 — The Spectral Radius Is a Lower Bound for Every Norm

Let  $\mathbf{A} \in M_n(\mathbb{C})$ .

- (a) If  $\|\cdot\|$  is the operator norm on  $M_n(\mathbb{C})$  induced by a norm on  $\mathbb{C}^n$  (Definition 15.3.2), then  $\rho(\mathbf{A}) \leq \|\mathbf{A}\|$ .
- (b) More generally,  $\rho(\mathbf{A}) \leq \|\mathbf{A}\|$  for **every** matrix norm  $\|\cdot\|$  on  $M_n(\mathbb{C})$  (Definition 15.3.1).

*Idea.* For (a) there is nothing to do but write down the eigenvalue equation: an eigenvector is a vector on which  $\mathbf{A}$  acts by multiplication by  $\lambda$ , so the stretching factor  $|\lambda|$  is one of the factors the maximum in Definition 15.3.2 ranges over. For (b) there is no vector to feed the norm, only matrices — so manufacture one: fill an  $n \times n$  matrix with copies of the eigenvector, and the eigenvalue equation becomes a matrix equation that submultiplicativity can be applied to.

*Proof.*

- (a) Let  $\lambda \in \text{spec}(\mathbf{A})$  and let  $\mathbf{x} \neq \mathbf{0}$  with  $\mathbf{Ax} = \lambda\mathbf{x}$ . Then  $\|\mathbf{Ax}\| = |\lambda| \|\mathbf{x}\|$  by homogeneity, while  $\|\mathbf{Ax}\| \leq \|\mathbf{A}\| \|\mathbf{x}\|$  by Theorem 15.3.2 (a). Since  $\mathbf{x} \neq \mathbf{0}$  we have  $\|\mathbf{x}\| > 0$  and may divide, giving  $|\lambda| \leq \|\mathbf{A}\|$ . This holds for every eigenvalue, so it holds for the largest modulus.
- (b) Let  $\lambda$  and  $\mathbf{x}$  be as above, and let  $\mathbf{X} \in M_n(\mathbb{C})$  be the matrix all of whose  $n$  columns equal  $\mathbf{x}$ . Then  $\mathbf{X} \neq \mathbf{0}$ , so  $\|\mathbf{X}\| > 0$ . Computing column by column,  $\mathbf{AX}$  has every column equal to  $\mathbf{Ax} = \lambda\mathbf{x}$ , that is,

$$\mathbf{AX} = \lambda\mathbf{X}.$$

Taking norms and using homogeneity on the left and submultiplicativity on the right,

$$|\lambda| \|\mathbf{X}\| = \|\mathbf{AX}\| \leq \|\mathbf{A}\| \|\mathbf{X}\|.$$

Dividing by  $\|\mathbf{X}\| > 0$  gives  $|\lambda| \leq \|\mathbf{A}\|$ , and again the largest modulus obeys the same bound. This proves the theorem. ■

This is the inequality Chapter 9 §10 promised: the spectral radius never exceeds the norm of  $\mathbf{A}$  induced by any vector norm. Part (b) says more than was promised, and costs one extra line.

The immediate use is that every computable norm is a computable *upper bound* for every eigenvalue — and two of the three formulas of Theorem 15.3.3 need no arithmetic beyond adding up moduli.

### Corollary 15.4.2 — Cheap Bounds on the Spectrum

Let  $\mathbf{A} \in M_n(\mathbb{C})$ . Then every eigenvalue  $\lambda$  of  $\mathbf{A}$  satisfies

$$|\lambda| \leq \rho(\mathbf{A}) \leq \min\{\|\mathbf{A}\|_1, \|\mathbf{A}\|_\infty, \|\mathbf{A}\|_F\},$$

that is,  $\rho(\mathbf{A})$  is at most the largest absolute column sum, at most the largest absolute row sum, and at most  $\|\mathbf{A}\|_F$ .

*Proof.* The first two are operator norms (Theorem 15.3.3 (a), (b)), so Theorem 15.4.1 (a) applies;  $\|\cdot\|_F$  is a matrix norm (Example (The Frobenius norm is submultiplicative)), so Theorem 15.4.1 (b) applies. A minimum of upper bounds is an upper bound. ■

*Remark.* The bound  $\rho(\mathbf{A}) \leq \|\mathbf{A}\|_2 = \sigma_1(\mathbf{A})$  is the same crude inequality  $|\lambda| \leq \sigma_1$  that Chapter 12 §08 obtained directly from  $\mathbf{A}\mathbf{x} = \lambda\mathbf{x}$ . It is now one instance of a general principle rather than an isolated computation, and Theorem 15.4.1 (a) is that computation with the norm left unnamed.

#### Example. (Bounding a spectrum without finding it)

Let  $\mathbf{A} = \frac{1}{10} \begin{pmatrix} 2 & 5 & 1 \\ 3 & 1 & 4 \\ 1 & 2 & 5 \end{pmatrix}$ . Show that  $\rho(\mathbf{A}) \leq \frac{4}{5}$ , and hence that  $\mathbf{A}^m \rightarrow 0$ .

*Solution.* Every row of  $\mathbf{A}$  sums to  $\frac{8}{10} = \frac{4}{5}$ , so  $\|\mathbf{A}\|_\infty = \frac{4}{5}$  by Theorem 15.3.3 (b), and  $\rho(\mathbf{A}) \leq \frac{4}{5}$  by Corollary 15.4.2. Since  $\frac{4}{5} < 1$ , every eigenvalue has modulus less than 1 and  $\mathbf{A}^m \rightarrow 0$  by Theorem 9.10.2. No characteristic polynomial was computed.

The column sums are  $\frac{6}{10}$ ,  $\frac{8}{10}$  and 1, so  $\|\mathbf{A}\|_1 = 1$  and the column bound gives nothing. It is worth computing both and keeping the smaller. In fact  $\rho(\mathbf{A}) = \frac{4}{5}$  exactly, because the constant row sums make  $\mathbf{1}$  an eigenvector with eigenvalue  $\frac{4}{5}$ , so the row bound here is attained.

*Quick check.* Theorem 15.4.1 says  $\rho(\mathbf{A}) \leq \|\mathbf{A}\|$  for every matrix norm. Why does it not follow that  $\rho$  is the smallest matrix norm?

*Solution.* Because  $\rho$  is not a matrix norm at all:  $\rho$  vanishes on every non-zero nilpotent matrix, and it fails both subadditivity and submultiplicativity, as the warning above showed. The correct statement is that  $\rho(\mathbf{A})$  is a lower bound for the *values*  $\|\mathbf{A}\|$ , one matrix at a time, and the next theorem shows it is the greatest such lower bound.

### 15.4.3 A norm as close to the spectral radius as we like

[Theorem 15.4.1](#) leaves open how large the gap can be. It can be enormous for a fixed norm:  $\mathbf{J}_2(0)$  has  $\rho = 0$  and  $\|\mathbf{J}_2(0)\|_2 = 1$ , and multiplying the corner entry by  $10^6$  leaves  $\rho = 0$  while the norm becomes  $10^6$ . But the norm may be chosen after the matrix, and then the gap can be made as small as we please.

#### Theorem 15.4.3 — Norms Close to the Spectral Radius

Let  $\mathbf{A} \in M_n(\mathbb{C})$  and let  $\varepsilon > 0$ . Then there is a norm  $\|\cdot\|_t$  on  $\mathbb{C}^n$  whose induced operator norm satisfies

$$\rho(\mathbf{A}) \leq \|\mathbf{A}\|_t \leq \rho(\mathbf{A}) + \varepsilon.$$

*Idea.* Schur-triangularize and then squash. ① By [Corollary 11.1.3](#),  $\mathbf{A} = \mathbf{U}\mathbf{T}\mathbf{U}^*$  with  $\mathbf{T}$  upper triangular carrying the eigenvalues on its diagonal; the row sums of  $\mathbf{T}$  are the diagonal entry plus the off-diagonal junk. ② Conjugating  $\mathbf{T}$  by  $\mathbf{D}_t = \text{diag}(1, t, \dots, t^{n-1})$  multiplies the entry in position  $(i, j)$  by  $t^{i-j}$ , which leaves the diagonal alone and, for large  $t$ , shrinks everything strictly above it, because  $i - j < 0$  there. ③ Choose  $t$  so that every off-diagonal row sum is below  $\varepsilon$ ; the largest row sum is then below  $\rho(\mathbf{A}) + \varepsilon$ , and the largest row sum is a norm we can compute, by [Theorem 15.3.3](#) (b). ④ Transport that norm back to  $\mathbb{C}^n$  along the invertible matrix  $\mathbf{D}_t\mathbf{U}^*$ .

*Proof.* By [Corollary 11.1.3](#) there are a unitary  $\mathbf{U}$  and an upper triangular  $\mathbf{T} = (t_{ij})$  with  $\mathbf{A} = \mathbf{U}\mathbf{T}\mathbf{U}^*$  and  $t_{ii} = \lambda_i$ , the eigenvalues of  $\mathbf{A}$ . For  $t \geq 1$  put  $\mathbf{D}_t = \text{diag}(1, t, t^2, \dots, t^{n-1})$ , an invertible matrix, and  $\mathbf{T}_t = \mathbf{D}_t\mathbf{T}\mathbf{D}_t^{-1}$ . Multiplying out,

$$(\mathbf{T}_t)_{ij} = t^{i-1} t_{ij} t^{-(j-1)} = t^{i-j} t_{ij}.$$

So  $\mathbf{T}_t$  is again upper triangular with the same diagonal  $\lambda_1, \dots, \lambda_n$ , while for  $i < j$  the entry is scaled by  $t^{i-j} \leq t^{-1}$ , using  $t \geq 1$  and  $j - i \geq 1$ . Let  $M = \max_{i < j} |t_{ij}|$  if  $n \geq 2$ , and  $M = 0$  if  $n = 1$ . Then the  $i$ -th absolute row sum of  $\mathbf{T}_t$  is at most

$$|\lambda_i| + \sum_{j>i} t^{-1} |t_{ij}| \leq \rho(\mathbf{A}) + \frac{(n-1)M}{t}.$$

Choose  $t = \max(1, (n-1)M/\varepsilon)$ . Then every absolute row sum of  $\mathbf{T}_t$  is at most  $\rho(\mathbf{A}) + \varepsilon$ , so  $\|\mathbf{T}_t\|_\infty \leq \rho(\mathbf{A}) + \varepsilon$  by [Theorem 15.3.3](#) (b).

Now define, for  $\mathbf{x} \in \mathbb{C}^n$ ,

$$\|\mathbf{x}\|_t := \|\mathbf{D}_t\mathbf{U}^*\mathbf{x}\|_\infty.$$

This is a norm: it is non-negative, homogeneous and subadditive because  $\|\cdot\|_\infty$  is and  $\mathbf{x} \mapsto \mathbf{D}_t\mathbf{U}^*\mathbf{x}$  is linear, and it vanishes only at  $\mathbf{x} = \mathbf{0}$  because  $\mathbf{D}_t\mathbf{U}^*$  is invertible.

It remains to compute the induced operator norm. Put  $\mathbf{Q} = \mathbf{D}_t \mathbf{U}^*$ , so  $\|\mathbf{x}\|_t = \|\mathbf{Q}\mathbf{x}\|_\infty$ . As  $\mathbf{x}$  runs over  $\{\|\mathbf{x}\|_t = 1\}$ , the vector  $\mathbf{y} = \mathbf{Q}\mathbf{x}$  runs over  $\{\|\mathbf{y}\|_\infty = 1\}$ , since  $\mathbf{Q}$  is a bijection. Moreover  $\mathbf{Q}\mathbf{A}\mathbf{Q}^{-1} = \mathbf{D}_t \mathbf{U}^* \mathbf{A} \mathbf{U} \mathbf{D}_t^{-1} = \mathbf{D}_t \mathbf{T} \mathbf{D}_t^{-1} = \mathbf{T}_t$ , so

$$\|\mathbf{A}\mathbf{x}\|_t = \|\mathbf{Q}\mathbf{A}\mathbf{x}\|_\infty = \|\mathbf{T}_t \mathbf{y}\|_\infty.$$

Taking maxima over the two matching sets gives  $\|\mathbf{A}\|_t = \|\mathbf{T}_t\|_\infty \leq \rho(\mathbf{A}) + \varepsilon$ . The lower bound  $\rho(\mathbf{A}) \leq \|\mathbf{A}\|_t$  is [Theorem 15.4.1](#) (a). This proves the theorem. ■

Read together, the last two theorems say

$$\rho(\mathbf{A}) = \inf\{\|\mathbf{A}\| : \|\cdot\| \text{ an induced matrix norm}\}.$$

*Remark.* The infimum need not be attained. If  $\mathbf{A} \neq 0$  is nilpotent then  $\rho(\mathbf{A}) = 0$  while  $\|\mathbf{A}\| > 0$  for every norm, so no norm achieves the value. At the other extreme, for a normal  $\mathbf{A}$  the spectral norm achieves it exactly:  $\|\mathbf{A}\|_2 = \rho(\mathbf{A})$ , as computed at the start of this section. The gap between  $\rho$  and the norms is a measure of how far  $\mathbf{A}$  is from being normal, which is the theme Chapter 11 §11 opened with the departure from normality.

## 15.4.4 Gelfand's formula

The infimum above is over norms, and to use it we must choose one. There is a second description of  $\rho(\mathbf{A})$  that needs no choice: fix any norm at all, and let the powers of  $\mathbf{A}$  tell you the answer.

### Theorem 15.4.4 — Gelfand's Formula

Let  $\mathbf{A} \in M_n(\mathbb{C})$  and let  $\|\cdot\|$  be any matrix norm on  $M_n(\mathbb{C})$ . Then

$$\lim_{k \rightarrow \infty} \|\mathbf{A}^k\|^{1/k} = \rho(\mathbf{A}).$$

In particular the limit exists and does not depend on which matrix norm is used.

*Idea.* Two inequalities, one for each direction, and they come from opposite ends of the chapter. ① The lower bound  $\rho(\mathbf{A}) \leq \|\mathbf{A}^k\|^{1/k}$  is [Theorem 15.4.1](#) applied to  $\mathbf{A}^k$ , once we know  $\rho(\mathbf{A}^k) = \rho(\mathbf{A})^k$  — which is the Spectral Mapping Theorem. This holds for every  $k$ , with no limit taken. ② For the upper bound, scale  $\mathbf{A}$  down by a whisker more than  $\rho(\mathbf{A})$ . The scaled matrix has spectral radius below 1, so Chapter 9 §10's criterion makes its powers tend to 0, and once they are below 1 in norm the inequality we want has been written down. [Theorem 15.4.3](#) says the same decay is geometric, which is what the corollary below extracts.

*Proof. Lower bound.* Fix  $k \geq 1$ . By [Theorem 8.10.1](#) applied to  $q(x) = x^k$ ,  $\text{spec}(\mathbf{A}^k) = \{\lambda^k : \lambda \in \text{spec}(\mathbf{A})\}$ , so

$$\rho(\mathbf{A}^k) = \max_{\lambda \in \text{spec}(\mathbf{A})} |\lambda|^k = \rho(\mathbf{A})^k.$$

By [Theorem 15.4.1](#) (b) applied to  $\mathbf{A}^k$ ,  $\rho(\mathbf{A})^k = \rho(\mathbf{A}^k) \leq \|\mathbf{A}^k\|$ . Taking  $k$ -th roots of these non-negative reals,

$$\rho(\mathbf{A}) \leq \|\mathbf{A}^k\|^{1/k} \quad \text{for every } k \geq 1.$$

**Upper bound.** Let  $\varepsilon > 0$ , put  $s = \rho(\mathbf{A}) + \varepsilon > 0$  and  $\mathbf{B} = s^{-1}\mathbf{A}$ . Every eigenvalue of  $\mathbf{B}$  is  $\lambda/s$  for some  $\lambda \in \text{spec}(\mathbf{A})$  ([Theorem 8.10.1](#) with  $q(x) = x/s$ ), so

$$\rho(\mathbf{B}) = \frac{\rho(\mathbf{A})}{\rho(\mathbf{A}) + \varepsilon} < 1.$$

By [Theorem 9.10.2](#),  $\mathbf{B}^k \rightarrow 0$  entrywise. Entrywise convergence of matrices is convergence in **every** norm on  $M_n(\mathbb{C})$ , by [Corollary 15.2.4](#), so  $\|\mathbf{B}^k\| \rightarrow 0$ . A real sequence tending to 0 is eventually below 1: there is  $K$  with  $\|\mathbf{B}^k\| \leq 1$  for all  $k \geq K$ . Since  $\mathbf{B}^k = s^{-k}\mathbf{A}^k$ , homogeneity gives  $\|\mathbf{A}^k\| \leq s^k$ , that is,

$$\|\mathbf{A}^k\|^{1/k} \leq \rho(\mathbf{A}) + \varepsilon \quad (k \geq K).$$

**Conclusion.** Combining, for every  $\varepsilon > 0$  there is  $K$  with

$$\rho(\mathbf{A}) \leq \|\mathbf{A}^k\|^{1/k} \leq \rho(\mathbf{A}) + \varepsilon \quad (k \geq K),$$

which is the definition of  $\|\mathbf{A}^k\|^{1/k} \rightarrow \rho(\mathbf{A})$ . The independence of the norm is automatic, since the limit equals  $\rho(\mathbf{A})$  whichever matrix norm was fixed at the start. This proves the theorem. ■

*Remark. Which analysis was used.* The lower bound used none. The upper bound used [Theorem 9.10.2](#), which is a theorem of this book, and [Corollary 15.2.4](#), likewise; the only analysis is that a real sequence tending to 0 is eventually less than 1, and that a sequence trapped between  $\rho(\mathbf{A})$  and  $\rho(\mathbf{A}) + \varepsilon$  for every  $\varepsilon > 0$  converges to  $\rho(\mathbf{A})$ . Neither is an extra import: both are the definition of a limit of a real sequence, read forwards and backwards. Nothing from the list (A1)–(A4) of the chapter introduction is needed here, and nothing outside it is used.

The formula is a genuine computation rule as well as a theorem. For a fixed  $k$  the number  $\|\mathbf{A}^k\|^{1/k}$  is an *upper* estimate of  $\rho(\mathbf{A})$  — that is the lower bound above, read the other way — and the estimates improve as  $k$  grows.

**Example. (Watching the limit converge)**

Let  $\mathbf{A} = \begin{pmatrix} 1/2 & 10 \\ 0 & 1/3 \end{pmatrix}$ . Compute  $\|\mathbf{A}^k\|_\infty$  exactly and confirm Gelfand's formula.

*Solution.*  $\mathbf{A}$  is upper triangular, so  $\rho(\mathbf{A}) = \max(\frac{1}{2}, \frac{1}{3}) = \frac{1}{2}$  and the powers are upper triangular with diagonal  $2^{-k}, 3^{-k}$ . For the corner entry, induction on  $k$  gives

$$(\mathbf{A}^k)_{12} = 10 \sum_{j=0}^{k-1} 2^{-j} 3^{-(k-1-j)} = 60(2^{-k} - 3^{-k}),$$

the last step by summing the geometric series with ratio  $3/2$ . The two absolute row sums are  $2^{-k} + 60(2^{-k} - 3^{-k})$  and  $3^{-k}$ , the first being the larger, so by [Theorem 15.3.3](#) (b)

$$\|\mathbf{A}^k\|_\infty = 61 \cdot 2^{-k} - 60 \cdot 3^{-k}.$$

At  $k = 1$  this is 10.5, which overestimates  $\rho(\mathbf{A}) = \frac{1}{2}$  by a factor of 21; the  $k$ -th roots run 10.5, 2.93, 1.75, 1.32, 1.11, 0.98, ... and settle toward  $\frac{1}{2}$ , which is what [Theorem 15.4.4](#) guarantees. The convergence is slow, and it has to be: the single number  $\|\mathbf{A}\|_\infty$  cannot know that the entry 10 sits above the diagonal rather than on it.

## 15.4.5 What is new, and what is not

### Corollary 15.4.5 — Powers Tending to Zero, with a Rate

Let  $\mathbf{A} \in M_n(\mathbb{C})$  and let  $\|\cdot\|$  be any matrix norm on  $M_n(\mathbb{C})$ .

- (a)  $\mathbf{A}^k \rightarrow 0$  if and only if  $\rho(\mathbf{A}) < 1$ , and this holds if and only if  $\|\mathbf{A}^k\| \rightarrow 0$ .
- (b) If  $\rho(\mathbf{A}) < 1$ , then for every  $q$  with  $\rho(\mathbf{A}) < q < 1$  there is a constant  $C \geq 1$ , depending on  $\mathbf{A}$ , on  $q$  and on the norm, such that  $\|\mathbf{A}^k\| \leq C q^k$  for **every**  $k \geq 0$ .

*Idea.* Part (a) is [Theorem 9.10.2](#) translated into norms by [Corollary 15.2.4](#); nothing is proved that was not proved in Chapter 9. Part (b) is the new content, and it is [Theorem 15.4.3](#) doing what it was built for: choose a norm in which  $\mathbf{A}$  itself is already smaller than  $q$ , take powers there, and pay one fixed constant to return to the norm we were given.

*Proof.*

- (a) The first equivalence is [Theorem 9.10.2](#), since  $\rho(\mathbf{A}) < 1$  says exactly that every eigenvalue has modulus less than 1. The second is [Corollary 15.2.4](#): entrywise convergence of matrices and convergence in the norm  $\|\cdot\|$  are the same thing.
- (b) Let  $\varepsilon = q - \rho(\mathbf{A}) > 0$ . By [Theorem 15.4.3](#) there is an induced matrix norm  $\|\cdot\|_t$  on  $M_n(\mathbb{C})$  with  $\|\mathbf{A}\|_t \leq \rho(\mathbf{A}) + \varepsilon = q$ . Submultiplicativity ([Theorem 15.3.2](#) (d)) gives  $\|\mathbf{A}^k\|_t \leq \|\mathbf{A}\|_t^k \leq q^k$  for every  $k \geq 0$ . By [Theorem 15.2.2](#), applied to the two norms  $\|\cdot\|$  and  $\|\cdot\|_t$  on the finite-dimensional space  $M_n(\mathbb{C})$ , there is  $C_0 > 0$  with  $\|\mathbf{M}\| \leq C_0 \|\mathbf{M}\|_t$  for every  $\mathbf{M}$ . Taking  $C = \max(C_0, 1)$ ,

$$\|\mathbf{A}^k\| \leq C_0 \|\mathbf{A}^k\|_t \leq C q^k.$$

This proves the corollary. ■

It is worth being exact about the accounting. Part (a) is **not** new. Chapter 9 §10 proved that  $\mathbf{A}^m \rightarrow 0$  if and only if every eigenvalue has modulus less than 1, and it proved it with the Jordan form and no norms whatsoever; all that has happened here is that the conclusion has been restated in a language where “small” is a number. What is new is part (b): a **rate**. Chapter 9 could say that the powers die; it could not say how fast, because it had nothing to measure a matrix with. Now every decaying matrix comes with a geometric envelope  $Cq^k$  for every  $q$  above its spectral radius, and that envelope is what later sections integrate, sum and differentiate.

Two honest limitations of the envelope deserve saying out loud.

*Warning.*  $\rho(\mathbf{A}) < 1$  **promises eventual decay, not immediate decay**. The constant  $C$  in Corollary 15.4.5 (b) is generally larger than 1 and grows as  $q$  approaches  $\rho(\mathbf{A})$ . For

$$\mathbf{A} = \begin{pmatrix} 9/10 & 10 \\ 0 & 9/10 \end{pmatrix}, \quad \mathbf{A}^k = \begin{pmatrix} (9/10)^k & 10k(9/10)^{k-1} \\ 0 & (9/10)^k \end{pmatrix},$$

we have  $\rho(\mathbf{A}) = 9/10 < 1$ , yet  $\|\mathbf{A}^k\|_\infty = (9/10)^{k-1}(9/10 + 10k)$  **rises** from 10.9 at  $k = 1$  to about 39.13 at  $k = 9$  before beginning its decay, and does not fall below 1 until  $k = 63$ . A transient hump of this kind is invisible to the spectral radius and is exactly what the constant  $C$  is paying for.

*Quick check.* For which matrices can one take  $C = 1$  and  $q = \rho(\mathbf{A})$  in the bound of Corollary 15.4.5 (b), with the spectral norm?

*Solution.* For normal  $\mathbf{A}$ , among others. If  $\mathbf{A}$  is normal then so is  $\mathbf{A}^k$  (its adjoint is  $(\mathbf{A}^*)^k$ , and powers of commuting matrices commute), so  $\|\mathbf{A}^k\|_2 = \rho(\mathbf{A}^k) = \rho(\mathbf{A})^k$  by the computation at the start of this section together with Theorem 8.10.1. There is then no transient at all: the norm of the powers is exactly geometric. The hump in the warning is possible only because that matrix is far from normal, with departure from normality 10 in the sense of Definition 11.11.1.

## 15.4.6 Exercises

### 15.4.6.1 A. Check your understanding

#### Exercise A1.

- (a) Define the **spectral radius**, and say why the maximum in the definition exists.
- (b) State Theorem 15.4.1 and Theorem 15.4.4.
- (c) Determine whether the following statement is correct, and justify your answer:  $\rho$

is a matrix norm on  $M_n(\mathbb{C})$ .

- (d) Determine whether the following statement is correct, and justify your answer: if  $\rho(\mathbf{A}) < 1$  then  $\|\mathbf{A}\|_2 < 1$ .
- (e) Chapter 9 §10 already proved that  $\mathbf{A}^m \rightarrow 0$  exactly when  $\rho(\mathbf{A}) < 1$ . What does [Corollary 15.4.5](#) add to that?

*Solution.*

- (a)  $\rho(\mathbf{A}) = \max\{|\lambda| : \lambda \in \text{spec}(\mathbf{A})\}$  ([Definition 15.4.1](#)). The set  $\text{spec}(\mathbf{A})$  is finite and non-empty — it is the root set of  $p_{\mathbf{A}}$ , non-empty by [Theorem 5.5.1](#) — so the maximum of the finitely many moduli exists.
- (b) For every matrix norm on  $M_n(\mathbb{C})$ ,  $\rho(\mathbf{A}) \leq \|\mathbf{A}\|$ ; and for every matrix norm,  $\|\mathbf{A}^k\|^{1/k} \rightarrow \rho(\mathbf{A})$  as  $k \rightarrow \infty$ .
- (c) Incorrect.  $\rho$  vanishes on non-zero nilpotent matrices, so it is not even a norm, and it is neither subadditive nor submultiplicative: see the warning after [Definition 15.4.1](#).
- (d) Incorrect.  $\mathbf{A} = \begin{pmatrix} 0 & 10 \\ 0 & 0 \end{pmatrix}$  has  $\rho(\mathbf{A}) = 0$  and  $\|\mathbf{A}\|_2 = 10$ . The correct statement is that *some* norm has  $\|\mathbf{A}\| < 1$ , by [Theorem 15.4.3](#).
- (e) Nothing about the criterion, which is [Theorem 9.10.2](#) restated; what it adds is the quantitative envelope  $\|\mathbf{A}^k\| \leq Cq^k$  for every  $q \in (\rho(\mathbf{A}), 1)$ , a rate of decay rather than the bare fact of it.

### 15.4.6.2 B. Practice

#### Exercise B1: Four numbers again.

Let  $\mathbf{A} = \begin{pmatrix} 0 & 2 \\ 1/4 & 0 \end{pmatrix}$ . Compute  $\rho(\mathbf{A})$ ,  $\|\mathbf{A}\|_1$ ,  $\|\mathbf{A}\|_\infty$  and  $\|\mathbf{A}\|_2$ , and verify [Theorem 15.4.1](#) in each case.

*Solution.*  $p_{\mathbf{A}}(x) = x^2 - \frac{1}{2}$ , so  $\text{spec}(\mathbf{A}) = \{\pm 1/\sqrt{2}\}$  and  $\rho(\mathbf{A}) = 1/\sqrt{2} \approx 0.707$ .

The absolute column sums are  $\frac{1}{4}$  and 2, so  $\|\mathbf{A}\|_1 = 2$ ; the absolute row sums are 2 and  $\frac{1}{4}$ , so  $\|\mathbf{A}\|_\infty = 2$ . For the spectral norm,  $\mathbf{A}^\top \mathbf{A} = \text{diag}(\frac{1}{16}, 4)$ , whose eigenvalues are  $\frac{1}{16}$  and 4; hence  $\sigma_1 = 2$  and  $\|\mathbf{A}\|_2 = 2$  by [Theorem 15.3.3](#) (c).

So  $\rho(\mathbf{A}) = 0.707\ldots \leq 2$  in all three cases, and the gap is a factor of nearly 3 even though  $\mathbf{A}$  is a very small matrix. The reason is the mismatch between the two off-diagonal entries:  $\mathbf{A}$  is far from normal.

#### Exercise B2: Gelfand exactly.

With  $\mathbf{A}$  as in [Exercise \(B1: Four numbers again\)](#), compute  $\mathbf{A}^k$  for every  $k$  and evaluate  $\lim_k \|\mathbf{A}^k\|_\infty^{1/k}$  directly.

*Solution.*  $\mathbf{A}^2 = \frac{1}{2}\mathbf{I}$ , so by induction  $\mathbf{A}^{2l} = 2^{-l}\mathbf{I}$  and  $\mathbf{A}^{2l+1} = 2^{-l}\mathbf{A}$ . Hence

$$\|\mathbf{A}^{2l}\|_\infty = 2^{-l}, \quad \|\mathbf{A}^{2l+1}\|_\infty = 2^{-l} \cdot 2 = 2^{1-l}.$$



For even  $k = 2l$ ,  $\|\mathbf{A}^k\|_\infty^{1/k} = (2^{-l})^{1/(2l)} = 2^{-1/2}$ , exactly  $\rho(\mathbf{A})$  for every  $l$ . For odd  $k = 2l + 1$ ,  $\|\mathbf{A}^k\|_\infty^{1/k} = 2^{(1-l)/(2l+1)}$ , and  $(1-l)/(2l+1) \rightarrow -\frac{1}{2}$ , so this too tends to  $2^{-1/2}$ . Both subsequences converge to  $1/\sqrt{2} = \rho(\mathbf{A})$ , confirming [Theorem 15.4.4](#).

### Exercise B3: Deciding convergence without eigenvalues.

For each matrix, decide whether  $\mathbf{A}^m \rightarrow 0$ , using only [Corollary 15.4.2](#) and [Theorem 9.10.2](#).

$$(a) \quad \mathbf{A} = \frac{1}{4} \begin{pmatrix} 1 & 2 & 0 \\ 0 & 1 & 1 \\ 2 & 0 & 1 \end{pmatrix}.$$

$$(b) \quad \mathbf{A} = \begin{pmatrix} 1/2 & 3 \\ 0 & 2 \end{pmatrix}.$$

*Solution.*

- (a) The absolute row sums are  $\frac{3}{4}$ ,  $\frac{1}{2}$  and  $\frac{3}{4}$ , so  $\|\mathbf{A}\|_\infty = \frac{3}{4}$  and  $\rho(\mathbf{A}) \leq \frac{3}{4} < 1$ . By [Theorem 9.10.2](#),  $\mathbf{A}^m \rightarrow 0$ .
- (b) Here the row sums are  $\frac{7}{2}$  and 2 and the column sums are  $\frac{1}{2}$  and 5, so the bounds give only  $\rho(\mathbf{A}) \leq \frac{7}{2}$  and decide nothing. But the bound is an upper bound, and here the matrix is triangular, so the eigenvalues are visible:  $\frac{1}{2}$  and 2, giving  $\rho(\mathbf{A}) = 2 > 1$ . By [Theorem 9.10.2](#) the powers do **not** tend to 0. The moral is that [Corollary 15.4.2](#) can only ever certify convergence, never rule it out.

### 15.4.6.3 C. Going deeper

#### Exercise C1: A norm that sees the decay.

Let  $\mathbf{A} \in M_n(\mathbb{C})$ .

- (a) Prove that if  $\|\mathbf{A}\| < 1$  for **some** matrix norm, then  $\mathbf{A}^k \rightarrow 0$ .
- (b) Give a matrix with  $\mathbf{A}^k \rightarrow 0$  and  $\|\mathbf{A}\|_2 > 1$ , so that the converse of (a) fails for a fixed norm.
- (c) Prove that the converse of (a) nevertheless holds if the norm is allowed to depend on  $\mathbf{A}$ .

*Solution.*

- (a) Submultiplicativity gives  $\|\mathbf{A}^k\| \leq \|\mathbf{A}\|^k$  by induction on  $k$ . With  $r = \|\mathbf{A}\| < 1$  the right-hand side tends to 0, so  $\|\mathbf{A}^k\| \rightarrow 0$ , and by [Corollary 15.2.4](#) this is the same as  $\mathbf{A}^k \rightarrow 0$  entrywise.
- (b)  $\mathbf{A} = \begin{pmatrix} 0 & 10 \\ 0 & 0 \end{pmatrix}$  satisfies  $\mathbf{A}^2 = 0$ , so certainly  $\mathbf{A}^k \rightarrow 0$ ; but  $\|\mathbf{A}\|_2 = 10 > 1$ . The same happens for  $\begin{pmatrix} 1/2 & 10 \\ 0 & 1/2 \end{pmatrix}$ , which is not even nilpotent.
- (c) Suppose  $\mathbf{A}^k \rightarrow 0$ . By [Theorem 9.10.2](#),  $\rho(\mathbf{A}) < 1$ . Apply [Theorem 15.4.3](#) with  $\varepsilon = \frac{1}{2}(1 - \rho(\mathbf{A})) > 0$ : it produces an induced matrix norm  $\|\cdot\|_t$  with

$$\|\mathbf{A}\|_t \leq \rho(\mathbf{A}) + \varepsilon = \frac{1}{2}(1 + \rho(\mathbf{A})) < 1.$$

So a norm witnessing the decay always exists, though it must be chosen after  $\mathbf{A}$ .

### Exercise C2: Spectral radius zero.

Let  $\mathbf{A} \in M_n(\mathbb{C})$ .

- (a) Prove that  $\rho(\mathbf{A}) = 0$  if and only if  $\mathbf{A}$  is nilpotent.
- (b) Deduce that for a nilpotent  $\mathbf{A}$  and any matrix norm,  $\|\mathbf{A}^k\|^{1/k} \rightarrow 0$ , and verify this directly for  $\mathbf{A} = \mathbf{J}_n(0)$ .
- (c) Explain why (a) shows that no inequality of the form  $\|\mathbf{A}\| \leq c\rho(\mathbf{A})$ , with  $c$  depending only on  $n$ , can hold.

*Solution.*

- (a) ( $\Rightarrow$ ) If  $\rho(\mathbf{A}) = 0$  then every eigenvalue is 0, so  $p_{\mathbf{A}}(x) = x^n$  because  $p_{\mathbf{A}}$  is monic of degree  $n$  and splits over  $\mathbb{C}$  with all roots 0 (Theorem 5.5.1). By the Cayley–Hamilton theorem (Theorem 8.6.1),  $\mathbf{A}^n = p_{\mathbf{A}}(\mathbf{A}) = 0$ , so  $\mathbf{A}$  is nilpotent (Definition 9.2.1).
- ( $\Leftarrow$ ) If  $\mathbf{A}^d = 0$  and  $\mathbf{A}\mathbf{x} = \lambda\mathbf{x}$  with  $\mathbf{x} \neq \mathbf{0}$ , then  $\mathbf{0} = \mathbf{A}^d\mathbf{x} = \lambda^d\mathbf{x}$ , so  $\lambda^d = 0$  and  $\lambda = 0$ . Hence  $\text{spec}(\mathbf{A}) = \{0\}$  and  $\rho(\mathbf{A}) = 0$ .
- (b) By Theorem 15.4.4 the limit is  $\rho(\mathbf{A}) = 0$ . Directly for  $\mathbf{A} = \mathbf{J}_n(0)$ :  $\mathbf{A}^k = 0$  for  $k \geq n$ , so  $\|\mathbf{A}^k\|^{1/k} = 0$  for all  $k \geq n$ , and the sequence is eventually constant at 0.
- (c) Such an inequality would force  $\|\mathbf{A}\| = 0$ , hence  $\mathbf{A} = 0$ , for every nilpotent  $\mathbf{A}$ . But  $\mathbf{J}_2(0) \neq 0$  is nilpotent. So the two quantities cannot be compared in that direction at all; Theorem 15.4.3 is the best available substitute, and it buys its conclusion by changing the norm.

### Exercise C3: When the infimum is attained.

Call  $\mathbf{A} \in M_n(\mathbb{C})$  **radius-attaining** if  $\|\mathbf{A}\| = \rho(\mathbf{A})$  for some matrix norm.

- (a) Prove that every normal  $\mathbf{A}$  is radius-attaining.
- (b) Prove that a non-zero nilpotent  $\mathbf{A}$  is not.
- (c) Deduce that  $\begin{pmatrix} 1 & 1 \\ 0 & 1 \end{pmatrix}$  is not radius-attaining, even though it is invertible.

*Hint for (c): consider  $\mathbf{A} - \mathbf{I}$ .*

*Solution.*

- (a) Let  $\mathbf{A}$  be normal, so  $\mathbf{A} = \mathbf{U}\mathbf{D}\mathbf{U}^*$  with  $\mathbf{U}$  unitary and  $\mathbf{D} = \text{diag}(\lambda_1, \dots, \lambda_n)$  (Corollary 11.4.3). Then  $\mathbf{A}^*\mathbf{A} = \mathbf{U}\text{diag}(|\lambda_i|^2)\mathbf{U}^*$ , so the singular values of  $\mathbf{A}$  are the  $|\lambda_i|$  in decreasing order and  $\|\mathbf{A}\|_2 = \sigma_1 = \max_i |\lambda_i| = \rho(\mathbf{A})$  by Theorem 15.3.3 (c).
- (b) If  $\mathbf{A} \neq 0$  is nilpotent then  $\rho(\mathbf{A}) = 0$  by Exercise (C2: Spectral radius zero) (a), while  $\|\mathbf{A}\| > 0$  for every norm, since a norm vanishes only at 0 (Definition 15.1.1).

So no norm can give  $\|\mathbf{A}\| = \rho(\mathbf{A})$ .

- (c) Put  $\mathbf{A} = \begin{pmatrix} 1 & 1 \\ 0 & 1 \end{pmatrix}$ , whose only eigenvalue is 1, so  $\rho(\mathbf{A}) = 1$ . Suppose some matrix norm had  $\|\mathbf{A}\| = 1$ . Then  $\|\mathbf{A}^k\| \leq \|\mathbf{A}\|^k = 1$  for every  $k$ , so the sequence  $(\mathbf{A}^k)$  is bounded in that norm, hence bounded entrywise by [Corollary 15.2.4](#), whose second sentence covers boundedness. But  $\mathbf{A}^k = \begin{pmatrix} 1 & k \\ 0 & 1 \end{pmatrix}$ , whose  $(1, 2)$ -entry is unbounded. Contradiction. (The hint points at the same phenomenon:  $\mathbf{A} - \mathbf{I}$  is a non-zero nilpotent, so  $\mathbf{A}$  is a unipotent matrix, and it is the nilpotent part that the spectral radius cannot see.)

## 15.5 Inverting a Perturbation of the Identity

Chapter 9 §10 summed the series  $\mathbf{I} + \mathbf{A} + \mathbf{A}^2 + \dots$  and identified its sum as  $(\mathbf{I} - \mathbf{A})^{-1}$ , under the exact hypothesis that every eigenvalue of  $\mathbf{A}$  has modulus less than 1. What it could not say is how large that inverse is. This section reruns the argument with a norm in hand. The hypothesis becomes cruder —  $\|\mathbf{A}\| < 1$  instead of  $\rho(\mathbf{A}) < 1$  — and the conclusion becomes quantitative, which is the trade this chapter keeps making. The payoff is immediate: the invertible matrices turn out to form an open set, inversion turns out to be continuous, and a first estimate appears for how far the solution of  $\mathbf{A}\mathbf{x} = \mathbf{b}$  moves when  $\mathbf{A}$  and  $\mathbf{b}$  are disturbed.

**Throughout,**  $\|\cdot\|$  is the operator norm on  $M_n(\mathbb{C})$  induced by a norm on  $\mathbb{C}^n$  ([Definition 15.3.2](#)), so that  $\|\mathbf{I}\| = 1$  and  $\|\mathbf{XY}\| \leq \|\mathbf{X}\|\|\mathbf{Y}\|$  by [Theorem 15.3.2](#) (c) and (d). Both properties are used in every proof below, and the first of them is not free: the closing warning shows what happens without it.

### 15.5.1 The geometric series, with a number attached

The scalar identity behind everything here is  $1 + r + r^2 + \dots = 1/(1 - r)$  for  $0 \leq r < 1$ . Chapter 9 §10 gave the matrix version of the *identity*. What a norm adds is the matrix version of the *inequality*  $|1 + r + \dots| \leq 1/(1 - r)$ : a ceiling on the inverse, expressed in the data.

#### Theorem 15.5.1 — Neumann Series with a Bound

Let  $\mathbf{A} \in M_n(\mathbb{C})$  with  $\|\mathbf{A}\| < 1$ , and write  $a = \|\mathbf{A}\|$ . Then:

- (a)  $\mathbf{I} - \mathbf{A}$  is invertible, and  $\sum_{k \geq 0} \mathbf{A}^k = (\mathbf{I} - \mathbf{A})^{-1}$ ;
- (b)  $\|(\mathbf{I} - \mathbf{A})^{-1}\| \leq \frac{1}{1 - a}$ ;
- (c) for every  $K \geq 0$ , the partial sum  $\mathbf{S}_K = \sum_{k=0}^K \mathbf{A}^k$  satisfies

$$\|(\mathbf{I} - \mathbf{A})^{-1} - \mathbf{S}_K\| \leq \frac{a^{K+1}}{1 - a};$$

- (d)  $\|(\mathbf{I} - \mathbf{A})^{-1}\| \geq \frac{1}{1 + a}$ .

*Idea.* Part (a) is not new work:  $\|\mathbf{A}\| < 1$  forces  $\rho(\mathbf{A}) < 1$  by [Theorem 15.4.1](#), which is exactly the hypothesis of Chapter 9's [Theorem 9.10.6](#). Everything else is the tail-

bounding move. Submultiplicativity turns  $\|\mathbf{A}^k\|$  into  $a^k$ , so each partial sum is bounded by a scalar geometric sum; the bound then has to be carried across the limit, and what carries it is the fact that a norm is continuous. For (c), the tail of the series factors as  $\mathbf{A}^{K+1}$  times the whole series again, so (b) bounds it. For (d), read  $\mathbf{I} = (\mathbf{I} - \mathbf{A})(\mathbf{I} - \mathbf{A})^{-1}$  through submultiplicativity.

*Proof.*

- (a) By [Theorem 15.4.1](#) (a),  $\rho(\mathbf{A}) \leq \|\mathbf{A}\| = a < 1$ , so every eigenvalue of  $\mathbf{A}$  has modulus less than 1. That is the hypothesis of [Theorem 9.10.6](#), which gives both the invertibility of  $\mathbf{I} - \mathbf{A}$  and the value of the sum.
- (b) Submultiplicativity ([Theorem 15.3.2](#) (d)) gives  $\|\mathbf{A}^k\| \leq a^k$  for every  $k \geq 1$ , by induction on  $k$ ; and  $\|\mathbf{A}^0\| = \|\mathbf{I}\| = 1 = a^0$  by part (c) of the same theorem. Hence, by subadditivity applied  $K$  times,

$$\|\mathbf{S}_K\| \leq \sum_{k=0}^K \|\mathbf{A}^k\| \leq \sum_{k=0}^K a^k = \frac{1 - a^{K+1}}{1 - a} \leq \frac{1}{1 - a},$$

the last step because  $a^{K+1} \geq 0$  and  $1 - a > 0$ . By (a) the sequence  $(\mathbf{S}_K)$  converges entrywise to  $(\mathbf{I} - \mathbf{A})^{-1}$ , so  $\|\mathbf{S}_K - (\mathbf{I} - \mathbf{A})^{-1}\| \rightarrow 0$  by [Corollary 15.2.4](#). Then [Lemma 15.1.1](#) gives

$$|\|\mathbf{S}_K\| - \|(\mathbf{I} - \mathbf{A})^{-1}\|| \leq \|\mathbf{S}_K - (\mathbf{I} - \mathbf{A})^{-1}\| \rightarrow 0,$$

so  $\|\mathbf{S}_K\| \rightarrow \|(\mathbf{I} - \mathbf{A})^{-1}\|$ . A non-strict inequality survives a limit, so a limit of numbers each at most  $1/(1 - a)$  is at most  $1/(1 - a)$ , which proves (b).

- (c) Fix  $K$ . For every  $M \geq 1$ , multiplying out and canceling,

$$\mathbf{S}_{K+M} - \mathbf{S}_K = \sum_{k=K+1}^{K+M} \mathbf{A}^k = \mathbf{A}^{K+1} \mathbf{S}_{M-1}.$$

Let  $M \rightarrow \infty$ . The left-hand side tends to  $(\mathbf{I} - \mathbf{A})^{-1} - \mathbf{S}_K$  by (a). The right-hand side tends to  $\mathbf{A}^{K+1}(\mathbf{I} - \mathbf{A})^{-1}$ , because

$$\|\mathbf{A}^{K+1} \mathbf{S}_{M-1} - \mathbf{A}^{K+1}(\mathbf{I} - \mathbf{A})^{-1}\| \leq \|\mathbf{A}^{K+1}\| \|\mathbf{S}_{M-1} - (\mathbf{I} - \mathbf{A})^{-1}\|,$$

which tends to 0 by (a) and [Corollary 15.2.4](#) again. A sequence has at most one limit, so

$$(\mathbf{I} - \mathbf{A})^{-1} - \mathbf{S}_K = \mathbf{A}^{K+1}(\mathbf{I} - \mathbf{A})^{-1}.$$

Taking norms and using submultiplicativity together with (b),

$$\|(\mathbf{I} - \mathbf{A})^{-1} - \mathbf{S}_K\| \leq \|\mathbf{A}^{K+1}\| \|(\mathbf{I} - \mathbf{A})^{-1}\| \leq \frac{a^{K+1}}{1 - a}.$$

- (d)  $1 = \|\mathbf{I}\| = \|(\mathbf{I} - \mathbf{A})(\mathbf{I} - \mathbf{A})^{-1}\| \leq \|\mathbf{I} - \mathbf{A}\| \|(\mathbf{I} - \mathbf{A})^{-1}\|$ , and  $\|\mathbf{I} - \mathbf{A}\| \leq \|\mathbf{I}\| + \|\mathbf{A}\| = 1 + a$  by subadditivity. Since  $1 + a > 0$  we may divide, which proves (d). This proves the theorem. ■

Parts (b) and (d) together pin the inverse between two numbers computed from  $a$  alone:

$$\frac{1}{1+a} \leq \|(\mathbf{I} - \mathbf{A})^{-1}\| \leq \frac{1}{1-a}.$$

As  $a$  approaches 1 the upper bound escapes to infinity while the lower one stays near  $\frac{1}{2}$ , and the gap is real rather than an artifact of the proof:  $\mathbf{A} = a\mathbf{I}$  attains the upper bound, and  $\mathbf{A} = -a\mathbf{I}$  attains the lower one.

### Example. (The bound and the truth)

Let

$$\mathbf{A} = \begin{pmatrix} 0 & 1/2 \\ 1/3 & 0 \end{pmatrix}.$$

Using the  $\infty$ -norm, check the hypothesis, compute  $(\mathbf{I} - \mathbf{A})^{-1}$  exactly, and compare it with the bound of [Theorem 15.5.1](#) (b). Then compare  $\mathbf{S}_3$  with the true inverse and with the tail bound of (c).

*Solution. The hypothesis.* The absolute row sums are  $\frac{1}{2}$  and  $\frac{1}{3}$ , so  $\|\mathbf{A}\|_\infty = \frac{1}{2} < 1$  by [Theorem 15.3.3](#) (b). So [Theorem 15.5.1](#) applies with  $a = \frac{1}{2}$ .

*The inverse.*  $\mathbf{I} - \mathbf{A} = \begin{pmatrix} 1 & -1/2 \\ -1/3 & 1 \end{pmatrix}$  has determinant  $1 - \frac{1}{6} = \frac{5}{6}$ , so by [Theorem 0.9.8](#)

$$(\mathbf{I} - \mathbf{A})^{-1} = \frac{6}{5} \begin{pmatrix} 1 & 1/2 \\ 1/3 & 1 \end{pmatrix} = \begin{pmatrix} 6/5 & 3/5 \\ 2/5 & 6/5 \end{pmatrix},$$

whose largest absolute row sum is  $\frac{6}{5} + \frac{3}{5} = \frac{9}{5} = 1.8$ . The bound is  $1/(1 - \frac{1}{2}) = 2$ . So the estimate is correct and loses only ten percent.

*The partial sum.* Here  $\mathbf{A}^2 = \frac{1}{6}\mathbf{I}$ , so  $\mathbf{A}^3 = \frac{1}{6}\mathbf{A}$  and

$$\mathbf{S}_3 = (1 + \frac{1}{6})(\mathbf{I} + \mathbf{A}) = \frac{7}{6} \begin{pmatrix} 1 & 1/2 \\ 1/3 & 1 \end{pmatrix}.$$

Hence  $(\mathbf{I} - \mathbf{A})^{-1} - \mathbf{S}_3 = (\frac{6}{5} - \frac{7}{6})(\mathbf{I} + \mathbf{A}) = \frac{1}{30}(\mathbf{I} + \mathbf{A})$ , with  $\infty$ -norm  $\frac{1}{30} \cdot \frac{3}{2} = \frac{1}{20} = 0.05$ . The bound of (c) with  $K = 3$  is  $(\frac{1}{2})^4/(1 - \frac{1}{2}) = \frac{1}{8} = 0.125$ . Again correct, and this time loose by a factor of 2.5 — the price of replacing four matrices by four copies of one number.

## 15.5.2 Which theorem is stronger

The two Neumann theorems must be compared honestly, because neither contains the other.

Chapter 9's [Theorem 9.10.6](#) assumes  $\rho(\mathbf{A}) < 1$  and concludes that the series converges, with sum  $(\mathbf{I} - \mathbf{A})^{-1}$ ; and its hypothesis is **sharp**, since the statement is an “if and only if”. [Theorem 15.5.1](#) assumes  $\|\mathbf{A}\| < 1$ , which by [Theorem 15.4.1](#) is a **strictly stronger** demand:  $\rho(\mathbf{A}) \leq \|\mathbf{A}\|$  always, and the inequality is often strict. For

$$\mathbf{A} = \begin{pmatrix} 0 & 2 \\ 0 & 0 \end{pmatrix}$$

we have  $\rho(\mathbf{A}) = 0$ , so Chapter 9's theorem applies and returns  $\sum_k \mathbf{A}^k = \mathbf{I} + \mathbf{A}$ ; but  $\|\mathbf{A}\| = 2$  in each of the 1-, 2- and  $\infty$ -norms, so this section's theorem says nothing at all about it.

In the other direction, when this section's theorem does apply it says strictly more: a ceiling on the inverse, a rate at which the partial sums approach it, and a floor as well. Chapter 9 could not have stated any of the three, having no way to measure a matrix.

There is a third thing to say, and it dissolves most of the apparent conflict. If  $\rho(\mathbf{A}) < 1$ , then by [Theorem 15.4.3](#) there is a norm  $\|\cdot\|_t$  with  $\|\mathbf{A}\|_t \leq \rho(\mathbf{A}) + \varepsilon < 1$ , and [Theorem 15.5.1](#) applies **in that norm**. For the matrix above, taking  $\mathbf{D} = \text{diag}(1, 4)$  and  $\|\mathbf{x}\|_t = \|\mathbf{D}\mathbf{x}\|_\infty$  gives  $\|\mathbf{A}\|_t = \|\mathbf{D}\mathbf{A}\mathbf{D}^{-1}\|_\infty = \frac{1}{2}$ , so the bound reads  $\|(\mathbf{I} - \mathbf{A})^{-1}\|_t \leq 2$ ; and indeed  $\mathbf{D}(\mathbf{I} + \mathbf{A})\mathbf{D}^{-1} = \begin{pmatrix} 1 & 1/2 \\ 0 & 1 \end{pmatrix}$  has  $\infty$ -norm  $\frac{3}{2} \leq 2$ . So the quantitative theorem can always be brought to bear on a matrix Chapter 9 accepts — at the cost of changing the norm, and the bound it then produces is a bound in that new norm, which may be a strange unit of measurement.

*Quick check.* Both theorems need a hypothesis. Which of the two hypotheses,  $\rho(\mathbf{A}) < 1$  and  $\|\mathbf{A}\| < 1$ , can be checked by looking at the entries of  $\mathbf{A}$  and doing no linear algebra?

*Solution.* The second, for a suitable norm:  $\|\mathbf{A}\|_1$  and  $\|\mathbf{A}\|_\infty$  are the largest absolute column and row sums ([Theorem 15.3.3](#)), read straight off the entries. Deciding  $\rho(\mathbf{A}) < 1$  requires the eigenvalues, that is, the roots of a degree- $n$  polynomial. This is the practical reason the cruder hypothesis is worth having: it is the one a reader can verify at a glance, and [Corollary 15.4.2](#) is the same observation used in the other direction.

*Warning.* **The hypothesis  $\|\mathbf{A}\| < 1$  is sufficient and never necessary.** A matrix can be as far from  $\mathbf{I}$  as one likes and still have  $\mathbf{I} - \mathbf{A}$  invertible:  $\mathbf{A} = 5\mathbf{I}$  gives  $(\mathbf{I} - \mathbf{A})^{-1} = -\frac{1}{4}\mathbf{I}$ . What fails when  $\|\mathbf{A}\| \geq 1$  is not the invertibility but the *argument*, and with it the bound. Read  $\|\mathbf{A}\| < 1$  as “ $\mathbf{A}$  is a small perturbation”, not as a criterion.

### 15.5.3 The invertible matrices form an open set

The theorem above concerns perturbations of  $\mathbf{I}$ . Every invertible matrix can be moved to  $\mathbf{I}$  by multiplication, so the general statement follows at once, and it is the statement several earlier arguments wanted: a matrix near an invertible matrix is invertible, and its inverse is near.

#### Corollary 15.5.2 — Invertibility Is an Open Condition, and Inversion Is Continuous

Let  $\mathbf{B} \in M_n(\mathbb{C})$  be invertible and let  $\mathbf{E} \in M_n(\mathbb{C})$  satisfy

$$\|\mathbf{E}\| < \frac{1}{\|\mathbf{B}^{-1}\|}.$$

Write  $r = \|\mathbf{B}^{-1}\|\|\mathbf{E}\| < 1$ . Then:

- (a)  $\mathbf{B} + \mathbf{E}$  is invertible, and  $\|(\mathbf{B} + \mathbf{E})^{-1}\| \leq \frac{\|\mathbf{B}^{-1}\|}{1 - r}$ ;
- (b)  $\|(\mathbf{B} + \mathbf{E})^{-1} - \mathbf{B}^{-1}\| \leq \frac{\|\mathbf{B}^{-1}\|^2\|\mathbf{E}\|}{1 - r}$ .

Consequently  $\text{GL}_n(\mathbb{C})$  is an open subset of  $M_n(\mathbb{C})$ : around every invertible  $\mathbf{B}$  the ball

of radius  $1/\|\mathbf{B}^{-1}\|$  consists of invertible matrices. And inversion is continuous on it: if  $\mathbf{B}_m \rightarrow \mathbf{B}$  with  $\mathbf{B}$  invertible, then  $\mathbf{B}_m$  is invertible for all large  $m$  and  $\mathbf{B}_m^{-1} \rightarrow \mathbf{B}^{-1}$ .

*Idea.* Factor the perturbed matrix as  $\mathbf{B} + \mathbf{E} = \mathbf{B}(\mathbf{I} + \mathbf{B}^{-1}\mathbf{E})$  — legitimate because  $\mathbf{B}$  is invertible — and notice that  $\|\mathbf{B}^{-1}\mathbf{E}\| \leq r < 1$ , so the second factor is a perturbation of the identity of exactly the kind [Theorem 15.5.1](#) handles, with  $\mathbf{A} = -\mathbf{B}^{-1}\mathbf{E}$ . For (b), the difference of two inverses always factors as  $\mathbf{X}^{-1} - \mathbf{Y}^{-1} = \mathbf{X}^{-1}(\mathbf{Y} - \mathbf{X})\mathbf{Y}^{-1}$ , and both inverses are now bounded.

*Proof.* Put  $\mathbf{A} = -\mathbf{B}^{-1}\mathbf{E}$ . By submultiplicativity,  $\|\mathbf{A}\| \leq \|\mathbf{B}^{-1}\|\|\mathbf{E}\| = r < 1$ .

- (a) Since  $\mathbf{B}$  is invertible,  $\mathbf{B} + \mathbf{E} = \mathbf{B}(\mathbf{I} + \mathbf{B}^{-1}\mathbf{E}) = \mathbf{B}(\mathbf{I} - \mathbf{A})$ . By [Theorem 15.5.1](#) (a) the factor  $\mathbf{I} - \mathbf{A}$  is invertible, so  $\mathbf{B} + \mathbf{E}$  is a product of invertible matrices and is invertible, with

$$(\mathbf{B} + \mathbf{E})^{-1} = (\mathbf{I} - \mathbf{A})^{-1}\mathbf{B}^{-1}.$$

By submultiplicativity and [Theorem 15.5.1](#) (b),

$$\|(\mathbf{B} + \mathbf{E})^{-1}\| \leq \|(\mathbf{I} - \mathbf{A})^{-1}\| \|\mathbf{B}^{-1}\| \leq \frac{\|\mathbf{B}^{-1}\|}{1 - \|\mathbf{A}\|} \leq \frac{\|\mathbf{B}^{-1}\|}{1 - r},$$

the last step because  $\|\mathbf{A}\| \leq r$  and  $t \mapsto 1/(1 - t)$  increases on  $[0, 1)$ .

- (b) Multiplying the identity  $\mathbf{B} - (\mathbf{B} + \mathbf{E}) = -\mathbf{E}$  on the left by  $(\mathbf{B} + \mathbf{E})^{-1}$  and on the right by  $\mathbf{B}^{-1}$  gives

$$(\mathbf{B} + \mathbf{E})^{-1} - \mathbf{B}^{-1} = -(\mathbf{B} + \mathbf{E})^{-1}\mathbf{E}\mathbf{B}^{-1}.$$

Taking norms and applying submultiplicativity twice and then (a),

$$\|(\mathbf{B} + \mathbf{E})^{-1} - \mathbf{B}^{-1}\| \leq \frac{\|\mathbf{B}^{-1}\|}{1 - r} \|\mathbf{E}\| \|\mathbf{B}^{-1}\| = \frac{\|\mathbf{B}^{-1}\|^2 \|\mathbf{E}\|}{1 - r}.$$

For the two consequences: openness is the statement of (a) with  $\mathbf{E}$  ranging over the ball of radius  $1/\|\mathbf{B}^{-1}\|$ . For continuity, let  $\mathbf{B}_m \rightarrow \mathbf{B}$  with  $\mathbf{B}$  invertible, and put  $\mathbf{E}_m = \mathbf{B}_m - \mathbf{B}$ . By [Corollary 15.2.4](#),  $\|\mathbf{E}_m\| \rightarrow 0$ , so  $\|\mathbf{E}_m\| < 1/(2\|\mathbf{B}^{-1}\|)$  for all large  $m$ ; for those  $m$ , part (a) applies with  $r \leq \frac{1}{2}$  and  $\mathbf{B}_m$  is invertible, and part (b) gives

$$\|\mathbf{B}_m^{-1} - \mathbf{B}^{-1}\| \leq 2\|\mathbf{B}^{-1}\|^2 \|\mathbf{E}_m\| \rightarrow 0.$$

This proves the corollary. ■

### Example. (How far an invertible matrix can be pushed)

Take  $\mathbf{B} = \begin{pmatrix} 2 & 1 \\ 0 & 2 \end{pmatrix}$  and work with  $\|\cdot\|_\infty$ . Then  $\mathbf{B}^{-1} = \begin{pmatrix} \frac{1}{2} & -\frac{1}{4} \\ 0 & \frac{1}{2} \end{pmatrix}$ , whose largest absolute row sum is  $\frac{1}{2} + \frac{1}{4} = \frac{3}{4}$ , so  $\|\mathbf{B}^{-1}\|_\infty = \frac{3}{4}$ . [Corollary 15.5.2](#) therefore keeps  $\mathbf{B} + \mathbf{E}$  invertible for every  $\mathbf{E}$  with

$$\|\mathbf{E}\|_\infty < \frac{1}{\|\mathbf{B}^{-1}\|_\infty} = \frac{4}{3}.$$

For a perturbation of size  $\|\mathbf{E}\|_\infty \leq \frac{1}{2}$  we get  $r = \frac{3}{4} \cdot \frac{1}{2} = \frac{3}{8}$ , and part (a) bounds

$$\|(\mathbf{B} + \mathbf{E})^{-1}\|_\infty \leq \frac{3/4}{1 - 3/8} = \frac{6}{5}.$$

The bound is honest rather than tight: pushing every entry of  $\mathbf{E}$  to  $\pm \frac{1}{4}$  gives at worst  $\|(\mathbf{B} + \mathbf{E})^{-1}\|_\infty = \frac{12}{11} \approx 1.09$ , against the promised 1.2. What the corollary buys is not the sharpest constant but a radius inside which no invertibility check is needed at all.

Part (a) has a reading worth recording, since the relative-error theorem of §08 is the same sentence with the constant named: **the distance from an invertible  $\mathbf{B}$  to the nearest singular matrix is at least  $1/\|\mathbf{B}^{-1}\|$** . A large  $\|\mathbf{B}^{-1}\|$  is therefore not an abstract complaint; it says that  $\mathbf{B}$  is close, in the literal sense of the norm, to a matrix that cannot be inverted at all.

*Remark.* Chapter 11 §01 proved that every matrix has diagonalizable matrices arbitrarily close to it (Corollary 11.1.6) and used the word “close” before this chapter existed. Corollary 15.5.2 is the first of the two facts that argument needs; the second is the continuity of the eigenvalues, and the two are put together in §07.

## 15.5.4 Perturbing a solve

The question the rest of the chapter circles is this. A linear system  $\mathbf{Ax} = \mathbf{b}$  is solved; the data  $\mathbf{A}$  and  $\mathbf{b}$  were only known approximately; how wrong can  $\mathbf{x}$  be? With Corollary 15.5.2 available the answer is three lines, and it already contains the quantity §08 will name.

### Theorem 15.5.3 — Relative Error in a Perturbed Solve

Let  $\mathbf{A} \in M_n(\mathbb{C})$  be invertible, let  $\mathbf{b} \neq \mathbf{0}$ , and let  $\mathbf{x}$  be the solution of  $\mathbf{Ax} = \mathbf{b}$ . Let  $\mathbf{E} \in M_n(\mathbb{C})$  and  $\mathbf{f} \in \mathbb{C}^n$  be perturbations with

$$r = \|\mathbf{A}^{-1}\| \|\mathbf{E}\| < 1,$$

and let  $\tilde{\mathbf{x}}$  be the solution of  $(\mathbf{A} + \mathbf{E})\tilde{\mathbf{x}} = \mathbf{b} + \mathbf{f}$ , which exists and is unique by Corollary 15.5.2. Then

$$\frac{\|\tilde{\mathbf{x}} - \mathbf{x}\|}{\|\mathbf{x}\|} \leq \frac{\|\mathbf{A}\| \|\mathbf{A}^{-1}\|}{1 - r} \left( \frac{\|\mathbf{E}\|}{\|\mathbf{A}\|} + \frac{\|\mathbf{f}\|}{\|\mathbf{b}\|} \right).$$

*Idea.* Subtract the two equations to isolate  $\mathbf{A}(\tilde{\mathbf{x}} - \mathbf{x})$ ; what is left on the other side is  $\mathbf{f} - \mathbf{E}\tilde{\mathbf{x}}$ . Apply  $\mathbf{A}^{-1}$  and take norms. The one awkward term is  $\|\tilde{\mathbf{x}}\|$ , which is not data; write  $\tilde{\mathbf{x}} = \mathbf{x} + (\tilde{\mathbf{x}} - \mathbf{x})$  and move the resulting copy of  $\|\tilde{\mathbf{x}} - \mathbf{x}\|$  to the left, which is where the factor  $1/(1 - r)$  comes from. Finally  $\|\mathbf{b}\| \leq \|\mathbf{A}\| \|\mathbf{x}\|$  converts  $1/\|\mathbf{x}\|$  into  $\|\mathbf{A}\|/\|\mathbf{b}\|$ , and the two perturbations appear only through their relative sizes.

*Proof.* Write  $\mathbf{d} = \tilde{\mathbf{x}} - \mathbf{x}$ . Subtracting  $\mathbf{Ax} = \mathbf{b}$  from  $(\mathbf{A} + \mathbf{E})\tilde{\mathbf{x}} = \mathbf{b} + \mathbf{f}$  gives  $\mathbf{A}\tilde{\mathbf{x}} + \mathbf{E}\tilde{\mathbf{x}} - \mathbf{Ax} = \mathbf{f}$ , that is,  $\mathbf{Ad} = \mathbf{f} - \mathbf{E}\tilde{\mathbf{x}}$ . Since  $\mathbf{A}$  is invertible,  $\mathbf{d} = \mathbf{A}^{-1}(\mathbf{f} - \mathbf{E}\tilde{\mathbf{x}})$ , and Theorem 15.3.2 (a) with subadditivity gives

$$\|\mathbf{d}\| \leq \|\mathbf{A}^{-1}\| (\|\mathbf{f}\| + \|\mathbf{E}\| \|\tilde{\mathbf{x}}\|).$$



Now  $\|\tilde{\mathbf{x}}\| \leq \|\mathbf{x}\| + \|\mathbf{d}\|$ , so

$$\|\mathbf{d}\| \leq \|\mathbf{A}^{-1}\| \|\mathbf{f}\| + \|\mathbf{A}^{-1}\| \|\mathbf{E}\| \|\mathbf{x}\| + r \|\mathbf{d}\|,$$

using  $\|\mathbf{A}^{-1}\| \|\mathbf{E}\| = r$  on the last term. Since  $r < 1$  we may subtract  $r \|\mathbf{d}\|$  and divide by  $1 - r > 0$ :

$$\|\mathbf{d}\| \leq \frac{\|\mathbf{A}^{-1}\|}{1 - r} (\|\mathbf{f}\| + \|\mathbf{E}\| \|\mathbf{x}\|).$$

Divide by  $\|\mathbf{x}\|$ , which is non-zero because  $\mathbf{b} \neq \mathbf{0}$ :

$$\frac{\|\mathbf{d}\|}{\|\mathbf{x}\|} \leq \frac{\|\mathbf{A}^{-1}\|}{1 - r} \left( \frac{\|\mathbf{f}\|}{\|\mathbf{x}\|} + \|\mathbf{E}\| \right).$$

Finally  $\|\mathbf{b}\| = \|\mathbf{A}\mathbf{x}\| \leq \|\mathbf{A}\| \|\mathbf{x}\|$ , so  $1/\|\mathbf{x}\| \leq \|\mathbf{A}\|/\|\mathbf{b}\|$ . Substituting this in the first term and writing  $\|\mathbf{E}\| = \|\mathbf{A}\| \cdot \|\mathbf{E}\|/\|\mathbf{A}\|$  in the second gives the stated inequality. This proves the theorem. ■

The shape of the conclusion is the point. On the right, the two perturbations appear only as **relative** sizes,  $\|\mathbf{E}\|/\|\mathbf{A}\|$  and  $\|\mathbf{f}\|/\|\mathbf{b}\|$ ; on the left is the relative size of the error. Between them stands one number built from  $\mathbf{A}$  alone,

$$\|\mathbf{A}\| \|\mathbf{A}^{-1}\|,$$

inflated slightly by  $1/(1 - r)$ , which is close to 1 whenever the perturbation of  $\mathbf{A}$  is small. That number is the condition number of §08. Everything that section does is to give it a name, compute it for the 2-norm, and show that the bound is attained.

#### Example. ( $\mathbf{A}$ perturbation of the matrix)

Let

$$\mathbf{A} = \begin{pmatrix} 2 & 1 \\ 1 & 1 \end{pmatrix}, \quad \mathbf{b} = \begin{pmatrix} 3 \\ 2 \end{pmatrix}, \quad \mathbf{E} = \begin{pmatrix} 0 & 0 \\ 0 & 1/10 \end{pmatrix}, \quad \mathbf{f} = \mathbf{0}.$$

Solve both systems in the  $\infty$ -norm, compute the relative error, and compare it with [Theorem 15.5.3](#).

*Solution. The two solutions.*  $\det \mathbf{A} = 1$ , so  $\mathbf{A}^{-1} = \begin{pmatrix} 1 & -1 \\ -1 & 2 \end{pmatrix}$  by [Theorem 0.9.8](#), and  $\mathbf{x} = \mathbf{A}^{-1}\mathbf{b} = (3 - 2, -3 + 4) = (1, 1)$ . For the perturbed matrix,  $\mathbf{A} + \mathbf{E} = \begin{pmatrix} 2 & 1 \\ 1 & 11/10 \end{pmatrix}$  has determinant  $\frac{22}{10} - 1 = \frac{6}{5}$ , so

$$\tilde{\mathbf{x}} = \frac{5}{6} \begin{pmatrix} 11/10 & -1 \\ -1 & 2 \end{pmatrix} \begin{pmatrix} 3 \\ 2 \end{pmatrix} = \frac{5}{6} \begin{pmatrix} 13/10 \\ 1 \end{pmatrix} = \begin{pmatrix} 13/12 \\ 5/6 \end{pmatrix}.$$

*The error.*  $\tilde{\mathbf{x}} - \mathbf{x} = (\frac{1}{12}, -\frac{1}{6})$ , so  $\|\tilde{\mathbf{x}} - \mathbf{x}\|_{\infty} = \frac{1}{6}$  and, since  $\|\mathbf{x}\|_{\infty} = 1$ , the relative error is  $\frac{1}{6} \approx 0.167$ .

*The bound.* The absolute row sums give  $\|\mathbf{A}\|_{\infty} = 3$  and  $\|\mathbf{A}^{-1}\|_{\infty} = 3$ , so  $\|\mathbf{A}\|_{\infty} \|\mathbf{A}^{-1}\|_{\infty} = 9$ . Also  $\|\mathbf{E}\|_{\infty} = \frac{1}{10}$  and  $r = 3 \cdot \frac{1}{10} = \frac{3}{10} < 1$ . With  $\mathbf{f} = \mathbf{0}$  the bound is

$$\frac{9}{1 - \frac{3}{10}} \cdot \frac{1/10}{3} = \frac{9}{7/10} \cdot \frac{1}{30} = \frac{3}{7} \approx 0.429.$$

So  $0.167 \leq 0.429$ : correct, and loose by a factor of about 2.6. Note the accounting. The matrix was disturbed by 3.3% of its size and the answer moved by 16.7%, an amplification of about 5; the theorem promised at most  $9/(1 - 0.3) \approx 12.9$ .

**Warning.** All four bounds of this section need  $\|\mathbf{I}\| = 1$ , which is a property of induced norms only. The Frobenius norm is submultiplicative (Example (The Frobenius norm is submultiplicative)) but has  $\|\mathbf{I}_n\|_F = \sqrt{n}$ . Take  $\mathbf{A} = 0$  in  $M_2(\mathbb{C})$ : then  $\|\mathbf{A}\|_F = 0 < 1$ , while

$$\|(\mathbf{I} - \mathbf{A})^{-1}\|_F = \|\mathbf{I}_2\|_F = \sqrt{2} > 1 = \frac{1}{1 - \|\mathbf{A}\|_F},$$

so Theorem 15.5.1 (b) is false for it. The repair is easy — Exercise (C3: Without  $\|\mathbf{I}\| = 1$ ) — but it is not the same inequality, and using the Frobenius norm here is the most common way to arrive at a wrong constant.

## 15.5.5 Exercises

### 15.5.5.1 A. Check your understanding

#### Exercise A1.

- State Theorem 15.5.1 in full, including which kind of norm it assumes.
- Chapter 9 §10 already proved that  $\sum_k \mathbf{A}^k$  converges to  $(\mathbf{I} - \mathbf{A})^{-1}$  when  $\rho(\mathbf{A}) < 1$ . Say precisely what this section adds, and what it gives up.
- Determine whether the following statement is correct, and justify your answer: if  $\mathbf{I} - \mathbf{A}$  is invertible then  $\|\mathbf{A}\| < 1$ .
- Where in the proof of Theorem 15.5.1 is  $\|\mathbf{I}\| = 1$  used?
- State what Corollary 15.5.2 says about the distance from an invertible matrix to the singular matrices.

*Solution.*

- If  $\|\cdot\|$  is the operator norm induced by a vector norm and  $\|\mathbf{A}\| = a < 1$ , then  $\mathbf{I} - \mathbf{A}$  is invertible with  $\sum_{k \geq 0} \mathbf{A}^k = (\mathbf{I} - \mathbf{A})^{-1}$ ; moreover  $\|(\mathbf{I} - \mathbf{A})^{-1}\| \leq 1/(1 - a)$ , the partial sums satisfy  $\|(\mathbf{I} - \mathbf{A})^{-1} - \mathbf{S}_K\| \leq a^{K+1}/(1 - a)$ , and  $\|(\mathbf{I} - \mathbf{A})^{-1}\| \geq 1/(1 + a)$ .
- It adds three numbers: a ceiling on the inverse, a geometric rate for the partial sums, and a floor. It gives up sharpness of the hypothesis:  $\rho(\mathbf{A}) < 1$  is necessary and sufficient for convergence, whereas  $\|\mathbf{A}\| < 1$  is only sufficient, and strictly stronger by Theorem 15.4.1.
- Incorrect.  $\mathbf{A} = 5\mathbf{I}$  has  $\mathbf{I} - \mathbf{A} = -4\mathbf{I}$ , invertible, while  $\|\mathbf{A}\| = 5$  in every induced norm. Invertibility of  $\mathbf{I} - \mathbf{A}$  says only  $1 \notin \text{spec}(\mathbf{A})$ .
- Twice, and both times to make a geometric sum start at 1: in (b), where  $\|\mathbf{A}^0\| = \|\mathbf{I}\| = 1$  is the  $k = 0$  term of  $\sum_k a^k$ ; and in (d), where  $1 = \|\mathbf{I}\|$  is the left-hand side of the estimate.
- It is at least  $1/\|\mathbf{B}^{-1}\|$ : every  $\mathbf{E}$  with  $\|\mathbf{E}\| < 1/\|\mathbf{B}^{-1}\|$  leaves  $\mathbf{B} + \mathbf{E}$  invertible.

## 15.5.5.2 B. Practice

**Exercise B1: Check, sum, compare.**

Let  $\mathbf{A} = \begin{pmatrix} 1/4 & 1/4 \\ 0 & 1/2 \end{pmatrix}$ .

- (a) Verify the hypothesis of [Theorem 15.5.1](#) for  $\|\cdot\|_\infty$ .
- (b) Compute  $(\mathbf{I} - \mathbf{A})^{-1}$  exactly and compare  $\|(\mathbf{I} - \mathbf{A})^{-1}\|_\infty$  with the bound of (b) and with the lower bound of (d).

*Solution.*

- (a) The absolute row sums are  $\frac{1}{2}$  and  $\frac{1}{2}$ , so  $\|\mathbf{A}\|_\infty = \frac{1}{2} < 1$  by [Theorem 15.3.3](#) (b).
- (b)  $\mathbf{I} - \mathbf{A} = \begin{pmatrix} 3/4 & -1/4 \\ 0 & 1/2 \end{pmatrix}$ , upper triangular with determinant  $\frac{3}{8}$ , so by [Theorem 0.9.8](#)

$$(\mathbf{I} - \mathbf{A})^{-1} = \frac{8}{3} \begin{pmatrix} 1/2 & 1/4 \\ 0 & 3/4 \end{pmatrix} = \begin{pmatrix} 4/3 & 2/3 \\ 0 & 2 \end{pmatrix}.$$

The absolute row sums are 2 and 2, so  $\|(\mathbf{I} - \mathbf{A})^{-1}\|_\infty = 2$ . The upper bound is  $1/(1 - \frac{1}{2}) = 2$ , attained exactly; the lower bound is  $1/(1 + \frac{1}{2}) = \frac{2}{3}$ , comfortably satisfied. The upper bound is attained because both rows of  $\mathbf{A}$  have the same absolute sum and no cancellation occurs anywhere in the geometric series.

**Exercise B2: A perturbed right-hand side.**

Let  $\mathbf{A} = \begin{pmatrix} 3 & 1 \\ 1 & 2 \end{pmatrix}$ ,  $\mathbf{b} = (4, 3)$  and  $\mathbf{f} = (1/10, 0)$ , with  $\mathbf{E} = 0$ . Solve both systems, compute the relative error in the  $\infty$ -norm, and compare with [Theorem 15.5.3](#).

*Solution.*  $\det \mathbf{A} = 5$ , so  $\mathbf{A}^{-1} = \frac{1}{5} \begin{pmatrix} 2 & -1 \\ -1 & 3 \end{pmatrix}$  by [Theorem 0.9.8](#), and  $\mathbf{x} = \mathbf{A}^{-1}\mathbf{b} = \frac{1}{5}(8 - 3, -4 + 9) = (1, 1)$ .

With  $\mathbf{b} + \mathbf{f} = (\frac{41}{10}, 3)$  we get  $\tilde{\mathbf{x}} = \frac{1}{5}(\frac{82}{10} - 3, -\frac{41}{10} + 9) = (\frac{52}{50}, \frac{49}{50})$ . Hence  $\tilde{\mathbf{x}} - \mathbf{x} = (\frac{2}{50}, -\frac{1}{50})$  and  $\|\tilde{\mathbf{x}} - \mathbf{x}\|_\infty = \frac{1}{25} = 0.04$ , a relative error of 0.04 since  $\|\mathbf{x}\|_\infty = 1$ .

For the bound:  $\|\mathbf{A}\|_\infty = 4$ ,  $\|\mathbf{A}^{-1}\|_\infty = \frac{4}{5}$ , so  $\|\mathbf{A}\|_\infty \|\mathbf{A}^{-1}\|_\infty = \frac{16}{5}$ . Here  $\mathbf{E} = 0$ , so  $r = 0$ , and  $\|\mathbf{f}\|_\infty / \|\mathbf{b}\|_\infty = \frac{1/10}{4} = \frac{1}{40}$ . The bound is  $\frac{16}{5} \cdot \frac{1}{40} = \frac{2}{25} = 0.08$ . So  $0.04 \leq 0.08$ , loose by a factor of exactly 2.

**Exercise B3: Staying invertible.**

Let  $\mathbf{B} = \begin{pmatrix} 2 & 0 \\ 0 & 1/2 \end{pmatrix}$ . Using  $\|\cdot\|_\infty$ , find a radius  $\delta > 0$  such that every  $\mathbf{E}$  with  $\|\mathbf{E}\|_\infty < \delta$  leaves  $\mathbf{B} + \mathbf{E}$  invertible, and bound  $\|(\mathbf{B} + \mathbf{E})^{-1}\|_\infty$  for  $\|\mathbf{E}\|_\infty \leq \frac{1}{4}$ . Is your  $\delta$  the largest possible?

*Solution.*  $\mathbf{B}^{-1} = \text{diag}(\frac{1}{2}, 2)$ , so  $\|\mathbf{B}^{-1}\|_\infty = 2$  and [Corollary 15.5.2](#) gives  $\delta = 1/\|\mathbf{B}^{-1}\|_\infty = \frac{1}{2}$ .

For  $\|\mathbf{E}\|_\infty \leq \frac{1}{4}$  we get  $r = 2 \cdot \frac{1}{4} = \frac{1}{2}$ , so part (a) of the corollary gives

$$\|(\mathbf{B} + \mathbf{E})^{-1}\|_\infty \leq \frac{2}{1 - \frac{1}{2}} = 4.$$

It is the largest possible here. Taking  $\mathbf{E} = \text{diag}(0, -\frac{1}{2})$  gives  $\|\mathbf{E}\|_\infty = \frac{1}{2}$  and  $\mathbf{B} + \mathbf{E} = \text{diag}(2, 0)$ , which is singular. So no radius larger than  $\frac{1}{2}$  works, and the estimate  $1/\|\mathbf{B}^{-1}\|$  is attained by this  $\mathbf{B}$ . (It is not attained by every  $\mathbf{B}$  in every norm; [Exercise \(C2: Distance to the singular matrices\)](#) returns to this.)

### 15.5.5.3 C. Going deeper

#### Exercise C1: The inverse to first order.

Let  $\|\mathbf{A}\| < 1$ .

- (a) Prove that  $\|(\mathbf{I} - \mathbf{A})^{-1} - \mathbf{I} - \mathbf{A}\| \leq \frac{\|\mathbf{A}\|^2}{1 - \|\mathbf{A}\|}$ .
- (b) Deduce that for a fixed  $\mathbf{M} \in M_n(\mathbb{C})$  and small real  $t$ ,

$$(\mathbf{I} - t\mathbf{M})^{-1} = \mathbf{I} + t\mathbf{M} + \mathbf{R}(t), \quad \|\mathbf{R}(t)\|/|t| \rightarrow 0$$

as  $t \rightarrow 0$ .

*Hint for (a): this is [Theorem 15.5.1](#) (c) with a particular  $K$ .*

*Solution.*

- (a) Take  $K = 1$  in [Theorem 15.5.1](#) (c):  $\mathbf{S}_1 = \mathbf{I} + \mathbf{A}$ , so

$$\|(\mathbf{I} - \mathbf{A})^{-1} - \mathbf{I} - \mathbf{A}\| \leq \frac{\|\mathbf{A}\|^{1+1}}{1 - \|\mathbf{A}\|} = \frac{\|\mathbf{A}\|^2}{1 - \|\mathbf{A}\|}.$$

- (b) If  $\mathbf{M} = 0$  the claim is trivial, so assume  $\mathbf{M} \neq 0$  and let  $|t| < 1/(2\|\mathbf{M}\|)$ . Then  $\|t\mathbf{M}\| = |t|\|\mathbf{M}\| < \frac{1}{2}$ , so (a) applies with  $\mathbf{A} = t\mathbf{M}$  and gives, with  $\mathbf{R}(t) = (\mathbf{I} - t\mathbf{M})^{-1} - \mathbf{I} - t\mathbf{M}$ ,

$$\|\mathbf{R}(t)\| \leq \frac{t^2\|\mathbf{M}\|^2}{1 - |t|\|\mathbf{M}\|} \leq 2t^2\|\mathbf{M}\|^2,$$

using  $1 - |t|\|\mathbf{M}\| > \frac{1}{2}$ . Hence  $\|\mathbf{R}(t)\|/|t| \leq 2|t|\|\mathbf{M}\|^2 \rightarrow 0$  as  $t \rightarrow 0$ . So  $t \mapsto (\mathbf{I} - t\mathbf{M})^{-1}$  is differentiable at  $t = 0$  with derivative  $\mathbf{M}$ , which is the case  $\mathbf{A}(t) = \mathbf{I} - t\mathbf{M}$  of the general formula for differentiating an inverse, proved in §06.

#### Exercise C2: Distance to the singular matrices.

Let  $\mathbf{B} \in M_n(\mathbb{C})$  be invertible and let  $\|\cdot\|$  be an induced matrix norm.

- (a) Prove that every singular  $\mathbf{S} \in M_n(\mathbb{C})$  satisfies  $\|\mathbf{B} - \mathbf{S}\| \geq 1/\|\mathbf{B}^{-1}\|$ .
- (b) For  $\mathbf{B} = \text{diag}(1, \varepsilon)$  with  $0 < \varepsilon < 1$  and the 2-norm, exhibit a singular  $\mathbf{S}$  attaining

the bound.

- (c) Hence say what  $\|\mathbf{B}^{-1}\|_2$  measures.

*Solution.*

- (a) Suppose  $\|\mathbf{B} - \mathbf{S}\| < 1/\|\mathbf{B}^{-1}\|$ . Apply [Corollary 15.5.2](#) with  $\mathbf{E} = \mathbf{S} - \mathbf{B}$ : since  $\|\mathbf{E}\| < 1/\|\mathbf{B}^{-1}\|$ , the matrix  $\mathbf{B} + \mathbf{E} = \mathbf{S}$  is invertible, contradicting the assumption that  $\mathbf{S}$  is singular. Hence  $\|\mathbf{B} - \mathbf{S}\| \geq 1/\|\mathbf{B}^{-1}\|$  for every singular  $\mathbf{S}$ .
- (b)  $\mathbf{B}^{-1} = \text{diag}(1, 1/\varepsilon)$ , a normal matrix, so its singular values are the moduli of its eigenvalues and  $\|\mathbf{B}^{-1}\|_2 = 1/\varepsilon$  by [Theorem 15.3.3](#) (c). The bound is therefore  $\varepsilon$ . Take  $\mathbf{S} = \text{diag}(1, 0)$ , which is singular; then  $\mathbf{B} - \mathbf{S} = \text{diag}(0, \varepsilon)$  and  $\|\mathbf{B} - \mathbf{S}\|_2 = \varepsilon$ . So the bound is attained.
- (c) The reciprocal  $1/\|\mathbf{B}^{-1}\|_2$  is a lower bound for the distance from  $\mathbf{B}$  to the singular matrices, attained in the example. So  $\|\mathbf{B}^{-1}\|_2$  is large exactly when  $\mathbf{B}$  is close to being singular, which is the geometric content behind the condition number of §08.

**Exercise C3: Without  $\|\mathbf{I}\| = 1$ .**

Let  $\|\cdot\|$  be any matrix norm on  $M_n(\mathbb{C})$  in the sense of [Definition 15.3.1](#), not necessarily induced, and let  $\|\mathbf{A}\| < 1$ .

- (a) Prove that  $\mathbf{I} - \mathbf{A}$  is invertible and that

$$\|(\mathbf{I} - \mathbf{A})^{-1}\| \leq \|\mathbf{I}\| + \frac{\|\mathbf{A}\|}{1 - \|\mathbf{A}\|}.$$

- (b) Verify that this bound is correct for the Frobenius norm on  $M_2(\mathbb{C})$  with  $\mathbf{A} = 0$ , where the bound of [Theorem 15.5.1](#) (b) fails.
- (c) Explain why the argument of (a) cannot be improved to  $1/(1 - \|\mathbf{A}\|)$  for a general matrix norm.

*Solution.*

- (a) By [Theorem 15.4.1](#) (b), which holds for every matrix norm,  $\rho(\mathbf{A}) \leq \|\mathbf{A}\| < 1$ ; so [Theorem 9.10.6](#) applies and gives both the invertibility of  $\mathbf{I} - \mathbf{A}$  and  $\mathbf{S}_K \rightarrow (\mathbf{I} - \mathbf{A})^{-1}$ . Submultiplicativity gives  $\|\mathbf{A}^k\| \leq \|\mathbf{A}\|^k$  for  $k \geq 1$ , so separating the term  $k = 0$ ,

$$\|\mathbf{S}_K\| \leq \|\mathbf{I}\| + \sum_{k=1}^K \|\mathbf{A}\|^k \leq \|\mathbf{I}\| + \frac{\|\mathbf{A}\|}{1 - \|\mathbf{A}\|}.$$

As in the proof of [Theorem 15.5.1](#) (b), [Corollary 15.2.4](#) and [Lemma 15.1.1](#) let the bound pass to the limit.

- (b) With  $\mathbf{A} = 0$ , the right-hand side is  $\|\mathbf{I}_2\|_F + 0 = \sqrt{2}$ , and the left-hand side is  $\|\mathbf{I}_2\|_F = \sqrt{2}$ . The bound holds, with equality; the bound  $1/(1 - 0) = 1$  of [Theorem 15.5.1](#) (b) does not.
- (c) The case  $\mathbf{A} = 0$  already decides it: the left-hand side is  $\|\mathbf{I}\|$ , so any valid bound must be at least  $\|\mathbf{I}\|$ , and  $1/(1 - \|\mathbf{A}\|) = 1$  there. A norm with  $\|\mathbf{I}\| > 1$  therefore falsifies the sharper form at once, and by [Theorem 15.3.2](#) (c) an induced norm

never has  $\|\mathbf{I}\| > 1$ . This is exactly the difference between a matrix norm and an induced one.

## 15.6 Bounds for the Exponential, and Two Derivatives

Chapter 9 §9 built  $e^{\mathbf{A}}$  out of a series and closed with a promise: “Chapter 15 will return to  $e^{\mathbf{A}}$  with norms available, which give quantitative bounds instead of exact formulas.” That is what this section does. It does not rebuild the exponential; [Definition 9.9.2](#) and [Theorem 9.9.3](#) stand as they are, and the Jordan form is not touched. What is new is that the series now has a size, so its sum has a ceiling, its truncation has an error bound, and its behavior as  $t \rightarrow \infty$  has a rate.

The second half takes the other step Chapter 9 could not: it differentiates. The derivative of the determinant and the derivative of the inverse are both needed later in the book, and both require one fact from calculus that linear algebra does not supply. That fact is stated before it is used.

**Throughout,**  $\|\cdot\|$  is the operator norm on  $M_n(\mathbb{C})$  induced by a norm on  $\mathbb{C}^n$  ([Definition 15.3.2](#)), so  $\|\mathbf{I}\| = 1$  and  $\|\mathbf{XY}\| \leq \|\mathbf{X}\|\|\mathbf{Y}\|$  ([Theorem 15.3.2](#) (c), (d)).

### 15.6.1 How big is the exponential

Chapter 9 §10’s criteria answered *whether*; the norm answers *how much*. The bound to aim at is the one the scalar case suggests:  $|e^z| \leq e^{|z|}$ , because the series has non-negative terms once every term is replaced by its modulus. The matrix proof is the same sentence with submultiplicativity in place of  $|zw| = |z||w|$ .

One scalar fact is needed and is not on the chapter’s list of imports. We name it here rather than let it in quietly.

*Remark. A quoted fact.* For every real  $s \geq 0$  the series  $\sum_{m \geq 0} s^m/m!$  converges, with sum  $e^s$ . This is fact (A1) of Chapter 9 §9, quoted there and quoted again here; this book does not prove it. Because its terms are non-negative, its partial sums increase to  $e^s$ , so every partial sum is at most  $e^s$  — that last step is **(A2), monotone convergence**, from the list in this chapter’s introduction.

#### Theorem 15.6.1 — Bounds for the Exponential

Let  $\mathbf{A} \in M_n(\mathbb{C})$  and write  $a = \|\mathbf{A}\|$ . Then

- (a)  $\|e^{\mathbf{A}}\| \leq e^a$ ;
- (b)  $\|e^{\mathbf{A}} - \mathbf{I}\| \leq e^a - 1$ ;
- (c)  $\|e^{\mathbf{A}} - \mathbf{I} - \mathbf{A}\| \leq e^a - 1 - a \leq \frac{a^2 e^a}{2}$ .

In particular  $\|e^{t\mathbf{A}}\| \leq e^{|t|a}$  for every real  $t$ .

*Idea.* All three are the same computation stopped at a different place: bound the norm of a partial sum by the corresponding partial sum of  $\sum s^m/m!$ , then let the partial sums converge. The only thing that needs care is the passage to the limit, and [Lemma 15.1.1](#) does it, exactly as in [Theorem 15.5.1](#). For the second inequality in (c), compare  $m!$  with  $2(m-2)!$ .

*Proof.* Write  $\mathbf{S}_M = \sum_{m=0}^M \mathbf{A}^m/m!$ . By [Theorem 9.9.3](#) the sequence  $(\mathbf{S}_M)$  converges entrywise to  $e^{\mathbf{A}}$ , hence  $\|\mathbf{S}_M - e^{\mathbf{A}}\| \rightarrow 0$  by [Corollary 15.2.4](#), hence  $\|\mathbf{S}_M\| \rightarrow \|e^{\mathbf{A}}\|$  by [Lemma 15.1.1](#). The same three steps apply to  $\mathbf{S}_M - \mathbf{I}$  and to  $\mathbf{S}_M - \mathbf{I} - \mathbf{A}$ . So in each part it suffices to bound the partial sums, uniformly in  $M$ .

- (a) Submultiplicativity gives  $\|\mathbf{A}^m\| \leq a^m$  for  $m \geq 1$ , by induction, and  $\|\mathbf{A}^0\| = \|\mathbf{I}\| = 1$ . By subadditivity and the quoted fact above,

$$\|\mathbf{S}_M\| \leq \sum_{m=0}^M \frac{\|\mathbf{A}^m\|}{m!} \leq \sum_{m=0}^M \frac{a^m}{m!} \leq e^a.$$

Letting  $M \rightarrow \infty$  gives  $\|e^{\mathbf{A}}\| \leq e^a$ .

- (b) The same estimate with the term  $m = 0$  removed:

$$\|\mathbf{S}_M - \mathbf{I}\| \leq \sum_{m=1}^M \frac{a^m}{m!} \leq e^a - 1,$$

since the partial sums of  $\sum_{m \geq 1} a^m/m!$  increase to  $e^a - 1$ .

- (c) Removing the terms  $m = 0$  and  $m = 1$  in the same way gives

$$\|e^{\mathbf{A}} - \mathbf{I} - \mathbf{A}\| \leq \sum_{m \geq 2} \frac{a^m}{m!} = e^a - 1 - a.$$

For the second inequality, note that  $m! = m(m-1)(m-2)! \geq 2(m-2)!$  for every  $m \geq 2$ , so  $a^m/m! \leq \frac{1}{2}a^2 \cdot a^{m-2}/(m-2)!$ . Summing over  $m \geq 2$  and substituting  $l = m - 2$ ,

$$\sum_{m \geq 2} \frac{a^m}{m!} \leq \frac{a^2}{2} \sum_{l \geq 0} \frac{a^l}{l!} = \frac{a^2 e^a}{2}.$$

The final claim is (a) applied to  $t\mathbf{A}$ , whose norm is  $|t|a$  by homogeneity. This proves the theorem. ■

Part (c) is the statement that  $e^{\mathbf{A}}$  agrees with  $\mathbf{I} + \mathbf{A}$  to second order in  $\|\mathbf{A}\|$ , with an explicit constant. It is the bound one uses when  $\mathbf{A}$  is a small perturbation, and it is the counterpart, for the exponential, of [Theorem 15.5.1](#) (c).

**Example. (Where the bound is exact, and where it is useless)**

Compare  $\|e^{\mathbf{A}}\|_2$  with  $e^{\|\mathbf{A}\|_2}$  for

- (a)  $\mathbf{A} = \text{diag}(2, 1)$ ;  
 (b)  $\mathbf{A} = \begin{pmatrix} 0 & 1 \\ -1 & 0 \end{pmatrix}$ , and more generally  $t\mathbf{A}$  for  $t > 0$ .

*Solution.*

- (a)  $\mathbf{A}$  is Hermitian, hence normal, so its singular values are the moduli of its eigenvalues and  $\|\mathbf{A}\|_2 = 2$  by [Theorem 15.3.3](#) (c). By [Definition 9.9.1](#) (or directly from the series, entry by entry)  $e^{\mathbf{A}} = \text{diag}(e^2, e)$ , again normal, so  $\|e^{\mathbf{A}}\|_2 = e^2$ . The bound  $e^{\|\mathbf{A}\|_2} = e^2$  is attained exactly. The same happens for every  $\mathbf{A} = c\mathbf{I}$  with  $c > 0$ , so no constant smaller than 1 can be inserted in [Theorem 15.6.1](#) (a).  
 (b)  $\mathbf{A}^2 = -\mathbf{I}$ , so  $\mathbf{A}^3 = -\mathbf{A}$ ,  $\mathbf{A}^4 = \mathbf{I}$ , and grouping the series by parity gives, using fact (A1) of Chapter 9 §09 for the series of cos and sin and its (A3) to rearrange a convergent series into its even and odd parts,

$$e^{t\mathbf{A}} = (\cos t)\mathbf{I} + (\sin t)\mathbf{A} = \begin{pmatrix} \cos t & \sin t \\ -\sin t & \cos t \end{pmatrix},$$

a real orthogonal matrix. Hence  $\|e^{t\mathbf{A}}\|_2 = 1$  for every  $t$ , while  $\|t\mathbf{A}\|_2 = t$  and the bound reads  $e^t$ . At  $t = 10$  the bound is about 22026 and the truth is 1.

**Warning.** The bound sees only the size of  $\mathbf{A}$ , never the direction of its spectrum. Part (b) above is the extreme case:  $\|e^{t\mathbf{A}}\|$  stays at 1 while  $e^{t\|\mathbf{A}\|}$  runs away. The reason is that [Theorem 15.6.1](#) throws away every cancellation in the series at the first step, when  $\|\mathbf{A}^m\|$  is replaced by  $\|\mathbf{A}\|^m$ . Any statement about  $e^{t\mathbf{A}}$  for large  $t$  must come from the eigenvalues instead, which is the subject of the next subsection.

**Quick check.** Why does [Theorem 15.6.1](#) (a) fail for the Frobenius norm, and what is the correct statement for it?

*Solution.* Because  $\|\mathbf{I}_n\|_F = \sqrt{n}$ , so already at  $\mathbf{A} = 0$  the claim reads  $\sqrt{n} \leq 1$ . Repeating the proof with the term  $m = 0$  kept separate gives the correct version for any matrix norm:  $\|e^{\mathbf{A}}\| \leq \|\mathbf{I}\| + e^{\|\mathbf{A}\|} - 1$ . This is the same correction as in [Exercise \(C3: Without  \$\|\mathbf{I}\| = 1\$ \)](#), and it comes from the same place: only an induced norm has  $\|\mathbf{I}\| = 1$ .

## 15.6.2 Decay in the stable case

The warning above says the bound  $e^{t\|\mathbf{A}\|}$  is the wrong instrument for large  $t$ . The right one reads the eigenvalues, and the bridge to the discrete theory of §04 is the observation that the exponential turns the *real part* of an eigenvalue into a *modulus*.

### Lemma 15.6.2 — The Spectrum of the Exponential

Let  $\mathbf{A} \in M_n(\mathbb{C})$ . Then  $\text{spec}(e^{\mathbf{A}}) = \{e^\lambda : \lambda \in \text{spec}(\mathbf{A})\}$ , and

$$\rho(e^{\mathbf{A}}) = e^\mu, \quad \mu = \max\{\text{Re } \lambda : \lambda \in \text{spec}(\mathbf{A})\}.$$



*Idea.* The eigenvalues of  $e^{\mathbf{A}}$  are the numbers  $e^\lambda$  for  $\lambda \in \text{spec}(\mathbf{A})$ , which the Jordan form makes visible. Taking moduli turns  $e^\lambda$  into  $e^{\text{Re } \lambda}$ , and the largest of finitely many such numbers is the one with the largest exponent.

*Proof.* By [Theorem 9.3.1](#) there are an invertible  $\mathbf{P}$  and a Jordan matrix  $\mathbf{J}$  with  $\mathbf{A} = \mathbf{PJP}^{-1}$ , and the diagonal of  $\mathbf{J}$  lists the eigenvalues  $\lambda_1, \dots, \lambda_n$  of  $\mathbf{A}$  with multiplicity. By [Theorem 9.9.4](#) (a),  $e^{\mathbf{J}} = e^{\mathbf{P}^{-1}\mathbf{A}\mathbf{P}} = \mathbf{P}^{-1}e^{\mathbf{A}}\mathbf{P}$ . By [Theorem 9.9.3](#) each block  $e^{\mathbf{J}_k(\lambda)}$  is upper triangular with  $e^\lambda$  on its diagonal, so  $e^{\mathbf{J}}$  is upper triangular with diagonal  $e^{\lambda_1}, \dots, e^{\lambda_n}$ ; hence  $\text{spec}(e^{\mathbf{J}}) = \{e^{\lambda_i}\}$  by [Theorem 8.7.4](#) (b). Similar matrices have the same characteristic polynomial ([Theorem 6.10.3](#)), so  $\text{spec}(e^{\mathbf{A}}) = \text{spec}(e^{\mathbf{J}})$ , which is the first claim.

For the second,  $|e^\lambda| = e^{\text{Re } \lambda}$  by fact (A1) of Chapter 9 §09, and  $s \mapsto e^s$  is non-decreasing on  $\mathbb{R}$  — for  $h \geq 0$ ,  $e^{s+h} = e^s e^h \geq e^s$  by fact (A1) of Chapter 9 §09 — so the largest of the finitely many numbers  $e^{\text{Re } \lambda_i}$  is  $e^\mu$ . This proves the lemma. ■

Chapter 11 §10 called  $\mathbf{A}$  **stable** when every  $\lambda \in \text{spec}(\mathbf{A})$  has  $\text{Re } \lambda < 0$  ([Definition 11.10.2](#)). In the language of the lemma,  $\mathbf{A}$  is stable exactly when  $\rho(e^{\mathbf{A}}) < 1$  — the continuous-time condition and Chapter 9 §10's discrete-time condition are the same condition, read through the exponential.

### Theorem 15.6.3 — Exponential Decay of a Stable System

Let  $\mathbf{A} \in M_n(\mathbb{C})$ , let  $\mu = \max\{\text{Re } \lambda : \lambda \in \text{spec}(\mathbf{A})\}$ , and let  $\nu > \mu$  be real. Then there is a constant  $C \geq 1$ , depending on  $\mathbf{A}$ , on  $\nu$  and on the norm, with

$$\|e^{t\mathbf{A}}\| \leq C e^{\nu t} \quad \text{for every } t \geq 0.$$

In particular, if  $\mathbf{A}$  is **stable** then  $\nu$  may be taken negative, and every solution of  $\mathbf{x}' = \mathbf{A}\mathbf{x}$  satisfies

$$\|\mathbf{x}(t)\| \leq C e^{\nu t} \|\mathbf{x}(0)\| \xrightarrow{t \rightarrow \infty} 0.$$

*Idea.* Sample the flow at integer times and use §04. The matrix  $\mathbf{M} = e^{-\nu} e^{\mathbf{A}}$  has  $\rho(\mathbf{M}) = e^{\mu-\nu} < 1$  by [Lemma 15.6.2](#), so [Corollary 15.4.5](#) bounds all of its powers by one constant; unwinding the powers gives  $\|e^{k\mathbf{A}}\| \leq C_0 e^{\nu k}$  at every integer  $k$ . Between the integers,  $e^{t\mathbf{A}} = e^{(t-k)\mathbf{A}} e^{k\mathbf{A}}$  with  $0 \leq t - k < 1$ , and the crude bound [Theorem 15.6.1](#) (a) is more than enough to cover a gap of length 1. The whole point is that the crude bound is applied only on a bounded interval, where crudeness costs a constant and not a rate.

*Proof.* Put  $\mathbf{M} = e^{-\nu} e^{\mathbf{A}}$ . Multiplying a matrix by a scalar  $c$  multiplies every eigenvalue by  $c$ , so by [Lemma 15.6.2](#),

$$\rho(\mathbf{M}) = e^{-\nu} \rho(e^{\mathbf{A}}) = e^{-\nu} e^\mu = e^{\mu-\nu} < 1,$$

using  $\mu - \nu < 0$ . Fix any  $q$  with  $\rho(\mathbf{M}) < q < 1$ . By [Corollary 15.4.5](#) (b) there is  $C_0 \geq 1$  with  $\|\mathbf{M}^k\| \leq C_0 q^k \leq C_0$  for every integer  $k \geq 0$ .

Since  $\mathbf{A}$  commutes with itself, [Theorem 9.9.4](#) (d) gives  $e^{(s+u)\mathbf{A}} = e^{s\mathbf{A}} e^{u\mathbf{A}}$  for all real  $s, u$ ;

in particular  $(e^{\mathbf{A}})^k = e^{k\mathbf{A}}$  by induction on  $k$ . Hence  $\mathbf{M}^k = e^{-\nu k} e^{k\mathbf{A}}$ , and

$$\|e^{k\mathbf{A}}\| = e^{\nu k} \|\mathbf{M}^k\| \leq C_0 e^{\nu k} \quad (k = 0, 1, 2, \dots).$$

Now let  $t \geq 0$  and let  $k$  be the integer part of  $t$ , so that  $k \geq 0$  and  $0 \leq t - k < 1$ . Then  $e^{t\mathbf{A}} = e^{(t-k)\mathbf{A}} e^{k\mathbf{A}}$ , so by submultiplicativity and [Theorem 15.6.1](#) (a),

$$\|e^{t\mathbf{A}}\| \leq \|e^{(t-k)\mathbf{A}}\| \|e^{k\mathbf{A}}\| \leq e^{(t-k)\|\mathbf{A}\|} C_0 e^{\nu k} \leq C_0 e^{\|\mathbf{A}\|} e^{\nu k},$$

using  $0 \leq t - k < 1$  in the last step. Finally  $e^{\nu k} = e^{\nu t} e^{-\nu(t-k)} \leq e^{\nu t} e^{|\nu|}$ , again because  $0 \leq t - k < 1$ . So the constant

$$C = C_0 e^{\|\mathbf{A}\| + |\nu|} \geq 1$$

works for every  $t \geq 0$ .

For the last statement, suppose  $\mathbf{A}$  is stable, so  $\mu < 0$ , and choose  $\nu \in (\mu, 0)$ . By [Theorem 9.9.6](#) the unique solution with the given initial value is  $\mathbf{x}(t) = e^{t\mathbf{A}} \mathbf{x}(0)$ , so  $\|\mathbf{x}(t)\| \leq \|e^{t\mathbf{A}}\| \|\mathbf{x}(0)\| \leq C e^{\nu t} \|\mathbf{x}(0)\|$  by [Theorem 15.3.2](#) (a). Since  $\nu < 0$ ,  $e^{\nu t} \rightarrow 0$  as  $t \rightarrow \infty$  — this is fact (A5) of Chapter 9 §09 with  $j = 0$ , quoted there and quoted again here — so  $\|\mathbf{x}(t)\| \rightarrow 0$ , and by [Corollary 15.2.4](#) the vectors  $\mathbf{x}(t)$  tend to  $\mathbf{0}$ . This proves the theorem. ■

### Example. (Decay after a detour)

Let  $\mathbf{A} = \begin{pmatrix} -1 & 5 \\ 0 & -1 \end{pmatrix}$ . Find  $\mu$ , compute  $\|e^{t\mathbf{A}}\|_\infty$  exactly, and find the smallest constant  $C$  that works in [Theorem 15.6.3](#) for  $\nu = -\frac{1}{2}$ .

*Solution. The eigenvalues.*  $\mathbf{A}$  is upper triangular with both diagonal entries  $-1$ , so  $\text{spec}(\mathbf{A}) = \{-1\}$  ([Theorem 8.7.4](#) (b)) and  $\mu = -1 < 0$ : the matrix is stable.

*The exponential.* Write  $\mathbf{A} = -\mathbf{I} + 5\mathbf{N}$  with  $\mathbf{N} = \begin{pmatrix} 0 & 1 \\ 0 & 0 \end{pmatrix}$ . The two summands commute and  $\mathbf{N}^2 = \mathbf{0}$ , so [Corollary 9.9.5](#) gives

$$e^{t\mathbf{A}} = e^{-t}(\mathbf{I} + 5t\mathbf{N}) = e^{-t} \begin{pmatrix} 1 & 5t \\ 0 & 1 \end{pmatrix}.$$

The absolute row sums are  $e^{-t}(1 + 5t)$  and  $e^{-t}$ , so  $\|e^{t\mathbf{A}}\|_\infty = e^{-t}(1 + 5t)$  for  $t \geq 0$  by [Theorem 15.3.3](#) (b).

*The transient.* The function  $t \mapsto e^{-t}(1 + 5t)$  has derivative  $e^{-t}(4 - 5t)$ , so it **ris**es from 1 at  $t = 0$  to  $5e^{-4/5} \approx 2.247$  at  $t = \frac{4}{5}$ , and only then decays. A stable matrix does not decay from the start.

*The constant.* We need the smallest  $C$  with  $e^{-t}(1 + 5t) \leq C e^{-t/2}$  for all  $t \geq 0$ , that is,  $C = \max_{t \geq 0} (1 + 5t)e^{-t/2}$ . The derivative of  $(1 + 5t)e^{-t/2}$  is  $e^{-t/2}(5 - \frac{1}{2}(1 + 5t))$ , which is positive for  $t < \frac{9}{5}$  and negative for  $t > \frac{9}{5}$ , so by fact (A6) of the chapter introduction the maximum is at  $t = \frac{9}{5}$  and

$$C = 10 e^{-9/10} \approx 4.066.$$

The rate  $e^{-t/2}$  is honest and the constant pays for the hump.

*Remark.* **What this settles from Chapter 11 §10, and what it does not.** That section introduced the Lyapunov equation  $\mathbf{A}\mathbf{X} + \mathbf{X}\mathbf{A}^* = -\mathbf{Q}$  and the word *stable* (Definition 11.10.2). It then stated, without proof, that a stable  $\mathbf{A}$  and a positive definite  $\mathbf{Q}$  produce a positive definite solution  $\mathbf{X}$ , and conversely; it assigned the positivity half to Chapter 12, and assigned to this chapter “the consequence for differential equations — that a stable  $\mathbf{A}$  makes every solution of  $\dot{\mathbf{x}} = \mathbf{A}\mathbf{x}$  decay, with  $\mathbf{x}^*\mathbf{X}\mathbf{x}$  as the quantity that measures the decay”.

Theorem 15.6.3 delivers the decay, with a rate, and does so without the Lyapunov equation at all. What it does not deliver is the existence of a positive definite  $\mathbf{X}$  for a stable  $\mathbf{A}$ : that statement is not proved anywhere in this book, and nothing here depends on it. What can be said cheaply is the other half of the connection: **any** positive definite  $\mathbf{X}$  satisfying  $\mathbf{A}^*\mathbf{X} + \mathbf{X}\mathbf{A} = -\mathbf{Q}$  with  $\mathbf{Q} \succ 0$  does measure the decay, in the sense that  $\mathbf{x}^*\mathbf{X}\mathbf{x}$  is strictly decreasing along every non-zero solution of  $\mathbf{x}' = \mathbf{A}\mathbf{x}$ . That is Exercise (C3: A quantity that measures the decay), and it is three lines once the product rule below is available. Note the transpose: the equation the decay computation wants is Definition 11.10.2 for the pair  $(\mathbf{A}^*, \mathbf{Q})$ , and  $\mathbf{A}$  is stable if and only if  $\mathbf{A}^*$  is, since the eigenvalues of  $\mathbf{A}^*$  are the conjugates of those of  $\mathbf{A}$ .

### 15.6.3 Differentiating a matrix-valued function

A matrix whose entries vary — a family  $\mathbf{A}(t)$ , a curve in  $M_n(F)$  — is differentiated entrywise, exactly as Chapter 9 §09 differentiated  $t \mapsto e^{t\mathbf{A}}$ . The two derivatives this book needs later, of det and of inversion, are algebraic facts with one analytic ingredient, and the ingredient is named first.

#### Definition 15.6.1 — Derivative of a Matrix-Valued Function

Let  $I \subseteq \mathbb{R}$  be an open interval and  $\mathbf{A}: I \rightarrow M_{m \times n}(\mathbb{C})$ , written  $\mathbf{A}(t) = (a_{ij}(t))$ . Call  $\mathbf{A}$  **differentiable** at  $t_0 \in I$  if every entry function  $a_{ij}: I \rightarrow \mathbb{C}$  is differentiable at  $t_0$ , and then set

$$\mathbf{A}'(t_0) := (a'_{ij}(t_0)) \in M_{m \times n}(\mathbb{C}).$$

The prime here always means the derivative in  $t$ ; no dual map (Definition 4.4.1) appears anywhere in this section, so the notation cannot collide. By Corollary 15.2.4 the definition can be restated without coordinates:  $\mathbf{A}$  is differentiable at  $t_0$  with derivative  $\mathbf{B}$  exactly when

$$\left\| \frac{\mathbf{A}(t) - \mathbf{A}(t_0)}{t - t_0} - \mathbf{B} \right\| \rightarrow 0 \quad (t \rightarrow t_0),$$

in any norm at all. Both readings are used below, the entrywise one to import calculus and the norm one to take limits.

*Remark.* **The calculus this section quotes.** Let  $f, g: I \rightarrow \mathbb{C}$  be differentiable at  $t_0$ .

- (D1) (*Linearity and the product rule.*)  $f+g$  and  $fg$  are differentiable at  $t_0$ , with  $(f+g)' = f' + g'$  and  $(fg)' = f'g + fg'$ ; a constant function has derivative 0.
- (D2) (*Differentiable implies continuous.*)  $f$  is continuous at  $t_0$ .

Neither is proved here. (D1) is Chapter 9 §09's fact (A3), read for complex-valued functions of a real variable; (D2) is one more standard fact of the same kind, quoted here for the first time in this book. Both are of a piece with Chapter 13 §06's Taylor assumption: standard one-variable calculus, quoted so that the theorems below have hypotheses to stand on. Everything that follows them is linear algebra.

### Lemma 15.6.4 — Product Rule for Matrices

Let  $\mathbf{A}: I \rightarrow M_{m \times n}(\mathbb{C})$  and  $\mathbf{B}: I \rightarrow M_{n \times p}(\mathbb{C})$  be differentiable at  $t_0$ . Then so is  $t \mapsto \mathbf{A}(t)\mathbf{B}(t)$ , with

$$(\mathbf{AB})'(t_0) = \mathbf{A}'(t_0)\mathbf{B}(t_0) + \mathbf{A}(t_0)\mathbf{B}'(t_0).$$

*Proof.* Fix a position  $(i, k)$ . The entry function is  $t \mapsto \sum_{j=1}^n a_{ij}(t)b_{jk}(t)$ , a finite sum of products of functions differentiable at  $t_0$ . By (D1) it is differentiable at  $t_0$  with derivative

$$\sum_{j=1}^n (a'_{ij}(t_0)b_{jk}(t_0) + a_{ij}(t_0)b'_{jk}(t_0)),$$

which is the  $(i, k)$ -entry of  $\mathbf{A}'(t_0)\mathbf{B}(t_0) + \mathbf{A}(t_0)\mathbf{B}'(t_0)$ . Since this holds at every position, Definition 15.6.1 gives the claim. Note that the order of the factors is preserved throughout: matrices do not commute, and the product rule must not be written  $2\mathbf{AA}'$  even for  $\mathbf{B} = \mathbf{A}$ . This proves the lemma. ■

## 15.6.4 The derivative of the inverse

Nothing so far says that  $t \mapsto \mathbf{A}(t)^{-1}$  is even defined near  $t_0$ , let alone differentiable. Both come from §05.

### Theorem 15.6.5 — Derivative of the Inverse

Let  $\mathbf{A}: I \rightarrow M_n(\mathbb{C})$  be differentiable at  $t_0 \in I$ , with  $\mathbf{A}(t_0)$  invertible. Then  $\mathbf{A}(t)$  is invertible for all  $t$  in some open interval around  $t_0$ , the function  $t \mapsto \mathbf{A}(t)^{-1}$  is differentiable at  $t_0$ , and

$$(\mathbf{A}^{-1})'(t_0) = -\mathbf{A}(t_0)^{-1}\mathbf{A}'(t_0)\mathbf{A}(t_0)^{-1}.$$

*Idea.* The formula is what one gets by differentiating  $\mathbf{AA}^{-1} = \mathbf{I}$  with Lemma 15.6.4 and solving — but that argument presupposes what has to be proved, namely that  $\mathbf{A}^{-1}$  is differentiable. So run it in the honest order. The difference of two inverses factors as  $\mathbf{X}^{-1} - \mathbf{Y}^{-1} = \mathbf{X}^{-1}(\mathbf{Y} - \mathbf{X})\mathbf{Y}^{-1}$ ; divide by  $t - t_0$  and the middle factor becomes a difference quotient, which converges by hypothesis, while the outer factor converges because inversion is continuous (Corollary 15.5.2). The limit of the product is the product of the limits, and that is the formula.

*Proof.* Write  $\mathbf{A}_0 = \mathbf{A}(t_0)$ . By (D2) every entry of  $\mathbf{A}$  is continuous at  $t_0$ , so  $\mathbf{A}(t) \rightarrow \mathbf{A}_0$  entrywise as  $t \rightarrow t_0$ , hence  $\|\mathbf{A}(t) - \mathbf{A}_0\| \rightarrow 0$  by [Corollary 15.2.4](#). By [Corollary 15.5.2](#) there is therefore an open interval  $J \ni t_0$  on which  $\|\mathbf{A}(t) - \mathbf{A}_0\| < 1/\|\mathbf{A}_0^{-1}\|$ , so that  $\mathbf{A}(t)$  is invertible for  $t \in J$ , and on which  $\|\mathbf{A}(t)^{-1} - \mathbf{A}_0^{-1}\| \rightarrow 0$  as  $t \rightarrow t_0$ .

Let  $t \in J$  with  $t \neq t_0$ . Multiplying  $\mathbf{A}_0 - \mathbf{A}(t)$  on the left by  $\mathbf{A}(t)^{-1}$  and on the right by  $\mathbf{A}_0^{-1}$ ,

$$\mathbf{A}(t)^{-1} - \mathbf{A}_0^{-1} = -\mathbf{A}(t)^{-1}(\mathbf{A}(t) - \mathbf{A}_0)\mathbf{A}_0^{-1}.$$

Divide by  $t - t_0$ :

$$\frac{\mathbf{A}(t)^{-1} - \mathbf{A}_0^{-1}}{t - t_0} = -\mathbf{A}(t)^{-1} \frac{\mathbf{A}(t) - \mathbf{A}_0}{t - t_0} \mathbf{A}_0^{-1}.$$

As  $t \rightarrow t_0$  the middle factor tends to  $\mathbf{A}'(t_0)$ , by [Definition 15.6.1](#). The left factor tends to  $\mathbf{A}_0^{-1}$ , by the first paragraph. Products of convergent matrix families converge to the product of the limits: if  $\|\mathbf{X}(t) - \mathbf{X}\| \rightarrow 0$  and  $\|\mathbf{Y}(t) - \mathbf{Y}\| \rightarrow 0$  then

$$\|\mathbf{X}(t)\mathbf{Y}(t) - \mathbf{XY}\| \leq \|\mathbf{X}(t) - \mathbf{X}\|\|\mathbf{Y}(t)\| + \|\mathbf{X}\|\|\mathbf{Y}(t) - \mathbf{Y}\|,$$

and  $\|\mathbf{Y}(t)\|$  is bounded near  $t_0$  because  $\|\mathbf{Y}(t)\| \leq \|\mathbf{Y}\| + \|\mathbf{Y}(t) - \mathbf{Y}\|$ . Applying this twice, the right-hand side converges to  $-\mathbf{A}_0^{-1}\mathbf{A}'(t_0)\mathbf{A}_0^{-1}$ . Hence the difference quotient on the left converges, which by [Definition 15.6.1](#) is exactly the differentiability of  $t \mapsto \mathbf{A}(t)^{-1}$  at  $t_0$ , with the stated derivative. This proves the theorem. ■

*Warning.* The two copies of  $\mathbf{A}^{-1}$  sit on either side and cannot be combined. The scalar formula  $(1/a)' = -a'/a^2$  tempts one to write  $-\mathbf{A}^{-2}\mathbf{A}'$ , and that is wrong unless  $\mathbf{A}'(t_0)$  commutes with  $\mathbf{A}(t_0)$ . For  $\mathbf{A}(t) = \begin{pmatrix} 1 & t \\ 0 & 2 \end{pmatrix}$  at  $t = 0$ : we have  $\mathbf{A}(0)^{-1} = \text{diag}(1, \frac{1}{2})$  and  $\mathbf{A}'(0) = \begin{pmatrix} 0 & 1 \\ 0 & 0 \end{pmatrix}$ , so the correct answer is  $-\text{diag}(1, \frac{1}{2})\begin{pmatrix} 0 & 1 \\ 0 & 0 \end{pmatrix}\text{diag}(1, \frac{1}{2}) = \begin{pmatrix} 0 & -1/2 \\ 0 & 0 \end{pmatrix}$ , whereas  $-\mathbf{A}(0)^{-2}\mathbf{A}'(0) = -\begin{pmatrix} 1 & 0 \\ 0 & 1/4 \end{pmatrix}\begin{pmatrix} 0 & 1 \\ 0 & 0 \end{pmatrix} = \begin{pmatrix} 0 & -1 \\ 0 & 0 \end{pmatrix}$ .

## 15.6.5 Jacobi's formula

The determinant is a polynomial in the entries, so differentiating it needs no analysis beyond (D1); the work is in recognizing the answer.

### Theorem 15.6.6 — Jacobi's Formula

Let  $\mathbf{A}: I \rightarrow M_n(\mathbb{C})$  be differentiable at  $t_0 \in I$ . Then  $t \mapsto \det \mathbf{A}(t)$  is differentiable at  $t_0$ , with

$$\left. \frac{d}{dt} \right|_{t_0} \det \mathbf{A}(t) = \text{tr}(\text{adj } \mathbf{A}(t_0) \mathbf{A}'(t_0)).$$

If moreover  $\mathbf{A}(t_0)$  is invertible, this equals  $\det \mathbf{A}(t_0) \text{tr}(\mathbf{A}(t_0)^{-1} \mathbf{A}'(t_0))$ .

*Idea.* Use the Leibniz formula, which writes  $\det$  as a sum of  $n!$  products of  $n$  entries, one from each column. The product rule differentiates each such product into  $n$  terms, the  $j$ -th of which has column  $j$  differentiated and the rest untouched. Collecting by  $j$  gives  $n$  determinants, the  $j$ -th being  $\mathbf{A}$  with its  $j$ -th column replaced by  $\mathbf{a}'_j$ . Expanding that

determinant along column  $j$  (Theorem 6.6.1) produces  $\sum_i a'_{ij} C_{ij}$ , and summing over  $j$  gives a double sum which is precisely a trace against the adjugate.

*Proof.* For  $n = 1$  the statement reads  $\frac{d}{dt}\big|_{t_0} a_{11}(t) = a'_{11}(t_0)$ , since  $\text{adj } \mathbf{A} = (1)$  by Definition 6.6.2, so assume  $n \geq 2$ ; that is what lets us speak of cofactors. Write  $\mathbf{A}(t) = (a_{ij}(t))$  with columns  $\mathbf{a}_1(t), \dots, \mathbf{a}_n(t)$ , and write  $C_{ij}$  for the cofactors of  $\mathbf{A}(t_0)$  (Definition 6.6.1). By Theorem 6.4.3,

$$\det \mathbf{A}(t) = \sum_{\sigma \in S_n} \text{sgn}(\sigma) a_{\sigma(1)1}(t) \cdots a_{\sigma(n)n}(t),$$

a finite sum of products of  $n$  functions each differentiable at  $t_0$ . By (D1), applied  $n - 1$  times to each product and then to the sum,  $\det \mathbf{A}$  is differentiable at  $t_0$  with

$$\frac{d}{dt}\bigg|_{t_0} \det \mathbf{A} = \sum_{j=1}^n \sum_{\sigma \in S_n} \text{sgn}(\sigma) a'_{\sigma(j)j}(t_0) \prod_{l \neq j} a_{\sigma(l)l}(t_0),$$

where the outer sum collects, for each  $j$ , the terms in which the  $j$ -th factor was the one differentiated. By Theorem 6.4.3 read backwards, the inner sum is the determinant of the matrix  $\mathbf{A}_j$  obtained from  $\mathbf{A}(t_0)$  by replacing its  $j$ -th column with  $\mathbf{a}'_j(t_0)$ . Expanding  $\det \mathbf{A}_j$  along that column with Theorem 6.6.1 (a), and using that the cofactors of column  $j$  do not involve column  $j$  and are therefore the cofactors  $C_{ij}$  of  $\mathbf{A}(t_0)$  itself,

$$\det \mathbf{A}_j = \sum_{i=1}^n a'_{ij}(t_0) C_{ij}.$$

Summing over  $j$ ,

$$\frac{d}{dt}\bigg|_{t_0} \det \mathbf{A} = \sum_{j=1}^n \sum_{i=1}^n C_{ij} a'_{ij}(t_0).$$

It remains to recognize the right-hand side. By Definition 6.6.2,  $(\text{adj } \mathbf{A}(t_0))_{ji} = C_{ij}$ , so

$$\text{tr}(\text{adj } \mathbf{A}(t_0) \mathbf{A}'(t_0)) = \sum_{j=1}^n \sum_{i=1}^n (\text{adj } \mathbf{A}(t_0))_{ji} a'_{ij}(t_0),$$

which is the same double sum. For the last claim, if  $\mathbf{A}(t_0)$  is invertible then  $\text{adj } \mathbf{A}(t_0) = \det \mathbf{A}(t_0) \mathbf{A}(t_0)^{-1}$  by Theorem 6.6.2, and the trace is linear (Theorem 0.9.6), so the scalar  $\det \mathbf{A}(t_0)$  comes out in front. This proves the theorem. ■

### Corollary 15.6.7 — The Determinant's Derivative at the Identity Is the Trace

For every  $\mathbf{B} \in M_n(\mathbb{C})$ ,

$$\frac{d}{dt}\bigg|_{t=0} \det(\mathbf{I} + t\mathbf{B}) = \text{tr } \mathbf{B}.$$

*Proof.* Apply Theorem 15.6.6 to  $\mathbf{A}(t) = \mathbf{I} + t\mathbf{B}$ , which is differentiable with  $\mathbf{A}'(t) = \mathbf{B}$  and  $\mathbf{A}(0) = \mathbf{I}$ . Since  $\mathbf{I}$  is invertible with  $\mathbf{I}^{-1} = \mathbf{I}$  and  $\det \mathbf{I} = 1$ , the second form of the

formula gives  $\text{tr}(\mathbf{IB}) = \text{tr } \mathbf{B}$ .

■

So the determinant, which is a product of  $n$  numbers, has the trace, a sum of  $n$  numbers, as its linearization at  $\mathbf{I}$ . That is the honest source of the identity  $\det e^{\mathbf{A}} = e^{\text{tr } \mathbf{A}}$  of [Theorem 9.9.4](#) (e), and [Exercise \(C1: The determinant of the exponential, again\)](#) reproves that identity along these lines.

**Example. (Both derivatives on one family)**

Let  $\mathbf{A}(t) = \begin{pmatrix} 1+t & t^2 \\ t & 1 \end{pmatrix}$ .

- (a) Compute  $\frac{d}{dt} \det \mathbf{A}(t)$  directly and by [Theorem 15.6.6](#).
- (b) Compute  $(\mathbf{A}^{-1})'(0)$  by [Theorem 15.6.5](#), and check it against a direct expansion.

*Solution.*

- (a) *Directly.*  $\det \mathbf{A}(t) = (1+t) \cdot 1 - t^2 \cdot t = 1 + t - t^3$ , so the derivative is  $1 - 3t^2$ .  
*By the formula.*  $\text{adj } \mathbf{A}(t) = \begin{pmatrix} 1 & -t^2 \\ -t & 1+t \end{pmatrix}$  ([Definition 6.6.2](#)) and  $\mathbf{A}'(t) = \begin{pmatrix} 1 & 2t \\ 1 & 0 \end{pmatrix}$ , so

$$\text{adj } \mathbf{A}(t) \mathbf{A}'(t) = \begin{pmatrix} 1-t^2 & 2t \\ 1 & -2t^2 \end{pmatrix},$$

whose trace is  $1 - t^2 - 2t^2 = 1 - 3t^2$ . The two agree.

- (b)  $\mathbf{A}(0) = \mathbf{I}$  and  $\mathbf{A}'(0) = \begin{pmatrix} 1 & 0 \\ 1 & 0 \end{pmatrix}$ , so [Theorem 15.6.5](#) gives

$$(\mathbf{A}^{-1})'(0) = -\mathbf{I} \begin{pmatrix} 1 & 0 \\ 1 & 0 \end{pmatrix} \mathbf{I} = \begin{pmatrix} -1 & 0 \\ -1 & 0 \end{pmatrix}.$$

*Check.*  $\mathbf{A}(t)^{-1} = \frac{1}{1+t-t^3} \begin{pmatrix} 1 & -t^2 \\ -t & 1+t \end{pmatrix}$ . Near  $t = 0$  the scalar factor is  $1 - t + (\text{terms in } t^2)$ , so

$$\mathbf{A}(t)^{-1} = \begin{pmatrix} 1-t & 0 \\ -t & 1 \end{pmatrix} + (\text{terms in } t^2),$$

whose derivative at 0 is  $\begin{pmatrix} -1 & 0 \\ -1 & 0 \end{pmatrix}$ , as predicted. The formula did the work that the quotient rule would otherwise have had to do four times.

*Quick check.* [Theorem 15.6.6](#) holds with no invertibility hypothesis, yet the second form of the formula has one. Where exactly is invertibility used, and what does the formula say at a  $t_0$  where  $\det \mathbf{A}(t_0) = 0$ ?

*Solution.* Only in the last line, where  $\text{adj } \mathbf{A}(t_0)$  is rewritten as  $\det \mathbf{A}(t_0) \mathbf{A}(t_0)^{-1}$ . The adjugate itself is defined for every square matrix ([Definition 6.6.2](#)), so the first form survives. At a  $t_0$  with  $\det \mathbf{A}(t_0) = 0$  it still gives a number: for  $\mathbf{A}(t) = \text{diag}(t, t)$  at  $t_0 = 0$  we get  $\text{adj } \mathbf{A}(0) = 0$ , so the derivative of  $\det \mathbf{A}(t) = t^2$  is 0 there, which is correct. The second form would read “0 times something undefined”.

## 15.6.6 Exercises

### 15.6.6.1 A. Check your understanding

#### Exercise A1.

- State the three bounds of [Theorem 15.6.1](#), and say which property of the norm each one needs.
- Determine whether the following statement is correct, and justify your answer: if  $\|\mathbf{A}\|_2$  is large then  $\|e^{\mathbf{A}}\|_2$  is large.
- State [Theorem 15.6.3](#), and say what the constant  $C$  is paying for.
- State Jacobi's formula ([Theorem 15.6.6](#)), and name the fact from calculus its proof quotes.
- Write down the derivative of  $t \mapsto \mathbf{A}(t)^{-1}$  and explain in one sentence why it is not  $-\mathbf{A}^{-2}\mathbf{A}'$ .

*Solution.*

- $\|e^{\mathbf{A}}\| \leq e^{\|\mathbf{A}\|}$ ,  $\|e^{\mathbf{A}} - \mathbf{I}\| \leq e^{\|\mathbf{A}\|} - 1$ , and  $\|e^{\mathbf{A}} - \mathbf{I} - \mathbf{A}\| \leq e^{\|\mathbf{A}\|} - 1 - \|\mathbf{A}\| \leq \|\mathbf{A}\|^2 e^{\|\mathbf{A}\|} / 2$ . All three need subadditivity (to bound a sum term by term) and submultiplicativity (to get  $\|\mathbf{A}^m\| \leq \|\mathbf{A}\|^m$ ); the first also needs  $\|\mathbf{I}\| = 1$ , which is what makes the  $m = 0$  term equal 1.
- Incorrect. For  $\mathbf{A} = t \begin{pmatrix} 0 & 1 \\ -1 & 0 \end{pmatrix}$  we have  $\|\mathbf{A}\|_2 = t$ , arbitrarily large, while  $e^{\mathbf{A}}$  is a rotation matrix and  $\|e^{\mathbf{A}}\|_2 = 1$  ([Example \(Where the bound is exact, and where it is useless\)](#)).
- If  $\mu$  is the largest real part of an eigenvalue of  $\mathbf{A}$  and  $\nu > \mu$ , there is  $C \geq 1$  with  $\|e^{t\mathbf{A}}\| \leq Ce^{\nu t}$  for all  $t \geq 0$ . The constant pays for the transient:  $\|e^{t\mathbf{A}}\|$  may first grow, as in [Example \(Decay after a detour\)](#), and it also pays for the gap between  $\nu$  and  $\mu$ .
- $\frac{d}{dt} \det \mathbf{A}(t) = \text{tr}(\text{adj } \mathbf{A}(t) \mathbf{A}'(t))$ , for a matrix-valued  $\mathbf{A}$  differentiable at  $t$ . The quoted fact is (D1): the linearity of differentiation and the product rule for complex-valued functions of a real variable.
- $(\mathbf{A}^{-1})' = -\mathbf{A}^{-1}\mathbf{A}'\mathbf{A}^{-1}$ . The two copies of  $\mathbf{A}^{-1}$  sit on opposite sides of  $\mathbf{A}'$ , and matrices do not commute, so they may not be collected into  $\mathbf{A}^{-2}$  unless  $\mathbf{A}'$  commutes with  $\mathbf{A}$ .

### 15.6.6.2 B. Practice

#### Exercise B1: Bound against truth.

For each matrix, compute  $\|e^{\mathbf{A}}\|_{\infty}$  exactly and compare with  $e^{\|\mathbf{A}\|_{\infty}}$ .

- $\mathbf{A} = \begin{pmatrix} 0 & 3 \\ 0 & 0 \end{pmatrix}$ .
- $\mathbf{A} = \text{diag}(1, -4)$ .

*Solution.*



- (a)  $\mathbf{A}^2 = 0$ , so the series stops:  $e^{\mathbf{A}} = \mathbf{I} + \mathbf{A} = \begin{pmatrix} 1 & 3 \\ 0 & 1 \end{pmatrix}$ , with  $\|e^{\mathbf{A}}\|_{\infty} = 4$ . Also  $\|\mathbf{A}\|_{\infty} = 3$ , so the bound is  $e^3 \approx 20.09$ . Correct, and loose by a factor of about 5.
- (b)  $e^{\mathbf{A}} = \text{diag}(e, e^{-4})$ , so  $\|e^{\mathbf{A}}\|_{\infty} = e \approx 2.718$ . Here  $\|\mathbf{A}\|_{\infty} = 4$  and the bound is  $e^4 \approx 54.6$ . The bound is governed by the entry of largest modulus,  $-4$ , while the exponential is governed by the largest **real part**, 1. This is the discrepancy [Theorem 15.6.3](#) exists to remove.

### Exercise B2: A rate for a stable matrix.

Let  $\mathbf{A} = \begin{pmatrix} -2 & 4 \\ 0 & -2 \end{pmatrix}$ .

- (a) Compute  $\mu$  and  $e^{t\mathbf{A}}$ , and give  $\|e^{t\mathbf{A}}\|_{\infty}$  for  $t \geq 0$ .
- (b) Find the largest value of  $\|e^{t\mathbf{A}}\|_{\infty}$  on  $t \geq 0$ .
- (c) Find the smallest  $C$  that works in [Theorem 15.6.3](#) for  $\nu = -1$ .

*Solution.*

- (a)  $\mathbf{A}$  is upper triangular with diagonal  $-2, -2$ , so  $\text{spec}(\mathbf{A}) = \{-2\}$  ([Theorem 8.7.4](#) (b)) and  $\mu = -2$ . Writing  $\mathbf{A} = -2\mathbf{I} + 4\mathbf{N}$  with  $\mathbf{N} = \begin{pmatrix} 0 & 1 \\ 0 & 0 \end{pmatrix}$ , the summands commute and  $\mathbf{N}^2 = 0$ , so by [Corollary 9.9.5](#)

$$e^{t\mathbf{A}} = e^{-2t} \begin{pmatrix} 1 & 4t \\ 0 & 1 \end{pmatrix}, \quad \|e^{t\mathbf{A}}\|_{\infty} = e^{-2t}(1 + 4t).$$

- (b) The derivative of  $e^{-2t}(1 + 4t)$  is  $e^{-2t}(4 - 2 - 8t) = e^{-2t}(2 - 8t)$ , which is positive for  $t < \frac{1}{4}$  and negative after, so by fact (A6) the maximum is at  $t = \frac{1}{4}$ . The value there is  $2e^{-1/2} \approx 1.213$ .
- (c) We need the smallest  $C$  with  $e^{-2t}(1 + 4t) \leq Ce^{-t}$ , that is,  $C = \max_{t \geq 0} (1 + 4t)e^{-t}$ . Its derivative is  $e^{-t}(4 - 1 - 4t) = e^{-t}(3 - 4t)$ , positive for  $t < \frac{3}{4}$  and negative after, so by fact (A6) the maximum is at  $t = \frac{3}{4}$  and  $C = 4e^{-3/4} \approx 1.890$ .

### Exercise B3: Two derivatives.

Let  $\mathbf{A}(t) = \begin{pmatrix} \cos t & -\sin t \\ \sin t & \cos t \end{pmatrix}$ .

- (a) Compute  $\frac{d}{dt} \det \mathbf{A}(t)$  by [Theorem 15.6.6](#) and check it against the exact value of  $\det \mathbf{A}(t)$ .
- (b) Compute  $(\mathbf{A}^{-1})'(t)$  by [Theorem 15.6.5](#) and check it by differentiating  $\mathbf{A}(t)^{-1}$  directly.

*Solution.*

- (a)  $\det \mathbf{A}(t) = \cos^2 t + \sin^2 t = 1$ , so the derivative is 0. By the formula:  $\text{adj } \mathbf{A}(t) = \begin{pmatrix} \cos t & \sin t \\ -\sin t & \cos t \end{pmatrix}$  and  $\mathbf{A}'(t) = \begin{pmatrix} -\sin t & -\cos t \\ \cos t & -\sin t \end{pmatrix}$ , so

$$\text{adj } \mathbf{A}(t) \mathbf{A}'(t) = \begin{pmatrix} 0 & -1 \\ 1 & 0 \end{pmatrix},$$

whose trace is 0. The two agree. (The diagonal entries are  $-\cos t \sin t + \sin t \cos t$  and  $\sin t \cos t - \cos t \sin t$ , both zero.)

- (b) Since  $\det \mathbf{A}(t) = 1$ ,  $\mathbf{A}(t)^{-1} = \operatorname{adj} \mathbf{A}(t) = \begin{pmatrix} \cos t & \sin t \\ -\sin t & \cos t \end{pmatrix}$ , whose entrywise derivative is  $\begin{pmatrix} -\sin t & \cos t \\ -\cos t & -\sin t \end{pmatrix}$ . By the formula, using the product  $\operatorname{adj} \mathbf{A}(t) \mathbf{A}'(t)$  computed in (a),

$$-\mathbf{A}^{-1} \mathbf{A}' \mathbf{A}^{-1} = - \begin{pmatrix} 0 & -1 \\ 1 & 0 \end{pmatrix} \begin{pmatrix} \cos t & \sin t \\ -\sin t & \cos t \end{pmatrix} = \begin{pmatrix} -\sin t & \cos t \\ -\cos t & -\sin t \end{pmatrix}.$$

The two agree.

### 15.6.6.3 C. Going deeper

#### Exercise C1: The determinant of the exponential, again.

Let  $\mathbf{A} \in M_n(\mathbb{C})$  and put  $\varphi(t) = \det e^{t\mathbf{A}}$ .

- (a) Prove that  $\varphi$  is differentiable with  $\varphi'(t) = (\operatorname{tr} \mathbf{A})\varphi(t)$ .  
 (b) Hence deduce  $\det e^{\mathbf{A}} = e^{\operatorname{tr} \mathbf{A}}$ , recovering [Theorem 9.9.4](#) (e).

*Hint for (b): consider  $\psi(t) = e^{-t \operatorname{tr} \mathbf{A}} \varphi(t)$ .*

*Solution.*

- (a) The function  $t \mapsto e^{t\mathbf{A}}$  is differentiable with derivative  $\mathbf{A}e^{t\mathbf{A}}$  ([Theorem 9.9.4](#) (b)), and  $e^{t\mathbf{A}}$  is invertible with inverse  $e^{-t\mathbf{A}}$  ([Theorem 9.9.4](#) (c)). So [Theorem 15.6.6](#) applies in its second form:

$$\varphi'(t) = \det(e^{t\mathbf{A}}) \operatorname{tr}(e^{-t\mathbf{A}} \mathbf{A} e^{t\mathbf{A}}).$$

By [Theorem 0.9.6](#) (3) with the two factors  $e^{-t\mathbf{A}}$  and  $\mathbf{A}e^{t\mathbf{A}}$ , we get  $\operatorname{tr}(e^{-t\mathbf{A}} \mathbf{A} e^{t\mathbf{A}}) = \operatorname{tr}(\mathbf{A} e^{t\mathbf{A}} e^{-t\mathbf{A}}) = \operatorname{tr} \mathbf{A}$ , giving  $\varphi'(t) = (\operatorname{tr} \mathbf{A})\varphi(t)$ .

- (b) Put  $c = \operatorname{tr} \mathbf{A}$  and  $\psi(t) = e^{-ct} \varphi(t)$ . By the product rule (D1) and (a),

$$\psi'(t) = -c e^{-ct} \varphi(t) + e^{-ct} c \varphi(t) = 0$$

for every  $t$ , where  $\frac{d}{dt} e^{-ct} = -c e^{-ct}$  is [Theorem 9.9.4](#) (b) in the case  $n = 1$ . A differentiable function with vanishing derivative is constant — fact (A4) of Chapter 9 §09 — so  $\psi(t) = \psi(0) = \det(e^0) = \det \mathbf{I} = 1$ . Hence  $\varphi(t) = e^{ct}$ , and  $t = 1$  gives  $\det e^{\mathbf{A}} = e^{\operatorname{tr} \mathbf{A}}$ .

#### Exercise C2: Comparing two exponentials.

Let  $\mathbf{A}, \mathbf{B} \in M_n(\mathbb{C})$  with  $\mathbf{AB} = \mathbf{BA}$ .

- (a) Prove that  $\|e^{\mathbf{A}} - e^{\mathbf{B}}\| \leq e^{\|\mathbf{B}\|} (e^{\|\mathbf{A}-\mathbf{B}\|} - 1)$ .  
 (b) Deduce that  $\mathbf{M} \mapsto e^{\mathbf{M}}$  is continuous at every  $\mathbf{M}$  that commutes with a neighborhood of itself, and identify those  $\mathbf{M}$ : they are exactly the scalar matrices  $c\mathbf{I}$ .  
 (c) Say exactly where the commuting hypothesis is used, and why the argument cannot simply be dropped.

*Solution.*

- (a) Since  $\mathbf{A}$  and  $\mathbf{B}$  commute, so do  $\mathbf{B}$  and  $\mathbf{A} - \mathbf{B}$ . By [Theorem 9.9.4](#) (d),  $e^{\mathbf{A}} = e^{\mathbf{B} + (\mathbf{A} - \mathbf{B})} = e^{\mathbf{B}} e^{\mathbf{A} - \mathbf{B}}$ . Hence

$$e^{\mathbf{A}} - e^{\mathbf{B}} = e^{\mathbf{B}}(e^{\mathbf{A} - \mathbf{B}} - \mathbf{I}),$$

and submultiplicativity with [Theorem 15.6.1](#) (a) and (b) gives

$$\|e^{\mathbf{A}} - e^{\mathbf{B}}\| \leq \|e^{\mathbf{B}}\| \|e^{\mathbf{A} - \mathbf{B}} - \mathbf{I}\| \leq e^{\|\mathbf{B}\|} (e^{\|\mathbf{A} - \mathbf{B}\|} - 1).$$

- (b) Take  $\mathbf{B} = c\mathbf{I}$ , which commutes with every matrix. For  $\mathbf{A}$  with  $\|\mathbf{A} - c\mathbf{I}\| = \delta$ , part (a) gives  $\|e^{\mathbf{A}} - e^{c\mathbf{I}}\| \leq e^{|c|} (e^{\delta} - 1)$ , and  $e^{\delta} - 1 \rightarrow 0$  as  $\delta \rightarrow 0$ , because  $t \mapsto e^t$  is differentiable by fact (A2) of Chapter 9 §09 and so continuous at 0. So  $e^{\mathbf{A}} \rightarrow e^{c\mathbf{I}}$  as  $\mathbf{A} \rightarrow c\mathbf{I}$ .
- (c) Only in the first line, where  $e^{\mathbf{B} + (\mathbf{A} - \mathbf{B})}$  is split as  $e^{\mathbf{B}} e^{\mathbf{A} - \mathbf{B}}$ ; [Theorem 9.9.4](#) (d) has commutation as a hypothesis, and Chapter 9 §09 exhibits matrices with  $e^{\mathbf{A} + \mathbf{B}} \neq e^{\mathbf{A}} e^{\mathbf{B}}$ . Without it there is no factorization to take norms of, and a different argument is needed. (The inequality itself is in fact true in general, but proving it needs an integral representation of  $e^{\mathbf{A}} - e^{\mathbf{B}}$ , which this book does not have.)

### Exercise C3: A quantity that measures the decay.

Let  $\mathbf{A}, \mathbf{X}, \mathbf{Q} \in M_n(\mathbb{C})$  with  $\mathbf{X}$  and  $\mathbf{Q}$  positive definite ([Definition 12.1.1](#)) and

$$\mathbf{A}^* \mathbf{X} + \mathbf{X} \mathbf{A} = -\mathbf{Q}.$$

Let  $\mathbf{x}: \mathbb{R} \rightarrow \mathbb{C}^n$  be differentiable with  $\mathbf{x}' = \mathbf{A}\mathbf{x}$ , and put  $V(t) = \mathbf{x}(t)^* \mathbf{X} \mathbf{x}(t)$ .

- (a) Prove that  $V$  is real-valued and differentiable, with  $V'(t) = \mathbf{x}(t)^* (\mathbf{A}^* \mathbf{X} + \mathbf{X} \mathbf{A}) \mathbf{x}(t)$ .
- (b) Deduce that if  $\mathbf{x}(0) \neq \mathbf{0}$ , then  $V$  is strictly decreasing.
- (c) Explain how this relates to the Lyapunov equation of [Definition 11.10.2](#), whose unknown sits between  $\mathbf{A}$  and  $\mathbf{A}^*$  the other way round.

*Hint for (a):*  $\mathbf{x}^*$  is a  $1 \times n$  matrix-valued function, and  $\overline{f'} = \overline{f'}$  for a differentiable  $f: \mathbb{R} \rightarrow \mathbb{C}$ .

*Solution.*

- (a)  $V(t)$  is a  $1 \times 1$  matrix, and  $V(t)^* = \mathbf{x}^* \mathbf{X}^* \mathbf{x} = \mathbf{x}^* \mathbf{X} \mathbf{x} = V(t)$  since  $\mathbf{X}$  is Hermitian, so  $V(t)$  is real. Each entry of  $t \mapsto \mathbf{x}(t)^*$  is the conjugate of an entry of  $\mathbf{x}$ , hence differentiable with  $(\mathbf{x}^*)' = (\mathbf{x}')^* = (\mathbf{A}\mathbf{x})^* = \mathbf{x}^* \mathbf{A}^*$ . By [Lemma 15.6.4](#), applied twice to the product  $\mathbf{x}^* \cdot \mathbf{X} \cdot \mathbf{x}$  and using  $\mathbf{X}' = 0$ ,

$$V'(t) = \mathbf{x}^* \mathbf{A}^* \mathbf{X} \mathbf{x} + \mathbf{x}^* \mathbf{X} \mathbf{A} \mathbf{x} = \mathbf{x}^* (\mathbf{A}^* \mathbf{X} + \mathbf{X} \mathbf{A}) \mathbf{x}.$$

- (b) By hypothesis  $V'(t) = -\mathbf{x}(t)^* \mathbf{Q} \mathbf{x}(t)$ . If  $\mathbf{x}(t_1) = \mathbf{0}$  for some  $t_1$ , then by [Theorem 9.9.6](#)  $\mathbf{x}(t) = e^{t\mathbf{A}} \mathbf{x}(0)$  for every  $t$ , and  $e^{t_1 \mathbf{A}}$  is invertible by [Theorem 9.9.4](#) (c), so  $\mathbf{x}(0) = \mathbf{0}$ , contradicting the hypothesis. So  $\mathbf{x}(t) \neq \mathbf{0}$  for every  $t$ , and  $\mathbf{Q} \succ 0$  gives  $\mathbf{x}(t)^* \mathbf{Q} \mathbf{x}(t) > 0$ , hence  $V'(t) < 0$  for every  $t$ . A differentiable real function with everywhere negative derivative is strictly decreasing; that last step is the mean value theorem, one more fact of one-variable calculus quoted here and not proved.

- (c) [Definition 11.10.2](#) writes the equation as  $\mathbf{A}\mathbf{X} + \mathbf{X}\mathbf{A}^* = -\mathbf{Q}$ ; the equation used here is that one for the pair  $(\mathbf{A}^*, \mathbf{Q})$ . The two are interchanged by replacing  $\mathbf{A}$  with  $\mathbf{A}^*$ , which changes nothing about stability, since the eigenvalues of  $\mathbf{A}^*$  are the complex conjugates of those of  $\mathbf{A}$  and conjugation preserves real parts. So the decay of  $V$  is exactly the statement Chapter 11 §10 pointed forward to, for whichever of the two conventions one adopts — granted a positive definite solution  $\mathbf{X}$ , whose existence is not proved in this book.

## 15.7 Eigenvalues Move Continuously

Chapter 11 §01 proved that every complex matrix has diagonalizable matrices arbitrarily close to it ([Corollary 11.1.6](#)), then admitted that the word “close” was doing work no definition supported, and handed the repair to this chapter. The repair is now in place: a norm on  $M_n(\mathbb{C})$  says how close two matrices are, and [Corollary 15.2.4](#) says that it does not matter which norm. This section asks the question that could not even be posed before that point. If the matrix moves a little, how far do its eigenvalues move?

The answer has two halves, and the second is the one worth remembering. The eigenvalues do move only a little — provided they are read as an unordered list, which is the only way they can be read. But “a little” is not proportional to the movement of the matrix, and for a matrix with a large Jordan block it is spectacularly larger.

**Throughout, matrices are complex**, as in Section 4, so that an  $n \times n$  matrix always has exactly  $n$  eigenvalues counted with algebraic multiplicity ([Corollary 5.5.5](#)).

### 15.7.1 From matrices to polynomials

The eigenvalues of  $\mathbf{A}$  are the roots of  $p_{\mathbf{A}}$  ([Theorem 8.2.1](#)), with algebraic multiplicity meaning multiplicity as a root ([Definition 8.3.1](#)). So the question splits in two: do the coefficients of  $p_{\mathbf{A}}$  move a little when  $\mathbf{A}$  does, and do the roots of a polynomial move a little when its coefficients do? The first half is easy and we do it now; the whole difficulty is in the second.

The reason the first half is easy is that everything in sight is a polynomial in the entries. It is worth isolating that observation once, because it will be used four times in this section.

#### Lemma 15.7.1 — Polynomial Maps of the Entries Are Continuous

Let  $\Phi: M_{p \times q}(\mathbb{C}) \rightarrow M_{r \times s}(\mathbb{C})$  be a map each of whose output entries is given by a fixed polynomial, with complex coefficients, in the  $pq$  entries of the input. Fix any norm on the source and any norm on the target. Then  $\Phi$  is **continuous**: whenever  $\|\mathbf{A}_k - \mathbf{A}\| \rightarrow 0$  we have  $\|\Phi(\mathbf{A}_k) - \Phi(\mathbf{A})\| \rightarrow 0$ , and equivalently, for every  $\mathbf{A}$  and every  $\varepsilon > 0$  there is  $\delta > 0$  such that  $\|\mathbf{B} - \mathbf{A}\| < \delta$  forces  $\|\Phi(\mathbf{B}) - \Phi(\mathbf{A})\| < \varepsilon$ . The same holds for a map into  $\mathbb{C}$ .

*Idea.* Two reductions and one piece of arithmetic. Convergence in any norm is entrywise convergence, by [Corollary 15.2.4](#), in the source and in the target alike; so the claim is about  $pq$  complex sequences producing  $rs$  complex sequences. That reduces it to the statement that a polynomial in finitely many complex variables respects limits, which is

the algebra of limits.

*Proof.* Suppose  $\|\mathbf{A}_k - \mathbf{A}\| \rightarrow 0$ . By [Corollary 15.2.4](#), each entry  $(\mathbf{A}_k)_{ij} \rightarrow a_{ij}$ . Sums and products of convergent complex sequences converge to the sum and the product of the limits: for sums this is [Theorem 0.6.4](#), and for products it follows from

$$|z_k w_k - zw| \leq |z_k| |w_k - w| + |w| |z_k - z|$$

together with the fact that a convergent sequence is bounded. A polynomial is built from its variables and constants by finitely many sums and products, so each entry of  $\Phi(\mathbf{A}_k)$  converges to the corresponding entry of  $\Phi(\mathbf{A})$ . Applying [Corollary 15.2.4](#) in  $M_{r \times s}(\mathbb{C})$  gives  $\|\Phi(\mathbf{A}_k) - \Phi(\mathbf{A})\| \rightarrow 0$ .

For the equivalence of the two formulations, suppose the  $\varepsilon$ - $\delta$  statement failed at some  $\mathbf{A}$  for some  $\varepsilon > 0$ . Then for each  $k \geq 1$  the value  $\delta = 1/k$  is not good enough, so there is  $\mathbf{B}_k$  with  $\|\mathbf{B}_k - \mathbf{A}\| < 1/k$  and  $\|\Phi(\mathbf{B}_k) - \Phi(\mathbf{A})\| \geq \varepsilon$ . The sequence  $(\mathbf{B}_k)$  contradicts the sequential statement just proved. This proves the lemma. ■

Now the first half of the question is answered. Write  $p_{\mathbf{A}}(x) = x^n + c_{n-1}(\mathbf{A})x^{n-1} + \cdots + c_0(\mathbf{A})$ , which is legitimate because  $p_{\mathbf{A}}$  is monic of degree  $n$  ([Theorem 6.10.2](#)). Each  $c_j$  is a polynomial in the entries of  $\mathbf{A}$ : the Leibniz sum of [Definition 6.10.1](#) expresses  $\det(x\mathbf{I} - \mathbf{A})$  as a signed sum of products of the entries of  $x\mathbf{I} - \mathbf{A}$ , each of which is  $x - a_{jj}$  or  $-a_{ij}$ , and collecting the powers of  $x$  leaves every coefficient a polynomial with integer coefficients in the  $a_{ij}$ . So  $\mathbf{A} \mapsto (c_0(\mathbf{A}), \dots, c_{n-1}(\mathbf{A}))$  is continuous by [Lemma 15.7.1](#).

That leaves the second half: roots against coefficients. Before stating what is true there, we settle what cannot be true.

## 15.7.2 Why the answer must be about the multiset

The statement a reader would write down first is “each eigenvalue is a continuous function of the matrix”. It is not merely unproved; it is meaningless as written, because the eigenvalues come with no numbering — and the natural repair, “there is a way of numbering them that makes each one continuous”, is false. One two-by-two family shows it, and the failure is the ordinary square root.

### Proposition 15.7.2 — There Is No Continuous Choice of a Single Eigenvalue

For  $t \in \mathbb{C}$  put

$$\mathbf{A}(t) = \begin{pmatrix} 0 & 1 \\ t & 0 \end{pmatrix},$$

so that  $p_{\mathbf{A}(t)}(x) = x^2 - t$  and the eigenvalues of  $\mathbf{A}(t)$  are the two square roots of  $t$ . Then there is **no** continuous function  $\lambda: D \rightarrow \mathbb{C}$  on the closed unit disc  $D = \{t \in \mathbb{C} : |t| \leq 1\}$  such that  $\lambda(t)$  is an eigenvalue of  $\mathbf{A}(t)$  for every  $t \in D$ . Consequently there is no continuous  $\Lambda: M_2(\mathbb{C}) \rightarrow \mathbb{C}$  with  $\Lambda(\mathbf{B}) \in \text{spec}(\mathbf{B})$  for every  $\mathbf{B}$ .

*Idea.* Walk  $t$  once around the unit circle. The two roots  $\pm\sqrt{t}$  trade places on the way round, so a choice that is continuous along the walk comes home as the *other* root. The bookkeeping is done by dividing the chosen root by one fixed root, which turns the walk into a continuous function with only the two values 1 and  $-1$ ; on an interval such a function cannot jump, and it must.

*Proof.* The characteristic polynomial is  $\det\begin{pmatrix} x & -1 \\ -t & x \end{pmatrix} = x^2 - t$ , so  $\mu$  is an eigenvalue of  $\mathbf{A}(t)$  if and only if  $\mu^2 = t$  (Theorem 8.2.1).

Suppose such a  $\lambda$  exists, so  $\lambda(t)^2 = t$  for all  $|t| \leq 1$ . Define  $g: [0, 2\pi] \rightarrow \mathbb{C}$  by

$$g(\theta) = \lambda(e^{i\theta}) e^{-i\theta/2}.$$

It is continuous, being a composite and a product of continuous functions — the elementary algebra of limits recorded in the chapter introduction, applied to  $\theta \mapsto e^{i\theta}$ , to  $\lambda$ , and to  $\theta \mapsto e^{-i\theta/2}$ . Squaring,

$$g(\theta)^2 = \lambda(e^{i\theta})^2 e^{-i\theta} = e^{i\theta} e^{-i\theta} = 1,$$

so  $g(\theta) \in \{1, -1\}$  for every  $\theta$ . In particular  $g$  is real-valued, and a continuous real-valued function on an interval taking only the values 1 and  $-1$  is constant: if it took both, the intermediate value theorem would produce a  $\theta$  with  $g(\theta) = 0$ . Hence  $g(2\pi) = g(0)$ .

But  $e^{i \cdot 0} = e^{i \cdot 2\pi} = 1$ , so  $g(0) = \lambda(1)$  while  $g(2\pi) = \lambda(1)e^{-i\pi} = -\lambda(1)$ . Therefore  $\lambda(1) = -\lambda(1)$ , that is  $\lambda(1) = 0$ , contradicting  $\lambda(1)^2 = 1$ . So no such  $\lambda$  exists.

For the last sentence, if  $\Lambda$  were such a function then  $t \mapsto \Lambda(\mathbf{A}(t))$  would be continuous, because  $t \mapsto \mathbf{A}(t)$  is continuous by Lemma 15.7.1, and it would select an eigenvalue of  $\mathbf{A}(t)$  for each  $t$ . This proves the proposition. ■

The intermediate value theorem is quoted here, and it is the only place in this chapter that uses it; it is fact (A5) of the chapter introduction, borrowed and not proved.

Two things are worth reading off the argument. First, the obstruction is not exotic:  $\mathbf{A}(0) = \mathbf{J}_2(0)$  is the standard nilpotent matrix, and the trouble happens exactly at the parameter where the two eigenvalues collide. Second, nothing is wrong with the pair  $\{\sqrt{t}, -\sqrt{t}\}$ , which shrinks to  $\{0, 0\}$  as  $t \rightarrow 0$  in any reasonable sense. It is only the attempt to call one of them “the first” that fails. This is why the correct statement speaks of the roots as a **multiset**: a list with multiplicities and no order.

### 15.7.3 The imported fact

Here is what is true. It is the one statement in this chapter that the book neither proves nor can prove with the tools it has built.

**Theorem 15.7.3 — Continuous Dependence of the Roots on the Coefficients**

Let  $n \geq 1$  and let

$$p(x) = x^n + c_{n-1}x^{n-1} + \cdots + c_1x + c_0 \in \mathbb{C}[x]$$

be monic, with roots  $\mu_1, \dots, \mu_n \in \mathbb{C}$  listed with multiplicity ([Corollary 5.5.5](#)). Let  $\varepsilon > 0$ . Then there is  $\delta > 0$  with the following property. If

$$q(x) = x^n + b_{n-1}x^{n-1} + \cdots + b_1x + b_0$$

is monic of the **same degree**  $n$  and  $|b_j - c_j| < \delta$  for every  $j$ , then the roots  $\nu_1, \dots, \nu_n$  of  $q$ , listed with multiplicity, can be **numbered** so that

$$|\nu_i - \mu_i| < \varepsilon \quad (i = 1, \dots, n).$$

**This book does not prove this theorem, and quotes it.** A proof needs complex analysis: one encircles each root of  $p$  by a small circle on which  $p$  does not vanish, counts the roots inside by a contour integral, and observes that an integer-valued quantity depending continuously on the coefficients cannot change. Nothing of that machinery is available here, and manufacturing a substitute would take a chapter. The statement is therefore recorded in the manner this book has used before — Chapter 13 §06 with Taylor’s theorem, Chapter 10 with the theorem of Abel and Ruffini: named at the point of use, never buried inside a proof that presents itself as self-contained, and never strengthened in passing. What *can* be proved with the tools at hand is that the numbering in the statement has to be chosen afresh for each  $q$ , and that was [Proposition 15.7.2](#).

Two clauses of the statement carry weight. **Monic of the same degree** is essential: the roots of  $\delta x^2 + x$  are 0 and  $-1/\delta$ , which runs off to infinity as  $\delta \rightarrow 0$ , while the limit polynomial  $x$  has the single root 0. Allowing the leading coefficient to vanish destroys everything, and monic polynomials of a fixed degree are exactly the family where it cannot. And **numbered so that** is the multiset formulation: the permutation matching up the two lists depends on  $q$ , and [Proposition 15.7.2](#) says it must.

## 15.7.4 Eigenvalues, and the spectral radius

Now the two halves are assembled.

### Corollary 15.7.4 — Eigenvalues Depend Continuously on the Matrix

Let  $\mathbf{A} \in M_n(\mathbb{C})$  with eigenvalues  $\lambda_1, \dots, \lambda_n$  listed with algebraic multiplicity, let  $\|\cdot\|$  be any norm on  $M_n(\mathbb{C})$ , and let  $\varepsilon > 0$ . Then there is  $\delta > 0$  such that every  $\mathbf{B} \in M_n(\mathbb{C})$  with  $\|\mathbf{B} - \mathbf{A}\| < \delta$  has eigenvalues  $\mu_1, \dots, \mu_n$ , listed with algebraic multiplicity, which can be numbered so that  $|\mu_i - \lambda_i| < \varepsilon$  for every  $i$ .

*Idea.* The characteristic polynomial’s coefficients are polynomials in the entries, so they move continuously with  $\mathbf{A}$ ; the quoted root theorem then moves the roots. The only work is composing the two estimates, choosing the tolerance for the coefficients small enough that the roots come out within  $\varepsilon$ .



*Proof.* By Theorem 6.10.2 the polynomials  $p_{\mathbf{A}}$  and  $p_{\mathbf{B}}$  are monic of degree  $n$ , and by Theorem 8.2.1 together with Definition 8.3.1 their root lists with multiplicity are the eigenvalue lists with algebraic multiplicity. Let  $\delta' > 0$  be the number supplied by Theorem 15.7.3 for  $p = p_{\mathbf{A}}$  and this  $\varepsilon$ . The map  $\mathbf{B} \mapsto (c_0(\mathbf{B}), \dots, c_{n-1}(\mathbf{B}))$  is continuous by Lemma 15.7.1, as noted after that lemma, so there is  $\delta > 0$  such that  $\|\mathbf{B} - \mathbf{A}\| < \delta$  forces  $|c_j(\mathbf{B}) - c_j(\mathbf{A})| < \delta'$  for every  $j$ . For such a  $\mathbf{B}$ , Theorem 15.7.3 applies to  $q = p_{\mathbf{B}}$ . This proves the corollary. ■

The corollary says exactly as much as Theorem 15.7.3 and no more: the numbering is not canonical and depends on  $\mathbf{B}$ . Any quantity that ignores the numbering, however, inherits honest continuity, and the spectral radius of Section 4 is the first example.

*Quick check.* Does Corollary 15.7.4 make  $\rho: M_n(\mathbb{C}) \rightarrow \mathbb{R}$  continuous, given that no individual eigenvalue is a continuous function of  $\mathbf{A}$ ?

*Solution.* Yes. Let  $\varepsilon > 0$  and take  $\delta$  from Corollary 15.7.4. If  $\|\mathbf{B} - \mathbf{A}\| < \delta$ , fix a numbering with  $|\mu_i - \lambda_i| < \varepsilon$ . Then  $|\mu_i| \leq |\lambda_i| + \varepsilon \leq \rho(\mathbf{A}) + \varepsilon$  for each  $i$  by Definition 15.4.1, so  $\rho(\mathbf{B}) \leq \rho(\mathbf{A}) + \varepsilon$ ; the same numbering read the other way gives  $|\lambda_i| \leq |\mu_i| + \varepsilon \leq \rho(\mathbf{B}) + \varepsilon$  for each  $i$ , hence  $\rho(\mathbf{A}) \leq \rho(\mathbf{B}) + \varepsilon$ , so  $|\rho(\mathbf{B}) - \rho(\mathbf{A})| \leq \varepsilon$ . The argument never names an individual eigenvalue, which is why it survives:  $\rho$  is a function of the multiset alone.

## 15.7.5 The limit argument, made routine

Chapter 11 §01 called Corollary 11.1.6 “the tool behind a standard strategy: *prove it for diagonalizable matrices, then take a limit*”, and said that this chapter, “where matrices acquire norms and limits can be spoken of properly, runs arguments of exactly this shape”. Here is the shape. It costs two corollaries, and afterwards the strategy is three lines every time.

### Corollary 15.7.5 — Diagonalizable Matrices Are Dense

Let  $\|\cdot\|$  be any norm on  $M_n(\mathbb{C})$ . For every  $\mathbf{A} \in M_n(\mathbb{C})$  and every  $\varepsilon > 0$  there is a **diagonalizable**  $\mathbf{A}' \in M_n(\mathbb{C})$  with  $\|\mathbf{A} - \mathbf{A}'\| < \varepsilon$ . Equivalently, every complex square matrix is the limit of a sequence of diagonalizable matrices, in every norm.

*Idea.* Chapter 11 produced a diagonalizable matrix near  $\mathbf{A}$  with the *entries* close. “Close in every norm” is now the same thing, by Corollary 15.2.4, so the only task is to convert one tolerance into the other.

*Proof.* Let  $N_\infty(\mathbf{M}) = \max_{i,j} |m_{ij}|$ , which is a norm on  $M_n(\mathbb{C})$ : it is the  $\infty$ -norm of Example (The 1-, 2- and  $\infty$ -norms on  $F^n$ ) read on the list of  $n^2$  entries. By Theorem 15.2.2 there is  $C > 0$  with  $\|\mathbf{M}\| \leq C N_\infty(\mathbf{M})$  for every  $\mathbf{M}$ . Apply Corollary 11.1.6 with the tolerance  $\varepsilon/C$ : it supplies a diagonalizable  $\mathbf{A}'$  with  $|a_{ij} - a'_{ij}| < \varepsilon/C$  for all  $i, j$ , hence with  $N_\infty(\mathbf{A} - \mathbf{A}') < \varepsilon/C$ , a maximum of finitely many numbers each below that bound. Therefore  $\|\mathbf{A} - \mathbf{A}'\| \leq C N_\infty(\mathbf{A} - \mathbf{A}') < \varepsilon$ . Taking  $\varepsilon = 1/k$  gives a sequence  $\mathbf{A}_k$  of diagonalizable matrices with  $\|\mathbf{A} - \mathbf{A}_k\| \rightarrow 0$ . This proves the corollary. ■



### Corollary 15.7.6 — Proving an Identity by Density

Let  $\Phi: M_n(\mathbb{C}) \rightarrow M_{r \times s}(\mathbb{C})$  be a map each of whose output entries is a polynomial in the entries of the input. If  $\Phi(\mathbf{A}) = 0$  for every **diagonalizable**  $\mathbf{A} \in M_n(\mathbb{C})$ , then  $\Phi(\mathbf{A}) = 0$  for every  $\mathbf{A} \in M_n(\mathbb{C})$ .

*Idea.* Both sides of the identity are continuous in  $\mathbf{A}$ , and they agree on the diagonalizable matrices, which are dense. A continuous function is determined by its values on a dense set, so they agree everywhere.

*Proof.* Fix  $\mathbf{A}$  and a norm on each side. By [Corollary 15.7.5](#) choose diagonalizable  $\mathbf{A}_k$  with  $\|\mathbf{A} - \mathbf{A}_k\| \rightarrow 0$ . By [Lemma 15.7.1](#),  $\|\Phi(\mathbf{A}_k) - \Phi(\mathbf{A})\| \rightarrow 0$ . Since  $\Phi(\mathbf{A}_k) = 0$  by hypothesis, that says  $\|\Phi(\mathbf{A})\| \rightarrow 0$ ; but  $\|\Phi(\mathbf{A})\|$  does not depend on  $k$ , so  $\|\Phi(\mathbf{A})\| = 0$  and  $\Phi(\mathbf{A}) = 0$  by the positivity clause of [Definition 15.1.1](#). This proves the corollary. ■

The hypothesis “each output entry is a polynomial in the input entries” is the one to check, and it is usually free: any expression built from  $\mathbf{A}$  by addition, multiplication, transposition, conjugate transposition of a real matrix, determinants, adjugates, traces and the coefficients of  $p_{\mathbf{A}}$  qualifies. What does *not* qualify is anything built from an inverse, an eigenvector or a choice of ordering. Here is the strategy in action, on the theorem that Chapter 8 proved the long way.

#### Example. (Cayley–Hamilton, by a limit)

Prove that  $p_{\mathbf{A}}(\mathbf{A}) = 0$  for every  $\mathbf{A} \in M_n(\mathbb{C})$ , using density rather than the adjugate identity of [Theorem 6.6.2](#).

*Solution.* Put  $\Phi(\mathbf{A}) = p_{\mathbf{A}}(\mathbf{A})$ . Each entry of  $\Phi(\mathbf{A})$  is a polynomial in the entries of  $\mathbf{A}$ : the coefficients  $c_j(\mathbf{A})$  are, as noted after [Lemma 15.7.1](#), and  $\Phi(\mathbf{A}) = \mathbf{A}^n + c_{n-1}(\mathbf{A})\mathbf{A}^{n-1} + \cdots + c_0(\mathbf{A})\mathbf{I}$  is then built from them by products and sums.

Let  $\mathbf{A}$  be diagonalizable, say  $\mathbf{A} = \mathbf{P}\mathbf{D}\mathbf{P}^{-1}$  with  $\mathbf{D} = \text{diag}(d_1, \dots, d_n)$ . By [Theorem 8.4.3](#),  $p_{\mathbf{A}}(\mathbf{A}) = \mathbf{P} \text{diag}(p_{\mathbf{A}}(d_1), \dots, p_{\mathbf{A}}(d_n)) \mathbf{P}^{-1}$ . Each  $d_i$  is an eigenvalue of  $\mathbf{A}$ , hence a root of  $p_{\mathbf{A}}$  ([Theorem 8.2.1](#)), so every diagonal entry is 0 and  $\Phi(\mathbf{A}) = 0$ .

By [Corollary 15.7.6](#),  $\Phi(\mathbf{A}) = 0$  for every  $\mathbf{A} \in M_n(\mathbb{C})$ , which is [Theorem 8.6.1](#) over  $\mathbb{C}$ .

That is the whole strategy: check the identity where the matrix is diagonal in some basis, where everything is a scalar computation, and let density carry it to the rest. It proves nothing new here — [Theorem 8.6.1](#) holds over every field, and the density argument is confined to  $\mathbb{C}$  — but the three lines at the end are the part to transfer. [Exercise \(C2: Density of the invertible matrices, and  \$\mathbf{AB}\$  versus  \$\mathbf{BA}\$ \)](#) runs the same argument with a different dense set.

## 15.7.6 The warning: continuous is not Lipschitz

Everything so far has been qualitative. The quantitative question is whether there is a constant  $L$  with

$$(\text{eigenvalue movement}) \leq L \|\mathbf{B} - \mathbf{A}\|,$$

which is what “continuous” is usually imagined to mean and what a numerical analyst actually needs. There is not, and the failure is severe.

**Example. (A Jordan block’s eigenvalues move like  $\varepsilon^{1/k}$ )**

Let  $k \geq 2$ , let  $\varepsilon > 0$ , and perturb the nilpotent Jordan block in its bottom left corner:

$$\mathbf{A} = \mathbf{J}_k(0), \quad \mathbf{A}_\varepsilon = \mathbf{J}_k(0) + \varepsilon \mathbf{E}_{k1}.$$

Find the eigenvalues of  $\mathbf{A}_\varepsilon$  and compare their distance from  $\text{spec}(\mathbf{A}) = \{0\}$  with  $\|\mathbf{A}_\varepsilon - \mathbf{A}\|$ .

*Solution.* The characteristic polynomial is  $x^k - \varepsilon$ . Write  $\mathbf{M} = x\mathbf{I} - \mathbf{A}_\varepsilon$ , so  $\mathbf{M}$  has  $x$  on the diagonal,  $-1$  on the superdiagonal,  $-\varepsilon$  in position  $(k, 1)$ , and zeros elsewhere. Expand along the first column (Theorem 6.6.1 (a)), whose only non-zero entries are  $m_{11} = x$  and  $m_{k1} = -\varepsilon$ .

Deleting row 1 and column 1 leaves the  $(k-1) \times (k-1)$  upper triangular matrix with  $x$  on the diagonal, of determinant  $x^{k-1}$  (Theorem 6.4.6). Deleting row  $k$  and column 1 leaves the matrix with rows  $1, \dots, k-1$  and columns  $2, \dots, k$  of  $\mathbf{M}$ ; its  $(i, j)$  entry is  $m_{i,j+1}$ , which is  $-1$  when  $j = i$  and  $x$  when  $i = j+1$ , so it is lower triangular with  $-1$  all along its diagonal and determinant  $(-1)^{k-1}$ . The cofactor signs are  $(-1)^{1+1} = 1$  and  $(-1)^{k+1}$ , so

$$\det \mathbf{M} = x \cdot x^{k-1} + (-\varepsilon)(-1)^{k+1}(-1)^{k-1} = x^k - \varepsilon,$$

since  $(-1)^{k+1}(-1)^{k-1} = (-1)^{2k} = 1$ .

*The eigenvalues.* They are the  $k$ -th roots of  $\varepsilon$ , namely  $\varepsilon^{1/k} e^{2\pi i j/k}$  for  $j = 0, \dots, k-1$ , each of modulus  $\varepsilon^{1/k}$ . For  $k = 2$  this is the pair  $\pm\sqrt{\varepsilon}$ , and  $\mathbf{A}_\varepsilon$  is the matrix  $\mathbf{A}(\varepsilon)$  of Proposition 15.7.2.

*The comparison.* The perturbation  $\mathbf{E} = \varepsilon \mathbf{E}_{k1}$  has a single non-zero entry, so its largest absolute row sum and largest absolute column sum are both  $\varepsilon$ , and  $\mathbf{E}^* \mathbf{E} = \varepsilon^2 \mathbf{E}_{11}$  has largest eigenvalue  $\varepsilon^2$ ; hence  $\|\mathbf{E}\|_1 = \|\mathbf{E}\|_\infty = \|\mathbf{E}\|_2 = \varepsilon$  by Theorem 15.3.3. Every eigenvalue of  $\mathbf{A}$  is 0, so the eigenvalues have moved a distance  $\varepsilon^{1/k}$  while the matrix moved a distance  $\varepsilon$ . The ratio is

$$\frac{\varepsilon^{1/k}}{\varepsilon} = \varepsilon^{-(k-1)/k} \xrightarrow{\varepsilon \rightarrow 0^+} \infty.$$

*A by-product worth keeping.* The  $k$  roots  $\varepsilon^{1/k} e^{2\pi i j/k}$  are **distinct** for every  $\varepsilon > 0$ , so  $\mathbf{A}_\varepsilon$  has  $k$  distinct eigenvalues and is diagonalizable by Corollary 8.4.2. A single nilpotent Jordan block, which is as far from diagonalizable as a matrix gets, therefore has diagonalizable matrices arbitrarily close to it, and here they are written down. That is Corollary 15.7.5 made concrete for one family.

Put numbers on it. For  $k = 2$  and  $\varepsilon = 10^{-6}$  the matrix moves by one part in a million and the eigenvalues move to  $\pm 10^{-3}$ : an amplification of a thousand. For  $k = 10$  and  $\varepsilon = 10^{-10}$  the matrix moves by  $10^{-10}$  and the eigenvalues land on the circle of radius  $10^{-1}$ , an amplification of a billion. No constant  $L$  survives this, so the continuity of Corollary 15.7.4 cannot be upgraded to a Lipschitz bound, and the  $\delta$  it produces genuinely depends on  $\mathbf{A}$  and not only on  $n$ .

*Warning.* **Continuous does not mean stable.** [Corollary 15.7.4](#) guarantees that a small enough perturbation moves the eigenvalues a little, and says nothing whatever about how small “small enough” is. For  $\mathbf{J}_{10}(0)$  a perturbation of size  $10^{-10}$  — the kind of change one would call negligible in data recorded to ten digits — scatters the eigenvalues over a circle of radius 0.1. Reading a printed eigenvalue of a matrix with a large Jordan block as a property of the matrix, rather than of the matrix plus whatever noise reached it, is the mistake this example exists to prevent.

The opposite extreme is worth seeing beside it. Take the Hermitian  $\mathbf{A} = \text{diag}(1, 2)$  and the Hermitian perturbation  $\mathbf{E} = \varepsilon(\mathbf{E}_{12} + \mathbf{E}_{21})$ , of norm  $\|\mathbf{E}\|_2 = \varepsilon$ . The matrix  $\mathbf{A} + \mathbf{E} = \begin{pmatrix} 1 & \varepsilon \\ \varepsilon & 2 \end{pmatrix}$  has trace 3 and determinant  $2 - \varepsilon^2$ , so its eigenvalues are

$$\frac{3 \pm \sqrt{1 + 4\varepsilon^2}}{2},$$

and since  $\sqrt{1+u} \leq 1+u/2$  for  $u \geq 0$ , each of them lies within  $\varepsilon^2$  of the corresponding eigenvalue of  $\mathbf{A}$ . Here the eigenvalues move *less* than the matrix does, by a whole order in  $\varepsilon$ . That is not an accident of the example: Chapter 16 proves that the eigenvalues of a Hermitian matrix move by at most  $\|\mathbf{E}\|_2$  under a Hermitian perturbation, and Chapter 19 proves the theorem of Bauer and Fike, which bounds the movement for a diagonalizable matrix by  $\|\mathbf{E}\|_2$  times a factor measuring how far its eigenvector basis is from orthonormal.

So the exponent  $1/k$  in [Example \(A Jordan block’s eigenvalues move like  \$\varepsilon^{1/k}\$ \)](#) is not about size; it is about shape. It is decided by the largest Jordan block at the eigenvalue ([Theorem 9.3.1](#)), and block size is precisely what Chapter 11 §11 was measuring when [Example \(The defect of a Jordan block\)](#) computed

$$\Delta(\mathbf{J}_k(\lambda)) = \sqrt{k-1}$$

for the departure from normality of [Definition 11.11.1](#). The two statements match at both ends:  $\Delta = 0$  exactly for the normal matrices, whose eigenvalues are the stable ones, and  $\Delta$  grows with the block that makes them fragile. They do not match in between, and it would be wrong to read  $\Delta$  as a formula for the exponent —  $\Delta$  is one number for the whole matrix, so it cannot report which eigenvalue carries which block, and [Example \(The defect of a Jordan block\)](#) records that it does not depend on  $\lambda$  at all. What  $\Delta(\mathbf{A})$  does is tell you whether the question needs asking at all.

## 15.7.7 Exercises

### 15.7.7.1 A. Check your understanding

#### Exercise A1.

- State [Theorem 15.7.3](#), paying attention to the two hypotheses on  $q$ .
- Explain in one sentence why the theorem is stated for the multiset of roots rather than for a list of  $n$  individual roots.
- Decide whether the following is correct, and justify your answer: “the function sending  $\mathbf{A} \in M_2(\mathbb{C})$  to its eigenvalue of largest modulus is continuous”.
- For the  $k \times k$  Jordan block, how far do the eigenvalues move under a perturbation of norm  $\varepsilon$ , and what does that say about a Lipschitz bound?

*Solution.*

- (a) See [Theorem 15.7.3](#). The two hypotheses are that  $q$  is **monic** and of the **same degree**  $n$  as  $p$ ; without them the roots can escape to infinity, as  $\delta x^2 + x$  shows.
- (b) Because no continuous choice of an individual root exists: [Proposition 15.7.2](#) shows that following one square root of  $t$  once around the unit circle returns the other one, so the matching numbering must be allowed to depend on  $q$ .
- (c) Incorrect as stated, because a matrix can have two eigenvalues of equal largest modulus and then the phrase does not name a number; and incorrect even where it does name a number, since  $\mathbf{A}(t)$  of [Proposition 15.7.2](#) has the two eigenvalues  $\pm\sqrt{t}$  of equal modulus for every  $t \neq 0$ . The modulus itself, that is  $\rho$ , is continuous — see the Quick check.
- (d) They move a distance  $\varepsilon^{1/k}$  ([Example \(A Jordan block's eigenvalues move like  \$\varepsilon^{1/k}\$ \)](#)), so the ratio of eigenvalue movement to matrix movement is  $\varepsilon^{-(k-1)/k}$ , which is unbounded as  $\varepsilon \rightarrow 0^+$ . No Lipschitz constant exists.

### 15.7.7.2 B. Practice

#### Exercise B1: Two perturbations of the same size.

Let  $\varepsilon = 10^{-8}$ . Compute the eigenvalues of

$$\mathbf{B} = \begin{pmatrix} 0 & 1 \\ \varepsilon & 0 \end{pmatrix} \quad \text{and} \quad \mathbf{C} = \begin{pmatrix} \varepsilon & 1 \\ 0 & 0 \end{pmatrix},$$

each of which differs from  $\mathbf{J}_2(0)$  by a matrix of 2-norm  $\varepsilon$ . Hence explain why the constant in [Corollary 15.7.4](#) cannot depend on  $\|\mathbf{B} - \mathbf{A}\|$  alone.

*Solution.* For  $\mathbf{B}$ :  $p_{\mathbf{B}}(x) = x^2 - \varepsilon$  by [Example \(A Jordan block's eigenvalues move like  \$\varepsilon^{1/k}\$ \)](#) with  $k = 2$ , so the eigenvalues are  $\pm 10^{-4}$ .

For  $\mathbf{C}$ : the matrix is upper triangular, so its eigenvalues are its diagonal entries  $\varepsilon = 10^{-8}$  and 0 ([Theorem 8.7.4](#) (b)).

Both perturbations have 2-norm  $\varepsilon = 10^{-8}$ , since each is  $\varepsilon$  times a single matrix unit. The eigenvalues of  $\mathbf{B}$  have moved  $10^{-4}$  from  $\text{spec}(\mathbf{J}_2(0)) = \{0\}$ , those of  $\mathbf{C}$  only  $10^{-8}$ : a factor of  $10^4$  between two perturbations of exactly the same size. So the movement is not a function of  $\|\mathbf{E}\|$ , and the  $\delta$  of [Corollary 15.7.4](#) cannot be computed from  $\varepsilon$  and  $n$  alone.

#### Exercise B2: A diagonalizable neighbor, explicitly.

Let  $\mathbf{A} = \begin{pmatrix} 2 & 1 \\ 0 & 2 \end{pmatrix}$  and let  $\varepsilon > 0$ .

- (a) Show that  $\mathbf{A}$  is not diagonalizable.
- (b) Exhibit a diagonalizable  $\mathbf{A}'$  with  $\|\mathbf{A} - \mathbf{A}'\|_2 < \varepsilon$ , and say which theorem certifies that your  $\mathbf{A}'$  is diagonalizable.
- (c) What are the eigenvalues of your  $\mathbf{A}'$ , and how far have they moved?

*Solution.*

- (a) The only eigenvalue is 2, with algebraic multiplicity 2 ([Theorem 8.7.4 \(b\)](#)). But  $\mathbf{A} - 2\mathbf{I} = \begin{pmatrix} 0 & 1 \\ 0 & 0 \end{pmatrix}$  has rank 1, so  $g_{\mathbf{A}}(2) = 1 < 2 = a_{\mathbf{A}}(2)$  and  $\mathbf{A}$  is not diagonalizable by [Theorem 8.4.1 \(e\)](#).
- (b) Take  $\delta$  with  $0 < \delta < \varepsilon$  and  $\mathbf{A}' = \begin{pmatrix} 2 & 1 \\ 0 & 2+\delta \end{pmatrix}$ . It is upper triangular with distinct diagonal entries 2 and  $2 + \delta$ , hence has two distinct eigenvalues ([Theorem 8.7.4 \(b\)](#)) and is diagonalizable by [Corollary 8.4.2](#). And  $\mathbf{A} - \mathbf{A}' = -\delta\mathbf{E}_{22}$ , so  $\|\mathbf{A} - \mathbf{A}'\|_2 = \delta < \varepsilon$  by [Theorem 15.3.3 \(c\)](#).
- (c) They are 2 and  $2 + \delta$ . One eigenvalue has not moved and the other has moved  $\delta$ , which is exactly  $\|\mathbf{A} - \mathbf{A}'\|_2$ : here the movement is Lipschitz with constant 1. The failure in [Example \(A Jordan block's eigenvalues move like  \$\varepsilon^{1/k}\$ \)](#) needed a perturbation *below* the diagonal, which is the direction a triangular form cannot absorb.

### Exercise B3: The determinant and the trace, by density.

Let  $\mathbf{A}, \mathbf{B} \in M_n(\mathbb{C})$ . Use [Corollary 15.7.6](#) to prove that  $\text{tr}(\mathbf{AB}) = \text{tr}(\mathbf{BA})$  — after first checking that the map in question satisfies the hypothesis of that corollary — and then explain why this is a bad way to prove it.

*Solution.* Fix  $\mathbf{B}$  and put  $\Phi(\mathbf{A}) = \text{tr}(\mathbf{AB}) - \text{tr}(\mathbf{BA})$ , a  $1 \times 1$  matrix whose single entry is  $\sum_{i,j} (a_{ij}b_{ji} - b_{ij}a_{ji})$ , a polynomial in the entries of  $\mathbf{A}$ ; so the hypothesis of [Corollary 15.7.6](#) holds.

Let  $\mathbf{A} = \mathbf{PDP}^{-1}$  be diagonalizable,  $\mathbf{D} = \text{diag}(d_1, \dots, d_n)$ , and put  $\mathbf{X} = \mathbf{P}^{-1}\mathbf{BP}$ . By [Theorem 3.7.5](#), similar matrices have equal traces, and  $\mathbf{AB} = \mathbf{P}(\mathbf{DX})\mathbf{P}^{-1}$  while  $\mathbf{BA} = \mathbf{P}(\mathbf{XD})\mathbf{P}^{-1}$ , so  $\text{tr}(\mathbf{AB}) = \text{tr}(\mathbf{DX})$  and  $\text{tr}(\mathbf{BA}) = \text{tr}(\mathbf{XD})$ . For a diagonal  $\mathbf{D}$  both of those are  $\sum_i d_i x_{ii}$ , by inspection of the diagonal entries of the two products. Hence  $\Phi(\mathbf{A}) = 0$  for every diagonalizable  $\mathbf{A}$ , and [Corollary 15.7.6](#) extends it to every  $\mathbf{A}$ .

It is a bad way to prove it because the identity is a two-line computation from the definition,  $\text{tr}(\mathbf{AB}) = \sum_{i,j} a_{ij}b_{ji} = \text{tr}(\mathbf{BA})$ , valid over every field; the density argument needs  $\mathbb{C}$ , an imported theorem and two corollaries, and delivers less. A density argument earns its keep only when the diagonalizable case is genuinely easier, as in [Example \(Cayley–Hamilton, by a limit\)](#).

### 15.7.7.3 C. Going deeper

#### Exercise C1: How close to singular a nearby eigenvalue forces you to be.

Let  $\mathbf{A} \in M_n(\mathbb{C})$ , let  $\|\cdot\|$  be an operator norm on  $M_n(\mathbb{C})$  ([Definition 15.3.2](#)) and let  $\mathbf{E} \in M_n(\mathbb{C})$  with  $\mathbf{E} \neq 0$ .

- (a) Suppose  $\mu$  is an eigenvalue of  $\mathbf{A} + \mathbf{E}$  and  $\mu \notin \text{spec}(\mathbf{A})$ . Prove that

$$\|(\mathbf{A} - \mu\mathbf{I})^{-1}\| \geq \frac{1}{\|\mathbf{E}\|}.$$

- (b) Deduce that if  $\|(\mathbf{A} - z\mathbf{I})^{-1}\| \leq M$  for every  $z$  with  $|z - \lambda| = r$ , then no eigenvalue of  $\mathbf{A} + \mathbf{E}$  lies on that circle as soon as  $\|\mathbf{E}\| < 1/M$ .

*Hint: for (a), factor  $\mathbf{A} + \mathbf{E} - \mu\mathbf{I}$  and use Theorem 15.5.1.*

*Solution.*

- (a) Since  $\mu \notin \text{spec}(\mathbf{A})$ , the matrix  $\mathbf{A} - \mu\mathbf{I}$  is invertible (Theorem 8.2.1). Write

$$\mathbf{A} + \mathbf{E} - \mu\mathbf{I} = (\mathbf{A} - \mu\mathbf{I})(\mathbf{I} + (\mathbf{A} - \mu\mathbf{I})^{-1}\mathbf{E}).$$

The left side is singular, because  $\mu$  is an eigenvalue of  $\mathbf{A} + \mathbf{E}$ . The first factor on the right is invertible, so the second factor is singular: a product with an invertible left factor is invertible as soon as the right factor is. Put  $\mathbf{X} = -(\mathbf{A} - \mu\mathbf{I})^{-1}\mathbf{E}$ , so  $\mathbf{I} - \mathbf{X}$  is singular. By Theorem 15.5.1,  $\|\mathbf{X}\| < 1$  would make  $\mathbf{I} - \mathbf{X}$  invertible; hence  $\|\mathbf{X}\| \geq 1$ . Submultiplicativity (Theorem 15.3.2) gives

$$1 \leq \|\mathbf{X}\| \leq \|(\mathbf{A} - \mu\mathbf{I})^{-1}\| \|\mathbf{E}\|,$$

and dividing by  $\|\mathbf{E}\| > 0$  gives the claim.

- (b) Suppose  $\mu$  is an eigenvalue of  $\mathbf{A} + \mathbf{E}$  with  $|\mu - \lambda| = r$ . Then  $\|(\mathbf{A} - \mu\mathbf{I})^{-1}\| \leq M$  by hypothesis, and in particular  $\mathbf{A} - \mu\mathbf{I}$  is invertible, so  $\mu \notin \text{spec}(\mathbf{A})$  and part (a) applies:  $M \geq 1/\|\mathbf{E}\|$ , that is  $\|\mathbf{E}\| \geq 1/M$ . Contrapositively,  $\|\mathbf{E}\| < 1/M$  leaves no eigenvalue of  $\mathbf{A} + \mathbf{E}$  on the circle.

### Exercise C2: Density of the invertible matrices, and $\mathbf{AB}$ versus $\mathbf{BA}$ .

- (a) Prove that for every  $\mathbf{A} \in M_n(\mathbb{C})$  and every  $\varepsilon > 0$  there is an invertible  $\mathbf{A}'$  with  $\|\mathbf{A} - \mathbf{A}'\| < \varepsilon$ , in any norm. *Hint: consider  $\mathbf{A} - t\mathbf{I}$ .*
- (b) Prove a version of Corollary 15.7.6 with “invertible” in place of “diagonalizable”.
- (c) Hence give a second proof that  $p_{\mathbf{AB}} = p_{\mathbf{BA}}$  for all  $\mathbf{A}, \mathbf{B} \in M_n(\mathbb{C})$ , a statement first proved in Theorem 8.10.4 (b).

*Solution.*

- (a) The matrix  $\mathbf{A} - t\mathbf{I}$  is singular exactly when  $t \in \text{spec}(\mathbf{A})$  (Theorem 8.2.1), and  $\text{spec}(\mathbf{A})$  has at most  $n$  elements. Let  $N_\infty$  and  $C$  be as in the proof of Corollary 15.7.5, so  $\|\mathbf{M}\| \leq C N_\infty(\mathbf{M})$ . Choose  $t \neq 0$  with  $|t| < \varepsilon/C$  and  $t \notin \text{spec}(\mathbf{A})$ , possible because the disc of radius  $\varepsilon/C$  is infinite and only finitely many of its points are forbidden. Then  $\mathbf{A}' = \mathbf{A} - t\mathbf{I}$  is invertible and  $N_\infty(\mathbf{A} - \mathbf{A}') = |t|$ , so  $\|\mathbf{A} - \mathbf{A}'\| < \varepsilon$ .
- (b) Verbatim the proof of Corollary 15.7.6, with the sequence of diagonalizable matrices replaced by the sequence of invertible matrices supplied by (a) with  $\varepsilon = 1/k$ .
- (c) Fix  $\mathbf{B}$  and set  $\Phi(\mathbf{A}) = (c_0(\mathbf{AB}) - c_0(\mathbf{BA}), \dots, c_{n-1}(\mathbf{AB}) - c_{n-1}(\mathbf{BA}))$ , a row of  $n$  entries. Each entry of  $\mathbf{AB}$  and of  $\mathbf{BA}$  is a polynomial in the entries of  $\mathbf{A}$ , and each  $c_j$  is a polynomial in the entries of its argument, so  $\Phi$  satisfies the hypothesis of (b). For invertible  $\mathbf{A}$  we have  $\mathbf{BA} = \mathbf{A}^{-1}(\mathbf{AB})\mathbf{A}$ , so  $\mathbf{AB}$  and  $\mathbf{BA}$  are similar and  $p_{\mathbf{AB}} = p_{\mathbf{BA}}$  by Theorem 6.10.3; that is exactly  $\Phi(\mathbf{A}) = 0$ , and it is Exercise (B3:  $\mathbf{AB}$  and  $\mathbf{BA}$ ) (a). By (b),  $\Phi(\mathbf{A}) = 0$  for every  $\mathbf{A}$ , and two monic polynomials of degree  $n$  with the same lower coefficients are equal.

**Exercise C3: Distinct eigenvalues is an open condition.**

Let  $\mathbf{A} \in M_n(\mathbb{C})$  have  $n$  **distinct** eigenvalues  $\lambda_1, \dots, \lambda_n$ , and let  $\|\cdot\|$  be a norm on  $M_n(\mathbb{C})$ .

- (a) Prove that there is  $\delta > 0$  such that every  $\mathbf{B}$  with  $\|\mathbf{B} - \mathbf{A}\| < \delta$  also has  $n$  distinct eigenvalues, and is therefore diagonalizable.
- (b) Deduce that the set of matrices in  $M_n(\mathbb{C})$  with a repeated eigenvalue contains no ball, that is, that its complement is dense.
- (c) Is the set of diagonalizable matrices open? Justify your answer.

*Solution.*

- (a) Put  $2\varepsilon = \min_{i \neq j} |\lambda_i - \lambda_j|$ , which is positive because the  $\lambda_i$  are distinct and there are finitely many pairs. Let  $\delta$  be the number that [Corollary 15.7.4](#) supplies for this  $\varepsilon$ . If  $\|\mathbf{B} - \mathbf{A}\| < \delta$ , number the eigenvalues of  $\mathbf{B}$  so that  $|\mu_i - \lambda_i| < \varepsilon$  for every  $i$ . For  $i \neq j$ , [Theorem 0.6.4](#) gives

$$|\mu_i - \mu_j| \geq |\lambda_i - \lambda_j| - |\mu_i - \lambda_i| - |\mu_j - \lambda_j| > 2\varepsilon - \varepsilon - \varepsilon = 0,$$

so the  $\mu_i$  are pairwise distinct. There are  $n$  of them, so  $\mathbf{B}$  is diagonalizable by [Corollary 8.4.2](#).

- (b) Let  $\mathbf{A}$  be arbitrary and  $\varepsilon > 0$ . By [Exercise \(C3: Distinct eigenvalues are dense, but not over the reals\)](#) (a) there is  $\mathbf{A}'$  with  $n$  distinct eigenvalues and  $|a_{ij} - a'_{ij}| < \varepsilon/C$  for all  $i, j$ , with  $C$  as in the proof of [Corollary 15.7.5](#); hence  $\|\mathbf{A} - \mathbf{A}'\| < \varepsilon$ . So every ball meets the set of matrices with  $n$  distinct eigenvalues, and no ball is contained in its complement.
- (c) No. The zero matrix is diagonalizable, and in  $M_2(\mathbb{C})$  every ball around it contains  $\varepsilon \mathbf{J}_2(0)$ , whose only eigenvalue is 0, with eigenspace of dimension 1 and algebraic multiplicity 2; so it is not diagonalizable by [Theorem 8.4.1](#) (e). Dense ([Corollary 15.7.5](#)) and open are independent properties, and the diagonalizable matrices are the first without being the second.

## 15.8 How Wrong Can the Answer Be

Chapter 2 solved  $\mathbf{Ax} = \mathbf{b}$  and Chapter 10 solved it when it had no solution. Neither chapter could ask the question a user of either method asks first: the vector  $\mathbf{b}$  is a measurement, so it is wrong in the last few digits, and the answer  $\mathbf{x}$  is then wrong too — by how much? The question needs a way to compare the size of an error with the size of the thing it corrupts, and that is what the last seven sections have built.

The answer turns out to be one number attached to  $\mathbf{A}$ , independent of  $\mathbf{b}$ , and it is the last object this chapter defines. It also carries a warning, and the warning is more useful than the number.

**Throughout, matrices are complex**, as in Sections 4 to 7, and  $\|\cdot\|$  is a norm on  $\mathbb{C}^n$  together with the operator norm it induces on  $M_n(\mathbb{C})$  ([Definition 15.3.2](#)) — the standing convention of Section 5, whose results are used here throughout. A real matrix is read inside  $M_n(\mathbb{C})$ , which changes none of the three norms we actually compute with, since [Theorem 15.3.3](#) gives the same



three formulas over either field.

## 15.8.1 A recurring product

Let  $\mathbf{A} \in M_n(\mathbb{C})$  be invertible, let  $\mathbf{b} \neq \mathbf{0}$ , and let  $\mathbf{x}$  solve  $\mathbf{Ax} = \mathbf{b}$ . Suppose the data arrives as  $\mathbf{b} + \Delta\mathbf{b}$  instead, so the computed answer solves  $\mathbf{A}\tilde{\mathbf{x}} = \mathbf{b} + \Delta\mathbf{b}$ . Subtracting the two equations,  $\mathbf{A}(\tilde{\mathbf{x}} - \mathbf{x}) = \Delta\mathbf{b}$ , so

$$\tilde{\mathbf{x}} - \mathbf{x} = \mathbf{A}^{-1}\Delta\mathbf{b}, \quad \|\tilde{\mathbf{x}} - \mathbf{x}\| \leq \|\mathbf{A}^{-1}\| \|\Delta\mathbf{b}\|$$

by [Theorem 15.3.2](#) (a). That is an *absolute* bound, and absolute bounds are the wrong currency: whether an error of  $10^{-3}$  is a catastrophe depends on whether the answer is of size 1 or of size  $10^6$ . Divide by  $\|\mathbf{x}\|$ , and use  $\|\mathbf{b}\| = \|\mathbf{Ax}\| \leq \|\mathbf{A}\|\|\mathbf{x}\|$ , that is  $1/\|\mathbf{x}\| \leq \|\mathbf{A}\|/\|\mathbf{b}\|$ :

$$\frac{\|\tilde{\mathbf{x}} - \mathbf{x}\|}{\|\mathbf{x}\|} \leq \|\mathbf{A}^{-1}\| \|\Delta\mathbf{b}\| \frac{\|\mathbf{A}\|}{\|\mathbf{b}\|} = \|\mathbf{A}\| \|\mathbf{A}^{-1}\| \frac{\|\Delta\mathbf{b}\|}{\|\mathbf{b}\|}.$$

The relative error in the answer is at most the relative error in the data, multiplied by  $\|\mathbf{A}\| \|\mathbf{A}^{-1}\|$ . That product depends only on  $\mathbf{A}$ . It will reappear in every estimate in this section, so we name it.

*The condition number is the worst factor by which  $\mathbf{A}$  can magnify a relative error.*

### Definition 15.8.1 — Condition Number

Let  $\|\cdot\|$  be a norm on  $\mathbb{C}^n$ , and write  $\|\cdot\|$  also for the operator norm it induces on  $M_n(\mathbb{C})$ . For  $\mathbf{A} \in M_n(\mathbb{C})$ , the **condition number** of  $\mathbf{A}$  with respect to that norm is

$$\kappa(\mathbf{A}) := \begin{cases} \|\mathbf{A}\| \|\mathbf{A}^{-1}\| & \text{if } \mathbf{A} \text{ is invertible,} \\ \infty & \text{if } \mathbf{A} \text{ is singular.} \end{cases}$$

We write  $\kappa_1, \kappa_2, \kappa_\infty$  for the condition numbers with respect to  $\|\cdot\|_1, \|\cdot\|_2, \|\cdot\|_\infty$ ;  $\kappa_2$  is also called the **spectral condition number**. For a rectangular  $\mathbf{A} \in M_{m \times n}(\mathbb{C})$  with  $m \geq n$  and  $\text{rank } \mathbf{A} = n$ , we define

$$\kappa_2(\mathbf{A}) := \frac{\sigma_1(\mathbf{A})}{\sigma_n(\mathbf{A})},$$

the ratio of the largest to the smallest singular value ([Definition 12.8.1](#)).

Clause by clause. The number is attached to  **$\mathbf{A}$  and to a norm**:  $\kappa_1(\mathbf{A})$ ,  $\kappa_2(\mathbf{A})$  and  $\kappa_\infty(\mathbf{A})$  are three different numbers, and a statement like “ $\kappa(\mathbf{A}) = 400$ ” is incomplete until the norm is named. They cannot disagree wildly — [Theorem 15.2.2](#) traps them within dimension-dependent factors of each other, and [Exercise \(C1: Changing the norm changes  \$\kappa\$ , but not much\)](#) makes that precise — so the *order of magnitude* of  $\kappa$ , which is all anyone uses, does not depend on the choice. The value  $\infty$  for a singular matrix is a convention, and the right one: a singular  $\mathbf{A}$  either makes  $\mathbf{Ax} = \mathbf{b}$  unsolvable or makes its solution set a whole coset  $\mathbf{x} + \text{nul}(\mathbf{A})$  with  $\text{nul}(\mathbf{A}) \neq \{\mathbf{0}\}$  ([Theorem 2.1.3](#)), so no finite bound on the error can hold. Finally, the square case and the rectangular case agree where they overlap: [Proposition 15.8.2](#) (d) shows  $\kappa_2(\mathbf{A}) = \sigma_1/\sigma_n$  for an invertible square  $\mathbf{A}$  as well.

Before the examples, a small lemma that says what  $\kappa$  is measuring. It is the one geometric picture in this section, and it is worth carrying.



### Lemma 15.8.1 — The Inverse Measures the Smallest Stretch

Let  $\mathbf{A} \in M_n(\mathbb{C})$  be invertible and let  $\|\cdot\|$  be a norm on  $\mathbb{C}^n$ . Then

$$\|\mathbf{A}^{-1}\| = \frac{1}{\min\{\|\mathbf{A}\mathbf{x}\| : \|\mathbf{x}\| = 1\}}, \quad \kappa(\mathbf{A}) = \frac{\max\{\|\mathbf{A}\mathbf{x}\| : \|\mathbf{x}\| = 1\}}{\min\{\|\mathbf{A}\mathbf{x}\| : \|\mathbf{x}\| = 1\}}.$$

*Idea.* Both maxima and the minimum exist by compactness, which is [Lemma 15.3.1](#) and the same argument with a minimum. After that it is one substitution: as  $\mathbf{x}$  runs over the non-zero vectors so does  $\mathbf{y} = \mathbf{A}\mathbf{x}$ , and the ratio  $\|\mathbf{A}^{-1}\mathbf{y}\|/\|\mathbf{y}\|$  is the reciprocal of  $\|\mathbf{A}\mathbf{x}\|/\|\mathbf{x}\|$ .

*Proof.* The function  $\mathbf{x} \mapsto \|\mathbf{A}\mathbf{x}\|$  is continuous on the unit sphere  $S$ , which is compact, so it attains a maximum and a minimum there ([Lemma 15.3.1](#) for the maximum; the same proof, or the extreme value theorem (A4) applied to the same continuous function, for the minimum). Write  $m = \min_{\mathbf{x} \in S} \|\mathbf{A}\mathbf{x}\|$ . Since  $\mathbf{A}$  is invertible,  $\mathbf{A}\mathbf{x} \neq \mathbf{0}$  for  $\mathbf{x} \in S$ , so  $m > 0$ .

By [Definition 15.3.2](#) and homogeneity,  $\|\mathbf{A}^{-1}\| = \max\{\|\mathbf{A}^{-1}\mathbf{y}\|/\|\mathbf{y}\| : \mathbf{y} \neq \mathbf{0}\}$ . The map  $\mathbf{y} \mapsto \mathbf{A}^{-1}\mathbf{y}$  is a bijection of  $\mathbb{C}^n \setminus \{\mathbf{0}\}$  onto itself, so substituting  $\mathbf{y} = \mathbf{A}\mathbf{x}$  with  $\mathbf{x} \neq \mathbf{0}$ ,

$$\|\mathbf{A}^{-1}\| = \max_{\mathbf{x} \neq \mathbf{0}} \frac{\|\mathbf{x}\|}{\|\mathbf{A}\mathbf{x}\|} = \frac{1}{\min_{\mathbf{x} \neq \mathbf{0}} \frac{\|\mathbf{A}\mathbf{x}\|}{\|\mathbf{x}\|}} = \frac{1}{m},$$

the last step by homogeneity again, which lets the minimum over  $\mathbf{x} \neq \mathbf{0}$  be taken over  $S$ . Multiplying by  $\|\mathbf{A}\| = \max_{\mathbf{x} \in S} \|\mathbf{A}\mathbf{x}\|$  gives the second formula. This proves the lemma. ■

So  $\kappa(\mathbf{A})$  is the **distortion** of  $\mathbf{A}$ : feed  $\mathbf{A}$  the unit sphere and it comes out as a lopsided surface, and  $\kappa$  is the ratio of its longest radius to its shortest. A matrix with  $\kappa = 1$  carries the sphere to a sphere. A matrix with  $\kappa = 10^6$  flattens it into something a million times longer than it is wide, and a direction of  $\mathbf{b}$  that lands along the short axis has its errors magnified by a million on the way back.

Some values, simplest first.

- **The identity, and any unitary matrix.**  $\|\mathbf{I}\| = 1 = \|\mathbf{I}^{-1}\|$ , so  $\kappa(\mathbf{I}) = 1$  in every norm. For unitary  $\mathbf{U}$ ,  $\|\mathbf{U}\mathbf{x}\|_2 = \|\mathbf{x}\|_2$  for every  $\mathbf{x}$  ([Theorem 10.7.3](#)), so every stretch is 1 and  $\kappa_2(\mathbf{U}) = 1$  by [Lemma 15.8.1](#). These are the best-conditioned matrices there are.
- **A diagonal matrix.** For  $\mathbf{D} = \text{diag}(d_1, \dots, d_n)$  with all  $d_i \neq 0$ ,  $\|\mathbf{D}\|_2 = \max_i |d_i|$  and  $\|\mathbf{D}^{-1}\|_2 = 1/\min_i |d_i|$ , so  $\kappa_2(\mathbf{D})$  is the ratio of the largest to the smallest modulus. Thus  $\kappa_2(\text{diag}(1, 10^{-6})) = 10^6$ .
- **A small matrix that is perfectly conditioned.**  $\mathbf{A} = 10^{-6}\mathbf{I}$  has  $\|\mathbf{A}\|_2 = 10^{-6}$ , a tiny norm, and  $\kappa_2(\mathbf{A}) = 1$ . Being small and being ill-conditioned are unrelated.
- **A singular matrix**, the degenerate case:  $\kappa = \infty$  by convention, consistent with [Lemma 15.8.1](#), where the minimum stretch is 0.

**Example.** (Three condition numbers of one matrix)

Compute  $\kappa_\infty(\mathbf{A})$  and  $\kappa_2(\mathbf{A})$  for  $\mathbf{A} = \begin{pmatrix} 2 & 1 \\ 1 & 1 \end{pmatrix}$ .

*Solution.*  $\det \mathbf{A} = 1$ , so  $\mathbf{A}^{-1} = \begin{pmatrix} 1 & -1 \\ -1 & 2 \end{pmatrix}$ . The largest absolute row sums are 3 for  $\mathbf{A}$  and 3 for  $\mathbf{A}^{-1}$ , so by [Theorem 15.3.3](#) (b),  $\kappa_\infty(\mathbf{A}) = 3 \cdot 3 = 9$ .

For  $\kappa_2$ :  $\mathbf{A}$  is real symmetric with  $\operatorname{tr} \mathbf{A} = 3$  and  $\det \mathbf{A} = 1$ , so its eigenvalues are  $(3 \pm \sqrt{5})/2$ , both positive, and the singular values of a symmetric matrix with positive eigenvalues are those eigenvalues (its  $\mathbf{A}^* \mathbf{A} = \mathbf{A}^2$  has eigenvalues their squares). Hence

$$\kappa_2(\mathbf{A}) = \frac{3 + \sqrt{5}}{3 - \sqrt{5}} = \frac{(3 + \sqrt{5})^2}{4} = \frac{7 + 3\sqrt{5}}{2} \approx 6.854.$$

Two different norms, two different numbers, the same order of magnitude — which is the general picture.

**Warning.** **A large  $\kappa$  is not a large determinant defect, and a small determinant is not ill-conditioning.** The matrix  $10^{-6}\mathbf{I}_{10}$  has determinant  $10^{-60}$  and  $\kappa_2 = 1$ ; the matrix  $\operatorname{diag}(10^6, 10^{-6})$  has determinant 1 and  $\kappa_2 = 10^{12}$ . The determinant scales with  $c^n$  under  $\mathbf{A} \mapsto c\mathbf{A}$  while  $\kappa$  does not move at all ([Proposition 15.8.2](#) (b)), so no threshold on  $|\det \mathbf{A}|$  can detect ill-conditioning. This is the most common wrong test, and it fails in both directions.

The basic properties are all one line each, and the third is the reason  $\kappa$  behaves well under the factorizations of Chapters 10 and 12.

### Proposition 15.8.2 — Properties of the Condition Number

Let  $\mathbf{A}, \mathbf{B} \in M_n(\mathbb{C})$  be invertible and let  $c \in \mathbb{C} \setminus \{0\}$ . With respect to any operator norm:

- (a)  $\kappa(\mathbf{A}) \geq 1$ ;
- (b)  $\kappa(c\mathbf{A}) = \kappa(\mathbf{A})$ ;
- (c)  $\kappa(\mathbf{AB}) \leq \kappa(\mathbf{A})\kappa(\mathbf{B})$ ;
- (d)  $\kappa_2(\mathbf{A}) = \sigma_1(\mathbf{A})/\sigma_n(\mathbf{A})$ , and  $\kappa_2(\mathbf{UAV}) = \kappa_2(\mathbf{A})$  for all unitary  $\mathbf{U}, \mathbf{V}$ .

*Proof.*

- (a)  $1 = \|\mathbf{I}\| = \|\mathbf{AA}^{-1}\| \leq \|\mathbf{A}\| \|\mathbf{A}^{-1}\| = \kappa(\mathbf{A})$ , using [Theorem 15.3.2](#) (c) and (d).
- (b)  $(c\mathbf{A})^{-1} = c^{-1}\mathbf{A}^{-1}$ , so by homogeneity  $\|c\mathbf{A}\| \|(c\mathbf{A})^{-1}\| = |c| |c|^{-1} \|\mathbf{A}\| \|\mathbf{A}^{-1}\|$ .
- (c)  $(\mathbf{AB})^{-1} = \mathbf{B}^{-1}\mathbf{A}^{-1}$ , so submultiplicativity ([Theorem 15.3.2](#) (d)) gives  $\|\mathbf{AB}\| \|\mathbf{B}^{-1}\mathbf{A}^{-1}\| \leq \|\mathbf{A}\| \|\mathbf{B}\| \|\mathbf{B}^{-1}\| \|\mathbf{A}^{-1}\|$ .
- (d) By [Theorem 15.3.3](#) (c),  $\|\mathbf{A}\|_2 = \sigma_1(\mathbf{A})$ . By [Lemma 15.8.1](#),  $\|\mathbf{A}^{-1}\|_2 = 1/m$  with  $m = \min\{\|\mathbf{Ax}\|_2 : \|\mathbf{x}\|_2 = 1\}$ , so it remains to see  $m = \sigma_n(\mathbf{A})$ . Write  $\mathbf{A} = \mathbf{U}\Sigma\mathbf{V}^*$  ([Theorem 12.8.1](#)) and  $\mathbf{y} = \mathbf{V}^*\mathbf{x}$ ; unitary matrices preserve  $\|\cdot\|_2$  ([Theorem 10.7.3](#)), so  $\|\mathbf{x}\|_2 = \|\mathbf{y}\|_2$  and  $\|\mathbf{Ax}\|_2 = \|\Sigma\mathbf{y}\|_2$ . Then

$$\|\Sigma\mathbf{y}\|_2^2 = \sum_{i=1}^n \sigma_i^2 |y_i|^2 \geq \sigma_n^2 \sum_{i=1}^n |y_i|^2 = \sigma_n^2$$

for  $\|\mathbf{y}\|_2 = 1$ , with equality at  $\mathbf{y} = \mathbf{e}_n$ , that is at  $\mathbf{x} = \mathbf{v}_n$ , the last column of  $\mathbf{V}$ . So  $m = \sigma_n$ . For the last claim, if  $\mathbf{A} = \mathbf{U}_1 \mathbf{\Sigma} \mathbf{V}_1^*$  then  $\mathbf{U} \mathbf{A} \mathbf{V} = (\mathbf{U} \mathbf{U}_1) \mathbf{\Sigma} (\mathbf{V}^* \mathbf{V}_1)^*$ , and a product of unitary matrices is unitary, so this is a singular value decomposition of  $\mathbf{U} \mathbf{A} \mathbf{V}$  with the same  $\mathbf{\Sigma}$ : unitary factors do not change the singular values. This proves the proposition. ■

Part (d) is the reason  $\kappa_2$  is the condition number of choice in theory: it is invariant under the unitary changes of coordinates that Chapters 10 to 12 use throughout, so it is a property of the linear map and not of the basis it is written in.

## 15.8.2 What the condition number predicts

The computation that opened the section is now a theorem, together with the version where  $\mathbf{A}$  itself, and not only  $\mathbf{b}$ , carries the error. Both halves matter in practice: the entries of  $\mathbf{A}$  are measurements too. Section 5 already proved the two perturbations together, in [Theorem 15.5.3](#), and its constant  $\|\mathbf{A}\| \|\mathbf{A}^{-1}\|$  is the product we have just named; what is added here is that the first half is sharp on its own, and that it needs no smallness hypothesis at all.

### Theorem 15.8.3 — Relative Error Under Perturbation of the Data

Let  $\mathbf{A} \in M_n(\mathbb{C})$  be invertible, let  $\mathbf{b} \neq \mathbf{0}$ , and let  $\mathbf{A}\mathbf{x} = \mathbf{b}$ .

(a) **(Perturbing  $\mathbf{b}$ .)** If  $\mathbf{A}\tilde{\mathbf{x}} = \mathbf{b} + \Delta\mathbf{b}$ , then

$$\frac{\|\tilde{\mathbf{x}} - \mathbf{x}\|}{\|\mathbf{x}\|} \leq \kappa(\mathbf{A}) \frac{\|\Delta\mathbf{b}\|}{\|\mathbf{b}\|},$$

and there are  $\mathbf{b}$  and  $\Delta\mathbf{b}$  for which equality holds.

(b) **(Perturbing  $\mathbf{A}$ .)** Let  $\mathbf{E} \in M_n(\mathbb{C})$  with  $r := \kappa(\mathbf{A}) \|\mathbf{E}\| / \|\mathbf{A}\| < 1$ . Then  $\mathbf{A} + \mathbf{E}$  is invertible; and if  $(\mathbf{A} + \mathbf{E})\tilde{\mathbf{x}} = \mathbf{b}$ , then

$$\frac{\|\tilde{\mathbf{x}} - \mathbf{x}\|}{\|\tilde{\mathbf{x}}\|} \leq \kappa(\mathbf{A}) \frac{\|\mathbf{E}\|}{\|\mathbf{A}\|}, \quad \frac{\|\tilde{\mathbf{x}} - \mathbf{x}\|}{\|\mathbf{x}\|} \leq \frac{r}{1 - r}.$$

*Idea.* For (a), the displayed computation at the head of the section, plus a choice of the two vectors that makes both inequalities in it equalities — and both are attained, because [Lemma 15.3.1](#) says an operator norm is a maximum and not a supremum. For (b), factor  $\mathbf{A} + \mathbf{E} = \mathbf{A}(\mathbf{I} + \mathbf{A}^{-1}\mathbf{E})$  so that [Theorem 15.5.1](#) decides invertibility, then subtract the two equations. Subtracting gives  $\tilde{\mathbf{x}} - \mathbf{x}$  in terms of  $\tilde{\mathbf{x}}$ , which is why the clean bound is relative to  $\tilde{\mathbf{x}}$ ; subadditivity of the norm converts it to a bound relative to  $\mathbf{x}$  at the cost of the factor  $1/(1 - r)$ , and that factor is where “to first order” hides.

*Proof.* (a). Subtracting  $\mathbf{A}\mathbf{x} = \mathbf{b}$  from  $\mathbf{A}\tilde{\mathbf{x}} = \mathbf{b} + \Delta\mathbf{b}$  gives  $\mathbf{A}(\tilde{\mathbf{x}} - \mathbf{x}) = \Delta\mathbf{b}$ , so  $\tilde{\mathbf{x}} - \mathbf{x} = \mathbf{A}^{-1}\Delta\mathbf{b}$  and  $\|\tilde{\mathbf{x}} - \mathbf{x}\| \leq \|\mathbf{A}^{-1}\| \|\Delta\mathbf{b}\|$  by [Theorem 15.3.2](#) (a). Also  $\|\mathbf{b}\| = \|\mathbf{A}\mathbf{x}\| \leq \|\mathbf{A}\| \|\mathbf{x}\|$ ,

and  $\mathbf{x} \neq \mathbf{0}$  because  $\mathbf{b} \neq \mathbf{0}$ , so  $1/\|\mathbf{x}\| \leq \|\mathbf{A}\|/\|\mathbf{b}\|$ . Multiplying the two estimates,

$$\frac{\|\tilde{\mathbf{x}} - \mathbf{x}\|}{\|\mathbf{x}\|} \leq \|\mathbf{A}^{-1}\| \|\Delta \mathbf{b}\| \cdot \frac{\|\mathbf{A}\|}{\|\mathbf{b}\|} = \kappa(\mathbf{A}) \frac{\|\Delta \mathbf{b}\|}{\|\mathbf{b}\|}.$$

For equality, use [Lemma 15.3.1](#) twice. Choose a unit vector  $\mathbf{x}$  with  $\|\mathbf{Ax}\| = \|\mathbf{A}\|$  and put  $\mathbf{b} = \mathbf{Ax}$ , so that  $\|\mathbf{b}\| = \|\mathbf{A}\| \|\mathbf{x}\|$  and the second estimate is an equality. Choose a unit vector  $\mathbf{y}$  with  $\|\mathbf{A}^{-1}\mathbf{y}\| = \|\mathbf{A}^{-1}\|$  and put  $\Delta \mathbf{b} = \mathbf{y}$ , so that the first estimate is an equality. Then

$$\frac{\|\tilde{\mathbf{x}} - \mathbf{x}\|}{\|\mathbf{x}\|} \bigg/ \frac{\|\Delta \mathbf{b}\|}{\|\mathbf{b}\|} = \frac{\|\mathbf{A}^{-1}\|}{1} \cdot \frac{\|\mathbf{A}\|}{1} = \kappa(\mathbf{A}).$$

(b). Write  $\mathbf{A} + \mathbf{E} = \mathbf{A}(\mathbf{I} + \mathbf{A}^{-1}\mathbf{E})$ . By [Theorem 15.3.2](#) (d),

$$\|\mathbf{A}^{-1}\mathbf{E}\| \leq \|\mathbf{A}^{-1}\| \|\mathbf{E}\| = \kappa(\mathbf{A}) \frac{\|\mathbf{E}\|}{\|\mathbf{A}\|} = r < 1,$$

so  $\mathbf{I} - (-\mathbf{A}^{-1}\mathbf{E})$  is invertible by [Theorem 15.5.1](#), and  $\mathbf{A} + \mathbf{E}$  is a product of two invertible matrices, hence invertible. (This is the openness of  $\text{GL}_n(F)$  recorded in [Corollary 15.5.2](#), with the radius made explicit.)

Since  $\mathbf{b} \neq \mathbf{0}$  we have  $\tilde{\mathbf{x}} = (\mathbf{A} + \mathbf{E})^{-1}\mathbf{b} \neq \mathbf{0}$ . Subtracting  $\mathbf{Ax} = \mathbf{b}$  from  $(\mathbf{A} + \mathbf{E})\tilde{\mathbf{x}} = \mathbf{b}$  gives  $\mathbf{A}\tilde{\mathbf{x}} - \mathbf{Ax} = -\mathbf{E}\tilde{\mathbf{x}}$ , hence

$$\tilde{\mathbf{x}} - \mathbf{x} = -\mathbf{A}^{-1}\mathbf{E}\tilde{\mathbf{x}}, \quad \|\tilde{\mathbf{x}} - \mathbf{x}\| \leq \|\mathbf{A}^{-1}\| \|\mathbf{E}\| \|\tilde{\mathbf{x}}\| = r \|\tilde{\mathbf{x}}\|,$$

which is the first bound. For the second,  $\|\tilde{\mathbf{x}}\| \leq \|\mathbf{x}\| + \|\tilde{\mathbf{x}} - \mathbf{x}\|$  since  $\|\cdot\|$  is subadditive ([Definition 15.1.1](#)), so  $\|\tilde{\mathbf{x}} - \mathbf{x}\| \leq r\|\mathbf{x}\| + r\|\tilde{\mathbf{x}} - \mathbf{x}\|$ ; as  $r < 1$  we may divide by  $1 - r > 0$  to get  $\|\tilde{\mathbf{x}} - \mathbf{x}\| \leq \frac{r}{1-r}\|\mathbf{x}\|$ . This proves the theorem. ■

The two bounds in (b) say the same thing differently. The first is exact and needs nothing but invertibility, but it measures the error against the *computed* answer. The second measures it against the true answer, at the cost of the factor  $1/(1-r)$  — which is  $1 + r + r^2 + \cdots$ , so for a small  $r$  it is 1 to first order and the two bounds agree. That is all “to first order in the perturbation of  $\mathbf{A}$ ” ever means here, and it is now a stated inequality rather than a gesture. Adding the two halves, with the factor  $1/(1-r)$  covering both, recovers [Theorem 15.5.3](#) exactly.

*Quick check.* Where in the proof of (a) is the hypothesis  $\mathbf{b} \neq \mathbf{0}$  used, and what goes wrong without it?

*Solution.* It is used to know  $\mathbf{x} \neq \mathbf{0}$ , so that  $\|\mathbf{x}\|$  may be divided by, and to know  $\|\mathbf{b}\| \neq 0$ , so that the relative perturbation  $\|\Delta \mathbf{b}\|/\|\mathbf{b}\|$  means anything. With  $\mathbf{b} = \mathbf{0}$  the true solution is  $\mathbf{x} = \mathbf{0}$  and *every* non-zero  $\Delta \mathbf{b}$  produces an infinite relative error. Relative error is only defined against a non-zero reference.

Now a case where the bound of (a) is not merely an inequality. The point of computing the example exactly is to see that  $\kappa$  is not a pessimistic overestimate manufactured by the inequalities used to reach it: it is achieved.

**Example. (A system where the bound is attained)**

Let

$$\mathbf{A} = \begin{pmatrix} 1 & 1 \\ 1 & 1.01 \end{pmatrix}, \quad \mathbf{b} = \begin{pmatrix} 2 \\ 2.01 \end{pmatrix},$$

whose solution is  $\mathbf{x} = (1, 1)$ . Compute  $\kappa_\infty(\mathbf{A})$ ; then perturb  $\mathbf{b}$  to  $\mathbf{b} + \Delta\mathbf{b}$  with  $\Delta\mathbf{b} = (0.01, -0.01)$ , solve again, and compare the two relative errors.

*Solution.* Work with exact fractions:  $1.01 = \frac{101}{100}$  and  $\det \mathbf{A} = \frac{101}{100} - 1 = \frac{1}{100}$ , so

$$\mathbf{A}^{-1} = 100 \begin{pmatrix} \frac{101}{100} & -1 \\ -1 & 1 \end{pmatrix} = \begin{pmatrix} 101 & -100 \\ -100 & 100 \end{pmatrix}.$$

The largest absolute row sums are  $\frac{201}{100}$  for  $\mathbf{A}$  and 201 for  $\mathbf{A}^{-1}$ , so by [Theorem 15.3.3](#) (b),

$$\kappa_\infty(\mathbf{A}) = \frac{201}{100} \cdot 201 = \frac{40401}{100} = 404.01.$$

Check that  $\mathbf{Ax} = \mathbf{b}$ : the two rows give  $1 + 1 = 2$  and  $1 + \frac{101}{100} = \frac{201}{100}$ . Now

$$\tilde{\mathbf{x}} - \mathbf{x} = \mathbf{A}^{-1} \Delta\mathbf{b} = \frac{1}{100} \begin{pmatrix} 101 + 100 \\ -100 - 100 \end{pmatrix} = \begin{pmatrix} 2.01 \\ -2 \end{pmatrix},$$

so  $\tilde{\mathbf{x}} = (3.01, -1)$ . Verify directly:  $3.01 - 1 = 2.01$  and  $3.01 - 1.01 = 2$ , which is  $\mathbf{b} + \Delta\mathbf{b} = (2.01, 2)$ .

The two relative errors, in  $\|\cdot\|_\infty$ :

$$\begin{aligned} \frac{\|\Delta\mathbf{b}\|_\infty}{\|\mathbf{b}\|_\infty} &= \frac{1/100}{201/100} = \frac{1}{201} \approx 0.50\%, \\ \frac{\|\tilde{\mathbf{x}} - \mathbf{x}\|_\infty}{\|\mathbf{x}\|_\infty} &= \frac{201/100}{1} = \frac{201}{100} = 201\%. \end{aligned}$$

Their ratio is  $\frac{201}{100} \cdot 201 = 404.01$ , which is exactly  $\kappa_\infty(\mathbf{A})$ . A perturbation of half a percent in the data has turned the answer  $(1, 1)$  into  $(3.01, -1)$ , and the bound of [Theorem 15.8.3](#) (a) is attained, not approached.

Geometrically the example is [Lemma 15.8.1](#) at work: the two rows of  $\mathbf{A}$  are nearly parallel, so the two lines whose intersection is  $\mathbf{x}$  cross at a very small angle, and moving either line slightly slides the crossing point a long way. Ill-conditioning is that picture, in any dimension.

*Warning.* **The condition number bounds the error; it does not produce it.** [Theorem 15.8.3](#) is a worst case over all perturbations  $\Delta\mathbf{b}$  of a given size. A particular  $\Delta\mathbf{b}$  may do far less damage: in the example above,  $\Delta\mathbf{b} = (0.01, 0.01)$  gives  $\tilde{\mathbf{x}} - \mathbf{x} = \mathbf{A}^{-1} \Delta\mathbf{b} = (0.01, 0)$ , a relative error of 1% against a relative perturbation of 0.5% — a magnification of 2, not 404. A large  $\kappa$  says that *some* direction is dangerous, not that yours is.

### 15.8.3 Why the normal equations are worse

Chapter 10 §04 solved the least-squares problem through the normal equations  $\mathbf{A}^* \mathbf{A} \mathbf{x} = \mathbf{A}^* \mathbf{b}$  (Theorem 10.4.2), and Chapter 10 §08 then recommended against actually forming that system. Its words were: “The deeper reason to prefer it is not the operation count: forming  $\mathbf{A}^* \mathbf{A}$  squares the sensitivity of the problem to rounding, while  $(\dagger)$  never forms that product at all. Chapter 23 makes ‘sensitivity’ precise and proves the comparison; here we record only the practice.” The word “sensitivity” now has a meaning, and the word “squares” is now a theorem.

#### Theorem 15.8.4 — The Normal Equations Square the Condition Number

Let  $m \geq n \geq 1$  and let  $\mathbf{A} \in M_{m \times n}(\mathbb{C})$  have rank  $n$ . Then  $\mathbf{A}^* \mathbf{A} \in M_n(\mathbb{C})$  is invertible, and

$$\kappa_2(\mathbf{A}^* \mathbf{A}) = \kappa_2(\mathbf{A})^2.$$

*Idea.* An SVD of  $\mathbf{A}$  is also, after one multiplication, an eigenvalue decomposition of  $\mathbf{A}^* \mathbf{A}$ : the singular values get squared and nothing else happens. Both condition numbers are then ratios of the same list of numbers, one list being the squares of the other.

*Proof.* Write  $\mathbf{A} = \mathbf{U} \mathbf{\Sigma} \mathbf{V}^*$  with  $\mathbf{U} \in M_m(\mathbb{C})$ ,  $\mathbf{V} \in M_n(\mathbb{C})$  unitary and  $\mathbf{\Sigma} \in M_{m \times n}(\mathbb{R})$  carrying  $\sigma_1 \geq \cdots \geq \sigma_n \geq 0$  (Theorem 12.8.1). Since  $\text{rank } \mathbf{A} = n$ , exactly  $n$  singular values are non-zero (Theorem 12.8.1), so  $\sigma_n > 0$ . Using  $\mathbf{U}^* \mathbf{U} = \mathbf{I}_m$ ,

$$\mathbf{A}^* \mathbf{A} = \mathbf{V} \mathbf{\Sigma}^* \mathbf{U}^* \mathbf{U} \mathbf{\Sigma} \mathbf{V}^* = \mathbf{V} (\mathbf{\Sigma}^* \mathbf{\Sigma}) \mathbf{V}^*, \quad \mathbf{\Sigma}^* \mathbf{\Sigma} = \text{diag}(\sigma_1^2, \dots, \sigma_n^2).$$

So  $\mathbf{A}^* \mathbf{A}$  is unitarily similar to a diagonal matrix with the positive entries  $\sigma_i^2$ ; it is therefore invertible, Hermitian and positive definite, with eigenvalues  $\sigma_1^2 \geq \cdots \geq \sigma_n^2 > 0$ .

The singular values of  $\mathbf{A}^* \mathbf{A}$  are those same numbers. Indeed the displayed identity *is* a singular value decomposition of  $\mathbf{A}^* \mathbf{A}$ , with both unitary factors equal to  $\mathbf{V}$  and the diagonal entries already in decreasing order; alternatively,  $(\mathbf{A}^* \mathbf{A})^* (\mathbf{A}^* \mathbf{A}) = \mathbf{V} \text{diag}(\sigma_i^4) \mathbf{V}^*$  has eigenvalues  $\sigma_i^4$ , whose non-negative square roots are the  $\sigma_i^2$  (Definition 12.8.1). Hence by Definition 15.8.1 and Proposition 15.8.2 (d),

$$\kappa_2(\mathbf{A}^* \mathbf{A}) = \frac{\sigma_1^2}{\sigma_n^2} = \left( \frac{\sigma_1}{\sigma_n} \right)^2 = \kappa_2(\mathbf{A})^2.$$

This proves the theorem. ■

#### Example. (Squaring, on a $3 \times 2$ example)

Let  $\mathbf{A} = \begin{pmatrix} 1 & 1 \\ 0.1 & 0 \\ 0 & 0.1 \end{pmatrix}$ . Compute  $\kappa_2(\mathbf{A})$  and  $\kappa_2(\mathbf{A}^\top \mathbf{A})$ .

*Solution.*  $\mathbf{A}^\top \mathbf{A} = \begin{pmatrix} 1.01 & 1 \\ 1 & 1.01 \end{pmatrix}$ . It sends  $(1, 1)$  to  $2.01(1, 1)$  and  $(1, -1)$  to  $0.01(1, -1)$ , so

its eigenvalues are  $\frac{201}{100}$  and  $\frac{1}{100}$ . Hence  $\sigma_1(\mathbf{A}) = \sqrt{201}/10$  and  $\sigma_2(\mathbf{A}) = 1/10$ , so

$$\kappa_2(\mathbf{A}) = \sqrt{201} \approx 14.18, \quad \kappa_2(\mathbf{A}^\top \mathbf{A}) = \frac{201/100}{1/100} = 201 = (\sqrt{201})^2.$$

A matrix whose conditioning is unremarkable becomes, on forming  $\mathbf{A}^\top \mathbf{A}$ , a matrix whose conditioning is fourteen times worse.

**What is proved here, and what is not.** [Theorem 15.8.4](#) is a statement about matrices: the matrix one *forms* when one writes down the normal equations has the square of the condition number of the matrix one started with. Combined with [Theorem 15.8.3](#), it says that the linear system  $\mathbf{A}^* \mathbf{A} \mathbf{x} = \mathbf{A}^* \mathbf{b}$ , considered as a problem in its own data, is more sensitive than  $\mathbf{A}$  itself by a factor  $\kappa_2(\mathbf{A})$  — and the QR route  $\mathbf{R} \mathbf{x} = \mathbf{Q}^* \mathbf{b}$  of Chapter 10 §08 never forms  $\mathbf{A}^* \mathbf{A}$ , so it never incurs that factor. Indeed the triangular matrix it does solve with is no worse conditioned than  $\mathbf{A}$ : from  $\mathbf{A} = \mathbf{Q} \mathbf{R}$  with  $\mathbf{Q}^* \mathbf{Q} = \mathbf{I}_n$  ([Theorem 10.8.1](#)) we get  $\mathbf{A}^* \mathbf{A} = \mathbf{R}^* \mathbf{Q}^* \mathbf{Q} \mathbf{R} = \mathbf{R}^* \mathbf{R}$ , so  $\mathbf{A}$  and  $\mathbf{R}$  have the same singular values ([Definition 12.8.1](#)) and  $\kappa_2(\mathbf{R}) = \kappa_2(\mathbf{A})$  exactly.

Three things are *not* proved here, and all three belong to Chapter 23.

- **The conditioning of the least-squares problem itself** is not  $\kappa_2(\mathbf{A})$ . The map  $\mathbf{b} \mapsto \mathbf{x}^+$  from data to least-squares solution has a sensitivity that depends on  $\kappa_2(\mathbf{A})$ , on the residual  $\|\mathbf{A} \mathbf{x}^+ - \mathbf{b}\|$  and on the angle between  $\mathbf{b}$  and  $\text{col}(\mathbf{A})$ ; when the residual is large, a term in  $\kappa_2(\mathbf{A})^2$  appears even for the QR route. The bound above is about the *linear systems being solved*, not about the underlying fitting problem.
- **Nothing here is about arithmetic.** A condition number describes a problem, and an error bound of the form “relative error  $\leq \kappa \cdot$  (relative perturbation)” only becomes a statement about a computation once one knows how large a perturbation the computation itself introduces. That requires a model of floating-point arithmetic and a backward error analysis, neither of which exists in this book yet.
- **Hence the comparison of the two algorithms** — that a stable implementation of the QR route loses about  $\log_{10} \kappa_2(\mathbf{A})$  decimal digits while forming and solving the normal equations loses about  $2 \log_{10} \kappa_2(\mathbf{A})$  — is still Chapter 23’s to prove. What this section supplies is the number that appears in both statements, and the reason a square appears in one of them.

## 15.8.4 Conditioning is a property of the problem

This is the last warning of the chapter, and it is the one readers most often need.

*Warning.* **Conditioning belongs to the problem; stability belongs to the algorithm.**  $\kappa(\mathbf{A})$  is computed from  $\mathbf{A}$  alone. It knows nothing about how anyone intends to solve  $\mathbf{A} \mathbf{x} = \mathbf{b}$ , and it does not change if the method changes. Two consequences, both regularly got wrong:

- **A large  $\kappa(\mathbf{A})$  does not mean a computed answer is wrong.** It means that the answer is not determined to high relative accuracy by data of limited accuracy — a statement about the information available, not about the arithmetic performed. If the data are exact and the arithmetic is exact, the answer is exact however large  $\kappa$  is. And even in inexact arithmetic, a good algorithm returns the exact answer to a problem whose data differ from yours by a rounding error, which is the most anyone can ask of it.

- **A small  $\kappa(\mathbf{A})$  does not make an algorithm trustworthy.** An unstable method can wreck a perfectly conditioned problem all by itself, by introducing errors far larger than the data warrant; [Exercise \(C3: A well-conditioned problem and a bad method\)](#) gives an elementary example in which  $\kappa_\infty$  is below 5 and a computed coordinate comes out as 0 instead of 1.

Chapter 23 draws the distinction properly, defines backward stability, and proves which of the algorithms in this book have it. Until then, read  $\kappa$  as an upper bound on what any method can promise, and never as a verdict on a particular answer.

### 15.8.5 What this chapter has settled

Nine chapters wrote promissory notes that only a norm could pay. Here is the ledger, with the section that closed each entry.

- **Chapter 10 §01** said that not every norm comes from an inner product and that the parallelogram law is the test. Section 1 proved it, in [Theorem 15.1.3](#), building on the  $\mathbb{R}^2$  case of [Exercise \(C1: The parallelogram law characterizes inner-product norms\)](#).
- **Chapter 9 §10** promised that this chapter would prove  $\rho(\mathbf{A}) \leq \|\mathbf{A}\|$  for the norm induced by any vector norm, and would recover  $\rho(\mathbf{A})$  as a limit built from the norms of the powers. Section 4 did both: [Theorem 15.4.1](#) and Gelfand's formula [Theorem 15.4.4](#). The same section also gave  $\rho$  the labeled definition [Definition 15.4.1](#) that Chapter 9 used only in prose, and [Corollary 15.4.5](#) restored Chapter 9's convergence criterion with a rate attached. And Chapter 9's entrywise convergence turned out to be *the* convergence, by [Corollary 15.2.4](#).
- **Chapter 12 §08 and §12** asked, twice, for a name for  $\max\{\|\mathbf{A}\mathbf{x}\| : \|\mathbf{x}\| = 1\}$ . Section 3 supplied it — [Definition 15.3.2](#) — and separated the name from the computation  $\|\mathbf{A}\|_2 = \sigma_1(\mathbf{A})$ , which is [Theorem 15.3.3](#) (c); [Example \(The oblique projection, finally measured\)](#) measured Chapter 12 §12's oblique projection. The spectral norm that Chapter 12 §10 needed in order even to *state* the second half of its approximation theorem now exists; the theorem itself waits for Chapter 16, which proves it from the min–max description of the singular values. The version covering every unitarily invariant norm at once is Chapter 20's.
- **Chapter 9 §09** promised a return to  $e^{\mathbf{A}}$  with norms available, giving quantitative bounds in place of exact formulas. Section 6 gave them, in [Theorem 15.6.1](#), and its [Theorem 15.6.3](#) is the statement Chapter 11 §10 pointed at when it said that the consequence of stability for  $\dot{\mathbf{x}} = \mathbf{A}\mathbf{x}$  belonged to Chapter 15.
- **Chapter 11 §01** said that this chapter, “where matrices acquire norms and limits can be spoken of properly, runs arguments of exactly this shape”. Section 7 ran one: [Corollary 15.7.5](#) and [Corollary 15.7.6](#) turn “prove it for diagonalizable matrices, then take a limit” into three lines, and [Example \(Cayley–Hamilton, by a limit\)](#) is the demonstration.
- **Chapter 10 §08** deferred its reason for preferring **QR** to the normal equations. [Theorem 15.8.4](#) supplies the matrix half of it; the arithmetic half is Chapter 23's, as listed above.

What the chapter has *not* done is equally worth recording. It has not bounded how far an individual eigenvalue moves under a perturbation — [Example \(A Jordan block's eigenvalues move like  \$\varepsilon^{1/k}\$ \)](#) shows there is no bound of the obvious kind, and the bounds that do exist, Weyl's and Bauer–Fike's, are Chapters 16 and 19. It has not said which norms are invariant under unitary multiplication, which is Chapter 20. And it has not touched arithmetic. The condition number is the bridge to that last subject: it is the exact point at which linear algebra



stops and numerical analysis begins.

## 15.8.6 Exercises

### 15.8.6.1 A. Check your understanding

#### Exercise A1.

- Define  $\kappa(\mathbf{A})$ , naming what it depends on besides  $\mathbf{A}$ .
- Explain why  $\kappa(\mathbf{A}) \geq 1$  for every invertible  $\mathbf{A}$ .
- Decide whether each statement is correct, and justify: (i) “if  $\kappa(\mathbf{A}) = 10^8$  then any computed solution of  $\mathbf{Ax} = \mathbf{b}$  has at most a few correct digits”; (ii) “if  $\kappa(\mathbf{A})$  is small then any method of solving  $\mathbf{Ax} = \mathbf{b}$  is reliable”.
- State what  $\kappa_2(\mathbf{A}^*\mathbf{A})$  is in terms of  $\kappa_2(\mathbf{A})$ , and say what that has to do with the QR factorization.

*Solution.*

- See [Definition 15.8.1](#):  $\kappa(\mathbf{A}) = \|\mathbf{A}\|\|\mathbf{A}^{-1}\|$ , with  $\kappa(\mathbf{A}) = \infty$  for singular  $\mathbf{A}$ . Besides  $\mathbf{A}$  it depends on the **choice of vector norm**, through the operator norm it induces;  $\kappa_1$ ,  $\kappa_2$  and  $\kappa_\infty$  are three different numbers.
- $1 = \|\mathbf{I}\| = \|\mathbf{AA}^{-1}\| \leq \|\mathbf{A}\|\|\mathbf{A}^{-1}\|$ , by [Theorem 15.3.2](#) (c) and (d). See [Proposition 15.8.2](#) (a).
- (i) Incorrect.  $\kappa$  bounds the effect of a perturbation of the data; with exact data and exact arithmetic the answer is exact. It says that the answer *cannot be determined* to high relative accuracy from inaccurate data, which is a statement about the problem, not a verdict on a computed answer. (ii) Incorrect. Conditioning is a property of the problem and stability a property of the algorithm; a small  $\kappa$  leaves an unstable method free to introduce errors of its own.
- $\kappa_2(\mathbf{A}^*\mathbf{A}) = \kappa_2(\mathbf{A})^2$  for  $\mathbf{A}$  of full column rank ([Theorem 15.8.4](#)). The normal equations form  $\mathbf{A}^*\mathbf{A}$  and so solve a system with the squared condition number, while the route  $\mathbf{Rx} = \mathbf{Q}^*\mathbf{b}$  of Chapter 10 §08 never forms that product and keeps  $\kappa_2(\mathbf{R}) = \kappa_2(\mathbf{A})$ .

### 15.8.6.2 B. Practice

#### Exercise B1: Three condition numbers.

For  $\mathbf{A} = \begin{pmatrix} 3 & 0 \\ 0 & -\frac{1}{2} \end{pmatrix}$  and  $\mathbf{B} = \begin{pmatrix} 1 & 2 \\ 0 & 1 \end{pmatrix}$ , compute  $\kappa_\infty$  and  $\kappa_2$  of each. *Hint:  $\mathbf{B}^*\mathbf{B}$  has trace 6 and determinant 1.*

*Solution.*  $\mathbf{A}^{-1} = \text{diag}(\frac{1}{3}, -2)$ . Largest absolute row sums: 3 for  $\mathbf{A}$ , 2 for  $\mathbf{A}^{-1}$ , so  $\kappa_\infty(\mathbf{A}) = 6$  by [Theorem 15.3.3](#) (b). For  $\kappa_2$ :  $\mathbf{A}^*\mathbf{A} = \text{diag}(9, \frac{1}{4})$ , so  $\sigma_1 = 3$ ,  $\sigma_2 = \frac{1}{2}$  and  $\kappa_2(\mathbf{A}) = 6$  by [Proposition 15.8.2](#) (d). The two agree here because  $\mathbf{A}$  is diagonal.

$\mathbf{B}^{-1} = \begin{pmatrix} 1 & -2 \\ 0 & 1 \end{pmatrix}$ , and both have largest absolute row sum 3, so  $\kappa_\infty(\mathbf{B}) = 9$ . For  $\kappa_2$ :

$\mathbf{B}^*\mathbf{B} = \begin{pmatrix} 1 & 2 \\ 2 & 5 \end{pmatrix}$  has trace 6 and determinant 1, so its eigenvalues are  $3 \pm 2\sqrt{2}$ , giving

$$\kappa_2(\mathbf{B}) = \sqrt{\frac{3+2\sqrt{2}}{3-2\sqrt{2}}} = 3 + 2\sqrt{2} \approx 5.828,$$

using  $(3 - 2\sqrt{2})(3 + 2\sqrt{2}) = 1$ , so that the ratio is  $(3 + 2\sqrt{2})^2$ .

### Exercise B2: A perturbation, measured against the bound.

Let  $\mathbf{A} = \begin{pmatrix} 1 & 1 \\ 1 & 1.01 \end{pmatrix}$  and  $\mathbf{b} = (2, 2.01)$  as in [Example \(A system where the bound is attained\)](#), and take  $\Delta\mathbf{b} = (0, 0.01)$ .

- Solve  $\mathbf{A}\tilde{\mathbf{x}} = \mathbf{b} + \Delta\mathbf{b}$  exactly.
- Compute both relative errors in  $\|\cdot\|_\infty$  and the factor between them, and compare with  $\kappa_\infty(\mathbf{A}) = 404.01$ .

*Solution.*

- With  $\mathbf{A}^{-1} = \begin{pmatrix} 101 & -100 \\ -100 & 100 \end{pmatrix}$  from [Example \(A system where the bound is attained\)](#) and  $\Delta\mathbf{b} = (0, \frac{1}{100})$ ,

$$\tilde{\mathbf{x}} - \mathbf{x} = \mathbf{A}^{-1}\Delta\mathbf{b} = \begin{pmatrix} -1 \\ 1 \end{pmatrix}, \quad \tilde{\mathbf{x}} = \begin{pmatrix} 0 \\ 2 \end{pmatrix}.$$

Check:  $0 + 2 = 2$  and  $0 + 2 \cdot \frac{101}{100} = \frac{202}{100} = 2.02 = \mathbf{b} + \Delta\mathbf{b}$  in the second coordinate.

- $\|\Delta\mathbf{b}\|_\infty / \|\mathbf{b}\|_\infty = \frac{1/100}{201/100} = \frac{1}{201}$ , and  $\|\tilde{\mathbf{x}} - \mathbf{x}\|_\infty / \|\mathbf{x}\|_\infty = 1$ . The factor is 201, which is below  $\kappa_\infty(\mathbf{A}) = 404.01$  by a factor of about 2. The bound holds, as [Theorem 15.8.3](#) (a) requires, and is not attained: this  $\Delta\mathbf{b}$  is not the worst direction, which [Example \(A system where the bound is attained\)](#) found.

### Exercise B3: Squaring, again.

Let  $\mathbf{A} = \begin{pmatrix} 1 & 0 \\ 0 & \frac{1}{2} \\ 0 & 0 \end{pmatrix}$ .

- Compute the singular values of  $\mathbf{A}$  and  $\kappa_2(\mathbf{A})$ .
- Compute  $\mathbf{A}^\top\mathbf{A}$  and  $\kappa_2(\mathbf{A}^\top\mathbf{A})$  directly, and confirm [Theorem 15.8.4](#).
- Does  $\kappa_\infty$  also square? Compute  $\kappa_\infty(\mathbf{A}^\top\mathbf{A})$  and compare with  $\kappa_\infty$  of the square matrix  $\begin{pmatrix} 1 & 0 \\ 0 & \frac{1}{2} \end{pmatrix}$ .

*Solution.*

- $\mathbf{A}^\top\mathbf{A} = \text{diag}(1, \frac{1}{4})$ , whose eigenvalues are 1 and  $\frac{1}{4}$ ; so  $\sigma_1(\mathbf{A}) = 1$ ,  $\sigma_2(\mathbf{A}) = \frac{1}{2}$  and  $\kappa_2(\mathbf{A}) = 2$ .
- $\mathbf{A}^\top\mathbf{A} = \text{diag}(1, \frac{1}{4})$  is diagonal with positive entries, so its singular values are 1 and  $\frac{1}{4}$  and  $\kappa_2(\mathbf{A}^\top\mathbf{A}) = 4 = 2^2 = \kappa_2(\mathbf{A})^2$ , as [Theorem 15.8.4](#) says.

- (c) For a diagonal matrix every  $\kappa_p$  is the ratio of the largest to the smallest modulus on the diagonal, so  $\kappa_\infty(\mathbf{A}^\top \mathbf{A}) = 4$  while  $\kappa_\infty(\text{diag}(1, \frac{1}{2})) = 2$ ; here the squaring happens in  $\kappa_\infty$  too. It does not in general: [Theorem 15.8.4](#) is stated for  $\kappa_2$ , and the proof uses singular values, which only  $\|\cdot\|_2$  sees. [Exercise \(C1: Changing the norm changes  \$\kappa\$ , but not much\)](#) bounds how far the other condition numbers can stray.

### 15.8.6.3 C. Going deeper

#### Exercise C1: Changing the norm changes $\kappa$ , but not much.

Let  $\|\cdot\|_a$  and  $\|\cdot\|_b$  be norms on  $\mathbb{C}^n$  with  $c\|\mathbf{v}\|_b \leq \|\mathbf{v}\|_a \leq C\|\mathbf{v}\|_b$  for all  $\mathbf{v}$ , as supplied by [Theorem 15.2.2](#), and write  $\kappa_a, \kappa_b$  for the corresponding condition numbers.

- (a) Prove that  $\|\mathbf{A}\|_a \leq (C/c)\|\mathbf{A}\|_b$  for every  $\mathbf{A} \in M_n(\mathbb{C})$ .  
 (b) Deduce that  $\kappa_a(\mathbf{A}) \leq (C/c)^2 \kappa_b(\mathbf{A})$  and  $\kappa_b(\mathbf{A}) \leq (C/c)^2 \kappa_a(\mathbf{A})$  for every invertible  $\mathbf{A}$ .  
 (c) Using  $\|\mathbf{v}\|_\infty \leq \|\mathbf{v}\|_2 \leq \sqrt{n}\|\mathbf{v}\|_\infty$ , bound  $\kappa_2$  in terms of  $\kappa_\infty$  for  $n = 2$ , and check the bound on the matrix  $\mathbf{B} = \begin{pmatrix} 1 & 2 \\ 0 & 1 \end{pmatrix}$  of [Exercise \(B1: Three condition numbers\)](#).

*Solution.*

- (a) Let  $\mathbf{x} \neq \mathbf{0}$ . Then

$$\frac{\|\mathbf{Ax}\|_a}{\|\mathbf{x}\|_a} \leq \frac{C\|\mathbf{Ax}\|_b}{c\|\mathbf{x}\|_b} \leq \frac{C}{c} \|\mathbf{A}\|_b,$$

using the two inequalities and [Theorem 15.3.2](#) (a) for  $\|\cdot\|_b$ . Taking the maximum over  $\mathbf{x} \neq \mathbf{0}$ , which is  $\|\mathbf{A}\|_a$  by [Definition 15.3.2](#) and homogeneity, gives the claim.

- (b) Apply (a) to  $\mathbf{A}$  and to  $\mathbf{A}^{-1}$  and multiply:  $\kappa_a(\mathbf{A}) = \|\mathbf{A}\|_a \|\mathbf{A}^{-1}\|_a \leq (C/c)^2 \|\mathbf{A}\|_b \|\mathbf{A}^{-1}\|_b = (C/c)^2 \kappa_b(\mathbf{A})$ . The reverse follows by symmetry, since the hypothesis also reads  $C^{-1}\|\mathbf{v}\|_a \leq \|\mathbf{v}\|_b \leq c^{-1}\|\mathbf{v}\|_a$ , whose ratio of constants is again  $C/c$ .  
 (c) With  $\|\cdot\|_a = \|\cdot\|_2$  and  $\|\cdot\|_b = \|\cdot\|_\infty$ , we have  $c = 1$  and  $C = \sqrt{n} = \sqrt{2}$ , so  $\kappa_2(\mathbf{A}) \leq 2\kappa_\infty(\mathbf{A})$  and  $\kappa_\infty(\mathbf{A}) \leq 2\kappa_2(\mathbf{A})$ . For  $\mathbf{A} = \begin{pmatrix} 1 & 2 \\ 0 & 1 \end{pmatrix}$ :  $\kappa_2 = 3 + 2\sqrt{2} \approx 5.83$  and  $\kappa_\infty = 9$ , and indeed  $5.83 \leq 18$  and  $9 \leq 11.66$ . Both bounds hold, with room to spare; the point is only that no choice of norm can turn a well-conditioned matrix into an ill-conditioned one.

#### Exercise C2: The condition number is the reciprocal distance to singularity.

Let  $\mathbf{A} \in M_n(\mathbb{C})$  be invertible.

- (a) Prove that every singular  $\mathbf{B} \in M_n(\mathbb{C})$  satisfies

$$\frac{\|\mathbf{A} - \mathbf{B}\|}{\|\mathbf{A}\|} \geq \frac{1}{\kappa(\mathbf{A})}.$$

- (b) For the 2-norm, prove that equality is achieved: exhibit a singular  $\mathbf{B}$  with  $\|\mathbf{A} - \mathbf{B}\|_2 = \sigma_n(\mathbf{A})$ . *Hint: use a singular value decomposition and delete one term.*  
 (c) Hence interpret  $1/\kappa_2(\mathbf{A})$  in one sentence, and say what it gives for the matrix of

Example (A system where the bound is attained).

*Solution.*

- (a) Put  $\mathbf{E} = \mathbf{B} - \mathbf{A}$ , so  $\mathbf{A} + \mathbf{E} = \mathbf{B}$  is singular. If  $\|\mathbf{A}^{-1}\mathbf{E}\| < 1$  then  $\mathbf{A} + \mathbf{E} = \mathbf{A}(\mathbf{I} + \mathbf{A}^{-1}\mathbf{E})$  would be invertible, by [Theorem 15.5.1](#) and the fact that a product of invertible matrices is invertible; so  $\|\mathbf{A}^{-1}\mathbf{E}\| \geq 1$ . By [Theorem 15.3.2](#) (d),  $1 \leq \|\mathbf{A}^{-1}\| \|\mathbf{E}\|$ , that is  $\|\mathbf{E}\| \geq 1/\|\mathbf{A}^{-1}\|$ . Dividing by  $\|\mathbf{A}\|$ ,

$$\frac{\|\mathbf{A} - \mathbf{B}\|}{\|\mathbf{A}\|} = \frac{\|\mathbf{E}\|}{\|\mathbf{A}\|} \geq \frac{1}{\|\mathbf{A}\| \|\mathbf{A}^{-1}\|} = \frac{1}{\kappa(\mathbf{A})}.$$

- (b) Let  $\mathbf{A} = \mathbf{U}\mathbf{\Sigma}\mathbf{V}^*$  ([Theorem 12.8.1](#)) with  $\mathbf{\Sigma} = \text{diag}(\sigma_1, \dots, \sigma_n)$ , and put

$$\mathbf{B} = \mathbf{U} \text{diag}(\sigma_1, \dots, \sigma_{n-1}, 0) \mathbf{V}^*.$$

Then  $\mathbf{B}$  is singular, since  $\text{diag}(\sigma_1, \dots, \sigma_{n-1}, 0)$  is and  $\mathbf{U}, \mathbf{V}$  are invertible; and  $\mathbf{A} - \mathbf{B} = \mathbf{U} \text{diag}(0, \dots, 0, \sigma_n) \mathbf{V}^*$ , whose singular values are those of  $\text{diag}(0, \dots, 0, \sigma_n)$  — for any  $\mathbf{M}$ ,  $(\mathbf{U}\mathbf{M}\mathbf{V}^*)^*(\mathbf{U}\mathbf{M}\mathbf{V}^*) = \mathbf{V}\mathbf{M}^*\mathbf{M}\mathbf{V}^*$  is similar to  $\mathbf{M}^*\mathbf{M}$ , so unitary factors leave the singular values alone, invertible or not — the largest of them being  $\sigma_n$ ; so  $\|\mathbf{A} - \mathbf{B}\|_2 = \sigma_n$  by [Theorem 15.3.3](#) (c). Since  $\|\mathbf{A}\|_2 = \sigma_1$ , the ratio is  $\sigma_n/\sigma_1 = 1/\kappa_2(\mathbf{A})$ , matching (a) with equality.

- (c)  $1/\kappa_2(\mathbf{A})$  is the relative distance, in the 2-norm, from  $\mathbf{A}$  to the nearest singular matrix. For  $\mathbf{A} = \begin{pmatrix} 1 & 1 \\ 1 & 1.01 \end{pmatrix}$ , whose  $\kappa_2$  is the ratio of the eigenvalues  $(201 \pm \sqrt{40001})/200$  of the symmetric  $\mathbf{A}$ , one gets  $\kappa_2(\mathbf{A}) \approx 402.0$ , so a relative perturbation of about 0.25% suffices to make  $\mathbf{A}$  singular. Ill-conditioned means *nearly singular*, in exactly this quantitative sense.

### Exercise C3: A well-conditioned problem and a bad method.

Consider the two-by-two system

$$\begin{aligned} 10^{-15}x_1 + x_2 &= 1, \\ x_1 + x_2 &= 2, \end{aligned}$$

with coefficient matrix  $\mathbf{A} = \begin{pmatrix} 10^{-15} & 1 \\ 1 & 1 \end{pmatrix}$ .

- Show that  $\kappa_\infty(\mathbf{A}) < 5$ , so that the problem is well conditioned.
- Solve the system exactly.
- Eliminate  $x_1$  using the **first** equation as the pivot row, rounding every intermediate quantity to 12 significant decimal digits, and compare the result with (b). Then do the same after exchanging the two equations, and say what the exchange achieved.
- What does this show about the relationship between  $\kappa(\mathbf{A})$  and the accuracy of a computed answer?

*Solution.*

- (a) Write  $\varepsilon = 10^{-15}$ . Then  $\det \mathbf{A} = \varepsilon - 1 \neq 0$  and

$$\mathbf{A}^{-1} = \frac{1}{\varepsilon - 1} \begin{pmatrix} 1 & -1 \\ -1 & \varepsilon \end{pmatrix}.$$

The largest absolute row sum of  $\mathbf{A}$  is 2, and that of  $\mathbf{A}^{-1}$  is  $2/(1 - \varepsilon)$ , so by [Theorem 15.3.3](#) (b),

$$\kappa_{\infty}(\mathbf{A}) = \frac{4}{1 - \varepsilon} < 5.$$

- (b) Subtracting the first equation from the second gives  $(1 - \varepsilon)x_1 = 1$ , so

$$x_1 = \frac{1}{1 - \varepsilon} = 1 + 10^{-15} + \dots, \quad x_2 = 1 - \varepsilon x_1 = \frac{1 - 2\varepsilon}{1 - \varepsilon} = 1 - 10^{-15} + \dots$$

Both coordinates are 1 to fourteen decimal places.

- (c) *Without the exchange.* The multiplier is  $1/\varepsilon = 10^{15}$ , and the second equation becomes

$$(1 - 10^{15})x_2 = 2 - 10^{15}.$$

Both sides need fifteen significant digits to be recorded exactly, so at twelve digits both round to  $-1.00000000000 \times 10^{15}$ , and the computed  $x_2$  is exactly 1 — accurate, as it happens. Back-substitution into the first equation then gives

$$x_1 = \frac{1 - x_2}{\varepsilon} = \frac{0}{10^{-15}} = 0,$$

against a true value of  $1 + 10^{-15}$ : every digit is wrong. The damage was done by subtracting two numbers that agree in all twelve retained digits, so that the difference retains none of them.

*With the exchange.* Pivoting on  $x_1 + x_2 = 2$ , the multiplier is  $\varepsilon = 10^{-15}$  and the second equation becomes  $(1 - \varepsilon)x_2 = 1 - 2\varepsilon$ . At twelve digits both coefficients round to 1.00000000000, giving  $x_2 = 1.00000000000$ , and back-substitution gives  $x_1 = 2 - x_2 = 1.00000000000$ . Both agree with (b) to all twelve digits. The exchange achieved one thing: it made the multiplier at most 1 in modulus, so nothing in the elimination was amplified, and no large quantity was ever created to be subtracted from a comparable one. This is why every practical elimination scheme chooses the largest available pivot.

- (d) The conditioning of the problem is excellent,  $\kappa_{\infty}(\mathbf{A}) < 5$ , and it is the same for both computations, since it does not mention a method. Yet one method returns  $(0, 1)$  and the other returns the right answer. Conditioning bounds what *any* method can promise; whether a given method keeps that promise is stability, and it is Chapter 23's subject.

# Chapter 16

## Variational Principles and Interlacing

Since Chapter 12 the book has written  $\lambda_1(\mathbf{A}) \geq \cdots \geq \lambda_n(\mathbf{A})$  for the eigenvalues of a Hermitian matrix without ever saying what the middle ones *are*. The two at the ends have an answer. Chapter 10 observed that  $\langle T\mathbf{v}, \mathbf{v} \rangle$  is real for a self-adjoint operator, and Chapter 12 §05 turned that into [Lemma 12.5.2](#): on the unit sphere,  $\langle \mathbf{A}\mathbf{x}, \mathbf{x} \rangle$  runs from  $\lambda_n(\mathbf{A})$  up to  $\lambda_1(\mathbf{A})$ , and both ends are attained at eigenvectors. Everything between the ends was left blank, twice and deliberately. Chapter 12 §05 said it in one sentence: the intermediate eigenvalues “have no description as a plain maximum over the whole unit sphere”. Chapter 12 §12 said where the blank would be filled: “Chapter 16 characterizes the eigenvalues of a Hermitian matrix as maxima and minima of  $\langle \mathbf{A}\mathbf{x}, \mathbf{x} \rangle$  — at which point the test becomes the theory.”

This is that chapter, and the device that fills the blank is a **min–max**. One maximum over the unit sphere is not enough information to see  $\lambda_2$ ; a maximum over a *subspace*, followed by a minimum over all subspaces of the right dimension, is exactly enough. That is the Courant–Fischer theorem, and its proof is one dimension count repeated: two subspaces of  $F^n$  whose dimensions add to more than  $n$  must share a non-zero vector. Once every eigenvalue is a maximum and a minimum of a quantity that does not mention an eigenvector, eigenvalues of *different* matrices become comparable, and the rest of the chapter is the harvest. How far the eigenvalues move when the matrix moves (Weyl). How they move when a row and its column are deleted, or a rank-one matrix is added (Cauchy interlacing, and the secular equation). What sums of the largest eigenvalues optimize (Poincaré, Ky Fan), and what arbitrary index sets do (Wielandt, Lidskii). What the diagonal of a Hermitian matrix can be, given its spectrum (Schur). The same theory for singular values, which finally proves the approximation theorem Chapter 12 §10 could only state. Inertia, re-proved without a basis and then made additive across a Schur complement. And a first look at which scalar functions respect the Loewner order.

**Every matrix whose eigenvalues are indexed in this chapter is Hermitian, and the field is  $\mathbb{R}$  or  $\mathbb{C}$ .** This is not a convenience. The index  $i$  in  $\lambda_i(\mathbf{A})$  presupposes that the eigenvalues are real and can be sorted, which is [Theorem 11.2.1](#), and every eigenvalue inequality below is false without it — §04 exhibits a non-Hermitian matrix with a  $1 \times 1$  principal submatrix lying outside the convex hull of the whole matrix’s spectrum. Each section states the hypothesis once and keeps it.

**Eigenvalues are indexed decreasingly,  $\lambda_1(\mathbf{A}) \geq \lambda_2(\mathbf{A}) \geq \cdots \geq \lambda_n(\mathbf{A})$ ,** repeated according

to multiplicity, as Chapter 12 §05 fixed it. Many sources index the min–max theorem the other way, and the two statements differ by  $i \leftrightarrow n + 1 - i$ ; §02 gives the increasing form once, as a remark, so that a reader who meets it elsewhere is not lost.

## 16.0.1 The analysis this chapter imports

None that is new. Chapter 15 named everything it borrowed from analysis in a numbered list (A1)–(A6) in its introduction, and this chapter uses that list and nothing else, citing an item by the same name at the point of use — “by the extreme value theorem, fact (A4) of Chapter 15’s introduction”. Compactness of the unit sphere (A3) and the extreme value theorem (A4) are the two that a reader will expect to see, since a min–max is an optimization and optimizations need their extrema to exist.

It is worth saying in advance that the expectation is not always met. In §§01–02, where the maxima and minima are first written down, **no analytic fact is invoked**: the spectral theorem is already available, and it hands over a maximizer explicitly, so the extreme value theorem is never called on to prove that one exists. That is not the same as saying no analysis lies underneath. The spectral theorem rests on the fundamental theorem of algebra, and Chapter 5 §05 proves that with the extreme value theorem on a closed disc; the analysis was paid for there, once, and is not paid again here. Chapter 11 §02 avoided that theorem on purpose and said so; §§01–02 below return to the remark and settle the account. Where a later section does need an item of the list, it names it. Nothing is added to the list.

## 16.0.2 What you need

- **Chapter 1 §07** — the dimension formula  $\dim(U + W) = \dim U + \dim W - \dim(U \cap W)$ , [Theorem 1.7.3](#). It is the engine of every min–max proof in the chapter, and §02 packages it as a lemma.
- **Chapter 6 and Chapter 7** — principal submatrices ([Definition 6.7.1](#)) and Schur complements: [Definition 7.3.1](#), [Theorem 7.3.2](#), [Theorem 7.3.1](#), [Corollary 7.3.6](#). §§04, 05 and 10 lean on them.
- **Chapter 8** — the trace and determinant as sum and product of the eigenvalues, [Theorem 8.3.3](#).
- **Chapter 10** — inner products, orthonormal bases and the adjoint: [Definition 10.1.1](#), [Definition 10.1.3](#), [Theorem 10.1.5](#), [Theorem 10.2.1](#), [Definition 10.3.1](#), [Theorem 10.3.2](#), [Corollary 10.2.3](#), [Definition 10.6.2](#), [Theorem 10.6.3](#), [Proposition 10.6.7](#).
- **Chapter 11** — the spectral theorems, in both the operator and the matrix form: [Theorem 11.2.1](#), [Theorem 11.4.2](#), [Corollary 11.4.3](#), [Theorem 11.5.1](#), [Corollary 11.5.2](#), [Corollary 11.5.4](#), and the functional calculus [Definition 11.8.2](#), [Theorem 11.8.3](#).
- **Chapter 12** — positivity, the Loewner order and the singular value decomposition: [Definition 12.1.1](#), [Theorem 12.1.4](#), [Theorem 12.1.5](#), [Theorem 12.2.1](#), [Definition 12.5.1](#), [Proposition 12.5.1](#), [Lemma 12.5.2](#), [Theorem 12.5.3](#), [Proposition 12.5.4](#), [Definition 12.8.1](#), [Theorem 12.8.1](#), [Theorem 12.8.3](#), [Definition 12.10.1](#), [Lemma 12.10.2](#), [Theorem 12.10.3](#); from §04 the generalized eigenvalues ([Definition 12.4.1](#), [Theorem 12.4.2](#)), and from §06 Hadamard’s inequality and the block criterion ([Theorem 12.6.4](#), [Theorem 12.6.1](#)), each of which gets a second proof here.
- **Chapter 13 §§02, 05** — congruence ([Definition 13.2.1](#)), inertia and signature: [Theorem 13.5.1](#), [Definition 13.5.1](#), [Theorem 13.5.3](#).
- **Chapter 15** — the way a matrix is measured, and the questions left open: [Defini-](#)

tion 15.3.2, Theorem 15.3.2, and Theorem 15.3.3, whose part (c) is the formula  $\|\mathbf{A}\|_2 = \sigma_1(\mathbf{A})$ ; then Definition 15.4.1, Theorem 15.4.1, Definition 15.8.1, and the imported-analysis list of the Chapter 15 introduction.

### 16.0.3 Roadmap

- **The Rayleigh quotient.**  $R_{\mathbf{A}}(\mathbf{x}) = \langle \mathbf{A}\mathbf{x}, \mathbf{x} \rangle / \langle \mathbf{x}, \mathbf{x} \rangle$  promoted from a test to an object: its range is exactly  $[\lambda_n, \lambda_1]$ , its critical points on the unit sphere are exactly the unit eigenvectors, and the first description of  $\lambda_k$  — maximize after deflating by the top eigenvectors — is correct but circular.
- **The min–max theorem.** Courant–Fischer:  $\lambda_k$  is a minimum over subspaces of a maximum over vectors, and equally a maximum over subspaces of a minimum. No eigenvector is named, so two matrices can now be compared eigenvalue by eigenvalue.
- **How far eigenvalues move.** Weyl’s inequalities, and the Lipschitz bound  $|\lambda_i(\mathbf{A} + \mathbf{E}) - \lambda_i(\mathbf{A})| \leq \|\mathbf{E}\|_2$  that Chapter 15 §07 promised.
- **Deleting a row and a column.** Cauchy interlacing: the spectrum of a principal submatrix is caught between consecutive eigenvalues of the whole matrix. When the interlacing is strict, which settles a promise of Chapter 10: the zeros of consecutive orthogonal polynomials interlace strictly. And why none of it survives without symmetry.
- **Adding a rank-one matrix.** Interlacing again, the secular equation that locates the new eigenvalues between the old, and the shifts that add up to  $\tau\|\mathbf{v}\|^2$ .
- **Subspaces and sums.** Poincaré’s separation theorem, and Ky Fan’s: the sum of the  $k$  largest eigenvalues is a maximum of a trace over orthonormal  $k$ -frames, hence a convex function of the matrix. Chapter 12’s capture bound turns out to be one case of it.
- **Arbitrary index sets.** The Wielandt min–max over a flag of subspaces, and Lidskii’s inequality. It contains Ky Fan’s subadditivity and the Weyl inequalities with one index at an end — but not all of Weyl’s, which is worth seeing.
- **The diagonal and the spectrum.** Majorization; Schur’s theorem and Horn’s converse, which together say that the diagonals of the Hermitian matrices with a given spectrum are exactly the vectors that spectrum majorizes.
- **The same theory for singular values.** The min–max for  $\sigma_k$ , perturbation with no hypothesis on the perturbation at all, the spectral-norm form of Eckart–Young, interlacing when a row or column is deleted, and the distance from an invertible matrix to the nearest singular one — which is what Chapter 15’s condition number was measuring all along.
- **Inertia, again and additively.** A second proof of Sylvester’s law with no basis in it, now over  $\mathbb{C}$  as well; how far a rank- $r$  perturbation can move the inertia; and Haynsworth’s additivity across a Schur complement.
- **Which functions respect the order.** Operator monotone and operator convex functions, why the square root behaves and the square does not, and a statement of the classification that Chapter 20 proves.

### 16.0.4 Named moves

- **Count dimensions to force an intersection.** Two subspaces of an  $n$ -dimensional space whose dimensions add to more than  $n$  meet in a non-zero vector. This one line proves Courant–Fischer and Weyl directly, and Cauchy interlacing and Poincaré through Courant–Fischer. It is the chapter’s single technique, and everything else is bookkeeping around it.
- **Prove a min–max by two inequalities of different characters.** One direction ex-



hibits a **witness** subspace — usually a span of eigenvectors — and computes on it. The other takes an **arbitrary** subspace and intersects it with a span of eigenvectors from the other end. Neither half resembles the other, and recognizing which half is which is most of the work in reading any proof below.

- **Apply the result to  $-\mathbf{A}$ .** The two halves of Courant–Fischer, the two families of Weyl inequalities and the top- and bottom- $k$  trace theorems are each other’s images under  $\mathbf{A} \mapsto -\mathbf{A}$ , which reverses the order and sends  $\lambda_i$  to  $-\lambda_{n+1-i}$ . Deriving the second statement this way is a genuine argument, not a wave of the hand, and it is written out each time.
- **Compress, then interlace.** Restricting the quadratic form to a subspace produces a smaller Hermitian matrix whose eigenvalues are trapped by the original’s. Deleting rows and columns, adding a rank-one matrix and cutting a singular value decomposition short are all this move.
- **Perturb and compare, eigenvalue by eigenvalue.** Chapter 15’s move, now applied one index at a time rather than to the spectrum as a whole.
- **Reduce singular values to eigenvalues.** Either through  $\mathbf{A}^*\mathbf{A}$ , which is shortest, or through the Hermitian dilation, which is symmetric in  $\mathbf{A}$  and  $\mathbf{A}^*$ . Both appear, and each section says which it uses and why.
- **Majorize.** When the right comparison is between *sums* of the largest entries rather than individual ones, compare the partial sums of the decreasing rearrangements. The diagonal of a Hermitian matrix against its spectrum (§08), and the Ky Fan and Lidskii sums (§§06–07), are all of this type.

## 16.1 The Rayleigh Quotient

Chapter 12 used the number  $\langle \mathbf{A}\mathbf{x}, \mathbf{x} \rangle$  as a *test*: positive for every non-zero  $\mathbf{x}$  means positive definite, non-negative means positive semidefinite, and that was the end of it. Chapter 12 §05 then noticed that the test carries more information than the verdict — normalized, it ranges over the interval between the smallest and largest eigenvalues — and stopped there. This section makes the normalized quantity an object with a name, finds its range exactly, finds the vectors where it is flat, and extracts a first description of every eigenvalue as a maximum. The description is correct and unsatisfying, and saying precisely why is the business of §02.

**Throughout,  $F$  is  $\mathbb{R}$  or  $\mathbb{C}$ , every matrix is Hermitian, and its eigenvalues are indexed decreasingly,  $\lambda_1(\mathbf{A}) \geq \cdots \geq \lambda_n(\mathbf{A})$ , repeated according to multiplicity.** The indexing presupposes that the eigenvalues are real, which is [Theorem 11.2.1](#). Inner products are the standard ones of  $F^n$ , linear in the first slot.

### 16.1.1 From a test to an object

Look at what [Lemma 12.5.2](#) actually says. For a unit vector  $\mathbf{x}$ , the number  $\langle \mathbf{A}\mathbf{x}, \mathbf{x} \rangle$  lies between  $\lambda_n(\mathbf{A})$  and  $\lambda_1(\mathbf{A})$ , and it reaches both ends. The restriction to unit vectors is a nuisance: doubling  $\mathbf{x}$  quadruples  $\langle \mathbf{A}\mathbf{x}, \mathbf{x} \rangle$ , so for  $\mathbf{A} \neq 0$  the raw quantity has no bounded range at all and every statement about it must carry the clause “for  $\|\mathbf{x}\| = 1$ ” along. Dividing by  $\|\mathbf{x}\|^2$  removes the nuisance permanently, and the resulting expression occurs often enough in what follows to deserve a name of its own.

*The Rayleigh quotient measures how much a Hermitian matrix stretches a vector along its own direction, on a scale where the vector’s length has been divided out.*

**Definition 16.1.1 — Rayleigh Quotient**

Let  $F$  be  $\mathbb{R}$  or  $\mathbb{C}$  and let  $\mathbf{A} \in M_n(F)$  be **Hermitian**. The **Rayleigh quotient** of  $\mathbf{A}$  is the function

$$R_{\mathbf{A}}(\mathbf{x}) := \frac{\langle \mathbf{A}\mathbf{x}, \mathbf{x} \rangle}{\langle \mathbf{x}, \mathbf{x} \rangle},$$

defined at **every**  $\mathbf{x} \in F^n$  with  $\mathbf{x} \neq \mathbf{0}$ , and at no other vector.

For a self-adjoint operator  $T$  on a finite-dimensional inner product space  $V$  over  $F$ , the same formula  $R_T(\mathbf{v}) = \langle T\mathbf{v}, \mathbf{v} \rangle / \langle \mathbf{v}, \mathbf{v} \rangle$ , for  $\mathbf{v} \neq \mathbf{0}$ , defines the Rayleigh quotient of  $T$ .

Clause by clause. The word **Hermitian** is what makes the values real, and so what makes it sensible to ask for a maximum: by [Proposition 10.6.7](#) (a),  $\langle \mathbf{A}\mathbf{x}, \mathbf{x} \rangle \in \mathbb{R}$  for every  $\mathbf{x}$ , even over  $\mathbb{C}$ . The condition  $\mathbf{x} \neq \mathbf{0}$  is what makes the fraction mean anything:  $\langle \mathbf{x}, \mathbf{x} \rangle = \|\mathbf{x}\|^2 > 0$  exactly for  $\mathbf{x} \neq \mathbf{0}$ , by positive definiteness of the inner product ([Definition 10.1.1](#)), and there is no value to assign at  $\mathbf{0}$  — the numerator vanishes there too, and the quotient  $0/0$  would have to be given a value by hand, which no theorem below would thank us for. So  $R_{\mathbf{A}}$  is a real-valued function on  $F^n \setminus \{\mathbf{0}\}$ .

**Well-definedness has a second half, and it is the useful one.** For a scalar  $c \neq 0$ ,

$$\begin{aligned}\langle \mathbf{A}(c\mathbf{x}), c\mathbf{x} \rangle &= c\bar{c} \langle \mathbf{A}\mathbf{x}, \mathbf{x} \rangle = |c|^2 \langle \mathbf{A}\mathbf{x}, \mathbf{x} \rangle, \\ \langle c\mathbf{x}, c\mathbf{x} \rangle &= |c|^2 \langle \mathbf{x}, \mathbf{x} \rangle,\end{aligned}$$

so the factor  $|c|^2$  cancels and

$$R_{\mathbf{A}}(c\mathbf{x}) = R_{\mathbf{A}}(\mathbf{x}) \quad \text{for every } c \neq 0.$$

The Rayleigh quotient is therefore constant on each line through the origin with the origin removed. It is really a function on the set of lines, and the cheapest way to evaluate it is to normalize: for a **unit** vector  $\mathbf{x}$ ,

$$R_{\mathbf{A}}(\mathbf{x}) = \langle \mathbf{A}\mathbf{x}, \mathbf{x} \rangle.$$

Every statement below may be read either way — over all non-zero vectors, or over the unit sphere — and we shall move between the two without comment.

**Example. (A diagonal matrix, and the whole chapter in miniature)**

Let  $\mathbf{A} = \text{diag}(d_1, \dots, d_n)$  with real  $d_i$ . Compute  $R_{\mathbf{A}}$ .

*Solution.* For  $\mathbf{x} = (x_1, \dots, x_n) \neq \mathbf{0}$  we have  $\mathbf{A}\mathbf{x} = (d_1x_1, \dots, d_nx_n)$ , so

$$R_{\mathbf{A}}(\mathbf{x}) = \frac{\sum_i d_i |x_i|^2}{\sum_i |x_i|^2} = \sum_i w_i d_i, \quad w_i = \frac{|x_i|^2}{\|\mathbf{x}\|^2}.$$

The weights  $w_i$  are non-negative and sum to 1. So  $R_{\mathbf{A}}(\mathbf{x})$  is a **weighted average of the diagonal entries**, and  $\mathbf{x}$  is nothing but the recipe for the weights.

That one line is the entire chapter in miniature, and it is worth pausing on. A weighted average of  $d_1, \dots, d_n$  can be any number between the smallest and the largest, and nothing else; it equals the largest only when all the weight sits on the largest entries; and to force it down to the second largest one must **forbid** the weight from sitting on the first. Every theorem in this chapter is

one of those three sentences, stated for a general Hermitian matrix, where the diagonal entries are replaced by the eigenvalues and the coordinates  $x_i$  by the coefficients of  $\mathbf{x}$  in an orthonormal eigenbasis. The spectral theorem is what turns the general case into this one.

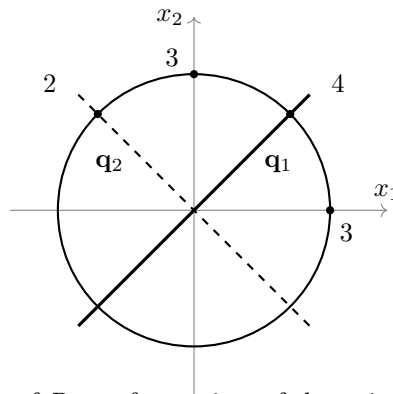
**Example. (A symmetric matrix on the unit circle)**

Let  $\mathbf{A} = \begin{pmatrix} 3 & 1 \\ 1 & 3 \end{pmatrix} \in M_2(\mathbb{R})$ . Describe  $R_{\mathbf{A}}$  on the unit circle, and find its largest and smallest values.

*Solution.* The eigenvalues are 4 and 2, with orthogonal eigenvectors  $\mathbf{q}_1 = \frac{1}{\sqrt{2}}(1, 1)$  and  $\mathbf{q}_2 = \frac{1}{\sqrt{2}}(1, -1)$ : indeed  $\mathbf{A}(1, 1) = (4, 4)$  and  $\mathbf{A}(1, -1) = (2, -2)$ . Writing a unit vector as  $\mathbf{x} = (\cos \theta, \sin \theta)$ ,

$$R_{\mathbf{A}}(\mathbf{x}) = \langle \mathbf{A}\mathbf{x}, \mathbf{x} \rangle = 3 \cos^2 \theta + 2 \cos \theta \sin \theta + 3 \sin^2 \theta = 3 + \sin 2\theta.$$

So the values run over  $[2, 4]$ ; the maximum 4 occurs at  $\theta = \pi/4$  and  $\theta = 5\pi/4$ , that is along the line through  $\mathbf{q}_1$ , and the minimum 2 at  $\theta = 3\pi/4$  and  $7\pi/4$ , along the line through  $\mathbf{q}_2$ . At  $\theta = 0$  and  $\theta = \pi/2$  the value is 3, the average of the two eigenvalues, which is also  $a_{11} = a_{22}$ .



The values of  $R_{\mathbf{A}}$  at four points of the unit circle, for the symmetric matrix with diagonal entries 3 and off-diagonal entries 1. It is largest along the eigenline through  $\mathbf{q}_1$ , smallest along the one through  $\mathbf{q}_2$ , and takes every value in between

**Example. (Two matrices whose quotient is forced)**

Two further cases, at opposite extremes.

- (a) **The identity.**  $R_{\mathbf{I}}(\mathbf{x}) = \langle \mathbf{x}, \mathbf{x} \rangle / \langle \mathbf{x}, \mathbf{x} \rangle = 1$  for every  $\mathbf{x} \neq \mathbf{0}$ . The quotient is constant, and the constant is the only eigenvalue. More generally  $R_{c\mathbf{I}} \equiv c$  for real  $c$ .
- (b) **An orthogonal projection.** Let  $\mathbf{P} \in M_n(F)$  be the matrix of the orthogonal projection onto a subspace  $U$ , so  $\mathbf{P}^2 = \mathbf{P}$  and  $\mathbf{P}^* = \mathbf{P}$  (Theorem 12.12.1). Then  $\langle \mathbf{P}\mathbf{x}, \mathbf{x} \rangle = \langle \mathbf{P}^2\mathbf{x}, \mathbf{x} \rangle = \langle \mathbf{P}\mathbf{x}, \mathbf{P}\mathbf{x} \rangle = \|\mathbf{P}\mathbf{x}\|^2$ , so

$$R_{\mathbf{P}}(\mathbf{x}) = \frac{\|\mathbf{P}\mathbf{x}\|^2}{\|\mathbf{x}\|^2} \in [0, 1],$$

the fraction of  $\mathbf{x}$  that survives the projection. Since  $\|\mathbf{x}\|^2 = \|\mathbf{P}\mathbf{x}\|^2 + \|\mathbf{x} - \mathbf{P}\mathbf{x}\|^2$  by [Theorem 10.1.3](#), it equals 1 exactly when  $\mathbf{P}\mathbf{x} = \mathbf{x}$ , that is on  $U \setminus \{\mathbf{0}\}$ , and 0 exactly when  $\mathbf{P}\mathbf{x} = \mathbf{0}$ , that is on  $U^\perp \setminus \{\mathbf{0}\}$  — matching the eigenvalues 1 and 0 of  $\mathbf{P}$ .

Case (a) is the degenerate one, and it earns its place: when all the eigenvalues coincide the interval of values collapses to a point, every non-zero vector is an eigenvector, and every theorem below becomes an equality between two copies of the same number. A statement that fails to survive  $\mathbf{A} = c\mathbf{I}$  has been stated wrongly.

**Example. (Change one word: drop Hermitian)**

Take  $\mathbf{A} = \begin{pmatrix} 0 & 1 \\ 0 & 0 \end{pmatrix} \in M_2(\mathbb{C})$ , which is not Hermitian, and form  $\langle \mathbf{A}\mathbf{x}, \mathbf{x} \rangle / \langle \mathbf{x}, \mathbf{x} \rangle$  anyway.

At  $\mathbf{x} = (1, i)$  the numerator is  $\langle (i, 0), (1, i) \rangle = i \cdot \bar{1} = i$  and the denominator is 2, so the quotient is  $i/2$ , which is not real. The clause that fails is the word **Hermitian** in [Definition 16.1.1](#), and it fails at the first step: [Proposition 10.6.7](#) (a) is what made the numerator real, and its hypothesis is gone.

Over  $\mathbb{R}$  the failure is quieter and worse. For  $\mathbf{A} = \begin{pmatrix} 0 & -1 \\ 1 & 0 \end{pmatrix} \in M_2(\mathbb{R})$  and real  $\mathbf{x} = (x_1, x_2)$ ,

$$\langle \mathbf{A}\mathbf{x}, \mathbf{x} \rangle = -x_2x_1 + x_1x_2 = 0,$$

so the quotient is real, and identically 0. It is perfectly well behaved and tells us nothing:  $\mathbf{A}$  has no real eigenvalue at all, and there is no interval  $[\lambda_n, \lambda_1]$  for the quotient to describe.

Why this definition, then. Dropping “Hermitian” costs the reality of the values, hence any notion of largest; the real  $2 \times 2$  example above shows that it also destroys the link with the spectrum even when the values happen to stay real. Dropping the denominator costs scale invariance, and with it the possibility of writing max without a side condition. Keeping the denominator but allowing  $\mathbf{x} = \mathbf{0}$  costs nothing but sense.

*Warning.*  $R_{\mathbf{A}}(\mathbf{x})$  and  $\langle \mathbf{A}\mathbf{x}, \mathbf{x} \rangle$  are the same number only for unit  $\mathbf{x}$ . The bound  $\langle \mathbf{A}\mathbf{x}, \mathbf{x} \rangle \leq \lambda_1(\mathbf{A})$  of [Lemma 12.5.2](#) is stated for unit vectors and is false without that hypothesis: for  $\mathbf{A} = \mathbf{I}_2$  and  $\mathbf{x} = (10, 0)$  it would read  $100 \leq 1$ . The bound  $R_{\mathbf{A}}(\mathbf{x}) \leq \lambda_1(\mathbf{A})$ , by contrast, holds for every  $\mathbf{x} \neq \mathbf{0}$ . Whenever a maximum over “all  $\mathbf{x}$ ” appears below, it is a maximum of the **quotient**; the un-normalized form of the same statement always carries  $\|\mathbf{x}\| = 1$  with it.

## 16.1.2 What values it takes

[Lemma 12.5.2](#) gives the two ends of the range. The following proposition adds that nothing in between is missing, which is a smaller statement than it looks and is worth proving explicitly rather than quoting a continuity argument.

### Proposition 16.1.1 — The Range of the Rayleigh Quotient

Let  $\mathbf{A} \in M_n(F)$  be Hermitian,  $n \geq 1$ . Then

$$\{R_{\mathbf{A}}(\mathbf{x}) : \mathbf{x} \in F^n, \mathbf{x} \neq \mathbf{0}\} = [\lambda_n(\mathbf{A}), \lambda_1(\mathbf{A})],$$

the whole closed interval. In particular

$$\lambda_1(\mathbf{A}) = \max_{\mathbf{x} \neq \mathbf{0}} R_{\mathbf{A}}(\mathbf{x}), \quad \lambda_n(\mathbf{A}) = \min_{\mathbf{x} \neq \mathbf{0}} R_{\mathbf{A}}(\mathbf{x}),$$

and both are attained, at unit eigenvectors for  $\lambda_1(\mathbf{A})$  and  $\lambda_n(\mathbf{A})$ .

*Idea.* Two inclusions of quite different weights. ( $\subseteq$ ) is [Lemma 12.5.2](#) with the normalization divided out, and costs one line. ( $\supseteq$ ) asks for a vector realizing each intermediate value; the natural instinct is to move continuously along the sphere from a top eigenvector to a bottom one and quote the intermediate value theorem, but there is no need to import anything. Mixing a top eigenvector and a bottom eigenvector in the proportion  $t : 1 - t$  puts weight  $t$  on  $\lambda_1$  and  $1 - t$  on  $\lambda_n$ , and the weighted average of [Example \(A diagonal matrix, and the whole chapter in miniature\)](#) sweeps the interval as  $t$  does.

*Proof.* By the spectral theorem for matrices — [Corollary 11.4.3](#) over  $\mathbb{C}$ , [Corollary 11.5.2](#) over  $\mathbb{R}$  — there is an orthonormal basis  $(\mathbf{q}_1, \dots, \mathbf{q}_n)$  of  $F^n$  with  $\mathbf{A}\mathbf{q}_i = \lambda_i(\mathbf{A})\mathbf{q}_i$  for every  $i$ .

( $\subseteq$ ) Let  $\mathbf{x} \neq \mathbf{0}$  and put  $\mathbf{u} = \mathbf{x}/\|\mathbf{x}\|$ , a unit vector. By the scale invariance established above,  $R_{\mathbf{A}}(\mathbf{x}) = R_{\mathbf{A}}(\mathbf{u}) = \langle \mathbf{A}\mathbf{u}, \mathbf{u} \rangle$ , which lies in  $[\lambda_n(\mathbf{A}), \lambda_1(\mathbf{A})]$  by [Lemma 12.5.2](#).

( $\supseteq$ ) Let  $\mu \in [\lambda_n(\mathbf{A}), \lambda_1(\mathbf{A})]$ . If  $\lambda_1(\mathbf{A}) = \lambda_n(\mathbf{A})$  the interval is the single point  $\mu = \lambda_1(\mathbf{A}) = R_{\mathbf{A}}(\mathbf{q}_1)$ , and we are done. Otherwise put

$$t = \frac{\mu - \lambda_n(\mathbf{A})}{\lambda_1(\mathbf{A}) - \lambda_n(\mathbf{A})} \in [0, 1], \quad \mathbf{x} = \sqrt{t}\mathbf{q}_1 + \sqrt{1-t}\mathbf{q}_n,$$

which is legitimate because the denominator is non-zero and both square roots are of non-negative reals. Note  $n \geq 2$  here, since  $\lambda_1 \neq \lambda_n$ , so  $\mathbf{q}_1$  and  $\mathbf{q}_n$  are distinct members of an orthonormal list. By [Theorem 10.2.1](#) (c),

$$\|\mathbf{x}\|^2 = t + (1-t) = 1,$$

so in particular  $\mathbf{x} \neq \mathbf{0}$ . Since  $\mathbf{A}\mathbf{x} = \sqrt{t}\lambda_1(\mathbf{A})\mathbf{q}_1 + \sqrt{1-t}\lambda_n(\mathbf{A})\mathbf{q}_n$ , [Theorem 10.2.1](#) (b) gives

$$R_{\mathbf{A}}(\mathbf{x}) = \langle \mathbf{A}\mathbf{x}, \mathbf{x} \rangle = t\lambda_1(\mathbf{A}) + (1-t)\lambda_n(\mathbf{A}) = \mu,$$

the last equality by the choice of  $t$ . So every point of the interval is a value.

For the attainment,  $R_{\mathbf{A}}(\mathbf{q}_1) = \lambda_1(\mathbf{A})$  and  $R_{\mathbf{A}}(\mathbf{q}_n) = \lambda_n(\mathbf{A})$ . This proves the proposition. ■

So the range is an **interval**, not a finite set, even though  $\mathbf{A}$  has at most  $n$  eigenvalues. That single fact is responsible for most of the confusion this chapter has to clear up, and the warning below spells out why.

*Quick check.* For  $\mathbf{A} = \text{diag}(6, 3, 0)$ , write down two unit vectors with Rayleigh quotient 4, each having one zero coordinate, in different positions.

*Solution.* Mixing  $\mathbf{e}_1$  and  $\mathbf{e}_3$  as in the proof:  $4 = 6t + 0(1 - t)$  gives  $t = 2/3$  and  $\mathbf{x} = (\sqrt{2/3}, 0, \sqrt{1/3})$ , for which  $R_{\mathbf{A}}(\mathbf{x}) = 6 \cdot \frac{2}{3} + 0 \cdot \frac{1}{3} = 4$ . Mixing  $\mathbf{e}_1$  and  $\mathbf{e}_2$  instead:  $4 = 6t + 3(1 - t)$  gives  $t = 1/3$  and  $\mathbf{y} = (\sqrt{1/3}, \sqrt{2/3}, 0)$ , for which  $R_{\mathbf{A}}(\mathbf{y}) = 6 \cdot \frac{1}{3} + 3 \cdot \frac{2}{3} = 4$ . A value of  $R_{\mathbf{A}}$  is realized by many vectors, and they need not resemble each other.

### 16.1.3 Where the quotient is flat

The maximum and the minimum are attained at eigenvectors. There is a converse, and it identifies the eigenvectors among all vectors by a condition that does not mention eigenvalues: they are the places where  $R_{\mathbf{A}}$  does not move to first order. Everything needed is one algebraic identity, exact for every  $t$ , so no calculus enters.

#### Proposition 16.1.2 — Critical Points of the Rayleigh Quotient

Let  $\mathbf{A} \in M_n(F)$  be Hermitian, let  $\mathbf{x} \in F^n$  be a **unit** vector and put  $\mu = R_{\mathbf{A}}(\mathbf{x}) = \langle \mathbf{A}\mathbf{x}, \mathbf{x} \rangle$ . For every  $\mathbf{v} \in F^n$  and every  $t \in \mathbb{R}$  with  $\mathbf{x} + t\mathbf{v} \neq \mathbf{0}$ ,

$$\begin{aligned} R_{\mathbf{A}}(\mathbf{x} + t\mathbf{v}) - \mu \\ = \frac{2t \operatorname{Re} \langle \mathbf{A}\mathbf{x} - \mu\mathbf{x}, \mathbf{v} \rangle + t^2 (\langle \mathbf{A}\mathbf{v}, \mathbf{v} \rangle - \mu \|\mathbf{v}\|^2)}{\|\mathbf{x} + t\mathbf{v}\|^2}. \end{aligned} \quad (16.1.1)$$

Moreover:

- (a)  $\operatorname{Re} \langle \mathbf{A}\mathbf{x} - \mu\mathbf{x}, \mathbf{v} \rangle = 0$  for every  $\mathbf{v} \in F^n$  **if and only if**  $\mathbf{A}\mathbf{x} = \mu\mathbf{x}$ ; that is, if and only if  $\mathbf{x}$  is an eigenvector of  $\mathbf{A}$ , necessarily for the eigenvalue  $\mu$ .
- (b) If  $R_{\mathbf{A}}(\mathbf{x}) \geq R_{\mathbf{A}}(\mathbf{y})$  for every  $\mathbf{y} \neq \mathbf{0}$ , then the condition in (a) holds. The same is true if  $R_{\mathbf{A}}(\mathbf{x}) \leq R_{\mathbf{A}}(\mathbf{y})$  for every  $\mathbf{y} \neq \mathbf{0}$ .

*Idea.* Expand the numerator and the denominator of  $R_{\mathbf{A}}(\mathbf{x} + t\mathbf{v})$  in powers of  $t$  and subtract  $\mu$  times the denominator from the numerator. The constant terms cancel by the definition of  $\mu$ , and what is left is exactly Eq. 16.1.1. The coefficient of  $t$  is the object of interest: (a) says it vanishes in every direction precisely at eigenvectors, and (b) says that at a maximum it must vanish, since otherwise a small step in the right direction would increase the quotient — the  $t^2$  term cannot interfere, because for small  $t$  it is smaller than the  $t$  term.

*Proof. The identity.* Expanding by sesquilinearity,

$$\begin{aligned} \langle \mathbf{A}(\mathbf{x} + t\mathbf{v}), \mathbf{x} + t\mathbf{v} \rangle &= \langle \mathbf{A}\mathbf{x}, \mathbf{x} \rangle + t\langle \mathbf{A}\mathbf{x}, \mathbf{v} \rangle + t\langle \mathbf{A}\mathbf{v}, \mathbf{x} \rangle + t^2\langle \mathbf{A}\mathbf{v}, \mathbf{v} \rangle \\ &= \mu + 2t \operatorname{Re} \langle \mathbf{A}\mathbf{x}, \mathbf{v} \rangle + t^2\langle \mathbf{A}\mathbf{v}, \mathbf{v} \rangle, \end{aligned}$$

where the second line used  $\langle \mathbf{A}\mathbf{v}, \mathbf{x} \rangle = \langle \mathbf{v}, \mathbf{A}\mathbf{x} \rangle = \overline{\langle \mathbf{A}\mathbf{x}, \mathbf{v} \rangle}$ , the first equality because  $\mathbf{A}$  is Hermitian and the second by conjugate symmetry (Definition 10.1.1), together with  $\langle \mathbf{A}\mathbf{x}, \mathbf{x} \rangle = \mu$  for the unit vector  $\mathbf{x}$ . The same computation with  $\mathbf{A}$  replaced by  $\mathbf{I}$ , where the constant term is  $\|\mathbf{x}\|^2 = 1$ , gives

$$\|\mathbf{x} + t\mathbf{v}\|^2 = 1 + 2t \operatorname{Re} \langle \mathbf{x}, \mathbf{v} \rangle + t^2\|\mathbf{v}\|^2.$$

Since  $\mu$  is real,  $\mu \operatorname{Re}\langle \mathbf{x}, \mathbf{v} \rangle = \operatorname{Re}\langle \mu \mathbf{x}, \mathbf{v} \rangle$ . Subtracting  $\mu \|\mathbf{x} + t\mathbf{v}\|^2$  from the first display therefore leaves

$$2t \operatorname{Re}\langle \mathbf{A}\mathbf{x} - \mu \mathbf{x}, \mathbf{v} \rangle + t^2(\langle \mathbf{A}\mathbf{v}, \mathbf{v} \rangle - \mu \|\mathbf{v}\|^2),$$

and dividing by  $\|\mathbf{x} + t\mathbf{v}\|^2 \neq 0$  gives Eq. 16.1.1.

- (a) Put  $\mathbf{w} = \mathbf{A}\mathbf{x} - \mu \mathbf{x}$ . ( $\Leftarrow$ ) If  $\mathbf{A}\mathbf{x} = \mu \mathbf{x}$  then  $\mathbf{w} = \mathbf{0}$  and every  $\langle \mathbf{w}, \mathbf{v} \rangle$  vanishes. ( $\Rightarrow$ ) Take  $\mathbf{v} = \mathbf{w}$ : the hypothesis gives  $\operatorname{Re}\langle \mathbf{w}, \mathbf{w} \rangle = \|\mathbf{w}\|^2 = 0$ , so  $\mathbf{w} = \mathbf{0}$  by positive definiteness, that is  $\mathbf{A}\mathbf{x} = \mu \mathbf{x}$ . Since  $\mathbf{x} \neq \mathbf{0}$ , this exhibits  $\mathbf{x}$  as an eigenvector for the eigenvalue  $\mu$ .
- (b) Suppose  $\mathbf{x}$  maximizes  $R_{\mathbf{A}}$  and, for a contradiction, that  $c := 2 \operatorname{Re}\langle \mathbf{A}\mathbf{x} - \mu \mathbf{x}, \mathbf{v} \rangle \neq 0$  for some  $\mathbf{v}$ ; then  $\mathbf{v} \neq \mathbf{0}$ . Put  $d = \langle \mathbf{A}\mathbf{v}, \mathbf{v} \rangle - \mu \|\mathbf{v}\|^2$ , and let  $t$  be any real number with

$$0 < |t| < \frac{1}{2\|\mathbf{v}\|}, \quad |t| |d| < \frac{1}{2}|c|, \quad t \text{ of the same sign as } c,$$

which is possible because the two upper bounds are positive. The first bound gives  $\|\mathbf{x} + t\mathbf{v}\| \geq \|\mathbf{x}\| - |t|\|\mathbf{v}\| > 1 - \frac{1}{2} > 0$ , by the triangle inequality (Corollary 10.1.6), so  $\mathbf{x} + t\mathbf{v} \neq \mathbf{0}$  and the identity applies. The second gives  $|td| < |c|/2$ , so  $c + td$  has the same sign as  $c$ , and hence  $t(c + td) > 0$ . By Eq. 16.1.1,

$$R_{\mathbf{A}}(\mathbf{x} + t\mathbf{v}) - \mu = \frac{t(c + td)}{\|\mathbf{x} + t\mathbf{v}\|^2} > 0,$$

contradicting the maximality of  $R_{\mathbf{A}}(\mathbf{x})$ . So  $c = 0$  for every  $\mathbf{v}$ . For a minimizer, apply what has just been proved to  $-\mathbf{A}$ , whose Rayleigh quotient is  $-R_{\mathbf{A}}$ , so that a minimizer of  $R_{\mathbf{A}}$  is a maximizer of  $R_{-\mathbf{A}}$ ; the condition in (a) for  $-\mathbf{A}$  and  $-\mu$  is the same condition. This proves the proposition. ■

Call a unit vector  $\mathbf{x}$  a **critical point** of  $R_{\mathbf{A}}$  when the condition in (a) holds, that is, when the coefficient of  $t$  in Eq. 16.1.1 vanishes in every direction  $\mathbf{v}$  — when  $R_{\mathbf{A}}$  moves away from  $R_{\mathbf{A}}(\mathbf{x})$  only at second order. Parts (a) and (b) then say: *the critical points of  $R_{\mathbf{A}}$  on the unit sphere are exactly the unit eigenvectors of  $\mathbf{A}$ , the critical values are exactly the eigenvalues, and every maximizer and every minimizer is among them.* That the eigenvalues are all realized as critical values is a direct computation:  $R_{\mathbf{A}}(\mathbf{q}_i) = \langle \lambda_i \mathbf{q}_i, \mathbf{q}_i \rangle = \lambda_i$ , and each  $\mathbf{q}_i$  is a unit eigenvector, hence a critical point by (b).

*Remark.* A reader who knows calculus will recognize Eq. 16.1.1 as saying that the directional derivative of  $R_{\mathbf{A}}$  at a unit vector  $\mathbf{x}$  in the direction  $\mathbf{v}$  is  $2 \operatorname{Re}\langle \mathbf{A}\mathbf{x} - \mu \mathbf{x}, \mathbf{v} \rangle$ , so that “critical point” has its usual meaning of a vanishing gradient. Nothing above uses derivatives: the identity Eq. 16.1.1 is exact for every  $t$ , not an approximation, and the argument in (b) compares two explicit numbers. This matters for the accounting in the next remark.

*Remark. The promise from Chapter 11 §02, and what it cost.* Chapter 11 §02 proved that a real self-adjoint operator has an eigenvalue by an algebraic route, and then described the alternative it was declining: “the function  $\mathbf{v} \mapsto \langle T\mathbf{v}, \mathbf{v} \rangle$  is real-valued and continuous on the unit sphere of  $V$ , which is closed and bounded, so it attains a maximum, and a vector where it does so turns out to be an eigenvector for the largest eigenvalue. That argument is shorter to state but it is analysis, not algebra, and it needs the extreme value theorem; we keep to the algebraic route so that nothing in



this chapter depends on it. The maximization picture returns in Chapter 16, where the Rayleigh quotient and the Courant–Fischer theorem describe every eigenvalue of a self-adjoint operator by optimization.”

The picture has now returned, and the account can be settled exactly. The clause “a vector where it does so turns out to be an eigenvector” is [Proposition 16.1.2](#) (b) and (a), and that the eigenvalue is the *largest* one is [Proposition 16.1.1](#); neither invokes an analytic fact. What Chapter 11 would have had to import is only the *existence* of the maximizer, which is the extreme value theorem — fact (A4) of Chapter 15’s introduction — applied to a continuous function on the unit sphere, compact by fact (A3). This chapter does not import it, because it does not have to: the spectral theorem is available here and it hands over the maximizer by name, namely  $\mathbf{q}_1$ , in [Proposition 16.1.1](#). So the optimization description of eigenvalues in this chapter is a **consequence** of the spectral theorem, not an independent proof of it. One honest qualification: the spectral theorem itself rests on the fundamental theorem of algebra, which Chapter 5 §05 proved with the extreme value theorem on a disc. So the analysis is not absent from the foundations — it was spent once, there, and nothing here spends it again. Section 2 completes the promise for the intermediate eigenvalues, on the same terms.

## 16.1.4 Why the second eigenvalue is not a second maximum

**Warning.**  $\lambda_2(\mathbf{A})$  is not “the second largest value of  $R_{\mathbf{A}}$ ”, and there is no sense in which it could be. Take  $\mathbf{A} = \text{diag}(3, 1, -1)$ , so  $\lambda_1 = 3$ ,  $\lambda_2 = 1$ ,  $\lambda_3 = -1$ . By [Proposition 16.1.1](#) the set of values of  $R_{\mathbf{A}}$  is the whole interval  $[-1, 3]$ . Three things go wrong at once.

- **There is no second largest value.** The values below 3 form the interval  $[-1, 3)$ , which has no largest element. So the phrase does not describe a number.
- **The value  $\lambda_2$  is attained, but so is every other value, and not only at eigenvectors.** At  $\mathbf{x} = \frac{1}{\sqrt{2}}(1, 0, 1)$  the quotient is  $\frac{1}{2}(3) + \frac{1}{2}(-1) = 1 = \lambda_2(\mathbf{A})$ , and  $\mathbf{x}$  is not an eigenvector of  $\mathbf{A}$ : it mixes the eigenvectors  $\mathbf{e}_1$  and  $\mathbf{e}_3$  of the first and the *third* eigenvalues, and  $\lambda_2$  plays no part in it.
- **Removing a direction does not help by itself.** At  $\mathbf{y} = \frac{1}{\sqrt{2}}(1, 1, 0)$ , which is orthogonal to the eigenvector  $\mathbf{e}_3$  of  $\lambda_3$ , the quotient is  $\frac{1}{2}(3) + \frac{1}{2}(1) = 2$ , strictly between  $\lambda_1$  and  $\lambda_2$ . Constraining  $\mathbf{x}$  to a subspace changes the answer, and which subspace is everything.

Whatever describes  $\lambda_2$  must be a maximum over something smaller than the sphere.

The last bullet says what to try. Cutting the sphere down by orthogonality to the top eigenvector does work, and the proof is [Example \(A diagonal matrix, and the whole chapter in miniature\)](#) again, with the top entry removed from the average.

### Proposition 16.1.3 — Eigenvalues by Deflation

Let  $\mathbf{A} \in M_n(F)$  be Hermitian and let  $(\mathbf{q}_1, \dots, \mathbf{q}_n)$  be an orthonormal basis of  $F^n$  with



$\mathbf{A}\mathbf{q}_i = \lambda_i(\mathbf{A})\mathbf{q}_i$  for every  $i$ . Then for every  $k$  with  $1 \leq k \leq n$ ,

$$\lambda_k(\mathbf{A}) = \max\{R_{\mathbf{A}}(\mathbf{x}) : \mathbf{x} \neq \mathbf{0}, \mathbf{x} \perp \mathbf{q}_1, \dots, \mathbf{q}_{k-1}\},$$

the maximum being attained at  $\mathbf{x} = \mathbf{q}_k$ . For  $k = 1$  the list of constraints is empty and the statement is [Proposition 16.1.1](#).

*Idea.* The constraints do one thing: they delete the first  $k - 1$  coordinates in the eigenbasis. What is left is the weighted average of [Example \(A diagonal matrix, and the whole chapter in miniature\)](#) over the remaining eigenvalues  $\lambda_k \geq \dots \geq \lambda_n$ , whose largest member is  $\lambda_k$ .

*Proof.* Let  $W = \{\mathbf{x} \in F^n : \langle \mathbf{x}, \mathbf{q}_i \rangle = 0 \text{ for } 1 \leq i \leq k - 1\}$ . By [Theorem 10.2.1](#) (a) every  $\mathbf{x} \in F^n$  satisfies  $\mathbf{x} = \sum_{i=1}^n \langle \mathbf{x}, \mathbf{q}_i \rangle \mathbf{q}_i$ , so  $\mathbf{x} \in W$  exactly when  $\mathbf{x} \in \text{span}(\mathbf{q}_k, \dots, \mathbf{q}_n)$ .

Let  $\mathbf{x} \in W$  with  $\mathbf{x} \neq \mathbf{0}$ , and write  $c_i = \langle \mathbf{x}, \mathbf{q}_i \rangle$ , so that  $\mathbf{x} = \sum_{i \geq k} c_i \mathbf{q}_i$ . Then  $\mathbf{A}\mathbf{x} = \sum_{i \geq k} \lambda_i(\mathbf{A})c_i \mathbf{q}_i$ , and [Theorem 10.2.1](#) (b), (c) give

$$\langle \mathbf{A}\mathbf{x}, \mathbf{x} \rangle = \sum_{i \geq k} \lambda_i(\mathbf{A})|c_i|^2, \quad \|\mathbf{x}\|^2 = \sum_{i \geq k} |c_i|^2 > 0.$$

Hence  $R_{\mathbf{A}}(\mathbf{x})$  is a weighted average of  $\lambda_k(\mathbf{A}), \dots, \lambda_n(\mathbf{A})$  with non-negative weights  $|c_i|^2 / \|\mathbf{x}\|^2$  summing to 1. The largest of those eigenvalues is  $\lambda_k(\mathbf{A})$ , because the list is indexed decreasingly, so  $R_{\mathbf{A}}(\mathbf{x}) \leq \lambda_k(\mathbf{A})$ .

Finally  $\mathbf{q}_k \in W$ , since the  $\mathbf{q}_i$  are orthonormal, and  $R_{\mathbf{A}}(\mathbf{q}_k) = \langle \lambda_k(\mathbf{A})\mathbf{q}_k, \mathbf{q}_k \rangle = \lambda_k(\mathbf{A})$ . So the bound is attained and the supremum is a maximum. This proves the proposition. ■

*Quick check.* In [Proposition 16.1.3](#), does the value of the maximum depend on which orthonormal eigenbasis  $(\mathbf{q}_1, \dots, \mathbf{q}_n)$  is chosen? Does the *maximizing vector*?

*Solution.* The value does not: it is  $\lambda_k(\mathbf{A})$ , a number attached to  $\mathbf{A}$  alone. The maximizing vector generally does, and so does the constraint set  $W$ . For  $\mathbf{A} = \text{diag}(1, 1, 0)$  and  $k = 2$ , one admissible eigenbasis is  $(\mathbf{e}_1, \mathbf{e}_2, \mathbf{e}_3)$ , giving  $W = \text{span}(\mathbf{e}_2, \mathbf{e}_3)$  and maximizer  $\mathbf{e}_2$ ; another is  $(\frac{1}{\sqrt{2}}(1, 1, 0), \frac{1}{\sqrt{2}}(1, -1, 0), \mathbf{e}_3)$ , giving a different  $W$  and the maximizer  $\frac{1}{\sqrt{2}}(1, -1, 0)$ . Both maxima equal  $\lambda_2(\mathbf{A}) = 1$ .

## 16.1.5 What is still wrong

[Proposition 16.1.3](#) is a complete answer to the question “what is  $\lambda_k(\mathbf{A})$ , as an optimization?” and it is nearly useless. The reason is visible in its statement: the constraint set is built from  $\mathbf{q}_1, \dots, \mathbf{q}_{k-1}$ , which are eigenvectors of  $\mathbf{A}$ . To use the formula one must already know the eigenvectors, and to know the eigenvectors one must already have solved the problem the formula claims to describe. It is a description of  $\lambda_k(\mathbf{A})$  in terms of  $\mathbf{A}$ ’s own answer.

The cost is not merely aesthetic. The whole point of turning an eigenvalue into an optimization is to compare the eigenvalues of **different** matrices: to say that  $\mathbf{A} \succeq \mathbf{B}$  forces  $\lambda_i(\mathbf{A}) \geq \lambda_i(\mathbf{B})$ , or that perturbing  $\mathbf{A}$  moves  $\lambda_i$  only a little. A comparison needs the two optimizations to run over

the *same* set. Deflation gives  $\lambda_i(\mathbf{A})$  as a maximum over the orthogonal complement of  $\mathbf{A}$ 's top eigenvectors, and  $\lambda_i(\mathbf{B})$  as a maximum over the orthogonal complement of  $\mathbf{B}$ 's top eigenvectors, and those two subspaces have nothing to do with each other. Nothing can be concluded.

Chapter 12 §12 promised that “Chapter 16 characterizes the eigenvalues of a Hermitian matrix as maxima and minima of  $\langle \mathbf{A}\mathbf{x}, \mathbf{x} \rangle$  — at which point the test becomes the theory”. [Proposition 16.1.1](#) and [Proposition 16.1.3](#) are the first installment of that, and they show that the promise can be kept. The test has already become a statement about a single number:  $\mathbf{A} \succ 0$  exactly when  $\lambda_n(\mathbf{A}) = \min_{\mathbf{x} \neq \mathbf{0}} R_{\mathbf{A}}(\mathbf{x})$  is positive, and  $\mathbf{A} \succeq 0$  exactly when it is non-negative, which is the eigenvalue criterion of [Theorem 12.1.5](#) and [Theorem 12.1.4](#) read through [Proposition 16.1.1](#). The rest of it is the removal of the eigenvectors from the statement, which §02 achieves by a device that looks extravagant and is exactly right: keep the maximum over a subspace, but let the subspace vary over **all** subspaces of the appropriate dimension, and take the minimum of what comes out.

## 16.1.6 Exercises

### 16.1.6.1 A. Check your understanding

#### Exercise A1.

- State [Definition 16.1.1](#), naming both hypotheses, and say what each one is for.
- Prove that  $R_{\mathbf{A}}(c\mathbf{x}) = R_{\mathbf{A}}(\mathbf{x})$  for every scalar  $c \neq 0$ , and explain why this makes  $R_{\mathbf{A}}$  a function on lines.
- Determine whether the following is correct, and justify your answer: if  $\mathbf{A} \in M_3(\mathbb{R})$  is symmetric with three distinct eigenvalues, then  $R_{\mathbf{A}}$  takes exactly three values.
- State [Proposition 16.1.3](#), and say in one sentence why it is not a satisfactory description of  $\lambda_k(\mathbf{A})$ .
- Give a Hermitian  $\mathbf{A}$  and a vector  $\mathbf{x}$  that is **not** an eigenvector of  $\mathbf{A}$  but has  $R_{\mathbf{A}}(\mathbf{x})$  equal to an eigenvalue of  $\mathbf{A}$ .

*Solution.*

- Let  $F$  be  $\mathbb{R}$  or  $\mathbb{C}$  and  $\mathbf{A} \in M_n(F)$  Hermitian. For  $\mathbf{x} \neq \mathbf{0}$ , the Rayleigh quotient is  $R_{\mathbf{A}}(\mathbf{x}) = \langle \mathbf{A}\mathbf{x}, \mathbf{x} \rangle / \langle \mathbf{x}, \mathbf{x} \rangle$ . Hermitian makes the numerator real ([Proposition 10.6.7](#) (a)) and ties the values to the spectrum;  $\mathbf{x} \neq \mathbf{0}$  makes the denominator non-zero ([Definition 10.1.1](#)).
- $\langle \mathbf{A}(c\mathbf{x}), c\mathbf{x} \rangle = c\bar{c}\langle \mathbf{A}\mathbf{x}, \mathbf{x} \rangle = |c|^2\langle \mathbf{A}\mathbf{x}, \mathbf{x} \rangle$  and likewise  $\langle c\mathbf{x}, c\mathbf{x} \rangle = |c|^2\langle \mathbf{x}, \mathbf{x} \rangle$ ; the factor  $|c|^2 \neq 0$  cancels. So  $R_{\mathbf{A}}$  is constant on  $\{c\mathbf{x} : c \neq 0\}$ , the punctured line through  $\mathbf{x}$ , and its value depends only on which line  $\mathbf{x}$  lies on.
- Incorrect. By [Proposition 16.1.1](#) the set of values is the whole of the interval  $[\lambda_3(\mathbf{A}), \lambda_1(\mathbf{A})]$ , which is infinite whenever  $\lambda_1(\mathbf{A}) \neq \lambda_3(\mathbf{A})$ , and distinct eigenvalues guarantee that.
- With  $(\mathbf{q}_1, \dots, \mathbf{q}_n)$  an orthonormal eigenbasis ordered so that  $\mathbf{A}\mathbf{q}_i = \lambda_i(\mathbf{A})\mathbf{q}_i$ ,  $\lambda_k(\mathbf{A})$  is the maximum of  $R_{\mathbf{A}}$  over the non-zero vectors orthogonal to  $\mathbf{q}_1, \dots, \mathbf{q}_{k-1}$ . It is unsatisfactory because the constraint set is built from eigenvectors of  $\mathbf{A}$ , so the formula presupposes the answer and cannot compare two different matrices.
- $\mathbf{A} = \text{diag}(3, 1, -1)$  and  $\mathbf{x} = (1, 0, 1)$ :  $R_{\mathbf{A}}(\mathbf{x}) = (3 - 1)/2 = 1 = \lambda_2(\mathbf{A})$ , while

$\mathbf{Ax} = (3, 0, -1)$  is not a multiple of  $\mathbf{x}$ . (Any mixture of a top and a bottom eigenvector in the right proportion does the same job.)

### 16.1.6.2 B. Practice

#### Exercise B1: Locating four values.

Let

$$\mathbf{A} = \begin{pmatrix} 2 & 2 & 0 \\ 2 & 3 & 2 \\ 0 & 2 & 4 \end{pmatrix} \in M_3(\mathbb{R}).$$

Its eigenvalues are 6, 3, 0, with eigenvectors  $(1, 2, 2)$ ,  $(2, 1, -2)$  and  $(2, -2, 1)$  respectively; you may take this as given. Compute  $R_{\mathbf{A}}(\mathbf{x})$  for  $\mathbf{x} = \mathbf{e}_1$ ,  $\mathbf{x} = (1, 1, 1)$ ,  $\mathbf{x} = (1, 1, 0)$  and  $\mathbf{x} = (1, 0, -1)$ , and check each value against [Proposition 16.1.1](#). Hence exhibit a non-eigenvector at which  $R_{\mathbf{A}}$  equals an eigenvalue.

*Solution.* The three given vectors are pairwise orthogonal, each of norm 3, so  $\mathbf{q}_1 = \frac{1}{3}(1, 2, 2)$ ,  $\mathbf{q}_2 = \frac{1}{3}(2, 1, -2)$ ,  $\mathbf{q}_3 = \frac{1}{3}(2, -2, 1)$  is an orthonormal eigenbasis with  $\lambda_1 = 6$ ,  $\lambda_2 = 3$ ,  $\lambda_3 = 0$ . By [Proposition 16.1.1](#) every value must lie in  $[0, 6]$ .

For  $\mathbf{x} = \mathbf{e}_1$ :  $\mathbf{Ax} = (2, 2, 0)$ , so  $\langle \mathbf{Ax}, \mathbf{x} \rangle = 2$  and  $\|\mathbf{x}\|^2 = 1$ , giving  $R_{\mathbf{A}}(\mathbf{e}_1) = 2$ . (This is the diagonal entry  $a_{11}$ , as it must be for a standard basis vector.)

For  $\mathbf{x} = (1, 1, 1)$ :  $\mathbf{Ax} = (4, 7, 6)$  and  $\langle \mathbf{Ax}, \mathbf{x} \rangle = 4 + 7 + 6 = 17$ , while  $\|\mathbf{x}\|^2 = 3$ ; so  $R_{\mathbf{A}}(\mathbf{x}) = 17/3 \approx 5.67$ .

For  $\mathbf{x} = (1, 1, 0)$ :  $\mathbf{Ax} = (4, 5, 2)$  and  $\langle \mathbf{Ax}, \mathbf{x} \rangle = 9$ ,  $\|\mathbf{x}\|^2 = 2$ ; so  $R_{\mathbf{A}}(\mathbf{x}) = 9/2$ .

For  $\mathbf{x} = (1, 0, -1)$ :  $\mathbf{Ax} = (2, 0, -4)$  and  $\langle \mathbf{Ax}, \mathbf{x} \rangle = 2 + 4 = 6$ ,  $\|\mathbf{x}\|^2 = 2$ ; so  $R_{\mathbf{A}}(\mathbf{x}) = 3$ .

All four values lie in  $[0, 6]$ . The last one equals  $\lambda_2(\mathbf{A}) = 3$ , yet  $\mathbf{A}(1, 0, -1) = (2, 0, -4)$  is not a multiple of  $(1, 0, -1)$ , so  $(1, 0, -1)$  is not an eigenvector. This is the trap of the warning above, in a concrete matrix.

#### Exercise B2: A prescribed value.

With  $\mathbf{A}$  and the orthonormal eigenbasis of [Exercise \(B1: Locating four values\)](#), use the construction in the proof of [Proposition 16.1.1](#) to produce a unit vector  $\mathbf{x}$  with  $R_{\mathbf{A}}(\mathbf{x}) = 2$ . Then find, by mixing  $\mathbf{q}_1$  and  $\mathbf{q}_2$  in equal proportion, a vector with integer entries whose Rayleigh quotient is  $9/2$ .

*Solution.* For the first,  $\mu = 2$  with  $\lambda_1 = 6$  and  $\lambda_3 = 0$  gives  $t = (2 - 0)/(6 - 0) = 1/3$ , so

$$\mathbf{x} = \sqrt{\frac{1}{3}} \mathbf{q}_1 + \sqrt{\frac{2}{3}} \mathbf{q}_3,$$

a unit vector with  $R_{\mathbf{A}}(\mathbf{x}) = \frac{1}{3} \cdot 6 + \frac{2}{3} \cdot 0 = 2$ .

For the second,  $\mathbf{y} = \mathbf{q}_1 + \mathbf{q}_2 = \frac{1}{3}((1, 2, 2) + (2, 1, -2)) = (1, 1, 0)$ , whose Rayleigh quotient is the average  $\frac{1}{2}(6) + \frac{1}{2}(3) = 9/2$  — matching the direct computation in [Exercise \(B1: Locating four values\)](#). Equal proportions of two orthonormal eigenvectors give the plain average of their eigenvalues.

**Exercise B3: The quotient of a projection.**

Let  $U = \text{span}\{(1, 1, 0), (0, 0, 1)\} \subseteq \mathbb{R}^3$  and let  $\mathbf{P}$  be the matrix of the orthogonal projection onto  $U$ .

- (a) Compute  $\mathbf{P}$ , and list the eigenvalues of  $\mathbf{P}$  with multiplicity.
- (b) Compute  $R_{\mathbf{P}}(\mathbf{x})$  for  $\mathbf{x} = \mathbf{e}_1$ ,  $\mathbf{x} = (1, -1, 0)$  and  $\mathbf{x} = (1, -1, 1)$ , and check each against [Proposition 16.1.1](#).

*Solution.*

- (a) The two spanning vectors are orthogonal, of norms  $\sqrt{2}$  and 1, so  $\mathbf{u}_1 = \frac{1}{\sqrt{2}}(1, 1, 0)$  and  $\mathbf{u}_2 = (0, 0, 1)$  form an orthonormal basis of  $U$  and  $\mathbf{P} = \mathbf{u}_1\mathbf{u}_1^\top + \mathbf{u}_2\mathbf{u}_2^\top$ , that is

$$\mathbf{P} = \begin{pmatrix} 1/2 & 1/2 & 0 \\ 1/2 & 1/2 & 0 \\ 0 & 0 & 1 \end{pmatrix}.$$

Its eigenvalues are 1 with multiplicity 2 (on  $U$ ) and 0 with multiplicity 1 (on  $U^\perp = \text{span}\{(1, -1, 0)\}$ ), so  $\lambda_1 = \lambda_2 = 1$  and  $\lambda_3 = 0$ .

- (b) By [Example \(Two matrices whose quotient is forced\)](#) (b),  $R_{\mathbf{P}}(\mathbf{x}) = \|\mathbf{P}\mathbf{x}\|^2 / \|\mathbf{x}\|^2$ . For  $\mathbf{x} = \mathbf{e}_1$ :  $\mathbf{P}\mathbf{e}_1 = (\frac{1}{2}, \frac{1}{2}, 0)$ , so  $\|\mathbf{P}\mathbf{x}\|^2 = \frac{1}{2}$  and  $R_{\mathbf{P}}(\mathbf{e}_1) = \frac{1}{2}$ . For  $\mathbf{x} = (1, -1, 0)$ :  $\mathbf{P}\mathbf{x} = \mathbf{0}$ , so  $R_{\mathbf{P}}(\mathbf{x}) = 0$ . For  $\mathbf{x} = (1, -1, 1)$ :  $\mathbf{P}\mathbf{x} = (0, 0, 1)$ , so  $\|\mathbf{P}\mathbf{x}\|^2 = 1$  and  $\|\mathbf{x}\|^2 = 3$ , giving  $R_{\mathbf{P}}(\mathbf{x}) = 1/3$ . All three lie in  $[\lambda_3, \lambda_1] = [0, 1]$ , and the middle one attains the lower end, as it must, being a non-zero vector of  $U^\perp$ .

**16.1.6.3 C. Going deeper****Exercise C1: Why the quotient estimates eigenvalues so well.**

Let  $\mathbf{A} \in M_n(F)$  be Hermitian with orthonormal eigenbasis  $(\mathbf{q}_1, \dots, \mathbf{q}_n)$ ,  $\mathbf{A}\mathbf{q}_i = \lambda_i(\mathbf{A})\mathbf{q}_i$ .

- (a) Prove that for every unit vector  $\mathbf{x}$ ,

$$\lambda_1(\mathbf{A}) - R_{\mathbf{A}}(\mathbf{x}) \leq (\lambda_1(\mathbf{A}) - \lambda_n(\mathbf{A})) \|\mathbf{x} - \mathbf{q}_1\|^2.$$

- (b) Deduce that if  $\|\mathbf{x} - \mathbf{q}_1\| \leq \varepsilon$  then  $R_{\mathbf{A}}(\mathbf{x})$  approximates  $\lambda_1(\mathbf{A})$  with error at most  $(\lambda_1 - \lambda_n)\varepsilon^2$ : an error in the eigenvector of size  $\varepsilon$  costs only  $\varepsilon^2$  in the eigenvalue.
- (c) Show that the exponent 2 cannot be improved, by computing both sides for  $\mathbf{A} = \text{diag}(1, 0)$  and  $\mathbf{x} = (\cos \varepsilon, \sin \varepsilon)$ .

*Hint for (a): expand  $\mathbf{x}$  in the eigenbasis and compare the two sums term by term.*

*Solution.*

- (a) Write  $c_i = \langle \mathbf{x}, \mathbf{q}_i \rangle$ , so  $\sum_i |c_i|^2 = 1$  by [Theorem 10.2.1](#) (c). As in [Proposition 16.1.3](#),  $R_{\mathbf{A}}(\mathbf{x}) = \sum_i \lambda_i(\mathbf{A})|c_i|^2$ , so

$$\lambda_1(\mathbf{A}) - R_{\mathbf{A}}(\mathbf{x}) = \sum_{i=1}^n (\lambda_1(\mathbf{A}) - \lambda_i(\mathbf{A}))|c_i|^2 \leq (\lambda_1(\mathbf{A}) - \lambda_n(\mathbf{A})) \sum_{i \geq 2} |c_i|^2,$$

where the  $i = 1$  term was dropped because it is 0 and each remaining factor  $\lambda_1 - \lambda_i$  was enlarged to  $\lambda_1 - \lambda_n$ , legitimate since  $|c_i|^2 \geq 0$  and the eigenvalues decrease. On the other side, again by [Theorem 10.2.1 \(c\)](#),

$$\|\mathbf{x} - \mathbf{q}_1\|^2 = |c_1 - 1|^2 + \sum_{i \geq 2} |c_i|^2 \geq \sum_{i \geq 2} |c_i|^2.$$

Combining the two displays gives the inequality.

- (b) Immediate from (a), since the right-hand side is increasing in  $\|\mathbf{x} - \mathbf{q}_1\|$ .  
 (c) Here  $\lambda_1 = 1$ ,  $\lambda_2 = 0$  and  $\mathbf{q}_1 = \mathbf{e}_1$ . The vector  $\mathbf{x} = (\cos \varepsilon, \sin \varepsilon)$  is a unit vector with

$$R_{\mathbf{A}}(\mathbf{x}) = \cos^2 \varepsilon, \quad \lambda_1 - R_{\mathbf{A}}(\mathbf{x}) = \sin^2 \varepsilon,$$

while  $\|\mathbf{x} - \mathbf{e}_1\|^2 = (\cos \varepsilon - 1)^2 + \sin^2 \varepsilon = 2 - 2 \cos \varepsilon$ . The bound in (a) reads  $\sin^2 \varepsilon \leq 2 - 2 \cos \varepsilon$ , which is true, and the ratio of the two sides is

$$\frac{\sin^2 \varepsilon}{2 - 2 \cos \varepsilon} = \frac{(1 - \cos \varepsilon)(1 + \cos \varepsilon)}{2(1 - \cos \varepsilon)} = \frac{1 + \cos \varepsilon}{2},$$

which tends to 1 as  $\varepsilon \rightarrow 0$ . So the error really is of order  $\varepsilon^2$ , not smaller, and the constant in (a) is sharp in the limit. This second-order accuracy is why the Rayleigh quotient is the standard way to read an eigenvalue off an approximate eigenvector, and Chapter 23 builds an iteration on it.

### Exercise C2: The equality case at the top.

Let  $\mathbf{A} \in M_n(F)$  be Hermitian and let  $E_{\lambda_1}(\mathbf{A}) = \ker(\mathbf{A} - \lambda_1(\mathbf{A})\mathbf{I})$  be the eigenspace of the largest eigenvalue.

- (a) Prove that if  $\mathbf{x} \neq \mathbf{0}$  and  $R_{\mathbf{A}}(\mathbf{x}) = \lambda_1(\mathbf{A})$ , then  $\mathbf{x} \in E_{\lambda_1}(\mathbf{A})$ .  
 (b) Deduce that  $\{\mathbf{x} \neq \mathbf{0} : R_{\mathbf{A}}(\mathbf{x}) = \lambda_1(\mathbf{A})\} \cup \{\mathbf{0}\} = E_{\lambda_1}(\mathbf{A})$ .  
 (c) Show that the corresponding statement for an intermediate eigenvalue is **false**, by naming a matrix, an index  $k$  with  $1 < k < n$ , and a vector.

*Solution.*

- (a) We may assume  $\|\mathbf{x}\| = 1$ , by scale invariance and because  $E_{\lambda_1}(\mathbf{A})$  is a subspace. Write  $c_i = \langle \mathbf{x}, \mathbf{q}_i \rangle$  for an orthonormal eigenbasis as above, so  $\sum_i |c_i|^2 = 1$  and  $R_{\mathbf{A}}(\mathbf{x}) = \sum_i \lambda_i(\mathbf{A}) |c_i|^2$ . Then

$$0 = \lambda_1(\mathbf{A}) - R_{\mathbf{A}}(\mathbf{x}) = \sum_{i=1}^n (\lambda_1(\mathbf{A}) - \lambda_i(\mathbf{A})) |c_i|^2,$$

a sum of non-negative terms. Every term is therefore 0, so  $c_i = 0$  whenever  $\lambda_i(\mathbf{A}) < \lambda_1(\mathbf{A})$ . Hence  $\mathbf{x} = \sum_{i: \lambda_i = \lambda_1} c_i \mathbf{q}_i$ , a combination of eigenvectors for  $\lambda_1(\mathbf{A})$ , so  $\mathbf{x} \in E_{\lambda_1}(\mathbf{A})$ .

- (b) ( $\subseteq$ ) is (a), together with  $\mathbf{0} \in E_{\lambda_1}(\mathbf{A})$ . ( $\supseteq$ ) If  $\mathbf{0} \neq \mathbf{x} \in E_{\lambda_1}(\mathbf{A})$  then  $\mathbf{A}\mathbf{x} = \lambda_1(\mathbf{A})\mathbf{x}$ , so  $R_{\mathbf{A}}(\mathbf{x}) = \lambda_1(\mathbf{A}) \langle \mathbf{x}, \mathbf{x} \rangle / \langle \mathbf{x}, \mathbf{x} \rangle = \lambda_1(\mathbf{A})$ .  
 (c) Take  $\mathbf{A} = \text{diag}(3, 1, -1)$ ,  $k = 2$  and  $\mathbf{x} = (1, 0, 1)$ . Then  $R_{\mathbf{A}}(\mathbf{x}) = 1 = \lambda_2(\mathbf{A})$ , but  $\mathbf{A}\mathbf{x} = (3, 0, -1)$  is not a multiple of  $\mathbf{x}$ , so  $\mathbf{x} \notin E_{\lambda_2}(\mathbf{A}) = \text{span}\{\mathbf{e}_2\}$ . The argument

in (a) breaks because the terms  $(\lambda_2 - \lambda_i)|c_i|^2$  are no longer all of the same sign: the  $i = 1$  term is negative and cancels against the  $i = 3$  term instead of forcing both to vanish.

### Exercise C3: What the quotient cannot see.

Let  $\mathbf{M} \in M_n(F)$  be an arbitrary matrix, not assumed Hermitian, and put

$$\mathbf{A} = \frac{1}{2}(\mathbf{M} + \mathbf{M}^*), \quad \mathbf{B} = \frac{1}{2}(\mathbf{M} - \mathbf{M}^*),$$

so that  $\mathbf{M} = \mathbf{A} + \mathbf{B}$  with  $\mathbf{A}^* = \mathbf{A}$  and  $\mathbf{B}^* = -\mathbf{B}$ .

- (a) Prove that  $\operatorname{Re}\langle \mathbf{M}\mathbf{x}, \mathbf{x} \rangle = \langle \mathbf{A}\mathbf{x}, \mathbf{x} \rangle$  for every  $\mathbf{x} \in F^n$ .
- (b) Deduce that over  $\mathbb{R}$ , the function  $\mathbf{x} \mapsto \langle \mathbf{M}\mathbf{x}, \mathbf{x} \rangle$  determines  $\mathbf{A}$  and nothing whatever about  $\mathbf{B}$ . Give two distinct real  $2 \times 2$  matrices with the same such function.
- (c) Over  $\mathbb{C}$ , determine whether  $\langle \mathbf{M}\mathbf{x}, \mathbf{x} \rangle = 0$  for all  $\mathbf{x}$  forces  $\mathbf{M} = 0$ . Justify your answer.

*Solution.*

- (a) By conjugate symmetry and the definition of the adjoint,  $\langle \mathbf{M}^*\mathbf{x}, \mathbf{x} \rangle = \langle \mathbf{x}, \mathbf{M}\mathbf{x} \rangle = \overline{\langle \mathbf{M}\mathbf{x}, \mathbf{x} \rangle}$ . Hence

$$\langle \mathbf{A}\mathbf{x}, \mathbf{x} \rangle = \frac{1}{2}(\langle \mathbf{M}\mathbf{x}, \mathbf{x} \rangle + \overline{\langle \mathbf{M}\mathbf{x}, \mathbf{x} \rangle}) = \operatorname{Re}\langle \mathbf{M}\mathbf{x}, \mathbf{x} \rangle.$$

- (b) Over  $\mathbb{R}$  every inner product value is real, so (a) reads  $\langle \mathbf{M}\mathbf{x}, \mathbf{x} \rangle = \langle \mathbf{A}\mathbf{x}, \mathbf{x} \rangle$  for all  $\mathbf{x}$ : the function sees only  $\mathbf{A}$ . It determines  $\mathbf{A}$  completely, since two symmetric matrices with the same quadratic form are equal ([Theorem 11.2.5](#) applied to their difference). It sees nothing of  $\mathbf{B}$ : for

$$\mathbf{M}_1 = \begin{pmatrix} 0 & 0 \\ 0 & 0 \end{pmatrix}, \quad \mathbf{M}_2 = \begin{pmatrix} 0 & -1 \\ 1 & 0 \end{pmatrix},$$

both functions are identically 0, since  $\langle \mathbf{M}_2\mathbf{x}, \mathbf{x} \rangle = -x_2x_1 + x_1x_2 = 0$ , yet  $\mathbf{M}_1 \neq \mathbf{M}_2$ . This is [Exercise \(C3: What the real quadratic form does determine\)](#) in matrix language.

- (c) Yes, over  $\mathbb{C}$  it does force  $\mathbf{M} = 0$ . This is [Theorem 11.2.6](#), and the reason the complex theory is the cleaner one: a complex quadratic form determines its matrix, so nothing is invisible. Concretely, (a) gives  $\mathbf{A} = 0$ , and applying the hypothesis to  $i\mathbf{M}$  — or expanding  $\langle \mathbf{M}(\mathbf{x} + \mathbf{y}), \mathbf{x} + \mathbf{y} \rangle$  and  $\langle \mathbf{M}(\mathbf{x} + i\mathbf{y}), \mathbf{x} + i\mathbf{y} \rangle$  — kills  $\mathbf{B}$  as well. The real  $2 \times 2$  matrix  $\mathbf{M}_2$  of (b) shows that the same statement over  $\mathbb{R}$  is false, so this is genuinely a fact about the field.

## 16.2 The Min–Max Theorem

Section 1 ended with a description of  $\lambda_k(\mathbf{A})$  that works and cannot be used: a maximum of the Rayleigh quotient over the orthogonal complement of  $\mathbf{A}$ 's own top eigenvectors. The defect is that the constraint set is built from the answer. This section repairs it by a move that

looks like an extravagance and is exactly the right one: keep the maximum over a subspace, stop insisting that the subspace be the right one, and let it range over **all** subspaces of the appropriate dimension. The minimum of what comes out is again  $\lambda_k(\mathbf{A})$ , and no eigenvector appears anywhere in the statement. That is the Courant–Fischer theorem, and everything in the rest of the chapter is a consequence of it.

**Throughout,  $F$  is  $\mathbb{R}$  or  $\mathbb{C}$  and every matrix is Hermitian.** The eigenvalues of a Hermitian  $\mathbf{A} \in M_n(F)$  are real ([Theorem 11.2.1](#)) and are **indexed decreasingly**,

$$\lambda_1(\mathbf{A}) \geq \lambda_2(\mathbf{A}) \geq \dots \geq \lambda_n(\mathbf{A}),$$

each repeated according to multiplicity, so that the list has exactly  $n$  entries. This is the book’s convention, fixed in Chapter 12 §05, and it is used without exception below. Many sources index the theorem of this section the other way round; the remark after the theorem gives that form, once.

## 16.2.1 Two subspaces that have to meet

One dimension count does all the work in this chapter, and it is worth a label of its own, because four later proofs cite it.

*Two subspaces of an  $n$ -dimensional space cannot both be large and disjoint: if their dimensions add to more than  $n$ , they share a non-zero vector.*

### Lemma 16.2.1 — Large Subspaces Intersect

Let  $V$  be a vector space with  $\dim V = n$  over any field, and let  $U$  and  $W$  be subspaces of  $V$  with

$$\dim U + \dim W > n.$$

Then  $U \cap W \neq \{\mathbf{0}\}$ . More precisely,  $\dim(U \cap W) \geq \dim U + \dim W - n$ .

*Proof.* By [Theorem 1.7.3](#),

$$\dim(U + W) = \dim U + \dim W - \dim(U \cap W).$$

Both  $U$  and  $W$  are finite-dimensional, being subspaces of the  $n$ -dimensional  $V$  ([Theorem 1.6.7](#));  $U + W$  is a subspace of  $V$  ([Theorem 1.7.1](#)), so the same theorem gives  $\dim(U + W) \leq n$ , and rearranging gives

$$\dim(U \cap W) \geq \dim U + \dim W - n,$$

which is  $> 0$  by hypothesis. A subspace of positive dimension contains a non-zero vector. This proves the lemma. ■

*Remark.* The bound is sharp, and the hypothesis cannot be weakened to  $\dim U + \dim W = n$ : two distinct lines in  $\mathbb{R}^2$  meet only at  $\mathbf{0}$ , and so do a line and a plane in  $\mathbb{R}^3$  chosen in general position. What makes the count useful is that it needs nothing about the two subspaces beyond their dimensions — in every application below, one of them is a span of eigenvectors that we choose and the other is an arbitrary competitor we know

nothing about.

## 16.2.2 The quotient on a subspace

Before quantifying over subspaces, we must know that the inner maximum and minimum exist. They do, and for an algebraic reason: restricting the quadratic form to a subspace produces a smaller Hermitian matrix, to which [Proposition 16.1.1](#) applies verbatim.

### Lemma 16.2.2 — The Rayleigh Quotient on a Subspace

Let  $\mathbf{A} \in M_n(F)$  be Hermitian and let  $W \subseteq F^n$  be a subspace with  $\dim W = m \geq 1$ . Let  $\mathbf{Q} \in M_{n \times m}(F)$  be a matrix whose columns form an orthonormal basis of  $W$ , and put  $\mathbf{B} = \mathbf{Q}^* \mathbf{A} \mathbf{Q} \in M_m(F)$ . Then  $\mathbf{B}$  is Hermitian and

$$\{R_{\mathbf{A}}(\mathbf{x}) : \mathbf{0} \neq \mathbf{x} \in W\} = [\lambda_m(\mathbf{B}), \lambda_1(\mathbf{B})].$$

In particular  $\max_{\mathbf{0} \neq \mathbf{x} \in W} R_{\mathbf{A}}(\mathbf{x})$  and  $\min_{\mathbf{0} \neq \mathbf{x} \in W} R_{\mathbf{A}}(\mathbf{x})$  exist and are attained.

*Idea.* Coordinates on  $W$ . Writing a vector of  $W$  as  $\mathbf{Q}\mathbf{c}$  turns  $\langle \mathbf{A}\mathbf{x}, \mathbf{x} \rangle$  into  $\langle \mathbf{B}\mathbf{c}, \mathbf{c} \rangle$  and, because the columns of  $\mathbf{Q}$  are orthonormal, turns  $\langle \mathbf{x}, \mathbf{x} \rangle$  into  $\langle \mathbf{c}, \mathbf{c} \rangle$  with no distortion. So the Rayleigh quotient of  $\mathbf{A}$  on  $W$  is the Rayleigh quotient of  $\mathbf{B}$  on  $F^m$ , and [Proposition 16.1.1](#) answers the question there.

*Proof.* Such a  $\mathbf{Q}$  exists:  $W$  has an orthonormal basis by [Theorem 10.2.2](#), and we take its members as columns. Orthonormality of the columns says exactly  $\mathbf{Q}^* \mathbf{Q} = \mathbf{I}_m$ . Then  $\mathbf{B}^* = \mathbf{Q}^* \mathbf{A}^* \mathbf{Q} = \mathbf{Q}^* \mathbf{A} \mathbf{Q} = \mathbf{B}$ , so  $\mathbf{B}$  is Hermitian and [Proposition 16.1.1](#) applies to it.

The map  $\mathbf{c} \mapsto \mathbf{Q}\mathbf{c}$  sends  $F^m$  onto  $W$ , since the columns of  $\mathbf{Q}$  span  $W$ , and it is injective, since  $\mathbf{Q}\mathbf{c} = \mathbf{0}$  gives  $\mathbf{c} = \mathbf{Q}^* \mathbf{Q}\mathbf{c} = \mathbf{0}$ . So it is a bijection  $F^m \rightarrow W$  carrying  $\mathbf{0}$  to  $\mathbf{0}$ , hence a bijection from the non-zero vectors of  $F^m$  onto those of  $W$ . For  $\mathbf{c} \neq \mathbf{0}$ , writing the standard inner product as  $\langle \mathbf{u}, \mathbf{v} \rangle = \mathbf{v}^* \mathbf{u}$ ,

$$\begin{aligned} \langle \mathbf{A}\mathbf{Q}\mathbf{c}, \mathbf{Q}\mathbf{c} \rangle &= \mathbf{c}^* \mathbf{Q}^* \mathbf{A} \mathbf{Q} \mathbf{c} = \langle \mathbf{B}\mathbf{c}, \mathbf{c} \rangle, \\ \langle \mathbf{Q}\mathbf{c}, \mathbf{Q}\mathbf{c} \rangle &= \mathbf{c}^* \mathbf{Q}^* \mathbf{Q} \mathbf{c} = \langle \mathbf{c}, \mathbf{c} \rangle, \end{aligned}$$

so  $R_{\mathbf{A}}(\mathbf{Q}\mathbf{c}) = R_{\mathbf{B}}(\mathbf{c})$ . Therefore the two value sets  $\{R_{\mathbf{A}}(\mathbf{x}) : \mathbf{0} \neq \mathbf{x} \in W\}$  and  $\{R_{\mathbf{B}}(\mathbf{c}) : \mathbf{0} \neq \mathbf{c} \in F^m\}$  are equal, and the second is  $[\lambda_m(\mathbf{B}), \lambda_1(\mathbf{B})]$  with both ends attained, by [Proposition 16.1.1](#). This proves the lemma. ■

*Remark.* **Which analysis was invoked: none.** The existence of a maximum of a function over an infinite set is normally an analytic fact — the set of unit vectors of  $W$  is compact, by fact (A3) of Chapter 15's introduction, and  $R_{\mathbf{A}}$  is continuous on it, so the extreme value theorem, fact (A4), applies. That route is available and correct. It is not needed. The spectral theorem, already proved in Chapter 11, produces the maximizing vector by name, and [Lemma 16.2.2](#) inherits that. Nothing in this section, or in §01, uses an item of Chapter 15's imported-analysis list. The analysis has not vanished, only



moved: the spectral theorem rests on the fundamental theorem of algebra, whose proof in Chapter 5 §05 uses the extreme value theorem on a closed disc.

### 16.2.3 Every eigenvalue is a min-max

#### Theorem 16.2.3 — Courant–Fischer Min-Max Theorem

Let  $F$  be  $\mathbb{R}$  or  $\mathbb{C}$ , let  $\mathbf{A} \in M_n(F)$  be Hermitian and let  $1 \leq k \leq n$ . Then

$$\begin{aligned}\lambda_k(\mathbf{A}) &= \min_{\dim W = n-k+1} \max_{\mathbf{0} \neq \mathbf{x} \in W} R_{\mathbf{A}}(\mathbf{x}) \\ &= \max_{\dim W = k} \min_{\mathbf{0} \neq \mathbf{x} \in W} R_{\mathbf{A}}(\mathbf{x}),\end{aligned}$$

where in each line  $W$  runs over **all** subspaces of  $F^n$  of the stated dimension. All four extrema are attained.

*Idea.* Read the second line first, because it is the one with a slogan:  $\lambda_k$  is the best value you can guarantee on a  $k$ -dimensional subspace. On a  $k$ -dimensional subspace the quotient has a smallest value, which is what that subspace guarantees; the span of the top  $k$  eigenvectors guarantees  $\lambda_k$ , and no other  $k$ -dimensional subspace guarantees more.

The proof is three steps, and the first two have entirely different characters. **Step 1** exhibits a **witness** subspace and computes on it, giving one inequality. **Step 2** takes an **arbitrary** competitor subspace and shows it cannot beat the witness; the only thing known about a competitor is its dimension, so the only available tool is [Lemma 16.2.1](#), which forces the competitor to meet a span of eigenvectors from the other end of the list. On that shared non-zero vector the quotient is pinned, and the inequality follows. **Step 3** obtains the second line from the first by applying it to  $-\mathbf{A}$ , which reverses the order of the eigenvalues and turns every maximum into a minimum.

*Proof.* By [Corollary 11.4.3](#) over  $\mathbb{C}$ , or [Corollary 11.5.2](#) over  $\mathbb{R}$ , fix an orthonormal basis  $(\mathbf{q}_1, \dots, \mathbf{q}_n)$  of  $F^n$  with  $\mathbf{A}\mathbf{q}_i = \lambda_i(\mathbf{A})\mathbf{q}_i$  for every  $i$ . Abbreviate  $\lambda_i = \lambda_i(\mathbf{A})$  and put

$$U_j = \text{span}(\mathbf{q}_1, \dots, \mathbf{q}_j), \quad L_j = \text{span}(\mathbf{q}_j, \dots, \mathbf{q}_n),$$

so that  $\dim U_j = j$  and  $\dim L_j = n - j + 1$ , the lists being orthonormal and hence independent. By [Lemma 16.2.2](#) the inner extrema below all exist.

*Claim.* For every  $j$ :  $R_{\mathbf{A}}(\mathbf{x}) \geq \lambda_j$  for all  $\mathbf{0} \neq \mathbf{x} \in U_j$ , and  $R_{\mathbf{A}}(\mathbf{x}) \leq \lambda_j$  for all  $\mathbf{0} \neq \mathbf{x} \in L_j$ . Both bounds are attained at  $\mathbf{q}_j$ .

*Proof.* Let  $\mathbf{0} \neq \mathbf{x} \in U_j$  and write  $c_i = \langle \mathbf{x}, \mathbf{q}_i \rangle$ , so  $\mathbf{x} = \sum_{i \leq j} c_i \mathbf{q}_i$  by [Theorem 10.2.1](#) (a) and the definition of  $U_j$ . Then  $\mathbf{A}\mathbf{x} = \sum_{i \leq j} \lambda_i c_i \mathbf{q}_i$ , and [Theorem 10.2.1](#) (b), (c) give

$$R_{\mathbf{A}}(\mathbf{x}) = \frac{\sum_{i \leq j} \lambda_i |c_i|^2}{\sum_{i \leq j} |c_i|^2},$$

a weighted average of  $\lambda_1, \dots, \lambda_j$  with non-negative weights summing to 1. Since the list is indexed decreasingly, the smallest of those eigenvalues is  $\lambda_j$ , so the average is  $\geq \lambda_j$ . The statement for  $L_j$  is [Proposition 16.1.3](#):  $L_j$  is precisely the set of vectors orthogonal to  $\mathbf{q}_1, \dots, \mathbf{q}_{j-1}$ , and the maximum of  $R_{\mathbf{A}}$  there is  $\lambda_j$ . In both cases  $R_{\mathbf{A}}(\mathbf{q}_j) = \lambda_j$ . ■

**Step 1. The witness for the first line.** For each subspace  $W$  of dimension  $n - k + 1$  write  $f(W) = \max_{\mathbf{0} \neq \mathbf{x} \in W} R_{\mathbf{A}}(\mathbf{x})$ , which exists by [Lemma 16.2.2](#). The outer minimum of  $f$  over all such  $W$  is not yet known to exist — there are infinitely many  $W$  — so we show that  $\lambda_k$  is a value of  $f$  and a lower bound for all its values; it is then the least value, which is what the minimum means. For the first half take  $W = L_k$ , of dimension  $n - k + 1$ : by the claim,  $f(L_k) = \lambda_k$ .

**Step 2. Every competitor is at least as large.** Let  $W \subseteq F^n$  be any subspace with  $\dim W = n - k + 1$ . Then

$$\dim W + \dim U_k = (n - k + 1) + k = n + 1 > n,$$

so by [Lemma 16.2.1](#) there is a vector  $\mathbf{x} \neq \mathbf{0}$  lying in  $W \cap U_k$ . Being in  $U_k$ , it satisfies  $R_{\mathbf{A}}(\mathbf{x}) \geq \lambda_k$  by the claim; being in  $W$ , it is one of the vectors the inner maximum ranges over. Therefore

$$\max_{\mathbf{0} \neq \mathbf{y} \in W} R_{\mathbf{A}}(\mathbf{y}) \geq R_{\mathbf{A}}(\mathbf{x}) \geq \lambda_k.$$

This holds for every competitor  $W$ , so  $f(W) \geq \lambda_k$  for all of them. With Step 1,  $\lambda_k$  is a value of  $f$  and a lower bound for every value, so it is the least value: the minimum exists, equals  $\lambda_k$ , and is attained at  $W = L_k$ . This proves the first line.

**Step 3. The second line, from the first applied to  $-\mathbf{A}$ .** The matrix  $-\mathbf{A}$  is Hermitian, its eigenvalues are the numbers  $-\lambda_i$ , and sorting those decreasingly reverses the list:

$$\lambda_j(-\mathbf{A}) = -\lambda_{n+1-j}(\mathbf{A}) \quad (1 \leq j \leq n).$$

Also  $R_{-\mathbf{A}}(\mathbf{x}) = -R_{\mathbf{A}}(\mathbf{x})$  for every  $\mathbf{x} \neq \mathbf{0}$ , directly from the definition. Since negation turns a maximum of a set of reals into the minimum of the negated set, and a minimum into a maximum,

$$\begin{aligned} \max_{\mathbf{0} \neq \mathbf{x} \in W} R_{-\mathbf{A}}(\mathbf{x}) &= - \min_{\mathbf{0} \neq \mathbf{x} \in W} R_{\mathbf{A}}(\mathbf{x}), \\ \min_{\dim W=m} (-f(W)) &= - \max_{\dim W=m} f(W) \end{aligned}$$

for any real-valued  $f$  on subspaces, where each side exists exactly when the other does. Apply the first line, already proved, to  $-\mathbf{A}$  with index  $j$ :

$$\begin{aligned} -\lambda_{n+1-j}(\mathbf{A}) &= \lambda_j(-\mathbf{A}) = \min_{\dim W=n-j+1} \max_{\mathbf{0} \neq \mathbf{x} \in W} R_{-\mathbf{A}}(\mathbf{x}) \\ &= - \max_{\dim W=n-j+1} \min_{\mathbf{0} \neq \mathbf{x} \in W} R_{\mathbf{A}}(\mathbf{x}). \end{aligned}$$

Now substitute  $j = n + 1 - k$ , so that  $n - j + 1 = k$  and  $n + 1 - j = k$ . Multiplying by  $-1$  gives

$$\lambda_k(\mathbf{A}) = \max_{\dim W=k} \min_{\mathbf{0} \neq \mathbf{x} \in W} R_{\mathbf{A}}(\mathbf{x}),$$

with the maximum attained, the attainment transported from Step 1 by the same negation; explicitly it occurs at  $W = U_k$ , which is the subspace  $L_{n+1-k}$  of the eigenbasis of  $-\mathbf{A}$  taken in its own decreasing order. This proves the theorem. ■

The theorem removes the eigenvectors from the description of  $\lambda_k(\mathbf{A})$ . What is left mentions only  $\mathbf{A}$ , through  $R_{\mathbf{A}}$ , and the subspaces of  $F^n$ , which are the same for every matrix. Two Hermitian matrices of the same size can therefore be compared term by term, and that is the whole reason this chapter exists.

**Remark. The increasing-index form.** If the eigenvalues are listed *increasingly* as  $\mu_1 \leq \mu_2 \leq \dots \leq \mu_n$ , then  $\mu_j = \lambda_{n+1-j}(\mathbf{A})$ , and substituting  $k = n + 1 - j$  in [Theorem 16.2.3](#) turns it into

$$\begin{aligned} \mu_j &= \min_{\dim W=j} \max_{\mathbf{0} \neq \mathbf{x} \in W} R_{\mathbf{A}}(\mathbf{x}) \\ &= \max_{\dim W=n-j+1} \min_{\mathbf{0} \neq \mathbf{x} \in W} R_{\mathbf{A}}(\mathbf{x}). \end{aligned}$$

The pairing of min max with max min is unchanged; only the dimensions move. Many books state the theorem this way, so a reader comparing sources should check which convention is in force before comparing formulas. This book uses the decreasing one everywhere, and this is the only place the other appears.

**Warning. Both the order of the two extrema and the dimension attached to each are load-bearing.** Neither

$$\max_{\dim W=n-k+1} \min_{\mathbf{0} \neq \mathbf{x} \in W} R_{\mathbf{A}}(\mathbf{x}) \quad \text{nor} \quad \min_{\dim W=k} \max_{\mathbf{0} \neq \mathbf{x} \in W} R_{\mathbf{A}}(\mathbf{x})$$

equals  $\lambda_k(\mathbf{A})$ . Each of them is, by [Theorem 16.2.3](#) itself, equal to  $\lambda_{n+1-k}(\mathbf{A})$ : the first is the second line of the theorem at index  $n - k + 1$ , and the second is the first line at the index  $j$  with  $n - j + 1 = k$ , namely  $j = n - k + 1$ . So swapping either the order or the dimension returns the eigenvalue counted from the *other end* of the list.

For a witness take  $\mathbf{A} = \text{diag}(2, 0)$  and  $k = 1$ , so that  $\lambda_1(\mathbf{A}) = 2$ . The first false expression maximizes over the single subspace  $W = F^2$  the minimum of  $R_{\mathbf{A}}$ , which is 0. The second minimizes over lines the maximum on the line, which is the value of  $R_{\mathbf{A}}$  on that line, and the smallest such value is again 0. Both return  $\lambda_2(\mathbf{A}) = 0$  instead of 2. The mistake is invisible when  $n = 2k - 1$ , since then  $n + 1 - k = k$ , and this is exactly why it survives in the wild.

**Quick check.** In the first line of [Theorem 16.2.3](#), why is the dimension  $n - k + 1$  rather than  $k$ ? Check your reason against  $k = 1$ .

**Solution.** The competitor  $W$  must be forced to meet  $U_k = \text{span}(\mathbf{q}_1, \dots, \mathbf{q}_k)$ , which has dimension  $k$ ; by [Lemma 16.2.1](#) that needs  $\dim W > n - k$ , and  $n - k + 1$  is the smallest such dimension. It must not be larger either: competitors of dimension  $n - k + 2$  would be forced to meet  $U_{k-1}$  as well, and by the theorem itself the min-max over them is  $\lambda_{k-1}(\mathbf{A})$ , the wrong eigenvalue.

At  $k = 1$  the dimension is  $n - 1 + 1 = n$ , so the only competitor is  $W = F^n$ , and the statement reads  $\lambda_1(\mathbf{A}) = \max_{\mathbf{x} \neq 0} R_{\mathbf{A}}(\mathbf{x})$ , which is [Proposition 16.1.1](#). With the dimension  $k = 1$  instead, we would be computing min over lines of the single value of  $R_{\mathbf{A}}$  on the line, which is  $\lambda_n(\mathbf{A})$  — the wrong end.

**Example. (Both forms on one matrix)**

Let

$$\mathbf{A} = \begin{pmatrix} 2 & 2 & 0 \\ 2 & 3 & 2 \\ 0 & 2 & 4 \end{pmatrix} \in M_3(\mathbb{R}),$$

whose eigenvalues are 6, 3, 0, with orthonormal eigenvectors

$$\mathbf{q}_1 = \frac{1}{3}(1, 2, 2), \quad \mathbf{q}_2 = \frac{1}{3}(2, 1, -2), \quad \mathbf{q}_3 = \frac{1}{3}(2, -2, 1).$$

Verify both lines of [Theorem 16.2.3](#) for  $k = 2$  by computing the witness and two competitors in each case.

*Solution.* Throughout we use [Lemma 16.2.2](#): for a subspace  $W$  with orthonormal basis the columns of  $\mathbf{Q}$ , the values of  $R_{\mathbf{A}}$  on  $W$  fill the interval between the smallest and largest eigenvalues of the  $2 \times 2$  matrix  $\mathbf{B} = \mathbf{Q}^T \mathbf{A} \mathbf{Q}$ .

**Max–min form**,  $\dim W = k = 2$ . The witness is  $U_2 = \text{span}(\mathbf{q}_1, \mathbf{q}_2)$ ; since  $\mathbf{q}_1, \mathbf{q}_2$  are eigenvectors,  $\mathbf{B} = \text{diag}(6, 3)$  and the minimum of  $R_{\mathbf{A}}$  on  $U_2$  is  $3 = \lambda_2(\mathbf{A})$ . Two competitors:

- $W = \text{span}(\mathbf{e}_1, \mathbf{e}_2)$ , where  $\mathbf{B}$  is the top-left  $2 \times 2$  block  $\begin{pmatrix} 2 & 2 \\ 2 & 3 \end{pmatrix}$ , with eigenvalues  $\frac{1}{2}(5 \pm \sqrt{17})$ . Its minimum is  $\frac{1}{2}(5 - \sqrt{17}) \approx 0.438$ , well below 3.
- $W = \text{span}(\mathbf{e}_1, \mathbf{e}_3)$ , where  $\mathbf{B} = \text{diag}(2, 4)$  because  $a_{13} = 0$ . Its minimum is 2, again below 3.

**Min–max form**,  $\dim W = n - k + 1 = 2$  as well, since  $n = 3$ . The witness is  $L_2 = \text{span}(\mathbf{q}_2, \mathbf{q}_3)$ , where  $\mathbf{B} = \text{diag}(3, 0)$  and the maximum is  $3 = \lambda_2(\mathbf{A})$ . The same two competitors now go the other way:

- on  $\text{span}(\mathbf{e}_1, \mathbf{e}_2)$  the maximum is  $\frac{1}{2}(5 + \sqrt{17}) \approx 4.562 > 3$ ;
- on  $\text{span}(\mathbf{e}_1, \mathbf{e}_3)$  the maximum is  $4 > 3$ .

So each competitor overshoots in the min–max form and undershoots in the max–min form, and the two witnesses pin  $\lambda_2(\mathbf{A}) = 3$  from both sides. Notice that the competitor  $\text{span}(\mathbf{e}_1, \mathbf{e}_2)$  is the coordinate plane obtained by deleting the third row and column, and its two eigenvalues  $0.438 \dots$  and  $4.562 \dots$  are caught between consecutive eigenvalues  $0 \leq 0.438 \leq 3 \leq 4.562 \leq 6$  of  $\mathbf{A}$ . That is no accident; it is the interlacing theorem of §04.

## 16.2.4 The two debts this pays at once

### Corollary 16.2.4 — The Loewner Order Compares Every Eigenvalue

Let  $\mathbf{A}, \mathbf{B} \in M_n(F)$  be Hermitian with  $\mathbf{A} \succeq \mathbf{B}$ . Then

$$\lambda_i(\mathbf{A}) \geq \lambda_i(\mathbf{B}) \quad \text{for every } i = 1, \dots, n.$$

*Idea.*  $\mathbf{A} \succeq \mathbf{B}$  says the Rayleigh quotient of  $\mathbf{A}$  sits above that of  $\mathbf{B}$  at every vector. Courant–Fischer builds each  $\lambda_i$  out of the Rayleigh quotient by a minimum and then a maximum, and both operations preserve “everywhere above”. So the domination passes straight through to every eigenvalue.

*Proof.* By Definition 12.5.1,  $\mathbf{A} \succeq \mathbf{B}$  means  $\mathbf{A} - \mathbf{B} \succeq 0$ , so  $\langle (\mathbf{A} - \mathbf{B})\mathbf{x}, \mathbf{x} \rangle \geq 0$  for every  $\mathbf{x}$ . Dividing by  $\langle \mathbf{x}, \mathbf{x} \rangle > 0$  for  $\mathbf{x} \neq \mathbf{0}$ ,

$$R_{\mathbf{A}}(\mathbf{x}) \geq R_{\mathbf{B}}(\mathbf{x}) \quad \text{for every } \mathbf{x} \neq \mathbf{0}.$$

Fix  $i$  and let  $W$  be any subspace of  $F^n$  with  $\dim W = i$ . Let  $\mathbf{x}_1 \in W \setminus \{\mathbf{0}\}$  attain  $\min_{\mathbf{0} \neq \mathbf{x} \in W} R_{\mathbf{A}}(\mathbf{x})$ , which exists by Lemma 16.2.2. Then

$$\min_{\mathbf{0} \neq \mathbf{x} \in W} R_{\mathbf{A}}(\mathbf{x}) = R_{\mathbf{A}}(\mathbf{x}_1) \geq R_{\mathbf{B}}(\mathbf{x}_1) \geq \min_{\mathbf{0} \neq \mathbf{x} \in W} R_{\mathbf{B}}(\mathbf{x}).$$

So the function  $W \mapsto \min_{\mathbf{0} \neq \mathbf{x} \in W} R_{\mathbf{A}}(\mathbf{x})$  dominates the corresponding function for  $\mathbf{B}$  at every  $W$ , and therefore so do their maxima over the subspaces of dimension  $i$ . By Theorem 16.2.3 applied to  $\mathbf{A}$  and to  $\mathbf{B}$ , those maxima are  $\lambda_i(\mathbf{A})$  and  $\lambda_i(\mathbf{B})$ . This proves the corollary. ■

#### Example. (Checking the corollary on a pair)

Let  $\mathbf{A} = \begin{pmatrix} 4 & 1 \\ 1 & 3 \end{pmatrix}$  and  $\mathbf{B} = \begin{pmatrix} 2 & 1 \\ 1 & 2 \end{pmatrix}$ . Then  $\mathbf{A} - \mathbf{B} = \text{diag}(2, 1)$ , which is positive semidefinite, so  $\mathbf{A} \succeq \mathbf{B}$ . The eigenvalues of  $\mathbf{B}$  are 3 and 1, from trace 4 and determinant 3; those of  $\mathbf{A}$  are  $\frac{7 \pm \sqrt{5}}{2} \approx 4.618, 2.382$ , from trace 7 and determinant 11. So  $\lambda_1(\mathbf{A}) \approx 4.618 \geq 3 = \lambda_1(\mathbf{B})$  and  $\lambda_2(\mathbf{A}) \approx 2.382 \geq 1 = \lambda_2(\mathbf{B})$ , as the corollary demands. Note that the two gaps,  $\frac{1 + \sqrt{5}}{2} \approx 1.618$  and  $\frac{5 - \sqrt{5}}{2} \approx 1.382$ , are different: the corollary orders the eigenvalues, and says nothing about moving them by the same amount.

This settles an account opened in Chapter 12 §05. Discussing Theorem 12.5.3 (b), which proved only  $\lambda_1(\mathbf{A}) \geq \lambda_1(\mathbf{B})$  and  $\lambda_n(\mathbf{A}) \geq \lambda_n(\mathbf{B})$ , that section said: “The full statement is that  $\lambda_i(\mathbf{A}) \geq \lambda_i(\mathbf{B})$  for **every**  $i$ , not only for  $i = 1$  and  $i = n$ ; but the intermediate eigenvalues have no description as a plain maximum over the whole unit sphere, and the min–max description that does the job is the Courant–Fischer theorem of Chapter 16.” Corollary 16.2.4 is that full statement, and the diagnosis was exact: what was missing was not an extra trick but a description of  $\lambda_i$  that two different matrices could be measured against. The proof above is short, and all of its weight is carried by Theorem 16.2.3.

**Corollary 16.2.5 — Courant–Fischer for a Self-Adjoint Operator**

Let  $V$  be a finite-dimensional inner product space over  $F = \mathbb{R}$  or  $F = \mathbb{C}$ , with  $\dim V = n \geq 1$ , and let  $T \in \mathcal{L}(V)$  be self-adjoint, with eigenvalues  $\lambda_1(T) \geq \cdots \geq \lambda_n(T)$  repeated according to multiplicity. Then for every  $k$  with  $1 \leq k \leq n$ ,

$$\begin{aligned}\lambda_k(T) &= \min_{\dim U = n-k+1} \max_{\mathbf{0} \neq \mathbf{v} \in U} \frac{\langle T\mathbf{v}, \mathbf{v} \rangle}{\langle \mathbf{v}, \mathbf{v} \rangle} \\ &= \max_{\dim U = k} \min_{\mathbf{0} \neq \mathbf{v} \in U} \frac{\langle T\mathbf{v}, \mathbf{v} \rangle}{\langle \mathbf{v}, \mathbf{v} \rangle},\end{aligned}$$

where  $U$  runs over the subspaces of  $V$  of the stated dimension.

*Idea.* Transport the theorem along coordinates. An orthonormal basis of eigenvectors turns  $V$  into  $F^n$  without changing any inner product, turns  $T$  into a real diagonal matrix, and matches subspaces with subspaces of the same dimension. There is nothing to prove beyond checking that each of those three things is true.

*Proof.* By [Proposition 10.6.7](#) (c) a self-adjoint operator is normal, so [Theorem 11.4.2](#) over  $\mathbb{C}$ , and [Theorem 11.5.1](#) over  $\mathbb{R}$ , give an orthonormal basis  $\mathcal{E} = (\mathbf{e}_1, \dots, \mathbf{e}_n)$  of  $V$  consisting of eigenvectors of  $T$ ; relabel it so that  $T\mathbf{e}_i = \lambda_i(T)\mathbf{e}_i$  with the eigenvalues in decreasing order, which is possible because they are real by [Theorem 11.2.1](#). In particular the eigenvalue list of  $T$ , repeated according to multiplicity, has exactly  $n$  entries.

Let  $\Phi: V \rightarrow F^n$ ,  $\Phi(\mathbf{v}) = [\mathbf{v}]_{\mathcal{E}}$ , be the coordinate map, an isomorphism by [Corollary 3.5.5](#). By [Theorem 10.2.1](#) (b),  $\langle \Phi\mathbf{u}, \Phi\mathbf{v} \rangle = \langle \mathbf{u}, \mathbf{v} \rangle$  for all  $\mathbf{u}, \mathbf{v} \in V$ , so  $\Phi$  is an isometric isomorphism in the sense of [Corollary 10.2.4](#). Put  $\mathbf{A} = [T]_{\mathcal{E}}^{\mathcal{E}} = \text{diag}(\lambda_1(T), \dots, \lambda_n(T))$ , a real diagonal matrix, hence Hermitian, with  $\lambda_i(\mathbf{A}) = \lambda_i(T)$  for every  $i$ . By [Theorem 3.6.1](#),  $\Phi(T\mathbf{v}) = \mathbf{A}\Phi(\mathbf{v})$ .

Consequently, for  $\mathbf{v} \neq \mathbf{0}$ ,

$$\frac{\langle T\mathbf{v}, \mathbf{v} \rangle}{\langle \mathbf{v}, \mathbf{v} \rangle} = \frac{\langle \mathbf{A}\Phi\mathbf{v}, \Phi\mathbf{v} \rangle}{\langle \Phi\mathbf{v}, \Phi\mathbf{v} \rangle} = R_{\mathbf{A}}(\Phi\mathbf{v}),$$

and  $\Phi\mathbf{v} \neq \mathbf{0}$  because  $\Phi$  is injective. Finally  $U \mapsto \Phi(U)$  is a bijection from the subspaces of  $V$  to the subspaces of  $F^n$  preserving dimension, by [Theorem 3.5.2](#), and it matches the non-zero vectors of  $U$  with those of  $\Phi(U)$ . So each of the two displayed expressions equals the corresponding expression for  $\mathbf{A}$  in [Theorem 16.2.3](#), which is  $\lambda_k(\mathbf{A}) = \lambda_k(T)$ . This proves the corollary. ■

This is the statement Chapter 11 §02 promised when it wrote that “the maximization picture returns in Chapter 16, where the Rayleigh quotient and the Courant–Fischer theorem describe every eigenvalue of a self-adjoint operator by optimization”. The picture has returned and the description is complete: every eigenvalue, not merely the extreme two, and stated for an operator on an abstract inner product space, which is the setting Chapter 11 worked in. The remark in §01 records what it cost — nothing. Chapter 11 declined the maximization route because it would have needed the extreme value theorem to know that a maximum exists; here the spectral theorem supplies the maximizing vectors explicitly, so the min–max description is a *consequence* of Chapter 11 rather than an alternative to it, and no analysis has been imported.

What has been bought is comparison. Two Hermitian matrices are now comparable eigenvalue by eigenvalue whenever their Rayleigh quotients are comparable vector by vector, and [Corollary 16.2.4](#) is the first instance. Section 3 takes the same idea and applies it to  $\mathbf{A}$  and  $\mathbf{A} + \mathbf{E}$ , where the quotients differ by at most  $\|\mathbf{E}\|_2$ , and gets the perturbation bound that Chapter 15 could not state.

## 16.2.5 Exercises

### 16.2.5.1 A. Check your understanding

#### Exercise A1.

- (a) State both lines of [Theorem 16.2.3](#), being careful about which dimension goes with which line.
- (b) State [Lemma 16.2.1](#), and say at which point of the proof of [Theorem 16.2.3](#) it is used and what it is used on.
- (c) Determine whether the following is correct, and justify your answer:  $\lambda_k(\mathbf{A}) = \min_{\dim W=k} \max_{\mathbf{0} \neq \mathbf{x} \in W} R_{\mathbf{A}}(\mathbf{x})$ .
- (d) Write the first line of [Theorem 16.2.3](#) in the increasing-index convention  $\mu_1 \leq \dots \leq \mu_n$ .
- (e) Explain in one or two sentences why [Theorem 12.5.3](#) (b) could not be extended to intermediate eigenvalues by the argument Chapter 12 used, and what [Theorem 16.2.3](#) supplies instead.

*Solution.*

- (a) For Hermitian  $\mathbf{A} \in M_n(F)$  and  $1 \leq k \leq n$ :  $\lambda_k(\mathbf{A}) = \min \max$  over subspaces of dimension  $n - k + 1$  of the Rayleigh quotient, and  $= \max \min$  over subspaces of dimension  $k$ . The large dimension goes with the minimum of maxima; the small one with the maximum of minima.
- (b) If  $U, W$  are subspaces of an  $n$ -dimensional space with  $\dim U + \dim W > n$ , then  $U \cap W \neq \{\mathbf{0}\}$ . It is used in Step 2, applied to an arbitrary competitor  $W$  of dimension  $n - k + 1$  and the span  $U_k$  of the top  $k$  eigenvectors, whose dimensions add to  $n + 1$ .
- (c) Incorrect. By [Theorem 16.2.3](#) itself, that expression is the first line evaluated at the index  $j$  with  $n - j + 1 = k$ , so it equals  $\lambda_{n-k+1}(\mathbf{A})$ . For  $\mathbf{A} = \text{diag}(2, 0)$  and  $k = 1$  it returns 0, not  $\lambda_1(\mathbf{A}) = 2$ .
- (d)  $\mu_j = \min_{\dim W=j} \max_{\mathbf{0} \neq \mathbf{x} \in W} R_{\mathbf{A}}(\mathbf{x})$ , obtained by setting  $k = n + 1 - j$ .
- (e) Chapter 12's argument evaluated the quadratic form at a vector that is extremal for the *other* matrix, which works only when the extremal vector is characterized by a plain maximum over the whole sphere — true for  $\lambda_1$  and  $\lambda_n$  and for no other index, since the range of  $R_{\mathbf{A}}$  is an interval ([Proposition 16.1.1](#)). [Theorem 16.2.3](#) supplies a description of every  $\lambda_i$  as an optimization over a family of subspaces that does not depend on the matrix, so the same subspace can be fed to both matrices; that is exactly what the proof of [Corollary 16.2.4](#) does.

### 16.2.5.2 B. Practice

**Exercise B1: Both forms on a diagonal matrix.**

Let  $\mathbf{A} = \text{diag}(5, 2, -1) \in M_3(\mathbb{R})$  and  $k = 2$ .

- (a) Give the witness subspace for the max-min form, compute the minimum of  $R_{\mathbf{A}}$  on it, and check the value against [Theorem 16.2.3](#).
- (b) Do the same for the min-max form.
- (c) Put  $\mathbf{u} = \frac{1}{\sqrt{2}}(0, 1, 1)$ . Compute the minimum of  $R_{\mathbf{A}}$  over the non-zero vectors of  $\text{span}(\mathbf{e}_1, \mathbf{u})$ , and over those of  $\text{span}(\mathbf{e}_2, \mathbf{e}_3)$ , and verify that neither subspace beats the witness in (a).

*Solution.*

- (a) The eigenvalues are  $\lambda_1 = 5$ ,  $\lambda_2 = 2$ ,  $\lambda_3 = -1$  with eigenvectors  $\mathbf{e}_1, \mathbf{e}_2, \mathbf{e}_3$ . The witness is  $U_2 = \text{span}(\mathbf{e}_1, \mathbf{e}_2)$ , of dimension  $k = 2$ . There  $\mathbf{B} = \text{diag}(5, 2)$ , so by [Lemma 16.2.2](#) the values of  $R_{\mathbf{A}}$  fill  $[2, 5]$  and the minimum is  $2 = \lambda_2(\mathbf{A})$ .
- (b) Here  $n - k + 1 = 2$  as well, and the witness is  $L_2 = \text{span}(\mathbf{e}_2, \mathbf{e}_3)$ , where  $\mathbf{B} = \text{diag}(2, -1)$ ; the values fill  $[-1, 2]$  and the maximum is  $2 = \lambda_2(\mathbf{A})$ .
- (c) For  $W = \text{span}(\mathbf{e}_1, \mathbf{u})$  the two spanning vectors are orthonormal, and

$$\mathbf{B} = \begin{pmatrix} 5 & 0 \\ 0 & \frac{1}{2}(2-1) \end{pmatrix} = \text{diag}\left(5, \frac{1}{2}\right),$$

the off-diagonal entry vanishing because  $\mathbf{A}\mathbf{e}_1 = 5\mathbf{e}_1$  is orthogonal to  $(0, 1, 1)$ . So the minimum is  $\frac{1}{2} < 2$ . For  $W = \text{span}(\mathbf{e}_2, \mathbf{e}_3)$  we get  $\mathbf{B} = \text{diag}(2, -1)$  and minimum  $-1 < 2$ . Neither competitor reaches  $\lambda_2(\mathbf{A}) = 2$ , as [Theorem 16.2.3](#) requires.

**Exercise B2: Shifting by a multiple of the identity.**

Let  $\mathbf{A} \in M_n(F)$  be Hermitian and let  $c \in \mathbb{R}$ . Prove that  $\lambda_k(\mathbf{A} + c\mathbf{I}) = \lambda_k(\mathbf{A}) + c$  for every  $k$ , using the first line of [Theorem 16.2.3](#) for both matrices.

*Solution.* The matrix  $\mathbf{A} + c\mathbf{I}$  is Hermitian, since  $c$  is real. For every  $\mathbf{x} \neq \mathbf{0}$ ,

$$R_{\mathbf{A}+c\mathbf{I}}(\mathbf{x}) = \frac{\langle \mathbf{A}\mathbf{x}, \mathbf{x} \rangle + c\langle \mathbf{x}, \mathbf{x} \rangle}{\langle \mathbf{x}, \mathbf{x} \rangle} = R_{\mathbf{A}}(\mathbf{x}) + c.$$

Adding a constant to a real-valued function adds it to every maximum and every minimum of that function over any non-empty set. So for each subspace  $W$ ,

$$\max_{\mathbf{0} \neq \mathbf{x} \in W} R_{\mathbf{A}+c\mathbf{I}}(\mathbf{x}) = \left( \max_{\mathbf{0} \neq \mathbf{x} \in W} R_{\mathbf{A}}(\mathbf{x}) \right) + c,$$

and taking the minimum over all  $W$  of dimension  $n - k + 1$  adds  $c$  again. By [Theorem 16.2.3](#) applied to  $\mathbf{A} + c\mathbf{I}$  on the left and to  $\mathbf{A}$  on the right,  $\lambda_k(\mathbf{A} + c\mathbf{I}) = \lambda_k(\mathbf{A}) + c$ .

**Exercise B3: Comparing two matrices.**



Let

$$\mathbf{A} = \begin{pmatrix} 3 & 1 \\ 1 & 3 \end{pmatrix}, \quad \mathbf{B} = \begin{pmatrix} 2 & 0 \\ 0 & 1 \end{pmatrix}.$$

Verify that  $\mathbf{A} \succeq \mathbf{B}$ , compute both eigenvalue lists, and check [Corollary 16.2.4](#). Check whether  $\lambda_1(\mathbf{A}) - \lambda_1(\mathbf{B}) \leq \lambda_1(\mathbf{A} - \mathbf{B})$  holds for this pair. (Whether it holds for every pair is a question for Section 3.)

*Solution.*  $\mathbf{A} - \mathbf{B} = \begin{pmatrix} 1 & 1 \\ 1 & 2 \end{pmatrix}$  is symmetric with leading principal minors  $1 > 0$  and  $\det = 2 - 1 = 1 > 0$ , so  $\mathbf{A} - \mathbf{B} \succ 0$  by [Theorem 12.1.5](#); in particular  $\mathbf{A} \succeq \mathbf{B}$ .

The eigenvalues:  $\mathbf{A}$  has  $p_{\mathbf{A}}(x) = (x-3)^2 - 1$ , so  $\lambda_1(\mathbf{A}) = 4$ ,  $\lambda_2(\mathbf{A}) = 2$ ; and  $\mathbf{B}$  is diagonal with  $\lambda_1(\mathbf{B}) = 2$ ,  $\lambda_2(\mathbf{B}) = 1$ . Indeed  $4 \geq 2$  and  $2 \geq 1$ , as [Corollary 16.2.4](#) requires.

For the last question,  $\mathbf{A} - \mathbf{B}$  has trace 3 and determinant 1, so its eigenvalues are  $\frac{1}{2}(3 \pm \sqrt{5})$  and  $\lambda_1(\mathbf{A} - \mathbf{B}) = \frac{1}{2}(3 + \sqrt{5}) \approx 2.618$ . The inequality asks whether  $4 - 2 \leq 2.618$ , which holds. (It is not an accident: it is the simplest of the perturbation inequalities of §03, which carry Weyl's name. Note that [Corollary 16.2.4](#) alone gives no upper bound on the gap  $\lambda_i(\mathbf{A}) - \lambda_i(\mathbf{B})$ .)

### 16.2.5.3 C. Going deeper

#### Exercise C1: The definite generalized eigenvalue problem.

Let  $\mathbf{A}, \mathbf{B} \in M_n(F)$  be Hermitian with  $\mathbf{A} \succ 0$ , and recall [Definition 12.4.1](#):  $\lambda$  is a generalized eigenvalue of the pair  $(\mathbf{A}, \mathbf{B})$  when  $\mathbf{B}\mathbf{x} = \lambda\mathbf{A}\mathbf{x}$  for some  $\mathbf{x} \neq \mathbf{0}$ . Define the **generalized Rayleigh quotient**

$$R_{\mathbf{A}, \mathbf{B}}(\mathbf{x}) = \frac{\langle \mathbf{B}\mathbf{x}, \mathbf{x} \rangle}{\langle \mathbf{A}\mathbf{x}, \mathbf{x} \rangle} \quad (\mathbf{x} \neq \mathbf{0}).$$

- (a) Prove that  $R_{\mathbf{A}, \mathbf{B}}$  is well defined and real-valued.
- (b) Put  $\mathbf{C} = \mathbf{A}^{-1/2}\mathbf{B}\mathbf{A}^{-1/2}$ . Prove that  $\mathbf{C}$  is Hermitian and that  $\det(\mathbf{B} - \lambda\mathbf{A}) = \det(\mathbf{A})\det(\mathbf{C} - \lambda\mathbf{I})$ , so that the generalized eigenvalues of  $(\mathbf{A}, \mathbf{B})$ , with multiplicity, are exactly the eigenvalues of  $\mathbf{C}$ . Write them  $d_1 \geq \cdots \geq d_n$ .
- (c) Prove that for every  $k$ ,

$$d_k = \min_{\dim W = n-k+1} \max_{\mathbf{0} \neq \mathbf{x} \in W} R_{\mathbf{A}, \mathbf{B}}(\mathbf{x}).$$

*Hint for (c): the substitution  $\mathbf{x} = \mathbf{A}^{-1/2}\mathbf{y}$ .*

*Solution.*

- (a) By [Theorem 12.2.1](#) and [Theorem 12.1.5](#),  $\mathbf{A} \succ 0$  has a positive definite square root  $\mathbf{A}^{1/2}$ , which is invertible; write  $\mathbf{A}^{-1/2}$  for its inverse, again Hermitian and positive definite. For  $\mathbf{x} \neq \mathbf{0}$ ,  $\langle \mathbf{A}\mathbf{x}, \mathbf{x} \rangle > 0$  by definiteness, so the denominator is a non-zero real; the numerator is real by [Proposition 10.6.7](#) (a). So the quotient is a well-defined real number.
- (b)  $\mathbf{C}^* = \mathbf{A}^{-1/2}\mathbf{B}^*\mathbf{A}^{-1/2} = \mathbf{C}$ . Since  $\mathbf{A}^{1/2}\mathbf{C}\mathbf{A}^{1/2} = \mathbf{B}$  and  $\mathbf{A}^{1/2}\mathbf{A}^{1/2} = \mathbf{A}$ ,

$$\mathbf{B} - \lambda\mathbf{A} = \mathbf{A}^{1/2}(\mathbf{C} - \lambda\mathbf{I})\mathbf{A}^{1/2},$$

so taking determinants and using [Theorem 6.5.4](#) gives

$$\det(\mathbf{B} - \lambda\mathbf{A}) = \det(\mathbf{A}^{1/2})^2 \det(\mathbf{C} - \lambda\mathbf{I}) = \det(\mathbf{A}) \det(\mathbf{C} - \lambda\mathbf{I}).$$

Since  $\det \mathbf{A} \neq 0$ , the two polynomials in  $\lambda$  have the same roots with the same multiplicities. By [Theorem 12.4.2](#) (b) the generalized eigenvalues of  $(\mathbf{A}, \mathbf{B})$  are the roots of  $\det(\mathbf{B} - \lambda\mathbf{A})$ , so they are the eigenvalues of  $\mathbf{C}$ , all real because  $\mathbf{C}$  is Hermitian.

- (c) Put  $\mathbf{S} = \mathbf{A}^{-1/2}$ , which is Hermitian and invertible. For  $\mathbf{y} \neq \mathbf{0}$  set  $\mathbf{x} = \mathbf{S}\mathbf{y} \neq \mathbf{0}$ . Then

$$\langle \mathbf{B}\mathbf{x}, \mathbf{x} \rangle = \langle \mathbf{SBSy}, \mathbf{y} \rangle = \langle \mathbf{C}\mathbf{y}, \mathbf{y} \rangle, \quad \langle \mathbf{A}\mathbf{x}, \mathbf{x} \rangle = \langle \mathbf{SASy}, \mathbf{y} \rangle = \langle \mathbf{y}, \mathbf{y} \rangle,$$

using  $\mathbf{SAS} = \mathbf{A}^{-1/2}\mathbf{A}\mathbf{A}^{-1/2} = \mathbf{I}$ . Hence  $R_{\mathbf{A},\mathbf{B}}(\mathbf{S}\mathbf{y}) = R_{\mathbf{C}}(\mathbf{y})$ . The map  $\mathbf{y} \mapsto \mathbf{S}\mathbf{y}$  is an isomorphism of  $F^n$ , so  $W \mapsto \mathbf{S}^{-1}W$  is a bijection of the subspaces of dimension  $n - k + 1$  onto themselves, matching non-zero vectors with non-zero vectors. Therefore

$$\max_{\mathbf{0} \neq \mathbf{x} \in W} R_{\mathbf{A},\mathbf{B}}(\mathbf{x}) = \max_{\mathbf{0} \neq \mathbf{y} \in \mathbf{S}^{-1}W} R_{\mathbf{C}}(\mathbf{y}),$$

and minimizing both sides over all  $W$  of dimension  $n - k + 1$  gives, by [Theorem 16.2.3](#) applied to  $\mathbf{C}$ , the value  $\lambda_k(\mathbf{C}) = d_k$ .

### Exercise C2: How much weaker the eigenvalue comparison is.

- Give Hermitian  $\mathbf{A}, \mathbf{B} \in M_2(\mathbb{R})$  with  $\lambda_i(\mathbf{A}) \geq \lambda_i(\mathbf{B})$  for  $i = 1, 2$  but  $\mathbf{A} \not\geq \mathbf{B}$ . So the converse of [Corollary 16.2.4](#) is false.
- Prove that if  $\mathbf{A}$  and  $\mathbf{B}$  are Hermitian with  $\lambda_i(\mathbf{A}) \geq \lambda_i(\mathbf{B})$  for every  $i$  and  $\text{tr } \mathbf{A} = \text{tr } \mathbf{B}$ , then  $\lambda_i(\mathbf{A}) = \lambda_i(\mathbf{B})$  for every  $i$ .
- Prove that if  $\mathbf{A} \succeq \mathbf{B}$  and  $\text{tr } \mathbf{A} = \text{tr } \mathbf{B}$ , then  $\mathbf{A} = \mathbf{B}$ .

*Solution.*

- Take  $\mathbf{A} = \text{diag}(2, 0)$  and  $\mathbf{B} = \text{diag}(0, 1)$ . Then  $\lambda_1(\mathbf{A}) = 2 \geq 1 = \lambda_1(\mathbf{B})$  and  $\lambda_2(\mathbf{A}) = 0 \geq 0 = \lambda_2(\mathbf{B})$ . But  $\mathbf{A} - \mathbf{B} = \text{diag}(2, -1)$  has the eigenvalue  $-1 < 0$ , so  $\mathbf{A} - \mathbf{B} \not\geq \mathbf{0}$  by [Theorem 12.1.4](#) and  $\mathbf{A} \not\geq \mathbf{B}$ . Ordering the two spectra says nothing about how the eigenvectors are placed relative to each other, and the Loewner order sees both.
- By [Theorem 8.3.3](#) the trace is the sum of the eigenvalues with multiplicity, so

$$0 = \text{tr } \mathbf{A} - \text{tr } \mathbf{B} = \sum_{i=1}^n (\lambda_i(\mathbf{A}) - \lambda_i(\mathbf{B})).$$

Every summand is  $\geq 0$  by hypothesis, and a sum of non-negative reals is zero only if each is zero. Hence  $\lambda_i(\mathbf{A}) = \lambda_i(\mathbf{B})$  for every  $i$ .

- Put  $\mathbf{C} = \mathbf{A} - \mathbf{B}$ , Hermitian with  $\mathbf{C} \succeq \mathbf{0}$ , so every eigenvalue of  $\mathbf{C}$  is  $\geq 0$  by [Theorem 12.1.4](#). Also  $\text{tr } \mathbf{C} = \text{tr } \mathbf{A} - \text{tr } \mathbf{B} = 0$ , which by [Theorem 8.3.3](#) is the sum of those eigenvalues; a sum of non-negative reals vanishing forces each to vanish, so every eigenvalue of  $\mathbf{C}$  is 0. By [Corollary 11.4.3](#) (or [Corollary 11.5.2](#)),  $\mathbf{C} = \mathbf{U}\mathbf{D}\mathbf{U}^*$  with  $\mathbf{D}$  the diagonal matrix of eigenvalues, which is  $\mathbf{0}$ ; hence  $\mathbf{C} = \mathbf{0}$  and  $\mathbf{A} = \mathbf{B}$ . (One may also finish with (b) and [Corollary 16.2.4](#), which give the equality of the

two spectra but not yet of the matrices; the trace argument on  $\mathbf{C}$  is what closes the gap.)

### Exercise C3: Does enlarging the field change the answer?.

Let  $\mathbf{A} \in M_n(\mathbb{R})$  be symmetric. It may be read as an element of  $M_n(\mathbb{C})$ , where it is Hermitian, and then [Theorem 16.2.3](#) minimizes over a much larger collection of subspaces.

- (a) Prove that the eigenvalue lists of  $\mathbf{A}$  over  $\mathbb{R}$  and over  $\mathbb{C}$ , with multiplicity, are the same.
- (b) Deduce that for every  $k$ ,

$$\min_{\substack{W \subseteq \mathbb{R}^n \\ \dim W = n-k+1}} \max_{\mathbf{0} \neq \mathbf{x} \in W} R_{\mathbf{A}}(\mathbf{x}) = \min_{\substack{W \subseteq \mathbb{C}^n \\ \dim W = n-k+1}} \max_{\mathbf{0} \neq \mathbf{x} \in W} R_{\mathbf{A}}(\mathbf{x}).$$

- (c) Show that the second minimum really does range over strictly more subspaces, by exhibiting a one-dimensional subspace of  $\mathbb{C}^2$  that contains no non-zero real vector.

*Solution.*

- (a) By [Corollary 11.5.2](#) there is an orthogonal  $\mathbf{Q} \in M_n(\mathbb{R})$  and a real diagonal  $\mathbf{D}$  with  $\mathbf{A} = \mathbf{Q}\mathbf{D}\mathbf{Q}^\top$ , and the diagonal of  $\mathbf{D}$  is the eigenvalue list of  $\mathbf{A}$  over  $\mathbb{R}$  with multiplicity. Read over  $\mathbb{C}$ , the same  $\mathbf{Q}$  is unitary, since  $\mathbf{Q}^* = \overline{\mathbf{Q}}^\top = \mathbf{Q}^\top = \mathbf{Q}^{-1}$ , and the same equation  $\mathbf{A} = \mathbf{Q}\mathbf{D}\mathbf{Q}^*$  holds. So  $\mathbf{A}$  is unitarily similar over  $\mathbb{C}$  to the same  $\mathbf{D}$ , and its complex eigenvalue list with multiplicity is again the diagonal of  $\mathbf{D}$ .
- (b) Apply [Theorem 16.2.3](#) twice: once with  $F = \mathbb{R}$ , where the left-hand side equals  $\lambda_k(\mathbf{A})$  computed over  $\mathbb{R}$ , and once with  $F = \mathbb{C}$ , where the right-hand side equals  $\lambda_k(\mathbf{A})$  computed over  $\mathbb{C}$ . By (a) these are the same number.
- (c) Take  $W = \text{span}_{\mathbb{C}}\{(1, i)\} \subseteq \mathbb{C}^2$ . A non-zero element of  $W$  is  $c(1, i) = (c, ci)$  with  $c \neq 0$ ; for it to be real we would need  $c \in \mathbb{R}$  and  $ci \in \mathbb{R}$ , and  $c \neq 0$  makes the second impossible. So  $W$  is not the complex span of any real subspace.

The lesson is worth stating. Enlarging the field adds competitors to the minimization, which can only push the minimum **down**, and adds vectors to each inner maximization, which can only push those maxima **up**. The two effects cancel exactly, because both the witness subspace and the bound of Step 2 in the proof of [Theorem 16.2.3](#) survive the enlargement unchanged: the witness is spanned by real eigenvectors, and the dimension count of [Lemma 16.2.1](#) holds over any field.

## 16.3 How Far Eigenvalues Move

Chapter 15 §07 asked what happens to the eigenvalues of a matrix when the matrix is nudged, and came back with a discouraging answer: they move continuously, and nothing better. [Example \(A Jordan block's eigenvalues move like  \$\varepsilon^{1/k}\$ \)](#) exhibits a perturbation of size  $\varepsilon$  that moves every eigenvalue a distance  $\varepsilon^{1/k}$ , so no Lipschitz constant survives. That section also promised that the Hermitian case is different, and left the proof here.

The reason the Hermitian case is different is now available. [Theorem 16.2.3](#) describes  $\lambda_k(\mathbf{A})$  as an optimization over subspaces, and an optimization over subspaces reacts to a change in  $\mathbf{A}$  the way any optimization reacts to a change in the objective: by at most the change in the objective. Turning that sentence into inequalities is the business of this section, and the inequalities are sharper than the sentence suggests, because the index may be shifted as well.

**Throughout this section the field is  $\mathbb{R}$  or  $\mathbb{C}$ , and every matrix whose eigenvalues are written down is Hermitian**, that is,  $\mathbf{A}^* = \mathbf{A}$  with  $\mathbf{A} \in M_n(F)$  and  $n \geq 1$ . Its eigenvalues are real and are always indexed **decreasingly**,  $\lambda_1(\mathbf{A}) \geq \lambda_2(\mathbf{A}) \geq \cdots \geq \lambda_n(\mathbf{A})$ , as they have been since Chapter 12. Sums of Hermitian matrices are Hermitian, so  $\mathbf{A} + \mathbf{B}$  is again a matrix this section may write  $\lambda_i$  of.

### 16.3.1 Two small facts about eigenvalue lists

One small fact is used in every proof below, and it is the reason the index bookkeeping works out. If a vector lies in the span of the *bottom* eigenvectors of  $\mathbf{A}$ , its Rayleigh quotient cannot exceed the largest eigenvalue in that span; if it lies in the span of the *top* ones, the quotient cannot fall below the smallest eigenvalue there.

#### Lemma 16.3.1 — The Quotient on a Span of Eigenvectors

Let  $\mathbf{A} \in M_n(F)$  be Hermitian and let  $(\mathbf{u}_1, \dots, \mathbf{u}_n)$  be an orthonormal basis of  $F^n$  with  $\mathbf{A}\mathbf{u}_p = \lambda_p(\mathbf{A})\mathbf{u}_p$  for every  $p$ . Fix  $i$  with  $1 \leq i \leq n$ .

- (a) If  $\mathbf{x} \neq \mathbf{0}$  lies in  $\text{span}\{\mathbf{u}_i, \mathbf{u}_{i+1}, \dots, \mathbf{u}_n\}$ , then  $R_{\mathbf{A}}(\mathbf{x}) \leq \lambda_i(\mathbf{A})$ .
- (b) If  $\mathbf{x} \neq \mathbf{0}$  lies in  $\text{span}\{\mathbf{u}_1, \mathbf{u}_2, \dots, \mathbf{u}_i\}$ , then  $R_{\mathbf{A}}(\mathbf{x}) \geq \lambda_i(\mathbf{A})$ .

*Proof.* Such a basis exists by the spectral theorem: [Corollary 11.4.3](#) over  $\mathbb{C}$ , where a Hermitian matrix is normal and its eigenvalues may be listed in any prescribed order, and [Corollary 11.5.2](#) over  $\mathbb{R}$ , where Hermitian means symmetric.

- (a) Write  $\mathbf{x} = \sum_{p \geq i} c_p \mathbf{u}_p$ , with the  $c_p$  not all zero. Since the  $\mathbf{u}_p$  are orthonormal,

$$\begin{aligned} \langle \mathbf{A}\mathbf{x}, \mathbf{x} \rangle &= \left\langle \sum_{p \geq i} c_p \lambda_p(\mathbf{A}) \mathbf{u}_p, \sum_{q \geq i} c_q \mathbf{u}_q \right\rangle = \sum_{p \geq i} \lambda_p(\mathbf{A}) |c_p|^2, \\ \|\mathbf{x}\|^2 &= \sum_{p \geq i} |c_p|^2. \end{aligned}$$

Every index in these sums satisfies  $p \geq i$ , so  $\lambda_p(\mathbf{A}) \leq \lambda_i(\mathbf{A})$  by the decreasing indexing, and therefore  $\langle \mathbf{A}\mathbf{x}, \mathbf{x} \rangle \leq \lambda_i(\mathbf{A}) \|\mathbf{x}\|^2$ . Since  $\mathbf{x} \neq \mathbf{0}$  we have  $\|\mathbf{x}\|^2 > 0$  and may divide, which gives  $R_{\mathbf{A}}(\mathbf{x}) \leq \lambda_i(\mathbf{A})$  by [Definition 16.1.1](#).

- (b) The same computation with  $\mathbf{x} = \sum_{p \leq i} c_p \mathbf{u}_p$  gives  $\langle \mathbf{A}\mathbf{x}, \mathbf{x} \rangle = \sum_{p \leq i} \lambda_p(\mathbf{A}) |c_p|^2$ , and now every index satisfies  $p \leq i$ , so  $\lambda_p(\mathbf{A}) \geq \lambda_i(\mathbf{A})$  and the inequality reverses. This proves the lemma. ■

The two halves are the two ways of using an eigenvector span: (a) caps a quotient from above, (b) props it up from below. The proof of Weyl's inequalities needs only (a), applied to a single

vector for two different matrices at once; (b) is the form in which the same fact enters the witness half of a min–max argument.

A second small fact converts statements about the top of an eigenvalue list into statements about the bottom. Negating a matrix negates its eigenvalues, and negation reverses the order of real numbers.

### Lemma 16.3.2 — Negation Reverses the List

Let  $\mathbf{A} \in M_n(F)$  be Hermitian. Then  $-\mathbf{A}$  is Hermitian and

$$\lambda_p(-\mathbf{A}) = -\lambda_{n+1-p}(\mathbf{A}) \quad (1 \leq p \leq n).$$

*Proof.*  $(-\mathbf{A})^* = -\mathbf{A}^* = -\mathbf{A}$ , so  $-\mathbf{A}$  is Hermitian. By the spectral theorem quoted in the proof of Lemma 16.3.1,  $\mathbf{A} = \mathbf{U}\mathbf{D}\mathbf{U}^*$  with  $\mathbf{D} = \text{diag}(\lambda_1(\mathbf{A}), \dots, \lambda_n(\mathbf{A}))$  and  $\mathbf{U}$  unitary, whence  $-\mathbf{A} = \mathbf{U}(-\mathbf{D})\mathbf{U}^*$ . So the eigenvalue list of  $-\mathbf{A}$ , with multiplicity, is  $-\lambda_1(\mathbf{A}), \dots, -\lambda_n(\mathbf{A})$ . Multiplying by  $-1$  reverses the order of real numbers, so listing these decreasingly gives

$$-\lambda_n(\mathbf{A}) \geq -\lambda_{n-1}(\mathbf{A}) \geq \dots \geq -\lambda_1(\mathbf{A}),$$

whose  $p$ -th entry is  $-\lambda_{n+1-p}(\mathbf{A})$ . This proves the lemma. ■

## 16.3.2 The two families of Weyl inequalities

Here is the question. Given the eigenvalues of  $\mathbf{A}$  and the eigenvalues of  $\mathbf{B}$ , separately, what can be said about the eigenvalues of  $\mathbf{A} + \mathbf{B}$ ? Nothing exact:  $\text{diag}(1, 0)$  and  $\text{diag}(0, 1)$  have the same eigenvalue list as  $\text{diag}(1, 0)$  and  $\text{diag}(1, 0)$ , while the two sums  $\mathbf{I}$  and  $\text{diag}(2, 0)$  do not. But a great deal inexactly, and the shape of the answer is dictated by the index arithmetic.

### Theorem 16.3.3 — Weyl's Inequalities

Let  $\mathbf{A}, \mathbf{B} \in M_n(F)$  be Hermitian and let  $1 \leq i, j \leq n$ .

(a) If  $i + j - 1 \leq n$ , then

$$\lambda_{i+j-1}(\mathbf{A} + \mathbf{B}) \leq \lambda_i(\mathbf{A}) + \lambda_j(\mathbf{B}).$$

(b) If  $i + j - n \geq 1$ , then

$$\lambda_{i+j-n}(\mathbf{A} + \mathbf{B}) \geq \lambda_i(\mathbf{A}) + \lambda_j(\mathbf{B}).$$

*Idea.* For (a), read  $\lambda_{i+j-1}(\mathbf{A} + \mathbf{B})$  through the **max–min** half of Theorem 16.2.3: it is the best guaranteed value of  $R_{\mathbf{A}+\mathbf{B}}$  on a subspace of dimension  $i + j - 1$ . So take an arbitrary such subspace  $W$  and find one bad vector inside it — a vector on which  $R_{\mathbf{A}}$  is capped by  $\lambda_i(\mathbf{A})$  and  $R_{\mathbf{B}}$  is capped by  $\lambda_j(\mathbf{B})$  simultaneously. The span of the bottom  $n - i + 1$  eigenvectors of  $\mathbf{A}$  is a *subspace* on which  $R_{\mathbf{A}}$  is capped by  $\lambda_i(\mathbf{A})$ , and the span

of the bottom  $n - j + 1$  eigenvectors of  $\mathbf{B}$  is one on which  $R_{\mathbf{B}}$  is capped by  $\lambda_j(\mathbf{B})$ . (The set of *all* vectors capped by  $\lambda_i(\mathbf{A})$  is larger and is not a subspace; the spans are what the dimension count can use.) Three subspaces must meet, and the dimensions have been chosen so that they do — twice, by [Lemma 16.2.1](#). Since  $R_{\mathbf{A}+\mathbf{B}} = R_{\mathbf{A}} + R_{\mathbf{B}}$ , that one vector finishes the argument.

Part (b) is not a new proof. Negating a Hermitian matrix reverses its eigenvalue list ([Lemma 16.3.2](#)), so (a) applied to  $-\mathbf{A}$  and  $-\mathbf{B}$  at the reflected indices  $n + 1 - i$  and  $n + 1 - j$  says exactly (b), once every index is translated back.

*Proof.* (a) Put  $k = i + j - 1$ , so  $1 \leq k \leq n$ . By the spectral theorem choose an orthonormal basis  $(\mathbf{u}_1, \dots, \mathbf{u}_n)$  of eigenvectors of  $\mathbf{A}$  with  $\mathbf{A}\mathbf{u}_p = \lambda_p(\mathbf{A})\mathbf{u}_p$ , and an orthonormal basis  $(\mathbf{v}_1, \dots, \mathbf{v}_n)$  of eigenvectors of  $\mathbf{B}$  with  $\mathbf{B}\mathbf{v}_q = \lambda_q(\mathbf{B})\mathbf{v}_q$ . Set

$$U = \text{span}\{\mathbf{u}_i, \dots, \mathbf{u}_n\}, \quad V = \text{span}\{\mathbf{v}_j, \dots, \mathbf{v}_n\},$$

so that  $\dim U = n - i + 1$  and  $\dim V = n - j + 1$ , the two lists being independent as parts of a basis.

Let  $W \leq F^n$  be **any** subspace with  $\dim W = k$ . Then

$$\dim W + \dim U = (i + j - 1) + (n - i + 1) = n + j > n,$$

since  $j \geq 1$ , so  $W \cap U \neq \{\mathbf{0}\}$  by [Lemma 16.2.1](#). More than that is needed, namely how large  $W \cap U$  is, and the dimension formula [Theorem 1.7.3](#) supplies it:

$$\dim(W \cap U) = \dim W + \dim U - \dim(W + U) \geq (n + j) - n = j,$$

because  $W + U$  is a subspace of  $F^n$ . Therefore

$$\dim(W \cap U) + \dim V \geq j + (n - j + 1) = n + 1 > n,$$

and a second application of [Lemma 16.2.1](#), to the subspaces  $W \cap U$  and  $V$ , produces a vector

$$\mathbf{x} \in W \cap U \cap V, \quad \mathbf{x} \neq \mathbf{0}.$$

Now evaluate. Since  $\mathbf{x} \in U$ , part (a) of [Lemma 16.3.1](#) gives  $R_{\mathbf{A}}(\mathbf{x}) \leq \lambda_i(\mathbf{A})$ ; since  $\mathbf{x} \in V$ , the same part applied to  $\mathbf{B}$  gives  $R_{\mathbf{B}}(\mathbf{x}) \leq \lambda_j(\mathbf{B})$ . The inner product is additive in its first slot, so  $\langle (\mathbf{A} + \mathbf{B})\mathbf{x}, \mathbf{x} \rangle = \langle \mathbf{A}\mathbf{x}, \mathbf{x} \rangle + \langle \mathbf{B}\mathbf{x}, \mathbf{x} \rangle$  and hence  $R_{\mathbf{A}+\mathbf{B}}(\mathbf{x}) = R_{\mathbf{A}}(\mathbf{x}) + R_{\mathbf{B}}(\mathbf{x})$ . Since  $\mathbf{x}$  is a non-zero vector of  $W$ ,

$$\min_{\mathbf{0} \neq \mathbf{y} \in W} R_{\mathbf{A}+\mathbf{B}}(\mathbf{y}) \leq R_{\mathbf{A}+\mathbf{B}}(\mathbf{x}) \leq \lambda_i(\mathbf{A}) + \lambda_j(\mathbf{B}).$$

The subspace  $W$  of dimension  $k$  was arbitrary, so the maximum of the left-hand side over all such  $W$  obeys the same bound; and that maximum is  $\lambda_k(\mathbf{A} + \mathbf{B})$  by the max-min half of [Theorem 16.2.3](#). This proves (a).

(b) Suppose  $i + j - n \geq 1$  and set

$$i' = n + 1 - i, \quad j' = n + 1 - j,$$

both of which lie in  $\{1, \dots, n\}$ . Then

$$i' + j' - 1 = 2n + 1 - i - j \leq n$$

precisely because  $i + j - n \geq 1$ , so part (a) applies to the Hermitian matrices  $-\mathbf{A}$  and  $-\mathbf{B}$  at the indices  $i', j'$ :

$$\lambda_{i'+j'-1}(-\mathbf{A} - \mathbf{B}) \leq \lambda_{i'}(-\mathbf{A}) + \lambda_{j'}(-\mathbf{B}).$$

Translate each term with [Lemma 16.3.2](#). On the right,  $\lambda_{i'}(-\mathbf{A}) = -\lambda_{n+1-i'}(\mathbf{A}) = -\lambda_i(\mathbf{A})$ , and likewise  $\lambda_{j'}(-\mathbf{B}) = -\lambda_j(\mathbf{B})$ . On the left,  $-\mathbf{A} - \mathbf{B} = -(\mathbf{A} + \mathbf{B})$  with  $\mathbf{A} + \mathbf{B}$  Hermitian, and  $i' + j' - 1 = 2n + 1 - i - j$ , so

$$\lambda_{i'+j'-1}(-(\mathbf{A} + \mathbf{B})) = -\lambda_{n+1-(2n+1-i-j)}(\mathbf{A} + \mathbf{B}) = -\lambda_{i+j-n}(\mathbf{A} + \mathbf{B}).$$

The displayed inequality therefore reads  $-\lambda_{i+j-n}(\mathbf{A} + \mathbf{B}) \leq -\lambda_i(\mathbf{A}) - \lambda_j(\mathbf{B})$ . Multiplying by  $-1$  reverses it and gives (b). This proves the theorem. ■

Two remarks on the bookkeeping, because it is the whole content. First, the index on the left is *not*  $i$  or  $j$  but  $i + j - 1$ , so a fixed eigenvalue  $\lambda_k(\mathbf{A} + \mathbf{B})$  receives  $k$  upper bounds from (a), one for each way of splitting  $k = i + j - 1$ , and the best of them is their minimum. Second, (a) only ever bounds from above and (b) only from below. To trap one eigenvalue  $\lambda_i(\mathbf{A} + \mathbf{B})$  from both sides with the index of  $\mathbf{A}$  unchanged, take  $j = 1$  in (a) and  $j = n$  in (b); that is the next corollary.

*Quick check.* Take  $i = j = 1$  in (a) and  $i = j = n$  in (b). What do the two inequalities say, and which earlier result are they?

*Solution.* They say  $\lambda_1(\mathbf{A} + \mathbf{B}) \leq \lambda_1(\mathbf{A}) + \lambda_1(\mathbf{B})$  and  $\lambda_n(\mathbf{A} + \mathbf{B}) \geq \lambda_n(\mathbf{A}) + \lambda_n(\mathbf{B})$ . For  $i = j = n$  the index on the left of (b) is  $n + n - n = n$ , as required. These are the statements that the largest eigenvalue is subadditive and the smallest superadditive, which one can also read straight off [Lemma 12.5.2](#):  $\langle (\mathbf{A} + \mathbf{B})\mathbf{x}, \mathbf{x} \rangle = \langle \mathbf{A}\mathbf{x}, \mathbf{x} \rangle + \langle \mathbf{B}\mathbf{x}, \mathbf{x} \rangle \leq \lambda_1(\mathbf{A}) + \lambda_1(\mathbf{B})$  for every unit  $\mathbf{x}$ . Weyl's inequalities are what happens when this easy observation is run at every index instead of only at the two ends.

**Warning. The eigenvalues are not subadditive index by index.** It is tempting to read Weyl as  $\lambda_i(\mathbf{A} + \mathbf{B}) \leq \lambda_i(\mathbf{A}) + \lambda_i(\mathbf{B})$ , and that statement is false. Take  $\mathbf{A} = \text{diag}(1, 0)$  and  $\mathbf{B} = \text{diag}(0, 1)$ , so that  $\lambda(\mathbf{A}) = \lambda(\mathbf{B}) = (1, 0)$  and  $\mathbf{A} + \mathbf{B} = \mathbf{I}_2$  with  $\lambda(\mathbf{A} + \mathbf{B}) = (1, 1)$ . At  $i = 2$  the proposed inequality reads  $1 \leq 0 + 0$ . The reverse reading  $\lambda_i(\mathbf{A} + \mathbf{B}) \geq \lambda_i(\mathbf{A}) + \lambda_i(\mathbf{B})$  fails on the same pair at  $i = 1$ , where it reads  $1 \geq 2$ . Only the shifted indices  $i + j - 1$  and  $i + j - n$  are correct.

Here is the theorem in full on one pair of matrices, so that the index arithmetic can be seen doing its work.

#### Example. (Both families on one pair)

Let

$$\mathbf{A} = \begin{pmatrix} 1 & 1 & 2 \\ 1 & 1 & 2 \\ 2 & 2 & 0 \end{pmatrix}, \quad \mathbf{B} = \begin{pmatrix} 3 & 2 & 0 \\ 2 & 3 & 0 \\ 0 & 0 & 0 \end{pmatrix}.$$

Find the eigenvalues of  $\mathbf{A}$ ,  $\mathbf{B}$  and  $\mathbf{A} + \mathbf{B}$ , and check every inequality of [Theorem 16.3.3](#).

*Solution.* All three matrices are real symmetric. Their characteristic polynomials factor over  $\mathbb{Z}$ :

$$\begin{aligned} p_{\mathbf{A}}(x) &= x(x-4)(x+2), \\ p_{\mathbf{B}}(x) &= x(x-5)(x-1), \\ p_{\mathbf{A}+\mathbf{B}}(x) &= (x-8)(x-1)(x+1), \end{aligned}$$

so, in decreasing order,

$$\lambda(\mathbf{A}) = (4, 0, -2), \quad \lambda(\mathbf{B}) = (5, 1, 0), \quad \lambda(\mathbf{A} + \mathbf{B}) = (8, 1, -1).$$

For  $\mathbf{A}$  one can see the list without computing: the rows  $(1, 1, 2)$  and  $(1, 1, 2)$  are equal, so  $\text{rank } \mathbf{A} = 2$  and 0 is an eigenvalue;  $\text{tr } \mathbf{A} = 2$  and  $\text{tr}(\mathbf{A}^2) = 20$  then force the other two to be 4 and  $-2$ .

With  $n = 3$ , family (a) applies when  $i + j \leq 4$  and family (b) when  $i + j \geq 4$ .

| family (a): |                                                                                               |
|-------------|-----------------------------------------------------------------------------------------------|
| $(i, j)$    | $\lambda_{i+j-1}(\mathbf{A} + \mathbf{B}) \leq \lambda_i(\mathbf{A}) + \lambda_j(\mathbf{B})$ |
| (1, 1)      | $8 \leq 4 + 5 = 9$                                                                            |
| (1, 2)      | $1 \leq 4 + 1 = 5$                                                                            |
| (2, 1)      | $1 \leq 0 + 5 = 5$                                                                            |
| (1, 3)      | $-1 \leq 4 + 0 = 4$                                                                           |
| (2, 2)      | $-1 \leq 0 + 1 = 1$                                                                           |
| (3, 1)      | $-1 \leq -2 + 5 = 3$                                                                          |
| family (b): |                                                                                               |
| $(i, j)$    | $\lambda_{i+j-3}(\mathbf{A} + \mathbf{B}) \geq \lambda_i(\mathbf{A}) + \lambda_j(\mathbf{B})$ |
| (1, 3)      | $8 \geq 4 + 0 = 4$                                                                            |
| (3, 1)      | $8 \geq -2 + 5 = 3$                                                                           |
| (2, 2)      | $8 \geq 0 + 1 = 1$                                                                            |
| (2, 3)      | $1 \geq 0 + 0 = 0$                                                                            |
| (3, 2)      | $1 \geq -2 + 1 = -1$                                                                          |
| (3, 3)      | $-1 \geq -2 + 0 = -2$                                                                         |

Every inequality holds, and every one of them is strict. At  $(1, 1)$  one can see why. Equality there would mean a unit  $\mathbf{x}$  with  $R_{\mathbf{A}+\mathbf{B}}(\mathbf{x}) = \lambda_1(\mathbf{A}) + \lambda_1(\mathbf{B})$ ; since  $R_{\mathbf{A}}(\mathbf{x}) \leq \lambda_1(\mathbf{A})$  and  $R_{\mathbf{B}}(\mathbf{x}) \leq \lambda_1(\mathbf{B})$  by [Lemma 16.3.1](#) (a) at  $i = 1$ , both would have to be equalities, and the computation in that proof shows that equality in  $\sum_p \lambda_p |c_p|^2 \leq \lambda_1 \sum_p |c_p|^2$  forces  $c_p = 0$  at every  $p$  with  $\lambda_p < \lambda_1$ . So  $\mathbf{x}$  would lie in the  $\lambda_1$ -eigenspace of  $\mathbf{A}$  and in that of  $\mathbf{B}$ . Those are the lines through  $(1, 1, 1)$  and through  $(1, 1, 0)$  — as  $\mathbf{A}(1, 1, 1) = 4(1, 1, 1)$  and  $\mathbf{B}(1, 1, 0) = 5(1, 1, 0)$ , and both eigenvalues are simple — and two distinct lines meet only at  $\mathbf{0}$ .

### 16.3.3 Perturbing one matrix

Setting  $j = 1$  in family (a) and  $j = n$  in family (b) leaves the index on the left equal to  $i$ . That is the case in which Weyl becomes a statement about *perturbation*: one matrix, moved.



**Corollary 16.3.4 — Perturbation, Two-Sided**

Let  $\mathbf{A}, \mathbf{B} \in M_n(F)$  be Hermitian. Then for every  $i$ ,

$$\lambda_i(\mathbf{A}) + \lambda_n(\mathbf{B}) \leq \lambda_i(\mathbf{A} + \mathbf{B}) \leq \lambda_i(\mathbf{A}) + \lambda_1(\mathbf{B}).$$

*Proof.* For the upper bound take  $j = 1$  in [Theorem 16.3.3](#) (a): the hypothesis  $i + 1 - 1 = i \leq n$  holds, and the conclusion is  $\lambda_i(\mathbf{A} + \mathbf{B}) \leq \lambda_i(\mathbf{A}) + \lambda_1(\mathbf{B})$ . For the lower bound take  $j = n$  in [Theorem 16.3.3](#) (b): the hypothesis  $i + n - n = i \geq 1$  holds, and the conclusion is  $\lambda_i(\mathbf{A} + \mathbf{B}) \geq \lambda_i(\mathbf{A}) + \lambda_n(\mathbf{B})$ . This proves the corollary. ■

In words: adding  $\mathbf{B}$  shifts each eigenvalue of  $\mathbf{A}$ , in order, by an amount trapped between the smallest and the largest eigenvalue of  $\mathbf{B}$ . Taking  $\mathbf{B} \succeq 0$ , so that  $\lambda_n(\mathbf{B}) \geq 0$  by [Theorem 12.1.4](#) (b), the lower bound becomes  $\lambda_i(\mathbf{A} + \mathbf{B}) \geq \lambda_i(\mathbf{A})$  — which is [Corollary 16.2.4](#) again, now as a special case of Weyl's inequalities.

To turn [Corollary 16.3.4](#) into a bound of the kind Chapter 15 wanted, the two numbers  $\lambda_1(\mathbf{B})$  and  $\lambda_n(\mathbf{B})$  must be replaced by one, and the right one is a norm.

**Lemma 16.3.5 — The Spectral Norm of a Hermitian Matrix**

Let  $\mathbf{E} \in M_n(F)$  be Hermitian. Then

$$\|\mathbf{E}\|_2 = \max_{1 \leq i \leq n} |\lambda_i(\mathbf{E})| = \max(\lambda_1(\mathbf{E}), -\lambda_n(\mathbf{E})).$$

*Proof.* By the spectral theorem,  $\mathbf{E} = \mathbf{U}\mathbf{D}\mathbf{U}^*$  with  $\mathbf{U}$  unitary and  $\mathbf{D} = \text{diag}(\lambda_1(\mathbf{E}), \dots, \lambda_n(\mathbf{E}))$  real. Then

$$\mathbf{E}^* \mathbf{E} = \mathbf{E}^2 = \mathbf{U}\mathbf{D}^2\mathbf{U}^*,$$

so the eigenvalues of  $\mathbf{E}^* \mathbf{E}$  are the numbers  $\lambda_i(\mathbf{E})^2$ , and the singular values of  $\mathbf{E}$ , which are their non-negative square roots listed decreasingly ([Definition 12.8.1](#)), are the numbers  $|\lambda_i(\mathbf{E})|$  listed decreasingly. Hence  $\sigma_1(\mathbf{E}) = \max_i |\lambda_i(\mathbf{E})|$ , and  $\|\mathbf{E}\|_2 = \sigma_1(\mathbf{E})$  by [Theorem 15.3.3](#) (c).

For the second equality,  $\lambda_n(\mathbf{E}) \leq \lambda_i(\mathbf{E}) \leq \lambda_1(\mathbf{E})$  for every  $i$  gives  $|\lambda_i(\mathbf{E})| \leq \max(|\lambda_1(\mathbf{E})|, |\lambda_n(\mathbf{E})|)$ , so the maximum is attained at one end of the list. Now  $\lambda_1(\mathbf{E}) \leq |\lambda_1(\mathbf{E})|$  and  $-\lambda_n(\mathbf{E}) \leq |\lambda_n(\mathbf{E})|$ , which gives one inequality between the two expressions. Conversely,  $|\lambda_1(\mathbf{E})|$  is either  $\lambda_1(\mathbf{E})$  or  $-\lambda_1(\mathbf{E}) \leq -\lambda_n(\mathbf{E})$ , and  $|\lambda_n(\mathbf{E})|$  is either  $-\lambda_n(\mathbf{E})$  or  $\lambda_n(\mathbf{E}) \leq \lambda_1(\mathbf{E})$ ; in every case it is at most  $\max(\lambda_1(\mathbf{E}), -\lambda_n(\mathbf{E}))$ . This proves the lemma. ■

**Corollary 16.3.6 — Hermitian Eigenvalues Are 1-Lipschitz**

Let  $\mathbf{A}, \mathbf{E} \in M_n(F)$  be Hermitian. Then

$$|\lambda_i(\mathbf{A} + \mathbf{E}) - \lambda_i(\mathbf{A})| \leq \|\mathbf{E}\|_2 \quad \text{for every } i = 1, \dots, n.$$

*Proof.* By [Corollary 16.3.4](#) applied with  $\mathbf{B} = \mathbf{E}$ ,

$$\lambda_n(\mathbf{E}) \leq \lambda_i(\mathbf{A} + \mathbf{E}) - \lambda_i(\mathbf{A}) \leq \lambda_1(\mathbf{E}).$$

By [Lemma 16.3.5](#),  $\lambda_1(\mathbf{E}) \leq \|\mathbf{E}\|_2$  and  $-\lambda_n(\mathbf{E}) \leq \|\mathbf{E}\|_2$ , that is,  $-\|\mathbf{E}\|_2 \leq \lambda_n(\mathbf{E})$ . So the difference lies in  $[-\|\mathbf{E}\|_2, \|\mathbf{E}\|_2]$ , which is the assertion. This proves the corollary. ■

This is the sentence Chapter 15 §07 promised: **the eigenvalues of a Hermitian matrix move by at most  $\|\mathbf{E}\|_2$  under a Hermitian perturbation.** It is worth being precise about what it adds to [Corollary 15.7.4](#), the continuity statement of that section, because the gain is not continuity — that was already had — but a constant.

- [Corollary 15.7.4](#) says: for each  $\mathbf{A}$  and each  $\varepsilon > 0$  there is a  $\delta > 0$  such that every matrix within  $\delta$  of  $\mathbf{A}$  has eigenvalues that can be numbered to lie within  $\varepsilon$  of those of  $\mathbf{A}$ . The  $\delta$  depends on  $\mathbf{A}$ , and [Example \(A Jordan block’s eigenvalues move like  \$\varepsilon^{1/k}\$ \)](#) shows that it must.
- [Corollary 16.3.6](#) says, for Hermitian  $\mathbf{A}$  and Hermitian perturbations measured in  $\|\cdot\|_2$ :  $\delta = \varepsilon$  works, for every  $\mathbf{A}$  at once, and the numbering is simply the decreasing order. The movement is bounded by the perturbation itself, with **Lipschitz constant 1**.

The constant does not depend on the size  $n$  of the matrices, it does not depend on how ill-conditioned  $\mathbf{A}$  is, and it does not depend on how close together the eigenvalues of  $\mathbf{A}$  are — three quantities that every reader of Chapter 15 has learned to expect in an error bound. It also cannot be improved:  $\mathbf{E} = t\mathbf{I}$  with  $t \in \mathbb{R}$  gives  $\lambda_i(\mathbf{A} + t\mathbf{I}) = \lambda_i(\mathbf{A}) + t$  for every  $i$ , while  $\|t\mathbf{I}\|_2 = |t|$ , so the inequality is an equality at every index. Finally, the eigenvalues are matched **by index**: the  $i$ -th of  $\mathbf{A} + \mathbf{E}$  is compared with the  $i$ -th of  $\mathbf{A}$ , not with whichever eigenvalue of  $\mathbf{A}$  happens to be nearest, and no search for a good numbering is needed.

### Corollary 16.3.7 — The Hermitian Eigenvalue Problem Is Perfectly Conditioned in Absolute Terms

Let  $\mathbf{A}, \mathbf{E} \in M_n(F)$  be Hermitian and write  $\lambda(\mathbf{A}) = (\lambda_1(\mathbf{A}), \dots, \lambda_n(\mathbf{A}))$  for the decreasing eigenvalue list. Then

$$\|\lambda(\mathbf{A} + \mathbf{E}) - \lambda(\mathbf{A})\|_\infty \leq \|\mathbf{E}\|_2,$$

and the constant 1 is the smallest one for which this holds.

*Proof.* The  $\infty$ -norm of a vector is the largest modulus of its entries, so the left-hand side is  $\max_i |\lambda_i(\mathbf{A} + \mathbf{E}) - \lambda_i(\mathbf{A})|$ , and the inequality is [Corollary 16.3.6](#) read over all  $i$  at once. For the last clause,  $\mathbf{E} = t\mathbf{I}$  with  $t > 0$  makes both sides equal to  $t$ , so no constant below 1 can work. This proves the corollary. ■

Chapter 15 §08 measured a linear system by  $\kappa(\mathbf{A}) = \|\mathbf{A}\| \|\mathbf{A}^{-1}\|$  ([Definition 15.8.1](#)), the worst factor by which  $\mathbf{A}$  magnifies a relative error in the data. The corollary above is the corresponding statement for a different problem — not “solve  $\mathbf{A}\mathbf{x} = \mathbf{b}$ ” but “report the eigenvalues of  $\mathbf{A}$ ” — and the factor there is 1, in absolute rather than relative terms. The difference between

absolute and relative matters here, and it is worth being exact.  $\mathbf{A} = \text{diag}(10^6, 10^{-6})$  is Hermitian with  $\kappa_2(\mathbf{A}) = 10^{12}$ , so solving with it is a delicate business, and yet its eigenvalues move by at most  $\|\mathbf{E}\|_2$  under any Hermitian  $\mathbf{E}$ , exactly as those of every other Hermitian matrix do. **Ill-conditioning of the linear system says nothing about the *absolute* conditioning of the eigenvalues.** In *relative* terms the two numbers meet again. The Hermitian perturbation  $\mathbf{E} = -10^{-6} \mathbf{e}_2 \mathbf{e}_2^\top$  has relative size  $\|\mathbf{E}\|_2 / \|\mathbf{A}\|_2 = 10^{-12}$  and sends the eigenvalue  $10^{-6}$  to 0, a relative change of 100%: an amplification of exactly  $10^{12} = \kappa_2(\mathbf{A})$ . In general, for an invertible Hermitian  $\mathbf{A}$ , [Corollary 16.3.6](#) bounds the relative change of the eigenvalue of smallest modulus by  $\|\mathbf{E}\|_2 / \sigma_n(\mathbf{A})$ , which is  $\kappa_2(\mathbf{A})$  times the relative size of  $\mathbf{E}$ , and a rank-one  $\mathbf{E}$  along its eigenvector attains it. So for Hermitian matrices,  $\kappa_2$  *is* the relative condition number of the smallest eigenvalue.

The contrast that makes the corollary worth having is with [Example \(A Jordan block's eigenvalues move like  \$\varepsilon^{1/k}\$ \)](#). There  $\mathbf{A} = \mathbf{J}_k(0)$  and  $\mathbf{E} = \varepsilon \mathbf{E}_{k1}$ , and the eigenvalues move a distance  $\varepsilon^{1/k}$  while  $\|\mathbf{E}\|_2 = \varepsilon$ , an amplification of  $\varepsilon^{-(k-1)/k}$  that is unbounded as  $\varepsilon \rightarrow 0^+$ . Neither matrix there is Hermitian, and [Corollary 16.3.6](#) explains which hypothesis is doing the work: for  $k = 2$  the perturbed matrix  $\begin{pmatrix} 0 & 1 \\ \varepsilon & 0 \end{pmatrix}$  has the perfectly real eigenvalues  $\pm\sqrt{\varepsilon}$ , so the conclusion of [Corollary 16.3.6](#) can be *stated* for it, and it is false —  $\sqrt{\varepsilon} > \varepsilon$  for  $0 < \varepsilon < 1$ , by a factor of 1000 at  $\varepsilon = 10^{-6}$ .

**Warning.** Both matrices must be Hermitian, and “small” is no substitute. If  $\mathbf{A}$  is not Hermitian, the corollary fails even when  $\mathbf{E}$  is: take  $\mathbf{A} = \mathbf{J}_2(0)$  and the Hermitian  $\mathbf{E} = \varepsilon \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}$ , with  $\|\mathbf{E}\|_2 = \varepsilon$ . Then  $\mathbf{A} + \mathbf{E} = \begin{pmatrix} 0 & 1+\varepsilon \\ \varepsilon & 0 \end{pmatrix}$  has the real eigenvalues  $\pm\sqrt{\varepsilon(1+\varepsilon)}$ , which have moved from 0 by more than  $\varepsilon$ . If  $\mathbf{E}$  is not Hermitian, the conclusion usually cannot even be stated:  $\mathbf{A} = \mathbf{0}$  is Hermitian and  $\mathbf{E} = \begin{pmatrix} 0 & -1 \\ 1 & 0 \end{pmatrix}$  has  $\|\mathbf{E}\|_2 = 1$ , but  $\mathbf{A} + \mathbf{E}$  has eigenvalues  $\pm i$ , so there is no decreasing list  $\lambda_i(\mathbf{A} + \mathbf{E})$  of real numbers to compare with anything. A bound for a diagonalizable  $\mathbf{A}$  and an arbitrary  $\mathbf{E}$  does exist — the theorem of Bauer and Fike, in Chapter 19 — and it carries an extra factor that measures how far the eigenvector basis of  $\mathbf{A}$  is from orthonormal.

### Example. (Locating eigenvalues without recomputing them)

Let

$$\mathbf{A} = \begin{pmatrix} 2 & 1 & 0 \\ 1 & 2 & 1 \\ 0 & 1 & 2 \end{pmatrix}, \quad \mathbf{E} = \frac{1}{10}(\mathbf{E}_{13} + \mathbf{E}_{31}).$$

Use [Corollary 16.3.6](#) to locate the eigenvalues of  $\mathbf{A} + \mathbf{E}$ , then compute them exactly and compare.

*Solution. The unperturbed spectrum.* Expanding along the first row,  $p_{\mathbf{A}}(x) = (x - 2)((x - 2)^2 - 2)$ , so

$$\lambda(\mathbf{A}) = (2 + \sqrt{2}, 2, 2 - \sqrt{2}) \approx (3.41421, 2, 0.58579).$$

*The size of the perturbation.*  $\mathbf{E}$  is real symmetric with  $\mathbf{E}^2 = \frac{1}{100}(\mathbf{E}_{11} + \mathbf{E}_{33})$ , whose eigenvalues are  $\frac{1}{100}, \frac{1}{100}, 0$ ; so the singular values of  $\mathbf{E}$  are  $\frac{1}{10}, \frac{1}{10}, 0$  and  $\|\mathbf{E}\|_2 = \frac{1}{10}$  by [Theorem 15.3.3](#) (c). (One can also read this off [Lemma 16.3.5](#): the eigenvalues of  $\mathbf{E}$  are  $\frac{1}{10}, 0, -\frac{1}{10}$ .)

*The prediction.* By [Corollary 16.3.6](#) each eigenvalue of  $\mathbf{A} + \mathbf{E}$  lies within  $\frac{1}{10}$  of the

corresponding eigenvalue of  $\mathbf{A}$ :

$$\lambda_1 \in [3.31421, 3.51421], \quad \lambda_2 \in [1.9, 2.1], \quad \lambda_3 \in [0.48579, 0.68579].$$

In particular  $\mathbf{A} + \mathbf{E}$  is positive definite, since  $\lambda_3 > 0$ , and its three eigenvalues are distinct, since the three intervals are disjoint.

*The check.* The characteristic polynomial of  $\mathbf{A} + \mathbf{E}$  factors as

$$p_{\mathbf{A}+\mathbf{E}}(x) = \left(x - \frac{19}{10}\right) \left(x^2 - \frac{41}{10}x + \frac{11}{5}\right),$$

so

$$\begin{aligned} \lambda(\mathbf{A} + \mathbf{E}) &= \left(\frac{41+3\sqrt{89}}{20}, \frac{19}{10}, \frac{41-3\sqrt{89}}{20}\right) \\ &\approx (3.46510, 1.9, 0.63490). \end{aligned}$$

All three lie in the predicted intervals. The middle one lies at the very end of its interval:  $\lambda_2$  has moved by exactly  $\frac{1}{10} = \|\mathbf{E}\|_2$ . That is not luck. The vector  $(1, 0, -1)$  is an eigenvector of  $\mathbf{A}$  for 2 and of  $\mathbf{E}$  for  $-\frac{1}{10}$ , hence an eigenvector of  $\mathbf{A} + \mathbf{E}$  for  $2 - \frac{1}{10}$ ; a shared eigenvector is exactly what makes the bound tight.

## 16.3.4 Perturbations of small rank

The last corollary trades size for rank. A perturbation of rank  $r$  need not be small in norm at all, but it can only disturb  $r$  directions, and Weyl converts that into a shift of the index by  $r$ .

### Corollary 16.3.8 — A Rank- $r$ Perturbation Shifts the Index by $r$

Let  $\mathbf{A}, \mathbf{B} \in M_n(F)$  be Hermitian, let  $r \geq 0$  be an integer, and suppose  $\text{rank } \mathbf{B} \leq r$ . Then

$$\begin{aligned} \lambda_{i+r}(\mathbf{A} + \mathbf{B}) &\leq \lambda_i(\mathbf{A}) & (1 \leq i, i+r \leq n), \\ \lambda_i(\mathbf{A}) &\leq \lambda_{i-r}(\mathbf{A} + \mathbf{B}) & (i \leq n, i-r \geq 1). \end{aligned}$$

*Idea.* A Hermitian matrix of rank at most  $r$  has at most  $r$  non-zero eigenvalues, so once one walks  $r+1$  places down its decreasing list one is certainly at a non-positive number, and once one stops  $r$  places short of the bottom one is certainly at a non-negative number. Feed those two facts into the two Weyl families at  $j = r+1$  and  $j = n-r$ .

*Proof.* By the spectral theorem  $\mathbf{B} = \mathbf{U}\mathbf{D}\mathbf{U}^*$  with  $\mathbf{D} = \text{diag}(\lambda_1(\mathbf{B}), \dots, \lambda_n(\mathbf{B}))$  and  $\mathbf{U}$  unitary, so  $\text{rank } \mathbf{B} = \text{rank } \mathbf{D}$  is the number of non-zero  $\lambda_q(\mathbf{B})$ , which is at most  $r$  by hypothesis.

*Two facts about the list of  $\mathbf{B}$ .* If  $\lambda_{r+1}(\mathbf{B}) > 0$ , then  $\lambda_1(\mathbf{B}), \dots, \lambda_{r+1}(\mathbf{B})$  are all positive by the decreasing indexing, giving  $r+1$  non-zero eigenvalues, a contradiction; hence

$$\lambda_{r+1}(\mathbf{B}) \leq 0 \quad \text{whenever } r+1 \leq n.$$

If  $\lambda_{n-r}(\mathbf{B}) < 0$ , then  $\lambda_{n-r}(\mathbf{B}), \dots, \lambda_n(\mathbf{B})$  are all negative, again  $r+1$  non-zero eigenvalues and a contradiction; hence

$$\lambda_{n-r}(\mathbf{B}) \geq 0 \quad \text{whenever } n-r \geq 1.$$

*First inequality.* Assume  $i \geq 1$  and  $i + r \leq n$ ; then  $r + 1 \leq n$ , so the index  $j = r + 1$  is legitimate and  $i + j - 1 = i + r \leq n$ . By [Theorem 16.3.3](#) (a) and the first fact,

$$\lambda_{i+r}(\mathbf{A} + \mathbf{B}) \leq \lambda_i(\mathbf{A}) + \lambda_{r+1}(\mathbf{B}) \leq \lambda_i(\mathbf{A}).$$

*Second inequality.* Assume  $i \leq n$  and  $i - r \geq 1$ ; then  $n - r \geq n - i + 1 \geq 1$ , so the index  $j = n - r$  is legitimate, and  $i + j - n = i - r \geq 1$ . By [Theorem 16.3.3](#) (b) and the second fact,

$$\lambda_{i-r}(\mathbf{A} + \mathbf{B}) \geq \lambda_i(\mathbf{A}) + \lambda_{n-r}(\mathbf{B}) \geq \lambda_i(\mathbf{A}).$$

This proves the corollary. ■

Read together, the two inequalities say that the two eigenvalue lists interlace with a lag of  $r$ :

$$\lambda_{i+r}(\mathbf{A} + \mathbf{B}) \leq \lambda_i(\mathbf{A}) \leq \lambda_{i-r}(\mathbf{A} + \mathbf{B}),$$

whenever all three indices make sense. For  $r = 0$  this collapses to  $\lambda_i(\mathbf{A} + \mathbf{B}) = \lambda_i(\mathbf{A})$ , correctly, since a Hermitian matrix of rank 0 is  $\mathbf{0}$ . For  $r = 1$  it is the statement that adding a Hermitian rank-one matrix moves each eigenvalue by at most one place in the list. That case is the subject of Section 5, and Section 9 returns to low-rank perturbations when it finishes the approximation theorem that Chapter 12 §10 could only state in the Frobenius norm.

*Remark.* The bound is about the *index*, not the *value*. A rank-one  $\mathbf{B}$  can be enormous:  $\mathbf{B} = 10^6 \mathbf{e}_1 \mathbf{e}_1^*$  has rank 1, and  $\lambda_1(\mathbf{A} + \mathbf{B})$  lies within  $\|\mathbf{A}\|_2$  of  $10^6$ , by [Corollary 16.3.6](#) with the roles of the two matrices exchanged. What [Corollary 16.3.8](#) guarantees is that the others are trapped all the same:  $\lambda_{k+1}(\mathbf{A}) \leq \lambda_k(\mathbf{A} + \mathbf{B}) \leq \lambda_{k-1}(\mathbf{A})$  for  $2 \leq k \leq n - 1$ , however large the one that escapes.

## 16.3.5 Exercises

### 16.3.5.1 A. Check your understanding

#### Exercise A1.

- State both families of [Theorem 16.3.3](#), with their index conditions, and say why the condition  $i + j - 1 \leq n$  cannot be dropped from the first.
- Determine whether the following is correct, and justify your answer: for Hermitian  $\mathbf{A}, \mathbf{B}$ ,  $\lambda_2(\mathbf{A} + \mathbf{B}) \leq \lambda_2(\mathbf{A}) + \lambda_2(\mathbf{B})$ .
- [Corollary 15.7.4](#) already said that eigenvalues move continuously. What does [Corollary 16.3.6](#) add?
- Explain why [Corollary 16.3.6](#) cannot be applied to  $\mathbf{A} = \mathbf{J}_2(0)$  and  $\mathbf{E} = \varepsilon \mathbf{E}_{21}$ , and what goes wrong if one applies it anyway.
- If  $\mathbf{B}$  is Hermitian of rank 3 and  $\mathbf{A}$  is Hermitian of size 10, which eigenvalue of  $\mathbf{A} + \mathbf{B}$  is guaranteed to be at most  $\lambda_4(\mathbf{A})$ ?

*Solution.*

- See [Theorem 16.3.3](#). Family (a) needs  $i + j - 1 \leq n$  and family (b) needs  $i + j - n \geq 1$ .

Without the first condition the symbol  $\lambda_{i+j-1}(\mathbf{A} + \mathbf{B})$  names nothing: a matrix of size  $n$  has only  $n$  eigenvalues, and there is no  $\lambda_{n+1}$ .

- (b) Incorrect. With  $\mathbf{A} = \text{diag}(1, 0)$  and  $\mathbf{B} = \text{diag}(0, 1)$  it reads  $\lambda_2(\mathbf{I}_2) = 1 \leq 0 + 0$ . The correct instance of [Theorem 16.3.3](#) (a) at  $i = j = 2$  needs  $i + j - 1 = 3 \leq n$ , so it says nothing when  $n = 2$ ; when  $n \geq 3$  it bounds  $\lambda_3$ , not  $\lambda_2$ .
- (c) A constant. [Corollary 15.7.4](#) gives, for each  $\mathbf{A}$  and  $\varepsilon$ , some unspecified  $\delta$  depending on  $\mathbf{A}$ ; [Corollary 16.3.6](#) gives  $\delta = \varepsilon$  for every  $\mathbf{A}$  simultaneously, that is, a Lipschitz constant of 1, independent of  $n$ , of the conditioning of  $\mathbf{A}$  and of the gaps between its eigenvalues. It also matches the eigenvalues by index rather than as a multiset.
- (d) Neither matrix is Hermitian, and the corollary's hypothesis is that both are. Applying it anyway predicts movement at most  $\|\mathbf{E}\|_2 = \varepsilon$ , whereas the eigenvalues are  $\pm\sqrt{\varepsilon}$  and have moved a distance  $\sqrt{\varepsilon}$ , which exceeds  $\varepsilon$  for every  $\varepsilon \in (0, 1)$  — see [Example \(A Jordan block's eigenvalues move like  \$\varepsilon^{1/k}\$ \)](#).
- (e)  $\lambda_7(\mathbf{A} + \mathbf{B})$ , and hence every later one, by [Corollary 16.3.8](#) with  $i = 4$  and  $r = 3$ .

### 16.3.5.2 B. Practice

#### Exercise B1: A small table.

Let  $\mathbf{A} = \begin{pmatrix} 0 & 4 \\ 4 & 0 \end{pmatrix}$  and  $\mathbf{B} = \begin{pmatrix} 3 & 0 \\ 0 & -3 \end{pmatrix}$ . Compute the eigenvalues of  $\mathbf{A}$ ,  $\mathbf{B}$  and  $\mathbf{A} + \mathbf{B}$ , write out every inequality of [Theorem 16.3.3](#) for this pair, and verify [Corollary 16.3.6](#) with  $\mathbf{E} = \mathbf{B}$ .

*Solution.*  $p_{\mathbf{A}}(x) = x^2 - 16$ , so  $\lambda(\mathbf{A}) = (4, -4)$ ;  $\mathbf{B}$  is diagonal, so  $\lambda(\mathbf{B}) = (3, -3)$ ; and  $\mathbf{A} + \mathbf{B} = \begin{pmatrix} 3 & 4 \\ 4 & -3 \end{pmatrix}$  has  $\text{tr} = 0$  and  $\det = -25$ , so  $p_{\mathbf{A}+\mathbf{B}}(x) = x^2 - 25$  and  $\lambda(\mathbf{A} + \mathbf{B}) = (5, -5)$ .

Family (a), for  $i + j \leq 3$ :

$$\begin{aligned} (1, 1) : 5 &\leq 4 + 3 = 7, \\ (1, 2) : -5 &\leq 4 - 3 = 1, \\ (2, 1) : -5 &\leq -4 + 3 = -1. \end{aligned}$$

Family (b), for  $i + j \geq 3$ :

$$\begin{aligned} (1, 2) : 5 &\geq 4 - 3 = 1, \\ (2, 1) : 5 &\geq -4 + 3 = -1, \\ (2, 2) : -5 &\geq -4 - 3 = -7. \end{aligned}$$

All six hold. For the last part,  $\mathbf{B}$  is Hermitian with eigenvalues  $\pm 3$ , so  $\|\mathbf{B}\|_2 = 3$  by [Lemma 16.3.5](#), and

$$\begin{aligned} |\lambda_1(\mathbf{A} + \mathbf{B}) - \lambda_1(\mathbf{A})| &= |5 - 4| = 1 \leq 3, \\ |\lambda_2(\mathbf{A} + \mathbf{B}) - \lambda_2(\mathbf{A})| &= |-5 + 4| = 1 \leq 3, \end{aligned}$$

as [Corollary 16.3.6](#) requires.

#### Exercise B2: An interval for each eigenvalue.

Let  $\mathbf{A} = \begin{pmatrix} 2 & 1 & 0 \\ 1 & 2 & 1 \\ 0 & 1 & 2 \end{pmatrix}$ , whose eigenvalues are  $2 \pm \sqrt{2}$  and 2, and let  $\mathbf{E} = \frac{1}{100}\mathbf{J}$ , where

$\mathbf{J} \in M_3(\mathbb{R})$  is the all-ones matrix.

- (a) Compute  $\|\mathbf{E}\|_2$  and give an interval containing each eigenvalue of  $\mathbf{A} + \mathbf{E}$ .
- (b) Deduce that  $\mathbf{A} + \mathbf{E}$  is positive definite.

*Solution.*

- (a)  $\mathbf{J}$  is real symmetric with  $\mathbf{J}^2 = 3\mathbf{J}$ , so its eigenvalues satisfy  $\lambda^2 = 3\lambda$  and are 3 and 0; since  $\text{rank } \mathbf{J} = 1$  the list is  $(3, 0, 0)$ . Hence  $\mathbf{E}$  has eigenvalues  $(\frac{3}{100}, 0, 0)$  and  $\|\mathbf{E}\|_2 = \frac{3}{100}$  by [Lemma 16.3.5](#). By [Corollary 16.3.6](#),

$$\lambda_1(\mathbf{A} + \mathbf{E}) \in [3.38421, 3.44421],$$

$$\lambda_2(\mathbf{A} + \mathbf{E}) \in [1.97, 2.03],$$

$$\lambda_3(\mathbf{A} + \mathbf{E}) \in [0.55579, 0.61579],$$

the centers being  $2 + \sqrt{2} \approx 3.41421$ , 2 and  $2 - \sqrt{2} \approx 0.58579$ .

- (b) Every interval lies strictly to the right of 0, so all three eigenvalues of the Hermitian matrix  $\mathbf{A} + \mathbf{E}$  are positive and  $\mathbf{A} + \mathbf{E} \succ 0$  by [Theorem 12.1.5](#) (b). No characteristic polynomial was computed. (For the record, the exact eigenvalues are 2 and  $\frac{403 \pm \sqrt{81609}}{200}$ , that is 2, 3.44336... and 0.58664...; the middle one is unmoved because  $(1, 0, -1)$  is an eigenvector of both  $\mathbf{A}$  and  $\mathbf{J}$ .)

### Exercise B3: What a low-rank update can do.

Let  $\mathbf{A} \in M_5(\mathbb{C})$  be Hermitian with  $\lambda(\mathbf{A}) = (5, 4, 3, 2, 1)$ , and let  $\mathbf{B} \in M_5(\mathbb{C})$  be Hermitian with  $\text{rank } \mathbf{B} \leq 2$ . Find the best bounds that [Corollary 16.3.8](#) gives for  $\lambda_3(\mathbf{A} + \mathbf{B})$ , and give examples showing that each is attained.

*Solution.* Take  $r = 2$ . The first inequality reads  $\lambda_{i+2}(\mathbf{A} + \mathbf{B}) \leq \lambda_i(\mathbf{A})$ , and it mentions  $\lambda_3(\mathbf{A} + \mathbf{B})$  only at  $i = 1$ , giving  $\lambda_3(\mathbf{A} + \mathbf{B}) \leq \lambda_1(\mathbf{A}) = 5$ . The second reads  $\lambda_i(\mathbf{A}) \leq \lambda_{i-2}(\mathbf{A} + \mathbf{B})$ , and it mentions  $\lambda_3(\mathbf{A} + \mathbf{B})$  only at  $i = 5$ , giving  $\lambda_3(\mathbf{A} + \mathbf{B}) \geq \lambda_5(\mathbf{A}) = 1$ . So the corollary gives exactly  $\lambda_3(\mathbf{A} + \mathbf{B}) \in [1, 5]$ .

Both ends are attained. Write  $\mathbf{A} = \mathbf{U} \text{diag}(5, 4, 3, 2, 1) \mathbf{U}^*$  with  $\mathbf{U}$  unitary ([Corollary 11.4.3](#)) and let  $\mathbf{q}_1, \dots, \mathbf{q}_5$  be its columns, an orthonormal eigenbasis.

For the upper bound take  $\mathbf{B} = 4(\mathbf{q}_4 \mathbf{q}_4^* + \mathbf{q}_5 \mathbf{q}_5^*)$ , which is Hermitian of rank 2. It adds 4 to the fourth and fifth eigenvalues and leaves the rest alone, so  $\mathbf{A} + \mathbf{B} = \mathbf{U} \text{diag}(5, 4, 3, 6, 5) \mathbf{U}^*$ , whose eigenvalue list sorted decreasingly is  $(6, 5, 5, 4, 3)$ . Hence  $\lambda_3(\mathbf{A} + \mathbf{B}) = 5 = \lambda_1(\mathbf{A})$ .

For the lower bound take  $\mathbf{B} = -4(\mathbf{q}_1 \mathbf{q}_1^* + \mathbf{q}_2 \mathbf{q}_2^*)$ , again Hermitian of rank 2. Then  $\mathbf{A} + \mathbf{B} = \mathbf{U} \text{diag}(1, 0, 3, 2, 1) \mathbf{U}^*$ , sorted  $(3, 2, 1, 1, 0)$ , so  $\lambda_3(\mathbf{A} + \mathbf{B}) = 1 = \lambda_5(\mathbf{A})$ .

### 16.3.5.3 C. Going deeper

#### Exercise C1: The Loewner order through Weyl.

Let  $\mathbf{A}, \mathbf{B} \in M_n(F)$  be Hermitian.

- (a) Prove that if  $\mathbf{A} \succeq \mathbf{B}$  then  $\lambda_i(\mathbf{A}) \geq \lambda_i(\mathbf{B})$  for every  $i$ , as a consequence of [Corollary 16.3.4](#).
- (b) Hence deduce that  $|\lambda_i(\mathbf{A}) - \lambda_i(\mathbf{B})| \leq \|\mathbf{A} - \mathbf{B}\|_2$  for every  $i$ , without invoking [Corollary 16.3.6](#).

*Solution.*

- (a) Put  $\mathbf{C} = \mathbf{A} - \mathbf{B}$ , which is Hermitian, and  $\mathbf{C} \succeq 0$  by hypothesis and [Definition 12.5.1](#). By [Theorem 12.1.4](#) (b) every eigenvalue of  $\mathbf{C}$  is  $\geq 0$ , in particular  $\lambda_n(\mathbf{C}) \geq 0$ . Applying [Corollary 16.3.4](#) to  $\mathbf{B}$  and  $\mathbf{C}$ ,

$$\lambda_i(\mathbf{A}) = \lambda_i(\mathbf{B} + \mathbf{C}) \geq \lambda_i(\mathbf{B}) + \lambda_n(\mathbf{C}) \geq \lambda_i(\mathbf{B}).$$

This is [Corollary 16.2.4](#).

- (b) Put  $\mathbf{C} = \mathbf{A} - \mathbf{B}$  again, now with no sign hypothesis, and let  $t = \|\mathbf{C}\|_2$ . By [Lemma 16.3.5](#),  $-t \leq \lambda_n(\mathbf{C})$  and  $\lambda_1(\mathbf{C}) \leq t$ , so  $t\mathbf{I} - \mathbf{C} \succeq 0$  and  $\mathbf{C} + t\mathbf{I} \succeq 0$  by [Theorem 12.1.4](#) (b) applied to the Hermitian matrices  $t\mathbf{I} - \mathbf{C}$  and  $\mathbf{C} + t\mathbf{I}$ , whose eigenvalues are  $t - \lambda_q(\mathbf{C})$  and  $\lambda_q(\mathbf{C}) + t$ . Hence  $\mathbf{B} + t\mathbf{I} \succeq \mathbf{A} \succeq \mathbf{B} - t\mathbf{I}$ , and part (a) gives

$$\begin{aligned} \lambda_i(\mathbf{B}) + t &= \lambda_i(\mathbf{B} + t\mathbf{I}) \geq \lambda_i(\mathbf{A}), \\ \lambda_i(\mathbf{A}) &\geq \lambda_i(\mathbf{B} - t\mathbf{I}) = \lambda_i(\mathbf{B}) - t, \end{aligned}$$

using  $\lambda_i(\mathbf{M} + s\mathbf{I}) = \lambda_i(\mathbf{M}) + s$  for real  $s$ , which holds because  $\mathbf{M}$  and  $\mathbf{M} + s\mathbf{I}$  have the same eigenvectors with eigenvalues shifted by  $s$ , and shifting by a constant preserves the decreasing order. Rearranged, this is  $|\lambda_i(\mathbf{A}) - \lambda_i(\mathbf{B})| \leq t$ , as claimed.

### Exercise C2: The largest eigenvalue is a convex function.

Let  $\mathbf{A}, \mathbf{B} \in M_n(F)$  be Hermitian and let  $t \in [0, 1]$ .

- (a) Prove that  $\lambda_1(t\mathbf{A} + (1-t)\mathbf{B}) \leq t\lambda_1(\mathbf{A}) + (1-t)\lambda_1(\mathbf{B})$ .
- (b) Deduce that for every  $c \in \mathbb{R}$  the set  $\{\mathbf{M} \in M_n(F) : \mathbf{M}^* = \mathbf{M}, \lambda_1(\mathbf{M}) \leq c\}$  is convex.
- (c) Show that  $\lambda_2$  is **not** convex in this sense, by producing two Hermitian  $2 \times 2$  matrices and a value of  $t$  where the inequality fails.

*Solution.*

- (a) For  $s \geq 0$  and Hermitian  $\mathbf{M}$  we have  $\lambda_1(s\mathbf{M}) = s\lambda_1(\mathbf{M})$ , since  $s\mathbf{M} = \mathbf{U}(s\mathbf{D})\mathbf{U}^*$  and multiplying a decreasing list of reals by  $s \geq 0$  keeps it decreasing. By [Theorem 16.3.3](#) (a) at  $i = j = 1$ , applied to the Hermitian matrices  $t\mathbf{A}$  and  $(1-t)\mathbf{B}$ ,

$$\lambda_1(t\mathbf{A} + (1-t)\mathbf{B}) \leq \lambda_1(t\mathbf{A}) + \lambda_1((1-t)\mathbf{B}) = t\lambda_1(\mathbf{A}) + (1-t)\lambda_1(\mathbf{B}).$$

- (b) Let  $\mathbf{A}, \mathbf{B}$  lie in the set and let  $t \in [0, 1]$ . The matrix  $t\mathbf{A} + (1-t)\mathbf{B}$  is Hermitian, and by (a) its largest eigenvalue is at most  $tc + (1-t)c = c$ . So the set contains the whole segment between any two of its points.
- (c) Take  $\mathbf{A} = \text{diag}(1, 0)$ ,  $\mathbf{B} = \text{diag}(0, 1)$  and  $t = \frac{1}{2}$ . Then  $\lambda_2(\mathbf{A}) = \lambda_2(\mathbf{B}) = 0$ , while  $t\mathbf{A} + (1-t)\mathbf{B} = \frac{1}{2}\mathbf{I}_2$  has  $\lambda_2 = \frac{1}{2}$ . So  $\frac{1}{2} \leq \frac{1}{2} \cdot 0 + \frac{1}{2} \cdot 0 = 0$  is false. The reason is visible in [Theorem 16.3.3](#): at  $i = j = 2$  the index on the left is 3, not 2.



**Exercise C3: Sharpness of the whole family.**

Fix  $n$  and indices  $i, j$  with  $i + j - 1 \leq n$ . Construct Hermitian  $\mathbf{A}, \mathbf{B} \in M_n(\mathbb{R})$ , both diagonal, for which

$$\lambda_{i+j-1}(\mathbf{A} + \mathbf{B}) = \lambda_i(\mathbf{A}) + \lambda_j(\mathbf{B}).$$

*Hint: with diagonal matrices every quantity in sight is a sorting problem; arrange for the  $(i + j - 1)$ -st largest entry of the sum to sit in a coordinate where  $\mathbf{A}$  contributes its  $i$ -th largest and  $\mathbf{B}$  its  $j$ -th largest.*

*Solution.* Let  $k = i + j - 1$  and let  $\mathbf{A} = \text{diag}(a_1, \dots, a_n)$  and  $\mathbf{B} = \text{diag}(b_1, \dots, b_n)$  with

$$a_p = \begin{cases} 1 & p \leq i - 1, \\ 0 & p \geq i, \end{cases} \quad b_p = \begin{cases} 0 & p \leq i - 1, \\ 1 & i \leq p \leq k, \\ 0 & p > k. \end{cases}$$

Both are diagonal with entries 0 and 1, hence Hermitian, and their diagonals are already their eigenvalue lists up to sorting. The matrix  $\mathbf{A}$  has  $i - 1$  entries equal to 1, so  $\lambda_i(\mathbf{A}) = 0$ . The matrix  $\mathbf{B}$  has  $k - i + 1 = j$  entries equal to 1, so  $\lambda_j(\mathbf{B}) = 1$ . The sum is  $\mathbf{A} + \mathbf{B} = \text{diag}(1, \dots, 1, 0, \dots, 0)$  with exactly  $k$  ones, since the ones of  $\mathbf{A}$  occupy coordinates  $1, \dots, i - 1$  and those of  $\mathbf{B}$  occupy  $i, \dots, k$ , and these two sets of coordinates do not overlap. Hence  $\lambda_k(\mathbf{A} + \mathbf{B}) = 1 = \lambda_i(\mathbf{A}) + \lambda_j(\mathbf{B})$ .

Every inequality of the first family is therefore attained, for every admissible pair  $(i, j)$ , already by 0–1 diagonal matrices. Applying [Lemma 16.3.2](#) to  $-\mathbf{A}$  and  $-\mathbf{B}$ , as in the proof of [Theorem 16.3.3](#) (b), turns this family of equalities into one showing that the second family is attained as well.

## 16.4 Deleting a Row and a Column

Take a Hermitian matrix, cross out the last row and the last column, and look at what is left. It is again Hermitian, one size smaller, and it has its own eigenvalues. Two lists of numbers are now in play,  $n$  of them and  $n - 1$  of them, and they cannot be independent of each other: the smaller matrix is the larger one restricted to a hyperplane, and [Theorem 16.2.3](#) describes every eigenvalue by what happens on subspaces. Restricting the supply of subspaces can only make the description worse in one direction and better in the other, and the exact bookkeeping is this section's subject.

The answer is as tight as it could be. The  $n - 1$  eigenvalues of the corner fall into the  $n - 1$  gaps of the eigenvalues of the whole, one to a gap. Nothing stronger holds: every interlacing pattern actually occurs (Exercise C2 proves the case  $n = 2$ ).

**Throughout this section the field is  $\mathbb{R}$  or  $\mathbb{C}$ , and every matrix whose eigenvalues are written down is Hermitian**, with eigenvalues real and indexed **decreasingly**. For an index set  $I \subseteq \{1, \dots, n\}$  we write  $\mathbf{A}_{I,I}$  for the **principal submatrix** of  $\mathbf{A}$  on the rows and columns indexed by  $I$ , as in [Definition 6.7.1](#). A principal submatrix of a Hermitian matrix is Hermitian, since  $(\mathbf{A}_{I,I})^* = (\mathbf{A}^*)_{I,I} = \mathbf{A}_{I,I}$ .

## 16.4.1 A matrix restricted to a coordinate hyperplane

Deleting row  $m$  and column  $m$  is not an arbitrary mutilation of  $\mathbf{A}$ . It is what  $\mathbf{A}$  looks like to vectors whose  $m$ -th coordinate is zero, and that is the only observation the proof needs.

### Lemma 16.4.1 — Deleting a Row and a Column Is Restricting to a Hyperplane

Let  $\mathbf{A} \in M_n(F)$  be Hermitian with  $n \geq 2$ , fix  $m$  with  $1 \leq m \leq n$ , put  $I = \{1, \dots, n\} \setminus \{m\}$  and  $\mathbf{B} = \mathbf{A}_{I,I} \in M_{n-1}(F)$ . Let

$$H = \{\mathbf{x} \in F^n : x_m = 0\},$$

and let  $\varphi: H \rightarrow F^{n-1}$  delete the  $m$ -th coordinate. Then  $\varphi$  is a linear isomorphism, and for every  $\mathbf{x} \in H$ ,

$$\langle \mathbf{A}\mathbf{x}, \mathbf{x} \rangle = \langle \mathbf{B}\varphi(\mathbf{x}), \varphi(\mathbf{x}) \rangle, \quad \|\mathbf{x}\| = \|\varphi(\mathbf{x})\|.$$

In particular  $R_{\mathbf{A}}(\mathbf{x}) = R_{\mathbf{B}}(\varphi(\mathbf{x}))$  for every  $\mathbf{x} \in H$  with  $\mathbf{x} \neq \mathbf{0}$ .

*Proof.* The map  $\varphi$  is linear, being given by deleting a coordinate, and it is bijective from  $H$  to  $F^{n-1}$  because inserting 0 in position  $m$  inverts it. Write  $I = \{i_1 < \dots < i_{n-1}\}$ , so that  $\varphi(\mathbf{x})_r = x_{i_r}$  and  $(\mathbf{B})_{rs} = a_{i_r i_s}$  by Definition 6.7.1.

Let  $\mathbf{x} \in H$ . Expanding the inner product of  $F^n$ , which is linear in the first slot and conjugate-linear in the second,

$$\langle \mathbf{A}\mathbf{x}, \mathbf{x} \rangle = \sum_{p=1}^n \sum_{q=1}^n a_{pq} x_q \overline{x_p}.$$

Every term with  $p = m$  or  $q = m$  vanishes, because  $x_m = 0$ . The remaining terms are indexed by  $p, q \in I$ , say  $p = i_r$  and  $q = i_s$ , and there the summand is  $a_{i_r i_s} \varphi(\mathbf{x})_s \overline{\varphi(\mathbf{x})_r} = (\mathbf{B})_{rs} \varphi(\mathbf{x})_s \overline{\varphi(\mathbf{x})_r}$ . Summing over  $r, s$  gives  $\langle \mathbf{B}\varphi(\mathbf{x}), \varphi(\mathbf{x}) \rangle$ . The same deletion of a zero coordinate gives  $\|\mathbf{x}\|^2 = \sum_{p \neq m} |x_p|^2 = \|\varphi(\mathbf{x})\|^2$ . Dividing the first identity by the second, which is positive when  $\mathbf{x} \neq \mathbf{0}$ , gives the statement about the Rayleigh quotients (Definition 16.1.1). This proves the lemma. ■

Because  $\varphi$  is an isomorphism, it matches the  $k$ -dimensional subspaces of  $H$  with the  $k$ -dimensional subspaces of  $F^{n-1}$ , for every  $k$ . So a statement about  $R_{\mathbf{B}}$  on subspaces of  $F^{n-1}$  is a statement about  $R_{\mathbf{A}}$  on subspaces of  $H$ , and the only thing lost in passing from  $\mathbf{A}$  to  $\mathbf{B}$  is the subspaces of  $F^n$  that are not inside  $H$ .

## 16.4.2 Interlacing

### Theorem 16.4.2 — Cauchy's Interlacing Theorem

Let  $\mathbf{A} \in M_n(F)$  be Hermitian with  $n \geq 2$ , let  $I \subseteq \{1, \dots, n\}$  with  $|I| = n - 1$ , and let  $\mathbf{B} = \mathbf{A}_{I,I}$ . Then

$$\lambda_k(\mathbf{A}) \geq \lambda_k(\mathbf{B}) \geq \lambda_{k+1}(\mathbf{A}) \quad (1 \leq k \leq n - 1),$$

that is,

$$\lambda_1(\mathbf{A}) \geq \lambda_1(\mathbf{B}) \geq \lambda_2(\mathbf{A}) \geq \lambda_2(\mathbf{B}) \geq \dots \geq \lambda_{n-1}(\mathbf{B}) \geq \lambda_n(\mathbf{A}).$$

*Idea.* Both halves of [Theorem 16.2.3](#) are used, one for each inequality, and each time the move is the same: a subspace of  $H$  is also a subspace of  $F^n$ .

① For  $\lambda_k(\mathbf{B}) \leq \lambda_k(\mathbf{A})$  use the **max–min** form. The best  $k$ -dimensional subspace for  $\mathbf{B}$  lives inside  $H$ ; transplanted to  $F^n$  it is one competitor among more, so the maximum over all of them is at least as large.

② For  $\lambda_{k+1}(\mathbf{A}) \leq \lambda_k(\mathbf{B})$  use the **min–max** form, and count dimensions. The best subspace for  $\mathbf{B}$  there has dimension  $(n - 1) - k + 1 = n - k$ , and  $n - k = n - (k + 1) + 1$  is exactly the dimension the min–max form for  $\lambda_{k+1}(\mathbf{A})$  asks for. So the same subspace, viewed in  $F^n$ , is admissible for  $\mathbf{A}$  at the index  $k + 1$ .

The dimension count that makes min–max work at all is [Lemma 16.2.1](#), and it has already been spent inside [Theorem 16.2.3](#); nothing more is needed here than the coincidence of the two numbers in ②.

*Proof.* Let  $m$  be the index not in  $I$ , and let  $H$  and  $\varphi$  be as in [Lemma 16.4.1](#), so that  $R_{\mathbf{A}}(\mathbf{x}) = R_{\mathbf{B}}(\varphi(\mathbf{x}))$  for every non-zero  $\mathbf{x} \in H$ . Both  $\mathbf{A}$  and  $\mathbf{B}$  are Hermitian, so [Theorem 16.2.3](#) applies to each, to  $\mathbf{A}$  with  $n$  and to  $\mathbf{B}$  with  $n - 1$  in the role of the size. Fix  $k$  with  $1 \leq k \leq n - 1$ .

( $\lambda_k(\mathbf{B}) \leq \lambda_k(\mathbf{A})$ ) By the max–min form of [Theorem 16.2.3](#) for  $\mathbf{B}$ , there is a subspace  $W' \leq F^{n-1}$  with  $\dim W' = k$  and

$$\lambda_k(\mathbf{B}) = \min_{\mathbf{0} \neq \mathbf{y} \in W'} R_{\mathbf{B}}(\mathbf{y}).$$

Put  $W = \varphi^{-1}(W') \leq H \leq F^n$ . Since  $\varphi$  is an isomorphism,  $\dim W = k$ , and  $\varphi$  carries the non-zero vectors of  $W$  onto the non-zero vectors of  $W'$ , so

$$\min_{\mathbf{0} \neq \mathbf{x} \in W} R_{\mathbf{A}}(\mathbf{x}) = \min_{\mathbf{0} \neq \mathbf{y} \in W'} R_{\mathbf{B}}(\mathbf{y}) = \lambda_k(\mathbf{B}).$$

Now  $W$  is one of the  $k$ -dimensional subspaces of  $F^n$  over which the max–min form for  $\mathbf{A}$  takes its maximum, so  $\lambda_k(\mathbf{A}) \geq \lambda_k(\mathbf{B})$ .

( $\lambda_{k+1}(\mathbf{A}) \leq \lambda_k(\mathbf{B})$ ) By the min–max form of [Theorem 16.2.3](#) for  $\mathbf{B}$ , whose matrices have size  $n - 1$ , there is a subspace  $W' \leq F^{n-1}$  with  $\dim W' = (n - 1) - k + 1 = n - k$  and

$$\lambda_k(\mathbf{B}) = \max_{\mathbf{0} \neq \mathbf{y} \in W'} R_{\mathbf{B}}(\mathbf{y}).$$

Put  $W = \varphi^{-1}(W') \leq F^n$ , of dimension  $n - k$ . Since  $1 \leq k + 1 \leq n$  and  $n - (k + 1) + 1 = n - k = \dim W$ , the subspace  $W$  is admissible in the min–max form for  $\lambda_{k+1}(\mathbf{A})$ , and that form takes the minimum over all such subspaces. Hence

$$\lambda_{k+1}(\mathbf{A}) \leq \max_{\mathbf{0} \neq \mathbf{x} \in W} R_{\mathbf{A}}(\mathbf{x}) = \max_{\mathbf{0} \neq \mathbf{y} \in W'} R_{\mathbf{B}}(\mathbf{y}) = \lambda_k(\mathbf{B}).$$

This proves the theorem. ■

The picture to keep is one of nesting. The interval  $[\lambda_n(\mathbf{A}), \lambda_1(\mathbf{A})]$  contains the interval  $[\lambda_{n-1}(\mathbf{B}), \lambda_1(\mathbf{B})]$ , and inside it the two lists alternate, an eigenvalue of  $\mathbf{B}$  between each consecutive pair of eigenvalues of  $\mathbf{A}$ . Deleting a row and a column can only shrink the spectrum inward, never spread it out.

*Quick check.* Both inequalities in [Theorem 16.4.2](#) can be equalities. Give a matrix  $\mathbf{A}$  for which  $\lambda_1(\mathbf{A}) = \lambda_1(\mathbf{B})$ , and one for which  $\lambda_1(\mathbf{B}) = \lambda_2(\mathbf{A})$ .

*Solution.* For the first,  $\mathbf{A} = \mathbf{I}_2$  with  $I = \{1\}$ : then  $\mathbf{B} = (1)$  and  $\lambda_1(\mathbf{A}) = \lambda_1(\mathbf{B}) = 1$ . For the second,  $\mathbf{A} = \text{diag}(1, 0)$  with  $I = \{2\}$ : then  $\mathbf{B} = (0)$ , and  $\lambda_1(\mathbf{B}) = 0 = \lambda_2(\mathbf{A})$ . Nothing in the theorem forces a gap; when a gap is forced is exactly the question settled later in this section.

Deleting one index at a time and iterating gives the general statement.

### Corollary 16.4.3 — Deleting $m$ Rows and the Same $m$ Columns

Let  $\mathbf{A} \in M_n(F)$  be Hermitian, let  $0 \leq m \leq n-1$ , let  $I \subseteq \{1, \dots, n\}$  with  $|I| = n-m$ , and let  $\mathbf{B} = \mathbf{A}_{I,I}$ . Then

$$\lambda_i(\mathbf{A}) \geq \lambda_i(\mathbf{B}) \geq \lambda_{i+m}(\mathbf{A}) \quad (1 \leq i \leq n-m).$$

*Proof.* Induction on  $m$ . For  $m = 0$  we have  $\mathbf{B} = \mathbf{A}$  and both inequalities are equalities.

Let  $m \geq 1$  and suppose the statement holds for  $m-1$ , for every Hermitian matrix. Choose an index  $j \notin I$  and put  $I' = I \cup \{j\}$ , so  $|I'| = n-m+1$ , and  $\mathbf{B}' = \mathbf{A}_{I',I'} \in M_{n-m+1}(F)$ , which is Hermitian. Since  $I \subseteq I'$ , taking the principal submatrix on  $I$  of  $\mathbf{B}'$  means keeping the rows and columns of  $\mathbf{A}$  indexed by  $I$ , so  $(\mathbf{B}')_{I,I} = \mathbf{B}$  after relabeling the indices of  $I$  inside  $I'$ ; and  $|I| = |I'| - 1$ .

Fix  $i$  with  $1 \leq i \leq n-m$ . By the inductive hypothesis applied to  $\mathbf{A}$  and  $\mathbf{B}'$ ,

$$\lambda_i(\mathbf{A}) \geq \lambda_i(\mathbf{B}') \geq \lambda_{i+m-1}(\mathbf{A}),$$

which is available because  $i \leq n-m < n-m+1$ . By [Theorem 16.4.2](#) applied to  $\mathbf{B}'$  and its principal submatrix  $\mathbf{B}$ , which is one size smaller,

$$\lambda_k(\mathbf{B}') \geq \lambda_k(\mathbf{B}) \geq \lambda_{k+1}(\mathbf{B}') \quad (1 \leq k \leq n-m).$$

Combining,  $\lambda_i(\mathbf{B}) \leq \lambda_i(\mathbf{B}') \leq \lambda_i(\mathbf{A})$ , which is the first inequality. For the second, take  $k = i$  in the display above and then use the inductive hypothesis at the index  $i+1$ , legitimate since  $i+1 \leq n-m+1$ :

$$\lambda_i(\mathbf{B}) \geq \lambda_{i+1}(\mathbf{B}') \geq \lambda_{(i+1)+m-1}(\mathbf{A}) = \lambda_{i+m}(\mathbf{A}).$$

This proves the corollary. ■

The extreme case  $m = n - 1$  is worth writing down, because it recovers a fact about diagonal entries that also follows from Chapter 12's description of the extreme eigenvalues.

**Example. (The diagonal lies inside the spectrum)**

Let  $\mathbf{A} \in M_n(F)$  be Hermitian. Show that  $\lambda_n(\mathbf{A}) \leq a_{ii} \leq \lambda_1(\mathbf{A})$  for every  $i$ .

*Solution.* Take  $I = \{i\}$ , so  $m = n - 1$  and  $\mathbf{B} = \mathbf{A}_{I,I} = (a_{ii})$ , a  $1 \times 1$  Hermitian matrix whose single eigenvalue is  $a_{ii}$  — real, as a diagonal entry of a Hermitian matrix must be. [Corollary 16.4.3](#) at  $i = 1$  gives

$$\lambda_1(\mathbf{A}) \geq \lambda_1(\mathbf{B}) = a_{ii} \geq \lambda_{1+(n-1)}(\mathbf{A}) = \lambda_n(\mathbf{A}).$$

The same conclusion follows from [Lemma 12.5.2](#) applied to the unit vector  $\mathbf{e}_i$ , since  $\langle \mathbf{A}\mathbf{e}_i, \mathbf{e}_i \rangle = a_{ii}$ . What interlacing adds is the version for larger index sets: any  $k$  diagonal entries are the diagonal of a  $k \times k$  principal submatrix, whose  $i$ -th eigenvalue is trapped between  $\lambda_{i+n-k}(\mathbf{A})$  and  $\lambda_i(\mathbf{A})$ . How much more can be said about the diagonal of a Hermitian matrix, given its spectrum, is the subject of Section 8.

### 16.4.3 When is the interlacing strict?

[Theorem 16.4.2](#) allows equalities, and the Quick check produced them. Deciding when they occur is the delicate part of the subject, and it is easiest to see when the deleted index is the last one, so that  $\mathbf{A}$  is presented as  $\mathbf{B}$  with a border added.

**Theorem 16.4.4 — Bordering a Hermitian Matrix**

Let  $\mathbf{B} \in M_{n-1}(F)$  be Hermitian with  $n \geq 2$ , let  $\mathbf{b} \in F^{n-1}$ , let  $c \in \mathbb{R}$ , and let

$$\mathbf{A} = \begin{pmatrix} \mathbf{B} & \mathbf{b} \\ \mathbf{b}^* & c \end{pmatrix} \in M_n(F),$$

which is Hermitian and has  $\mathbf{A}_{I,I} = \mathbf{B}$  for  $I = \{1, \dots, n-1\}$ . For an eigenvalue  $\nu$  of  $\mathbf{B}$ , write  $P_\nu$  for the orthogonal projection of  $F^{n-1}$  onto the eigenspace  $E_\nu(\mathbf{B})$ , and  $m_\nu = \dim E_\nu(\mathbf{B})$ .

- (a) If  $P_\nu \mathbf{b} \neq \mathbf{0}$ , then  $\nu$  is a root of  $p_{\mathbf{A}}$  of multiplicity **exactly**  $m_\nu - 1$ ; in particular, if  $m_\nu = 1$  then  $\nu$  is not an eigenvalue of  $\mathbf{A}$ .
- (b) If  $P_\nu \mathbf{b} = \mathbf{0}$ , then  $\nu$  is a root of  $p_{\mathbf{A}}$  of multiplicity **at least**  $m_\nu$ .
- (c) The interlacing  $\lambda_1(\mathbf{A}) \geq \lambda_1(\mathbf{B}) \geq \dots \geq \lambda_{n-1}(\mathbf{B}) \geq \lambda_n(\mathbf{A})$  is strict at every one of its  $2(n-1)$  inequalities if and only if  $\mathbf{B}$  has  $n-1$  **distinct** eigenvalues **and**  $P_\nu \mathbf{b} \neq \mathbf{0}$  for every eigenvalue  $\nu$  of  $\mathbf{B}$ .

*Idea.* Write the characteristic polynomial of  $\mathbf{A}$  down. With  $\mathbf{B}$  diagonalized, a Schur complement turns  $\det(x\mathbf{I} - \mathbf{A})$  into

$$\left( \prod_k (x - \mu_k) \right) \left( (x - c) - \sum_k \frac{|\tilde{b}_k|^2}{x - \mu_k} \right),$$

and clearing the denominators leaves a polynomial identity in which each  $\mu_k$  appears with one factor fewer than one might expect — provided  $P_\nu \mathbf{b} \neq \mathbf{0}$ , that is, provided the  $|\tilde{b}_k|^2$  over the group belonging to that eigenvalue do not all vanish (for a repeated eigenvalue the individual  $\tilde{b}_k$  depend on the choice of basis, but their sum does not). Part (c) then costs almost nothing: strict interlacing means the combined list of  $2n - 1$  numbers is strictly decreasing, hence has no repetitions at all, and (a) and (b) say exactly which repetitions are possible.

**Proof. Step 1: diagonalize the corner.** By the spectral theorem (Corollary 11.4.3 over  $\mathbb{C}$ , Corollary 11.5.2 over  $\mathbb{R}$ ) write  $\mathbf{B} = \mathbf{U}\mathbf{D}\mathbf{U}^*$  with  $\mathbf{U} \in M_{n-1}(F)$  unitary and  $\mathbf{D} = \text{diag}(\mu_1, \dots, \mu_{n-1})$  carrying the eigenvalues of  $\mathbf{B}$  in decreasing order, the columns  $\mathbf{u}_1, \dots, \mathbf{u}_{n-1}$  of  $\mathbf{U}$  being an orthonormal basis of eigenvectors. Let  $\mathbf{V} = \begin{pmatrix} \mathbf{U} & \mathbf{0} \\ \mathbf{0}^* & 1 \end{pmatrix} \in M_n(F)$ , which is unitary because  $\mathbf{V}^*\mathbf{V} = \mathbf{I}_n$ . Multiplying in blocks (Theorem 7.1.1) gives

$$\mathbf{V}^*\mathbf{A}\mathbf{V} = \begin{pmatrix} \mathbf{D} & \tilde{\mathbf{b}} \\ \tilde{\mathbf{b}}^* & c \end{pmatrix}, \quad \tilde{\mathbf{b}} = \mathbf{U}^*\mathbf{b}.$$

Similar matrices have the same characteristic polynomial (Theorem 6.10.3), so we may compute with the right-hand side. Moreover  $\tilde{b}_k = (\mathbf{U}^*\mathbf{b})_k = \langle \mathbf{b}, \mathbf{u}_k \rangle$ , and the  $\mathbf{u}_k$  with  $\mu_k = \nu$  form an orthonormal basis of  $E_\nu(\mathbf{B})$ : they lie in it, and a vector  $\mathbf{x} = \sum_k c_k \mathbf{u}_k$  with  $\mathbf{B}\mathbf{x} = \nu\mathbf{x}$  has  $(\mu_k - \nu)c_k = 0$  for every  $k$ , so  $c_k = 0$  unless  $\mu_k = \nu$ . Hence exactly  $m_\nu$  of the  $\mu_k$  equal  $\nu$ , and

$$P_\nu \mathbf{b} = \sum_{k:\mu_k=\nu} \langle \mathbf{b}, \mathbf{u}_k \rangle \mathbf{u}_k \quad \text{and} \quad \|P_\nu \mathbf{b}\|^2 = \sum_{k:\mu_k=\nu} |\tilde{b}_k|^2.$$

In particular  $P_\nu \mathbf{b} = \mathbf{0}$  exactly when  $\tilde{b}_k = 0$  for every  $k$  with  $\mu_k = \nu$ .

**Step 2: the characteristic polynomial.** We claim that, as polynomials in  $x$ ,

$$\begin{aligned} p_{\mathbf{A}}(x) &= (x - c) \prod_{k=1}^{n-1} (x - \mu_k) \\ &\quad - \sum_{k=1}^{n-1} |\tilde{b}_k|^2 \prod_{l \neq k} (x - \mu_l). \end{aligned} \tag{16.4.1}$$

*Claim.* Eq. 16.4.1 holds.

*Proof.* Fix a scalar  $x_0$  that is not an eigenvalue of  $\mathbf{B}$ . Then  $x_0\mathbf{I}_{n-1} - \mathbf{D}$  is invertible, being diagonal with non-zero diagonal entries, so [Theorem 7.3.2](#) (a), applied to

$$\mathbf{M} = x_0\mathbf{I}_n - \mathbf{V}^*\mathbf{A}\mathbf{V} = \begin{pmatrix} x_0\mathbf{I}_{n-1} - \mathbf{D} & -\tilde{\mathbf{b}} \\ -\tilde{\mathbf{b}}^* & x_0 - c \end{pmatrix}$$

with the invertible block in the top left, gives  $\det \mathbf{M} = \det(x_0\mathbf{I} - \mathbf{D}) \cdot \det(\mathbf{M}/(x_0\mathbf{I} - \mathbf{D}))$ , where the Schur complement is the  $1 \times 1$  matrix

$$(x_0 - c) - (-\tilde{\mathbf{b}}^*)(x_0\mathbf{I} - \mathbf{D})^{-1}(-\tilde{\mathbf{b}}) = (x_0 - c) - \sum_k \frac{|\tilde{b}_k|^2}{x_0 - \mu_k}.$$

Multiplying out  $\det(x_0\mathbf{I} - \mathbf{D}) = \prod_k (x_0 - \mu_k)$  against this scalar, and canceling one factor  $(x_0 - \mu_k)$  in the  $k$ -th summand, gives exactly the right-hand side of [Eq. 16.4.1](#) evaluated at  $x_0$ . So the two sides of [Eq. 16.4.1](#), which are polynomials of degree  $n$ , agree at every scalar outside the finite set  $\text{spec}(\mathbf{B})$ . Their difference is therefore a polynomial of degree at most  $n$  with infinitely many roots in  $F$ , and a non-zero polynomial of degree at most  $n$  has at most  $n$  roots ([Corollary 5.1.4](#)); so the difference is zero. ■

**Step 3: multiplicities.** Let  $\nu_1 > \dots > \nu_p$  be the distinct eigenvalues of  $\mathbf{B}$ , with multiplicities  $m_1, \dots, m_p$  summing to  $n - 1$ , and put  $\beta_t = \|P_{\nu_t} \mathbf{b}\|^2$ . Grouping equal  $\mu_k$  in [Eq. 16.4.1](#) — the product  $\prod_{l \neq k} (x - \mu_l)$  is the same for all  $k$  in one group, and the coefficients  $|\tilde{b}_k|^2$  add up to  $\beta_t$  by Step 1 — turns it into

$$p_{\mathbf{A}}(x) = (x - c) \prod_{t=1}^p (x - \nu_t)^{m_t} - \sum_{t=1}^p \beta_t (x - \nu_t)^{m_t-1} \prod_{s \neq t} (x - \nu_s)^{m_s}.$$

Fix  $t$ . Every term on the right is divisible by  $(x - \nu_t)^{m_t-1}$ , so  $p_{\mathbf{A}}(x) = (x - \nu_t)^{m_t-1} g(x)$  with

$$g(x) = (x - c)(x - \nu_t) \prod_{s \neq t} (x - \nu_s)^{m_s} - \beta_t \prod_{s \neq t} (x - \nu_s)^{m_s} - \sum_{u \neq t} \beta_u (x - \nu_t)(x - \nu_u)^{m_u-1} \prod_{s \neq u, t} (x - \nu_s)^{m_s}.$$

Evaluate at  $x = \nu_t$ . The first term vanishes because of its factor  $(x - \nu_t)$ , and so does every summand of the third, for the same reason. What is left is

$$g(\nu_t) = -\beta_t \prod_{s \neq t} (\nu_t - \nu_s)^{m_s},$$

and the product is non-zero because the  $\nu_s$  are distinct. So  $g(\nu_t) \neq 0$  exactly when  $\beta_t \neq 0$ , that is, exactly when  $P_{\nu_t} \mathbf{b} \neq \mathbf{0}$ . In that case  $\nu_t$  is a root of  $p_{\mathbf{A}}$  of multiplicity exactly  $m_t - 1$ , which proves (a); in the other case  $(x - \nu_t)$  divides  $g$ , so the multiplicity is at least  $m_t$ , which proves (b).

**Step 4: strictness.** ( $\Leftarrow$ ) Suppose  $\mathbf{B}$  has  $n - 1$  distinct eigenvalues, so every  $m_t = 1$ , and  $P_\nu \mathbf{b} \neq \mathbf{0}$  for each of them. By (a), no eigenvalue of  $\mathbf{B}$  is an eigenvalue of  $\mathbf{A}$ . [Theorem 16.4.2](#) gives  $\lambda_k(\mathbf{A}) \geq \lambda_k(\mathbf{B}) \geq \lambda_{k+1}(\mathbf{A})$  for each  $k$ ; since  $\lambda_k(\mathbf{B})$  is an eigenvalue of  $\mathbf{B}$  and  $\lambda_k(\mathbf{A})$ ,  $\lambda_{k+1}(\mathbf{A})$  are eigenvalues of  $\mathbf{A}$ , neither inequality can be an equality. So all  $2(n - 1)$  of them are strict.

( $\Rightarrow$ ) Suppose all of them are strict, that is,

$$\lambda_1(\mathbf{A}) > \lambda_1(\mathbf{B}) > \lambda_2(\mathbf{A}) > \cdots > \lambda_{n-1}(\mathbf{B}) > \lambda_n(\mathbf{A}).$$

Reading two steps at a time gives  $\lambda_k(\mathbf{B}) > \lambda_{k+1}(\mathbf{A}) > \lambda_{k+1}(\mathbf{B})$  for  $1 \leq k \leq n - 2$ , so the eigenvalues of  $\mathbf{B}$  are distinct. And the displayed chain is strictly decreasing, so its  $2n - 1$  entries are pairwise distinct; in particular no eigenvalue of  $\mathbf{B}$  is an eigenvalue of  $\mathbf{A}$ . By (b), read as a contrapositive,  $P_\nu \mathbf{b} \neq \mathbf{0}$  for every eigenvalue  $\nu$  of  $\mathbf{B}$ . This proves the theorem. ■

*Remark.* Part (b) is deliberately one-sided. When  $P_\nu \mathbf{b} = \mathbf{0}$  the multiplicity of  $\nu$  in  $p_{\mathbf{A}}$  may be  $m_\nu$  or  $m_\nu + 1$  — never more, since by Cauchy interlacing  $q$  equal eigenvalues  $\nu$  of  $\mathbf{A}$  squeeze  $q - 1$  eigenvalues of  $\mathbf{B}$  to  $\nu$  — and both occur:  $\mathbf{B} = \text{diag}(3, 1)$ ,  $\mathbf{b} = (0, 1)$ ,  $c = 0$  gives  $p_{\mathbf{A}}(x) = (x - 3)(x^2 - x - 1)$ , where 3 keeps multiplicity  $1 = m_3$ ; while  $\mathbf{B} = (\mu)$ ,  $\mathbf{b} = \mathbf{0}$ ,  $c = \mu$  gives  $\mathbf{A} = \mu \mathbf{I}_2$ , where  $\mu$  has multiplicity  $2 = m_\mu + 1$ . Deciding between the two needs the value of  $c$  as well, and [Theorem 16.4.4](#) does not attempt it; what it does settle completely is the strictness question (c), which is what interlacing asks.

**Warning. Strictness needs both clauses, and a non-zero border is not enough.** It is tempting to say that the interlacing is strict as soon as  $\mathbf{b} \neq \mathbf{0}$ , or as soon as  $\mathbf{b}$  meets every eigenspace of  $\mathbf{B}$ . The first is wrong even when the eigenvalues of  $\mathbf{B}$  are distinct (Exercise B3(a)); the second fails exactly when  $\mathbf{B}$  has a repeated eigenvalue. Take  $\mathbf{B} = 2\mathbf{I}_2$ ,  $\mathbf{b} = (1, 1)$  and  $c = 2$ . Here  $\mathbf{b}$  has a non-zero component in the one eigenspace  $E_2(\mathbf{B}) = F^2$ , and yet  $\lambda_1(\mathbf{B}) = \lambda_2(\mathbf{B}) = 2$ , so the middle of the interlacing chain reads  $2 \geq \lambda_2(\mathbf{A}) \geq 2$  and  $\lambda_2(\mathbf{A}) = 2$  with no freedom at all. A repeated eigenvalue of  $\mathbf{B}$  is passed on to  $\mathbf{A}$ , by [Theorem 16.4.4](#) (a) with  $m_\nu = 2$ .

To apply [Theorem 16.4.4](#) when the deleted index is not the last one, permute. Let  $I = \{i_1 < \cdots < i_{n-1}\} = \{1, \dots, n\} \setminus \{m\}$ , and let  $\mathbf{P}$  be the matrix with columns  $\mathbf{e}_{i_1}, \dots, \mathbf{e}_{i_{n-1}}, \mathbf{e}_m$ , a permutation matrix. Its columns are orthonormal, so  $\mathbf{P}^\top \mathbf{P} = \mathbf{I}$  and  $\mathbf{P}$  is unitary ([Lemma 2.6.6](#) (b)); hence  $\mathbf{P}^\top \mathbf{A} \mathbf{P}$  is Hermitian and similar to  $\mathbf{A}$ , with the same eigenvalues. Its  $(r, s)$  entry is  $\mathbf{e}_{i_r}^\top \mathbf{A} \mathbf{e}_{i_s} = a_{i_r i_s}$  for  $r, s \leq n - 1$ , so its leading  $(n - 1) \times (n - 1)$  corner is  $\mathbf{A}_{I, I}$ , and its border is the column  $(a_{i_1 m}, \dots, a_{i_{n-1} m})$  with corner entry  $a_{mm}$ . So  $\mathbf{P}^\top \mathbf{A} \mathbf{P}$  is a bordered matrix with corner  $\mathbf{A}_{I, I}$ , and [Theorem 16.4.4](#) applies to it verbatim.

The strictness criterion pays a debt from Chapter 10. The solution of [Exercise \(C1: Interlacing roots\)](#) proved by hand that the roots of consecutive orthogonal polynomials interlace for degrees 1, 2, 3, and then said: “The general case is not proved this way. The coefficients  $a_k$  and  $b_k$  assemble into a symmetric tridiagonal matrix whose characteristic polynomial is  $p_k$ , and the interlacing then follows from the interlacing theorem for the eigenvalues of a symmetric matrix and its leading submatrices, which is Chapter 16’s business.” Two things in that sentence were asserted, not proved: that the characteristic polynomial is  $p_k$ , and that the interlacing is **strict**, which is what “exactly one root strictly between” demands. Plain [Theorem 16.4.2](#) gives only  $\geq$ .



Strictness needs [Theorem 16.4.4](#) (c), and (c) needs one fact about tridiagonal matrices.

### Lemma 16.4.5 — Eigenvectors of an Unreduced Tridiagonal Matrix

Let  $\mathbf{T} \in M_k(\mathbb{R})$  be symmetric and tridiagonal, with every entry  $t_{i,i+1} = t_{i+1,i}$  ( $1 \leq i \leq k-1$ ) **non-zero**. Then every eigenvector  $\mathbf{u}$  of  $\mathbf{T}$  has  $u_k \neq 0$ . Consequently every eigenspace of  $\mathbf{T}$  has dimension 1, so  $\mathbf{T}$  has  $k$  distinct eigenvalues.

*Proof.* Let  $\mathbf{T}\mathbf{u} = \nu\mathbf{u}$  and suppose  $u_k = 0$ ; we show  $\mathbf{u} = \mathbf{0}$ . If  $k = 1$  this is immediate. Otherwise row  $k$  of  $\mathbf{T}\mathbf{u} = \nu\mathbf{u}$  reads  $t_{k,k-1}u_{k-1} + t_{kk}u_k = \nu u_k$ , so  $t_{k,k-1}u_{k-1} = 0$  and  $u_{k-1} = 0$ , since  $t_{k,k-1} \neq 0$ . Now suppose  $u_i = u_{i+1} = 0$  for some  $2 \leq i \leq k-1$ . Row  $i$  reads  $t_{i,i-1}u_{i-1} + t_{ii}u_i + t_{i,i+1}u_{i+1} = \nu u_i$ , so  $t_{i,i-1}u_{i-1} = 0$  and  $u_{i-1} = 0$ . Walking up from  $i = k-1$  to  $i = 2$  gives  $u_{k-2} = \cdots = u_1 = 0$ , so  $\mathbf{u} = \mathbf{0}$ , and  $\mathbf{u}$  is not an eigenvector.

If some eigenspace  $E_\nu(\mathbf{T})$  contained two independent vectors  $\mathbf{u}, \mathbf{u}'$ , then  $u'_k\mathbf{u} - u_k\mathbf{u}'$  would be a non-zero vector of  $E_\nu(\mathbf{T})$  — non-zero by independence, since  $u_k \neq 0$  — with last coordinate 0, which we have just excluded. So each eigenspace is a line. Since  $\mathbf{T}$  is real symmetric, its eigenspaces have dimensions adding up to  $k$  ([Corollary 11.5.2](#)), so there are  $k$  of them, that is,  $k$  distinct eigenvalues. This proves the lemma. ■

### Corollary 16.4.6 — Zeros of Orthogonal Polynomials Interlace

Let  $(p_k)$  be the monic orthogonal polynomial sequence of an inner product on  $\mathbb{R}[x]$  to which [Theorem 10.10.3](#) applies, with coefficients  $a_k$  and  $b_k > 0$ ; for the node inner product on  $\mathbb{R}[x]_{\leq n}$  read everything below for  $k+1 \leq n$ . For  $k \geq 1$  let  $\mathbf{T}_k$  (the **Jacobi matrix** of the sequence) be the  $k \times k$  real symmetric tridiagonal matrix with diagonal  $a_0, a_1, \dots, a_{k-1}$  and entries  $\sqrt{b_1}, \dots, \sqrt{b_{k-1}}$  just above and just below it.

- (a)  $p_{\mathbf{T}_k} = p_k$ ; so the roots of  $p_k$  are the eigenvalues of  $\mathbf{T}_k$ , and they are real and distinct.
- (b) If  $r_1 > \cdots > r_{k+1}$  are the roots of  $p_{k+1}$  and  $s_1 > \cdots > s_k$  those of  $p_k$ , then

$$r_1 > s_1 > r_2 > s_2 > \cdots > s_k > r_{k+1}.$$

*Idea.* The polynomials  $p_k$  are the characteristic polynomials of a nested family of symmetric tridiagonal matrices, each the leading corner of the next. So their zeros are eigenvalues of a matrix and of its corner, and Cauchy interlacing applies at once — non-strictly. Strictness is the bordered-matrix criterion, and both of its clauses come from one fact about tridiagonal matrices with non-zero off-diagonal entries: an eigenvector can never end in a zero.

*Proof.*

- (a) Put  $D_k(x) = \det(x\mathbf{I}_k - \mathbf{T}_k)$  and  $D_0 = 1$ . For each real  $x_0$ , the matrix  $x_0\mathbf{I}_k - \mathbf{T}_k$  is tridiagonal in the sense of Chapter 6, with diagonal entries  $x_0 - a_{i-1}$  and off-diagonal entries  $-\sqrt{b_i}$  in both positions  $(i, i+1)$  and  $(i+1, i)$ , so [Theorem 6.8.3](#)

gives  $D_1(x_0) = x_0 - a_0$  and

$$\begin{aligned} D_{k+1}(x_0) &= (x_0 - a_k)D_k(x_0) - (-\sqrt{b_k})^2 D_{k-1}(x_0) \\ &= (x_0 - a_k)D_k(x_0) - b_k D_{k-1}(x_0) \end{aligned}$$

for  $k \geq 1$ . Here  $D_k(x_0) = \det(x_0 \mathbf{I} - \mathbf{T}_k)$  is the characteristic polynomial evaluated at  $x_0$  (Lemma 6.10.1), and two real polynomials that agree at every real number are equal (Theorem 5.1.5 (b)), so these identities hold as polynomials. They are the recurrence of Theorem 10.10.3 with the same starting values  $p_0 = 1$ ,  $p_1 = x - a_0$ , so  $D_k = p_k$  for every  $k$  by induction. The roots of  $p_{\mathbf{T}_k}$  are the eigenvalues of  $\mathbf{T}_k$  (Theorem 8.2.1), which are real because  $\mathbf{T}_k$  is real symmetric (Theorem 11.2.1) and distinct by Lemma 16.4.5, whose hypothesis holds because every  $\sqrt{b_i} > 0$ .

(b) Write  $\mathbf{T}_{k+1}$  in bordered form:

$$\mathbf{T}_{k+1} = \begin{pmatrix} \mathbf{T}_k & \sqrt{b_k} \mathbf{e}_k \\ \sqrt{b_k} \mathbf{e}_k^\top & a_k \end{pmatrix}, \quad \mathbf{e}_k \in \mathbb{R}^k.$$

The border of  $\mathbf{T}_{k+1}$  is  $\mathbf{w} = \sqrt{b_k} \mathbf{e}_k$ , whose  $k$ -th entry is  $\sqrt{b_k}$ . We call it  $\mathbf{w}$  rather than  $\mathbf{b}$  because  $b_k$  already names a recurrence coefficient. We check the two clauses of Theorem 16.4.4 (c) for  $\mathbf{B} = \mathbf{T}_k$ , with  $\mathbf{w}$  in place of  $\mathbf{b}$ . By Lemma 16.4.5,  $\mathbf{T}_k$  has  $k$  distinct eigenvalues. For each eigenvalue  $\nu$ , the eigenspace  $E_\nu(\mathbf{T}_k)$  is spanned by one unit vector  $\mathbf{u}$ , so  $P_\nu \mathbf{w} = \langle \mathbf{w}, \mathbf{u} \rangle \mathbf{u} = \sqrt{b_k} u_k \mathbf{u}$ , which is non-zero because  $b_k > 0$  and  $u_k \neq 0$  by the same lemma. So Theorem 16.4.4 (c) applies, and the eigenvalues of  $\mathbf{T}_{k+1}$  and of  $\mathbf{T}_k$  interlace strictly. By (a) these are the roots of  $p_{k+1}$  and of  $p_k$ . This proves the corollary. ■

Part (a) gives a second proof, with no integrals, that the roots of  $p_k$  are real and simple — half of Theorem 10.10.5. The other half, that they lie inside the interval of the weight, is not visible in the matrix and still needs the argument of Chapter 10. For the monic Legendre polynomials of Example (The monic Legendre polynomials, continued),  $a_k = 0$ ,  $b_1 = \frac{1}{3}$  and  $b_2 = \frac{4}{15}$ , so

$$\mathbf{T}_3 = \begin{pmatrix} 0 & 1/\sqrt{3} & 0 \\ 1/\sqrt{3} & 0 & 2/\sqrt{15} \\ 0 & 2/\sqrt{15} & 0 \end{pmatrix},$$

and expanding along the first row,

$$p_{\mathbf{T}_3}(x) = x^3 - \left(\frac{1}{3} + \frac{4}{15}\right)x = x^3 - \frac{3}{5}x = p_3.$$

Its leading corner  $\mathbf{T}_2$  has characteristic polynomial  $x^2 - \frac{1}{3} = p_2$  and unit eigenvectors  $\frac{1}{\sqrt{2}}(1, \pm 1)$ , both with non-zero last coordinate, so the border  $\frac{2}{\sqrt{15}}\mathbf{e}_2$  meets both eigenspaces. The prediction is strict interlacing, and indeed

$$\sqrt{3/5} > 1/\sqrt{3} > 0 > -1/\sqrt{3} > -\sqrt{3/5},$$

that is  $0.7746 > 0.5774 > 0 > -0.5774 > -0.7746$ , where the strict comparisons  $3/5 > 1/3$  and  $1/\sqrt{3} > 0$  need no decimals.

## 16.4.4 A worked example

**Example. (One matrix and all three of its corners)**

Let

$$\mathbf{A} = \begin{pmatrix} 2 & 1 & 0 \\ 1 & 2 & 1 \\ 0 & 1 & 2 \end{pmatrix}.$$

Compute the eigenvalues of  $\mathbf{A}$  and of each of its three  $2 \times 2$  principal submatrices, exhibit the interlacing in each case, and explain the pattern of strict and non-strict inequalities using [Theorem 16.4.4](#).

*Solution.* The eigenvalues of  $\mathbf{A}$ . Expanding  $\det(x\mathbf{I} - \mathbf{A})$  along the first row,

$$\begin{aligned} p_{\mathbf{A}}(x) &= (x-2)[(x-2)^2 - 1] + 1 \cdot [-(x-2)] \\ &= (x-2)[(x-2)^2 - 2], \end{aligned}$$

so

$$\lambda(\mathbf{A}) = (2 + \sqrt{2}, 2, 2 - \sqrt{2}) \approx (3.41421, 2, 0.58579).$$

The three corners. With  $I$  the index set kept,

$$\mathbf{A}_{\{1,2\},\{1,2\}} = \begin{pmatrix} 2 & 1 \\ 1 & 2 \end{pmatrix}, \quad \mathbf{A}_{\{1,3\},\{1,3\}} = \begin{pmatrix} 2 & 0 \\ 0 & 2 \end{pmatrix}, \quad \mathbf{A}_{\{2,3\},\{2,3\}} = \begin{pmatrix} 2 & 1 \\ 1 & 2 \end{pmatrix}.$$

A matrix  $\begin{pmatrix} 2 & 1 \\ 1 & 2 \end{pmatrix}$  has trace 4 and determinant 3, so its eigenvalues are 3 and 1; and  $2\mathbf{I}_2$  has the eigenvalue 2 twice.

The interlacing. In all three cases [Theorem 16.4.2](#) is confirmed:

$$\begin{aligned} I = \{1, 2\} : \quad & 3.41421 > 3 > 2 > 1 > 0.58579, \\ I = \{1, 3\} : \quad & 3.41421 > 2 = 2 = 2 > 0.58579, \\ I = \{2, 3\} : \quad & 3.41421 > 3 > 2 > 1 > 0.58579. \end{aligned}$$

The first and third are strict everywhere; the second has two equalities in the middle. The exact comparisons need no decimals:  $2 + \sqrt{2} > 3$  is  $\sqrt{2} > 1$ , and  $1 > 2 - \sqrt{2}$  is  $\sqrt{2} > 1$  again.

*Why.* Read  $\mathbf{A}$  as a bordered matrix in each case, using the permutation remark above.

For  $I = \{1, 2\}$  the deleted index is already the last one, so  $\mathbf{B} = \begin{pmatrix} 2 & 1 \\ 1 & 2 \end{pmatrix}$ ,  $\mathbf{b} = (0, 1)$  and  $c = 2$ . The eigenvalues 3 and 1 of  $\mathbf{B}$  are distinct, with unit eigenvectors  $\frac{1}{\sqrt{2}}(1, 1)$  and  $\frac{1}{\sqrt{2}}(1, -1)$ . Both inner products with  $\mathbf{b}$  are non-zero:

$$\langle \mathbf{b}, \frac{1}{\sqrt{2}}(1, 1) \rangle = \frac{1}{\sqrt{2}}, \quad \langle \mathbf{b}, \frac{1}{\sqrt{2}}(1, -1) \rangle = -\frac{1}{\sqrt{2}}.$$

So both clauses of [Theorem 16.4.4](#) (c) hold and the interlacing is strict, as computed. The case  $I = \{2, 3\}$  is the same: moving index 1 to the end gives the corner  $\begin{pmatrix} 2 & 1 \\ 1 & 2 \end{pmatrix}$  again, with border  $(a_{21}, a_{31}) = (1, 0)$ , whose inner products with the two unit eigenvectors  $\frac{1}{\sqrt{2}}(1, \pm 1)$  are both  $\frac{1}{\sqrt{2}}$ , non-zero.

For  $I = \{1, 3\}$  the deleted index is 2. Conjugating by the permutation matrix that sends the coordinate order  $(1, 2, 3)$  to  $(1, 3, 2)$  turns  $\mathbf{A}$  into

$$\begin{pmatrix} 2 & 0 & 1 \\ 0 & 2 & 1 \\ 1 & 1 & 2 \end{pmatrix}, \quad \mathbf{B} = 2\mathbf{I}_2, \quad \mathbf{b} = (1, 1), \quad c = 2.$$

Here  $\mathbf{b} \neq \mathbf{0}$  and  $P_2\mathbf{b} = \mathbf{b} \neq \mathbf{0}$ , so the first clause of [Theorem 16.4.4](#) (c) is the one that fails:  $\mathbf{B}$  has the repeated eigenvalue 2. Part (a) with  $m_2 = 2$  then says that 2 is an eigenvalue of  $\mathbf{A}$  of multiplicity exactly 1, which is visible in  $p_{\mathbf{A}}$ , and the interlacing collapses at the middle to  $2 = 2 = 2$ . This is exactly the situation of the warning above, occurring inside a perfectly ordinary integer matrix.

### 16.4.5 Inertia under compression

Interlacing controls not only where the eigenvalues are but how many of them have each sign, and that is a statement about inertia.

#### Corollary 16.4.7 — A Principal Submatrix Keeps Most of the Inertia

Let  $\mathbf{A} \in M_n(F)$  be Hermitian, let  $0 \leq m \leq n - 1$ , let  $|I| = n - m$  and  $\mathbf{B} = \mathbf{A}_{I,I}$ . Write  $n_+(\mathbf{M})$  and  $n_-(\mathbf{M})$  for the number of positive and of negative eigenvalues of a Hermitian  $\mathbf{M}$ , counted with multiplicity. Then

$$n_+(\mathbf{B}) \leq n_+(\mathbf{A}) \leq n_+(\mathbf{B}) + m, \quad n_-(\mathbf{B}) \leq n_-(\mathbf{A}) \leq n_-(\mathbf{B}) + m.$$

In particular  $\mathbf{A} \succeq 0$  implies  $\mathbf{B} \succeq 0$ , and  $\mathbf{A} \succ 0$  implies  $\mathbf{B} \succ 0$ .

*Proof.* Write  $r = n_+(\mathbf{B})$  and  $s = n_+(\mathbf{A})$ , and abbreviate  $\lambda_i = \lambda_i(\mathbf{A})$ ,  $\mu_i = \lambda_i(\mathbf{B})$ . By [Corollary 16.4.3](#),  $\lambda_i \geq \mu_i \geq \lambda_{i+m}$  for  $1 \leq i \leq n - m$ .

( $n_+(\mathbf{B}) \leq n_+(\mathbf{A})$ ) For  $i \leq r$  we have  $\mu_i > 0$ , hence  $\lambda_i \geq \mu_i > 0$ . So  $\mathbf{A}$  has at least  $r$  positive eigenvalues.

( $n_+(\mathbf{A}) \leq n_+(\mathbf{B}) + m$ ) If  $r + m \geq n$  there is nothing to prove, since  $s \leq n$ . Otherwise  $r + 1 \leq n - m$ , so the index  $i = r + 1$  is admissible and  $\mu_{r+1} \leq 0$  by the definition of  $r$ ; hence  $\lambda_{r+1+m} \leq \mu_{r+1} \leq 0$ , and the eigenvalues  $\lambda_{r+m+1}, \dots, \lambda_n$  are all  $\leq 0$  by the decreasing indexing. So  $s \leq r + m$ .

For the two statements about  $n_-$ , apply what has just been proved to  $-\mathbf{A}$ , whose principal submatrix on  $I$  is  $-\mathbf{B}$ . By [Lemma 16.3.2](#) the eigenvalue lists are negated and reversed, so  $n_+(-\mathbf{A}) = n_-(\mathbf{A})$  and  $n_+(-\mathbf{B}) = n_-(\mathbf{B})$ , and the two displayed chains are the same statement.

Finally,  $\mathbf{A} \succeq 0$  means every eigenvalue of  $\mathbf{A}$  is  $\geq 0$  ([Theorem 12.1.4](#) (b)), that is,  $n_-(\mathbf{A}) = 0$ ; then  $n_-(\mathbf{B}) \leq n_-(\mathbf{A}) = 0$ , so  $\mathbf{B} \succeq 0$ . And  $\mathbf{A} \succ 0$  means  $n_+(\mathbf{A}) = n$  ([Theorem 12.1.5](#) (b)); then  $n_+(\mathbf{B}) \geq n_+(\mathbf{A}) - m = n - m$ , which is the size of  $\mathbf{B}$ , so every eigenvalue of  $\mathbf{B}$  is positive and  $\mathbf{B} \succ 0$ . This proves the corollary. ■

The last two sentences are a second proof of something Chapter 12 already knew. That a principal submatrix of a positive semidefinite matrix is positive semidefinite is the step (a)  $\Rightarrow$  (e) in the proof of [Theorem 12.1.4](#), where it comes from the observation that a principal submatrix is what the quadratic form does to vectors supported on the corresponding coordinates — one line, needing no interlacing. What is new here is the quantitative form. Sylvester’s law of inertia ([Theorem 13.5.1](#)) says that the triple of counts is an invariant of a real symmetric matrix up to congruence, and [Theorem 13.5.3](#) identifies Chapter 13’s counts with the eigenvalue counts  $n_+$  and  $n_-$  used here; the corollary says that deleting  $m$  rows and the matching  $m$  columns can destroy at most  $m$  positive and at most  $m$  negative eigenvalues, and create none of either. Section 10 gives a second proof of Sylvester’s law from Courant–Fischer, and bounds how far a low-rank perturbation can move the inertia.

### 16.4.6 Interlacing is a Hermitian phenomenon

Everything above rests on [Theorem 16.2.3](#), which rests on the quadratic form being real. Without the Hermitian hypothesis, nothing survives — not a weakened interlacing, not even a bound.

#### Example. (A corner outside the spectrum)

Let

$$\mathbf{A} = \begin{pmatrix} 5 & 1 \\ -8 & -1 \end{pmatrix}.$$

Show that  $\mathbf{A}$  has real eigenvalues, so that the conclusion of [Theorem 16.4.2](#) can be stated for it, and that the conclusion is false.

*Solution.*  $\operatorname{tr} \mathbf{A} = 4$  and  $\det \mathbf{A} = 5(-1) - 1(-8) = 3$ , so  $p_{\mathbf{A}}(x) = x^2 - 4x + 3 = (x-3)(x-1)$  and the eigenvalues are 3 and 1, both real. The two  $1 \times 1$  principal submatrices are  $\mathbf{A}_{\{1\},\{1\}} = (5)$  and  $\mathbf{A}_{\{2\},\{2\}} = (-1)$ , with eigenvalues 5 and  $-1$ .

Interlacing would require  $3 \geq 5 \geq 1$  in the first case and  $3 \geq -1 \geq 1$  in the second. Both fail, and they fail on opposite sides: the diagonal entries 5 and  $-1$  lie outside the interval  $[1, 3]$ , which is the convex hull of the spectrum. For a Hermitian matrix that is impossible, by [Example \(The diagonal lies inside the spectrum\)](#).

*Remark.* The failure is total, not marginal. For any  $a \in F$ , the matrix

$$\mathbf{A}_a = \begin{pmatrix} a & 1 \\ -(a-1)(a-3) & 4-a \end{pmatrix}$$

has trace 4 and determinant  $a(4-a) + (a-1)(a-3) = 4a - a^2 + a^2 - 4a + 3 = 3$ , hence the same characteristic polynomial  $(x-3)(x-1)$  as above. Its  $(1,1)$  principal submatrix is  $(a)$ . So a general matrix with spectrum  $\{1, 3\}$  can have any prescribed number whatsoever as its  $(1,1)$  entry: the eigenvalue of a single principal submatrix is subject to no constraint at all. (The two diagonal entries together still add up to the trace, 4; that constraint holds for every matrix and has nothing to do with interlacing.) The choice  $a = 5$  recovers  $\mathbf{A}$ , and  $a = 0$  gives  $\begin{pmatrix} 0 & 1 \\ -3 & 4 \end{pmatrix}$ .

*Warning.* **Principal means the same index set on both sides.** Interlacing says nothing about a submatrix obtained by deleting row  $i$  and column  $j$  with  $i \neq j$ , even for a Hermitian  $\mathbf{A}$ . Take  $\mathbf{A} = \begin{pmatrix} -10 & 1 \\ 1 & -10 \end{pmatrix}$ , whose eigenvalues are  $-9$  and  $-11$ . Deleting

row 1 and column 2 leaves the  $1 \times 1$  matrix (1), and 1 is nowhere near  $[-11, -9]$ . The submatrix in [Theorem 16.4.2](#) is  $\mathbf{A}_{I,I}$ , and the two  $I$ 's are not decoration.

## 16.4.7 Exercises

### 16.4.7.1 A. Check your understanding

#### Exercise A1.

- State [Theorem 16.4.2](#), and say which hypothesis on  $\mathbf{A}$  and which hypothesis on the submatrix are used.
- Let  $\mathbf{A} \in M_6(\mathbb{C})$  be Hermitian and let  $\mathbf{B} = \mathbf{A}_{I,I}$  with  $|I| = 4$ . Which eigenvalues of  $\mathbf{A}$  bound  $\lambda_3(\mathbf{B})$  above and below?
- Determine whether the following is correct, and justify your answer: if  $\mathbf{b} \neq \mathbf{0}$  then the interlacing between  $\begin{pmatrix} \mathbf{B} & \mathbf{b} \\ \mathbf{b}^* & c \end{pmatrix}$  and  $\mathbf{B}$  is strict.
- Explain why a principal submatrix of a positive semidefinite matrix is positive semidefinite, in one sentence, without using interlacing.
- Give a  $2 \times 2$  matrix with real eigenvalues whose  $(1, 1)$  entry is larger than both of them.

*Solution.*

- See [Theorem 16.4.2](#). The matrix  $\mathbf{A}$  must be **Hermitian**, so that [Theorem 16.2.3](#) applies and the eigenvalues are real and can be put in order; and  $\mathbf{B}$  must be a **principal** submatrix,  $\mathbf{B} = \mathbf{A}_{I,I}$  with the same index set on rows and columns, so that [Lemma 16.4.1](#) identifies  $R_{\mathbf{B}}$  with  $R_{\mathbf{A}}$  restricted to a coordinate hyperplane.
- Here  $n = 6$  and  $m = 2$ , so [Corollary 16.4.3](#) gives  $\lambda_3(\mathbf{A}) \geq \lambda_3(\mathbf{B}) \geq \lambda_5(\mathbf{A})$ .
- Incorrect. The warning after [Theorem 16.4.4](#) gives  $\mathbf{B} = 2\mathbf{I}_2$ ,  $\mathbf{b} = (1, 1)$ ,  $c = 2$ , where  $\mathbf{b} \neq \mathbf{0}$  and the interlacing is not strict, because  $\mathbf{B}$  has a repeated eigenvalue. The correct criterion is [Theorem 16.4.4](#) (c), which needs distinct eigenvalues of  $\mathbf{B}$  as well as  $P_\nu \mathbf{b} \neq \mathbf{0}$  for each  $\nu$ .
- Because  $\langle \mathbf{B}\mathbf{y}, \mathbf{y} \rangle = \langle \mathbf{A}\mathbf{x}, \mathbf{x} \rangle \geq 0$ , where  $\mathbf{x} \in F^n$  is  $\mathbf{y}$  placed in the coordinates of  $I$  and 0 elsewhere ([Lemma 16.4.1](#)); this is how [Theorem 12.1.4](#) obtained its clause (e).
- $\begin{pmatrix} 5 & 1 \\ -8 & -1 \end{pmatrix}$ , with eigenvalues 3 and 1 and  $(1, 1)$  entry 5 ([Example \(A corner outside the spectrum\)](#)). It is necessarily non-Hermitian, by [Example \(The diagonal lies inside the spectrum\)](#).

### 16.4.7.2 B. Practice

#### Exercise B1: Interlacing in integers.

Let

$$\mathbf{A} = \begin{pmatrix} 0 & 1 & 2 \\ 1 & 0 & 2 \\ 2 & 2 & 3 \end{pmatrix}.$$

Compute the eigenvalues of  $\mathbf{A}$  and of all three of its  $2 \times 2$  principal submatrices, and

exhibit the interlacing in each case. Hence say, for each corner, whether the interlacing is strict.

*Solution. The eigenvalues of  $\mathbf{A}$ .* The vector  $(1, -1, 0)$  satisfies  $\mathbf{A}(1, -1, 0) = (-1, 1, 0)$ , so  $-1$  is an eigenvalue. Dividing,  $p_{\mathbf{A}}(x) = x^3 - 3x^2 - 9x - 5 = (x - 5)(x + 1)^2$ , so

$$\lambda(\mathbf{A}) = (5, -1, -1).$$

(As a check,  $\text{tr } \mathbf{A} = 3 = 5 - 1 - 1$ , and expanding along the first row  $\det \mathbf{A} = 0 - 1(3 - 4) + 2(2 - 0) = 5 = 5 \cdot (-1) \cdot (-1)$ .)

*The corners.*

$$\mathbf{A}_{\{1,2\},\{1,2\}} = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}, \quad \mathbf{A}_{\{1,3\},\{1,3\}} = \mathbf{A}_{\{2,3\},\{2,3\}} = \begin{pmatrix} 0 & 2 \\ 2 & 3 \end{pmatrix}.$$

The first has trace 0 and determinant  $-1$ , so its eigenvalues are 1 and  $-1$ . The second has trace 3 and determinant  $-4$ , so its characteristic polynomial is  $x^2 - 3x - 4 = (x - 4)(x + 1)$  and its eigenvalues are 4 and  $-1$ .

*Interlacing.*

$$I = \{1, 2\} : \quad 5 > 1 > -1 = -1 = -1,$$

$$I = \{1, 3\} : \quad 5 > 4 > -1 = -1 = -1,$$

$$I = \{2, 3\} : \quad 5 > 4 > -1 = -1 = -1.$$

No corner gives strict interlacing, and the reason is on the side of  $\mathbf{A}$ : whichever index is deleted,  $\lambda_2(\mathbf{A}) = \lambda_3(\mathbf{A}) = -1$ , so the chain  $\lambda_2(\mathbf{A}) \geq \lambda_2(\mathbf{B}) \geq \lambda_3(\mathbf{A})$  squeezes  $\lambda_2(\mathbf{B})$  to  $-1$ . Strict interlacing would make the combined chain strictly decreasing, and so would force  $\mathbf{A}$  to have three distinct eigenvalues. [Theorem 16.4.4](#) sees the same thing from the side of the corner. For  $I = \{1, 2\}$  the border is  $\mathbf{b} = (2, 2)$ , orthogonal to  $(1, -1)$ , which spans  $E_{-1}(\mathbf{B})$ ; so  $P_{-1}\mathbf{b} = \mathbf{0}$ , and part (b) predicts that  $-1$  is an eigenvalue of  $\mathbf{A}$ . For  $I = \{1, 3\}$  the border is  $(a_{12}, a_{32}) = (1, 2)$ , orthogonal to  $(2, -1)$ , which spans  $E_{-1}$  of  $\begin{pmatrix} 0 & 2 \\ 2 & 3 \end{pmatrix}$ , with the same conclusion.

### Exercise B2: Bounds without a characteristic polynomial.

Let  $\mathbf{A} \in M_4(\mathbb{R})$  be symmetric with

$$\mathbf{A} = \begin{pmatrix} 7 & * & * & * \\ * & 2 & * & * \\ * & * & -3 & * \\ * & * & * & 1 \end{pmatrix},$$

the entries marked  $*$  being unknown.

- What do [Corollary 16.4.3](#) and [Example \(The diagonal lies inside the spectrum\)](#) say about  $\lambda_1(\mathbf{A})$  and  $\lambda_4(\mathbf{A})$ ?
- Suppose in addition that  $\mathbf{A}_{\{1,2\},\{1,2\}} = \begin{pmatrix} 7 & 4 \\ 4 & 2 \end{pmatrix}$ . Improve the lower bound on  $\lambda_1(\mathbf{A})$ , and give a lower bound on  $\lambda_2(\mathbf{A})$ .

*Solution.*

- Each diagonal entry is a  $1 \times 1$  principal submatrix, so  $\lambda_1(\mathbf{A}) \geq \max\{7, 2, -3, 1\} = 7$

and  $\lambda_4(\mathbf{A}) \leq \min\{7, 2, -3, 1\} = -3$ . In particular  $\mathbf{A}$  is indefinite whatever the unknown entries are.

- (b) The submatrix  $\mathbf{B} = \begin{pmatrix} 7 & 4 \\ 4 & 2 \end{pmatrix}$  has trace 9 and determinant  $14 - 16 = -2$ , so its characteristic polynomial is  $x^2 - 9x - 2$  and its eigenvalues are

$$\lambda_1(\mathbf{B}) = \frac{9+\sqrt{89}}{2} \approx 9.2170, \quad \lambda_2(\mathbf{B}) = \frac{9-\sqrt{89}}{2} \approx -0.2170.$$

Here  $n = 4$  and  $m = 2$ , and the left-hand inequality  $\lambda_i(\mathbf{A}) \geq \lambda_i(\mathbf{B})$  of [Corollary 16.4.3](#) at  $i = 1, 2$  gives

$$\lambda_1(\mathbf{A}) \geq \frac{9+\sqrt{89}}{2} \approx 9.2170, \quad \lambda_2(\mathbf{A}) \geq \frac{9-\sqrt{89}}{2} \approx -0.2170.$$

The first improves the bound 7 from (a). The second is new: the diagonal entries alone say nothing about  $\lambda_2(\mathbf{A})$ . The right-hand inequalities  $\lambda_i(\mathbf{B}) \geq \lambda_{i+2}(\mathbf{A})$  give  $\lambda_3(\mathbf{A}) \leq 9.2170$ , which is true but of little use, and  $\lambda_4(\mathbf{A}) \leq -0.2170$ , which is weaker than the bound  $\lambda_4(\mathbf{A}) \leq -3$  of (a). Each principal submatrix contributes what it can, and one keeps the best of the bounds.

### Exercise B3: Deciding strictness.

For each of the following bordered matrices  $\mathbf{A} = \begin{pmatrix} \mathbf{B} & \mathbf{b} \\ \mathbf{b}^\top & c \end{pmatrix}$ , decide without computing  $p_{\mathbf{A}}$  whether the interlacing between  $\mathbf{A}$  and  $\mathbf{B}$  is strict, and justify with [Theorem 16.4.4](#).

- (a)  $\mathbf{B} = \text{diag}(3, 1)$ ,  $\mathbf{b} = (0, 1)$ ,  $c = 0$ .
- (b)  $\mathbf{B} = \text{diag}(3, 1)$ ,  $\mathbf{b} = (2, -1)$ ,  $c = 5$ .
- (c)  $\mathbf{B} = \begin{pmatrix} 4 & 0 \\ 0 & 4 \end{pmatrix}$ ,  $\mathbf{b} = (1, 2)$ ,  $c = 0$ .

*Solution.* In each case  $\mathbf{B}$  is diagonal, so the standard basis vectors are an orthonormal eigenbasis and  $P_\nu \mathbf{b}$  is just the part of  $\mathbf{b}$  in the coordinates where the diagonal entry equals  $\nu$ .

- (a) Not strict. The eigenvalues 3, 1 of  $\mathbf{B}$  are distinct, but  $P_3 \mathbf{b} = (0, 0) = \mathbf{0}$ . By [Theorem 16.4.4](#) (b), 3 is an eigenvalue of  $\mathbf{A}$ , so one of the inequalities  $\lambda_1(\mathbf{A}) \geq 3 \geq \lambda_2(\mathbf{A})$  is an equality. (Indeed  $p_{\mathbf{A}}(x) = (x-3)(x^2-x-1)$ .)
- (b) Strict. The eigenvalues 3, 1 are distinct and  $P_3 \mathbf{b} = (2, 0) \neq \mathbf{0}$ ,  $P_1 \mathbf{b} = (0, -1) \neq \mathbf{0}$ . Both clauses of [Theorem 16.4.4](#) (c) hold. (Indeed  $p_{\mathbf{A}}(x) = (x-2)(x^2-7x+4)$ , with roots  $\frac{7+\sqrt{33}}{2} \approx 6.372$ , 2 and  $\frac{7-\sqrt{33}}{2} \approx 0.628$ , and  $6.372 > 3 > 2 > 1 > 0.628$ .)
- (c) Not strict.  $P_4 \mathbf{b} = \mathbf{b} = (1, 2) \neq \mathbf{0}$ , so the second clause holds, but  $\mathbf{B}$  has the repeated eigenvalue 4 and the first clause fails. By [Theorem 16.4.4](#) (a) with  $m_4 = 2$ , the number 4 is an eigenvalue of  $\mathbf{A}$  of multiplicity exactly 1, and the middle of the chain reads  $4 \geq \lambda_2(\mathbf{A}) \geq 4$ . (Indeed  $p_{\mathbf{A}}(x) = (x-5)(x-4)(x+1)$ .)

### 16.4.7.3 C. Going deeper

#### Exercise C1: Rank under compression.

Let  $\mathbf{A} \in M_n(F)$  be Hermitian of rank  $r$ , and let  $\mathbf{B} = \mathbf{A}_{I,I}$  with  $|I| = n - m$ .

- (a) Prove that  $\text{rank } \mathbf{B} \geq r - 2m$ .



- (b) Show that the bound is attained, by exhibiting for each  $n$  and each  $m \leq n/2$  a Hermitian  $\mathbf{A}$  of rank  $2m$  with a principal submatrix of size  $n - m$  and rank 0.

*Solution.*

- (a) For a Hermitian matrix  $\mathbf{M}$ , diagonalizing by the spectral theorem gives  $\text{rank } \mathbf{M} = n_+(\mathbf{M}) + n_-(\mathbf{M})$ , the total number of non-zero eigenvalues. By [Corollary 16.4.7](#),  $n_+(\mathbf{B}) \geq n_+(\mathbf{A}) - m$  and  $n_-(\mathbf{B}) \geq n_-(\mathbf{A}) - m$ . Adding,

$$\text{rank } \mathbf{B} = n_+(\mathbf{B}) + n_-(\mathbf{B}) \geq (n_+(\mathbf{A}) + n_-(\mathbf{A})) - 2m = r - 2m.$$

- (b) Let  $m \leq n/2$  and let  $I = \{m+1, m+2, \dots, n\}$ , of size  $n - m$ . Define  $\mathbf{A}$  by  $a_{i, m+i} = a_{m+i, i} = 1$  for  $1 \leq i \leq m$ , and all other entries 0. Then  $\mathbf{A}$  is real symmetric. It is a direct sum of  $m$  copies of  $\begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}$ , acting on the coordinate pairs  $\{i, m+i\}$ , together with zeros on the remaining coordinates, so its eigenvalues are  $\pm 1$  each  $m$  times and 0 otherwise; hence  $\text{rank } \mathbf{A} = 2m$  and  $n_+(\mathbf{A}) = n_-(\mathbf{A}) = m$ . Every non-zero entry of  $\mathbf{A}$  has one index in  $\{1, \dots, m\}$ , which  $I$  omits, so  $\mathbf{A}_{I,I} = \mathbf{0}$  and  $\text{rank } \mathbf{A}_{I,I} = 0 = \text{rank } \mathbf{A} - 2m$ . The bound in (a) is therefore sharp for every  $m$ .

### Exercise C2: The converse, in the smallest case.

[Theorem 16.4.2](#) says that the spectrum of a corner interlaces the spectrum of the whole. This exercise asks whether every interlacing pattern occurs.

- (a) Let  $\lambda_1 > \mu_1 > \lambda_2$  be real numbers. Find  $b \in \mathbb{R}$  and  $c \in \mathbb{R}$  such that  $\mathbf{A} = \begin{pmatrix} \mu_1 & b \\ b & c \end{pmatrix}$  has eigenvalues  $\lambda_1$  and  $\lambda_2$ .
- (b) Deduce that for  $n = 2$  the interlacing condition of [Theorem 16.4.2](#) is not merely necessary but also sufficient.
- (c) Explain why the strict hypothesis  $\lambda_1 > \mu_1 > \lambda_2$  may be relaxed to  $\lambda_1 \geq \mu_1 \geq \lambda_2$ .

*Hint: match the trace and the determinant.*

*Solution.*

- (a) A real symmetric  $2 \times 2$  matrix with eigenvalues  $\lambda_1, \lambda_2$  must have  $\text{tr } \mathbf{A} = \lambda_1 + \lambda_2$  and  $\det \mathbf{A} = \lambda_1 \lambda_2$ . The first forces

$$c = \lambda_1 + \lambda_2 - \mu_1,$$

and the second then forces  $\mu_1 c - b^2 = \lambda_1 \lambda_2$ , that is,

$$b^2 = \mu_1(\lambda_1 + \lambda_2 - \mu_1) - \lambda_1 \lambda_2 = -(\mu_1 - \lambda_1)(\mu_1 - \lambda_2) = (\lambda_1 - \mu_1)(\mu_1 - \lambda_2),$$

where the middle equality is the expansion  $-(\mu_1^2 - (\lambda_1 + \lambda_2)\mu_1 + \lambda_1 \lambda_2)$ . The hypothesis  $\lambda_1 > \mu_1 > \lambda_2$  makes both factors positive, so  $b = \sqrt{(\lambda_1 - \mu_1)(\mu_1 - \lambda_2)}$  is a real number. With this  $b$  and  $c$ , the matrix  $\mathbf{A}$  is real symmetric with the prescribed trace and determinant, hence with characteristic polynomial  $x^2 - (\lambda_1 + \lambda_2)x + \lambda_1 \lambda_2 = (x - \lambda_1)(x - \lambda_2)$ , hence with eigenvalues  $\lambda_1$  and  $\lambda_2$ .

For example  $\lambda_1 = 4, \mu_1 = 2, \lambda_2 = 1$  gives  $c = 3$  and  $b^2 = 2$ , that is  $\mathbf{A} = \begin{pmatrix} 2 & \sqrt{2} \\ \sqrt{2} & 3 \end{pmatrix}$ , whose trace is 5 and determinant  $6 - 2 = 4$ , as required.

- (b) Necessity is [Theorem 16.4.2](#) with  $n = 2$  and  $I = \{1\}$ . Sufficiency is part (a): given any  $\lambda_1 > \mu_1 > \lambda_2$ , a Hermitian  $\mathbf{A}$  with spectrum  $\{\lambda_1, \lambda_2\}$  and  $\mathbf{A}_{\{1\},\{1\}} = (\mu_1)$  exists.
- (c) If  $\mu_1 = \lambda_1$  or  $\mu_1 = \lambda_2$  then the formula gives  $b = 0$ , and  $\mathbf{A} = \text{diag}(\mu_1, \lambda_1 + \lambda_2 - \mu_1)$  is diagonal with the two entries  $\lambda_1$  and  $\lambda_2$  in one order or the other. So the construction still works, with a zero border; consistently with [Theorem 16.4.4](#), a zero border is exactly what makes the interlacing non-strict.

### Exercise C3: How far apart can the extremes be.

Let  $\mathbf{A} \in M_n(F)$  be Hermitian.

- (a) Prove that  $\lambda_1(\mathbf{A}) - \lambda_n(\mathbf{A}) \geq \max_{i,j} |a_{ii} - a_{jj}|$ .
- (b) Prove that  $\lambda_1(\mathbf{A}) - \lambda_n(\mathbf{A}) \geq 2 \max_{i \neq j} |a_{ij}|$ .

*Hint for (b): look at a  $2 \times 2$  principal submatrix.*

*Solution.*

- (a) By [Example \(The diagonal lies inside the spectrum\)](#),  $\lambda_n(\mathbf{A}) \leq a_{kk} \leq \lambda_1(\mathbf{A})$  for every  $k$ . So for any  $i, j$ , both  $a_{ii} - a_{jj}$  and  $a_{jj} - a_{ii}$  are at most  $\lambda_1(\mathbf{A}) - \lambda_n(\mathbf{A})$ , which gives the bound after taking the maximum over  $i, j$ .
- (b) Fix  $i \neq j$  and let  $I = \{i, j\}$ , so that

$$\mathbf{B} = \mathbf{A}_{I,I} = \begin{pmatrix} a_{ii} & a_{ij} \\ \overline{a_{ij}} & a_{jj} \end{pmatrix}$$

is Hermitian of size 2. Its trace is  $a_{ii} + a_{jj}$  and its determinant is  $a_{ii}a_{jj} - |a_{ij}|^2$ , so its eigenvalues are

$$\frac{a_{ii} + a_{jj}}{2} \pm \sqrt{\left(\frac{a_{ii} - a_{jj}}{2}\right)^2 + |a_{ij}|^2},$$

and their difference is  $2\sqrt{((a_{ii} - a_{jj})/2)^2 + |a_{ij}|^2} \geq 2|a_{ij}|$ . Now apply [Corollary 16.4.3](#) with  $m = n - 2$ : it gives  $\lambda_1(\mathbf{A}) \geq \lambda_1(\mathbf{B})$  and  $\lambda_2(\mathbf{B}) \geq \lambda_{2+(n-2)}(\mathbf{A}) = \lambda_n(\mathbf{A})$ , so

$$\lambda_1(\mathbf{A}) - \lambda_n(\mathbf{A}) \geq \lambda_1(\mathbf{B}) - \lambda_2(\mathbf{B}) \geq 2|a_{ij}|.$$

Taking the maximum over the pairs  $i \neq j$  finishes the proof. (For  $n = 2$  the step with  $m = 0$  is the identity  $\mathbf{B} = \mathbf{A}$ , and the conclusion is the computation itself.)

## 16.5 Adding a Rank-One Matrix

Section 4 deleted a row and the matching column and found that the spectrum of the smaller matrix is pinned between the eigenvalues of the larger one. This section makes the opposite kind of change: it leaves the size alone and adds the simplest non-zero Hermitian matrix there is, a positive multiple of  $\mathbf{v}\mathbf{v}^*$ . The answer has the same shape — the new eigenvalues interlace the old ones — but here we can say more than that they interlace. Under one extra hypothesis the new eigenvalues are the roots of a single explicit scalar equation in one variable, and the

graph of that function shows at a glance where every one of them sits.

**Throughout this section**  $F = \mathbb{R}$  or  $F = \mathbb{C}$ , **the matrix**  $\mathbf{A} \in M_n(F)$  **is Hermitian, and its eigenvalues are indexed decreasingly**,  $\lambda_1(\mathbf{A}) \geq \cdots \geq \lambda_n(\mathbf{A})$ , as everywhere in this chapter. We fix  $\mathbf{v} \in F^n$  and a real number  $\tau$ , and study

$$\tilde{\mathbf{A}} := \mathbf{A} + \tau \mathbf{v} \mathbf{v}^*.$$

The added term is Hermitian, since  $(\mathbf{v} \mathbf{v}^*)^* = \mathbf{v} \mathbf{v}^*$  and  $\tau$  is real, so  $\tilde{\mathbf{A}}$  is Hermitian too and its eigenvalues are indexed the same way. The added term has rank 1 when  $\tau \neq 0$  and  $\mathbf{v} \neq \mathbf{0}$ , and is the zero matrix otherwise. For  $\tau > 0$  it is positive semidefinite, because

$$\langle \tau \mathbf{v} \mathbf{v}^* \mathbf{x}, \mathbf{x} \rangle = \tau \mathbf{x}^* \mathbf{v} \mathbf{v}^* \mathbf{x} = \tau |\mathbf{v}^* \mathbf{x}|^2 \geq 0$$

for every  $\mathbf{x}$ , which is clause (P2) of [Definition 12.1.1](#).

## 16.5.1 One bump, every eigenvalue up

Adding a positive semidefinite matrix can only raise the quadratic form, so no eigenvalue can go down. The content of the theorem is the ceiling: however large  $\tau$  is, the  $i$ -th new eigenvalue cannot climb past the  $(i - 1)$ -st old one. All the room created by the bump is spent at the top.

### Theorem 16.5.1 — Interlacing for a Rank-One Update

Let  $\mathbf{A} \in M_n(F)$  be Hermitian, let  $\mathbf{v} \in F^n$ , let  $\tau \geq 0$ , and put  $\tilde{\mathbf{A}} = \mathbf{A} + \tau \mathbf{v} \mathbf{v}^*$ . Then

$$\lambda_i(\tilde{\mathbf{A}}) \geq \lambda_i(\mathbf{A}) \geq \lambda_{i+1}(\tilde{\mathbf{A}}) \quad (1 \leq i \leq n - 1),$$

and  $\lambda_n(\tilde{\mathbf{A}}) \geq \lambda_n(\mathbf{A})$ . Written as one chain,

$$\begin{aligned} \lambda_1(\tilde{\mathbf{A}}) &\geq \lambda_1(\mathbf{A}) \geq \lambda_2(\tilde{\mathbf{A}}) \geq \lambda_2(\mathbf{A}) \\ &\geq \cdots \geq \lambda_n(\tilde{\mathbf{A}}) \geq \lambda_n(\mathbf{A}). \end{aligned}$$

*Idea.* The two halves have different characters and different sources. The left-hand inequalities say the eigenvalues go **up**, and that is a statement about the quadratic form: on every vector,  $\tilde{\mathbf{A}}$  scores at least what  $\mathbf{A}$  scores, so every max–min it competes in can only improve. The right-hand inequalities say the eigenvalues go up **by at most one index**, and that has nothing to do with the sign of  $\tau$ : it is the rank of the perturbation doing the work, and [Corollary 16.3.8](#) has already isolated exactly that.

*Proof.* If  $\tau = 0$  or  $\mathbf{v} = \mathbf{0}$  then  $\tilde{\mathbf{A}} = \mathbf{A}$  and every inequality is an equality, so assume  $\tau > 0$  and  $\mathbf{v} \neq \mathbf{0}$ .

For the inequalities  $\lambda_i(\tilde{\mathbf{A}}) \geq \lambda_i(\mathbf{A})$ , fix  $i$  and let  $\mathbf{x} \neq \mathbf{0}$ . By the computation above,

$$R_{\tilde{\mathbf{A}}}(\mathbf{x}) = R_{\mathbf{A}}(\mathbf{x}) + \frac{\tau |\mathbf{v}^* \mathbf{x}|^2}{\|\mathbf{x}\|^2} \geq R_{\mathbf{A}}(\mathbf{x}),$$

using [Definition 16.1.1](#) and  $\tau \geq 0$ . Multiplying through by  $\|\mathbf{x}\|^2$ , this says  $\langle \tilde{\mathbf{A}} \mathbf{x}, \mathbf{x} \rangle \geq \langle \mathbf{A} \mathbf{x}, \mathbf{x} \rangle$  for every  $\mathbf{x}$ , that is,  $\tilde{\mathbf{A}} \succeq \mathbf{A}$  in the Loewner order of [Definition 12.5.1](#). So [Corollary 16.2.4](#) gives  $\lambda_i(\tilde{\mathbf{A}}) \geq \lambda_i(\mathbf{A})$  for every  $i$ .

For the inequalities  $\lambda_i(\mathbf{A}) \geq \lambda_{i+1}(\tilde{\mathbf{A}})$ , note that  $\text{rank}(\tau \mathbf{v} \mathbf{v}^*) = 1$ , because every column of  $\mathbf{v} \mathbf{v}^*$  is a multiple of  $\mathbf{v} \neq \mathbf{0}$  and at least one of them is non-zero. So [Corollary 16.3.8](#) applies with  $\mathbf{B} = \tau \mathbf{v} \mathbf{v}^*$  and  $r = 1$ , and its first inequality reads

$$\lambda_{i+1}(\mathbf{A} + \tau \mathbf{v} \mathbf{v}^*) \leq \lambda_i(\mathbf{A})$$

for  $1 \leq i \leq n - 1$ . This proves the theorem. ■

Two remarks on the proof. The first half used nothing about the rank: it is [Corollary 16.2.4](#), the general fact that  $\tilde{\mathbf{A}} \succeq \mathbf{A}$  forces  $\lambda_i(\tilde{\mathbf{A}}) \geq \lambda_i(\mathbf{A})$  for every  $i$ , specialized to a rank-one increment. The second half used nothing about the sign: the eigenvalues of  $\mathbf{A} + \mathbf{B}$  and of  $\mathbf{A}$  are never more than  $\text{rank } \mathbf{B}$  indices apart. Interlacing is what happens when both restrictions apply at once.

The secular equation of the next subsection gives a sharper, **strict** version of the theorem — it locates each new eigenvalue exactly, strictly between consecutive old ones — but only under extra hypotheses (a diagonal  $\mathbf{A}$  with distinct entries, and a  $\mathbf{v}$  with no zero coordinate), so it is not a second proof of the general statement. The other short route to the general theorem is Cauchy interlacing ([Theorem 16.4.2](#)) applied to a suitable bordered matrix.

### Example. (A bump on a diagonal matrix)

Let  $\mathbf{A} = \text{diag}(4, 1)$ ,  $\mathbf{v} = (1, 2)$  and  $\tau = 1$ . Compute  $\tilde{\mathbf{A}}$  and its eigenvalues, and check [Theorem 16.5.1](#).

*Solution.* Here  $\mathbf{v} \mathbf{v}^T = \begin{pmatrix} 1 & 2 \\ 2 & 4 \end{pmatrix}$ , so

$$\tilde{\mathbf{A}} = \begin{pmatrix} 4 & 0 \\ 0 & 1 \end{pmatrix} + \begin{pmatrix} 1 & 2 \\ 2 & 4 \end{pmatrix} = \begin{pmatrix} 5 & 2 \\ 2 & 5 \end{pmatrix}.$$

Its trace is 10 and its determinant is  $25 - 4 = 21$ , so its characteristic polynomial is  $x^2 - 10x + 21 = (x - 7)(x - 3)$  and  $\lambda_1(\tilde{\mathbf{A}}) = 7$ ,  $\lambda_2(\tilde{\mathbf{A}}) = 3$ . The chain of the theorem reads

$$7 \geq 4 \geq 3 \geq 1,$$

and all four inequalities are strict. Note the accounting: the two eigenvalues rose by 3 and by 2, a total of  $5 = \|\mathbf{v}\|^2$ . That is no coincidence, and the last result of this section says why.

*Quick check.* The matrix  $\mathbf{A} = \text{diag}(9, 5, 2)$  is bumped by  $\tau \mathbf{v} \mathbf{v}^*$  with  $\tau > 0$ . Without knowing  $\mathbf{v}$  or  $\tau$ , what is the largest possible value of  $\lambda_3(\mathbf{A} + \tau \mathbf{v} \mathbf{v}^*)$ , and what is the smallest?

*Solution.* By [Theorem 16.5.1](#),  $\lambda_2(\mathbf{A}) \geq \lambda_3(\tilde{\mathbf{A}}) \geq \lambda_3(\mathbf{A})$ , that is,  $5 \geq \lambda_3(\tilde{\mathbf{A}}) \geq 2$ . Both ends are reached:  $\mathbf{v} = \mathbf{e}_3$  with  $\tau$  large gives  $\tilde{\mathbf{A}} = \text{diag}(9, 5, 2 + \tau)$ , whose third eigenvalue is 5 as soon as  $\tau \geq 3$ ; and  $\mathbf{v} = \mathbf{e}_1$  gives  $\tilde{\mathbf{A}} = \text{diag}(9 + \tau, 5, 2)$ , whose third eigenvalue is 2. So the smallest eigenvalue can be pushed up as far as the *second* smallest of  $\mathbf{A}$  and no further, no matter how big  $\tau$  is.

## 16.5.2 Where the new eigenvalues actually are

Interlacing brackets each new eigenvalue in an interval. To pin it down we need an equation, and there is a cheap one: an eigenvalue of  $\tilde{\mathbf{A}}$  is a number  $\lambda$  making  $\tilde{\mathbf{A}} - \lambda\mathbf{I}$  singular, and  $\tilde{\mathbf{A}} - \lambda\mathbf{I} = (\mathbf{A} - \lambda\mathbf{I}) + \tau\mathbf{v}\mathbf{v}^*$  is a rank-one update of a matrix whose determinant we know. Chapter 7 computed the determinant of exactly such an update.

*Divide the characteristic polynomial of the bumped matrix by that of the original one; what is left is a sum of  $n$  simple fractions.*

### Definition 16.5.1 — Secular function

Let  $\mathbf{A} = \text{diag}(d_1, \dots, d_n)$  with  $d_1, \dots, d_n \in \mathbb{R}$ , let  $\mathbf{v} \in F^n$  and let  $\tau \in \mathbb{R}$ . The **secular function** of the update  $\mathbf{A} + \tau\mathbf{v}\mathbf{v}^*$  is

$$f(\lambda) := 1 + \tau \sum_{i=1}^n \frac{|v_i|^2}{d_i - \lambda},$$

defined for every real  $\lambda$  **other than**  $d_1, \dots, d_n$ . The equation  $f(\lambda) = 0$  is the **secular equation**.

In words:  $f$  is the constant 1 plus one simple pole at each diagonal entry of  $\mathbf{A}$ , weighted by  $\tau$  times the squared modulus of the matching coordinate of  $\mathbf{v}$ . The name is inherited from celestial mechanics, where equations of this shape governed slow (“secular”) drifts in planetary orbits.

### Theorem 16.5.2 — Eigenvalues from the Secular Equation

Let  $\mathbf{A} = \text{diag}(d_1, \dots, d_n)$  with

$$d_1 > d_2 > \dots > d_n$$

**strictly decreasing** and real, let  $\tau > 0$ , and let  $\mathbf{v} \in F^n$  have **every** coordinate non-zero. Write  $\tilde{\mathbf{A}} = \mathbf{A} + \tau\mathbf{v}\mathbf{v}^*$  and let  $f$  be its secular function. Then:

- (a) no  $d_i$  is an eigenvalue of  $\tilde{\mathbf{A}}$ ;
- (b) a real number  $\lambda \notin \{d_1, \dots, d_n\}$  is an eigenvalue of  $\tilde{\mathbf{A}}$  if and only if  $f(\lambda) = 0$ ;
- (c)  $f$  has exactly  $n$  roots  $\mu_1 > \dots > \mu_n$ , one in each of the intervals

$$(d_1, \infty), (d_2, d_1), \dots, (d_n, d_{n-1}),$$

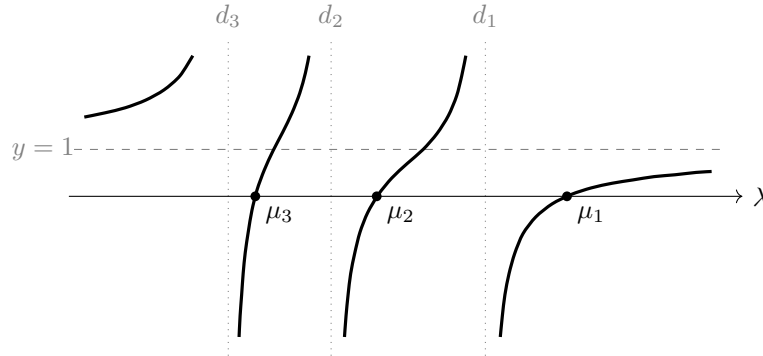
and none below  $d_n$ ; these are the eigenvalues of  $\tilde{\mathbf{A}}$ , each simple, and

$$\mu_1 > d_1 > \mu_2 > d_2 > \dots > \mu_n > d_n;$$

- (d) for each  $j$ , the vector  $(\mathbf{A} - \mu_j\mathbf{I})^{-1}\mathbf{v}$  is an eigenvector of  $\tilde{\mathbf{A}}$  for  $\mu_j$ .

*Idea.* Part (b) is one line of Chapter 7 once (a) has cleared the denominators out of the way. Part (c) is a picture. Each term  $|v_i|^2/(d_i - \lambda)$  increases with  $\lambda$  on any interval that avoids  $d_i$ , so on each of the  $n + 1$  open intervals into which the poles cut the line  $f$  climbs steadily from left to right. Just to the left of each pole  $f$  shoots up to  $+\infty$ ;

just to the right it starts again from  $-\infty$ ; and far out in either direction it settles at 1. Draw that, and the roots count themselves. The proof turns the picture into a count that needs no limits at all: the  $n$  eigenvalues turn out to be distinct, there are exactly  $n$  intervals where a root can live, and each holds at most one.



The dotted verticals are the poles, at the old eigenvalues  $d_3 < d_2 < d_1$ ; the dashed horizontal is  $y = 1$ , the level  $f$  approaches at both ends. On the branch left of  $d_3$  every term of the sum is positive, so the curve stays above 1 and never meets the axis. Each of the other three branches runs from  $-\infty$  to  $+\infty$  or from  $-\infty$  to 1, increasing throughout, so each crosses the axis exactly once. Three branches, three roots, one strictly inside each gap and one above the top pole.

*Proof.* (a) Suppose  $\tilde{\mathbf{A}}\mathbf{x} = d_i\mathbf{x}$  for some  $\mathbf{x} \neq \mathbf{0}$ . Rearranging,  $(\mathbf{A} - d_i\mathbf{I})\mathbf{x} = -\tau(\mathbf{v}^*\mathbf{x})\mathbf{v}$ . The matrix  $\mathbf{A} - d_i\mathbf{I}$  is diagonal with  $i$ -th diagonal entry 0, so the  $i$ -th coordinate of the left-hand side is 0; the  $i$ -th coordinate of the right-hand side is  $-\tau(\mathbf{v}^*\mathbf{x})v_i$ . Since  $\tau \neq 0$  and  $v_i \neq 0$ , this forces  $\mathbf{v}^*\mathbf{x} = 0$ , and then  $(\mathbf{A} - d_i\mathbf{I})\mathbf{x} = \mathbf{0}$ . The diagonal entries  $d_j$  are distinct, so the only diagonal entry of  $\mathbf{A} - d_i\mathbf{I}$  that vanishes is the  $i$ -th, and therefore  $\mathbf{x} = c\mathbf{e}_i$  for some scalar  $c$ . But then  $0 = \mathbf{v}^*\mathbf{x} = c\bar{v}_i$ , and  $v_i \neq 0$  gives  $c = 0$ , so  $\mathbf{x} = \mathbf{0}$  — a contradiction.

(b) Let  $\lambda \notin \{d_1, \dots, d_n\}$ . Then  $\mathbf{A} - \lambda\mathbf{I}$  is diagonal with non-zero diagonal entries, hence invertible, with

$$(\mathbf{A} - \lambda\mathbf{I})^{-1} = \text{diag}\left(\frac{1}{d_1 - \lambda}, \dots, \frac{1}{d_n - \lambda}\right).$$

Apply the determinant statement of [Corollary 7.3.6](#) — itself a  $1 \times 1$  case of [Theorem 7.3.2](#) — to the invertible matrix  $\mathbf{A} - \lambda\mathbf{I}$ , with the column  $\tau\mathbf{v}$  and the row  $\mathbf{v}^*$  — that is, with  $\tau\mathbf{v}$  as its  $\mathbf{u}$  and  $\bar{\mathbf{v}}$  in place of its  $\mathbf{v}$ , so that its  $\mathbf{v}^\top$  is our  $\mathbf{v}^*$ :

$$\begin{aligned} \det(\tilde{\mathbf{A}} - \lambda\mathbf{I}) &= \det((\mathbf{A} - \lambda\mathbf{I}) + (\tau\mathbf{v})\mathbf{v}^*) \\ &= \det(\mathbf{A} - \lambda\mathbf{I}) (1 + \tau\mathbf{v}^*(\mathbf{A} - \lambda\mathbf{I})^{-1}\mathbf{v}). \end{aligned}$$

The scalar in the second factor is  $\sum_i |v_i|^2 / (d_i - \lambda)$ , since  $\mathbf{v}^*\mathbf{D}\mathbf{v} = \sum_i \bar{v}_i d_{ii} v_i$  for diagonal  $\mathbf{D}$ . So

$$\det(\tilde{\mathbf{A}} - \lambda\mathbf{I}) = \left( \prod_{i=1}^n (d_i - \lambda) \right) f(\lambda), \quad (*)$$

and the product is non-zero. Hence  $\det(\tilde{\mathbf{A}} - \lambda\mathbf{I}) = 0$  if and only if  $f(\lambda) = 0$ . Finally  $\det(\lambda\mathbf{I} - \tilde{\mathbf{A}}) = (-1)^n \det(\tilde{\mathbf{A}} - \lambda\mathbf{I})$ , since negating all  $n$  rows of a matrix multiplies its determinant by  $(-1)^n$ , by  $n$  applications of [Theorem 6.5.1 \(b\)](#); so the two determinants

vanish together, and by [Theorem 6.10.4](#) (a) this happens exactly when  $\lambda$  is an eigenvalue of  $\tilde{\mathbf{A}}$ .

(c) Let  $J$  be one of the  $n+1$  open intervals into which  $d_1, \dots, d_n$  cut the real line, and let  $\lambda < \lambda'$  lie in  $J$ . For each  $i$ , the numbers  $d_i - \lambda$  and  $d_i - \lambda'$  are non-zero and have the same sign, because  $d_i \notin J$ ; hence

$$\frac{1}{d_i - \lambda'} - \frac{1}{d_i - \lambda} = \frac{\lambda' - \lambda}{(d_i - \lambda)(d_i - \lambda')} > 0.$$

Multiplying by  $\tau|v_i|^2 > 0$  and summing over  $i$  gives  $f(\lambda') > f(\lambda)$ . So  $f$  is **strictly increasing** on each such interval, and in particular has at most one root in each.

Next, every eigenspace of  $\tilde{\mathbf{A}}$  is one-dimensional. Let  $\tilde{\mathbf{A}}\mathbf{x} = \mu\mathbf{x}$ . By (a),  $\mu \notin \{d_1, \dots, d_n\}$ , so  $\mathbf{A} - \mu\mathbf{I}$  is invertible, and rearranging as in (a) gives  $(\mathbf{A} - \mu\mathbf{I})\mathbf{x} = -\tau(\mathbf{v}^*\mathbf{x})\mathbf{v}$ , that is,

$$\mathbf{x} = -\tau(\mathbf{v}^*\mathbf{x})(\mathbf{A} - \mu\mathbf{I})^{-1}\mathbf{v}.$$

So  $\ker(\tilde{\mathbf{A}} - \mu\mathbf{I}) \subseteq \text{span}((\mathbf{A} - \mu\mathbf{I})^{-1}\mathbf{v})$ , a space of dimension 1. Now write  $\tilde{\mathbf{A}} = \mathbf{U}\mathbf{D}\mathbf{U}^*$  with  $\mathbf{U}$  unitary and  $\mathbf{D}$  diagonal carrying the eigenvalue list of  $\tilde{\mathbf{A}}$  with multiplicity ([Corollary 11.4.3](#), or [Corollary 11.5.2](#) over  $\mathbb{R}$ ). Since  $\mathbf{U}$  is invertible,  $\ker(\tilde{\mathbf{A}} - \mu\mathbf{I}) = \mathbf{U}\ker(\mathbf{D} - \mu\mathbf{I})$ , and  $\dim \ker(\mathbf{D} - \mu\mathbf{I})$  is the number of diagonal entries of  $\mathbf{D}$  equal to  $\mu$ . That number is therefore at most 1: the  $n$  eigenvalues of  $\tilde{\mathbf{A}}$  are **distinct**, and each is simple.

Now count. Call the eigenvalues  $\mu_1 > \dots > \mu_n$ . By (b) each is a root of  $f$ , and by (a) none equals a  $d_i$ , so each lies in one of the  $n+1$  open intervals. None lies in  $(-\infty, d_n)$ : there every  $d_i - \lambda$  is positive, so every term of the sum is positive and  $f(\lambda) > 1$ . That leaves  $n$  intervals, each containing at most one root of  $f$ , to accommodate  $n$  distinct roots — so each of the intervals  $(d_1, \infty), (d_2, d_1), \dots, (d_n, d_{n-1})$  contains exactly one of them, and since they are listed in decreasing order,  $\mu_1 \in (d_1, \infty)$  and  $\mu_{i+1} \in (d_{i+1}, d_i)$ . Conversely every root of  $f$  is an eigenvalue by (b), so  $f$  has no roots besides these  $n$ . Reading the intervals in decreasing order gives the displayed chain.

(d) Fix  $j$  and put  $\mathbf{x} = (\mathbf{A} - \mu_j\mathbf{I})^{-1}\mathbf{v}$ , which is defined because  $\mu_j \notin \{d_1, \dots, d_n\}$  and non-zero because  $\mathbf{v} \neq \mathbf{0}$ . As in (b),

$$\tau\mathbf{v}^*(\mathbf{A} - \mu_j\mathbf{I})^{-1}\mathbf{v} = \tau \sum_i \frac{|v_i|^2}{d_i - \mu_j} = f(\mu_j) - 1 = -1.$$

Therefore

$$\begin{aligned} (\tilde{\mathbf{A}} - \mu_j\mathbf{I})\mathbf{x} &= (\mathbf{A} - \mu_j\mathbf{I})\mathbf{x} + \tau\mathbf{v}(\mathbf{v}^*\mathbf{x}) \\ &= \mathbf{v} + \mathbf{v}(\tau\mathbf{v}^*(\mathbf{A} - \mu_j\mathbf{I})^{-1}\mathbf{v}) = \mathbf{v} - \mathbf{v} = \mathbf{0}. \end{aligned}$$

This proves the theorem. ■

Identity (\*) is worth keeping on its own: it says that the characteristic polynomial of the bumped matrix is the characteristic polynomial of the original one multiplied by the secular function. The poles of  $f$  cancel the roots of the product exactly, and what survives is a polynomial of the same degree with its roots shifted one gap to the right.

**Example. (A three-by-three update, found from its secular equation)**

Let

$$\mathbf{A} = \text{diag}(7, 1, -3), \quad \mathbf{v} = (2, 1, 2), \quad \tau = 1.$$

Write down  $\tilde{\mathbf{A}} = \mathbf{A} + \mathbf{v}\mathbf{v}^\top$  and its secular function, find all three eigenvalues from the secular equation, and confirm them independently.

*Solution.* The matrix  $\mathbf{v}\mathbf{v}^\top$  has  $(i, j)$  entry  $v_i v_j$ , so

$$\tilde{\mathbf{A}} = \begin{pmatrix} 7 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & -3 \end{pmatrix} + \begin{pmatrix} 4 & 2 & 4 \\ 2 & 1 & 2 \\ 4 & 2 & 4 \end{pmatrix} = \begin{pmatrix} 11 & 2 & 4 \\ 2 & 2 & 2 \\ 4 & 2 & 1 \end{pmatrix}.$$

The hypotheses of [Theorem 16.5.2](#) hold:  $7 > 1 > -3$  are distinct,  $\tau = 1 > 0$ , and no coordinate of  $\mathbf{v}$  vanishes.

*The secular function.* With  $|v_i|^2 = 4, 1, 4$ ,

$$f(\lambda) = 1 + \frac{4}{7 - \lambda} + \frac{1}{1 - \lambda} + \frac{4}{-3 - \lambda}.$$

*The roots.* Multiply through by  $(7 - \lambda)(1 - \lambda)(-3 - \lambda)$ , which by  $(*)$  turns  $f$  into  $\det(\tilde{\mathbf{A}} - \lambda\mathbf{I})$ :

$$\begin{aligned} & (7 - \lambda)(1 - \lambda)(-3 - \lambda) + 4(1 - \lambda)(-3 - \lambda) \\ & + (7 - \lambda)(-3 - \lambda) + 4(7 - \lambda)(1 - \lambda) \\ & = -(\lambda - 13)(\lambda - 2)(\lambda + 1). \end{aligned}$$

So the eigenvalues are  $\mu_1 = 13$ ,  $\mu_2 = 2$ ,  $\mu_3 = -1$ . They sit where the theorem promised:

$$13 > 7 > 2 > 1 > -1 > -3.$$

It is quicker to check the three roots than to expand the product. At  $\lambda = 13$ ,

$$f(13) = 1 - \frac{4}{6} - \frac{1}{12} - \frac{4}{16} = \frac{12-8-1-3}{12} = 0;$$

at  $\lambda = 2$ ,  $f(2) = 1 + \frac{4}{5} - 1 - \frac{4}{5} = 0$ ; at  $\lambda = -1$ ,  $f(-1) = 1 + \frac{1}{2} + \frac{1}{2} - 2 = 0$ .

*An independent confirmation.* Part (d) of the theorem produces eigenvectors without any further elimination:  $(\mathbf{A} - \mu\mathbf{I})^{-1}\mathbf{v}$  is diagonal applied to  $\mathbf{v}$ . For  $\mu = 13$ ,  $\mathbf{A} - 13\mathbf{I} = \text{diag}(-6, -12, -16)$  and

$$(\mathbf{A} - 13\mathbf{I})^{-1}\mathbf{v} = \left(-\frac{1}{3}, -\frac{1}{12}, -\frac{1}{8}\right) \sim (8, 2, 3)$$

after scaling by  $-24$ . Multiplying out,  $\tilde{\mathbf{A}}(8, 2, 3) = (88 + 4 + 12, 16 + 4 + 6, 32 + 4 + 3) = (104, 26, 39) = 13(8, 2, 3)$ . The same recipe gives  $(2, -5, -2)$  for  $\mu = 2$  and  $(1, 2, -4)$  for  $\mu = -1$ , and both check by one matrix-vector product. Finally the trace:  $11 + 2 + 1 = 14 = 13 + 2 - 1$ , as [Corollary 11.1.4](#) requires.

**Warning.** Both extra hypotheses of [Theorem 16.5.2](#) earn their place, and the sign of  $\tau$  reverses the picture.

*A zero coordinate deflates.* Take  $\mathbf{A} = \text{diag}(2, 1, 0)$  and  $\mathbf{v} = (1, 0, 1)$ , so  $v_2 = 0$ . Then  $\mathbf{e}_2$  is an eigenvector of  $\tilde{\mathbf{A}} = \mathbf{A} + \mathbf{v}\mathbf{v}^\top$  with eigenvalue  $1 = d_2$ , because  $\mathbf{v}^*\mathbf{e}_2 = 0$ . So



conclusion (a) fails. The secular function  $1 + 1/(2 - \lambda) - 1/\lambda$  has only two poles and only two roots,  $2 \pm \sqrt{2}$ ; the third eigenvalue, 1, is invisible to it. The eigenvalues still interlace,  $2 + \sqrt{2} \geq 2 \geq 1 \geq 2 - \sqrt{2} \geq 0$ , but not strictly.

*A repeated  $d_i$  deflates too.* If  $d_1 = d_2$ , any vector in  $\text{span}(\mathbf{e}_1, \mathbf{e}_2)$  orthogonal to  $\mathbf{v}$  is an eigenvector of  $\tilde{\mathbf{A}}$  for  $d_1$ , for the same reason.

*Negative  $\tau$  reverses the interlacing.* For  $\tau < 0$  the matrix  $\tau \mathbf{v} \mathbf{v}^*$  is negative semidefinite and every eigenvalue goes **down**: applying [Theorem 16.5.1](#) to  $\tilde{\mathbf{A}}$  and  $-\tau > 0$  gives  $\lambda_i(\mathbf{A}) \geq \lambda_i(\tilde{\mathbf{A}}) \geq \lambda_{i+1}(\mathbf{A})$ . Reading the chain of the theorem in the wrong direction is the standard way to get this backwards.

### 16.5.3 The shifts add up

Interlacing says where each eigenvalue may go. One number says how far they all go together, and it costs a line.

#### Corollary 16.5.3 — Total Eigenvalue Shift

Let  $\mathbf{A} \in M_n(F)$  be Hermitian,  $\mathbf{v} \in F^n$ ,  $\tau \in \mathbb{R}$ , and  $\tilde{\mathbf{A}} = \mathbf{A} + \tau \mathbf{v} \mathbf{v}^*$ . Then

$$\sum_{i=1}^n (\lambda_i(\tilde{\mathbf{A}}) - \lambda_i(\mathbf{A})) = \tau \|\mathbf{v}\|^2.$$

*Proof.* By [Theorem 0.9.6](#) (3) applied to the  $n \times 1$  matrix  $\mathbf{v}$  and the  $1 \times n$  matrix  $\mathbf{v}^*$ ,

$$\text{tr}(\mathbf{v} \mathbf{v}^*) = \text{tr}(\mathbf{v}^* \mathbf{v}) = \|\mathbf{v}\|^2.$$

Hence  $\text{tr} \tilde{\mathbf{A}} = \text{tr} \mathbf{A} + \tau \|\mathbf{v}\|^2$ , by part (1) of the same theorem. Both  $\mathbf{A}$  and  $\tilde{\mathbf{A}}$  are Hermitian, so [Corollary 11.1.4](#) expresses each trace as the sum of the eigenvalues with multiplicity, and subtracting the two expressions gives the claim. ■

So the whole budget for the movement is  $\tau \|\mathbf{v}\|^2$ , and for  $\tau \geq 0$  [Theorem 16.5.1](#) says how it may be spent: each eigenvalue takes a non-negative share, and no share except the top one is large enough to carry an eigenvalue past its old upper neighbor. In [Example \(A bump on a diagonal matrix\)](#) the budget 5 was split as  $3 + 2$ ; in [Example \(A three-by-three update, found from its secular equation\)](#) the budget  $\|\mathbf{v}\|^2 = 9$  was split as  $6 + 1 + 2$ .

*Remark.* The secular equation of [Theorem 16.5.2](#) is not only a proof device. Because  $f$  is increasing between consecutive poles and the bracketing interval is known in advance, each root can be located by bisection on an interval that is guaranteed to contain exactly one of them — a far better-behaved problem than finding the roots of a characteristic polynomial. The secular equation is the engine of the divide-and-conquer eigenvalue algorithms for Hermitian matrices, which split the matrix into two halves joined by a rank-one term and solve one secular equation to merge their spectra. The deflation observed in the warning above, where a zero coordinate of  $\mathbf{v}$  hands over an eigenvalue for free, is the same phenomenon that makes the algorithm fast in practice.

## 16.5.4 Exercises

### 16.5.4.1 A. Check your understanding

#### Exercise A1.

- State [Theorem 16.5.1](#), including the hypothesis on  $\tau$ .
- Write down the secular function of  $\mathbf{A} + 2\mathbf{v}\mathbf{v}^\top$  for  $\mathbf{A} = \text{diag}(6, 2, -1)$  and  $\mathbf{v} = (1, 3, 1)$ .
- Explain in one sentence why no  $d_i$  is an eigenvalue of  $\mathbf{A} + \tau\mathbf{v}\mathbf{v}^*$  under the hypotheses of [Theorem 16.5.2](#).
- Determine whether the following is correct, and justify your answer: if  $\tau < 0$  then  $\lambda_i(\mathbf{A} + \tau\mathbf{v}\mathbf{v}^*) \geq \lambda_i(\mathbf{A})$  for every  $i$ .
- A Hermitian  $4 \times 4$  matrix  $\mathbf{A}$  has eigenvalues 8, 5, 5, 0. What does [Corollary 16.5.3](#) say about  $\mathbf{A} + 3\mathbf{v}\mathbf{v}^*$  when  $\|\mathbf{v}\| = 2$ ?

*Solution.*

- For Hermitian  $\mathbf{A} \in M_n(F)$ ,  $\mathbf{v} \in F^n$  and  $\tau \geq 0$ , the eigenvalues of  $\tilde{\mathbf{A}} = \mathbf{A} + \tau\mathbf{v}\mathbf{v}^*$  satisfy  $\lambda_i(\tilde{\mathbf{A}}) \geq \lambda_i(\mathbf{A}) \geq \lambda_{i+1}(\tilde{\mathbf{A}})$  for  $1 \leq i \leq n-1$ , together with  $\lambda_n(\tilde{\mathbf{A}}) \geq \lambda_n(\mathbf{A})$ . The hypothesis  $\tau \geq 0$  is what makes the perturbation positive semidefinite; without it only the second family of inequalities survives.
- $f(\lambda) = 1 + 2\left(\frac{1}{6-\lambda} + \frac{9}{2-\lambda} + \frac{1}{-1-\lambda}\right)$ .
- If  $\tilde{\mathbf{A}}\mathbf{x} = d_i\mathbf{x}$ , comparing  $i$ -th coordinates in  $(\mathbf{A} - d_i\mathbf{I})\mathbf{x} = -\tau(\mathbf{v}^*\mathbf{x})\mathbf{v}$  forces  $\mathbf{v}^*\mathbf{x} = 0$  because  $v_i \neq 0$ ; then  $\mathbf{x}$  is a multiple of  $\mathbf{e}_i$ , and  $\mathbf{v}^*\mathbf{x} = 0$  forces that multiple to be 0.
- Incorrect; the inequality goes the other way. For  $\tau < 0$  the added matrix is negative semidefinite, and  $\lambda_i(\tilde{\mathbf{A}}) \geq \lambda_i(\mathbf{A} + \tau\mathbf{v}\mathbf{v}^*)$  for every  $i$ . Concretely,  $\mathbf{A} = \text{diag}(1, 0)$ ,  $\tau = -1$ ,  $\mathbf{v} = \mathbf{e}_1$  gives  $\tilde{\mathbf{A}} = \text{diag}(0, 0)$  and  $\lambda_1$  drops from 1 to 0.
- It says the four eigenvalues of  $\mathbf{A} + 3\mathbf{v}\mathbf{v}^*$  sum to  $8 + 5 + 5 + 0 + 3 \cdot 4 = 30$ , that is, 12 more than those of  $\mathbf{A}$ . Combined with [Theorem 16.5.1](#), which caps  $\lambda_2$  at 8,  $\lambda_3$  at 5 and  $\lambda_4$  at 5, it forces  $\lambda_1(\mathbf{A} + 3\mathbf{v}\mathbf{v}^*) \geq 30 - 18 = 12$ .

### 16.5.4.2 B. Practice

#### Exercise B1: A secular equation, end to end.

Let  $\mathbf{A} = \text{diag}(7, 0, -3)$ ,  $\mathbf{v} = (3, 2, 2)$  and  $\tau = 1$ .

- Write down  $\tilde{\mathbf{A}} = \mathbf{A} + \mathbf{v}\mathbf{v}^\top$  and the secular function  $f$ .
- Verify that 21, 2 and  $-2$  are roots of  $f$ , and explain why these are all the eigenvalues of  $\tilde{\mathbf{A}}$ .
- Exhibit the interlacing chain of [Theorem 16.5.1](#) for this pair.

*Solution.*

- (a)  $\mathbf{v}\mathbf{v}^\top$  has entries  $v_i v_j$ , so

$$\tilde{\mathbf{A}} = \begin{pmatrix} 7 & 0 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & -3 \end{pmatrix} + \begin{pmatrix} 9 & 6 & 6 \\ 6 & 4 & 4 \\ 6 & 4 & 4 \end{pmatrix} = \begin{pmatrix} 16 & 6 & 6 \\ 6 & 4 & 4 \\ 6 & 4 & 1 \end{pmatrix},$$

$$\text{and } f(\lambda) = 1 + \frac{9}{7-\lambda} + \frac{4}{0-\lambda} + \frac{4}{-3-\lambda}.$$

- (b) At  $\lambda = 21$ :  $1 - \frac{9}{14} - \frac{4}{21} - \frac{4}{24} = 1 - \frac{9}{14} - \frac{4}{21} - \frac{1}{6}$ . Over 42 this is  $(42 - 27 - 8 - 7)/42 = 0$ . At  $\lambda = 2$ :  $1 + \frac{9}{5} - 2 - \frac{4}{5} = (5 + 9 - 10 - 4)/5 = 0$ . At  $\lambda = -2$ :  $1 + 1 + 2 - 4 = 0$ . The hypotheses of [Theorem 16.5.2](#) hold —  $7 > 0 > -3$  are distinct,  $\tau = 1 > 0$ , and every coordinate of  $\mathbf{v}$  is non-zero — so by part (c) the secular function has exactly three roots and they are the eigenvalues. We have found three distinct roots, so there are no others.
- (c)  $21 \geq 7 \geq 2 \geq 0 \geq -2 \geq -3$ , all strict, as part (c) of [Theorem 16.5.2](#) predicts.

### Exercise B2: Reading the chain backwards.

Let  $\mathbf{B} = \begin{pmatrix} 5 & 2 \\ 2 & 5 \end{pmatrix}$  and  $\mathbf{v} = (1, 2)$ .

- (a) Compute the eigenvalues of  $\mathbf{B}$  and of  $\mathbf{B} - \mathbf{v}\mathbf{v}^\top$ .
- (b) State which chain of inequalities [Theorem 16.5.1](#) gives here, being careful about which matrix plays the role of  $\mathbf{A}$ .

*Solution.*

- (a)  $\mathbf{B}$  has trace 10 and determinant 21, so its characteristic polynomial is  $x^2 - 10x + 21 = (x - 7)(x - 3)$  and its eigenvalues are 7 and 3. Also

$$\mathbf{B} - \mathbf{v}\mathbf{v}^\top = \begin{pmatrix} 5 & 2 \\ 2 & 5 \end{pmatrix} - \begin{pmatrix} 1 & 2 \\ 2 & 4 \end{pmatrix} = \begin{pmatrix} 4 & 0 \\ 0 & 1 \end{pmatrix},$$

with eigenvalues 4 and 1.

- (b) Put  $\mathbf{A} = \mathbf{B} - \mathbf{v}\mathbf{v}^\top = \text{diag}(4, 1)$  and  $\tau = 1$ , so that  $\mathbf{B} = \mathbf{A} + \tau\mathbf{v}\mathbf{v}^\top$  is the *bumped* matrix. The theorem then reads  $\lambda_1(\mathbf{B}) \geq \lambda_1(\mathbf{A}) \geq \lambda_2(\mathbf{B}) \geq \lambda_2(\mathbf{A})$ , that is,  $7 \geq 4 \geq 3 \geq 1$ . Taking  $\mathbf{B}$  for  $\mathbf{A}$  and  $\tau = -1$  instead would give the reversed chain  $\lambda_i(\mathbf{B}) \geq \lambda_i(\mathbf{B} - \mathbf{v}\mathbf{v}^\top) \geq \lambda_{i+1}(\mathbf{B})$ , namely  $7 \geq 4 \geq 3 \geq 1$  again: the same four numbers, read from the other side.

### Exercise B3: The missing eigenvalue.

Let  $\mathbf{A} = \text{diag}(5, 3, 0)$ ,  $\mathbf{v} = (2, 2, 2)$  and  $\tau = 1$ . It is given that two of the eigenvalues of  $\tilde{\mathbf{A}} = \mathbf{A} + \mathbf{v}\mathbf{v}^\top$  are 15 and 4. Find the third without computing a characteristic polynomial, and then confirm it from the secular equation.

*Solution.* By [Corollary 16.5.3](#) the eigenvalues of  $\tilde{\mathbf{A}}$  sum to

$$(5 + 3 + 0) + \|\mathbf{v}\|^2 = 8 + 12 = 20.$$

Hence the third eigenvalue is  $20 - 15 - 4 = 1$ .

For the confirmation,  $|v_i|^2 = 4$  for each  $i$ , so

$$f(\lambda) = 1 + \frac{4}{5-\lambda} + \frac{4}{3-\lambda} + \frac{4}{-\lambda},$$

and  $f(1) = 1 + 1 + 2 - 4 = 0$ . The hypotheses of [Theorem 16.5.2](#) hold, and 1 lies in the interval  $(0, 3)$  where part (c) puts the third root. As a check on the other two,  $15 > 5$  and  $4 \in (3, 5)$ , exactly as the theorem requires.

### 16.5.4.3 C. Going deeper

#### Exercise C1: What a zero coordinate does.

Let  $\mathbf{A} = \text{diag}(d_1, \dots, d_n)$  with the  $d_i$  real, let  $\tau \in \mathbb{R}$  and let  $\mathbf{v} \in F^n$ .

- (a) Prove that if  $v_i = 0$  then  $\mathbf{e}_i$  is an eigenvector of  $\mathbf{A} + \tau\mathbf{v}\mathbf{v}^*$  with eigenvalue  $d_i$ .
- (b) Suppose  $\tau > 0$ , the  $d_i$  are strictly decreasing, and exactly one coordinate  $v_k$  vanishes. Show that the remaining  $n - 1$  eigenvalues of  $\mathbf{A} + \tau\mathbf{v}\mathbf{v}^*$  are the roots of the function obtained from the secular function by deleting the  $k$ -th term. *Hint: restrict to the coordinate subspace spanned by the  $\mathbf{e}_i$  with  $i \neq k$ .*
- (c) Illustrate both parts with  $\mathbf{A} = \text{diag}(2, 1, 0)$ ,  $\mathbf{v} = (1, 0, 1)$  and  $\tau = 1$ .

*Solution.*

- (a)  $\mathbf{v}^*\mathbf{e}_i = \overline{v_i} = 0$ , so  $(\mathbf{A} + \tau\mathbf{v}\mathbf{v}^*)\mathbf{e}_i = \mathbf{A}\mathbf{e}_i + \tau\mathbf{v}(\mathbf{v}^*\mathbf{e}_i) = d_i\mathbf{e}_i$ .
- (b) Let  $U = \text{span}(\mathbf{e}_i : i \neq k)$ , a subspace of dimension  $n - 1$ . Every column of  $\mathbf{A}$  other than the  $k$ -th lies in  $U$ , and  $\mathbf{v} \in U$  because  $v_k = 0$ ; hence  $(\mathbf{A} + \tau\mathbf{v}\mathbf{v}^*)U \subseteq U$ . In the ordered basis  $(\mathbf{e}_i)_{i \neq k}$  of  $U$ , the restriction of  $\mathbf{A} + \tau\mathbf{v}\mathbf{v}^*$  to  $U$  has matrix  $\mathbf{A}' + \tau\mathbf{v}'(\mathbf{v}')^*$ , where  $\mathbf{A}' = \text{diag}(d_i : i \neq k)$  and  $\mathbf{v}'$  is  $\mathbf{v}$  with its  $k$ -th coordinate removed. Since  $U \oplus \text{span}(\mathbf{e}_k) = F^n$  and  $\text{span}(\mathbf{e}_k)$  is also invariant by (a), the eigenvalues of  $\mathbf{A} + \tau\mathbf{v}\mathbf{v}^*$  are those of the restriction to  $U$  together with  $d_k$ . Now  $\mathbf{A}'$  has strictly decreasing diagonal entries,  $\tau > 0$ , and  $\mathbf{v}'$  has no zero coordinate, so [Theorem 16.5.2](#) applies to  $\mathbf{A}' + \tau\mathbf{v}'(\mathbf{v}')^*$ : its  $n - 1$  eigenvalues are exactly the roots of its secular function. That function is the secular function of the original update with the  $k$ -th term deleted, because that term was  $\tau|v_k|^2/(d_k - \lambda) = 0$  anyway.
- (c) Here  $v_2 = 0$ , so  $\mathbf{e}_2$  is an eigenvector with eigenvalue  $d_2 = 1$  by (a). By (b) the other two eigenvalues are the roots of

$$1 + \frac{1}{2-\lambda} + \frac{1}{-\lambda} = 0,$$

that is, of  $\lambda(2-\lambda) + \lambda - (2-\lambda) = 0$ , i.e.  $\lambda^2 - 4\lambda + 2 = 0$ , with roots  $2 \pm \sqrt{2}$ . So the spectrum is  $\{2 + \sqrt{2}, 1, 2 - \sqrt{2}\}$ . Directly,

$$\mathbf{A} + \mathbf{v}\mathbf{v}^\top = \begin{pmatrix} 3 & 0 & 1 \\ 0 & 1 & 0 \\ 1 & 0 & 1 \end{pmatrix},$$

whose second row and column confirm the eigenvalue 1, and whose remaining  $2 \times 2$  block  $\begin{pmatrix} 3 & 1 \\ 1 & 1 \end{pmatrix}$  has trace 4 and determinant 2.

**Exercise C2: Only one eigenvalue can escape.**

Let  $\mathbf{A} \in M_n(F)$  be Hermitian,  $\mathbf{v} \in F^n$ ,  $\tau \in \mathbb{R}$ , and  $\tilde{\mathbf{A}} = \mathbf{A} + \tau \mathbf{v} \mathbf{v}^*$ .

- (a) Prove that **at most one** eigenvalue of  $\tilde{\mathbf{A}}$  lies outside the closed interval  $[\lambda_n(\mathbf{A}), \lambda_1(\mathbf{A})]$ , and say which one it is and on which side, according to the sign of  $\tau$ .
- (b) Prove that the escaping eigenvalue moves by at most  $|\tau| \|\mathbf{v}\|^2$ , and show by examples with  $n = 2$  that it may move by exactly that amount, and that it may not escape at all.

*Solution.*

- (a) Suppose first  $\tau \geq 0$ . By [Theorem 16.5.1](#),  $\lambda_n(\tilde{\mathbf{A}}) \geq \lambda_n(\mathbf{A})$ , so no eigenvalue of  $\tilde{\mathbf{A}}$  falls below the interval; and for  $i \geq 2$ ,  $\lambda_i(\tilde{\mathbf{A}}) \leq \lambda_{i-1}(\mathbf{A}) \leq \lambda_1(\mathbf{A})$ , so only  $\lambda_1(\tilde{\mathbf{A}})$  can rise above it. So at most one eigenvalue escapes, namely the largest, and it escapes upwards.  
For  $\tau < 0$ , the warning above gives  $\lambda_i(\mathbf{A}) \geq \lambda_i(\tilde{\mathbf{A}}) \geq \lambda_{i+1}(\mathbf{A})$  for  $i \leq n-1$  and  $\lambda_n(\mathbf{A}) \geq \lambda_n(\tilde{\mathbf{A}})$ . Hence  $\lambda_1(\tilde{\mathbf{A}}) \leq \lambda_1(\mathbf{A})$ , and  $\lambda_i(\tilde{\mathbf{A}}) \geq \lambda_{i+1}(\mathbf{A}) \geq \lambda_n(\mathbf{A})$  for  $i \leq n-1$ . So only  $\lambda_n(\tilde{\mathbf{A}})$  can escape, and it escapes downwards.
- (b) Take  $\tau > 0$ ; the case  $\tau < 0$  is the mirror image. By [Theorem 16.5.1](#) every shift  $\lambda_i(\tilde{\mathbf{A}}) - \lambda_i(\mathbf{A})$  is  $\geq 0$ , and by [Corollary 16.5.3](#) they sum to  $\tau \|\mathbf{v}\|^2$ . So each shift, in particular that of  $\lambda_1$ , is at most  $\tau \|\mathbf{v}\|^2$ .

For equality, let  $\mathbf{A} = \text{diag}(2, 1)$ ,  $\mathbf{v} = \mathbf{e}_1$ ,  $\tau = 3$ : then  $\tilde{\mathbf{A}} = \text{diag}(5, 1)$ , and  $\lambda_1$  moves from 2 to 5, by exactly  $\tau \|\mathbf{v}\|^2 = 3$ . For no escape, let  $\mathbf{v} = \mathbf{e}_2$  and  $\tau = 1$ : then  $\tilde{\mathbf{A}} = \text{diag}(2, 2)$ , whose eigenvalues both lie in  $[1, 2]$ . The whole budget went to  $\lambda_2$ , which [Theorem 16.5.1](#) caps at  $\lambda_1(\mathbf{A}) = 2$ .

**Exercise C3: Bumping along an eigenvector.**

Let  $\mathbf{A} \in M_n(F)$  be Hermitian with an orthonormal basis of eigenvectors  $\mathbf{u}_1, \dots, \mathbf{u}_n$ , where  $\mathbf{A}\mathbf{u}_i = \lambda_i(\mathbf{A})\mathbf{u}_i$ .

- (a) Fix  $k$  and  $\tau \in \mathbb{R}$ . Prove that the eigenvalues of  $\mathbf{A} + \tau \mathbf{u}_k \mathbf{u}_k^*$ , listed with multiplicity but not necessarily in order, are  $\lambda_k(\mathbf{A}) + \tau$  together with  $\lambda_i(\mathbf{A})$  for  $i \neq k$ .
- (b) Deduce that  $\mathbf{A} - \lambda_1(\mathbf{A})\mathbf{u}_1 \mathbf{u}_1^*$  has eigenvalue list  $0, \lambda_2(\mathbf{A}), \dots, \lambda_n(\mathbf{A})$ . (This is the **deflation** step: it removes the top eigenvalue and leaves the rest untouched, which is what makes it possible to find eigenvalues one at a time.)
- (c) Explain why (a) does not contradict part (c) of [Theorem 16.5.2](#).

*Solution.*

- (a) For  $i \neq k$ , orthonormality gives  $\mathbf{u}_k^* \mathbf{u}_i = 0$ , so

$$(\mathbf{A} + \tau \mathbf{u}_k \mathbf{u}_k^*) \mathbf{u}_i = \mathbf{A} \mathbf{u}_i + \tau \mathbf{u}_k (\mathbf{u}_k^* \mathbf{u}_i) = \lambda_i(\mathbf{A}) \mathbf{u}_i.$$

For  $i = k$ ,  $\mathbf{u}_k^* \mathbf{u}_k = 1$  gives  $(\mathbf{A} + \tau \mathbf{u}_k \mathbf{u}_k^*) \mathbf{u}_k = (\lambda_k(\mathbf{A}) + \tau) \mathbf{u}_k$ . So  $(\mathbf{u}_1, \dots, \mathbf{u}_n)$  is a basis of  $F^n$  consisting of eigenvectors of  $\mathbf{A} + \tau \mathbf{u}_k \mathbf{u}_k^*$ , with the stated eigenvalues. A basis of eigenvectors exhibits the whole eigenvalue list with multiplicity.

- (b) Take  $k = 1$  and  $\tau = -\lambda_1(\mathbf{A})$  in (a).

- (c) There is no contradiction, because the hypothesis of [Theorem 16.5.2](#) that fails here fails as badly as possible. Writing everything in the orthonormal eigenbasis, the update vector is  $\mathbf{u}_k$ , whose coordinate vector is  $\mathbf{e}_k$ : **all but one** of its coordinates are zero. Part (a) is the extreme case of [Exercise \(C1: What a zero coordinate does\)](#) (a), with  $n - 1$  coordinates deflating at once, and only one eigenvalue is left to move.

## 16.6 Subspaces and Sums of Eigenvalues

Section 4 deleted a row and the matching column, which is a coordinate-dependent operation, and got a coordinate-free conclusion: the spectra interlace. That is a hint. This section replaces “delete some coordinates” by “restrict to a subspace”, which is the same operation with the accident of a basis removed, and the interlacing survives in a stronger form. Then it asks a question Courant–Fischer cannot answer one eigenvalue at a time: not “what is  $\lambda_2$ ?” but “what is  $\lambda_1 + \lambda_2$ ?” That sum turns out to be a maximum over a set of subspaces, and the answer discharges a promise Chapter 12 made about singular values.

Throughout this section  $F = \mathbb{R}$  or  $F = \mathbb{C}$ , every matrix whose eigenvalues are indexed is Hermitian, and the indexing is decreasing,  $\lambda_1(\mathbf{A}) \geq \cdots \geq \lambda_n(\mathbf{A})$ .

### 16.6.1 Restricting a matrix to a subspace

A principal submatrix of  $\mathbf{A}$  records what the quadratic form  $\mathbf{x} \mapsto \langle \mathbf{A}\mathbf{x}, \mathbf{x} \rangle$  does to vectors supported on a few coordinates. Nothing in that description needs the coordinates to be the standard ones: any subspace  $W$  and any orthonormal basis of it will do, and the matrix that results plays the same role.

#### Definition 16.6.1 — Compression to a subspace

Let  $\mathbf{A} \in M_n(F)$ , let  $1 \leq k \leq n$ , and let  $\mathbf{q}_1, \dots, \mathbf{q}_k$  be an orthonormal list in  $F^n$ , so that the matrix  $\mathbf{Q} \in M_{n \times k}(F)$  with these columns satisfies  $\mathbf{Q}^* \mathbf{Q} = \mathbf{I}_k$ . The **compression** of  $\mathbf{A}$  to  $W = \text{span}(\mathbf{q}_1, \dots, \mathbf{q}_k)$  with respect to this basis is the  $k \times k$  matrix

$$\mathbf{Q}^* \mathbf{A} \mathbf{Q}, \quad (\mathbf{Q}^* \mathbf{A} \mathbf{Q})_{ij} = \langle \mathbf{A} \mathbf{q}_j, \mathbf{q}_i \rangle.$$

The entry formula is just  $(\mathbf{Q}^* \mathbf{A} \mathbf{Q})_{ij} = \mathbf{q}_i^* \mathbf{A} \mathbf{q}_j$ , and  $\mathbf{q}_i^* \mathbf{y} = \langle \mathbf{y}, \mathbf{q}_i \rangle$  for the standard inner product. If  $\mathbf{A}$  is Hermitian so is  $\mathbf{Q}^* \mathbf{A} \mathbf{Q}$ , since  $(\mathbf{Q}^* \mathbf{A} \mathbf{Q})^* = \mathbf{Q}^* \mathbf{A}^* \mathbf{Q} = \mathbf{Q}^* \mathbf{A} \mathbf{Q}$ .

**The eigenvalues do not depend on the basis chosen.** Let  $(\mathbf{q}'_1, \dots, \mathbf{q}'_k)$  be another orthonormal basis of the same  $W$ , with matrix  $\mathbf{Q}'$ . Each  $\mathbf{q}'_j$  lies in  $W = \text{col}(\mathbf{Q})$ , so  $\mathbf{Q}' = \mathbf{Q} \mathbf{U}$  for some  $\mathbf{U} \in M_k(F)$ , and then  $\mathbf{I}_k = (\mathbf{Q}')^* \mathbf{Q}' = \mathbf{U}^* \mathbf{Q}^* \mathbf{Q} \mathbf{U} = \mathbf{U}^* \mathbf{U}$ , so  $\mathbf{U}$  is unitary. Hence

$$(\mathbf{Q}')^* \mathbf{A} \mathbf{Q}' = \mathbf{U}^* (\mathbf{Q}^* \mathbf{A} \mathbf{Q}) \mathbf{U},$$

a unitary similarity, which leaves the eigenvalues alone. So it is legitimate to speak of *the* eigenvalues of the compression of  $\mathbf{A}$  to  $W$ , and we do.

Three cases to keep in mind.

- **Coordinate subspaces.** If  $\mathbf{q}_j = \mathbf{e}_{i_j}$  for indices  $i_1 < \cdots < i_k$ , then  $(\mathbf{Q}^* \mathbf{A} \mathbf{Q})_{lj} = a_{i_l i_j}$ : the compression is the principal submatrix  $\mathbf{A}_{I,I}$  of Definition 6.7.1 on the index set  $I = \{i_1, \dots, i_k\}$ . Compression generalizes “delete rows and the matching columns”.
- $k = n$ . Then  $\mathbf{Q}$  is a unitary matrix and  $\mathbf{Q}^* \mathbf{A} \mathbf{Q}$  is unitarily similar to  $\mathbf{A}$ : nothing is lost and nothing is learned.
- $k = 1$ . Then  $\mathbf{Q}^* \mathbf{A} \mathbf{Q}$  is the  $1 \times 1$  matrix  $\langle \mathbf{A} \mathbf{q}_1, \mathbf{q}_1 \rangle = R_{\mathbf{A}}(\mathbf{q}_1)$ , the Rayleigh quotient of Definition 16.1.1 at a unit vector. Everything below is a statement about  $k$  Rayleigh quotients at once.

Drop the orthonormality and the notion collapses. With  $\mathbf{A} = \text{diag}(1, 0)$  and the single column  $\mathbf{Q} = (2, 0)^\top$ , which has  $\mathbf{Q}^* \mathbf{Q} = (4) \neq \mathbf{I}_1$ , the “compression” is the  $1 \times 1$  matrix  $(4)$ , whose eigenvalue 4 is larger than every eigenvalue of  $\mathbf{A}$ . The clause that fails is  $\mathbf{Q}^* \mathbf{Q} = \mathbf{I}_k$ , and with it every conclusion of this section.

## 16.6.2 Interlacing without coordinates

### Theorem 16.6.1 — Poincaré Separation Theorem

Let  $\mathbf{A} \in M_n(F)$  be Hermitian, let  $1 \leq k \leq n$ , and let  $\mathbf{B} = \mathbf{Q}^* \mathbf{A} \mathbf{Q}$  be the compression of  $\mathbf{A}$  to a  $k$ -dimensional subspace  $W \subseteq F^n$ , as in Definition 16.6.1. Then

$$\lambda_i(\mathbf{A}) \geq \lambda_i(\mathbf{B}) \geq \lambda_{i+n-k}(\mathbf{A}) \quad (1 \leq i \leq k).$$

*Idea.* Both inequalities come from the same observation, used twice. The map  $\mathbf{y} \mapsto \mathbf{Q}\mathbf{y}$  carries  $F^k$  isometrically onto  $W$ , and it carries the Rayleigh quotient of  $\mathbf{B}$  to the Rayleigh quotient of  $\mathbf{A}$ . So each of the two Courant–Fischer descriptions of  $\lambda_i(\mathbf{B})$  — a maximum of minima, and a minimum of maxima — is the corresponding description of  $\lambda_i(\mathbf{A})$  with the competing subspaces restricted to those lying inside  $W$ . A maximum over fewer competitors is smaller; a minimum over fewer competitors is larger. That is the whole proof; the only care needed is in matching the dimensions, and it is there that  $n - k$  appears.

*Proof. Step 1: transport.* Throughout,  $R_{\mathbf{A}}$  and  $R_{\mathbf{B}}$  are the Rayleigh quotients of Definition 16.1.1. For  $\mathbf{0} \neq \mathbf{y} \in F^k$  we have  $\mathbf{Q}\mathbf{y} \neq \mathbf{0}$ , since  $\|\mathbf{Q}\mathbf{y}\|^2 = \mathbf{y}^* \mathbf{Q}^* \mathbf{Q} \mathbf{y} = \|\mathbf{y}\|^2$  and  $\mathbf{y} \neq \mathbf{0}$ . The same computation with  $\mathbf{A}$  inserted gives

$$\langle \mathbf{B}\mathbf{y}, \mathbf{y} \rangle = \mathbf{y}^* \mathbf{Q}^* \mathbf{A} \mathbf{Q} \mathbf{y} = \langle \mathbf{A}(\mathbf{Q}\mathbf{y}), \mathbf{Q}\mathbf{y} \rangle,$$

so  $R_{\mathbf{B}}(\mathbf{y}) = R_{\mathbf{A}}(\mathbf{Q}\mathbf{y})$ . Moreover  $\mathbf{y} \mapsto \mathbf{Q}\mathbf{y}$  is injective and linear, so it maps the subspaces of  $F^k$  of dimension  $j$  bijectively onto the subspaces of  $W$  of dimension  $j$ , for every  $j$ . Consequently, for each  $j$ ,

$$\{ R_{\mathbf{B}}(\mathbf{y}) : \mathbf{0} \neq \mathbf{y} \in S \} = \{ R_{\mathbf{A}}(\mathbf{x}) : \mathbf{0} \neq \mathbf{x} \in \mathbf{Q}S \}$$

for every subspace  $S \subseteq F^k$ .

**Step 2: the upper bound.** By the max–min form of Theorem 16.2.3 applied to

$\mathbf{B} \in M_k(F)$ ,

$$\begin{aligned}\lambda_i(\mathbf{B}) &= \max_{\substack{S \subseteq F^k \\ \dim S=i}} \min_{\mathbf{0} \neq \mathbf{y} \in S} R_{\mathbf{B}}(\mathbf{y}) \\ &= \max_{\substack{U \subseteq W \\ \dim U=i}} \min_{\mathbf{0} \neq \mathbf{x} \in U} R_{\mathbf{A}}(\mathbf{x}),\end{aligned}$$

the second equality by Step 1. The subspaces  $U \subseteq W$  of dimension  $i$  form a subfamily of the subspaces  $U \subseteq F^n$  of dimension  $i$ , and enlarging the family over which a maximum is taken cannot decrease it. So by the max–min form of [Theorem 16.2.3](#) applied to  $\mathbf{A}$ , the right-hand side is at most  $\lambda_i(\mathbf{A})$ .

**Step 3: the lower bound.** By the min–max form of [Theorem 16.2.3](#) applied to  $\mathbf{B} \in M_k(F)$ , with the complementary dimension  $k - i + 1$ ,

$$\begin{aligned}\lambda_i(\mathbf{B}) &= \min_{\substack{S \subseteq F^k \\ \dim S=k-i+1}} \max_{\mathbf{0} \neq \mathbf{y} \in S} R_{\mathbf{B}}(\mathbf{y}) \\ &= \min_{\substack{U \subseteq W \\ \dim U=k-i+1}} \max_{\mathbf{0} \neq \mathbf{x} \in U} R_{\mathbf{A}}(\mathbf{x}),\end{aligned}$$

again by Step 1. Put  $l = i + n - k$ , so that  $1 \leq l \leq n$  and  $n - l + 1 = k - i + 1$ . The min–max form of [Theorem 16.2.3](#) applied to  $\mathbf{A}$  therefore reads

$$\lambda_l(\mathbf{A}) = \min_{\substack{U \subseteq F^n \\ \dim U=k-i+1}} \max_{\mathbf{0} \neq \mathbf{x} \in U} R_{\mathbf{A}}(\mathbf{x}),$$

a minimum over a **larger** family than the one above, hence no larger. So  $\lambda_i(\mathbf{B}) \geq \lambda_{i+n-k}(\mathbf{A})$ . This proves the theorem. ■

*Remark.* Taking  $W$  to be the coordinate hyperplane  $\text{span}(\mathbf{e}_j : j \neq l)$  makes  $\mathbf{B}$  the principal submatrix obtained by deleting row  $l$  and column  $l$ , and  $k = n - 1$  turns the conclusion into  $\lambda_i(\mathbf{A}) \geq \lambda_i(\mathbf{B}) \geq \lambda_{i+1}(\mathbf{A})$ . That is [Theorem 16.4.2](#). What the present theorem adds is that the coordinates played no part, and that deleting  $n - k$  dimensions at once costs exactly  $n - k$  indices.

### Example. ( $\mathbf{A}$ compression of a diagonal matrix)

Let  $\mathbf{A} = \text{diag}(5, 3, -2)$  and let

$$\mathbf{q}_1 = \frac{1}{3}(2, 2, 1), \quad \mathbf{q}_2 = \frac{1}{3}(-2, 1, 2).$$

Check that  $(\mathbf{q}_1, \mathbf{q}_2)$  is orthonormal, compute the compression  $\mathbf{B}$  of  $\mathbf{A}$  to  $W = \text{span}(\mathbf{q}_1, \mathbf{q}_2)$  and its eigenvalues, and verify [Theorem 16.6.1](#).

*Solution. Orthonormality.*  $\|\mathbf{q}_1\|^2 = (4 + 4 + 1)/9 = 1$  and  $\|\mathbf{q}_2\|^2 = (4 + 1 + 4)/9 = 1$ , while  $9\langle \mathbf{q}_1, \mathbf{q}_2 \rangle = -4 + 2 + 2 = 0$ .

*The compression.* Multiplying by the diagonal matrix,  $3\mathbf{A}\mathbf{q}_1 = (10, 6, -2)$  and  $3\mathbf{A}\mathbf{q}_2 =$



$(-10, 3, -4)$ . Hence

$$\begin{aligned} b_{11} &= \frac{1}{9}(2, 2, 1) \cdot (10, 6, -2) = \frac{20+12-2}{9} = \frac{10}{3}, \\ b_{21} &= \frac{1}{9}(-2, 1, 2) \cdot (10, 6, -2) = \frac{-20+6-4}{9} = -2, \\ b_{22} &= \frac{1}{9}(-2, 1, 2) \cdot (-10, 3, -4) = \frac{20+3-8}{9} = \frac{5}{3}, \end{aligned}$$

and  $b_{12} = b_{21}$  because  $\mathbf{B}$  is symmetric. So

$$\mathbf{B} = \begin{pmatrix} 10/3 & -2 \\ -2 & 5/3 \end{pmatrix}.$$

*Its eigenvalues.*  $\operatorname{tr} \mathbf{B} = 5$  and  $\det \mathbf{B} = \frac{50}{9} - 4 = \frac{14}{9}$ , so the characteristic polynomial is  $x^2 - 5x + \frac{14}{9}$ , that is  $9x^2 - 45x + 14$ . Its discriminant is  $2025 - 504 = 1521 = 39^2$ , so the roots are  $(45 \pm 39)/18$ , namely  $\lambda_1(\mathbf{B}) = \frac{14}{3}$  and  $\lambda_2(\mathbf{B}) = \frac{1}{3}$ .

*The separation.* Here  $n = 3$  and  $k = 2$ , so the theorem says  $\lambda_i(\mathbf{A}) \geq \lambda_i(\mathbf{B}) \geq \lambda_{i+1}(\mathbf{A})$  for  $i = 1, 2$ :

$$5 \geq \frac{14}{3} \geq 3 \geq \frac{1}{3} \geq -2,$$

and every inequality is strict. Note also  $\operatorname{tr} \mathbf{B} = 5 \leq 5 + 3 = \lambda_1(\mathbf{A}) + \lambda_2(\mathbf{A})$ ; the next theorem says that this is no accident and that 8 is the best possible ceiling.

*Quick check.* Let  $\mathbf{A}$  be Hermitian  $n \times n$  and let  $\mathbf{B}$  be its compression to a  $k$ -dimensional subspace. Which single inequality of [Theorem 16.6.1](#) says that a compression of a positive semidefinite matrix is positive semidefinite?

*Solution.* The case  $i = k$  of the lower bound:  $\lambda_k(\mathbf{B}) \geq \lambda_{k+n-k}(\mathbf{A}) = \lambda_n(\mathbf{A})$ . If  $\mathbf{A} \succeq 0$  then  $\lambda_n(\mathbf{A}) \geq 0$  by [Theorem 12.1.4](#) (b), so every eigenvalue of  $\mathbf{B}$  is  $\geq 0$ , and  $\mathbf{B}$  is Hermitian, so  $\mathbf{B} \succeq 0$  by the same theorem. (A direct proof is also one line:  $\langle \mathbf{B}\mathbf{y}, \mathbf{y} \rangle = \langle \mathbf{A}\mathbf{Q}\mathbf{y}, \mathbf{Q}\mathbf{y} \rangle \geq 0$ . The point of the check is to locate the statement inside the interlacing.)

### 16.6.3 The top eigenvalues, summed

Chapter 12 described  $\lambda_1$  as a maximum over the unit sphere ([Lemma 12.5.2](#)), and Section 1 pointed out that  $\lambda_2$  is not the “second largest value” of anything. But  $\lambda_1 + \lambda_2$  is a maximum again — over pairs of orthonormal vectors rather than single ones. The pattern continues for every  $k$ , and it is the sum, never the individual eigenvalue, that behaves.

#### Theorem 16.6.2 — Ky Fan Maximum Principle

Let  $\mathbf{A} \in M_n(F)$  be Hermitian and let  $1 \leq k \leq n$ . Then

$$\begin{aligned} \sum_{i=1}^k \lambda_i(\mathbf{A}) &= \max \left\{ \sum_{i=1}^k \langle \mathbf{A}\mathbf{q}_i, \mathbf{q}_i \rangle : \mathbf{q}_1, \dots, \mathbf{q}_k \text{ orthonormal} \right\} \\ &= \max \{ \operatorname{tr}(\mathbf{Q}^* \mathbf{A} \mathbf{Q}) : \mathbf{Q} \in M_{n \times k}(F), \mathbf{Q}^* \mathbf{Q} = \mathbf{I}_k \}, \end{aligned}$$

and the maximum is attained by  $\mathbf{Q} = (\mathbf{u}_1 \dots \mathbf{u}_k)$  for any orthonormal basis  $(\mathbf{u}_1, \dots, \mathbf{u}_n)$

of  $F^n$  with  $\mathbf{A}\mathbf{u}_i = \lambda_i(\mathbf{A})\mathbf{u}_i$ .

*Idea.* The two right-hand sides are the same number written twice, since the diagonal entries of  $\mathbf{Q}^*\mathbf{A}\mathbf{Q}$  are exactly the  $k$  numbers  $\langle \mathbf{A}\mathbf{q}_i, \mathbf{q}_i \rangle$ . That reformulation is the whole idea: the quantity to be maximized is the **trace of a compression**, and [Theorem 16.6.1](#) already bounds every eigenvalue of a compression by the matching eigenvalue of  $\mathbf{A}$ . Add  $k$  such bounds and read the left-hand side as a trace. The maximizing  $\mathbf{Q}$  is the one that makes the compression diagonal with the top eigenvalues on it, and there is no cleverness in guessing it.

*Proof.* Write  $\mathbf{B} = \mathbf{Q}^*\mathbf{A}\mathbf{Q}$  for a matrix  $\mathbf{Q}$  with orthonormal columns  $\mathbf{q}_1, \dots, \mathbf{q}_k$ . Its diagonal entries are  $b_{ii} = \langle \mathbf{A}\mathbf{q}_i, \mathbf{q}_i \rangle$  by [Definition 16.6.1](#), so

$$\operatorname{tr} \mathbf{B} = \sum_{i=1}^k \langle \mathbf{A}\mathbf{q}_i, \mathbf{q}_i \rangle,$$

and the two sets over which the maxima are taken consist of the same numbers. It therefore suffices to prove the statement for  $\operatorname{tr}(\mathbf{Q}^*\mathbf{A}\mathbf{Q})$ .

**The bound.** Let  $\mathbf{Q}^*\mathbf{Q} = \mathbf{I}_k$  and put  $\mathbf{B} = \mathbf{Q}^*\mathbf{A}\mathbf{Q}$ , a Hermitian matrix in  $M_k(F)$ . By [Corollary 11.1.4](#),  $\operatorname{tr} \mathbf{B} = \sum_{i=1}^k \lambda_i(\mathbf{B})$ . By [Theorem 16.6.1](#),  $\lambda_i(\mathbf{B}) \leq \lambda_i(\mathbf{A})$  for every  $1 \leq i \leq k$ . Summing these  $k$  inequalities,

$$\operatorname{tr}(\mathbf{Q}^*\mathbf{A}\mathbf{Q}) = \sum_{i=1}^k \lambda_i(\mathbf{B}) \leq \sum_{i=1}^k \lambda_i(\mathbf{A}).$$

**Attainment.** Since  $\mathbf{A}$  is Hermitian, [Corollary 11.4.3](#) (for  $F = \mathbb{C}$ ) or [Corollary 11.5.2](#) (for  $F = \mathbb{R}$ ) provides an orthonormal basis  $(\mathbf{u}_1, \dots, \mathbf{u}_n)$  of  $F^n$  of eigenvectors, in any prescribed order; take the order with  $\mathbf{A}\mathbf{u}_i = \lambda_i(\mathbf{A})\mathbf{u}_i$ . Let  $\mathbf{Q}$  have columns  $\mathbf{u}_1, \dots, \mathbf{u}_k$ ; these are orthonormal, so  $\mathbf{Q}^*\mathbf{Q} = \mathbf{I}_k$ . Then

$$\langle \mathbf{A}\mathbf{u}_i, \mathbf{u}_i \rangle = \lambda_i(\mathbf{A})\langle \mathbf{u}_i, \mathbf{u}_i \rangle = \lambda_i(\mathbf{A}),$$

$$\text{so } \operatorname{tr}(\mathbf{Q}^*\mathbf{A}\mathbf{Q}) = \sum_{i=1}^k \lambda_i(\mathbf{A}).$$

The bound is therefore attained, so the supremum is a maximum and equals  $\sum_{i=1}^k \lambda_i(\mathbf{A})$ . This proves the theorem. ■

For  $k = 1$  this is [Lemma 12.5.2](#) of Chapter 12:  $\lambda_1$  is the largest value of  $\langle \mathbf{A}\mathbf{x}, \mathbf{x} \rangle$  on the unit sphere. For  $k = n$  it degenerates in an instructive way: every  $\mathbf{Q}$  is unitary, every  $\mathbf{Q}^*\mathbf{A}\mathbf{Q}$  is unitarily similar to  $\mathbf{A}$ , and the “maximum” is attained everywhere, at the common value  $\operatorname{tr} \mathbf{A}$ . The content is in between.

**Example. (Which pairs of directions capture the most)**

Let

$$\mathbf{A} = \begin{pmatrix} 2 & 1 & 0 \\ 1 & 2 & 0 \\ 0 & 0 & -1 \end{pmatrix}.$$

Find  $\max\{\operatorname{tr}(\mathbf{Q}^\top \mathbf{A} \mathbf{Q}) : \mathbf{Q}^\top \mathbf{Q} = \mathbf{I}_2\}$  and a maximizer, and compare the values at  $\mathbf{Q} = (\mathbf{e}_1 \ \mathbf{e}_2)$  and  $\mathbf{Q} = (\mathbf{e}_1 \ \mathbf{e}_3)$ .

*Solution.* The eigenvalues:  $\mathbf{e}_3$  is an eigenvector for  $-1$ , and the top-left block  $\begin{pmatrix} 2 & 1 \\ 1 & 2 \end{pmatrix}$  has eigenvectors  $(1, 1)$  and  $(1, -1)$  for  $3$  and  $1$ . So  $\lambda_1 = 3$ ,  $\lambda_2 = 1$ ,  $\lambda_3 = -1$ , with orthonormal eigenvectors  $\mathbf{u}_1 = \frac{1}{\sqrt{2}}(1, 1, 0)$ ,  $\mathbf{u}_2 = \frac{1}{\sqrt{2}}(1, -1, 0)$ ,  $\mathbf{u}_3 = \mathbf{e}_3$ . By [Theorem 16.6.2](#) the maximum is  $3 + 1 = 4$ , attained at  $\mathbf{Q} = (\mathbf{u}_1 \ \mathbf{u}_2)$ .

At  $\mathbf{Q} = (\mathbf{e}_1 \ \mathbf{e}_2)$  the trace is  $a_{11} + a_{22} = 4$ : this is also a maximizer. That is no accident —  $\operatorname{span}(\mathbf{e}_1, \mathbf{e}_2) = \operatorname{span}(\mathbf{u}_1, \mathbf{u}_2)$ , and the trace of a compression, like its eigenvalues, depends only on the subspace. At  $\mathbf{Q} = (\mathbf{e}_1 \ \mathbf{e}_3)$  the trace is  $2 + (-1) = 1$ , well below the ceiling, as [Theorem 16.6.2](#) requires.

The simplest competitors are the standard basis vectors, and they give a statement about the diagonal of  $\mathbf{A}$  for free.

### Corollary 16.6.3 — Partial Sums of the Diagonal

Let  $\mathbf{A} \in M_n(F)$  be Hermitian with diagonal entries  $a_{11}, \dots, a_{nn}$ , and let  $1 \leq i_1 < \dots < i_k \leq n$ . Then

$$\sum_{j=1}^k a_{i_j i_j} \leq \sum_{i=1}^k \lambda_i(\mathbf{A}),$$

with equality for  $k = n$ .

*Proof.* The list  $(\mathbf{e}_{i_1}, \dots, \mathbf{e}_{i_k})$  is orthonormal, and  $\langle \mathbf{A} \mathbf{e}_i, \mathbf{e}_i \rangle = a_{ii}$ . Apply [Theorem 16.6.2](#). For  $k = n$  both sides equal  $\operatorname{tr} \mathbf{A}$ , by [Corollary 11.1.4](#). ■

*Quick check.* Can a Hermitian  $3 \times 3$  matrix with diagonal entries  $5, 1, 0$  have eigenvalues  $4, 2, 0$ ? Can it have eigenvalues  $6, 0, 0$ ?

*Solution.* Not the first: the case  $k = 1$  of [Corollary 16.6.3](#) says every diagonal entry is at most  $\lambda_1$ , and  $5 > 4$ . The second passes every test —  $5 \leq 6$ ,  $5 + 1 \leq 6$ ,  $6 = 6$  — and is realized: with  $\mathbf{v} = (\sqrt{5}, 1, 0)$ , the matrix  $\mathbf{v} \mathbf{v}^\top$  has diagonal  $5, 1, 0$ ; it sends  $\mathbf{v}$  to  $\|\mathbf{v}\|^2 \mathbf{v} = 6\mathbf{v}$  and kills the plane orthogonal to  $\mathbf{v}$ , so its eigenvalues are  $6, 0, 0$ . Whether passing every test is always enough is the converse question, which Section 8 takes up.

## 16.6.4 The promise from Chapter 12

Chapter 12 §10 proved a bound on how much of a matrix an orthonormal list can capture ([Lemma 12.10.2](#)), and said of it: it “is the extreme case of a family of statements about sums of the largest eigenvalues of a Hermitian matrix; Chapter 16 develops the family through the Courant–Fischer min–max theorem”. [Theorem 16.6.2](#) is that family. Here is the lemma, recovered from it in four lines.

**Corollary 16.6.4 — The Capture Bound as a Case of Ky Fan**

Let  $\mathbf{A} \in M_{m \times n}(F)$  with singular values  $\sigma_1 \geq \cdots \geq \sigma_p$ ,  $p = \min(m, n)$ , extended by  $\sigma_i = 0$  for  $p < i \leq m$ , and let  $\mathbf{q}_1, \dots, \mathbf{q}_s$  be an orthonormal list in  $F^m$ . Then

$$\sum_{j=1}^s \|\mathbf{A}^* \mathbf{q}_j\|^2 \leq \sum_{i=1}^s \sigma_i^2.$$

*Proof.* If  $s = 0$  both sides are empty sums, equal to 0; so assume  $s \geq 1$ . Put  $\mathbf{M} = \mathbf{A}\mathbf{A}^* \in M_m(F)$ , which is Hermitian. For each  $j$ ,

$$\|\mathbf{A}^* \mathbf{q}_j\|^2 = (\mathbf{A}^* \mathbf{q}_j)^* (\mathbf{A}^* \mathbf{q}_j) = \mathbf{q}_j^* \mathbf{M} \mathbf{q}_j = \langle \mathbf{M} \mathbf{q}_j, \mathbf{q}_j \rangle.$$

Fix a singular value decomposition  $\mathbf{A} = \mathbf{U}\mathbf{\Sigma}\mathbf{V}^*$  (Theorem 12.8.1). Since  $\mathbf{V}^*\mathbf{V} = \mathbf{I}_n$ ,

$$\mathbf{M} = \mathbf{U}\mathbf{\Sigma}\mathbf{V}^*\mathbf{V}\mathbf{\Sigma}^*\mathbf{U}^* = \mathbf{U}(\mathbf{\Sigma}\mathbf{\Sigma}^*)\mathbf{U}^*,$$

and  $\mathbf{\Sigma}\mathbf{\Sigma}^*$  is the  $m \times m$  diagonal matrix with diagonal  $\sigma_1^2, \dots, \sigma_p^2$  followed by  $m - p$  zeros — that is,  $\sigma_1^2 \geq \cdots \geq \sigma_m^2$  in the extended notation. So this display is a unitary diagonalization of  $\mathbf{M}$ , and Corollary 11.4.3 identifies its diagonal as the eigenvalue list:  $\lambda_i(\mathbf{M}) = \sigma_i^2$  for  $1 \leq i \leq m$ . Now Theorem 16.6.2 applied to  $\mathbf{M}$  with  $k = s$  gives

$$\sum_{j=1}^s \langle \mathbf{M} \mathbf{q}_j, \mathbf{q}_j \rangle \leq \sum_{i=1}^s \lambda_i(\mathbf{M}) = \sum_{i=1}^s \sigma_i^2,$$

which is the claim. ■

This is Lemma 12.10.2 exactly, hypotheses and conclusion, so the debt is paid. Two things are worth saying about the shape of the payment.

First, Chapter 12 had to prove the inequality by hand, weight by weight, because it had no min–max theorem; the argument there is a careful estimate on a weighted sum of a decreasing sequence. Here the same estimate is invisible: it has been absorbed into Theorem 16.2.3, whose proof is a dimension count, and passed on through Theorem 16.6.1.

Second, the words “extreme case”. Strictly, the lemma is a *special* case of the family rather than an extreme one: it is the instance in which the Hermitian matrix is a product  $\mathbf{A}\mathbf{A}^*$ , so that every eigenvalue is a squared singular value. Its crudest instance,  $s = 1$ , says  $\|\mathbf{A}^* \mathbf{q}\|^2 \leq \sigma_1^2$  for a unit  $\mathbf{q}$ , that is,  $\|\mathbf{A}^*\|_2 = \sigma_1$ . The same one line of Ky Fan, run with the other product — Theorem 16.6.2 with  $k = 1$  applied to the Hermitian matrix  $\mathbf{A}^* \mathbf{A} \in M_n(F)$  — whose largest eigenvalue is  $\sigma_1^2$  by Definition 12.8.1 — gives

$$\max_{\|\mathbf{x}\|=1} \|\mathbf{A}\mathbf{x}\|^2 = \max_{\|\mathbf{x}\|=1} \langle \mathbf{A}^* \mathbf{A} \mathbf{x}, \mathbf{x} \rangle = \lambda_1(\mathbf{A}^* \mathbf{A}) = \sigma_1^2.$$

So the largest singular value is the largest stretching factor, which is Theorem 15.3.3 (c),  $\|\mathbf{A}\|_2 = \sigma_1(\mathbf{A})$ . Chapter 15 proved that from the singular value decomposition; here it is one line of Ky Fan.

## 16.6.5 Sums are convex; individual eigenvalues are not

### Corollary 16.6.5 — Subadditivity of the Top Partial Sums

Let  $\mathbf{A}, \mathbf{B} \in M_n(F)$  be Hermitian and let  $1 \leq k \leq n$ . Then

$$\sum_{i=1}^k \lambda_i(\mathbf{A} + \mathbf{B}) \leq \sum_{i=1}^k \lambda_i(\mathbf{A}) + \sum_{i=1}^k \lambda_i(\mathbf{B}).$$

*Proof.* By [Theorem 16.6.2](#) applied to the Hermitian matrix  $\mathbf{A} + \mathbf{B}$ , there is a  $\mathbf{Q} \in M_{n \times k}(F)$  with  $\mathbf{Q}^* \mathbf{Q} = \mathbf{I}_k$  attaining the maximum, so that

$$\sum_{i=1}^k \lambda_i(\mathbf{A} + \mathbf{B}) = \text{tr}(\mathbf{Q}^*(\mathbf{A} + \mathbf{B})\mathbf{Q}).$$

By [Theorem 0.9.6](#) (1) the right-hand side is  $\text{tr}(\mathbf{Q}^* \mathbf{A} \mathbf{Q}) + \text{tr}(\mathbf{Q}^* \mathbf{B} \mathbf{Q})$ . This one  $\mathbf{Q}$  is a competitor in each of the two maximization problems of [Theorem 16.6.2](#), for  $\mathbf{A}$  and for  $\mathbf{B}$ , so each summand is at most the corresponding maximum. Adding the two bounds proves the corollary. ■

Write  $s_k(\mathbf{A}) = \sum_{i \leq k} \lambda_i(\mathbf{A})$  for the function just bounded. Two properties have now been proved: it is **subadditive**,  $s_k(\mathbf{A} + \mathbf{B}) \leq s_k(\mathbf{A}) + s_k(\mathbf{B})$ , and it is **positively homogeneous**,  $s_k(t\mathbf{A}) = ts_k(\mathbf{A})$  for  $t \geq 0$ , since multiplying a Hermitian matrix by  $t \geq 0$  multiplies each eigenvalue by  $t$  without disturbing the order. Together these say that for  $t \in [0, 1]$ ,

$$s_k(t\mathbf{A} + (1-t)\mathbf{B}) \leq ts_k(\mathbf{A}) + (1-t)s_k(\mathbf{B}) :$$

$s_k$  is a **convex** function on the real vector space of Hermitian matrices. That is exactly what one expects of a maximum of linear functions, which is what [Theorem 16.6.2](#) says  $s_k$  is: each fixed  $\mathbf{Q}$  makes  $\mathbf{A} \mapsto \text{tr}(\mathbf{Q}^* \mathbf{A} \mathbf{Q})$  linear, and  $s_k$  is the pointwise maximum of this family. Chapter 17 studies convex functions in general, and the observation that a pointwise supremum of linear functions is convex is one of its basic facts.

### Corollary 16.6.6 — The Bottom Partial Sums as a Minimum

Let  $\mathbf{A} \in M_n(F)$  be Hermitian and let  $1 \leq k \leq n$ . Then

$$\sum_{i=n-k+1}^n \lambda_i(\mathbf{A}) = \min \{ \text{tr}(\mathbf{Q}^* \mathbf{A} \mathbf{Q}) : \mathbf{Q}^* \mathbf{Q} = \mathbf{I}_k \},$$

the minimum being attained at the bottom  $k$  eigenvectors.

*Proof.* The eigenvalues of  $-\mathbf{A}$ , with multiplicity, are the negatives of those of  $\mathbf{A}$ , and negation reverses order, so

$$\lambda_i(-\mathbf{A}) = -\lambda_{n+1-i}(\mathbf{A}) \quad (1 \leq i \leq n).$$

Apply [Theorem 16.6.2](#) to  $-\mathbf{A}$ . Its left-hand side is

$$\sum_{i=1}^k \lambda_i(-\mathbf{A}) = - \sum_{i=1}^k \lambda_{n+1-i}(\mathbf{A}) = - \sum_{i=n-k+1}^n \lambda_i(\mathbf{A}),$$

the last step re-indexing the sum. Its right-hand side is

$$\max_{\mathbf{Q}^* \mathbf{Q} = \mathbf{I}_k} \operatorname{tr}(\mathbf{Q}^*(-\mathbf{A})\mathbf{Q}) = - \min_{\mathbf{Q}^* \mathbf{Q} = \mathbf{I}_k} \operatorname{tr}(\mathbf{Q}^* \mathbf{A} \mathbf{Q}),$$

using [Theorem 0.9.6](#) (1) and the fact that maximizing  $-g$  is minimizing  $g$ . Negating both sides gives the claim, and the maximizer for  $-\mathbf{A}$  is a list of top eigenvectors of  $-\mathbf{A}$ , which are bottom eigenvectors of  $\mathbf{A}$ . ■

*Warning. Only the sums are convex. A single eigenvalue  $\lambda_k$  with  $1 < k < n$  is neither convex nor concave in  $\mathbf{A}$ .* Both failures happen among  $3 \times 3$  diagonal matrices, where  $\lambda_2$  is just the middle entry; for general  $n$  and  $1 < k < n$ , take the direct sum of each witness with  $2\mathbf{I}_{k-2}$  above and  $-\mathbf{I}_{n-k-1}$  below, which moves the middle entry to position  $k$  without changing the computation.

*Not convex.* Take  $\mathbf{A} = \operatorname{diag}(1, 0, 0)$  and  $\mathbf{B} = \operatorname{diag}(0, 1, 0)$ , so  $\lambda_2(\mathbf{A}) = \lambda_2(\mathbf{B}) = 0$ . Their midpoint is  $\frac{1}{2}(\mathbf{A} + \mathbf{B}) = \operatorname{diag}(\frac{1}{2}, \frac{1}{2}, 0)$ , with  $\lambda_2 = \frac{1}{2}$ . Convexity would require  $\frac{1}{2} \leq 0$ .

*Not concave.* Take  $\mathbf{A} = \operatorname{diag}(1, 1, 0)$  and  $\mathbf{B} = \operatorname{diag}(1, 0, 1)$ , so  $\lambda_2(\mathbf{A}) = \lambda_2(\mathbf{B}) = 1$ . Their midpoint is  $\operatorname{diag}(1, \frac{1}{2}, \frac{1}{2})$ , with  $\lambda_2 = \frac{1}{2}$ . Concavity would require  $\frac{1}{2} \geq 1$ .

The partial sum  $s_2$  satisfies the **convexity** inequality on both pairs, as it must: in the first pair its values at the two matrices and their midpoint are  $1, 1, 1$ , and in the second  $2, 2, \frac{3}{2}$ . The moral is that  $\lambda_k = s_k - s_{k-1}$  is a **difference** of two convex functions, and a difference of convex functions has, in general, no convexity property at all. This is why the sharpest statements in this chapter are about sums, and why a statement about a single intermediate eigenvalue usually has to be proved as a statement about two sums.

## 16.6.6 Exercises

### 16.6.6.1 A. Check your understanding

#### Exercise A1.

- Define the compression of  $\mathbf{A}$  to a subspace, and say what has to be checked before the phrase “the eigenvalues of the compression” makes sense.
- State [Theorem 16.6.2](#), including where the maximum is attained.
- Explain why the maximum in [Theorem 16.6.2](#) does not need an appeal to compactness.
- Is  $\mathbf{A} \mapsto \lambda_1(\mathbf{A})$  convex on the Hermitian matrices? Is  $\mathbf{A} \mapsto \lambda_n(\mathbf{A})$ ? Justify each answer.
- State [Theorem 16.6.1](#) and say which case of it is [Theorem 16.4.2](#).

*Solution.*

- (a) For an orthonormal list  $\mathbf{q}_1, \dots, \mathbf{q}_k$  with matrix  $\mathbf{Q}$ , the compression of  $\mathbf{A}$  to  $W = \text{span}(\mathbf{q}_1, \dots, \mathbf{q}_k)$  is  $\mathbf{Q}^* \mathbf{A} \mathbf{Q}$ , the  $k \times k$  matrix with entries  $\langle \mathbf{A} \mathbf{q}_j, \mathbf{q}_i \rangle$ . What must be checked is that changing the orthonormal basis of  $W$  changes  $\mathbf{Q}^* \mathbf{A} \mathbf{Q}$  only by a unitary similarity, so the eigenvalues depend on  $W$  alone.
- (b) For Hermitian  $\mathbf{A} \in M_n(F)$  and  $1 \leq k \leq n$ ,  $\sum_{i \leq k} \lambda_i(\mathbf{A})$  is the maximum of  $\text{tr}(\mathbf{Q}^* \mathbf{A} \mathbf{Q})$  over all  $\mathbf{Q} \in M_{n \times k}(F)$  with  $\mathbf{Q}^* \mathbf{Q} = \mathbf{I}_k$ , equivalently the maximum of  $\sum_{i \leq k} \langle \mathbf{A} \mathbf{q}_i, \mathbf{q}_i \rangle$  over orthonormal lists. It is attained at  $\mathbf{Q} = (\mathbf{u}_1 \dots \mathbf{u}_k)$  for an orthonormal eigenbasis ordered with  $\mathbf{A} \mathbf{u}_i = \lambda_i(\mathbf{A}) \mathbf{u}_i$ .
- (c) Because the proof exhibits both a bound valid for every competitor and a competitor achieving it. A supremum that is achieved is a maximum, whatever the topology of the competing set.
- (d) Incorrect. Take  $\mathbf{A} = \text{diag}(1, 0, 0)$  and  $\mathbf{B} = \text{diag}(0, 1, 0)$  as in the warning:  $\lambda_2(\mathbf{A}) = \lambda_2(\mathbf{B}) = 0$  while  $\mathbf{A} + \mathbf{B} = \text{diag}(1, 1, 0)$  has  $\lambda_2 = 1$ . What is true is the statement about sums, [Corollary 16.6.5](#): here  $\lambda_1 + \lambda_2$  of the sum is 2, and  $(1+0) + (1+0) = 2$ .
- (e) If  $\mathbf{B}$  is the compression of the Hermitian  $\mathbf{A} \in M_n(F)$  to a  $k$ -dimensional subspace, then  $\lambda_i(\mathbf{A}) \geq \lambda_i(\mathbf{B}) \geq \lambda_{i+n-k}(\mathbf{A})$  for  $1 \leq i \leq k$ . Cauchy interlacing is the case  $k = n - 1$  with the subspace a coordinate hyperplane.

### 16.6.6.2 B. Practice

#### Exercise B1: A compression, and two bounds.

Let  $\mathbf{A} = \text{diag}(5, 0, -2)$  and let  $\mathbf{q}_1 = \frac{1}{3}(1, 2, 2)$ ,  $\mathbf{q}_2 = \frac{1}{3}(2, 1, -2)$ .

- (a) Verify that  $(\mathbf{q}_1, \mathbf{q}_2)$  is orthonormal and compute the compression  $\mathbf{B}$  of  $\mathbf{A}$  to  $\text{span}(\mathbf{q}_1, \mathbf{q}_2)$ .
- (b) Find the eigenvalues of  $\mathbf{B}$  and verify [Theorem 16.6.1](#).
- (c) Compare  $\text{tr } \mathbf{B}$  with the bound of [Theorem 16.6.2](#), and say how far this subspace is from optimal.

*Solution.*

- (a)  $\|\mathbf{q}_1\|^2 = (1+4+4)/9 = 1$ ,  $\|\mathbf{q}_2\|^2 = (4+1+4)/9 = 1$  and  $9\langle \mathbf{q}_1, \mathbf{q}_2 \rangle = 2+2-4 = 0$ . Next  $3\mathbf{A}\mathbf{q}_1 = (5, 0, -4)$  and  $3\mathbf{A}\mathbf{q}_2 = (10, 0, 4)$ , so

$$\begin{aligned} b_{11} &= \frac{1}{9}(1, 2, 2) \cdot (5, 0, -4) = \frac{5-8}{9} = -\frac{1}{3}, \\ b_{12} &= \frac{1}{9}(1, 2, 2) \cdot (10, 0, 4) = \frac{10+8}{9} = 2, \\ b_{22} &= \frac{1}{9}(2, 1, -2) \cdot (10, 0, 4) = \frac{20-8}{9} = \frac{4}{3}, \end{aligned}$$

$$\text{so } \mathbf{B} = \begin{pmatrix} -1/3 & 2 \\ 2 & 4/3 \end{pmatrix}.$$

- (b)  $\text{tr } \mathbf{B} = 1$  and  $\det \mathbf{B} = -\frac{4}{9} - 4 = -\frac{40}{9}$ , so  $9x^2 - 9x - 40 = 0$ , with discriminant  $81 + 1440 = 1521 = 39^2$  and roots  $(9 \pm 39)/18$ . Hence  $\lambda_1(\mathbf{B}) = \frac{8}{3}$  and  $\lambda_2(\mathbf{B}) = -\frac{5}{3}$ . The separation  $\lambda_i(\mathbf{A}) \geq \lambda_i(\mathbf{B}) \geq \lambda_{i+1}(\mathbf{A})$  reads

$$5 \geq \frac{8}{3} \geq 0 \geq -\frac{5}{3} \geq -2,$$

all strict.

- (c)  $\text{tr } \mathbf{B} = 1$ , while [Theorem 16.6.2](#) gives  $\lambda_1(\mathbf{A}) + \lambda_2(\mathbf{A}) = 5 + 0 = 5$  as the maximum, attained at  $\text{span}(\mathbf{e}_1, \mathbf{e}_2)$ . So this subspace scores 1 out of a possible 5; it is far from optimal, and one can see why. Between them  $\mathbf{q}_1$  and  $\mathbf{q}_2$  put weight  $\frac{4}{9} + \frac{4}{9} = \frac{8}{9}$  on the third coordinate, where  $\mathbf{A}$  is  $-2$ , and only  $\frac{1}{9} + \frac{4}{9} = \frac{5}{9}$  on the first, where it is 5: indeed  $5 \cdot \frac{5}{9} + 0 - 2 \cdot \frac{8}{9} = 1$ .

### Exercise B2: Reading the sums off the spectrum.

A Hermitian  $\mathbf{A} \in M_4(\mathbb{R})$  has eigenvalues 6, 4, 1,  $-3$ .

- (a) Compute  $\max\{\text{tr}(\mathbf{Q}^\top \mathbf{A} \mathbf{Q}) : \mathbf{Q}^\top \mathbf{Q} = \mathbf{I}_k\}$  for  $k = 1, 2, 3, 4$ .  
 (b) Compute the corresponding minima, using [Corollary 16.6.6](#).  
 (c) For  $\mathbf{A} = \text{diag}(6, 4, 1, -3)$  and  $\mathbf{q}_1 = \frac{1}{\sqrt{2}}(1, 1, 0, 0)$ ,  $\mathbf{q}_2 = \frac{1}{\sqrt{2}}(0, 0, 1, 1)$ , compute  $\text{tr}(\mathbf{Q}^\top \mathbf{A} \mathbf{Q})$  and check it against (a) and (b).

*Solution.*

- (a) By [Theorem 16.6.2](#) the maxima are the partial sums from the top: 6, 10, 11, 8.  
 (b) By [Corollary 16.6.6](#) the minima are the partial sums from the bottom:  $-3$ ,  $-2$ , 2, 8. The two agree at  $k = 4$ , as they must, both being  $\text{tr } \mathbf{A}$ .  
 (c)  $\langle \mathbf{A} \mathbf{q}_1, \mathbf{q}_1 \rangle = \frac{1}{2}(6 + 4) = 5$  and  $\langle \mathbf{A} \mathbf{q}_2, \mathbf{q}_2 \rangle = \frac{1}{2}(1 - 3) = -1$ , so  $\text{tr}(\mathbf{Q}^\top \mathbf{A} \mathbf{Q}) = 4$ . Indeed  $-2 \leq 4 \leq 10$ , as (a) and (b) require for  $k = 2$ .

### Exercise B3: Subadditivity, and how loose it can be.

Let

$$\mathbf{A} = \begin{pmatrix} 1 & 1 \\ 1 & 1 \end{pmatrix}, \quad \mathbf{B} = \begin{pmatrix} 1 & -1 \\ -1 & 1 \end{pmatrix}.$$

- (a) Compute the eigenvalues of  $\mathbf{A}$ , of  $\mathbf{B}$  and of  $\mathbf{A} + \mathbf{B}$ .  
 (b) Verify [Corollary 16.6.5](#) for  $k = 1$  and  $k = 2$ , and say for which  $k$  it is strict.  
 (c) Give a pair of Hermitian  $2 \times 2$  matrices for which the  $k = 1$  inequality is an equality.

*Solution.*

- (a)  $\mathbf{A}$  has trace 2 and determinant 0, so its eigenvalues are 2, 0; the same for  $\mathbf{B}$ . And  $\mathbf{A} + \mathbf{B} = 2\mathbf{I}_2$  has eigenvalues 2, 2.  
 (b) For  $k = 1$ :  $\lambda_1(\mathbf{A} + \mathbf{B}) = 2$  and  $\lambda_1(\mathbf{A}) + \lambda_1(\mathbf{B}) = 4$ , so  $2 \leq 4$ , strictly. For  $k = 2$ :  $4 \leq 2 + 2 = 4$ , an equality — as it must be, since for  $k = n$  both sides are traces and [Theorem 0.9.6](#) (1) makes the trace additive.  
 (c) Take  $\mathbf{A} = \mathbf{B} = \text{diag}(1, 0)$ . Then  $\mathbf{A} + \mathbf{B} = \text{diag}(2, 0)$  and  $\lambda_1(\mathbf{A} + \mathbf{B}) = 2 = 1 + 1$ . More generally equality holds at  $k = 1$  whenever  $\mathbf{A}$  and  $\mathbf{B}$  have a common top eigenvector, because then that one vector attains all three maxima at once.

#### 16.6.6.3 C. Going deeper



**Exercise C1: The Loewner order and the partial sums.**

Let  $\mathbf{A}, \mathbf{B} \in M_n(F)$  be Hermitian.

- (a) Prove that  $\mathbf{A} \succeq \mathbf{B}$  implies  $\sum_{i \leq k} \lambda_i(\mathbf{A}) \geq \sum_{i \leq k} \lambda_i(\mathbf{B})$  for every  $k$ . *Hint: compare the two maximization problems of Theorem 16.6.2 competitor by competitor.*
- (b) Show that the converse fails: give Hermitian  $\mathbf{A}, \mathbf{B}$  satisfying all  $n$  of these inequalities with  $\mathbf{A} \not\succeq \mathbf{B}$ .

*Solution.*

- (a) Let  $\mathbf{Q}^* \mathbf{Q} = \mathbf{I}_k$  with columns  $\mathbf{q}_1, \dots, \mathbf{q}_k$ . Since  $\mathbf{A} - \mathbf{B} \succeq 0$ , clause (P2) of Definition 12.1.1 gives  $\langle (\mathbf{A} - \mathbf{B})\mathbf{q}_i, \mathbf{q}_i \rangle \geq 0$  for each  $i$ , that is,  $\langle \mathbf{A}\mathbf{q}_i, \mathbf{q}_i \rangle \geq \langle \mathbf{B}\mathbf{q}_i, \mathbf{q}_i \rangle$ . Summing over  $i$ ,

$$\operatorname{tr}(\mathbf{Q}^* \mathbf{A} \mathbf{Q}) \geq \operatorname{tr}(\mathbf{Q}^* \mathbf{B} \mathbf{Q}).$$

Now take  $\mathbf{Q}$  to be a maximizer for  $\mathbf{B}$  in Theorem 16.6.2. Then

$$\sum_{i \leq k} \lambda_i(\mathbf{B}) = \operatorname{tr}(\mathbf{Q}^* \mathbf{B} \mathbf{Q}) \leq \operatorname{tr}(\mathbf{Q}^* \mathbf{A} \mathbf{Q}) \leq \sum_{i \leq k} \lambda_i(\mathbf{A}),$$

the last step because  $\mathbf{Q}$  is one competitor in the maximization for  $\mathbf{A}$ .

- (b) Take  $\mathbf{A} = \operatorname{diag}(2, 0)$  and  $\mathbf{B} = \operatorname{diag}(1, 1)$ . The partial sums are  $2 \geq 1$  for  $k = 1$  and  $2 \geq 2$  for  $k = 2$ . But  $\mathbf{A} - \mathbf{B} = \operatorname{diag}(1, -1)$  has a negative eigenvalue, so  $\mathbf{A} - \mathbf{B} \not\succeq 0$  by Theorem 12.1.4 (b), and  $\mathbf{A} \not\succeq \mathbf{B}$ . Concretely,  $\langle (\mathbf{A} - \mathbf{B})\mathbf{e}_2, \mathbf{e}_2 \rangle = -1 < 0$ .

**Exercise C2: Partial sums that behave like a norm.**

For positive semidefinite  $\mathbf{A} \in M_n(F)$  and  $1 \leq k \leq n$ , consider  $s_k(\mathbf{A}) = \sum_{i \leq k} \lambda_i(\mathbf{A})$ , the partial sum of this section, now restricted to the positive semidefinite matrices.

- (a) Prove that  $N_k(\mathbf{A}) \geq 0$ , with equality if and only if  $\mathbf{A} = \mathbf{0}$ .
- (b) Prove that  $N_k(c\mathbf{A}) = cN_k(\mathbf{A})$  for every real  $c \geq 0$ , and that  $N_k(\mathbf{A} + \mathbf{B}) \leq N_k(\mathbf{A}) + N_k(\mathbf{B})$  for positive semidefinite  $\mathbf{A}, \mathbf{B}$ .
- (c) The positive semidefinite matrices form a **cone**, not a vector space, so a norm cannot be defined on them. Deduce instead that  $s_k$  is positive, positively homogeneous and subadditive on this cone. Which step of your argument for (a) uses positive semidefiniteness, and what goes wrong without it?

*Solution.*

- (a) By Theorem 12.1.4 (b) every eigenvalue of  $\mathbf{A}$  is  $\geq 0$ , so  $N_k(\mathbf{A}) \geq \lambda_1(\mathbf{A}) \geq 0$  for  $k \geq 1$ . If  $N_k(\mathbf{A}) = 0$  then in particular  $\lambda_1(\mathbf{A}) = 0$ , so every eigenvalue is 0; by Corollary 11.4.3 (or Corollary 11.5.2)  $\mathbf{A} = \mathbf{U}\mathbf{0}\mathbf{U}^* = \mathbf{0}$ . Conversely  $N_k(\mathbf{0}) = 0$ .
- (b) Multiplying a Hermitian matrix by  $c \geq 0$  multiplies each eigenvalue by  $c$  and preserves the decreasing order, so  $\lambda_i(c\mathbf{A}) = c\lambda_i(\mathbf{A})$  and the first claim follows by summing. The second is Corollary 16.6.5, which needs no positivity at all.
- (c) Positivity with  $N_k(\mathbf{A}) = 0$  only for  $\mathbf{A} = \mathbf{0}$  is (a); homogeneity under scalars  $c \geq 0$  is (b); subadditivity is (b). So  $N_k$  behaves as a norm wherever it is defined. It cannot be extended to all Hermitian matrices by the same formula: for  $\mathbf{A} = \operatorname{diag}(0, -1)$

and  $k = 1$  we get  $N_1(\mathbf{A}) = 0$  with  $\mathbf{A} \neq \mathbf{0}$ , so the positivity clause fails, and for  $c = -1$  the value  $N_1(-\mathbf{A}) = 1$  differs from  $|c|N_1(\mathbf{A}) = 0$ . The repair is to apply  $N_k$  to the singular values instead of the eigenvalues, which is what Chapter 20 does.

### Exercise C3: How much of a matrix a subspace holds.

Let  $\mathbf{A} \in M_{m \times n}(F)$  with singular values  $\sigma_1 \geq \cdots \geq \sigma_p$  extended by zeros as in [Corollary 16.6.4](#), let  $S \subseteq F^m$  have dimension  $k$ , and let  $\mathbf{Q}$  have as columns an orthonormal basis of  $S$ , so that  $P_S = \mathbf{Q}\mathbf{Q}^*$  by [Theorem 10.3.3](#) (a).

- (a) Prove that  $\|P_S \mathbf{A}\|_F^2 = \text{tr}(\mathbf{Q}^* \mathbf{A} \mathbf{A}^* \mathbf{Q})$ .
- (b) Deduce from [Theorem 16.6.2](#) that  $\|P_S \mathbf{A}\|_F^2 \leq \sigma_1^2 + \cdots + \sigma_k^2$ , with equality when  $S$  is spanned by the first  $k$  left singular vectors.
- (c) Hence recover the value  $\sum_{i>k} \sigma_i^2$  of the minimum in [Corollary 12.10.4](#). *Hint:  $\mathbf{a} - P_S \mathbf{a}$  is orthogonal to  $P_S \mathbf{a}$ .*

*Solution.*

- (a) The Frobenius norm satisfies  $\|\mathbf{X}\|_F^2 = \text{tr}(\mathbf{X}^* \mathbf{X})$ , because the  $(j, j)$  entry of  $\mathbf{X}^* \mathbf{X}$  is the squared length of the  $j$ -th column of  $\mathbf{X}$ . With  $P_S = \mathbf{Q}\mathbf{Q}^*$ , which is Hermitian and satisfies  $P_S^2 = \mathbf{Q}(\mathbf{Q}^* \mathbf{Q})\mathbf{Q}^* = P_S$ ,

$$\|P_S \mathbf{A}\|_F^2 = \text{tr}(\mathbf{A}^* P_S^* P_S \mathbf{A}) = \text{tr}(\mathbf{A}^* \mathbf{Q}\mathbf{Q}^* \mathbf{A}) = \text{tr}(\mathbf{Q}^* \mathbf{A} \mathbf{A}^* \mathbf{Q}),$$

the last step by [Theorem 0.9.6](#) (3) applied to  $\mathbf{A}^* \mathbf{Q}$  and  $\mathbf{Q}^* \mathbf{A}$ .

- (b) The matrix  $\mathbf{M} = \mathbf{A} \mathbf{A}^*$  is Hermitian with  $\lambda_i(\mathbf{M}) = \sigma_i^2$ , as computed in the proof of [Corollary 16.6.4](#). By [Theorem 16.6.2](#) with  $k$  columns,

$$\text{tr}(\mathbf{Q}^* \mathbf{M} \mathbf{Q}) \leq \sum_{i \leq k} \lambda_i(\mathbf{M}) = \sum_{i \leq k} \sigma_i^2,$$

with equality when the columns of  $\mathbf{Q}$  are top eigenvectors of  $\mathbf{M}$ . From  $\mathbf{A} = \mathbf{U} \mathbf{\Sigma} \mathbf{V}^*$  the columns  $\mathbf{u}_1, \dots, \mathbf{u}_m$  of  $\mathbf{U}$  are exactly such an eigenbasis, so equality holds for  $S = \text{span}(\mathbf{u}_1, \dots, \mathbf{u}_k)$ .

- (c) Write  $\mathbf{a}_1, \dots, \mathbf{a}_n$  for the columns of  $\mathbf{A}$ . For each  $j$  the vectors  $P_S \mathbf{a}_j$  and  $\mathbf{a}_j - P_S \mathbf{a}_j$  are orthogonal, so [Theorem 10.1.3](#) gives  $\|\mathbf{a}_j\|^2 = \|P_S \mathbf{a}_j\|^2 + \|\mathbf{a}_j - P_S \mathbf{a}_j\|^2$ . Summing over  $j$ ,

$$\sum_j d(\mathbf{a}_j, S)^2 = \|\mathbf{A}\|_F^2 - \|P_S \mathbf{A}\|_F^2.$$

By [Lemma 12.10.1](#),  $\|\mathbf{A}\|_F^2 = \sum_i \sigma_i^2$ . By (b) the subtracted term is at most  $\sum_{i \leq k} \sigma_i^2$ , with equality for  $S = \text{span}(\mathbf{u}_1, \dots, \mathbf{u}_k)$ . Hence the total squared distance is at least  $\sum_{i>k} \sigma_i^2$ , and equals it for that  $S$ ; a subspace of smaller dimension  $k' < k$  does no better, since the same argument bounds what it captures by  $\sum_{i \leq k'} \sigma_i^2 \leq \sum_{i \leq k} \sigma_i^2$  — which is [Corollary 12.10.4](#), now read as a statement about a maximum of traces rather than a minimum of distances.

## 16.7 Arbitrary Index Sets

The Courant–Fischer theorem describes one eigenvalue at a time, and Ky Fan’s theorem describes the sum of the top  $k$ . Between those two lies every other sum, for instance  $\lambda_2 + \lambda_4$ , and §06 ended by warning that the sharpest statements about eigenvalues tend to be statements about sums. This section describes the sum over an **arbitrary** index set as a max–min, which is Wielandt’s theorem. It then uses that description to compare two Hermitian matrices across any index set at once, which is Lidskii’s inequality. The price of the generality is that single subspaces are replaced by nested chains of them, and the proof takes some care.

**Throughout,  $F = \mathbb{R}$  or  $F = \mathbb{C}$ , every matrix whose eigenvalues are indexed is Hermitian, and the indexing is decreasing,  $\lambda_1(\mathbf{A}) \geq \cdots \geq \lambda_n(\mathbf{A})$ , with multiplicity. An index set of size  $k$  is a list of integers  $1 \leq i_1 < i_2 < \cdots < i_k \leq n$ .**

### 16.7.1 What adding up the min–max gives

Here is the question that drives the section. Let  $\mathbf{A}$  and  $\mathbf{B}$  be Hermitian and fix an index set. How large can

$$\sum_{j=1}^k \lambda_{i_j}(\mathbf{A}) - \sum_{j=1}^k \lambda_{i_j}(\mathbf{B})$$

be, in terms of  $\mathbf{A} - \mathbf{B}$ ? One answer is immediate. By [Corollary 16.3.4](#), applied to  $\mathbf{B}$  and the Hermitian matrix  $\mathbf{A} - \mathbf{B}$ , each single difference satisfies  $\lambda_i(\mathbf{A}) - \lambda_i(\mathbf{B}) \leq \lambda_1(\mathbf{A} - \mathbf{B})$ . Adding  $k$  of these gives the bound  $k \lambda_1(\mathbf{A} - \mathbf{B})$ .

That bound is poor, and a rank-one example shows by how much. Let  $\mathbf{A} = \mathbf{B} + \mathbf{u}\mathbf{u}^*$  with  $\|\mathbf{u}\| = 1$ . By [Theorem 16.5.1](#) every shift  $\lambda_i(\mathbf{A}) - \lambda_i(\mathbf{B})$  is  $\geq 0$ , and by [Corollary 16.5.3](#) the shifts add up to exactly  $\|\mathbf{u}\|^2 = 1$ . So **every** partial sum of shifts is at most 1, whatever the index set. The eigenvalues of  $\mathbf{u}\mathbf{u}^*$  are  $1, 0, \dots, 0$ , so  $1 = \lambda_1 + \cdots + \lambda_k$  of  $\mathbf{A} - \mathbf{B}$  for every  $k$ . The summed Weyl bound says  $k$ , which is wrong by a factor of  $k$ .

The bound we would like is therefore

$$\sum_{j=1}^k \lambda_{i_j}(\mathbf{A}) - \sum_{j=1}^k \lambda_{i_j}(\mathbf{B}) \leq \sum_{j=1}^k \lambda_j(\mathbf{A} - \mathbf{B}),$$

and the obstruction is visible in the proof of Weyl’s inequality. That proof finds, for each index, **one** good vector. Summing  $k$  such bounds uses  $k$  unrelated vectors, and the only control over  $\sum_j \langle (\mathbf{A} - \mathbf{B})\mathbf{x}_j, \mathbf{x}_j \rangle$  for unrelated unit vectors is  $k \lambda_1(\mathbf{A} - \mathbf{B})$ . If the  $\mathbf{x}_j$  were **orthonormal**, [Theorem 16.6.2](#) would bound the same sum by  $\lambda_1 + \cdots + \lambda_k$  of  $\mathbf{A} - \mathbf{B}$ . So we need a single orthonormal list  $\mathbf{x}_1, \dots, \mathbf{x}_k$  that is simultaneously good for  $\mathbf{A}$  and good for  $\mathbf{B}$ :

$$\sum_j \langle \mathbf{A}\mathbf{x}_j, \mathbf{x}_j \rangle \geq \sum_j \lambda_{i_j}(\mathbf{A}), \quad \sum_j \langle \mathbf{B}\mathbf{x}_j, \mathbf{x}_j \rangle \leq \sum_j \lambda_{i_j}(\mathbf{B}).$$

The first requirement is easy to meet: it will hold whenever each  $\mathbf{x}_j$  lies in the span  $W_j$  of the top  $i_j$  eigenvectors of  $\mathbf{A}$ . Those spans are nested,  $W_1 \subset W_2 \subset \cdots \subset W_k$ , and from  $\mathbf{B}$ ’s point of view they are an arbitrary nested chain of subspaces. The second requirement is then a question about  $\mathbf{B}$  alone: **inside any nested chain of subspaces of these dimensions, is there an orthonormal list whose sum for  $\mathbf{B}$  is small?** Wielandt’s theorem answers yes, and the chain needs a name.

## 16.7.2 Flags and adapted lists

Chapter 8 §07 called a chain  $U_1 \subset U_2 \subset \cdots \subset U_n = V$  with  $\dim U_k = k$  a **flag**, and a basis whose first  $k$  members span  $U_k$  a basis adapted to it. We need chains that skip dimensions, and lists that meet each member of the chain once.

*A flag of type  $(i_1, \dots, i_k)$  is a nested chain of subspaces with prescribed dimensions; an adapted list takes its  $j$ -th vector from the  $j$ -th subspace and is orthonormal.*

### Definition 16.7.1 — Flags of a given type, and adapted lists

Let  $F = \mathbb{R}$  or  $\mathbb{C}$ , and let  $1 \leq i_1 < \cdots < i_k \leq n$  be an index set. A **flag of type**  $(i_1, \dots, i_k)$  in  $F^n$  is a list  $\mathcal{W} = (W_1, \dots, W_k)$  of subspaces of  $F^n$  with

$$W_1 \subseteq W_2 \subseteq \cdots \subseteq W_k \quad \text{and} \quad \dim W_j = i_j \quad \text{for every } j.$$

A list  $(\mathbf{x}_1, \dots, \mathbf{x}_k)$  of vectors of  $F^n$  is **adapted to**  $\mathcal{W}$  if it is **orthonormal** and  $\mathbf{x}_j \in W_j$  for **every**  $j$ .

In words. The subspaces are nested, and the  $j$ -th one has dimension equal to the  $j$ -th **index**  $i_j$ , not to  $j$ ; since the indices strictly increase, so do the dimensions, and each inclusion is proper. An adapted list has exactly one vector per subspace; the  $j$ -th vector may be taken anywhere in  $W_j$ , including in the smaller  $W_1, \dots, W_{j-1}$ , but the whole list must be orthonormal.

**Every flag has an adapted list.** Choose a basis of  $W_1$ , extend it to a basis of  $W_2$  ([Theorem 1.6.5](#)), then to  $W_3$ , and so on up to  $W_k$ . This gives a basis  $\mathbf{v}_1, \dots, \mathbf{v}_{i_k}$  of  $W_k$  whose first  $i_j$  members span  $W_j$ . Applying [Theorem 10.2.2](#) yields an orthonormal list  $\mathbf{e}_1, \dots, \mathbf{e}_{i_k}$  with  $\text{span}(\mathbf{e}_1, \dots, \mathbf{e}_{i_j}) = W_j$  for each  $j$ , and  $(\mathbf{e}_{i_1}, \dots, \mathbf{e}_{i_k})$  is adapted to  $\mathcal{W}$ .

Four cases to keep in mind.

- **One index.** For  $k = 1$  a flag is a single subspace  $W_1$  of dimension  $i_1$ , and an adapted list is a single unit vector in it.
- **An initial segment.** For the index set  $(1, 2, \dots, k)$ ,  $W_j$  has dimension  $j$ . An adapted list is then an orthonormal basis of  $W_k$  whose first  $j$  vectors span  $W_j$ , since  $j$  orthonormal vectors in the  $j$ -dimensional  $W_j$  span it. With  $k = n$  this is Chapter 8's complete flag.
- **The eigenflag.** Let  $(\mathbf{u}_1, \dots, \mathbf{u}_n)$  be an orthonormal basis of  $F^n$  with  $\mathbf{A}\mathbf{u}_p = \lambda_p(\mathbf{A})\mathbf{u}_p$ , and put  $W_j = \text{span}(\mathbf{u}_1, \dots, \mathbf{u}_{i_j})$ . This is a flag of type  $(i_1, \dots, i_k)$ , and  $(\mathbf{u}_{i_1}, \dots, \mathbf{u}_{i_k})$  is adapted to it, since  $\mathbf{u}_{i_j}$  is among the vectors spanning  $W_j$ .
- **The degenerate case**  $k = n$ . Then  $i_j = j$  is forced,  $W_n = F^n$ , and an adapted list is an orthonormal basis of  $F^n$  whose first  $j$  vectors span  $W_j$ .

A minimal change breaks adaptedness. For the eigenflag of type  $(1, 3)$  in  $F^3$ , namely  $\text{span}(\mathbf{u}_1) \subset F^3$ , the list  $(\mathbf{u}_1, \mathbf{u}_3)$  is adapted. Swap the two vectors to get  $(\mathbf{u}_3, \mathbf{u}_1)$ . It is still orthonormal and its members still lie in  $W_2 = F^3$ , but the clause  $\mathbf{x}_1 \in W_1$  fails:  $\mathbf{u}_3 \notin \text{span}(\mathbf{u}_1)$ .

Why this definition? Orthonormality is what makes  $\sum_j \langle \mathbf{A}\mathbf{x}_j, \mathbf{x}_j \rangle$  the trace of a compression, which is what [Theorem 16.6.2](#) controls. The nesting will let us orthonormalize a list by Gram–Schmidt without moving any vector out of its own subspace, since Gram–Schmidt keeps the  $j$ -th vector in the span of the first  $j$ .

*Quick check.* In  $\mathbb{R}^3$ , take the flag  $W_1 = \text{span}(\mathbf{e}_1) \subset W_2 = \text{span}(\mathbf{e}_1, \mathbf{e}_2, \mathbf{e}_3)$ , of type  $(1, 3)$ . Which of these lists are adapted to it: (a)  $(\mathbf{e}_1, \mathbf{e}_3)$ ; (b)  $(\mathbf{e}_3, \mathbf{e}_1)$ ; (c)  $(\mathbf{e}_1, \mathbf{e}_1 + \mathbf{e}_2)$ ; (d)  $(-\mathbf{e}_1, \frac{1}{\sqrt{2}}(\mathbf{e}_2 - \mathbf{e}_3))$ ?

*Solution.*

- (a) Yes: orthonormal,  $\mathbf{e}_1 \in W_1$  and  $\mathbf{e}_3 \in W_2$ . (b) No:  $\mathbf{x}_1 = \mathbf{e}_3$  is not in  $W_1$ . (c) No: the list is not orthonormal, since  $\langle \mathbf{e}_1, \mathbf{e}_1 + \mathbf{e}_2 \rangle = 1$  and  $\|\mathbf{e}_1 + \mathbf{e}_2\| = \sqrt{2}$ . (d) Yes:  $-\mathbf{e}_1 \in W_1$ , the second vector has length 1 and is orthogonal to  $\mathbf{e}_1$ , and every vector lies in  $W_2 = \mathbb{R}^3$ .

### 16.7.3 Wielandt's minimax principle

The eigenflag guarantees every adapted list a large sum, and no flag of the same type can guarantee more. That is the statement; the second half is the hard one.

#### Theorem 16.7.1 — Wielandt's Minimax Principle

Let  $\mathbf{A} \in M_n(F)$  be Hermitian, with  $F = \mathbb{R}$  or  $\mathbb{C}$ , and let  $1 \leq i_1 < \dots < i_k \leq n$ .

- (a) For **every** flag  $\mathcal{W}$  of type  $(i_1, \dots, i_k)$  there is a list  $(\mathbf{x}_1, \dots, \mathbf{x}_k)$  adapted to  $\mathcal{W}$  with

$$\sum_{j=1}^k \langle \mathbf{A}\mathbf{x}_j, \mathbf{x}_j \rangle \leq \sum_{j=1}^k \lambda_{i_j}(\mathbf{A}).$$

- (b) Let  $(\mathbf{u}_1, \dots, \mathbf{u}_n)$  be an orthonormal basis with  $\mathbf{A}\mathbf{u}_p = \lambda_p(\mathbf{A})\mathbf{u}_p$  for every  $p$ , and let  $\mathcal{W}^{\mathbf{A}}$  be the flag  $W_j = \text{span}(\mathbf{u}_1, \dots, \mathbf{u}_{i_j})$ . Then **every** list  $(\mathbf{x}_1, \dots, \mathbf{x}_k)$  adapted to  $\mathcal{W}^{\mathbf{A}}$  satisfies

$$\sum_{j=1}^k \langle \mathbf{A}\mathbf{x}_j, \mathbf{x}_j \rangle \geq \sum_{j=1}^k \lambda_{i_j}(\mathbf{A}),$$

with equality for  $\mathbf{x}_j = \mathbf{u}_{i_j}$ .

- (c) Consequently

$$\sum_{j=1}^k \lambda_{i_j}(\mathbf{A}) = \max_{\mathcal{W}} \min_{(\mathbf{x}_1, \dots, \mathbf{x}_k)} \sum_{j=1}^k \langle \mathbf{A}\mathbf{x}_j, \mathbf{x}_j \rangle,$$

where  $\mathcal{W}$  runs over all flags of type  $(i_1, \dots, i_k)$  in  $F^n$ , and for each  $\mathcal{W}$  the list runs over all lists adapted to  $\mathcal{W}$ . The inner minimum exists for every  $\mathcal{W}$ , and the outer maximum is attained at  $\mathcal{W}^{\mathbf{A}}$ .

*Idea.* Two inequalities of different characters, as in Theorem 16.2.3. In both, the sum  $\sum_j \langle \mathbf{A}\mathbf{x}_j, \mathbf{x}_j \rangle$  is the trace of the compression  $\mathbf{B}$  of  $\mathbf{A}$  to the span of the list, so it is  $\sum_j \lambda_j(\mathbf{B})$ . Both parts therefore come down to comparing  $\lambda_j(\mathbf{B})$  with  $\lambda_{i_j}(\mathbf{A})$ , one  $j$  at a time, by Courant–Fischer applied to the small matrix  $\mathbf{B}$ .

For (b), the first  $j$  vectors of the list lie in  $\text{span}(\mathbf{u}_1, \dots, \mathbf{u}_{i_j})$ , where  $R_{\mathbf{A}} \geq \lambda_{i_j}(\mathbf{A})$ , so  $\lambda_j(\mathbf{B}) \geq \lambda_{i_j}(\mathbf{A})$ .

For (a), we need a list whose span  $X$  meets each bottom span  $L_{i_j} = \text{span}(\mathbf{u}_{i_j}, \dots, \mathbf{u}_n)$

in dimension at least  $k - j + 1$ , since  $R_{\mathbf{A}} \leq \lambda_{i_j}(\mathbf{A})$  there. For a **single**  $j$ , [Lemma 16.2.1](#) gives such a meeting. For all  $j$  at once, with one  $X$ , it does not: the vectors chosen for different  $j$  can collide, and the example after the proof shows that they do. The remedy is to count in coordinates. Record, for each vector, the first eigenvector coordinate at which it is non-zero; vectors with different first coordinates are independent; and a greedy choice of these first coordinates achieves every count at once.

**Step roadmap.** A preliminary claim turns bounds on  $R_{\mathbf{A}}$  over subspaces into bounds on the eigenvalues of a compression. **Step 1** proves (b). **Step 2** introduces leading indices and makes the greedy selection. **Step 3** orthonormalizes the selected vectors and proves (a). **Step 4** proves that the inner minimum exists, which is the only place analysis enters, and deduces (c).

*Proof.* Let  $(\mathbf{u}_1, \dots, \mathbf{u}_n)$  be an orthonormal basis of  $F^n$  with  $\mathbf{A}\mathbf{u}_p = \lambda_p(\mathbf{A})\mathbf{u}_p$  for every  $p$ ; such a basis exists by [Corollary 11.4.3](#) over  $\mathbb{C}$  or [Corollary 11.5.2](#) over  $\mathbb{R}$ . For part (b) take the basis named there; for part (a) any such basis will do. Abbreviate  $\lambda_p = \lambda_p(\mathbf{A})$ . For  $1 \leq p \leq n$  put

$$L_p = \text{span}(\mathbf{u}_p, \mathbf{u}_{p+1}, \dots, \mathbf{u}_n), \quad L_{n+1} = \{\mathbf{0}\}.$$

By [Lemma 16.3.1](#),  $R_{\mathbf{A}}(\mathbf{x}) \leq \lambda_p$  for every non-zero  $\mathbf{x} \in L_p$ , and  $R_{\mathbf{A}}(\mathbf{x}) \geq \lambda_p$  for every non-zero  $\mathbf{x} \in \text{span}(\mathbf{u}_1, \dots, \mathbf{u}_p)$ .

*Claim.* **Claim 1 (compression bounds).** Let  $(\mathbf{y}_1, \dots, \mathbf{y}_k)$  be orthonormal, let  $Y$  be its span, let  $\mathbf{Q} \in M_{n \times k}(F)$  have columns  $\mathbf{y}_1, \dots, \mathbf{y}_k$ , and let  $\mathbf{B} = \mathbf{Q}^* \mathbf{A} \mathbf{Q}$  be the compression of [Definition 16.6.1](#). Let  $1 \leq m \leq k$  and  $c \in \mathbb{R}$ .

- (i) If  $S \subseteq Y$  is a subspace with  $\dim S = m$  and  $R_{\mathbf{A}}(\mathbf{x}) \geq c$  for every non-zero  $\mathbf{x} \in S$ , then  $\lambda_m(\mathbf{B}) \geq c$ .
- (ii) If  $S \subseteq Y$  is a subspace with  $\dim S = k - m + 1$  and  $R_{\mathbf{A}}(\mathbf{x}) \leq c$  for every non-zero  $\mathbf{x} \in S$ , then  $\lambda_m(\mathbf{B}) \leq c$ .

Moreover  $\sum_{j=1}^k \langle \mathbf{A} \mathbf{y}_j, \mathbf{y}_j \rangle = \text{tr } \mathbf{B} = \sum_{j=1}^k \lambda_j(\mathbf{B})$ .

*Proof.* The matrix  $\mathbf{B}$  is Hermitian, and the map  $\mathbf{c} \mapsto \mathbf{Q}\mathbf{c}$  is a linear bijection  $F^k \rightarrow Y$  with  $R_{\mathbf{B}}(\mathbf{c}) = R_{\mathbf{A}}(\mathbf{Q}\mathbf{c})$  for  $\mathbf{c} \neq \mathbf{0}$ , as computed in the proof of [Lemma 16.2.2](#). Let  $S' = \{\mathbf{c} \in F^k : \mathbf{Q}\mathbf{c} \in S\}$ ; it is a subspace, and  $\mathbf{c} \mapsto \mathbf{Q}\mathbf{c}$  maps it bijectively onto  $S$ , so  $\dim S' = \dim S$ .

- (i) By the max–min line of [Theorem 16.2.3](#) for  $\mathbf{B} \in M_k(F)$ ,  $\lambda_m(\mathbf{B})$  is the largest of the numbers  $\min_{\mathbf{0} \neq \mathbf{c} \in T} R_{\mathbf{B}}(\mathbf{c})$  over subspaces  $T \subseteq F^k$  of dimension  $m$ . Taking  $T = S'$ ,

$$\lambda_m(\mathbf{B}) \geq \min_{\mathbf{0} \neq \mathbf{c} \in S'} R_{\mathbf{B}}(\mathbf{c}) = \min_{\mathbf{0} \neq \mathbf{x} \in S} R_{\mathbf{A}}(\mathbf{x}) \geq c.$$

- (ii) By the min–max line of [Theorem 16.2.3](#) for  $\mathbf{B}$ , with  $T = S'$  of dimension  $k - m + 1$ ,

$$\lambda_m(\mathbf{B}) \leq \max_{\mathbf{0} \neq \mathbf{c} \in S'} R_{\mathbf{B}}(\mathbf{c}) = \max_{\mathbf{0} \neq \mathbf{x} \in S} R_{\mathbf{A}}(\mathbf{x}) \leq c.$$

Finally, the diagonal entries of  $\mathbf{B}$  are  $\langle \mathbf{A}\mathbf{y}_j, \mathbf{y}_j \rangle$  by [Definition 16.6.1](#), and the trace of the Hermitian  $\mathbf{B}$  is the sum of its eigenvalues by [Corollary 11.1.4](#). ■

**Step 1. The eigenflag guarantees the sum: part (b).** Let  $(\mathbf{x}_1, \dots, \mathbf{x}_k)$  be adapted to  $\mathcal{W}^{\mathbf{A}}$ , so  $\mathbf{x}_j \in W_j = \text{span}(\mathbf{u}_1, \dots, \mathbf{u}_{i_j})$ . Fix  $j$  and put  $S = \text{span}(\mathbf{x}_1, \dots, \mathbf{x}_j)$ . The list is orthonormal, hence independent, so  $\dim S = j$ . Each  $\mathbf{x}_m$  with  $m \leq j$  lies in  $W_m \subseteq W_j$ , so  $S \subseteq W_j$ , where  $R_{\mathbf{A}} \geq \lambda_{i_j}$ . By Claim 1 (i) with  $m = j$ ,  $\lambda_j(\mathbf{B}) \geq \lambda_{i_j}$ , where  $\mathbf{B}$  is the compression to the span of the list. Summing over  $j$  and using the last line of Claim 1,

$$\sum_{j=1}^k \langle \mathbf{A}\mathbf{x}_j, \mathbf{x}_j \rangle = \sum_{j=1}^k \lambda_j(\mathbf{B}) \geq \sum_{j=1}^k \lambda_{i_j}.$$

For  $\mathbf{x}_j = \mathbf{u}_{i_j}$  each term is  $\langle \mathbf{A}\mathbf{u}_{i_j}, \mathbf{u}_{i_j} \rangle = \lambda_{i_j}$ , so equality holds. This proves (b).

**Step 2. Leading indices, and a selection.** Let  $\mathcal{W} = (W_1, \dots, W_k)$  be an arbitrary flag of type  $(i_1, \dots, i_k)$ . For  $\mathbf{x} \in F^n$  write  $c_p(\mathbf{x}) = \langle \mathbf{x}, \mathbf{u}_p \rangle$ , so that  $\mathbf{x} = \sum_p c_p(\mathbf{x})\mathbf{u}_p$  by [Theorem 10.2.1](#) (a). Then  $\mathbf{x} \in L_p$  exactly when  $c_1(\mathbf{x}) = \dots = c_{p-1}(\mathbf{x}) = 0$ . The **leading index**  $\ell(\mathbf{x})$  of a non-zero  $\mathbf{x}$  is the smallest  $p$  with  $c_p(\mathbf{x}) \neq 0$ , so that

$$\mathbf{x} \in L_p \iff \ell(\mathbf{x}) \geq p \quad (\mathbf{x} \neq \mathbf{0}).$$

For a subspace  $V \subseteq F^n$ , let  $J(V)$  be the set of  $p \in \{1, \dots, n\}$  with  $V \cap L_{p+1} \neq V \cap L_p$ . These are the indices at which the chain  $V = V \cap L_1 \supseteq V \cap L_2 \supseteq \dots \supseteq V \cap L_{n+1} = \{\mathbf{0}\}$  gets strictly smaller.

*Claim.* **Claim 2 (the jumps count the dimension).** For every subspace  $V \subseteq F^n$  and every  $1 \leq p \leq n+1$ ,

$$\dim(V \cap L_p) = |J(V) \cap \{p, p+1, \dots, n\}|.$$

In particular  $|J(V)| = \dim V$ . Moreover, for each  $p \in J(V)$  there is  $\mathbf{x} \in V$  with  $\ell(\mathbf{x}) = p$ .

*Proof.* Fix  $p \leq n$ . The map  $\varphi_p: V \cap L_p \rightarrow F$ ,  $\mathbf{x} \mapsto c_p(\mathbf{x})$ , is linear. A vector of  $V \cap L_p$  already has  $c_1 = \cdots = c_{p-1} = 0$ , so it lies in  $L_{p+1}$  exactly when also  $c_p = 0$ ; hence  $\ker \varphi_p = V \cap L_{p+1}$ . By [Theorem 3.3.1](#),

$$\dim(V \cap L_p) - \dim(V \cap L_{p+1}) = \text{rank } \varphi_p,$$

and  $\text{rank } \varphi_p \leq \dim F = 1$ . The rank is 1 exactly when  $\ker \varphi_p \neq V \cap L_p$ , that is, when  $p \in J(V)$ . Adding these equations for  $p, p+1, \dots, n$ , the left side telescopes to  $\dim(V \cap L_p) - \dim(V \cap L_{n+1}) = \dim(V \cap L_p)$ , and the right side counts the elements of  $J(V)$  that are  $\geq p$ . This proves the formula, which for  $p = n+1$  reads  $0 = 0$ . At  $p = 1$ ,  $L_1 = F^n$  gives  $|J(V)| = \dim V$ . Finally, if  $p \in J(V)$ , pick  $\mathbf{x} \in (V \cap L_p) \setminus L_{p+1}$ ; then  $c_1(\mathbf{x}) = \cdots = c_{p-1}(\mathbf{x}) = 0 \neq c_p(\mathbf{x})$ , so  $\ell(\mathbf{x}) = p$ . ■

*Claim.* **Claim 3 (distinct leading indices force independence).** Non-zero vectors  $\mathbf{x}_1, \dots, \mathbf{x}_r$  whose leading indices are pairwise distinct are linearly independent.

*Proof.* Suppose  $\sum_m a_m \mathbf{x}_m = \mathbf{0}$  with not all  $a_m$  zero. Among the  $m$  with  $a_m \neq 0$ , let  $m_0$  be the one with the smallest leading index,  $q = \ell(\mathbf{x}_{m_0})$ . Apply  $c_q$ . Every other  $\mathbf{x}_m$  with  $a_m \neq 0$  has  $\ell(\mathbf{x}_m) > q$ , so  $c_q(\mathbf{x}_m) = 0$ . Hence  $0 = a_{m_0} c_q(\mathbf{x}_{m_0})$ , a product of two non-zero numbers, which is a contradiction. ■

*Claim.* **Claim 4 (a greedy selection).** There are non-zero vectors  $\mathbf{x}_1, \dots, \mathbf{x}_k$  with  $\mathbf{x}_m \in W_m$ , with pairwise distinct leading indices  $q_m = \ell(\mathbf{x}_m)$ , and such that for every  $j$ ,

$$\#\{m : q_m \geq i_j\} \geq k - j + 1.$$

*Proof.* Choose  $q_1, \dots, q_k$  in turn:  $q_m$  is the **largest** element of  $J(W_m)$  not among  $q_1, \dots, q_{m-1}$ . This is possible, because  $J(W_m)$  has  $\dim W_m = i_m \geq m$  elements by Claim 2, so at most  $m-1$  of them are used. By the last part of Claim 2, pick  $\mathbf{x}_m \in W_m$  with  $\ell(\mathbf{x}_m) = q_m$ . The  $q_m$  are distinct by construction.

Fix  $j$ , and for  $m \geq j-1$  let  $N_m$  be the number of  $l \leq m$  with  $q_l \geq i_j$ . We show  $N_m \geq m - j + 1$  by induction on  $m$ . For  $m = j-1$  this says  $N_{j-1} \geq 0$ , which is true. Let  $m \geq j$  and assume  $N_{m-1} \geq m - j$ .

*Case 1:*  $q_m \geq i_j$ . Then  $N_m = N_{m-1} + 1 \geq m - j + 1$ .

*Case 2:*  $q_m < i_j$ . By Claim 2,  $J(W_m)$  has  $i_m$  elements, of which at most  $i_j - 1$  are smaller than  $i_j$ . So at least  $i_m - i_j + 1$  of them are  $\geq i_j$ , and  $i_m - i_j \geq m - j$  because the indices are strictly increasing integers. Every element of  $J(W_m)$  that is  $\geq i_j$  exceeds  $q_m$ , so by the choice of  $q_m$  as the largest unused element, each of them is already among  $q_1, \dots, q_{m-1}$ . These are at least  $i_m - i_j + 1 \geq m - j + 1$  of the earlier  $q_l$ , all  $\geq i_j$ , so  $N_{m-1} \geq m - j + 1$ . Hence  $N_m = N_{m-1} \geq m - j + 1$ .

At  $m = k$  this gives  $N_k \geq k - j + 1$ , which is the claim. ■



**Step 3. An adapted list with a small sum: part (a).** Take  $\mathbf{x}_1, \dots, \mathbf{x}_k$  from Claim 4. They are linearly independent by Claim 3, so [Theorem 10.2.2](#) applies and produces an orthonormal list  $(\mathbf{y}_1, \dots, \mathbf{y}_k)$  with  $\text{span}(\mathbf{y}_1, \dots, \mathbf{y}_j) = \text{span}(\mathbf{x}_1, \dots, \mathbf{x}_j)$  for each  $j$ . Since  $\mathbf{x}_m \in W_m \subseteq W_j$  for  $m \leq j$ , the vector  $\mathbf{y}_j$  lies in  $W_j$ . So  $(\mathbf{y}_1, \dots, \mathbf{y}_k)$  is adapted to  $\mathcal{W}$ .

Let  $X = \text{span}(\mathbf{y}_1, \dots, \mathbf{y}_k) = \text{span}(\mathbf{x}_1, \dots, \mathbf{x}_k)$ , and let  $\mathbf{B}$  be the compression of  $\mathbf{A}$  to it with respect to the  $\mathbf{y}_j$ . Fix  $j$ . By Claim 4 at least  $k - j + 1$  of the  $\mathbf{x}_m$  have  $q_m \geq i_j$ ; each of them lies in  $L_{i_j}$ , and any  $k - j + 1$  of them are independent by Claim 3. Let  $S$  be the span of  $k - j + 1$  of them, so  $S \subseteq X \cap L_{i_j}$  and  $\dim S = k - j + 1$ . On  $S \subseteq L_{i_j}$  we have  $R_{\mathbf{A}} \leq \lambda_{i_j}$ , so Claim 1 (ii) with  $m = j$  gives  $\lambda_j(\mathbf{B}) \leq \lambda_{i_j}$ . Summing over  $j$  and using the last line of Claim 1,

$$\sum_{j=1}^k \langle \mathbf{A} \mathbf{y}_j, \mathbf{y}_j \rangle = \sum_{j=1}^k \lambda_j(\mathbf{B}) \leq \sum_{j=1}^k \lambda_{i_j}.$$

This proves (a).

**Step 4. The minimum exists, and the max–min formula: part (c).** Fix a flag  $\mathcal{W}$ . Identify a list  $(\mathbf{x}_1, \dots, \mathbf{x}_k)$  with the column of  $F^{nk}$  obtained by stacking the  $\mathbf{x}_j$ ; its Euclidean norm is  $(\sum_j \|\mathbf{x}_j\|^2)^{1/2}$ . Let  $\mathcal{O}$  be the set of lists adapted to  $\mathcal{W}$ . It is non-empty, by the paragraph after [Definition 16.7.1](#). It is bounded, since every element has norm  $\sqrt{k}$ . And it contains the limit of each of its convergent sequences. Indeed, the conditions  $\langle \mathbf{x}_i, \mathbf{x}_j \rangle = \delta_{ij}$  are equations between sums of products of the entries and their conjugates, which pass to limits. Each condition  $\mathbf{x}_j \in W_j$  says  $\mathbf{M}_j \mathbf{x}_j = \mathbf{0}$  for a matrix  $\mathbf{M}_j$  whose null space is  $W_j$ , for instance the matrix whose rows are the conjugate transposes of a basis of  $W_j^\perp$  (the zero matrix if  $W_j = F^n$ ), and this passes to limits too. So by fact (A3) of Chapter 15’s introduction, every sequence in  $\mathcal{O}$  has a subsequence converging to a point of  $\mathcal{O}$ . The function  $(\mathbf{x}_1, \dots, \mathbf{x}_k) \mapsto \sum_j \langle \mathbf{A} \mathbf{x}_j, \mathbf{x}_j \rangle$  is built from the real and imaginary parts of the entries by sums and products, so it is continuous, and it is real-valued since  $\mathbf{A}$  is Hermitian. By fact (A4) it attains a minimum on  $\mathcal{O}$ .

Call that minimum  $v(\mathcal{W})$ . By (a), some adapted list has sum  $\leq \sum_j \lambda_{i_j}$ , so  $v(\mathcal{W}) \leq \sum_j \lambda_{i_j}$  for every  $\mathcal{W}$ . By (b), every list adapted to  $\mathcal{W}^{\mathbf{A}}$  has sum  $\geq \sum_j \lambda_{i_j}$ , with equality at  $(\mathbf{u}_{i_1}, \dots, \mathbf{u}_{i_k})$ , so  $v(\mathcal{W}^{\mathbf{A}}) = \sum_j \lambda_{i_j}$ . Hence the largest value of  $v(\mathcal{W})$  over all flags of the type exists, is attained at  $\mathcal{W}^{\mathbf{A}}$ , and equals  $\sum_j \lambda_{i_j}$ . This proves (c), and the theorem. ■

*Remark. Where the analysis went.* Parts (a) and (b) are pure algebra: a dimension count, Gram–Schmidt, and Courant–Fischer for a small matrix. The only analytic input is in Step 4, facts (A3) and (A4) of Chapter 15’s introduction. This is the first time in the chapter that an item of that list is used rather than merely mentioned, and it is used only so that the word “min” in (c) refers to a value that is attained. Every later *result* in this section uses (a) and (b) alone, so no later result depends on analysis; only the exercises that quote (c), such as A1(b), lean on the attained minimum.

Two index sets give back what we already have. For  $k = 1$  a flag is a single subspace  $W$  of dimension  $i_1$ , an adapted list is a unit vector  $\mathbf{x} \in W$ , and  $\langle \mathbf{A} \mathbf{x}, \mathbf{x} \rangle = R_{\mathbf{A}}(\mathbf{x})$ . So (c) reads  $\lambda_{i_1}(\mathbf{A}) = \max_{\dim W = i_1} \min_{\mathbf{0} \neq \mathbf{x} \in W} R_{\mathbf{A}}(\mathbf{x})$ , the max–min line of [Theorem 16.2.3](#). For the initial segment  $(1, \dots, k)$ , an adapted list is an orthonormal basis of the  $k$ -dimensional  $W_k$ , and conversely every orthonormal basis  $(\mathbf{x}_1, \dots, \mathbf{x}_k)$  of a  $k$ -dimensional subspace is adapted to the flag

$\text{span}(\mathbf{x}_1) \subset \text{span}(\mathbf{x}_1, \mathbf{x}_2) \subset \dots$ . The sum  $\sum_j \langle \mathbf{A}\mathbf{x}_j, \mathbf{x}_j \rangle$  is the trace of the compression to  $W_k$ . That trace does not depend on the orthonormal basis, because §06 showed that a change of orthonormal basis changes the compression by a unitary similarity. So the inner minimum is constant, and (c) says that  $\sum_{j \leq k} \lambda_j(\mathbf{A})$  is the largest trace of a compression to a  $k$ -dimensional subspace, which is [Theorem 16.6.2](#). What is new is everything in between: index sets with gaps, such as  $(2, 4)$ , and index sets that do not start at 1.

**Example. (The selection at work, and why it is needed)**

Let  $\mathbf{A} = \text{diag}(4, 2, 1, -1) \in M_4(\mathbb{R})$ , take the index set  $(2, 4)$ , and take the flag

$$W_1 = \text{span}((1, 1, 0, 0), \mathbf{e}_4) \subset W_2 = \mathbb{R}^4,$$

of type  $(2, 4)$ . Show that the naive attempt, choosing  $\mathbf{x}_j \in W_j \cap L_{i_j}$  for each  $j$  separately, cannot produce an adapted list. Then follow the proof of part (a) to produce one with sum at most  $\lambda_2(\mathbf{A}) + \lambda_4(\mathbf{A})$ .

*Solution.* Here the eigenvectors are  $\mathbf{u}_p = \mathbf{e}_p$ , so  $L_p = \text{span}(\mathbf{e}_p, \dots, \mathbf{e}_4)$  is the set of vectors whose first  $p - 1$  entries vanish, and the target is  $\lambda_2 + \lambda_4 = 2 + (-1) = 1$ .

*The naive attempt.* A vector  $a(1, 1, 0, 0) + b\mathbf{e}_4$  of  $W_1$  lies in  $L_2$  exactly when its first entry  $a$  is 0, so  $W_1 \cap L_2 = \text{span}(\mathbf{e}_4)$ . Also  $W_2 \cap L_4 = L_4 = \text{span}(\mathbf{e}_4)$ . Both vectors would have to be multiples of  $\mathbf{e}_4$ , and two such vectors are never orthonormal. Each of the two intersections is non-zero, as [Lemma 16.2.1](#) promises, but the two choices collide.

*The selection.* The vectors of  $W_1$  have the form  $(a, a, 0, b)$ . They lie in  $L_2$ ,  $L_3$  or  $L_4$  exactly when  $a = 0$ , and in  $L_5 = \{\mathbf{0}\}$  only when also  $b = 0$ . So the dimensions of  $W_1 \cap L_p$  for  $p = 1, \dots, 5$  are 2, 1, 1, 1, 0, and the chain drops at  $p = 1$  and  $p = 4$ :  $J(W_1) = \{1, 4\}$ . For  $W_2 = \mathbb{R}^4$  the dimensions are 4, 3, 2, 1, 0, so  $J(W_2) = \{1, 2, 3, 4\}$ . The greedy rule takes  $q_1 = \max J(W_1) = 4$ , with  $\mathbf{x}_1 = \mathbf{e}_4 \in W_1$ . It then takes  $q_2 = \max(J(W_2) \setminus \{4\}) = 3$ , with  $\mathbf{x}_2 = \mathbf{e}_3$ .

*The counts of Claim 4.* For  $j = 1$ ,  $i_1 = 2$ , and both  $q_1 = 4$  and  $q_2 = 3$  are  $\geq 2$ : the count is  $2 \geq 2$ . For  $j = 2$ ,  $i_2 = 4$ , and only  $q_1 = 4$  is  $\geq 4$ : the count is  $1 \geq 1$ . Notice that the requirement for  $j = 2$  is met by  $\mathbf{x}_1$ , not by  $\mathbf{x}_2$ ; insisting that  $\mathbf{x}_2$  itself lie in  $L_4$  is exactly the naive attempt that failed.

*The list and its sum.*  $(\mathbf{e}_4, \mathbf{e}_3)$  is already orthonormal, so Gram–Schmidt leaves it alone, and it is adapted:  $\mathbf{e}_4 \in W_1$ ,  $\mathbf{e}_3 \in W_2$ . Its sum is  $\langle \mathbf{A}\mathbf{e}_4, \mathbf{e}_4 \rangle + \langle \mathbf{A}\mathbf{e}_3, \mathbf{e}_3 \rangle = -1 + 1 = 0 \leq 1$ . The compression to  $X = \text{span}(\mathbf{e}_4, \mathbf{e}_3)$  is  $\text{diag}(-1, 1)$ , with eigenvalues  $1 \leq \lambda_2(\mathbf{A}) = 2$  and  $-1 \leq \lambda_4(\mathbf{A}) = -1$ , as Step 3 predicts one index at a time.

*Quick check.* Suppose “orthonormal” is weakened to “each  $\mathbf{x}_j$  is a unit vector” in the definition of an adapted list. Show that the formula (c) of [Theorem 16.7.1](#) remains true. Why is this weaker version useless for comparing two matrices?

*Solution.* With the  $\mathbf{x}_j$  chosen independently, the inner minimum splits as  $\sum_j \min_{\mathbf{0} \neq \mathbf{x} \in W_j} R_{\mathbf{A}}(\mathbf{x})$ , one term per subspace. By [Theorem 16.2.3](#) each term is at most  $\lambda_{i_j}(\mathbf{A})$ , since  $\dim W_j = i_j$ , and the eigenflag makes every term equal to  $\lambda_{i_j}(\mathbf{A})$  at once. So the max–min is still  $\sum_j \lambda_{i_j}(\mathbf{A})$ ; it is just Courant–Fischer added up. But the lists it produces are not orthonormal, so [Theorem 16.6.2](#) cannot be applied to them. Comparing  $\mathbf{A}$  with  $\mathbf{B}$  through

such lists gives only the summed Weyl bound  $k\lambda_1(\mathbf{A} - \mathbf{B})$  of the first subsection. The whole content of Wielandt's theorem is that orthonormality costs nothing.

## 16.7.4 Lidskii's inequality

The plan of the first subsection can now be carried out.

### Theorem 16.7.2 — Lidskii's Inequality

Let  $\mathbf{A}, \mathbf{B} \in M_n(F)$  be Hermitian, with  $F = \mathbb{R}$  or  $\mathbb{C}$ . Then for every index set  $1 \leq i_1 < \dots < i_k \leq n$ ,

$$\sum_{j=1}^k \lambda_{i_j}(\mathbf{A}) - \sum_{j=1}^k \lambda_{i_j}(\mathbf{B}) \leq \sum_{j=1}^k \lambda_j(\mathbf{A} - \mathbf{B}).$$

*Idea.* Find one orthonormal list that is good for both matrices. Take the eigenflag of  $\mathbf{A}$ . Part (b) of Theorem 16.7.1 says that **every** list adapted to it is good for  $\mathbf{A}$ . Part (a), applied to  $\mathbf{B}$  and this same flag, produces **one** adapted list that is good for  $\mathbf{B}$ . Subtract the two sums. What remains is  $\sum_j \langle (\mathbf{A} - \mathbf{B})\mathbf{x}_j, \mathbf{x}_j \rangle$  over an orthonormal list, which Theorem 16.6.2 bounds by the top  $k$  eigenvalues of  $\mathbf{A} - \mathbf{B}$ .

*Proof.* Let  $(\mathbf{u}_1, \dots, \mathbf{u}_n)$  be an orthonormal basis of  $F^n$  with  $\mathbf{A}\mathbf{u}_p = \lambda_p(\mathbf{A})\mathbf{u}_p$ , which exists by Corollary 11.4.3 or Corollary 11.5.2, and let  $\mathcal{W}^{\mathbf{A}}$  be the flag  $W_j = \text{span}(\mathbf{u}_1, \dots, \mathbf{u}_{i_j})$ , of type  $(i_1, \dots, i_k)$ . By Theorem 16.7.1 (a) applied to the Hermitian matrix  $\mathbf{B}$  and the flag  $\mathcal{W}^{\mathbf{A}}$ , there is a list  $(\mathbf{x}_1, \dots, \mathbf{x}_k)$  adapted to  $\mathcal{W}^{\mathbf{A}}$  with

$$\sum_{j=1}^k \langle \mathbf{B}\mathbf{x}_j, \mathbf{x}_j \rangle \leq \sum_{j=1}^k \lambda_{i_j}(\mathbf{B}).$$

By Theorem 16.7.1 (b) applied to  $\mathbf{A}$ , the same list satisfies

$$\sum_{j=1}^k \langle \mathbf{A}\mathbf{x}_j, \mathbf{x}_j \rangle \geq \sum_{j=1}^k \lambda_{i_j}(\mathbf{A}).$$

Subtracting the first inequality from the second, and using that the inner product is linear in its first slot,

$$\sum_{j=1}^k \lambda_{i_j}(\mathbf{A}) - \sum_{j=1}^k \lambda_{i_j}(\mathbf{B}) \leq \sum_{j=1}^k \langle (\mathbf{A} - \mathbf{B})\mathbf{x}_j, \mathbf{x}_j \rangle.$$

The matrix  $\mathbf{A} - \mathbf{B}$  is Hermitian and  $\mathbf{x}_1, \dots, \mathbf{x}_k$  are orthonormal, so Theorem 16.6.2 bounds the right-hand side by  $\sum_{j=1}^k \lambda_j(\mathbf{A} - \mathbf{B})$ . This proves the theorem. ■

For the rank-one example of the first subsection, the right-hand side is  $\lambda_1 + \dots + \lambda_k$  of  $\mathbf{u}\mathbf{u}^*$ , which is 1 for every  $k$ : the theorem gives exactly the bound that the interlacing argument found there, and it gives it for every Hermitian  $\mathbf{A} - \mathbf{B}$ , whatever its rank. One feature of the statement

deserves a warning.

*Warning.* The left side compares the two spectra at the same indices; the right side always uses the top  $k$  eigenvalues of  $\mathbf{A} - \mathbf{B}$ , whatever the index set. The tempting “index-by-index” version

$$\lambda_i(\mathbf{A}) - \lambda_i(\mathbf{B}) \leq \lambda_i(\mathbf{A} - \mathbf{B})$$

is false. Take  $\mathbf{A} = \text{diag}(1, 1)$  and  $\mathbf{B} = \text{diag}(1, 0)$ , so  $\mathbf{A} - \mathbf{B} = \text{diag}(0, 1)$  with  $\lambda_1(\mathbf{A} - \mathbf{B}) = 1$  and  $\lambda_2(\mathbf{A} - \mathbf{B}) = 0$ . At  $i = 2$  the left side is  $1 - 0 = 1$ , which exceeds  $\lambda_2(\mathbf{A} - \mathbf{B}) = 0$ . Lidskii’s bound is  $\lambda_1(\mathbf{A} - \mathbf{B}) = 1$ , and it is attained.

**Example. (All seven index sets)**

Let

$$\mathbf{A} = \begin{pmatrix} 0 & 2 & 0 \\ 2 & 1 & 2 \\ 0 & 2 & 2 \end{pmatrix}, \quad \mathbf{B} = \text{diag}(0, 2, -1).$$

Find the three spectra, check [Theorem 16.7.2](#) for every index set, and compare with the summed Weyl bound  $k \lambda_1(\mathbf{A} - \mathbf{B})$ .

*Solution. The spectra.* The vectors  $(1, 2, 2)$ ,  $(2, 1, -2)$ ,  $(2, -2, 1)$  are eigenvectors of  $\mathbf{A}$ : for instance  $\mathbf{A}(1, 2, 2) = (4, 2 + 2 + 4, 4 + 4) = 4(1, 2, 2)$ , and the other two give the eigenvalues 1 and  $-2$  in the same way. So  $\lambda(\mathbf{A}) = (4, 1, -2)$ , writing  $\lambda(\mathbf{A})$  for the vector of eigenvalues in decreasing order. Sorting the diagonal of  $\mathbf{B}$  gives  $\lambda(\mathbf{B}) = (2, 0, -1)$ . Next,

$$\mathbf{A} - \mathbf{B} = \begin{pmatrix} 0 & 2 & 0 \\ 2 & -1 & 2 \\ 0 & 2 & 3 \end{pmatrix}$$

has eigenvectors  $(1, 2, 4)$ ,  $(-2, -1, 1)$ ,  $(2, -3, 1)$ . For example  $(\mathbf{A} - \mathbf{B})(1, 2, 4) = (4, 2 - 2 + 8, 4 + 12) = 4(1, 2, 4)$ . So  $\lambda(\mathbf{A} - \mathbf{B}) = (4, 1, -3)$ , whose running totals from the top are 4, 5, 2.

*The inequalities.* The differences at equal indices are  $\lambda_i(\mathbf{A}) - \lambda_i(\mathbf{B}) = 2, 1, -1$  for  $i = 1, 2, 3$ .

| index set                | left side | Lidskii bound | summed Weyl bound |
|--------------------------|-----------|---------------|-------------------|
| $(1), (2), (3)$          | 2, 1, -1  | 4             | 4                 |
| $(1, 2), (1, 3), (2, 3)$ | 3, 1, 0   | 5             | 8                 |
| $(1, 2, 3)$              | 2         | 2             | 12                |

Every inequality holds. For  $k = 3$  it is an equality, and must be: both sides are then  $\text{tr } \mathbf{A} - \text{tr } \mathbf{B} = \text{tr}(\mathbf{A} - \mathbf{B})$ , by [Corollary 11.1.4](#) and [Theorem 0.9.6](#) (1). The summed Weyl bound grows like  $k \lambda_1(\mathbf{A} - \mathbf{B})$ , while Lidskii’s bound can even decrease, because the eigenvalue  $-3$  of  $\mathbf{A} - \mathbf{B}$  enters the sum as soon as  $k = 3$ .

The index sets that give the results we already have are the two families from the discussion of Wielandt’s theorem, now at the level of two matrices.

**Corollary 16.7.3 — Weyl and Ky Fan as Cases of Lidskii**

Let  $\mathbf{A}, \mathbf{B} \in M_n(F)$  be Hermitian.

- (a) **(One index.)** For every  $i$ ,

$$\lambda_n(\mathbf{A} - \mathbf{B}) \leq \lambda_i(\mathbf{A}) - \lambda_i(\mathbf{B}) \leq \lambda_1(\mathbf{A} - \mathbf{B}),$$

and in particular  $|\lambda_i(\mathbf{A}) - \lambda_i(\mathbf{B})| \leq \|\mathbf{A} - \mathbf{B}\|_2$ .

- (b) **(An initial segment.)** For every  $k$ ,

$$\sum_{i=1}^k \lambda_i(\mathbf{A}) - \sum_{i=1}^k \lambda_i(\mathbf{B}) \leq \sum_{i=1}^k \lambda_i(\mathbf{A} - \mathbf{B}).$$

*Idea.* Both are [Theorem 16.7.2](#) with a particular index set, the lower bound in (a) coming from the same theorem with  $\mathbf{A}$  and  $\mathbf{B}$  exchanged.

*Proof.*

- (a) The upper bound is [Theorem 16.7.2](#) with  $k = 1$  and the index set  $(i)$ . Exchanging  $\mathbf{A}$  and  $\mathbf{B}$  in the same case gives  $\lambda_i(\mathbf{B}) - \lambda_i(\mathbf{A}) \leq \lambda_1(\mathbf{B} - \mathbf{A})$ . By [Lemma 16.3.2](#) with  $p = 1$ ,  $\lambda_1(\mathbf{B} - \mathbf{A}) = \lambda_1(-(\mathbf{A} - \mathbf{B})) = -\lambda_n(\mathbf{A} - \mathbf{B})$ , and multiplying by  $-1$  gives the lower bound. So  $\lambda_i(\mathbf{A}) - \lambda_i(\mathbf{B})$  lies between  $\lambda_n(\mathbf{A} - \mathbf{B})$  and  $\lambda_1(\mathbf{A} - \mathbf{B})$ , and its absolute value is at most  $\max(\lambda_1(\mathbf{A} - \mathbf{B}), -\lambda_n(\mathbf{A} - \mathbf{B}))$ , which is  $\|\mathbf{A} - \mathbf{B}\|_2$  by [Lemma 16.3.5](#).
- (b) This is [Theorem 16.7.2](#) with the index set  $(1, 2, \dots, k)$ .

■

Read with  $\mathbf{A} = \mathbf{X} + \mathbf{Y}$  and  $\mathbf{B} = \mathbf{X}$ , part (a) is [Corollary 16.3.4](#) for the pair  $\mathbf{X}, \mathbf{Y}$ ; read with  $\mathbf{B} = \mathbf{Y}$  instead, it is the same corollary with  $\mathbf{X}$  and  $\mathbf{Y}$  exchanged. Together these are the cases of [Theorem 16.3.3](#) in which one of the two indices on the right is 1 (in part (a) of that theorem) or  $n$  (in its part (b)); applied to negated pairs, the one-index case gives a few more, as the remark below explains. The last clause of part (a) is [Corollary 16.3.6](#). Part (b), read the same way, is [Corollary 16.6.5](#). Here is the hierarchy, one row per index set.

| index set                   | <a href="#">Theorem 16.7.1</a> gives          | <a href="#">Theorem 16.7.2</a> gives                                                                                                       |
|-----------------------------|-----------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------|
| $(i)$ , one index           | <a href="#">Theorem 16.2.3</a> , max–min line | <a href="#">Theorem 16.3.3</a> with one index 1 or $n$ (and, through negation, with $i + j - 1 = n$ );<br><a href="#">Corollary 16.3.6</a> |
| $(1, \dots, k)$             | <a href="#">Theorem 16.6.2</a>                | <a href="#">Corollary 16.6.5</a>                                                                                                           |
| any other, such as $(2, 4)$ | new                                           | new                                                                                                                                        |

*Remark.* **Not all of Weyl’s inequalities are cases of Lidskii’s.** To test this fairly, [Theorem 16.7.2](#) must be allowed every pair it can see: all pairs of matrices among  $\pm\mathbf{X}$ ,  $\pm\mathbf{Y}$ ,  $\pm(\mathbf{X} + \mathbf{Y})$  whose difference is another of the six. Negation matters. With it, the one-index case already reaches some Weyl inequalities with both indices inside: applied to  $\mathbf{A} = -\mathbf{X}$ ,  $\mathbf{B} = \mathbf{Y}$ , whose difference is  $-(\mathbf{X} + \mathbf{Y})$ , and read through [Lemma 16.3.2](#), it

gives  $\lambda_n(\mathbf{X} + \mathbf{Y}) \leq \lambda_{n+1-i}(\mathbf{X}) + \lambda_i(\mathbf{Y})$  for every  $i$ , which is exactly the Weyl inequality (a) with  $i + j - 1 = n$ . So a small example that ignores negation can look like a counterexample and not be one.

Even with every signed pair allowed, the family does not contain all of Weyl's. Take  $n = 4$  and the candidate spectra

$$\begin{aligned}\lambda(\mathbf{X}) &= (2, 0, 0, -1), & \lambda(\mathbf{Y}) &= (2, 0, 0, 0), \\ \lambda(\mathbf{X} + \mathbf{Y}) &= (2, 1, 1, -1).\end{aligned}$$

The totals agree,  $1 + 2 = 3$ , and a direct check of every index set shows that all twelve signed pairs satisfy every inequality of [Theorem 16.7.2](#). Yet Weyl's inequality with  $i = j = 2$  demands  $\lambda_3(\mathbf{X} + \mathbf{Y}) \leq \lambda_2(\mathbf{X}) + \lambda_2(\mathbf{Y}) = 0$ , and here  $\lambda_3(\mathbf{X} + \mathbf{Y}) = 1$ . Since [Theorem 16.3.3](#) is a theorem, no Hermitian  $\mathbf{X}, \mathbf{Y}$  have these spectra: the lists are ruled out by Weyl and not by Lidskii.

*Remark. The majorization form.* Section 8 introduces the language that turns this family of inequalities into a single statement. We record the translation here, together with exactly what has been proved. For a real vector  $\mathbf{x} \in \mathbb{R}^n$ , the sum of its  $k$  largest entries is the largest value of  $\sum_{i \in S} x_i$  over the sets  $S$  of  $k$  positions: sorting the entries shows that the  $k$  largest do best. Apply this to  $\mathbf{x} = \lambda(\mathbf{A}) - \lambda(\mathbf{B})$ , whose  $i$ -th entry is  $\lambda_i(\mathbf{A}) - \lambda_i(\mathbf{B})$  and which need not be decreasing. Then [Theorem 16.7.2](#), over all index sets of size  $k$ , says that the  $k$  largest entries of  $\lambda(\mathbf{A}) - \lambda(\mathbf{B})$  add up to at most  $\sum_{j \leq k} \lambda_j(\mathbf{A} - \mathbf{B})$ , for every  $k$ . The two vectors also have the same total,  $\text{tr } \mathbf{A} - \text{tr } \mathbf{B} = \text{tr}(\mathbf{A} - \mathbf{B})$ . In the notation of Section 8 this is

$$\lambda(\mathbf{A}) - \lambda(\mathbf{B}) \prec \lambda(\mathbf{A} - \mathbf{B}).$$

**Proved here:** every inequality in this majorization, namely [Theorem 16.7.2](#) for every index set, and the equality of totals, which is the trace. **Not proved here, and left to Chapter 20:** the general consequences of a majorization  $\mathbf{x} \prec \mathbf{y}$ , namely  $\sum_i \phi(x_i) \leq \sum_i \phi(y_i)$  for every convex  $\phi: \mathbb{R} \rightarrow \mathbb{R}$ , together with the matching statement for the unitarily invariant norms of that chapter. Applied to the display above, they bound  $\sum_i \phi(\lambda_i(\mathbf{A}) - \lambda_i(\mathbf{B}))$  by  $\sum_i \phi(\lambda_i(\mathbf{A} - \mathbf{B}))$  for every convex  $\phi$ . One case,  $\phi(t) = |t|$ , follows from this section directly, and exercise C2 proves it. Nothing in this chapter uses the general case.

## 16.7.5 What the flags bought

Courant–Fischer turned each eigenvalue into an optimization that two matrices can share, and the comparison theorems of §§02–05 followed one index at a time. Ky Fan did the same for the top partial sums. Wielandt's theorem does it for every sum, at the price of optimizing over flags instead of subspaces, and Lidskii's inequality is what the shared optimization yields: the eigenvalue shifts  $\lambda_i(\mathbf{A}) - \lambda_i(\mathbf{B})$ , summed over **any**  $k$  indices, never exceed the top  $k$  eigenvalues of  $\mathbf{A} - \mathbf{B}$ . The single idea behind it is the one the first subsection isolated. A comparison of sums needs one orthonormal list that serves both matrices, and a flag is exactly the structure that lets such a list be found by counting.

## 16.7.6 Exercises

### 16.7.6.1 A. Check your understanding

#### Exercise A1.

- Define a flag of type  $(i_1, \dots, i_k)$  in  $F^n$  and a list adapted to it.
- State the max–min formula of [Theorem 16.7.1](#), and name the index sets for which it reduces to [Theorem 16.2.3](#) and to [Theorem 16.6.2](#).
- Determine whether the following is true, and justify your answer: if  $(\mathbf{x}_1, \dots, \mathbf{x}_k)$  is adapted to a flag  $(W_1, \dots, W_k)$ , then  $\text{span}(\mathbf{x}_1, \dots, \mathbf{x}_j) = W_j$  for every  $j$ .
- Determine whether the following is true for all Hermitian  $\mathbf{A}, \mathbf{B} \in M_n(F)$ , and justify your answer:  $\lambda_2(\mathbf{A}) - \lambda_2(\mathbf{B}) \leq \lambda_2(\mathbf{A} - \mathbf{B})$ .
- In the proof of [Theorem 16.7.2](#), which part of [Theorem 16.7.1](#) is applied to  $\mathbf{A}$ , which to  $\mathbf{B}$ , and with which flag?

*Solution.*

- For an index set  $1 \leq i_1 < \dots < i_k \leq n$ , a flag of that type is a list  $(W_1, \dots, W_k)$  of subspaces of  $F^n$  with  $W_1 \subseteq \dots \subseteq W_k$  and  $\dim W_j = i_j$ . A list  $(\mathbf{x}_1, \dots, \mathbf{x}_k)$  is adapted to it if it is orthonormal and  $\mathbf{x}_j \in W_j$  for every  $j$ .
- $\sum_j \lambda_{i_j}(\mathbf{A})$  is the maximum over flags  $\mathcal{W}$  of type  $(i_1, \dots, i_k)$  of the minimum of  $\sum_j \langle \mathbf{A}\mathbf{x}_j, \mathbf{x}_j \rangle$  over lists adapted to  $\mathcal{W}$ . For  $k = 1$ , with index set  $(i)$ , it is the max–min line of Courant–Fischer for  $\lambda_i$ . For the initial segment  $(1, \dots, k)$  it is Ky Fan’s theorem for  $\sum_{j \leq k} \lambda_j$ .
- False in general. The span of  $j$  orthonormal vectors has dimension  $j$ , while  $\dim W_j = i_j$ , which exceeds  $j$  unless  $i_j = j$ . For instance, with  $k = 1$  and the flag  $W_1 = \mathbb{R}^2$  of type  $(2)$ , the adapted list  $(\mathbf{e}_1)$  spans a line. What is true is  $\text{span}(\mathbf{x}_1, \dots, \mathbf{x}_j) \subseteq W_j$ .
- False. With  $\mathbf{A} = \text{diag}(1, 1)$  and  $\mathbf{B} = \text{diag}(1, 0)$ , the left side is  $1 - 0 = 1$  and  $\lambda_2(\mathbf{A} - \mathbf{B}) = \lambda_2(\text{diag}(0, 1)) = 0$ . The correct bound, from [Theorem 16.7.2](#) with  $k = 1$ , is  $\lambda_1(\mathbf{A} - \mathbf{B})$ .
- The flag is the eigenflag  $\mathcal{W}^{\mathbf{A}}$  of  $\mathbf{A}$  for the given index set. Part (b) is applied to  $\mathbf{A}$ : every list adapted to  $\mathcal{W}^{\mathbf{A}}$  has  $\mathbf{A}$ -sum at least  $\sum_j \lambda_{i_j}(\mathbf{A})$ . Part (a) is applied to  $\mathbf{B}$  with the same flag: some adapted list has  $\mathbf{B}$ -sum at most  $\sum_j \lambda_{i_j}(\mathbf{B})$ .

### 16.7.6.2 B. Practice

#### Exercise B1: Running the selection.

Let  $\mathbf{A} = \text{diag}(4, 1, -1) \in M_3(\mathbb{R})$ , take the index set  $(1, 3)$ , and let  $\mathcal{W}$  be the flag  $W_1 = \text{span}((1, 1, 1)) \subset W_2 = \mathbb{R}^3$ .

- Compute  $\lambda_1(\mathbf{A}) + \lambda_3(\mathbf{A})$ , and show directly that every list adapted to the eigenflag  $\text{span}(\mathbf{e}_1) \subset \mathbb{R}^3$  has sum at least this number.
- Follow the proof of [Theorem 16.7.1](#) (a) for  $\mathcal{W}$ : find  $J(W_1)$  and  $J(W_2)$ , select  $\mathbf{x}_1, \mathbf{x}_2$ , apply Gram–Schmidt, and compute  $\sum_j \langle \mathbf{A}\mathbf{y}_j, \mathbf{y}_j \rangle$ .



- (c) Compute the compression  $\mathbf{B}$  of  $\mathbf{A}$  to  $\text{span}(\mathbf{y}_1, \mathbf{y}_2)$  and its eigenvalues, and check  $\lambda_j(\mathbf{B}) \leq \lambda_{i_j}(\mathbf{A})$  for  $j = 1, 2$ .
- (d) Find the minimum of  $\sum_j \langle \mathbf{A}\mathbf{x}_j, \mathbf{x}_j \rangle$  over all lists adapted to  $\mathcal{W}$ . Is it the value found in (b)?

*Solution.*

- (a) The eigenvalues are  $4, 1, -1$  with eigenvectors  $\mathbf{e}_1, \mathbf{e}_2, \mathbf{e}_3$ , so  $\lambda_1 + \lambda_3 = 3$ , and  $\text{span}(\mathbf{e}_1) \subset \mathbb{R}^3$  is the eigenflag of type  $(1, 3)$ . An adapted list has  $\mathbf{x}_1 = \pm \mathbf{e}_1$ , the unit vectors of the line, so  $\langle \mathbf{A}\mathbf{x}_1, \mathbf{x}_1 \rangle = 4$ . Its  $\mathbf{x}_2$  is a unit vector orthogonal to  $\mathbf{e}_1$ , so  $\mathbf{x}_2 = (0, b, c)$  with  $b^2 + c^2 = 1$ , and  $\langle \mathbf{A}\mathbf{x}_2, \mathbf{x}_2 \rangle = b^2 - c^2 \geq -1$ . The sum is at least  $4 - 1 = 3$ , with equality at  $\mathbf{x}_2 = \pm \mathbf{e}_3$ , as part (b) of the theorem says.
- (b) Here  $L_p = \text{span}(\mathbf{e}_p, \dots, \mathbf{e}_3)$ . The line  $W_1$  is spanned by  $\mathbf{w} = (1, 1, 1)$ , whose first entry is non-zero, so  $W_1 \cap L_2 = \{\mathbf{0}\}$  and  $J(W_1) = \{1\}$ . For  $W_2 = \mathbb{R}^3$ ,  $J(W_2) = \{1, 2, 3\}$ . The greedy rule gives  $q_1 = 1$  with  $\mathbf{x}_1 = \mathbf{w}$ , and  $q_2 = \max\{2, 3\} = 3$  with  $\mathbf{x}_2 = \mathbf{e}_3$ . The counts of Claim 4 hold: for  $j = 1$ , both  $q$ 's are  $\geq i_1 = 1$ ; for  $j = 2$ ,  $q_2 = 3 \geq i_2 = 3$ . Gram-Schmidt gives  $\mathbf{y}_1 = \frac{1}{\sqrt{3}}(1, 1, 1)$  and

$$\mathbf{e}_3 - \langle \mathbf{e}_3, \mathbf{y}_1 \rangle \mathbf{y}_1 = \frac{1}{3}(-1, -1, 2), \quad \mathbf{y}_2 = \frac{1}{\sqrt{6}}(-1, -1, 2).$$

Then  $\langle \mathbf{A}\mathbf{y}_1, \mathbf{y}_1 \rangle = \frac{1}{3}(4 + 1 - 1) = \frac{4}{3}$  and  $\langle \mathbf{A}\mathbf{y}_2, \mathbf{y}_2 \rangle = \frac{1}{6}(4 + 1 - 4) = \frac{1}{6}$ , so the sum is  $\frac{5}{2} \leq 3$ , as the theorem requires.

- (c) The off-diagonal entry is  $\langle \mathbf{A}\mathbf{y}_2, \mathbf{y}_1 \rangle = \frac{1}{\sqrt{18}}(-4 - 1 - 2) = -\frac{7}{3\sqrt{2}}$ , so

$$\mathbf{B} = \begin{pmatrix} 4/3 & -7/(3\sqrt{2}) \\ -7/(3\sqrt{2}) & 1/6 \end{pmatrix}, \quad \text{tr } \mathbf{B} = \frac{3}{2}, \quad \det \mathbf{B} = \frac{2}{9} - \frac{49}{18} = -\frac{5}{2}.$$

The characteristic polynomial is  $t^2 - \frac{3}{2}t - \frac{5}{2}$ , that is  $\frac{1}{2}(2t - 5)(t + 1)$ , so  $\lambda_1(\mathbf{B}) = \frac{5}{2} \leq 4 = \lambda_1(\mathbf{A})$  and  $\lambda_2(\mathbf{B}) = -1 \leq -1 = \lambda_3(\mathbf{A})$ . The second is an equality, as it must be: [Theorem 16.6.1](#) with  $n = 3$ ,  $k = 2$  gives the reverse bound  $\lambda_2(\mathbf{B}) \geq \lambda_3(\mathbf{A})$ .

- (d) An adapted list has  $\mathbf{x}_1 = \pm \mathbf{y}_1$ , contributing  $\frac{4}{3}$ , and  $\mathbf{x}_2$  is any unit vector of  $\mathbf{w}^\perp$ . By [Lemma 16.2.2](#) the smallest value of  $\langle \mathbf{A}\mathbf{x}_2, \mathbf{x}_2 \rangle$  there is the smallest eigenvalue of the compression of  $\mathbf{A}$  to  $\mathbf{w}^\perp$ . With the orthonormal basis  $\frac{1}{\sqrt{2}}(1, -1, 0)$ ,  $\frac{1}{\sqrt{6}}(1, 1, -2)$  of  $\mathbf{w}^\perp$ , that compression is

$$\begin{pmatrix} 5/2 & \sqrt{3}/2 \\ \sqrt{3}/2 & 1/6 \end{pmatrix},$$

with trace  $\frac{8}{3}$ , determinant  $\frac{5}{12} - \frac{3}{4} = -\frac{1}{3}$ , and characteristic polynomial  $\frac{1}{3}(3t^2 - 8t - 1)$ . Its smaller eigenvalue is  $\frac{1}{3}(4 - \sqrt{19})$ . So the minimum over lists adapted to  $\mathcal{W}$  is

$$\frac{4}{3} + \frac{1}{3}(4 - \sqrt{19}) = \frac{1}{3}(8 - \sqrt{19}) \approx 1.214.$$

This is smaller than  $\frac{3}{2}$ . The construction in the proof finds an adapted list below the bound 3, which is all part (a) asks; it need not find the best one. Both numbers are below 3, the value at the eigenflag, as part (c) requires.

### Exercise B2: A complex pair.



Let

$$\mathbf{A} = \begin{pmatrix} 2 & 1+i \\ 1-i & 1 \end{pmatrix}, \quad \mathbf{B} = \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix} \quad \text{in } M_2(\mathbb{C}).$$

- Check that  $\mathbf{A}$  and  $\mathbf{B}$  are Hermitian, and compute  $\lambda(\mathbf{A})$ ,  $\lambda(\mathbf{B})$  and  $\lambda(\mathbf{A} - \mathbf{B})$ .
- Verify [Theorem 16.7.2](#) for all three index sets, and say which inequality is an equality and why it had to be.
- Verify both bounds of [Corollary 16.7.3](#) (a), and explain why here every  $\lambda_i(\mathbf{A}) - \lambda_i(\mathbf{B})$  had to be non-negative.

*Solution.*

- The diagonal of  $\mathbf{A}$  is real and  $a_{21} = 1 - i = \overline{a_{12}}$ , so  $\mathbf{A}^* = \mathbf{A}$ ;  $\mathbf{B}$  is real diagonal. For  $\mathbf{A}$ ,  $\text{tr } \mathbf{A} = 3$  and  $\det \mathbf{A} = 2 - (1+i)(1-i) = 2 - 2 = 0$ , so  $\lambda(\mathbf{A}) = (3, 0)$ . Next,  $\lambda(\mathbf{B}) = (1, -1)$ . Finally

$$\mathbf{A} - \mathbf{B} = \begin{pmatrix} 1 & 1+i \\ 1-i & 2 \end{pmatrix}$$

has trace 3 and determinant  $2 - 2 = 0$ , so  $\lambda(\mathbf{A} - \mathbf{B}) = (3, 0)$ .

- The differences at equal indices are  $3 - 1 = 2$  and  $0 - (-1) = 1$ . For (1):  $2 \leq 3$ . For (2):  $1 \leq 3$ . For (1, 2):  $3 \leq 3 + 0 = 3$ , an equality. It had to be one, since for the full index set both sides equal  $\text{tr } \mathbf{A} - \text{tr } \mathbf{B} = \text{tr}(\mathbf{A} - \mathbf{B})$ .
- The bounds are  $\lambda_2(\mathbf{A} - \mathbf{B}) = 0 \leq 2 \leq 3$  and  $0 \leq 1 \leq 3$ . The lower bound  $\lambda_n(\mathbf{A} - \mathbf{B}) = 0$  is non-negative because  $\mathbf{A} - \mathbf{B} \succeq 0$ : it is Hermitian with eigenvalues  $3, 0 \geq 0$  ([Theorem 12.1.4](#)). So  $\mathbf{A} \succeq \mathbf{B}$ , and [Corollary 16.2.4](#) already forces  $\lambda_i(\mathbf{A}) \geq \lambda_i(\mathbf{B})$  for each  $i$ .

### Exercise B3: The lower bound.

Let  $\mathbf{A}, \mathbf{B} \in M_n(F)$  be Hermitian and  $1 \leq i_1 < \cdots < i_k \leq n$ . Prove that

$$\sum_{j=1}^k \lambda_{i_j}(\mathbf{A}) - \sum_{j=1}^k \lambda_{i_j}(\mathbf{B}) \geq \sum_{j=n-k+1}^n \lambda_j(\mathbf{A} - \mathbf{B}),$$

that is, the shifts over any  $k$  indices add up to at least the  $k$  **smallest** eigenvalues of  $\mathbf{A} - \mathbf{B}$ . Check the result on [Example \(All seven index sets\)](#).

*Solution.* Apply [Theorem 16.7.2](#) with the roles of  $\mathbf{A}$  and  $\mathbf{B}$  exchanged:

$$\sum_{j=1}^k \lambda_{i_j}(\mathbf{B}) - \sum_{j=1}^k \lambda_{i_j}(\mathbf{A}) \leq \sum_{j=1}^k \lambda_j(\mathbf{B} - \mathbf{A}).$$

Since  $\mathbf{B} - \mathbf{A} = -(\mathbf{A} - \mathbf{B})$ , [Lemma 16.3.2](#) gives  $\lambda_j(\mathbf{B} - \mathbf{A}) = -\lambda_{n+1-j}(\mathbf{A} - \mathbf{B})$ . As  $j$  runs over  $1, \dots, k$ , the index  $n+1-j$  runs over  $n, \dots, n-k+1$ , so the right-hand side is  $-\sum_{j=n-k+1}^n \lambda_j(\mathbf{A} - \mathbf{B})$ . Multiplying the inequality by  $-1$  gives the claim.

In [Example \(All seven index sets\)](#),  $\lambda(\mathbf{A} - \mathbf{B}) = (4, 1, -3)$ , whose bottom partial sums are  $-3, -2, 2$  for  $k = 1, 2, 3$ . The left sides are  $2, 1, -1$  for  $k = 1$ , then  $3, 1, 0$  for  $k = 2$ , then  $2$  for  $k = 3$ . Each is at least the corresponding bottom sum.

## 16.7.6.3 C. Going deeper

**Exercise C1: A second proof of Lidskii's inequality.**

Let  $\mathbf{A}, \mathbf{B} \in M_n(F)$  be Hermitian, put  $\mathbf{C} = \mathbf{A} - \mathbf{B}$ , and fix an index set  $1 \leq i_1 < \cdots < i_k \leq n$ .

- Let  $c \in \mathbb{R}$ . Show that replacing  $\mathbf{A}$  by  $\mathbf{A} - c\mathbf{I}$ , with  $\mathbf{B}$  unchanged, lowers both sides of [Theorem 16.7.2](#) by  $kc$ . Deduce that it suffices to prove the inequality when  $\lambda_k(\mathbf{C}) = 0$ .
- Assume  $\lambda_k(\mathbf{C}) = 0$ . Write  $\mathbf{C} = \mathbf{U}\mathbf{D}\mathbf{U}^*$  with  $\mathbf{U}$  unitary and  $\mathbf{D} = \text{diag}(\lambda_1(\mathbf{C}), \dots, \lambda_n(\mathbf{C}))$ , and put  $\mathbf{C}_+ = \mathbf{U}\mathbf{D}_+\mathbf{U}^*$ , where  $\mathbf{D}_+$  replaces each negative diagonal entry of  $\mathbf{D}$  by 0. Prove that  $\mathbf{C}_+ \succeq \mathbf{C}$ ,  $\mathbf{C}_+ \succeq 0$  and  $\text{tr } \mathbf{C}_+ = \sum_{j \leq k} \lambda_j(\mathbf{C})$ .
- Prove that  $\lambda_i(\mathbf{A}) \leq \lambda_i(\mathbf{B} + \mathbf{C}_+)$  and  $\lambda_i(\mathbf{B} + \mathbf{C}_+) \geq \lambda_i(\mathbf{B})$  for every  $i$ , and deduce [Theorem 16.7.2](#).
- Which of the ingredients of this proof were available before this section? What does this proof not give that [Theorem 16.7.1](#) does?

*Hint:* [Corollary 16.2.4](#).

*Solution.*

- For every Hermitian  $\mathbf{M}$  and real  $c$ ,  $\lambda_i(\mathbf{M} - c\mathbf{I}) = \lambda_i(\mathbf{M}) - c$  for all  $i$ . Indeed, if  $\mathbf{M} = \mathbf{U}\mathbf{D}\mathbf{U}^*$  with  $\mathbf{U}$  unitary and  $\mathbf{D}$  real diagonal ([Corollary 11.4.3](#), or [Corollary 11.5.2](#) over  $\mathbb{R}$ ), then  $\mathbf{M} - c\mathbf{I} = \mathbf{U}(\mathbf{D} - c\mathbf{I})\mathbf{U}^*$ , and subtracting  $c$  from every diagonal entry keeps the order. Replacing  $\mathbf{A}$  by  $\mathbf{A} - c\mathbf{I}$  lowers each  $\lambda_{i_j}(\mathbf{A})$  by  $c$ , so the left side drops by  $kc$ . It also replaces  $\mathbf{C}$  by  $\mathbf{C} - c\mathbf{I}$ , so each  $\lambda_j(\mathbf{C})$  drops by  $c$  and the right side drops by  $kc$ . So the inequality for  $(\mathbf{A} - c\mathbf{I}, \mathbf{B})$  is equivalent to the one for  $(\mathbf{A}, \mathbf{B})$ . With  $c = \lambda_k(\mathbf{C})$ , the new difference  $\mathbf{C} - c\mathbf{I}$  has  $\lambda_k = 0$ .
- $\mathbf{C}_+ - \mathbf{C} = \mathbf{U}(\mathbf{D}_+ - \mathbf{D})\mathbf{U}^*$ , and  $\mathbf{D}_+ - \mathbf{D}$  is diagonal with entries  $\max(-\lambda_i(\mathbf{C}), 0) \geq 0$ . So  $\mathbf{C}_+ - \mathbf{C}$  is Hermitian with non-negative eigenvalues, hence  $\mathbf{C}_+ - \mathbf{C} \succeq 0$  by [Theorem 12.1.4](#), that is,  $\mathbf{C}_+ \succeq \mathbf{C}$ . In the same way  $\mathbf{C}_+ \succeq 0$ , since the entries of  $\mathbf{D}_+$  are  $\geq 0$ . For the trace,  $\text{tr } \mathbf{C}_+ = \text{tr } \mathbf{D}_+ = \sum_i \max(\lambda_i(\mathbf{C}), 0)$  by [Theorem 0.9.6 \(3\)](#). For  $i \leq k$ ,  $\lambda_i(\mathbf{C}) \geq \lambda_k(\mathbf{C}) = 0$ , so the term is  $\lambda_i(\mathbf{C})$ . For  $i > k$ ,  $\lambda_i(\mathbf{C}) \leq 0$ , so the term is 0. Hence  $\text{tr } \mathbf{C}_+ = \sum_{j \leq k} \lambda_j(\mathbf{C})$ .
- $\mathbf{A} = \mathbf{B} + \mathbf{C} \preceq \mathbf{B} + \mathbf{C}_+$  by (b), and  $\mathbf{B} + \mathbf{C}_+ \succeq \mathbf{B}$  because  $\mathbf{C}_+ \succeq 0$ . [Corollary 16.2.4](#) turns both into the stated eigenvalue inequalities. Therefore

$$\begin{aligned} \sum_{j=1}^k (\lambda_{i_j}(\mathbf{A}) - \lambda_{i_j}(\mathbf{B})) &\leq \sum_{j=1}^k (\lambda_{i_j}(\mathbf{B} + \mathbf{C}_+) - \lambda_{i_j}(\mathbf{B})) \\ &\leq \sum_{i=1}^n (\lambda_i(\mathbf{B} + \mathbf{C}_+) - \lambda_i(\mathbf{B})), \end{aligned}$$

the second step adding the  $n - k$  omitted terms, each  $\geq 0$ . The last sum is  $\text{tr}(\mathbf{B} + \mathbf{C}_+) - \text{tr } \mathbf{B} = \text{tr } \mathbf{C}_+$ , by [Corollary 11.1.4](#) and [Theorem 0.9.6 \(1\)](#), and  $\text{tr } \mathbf{C}_+ = \sum_{j \leq k} \lambda_j(\mathbf{C})$  by (b).

- Everything used here was available after §02: the spectral theorem, the characterization of positive semidefiniteness by eigenvalues, the trace, and [Corollary 16.2.4](#). The proof never looks at a flag. What it does not give is the variational description

itself. [Theorem 16.7.1](#) expresses  $\sum_j \lambda_{i_j}(\mathbf{A})$  as an optimization involving  $\mathbf{A}$  alone, and like Courant–Fischer and Ky Fan before it, such a description can be reused for other comparisons. This proof is a clever route to one consequence of that description.

### Exercise C2: The total absolute shift.

Let  $\mathbf{A}, \mathbf{B} \in M_n(F)$  be Hermitian.

(a) Prove that

$$\sum_{i=1}^n |\lambda_i(\mathbf{A}) - \lambda_i(\mathbf{B})| \leq \sum_{i=1}^n |\lambda_i(\mathbf{A} - \mathbf{B})|.$$

(b) Show that equality holds when  $\mathbf{A} \succeq \mathbf{B}$ . Compute both sides for [Example \(All seven index sets\)](#), and compare with the bound  $n\|\mathbf{A} - \mathbf{B}\|_2$  obtained by adding the bounds of [Corollary 16.3.6](#).

*Hint for (a): use the set of indices at which  $\lambda_i(\mathbf{A}) > \lambda_i(\mathbf{B})$ .*

*Solution.*

(a) Write  $d_i = \lambda_i(\mathbf{A}) - \lambda_i(\mathbf{B})$  and  $\mu_j = \lambda_j(\mathbf{A} - \mathbf{B})$ . Let  $S = \{i : d_i > 0\}$  and  $k = |S|$ . Then

$$\sum_{i=1}^n |d_i| = \sum_{i \in S} d_i - \sum_{i \notin S} d_i = 2 \sum_{i \in S} d_i - \sum_{i=1}^n d_i.$$

The total is  $\sum_i d_i = \text{tr } \mathbf{A} - \text{tr } \mathbf{B} = \text{tr}(\mathbf{A} - \mathbf{B}) = \sum_j \mu_j$ , by [Corollary 11.1.4](#) and [Theorem 0.9.6](#) (1). If  $k \geq 1$ , [Theorem 16.7.2](#) with the elements of  $S$  as index set gives  $\sum_{i \in S} d_i \leq \sum_{j \leq k} \mu_j$ ; if  $k = 0$  both sides of this inequality are the empty sum 0. In either case

$$\sum_{i=1}^n |d_i| \leq 2 \sum_{j \leq k} \mu_j - \sum_{j=1}^n \mu_j = \sum_{j \leq k} \mu_j - \sum_{j > k} \mu_j \leq \sum_{j=1}^n |\mu_j|,$$

the last step because  $\pm \mu_j \leq |\mu_j|$  for each  $j$ .

(b) If  $\mathbf{A} \succeq \mathbf{B}$  then every  $d_i \geq 0$  by [Corollary 16.2.4](#), and every  $\mu_j \geq 0$  by [Theorem 12.1.4](#). Both sides then equal their plain sums, and those are equal, both being  $\text{tr}(\mathbf{A} - \mathbf{B})$ .

In [Example \(All seven index sets\)](#), the  $d_i$  are 2, 1, −1, so the left side is 4; the  $\mu_j$  are 4, 1, −3, so the right side is 8. By [Lemma 16.3.5](#),  $\|\mathbf{A} - \mathbf{B}\|_2 = \max(4, 3) = 4$ , and adding the three bounds of [Corollary 16.3.6](#) gives  $3 \cdot 4 = 12$ . The bound of (a) is never worse than this one: each  $|\mu_j| \leq \|\mathbf{A} - \mathbf{B}\|_2$ , so the right side of (a) is at most  $n\|\mathbf{A} - \mathbf{B}\|_2$ , and it is often much smaller.

## 16.8 The Diagonal and the Spectrum

A Hermitian matrix carries two lists of  $n$  real numbers: its diagonal entries, read off at a glance, and its eigenvalues, computed with effort. Since Chapter 8 we have known that the two lists have the same sum, the trace. This section asks what else the eigenvalues force on the diagonal, and answers the question completely. The answer needs a name, **majorization**. One half of it, Schur's theorem, is §06's bound on partial sums of the diagonal in the new language. The other half, Horn's theorem, says that nothing more is forced; it is proved by an induction driven by a single  $2 \times 2$  rotation. The section ends with a short second proof of Hadamard's determinant inequality.

**Throughout, the field is  $\mathbb{R}$  or  $\mathbb{C}$ , and every matrix whose eigenvalues are indexed is Hermitian**, with its eigenvalues real and listed **decreasingly** with multiplicity,  $\lambda_1(\mathbf{A}) \geq \cdots \geq \lambda_n(\mathbf{A})$ . Two vectors in  $\mathbb{R}^n$  are attached to such an  $\mathbf{A} \in M_n(F)$ :

$$\begin{aligned}\mathbf{d}(\mathbf{A}) &= (a_{11}, a_{22}, \dots, a_{nn}), \\ \boldsymbol{\lambda}(\mathbf{A}) &= (\lambda_1(\mathbf{A}), \lambda_2(\mathbf{A}), \dots, \lambda_n(\mathbf{A})).\end{aligned}$$

The first is the **diagonal vector**, in the order the entries sit on the diagonal; the second is the **eigenvalue vector**, always in decreasing order. The diagonal entries are real because  $\mathbf{A}^* = \mathbf{A}$  gives  $a_{ii} = \overline{a_{ii}}$ , and the eigenvalues are real by [Theorem 11.2.1](#). The bold  $\mathbf{d}(\mathbf{A})$  is a vector, not the diagonal matrix  $\text{diag}(a_{11}, \dots, a_{nn})$ .

Two facts about a Hermitian  $\mathbf{A}$  are used below. The spectral theorem ([Corollary 11.4.3](#) over  $\mathbb{C}$ , [Corollary 11.5.2](#) over  $\mathbb{R}$ ) makes  $\mathbf{A}$  similar to the real diagonal matrix of its eigenvalues, so its characteristic polynomial splits as  $\prod_i (x - \lambda_i(\mathbf{A}))$ . Then [Theorem 8.3.3](#) gives

$$\text{tr } \mathbf{A} = \sum_{i=1}^n \lambda_i(\mathbf{A}), \quad \det \mathbf{A} = \prod_{i=1}^n \lambda_i(\mathbf{A}).$$

### 16.8.1 What the diagonal already knows

Start with the smallest interesting case. The matrix

$$\mathbf{A} = \begin{pmatrix} 2 & 1 \\ 1 & 2 \end{pmatrix}$$

has  $\mathbf{d}(\mathbf{A}) = (2, 2)$  and eigenvalues 3 and 1, with eigenvectors  $(1, 1)$  and  $(1, -1)$ , so  $\boldsymbol{\lambda}(\mathbf{A}) = (3, 1)$ . The totals agree,  $2 + 2 = 4 = 3 + 1$ . But the two lists are not alike: the diagonal is perfectly even, while the spectrum is spread out.

A larger example shows the pattern better. The tridiagonal matrix

$$\mathbf{A} = \begin{pmatrix} 3 & 2 & 0 \\ 2 & 4 & 2 \\ 0 & 2 & 5 \end{pmatrix}$$

has eigenvectors  $(1, 2, 2)$ ,  $(-2, -1, 2)$  and  $(2, -2, 1)$ , with eigenvalues 7, 4 and 1. To check the first,  $\mathbf{A}(1, 2, 2) = (3 + 4, 2 + 8 + 4, 4 + 10) = (7, 14, 14)$ ; the other two are checked the same way. So  $\boldsymbol{\lambda}(\mathbf{A}) = (7, 4, 1)$ , while  $\mathbf{d}(\mathbf{A}) = (3, 4, 5)$ . Sort the diagonal decreasingly to  $(5, 4, 3)$  and compare the running totals of the two lists:

| $k$                          | 1 | 2  | 3  |
|------------------------------|---|----|----|
| largest $k$ diagonal entries | 5 | 9  | 12 |
| largest $k$ eigenvalues      | 7 | 11 | 12 |

Each running total of the eigenvalues is at least the matching running total of the diagonal, and at  $k = n$  the two are equal. None of this is new. [Corollary 16.6.3](#) says that any  $k$  diagonal entries add up to at most  $\lambda_1(\mathbf{A}) + \cdots + \lambda_k(\mathbf{A})$ , with equality at  $k = n$ , and the  $k$  largest diagonal entries are one such choice. What is new is the point of view: to ask which diagonals are **possible**, the corollary's list of inequalities has to be read as a single relation between two vectors.

Two features of this comparison deserve notice. First, the diagonal had to be **sorted**, and this is forced. For a permutation matrix  $\mathbf{P}_\sigma$  ([Definition 2.6.2](#)),  $\mathbf{P}_\sigma^\top = \mathbf{P}_\sigma^{-1}$  by [Lemma 2.6.6](#) (b), so  $\mathbf{P}_\sigma^\top \mathbf{A} \mathbf{P}_\sigma$  is Hermitian with the same eigenvalues as  $\mathbf{A}$ , and its  $(j, j)$  entry is  $\mathbf{e}_{\sigma(j)}^\top \mathbf{A} \mathbf{e}_{\sigma(j)} = a_{\sigma(j)\sigma(j)}$ . Relabeling the basis shuffles the diagonal and fixes the spectrum, so any true relation between  $\mathbf{d}(\mathbf{A})$  and  $\boldsymbol{\lambda}(\mathbf{A})$  must ignore the order of the diagonal. Second, the comparison is between **running totals from the top**, and it measures how unevenly a fixed total is spread.

## 16.8.2 Majorization

*One vector is majorized by another when the two have the same total but the second piles more of it onto its largest entries: every “top  $k$ ” sum of the second is at least that of the first.*

The sorting step gets a notation first. For  $\mathbf{x} = (x_1, \dots, x_n) \in \mathbb{R}^n$ , the **decreasing rearrangement**  $\mathbf{x}^\downarrow = (x_1^\downarrow, \dots, x_n^\downarrow)$  is the vector with the same entries, repeated as often as in  $\mathbf{x}$ , listed so that  $x_1^\downarrow \geq x_2^\downarrow \geq \cdots \geq x_n^\downarrow$ . For example,  $(0, 3, 1, 3)^\downarrow = (3, 3, 1, 0)$ . In this notation,  $\boldsymbol{\lambda}(\mathbf{A}) = \boldsymbol{\lambda}(\mathbf{A})^\downarrow$  by convention, while  $\mathbf{d}(\mathbf{A})$  usually needs sorting.

### Definition 16.8.1 — Majorization

Let  $n \geq 1$  and let  $\mathbf{x}, \mathbf{y} \in \mathbb{R}^n$  be **real** vectors with the **same number** of entries. We say that  $\mathbf{x}$  is **majorized by**  $\mathbf{y}$ , and write  $\mathbf{x} \prec \mathbf{y}$ , if

$$\begin{aligned} \text{(M1)} \quad & \sum_{i=1}^k x_i^\downarrow \leq \sum_{i=1}^k y_i^\downarrow \text{ for every } k = 1, \dots, n-1; \text{ and} \\ \text{(M2)} \quad & \sum_{i=1}^n x_i = \sum_{i=1}^n y_i \text{ (equal totals).} \end{aligned}$$

We say that  $\mathbf{x}$  is **weakly majorized by**  $\mathbf{y}$ , and write  $\mathbf{x} \prec_w \mathbf{y}$ , if the inequality in (M1) holds for **every**  $k = 1, \dots, n$ , including  $k = n$ , and (M2) is dropped.

In words, clause by clause. (M1) sorts both vectors and compares their **running totals from the top**: the largest entry of  $\mathbf{x}$  is at most the largest entry of  $\mathbf{y}$ , the two largest of  $\mathbf{x}$  add up to at most the two largest of  $\mathbf{y}$ , and so on up to  $k = n-1$ . The word **every** matters: one failing  $k$  refutes  $\mathbf{x} \prec \mathbf{y}$ . (M2) says the totals are equal; a sum does not care about order, so no sorting is needed there. Weak majorization keeps (M1), adds its case  $k = n$ , which reads  $\sum_i x_i \leq \sum_i y_i$ , and asks nothing more. Comparing the definitions:  $\mathbf{x} \prec \mathbf{y}$  holds **exactly when**  $\mathbf{x} \prec_w \mathbf{y}$  and the totals are equal.

There is an equivalent way to read (M1) from the bottom. Subtracting the  $k$ -th running total from the equal totals of (M2) gives

$$\sum_{i=k+1}^n x_i^\downarrow \geq \sum_{i=k+1}^n y_i^\downarrow \quad (k = 1, \dots, n-1),$$

so under (M2), the  $n - k$  **smallest** entries of  $\mathbf{x}$  add up to at least the  $n - k$  smallest entries of  $\mathbf{y}$ . The vector  $\mathbf{y}$  is more extreme at both ends: its top is heavier and its bottom is lighter.

The definition makes sense because  $\mathbf{x}^\downarrow$  is determined by  $\mathbf{x}$ : when two entries are equal, it does not matter which is written first. A consequence worth recording: **the relation sees only which numbers occur in each vector and how often, not where they sit.** If  $\mathbf{x}'$  is a rearrangement of  $\mathbf{x}$  and  $\mathbf{y}'$  of  $\mathbf{y}$ , then  $\mathbf{x} \prec \mathbf{y}$  if and only if  $\mathbf{x}' \prec \mathbf{y}'$ , because the rearrangements have the same decreasing rearrangements and the same totals.

**Example. (A chain, the two extremes, and size one)**

Check each claim against [Definition 16.8.1](#).

- (a)  $(1, 1, 1) \prec (2, 1, 0) \prec (3, 0, 0)$ .
- (b) For every  $\mathbf{x} \in \mathbb{R}^n$  with mean  $m = \frac{1}{n} \sum_i x_i$ , the **flat vector**  $m\mathbf{1} = (m, \dots, m)$  satisfies  $m\mathbf{1} \prec \mathbf{x}$ .
- (c) For every  $\mathbf{x} \in \mathbb{R}^n$  with **non-negative** entries and total  $s = \sum_i x_i$ ,  $\mathbf{x} \prec (s, 0, \dots, 0)$ .
- (d) For  $n = 1$ ,  $\mathbf{x} \prec \mathbf{y}$  if and only if  $\mathbf{x} = \mathbf{y}$ .

*Solution.*

- (a) All three vectors are already decreasing, and all have total 3, so (M2) holds for both pairs. Their running totals are

$$(1, 1, 1): 1, 2, 3; \quad (2, 1, 0): 2, 3, 3; \quad (3, 0, 0): 3, 3, 3.$$

At  $k = 1$  and  $k = 2$  the first row is at most the second ( $1 \leq 2, 2 \leq 3$ ), and the second is at most the third ( $2 \leq 3, 3 \leq 3$ ). So (M1) holds for both pairs.

- (b) The totals are  $nm = \sum_i x_i$ , so (M2) holds. For (M1), fix  $k$  and let  $a = \frac{1}{k} \sum_{i \leq k} x_i^\downarrow$  be the average of the  $k$  largest entries. Each of these is at least  $x_k^\downarrow$ , so  $a \geq x_k^\downarrow$ . Every remaining entry  $x_j^\downarrow$  with  $j > k$  is at most  $x_k^\downarrow$ , hence at most  $a$ . Therefore

$$nm = \sum_{i \leq k} x_i^\downarrow + \sum_{j > k} x_j^\downarrow \leq ka + (n - k)a = na,$$

so  $m \leq a$ , and multiplying by  $k$  gives  $km \leq \sum_{i \leq k} x_i^\downarrow$ . The left side is the  $k$ -th running total of  $m\mathbf{1}$ . This is (M1).

- (c) The totals are both  $s$ . Every running total of  $(s, 0, \dots, 0)$  equals  $s$ , and  $\sum_{i \leq k} x_i^\downarrow \leq s$  because the omitted entries  $x_{k+1}^\downarrow, \dots, x_n^\downarrow$  are  $\geq 0$ . This is (M1). Non-negativity is really needed:  $(1, -1)$  has total 0, but  $(1, -1) \prec (0, 0)$  fails at  $k = 1$ , since  $1 > 0$ .
- (d) For  $n = 1$  there are no values of  $k$  in (M1), since  $k$  runs from 1 to 0. So only (M2) remains, and it says  $x_1 = y_1$ .

Parts (b) and (c) are the two ends of the order: among vectors with a given total, the flat vector is the most evenly spread, and among non-negative ones, the vector that puts everything in one entry is the least. Part (d) is the degenerate case: with a single entry there is nothing to spread, and the relation collapses to equality.

The chain in (a) might suggest that any two vectors with the same total can be ranked this way. They cannot.

**Example. (Neither majorizes the other)**

Show that  $\mathbf{x} = (4, 1, 1)$  and  $\mathbf{y} = (3, 3, 0)$  have the same total, that neither  $\mathbf{x} \prec \mathbf{y}$  nor  $\mathbf{y} \prec \mathbf{x}$ , and that both are majorized by  $(4, 2, 0)$ .

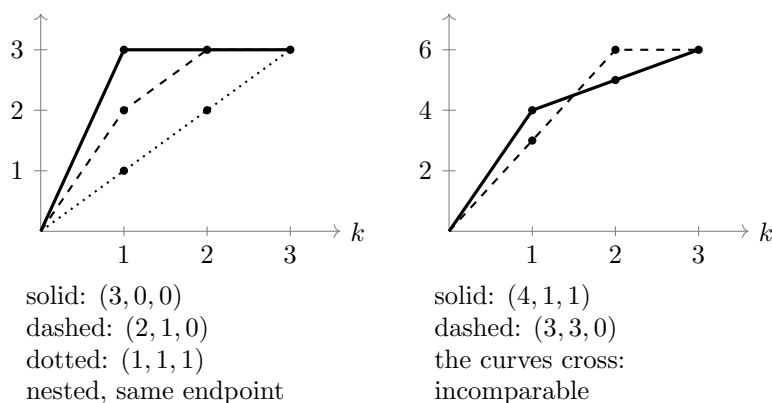
*Solution.* Both totals are 6, and both vectors are already decreasing. Their running totals are

$$\mathbf{x}: 4, 5, 6; \quad \mathbf{y}: 3, 6, 6.$$

At  $k = 1$ ,  $4 > 3$ , so (M1) fails for  $\mathbf{x} \prec \mathbf{y}$ . At  $k = 2$ ,  $6 > 5$ , so (M1) fails for  $\mathbf{y} \prec \mathbf{x}$ . The vector  $(4, 2, 0)$  has total 6 and running totals 4, 6, 6, which are at least both 4, 5 and 3, 6 at  $k = 1, 2$ . So  $\mathbf{x} \prec (4, 2, 0)$  and  $\mathbf{y} \prec (4, 2, 0)$ .

This is the non-example by minimal change. Start from the comparable pair  $(4, 1, 1) \prec (4, 2, 0)$ , move one unit in the second vector from its top entry to its middle entry to get  $(3, 3, 0)$ , and the pair becomes incomparable. (M2) still holds; what fails is (M1), at a **different**  $k$  in each direction. Spreading is not a single number.

The picture makes this visible. Plot the running totals of a sorted vector against  $k$ , joined by segments, starting from  $(0, 0)$ . Then  $\mathbf{x} \prec \mathbf{y}$  says that the curve of  $\mathbf{y}$  lies on or above the curve of  $\mathbf{x}$  and ends at the same height. Two curves can also cross.



**Why this definition.** Each clause earns its place, and changing any one of them changes the subject.

- *Sorting.* Relabeling a basis shuffles  $\mathbf{d}(\mathbf{A})$  and fixes  $\lambda(\mathbf{A})$ , so a relation comparing unsorted entries would answer a question the matrix does not pose.
- *Running totals from the top, with  $\leq$ .* These are the quantities that [Theorem 16.6.2](#) and [Corollary 16.6.3](#) control: the top  $k$  eigenvalues sum to a maximum. Totals from the bottom add nothing new, as shown above, because under (M2) they are the same information.
- *Equal totals.* This is the trace. Dropping it gives weak majorization, the right notion when there is no conservation law to enforce:  $(1, 0) \prec_w (2, 0)$ , but  $(1, 0) \not\prec (2, 0)$ , because the totals differ.

**Warning.**  $\mathbf{x} \prec \mathbf{y}$  does not mean  $x_i \leq y_i$ , not even after sorting both vectors. In  $(1, 1, 1) \prec (2, 1, 0)$ , the third entries are 1 and 0, and  $1 > 0$ . This is no accident: subtracting (M1) at  $k = n - 1$  from (M2) gives  $x_n^\downarrow \geq y_n^\downarrow$  whenever  $\mathbf{x} \prec \mathbf{y}$ , so at the

bottom the inequality always runs the other way. Majorization compares **sums of the largest entries**, not entries one at a time.

*Quick check.* Arrange the five vectors  $(2, 1, 1, 0)$ ,  $(4, 0, 0, 0)$ ,  $(1, 1, 1, 1)$ ,  $(3, 1, 0, 0)$  and  $(2, 2, 0, 0)$  in a chain  $\mathbf{x}^{(1)} \prec \mathbf{x}^{(2)} \prec \dots \prec \mathbf{x}^{(5)}$ .

*Solution.* All five are decreasing and all have total 4. Their running totals at  $k = 1, 2, 3$  are

$$\begin{array}{ll} (1, 1, 1, 1): & 1, 2, 3; \\ (2, 1, 1, 0): & 2, 3, 4; \\ (2, 2, 0, 0): & 2, 4, 4; \\ (3, 1, 0, 0): & 3, 4, 4; \\ (4, 0, 0, 0): & 4, 4, 4. \end{array}$$

Read in this order, each triple is entrywise at most the next, so

$$(1, 1, 1, 1) \prec (2, 1, 1, 0) \prec (2, 2, 0, 0) \prec (3, 1, 0, 0) \prec (4, 0, 0, 0).$$

The ends are the flat vector and the concentrated vector of [Example \(A chain, the two extremes, and size one\)](#) (b) and (c), as they must be. These five happen to be comparable in pairs, but [Example \(Neither majorizes the other\)](#) shows that this is not always so.

### 16.8.3 Majorization is an order

The first small result is that  $\prec$  behaves like an order, and it says precisely how it falls short of being a partial order on  $\mathbb{R}^n$ .

#### Proposition 16.8.1 — Basic Properties of Majorization

Let  $\mathbf{x}, \mathbf{y}, \mathbf{z} \in \mathbb{R}^n$ .

- (a)  $\mathbf{x} \prec \mathbf{x}$ .
- (b) If  $\mathbf{x} \prec \mathbf{y}$  and  $\mathbf{y} \prec \mathbf{z}$ , then  $\mathbf{x} \prec \mathbf{z}$ .
- (c) If  $\mathbf{x} \prec \mathbf{y}$  and  $\mathbf{y} \prec \mathbf{x}$ , then  $\mathbf{x}^\downarrow = \mathbf{y}^\downarrow$ ; that is,  $\mathbf{y}$  is a rearrangement of  $\mathbf{x}$ .

Statements (a) and (b) hold with  $\prec_w$  in place of  $\prec$ .

*Proof.* For  $k = 1, \dots, n$ , write  $X_k = \sum_{i \leq k} x_i^\downarrow$ , and define  $Y_k$  and  $Z_k$  in the same way. Since  $X_n = \sum_i x_i$ , [Definition 16.8.1](#) says that  $\mathbf{x} \prec \mathbf{y}$  means  $X_k \leq Y_k$  for  $k < n$  and  $X_n = Y_n$ .

- (a)  $X_k \leq X_k$  for every  $k$ , and  $X_n = X_n$ .
- (b) For  $k < n$ ,  $X_k \leq Y_k \leq Z_k$ . At  $k = n$ ,  $X_n = Y_n = Z_n$ . Hence  $\mathbf{x} \prec \mathbf{z}$ .
- (c) The hypotheses give  $X_k \leq Y_k$  and  $Y_k \leq X_k$  for  $k < n$ , and  $X_n = Y_n$ , so  $X_k = Y_k$  for every  $k = 1, \dots, n$ . Put  $X_0 = Y_0 = 0$ . Then for each  $k$ ,

$$x_k^\downarrow = X_k - X_{k-1} = Y_k - Y_{k-1} = y_k^\downarrow,$$

so  $\mathbf{x}^\downarrow = \mathbf{y}^\downarrow$ . Two vectors with the same decreasing rearrangement contain the same numbers, each the same number of times, so each is a rearrangement of the other.



For  $\prec_w$ , the arguments for (a) and (b) are the same with every  $k \leq n$  treated like  $k < n$ . This proves the proposition. ■

So  $\prec$  is reflexive and transitive. A relation with these two properties is called a **preorder**. Part (c) is antisymmetry up to rearrangement, and no more:  $(1, 0) \prec (0, 1) \prec (1, 0)$ , yet  $(1, 0) \neq (0, 1)$ . So  $\prec$  is **not** a partial order on  $\mathbb{R}^n$ . Restricted to **decreasing** vectors, where a rearrangement is the vector itself, (c) is exact antisymmetry, and there  $\prec$  is a partial order. By [Example \(Neither majorizes the other\)](#) it is **not total**, not even among non-negative decreasing vectors with the same total. Like the Loewner order of Chapter 12 ([Proposition 12.5.1](#)), it leaves many pairs uncomparable.

## 16.8.4 Schur's theorem

The running totals in the  $3 \times 3$  example were not a coincidence. Restated in the language of majorization, §06's [Corollary 16.6.3](#) says that for every Hermitian matrix the diagonal is more evenly spread than the spectrum:

### Theorem 16.8.2 — Schur's Majorization Theorem

Let  $F = \mathbb{R}$  or  $F = \mathbb{C}$ , and let  $\mathbf{A} \in M_n(F)$  be Hermitian. Then

$$\mathbf{d}(\mathbf{A}) \prec \lambda(\mathbf{A}).$$

Explicitly, if  $\mathbf{d}(\mathbf{A})^\downarrow = (a_1^\downarrow, \dots, a_n^\downarrow)$  lists the diagonal entries of  $\mathbf{A}$  decreasingly, then

$$\begin{aligned} a_1^\downarrow + \dots + a_k^\downarrow &\leq \lambda_1(\mathbf{A}) + \dots + \lambda_k(\mathbf{A}) \quad (1 \leq k \leq n-1), \\ a_1^\downarrow + \dots + a_n^\downarrow &= \lambda_1(\mathbf{A}) + \dots + \lambda_n(\mathbf{A}). \end{aligned}$$

*Proof.* Both vectors lie in  $\mathbb{R}^n$ , as noted at the start of the section. Fix  $k$  with  $1 \leq k \leq n$ . List the positions  $1, \dots, n$  in an order in which the diagonal entries decrease, breaking ties arbitrarily, and let  $i_1 < \dots < i_k$  be the first  $k$  positions of that list, written in increasing order. Then  $a_{i_1 i_1}, \dots, a_{i_k i_k}$  are  $a_1^\downarrow, \dots, a_k^\downarrow$  in some order, so [Corollary 16.6.3](#) gives

$$a_1^\downarrow + \dots + a_k^\downarrow = \sum_{j=1}^k a_{i_j i_j} \leq \lambda_1(\mathbf{A}) + \dots + \lambda_k(\mathbf{A}),$$

with equality when  $k = n$ . For  $k \leq n-1$  this is (M1). For  $k = n$  it is (M2), since a sum does not depend on the order of its terms. This proves the theorem. ■

So Schur's theorem is [Corollary 16.6.3](#) restated. What the restatement adds is a relation between two vectors, and a relation has a converse that can be asked about. Read from the bottom, (M1) says that the  $n-k$  smallest diagonal entries add up to **at least** the  $n-k$  smallest eigenvalues. At the two ends of [Example \(A chain, the two extremes, and size one\)](#), a **diagonal** matrix has  $\mathbf{d}(\mathbf{A})$  a rearrangement of  $\lambda(\mathbf{A})$ , the most spread a diagonal can be; a **constant** diagonal  $t$  is the flat vector  $t\mathbf{1}$ , where Schur's theorem says only that  $nt = \lambda_1(\mathbf{A}) + \dots + \lambda_n(\mathbf{A})$ .

**Example. (Coordinate vectors against eigenvectors)**

For the matrix

$$\mathbf{A} = \begin{pmatrix} 3 & 2 & 0 \\ 2 & 4 & 2 \\ 0 & 2 & 5 \end{pmatrix}$$

from the start of the section, whose running totals were tabulated there, exhibit for  $k = 2$  the coordinate pair that the proof of [Corollary 16.6.3](#) feeds to [Theorem 16.6.2](#) and an orthonormal pair that attains the maximum in [Theorem 16.6.2](#). Account for the difference between the two sums.

*Solution.* We found  $\lambda(\mathbf{A}) = (7, 4, 1)$ , with unit eigenvectors  $\mathbf{u}_1 = \frac{1}{3}(1, 2, 2)$ ,  $\mathbf{u}_2 = \frac{1}{3}(-2, -1, 2)$  and  $\mathbf{u}_3 = \frac{1}{3}(2, -2, 1)$ . They are pairwise orthogonal (for instance  $\langle (1, 2, 2), (-2, -1, 2) \rangle = -2 - 2 + 4 = 0$ ), and each vector in parentheses has length 3.

For  $k = 2$  the proof uses the two largest diagonal entries,  $a_{33} = 5$  and  $a_{22} = 4$ , so the coordinate pair is  $(\mathbf{e}_2, \mathbf{e}_3)$ , and  $\langle \mathbf{A}\mathbf{e}_2, \mathbf{e}_2 \rangle + \langle \mathbf{A}\mathbf{e}_3, \mathbf{e}_3 \rangle = 9$ . The pair  $(\mathbf{u}_1, \mathbf{u}_2)$  gives  $\langle \mathbf{A}\mathbf{u}_1, \mathbf{u}_1 \rangle + \langle \mathbf{A}\mathbf{u}_2, \mathbf{u}_2 \rangle = 7 + 4 = 11$ , since  $\langle \mathbf{A}\mathbf{u}_i, \mathbf{u}_i \rangle = \lambda_i \|\mathbf{u}_i\|^2 = \lambda_i$ . The coordinate pair falls short of the eigenvector pair by 2, and that shortfall is exactly the slack in Schur's theorem at  $k = 2$ . Ky Fan's maximum is attained by eigenvectors, and the diagonal only ever sees coordinate vectors.

*Quick check.* A Hermitian  $3 \times 3$  matrix  $\mathbf{A}$  has diagonal entries 4,  $-1$ , 2. Without knowing anything else about  $\mathbf{A}$ , what does [Theorem 16.8.2](#) say about  $\lambda_1(\mathbf{A})$ , about  $\lambda_1(\mathbf{A}) + \lambda_2(\mathbf{A})$ , and about  $\lambda_3(\mathbf{A})$ ?

*Solution.* Sorted,  $\mathbf{d}(\mathbf{A})^\downarrow = (4, 2, -1)$ , with running totals 4, 6, 5. So  $\lambda_1(\mathbf{A}) \geq 4$ ,  $\lambda_1(\mathbf{A}) + \lambda_2(\mathbf{A}) \geq 6$ , and  $\lambda_1(\mathbf{A}) + \lambda_2(\mathbf{A}) + \lambda_3(\mathbf{A}) = 5$ . Subtracting the middle inequality from the last equality gives  $\lambda_3(\mathbf{A}) \leq 5 - 6 = -1$ . In particular  $\mathbf{A}$  is indefinite, since  $\lambda_1(\mathbf{A}) > 0 > \lambda_3(\mathbf{A})$ , and that was visible without computing any eigenvalue.

## 16.8.5 Horn's converse

Schur's theorem gives a **necessary** condition on a diagonal. Is it also sufficient? Given  $\mathbf{d} \prec \lambda$ , is there a Hermitian matrix with eigenvalues  $\lambda_1, \dots, \lambda_n$  and diagonal entries  $d_1, \dots, d_n$ , in the given order? For  $n = 2$  a rotation answers the question. Turning the eigenbasis of  $\text{diag}(\lambda_1, \lambda_2)$  moves its first diagonal entry through every value between  $\lambda_2$  and  $\lambda_1$ , and the trace fixes the second entry.

### Lemma 16.8.3 — The Two-by-Two Rotation

Let  $\lambda_1 \geq \lambda_2$  and  $t$  be real numbers with  $\lambda_2 \leq t \leq \lambda_1$ . Put  $b = \sqrt{(\lambda_1 - t)(t - \lambda_2)}$ . Then there is an orthogonal  $\mathbf{R} \in \text{O}(2)$  with

$$\mathbf{R} \text{diag}(\lambda_1, \lambda_2) \mathbf{R}^\top = \begin{pmatrix} t & b \\ b & \lambda_1 + \lambda_2 - t \end{pmatrix}.$$

*Idea.* If the first row of  $\mathbf{R}$  is  $(c, -s)$  with  $c^2 + s^2 = 1$ , the  $(1, 1)$  entry of  $\mathbf{R} \operatorname{diag}(\lambda_1, \lambda_2) \mathbf{R}^\top$  is  $c^2 \lambda_1 + s^2 \lambda_2$ , an average of  $\lambda_1$  and  $\lambda_2$  with weights  $c^2$  and  $s^2$ . Every  $t$  between them is such an average, and solving for the weights gives  $c$  and  $s$ .

*Proof.* If  $\lambda_1 = \lambda_2$ , then  $t = \lambda_1 = \lambda_2$  and  $b = 0$ , and  $\mathbf{R} = \mathbf{I}$  works. Otherwise  $\lambda_1 - \lambda_2 > 0$ , and

$$c = \sqrt{\frac{t - \lambda_2}{\lambda_1 - \lambda_2}}, \quad s = \sqrt{\frac{\lambda_1 - t}{\lambda_1 - \lambda_2}}$$

are real numbers, because both radicands are  $\geq 0$  by the hypothesis on  $t$ . They satisfy  $c^2 + s^2 = 1$ . Let  $\mathbf{R} = \begin{pmatrix} c & -s \\ s & c \end{pmatrix}$ . Then  $\mathbf{R}^\top \mathbf{R} = (c^2 + s^2) \mathbf{I} = \mathbf{I}$ , so  $\mathbf{R} \in O(2)$ . Multiplying out,

$$\mathbf{R} \operatorname{diag}(\lambda_1, \lambda_2) \mathbf{R}^\top = \begin{pmatrix} c^2 \lambda_1 + s^2 \lambda_2 & cs(\lambda_1 - \lambda_2) \\ cs(\lambda_1 - \lambda_2) & s^2 \lambda_1 + c^2 \lambda_2 \end{pmatrix}.$$

Since  $s^2 = 1 - c^2$ , the  $(1, 1)$  entry is  $\lambda_2 + c^2(\lambda_1 - \lambda_2) = \lambda_2 + (t - \lambda_2) = t$ . The two diagonal entries add up to  $(c^2 + s^2)(\lambda_1 + \lambda_2) = \lambda_1 + \lambda_2$ , so the  $(2, 2)$  entry is  $\lambda_1 + \lambda_2 - t$ . Finally  $cs(\lambda_1 - \lambda_2) = \sqrt{(t - \lambda_2)(\lambda_1 - t)} = b$ . This proves the lemma. ■

This is the converse for  $n = 2$ : if  $\lambda_1 \geq \lambda_2$  and  $(d_1, d_2) \prec (\lambda_1, \lambda_2)$ , then  $d_1, d_2 \leq d_1^\downarrow \leq \lambda_1$  by (M1), so by (M2)  $d_1 = \lambda_1 + \lambda_2 - d_2 \geq \lambda_2$ , so  $t = d_1$  is allowed. For larger  $n$ , the lemma is used once, on a well-chosen pair of eigenvalues, to put  $d_1$  in the corner, and induction does the rest.

### Theorem 16.8.4 — Horn's Theorem

Let  $n \geq 1$ , and let  $\mathbf{d} = (d_1, \dots, d_n)$  and  $\boldsymbol{\lambda} = (\lambda_1, \dots, \lambda_n)$  be vectors in  $\mathbb{R}^n$  with  $\mathbf{d} \prec \boldsymbol{\lambda}$ . Then there is an orthogonal  $\mathbf{Q} \in O(n)$  such that the matrix

$$\mathbf{A} = \mathbf{Q} \operatorname{diag}(\lambda_1, \dots, \lambda_n) \mathbf{Q}^\top$$

has diagonal entries  $d_1, d_2, \dots, d_n$ , **in this order**. So  $\mathbf{A}$  is a real symmetric matrix with  $\mathbf{d}(\mathbf{A}) = \mathbf{d}$  and  $\boldsymbol{\lambda}(\mathbf{A}) = \boldsymbol{\lambda}^\downarrow$ .

*Idea.* Induction on  $n$ , one diagonal entry at a time.

① Permutation matrices are orthogonal and shuffle diagonal entries, so it is enough to treat  $\mathbf{d}$  and  $\boldsymbol{\lambda}$  both decreasing. ② The top and bottom clauses of majorization put  $d_1$  between  $\lambda_n$  and  $\lambda_1$ , hence between two **consecutive** eigenvalues,  $\lambda_k \geq d_1 \geq \lambda_{k+1}$ . The lemma rotates that pair so that  $d_1$  sits in the corner, with  $\eta = \lambda_k + \lambda_{k+1} - d_1$  next to it. ③ The Claim: what is left over, the diagonal  $d_2, \dots, d_n$  and the spectrum with  $\lambda_k, \lambda_{k+1}$  replaced by  $\eta$ , satisfies the hypothesis again, in size  $n - 1$ . ④ Induction rotates the lower  $(n - 1) \times (n - 1)$  block into place without touching the corner.

Step ③ is where the choice of  $k$  matters.

*Proof.* For every  $\mathbf{Q} \in O(n)$ , the matrix  $\mathbf{A} = \mathbf{Q} \operatorname{diag}(\lambda_1, \dots, \lambda_n) \mathbf{Q}^\top$  is real and symmetric, and by Corollary 11.5.2 its eigenvalues, with multiplicity, are  $\lambda_1, \dots, \lambda_n$ ; listed decreasingly they form  $\boldsymbol{\lambda}^\downarrow$ . So only the diagonal needs work. We use two facts repeatedly. A product of orthogonal matrices is orthogonal, since  $(\mathbf{X}\mathbf{Y})^\top \mathbf{X}\mathbf{Y} = \mathbf{Y}^\top \mathbf{X}^\top \mathbf{X}\mathbf{Y} = \mathbf{I}$ . And

for a permutation  $\pi \in S_n$  and any  $\mathbf{M} \in M_n(\mathbb{R})$ , [Definition 2.6.2](#) gives

$$(\mathbf{P}_\pi^\top \mathbf{M} \mathbf{P}_\pi)_{ij} = \mathbf{e}_{\pi(i)}^\top \mathbf{M} \mathbf{e}_{\pi(j)} = m_{\pi(i)\pi(j)}, \quad (\dagger)$$

where  $\mathbf{P}_\pi$  is orthogonal by [Lemma 2.6.6](#) (b).

**Step 1: reduction to decreasing vectors.** Suppose the theorem holds whenever  $\mathbf{d}$  and  $\lambda$  are both decreasing, and let  $\mathbf{d} \prec \lambda$  be arbitrary. Majorization sees only rearrangements (as noted after [Definition 16.8.1](#)), so  $\mathbf{d}^\downarrow \prec \lambda^\downarrow$ , and there is  $\mathbf{Q}_0 \in O(n)$  such that  $\mathbf{B} = \mathbf{Q}_0 \text{diag}(\lambda^\downarrow) \mathbf{Q}_0^\top$  has diagonal  $\mathbf{d}^\downarrow$ . Here  $\text{diag}(\lambda^\downarrow)$  is the diagonal matrix with diagonal  $\lambda_1^\downarrow, \dots, \lambda_n^\downarrow$ . Since  $\lambda^\downarrow$  is a rearrangement of  $\lambda$ , and  $\mathbf{d}^\downarrow$  of  $\mathbf{d}$ , there are permutations  $\sigma, \tau \in S_n$  with  $\lambda_{\sigma(i)} = \lambda_i^\downarrow$  and  $d_{\tau(j)}^\downarrow = d_j$  for all  $i, j$ . By  $(\dagger)$ ,  $\mathbf{P}_\sigma^\top \text{diag}(\lambda_1, \dots, \lambda_n) \mathbf{P}_\sigma = \text{diag}(\lambda^\downarrow)$ , and the  $(j, j)$  entry of  $\mathbf{P}_\tau^\top \mathbf{B} \mathbf{P}_\tau$  is  $b_{\tau(j)\tau(j)} = d_{\tau(j)}^\downarrow = d_j$ . So  $\mathbf{Q} = \mathbf{P}_\tau^\top \mathbf{Q}_0 \mathbf{P}_\sigma^\top$  is orthogonal and

$$\begin{aligned} \mathbf{Q} \text{diag}(\lambda_1, \dots, \lambda_n) \mathbf{Q}^\top &= \mathbf{P}_\tau^\top \mathbf{Q}_0 \text{diag}(\lambda^\downarrow) \mathbf{Q}_0^\top \mathbf{P}_\tau \\ &= \mathbf{P}_\tau^\top \mathbf{B} \mathbf{P}_\tau \end{aligned}$$

has diagonal  $\mathbf{d}$ . It therefore suffices to prove the theorem for decreasing  $\mathbf{d}$  and  $\lambda$ , and we do so by induction on  $n$ .

**Step 2: the corner.** For  $n = 1$ ,  $\mathbf{d} \prec \lambda$  means  $d_1 = \lambda_1$  ([Example \(A chain, the two extremes, and size one\)](#) (d)), and  $\mathbf{Q} = (1)$  works. Let  $n \geq 2$ , assume the theorem for  $n - 1$  (for vectors in any order, by Step 1), and let  $\mathbf{d}$  and  $\lambda$  be decreasing with  $\mathbf{d} \prec \lambda$ . By (M1) at  $k = 1$ ,  $d_1 \leq \lambda_1$ . Subtracting (M1) at  $k = n - 1$  from (M2) gives  $d_n \geq \lambda_n$ , so  $d_1 \geq d_n \geq \lambda_n$ . Let  $k$  be the smallest index in  $\{1, \dots, n - 1\}$  with  $\lambda_{k+1} \leq d_1$ ; one exists, since  $\lambda_n \leq d_1$ . Then  $\lambda_k \geq d_1$ : for  $k = 1$  this is  $d_1 \leq \lambda_1$ , and for  $k \geq 2$  the minimality of  $k$  gives  $\lambda_k > d_1$ . So

$$\lambda_k \geq d_1 \geq \lambda_{k+1}.$$

By [Lemma 16.8.3](#) with  $t = d_1$ , there are  $\mathbf{R} \in O(2)$  and a real  $b$  with

$$\mathbf{R} \text{diag}(\lambda_k, \lambda_{k+1}) \mathbf{R}^\top = \begin{pmatrix} d_1 & b \\ b & \eta \end{pmatrix}, \quad \eta = \lambda_k + \lambda_{k+1} - d_1.$$

Moreover  $\lambda_{k+1} \leq \eta \leq \lambda_k$ , since  $\eta - \lambda_{k+1} = \lambda_k - d_1 \geq 0$  and  $\lambda_k - \eta = d_1 - \lambda_{k+1} \geq 0$ .

**Step 3: the leftover data.** Put

$$\begin{aligned} \mathbf{d}' &= (d_2, \dots, d_n), \\ \lambda' &= (\eta, \lambda_1, \dots, \lambda_{k-1}, \lambda_{k+2}, \dots, \lambda_n), \end{aligned}$$

two vectors in  $\mathbb{R}^{n-1}$ . So  $\lambda'$  is  $\lambda$  with the pair  $\lambda_k, \lambda_{k+1}$  replaced by the single number  $\eta$ , written first.

*Claim.*  $\mathbf{d}' \prec \boldsymbol{\lambda}'$ .

*Proof.* By Step 2,  $\lambda_{k-1} \geq \lambda_k \geq \eta \geq \lambda_{k+1} \geq \lambda_{k+2}$ , for those of these indices that exist. So

$$(\mu_1, \dots, \mu_{n-1}) = (\lambda_1, \dots, \lambda_{k-1}, \eta, \lambda_{k+2}, \dots, \lambda_n)$$

is  $(\boldsymbol{\lambda}')^\downarrow$ , and  $\mathbf{d}'$  is already decreasing. For (M2),  $\sum_i d'_i = \sum_i d_i - d_1 = \sum_i \lambda_i - d_1 = \sum_i \lambda'_i$ , by (M2) for  $\mathbf{d} \prec \boldsymbol{\lambda}$  and the definition of  $\eta$ . For (M1), fix  $m$  with  $1 \leq m \leq n-2$ ; when  $n=2$  there is nothing to check.

*Case*  $m \leq k-1$ . Then  $\mu_i = \lambda_i$  for  $i \leq m$ . For each such  $i$ ,  $d_{i+1} \leq d_1 \leq \lambda_k \leq \lambda_i$ , because  $\mathbf{d}$  and  $\boldsymbol{\lambda}$  are decreasing and  $i < k$ . Summing over  $i \leq m$  gives  $d_2 + \dots + d_{m+1} \leq \mu_1 + \dots + \mu_m$ .

*Case*  $m \geq k$ . The first  $m$  entries of  $(\boldsymbol{\lambda}')^\downarrow$  are  $\lambda_1, \dots, \lambda_{k-1}$ , then  $\eta$ , then  $\lambda_{k+2}, \dots, \lambda_{m+1}$ . So

$$\begin{aligned} \mu_1 + \dots + \mu_m &= (\lambda_1 + \dots + \lambda_{m+1}) - \lambda_k - \lambda_{k+1} + \eta \\ &= (\lambda_1 + \dots + \lambda_{m+1}) - d_1. \end{aligned}$$

Since  $m+1 \leq n-1$ , (M1) for  $\mathbf{d} \prec \boldsymbol{\lambda}$  applies at  $m+1$  and gives

$$\begin{aligned} d_2 + \dots + d_{m+1} &= (d_1 + \dots + d_{m+1}) - d_1 \\ &\leq (\lambda_1 + \dots + \lambda_{m+1}) - d_1 = \mu_1 + \dots + \mu_m. \end{aligned}$$

In both cases (M1) holds, which proves the claim. ■

**Step 4: assembly.** Let  $\sigma \in S_n$  list the indices  $k, k+1$  first and the others after them in increasing order:  $\sigma(1) = k$ ,  $\sigma(2) = k+1$ , and  $\sigma(3), \dots, \sigma(n)$  are  $1, \dots, k-1, k+2, \dots, n$ . By (†),

$$\mathbf{P}_\sigma^\top \text{diag}(\lambda_1, \dots, \lambda_n) \mathbf{P}_\sigma = \begin{pmatrix} \text{diag}(\lambda_k, \lambda_{k+1}) & \mathbf{0} \\ \mathbf{0} & \mathbf{D}_1 \end{pmatrix},$$

where  $\mathbf{D}_1 = \text{diag}(\lambda_1, \dots, \lambda_{k-1}, \lambda_{k+2}, \dots, \lambda_n) \in M_{n-2}(\mathbb{R})$ ; when  $n=2$  the blocks involving  $\mathbf{D}_1$  are absent. Let  $\mathbf{R}_1 = \begin{pmatrix} \mathbf{R} & \mathbf{0} \\ \mathbf{0} & \mathbf{I}_{n-2} \end{pmatrix}$ . Multiplying blocks and using Step 2,

$$\mathbf{M} = \mathbf{R}_1 \mathbf{P}_\sigma^\top \text{diag}(\lambda_1, \dots, \lambda_n) \mathbf{P}_\sigma \mathbf{R}_1^\top = \begin{pmatrix} d_1 & b & \mathbf{0} \\ b & \eta & \mathbf{0} \\ \mathbf{0} & \mathbf{0} & \mathbf{D}_1 \end{pmatrix} = \begin{pmatrix} d_1 & b \mathbf{e}_1^\top \\ b \mathbf{e}_1 & \mathbf{E} \end{pmatrix},$$

where  $\mathbf{e}_1$  is the first standard basis vector of  $\mathbb{R}^{n-1}$  and  $\mathbf{E} = \text{diag}(\eta, \lambda_1, \dots, \lambda_{k-1}, \lambda_{k+2}, \dots, \lambda_n)$  is the diagonal matrix of  $\boldsymbol{\lambda}'$ . By the Claim and the inductive hypothesis, there is  $\mathbf{W} \in O(n-1)$  such that  $\mathbf{W} \mathbf{E} \mathbf{W}^\top$  has diagonal entries  $d_2, \dots, d_n$ . Let  $\mathbf{W}_1 = \begin{pmatrix} 1 & \mathbf{0}^\top \\ \mathbf{0} & \mathbf{W} \end{pmatrix}$ . Then

$$\mathbf{W}_1 \mathbf{M} \mathbf{W}_1^\top = \begin{pmatrix} d_1 & b (\mathbf{W} \mathbf{e}_1)^\top \\ b \mathbf{W} \mathbf{e}_1 & \mathbf{W} \mathbf{E} \mathbf{W}^\top \end{pmatrix},$$

whose diagonal entries are  $d_1, d_2, \dots, d_n$ . The block matrices  $\mathbf{R}_1$  and  $\mathbf{W}_1$  are orthogonal, since  $\mathbf{R}_1^\top \mathbf{R}_1$  and  $\mathbf{W}_1^\top \mathbf{W}_1$  are block diagonal with blocks  $\mathbf{R}^\top \mathbf{R} = \mathbf{I}$ ,  $\mathbf{I}$ , and  $1$ ,  $\mathbf{W}^\top \mathbf{W} = \mathbf{I}$ . So  $\mathbf{Q} = \mathbf{W}_1 \mathbf{R}_1 \mathbf{P}_\sigma^\top$  is orthogonal, and  $\mathbf{Q} \text{diag}(\lambda_1, \dots, \lambda_n) \mathbf{Q}^\top = \mathbf{W}_1 \mathbf{M} \mathbf{W}_1^\top$ . This completes the induction and proves the theorem.

Over  $\mathbb{C}$  nothing more is needed, because an orthogonal matrix is unitary and a real symmetric matrix is Hermitian. Together with Schur's theorem, this settles the question the section began with. ■

### Theorem 16.8.5 — Schur–Horn Theorem

Let  $F = \mathbb{R}$  or  $F = \mathbb{C}$ , and let  $\mathbf{d}, \boldsymbol{\lambda} \in \mathbb{R}^n$ . There is a Hermitian  $\mathbf{A} \in M_n(F)$  with  $\boldsymbol{\lambda}(\mathbf{A}) = \boldsymbol{\lambda}^\downarrow$  and  $\mathbf{d}(\mathbf{A}) = \mathbf{d}$  if and only if  $\mathbf{d} \prec \boldsymbol{\lambda}$ . In words: the diagonals of the Hermitian matrices with spectrum  $\boldsymbol{\lambda}$  are **exactly** the vectors majorized by  $\boldsymbol{\lambda}$ , and real symmetric matrices already give all of them.

*Proof.* ( $\Rightarrow$ ) By Theorem 16.8.2,  $\mathbf{d} = \mathbf{d}(\mathbf{A}) \prec \boldsymbol{\lambda}(\mathbf{A}) = \boldsymbol{\lambda}^\downarrow$ . Since majorization sees only rearrangements,  $\mathbf{d} \prec \boldsymbol{\lambda}$ .

( $\Leftarrow$ ) Theorem 16.8.4 gives a real symmetric  $\mathbf{A} \in M_n(\mathbb{R}) \subseteq M_n(F)$  with  $\mathbf{d}(\mathbf{A}) = \mathbf{d}$  and  $\boldsymbol{\lambda}(\mathbf{A}) = \boldsymbol{\lambda}^\downarrow$ , and a real symmetric matrix is Hermitian. ■

### Example. (Building a matrix from its diagonal and spectrum)

Find a real symmetric matrix with eigenvalues 6, 1, 0 and diagonal (4, 2, 1).

*Solution.* Both vectors are decreasing, and  $(4, 2, 1) \prec (6, 1, 0)$ : the running totals are 4, 6 against 6, 7, and both totals are 7. Follow the proof of Theorem 16.8.4.

*The corner.*  $d_1 = 4$  lies between  $\lambda_1 = 6$  and  $\lambda_2 = 1$ , so  $k = 1$ ,  $\eta = 6 + 1 - 4 = 3$ , and Lemma 16.8.3 gives  $b = \sqrt{(6-4)(4-1)} = \sqrt{6}$ . After the first rotation the matrix is

$$\mathbf{M} = \begin{pmatrix} 4 & \sqrt{6} & 0 \\ \sqrt{6} & 3 & 0 \\ 0 & 0 & 0 \end{pmatrix}.$$

*The leftover.*  $\mathbf{d}' = (2, 1)$  and  $\boldsymbol{\lambda}' = (3, 0)$ , with  $2 \leq 3$  and  $2 + 1 = 3 + 0$ , so  $\mathbf{d}' \prec \boldsymbol{\lambda}'$ , as the Claim promises. The lemma with  $t = 2$  gives  $c = \sqrt{2/3}$ ,  $s = \sqrt{1/3}$ , and

$$\mathbf{W} \operatorname{diag}(3, 0) \mathbf{W}^\top = \begin{pmatrix} 2 & \sqrt{2} \\ \sqrt{2} & 1 \end{pmatrix}, \quad \mathbf{W} \mathbf{e}_1 = (c, s).$$

*Assembly.* The first row becomes  $(4, \sqrt{6}c, \sqrt{6}s) = (4, 2, \sqrt{2})$ , so

$$\mathbf{A} = \begin{pmatrix} 4 & 2 & \sqrt{2} \\ 2 & 2 & \sqrt{2} \\ \sqrt{2} & \sqrt{2} & 1 \end{pmatrix}.$$

Check directly:  $\mathbf{A}(3, 2, \sqrt{2}) = (18, 12, 6\sqrt{2}) = 6(3, 2, \sqrt{2})$ ,  $\mathbf{A}(-\sqrt{2}, \sqrt{2}, 1) = (-\sqrt{2}, \sqrt{2}, 1)$ , and  $\mathbf{A}(0, -1, \sqrt{2}) = (0, 0, 0)$ . Three eigenvectors with distinct eigenvalues 6, 1, 0 give the whole spectrum, and the diagonal is (4, 2, 1).

## 16.8.6 Hadamard's inequality from the diagonal

Chapter 12 §06 proved [Theorem 12.6.4](#): for a positive semidefinite  $\mathbf{A}$ ,  $\det \mathbf{A} \leq a_{11}a_{22} \cdots a_{nn}$ , with equality if and only if  $\mathbf{A}$  is diagonal or some  $a_{ii} = 0$ . That proof inducts on the size through Fischer's inequality ([Corollary 12.6.3](#)). A much shorter route scales  $\mathbf{A}$  so that its diagonal is flat, and then compares the product of the eigenvalues with their sum. That comparison is the inequality between the arithmetic and geometric means, which the book has not yet proved; it is pure algebra, and it comes first.

### Lemma 16.8.6 — The Arithmetic–Geometric Mean Inequality

Let  $n \geq 1$  and let  $t_1, \dots, t_n$  be **non-negative** real numbers. Then

$$t_1 t_2 \cdots t_n \leq \left( \frac{t_1 + t_2 + \cdots + t_n}{n} \right)^n,$$

with equality if and only if  $t_1 = t_2 = \cdots = t_n$ .

*Idea.* Scale so that the mean is 1. If the numbers are not all 1, one of them lies above 1 and another below. Merge that pair into the single number  $t_j + t_k - 1$ . This leaves  $n - 1$  non-negative numbers with mean 1, so induction applies to them. The merge can only increase the product, because

$$(t_j + t_k - 1) - t_j t_k = (t_j - 1)(1 - t_k),$$

and both factors are positive. So the original product is at most, and when it is non-zero strictly less than, a product that induction bounds by 1.

*Proof. Step 1: mean 1.* We prove by induction on  $n \geq 1$ : if  $t_1, \dots, t_n \geq 0$  and  $t_1 + \cdots + t_n = n$ , then  $t_1 \cdots t_n \leq 1$ , with equality only when every  $t_i = 1$ . For  $n = 1$  the hypothesis says  $t_1 = 1$ , and the product is 1.

Let  $n \geq 2$  and assume the claim for  $n - 1$  numbers. If every  $t_i = 1$ , the product is 1. Otherwise some  $t_j > 1$ : if every  $t_i \leq 1$  with at least one strictly smaller, the sum would be less than  $n$ . Similarly some  $t_k < 1$ , since if every  $t_i \geq 1$  with at least one strictly larger, the sum would exceed  $n$ . Then  $j \neq k$ . Neither the sum nor the product depends on the order of the numbers, so we may relabel so that  $j = n - 1$  and  $k = n$ . Put

$$s = t_{n-1} + t_n - 1.$$

Then  $s > 0$ , since  $t_{n-1} > 1$  and  $t_n \geq 0$ . The  $n - 1$  numbers  $t_1, \dots, t_{n-2}, s$  are non-negative and add up to  $n - 1$ . So by the inductive hypothesis  $ps \leq 1$ , where  $p = t_1 \cdots t_{n-2} \geq 0$  (with  $p = 1$  when  $n = 2$ ). Moreover

$$s - t_{n-1}t_n = (t_{n-1} - 1)(1 - t_n) > 0.$$

If  $p = 0$ , then  $t_1 \cdots t_n = 0 < 1$ . If  $p > 0$ , then  $t_1 \cdots t_n = p t_{n-1} t_n < p s \leq 1$ . In either

case the product is strictly less than 1 when the  $t_i$  are not all 1. This completes the induction.

**Step 2: the general case.** Let  $m = \frac{1}{n}(t_1 + \cdots + t_n) \geq 0$ . If  $m = 0$ , then every  $t_i = 0$ , since non-negative numbers with sum 0 are all 0. Both sides are then 0, and the  $t_i$  are all equal. If  $m > 0$ , the numbers  $t_i/m$  are non-negative with sum  $n$ . By Step 1,

$$\frac{t_1}{m} \cdots \frac{t_n}{m} \leq 1, \quad \text{that is,} \quad t_1 \cdots t_n \leq m^n,$$

with equality if and only if every  $t_i/m = 1$ , that is, every  $t_i = m$ . Numbers that are all equal to some  $c$  have mean  $m = c$ , so “every  $t_i = m$ ” is the same as “all the  $t_i$  are equal”. This proves the lemma. ■

In the language of this section, the lemma says that among non-negative vectors with a fixed total, the **flat** vector, which is the bottom of the majorization order ([Example \(A chain, the two extremes, and size one\)](#) (b)), has the largest product.

### Corollary 16.8.7 — Hadamard’s Inequality, Second Proof

Let  $F = \mathbb{R}$  or  $F = \mathbb{C}$ , and let  $\mathbf{A} \in M_n(F)$  with  $\mathbf{A} \succeq 0$ . Then

$$\det \mathbf{A} \leq a_{11}a_{22} \cdots a_{nn},$$

with equality if and only if  $\mathbf{A}$  is diagonal or some  $a_{ii} = 0$ .

This is [Theorem 12.6.4](#) (a) again, with the same hypothesis and the same equality condition. What is new is the proof.

*Idea.* Suppose first that every diagonal entry equals 1. Then  $\text{tr } \mathbf{A} = n$ , so the eigenvalues, which are non-negative, add up to  $n$ , and [Lemma 16.8.6](#) gives  $\det \mathbf{A} = \prod_i \lambda_i(\mathbf{A}) \leq 1 = \prod_i a_{ii}$ . A general diagonal is made flat by a diagonal congruence  $\mathbf{A} \mapsto \mathbf{SAS}$  with  $\mathbf{S} = \text{diag}(a_{11}^{-1/2}, \dots, a_{nn}^{-1/2})$ . This congruence preserves positivity and divides both sides of the inequality by the same number  $a_{11} \cdots a_{nn}$ . Zero diagonal entries, where  $\mathbf{S}$  does not exist, are handled first and separately.

*Proof.* Each  $a_{ii} = \langle \mathbf{A}\mathbf{e}_i, \mathbf{e}_i \rangle \geq 0$  by [Definition 12.1.1](#), and each  $\lambda_i(\mathbf{A}) \geq 0$  by [Theorem 12.1.4](#) (b).

*Case 1: some  $a_{ii} = 0$ .* The right-hand side is 0. Suppose  $\det \mathbf{A} \neq 0$ . By [Theorem 8.3.3](#),  $\det \mathbf{A} = \prod_j \lambda_j(\mathbf{A})$ , so every  $\lambda_j(\mathbf{A})$  is non-zero, hence positive. By [Theorem 12.1.5](#) (b),  $\mathbf{A} \succ 0$ , so  $a_{ii} = \langle \mathbf{A}\mathbf{e}_i, \mathbf{e}_i \rangle > 0$ , a contradiction. Hence  $\det \mathbf{A} = 0$ , and the inequality holds with equality, as the statement says it should when some  $a_{ii} = 0$ .

*Case 2: every  $a_{ii} > 0$ .* Let  $\mathbf{S} = \text{diag}(a_{11}^{-1/2}, \dots, a_{nn}^{-1/2})$ , a real diagonal matrix with non-zero diagonal, so  $\mathbf{S}^* = \mathbf{S}$  and  $\mathbf{S}$  is invertible. Put  $\mathbf{B} = \mathbf{S}^*\mathbf{A}\mathbf{S} = \mathbf{SAS}$ . By [Proposition 12.1.2](#) (a),  $\mathbf{B} \succeq 0$ . Its entries are  $b_{ij} = a_{ij}/\sqrt{a_{ii}a_{jj}}$ , so  $b_{ii} = 1$  for every  $i$ , and  $\text{tr } \mathbf{B} = n$ . By [Theorem 8.3.3](#),

$$\lambda_1(\mathbf{B}) + \cdots + \lambda_n(\mathbf{B}) = n.$$



The  $\lambda_i(\mathbf{B})$  are non-negative by [Theorem 12.1.4](#) (b). So [Lemma 16.8.6](#) and [Theorem 8.3.3](#) give

$$\det \mathbf{B} = \prod_{i=1}^n \lambda_i(\mathbf{B}) \leq \left(\frac{n}{n}\right)^n = 1,$$

with equality if and only if all  $\lambda_i(\mathbf{B})$  are equal, hence all equal to 1. By [Theorem 6.5.4](#), and since  $\det \mathbf{S} = (a_{11} \cdots a_{nn})^{-1/2}$ ,

$$\det \mathbf{B} = (\det \mathbf{S})^2 \det \mathbf{A} = \frac{\det \mathbf{A}}{a_{11}a_{22} \cdots a_{nn}}.$$

Multiplying  $\det \mathbf{B} \leq 1$  by the positive number  $a_{11} \cdots a_{nn}$  gives  $\det \mathbf{A} \leq a_{11} \cdots a_{nn}$ .

For equality in Case 2: equality holds exactly when every  $\lambda_i(\mathbf{B}) = 1$ . If so, the spectral theorem ([Corollary 11.4.3](#), or [Corollary 11.5.2](#) over  $\mathbb{R}$ ) gives  $\mathbf{B} = \mathbf{U}\mathbf{I}\mathbf{U}^* = \mathbf{I}$  for some unitary (or orthogonal)  $\mathbf{U}$ . Then  $\mathbf{A} = \mathbf{S}^{-1}\mathbf{B}\mathbf{S}^{-1} = \mathbf{S}^{-2} = \text{diag}(a_{11}, \dots, a_{nn})$  is diagonal. Conversely, if  $\mathbf{A}$  is diagonal, then  $\mathbf{B} = \mathbf{S}\mathbf{A}\mathbf{S} = \mathbf{I}$ , and every  $\lambda_i(\mathbf{B}) = 1$ . So in Case 2, equality holds if and only if  $\mathbf{A}$  is diagonal. Together with Case 1, this is the stated equality condition. This proves the corollary. ■

For the matrix of [Example \(Coordinate vectors against eigenvectors\)](#), which is positive definite since its eigenvalues are 7, 4, 1, the inequality reads  $\det \mathbf{A} = 7 \cdot 4 \cdot 1 = 28 \leq 60 = 3 \cdot 4 \cdot 5$ .

The two proofs show different things. The first proves more along the way: Fischer's inequality for blocks of any sizes. The second shows that once the diagonal is scaled flat, the trace and the arithmetic–geometric mean inequality suffice. Of  $\mathbf{d}(\mathbf{B}) \prec \boldsymbol{\lambda}(\mathbf{B})$  it uses only clause (M2), which for a matrix is just the trace.

*Remark.* A proof by majorization in earnest would use all of  $\mathbf{d}(\mathbf{A}) \prec \boldsymbol{\lambda}(\mathbf{A})$ , without scaling, together with the fact that for vectors with **non-negative** entries,  $\mathbf{x} \prec \mathbf{y}$  implies  $x_1 \cdots x_n \geq y_1 \cdots y_n$  (the product is **Schur-concave**). That fact is true but **not proved in this book yet**; Chapter 20 proves it, for the whole class of functions that reverse majorization.

Majorization is the language this chapter's eigenvalue inequalities were already using, and the Schur–Horn theorem shows that it fits the diagonal problem exactly. The same language compresses the inequality of Section 7 for arbitrary index sets, [Theorem 16.7.2](#), into one line. The  $k$  largest entries of the vector  $\boldsymbol{\lambda}(\mathbf{A}) - \boldsymbol{\lambda}(\mathbf{B})$  add up to the largest value of  $\sum_j (\lambda_{i_j}(\mathbf{A}) - \lambda_{i_j}(\mathbf{B}))$  over index sets of size  $k$ , and the totals are  $\text{tr } \mathbf{A} - \text{tr } \mathbf{B} = \text{tr}(\mathbf{A} - \mathbf{B})$ . So a bound of every such sum by  $\sum_{j \leq k} \lambda_j(\mathbf{A} - \mathbf{B})$  says exactly that

$$\boldsymbol{\lambda}(\mathbf{A}) - \boldsymbol{\lambda}(\mathbf{B}) \prec \boldsymbol{\lambda}(\mathbf{A} - \mathbf{B}).$$

The diagonal of a Hermitian matrix and the eigenvalues of a difference are both controlled by one ordering of vectors in  $\mathbb{R}^n$ .

## 16.8.7 Exercises

### 16.8.7.1 A. Check your understanding

**Exercise A1.**

- (a) State the definitions of  $\mathbf{x} \prec \mathbf{y}$  and  $\mathbf{x} \prec_w \mathbf{y}$  for  $\mathbf{x}, \mathbf{y} \in \mathbb{R}^n$ .
- (b) Decide whether  $(2, 1, 1) \prec (2, 2, 0)$ , and whether  $(2, 2, 0) \prec (2, 1, 1)$ .
- (c) True or false: if  $\mathbf{x} \prec \mathbf{y}$ , then  $x_1 \leq y_1$ . Justify your answer.
- (d) State [Theorem 16.8.2](#). Which result of §06 is it a restatement of?
- (e) State [Theorem 16.8.5](#). Which half is proved by induction on  $n$ , and which lemma does each step of that induction use?

*Solution.*

- (a)  $\mathbf{x} \prec \mathbf{y}$  means that  $\sum_{i \leq k} x_i^\downarrow \leq \sum_{i \leq k} y_i^\downarrow$  for every  $k = 1, \dots, n-1$ , and  $\sum_{i=1}^n x_i = \sum_{i=1}^n y_i$  ([Definition 16.8.1](#)).  $\mathbf{x} \prec_w \mathbf{y}$  means that the same inequality holds for every  $k = 1, \dots, n$ , with no condition on the totals.
- (b) Both vectors are decreasing with total 4. Their running totals at  $k = 1, 2$  are 2, 3 and 2, 4. Since  $2 \leq 2$  and  $3 \leq 4$ ,  $(2, 1, 1) \prec (2, 2, 0)$ . The reverse fails at  $k = 2$ , since  $4 > 3$ .
- (c) False. The definition compares **sorted** vectors, so the unsorted first entries can go either way. Take  $\mathbf{x} = (1, 0)$  and  $\mathbf{y} = (0, 1)$ . Both have decreasing rearrangement  $(1, 0)$ , so (M1) at  $k = 1$  reads  $1 \leq 1$ , and both totals are 1; hence  $\mathbf{x} \prec \mathbf{y}$ . But  $x_1 = 1 > 0 = y_1$ . What is true is  $x_1^\downarrow \leq y_1^\downarrow$ , which is (M1) at  $k = 1$ .
- (d) For a Hermitian  $\mathbf{A} \in M_n(F)$ ,  $F = \mathbb{R}$  or  $\mathbb{C}$ ,  $\mathbf{d}(\mathbf{A}) \prec \boldsymbol{\lambda}(\mathbf{A})$ . It restates [Corollary 16.6.3](#): applied to the positions of the  $k$  largest diagonal entries, the corollary gives (M1), and its equality at  $k = n$  gives (M2).
- (e) For  $\mathbf{d}, \boldsymbol{\lambda} \in \mathbb{R}^n$ , there is a Hermitian matrix over  $\mathbb{R}$  or  $\mathbb{C}$  with diagonal  $\mathbf{d}$ , in the given order, and eigenvalues  $\lambda_1, \dots, \lambda_n$  if and only if  $\mathbf{d} \prec \boldsymbol{\lambda}$ . The half “ $\mathbf{d} \prec \boldsymbol{\lambda}$  implies such a matrix exists” is Horn’s theorem, [Theorem 16.8.4](#), proved by induction on  $n$ . Each step uses [Lemma 16.8.3](#), applied to a pair of consecutive eigenvalues  $\lambda_k \geq d_1 \geq \lambda_{k+1}$ , to put  $d_1$  in the corner. The other half is Schur’s theorem.

**16.8.7.2 B. Practice****Exercise B1: Determine which pairs are comparable.**

For each pair, determine whether  $\mathbf{x} \prec \mathbf{y}$  and whether  $\mathbf{x} \prec_w \mathbf{y}$ . Where a relation fails, name the exact clause and the value of  $k$  at which it fails. Justify your answer.

- (a)  $\mathbf{x} = (2, 2, 1, 1)$ ,  $\mathbf{y} = (3, 2, 1, 0)$ .
- (b)  $\mathbf{x} = (0, 3, 1)$ ,  $\mathbf{y} = (4, 0, 0)$ .
- (c)  $\mathbf{x} = (1, 1, 0)$ ,  $\mathbf{y} = (2, 1, 0)$ .
- (d)  $\mathbf{x} = (-1, 2, 2)$ ,  $\mathbf{y} = (3, 0, 0)$ .
- (e)  $\mathbf{x} = (3, 1, 1, 1)$ ,  $\mathbf{y} = (2, 2, 2, 0)$ ; decide also whether  $\mathbf{y} \prec \mathbf{x}$ .

*Solution.* In each case, sort first, then compare running totals.

- (a) Both are decreasing with total 6. Running totals: 2, 4, 5, 6 and 3, 5, 6, 6. Every entry of the first list is at most the matching entry of the second, so  $\mathbf{x} \prec \mathbf{y}$  and

hence  $\mathbf{x} \prec_w \mathbf{y}$ .

- (b)  $\mathbf{x}^\downarrow = (3, 1, 0)$  with running totals 3, 4, 4, and  $\mathbf{y}$  has running totals 4, 4, 4. The totals agree and  $3 \leq 4$ ,  $4 \leq 4$ , so  $\mathbf{x} \prec \mathbf{y}$  and  $\mathbf{x} \prec_w \mathbf{y}$ . This is [Example \(A chain, the two extremes, and size one\)](#) (c), applied after sorting.
- (c) Running totals 1, 2, 2 and 2, 3, 3. Every inequality holds, including at  $k = 3$ , so  $\mathbf{x} \prec_w \mathbf{y}$ . But the totals are  $2 \neq 3$ , so **(M2) fails** and  $\mathbf{x} \not\prec \mathbf{y}$ .
- (d)  $\mathbf{x}^\downarrow = (2, 2, -1)$  with running totals 2, 4, 3, and  $\mathbf{y}$  has running totals 3, 3, 3. The totals agree, but at  $k = 2$ ,  $4 > 3$ , so **(M1) fails at  $k = 2$** . Neither  $\mathbf{x} \prec \mathbf{y}$  nor  $\mathbf{x} \prec_w \mathbf{y}$  holds.
- (e) Both are decreasing with total 6. Running totals: 3, 4, 5, 6 and 2, 4, 6, 6. At  $k = 1$ ,  $3 > 2$ , so **(M1) fails at  $k = 1$**  and  $\mathbf{x} \not\prec \mathbf{y}$ , and also  $\mathbf{x} \not\prec_w \mathbf{y}$ . For  $\mathbf{y} \prec \mathbf{x}$ , the inequalities  $2 \leq 3$  and  $4 \leq 4$  hold, but  $6 > 5$  at  $k = 3$ , so **(M1) fails at  $k = 3$** . The pair is incomparable, like [Example \(Neither majorizes the other\)](#).

### Exercise B2: Schur's theorem for one matrix.

Let

$$\mathbf{A} = \begin{pmatrix} 3 & 2 & 2 \\ 2 & 4 & 0 \\ 2 & 0 & 7 \end{pmatrix}.$$

- (a) Verify that  $(2, 1, 4)$ ,  $(-1, -2, 1)$  and  $(-3, 2, 1)$  are eigenvectors of  $\mathbf{A}$ , and find  $\lambda(\mathbf{A})$ .
- (b) Write down  $\mathbf{d}(\mathbf{A})$  and verify every clause of  $\mathbf{d}(\mathbf{A}) \prec \lambda(\mathbf{A})$ .
- (c) Hence verify Hadamard's inequality ([Theorem 12.6.4](#)) for  $\mathbf{A}$ .

*Solution.*

- (a) Multiply out:

$$\begin{aligned} \mathbf{A}(2, 1, 4) &= (6 + 2 + 8, 4 + 4, 4 + 28) = (16, 8, 32) = 8(2, 1, 4), \\ \mathbf{A}(-1, -2, 1) &= (-3 - 4 + 2, -2 - 8, -2 + 7) = (-5, -10, 5) = 5(-1, -2, 1), \\ \mathbf{A}(-3, 2, 1) &= (-9 + 4 + 2, -6 + 8, -6 + 7) = (-3, 2, 1) = 1(-3, 2, 1). \end{aligned}$$

Three distinct eigenvalues of a  $3 \times 3$  matrix account for the whole spectrum, so  $\lambda(\mathbf{A}) = (8, 5, 1)$ .

- (b)  $\mathbf{d}(\mathbf{A}) = (3, 4, 7)$ , and  $\mathbf{d}(\mathbf{A})^\downarrow = (7, 4, 3)$ . (M1): at  $k = 1$ ,  $7 \leq 8$ ; at  $k = 2$ ,  $7 + 4 = 11 \leq 13 = 8 + 5$ . (M2):  $3 + 4 + 7 = 14 = 8 + 5 + 1$ . Hence  $\mathbf{d}(\mathbf{A}) \prec \lambda(\mathbf{A})$ , as [Theorem 16.8.2](#) requires.
- (c) All eigenvalues are positive, so  $\mathbf{A} \succ 0$  by [Theorem 12.1.5](#) (b), and [Corollary 16.8.7](#) applies. By [Theorem 8.3.3](#),  $\det \mathbf{A} = 8 \cdot 5 \cdot 1 = 40$ , while  $a_{11}a_{22}a_{33} = 3 \cdot 4 \cdot 7 = 84$ . Hence  $40 \leq 84$ . The inequality is strict, as it must be, since  $\mathbf{A}$  is not diagonal and no diagonal entry is 0.

### Exercise B3: Which diagonals are ruled out?.

A Hermitian  $3 \times 3$  matrix has eigenvalues 5, 1, 0. For each proposed list of diagonal entries, determine whether [Theorem 16.8.2](#) rules it out. Justify your answer.

- (a) 4, 2, 0.
- (b) 6, 0, 0.
- (c) 3, 3, 1.
- (d) 2, 2, 2.
- (e) -1, 4, 3.

For the lists that are not ruled out, what does [Theorem 16.8.5](#) add?

*Solution.* Here  $\lambda = (5, 1, 0)$ , with running totals 5, 6 and total 6.

- (a) Running totals 4, 6, total 6. Since  $4 \leq 5$  and  $6 \leq 6$ , the list is **not** ruled out.
- (b) At  $k = 1$ ,  $6 > 5$ . **Ruled out.** No diagonal entry can exceed  $\lambda_1$ .
- (c) The total is  $7 \neq 6$ . **Ruled out** by (M2): the trace would be wrong.
- (d) This is the flat vector with total 6, which is majorized by every vector with total 6 ([Example \(A chain, the two extremes, and size one\)](#) (b)). **Not** ruled out.
- (e) Sorted, it is  $(4, 3, -1)$ , with running totals 4, 7 and total 6. At  $k = 2$ ,  $7 > 6$ . **Ruled out.** Independently,  $-1 < 0 = \lambda_3$  contradicts the bottom-end inequality of the warning.

Schur's theorem is only a necessary condition, so for (a) and (d) it shows only that nothing forbids these diagonals. [Theorem 16.8.5](#) adds that they occur: for each of them there is a real symmetric matrix with eigenvalues 5, 1, 0 and that diagonal.

### 16.8.7.3 C. Going deeper

#### Exercise C1: Majorization squeezes the range.

- (a) Prove that if  $\mathbf{x} \prec \mathbf{y}$ , then  $y_n^\downarrow \leq x_n^\downarrow \leq x_1^\downarrow \leq y_1^\downarrow$ .
- (b) Deduce that every diagonal entry of a Hermitian  $\mathbf{A}$  lies in  $[\lambda_n(\mathbf{A}), \lambda_1(\mathbf{A})]$ .
- (c) Show that the conclusion of (b), together with equal totals, does **not** imply majorization. Prove that no Hermitian matrix has eigenvalues 4, 1, 1, 0 and diagonal entries 3, 3, 0, 0, although every one of these entries lies in  $[0, 4]$  and the totals agree.

*Solution.*

- (a) The inequality  $x_1^\downarrow \leq y_1^\downarrow$  is (M1) at  $k = 1$  (for  $n = 1$ ,  $\mathbf{x} = \mathbf{y}$  by [Example \(A chain, the two extremes, and size one\)](#) (d), and there is nothing to prove). The middle inequality holds because  $\mathbf{x}^\downarrow$  is decreasing. For the left inequality, subtract (M1) at  $k = n - 1$  from (M2):

$$\begin{aligned} x_n^\downarrow &= \sum_{i=1}^n x_i - \sum_{i=1}^{n-1} x_i^\downarrow \\ &\geq \sum_{i=1}^n y_i - \sum_{i=1}^{n-1} y_i^\downarrow = y_n^\downarrow. \end{aligned}$$

- (b) By [Theorem 16.8.2](#),  $\mathbf{d}(\mathbf{A}) \prec \lambda(\mathbf{A})$ . Apply (a) with  $\mathbf{x} = \mathbf{d}(\mathbf{A})$  and  $\mathbf{y} = \lambda(\mathbf{A})$ , using  $\lambda(\mathbf{A})^\downarrow = \lambda(\mathbf{A})$ . Every diagonal entry lies between the smallest and the

largest diagonal entry, which (a) places between  $\lambda_n(\mathbf{A})$  and  $\lambda_1(\mathbf{A})$ . (This also follows from [Lemma 12.5.2](#) at  $\mathbf{x} = \mathbf{e}_i$ . The point of (a) is that it is a property of majorization itself.)

- (c) The vector  $(3, 3, 0, 0)$  has running totals 3, 6, 6, and  $(4, 1, 1, 0)$  has 4, 5, 6. At  $k = 2$ ,  $6 > 5$ , so  $(3, 3, 0, 0) \not\prec (4, 1, 1, 0)$ , and by [Theorem 16.8.2](#) no Hermitian matrix has this diagonal and this spectrum. Yet  $0 \leq 3, 3, 0, 0 \leq 4$ , and both totals are 6. The range condition controls only the running totals at  $k = 1$  and  $k = n - 1$ , while majorization constrains the middle ones too, which is why the example needs  $n \geq 4$ .

### Exercise C2: A weaker determinant bound.

Let  $\mathbf{A} \in M_n(F)$  with  $\mathbf{A} \succeq 0$ .

- (a) Prove that  $\det \mathbf{A} \leq \left(\frac{1}{n} \operatorname{tr} \mathbf{A}\right)^n$ , with equality if and only if  $\mathbf{A} = c\mathbf{I}$  for some real  $c \geq 0$ .  
 (b) Prove that  $a_{11} \cdots a_{nn} \leq \left(\frac{1}{n} \operatorname{tr} \mathbf{A}\right)^n$ . Deduce that Hadamard's inequality ([Theorem 12.6.4](#)) always implies the bound in (a).  
 (c) For

$$\mathbf{A} = \begin{pmatrix} 2 & 0 & 2 \\ 0 & 5 & 2 \\ 2 & 2 & 6 \end{pmatrix},$$

whose eigenvectors are  $(1, 2, 3)$ ,  $(1, -2, 1)$  and  $(-4, -1, 2)$ , compute the three numbers  $\det \mathbf{A}$ ,  $a_{11}a_{22}a_{33}$  and  $\left(\frac{1}{3} \operatorname{tr} \mathbf{A}\right)^3$ .

*Solution.*

- (a) The eigenvalues  $\lambda_i(\mathbf{A})$  are non-negative by [Theorem 12.1.4](#) (b). By [Theorem 8.3.3](#) and [Lemma 16.8.6](#),

$$\det \mathbf{A} = \prod_i \lambda_i(\mathbf{A}) \leq \left(\frac{1}{n} \sum_i \lambda_i(\mathbf{A})\right)^n = \left(\frac{1}{n} \operatorname{tr} \mathbf{A}\right)^n.$$

Equality holds if and only if all  $\lambda_i(\mathbf{A})$  equal a common value  $c \geq 0$ . By the spectral theorem this happens if and only if  $\mathbf{A} = \mathbf{U}(c\mathbf{I})\mathbf{U}^* = c\mathbf{I}$ . Conversely,  $\mathbf{A} = c\mathbf{I}$  gives equality directly.

- (b) The  $a_{ii}$  are non-negative (as in the proof of [Corollary 16.8.7](#)) and add up to  $\operatorname{tr} \mathbf{A}$ . So [Lemma 16.8.6](#) gives  $\prod_i a_{ii} \leq \left(\frac{1}{n} \operatorname{tr} \mathbf{A}\right)^n$ . Combined with [Corollary 16.8.7](#), this gives  $\det \mathbf{A} \leq \prod_i a_{ii} \leq \left(\frac{1}{n} \operatorname{tr} \mathbf{A}\right)^n$ , so the Hadamard bound always sits between  $\det \mathbf{A}$  and the bound of (a). The bound in (a) sees only the total of the diagonal, while Hadamard's bound sees each entry, and so it is never worse.  
 (c) Check the eigenvectors:  $\mathbf{A}(1, 2, 3) = (2 + 6, 10 + 6, 2 + 4 + 18) = (8, 16, 24)$ ;  $\mathbf{A}(1, -2, 1) = (2 + 2, -10 + 2, 2 - 4 + 6) = (4, -8, 4)$ ;  $\mathbf{A}(-4, -1, 2) = (-8 +$

4,  $-5 + 4$ ,  $-8 - 2 + 12$ ) =  $(-4, -1, 2)$ . So  $\lambda(\mathbf{A}) = (8, 4, 1)$ , and  $\mathbf{A} \succ 0$ . Then

$$\det \mathbf{A} = 8 \cdot 4 \cdot 1 = 32, \quad a_{11}a_{22}a_{33} = 2 \cdot 5 \cdot 6 = 60,$$

$$\left(\frac{\operatorname{tr} \mathbf{A}}{3}\right)^3 = \left(\frac{13}{3}\right)^3 = \frac{2197}{27} \approx 81.4,$$

and  $32 \leq 60 \leq 81.4$ , with both inequalities strict.

### Exercise C3: The possible diagonals form a convex set.

For  $\mathbf{x} \in \mathbb{R}^n$  and  $1 \leq k \leq n$ , write  $s_k(\mathbf{x}) = x_1^\downarrow + \cdots + x_k^\downarrow$ .

- (a) Prove that  $s_k(\mathbf{x}) = \max\{\sum_{i \in I} x_i : I \subseteq \{1, \dots, n\}, |I| = k\}$ .
- (b) Deduce that  $s_k(\mathbf{x} + \mathbf{y}) \leq s_k(\mathbf{x}) + s_k(\mathbf{y})$  for all  $\mathbf{x}, \mathbf{y} \in \mathbb{R}^n$ .
- (c) Let  $\lambda \in \mathbb{R}^n$ . Prove that if  $\mathbf{d} \prec \lambda$  and  $\mathbf{d}' \prec \lambda$ , then  $t\mathbf{d} + (1-t)\mathbf{d}' \prec \lambda$  for every  $0 \leq t \leq 1$ .
- (d) Hence show that if  $\mathbf{A}$  and  $\mathbf{B}$  are Hermitian  $n \times n$  matrices with the same eigenvalues, then for every  $0 \leq t \leq 1$  there is a real symmetric matrix with those eigenvalues and diagonal  $t\mathbf{d}(\mathbf{A}) + (1-t)\mathbf{d}(\mathbf{B})$ .

*Hint for (a): if  $v_1 \geq \cdots \geq v_k$  are the entries  $x_i$ ,  $i \in I$ , in decreasing order, count how many entries of  $\mathbf{x}$  are at least  $v_j$ .*

*Solution.*

- (a) Let  $|I| = k$ , and list the entries  $x_i$ ,  $i \in I$ , decreasingly as  $v_1 \geq \cdots \geq v_k$ . For each  $j \leq k$ , the entries  $v_1, \dots, v_j$  sit at  $j$  different positions of  $\mathbf{x}$  and are all  $\geq v_j$ . If  $x_j^\downarrow < v_j$ , then, since  $\mathbf{x}^\downarrow$  is decreasing, only  $x_1^\downarrow, \dots, x_{j-1}^\downarrow$  could be  $\geq v_j$ , which is fewer than  $j$  entries. Hence  $v_j \leq x_j^\downarrow$  for every  $j$ , and summing gives  $\sum_{i \in I} x_i \leq s_k(\mathbf{x})$ . The bound is attained by the positions of the  $k$  largest entries, chosen as in the proof of [Theorem 16.8.2](#). This proves (a).
- (b) Let  $I$  be a set of  $k$  positions at which  $\mathbf{x} + \mathbf{y}$  attains the maximum in (a). Applying (a) three times,

$$s_k(\mathbf{x} + \mathbf{y}) = \sum_{i \in I} x_i + \sum_{i \in I} y_i \leq s_k(\mathbf{x}) + s_k(\mathbf{y}).$$

- (c) Put  $\mathbf{z} = t\mathbf{d} + (1-t)\mathbf{d}'$ . Its total is  $t \sum_i \lambda_i + (1-t) \sum_i \lambda_i = \sum_i \lambda_i$ , by (M2) for both hypotheses. For a scalar  $c \geq 0$ , multiplying by  $c$  does not change the order of the entries, so  $s_k(c\mathbf{x}) = c s_k(\mathbf{x})$ . For  $1 \leq k \leq n-1$ , (b) and (M1) for both hypotheses give

$$s_k(\mathbf{z}) \leq t s_k(\mathbf{d}) + (1-t) s_k(\mathbf{d}') \leq t s_k(\lambda) + (1-t) s_k(\lambda) = s_k(\lambda).$$

So  $\mathbf{z} \prec \lambda$ .

- (d) Let  $\lambda = \lambda(\mathbf{A}) = \lambda(\mathbf{B})$ . By [Theorem 16.8.2](#),  $\mathbf{d}(\mathbf{A}) \prec \lambda$  and  $\mathbf{d}(\mathbf{B}) \prec \lambda$ . By (c),  $t\mathbf{d}(\mathbf{A}) + (1-t)\mathbf{d}(\mathbf{B}) \prec \lambda$ . By [Theorem 16.8.4](#) there is a real symmetric matrix with eigenvalues  $\lambda$  and exactly this diagonal. So among Hermitian matrices with a fixed spectrum, the set of diagonals contains the segment between any two of its points: it is **convex**.

## 16.9 The Same Theory for Singular Values

Everything so far in this chapter has been about Hermitian matrices, and it had to be: a general square matrix need not have real eigenvalues to order, and a rectangular matrix has none at all. But every matrix has singular values, they are real and non-negative, and Chapter 12 listed them decreasingly from the moment it defined them. This section carries the apparatus across — a min–max description, a perturbation bound, an interlacing theorem — and then spends the rest of its length paying debts that earlier chapters recorded and could not settle.

**Throughout,**  $F = \mathbb{R}$  or  $F = \mathbb{C}$ ,  $\mathbf{A} \in M_{m \times n}(F)$ ,  $p = \min(m, n)$ , and  $\sigma_1(\mathbf{A}) \geq \cdots \geq \sigma_p(\mathbf{A}) \geq 0$  are the singular values of  $\mathbf{A}$  (Definition 12.8.1). Nothing here is assumed Hermitian and nothing is assumed square; where a matrix must be square, or invertible, the statement says so. Vector norms are Euclidean, and  $\|\mathbf{A}\|_2$  is the spectral norm of Definition 15.3.2, which Theorem 15.3.3 (c) computes as  $\sigma_1(\mathbf{A})$ .

**One convention, stated once and used everywhere below.** We set

$$\sigma_i(\mathbf{A}) := 0 \quad \text{for } p < i \leq n.$$

This is not new: Theorem 12.8.2 (a) already makes it, and records what it buys. With it in force, if  $\mathbf{A} = \mathbf{U}\mathbf{\Sigma}\mathbf{V}^*$  is a singular value decomposition (Theorem 12.8.1) with columns  $\mathbf{v}_1, \dots, \mathbf{v}_n$  of  $\mathbf{V}$ , then

$$\mathbf{A}^* \mathbf{A} \mathbf{v}_i = \sigma_i(\mathbf{A})^2 \mathbf{v}_i \quad (1 \leq i \leq n), \quad (16.9.1)$$

so that  $\mathbf{v}_1, \dots, \mathbf{v}_n$  is an orthonormal eigenbasis of  $\mathbf{A}^* \mathbf{A}$  whose eigenvalues  $\sigma_1^2 \geq \cdots \geq \sigma_n^2 \geq 0$  are already sorted. In the notation of this chapter,

$$\lambda_i(\mathbf{A}^* \mathbf{A}) = \sigma_i(\mathbf{A})^2 \quad (1 \leq i \leq n).$$

Every statement in this section is that identity, pushed through a theorem about Hermitian matrices.

### 16.9.1 The stretch ratio is a Rayleigh quotient

Chapter 12 introduced the singular values by an algebraic route — take  $\mathbf{A}^* \mathbf{A}$ , diagonalize it, take square roots. Their geometric meaning is stretching: how much longer  $\mathbf{A}\mathbf{x}$  is than  $\mathbf{x}$ . The bridge between the two descriptions is one line of algebra, and it is the only computation this section needs before the theorems start.

For  $\mathbf{x} \neq \mathbf{0}$  in  $F^n$ , the matrix  $\mathbf{A}^* \mathbf{A}$  is Hermitian, so its Rayleigh quotient (Definition 16.1.1) is defined, and

$$R_{\mathbf{A}^* \mathbf{A}}(\mathbf{x}) = \frac{\langle \mathbf{A}^* \mathbf{A} \mathbf{x}, \mathbf{x} \rangle}{\langle \mathbf{x}, \mathbf{x} \rangle} = \frac{\mathbf{x}^* \mathbf{A}^* \mathbf{A} \mathbf{x}}{\mathbf{x}^* \mathbf{x}} = \frac{(\mathbf{A} \mathbf{x})^* (\mathbf{A} \mathbf{x})}{\mathbf{x}^* \mathbf{x}} = \frac{\|\mathbf{A} \mathbf{x}\|^2}{\|\mathbf{x}\|^2}. \quad (16.9.2)$$

*The Rayleigh quotient of  $\mathbf{A}^* \mathbf{A}$  is the squared stretching factor of  $\mathbf{A}$ .*

So a statement about the Rayleigh quotient of  $\mathbf{A}^* \mathbf{A}$  is a statement about the function  $\mathbf{x} \mapsto \|\mathbf{A}\mathbf{x}\|/\|\mathbf{x}\|$ , and conversely. Nothing is lost in passing between them, because  $t \mapsto \sqrt{t}$  is a

strictly increasing bijection of  $[0, \infty)$  onto itself: it carries maxima to maxima and minima to minima, over any set whatever.

Before the min–max theorem, it is worth writing down what the singular value decomposition says about this function in coordinates. The computation is three lines, and it is used four times below — for the witness subspaces in every proof of the section — so it gets a label.

### Lemma 16.9.1 — Stretching in the Right Singular Basis

Let  $\mathbf{A} \in M_{m \times n}(F)$  have a singular value decomposition  $\mathbf{A} = \mathbf{U}\mathbf{\Sigma}\mathbf{V}^*$  (Theorem 12.8.1), with  $\mathbf{v}_1, \dots, \mathbf{v}_n$  the columns of  $\mathbf{V}$ , and write  $\sigma_i = \sigma_i(\mathbf{A})$  with the convention above. Then for  $\mathbf{x} = \sum_{i=1}^n c_i \mathbf{v}_i$ ,

$$\|\mathbf{Ax}\|^2 = \sum_{i=1}^n \sigma_i^2 |c_i|^2, \quad \|\mathbf{x}\|^2 = \sum_{i=1}^n |c_i|^2.$$

Consequently, for each  $1 \leq k \leq n$ :

- (a) on the **tail** subspace  $T_k = \text{span}(\mathbf{v}_k, \dots, \mathbf{v}_n)$ , of dimension  $n - k + 1$ ,

$$\max \left\{ \frac{\|\mathbf{Ax}\|}{\|\mathbf{x}\|} : \mathbf{0} \neq \mathbf{x} \in T_k \right\} = \sigma_k,$$

attained at  $\mathbf{x} = \mathbf{v}_k$ ;

- (b) on the **head** subspace  $H_k = \text{span}(\mathbf{v}_1, \dots, \mathbf{v}_k)$ , of dimension  $k$ ,

$$\min \left\{ \frac{\|\mathbf{Ax}\|}{\|\mathbf{x}\|} : \mathbf{0} \neq \mathbf{x} \in H_k \right\} = \sigma_k,$$

attained at  $\mathbf{x} = \mathbf{v}_k$ .

*Idea.* In the right singular basis  $\mathbf{A}$  does nothing but scale: it sends  $\mathbf{v}_i$  to  $\sigma_i \mathbf{u}_i$ , and the  $\mathbf{u}_i$  are orthonormal. So measuring  $\mathbf{Ax}$  is Pythagoras applied to the scaled coordinates, and every claim about how much  $\mathbf{A}$  can stretch a subspace becomes a claim about a weighted sum of  $|c_i|^2$ .

*Proof.* By Eq. 16.9.1 the list  $(\mathbf{v}_1, \dots, \mathbf{v}_n)$  is an orthonormal basis of  $F^n$  consisting of eigenvectors of  $\mathbf{A}^* \mathbf{A}$ , with  $\mathbf{A}^* \mathbf{A} \mathbf{v}_i = \sigma_i^2 \mathbf{v}_i$ . Hence, for  $\mathbf{x} = \sum_i c_i \mathbf{v}_i$ , using Eq. 16.9.2 in the form  $\|\mathbf{Ax}\|^2 = \mathbf{x}^* \mathbf{A}^* \mathbf{A} \mathbf{x}$  and expanding with orthonormality,

$$\|\mathbf{Ax}\|^2 = \left( \sum_i c_i \mathbf{v}_i \right)^* \left( \sum_j c_j \sigma_j^2 \mathbf{v}_j \right) = \sum_{i=1}^n \sigma_i^2 |c_i|^2,$$

the cross terms vanishing because  $\mathbf{v}_i^* \mathbf{v}_j = \delta_{ij}$ . The second identity is the same expansion with  $\mathbf{A}$  removed.

- (a) For  $\mathbf{0} \neq \mathbf{x} \in T_k$  we have  $c_i = 0$  for  $i < k$ , so  $\|\mathbf{Ax}\|^2 = \sum_{i \geq k} \sigma_i^2 |c_i|^2 \leq \sigma_k^2 \sum_{i \geq k} |c_i|^2 = \sigma_k^2 \|\mathbf{x}\|^2$ , since  $\sigma_i \leq \sigma_k$  for  $i \geq k$  and the singular values are non-negative. Taking square roots gives  $\|\mathbf{Ax}\|/\|\mathbf{x}\| \leq \sigma_k$ , and  $\mathbf{x} = \mathbf{v}_k$  gives  $\|\mathbf{A}\mathbf{v}_k\| = \sigma_k = \sigma_k \|\mathbf{v}_k\|$ , so the bound is attained.
- (b) For  $\mathbf{0} \neq \mathbf{x} \in H_k$  we have  $c_i = 0$  for  $i > k$ , so  $\|\mathbf{Ax}\|^2 = \sum_{i \leq k} \sigma_i^2 |c_i|^2 \geq \sigma_k^2 \sum_{i \leq k} |c_i|^2 = \sigma_k^2 \|\mathbf{x}\|^2$ , since now  $\sigma_i \geq \sigma_k$  for  $i \leq k$ ; and  $\mathbf{x} = \mathbf{v}_k$  attains equality. This proves the lemma.



In words: the tail subspace is a place where  $\mathbf{A}$  stretches **no more** than  $\sigma_k$ , and the head subspace is a place where it stretches **no less**. The min–max theorem says that these two subspaces are optimal.

## 16.9.2 Each singular value as an optimization

### Theorem 16.9.2 — Min–Max for Singular Values

Let  $\mathbf{A} \in M_{m \times n}(F)$  and let  $1 \leq k \leq n$ , with  $\sigma_k(\mathbf{A}) = 0$  read by convention when  $k > p$ . Then

$$\begin{aligned}\sigma_k(\mathbf{A}) &= \min_{\substack{W \leq F^n \\ \dim W = n-k+1}} \max_{\mathbf{0} \neq \mathbf{x} \in W} \frac{\|\mathbf{Ax}\|}{\|\mathbf{x}\|} \\ &= \max_{\substack{W \leq F^n \\ \dim W = k}} \min_{\mathbf{0} \neq \mathbf{x} \in W} \frac{\|\mathbf{Ax}\|}{\|\mathbf{x}\|}.\end{aligned}$$

The outer minimum is attained at  $W = \text{span}(\mathbf{v}_k, \dots, \mathbf{v}_n)$  and the outer maximum at  $W = \text{span}(\mathbf{v}_1, \dots, \mathbf{v}_k)$ , for the right singular vectors of any singular value decomposition of  $\mathbf{A}$ .

*Idea.* There is nothing to discover: [Eq. 16.9.2](#) turns the objective function into the Rayleigh quotient of the Hermitian matrix  $\mathbf{A}^* \mathbf{A}$ , [Theorem 16.2.3](#) applies to that matrix, and the square root — increasing, hence order-preserving — moves through the minima and maxima without changing which subspace attains them.

*Proof.* Put  $\mathbf{B} = \mathbf{A}^* \mathbf{A} \in M_n(F)$ , which is Hermitian, and fix  $1 \leq k \leq n$ . By [Eq. 16.9.1](#),  $\lambda_k(\mathbf{B}) = \sigma_k(\mathbf{A})^2$ . By [Theorem 16.2.3](#) applied to  $\mathbf{B}$ ,

$$\sigma_k(\mathbf{A})^2 = \lambda_k(\mathbf{B}) = \min_{\dim W = n-k+1} \max_{\mathbf{0} \neq \mathbf{x} \in W} R_{\mathbf{B}}(\mathbf{x}),$$

and by [Eq. 16.9.2](#),  $R_{\mathbf{B}}(\mathbf{x}) = (\|\mathbf{Ax}\|/\|\mathbf{x}\|)^2$  for every  $\mathbf{x} \neq \mathbf{0}$ .

Fix a subspace  $W$ . The function  $\mathbf{x} \mapsto \|\mathbf{Ax}\|/\|\mathbf{x}\|$  takes only non-negative values, and  $t \mapsto t^2$  is strictly increasing on  $[0, \infty)$ ; so a vector of  $W$  maximizes  $R_{\mathbf{B}}$  over  $W$  exactly when it maximizes  $\|\mathbf{Ax}\|/\|\mathbf{x}\|$ , and

$$\max_{\mathbf{0} \neq \mathbf{x} \in W} R_{\mathbf{B}}(\mathbf{x}) = \left( \max_{\mathbf{0} \neq \mathbf{x} \in W} \frac{\|\mathbf{Ax}\|}{\|\mathbf{x}\|} \right)^2.$$

(Both maxima exist: by homogeneity the quotient may be taken over the unit sphere of  $W$ , which is compact by fact (A3) of Chapter 15's introduction, and the function is continuous, so the extreme value theorem — fact (A4) of Chapter 15's introduction — applies. This is the same appeal [Theorem 16.2.3](#) makes.) Taking square roots, and using once more that  $t \mapsto \sqrt{t}$  is increasing, so that it commutes with a minimum over the family of subspaces,

$$\sigma_k(\mathbf{A}) = \sqrt{\lambda_k(\mathbf{B})} = \min_{\dim W = n-k+1} \max_{\mathbf{0} \neq \mathbf{x} \in W} \frac{\|\mathbf{Ax}\|}{\|\mathbf{x}\|}.$$

The max–min form follows from the second equality of [Theorem 16.2.3](#) by the same two steps.

For the attainment claims, [Lemma 16.9.1](#) (a) says that the tail subspace  $T_k$ , of dimension  $n - k + 1$ , has inner maximum exactly  $\sigma_k(\mathbf{A})$ , which is therefore the outer minimum; and part (b) says the head subspace  $H_k$ , of dimension  $k$ , has inner minimum exactly  $\sigma_k(\mathbf{A})$ . This proves the theorem. ■

Two things are worth noticing. First, the subspaces  $W$  live in  $F^n$ , the **domain**, whatever  $m$  is; the theorem compares two matrices with the same number of columns without caring how many rows they have. Second, the extremal subspaces are named, and they are spans of right singular vectors: the theorem is not only an identity but a recipe.

There is a second route to the same theorem, and it is worth recording because it converts a statement about a rectangular matrix into a statement about a Hermitian one, which is what the later chapters will want.

### Proposition 16.9.3 — The Hermitian Dilation

Let  $\mathbf{A} \in M_{m \times n}(F)$  and put

$$\mathcal{H}(\mathbf{A}) := \begin{pmatrix} \mathbf{0} & \mathbf{A} \\ \mathbf{A}^* & \mathbf{0} \end{pmatrix} \in M_{m+n}(F).$$

Then  $\mathcal{H}(\mathbf{A})$  is Hermitian, and its eigenvalues, with multiplicity, are

$$\sigma_1(\mathbf{A}), \dots, \sigma_p(\mathbf{A}), -\sigma_1(\mathbf{A}), \dots, -\sigma_p(\mathbf{A}), \underbrace{0, \dots, 0}_{m+n-2p}.$$

*Idea.* Guess the eigenvectors from the SVD. The pair  $(\mathbf{u}_i, \mathbf{v}_i)$ , stacked, is sent by  $\mathcal{H}(\mathbf{A})$  to  $(\mathbf{A}\mathbf{v}_i, \mathbf{A}^*\mathbf{u}_i) = \sigma_i(\mathbf{u}_i, \mathbf{v}_i)$ ; flipping the sign of the second half flips the sign of the eigenvalue. That accounts for  $2p$  orthonormal eigenvectors, and what is left over is a kernel whose dimension is a count.

*Proof.* Writing out the conjugate transpose block by block,  $\mathcal{H}(\mathbf{A})^* = \begin{pmatrix} \mathbf{0} & (\mathbf{A}^*)^* \\ \mathbf{A}^* & \mathbf{0} \end{pmatrix} = \mathcal{H}(\mathbf{A})$ , so  $\mathcal{H}(\mathbf{A})$  is Hermitian.

Fix a singular value decomposition  $\mathbf{A} = \mathbf{U}\mathbf{\Sigma}\mathbf{V}^*$  ([Theorem 12.8.1](#)), with columns  $\mathbf{u}_1, \dots, \mathbf{u}_m$  of  $\mathbf{U}$  and  $\mathbf{v}_1, \dots, \mathbf{v}_n$  of  $\mathbf{V}$ . Since  $\mathbf{V}^*\mathbf{v}_i = \mathbf{e}_i$  and  $\mathbf{\Sigma}\mathbf{e}_i = \sigma_i\mathbf{e}_i$  for  $i \leq p$  while  $\mathbf{\Sigma}\mathbf{e}_j = \mathbf{0}$  for  $p < j \leq n$ , and symmetrically for  $\mathbf{A}^* = \mathbf{V}\mathbf{\Sigma}^*\mathbf{U}^*$ ,

$$\mathbf{A}\mathbf{v}_i = \sigma_i\mathbf{u}_i, \quad \mathbf{A}^*\mathbf{u}_i = \sigma_i\mathbf{v}_i \quad (i \leq p), \quad \mathbf{A}\mathbf{v}_j = \mathbf{0}, \quad \mathbf{A}^*\mathbf{u}_j = \mathbf{0} \quad (j > p).$$

For  $i \leq p$  put

$$\mathbf{w}_i^\pm = \frac{1}{\sqrt{2}} \begin{pmatrix} \mathbf{u}_i \\ \pm \mathbf{v}_i \end{pmatrix} \in F^{m+n}.$$

Then  $\mathcal{H}(\mathbf{A})\mathbf{w}_i^\pm = \frac{1}{\sqrt{2}}(\pm \mathbf{A}\mathbf{v}_i, \mathbf{A}^*\mathbf{u}_i) = \frac{1}{\sqrt{2}}(\pm \sigma_i\mathbf{u}_i, \sigma_i\mathbf{v}_i) = \pm \sigma_i \mathbf{w}_i^\pm$ . Likewise  $\mathcal{H}(\mathbf{A})(\mathbf{u}_j, \mathbf{0}) = (\mathbf{0}, \mathbf{A}^*\mathbf{u}_j) = \mathbf{0}$  for  $j > p$ , and  $\mathcal{H}(\mathbf{A})(\mathbf{0}, \mathbf{v}_j) = (\mathbf{A}\mathbf{v}_j, \mathbf{0}) = \mathbf{0}$  for  $j > p$ .

These  $2p + (m - p) + (n - p) = m + n$  vectors are orthonormal: the  $\mathbf{w}_i^\pm$  are unit vectors because  $\mathbf{u}_i$  and  $\mathbf{v}_i$  are, and any two distinct vectors in the list pair to 0, since  $(\mathbf{w}_i^+)^* \mathbf{w}_i^- = \frac{1}{2}(1 - 1) = 0$  and all other pairs have orthogonal  $\mathbf{u}$ -parts or orthogonal  $\mathbf{v}$ -parts. So they form an orthonormal eigenbasis of  $F^{m+n}$  with the stated eigenvalues, which are therefore the eigenvalue list of  $\mathcal{H}(\mathbf{A})$  with multiplicity. This proves the proposition. ■

Applying [Theorem 16.2.3](#) to  $\mathcal{H}(\mathbf{A})$  instead of  $\mathbf{A}^* \mathbf{A}$  gives [Theorem 16.9.2](#) again, with subspaces of  $F^{m+n}$  in place of  $F^n$ . We took the shorter road; the dilation is recorded because it turns *any* theorem about Hermitian eigenvalues into one about singular values, which Section 10's exercises and Chapter 19 both exploit.

### 16.9.3 Perturbation, with no hypothesis at all

For Hermitian matrices, [Corollary 16.3.6](#) says that  $|\lambda_i(\mathbf{A} + \mathbf{E}) - \lambda_i(\mathbf{A})| \leq \|\mathbf{E}\|_2$  — provided  $\mathbf{A}$  and  $\mathbf{E}$  are Hermitian. That proviso is not decoration: Chapter 15's [Example \(A Jordan block's eigenvalues move like  \$\varepsilon^{1/k}\$ \)](#) shows the conclusion failing spectacularly without it. The corresponding statement for singular values needs no proviso whatsoever.

#### Theorem 16.9.4 — Weyl's Inequalities for Singular Values

Let  $\mathbf{A}, \mathbf{B} \in M_{m \times n}(F)$  and let  $i, j \geq 1$  with  $i + j - 1 \leq n$ . Then

$$\sigma_{i+j-1}(\mathbf{A} + \mathbf{B}) \leq \sigma_i(\mathbf{A}) + \sigma_j(\mathbf{B}).$$

*Idea.* Put  $k = i + j - 1$ . The min-max form of [Theorem 16.9.2](#) bounds  $\sigma_k(\mathbf{A} + \mathbf{B})$  from above as soon as we produce **one** subspace of dimension  $n - k + 1$  on which  $\mathbf{A} + \mathbf{B}$  stretches by at most  $\sigma_i(\mathbf{A}) + \sigma_j(\mathbf{B})$ . Each summand supplies a subspace where it is individually small — the tail subspaces of [Lemma 16.9.1](#) — and the dimensions are chosen so that the two subspaces must overlap in something of the required size. On the overlap the triangle inequality finishes it.

*Proof.* Put  $k = i + j - 1 \leq n$ . Let  $T = \text{span}(\mathbf{v}_i, \dots, \mathbf{v}_n)$  for the right singular vectors of  $\mathbf{A}$  and  $T' = \text{span}(\mathbf{v}'_j, \dots, \mathbf{v}'_n)$  for those of  $\mathbf{B}$ , so that  $\dim T = n - i + 1$ ,  $\dim T' = n - j + 1$ , and by [Lemma 16.9.1](#) (a),

$$\|\mathbf{A}\mathbf{x}\| \leq \sigma_i(\mathbf{A})\|\mathbf{x}\| \quad (\mathbf{x} \in T), \quad \|\mathbf{B}\mathbf{x}\| \leq \sigma_j(\mathbf{B})\|\mathbf{x}\| \quad (\mathbf{x} \in T').$$

By the dimension formula ([Theorem 1.7.3](#)) and  $\dim(T + T') \leq n$ ,

$$\begin{aligned} \dim(T \cap T') &\geq (n - i + 1) + (n - j + 1) - n \\ &= n - (i + j - 1) + 1 = n - k + 1, \end{aligned}$$

which is at least 1 because  $k \leq n$ . Choose a subspace  $W \subseteq T \cap T'$  with  $\dim W = n - k + 1$  exactly; this is possible because every subspace has a subspace of each smaller dimension, obtained by truncating a basis. For  $\mathbf{0} \neq \mathbf{x} \in W$ , the triangle inequality ([Corollary 10.1.6](#)) gives

$$\|(\mathbf{A} + \mathbf{B})\mathbf{x}\| \leq \|\mathbf{A}\mathbf{x}\| + \|\mathbf{B}\mathbf{x}\| \leq (\sigma_i(\mathbf{A}) + \sigma_j(\mathbf{B}))\|\mathbf{x}\|.$$

Hence the inner maximum over  $W$  is at most  $\sigma_i(\mathbf{A}) + \sigma_j(\mathbf{B})$ , and since  $\sigma_k(\mathbf{A} + \mathbf{B})$  is the **minimum** of that inner maximum over all subspaces of dimension  $n - k + 1$  (Theorem 16.9.2), we get  $\sigma_k(\mathbf{A} + \mathbf{B}) \leq \sigma_i(\mathbf{A}) + \sigma_j(\mathbf{B})$ . This proves the theorem. ■

### Corollary 16.9.5 — Singular Values Are Perfectly Conditioned

Let  $\mathbf{A}, \mathbf{E} \in M_{m \times n}(F)$ . Then

$$|\sigma_i(\mathbf{A} + \mathbf{E}) - \sigma_i(\mathbf{A})| \leq \|\mathbf{E}\|_2 \quad \text{for every } 1 \leq i \leq n.$$

No hypothesis is placed on  $\mathbf{A}$  or on  $\mathbf{E}$ .

*Idea.* Put  $j = 1$  in the Weyl inequality for singular values: one index passes through unchanged, and the other matrix contributes only its largest singular value, which is  $\|\mathbf{E}\|_2$ . That bounds  $\sigma_i(\mathbf{A} + \mathbf{E})$  from above; writing  $\mathbf{A} = (\mathbf{A} + \mathbf{E}) + (-\mathbf{E})$  and doing it again bounds it from below.

*Proof.* Take  $j = 1$  in Theorem 16.9.4, so that  $i + j - 1 = i \leq n$ :

$$\sigma_i(\mathbf{A} + \mathbf{E}) \leq \sigma_i(\mathbf{A}) + \sigma_1(\mathbf{E}) = \sigma_i(\mathbf{A}) + \|\mathbf{E}\|_2,$$

using Theorem 15.3.3 (c) for the last equality. Now apply the same inequality with  $\mathbf{A} + \mathbf{E}$  in place of  $\mathbf{A}$  and  $-\mathbf{E}$  in place of  $\mathbf{E}$ , noting  $\|-\mathbf{E}\|_2 = \|\mathbf{E}\|_2$  by homogeneity (Theorem 15.3.2 (b)):

$$\sigma_i(\mathbf{A}) = \sigma_i((\mathbf{A} + \mathbf{E}) + (-\mathbf{E})) \leq \sigma_i(\mathbf{A} + \mathbf{E}) + \|\mathbf{E}\|_2.$$

The two inequalities together are the claim. ■

This is a Lipschitz bound with constant 1, exactly as Corollary 16.3.6 is for Hermitian eigenvalues, and with no dependence on  $n$ , on the conditioning of  $\mathbf{A}$ , or on the gaps between the singular values. What is new is the absence of hypotheses. It is worth seeing the contrast on the very matrix Chapter 15 used to show that eigenvalues have no such bound.

#### Example. (The Jordan block again, measured by its singular values)

Let  $k \geq 2$ , let  $0 < \varepsilon < 1$ , and let

$$\mathbf{A} = \mathbf{J}_k(0), \quad \mathbf{A}_\varepsilon = \mathbf{J}_k(0) + \varepsilon \mathbf{E}_{k1}$$

be the matrices of Example (A Jordan block's eigenvalues move like  $\varepsilon^{1/k}$ ). Compute the singular values of both, and compare how far they move with how far the eigenvalues move.

*Solution.* The columns of  $\mathbf{A}_\varepsilon$  are  $\varepsilon \mathbf{e}_k$  in position 1, and  $\mathbf{e}_{j-1}$  in position  $j$  for  $2 \leq j \leq k$ .

These  $k$  columns are pairwise orthogonal, with norms  $\varepsilon, 1, \dots, 1$ , so

$$\mathbf{A}_\varepsilon^* \mathbf{A}_\varepsilon = \text{diag}(\varepsilon^2, 1, \dots, 1),$$

whose eigenvalues in decreasing order are 1 with multiplicity  $k - 1$  and then  $\varepsilon^2$ , since  $\varepsilon < 1$ . By [Definition 12.8.1](#) the singular values of  $\mathbf{A}_\varepsilon$  are

$$\sigma_1 = \dots = \sigma_{k-1} = 1, \quad \sigma_k = \varepsilon.$$

Setting  $\varepsilon = 0$  in the same computation, the singular values of  $\mathbf{A} = \mathbf{J}_k(0)$  are  $1, \dots, 1, 0$ .

So every singular value moves by at most  $\varepsilon$ , and exactly one of them — the last — moves by exactly  $\varepsilon$ . The perturbation has  $\|\varepsilon \mathbf{E}_{k1}\|_2 = \varepsilon$ , computed in [Example \(A Jordan block's eigenvalues move like  \$\varepsilon^{1/k}\$ \)](#), so [Corollary 16.9.5](#) is attained here and not merely satisfied.

Meanwhile [Example \(A Jordan block's eigenvalues move like  \$\varepsilon^{1/k}\$ \)](#) computed that the **eigenvalues** of the same two matrices are 0 (with multiplicity  $k$ ) and the  $k$ -th roots of  $\varepsilon$ , all of modulus  $\varepsilon^{1/k}$ . For  $k = 10$  and  $\varepsilon = 10^{-10}$ : the singular values move by  $10^{-10}$ , and the eigenvalues move by  $10^{-1}$ , a factor of a billion more. The same matrix, the same perturbation, two spectra — one of them stable and one of them not.

This is not a paradox. The singular values of  $\mathbf{A}$  are the square roots of the eigenvalues of the Hermitian matrix  $\mathbf{A}^* \mathbf{A}$ , and Hermitian eigenvalues are stable; the eigenvalues of  $\mathbf{A}$  itself are roots of a polynomial, and a repeated root is fragile. The instability lives in the passage from  $\mathbf{A}$  to its eigenvalues, not in  $\mathbf{A}$  — which is why a question that can be phrased in singular values should be.

*Warning.* **Weyl's inequality does not say**  $\sigma_k(\mathbf{A} + \mathbf{B}) \leq \sigma_k(\mathbf{A}) + \sigma_k(\mathbf{B})$ . The index bookkeeping  $i + j - 1$  is the content, not a technicality. Take

$$\mathbf{A} = \text{diag}(1, 0), \quad \mathbf{B} = \text{diag}(0, 1).$$

Then  $\sigma(\mathbf{A}) = \sigma(\mathbf{B}) = (1, 0)$ , while  $\mathbf{A} + \mathbf{B} = \mathbf{I}_2$  has  $\sigma(\mathbf{I}_2) = (1, 1)$ . So  $\sigma_2(\mathbf{A} + \mathbf{B}) = 1$  while  $\sigma_2(\mathbf{A}) + \sigma_2(\mathbf{B}) = 0$ . What [Theorem 16.9.4](#) does give here, with  $i = j = 1$ , is  $\sigma_1(\mathbf{A} + \mathbf{B}) \leq \sigma_1(\mathbf{A}) + \sigma_1(\mathbf{B}) = 2$ , which holds. Only  $i = 1$  or  $j = 1$  lets an index pass through unchanged, which is why [Corollary 16.9.5](#) is the case  $j = 1$ .

## 16.9.4 The best approximation of low rank

Chapter 12 §10 proved that the truncated singular value decomposition  $\mathbf{A}_k$  of [Definition 12.10.1](#) is the closest matrix of rank at most  $k$  to  $\mathbf{A}$  in the Frobenius norm ([Theorem 12.10.3](#)), and then wrote, in the remark that closes its main proof:

The same matrix  $\mathbf{A}_k$  is also optimal in the **spectral norm**  $\|\cdot\|_2$ , where the minimum value is  $\sigma_{k+1}$  rather than  $(\sum_{i>k} \sigma_i^2)^{1/2}$ . We cannot state that here, because the spectral norm is defined only in Chapter 15.

Chapter 15 supplied the norm and then, in its closing ledger, handed the theorem to this chapter: “the theorem itself waits for Chapter 16, which proves it from the min–max description of the singular values.” That description now exists. Here is the theorem.

**Theorem 16.9.6 — Eckart–Young Theorem, Spectral Case**

Let  $\mathbf{A} \in M_{m \times n}(F)$  and let  $0 \leq k < n$ . Then for **every**  $\mathbf{B} \in M_{m \times n}(F)$  with  $\text{rank } \mathbf{B} \leq k$ ,

$$\|\mathbf{A} - \mathbf{B}\|_2 \geq \sigma_{k+1}(\mathbf{A}),$$

and equality holds for the truncation  $\mathbf{B} = \mathbf{A}_k$  of [Definition 12.10.1](#). In words:  $\mathbf{A}_k$  minimizes  $\|\mathbf{A} - \mathbf{B}\|_2$  over all matrices of rank at most  $k$ , and the minimum value is  $\sigma_{k+1}(\mathbf{A})$ .

*Idea.* A matrix  $\mathbf{B}$  of rank at most  $k$  has a large null space — dimension at least  $n - k$  — and on that null space  $\mathbf{A} - \mathbf{B}$  acts exactly as  $\mathbf{A}$  does. The head subspace  $H_{k+1} = \text{span}(\mathbf{v}_1, \dots, \mathbf{v}_{k+1})$  is a place where  $\mathbf{A}$  stretches by at least  $\sigma_{k+1}$ , and it has dimension  $k + 1$ . Two subspaces of  $F^n$  of dimensions  $n - k$  and  $k + 1$  cannot avoid each other, by [Lemma 16.2.1](#), and a vector in both settles the matter in one line.

*Proof. The lower bound.* Let  $\text{rank } \mathbf{B} \leq k$ . By Rank–Nullity for matrices ([Theorem 2.5.7](#)),  $\dim \text{nul}(\mathbf{B}) = n - \text{rank } \mathbf{B} \geq n - k$ . Let  $H = \text{span}(\mathbf{v}_1, \dots, \mathbf{v}_{k+1})$ , where  $\mathbf{v}_1, \dots, \mathbf{v}_n$  are the right singular vectors of a fixed singular value decomposition of  $\mathbf{A}$ ; then  $\dim H = k + 1$ . Since

$$\dim \text{nul}(\mathbf{B}) + \dim H \geq (n - k) + (k + 1) = n + 1 > n,$$

[Lemma 16.2.1](#) gives a vector  $\mathbf{x} \neq \mathbf{0}$  with  $\mathbf{x} \in \text{nul}(\mathbf{B}) \cap H$ . For that  $\mathbf{x}$ ,  $\mathbf{B}\mathbf{x} = \mathbf{0}$ , so  $(\mathbf{A} - \mathbf{B})\mathbf{x} = \mathbf{A}\mathbf{x}$ , and by [Lemma 16.9.1](#) (b) applied with  $k + 1$  in place of  $k$ ,

$$\|(\mathbf{A} - \mathbf{B})\mathbf{x}\| = \|\mathbf{A}\mathbf{x}\| \geq \sigma_{k+1}(\mathbf{A}) \|\mathbf{x}\|.$$

Since  $\mathbf{x} \neq \mathbf{0}$ , dividing by  $\|\mathbf{x}\|$  and using [Theorem 15.3.2](#) (a), which says  $\|\mathbf{A} - \mathbf{B}\|_2$  is the largest such ratio, gives  $\|\mathbf{A} - \mathbf{B}\|_2 \geq \sigma_{k+1}(\mathbf{A})$ .

*Equality at the truncation.* If  $k \geq p$ , then  $\mathbf{A}_k = \mathbf{A}$  by [Definition 12.10.1](#) and  $\sigma_{k+1}(\mathbf{A}) = 0$  by our convention, so equality holds trivially; assume  $k < p$ . By [Definition 12.10.1](#),  $\mathbf{A}_k = \sum_{i \leq k} \sigma_i \mathbf{u}_i \mathbf{v}_i^*$  has rank  $\min(k, \text{rank } \mathbf{A}) \leq k$ , and

$$\mathbf{A} - \mathbf{A}_k = \sum_{i > k} \sigma_i \mathbf{u}_i \mathbf{v}_i^* = \mathbf{U} \Sigma^{(k)} \mathbf{V}^*,$$

where  $\Sigma^{(k)}$  is  $\Sigma$  with its first  $k$  diagonal entries replaced by 0. Since  $\mathbf{U}$  is unitary,  $(\mathbf{A} - \mathbf{A}_k)^*(\mathbf{A} - \mathbf{A}_k) = \mathbf{V} (\Sigma^{(k)})^* \Sigma^{(k)} \mathbf{V}^*$ , which is similar to the diagonal matrix  $(\Sigma^{(k)})^* \Sigma^{(k)}$ ; its eigenvalues are therefore  $\sigma_{k+1}^2, \dots, \sigma_p^2$  together with zeros. By [Definition 12.8.1](#) the singular values of  $\mathbf{A} - \mathbf{A}_k$  are  $\sigma_{k+1} \geq \sigma_{k+2} \geq \dots$  followed by zeros, the largest being  $\sigma_{k+1}$ . Hence  $\|\mathbf{A} - \mathbf{A}_k\|_2 = \sigma_{k+1}(\mathbf{A})$  by [Theorem 15.3.3](#) (c). This proves the theorem. ■

So the *minimizer* is the same matrix in both norms, and only the *minimum value* changes:  $\sigma_{k+1}$  for the spectral norm,  $(\sum_{i > k} \sigma_i^2)^{1/2}$  for the Frobenius norm. That the same  $\mathbf{A}_k$  works for both is not a coincidence — it is optimal in every unitarily invariant norm at once, which is Chapter 20's theorem and needs machinery neither this section nor Chapter 12 has.

**Example. (One matrix, two norms)**

Let

$$\mathbf{A} = \begin{pmatrix} 3 & 1 & 0 \\ 1 & 3 & 0 \\ 0 & 0 & 1 \end{pmatrix} \in M_3(\mathbb{R}).$$

Find the best approximations of rank 1 and rank 2, and compute the error in both the spectral and the Frobenius norm.

*Solution.*  $\mathbf{A}$  is symmetric and block diagonal. The  $2 \times 2$  block  $\begin{pmatrix} 3 & 1 \\ 1 & 3 \end{pmatrix}$  has trace 6 and determinant 8, hence eigenvalues 4 and 2, with orthonormal eigenvectors  $\frac{1}{\sqrt{2}}(1, 1)$  and  $\frac{1}{\sqrt{2}}(1, -1)$ ; the third eigenvalue is 1, with eigenvector  $\mathbf{e}_3$ . All three eigenvalues are positive, so  $\mathbf{A} \succ 0$  and its singular values are those eigenvalues ([Theorem 12.1.5](#);  $\mathbf{A}^* \mathbf{A} = \mathbf{A}^2$  has eigenvalues their squares):

$$\sigma_1 = 4, \quad \sigma_2 = 2, \quad \sigma_3 = 1,$$

with  $\mathbf{u}_i = \mathbf{v}_i$  the unit eigenvectors  $\frac{1}{\sqrt{2}}(1, 1, 0)$ ,  $\frac{1}{\sqrt{2}}(1, -1, 0)$ ,  $\mathbf{e}_3$ .

Truncating,

$$\mathbf{A}_1 = 4\mathbf{v}_1\mathbf{v}_1^\top = \begin{pmatrix} 2 & 2 & 0 \\ 2 & 2 & 0 \\ 0 & 0 & 0 \end{pmatrix}, \quad \mathbf{A}_2 = \begin{pmatrix} 3 & 1 & 0 \\ 1 & 3 & 0 \\ 0 & 0 & 0 \end{pmatrix},$$

the second because  $4\mathbf{v}_1\mathbf{v}_1^\top + 2\mathbf{v}_2\mathbf{v}_2^\top$  restores the  $2 \times 2$  block exactly. The error matrices are

$$\mathbf{A} - \mathbf{A}_1 = \begin{pmatrix} 1 & -1 & 0 \\ -1 & 1 & 0 \\ 0 & 0 & 1 \end{pmatrix}, \quad \mathbf{A} - \mathbf{A}_2 = \text{diag}(0, 0, 1).$$

For  $k = 1$ :  $\|\mathbf{A} - \mathbf{A}_1\|_2 = \sigma_2 = 2$  and  $\|\mathbf{A} - \mathbf{A}_1\|_F = \sqrt{\sigma_2^2 + \sigma_3^2} = \sqrt{5} \approx 2.236$ . Check the first directly:  $\mathbf{A} - \mathbf{A}_1$  is symmetric with the  $2 \times 2$  block  $\begin{pmatrix} 1 & -1 \\ -1 & 1 \end{pmatrix}$ , of eigenvalues 2 and 0, and the entry 1, so its largest eigenvalue in modulus is 2. Check the second directly: the sum of the squares of the five non-zero entries is  $4 \cdot 1 + 1 = 5$ .

For  $k = 2$ : both norms of  $\text{diag}(0, 0, 1)$  equal  $1 = \sigma_3$ , and the two formulas agree because only one singular value is discarded.

*Quick check.* A matrix has singular values 6, 3, 2, 1. What is the smallest spectral-norm error of a rank-2 approximation, and how does it compare with the Frobenius-norm error found in [Exercise \(B1: Errors from the singular values alone\)](#)?

*Solution.* By [Theorem 16.9.6](#) with  $k = 2$ , the minimum is  $\sigma_3 = 2$ . By [Theorem 12.10.3](#) the Frobenius minimum is  $\sqrt{\sigma_3^2 + \sigma_4^2} = \sqrt{5} \approx 2.236$ . The Frobenius error is the larger of the two, as it must be: it counts every discarded singular value, while the spectral error counts only the largest one.

## 16.9.5 How far an invertible matrix is from being singular

Take  $k = n - 1$  in [Theorem 16.9.6](#), with  $\mathbf{A}$  square. A matrix of rank at most  $n - 1$  is precisely a singular matrix, so the theorem computes the distance from  $\mathbf{A}$  to the set of singular matrices. That is the geometric content of Chapter 15's condition number, and it is the best payoff in this chapter.

### Corollary 16.9.7 — The Distance to Singularity

Let  $\mathbf{A} \in M_n(F)$  be invertible. Then

$$\min\{\|\mathbf{A} - \mathbf{B}\|_2 : \mathbf{B} \in M_n(F) \text{ singular}\} = \sigma_n(\mathbf{A}) = \frac{1}{\|\mathbf{A}^{-1}\|_2},$$

the minimum being attained at  $\mathbf{B} = \mathbf{A}_{n-1}$ . Consequently the **relative** distance from  $\mathbf{A}$  to the nearest singular matrix is

$$\frac{\min\{\|\mathbf{A} - \mathbf{B}\|_2 : \mathbf{B} \text{ singular}\}}{\|\mathbf{A}\|_2} = \frac{\sigma_n(\mathbf{A})}{\sigma_1(\mathbf{A})} = \frac{1}{\kappa_2(\mathbf{A})},$$

the reciprocal of the spectral condition number of [Definition 15.8.1](#).

*Idea.* A singular matrix has rank at most  $n - 1$ , so this is the spectral Eckart–Young theorem with  $k = n - 1$ , and the answer is the next singular value,  $\sigma_n$ . The formula  $1/\|\mathbf{A}^{-1}\|_2$  is the same number read through the inverse, whose largest singular value is  $1/\sigma_n$ .

*Proof.* A matrix  $\mathbf{B} \in M_n(F)$  is singular if and only if  $\text{rank } \mathbf{B} \leq n - 1$  ([Theorem 2.4.4](#)). So the set over which the minimum is taken is exactly the set appearing in [Theorem 16.9.6](#) with  $k = n - 1$ , and that theorem gives the value  $\sigma_{(n-1)+1}(\mathbf{A}) = \sigma_n(\mathbf{A})$ , attained at  $\mathbf{A}_{n-1}$ . The matrix  $\mathbf{A}_{n-1}$  is singular because its rank is  $\min(n - 1, \text{rank } \mathbf{A}) = n - 1 < n$ .

For the second expression: since  $\text{rank } \mathbf{A} = n$ , exactly  $n$  singular values of  $\mathbf{A}$  are non-zero ([Theorem 12.8.1](#)), so  $\sigma_n(\mathbf{A}) > 0$ . By [Definition 15.8.1](#) and [Proposition 15.8.2](#) (d),  $\|\mathbf{A}\|_2 \|\mathbf{A}^{-1}\|_2 = \kappa_2(\mathbf{A}) = \sigma_1(\mathbf{A})/\sigma_n(\mathbf{A})$ , and  $\|\mathbf{A}\|_2 = \sigma_1(\mathbf{A})$  by [Theorem 15.3.3](#) (c); dividing,  $\|\mathbf{A}^{-1}\|_2 = 1/\sigma_n(\mathbf{A})$ . Dividing the minimum  $\sigma_n(\mathbf{A})$  by  $\|\mathbf{A}\|_2 = \sigma_1(\mathbf{A})$  gives  $\sigma_n/\sigma_1 = 1/\kappa_2(\mathbf{A})$ , the last display. This proves the corollary. ■

**What this settles.** Chapter 15 defined  $\kappa(\mathbf{A}) = \|\mathbf{A}\| \|\mathbf{A}^{-1}\|$  as the worst factor by which  $\mathbf{A}$  magnifies a relative error, proved that the factor is attained, and described it geometrically through [Lemma 15.8.1](#) as the ratio of the largest to the smallest stretch  $\|\mathbf{A}\mathbf{x}\|$  over unit vectors  $\mathbf{x}$ . That is a statement about what  $\mathbf{A}$  *does*. [Corollary 16.9.7](#) is a statement about where  $\mathbf{A}$  *is*: it says that  $1/\kappa_2(\mathbf{A})$  measures the distance from  $\mathbf{A}$  to the boundary of the invertible matrices, relative to the size of  $\mathbf{A}$ . A matrix with  $\kappa_2 = 10^6$  is not merely a matrix that magnifies errors by a million; it is a matrix that a relative perturbation of one part in a million can make singular. “Ill-conditioned” and “nearly singular” are the same property, and this is the theorem that identifies them.



Two earlier exercises reached for this. [Exercise \(C1: How far is a matrix from being singular?\)](#) found the same distance in the Frobenius norm — also  $\sigma_n$ , since only one singular value is discarded when  $k = n - 1$ . [Exercise \(C2: The condition number is the reciprocal distance to singularity\)](#) obtained the spectral statement by hand, with a Neumann-series lower bound and the truncation as witness. Nothing above rests on either. What is new is that the distance to singularity is now the case  $k = n - 1$  of one theorem covering every rank, proved by a dimension count and stated as an attained minimum.

**Example. (The nearest singular matrix, exactly and approximately)**

- (a) For the matrix  $\mathbf{A}$  of [Example \(One matrix, two norms\)](#), find the nearest singular matrix and the relative distance to it.
- (b) Do the same, to three significant figures, for  $\mathbf{C} = \begin{pmatrix} 1 & 1 \\ 1 & 1.01 \end{pmatrix}$ , the ill-conditioned matrix of [Example \(A system where the bound is attained\)](#).

*Solution.*

- (a) The singular values are 4, 2, 1, so by [Corollary 16.9.7](#) the nearest singular matrix is at distance  $\sigma_3 = 1$ , and it is

$$\mathbf{A}_2 = \begin{pmatrix} 3 & 1 & 0 \\ 1 & 3 & 0 \\ 0 & 0 & 0 \end{pmatrix},$$

computed in [Example \(One matrix, two norms\)](#); it is singular because its last row is zero. Here  $\kappa_2(\mathbf{A}) = 4/1 = 4$ , so the relative distance is  $\frac{1}{4}$ : a perturbation of a quarter of the size of  $\mathbf{A}$  is needed, and  $\mathbf{A}$  is very well conditioned.

- (b)  $\mathbf{C}$  is symmetric with  $\text{tr } \mathbf{C} = \frac{201}{100}$  and  $\det \mathbf{C} = \frac{1}{100}$ , so its eigenvalues are

$$\frac{201 \pm \sqrt{201^2 - 400}}{200} = \frac{201 \pm \sqrt{40001}}{200},$$

both positive, hence equal to the singular values:  $\sigma_1 \approx 2.00501$  and  $\sigma_2 \approx 0.0049875$ . So  $\kappa_2(\mathbf{C}) \approx 402.0$  and the nearest singular matrix is at distance  $\sigma_2 \approx 0.00499$ , a **relative** distance of  $1/\kappa_2(\mathbf{C}) \approx 0.00249$ , about a quarter of one percent. The nearest singular matrix itself is  $\mathbf{C}_1 = \sigma_1 \mathbf{v}_1 \mathbf{v}_1^\top$ , which to five decimals is

$$\mathbf{C}_1 \approx \begin{pmatrix} 0.99749 & 1.00249 \\ 1.00249 & 1.00752 \end{pmatrix};$$

it is singular, and each entry differs from the corresponding entry of  $\mathbf{C}$  by about 0.0025. Changing  $\mathbf{C}$  in its third decimal place destroys its invertibility. That is what  $\kappa_2 \approx 402$  means.

**Warning. The distance to singularity is not measured by the determinant.** Chapter 15 §08 warned that no threshold on  $|\det \mathbf{A}|$  detects ill-conditioning, and [Corollary 16.9.7](#) says exactly why: the quantity that decides nearness to singularity is  $\sigma_n$ , and  $|\det \mathbf{A}| = \sigma_1 \sigma_2 \cdots \sigma_n$  is a product in which a tiny  $\sigma_n$  can be hidden by large partners, or a healthy  $\sigma_n$  buried by many small ones. For  $\mathbf{A} = 10^{-6} \mathbf{I}_{10}$  all singular values are  $10^{-6}$ , so  $|\det \mathbf{A}| = 10^{-60}$  while the relative distance to singularity is  $1/\kappa_2 = 1$ , the largest it can ever be. For  $\mathbf{A} = \text{diag}(10^6, 10^{-6})$ ,  $|\det \mathbf{A}| = 1$  while the relative distance is  $10^{-12}$ . Only the *smallest* singular value knows.

## 16.9.6 Deleting a row

The last transfer is Cauchy interlacing. For a Hermitian matrix the operation was deleting a row **and** the matching column; for a general matrix there is no matching, and deleting a single row is enough.

### Theorem 16.9.8 — Interlacing Under Deletion of a Row or a Column

Let  $\mathbf{A} \in M_{m \times n}(F)$  with  $m \geq 2$ , and let  $\mathbf{B} \in M_{(m-1) \times n}(F)$  be obtained from  $\mathbf{A}$  by deleting one row. Then

$$\sigma_i(\mathbf{B}) \leq \sigma_i(\mathbf{A}) \quad (1 \leq i \leq n), \quad \sigma_{i+1}(\mathbf{A}) \leq \sigma_i(\mathbf{B}) \quad (1 \leq i \leq n-1),$$

singular values beyond the  $p$ -th being read as 0. If instead  $n \geq 2$  and  $\mathbf{B} \in M_{m \times (n-1)}(F)$  is obtained from  $\mathbf{A}$  by deleting one **column**, the same two chains hold, both over  $1 \leq i \leq n-1$ ; written out,  $\sigma_1(\mathbf{A}) \geq \sigma_1(\mathbf{B}) \geq \sigma_2(\mathbf{A}) \geq \cdots \geq \sigma_{n-1}(\mathbf{B}) \geq \sigma_n(\mathbf{A})$ .

*Idea.* Deleting a row removes one term from the sum  $\|\mathbf{A}\mathbf{x}\|^2 = \sum_{\text{rows}} |\text{row} \cdot \mathbf{x}|^2$ . So  $\|\mathbf{B}\mathbf{x}\| \leq \|\mathbf{A}\mathbf{x}\|$  for every  $\mathbf{x}$ , which gives the first chain from the min-max formula at once, because both matrices are compared over the same subspaces of  $F^n$ . For the second chain, the two norms **agree** on the hyperplane where the deleted row vanishes; intersecting a good subspace for  $\mathbf{B}$  with that hyperplane costs one dimension, and one dimension is exactly one index.

*Proof.* Say row  $r$  of  $\mathbf{A}$  is deleted, and write  $\mathbf{r}^* \in M_{1 \times n}(F)$  for that row. Summing squares of entries coordinate by coordinate,

$$\|\mathbf{A}\mathbf{x}\|^2 = \|\mathbf{B}\mathbf{x}\|^2 + |\mathbf{r}^*\mathbf{x}|^2 \quad (\mathbf{x} \in F^n), \quad (16.9.3)$$

since the entries of  $\mathbf{A}\mathbf{x}$  are those of  $\mathbf{B}\mathbf{x}$  together with  $\mathbf{r}^*\mathbf{x}$ .

*First chain.* By Eq. 16.9.3,  $\|\mathbf{B}\mathbf{x}\|/\|\mathbf{x}\| \leq \|\mathbf{A}\mathbf{x}\|/\|\mathbf{x}\|$  for every  $\mathbf{x} \neq \mathbf{0}$ . Both  $\mathbf{A}$  and  $\mathbf{B}$  have  $n$  columns, so Theorem 16.9.2 describes  $\sigma_i(\mathbf{A})$  and  $\sigma_i(\mathbf{B})$  as a minimum over the *same* family of subspaces of  $F^n$ , of a maximum of these two objective functions. A pointwise smaller function has a smaller maximum on each subspace and hence a smaller minimum over the family, so  $\sigma_i(\mathbf{B}) \leq \sigma_i(\mathbf{A})$  for every  $i \leq n$ .

*Second chain.* Fix  $1 \leq i \leq n-1$ . Let  $\mathbf{w}_1, \dots, \mathbf{w}_n$  be the right singular vectors of a singular value decomposition of  $\mathbf{B}$ , and put  $T = \text{span}(\mathbf{w}_i, \dots, \mathbf{w}_n)$ , of dimension  $n-i+1$ ; by Lemma 16.9.1 (a) applied to  $\mathbf{B}$ ,

$$\|\mathbf{B}\mathbf{x}\| \leq \sigma_i(\mathbf{B})\|\mathbf{x}\| \quad (\mathbf{x} \in T).$$

Let  $H = \{\mathbf{x} \in F^n : \mathbf{r}^*\mathbf{x} = 0\} = \text{nul}(\mathbf{r}^*)$ . Since  $\mathbf{r}^*$  has at most one non-zero row,  $\text{rank } \mathbf{r}^* \leq 1$ , so  $\dim H \geq n-1$  by Rank-Nullity for matrices (Theorem 2.5.7). By the dimension formula (Theorem 1.7.3),

$$\dim(T \cap H) \geq (n-i+1) + (n-1) - n = n-i \geq 1.$$

Choose  $W \subseteq T \cap H$  with  $\dim W = n - i = n - (i + 1) + 1$ . For  $\mathbf{0} \neq \mathbf{x} \in W$  we have  $\mathbf{r}^* \mathbf{x} = 0$ , so Eq. 16.9.3 gives  $\|\mathbf{A}\mathbf{x}\| = \|\mathbf{B}\mathbf{x}\| \leq \sigma_i(\mathbf{B})\|\mathbf{x}\|$ . Hence the inner maximum over  $W$  is at most  $\sigma_i(\mathbf{B})$ , and  $\sigma_{i+1}(\mathbf{A})$ , being the minimum of that inner maximum over all subspaces of dimension  $n - (i + 1) + 1$  (Theorem 16.9.2), is at most  $\sigma_i(\mathbf{B})$ .

*Columns.* Deleting column  $c$  of  $\mathbf{A}$  is deleting row  $c$  of  $\mathbf{A}^*$ . Taking conjugate transposes in  $\mathbf{A} = \mathbf{U}\mathbf{\Sigma}\mathbf{V}^*$  gives  $\mathbf{A}^* = \mathbf{V}\mathbf{\Sigma}^*\mathbf{U}^*$ , in which  $\mathbf{\Sigma}^*$  has the same diagonal  $\sigma_1, \dots, \sigma_p$ ; so by Theorem 12.8.2,  $\sigma_i(\mathbf{A}^*) = \sigma_i(\mathbf{A})$  for every  $i \leq p$ , and the same holds for  $\mathbf{B}$  and  $\mathbf{B}^*$ . Both  $\mathbf{A}^*$  and  $\mathbf{B}^*$  have  $m$  columns, so the row statement applied to them gives  $\sigma_i(\mathbf{B}) \leq \sigma_i(\mathbf{A})$  for  $i \leq m$  and  $\sigma_{i+1}(\mathbf{A}) \leq \sigma_i(\mathbf{B})$  for  $i \leq m - 1$ . The remaining indices are free: if  $i > m$  then  $i$  exceeds  $\min(m, n - 1)$ , so  $\sigma_i(\mathbf{B}) = 0$  and the first inequality is trivial; and if  $i \geq m$  then  $i + 1 > p$ , so  $\sigma_{i+1}(\mathbf{A}) = 0$  and the second is trivial. Hence both chains hold on the stated ranges. This proves the theorem. ■

**Example. (Interlacing on a  $3 \times 2$  matrix)**

Let  $\mathbf{A} = \begin{pmatrix} 1 & 0 \\ 0 & 1 \\ 1 & 1 \end{pmatrix}$ , the matrix of Example (The best rank-one approximation of a  $3 \times 2$  matrix). Verify Theorem 16.9.8 for each of the three rows that could be deleted.

*Solution.* Example (The best rank-one approximation of a  $3 \times 2$  matrix) computed  $\mathbf{A}^\top \mathbf{A} = \begin{pmatrix} 2 & 1 \\ 1 & 2 \end{pmatrix}$ , with eigenvalues 3 and 1, so  $\sigma_1(\mathbf{A}) = \sqrt{3} \approx 1.732$  and  $\sigma_2(\mathbf{A}) = 1$ .

*Deleting row 1 or row 2.* The remaining matrix is  $\begin{pmatrix} 0 & 1 \\ 1 & 1 \end{pmatrix}$  or  $\begin{pmatrix} 1 & 0 \\ 1 & 1 \end{pmatrix}$ . In both cases  $\mathbf{B}^\top \mathbf{B} = \begin{pmatrix} 1 & 1 \\ 1 & 2 \end{pmatrix}$ , with trace 3 and determinant 1, hence eigenvalues  $(3 \pm \sqrt{5})/2$ . Their square roots are  $(1 + \sqrt{5})/2 \approx 1.618$  and  $(\sqrt{5} - 1)/2 \approx 0.618$  — the two are reciprocal, since  $(3 + \sqrt{5})/2 \cdot (3 - \sqrt{5})/2 = 1$ . The interlacing reads

$$1.732 \geq 1.618 \geq 1 \geq 0.618,$$

that is  $\sigma_1(\mathbf{A}) \geq \sigma_1(\mathbf{B}) \geq \sigma_2(\mathbf{A}) \geq \sigma_2(\mathbf{B})$ , as the theorem requires.

*Deleting row 3.* The remaining matrix is  $\mathbf{I}_2$ , with  $\sigma_1 = \sigma_2 = 1$ , and the chain reads  $1.732 \geq 1 \geq 1 \geq 1$ . Here the second inequality of the first chain and the inequality of the second chain are both equalities: deleting the row that carried all the interaction between the two columns leaves a perfectly conditioned matrix.

Everything in this section came from one move: a singular value is the square root of an eigenvalue of a Hermitian matrix, so the min–max theory transfers wholesale. What the transfer bought is worth listing. The spectral half of the Eckart–Young theorem, which Chapter 12 could state only in the Frobenius norm, is now proved. The condition number of Chapter 15 has a geometric meaning:  $1/\kappa_2(\mathbf{A})$  is how far, relatively,  $\mathbf{A}$  sits from the nearest singular matrix. And singular values turned out to be perfectly conditioned with no hypothesis at all, which the Jordan block showed eigenvalues are not. The Hermitian dilation  $\mathcal{H}(\mathbf{A})$  is the other bridge from singular values to Hermitian eigenvalues; Section 10’s exercises use it, and Chapter 19 leans on it for perturbation bounds.

## 16.9.7 Exercises

### 16.9.7.1 A. Check your understanding

#### Exercise A1.

- State the min–max description of  $\sigma_k(\mathbf{A})$ , naming the dimension of the subspaces in each of the two forms.
- Explain in one sentence why the identity  $\|\mathbf{A}\mathbf{x}\|^2/\|\mathbf{x}\|^2 = R_{\mathbf{A}^*\mathbf{A}}(\mathbf{x})$  is what makes the whole section possible.
- State the bound of [Corollary 16.9.5](#) and say precisely which hypotheses it needs, comparing with [Corollary 16.3.6](#).
- What is  $\min\{\|\mathbf{A} - \mathbf{B}\|_2 : \text{rank } \mathbf{B} \leq k\}$ , and what is the corresponding Frobenius-norm minimum? Which of the two is larger, and why?
- Decide whether the statement is correct, and justify: “a square matrix with a very small determinant is close to a singular matrix”.

*Solution.*

- See [Theorem 16.9.2](#):  $\sigma_k(\mathbf{A})$  is the minimum, over subspaces  $W \leq F^n$  of dimension  $n - k + 1$ , of  $\max\{\|\mathbf{A}\mathbf{x}\|/\|\mathbf{x}\| : \mathbf{0} \neq \mathbf{x} \in W\}$ ; equivalently the maximum, over subspaces of dimension  $k$ , of the corresponding minimum.
- It converts a statement about the rectangular matrix  $\mathbf{A}$ , which has no eigenvalues, into a statement about the Hermitian matrix  $\mathbf{A}^*\mathbf{A}$ , which has the whole of Sections 1 to 8 available; and the conversion loses nothing, because the square root is increasing.
- $|\sigma_i(\mathbf{A} + \mathbf{E}) - \sigma_i(\mathbf{A})| \leq \|\mathbf{E}\|_2$ , for **arbitrary**  $\mathbf{A}, \mathbf{E} \in M_{m \times n}(F)$  — no symmetry, no smallness, no squareness. [Corollary 16.3.6](#) is the same bound for eigenvalues, but it requires both  $\mathbf{A}$  and  $\mathbf{E}$  to be Hermitian, and [Example \(A Jordan block’s eigenvalues move like  \$\epsilon^{1/k}\$ \)](#) shows the bound is false without that.
- $\sigma_{k+1}(\mathbf{A})$  in the spectral norm ([Theorem 16.9.6](#)) and  $(\sum_{i>k} \sigma_i^2)^{1/2}$  in the Frobenius norm ([Theorem 12.10.3](#)). The Frobenius value is the larger, since  $\sum_{i>k} \sigma_i^2 \geq \sigma_{k+1}^2$ ; it charges for every discarded singular value, the spectral norm only for the biggest.
- Incorrect. Nearness to singularity is measured by  $\sigma_n$ , not by  $\det$  ([Corollary 16.9.7](#)). The matrix  $10^{-6}\mathbf{I}_{10}$  has determinant  $10^{-60}$  and relative distance 1 to the singular matrices, the largest possible value. The determinant is the product of all the singular values and cannot report the smallest one.

### 16.9.7.2 B. Practice

#### Exercise B1: Two norms, one list of singular values.

A matrix  $\mathbf{A} \in M_{5 \times 4}(\mathbb{R})$  has singular values 8, 4, 4, 1.

- Compute  $\min\{\|\mathbf{A} - \mathbf{B}\|_2 : \text{rank } \mathbf{B} \leq k\}$  for  $k = 0, 1, 2, 3$ .
- Compute the corresponding Frobenius minima and say for which  $k$  the two agree.

- (c) Compute  $\kappa_2(\mathbf{A})$  and the relative distance from  $\mathbf{A}$  to the set of matrices of rank at most 3.

*Solution.*

- (a) By [Theorem 16.9.6](#) the minimum is  $\sigma_{k+1}$ , so the four values are 8, 4, 4, 1 for  $k = 0, 1, 2, 3$ .  
 (b) By [Theorem 12.10.3](#) the minimum is  $(\sum_{i>k} \sigma_i^2)^{1/2}$ :

$$\sqrt{64 + 16 + 16 + 1} = \sqrt{97}, \quad \sqrt{33}, \quad \sqrt{17}, \quad 1$$

for  $k = 0, 1, 2, 3$ , approximately 9.849, 5.745, 4.123, 1. They agree only at  $k = 3$ , where exactly one singular value is discarded.

- (c)  $\mathbf{A}$  has rank 4 and full column rank, so [Definition 15.8.1](#) gives  $\kappa_2(\mathbf{A}) = \sigma_1/\sigma_4 = 8$ . The distance to the matrices of rank at most 3 is  $\sigma_4 = 1$  and  $\|\mathbf{A}\|_2 = 8$ , so the relative distance is  $\frac{1}{8} = 1/\kappa_2(\mathbf{A})$  — the rectangular analogue of [Corollary 16.9.7](#), and for the same reason.

### Exercise B2: The nearest singular matrix.

Let  $\mathbf{A} = \begin{pmatrix} 2 & 1 \\ 1 & 1 \end{pmatrix}$ , whose condition number  $\kappa_2(\mathbf{A}) = \frac{7+3\sqrt{5}}{2} \approx 6.854$  was computed in [Example \(Three condition numbers of one matrix\)](#).

- (a) Find the singular values of  $\mathbf{A}$  and the distance from  $\mathbf{A}$  to the nearest singular matrix.  
 (b) Write down a singular matrix at exactly that distance, and check that it is singular.  
 (c) Compute the relative distance and compare it with  $1/\kappa_2(\mathbf{A})$ .

*Solution.*

- (a)  $\mathbf{A}$  is symmetric with  $\text{tr } \mathbf{A} = 3$  and  $\det \mathbf{A} = 1$ , so its eigenvalues are  $(3 \pm \sqrt{5})/2$ , both positive; hence these are also its singular values,

$$\sigma_1 = \frac{3+\sqrt{5}}{2} \approx 2.618, \quad \sigma_2 = \frac{3-\sqrt{5}}{2} \approx 0.382.$$

By [Corollary 16.9.7](#) the distance to the nearest singular matrix is  $\sigma_2 = (3 - \sqrt{5})/2$ .

- (b) The witness is  $\mathbf{A}_1 = \sigma_1 \mathbf{v}_1 \mathbf{v}_1^\top$ , where  $\mathbf{v}_1$  is a unit eigenvector for  $\sigma_1$ . From  $(2 - \sigma_1)x_1 + x_2 = 0$  we may take  $\mathbf{v}_1$  parallel to  $(2, \sqrt{5} - 1)$ , whose squared length is  $4 + (6 - 2\sqrt{5}) = 10 - 2\sqrt{5}$ . Hence

$$\mathbf{A}_1 = \frac{3 + \sqrt{5}}{2} \cdot \frac{1}{10 - 2\sqrt{5}} \begin{pmatrix} 4 & 2(\sqrt{5} - 1) \\ 2(\sqrt{5} - 1) & 6 - 2\sqrt{5} \end{pmatrix} = \begin{pmatrix} 1 + \frac{2\sqrt{5}}{5} & \frac{1}{2} + \frac{3\sqrt{5}}{10} \\ \frac{1}{2} + \frac{3\sqrt{5}}{10} & \frac{1}{2} + \frac{\sqrt{5}}{10} \end{pmatrix},$$

approximately  $\begin{pmatrix} 1.894 & 1.171 \\ 1.171 & 0.724 \end{pmatrix}$ . It is singular: it is a scalar multiple of  $\mathbf{v}_1 \mathbf{v}_1^\top$ , which has rank 1, so its determinant is 0. (Directly:  $1.894 \times 0.724 - 1.171^2 = 0$  to three decimals.)

(c)  $\|\mathbf{A}\|_2 = \sigma_1$ , so the relative distance is

$$\frac{\sigma_2}{\sigma_1} = \frac{3 - \sqrt{5}}{3 + \sqrt{5}} = \frac{(3 - \sqrt{5})^2}{4} = \frac{7 - 3\sqrt{5}}{2} \approx 0.1459,$$

using  $(3 - \sqrt{5})(3 + \sqrt{5}) = 4$ . And  $1/\kappa_2(\mathbf{A}) = 2/(7 + 3\sqrt{5}) = (7 - 3\sqrt{5})/2$ , since  $(7 + 3\sqrt{5})(7 - 3\sqrt{5}) = 49 - 45 = 4$ . The two agree, as [Corollary 16.9.7](#) says.

### Exercise B3: Interlacing by hand.

Let  $\mathbf{A} = \begin{pmatrix} 2 & 0 \\ 0 & 1 \\ 0 & 2 \end{pmatrix}$ .

- Compute the singular values of  $\mathbf{A}$ .
- Compute the singular values of each of the three matrices obtained by deleting one row, and check [Theorem 16.9.8](#) in each case.

*Solution.*

- $\mathbf{A}^\top \mathbf{A} = \begin{pmatrix} 4 & 0 \\ 0 & 5 \end{pmatrix}$ , so the eigenvalues are 5 and 4 and  $\sigma_1(\mathbf{A}) = \sqrt{5} \approx 2.236$ ,  $\sigma_2(\mathbf{A}) = 2$ .
- Deleting row 1 leaves  $\begin{pmatrix} 0 & 1 \\ 0 & 2 \end{pmatrix}$ , with  $\mathbf{B}^\top \mathbf{B} = \text{diag}(0, 5)$ , so  $\sigma_1 = \sqrt{5}$ ,  $\sigma_2 = 0$ ; the chain reads  $\sqrt{5} \geq \sqrt{5} \geq 2 \geq 0$ .

Deleting row 2 leaves  $\begin{pmatrix} 2 & 0 \\ 0 & 2 \end{pmatrix}$ , with singular values 2, 2; the chain reads  $\sqrt{5} \geq 2 \geq 2 \geq 2$ .

Deleting row 3 leaves  $\begin{pmatrix} 2 & 0 \\ 0 & 1 \end{pmatrix}$ , with singular values 2, 1; the chain reads  $\sqrt{5} \geq 2 \geq 2 \geq 1$ .

All three satisfy  $\sigma_1(\mathbf{A}) \geq \sigma_1(\mathbf{B}) \geq \sigma_2(\mathbf{A}) \geq \sigma_2(\mathbf{B})$ , as [Theorem 16.9.8](#) requires.

### 16.9.7.3 C. Going deeper

#### Exercise C1: Rank cannot drop suddenly.

Let  $\mathbf{A} \in M_{m \times n}(F)$  have rank  $r \geq 1$ .

- Prove that every  $\mathbf{B} \in M_{m \times n}(F)$  with  $\|\mathbf{A} - \mathbf{B}\|_2 < \sigma_r(\mathbf{A})$  has rank  $\mathbf{B} \geq r$ .
- Deduce that for every  $\mathbf{A}$  there is an  $\varepsilon > 0$  such that every  $\mathbf{B}$  within  $\varepsilon$  of  $\mathbf{A}$  in the spectral norm has rank  $\mathbf{B} \geq \text{rank } \mathbf{A}$ .
- Show that the reverse inequality can fail however small  $\varepsilon$  is: give matrices  $\mathbf{A}$  and  $\mathbf{B}$  with  $\|\mathbf{A} - \mathbf{B}\|_2$  as small as we like and rank  $\mathbf{B} > \text{rank } \mathbf{A}$ .

*Hint: for (a), argue by contradiction with [Theorem 16.9.6](#).*

*Solution.*

- Suppose rank  $\mathbf{B} \leq r - 1$ . By [Theorem 16.9.6](#) with  $k = r - 1$ ,

$$\|\mathbf{A} - \mathbf{B}\|_2 \geq \sigma_r(\mathbf{A}),$$

contradicting the hypothesis. Hence rank  $\mathbf{B} \geq r$ .

- (b) If  $\mathbf{A} = 0$  any  $\varepsilon > 0$  works, since then  $r = 0$  and every  $\mathbf{B}$  has rank at least 0. Otherwise take  $\varepsilon = \sigma_r(\mathbf{A})$ , which is  $> 0$  because  $\mathbf{A}$  has exactly  $r = \text{rank } \mathbf{A}$  non-zero singular values ([Theorem 12.8.3](#)). Then (a) applies.
- (c) Take  $\mathbf{A} = \text{diag}(1, 0)$ , of rank 1, and  $\mathbf{B} = \text{diag}(1, \delta)$  with  $\delta > 0$ . Then  $\mathbf{A} - \mathbf{B} = \text{diag}(0, -\delta)$ , so  $\|\mathbf{A} - \mathbf{B}\|_2 = \delta$ , as small as we like, while  $\text{rank } \mathbf{B} = 2 > 1 = \text{rank } \mathbf{A}$ . Rank can jump **up** under an arbitrarily small perturbation, and never jumps down under a small one: that asymmetry is exactly (a), and it is why numerical rank is defined by a threshold on the singular values rather than by exact vanishing.

### Exercise C2: The dilation, and a second proof of the perturbation bound.

Let  $\mathbf{A}, \mathbf{E} \in M_{m \times n}(F)$  and let  $\mathcal{H}(\cdot)$  be the Hermitian dilation of [Proposition 16.9.3](#).

- (a) Prove that  $\mathcal{H}(\mathbf{A} + \mathbf{E}) = \mathcal{H}(\mathbf{A}) + \mathcal{H}(\mathbf{E})$  and that  $\mathcal{H}(\mathbf{E})$  is Hermitian.
- (b) Prove that  $\|\mathcal{H}(\mathbf{E})\|_2 = \|\mathbf{E}\|_2$ .
- (c) Hence give a second proof of [Corollary 16.9.5](#) for  $1 \leq i \leq p$ , using [Corollary 16.3.6](#), and say in one sentence why this route explains the absence of hypotheses on  $\mathbf{E}$ .

*Hint: for (b), use [Proposition 16.9.3](#) and [Theorem 15.3.3](#) (c).*

*Solution.*

- (a) Both claims are immediate from the block form:  $\mathcal{H}$  is built entrywise from  $\mathbf{A}$  and  $\mathbf{A}^*$ , and both  $\mathbf{A} \mapsto \mathbf{A}$  and  $\mathbf{A} \mapsto \mathbf{A}^*$  are additive, so

$$\mathcal{H}(\mathbf{A} + \mathbf{E}) = \begin{pmatrix} \mathbf{0} & \mathbf{A} + \mathbf{E} \\ \mathbf{A}^* + \mathbf{E}^* & \mathbf{0} \end{pmatrix} = \mathcal{H}(\mathbf{A}) + \mathcal{H}(\mathbf{E}).$$

That  $\mathcal{H}(\mathbf{E})$  is Hermitian was proved in [Proposition 16.9.3](#), which places no hypothesis on its argument.

- (b) Write  $\mathbf{M} = \mathcal{H}(\mathbf{E})$ , which is Hermitian by (a). Then  $\mathbf{M}^* \mathbf{M} = \mathbf{M}^2$ , whose eigenvalues are the squares of those of  $\mathbf{M}$  (apply [Corollary 11.4.3](#) to  $\mathbf{M}$  and square the diagonal). By [Proposition 16.9.3](#) the eigenvalues of  $\mathbf{M}$  are  $\pm \sigma_i(\mathbf{E})$  for  $i \leq p$  together with zeros, so the eigenvalues of  $\mathbf{M}^* \mathbf{M}$  are the numbers  $\sigma_i(\mathbf{E})^2$ , each twice, together with zeros. The largest is  $\sigma_1(\mathbf{E})^2$ , so  $\sigma_1(\mathbf{M}) = \sigma_1(\mathbf{E})$  by [Definition 12.8.1](#), and [Theorem 15.3.3](#) (c) turns this into  $\|\mathcal{H}(\mathbf{E})\|_2 = \|\mathbf{E}\|_2$ .
- (c) Let  $1 \leq i \leq p$ . By [Proposition 16.9.3](#) the largest  $p$  eigenvalues of  $\mathcal{H}(\mathbf{A})$  are  $\sigma_1(\mathbf{A}) \geq \dots \geq \sigma_p(\mathbf{A})$ , all of the remaining ones being  $\leq 0$ ; so  $\lambda_i(\mathcal{H}(\mathbf{A})) = \sigma_i(\mathbf{A})$ , and likewise  $\lambda_i(\mathcal{H}(\mathbf{A} + \mathbf{E})) = \sigma_i(\mathbf{A} + \mathbf{E})$ . By (a),  $\mathcal{H}(\mathbf{A} + \mathbf{E}) = \mathcal{H}(\mathbf{A}) + \mathcal{H}(\mathbf{E})$  with  $\mathcal{H}(\mathbf{A})$  and  $\mathcal{H}(\mathbf{E})$  both Hermitian, so [Corollary 16.3.6](#) applies and gives

$$|\sigma_i(\mathbf{A} + \mathbf{E}) - \sigma_i(\mathbf{A})| = |\lambda_i(\mathcal{H}(\mathbf{A} + \mathbf{E})) - \lambda_i(\mathcal{H}(\mathbf{A}))| \leq \|\mathcal{H}(\mathbf{E})\|_2 = \|\mathbf{E}\|_2.$$

The route explains the missing hypotheses: [Corollary 16.3.6](#) demands that the perturbation be Hermitian, and  $\mathcal{H}(\mathbf{E})$  is Hermitian for every  $\mathbf{E}$  whatsoever. Passing to the dilation manufactures the hypothesis instead of assuming it.



## 16.10 Inertia, Again and Additively

Chapter 13 attached three numbers to a real symmetric matrix — how many positive, how many negative, how many zero entries appear on the diagonal after a congruence — and proved that the three do not depend on which congruence was used. The proof there was elementary and self-contained: an orthogonal basis for the form, two subspaces, a dimension count. This section proves the same theorem a second time, from the min–max description of the eigenvalues, and then shows what the second proof buys that the first could not: the three numbers become **additive across a Schur complement**.

**Throughout,  $F = \mathbb{R}$  or  $F = \mathbb{C}$ , and every matrix whose eigenvalues or inertia are named is Hermitian** — real symmetric when  $F = \mathbb{R}$ . Its eigenvalues are real and are indexed decreasingly,  $\lambda_1(\mathbf{A}) \geq \cdots \geq \lambda_n(\mathbf{A})$ , as everywhere in this chapter. **Congruence** means the operation  $\mathbf{A} \mapsto \mathbf{P}^* \mathbf{A} \mathbf{P}$  with  $\mathbf{P}$  invertible, which sends Hermitian matrices to Hermitian matrices; over  $\mathbb{R}$  the star is the plain transpose and this is exactly the relation  $\mathbf{A} \simeq \mathbf{B}$  of [Definition 13.2.1](#), and Chapter 12 already used the complex version in [Proposition 12.1.2](#).

### 16.10.1 A symbol for the triple

Chapter 13 §05 named the three counts and proved them well defined, and [Definition 13.5.1](#) records the words: the triple  $(n_+, n_-, n_0)$  is the **inertia** and the pair  $(n_+, n_-)$  the **signature**. What it did not do is give the triple a symbol, because it never needed to write an equation between two triples. This section writes several, so the symbol is worth the line it costs.

#### Definition 16.10.1 — The Inertia Triple

Let  $\mathbf{A} \in M_n(F)$  be Hermitian, with eigenvalues  $\lambda_1(\mathbf{A}) \geq \cdots \geq \lambda_n(\mathbf{A})$  listed **with multiplicity**. Write  $n_+(\mathbf{A})$ ,  $n_-(\mathbf{A})$  and  $n_0(\mathbf{A})$  for the numbers of indices  $i$  with  $\lambda_i(\mathbf{A}) > 0$ , with  $\lambda_i(\mathbf{A}) < 0$  and with  $\lambda_i(\mathbf{A}) = 0$ , and set

$$\text{In}(\mathbf{A}) := (n_+(\mathbf{A}), n_-(\mathbf{A}), n_0(\mathbf{A})) \in \mathbb{N}^3.$$

Triples are added **entrywise**.

Clause by clause. Every eigenvalue is counted, so  $n_+ + n_- + n_0 = n$  always. The counts are *with multiplicity*:  $\text{In}(\mathbf{I}_n) = (n, 0, 0)$ , not  $(1, 0, 0)$ . And  $n_0(\mathbf{A}) = \dim \text{nul}(\mathbf{A})$ , because a Hermitian matrix is unitarily diagonalizable ([Corollary 11.4.3](#), or [Corollary 11.5.2](#) when  $F = \mathbb{R}$ ), so the eigenspace for 0 has dimension equal to the multiplicity of 0 in the list; equivalently  $n_+ + n_- = \text{rank } \mathbf{A}$ .

For  $F = \mathbb{R}$  this is not a new object. [Theorem 13.5.3](#) says precisely that the numbers of positive, negative and zero *eigenvalues* of a real symmetric  $\mathbf{A}$  are the numbers of positive, negative and zero *diagonal entries* after any congruence to diagonal form — that is, the inertia of [Definition 13.5.1](#). So  $\text{In}(\mathbf{A})$  is the old triple, freshly abbreviated.

A few values, and the dictionary with definiteness.  $\text{In}(\mathbf{0}_n) = (0, 0, n)$ , and  $\text{In}(\text{diag}(1, -1)) = (1, 1, 0)$ . By [Theorem 12.1.4](#),  $\mathbf{A} \succeq 0$  exactly when every eigenvalue is  $\geq 0$ , that is when  $n_-(\mathbf{A}) = 0$ ; by [Theorem 12.1.5](#),  $\mathbf{A} \succ 0$  exactly when  $\text{In}(\mathbf{A}) = (n, 0, 0)$ . Definiteness is the extreme case of inertia, and everything below degenerates to a known statement about definiteness when the



triple is extreme.

## 16.10.2 Sylvester's law, proved again

The description of  $n_+$  that this chapter can supply is not a count of diagonal entries at all. It is a maximum over subspaces, and it mentions no basis, no diagonalization and no matrix.

### Theorem 16.10.1 — Sylvester's Law of Inertia, Second Proof

Let  $\mathbf{A} \in M_n(F)$  be Hermitian. Call a subspace  $W \leq F^n$  **A-positive** if  $\langle \mathbf{A}\mathbf{x}, \mathbf{x} \rangle > 0$  for every  $\mathbf{x} \in W$  with  $\mathbf{x} \neq \mathbf{0}$ . Then:

- (a)  $n_+(\mathbf{A}) = \max\{\dim W : W \text{ is } \mathbf{A}\text{-positive}\}$ , and the maximum is attained at the span of an orthonormal family of eigenvectors for the positive eigenvalues of  $\mathbf{A}$ ;
- (b) for every invertible  $\mathbf{P} \in M_n(F)$ , the matrix  $\mathbf{P}^*\mathbf{A}\mathbf{P}$  is Hermitian and

$$\text{In}(\mathbf{P}^*\mathbf{A}\mathbf{P}) = \text{In}(\mathbf{A}).$$

Over  $\mathbb{R}$ , part (b) is Sylvester's Law of Inertia, [Theorem 13.5.1](#).

*Idea.* For (a), read [Theorem 16.2.3](#) through a sign. Its max–min form says  $\lambda_k(\mathbf{A})$  is the largest value that  $R_{\mathbf{A}}$  can be *guaranteed* to exceed on a  $k$ -dimensional subspace; and “ $W$  is  $\mathbf{A}$ -positive” says exactly that  $R_{\mathbf{A}}$  is guaranteed to be positive on  $W$ . So  $\mathbf{A}$ -positive subspaces of dimension  $k$  exist precisely when  $\lambda_k(\mathbf{A}) > 0$ , which is precisely when  $k \leq n_+$ . For (b), the right-hand side of (a) is defined from the quadratic form alone, and a congruence is a change of variables inside that form:  $\mathbf{x} \mapsto \mathbf{P}\mathbf{x}$  matches up the two families of positive subspaces, dimension for dimension.

*Proof.* Throughout, note that  $\langle \mathbf{A}\mathbf{x}, \mathbf{x} \rangle = \mathbf{x}^*\mathbf{A}\mathbf{x}$  is real for Hermitian  $\mathbf{A}$  ([Proposition 10.6.7](#) (a), or directly:  $\overline{\mathbf{x}^*\mathbf{A}\mathbf{x}} = \mathbf{x}^*\mathbf{A}^*\mathbf{x} = \mathbf{x}^*\mathbf{A}\mathbf{x}$ ), and that for  $\mathbf{x} \neq \mathbf{0}$  the sign of  $\langle \mathbf{A}\mathbf{x}, \mathbf{x} \rangle$  is the sign of the Rayleigh quotient  $R_{\mathbf{A}}(\mathbf{x}) = \langle \mathbf{A}\mathbf{x}, \mathbf{x} \rangle / \langle \mathbf{x}, \mathbf{x} \rangle$  of [Definition 16.1.1](#), since  $\langle \mathbf{x}, \mathbf{x} \rangle = \|\mathbf{x}\|^2 > 0$ . So  $W$  is  $\mathbf{A}$ -positive if and only if  $R_{\mathbf{A}}(\mathbf{x}) > 0$  for every  $\mathbf{x} \in W \setminus \{\mathbf{0}\}$ .

(a), *the lower bound.* Put  $k = n_+(\mathbf{A})$ . If  $k = 0$  the zero subspace is  $\mathbf{A}$ -positive and there is nothing to prove, so assume  $k \geq 1$ . By the spectral theorem ([Corollary 11.4.3](#), or [Corollary 11.5.2](#) when  $F = \mathbb{R}$ ) there is an orthonormal basis  $\mathbf{q}_1, \dots, \mathbf{q}_n$  of  $F^n$  with  $\mathbf{A}\mathbf{q}_i = \lambda_i(\mathbf{A})\mathbf{q}_i$ , the eigenvalues in decreasing order. Put  $W_0 = \text{span}(\mathbf{q}_1, \dots, \mathbf{q}_k)$ , of dimension  $k$ . For  $\mathbf{0} \neq \mathbf{x} = \sum_{i \leq k} c_i \mathbf{q}_i$ , expanding by orthonormality,

$$\langle \mathbf{A}\mathbf{x}, \mathbf{x} \rangle = \sum_{i=1}^k \lambda_i(\mathbf{A})|c_i|^2 \geq \lambda_k(\mathbf{A}) \sum_{i=1}^k |c_i|^2 = \lambda_k(\mathbf{A})\|\mathbf{x}\|^2 > 0,$$

because  $\lambda_i(\mathbf{A}) \geq \lambda_k(\mathbf{A})$  for  $i \leq k$  and  $\lambda_k(\mathbf{A}) > 0$ , the latter since  $k = n_+(\mathbf{A})$  counts the positive eigenvalues of a decreasing list. So  $W_0$  is  $\mathbf{A}$ -positive of dimension  $n_+(\mathbf{A})$ .

(a), *the upper bound.* Let  $W$  be  $\mathbf{A}$ -positive with  $d = \dim W \geq 1$ . The unit sphere  $S_W = \{\mathbf{x} \in W : \|\mathbf{x}\| = 1\}$  is a closed bounded subset of  $F^n$ , hence compact by fact (A3) of Chapter 15's introduction, and it is non-empty because  $d \geq 1$ . The function  $\mathbf{x} \mapsto R_{\mathbf{A}}(\mathbf{x})$  is continuous on it, so by the extreme value theorem — fact (A4) of Chapter

15's introduction — it attains a minimum  $m = R_{\mathbf{A}}(\mathbf{x}_0)$  there, and  $m > 0$  because  $\mathbf{x}_0 \in W$  is non-zero. By homogeneity of  $R_{\mathbf{A}}$ , the same  $m$  is the minimum over all non-zero  $\mathbf{x} \in W$ . Now  $W$  is one competitor in the max-min form of [Theorem 16.2.3](#), so

$$\lambda_d(\mathbf{A}) = \max_{\dim W'=d} \min_{\mathbf{0} \neq \mathbf{x} \in W'} R_{\mathbf{A}}(\mathbf{x}) \geq m > 0.$$

A decreasing list with  $\lambda_d > 0$  has at least  $d$  positive entries, so  $d \leq n_+(\mathbf{A})$ . Together with the lower bound this proves (a).

(b). Put  $\mathbf{B} = \mathbf{P}^* \mathbf{A} \mathbf{P}$ . Then  $\mathbf{B}^* = \mathbf{P}^* \mathbf{A}^* \mathbf{P} = \mathbf{B}$ , so  $\mathbf{B}$  is Hermitian. For every  $\mathbf{x} \in F^n$ ,

$$\langle \mathbf{B}\mathbf{x}, \mathbf{x} \rangle = \mathbf{x}^* \mathbf{P}^* \mathbf{A} \mathbf{P} \mathbf{x} = (\mathbf{P}\mathbf{x})^* \mathbf{A} (\mathbf{P}\mathbf{x}) = \langle \mathbf{A}(\mathbf{P}\mathbf{x}), \mathbf{P}\mathbf{x} \rangle.$$

Since  $\mathbf{P}$  is invertible,  $\mathbf{x} \mapsto \mathbf{P}\mathbf{x}$  is an isomorphism of  $F^n$ ; its restriction to a subspace  $W$  is an isomorphism  $W \rightarrow \mathbf{P}W$ , which carries a basis to a basis ([Theorem 3.5.2](#)), so  $\dim \mathbf{P}W = \dim W$ . And  $W \mapsto \mathbf{P}W$  is a bijection on subspaces, with inverse  $W' \mapsto \mathbf{P}^{-1}W'$ . Moreover  $W$  is  $\mathbf{B}$ -positive if and only if  $\mathbf{P}W$  is  $\mathbf{A}$ -positive: the non-zero vectors of  $\mathbf{P}W$  are exactly the vectors  $\mathbf{P}\mathbf{x}$  with  $\mathbf{0} \neq \mathbf{x} \in W$ , and the display says the two quadratic forms agree at matched vectors. Hence the two sets of dimensions appearing in (a) coincide, and (a) gives  $n_+(\mathbf{B}) = n_+(\mathbf{A})$ .

Applying this to  $-\mathbf{A}$ , whose eigenvalues are the negatives of those of  $\mathbf{A}$  and whose congruence by  $\mathbf{P}$  is  $-\mathbf{B}$ , gives  $n_-(\mathbf{B}) = n_+(-\mathbf{B}) = n_+(-\mathbf{A}) = n_-(\mathbf{A})$ . Finally  $n_0 = n - n_+ - n_-$  for both. This proves the theorem. ■

**What each proof gives.** Over  $\mathbb{R}$  the two theorems say the same thing, since a congruence is exactly a change of basis for the underlying form ([Theorem 13.2.1](#)). They are not interchangeable.

Chapter 13's proof uses only the existence of an orthogonal basis for the form, the invariance of rank under congruence and the dimension formula. It mentions no inner product, no eigenvalue and no limit — in Chapter 13 the eigenvalues enter only afterwards, in [Theorem 13.5.3](#) — and it is the proof to give for a symmetric bilinear form on an abstract real vector space, where no inner product has been chosen and the words “Hermitian matrix” have no meaning until one is.

The proof above costs the spectral theorem and one appeal to compactness. In exchange it gives three things. First, a **description of  $n_+$  with no basis in it at all** — a maximum over subspaces — together with an explicit optimal subspace, which is what [Exercise \(C1: An intrinsic description of  \$n\_+\$ \)](#) asked the reader to establish by hand from an orthogonal basis; here it falls out of the min-max, with the eigenvalues attached. Second, it works **over  $\mathbb{C}$** , for Hermitian matrices and the congruence  $\mathbf{A} \mapsto \mathbf{P}^* \mathbf{A} \mathbf{P}$ , which [Theorem 13.5.1](#) does not cover. Third, it keeps the eigenvalues in view, so the perturbation theory of Section 3 applies to the inertia directly; that is the next theorem. Haynsworth's theorem at the end of the section leans on the second gain: for complex Hermitian matrices, only this proof supplies the law of inertia it needs.

The companions of part (a) are worth stating, because one of them is a trap.

### Corollary 16.10.2 — What Each Count Is a Maximum Of

Let  $\mathbf{A} \in M_n(F)$  be Hermitian.

- (a)  $n_-(\mathbf{A}) = \max\{\dim W : \langle \mathbf{A}\mathbf{x}, \mathbf{x} \rangle < 0 \text{ for every } \mathbf{0} \neq \mathbf{x} \in W\}$ ;
- (b)  $n_+(\mathbf{A}) + n_0(\mathbf{A}) = \max\{\dim W : \langle \mathbf{A}\mathbf{x}, \mathbf{x} \rangle \geq 0 \text{ for every } \mathbf{x} \in W\}$ , and symmetrically  $n_-(\mathbf{A}) + n_0(\mathbf{A})$  is the largest dimension of a subspace on which  $\langle \mathbf{A}\mathbf{x}, \mathbf{x} \rangle \leq 0$ ;
- (c)  $n_0(\mathbf{A}) = \dim \text{nul}(\mathbf{A}) = n - n_+(\mathbf{A}) - n_-(\mathbf{A})$ .

*Idea.* Each claim has a witness and a competitor, as in Courant–Fischer. The witness is the span of the eigenvectors with the right sign; any competing subspace that is too large must meet the span of the eigenvectors of the wrong sign, and on that intersection the form has the wrong sign. For (c), no maximum is involved at all — the kernel is read off the eigenvalues.

*Proof.*

- (a) Apply [Theorem 16.10.1](#) (a) to  $-\mathbf{A}$ : its eigenvalues are the negatives of those of  $\mathbf{A}$ , so  $n_+(-\mathbf{A}) = n_-(\mathbf{A})$ , and a subspace is  $(-\mathbf{A})$ -positive exactly when  $\langle \mathbf{A}\mathbf{x}, \mathbf{x} \rangle < 0$  on its non-zero vectors.
- (b) Fix, as in the previous proof, an orthonormal eigenbasis  $\mathbf{q}_1, \dots, \mathbf{q}_n$  of  $\mathbf{A}$  with  $\mathbf{A}\mathbf{q}_i = \lambda_i(\mathbf{A})\mathbf{q}_i$  ([Corollary 11.4.3](#)). Let  $t = n_+(\mathbf{A}) + n_0(\mathbf{A})$ , so that  $\lambda_i(\mathbf{A}) \geq 0$  exactly for  $i \leq t$ . On  $W_1 = \text{span}(\mathbf{q}_1, \dots, \mathbf{q}_t)$  the same expansion gives  $\langle \mathbf{A}\mathbf{x}, \mathbf{x} \rangle = \sum_{i \leq t} \lambda_i(\mathbf{A})|c_i|^2 \geq 0$ , so the maximum is at least  $t$ . Conversely let  $W$  satisfy  $\langle \mathbf{A}\mathbf{x}, \mathbf{x} \rangle \geq 0$  throughout and suppose  $\dim W > t$ . Put  $W_2 = \text{span}(\mathbf{q}_{t+1}, \dots, \mathbf{q}_n)$ , of dimension  $n - t = n_-(\mathbf{A})$ , on whose non-zero vectors  $\langle \mathbf{A}\mathbf{x}, \mathbf{x} \rangle = \sum_{i > t} \lambda_i(\mathbf{A})|c_i|^2 < 0$ , every  $\lambda_i(\mathbf{A})$  with  $i > t$  being negative. Since  $\dim W + \dim W_2 > t + (n - t) = n$ , [Lemma 16.2.1](#) produces  $\mathbf{0} \neq \mathbf{x} \in W \cap W_2$ , for which  $\langle \mathbf{A}\mathbf{x}, \mathbf{x} \rangle$  is both  $\geq 0$  and  $< 0$ . This is impossible, so  $\dim W \leq t$ . The statement with  $\leq$  follows by applying this one to  $-\mathbf{A}$ .
- (c) The first equality was recorded after [Definition 16.10.1](#), and the second is  $n_+ + n_- + n_0 = n$ .

■

**Warning.**  $n_0$  is not the largest dimension of a subspace on which the form vanishes. Parts (a) and (b) describe  $n_-$ ,  $n_+$  and their sums with  $n_0$  as maxima over subspaces;  $n_0$  itself has no such description, and guessing one is the standard error here. Take  $\mathbf{A} = \text{diag}(1, -1)$ , so  $\text{In}(\mathbf{A}) = (1, 1, 0)$  and  $n_0(\mathbf{A}) = 0$ . On the line  $W = \text{span}((1, 1))$ ,

$$\langle \mathbf{A}\mathbf{x}, \mathbf{x} \rangle = x_1^2 - x_2^2 = 0 \quad \text{for every } \mathbf{x} \in W,$$

so a subspace of dimension 1 carries the identically zero form while  $n_0(\mathbf{A}) = 0$ . What  $n_0$  is is  $\dim \text{nul}(\mathbf{A})$ , by (c) — a kernel, not a maximum. The largest dimension of a subspace on which the form vanishes identically is a different invariant. For a real symmetric **invertible**  $\mathbf{A}$  it is the Witt index of Chapter 13 §07, which that section defines only for non-degenerate forms and bounds by  $n/2$ . For a singular  $\mathbf{A}$  the bound fails — for  $\mathbf{A} = 0$  the whole space qualifies — and in general the answer is  $n_0 + \min(n_+, n_-)$ , which [Exercise \(C1: How large can a subspace be on which the form vanishes?\)](#) computes.

*Quick check.* A Hermitian  $\mathbf{A} \in M_5(F)$  has  $\text{In}(\mathbf{A}) = (2, 2, 1)$ . What is the largest dimension of a subspace on which  $\langle \mathbf{A}\mathbf{x}, \mathbf{x} \rangle > 0$  for all non-zero  $\mathbf{x}$ , and the largest on which  $\langle \mathbf{A}\mathbf{x}, \mathbf{x} \rangle \geq 0$  for all  $\mathbf{x}$ ?

*Solution.* 2 and 3. The first is  $n_+(\mathbf{A}) = 2$  by [Theorem 16.10.1](#) (a); the second is  $n_+(\mathbf{A}) + n_0(\mathbf{A}) = 2 + 1 = 3$  by [Corollary 16.10.2](#) (b). The kernel direction may be added to a positive subspace without breaking  $\geq 0$ , but it does break  $> 0$  — which is the whole difference between the two numbers.

### 16.10.3 Inertia under a perturbation of small rank

A congruence leaves the inertia alone and does nothing else: it can move the eigenvalues anywhere compatible with their signs. Chapter 13 §05's warning displayed the pair

$$\mathbf{A} = \text{diag}(1, -1), \quad \mathbf{P}^\top \mathbf{A} \mathbf{P} = \text{diag}(4, -9) \quad (\mathbf{P} = \text{diag}(2, 3)),$$

where  $\lambda_1$  rose from 1 to 4 and  $\lambda_2$  fell from  $-1$  to  $-9$ . There is no interlacing here:  $\lambda_1(\mathbf{A}) \geq \lambda_1(\mathbf{P}^\top \mathbf{A} \mathbf{P})$  already fails. A congruence *is* a perturbation,  $\mathbf{P}^* \mathbf{A} \mathbf{P} = \mathbf{A} + \mathbf{E}$  with  $\mathbf{E} = \mathbf{P}^* \mathbf{A} \mathbf{P} - \mathbf{A}$ , but  $\mathbf{E}$  can have full rank, and then the rank bound of Section 3 says nothing. When the perturbation has small rank, it does.

#### Theorem 16.10.3 — Inertia Moves by at Most the Rank

Let  $\mathbf{A}, \mathbf{E} \in M_n(F)$  be Hermitian with  $\text{rank } \mathbf{E} \leq r$ . Then

$$|n_+(\mathbf{A} + \mathbf{E}) - n_+(\mathbf{A})| \leq r, \quad |n_-(\mathbf{A} + \mathbf{E}) - n_-(\mathbf{A})| \leq r,$$

and  $|n_0(\mathbf{A} + \mathbf{E}) - n_0(\mathbf{A})| \leq r$  as well.

*Idea.* [Corollary 16.3.8](#) says that adding a matrix of rank  $r$  shifts each eigenvalue by at most  $r$  places in the ordered list. A count of positive eigenvalues is a count of places, so it too can move by at most  $r$ .

*Proof.* Put  $s = n_+(\mathbf{A} + \mathbf{E})$ . If  $s \leq r$  then certainly  $n_+(\mathbf{A}) \geq 0 \geq s - r$ . Otherwise  $i := s - r$  satisfies  $1 \leq i$  and  $i + r = s \leq n$ , so [Corollary 16.3.8](#) applies with this  $i$  and gives

$$\lambda_s(\mathbf{A} + \mathbf{E}) = \lambda_{i+r}(\mathbf{A} + \mathbf{E}) \leq \lambda_i(\mathbf{A}) = \lambda_{s-r}(\mathbf{A}).$$

Since  $s = n_+(\mathbf{A} + \mathbf{E})$  we have  $\lambda_s(\mathbf{A} + \mathbf{E}) > 0$ , hence  $\lambda_{s-r}(\mathbf{A}) > 0$ , hence  $n_+(\mathbf{A}) \geq s - r$ . In either case

$$n_+(\mathbf{A} + \mathbf{E}) - n_+(\mathbf{A}) \leq r.$$

For the opposite inequality, write  $\mathbf{A} = (\mathbf{A} + \mathbf{E}) + (-\mathbf{E})$ , where  $-\mathbf{E}$  is Hermitian with  $\text{rank}(-\mathbf{E}) = \text{rank } \mathbf{E} \leq r$ , and apply what was just proved with  $\mathbf{A} + \mathbf{E}$  in the role of  $\mathbf{A}$ :  $n_+(\mathbf{A}) - n_+(\mathbf{A} + \mathbf{E}) \leq r$ . The two together give the first claim.

The second claim is the first applied to the Hermitian pair  $-\mathbf{A}, -\mathbf{E}$ , using  $n_-(\mathbf{M}) = n_+(-\mathbf{M})$ . For the third, adding the first two would only give  $2r$ ; the sharp bound comes from kernels instead. By [Corollary 16.10.2](#) (c),  $n_0(\mathbf{M}) = \dim \text{nul}(\mathbf{M})$  for every Hermitian  $\mathbf{M}$ . A vector killed by both  $\mathbf{A}$  and  $\mathbf{E}$  is killed by  $\mathbf{A} + \mathbf{E}$ , so  $\text{nul}(\mathbf{A}) \cap \text{nul}(\mathbf{E}) \subseteq \text{nul}(\mathbf{A} + \mathbf{E})$ ; and  $\dim \text{nul}(\mathbf{E}) = n - \text{rank } \mathbf{E} \geq n - r$  by [Theorem 3.3.1](#). The sharper form of [Lemma 16.2.1](#) then gives

$$n_0(\mathbf{A} + \mathbf{E}) \geq \dim(\text{nul}(\mathbf{A}) \cap \text{nul}(\mathbf{E})) \geq n_0(\mathbf{A}) + (n - r) - n = n_0(\mathbf{A}) - r.$$

Applied to  $\mathbf{A} = (\mathbf{A} + \mathbf{E}) + (-\mathbf{E})$  the same argument gives  $n_0(\mathbf{A}) \geq n_0(\mathbf{A} + \mathbf{E}) - r$ , and

the two together are the claim. The bound is attained:  $\mathbf{A} = \mathbf{I}_r \oplus 0$  and  $\mathbf{E} = -\mathbf{I}_r \oplus 0$  give  $n_0(\mathbf{A}) = n - r$  and  $n_0(\mathbf{A} + \mathbf{E}) = n$ . This proves the theorem. ■

**Example. ( $\mathbf{A}$  rank-one perturbation, three ways)**

Let  $\mathbf{A} = \text{diag}(3, -1, -2)$  and  $\mathbf{v} = (1, 1, 0)$ , and let  $\mathbf{E}_t = t\mathbf{v}\mathbf{v}^\top$ , which has rank 1 for  $t \neq 0$ . Find  $\text{In}(\mathbf{A} + \mathbf{E}_t)$  for  $t = 2$ ,  $t = \frac{3}{2}$  and  $t = -5$ , and check [Theorem 16.10.3](#).

*Solution.*  $\text{In}(\mathbf{A}) = (1, 2, 0)$ , read off the diagonal. Since  $\mathbf{v}$  has third entry 0,

$$\mathbf{A} + \mathbf{E}_t = \begin{pmatrix} 3+t & t & 0 \\ t & t-1 & 0 \\ 0 & 0 & -2 \end{pmatrix} = \begin{pmatrix} 3+t & t \\ t & t-1 \end{pmatrix} \oplus (-2),$$

and the eigenvalue list of a block diagonal matrix is the two lists together (its characteristic polynomial is the product of the two, by [Theorem 6.8.4](#)). The entry  $-2$  contributes one negative eigenvalue for every  $t$ . The  $2 \times 2$  block has

$$\det = (3+t)(t-1) - t^2 = 2t - 3, \quad \text{tr} = 2t + 2,$$

and since the determinant of a real symmetric  $2 \times 2$  matrix is the product of its two real eigenvalues and the trace is their sum, the eigenvalues both have the sign of the trace when the determinant is positive, have opposite signs when it is negative, and are 0 and  $\text{tr}$  when it is zero.

- $t = 2$ :  $\det = 1 > 0$ ,  $\text{tr} = 6 > 0$ , so the block contributes  $(2, 0, 0)$  and  $\text{In}(\mathbf{A} + \mathbf{E}_2) = (2, 1, 0)$ .
- $t = \frac{3}{2}$ :  $\det = 0$ ,  $\text{tr} = 5$ , so the block contributes  $(1, 0, 1)$  and  $\text{In}(\mathbf{A} + \mathbf{E}_{3/2}) = (1, 1, 1)$ .
- $t = -5$ :  $\det = -13 < 0$ , so the block contributes  $(1, 1, 0)$  and  $\text{In}(\mathbf{A} + \mathbf{E}_{-5}) = (1, 2, 0)$ , unchanged.

Against  $\text{In}(\mathbf{A}) = (1, 2, 0)$ : at  $t = 2$  one negative direction has become positive, so  $n_+$  and  $n_-$  each move by  $1 = r$  and the bound of [Theorem 16.10.3](#) is attained; at  $t = \frac{3}{2}$ , exactly on the way between, the direction passes through 0, so  $n_-$  and  $n_0$  each move by 1. No rank-one perturbation of this  $\mathbf{A}$  can reach  $(3, 0, 0)$ , which would need  $n_+$  to move by 2.

## 16.10.4 Additivity across a Schur complement

Chapter 7 built the Schur complement out of block elimination and proved two things about it:  $\det \mathbf{M} = \det \mathbf{A} \cdot \det(\mathbf{M}/\mathbf{A})$  ([Theorem 7.3.2](#)) and  $\text{rank } \mathbf{M} = k + \text{rank}(\mathbf{M}/\mathbf{A})$  ([Theorem 7.3.3](#)). Chapter 12 then used the same elimination on Hermitian matrices — in §01, observing that it is the block **LDU** factorization written as a congruence, and in §06, to prove [Theorem 12.6.1](#) — and each time read off a statement about positive definiteness only. Haynsworth's theorem is what that congruence is worth in general: the rank does not merely add, the *signs* add.

**Theorem 16.10.4 — Haynsworth's Inertia Additivity Formula**

Let  $1 \leq k \leq n-1$  and let

$$\mathbf{M} = \begin{pmatrix} \mathbf{A} & \mathbf{B} \\ \mathbf{B}^* & \mathbf{D} \end{pmatrix} \in M_n(F)$$

be Hermitian, with  $\mathbf{A} \in M_k(F)$  **invertible**,  $\mathbf{B} \in M_{k \times (n-k)}(F)$  and  $\mathbf{D} \in M_{n-k}(F)$ . Then  $\mathbf{A}$  and the Schur complement  $\mathbf{M}/\mathbf{A} = \mathbf{D} - \mathbf{B}^* \mathbf{A}^{-1} \mathbf{B}$  (Definition 7.3.1) are Hermitian, and

$$\text{In}(\mathbf{M}) = \text{In}(\mathbf{A}) + \text{In}(\mathbf{M}/\mathbf{A}).$$

*Idea.* Block elimination factors  $\mathbf{M}$  as  $\mathbf{L}(\mathbf{A} \oplus \mathbf{M}/\mathbf{A})\mathbf{U}$  with  $\mathbf{L}$  lower and  $\mathbf{U}$  upper unit block triangular (Theorem 7.3.1). For a **Hermitian**  $\mathbf{M}$  with  $\mathbf{A}$  Hermitian, the two outer factors are conjugate transposes of each other, so the factorization is a congruence and not merely a factorization. Theorem 16.10.1 (b) then says the inertia of  $\mathbf{M}$  is the inertia of  $\mathbf{A} \oplus \mathbf{M}/\mathbf{A}$ , and the inertia of a block diagonal matrix is the sum of the inertias of its blocks, because its characteristic polynomial is the product.

*Proof.* Since  $\mathbf{M}^* = \mathbf{M}$ , comparing blocks gives  $\mathbf{A}^* = \mathbf{A}$  and  $\mathbf{D}^* = \mathbf{D}$ . As  $\mathbf{A}$  is Hermitian and invertible,  $(\mathbf{A}^{-1})^* = (\mathbf{A}^*)^{-1} = \mathbf{A}^{-1}$ , so  $\mathbf{A}^{-1}$  is Hermitian and

$$(\mathbf{M}/\mathbf{A})^* = \mathbf{D}^* - \mathbf{B}^*(\mathbf{A}^{-1})^* \mathbf{B} = \mathbf{D} - \mathbf{B}^* \mathbf{A}^{-1} \mathbf{B} = \mathbf{M}/\mathbf{A}.$$

**Step 1: the elimination is a congruence.** By Theorem 7.3.1 (a), applied with  $\mathbf{C} = \mathbf{B}^*$ ,

$$\mathbf{M} = \mathbf{L}(\mathbf{A} \oplus (\mathbf{M}/\mathbf{A}))\mathbf{U}, \quad \mathbf{L} = \begin{pmatrix} \mathbf{I}_k & \mathbf{0} \\ \mathbf{B}^* \mathbf{A}^{-1} & \mathbf{I}_{n-k} \end{pmatrix}, \quad \mathbf{U} = \begin{pmatrix} \mathbf{I}_k & \mathbf{A}^{-1} \mathbf{B} \\ \mathbf{0} & \mathbf{I}_{n-k} \end{pmatrix}.$$

Now  $\mathbf{U}^*$  has blocks  $\mathbf{I}_k, \mathbf{0}$  on top and  $(\mathbf{A}^{-1} \mathbf{B})^* = \mathbf{B}^* \mathbf{A}^{-1}$ ,  $\mathbf{I}_{n-k}$  below, using that  $\mathbf{A}^{-1}$  is Hermitian; that is,  $\mathbf{U}^* = \mathbf{L}$ . So

$$\mathbf{M} = \mathbf{U}^*(\mathbf{A} \oplus (\mathbf{M}/\mathbf{A}))\mathbf{U}.$$

The matrix  $\mathbf{U}$  is invertible, with inverse  $\mathbf{P} := \begin{pmatrix} \mathbf{I}_k & -\mathbf{A}^{-1} \mathbf{B} \\ \mathbf{0} & \mathbf{I}_{n-k} \end{pmatrix}$ , as multiplying out confirms. Multiplying the display by  $\mathbf{P}^*$  on the left and  $\mathbf{P}$  on the right, and using  $\mathbf{P}^* \mathbf{U}^* = (\mathbf{U} \mathbf{P})^* = \mathbf{I}_n$ ,

$$\mathbf{P}^* \mathbf{M} \mathbf{P} = \mathbf{A} \oplus (\mathbf{M}/\mathbf{A}). \quad (16.10.1)$$

**Step 2: apply the law of inertia.**  $\mathbf{P}$  is invertible, so Theorem 16.10.1 (b) gives

$$\text{In}(\mathbf{M}) = \text{In}(\mathbf{P}^* \mathbf{M} \mathbf{P}) = \text{In}(\mathbf{A} \oplus (\mathbf{M}/\mathbf{A})).$$

**Step 3: inertia is additive over a block diagonal sum.** Let  $\mathbf{X} \in M_k(F)$  and  $\mathbf{Y} \in M_{n-k}(F)$  be Hermitian. The matrix  $x\mathbf{I}_n - (\mathbf{X} \oplus \mathbf{Y})$  is block diagonal with blocks  $x\mathbf{I}_k - \mathbf{X}$  and  $x\mathbf{I}_{n-k} - \mathbf{Y}$ , so by Theorem 6.8.4 its determinant is  $\det(x\mathbf{I}_k - \mathbf{X}) \det(x\mathbf{I}_{n-k} - \mathbf{Y})$ ; that is,  $p_{\mathbf{X} \oplus \mathbf{Y}} = p_{\mathbf{X}} p_{\mathbf{Y}}$ . The eigenvalue list of a Hermitian matrix with multiplicity is the list of roots of its characteristic polynomial with multiplicity, so the list for  $\mathbf{X} \oplus \mathbf{Y}$  is the

two lists concatenated. Counting positive, negative and zero entries gives  $\text{In}(\mathbf{X} \oplus \mathbf{Y}) = \text{In}(\mathbf{X}) + \text{In}(\mathbf{Y})$ .

Applying Step 3 with  $\mathbf{X} = \mathbf{A}$  and  $\mathbf{Y} = \mathbf{M}/\mathbf{A}$  to the conclusion of Step 2 proves the theorem. ■

Three remarks. First, [Eq. 16.10.1](#) is the cleanest illustration in the book of what a Schur complement *is*: the block elimination of Chapter 7, performed with the same matrix on both sides, is a congruence, and  $\mathbf{M}/\mathbf{A}$  is simply what the congruence leaves in the lower corner. For Hermitian  $\mathbf{M}$ , Chapter 7's determinant and rank formulas are both shadows of this one identity. Second, taking ranks in [Eq. 16.10.1](#) recovers [Theorem 7.3.3](#) for Hermitian  $\mathbf{M}$ , since  $n_+ + n_- = \text{rank}$ ; Haynsworth refines the rank formula by splitting each rank into its two signs. Third, the theorem needs  $\mathbf{A}$  invertible and nothing else — no definiteness anywhere, which is what separates it from Chapter 12 §06's test.

**Example. (The Chapter 13 example, through a Schur complement)**

Find  $\text{In}(\mathbf{M})$  for

$$\mathbf{M} = \begin{pmatrix} 1 & 1 & 2 \\ 1 & 2 & 3 \\ 2 & 3 & 1 \end{pmatrix},$$

the matrix of [Example \(A signature by symmetric elimination\)](#), using [Theorem 16.10.4](#) with the leading  $2 \times 2$  block.

*Solution.* Take  $\mathbf{A} = \begin{pmatrix} 1 & 1 \\ 1 & 2 \end{pmatrix}$ ,  $\mathbf{B} = (2, 3)^\top$  and  $\mathbf{D} = (1)$ . Then  $\det \mathbf{A} = 1 \neq 0$ , so  $\mathbf{A}$  is invertible with  $\mathbf{A}^{-1} = \begin{pmatrix} 2 & -1 \\ -1 & 1 \end{pmatrix}$ . Its leading principal minors are 1 and 1, both positive, so  $\mathbf{A} \succ 0$  by [Theorem 12.1.5](#) and  $\text{In}(\mathbf{A}) = (2, 0, 0)$ .

For the Schur complement,  $\mathbf{B}^\top \mathbf{A}^{-1} = (2, 3) \begin{pmatrix} 2 & -1 \\ -1 & 1 \end{pmatrix} = (1, 1)$ , so  $\mathbf{B}^\top \mathbf{A}^{-1} \mathbf{B} = (1, 1) \cdot (2, 3) = 5$  and

$$\mathbf{M}/\mathbf{A} = 1 - 5 = -4, \quad \text{In}(\mathbf{M}/\mathbf{A}) = (0, 1, 0).$$

Hence  $\text{In}(\mathbf{M}) = (2, 0, 0) + (0, 1, 0) = (2, 1, 0)$ .

Two checks. The congruence of [Eq. 16.10.1](#) is given by  $\mathbf{A}^{-1} \mathbf{B} = (1, 1)^\top$ , hence

$$\mathbf{P} = \begin{pmatrix} 1 & 0 & -1 \\ 0 & 1 & -1 \\ 0 & 0 & 1 \end{pmatrix}, \quad \mathbf{P}^\top \mathbf{M} \mathbf{P} = \begin{pmatrix} 1 & 1 & 0 \\ 1 & 2 & 0 \\ 0 & 0 & -4 \end{pmatrix},$$

which is  $\mathbf{A} \oplus (-4)$  as the theorem says; and [Example \(A signature by symmetric elimination\)](#) reached  $\text{diag}(1, 1, -4)$  by paired row and column operations, finding the same  $-4$ . The Schur complement of the leading block *is* the last pivot of the symmetric elimination. The eigenvalues were never needed: they are the roots of  $x^3 - 4x^2 - 9x + 4$ , which [Example \(A signature by symmetric elimination\)](#) already noted are irrational.

**Example. (A  $4 \times 4$  inertia in two lines)**



Find  $\text{In}(\mathbf{N})$  for

$$\mathbf{N} = \begin{pmatrix} 1 & 0 & 1 & 2 \\ 0 & 1 & 1 & -1 \\ 1 & 1 & 1 & 0 \\ 2 & -1 & 0 & 3 \end{pmatrix}.$$

*Solution.* The leading  $2 \times 2$  block is  $\mathbf{A} = \mathbf{I}_2$ , which is invertible with  $\text{In}(\mathbf{A}) = (2, 0, 0)$  and  $\mathbf{A}^{-1} = \mathbf{I}_2$ . With

$$\mathbf{B} = \begin{pmatrix} 1 & 2 \\ 1 & -1 \end{pmatrix}, \quad \mathbf{D} = \begin{pmatrix} 1 & 0 \\ 0 & 3 \end{pmatrix},$$

the Schur complement needs no inversion at all:

$$\mathbf{N}/\mathbf{A} = \mathbf{D} - \mathbf{B}^\top \mathbf{B} = \begin{pmatrix} 1 & 0 \\ 0 & 3 \end{pmatrix} - \begin{pmatrix} 2 & 1 \\ 1 & 5 \end{pmatrix} = \begin{pmatrix} -1 & -1 \\ -1 & -2 \end{pmatrix}.$$

This  $2 \times 2$  matrix has determinant  $2 - 1 = 1 > 0$  and trace  $-3 < 0$ , so both its eigenvalues are negative and  $\text{In}(\mathbf{N}/\mathbf{A}) = (0, 2, 0)$ . By [Theorem 16.10.4](#),

$$\text{In}(\mathbf{N}) = (2, 0, 0) + (0, 2, 0) = (2, 2, 0).$$

In particular  $\mathbf{N}$  is invertible and indefinite. The characteristic polynomial of  $\mathbf{N}$  is  $x^4 - 6x^3 + 5x^2 + 8x + 1$ , which has no rational root; the four eigenvalues were never computed, and the signature came from one  $2 \times 2$  determinant and one trace.

**Warning.**  $\mathbf{A}$  must be invertible, and the off-diagonal block is not a spectator. Take  $\mathbf{M} = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}$ , with  $k = 1$ . Neither diagonal block is invertible, so no Schur complement exists for this partition and [Theorem 16.10.4](#) does not apply — and its conclusion would be false if one tried to read the blocks directly:  $\text{In}(\mathbf{M}) = (1, 1, 0)$ , since the eigenvalues are  $\pm 1$ , while the two diagonal blocks contribute  $(0, 0, 1) + (0, 0, 1) = (0, 0, 2)$ . Even when  $\mathbf{A}$  is invertible, the diagonal blocks alone decide nothing: for  $\mathbf{M}_b = \begin{pmatrix} 1 & b \\ b & 1 \end{pmatrix}$  with  $b$  real, the Schur complement is  $1 - b^2$ , so  $\text{In}(\mathbf{M}_b)$  is  $(2, 0, 0)$ ,  $(1, 0, 1)$  or  $(1, 1, 0)$  according as  $|b| < 1$ ,  $= 1$  or  $> 1$ . All the information is in the coupling.

The positive definite case of Haynsworth is a theorem Chapter 12 proved by hand, and it is now one line.

### Corollary 16.10.5 — Block Definiteness, a Second Proof

Let  $1 \leq k \leq n - 1$  and let  $\mathbf{M} = \begin{pmatrix} \mathbf{A} & \mathbf{B} \\ \mathbf{B}^* & \mathbf{D} \end{pmatrix} \in M_n(F)$  be Hermitian with  $\mathbf{A} \in M_k(F)$  and  $\mathbf{A} \succ 0$ . Then

$$\mathbf{M} \succ 0 \iff \mathbf{M}/\mathbf{A} \succ 0, \quad \mathbf{M} \succeq 0 \iff \mathbf{M}/\mathbf{A} \succeq 0.$$

*Proof.*  $\mathbf{A} \succ 0$  is invertible ([Theorem 12.1.5](#)) with  $\text{In}(\mathbf{A}) = (k, 0, 0)$ , so [Theorem 16.10.4](#) gives

$$\text{In}(\mathbf{M}) = (k, 0, 0) + \text{In}(\mathbf{M}/\mathbf{A}).$$

Comparing entries,  $n_+(\mathbf{M}) = k + n_+(\mathbf{M}/\mathbf{A})$  and  $n_-(\mathbf{M}) = n_-(\mathbf{M}/\mathbf{A})$ , while  $\mathbf{M}/\mathbf{A} \in M_{n-k}(F)$ .

Now  $\mathbf{M} \succ 0$  means  $n_+(\mathbf{M}) = n$  ([Theorem 12.1.5](#)), that is  $n_+(\mathbf{M}/\mathbf{A}) = n - k$ , that is



$\mathbf{M}/\mathbf{A} \succ 0$ . And  $\mathbf{M} \succeq 0$  means  $n_-(\mathbf{M}) = 0$  (Theorem 12.1.4), that is  $n_-(\mathbf{M}/\mathbf{A}) = 0$ , that is  $\mathbf{M}/\mathbf{A} \succeq 0$ . This proves the corollary. ■

This is Theorem 12.6.1 (b). Chapter 12 §06 proved it with the very same congruence, and then read positivity directly off the split quadratic form  $\mathbf{y}_1^* \mathbf{A} \mathbf{y}_1 + \mathbf{y}_2^* (\mathbf{M}/\mathbf{A}) \mathbf{y}_2$ , with no eigenvalue in sight. The two proofs share their first step and differ in the second: Chapter 12's is shorter and needs nothing beyond Proposition 12.1.2, while the one above reads the verdict off a count, and so shows the definite case as the extreme instance of an equality that holds for every Hermitian  $\mathbf{M}$  with  $\mathbf{A}$  invertible, definite or not.

## 16.10.5 Exercises

### 16.10.5.1 A. Check your understanding

#### Exercise A1.

- (a) Define  $\text{In}(\mathbf{A})$  for a Hermitian  $\mathbf{A}$ , and say what  $n_+ + n_-$  equals.
- (b) State the subspace description of  $n_+(\mathbf{A})$  and name the subspace that attains it.
- (c) Determine whether the following is correct, and justify: “ $n_0(\mathbf{A})$  is the largest dimension of a subspace on which  $\langle \mathbf{A}\mathbf{x}, \mathbf{x} \rangle$  vanishes identically.”
- (d) State Theorem 16.10.4, with all hypotheses.
- (e) A congruence preserves the inertia of a Hermitian matrix. Does it preserve the eigenvalues, the trace, or the rank? Justify each answer.

*Solution.*

- (a)  $\text{In}(\mathbf{A}) = (n_+, n_-, n_0)$ , the numbers of positive, negative and zero eigenvalues of  $\mathbf{A}$  counted with multiplicity (Definition 16.10.1). Their sum is  $n$ , and  $n_+ + n_- = \text{rank } \mathbf{A}$ , since  $n_0 = \dim \text{nul}(\mathbf{A})$  by Corollary 16.10.2 (c).
- (b)  $n_+(\mathbf{A})$  is the largest dimension of a subspace  $W$  with  $\langle \mathbf{A}\mathbf{x}, \mathbf{x} \rangle > 0$  for every non-zero  $\mathbf{x} \in W$  (Theorem 16.10.1 (a)), attained at the span of orthonormal eigenvectors for the positive eigenvalues, that is at  $\text{span}(\mathbf{q}_1, \dots, \mathbf{q}_{n_+})$  in an eigenbasis ordered decreasingly.
- (c) Incorrect. For  $\mathbf{A} = \text{diag}(1, -1)$  we have  $n_0(\mathbf{A}) = 0$ , yet  $\langle \mathbf{A}\mathbf{x}, \mathbf{x} \rangle = x_1^2 - x_2^2$  vanishes identically on the line spanned by  $(1, 1)$ . The correct description is  $n_0(\mathbf{A}) = \dim \text{nul}(\mathbf{A})$ .
- (d) If  $\mathbf{M} = \begin{pmatrix} \mathbf{A} & \mathbf{B} \\ \mathbf{B}^* & \mathbf{D} \end{pmatrix} \in M_n(F)$  is Hermitian with  $\mathbf{A} \in M_k(F)$  invertible and  $1 \leq k \leq n-1$ , then  $\text{In}(\mathbf{M}) = \text{In}(\mathbf{A}) + \text{In}(\mathbf{M}/\mathbf{A})$ , where  $\mathbf{M}/\mathbf{A} = \mathbf{D} - \mathbf{B}^* \mathbf{A}^{-1} \mathbf{B}$ .
- (e) Rank: yes, since  $\text{rank} = n_+ + n_-$  is determined by the inertia (and independently, congruence by an invertible matrix preserves rank by Theorem 2.5.9). Eigenvalues: no —  $\text{diag}(1, -1)$  and  $\text{diag}(4, -9)$  are congruent by  $\mathbf{P} = \text{diag}(2, 3)$ . Trace: no, by the same pair, whose traces are 0 and  $-5$ .

### 16.10.5.2 B. Practice

**Exercise B1: Two inertias by Schur complement.**

Find the inertia of each of the following, using [Theorem 16.10.4](#) with the leading  $2 \times 2$  block, and compute no eigenvalues.

$$(a) \mathbf{M} = \begin{pmatrix} 2 & 1 & 1 \\ 1 & 1 & 0 \\ 1 & 0 & -1 \end{pmatrix}.$$

$$(b) \mathbf{N} = \begin{pmatrix} 1 & 0 & 1 & 1 \\ 0 & -1 & 1 & 0 \\ 1 & 1 & 2 & 0 \\ 1 & 0 & 0 & 1 \end{pmatrix}.$$

*Solution.*

- (a)  $\mathbf{A} = \begin{pmatrix} 2 & 1 \\ 1 & 1 \end{pmatrix}$  has leading principal minors 2 and 1, both positive, so  $\mathbf{A} \succ 0$  ([Theorem 12.1.5](#)) and  $\text{In}(\mathbf{A}) = (2, 0, 0)$ ; also  $\mathbf{A}^{-1} = \begin{pmatrix} 1 & -1 \\ -1 & 2 \end{pmatrix}$ . With  $\mathbf{B} = (1, 0)^\top$  and  $\mathbf{D} = (-1)$ ,

$$\mathbf{B}^\top \mathbf{A}^{-1} \mathbf{B} = (1, 0) \begin{pmatrix} 1 & -1 \\ -1 & 2 \end{pmatrix} \begin{pmatrix} 1 \\ 0 \end{pmatrix} = 1,$$

so  $\mathbf{M}/\mathbf{A} = -1 - 1 = -2$  and  $\text{In}(\mathbf{M}/\mathbf{A}) = (0, 1, 0)$ . Hence  $\text{In}(\mathbf{M}) = (2, 1, 0)$ . (Consistency check:  $\det \mathbf{M} = \det \mathbf{A} \cdot (\mathbf{M}/\mathbf{A}) = 1 \cdot (-2) = -2$  by [Theorem 7.3.2](#), and a  $3 \times 3$  matrix with inertia  $(2, 1, 0)$  has negative determinant.)

- (b)  $\mathbf{A} = \text{diag}(1, -1)$  is invertible with  $\mathbf{A}^{-1} = \mathbf{A}$  and  $\text{In}(\mathbf{A}) = (1, 1, 0)$ . With

$$\mathbf{B} = \begin{pmatrix} 1 & 1 \\ 1 & 0 \end{pmatrix}, \quad \mathbf{D} = \begin{pmatrix} 2 & 0 \\ 0 & 1 \end{pmatrix},$$

we get  $\mathbf{A}^{-1} \mathbf{B} = \begin{pmatrix} 1 & 1 \\ -1 & 0 \end{pmatrix}$  and

$$\mathbf{B}^\top \mathbf{A}^{-1} \mathbf{B} = \begin{pmatrix} 1 & 1 \\ 1 & 0 \end{pmatrix} \begin{pmatrix} 1 & 1 \\ -1 & 0 \end{pmatrix} = \begin{pmatrix} 0 & 1 \\ 1 & 1 \end{pmatrix},$$

so

$$\mathbf{N}/\mathbf{A} = \begin{pmatrix} 2 & 0 \\ 0 & 1 \end{pmatrix} - \begin{pmatrix} 0 & 1 \\ 1 & 1 \end{pmatrix} = \begin{pmatrix} 2 & -1 \\ -1 & 0 \end{pmatrix},$$

whose determinant is  $-1 < 0$ , so its two eigenvalues have opposite signs and  $\text{In}(\mathbf{N}/\mathbf{A}) = (1, 1, 0)$ . Hence  $\text{In}(\mathbf{N}) = (1, 1, 0) + (1, 1, 0) = (2, 2, 0)$ .

**Exercise B2: How much can a rank-two perturbation do?.**

Let  $\mathbf{A} \in M_6(\mathbb{R})$  be symmetric with  $\text{In}(\mathbf{A}) = (4, 1, 1)$ , and let  $\mathbf{E}$  be symmetric with  $\text{rank } \mathbf{E} = 2$ .

- Which values of  $n_+(\mathbf{A} + \mathbf{E})$  does [Theorem 16.10.3](#) allow?
- Show that the two extreme values are both achieved, for suitable  $\mathbf{A}$  and  $\mathbf{E}$ .
- Can  $\mathbf{A} + \mathbf{E}$  be negative definite? Justify.

*Solution.*

- (a) By [Theorem 16.10.3](#) with  $r = 2$ ,  $|n_+(\mathbf{A} + \mathbf{E}) - 4| \leq 2$ , and  $n_+ \leq 6$  always, so  $n_+(\mathbf{A} + \mathbf{E}) \in \{2, 3, 4, 5, 6\}$ .
- (b) Take  $\mathbf{A} = \text{diag}(1, 1, 1, 1, -1, 0)$ , which has inertia  $(4, 1, 1)$ . With  $\mathbf{E} = \text{diag}(0, 0, 0, 0, 2, 1)$ , of rank 2, we get  $\mathbf{A} + \mathbf{E} = \mathbf{I}_6$ , so  $n_+ = 6$ . With  $\mathbf{E} = \text{diag}(-2, -2, 0, 0, 0, 0)$ , of rank 2, we get  $\mathbf{A} + \mathbf{E} = \text{diag}(-1, -1, 1, 1, -1, 0)$ , so  $n_+ = 2$ .
- (c) No. Negative definiteness would mean  $n_+(\mathbf{A} + \mathbf{E}) = 0$ , and  $|0 - 4| = 4 > 2$  contradicts [Theorem 16.10.3](#).

### Exercise B3: A bordered matrix.

Let  $\mathbf{D} \in M_n(\mathbb{R})$  be symmetric and invertible, let  $\mathbf{b} \in \mathbb{R}^n$  and  $c \in \mathbb{R}$ , and put  $\mathbf{M} = \begin{pmatrix} \mathbf{D} & \mathbf{b} \\ \mathbf{b}^\top & c \end{pmatrix}$ .

- (a) Express  $\text{In}(\mathbf{M})$  in terms of  $\text{In}(\mathbf{D})$  and the scalar  $s = c - \mathbf{b}^\top \mathbf{D}^{-1} \mathbf{b}$ .
- (b) Take  $\mathbf{D} = \text{diag}(1, -1)$  and  $\mathbf{b} = (1, 1)$ . Compute  $\text{In}(\mathbf{M})$  for  $c = 0$ ,  $c = 1$  and  $c = -1$ .
- (c) For which  $c$  is  $\mathbf{M}$  singular?

*Solution.*

- (a) By [Theorem 16.10.4](#) with  $\mathbf{A} = \mathbf{D}$ , and  $\mathbf{M}/\mathbf{D} = s$  a  $1 \times 1$  matrix,

$$\text{In}(\mathbf{M}) = \text{In}(\mathbf{D}) + \begin{cases} (1, 0, 0) & s > 0, \\ (0, 1, 0) & s < 0, \\ (0, 0, 1) & s = 0. \end{cases}$$

- (b) Here  $\mathbf{D}^{-1} = \text{diag}(1, -1) = \mathbf{D}$ , so  $\mathbf{b}^\top \mathbf{D}^{-1} \mathbf{b} = 1 - 1 = 0$  and  $s = c$ . Also  $\text{In}(\mathbf{D}) = (1, 1, 0)$ . Hence  $\text{In}(\mathbf{M})$  is  $(1, 1, 1)$  for  $c = 0$ ,  $(2, 1, 0)$  for  $c = 1$ , and  $(1, 2, 0)$  for  $c = -1$ . The case  $c = 0$  recovers [Exercise \(C3: Inertia of a block matrix\)](#) (c), which found the same triple by exhibiting the congruence directly.
- (c)  $\mathbf{M}$  is singular exactly when  $n_0(\mathbf{M}) \geq 1$ , which by (a) happens exactly when  $s = 0$ , since  $\mathbf{D}$  is invertible and so contributes  $n_0(\mathbf{D}) = 0$ . Here that is  $c = 0$ .

### 16.10.5.3 C. Going deeper

#### Exercise C1: How large can a subspace be on which the form vanishes?.

Let  $\mathbf{A} \in M_n(F)$  be Hermitian with  $\text{In}(\mathbf{A}) = (n_+, n_-, n_0)$ . Call a subspace  $W \leq F^n$  **totally isotropic** for  $\mathbf{A}$  if  $\langle \mathbf{A}\mathbf{x}, \mathbf{x} \rangle = 0$  for every  $\mathbf{x} \in W$ ; for real symmetric  $\mathbf{A}$  this is the notion of Chapter 13 §07, in the form that section derives from polarization.

- (a) Prove that every totally isotropic  $W$  satisfies  $\dim W \leq n_0 + \min(n_+, n_-)$ .
- (b) Construct a totally isotropic subspace of dimension exactly  $n_0 + \min(n_+, n_-)$ .
- (c) Deduce that the largest dimension of a totally isotropic subspace equals  $n_0$  if and only if  $\mathbf{A}$  is positive semidefinite or negative semidefinite.

*Hint: for (a), a totally isotropic subspace satisfies both inequalities of Corollary 16.10.2 (b). For (b), pair up a positive eigenvector with a negative one.*

*Solution.*

- (a) A totally isotropic  $W$  satisfies  $\langle \mathbf{Ax}, \mathbf{x} \rangle \geq 0$  throughout and also  $\langle \mathbf{Ax}, \mathbf{x} \rangle \leq 0$  throughout. By Corollary 16.10.2 (b), the first gives  $\dim W \leq n_+ + n_0$  and the second  $\dim W \leq n_- + n_0$ . Hence  $\dim W \leq n_0 + \min(n_+, n_-)$ .
- (b) Fix an orthonormal eigenbasis  $\mathbf{q}_1, \dots, \mathbf{q}_n$  with  $\mathbf{A}\mathbf{q}_i = \lambda_i \mathbf{q}_i$  and  $\lambda_1 \geq \dots \geq \lambda_n$  (Corollary 11.4.3, or Corollary 11.5.2 when  $F = \mathbb{R}$ ), so that  $\lambda_i > 0$  for  $i \leq n_+$ ,  $\lambda_i = 0$  for  $n_+ < i \leq n_+ + n_0$ , and  $\lambda_i < 0$  for  $i > n_+ + n_0$ . Put  $t = \min(n_+, n_-)$  and, for  $1 \leq j \leq t$ , let  $j' = n_+ + 1 - j$ . Then  $j \leq n_+$  and  $j' \geq n_+ + 1 - n_- > n_+ + n_0$ , so  $\lambda_j > 0 > \lambda_{j'}$  and the indices  $1, \dots, t, 1', \dots, t'$  are  $2t$  distinct indices, none of them in the zero block. Put

$$\mathbf{x}_j = \frac{\mathbf{q}_j}{\sqrt{\lambda_j}} + \frac{\mathbf{q}_{j'}}{\sqrt{-\lambda_{j'}}} \quad (1 \leq j \leq t),$$

and let  $W$  be spanned by  $\mathbf{x}_1, \dots, \mathbf{x}_t$  together with the  $n_0$  vectors  $\mathbf{q}_i$  with  $\lambda_i = 0$ . These  $t + n_0$  vectors are independent, since they are non-zero combinations of pairwise disjoint sets of vectors from a basis.

For  $\mathbf{x} \in W$  write  $\mathbf{x} = \sum_j c_j \mathbf{x}_j + \mathbf{z}$  with  $\mathbf{Az} = \mathbf{0}$ . Then  $\mathbf{Ax} = \sum_j c_j \mathbf{Ax}_j$  with  $\mathbf{Ax}_j = \sqrt{\lambda_j} \mathbf{q}_j - \sqrt{-\lambda_{j'}} \mathbf{q}_{j'}$ , and

$$\langle \mathbf{Ax}, \mathbf{x} \rangle = \mathbf{x}^* \mathbf{Ax} = \sum_{j,k} \overline{c_k} c_j \mathbf{x}_k^* \mathbf{Ax}_j + \sum_j c_j \mathbf{z}^* \mathbf{Ax}_j.$$

Each  $\mathbf{z}^* \mathbf{Ax}_j = 0$ , because  $\mathbf{z}$  lies in the span of eigenvectors orthogonal to  $\mathbf{q}_j$  and  $\mathbf{q}_{j'}$ . For  $k \neq j$ ,  $\mathbf{x}_k^* \mathbf{Ax}_j = 0$ , because  $\mathbf{x}_k$  and  $\mathbf{Ax}_j$  are combinations of disjoint sets of orthonormal vectors. And

$$\mathbf{x}_j^* \mathbf{Ax}_j = \frac{\sqrt{\lambda_j}}{\sqrt{\lambda_j}} - \frac{\sqrt{-\lambda_{j'}}}{\sqrt{-\lambda_{j'}}} = 1 - 1 = 0.$$

Hence  $\langle \mathbf{Ax}, \mathbf{x} \rangle = 0$  for every  $\mathbf{x} \in W$ , and  $W$  is totally isotropic of dimension  $n_0 + \min(n_+, n_-)$ .

- (c) By (a) and (b) the largest totally isotropic dimension is  $n_0 + \min(n_+, n_-)$ , which equals  $n_0$  exactly when  $\min(n_+, n_-) = 0$ , that is when  $n_- = 0$  or  $n_+ = 0$ , that is when  $\mathbf{A} \succeq 0$  or  $-\mathbf{A} \succeq 0$  (Theorem 12.1.4).

### Exercise C2: The inertia of a Hermitian dilation.

Let  $\mathbf{A} \in M_{m \times n}(F)$  have rank  $r$ , and let  $\mathcal{H}(\mathbf{A}) = \begin{pmatrix} \mathbf{0} & \mathbf{A} \\ \mathbf{A}^* & \mathbf{0} \end{pmatrix} \in M_{m+n}(F)$  be the Hermitian dilation of Proposition 16.9.3.

- (a) Prove that  $\text{In}(\mathcal{H}(\mathbf{A})) = (r, r, m + n - 2r)$ .
- (b) Deduce a formula for  $\text{rank } \mathbf{A}$  in terms of the inertia of  $\mathcal{H}(\mathbf{A})$ , and check that it gives  $\text{rank } \mathbf{0} = 0$ .
- (c) Verify (a) for  $\mathbf{A} = \begin{pmatrix} 1 & 0 & 2 \\ 0 & 1 & 1 \end{pmatrix}$ .

*Solution.*

- (a) By [Proposition 16.9.3](#) the eigenvalue list of  $\mathcal{H}(\mathbf{A})$  with multiplicity consists of  $\pm\sigma_i(\mathbf{A})$  for  $1 \leq i \leq p = \min(m, n)$ , together with  $m + n - 2p$  zeros. Exactly  $r$  of the  $\sigma_i(\mathbf{A})$  are non-zero ([Theorem 12.8.3](#)), so the positive entries are the  $r$  numbers  $\sigma_1, \dots, \sigma_r$  and the negative entries are their negatives, giving  $n_+ = n_- = r$ . The rest are zero:  $(m + n) - 2r$  of them. Hence  $\text{In}(\mathcal{H}(\mathbf{A})) = (r, r, m + n - 2r)$ .
- (b)  $\text{rank } \mathbf{A} = n_+(\mathcal{H}(\mathbf{A})) = \frac{1}{2}(\text{rank } \mathcal{H}(\mathbf{A}))$ , using  $n_+ + n_- = \text{rank}$ . For  $\mathbf{A} = \mathbf{0}$  the dilation is the zero matrix of size  $m + n$ , whose inertia is  $(0, 0, m + n)$ , so the formula returns 0.
- (c) Here  $m = 2$ ,  $n = 3$  and  $\text{rank } \mathbf{A} = 2$ , since the first two columns are  $\mathbf{e}_1, \mathbf{e}_2$ . Also

$$\mathbf{A}\mathbf{A}^\top = \begin{pmatrix} 5 & 2 \\ 2 & 2 \end{pmatrix},$$

of trace 7 and determinant 6, hence eigenvalues 6 and 1; so  $\sigma_1(\mathbf{A}) = \sqrt{6}$  and  $\sigma_2(\mathbf{A}) = 1$ . By [Proposition 16.9.3](#) the eigenvalues of the  $5 \times 5$  matrix  $\mathcal{H}(\mathbf{A})$  are  $\pm\sqrt{6}, \pm 1$  and one 0, so  $\text{In}(\mathcal{H}(\mathbf{A})) = (2, 2, 1)$ , which is  $(r, r, m + n - 2r) = (2, 2, 5 - 4)$ .

### Exercise C3: Inertia from the leading principal minors.

Let  $\mathbf{M} \in M_n(F)$  be Hermitian and suppose that every leading principal submatrix  $\mathbf{M}_k \in M_k(F)$  (rows and columns  $1, \dots, k$ ) is **invertible**, for  $k = 1, \dots, n$ . Write  $d_k = \det \mathbf{M}_k$  and  $d_0 = 1$ .

- (a) Prove that for  $2 \leq k \leq n$  the Schur complement  $\mathbf{M}_k/\mathbf{M}_{k-1}$  is the scalar  $d_k/d_{k-1}$ .
- (b) Deduce that  $\text{In}(\mathbf{M}) = (n_+, n_-, 0)$ , where  $n_-$  is the number of indices  $k \in \{1, \dots, n\}$  with  $d_k/d_{k-1} < 0$  — that is, the number of **sign changes** in the sequence  $d_0, d_1, \dots, d_n$ .
- (c) Apply this to the matrix  $\mathbf{M}$  of [Example \(The Chapter 13 example, through a Schur complement\)](#) and compare.

*Hint: for (b), induct on  $k$  using [Theorem 16.10.4](#).*

*Solution.*

- (a) Write  $\mathbf{M}_k = \begin{pmatrix} \mathbf{M}_{k-1} & \mathbf{b} \\ \mathbf{b}^* & c \end{pmatrix}$ . Since  $\mathbf{M}_{k-1}$  is invertible,  $\mathbf{M}_k/\mathbf{M}_{k-1}$  is defined and is a  $1 \times 1$  matrix, and [Theorem 7.3.2](#) (a) gives  $d_k = \det \mathbf{M}_k = \det \mathbf{M}_{k-1} \cdot (\mathbf{M}_k/\mathbf{M}_{k-1}) = d_{k-1}(\mathbf{M}_k/\mathbf{M}_{k-1})$ . As  $d_{k-1} \neq 0$ , dividing gives  $\mathbf{M}_k/\mathbf{M}_{k-1} = d_k/d_{k-1}$ .
- (b) Induct on  $k$ . For  $k = 1$ ,  $\mathbf{M}_1 = (d_1)$  is a non-zero real scalar (real because  $\mathbf{M}$  is Hermitian), and  $\text{In}(\mathbf{M}_1)$  is  $(1, 0, 0)$  or  $(0, 1, 0)$  according to the sign of  $d_1/d_0 = d_1$ . For the step,  $\mathbf{M}_{k-1}$  is invertible, so [Theorem 16.10.4](#) applies to  $\mathbf{M}_k$  with  $\mathbf{A} = \mathbf{M}_{k-1}$ :

$$\text{In}(\mathbf{M}_k) = \text{In}(\mathbf{M}_{k-1}) + \text{In}(d_k/d_{k-1}),$$

and the last term is  $(1, 0, 0)$  or  $(0, 1, 0)$  according to the sign of the ratio, which is non-zero. Summing from  $k = 1$  to  $n$ ,

$$\text{In}(\mathbf{M}) = \text{In}(\mathbf{M}_n) = (\#\{k : d_k/d_{k-1} > 0\}, \#\{k : d_k/d_{k-1} < 0\}, 0),$$

and  $d_k/d_{k-1} < 0$  says exactly that  $d_{k-1}$  and  $d_k$  have opposite signs. (Taking

$n_- = 0$  here recovers the leading-minor criterion of [Theorem 12.1.5](#): all  $d_k > 0$  if and only if  $\mathbf{M} \succ 0$ .)

- (c) For  $\mathbf{M} = \begin{pmatrix} 1 & 1 & 2 \\ 1 & 2 & 3 \\ 2 & 3 & 1 \end{pmatrix}$  the leading minors are  $d_1 = 1$ ,  $d_2 = 2 - 1 = 1$  and  $d_3 = -4$ , all non-zero, so (b) applies. The sequence  $d_0, d_1, d_2, d_3 = 1, 1, 1, -4$  has exactly one sign change, at the last step. Hence  $n_- = 1$ ,  $n_+ = 2$  and  $\text{In}(\mathbf{M}) = (2, 1, 0)$ , agreeing with [Example \(The Chapter 13 example, through a Schur complement\)](#) and with [Example \(A signature by symmetric elimination\)](#).

## 16.11 Which Functions Respect the Order

Chapter 12 §05 ended on two facts that sit awkwardly together. If  $\mathbf{A} \succeq \mathbf{B} \succeq 0$ , then  $\mathbf{A}^{1/2} \succeq \mathbf{B}^{1/2}$  ([Proposition 12.5.4](#)); but  $\mathbf{A}^2 \succeq \mathbf{B}^2$  can fail ([Example \(Squaring does not preserve the Loewner order\)](#)). For numbers, both  $t \mapsto t^{1/2}$  and  $t \mapsto t^2$  are increasing on  $[0, \infty)$ , and both preserve inequalities. This section names the property that separates them and proves it for every function the book can reach. It then compares that property with what §02's min-max theorem already gives for *every* increasing function, and states without proof the classification that settles the question. Since this is the last section of the chapter, it ends by summing up the chapter's moves and listing the promises the chapter has paid.

**Throughout,  $F = \mathbb{R}$  or  $F = \mathbb{C}$ , and every matrix whose eigenvalues are indexed is Hermitian**, with eigenvalues  $\lambda_1(\mathbf{A}) \geq \dots \geq \lambda_n(\mathbf{A})$  indexed decreasingly and repeated according to multiplicity. The letter  $I$  denotes an **interval** of real numbers (open, closed or half-open, bounded or not). It is plain italic, and it is not the bold identity matrix  $\mathbf{I}$ . A function  $f: I \rightarrow \mathbb{R}$  is **increasing** when  $s \leq t$  implies  $f(s) \leq f(t)$ ; strictness is not required. Every result below holds for either choice of  $F$ , with the same proof. No analysis is used anywhere in this section except in the last part, which states a theorem without proving it.

### 16.11.1 Functions of a Hermitian matrix

Chapter 11 §08 already applies a function to a matrix, and nothing needs rebuilding. Let  $\mathbf{A} \in M_n(F)$  be Hermitian. The operator  $\mathbf{x} \mapsto \mathbf{A}\mathbf{x}$  on  $F^n$  has an orthonormal basis of eigenvectors ([Corollary 11.4.3](#), [Corollary 11.5.2](#)), so it has a spectral resolution

$$\mathbf{A} = \mu_1 \mathbf{P}_1 + \dots + \mu_k \mathbf{P}_k$$

with  $\mu_1, \dots, \mu_k$  the **distinct** eigenvalues and  $\mathbf{P}_j$  the orthogonal projection onto the eigenspace of  $\mu_j$  ([Definition 11.8.1](#), [Theorem 11.8.1](#)). For any function  $f$  defined on  $\text{spec}(\mathbf{A})$ , [Definition 11.8.2](#) sets

$$f(\mathbf{A}) := f(\mu_1) \mathbf{P}_1 + \dots + f(\mu_k) \mathbf{P}_k.$$

In this section  $f$  is always defined on an interval  $I$  containing  $\text{spec}(\mathbf{A})$ , and  $f(\mathbf{A})$  is formed from the restriction of  $f$  to the spectrum: the values of  $f$  off the spectrum play no part. Four facts from [Theorem 11.8.3](#) are used constantly.

- **Polynomials give the expected answer** (part (a)):  $t \mapsto \alpha + \beta t$  gives  $\alpha \mathbf{I} + \beta \mathbf{A}$ , and  $t \mapsto t^2$  gives  $\mathbf{A}^2$ .
- **Sums and products go through** (part (b)):  $(f + g)(\mathbf{A}) = f(\mathbf{A}) + g(\mathbf{A})$ ,  $(cf)(\mathbf{A}) = c f(\mathbf{A})$  and  $(fg)(\mathbf{A}) = f(\mathbf{A})g(\mathbf{A})$ .

- **Real-valued  $f$  gives a Hermitian  $f(\mathbf{A})$**  (part (c)). This is what lets  $f(\mathbf{A})$  be compared with anything in the Loewner order.
- **The eigenvectors do not change** (part (d)):  $f(\mathbf{A})$  has the same orthonormal eigenbasis as  $\mathbf{A}$ .

Two instances have names already. If  $0 \notin \text{spec}(\mathbf{A})$ , then  $t \mapsto 1/t$  gives  $\mathbf{A}^{-1}$ , as Chapter 11 §08 records right after [Definition 11.8.2](#). If  $\mathbf{A} \succeq 0$ , then  $t \mapsto t^{1/2}$  gives the positive square root  $\mathbf{A}^{1/2}$  of [Theorem 12.2.1](#), which Chapter 12 §02 constructed in exactly this way.

## 16.11.2 Operator monotone and operator convex functions

Chapter 12 §05 introduced the Loewner order  $\mathbf{A} \succeq \mathbf{B}$ , meaning  $\mathbf{A} - \mathbf{B} \succeq 0$  ([Definition 12.5.1](#)). With the functional calculus available, the natural question is which functions  $f$  carry  $\mathbf{A} \succeq \mathbf{B}$  to  $f(\mathbf{A}) \succeq f(\mathbf{B})$ .

The obvious guess is “the increasing ones”, since that is the answer for numbers. It is wrong. The function  $t \mapsto t^2$  is increasing on  $[0, \infty)$ , and [Example \(Squaring does not preserve the Loewner order\)](#) gives Hermitian  $2 \times 2$  matrices  $\mathbf{A} \succeq \mathbf{B} \succeq 0$  with  $\mathbf{A}^2 \not\succeq \mathbf{B}^2$ . So the property we want is stronger than being increasing, and it needs its own name.

*A function is operator monotone when applying it to Hermitian matrices never breaks the Loewner order, at every size at once.*

### Definition 16.11.1 — Operator Monotone and Operator Convex Functions

Let  $I \subseteq \mathbb{R}$  be an interval and let  $f: I \rightarrow \mathbb{R}$ .

- (a)  $f$  is **operator monotone on  $I$**  if, for every size  $n \geq 1$  and every pair of Hermitian matrices  $\mathbf{A}, \mathbf{B} \in M_n(F)$  whose spectra lie in  $I$ ,

$$\mathbf{A} \succeq \mathbf{B} \implies f(\mathbf{A}) \succeq f(\mathbf{B}).$$

- (b)  $f$  is **operator convex on  $I$**  if, for every size  $n \geq 1$ , every pair of Hermitian matrices  $\mathbf{A}, \mathbf{B} \in M_n(F)$  whose spectra lie in  $I$ , and every  $\theta \in [0, 1]$ ,

$$f(\theta \mathbf{A} + (1 - \theta) \mathbf{B}) \preceq \theta f(\mathbf{A}) + (1 - \theta) f(\mathbf{B}).$$

It is **operator concave on  $I$**  if  $-f$  is operator convex on  $I$ .

Here  $f(\mathbf{A})$  is the matrix of [Definition 11.8.2](#).

In words: the first condition takes any two Hermitian matrices of the same size with spectra inside the domain of  $f$ . If the first dominates the second, applying  $f$  must preserve that. The second condition is the matrix version of “the chord lies above the graph”. Both are infinitely many conditions, one for each size  $n$ . A function that passes the test for  $2 \times 2$  matrices has not yet passed it for  $3 \times 3$  ones. The clause “for every  $n$ ” is why the notion is hard, and also why it is interesting.

**Everything in the definition makes sense.** Since  $\text{spec}(\mathbf{A}) \subseteq I$ , the matrix  $f(\mathbf{A})$  is defined, and it is Hermitian because  $f$  is real-valued (Theorem 11.8.3 (c)). So  $f(\mathbf{A}) \succeq f(\mathbf{B})$  is a statement about two Hermitian matrices, as Definition 12.5.1 requires. In (b) one more thing needs checking: the spectrum of  $\mathbf{C} = \theta\mathbf{A} + (1 - \theta)\mathbf{B}$  must lie in  $I$ , or  $f(\mathbf{C})$  would not be defined. The matrix  $\mathbf{C}$  is Hermitian. Let  $\mathbf{x}$  be a unit eigenvector of  $\mathbf{C}$  for the eigenvalue  $\mu$ . Then

$$\mu = \langle \mathbf{C}\mathbf{x}, \mathbf{x} \rangle = \theta \langle \mathbf{A}\mathbf{x}, \mathbf{x} \rangle + (1 - \theta) \langle \mathbf{B}\mathbf{x}, \mathbf{x} \rangle.$$

By Lemma 12.5.2,  $\langle \mathbf{A}\mathbf{x}, \mathbf{x} \rangle$  lies in  $[\lambda_n(\mathbf{A}), \lambda_1(\mathbf{A})]$ . That interval is contained in  $I$ , because its endpoints are eigenvalues of  $\mathbf{A}$  and  $I$  is an interval. Likewise  $\langle \mathbf{B}\mathbf{x}, \mathbf{x} \rangle \in I$ . So  $\mu$  is a convex combination of two points of the interval  $I$ , and it therefore lies in  $I$ .

The smallest size already says something, and the proof is three lines.

### Proposition 16.11.1 — Operator Monotone Functions Are Increasing

If  $f$  is operator monotone on  $I$ , then  $f$  is increasing on  $I$ .

*Proof.* Let  $s \leq t$  in  $I$ , and take  $n = 1$ ,  $\mathbf{A} = (t)$ ,  $\mathbf{B} = (s)$ . These are Hermitian, their spectra  $\{t\}$  and  $\{s\}$  lie in  $I$ , and  $\mathbf{A} - \mathbf{B} = (t - s) \succeq 0$ . Hence  $f(\mathbf{A}) \succeq f(\mathbf{B})$ , and since  $f(\mathbf{A}) = (f(t))$  and  $f(\mathbf{B}) = (f(s))$ , this says  $f(t) - f(s) \geq 0$ . This shows that  $f$  is increasing. ■

The converse is false, and the failure is the reason this section exists.

**Warning. Increasing is not enough.** “Increasing” is exactly the case  $n = 1$  of the definition, and sizes  $n \geq 2$  are where all the new content lies. The function  $t \mapsto t^2$  is increasing on  $[0, \infty)$  and is **not** operator monotone there. The  $2 \times 2$  matrices  $\mathbf{A} = \begin{pmatrix} 2 & 1 \\ 1 & 1 \end{pmatrix}$  and  $\mathbf{B} = \begin{pmatrix} 1 & 1 \\ 1 & 1 \end{pmatrix}$  of Example (Squaring does not preserve the Loewner order) satisfy  $\mathbf{A} \succeq \mathbf{B} \succeq 0$ , yet  $\mathbf{A}^2 - \mathbf{B}^2$  has negative determinant. Chapter 12 §05 traced the failure to non-commutativity: the number proof factors  $a^2 - b^2$  as  $(a - b)(a + b)$ , and for matrices that factorization needs  $\mathbf{AB} = \mathbf{BA}$ .

*Quick check.* Is  $t \mapsto t^3$  operator monotone on  $\mathbb{R}$  when only  $n = 1$  is tested? What does the answer tell you about  $n = 2$ ?

*Solution.* For  $n = 1$  the condition says only that  $f$  is increasing, as the proof of Proposition 16.11.1 shows, and  $t \mapsto t^3$  is increasing on  $\mathbb{R}$ . So it passes at  $n = 1$ . That tells us nothing about  $n = 2$ , where the test is genuinely new. In fact  $t \mapsto t^3$  fails at  $n = 2$  (exercise C3).

## 16.11.3 The examples we can prove

Four families are within reach, and the first is the simplest. It includes a degenerate case, the constant functions, which respect every order because they flatten everything to one matrix.



### Proposition 16.11.2 — Increasing Affine Functions

Let  $\alpha, \beta \in \mathbb{R}$  with  $\beta \geq 0$ . Then  $f(t) = \alpha + \beta t$  is operator monotone on  $\mathbb{R}$ . Moreover, every affine function  $t \mapsto \alpha + \beta t$ , with  $\beta$  of either sign, is both operator convex and operator concave on  $\mathbb{R}$ .

*Proof.* By [Theorem 11.8.3](#) (a),  $f(\mathbf{A}) = \alpha \mathbf{I} + \beta \mathbf{A}$  for every Hermitian  $\mathbf{A}$ . Let  $\mathbf{A} \succeq \mathbf{B}$  be Hermitian of the same size. Then  $f(\mathbf{A}) - f(\mathbf{B}) = \beta(\mathbf{A} - \mathbf{B})$ , and for every  $\mathbf{x}$ ,

$$\langle \beta(\mathbf{A} - \mathbf{B})\mathbf{x}, \mathbf{x} \rangle = \beta \langle (\mathbf{A} - \mathbf{B})\mathbf{x}, \mathbf{x} \rangle \geq 0,$$

a product of two non-negative numbers. Hence  $f(\mathbf{A}) \succeq f(\mathbf{B})$ . For the second statement, with  $\mathbf{C} = \theta \mathbf{A} + (1 - \theta)\mathbf{B}$ ,

$$\begin{aligned} f(\mathbf{C}) &= \alpha \mathbf{I} + \beta \theta \mathbf{A} + \beta(1 - \theta)\mathbf{B} \\ &= \theta(\alpha \mathbf{I} + \beta \mathbf{A}) + (1 - \theta)(\alpha \mathbf{I} + \beta \mathbf{B}) = \theta f(\mathbf{A}) + (1 - \theta)f(\mathbf{B}), \end{aligned}$$

so the inequality in [Definition 16.11.1](#) (b) holds with equality, for  $f$  and for  $-f$  alike. ■

The second example is the one that behaves like the square root:  $t \mapsto -1/t$ . For numbers,  $a \geq b > 0$  gives  $-1/a \geq -1/b$  because one may divide by the product  $ab$ . For matrices that division is unavailable, since  $\mathbf{A}$  and  $\mathbf{B}$  need not commute. Chapter 12 §05 already found the way round.

### Proposition 16.11.3 — The Negative Reciprocal Is Operator Monotone

The function  $t \mapsto -1/t$  is operator monotone on  $(0, \infty)$ . Explicitly: if  $\mathbf{A}, \mathbf{B} \in M_n(F)$  are Hermitian with  $\mathbf{A} \succeq \mathbf{B} \succ 0$ , then  $\mathbf{A} \succ 0$  and

$$-\mathbf{A}^{-1} \succeq -\mathbf{B}^{-1}, \quad \text{that is,} \quad \mathbf{B}^{-1} \succeq \mathbf{A}^{-1}.$$

*Idea.* A congruence removes the non-commutativity. Conjugating by  $\mathbf{B}^{-1/2}$  turns  $\mathbf{B}$  into  $\mathbf{I}$  and  $\mathbf{A}$  into  $\mathbf{C} = \mathbf{B}^{-1/2} \mathbf{A} \mathbf{B}^{-1/2} \succeq \mathbf{I}$ . Now one of the two matrices is the identity, which commutes with everything, so the argument for numbers applies to the eigenvalues of  $\mathbf{C}$ : they are all  $\geq 1$ , so those of  $\mathbf{C}^{-1}$  are all  $\leq 1$ . Undoing the congruence carries  $\mathbf{I} \succeq \mathbf{C}^{-1}$  back to  $\mathbf{B}^{-1} \succeq \mathbf{A}^{-1}$ . That is exactly the proof of [Theorem 12.5.3](#) (d), so the proposition is that clause read as a statement about the function  $t \mapsto -1/t$ .

*Proof.* Let  $\mathbf{A} \succeq \mathbf{B}$  be Hermitian with spectra in  $(0, \infty)$ . By [Theorem 12.1.5](#) (b), a Hermitian matrix whose eigenvalues are all positive is positive definite, so  $\mathbf{B} \succ 0$ . By [Theorem 12.5.3](#) (d),  $\mathbf{A} \succ 0$  and  $\mathbf{B}^{-1} \succeq \mathbf{A}^{-1}$ . Put  $f(t) = -1/t$ . By [Definition 11.8.2](#) and the instance  $t \mapsto 1/t$  recalled above,  $f(\mathbf{A}) = -\mathbf{A}^{-1}$  and  $f(\mathbf{B}) = -\mathbf{B}^{-1}$ . Hence

$$f(\mathbf{A}) - f(\mathbf{B}) = \mathbf{B}^{-1} - \mathbf{A}^{-1} \succeq 0,$$

which is the assertion. The size  $n$  was arbitrary, so  $f$  is operator monotone on  $(0, \infty)$ . ■

Two things make the whitening work, and they are worth separating. First, once one of the two matrices is  $\mathbf{I}$ , the comparison  $\mathbf{C} \succeq \mathbf{I}$  can be settled eigenvalue by eigenvalue, because  $\mathbf{I}$  commutes with everything. Second, inversion gets along with congruence:  $(\mathbf{S}^* \mathbf{M} \mathbf{S})^{-1} = \mathbf{S}^{-1} \mathbf{M}^{-1} (\mathbf{S}^{-1})^*$  is again a congruence, and that is what carries the conclusion back from  $\mathbf{C}$  to  $\mathbf{A}$ . For a general  $f$  there is no such identity, since  $f(\mathbf{S}^* \mathbf{M} \mathbf{S})$  is usually not a congruence of  $f(\mathbf{M})$ , so the argument does not simply transfer to other functions.

To build more examples out of these, one needs to know which ways of combining functions preserve the property.

### Proposition 16.11.4 — Combining Operator Monotone Functions

Let  $f$  and  $g$  be operator monotone on  $I$ .

- (a) For every  $c \geq 0$  and every  $\alpha \in \mathbb{R}$ , the functions  $f + g$  and  $\alpha + cf$  are operator monotone on  $I$ .
- (b) For every  $s \in \mathbb{R}$ , the translate  $t \mapsto f(t + s)$  is operator monotone on the interval  $I - s = \{t - s : t \in I\}$ .

*Proof.*

- (a) Let  $\mathbf{A} \succeq \mathbf{B}$  be Hermitian of the same size with spectra in  $I$ . By Theorem 11.8.3 (b),  $(f + g)(\mathbf{A}) - (f + g)(\mathbf{B}) = (f(\mathbf{A}) - f(\mathbf{B})) + (g(\mathbf{A}) - g(\mathbf{B}))$ . Both brackets are  $\succeq 0$  by hypothesis, and for every  $\mathbf{x}$  the quadratic form of the sum is the sum of the two non-negative quadratic forms. So the sum is  $\succeq 0$  by Definition 12.1.1. Likewise, using (b) and (a) of the same theorem,  $(\alpha + cf)(\mathbf{A}) - (\alpha + cf)(\mathbf{B}) = c(f(\mathbf{A}) - f(\mathbf{B}))$ , whose quadratic form is  $c \geq 0$  times a non-negative one.
- (b) Let  $g(t) = f(t + s)$  on  $I - s$ , and let  $\mathbf{A}$  be Hermitian with spectrum in  $I - s$ , with spectral resolution  $\mathbf{A} = \sum_j \mu_j \mathbf{P}_j$ . By Theorem 11.8.1 (c) and (d),

$$\mathbf{A} + s\mathbf{I} = \sum_j (\mu_j + s) \mathbf{P}_j,$$

and the numbers  $\mu_j + s$  are distinct. The  $\mathbf{P}_j$  are non-zero self-adjoint idempotents with  $\mathbf{P}_i \mathbf{P}_j = \mathbf{0}$  for  $i \neq j$  and  $\sum_j \mathbf{P}_j = \mathbf{I}$  (Theorem 11.8.1 (b), (c)). So by Theorem 11.8.2 this is the spectral resolution of  $\mathbf{A} + s\mathbf{I}$ , whose spectrum  $\{\mu_j + s\}$  lies in  $I$ . By Definition 11.8.2, applied to each side,

$$f(\mathbf{A} + s\mathbf{I}) = \sum_j f(\mu_j + s) \mathbf{P}_j = g(\mathbf{A}).$$

Now if  $\mathbf{A} \succeq \mathbf{B}$  with spectra in  $I - s$ , then  $(\mathbf{A} + s\mathbf{I}) - (\mathbf{B} + s\mathbf{I}) = \mathbf{A} - \mathbf{B} \succeq 0$ , and both shifted matrices have spectra in  $I$ . Hence  $g(\mathbf{A}) = f(\mathbf{A} + s\mathbf{I}) \succeq f(\mathbf{B} + s\mathbf{I}) = g(\mathbf{B})$ . ■

Combining the last two propositions produces a whole one-parameter family. It will reappear

in the final part of the section as the building block of every operator monotone function.

### Corollary 16.11.5 — Shifted Reciprocals

For every  $s \geq 0$ , the function  $t \mapsto -1/(t+s)$  is operator monotone on  $(0, \infty)$ . For every  $s > 0$ , so is

$$t \mapsto \frac{t}{t+s} = 1 - \frac{s}{t+s}.$$

*Proof.* By Proposition 16.11.3 and Proposition 16.11.4 (b),  $t \mapsto -1/(t+s)$  is operator monotone on  $(0, \infty) - s = (-s, \infty)$ . Every Hermitian matrix with spectrum in  $(0, \infty)$  has spectrum in  $(-s, \infty)$ , so it is in particular operator monotone on  $(0, \infty)$ . For the second function,  $t/(t+s) = 1 + s \cdot (-1/(t+s))$  with  $s > 0$ , so Proposition 16.11.4 (a) applies. ■

The fourth example is the one that started the section, and it is already proved. By Proposition 12.5.4,  $\mathbf{A} \succeq \mathbf{B} \succeq 0$  implies  $\mathbf{A}^{1/2} \succeq \mathbf{B}^{1/2}$ . A Hermitian matrix has spectrum in  $[0, \infty)$  exactly when it is  $\succeq 0$  (Theorem 12.1.4 (b)), and  $\mathbf{A}^{1/2}$  is  $f(\mathbf{A})$  for  $f(t) = t^{1/2}$ . So Chapter 12 §05's proposition says precisely that  $t \mapsto t^{1/2}$  is **operator monotone on**  $[0, \infty)$ . It is not re-proved here. Its proof was a different kind of argument, an eigenvector of  $\mathbf{A}^{1/2} - \mathbf{B}^{1/2}$  pushed to a contradiction, and it too never needs  $\mathbf{A}$  and  $\mathbf{B}$  to commute.

**A non-example by minimal change.** The function  $t \mapsto t = t^1$  is operator monotone on  $[0, \infty)$  by Proposition 16.11.2. Raise the exponent from 1 to 2. The function is still increasing, still a polynomial, and still satisfies the case  $n = 1$ . What fails is the clause “for every  $n$ ”, already at  $n = 2$ , with the witness of Example (Squaring does not preserve the Loewner order). Lowering the exponent from 1 to  $\frac{1}{2}$  instead keeps the property. Among the powers, then, the dividing line lies somewhere around the exponent 1. Chapter 12 §05 said where it lies exactly, and the last part of this section returns to that.

**Why “for every  $n$ ”.** Each size imposes its own condition, and the conditions differ. At  $n = 1$ ,  $t \mapsto t^2$  passes on  $[0, \infty)$  and at  $n = 2$  it fails. A definition at one fixed size would admit functions that break at the next size, and it would not say what the functional calculus does to matrices in general. Requiring every size at once makes the class small enough to classify completely, which is what Loewner's theorem at the end of this section does. The name comes from the same place as the name of the order: both are Loewner's.

## 16.11.4 Eigenvalues move monotonically for every increasing function

Operator monotonicity is a strong requirement, and a weaker one comes for free. §02 proved that the whole ordered list of eigenvalues respects the Loewner order: by Corollary 16.2.4,  $\mathbf{A} \succeq \mathbf{B}$  implies  $\lambda_i(\mathbf{A}) \geq \lambda_i(\mathbf{B})$  for **every**  $i$ . That paid the debt Chapter 12 §05 recorded after Theorem 12.5.3, and it holds with no hypothesis on commuting. Composing with an increasing  $f$  keeps it, once one knows what  $f$  does to the ordered list of eigenvalues.

**Lemma 16.11.6 — The Eigenvalue List of an Increasing Function**

Let  $\mathbf{A} \in M_n(F)$  be Hermitian with  $\text{spec}(\mathbf{A}) \subseteq I$ , and let  $f: I \rightarrow \mathbb{R}$  be increasing. Then  $f(\mathbf{A})$  is Hermitian and

$$\lambda_i(f(\mathbf{A})) = f(\lambda_i(\mathbf{A})) \quad (i = 1, \dots, n).$$

*Proof.* The matrix  $f(\mathbf{A})$  is Hermitian by [Theorem 11.8.3](#) (c). By [Corollary 11.4.3](#) (over  $\mathbb{C}$ ) or [Corollary 11.5.2](#) (over  $\mathbb{R}$ ), there is an orthonormal basis  $(\mathbf{q}_1, \dots, \mathbf{q}_n)$  of  $F^n$  with  $\mathbf{A}\mathbf{q}_i = \lambda_i(\mathbf{A})\mathbf{q}_i$ . Let  $\mathbf{A} = \sum_j \mu_j \mathbf{P}_j$  be the spectral resolution, and fix  $i$ . Exactly one of the distinct eigenvalues equals  $\lambda_i(\mathbf{A})$ ; call it  $\mu_{j_0}$ . Then  $\mathbf{q}_i$  lies in the eigenspace of  $\mu_{j_0}$ , which is the image of  $\mathbf{P}_{j_0}$ , so  $\mathbf{P}_{j_0}\mathbf{q}_i = \mathbf{q}_i$ . For  $j \neq j_0$ , the eigenspace of  $\mu_{j_0}$  is orthogonal to that of  $\mu_j$  ([Theorem 11.8.1](#) (a)), so  $\mathbf{P}_j\mathbf{q}_i = \mathbf{0}$ . By [Definition 11.8.2](#),

$$f(\mathbf{A})\mathbf{q}_i = \sum_j f(\mu_j)\mathbf{P}_j\mathbf{q}_i = f(\mu_{j_0})\mathbf{q}_i = f(\lambda_i(\mathbf{A}))\mathbf{q}_i.$$

Thus  $(\mathbf{q}_1, \dots, \mathbf{q}_n)$  is an orthonormal basis of eigenvectors of  $f(\mathbf{A})$ , and  $f(\lambda_1(\mathbf{A})), \dots, f(\lambda_n(\mathbf{A}))$  is its list of eigenvalues with multiplicity. Since  $\lambda_1(\mathbf{A}) \geq \dots \geq \lambda_n(\mathbf{A})$  and  $f$  is increasing, this list is already in decreasing order. A decreasing list of all the eigenvalues with multiplicity is the list  $\lambda_1(f(\mathbf{A})) \geq \dots \geq \lambda_n(f(\mathbf{A}))$ , so the two agree term by term. ■

The lemma needs  $f$  increasing only for the order of the list. For a general  $f$ , the numbers  $f(\lambda_i(\mathbf{A}))$  are still the eigenvalues of  $f(\mathbf{A})$ , but they may have to be re-sorted. Two lines now give the eigenvalue-level statement.

**Proposition 16.11.7 — Every Increasing Function Is Monotone on Eigenvalues**

Let  $f: I \rightarrow \mathbb{R}$  be increasing, and let  $\mathbf{A}, \mathbf{B} \in M_n(F)$  be Hermitian with spectra in  $I$  and  $\mathbf{A} \succeq \mathbf{B}$ . Then

$$\lambda_i(f(\mathbf{A})) \geq \lambda_i(f(\mathbf{B})) \quad \text{for every } i = 1, \dots, n.$$

*Proof.* By [Corollary 16.2.4](#),  $\lambda_i(\mathbf{A}) \geq \lambda_i(\mathbf{B})$  for every  $i$ . Since  $f$  is increasing,  $f(\lambda_i(\mathbf{A})) \geq f(\lambda_i(\mathbf{B}))$ . By [Lemma 16.11.6](#), applied to  $\mathbf{A}$  and to  $\mathbf{B}$ , the two sides are  $\lambda_i(f(\mathbf{A}))$  and  $\lambda_i(f(\mathbf{B}))$ . ■

So the squaring map, which is not operator monotone, is nevertheless monotone on eigenvalues. The Chapter 12 witness shows both at once.

**Example. (The Squares Are Not Ordered, but Their Eigenvalues Are)**

Let  $\mathbf{A} = \begin{pmatrix} 2 & 1 \\ 1 & 1 \end{pmatrix}$  and  $\mathbf{B} = \begin{pmatrix} 1 & 1 \\ 1 & 1 \end{pmatrix}$ , the pair of [Example \(Squaring does not preserve the Loewner order\)](#), for which  $\mathbf{A} \succeq \mathbf{B} \succeq 0$  and  $\mathbf{A}^2 \not\succeq \mathbf{B}^2$ . Compute the eigenvalues of  $\mathbf{A}^2$  and  $\mathbf{B}^2$  and compare them index by index.

*Solution.* From [Example \(Squaring does not preserve the Loewner order\)](#),  $\mathbf{A}^2 = \begin{pmatrix} 5 & 3 \\ 2 & 2 \end{pmatrix}$  and  $\mathbf{B}^2 = \begin{pmatrix} 2 & 2 \\ 2 & 2 \end{pmatrix}$ . The first has trace 7 and determinant  $10 - 9 = 1$ , so its eigenvalues are the roots of  $x^2 - 7x + 1$ :

$$\lambda_1(\mathbf{A}^2) = \frac{7 + 3\sqrt{5}}{2} \approx 6.854, \quad \lambda_2(\mathbf{A}^2) = \frac{7 - 3\sqrt{5}}{2} \approx 0.146.$$

The second has trace 4 and determinant 0, so  $\lambda_1(\mathbf{B}^2) = 4$  and  $\lambda_2(\mathbf{B}^2) = 0$ . Index by index:  $\frac{7+3\sqrt{5}}{2} \geq 4$  because  $3\sqrt{5} \geq 1$ , and  $\frac{7-3\sqrt{5}}{2} \geq 0$  because  $49 \geq 45$ . So  $\lambda_i(\mathbf{A}^2) \geq \lambda_i(\mathbf{B}^2)$  for  $i = 1, 2$ , as [Proposition 16.11.7](#) predicts for the increasing function  $t \mapsto t^2$  on  $[0, \infty)$ . Nevertheless  $\mathbf{A}^2 - \mathbf{B}^2$  is not  $\succeq 0$ .

So for every increasing  $f$ , the eigenvalues of  $f(\mathbf{A})$  dominate those of  $f(\mathbf{B})$  one index at a time. What operator monotonicity adds is that the *matrices* are ordered, and the example shows that this is strictly more.

**Warning. Eigenvalues in order do not make matrices in order.** It is tempting to read [Proposition 16.11.7](#) as “almost” operator monotonicity. It is not, because  $\lambda_i(\mathbf{M}) \geq \lambda_i(\mathbf{N})$  for every  $i$  does **not** imply  $\mathbf{M} \succeq \mathbf{N}$ . Take  $\mathbf{M} = \text{diag}(1, 0)$  and  $\mathbf{N} = \text{diag}(0, 1)$ : they have the same eigenvalues  $1 \geq 0$ , yet  $\mathbf{M} - \mathbf{N} = \text{diag}(1, -1)$  is indefinite. The Loewner order sees the eigenvectors as well as the eigenvalues, and the eigenvalue list forgets them.

## 16.11.5 Operator convexity, and where the two notions part

Operator convexity is the second notion of [Definition 16.11.1](#), and one might expect it to behave like operator monotonicity. It does not: the function that fails to be operator monotone turns out to be operator convex.

### Proposition 16.11.8 — The Square Is Operator Convex

The function  $t \mapsto t^2$  is operator convex on  $\mathbb{R}$ . Explicitly: for Hermitian  $\mathbf{A}, \mathbf{B} \in M_n(F)$  and  $\theta \in [0, 1]$ ,

$$\begin{aligned} \theta \mathbf{A}^2 + (1 - \theta) \mathbf{B}^2 - (\theta \mathbf{A} + (1 - \theta) \mathbf{B})^2 \\ = \theta(1 - \theta)(\mathbf{A} - \mathbf{B})^2 \succeq 0. \end{aligned}$$

*Idea.* For numbers,  $\theta a^2 + (1 - \theta)b^2 - (\theta a + (1 - \theta)b)^2 = \theta(1 - \theta)(a - b)^2$ , which is a variance. The matrix version of the expansion produces  $\mathbf{AB} + \mathbf{BA}$  where the number version has  $2ab$ , and that is exactly the cross term of  $(\mathbf{A} - \mathbf{B})^2 = \mathbf{A}^2 - \mathbf{AB} - \mathbf{BA} + \mathbf{B}^2$ . So this time non-commutativity cancels itself out and costs nothing. The square of a Hermitian matrix is  $\succeq 0$ , which finishes the proof.

*Proof.* By [Theorem 11.8.3](#) (a), the function  $t \mapsto t^2$  sends a Hermitian matrix to its square, so the displayed identity is what [Definition 16.11.1](#) (b) requires. Expanding, and

keeping the order of every product,

$$\begin{aligned} (\theta \mathbf{A} + (1 - \theta) \mathbf{B})^2 &= \theta^2 \mathbf{A}^2 + \theta(1 - \theta)(\mathbf{AB} + \mathbf{BA}) \\ &\quad + (1 - \theta)^2 \mathbf{B}^2. \end{aligned}$$

Subtracting this from  $\theta \mathbf{A}^2 + (1 - \theta) \mathbf{B}^2$ , and using  $\theta - \theta^2 = \theta(1 - \theta) = (1 - \theta) - (1 - \theta)^2$ ,

$$\begin{aligned} \theta \mathbf{A}^2 + (1 - \theta) \mathbf{B}^2 - (\theta \mathbf{A} + (1 - \theta) \mathbf{B})^2 &= \theta(1 - \theta)(\mathbf{A}^2 - \mathbf{AB} - \mathbf{BA} + \mathbf{B}^2) \\ &= \theta(1 - \theta)(\mathbf{A} - \mathbf{B})^2. \end{aligned}$$

The matrix  $\mathbf{D} = \mathbf{A} - \mathbf{B}$  is Hermitian, so  $\mathbf{D}^2 = \mathbf{D}^* \mathbf{D} \succeq 0$  by [Theorem 12.1.4](#) (c). Since  $\theta(1 - \theta) \geq 0$ , the quadratic form of  $\theta(1 - \theta) \mathbf{D}^2$  is non-negative, which proves the inequality. ■

The contrast is now complete, and it runs both ways.

- $t \mapsto t^2$  is **operator convex but not operator monotone** on  $[0, \infty)$ : [Proposition 16.11.8](#) and [Example \(Squaring does not preserve the Loewner order\)](#).
- $t \mapsto -1/t$  is **operator monotone but not operator convex** on  $(0, \infty)$ : [Proposition 16.11.3](#), and at  $n = 1$  the scalar function  $-1/t$  is not even convex, since with  $a = 1$ ,  $b = 3$  and  $\theta = \frac{1}{2}$  the value at the midpoint is  $-\frac{1}{2}$ , above the chord's  $-\frac{2}{3}$ .

So neither notion implies the other. Each implies its scalar version (increasing, convex) by taking  $n = 1$ , and they strengthen those versions in different directions. Exercise C2 adds that  $t \mapsto 1/t$  is operator convex on  $(0, \infty)$ , so  $-1/t$  is operator monotone *and* operator concave there.

## 16.11.6 What is not proved here

The question the section opened with has a complete answer, and this chapter does not prove it. Here is the statement, so that the reader knows what the answer looks like and which of the facts above are pieces of it.

### Theorem 16.11.9 — Loewner's Theorem, Statement

Let  $f: (0, \infty) \rightarrow \mathbb{R}$ . The following are equivalent.

- (i)  $f$  is operator monotone on  $(0, \infty)$ .
- (ii) There are  $\alpha \in \mathbb{R}$ ,  $\beta \geq 0$  and a positive measure  $\nu$  on  $[0, \infty)$  with  $\int_{[0, \infty)} \frac{d\nu(s)}{1+s^2} < \infty$ , such that for every  $t > 0$

$$\begin{aligned} f(t) &= \alpha + \beta t \\ &\quad + \int_{[0, \infty)} \left( \frac{s}{1+s^2} - \frac{1}{t+s} \right) d\nu(s). \end{aligned}$$

- (iii)  $f$  is the restriction to  $(0, \infty)$  of a function that is analytic on  $\mathbb{C} \setminus (-\infty, 0]$  and sends every point of the upper half-plane  $\{z : \text{Im } z > 0\}$  to a point with non-negative imaginary part.

We do not prove this theorem here. Its proof needs complex analysis and some measure theory, which lie outside Chapter 15's list (A1)–(A6); Chapter 20 proves it. Nothing in this chapter depends on it.

Read (ii) against what was proved above. The functions it superposes are  $t \mapsto -1/(t+s)$  for  $s \geq 0$ , the translates of  $-1/t$  that [Corollary 16.11.5](#) showed to be operator monotone. The constants  $s/(1+s^2)$  do not depend on  $t$  and are there only to make the integral converge. The term  $\alpha + \beta t$  is [Proposition 16.11.2](#). So the theorem says that **the examples of this section are, in a precise sense, all the examples there are**: every operator monotone function on  $(0, \infty)$  is an increasing affine function plus a non-negative superposition of shifted reciprocals. The easy direction, (ii)  $\Rightarrow$  (i), is essentially [Proposition 16.11.4](#) applied to an integral rather than a finite sum. What this book lacks is the analysis that carries the Loewner order through an integral, which is not on the list (A1)–(A6) in Chapter 15's introduction. The hard direction, (i)  $\Rightarrow$  (ii), is the classification itself.

Condition (iii) answers the opening question in one line. For  $z$  in the upper half-plane,  $z^{1/2}$  (the branch that is positive on  $(0, \infty)$ ) halves the argument of  $z$ , so its argument stays between 0 and  $\pi/2$  and  $z^{1/2}$  stays in the upper half-plane. The square doubles the argument, which can pass  $\pi$ . **The square root keeps the upper half-plane, and the square folds it over.**

*Quick check.* Find a point  $z$  of the upper half-plane whose square has negative imaginary part.

*Solution.* Take  $z = -1 + i$ , with  $\operatorname{Im} z = 1 > 0$ . Then  $z^2 = 1 - 2i + i^2 = -2i$ , with imaginary part  $-2 < 0$ . So  $z \mapsto z^2$  fails condition (iii) of [Theorem 16.11.9](#), consistently with [Example \(Squaring does not preserve the Loewner order\)](#).

The same test settles the powers  $t \mapsto t^p$ . For  $0 < p < 1$ , the function  $z^p$  multiplies the argument by  $p$  and keeps the upper half-plane, and indeed for  $t > 0$  (a standard integral, not proved here)

$$t^p = \frac{\sin p\pi}{\pi} \int_0^\infty s^{p-1} \frac{t}{t+s} ds,$$

a positive superposition of the functions  $t/(t+s)$  of [Corollary 16.11.5](#). For  $p > 1$ , the argument of  $z^p$  passes  $\pi$  for some  $z$  in the upper half-plane, and the test fails. This is the dividing line at exponent 1 that Chapter 12 §05 announced. Both halves, that  $t^p$  is operator monotone on  $(0, \infty)$  for every  $p \in (0, 1)$  and that it is not for any  $p > 1$ , are proved in Chapter 20, with Loewner's theorem. What is proved so far is the case  $p = \frac{1}{2}$  ([Proposition 12.5.4](#)) and, by applying it repeatedly, the cases  $p = 1/2^k$  (exercise C1).

## 16.11.7 Summary and transfer

The chapter began with a list of eigenvalues that the book had been able to *order* since Chapter 12 without being able to *describe*. It ends with each eigenvalue described as an optimization, and with a set of comparison theorems that follow from that description. Five moves did the work, and each transfers beyond this chapter.

- **The dimension count.** Two subspaces of an  $n$ -dimensional space whose dimensions add to more than  $n$  share a non-zero vector. This one line drives Courant–Fischer ([Theorem 16.2.3](#)) and Weyl's inequalities (§03) directly, and through Courant–Fischer everything after them. *Transfer*: whenever an extremal problem over vectors has to be compared with a fixed subspace, count dimensions before computing anything.



- **Min–max.** Describe a quantity as a minimum over subspaces of a maximum over vectors, and prove it with two inequalities of different characters: one exhibits a witness subspace and computes, and the other takes an arbitrary subspace and intersects it with a span of eigenvectors. Applying the result to  $-\mathbf{A}$  gives the dual form for free. *Transfer:* a description that names no eigenvector is what makes two different matrices comparable. The same shape describes singular values (§09) and inertia (§10).
- **Compress and interlace.** Restricting a Hermitian form to a subspace gives a smaller Hermitian matrix whose eigenvalues are trapped by the original's: compressing to a  $k$ -dimensional subspace puts  $\lambda_i$  of the piece between  $\lambda_i$  and  $\lambda_{i+n-k}$  of the whole, which are consecutive when  $k = n - 1$ . Deleting a row and column (§04), compressing to an orthonormal frame (§06) and deleting a row or a column of a rectangular matrix (§09) are all this move; adding a rank-one term (§05) reaches the same interlaced picture by a different route, the Loewner order together with Weyl's rank bound. *Transfer:* to bound the spectrum of a piece, look for the whole of which it is a compression.
- **Perturb in order.** Compare  $\lambda_i(\mathbf{A} + \mathbf{E})$  with  $\lambda_i(\mathbf{A})$  index by index, never “each eigenvalue with its nearest neighbor”. For Hermitian perturbations the Lipschitz constant is 1 in the spectral norm, independent of  $n$ , of the gaps and of the eigenvectors (§03). For singular values it needs no hypothesis on  $\mathbf{E}$  at all (§09). *Transfer:* where the Jordan block of Chapter 15 §07 showed that eigenvalues of general matrices can move like  $\varepsilon^{1/k}$ , the Hermitian and singular-value problems are perfectly conditioned. Reduce a question to one of them whenever possible.
- **Majorize.** When the right comparison is between *sums* of the largest entries rather than individual ones, compare partial sums of decreasing rearrangements. The diagonal of a Hermitian matrix is majorized by its spectrum, and by the Schur–Horn theorem majorization describes the possible diagonals *exactly* (§08); and the Ky Fan and Lidskii sums (§§06–07) are of the same type. *Transfer:* a family of partial-sum inequalities indexed by  $k$ , together with equality of the totals, is a majorization in disguise, and writing it as one lets the whole family be used at once.

Two supporting moves recur under these five: **apply the result to  $-\mathbf{A}$** , which turns every max–min into a min–max and every top- $k$  statement into a bottom- $k$  one, and **reduce singular values to eigenvalues**, through  $\mathbf{A}^*\mathbf{A}$  or the Hermitian dilation. This section also reused a move from Chapter 12 §05, **whiten, then compare**: use a congruence to make one of the two matrices  $\mathbf{I}$ . It works for  $t \mapsto -1/t$  because inversion turns a congruence into a congruence, and not for a general  $f$ .

The chapter also paid a series of promises made in earlier chapters, some of them long ago.

| The promise                                                                                                                                                                                     | Made in        | Paid by                                                                                   |
|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------|-------------------------------------------------------------------------------------------|
| The maximization route to eigenvalues “returns in Chapter 16, where the Rayleigh quotient and the Courant–Fischer theorem describe every eigenvalue of a self-adjoint operator by optimization” | Chapter 11 §02 | §01 (The Rayleigh quotient) and §02 (The min–max theorem), <a href="#">Theorem 16.2.3</a> |



| The promise                                                                                                                                                                              | Made in                           | Paid by                                                                                                              |
|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------|----------------------------------------------------------------------------------------------------------------------|
| “Chapter 16 characterizes the eigenvalues of a Hermitian matrix as maxima and minima of $\langle \mathbf{A}\mathbf{x}, \mathbf{x} \rangle$ — at which point the test becomes the theory” | Chapter 12 §12                    | §01 and §02, <a href="#">Theorem 16.2.3</a>                                                                          |
| $\mathbf{A} \succeq \mathbf{B}$ gives $\lambda_i(\mathbf{A}) \geq \lambda_i(\mathbf{B})$ for <b>every</b> $i$ , not only $i = 1$ and $i = n$ as in <a href="#">Theorem 12.5.3</a> (b)    | Chapter 12 §05                    | §02, <a href="#">Corollary 16.2.4</a>                                                                                |
| “The eigenvalues of a Hermitian matrix move by at most $\ \mathbf{E}\ _2$ under a Hermitian perturbation”                                                                                | Chapter 15 §07                    | §03 (How far eigenvalues move), Weyl’s perturbation bound                                                            |
| <a href="#">Lemma 12.10.2</a> is “the extreme case of a family of statements about sums of the largest eigenvalues”                                                                      | Chapter 12 §10                    | §06 (Subspaces and Sums of Eigenvalues), Ky Fan’s theorem, of which it is a special case                             |
| The second half of the approximation theorem, in the spectral norm                                                                                                                       | Chapter 12 §10 and Chapter 15 §08 | §09 (The same theory for singular values), Eckart–Young in the spectral norm                                         |
| Interlacing of the zeros of consecutive orthogonal polynomials, “Chapter 16’s business”                                                                                                  | Chapter 10 §10                    | §04 (Deleting a row and a column): strict interlacing, through the bordered-matrix criterion                         |
| The operator monotone functions named in Chapter 12 §05                                                                                                                                  | Chapter 12 §05                    | §11: definition and first examples; $t^s$ , $\log t$ and Loewner’s theorem remain Chapter 20’s, as that section said |

Four earlier theorems also received second proofs from the variational side, each of which shows something the first proof did not. The positivity of principal submatrices of a positive semidefinite matrix now follows from interlacing (§04, Deleting a row and a column). Hadamard’s inequality [Theorem 12.6.4](#) follows from AM–GM after a diagonal scaling, with Schur’s theorem supplying the equal totals (§08, The diagonal and the spectrum). Sylvester’s law of inertia [Theorem 13.5.1](#) is now a statement about the dimensions of subspaces, with no basis in it (§10, Inertia, again and additively). And the block test [Theorem 12.6.1](#) is now a consequence of inertia additivity (§10).

Some things the chapter did not do. Bounds for the eigenvalues of non-Hermitian matrices under perturbation, beginning with Bauer–Fike, are Chapter 19’s. Chapter 20 has Loewner’s theorem with the powers  $t^p$ , Schur-concave functions, the strongest form of Lidskii’s inequality, and Eckart–Young in every unitarily invariant norm. The variational principle itself, though, is now a tool the book can use without comment: an eigenvalue of a Hermitian matrix is a min–max, and comparison theorems follow from that.

## 16.11.8 Exercises

### 16.11.8.1 A. Check your understanding

#### Exercise A1.

- Define “ $f$  is operator monotone on  $I$ ”, and say what the clause “for every  $n$ ” ranges over.
- Explain why an operator monotone function is increasing, and name an increasing function that is not operator monotone.
- Decide whether the following is correct, and justify your answer: if  $f: I \rightarrow \mathbb{R}$  is increasing and  $\mathbf{A} \succeq \mathbf{B}$  are Hermitian with spectra in  $I$ , then  $\lambda_i(f(\mathbf{A})) \geq \lambda_i(f(\mathbf{B}))$  for every  $i$ .
- Decide whether the following is correct, and justify your answer: if  $\mathbf{A}, \mathbf{B}$  are Hermitian of the same size and  $\lambda_i(\mathbf{A}) \geq \lambda_i(\mathbf{B})$  for every  $i$ , then  $\mathbf{A} \succeq \mathbf{B}$ .
- Which of  $t \mapsto t^2$  and  $t \mapsto -1/t$  is operator monotone on  $(0, \infty)$ , and which is operator convex there?
- What does this section prove about Loewner’s theorem?

*Solution.*

- For every  $n \geq 1$  and all Hermitian  $\mathbf{A}, \mathbf{B} \in M_n(F)$  with spectra in  $I$ ,  $\mathbf{A} \succeq \mathbf{B}$  implies  $f(\mathbf{A}) \succeq f(\mathbf{B})$ , where  $f(\mathbf{A})$  is formed by [Definition 11.8.2](#) ([Definition 16.11.1](#) (a)). The clause ranges over all matrix sizes: the implication must hold for  $1 \times 1, 2 \times 2, 3 \times 3, \dots$  matrices simultaneously.
- Taking  $n = 1$  gives  $f(t) \geq f(s)$  whenever  $t \geq s$  in  $I$  ([Proposition 16.11.1](#)). The function  $t \mapsto t^2$  on  $[0, \infty)$  is increasing and is not operator monotone ([Example \(Squaring does not preserve the Loewner order\)](#)).
- Correct. This is [Proposition 16.11.7](#): [Corollary 16.2.4](#) orders the eigenvalues of  $\mathbf{A}$  and  $\mathbf{B}$  index by index,  $f$  preserves each inequality, and [Lemma 16.11.6](#) identifies  $f(\lambda_i(\mathbf{A}))$  with  $\lambda_i(f(\mathbf{A}))$ .
- Incorrect.  $\text{diag}(1, 0)$  and  $\text{diag}(0, 1)$  have the same eigenvalues, but their difference  $\text{diag}(1, -1)$  is indefinite, so neither dominates the other.
- $t \mapsto -1/t$  is operator monotone on  $(0, \infty)$  ([Proposition 16.11.3](#)) and is not operator convex, since already the scalar function is not convex: at 1, 3 and their midpoint 2,  $-\frac{1}{2} > \frac{1}{2}(-1 - \frac{1}{3}) = -\frac{2}{3}$ .  $t \mapsto t^2$  is operator convex ([Proposition 16.11.8](#)) and not operator monotone on  $(0, \infty)$ . The witness of [Example \(Squaring does not preserve the Loewner order\)](#) has 0 in the spectrum of  $\mathbf{B}$ , but the pair  $\mathbf{A} = \begin{pmatrix} 3 & 1 \\ 1 & 1 \end{pmatrix} \succeq \mathbf{B} = \begin{pmatrix} 2 & 1 \\ 1 & 1 \end{pmatrix} \succ 0$  works:  $\mathbf{A}^2 - \mathbf{B}^2 = \begin{pmatrix} 5 & 1 \\ 1 & 0 \end{pmatrix}$  has determinant  $-1$ .
- Only the statement ([Theorem 16.11.9](#)) and the fact that its building blocks  $t \mapsto -1/(t+s)$  are operator monotone ([Corollary 16.11.5](#)). The theorem itself is proved in Chapter 20.

### 16.11.8.2 B. Practice

#### Exercise B1: Which are operator monotone?.

Determine which of the following functions are operator monotone on the interval given.

Justify your answer.

- (i)  $f(t) = 5 + 4t$  on  $\mathbb{R}$ .
- (ii)  $f(t) = 3 - 2t$  on  $\mathbb{R}$ .
- (iii)  $f(t) = 2 - \frac{3}{t+1}$  on  $(0, \infty)$ .
- (iv)  $f(t) = \frac{t}{t+2}$  on  $(0, \infty)$ .
- (v)  $f(t) = t^2$  on  $[0, \infty)$ .

*Solution.*

- (i) Operator monotone, by [Proposition 16.11.2](#) with  $\beta = 4 \geq 0$ .
- (ii) Not operator monotone. It is not increasing, since  $f(1) = 1 < 3 = f(0)$ , so it fails at  $n = 1$  with  $\mathbf{A} = (1) \succeq \mathbf{B} = (0)$ ; compare [Proposition 16.11.1](#).
- (iii) Operator monotone. By [Corollary 16.11.5](#) with  $s = 1$ ,  $t \mapsto -1/(t+1)$  is operator monotone on  $(0, \infty)$ , and  $f = 2 + 3 \cdot (-1/(t+1))$  with  $3 \geq 0$ , so [Proposition 16.11.4](#) (a) applies.
- (iv) Operator monotone, by [Corollary 16.11.5](#) with  $s = 2 > 0$ .
- (v) Not operator monotone: [Example \(Squaring does not preserve the Loewner order\)](#) gives  $\mathbf{A} \succeq \mathbf{B} \succeq 0$ , with spectra  $\{(3 \pm \sqrt{5})/2\}$  and  $\{0, 2\}$  in  $[0, \infty)$ , and  $\mathbf{A}^2 \not\succeq \mathbf{B}^2$ .

### Exercise B2: One pair, three functions.

Let

$$\mathbf{A} = \begin{pmatrix} 3 & 1 \\ 1 & 1 \end{pmatrix}, \quad \mathbf{B} = \begin{pmatrix} 2 & 1 \\ 1 & 1 \end{pmatrix}.$$

- (a) Show that  $\mathbf{A} \succeq \mathbf{B} \succ 0$ .
- (b) Compute  $\mathbf{A}^{-1}$  and  $\mathbf{B}^{-1}$ , and verify directly that  $\mathbf{B}^{-1} \succeq \mathbf{A}^{-1}$ , as [Proposition 16.11.3](#) predicts.
- (c) Show that  $\mathbf{A}^2 \not\succeq \mathbf{B}^2$  by exhibiting a vector  $\mathbf{x}$  with  $\mathbf{x}^\top (\mathbf{A}^2 - \mathbf{B}^2) \mathbf{x} < 0$ .
- (d) Compute the eigenvalues of  $\mathbf{A}^2$  and  $\mathbf{B}^2$ , and verify that  $\lambda_i(\mathbf{A}^2) \geq \lambda_i(\mathbf{B}^2)$  for  $i = 1, 2$ . Hence explain which result of this section (c) and (d) illustrate together.

*Solution.*

- (a)  $\mathbf{A} - \mathbf{B} = \text{diag}(1, 0)$ , a diagonal matrix with non-negative entries, so  $\mathbf{A} - \mathbf{B} \succeq 0$  and  $\mathbf{A} \succeq \mathbf{B}$ . The leading principal minors of  $\mathbf{B}$  are 2 and  $2 - 1 = 1$ , both positive, so  $\mathbf{B} \succ 0$  by Sylvester's criterion ([Theorem 12.1.5](#) (d)).
- (b)  $\det \mathbf{A} = 2$  and  $\det \mathbf{B} = 1$ , so

$$\mathbf{A}^{-1} = \frac{1}{2} \begin{pmatrix} 1 & -1 \\ -1 & 3 \end{pmatrix}, \quad \mathbf{B}^{-1} = \begin{pmatrix} 1 & -1 \\ -1 & 2 \end{pmatrix}, \quad \mathbf{B}^{-1} - \mathbf{A}^{-1} = \frac{1}{2} \begin{pmatrix} 1 & -1 \\ -1 & 1 \end{pmatrix}.$$

For every  $\mathbf{x} \in \mathbb{R}^2$ ,  $\mathbf{x}^\top (\mathbf{B}^{-1} - \mathbf{A}^{-1}) \mathbf{x} = \frac{1}{2}(x_1 - x_2)^2 \geq 0$ , so  $\mathbf{B}^{-1} \succeq \mathbf{A}^{-1}$ .

- (c)  $\mathbf{A}^2 = \begin{pmatrix} 10 & 4 \\ 4 & 2 \end{pmatrix}$  and  $\mathbf{B}^2 = \begin{pmatrix} 5 & 3 \\ 3 & 2 \end{pmatrix}$ , so  $\mathbf{A}^2 - \mathbf{B}^2 = \begin{pmatrix} 5 & 1 \\ 1 & 0 \end{pmatrix}$ , with determinant  $-1 < 0$ . With  $\mathbf{x} = (1, -3)$ ,

$$\mathbf{x}^\top (\mathbf{A}^2 - \mathbf{B}^2) \mathbf{x} = 5 + 2 \cdot (1)(-3) + 0 = -1 < 0.$$

- (d)  $\mathbf{A}^2$  has trace 12 and determinant  $(\det \mathbf{A})^2 = 4$ , so its eigenvalues are  $6 \pm \sqrt{32} = 6 \pm 4\sqrt{2}$ , approximately 11.657 and 0.343.  $\mathbf{B}^2$  has trace 7 and determinant 1, so its eigenvalues are  $(7 \pm 3\sqrt{5})/2$ , approximately 6.854 and 0.146. For  $i = 1$ ,  $12 + 8\sqrt{2} \geq 7 + 3\sqrt{5}$  because  $8\sqrt{2} > 3\sqrt{5}$ . For  $i = 2$ ,  $12 - 8\sqrt{2} \geq 7 - 3\sqrt{5}$  is  $5 + 3\sqrt{5} \geq 8\sqrt{2}$ . Both sides are positive, and squaring gives  $70 + 30\sqrt{5} \geq 128$ , that is  $\sqrt{5} \geq \frac{29}{15}$ , which holds because  $5 \geq \frac{841}{225}$ . Hence (c) and (d) together illustrate [Proposition 16.11.7](#) and the warning after it: the increasing function  $t \mapsto t^2$  orders the eigenvalues but not the matrices.

### Exercise B3: The convexity identity in numbers.

With  $\mathbf{A}$  and  $\mathbf{B}$  as in B2 and  $\theta = \frac{1}{2}$ , compute

$$\mathbf{M} = \frac{1}{2}\mathbf{A}^2 + \frac{1}{2}\mathbf{B}^2 - \left(\frac{1}{2}\mathbf{A} + \frac{1}{2}\mathbf{B}\right)^2$$

directly, confirm that  $\mathbf{M} = \frac{1}{4}(\mathbf{A} - \mathbf{B})^2$ , and hence conclude that  $\left(\frac{1}{2}(\mathbf{A} + \mathbf{B})\right)^2 \preceq \frac{1}{2}(\mathbf{A}^2 + \mathbf{B}^2)$ .

*Solution.* From B2,  $\frac{1}{2}\mathbf{A}^2 + \frac{1}{2}\mathbf{B}^2 = \frac{1}{2}\begin{pmatrix} 15 & 7 \\ 7 & 4 \end{pmatrix} = \begin{pmatrix} 15/2 & 7/2 \\ 7/2 & 2 \end{pmatrix}$ . Next  $\frac{1}{2}(\mathbf{A} + \mathbf{B}) = \begin{pmatrix} 5/2 & 1 \\ 1 & 1 \end{pmatrix}$ , whose square is

$$\begin{pmatrix} \frac{25}{4} + 1 & \frac{5}{2} + 1 \\ \frac{5}{2} + 1 & 1 + 1 \end{pmatrix} = \begin{pmatrix} \frac{29}{4} & \frac{7}{2} \\ \frac{7}{2} & 2 \end{pmatrix}.$$

Hence  $\mathbf{M} = \begin{pmatrix} 15/2 - 29/4 & 0 \\ 0 & 0 \end{pmatrix} = \text{diag}(\frac{1}{4}, 0)$ . On the other side,  $\mathbf{A} - \mathbf{B} = \text{diag}(1, 0)$ , so  $\frac{1}{4}(\mathbf{A} - \mathbf{B})^2 = \text{diag}(\frac{1}{4}, 0) = \mathbf{M}$ , as [Proposition 16.11.8](#) says with  $\theta(1 - \theta) = \frac{1}{4}$ . Since  $\text{diag}(\frac{1}{4}, 0)$  is diagonal with non-negative entries,  $\mathbf{M} \succeq 0$ , which is the displayed inequality.

### 16.11.8.3 C. Going deeper

#### Exercise C1: Iterated square roots.

- Let  $\mathbf{A} \succeq \mathbf{B} \succeq 0$ . Prove that  $\mathbf{A}^{1/4} \succeq \mathbf{B}^{1/4}$ , where  $\mathbf{A}^{1/4}$  is  $f(\mathbf{A})$  for  $f(t) = t^{1/4}$  on  $[0, \infty)$ .
- Deduce that  $t \mapsto t^{1/2^k}$  is operator monotone on  $[0, \infty)$  for every integer  $k \geq 0$ .
- Explain why the argument of (b) cannot reach  $t \mapsto t^{3/4}$ , and say where the operator monotonicity of  $t^p$  for every  $p \in (0, 1)$  is proved.

*Hint for (a): the positive square root is unique.*

*Solution.*

- Let  $g(t) = t^{1/4}$  on  $[0, \infty)$ . Since  $\mathbf{A} \succeq 0$ , its spectrum lies in  $[0, \infty)$  ([Theorem 12.1.4](#) (b)), and  $g(\mathbf{A}) = \sum_j \mu_j^{1/4} \mathbf{P}_j$  is Hermitian with non-negative eigenvalues by [Theorem 11.8.3](#) (c) and (d), so  $g(\mathbf{A}) \succeq 0$ . By [Theorem 11.8.3](#) (b),  $g(\mathbf{A})^2 = (g^2)(\mathbf{A})$ , and  $g^2(t) = t^{1/2}$ , so  $g(\mathbf{A})^2 = \mathbf{A}^{1/2}$ . By the uniqueness in [Theorem 12.1.4](#) (d), applied to  $\mathbf{A}^{1/2} \succeq 0$ ,  $g(\mathbf{A})$  is the positive square root of  $\mathbf{A}^{1/2}$ :  $\mathbf{A}^{1/4} = (\mathbf{A}^{1/2})^{1/2}$ . The same holds for  $\mathbf{B}$ . Now [Proposition 12.5.4](#) applied to  $\mathbf{A} \succeq \mathbf{B} \succeq 0$  gives  $\mathbf{A}^{1/2} \succeq \mathbf{B}^{1/2}$ , and  $\mathbf{B}^{1/2} \succeq 0$  by [Theorem 12.2.1](#). Applying [Proposition 12.5.4](#)

again, to  $\mathbf{A}^{1/2} \succeq \mathbf{B}^{1/2} \succeq 0$ , gives  $(\mathbf{A}^{1/2})^{1/2} \succeq (\mathbf{B}^{1/2})^{1/2}$ , that is  $\mathbf{A}^{1/4} \succeq \mathbf{B}^{1/4}$ .

- (b) Induction on  $k$ . For  $k = 0$  the function is  $t \mapsto t$ , operator monotone by [Proposition 16.11.2](#). Suppose  $\mathbf{A}^{1/2^k} \succeq \mathbf{B}^{1/2^k}$  whenever  $\mathbf{A} \succeq \mathbf{B} \succeq 0$ . The argument of (a), with  $t^{1/2^{k+1}}$  in place of  $t^{1/4}$  and  $t^{1/2^k}$  in place of  $t^{1/2}$ , shows that  $\mathbf{A}^{1/2^{k+1}} = (\mathbf{A}^{1/2^k})^{1/2}$ , and similarly for  $\mathbf{B}$ . Both  $\mathbf{A}^{1/2^k}$  and  $\mathbf{B}^{1/2^k}$  are  $\succeq 0$ , so [Proposition 12.5.4](#) applied to  $\mathbf{A}^{1/2^k} \succeq \mathbf{B}^{1/2^k} \succeq 0$  gives  $\mathbf{A}^{1/2^{k+1}} \succeq \mathbf{B}^{1/2^{k+1}}$ .
- (c) Each step of (b) takes a square root, which halves the exponent. Starting from 1, the exponents reached are exactly  $1, \frac{1}{2}, \frac{1}{4}, \dots$ , and  $\frac{3}{4}$  is not among them. The operator monotonicity of  $t^p$  for every  $p \in (0, 1)$  needs a new idea, an integral representation of the kind in [Theorem 16.11.9](#), and it is proved in Chapter 20.

### Exercise C2: The reciprocal is operator convex.

- (a) Let  $\mathbf{G} \in M_n(F)$  with  $\mathbf{G} \succ 0$ . Prove that the  $2n \times 2n$  matrix  $\begin{pmatrix} \mathbf{G} & \mathbf{I}_n \\ \mathbf{I}_n & \mathbf{G}^{-1} \end{pmatrix}$  is Hermitian and  $\succeq 0$ .

- (b) Let  $\mathbf{A}, \mathbf{B} \succ 0$  be of the same size and  $\theta \in [0, 1]$ . Prove that

$$(\theta \mathbf{A} + (1 - \theta) \mathbf{B})^{-1} \succeq \theta \mathbf{A}^{-1} + (1 - \theta) \mathbf{B}^{-1},$$

that is,  $t \mapsto 1/t$  is operator convex on  $(0, \infty)$ .

- (c) Deduce that  $t \mapsto -1/t$  is both operator monotone and operator concave on  $(0, \infty)$ .

*Hint: Schur complements.*

*Solution.*

- (a)  $\mathbf{G}$  is Hermitian, and so is  $\mathbf{G}^{-1}$ , since  $(\mathbf{G}^{-1})^* = (\mathbf{G}^*)^{-1} = \mathbf{G}^{-1}$ . The off-diagonal blocks are  $\mathbf{I}_n$  and  $\mathbf{I}_n^* = \mathbf{I}_n$ , so the block matrix  $\mathbf{M}_{\mathbf{G}}$  is Hermitian. Its corner  $\mathbf{G}$  is  $\succ 0$ , and its Schur complement is

$$\mathbf{M}_{\mathbf{G}}/\mathbf{G} = \mathbf{G}^{-1} - \mathbf{I}_n^* \mathbf{G}^{-1} \mathbf{I}_n = \mathbf{0} \succeq 0.$$

By [Theorem 12.6.1](#) (b),  $\mathbf{M}_{\mathbf{G}} \succeq 0$ .

- (b) Put  $\mathbf{C} = \theta \mathbf{A} + (1 - \theta) \mathbf{B}$ . For  $\mathbf{x} \neq \mathbf{0}$ ,  $\langle \mathbf{C} \mathbf{x}, \mathbf{x} \rangle = \theta \langle \mathbf{A} \mathbf{x}, \mathbf{x} \rangle + (1 - \theta) \langle \mathbf{B} \mathbf{x}, \mathbf{x} \rangle$ . Both inner products are  $> 0$ , and  $\theta$  and  $1 - \theta$  are  $\geq 0$  and not both zero, so the sum is  $> 0$ . Thus  $\mathbf{C} \succ 0$ . By (a),  $\mathbf{M}_{\mathbf{A}} \succeq 0$  and  $\mathbf{M}_{\mathbf{B}} \succeq 0$ , so for every  $\mathbf{y} \in F^{2n}$  the quadratic form of

$$\theta \mathbf{M}_{\mathbf{A}} + (1 - \theta) \mathbf{M}_{\mathbf{B}} = \begin{pmatrix} \mathbf{C} & \mathbf{I}_n \\ \mathbf{I}_n & \theta \mathbf{A}^{-1} + (1 - \theta) \mathbf{B}^{-1} \end{pmatrix}$$

is  $\theta \langle \mathbf{M}_{\mathbf{A}} \mathbf{y}, \mathbf{y} \rangle + (1 - \theta) \langle \mathbf{M}_{\mathbf{B}} \mathbf{y}, \mathbf{y} \rangle \geq 0$ , and this block matrix is  $\succeq 0$ . Its corner  $\mathbf{C}$  is  $\succ 0$ , so [Theorem 12.6.1](#) (b) gives

$$\theta \mathbf{A}^{-1} + (1 - \theta) \mathbf{B}^{-1} - \mathbf{C}^{-1} \succeq 0,$$

which is the claim. Since  $t \mapsto 1/t$  sends a positive definite matrix to its inverse ([Definition 11.8.2](#)), and a Hermitian matrix has spectrum in  $(0, \infty)$  exactly when it is  $\succ 0$  ([Theorem 12.1.5](#) (b)), this is [Definition 16.11.1](#) (b) for  $f(t) = 1/t$ .

- (c) Operator monotone by [Proposition 16.11.3](#). Operator concave by definition, because its negative  $t \mapsto 1/t$  is operator convex by (b).

**Exercise C3: The cube needs its own witness.**

- (a) With  $\mathbf{A}$  and  $\mathbf{B}$  as in B2, show that  $\mathbf{A}^3 \not\preceq \mathbf{B}^3$ . Conclude that  $t \mapsto t^3$  is not operator monotone on  $[0, \infty)$ .
- (b) One might hope to deduce (a) from the failure of  $t \mapsto t^2$ . Suppose  $t \mapsto t^3$  were operator monotone on  $[0, \infty)$ . Applying it and  $t \mapsto t^{1/2}$  ([Proposition 12.5.4](#)) to both sides of  $\mathbf{A} \succeq \mathbf{B} \succeq 0$ , repeatedly and in any order, yields inequalities  $\mathbf{A}^q \succeq \mathbf{B}^q$  for certain exponents  $q$ . Show that every such  $q$  has the form  $3^j/2^k$  with integers  $j, k \geq 0$ , and that 2 never has this form. Explain why this means (a) needed a witness of its own.

*Solution.*

- (a) From B2,  $\mathbf{A} \succeq \mathbf{B} \succeq 0$ , with spectra in  $(0, \infty)$ . Multiplying the squares of B2 once more,

$$\mathbf{A}^3 = \begin{pmatrix} 34 & 14 \\ 14 & 6 \end{pmatrix}, \quad \mathbf{B}^3 = \begin{pmatrix} 13 & 8 \\ 8 & 5 \end{pmatrix}, \quad \mathbf{A}^3 - \mathbf{B}^3 = \begin{pmatrix} 21 & 6 \\ 6 & 1 \end{pmatrix}.$$

The determinant is  $21 - 36 = -15 < 0$ , so the difference is indefinite. Explicitly, with  $\mathbf{x} = (1, -5)$ ,

$$\mathbf{x}^\top (\mathbf{A}^3 - \mathbf{B}^3) \mathbf{x} = 21 + 2 \cdot 6 \cdot (-5) + 25 = -14 < 0.$$

By [Theorem 11.8.3](#) (a),  $t \mapsto t^3$  sends  $\mathbf{A}$  to  $\mathbf{A}^3$ , so it is not operator monotone on  $[0, \infty)$ .

- (b) Start from  $\mathbf{A} = \mathbf{A}^{3^0/2^0}$ . Taking the square root of  $\mathbf{A}^q$  gives  $\mathbf{A}^{q/2}$ , and cubing it gives  $\mathbf{A}^{3q}$ , by [Theorem 11.8.3](#) (b) and the uniqueness of the positive square root, as in C1 (a); likewise for  $\mathbf{B}$ . If  $q = 3^j/2^k$ , the results are  $3^j/2^{k+1}$  and  $3^{j+1}/2^k$ , of the same form. By induction on the number of steps, every exponent reached has the form  $3^j/2^k$ . If  $2 = 3^j/2^k$ , then  $3^j = 2^{k+1}$ , and the left side is odd while the right side is even, since  $k+1 \geq 1$ . So no exponent reached equals 2. The hypothetical operator monotonicity of the cube therefore cannot be converted, by these means, into that of the square. The failure of the square does not formally imply the failure of the cube, and (a) had to find a pair of matrices of its own.